数学猜想与开放问题

第二卷：第一卷之外的补充目录

80,537

条经规则筛查保留的补充记录

79,521 条题名与摘录 / 1,016 条形式化研究声明

对照第一卷 17,614 条记录筛查；并附来源、状态、排除日志与机器可读数据

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整理日期：2026 年 10 月 25 日（Asia/Shanghai）

“剩余”以本次已取得语料及筛查规则为边界，不是人类数学猜想的剩余全集。

01 本卷收录了什么

本卷承接第一卷的 17,614 条记录，继续搜集其他公开来源。正文逐条列出筛查后保留的 80,537 条补充记录。这个数能够核对到具体来源和编号，但不等于 80,537 个已确认彼此独立、且第一卷绝未涉及的数学猜想。

组成	条数	读者实际得到的内容
MathDB 公开目录摘录	79,521	题名、来源原样摘录、原编号、状态标签与页面链接；不能保证包含完整条件或结论。
Formal Conjectures 研究声明	1,016	原注释、Lean 声明节选、模块背景和官方源代码链接；研究已解决项也单独标明。
正文合计	80,537	80,537 个不同来源主键。不是全球总数，也不是数学语义等价类的精确数。

“全部剩余”的可实现边界

全球没有一份经过统一登记、语义去重并持续核验的猜想总账。未发表内容、语言差异、一个问题的多个版本及新出现的文献都可能超出本次观察。本卷完整列出的是本次取得的两组语料经列明规则筛查后的保留记录；不能证明数学界其他未收录猜想已不存在。

与第一卷一起使用时

第一卷适合查阅所收录问题的 statement 字段；第二卷主要扩大检索入口。两卷条数可以用于核对文档规模，不能相加后称为“人类猜想总数”。若用于研究选题，请打开单条记录的来源页核对完整表述、最新进展和原文献。

02 如何排除第一卷内容

排除基准为第一卷全部 17,614 条 published=true 记录；另外利用原数据中能够映射回这些主条目的归档别名。按以下顺序处理，前一步排除的记录不再重复计入后一步。

筛查阶段	排除数	规则含义
Formal Conjectures 预筛	3,190	整体排除已覆盖的 Erdős、Green、Kourovka、千禧年、Hilbert 来源族；无法稳妥定位声明者也暂不纳入。
编号、题名或表述匹配	5,279	匹配固定编号、规范化题名、规范化全文或可靠的表述前缀。
文本近似重合，暂不刊入	2,032	标题加权的词/二元词 TF-IDF；与基准最相似记录的余弦值达到 0.58。
卷内同文本筛查	240	MathDB 同摘录文本，或同源形式声明；保留一条并保存替代编号。
追加第一卷匹配	51	形式模块题名、同一 OEIS 序列族，以及人工识别的双曲群剩余有限性问题族。
类型或提取歧义排除	172	164 条 MathDB 的 theorem 类型；8 条无法唯一定位声明的形式记录。
本卷来源之间的重合	15	目录题名/摘录与形式模块匹配，或同一 OEIS 题名族；优先保留形式资料。

可复算等式：87,105 + 4,411 - 3,190 - 5,279 - 2,032 - 240 - 51 - 172 - 15 = 80,537。相似阈值只是筛查工具，既可能漏掉不同表述的同一问题，也可能暂时排除真正不同的变体。不能从“未匹配”推断“数学上已确认不同”。

03 标签、状态与计数限制

来源标签	保留数	含义
MathDB: conjecture	66,658	来源自标为猜想；未经本报告逐条确认其历史、正确性或独立性。
MathDB: open_problem	12,863	来源自标为开放问题；可能是构造、分类、最优值或其他研究问题。
Formal: research open	492	来源元数据标记为开放的研究声明。
Formal: research solved	524	与研究问题相关的已解决声明，也可能是特殊情形、等价式或相关定理；不计为仍未解决的猜想。

MathDB 的状态标签分布

open	claimed_progress	claimed_solved	solved	refuted	partial	special_case
77,731	390	874	283	232	9	2

claimed_solved 只表示有人声称已解决，不能改写为“已证明”。MathDB 明示其平台不验证数学内容；Formal Conjectures 也提示形式化表述可能误表原问题。形式声明中出现 theorem 或存在 Lean 文件，不等于具有已核验的完整证明。[S4, S5]

MathDB 的 17 个学科分组采用 OPDP 汇编附带的 category_label，主要用于检索，属于其分类结果，不是本报告进行的专家分类。Formal Conjectures 则按官方来源集合分组。

04 来源、归属与复用说明

S1 Alejandro Zarzuelo Urdiales / OPDP difficulty atlas，v1.8 数据包

https://github.com/alejandrozu/ulam-opdp-difficulty-atlas

本卷从公开发布的数据包提取 87,105 条 MathDB 目录项；不采用该项目的 AI 难度评分。

S2 OPDP MathDB 扩展记录与快照说明

https://github.com/alejandrozu/ulam-opdp-difficulty-atlas/blob/main/docs/OPDP_v1.7_APPEND_ONLY_NOTES.md

MathDB 来源清单冻结于 2026-09-08，excerpt 字段不保证是完整命题。

S3 OPDP Attribution and provenance notice

https://github.com/alejandrozu/ulam-opdp-difficulty-atlas/blob/main/NOTICE.md

MathDB 数据来源及可用逐条引用信息随记录保留；原始数学材料仍按各自署名与权利说明处理。

S4 MathDB Terms of Service

https://mathdb.com/terms

MathDB 贡献内容（论文除外）声明按 CC BY 4.0 许可；平台明确不对数学内容或声称状态作真实性认证。

S5 Formal Conjectures：官方项目说明

https://google-deepmind.github.io/formal-conjectures/about/

软件 Apache 2.0；其他材料 CC BY 4.0；各文件中的第三方权利说明仍适用。

S6 Google DeepMind / Formal Conjectures 公开源代码库

https://github.com/google-deepmind/formal-conjectures

仅使用 research open / research solved 元数据；代码与注释来自本次下载快照。

05 快照日期与阅读方法

本卷整理日期为 2026-10-25（Asia/Shanghai）。MathDB 数据实际冻结日期为 2026-09-08；Formal Conjectures 元数据与源码为本次检索下载快照。它们不是所有平台在整理日的同步全量状态。

本报告新增的工作包括：第一卷比对、规则筛查、分组、中文导读、排版、页码索引和审计数据；原始数学题名、摘录与代码的归属仍为原作者、MathDB 贡献者及 Formal Conjectures 贡献者。按用户要求，两卷封面注明“由中国跨学科学者 沈阳 124739259@qq.com 调度 AI 生成”。该说明不替代逐条原始来源署名。

如何查找

用 PDF 书签进入来源集合或学科章节；用全文搜索查找题名、MathDB 编号或形式声明名。每条记录有 V2- 六位本卷编号，原来源编号同时保留。链接“MathDB 页面”或“官方代码”指向来源；“原文线索”来自来源字段，未逐一核验。

如何理解正文

MathDB 正文始终标为“摘录”，忠实保留已有文字及省略符号，不补写被截断的公式。英文、LaTeX 和 Lean 源文本保留原语；为了避免改写数学含义，不把公式自动翻译为新表述。Formal Conjectures 中的模块背景只在同模块首次出现时刊出；后续声明可回看该模块，并查阅嵌入的原始 .lean 文件及依赖。

PDF 内嵌审计包

附件 volume2_records_and_audit.zip 包含全部保留记录（JSONL）、原始可用字段、排除日志、来源快照哈希、页码映射与筛查脚本；另含所用形式模块及许可证。若阅读器不显示附件，请使用支持 PDF 附件的桌面阅读器。JSONL 保存原始 Unicode；正文中极少数缺字用 [U+编码] 明示，不暗中删去。

没有读取付费完整 MathDB 数据包，本卷 MathDB 内容不是全文恢复版。含省略号的摘录无法独立承担严谨的数学定义和证明；应返回原论文或完整页面。

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A 形式化研究资料：1,016 条

以下是来源的研究声明及相关背景；同一个数学问题可能拆成多个声明。open / solved 是来源标签，proof 标记不代表本报告重新运行了证明核验。

A01 Books | 26 条

Convergence of the Borwein Series with Sinusoidal Coefficient

MODULE CONTEXT: # Convergence of the Borwein Series with Sinusoidal Coefficient *References:* - [MathWorld, Harmonic Series](https://mathworld.wolfram.com/HarmonicSeries.html) - Borwein, J.; Bailey, D.; Girgensohn, R. *Experimentation in Mathematics: Computational Paths to Discovery*, A K Peters, 2004, p. 56.

V2-000001 | Convergence of the Borwein Series with Sinusoidal Coefficient | borwein_sine_series

research solved | proof annotation: True | 官方代码 | 原文线索

Does the series $$ \sum_{n=1}^{\infty} \frac{\left(\frac{2}{3} + \frac{1}{3}\sin n\right)^n}{n} $$ converge? After computing approximately 10^7 terms, the partial sums approximate 2.163. See https://arxiv.org/abs/2007.11017 for a proof of the convergence, relying on an irrationality measure for pi. Also see https://github.com/AxiomMath/gdm-formal-conjectures/blob/main/docs/BorweinSineSeries.md for a partial formalization of the conjecture, conditional on such an irrationality measure of pi (cf https://arxiv.org/abs/1912.06345).

theorem borwein_sine_series :
Summable fun n : ℕ+ ↦ ((2 / 3 + 1 / 3 * Real.sin (n : ℝ)) ^ (n : ℕ)) / (n : ℝ)

FC-BorweinSineSeries.borwein_sine_series

Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers Chapter 10 of the book collects open questions. This file formalizes Problems 10.1, 10.2, 10.3 and the unnumbered conjecture by Waldschmidt. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Har19] Hardy, Gr H. "A problem of Diophantine approximation." J. Indian Math. Soc 11 (1919): 162-166. - [Kok45] Koksma, J. F. "Sur la théorie métrique des approximations diophantiques." Indag. Math 7 (1945): 54-70. - [Mah53] Mahler, Kurt. "On the approximation of logarithms of algebraic numbers." Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences 245.898 (1953): 371-398. - [Wal03](http://webusers.imj-prg.fr/~michel.waldschmidt/articles/pdf/Cetraro.pdf) Waldschmidt, Michel. "Linear independence measures for logarithms of algebraic numbers." Diophantine Approximation: Lectures given at the CIME Summer School held in Cetraro, Italy, June 28-July 6, 2000. Berlin, Heidelberg: Springer Berlin Heidelberg, 2003. 249-344.

V2-000002 | Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers | problem_10_1

research open | proof annotation: False | 官方代码 | 原文线索

Problem 10.1. Are there a transcendental number α and a positive real number ξ such that $\lVert \xi \alpha^n \rVert$ tends to~0 as~n tends to infinity? [Har19] (Trivial for $|\alpha| < 1$)

theorem problem_10_1 :
∃ (α ξ : ℝ), 1 < |α| ∧ Transcendental ℚ α ∧ 0 < ξ ∧
Filter.Tendsto (fun n : ℕ ↦ distToNearestInt (ξ * α ^ n)) Filter.atTop (nhds 0)

FC-Bugeaud01.problem_10_1

V2-000003 | Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers | problem_10_2

research open | proof annotation: False | 官方代码 | 原文线索

Problem 10.2. To prove that $\lVert e^n \rVert$ does not tend to 0 as n tends to infinity.

theorem problem_10_2 :
¬ Filter.Tendsto (fun n : ℕ ↦ distToNearestInt (Real.exp n)) Filter.atTop (nhds 0)

FC-Bugeaud01.problem_10_2

V2-000004 | Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers | problem_10_3

research open | proof annotation: False | 官方代码 | 原文线索

Problem 10.3. To prove that there exists a positive real number~c such that $\lVert e^n \rVert > e^{-cn}$, for every~$n \ge 1$. Posed by Mahler [Mah53].

theorem problem_10_3 :
∃ c : ℝ, 0 < c ∧ ∀ n : ℕ, 1 ≤ n → Real.exp (-c * n) < distToNearestInt (Real.exp n)

FC-Bugeaud01.problem_10_3

V2-000005 | Bugeaud Collection of Conjectures and Open Questions: Fractional Parts of Powers | waldschmidt

research open | proof annotation: False | 官方代码 | 原文线索

Waldschmidt [Wal03] conjectured that a stronger result holds, namely that there exists a positive real number~c such that $\lVert e^n \rVert > n^{-c}$ for every~$n \ge 2$. This is supported by metrical results [Kok45]. Note: the bound n^{-c} equals 1 when $n = 1$ for all c, while the distance to the nearest integer is always at most $1/2$, so the conjecture must start at $n \ge 2$.

theorem waldschmidt :
∃ c : ℝ, 0 < c ∧ ∀ n : ℕ, 2 ≤ n → (n : ℝ) ^ (-c) < distToNearestInt (Real.exp n)

FC-Bugeaud01.waldschmidt

Bugeaud Collection of Conjectures and Open Questions: Automatic Expansions in Two Bases

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Automatic Expansions in Two Bases Problem 10.32 asks whether some irrational real number has automatic expansions in two multiplicatively independent bases. The answer is no. Adamczewski and Faverjon [AF20] proved it with the multivariate Mahler method, in a form that is stronger than the question asked: the base-b digit sequence may be k-automatic for an arbitrary $k \ge 2$, not only for $k = b$. The definitions used here are in `FormalConjecturesForMathlib`: `Nat.MultiplicativelyIndependent` and `Nat.MultiplicativelyIndependentFamily`, `AutomaticSequence.IsAutomatic`, `AutomaticSequence.IsAutomaticInBase` and `AutomaticSequence.IsBroadlyAutomaticInBase`. *References:* - [AF20](https://arxiv.org/abs/2012.08283) Adamczewski, Boris, and Colin Faverjon. "Mahler's method in several variables and finite automata." arXiv:2012.08283 (2020). Appendix A, Conjectures A.2 and A.3, both proved there from Part (i) of Theorem 1.1. - [AS03] Allouche, Jean-Paul, and Jeffrey Shallit. "Automatic Sequences: Theory, Applications, Generalizations." Cambridge University Press, 2003. Chapter 13, Open Problems 7, p. 403, is the problem that Problem 10.32 modifies. - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Cob69] Cobham, Alan. "On the base-dependence of sets of numbers recognizable by finite automata." Mathematical Systems Theory 3 (1969): 186-192. - [LvdP82] Loxton, John H., and Alfred J. van der Poorten. "Arithmetic properties of the solutions of a class of functional equations." Journal für die reine und angewandte Mathematik 330 (1982): 159-172. Its transcendence method was the first attack on the problem; Nishioka later reported a gap in its main proof.

V2-000006 | Bugeaud Collection of Conjectures and Open Questions: Automatic Expansions in Two Bases | adamczewski_faverjon

research solved | proof annotation: False | 官方代码 | 原文线索

Adamczewski-Faverjon [AF20, Conjecture A.3], proved there from Part (i) of Theorem 1.1. Let $b_1, \dots, b_r$ be multiplicatively independent positive integers, and for every i let ξ_i be a real number that is automatic in base b_i. Then $\xi_1, \dots, \xi_r$ are algebraically independent over $\mathbb{Q}$, unless one of them is rational. Two points of fidelity. The hypothesis is the *joint* multiplicative independence of $b_1, \dots, b_r$ that [AF20] defines on p. 3 (no nonzero integer tuple (n_i) has $\prod_i b_i^{n_i} = 1$), not the weaker pairwise condition; it is what the proof feeds to Part (i) of Theorem 1.1, via the multiplicative independence of $1/b_1, \dots, 1/b_r$. Automaticity is the broad notion that the proof of Conjecture A.3 uses: the base-b_i digits of ξ_i are k-automatic for some $k \ge 2$, which need not be b_i.

```
theorem adamczewski_faverjon {r : ℕ} (b : Fin r → ℕ) (hb : ∀ i, 2 ≤ b i)
(hind : Nat.MultiplicativelyIndependentFamily b)
(ξ : Fin r → ℝ) (hauto : ∀ i, IsBroadlyAutomaticInBase (b i) (ξ i))
(hirr : ∀ i, Irrational (ξ i)) :
AlgebraicIndependent ℚ ξ
```

FC-Bugeaud32.adamczewski_faverjon

V2-000007 | Bugeaud Collection of Conjectures and Open Questions: Automatic Expansions in Two Bases | problem_10_32

research solved | proof annotation: False | 官方代码 | 原文线索

Problem 10.32. Is there an irrational real number whose expansions in two multiplicatively independent bases can both be generated by a finite automaton? The answer is no, by [AF20].

```
theorem problem_10_32 :
¬ ∃ ξ : ℝ, Irrational ξ ∧ ∃ b₁ b₂ : ℕ, 2 ≤ b₁ ∧ 2 ≤ b₂ ∧
Nat.MultiplicativelyIndependent b₁ b₂ ∧
IsAutomaticInBase b₁ ξ ∧ IsAutomaticInBase b₂ ξ
```

FC-Bugeaud32.problem_10_32

Bugeaud Collection of Conjectures and Open Questions: Spectrum of Sequence

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Spectrum of Sequence *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Men73] Mendès France, Michel. "Les ensembles de Bésineau." Séminaire Delange-Pisot-Poitou 15.1 (1973): 1-6. - [Özc26] Özcan, Hikmet Burak. "The spectrum of $(\xi\alpha^n)$ can be uncountable." [arXiv:2609.07714](https://arxiv.org/abs/2609.07714) (2026). ## Counterexample Problem 10.4 is false. Özcan [Özc26] proves that for every real $\alpha > 1$ there are $2^{\aleph_0}$ reals $\xi > 0$ whose spectrum contains one and the same uncountable set. The counterexample below is an independent, self-contained instance of this for the base $\alpha = 64$, where the digits of ξ can be prescribed directly. For a bit sequence $u \in \{0, 1\}^{\mathbb{N}}$ put $$\theta_u = \sum_{i \ge 0} (1 + u_i) \, 8^{-e(i)}, \qquad e(i) = 4^{i+1} + i + 6.$$ The map $u \mapsto \theta_u$ is injective with values in $(0, 1)$. Split the indices into the blocks $[5 m_k, 6 m_k)$ with $m_k = 20 \cdot 8^k$; these are pairwise disjoint. Block number $k = \langle L, c \rangle$, where $\langle \cdot, \cdot \rangle$ is the Cantor pairing, is reserved for the bit string c of length L. On that block the base-64 digits of ξ are chosen so that $\{\xi 64^n\}$ shadows $\{n \theta\}$ for every θ whose first L bits are c. This is possible because one base-64 digit pins down $\{\xi 64^n\}$ up to $2/64$, and because $k \le 4^{L+1}$ forces the block to sit far to the left of the precision $8^{-e(L)}$ of the truncation of θ, so that n times the truncation error stays below $1/512$ on the block. Consequently $\{\xi 64^n - n \theta_u\} < 1/10$ for all n in the block attached to the first L bits of u, for every L. That block is the last sixth of $[0, 6 m_k)$, so at the times $N = 6 m_k - 6$ at most a proportion $5/6$ of the points $\{\xi 64^n - n\theta_u\}$ with $n < N$ lies in $[1/10, 1]$, whereas uniform distribution modulo one would force the proportion $9/10$. Hence every irrational θ_u lies in the spectrum of $(\xi 64^n)$, and since $\{\theta_u\}$ is uncountable the spectrum cannot be countable.

V2-000008 | Bugeaud Collection of Conjectures and Open Questions: Spectrum of Sequence | spectrum_xi_alpha_pow_countable

research solved | proof annotation: True | 官方代码 | 原文线索

Problem 10.4. Let ξ be a non-zero real number and $\alpha > 1$ be a real number. Is the spectrum of the sequence $(\xi \alpha^n)_{n \ge 1}$ at most countable? Posed by Mendès France [Men73]. The answer is no. Özcan [Özc26] disproved this for every $\alpha > 1$; the counterexample formalised here takes $\alpha = 64$ and the real number $\xi = $ `xiVal`, whose spectrum contains the uncountable set of irrational numbers of the form `thetaSeq u`.

theorem spectrum_xi_alpha_pow_countable :
¬ ∀ (ξ : ℝ), ξ ≠ 0 → ∀ (α : ℝ), 1 < α → (Spectrum (fun n => ξ * α ^ n)).Countable

FC-Bugeaud04.spectrum_xi_alpha_pow_countable

Bugeaud Collection of Conjectures and Open Questions: Lacunary Sequences in Real Number Fields

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Lacunary Sequences in Real Number Fields The following problems were proposed and discussed by Dubickas as Conjecture 2 in [Dub09]. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Dub09] Dubickas, Artūras. "An approximation property of lacunary sequences." Israel Journal of Mathematics 170.1 (2009): 95-111.

V2-000009 | Bugeaud Collection of Conjectures and Open Questions: Lacunary Sequences in Real Number Fields | problem_10_5

research solved | proof annotation: True | 官方代码 | 原文线索

Problem 10.5 (first part). Let $\mathbb{K}$ be a real number field. Then, for any $\varepsilon > 0$, there exists a lacunary sequence $(t_n)_{n \ge 1}$ of positive numbers in $\mathbb{K}$ such that $$\limsup_{n \to \infty} \{\xi t_n\} \ge 1 - \varepsilon,$$ for any real number ξ not in $\mathbb{K}$. This follows from the stronger `problem_10_5_moreover` via `problem_10_5_of_moreover`. A Lean 4 formalization by Kenta Kitamura ([KitaKen1 on GitHub](https://github.com/KitaKen1)) is available at [KitaKen1/bugeaud-10-5-lacunary](https://github.com/KitaKen1/bugeaud-10-5-lacunary).

theorem problem_10_5 (K : IntermediateField ℚ ℝ) [FiniteDimensional ℚ K]
{ε : ℝ} (hε : 0 < ε) :
∃ t : ℕ → K, (∀ n, 0 < (t n : ℝ)) ∧
IsLacunaryReal (fun k => (t k : ℝ)) ∧
∀ ξ : ℝ, ξ ∉ K →
(1 - ε) ≤ limsup (fun n => Int.fract (ξ * (t n : ℝ))) atTop

FC-Bugeaud05.problem_10_5

V2-000010 | Bugeaud Collection of Conjectures and Open Questions: Lacunary Sequences in Real Number Fields | problem_10_5_moreover

research solved | proof annotation: True | 官方代码 | 原文线索

Problem 10.5 ("moreover" clause). With the same hypotheses as `problem_10_5`, the sequence (t_n) can be chosen so that, for any real ξ not in $\mathbb{K}$, each subinterval of $[0, 1]$ of length ε contains a limit point of the sequence $(\{\xi t_n\})_{n \ge 1}$. This is strictly stronger than `problem_10_5`: the limsup bound is the special case at the subinterval $[1 - \varepsilon, 1]$. A Lean 4 formalization by Kenta Kitamura ([KitaKen1 on GitHub](https://github.com/KitaKen1)) proves this statement using the explicit positive integer sequence $$t_{Jk+r} = (J+r)2^k \qquad (k \ge 0,\ 0 \le r < J).$$ Here Q is a positive integer chosen so that $Q > 1/\varepsilon$, and $J = 10Q!$. Since $\mathbb{K}$ contains $\mathbb{Q}$, every term belongs to $\mathbb{K}$. The proof verifies that this sequence is lacunary and has the required cluster-point property simultaneously for every $\xi \notin \mathbb{K}$. See [KitaKen1/bugeaud-10-5-lacunary](https://github.com/KitaKen1/bugeaud-10-5-lacunary).

theorem problem_10_5_moreover (K : IntermediateField ℚ ℝ) [FiniteDimensional ℚ K]
{ε : ℝ} (hε : 0 < ε) :
∃ t : ℕ → K, (∀ n, 0 < (t n : ℝ)) ∧
IsLacunaryReal (fun k => (t k : ℝ)) ∧
∀ ξ : ℝ, ξ ∉ K → ∀ a, a ∈ Set.Icc (0 : ℝ) (1 - ε) →
∃ y ∈ Set.Icc a (a + ε),
MapClusterPt y atTop (fun n => Int.fract (ξ * (t n : ℝ)))

FC-Bugeaud05.problem_10_5_moreover

Bugeaud Collection of Conjectures and Open Questions: Simultaneously Small Entropies

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Simultaneously Small Entropies A real number can be described by its continued fraction expansion and by its expansion in an integer base. Problem 10.53 asks for an irrational number for which both descriptions have small block complexity. Temur [Tem26] answered it, in the stronger form that asks for every base at once, by an explicit construction. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Tem26](https://arxiv.org/abs/2609.16362) Temur, Faruk. "Simultaneously small continued fraction entropy and base b entropy." arXiv preprint arXiv:2609.16362 (2026). - [BBFKW10] Broderick, Ryan, Yann Bugeaud, Lior Fishman, Dmitry Kleinbock, and Barak Weiss. "Schmidt's game, fractals, and numbers normal to no base." Mathematical Research Letters 17.2 (2010): 307-321.

V2-000011 | Bugeaud Collection of Conjectures and Open Questions: Simultaneously Small Entropies | problem_10_53

research solved | proof annotation: False | 官方代码 | 原文线索

Problem 10.53. There is an irrational real number ξ such that $E(\xi) < \log 2$ and $E(\xi, b) < \log b$ for some integer $b \ge 2$. Answered by Temur [Tem26].

```
theorem problem_10_53 :
∃ ξ : ℝ, Irrational ξ ∧ cfEntropy ξ < (Real.log 2 : EReal) ∧
∃ b : ℕ, 2 ≤ b ∧ baseEntropy b ξ < (Real.log b : EReal)
```

FC-Bugeaud53.problem_10_53

Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One Problem 10.6 asks for a "very rapidly increasing" sequence $(m_n)_{n \ge 1}$ of positive integers with $(\{\xi m_n\})_{n \ge 1}$ dense modulo one for every irrational ξ. The informal phrase admits several readings, and they are not equivalent: * a lower bound on the size of the terms, $m_n \ge f(n)$; * an upper bound on the counting function of $\{m_n\}$; * a lower bound on the consecutive ratios, $m_{n+1}/m_n \ge 1 + c/\log n$. The first two readings are vacuous, and `exists_dense_of_growth_and_sparsity` below says so: a sequence built from runs of consecutive integers is dense modulo one for every irrational ξ by an elementary argument, and the runs may be placed so that its terms dominate any prescribed f while its counting function stays below $(\log\log N)^2$. In the subexponential range the growth half is a special case of [Bos83, Thm. 1.5], which produces such a sequence in any prescribed subexponential corridor and in the stronger uniform-distribution form; the sparsity half sits below the $\log N$ floor that every sublacunary sequence obeys, and so below the counting function of [Kat16, Cor. 4.10]. Consequently `problem_10_6_variant_2`, which asks only for intermediate growth, is not open. The third reading is `IsGenuinelySublacunary`, and it is the only one with content: it implies the first (`hasIntermediateGrowth_of_isGenuinelySublacunary`), and it is not answered by runs, since inside a run $m_{n+1} - m_n = 1$. That reading is `problem_10_6_variant_1`, which remains open; [Kat16, Cor. 4.9] gives a universally densifying sequence of multiplicative shape $\{2^n 3^e\}$ whose ratio behaviour reduces the question to a Diophantine statement about $\log_2 3$. The three statements that need a long proof carry a `formal_proof` link to the version of this file that contains it: the construction by runs, its counting function, and the growth estimate behind `hasIntermediateGrowth_of_isGenuinelySublacunary`. *References:* - [Bos83] Boshernitzan, Michael D. "Homogeneously distributed sequences and Poincaré sequences of integers of sublacunary growth." Monatshefte für Mathematik 96.3 (1983): 173-181. - [Bos94] Boshernitzan, Michael D. "Density modulo 1 of dilations of sublacunary sequences." Advances in Mathematics 108.1 (1994): 104-117. - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Fur67] Furstenberg, H. "Disjointness in ergodic theory, minimal sets, and a problem in diophantine approximation". Math. Systems Theory 1, 1-49 (1967). - [Kat16] Katz, Asaf. "Generalizations of Furstenberg's Diophantine result." arXiv:1607.00670 (2016). - [Mat80] de Mathan, Bernard. "Numbers contravening a condition in density modulo 1." Acta Mathematica Hungarica 36.3-4 (1980): 237-241. - [Pol79] Pollington, Andrew Douglas. "On the density of sequence $\{n_ {k}\xi\} $." Illinois Journal of Mathematics 23.4 (1979): 511-515.

V2-000012 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | boshernitzan

research solved | proof annotation: False | 官方代码

Boshernitzan's theorem [Bos94]. Given a real sublacunary sequence r, the set of real numbers ξ for which $(\{\xi r_n\})_{n \ge 1}$ is *not* dense modulo one has Hausdorff dimension zero.

```
theorem boshernitzan (r : ℕ → ℝ) (hr : ∀ n, 0 < r n) (hunb : ¬ BddAbove (Set.range r))
(hsub : Tendsto (fun n => r (n + 1) / r n) atTop (nhds 1)) :
dimH {ξ : ℝ | ¬ Dense (Set.range fun n => (↑(ξ * r n) : AddCircle (1 : ℝ)))} = 0
```

FC-Bugeaud06.boshernitzan

V2-000013 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | exists_dense_of_growth

research solved | proof annotation: True | 官方代码

A lower bound on the size of the terms cannot make Problem 10.6 nontrivial. For every growth demand `f` there is a strictly increasing sequence dominating `f` at every index which is dense modulo one for every irrational ξ. In the subexponential range this is a special case of [Bos83, Thm. 1.5], which produces such a sequence inside any prescribed corridor $[n_k, m_k)$ with $m_k - n_k \to \infty$ and $m_k / n_k \to 1$, and in the stronger form of uniform distribution; the elementary construction of the linked proof is uniform in `f` and has no growth ceiling.

```
theorem exists_dense_of_growth (f : ℕ → ℕ) :
∃ m : ℕ → ℕ, StrictMono m ∧ (∀ n, f n ≤ m n) ∧
∀ ξ : ℝ, Irrational ξ →
Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ)))
```

FC-Bugeaud06.exists_dense_of_growth

V2-000014 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | exists_dense_of_growth_and_sparsity

research solved | proof annotation: True | 官方代码

Neither can an upper bound on the counting function, even together with the growth demand. The same sequence dominates an arbitrary `f` *and* has counting function below $(\log\log N)^2$ - an exponential below the $\log N$ floor that every sublacunary sequence obeys, and so below the counting function of [Kat16, Cor. 4.10]. So a reading of "very rapidly increasing" that constrains only the size of the terms or the sparsity of their range leaves Problem 10.6 with no content.

```
theorem exists_dense_of_growth_and_sparsity (f : ℕ → ℕ) :
∃ m : ℕ → ℕ, StrictMono m ∧ (∀ n, f n ≤ m n) ∧
(∀ᶠ N : ℕ in atTop, (countingFn m N : ℝ) ≤ (Real.log (Real.log N)) ^ 2) ∧
∀ ξ : ℝ, Irrational ξ →
Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ)))
```

FC-Bugeaud06.exists_dense_of_growth_and_sparsity

V2-000015 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | furstenberg_two_three

research solved | proof annotation: False | 官方代码

Furstenberg's theorem [Fur67] (the $\times 2, \times 3$ case). For every irrational number ξ, the two-parameter family $(\{\xi \, 2^m 3^n\})_{m, n \ge 1}$ is dense modulo one.

```
theorem furstenberg_two_three (ξ : ℝ) (hξ : Irrational ξ) :
Dense {x : AddCircle (1 : ℝ) |
∃ m n : ℕ, 0 < m ∧ 0 < n ∧ x = ↑(ξ * (2 ^ m * 3 ^ n : ℕ))}
```

FC-Bugeaud06.furstenberg_two_three

V2-000016 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | pollington_de_mathan

research solved | proof annotation: False | 官方代码

The **Pollington-de Mathan theorem** [Pol79][Mat80]. For every lacunary sequence $(m_n)_{n \ge 1}$ of positive integers, the set of real numbers ξ for which $(\{\xi m_n\})_{n \ge 1}$ is *not* dense modulo one has full Hausdorff dimension.

```
theorem pollington_de_mathan (m : ℕ → ℕ) (hm : ∀ n, 0 < m n) (hlac : IsLacunary m) :
dimH {ξ : ℝ | ¬ Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ)))} = 1
```

FC-Bugeaud06.pollington_de_mathan

V2-000017 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | problem_10_6_variant_1

research open | proof annotation: False | 官方代码

Problem 10.6. Find a very rapidly increasing sequence $(m_n)_{n \ge 1}$ of positive integers such that $(\{\xi m_n\})_{n \ge 1}$ is dense modulo one for every irrational number ξ. Note: Furstenberg's 2^m3^n is sublacunary but requires two parameters. This is the reading of "very rapidly increasing" as a lower bound on the consecutive ratios, and it is the one that remains open. It is not answered by runs of consecutive integers, since inside a run $m_{n+1} - m_n = 1$; and by `not_isLacunary_of_dense` any answer is pinched between a decaying ratio floor and the failure of lacunarity. [Kat16, Cor. 4.9] gives a universally densifying sequence of the multiplicative shape $\{2^n 3^e\}$, for which the ratio condition reduces to a Diophantine statement about the differences of the exponents e times $\log_2 3$ modulo one.

```
theorem problem_10_6_variant_1 :
∃ m : ℕ → ℕ,
StrictMono m ∧
IsGenuinelySublacunary m ∧
∀ ξ : ℝ, Irrational ξ →
Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ)))
```

FC-Bugeaud06.problem_10_6_variant_1

V2-000018 | Bugeaud Collection of Conjectures and Open Questions: Rapidly Increasing Sequences Dense Modulo One | problem_10_6_variant_2

research solved | proof annotation: False | 官方代码

Problem 10.6, intermediate-growth variant. This asks only for a lower bound on the size of the terms, and under that reading the problem is not open: runs of consecutive integers answer it, by `exists_dense_of_growth` applied to $f(n) = \lceil e^n \rceil$. The witness below is exhibited at $\alpha = 1/2$; the same sequence works for every $\alpha < 1$, since $\exp(n^\alpha) \le \exp(n) \le f(n)$ there. The result is not new: [Bos83, Thm. 1.5] gives a sequence of any prescribed subexponential growth that is uniformly distributed - not merely dense - modulo one for every irrational dilate.

```
theorem problem_10_6_variant_2 :
∃ m : ℕ → ℕ,
StrictMono m ∧
(∃ α : ℝ, 0 < α ∧ α < 1 ∧ HasIntermediateGrowth α m) ∧
∀ ξ : ℝ, Irrational ξ →
Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ)))
```

FC-Bugeaud06.problem_10_6_variant_2

Bugeaud Collection of Conjectures and Open Questions: Pisot orbits on the Cantor set

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Pisot orbits on the Cantor set Problem 10.61, proposed by Mendès France [MF67, Problème 1, p. 41]. For a Pisot number $\alpha > 2$ put $$C(\alpha) = \left\{ (\alpha - 1) \sum_{k \ge 1} \varepsilon_k \alpha^{-k} : \varepsilon_k \in \{0, 1\} \right\}.$$ The problem asks to prove that $(\xi \alpha^n)_{n \ge 1}$ is not uniformly distributed modulo one for any $\xi \in C(\alpha)$. In the language of [MF67] this reads $C(\alpha) \cap B(\alpha) = \emptyset$, where $B(\alpha)$ is the set of ξ for which $(\xi \alpha^n)$ is uniformly distributed modulo one. The problem is open. It is easy for an integer base and settled for two quadratic α; the general Pisot case is what remains. [Ste26] reduces it to a statement about shift-invariant measures of the full two-shift, and proves that every true instance admits a finite certificate, so the difficulty is uniformity in α rather than any single α. The variants below state the criterion and the instances of [Ste26] that concern Problem 10.61 directly; the reduction itself is not stated here, since it needs the symbolic model. Deleting the term $n = 0$ does not change uniform distribution modulo one, so the statements indexed from $n = 1$ agree with the ones of [Ste26], which are indexed from $n = 0$. Where an orbit is confined away from an interval the stronger form, for every $n \ge 0$, is stated. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10, Problem 10.61, p. 222. - [MF67] Mendès France, Michel. "Nombres normaux. Applications aux fonctions pseudo-aléatoires." Journal d'Analyse Mathématique 20 (1967): 1-56. - [Kok35] Koksma, Jurjen F. "Ein mengentheoretischer Satz über die Gleichverteilung modulo Eins." Compositio Mathematica 2 (1935): 250-258. - [Ste26] Stephan, Ralf. "Criteria for the non-equidistribution of $(\xi \alpha^n)$ on the Cantor set $C(\alpha)$." Preprint, 2026. https://doi.org/10.13140/RG.2.2.13923.52001

V2-000019 | Bugeaud Collection of Conjectures and Open Questions: Pisot orbits on the Cantor set | problem_10_61

research open | proof annotation: False | 官方代码 | 原文线索

Problem 10.61. Let $\alpha > 2$ be a Pisot number. For every $\xi \in C(\alpha)$ the sequence $(\xi \alpha^n)_{n \ge 1}$ is not uniformly distributed modulo one.

```
theorem problem_10_61 (α : ℝ) (hα : IsPisot α) (hα2 : 2 < α) :
∀ ξ ∈ pisotCantorSet α, ¬ IsEquidistributedModuloOne fun n : ℕ => ξ * α ^ (n + 1)
```

FC-Bugeaud61.problem_10_61

Bugeaud Collection of Conjectures and Open Questions: Confined Powers of Non-Pisot Numbers

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Confined Powers of Non-Pisot Numbers *References:* - [Bug12a] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Bug12b] Bugeaud, Yann, and Nikolay Moshchevitin. "On fractional parts of powers of real numbers close to 1." Mathematische Zeitschrift 271.3 (2012): 627-637.

V2-000020 | Bugeaud Collection of Conjectures and Open Questions: Confined Powers of Non-Pisot Numbers | problem_10_7

research open | proof annotation: False | 官方代码

Problem 10.7. Let ε be a positive real number. Are there arbitrarily large real numbers α such that α is not a Pisot number and all the fractional parts $\{\alpha^n\}$, $n \ge 1$, are lying in an interval of length ε / α? [Bug12b]

```
theorem problem_10_7 :
∀ ε : ℝ, 0 < ε → ∀ M : ℝ, ∃ α : ℝ, M < α ∧ ¬ IsPisot α ∧
∃ c : ℝ, ∀ n : ℕ, 1 ≤ n → Int.fract (α ^ n) ∈ Set.Icc c (c + ε / α)
```

FC-Bugeaud07.problem_10_7

Bugeaud Collection of Conjectures and Open Questions: p-adic Littlewood Conjecture

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: p-adic Littlewood Conjecture This is the p-adic analogue of the Littlewood conjecture, posed by de Mathan and Teulié. A liminf-based formulation also appears in the file `FormalConjectures/Wikipedia/LittlewoodConjecture.lean` as `padic_littlewood_conjecture`. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [dMT04] de Mathan, Bernard, and Olivier Teulié. "Problèmes diophantiens simultanés." Monatshefte für Mathematik 143.3 (2004): 229-245. - [EK07] Einsiedler, Manfred, and Dmitry Kleinbock. "Measure rigidity and p-adic Littlewood-type problems." Compositio Mathematica 143.3 (2007): 689-702.

V2-000021 | Bugeaud Collection of Conjectures and Open Questions: p-adic Littlewood Conjecture | problem_10_8

research open | proof annotation: False | 官方代码

Problem 10.8 (p-adic Littlewood conjecture). For every real number ξ and every prime number p, $$\inf_{q \ge 1} q \cdot \lVert q \xi \rVert \cdot |q|_p = 0,$$ where $\lVert \cdot \rVert$ denotes the distance to the nearest integer and $|\cdot|_p$ denotes the p-adic absolute value. Posed by de Mathan and Teulié [dMT04].

```
theorem problem_10_8 (ξ : ℝ) (p : ℕ) (hp : p.Prime) :
sInf {x : ℝ | ∃ q : ℕ, 1 ≤ q ∧
x = q * padicNorm p q * distToNearestInt (q * ξ)} = 0
```

FC-Bugeaud08.problem_10_8

Bugeaud Collection of Conjectures and Open Questions: Mahler's Z-numbers

MODULE CONTEXT: # Bugeaud Collection of Conjectures and Open Questions: Mahler's Z-numbers See also FormalConjectures/Wikipedia/Mahler32.lean. *References:* - [Bug12] Bugeaud, Yann. "Distribution modulo one and Diophantine approximation." Vol. 193. Cambridge University Press, 2012. Chapter 10. - [Mah68] Mahler, Kurt. "An unsolved problem on the powers of 3/2." Journal of the Australian Mathematical Society 8.2 (1968): 313-321. - [FLP95] Flatto, Leopold, Jeffrey C. Lagarias, and Andrew D. Pollington. "On the range of fractional parts $\{\xi(p/q)^n\}$." Acta Arithmetica 70.2 (1995): 125-147.

V2-000022 | Bugeaud Collection of Conjectures and Open Questions: Mahler's Z-numbers | problem_10_9

research open | proof annotation: False | 官方代码

Problem 10.9. There are no real numbers ξ such that $0 \le \{\xi (3/2)^n\} < 1/2$ for every positive integer n, i.e. no Z-number exists. Posed by Mahler [Mah68].

```
theorem problem_10_9 : type_of% Mahler32.mahler_conjecture
```

FC-Bugeaud09.problem_10_9

Equidistributed Sequences

MODULE CONTEXT: # Equidistributed Sequences Corollary 4.2 of Chapter 1 states that the sequence $(x^n), n = 1, 2, ... ,$ is equidistributed modulo 1 for almost all x > 1. And a little bit further down: "one does not know whether sequences such as (e^n), $(π^n)$, or even $((\frac 3 2)^n)$" are equidistributed modulo 1 or not. *References:* - [Uniform Distribution of Sequences](https://store.doverpublications.com/products/9780486149998) by *L. Kuipers* and *H. Niederreiter*, 1974 - [Wikipedia](https://en.wikipedia.org/wiki/Equidistributed_sequence) - [Mat80] de Mathan, Bernard. "Numbers contravening a condition in density modulo 1." Acta Mathematica Hungarica 36.3-4 (1980): 237-241. - [Pol79] Pollington, Andrew Douglas. "On the density of sequence $\{n_ {k}\xi\} $." Illinois Journal of Mathematics 23.4 (1979): 511-515.

V2-000023 | Equidistributed Sequences | isAccumulationPoint_three_halves_pow

research open | proof annotation: False | 官方代码 | 原文线索

Find an accumulation point of the sequence `(3/2)^n` modulo `1`.

```
theorem isAccumulationPoint_three_halves_pow :
IsAccumulationPoint isAccumulationPoint_three_halves_pow.answer
(fun n => Int.fract <| (3 / 2 : ℝ)^n)
```

FC-Equidistribution.isAccumulationPoint_three_halves_pow

V2-000024 | Equidistributed Sequences | isAccumulationPoint_three_halves_pow_infinite

research solved | proof annotation: False | 官方代码 | 原文线索

The sequence `(3/2)^n` has infinitely many accumulation points modulo `1`.

```
theorem isAccumulationPoint_three_halves_pow_infinite :
{x | IsAccumulationPoint x (fun n => Int.fract <| (3 / 2 : ℝ)^n)}.Infinite
```

FC-Equidistribution.isAccumulationPoint_three_halves_pow_infinite

V2-000025 | Equidistributed Sequences | isEquidistributedModuloOne_three_halves_pow

research open | proof annotation: False | 官方代码 | 原文线索

The sequence `(3/2)^n` is equidistributed modulo `1`.

```
theorem isEquidistributedModuloOne_three_halves_pow :
IsEquidistributedModuloOne (fun n => (3 / 2 : ℝ)^n)
```

FC-Equidistribution.isEquidistributedModuloOne_three_halves_pow

V2-000026 | Equidistributed Sequences | isEquidistributedModuloOne_transcendental_three_halves_pow

research solved | proof annotation: False | 官方代码 | 原文线索

It is not true that for every transcendental number `x` the sequence `x * (3 / 2) ^ n` is equidistributed modulo `1`. The sequence `(3 / 2) ^ n` is lacunary, so by the theorem of Pollington [Pol79] and de Mathan [Mat80] the set of real numbers `x` for which `x * (3 / 2) ^ n` is not even dense modulo `1` has Hausdorff dimension `1`. This set is uncountable, so it contains transcendental numbers. Alternatively, an elementary nested-interval construction (see issue #5003) gives a Cantor set of `x` with `Int.fract (x * (3 / 2) ^ (8 * j)) ∈ [0, 1 / 10]` for all `j`, which already rules out equidistribution for uncountably many, hence some transcendental, `x`. By Koksma's metric theorem (Kuipers-Niederreiter, Chapter 1, Section 4), the sequence `x * (3 / 2) ^ n` is equidistributed modulo `1` for almost all `x`.

```
theorem isEquidistributedModuloOne_transcendental_three_halves_pow :
¬ ∀ x : ℝ, Transcendental ℚ x →
IsEquidistributedModuloOne (fun n ↦ x * (3 / 2 : ℝ) ^ n)
```

FC-Equidistribution.isEquidistributedModuloOne_transcendental_three_halves_pow

A02 MathOverflow | 21 条

Optimal monotone families for the discrete isoperimetric inequality

MODULE CONTEXT: # Optimal monotone families for the discrete isoperimetric inequality *References:* - [mathoverflow/10799](https://mathoverflow.net/questions/10799) asked by user [*Gil Kalai*](https://mathoverflow.net/users/1532/gil-kalai) - [Optimal Monotone Families for the Discrete Isoperimetric Inequality](https://gilkalai.wordpress.com/ai/optimal-monotone-families-for-the-discrete-isoperimetric-inequality/) by *Gil Kalai* (2026), a Polymath project with AI agents - [An Isoperimetric Inequality for the Hamming Cube and Integrality Gaps in Bounded-Degree Graphs](https://arxiv.org/abs/math/0603218) by *Jeff Kahn* and *Gil Kalai* (2006) - [A Proof of the Kahn-Kalai Conjecture](https://arxiv.org/abs/2203.17207) by *Jinyoung Park* and *Huy Tuan Pham* (2022)

V2-000027 | Optimal monotone families for the discrete isoperimetric inequality | mathoverflow_10799

research solved | proof annotation: True | 官方代码 | 原文线索

Problem: For every monotone increasing family F, given an interval $[s,t]$ of real numbers so that $t/s > 1000 \log n$ we have some p in the interval $[s,t]$ so that F is optimal with respect to μ_p. This was a "missing lemma" in the work of Kahn and Kalai on threshold behavior of monotone properties. The related [Kahn-Kalai conjecture](https://arxiv.org/abs/math/0603218) was [settled by Park and Pham](https://arxiv.org/abs/2203.17207). **This conjecture is false** without the additional assumption $\mu_t(F) = 1/2$. A counterexample was found by Shlomo Perles (April 7, 2026).

```
theorem mathoverflow_10799 :
¬ ∀ (n : ℕ) (_ : 2 ≤ n)
(F : Finset (Finset (Fin n))) (_ : IsMonotoneIncreasing F)
(s t : ℝ) (_ : 0 < s) (_ : s ≤ t) (_ : t < 1)
(_ : t / s > 1000 * Real.log n),
∃ p, s ≤ p ∧ p ≤ t ∧ IsOptimal p F
```

FC-Mathoverflow10799.mathoverflow_10799

Mathoverflow 17560

MODULE CONTEXT: # Mathoverflow 17560 *Reference:* [mathoverflow/17560](https://mathoverflow.net/questions/17560) asked by user [Alon-Amit](https://mathoverflow.net/users/25/alon-amit)

V2-000028 | Mathoverflow 17560 | mathoverflow_17560

research open | proof annotation: False | 官方代码 | 原文线索

If 2^x and 3^x are integers, then x must be an integer.

```
theorem mathoverflow_17560 {x : ℝ} (hx : ∃ m : ℕ, (2 : ℝ) ^ x = m) (hx' : ∃ m : ℕ, (3 : ℝ) ^ x = m) :
∃ m : ℕ, x = m
```

FC-Mathoverflow17560.mathoverflow_17560

Mathoverflow 1973

MODULE CONTEXT: # Mathoverflow 1973 Does the 6-sphere S^6 admit the structure of a complex manifold? *References:* - [mathoverflow/1973](https://mathoverflow.net/questions/1973/), asked by user [*Fetchinson0234*](https://mathoverflow.net/users/41312/victor-ramos). - [Al26] L. Alpöge, [*A compact complex threefold fibred by tori over the projective line, and the six-sphere*](https://alpo.ge/s6.pdf) (2026), originally [shared on X](https://x.com/__alpoge__/status/2091639597193368014).

V2-000029 | Mathoverflow 1973 | mathoverflow_1973

research solved | proof annotation: True | 官方代码 | 原文线索

Does the 6-sphere admit a complex structure, i.e. an atlas of holomorphically compatible charts relating it to `EuclideanSpace ℂ (Fin 3)`? This is known as the Hopf Problem. The answer is yes, see [Al26]. Formalisation of the proof by Boris Alexeev.

```
theorem mathoverflow_1973 :
∃ atlas : ChartedSpace (EuclideanSpace ℂ (Fin 3)) (unitSphere 6),
IsManifold 𝓘(ℂ, EuclideanSpace ℂ (Fin 3)) 1 (unitSphere 6)
```

FC-Mathoverflow1973.mathoverflow_1973

Mathoverflow 21003

MODULE CONTEXT: # Mathoverflow 21003 Is there any polynomial $f(x, y) \in \mathbb{Q}[x, y]$ such that $f : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is a bijection? *Reference:* [mathoverflow/21003](https://mathoverflow.net/questions/21003) asked by user [*Z.H.*](https://mathoverflow.net/users/5098/z-h)

V2-000030 | Mathoverflow 21003 | mathoverflow_21003

research open | proof annotation: False | 官方代码 | 原文线索

Is there any polynomial $f(x, y) \in \mathbb{Q}[x, y]$ such that $f : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is a bijection?

```
theorem mathoverflow_21003 :
∃ f : MvPolynomial (Fin 2) ℚ, Function.Bijective fun x ↦ f.eval x
```

FC-Mathoverflow21003.mathoverflow_21003

Mathoverflow 235893

MODULE CONTEXT: # Mathoverflow 235893 *Reference:* [mathoverflow/235893](https://mathoverflow.net/questions/235893) asked by user [*Willie Wong*](https://mathoverflow.net/users/3948/willie-wong)

V2-000031 | Mathoverflow 235893 | mathoverflow_235893

research open | proof annotation: False | 官方代码 | 原文线索

Assume for $n>1$, $f:\mathbb{R}^n\to\mathbb{R}^n$ is a bijection, where $\mathbb{R}^n$ is equipped with the standard topology. Does the connectedness of (the induced power set map) f imply that of f^{-1}?

```
theorem mathoverflow_235893 :
∀ n > 1, ∀ (f : ℝ^n ≃ ℝ^n), IsConnectedMap f → IsConnectedMap f.symm
```

FC-Mathoverflow235893.mathoverflow_235893

V2-000032 | Mathoverflow 235893 | mathoverflow_260589

research solved | proof annotation: False | 官方代码 | 原文线索

There exists a connected bijection $\mathbb{R} \to \mathbb{R}^2$ where the inverse is not connected, proven in [mathoverflow/260589](https://mathoverflow.net/questions/260589) by user [Gro-Tsen](https://mathoverflow.net/users/17064/gro-tsen).

```
theorem mathoverflow_260589 :
∃ f : ℝ ≃ ℝ^2, IsConnectedMap f ∧ ¬ IsConnectedMap f.symm
```

FC-Mathoverflow235893.mathoverflow_260589

Mathoverflow 31809

MODULE CONTEXT: # Mathoverflow 31809 Source: [Mathoverflow/31809](https://mathoverflow.net/questions/31809/pre-triangulated-category-that-isnt-triangulated) *References:* - [CLLZ26] X.-W. Chen, J. Liu, X.-S. Lu and C. Zhang, "A pre-triangulated category which is not triangulated", [arXiv:2608.09777](https://arxiv.org/abs/2608.09777)

V2-000033 | Mathoverflow 31809 | mathoverflow_31809

research solved | proof annotation: False | 官方代码 | 原文线索

Does there exist a category that is pretriangulated but not triangulated? Yes. Theorem 1.2 of [CLLZ26] twists the triangulated structure on the category of finitely generated projective modules over the preprojective algebra of type A_5 over $\mathbb{F}_2$. The result is pretriangulated but does not satisfy the octahedral axiom.

```
theorem mathoverflow_31809 : ¬ (∀ (C : Type*) [Category C] [Preadditive C]
[HasZeroObject C] [HasShift C ℤ] [∀ (n : ℤ), (shiftFunctor C n).Additive]
[Pretriangulated C], IsTriangulated C)
```

FC-Mathoverflow31809.mathoverflow_31809

Mathoverflow 339137

MODULE CONTEXT: # Mathoverflow 339137 Why do polynomials with coefficients 0,1 like to have only factors with 0,1 coefficients? *Reference:* [mathoverflow/339137](https://mathoverflow.net/questions/339137) asked by user [*Sil*](https://mathoverflow.net/users/136794/sil)

V2-000034 | Mathoverflow 339137 | mathoverflow_339137

research open | proof annotation: False | 官方代码 | 原文线索

Let $P(x), Q(x) \in \mathbb{R}[x]$ be two monic polynomials with non-negative coefficients. If $R(x) = P(x)Q(x)$ is a $0,1$ polynomial (coefficients only from $\{0,1\}$), then $P(x)$ and $Q(x)$ are also $0, 1$ polynomials.

```
theorem mathoverflow_339137 (P Q R : ℝ[X]) (hP: P.Monic) (hQ : Q.Monic)
(hp : ∀ c ∈ P.coeffs, 0 ≤ c) (hq : ∀ c ∈ Q.coeffs, 0 ≤ c)
(h : R = P * Q) (hR : IsZeroOne R) :
IsZeroOne P ∧ IsZeroOne Q
```

FC-Mathoverflow339137.mathoverflow_339137

Mathoverflow 34145

MODULE CONTEXT: # Mathoverflow 34145 Can the unit square be covered by $1/k$-by-$1/(k+1)$ rectangles (across $1 \le k$ natural)? I am deliberately not requiring that the rotations can only be $0^\circ, 90^\circ, 180^\circ, \text{ or } 270^\circ$. Because of indexing, since `n : ℕ` starts at 0, we change the side lengths to $1 / (n + 1)$ and $1 / (n + 2)$, so that the first rectangle is $1/1$ by $1/2$, the second is $1/2$ by $1/3$, etc. *Reference:* [mathoverflow/34145](https://mathoverflow.net/q/34145) asked by user [*Kaveh*](https://mathoverflow.net/users/7507/kaveh)

V2-000035 | Mathoverflow 34145 | rectangles_cover_unit_square

research open | proof annotation: False | 官方代码 | 原文线索

Can a unit square be covered by rectangles of width `1 / (n + 1)` and height `1 / (n + 2)`?

```
theorem rectangles_cover_unit_square :
∃ c : Configuration, ∀ p ∈ unitSquare, ∃ n, p ∈ (c.rect n).toSet
```

FC-Mathoverflow34145.rectangles_cover_unit_square

V2-000036 | Mathoverflow 34145 | rectangles_pack_square_133_div_132

research solved | proof annotation: False | 官方代码 | 原文线索

It is known that packing the rectangles into a square of side length `133/132` is possible. Reference: https://www.sciencedirect.com/science/article/pii/0097316594901163

```
theorem rectangles_pack_square_133_div_132 :
(∃ c : Configuration,
(∀ n, (c.rect n).toSet ⊆ Rectangle.toSet ⟨133/132, 133/132, (0, 0), 0⟩) ∧
c.IsPacking)
```

FC-Mathoverflow34145.rectangles_pack_square_133_div_132

V2-000037 | Mathoverflow 34145 | rectangles_pack_square_501_div_500

research solved | proof annotation: False | 官方代码 | 原文线索

It is known that packing the rectangles into a square of side length `501/500` is possible. Reference: https://www.sciencedirect.com/science/article/pii/S0167506008706009

```
theorem rectangles_pack_square_501_div_500 :
(∃ c : Configuration,
(∀ n, (c.rect n).toSet ⊆ Rectangle.toSet (501/500, 501/500, (0, 0), 0)) ∧
c.IsPacking)
```

FC-Mathoverflow34145.rectangles_pack_square_501_div_500

V2-000038 | Mathoverflow 34145 | rectangles_pack_unit_square

research open | proof annotation: False | 官方代码 | 原文线索

Equivalently, can a unit square be packed with rectangles of width `1 / (n + 1)` and height `1 / (n + 2)`?

```
theorem rectangles_pack_unit_square :
∃ c : Configuration, (∀ n, (c.rect n).toSet ⊆ unitSquare) ∧ c.IsPacking
```

FC-Mathoverflow34145.rectangles_pack_unit_square

Mathoverflow 347178

MODULE CONTEXT: # Mathoverflow 347178 *Reference:* [mathoverflow/347178](https://mathoverflow.net/questions/347178) asked by user [*Biagio Ricceri*](https://mathoverflow.net/users/149235/biagio-ricceri)

V2-000039 | Mathoverflow 347178 | mathoverflow_347178

research solved | proof annotation: True | 官方代码 | 原文线索

Let $f : \mathbb R^n \to \mathbb R, n \geq 2$ be a C^1 function. Is it true that $$\sup_{x \in \mathbb R^n}f(x) = \sup_{x\in \mathbb R^n} f(x+\nabla f(x))$$? Answer: No. A counterexample in $\mathbb R^2$ is recorded in the linked formal proof.

```
theorem mathoverflow_347178 :
¬ ∀ᵉ (n ≥ 2) (f : ℝ^n → ℝ) (_ : ContDiff ℝ 1 f),
(BddAbove (range f) ↔ BddAbove (range (fun x ↦ f (x + gradient f x)))) ∧
(⨆ x, (f x : EReal)) = ⨆ x, (f (x + gradient f x) : EReal)
```

FC-Mathoverflow347178.mathoverflow_347178

Are prime numbers among sums of prime numbers distributed as $\frac n{2\ln(n)}$?

MODULE CONTEXT: # Are prime numbers among sums of prime numbers distributed as $\frac n{2\ln(n)}$? *Reference:* [mathoverflow.net/questions/434111](https://mathoverflow.net/questions/434111/are-prime-numbers-among-sums-of-prime-numbers-distributed-as-frac-n2-lnn) [Me18] Meštrović, R., *Curious Conjectures on the Distribution of Primes Among the Sums of the First `2n` Primes*, [arXiv:1804.04198](https://arxiv.org/abs/1804.04198) (2018), Conjecture 3.3.

V2-000040 | Are prime numbers among sums of prime numbers distributed as $\frac n{2\ln(n)}$? | restricted_prime_number_theorem

research open | proof annotation: False | 官方代码 | 原文线索

The conjecture claims that $\pi_n\sim\frac n{2\ln(n)}$. In other words, primes are distributed among the much sparser sequence $(S_n)_n$ with essentially the same density as in the positive integers, up to a factor of 2. [MathOverflow 434111](https://mathoverflow.net/questions/434111/are-prime-numbers-among-sums-of-prime-numbers-distributed-as-frac-n2-lnn). [Me18] Meštrović, R., *Curious Conjectures on the Distribution of Primes Among the Sums of the First `2n` Primes*, [arXiv:1804.04198](https://arxiv.org/abs/1804.04198) (2018).

```
theorem restricted_prime_number_theorem :
((fun n : ℕ => (piRestricted n : ℝ)) ~[atTop] (fun n : ℕ => (n : ℝ) / (2 * Real.log n)))
```

FC-MathOverflow434111.restricted_prime_number_theorem

Mathoverflow 486451

MODULE CONTEXT: # Mathoverflow 486451 *Reference:* [mathoverflow/486451](https://mathoverflow.net/questions/486451) asked by user [*Junyan Xu*](https://mathoverflow.net/users/3332/junyan-xu)

V2-000041 | Mathoverflow 486451 | exists_semiring_unique_left_maximal_not_unique_right_maximal

research solved | proof annotation: False | 官方代码 | 原文线索

There exists a semiring with a unique left maximal ideal but more than one right maximal ideals.

```
theorem exists_semiring_unique_left_maximal_not_unique_right_maximal :
∃ (R : Type) (_ : Semiring R), (∃! I : Ideal R, I.IsMaximal) ∧
∃ I J : Ideal Rᵐᵒᵖ, I.IsMaximal ∧ J.IsMaximal ∧ I ≠ J
```

FC-Mathoverflow486451.exists_semiring_unique_left_maximal_not_unique_right_maximal

V2-000042 | Mathoverflow 486451 | exists_semiring_unique_left_right_maximal_ne

research solved | proof annotation: True | 官方代码 | 原文线索

There exists a semiring with a unique left maximal ideal and a unique right maximal ideal which are not the same as sets. This has been shown by Goran Žužić and Moritz Firsching using an experimental pipeline: An example is the monoid algebra of the monoid of maps from $\mathbb{N}$ to $\mathbb{N}$ over $\mathbb{N}$.

```
theorem exists_semiring_unique_left_right_maximal_ne :
∃ (R : Type) (_ : Semiring R) (hI : ∃! I : Ideal R, I.IsMaximal)
(hJ : ∃! J : Ideal Rᵐᵒᵖ, J.IsMaximal),
(hI.choose : Set R) ≠ MulOpposite.op ⁻¹' hJ.choose
```

FC-Mathoverflow486451.exists_semiring_unique_left_right_maximal_ne

Mathoverflow 507128

MODULE CONTEXT: # Mathoverflow 507128 *Reference:* [mathoverflow/507128](https://mathoverflow.net/questions/507128/embeddability-order-on-picard-groups) asked by user [*Junyan Xu*](https://mathoverflow.net/users/3332/junyan-xu)

V2-000043 | Mathoverflow 507128 | exists_isFractionRing_self_ideal_ne_top_invertible

research solved | proof annotation: True | 官方代码 | 原文线索

There exists a proper ideal `I` in a (commutative) total ring `R` of fractions that is an invertible module. If `I ⊊ R` is such an example, `I` must have infinite order in the Picard group. Moreover, `R` must not be Noetherian (otherwise it must be semi-local and therefore have trivial Picard group). The linked formal proof gives the following explicit construction. Let `D = ℂ[X, Y] / (Y² - X³)` be the coordinate ring of the cuspidal cubic. Let `P = (X - 1, Y - 1)`, which is an invertible ideal. Let `M` be the direct sum of the evaluation fibres at cusp parameters `r ≠ 1`. Form the idealization `R = D ⋉ M`. The desired ideal is the range of the multiplication map `R ⊗[D] P → R`.

```
theorem exists_isFractionRing_self_ideal_ne_top_invertible :
∃ (R : Type) (_ : CommRing R) (_ : IsFractionRing R R) (I : Ideal R),
I ≠ ⊤ ∧ Module.Invertible R I
```

FC-Mathoverflow507128.exists_isFractionRing_self_ideal_ne_top_invertible

MathOverflow 508539

MODULE CONTEXT: # MathOverflow 508539 Edge-vertex inequality $EV \geq 4^{n-1}$. *References:* - [MathOverflow 508539](https://mathoverflow.net/questions/508539), asked by [Paata Ivanisvili](https://mathoverflow.net/users/50901/paata-ivanisvili). - [DIRX26] Durcik, P., Ivanisvili, P., Roos, J., and Xie, X., *Sharp isoperimetric inequalities on the Hamming cube II: The critical exponent* (2026), [arXiv:2602.20462](https://arxiv.org/abs/2602.20462), inequality (1.2).

V2-000044 | MathOverflow 508539 | mathoverflow_508539

research solved | proof annotation: False | 官方代码 | 原文线索

For any set $A \subset \{0,1\}^n$ of half size 2^{n-1}, we have $EV \geq 4^{n-1}$, where E denotes the edge boundary (the number of edges between A and its complement), and V denotes the inner vertex boundary (the number of points in A having a neighbor outside A). This is the half-size case of [DIRX26, (1.2)]. The hypothesis $n > 0$ is necessary: in dimension zero the cardinality hypothesis allows the whole cube, whose boundaries are empty.

```
theorem mathoverflow_508539 {n : ℕ} (A : Finset (Fin n → Bool))
(hA : #A = 2 ^ (n - 1)) (hn : 0 < n) :
4 ^ (n - 1) ≤ #((hypercube n).edgeBoundary A) *
#((hypercube n).innerVertexBoundary A)
```

FC-Mathoverflow508539.mathoverflow_508539

Mathoverflow 75792

MODULE CONTEXT: # Mathoverflow 75792 Various questions about integer complexity, which is the minimum number of `1`s needed to express a natural number using addition, multiplication, and parentheses. Let `‖n‖` denote the integer complexity of `n > 0`. * It is known that `‖3^n‖ = 3n` for `n > 0`. * Is it true that `‖2^n‖ = 2n` for `n > 0`? * The corresponding conjecture for `5` is false, because `5^6 = 15625 = 1 + 2^3 * 3^2 * (1 + 2^3 * 3^3)`! We have chosen to formalise this using an inductive type. *References:* - [mathoverflow/75792](https://mathoverflow.net/a/75792) by user [Harry Altman](https://mathoverflow.net/users/5583) - http://arxiv.org/abs/1203.6462 by Jānis Iraids, Kaspars Balodis, Juris Čerņenoks, Mārtiņš Opmanis, Rihards Opmanis, Kārlis Podnieks - http://arxiv.org/abs/1207.4841 by Harry Altman, Joshua Zelinsky - https://oeis.org/A5245 : Mahler-Popken complexity.

V2-000045 | Mathoverflow 75792 | complexity_five_pow

research solved | proof annotation: False | 官方代码 | 原文线索

Is `5n` the complexity of `5^n` for `0 < n`? Answer: No.

```
theorem complexity_five_pow : ¬ ∀ n : ℕ, 0 < n → complexity (5 ^ n) = 5 * n
```

FC-Mathoverflow75792.complexity_five_pow

V2-000046 | Mathoverflow 75792 | complexity_three_pow

research solved | proof annotation: False | 官方代码 | 原文线索

Is `3n` the complexity of `3^n` for `0 < n`? Answer: Yes, by John Selfridge. Reference: https://arxiv.org/abs/1207.4841

```
theorem complexity_three_pow : ∀ n : ℕ, 0 < n → complexity (3 ^ n) = 3 * n
```

FC-Mathoverflow75792.complexity_three_pow

V2-000047 | Mathoverflow 75792 | complexity_two_pow

research open | proof annotation: False | 官方代码 | 原文线索

Is `2n` the complexity of `2^n` for `0 < n`?

```
theorem complexity_two_pow : ∀ n : ℕ, 0 < n → complexity (2 ^ n) = 2 * n
```

FC-Mathoverflow75792.complexity_two_pow

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Expansion of g.f. $(1+x)(3+x)/(1+6x^2+x^4)$

MODULE CONTEXT: # Expansion of g.f. $(1+x)(3+x)/(1+6x^2+x^4)$ This sequence is defined by the linear recurrence relation $a(n) = -6 a(n-2) - a(n-4)$ for $n \ge 4$, with initial values $a(0)=3$, $a(1)=4$, $a(2)=-17$, $a(3)=-24$. *References:* - [A100434](https://oeis.org/A100434)

V2-000048 | Expansion of g.f. $(1+x)(3+x)/(1+6x^2+x^4)$ | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture from Creighton Dement (A100434): Let the auxiliary sequences c, d, e, f, g, b be defined as specified. Then for all $n \ge 0$, $c(n) + d(n) = b(n)$. **Proof outline**: strong two-step induction on the paired recurrences. The even/odd cases are coupled through closed identities between consecutive terms of the relevant pair of sequences (here c, d), and the induction step is discharged by `rfl`-level unfolding of the definitions plus linear arithmetic. (Numerically verified for $n < 600$ before formalization.)

```
theorem conjecture1 (n : ℕ) : c n + d n = b n
```

FC-OeisA100434.conjecture1

V2-000049 | Expansion of g.f. $(1+x)(3+x)/(1+6x^2+x^4)$ | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture from Creighton Dement (A100434): Let the auxiliary sequences c, d, e, f, g, b be defined as specified. Then for all $n \ge 0$, $e(n) + f(n) = b(n)$. **Proof outline**: strong two-step induction on the paired recurrences. The even/odd cases are coupled through closed identities between consecutive terms of the relevant pair of sequences (here e, f), and the induction step is discharged by `rfl`-level unfolding of the definitions plus linear arithmetic. (Numerically verified for $n < 600$ before formalization.)

```
theorem conjecture2 (n : ℕ) : e n + f n = b n
```

FC-OeisA100434.conjecture2

V2-000050 | Expansion of g.f. $(1+x)(3+x)/(1+6x^2+x^4)$ | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture from Creighton Dement (A100434): Let the auxiliary sequences c, d, e, f, g, b be defined as specified. Then for all $n \ge 0$, $g(n) + a(n) = b(n)$. **Proof outline**: strong two-step induction on the paired recurrences. The even/odd cases are coupled through closed identities between consecutive terms of the relevant pair of sequences (here g, a), and the induction step is discharged by `rfl`-level unfolding of the definitions plus linear arithmetic. (Numerically verified for $n < 600$ before formalization.)

```
theorem conjecture3 (n : ℕ) : g n + a n = b n
```

FC-OeisA100434.conjecture3

Conjectures associated with A100474

MODULE CONTEXT: # Conjectures associated with A100474 $a(1) = 1$; $a(n)$ is the smallest integer such that $a(n) + a(n-1)$ has the first n distinct prime factors not used before in this construction. *References:* - [A100474](https://oeis.org/A100474)

V2-000051 | Conjectures associated with A100474 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

After $a(2) = 5$, is there another prime?

```
theorem conjecture : ∃ n > 2, (a n).Prime
```

FC-OeisA100474.conjecture

V2-000052 | Conjectures associated with A100474 | next_semiprime

research solved | proof annotation: True | 官方代码 | 原文线索

What is the next semiprime in the sequence after $a(11)$?

```
theorem next_semiprime :
next_semiprime.answer =
a (sInf {n : ℕ | 11 < n ∧ (a n).IsSemiprime})
```

FC-OeisA100474.next_semiprime

Prime-th recurrence with reversal at each step

MODULE CONTEXT: # Prime-th recurrence with reversal at each step $$a(n) = \operatorname{reversal}(p_{a(n-1)})$$ with $a(0)=1$, where p_k is the k-th prime number. *References:* - [A100475](https://oeis.org/A100475)

V2-000053 | Prime-th recurrence with reversal at each step | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Starting at a positive value other than $a(0) = 1$, does this sequence ever go into a loop? The positivity hypothesis is required because the source recurrence uses the one-based prime index `p₁ = 2`; the `x = 0` branch above is only an artifact of making `aStartAt` total on `ℕ`.

```
theorem conjecture :
∃ x : ℕ, 0 < x ∧ x ≠ 1 ∧ IsUltimatelyPeriodic (aStartAt x)
```

FC-OeisA100475.conjecture

V2-000054 | Prime-th recurrence with reversal at each step | conjecture_with_zero

research solved | proof annotation: False | 官方代码 | 原文线索

If zero is admitted as a starting value, then a start other than 1 does go into a loop: the sequence starting at zero is constant. This records the degenerate answer created by totalizing the one-based prime recurrence at index zero.

```
theorem conjecture_with_zero :
∃ x : ℕ, x ≠ 1 ∧ IsUltimatelyPeriodic (aStartAt x)
```

FC-OeisA100475.conjecture_with_zero

Pentanacci π sequence

MODULE CONTEXT: # Pentanacci π sequence Start with $a(1)=a(2)=a(3)=a(4)=a(5)=1$ and for $n>5$, $a(n) = \pi(\sum_{j=1}^5 a(n-j))$ where $\pi = A000720$. *References:* - [A100478](https://oeis.org/A100478)

V2-000055 | Pentanacci π sequence | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Starting with other values of $a(1)$, $a(2)$, $a(3)$, $a(4)$, $a(5)$ what behaviors are possible? Does the sequence always stick at a single integer after some point, or can it go into a loop, or is there a third pattern? KitaKen1 showed the sequence either becomes constant or goes into a loop, see https://github.com/KitaKen1/oeis-a100478-eventual-periodicity/blob/642eed0ffee26415528ab8c48c5181826be04860/lean/OeisA100478FC.lean#L158-L168. tadamcz showed the sequence always becomes constant, see https://tadamcz.com/fc-review-results/aea251bb26/#/f/OEIS/100478.

```
theorem conjecture:
∀ v : Fin 5 → ℕ, (∀ i, 0 < v i) → ∃ c ∈ ({66, 67, 68, 70, 71, 72} : Finset ℕ),
∀ᶠ n in Filter.atTop, aGeneral v n = c
```

FC-OeisA100478.conjecture

Conjectures associated with A100800

MODULE CONTEXT: # Conjectures associated with A100800 Let $f(n) = n + \text{sum of the digits of } n$. If $f(n)$ is multiple of n then $a(n)= f(n)$ else $a(n) = f(f(f(n)))\dots$ until one gets a multiple of n; $a(n) = 0$ if no such number exists. *References:* - [A100800](https://oeis.org/A100800)

V2-000056 | Conjectures associated with A100800 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

A100800 Conjecture: No term is zero.

```
theorem conjecture : ∀ (n : ℕ), n ≠ 0 → a n ≠ 0
```

FC-OeisA100800.conjecture

Conjectures associated with A101779

MODULE CONTEXT: # Conjectures associated with A101779 $a(n)$ is the least k such that all of $k, 2k+1, 3k+2, ..., nk+n-1$ are primes, or 0 if no such k is found. It is conjectured k always exists. *References:* - [A101779](https://oeis.org/A101779)

V2-000057 | Conjectures associated with A101779 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured k always exists.

```
theorem conjecture : ∀ (n : ℕ), 1 ≤ n → ∃ k : ℕ, Ak n k
```

FC-OeisA101779.conjecture

Conjectures associated with A102371

MODULE CONTEXT: # Conjectures associated with A102371 The sequence $a(n)$ is defined by the recurrence relation $a(1)=1$, and for $n>1$, $a(n) = a(n-1) \operatorname{XOR} (a(n-1) + n)$. The conjecture asks if $a(n) = 2^n - 1 - \operatorname{A105033}(n-1)$ for $n \ge 1$. *References:* - [A102371](https://oeis.org/A102371)

V2-000058 | Conjectures associated with A102371 | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Do we have $a(n) = 2^n - 1 - \operatorname{A105033}(n-1)$ for $n \ge 1$? Note: The OEIS comment suggests $n-1$ which means this applies at least for $n \ge 1$, as we assume $\operatorname{A105033}(\mathbb{N})$ is defined on $\mathbb{N}$. We include the case $n=1$ which relies on $A105033(0)$, which is 0.

```
theorem conjecture : ∀ n : ℕ, 0 < n → a n = 2^n - 1 - OeisA105033.a (n - 1)
```

FC-OeisA102371.conjecture

Floor of sum of $\{n/k\}$

MODULE CONTEXT: # Floor of sum of $\{n/k\}$ Given n, sum all division remainders $\{n/k\}$, with $k=1,\dots,n$. The value $a(n)$ is given by the floor of that sum. Note that $\{x\}:=x-[x]$. Conjecture: a(n) ~ (1-EulerGamma)n. *References:* - [A102722](https://oeis.org/A102722)

V2-000059 | Floor of sum of $\{n/k\}$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

A102722 Conjecture: $a(n) \sim (1-\gamma)n$.

```
theorem conjecture :
(fun n : ℕ => (a n : ℝ)) ~[atTop]
(fun n : ℕ => (1 - Real.eulerMascheroniConstant) * (n : ℝ))
```

FC-OeisA102722.conjecture

$a(0) = 1$, $a(n) = a(n-1)a(n-1) + 2$

MODULE CONTEXT: # $a(0) = 1$, $a(n) = a(n-1)a(n-1) + 2$ *References:* - [A102847](https://oeis.org/A102847)

V2-000060 | $a(0) = 1$, $a(n) = a(n-1)a(n-1) + 2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Prime for $a(1) = 3$, $a(2) = 11$, $a(4) = 15131$; semiprime for $a(3) = 123 = 3 * 41$, $a(5) = 228947163 = 3 * 76315721$. $a(6)$, added by Jonathan Vos Post, has 4 prime factors. $a(7) = 41 * 811^2 * 106693969 * 317171188688357726699 * 827223692554099605444017244976l$. When is the next prime in the sequence?

theorem conjecture : conjecture.answer = sInf {n : ℕ | 4 < n ∧ (a n).Prime}

FC-OeisA102847.conjecture

Number of decompositions of $2n+1$ into $2p+q$, where p and q are both odd primes

MODULE CONTEXT: # Number of decompositions of $2n+1$ into $2p+q$, where p and q are both odd primes *References:* - [A103151](https://oeis.org/A103151)

V2-000061 | Number of decompositions of $2n+1$ into $2p+q$, where p and q are both odd primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: all items for $n \ge 4$ are greater than or equal to 1. This is a stronger conjecture than the Goldbach conjecture.

theorem conjecture (n : ℕ) (hn : n ≥ 4) : a n ≥ 1

FC-OeisA103151.conjecture

Fibonacci transform satisfying $|a(n)| = F(n+1)$

MODULE CONTEXT: # Fibonacci transform satisfying $|a(n)| = F(n+1)$ The sequence $a(n)$ satisfies the linear recurrence relation: $$a(n) = 3a(n-1) - 4a(n-2) + 2a(n-3) - a(n-4)$$ with initial terms $a(0)=0, a(1)=1, a(2)=1, a(3)=0$. The sequence takes values in $\mathbb{Z}$. *References:* - [A103311](https://oeis.org/A103311) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000062 | Fibonacci transform satisfying $|a(n)| = F(n+1)$ | a_abs_eq_fib

research solved | proof annotation: True | 官方代码 | 原文线索

Fibonacci transform satisfying $|a(n)| = F(n+1)$. Conjecture: all elements in absolute value are Fibonacci numbers. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_abs_eq_fib (n : ℕ) : ∃ m : ℕ, Int.natAbs (a n) = Nat.fib m

FC-OeisA103311.a_abs_eq_fib

$a(n) = 3a(n-1) + a(n-2) - 3a(n-3)$

MODULE CONTEXT: # $a(n) = 3a(n-1) + a(n-2) - 3a(n-3)$ *References:* - [A103425](https://oeis.org/A103425)

V2-000063 | $a(n) = 3a(n-1) + a(n-2) - 3a(n-3)$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The current sequence contains primes, including $3, 5, 41, 21523361$. Is there an (a, b, c) weighted tribonacci sequence with a, b, c relatively prime which is prime-free? Yes: take $(a, b, c) = (1, 1, -1)$ and the constant sequence $x(n) = 4$. The linked Lean proof is by Kenta Kitamura.

theorem conjecture :
∃ (a b c : ℤ) (x : ℕ → ℤ),
Nat.gcd (Int.gcd a b) c.natAbs = 1 ∧
IsWeightedTribonacci a b c x ∧
∀ n, ¬ (x n).natAbs.Prime

FC-OeisA103425.conjecture

Smallest power with base>1 and exponent n without digit 0

MODULE CONTEXT: # Smallest power with base>1 and exponent n without digit 0 For statistical reasons it is conjectured that the sequence is finite. Also it is conjectured that $a(40)$ does not exist (i.e. the sequence is empty for $n=40$). *References:* - [A103662](https://oeis.org/A103662)

V2-000064 | Smallest power with base>1 and exponent n without digit 0 | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

For statistical reasons it is conjectured that the sequence is finite. Finite means here that for some n, no power b^n with base $b > 1$ has a zeroless decimal representation, which in our definition results in $a(n) = 0$. This holds for $n = 2500$ ([source](https://tadamcz.com/fc-review-results/aea251bb26/#/f/OEIS/103662)). Take any integer $b > 1$ and let $N = b^{2500}$. Since $b \ge 2$, we have $N \ge 2^{2500} > 10^5$, so N has at least six decimal digits. We split into four cases depending on the divisibility of b by 2 and 5: - **If $10 \mid b$:** Then $10 \mid N$, so the units digit of N is 0. - **If b is odd and $5 \nmid b$:** Since b is odd and the Carmichael function satisfies $\lambda(16) = 4 \mid 2500$, we have $N \equiv 1 \pmod{16}$. Since $5 \nmid b$ and $\varphi(5^5) = 2500$, Euler's totient theorem gives $N \equiv 1 \pmod{3125}$. By the Chinese Remainder Theorem, $N \equiv 1 \pmod{50000}$ (as $16 \cdot 3125 = 50000$), so the tens digit of N is 0. - **If b is odd and $5 \mid b$:** As above, $N \equiv 1 \pmod{16}$. Since $5 \mid b$ and $2500 \ge 4$, we also have $5^4 = 625 \mid N$, so $N \equiv 0 \pmod{625}$. Since $16 \cdot 625 = 10^4$ and $625 \equiv 1 \pmod{16}$, the Chinese Remainder Theorem gives $N \equiv 625 \pmod{10^4}$. Because $N > 10^4$, the last four digits of N are 0625, so its thousands digit is 0. - **If b is even and $5 \nmid b$:** Since $2 \mid b$ and $2500 \ge 5$, we have $2^5 = 32 \mid N$, so $N \equiv 0 \pmod{32}$. Since $5 \nmid b$, Euler's theorem again gives $N \equiv 1 \pmod{3125}$. Because $32 \cdot 3125 = 10^5$ and $9376 = 293 \cdot 32 = 3 \cdot 3125 + 1$, the Chinese Remainder Theorem gives $N \equiv 9376 \pmod{10^5}$. Since $N > 10^5$, the last five digits of N are 09376, so its ten-thousands digit is 0. In every case, the decimal representation of N contains the digit 0, so $a(2500) = 0$.

theorem conjecture : ∃ n : ℕ, a n = 0

FC-OeisA103662.conjecture

$a(n) = [x^{2n}] \left(\frac{1 + x}{1 - x}\right)^n$

MODULE CONTEXT: # $a(n) = [x^{2n}] \left(\frac{1 + x}{1 - x}\right)^n$ The sequence is given by the combinatorial identity: $a(n) = \sum_{k = 0}^n \binom{n}{k} \binom{2n+k-1}{n-1}$ with $a(0) = 1$. *References:* - [A103885](https://oeis.org/A103885)

V2-000065 | $a(n) = [x^{2n}] \left(\frac{1 + x}{1 - x}\right)^n$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The recurrence given below can be rewritten in the form $$(2n+1)(2n+2)P(2,n)a(n+1) - (2n-1)(2n-2)P(2,-n)a(n-1) = Q(2,n^2)a(n),$$ where the polynomial $Q(2,n) = 4(55n^2 - 34n + 3)$ and the polynomial $P(2,n) = 5n^2 - 5n + 1$ satisfies the symmetry condition $P(2,n) = P(2,1-n)$ and has real zeros. More generally, for fixed $m = 1,2,3, \ldots$, we conjecture that the sequence $b(n) := a(mn)$ satisfies a recurrence of the form $$(\prod_{k = 1}^{2m} (2mn + k))P(2m,n)b(n+1) + (-1)^m(\prod_{k = 1}^{2*m} (2mn - k)) P(2m,-n)b(n-1) = Q(2m,n^2)b(n),$$ where the polynomials $P(2m,n)$ and $Q(2m,n)$ have degree $2m$. Conjecturally, the polynomial $P(2m,n) = P(2m,1-n)$ and has real zeros in the interval [0, 1]. The $4m$ zeros of the polynomial $Q(2m,n^2)$ seem to belong to the interval $[-1, 1]$ and $4m - 2$ of these zeros appear to be approximated by the rational numbers $\pm k/(3m)$, where $1 \le k \le 3m - 2$, k not a multiple of 3.

theorem conjecture (m : ℕ) (hm : 1 ≤ m) :
∃ (P Q : Polynomial ℝ),
-- P and Q have degree 2m
P.degree = (2 * m : ℕ) ∧ Q.degree = (2 * m : ℕ) ∧
-- The recurrence relation holds for all n >= 1
(∀ (n : ℕ) (hn : 1 ≤ n),
(prodFactorPlus m n * P.eval (n : ℝ)) * (aSubsequenceReal m (n + 1)) +

((-1 : ℝ) ^ m * prodFactorMinus m n * P.eval (-(n : ℝ))) * (aSubsequenceReal m (n - 1)) =

(Q.eval ((n : ℝ)^2)) * (aSubsequenceReal m n)) ∧

-- P symmetry: P(x) = P(1-x)
(∀ x : ℝ, P.eval x = P.eval (1 - x)) ∧

-- P has real zeros in [0, 1]: all complex zeros are real and in [0, 1]
(∀ z : ℂ, (P.map (algebraMap ℝ ℂ)).eval z = 0 → z.im = 0 ∧ z.re ∈ (Set.Icc 0 1)) ∧

-- Q zero properties: The zeros of Q(x^2) are real and in [-1, 1].
(∀ z : ℂ, (Q.map (algebraMap ℝ ℂ)).eval (z^2) = 0 → z.im = 0 ∧ z.re ∈ (Set.Icc (-1) 1))

FC-OeisA103885.conjecture

Number of zeros in ternary representation of 2^n

MODULE CONTEXT: # Number of zeros in ternary representation of 2^n *References:* - [A104320](https://oeis.org/A104320)

V2-000066 | Number of zeros in ternary representation of 2^n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture from N. J. A. Sloane: $a(n) > 0$ for $n > 15$.

theorem conjecture : ∀ n : ℕ, 15 < n → a n > 0

FC-OeisA104320.conjecture

Array read by upward antidiagonals

MODULE CONTEXT: # Array read by upward antidiagonals Array read by upward antidiagonals: row n ($n \ge 0$) contains the numbers $m^2 - n^2$, $m \ge n+1$. *References:* - [A105020](https://oeis.org/A105020)

V2-000067 | Array read by upward antidiagonals | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

A "Goldbach Conjecture" for this sequence: when there are n terms between consecutive odd integers $2n+1$ and $2n+3$ for $n > 0$, at least one will be the product of 2 primes (not necessarily distinct). Example: $n=3$ for consecutive odd integers $a(7) = 7$ and $a(11) = 9$ and of the 3 sequence entries $a(8) = 12$, $a(9) = 15$ and $a(10) = 16$ between them, one is the product of 2 primes $a(9) = 15=3*5$. - _Michael Hiebl_, Jul 15 2007

```
theorem conjecture :
∀ (n i j : ℕ), 1 ≤ n →
a i = 2 * n + 1 →
a j = 2 * n + 3 →
j = i + n + 1 →
∃ (k : ℕ),
i < k ∧
k < j ∧
(a k).IsSemiprime
```

FC-OeisA105020.conjecture

Conjectures associated with A105210

MODULE CONTEXT: # Conjectures associated with A105210 $a(1) = 393$; for $n > 1$, $a(n) = a(n-1)$ + 1 + sum of distinct prime factors of $a(n-1)$ that are $< a(n-1)$. *References:* - [A105210](https://oeis.org/A105210)

V2-000068 | Conjectures associated with A105210 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

This suggests that there may be infinitely many different (non-merging) sequences obtained by choosing different starting values.

```
theorem conjecture :
∃ K : Set ℕ,
Set.Infinite K ∧
(∀ k ∈ K, 1 ≤ k) ∧
(∀ j k : ℕ,
j ∈ K → k ∈ K → j ≠ k →
sequenceSet j ∩ sequenceSet k = ∅)
```

FC-OeisA105210.conjecture

V2-000069 | Conjectures associated with A105210 | conjecture_disjoint_starting_values

research open | proof annotation: False | 官方代码 | 原文线索

Cormier and Selfridge found 5 starting values for which the sequences appear to not merge. The sequences were checked up to 10^8.

```
theorem conjecture_disjoint_starting_values :
∀ j k : ℕ,
j ∈ ({1, 393, 412, 668, 932} : Set ℕ) →
k ∈ ({1, 393, 412, 668, 932} : Set ℕ) →
j ≠ k →
sequenceSet j ∩ sequenceSet k = ∅
```

FC-OeisA105210.conjecture_disjoint_starting_values

Indicator sequence for 5 Fibonacci numbers with n digits

MODULE CONTEXT: # Indicator sequence for 5 Fibonacci numbers with n digits $a(n) = 1$ if exactly 5 Fibonacci numbers exist with exactly n digits, otherwise 0. For the partial sums $S(n) = \sum_{k=1}^n a(k)$, it is conjectured that $\beta-2 < S(n)-\alpha n < \beta-1$, where $\alpha = \log(10)/\log(\phi) - 4$ and $\beta = \log(5)/(2\log(\phi)) - 1$. *References:* - [A105565](https://oeis.org/A105565)

V2-000070 | Indicator sequence for 5 Fibonacci numbers with n digits | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $\beta-2 < S(n)-\alpha n < \beta-1$. The constants α and β are as defined in the formula section. Solved by OpenAI Codex, prompted by Adam Haig. A complete Lean 4 proof is linked by the `formal_proof` attribute below.

```
theorem conjecture (n : ℕ) (hn : 1 ≤ n) :
betaConst - 2 < s n - alphaConst * (n : Real) ∧
s n - alphaConst * (n : Real) < betaConst - 1
```

FC-OeisA105565.conjecture

Triangular matchstick numbers in the class of prime numbers

MODULE CONTEXT: # Triangular matchstick numbers in the class of prime numbers $a(n) = \sum_{k = n}^{2n} p_k$, where p_k is the k-th prime. *References:* - [A105720](https://oeis.org/A105720)

V2-000071 | Triangular matchstick numbers in the class of prime numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Terms are squares at only(?) three values of $n = 3, 6, 4072$: corresponding terms are 6^2, 13^2, and 15735^2.

```
theorem conjecture :
∀ n : ℕ, 0 < n → (IsSquare (a n) ↔ (n = 3 ∨ n = 6 ∨ n = 4072))
```

FC-OeisA105720.conjecture

Imaginary part of $\prod_{k=0}^n (1 + k \cdot i)$, $i = \sqrt{-1}$

MODULE CONTEXT: # Imaginary part of $\prod_{k=0}^n (1 + k \cdot i)$, $i = \sqrt{-1}$ *References:* - [A105751](https://oeis.org/A105751)

V2-000072 | Imaginary part of $\prod_{k=0}^n (1 + k \cdot i)$, $i = \sqrt{-1}$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Moll's conjecture 5.5 extends to this sequence and takes the form: (i) the 2-adic valuation $\nu_2(a(n)) \sim n/4$ as n -> oo.

theorem conjecture :
Tendsto (fun n ↦ (4 : ℚ) * (padicValInt 2 (a n) : ℚ) / (n : ℚ)) atTop (nhds 1)

FC-OeisA105751.conjecture

Fibonacci-Collatz sequence

MODULE CONTEXT: # Fibonacci-Collatz sequence Fibonacci-Collatz sequence: $a(1)=1, a(2)=2$; for $n > 2$, let $\mathrm{fib} = a(n-1) + a(n-2)$; if fib is odd then $a(n) = 3 \cdot \mathrm{fib} + 1$ else $a(n) = \mathrm{fib}/2$. *References:* - [A105801](https://oeis.org/A105801)

V2-000073 | Fibonacci-Collatz sequence | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: for every $k > 0$ there is an index m such that all the $a(n)$ with $n > m$ have the same residue $\bmod 3^k$.

theorem conjecture :
∀ k : ℕ, 0 < k → ∃ m : ℕ, ∀ n : ℕ, m < n → a n ≡ a (m + 1) [MOD (3^k)]

FC-OeisA105801.conjecture

Sum of squares of nonacci numbers

MODULE CONTEXT: # Sum of squares of nonacci numbers Sum of squares of nonacci numbers (Fibonacci 9-step numbers). *References:* - [A107247](https://oeis.org/A107247)

V2-000074 | Sum of squares of nonacci numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Primes in this sequence include: $a(8) = 2$, which is next?

theorem conjecture :
conjecture.answer = a (sInf {n : ℕ | 8 < n ∧ (a n).Prime})

FC-OeisA107247.conjecture

$a(n) = \sum_{i=0}^n \binom{2n-i}{n+i}$

MODULE CONTEXT: # $a(n) = \sum_{i=0}^n \binom{2n-i}{n+i}$ Alternatively the sequence `a` can be defined as $a(n) = \sum_{k=0}^n \binom{n+k-1}{k} F(n-k+1)$, where $F(m)$ is the m-th Fibonacci number. We formalize a conjecture about the number of words of length n in a set X being related to this sequence. *References:* - [A108081](https://oeis.org/A108081)

V2-000075 | $a(n) = \sum_{i=0}^n \binom{2n-i}{n+i}$ | count_words_in_x_is_a_shifted

research solved | proof annotation: True | 官方代码 | 原文线索

"The number of words of length n for $n \le 12$ is given by $a(n+1)$. Is this always true?" Formalized as $|X_n| = a(n-1)$ for $n \ge 1$, because the sequence values $a(0)=1, a(1)=2, a(2)=7$ match the examples given for word lengths $n=1, 2, 3$ respectively. Solved: a Lean 4 proof, derived from the Atlas proofs in [facebookresearch/atlas-lean](https://github.com/facebookresearch/atlas-lean), is linked in `formal_proof`.

theorem count_words_in_x_is_a_shifted (n : ℕ) :
n ≥ 1 → Set.ncard (xN n) = a (n - 1)

FC-OeisA108081.count_words_in_x_is_a_shifted

Riesel Problem

MODULE CONTEXT: # Riesel Problem Riesel problem: let $k=2n-1$; then $a(n)$ is the smallest $m \ge 1$ such that $k \cdot 2^m-1$ is prime, or -1 if no such prime exists. *References:* - [A108129](https://oeis.org/A108129)

V2-000076 | Riesel Problem | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured that the integer $k = 509203$ is the smallest Riesel number, that is, the first n such that $a(n) = -1$ is 254602.

theorem conjecture :
a 254602 = -1 ∧ (∀ n : ℕ, 1 ≤ n ∧ n < 254602 → a n ≠ -1)

FC-OeisA108129.conjecture

$a(n) = 16n^2 + 1$

MODULE CONTEXT: # $a(n) = 16n^2 + 1$ *References:* - [A108211](https://oeis.org/A108211)

V2-000077 | $a(n) = 16n^2 + 1$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $$a(n) = \left\lfloor \frac{1}{\frac{1}{4n} - \log(2) + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}} \right\rfloor.$$ **Proof sketch** (certificate style; the kernel-checked development lives at the `formal_proof` permalink below). Write $T(n) = \log 2 - (H(2n) - H(n))$ for the harmonic tail defect. The proof sandwiches $T(n)$ between two explicit telescoping bounds - $h(n)$ from below and $h(n) + 60/(4n+1)^7$ from above, where h telescopes a degree-7 rational certificate. The two resulting inequalities reduce to polynomial coefficient-nonnegativity facts discharged by elementary tactics, after which the reciprocal lands in $[16n^2 + 1,\, 16n^2 + 2)$ and the floor evaluates exactly.

```
theorem conjecture (n : ℕ) (hn : n > 0) :
(a n : ℝ) =
(⌊ 1 / ((4 * n : ℝ)⁻¹ - log 2 + ∑ k ∈ (Finset.Icc (n + 1) (2 * n)), (k : ℝ)⁻¹) ⌋ : ℝ)
```

FC-OeisA108211.conjecture

Digital sum of the Fermat number $2^{2^n} + 1$

MODULE CONTEXT: # Digital sum of the Fermat number $2^{2^n} + 1$ `a n` is the digital sum of the Fermat number $2^{2^n} + 1$. The conjecture asks if there are any prime numbers in this sequence beyond $n=11$. *References:* - [A108301](https://oeis.org/A108301)

V2-000078 | Digital sum of the Fermat number $2^{2^n} + 1$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$a(0)$, $a(1)$, $a(5)$, $a(6)$, $a(7)$ and $a(11)$ are primes. Are there any more?

```
theorem conjecture : ∃ n > 11, (a n).Prime
```

FC-OeisA108301.conjecture

A108306: Expansion of $(3x+1)/(1-3x-3x^2)$

MODULE CONTEXT: # A108306: Expansion of $(3x+1)/(1-3x-3x^2)$ This sequence satisfies the linear recurrence relation $a(0)=1$, $a(1)=6$, and $a(n) = 3a(n-1) + 3a(n-2)$ for $n \ge 2$. *References:* - [A108306](https://oeis.org/A108306)

V2-000079 | A108306: Expansion of $(3x+1)/(1-3x-3x^2)$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The conjecture: The INVERT transform of a sequence starting $(1, a, ab, ab^2, ab^3, \ldots)$ is equivalent to extracting the upper left terms of powers of the 2x2 matrix [(1,a); (1,b)].

```
theorem conjecture (a_val b_val n : ℕ) :
invertSeqD a_val b_val n = (genMatrix a_val b_val ^ n) 0 0
```

FC-OeisA108306.conjecture

Numbers n such that $\phi(n) = \phi(n + \phi(n))$

MODULE CONTEXT: # Numbers n such that $\phi(n) = \phi(n + \phi(n))$ *References:* - [A108569](https://oeis.org/A108569)

V2-000080 | Numbers n such that $\phi(n) = \phi(n + \phi(n))$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: Except for the first term all terms are even.

```
theorem conjecture : ∀ n, 0 < n → Even (a n)
```

FC-OeisA108569.conjecture

Numbers n such that the perfect deficiency of n is ≤ 10.

MODULE CONTEXT: # Numbers n such that the perfect deficiency of n is ≤ 10. The perfect deficiency of n (A109883) is the remainder after greedily subtracting from n its divisors in increasing order, skipping any divisor larger than the current remainder. *References:* - [A108864](https://oeis.org/A108864) - [A109883](https://oeis.org/A109883)

V2-000081 | Numbers n such that the perfect deficiency of n is ≤ 10. | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Is 1155 the last odd number in this sequence? (1155 is the 59th term starting from 1, corresponding to $a(58) = 1155$).

```
theorem conjecture :
∀ n > 58, Even (a n)
```

FC-OeisA108864.conjecture

Numerator of $\sum_{k=1}^n 2^k/k$.

MODULE CONTEXT: # Numerator of $\sum_{k=1}^n 2^k/k$. Conjecture: for $n > 3$, $\textrm{numerator}(-2/n + \sum_{k=1}^{n} \frac{2^k}{k}) == 0 (\textrm{mod} n^2)$ if and only if n is prime. *References:* - [A108866](https://oeis.org/A108866)

V2-000082 | Numerator of $\sum_{k=1}^n 2^k/k$. | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: for $n > 3$, $\textrm{numerator}(-2/n + \sum_{k=1}^{n} \frac{2^k}{k}) == 0 (\textrm{mod} n^2)$ if and only if n is prime.

```
theorem conjecture {n : ℕ} (hn : n > 3) :
(ratExpression n).num ≡ 0 [ZMOD (n^2 : ℤ)] ↔ n.Prime
```

FC-OeisA108866.conjecture

Fractional parts of sums of reciprocals of Catalan numbers $C(n)$

MODULE CONTEXT: # Fractional parts of sums of reciprocals of Catalan numbers $C(n)$ Catalan numbers $C(n) = \frac{1}{n+1}\binom{2n}{n}$. The sum $\sum_{i=j}^k \frac{1}{a(i)}$ of reciprocals of Catalan numbers. *References:* - [A000108](https://oeis.org/A000108) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000083 | Fractional parts of sums of reciprocals of Catalan numbers $C(n)$ | catalanReciprocalSum_fracPart_inj

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: All the rational numbers $\sum_{i=j}^k \frac{1}{a(i)}$ with $0 < \min\{2,k\} \le j \le k$ have pairwise distinct fractional parts. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem catalanReciprocalSum_fracPart_inj : ∀ {j₁ k₁ j₂ k₂ : ℕ},
IndexCond j₁ k₁ → IndexCond j₂ k₂ →
(j₁, k₁) ≠ (j₂, k₂) →
fracPart (catalanReciprocalSum j₁ k₁) ≠ fracPart (catalanReciprocalSum j₂ k₂)
```

FC-OeisA108.catalanReciprocalSum_fracPart_inj

Numerator of $\binom{6n-2}{2n} / \left(2 \binom{4n-1}{2n}\right)$

MODULE CONTEXT: # Numerator of $\binom{6n-2}{2n} / \left(2 \binom{4n-1}{2n}\right)$ The sequence $a(n)$ is the numerator of $\binom{6n-2}{2n} / \left(2 \binom{4n-1}{2n}\right)$. The OEIS entry records the conjecture that this ratio equals $\text{A005156}(n) / \text{A005156}(n-1)$, where $\text{A005156}(n)$ is the number of $(2n+1) \times (2n+1)$ vertically symmetric alternating sign matrices (VSASMs). This follows from Robbins's VSASM product formula, proved by Kuperberg (2002). *References:* - [A109074](https://oeis.org/A109074) - [A005156](https://oeis.org/A005156) - G. Kuperberg, "Symmetry classes of alternating-sign matrices under one roof," [arXiv:math/0008184](https://arxiv.org/abs/math/0008184), *Ann. of Math.* **156** (2002), 835-866.

V2-000084 | Numerator of $\binom{6n-2}{2n} / \left(2 \binom{4n-1}{2n}\right)$ | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

"It is conjectured that $\binom{6n-2}{2n} / \left(2 \binom{4n-1}{2n}\right) = \text{A005156}(n+1)/\text{A005156}(n)$." The OEIS comment indexes A005156 from 1; for the 0-indexed sequence `b` this states `frac (n + 1) = b (n + 1) / b n`. This holds by the product formula $$\text{A005156}(n) = \frac{1}{2^n} \prod_{k=1}^{n} \frac{(6k-2)!\,(2k-1)!}{(4k-1)!\,(4k-2)!}$$ conjectured by Robbins and proved by Kuperberg (2002).

```
theorem conjecture (n : ℕ) :
frac (n + 1) = (b (n + 1) : ℚ) / (b n : ℚ)
```

FC-OeisA109074.conjecture

Conjectures associated with A109227

MODULE CONTEXT: # Conjectures associated with A109227 Binary strings that have 1's where primes occur, 0's elsewhere and every term ends with the n-th prime index. Conjecture: $a(2)$ and $a(121)$ are primes. Are there any more? *References:* - [A109227](https://oeis.org/A109227)

V2-000085 | Conjectures associated with A109227 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(2)$ and $a(121)$ are primes. Are there any more?

```
theorem conjecture :
∃ n > 0, n ≠ 2 ∧ n ≠ 121 ∧ (a n).Prime
```

FC-OeisA109227.conjecture

Conjectures associated with A109671

MODULE CONTEXT: # Conjectures associated with A109671 $a(1)=1$; thereafter, $a(2n)=a(n)$, $a(2n+1)$ is the smallest positive number such that $|a(2n+1)-a(2n-1)|=a(n)$. Conjecture: Does the sequence contain every positive integer? *References:* - [A109671](https://oeis.org/A109671)

V2-000086 | Conjectures associated with A109671 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Does the sequence contain every positive integer (cf. A169741)?

```
theorem conjecture :
∀ m : ℕ, 0 < m → ∃ n : ℕ, 0 < n ∧ a n = m
```

FC-OeisA109671.conjecture

Conjectures associated with A109845

MODULE CONTEXT: # Conjectures associated with A109845 $a(1) = 2$; $a(2n)$ = lcm of all previous terms + 1; $a(2n+1)$ = lcm of all previous terms - 1. *References:* - [A109845](https://oeis.org/A109845)

V2-000087 | Conjectures associated with A109845 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: There are infinitely many primes in this sequence.

```
theorem conjecture : Set.Infinite {n : ℕ | (a n).Prime}
```

FC-OeisA109845.conjecture

Conjectures associated with A109905

MODULE CONTEXT: # Conjectures associated with A109905 $a(n)$ is the greatest prime of the form $k(n-k)+1$, where k can take values from 1 to $\lfloor n/2 \rfloor$. $a(n)=0$ if no such prime exists. *References:* - [A109905](https://oeis.org/A109905)

V2-000088 | Conjectures associated with A109905 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$a(n) = 0$ for $n = 1$, 6, 30 and 54. Are there any others?

theorem conjecture :
∃ n : ℕ, n > 0 ∧ a n = 0 ∧ n ∉ ({1, 6, 30, 54} : Finset ℕ)

FC-OeisA109905.conjecture

Conjectures associated with A109908

MODULE CONTEXT: # Conjectures associated with A109908 $a(n)$ = greatest prime of the form $k(n-k)-1$, or 0 if no such prime exists. *References:* - [A109908](https://oeis.org/A109908)

V2-000089 | Conjectures associated with A109908 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) > 0$ for $n > 3$.

theorem conjecture : ∀ n > 3, a n > 0

FC-OeisA109908.conjecture

Conjectures associated with A109909

MODULE CONTEXT: # Conjectures associated with A109909 $a(n)$ = number of primes of the form $k(n-k)-1$. *References:* - [A109909](https://oeis.org/A109909)

V2-000090 | Conjectures associated with A109909 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) > 0$ for $n > 3$.

theorem conjecture : ∀ n > 3, a n > 0

FC-OeisA109909.conjecture

Number of symbols '*' and '^' to write the canonical prime factorization of n

MODULE CONTEXT: # Number of symbols '*' and '^' to write the canonical prime factorization of n The canonical prime factorization is $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$. The written form is $p_1^{\wedge} e_1 * p_2^{\wedge} e_2 * \cdots * p_k^{\wedge} e_k$, where the $\wedge$ appears only if $e_i > 1$. $a(n) = (\text{number of distinct prime factors}) - 1 +$ $(\text{number of distinct prime factors with exponent } > 1)$. *References:* - [A110475](https://oeis.org/A110475)

V2-000091 | Number of symbols '*' and '^' to write the canonical prime factorization of n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured that $1,2,3,4,5,6,7,9,11$ are the only positive integers which cannot be represented as the sum of two elements of indices n such that $a(n) = 1$.

theorem conjecture :
∀ m > 0, m ∉ exceptionalSet ↔ ∃ x y : ℕ, a x = 1 ∧ a y = 1 ∧ m = x + y

FC-OeisA110475.conjecture

Smallest $m > 0$ such that there are no primes between nm and $n(m+1)$ inclusive.

MODULE CONTEXT: # Smallest $m > 0$ such that there are no primes between nm and $n(m+1)$ inclusive. Sierpinski's conjecture (1958) is precisely that a(n) >= n for all n. *References:* - [A110835](https://oeis.org/A110835)

V2-000092 | Smallest $m > 0$ such that there are no primes between nm and $n(m+1)$ inclusive. | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Sierpinski's conjecture (1958) is precisely that $a(n) >= n$ for all n.

theorem conjecture : ∀ n > 0, a n ≥ n

FC-OeisA110835.conjecture

Conjectures associated with A110854

MODULE CONTEXT: # Conjectures associated with A110854 $a(n) = \mathrm{prime}(2n+2) - \mathrm{prime}(2n+1) - \mathrm{prime}(2n) + \mathrm{prime}(2n-1)$, where $\mathrm{prime}(k)$ is the k-th prime number. *References:* - [A110854](https://oeis.org/A110854)

V2-000093 | Conjectures associated with A110854 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Do the absolute values cover A004275? A004275 is 1 together with the nonnegative even numbers. The conjecture asks whether every member of A004275 occurs as $|a(n)|$ for some term of the sequence.

```
theorem conjecture :
∀ d : ℕ, (d = 1 ∨ Even d) → ∃ n > 0, d = (a n).natAbs
```

FC-OeisA110854.conjecture

Integer part of $\mathrm{prime}(n)/\pi(n)$

MODULE CONTEXT: # Integer part of $\mathrm{prime}(n)/\pi(n)$ Here $\mathrm{prime}(n)$ is the n-th prime number, and $\pi(n)$ is the prime-counting function. *References:* - [A111114](https://oeis.org/A111114)

V2-000094 | Integer part of $\mathrm{prime}(n)/\pi(n)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: As $n \rightarrow \infty$, there are infinitely many n's such that $a(n)$ is greater than $a(n+1)$.

```
theorem conjecture : ∃ᶠ n in atTop, a n > a (n + 1)
```

FC-OeisA111114.conjecture

Number of refactorable numbers (A033950) $\le 10^n$

MODULE CONTEXT: # Number of refactorable numbers (A033950) $\le 10^n$ A number k is refactorable if its number of divisors, $\tau(k)$, divides k. *References:* - [A111291](https://oeis.org/A111291) - [Co99] Colton, S., *Refactorable numbers - a machine invention*. J. Integer Seq. 2 (1999), Article 99.1.2. - [Ze02] Zelinsky, J., *Tau numbers: a partial proof of a conjecture and other results*. J. Integer Seq. 5 (2002), Article 02.2.8. - [Sp85] Spiro, C., *How often is the number of divisors of n a divisor of n?* J. Number Theory 21 (1985), 81--100.

V2-000095 | Number of refactorable numbers (A033950) $\le 10^n$ | colton_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Colton's conjecture [Co99] as stated by Zelinsky [Ze02]: for every n, the number of refactorable numbers $\le n$ is at least half the number of primes $\le n$, i.e. $\pi(n) \le 2\,T(n)$. Zelinsky [Ze02] proved this for all sufficiently large n (see `conjecture`), with an explicit bound of $7.42 \cdot 10^{13}$ beyond which it holds; the remaining range is open.

```
theorem colton_conjecture : ∀ n : ℕ, Nat.primeCounting n ≤ 2 * countRefactorableNat n
```

FC-OeisA111291.colton_conjecture

V2-000096 | Number of refactorable numbers (A033950) $\le 10^n$ | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Simon Colton conjectures that the number of refactorable numbers less than x is at least $\frac{x}{2\log x}$. This is an asymptotic claim, so we state it for sufficiently large x. In this form it is a theorem: Zelinsky [Ze02, Theorem 7] proved that for every k the number of refactorable numbers $\le n$ exceeds $k \pi(n)$ for all sufficiently large n, and $\pi(n) \sim n / \log n$. It also follows from Spiro's asymptotic [Sp85], by which the count is $\frac{x}{\sqrt{\log x}} (\log \log x)^{-1 + o(1)}$. See `colton_conjecture` for the pointwise conjecture that remains open.

```
theorem conjecture : ∀ᶠ x in Filter.atTop,
(countRefactorable x : ℝ) ≥ x / (2 * Real.log x)
```

FC-OeisA111291.conjecture

Sequence related to NOR bracketings

MODULE CONTEXT: # Sequence related to NOR bracketings $$a(n) = \sum_{j=0}^{n-1} (-1)^j \binom{2j}{j} \binom{2n-j-2}{n-j-1}$$ *References:* - [A112521](https://oeis.org/A112521)

V2-000097 | Sequence related to NOR bracketings | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Starting with $n=1$, $a(n)$ is the main diagonal of the array $T(n, k)$.

```
theorem conjecture : ∀ (n : ℕ), n ≥ 1 → (a n : ℤ) = T n n
```

FC-OeisA112521.conjecture

A generalized Stern sequence

MODULE CONTEXT: # A generalized Stern sequence A112970: A generalized Stern sequence, defined by the recurrence relations: $a(2n+1) = a(n)$ and $a(2n) = a(n) + a(n-2)$ with $a(0)=1$, $a(1)=1$ and $a(n)=0$ for $n \le -1$. *References:* - [A112970](https://oeis.org/A112970)

V2-000098 | A generalized Stern sequence | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(2^n)=a(2^(n+1)+1)=\textrm{A033638}(n)$. This formalizes the equality $a(2^n) = a(2^(n+1)+1)$.

```
theorem conjecture1 (n : ℕ) : a (2^n) = a (2^(n + 1) + 1)
```

FC-OeisA112970.conjecture1

V2-000099 | A generalized Stern sequence | conjecture1_value

research solved | proof annotation: True | 官方代码 | 原文线索

Second part of the conjecture, $a(2^n)=\textrm{A033638}(n)=\lfloor n^2 / 4 \rfloor + 1$. It holds by induction on n, since $a(2^{n+1}) = a(2^n) + a(2^n - 2)$ for $n \ge 1$ and $a(2^m - 2) = \lfloor (m+1)/2 \rfloor$ for $m \ge 1$.

theorem conjecture1_value (n : ℕ) : a (2^n) = n ^ 2 / 4 + 1

FC-OeisA112970.conjecture1_value

V2-000100 | A generalized Stern sequence | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(2^n-1)=a(3*2^n-1)=1$. This formalizes the equality part $a(2^n-1) = a(3*2^n-1)$.

theorem conjecture2 (n : ℕ) : a (2^n - 1) = a (3 * 2^n - 1)

FC-OeisA112970.conjecture2

V2-000101 | A generalized Stern sequence | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(2^n-1)=a(3*2^n-1)=1$. This formalizes the value part $a(2^n-1)=1$.

theorem conjecture3 (n : ℕ) : a (2^n - 1) = 1

FC-OeisA112970.conjecture3

Number of digits of n raised to the power of the sum of the digits of n

MODULE CONTEXT: # Number of digits of n raised to the power of the sum of the digits of n *References:* - [A113010](https://oeis.org/A113010)

V2-000102 | Number of digits of n raised to the power of the sum of the digits of n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$n=1$ and 32 are two fixed points. Are there any others?

theorem conjecture : ∃ n : ℕ, a n = n ∧ n ≠ 1 ∧ n ≠ 32

FC-OeisA113010.conjecture

Number of digits of n raised to the power of the digital root of n

MODULE CONTEXT: # Number of digits of n raised to the power of the digital root of n *References:* - [A113019](https://oeis.org/A113019)

V2-000103 | Number of digits of n raised to the power of the digital root of n | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

$n=1$ and 32 are fixed points. Are there any others? Yes: 9^9 = 387420489 is also a fixed point. - [Kenta Kitamura](https://oeis.org/wiki/User:Kenta_Kitamura), Aug 14 2026

theorem conjecture : ¬ ∀ n : ℕ, a n = n → n = 1 ∨ n = 32

FC-OeisA113019.conjecture

Smallest number m such that $2^n - m$ and $2^n + m$ are primes

MODULE CONTEXT: # Smallest number m such that $2^n - m$ and $2^n + m$ are primes *References:* - [A113213](https://oeis.org/A113213)

V2-000104 | Smallest number m such that $2^n - m$ and $2^n + m$ are primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) = O(n^3)$. The source defines $a(n)$ as the least m with $2^n - m$ and $2^n + m$ prime, so it implicitly asserts that such an m exists. Since `a n = 0` when no such m exists, the existence of a prime pair is stated explicitly for all sufficiently large n.

theorem conjecture :
(∀ᶠ n : ℕ in Filter.atTop, ∃ m, (2 ^ n - m).Prime ∧ (2 ^ n + m).Prime) ∧
(fun n : ℕ => (a n : ℝ)) =O[Filter.atTop] (fun n : ℕ => (n ^ 3 : ℝ))

FC-OeisA113213.conjecture

Expansion of g.f. $-(1 - 48x^2 - 256x^3) / ((1 - 4x)(1 + 4x)(1 + 4x + 16x^2))$

MODULE CONTEXT: # Expansion of g.f. $-(1 - 48x^2 - 256x^3) / ((1 - 4x)(1 + 4x)(1 + 4x + 16x^2))$ Corresponds to m = 4 in a family of 4th-order linear recurrence sequences This sequence is defined by the linear recurrence relation with signature $(-4, 0, 64, 256)$ and initial values $a(0) = -1, a(1) = 4, a(2) = 32, a(3) = 64$. The recurrence is $a(n) = -4 a(n-1) + 64 a(n-3) + 256 a(n-4)$. *References:* - [A113250](https://oeis.org/A113250)

V2-000105 | Expansion of g.f. $-(1 - 48x^2 - 256x^3) / ((1 - 4x)(1 + 4x)(1 + 4x + 16x^2))$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(m, 2n+1)$ is a perfect square for all m (see A113249). Specialized to $m = 4$, which is A113250.

theorem conjecture : ∀ n : ℕ, IsSquare (a (2 * n + 1))

FC-OeisA113250.conjecture

Conjectures associated with A113252

MODULE CONTEXT: # Conjectures associated with A113252 Corresponds to m = 6 in a family of 4th order linear recurrence sequences This sequence is defined by the linear recurrence relation a(n) = -4 a(n-1) + 144 a(n-3) + 1296 a(n-4) for n > 3. Initial values are a(0) = -1, a(1) = 4, a(2) = 92, a(3) = 784. *References:* - [A113252](https://oeis.org/A113252)

V2-000106 | Conjectures associated with A113252 | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(m, 2n+1)$ is a perfect square for all m, n (see A113249). Specialized for $m=6$, which is A113252.

theorem conjecture : ∀ n : ℕ, IsSquare (a (2 * n + 1))

FC-OeisA113252.conjecture

Square odd-indexed terms in the recurrence $a(n) = 8a(n-1) - 8a(n-2) + 8a(n-3) - a(n-4)$

MODULE CONTEXT: # Square odd-indexed terms in the recurrence $a(n) = 8a(n-1) - 8a(n-2) + 8a(n-3) - a(n-4)$ A113254: Corresponds to $m = 8$ in a family of 4th-order linear recurrence sequences. The sequence $a(n)$ is defined by the initial conditions $a(0)=-1, a(1)=4, a(2)=176, a(3)=3136$, and the linear recurrence relation $a(n) = -4 * a (n-1) + 256 * a (n-3) + 4096 * a (n-4)$ for $n \ge 4$. *References:* - [A113254](https://oeis.org/A113254) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000107 | Square odd-indexed terms in the recurrence $a(n) = 8a(n-1) - 8a(n-2) + 8a(n-3) - a(n-4)$ | a_odd_is_square

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(8, 2n+1)$ is a perfect square for all n (see A113249). A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_odd_is_square : ∀ n : ℕ, IsSquare (a (2 * n + 1))

FC-OeisA113254.a_odd_is_square

Conjectures associated with A113255

MODULE CONTEXT: # Conjectures associated with A113255 Corresponds to m = 9 in a family of 4th-order linear recurrence sequences This sequence is defined by the recurrence relation $a(n) = -4 a(n-1) + 324 a(n-3) + 6561 a(n-4)$ for n > 3. Initial values are $a(0) = -1, a(1) = 4, a(2) = 227, a(3) = 5329$. *References:* - [A113255](https://oeis.org/A113255)

V2-000108 | Conjectures associated with A113255 | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(m, 2n+1)$ is a perfect square for all m, n (see A113249). Specialized for $m=9$, which is A113255.

theorem conjecture : ∀ n : ℕ, IsSquare (a (2 * n + 1))

FC-OeisA113255.conjecture

Ascending descending base exponent transform of squares

MODULE CONTEXT: # Ascending descending base exponent transform of squares a n is $\sum_{i=1}^n (i^2)^((n-i+1)^2)$. *References:* - [A113257](https://oeis.org/A113257)

V2-000109 | Ascending descending base exponent transform of squares | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

The smallest prime in this sequence is $a(2) = 5$. What is the next prime?

theorem conjecture1 :
conjecture1.answer = a (sInf {n : ℕ | 2 < n ∧ (a n).Prime})

FC-OeisA113257.conjecture1

V2-000110 | Ascending descending base exponent transform of squares | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

What is the first square value after 1?

theorem conjecture2 :
conjecture2.answer = a (sInf {n : ℕ | 1 < n ∧ IsSquare (a n)})

FC-OeisA113257.conjecture2

Ascending descending base exponent transform of factorials

MODULE CONTEXT: # Ascending descending base exponent transform of factorials *References:* - [A113258](https://oeis.org/A113258)

V2-000111 | Ascending descending base exponent transform of factorials | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Is there a nontrivial power after $a(4) = 5^3$?

theorem conjecture :
∃ n > 4, ∃ b > 1, ∃ e > 1, a n = b ^ e

FC-OeisA113258.conjecture

Ascending descending base exponent transform of 2^n

MODULE CONTEXT: # Ascending descending base exponent transform of 2^n *References:* - [A113271](https://oeis.org/A113271)

V2-000112 | Ascending descending base exponent transform of 2^n | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

The first prime terms in this (always odd) sequence are $a(1) = 3$, $a(3) = 41$, and $a(4) = 593$. What is the next prime? The OEIS comment currently says `$a(5) = 543$`, but this conflicts with its defining formula, b-file, and examples: the actual index-five term is the composite number `135457`. Consequently the search for the next prime must begin after index `4`, not after index `5`.

```
theorem conjecture1 :
conjecture1.answer = a (sInf {n : ℕ | 4 < n ∧ (a n).Prime})
```

FC-OeisA113271.conjecture1

Number of prime powers $q<=n$ such that also $q+2$ is a prime power

MODULE CONTEXT: # Number of prime powers $q<=n$ such that also $q+2$ is a prime power *References:* - [A113609](https://oeis.org/A113609)

V2-000113 | Number of prime powers $q<=n$ such that also $q+2$ is a prime power | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

(25,27) is the smallest pair of prime powers (q,q+2) such that both q and q+2 are not primes, conjecture: there are more (but not < 10^6).

```
theorem conjecture :
∃ q ≥ 1000000,
IsOeisPrimePower q ∧ IsOeisPrimePower (q + 2) ∧
¬ q.Prime ∧ ¬ (q + 2).Prime
```

FC-OeisA113609.conjecture

Difference between first odd semiprime $> 2^n$ and 2^n

MODULE CONTEXT: # Difference between first odd semiprime $> 2^n$ and 2^n *References:* - [A114137](https://oeis.org/A114137)

V2-000114 | Difference between first odd semiprime $> 2^n$ and 2^n | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

In this powers of 2 sequence, does 1 occur infinitely often?

```
theorem conjecture1 :
Set.Infinite {n : ℕ | a n = 1}
```

FC-OeisA114137.conjecture1

V2-000115 | Difference between first odd semiprime $> 2^n$ and 2^n | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Does every odd number occur?

```
theorem conjecture2 :
∀ k : ℕ, Odd k → ∃ n : ℕ, a n = k
```

FC-OeisA114137.conjecture2

Largest odd divisor of $a(n-1) + \textrm{prime}(n)$

MODULE CONTEXT: # Largest odd divisor of $a(n-1) + \textrm{prime}(n)$ $a(0)=0$; thereafter $a(n)$ = largest odd divisor of $a(n-1) + \textrm{prime}(n)$. *References:* - [A114216](https://oeis.org/A114216)

V2-000116 | Largest odd divisor of $a(n-1) + \textrm{prime}(n)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Is $a(33900)$ the last term equal to 1?

```
theorem conjecture :
∀ n > 33900, a n ≠ 1
```

FC-OeisA114216.conjecture

Numerator of $\zeta(4n)/\zeta(2n)^2$ (with $a(0)=2$ instead of -2)

MODULE CONTEXT: # Numerator of $\zeta(4n)/\zeta(2n)^2$ (with $a(0)=2$ instead of -2) The ratio $\zeta(4n)/\zeta(2n)^2$ for $n \ge 1$ is the rational number $$ Q_n = -2 \frac{B_{4n}}{B_{2n}^2 \binom{4n}{2n}} $$ where B_k is the k-th Bernoulli number. The sequence $a(n)$ is the numerator of Q_n, with $a(0)$ defined as 2. *References:* - [A114362](https://oeis.org/A114362)

V2-000117 | Numerator of $\zeta(4n)/\zeta(2n)^2$ (with $a(0)=2$ instead of -2) | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: if an integer $n > 1$ is odd, then $\zeta(2n)/\zeta(n)^2$ is irrational. Cf. W. Kohnen (link) and my conjecture in A348829. - Thomas Ordowski, Jan 05 2022

theorem conjecture1 (n : ℕ) (hn_gt_one : 1 < n) (hn_odd : Odd n) :
Irrational ((riemannZeta (2 * n : ℂ) / (riemannZeta (n : ℂ)) ^ 2).re)

FC-OeisA114362.conjecture1

V2-000118 | Numerator of $\zeta(4n)/\zeta(2n)^2$ (with $a(0)=2$ instead of -2) | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $\frac{1 - t(n)}{1 + t(n)} = \frac{1}{2^n} + \frac{1}{3^n} + \frac{1}{5^n} + \frac{1}{7^n} + O(\frac{1}{11^n})$, where $t(n) = \zeta(2n)/\zeta(n)^2$. Cf. A348829. - Thomas Ordowski, Nov 13 2022

theorem conjecture2 :
(fun n : ℕ => (1 - t n) / (1 + t n) -
(1 / (2:ℝ)^n + 1 / (3:ℝ)^n + 1 / (5:ℝ)^n + 1 / (7:ℝ)^n))
=O[atTop] (fun n : ℕ => 1 / (11:ℝ)^n)

FC-OeisA114362.conjecture2

$a(n) = 2^(2^n)$

MODULE CONTEXT: # $a(n) = 2^(2^n)$ *References:* - [A001146](https://oeis.org/A001146)

V2-000119 | $a(n) = 2^(2^n)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

I conjecture that { $a(n)$; $n>1$ } are the numbers such that n^4-1 divides 2^n-1, intersection of A247219 and A247165. - M. F. Hasler, Jul 25 2015 This formalizes the reverse direction.

theorem conjecture :
∀ k : ℕ, ((k^4 - 1) : ℕ) | (2^k - 1 : ℕ) → k > 1 → ∃ n : ℕ, 2 ≤ n ∧ k = a n

FC-OeisA1146.conjecture

V2-000120 | $a(n) = 2^(2^n)$ | divisibility_fact

research solved | proof annotation: False | 官方代码 | 原文线索

The forward direction: if $k = a(n)$ for $n \ge 2$, then $k^4 - 1$ divides $2^k - 1$.

theorem divisibility_fact (n : ℕ) (hn : 2 ≤ n) : (((a n)^4 - 1) : ℕ) | (2^(a n) - 1 : ℕ)

FC-OeisA1146.divisibility_fact

Each term is previous term plus floor of harmonic mean of two previous terms.

MODULE CONTEXT: # Each term is previous term plus floor of harmonic mean of two previous terms. $a(1) = 1, a(2) = 2$ and $a(n) = a(n-1) + \lfloor \frac{2 a(n-1) a(n-2)}{a(n-1) + a(n-2)} \rfloor$ for $n \ge 3$. *References:* - [A114831](https://oeis.org/A114831)

V2-000121 | Each term is previous term plus floor of harmonic mean of two previous terms. | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture based on OEIS A114831: What is this sequence, asymptotically? If the limit exists, the ratio of consecutive terms must tend to $\sqrt{3}$: $$ \lim_{n \to \infty} \frac{a(n+1)}{a(n)} = \sqrt{3}. $$ That's because $a(n)$ is positive, monotonically increasing ($a(n) > a(n-1)$) and $a(n+2) \geq a(n+1) + a(n)$. So $a(n)$ grows exponentially, at least as fast as the Fibonnaci numbers. Assuming $\frac{a(n+1)}{a(n)}$ tend to a limit L, solving for L in the definition of $a(n)$ gives $L=\sqrt{3}$.

theorem conjecture3 :
Tendsto (fun n ↦ (a (n + 1) : ℝ) / (a n : ℝ)) atTop (nhds (Real.sqrt 3))

FC-OeisA114831.conjecture3

Partial sums of $\binom{2n}{n}^2$

MODULE CONTEXT: # Partial sums of $\binom{2n}{n}^2$ $$a(n) = \sum_{k=0}^n \binom{2k}{k}^2$$ *References:* - [A115257](https://oeis.org/A115257)

V2-000122 | Partial sums of $\binom{2n}{n}^2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: For any positive integer n, the polynomials Sum_{k=0}^n binomial(2k,k)^2*x^k and Sum_{k=0}^n binomial(2k,k)^2*x^k/(k+1) are irreducible over the field of rational numbers. - Zhi-Wei Sun, Mar 23 2013

theorem conjecture :
∀ (n : ℕ), 1 ≤ n → Irreducible (polyP n) ∧ Irreducible (polyQ n)

FC-OeisA115257.conjecture

$a(n)$ = the number of values of $k <= 10^n$ such that $\sqrt{k(k+1)(k+2)(k+3)+1}$ is prime

MODULE CONTEXT: # $a(n)$ = the number of values of $k <= 10^n$ such that $\sqrt{k(k+1)(k+2)(k+3)+1}$ is prime Since $\sqrt{k(k+1)(k+2)(k+3)+1} = k^2 + 3k + 1$, $a(n) = \#\{k \in \mathbb{N} \mid 1 \le k \le 10^n \land (k^2 + 3k + 1) \text{ is prime} \}.$ *References:* - [A115366](https://oeis.org/A115366)

V2-000123 | $a(n)$ = the number of values of $k <= 10^n$ such that $\sqrt{k(k+1)(k+2)(k+3)+1}$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n)/A006880(n) \rightarrow 1.77...$ where A006880(n) is the number of primes $\le 10^n$.

```
theorem conjecture :
∃ L : ℝ,
Tendsto (fun n : ℕ => (a n : ℝ) / (Nat.primeCounting' (10 ^ n) : ℝ)) atTop (nhds L) ∧
1.77 ≤ L ∧ L ≤ 1.78
```

FC-OeisA115366.conjecture

$a(n)$ is the integer whose decimal digits are the first $n+1$ decimal digits of π

MODULE CONTEXT: # $a(n)$ is the integer whose decimal digits are the first $n+1$ decimal digits of π *References:* - [A011545](https://oeis.org/A011545)

V2-000124 | $a(n)$ is the integer whose decimal digits are the first $n+1$ decimal digits of π | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Wolfgang Haken (1977) conjectured that no term of this sequence is a perfect square, and estimated the probability that this conjecture is false to be smaller than 10^-9.

```
theorem conjecture1 : ∀ n, ¬ IsSquare (a n)
```

FC-OeisA11545.conjecture1

V2-000125 | $a(n)$ is the integer whose decimal digits are the first $n+1$ decimal digits of π | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Number of collisions occurring in a system consisting of an infinitely massive, rigid wall at the origin, a ball with mass m stationary at position $x_1 > 0$, and a ball with mass $(10^2n)m$ at position $x_2 > x_1$ and rolling toward the origin, assuming perfectly elastic collisions and no friction. Strictly speaking, this property, which is equivalent to the statement that the interval $(m\pi, \pi/\textrm{arctan}(1/m))$ does not contain an integer for all $m = 10^n$, is not known to be true for sure. In other words, we do not know for certain that A332045 does not contain a power of 10.

```
theorem conjecture2 :
∀ n : ℕ, ¬ ∃ (k : ℤ),
(Real.pi * (10 : ℝ) ^ n.cast < k.cast) ∧
(k.cast < Real.pi / Real.arctan (1 / (10 : ℝ) ^ n.cast))
```

FC-OeisA11545.conjecture2

Sum of squares of divisors of n

MODULE CONTEXT: # Sum of squares of divisors of n *References:* - [A001157](https://oeis.org/A001157)

V2-000126 | Sum of squares of divisors of n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: For each k = 2,3,..., all the rational numbers $\frac{\sigma_k(n)}{n^k} = \sum_{d|n} \frac{1}{d^k}$ (n = 1,2,3,...) have pairwise distinct fractional parts. - Zhi-Wei Sun, Oct 15 2015

```
theorem conjecture :
∀ k : ℕ, 2 ≤ k →
∀ n₁ n₂ : ℕ, 0 < n₁ → 0 < n₂ → n₁ ≠ n₂ →
Int.fract (↑((sigma k) n₁) / (↑n₁ ^ k : ℚ)) ≠
Int.fract (↑((sigma k) n₂) / (↑n₂ ^ k : ℚ))
```

FC-OeisA1157.conjecture

$a(n) = \sum_{j=1}^{n} (3^j + (-2)^j)$

MODULE CONTEXT: # $a(n) = \sum_{j=1}^{n} (3^j + (-2)^j)$ *References:* - [A116150](https://oeis.org/A116150)

V2-000127 | $a(n) = \sum_{j=1}^{n} (3^j + (-2)^j)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

First primes are $a(11) = 264353$ and $a(17) = 193622861$. Additional primes: $a(71)$, $a(91)$, $a(431)$. What is the next prime?

```
theorem conjecture : conjecture.answer = a (sInf {n : ℕ | 431 < n ∧ (a n).Prime})
```

FC-OeisA116150.conjecture

Determinants of 2 X 2 matrices of non-overlapping blocks of 4 consecutive primes

MODULE CONTEXT: # Determinants of 2 X 2 matrices of non-overlapping blocks of 4 consecutive primes $a(n) = p_{4n-3}p_{4n} - p_{4n-2}p_{4n-1}$ where p_k is the k-th prime number (1-indexed). *References:* - [A117027](https://oeis.org/A117027)

V2-000128 | Determinants of 2 X 2 matrices of non-overlapping blocks of 4 consecutive primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

This suggests the ratio is approaching a limit close to 0.87. Formalized as: The sequence of ratios $P(N)/Neg(N)$ converges to a limit L, and L is in the interval (0.8, 0.9).

theorem conjecture :
∃ L : ℝ, Tendsto ratioSeq atTop (nhds L) ∧ 0.8 < L ∧ L < 0.9

FC-OeisA117027.conjecture

Number of primes in n-th row of triangle $k^2 - k + p_n$

MODULE CONTEXT: # Number of primes in n-th row of triangle $k^2 - k + p_n$ $a(n)$ is the number of primes in the n-th row of the triangle $T(n, k) = k^2 - k + p_n$ for $1 \le k \le n$, where p_n is the n-th prime ($p_1=2, p_2=3, \dots$). *References:* - [A117531](https://oeis.org/A117531)

V2-000129 | Number of primes in n-th row of triangle $k^2 - k + p_n$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) < n$ for $n > 13$.

theorem conjecture (n : ℕ) (h : n > 13) : a n < n

FC-OeisA117531.conjecture

Least k such that cyclotomic polynomial $\Phi_k(n)$ is prime

MODULE CONTEXT: # Least k such that cyclotomic polynomial $\Phi_k(n)$ is prime $a(n) = \min \{k \in \mathbb{N} \mid 0 < k \wedge \text{Prime}(|\Phi_k(n)|) \}$, where $\Phi_k(n)$ is the k-th cyclotomic polynomial evaluated at n. *References:* - [A117545](https://oeis.org/A117545)

V2-000130 | Least k such that cyclotomic polynomial $\Phi_k(n)$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Is $a(n)$ defined for all $n \ge 1$? That is, for every $n \ge 1$, does there exist $k > 0$ such that $|\Phi_k(n)|$ is prime?

theorem conjecture (n : ℕ) (hn : 0 < n) :
∃ k > 0, ((Polynomial.cyclotomic k ℤ).eval (n : ℤ)).natAbs.Prime

FC-OeisA117545.conjecture

Sum of Fermat number and Mersenne number minus 1: $2^{2^n} + 2^n - 1$

MODULE CONTEXT: # Sum of Fermat number and Mersenne number minus 1: $2^{2^n} + 2^n - 1$ Define $F(n) = 2^{2^n} + 1$ (the n-th Fermat number) and $M(n) = 2^n - 1$ (the n-th Mersenne number). Then $a(n) = F(n) + M(n) - 1 = 2^{2^n} + 2^n - 1$. *References:* - [A119563](https://oeis.org/A119563)

V2-000131 | Sum of Fermat number and Mersenne number minus 1: $2^{2^n} + 2^n - 1$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The first 5 entries are primes. Are there infinitely many primes in this sequence?

theorem conjecture : Set.Infinite {n : ℕ | (a n).Prime}

FC-OeisA119563.conjecture

Least $k \ge 1$ such that $2 \cdot n^k - 1$ is prime

MODULE CONTEXT: # Least $k \ge 1$ such that $2 \cdot n^k - 1$ is prime $a(n) = \min \{k \ge 1 \mid \text{Prime}(2 \cdot n^k - 1)\}$ for $n \ge 2$. *References:* - [A119591](https://oeis.org/A119591)

V2-000132 | Least $k \ge 1$ such that $2 \cdot n^k - 1$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Is $a(n)$ defined for all $n \ge 2$? That is, does there exist $k > 0$ such that $2 \cdot n^k - 1$ is prime?

theorem conjecture (n : ℕ) (hn : 2 ≤ n) : ∃ k > 0, (2 * n ^ k - 1).Prime

FC-OeisA119591.conjecture

Half-Fibonacci sequence

MODULE CONTEXT: # Half-Fibonacci sequence $a(0) = 1, a(1) = 3$; for $n \ge 2$, $a(n) = f(a(n-1)) + f(a(n-2))$ where $f(x) = x/2$ if x is even and $f(x) = x$ if x is odd. *References:* - [A120424](https://oeis.org/A120424)

V2-000133 | Half-Fibonacci sequence | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (1): The natural density of even terms in the sequence is $1/2$.

theorem conjecture1 :
Filter.Tendsto (fun n : ℕ => ((Finset.filter (fun k => a k % 2 = 0) (Finset.range n)).card
: ℝ) / (n : ℝ))
Filter.atTop (nhds (1 / 2 : ℝ))

FC-OeisA120424.conjecture1

V2-000134 | Half-Fibonacci sequence | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (2): There are infinitely many consecutive pairs that differ by 1.

theorem conjecture2 :
Set.Infinite {n : ℕ | a (n + 1) = a n + 1 ∨ a n = a (n + 1) + 1}

FC-OeisA120424.conjecture2

Ratio of product of GCDs to product of factorials of floor divisions

MODULE CONTEXT: # Ratio of product of GCDs to product of factorials of floor divisions $$a(n) = \frac{\prod_{j=1}^n \prod_{k=1}^n \gcd(j,k)}{\prod_{k=1}^n (\lfloor n/k \rfloor!)^k}$$ *References:* - [A129365](https://oeis.org/A129365)

V2-000135 | Ratio of product of GCDs to product of factorials of floor divisions | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (1): $a(n)$ is always an integer (the denominator divides the numerator). Proof summary: Let $c(n) = \prod_{j=1}^n \prod_{k=1}^n (j,k)$ be the numerator. Observe that $$\textrm{ord}(c(n), p) = \left\lfloor \frac{n}{p} \right\rfloor^2 + \left\lfloor \frac{n}{p^2} \right\rfloor^2 + ...$$ proven similar to Legendre's formula for $\textrm{ord}(n!, p)$. Next let $d(n) = \prod_{k=1}^n (\lfloor n/k \rfloor!)^k$ be the denominator. Observe that $$\textrm{ord}(d(n), p) = \sum_{k=1}^n k\,\textrm{ord}(\lfloor n/k \rfloor, p) = \sum_{k=1}^n \sum_{r \geq 1} k \left\lfloor \frac{\lfloor n/p^r \rfloor}{k} \right\rfloor.$$ Combining the two gives: $$\textrm{ord}(a(n), p) = \sum_{r \geq 1} \sum_{k=1}^n \left(\left\lfloor \frac{n}{p^r} \right\rfloor \textrm{mod } k\right).$$ Defining $b(m) = \sum_{k=1}^m m \textrm{ mod } k$, we obtain $\textrm{ord}(a(n), p) = \sum_{r \geq 1} b(\lfloor n/p^r \rfloor)$. As $b(m) \geq 0$, $\textrm{ord}(a(n), p)$ is non-negative for any p, so $a(n)$ is an integer.

```
theorem conjecture1 (n : ℕ) (hn : 0 < n) :
((Finset.Icc 1 n).prod fun k => (n / k).factorial ^ k) |
((Finset.Icc 1 n).prod fun j => (Finset.Icc 1 n).prod fun k => Nat.gcd j k)
```

FC-OeisA129365.conjecture1

V2-000136 | Ratio of product of GCDs to product of factorials of floor divisions | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (2): If p is a prime, then $p \mid a(n)$ if and only if $p \le n/3$. Proof summary: We use the formula shown above: $\textrm{ord}(a(n), p) = \sum_{r \geq 1} b(\lfloor n/p^r \rfloor)$ where $b(m) = \sum_{k=1}^m m \textrm{ mod } k$. If $p \mid a(n)$ then $b(\lfloor n/p^r \rfloor) ≥ 1$ for some $r ≥ 1$. Since b(m) = 0$ for $m \leq 2$, we must have $n / p^r \geq 3$. If the inequality holds for $r \geq 1$, it must be true for $r = 1$. Hence $n \geq 3p$, proving the conjecture.

```
theorem conjecture2 (n p : ℕ) (hn : 0 < n) (hp : p.Prime) :
(∃ m : ℕ, a n = m ∧ p ∣ m) ↔ p ≤ n / 3
```

FC-OeisA129365.conjecture2

V2-000137 | Ratio of product of GCDs to product of factorials of floor divisions | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (3): For each positive integer n, prime p, and $0 \le k < p$, $\mathrm{ord}_p(a(np)) = \mathrm{ord}_p(a(np + k))$. Proof summary: This follows from the formula shown above: $\textrm{ord}(a(n), p) = \sum_{r \geq 1} b(\lfloor n/p^r \rfloor)$ where $b(m) = \sum_{k=1}^m m \textrm{ mod } k$. Since $k < p$, we have $\lfloor (np + k)/p^r \rfloor = \lfloor np/p^r \rfloor$ for every $r \geq 1$, proving the claim.

```
theorem conjecture3 (n p k : ℕ) (hn : 0 < n) (hp : p.Prime) (hk : k < p) :
padicValRat p (a (n * p)) = padicValRat p (a (n * p + k))
```

FC-OeisA129365.conjecture3

V2-000138 | Ratio of product of GCDs to product of factorials of floor divisions | conjecture4

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (4): Let $b(n) = \mathrm{A004125}(n) = \sum_{k=1}^n (n \bmod k)$. Then $\mathrm{ord}_p(a(np)) = \sum_{i \ge 0} b(\lfloor n/p^i \rfloor)$. Proof summary: This follows directly from the formula shown at the beginning: $\textrm{ord}(a(n), p) = \sum_{r \geq 1} b(\lfloor n/p^r \rfloor)$ where $b(m) = \sum_{k=1}^m m \textrm{ mod } k$.

```
theorem conjecture4 (n p : ℕ) (hn : 0 < n) (hp : p.Prime) :
padicValRat p (a (n * p)) = ∑' i : ℕ, (b (n / p ^ i) : ℤ)
```

FC-OeisA129365.conjecture4

Odious primes minus evil primes among first n primes

MODULE CONTEXT: # Odious primes minus evil primes among first n primes $a(n)$ is the number of primes with odd binary weight (odious primes) among the first n primes minus the number with even binary weight (evil primes). *References:* - [A130911](https://oeis.org/A130911)

V2-000139 | Odious primes minus evil primes among first n primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Shevelev conjectures that $a(n) \ge 0$ for $n > 3$.

```
theorem conjecture (n : ℕ) (hn : 3 < n) : a n ≥ 0
```

FC-OeisA130911.conjecture

Recurrence involving LCM: $a(n) = x(n+1)/x(n) - 2$

MODULE CONTEXT: # Recurrence involving LCM: $a(n) = x(n+1)/x(n) - 2$ $a(n) = x(n+1)/x(n) - 2$ where $x(1)=1$ and $x(n) = 2 x(n-1) + \operatorname{lcm}(x(n-1),n)$ for $n > 1$. *References:* - [A135508](https://oeis.org/A135508)

V2-000140 | Recurrence involving LCM: $a(n) = x(n+1)/x(n) - 2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: For prime p such that $p-2$ is not a prime, $a(p-1) = p$. - _Bill McEachen_, Sep 26 2025

```
theorem conjecture (p : ℕ) (hp : p.Prime) (hp_twin : ¬ (p - 2).Prime) : a (p - 1) = p
```

FC-OeisA135508.conjecture

Lesser of twin primes

MODULE CONTEXT: # Lesser of twin primes Primes p such that $p+2$ is also prime. *References:* - [A001359](https://oeis.org/A001359)

V2-000141 | Lesser of twin primes | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Primes p_k such that $p_k! \equiv 1 \pmod{p_{k+1}}$ with the exception of $p_{991} = 7841$ and other unknown primes p_k for which $(p_k+1)(p_k+2)\cdots(p_{k+1}-2) \equiv 1 \pmod{p_{k+1}}$ where $p_{k+1} - p_k > 2$. A formal proof is hosted at the `formal_proof` link: a Wilson-theorem argument with a fixed-divisor trial-division sieve and a verified length-7840 certificate.

```
theorem conjecture (k : ℕ) (hk : k > 1) :
let Pk := Nat.nth Nat.Prime (k - 1)
let Pk_succ := Nat.nth Nat.Prime k
let Congruence := Pk.factorial ≡ 1 [MOD Pk_succ]
let IsLesserTwinPrime := (Pk + 2).Prime
let Wk_prod : ℕ := ∏ i ∈ Finset.Icc (Pk + 1) (Pk_succ - 2), i
Congruence ↔ (IsLesserTwinPrime ∨ (k = 991) ∨ (Pk_succ - Pk > 2 ∧ Wk_prod ≡ 1 [MOD Pk_succ]))
```

FC-OeisA1359.conjecture

Number of Abelian cubes of length $3n$ over an alphabet of size 3

MODULE CONTEXT: # Number of Abelian cubes of length $3n$ over an alphabet of size 3 An Abelian cube is a string of the form $x x' x''$ with $|x| = |x'| = |x''|$ and x is a permutation of x' and x''. The number of Abelian cubes of length $3n$ over an alphabet of size 3 is given by $$a(n) = \sum_{k=0}^n \binom{n}{k}^3 \sum_{j=0}^k \binom{k}{j}^3.$$ *References:* - [A141057](https://oeis.org/A141057)

V2-000142 | Number of Abelian cubes of length $3n$ over an alphabet of size 3 | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: the supercongruences $a(n \cdot p^k) \equiv a(n \cdot p^{k-1}) \pmod{p^{3k}}$ hold for primes $p \ge 5$ and positive integers n and k.

```
theorem conjecture1 (p k n : ℕ) (hp : p.Prime) (h_p_ge_5 : 5 ≤ p) (h_k_pos : 1 ≤ k)
(h_n_pos : 1 ≤ n) :
(a (n * p ^ k) : ℤ) ≡ a (n * p ^ (k - 1)) [ZMOD (p ^ (3 * k))]
```

FC-OeisA141057.conjecture1

V2-000143 | Number of Abelian cubes of length $3n$ over an alphabet of size 3 | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (Peter Bala, 2022): The supercongruences $a(n \cdot p^k) \equiv a(n \cdot p^{k-1}) \pmod{p^{3k}}$ hold for the integer-indexed extension $a(n)$ for all $n \in \mathbb{Z} \setminus \{0\}$, primes $p \ge 5$, and $k \ge 1$. Proved in Lean by Jensen Kohlmeyer and Liam Kruer, extending the positive-index proof by Epoch Research.

```
theorem conjecture2 (p k : ℕ) (n : ℤ) (hp : p.Prime) (h_p_ge_5 : 5 ≤ p) (h_k_pos : 1 ≤ k)
(hn : n ≠ 0) :
aInt (n * (p ^ k : ℤ)) ≡ aInt (n * (p ^ (k - 1) : ℤ)) [ZMOD (p ^ (3 * k) : ℤ)]
```

FC-OeisA141057.conjecture2

Factorial distance to nearest square

MODULE CONTEXT: # Factorial distance to nearest square The sequence is defined as $$a(n) = \mathrm{round}\left(\frac{\mathrm{round}(\sqrt{n!})}{\left|(\mathrm{round}(\sqrt{n!}))^2 - n!\right|}\right)$$ for $n \ge 2$. *References:* - [A145355](https://oeis.org/A145355)

V2-000144 | Factorial distance to nearest square | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

This sequence suggests that the distance between a factorial and the closest power is tightly bounded.

```
theorem conjecture : ∃ C : ℕ, ∀ n : ℕ, 2 ≤ n → a n ≤ C
```

FC-OeisA145355.conjecture

Collatz step differences

MODULE CONTEXT: # Collatz step differences Differences in adjacent elements of the sequence quantifying the steps needed for n to converge to 1 in the Collatz Conjecture. $$a(n) = \mathrm{A006577}(n+1) - \mathrm{A006577}(n)$$ for $n > 0$. *References:* - [A153330](https://oeis.org/A153330)

V2-000145 | Collatz step differences | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1: More than half of the terms are 0. - _Ya-Ping Lu_, May 04 2024

```
theorem conjecture1 :
1 / 2 < Filter.atTop.liminf (fun n : ℕ ↦
(((Finset.Icc 1 n).filter (fun i ↦ a i = some 0)).card : ℝ) / (n : ℝ))
```

FC-OeisA153330.conjecture1

V2-000146 | Collatz step differences | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 2: 1, 6 and 16 appear only once and 3 appears twice in the sequence, i.e., $a(1) = 1$, $a(2) = 6$, $a(4) = a(5) = 3$, and $a(8) = 16$. - _Ya-Ping Lu_, May 04 2024

```
theorem conjecture2 :
indices 1 = {1} ∧
indices 6 = {2} ∧
indices 16 = {8} ∧
indices 3 = {4, 5}
```

FC-OeisA153330.conjecture2

V2-000147 | Collatz step differences | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 3 (Ya-Ping Lu, 2024): Except 1, 3 and 6, the absolute value of all terms can be written as $5x + 8y$ for $x, y \in \mathbb{N}$. (Note: in the OEIS comment, "x and y are integers" means x and y have the same sign, i.e., $|v| = 5x + 8y$ with $x, y \ge 0$, since every integer is a $\mathbb{Z}$-linear combination of 5 and 8).

```
theorem conjecture3 (n : ℕ) (v : ℤ) (hn : 0 < n) (ha : a n = some v)
(hv : v ≠ 1 ∧ v ≠ 3 ∧ v ≠ 6) :
∃ x y : ℕ, v.natAbs = 5 * x + 8 * y
```

FC-OeisA153330.conjecture3

V2-000148 | Collatz step differences | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4 (Ya-Ping Lu, 2024): For an m that appears in the sequence, the ratio of the number of terms with value m to that of $-m$ approaches 1 as $n \to \infty$, for any $|m| \notin \{1, 3, 6, 16\}$.

```
theorem conjecture4 (m : ℤ) (hm : m.natAbs ∉ ({1, 3, 6, 16} : Finset ℕ))
(hocc : ∃ n, 0 < n ∧ a n = some m) :
Filter.atTop.Tendsto
(fun n : ℕ ↦
(((Finset.Icc 1 n).filter (fun i ↦ a i = some m)).card : ℝ) /
(((Finset.Icc 1 n).filter (fun i ↦ a i = some (-m))).card : ℝ))
(nhds 1)
```

FC-OeisA153330.conjecture4

Representations as $p + 2^x + 7 \cdot 2^y$ with $p \equiv 5 \pmod 6$

MODULE CONTEXT: # Representations as $p + 2^x + 7 \cdot 2^y$ with $p \equiv 5 \pmod 6$ Number of ways to write the n-th positive odd integer in the form $p + 2^x + 7 \cdot 2^y$ with p a prime congruent to $5 \bmod 6$ and x, y positive integers. $$a(n) = \left|\left\{(p, x, y) : p + 2^x + 7 \cdot 2^y = 2n - 1 \text{ with } p \text{ prime}, p \equiv 5 \pmod 6, x, y \in \mathbb{Z}^+\right\}\right|.$$ *References:* - [A157225](https://oeis.org/A157225) - Z.-W. Sun, "Mixed sums of primes and other terms", arXiv preprint [arXiv:0901.3075](https://arxiv.org/abs/0901.3075) [math.NT], 2009.

V2-000149 | Representations as $p + 2^x + 7 \cdot 2^y$ with $p \equiv 5 \pmod 6$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

On Feb. 24, 2009, Zhi-Wei Sun conjectured that $a(n) = 0$ if and only if $n < 11$ or $n \in \{13, 16, 992\}$; in other words, except for $25, 31, 1983$, any odd integer greater than 20 can be written as the sum of a prime congruent to $5 \bmod 6$, a positive power of 2 and seven times a positive power of 2. Answer: false, for n = 716993899 we have a(n) = 0. See T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, [arxiv/2608.11941](https://arxiv.org/pdf/2608.11941).

```
theorem conjecture :
¬ ∀ n : ℕ, 0 < n → (a n = 0 ↔ n < 11 ∨ n = 13 ∨ n = 16 ∨ n = 992)
```

FC-OeisA157225.conjecture

Representations as $p + 2^x + 11 \cdot 2^y$ with $p \equiv 1 \pmod 6$

MODULE CONTEXT: # Representations as $p + 2^x + 11 \cdot 2^y$ with $p \equiv 1 \pmod 6$ Number of ways to write the n-th positive odd integer in the form $p + 2^x + 11 \cdot 2^y$ with p a prime congruent to $1 \bmod 6$ and x, y positive integers. $$a(n) = \left|\left\{(p, x, y) : p + 2^x + 11 \cdot 2^y = 2n - 1 \text{ with } p \text{ prime}, p \equiv 1 \pmod 6, x, y \in \mathbb{Z}^+\right\}\right|.$$ *References:* - [A157237](https://oeis.org/A157237) - Z.-W. Sun, "Mixed sums of primes and other terms", arXiv preprint [arXiv:0901.3075](https://arxiv.org/abs/0901.3075) [math.NT], 2009.

V2-000150 | Representations as $p + 2^x + 11 \cdot 2^y$ with $p \equiv 1 \pmod 6$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

On Feb. 24, 2009, Zhi-Wei Sun conjectured that $a(n) = 0$ if and only if $n < 16$ or $n \in \{18, 21, 24, 51, 84, 1011, 59586\}$; in other words, except for $35, 41, 47, 101, 167, 2021, 119171$, any odd integer greater than 30 can be written as the sum of a prime congruent to $1 \bmod 6$, a positive power of 2 and eleven times a positive power of 2.

```
theorem conjecture (n : ℕ) (hn : 0 < n) :
a n = 0 ↔ n ≤ 15 ∨ n = 18 ∨ n = 21 ∨ n = 24 ∨ n = 51 ∨ n = 84 ∨ n = 1011 ∨ n = 59586
```

FC-OeisA157237.conjecture

Smallest m such that $n^3 + m^3 + 1$ is prime

MODULE CONTEXT: # Smallest m such that $n^3 + m^3 + 1$ is prime $a(n)$ is the smallest natural number $m \ge 1$ such that $n^3 + m^3 + 1$ is prime. *References:* - [A159829](https://oeis.org/A159829)

V2-000151 | Smallest m such that $n^3 + m^3 + 1$ is prime | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1: For any $k \ge 3$, there are infinitely many primes of the form $n^k + m^k$ for $n, m \ge 1$. - _Ulrich Krug_, 2009 Answer: No. - _Kenta Kitamura_, 2026

```
theorem conjecture1 :
¬ ∀ (k : ℕ), 3 ≤ k →
Set.Infinite {p : ℕ | ∃ n m : ℕ, 1 ≤ n ∧ 1 ≤ m ∧ p.Prime ∧
p = n ^ k + m ^ k}
```

FC-OeisA159829.conjecture1

V2-000152 | Smallest m such that $n^3 + m^3 + 1$ is prime | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 2: For any $k \ge 3$, there are infinitely many primes of the form $n^k + m^k + 1$ for $n, m \ge 1$. - _Ulrich Krug_, 2009

```
theorem conjecture2 (k : ℕ) (hk : 3 ≤ k) :
Set.Infinite {p : ℕ | ∃ n m : ℕ, 1 ≤ n ∧ 1 ≤ m ∧ p.Prime ∧ p = n ^ k + m ^ k + 1}
```

FC-OeisA159829.conjecture2

Ore's harmonic numbers

MODULE CONTEXT: # Ore's harmonic numbers A001599 lists the positive integers whose positive divisors have an integral harmonic mean. Equivalently, n belongs to the sequence when $\sigma_1(n) \mid n\sigma_0(n)$, where $\sigma_1(n)$ is the sum of the divisors of n and $\sigma_0(n)$ is their number. *References:* - [A001599](https://oeis.org/A001599)

V2-000153 | Ore's harmonic numbers | eq_one_of_odd_of_a

research open | proof annotation: False | 官方代码 | 原文线索

"Ore conjectured that 1 is the only odd harmonic number." *Reference:* [A001599](https://oeis.org/A001599)

```
theorem eq_one_of_odd_of_a {n : ℕ} (hn : A n) (hodd : Odd n) : n = 1
```

FC-OeisA1599.eq_one_of_odd_of_a

Number of ways to express n as sum of square, pentagonal, and hexagonal numbers

MODULE CONTEXT: # Number of ways to express n as sum of square, pentagonal, and hexagonal numbers $$a(n) = |\{(x, y, z) \in \mathbb{N}^3 : x^2 + p_5(y) + p_6(z) = n\}|$$ *References:* - [A160324](https://oeis.org/A160324)

V2-000154 | Number of ways to express n as sum of square, pentagonal, and hexagonal numbers | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

In April 2009, _Zhi-Wei Sun_ conjectured that $a(n) > 0$ for every $n = 0, 1, 2, 3, \dots$.

```
theorem conjecture1 (n : ℕ) : 0 < a n
```

FC-OeisA160324.conjecture1

V2-000155 | Number of ways to express n as sum of square, pentagonal, and hexagonal numbers | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

For each integer $m > 2$, any natural number n can be written in the form $p_{m+1}(x_1) + \cdots + p_{2m}(x_m)$ with $x_1, \dots, x_m$ nonnegative integers, where $p_k(x) = (k-2)x(x-1)/2 + x$ ($x=0,1,2,\dots$) are k-gonal numbers. - _Zhi-Wei Sun_, Aug 15 2009

```
theorem conjecture2 (m : ℕ) (hm : m > 2) (n : ℕ) :
∃ x : Fin m → ℕ, n = ∑ i : Fin m, polygonalNumber (m + (i : ℕ) + 1) (x i)
```

FC-OeisA160324.conjecture2

V2-000156 | Number of ways to express n as sum of square, pentagonal, and hexagonal numbers | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

The sequence contains every positive integer. - _Zhi-Wei Sun_, Sep 04 2009

```
theorem conjecture3 (k : ℕ) (hk : 0 < k) : ∃ n : ℕ, a n = k
```

FC-OeisA160324.conjecture3

V2-000157 | Number of ways to express n as sum of square, pentagonal, and hexagonal numbers | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (Zhi-Wei Sun, Aug 21 2009): For any integer $m > 2$, each natural number n can be expressed as $p_{m+1}(x_1) + p_{m+2}(x_2) + p_{m+3}(x_3) + r$ with $x_1, x_2, x_3 \in \mathbb{N}$ and $r \in \{0, \dots, m-3\}$.

```
theorem conjecture4 (m : ℕ) (hm : 2 < m) (n : ℕ) :
∃ x1 x2 x3 r : ℕ, r ≤ m - 3 ∧
n = polygonalNumber (m + 1) x1 + polygonalNumber (m + 2) x2 + polygonalNumber (m + 3) x3
+ r
```

FC-OeisA160324.conjecture4

V2-000158 | Number of ways to express n as sum of square, pentagonal, and hexagonal numbers | conjecture5

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (Zhi-Wei Sun, Aug 21 2009): For each integer $m > 2$, all sufficiently large integers n can be expressed in the form $p_{m+1}(x_1) + p_{m+2}(x_2) + p_{m+3}(x_3)$ with $x_1, x_2, x_3 \in \mathbb{N}$.

```
theorem conjecture5 (m : ℕ) (hm : 2 < m) :
∀ᶠ n in Filter.atTop,
∃ x1 x2 x3 : ℕ, n = polygonalNumber (m + 1) x1 + polygonalNumber (m + 2) x2 +
polygonalNumber (m + 3) x3
```

FC-OeisA160324.conjecture5

Rowland-style prime-generating recurrence

MODULE CONTEXT: # Rowland-style prime-generating recurrence The sequence is defined by $a(1) = 2$ and for $n \ge 2$: $a(n) = a(n-1) + \gcd(n, a(n-1))$ if n is even, and $a(n) = a(n-1) + \gcd(n-2, a(n-1))$ if n is odd. *References:* - [A166944](https://oeis.org/A166944) - E. S. Rowland, "A natural prime-generating recurrence", *J. Integer Sequences* **11** (2008), Article 08.2.8. - V. Shevelev, "An infinite set of generators of primes based on the Rowland idea and conjectures concerning twin primes", arXiv preprint [arXiv:0910.4676](https://arxiv.org/abs/0910.4676) [math.NT], 2009.

V2-000159 | Rowland-style prime-generating recurrence | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: Every record of differences $a(n)-a(n-1)$ more than 5 is the greater of twin primes (A006512).

```
theorem conjecture (R : ℕ) (hR : 5 < R) (hrec : IsDifferenceRecord R) :
IsGreaterTwinPrime R
```

FC-OeisA166944.conjecture

Chua's Euclidean prime sequence

MODULE CONTEXT: # Chua's Euclidean prime sequence For a product n of the preceding terms, Chua's sequence chooses the least prime dividing $d + n / d$ for some divisor d of n. The open question is whether every prime occurs. *References:* - [OEIS A167604](https://oeis.org/A167604) - Andrew R. Booker, [A variant of the Euclid--Mullin sequence containing every prime](https://arxiv.org/abs/1605.08929)

V2-000160 | Chua's Euclidean prime sequence | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Does Chua's sequence contain every prime?

```
theorem conjecture :
∀ p : ℕ, p.Prime → ∃ n ≥ 1, a n = p
```

FC-OeisA167604.conjecture

Smallest index $k > n$ such that $(p_k+p_{k+1})/(p_n+p_{n+1})$ is an integer ≥ 2

MODULE CONTEXT: # Smallest index $k > n$ such that $(p_k+p_{k+1})/(p_n+p_{n+1})$ is an integer ≥ 2 *References:* - [A167918](https://oeis.org/A167918)

V2-000161 | Smallest index $k > n$ such that $(p_k+p_{k+1})/(p_n+p_{n+1})$ is an integer ≥ 2 | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $f(n, k) = 2$ for infinitely many cases, where $k = a(n)$. No hypothesis on the existence of $a(n)$ is needed: if no suitable $k > n$ exists then $a(n) = 0$ and $S(0) = 4 < 2 S(n)$, so such n cannot be witnesses.

```
theorem conjecture1 (M : ℕ) :
∃ n : ℕ, n ≥ M ∧ n > 0 ∧ S (a n) = 2 * S n
```

FC-OeisA167918.conjecture1

V2-000162 | Smallest index $k > n$ such that $(p_k+p_{k+1})/(p_n+p_{n+1})$ is an integer ≥ 2 | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Open problem: is the ratio $f(n, k)$ bounded, where $k = a(n)$? No hypothesis on the existence of $a(n)$ is needed: if no suitable $k > n$ exists then $a(n) = 0$ and $S(0) / S(n) = 4 / S(n) = 0$ for $n > 0$, so such n do not affect boundedness. When $a(n) \neq 0$ we have $S(n) \mid S(a(n))$ by definition, so the division is exact.

```
theorem conjecture2 : ∃ C : ℕ, ∀ n : ℕ, n > 0 → S (a n) / S n ≤ C
```

FC-OeisA167918.conjecture2

V2-000163 | Smallest index $k > n$ such that $(p_k+p_{k+1})/(p_n+p_{n+1})$ is an integer ≥ 2 | conjecture_infinite

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: the sequence is infinite, that is, for every $n \geq 1$ there is some $k > n$ with $S(n) \mid S(k)$, so that $a(n)$ is defined.

theorem conjecture_infinite (n : ℕ) (hn : n > 0) : ∃ k > n, S n | S k

FC-OeisA167918.conjecture_infinite

Denominators of $\sum_{k=1}^n \frac{1}{k 2^k}$

MODULE CONTEXT: # Denominators of $\sum_{k=1}^n \frac{1}{k 2^k}$ $a(n)$ is the denominator of the sum $$\sum_{i=1}^n \frac{1}{i} \binom{2n-i-1}{i-1}$$ *References:* - [A175386](https://oeis.org/A175386) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000164 | Denominators of $\sum_{k=1}^n \frac{1}{k 2^k}$ | a_ne_one

research solved | proof annotation: True | 官方代码 | 原文线索

We conjecture that $\sum_{i=1}^{n} \frac{1}{i} \binom{2n-i-1}{i-1}$ is not an integer for $n > 1$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_ne_one (n : ℕ) (hn : 1 < n) : a n ≠ 1

FC-OeisA175386.a_ne_one

Recurrence with fourth powers of binomial coefficients

MODULE CONTEXT: # Recurrence with fourth powers of binomial coefficients The sequence is defined by $a(1) = 2$, and for $n \ge 2$, $$(2n+1)^3 a(n) = 32n^3 a(n-1) + (21n^3 + 22n^2 + 8n + 1) \binom{2n-1}{n}^4.$$ *References:* - [A176477](https://oeis.org/A176477) - Z.-W. Sun, "Open Conjectures on Congruences", arXiv preprint [arXiv:0911.5665](https://arxiv.org/abs/0911.5665) [math.NT], 2009-2011. - T. Adamczewski, "OEIS Open: How many conjectures can language models turn into theorems?", [arXiv:2608.11941](https://arxiv.org/abs/2608.11941) [cs.AI], 2026.

V2-000165 | Recurrence with fourth powers of binomial coefficients | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Each term $a(n)$ is a positive integer. - _Zhi-Wei Sun_, Apr 06 2010 Proved in the OEIS Open project, see T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, [arXiv:2608.11941](https://arxiv.org/abs/2608.11941).

theorem conjecture1 (n : ℕ) (hn : 1 ≤ n) : (a n).den = 1 ∧ 0 < a n

FC-OeisA176477.conjecture1

V2-000166 | Recurrence with fourth powers of binomial coefficients | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

$a(n)$ is odd if and only if $n = 2, 2^2, 2^3, \dots$. - _Zhi-Wei Sun_, Apr 06 2010 Proved in the OEIS Open project, see T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, [arXiv:2608.11941](https://arxiv.org/abs/2608.11941).

theorem conjecture2 (n : ℕ) (hn : 1 ≤ n) :
((a n).den = 1 ∧ Odd (a n).num) ↔ ∃ m : ℕ, 1 ≤ m ∧ n = 2 ^ m

FC-OeisA176477.conjecture2

Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$

MODULE CONTEXT: # Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ The sequence is defined by $$a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2.$$ *References:* - [A179524](https://oeis.org/A179524) - Z.-W. Sun, "Open Conjectures on Congruences", arXiv preprint [arXiv:0911.5665](https://arxiv.org/abs/0911.5665) [math.NT], 2009-2011.

V2-000167 | Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

If p is a prime with $p \equiv 1, 9 \pmod{20}$ and $p = x^2 + 5y^2$ with x, y integers, then $\sum_{k=0}^{p-1} a(k) \equiv 4x^2 - 2p \pmod{p^2}$. - _Zhi-Wei Sun_, Jul 01 2010

theorem conjecture1 (p : ℕ) (hp : p.Prime)
(h_mod : (p : ℤ) ≡ 1 [ZMOD 20] ∨ (p : ℤ) ≡ 9 [ZMOD 20])
(x y : ℤ) (h_sq : (p : ℤ) = x ^ 2 + 5 * y ^ 2) :
∑ k ∈ Finset.range p, a k ≡ 4 * x ^ 2 - 2 * (p : ℤ) [ZMOD (p : ℤ) ^ 2]

FC-OeisA179524.conjecture1

V2-000168 | Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

If p is a prime with $p \equiv 3, 7 \pmod{20}$ and $2p = x^2 + 5y^2$ with x, y integers, then $\sum_{k=0}^{p-1} a(k) \equiv 2x^2 - 2p \pmod{p^2}$. - _Zhi-Wei Sun_, Jul 01 2010

theorem conjecture2 (p : ℕ) (hp : p.Prime)
(h_mod : (p : ℤ) ≡ 3 [ZMOD 20] ∨ (p : ℤ) ≡ 7 [ZMOD 20])
(x y : ℤ) (h_sq : 2 * (p : ℤ) = x ^ 2 + 5 * y ^ 2) :
∑ k ∈ Finset.range p, a k ≡ 2 * x ^ 2 - 2 * (p : ℤ) [ZMOD (p : ℤ) ^ 2]

FC-OeisA179524.conjecture2

V2-000169 | Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

If p is a prime with $p \equiv 11, 13, 17, 19 \pmod{20}$, then $\sum_{k=0}^{p-1} a(k) \equiv 0 \pmod{p^2}$. - _Zhi-Wei Sun_, Jul 01 2010

```
theorem conjecture3 (p : ℕ) (hp : p.Prime)
(h_mod : (p : ℤ) ≡ 11 [ZMOD 20] ∨ (p : ℤ) ≡ 13 [ZMOD 20] ∨
(p : ℤ) ≡ 17 [ZMOD 20] ∨ (p : ℤ) ≡ 19 [ZMOD 20]) :
∑ k ∈ Finset.range p, a k ≡ 0 [ZMOD (p : ℤ) ^ 2]
```

FC-OeisA179524.conjecture3

V2-000170 | Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

$\sum_{k=0}^{n-1}(20k+17)a(k) \equiv 0 \pmod n$ for all $n=1,2,3,\dots$. - _Zhi-Wei Sun_, Jul 01 2010

```
theorem conjecture4 (n : ℕ) (hn : 1 ≤ n) :
(n : ℤ) ∣ ∑ k ∈ Finset.range n, ((20 * (k : ℤ) + 17) * a k)
```

FC-OeisA179524.conjecture4

V2-000171 | Central binomial sum $a(n) = \sum_{k=0}^n (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ | conjecture5

research open | proof annotation: False | 官方代码 | 原文线索

$\sum_{k=0}^{p-1}(20k+17)a(k) \equiv p(10(-1/p)+7) \pmod{p^2}$ for any odd prime p. - _Zhi-Wei Sun_, Jul 01 2010

```
theorem conjecture5 (p : ℕ) [hp : Fact p.Prime] (_hp2 : p ≠ 2) :
(∑ k ∈ Finset.range p, ((20 * (k : ℤ) + 17) * a k)) ≡
(p : ℤ) * (10 * legendreSym p (-1) + 7) [ZMOD (p : ℤ) ^ 2]
```

FC-OeisA179524.conjecture5

Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$

MODULE CONTEXT: # Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$ The sequence is defined by $$a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k.$$ *References:* - [A179537](https://oeis.org/A179537) - Z.-W. Sun, "Open Conjectures on Congruences", arXiv preprint [arXiv:0911.5665](https://arxiv.org/abs/0911.5665) [math.NT], 2009-2011.

V2-000172 | Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

If p is a prime with $(p/7) = 1$ and $p = x^2 + 7y^2$ with x, y integers, then $\sum_{k=0}^{p-1} (-1)^k a(k) \equiv 4x^2 - 2p \pmod{p^2}$. - _Zhi-Wei Sun_, Jul 17 2010

```
theorem conjecture1 (p : ℕ) [Fact p.Prime] (hp7 : p ≠ 7)
(h_leg : letI : Fact (Nat.Prime 7) := ⟨by decide⟩; legendreSym 7 p = 1)
(x y : ℤ) (h_sq : (p : ℤ) = x ^ 2 + 7 * y ^ 2) :
(∑ k ∈ Finset.range p, (-1 : ℤ) ^ k * a k) ≡
4 * x ^ 2 - 2 * (p : ℤ) [ZMOD (p : ℤ) ^ 2]
```

FC-OeisA179537.conjecture1

V2-000173 | Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

If p is a prime with $(p/7) = -1$, then $\sum_{k=0}^{p-1} (-1)^k a(k) \equiv 0 \pmod{p^2}$. - _Zhi-Wei Sun_, Jul 17 2010

```
theorem conjecture2 (p : ℕ) [Fact p.Prime] (hp7 : p ≠ 7)
(h_leg : letI : Fact (Nat.Prime 7) := ⟨by decide⟩; legendreSym 7 p = -1) :
(∑ k ∈ Finset.range p, (-1 : ℤ) ^ k * a k) ≡ 0 [ZMOD (p : ℤ) ^ 2]
```

FC-OeisA179537.conjecture2

V2-000174 | Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$ | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

$\sum_{k=0}^{n-1} (42k + 37) (-1)^k a(k) \equiv 0 \pmod n$ for all $n \ge 1$. - _Zhi-Wei Sun_, Jul 17 2010

```
theorem conjecture3 (n : ℕ) (hn : 1 ≤ n) :
(n : ℤ) ∣ ∑ k ∈ Finset.range n, ((42 * (k : ℤ) + 37) * (-1 : ℤ) ^ k * a k)
```

FC-OeisA179537.conjecture3

V2-000175 | Central binomial sum $a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n-k}{k}^2 (-16)^k$ | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

$\sum_{k=0}^{p-1} (42k + 37) (-1)^k a(k) \equiv p(21(p/7) + 16) \pmod{p^2}$ for any prime $p \ne 7$. - _Zhi-Wei Sun_, Jul 17 2010

```
theorem conjecture4 (p : ℕ) [Fact p.Prime] (_hp7 : p ≠ 7) :
letI : Fact (Nat.Prime 7) := ⟨by decide⟩
(∑ k ∈ Finset.range p, ((42 * (k : ℤ) + 37) * (-1 : ℤ) ^ k * a k)) ≡
(p : ℤ) * (21 * legendreSym 7 p + 16) [ZMOD (p : ℤ) ^ 2]
```

FC-OeisA179537.conjecture4

Difference of digit sums in base 3 and base 2

MODULE CONTEXT: # Difference of digit sums in base 3 and base 2 Difference of sums of digits of n in ternary and in binary: $$a(n) = \sum \mathrm{digits}_3(n) - \sum \mathrm{digits}_2(n).$$ *References:* - [A180017](https://oeis.org/A180017)

V2-000176 | Difference of digit sums in base 3 and base 2 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

"This sequence is positive on average, since 1/log(3) > 1/log(4). Do all integers appear infinitely often?" - Charles R Greathouse IV, Feb 07 2013

```
theorem conjecture (z : ℤ) : Set.Infinite {n : ℕ | a n = z}
```

FC-OeisA180017.conjecture

Van Eck's sequence (OEIS A181391)

MODULE CONTEXT: # Van Eck's sequence (OEIS A181391) Van Eck's sequence starts with zero. If the current value has occurred before, the next value is the distance to its most recent earlier occurrence; otherwise the next value is zero. *References:* - [OEIS A181391](https://oeis.org/A181391) - N. J. A. Sloane, [Some Open Problems](https://neilsloane.com/doc/EMMay2016.pdf#page=14)

V2-000177 | Van Eck's sequence (OEIS A181391) | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

[OEIS A181391](https://oeis.org/A181391) conjectures: "every number eventually appears."

```
theorem conjecture : Function.Surjective a
```

FC-OeisA181391.conjecture

Sum of fourth powers of Fibonacci-like binomial coefficients

MODULE CONTEXT: # Sum of fourth powers of Fibonacci-like binomial coefficients The sequence is defined by $$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}^4.$$ *References:* - [A181546](https://oeis.org/A181546)

V2-000178 | Sum of fourth powers of Fibonacci-like binomial coefficients | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Given $F(n,L) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}^L$, then $\lim_{n\to\infty} F(n+1,L)/F(n,L) = (\mathrm{Fibonacci}(L)\sqrt{5} + \mathrm{Lucas}(L))/2$ for $L \ge 0$ where $\mathrm{Fibonacci}(n) = \mathrm{A000045}(n)$ and $\mathrm{Lucas}(n) = \mathrm{A000032}(n)$.

```
theorem conjecture (L : ℕ) :
Tendsto (fun n => (F (n + 1) L : ℝ) / (F n L : ℝ)) atTop (nhds (limitValue L))
```

FC-OeisA181546.conjecture

Squares of double factorials

MODULE CONTEXT: # Squares of double factorials Squares of double factorials: $a(n) = ((2n-1)!!)^2 = (1 \cdot 3 \cdot 5 \cdots (2n-1))^2$. *References:* - [A001818](https://oeis.org/A001818) - [She, Sun, Xia, *A novel permanent identity with applications*, Theorem 1.3(ii)](https://arxiv.org/abs/2208.12167) - [Yang, Zhang, *Sun-type determinant and permanent congruences*, Proposition 17](https://arxiv.org/abs/2605.19502)

V2-000179 | Squares of double factorials | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1: For any primitive $2n$-th root ζ of unity, the permanent of the $2n \times 2n$ matrix $[m(j,k)]_{j,k=1..2n}$ coincides with $a(n) = ((2n-1)!!)^2$, where $m(j,k)$ is $(1+\zeta^{j-k})/(1-\zeta^{j-k})$ if $j \neq k$, and 1 otherwise. - Zhi-Wei Sun, Dec 21 2021 Proved by She, Sun and Xia, Theorem 1.3(ii) in the reference above. Lean formalization by Wentao Li, with AI assistance.

```
theorem conjecture1 (n : ℕ) (hn : 1 ≤ n) :
∀ (ζ : ℂ), IsPrimitiveRoot ζ (2 * n) →
Matrix.permanent (fun (i j : Fin (2 * n)) =>
if i = j then
(1 : ℂ)
else
(1 + ζ ^ (i.val - j.val : ℤ)) / (1 - ζ ^ (i.val - j.val : ℤ))
) = (a n : ℂ)
```

FC-OeisA1818.conjecture1

V2-000180 | Squares of double factorials | conjecture2

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture 2: Let p be an odd prime. Then the permanent of the $(p-1) \times (p-1)$ matrix $[f(j,k)]_{j,k=1..p-1}$ is congruent to $a((p-1)/2) = ((p-2)!!)^2 \pmod{p^2}$, where $f(j,k)$ is $(j+k)/(j-k)$ if $j \neq k$, and $f(j,k) = 1$ otherwise. - Zhi-Wei Sun, Dec 22 2021

```
theorem conjecture2 {p : ℕ} (hp : p.Prime) (h_odd : p ≠ 2) :
let N : ℕ := p - 1
let R := ZMod (p ^ 2)
let Idx := Fin N
let M : Matrix Idx Idx R := fun i j => fEntry (i.val + 1) (j.val + 1)
(M.permanent : R) = (a ((p - 1) / 2) : R)
```

FC-OeisA1818.conjecture2

Product of two consecutive primes modulo the next prime

MODULE CONTEXT: # Product of two consecutive primes modulo the next prime The sequence is defined by $$a(n) = \mathrm{prime}(n) \cdot \mathrm{prime}(n+1) \bmod \mathrm{prime}(n+2),$$ where $\mathrm{prime}(k)$ is the k-th prime number ($\mathrm{prime}(1)=2$). *References:* - [A182126](https://oeis.org/A182126)

V2-000181 | Product of two consecutive primes modulo the next prime | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: For $x > 10^9$, the most frequent value in $a(n)$, $n=1\dots x$, has form $120k$.

```
theorem conjecture1 (x : ℕ) (hx : 10 ^ 9 < x) (v₀ : ℕ) (hv : IsMostFrequent x v₀) :
120 | v₀
```

FC-OeisA182126.conjecture1

V2-000182 | Product of two consecutive primes modulo the next prime | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Let $b = \mathrm{prime}(n+2) - \mathrm{prime}(n)$ and $c = \mathrm{prime}(n+2) - \mathrm{prime}(n+1)$. Conjecture: for $n > 61$, $a(n) = b \cdot c$. - _Charles R Greathouse IV_, May 11 2012

```
theorem conjecture2 (n : ℕ) (hn : 61 < n) :
let b := prime (n + 2) - prime n
let c := prime (n + 2) - prime (n + 1)
a n = b * c
```

FC-OeisA182126.conjecture2

V2-000183 | Product of two consecutive primes modulo the next prime | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

Are 2, 7, 11, 13, 29 the only primes in this sequence? - _Hugo Pfoertner_, Sep 22 2025

```
theorem conjecture3 (n : ℕ) (hn : 0 < n) :
(a n).Prime ↔ a n ∈ ([2, 7, 11, 13, 29] : List ℕ)
```

FC-OeisA182126.conjecture3

Recurrence with bitwise XOR

MODULE CONTEXT: # Recurrence with bitwise XOR The sequence is defined by $a(0) = 0$, $a(1) = 1$, and for $n \ge 0$, $$a(n+2) = (a(n+1) \mathbin{\mathrm{XOR}} (n+2)) - a(n),$$ where XOR is the bitwise exclusive-or operator on integers. *References:* - [A182510](https://oeis.org/A182510)

V2-000184 | Recurrence with bitwise XOR | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: the sequence contains 8 zeros.

```
theorem conjecture1 : Set.ncard {n : ℕ | a n = 0} = 8
```

FC-OeisA182510.conjecture1

V2-000185 | Recurrence with bitwise XOR | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: more positive terms than negative. As $n \to \infty$, the count of positive terms is greater than the count of negative terms. The requirement that n is large enough is needed, since the claim fails for $n = 100$.

```
theorem conjecture2 :
∀ᶠ n in Filter.atTop,
((Finset.range n).filter (fun k => a k < 0)).card <
((Finset.range n).filter (fun k => 0 < a k)).card
```

FC-OeisA182510.conjecture2

Number of odd primes between n^2 and $(n+1)^2$ with $(n/p) = 1$

MODULE CONTEXT: # Number of odd primes between n^2 and $(n+1)^2$ with $(n/p) = 1$ $a(n)$ is the number of odd primes p between n^2 and $(n+1)^2$ such that the Legendre symbol $\left(\frac{n}{p}\right) = 1$. *References:* - [A185150](https://oeis.org/A185150) - Z.-W. Sun, "Conjectures involving primes and quadratic forms", arXiv preprint [arXiv:1211.1588](https://arxiv.org/abs/1211.1588) [math.NT], 2012.

V2-000186 | Number of odd primes between n^2 and $(n+1)^2$ with $(n/p) = 1$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) > 0$ for all $n > 0$. - _Zhi-Wei Sun_, Dec 29 2012

```
theorem conjecture (n : ℕ) (hn : 0 < n) : 0 < a n
```

FC-OeisA185150.conjecture

Coefficients of $\prod_{k>0} (1 - x^k/k!)$

MODULE CONTEXT: # Coefficients of $\prod_{k>0} (1 - x^k/k!)$ The sequence $a(n)$ has exponential generating function $$E(x) = \prod_{k=1}^\infty \left(1 - \frac{x^k}{k!}\right),$$ so that $a(n) = n! [x^n] \prod_{k=1}^n \left(1 - \frac{x^k}{k!}\right)$. *References:* - [A185895](https://oeis.org/A185895)

V2-000187 | Coefficients of $\prod_{k>0} (1 - x^k/k!)$ | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

$a(n)$ differs in sign from $a(n-1)$ if and only if n is a triangular number (checked up to $n = 1225 = (50 \cdot 51)/2$). - _Peter Bala_, Mar 17 2022

```
theorem conjecture1 (n : ℕ) (hn : 0 < n) :
a n * a (n - 1) < 0 ↔ IsTriangular n
```

FC-OeisA185895.conjecture1

V2-000188 | Coefficients of $\prod_{k>0} (1 - x^k/k!)$ | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

The coefficients $c(n)$ of $A(x)^2 = (\sum_{n \ge 0} a(n) x^n)^2$ differ in sign from $c(n-1)$ if and only if n is a triangular number. - _Peter Bala_, Mar 17 2022 **Proof sketch** (the kernel-checked development lives at the `formal_proof` permalink below). Write $a(n)/n! = \sum_j (-1)^j L_j(n)$, where $L_j(n)$ is the sum of $\prod_{s \in S} 1/s!$ over all sets S of j distinct positive integers with sum n. A "down map" (lowering by one the least element of S above its first gap) gives $L_{k-1}(n) \le \frac{91}{100} L_k(n)$ for all $k \ge 1$ and $n \ge k(k+1)/2$. Hence the top nonzero layer $L_m(n)$ dominates, where $m(m+1)/2 \le n < (m+1)(m+2)/2$; this reproves `conjecture1` and gives $|a(n)| \ge \frac{9}{100} n! L_m(n)$. Since $c(n) = 2a(n) + \sum_{0<k<n} a(k)a(n-k)$, the sign of $c(n)$ equals that of $a(n)$ whenever $\sum_{0<k<n} |a(k)a(n-k)| < 2|a(n)|$. For $n \le 2500$, conjecture 2 is checked in the kernel from a certified table of $a(n)$; for $n > 2500$, explicit estimates (rational arithmetic and elementary bounds only) bound the cross sum by $0.18 \cdot n! L_m(n)$, which is less than $2|a(n)|$.

```
theorem conjecture2 (n : ℕ) (hn : 0 < n) :
c n * c (n - 1) < 0 ↔ IsTriangular n
```

FC-OeisA185895.conjecture2

V2-000189 | Coefficients of $\prod_{k>0} (1 - x^k/k!)$ | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

The Gauss congruences $a(n \cdot p^k) \equiv a(n \cdot p^{k-1}) \pmod{p^k}$ hold for all primes p and positive integers n and k. - _Peter Bala_, Mar 17 2022

```
theorem conjecture3 (p : ℕ) (hp : p.Prime) (n k : ℕ) (hn : 0 < n) (hk : 0 < k) :
a (n * p ^ k) ≡ a (n * p ^ (k - 1)) [ZMOD (p : ℤ) ^ k]
```

FC-OeisA185895.conjecture3

Size of smallest subset of $\{1, 2, \dots, n\}$ with distinct subset sums

MODULE CONTEXT: # Size of smallest subset of $\{1, 2, \dots, n\}$ with distinct subset sums Size of the smallest subset S of $T = \{1,2,3,\dots,n\}$ such that $S \cdot S$ contains T, where $S \cdot S$ is the set of all products of elements of S. *References:* - [A194806](https://oeis.org/A194806) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000190 | Size of smallest subset of $\{1, 2, \dots, n\}$ with distinct subset sums | a_div_prime_counting_bounded

research solved | proof annotation: True | 官方代码 | 原文线索

Is $a(n) / \pi(n)$ bounded as $n \to \infty$? - _Robert Israel_, Jan 09 2017 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_div_prime_counting_bounded : ∃ C : ℝ,
∀ n : ℕ, 2 ≤ n → (a n : ℝ) / (Nat.primeCounting n : ℝ) ≤ C
```

FC-OeisA194806.a_div_prime_counting_bounded

Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$

MODULE CONTEXT: # Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ Integral factorial ratio sequence: $$a(n) = \frac{(30n)! n!}{(15n)! (10n)! (6n)!}$$ *References:* - [A211417](https://oeis.org/A211417) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [Me11] R. Meštrović, *Wolstenholme's theorem: Its Generalizations and Extensions in the last hundred and fifty years (1862-2012)*, [arXiv:1111.3057](https://arxiv.org/abs/1111.3057) (2011).

V2-000191 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | a_dvd_five_mul_add_one

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: "a(n)/(5*n + 1) ... [is an] integer for all n (checked up to n = 1000)." - _Peter Bala_, Aug 28 2025

```
theorem a_dvd_five_mul_add_one (n : ℕ) :
(5 * (n : ℤ) + 1) | (a n : ℤ)
```

FC-OeisA211417.a_dvd_five_mul_add_one

V2-000192 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | a_dvd_three_mul_add_one

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: "a(n)/(3*n + 1) ... [is an] integer for all n (checked up to n = 1000)." - _Peter Bala_, Aug 28 2025

theorem a_dvd_three_mul_add_one (n : ℕ) :
(3 * (n : ℤ) + 1) | (a n : ℤ)

FC-OeisA211417.a_dvd_three_mul_add_one

V2-000193 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | forty_two_mul_a_dvd_product

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: "42*a(n)/((2*n + 1)*(3*n + 1)*(5*n + 1)) [is an] integer for all n (checked up to n = 1000)." - _Peter Bala_, Aug 28 2025

theorem forty_two_mul_a_dvd_product (n : ℕ) :
((2 * (n : ℤ) + 1) * (3 * (n : ℤ) + 1) * (5 * (n : ℤ) + 1)) |
42 * (a n : ℤ)

FC-OeisA211417.forty_two_mul_a_dvd_product

V2-000194 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | general_divisibility

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: "More generally, for r >= 1, we conjecture that there exists a constant D(r) such that D(r)*a(n)/Product_{i = 1..r, i coprime to 30} (30*n - i) is integral for all n." - _Peter Bala_, Aug 28 2025 This generalizes `thirty_mul_sub_one_dvd_a` (the $r = 1$ case where $D(1) = 1$). **Proof sketch** (kernel-checked development at the `formal_proof` permalink below, which proves the non-vacuous form with the explicit positive witness $D = (r!)^{r^2}$): Legendre-valuation analysis of the Landau step function $\Delta(x) = \lfloor 30x \rfloor + \lfloor x \rfloor - \lfloor 15x \rfloor - \lfloor 10x \rfloor - \lfloor 6x \rfloor$, whose values lie in $\{0, 1\}$. Three ingredients: a unit-class rigidity lemma ($\Delta_c = 1$ for every unit class c modulo 30), a witness-uniqueness bound for prime powers $p^k > r$ (at most one factor of the divisor product is divisible by p^k), and a uniform low-layer budget for $p^k \le r$ absorbed by the witness constant.

theorem general_divisibility (r : ℕ) (hr : 1 ≤ r) :
∃ D : ℤ, 0 < D ∧ ∀ n : ℕ, (divisorProduct n r) | (D * (a n : ℤ))

FC-OeisA211417.general_divisibility

V2-000195 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | seven_mul_a_dvd_two_mul_add_one

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: "7*a(n)/(2*n + 1) ... [is an] integer for all n (checked up to n = 1000)." - _Peter Bala_, Aug 28 2025

theorem seven_mul_a_dvd_two_mul_add_one (n : ℕ) :
(2 * (n : ℤ) + 1) | 7 * (a n : ℤ)

FC-OeisA211417.seven_mul_a_dvd_two_mul_add_one

V2-000196 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | supercongruence

research solved | proof annotation: True | 官方代码 | 原文线索

Supercongruence: "a(p^k) == a(p^(k-1)) (mod p^(3*k)) for any prime p >= 5 and any positive integer k." - _Peter Bala_, Jan 24 2020 More generally, "the congruences a(n*p^k) == a(n*p^(k-1)) (mod p^(3*k)) may hold for any prime p >= 5 and any positive integers n and k." The same OEIS comment (Peter Bala, Jan 24 2020) already sketches an informal proof of the supercongruence: write $a(n) = \binom{30n}{15n}\binom{15n}{5n} / \binom{6n}{n}$ and apply [Me11, eq. (39), p. 12]. Solved: a Lean 4 proof, derived from the Atlas proofs in [facebookresearch/atlas-lean](https://github.com/facebookresearch/atlas-lean), is linked in `formal_proof`.

theorem supercongruence (p k : ℕ) (hp : p.Prime) (hp5 : 5 ≤ p) (hk : 0 < k) :
(p : ℤ) ^ (3 * k) | ((a (p ^ k) : ℤ) - (a (p ^ (k - 1)) : ℤ))

FC-OeisA211417.supercongruence

V2-000197 | Integrality and supercongruences of the factorial ratio $\frac{(6n)! n!}{(3n)! (2n)!^2}$ | thirty_mul_sub_one_dvd_a

research solved | proof annotation: True | 官方代码 | 原文线索

It appears that $a(n)/(30n - 1)$ is integral for all n (checked up to $n = 1000$). - _Peter Bala_, Aug 28 2025 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem thirty_mul_sub_one_dvd_a (n : ℕ) : (30 * (n : ℤ) - 1) | (a n : ℤ)

FC-OeisA211417.thirty_mul_sub_one_dvd_a

A double sum of five binomial factors

MODULE CONTEXT: # A double sum of five binomial factors The sequence is defined by $a(n) =\sum_{j,k=0}^{n}\binom{n}{j}^2\binom{n}{k}^2\binom{n+j}{n}\binom{n+k}{n}\binom{j+k}{n}$. *References:* - [A220119](https://oeis.org/A220119) - [Li26] Wentao Li, [A proof of the OEIS A220119 conjecture](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-proofs/A220119/PROOF.md), 2026.

V2-000198 | A double sum of five binomial factors | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1: "Conjecture: For $n > 0$, $a(n)$ is always divisible by $(n+1) * (n+2)$." - F. Chapoton, Apr 23 2026. The restriction $n > 0$ is necessary since $a(0) = 1$. Proved and formalized in Lean by Wentao Li, with AI assistance; see [Li26].

theorem conjecture_1 (n : ℕ) (hn : 0 < n) : (n + 1) * (n + 2) | a n

FC-OeisA220119.conjecture_1

A nonlinear recurrence sequence

MODULE CONTEXT: # A nonlinear recurrence sequence For even n, $a(n+2)$ is the greatest integer such that $a(n+2)/a(n+1) < a(n+1)/a(n)$; for odd n, the least integer such that $a(n+2)/a(n+1) > a(n+1)/a(n)$; $a(0) = 4, a(1) = 16$. *References:* - [A022030](https://oeis.org/A022030)

V2-000199 | A nonlinear recurrence sequence | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) = 4 a(n-1) - a(n-3) + a(n-4)$. - Colin Barker, Feb 16 2012

```
theorem conjecture (n : ℕ) (hn : 4 ≤ n) :
a n = 4 * a (n - 1) - a (n - 3) + a (n - 4)
```

FC-OeisA22030.conjecture

Existence of integers k with $k^2 \operatorname{XOR} (k+1)^2 = (2n+1)^2$

MODULE CONTEXT: # Existence of integers k with $k^2 \operatorname{XOR} (k+1)^2 = (2n+1)^2$ a: $a(n) = \text{least } k \text{ such that } \sqrt{k^2 \operatorname{XOR} (k+1)^2} = 2n+1$, $a(n) = -1 \text{ if there is no such } k$. This is equivalent to finding the smallest $k \in \mathbb{N}$ such that $k^2 \oplus (k+1)^2 = (2n+1)^2$. We use the set infimum (sInf) to denote the least element of the set of natural numbers satisfying the condition. Since Mathlib's `sInf` on a subset of `ℕ` gives a result in `ℕ`, this definition is only completely faithful to the OEIS when the set is non-empty. The OEIS definition implies that the set of k's is non-empty for all n. *References:* - [A224515](https://oeis.org/A224515) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000200 | Existence of integers k with $k^2 \operatorname{XOR} (k+1)^2 = (2n+1)^2$ | exists_xor_sq_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) \ge 0$, i.e., for every n there exists k such that $\sqrt{k^2 \oplus (k+1)^2} = 2n+1$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem exists_xor_sq_eq (n : ℕ) : ∃ k : ℕ, Nat.xor (k ^ 2) ((k + 1) ^ 2) = (2 * n + 1) ^ 2
```

FC-OeisA224515.exists_xor_sq_eq

Number of squares $\bmod n$

MODULE CONTEXT: # Number of squares $\bmod n$ The number of squares modulo n. This is the cardinality of the set $\{k^2 \bmod n \mid k \in \{0, 1, \dots, n-1\}\}$. *References:* - [A000224](https://oeis.org/A000224)

V2-000201 | Number of squares $\bmod n$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$ if and only if n is an odd prime. - Thomas Ordowski, Jun 08 2017

```
theorem conjecture (n : ℕ) (hn : 1 < n) :
(n.Prime ∧ n ≠ 2) ↔ n ^ 2 ≡ 1 [MOD a n * (a n - 1)]
```

FC-OeisA224.conjecture

Representation of sequence terms by harmonic numbers $\lfloor \frac{1}{2H(n) - H(n^2+n-1) - \gamma} \rfloor$

MODULE CONTEXT: # Representation of sequence terms by harmonic numbers $\lfloor \frac{1}{2H(n) - H(n^2+n-1) - \gamma} \rfloor$ A227582: Expansion of $(2+3x+2x^2+2x^3+3x^4+x^5-x^6)/(1-2x+x^2-x^5+2x^6-x^7)$. The sequence satisfies the linear recurrence: $$a(n) = 2 a(n-1) - a(n-2) + a(n-5) - 2 a(n-6) + a(n-7)$$ with initial values $a(1) = 2, a(2) = 7, a(3) = 14, a(4) = 23, a(5) = 35, a(6) = 50, a(7) = 67$. The sequence is 1-indexed in OEIS, so $a(n)$ is the $(n-1)$-th term of the 0-indexed solution. *References:* - [A227582](https://oeis.org/A227582) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000202 | Representation of sequence terms by harmonic numbers $\lfloor \frac{1}{2H(n) - H(n^2+n-1) - \gamma} \rfloor$ | a_eq_floor_harmonic_expr

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (from A227581): $a(n) = \lfloor 1/(2 H(n) - H(n^2 + n - 1) - \gamma) \rfloor$, where H denotes harmonic numbers and γ denotes the Euler-Mascheroni constant. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_eq_floor_harmonic_expr (n : ℕ) (hn : 0 < n) :
a n = (Int.floor (1 / (2 * (↑(harmonic n) : ℝ) -
(↑(harmonic (n * n + n - 1)) : ℝ) - Real.eulerMascheroniConstant))).toNat
```

FC-OeisA227582.a_eq_floor_harmonic_expr

Eighth root of the generating function of an Apéry-like sequence

MODULE CONTEXT: # Eighth root of the generating function of an Apéry-like sequence The auxiliary sequence used for the Hankel matrix, defined as $$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$$ Determinant of the $(n+1) \times (n+1)$ Hankel-type matrix with (i,j)-entry equal to A005259$(i+j)$ for all $i,j = 0,\dots,n$. The entry function A005259 is taken to be $\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$. *References:* - [A228143](https://oeis.org/A228143) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A005259](https://oeis.org/A005259)

V2-000203 | Eighth root of the generating function of an Apéry-like sequence | exists_power_series_eighth_pow_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: if $A(x) = 1 + 48x + 161856 x^2 + \cdots$ denotes the o.g.f. then $A(x/3)^{1/8}$ has integer coefficients. - _Peter Bala_, Apr 22 2018 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem exists_power_series_eighth_pow_eq : ∃ C : PowerSeries ℤ,
(PowerSeries.map (Int.castRingHom ℚ)) (C ^ 8) = ogfAScaled
```

FC-OeisA228143.exists_power_series_eighth_pow_eq

Numbers n such that $n^2 + \pi(n)$ is prime

MODULE CONTEXT: # Numbers n such that $n^2 + \pi(n)$ is prime *References:* - [A228828](https://oeis.org/A228828)

V2-000204 | Numbers n such that $n^2 + \pi(n)$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: the sequence A228828 is infinite.

```
theorem conjecture : {a n | n}.Infinite
```

FC-OeisA228828.conjecture

Sum of two numbers with prime conditions

MODULE CONTEXT: # Sum of two numbers with prime conditions Number of ways to write $n = x+y$, for $x,y > 0$ such that $2^x + y$ is prime. Zhi-Wei Sun has offered a \$1000 prize for the first proof. *References:* - [A231201](https://oeis.org/A231201) - Zhi-Wei Sun, "Table of n, a(n) for n = 1..10000", "Write n = k + m with 2^k + m prime", a message to Number Theory List, Nov. 16, 2013, "On a^n+ bn modulo m", arXiv:1312.1166 [math.NT], 2013-2014, "Problems on combinatorial properties of primes", arXiv:1402.6641 [math.NT], 2014-2015.

V2-000205 | Sum of two numbers with prime conditions | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The conjecture for sequence A231201: for any $n > 1$, there exist $x, y > 0$ such that $n = x + y$ and $2^x + y$ is prime.

```
theorem conjecture (n : ℕ) (hn : 1 < n) : A n
```

FC-OeisA231201.conjecture

Representations with prime conditions

MODULE CONTEXT: # Representations with prime conditions Any integer $n > 1$ can be written as $x + y$ with $x, y > 0$ such that both $x + ny$ and $x^2 + ny^2$ are prime. Zhi-Wei Sun has offered a \$200 prize for the first proof. *References:* - [A232174](https://oeis.org/A232174) - Z.-W. Sun, "Conjectures on representations involving primes," in: M. Nathanson (ed.), Combinatorial and Additive Number Theory II: CANT, Springer Proc. in Math. & Stat., Vol. 220, Springer, 2017, pp. 279-310. https://arxiv.org/abs/1211.1588 - D.A. Cox, "Primes of the Form x² + ny²," John Wiley & Sons, 1989.

V2-000206 | Representations with prime conditions | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A232174): Any integer $n > 1$ can be written as $x + y$ with $x, y > 0$ such that both $x + ny$ and $x^2 + ny^2$ are prime.

```
theorem conjecture (n : ℕ) (hn : 1 < n) : A n
```

FC-OeisA232174.conjecture

Multiplicative order of 2 mod $2n+1$

MODULE CONTEXT: # Multiplicative order of 2 mod $2n+1$ The multiplicative order of 2 modulo $2n+1$. In other words, the least $m > 0$ such that $2n+1$ divides $2^m - 1$. *References:* - [A002326](https://oeis.org/A002326)

V2-000207 | Multiplicative order of 2 mod $2n+1$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

If p is an odd prime then $a((p^3-1)/2) = p \cdot a((p^2-1)/2)$. Because otherwise $a((p^3-1)/2) < p \cdot a((p^2-1)/2)$ iff $a((p^3-1)/2) = a((p-1)/2)$ for a prime p. Equivalently p^3 divides $2^{p-1}-1$, but no such prime p is known. - Thomas Ordowski, Feb 10 2014

```
theorem conjecture1 (p : ℕ) (hp : p.Prime) (hp_odd : p ≠ 2) :
a ((p ^ 3 - 1) / 2) = p * a ((p ^ 2 - 1) / 2)
```

FC-OeisA2326.conjecture1

V2-000208 | Multiplicative order of 2 mod $2n+1$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

A generalization of the previous conjecture: For each $k \ge 2$, if p is an odd prime then $a((p^{k+1}-1)/2) = p \cdot a((p^k-1)/2)$. Computer testing of this generalized conjecture shows that there is no counterexample for k and p both up to 1000. - [Ahmad J. Masad](https://oeis.org/wiki/User:Ahmad_J._Masad), Oct 17 2020 The two are in fact equivalent. Its $k = 2$ instance is `conjecture1`. Conversely, write $d = \operatorname{ord}_{p^2}(2)$. `conjecture1` for p says $v_p(2^d - 1) = 2$, and lifting the exponent for the odd prime p then gives $v_p(2^{d p^t} - 1) = 2 + t$, hence $\operatorname{ord}_{p^{2+t}}(2) = d p^t$ for every $t \ge 0$, which is every instance for that p.

```
theorem conjecture2 (k : ℕ) (hk : 2 ≤ k) (p : ℕ) (hp : p.Prime) (hp_odd : p ≠ 2) :
a ((p ^ (k + 1) - 1) / 2) = p * a ((p ^ k - 1) / 2)
```

FC-OeisA2326.conjecture2

Number of parts in the symmetric representation of $\sigma(n)$

MODULE CONTEXT: # Number of parts in the symmetric representation of $\sigma(n)$ Number of parts in the symmetric representation of $\sigma(n)$. $a(n)$ is 1 plus the number of pairs (d_k, d_{k+1}) of consecutive divisors of n such that d_{k+1} is odd and $d_{k+1} \ge 2 d_k$. The formula used is $1 + |\{(d_k, d_{k+1}) \in \text{consecutive pairs of divisors of } n \mid d_{k+1} \text{ is odd and } d_{k+1} \ge 2 d_k\}|$, which is a known characterization of the sequence. *References:* - [A237271](https://oeis.org/A237271) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000209 | Number of parts in the symmetric representation of $\sigma(n)$ | a_odd_prime_pow

research solved | proof annotation: True | 官方代码 | 原文线索

Theorem: "a(p^k) = k + 1, where p is an odd prime and k >= 0." - _Hartmut F. W. Hoft_, Dec 26 2016

```
theorem a_odd_prime_pow (p k : ℕ) (hp : p.Prime) (ho : Odd p) :
a (p ^ k) = k + 1
```

FC-OeisA237271.a_odd_prime_pow

V2-000210 | Number of parts in the symmetric representation of $\sigma(n)$ | conjecture_2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 2: "a(n) is the number of 2-dense sublists of divisors of n. We call '2-dense sublists of divisors of n' to the maximal sublists of divisors of n whose terms increase by a factor of at most 2." - _Omar E. Pol_, Jul 31 2025 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem conjecture_2 (n : ℕ) : a n = num2DenseSublists n
```

FC-OeisA237271.conjecture_2

V2-000211 | Number of parts in the symmetric representation of $\sigma(n)$ | conjecture_3

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture 3: "a(n) is the number of divisors p of n such that p is greater than twice the adjacent previous divisor of n. The divisors p give the n-th row of A379288." - _Omar E. Pol_, Aug 02 2025 Note: this is equivalent to `a_eq_num2DenseSublists` (Conjecture 2), since the divisors that start a new 2-dense sublist are exactly those greater than twice their predecessor (plus the smallest divisor).

```
theorem conjecture_3 (n : ℕ) :
a n = 1 + ((sortedDivisorsList n).zip (sortedDivisorsList n).tail).countP
fun pair => pair.snd > 2 * pair.fst :=
conjecture_2 n

/--
Conjecture 4: "a(A000290(n)) is odd." - _Omar E. Pol_, Oct 21 2025

That is, the number of parts in the symmetric representation of $\sigma(n^2)$ is always odd.
-/
@[category research solved, AMS 11,
formal_proof using lean4 at
"https://github.com/KitaKen1/oeis-a237271-square-hexagonal-parity/blob/430c09114d4ad3a3a2654b9816c8bfdc3cdf38de/lean/OeisA237271ParityFC.lean#L19-L24"]
theorem conjecture_4 (n : ℕ) (hn : 0 < n) : Odd (a (n ^ 2))
```

FC-OeisA237271.conjecture_3

V2-000212 | Number of parts in the symmetric representation of $\sigma(n)$ | conjecture_4

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 4: "a(A000290(n)) is odd." - _Omar E. Pol_, Oct 21 2025 That is, the number of parts in the symmetric representation of $\sigma(n^2)$ is always odd.

```
theorem conjecture_4 (n : ℕ) (hn : 0 < n) : Odd (a (n ^ 2))
```

FC-OeisA237271.conjecture_4

V2-000213 | Number of parts in the symmetric representation of $\sigma(n)$ | conjecture_5

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 5: "a(A000384(n)) is odd." - _Omar E. Pol_, Oct 21 2025 That is, the number of parts in the symmetric representation of $\sigma(n(2n-1))$ is always odd, where $n(2n-1)$ is the n-th hexagonal number.

theorem conjecture_5 (n : ℕ) (hn : 0 < n) : Odd (a (n * (2 * n - 1)))

FC-OeisA237271.conjecture_5

V2-000214 | Number of parts in the symmetric representation of $\sigma(n)$ | observation_carmichael

research solved | proof annotation: True | 官方代码 | 原文线索

Observation: "a(A002997(n)) >= 3, at least for 1 <= n <= 10000." - _Omar E. Pol_, Oct 21 2025 That is, $a(k) \ge 3$ for every Carmichael number k. A002997 is the sequence of Carmichael numbers: the composite numbers k such that $b^{k-1} \equiv 1 \pmod k$ for every b coprime to k. This is `IsCarmichael`, which also forces k to be composite. The observation holds for every odd composite k, so in particular for every Carmichael number. Write p for the least prime factor of k and $q = k/p$. The two smallest divisors of k are 1 and p, and the two largest are q and k, so $(1, p)$ and (q, k) are both consecutive pairs in the sorted divisor list, and they are distinct because $1 < q$. Both qualify: p and k are odd, $p \ge 2 \cdot 1$, and $k = pq \ge 3q > 2q$. Hence the count is at least 2 and $a(k) \ge 3$. A Carmichael number is odd: it is composite by the $b = 1$ case of `IsCarmichael`, so $k > 2$, and if k were even then the coprime base $k - 1 \equiv -1$ would give $k \mid (-1)^{k-1} - 1 = -2$.

theorem observation_carmichael (k : ℕ) (hk : IsCarmichael k) :
3 ≤ a k

FC-OeisA237271.observation_carmichael

Primitive roots of the form k² + 1

MODULE CONTEXT: # Primitive roots of the form k² + 1 Every prime p has a primitive root $0 < g < p$ of the form $k^2 + 1$, where k is an integer. Zhi-Wei Sun has offered a prize of RMB 2,000 for the first proof. *References:* - [A239957](https://oeis.org/A239957) - Z.-W. Sun, "New observations on primitive roots modulo primes," arXiv:1405.0290 [math.NT], 2014.

V2-000215 | Primitive roots of the form k² + 1 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A239957): Every prime p has a primitive root $0 < g < p$ of the form $k^2 + 1$, where k is an integer.

theorem conjecture (p : ℕ) (hp : p.Prime) : A p

FC-OeisA239957.conjecture

Cuban Primes

MODULE CONTEXT: # Cuban Primes OEIS A002407 lists the primes that are differences of two consecutive positive cubes. The sequence is conjectured to be infinite. *References:* - [OEIS A002407](https://oeis.org/A002407)

V2-000216 | Cuban Primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

This sequence is believed to be infinite.

theorem conjecture : {p : ℕ | A p}.Infinite

FC-OeisA2407.conjecture

Central trinomial coefficients

MODULE CONTEXT: # Central trinomial coefficients Central trinomial coefficients: largest coefficient of $(1 + x + x^2)^n$, which is the coefficient of x^n in the expansion of $(1 + x + x^2)^n$. *References:* - [A002426](https://oeis.org/A002426)

V2-000217 | Central trinomial coefficients | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

An integer $n > 3$ is prime if and only if $a(n) \equiv 1 \pmod{n^2}$. We have verified this for n up to $8 \cdot 10^5$, and proved that $a(p) \equiv 1 \pmod{p^2}$ for any prime $p > 3$ (cf. A277640). - Zhi-Wei Sun, Nov 30 2016

theorem conjecture (n : ℕ) (hn : 3 < n) :
n.Prime ↔ a n ≡ 1 [MOD n ^ 2]

FC-OeisA2426.conjecture

Signed digit sums in bases $b \ge 5$

MODULE CONTEXT: # Signed digit sums in bases $b \ge 5$ The sequence is defined by $$a(n) = \sum_{k=1}^n (-1)^{\operatorname{isprime}(k)} 10^k$$ where the sign is -1 if k is prime, and 1 if k is not prime. Generalizing the digit pattern observed in $a(n)$, it was conjectured on the OEIS entry that for any base $b \ge 5$ and any choice of signs ± 1, the absolute value of $\sum_{k=1}^n \pm b^k$ only contains digits in $\{0, 1, b-2, b-1\}$. *References:* - [A243106](https://oeis.org/A243106) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000218 | Signed digit sums in bases $b \ge 5$ | digits_sum_restricted

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: When expressed in base $b \ge 5$, the absolute value of any partial sum $\sum \pm b^k$ only contains digits belonging to $\{0, 1, b-2, b-1\}$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem digits_sum_restricted (b n : ℕ) (hb : b ≥ 5) :
∀ (σ : ℕ → Int) (hσ : ∀ k ∈ Icc 1 n, σ k = 1 ∨ σ k = -1),
let x : Int := (Icc 1 n).sum fun k ↦ σ k * (b : Int) ^ k;
∀ d ∈ (b.digits x.natAbs), d = 0 ∨ d = 1 ∨ d = b - 2 ∨ d = b - 1
```

FC-OeisA243106.digits_sum_restricted

Determinant of Hankel matrix of the first $2n-1$ prime numbers

MODULE CONTEXT: # Determinant of Hankel matrix of the first $2n-1$ prime numbers The determinant of the $n \times n$ Hankel matrix whose entries are the first $2n-1$ prime numbers. The matrix M has entries $M_{i, j} = p_{i+j}$ for $i, j \in \{0, \dots, n-1\}$, where $p_k = \mathrm{Nat.nth\;Nat.Prime} (k)$ is the k-th prime starting at $p_0=2$. $a(0)=1$ by convention. *References:* - [A024356](https://oeis.org/A024356)

V2-000219 | Determinant of Hankel matrix of the first $2n-1$ prime numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

"I conjecture that $a(4)$ is the only zero. - _Jon Perry_, Mar 22 2004" Stated as a biconditional: the claim that $a(4)$ is *the only* zero asserts both that $a(4) = 0$ and that no other index vanishes. A bare implication `a n = 0 → n = 4` would be satisfied vacuously by a sequence with no zero at all. The existence direction is certified by `a_4`; the uniqueness direction is the open part.

```
theorem conjecture : ∀ n : ℕ, a n = 0 ↔ n = 4
```

FC-OeisA24356.conjecture

Central factorial numbers: $((2n)!!)^2$

MODULE CONTEXT: # Central factorial numbers: $((2n)!!)^2$ Central factorial numbers: $a(n) = 4^n (n!)^2 = ((2n)!!)^2$. *References:* - [A002454](https://oeis.org/A002454) - [SSX22] [She, Y.-F., Sun, Z.-W., Xia, W., *A novel permanent identity with applications*, arXiv:2208.12167 (2022)](https://arxiv.org/abs/2208.12167)

V2-000220 | Central factorial numbers: $((2n)!!)^2$ | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Let ζ be a primitive $(2n+1)$-th root of unity. Then the permanent of the $2n \times 2n$ matrix $[m(j,k)]_{j,k=1..2n}$ is $a(n)/(2n+1) = ((2n)!!)^2/(2n+1)$, where $m(j,k)$ is 1 or $(1+\zeta^{j-k})/(1-\zeta^{j-k})$ according as $j = k$ or not. - Zhi-Wei Sun, Jun 26 2022 Proved by [SSX22], Theorem 1.3(ii): for odd $m > 1$ and ζ a primitive m-th root of unity, the permanent of $[c_{j,k}]_{1 \le j,k \le m-1}$ is $((m-1)!!)^2/m$. Take $m = 2n+1$; translating both matrix indices by one leaves $j - k$ unchanged. For $n = 0$ the matrix is empty and both sides are 1.

```
theorem conjecture (n : ℕ) :
let N : ℕ := 2 * n
let K : ℕ := N + 1
∀ (ζ : ℂ), IsPrimitiveRoot ζ K →
Matrix.permanent (fun (j k : Fin N) =>
if j = k then
(1 : ℂ)
else
let pow : ℤ := (j : ℤ) - (k : ℤ)
(1 + ζ ^ pow) / (1 - ζ ^ pow)
) = (a n : ℂ) / (K : ℂ)
```

FC-OeisA2454.conjecture

Smallest prime factors of $2^{2^n+2} + 3$

MODULE CONTEXT: # Smallest prime factors of $2^{2^n+2} + 3$ Smallest prime factor of $2^{2^n+2} + 3$. *References:* - [A248802](https://oeis.org/A248802) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000221 | Smallest prime factors of $2^{2^n+2} + 3$ | a_fifty_eight_mul_add_twenty_six_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 4: $a(58n+26) = 1399$ for $n \ge 0$ and when it is not covered by Conjectures 1-3. - _Chai Wah Wu_, Oct 21 2019 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_fifty_eight_mul_add_twenty_six_eq (n : ℕ) :
(¬CoveredByC1 (58 * n + 26) ∧ ¬CoveredByC2 (58 * n + 26) ∧
¬CoveredByC3 (58 * n + 26)) → a (58 * n + 26) = 1399
```

FC-OeisA248802.a_fifty_eight_mul_add_twenty_six_eq

V2-000222 | Smallest prime factors of $2^{2^n+2} + 3$ | a_ten_mul_add_two_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1: $a(10n+2) = 67$ for $n \ge 0$. - _Chai Wah Wu_, Oct 21 2019 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_ten_mul_add_two_eq (n : ℕ) : a (10 * n + 2) = 67
```

FC-OeisA248802.a_ten_mul_add_two_eq

Partitions of n into distinct non-squarefree parts for $n > 23$

MODULE CONTEXT: # Partitions of n into distinct non-squarefree parts for $n > 23$ a: Number of partitions of n into distinct parts that are not squarefree. This is the number of finite subsets of positive integers P such that $\sum_{k \in P} k = n$ and every element $k \in P$ is not squarefree. *References:* - [A256012](https://oeis.org/A256012) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000223 | Partitions of n into distinct non-squarefree parts for $n > 23$ | a_pos_of_gt

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) > 0$ for $n > 23$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_pos_of_gt (n : ℕ) (hn : n > 23) : a n > 0

FC-OeisA256012.a_pos_of_gt

Asymptotics of an inner sum formula for $a(n)$

MODULE CONTEXT: # Asymptotics of an inner sum formula for $a(n)$ The inner sum of the formula used in a: $$\sum_{\max(k-n+5, 0) \le j \le \min(k,4)} \binom{8-j}{j}\binom{2n-k+j-10}{k-j}$$ a: A total of n married couples, including a mathematician M and his wife, are to be seated at the $2n$ chairs around a circular table, with no man seated next to his wife. After the ladies are seated at every other chair, M is the first man allowed to choose one of the remaining chairs. The sequence gives the number of ways of seating the other men, with no man seated next to his wife, if M chooses the chair that is 9 seats clockwise from his wife's chair. $$a(n) = \begin{cases} 0 & \text{if } n \le 5 \cr \sum_{k=0}^{n-1}(-1)^k(n-k-1)! \sum_{\max(k-n+5, 0) \le j \le \min(k,4)} \binom{8-j}{j}\binom{2n-k+j-10}{k-j} & \text{if } n > 5 \end{cases}$$ *References:* - [A258667](https://oeis.org/A258667) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000224 | Asymptotics of an inner sum formula for $a(n)$ | a_is_equivalent_asymptoticTerm

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) \sim e^{-2} \cdot n! / (n-2)$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_is_equivalent_asymptoticTerm :
IsEquivalent atTop (fun n : ℕ => (a n : ℝ)) asymptoticTerm

FC-OeisA258667.a_is_equivalent_asymptoticTerm

A GCD-Driven Sequence with Universal Jumps

MODULE CONTEXT: # A GCD-Driven Sequence with Universal Jumps OEIS A260194 begins with three ones and then adds the greatest common divisor of the current term and the term two places earlier. The open question asks whether every positive integer occurs as an adjacent difference. *References:* - [OEIS A260194](https://oeis.org/A260194)

V2-000225 | A GCD-Driven Sequence with Universal Jumps | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Does every positive integer occur as a difference in this sequence?

theorem conjecture :
∀ d > 0, ∃ n, a (n + 1) = a n + d

FC-OeisA260194.conjecture

Recurrence for Rule 167 cellular automaton sequence

MODULE CONTEXT: # Recurrence for Rule 167 cellular automaton sequence Decimal representation of the middle column of the "Rule 167" elementary cellular automaton starting with a single ON (black) cell. *References:* - [A267581](https://oeis.org/A267581) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000226 | Recurrence for Rule 167 cellular automaton sequence | a_recurrence

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) = 2 a(n-1) + 1 - \lfloor (1/2)^{2^{n+1} \bmod n} \rfloor$ for $n \ge 2$. - _Andres Cicuttin_, Mar 29 2016 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_recurrence (n : ℕ) (hn : 2 ≤ n) : a n = 2 * a (n - 1) + 1 - oeisFloorTerm n

FC-OeisA267581.a_recurrence

Maximal run lengths in the second MSB of Tribonacci numbers $T(n)$

MODULE CONTEXT: # Maximal run lengths in the second MSB of Tribonacci numbers $T(n)$ The Tribonacci numbers T_n (A000073). $T_0=0, T_1=0, T_2=1$, and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$ for $n \ge 3$. Second most significant bit of the tribonacci number A000073(n). This is formalized by extracting the bit at position $\lfloor \log_2 T_n \rfloor - 1$. *References:* - [A271591](https://oeis.org/A271591) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A000073](https://oeis.org/A000073)

V2-000227 | Maximal run lengths in the second MSB of Tribonacci numbers $T(n)$ | maximal_run_lengths

research solved | proof annotation: True | 官方代码 | 原文线索

It is conjectured that after the first two 0's, the number of consecutive 0's is only 4 or 5, and the number of consecutive 1's is only 3 or 4 (tested up to $n = 10^4$). A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem maximal_run_lengths : (∀ n L, IsMaximalRun 0 n L → (L = 4 ∨ L = 5)) ∧
(∀ n L, IsMaximalRun 1 n L → (L = 3 ∨ L = 4))

FC-OeisA271591.maximal_run_lengths

Congruences $a(n+k) \equiv a(n) \pmod k$ for hypergeometric values ${}_2F_1(n, -n; ; -1)$

MODULE CONTEXT: # Congruences $a(n+k) \equiv a(n) \pmod k$ for hypergeometric values ${}_2F_1(n, -n; ; -1)$ a: $a(n) = \text{hypergeometric}([n, -n], [], -1)$. This is equivalent to the combinatorial sum: $$a(n) = \sum_{k=0}^n \binom{n}{k} \binom{n+k-1}{k} k!$$ The expression uses $\mathbb{N}$ arithmetic throughout, safely handling the subtraction via `Nat.pred`. *References:* - [A278070](https://oeis.org/A278070) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000228 | Congruences $a(n+k) \equiv a(n) \pmod k$ for hypergeometric values ${}_2F_1(n, -n; ; -1)$ | a_modeq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n+k) \equiv a(n) \pmod{k}$ for all n and k. - _Peter Bala_, Mar 12 2023 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_modeq : ∀ (n k : ℕ), Nat.ModEq k (a (n + k)) (a n)

FC-OeisA278070.a_modeq

The 1680-Conjecture

MODULE CONTEXT: # The 1680-Conjecture Any nonnegative integer can be written as $x^2 + y^2 + z^2 + w^2$ with x, y, z, w nonnegative integers such that $x^4 + 1680 y^3 z$ is a square. Zhi-Wei Sun has offered a prize of 1,680 RMB for the first proof. *References:* - [A280831](https://oeis.org/A280831) - Z.-W. Sun, "Refining Lagrange's four-square theorem," *J. Number Theory* **175** (2017), 167-190. - Z.-W. Sun, "Refining Lagrange's four-square theorem," arXiv:1604.06723 [math.NT], 2016.

V2-000229 | The 1680-Conjecture | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's 1680-Conjecture (A280831): Any nonnegative integer can be written as $x^2 + y^2 + z^2 + w^2$ with x, y, z, w nonnegative integers such that $x^4 + 1680 y^3 z$ is a square.

theorem conjecture (n : ℕ) : A n

FC-OeisA280831.conjecture

The cup-of-coffee sequence

MODULE CONTEXT: # The cup-of-coffee sequence The sequence starts with $1, 2$. Each later term is the least unused positive integer such that, for every prime, runs of consecutive multiples have length exactly two. The local rule requires prime factors that have just appeared to occur once more, and forbids prime factors that occurred in both of the preceding terms. *References:* - [A280864](https://oeis.org/A280864)

V2-000230 | The cup-of-coffee sequence | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjectured to be a permutation of the positive integers.

theorem conjecture :
∃ a : ℕ → ℕ, FollowsRule a ∧ ∀ k > 0, ∃ n, a n = k

FC-OeisA280864.conjecture

Sum of four squares with square conditions

MODULE CONTEXT: # Sum of four squares with square conditions Any integer $n \geq 0$ can be written as $x^2 + y^2 + z^2 + w^2$ with x, y, z, w nonnegative integers and $z \leq w$, such that both x and $x + 24y$ are squares. Zhi-Wei Sun has offered a \$2,400 prize for the first proof. *References:* - [A281976](https://oeis.org/A281976) - Z.-W. Sun, "Refining Lagrange's four-square theorem," *J. Number Theory* **175** (2017), 167-190. https://doi.org/10.1016/j.jnt.2016.11.008 - Z.-W. Sun, "Restricted sums of four squares," *arXiv:1701.05868* [math.NT], 2017. https://arxiv.org/abs/1701.05868

V2-000231 | Sum of four squares with square conditions | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A281976): Any integer $n \geq 0$ can be written as $x^2 + y^2 + z^2 + w^2$ with x, y, z, w nonnegative integers and $z \leq w$, such that both x and $x + 24y$ are squares.

theorem conjecture (n : ℕ) : A n

FC-OeisA281976.conjecture

Period of p-th powers modulo n

MODULE CONTEXT: # Period of p-th powers modulo n a: Period of cubes mod n. The n-th term $a(n)$ is the smallest positive integer T such that $\forall k \in \mathbb{N}$, $(k+T)^3 \equiv k^3 \pmod n$. The length of the minimal positive period of the sequence $k^p \pmod n$. $a_p(n) = \min \{ T \in \mathbb{N}^+ \mid \forall k \in \mathbb{N}, (k+T)^p \equiv k^p \pmod n \}$. *References:* - [A282779](https://oeis.org/A282779) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000232 | Period of p-th powers modulo n | periodOfPowerMod_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: let $a_p(n)$ be the length of the period of the sequence $k^p \bmod n$ where p is a prime, then $a_p(n) = n/p$ if $n \equiv 0 \pmod{p^2}$, else $a_p(n) = n$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem periodOfPowerMod_eq (p n : ℕ) (hp : Nat.Prime p) (hn : n > 0) :
periodOfPowerMod p n = if p ^ 2 | n then n / p else n
```

FC-OeisA282779.periodOfPowerMod_eq

Sum of a triangular number, a generalized pentagonal number, and a generalized heptagonal number

MODULE CONTEXT: # Sum of a triangular number, a generalized pentagonal number, and a generalized heptagonal number Any nonnegative integer can be written as $x(x+1)/2 + y(3y+1)/2 + z(5z+1)/2$ with x, y, z nonnegative integers. Zhi-Wei Sun offered a USD 135 prize for the first proof of this conjecture. Cao, Guo, Qiu, Feng and Gao proved it in [CGQFG26]. *References:* - [A287616](https://oeis.org/A287616) - Zhi-Wei Sun, "Universal sums of three quadratic polynomials", arXiv:1502.03056 [math.NT] - [CGQFG26] Y. Cao, D. Guo, R. Qiu, R. Feng and X.-S. Gao, "Every Nonnegative Integer Is a Sum of a Triangular, a Pentagonal, and a Heptagonal Number", [arXiv:2606.26035](https://arxiv.org/abs/2606.26035)

V2-000233 | Sum of a triangular number, a generalized pentagonal number, and a generalized heptagonal number | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A287616): Any nonnegative integer can be written as the sum of a triangular number $x(x+1)/2$, a generalized pentagonal number $y(3y+1)/2$, and a generalized heptagonal number $z(5z+1)/2$, where x, y, z are nonnegative integers. This was proved in [CGQFG26].

```
theorem conjecture (n : ℕ) : A n
```

FC-OeisA287616.conjecture

Combinatorial representation of the recurrence $a(n+2) = 2a(n+1) + 2a(n)$

MODULE CONTEXT: # Combinatorial representation of the recurrence $a(n+2) = 2a(n+1) + 2a(n)$ A028859 (OEIS): $a(n+2) = 2 \cdot a(n+1) + 2 \cdot a(n)$; $a(0) = 1$, $a(1) = 3$. *References:* - [A028859](https://oeis.org/A028859) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000234 | Combinatorial representation of the recurrence $a(n+2) = 2a(n+1) + 2a(n)$ | exists_finset_sequence

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: The sequence a(n) is also the number of compositions of n into positive integers such that adjacent parts and the largest part differ by at most 1. - _Gus Wiseman_, May 19 2020 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem exists_finset_sequence (n : ℕ) :
let L := n + 1
let Sequence := Fin L → ℕ
let S : Set Sequence :=
{σ : Sequence |
L > 0 ∧
(∀ i : Fin L, σ i > 0) ∧
let max_val := Finset.sup Finset.univ σ
(∀ k : ℕ, 1 ≤ k ∧ k ≤ max_val → ∃ i : Fin L, σ i = k) ∧
(∀ i j : Fin L, i < j → j.val ≠ i.val + 1 → σ i ≥ σ j)}
∃ (F : Finset Sequence), (F : Set Sequence) = S ∧ F.card = a n
```

FC-OeisA28859.exists_finset_sequence

Symmetry of digit sum differences $\operatorname{sign}(S_5(k) - S_1(k))$

MODULE CONTEXT: # Symmetry of digit sum differences $\operatorname{sign}(S_5(k) - S_1(k))$ a: $\mathrm{a}(n) = \sum_{k=0}^n \mathrm{sign}(\mathrm{A007953}(5k) - \mathrm{A007953}(k))$. $\mathrm{A007953}(n)$ is the digital sum of n in base 10. The sequence is non-negative, so the sum over $\mathbb{Z}$ is converted to $\mathbb{N}$. *References:* - [A289411](https://oeis.org/A289411) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A007953](https://oeis.org/A007953)

V2-000235 | Symmetry of digit sum differences $\operatorname{sign}(S_5(k) - S_1(k))$ | a_symm

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: For $k \ge 1$, let $m_k = 10^k / 2 - 1$. Then for $i = 0, \ldots, m_k$, we have $a(m_k - i) = a(m_k + i)$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_symm (k : ℕ) (hk : 0 < k) :
let m_k : ℕ := (10 ^ k) / 2 - 1
∀ i : ℕ, i ≤ m_k → a (m_k - i) = a (m_k + i)
```

FC-OeisA289411.a_symm

Coefficient of $(xyz)^n$ in $((x+y)(y+z)(z+x))^n$ equaling $\binom{2n}{n}^3$

MODULE CONTEXT: # Coefficient of $(xyz)^n$ in $((x+y)(y+z)(z+x))^n$ equaling $\binom{2n}{n}^3$ The sequence $a(n)$ is defined by $a(n) = \binom{2n}{n}^3$. *References:* - [A002897](https://oeis.org/A002897) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000236 | Coefficient of $(xyz)^n$ in $((x+y)(y+z)(z+x))^n$ equaling $\binom{2n}{n}^3$ | a_eq_coeff

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) = [x^n y^n z^n] (1+x+y+z)^{2n} (1+x+y-z)^n (1+x-y+z)^n$. - _Peter Bala_, Apr 10 2022 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_eq_coeff (n : ℕ) : (a n : ℤ) = MvPolynomial.coeff (xyzPowN n) (pPoly n)
```

FC-OeisA2897.a_eq_coeff

Density of ones in the Kolakoski sequence

MODULE CONTEXT: # Density of ones in the Kolakoski sequence The Kolakoski sequence is the self-describing sequence of ones and twos whose terms are the lengths of its successive runs. Its official indexing begins with $a(1) = 1$; here it is extended by $a(0) = 0$. *References:* - [A000002](https://oeis.org/A000002) - E. S. Kupin and E. S. Rowland, ["Bounds on the frequency of 1 in the Kolakoski word"](https://arxiv.org/abs/0809.2776), arXiv:0809.2776 [math.CO] (2008).

V2-000237 | Density of ones in the Kolakoski sequence | density_one

research open | proof annotation: False | 官方代码 | 原文线索

"It is an unsolved problem to show that the density of 1's is equal to 1/2." - [A000002](https://oeis.org/A000002)

```
theorem density_one : {n : ℕ | a n = 1}.HasDensity (1 / 2 : ℝ)
```

FC-OeisA2.density_one

Step difference bound $a(n+1) - a(n) \in \{1, 2\}$ in mass-redistribution cellular automata

MODULE CONTEXT: # Step difference bound $a(n+1) - a(n) \in \{1, 2\}$ in mass-redistribution cellular automata $a(n)$ is the number of steps needed to reach a stable configuration in the 1D cellular automaton initialized with one cell with mass n and based on the rule "each cell gives half of its mass, rounded down, to its right neighbor". The stable configuration is n cells with mass 1. *References:* - [A300997](https://oeis.org/A300997) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000238 | Step difference bound $a(n+1) - a(n) \in \{1, 2\}$ in mass-redistribution cellular automata | a_succ_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: It appears that the finite difference of this sequence only contains 1's and 2's. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_succ_eq : ∀ n : ℕ, 1 ≤ n → a (n + 1) = a n + 1 ∨ a (n + 1) = a n + 2
```

FC-OeisA300997.a_succ_eq

Sum of two squares and two central-binomial-type terms

MODULE CONTEXT: # Sum of two squares and two central-binomial-type terms A303639 counts the ways to write n as $a^2 + b^2 + \binom{2c+1}{c} + \binom{2d+1}{d}$ with a, b, c, d nonnegative integers, $a \le b$ and $c \le d$. *References:* - [A303639](https://oeis.org/A303639) - Z.-W. Sun, ["Restricted sums of four squares"](https://arxiv.org/abs/1701.05868), arXiv:1701.05868 [math.NT], 2017-2018. - Z.-W. Sun, ["Refining Lagrange's four-square theorem"](https://doi.org/10.1016/j.jnt.2016.11.008), *J. Number Theory* **175** (2017), 167-190. - S. Jeong, [A303639 Counterexample Project](https://github.com/DCLXAI/a303639-counterexample), search, certificates and independent verifiers, [DOI 10.5281/zenodo.21863025](https://doi.org/10.5281/zenodo.21863025). The counterexample is recorded as an approved comment on the OEIS entry (Aug 10 2026).

V2-000239 | Sum of two squares and two central-binomial-type terms | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A303639): any integer $n > 1$ can be written as $a^2 + b^2 + \binom{2c+1}{c} + \binom{2d+1}{d}$ with a, b, c, d nonnegative integers. Sun checked this for n up to $6 \cdot 10^8$. This is false: $n = 800322180$ admits no such representation, so $a(800322180) = 0$. Since $\binom{33}{16} > 800322180$, only $c, d \le 15$ are possible, and each of the resulting 136 remainders $800322180 - \binom{2c+1}{c} - \binom{2d+1}{d}$ is divisible by some prime $p \equiv 3 \pmod 4$ to an odd power, hence is not a sum of two squares by Fermat's two-square theorem. The counterexample is recorded as an approved comment on the OEIS entry; the Lean proof linked below formalises this argument, and was produced by Claude Opus 5 prompted by Sunsu Jeong ([DCLXAI](https://github.com/DCLXAI)).

theorem conjecture : ¬ ∀ n : ℕ, 1 < n → A n

FC-OeisA303639.conjecture

Sum of two squares, a power of 3, and a power of 5

MODULE CONTEXT: # Sum of two squares, a power of 3, and a power of 5 Any integer $n > 1$ can be written as $a^2 + b^2 + 3^c + 5^d$ where a, b, c, d are nonnegative integers. Zhi-Wei Sun has offered a \$3,500 prize for the first proof. *References:* - [A303656](https://oeis.org/A303656) - Z.-W. Sun, "Restricted sums of four squares," arXiv preprint: https://arxiv.org/abs/1701.05868v10 - Z.-W. Sun, "Refining Lagrange's four-square theorem," Journal of Number Theory: http://maths.nju.edu.cn/~zwsun/RefineFourSquareTh.pdf - Z.-W. Sun, "Restricted sums of three or four squares": http://maths.nju.edu.cn/~zwsun/Square-sum.pdf - Zhi-Wei Sun's 1-3-5 conjecture and variations: https://www.aimspress.com/aimspress-data/era/2020/2/PDF/1935-9179_2020_2_589.pdf

V2-000240 | Sum of two squares, a power of 3, and a power of 5 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Conjecture (A303656): Any integer $n > 1$ can be written as the sum of two squares, a power of 3, and a power of 5.

theorem conjecture (n : ℕ) (hn : 1 < n) : A n

FC-OeisA303656.conjecture

Maximality of $k = 43$ with restricted digit counts in bases $3 \le b < k$

MODULE CONTEXT: # Maximality of $k = 43$ with restricted digit counts in bases $3 \le b < k$ Numbers k such that the base b expansion of k for each $b = 3..k-1$ never contains more than two distinct digits. The sequence Numbers k such that the base b expansion of k for each $b = 3..k-1$ never contains more than two distinct digits. *References:* - [A306424](https://oeis.org/A306424) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000241 | Maximality of $k = 43$ with restricted digit counts in bases $3 \le b < k$ | forty_three_is_max

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: The sequence is finite, with 43 being the last term. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem forty_three_is_max : Condition 43 ∧ ∀ k : ℕ, 43 < k → ¬Condition k

FC-OeisA306424.forty_three_is_max

The 2-4-6-8 Conjecture

MODULE CONTEXT: # The 2-4-6-8 Conjecture Any integer $n > 0$ can be written as $\binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$ with w, x, y, z nonnegative integers. Zhi-Wei Sun has offered a $2,468 prize for the first proof (or $2,468 RMB for a counterexample). The conjecture has been verified for all n up to 1.2×10^{12} by Yaakov Baruch (March 2019). The conjecture is false: $n = 896315812331399$ has no such representation. The counterexample was found in the OEIS Open project, with a Lean proof. *References:* - [A306477](https://oeis.org/A306477) - [mathoverflow/323541](https://mathoverflow.net/questions/323541): Z.-W. Sun, "Positive integers written as C(w,2) + C(x,4) + C(y,6) + C(z,8) with w,x,y,z in {2,3,...},", Feb. 19, 2019. - T. Adamczewski, "OEIS Open: How many conjectures can language models turn into theorems?", [arXiv:2608.11941](https://arxiv.org/abs/2608.11941) [cs.AI], 2026.

V2-000242 | The 2-4-6-8 Conjecture | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Zhi-Wei Sun's 2-4-6-8 Conjecture (A306477): Any integer $n > 0$ can be written as $\binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$ for nonnegative integers w, x, y, z. This is false: $n = 896315812331399$ is a counterexample. See T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, [arXiv:2608.11941](https://arxiv.org/abs/2608.11941).

theorem conjecture : ¬ ∀ n : ℕ, 0 < n → A n

FC-OeisA306477.conjecture

Vanishing of bases with $b^n \equiv -1 \pmod{2n+1}$ for absolute Euler pseudoprimes

MODULE CONTEXT: # Vanishing of bases with $b^n \equiv -1 \pmod{2n+1}$ for absolute Euler pseudoprimes $a(n)$ is the number of natural bases $b < 2n+1$ such that $b^n \equiv -1 \pmod{2n+1}$. If $2n+1$ is an absolute Euler pseudoprime, then $a(n) = 0$. *References:* - [A307865](https://oeis.org/A307865) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000243 | Vanishing of bases with $b^n \equiv -1 \pmod{2n+1}$ for absolute Euler pseudoprimes | a_eq_zero_of_pseudoprime

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: if $2n+1$ is an absolute Euler pseudoprime, then $a(n) = 0$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_eq_zero_of_pseudoprime (h : IsAbsoluteEulerPseudoprime (2 * n + 1)) : a n = 0

FC-OeisA307865.a_eq_zero_of_pseudoprime

Four-square conjecture with powers of 2, 3, and 5

MODULE CONTEXT: # Four-square conjecture with powers of 2, 3, and 5 Any integer $n > 1$ can be written as $(2^a \cdot 3^b)^2 + (2^c \cdot 5^d)^2 + x^2 + y^2$ where a, b, c, d, x, y are nonnegative integers. Zhi-Wei Sun has offered a \$2,500 prize for the first proof. *References:* - [A308734](https://oeis.org/A308734) - Z.-W. Sun, "Refining Lagrange's four-square theorem," *J. Number Theory* **175** (2017), 167-190. https://doi.org/10.1016/j.jnt.2016.11.008 - Z.-W. Sun, "Restricted sums of four squares," *Int. J. Number Theory* **15** (2019), 1863-1893. - Z.-W. Sun, "Various Refinements of Lagrange's Four-Square Theorem," Westlake Number Theory Symposium, Nanjing University, China, 2020. - S. Banerjee, "On a conjecture of Sun about sums of restricted squares," *J. Number Theory* **256** (2024), 253-289.

V2-000244 | Four-square conjecture with powers of 2, 3, and 5 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zhi-Wei Sun's Four-Square Conjecture (A308734): Any integer $n > 1$ can be written as $(2^a \cdot 3^b)^2 + (2^c \cdot 5^d)^2 + x^2 + y^2$ for nonnegative integers a, b, c, d, x, y.

theorem conjecture (n : ℕ) (hn : 1 < n) : A n

FC-OeisA308734.conjecture

Characterization of Carmichael numbers via squarefree denominators $a(n)$

MODULE CONTEXT: # Characterization of Carmichael numbers via squarefree denominators $a(n)$ $a(n)$ is the denominator of $F(n) = \operatorname{num}(B_{n-1})/n + \operatorname{den}(B_{n-1})/n^2$. A composite number n has squarefree $a(n)$ if and only if n is a Carmichael number. *References:* - [A309132](https://oeis.org/A309132) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A027641](https://oeis.org/A027641) - [A027642](https://oeis.org/A027642)

V2-000245 | Characterization of Carmichael numbers via squarefree denominators $a(n)$ | carmichael_iff_squarefree_a

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: composite numbers n such that $a(n)$ is squarefree are only the Carmichael numbers (A002997). - _Thomas Ordowski_, Jul 15 2019 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem carmichael_iff_squarefree_a :
∀ (n : ℕ), (IsComposite n ∧ Squarefree (a n)) ↔ IsCarmichaelNumber n

FC-OeisA309132.carmichael_iff_squarefree_a

A binomial coefficient sum

MODULE CONTEXT: # A binomial coefficient sum A binomial coefficient sum: $$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left(\binom{n}{k} - \binom{n}{k-1} \right)^3$$ where $\binom{n}{-1} = 0$. *References:* - [A003161](https://oeis.org/A003161) - [Li26] Wentao Li, [A supercongruence for A003161](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/41f7b2ba762827a74f1651e76a2ad6d77be6a021/proofs/new-formalization/A003161/PROOF.md), 2026.

V2-000246 | A binomial coefficient sum | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: Let $b(n) = a(2n-1)$. Then the supercongruence $b(np^k) \equiv b(np^{k-1}) \pmod{p^{3k}}$ holds for positive integers n and k and all primes $p \ge 5$." - Peter Bala, Mar 20 2023. Lean formalization by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n k p : ℕ) (hn : 0 < n) (hk : 0 < k) (hp : p.Prime) (hp_ge : 5 ≤ p) :
(b (n * p ^ k) : ℤ) ≡ (b (n * p ^ (k - 1)) : ℤ) [ZMOD (p : ℤ) ^ (3 * k)]

FC-OeisA3161.conjecture

A binomial coefficient summation

MODULE CONTEXT: # A binomial coefficient summation A binomial coefficient summation: $a(n) = S(3, n) / S(1, n)$, where for a positive integer r we define $$S(r,n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left(\binom{n}{k} - \binom{n}{k-1} \right)^r$$ with $\binom{n}{-1} = 0$. *References:* - [A003162](https://oeis.org/A003162) - H. W. Gould, Problem E2384 (proposal), Amer. Math. Monthly, 81 (1974), 170-171 - [Li26] Wentao Li, [A supercongruence for A003162](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/41f7b2ba762827a74f1651e76a2ad6d77be6a021/proofs/new-formalization/A003162/PROOF.md), 2026.

V2-000247 | A binomial coefficient summation | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: Let $b(n) = a(2n-1)$. Then the supercongruence $b(np^k) \equiv b(np^{k-1}) \pmod{p^{3k}}$ holds for positive integers n and k and all primes $p \ge 5$." - Peter Bala, Mar 26 2023. Lean formalization by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture (n k p : ℕ) (hn : 0 < n) (hk : 0 < k) (hp : p.Prime) (hp_ge : 5 ≤ p) :
(b (n * p ^ k)).num ≡ (b (n * p ^ (k - 1))).num [ZMOD (p : ℤ) ^ (3 * k)]
```

FC-OeisA3162.conjecture

Nonnegativity of the Dirichlet square root of A046644

MODULE CONTEXT: # Nonnegativity of the Dirichlet square root of A046644 A317940 is the integer sequence whose value `a n` is the numerator of the rational sequence `f n`, where the Dirichlet convolution square of `f` is A046644. The auxiliary sequence A046644 is multiplicative and takes the value $2^{\operatorname{A005187}(e)}$ on a prime power p^e. The conjecture asks whether `f n` is nonnegative for every positive `n`. The proof reduces the problem to prime powers and constructs positive rational coefficients $c(e)$ satisfying $$\sum_{i=0}^e c(i)c(e-i)=2^{\operatorname{A005187}(e)}.$$ The coefficients are obtained from formal power series. A positive series A is defined by the first-order differential equation $A' = \frac12 D A$. A second explicitly defined series B satisfies $B' = D B$ and has the same constant term as A^2; uniqueness of the coefficient recurrence therefore gives $A^2=B$. After rescaling the coefficients by 4^e, this becomes the displayed prime-power convolution identity. Extending $c(e)$ multiplicatively over prime factorizations gives a positive arithmetic function `root` with `root * root = a046644`. Finally, the recursive definition of `f` is shown to be the unique Dirichlet square root with value one at 1, so `f = root` and every positive-index value of `f` is positive. This route was found by first exploiting the multiplicativity recorded in the OEIS entry, reducing the recurrence to the exponent of a single prime, and then recognizing the resulting coefficient identities as a formal-power-series differential equation. *References:* - [A317940](https://oeis.org/A317940) - [A046644](https://oeis.org/A046644) - [A005187](https://oeis.org/A005187)

V2-000248 | Nonnegativity of the Dirichlet square root of A046644 | f_nonnegative

research solved | proof annotation: True | 官方代码 | 原文线索

"No negative terms among the first 2^20 terms. Is the sequence nonnegative?" Equivalently, the rational Dirichlet square root `f` is nonnegative at every positive index. Informally, the proof constructs a strictly positive multiplicative square root prime-power by prime-power using formal power series, and then identifies it with the recursively defined sequence `f`. The proof was obtained by exploiting the multiplicativity of A046644 and reducing the problem to its values on prime powers. The resulting convolution recurrence for the prime-power coefficients was recognized as a formal-power-series identity. A positive coefficient sequence was constructed through a differential equation for the generating series, extended multiplicatively to all positive integers, and finally identified with the recursively defined Dirichlet square root by uniqueness.

```
theorem f_nonnegative (n : ℕ) (h : n > 0) : f n ≥ 0
```

FC-OeisA317940.f_nonnegative

Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1+x^{n+1})^{n+1}}$

MODULE CONTEXT: # Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1+x^{n+1})^{n+1}}$ Coefficients of G.f. $\sum_{n\ge 0} x^n \cdot \frac{(1 + x^n)^n}{(1 + x^{n+1})^{n+1}}$. The m-th term $a(m)$ is the coefficient of x^m, which is explicitly given by the sum: $$ a(m) = \sum_{n=0}^m \sum_{k=0}^n \binom{n}{k} (-1)^j \binom{n+j}{j},$$ where $j = \frac{m - n(k+1)}{n+1}$, and the term is zero unless j is a natural number. *References:* - [A323557](https://oeis.org/A323557) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000249 | Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1+x^{n+1})^{n+1}}$ | odd_a_implies_pronic

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Odd terms occur only at positions $n(n+1)$ for $n \ge 0$ (verified for initial 32600 terms). A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763). This statement is equivalent to `OeisA325046.odd_a_implies_pronic`. The two sequences differ only by the sign $(-1)^j$ on each summand, so they agree modulo 2 and have the same odd indices.

```
theorem odd_a_implies_pronic (m : ℕ) : Odd (a m) → ∃ n : ℕ, m = n * (n + 1)
```

FC-OeisA323557.odd_a_implies_pronic

Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1-x^{n+1})^{n+1}}$

MODULE CONTEXT: # Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1-x^{n+1})^{n+1}}$ Coefficients of G.f. $\sum_{n \ge 0} x^n \cdot \frac{(1 + x^n)^n}{(1 - x^{n+1})^{n+1}}$. The term $a(N)$ is the coefficient of x^N in the generating function. Expanding the terms, we get a formula for $a(N)$: $$a(N) = \sum_{n=0}^N \sum_{k=0}^n \mathbf{1}_{n + nk + (n+1)j = N} \binom{n}{k} \binom{n+j}{j}$$ where $j = \frac{N - n(k+1)}{n+1}$. *References:* - [A325046](https://oeis.org/A325046) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000250 | Pronic indices for odd coefficients of $\sum_{n \ge 0} x^n \frac{(1+x^n)^n}{(1-x^{n+1})^{n+1}}$ | odd_a_implies_pronic

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Odd terms occur only at positions $n(n+1)$ for $n \ge 0$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763). This statement is equivalent to `OeisA323557.odd_a_implies_pronic`. The two sequences differ only by the sign $(-1)^j$ on each summand, so they agree modulo 2 and have the same odd indices.

```
theorem odd_a_implies_pronic (N : ℕ) : a N % 2 = 1 → ∃ k : ℕ, N = k * (k + 1)
```

FC-OeisA325046.odd_a_implies_pronic

Tripling until the seed's digits reappear

MODULE CONTEXT: # Tripling until the seed's digits reappear Starting with the least positive integer not used previously, repeatedly triple it and stop the block at the first positive power whose decimal expansion contains the seed's digits. Then repeat from the new least missing positive integer. *References:* - [A331631](https://oeis.org/A331631)

V2-000251 | Tripling until the seed's digits reappear | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

[OEIS A331631](https://oeis.org/A331631) states: The sequence is conjectured to be a permutation of the positive integers (see the Crossrefs section).

```
theorem conjecture :
∃ a : ℕ → ℕ, FollowsRule a ∧ Function.Injective a ∧
∀ k > 0, ∃ n, a n = k
```

FC-OeisA331631.conjecture

Convergence of fraction numerators $a(n)/b(n)$ to e

MODULE CONTEXT: # Convergence of fraction numerators $a(n)/b(n)$ to e a: Numerators of a sequence of fractions converging to e. $$a(1) = 3, a(2) = 5$$ For $n > 2$: $$a(n) = \begin{cases} \left(\frac{n+2}{2}\right) a(n-1) - a(n-2) - \left(\frac{n-2}{2}\right) a(n-3) & \text{if } n \text{ is even} \cr 2 a(n-1) + n a(n-2) & \text{if } n \text{ is odd} \end{cases}$$ *References:* - [A340737](https://oeis.org/A340737) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [b](https://oeis.org/b)

V2-000252 | Convergence of fraction numerators $a(n)/b(n)$ to e | tendsto_exp_one

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $\lim_{n \to \infty} a(n) / b(n) = e$, where $b(n)$ is the companion denominator sequence. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem tendsto_exp_one :
Filter.Tendsto (fun n : ℕ => (a n : ℝ) / (b n : ℝ)) Filter.atTop (nhds (Real.exp 1))
```

FC-OeisA340737.tendsto_exp_one

Bounds on fractional parts $n r^2 - a(n)$ for $r = (2+\sqrt{5})/2$

MODULE CONTEXT: # Bounds on fractional parts $n r^2 - a(n)$ for $r = (2+\sqrt{5})/2$ $a(n) = \lfloor r \cdot \lfloor r \cdot n \rfloor \rfloor$, where $r = (2 + \sqrt{5})/2$. *References:* - [A341254](https://oeis.org/A341254) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000253 | Bounds on fractional parts $n r^2 - a(n)$ for $r = (2+\sqrt{5})/2$ | a_bounds

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $1/4 < n r^2 - a(n) < 3$ for $n \ge 1$, where $r = (2 + \sqrt{5})/2$. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_bounds (n : ℕ) (hn : 1 ≤ n) : (1 / 4 : ℝ) < (n : ℝ) * rSq - (a n : ℝ) ∧
(n : ℝ) * rSq - (a n : ℝ) < 3
```

FC-OeisA341254.a_bounds

Smallest number k such that $kn + 1$ is prime

MODULE CONTEXT: # Smallest number k such that $kn + 1$ is prime *References:* - [A34693](https://oeis.org/A34693)

V2-000254 | Smallest number k such that $kn + 1$ is prime | a_isBigO

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) = O(\log(n)\log(\log(n)))$.

theorem a_isBigO : (fun n ↦ (a n : ℝ)) =O[atTop] (fun n ↦ Real.log n * Real.log (Real.log n))

FC-OeisA34693.a_isBigO

V2-000255 | Smallest number k such that $kn + 1$ is prime | a_unbounded

research open | proof annotation: False | 官方代码 | 原文线索

Counter-conjecture to `a_isBigO`: $a(n) / (\log n \log \log n)$ is unbounded.

theorem a_unbounded : ¬BddAbove (Set.range fun n ↦ a n / (Real.log n * Real.log (Real.log n)))

FC-OeisA34693.a_unbounded

V2-000256 | Smallest number k such that $kn + 1$ is prime | exists_k

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: for every $n > 1$ there exists a number $k < n$ such that $nk + 1$ is a prime.

theorem exists_k {n : ℕ} (hn : 1 < n) : ∃ k < n, (n * k + 1).Prime

FC-OeisA34693.exists_k

V2-000257 | Smallest number k such that $kn + 1$ is prime | exists_k_best_possible

research solved | proof annotation: False | 官方代码 | 原文线索

The expression $1 + n^{0.74}$ does not work as an upper bound.

theorem exists_k_best_possible : ∃ n > (0 : ℕ), ∀ (k : ℕ),
k < 1 + (Real.nthRoot 100 n) ^ 74 → ¬(n * k + 1).Prime

FC-OeisA34693.exists_k_best_possible

V2-000258 | Smallest number k such that $kn + 1$ is prime | exists_k_stronger

research open | proof annotation: False | 官方代码 | 原文线索

A stronger conjecture: for every n there exists a number $k < 1 + n^{0.75}$ such that $nk + 1$ is a prime.

theorem exists_k_stronger {n : ℕ} (hn : 0 < n) : ∃ k : ℕ,
k < 1 + (Real.nthRoot 4 n) ^ 3 ∧ (n * k + 1).Prime

FC-OeisA34693.exists_k_stronger

Numerator of a sum involving binomial coefficients

MODULE CONTEXT: # Numerator of a sum involving binomial coefficients $a(n)$ is the numerator of $\sum_{k = 1}^n \frac{1}{k^3} \binom{n}{k}^2 \binom{n+k}{k}^2$ for $n \ge 1$ with $a(0) = 0$. *References:* - [A357513](https://oeis.org/A357513)

V2-000259 | Numerator of a sum involving binomial coefficients | a357513_supercongruence

research solved | proof annotation: True | 官方代码 | 原文线索

We have $a(p-1) \equiv 0 \pmod{p^4}$ for all primes $p \ge 3$ except $p=7$. proved by AlphaProof

theorem a357513_supercongruence (p : ℕ) (hp : p.Prime) (h_ge3 : p ≥ 3) (h_neq7 : p ≠ 7) :
(a (p - 1) : ℤ) ≡ 0 [ZMOD (p : ℤ) ^ 4]

FC-OeisA357513.a357513_supercongruence

V2-000260 | Numerator of a sum involving binomial coefficients | general_supercongruence

research open | proof annotation: False | 官方代码 | 原文线索

We conjecture that $u(p-1) == 0 (mod p^4)$ for all primes p, with a finite number of exceptions that depend on m.

theorem general_supercongruence (m : ℕ) : ∃ (exceptions : Finset ℕ), ∀ p, p.Prime →
p ∉ exceptions → u m (p - 1) = (0 : ZMod (p ^ 4))

FC-OeisA357513.general_supercongruence

Factor bounds for Fermat numbers

MODULE CONTEXT: # Factor bounds for Fermat numbers $a(n)$ is the minimum integer k such that the smallest prime factor of the n-th Fermat number exceeds $2^{2^n - k}$. *References:* - [A358684](https://oeis.org/A358684) - [SA22](https://doi.org/10.26493/2590-9770.1473.ec5) Lorenzo Sauras-Altuzarra, *Some properties of the factors of Fermat numbers*, Art Discrete Appl. Math. (2022).

V2-000261 | Factor bounds for Fermat numbers | oeis_358684_conjecture_0

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture: the dyadic valuation of A93179(n) - 1 does not exceed 2^n - a(n). A93179(n) is minFac(fermatNumber n), the smallest prime factor of the n-th Fermat number. The conjecture states that if P_n is the smallest prime factor of the n-th Fermat number, then $\nu_2(P_n - 1) \le 2^n - a(n)$. Substituting the definition of $a(n)$, this is equivalent to $\nu_2(P_n - 1) \le \lfloor \log_2(P_n) \rfloor$. This is Conjecture 3.4 in [SA22].

theorem oeis_358684_conjecture_0 (n : ℕ) :
padicValNat 2 (minFac (fermatNumber n) - 1) ≤ 2 ^ n - a n

FC-OeisA358684.oeis_358684_conjecture_0

A factorial ratio analogous to the Catalan numbers

MODULE CONTEXT: # A factorial ratio analogous to the Catalan numbers The sequence is defined by $a(n)=3(4n)!/(n!((n+1)!)^3)$ for $n \ge 0$. *References:* - [A361033](https://oeis.org/A361033) - [Li26] Wentao Li, [Integrality and parity of a factorial ratio](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-formalization/A361033/PROOF.md), 2026.

V2-000262 | A factorial ratio analogous to the Catalan numbers | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

The parity conjecture: "Conjecture: $a(n)$ is odd iff $n = 2^k - 1$ for some $k \ge 0$." - Peter Bala, Mar 01 2023. Proved and formalized in Lean by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture_1 (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k - 1
```

FC-OeisA361033.conjecture_1

Primes congruent to $\{3, 5, 6\} \pmod 7$

MODULE CONTEXT: # Primes congruent to $\{3, 5, 6\} \pmod 7$ Primes congruent to $3, 5, \text{ or } 6 \pmod 7$. *References:* - [A003625](https://oeis.org/A003625)

V2-000263 | Primes congruent to $\{3, 5, 6\} \pmod 7$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Represents primes p where the polynomial $x^2 + x + 2$ is irreducible over $\text{GF}(p)$. - _Federico Provvedi_, Jul 21 2018 Answer: true, the equivalence is classical (complete the square: $4(x^2+x+2) = (2x+1)^2 + 7$, then use quadratic reciprocity).

```
theorem conjecture (p : ℕ) (hp : p.Prime) :
A p ↔ Irreducible (X ^ 2 + X + 2 : (ZMod p)[X])
```

FC-OeisA3625.conjecture

Primality of continued fraction denominators $a(n)$ for $n \ge 3$

MODULE CONTEXT: # Primality of continued fraction denominators $a(n)$ for $n \ge 3$ Auxiliary sequence A051403, defined as $$\frac{(n+2) \sum_{k=0}^n k!}{2}$$ Denominator of the continued fraction $1/(2-3/(3-4/(4-5/(...(n-1)-n/(-2)))))$. The sequence is defined by the formula: $$a(n) = \frac{n^2 - 2}{\gcd(n^2 - 2, 2 \cdot A051403(n-3) + n \cdot A051403(n-4))}$$ The formula is valid for $n \ge 3$. *References:* - [A363102](https://oeis.org/A363102) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A051403](https://oeis.org/A051403)

V2-000264 | Primality of continued fraction denominators $a(n)$ for $n \ge 3$ | a_eq_one_or_prime

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: The sequence contains only 1's and primes. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_eq_one_or_prime : ∀ n : ℕ, 3 ≤ n → a n = 1 ∨ Nat.Prime (a n)
```

FC-OeisA363102.a_eq_one_or_prime

Realization of primes $p \equiv \pm 1 \pmod{10}$ by continued fraction denominators

MODULE CONTEXT: # Realization of primes $p \equiv \pm 1 \pmod{10}$ by continued fraction denominators $a(n)$ is the denominator of the finite continued fraction $$\frac{1}{2 - \frac{3}{3 - \frac{4}{4 - \frac{5}{\dots - \frac{n-1}{(n-1) - \frac{n}{-4}}}}}}$$ *References:* - [A363347](https://oeis.org/A363347) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000265 | Realization of primes $p \equiv \pm 1 \pmod{10}$ by continued fraction denominators | exists_a_eq_prime

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: The sequence contains all prime numbers which end with a 1 or 9 (i.e., primes $p \equiv 1$ or $9 \pmod{10}$). A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem exists_a_eq_prime :
∀ p : ℕ, (p.Prime ∧ (p ≡ 1 [MOD 10] ∨ p ≡ 9 [MOD 10])) → ∃ n : ℕ, a n = p
```

FC-OeisA363347.exists_a_eq_prime

A generating function with powers two and three

MODULE CONTEXT: # A generating function with powers two and three The sequence consists of the coefficients of $A(x)=1+x(A(x)^2-A(-x)^2)/2+x(A(x)^3+A(-x)^3)/2$. *References:* - [A368626](https://oeis.org/A368626) - [Li26] Wentao Li, [Generating-function bridge and parity for A368626](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/9d4a205da2a4094800f9c600a7b0b5223eda51ce/proofs/new-proofs/A368626/PROOF.md), 2026.

V2-000266 | A generating function with powers two and three | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd when $n = 2^k - 1$ for $k \ge 0$ and even elsewhere." - Paul D. Hanna, Jan 09 2024. Lean proof by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k - 1
```

FC-OeisA368626.conjecture

A generating function with powers two and four

MODULE CONTEXT: # A generating function with powers two and four The sequence consists of the coefficients of $A(x)=1+x(A(x)^2-A(-x)^2)/2+x(A(x)^4+A(-x)^4)/2$. *References:* - [A368628](https://oeis.org/A368628) - [Li26] Wentao Li, [Generating-function bridge and parity for A368628](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/b5eabe63ce276d12170d18f91b5b7327130c4cbe/proofs/new-formalization/A368628/PROOF.md), 2026.

V2-000267 | A generating function with powers two and four | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd when $n = (4^k - 1)/3$ for $k \ge 0$, and even elsewhere." - Paul D. Hanna, Jan 10 2024. Lean formalization by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = (4 ^ k - 1) / 3

FC-OeisA368628.conjecture

A generating function with powers two and four

MODULE CONTEXT: # A generating function with powers two and four The sequence consists of the coefficients of $A(x)=1+x(A(x)^2+A(-x)^2)/2+x(A(x)^4-A(-x)^4)/2$. *References:* - [A368629](https://oeis.org/A368629) - [Li26] Wentao Li, [Generating-function bridge and parity for A368629](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/9d4a205da2a4094800f9c600a7b0b5223eda51ce/proofs/new-proofs/A368629/PROOF.md), 2026.

V2-000268 | A generating function with powers two and four | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd when $n = 2^k - 1$ for $k \ge 0$, and even elsewhere." - Paul D. Hanna, Jan 10 2024. Lean proof by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k - 1

FC-OeisA368629.conjecture

A quadratic generating function with sign substitution

MODULE CONTEXT: # A quadratic generating function with sign substitution The sequence consists of the coefficients of $A(x)=1+2xA(x)^2-xA(-x)^2$. *References:* - [A368633](https://oeis.org/A368633) - [Li26] Wentao Li, [Catalan parity for A368633](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-formalization/A368633/PROOF.md), 2026.

V2-000269 | A quadratic generating function with sign substitution | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

The parity conjecture (Conjecture 1): "Conjecture: $a(n)$ is odd when $n = 2^k - 1$ for $k \ge 0$ and even elsewhere." - Paul D. Hanna, Jan 11 2024. Lean formalization by Wentao Li, with AI assistance; see [Li26].

theorem conjecture_1 (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n + 1 = 2 ^ k

FC-OeisA368633.conjecture_1

A generating function with signed square terms

MODULE CONTEXT: # A generating function with signed square terms The sequence consists of the coefficients of $A(x)=1+x(3A(x)^2-A(-x)^2)/2$. *References:* - [A368634](https://oeis.org/A368634) - [Li26] Wentao Li, [Generating-function bridge and parity for A368634](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/9d4a205da2a4094800f9c600a7b0b5223eda51ce/proofs/new-proofs/A368634/PROOF.md), 2026.

V2-000270 | A generating function with signed square terms | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd when $n = 2^k - 1$ for $k \ge 0$ and even elsewhere." - Paul D. Hanna, Jan 12 2024. Lean proof by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k - 1

FC-OeisA368634.conjecture

The generating function $A(x)=1+3xA(x)^2-2xA(-x)^2$

MODULE CONTEXT: # The generating function $A(x)=1+3xA(x)^2-2xA(-x)^2$ The sequence consists of the coefficients of $A(x)=1+3xA(x)^2-2xA(-x)^2$. *References:* - [A368635](https://oeis.org/A368635) - [Li26] Wentao Li, [Generating-function bridge and parity for A368635](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/9d4a205da2a4094800f9c600a7b0b5223eda51ce/proofs/new-formalization/A368635/PROOF.md), 2026.

V2-000271 | The generating function $A(x)=1+3xA(x)^2-2xA(-x)^2$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd when $n = 2^k - 1$ for $k \ge 0$, and even elsewhere." - Paul D. Hanna, Jan 12 2024. Lean proof by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n : ℕ) : Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k - 1

FC-OeisA368635.conjecture

Integrality of the factorial ratio $\frac{(12n+6)! (6n+9)!}{108 (4n+2)! (2n+3)! ((6n+5)!)^2}$

MODULE CONTEXT: # Integrality of the factorial ratio $\frac{(12n+6)! (6n+9)!}{108 (4n+2)! (2n+3)! ((6n+5)!)^2}$ The factorial ratio $$a(n) = \frac{(12n + 6)! \cdot (6n + 9)!}{108 \cdot (4n + 2)! \cdot (2n + 3)! \cdot ((6n + 5)!)^2}$$ It is conjectured that $a(n)$ are integers. *References:* - [A368692](https://oeis.org/A368692) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000272 | Integrality of the factorial ratio $\frac{(12n+6)! (6n+9)!}{108 (4n+2)! (2n+3)! ((6n+5)!)^2}$ | a_is_int

research solved | proof annotation: True | 官方代码 | 原文线索

It is conjectured here that $a(n)$ are integers. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem a_is_int (n : ℕ) : 108 * ((4 * n + 2)! * (2 * n + 3)! * ((6 * n + 5)!) ^ 2) | (12 * n + 6)! * (6 * n + 9)!

FC-OeisA368692.a_is_int

Home primes (OEIS A037274)

MODULE CONTEXT: # Home primes (OEIS A037274) Starting from an integer $n\geq 2$, list its prime factors in nondecreasing order with multiplicity, concatenate their decimal representations, and repeat. The home-prime conjecture says that this process always reaches a prime. For example, $$25 \longmapsto 55 \longmapsto 511 \longmapsto 773.$$ *References:* * [OEIS A037274](https://oeis.org/A037274) * M. Herman and J. Schiffman, *Investigating home primes and their families*, Mathematics Teacher 107 (2014), 606-614

V2-000273 | Home primes (OEIS A037274) | home_prime_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Every integer at least two reaches a home prime.

theorem home_prime_conjecture : ∀ n : ℕ, 2 ≤ n → ReachesPrime n

FC-OeisA37274.home_prime_conjecture

Unique realization of odd primes $p \notin \{3, 5\}$ by continued fraction values

MODULE CONTEXT: # Unique realization of odd primes $p \notin \{3, 5\}$ by continued fraction values $a(n)$ is the denominator of the finite continued fraction $$\frac{1}{2 - \frac{3}{3 - \frac{4}{4 - \frac{5}{\dots - \frac{n-1}{(n-1) - \frac{n}{n+4}}}}}}$$ *References:* - [A372761](https://oeis.org/A372761) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000274 | Unique realization of odd primes $p \notin \{3, 5\}$ by continued fraction values | exists_unique_a_eq_prime

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: Except for 3 and 5, all odd primes appear in the sequence once. - _Thomas Scheuerle_, May 11 2024 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem exists_unique_a_eq_prime :
∀ p : ℕ, Nat.Prime p ∧ p % 2 = 1 ∧ p ≠ 3 ∧ p ≠ 5 → ∃! n, n ≥ 3 ∧ a n = p

FC-OeisA372761.exists_unique_a_eq_prime

A cubic generating function with ternary substitution

MODULE CONTEXT: # A cubic generating function with ternary substitution The coefficients of the generating function $A(x)=x+x^2+(2A(x)^3+A(x^3))/3$, with constant term $a(0)=0$. *References:* - [A375439](https://oeis.org/A375439) - [A038754](https://oeis.org/A038754) - [Li26] Wentao Li, [Parity of A375439](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-proofs/A375439/PROOF.md), 2026.

V2-000275 | A cubic generating function with ternary substitution | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd iff n is in A038754, which consists of numbers of the form 3^k and $2*3^k$." - Paul D. Hanna, Aug 21 2024. Lean proof by Wentao Li, with AI assistance; see [Li26].

theorem conjecture (n : ℕ) (hn : 1 ≤ n) :
Odd (a n) ↔ ∃ k : ℕ, n = 3 ^ k ∨ n = 2 * 3 ^ k

FC-OeisA375439.conjecture

Number of primes $< n^3$

MODULE CONTEXT: # Number of primes $< n^3$ Number of primes strictly less than n^3. *References:* - [A038098](https://oeis.org/A038098)

V2-000276 | Number of primes $< n^3$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (i): for any integer $k > 2$, the sequence $\pi(n^k)/n^k$ ($n = 2, 3, \ldots$) is strictly decreasing, where $\pi(x)$ denotes the number of primes not exceeding x. - Zhi-Wei Sun, Oct 17 2015

theorem conjecture1 (k : ℕ) (hk : 2 < k) (n : ℕ) (hn : 2 ≤ n) :
(Nat.primeCounting ((n + 1) ^ k) : ℚ) / ((n + 1) ^ k : ℚ) <
(Nat.primeCounting (n ^ k) : ℚ) / (n ^ k : ℚ)

FC-OeisA38098.conjecture1

V2-000277 | Number of primes $< n^3$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (ii): all the numbers $\pi(n^2)/n^2$ ($n = 1, 2, 3, \ldots$) are pairwise distinct. Moreover, we have $\pi(n^2)/n^2 > \pi((n+1)^2)/(n+1)^2$ for all $n > 15646$. - Zhi-Wei Sun, Oct 17 2015

```
theorem conjecture2 :
(∀ m n : ℕ, 1 ≤ m → 1 ≤ n →
(Nat.primeCounting (m ^ 2) : ℚ) / (m ^ 2 : ℚ) =
(Nat.primeCounting (n ^ 2) : ℚ) / (n ^ 2 : ℚ) → m = n) ∧
(∀ n : ℕ, 15646 < n →
(Nat.primeCounting ((n + 1) ^ 2) : ℚ) / ((n + 1) ^ 2 : ℚ) <
(Nat.primeCounting (n ^ 2) : ℚ) / (n ^ 2 : ℚ))
```

FC-OeisA38098.conjecture2

Number of primes $< n^2$

MODULE CONTEXT: # Number of primes $< n^2$ Number of primes strictly less than n^2. *References:* - [A038107](https://oeis.org/A038107)

V2-000278 | Number of primes $< n^2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: all the numbers $\sum_{i=j}^k \frac{1}{a(i)}$ with $1 < j \le k$ have pairwise distinct fractional parts. - Zhi-Wei Sun, Sep 24 2015

```
theorem conjecture (j k j' k' : ℕ)
(hj : 1 < j) (hjk : j ≤ k) (hj' : 1 < j') (hj'k' : j' ≤ k')
(h_eq : Int.fract (∑ i ∈ Finset.Icc j k, (1 : ℝ) / (a i : ℝ)) =
Int.fract (∑ i ∈ Finset.Icc j' k', (1 : ℝ) / (a i : ℝ))) :
j = j' ∧ k = k'
```

FC-OeisA38107.conjecture

A logarithmic derivative with prime-divisible coefficients

MODULE CONTEXT: # A logarithmic derivative with prime-divisible coefficients Let $F(x)$ be the generating function of A381353, with $F(0)=1$, $[x]F(x)=1$, and $[x^n]F(x)^{p_n}=0$ for $n>1$, where p_n is the n-th prime. The sequence is defined by $\sum_{n\ge1}a(n)x^n=xF'(x)/F(x)$. *References:* - [A381355](https://oeis.org/A381355) - [A381353](https://oeis.org/A381353) - [Li26] Wentao Li, [Prime divisibility of logarithmic-derivative coefficients](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-proofs/A381355/PROOF.md), 2026.

V2-000279 | A logarithmic derivative with prime-divisible coefficients | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is divisible by $\operatorname{prime}(n)$ for $n > 1$." - Paul D. Hanna, Mar 11 2025. Proved and formalized in Lean by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture (n : ℕ) (hn : 1 < n) : (primeN n : ℤ) ∣ a n
```

FC-OeisA381355.conjecture

Periodicity of k-th prime factors in coupled nonlinear recurrence $a(n)$

MODULE CONTEXT: # Periodicity of k-th prime factors in coupled nonlinear recurrence $a(n)$ The sequence $a(n)$ is defined by $a(n) = a(n-1)b(n-2) + a(n-2)b(n-1)$ where $b(n) = a(n-1)b(n-2) - a(n-2)b(n-1)$, with $a(0) = b(0) = b(1) = 1$ and $a(1) = 2$. *References:* - [A382590](https://oeis.org/A382590) - [mathoverflow/490330](https://mathoverflow.net/questions/490330): B. Morga, "Peculiar family of recurrence formula where for $n>1$ if you take the n-th prime factor of each term, you get an eventually periodic sequence", Mar. 31, 2025.

V2-000280 | Periodicity of k-th prime factors in coupled nonlinear recurrence $a(n)$ | kthPrimeFactor_periodic

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: For any $k > 1$, if you take the k-th prime factor of each term, you get an eventually periodic sequence. - _Bryle Morga_, Mar 31 2025 Here the k-th prime factor of $a(n)$ is the k-th smallest distinct prime divisor of $|a(n)|$; for example the second prime factors of $a(5) = 18$, $a(6) = 20$ and $a(7) = 896$ are 3, 5 and 7. This was proved by Terence Tao in a [MathOverflow answer](https://mathoverflow.net/a/490348), using that $a(n)$ divides $a(n+3)$ for all n. Lean formalization by Wentao Li, with AI assistance.

```
theorem kthPrimeFactor_periodic : ∀ k : ℕ, k ≥ 2 → ∃ N₀ p : ℕ, p > 0 ∧
∀ n : ℕ, n ≥ N₀ → kthPrimeFactor k (a (n + p)) = kthPrimeFactor k (a n)
```

FC-OeisA382590.kthPrimeFactor_periodic

Lengths of the 2-dense sublists of divisors of n

MODULE CONTEXT: # Lengths of the 2-dense sublists of divisors of n Write the divisors of n in increasing order and split this list between consecutive divisors $d < d'$ with $d' > 2d$. The pieces are the 2-dense sublists of divisors of n. Row n of the triangle lists their lengths. Row n of A384225 lists the numbers of odd terms in them. *References:* - [A384222](https://oeis.org/A384222) - [A384225](https://oeis.org/A384225)

V2-000281 | Lengths of the 2-dense sublists of divisors of n | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture 1: row n is a palindromic composition of A000005(n)." A composition of $A000005(n)$ is a list of positive integers with sum $A000005(n)$, the number of divisors of n. Solved by Claude (Anthropic) prompted by otoinobuo. The map $d \mapsto n/d$ reverses the list of divisors and preserves the ratio of consecutive divisors.

```
theorem conjecture_1 (n : ℕ) (hn : 0 < n) :
(∀ k ∈ row n, 0 < k) ∧ (row n).sum = n.divisors.card ∧ (row n).reverse = row n
```

FC-OeisA384222.conjecture_1

V2-000282 | Lengths of the 2-dense sublists of divisors of n | conjecture_formula

research solved | proof annotation: True | 官方代码 | 原文线索

"T(n,k) = A384225(n,k) * A001511(n) = A384225(n,k) * A000005(n) / A001227(n). (conjectured)." - Omar E. Pol, Apr 13 2026 This formalizes the first equality. The ruler function is $A001511(n) = \nu_2(n) + 1$. Solved by Claude (Anthropic) prompted by otoinobuo. If d and $2d$ divide n, then they lie in the same 2-dense sublist. So each sublist is a union of sets $\{o, 2o, \dots, 2^{\nu_2(n)} o\}$ with o odd.

```
theorem conjecture_formula (n : ℕ) (hn : 0 < n) :
row n = (oddRow n).map ((padicValNat 2 n + 1) * ·)
```

FC-OeisA384222.conjecture_formula

Conjectures associated with A038552

MODULE CONTEXT: # Conjectures associated with A038552 A038552 lists the largest squarefree number k such that the imaginary quadratic field $\mathbb{Q}(\sqrt{-k})$ has class number n. The conjectures state that: 1. All terms are congruent to $19 \pmod{24}$. 2. This is also the largest absolute value of negative fundamental discriminant d for class number n. 3. For even n, if k is the largest odd number with $h(-k) = n$ and k' is the largest even number with $h(-k') = n$, then $k > k'$. Here $h(D)$ is the class number of the quadratic field with discriminant D, so k and k' are absolute values of negative fundamental discriminants, not radicands. The source states conjecture 2 in this form. The squarefree condition in the definition is needed for the maximum to exist, since $\mathbb{Q}(\sqrt{-k}) = \mathbb{Q}(\sqrt{-4k})$. Conjecture 2 is not a restatement of the definition. Both maxima range over the same imaginary quadratic fields, but they maximize different integers attached to those fields. A038552 uses the squarefree radicand k, whereas the discriminant of $\mathbb{Q}(\sqrt{-k})$ is $-k$ for $k \equiv 3 \pmod 4$ and $-4k$ otherwise. The map $k \mapsto |d|$ is not monotone: it sends 2 to 8 and 3 to 3. So conjecture 2 says that the largest term k satisfies $k \equiv 3 \pmod 4$, and that $4k' \le k$ for every $k' \equiv 1, 2 \pmod 4$ with class number n. *References:* - [Sta67] Stark, Harold M. "A complete determination of the complex quadratic fields of class-number one." Michigan Mathematical Journal 14.1 (1967): 1-27. - oeis.org/A038552

V2-000283 | Conjectures associated with A038552 | isA038552_eq_largestNegFundDisc

research open | proof annotation: False | 官方代码 | 原文线索

A038552 also gives the largest absolute value of negative fundamental discriminant for each class number.

```
theorem isA038552_eq_largestNegFundDisc {n k : ℕ} (h : IsA038552 n k) :
IsLargestNegFundDiscrForClassNumber (n := n) k
```

FC-OeisA38552.isA038552_eq_largestNegFundDisc

V2-000284 | Conjectures associated with A038552 | largestEven_lt_largestOdd_negFundDisc

research open | proof annotation: False | 官方代码 | 原文线索

For even class number n, let k be the largest odd number such that the quadratic field with discriminant $-k$ has class number n, and let k' be the largest even such number, when they exist. Then $k > k'$.

```
theorem largestEven_lt_largestOdd_negFundDisc {n k k' : ℕ} (hn : Even n)
(hk : IsGreatest {m : ℕ | Odd m ∧ IsFundamentalDiscr (-m : ℤ) ∧
classNumberOfDiscriminant (-m : ℤ) = n} k)
(hk' : IsGreatest {m : ℕ | Even m ∧ IsFundamentalDiscr (-m : ℤ) ∧
classNumberOfDiscriminant (-m : ℤ) = n} k') : k' < k
```

FC-OeisA38552.largestEven_lt_largestOdd_negFundDisc

V2-000285 | Conjectures associated with A038552 | mod_24_of_isA038552

research open | proof annotation: False | 官方代码 | 原文线索

All terms of A038552 are congruent to $19 \pmod{24}$.

```
theorem mod_24_of_isA038552 {n k : ℕ} (h : IsA038552 n k) : k % 24 = 19
```

FC-OeisA38552.mod_24_of_isA038552

V2-000286 | Conjectures associated with A038552 | starkHeegner_classNumberOne

research solved | proof annotation: False | 官方代码 | 原文线索

The Stark-Heegner theorem [Sta67] implies that the squarefree $k > 0$ such that $\mathbb{Q}(\sqrt{-k})$ has class number 1 are exactly $\{1, 2, 3, 7, 11, 19, 43, 67, 163\}$.

```
theorem starkHeegner_classNumberOne :
{k : ℕ | Squarefree k ∧ HasClassNumber k 1} = {1, 2, 3, 7, 11, 19, 43, 67, 163}
```

FC-OeisA38552.starkHeegner_classNumberOne

Smallest composite c such that $\textrm{primorial}(n) + c$ is prime

MODULE CONTEXT: # Smallest composite c such that $\textrm{primorial}(n) + c$ is prime *References:* - [A038771](https://oeis.org/A038771)

V2-000287 | Smallest composite c such that $\textrm{primorial}(n) + c$ is prime | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $\liminf_{n \to \infty} \frac{a(n)}{p_{n+1}^2} = 1 <$ $\limsup_{n \to \infty} \frac{a(n)}{p_{n+1}^2} = 2$. - Charles R Greathouse IV and Thomas Ordowski, Apr 24 2015

```
theorem conjecture1 :
let p_next_sq (n : ℕ) : ℝ := ((Nat.nth Nat.Prime n : ℝ)) ^ 2
let seq (n : ℕ) : ℝ := (a n : ℝ) / p_next_sq n
(liminf seq atTop = 1) ∧ (limsup seq atTop = 2)
```

FC-OeisA38771.conjecture1

V2-000288 | Smallest composite c such that $\textrm{primorial}(n) + c$ is prime | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

All the terms in this sequence have exactly two prime factors. This conjecture is true for the first 133 terms. - [Dmitry Kamenetsky](https://oeis.org/wiki/User:Dmitry_Kamenetsky), Jan 06 2019

```
theorem conjecture2 (n : ℕ) : (a n).IsSemiprime
```

FC-OeisA38771.conjecture2

A differential generating function

MODULE CONTEXT: # A differential generating function The generating function satisfies $A(x)=x+(xA(x)^2)'$. Its constant term is zero. Seiichi Manyama's recurrence is $a(1)=1$ and $a(n)=(n+1)\sum_{k=1}^{n-1}a(k)a(n-k)$ for $n>1$. *References:* - [A397588](https://oeis.org/A397588) - [Li26] Wentao Li, [Parity and divisibility for A397588](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-formalization/A397588/PROOF.md), 2026.

V2-000289 | A differential generating function | conjecture_parity

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n)$ is odd iff n is a power of 2 for $n \ge 1$." - Paul D. Hanna, Jul 03 2026. Lean formalization by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture_parity (n : ℕ) (hn : 1 ≤ n) :
Odd (a n) ↔ ∃ k : ℕ, n = 2 ^ k
```

FC-OeisA397588.conjecture_parity

V2-000290 | A differential generating function | three_dvd_a

research solved | proof annotation: True | 官方代码 | 原文线索

"$a(n)$ divisible by 3 for $n > 1$." - Paul D. Hanna, Jul 03 2026. Lean formalization by Wentao Li, with AI assistance; see [Li26].

```
theorem three_dvd_a (n : ℕ) (hn : 1 < n) : 3 ∣ a n
```

FC-OeisA397588.three_dvd_a

Maximum number of terms in a 2-dense sublist of divisors of n

MODULE CONTEXT: # Maximum number of terms in a 2-dense sublist of divisors of n Write the divisors of n in increasing order and split this list between consecutive divisors $d < d'$ with $d' > 2d$. The pieces are the 2-dense sublists of divisors of n. The sequence gives the maximum number of terms in such a sublist. *References:* - [A400194](https://oeis.org/A400194) - [A400195](https://oeis.org/A400195) - [A384222](https://oeis.org/A384222)

V2-000291 | Maximum number of terms in a 2-dense sublist of divisors of n | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: a(n) is the product of the ruler function of n and the maximum number of odd terms in a 2-dense sublist of divisors of n." "Conjecture: a(n) = A001511(n) * A400195(n)." The ruler function is $A001511(n) = \nu_2(n) + 1$. Solved by Claude (Anthropic) prompted by otoinobuo. If d and $2d$ divide n, then they lie in the same 2-dense sublist. So each sublist is a union of sets $\{o, 2o, \dots, 2^{\nu_2(n)} o\}$ with o odd.

```
theorem conjecture (n : ℕ) (hn : 0 < n) : a n = (padicValNat 2 n + 1) * b n
```

FC-OeisA400194.conjecture

Decimal concatenation of a cofactor and the largest prime factor

MODULE CONTEXT: # Decimal concatenation of a cofactor and the largest prime factor For a composite positive integer n, let p be its largest prime factor. The sequence value $a(n)$ concatenates the decimal representations of n/p and p, in that order. It fixes primes and one, and is extended here by $a(0)=0$. OEIS A397122 records the prime reached by iterating this map from each starting value, or -1 if no prime is reached. *References:* - [OEIS A401006](https://oeis.org/A401006) - [OEIS A397122](https://oeis.org/A397122)

V2-000292 | Decimal concatenation of a cofactor and the largest prime factor | eventually_prime

research open | proof annotation: False | 官方代码 | 原文线索

If we start at $n \ge 2$, and repeatedly apply the map $k \mapsto a(k)$, we will eventually reach a prime, which is then a fixed point. - [OEIS A401006](https://oeis.org/A401006)

```
theorem eventually_prime : ∀ n : ℕ, 2 ≤ n → ReachesPrime n
```

FC-OeisA401006.eventually_prime

The prime numbers

MODULE CONTEXT: # The prime numbers The n-th prime number p_n, where $p_1 = 2$. *References:* - [A000040](https://oeis.org/A000040)

V2-000293 | The prime numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture from Thomas Ordowski (2023): $\log \log a(n+1) - \log \log a(n) < 1/n$ for $n > 0$.

```
theorem conjecture (n : ℕ) (hn : 0 < n) :
Real.log (Real.log (a (n + 1) : ℝ)) - Real.log (Real.log (a n : ℝ)) < 1 / (n : ℝ)
```

FC-OeisA40.conjecture

No powers as partition numbers

MODULE CONTEXT: # No powers as partition numbers There are no partition numbers $a(k)$ of the form x^m, with x,m integers >1. *Reference:* [A41](https://oeis.org/A41)

V2-000294 | No powers as partition numbers | noPowerPartitionNumber

research open | proof annotation: False | 官方代码 | 原文线索

There are no partition numbers $a(k)$ of the form x^m, with x,m integers >1. See comment by Zhi-Wei Sun (Dec 02 2013).

```
theorem noPowerPartitionNumber : ∀ k, ¬IsPerfectPower (partitionNumber k)
```

FC-OeisA41.noPowerPartitionNumber

Least positive multiple of n in base 10 with digits 0 and 1

MODULE CONTEXT: # Least positive multiple of n in base 10 with digits 0 and 1 Least positive multiple of n that when written in base 10 uses only 0's and 1's. *References:* - [A004290](https://oeis.org/A004290)

V2-000295 | Least positive multiple of n in base 10 with digits 0 and 1 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is known that $a(10^k - 1) = (10^{9k} - 1) / 9$ for all k. Is $a(n) < a(10^k - 1)$ for all $n < 10^k - 1$? - David Radcliffe, Aug 01 2025

```
theorem conjecture (k : ℕ) :
∀ n < 10 ^ k - 1, a n < a (10 ^ k - 1)
```

FC-OeisA4290.conjecture

Denominators of coefficients in Stirling's expansion for $\log(\Gamma(z))$

MODULE CONTEXT: # Denominators of coefficients in Stirling's expansion for $\log(\Gamma(z))$ The n-th term is the denominator of $\frac{B_{2n}}{2n(2n-1)}$ where B_{2n} is the $2n$-th Bernoulli number. *References:* - [A046969](https://oeis.org/A046969)

V2-000296 | Denominators of coefficients in Stirling's expansion for $\log(\Gamma(z))$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture I: if $n > 2$, then $\frac{a(\text{A005382}(n))}{12}$ is prime, where A005382 is the sequence of primes p such that $2p-1$ is also prime. Since $\text{A005382}(1) = 2$, $\text{A005382}(2) = 3$ and $\text{A005382}(3) = 7$, this says that $\frac{a(p)}{12}$ is prime for every prime $p > 3$ such that $2p-1$ is also prime. - Lorenzo Sauras Altuzarra, Oct 13 2020

```
theorem conjecture1 (p : ℕ) (hp : p.Prime) (hp' : (2 * p - 1).Prime) (h3 : 3 < p) :
(a p / 12).Prime
```

FC-OeisA46969.conjecture1

V2-000297 | Denominators of coefficients in Stirling's expansion for $\log(\Gamma(z))$ | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture II: if $\frac{a(n)}{12}$ is prime, then $\frac{a(n-1)}{12} - (n-1)$, $\frac{a(n)}{12} - n$ and $\frac{a(n+2)}{12} - (n+2)$ are multiples of 6. - Lorenzo Sauras Altuzarra, Oct 13 2020 This is false for $n = 236791$.

```
theorem conjecture2 :
¬ ∀ (n : ℕ), 2 ≤ n → 12 ∣ a n → Nat.Prime (a n / 12) →
12 ∣ a (n - 1) → 12 ∣ a (n + 2) →
6 ∣ ((a (n - 1) / 12 : ℤ) - (n - 1 : ℤ)) ∧
6 ∣ ((a n / 12 : ℤ) - (n : ℤ)) ∧
6 ∣ ((a (n + 2) / 12 : ℤ) - (n + 2 : ℤ))
```

FC-OeisA46969.conjecture2

$a(n) = \sum_{k=1}^n (k^2 \bmod n)$

MODULE CONTEXT: # $a(n) = \sum_{k=1}^n (k^2 \bmod n)$ *References:* - [A048153](https://oeis.org/A048153)

V2-000298 | $a(n) = \sum_{k=1}^n (k^2 \bmod n)$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: $a(n) <= \frac{n^2-1}{2}$. - _Aspen A.M. Meissner_, Mar 06 2025"

theorem conjecture (n : ℕ) (hn : 1 ≤ n) : a n ≤ (n ^ 2 - 1) / 2

FC-OeisA48153.conjecture

Nearest integer to $n/\sqrt{2}$

MODULE CONTEXT: # Nearest integer to $n/\sqrt{2}$ Nearest integer to $n/\sqrt{2}$, defined by $\lfloor n/\sqrt{2} + 1/2 \rfloor$. *References:* - [A049473](https://oeis.org/A049473)

V2-000299 | Nearest integer to $n/\sqrt{2}$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Let $s(n) = \zeta(3) - \sum_{k=1}^n \frac{1}{k^3}$. Conjecture: for $n \ge 1$, $s(a(n)) < \frac{1}{n^2} < s(a(n)-1)$, and the difference sequence of A049473 consists solely of 0's and 1's, in positions given by the nonhomogeneous Beatty sequences A001954 and A001953, respectively. - Clark Kimberling, Oct 05 2014

theorem conjecture :
(∀ n : ℕ, 1 ≤ n → s (a n) < 1 / (n : ℝ) ^ 2 ∧ 1 / (n : ℝ) ^ 2 < s (a n - 1)) ∧
(∀ n : ℕ, 1 ≤ n →
let diff : ℕ := a n - a (n - 1)
(diff = 0 ↔ n - 1 ∈ A001954) ∧ (diff = 1 ↔ n - 1 ∈ A001953))

FC-OeisA49473.conjecture

Asymptotics of subsets of $\{1, 2, \dots, n\}$ with integer average

MODULE CONTEXT: # Asymptotics of subsets of $\{1, 2, \dots, n\}$ with integer average Number of nonempty subsets of $\{1, 2, 3, \dots, n\}$ whose elements have an integer average. Benoit Cloitre conjectured the asymptotic expansion: $$a(n) = \frac{2^{n+1}}{n} \left(1 + \frac{1}{n} + \frac{3}{n^2} + \frac{13}{n^3} + \frac{75}{n^4} + \frac{541}{n^5} + o\left(\frac{1}{n^5}\right)\right)$$ where the coefficients $1, 1, 3, 13, 75, 541, \dots$ are the Fubini numbers (preferential arrangements, A000670). *References:* - [A051293](https://oeis.org/A051293) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al. - [A000670](https://oeis.org/A000670)

V2-000300 | Asymptotics of subsets of $\{1, 2, \dots, n\}$ with integer average | tendsto_aReal_asymptotic

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: $a(n) = 2^{n+1}/n \cdot (1 + 1/n + 3/n^2 + 13/n^3 + 75/n^4 + 541/n^5 + o(1/n^5))$. - _Benoit Cloitre_, Oct 20 2002 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem tendsto_aReal_asymptotic :
Tendsto (fun n : ℕ =>
(aReal n - (2 ^ (n + 1) / (n : ℝ)) *
(1 + 1 / (n : ℝ) + 3 / (n : ℝ) ^ 2 + 13 / (n : ℝ) ^ 3 + 75 / (n : ℝ) ^ 4 + 541 / (n : ℝ) ^ 5)) /
(2 ^ (n + 1) / (n : ℝ) ^ 6)) atTop (nhds 0)

FC-OeisA51293.tendsto_aReal_asymptotic

Maximum exponent in the prime factorization of n

MODULE CONTEXT: # Maximum exponent in the prime factorization of n *References:* - [A051903](https://oeis.org/A051903)

V2-000301 | Maximum exponent in the prime factorization of n | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Are there composite numbers $n > 4$ such that $n \equiv a(n) \pmod{\phi(n)}$? - Thomas Ordowski, Dec 02 2019 This question is equivalent to Lehmer's totient problem `LehmerTotient.lehmer_totient`; a positive answer here falsifies the universal statement asked about in `Erdos828.erdos_828.variants.lehmer_conjecture`. Any composite n with $\phi(n) \mid n - 1$ is squarefree, so $a(n) = 1$ and the condition here holds. Conversely, let $n > 4$ be composite with $\phi(n) \mid n - a(n)$ and $e = a(n) \ge 2$, and pick p with $p^e \mid n$. Then $p^{e-1} \mid \phi(n) \mid n - e$ and $p^{e-1} \mid n$, so $p^{e-1} \mid e$, which forces $p = 2$ and $e = 2$. Now $n = 4m$ with m odd and squarefree, and $2\phi(m) \mid 4m - 2$ makes $\phi(m)$ odd, so $m = 1$ and $n = 4$. Hence $e = 1$ and the condition is $\phi(n) \mid n - 1$.

theorem conjecture1 :
∃ n : ℕ, 4 < n ∧ ¬ n.Prime ∧ n.totient | (n - a n)

FC-OeisA51903.conjecture1

V2-000302 | Maximum exponent in the prime factorization of n | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Are there odd numbers n such that $a(n) > 1$ and $n \equiv a(n) \pmod{\lambda(n)}$? (Equivalently, odd numbers n such that $a(n) > 1$ and $b^n \equiv b^{a(n)} \pmod n$ for all b.) - Thomas Ordowski, Dec 02 2019 The answer is no. Let $q^E \| n$ with $E = a(n) \ge 2$ and take $b = q + 1$: then $(q+1)^{n-E} \equiv 1 \pmod{q^E}$, so lifting the exponent gives $q^{E-1} \mid n - E$. Since also $q^{E-1} \mid n$, we get $q^{E-1} \mid E$, which is impossible for $q \ge 3$.

theorem conjecture2 :
∃ n : ℕ, Odd n ∧ 1 < a n ∧ ∀ b : ℕ, b ^ n ≡ b ^ (a n) [MOD n]

FC-OeisA51903.conjecture2

V2-000303 | Maximum exponent in the prime factorization of n | conjecture3

research solved | proof annotation: True | 官方代码 | 原文线索

Are there odd numbers n such that $a(n) > 1$ and $n \equiv a(n) \pmod{\operatorname{ord}_n(2)}$? (Equivalently, odd numbers n such that $a(n) > 1$ and $2^n \equiv 2^{a(n)} \pmod n$.) - Thomas Ordowski, Dec 02 2019 The answer is yes: $n = 7 \cdot 631 \cdot 881 \cdot 3511^2 \cdot 201961 = 96879631678643449 37$ is odd, $a(n) = 2$ and $2^n \equiv 2^2 \pmod n$. A square factor $q^2 \mid n$ forces q to be a Wieferich prime; 3511 works because $\operatorname{ord}_{3511^2}(2) = 1755$ is odd and $n \equiv 2 \pmod{1755}$.

theorem conjecture3 :
∃ n : ℕ, Odd n ∧ 1 < a n ∧ 2 ^ n ≡ 2 ^ (a n) [MOD n]

FC-OeisA51903.conjecture3

Apéry numbers

MODULE CONTEXT: # Apéry numbers Apéry numbers: $$a(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}$$ *References:* - [A005258](https://oeis.org/A005258)

V2-000304 | Apéry numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For each $n = 1, 2, 3, \dots$ the polynomial $a_n(x) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k} x^k$ is irreducible over the field of rational numbers. - Zhi-Wei Sun, Mar 21 2013

theorem conjecture (n : ℕ) (hn : 1 ≤ n) : Irreducible (aperyPoly n)

FC-OeisA5258.conjecture

Expansion of g.f. $(1-\sqrt{1-4x-4x^2})/(2(1+x))$

MODULE CONTEXT: # Expansion of g.f. $(1-\sqrt{1-4x-4x^2})/(2(1+x))$ The n-th term $a(n)$ is given by $$a(n) = \sum_{k=0}^{n-1} \frac{1}{k+1} \binom{2k}{k} \binom{k}{n-1-k}$$ *References:* - [A052709](https://oeis.org/A052709)

V2-000305 | Expansion of g.f. $(1-\sqrt{1-4x-4x^2})/(2(1+x))$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: for $n > 0$, $a(n)$ is also the number of sequences of length $n - 1$ covering an initial interval of positive integers and avoiding three terms $(\dots, x, \dots, y, \dots, z, \dots)$ such that $x \le y \le z$. - Gus Wiseman, Jun 17 2021

theorem conjecture (n : ℕ) (hn : 0 < n) [Fintype (sequencesCountedByA052709 n)] :
a n = Fintype.card (sequencesCountedByA052709 n)

FC-OeisA52709.conjecture

$a(n) = (\text{smallest prime} > n^2) - n^2$

MODULE CONTEXT: # $a(n) = (\text{smallest prime} > n^2) - n^2$ The difference between the smallest prime strictly greater than n^2 and n^2. *References:* - [A053000](https://oeis.org/A053000)

V2-000306 | $a(n) = (\text{smallest prime} > n^2) - n^2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) \le 1 + \phi(n)$ for $n > 0$. This improves on Oppermann's conjecture, which says $a(n) < n$. - Thomas Ordowski, Dec 17 2014

theorem conjecture (n : ℕ) (hn : 0 < n) : a n ≤ 1 + n.totient

FC-OeisA53000.conjecture

Concatenation of the next n numbers

MODULE CONTEXT: # Concatenation of the next n numbers $a(n)$ is the concatenation of the next n numbers: the integers from $\frac{(n-1)n}{2} + 1$ up to $\frac{n(n+1)}{2}$. *References:* - [A053067](https://oeis.org/A053067)

V2-000307 | Concatenation of the next n numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

"The second term is a prime. When is the next prime, if there is another? - _N. J. A. Sloane_, Dec 16 2016"

theorem conjecture :
conjecture.answer =
if h : ∃ n, 2 < n ∧ (a n).Prime then
some (sInf {n | 2 < n ∧ (a n).Prime})
else
none

FC-OeisA53067.conjecture

Catalan-Larcombe-French sequence

MODULE CONTEXT: # Catalan-Larcombe-French sequence The Catalan-Larcombe-French sequence defined by $a(0)=1$, $a(1)=8$, and $$n^2 a(n) = 8(3n^2 - 3n + 1) a(n-1) - 128(n-1)^2 a(n-2)$$ for $n \ge 2$. *References:* - [A053175](https://oeis.org/A053175) - [ZS18] [Zhu, B.-X., Sun, Z.-W., *Hankel-type determinants for some combinatorial sequences*, Int. J. Number Theory **14** (2018), 1265-1277](https://arxiv.org/abs/1609.06810)

V2-000308 | Catalan-Larcombe-French sequence | conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture: let $P(n)$ be the $(n+1) \times (n+1)$ Hankel-type determinant with (i,j)-entry equal to $a(i+j)$ for all $i,j = 0, \ldots, n$. Then $P(n)/2^{n(n+3)}$ is a positive odd integer. - Zhi-Wei Sun, Aug 14 2013 Proved by [ZS18], Theorem 1.2: $2^{-n(n+3)} |P_{i+j}|_{0 \le i,j \le n}$ is a positive odd integer for every $n \in \mathbb{N}$, where P_n are the Catalan-Larcombe-French numbers.

```
theorem conjecture (n : ℕ) :
let detP := (hankelMatrix n).det
let pow2 := (2 : ℤ) ^ (n * (n + 3))
pow2 ∣ detP ∧ 0 < detP / pow2 ∧ (detP / pow2) % 2 = 1
```

FC-OeisA53175.conjecture

Least m such that $\phi(m) = n!$

MODULE CONTEXT: # Least m such that $\phi(m) = n!$ The smallest positive integer m whose Euler totient equals $n!$. *References:* - [A055487](https://oeis.org/A055487)

V2-000309 | Least m such that $\phi(m) = n!$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: unless $n! + 1$ is prime (i.e., $n \in \text{A002981}$), $a(n) = p q$ where p is the least prime $> \sqrt{n!}$ such that $(p - 1) \mid n!$ and $q = \frac{n!}{p - 1} + 1$ is prime. - M. F. Hasler, Oct 04 2009 The conclusion $(p(n)).\text{Prime}$ asserts that such a prime p exists: if no prime satisfies the search conditions then `p n` is `sInf ∅ = 0`, which is not prime.

```
theorem conjecture (n : ℕ) (hn : 1 ≤ n) (h_not_prime : ¬ isFactorialPrime n) :
(p n).Prime ∧ a n = p n * q n
```

FC-OeisA55487.conjecture

Divisibility of $2^n + 1$ by n

MODULE CONTEXT: # Divisibility of $2^n + 1$ by n A56777 lists composite numbers n satisfying both $\varphi(n+12) = \varphi(n) + 12$ and $\sigma(n+12) = \sigma(n) + 12$. The conjectures state identities connecting A56777 and prime quadruples (A7530), as well as congruences satisfied by the members of A56777. *References:* - [A56777](https://oeis.org/A56777)

V2-000310 | Divisibility of $2^n + 1$ by n | comesFromPrimeQuadruple_of_a

research open | proof annotation: False | 官方代码 | 原文线索

All members of the sequence A56777 come from prime quadruples.

```
theorem comesFromPrimeQuadruple_of_a {n : ℕ} (h : A n) : ComesFromPrimeQuadruple n
```

FC-OeisA56777.comesFromPrimeQuadruple_of_a

Numerator of $1/\det(M)$ for $M[i,j] = 1/\operatorname{lcm}(i,j)$

MODULE CONTEXT: # Numerator of $1/\det(M)$ for $M[i,j] = 1/\operatorname{lcm}(i,j)$ Numerator of $1/\det(M)$ where M is the $n \times n$ matrix with $M[i,j] = 1/\operatorname{lcm}(i,j)$. *References:* - [A060841](https://oeis.org/A060841) - T. Adamczewski, "OEIS Open: How many conjectures can language models turn into theorems?", [arXiv:2608.11941](https://arxiv.org/abs/2608.11941) [cs.AI], 2026.

V2-000311 | Numerator of $1/\det(M)$ for $M[i,j] = 1/\operatorname{lcm}(i,j)$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

"Conjecture: $1/\det(M)$ is an integer only for n: 1 to 34, 36 and 38." - _Robert G. Wilson v_, Aug 02 2015

```
theorem conjecture1 :
∀ n : ℕ, 1 ≤ n → (((lcmMatrix n).det)⁻¹.den = 1 ↔ n ∈ integerDetN)
```

FC-OeisA60841.conjecture1

V2-000312 | Numerator of $1/\det(M)$ for $M[i,j] = 1/\operatorname{lcm}(i,j)$ | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

"All denominators are powers of two (A000079)." - _Robert G. Wilson v_, Aug 02 2015 This is false: for $n = 1807$ the denominator of $1/\det(M)$ is divisible by 3. See T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, [arXiv:2608.11941](https://arxiv.org/abs/2608.11941). The Lean disproof there uses the closed form $1/\det(M) = \prod_{k=1}^n k^2/\varphi(k)$ from the OEIS entry: https://github.com/epoch-research/LeanOpenProblems-results/blob/fd09021e79869476ef83cda231312f1a2a89c8d7/runs/oeis-full-50usd-ant-j0j0g4uzligm1k41/oeis_60841_conjecture_0/Submission/Spec.lean#L272

```
theorem conjecture2 :
¬ ∀ n : ℕ, 1 ≤ n → ∃ k : ℕ, ((lcmMatrix n).det)⁻¹.den = 2 ^ k
```

FC-OeisA60841.conjecture2

Number of different products of subsets of $\{1, 2, \dots, n\}$

MODULE CONTEXT: # Number of different products of subsets of $\{1, 2, \dots, n\}$ The number of distinct products (including the empty product 1) of any subset of $\{1, 2, \dots, n\}$. *References:* - [A060957](https://oeis.org/A060957) - [Li26] Wentao Li, [A counterexample to the A060957 interpolation conjecture](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/3be119e75ba5b31f034dc0c0b249975d17def711/proofs/new-proofs/A060957/PROOF.md), 2026.

V2-000313 | Number of different products of subsets of $\{1, 2, \dots, n\}$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: let $p \le n$ be prime. If m and $p^a m$ are two such products, then so is $p^k m$ for all $0 < k < a$. - Yan Sheng Ang, Feb 13 2020 This interpolation conjecture is false. Disproved by Wentao Li (2026), with AI assistance; see [Li26]. The counterexample has $p = 7 \cdot 2^{120} + 1$, $n = p \cdot 2^{189}$, $a = 6$, and $k = 5$.

```
theorem conjecture :
¬ ∀ (n p : ℕ), p.Prime → p ≤ n →
∀ (m a_exp : ℕ), m ∈ productsOfSubsets n → p ^ a_exp * m ∈ productsOfSubsets n →
∀ k : ℕ, 0 < k → k < a_exp → p ^ k * m ∈ productsOfSubsets n
```

FC-OeisA60957.conjecture

V2-000314 | Number of different products of subsets of $\{1, 2, \dots, n\}$ | counterexample

research solved | proof annotation: True | 官方代码 | 原文线索

An explicit counterexample from [Li26]: both endpoint products occur, but the intermediate product with exponent 5 does not. The definitions above specify every parameter, including `Counterexample.circuitM` as a finite product.

```
theorem counterexample :
Counterexample.circuitPrime.Prime ∧ Counterexample.circuitPrime ≤ Counterexample.circuitN ∧
Counterexample.circuitM ∈ productsOfSubsets Counterexample.circuitN ∧
Counterexample.circuitPrime ^ 6 * Counterexample.circuitM ∈
productsOfSubsets Counterexample.circuitN ∧
0 < (5 : ℕ) ∧ 5 < (6 : ℕ) ∧
Counterexample.circuitPrime ^ 5 * Counterexample.circuitM ∉
productsOfSubsets Counterexample.circuitN
```

FC-OeisA60957.counterexample

Reversible multiples $a(3^n) = 10^{3^{n-2}} - 1$ for powers of 3

MODULE CONTEXT: # Reversible multiples $a(3^n) = 10^{3^{n-2}} - 1$ for powers of 3 First multiple of n whose reverse is also divisible by n, or 0 if no such multiple exists. *References:* - [A062567](https://oeis.org/A062567) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000315 | Reversible multiples $a(3^n) = 10^{3^{n-2}} - 1$ for powers of 3 | a_three_pow_eq

research solved | proof annotation: True | 官方代码 | 原文线索

The conjecture that $a(3^n) = 10^{3^{n-2}} - 1$ for $n > 1$ was shown to be false for $4 < n < 21$ by Farideh Firoozbakht, who conjectured that for all $n > 4$, $a(3^n) \neq 10^{3^{n-2}} - 1$. This latter conjecture is proved here. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

```
theorem a_three_pow_eq (n : ℕ) :
2 ≤ n → (a (3 ^ n) = 10 ^ (3 ^ (n - 2)) - 1 ↔ n = 2 ∨ n = 3 ∨ n = 4)
```

FC-OeisA62567.a_three_pow_eq

Conjectures associated with A063880

MODULE CONTEXT: # Conjectures associated with A063880 A063880 lists numbers n such that $\sigma(n) = 2 \cdot \text{usigma}(n)$, where $\sigma(n)$ is the sum of all divisors and $\text{usigma}(n)$ is the sum of unitary divisors. Equivalently, these are numbers whose unitary and non-unitary divisors have equal sum. The conjectures state that all members satisfy $n \equiv 108 \pmod{216}$, and that all primitive terms (those whose proper divisors aren't in the sequence) are powerful numbers, with 108 being the only primitive term. *References:* - [A063880](https://oeis.org/A063880)

V2-000316 | Conjectures associated with A063880 | exists_primitive_of_a

research solved | proof annotation: True | 官方代码 | 原文线索

Non-primitive terms have the form $m \cdot s$ where m is primitive and s is squarefree with $\gcd(m, s) = 1$.

```
theorem exists_primitive_of_a {n : ℕ} (h : A n) :
∃ m s, IsPrimitiveTerm m ∧ Squarefree s ∧ m.Coprime s ∧ n = m * s
```

FC-OeisA63880.exists_primitive_of_a

V2-000317 | Conjectures associated with A063880 | mod_216_of_a

research open | proof annotation: False | 官方代码 | 原文线索

All members of the sequence satisfy $n \equiv 108 \pmod{216}$.

```
theorem mod_216_of_a {n : ℕ} (h : A n) : n % 216 = 108
```

FC-OeisA63880.mod_216_of_a

V2-000318 | Conjectures associated with A063880 | unique_primitive_108

research open | proof annotation: False | 官方代码 | 原文线索

108 is the only primitive term.

```
theorem unique_primitive_108 {n : ℕ} (h : IsPrimitiveTerm n) : n = 108
```

FC-OeisA63880.unique_primitive_108

Numerator - denominator in n-th harmonic number

MODULE CONTEXT: # Numerator - denominator in n-th harmonic number $a(n) = \text{numerator}(H_n) - \text{denominator}(H_n)$, where $H_n = 1 + 1/2 + \dots + 1/n$. *References:* - [A064169](https://oeis.org/A064169)

V2-000319 | Numerator - denominator in n-th harmonic number | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: for $n > 2$, n divides $a(n-2)$ if and only if n is a prime. Checked up to 20000. - _Amiram Eldar_ and _Thomas Ordowski_, Jul 27 2019 This is false for the composite number 16843^2.

theorem conjecture : ¬ ∀ (n : ℕ), 2 < n → ((n : ℤ) | a (n - 2) ↔ n.Prime)

FC-OeisA64169.conjecture

Integer part of area of a regular polygon with n sides each of length 1

MODULE CONTEXT: # Integer part of area of a regular polygon with n sides each of length 1 The area of a regular n-gon with side length 1 is given by $\frac{n}{4} \cot(\pi / n) = \frac{n}{4 \tan(\pi / n)}$. *References:* - [A064313](https://oeis.org/A064313)

V2-000320 | Integer part of area of a regular polygon with n sides each of length 1 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

"Usually (perhaps always?) $\lfloor n^2 / (4\pi) - \pi / 12 \rfloor$ for a polygon of circumference n. Note that the area of a circle with circumference C is $C^2 / (4\pi)$."

theorem conjecture (n : ℕ) (hn : 3 ≤ n) :
a n = (Int.floor ((n : ℝ) ^ 2 / (4 * Real.pi) - Real.pi / 12)).toNat

FC-OeisA64313.conjecture

Subword complexity of the morphism a → aab, b → b

MODULE CONTEXT: # Subword complexity of the morphism a → aab, b → b Let a_n be the number of distinct subwords (contiguous factors) of length n in the infinite word generated by the morphism $\sigma: a \mapsto aab, b \mapsto b$, starting from a. As shown in the references, a_n is given by the formula $$ a_n = \sum_{i=0}^{n} \min(2^i,n-i+1). $$ The conjectured generating function is: $$\sum_{n \geq 0} a_n x^n = \frac{1}{1-x} + \frac{x}{(1-x)^2}\left(\frac{1}{1-x} - \sum_{k \geq 1} x^{2^k + k - 1}\right).$$ *References:* - [A6697](https://oeis.org/A6697) - J.-P. Allouche and J. Shallit, "On the subword complexity of the fixed point of a → aab, b → b, and generalizations," arXiv:1605.02361 [math.CO], 2016. - N. J. A. Sloane and Simon Plouffe, *The Encyclopedia of Integer Sequences*, Academic Press, 1995.

V2-000321 | Subword complexity of the morphism a → aab, b → b | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (A6697): The generating function for the number of subwords of length n in the infinite word generated by $a \mapsto aab, b \mapsto b$ is $$\sum_{n \geq 0} a_n x^n = 1 + \frac{1}{1-x} + \frac{1}{(1-x)^2}\left(\frac{1}{1-x} - \sum_{k \geq 0} x^{2^{k+1} + k}\right).$$ Equivalently, a(n) equals the n-th coefficient of this generating function. However, the reference quoted in the OEIS sequence > J.-P. Allouche and J. Shallit, "On the subword complexity of the fixed point of a → aab, b → b, > and generalizations," arXiv:1605.02361 [math.CO], 2016. provides an explicit formula $$ a_n = \sum_{i=0}^{n} \min(2^i,n-i+1). $$ If one takes this as a definition of a(n) instead, it becomes straightforward to prove the conjecture. See https://github.com/AxiomMath/gdm-formal-conjectures/blob/main/docs/OeisA6697.md for a formal proof of the generating function using this definition. Hence, a formalization of [arXiv:1605.02361](https://arxiv.org/abs/1605.02361) would complete a formal proof as below.

theorem conjecture (n : ℕ) :
a n = coeff (R := ℚ) n
((1 - X)⁻¹ + X * (1 - X)⁻¹ ^ 2 * ((1 - X)⁻¹ - ∑' k, X ^ (2 ^ (k + 1) + k)))

FC-OeisA6697.conjecture

Decimal encoding of the prime factorization of n

MODULE CONTEXT: # Decimal encoding of the prime factorization of n If n has prime factorization $p_1^{e_1} \dots p_r^{e_r}$ with $p_1 < \dots < p_r$, then its decimal encoding is $p_1 e_1 \dots p_r e_r$. *References:* - [A067599](https://oeis.org/A067599)

V2-000322 | Decimal encoding of the prime factorization of n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

"$a(31) = a(177147) = 311$. Is there any solution to $a(n) = n$? - _Franklin T. Adams-Watters_, Dec 18 2006"

theorem conjecture :
∃ n : ℕ, 2 ≤ n ∧ a n = n

FC-OeisA67599.conjecture

Conjectures associated with A067720

MODULE CONTEXT: # Conjectures associated with A067720 A067720 lists numbers k such that $\varphi(k^2 + 1) = k \cdot \varphi(k + 1)$, where φ is Euler's totient function. The sequence exhibits a strong connection to primes: for almost all terms k, $k + 1$ is prime. The conjecture states that $k = 8$ is the only exception. *References:* - [A067720](https://oeis.org/A067720)

V2-000323 | Conjectures associated with A067720 | prime_add_one_of_a

research open | proof annotation: False | 官方代码 | 原文线索

For members of the sequence other than 8, we have $k + 1$ is prime.

```
theorem prime_add_one_of_a {k : ℕ} (h : A k) (hne : k ≠ 8) : (k + 1).Prime
```

FC-OeisA67720.prime_add_one_of_a

Euclid-Fortunate prime coverage

MODULE CONTEXT: # Euclid-Fortunate prime coverage OEIS A067836 starts with 2. Each later term is the gap from the product of all previous terms to the least prime strictly greater than that product plus one. *References:* - [OEIS A067836](https://oeis.org/A067836)

V2-000324 | Euclid-Fortunate prime coverage | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

[OEIS A067836](https://oeis.org/A067836) asks: "Do all primes occur in the sequence?"

```
theorem conjecture :
∀ p, p.Prime → ∃ n ≥ 1, a n = p
```

FC-OeisA67836.conjecture

Sum of $a(k)/k!$ over divisors equals harmonic number

MODULE CONTEXT: # Sum of $a(k)/k!$ over divisors equals harmonic number The sequence $a(n)$ satisfies $\sum_{k \mid n} \frac{a(k)}{k!} = \sum_{j=1}^n \frac{1}{j} = H_n$, where the sum on the left is over positive divisors k of n. By Möbius inversion, $$a(n) = n! \sum_{d \mid n} \mu(n/d) H_d$$ where $H_d = \sum_{j=1}^d \frac{1}{j}$ is the d-th harmonic number. *References:* - [A067857](https://oeis.org/A067857) - [A001221](https://oeis.org/A001221)

V2-000325 | Sum of $a(k)/k!$ over divisors equals harmonic number | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The terms are not all positive. The first negative one is $a(30) = -22690644647302814715858124800000$. Conjecture: $a(n) < 0$ if and only if A001221(n) is an odd number ≥ 3. This conjecture is false. A counterexample is the product of all primes at most 1000, which has 168 distinct prime factors and a negative sequence value. The counterexample and formal proof were developed by Codex (GPT-6), prompted by Samuel Schlesinger.

```
theorem conjecture :
¬ ∀ n : ℕ, 0 < n →
(a n < 0 ↔ Odd (cardDistinctFactors n) ∧ 3 ≤ cardDistinctFactors n)
```

FC-OeisA67857.conjecture

Number of times $n^2 + s^2$ is prime for positive integers $s < n$

MODULE CONTEXT: # Number of times $n^2 + s^2$ is prime for positive integers $s < n$ The sequence $a(n)$ counts the number of integers $s \in \{1, \dots, n-1\}$ such that $n^2 + s^2$ is prime: $$a(n) = \sum_{s=1}^{n-1} [\text{Prime}(n^2 + s^2)]$$ *References:* - [A069004](https://oeis.org/A069004) - [CE26] j2d9w5xtjn-png, [Counterexample to the upper bound](https://github.com/google-deepmind/formal-conjectures/pull/5453), 2026. - [Li26] Wentao Li, [Lean certificate for the A069004 counterexample](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/a16a2711b082ccab4ba941330148e7b7c78b11c1/proofs/new-formalization/A069004/PROOF.md), 2026.

V2-000326 | Number of times $n^2 + s^2$ is prime for positive integers $s < n$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) > 0$ for all $n > 1$.

```
theorem conjecture1 (n : ℕ) (hn : 1 < n) : 0 < a n
```

FC-OeisA69004.conjecture1

V2-000327 | Number of times $n^2 + s^2$ is prime for positive integers $s < n$ | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

Stronger conjecture: Let pi(n) be the prime counting function (A000720). Then pi(n) >= a(n) >= pi(n)/5 for n>1, with the following equalities: pi(2)=a(2), pi(10)=a(10) and a(12)=pi(12)/5. - T. D. Noe, Feb 26 2007 This conjunction is false. The upper bound fails at $n = 512720$: a Lean certificate proves $a(512720) \ge 42494 > 42493 = \pi(512720)$. The counterexample was recorded by j2d9w5xtjn-png [CE26]. Lean formalization by Wentao Li, with AI assistance, adapting Epoch Research's certificate attempt; see [Li26].

```
theorem conjecture2 : ¬ (
(∀ n : ℕ, 1 < n → Nat.primeCounting n ≥ a n) ∧
(∀ n : ℕ, 1 < n → 5 * a n ≥ Nat.primeCounting n) ∧
Nat.primeCounting 2 = a 2 ∧
Nat.primeCounting 10 = a 10 ∧
5 * a 12 = Nat.primeCounting 12)
```

FC-OeisA69004.conjecture2

Number of primes p such that $n^n \le p \le n^n + n^2$

MODULE CONTEXT: # Number of primes p such that $n^n \le p \le n^n + n^2$ The sequence $a(n)$ counts the number of prime numbers in the interval $[n^n, n^n + n^2]$: $$a(n) = |\{p \text{ prime} \mid n^n \le p \le n^n + n^2\}|$$ *References:* - [A069922](https://oeis.org/A069922)

V2-000328 | Number of primes p such that $n^n \le p \le n^n + n^2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Question: for any $n > 0$, is there at least one prime p such that $n^n \le p \le n^n + n^2$? In this case, that would be stronger than the Schinzel conjecture: "for $m > 1$ there's at least one prime p such that $m \le p \le m + \log(m)^2$" since $n^2 < \log(n^n)^2 = n^2 \log(n)^2$.

```
theorem conjecture (n : ℕ) (hn : 0 < n) : 1 ≤ a n
```

FC-OeisA69922.conjecture

Number of primes p such that $2^n \le p \le 2^n + \mathrm{prime}(n)$

MODULE CONTEXT: # Number of primes p such that $2^n \le p \le 2^n + \mathrm{prime}(n)$ The sequence $a(n)$ counts the number of primes p in the interval $[2^n, 2^n + p_n]$, where p_n is the n-th prime ($p_1 = 2, p_2 = 3, \dots$): $$a(n) = |\{p \text{ prime} \mid 2^n \le p \le 2^n + p_n\}|$$ for $n \ge 1$, and $a(0) = 0$. *References:* - [A069923](https://oeis.org/A069923)

V2-000329 | Number of primes p such that $2^n \le p \le 2^n + \mathrm{prime}(n)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For any $n > 0$, is there always at least one prime p such that $2^n \le p \le 2^n + \mathrm{prime}(n)$? (checked up to $n = 250$). In this case, that would be stronger than the Schinzel conjecture: "for $m > 1$ there's at least one prime p such that $m \le p \le m + \log(m)^2$" since, for $n > 2$, $\mathrm{prime}(n) < \log(2^n)^2 = n^2 \log(2)$.

```
theorem conjecture (n : ℕ) (hn : 0 < n) : 1 ≤ a n
```

FC-OeisA69923.conjecture

Value of n-th cyclotomic polynomial at n

MODULE CONTEXT: # Value of n-th cyclotomic polynomial at n The sequence $a(n) = \Phi_n(n)$ gives the value of the n-th cyclotomic polynomial evaluated at n: $$a(n) = |\Phi_n(n)|$$ *References:* - [A070518](https://oeis.org/A070518)

V2-000330 | Value of n-th cyclotomic polynomial at n | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$a(28341)$ is divisible by 283411^2. What is the next n such that $a(n)$ is not squarefree?

```
theorem conjecture :
conjecture.answer =
if h : ∃ n, 28341 < n ∧ ¬ Squarefree (a n) then
some (sInf {n | 28341 < n ∧ ¬ Squarefree (a n)})
else
none
```

FC-OeisA70518.conjecture

Recurrence $a(n+2) = |a(n+1)a(n) - a(n)a(n+1)|$ via concatenation

MODULE CONTEXT: # Recurrence $a(n+2) = |a(n+1)a(n) - a(n)a(n+1)|$ via concatenation The sequence starts with $a(1) = 0, a(2) = 1$. For $n \ge 1$, $$a(n+2) = |\text{concat}(a(n+1), a(n)) - \text{concat}(a(n), a(n+1))|$$ where concatenation is in decimal representation. *References:* - [A070823](https://oeis.org/A070823)

V2-000331 | Recurrence $a(n+2) = |a(n+1)a(n) - a(n)a(n+1)|$ via concatenation | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

$a(n) \equiv 0 \pmod 3$ if $n > 2$. Is $a(n)$ always of the form $2^j \cdot 3^k \cdot s$ where s is a squarefree number? Answer: False, $a(20)$ is divisible by 13^2 but not by 13^3.

```
theorem conjecture :
¬ ∀ n : ℕ, 2 < n →
a n ≡ 0 [MOD 3] ∧
∃ j k s : ℕ, a n = 2 ^ j * 3 ^ k * s ∧ Squarefree s
```

FC-OeisA70823.conjecture

Determinant of matrix with entries indicating primality of $i^2 + j^2$

MODULE CONTEXT: # Determinant of matrix with entries indicating primality of $i^2 + j^2$ The sequence $a(n)$ is the determinant of the $n \times n$ matrix M defined by $M(i,j) = 1$ if $i^2 + j^2$ is prime, and 0 otherwise, where $1 \le i, j \le n$. *References:* - [A071524](https://oeis.org/A071524)

V2-000332 | Determinant of matrix with entries indicating primality of $i^2 + j^2$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) = 0$ for no $n > 28$. - _Zhi-Wei Sun_, Aug 26 2013

```
theorem conjecture1 (n : ℕ) (hn : 28 < n) : a n ≠ 0
```

FC-OeisA71524.conjecture1

V2-000333 | Determinant of matrix with entries indicating primality of $i^2 + j^2$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (Generalization): For every $m \in \mathbb{N}$, the determinant `generalDet m n` is nonzero for all sufficiently large n. - _Zhi-Wei Sun_, Aug 26-27 2013

```
theorem conjecture2 (m : ℕ) : ∃ N : ℕ, ∀ n : ℕ, N < n → generalDet m n ≠ 0
```

FC-OeisA71524.conjecture2

Alternating sum of signs of powers of $3/2$

MODULE CONTEXT: # Alternating sum of signs of powers of $3/2$ The sequence $a(n) = -\sum_{k=1}^n (-1)^{\lfloor (3/2)^k \rfloor}$. *References:* - [A071532](https://oeis.org/A071532)

V2-000334 | Alternating sum of signs of powers of $3/2$ | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

"Is $a(n)>0$?" The answer is negative: $a(331523) = -1$. The restriction $n > 2$ excludes the initial zero $a(2) = 0$. The counterexample and formal proof were developed by Codex (GPT-6), prompted by Samuel Schlesinger.

```
theorem conjecture1 : ¬ ∀ n : ℕ, 2 < n → 0 < a n
```

FC-OeisA71532.conjecture1

V2-000335 | Alternating sum of signs of powers of $3/2$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

For n large enough, does $a(n) > \sqrt{n}$ always hold?

```
theorem conjecture2 :
∃ N : ℕ, ∀ n : ℕ, N ≤ n → (a n : ℝ) > Real.sqrt (n : ℝ)
```

FC-OeisA71532.conjecture2

V2-000336 | Alternating sum of signs of powers of $3/2$ | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: asymptotically, $a(n) \sim C \log(n)^2$ for some constant $C > 0$.

```
theorem conjecture3 :
∃ C : ℝ, 0 < C ∧ (fun n : ℕ ↦ (a n : ℝ)) ~[atTop] (fun n : ℕ ↦ C * Real.log (n : ℝ) ^ 2)
```

FC-OeisA71532.conjecture3

V2-000337 | Alternating sum of signs of powers of $3/2$ | conjecture3_value

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: the constant C in $a(n) \sim C \log(n)^2$ is approximately 1.4.

```
theorem conjecture3_value :
|conjecture3_value.answer - 1.4| < 0.1 ∧
(fun n : ℕ ↦ (a n : ℝ)) ~[atTop] (fun n : ℕ ↦ conjecture3_value.answer * Real.log (n : ℝ) ^ 2)
```

FC-OeisA71532.conjecture3_value

Smallest factorial containing exactly n 6's

MODULE CONTEXT: # Smallest factorial containing exactly n 6's The sequence $a(n)$ gives the smallest k such that the decimal expansion of $k!$ contains exactly n occurrences of the digit '6', or 0 if no such k exists. *References:* - [A072200](https://oeis.org/A072200)

V2-000338 | Smallest factorial containing exactly n 6's | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured that $a(24) = 0$ since no factorial less than 10000 contained just 24 sixes.

```
theorem conjecture : a 24 = 0
```

FC-OeisA72200.conjecture

Wolstenholme numbers

MODULE CONTEXT: # Wolstenholme numbers Wolstenholme numbers: numerator of $\sum_{k=1}^n \frac{1}{k^2}$. *References:* - [A007406](https://oeis.org/A007406)

V2-000339 | Wolstenholme numbers | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: for $n > 3$, $\gcd(n, a(n-1)) = \text{A089026}(n)$. - Amiram Eldar and Thomas Ordowski, Jul 28 2019

```
theorem conjecture (n : ℕ) (hn : 3 < n) :
Nat.gcd n (a (n - 1)) = OeisA89026.a n
```

FC-OeisA7406.conjecture

Sum of next n primes

MODULE CONTEXT: # Sum of next n primes The sum of the primes in the n-th row of the prime number triangle: $$a(n) = \sum_{i = 1 + n(n-1)/2}^{n + n(n-1)/2} p_i$$ with $a(0) = 0$. *References:* - [A007468](https://oeis.org/A007468)

V2-000340 | Sum of next n primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The only positive integer n such that $a(n)$ is a perfect square is $n=38$. - Carlos Eduardo Olivieri, Mar 09 2015

theorem conjecture (n : ℕ) (hn : 0 < n) : IsSquare (a n) ↔ n = 38

FC-OeisA7468.conjecture

Number of times n occurs as a binary sub-pattern of n^2

MODULE CONTEXT: # Number of times n occurs as a binary sub-pattern of n^2 The sequence $a(n)$ is the number of times the binary expansion of n appears as a contiguous sublist (infix) in the binary expansion of n^2. *References:* - [A076141](https://oeis.org/A076141)

V2-000341 | Number of times n occurs as a binary sub-pattern of n^2 | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Is $a(n) \le 1$ for all n?

theorem conjecture (n : ℕ) : a n ≤ 1

FC-OeisA76141.conjecture

Smallest x such that $\sigma(x) \bmod x = n$

MODULE CONTEXT: # Smallest x such that $\sigma(x) \bmod x = n$ The sequence $a(n)$ is the smallest positive integer x such that $\sigma_1(x) \bmod x = n$, or 0 if no such x exists. *References:* - [A076495](https://oeis.org/A076495)

V2-000342 | Smallest x such that $\sigma(x) \bmod x = n$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

At present, the 0 entry for $n = 5$ is only a conjecture. That is, it is conjectured that there is no positive integer x such that $\sigma_1(x) \bmod x = 5$.

theorem conjecture : a 5 = 0

FC-OeisA76495.conjecture

Trajectory of 103 under the Reverse and Add! operation in base 3

MODULE CONTEXT: # Trajectory of 103 under the Reverse and Add! operation in base 3 The sequence $a(n)$ is the trajectory of 103 under the Reverse and Add! operation carried out in base 3, written in base 10. $a(0) = 103$, and $a(n+1) = a(n) + \text{rev}_3(a(n))$. *References:* - [A077408](https://oeis.org/A077408)

V2-000343 | Trajectory of 103 under the Reverse and Add! operation in base 3 | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

103 is conjectured to be the smallest number such that the Reverse and Add! algorithm in base 3 does not lead to a palindrome. Its trajectory is conjectured to never reach a palindrome.

theorem conjecture (n : ℕ) : ¬ IsBasePalindrome 3 (a n)

FC-OeisA77408.conjecture

$a(1)=1, a(2)=1, a(n)=(2^{a(n-1)} + 1)/a(n-2)$

MODULE CONTEXT: # $a(1)=1, a(2)=1, a(n)=(2^{a(n-1)} + 1)/a(n-2)$ The sequence is defined by $a(1) = 1, a(2) = 1$, and for $n \ge 3$, $$a(n) = \frac{2^{a(n-1)} + 1}{a(n-2)}$$ *References:* - [A078590](https://oeis.org/A078590)

V2-000344 | $a(1)=1, a(2)=1, a(n)=(2^{a(n-1)} + 1)/a(n-2)$ | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Are all terms integers? No. Exact divisibility first fails at $n = 7$; the corresponding rational term has reduced denominator 19.

theorem conjecture :
¬ ∀ n : ℕ, 3 ≤ n → a (n - 2) | 2 ^ a (n - 1) + 1

FC-OeisA78590.conjecture

Smallest $m > 0$ such that $n \cdot 2^m + 1$ is prime

MODULE CONTEXT: # Smallest $m > 0$ such that $n \cdot 2^m + 1$ is prime The sequence $a(n)$ is the smallest positive integer m such that $n \cdot 2^m + 1$ is prime, or 0 if no such m exists. *References:* - [A078680](https://oeis.org/A078680)

V2-000345 | Smallest $m > 0$ such that $n \cdot 2^m + 1$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

There is a conjecture that the first zero is $n = 65536 = 2^{16}$ (which is equivalent to the statement that $2^{2^k} + 1$ is composite for $k > 4$). - _T. D. Noe_, Feb 25 2011

theorem conjecture :
a (2 ^ 16) = 0 ∧ ∀ n : ℕ, 1 ≤ n ∧ n < 2 ^ 16 → a n ≠ 0

FC-OeisA78680.conjecture

Least $k > 0$ such that $(k+1)(k+2)\cdots(k+n) + 1$ is prime

MODULE CONTEXT: # Least $k > 0$ such that $(k+1)(k+2)\cdots(k+n) + 1$ is prime The sequence $a(n)$ is the least positive integer k such that $(k+1)(k+2)\cdots(k+n) + 1$ is prime, if such k exists; otherwise $a(n) = 0$. *References:* - [A078729](https://oeis.org/A078729)

V2-000346 | Least $k > 0$ such that $(k+1)(k+2)\cdots(k+n) + 1$ is prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

$(k+1)(k+2)(k+3)(k+4) + 1 = (k^2 + 5k + 5)^2$, which is never prime. Hence $a(4) = 0$. Conjecture: $a(n) = 0$ if and only if $n = 4$.

```
theorem conjecture (n : ℕ) (hn : 0 < n) : a n = 0 ↔ n = 4
```

FC-OeisA78729.conjecture

Least prime $\ge n$

MODULE CONTEXT: # Least prime $\ge n$ Least prime $\ge n$ (version 1 of the "next prime" function). *References:* - [A007918](https://oeis.org/A007918)

V2-000347 | Least prime $\ge n$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

According to the "k-tuple" conjecture, $a(n)$ is the initial term of the lexicographically earliest increasing arithmetic progression of n primes; the corresponding common differences are given by A061558.

```
theorem conjecture1 (n : ℕ) (hn : 0 < n) :
a n = sInf { p0 : ℕ | ∃ d : ℕ, isApOfNPrimes n p0 d }
```

FC-OeisA7918.conjecture1

V2-000348 | Least prime $\ge n$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

If $n > 1$, then $a(n) < n^{n^{1/n}}$. - Thomas Ordowski, Feb 23 2023

```
theorem conjecture2 (n : ℕ) (hn : 1 < n) :
(a n : ℝ) < (n : ℝ) ^ ((n : ℝ) ^ (1 / (n : ℝ)))
```

FC-OeisA7918.conjecture2

$a(n) = \sum_{k=0}^n \binom{2k}{k}^3$

MODULE CONTEXT: # $a(n) = \sum_{k=0}^n \binom{2k}{k}^3$ The sequence $a(n) = 1 + \binom{2}{1}^3 + \binom{4}{2}^3 + \cdots + \binom{2n}{n}^3$: $$a(n) = \sum_{k=0}^n \binom{2k}{k}^3$$ *References:* - [A079727](https://oeis.org/A079727)

V2-000349 | $a(n) = \sum_{k=0}^n \binom{2k}{k}^3$ | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1 (Peter Bala, 2024): If prime p is in A003625 then $a(p^2) \equiv 8 + p^2 \pmod{p^3}$.

```
theorem conjecture1 (p : ℕ) (hp : OeisA3625.A p) :
a (p ^ 2) ≡ 8 + p ^ 2 [MOD p ^ 3]
```

FC-OeisA79727.conjecture1

V2-000350 | $a(n) = \sum_{k=0}^n \binom{2k}{k}^3$ | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 2 (Peter Bala, 2024): If prime p is in A003625 then $a(p(p-1)) \equiv p^2 \pmod{p^3}$.

```
theorem conjecture2 (p : ℕ) (hp : OeisA3625.A p) :
a (p * (p - 1)) ≡ p ^ 2 [MOD p ^ 3]
```

FC-OeisA79727.conjecture2

V2-000351 | $a(n) = \sum_{k=0}^n \binom{2k}{k}^3$ | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 3 (Peter Bala, 2024): If prime p is in A003625 then $a((p^2-1)/2) \equiv p^2 \pmod{p^4}$.

```
theorem conjecture3 (p : ℕ) (hp : OeisA3625.A p) :
a ((p ^ 2 - 1) / 2) ≡ p ^ 2 [MOD p ^ 4]
```

FC-OeisA79727.conjecture3

V2-000352 | $a(n) = \sum_{k=0}^n \binom{2k}{k}^3$ | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4 (Peter Bala, 2024): If n is a product of distinct primes from A003625 then $a((n-1)/2)$ is divisible by n^2.

```
theorem conjecture4 (S : Finset ℕ) (hS : ∀ p ∈ S, OeisA3625.A p) :
(∏ p ∈ S, p) ^ 2 ∣ a ((∏ p ∈ S, p - 1) / 2)
```

FC-OeisA79727.conjecture4

Number of prime powers strictly between n-th prime and $(n+1)$-th prime

MODULE CONTEXT: # Number of prime powers strictly between n-th prime and $(n+1)$-th prime The sequence $a(n)$ is the number of prime powers k strictly between the n-th prime p_n and the $(n+1)$-th prime p_{n+1}: $p_n < k < p_{n+1}$. *References:* - [A080101](https://oeis.org/A080101)

V2-000353 | Number of prime powers strictly between n-th prime and $(n+1)$-th prime | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured that $a(n) \le 2$ for all n.

theorem conjecture (n : ℕ) : a n ≤ 2

FC-OeisA80101.conjecture

Conjecture relating two characterizations of a set of integers.

MODULE CONTEXT: # Conjecture relating two characterizations of a set of integers. Informal Statement: For an integer $k \ge 2$, the following are equivalent: 1. The greatest common divisor of the binomial coefficients $\binom{2k}{k}, \binom{3k}{k}, \dots, \binom{(k+1)k}{k} = 1$. 2. Writing prime factorization of k as $k = \prod p_i^{e_i}$, and let $P = \max_i p_i^{e_i}$, one has $k / P > P$. This conjecture asserts that the sequence defined by 1. is obtained by taking 1 off each number in the sequence defined by 2. *References:* - [A80170](https://oeis.org/A80170) - [A51283](https://oeis.org/A51283)

V2-000354 | Conjecture relating two characterizations of a set of integers. | gcdCondition_iff_primePowerCondition

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: The gcd condition is equivalent to the prime power condition. This has been conjectured by Ralf Stephan. Both the natural-language proof and its Lean 4 formalization were carried out by the KLMM MechMath Agent Team; see the `formal_proof` attribute. *References:* - [Ralf Stephan, *Prove or Disprove. 100 Conjectures from the OEIS*, 2004, Conjecture 17 (arXiv:math/0409509)](https://arxiv.org/abs/math/0409509) - [Dakai Guo et al., *A Greatest Common Divisor Criterion of Certain Binomial Coefficients*, 2026 (arXiv:2606.22997)](https://arxiv.org/abs/2606.22997)

theorem gcdCondition_iff_primePowerCondition (k : ℕ) (hk : 2 ≤ k) :
A k ↔ B (k + 1)

FC-OeisA80170.gcdCondition_iff_primePowerCondition

Denominator of $\sum_{k=1}^n k^{\mu(k)}$

MODULE CONTEXT: # Denominator of $\sum_{k=1}^n k^{\mu(k)}$ The sequence $a(n)$ is the denominator of $\sum_{k=1}^n k^{\mu(k)}$, where μ is the Möbius function. *References:* - [A080326](https://oeis.org/A080326)

V2-000355 | Denominator of $\sum_{k=1}^n k^{\mu(k)}$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: $a(n) = \text{primorial}(n)$ for infinitely many n.

theorem conjecture : {n : ℕ | a n = primorial n}.Infinite

FC-OeisA80326.conjecture

Primes of the form 2^n + 2^i + 1

MODULE CONTEXT: # Primes of the form 2^n + 2^i + 1 There are infinite primes of the form $2^n + 2^i + 1$, with $0 < i < n$. See Wagstaff (2001) where this conjecture is posed. *References:* - [A81091](https://oeis.org/A81091) - Samuel S. Wagstaff, Jr., [Prime Numbers with a fixed number of one bits or zero bits in their binary representation](http://projecteuclid.org/euclid.em/999188636), Exp. Math. vol. 10, issue 2 (2001) 267.

V2-000356 | Primes of the form 2^n + 2^i + 1 | conjectureA81091

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (A81091): There are infinite primes of the form $2^n + 2^i + 1$, with $0 < i < n$.

theorem conjectureA81091 :
Set.Infinite {p : ℕ | A p}

FC-OeisA81091.conjectureA81091

Smallest palindrome with exactly n divisors

MODULE CONTEXT: # Smallest palindrome with exactly n divisors The sequence $a(n)$ is the smallest palindromic number with exactly n divisors, or 0 if no such number exists. *References:* - [A083753](https://oeis.org/A083753)

V2-000357 | Smallest palindrome with exactly n divisors | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

There are no palindromic numbers greater than 1 which are the fifth or higher power of a natural number.

theorem conjecture (m k : ℕ) (hm : 1 < m) (hpal : IsDecimalPalindrome m) (hk : 5 ≤ k) :
¬ ∃ x : ℕ, m = x ^ k

FC-OeisA83753.conjecture

Smallest prime p such that $p + n$ is an n-th power

MODULE CONTEXT: # Smallest prime p such that $p + n$ is an n-th power Smallest prime p such that $p + n$ is an n-th power, or 0 if no such number exists. That is, the smallest prime of the form $k^n - n$. *References:* - [A084046](https://oeis.org/A084046)

V2-000358 | Smallest prime p such that $p + n$ is an n-th power | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: if a(k) = 0 then k is an even square. This is false for $k = 27$. Every candidate $x^{27} - 27$ factors as $(x^9 - 3)(x^{18} + 3x^9 + 9)$, so $a(27) = 0$, but 27 is not an even square.

theorem conjecture : ¬ ∀ k : ℕ, a k = 0 → ∃ m : ℕ, k = (2 * m) ^ 2

FC-OeisA84046.conjecture

Prime numbers at the beginning of the 20th century

MODULE CONTEXT: # Prime numbers at the beginning of the 20th century The sequence consists of 1 and the prime numbers. Today 1 is no longer regarded as a prime. *References:* - [A008578](https://oeis.org/A008578)

V2-000359 | Prime numbers at the beginning of the 20th century | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

"Conjecture: the sequence contains exactly those k such that sigma(k) > k*BigOmega(k)." - Irina Gerasimova, Jun 08 2013 Here $\sigma(k)$ is the sum of the divisors of k and $\Omega(k)$ is the number of prime factors of k counted with multiplicity. Both sides are false for $k = 0$. Solved by Claude (Anthropic) prompted by otoinobuo. The proof shows $\sigma(k) < k P$ for the largest prime factor P of k and uses induction on k / P.

```
theorem conjecture (k : ℕ) : A k ↔ k * cardFactors k < sigma 1 k
```

FC-OeisA8578.conjecture

Smallest r such that (concatenation of n, r times) $\cdot 10 + 1$ is prime

MODULE CONTEXT: # Smallest r such that (concatenation of n, r times) $\cdot 10 + 1$ is prime $a(n)$ is the smallest r where (concatenation of n, r times with itself) $\cdot 10 + 1$ is a prime, or 0 if no such number exists. The number resulting from concatenating n, r times, is $n \cdot \sum_{i=0}^{r-1} (10^d)^i$, where d is the number of digits of n. *References:* - [A086766](https://oeis.org/A086766)

V2-000360 | Smallest r such that (concatenation of n, r times) $\cdot 10 + 1$ is prime | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest integer $m > 1$ such that $a(10^m)$ is nonzero? - _Farideh Firoozbakht_, Jan 07 2015

```
theorem conjecture1 :
conjecture1.answer =
if h : ∃ m, 1 < m ∧ a (10 ^ m) ≠ 0 then
some (sInf {m | 1 < m ∧ a (10 ^ m) ≠ 0})
else
none
```

FC-OeisA86766.conjecture1

V2-000361 | Smallest r such that (concatenation of n, r times) $\cdot 10 + 1$ is prime | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: If n is not of the form 10^m then $a(n)$ is nonzero. - _Farideh Firoozbakht_, Jan 07 2015

```
theorem conjecture2 (n : ℕ) (hn : 0 < n) (h : ∀ m : ℕ, n ≠ 10 ^ m) : a n ≠ 0
```

FC-OeisA86766.conjecture2

V2-000362 | Smallest r such that (concatenation of n, r times) $\cdot 10 + 1$ is prime | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest odd prime p such that $(10^{p^2}-1)/(10^p-1)$ is a prime number (and $a(10^{p-1})$ could be nonzero)? - _Farideh Firoozbakht_, Jan 07 2015

```
theorem conjecture3 :
conjecture3.answer =
if h : ∃ p, p.Prime ∧ 2 < p ∧ ((10 ^ (p ^ 2) - 1) / (10 ^ p - 1)).Prime then
some (sInf {p | p.Prime ∧ 2 < p ∧ ((10 ^ (p ^ 2) - 1) / (10 ^ p - 1)).Prime})
else
none
```

FC-OeisA86766.conjecture3

Binary representation of primes that divide a number, in decimal

MODULE CONTEXT: # Binary representation of primes that divide a number, in decimal The value $a(n)$ is given by $$ a(n) = \sum_{p \mid n, p \text{ prime}} 2^{\pi(p) - 1} $$ where $\pi(p) = \mathrm{primeCounting}(p)$ gives the 1-based index of the prime p. *References:* - [A087207](https://oeis.org/A087207)

V2-000363 | Binary representation of primes that divide a number, in decimal | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Starting at any n and iterating the map $n \mapsto a(n)$, we will always reach 0. - _Antti Karttunen_, Jun 18,20 2017

```
theorem conjecture (n : ℕ) : ∃ k : ℕ, (a^[k]) n = 0
```

FC-OeisA87207.conjecture

Expansion of $(1 - x)/(1 - 2 x + 3 x^2)$

MODULE CONTEXT: # Expansion of $(1 - x)/(1 - 2 x + 3 x^2)$ This sequence is the expansion of $(1 - x)/(1 - 2 x + 3 x^2)$ in powers of x. It satisfies the linear recurrence relation $a(n) = 2 a(n-1) - 3 a(n-2)$ for $n \ge 2$, with initial values $a(0)=1$ and $a(1)=1$. *References:* - [A087455](https://oeis.org/A087455)

V2-000364 | Expansion of $(1 - x)/(1 - 2 x + 3 x^2)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is an open question whether or not this sequence satisfies Benford's law [Berger-Hill, 2017; Arno Berger, email, Jan 06 2017]. - N. J. A. Sloane, Feb 08 2017

theorem conjecture : SatisfiesBenford a

FC-OeisA87455.conjecture

Smallest prime formed by concatenation $n, n-1, \dots, n-k$

MODULE CONTEXT: # Smallest prime formed by concatenation $n, n-1, \dots, n-k$ $a(n)$ is the smallest prime which has the form of the concatenation $n, n-1, n-2, \dots, n-k$ for some $k < n$, or 0 if no such prime exists. *References:* - [A087571](https://oeis.org/A087571)

V2-000365 | Smallest prime formed by concatenation $n, n-1, \dots, n-k$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: There are infinitely many composite numbers n such that $a(n)$ is nonzero.

theorem conjecture (M : ℕ) : ∃ n > M, 1 < n ∧ ¬ n.Prime ∧ a n ≠ 0

FC-OeisA87571.conjecture

Number of pairs of twin primes between n^2 and $(n+1)^2$

MODULE CONTEXT: # Number of pairs of twin primes between n^2 and $(n+1)^2$ $a(n)$ is the number of pairs of twin primes between n^2 and $(n+1)^2$. This counts the number of primes p such that p and $p+2$ are both prime, and the entire twin prime pair $(p, p+2)$ lies strictly between n^2 and $(n+1)^2$. That is, $n^2 < p$ and $p + 2 < (n+1)^2$. *References:* - [A091591](https://oeis.org/A091591)

V2-000366 | Number of pairs of twin primes between n^2 and $(n+1)^2$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

It is conjectured that $a(n)>0$ for all $n>122$. Proving this would also prove Legendre's conjecture that there is a prime between n^2 and $(n+1)^2$. - _T. D. Noe_, Feb 28 2007

theorem conjecture :
∀ n : ℕ, n > 122 → a n > 0

FC-OeisA91591.conjecture

Primality and primitive root property from divisibility $n \mid (a(n-1) + 2^{n-2})$

MODULE CONTEXT: # Primality and primitive root property from divisibility $n \mid (a(n-1) + 2^{n-2})$ $a(n) = \frac{2^{n-1}}{n!} \prod_{k=1}^{n-1} (2^k-1)$. The sequence $a(n)$ is composed of natural numbers, thus we define it as a function $\mathbb{N} \to \mathbb{N}$. *References:* - [A091669](https://oeis.org/A091669) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000367 | Primality and primitive root property from divisibility $n \mid (a(n-1) + 2^{n-2})$ | prime_and_primitive_root_of_dvd

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture (for $n > 2$): if $n \mid a(n-1) + 2^{n-2}$, then n is a prime with primitive root 2 (A001122). - _Amiram Eldar_ and _Thomas Ordowski_, Jan 19 2020 A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763).

theorem prime_and_primitive_root_of_dvd (n : ℕ) (hn : n > 2) : n | (a (n - 1) + 2 ^ (n - 2)) → Nat.Prime n ∧ IsPrimitiveRoot (2 : ZMod n) (Nat.totient n)

FC-OeisA91669.prime_and_primitive_root_of_dvd

Tug of war score between prime gap increases and decreases

MODULE CONTEXT: # Tug of war score between prime gap increases and decreases Score at stage n in "tug of war" between prime gap increases vs. prime gap decreases: start with score $= 0$ at $n = 1$ and at stage $k > 1$, increase (resp. decrease) the score by 1 if the k-th prime gap is greater (resp. less) than the previous prime gap. *References:* - [A092243](https://oeis.org/A092243)

V2-000368 | Tug of war score between prime gap increases and decreases | conjecture1

research open | proof annotation: False | 官方代码 | 原文线索

Is the score $a(n) > 0$ for some $n > 250000$?

theorem conjecture1 : ∃ n > 250000, a n > 0

FC-OeisA92243.conjecture1

V2-000369 | Tug of war score between prime gap increases and decreases | conjecture2

research open | proof annotation: False | 官方代码 | 原文线索

Is the score $a(n)$ bounded from below?

theorem conjecture2 : ∃ B : ℤ, ∀ n : ℕ, B ≤ a n

FC-OeisA92243.conjecture2

V2-000370 | Tug of war score between prime gap increases and decreases | conjecture3

research open | proof annotation: False | 官方代码 | 原文线索

Is the score $a(n)$ bounded from above?

theorem conjecture3 : ∃ B : ℤ, ∀ n : ℕ, a n ≤ B

FC-OeisA92243.conjecture3

V2-000371 | Tug of war score between prime gap increases and decreases | conjecture4

research open | proof annotation: False | 官方代码 | 原文线索

Is the score $a(n) > 0$ for infinitely many values of n?

theorem conjecture4 : Set.Infinite {n : ℕ | a n > 0}

FC-OeisA92243.conjecture4

V2-000372 | Tug of war score between prime gap increases and decreases | conjecture5

research open | proof annotation: False | 官方代码 | 原文线索

Is the score $a(n) < 0$ for infinitely many values of n?

theorem conjecture5 : Set.Infinite {n : ℕ | a n < 0}

FC-OeisA92243.conjecture5

Product of composite numbers in triangular intervals

MODULE CONTEXT: # Product of composite numbers in triangular intervals Product of all composite numbers between $n(n-1)/2+1$ and $n(n+1)/2$ (including boundaries), where $n(n-1)/2 = \binom{n}{2}$ and $n(n+1)/2 = \binom{n+1}{2}$. *References:* - [A093456](https://oeis.org/A093456)

V2-000373 | Product of composite numbers in triangular intervals | conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture: There are finitely many numbers such that $a(n)$ is not $\equiv 0 \pmod{a(n-1)}$. (Also mentioned in A093455.) In fact there are infinitely many such numbers.

theorem conjecture :
¬ Set.Finite {n : ℕ | 1 < n ∧ ¬ (a (n - 1) | a n)}

FC-OeisA93456.conjecture

$\gcd(\mathrm{numerator}(H_n), n!)$

MODULE CONTEXT: # $\gcd(\mathrm{numerator}(H_n), n!)$ $a(n) = \gcd(\mathrm{A001008}(n), n!)$, where $\mathrm{A001008}(n)$ is the numerator of the n-th harmonic number $H_n = \sum_{i=1}^n \frac{1}{i}$. *References:* - [A093818](https://oeis.org/A093818)

V2-000374 | $\gcd(\mathrm{numerator}(H_n), n!)$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: Every odd prime occurs as a term in the sequence.

theorem conjecture (p : ℕ) (hp : p.Prime) (hp_odd : p ≠ 2) : ∃ n > 0, a n = p

FC-OeisA93818.conjecture

Euclid-Mullin sequence

MODULE CONTEXT: # Euclid-Mullin sequence The Euclid-Mullin sequence starts with $a(1) = 2$. Each subsequent term is the smallest prime factor of one plus the product of all preceding terms. We extend the sequence by $a(0) = 1$ and write $b(n)$ for the product of the first n official terms. *References:* - [A000945](https://oeis.org/A000945) - [Mullin63] A. A. Mullin, ["Research Problem 8 (ii)"](https://doi.org/10.1090/S0002-9904-1963-11017-4), *Bull. Amer. Math. Soc.* **69** (1963), p. 737. - [Wagstaff93] S. S. Wagstaff, Jr., ["Computing Euclid's primes"](https://oeis.org/A000945/a000945_4.pdf), *Bull. Institute Combin. Applications* **8** (1993), pp. 23-32. - [CrandallPomerance01] R. Crandall and C. Pomerance, *Prime Numbers: A Computational Perspective*, Springer (2001), p. 6. - A. R. Booker, "A variant of the Euclid-Mullin sequence containing every prime," [arXiv:1605.08929](https://arxiv.org/abs/1605.08929), *Journal of Integer Sequences* **19** (2016), Article 16.6.4.

V2-000375 | Euclid-Mullin sequence | every_prime_occurs

research open | proof annotation: False | 官方代码 | 原文线索

"Does the sequence ... contain every prime? ... [It] was considered by Guy and Nowakowski and later by Shanks, [Wagstaff93] computed the sequence through the 43rd term. The computational problem inherent in continuing the sequence further is the enormous size of the numbers that must be factored. Already the number $a(1) \cdots a(43) + 1$ has 180 digits." - [CrandallPomerance01] See also [Mullin63].

theorem every_prime_occurs :
∀ p, p.Prime → ∃ n ≥ 1, a n = p

FC-OeisA945.every_prime_occurs

Recurrence $a(n) = (a(n-1) + a(n-2)) \pmod n$

MODULE CONTEXT: # Recurrence $a(n) = (a(n-1) + a(n-2)) \pmod n$ $a(0) = a(1) = 1$; $a(n) = (a(n-1) + a(n-2)) \pmod n$. *References:* - [A096535](https://oeis.org/A096535)

V2-000376 | Recurrence $a(n) = (a(n-1) + a(n-2)) \pmod n$ | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

All numbers appear infinitely often, i.e., for every number $k \ge 0$ and every frequency $f > 0$ there is an index i such that $a(i) = k$ is the f-th occurrence of k in the sequence. -_Klaus Brockhaus_, Aug 29 2006

```
theorem conjecture (k : ℕ) (N : ℕ) : ∃ i : ℕ, i > N ∧ a i = k
```

FC-OeisA96535.conjecture

Coefficients in a simultaneous approximation to two polylogarithms

MODULE CONTEXT: # Coefficients in a simultaneous approximation to two polylogarithms Coefficients in a simultaneous approximation to $\operatorname{Li}_2(-1)$ and $\operatorname{Li}_3(-1)$, given by $a(n)=\sum_{i,j=0}^{n}\binom{n}{i}^2\binom{n}{j}^2\binom{n+i}{n}\binom{i+j}{i}$. *References:* - [A098275](https://oeis.org/A098275) - [Li26] Wentao Li, [A proof of the OEIS A098275 conjecture](https://github.com/VictorLiwentao/lean-oeis-proofs/blob/0a23974e54b90afd1453ef02c27cf8f617bad940/proofs/new-proofs/A098275/PROOF.md), 2026.

V2-000377 | Coefficients in a simultaneous approximation to two polylogarithms | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1: "Conjecture: $a(n)$ is divisible by $n+1$." - F. Chapoton, Jan 28 2026. Proved and formalized in Lean by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture_1 (n : ℕ) : n + 1 ∣ a n
```

FC-OeisA98275.conjecture_1

A04 OpenQuantumProblems | 41 条

Open Quantum Problem 23: SIC-POVMs

MODULE CONTEXT: # Open Quantum Problem 23: SIC-POVMs ## Mathematical problem The OQP page presents three increasingly strong formulations of this problem. In this file we formalize the first one, closest to the physics terminology: existence of a symmetric informationally complete POVM in every finite dimension. A SIC-POVM in dimension d can be represented by a family of d^2 normalized vectors in $\mathbb{C}^d$ whose pairwise squared overlaps are all equal to $(d + 1)^{-1}$. We encode such a family as a map `Fin (d ^ 2) → StateVector d`. ## Background SIC-POVMs are a basic structure in finite-dimensional quantum information. They are closely related to equiangular lines, tight frames, quantum state reconstruction, and finite-dimensional measurement theory. The open problem asks whether such families exist in every dimension. ## What this file formalizes This file formalizes the existence problem for symmetric informationally complete POVMs through the predicate `HasSICPOVM d`. More precisely, it contains the following layers. ### Core API The main definitions formalized in this file are: - `StateVector d`: a state vector in `ℂ^d`; - `mkStateVector`: constructor from coordinates in the computational basis; - `IsNormalized ψ`: normalization predicate for a state vector; - `overlapSq φ ψ`: squared magnitude of the inner-product overlap; - `HasConstantOverlapSq c Φ`: constant pairwise squared-overlap condition; - `sicOverlapSq d`: the SIC overlap value `(d + 1)⁻¹`; - `IsSICFamily d Φ`: the predicate that a family of `d^2` vectors in `ℂ^d` is a SIC family; - `HasSICPOVM d`: existence of a SIC family in dimension `d`. In addition, the file includes explicit witness families and convenient constructors used in the low-dimensional benchmark cases: - `vec2`, `vec3`; - `qubitSICFamily`; - `hesseFamily`; - `bb84Family`. ### Complete open conjecture The main open theorem is: - `sicPOVMs`, expressing the conjecture that for every `d ≥ 1`, there exists a SIC-POVM in dimension `d`. ### Special cases The file also isolates several special cases: - solved low-dimensional benchmark cases: `hasSICPOVM_zero`, `hasSICPOVM_one`, `hasSICPOVM_two`, `hasSICPOVM_three`; - a negative benchmark result: `bb84Family_not_isSICFamily`, showing that the BB84 family in dimension `2` does not form a SIC family; - selected open benchmark dimensions: `hasSICPOVM_56`, `hasSICPOVM_58`, `hasSICPOVM_59`, `hasSICPOVM_60`, `hasSICPOVM_64`, `hasSICPOVM_68`, `hasSICPOVM_69`, `hasSICPOVM_70`, `hasSICPOVM_71`, `hasSICPOVM_72`, `hasSICPOVM_75`. ### Test lemmas The file includes the following test lemmas and benchmark-support statements: - `hasConstantOverlapSq_singleton`; - `sicOverlapSq_one`, `sicOverlapSq_two`, `sicOverlapSq_three`, `sicOverlapSq_pos`; - `isSICFamily_singleton_iff`, `isSICFamily_one_of_normalized`; - `qubitSICFamily_normalized`, `qubitSICFamily_pairwise`; - `hesseFamily_normalized`, `hesseFamily_pairwise`; - `bb84Family_normalized`. At present, these `@[category test, AMS 15 47 81]` results are included with placeholder proofs `by sorry`; they are intended to be proved in the next PR. ## References *Primary source list entry:* - IQOQI Vienna Open Quantum Problems, problem 23: https://oqp.iqoqi.oeaw.ac.at/sic-povms-and-zauners-conjecture - Formal Conjectures issue #1823: https://github.com/google-deepmind/formal-conjectures/issues/1823 ### Foundational references - J. M. Renes, R. Blume-Kohout, A. J. Scott, and M. C. Caves, *Symmetric informationally complete quantum measurements*, J. Math. Phys. 45, 2171-2180 (2004), arXiv:quant-ph/0310075. - G. Zauner, *Quantum Designs: Foundations of a Noncommutative Design Theory*, PhD thesis, University of Vienna (1999).

V2-000378 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_56

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 56.

theorem hasSICPOVM_56 : HasSICPOVM 56

FC-OpenQuantumProblem23.hasSICPOVM_56

V2-000379 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_58

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 58.

theorem hasSICPOVM_58 : HasSICPOVM 58

FC-OpenQuantumProblem23.hasSICPOVM_58

V2-000380 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_59

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 59.

theorem hasSICPOVM_59 : HasSICPOVM 59

FC-OpenQuantumProblem23.hasSICPOVM_59

V2-000381 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_60

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 60.

theorem hasSICPOVM_60 : HasSICPOVM 60

FC-OpenQuantumProblem23.hasSICPOVM_60

V2-000382 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_64

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 64.

theorem hasSICPOVM_64 : HasSICPOVM 64

FC-OpenQuantumProblem23.hasSICPOVM_64

V2-000383 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_68

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 68.

theorem hasSICPOVM_68 : HasSICPOVM 68

FC-OpenQuantumProblem23.hasSICPOVM_68

V2-000384 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_69

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 69.

theorem hasSICPOVM_69 : HasSICPOVM 69

FC-OpenQuantumProblem23.hasSICPOVM_69

V2-000385 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_70

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 70.

theorem hasSICPOVM_70 : HasSICPOVM 70

FC-OpenQuantumProblem23.hasSICPOVM_70

V2-000386 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_71

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 71.

theorem hasSICPOVM_71 : HasSICPOVM 71

FC-OpenQuantumProblem23.hasSICPOVM_71

V2-000387 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_72

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 72.

theorem hasSICPOVM_72 : HasSICPOVM 72

FC-OpenQuantumProblem23.hasSICPOVM_72

V2-000388 | Open Quantum Problem 23: SIC-POVMs | hasSICPOVM_75

research open | proof annotation: False | 官方代码 | 原文线索

Benchmark open subproblem: existence of a SIC-POVM in dimension 75.

theorem hasSICPOVM_75 : HasSICPOVM 75

FC-OpenQuantumProblem23.hasSICPOVM_75

V2-000389 | Open Quantum Problem 23: SIC-POVMs | sicPOVMs

research open | proof annotation: False | 官方代码 | 原文线索

Do SIC-POVMs exist in every finite dimension?

theorem sicPOVMs :
∀ d : ℕ, 1 ≤ d → HasSICPOVM d

FC-OpenQuantumProblem23.sicPOVMs

Open Quantum Problem 35: existence of absolutely maximally entangled pure states

MODULE CONTEXT: # Open Quantum Problem 35: existence of absolutely maximally entangled pure states **Problem:** For which numbers of parties n and local dimensions d does there exist a pure absolutely maximally entangled state ψ? A pure state ψ on n parties of local dimension d is called **absolutely maximally entangled (AME)** if, for every subset of at most half of the parties, the corresponding reduced density matrix is maximally mixed. *References:* - Open Quantum Problems, Problem 35: <https://oqp.iqoqi.oeaw.ac.at/existence-of-absolutely-maximally-entangled-pure-states> - Formal Conjectures issue #3452: <https://github.com/google-deepmind/formal-conjectures/issues/3452> - W. Helwig, W. Cui, A. Riera, J. I. Latorre, and H.-K. Lo, *Absolute Maximal Entanglement and Quantum Secret Sharing*, Phys. Rev. A 86, 052335 (2012), arXiv:1204.2289. - D. Goyeneche, D. Alsina, J. I. Latorre, A. Riera, and K. Życzkowski, *Absolutely Maximally Entangled states, combinatorial designs and multi-unitary matrices*, Phys. Rev. A 92, 032316 (2015), arXiv:1506.08857. - A. Higuchi and A. Sudbery, *How entangled can two couples get?*, Phys. Lett. A 273, 213-217 (2000), arXiv:quant-ph/0005013. - A. J. Scott, *Multipartite entanglement, quantum-error-correcting codes, and entangling power of quantum evolutions*, Phys. Rev. A 69, 052330 (2004), arXiv:quant-ph/0310137. - F. Huber, O. Gühne, and J. Siewert, *Absolutely maximally entangled states of seven qubits do not exist*, Phys. Rev. Lett. 118, 200502 (2017), arXiv:1608.06228. - F. Huber and M. Grassl, *Quantum Codes of Maximal Distance and Highly Entangled Subspaces*, Quantum 4, 284 (2020), arXiv:1907.07733. - S. A. Rather, A. Burchardt, W. Bruzda, G. Rajchel-Mieldzioć, A. Lakshminarayan, and K. Życzkowski, *Thirty-six entangled officers of Euler: Quantum solution to a classically impossible problem*, Phys. Rev. Lett. 128, 080507 (2022), arXiv:2104.05122. - G. Rajchel-Mieldzioć, R. Bistroń, A. Rico, A. Lakshminarayan, and K. Życzkowski, *Absolutely maximally entangled pure states of multipartite quantum systems*, arXiv:2508.04777 (2025). - S. Bevins and Y. Bidav, *Symmetry-guided constructions of absolutely maximally entangled states in five open cases*, [arXiv:2608.05781](https://arxiv.org/abs/2608.05781) (2026). - F. Shi, X. Zhang, Q. Zhao, and L. Li, *Complete Existence Classification of Seven-Partite Absolutely Maximally Entangled States*, [arXiv:2608.01011](https://arxiv.org/abs/2608.01011) (2026). This file formalizes the problem of determining for which pairs (n,d) there exists an absolutely maximally entangled pure state $\mathrm{AME}(n,d)$. We represent an n-partite state of local dimension d by the finite-dimensional Hilbert space `EuclideanSpace ℂ (Config n d)`, whose coordinates in the computational basis are amplitudes. The helper `mkStateVector` turns an amplitude function into such a state, and normalization is imposed explicitly via `IsNormalized`, i.e. via the ambient L^2 norm. The main reusable lemma is `reducedDensityFirst_of_completion`: if a state is a uniform superposition over the graph of an injective completion function `completion : Config m d → Config (n - m) d`, then the reduced state on the first m parties is maximally mixed. As demonstration, we show that the Bell states with $n=2$ and GHZ states with $n=3$ are AME states, and the GHZ state with $n=4$ is not an AME state.

V2-000390 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_10_10_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(10,10)$ state exist?

theorem ame_10_10_open :
ExistsAME 10 10

FC-OpenQuantumProblem35.ame_10_10_open

V2-000391 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_10_6_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(10,6)$ state exist?

theorem ame_10_6_open :
ExistsAME 10 6

FC-OpenQuantumProblem35.ame_10_6_open

V2-000392 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_11_10_open

research solved | proof annotation: True | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(11,10)$ state exist? Answer: Yes. A witness found by Kenta Kitamura (KitaKen1 on GitHub) is given here: https://github.com/KitaKen1/ame-11-10-lean

theorem ame_11_10_open :
ExistsAME 11 10

FC-OpenQuantumProblem35.ame_11_10_open

V2-000393 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_11_3_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(11,3)$ state exist?

theorem ame_11_3_open :
ExistsAME 11 3

FC-OpenQuantumProblem35.ame_11_3_open

V2-000394 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_11_4_open

research solved | proof annotation: True | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(11,4)$ state exist? Answer: `AME(11, 4)` exists. The graph state defined by the circulant matrix Gamma over $GF(4)$ with first row $(0, 0, 0, 1, ω, ω, ω, ω, 1, 0, 0)$ is an absolutely maximally entangled state of 11 ququarts. This result has been found by Moritz Firsching and Goran Žužić using an experimental pipeline Before, it was already known that there is a quantum code for `[11,0]]_5`, which corresponds to an `AME(11,4)` state (which is another approach to a solution).

theorem ame_11_4_open :
ExistsAME 11 4

FC-OpenQuantumProblem35.ame_11_4_open

V2-000395 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_11_5_open

research solved | proof annotation: True | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(11,5)$ state exist? The DeepMind prover agent has shown that such a state exists.

theorem ame_11_5_open :
ExistsAME 11 5

FC-OpenQuantumProblem35.ame_11_5_open

V2-000396 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_11_6_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(11,6)$ state exist?

theorem ame_11_6_open :
ExistsAME 11 6

FC-OpenQuantumProblem35.ame_11_6_open

V2-000397 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_12_10_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(12,10)$ state exist?

theorem ame_12_10_open :
ExistsAME 12 10

FC-OpenQuantumProblem35.ame_12_10_open

V2-000398 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_12_5_open

research solved | proof annotation: False | 官方代码 | 原文线索

Does an $\mathrm{AME}(12,5)$ state exist? The answer is yes. Bevins and Bidav construct an explicit Hermitian self-dual MDS code with parameters $[12,6,7]_{25}$, whose associated nonbinary stabilizer state is an $\mathrm{AME}(12,5)$ state.

theorem ame_12_5_open :
ExistsAME 12 5

FC-OpenQuantumProblem35.ame_12_5_open

V2-000399 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_12_6_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(12,6)$ state exist?

theorem ame_12_6_open :
ExistsAME 12 6

FC-OpenQuantumProblem35.ame_12_6_open

V2-000400 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_2_2_exists

research solved | proof annotation: False | 官方代码 | 原文线索

Source-backed benchmark statement: the Bell state witnesses the existence of an $\mathrm{AME}(2,2)$ state.

theorem ame_2_2_exists : ExistsAME 2 2

FC-OpenQuantumProblem35.ame_2_2_exists

V2-000401 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_2_exists

research solved | proof annotation: False | 官方代码 | 原文线索

The Bell state witnesses the existence of $\mathrm{AME}(2,d)$ for every local dimension $d \ge 2$.

theorem ame_2_exists {d : ℕ} (hd : 2 ≤ d) : ExistsAME 2 d

FC-OpenQuantumProblem35.ame_2_exists

V2-000402 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_3_2_exists

research solved | proof annotation: False | 官方代码 | 原文线索

Source-backed benchmark statement: the three-qubit GHZ state witnesses the existence of an $\mathrm{AME}(3,2)$ state.

theorem ame_3_2_exists : ExistsAME 3 2

FC-OpenQuantumProblem35.ame_3_2_exists

V2-000403 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_3_exists

research solved | proof annotation: False | 官方代码 | 原文线索

The 3-party GHZ state witnesses the existence of $\mathrm{AME}(3,d)$ for every local dimension $d \ge 2$.

theorem ame_3_exists {d : ℕ} (hd : 2 ≤ d) : ExistsAME 3 d

FC-OpenQuantumProblem35.ame_3_exists

V2-000404 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_4_2_not_exists

research solved | proof annotation: True | 官方代码 | 原文线索

Source-backed benchmark statement: no $\mathrm{AME}(4,2)$ state exists; see Higuchi--Sudbery (2000) and the OQP page.

theorem ame_4_2_not_exists : ¬ ExistsAME 4 2

FC-OpenQuantumProblem35.ame_4_2_not_exists

V2-000405 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_4_3_exists

research solved | proof annotation: True | 官方代码 | 原文线索

Source-backed benchmark statement: an $\mathrm{AME}(4,3)$ state exists; see Helwig et al. (2012) and Goyeneche et al. (2015).

theorem ame_4_3_exists : ExistsAME 4 3

FC-OpenQuantumProblem35.ame_4_3_exists

V2-000406 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_4_6_exists

research solved | proof annotation: False | 官方代码 | 原文线索

Source-backed benchmark statement: an $\mathrm{AME}(4,6)$ state exists; see Rather et al. (2022).

theorem ame_4_6_exists : ExistsAME 4 6

FC-OpenQuantumProblem35.ame_4_6_exists

V2-000407 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_5_2_exists

research solved | proof annotation: True | 官方代码 | 原文线索

Source-backed benchmark statement: an $\mathrm{AME}(5,2)$ state exists. This is one of the four qubit cases $n=2,3,5,6$; see the OQP page and Scott (2004).

theorem ame_5_2_exists : ExistsAME 5 2

FC-OpenQuantumProblem35.ame_5_2_exists

V2-000408 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_6_2_exists

research solved | proof annotation: True | 官方代码 | 原文线索

Source-backed benchmark statement: an $\mathrm{AME}(6,2)$ state exists. This is one of the four qubit cases $n=2,3,5,6$; see the OQP page and Scott (2004).

theorem ame_6_2_exists : ExistsAME 6 2

FC-OpenQuantumProblem35.ame_6_2_exists

V2-000409 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_7_10_open

research solved | proof annotation: False | 官方代码 | 原文线索

An $\mathrm{AME}(7,10)$ state exists, by the complete seven-party classification of Shi--Zhang--Zhao--Li (2026).

theorem ame_7_10_open :
ExistsAME 7 10

FC-OpenQuantumProblem35.ame_7_10_open

V2-000410 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_7_2_not_exists

research solved | proof annotation: False | 官方代码 | 原文线索

Source-backed benchmark statement: no $\mathrm{AME}(7,2)$ state exists; see Huber--Gühne--Siewert (2017) and the OQP page.

theorem ame_7_2_not_exists : ¬ ExistsAME 7 2

FC-OpenQuantumProblem35.ame_7_2_not_exists

V2-000411 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_7_6_open

research solved | proof annotation: False | 官方代码 | 原文线索

An $\mathrm{AME}(7,6)$ state exists, by the complete seven-party classification of Shi--Zhang--Zhao--Li (2026).

theorem ame_7_6_open :
ExistsAME 7 6

FC-OpenQuantumProblem35.ame_7_6_open

V2-000412 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_7_exists_iff

research solved | proof annotation: False | 官方代码 | 原文线索

For a physical local dimension $d \ge 2$, a seven-party AME state exists exactly when $d \ge 3$. The hypothesis excludes the degenerate dimension $d = 1$, for which the unique computational-basis state is AME under the definitions of this file. Shi--Zhang--Zhao--Li (2026) construct cyclic quadratic-phase states in every odd dimension and a coupled binary--odd-dimensional state in every dimension congruent to `2` modulo `4`; together with power-of-two constructions and the product property, this covers every `d ≥ 3`.

theorem ame_7_exists_iff (d : ℕ) (hd : 2 ≤ d) : ExistsAME 7 d ↔ 3 ≤ d

FC-OpenQuantumProblem35.ame_7_exists_iff

V2-000413 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_8_10_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(8,10)$ state exist?

theorem ame_8_10_open :
ExistsAME 8 10

FC-OpenQuantumProblem35.ame_8_10_open

V2-000414 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_8_4_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(8,4)$ state exist?

theorem ame_8_4_open :
ExistsAME 8 4

FC-OpenQuantumProblem35.ame_8_4_open

V2-000415 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_8_6_open

research open | proof annotation: False | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(8,6)$ state exist?

theorem ame_8_6_open :
ExistsAME 8 6

FC-OpenQuantumProblem35.ame_8_6_open

V2-000416 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_9_10_open

research solved | proof annotation: True | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(9,10)$ state exist? Solved: a Lean 4 proof, derived from the Atlas proofs in [facebookresearch/atlas-lean](https://github.com/facebookresearch/atlas-lean), is linked in `formal_proof`.

theorem ame_9_10_open :
ExistsAME 9 10

FC-OpenQuantumProblem35.ame_9_10_open

V2-000417 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | ame_9_6_open

research solved | proof annotation: True | 官方代码 | 原文线索

Open benchmark statement: does an $\mathrm{AME}(9,6)$ state exist? Answer: Yes. A witness found by Kenta Kitamura (KitaKen1 on GitHub) is given here: https://github.com/KitaKen1/ame-9-6-lean

theorem ame_9_6_open :
ExistsAME 9 6

FC-OpenQuantumProblem35.ame_9_6_open

V2-000418 | Open Quantum Problem 35: existence of absolutely maximally entangled pure states | oqp_35

research open | proof annotation: False | 官方代码 | 原文线索

Open Quantum Problem 35: classify all pairs (n,d) with $n \ge 2$ and $d \ge 2$ for which an $\mathrm{AME}(n,d)$ state exists.

theorem oqp_35 :
{nd : ℕ × ℕ | 2 ≤ nd.1 ∧ 2 ≤ nd.2 ∧ ExistsAME nd.1 nd.2} = oqp_35.answer

FC-OpenQuantumProblem35.oqp_35

A05 OptimizationConstants | 25 条

Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable

MODULE CONTEXT: # Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable *References:* - [Tao's Optimization Constant 14](https://teorth.github.io/optimizationproblems/constants/14a.html) - [LS1965] Lin, S. and Radó, T., [*Computer studies of Turing machine problems*](https://doi.org/10.1145/321264.321270). J. ACM **12** (1965), 196-212. - [B1983] Brady, A. H., [*The determination of Radó's noncomputable function $\Sigma(k)$ for four-state Turing machines*](https://docs.bbchallenge.org/papers/Brady1983.pdf). Math. Comp. 40 (1983), 647-665. - [BB2025] Blanchard, J. et al., *Determination of the fifth Busy Beaver value* (2025). [arXiv:2509.12337](https://arxiv.org/abs/2509.12337) - [YS2016] Yedidia, A. and Aaronson, S., *A relatively small Turing machine whose behavior is independent of set theory* (2016). [arXiv:1605.04343](https://arxiv.org/abs/1605.04343) - [O2016] O'Rear, S., [*metamath-turing-machines*](https://github.com/sorear/metamath-turing-machines) (2016). - [W2025] Wade, A. J., [*turing_machine_explorer*](https://codeberg.org/ajwade/turing_machine_explorer) (2025). - [G2026] Glazer, E., [*K230: a 230-state Turing machine for the consistency of ZFC*](https://github.com/elliotglazer/k230-rocq) (2026). [Announcement](https://x.com/ElliotGlazer/status/2112179703687856497) - [A2020] Aaronson, S., [*The Busy Beaver Frontier*](https://doi.org/10.1145/3427361.3427369). ACM SIGACT News **51** (2020), 32-54. This file defines the smallest n for which the value of $BB(n)$ is undecidable. More precisely, it formalizes the least number of states for which ZFC can neither prove nor disprove that a specific Turing machine halts. Its existence and known upper bounds follow from Gödel's second incompleteness theorem. We do this by modeling a version of first order logic, ZFC and Turing machines within Lean. The implementation of FOL and ZFC is taken from the [FormalizedLogicLean](https://github.com/FormalizedFormalLogic/Foundation) project. The implementation is copied and minimized from that project. It is not necessary to read it to understand the definition of `C14`. The copied definitions span the section `FormalizedFormalLogic`. Confidence in that implementation partly comes from the large body of proofs developed with it, including formalizations of independence theorems. This is followed by a model of Turing machines within the first-order logic framework, developed for Formal Conjectures in the section `BackgroundDefs`. The file then defines `C14` and states its known bounds and related conjectures. The known bounds use `sorry`; formalizing one would be a substantial achievement.

V2-000419 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_ge_4

research solved | proof annotation: False | 官方代码 | 原文线索

Proven by Lin and Radó in [LS1965] by computing $BB(3) = 21$.

theorem c14_ge_4 : 4 ≤ C14

FC-Constant14.c14_ge_4

V2-000420 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_ge_5

research solved | proof annotation: False | 官方代码 | 原文线索

An intermediate lower bound, proved by Brady in [B1983] by explicitly computing $BB(4) = 107$.

theorem c14_ge_5 : 5 ≤ C14

FC-Constant14.c14_ge_5

V2-000421 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_le_20

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 11 in [A2020] by Scott Aaronson

theorem c14_le_20 : C14 ≤ 20

FC-Constant14.c14_le_20

V2-000422 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_le_432

research solved | proof annotation: False | 官方代码 | 原文线索

An intermediate upper bound, proved by Wade in [W2025]. It refines the proof-enumerating machine construction from [O2016].

theorem c14_le_432 : C14 ≤ 432

FC-Constant14.c14_le_432

V2-000423 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_le_748

research solved | proof annotation: False | 官方代码 | 原文线索

An intermediate upper bound, proved by O'Rear in [O2016]. It uses a Turing machine that enumerates proofs in ZF.

theorem c14_le_748 : C14 ≤ 748

FC-Constant14.c14_le_748

V2-000424 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_le_7910

research solved | proof annotation: False | 官方代码 | 原文线索

The first known upper bound, proved by Yedidia and Aaronson in [YS2016]. Their construction reduces the independence problem to a graph-theoretic setting.

theorem c14_le_7910 : C14 ≤ 7910

FC-Constant14.c14_le_7910

V2-000425 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The current best known lower bound, proved by Blanchard et al. in [BB2025] by explcitily computing $BB(5) = 47176870$. Note $BB(5) = 47176870$ (as well as the values below) has been formally proven in Rocq as part of [BB2025], a formal proof of the declaration below would likely be much more difficult on a technical level than the proof in Rocq, which is based on dependent type theory.

theorem c14_lower_bound : 6 ≤ C14

FC-Constant14.c14_lower_bound

V2-000426 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_lower_bound_improved

research open | proof annotation: False | 官方代码 | 原文线索

Can the current best lower bound be improved?

theorem c14_lower_bound_improved : 6 < C14

FC-Constant14.c14_lower_bound_improved

V2-000427 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The current best known upper bound, from [G2026]. Glazer proved in Rocq, without axioms, that a certain 230-state Turing machine halts if and only if ZFC is inconsistent.

theorem c14_upper_bound : C14 ≤ 230

FC-Constant14.c14_upper_bound

V2-000428 | Tao's Optimization Constant 14 / Smallest n for which the value of $BB(n)$ is undecidable | c14_upper_bound_improved

research open | proof annotation: False | 官方代码 | 原文线索

Can the current best upper bound be improved?

theorem c14_upper_bound_improved : C14 < 230

FC-Constant14.c14_upper_bound_improved

Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets

MODULE CONTEXT: # Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets *References:* - [Tao's optimization constant 1a](https://teorth.github.io/optimizationproblems/constants/1a.html) - [M2010] Matolcsi, Máté, and Carlos Vinuesa. "Improved bounds on the supremum of autoconvolutions." Journal of mathematical analysis and applications 372.2 (2010): 439-447. [arXiv:0907.1379](https://arxiv.org/abs/0907.1379) - [Y2026] Yuksekgonul, Mert et al., "Learning to Discover at Test Time," 2026, [arXiv:2601.16175](https://arxiv.org/abs/2601.16175) - [T2026] Together AI, "EinsteinArena-new-SOTA: State-of-the-art results on open math problems," 2026, [GitHub](https://github.com/togethercomputer/EinsteinArena-new-SOTA) - [PBV2026] Piterbarg, Andrei, Jai Bajaj, and Derrick Vincent. "A multi-scale arcsine lower bound for the Sidon autocorrelation constant C_{1a}," 2026, [GitHub](https://github.com/AndreiPiterbarg/sidon-autocorrelation)

V2-000429 | Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets | c1a_eq

research open | proof annotation: False | 官方代码 | 原文线索

What is the exact value of the constant?

theorem c1a_eq : C1a = c1a_eq.answer

FC-Constant1a.c1a_eq

V2-000430 | Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets | c1a_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

A lower bound, proven by Matolcsi-Vinuesa in [M2010].

theorem c1a_lower_bound : 1.2748 ≤ C1a

FC-Constant1a.c1a_lower_bound

V2-000431 | Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets | c1a_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

An upper bound, proven by Yuksekgonul et al. in [Y2026].

theorem c1a_upper_bound : C1a ≤ 1.5029

FC-Constant1a.c1a_upper_bound

V2-000432 | Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets | mem_Ico_c1a

research open | proof annotation: False | 官方代码 | 原文线索

How can the best known upper bound $1.5028628587\ldots$ of [T2026] be improved? The step function published with [T2026] gives $1.50286285870\ldots$. The figure 1.502862 in the table of the first reference is this value truncated, so it is not a known upper bound.

theorem mem_Ico_c1a : mem_Ico_c1a.answer ∈ Set.Ico C1a 1.5028628587

FC-Constant1a.mem_Ico_c1a

V2-000433 | Tao's Optimization constant 1a / An autocorrelation constant related to Sidon sets | mem_Ioc_c1a

research open | proof annotation: False | 官方代码 | 原文线索

How can the best known lower bound 1.292 of [PBV2026] be improved?

theorem mem_Ioc_c1a : mem_Ioc_c1a.answer ∈ Set.Ioc 1.292 C1a

FC-Constant1a.mem_Ioc_c1a

Tao's Optimization Constant 7a / The Irrationality Measure of π

MODULE CONTEXT: # Tao's Optimization Constant 7a / The Irrationality Measure of π *References:* - [Tao's Optimization Constant 7a](https://teorth.github.io/optimizationproblems/constants/7a.html) - [D1842] Dirichlet, L. G. P., *Verallgemeinerung eines Satzes aus der Lehre von den Kettenbrüchen nebst einigen Anwendungen auf die Theorie der Zahlen*. Sitzungsberichte der Preussischen Akademie der Wissenschaften (1842), 93-95. - [M1953] Mahler, K., *On the approximation of π*. Nederl. Akad. Wetensch. Proc. Ser. A **56** = Indag. Math. **15** (1953), 30-42. - [R1955] Roth, K. F., [*Rational approximations to algebraic numbers*](https://doi.org/10.1112/S0025579300000644). Mathematika **2** (1955), 1-20. - [RV1993] Rhin, G. and Viola, C., [*On the irrationality measure of $\zeta(2)$*](https://doi.org/10.5802/aif.1322). Ann. Inst. Fourier **43** (1993), 85-109. - [ZZ2020] Zeilberger, D. and Zudilin, W., *The irrationality measure of π is at most $7.103205334137\ldots$*. Moscow J. Comb. Number Theory **9** (2020), 407-419. [arXiv:1912.06345](https://arxiv.org/abs/1912.06345) - [OAI2026] OpenAI, [*The irrationality exponent of π is 2*](https://github.com/openai/math/blob/main/preprints/The-irrationality-exponent-of-pi-is-2-September-24-2026/paper.pdf), 2026.

V2-000434 | Tao's Optimization Constant 7a / The Irrationality Measure of π | c7a_eq_two

research solved | proof annotation: True | 官方代码 | 原文线索

The irrationality measure of π is exactly 2, proved by OpenAI in [OAI2026].

theorem c7a_eq_two : C7a = 2

FC-Constant7a.c7a_eq_two

V2-000435 | Tao's Optimization Constant 7a / The Irrationality Measure of π | c7a_le_42

research solved | proof annotation: False | 官方代码 | 原文线索

The first known upper bound, proved by Mahler in [M1953]. This was the first proof that C_{7a} is finite, or equivalently that π is not a Liouville number.

theorem c7a_le_42 : C7a ≤ 42

FC-Constant7a.c7a_le_42

V2-000436 | Tao's Optimization Constant 7a / The Irrationality Measure of π | c7a_lt_14_797075

research solved | proof annotation: False | 官方代码 | 原文线索

An intermediate upper bound proved by Rhin and Viola in [RV1993]. It follows from an effective irrationality measure for $\zeta(2) = \pi^2 / 6$.

theorem c7a_lt_14_797075 : C7a < 14.797075

FC-Constant7a.c7a_lt_14_797075

V2-000437 | Tao's Optimization Constant 7a / The Irrationality Measure of π | c7a_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The best upper bound before [OAI2026], proved by Zeilberger and Zudilin in [ZZ2020].

theorem c7a_upper_bound : C7a < 7.103205334138

FC-Constant7a.c7a_upper_bound

V2-000438 | Tao's Optimization Constant 7a / The Irrationality Measure of π | irrationalityExponent_of_algebraic

research solved | proof annotation: False | 官方代码 | 原文线索

Every algebraic real number has irrationality exponent at most two. For irrational algebraic numbers, this is Roth's theorem [R1955].

theorem irrationalityExponent_of_algebraic {x : ℝ} (hx : IsAlgebraic ℚ x) : irrationalityExponent x ≤ 2

FC-Constant7a.irrationalityExponent_of_algebraic

Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$

MODULE CONTEXT: # Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ *References:* - [Tao's Optimization Constant 7b](https://teorth.github.io/optimizationproblems/constants/7b.html) - [D1842] Dirichlet, L. G. P., *Verallgemeinerung eines Satzes aus der Lehre von den Kettenbrüchen nebst einigen Anwendungen auf die Theorie der Zahlen*. Sitzungsberichte der Preussischen Akademie der Wissenschaften (1842), 93-95. - [C1976] Chudnovsky, G. V., *Algebraic independence of constants connected with the exponential and the elliptic functions*. Dokl. Akad. Nauk Ukrain. SSR Ser. A 1976, no. 8, 698-701. - [Bru2002] Bruiltet, S., [*D'une mesure d'approximation simultanée à une mesure d'irrationalité : le cas de $\Gamma(1/4)$ et $\Gamma(1/3)$*](https://doi.org/10.4064/aa104-3-3). Acta Arith. **104** (2002), 243-281.

V2-000439 | Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ | c7b_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The first and current best known lower bound, given by Dirichlet's theorem [D1842]. It applies because $\Gamma(1/4)$ is irrational by Chudnovsky's theorem [C1976].

```
theorem c7b_lower_bound : 2 ≤ C7b :=
Constant7a.two_le_irrationalityExponent irrational_gamma_one_div_four

/-- Can the current best lower bound be improved? -/
@[category research open, question, AMS 11 33]
theorem c7b_lower_bound_improved : 2 < C7b
```

FC-Constant7b.c7b_lower_bound

V2-000440 | Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ | c7b_lower_bound_improved

research open | proof annotation: False | 官方代码 | 原文线索

Can the current best lower bound be improved?

```
theorem c7b_lower_bound_improved : 2 < C7b
```

FC-Constant7b.c7b_lower_bound_improved

V2-000441 | Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ | c7b_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The first and current best known upper bound, proved by Bruiltet in [Bru2002]. It follows from an explicit lower bound on $|\Gamma(1/4) - p/q|$ for rational p/q of sufficiently large height.

```
theorem c7b_upper_bound : C7b ≤ 10 ^ 143
```

FC-Constant7b.c7b_upper_bound

V2-000442 | Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ | c7b_upper_bound_improved

research open | proof annotation: False | 官方代码 | 原文线索

Can the current best upper bound be improved?

```
theorem c7b_upper_bound_improved : C7b < 10 ^ 143
```

FC-Constant7b.c7b_upper_bound_improved

V2-000443 | Tao's Optimization Constant 7b / The Irrationality Measure of $\Gamma(1/4)$ | irrational_gamma_one_div_four

research solved | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/4)$ is irrational, as proven in [C1976].

```
theorem irrational_gamma_one_div_four : Irrational (Real.Gamma (1 / 4))
```

FC-Constant7b.irrational_gamma_one_div_four

A06 Other | 26 条

Beaver Math Olympiad (BMO)

MODULE CONTEXT: # Beaver Math Olympiad (BMO) The Beaver Math Olympiad (BMO) is a set of mathematical reformulations of the halting/nonhalting problem of specific Turing machines from all-0 tape. These problems came from studying small Busy Beaver values. Some problems are open and have a conjectured answer, some are open and don't have a conjectured answer, and, some are solved. Among these problems is the Collatz-like *Antihydra* problem which is open and coming from a 6-state Turing machine, and a testament to the difficulty of knowing the sixth Busy Beaver value. For some BMO problem, the equivalence between the mathematical formulation and the corresponding Turing machine non-termination has been formally proved in Rocq, we indicate it when done. *References:* - bbchallenge.org - [Beaver Math Olympiad wiki page](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad) - [Antihydra web page](https://bbchallenge.org/antihydra) - [Antihydra wiki page](https://wiki.bbchallenge.org/wiki/Antihydra)

V2-000444 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_1

research open | proof annotation: False | 官方代码 | 原文线索

[BMO#1](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#1._1RB1RE_1LC0RA_0RD1LB_---1RC_1LF1RE_0LB0LE_(bbch)) Let $(a_n)_{n \ge 1}$ and $(b_n)_{n \ge 1}$ be two sequences such that $(a_1, b_1) = (1, 2)$ and $$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n-b_n, 4b_n+2) & \text{if }a_n \ge b_n \cr (2a_n+1, b_n-a_n) & \text{if }a_n < b_n \end{cases}$$ for all positive integers n. Does there exist a positive integer i such that $a_i = b_i$? The first 10 values of (a_n, b_n) are $(1, 2), (3, 1), (2, 6), (5, 4), (1, 18), (3, 17), (7, 14), (15, 7), (8, 30), (17, 22)$. [BMO#1](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#1._1RB1RE_1LC0RA_0RD1LB_---1RC_1LF1RE_0LB0LE_(bbch)) is equivalent to asking whether the 6-state Turing machine [`1RB1RE_1LC0RA_0RD1LB_---1RC_1LF1RE_0LB0LE`](https://wiki.bbchallenge.org/wiki/1RB1RE_1LC0RA_0RD1LB_---1RC_1LF1RE_0LB0LE) halts or not. There is presently no consensus on whether the machine halts or not, hence the problem is tagged as a `question`. The machine was discovered by bbchallenge.org contributor Jason Yuen on June 25th 2024.

```
theorem beaver_math_olympiad_problem_1 :
∀ᵉ (a : ℕ → ℕ) (b : ℕ → ℕ)
(a_ini : a 0 = 1)
(a_rec : ∀ n, a (n + 1) = if b n ≤ a n then a n - b n else 2 * a n + 1)
(b_ini : b 0 = 2)
(b_rec : ∀ n, b (n + 1) = if b n ≤ a n then 4 * b n + 2 else b n - a n),
∃ i, a i = b i
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_1

V2-000445 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_2_antihydra

research open | proof annotation: False | 官方代码 | 原文线索

[BMO#2](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#2._Hydra_and_Antihydra) Antihydra is a sequence starting at 8, and iterating the function $$H(n) = \left\lfloor \frac {3n}2 \right\rfloor.$$ The conjecture states that the cumulative number of odd values in this sequence is never more than twice the cumulative number of even values. It is a relatively new open problem with, so it might be solvable, although seems quite hard because of its Collatz-like flavor. The underlying Collatz-like map has been studied independently in the past, see doi:[10.1017/S0017089508004655](https://doi.org/10.1017/S0017089508004655) (Corollary 4). It is equivalent to non-termination of the [`1RB1RA_0LC1LE_1LD1LC_1LA0LB_1LF1RE_---0RA`](https://wiki.bbchallenge.org/wiki/Antihydra) 6-state Turing machine (from all-0 tape). Note that the conjecture that the machine does not halt is based on [a probabilistic argument](https://wiki.bbchallenge.org/wiki/Antihydra#Trajectory). This machine and its mathematical reformulations were found by bbchallenge.org contributors mxdys and Rachel Hunter on June 28th 2024.

```
theorem beaver_math_olympiad_problem_2_antihydra
(a : ℕ → ℕ) (b : ℕ → ℤ)
(a_ini : a 0 = 8)
(a_rec : ∀ n, a (n + 1) = (3 * a n) / 2)
(b_ini : b 0 = 0)
(b_rec : ∀ n, b (n + 1) = if a n % 2 = 0 then b n + 2 else b n - 1) :
∀ n, b n ≥ 0
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_2_antihydra

V2-000446 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_3

research solved | proof annotation: True | 官方代码 | 原文线索

[BMO#3](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#3._1RB0RB3LA4LA2RA_2LB3RA---3RA4RB_(bbch)_and_1RB1RB3LA4LA2RA_2LB3RA---3RA4RB_(bbch)) Let $v_2(n)$ be the largest integer k such that 2^k divides n. Let $(a_n)_{n \ge 0}$ be a sequence such that $$a_n = \begin{cases} 2 & \text{if } n=0 \cr a_{n-1}+2^{v_2(a_{n-1})+2}-1 & \text{if } n \ge 1 \end{cases}$$ for all non-negative integers n. Is there an integer n such that $a_n=4^k$ for some positive integer k? [BMO#3](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#3._1RB0RB3LA4LA2RA_2LB3RA---3RA4RB_(bbch)_and_1RB1RB3LA4LA2RA_2LB3RA---3RA4RB_(bbch)) is equivalent to the non-termination of 2-state 5-symbol Turing machine [`1RB0RB3LA4LA2RA_2LB3RA---3RA4RB`](https://wiki.bbchallenge.org/wiki/1RB0RB3LA4LA2RA_2LB3RA---3RA4RB) (from all-0 tape). The machine was found and informally proven not to halt by bbchallenge.org contributor Daniel Yuan on June 18th 2024; see [Discord discussion](https://discord.com/channels/960643023006490684/1084047886494470185/1252634913220591728).

```
theorem beaver_math_olympiad_problem_3
(a : ℕ → ℕ)
(a_ini : a 0 = 2)
(a_rec : ∀ n, a (n + 1) = (a n) + 2 ^ ((padicValNat 2 (a n)) + 2) - 1) :
¬ (∃ n k, a n = 4 ^ k)
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_3

V2-000447 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_4

research solved | proof annotation: True | 官方代码 | 原文线索

[BMO#4](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#4._1RB3RB---1LB0LA_2LA4RA3LA4RB1LB_(bbch)) Bonnie the beaver was bored, so she tried to construct a sequence of integers $\{a_n\}_{n \ge 0}$. She first defined $a_0=2$, then defined a_{n+1} depending on a_n and n using the following rules: * If $a_n \equiv 0\text{ (mod 3)}$, then $a_{n+1}=\frac{a_n}{3}+2^n+1$. * If $a_n \equiv 2\text{ (mod 3)}$, then $a_{n+1}=\frac{a_n-2}{3}+2^n-1$. With these two rules alone, Bonnie calculates the first few terms in the sequence: $2, 0, 3, 6, 11, 18, 39, 78, 155, 306, \dots$. At this point, Bonnie plans to continue writing terms until a term becomes 1 (mod 3). If Bonnie sticks to her plan, will she ever finish? [BMO#4](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#4._1RB3RB---1LB0LA_2LA4RA3LA4RB1LB_(bbch)) is equivalent to the non-termination of 2-state 5-symbol Turing machine [`1RB3RB---1LB0LA_2LA4RA3LA4RB1LB`](https://wiki.bbchallenge.org/wiki/1RB3RB---1LB0LA_2LA4RA3LA4RB1LB) (from all-0 tape). The machine was informally proven not to halt bbchallenge.org contributor Daniel Yuan on July 19th 2024; see [sketched proof](https://wiki.bbchallenge.org/wiki/1RB3RB---1LB0LA_2LA4RA3LA4RB1LB) and [Discord discussion](https://discord.com/channels/960643023006490684/960643023530762343/1263666591900631210).

```
theorem beaver_math_olympiad_problem_4
(a : ℕ → ℕ)
(a_ini : a 0 = 2)
(a_rec : ∀ n, a (n+1)
= if a n % 3 = 0 then a n / 3 + 2 ^ n + 1 else (a n - 2) / 3 + 2 ^ n - 1) :
¬ (∃ n, a n % 3 = 1)
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_4

V2-000448 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_5

research open | proof annotation: False | 官方代码 | 原文线索

[BMO#5](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#5._1RB0LD_1LC0RA_1RA1LB_1LA1LE_1RF0LC_---0RE_(bbch)) Let $(a_n)_{n \ge 1}$ and $(b_n)_{n \ge 1}$ be two sequences such that $(a_1, b_1) = (0, 5)$ and $$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n+1, b_n-f(a_n)) & \text{if } b_n \ge f(a_n) \cr (a_n, 3b_n+a_n+5) & \text{if } b_n < f(a_n) \end{cases}$$ where $f(x)=10\cdot 2^x-1$ for all non-negative integers x. Does there exist a positive integer i such that $b_i = f(a_i)-1$? [BMO#5](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#5._1RB0LD_1LC0RA_1RA1LB_1LA1LE_1RF0LC_---0RE_(bbch)) is equivalent to asking whether the 6-state Turing machine [`1RB0LD_1LC0RA_1RA1LB_1LA1LE_1RF0LC_---0RE`](https://wiki.bbchallenge.org/wiki/1RB0LD_1LC0RA_1RA1LB_1LA1LE_1RF0LC_---0RE) halts or not. There is presently no consensus on whether the machine halts or not, hence the problem is tagged as a `question`. The machine was discovered by bbchallenge.org contributor mxdys on August 7th 2024. The correspondence between the machine's halting problem and the below reformulation has been proven in [Rocq](https://github.com/ccz181078/busycoq/blob/BB6/verify/1RB0LD_1LC0RA_1RA1LB_1LA1LE_1RF0LC_---0RE.v).

```
theorem beaver_math_olympiad_problem_5 :
∀ (a b f : ℕ → ℕ), ∀ᵉ (hf : f = fun x ↦ 10 * 2 ^ x - 1)
(a_ini : a 0 = 0) (b_ini : b 0 = 5)
(a_rec : ∀ n, a (n + 1) = if f (a n) ≤ b n then a n + 1 else a n)
(b_rec : ∀ n, b (n+1) = if f (a n) ≤ b n then b n - f (a n) else 3 * b n + a n + 5),
∃ i, b i = f (a i) - 1
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_5

V2-000449 | Beaver Math Olympiad (BMO) | beaver_math_olympiad_problem_8

research open | proof annotation: False | 官方代码 | 原文线索

[BMO#8](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#8._1RB0LD_0RC1RB_0RD0RA_1LE0RD_1LF---_0LA1LA_(bbch)) Let $(a_n)_{n \ge 1}$ and $(b_n)_{n \ge 1}$ be two sequences such that $(a_1, b_1) = (10, 12)$ and $$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n - \lfloor b_n/2 \rfloor - 3, 3 \lfloor (b_n+1)/2 \rfloor + 6) & \text{if } a_n > \lfloor b_n/2 \rfloor \cr (3 a_n + 5, b_n - 2 a_n) & \text{if } a_n \le \lfloor b_n/2 \rfloor \end{cases}$$ for all positive integers n. Does there exist a positive integer i such that $a_i = \lfloor b_i/2 \rfloor + 1$? [BMO#8](https://wiki.bbchallenge.org/wiki/Beaver_Math_Olympiad#8._1RB0LD_0RC1RB_0RD0RA_1LE0RD_1LF---_0LA1LA_(bbch)) is equivalent to asking whether the 6-state Turing machine [`1RB0LD_0RC1RB_0RD0RA_1LE0RD_1LF---_0LA1LA`](https://wiki.bbchallenge.org/wiki/1RB0LD_0RC1RB_0RD0RA_1LE0RD_1LF---_0LA1LA) halts or not. There is presently no consensus on whether the machine halts or not, hence the problem is tagged as a `question`.

```
theorem beaver_math_olympiad_problem_8 :
∀ᵉ (a : ℕ → ℤ) (b : ℕ → ℤ)
(a_ini : a 0 = 10)
(a_rec : ∀ n, a (n + 1) =
if b n / 2 < a n then a n - b n / 2 - 3 else 3 * a n + 5)
(b_ini : b 0 = 12)
(b_rec : ∀ n, b (n + 1) =
if b n / 2 < a n then 3 * ((b n + 1) / 2) + 6 else b n - 2 * a n),
∃ i, a i = b i / 2 + 1
```

FC-BeaverMathOlympiad.beaver_math_olympiad_problem_8

Equational Theories

MODULE CONTEXT: # Equational Theories *Reference:* [Equational Theories project site](https://teorth.github.io/equational_theories/implications/?677&finite)

V2-000450 | Equational Theories | Equation255_not_implies_Equation677

research solved | proof annotation: False | 官方代码 | 原文线索

Equation 255 does not imply Equation 677.

```
theorem Equation255_not_implies_Equation677 :
∃ (G : Type) (_ : Magma G), Equation255 G ∧ ¬ Equation677 G :=
⟨Fin 3, (![![1, 2, 0], ![2, 0, 1], ![0, 1, 2]]),
fun x ↦ by fin_cases x <;> rfl, of_decide_eq_false rfl⟩

/-- Equation 677 does not imply Equation 255. -/
@[category research solved, AMS 8]
theorem Equation677_not_implies_Equation255 :
∃ (G : Type) (_ : Magma G), Equation677 G ∧ ¬ Equation255 G
```

FC-EquationalTheories_677_255.Equation255_not_implies_Equation677

V2-000451 | Equational Theories | Equation677_not_implies_Equation255

research solved | proof annotation: False | 官方代码 | 原文线索

Equation 677 does not imply Equation 255.

```
theorem Equation677_not_implies_Equation255 :
∃ (G : Type) (_ : Magma G), Equation677 G ∧ ¬ Equation255 G
```

FC-EquationalTheories_677_255.Equation677_not_implies_Equation255

The Gerstenhaber problem for three commuting matrices

MODULE CONTEXT: # The Gerstenhaber problem for three commuting matrices Gerstenhaber proved in 1961 [Ger61] that the unital algebra generated by two commuting $n \times n$ matrices over a field has dimension at most n. It is an open problem, known as the **Gerstenhaber problem**, whether the same bound holds for three pairwise commuting matrices. The bound fails for four or more pairwise commuting matrices. *References:* - [Ger61] M. Gerstenhaber, *On dominance and varieties of commuting matrices*. Annals of Mathematics (2) 73 (1961), no. 2, 324-348. https://doi.org/10.2307/1970336 - [arXiv:2402.16334](https://arxiv.org/abs/2402.16334), M. Satriano and W. Zhang, *On the algebra generated by three commuting matrices: combinatorial cases*. - [arXiv:2006.08588](https://arxiv.org/abs/2006.08588), J. Holbrook and M. Omladič, *A computing strategy and programs to resolve the Gerstenhaber Problem for commuting triples of matrices*.

V2-000452 | The Gerstenhaber problem for three commuting matrices | exists_finrank_adjoin_quadruple_gt

research solved | proof annotation: False | 官方代码 | 原文线索

The analogue of Gerstenhaber's theorem fails for four pairwise commuting matrices: over any field there are four pairwise commuting 4×4 matrices generating a unital algebra of dimension greater than 4. The standard example is $e_{13}, e_{14}, e_{23}, e_{24}$, whose pairwise products all vanish, so the algebra they generate has dimension 5.

```
theorem exists_finrank_adjoin_quadruple_gt (K : Type*) [Field K] :
∃ A B C D : Matrix (Fin 4) (Fin 4) K,
({A, B, C, D} : Set (Matrix (Fin 4) (Fin 4) K)).Pairwise Commute ∧
4 < Module.finrank K (Algebra.adjoin K {A, B, C, D})
```

FC-Gerstenhaber.exists_finrank_adjoin_quadruple_gt

V2-000453 | The Gerstenhaber problem for three commuting matrices | finrank_adjoin_pair_le

research solved | proof annotation: False | 官方代码 | 原文线索

Gerstenhaber's theorem [Ger61]: if A and B are commuting $n \times n$ matrices over a field K, then the unital K-algebra $K[A, B]$ they generate has dimension at most n.

```
theorem finrank_adjoin_pair_le (A B : Matrix (Fin n) (Fin n) K) (hAB : Commute A B) :
Module.finrank K (Algebra.adjoin K {A, B}) ≤ n
```

FC-Gerstenhaber.finrank_adjoin_pair_le

V2-000454 | The Gerstenhaber problem for three commuting matrices | finrank_adjoin_triple_le

research open | proof annotation: False | 官方代码 | 原文线索

The Gerstenhaber problem: if A, B, and C are pairwise commuting $n \times n$ matrices over a field K, is the dimension of the unital K-algebra $K[A, B, C]$ they generate always at most n?

theorem finrank_adjoin_triple_le :
∀ (K : Type*) [Field K] (n : ℕ) (A B C : Matrix (Fin n) (Fin n) K),
({A, B, C} : Set (Matrix (Fin n) (Fin n) K)).Pairwise Commute →
Module.finrank K (Algebra.adjoin K {A, B, C}) ≤ n

FC-Gerstenhaber.finrank_adjoin_triple_le

The Rule 30 Prize Problems

MODULE CONTEXT: # The Rule 30 Prize Problems **Rule 30** is the elementary cellular automaton with local update $c' = l \oplus (c \lor r)$, the Boolean rule numbered 30 by Wolfram. Started from a single black cell on a bi-infinite row, its **center column** `t ↦ (state t) 0` looks random - it was long *Mathematica*'s default pseudorandom generator - yet nothing about that randomness is proven. In 2019 Wolfram offered the **Rule 30 Prizes** for three questions about it; we formalize the first two. * **Problem 1.** Is the center column non-periodic (never eventually periodic)? * **Problem 2.** Does each color occur on average equally often, i.e. does the running average of the values converge to $1/2$? Problem 3 - whether computing the n-th cell requires at least $O(n)$ work - is omitted. It is model-relative and has no canonical model-independent phrasing. All three are open. *References:* - [Announcing the Rule 30 Prizes](https://writings.stephenwolfram.com/2019/10/announcing-the-rule-30-prizes/), Stephen Wolfram, 2019. - [Rule 30 Prizes](https://rule30prize.org/). - [Wikipedia: Rule 30](https://en.wikipedia.org/wiki/Rule_30).

V2-000455 | The Rule 30 Prize Problems | centerColumn_frequency_half

research open | proof annotation: False | 官方代码 | 原文线索

Rule 30 Prize, Problem 2 (equal frequency). Each color occurs on average equally often in the center column: the set of times at which it is black has natural density $1/2$. This is Wolfram's phrasing that the discrete limit of $\mathrm{Total}[c[t]]/t$ as $t \to \infty$ is $1/2$.

theorem centerColumn_frequency_half :
{t : ℕ | centerColumn t}.HasDensity (1 / 2)

FC-Rule30.centerColumn_frequency_half

V2-000456 | The Rule 30 Prize Problems | centerColumn_not_eventually_periodic

research open | proof annotation: False | 官方代码 | 原文线索

Rule 30 Prize, Problem 1 (non-periodicity). The center column of Rule 30 is not eventually periodic: there is no positive period p and threshold N past which the column repeats with period p.

theorem centerColumn_not_eventually_periodic :
¬ ∃ p : ℕ, 0 < p ∧ ∃ N : ℕ, ∀ t : ℕ, N ≤ t → centerColumn (t + p) = centerColumn t

FC-Rule30.centerColumn_not_eventually_periodic

Schur's theorem on Galois groups of truncated exponential polynomials

MODULE CONTEXT: # Schur's theorem on Galois groups of truncated exponential polynomials *Reference:* (https://math.stackexchange.com/questions/2814220) *Reference* (https://mathoverflow.net/questions/477077)

V2-000457 | Schur's theorem on Galois groups of truncated exponential polynomials | schur_truncatedExp_galoisGroup_equiv

research solved | proof annotation: True | 官方代码 | 原文线索

Schur's Theorem (1924): Let `f_n(x) = ∑_{j=0}^n x^j/j!` be the `n`-th truncated exponential polynomial over `ℚ`. Then for `n ≥ 2`: - If `n ≡ 0 (mod 4)`, the Galois group of `f_n` is isomorphic to the alternating group `A_n` - If `n ≢ 0 (mod 4)`, the Galois group of `f_n` is isomorphic to the symmetric group `S_n`

theorem schur_truncatedExp_galoisGroup_equiv (n : ℕ) (hn : n ≥ 2) :
letI f := truncatedExp n
if n % 4 = 0 then
-- Galois group is alternating group A_n
Nonempty (f.Gal ≃* alternatingGroup (Fin n))
else
-- Galois group is symmetric group S_n
Nonempty (f.Gal ≃* Equiv.Perm (Fin n))

FC-SchurTruncatedExponential.schur_truncatedExp_galoisGroup_equiv

Smarandache-Wellin primes

MODULE CONTEXT: # Smarandache-Wellin primes The Smarandache-Wellin numbers are formed by concatenating the decimal expansions of the successive primes: $2, 23, 235, 2357, \ldots$. It is conjectured that infinitely many members of this sequence are prime. *References:* * [Smarandache Problems](https://fs.unm.edu/S-PROBL.HTM) * [OEIS A019518](https://oeis.org/A019518) * [OEIS A069151](https://oeis.org/A069151) * [OEIS A046035](https://oeis.org/A046035)

V2-000458 | Smarandache-Wellin primes | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

[Smarandache Problems](https://fs.unm.edu/S-PROBL.HTM) states: "There are infinitely many primes in the smarandache concatenated prime sequence."

theorem conjecture : {n : ℕ | (a n).Prime}.Infinite

FC-SmarandacheWellinPrime.conjecture

Suffix-prefix avoidance bound

MODULE CONTEXT: # Suffix-prefix avoidance bound Let A and B be sets of words of length n over an alphabet with q letters. If no (nonempty) suffix of any word in A coincides with a prefix of any word in B, then $$|A| \cdot |B| \leq \frac{q^{2n}}{en}.$$ *References:* - [X post by Dmitry Rybin](https://x.com/DmitryRybin1/status/2027278135847428577) - [Maximal sets of strings with no prefix-suffix overlap] (https://mathoverflow.net/questions/508648/maximal-sets-of-strings-with-no-prefix-suffix-overlap) by *Dmitry Rybin*, MathOverflow (2026) - [An isoperimetric inequality for word overlap](https://arxiv.org/abs/2602.20143) by *Dmitrii Zakharov* (2026)

V2-000459 | Suffix-prefix avoidance bound | suffix_prefix_avoidance_bound

research solved | proof annotation: False | 官方代码 | 原文线索

A and B are sets of words of length n over alphabet with $q \geq 1$ letters. No suffix of a word in A coincides with a prefix of a word in B. Then $|A| \cdot |B|$ is at most $\frac{q^{2n}}{en}$. This problem is from *Maximal sets of strings with no prefix-suffix overlap* and was proved in *An isoperimetric inequality for word overlap*.

```
theorem suffix_prefix_avoidance_bound
(A B : Finset (Fin n → Fin q))
(hq : 0 < q) (hn : 0 < n)
(h : IsSuffixPrefixAvoiding A B) :
(A.card : ℝ) * B.card ≤ (q : ℝ) ^ (2 * n) / (exp 1 * n)
```

FC-SuffixPrefixAvoidance.suffix_prefix_avoidance_bound

V2-000460 | Suffix-prefix avoidance bound | suffix_prefix_avoidance_weaker_bound

research solved | proof annotation: True | 官方代码 | 原文线索

A and B are sets of words of length n over alphabet with $q \geq 1$ letters. No suffix of a word in A coincides with a prefix of a word in B. Then $|A| \cdot |B|$ is at most $\frac{q^{2n}}{n}$.

```
theorem suffix_prefix_avoidance_weaker_bound
(A B : Finset (Fin n → Fin q))
(_hq : 0 < q) (hn : 0 < n)
(h : IsSuffixPrefixAvoiding A B) :
(A.card : ℚ) * B.card ≤ (q : ℚ) ^ (2 * n) / n
```

FC-SuffixPrefixAvoidance.suffix_prefix_avoidance_weaker_bound

The symbol length of $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p$

MODULE CONTEXT: # The symbol length of $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p$ For a field F and $\ell \ge 1$, the Milnor K-group $K^M_n(F)/\ell$ is generated by the symbols $\{a_1, \dots, a_n\}$ with $a_i \in F^\times$, and the *symbol length* of a class is the least number of symbols in a decomposition of the class as a sum of symbols. The symbol length of $K^M_n(F)/\ell$ is the supremum of the symbol lengths of its classes. When ℓ is invertible in F, the norm residue isomorphism theorem (the Bloch-Kato conjecture, [Voevodsky2011]) gives $K^M_n(F)/\ell \cong H^n(F, \mu_\ell^{\otimes n})$, so this is also the symbol length of Galois cohomology. The **symbol length problem** asks, for a given field F and integers n and ℓ, for a k such that the symbol length of every class in $K^M_n(F)/\ell$ is at most k, for example in terms of a dimension of the field. That a class is a sum of *some* number of symbols is part of the presentation of $K^M_n(F)$ recalled above, so the content lies entirely in the bound. In degree 2, over a field containing a primitive ℓ-th root of unity, Merkurjev and Suslin's theorem [MerkurjevSuslin1983] that the norm residue map is an isomorphism identifies $K^M_2(F)/\ell$ with the ℓ-torsion of the Brauer group and symbols with symbol algebras of degree ℓ; that is the form in which the question is usually studied, namely how many symbol algebras are needed to represent a central simple algebra of exponent ℓ. Becher and Hoffmann [BecherHoffmann2004] named the invariant λ_p in that degree and asked it for exactly the fields considered below (their Question 4.6). The general form stated here is Problem 2.1.3.12 of Krashen's PCMI notes; Krashen notes there that over complex function fields in at least three variables no bound is known beyond Matzri's in degree 2 and the case of ℓ a power of 2, which comes from the theory of quadratic forms, and that very little is known in degree 3 and higher [Krashen2024, §2.1.3.4]. This file states the problem for the rational function fields $\mathbb{C}(x_1, \dots, x_m)$ and a prime $\ell = p$: determine the symbol length of $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p$ as a function of m, n and p, where the symbol length is infinite if there is no bound. What is known: * in degree 0 every symbol is the empty product 1, so $K^M_0(F)/p = \mathbb{Z}/p$ has symbol length $p - 1$ for every field F; * in degree 1 every class of $K^M_1(F)/p = F^\times/(F^\times)^p$ is a single symbol, so the symbol length is at most 1; * $\mathbb{C}(x_1, \dots, x_m)$ has cohomological dimension m [Serre1997, Ch. II, §4.2], so $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p = 0$ for $n > m$ and the symbol length is 0; * $\mathbb{C}(x_1, \dots, x_m)$ is a C_m field (Tsen-Lang, [Lang1952]), so by Matzri's theorem [Matzri2016, Theorem 8.2] every class in $K^M_2(\mathbb{C}(x_1, \dots, x_m))/p^t$ is a sum of at most $t(p^{m-1} - 1)$ symbols when $m \ge 1$; * in degree 2 a tensor product of $m - 1$ symbol algebras of degree p over $\mathbb{C}(x_1, \dots, x_m)$ can be a division algebra, so the symbol length is at least $m - 1$ for $m \ge 1$ [BecherHoffmann2004, Proposition 4.5], by an argument going back to Nakayama. In degree 2, Becher and Hoffmann ask whether that lower bound is the answer, that is whether the symbol length is exactly $m - 1$ [BecherHoffmann2004, Question 4.6]. For $m = n = 2$, the smallest case not settled above, this asks whether every class of $K^M_2(\mathbb{C}(x_1, x_2))/p$ is a single symbol; they call it "a striking open question" and note that it holds for $p \le 3$ by a theorem of Artin [BecherHoffmann2004, Introduction]. Since $m - 1$ does not involve p, their question is an instance of the expectation recorded in [Krashen2024, §2.1.3.4] that a bound can be taken independent of ℓ. See also the survey [Krashen2016]. The definitions `MilnorK`, `MilnorK.symbol`, `MilnorK.grade` and `MilnorK.symbolLengthBounds` are in `FormalConjecturesForMathlib/FieldTheory/MilnorKTheory.lean`. *References:* - [Krashen2024] D. Krashen, *PCMI Summer School (not quite) notes*, notes for the minicourse *Field arithmetic and the complexity of algebraic objects*, PCMI Graduate Summer School on Motivic Homotopy Theory, 2024, draft of 24 July 2024,

[pdf](https://www.ias.edu/sites/default/files/pcmi-2024-07-24%20krashen.pdf). - [BecherHoffmann2004] K. J. Becher and D. W. Hoffmann, *Symbol lengths in Milnor K-theory*, Homology Homotopy Appl. 6 (2004), no. 1, 17-31, [doi:10.4310/HHA.2004.v6.n1.a3](https://doi.org/10.4310/HHA.2004.v6.n1.a3). - [MerkurjevSuslin1983] A. S. Merkurjev and A. A. Suslin, *K-cohomology of Severi-Brauer varieties and the norm residue homomorphism*, Izv. Akad. Nauk SSSR Ser. Mat. 46 (1982), 1011-1046; English translation Math. USSR-Izv. 21 (1983), no. 2, 307-340, [doi:10.1070/IM1983v021n02ABEH001793](https://doi.org/10.1070/IM1983v021n02ABEH001793). - [Krashen2016] D. Krashen, *Period and index, symbol lengths, and generic splittings in Galois cohomology*, Bull. London Math. Soc. 48 (2016), 985-1000, [doi:10.1112/blms/bdw060](https://doi.org/10.1112/blms/bdw060). - [Matzri2016] E. Matzri, *Symbol length in the Brauer group of a field*, Trans. Amer. Math. Soc. 368 (2016), 413-427, [doi:10.1090/tran/6326](https://doi.org/10.1090/tran/6326), [arXiv:1402.0332](https://arxiv.org/abs/1402.0332). - [Lang1952] S. Lang, *On quasi algebraic closure*, Ann. of Math. (2) 55 (1952), 373-390, [doi:10.2307/1969785](https://doi.org/10.2307/1969785). - [Serre1997] J.-P. Serre, *Galois cohomology*, Springer Monographs in Mathematics, Springer, 1997, [doi:10.1007/978-3-642-59141-9](https://doi.org/10.1007/978-3-642-59141-9). - [Voevodsky2011] V. Voevodsky, *On motivic cohomology with Z/l-coefficients*, Ann. of Math. (2) 174 (2011), 401-438, [doi:10.4007/annals.2011.174.1.11](https://doi.org/10.4007/annals.2011.174.1.11).

V2-000461 | The symbol length of $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p$ | symbol_length_problem

research open | proof annotation: False | 官方代码 | 原文线索

The symbol length problem for complex rational function fields [Krashen2024, Problem 2.1.3.12 and §2.1.3.4]: determine, as a function of m, n and the prime p, the symbol length of $K^M_n(\mathbb{C}(x_1, \dots, x_m))/p$, that is the least k such that every class is a sum of at most k symbols, or ∞ if there is no such k. The answer `sl m n p : ℕ∞` is characterised by: `k` is such a bound if and only if `sl m n p ≤ k`.

```
theorem symbol_length_problem :
∀ m n p : ℕ, p.Prime → ∀ k : ℕ,
k ∈ symbolLengthBounds (MvRatFunc (Fin m) ℂ) p n ↔ symbol_length_problem.answer m n p ≤ k
```

FC-SymbolLengthMilnorKTheory.symbol_length_problem

VCₙ dimension of convex sets in $\mathbb{R}^n$, $\mathbb{R}^{n+1}$, $\mathbb{R}^{n+2}$

MODULE CONTEXT: # VCₙ dimension of convex sets in $\mathbb{R}^n$, $\mathbb{R}^{n+1}$, $\mathbb{R}^{n+2}$ In the literature it is known that every convex set in $\mathbb{R}^2$ has VC dimension at most 3, and there exists a convex set in $\mathbb{R}^3$ with infinite VC dimension (even more strongly, which shatters an infinite set). This file states that every convex set in $\mathbb{R}^n$ has finite VCₙ dimension, constructs a convex set in $\mathbb{R}^{n+2}$ with infinite VCₙ dimension (even more strongly, which n-shatters an infinite set), and conjectures that every convex set in $\mathbb{R}^{n+1}$ has finite VCₙ dimension.

V2-000462 | VCₙ dimension of convex sets in $\mathbb{R}^n$, $\mathbb{R}^{n+1}$, $\mathbb{R}^{n+2}$ | exists_convex_rn_add_two_vc_n_forall_not_hasAddVCNDimAtMost

research solved | proof annotation: False | 官方代码

There exists a set of infinite VC_n dimension in $\mathbb R^{n + 2}$.

```
lemma exists_convex_rn_add_two_vc_n_forall_not_hasAddVCNDimAtMost (n : ℕ) :
∃ C : Set (Fin (n + 2) → ℝ), Convex ℝ C ∧ ∀ d, ¬ HasAddVCNDimAtMost C n d := sorry

/- ### Conjectures -/

/-- Not every convex set in $\mathbb R^3$ has
$\mathrm{VC}_2$ dimension at most 1. -/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-counterexample/blob/ad7ffff1514843c886633f5408c6456dfb2e2a49/formal-conjectures-v4.27.0/VCDimConvexCounterexample.lean"]
lemma hasAddVCNDimAtMost_two_one_of_convex_r3 :
¬ ∀ {C : Set ℝ³} (hC : Convex ℝ C),
HasAddVCNDimAtMost C 2 1 := sorry

/-- Every convex set in $\mathbb R^3$ has $\mathrm{VC}_2$ dimension at most 2. -/
@[category research open, AMS 5 52]
lemma hasAddVCNDimAtMost_two_two_of_convex_r3 {C : Set ℝ³} (hC : Convex ℝ C) :
HasAddVCNDimAtMost C 2 2 := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see
[Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).
-/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-finite-bound/blob/9d7685c7c7c8c8e82f47d59da5495a8d5374db99/lean/VCDimConvexBoundFC.lean#L14-L17"]
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of $n$.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
```

```
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all $n$, every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3
```

FC-VCDimConvex.exists_convex_rn_add_two_vc_n_forall_not_hasAddVCNDimAtMost

V2-000463 | VCₙ dimension of convex sets in ℝⁿ, ℝⁿ⁺¹, ℝⁿ⁺² | exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one

research open | proof annotation: False | 官方代码

For every $n \ge 1$ there exists some d such that every convex set in $\mathbb R^{n + 1}$ has VC_n dimension at most d. The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of n.

```
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all $n$, every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3
```

FC-VCDimConvex.exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one

V2-000464 | VCₙ dimension of convex sets in ℝⁿ, ℝⁿ⁺¹, ℝⁿ⁺² | exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one

research solved | proof annotation: True | 官方代码 | 原文线索

For every $n \ge 1$ there exists some d such that every convex set in $\mathbb R^{n + 1}$ has VC_n dimension at most d. This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see [Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).

```
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of $n$.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all $n$, every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3
```

FC-VCDimConvex.exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one

V2-000465 | VCₙ dimension of convex sets in ℝⁿ, ℝⁿ⁺¹, ℝⁿ⁺² | exists_infinite_convex_r3_shatters

research solved | proof annotation: False | 官方代码

There exists a set in $\mathbb R^3$ shattering an infinite set.

```
lemma exists_infinite_convex_r3_shatters :
∃ A C : Set ℝ³, A.Infinite ∧ Convex ℝ C ∧ Shatters {t +ᵥ C | t : ℝ³} A := sorry

/- ### What's not in the literature -/

/-- There exists a set of infinite $\mathrm{VC}_n$ dimension in $\mathbb R^{n + 2}$. -/
@[category research solved, AMS 5 52]
lemma exists_convex_rn_add_two_vc_n_forall_not_hasAddVCNDimAtMost (n : ℕ) :
∃ C : Set (Fin (n + 2) → ℝ), Convex ℝ C ∧ ∀ d, ¬ HasAddVCNDimAtMost C n d := sorry

/- ### Conjectures -/

/-- Not every convex set in $\mathbb R^3$ has
$\mathrm{VC}_2$ dimension at most 1. -/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-counterexample/blob/ad7ffff1514843c886633f5408c
6456dfb2e2a49/formal-conjectures-v4.27.0/VCDimConvexCounterexample.lean"]
lemma hasAddVCNDimAtMost_two_one_of_convex_r3 :
```

```
¬ ∀ {C : Set ℝ³} (hC : Convex ℝ C),
HasAddVCNDimAtMost C 2 1 := sorry

/-- Every convex set in $\mathbb R^3$ has $\mathrm{VC}_2$ dimension at most 2. -/
@[category research open, AMS 5 52]
lemma hasAddVCNDimAtMost_two_two_of_convex_r3 {C : Set ℝ³} (hC : Convex ℝ C) :
HasAddVCNDimAtMost C 2 2 := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see
[Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).
-/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-finite-bound/blob/9d7685c7c7c8c8e82f47d59da549
5a8d5374db99/lean/VCDimConvexBoundFC.lean#L14-L17"]
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of $n$.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all $n$, every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3
```

FC-VCDimConvex.exists_infinite_convex_r3_shatters

V2-000466 | VC_n dimension of convex sets in $\mathbb{R}^n$, $\mathbb{R}^{n+1}$, $\mathbb{R}^{n+2}$ | hasAddVCDimAtMost_three_of_convex_r2

research solved | proof annotation: False | 官方代码

Every convex set in $\mathbb R^2$ has VC dimension at most 3.

```
lemma hasAddVCDimAtMost_three_of_convex_r2 {C : Set ℝ²} (hC : Convex ℝ C) :
HasAddVCDimAtMost C 3 :=
sorry

/-- There exists a set in $\mathbb R^3$ shattering an infinite set. -/
@[category research solved, AMS 5 52]
lemma exists_infinite_convex_r3_shatters :
∃ A C : Set ℝ³, A.Infinite ∧ Convex ℝ C ∧ Shatters {t +ᵥ C | t : ℝ³} A := sorry

/- ### What's not in the literature -/

/-- There exists a set of infinite $\mathrm{VC}_n$ dimension in $\mathbb R^{n + 2}$. -/
@[category research solved, AMS 5 52]
lemma exists_convex_rn_add_two_vc_n_forall_not_hasAddVCNDimAtMost (n : ℕ) :
∃ C : Set (Fin (n + 2) → ℝ), Convex ℝ C ∧ ∀ d, ¬ HasAddVCNDimAtMost C n d := sorry

/- ### Conjectures -/

/-- Not every convex set in $\mathbb R^3$ has
$\mathrm{VC}_2$ dimension at most 1. -/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-counterexample/blob/ad7ffff1514843c886633f5408c
6456dfb2e2a49/formal-conjectures-v4.27.0/VCDimConvexCounterexample.lean"]
lemma hasAddVCNDimAtMost_two_one_of_convex_r3 :
¬ ∀ {C : Set ℝ³} (hC : Convex ℝ C),
HasAddVCNDimAtMost C 2 1 := sorry

/-- Every convex set in $\mathbb R^3$ has $\mathrm{VC}_2$ dimension at most 2. -/
@[category research open, AMS 5 52]
lemma hasAddVCNDimAtMost_two_two_of_convex_r3 {C : Set ℝ³} (hC : Convex ℝ C) :
HasAddVCNDimAtMost C 2 2 := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see
[Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).
-/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
```

"https://github.com/KitaKen1/vcdim-convex-finite-bound/blob/9d7685c7c7c8c8e82f47d59da5495a8d5374db99/lean/VCDimConvexBoundFC.lean#L14-L17"]
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some d such that every convex set in $\mathbb R^{n + 1}$ has
VC_n dimension at most d.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of n.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all n, every convex set in $\mathbb R^{n + 1}$ has
VC_n dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3

FC-VCDimConvex.hasAddVCNDimAtMost_three_of_convex_r2

V2-000467 | VCₙ dimension of convex sets in ℝⁿ, ℝⁿ⁺¹, ℝⁿ⁺² | hasAddVCNDimAtMost_n_two_of_convex_rn_add_one

research open | proof annotation: False | 官方代码

Is it true that, for all n, every convex set in $\mathbb R^{n + 1}$ has VC_n dimension at most 3?

lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3

FC-VCDimConvex.hasAddVCNDimAtMost_n_two_of_convex_rn_add_one

V2-000468 | VCₙ dimension of convex sets in ℝⁿ, ℝⁿ⁺¹, ℝⁿ⁺² | hasAddVCNDimAtMost_two_one_of_convex_r3

research solved | proof annotation: True | 官方代码

Not every convex set in $\mathbb R^3$ has VC_2 dimension at most 1.

lemma hasAddVCNDimAtMost_two_one_of_convex_r3 :
¬ ∀ {C : Set ℝ³} (hC : Convex ℝ C),
HasAddVCNDimAtMost C 2 1 := sorry

/-- Every convex set in $\mathbb R^3$ has VC_2 dimension at most 2. -/
@[category research open, AMS 5 52]
lemma hasAddVCNDimAtMost_two_two_of_convex_r3 {C : Set ℝ³} (hC : Convex ℝ C) :
HasAddVCNDimAtMost C 2 2 := sorry

/-- For every $n \ge 1$ there exists some d such that every convex set in $\mathbb R^{n + 1}$ has
VC_n dimension at most d.

This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see
[Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).
-/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-finite-bound/blob/9d7685c7c7c8c8e82f47d59da5495a8d5374db99/lean/VCDimConvexBoundFC.lean#L14-L17"]
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some d such that every convex set in $\mathbb R^{n + 1}$ has
VC_n dimension at most d.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of n.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all n, every convex set in $\mathbb R^{n + 1}$ has
VC_n dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ ⦃n : ℕ⦄, n ≠ 0 → ∀ ⦃C : Set (EuclideanSpace ℝ (Fin (n + 1)))⦄,
Convex ℝ C → HasAddVCNDimAtMost C n 3

FC-VCDimConvex.hasAddVCNDimAtMost_two_one_of_convex_r3

V2-000469 | VC_n dimension of convex sets in $\mathbb{R}^n$, $\mathbb{R}^{n+1}$, $\mathbb{R}^{n+2}$ | hasAddVCNDimAtMost_two_two_of_convex_r3

research open | proof annotation: False | 官方代码

Every convex set in $\mathbb R^3$ has VC_2 dimension at most 2.

```
lemma hasAddVCNDimAtMost_two_two_of_convex_r3 {C : Set ℝ³} (hC : Convex ℝ C) :
HasAddVCNDimAtMost C 2 2 := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

This holds with the explicit bound $d = 2^{8(n + 2)^n} - 1$; see
[Kitamura's Lean formalization](https://github.com/KitaKen1/vcdim-convex-finite-bound).
-/
@[category research solved, AMS 5 52,
formal_proof using lean4 at
"https://github.com/KitaKen1/vcdim-convex-finite-bound/blob/9d7685c7c7c8c8e82f47d59da549
5a8d5374db99/lean/VCDimConvexBoundFC.lean#L14-L17"]
lemma exists_hasAddVCNDimAtMost_n_of_convex_rn_add_one (n : ℕ) (hn : 1 ≤ n) :
∃ d : ℕ, ∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n d := sorry

/-- For every $n \ge 1$ there exists some $d$ such that every convex set in $\mathbb R^{n +
1}$ has
$\mathrm{VC}_n$ dimension at most $d$.

The bound can be taken to be $d = c^n$ for some constant $c \ge 2$ independent of $n$.
-/
@[category research open, AMS 5 52]
lemma exists_exponential_bound_hasAddVCNDimAtMost_n_of_convex_rn_add_one :
∃ c : ℕ, 2 ≤ c ∧ ∀ (n : ℕ) (hn : 1 ≤ n),
∀ C : Set (Fin (n + 1) → ℝ), Convex ℝ C → HasAddVCNDimAtMost C n (c ^ n) := sorry

/-- Is it true that, for all $n$, every convex set in $\mathbb R^{n + 1}$ has
$\mathrm{VC}_n$ dimension at most 3? -/
@[category research open, question, AMS 5 52]
lemma hasAddVCNDimAtMost_n_two_of_convex_rn_add_one :
∀ {{n : ℕ}}, n ≠ 0 → ∀ {{C : Set (EuclideanSpace ℝ (Fin (n + 1)))}},
Convex ℝ C → HasAddVCNDimAtMost C n 3
```

FC-VCDimConvex.hasAddVCNDimAtMost_two_two_of_convex_r3

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The Auslander-Reiten conjecture

MODULE CONTEXT: # The Auslander-Reiten conjecture **Auslander-Reiten conjecture.** Let Λ be an Artin algebra and let M be a finitely generated Λ-module with $\operatorname{Ext}^i_\Lambda(M, M) = 0$ and $\operatorname{Ext}^i_\Lambda(M, \Lambda) = 0$ for all $i > 0$. Then M is projective. An *Artin algebra* is a ring Λ that is an algebra over a commutative Artinian ring A and is finitely generated as an A-module. Auslander and Reiten [AR75] derived this from the generalized Nakayama conjecture. The conjecture for Artin algebras is false: an internal OpenAI model gave an explicit counterexample in September 2026 [OAI26], a finite-dimensional algebra over the field $\mathbb{F}_2(q, H_1, H_2)$ with a finite-dimensional non-projective module satisfying both vanishing conditions. The commutative version, obtained by replacing Λ with a commutative Noetherian ring R, is open and is what the commutative algebra literature calls the Auslander-Reiten conjecture. The two hypotheses are together equivalent to $\operatorname{Ext}^i_R(M, R \oplus M) = 0$ for all $i > 0$. Known cases of the commutative version include: R a locally excellent Cohen-Macaulay normal ring containing $\mathbb Q$ [HL04]; R a Gorenstein normal ring [Ar09]; R a Cohen-Macaulay normal ring [KOT22]; and, most generally, R any normal ring [Ki23]. Auslander, Ding and Solberg [ADS93] proved it for local complete intersections. The commutative version is open in general, already for commutative Noetherian local rings. *References:* - [AR75] M. Auslander, I. Reiten, *On a generalized version of the Nakayama conjecture*, Proc. Amer. Math. Soc. 52 (1975), 69-74. [PDF](https://www.ams.org/journals/proc/1975-052-01/S0002-9939-1975-0389977-6/S0002-9939-1975-0389977-6.pdf) - [Wikipedia, Nakayama conjecture](https://en.wikipedia.org/wiki/Nakayama_conjecture), on the generalized Nakayama conjecture of [AR75]. - [ADS93] M. Auslander, S. Ding, Ø. Solberg, *Liftings and weak liftings of modules*, J. Algebra 156 (1993), 273-317. - [HL04] C. Huneke, G. J. Leuschke, *On a conjecture of Auslander and Reiten*, J. Algebra 275 (2004), no. 2, 781-790. [arXiv:math/0305001](https://arxiv.org/abs/math/0305001) - [Ar09] T. Araya, *The Auslander-Reiten conjecture for Gorenstein rings*, Proc. Amer. Math. Soc. 137 (2009), 1941-1944. - [KOT22] K. Kimura, Y. Otake, R. Takahashi, *Maximal Cohen-Macaulay tensor products and vanishing of Ext modules*, Bull. Lond. Math. Soc. 54 (2022), no. 6, 2456-2468. - [Ki23] K. Kimura, *Auslander-Reiten conjecture for normal rings*, [arXiv:2304.03956](https://arxiv.org/abs/2304.03956). The statements of the conjecture formalised below follow this paper. - [OAI26] OpenAI, *An explicit counterexample to the Auslander-Reiten conjecture*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/An-explicit-counterexample-to-the-Auslander-Reiten-conjecture-September-23-2026/paper.pdf

V2-000470 | The Auslander-Reiten conjecture | auslander_reiten

research solved | proof annotation: True | 官方代码 | 原文线索

The Auslander-Reiten conjecture [AR75] is false. The conjecture states: let Λ be an Artin algebra and M a finitely generated Λ-module with $\operatorname{Ext}^i_\Lambda(M, \Lambda) = 0$ and $\operatorname{Ext}^i_\Lambda(M, M) = 0$ for all $i > 0$. Then M is projective. That Λ is an Artin algebra is the hypothesis that it is an algebra over some commutative Artinian ring `A` and is finitely generated as an `A`-module. This was disproved by an internal OpenAI model in September 2026 [OAI26]; see the linked Lean proof. The counterexample is a finite-dimensional algebra over the field $\mathbb{F}_2(q, H_1, H_2)$, which is the ring `A` there.

```
theorem auslander_reiten :
¬ ∀ (Λ : Type) [Ring Λ] (M : Type) [AddCommGroup M] [Module Λ M] [Module.Finite Λ M]
(A : Type) [CommRing A] [IsArtinianRing A] [Algebra A Λ] [Module.Finite A Λ],
(∀ i > 0, Subsingleton (Abelian.Ext (ModuleCat.of Λ M) (ModuleCat.of Λ Λ) i)) →
(∀ i > 0, Subsingleton (Abelian.Ext (ModuleCat.of Λ M) (ModuleCat.of Λ M) i)) →
Module.Projective Λ M
```

FC-AuslanderReiten.auslander_reiten

Decidability of reachability for branching vector addition systems

MODULE CONTEXT: # Decidability of reachability for branching vector addition systems A *branching vector addition system* (BVAS) of dimension `d` is given by a finite list of *axioms*, a finite list of *unary rules*, and a finite list of *binary rules*, each of which is a vector in `ℤ^d`. A *configuration* is a vector in `ℕ^d`, which here is represented as a vector `v ∈ ℤ^d` subject to `0 ≤ v`. The set of *reachable* configurations is defined inductively: * every axiom that is a valid configuration (i.e. lies in `ℕ^d`) is reachable; * if `v` is a reachable configuration, `r` is a unary rule and `v + r ∈ ℕ^d`, then `v + r` is reachable; * if `v₁, v₂` are both reachable configurations, `r` is a binary rule and `v₁ + v₂ + r ∈ ℕ^d`, then `v₁ + v₂ + r` is reachable. Modelling axioms as vectors in `ℤ^d` and only counting the non-negative ones as reachable yields the same set of reachable configurations as the usual definition in which axioms are required to lie in `ℕ^d`. Branching vector addition systems are distinguished from ordinary vector addition systems (VAS) by allowing binary rules. VAS reachability is known to be decidable. *References:* - [The covering and boundedness problems for branching vector addition systems](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSTTCS.2009.2317) (*Stéphane Demri, Marcin Jurdziński, Oded Lachish, Ranko Lazić*, FSTTCS 2009) Gives a similar definition of BVAS, and compares it to related equivalent definitions. - [On the Reachability Problem for Two-Dimensional Branching VASS](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2025.22) by *Clotilde Bizière, Thibault Hilaire, Jérôme Leroux, Grégoire Sutre*, MFCS 2025, which settles the two-dimensional case and states that "the decidability status of the reachability problem for BVASS remains open in higher dimensions". - [The General Vector Addition System Reachability Problem by Presburger Inductive Invariants](https://arxiv.org/abs/1009.1076) by *Jérôme Leroux* (2010), for the decidability of reachability for ordinary VAS. - [Solving the Reachability Problem for Branching Vector Addition Systems via Semilinear Inductive Invariants](https://arxiv.org/abs/2607.09558) by *Clotilde Bizière, Jérôme Leroux, Grégoire Sutre* (2026), a recent preprint claiming a positive resolution to the conjecture: reachability for branching vector addition systems is decidable.

V2-000471 | Decidability of reachability for branching vector addition systems | reachability_decidable

research solved | proof annotation: False | 官方代码 | 原文线索

The reachability problem for branching vector addition systems is decidable. That is, the predicate taking a dimension `d`, a branching vector addition system `b` of dimension `d` and a target vector `t ∈ ℤ^d`, and returning whether `t` is reachable in `b`, is a computable predicate. Note that the dimension is part of the input: a single algorithm must work for all dimensions. As of August 2026, the solution is quite recently announced and is not yet peer-reviewed.

```
theorem reachability_decidable :
ComputablePred fun p : Σ d : ℕ, Bvas d × (Fin d → ℤ) => p.2.1.Reachable p.2.2
```

FC-BranchingVAS.reachability_decidable

The burning number conjecture

MODULE CONTEXT: # The burning number conjecture *References:* * [BJR16] Bonato, A., Janssen, J., and Roshanbin, E. (2016). "How to burn a graph." *Internet Math.* 12, pp. 85--100. [arXiv:1507.06524](https://arxiv.org/abs/1507.06524) * [NT24] Norin, S. and Turcotte, J. (2024). "The burning number conjecture holds asymptotically." *J. Combin. Theory Ser. B* 168, pp. 208--235. [arXiv:2207.04035](https://arxiv.org/abs/2207.04035)

V2-000472 | The burning number conjecture | burning_number_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The burning number conjecture (Bonato-Janssen-Roshanbin [BJR16]). Every connected graph on `n` vertices can be burned in at most `⌈√n⌉` rounds: fires started at one new vertex per round, each spreading to all neighbouring vertices every round, can burn the whole graph in `⌈√n⌉` rounds.

```
theorem burning_number_conjecture {V : Type} [Fintype V] (G : SimpleGraph V)
(hc : G.Connected) :
burningNumber G ≤ ⌈Real.sqrt (Fintype.card V)⌉₊
```

FC-BurningNumberConjecture.burning_number_conjecture

The Catch-Up game and conjecture

MODULE CONTEXT: # The Catch-Up game and conjecture The game **Catch-Up** (Isaksen-Ismail-Brams-Nealen, 2015) is a two-player, perfect-information game played on a finite nonempty set `S` of positive integers. Each time a player removes a number from `S`, that number is added to the player's score. **Rules.** * The scores start at `0`. Player `p1` starts by removing **exactly one** number from `S`. * After the first move, players alternate turns. On a turn, the current player removes **one or more** numbers from `S`, one at a time, and must keep removing numbers until their score becomes **at least** the opponent's score; before the final pick they must remain **strictly behind**. * If the current player cannot catch up (in particular, even taking all remaining numbers would still leave them behind), the game ends immediately: the current player receives all remaining numbers. When `S` is empty, the player with higher score wins; equal scores give a draw. In this file we define: * `Player` and `Outcome`, * the recursive evaluator `value` (optimal play), * the conjecture `value_of_even_mul_succ_self_div_two`. ## Example For `S = {1,2,3,4}` one play is: `p1` takes `2`, `p2` takes `1` then `4`, and `p1` takes `3`, ending with scores `(5,5)`. ## References A. Isaksen, M. Ismail, S. J. Brams, A. Nealen, *Catch-Up: A Game in Which the Lead Alternates,* Game & Puzzle Design 1(2), 38-49 (2015).

V2-000473 | The Catch-Up game and conjecture | value_of_even_mul_succ_self_div_two

research open | proof annotation: False | 官方代码

Let $T_N = \sum_{k=1}^{N} k = \frac{N(N+1)}{2}$. If T_N is even (equivalently $N \equiv 0 \pmod 4$ or $N \equiv 3 \pmod 4$), then under optimal play the game `Catch-Up($\{1, \ldots, N\}$)` ends in a draw.

theorem value_of_even_mul_succ_self_div_two
(N : ℕ) (h_even : Even (N * (N + 1) / 2)) :
value (.Icc 1 N) = .draw

FC-CatchUp.value_of_even_mul_succ_self_div_two

Chvátal's Conjecture

MODULE CONTEXT: # Chvátal's Conjecture *References:* * [A Conjecture in Extremal Combinatorics](https://users.encs.concordia.ca/~chvatal/conjecture.html) * [Chvátal's Conjecture and Correlation Inequalities](https://arxiv.org/abs/1608.08954) * [A proof of Chvátal's conjecture via a sharp correlation inequality](https://arxiv.org/abs/2609.19123)

V2-000474 | Chvátal's Conjecture | exists_maximal_star

research solved | proof annotation: True | 官方代码 | 原文线索

If F is a decreasing family of sets of some finite type α, then there is some element x of α such that the family consisting of all members of F containing x is an intersecting subfamily of F with maximal cardinality.

theorem exists_maximal_star :
∀ F : Finset (Finset α), Decreasing F →
∃ x : α, ∀ G, G ⊆ F → Intersecting G → G.card ≤ { A ∈ F | x ∈ A }.card

FC-Chvatal.exists_maximal_star

Claude's Cycles

MODULE CONTEXT: # Claude's Cycles *Reference:* [Claude's Cycles](https://www-cs-faculty.stanford.edu/~knuth/papers/claude-cycles.pdf) by *Donald E. Knuth* (2026) Fix `m ≥ 2`. Consider the directed graph with vertex set `(ZMod m)³`, where from each vertex `(i, j, k)` there are directed arcs to `(i+1, j, k)`, `(i, j+1, k)`, and `(i, j, k+1)` (arithmetic mod `m`). The goal is to partition all `3m³` directed arcs into three edge-disjoint directed Hamiltonian cycles (each of length `m³`). Knuth describes an explicit construction, found by Claude (Anthropic), that achieves this decomposition for all odd `m ≥ 3`. The case `m = 2` is known to be impossible [Aub82]. The even case `m > 2` is also settled. Knuth's paper says so in its final section, added in the 14 April 2026 revision: "Breaking news: The problem for even values of m is no longer in doubt!" Ho Boon Suan's algorithm [Ho26] is proved correct for even `m ≥ 8` in [GPT26], and Aquino-Michaels [AM26] gives a decomposition for the even case that is simpler. The statement below starts at `m = 4`, which is below the range [GPT26] covers, so it also rests on the explicit solutions in [Kn26], whose header gives even `m ≥ 4`. ## References - [Knu26] D. E. Knuth, "Claude's Cycles" (2026). - [Aub82] J. Aubert, B. Schneider, "Graphes orientés indécomposables en circuits hamiltoniens", J. Combin. Theory Ser. B 32 (1982), 347-349. - [Ho26] Ho Boon Suan, closed-form construction for even `m`, <https://cs.stanford.edu/~knuth/even_closed_form.c> - [GPT26] A proof that [Ho26] yields three `m³`-cycles for every even `m ≥ 8`, 14 pages, <https://cs.stanford.edu/~knuth/even_closed_form_proof_final.pdf> - [AM26] K. Aquino-Michaels, "Completing Claude's cycles: Multi-agent structured exploration on an open combinatorial problem", <https://github.com/no-way-labs/residue> - [Kn26] Explicit even solutions for even `m ≥ 4`, recorded with [Knu26], <https://cs.stanford.edu/~knuth/even_solution.py> - [KM26] K. Morrison, a Lean formalisation of the odd case, <https://github.com/kim-em/KnuthClaudeLean>

V2-000475 | Claude's Cycles | cube_hamiltonian_arc_decomposition

research solved | proof annotation: True | 官方代码 | 原文线索

For odd `m > 1`, the cube digraph on `(ZMod m)³` has a Hamiltonian arc decomposition into three directed cycles [Knu26].

theorem cube_hamiltonian_arc_decomposition {m : ℕ} [NeZero m] (hm : Odd m) (hm' : 1 < m) :
HasHamiltonianArcDecomposition m

FC-ClaudesCycles.cube_hamiltonian_arc_decomposition

V2-000476 | Claude's Cycles | cube_hamiltonian_arc_decomposition_even

research solved | proof annotation: False | 官方代码 | 原文线索

For even `m > 2`, the cube digraph on `(ZMod m)³` has a Hamiltonian arc decomposition. Knuth records this as settled in the final section of [Knu26], by [Ho26] with the proof in [GPT26] for even `m ≥ 8`, and by [AM26] for the even case generally.

theorem cube_hamiltonian_arc_decomposition_even :
∀ᵉ (m : ℕ) (_ : NeZero m) (_ : Even m) (_ : 2 < m),
HasHamiltonianArcDecomposition m

FC-ClaudesCycles.cube_hamiltonian_arc_decomposition_even

V2-000477 | Claude's Cycles | cube_hamiltonian_arc_decomposition_impossible_m2

research solved | proof annotation: False | 官方代码 | 原文线索

The case `m = 2` is impossible: the cube digraph on `(ZMod 2)³` does not have a Hamiltonian arc decomposition [Aub82].

```
theorem cube_hamiltonian_arc_decomposition_impossible_m2 :
¬ HasHamiltonianArcDecomposition 2
```

FC-ClaudesCycles.cube_hamiltonian_arc_decomposition_impossible_m2

The S_3-conjecture (conjugacy classes of distinct sizes)

MODULE CONTEXT: # The S_3-conjecture (conjugacy classes of distinct sizes) *References:* * W. Zhou, I. Gorshkov, *On $\{2,3,5\}$-groups with conjugacy classes of distinct sizes*, [arXiv:2606.22244](https://arxiv.org/abs/2606.22244) (2026). * F. M. Markel, *Groups with many conjugate elements*, J. Algebra **26** (1973), 69-74. (Origin of the S_3-conjecture.) * R. Knörr, W. Lempken, B. Thielcke, *The S_3-conjecture for solvable groups*, Israel J. Math. **91** (1995), 61-76. * J. Zhang, *Finite groups with many conjugate elements*, J. Algebra **170** (1994), 608-624. * Z. Arad, M. Muzychuk, A. Oliver, *On groups with conjugacy classes of distinct sizes*, J. Algebra **280** (2004), 537-576. * [Conjugacy class](https://en.wikipedia.org/wiki/Conjugacy_class) A finite group in which distinct conjugacy classes have distinct cardinalities is called an *anti-homogeneous* group (or *ah-group*). The symmetric group S_3 is an ah-group: its three conjugacy classes have sizes 1, 2, and 3. Markel's **S_3-conjecture** (1973) asserts that, up to isomorphism, S_3 is the only nontrivial finite ah-group. The conjecture has been proved for all solvable groups (independently by Zhang and by Knörr-Lempken-Thielcke), but the general non-solvable case remains open.

V2-000478 | The S_3-conjecture (conjugacy classes of distinct sizes) | conjClassSizes_iff_sym_three

research open | proof annotation: False | 官方代码 | 原文线索

Markel's S_3-conjecture (1973): any nontrivial finite ah-group is isomorphic to S_3. The conjecture is open in general; it is known to be true for solvable groups.

```
theorem conjClassSizes_iff_sym_three
(G : Type) [Group G] [Fintype G] [Nontrivial G]
(h : HasDistinctConjClassSizes (G := G)) :
Nonempty (G ≃* Equiv.Perm (Fin 3))
```

FC-ConjugacyClassSizes.conjClassSizes_iff_sym_three

V2-000479 | The S_3-conjecture (conjugacy classes of distinct sizes) | conjClassSizes_iff_sym_three_solvable

research solved | proof annotation: False | 官方代码 | 原文线索

The S_3-conjecture holds for solvable groups: every nontrivial *solvable* finite ah-group is isomorphic to S_3. This was proved independently by Zhang (1994) and by Knörr-Lempken-Thielcke (1995).

```
theorem conjClassSizes_iff_sym_three_solvable
(G : Type) [Group G] [Fintype G] [Group.IsSolvable G] [Nontrivial G]
(h : HasDistinctConjClassSizes (G := G)) :
Nonempty (G ≃* Equiv.Perm (Fin 3))
```

FC-ConjugacyClassSizes.conjClassSizes_iff_sym_three_solvable

The Courtade-Kumar (most informative Boolean function) conjecture

MODULE CONTEXT: # The Courtade-Kumar (most informative Boolean function) conjecture Let X be uniform on the Boolean cube $\{0,1\}^n$. Independently of X, let the coordinates of Z be i.i.d. Bernoulli random variables with parameter $p \in [0, 1]$, and put $Y = X \oplus Z$, so that Y is the output of n independent binary symmetric channels with crossover probability p on input X. The **Courtade-Kumar conjecture** asserts that for every Boolean function $f : \{0,1\}^n \to \{0,1\}$, $I(f(X); Y) \le 1 - h(p)$, where mutual information and binary entropy are measured in bits. The usual restriction to $p \in [0, 1/2]$ is equivalent to the range used here. For $p > 1/2$, complementing every output bit replaces Z by its bitwise complement, which is i.i.d. Bernoulli($1 - p$) noise still independent of X; this bijective relabelling of the output preserves $I(f(X); Y)$, and $h(1 - p) = h(p)$. This file contains two formulations: * `CourtadeKumar.courtade_kumar` is the elementary finite-sum statement used by the external Lean proof and its comparator check. Its definitions and statement are retained unchanged. * `CourtadeKumar.variants.courtade_kumar_random_variable` is the random-variable formulation. Its product-law hypothesis specifies a uniform input and independent i.i.d. Bernoulli channel noise, and its conclusion is about the mutual information of the actual random variables. `CourtadeKumar.courtade_kumar_iff_random_variable` proves that the two formulations are equivalent, without using either research statement. Shannon entropy and mutual information of finite-valued random variables are defined in `FormalConjecturesForMathlib.InformationTheory.FiniteEntropy`. For $n \ge 1$, coordinate dictators attain the bound: $I(X_i; Y) = I(X_i; Y_i) = 1 - h(p)$, since the other output coordinates are independent of (X_i, Y_i). The test `mutualInfo_dictator` checks $n = 1$. *References:* - [CK14] T. A. Courtade and G. R. Kumar, *Which Boolean functions maximize mutual information on noisy inputs?*, IEEE Trans. Inf. Theory 60 (2014), no. 8, 4515-4525. Independent work on the conjecture from September 2026, in alphabetical order: - [CGJLMNW26] Z. Chen, A. Gohari, A. Javanmard, H. Lin, V. Mirrokni, C. Nair, D. P. Woodruff, *A Proof of the Most Informative Boolean Function Conjecture*, [arXiv:2609.24931](https://arxiv.org/abs/2609.24931). A Lean 4 formalisation of the proof is at https://github.com/dpwoodru/general-courtade-kumar-lean. - [KS26] D. Kramer and M. Saglam, *On the most informative Boolean function under noise*, manuscript, https://github.com/deepinfra/courtade-kumar. - [KT26] Vu Khac Ky and Tuan Tran, *Dictators are most informative*, [arXiv:2609.24184](https://arxiv.org/abs/2609.24184). See also their follow-up *Stability of the Courtade-Kumar inequality*, [arXiv:2609.31797](https://arxiv.org/abs/2609.31797). - [MB26] H. Mahdavifar and A. Beirami, *The Most Informative Bit and Beyond: A Proof of the Courtade-Kumar Conjecture and Multibit Extensions*, [arXiv:2609.26444](https://arxiv.org/abs/2609.26444). ## The elementary formulation The probability interpretations of `bsc`, `jointProb` and `mutualInfo` assume $0 \le p \le 1$, and `entropy q` is Shannon entropy when the masses q are nonnegative and sum to one; otherwise these definitions are just total real-valued expressions. ## The random-variable formulation ## API: equivalence of the formulations ## Tests of the definitions

V2-000480 | The Courtade-Kumar (most informative Boolean function) conjecture | courtade_kumar

research solved | proof annotation: True | 官方代码 | 原文线索

The Courtade-Kumar conjecture. For every n, every Boolean function f on $\{0,1\}^n$ and every crossover probability $p \in [0, 1]$, $I(f(X); Y) \le 1 - h(p)$.

```
theorem courtade_kumar (n : ℕ) (f : Cube n → Bool) (p : ℝ) (hp₀ : 0 ≤ p) (hp₁ : p ≤ 1) :
mutualInfo f p ≤ 1 - binEntropyBits p
```

FC-CourtadeKumar.courtade_kumar

V2-000481 | The Courtade-Kumar (most informative Boolean function) conjecture | courtade_kumar_random_variable

research solved | proof annotation: True | 官方代码 | 原文线索

The Courtade-Kumar conjecture, in random-variable form. The pushforward in the hypothesis is the joint distribution of (X, Z). The uniform measure on `Cube n` is the law of n i.i.d. Bernoulli($1/2$) bits. The product law makes the coordinates of Z i.i.d. Bernoulli(p) and makes the whole noise vector independent of X. In `bernoulliMeasure true false p`, `true` has probability p and denotes a flip. Conditional on X, $Y = X \oplus Z$ flips each input coordinate independently with probability p: this is the memoryless binary symmetric channel with crossover probability p from X to Y. Every Boolean function is allowed, including constants. All maps between these finite discrete types are measurable, so $f \circ X$ and Y are measurable. The conclusion is $I(f(X); Y) \le 1 - h(p)$ in bits, for the entire output vector Y. The linked proof proves the elementary statement `CourtadeKumar.courtade_kumar`, and `CourtadeKumar.courtade_kumar_iff_random_variable` transfers it to this formulation.

```
theorem courtade_kumar_random_variable :
∀ (n : ℕ) (f : Cube n → Bool) (p : ℝ) (hp₀ : 0 ≤ p) (hp₁ : p ≤ 1)
(Ω : Type) [MeasurableSpace Ω] (μ : Measure Ω) [IsProbabilityMeasure μ]
(X Z : Ω → Cube n),
Measurable X →
Measurable Z →
μ.map (fun ω => (X ω, Z ω)) =
(PMF.uniformOfFintype (Cube n)).toMeasure.prod
(Measure.pi fun _ : Fin n =>
ProbabilityTheory.bernoulliMeasure true false
((p, hp₀, hp₁) : unitInterval)) →
let Y : Ω → Cube n := fun ω => cubeXor (X ω) (Z ω)
InformationTheory.mutualInformationBits μ (f ∘ X) Y ≤ 1 - binEntropyBits p
```

FC-CourtadeKumar.variants.courtade_kumar_random_variable

De Giorgi's conjecture

MODULE CONTEXT: # De Giorgi's conjecture This file states a conjecture of De Giorgi about entire solutions to $Δ u + u - u^3 = 0$. The conjecture is a rigidity theorem: in spatial dimension $n ≤ 8$, the level sets of bounded solutions which satisfy $∂₁u > 0$ everywhere are hyperplanes. It has been shown that the condition $n ≤ 8$ is sharp. The main theorems are: - `DeGiorgi_le_eight`: the conjecture holds in dimension $n ≤ 8$. - `DeGiorgi_ge_nine`: the conclusion of the conjecture does not hold if $n ≥ 9$. The cases $1 ≤ n ≤ 8$ are also listed individually to enable partial solutions. The cases $1 ≤ n ≤ 3$ are solved, while $4 ≤ n ≤ 8$ remains open. ## Existing results - The case $n = 1$ trivially holds (u is injective since $∂_1 u > 0$). - The case $n = 2$ was proven by Ghoussoub and Gui. - The case $n = 3$ was proven by Ambrosio and Cabré. - The case $4 ≤ n ≤ 8$ was proven under an extra assumption by Savin. - The counterexample for $n ≥ 9$ was proven by Del Pino, Kowalczyk, and Wei. ## References * [Ghoussoub, Gui](https://doi.org/10.1007/s002080050196), Mathematische Annalen 311 (1998) proves the conjecture for $n = 2$. * [Ambrosio, Cabré](https://doi.org/10.1090/S0894-0347-00-00345-3), Journal of the American Mathematical Society 13 (2000) proves the conjecture for $n = 3$. * [Savin](https://doi.org/10.4007/annals.2009.169.41), Annals of Mathematics 169 (2009) proves the case $4 ≤ n ≤ 8$ under an additional assumption. * [Del Pino, Kowalczyk, Wei](http://dx.doi.org/10.4007/annals.2011.174.3.3), Annals of Mathematics 174 (2011) shows that the condition $n ≤ 8$ is sharp.

V2-000482 | De Giorgi's conjecture | DeGiorgi_eight

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 8$.

theorem DeGiorgi_eight : (DeGiorgi_conclusion 8)

FC-DeGiorgi.DeGiorgi_eight

V2-000483 | De Giorgi's conjecture | DeGiorgi_five

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 5$.

theorem DeGiorgi_five : (DeGiorgi_conclusion 5)

FC-DeGiorgi.DeGiorgi_five

V2-000484 | De Giorgi's conjecture | DeGiorgi_four

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 4$.

theorem DeGiorgi_four : (DeGiorgi_conclusion 4)

FC-DeGiorgi.DeGiorgi_four

V2-000485 | De Giorgi's conjecture | DeGiorgi_ge_nine

research solved | proof annotation: False | 官方代码 | 原文线索

In dimension $n ≥ 9$, the conclusion of De Giorgi's conjecture does not hold.

theorem DeGiorgi_ge_nine (hn : n ≥ 9) : ¬(DeGiorgi_conclusion n)

FC-DeGiorgi.DeGiorgi_ge_nine

V2-000486 | De Giorgi's conjecture | DeGiorgi_le_eight

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n ≤ 8$.

theorem DeGiorgi_le_eight (hn : n ≤ 8) : (DeGiorgi_conclusion n)

FC-DeGiorgi.DeGiorgi_le_eight

V2-000487 | De Giorgi's conjecture | DeGiorgi_one

research solved | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture trivially holds in dimension $n = 1$.

theorem DeGiorgi_one : (DeGiorgi_conclusion 1)

FC-DeGiorgi.DeGiorgi_one

V2-000488 | De Giorgi's conjecture | DeGiorgi_seven

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 7$.

theorem DeGiorgi_seven : (DeGiorgi_conclusion 7)

FC-DeGiorgi.DeGiorgi_seven

V2-000489 | De Giorgi's conjecture | DeGiorgi_six

research open | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 6$.

theorem DeGiorgi_six : (DeGiorgi_conclusion 6)

FC-DeGiorgi.DeGiorgi_six

V2-000490 | De Giorgi's conjecture | DeGiorgi_three

research solved | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 3$.

theorem DeGiorgi_three : (DeGiorgi_conclusion 3)

FC-DeGiorgi.DeGiorgi_three

V2-000491 | De Giorgi's conjecture | DeGiorgi_two

research solved | proof annotation: False | 官方代码 | 原文线索

De Giorgi's conjecture holds in dimension $n = 2$.

theorem DeGiorgi_two : (DeGiorgi_conclusion 2)

FC-DeGiorgi.DeGiorgi_two

DegreeSequencesTriangleFree

MODULE CONTEXT: Title: Degree sequences in triangle-free graphs Authors: P. Erdős, S. Fajtlowicz and W. Staton, Published in Discrete Mathematics 92 (1991) 85-88.

V2-000492 | DegreeSequencesTriangleFree | F_four_le

research solved | proof annotation: True | 官方代码

The smallest number of vertices of a triangle-free graph with chromatic number 4 and f=3 is at most 19.

theorem F_four_le : F 4 ≤ 19

FC-SimpleGraph.F_four_le

V2-000493 | DegreeSequencesTriangleFree | F_four_le_fifteen

research solved | proof annotation: True | 官方代码 | 原文线索

There is a triangle-free graph with chromatic number 4 and f=3 on 15 vertices: the Grötzsch graph with four more vertices, with degrees 2,2,2,3,3,3,4,4,4,5,5,5,6,6,6. This improves the bound 19 of Erdős, Fajtlowicz and Staton. An exhaustive computer search, not formalized, shows that 15 is optimal; see https://github.com/anatoliiohorodnyk/lean-fc-proofs/blob/fc4f1f1d43b54106b0f932518075fef57829d6fb/docs/EFS_F4.md.

theorem F_four_le_fifteen : F 4 ≤ 15

FC-SimpleGraph.F_four_le_fifteen

V2-000494 | DegreeSequencesTriangleFree | F_three

research solved | proof annotation: True | 官方代码

The smallest number of vertices of a triangle-free graph with chromatic number 3 and f=3 is 7.

theorem F_three : F 3 = 7

FC-SimpleGraph.F_three

V2-000495 | DegreeSequencesTriangleFree | theorem1

research solved | proof annotation: False | 官方代码

Theorem 1. If a triangle-free graph has `f = 2`, then it is bipartite, has minimum degree `1`, and its degree sequence is compact.

theorem theorem1 (G : SimpleGraph α) (h_conn: G.Connected) [DecidableRel G.Adj]
(h₁ : G.CliqueFree 3) (h₂ : degreeSequenceMultiplicity G = 2) :
G.IsBipartite ∧ G.minDegree = 1 ∧ HasCompactdegreeSequence G

FC-SimpleGraph.theorem1

V2-000496 | DegreeSequencesTriangleFree | theorem2

research solved | proof annotation: False | 官方代码

Theorem 2. Every triangle-free graph is an induced subgraph of one with `f = 3`.

theorem theorem2 (G : SimpleGraph α) [DecidableRel G.Adj] (h_conn: G.Connected)
(h : G.CliqueFree 3) :
∃ (β : Type*) (_ : Fintype β) (H : SimpleGraph β) (_ : DecidableRel H.Adj) (i : G ↪g H),
H.CliqueFree 3 ∧ degreeSequenceMultiplicity H = 3

FC-SimpleGraph.theorem2

The Eisenbud-Green-Harris conjecture

MODULE CONTEXT: # The Eisenbud-Green-Harris conjecture Let $S = k[x_1, \dots, x_n]$ with $x_1 > x_2 > \dots > x_n$ and let $I \subseteq S$ be a homogeneous ideal. The *Hilbert function* of I is $d \mapsto \dim_k I_d$. A *monomial ideal* is an ideal generated by monomials. A monomial ideal L is *lex* if for any two monomials u, v of the same degree with $u >_{\mathrm{lex}} v$ we have $u \in L$ whenever $v \in L$. Macaulay's theorem says that every homogeneous ideal has the same Hilbert function as a unique lex ideal. **Eisenbud-Green-Harris conjecture.** If I contains a regular sequence of forms of degrees $d_1 \le \dots \le d_c$, then there is a lex ideal L such that I has the same Hilbert function as $L + (x_1^{d_1}, \dots, x_c^{d_c})$. Eisenbud, Green and Harris proposed the conjecture in [EGH93] and [EGH96] to generalise a theorem of Castelnuovo and Cayley-Bacharach type theorems for zero-dimensional subschemes of $\mathbb P^m$. It refines Macaulay's theorem, which is the case $c = 0$. The conjecture is open in general. It holds when the regular sequence is monomial, by the Clements-Lindström theorem [CL69], and when the degrees grow quickly, namely when $d_i \ge \sum_{j < i} (d_j - 1)$ for all $i \ge 3$: this is due to Caviglia and De Stefani [CDS20], improving on Caviglia and Maclagan [CM08], who assumed the strict inequality. Other known cases are $c \le 2$, and $d_1 = \dots = d_c = 2$ with $c \le 5$ by Güntürkün and Hochster; [CDS20] lists these with references. The Lex-plus-powers conjecture, attributed to Charalambous and Evans, predicts that the ideal $L + (x_1^{d_1}, \dots, x_c^{d_c})$ above also has the largest graded Betti numbers among all ideals with that Hilbert function containing a regular sequence in those degrees. The Eisenbud-Green-Harris conjecture is its $i = 0$ case [CS18]. It is not formalised here. The Hilbert function `MvPolynomial.hilbertFunction`, lex ideals `MvPolynomial.IsLexIdeal` and the ideal of pure powers `MvPolynomial.purePowersIdeal` are defined in `FormalConjecturesForMathlib.RingTheory.MvPolynomial.LexPlusPowers`. *References:* - [EGH93] D. Eisenbud, M. Green, J. Harris, *Higher Castelnuovo theory*, in Journées de Géométrie Algébrique d'Orsay, Astérisque 218 (1993), 187-202. - [EGH96] D. Eisenbud, M. Green, J. Harris, *Cayley-Bacharach theorems and conjectures*, Bull. Amer. Math. Soc. 33 (1996), 295-324. [PDF](https://www.ams.org/journals/bull/1996-33-03/S0273-0979-96-00666-0/S0273-0979-96-00666-0.pdf) - [CS18] G. Caviglia, A. Sammartano, *On the Lex-plus-powers conjecture*, Adv. Math. 340 (2018), 1148-1163. [arXiv:1802.03035](https://arxiv.org/abs/1802.03035), Conjecture 1.1. - [CDS20] G. Caviglia, A. De Stefani, *The Eisenbud-Green-Harris conjecture for fast-growing degree sequences*, Trans. Amer. Math. Soc. 375 (2022), 2777-2792. [arXiv:2007.15467](https://arxiv.org/abs/2007.15467), Definition 2.1, Conjecture 2.3 and Theorem A. The statement formalised below is Conjecture 2.3. - [CM08] G. Caviglia, D. Maclagan, *Some cases of the Eisenbud-Green-Harris conjecture*, Math. Res. Lett. 15 (2008), 427-433. - [CL69] G. F. Clements, B. Lindström, *A generalization of a combinatorial theorem of Macaulay*, J. Combin. Theory 7 (1969), 230-238. - [FR12] C. A. Francisco, B. P. Richert, *Lex-plus-powers ideals*, in Syzygies and Hilbert functions, Lect. Notes Pure Appl. Math. 254, 113-144. [PDF](https://math.okstate.edu/people/chris/lpp-updated-final.pdf)

V2-000497 | The Eisenbud-Green-Harris conjecture | eisenbud_green_harris

research open | proof annotation: False | 官方代码 | 原文线索

The Eisenbud-Green-Harris conjecture. Let $I \subseteq k[x_1, \dots, x_n]$ be a homogeneous ideal containing a regular sequence of forms of degrees $d_1 \le \dots \le d_c$. Then there is a lex ideal L such that I has the same Hilbert function as $L + (x_1^{d_1}, \dots, x_c^{d_c})$.

```
theorem eisenbud_green_harris {c : ℕ} (hc : c ≤ n) (d : Fin c → ℕ) (hd : Monotone d)
(I : Ideal (MvPolynomial (Fin n) k))
(hI : I.IsHomogeneous (homogeneousSubmodule (Fin n) k))
(f : Fin c → MvPolynomial (Fin n) k) (hfd : ∀ i, (f i).IsHomogeneous (d i))
(hfI : ∀ i, f i ∈ I)
(hf : RingTheory.Sequence.IsRegular (MvPolynomial (Fin n) k) (List.ofFn f)) :
∃ L : Ideal (MvPolynomial (Fin n) k), IsLexIdeal L ∧
∀ e, hilbertFunction I e = hilbertFunction (L + purePowersIdeal hc d) e
```

FC-EisenbudGreenHarris.eisenbud_green_harris

Euclidean Ramsey configurations on a circle

MODULE CONTEXT: # Euclidean Ramsey configurations on a circle *Reference:* OpenAI, *A classification of finite Euclidean Ramsey configurations* (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-classification-of-finite-Euclidean-Ramsey-configurations-September-23-2026/paper.pdf

V2-000498 | Euclidean Ramsey configurations on a circle | at_most_five_circle_points_ramsey

research solved | proof annotation: False | 官方代码 | 原文线索

Every nonempty configuration of at most five distinct points on a circle is Euclidean Ramsey: every finite colouring of a suitable Euclidean space has a monochromatic copy with the same pairwise distances. Singletons and radius-zero circles are included.

```
theorem at_most_five_circle_points_ramsey {s : ℕ} (a : Fin s → E (Fin 2))
(hs : 0 < s) (hs5 : s ≤ 5) (ha : Function.Injective a)
(hsphere : ∃ c : E (Fin 2), ∃ r : ℝ, ∀ i, dist (a i) c = r) :
∀ colors : ℕ, 2 ≤ colors → ∃ D : ℕ, 1 ≤ D ∧
∀ c : E (Fin D) → Fin colors, ∃ b : Fin s → E (Fin D),
(∀ i j, dist (b i) (b j) = dist (a i) (a j)) ∧
∃ k : Fin colors, ∀ i, c (b i) = k
```

FC-EuclideanRamseyCircle.at_most_five_circle_points_ramsey

Is a finite locally free group scheme killed by its order?

MODULE CONTEXT: # Is a finite locally free group scheme killed by its order? A finite locally free group scheme of rank `n` is the group scheme analogue of a finite group of order `n`. Lagrange's theorem gives `g ^ n = 1` in a finite group of order `n`, and Grothendieck asked whether the `n`-th power map of such a group scheme is trivial. The answer is affirmative over a reduced base and for commutative group schemes, and negative in general. Which exponents kill every group scheme of a given rank is open. A group scheme is represented here by its coordinate Hopf algebra `A` over the base ring `R`. `A` is not assumed to be cocommutative, so the group scheme is not assumed to be commutative. The base is a ring rather than a scheme: a finite locally free group scheme is affine over its base, and being killed by `m` is local on the base, so nothing is lost. *References:* - [M. Demazure and A. Grothendieck, *Schémas en groupes* (SGA 3), re-edition by P. Gille and P. Polo](https://webusers.imj-prg.fr/~patrick.polo/SGA3/). Exp. VIII, Rem. 7.3.1 poses the question; an editors' note records that the re-edition drops the commutativity assumption there. Exp. VIIA, Prop. 8.5 settles the case of a field and Cor. 8.5.2 the case of a reduced base. - [R. Schoof, *Is a finite locally free group scheme killed by its order?*](https://reneschoof.github.io/schoof_oortAAG.pdf), a survey of the question and of the cases that are known, written before the counterexample - [J. Tate and F. Oort, *Group schemes of prime order*](https://www.numdam.org/item/ASENS_1970_4_3_1_1_0/), p. 4 for Deligne's theorem and p. 5, which records the general question as open in 1970 - [Mathlib, *A finite free group scheme of rank four that is not killed by four*](https://leanprover-community.github.io/mathlib4_docs/Counterexamples/GrothendieckPower.html), the counterexample

V2-000499 | Is a finite locally free group scheme killed by its order? | killed_by_rank

research solved | proof annotation: True | 官方代码 | 原文线索

Grothendieck's question: is every finite locally free group scheme of rank n, over every commutative ring, killed by n? The answer is no. There is a commutative Hopf algebra of rank 4 over a finite non-reduced ring whose fourth power map is not trivial; its eighth power map is trivial. The formalised counterexample lives in `Type`, so the statement quantifies over rings and algebras in `Type`.

```
theorem killed_by_rank :
¬ ∀ n : ℕ+, IsUniversalKillingExponent.{0, 0} n n
```

FC-FiniteLocallyFreeGroupSchemeExponent.killed_by_rank

V2-000500 | Is a finite locally free group scheme killed by its order? | optimal_exponent

research open | proof annotation: False | 官方代码 | 原文线索

Determine the function assigning to each positive integer n the least positive integer m such that every finite locally free group scheme of rank n, over every commutative ring, is killed by m. The value is 0 if there is no such m, since `sInf ∅ = 0` in `ℕ`. As in `killed_by_rank`, the rings and algebras are in `Type`. Grothendieck asked whether this least exponent is n itself. It is not, already for $n = 4$. If some positive exponent kills every group scheme of rank n, then these exponents are the multiples of the least one, so minimality for $\le$ agrees with minimality for divisibility.

```
theorem optimal_exponent :
∀ n, sInf ((↑) '' {m | IsUniversalKillingExponent.{0, 0} n m} : Set ℕ) = optimal_exponent.answer n
```

FC-FiniteLocallyFreeGroupSchemeExponent.optimal_exponent

Main conjecture on fusible numbers

MODULE CONTEXT: # Main conjecture on fusible numbers *References:* - [Fusible numbers and Peano Arithmetic](https://arxiv.org/abs/2003.14342), by Jeff Erickson, Gabriel Nivasch, and Junyan Xu. - [Fusible numbers and Peano Arithmetic](https://doi.org/10.46298/lmcs-18%283%3A6%292022), Logical Methods in Computer Science, Volume 18, Issue 3 (July 28, 2022).

V2-000501 | Main conjecture on fusible numbers | conj_7_1

research open | proof annotation: False | 官方代码 | 原文线索

If `x` is a fusible number and `y` is its successor, then the interval `[x + 1, y + 1)` can be divided into intervals `[ℓₙ, ℓₙ₊₁)`, such that the fusible numbers in `[ℓₙ, ℓₙ₊₁)` are obtained by fusing the `n + 1`st successor of `x` with a fusible number. This formalization differs from Conjecture 7.1 in the paper in four ways: (1) it is obtained from Conjecture 7.1 by plugging in `n + 1` into `n`, which simplifies the expressions and removes the need to assume `n ≥ 1`; (2) the `n + 1`st successor `s^(n+1)(x)` is replaced by the explicit value `x + (2 - 1 / 2 ^ n) * m`; (3) instead of defining `y` to be the successor of `x`, we assert that there is no fusible number strictly between `x` and `y`; (4) instead of using `∃ z, IsFusible z ∧ q = s^(n+1)(x) ~ z` we use the value of `z` determined by the equality, namely `z = 2 * q - 1 - s^(n+1)(x)`, and it is easy to see `z ∈ [x + 1 - m / 2 ^ n, x + 1)` as required.

```
theorem conj_7_1 (x y q : ℚ) (n : ℕ) (fus_x : IsFusible x) (fus_y : IsFusible y) (lt : x < y)
(nmem_Ioo : ∀ z, IsFusible z → z ∉ Set.Ioo x y) :
let m := y - x
let ℓ (n : ℕ) := y + 1 - m / 2 ^ n
IsFusible q → q ∈ Set.Ico (ℓ n) (ℓ (n + 1)) → IsFusible (2 * q - 1 - x - (2 - 1 / 2 ^ n) * m)
```

FC-FusibleNumber.conj_7_1

Gourevitch's series identity

MODULE CONTEXT: # Gourevitch's series identity *References:* - [About a New Kind of Ramanujan-Type Series](https://doi.org/10.1080/10586458.2003.10504518) by *Jesús Guillera* - [G2003] Guillera, Jesús. "About a new kind of Ramanujan-type series." Experimental Mathematics 12.4 (2003): 507-510. - [A2025] Au, Kam Cheong. "Wilf-Zeilberger seeds and non-trivial hypergeometric identities." Journal of Symbolic Computation 130 (2025): 102421. [arXiv:2312.14051](https://arxiv.org/abs/2312.14051)

V2-000502 | Gourevitch's series identity | gourevitch_series_identity

research solved | proof annotation: False | 官方代码 | 原文线索

The Gourevitch series identity: The following identity holds: $\sum_{n=0}^{\infty} \frac{1 + 14 n + 76 n^2 + 168 n^3}{2^{20 n}} \binom{2n}{n}^7 = \frac{32}{\pi^3}.$ This was originally conjectured in [G2003] by Guillera and proven in [A2025] by Au.

```
theorem gourevitch_series_identity :
∑' n : ℕ, ((1 + 14 * n + 76 * n ^ 2 + 168 * n ^ 3) / (2 ^ (20 * n)) : ℝ)
* Nat.centralBinom n ^ 7 = 32 / (Real.pi ^ 3)
```

FC-Gourevitch.gourevitch_series_identity

The Hall ratio conjecture (Harris 2019) - disproved

MODULE CONTEXT: # The Hall ratio conjecture (Harris 2019) - disproved *References:* * [Ha19] Harris, D. G. (2019). "Some results on chromatic number as a function of triangle count." *SIAM J. Discrete Math.* 33, pp. 546--563. * [DOW20] Dvořák, Z., Ossona de Mendez, P. and Wu, H. (2020). "1-subdivisions, the fractional chromatic number and the Hall ratio." *Combinatorica* 40, pp. 759--774. [arXiv:1812.07327](https://arxiv.org/abs/1812.07327) * [BLMNPV22] Blumenthal, A., Lidický, B., Martin, R. R., Norin, S., Pfender, F. and Volec, J. (2022). "Counterexamples to a conjecture of Harris on Hall ratio." *SIAM J. Discrete Math.* 36, pp. 1678--1686. [arXiv:1811.11116](https://arxiv.org/abs/1811.11116) * [SU97] Scheinerman, E. R. and Ullman, D. H. (1997). *Fractional Graph Theory.* Wiley.

V2-000503 | The Hall ratio conjecture (Harris 2019) - disproved | hall_ratio_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The Hall ratio conjecture (Harris 2019) - disproved. Harris conjectured that the fractional chromatic number is bounded by a constant multiple of the Hall ratio: there is a constant C with $\chi_f(G) \le C\,\rho(G)$ for every graph G. This is **false**: Dvořák, Ossona de Mendez and Wu [DOW20] and, independently, Blumenthal, Lidický, Martin, Norin, Pfender and Volec [BLMNPV22] constructed graphs with χ_f / ρ arbitrarily large.

```
theorem hall_ratio_conjecture : ¬
∃ C : ℝ, ∀ {V : Type} [Fintype V] [DecidableEq V] (G : SimpleGraph V),
G.fractionalChromaticNumber ≤ C * hallRatio G
```

FC-HallRatioConjecture.hall_ratio_conjecture

Hartshorne's conjecture on Vector Bundles

MODULE CONTEXT: # Hartshorne's conjecture on Vector Bundles *References:* * [Har1974] R. Hartshorne, [Varieties of small codimension in projective space](https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society-new-series/volume-80/issue-6/Varieties-of-small-codimension-in-projective-space/bams/1183535999.full). * [MO2010] [Evidences on Hartshorne's conjecture? References?](https://mathoverflow.net/questions/13990/evidences-on-hartshornes-conjecture-references)

V2-000504 | Hartshorne's conjecture on Vector Bundles | harthshorne_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

There are no indecomposable vector bundles of rank 2 on $\mathbb{P}^n$ for $n \ge 7$. This is Conjecture 6.3 in [Har1974].

```
theorem harthshorne_conjecture (n : ℕ) (hn : 7 ≤ n)
(𝓕 : VectorBundles ℙ(Fin (n + 1); Spec (.of ℂ)))
(h𝓕 : 𝓕.rank = 2) :
Nonempty (𝓕.Splitting (Fin 2))
```

FC-AlgebraicGeometry.Scheme.harthshorne_conjecture

Conjectures around homogeneous topological spaces

MODULE CONTEXT: # Conjectures around homogeneous topological spaces This file formalizes the notions of a homogeneous topological space and of (ω-)monolithic topological spaces, and states some open problems about homogeneous and monolithic compact spaces. *References:* * [Ar2013] Arhangeliski, Alexandr. "Selected old open problems in general topology." Buletinul Academiei de Ştiinţe a Republicii Moldova. Matematica 73.2-3 (2013): 37-46. https://www.math.md/files/basm/y2013-n2-3/y2013-n2-3-(pp37-46).pdf.pdf * [Ar1987] Arhangel'skii, A. V. "Topological homogeneity. Topological groups and their continuous images." Russian Mathematical Surveys 42.2 (1987): 83-131. https://doi.org/10.1070/RM1987v042n02ABEH001333

V2-000505 | Conjectures around homogeneous topological spaces | countablyMonolithicSpace_card_lt

research open | proof annotation: False | 官方代码 | 原文线索

Problem 16 in [Ar2013]: Is the cardinality of every homogeneous ω-monolithic compact hausdorff space not greater than 𝔠?

```
theorem countablyMonolithicSpace_card_lt :
∀ (X : Type) (_ : TopologicalSpace X), T2Space X → CompactSpace X →
HomogeneousSpace X → CountablyMonolithicSpace X → #X ≤ 𝔠
```

FC-Homogeneous.countablyMonolithicSpace_card_lt

V2-000506 | Conjectures around homogeneous topological spaces | firstCountableTopology_of_countablyMonolithicSpace

research open | proof annotation: False | 官方代码 | 原文线索

Problem 15 in [Ar2013]: Is every homogeneous ω-monolithic compact hausdorff space first countable?

```
theorem firstCountableTopology_of_countablyMonolithicSpace :
∀ (X : Type) (_ : TopologicalSpace X), T2Space X → CompactSpace X →
HomogeneousSpace X → CountablyMonolithicSpace X → FirstCountableTopology X
```

FC-Homogeneous.firstCountableTopology_of_countablyMonolithicSpace

V2-000507 | Conjectures around homogeneous topological spaces | homogeneousSpace_exists_inj_tendsto

research open | proof annotation: False | 官方代码 | 原文线索

Problem 13 in [Ar2013]: Is it true that every infinite homogeneous compact hausdorff space contains a non-trivial convergent sequence?

```
theorem homogeneousSpace_exists_inj_tendsto :
∀ (X : Type) (_ : TopologicalSpace X), ¬ Finite X → T2Space X → CompactSpace X →
HomogeneousSpace X → ∃ s : ℕ → X, s.Injective ∧ ∃ a : X, Tendsto s atTop (nhds a)
```

FC-Homogeneous.homogeneousSpace_exists_inj_tendsto

V2-000508 | Conjectures around homogeneous topological spaces | homogeneousSpace_exists_surjective

research open | proof annotation: False | 官方代码 | 原文线索

Problem 14 in [Ar2013]: Is it possible to represent an arbitrary compact hausdorff space as an image of a homogeneous compact space under a continuous mapping?

```
theorem homogeneousSpace_exists_surjective :
∀ (X : Type) (_ : TopologicalSpace X), T2Space X → CompactSpace X →
∃ (Y : Type) (_ : TopologicalSpace Y), T2Space Y ∧ CompactSpace Y ∧ HomogeneousSpace Y ∧
∃ f : Y → X, Continuous f ∧ f.Surjective
```

FC-Homogeneous.homogeneousSpace_exists_surjective

V2-000509 | Conjectures around homogeneous topological spaces | monolithicSpace_exists_nhds_generated_countable

research open | proof annotation: False | 官方代码 | 原文线索

Problem 17 in [Ar2013]: Is it true that every nonempty monolithic compact hausdorff space contains a point with a first countable neighborhood basis? Note: `Nonempty X` is required since the conclusion asserts the existence of a point.

```
theorem monolithicSpace_exists_nhds_generated_countable :
∀ (X : Type) (_ : TopologicalSpace X), T2Space X → CompactSpace X →
Nonempty X → MonolithicSpace X → ∃ x : X, (𝓝 x).IsCountablyGenerated
```

FC-Homogeneous.monolithicSpace_exists_nhds_generated_countable

Kaplansky's problem on the values of the u-invariant

MODULE CONTEXT: # Kaplansky's problem on the values of the u-invariant The u-invariant $u(F)$ of a field F is the largest dimension of an anisotropic quadratic form over F, or ∞ if there is no largest one; that is, $u(F) = n$ exactly when `IsGreatest (QuadraticForm.anisotropicDims F) n`. Following [Kaplansky1953] and [MerkurjevParimala2025, §5.1], all fields are of characteristic not 2. Kaplansky introduced the invariant (his $C(F)$) in [Kaplansky1953, p. 201], proved that $u(F) \ne 3$ (his Theorem 2), observed that every power of 2 occurs (his Theorem 3 gives $u(F((t))) = 2u(F)$), and conjectured that $u(F)$ is a power of 2 whenever it is finite [Kaplansky1953, p. 202]. It is classical that $u(F) \notin \{3, 5, 7\}$ [Lam2005, Proposition XI.6.8], [EKM2008, Corollary 36.4]. Merkurjev disproved Kaplansky's conjecture with a field of u-invariant 6 [Merkurjev1989] and then showed that every positive even integer is a u-invariant [Merkurjev1991]. Izhboldin constructed a field of u-invariant 9 [Izhboldin2001], and Vishik fields of u-invariant $2^r + 1$ for every $r \ge 3$ [Vishik2009]. A 2026 preprint of Karpenko constructs fields of u-invariant n for every n that is neither of the form $2^r - 1$ nor of the form $2^r - 3$ [Karpenko2026]; as of September 2026 it is not yet peer-reviewed. The problem (`u_invariant_values`) is to determine which integers are u-invariants of fields. The survey [MerkurjevParimala2025, §5.1] records the expectation that every odd integer ≥ 9 is a u-invariant (`u_invariant_values.variants.odd`). Given the results above, the problem is open exactly for the integers $2^r - 1$ and $2^r - 3$ with $r \ge 4$, the smallest of which are 13 and 15. *References:* - [Kaplansky1953] I. Kaplansky, *Quadratic forms*, J. Math. Soc. Japan 5 (1953), 200-207, [doi:10.2969/jmsj/00520200](https://doi.org/10.2969/jmsj/00520200). - [Lam2005] T. Y. Lam, *Introduction to quadratic forms over fields*, Grad. Stud. Math. 67, Amer. Math. Soc., 2005, [doi:10.1090/gsm/067](https://doi.org/10.1090/gsm/067). - [EKM2008] R. Elman, N. Karpenko, A. Merkurjev, *The algebraic and geometric theory of quadratic forms*, Amer. Math. Soc. Colloq. Publ. 56, Amer. Math. Soc., 2008, [doi:10.1090/coll/056](https://doi.org/10.1090/coll/056). - [Merkurjev1989] A. S. Merkurjev, *Kaplansky's conjecture in the theory of quadratic forms* (Russian), Zap. Nauchn. Sem. LOMI 175 (1989), 75-89; English transl. J. Soviet Math. 57 (1991), no. 6, 3489-3497, [doi:10.1007/BF01100118](https://doi.org/10.1007/BF01100118). - [Merkurjev1991] A. S. Merkurjev, *Simple algebras and quadratic forms* (Russian), Izv. Akad. Nauk SSSR Ser. Mat. 55 (1991), no. 1, 218-224; English transl. Math. USSR-Izv. 38 (1992), no. 1, 215-221, [doi:10.1070/IM1992v038n01ABEH002195](https://doi.org/10.1070/IM1992v038n01ABEH002195). - [Izhboldin2001] O. T. Izhboldin, *Fields of u-invariant 9*, Ann. of Math. (2) 154 (2001), no. 3, 529-587, [doi:10.2307/3062141](https://doi.org/10.2307/3062141). - [Vishik2009] A. Vishik, *Fields of u-invariant $2^r + 1$*, pp. 661-685 in Algebra, arithmetic, and geometry: in honor of Yu. I. Manin, Vol. II, Progr. Math. 270, Birkhäuser, 2009, [doi:10.1007/978-0-8176-4747-6_22](https://doi.org/10.1007/978-0-8176-4747-6_22). - [Karpenko2026] N. A. Karpenko, *Fields of any u-invariant but a 2-power minus 1 or 3*, preprint, 5 August 2026, [author's web page](https://sites.ualberta.ca/~karpenko/publ/u1or3-02.pdf). - [MerkurjevParimala2025] A. Merkurjev, R. Parimala, *Quadratic forms beyond arithmetic*, Notices Amer. Math. Soc. 72 (2025), no. 7, 711-718, [doi:10.1090/noti3192](https://doi.org/10.1090/noti3192).

V2-000510 | Kaplansky's problem on the values of the u-invariant | u_invariant_values

research open | proof annotation: False | 官方代码 | 原文线索

The values of the u-invariant, stated as in [MerkurjevParimala2025, §5.1]: which integers are u-invariants of fields (of characteristic not 2)? The question goes back to [Kaplansky1953], who conjectured that only powers of 2 occur (`u_invariant_values.variants.kaplansky_conjecture`). Every positive even integer is a u-invariant (`u_invariant_values.variants.even`), 3, 5 and 7 are not (`u_invariant_values.variants.not_three`, `u_invariant_values.variants.not_five`, `u_invariant_values.variants.not_seven`), and it is expected that every odd integer ≥ 9 is (`u_invariant_values.variants.odd`). Granting the preprint [Karpenko2026] (`u_invariant_values.variants.karpenko`), the answer is known for every n except $n = 2^r - 1$ and $n = 2^r - 3$ with $r \ge 4$.

theorem u_invariant_values :
∀ n, n ∈ u_invariant_values.answer ↔ IsUInvariant n

FC-KaplanskyUInvariant.u_invariant_values

Kotzig's Conjecture

MODULE CONTEXT: # Kotzig's Conjecture *Reference:* A. Kotzig, cited in A. Rosa, *On certain valuations of the vertices of a graph*, Theory of Graphs (Internat. Sympos., Rome, 1966), Gordon and Breach, 1967, pp. 349-355. Kotzig conjectured that for every n, the complete graph K_{2n+1} decomposes into copies of any n-edge tree via cyclic shifts of a single embedding. This is strictly stronger than Ringel's conjecture; see `Paper/RingelConjecture.lean`. The large-n case is proved by Montgomery-Pokrovskiy-Sudakov; see `Arxiv/2001.02665/RingelConjecture.lean`.

V2-000511 | Kotzig's Conjecture | kotzig_conjecture

research open | proof annotation: False | 官方代码

For any tree T with n edges, the complete graph K_{2n+1} decomposes into $2n+1$ edge-disjoint copies of T via cyclic shifts of a single embedding. The $2n+1$ copies are $f_0, f_1, \dots, f_{2n}$ where $f_i(v) = f_0(v) + i$ for all vertices v - each copy is obtained by adding $i \pmod{2n+1}$ to every vertex of the base copy. This is strictly stronger than `RingelConjecture.ringel_conjecture`.

theorem kotzig_conjecture {V : Type} [Finite V]
(T : SimpleGraph V) (hT : T.IsTree)
(n : ℕ) (hn : T.edgeSet.ncard = n) :
∃ f : Fin (2 * n + 1) → (V ↪ Fin (2 * n + 1)),
(∀ i v, f i v = f 0 v + i) ∧
Pairwise (fun i j => Disjoint (T.map (f i)).edgeSet (T.map (f j)).edgeSet) ∧
⨆ i, T.map (f i) = (⊤ : SimpleGraph (Fin (2 * n + 1)))

FC-KotzigConjecture.kotzig_conjecture

Kurepa's conjecture

MODULE CONTEXT: # Kurepa's conjecture *Reference:* [On the left factorial function !N](https://oeis.org/A3422), by *Đuro Kurepa* Math. Balkanica 1, p. 147-153, 1971

V2-000512 | Kurepa's conjecture | kurepa_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Kurepa's conjecture For all n, $$!n\not\equiv 0 \mod n$$ This appears as B44 "Sums of factorials." in [Unsolved Problems in Number Theory](https://doi.org/10.1007/978-0-387-26677-0) by *Richard K. Guy*

theorem kurepa_conjecture (n : ℕ) (h_n : 2 < n) : (!n : ℕ) % n ≠ 0

FC-Kurepa.kurepa_conjecture

Conjectures about Latin Squares

MODULE CONTEXT: # Conjectures about Latin Squares This file formalizes some conjectures and theorems around latin squares. *References:* * [Wa2011] Wanless, Ian. "Transversals in Latin Squares: A Survey." Surveys in Combinatorics 2011, R. Chapman, Ed. Cambridge University Press, 2011, pp. 403-437. https://users.monash.edu.au/~iwanless/papers/transurveyBCC.pdf * https://en.wikipedia.org/wiki/Problems_in_Latin_squares

V2-000513 | Conjectures about Latin Squares | growthRateZn

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 6.9 in [Wa2011]: $$ \lim_{\substack{n \to \infty \\ n \text{ odd}}} \frac{1}{n} \log(z_n / n!) = -1 $$ It is not even known if this limit exists. Note that $z_n = 0$ for even n (see `z_even`), so the limit must be restricted to odd n; here we parametrise odd n as $2k + 1$.

theorem growthRateZn :
Filter.Tendsto (fun k => (1 : ℝ) / (2 * k + 1) *
Real.log (z (2 * k + 1) / (2 * k + 1).factorial)) Filter.atTop
(nhds (-1))

FC-LatinSquare.growthRateZn

V2-000514 | Conjectures about Latin Squares | latinSquareNearTransversal

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 5.1 in [Wa2011]: Every latin square has a near-transversal

theorem latinSquareNearTransversal :
∀ (n : ℕ) (L : LatinSquare n), ∃ ρ σ, IsNearTransversal L ρ σ

FC-LatinSquare.latinSquareNearTransversal

V2-000515 | Conjectures about Latin Squares | latinSquareOrder11Transversal

research open | proof annotation: False | 官方代码 | 原文线索

The smallest odd number for which this conjecture is not known is 11.

theorem latinSquareOrder11Transversal :
∀ (L : LatinSquare 11), ∃ σ, IsTransversal L σ

FC-LatinSquare.latinSquareOrder11Transversal

V2-000516 | Conjectures about Latin Squares | maxTransversalsBound

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 7.2 in [Wa2011]: For all $n \geq 5$, $$ 15^{n/5} \leq T(n) \leq c^n \sqrt{n} \cdot n! $$ where $c = \sqrt{\frac{3 - \sqrt{3}}{6}} \cdot e^{\sqrt{3}/6}$

theorem maxTransversalsBound :
let c := Real.sqrt ((3 - Real.sqrt 3) / 6) * Real.exp (Real.sqrt 3 / 6)
∀ n : ℕ, n ≥ 5 →
(T n : ℝ) ∈ Set.Icc ((15 : ℝ) ^ ((n : ℝ) / 5)) (c ^ n * Real.sqrt n * n.factorial)

FC-LatinSquare.maxTransversalsBound

V2-000517 | Conjectures about Latin Squares | molsExistenceProblem

research open | proof annotation: False | 官方代码 | 原文线索

MOLS existence problem: determine exactly which orders `n` admit a complete set of `n - 1` mutually orthogonal latin squares. Equivalently, this asks for which orders affine planes of order `n` exist. Complete sets are known for prime-power orders; the smallest currently unresolved order is `12`.

theorem molsExistenceProblem : molsExistenceProblem.answer = {n : ℕ | HasCompleteMOLS n}

FC-LatinSquare.molsExistenceProblem

V2-000518 | Conjectures about Latin Squares | molsOrder12

research open | proof annotation: False | 官方代码 | 原文线索

The smallest unresolved case of the MOLS existence problem: whether there are `11` mutually orthogonal latin squares of order `12`.

theorem molsOrder12 : HasCompleteMOLS 12

FC-LatinSquare.molsOrder12

V2-000519 | Conjectures about Latin Squares | numTransversalsZn

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 6.7 in [Wa2011]: There exist real constants $0 < c_1 < c_2 < 1$ such that $$ c_1^n n! \leq z_n \leq c_2^n n! $$ for all odd $n \geq 3$.

theorem numTransversalsZn :
∃ᵉ (c₁ > (0 : ℝ)) (c₂ < (1 : ℝ)) (_ : c₁ < c₂),
∀ n ≥ 3, Odd n →
(z n : ℝ) ∈ Set.Icc (c₁ ^ n * n.factorial) (c₂ ^ n * n.factorial)

FC-LatinSquare.numTransversalsZn

V2-000520 | Conjectures about Latin Squares | oddOrderLatinSquareTransversal

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 3.2 in [Wa2011]: Each Latin square of odd order has at least one transversal.

theorem oddOrderLatinSquareTransversal :
∀ (n : ℕ), Odd n → ∀ (L : LatinSquare n), ∃ σ, IsTransversal L σ

FC-LatinSquare.oddOrderLatinSquareTransversal

V2-000521 | Conjectures about Latin Squares | oddOrderLeq9LatinSquareTransversal

research solved | proof annotation: False | 官方代码 | 原文线索

The conjecture is known to be true for $n \leq 9$.

theorem oddOrderLeq9LatinSquareTransversal :
∀ n ≤ 9, Odd n → ∀ (L : LatinSquare n), ∃ σ, IsTransversal L σ

FC-LatinSquare.oddOrderLeq9LatinSquareTransversal

Latin Tableau Conjecture

MODULE CONTEXT: # Latin Tableau Conjecture The Latin Tableau Conjecture states that the graph associated to any (finite) Young diagram (i.e., whose vertices are the cells of the diagram, with edges between cells in the same row or column) is CDS-colorable, meaning that there exists a proper coloring of the vertices of the graph such that for all k > 0, the number of vertices with color < k equals the maximum size of the union of k independent sets of the graph. *References:* * [The Latin Tableau Conjecture](https://www.combinatorics.org/ojs/index.php/eljc/article/view/v32i2p48)

V2-000522 | Latin Tableau Conjecture | LatinTableauConjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Latin Tableau Conjecture: If G is the simple graph of a Young diagram, then G is CDS-colorable.

theorem LatinTableauConjecture (μ : YoungDiagram) :
μ.toSimpleGraph.CDSColorable

FC-LatinTableau.SimpleGraph.LatinTableauConjecture

Lennon-Pittel conjecture on random stable matchings

MODULE CONTEXT: # Lennon-Pittel conjecture on random stable matchings *References:* - [On the Likely Number of Stable Marriages](https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=osu1194991095&disposition=inline) by *Craig Lennon*, PhD dissertation, The Ohio State University (2007). - [On the Likely Number of Solutions for the Stable Marriage Problem](https://doi.org/10.1017/S0963548308009607) by *Craig Lennon and Boris Pittel*, *Combinatorics, Probability and Computing* 18 (2009), 371-421. Let S_n be the number of stable matchings under uniformly random complete strict preferences.

V2-000523 | Lennon-Pittel conjecture on random stable matchings | lennon_pittel_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The [Lennon-Pittel conjecture](https://doi.org/10.1017/S0963548308009607) states that for every positive nonincreasing sequence $\varepsilon_n \to 0$, the probability that the number S_n of stable matchings is at least $\varepsilon_n \mathbb{E}[S_n]$ tends to one.

```
theorem lennon_pittel_conjecture
(ε : ℕ → ℝ) (hε_pos : ∀ n, 0 < ε n) (hε_antitone : Antitone ε)
(hε_tendsto : Tendsto ε atTop (𝓝 0)) :
Tendsto
(fun n ↦
(StableMarriage.uniformProfile n).toMeasure
{p | (StableMarriage.numStableMatchings p : ℝ) ≥
ε n * StableMarriage.expectedNumStableMatchings n})
atTop (𝓝 1)
```

FC-LennonPittel.lennon_pittel_conjecture

Monochromatic quantum graphs (inherited vertex colorings)

MODULE CONTEXT: # Monochromatic quantum graphs (inherited vertex colorings) This file studies the existence of *monochromatic quantum graphs*: edge-coloured, edge-weighted complete graphs whose perfect matchings induce vertex colourings, with the property that - every **non-monochromatic** inherited vertex colouring has total weight `0`, while - each of the `D` **monochromatic** colourings has total weight `1`. In the quantum-optics motivation, such a construction corresponds to generating high-dimensional multipartite GHZ-type states using probabilistic pair sources and linear optics (without additional resources), where interference patterns can be expressed as weighted sums over perfect matchings. ## Main questions (informal) - For `N = 4` and `D ≥ 4`, does there exist such a graph/weighting? - For even `N ≥ 6` and `D ≥ 3`, does there exist such a graph/weighting? ## Formalisation sketch A quantum graph with `N` vertices and `D` colours can be encoded by a weight function `W : EdgeN N D α → α` (for a coefficient domain `α`). For each assignment of vertex indices `ι : V N → Fin D`, we define a perfect-matching sum `pmSumN N D W ι` (a sum over perfect matchings, where each matching contributes the product of the corresponding edge weights determined by `ι`). The equation system `EqSystemN N D W` requires `pmSumN N D W ι = 1` iff `ι` is constant (all entries equal), and `0` otherwise. The open conjectures in this file ask for non-existence/existence of such `W` over various coefficient domains (e.g. `ℂ`, `ℝ`, `ℤ`, and restricted integer weights). ## References * [Krenn2017] M. Krenn, X. Gu, A. Zeilinger, "Quantum Experiments and Graphs: Multiparty States as Coherent Superpositions of Perfect Matchings", *Physical Review Letters* 119(24), 240403 (2017). * [MO2018] [Vertex coloring inherited from perfect matchings (motivated by quantum physics)](https://mathoverflow.net/questions/311325), MathOverflow question 311325. * [Gu2019] X. Gu, M. Erhard, A. Zeilinger, M. Krenn, "Quantum experiments and graphs II: Quantum interference, computation, and state generation", *PNAS* 116(10), 4147-4155 (2019). * [Krenn2019] [Questions on the Structure of Perfect Matchings inspired by Quantum Physics](https://arxiv.org/abs/1902.06023) by *M. Krenn, X. Gu, U. Soltész*, Proc. 2nd Croatian Combinatorial Days, 57-70 (2019). * [Chandran2022] [Edge-coloured graphs with only monochromatic perfect matchings and their connection to quantum physics](https://arxiv.org/abs/2202.05562) by *N. Chandran, S. Gajjala* (2022). * [Chandran2024] [Krenn-Gu conjecture for sparse graphs](https://arxiv.org/abs/2407.00303) by *N. Chandran, S. Gajjala, S. Illickan, M. Krenn*, MFCS 2024. * [Ki26] [A solver-free Lean 4 proof of the sharp bound $D \le N - 2$ for monochromatic quantum graph equation systems over integral domains](https://github.com/KitaKen1/monochromatic-quantum-graph-sharp-bound-lean/tree/6c5340384479dbb36129b2e0084449be2458cce2), commit `6c534038`. * [Ki26b] [A solver-free Lean 4 proof that the monochromatic quantum graph equation system has no solution for $N = 10$ and $D = 8$ over integral domains](https://github.com/KitaKen1/monochromatic-quantum-graph-n10-d8-lean/tree/a9309005c7a27a2615f8e7eebef7a1db017809ab), commit `a9309005`.

V2-000524 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d10

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 10$ and $D = 10$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? This follows from the $D = N$ case, see `eqSystem_no_solution_even_ge4_d_eq_n_explicit`.

theorem eqSystem10_no_solution_d10 :
¬ ∃ W : WeightsN 10 10 ℂ, EqSystemN 10 10 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d10

V2-000525 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

theorem eqSystem10_no_solution_d3 :
¬ ∃ W : WeightsN 10 3 ℂ, EqSystemN 10 3 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d3

V2-000526 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d3_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 10$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

theorem eqSystem10_no_solution_d3_int :
¬ ∃ W : WeightsN 10 3 ℤ, EqSystemN 10 3 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d3_int

V2-000527 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d3_real

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$?

theorem eqSystem10_no_solution_d3_real :
¬ ∃ W : WeightsN 10 3 ℝ, EqSystemN 10 3 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d3_real

V2-000528 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d3_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 10$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

theorem eqSystem10_no_solution_d3_trinary_int :
¬ ∃ W : WeightsN 10 3 ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 10 3 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d3_trinary_int

V2-000529 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d4

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

theorem eqSystem10_no_solution_d4 :
¬ ∃ W : WeightsN 10 4 ℂ, EqSystemN 10 4 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d4

V2-000530 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d5

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 5$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

theorem eqSystem10_no_solution_d5 :
¬ ∃ W : WeightsN 10 5 ℂ, EqSystemN 10 5 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d5

V2-000531 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d6

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 6$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

theorem eqSystem10_no_solution_d6 :
¬ ∃ W : WeightsN 10 6 ℂ, EqSystemN 10 6 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d6

V2-000532 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d7

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 10$ and $D = 7$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

theorem eqSystem10_no_solution_d7 :
¬ ∃ W : WeightsN 10 7 ℂ, EqSystemN 10 7 W

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d7

V2-000533 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d8

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 10$ and $D = 8$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The sharp bound $D \le N - 2$ of [Ki26] does not cover this equality case $D = N - 2$; it is settled over an integral domain in [Ki26b].

```
theorem eqSystem10_no_solution_d8 :
¬ ∃ W : WeightsN 10 8 ℂ, EqSystemN 10 8 W
```

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d8

V2-000534 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem10_no_solution_d9

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 10$ and $D = 9$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The sharp bound $D \le N - 2$ over an integral domain [Ki26] settles this: here $D = N - 1$.

```
theorem eqSystem10_no_solution_d9 :
¬ ∃ W : WeightsN 10 9 ℂ, EqSystemN 10 9 W
```

FC-MonochromaticQuantumGraph.eqSystem10_no_solution_d9

V2-000535 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem12_no_solution_d3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 12$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem12_no_solution_d3 :
¬ ∃ W : WeightsN 12 3 ℂ, EqSystemN 12 3 W
```

FC-MonochromaticQuantumGraph.eqSystem12_no_solution_d3

V2-000536 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem14_no_solution_d3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 14$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem14_no_solution_d3 :
¬ ∃ W : WeightsN 14 3 ℂ, EqSystemN 14 3 W
```

FC-MonochromaticQuantumGraph.eqSystem14_no_solution_d3

V2-000537 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem16_no_solution_d3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 16$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem16_no_solution_d3 :
¬ ∃ W : WeightsN 16 3 ℂ, EqSystemN 16 3 W
```

FC-MonochromaticQuantumGraph.eqSystem16_no_solution_d3

V2-000538 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_d4

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 4$ and $D = 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? This is the $D = N$ case, proved using `eqSystem_no_solution_even_ge4_d_eq_n_explicit`.

```
theorem eqSystem4_no_solution_d4 :
¬ ∃ W : WeightsN 4 4 ℂ, EqSystemN 4 4 W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_d4

V2-000539 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_ge4

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 4$ and all $D \geq 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The DeepMind prover agent has found a formal proof of this statement.

```
theorem eqSystem4_no_solution_ge4 :
∀ D : Nat, D ≥ 4 →
¬ ∃ W : WeightsN 4 D ℂ, EqSystemN 4 D W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_ge4

V2-000540 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_ge4_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 4$ and all $D \geq 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$? This follows from the solution of the complex version of the problem (see `eqSystem4_no_solution_ge4`).

```
theorem eqSystem4_no_solution_ge4_int :
∀ D : Nat, D ≥ 4 →
¬ ∃ W : WeightsN 4 D ℤ, EqSystemN 4 D W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_ge4_int

V2-000541 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_ge4_real

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 4$ and all $D \geq 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$? This follows from the solution of the complex version of the problem (see `eqSystem4_no_solution_ge4`)

```
theorem eqSystem4_no_solution_ge4_real :
∀ D : Nat, D ≥ 4 →
¬ ∃ W : WeightsN 4 D ℝ, EqSystemN 4 D W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_ge4_real

V2-000542 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_ge4_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 4$ and all $D \geq 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$? This follows from the complex version, see `eqSystem4_no_solution_ge4`.

```
theorem eqSystem4_no_solution_ge4_trinary_int :
∀ D : Nat, D ≥ 4 →
¬ ∃ W : WeightsN 4 D ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 4 D W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_ge4_trinary_int

V2-000543 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem4_no_solution_nnreal_ge4

research solved | proof annotation: False | 官方代码 | 原文线索

Bogdanov: for $N = 4$ and all $D \geq 4$, no solution exists over $\mathbb{R}_{\geq 0}$.

```
theorem eqSystem4_no_solution_nnreal_ge4 :
∀ D : Nat, D ≥ 4 →
¬ ∃ W : WeightsN 4 D ℝ≥0, EqSystemN 4 D W
```

FC-MonochromaticQuantumGraph.eqSystem4_no_solution_nnreal_ge4

V2-000544 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d3

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? A complete Lean 4 proof is available in the linked external certificate repository.

```
theorem eqSystem6_no_solution_d3 :
¬ ∃ W : WeightsN 6 3 ℂ, EqSystemN 6 3 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d3

V2-000545 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d3_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

```
theorem eqSystem6_no_solution_d3_int :
¬ ∃ W : WeightsN 6 3 ℤ, EqSystemN 6 3 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d3_int

V2-000546 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d3_real

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 6$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$?

```
theorem eqSystem6_no_solution_d3_real :
¬ ∃ W : WeightsN 6 3 ℝ, EqSystemN 6 3 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d3_real

V2-000547 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d3_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

```
theorem eqSystem6_no_solution_d3_trinary_int :
¬ ∃ W : WeightsN 6 3 ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 6 3 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d3_trinary_int

V2-000548 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d4

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 4$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem6_no_solution_d4 :
¬ ∃ W : WeightsN 6 4 ℂ, EqSystemN 6 4 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d4

V2-000549 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d5

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 5$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The sharp bound $D \le N - 2$ over an integral domain [Ki26] settles this: here $D = N - 1$.

```
theorem eqSystem6_no_solution_d5 :
¬ ∃ W : WeightsN 6 5 ℂ, EqSystemN 6 5 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d5

V2-000550 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d5_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 5$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

```
theorem eqSystem6_no_solution_d5_int :
¬ ∃ W : WeightsN 6 5 ℤ, EqSystemN 6 5 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d5_int

V2-000551 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d5_real

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 5$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$? The sharp bound $D \le N - 2$ over an integral domain [Ki26] settles this: here $D = N - 1$.

```
theorem eqSystem6_no_solution_d5_real :
¬ ∃ W : WeightsN 6 5 ℝ, EqSystemN 6 5 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d5_real

V2-000552 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d5_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 5$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

```
theorem eqSystem6_no_solution_d5_trinary_int :
¬ ∃ W : WeightsN 6 5 ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 6 5 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d5_trinary_int

V2-000553 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_d6

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and $D = 6$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? This follows from `eqSystem_no_solution_even_ge4_d_eq_n_explicit`.

```
theorem eqSystem6_no_solution_d6 :
¬ ∃ W : WeightsN 6 6 ℂ, EqSystemN 6 6 W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_d6

V2-000554 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_ge3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 6$ and all $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem6_no_solution_ge3 :
∀ D : Nat, D ≥ 3 →
¬ ∃ W : WeightsN 6 D ℂ, EqSystemN 6 D W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_ge3

V2-000555 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_ge3_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and all $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

```
theorem eqSystem6_no_solution_ge3_int :
∀ D : Nat, D ≥ 3 →
¬ ∃ W : WeightsN 6 D ℤ, EqSystemN 6 D W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_ge3_int

V2-000556 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_ge3_real

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 6$ and all $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$?

```
theorem eqSystem6_no_solution_ge3_real :
∀ D : Nat, D ≥ 3 →
¬ ∃ W : WeightsN 6 D ℝ, EqSystemN 6 D W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_ge3_real

V2-000557 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem6_no_solution_ge3_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 6$ and all $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

```
theorem eqSystem6_no_solution_ge3_trinary_int :
∀ D : Nat, D ≥ 3 →
¬ ∃ W : WeightsN 6 D ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 6 D W
```

FC-MonochromaticQuantumGraph.eqSystem6_no_solution_ge3_trinary_int

V2-000558 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem8_no_solution_d10

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 8$ and $D = 10$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The DeepMind prover agent has found a formal proof of this statement.

```
theorem eqSystem8_no_solution_d10 :
¬ ∃ W : WeightsN 8 10 ℂ, EqSystemN 8 10 W
```

FC-MonochromaticQuantumGraph.eqSystem8_no_solution_d10

V2-000559 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem8_no_solution_d3

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 8$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem8_no_solution_d3 :
¬ ∃ W : WeightsN 8 3 ℂ, EqSystemN 8 3 W
```

FC-MonochromaticQuantumGraph.eqSystem8_no_solution_d3

V2-000560 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem8_no_solution_d3_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 8$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

```
theorem eqSystem8_no_solution_d3_int :
¬ ∃ W : WeightsN 8 3 ℤ, EqSystemN 8 3 W
```

FC-MonochromaticQuantumGraph.eqSystem8_no_solution_d3_int

V2-000561 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem8_no_solution_d3_real

research open | proof annotation: False | 官方代码 | 原文线索

For $N = 8$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$?

```
theorem eqSystem8_no_solution_d3_real :
¬ ∃ W : WeightsN 8 3 ℝ, EqSystemN 8 3 W
```

FC-MonochromaticQuantumGraph.eqSystem8_no_solution_d3_real

V2-000562 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem8_no_solution_d3_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For $N = 8$ and $D = 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

```
theorem eqSystem8_no_solution_d3_trinary_int :
¬ ∃ W : WeightsN 8 3 ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN 8 3 W
```

FC-MonochromaticQuantumGraph.eqSystem8_no_solution_d3_trinary_int

V2-000563 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_even_ge4_d_eq_n_explicit

research solved | proof annotation: True | 官方代码 | 原文线索

For all even $N \geq 4$ and $D = N$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$? The DeepMind prover agent has found a formal proof for this statement.

```
theorem eqSystem_no_solution_even_ge4_d_eq_n_explicit :
∀ N : Nat, N ≥ 4 → Even N →
¬ ∃ W : WeightsN N N ℂ, EqSystemN N N W
```

FC-MonochromaticQuantumGraph.eqSystem_no_solution_even_ge4_d_eq_n_explicit

V2-000564 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_ge6_ge3

research open | proof annotation: False | 官方代码 | 原文线索

For all even $N \geq 6$ and $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{C}$?

```
theorem eqSystem_no_solution_ge6_ge3 :
∀ N D : Nat, N ≥ 6 → Even N → D ≥ 3 →
¬ ∃ W : WeightsN N D ℂ, EqSystemN N D W
```

FC-MonochromaticQuantumGraph.eqSystem_no_solution_ge6_ge3

V2-000565 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_ge6_ge3_int

research solved | proof annotation: True | 官方代码 | 原文线索

For all even $N \geq 6$ and $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$?

theorem eqSystem_no_solution_ge6_ge3_int :
∀ N D : Nat, N ≥ 6 → Even N → D ≥ 3 →
¬ ∃ W : WeightsN N D ℤ, EqSystemN N D W

FC-MonochromaticQuantumGraph.eqSystem_no_solution_ge6_ge3_int

V2-000566 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_ge6_ge3_real

research open | proof annotation: False | 官方代码 | 原文线索

For all even $N \geq 6$ and $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{R}$?

theorem eqSystem_no_solution_ge6_ge3_real :
∀ N D : Nat, N ≥ 6 → Even N → D ≥ 3 →
¬ ∃ W : WeightsN N D ℝ, EqSystemN N D W

FC-MonochromaticQuantumGraph.eqSystem_no_solution_ge6_ge3_real

V2-000567 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_ge6_ge3_trinary_int

research solved | proof annotation: True | 官方代码 | 原文线索

For all even $N \geq 6$ and $D \geq 3$, does there exist no solution to the monochromatic quantum graph equation system over $\mathbb{Z}$ with weights in $\{-1, 0, 1\}$?

theorem eqSystem_no_solution_ge6_ge3_trinary_int :
∀ N D : Nat, N ≥ 6 → Even N → D ≥ 3 →
¬ ∃ W : WeightsN N D ℤ,
(∀ e, W e = (-1 : ℤ) ∨ W e = 0 ∨ W e = 1) ∧
EqSystemN N D W

FC-MonochromaticQuantumGraph.eqSystem_no_solution_ge6_ge3_trinary_int

V2-000568 | Monochromatic quantum graphs (inherited vertex colorings) | eqSystem_no_solution_nnreal_even_ge6_ge3

research solved | proof annotation: False | 官方代码 | 原文线索

Bogdanov: for all even $N \geq 6$ and $D \geq 3$, no solution exists over $\mathbb{R}_{\geq 0}$.

theorem eqSystem_no_solution_nnreal_even_ge6_ge3 :
∀ N D : Nat,
N ≥ 6 → Even N → D ≥ 3 →
¬ ∃ W : WeightsN N D ℝ≥0, EqSystemN N D W

FC-MonochromaticQuantumGraph.eqSystem_no_solution_nnreal_even_ge6_ge3

Energy-critical NLS in three dimensions.

MODULE CONTEXT: # Energy-critical NLS in three dimensions. This file states an a priori estimate for the energy-critical nonlinear Schrödinger equation on $ℝ^3$. By 'soft' arguments, this bound implies global well-posedness in the energy space, scattering, asymptotic completeness, and uniform regularity (all of which are less elementary to state). This is remarked on p. 770 and p. 793 in the referenced article which first proved the a priori bound. The main theorem is: - `NLS_apriori`: a bound on the space-time L^10 norm in terms of the initial energy for local smooth solutions to the NLS equation. ## References * [Colliander, Keel, Staffilani, Takaoka, Tao](https://doi.org/10.4007/annals.2008.167.767), Annals of Mathematics 167 (2008) proves the a priori estimate, and thereby, global well-posedness.

V2-000569 | Energy-critical NLS in three dimensions. | NLS_apriori

research solved | proof annotation: False | 官方代码 | 原文线索

The $L^10_{t,x}$ a priori estimate for local schwartz solutions.

theorem NLS_apriori :
∃ f : ℝ → ℝ, ∀ I > 0, ∀ s : LocalSchwartzSolution I,
L10_local I s.u ≤ f (energy (s.u 0))

FC-NLSCritical.NLS_apriori

Parities of partition numbers

MODULE CONTEXT: # Parities of partition numbers There are several conjectures on parities of partition numbers. The most famous conjecture is the Parkin-Shanks conjecture, which states that the natural density of `n` where the partition number `p(n)` is even (resp. odd) exists and equals `1/2`. There are several related (weaker) results toward the conjecture. Kolberg showed that `p(n)` takes each parity infinitely often. Subbarao conjectured that every arithmetic progression `r (mod t)` contains infinitely many `n` with `p(n)` even and infinitely many `n` with `p(n)` odd; the even case was proved by Ono and the odd case by Radu. Quantitative lower bounds for the number of `n ≤ X` with `p(n)` even (resp. odd) have also been studied. The best known bounds are `≫ √X log log X` for even values (Bellaïche-Nicolas) and `≫ √X` for odd values (Zheng, in an appendix to Ono-Swaminathan). Ono-Swaminathan also proved that both parities occur infinitely often along the quadratic progressions `(D m² + 1) / 24`, and Ono conjectured the same for every nonconstant integer-valued polynomial that is eventually positive. *References:* * [On the distribution of parity in the partition function](https://www.jstor.org/stable/2003251), T. R. Parkin, D. Shanks, Math. Comp. 21 (1967), 466-480. * [Note on the parity of the partition function](https://doi.org/10.7146/math.scand.a-10584), O. Kolberg, Math. Scand. 7 (1959), 377-378. * Some remarks on the partition function, M. V. Subbarao, Amer. Math. Monthly 73 (1966), 851-854. * [Parity of the partition function in arithmetic progressions](https://doi.org/10.1515/crll.1996.472.1), K. Ono, J. Reine Angew. Math. 472 (1996), 1-16. * [Distribution of parity of the partition function in arithmetic progressions](https://doi.org/10.1016/S0019-3577(99)80014-7), S. Ahlgren, Indag. Math. (N.S.) 10 (1999), 173-181. * [Parity of the partition function](https://doi.org/10.1016/j.aim.2010.02.023), K. Ono, Adv. Math. 225 (2010), 349-366. * [A proof of Subbarao's conjecture](https://doi.org/10.1515/CRELLE.2011.165), C.-S. Radu, J. Reine Angew. Math. 672 (2012), 161-175. * [Parité des coefficients de formes modulaires](https://doi.org/10.1007/s11139-014-9645-9), J. Bellaïche, J.-L. Nicolas, Ramanujan J. 40 (2016), 1-44. * [Nonzero coefficients of half-integral weight modular forms mod ℓ](https://arxiv.org/abs/1704.07440), J. Bellaïche, B. Green, K. Soundararajan, Res. Math. Sci. 5 (2018), Paper No. 6. * [Parity of the partition function in quadratic progressions](https://arxiv.org/abs/2509.09553) (with an appendix by Q.-Y. Zheng), K. Ono, A. Swaminathan, arXiv:2509.09553. * [Cox] D. A. Cox, Primes of the Form $x^2 + ny^2$, 2nd ed., Wiley, 2013.

V2-000570 | Parities of partition numbers | kolberg

research solved | proof annotation: False | 官方代码 | 原文线索

`p(n)` is even for infinitely many `n` and odd for infinitely many `n`. (Kolberg)

```
theorem kolberg :
{n : ℕ | Even (partitionNumber n)}.Infinite ∧ {n : ℕ | Odd (partitionNumber n)}.Infinite
```

FC-PartitionParity.kolberg

V2-000571 | Parities of partition numbers | parkin_shanks

research open | proof annotation: False | 官方代码 | 原文线索

The natural density of `n` where the partition number `p(n)` is even (resp. odd) exists and equals `1/2`.

```
theorem parkin_shanks :
{n : ℕ | Even (partitionNumber n)}.HasDensity (1 / 2) ∧
{n : ℕ | Odd (partitionNumber n)}.HasDensity (1 / 2)
```

FC-PartitionParity.parkin_shanks

Pfister's problem on the Pythagoras number of $\mathbb{R}(X_1, \dots, X_n)$

MODULE CONTEXT: # Pfister's problem on the Pythagoras number of $\mathbb{R}(X_1, \dots, X_n)$ The Pythagoras number $p(K)$ of a field K is the least p such that every sum of squares in K is a sum of p squares, i.e. the least element of `pythagorasBounds K`. Artin's solution of Hilbert's 17th problem shows that a positive semidefinite rational function in $\mathbb{R}(X_1, \dots, X_n)$ is a sum of squares, and Pfister showed that 2^n squares suffice ([Pfister1967, Theorem 1]; for the field, Corollary 1 of Theorem 2 in [Pfister1971]). Since sums of squares are positive semidefinite, $p(\mathbb{R}(X_1, \dots, X_n)) \le 2^n$ [Pfister1995, p. 95]. Problem 1 of [Pfister1971, §4] asks for the true value of $p(\mathbb{R}(X_1, \dots, X_n))$. The question is often posed as: is $p(\mathbb{R}(X_1, \dots, X_n)) = 2^n$, i.e. is Pfister's bound optimal? See e.g. [Benoist2017, Question 0.2]. Problem 1 itself quotes the lower bound $n + 1 \le p(\mathbb{R}(X_1, \dots, X_n))$ from Cassels' theorem [Cassels1964]. For $n \ge 2$ the best known bounds are $n + 2 \le p(\mathbb{R}(X_1, \dots, X_n)) \le 2^n$ [Pfister1995, p. 97], where the lower bound follows from the Cassels-Ellison-Pfister theorem [CEP1971]. The bounds agree for $n = 2$, so the value is known for $n \le 2$ and open for every $n \ge 3$. The survey [MerkurjevParimala2025, §5.2] records the question as open even for $\mathbb{R}(X_1, X_2, X_3)$ (Question 5.7). *References:* - [Pfister1971] A. Pfister, *Sums of squares in real function fields*, pp. 297-300 in Actes du Congrès International des Mathématiciens, Tome 1 (Nice, 1970), Gauthier-Villars, 1971, [IMU scan](https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf). - [Pfister1967] A. Pfister, *Zur Darstellung definiter Funktionen als Summe von Quadraten*, Invent. Math. 4 (1967), 229-237, [doi:10.1007/BF01425382](https://doi.org/10.1007/BF01425382). - [Cassels1964] J. W. S. Cassels, *On the representation of rational functions as sums of squares*, Acta Arith. 9 (1964), 79-82, [doi:10.4064/aa-9-1-79-82](https://doi.org/10.4064/aa-9-1-79-82). - [Pfister1995] A. Pfister, *Quadratic forms with applications to algebraic geometry and topology*, London Math. Soc. Lecture Note Ser. 217, Cambridge University Press, 1995. - [CEP1971] J. W. S. Cassels, W. J. Ellison, A. Pfister, *On sums of squares and on elliptic curves over function fields*, J. Number Theory 3 (1971), 125-149, [doi:10.1016/0022-314X(71)90030-8](https://doi.org/10.1016/0022-314X(71)90030-8). - [Benoist2017] O. Benoist, *On Hilbert's 17th problem in low degree*, Algebra Number Theory 11 (2017), 929-959, [doi:10.2140/ant.2017.11.929](https://doi.org/10.2140/ant.2017.11.929), [arXiv:1602.07330](https://arxiv.org/abs/1602.07330). - [MerkurjevParimala2025] A. Merkurjev, R. Parimala, *Quadratic forms beyond arithmetic*, Notices Amer. Math. Soc. 72 (2025), no. 7, 711-718, [doi:10.1090/noti3192](https://doi.org/10.1090/noti3192).

V2-000572 | Pfister's problem on the Pythagoras number of $\mathbb{R}(X_1, \dots, X_n)$ | pfister_problem

research open | proof annotation: False | 官方代码 | 原文线索

Pfister's problem (Problem 1 of [Pfister1971, §4]): what is the true value of $p(\mathbb{R}(X_1, \dots, X_n))$, as a function of n? It is 1, 2, 4 for $n = 0, 1, 2$ (`pfister_problem.variants.zero`, `pfister_problem.variants.one`, `pfister_problem.variants.two`) and open for every $n \ge 3$, where only the bounds $n + 2 \le p(\mathbb{R}(X_1, \dots, X_n)) \le 2^n$ are known.

```
theorem pfister_problem :
∀ n, IsLeast (pythagorasBounds (MvRatFunc (Fin n) ℝ)) (pfister_problem.answer n)
```

FC-PfisterPythagorasNumber.pfister_problem

Prime Tuples Conjecture

MODULE CONTEXT: # Prime Tuples Conjecture *Reference:* [FLC07] Friedlander, J. B. and Luca, F. and Stoiciu, M., On the irrationality of a divisor function series. Integers (2007).

V2-000573 | Prime Tuples Conjecture | prime_tuples_conjecture

research open | proof annotation: False | 官方代码

For any `k ≥ 2`, let `a₁,...,aₖ` and `b₁,...,bₖ` be integers with `aᵢ > 0`. Suppose that for every prime `p` there exists an integer `n` such that `p ∤ ∏ i, (aᵢ n + bᵢ)`. Then there exist infinitely many `n` such that `aᵢ n + bᵢ` is prime for all `i`.

```
theorem prime_tuples_conjecture {k : ℕ} (hk : 2 ≤ k) (a : Fin k → ℕ+) (b : Fin k → ℕ)
(hab : ∀ p, p.Prime → ∃ n, ¬ p ∣ ∏ i, (a i * n + b i)) :
Set.Infinite {n | ∀ i : Fin k, (a i * n + b i).Prime}
```

FC-PrimeTuplesConjecture.prime_tuples_conjecture

Pudlák's conjectures on incompleteness in the finite domain

MODULE CONTEXT: # Pudlák's conjectures on incompleteness in the finite domain *References:* * [CR79] Cook, S. A. and Reckhow, R. A. (1979). "The relative efficiency of propositional proof systems." *J. Symbolic Logic* 44, pp. 36--50. * [HH88] Hartmanis, J. and Hemachandra, L. A. (1988). "Complexity classes without machines: on complete languages for UP." *Theoret. Comput. Sci.* 58, pp. 129--142. * [KP89] Krajíček, J. and Pudlák, P. (1989). "Propositional proof systems, the consistency of first order theories and the complexity of computations." *J. Symbolic Logic* 54, pp. 1063--1079. * [MP91] Megiddo, N. and Papadimitriou, C. H. (1991). "On total functions, existence theorems and computational complexity." *Theoret. Comput. Sci.* 81, pp. 317--324. * [Pa94] Papadimitriou, C. H. (1994). "On the complexity of the parity argument and other inefficient proofs of existence." *J. Comput. System Sci.* 48, pp. 498--532. * [KMT03] Köbler, J., Messner, J., and Torán, J. (2003). "Optimal proof systems imply complete sets for promise classes." *Inform. and Comput.* 184, pp. 71--92. * [Pu17] Pudlák, P. (2017). "Incompleteness in the finite domain." *Bull. Symbolic Logic* 23, pp. 405--441. [arXiv:1601.01487](https://arxiv.org/abs/1601.01487) * [Kh22] Khaniki, E. (2022). "New relations and separations of conjectures about incompleteness in the finite domain." *J. Symbolic Logic* 87, pp. 912--937. [arXiv:1904.01362](https://arxiv.org/abs/1904.01362)

V2-000574 | Pudlák's conjectures on incompleteness in the finite domain | np_inter_conp_no_complete_language

research open | proof annotation: False | 官方代码 | 原文线索

NP ∩ coNP has no complete language (Pudlák's conjecture `NP ∩ coNP` [Pu17]). No language in `NP ∩ coNP` is complete for that class under polynomial-time many-one reductions.

theorem np_inter_conp_no_complete_language :
¬∃ L : DecisionProblem, IsCompleteFor (NP ∩ coNP) L

FC-PudlakFiniteDomain.np_inter_conp_no_complete_language

V2-000575 | Pudlák's conjectures on incompleteness in the finite domain | sat_no_p_optimal_proof_system

research open | proof annotation: False | 官方代码 | 原文线索

SAT has no p-optimal proof system (Pudlák's conjecture `SAT` [Pu17]). No proof system for `SAT` p-simulates every other proof system for `SAT`. Since the existence of a p-optimal proof system depends only on the polynomial-time many-one degree of the language [KMT03], the conjecture is stated here for an arbitrary NP-complete language. Equivalently [Pu17], for every consistent polynomial-time axiomatized theory `T ⊇ S¹₂` there is a proof system for `SAT` whose soundness `T` cannot prove.

theorem sat_no_p_optimal_proof_system (L : DecisionProblem) (hL : IsCompleteFor NP L) :
¬∃ P : List Bool → List Bool, IsPOptimalProofSystem L P

FC-PudlakFiniteDomain.sat_no_p_optimal_proof_system

V2-000576 | Pudlák's conjectures on incompleteness in the finite domain | tfnp_no_complete_problem

research open | proof annotation: False | 官方代码 | 原文线索

TFNP has no complete problem (Megiddo-Papadimitriou [MP91]; Pudlák's conjecture `TFNP` [Pu17]). No total NP search problem is complete under polynomial-time many-one reductions of search problems. Equivalently [Pu17], for every consistent polynomial-time axiomatized theory `T ⊇ S¹₂` some total NP search problem is not polynomially reducible to any total NP search problem provably total in `T`.

theorem tfnp_no_complete_problem : ¬∃ T : TFNPProblem, T.IsComplete

FC-PudlakFiniteDomain.tfnp_no_complete_problem

V2-000577 | Pudlák's conjectures on incompleteness in the finite domain | up_no_complete_language

research open | proof annotation: False | 官方代码 | 原文线索

UP has no complete language (Hartmanis-Hemachandra [HH88]; Pudlák's conjecture `UP` [Pu17]). No language accepted by an unambiguous polynomial-time verifier is complete for `UP` under polynomial-time many-one reductions.

theorem up_no_complete_language : ¬∃ L : DecisionProblem, IsCompleteFor UP L

FC-PudlakFiniteDomain.up_no_complete_language

Equivalents to the Riemann Hypothesis

MODULE CONTEXT: # Equivalents to the Riemann Hypothesis This file contains formal statements that are known to be equivalent to the Riemann Hypothesis, following the survey [Co26], §4. The statement of the Riemann Hypothesis itself is in `FormalConjectures.Millennium.RiemannHypothesis`. *References:* * [Co26] A. Connes, [The Riemann Hypothesis: Past, Present and a Letter Through Time](https://arxiv.org/abs/2602.04022), 2026.

V2-000578 | Equivalents to the Riemann Hypothesis | beurling_nyman_criterion

research solved | proof annotation: False | 官方代码 | 原文线索

Beurling-Nyman Criterion. The Riemann Hypothesis is equivalent to the statement that the set of finite linear combination $\sum_{\nu = 1}^n c_{\nu}\rho_{\theta_{\nu}}$ is dense in $L^2(0, 1)$, when ranging over $0 < \theta_{\nu}\leq 1$ and $\sum_{\nu = 1}^n c_{\nu}\theta_{\nu} = 0$.

theorem beurling_nyman_criterion : RiemannHypothesis ↔
Dense { fractScalarInvSumL2 n c θ | (n) (c) (θ)
(hθ : ∀ ν, θ ν ∈ Set.Ioc 0 1) (h : ∑ ν, c ν * θ ν = 0) }

FC-RiemannHypothesis.beurling_nyman_criterion

V2-000579 | Equivalents to the Riemann Hypothesis | beurling_nyman_criterion_rhs

research open | proof annotation: False | 官方代码 | 原文线索

The statement that the Beurling-Nyman criterion holds.

theorem beurling_nyman_criterion_rhs :
Dense { fractScalarInvSumL2 n c θ | (n) (c) (θ)
(hθ : ∀ ν, θ ν ∈ Set.Ioc 0 1) (h : ∑ ν, c ν * θ ν = 0) }

FC-RiemannHypothesis.beurling_nyman_criterion_rhs

V2-000580 | Equivalents to the Riemann Hypothesis | lagarias_criterion

research solved | proof annotation: False | 官方代码 | 原文线索

Lagarias' Criterion. A refinement of Robin's criterion: the Riemann Hypothesis is equivalent to $$ \sigma(n) < H_n + e^{H_n}\log H_n $$ for all $n \geq 2$, where $H_n = 1 + 1/2 + \cdots + 1/n$ is the nth harmonic number. Note that [Co26] states this for all $n \geq 1$, but at $n = 1$ both sides equal 1; Lagarias' original formulation is the non-strict inequality for all $n \geq 1$, with equality only at $n = 1$.

theorem lagarias_criterion : RiemannHypothesis ↔
∀ n > 1, σ 1 n < harmonic n + exp (harmonic n) * log (harmonic n)

FC-RiemannHypothesis.lagarias_criterion

V2-000581 | Equivalents to the Riemann Hypothesis | lagarias_criterion_rhs

research open | proof annotation: False | 官方代码 | 原文线索

The statement that Lagarias' criterion holds.

```
theorem lagarias_criterion_rhs {n : ℕ} (hn : 1 < n) :
σ 1 n < harmonic n + exp (harmonic n) * log (harmonic n)
```

FC-RiemannHypothesis.lagarias_criterion_rhs

V2-000582 | Equivalents to the Riemann Hypothesis | li_criterion

research solved | proof annotation: False | 官方代码 | 原文线索

Li's Criterion. The Riemann Hypothesis is equivalent to the positivity of $$ \lambda_n := \sum_{\rho}\left(1 - \left(1 - \frac{1}{\rho}\right)^n\right) $$ for each $n > 0$.

```
theorem li_criterion : RiemannHypothesis ↔ ∀ n > 0, 0 < lambda n
```

FC-RiemannHypothesis.li_criterion

V2-000583 | Equivalents to the Riemann Hypothesis | li_criterion_rhs

research open | proof annotation: False | 官方代码 | 原文线索

The statement that Li's criterion holds.

```
theorem li_criterion_rhs {n : ℕ} (hn : 0 < n) : 0 < lambda n
```

FC-RiemannHypothesis.li_criterion_rhs

V2-000584 | Equivalents to the Riemann Hypothesis | robin_criterion

research solved | proof annotation: False | 官方代码 | 原文线索

Robin's Criterion. The Riemann Hypothesis is equivalent to $$ \sigma(n) < e ^ {\gamma} n\log\log n $$ for all $n > 5040$, where γ is the Euler-Mascheroni constant and $\sigma(n)$ is the sum of divisors.

```
theorem robin_criterion :
RiemannHypothesis ↔ ∀ n > 5040, σ 1 n < (exp eulerMascheroniConstant) * n * log (log n)
```

FC-RiemannHypothesis.robin_criterion

V2-000585 | Equivalents to the Riemann Hypothesis | robin_criterion_rhs

research open | proof annotation: False | 官方代码 | 原文线索

The statement that Robin's criterion holds.

```
theorem robin_criterion_rhs {n : ℕ} (hn : n > 5040) :
σ 1 n < exp eulerMascheroniConstant * n * log (log n)
```

FC-RiemannHypothesis.robin_criterion_rhs

V2-000586 | Equivalents to the Riemann Hypothesis | weil_positivity_criterion

research solved | proof annotation: False | 官方代码 | 原文线索

Weil's Positivity Criterion. The Riemann Hypothesis is equivalent to the non-positivity of Weil factor sums on the convolution of test functions. In [Co26] this is expressed as $$ \sum_v W_v(g\star g^*) \leq 0 $$ for all compactly-supported smooth functions $g : \mathbb{R}_+^{\times} \to \mathbb{C}$ such that the Mellin transform of g vanishes at $\pm i / 2$, where $g^*(x) := \bar{g}(x^{-1})$ and W_v are the multiplicative Weil factors defined in [Co26]. We formalise this as the corresponding additive statement (via $x = e^u$), since convolution in mathlib is additive, and restrict to real-valued test functions g, which suffices for the equivalence.

```
theorem weil_positivity_criterion :
RiemannHypothesis ↔ ∀ g : C_c(ℝ, ℝ), ContDiff ℝ ∞ g → IsTestFunction g →
weilArchimedean (g ⋆ fun x ↦ g (-x)) +
∑' p : Nat.Primes, weilNonArchimedean p (g ⋆ fun x ↦ g (-x)) ≤ 0
```

FC-RiemannHypothesis.weil_positivity_criterion

V2-000587 | Equivalents to the Riemann Hypothesis | weil_positivity_criterion_rhs

research open | proof annotation: False | 官方代码 | 原文线索

The statement that the Weil criterion holds.

```
theorem weil_positivity_criterion_rhs {g : C_c(ℝ, ℝ)} (hg : ContDiff ℝ ∞ g)
(h : IsTestFunction g) :
weilArchimedean (g ⋆ fun x ↦ g (-x)) +
∑' p : Nat.Primes, weilNonArchimedean p (g ⋆ fun x ↦ g (-x)) ≤ 0
```

FC-RiemannHypothesis.weil_positivity_criterion_rhs

Moments of the Riemann zeta function on the critical line

MODULE CONTEXT: # Moments of the Riemann zeta function on the critical line The moments $\frac{1}{T}\int_0^T |\zeta(1/2 + it)|^{2k}\,dt$ are conjectured to be asymptotic to $c_k (\log T)^{k^2}$ for some constants c_k. This is a theorem for $k = 1$ (Hardy-Littlewood, $c_1 = 1$) and $k = 2$ (Ingham, $c_2 = 1/(2\pi^2)$), and these are the only proven cases. Conrey-Ghosh and Conrey-Gonek conjectured $c_3 = 42 a_3 / 9!$ and $c_4 = 24024 a_4 / 16!$, with a_k the arithmetic factor $a_k = \prod_p (1 - 1/p)^{k^2} \sum_{m \ge 0} \binom{m+k-1}{m}^2 p^{-m}$; Keating-Snaith conjectured $c_k = a_k f_k$ with $f_k = \prod_{j=0}^{k-1} j!/(j+k)!$ from random matrix theory, recovering $g_k = (k^2)!\,f_k = 1, 2, 42, 24024$. The statements and constants below are taken from §3.4.5 of [Co26]. *References:* - [Co26] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022](https://arxiv.org/abs/2602.04022), §3.4.5. - [In26] A. E. Ingham, *Mean-value theorems in the theory of the Riemann zeta-function*, Proc. London Math. Soc. 27 (1926), 273-300. - [CG98] J. B. Conrey, A. Ghosh, *A conjecture for the sixth power moment of the Riemann zeta-function*, Internat. Math. Res. Notices 1998, no. 15, 775-780. - [CGo01] J. B. Conrey, S. Gonek, *High moments of the Riemann zeta-function*, Duke Math. J. 107 (2001), no. 3, 577-604. - [KS00] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2+it)$*, Commun. Math. Phys. 214 (2000), no. 1, 57-89.

V2-000588 | Moments of the Riemann zeta function on the critical line | moment_isEquiv

research open | proof annotation: False | 官方代码 | 原文线索

The asymptotic behaviour of the kth moment of the Riemann zeta function is conjectured to be $$ \frac{1}{T}\int_0^T |\zeta(1/2 + it)|^{2k}dt \sim c_k \log(T)^{k^2}, $$ for some constant c_k.

theorem moment_isEquiv (k : ℕ) : ∃ c > (0 : ℝ),
moment k ~[atTop] fun T ↦ c * T.log ^ (k ^ 2)

FC-RiemannZetaMoments.moment_isEquiv

V2-000589 | Moments of the Riemann zeta function on the critical line | moment_isEquiv₁

research solved | proof annotation: False | 官方代码 | 原文线索

The asymptotic behaviour of the first moment of the Riemann zeta function was proved by Hardy and Littlewood, where $c_1 = 1$ [Co26, §3.4.5].

theorem moment_isEquiv₁ : moment 1 ~[atTop] fun T ↦ T.log

FC-RiemannZetaMoments.moment_isEquiv₁

V2-000590 | Moments of the Riemann zeta function on the critical line | moment_isEquiv₂

research solved | proof annotation: False | 官方代码 | 原文线索

The asymptotic behaviour of the second moment of the Riemann zeta function was proved by Ingham [In26], where $c_2 = 1/(2\pi^2)$.

theorem moment_isEquiv₂ : moment 2 ~[atTop] fun T ↦ (1 / (2 * π ^ 2)) * T.log ^ 4

FC-RiemannZetaMoments.moment_isEquiv₂

V2-000591 | Moments of the Riemann zeta function on the critical line | moment_isEquiv₃

research open | proof annotation: False | 官方代码 | 原文线索

The constant in the asymptotic formula for the 3rd zeta moment is conjectured to be $42a_3/9!$ [CG98].

theorem moment_isEquiv₃ : moment 3 ~[atTop] fun T ↦ 42 * arithmeticFactor 3 / 9 ! * T.log ^ 9

FC-RiemannZetaMoments.moment_isEquiv₃

V2-000592 | Moments of the Riemann zeta function on the critical line | moment_isEquiv₄

research open | proof annotation: False | 官方代码 | 原文线索

The constant in the asymptotic formula for the 4th zeta moment is conjectured to be $24024a_4/16!$ [CGo01].

theorem moment_isEquiv₄ :
moment 4 ~[atTop] fun T ↦ 24024 * arithmeticFactor 4 / 16 ! * T.log ^ 16

FC-RiemannZetaMoments.moment_isEquiv₄

V2-000593 | Moments of the Riemann zeta function on the critical line | moment_isEquivₖ

research open | proof annotation: False | 官方代码 | 原文线索

The Keating-Snaith conjecture [KS00] for the kth moment of the Riemann zeta function, provides the value of c_k in the asymptotic formula [Co26, §3.4.5].

theorem moment_isEquivₖ (k : ℕ) : moment k ~[atTop]
fun T ↦ (arithmeticFactor k * ∏ j ∈ Finset.range k, (j ! : ℝ) / (j + k) !) *
T.log ^ (k ^ 2)

FC-RiemannZetaMoments.moment_isEquivₖ

Ringel's Conjecture

MODULE CONTEXT: # Ringel's Conjecture *Reference:* G. Ringel, *Problem 25*, in *Theory of Graphs and its Applications* (Proc. Sympos. Smolenice, 1963), Academia, Prague, 1964. Ringel's conjecture (1963): the complete graph K_{2n+1} decomposes into copies of any tree with n edges. It remains open; the case of all sufficiently large n is proved by Montgomery-Pokrovskiy-Sudakov, see `Arxiv/2001.02665/RingelConjecture.lean`.

V2-000594 | Ringel's Conjecture | ringel_conjecture

research open | proof annotation: False | 官方代码

For any tree T with n edges, the complete graph K_{2n+1} decomposes into $2n+1$ edge-disjoint copies of T. A "copy" of T is the image $T.\text{map}(f_i)$ of T under a vertex embedding $f_i : V \hookrightarrow \text{Fin}(2n+1)$; the copies are pairwise edge-disjoint and together cover every edge of K_{2n+1}.

```
theorem ringel_conjecture {V : Type} [Finite V]
(T : SimpleGraph V) (hT : T.IsTree)
(n : ℕ) (hn : T.edgeSet.ncard = n) :
∃ f : Fin (2 * n + 1) → (V ↪ Fin (2 * n + 1)),
Pairwise (fun i j => Disjoint (T.map (f i)).edgeSet (T.map (f j)).edgeSet) ∧
⨆ i, T.map (f i) = (⊤ : SimpleGraph (Fin (2 * n + 1)))
```

FC-RingelConjecture.ringel_conjecture

Serre's uniformity conjecture over the rationals

MODULE CONTEXT: # Serre's uniformity conjecture over the rationals Is there a bound C, independent of the non-CM elliptic curve $E/\mathbb{Q}$, such that the Galois action on $E[p]$ is surjective onto $\mathrm{GL}_2(\mathbb{F}_p)$ for every prime $p > C$? This is the $\mathbb{Q}$ case of Serre's question [Ser72], recalled in the introduction of [Lem17]. We express surjectivity without choosing a basis: every additive automorphism of $E[p]$ must be induced by an element of $G_{\mathbb{Q}}$. For prime p, these automorphisms are exactly the $\mathbb{F}_p$-linear automorphisms. The non-CM condition uses the classification of rational CM j-invariants. *References:* - [Ser72] J.-P. Serre, *Propriétés galoisiennes des points d'ordre fini des courbes elliptiques*. Inventiones Mathematicae 15 (1972), 259-331. https://doi.org/10.1007/BF01405086 - [Lem17] P. Lemos, *Serre's uniformity conjecture for elliptic curves with rational cyclic isogenies*, introduction. https://arxiv.org/abs/1702.01985

V2-000595 | Serre's uniformity conjecture over the rationals | serre_uniformity

research open | proof annotation: False | 官方代码 | 原文线索

Serre's uniformity question over $\mathbb{Q}$ [Ser72, Lem17]: is there a bound C such that every non-CM elliptic curve over $\mathbb{Q}$ has surjective mod-p Galois representation for every prime $p > C$?

```
theorem serre_uniformity :
∃ C : ℕ, ∀ (E : WeierstrassCurve ℚ) [E.IsElliptic], E.j ∉ cmJInvariants →
∀ p : ℕ, p.Prime → C < p → HasFullTorsionAction E p
```

FC-SerreUniformity.serre_uniformity

Shelah-Spencer's Zero-One Law for Sparse Random Graphs

MODULE CONTEXT: # Shelah-Spencer's Zero-One Law for Sparse Random Graphs *References*: * [Shelah and Spencer, Zero-one laws for sparse random graphs](https://doi.org/10.1090/S0894-0347-1988-0924703-8), Journal of the American Mathematical Society, 1(1), 97-115 (1988). * [Glebskii et al., Range and degree of realizability of formulas in the restricted predicate calculus](https://doi.org/10.1007/BF01071084), Cybernetics, 5(2), 142-154 (1969) (for $\alpha = 0$).

V2-000596 | Shelah-Spencer's Zero-One Law for Sparse Random Graphs | zeroOne_irrational

research solved | proof annotation: False | 官方代码 | 原文线索

If $0 < \alpha < 1$ is irrational, then for every first-order sentence φ in the language of graphs, the probability that $G(n, n^{-\alpha})$ satisfies φ tends to either zero or one as $n \to \infty$.

```
theorem zeroOne_irrational
(α : ℝ) (hα : 0 < α ∧ α < 1) (hirr : Irrational α) (φ : Language.graph.Sentence) :
Tendsto (fun n =>
sparseRandomGraph n α hα.1
{G | @Language.Sentence.Realize Language.graph (Fin n) G.structure φ}) atTop (𝓝 0) ∨
Tendsto (fun n =>
sparseRandomGraph n α hα.1
{G | @Language.Sentence.Realize Language.graph (Fin n) G.structure φ}) atTop (𝓝 1)
```

FC-ShelahSpencerZeroOneLaw.zeroOne_irrational

V2-000597 | Shelah-Spencer's Zero-One Law for Sparse Random Graphs | zeroOne_rational

research solved | proof annotation: False | 官方代码 | 原文线索

If $0 < \alpha < 1$ is rational, then there is a first-order sentence φ in the language of graphs whose probability in $G(n, n^{-\alpha})$ does not converge as $n \to \infty$.

```
theorem zeroOne_rational
(α : ℝ) (hα : 0 < α ∧ α < 1) (hrat : ¬Irrational α) :
∃ φ : Language.graph.Sentence,
∀ b : ENNReal,
¬ Tendsto (fun n =>
sparseRandomGraph n α hα.1
{G | @Language.Sentence.Realize Language.graph (Fin n) G.structure φ}) atTop (𝓝 b)
```

FC-ShelahSpencerZeroOneLaw.zeroOne_rational

Size, Order, and Connected Domination

MODULE CONTEXT: # Size, Order, and Connected Domination *References:* - [S. Mukwembi, _Size, order, and connected domination_, Canad. Math. Bull. 57 (2014), no. 1, 141-144](https://doi.org/10.4153/CMB-2013-020-5)

V2-000598 | Size, Order, and Connected Domination | mukwembi_theorem_2_1

research solved | proof annotation: True | 官方代码 | 原文线索

Theorem 2.1 of [S. Mukwembi, _Size, order, and connected domination_, Canad. Math. Bull. 57 (2014), no. 1, 141-144](https://doi.org/10.4153/CMB-2013-020-5) claims: if G is a connected triangle-free graph of order n and size m with connected domination number γ_c, then $$m \le \frac{(n - \gamma_c)^2}{4} + n - 1.$$ The claim is **false**: the 3-dimensional hypercube Q_3 is a counterexample, with $n = 8$, $m = 12$ and $\gamma_c = 4$, so the asserted bound reads $12 \le (8-4)^2/4 + 8 - 1 = 11$. The gap in the paper's proof (p. 143) is the unjustified assertion that there is an edge uv with $\gamma_c(G) \le \gamma_c(G - \{u, v\})$: in Q_3, removing any adjacent pair of vertices leaves a graph with connected domination number $2 < 4$. The corollaries of the paper (Corollary 2.2 and 2.3, on leaf numbers of triangle-free graphs) remain true; Corollary 2.2 is Graffiti.pc Conjecture 1.1, recorded as `WrittenOnTheWallII.GraphConjecture2.conjecture2`.

```
theorem mukwembi_theorem_2_1 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj],
G.Connected → G.CliqueFree 3 →
(G.edgeFinset.card : ℝ) ≤
((Fintype.card α : ℝ) - (G.connectedDominationNumber : ℝ)) ^ 2 / 4
+ (Fintype.card α : ℝ) - 1
```

FC-SizeOrderConnectedDomination.mukwembi_theorem_2_1

Strong Sensitivity Conjecture (`bs(f) ≤ s(f)^2`)

MODULE CONTEXT: # Strong Sensitivity Conjecture (`bs(f) ≤ s(f)^2`) This file formalizes the *strong* sensitivity conjecture, asserting: For every Boolean function `f : {0,1}^n → {0,1}`, `bs(f) ≤ s(f)^2`, where bs(f) denotes block sensitivity and s(f) denotes sensitivity. Huang's theorem proves a *quartic* upper bound, `bs(f) ≤ s(f)^4`, thereby resolving the most widely known form of the sensitivity conjecture. We now ask whether a stronger upper bound holds. Interestingly, the original paper of Nisan and Szegedy, where the sensitivity conjecture first appeared, already speculated that a *quadratic* upper bound might be the correct relation. On the lower bound side, Rubinstein (https://link.springer.com/article/10.1007/BF01200762) constructed Boolean functions exhibiting the first quadratic separation. The best gap known before [OAI26], due to Ambainis and Sun (https://arxiv.org/abs/1108.3494), was `bs(f) ≥ (2/3)·s(f)^2`. In September 2026 an internal OpenAI model disproved the quadratic bound [OAI26]: it constructs total Boolean functions with `bs(f) / s(f)^2` arbitrarily large. *References:* * [OAI26] [A superquadratic separation between sensitivity and block sensitivity](https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-superquadratic-separation-between-sensitivity-and-block-sensitivity-September-25-2026/paper.pdf) by OpenAI, OpenAI Math Release preprint (2026) * [Induced Subgraphs of Hypercubes and a Proof of the Sensitivity Conjecture](https://arxiv.org/abs/1907.00847) by Hao Huang (see Section 3, Concluding Remarks) * [Variations on the Sensitivity Conjecture](https://arxiv.org/abs/1011.0354) by Pooya Hatami, Raghav Kulkarni, and Denis Pankratov (see Question 3.1) * [On the Degree of Boolean Functions as Real Polynomials](https://link.springer.com/article/10.1007/BF01263419) by Noam Nisan, and Mario Szegedy (see Section 4, Open Problems)

V2-000599 | Strong Sensitivity Conjecture (`bs(f) ≤ s(f)^2`) | strong_sensitivity_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Strong Sensitivity Conjecture, for every Boolean function `f : {0,1}^n → {0,1}`, `bs(f) ≤ s(f)^2`. We call this the *strong* sensitivity conjecture because the original sensitivity conjecture only asked for a polynomial bound in terms of `s(f)`. Huang's celebrated result (often called the sensitivity theorem) gives a quartic bound, `bs(f) ≤ s(f)^4`, thereby settling the original conjecture. The quadratic bound is false. This was disproved by an internal OpenAI model in September 2026 [OAI26]; see the linked Lean proof.

```
theorem strong_sensitivity_conjecture :
¬ ∀ (n : ℕ) (f : (Fin n → Bool) → Bool), blockSensitivity f ≤ sensitivity f ^ 2
```

FC-StrongSensitivityConjecture.strong_sensitivity_conjecture

Tu-Deng Conjecture

MODULE CONTEXT: # Tu-Deng Conjecture *References:* * [Tu, Z., Deng, Y., *A conjecture about binary strings and its applications on constructing Boolean functions with optimal algebraic immunity*, Des. Codes Cryptogr. **60** (2011), 1-14](https://doi.org/10.1007/s10623-010-9413-9) * [IACR ePrint 2009/272](https://eprint.iacr.org/2009/272) * [LLX26] [Liu, R., Luo, H., Xie, T., *A Complete Proof for Tu-Deng Conjecture*, arXiv:2608.05187v2 (2026)](https://arxiv.org/abs/2608.05187v2) * [Cu26] [Cusick, T. W., *Proof of the TuDeng Conjecture*, arXiv:2608.14821 (2026)](https://arxiv.org/abs/2608.14821) * [Lean26] [A Lean 4 proof of the count form of the conjecture](https://github.com/ifeelok92/tu-deng-conjecture-lean-formal-proof/blob/3c00561415deedcdc56b9e803042c1d0f15f5d5b/TuDengFormal/FirstExitNat.lean#L529-L535), commit `3c005614`, theorem `TuDeng.FirstExit.tu_deng_conjecture_full` For an integer $k \ge 2$, identify the residues modulo $2^k - 1$ with the integers $0, 1, \dots, 2^k - 2$, and let $w(a)$ denote the binary weight of a (the number of ones in its binary expansion). The Tu-Deng conjecture states that for every residue $t \ne 0$, $$\#\{(a, b) : a + b \equiv t \pmod{2^k - 1},\ w(a) + w(b) \le k - 1\} \le 2^{k-1}.$$ Tu and Deng showed that this combinatorial statement implies that their constructions of Boolean functions (a bent class and a balanced class) achieve optimal algebraic immunity, which is the motivation for the conjecture. They verified the conjecture numerically for $k \le 29$ (Remark 3.1 of the ePrint version). The conjecture is no longer open: [LLX26] and, independently, [Cu26] give complete proofs. [Lean26] is a Lean 4 development of the same bound, stated for the count of natural-number representatives $x < 2^k - 1$ rather than for the set of residue pairs used below.

V2-000600 | Tu-Deng Conjecture | tu_deng_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The Tu-Deng conjecture. For $k \ge 2$ and a nonzero residue t modulo $2^k - 1$, there are at most 2^{k-1} pairs of residues (a, b) with $a + b = t$ whose binary weights (of their representatives in $0, \dots, 2^k - 2$) sum to at most $k - 1$. Proved in [LLX26] and independently in [Cu26]; see also the Lean development [Lean26].

```
theorem tu_deng_conjecture (k : ℕ) (hk : 2 ≤ k) (t : ZMod (2 ^ k - 1)) (ht : t ≠ 0) :
{p : ZMod (2 ^ k - 1) × ZMod (2 ^ k - 1) |
p.1 + p.2 = t ∧ binaryWeight p.1.val + binaryWeight p.2.val ≤ k - 1}.ncard
≤ 2 ^ (k - 1)
```

FC-TuDengConjecture.tu_deng_conjecture

Weak tiling problems

MODULE CONTEXT: # Weak tiling problems Problems 4.1, 4.2, and 4.3 from [arxiv/2506.23631](https://arxiv.org/abs/2506.23631). *Reference:* * [Geometric implications of weak tiling](https://arxiv.org/abs/2506.23631) See also `FormalConjectures.Wikipedia.Fuglede` for Fuglede's spectral set conjecture, which motivates the study of weak tilings.

V2-000601 | Weak tiling problems | problem_4_1

research solved | proof annotation: True | 官方代码 | 原文线索

Problem 4.1. Let $\Omega \subset \mathbb{R}$ be a finite union of intervals and ν a weak tiling measure for Ω. Must $\mathrm{supp}(\nu)$ have bounded density? The answer is positive: Kenta Kitamura (KitaKen1 on GitHub) has given a Lean proof ([weak-tiling-bounded-density](https://github.com/KitaKen1/weak-tiling-bounded-density)).

```
theorem problem_4_1 :
∀ (Ω : Set ℝ) (_ : IsFiniteUnionOfIntervals Ω)
(ν : Measure ℝ) (_ : IsWeakTilingMeasure Ω ν), HasBoundedDensity ν.support
```

FC-WeakTiling.problem_4_1

V2-000602 | Weak tiling problems | problem_4_2

research open | proof annotation: False | 官方代码 | 原文线索

Problem 4.2. Let $\Omega \subset \mathbb{R}$ be a finite union of three or more intervals. If Ω weakly tiles its complement, must it also tile its complement properly?

```
theorem problem_4_2 :
∀ (n : ℕ) (_ : 3 ≤ n) (Ω : Set ℝ)
(_ : IsUnionOfNIntervals n Ω) (ν : Measure ℝ) (_ : IsWeakTilingMeasure Ω ν),
∃ T : Set ℝ, IsProperTiling Ω T
```

FC-WeakTiling.problem_4_2

V2-000603 | Weak tiling problems | problem_4_3

research solved | proof annotation: True | 官方代码 | 原文线索

Problem 4.3. Let $\Omega \subset \mathbb{R}$ be a finite union of intervals and ν a weak tiling measure for Ω. Must ν be expressible as a convex combination of proper tiling measures? The answer is negative. An AI-assisted investigation by Kenta Kitamura produced a counterexample for $\Omega = (0,1) \cup (2,3) \cup (12,13) \cup (30,31)$. A finite computation shows that every convex combination of proper tiling measures for this Ω assigns equal mass to $\{7\}$ and $\{15\}$. However, the constructed weak tiling measure ν satisfies $\nu(\{7\}) = \frac{1}{2}$ and $\nu(\{15\}) = 0$, so ν cannot be expressed as a convex combination of proper tiling measures.

```
theorem problem_4_3 :
¬ ∀ (Ω : Set ℝ) (_ : IsFiniteUnionOfIntervals Ω)
(ν : Measure ℝ) (_ : IsWeakTilingMeasure Ω ν),
∃ (T : ℕ → Set ℝ) (c : ℕ → ℝ≥0), (∀ i, IsProperTiling Ω (T i)) ∧ ∑' i : ℕ, c i = 1 ∧
ν = Measure.sum
(fun i => (c i : ℝ≥0∞) • Measure.sum (fun t : T i => Measure.dirac (t : ℝ)))
```

FC-WeakTiling.problem_4_3

Conjectures about Weakly First Countable spaces

MODULE CONTEXT: # Conjectures about Weakly First Countable spaces This file formalizes the notion of a weakly first countable topological space and some conjectures around those. *References:* * [Ar2013] Arhangeliski, Alexandr. "Selected old open problems in general topology." Buletinul Academiei de Ştiinţe a Republicii Moldova. Matematica 73.2-3 (2013): 37-46. https://www.math.md/files/basm/y2013-n2-3/y2013-n2-3-(pp37-46).pdf.pdf * [Ya1976] Yakovlev, N. N. "On the theory of o-metrizable spaces." Doklady Akademii Nauk. Vol. 229. No. 6. Russian Academy of Sciences, 1976. https://www.mathnet.ru/links/016f74007f9f96fa3aadae05cbd98457/dan40570.pdf (in Russian)

V2-000604 | Conjectures about Weakly First Countable spaces | cardinalMk_le_continuum_of_weaklyFirstCountable_of_countableSouslinNumber

research open | proof annotation: False | 官方代码 | 原文线索

Problem 4 in [Ar2013]: If a Tychonoff weakly first-countable space has countable Souslin number, then does its cardinality not exceed the continuum?

```
theorem cardinalMk_le_continuum_of_weaklyFirstCountable_of_countableSouslinNumber :
∀ (X : Type) (_ : TopologicalSpace X), T35Space X →
WeaklyFirstCountableTopology X → HasCountableSouslinNumber X → #X ≤ 𝔠
```

FC-WeaklyFirstCountable.cardinalMk_le_continuum_of_weaklyFirstCountable_of_countableSouslinNumber

V2-000605 | Conjectures about Weakly First Countable spaces | existsWeaklyFirstCountableCompactBig

research open | proof annotation: False | 官方代码 | 原文线索

Problem 2 in [Ar2013]: Give an example in ZFC of a weakly first- countable compact Hausdorff space X such that $𝔠 < |X|$. Note: [Ar2013] uses a blanket convention that all spaces are Tychonoff and "compact" means compact Hausdorff.

```
theorem existsWeaklyFirstCountableCompactBig :
∃ (X : Type) (_ : TopologicalSpace X),
WeaklyFirstCountableTopology X ∧ CompactSpace X ∧ T2Space X ∧
ℵ < #X
```

FC-WeaklyFirstCountable.existsWeaklyFirstCountableCompactBig

V2-000606 | Conjectures about Weakly First Countable spaces | existsWeaklyFirstCountableCompactNotFirstCountable

research open | proof annotation: False | 官方代码 | 原文线索

Problem 3 in [Ar2013]: Give an example in ZFC of a weakly first- countable compact Hausdorff space which is not first countable. Note: [Ar2013] uses a blanket convention that all spaces are Tychonoff and "compact" means compact Hausdorff.

```
theorem existsWeaklyFirstCountableCompactNotFirstCountable :
∃ (X : Type) (_ : TopologicalSpace X),
WeaklyFirstCountableTopology X ∧ CompactSpace X ∧ T2Space X ∧
¬ FirstCountableTopology X
```

FC-WeaklyFirstCountable.existsWeaklyFirstCountableCompactNotFirstCountable

Zagier's Conjecture on Multiple Zeta Values

MODULE CONTEXT: # Zagier's Conjecture on Multiple Zeta Values *References:* - [Za94] Zagier, Don. "Values of zeta functions and their applications." First European Congress of Mathematics Paris, July 6-10, 1992: Vol. II: Invited Lectures (Part 2). Basel: Birkhäuser Basel, 1994. - [Co18] Combariza, Germán AG. "A few conjectures about the multiple zeta values." ACM Communications in Computer Algebra 52.1 (2018): 11-20. - [Te02] T. Terasoma. Mixed Tate motives and multiple zeta values. Invent. Math., 149(2):339-369, 2002. - [DG05] P. Deligne and A. Goncharov. Groupes fondamentaux motiviques de Tate mixte. Ann. Sci. Ecole Norm. Sup. (4), 38(1):1-56, 2005. - [OEIS A000931](https://oeis.org/A000931)

V2-000607 | Zagier's Conjecture on Multiple Zeta Values | zagier_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Zagier's conjecture The $\mathbb{Q}$-dimension of the vector space spanned by all multiple zeta values of weight n equals d_n, where d_n is the Zagier dimension sequence satisfying $d_0 = 1$, $d_1 = 0$, $d_2 = 1$, and $d_n = d_{n-2} + d_{n-3}$ for $n \geq 3$.

```
theorem zagier_conjecture :
∀ n : ℕ, Module.finrank ℚ (mzvSpanOfWeight n) = zagierDim n
```

FC-ZagierMZV.zagier_conjecture

V2-000608 | Zagier's Conjecture on Multiple Zeta Values | zagier_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Upper bound [Te02, DG05] The dimension of the $\mathbb{Q}$-vector space of MZVs of weight n is at most d_n.

```
theorem zagier_upper_bound :
∀ n : ℕ, Module.finrank ℚ (mzvSpanOfWeight n) ≤ zagierDim n
```

FC-ZagierMZV.zagier_upper_bound

A08 Wikipedia | 255 条

Non-Power-of-2 Almost Perfect Numbers Conjecture

MODULE CONTEXT: # Non-Power-of-2 Almost Perfect Numbers Conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Almost_perfect_number) - [mathworld](https://mathworld.wolfram.com/AlmostPerfectNumber.html) -

V2-000609 | Non-Power-of-2 Almost Perfect Numbers Conjecture | exists_almost_perfect_not_power_of_two

research open | proof annotation: False | 官方代码 | 原文线索

Non-Power-of-2 Almost Perfect Numbers Conjecture. Does there exist an almost perfect number that is not a power of 2?

theorem exists_almost_perfect_not_power_of_two :
∃ n : ℕ, AlmostPerfect n ∧ ¬ ∃ k : ℕ, n = 2^k

FC-AlmostPerfectNumbers.exists_almost_perfect_not_power_of_two

Alspach's conjecture on cycle decompositions of complete graphs (1981; proved 2014)

MODULE CONTEXT: # Alspach's conjecture on cycle decompositions of complete graphs (1981; proved 2014) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Alspach%27s_conjecture) * [Al81] Alspach, B. (1981). "Research problems, Problem 3." *Discrete Math.* 36, p. 333. * [BHP14] Bryant, D., Horsley, D. and Pettersson, W. (2014). "Cycle decompositions V: Complete graphs into cycles of arbitrary lengths." *Proc. London Math. Soc.* 108, pp. 1153--1192. [arXiv:1204.3709](https://arxiv.org/abs/1204.3709) * [GJKKO21] Glock, S., Joos, F., Kim, J., Kühn, D. and Osthus, D. (2021). "Resolution of the Oberwolfach problem." *J. Eur. Math. Soc.* 23, pp. 2511--2547. [arXiv:1806.04644](https://arxiv.org/abs/1806.04644)

V2-000610 | Alspach's conjecture on cycle decompositions of complete graphs (1981; proved 2014) | alspach_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Alspach's conjecture (1981), proved by Bryant, Horsley and Pettersson (2014). Let n be odd and let $m_1, \dots, m_t$ be integers with $3 \le m_i \le n$ and $m_1 + \dots + m_t = \binom{n}{2}$. Then the complete graph K_n decomposes into cycles of lengths $m_1, \dots, m_t$. (These conditions are clearly necessary: K_n has all degrees even only for odd n, and its edges must be split exactly among the cycles.)

theorem alspach_conjecture :
∀ (n : ℕ), Odd n → ∀ m : Multiset ℕ, (∀ x ∈ m, 3 ≤ x ∧ x ≤ n) → m.sum = n.choose 2 →
∃ C : Multiset (Cycle (completeGraph (Fin n))),
IsCycleDecomposition (completeGraph (Fin n)) C ∧ lengths C = m

FC-AlspachConjecture.alspach_conjecture

Andrica's conjecture

MODULE CONTEXT: # Andrica's conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Andrica%27s_conjecture) - [Luan Alberto Ferreira, *Real exponential sums over primes and prime gaps*](https://arxiv.org/abs/2307.08725)

V2-000611 | Andrica's conjecture | andrica_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Andrica's conjecture The inequality $\sqrt{p_{n+1}}-\sqrt{p_n} < 1$ holds for all n, where p_n is the n-th prime number.

theorem andrica_conjecture (n : ℕ) :
Real.sqrt ((n+1).nth Nat.Prime) - Real.sqrt (n.nth Nat.Prime) < 1

FC-Andrica.andrica_conjecture

Balanced prime conjecture

MODULE CONTEXT: # Balanced prime conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Balanced_prime) - [OEIS A6562](https://oeis.org/A6562)

V2-000612 | Balanced prime conjecture | balanced_primes

research open | proof annotation: False | 官方代码 | 原文线索

Let p_k be the k-th prime number. Are there infinitely many n such that $(p_n + p_{n+2}) / 2$ is prime?

theorem balanced_primes :
{n : ℕ | Prime ((Nat.nth Prime n + Nat.nth Prime (n + 2)) / 2)}.Infinite

FC-BalancedPrimes.balanced_primes

V2-000613 | Balanced prime conjecture | balanced_primes_order

research open | proof annotation: False | 官方代码 | 原文线索

Let p_k be the k-th prime number. Are there infinitely many n such that $p_n = \dfrac{\sum_{i = 1} ^ k p_{n - i} + p_{n + i}}{2*k}$?

theorem balanced_primes_order :
∀ k > 0, {n : ℕ | k ≤ n ∧ 2 * k * Nat.nth Prime n = ∑ i ∈ .Ioc 0 k,
((n - i).nth Prime + (n + i).nth Prime)}.Infinite

FC-BalancedPrimes.balanced_primes_order

Bateman-Horn Conjecture

MODULE CONTEXT: # Bateman-Horn Conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Bateman%E2%80%93Horn_conjecture)

V2-000614 | Bateman-Horn Conjecture | bateman_horn_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Bateman-Horn Conjecture Given a finite collection of distinct irreducible polynomials non-constant $f_1, f_2, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients that satisfy the Schinzel condition, the number of positive integers $n \leq x$ for which all polynomials f_i are simultaneously prime is asymptotic to: $$\frac{C}{D} \frac{x}{(\log x)^k}$$ where $D = \prod_i \deg(f_i)$ and C is the Bateman-Horn constant given by the convergent infinite product: $$C = \prod_{p\in\mathbb{P}} (1 - 1/p)^{-k} (1 - \omega_p/p)$$ Here ω_p is the number of residue classes modulo p for which at least one polynomial vanishes. The product is only conditionally convergent: it is the limit as $N \to \infty$ of the partial products over the primes $p < N$. The Schinzel condition ensures that for each prime p, there exists some integer n such that p does not divide the product $f_1(n) f_2(n) \dotsb f_k(n)$, which guarantees the infinite product converges to a positive value.

```
theorem bateman_horn_conjecture
(polys : Finset ℤ[X])
(h_nonempty : polys.Nonempty)
(h_irreducible : ∀ f ∈ polys, BunyakovskyCondition f)
(h_compat : SchinzelCondition polys) :
∃ C : ℝ, 0 < C ∧ Tendsto (BatemanHornPartialProduct polys) atTop (𝓝 C) ∧
(fun x : ℝ => (CountSimultaneousPrimes polys x : ℝ)) ~[atTop]
(fun x : ℝ => C / DegreesProduct polys * x / (Real.log x) ^ polys.card)
```

FC-BatemanHornConjecture.bateman_horn_conjecture

Beck-Fiala theorem and conjecture

MODULE CONTEXT: # Beck-Fiala theorem and conjecture Discrepancy of bounded-degree set systems. Given sets $S_1, \dots, S_m \subseteq [n]$ such that every element of $[n]$ belongs to at most t of the sets (the system has *degree* at most t), one seeks a colouring $\chi \colon [n] \to \{-1, +1\}$ making every set as balanced as possible, i.e. minimizing the *discrepancy* $\max_i \left|\sum_{j \in S_i} \chi(j)\right|$. The Beck-Fiala theorem (1981) states that every set system of degree at most $t \ge 1$ has discrepancy at most $2t - 1$. The Beck-Fiala conjecture asserts that the truth is much stronger: the discrepancy of a degree-t system is $O(\sqrt{t})$, with a constant independent of n, m and t. Despite considerable attention the bound $2t - 1$ has been improved only slightly: Bukh (2016) proved a bound of the form $2t - \log^* t$ (where $\log^*$ is the iterated logarithm), and Banaszczyk's vector balancing theorem yields $O(\sqrt{t \log n})$. The Komlós conjecture (see `KomlosConjecture.lean`) would imply the Beck-Fiala conjecture, since scaling the incidence vectors of a degree-t system by $1/\sqrt{t}$ produces vectors of Euclidean norm at most 1. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Beck%E2%80%93Fiala_theorem) - [J. Beck and T. Fiala, *"Integer-making" theorems*, Discrete Applied Mathematics **3** (1981), 1-8](https://doi.org/10.1016/0166-218X(81)90022-6) - [B. Bukh, *An improvement of the Beck-Fiala theorem*, Combinatorics, Probability and Computing **25** (2016), 380-398](https://doi.org/10.1017/S0963548315000140) - [W. Banaszczyk, *Balancing vectors and Gaussian measures of n-dimensional convex bodies*, Random Structures & Algorithms **12** (1998), 351-360](https://doi.org/10.1002/(SICI)1098-2418(199807)12:4%3C351::AID-RSA3%3E3.0.CO;2-S)

V2-000615 | Beck-Fiala theorem and conjecture | beck_fiala_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Beck-Fiala conjecture There exists a universal constant $C > 0$ such that every set system $S_1, \dots, S_m \subseteq [n]$ of degree at most t admits a colouring $\chi \colon [n] \to \{-1, +1\}$ with $\left|\sum_{j \in S_i} \chi(j)\right| \le C \sqrt{t}$ for every i.

```
theorem beck_fiala_conjecture :
∃ C : ℝ, 0 < C ∧ ∀ (n m t : ℕ) (S : Fin m → Finset (Fin n)),
(∀ j, (Finset.univ.filter fun i => j ∈ S i).card ≤ t) →
∃ χ : Fin n → ℝ, (∀ j, χ j = 1 ∨ χ j = -1) ∧
∀ i, |∑ j ∈ S i, χ j| ≤ C * Real.sqrt t
```

FC-BeckFiala.beck_fiala_conjecture

V2-000616 | Beck-Fiala theorem and conjecture | beck_fiala_theorem

research solved | proof annotation: True | 官方代码 | 原文线索

The Beck-Fiala theorem If $S_1, \dots, S_m \subseteq [n]$ is a set system of degree at most t, i.e. every $j \in [n]$ lies in at most t of the sets, and $t \ge 1$, then there is a colouring $\chi \colon [n] \to \{-1, +1\}$ with $\left|\sum_{j \in S_i} \chi(j)\right| \le 2t - 1$ for every i. The hypothesis $t \ge 1$ is necessary: a system of degree 0 consists of empty sets only, whose discrepancy is $0 > 2 \cdot 0 - 1$. [J. Beck and T. Fiala, *"Integer-making" theorems*, Discrete Applied Mathematics **3** (1981), 1-8.]

```
theorem beck_fiala_theorem (n m t : ℕ) (ht : 1 ≤ t) (S : Fin m → Finset (Fin n))
(hdeg : ∀ j, (Finset.univ.filter fun i => j ∈ S i).card ≤ t) :
∃ χ : Fin n → ℝ, (∀ j, χ j = 1 ∨ χ j = -1) ∧
∀ i, |∑ j ∈ S i, χ j| ≤ 2 * (t : ℝ) - 1
```

FC-BeckFiala.beck_fiala_theorem

Bloch and Landau constants

MODULE CONTEXT: # Bloch and Landau constants *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Bloch%27s_theorem_(complex_analysis)) - [CP96] Chen, H., Gauthier, P. M. "On Bloch's constant." Journal d'Analyse Mathématique 69 (1996), 275-291. - [AG37] Ahlfors, L. V., Grunsky, H. "Über die Blochsche Konstante." Mathematische Zeitschrift 42 (1937), 671-673. - [Ya95] Yanagihara, H. "On the locally univalent Bloch constant." Journal d'Analyse Mathématique 65 (1995), 1-17. - [Ra43] Rademacher, H. "On the Bloch-Landau Constant."" American Journal of Mathematics 65 (1943), 387-390. - [OptimizationConstants](https://teorth.github.io/optimizationproblems/constants/57c.html) - [Skin2009] Skinner, Brian. The univalent Bloch constant problem. Complex Variables and Elliptic Equations 54 (2009), no. 10, 951-955. - [MathWorld](https://mathworld.wolfram.com/BlochConstant.html) - [Bhowmik-Sen](https://www.cambridge.org/core/journals/canadian-mathematical-bulletin/article/improved-bloch-and-landau-constants-for-meromorphic-functions/FD465D1F2CEF7E8C62AFF16C3E89B7B4) - [Re2008] Rettinger, R. ["Bloch's constant is computable."](https://jucs.org/jucs_14_6/blochs_constant_is_computable/jucs_14_06_0896_0907_rettinger.pdf) Journal of Universal Computer Science 14 (2008), 896-907. In particular, a schlicht disk is the image of a subdomain under a biholomorphic restriction.

V2-000617 | Bloch and Landau constants | blochConstant_exact_value

research open | proof annotation: False | 官方代码 | 原文线索

Ahlfors and Grunsky also conjectured in [AG37] that this upper bound is the precise value of the Bloch constant.

```
theorem blochConstant_exact_value :
blochConstant = Real.Gamma (1 / 3) * Real.Gamma (11 / 12) /
(Real.Gamma (1 / 4) * Real.sqrt (1 + Real.sqrt 3))
```

FC-Bloch.blochConstant_exact_value

V2-000618 | Bloch and Landau constants | blochConstant_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

It is proved in [CP96] that the Bloch constant is bounded below by $\sqrt{3}/4 + 2 \times 10^{-4}$

```
theorem blochConstant_lower_bound : Real.sqrt 3 / 4 + 2 * 10 ^ (-4 : ℤ) ≤ blochConstant
```

FC-Bloch.blochConstant_lower_bound

V2-000619 | Bloch and Landau constants | blochConstant_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

It is proved in [AG37] that the Bloch constant is bounded above by $\frac{1}{\sqrt{1 + \sqrt{3}}}\frac{\Gamma(1/3) \Gamma(11/12)}{\Gamma(1/4)}$.

```
theorem blochConstant_upper_bound :
blochConstant ≤ Real.Gamma (1 / 3) * Real.Gamma (11 / 12) /
(Real.Gamma (1 / 4) * Real.sqrt (1 + Real.sqrt 3))
```

FC-Bloch.blochConstant_upper_bound

V2-000620 | Bloch and Landau constants | landauConstant_exact_value

research open | proof annotation: False | 官方代码 | 原文线索

In [Ra43], Rademacher says that he strongly believed that this upper bound is the precise value of the Landau constant.

```
theorem landauConstant_exact_value :
landauConstant = Real.Gamma (1 / 3) * Real.Gamma (5 / 6) / Real.Gamma (1 / 6)
```

FC-Bloch.landauConstant_exact_value

V2-000621 | Bloch and Landau constants | landauConstant_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

It is proved in [Ya95] that the Landau constant is bounded below by $0.5 + 10 ^ {-335}$.

```
theorem landauConstant_lower_bound : 0.5 + 10 ^ (-335 : ℤ) ≤ landauConstant
```

FC-Bloch.landauConstant_lower_bound

V2-000622 | Bloch and Landau constants | landauConstant_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

It is proved in [Ra43] that the Landau constant is bounded above by $\frac{1}{\sqrt{1 + \sqrt{3}}}\frac{\Gamma(1/3) \Gamma(5/6)}{\Gamma(1/6)}$.

```
theorem landauConstant_upper_bound :
landauConstant ≤ Real.Gamma (1 / 3) * Real.Gamma (5 / 6) / Real.Gamma (1 / 6)
```

FC-Bloch.landauConstant_upper_bound

V2-000623 | Bloch and Landau constants | univalentBlochConstant_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

It is proved in [Skin2009] that the Univalent Bloch constant is bounded below by 0.5708858.

```
theorem univalentBlochConstant_lower_bound : 0.5708858 ≤ univalentBlochConstant
```

FC-Bloch.univalentBlochConstant_lower_bound

V2-000624 | Bloch and Landau constants | univalentBlochConstant_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

The Univalent Bloch constant is trivially bounded above by the Bloch radius of the identity function, which is 1. This is the best upper bound we know according to [OptimizationConstants].

```
theorem univalentBlochConstant_upper_bound : univalentBlochConstant ≤ 1
```

FC-Bloch.univalentBlochConstant_upper_bound

Brennan's Conjecture

MODULE CONTEXT: # Brennan's Conjecture *Reference:* - [Wikipedia](https://en.wikipedia.org/wiki/Brennan_conjecture) - [arXiv:2409.15074](https://arxiv.org/abs/2409.15074) - [arXiv:2512.09330](https://arxiv.org/abs/2512.09330) - [OAI26a] OpenAI, *Brennan's conjecture and sharp inverse-square integral means*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Brennans-conjecture-and-sharp-inverse-square-integral-means-September-24-2026/paper.pdf

V2-000625 | Brennan's Conjecture | brennan_universalSpectrum

research solved | proof annotation: True | 官方代码 | 原文线索

Brennan's conjecture, part 1: $B(-2) = 1$. This was proved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof.

```
theorem brennan_universalSpectrum :
universalSpectrum (-2) = 1
```

FC-BrennanConjecture.brennan_universalSpectrum

V2-000626 | Brennan's Conjecture | brennan_universalSpectrumBounded

research open | proof annotation: False | 官方代码 | 原文线索

Brennan's conjecture, part 2: $B_b(-2) = 1$.

```
theorem brennan_universalSpectrumBounded :
universalSpectrumBounded (-2) = 1
```

FC-BrennanConjecture.brennan_universalSpectrumBounded

Brocard's Conjecture

MODULE CONTEXT: # Brocard's Conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Brocard%27s_conjecture) - [Luan Alberto Ferreira, *Real exponential sums over primes and prime gaps*](https://arxiv.org/abs/2307.08725)

V2-000627 | Brocard's Conjecture | brocard_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Brocard's Conjecture For every `n ≥ 2`, between the squares of the `n`-th and `(n+1)`-th primes, there are at least four prime numbers.

```
theorem brocard_conjecture (n : ℕ) (hn : 1 ≤ n) :
letI prev := n.nth Nat.Prime;
letI next := (n+1).nth Nat.Prime;
4 ≤ ((Ioo (prev^2) (next^2)).filter Nat.Prime).card
```

FC-Brocard.brocard_conjecture

Busy Beaver

MODULE CONTEXT: # Busy Beaver The Busy Beaver problem asks for the maximum number of steps that an n-state, 2-symbol Turing machine can take before halting, when started on an empty tape. *References:* - [The Busy Beaver Challenge](https://wiki.bbchallenge.org/wiki/Main_Page)

V2-000628 | Busy Beaver | BB_5

research solved | proof annotation: False | 官方代码 | 原文线索

The value of the Busy Beaver function for 5 states is 47176870.

```
theorem BB_5 : BB 5 = 47176870
```

FC-BusyBeaver.BB_5

V2-000629 | Busy Beaver | BB_6

research open | proof annotation: False | 官方代码 | 原文线索

Determine the value of the Busy Beaver function at n = 6.

```
theorem BB_6 : BB 6 = BB_6.answer
```

FC-BusyBeaver.BB_6

Catalan's conjecture and related Diophantine equations

MODULE CONTEXT: # Catalan's conjecture and related Diophantine equations *References:* - [Wikipedia - Catalan's conjecture](https://en.wikipedia.org/wiki/Catalan%27s_conjecture) - [arXiv:2507.12397](https://arxiv.org/abs/2507.12397) (Lebesgue-Nagell equation)

V2-000630 | Catalan's conjecture and related Diophantine equations | catalans_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The only natural number solution to the equation $x^a - y^b = 1$ such that $a, b > 1$ and $x, y > 0$ is given by $a = 2$, $b = 3$, $x = 3$, and $y = 2$.

```
theorem catalans_conjecture (a b x y : ℕ) (ha : 1 < a) (hb : 1 < b) (hx : 0 < x) (hy : 0 < y)
(heq : x ^ a - y ^ b = 1) : a = 2 ∧ b = 3 ∧ x = 3 ∧ y = 2
```

FC-Catalan.catalans_conjecture

V2-000631 | Catalan's conjecture and related Diophantine equations | pillais_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For positive integers a, b, and c, there are only finitely many positive solutions (x, y, m, n) to the equation $ax^n - by^m = c$ where $(m, n) \neq (2, 2)$ and $x, y > 1$.

```
theorem pillais_conjecture (a b c : ℕ) (ha : 0 < a) (hb : 0 < b) (hc : 0 < c) :
{ (x, y, m, n) : (ℕ × ℕ × ℕ × ℕ) |
1 < x ∧ 1 < y ∧ 1 < m ∧ 1 < n ∧ (m, n) ≠ (2, 2) ∧
a * x^n - b * y^m = c }.Finite
```

FC-Catalan.pillais_conjecture

V2-000632 | Catalan's conjecture and related Diophantine equations | lebesgue_nagell

research open | proof annotation: False | 官方代码 | 原文线索

Lebesgue-Nagell Equation Conjecture For any odd prime p, the only integer solutions (x, y) to the equation $x^2 - 2 = y^p$ are $(x, y) = (\pm 1, -1)$. *Reference:* Ethan Katz and Kyle Pratt, "On the Lebesgue-Nagell equation $x^2 - 2 = y^p$", [arXiv:2507.12397](https://arxiv.org/abs/2507.12397)

```
theorem lebesgue_nagell (p : ℕ) (hp : p.Prime) (hodd : Odd p) (x y : ℤ) :
x ^ 2 - 2 = y ^ p ↔ (x = 1 ∨ x = -1) ∧ y = -1
```

FC-LebesgueNagell.lebesgue_nagell

Černý Conjecture

MODULE CONTEXT: # Černý Conjecture A **synchronizing word** (also called a reset word) for a deterministic finite automaton (DFA) $M = (Q, \Sigma, \delta)$ is a word $w \in \Sigma^*$ such that reading w from any state always leads to the same single state - formally, $\exists p \in Q, \forall q \in Q, \delta^*(q, w) = p$. A DFA is called **synchronizing** if it admits at least one synchronizing word. The **Černý conjecture** asserts that every synchronizing DFA with n states has a synchronizing word of length at most $(n - 1)^2$. This bound is sharp: the family of Černý automata C_n witnesses it, requiring exactly $(n - 1)^2$ steps. **Status:** Open. The best known upper bound is $\left(\frac{7}{48} + \frac{2 \cdot 15625}{1597536}\right) n^3 + o(n^3) \approx 0.1654\,n^3$ (Shitov, 2019). The bound $(n - 1)^2$ has been verified for small n and for special classes of automata (e.g., Eulerian, aperiodic, cyclic automata). We use Mathlib's `DFA α σ` (from `Mathlib.Computability.DFA`), together with the auxiliary `DFA.IsSynchronizingWord` and `DFA.IsSynchronizing` predicates defined in `FormalConjecturesForMathlib.Computability.DFA`. *References:* - [Wikipedia: Synchronizing word](https://en.wikipedia.org/wiki/Synchronizing_word) - J. Černý, [*Poznámka k homogénnym experimentom s konečnými automatmi*](https://dml.cz/bitstream/handle/10338.dmlcz/126647/MathSlov_14-1964-3_2.pdf), Matematicko-fyzikálny časopis, Vol. 14 (1964), No. 3, 208--216. - Y. Shitov, *An improvement to a recent upper bound for synchronizing words of finite automata*, J. Autom. Lang. Comb. Vol. 24 (2019), 367--373.

V2-000633 | Černý Conjecture | cerny_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Černý Conjecture: Every synchronizing DFA with n states admits a synchronizing word of length at most $(n - 1)^2$.

```
theorem cerny_conjecture :
∀ {α : Type*} {σ : Type*} [Fintype σ] (M : DFA α σ) (hM : M.IsSynchronizing) ,
∃ w : List α, M.IsSynchronizingWord w ∧ w.length ≤ (Fintype.card σ - 1)^2
```

FC-CernyConjecture.cerny_conjecture

V2-000634 | Černý Conjecture | shitov_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Shitov's bound (2019): Every synchronizing DFA with n states admits a synchronizing word of length at most $\left(\frac{7}{48} + \frac{2 \cdot 15625}{1597536}\right) n^3 + o(n^3)$, where the $o(n^3)$ term is uniform over all alphabets. This is the best known upper bound towards the Černý conjecture.

```
theorem shitov_upper_bound :
∃ f : ℝ → ℝ, f =o[atTop] (fun n : ℝ => n ^ 3) ∧
∀ {α : Type*} {σ : Type*} [Fintype σ] (M : DFA α σ) (hM : M.IsSynchronizing) ,
∃ w : List α, M.IsSynchronizingWord w ∧
(w.length : ℝ) ≤ (7 / 48 + 2 * 15625 / 1597536) * (Fintype.card σ : ℝ) ^ 3
+ f (Fintype.card σ : ℝ)
```

FC-CernyConjecture.shitov_upper_bound

Congruent Number

MODULE CONTEXT: # Congruent Number A natural number n is called a congruent number if there exists a right triangle with rational sides a, b, and hypotenuse c such that the area of the triangle is $\frac{1}{2}ab = n$. *References:* - [Wikipedia (Congruent number)](https://en.wikipedia.org/wiki/Congruent_number) - [Wikipedia (Tunnell's theorem)](https://en.wikipedia.org/wiki/Tunnell%27s_theorem) - [Keith Conrad's note](https://kconrad.math.uconn.edu/blurbs/ugradnumthy/congnumber.pdf)

V2-000635 | Congruent Number | Tunnell_even

research solved | proof annotation: False | 官方代码 | 原文线索

Tunnell's theorem (necessary condition) for even squarefree congruent numbers.

```
theorem Tunnell_even (n : ℕ) (hsqf : Squarefree n) (heven : Even n) :
congruentNumber n → 2 * (C n).ncard = (D n).ncard
```

FC-CongruentNumber.Tunnell_even

V2-000636 | Congruent Number | Tunnell_even_converse

research open | proof annotation: False | 官方代码 | 原文线索

Tunnell's theorem (sufficient condition assuming BSD) for even squarefree congruent numbers.

```
theorem Tunnell_even_converse (n : ℕ) (hsqf : Squarefree n) (heven : Even n) :
2 * (C n).ncard = (D n).ncard → congruentNumber n
```

FC-CongruentNumber.Tunnell_even_converse

V2-000637 | Congruent Number | Tunnell_odd

research solved | proof annotation: False | 官方代码 | 原文线索

Tunnell's theorem (necessary condition) for odd squarefree congruent numbers.

```
theorem Tunnell_odd (n : ℕ) (hsqf : Squarefree n) (hodd : Odd n) :
congruentNumber n → 2 * (A n).ncard = (B n).ncard
```

FC-CongruentNumber.Tunnell_odd

V2-000638 | Congruent Number | Tunnell_odd_converse

research open | proof annotation: False | 官方代码 | 原文线索

Tunnell's theorem (sufficient condition assuming BSD) for odd squarefree congruent numbers.

```
theorem Tunnell_odd_converse (n : ℕ) (hsqf : Squarefree n) (hodd : Odd n) :
2 * (A n).ncard = (B n).ncard → congruentNumber n
```

FC-CongruentNumber.Tunnell_odd_converse

Babai-Seress Conjectures on the Diameter of Finite Groups

MODULE CONTEXT: # Babai-Seress Conjectures on the Diameter of Finite Groups *References:* - [Wikipedia, *Diameter (group theory)*](https://en.wikipedia.org/wiki/Diameter_(group_theory)) - [H. A. Helfgott and Á. Seress, *On the diameter of permutation groups*](https://arxiv.org/abs/1109.3550) - [L. Babai and Á. Seress, *On the diameter of permutation groups*, European Journal of Combinatorics 13 (1992), 231-243](https://doi.org/10.1016/S0195-6698(05)80029-0) This file contains two conjectures from the Babai-Seress paper: - **Conjecture 1.5**: $\operatorname{diam}(A_n) < n^C$ for some absolute constant C, where A_n is the alternating group on n elements. - **Conjecture 1.7**: $\operatorname{diam}(G) < (\log |G|)^C$ for some absolute constant C, where G ranges over all non-abelian finite simple groups. Conjecture 1.7 generalises Conjecture 1.5, since for $G = A_n$ we have $\log |A_n| \approx n \log n$, so a polylogarithmic bound in $|G|$ implies a polynomial bound in n.

V2-000639 | Babai-Seress Conjectures on the Diameter of Finite Groups | babai_seress_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Babai-Seress Conjecture (Conjecture 1.7): There exists an absolute constant C such that every finite simple non-abelian group G satisfies $$\operatorname{diam}(G) \leq (\log |G|)^C.$$ *Reference:* [L. Babai and Á. Seress, *On the diameter of permutation groups*, European Journal of Combinatorics 13 (1992), Conjecture 1.7](https://doi.org/10.1016/S0195-6698(05)80029-0)

```
theorem babai_seress_conjecture :
∃ C : ℕ,
∀ (G : Type) [Group G] [Fintype G] [IsSimpleGroup G],
(∃ a b : G, a * b ≠ b * a) →
(groupDiam G : ℝ) ≤ (Real.log (Fintype.card G : ℝ)) ^ C
```

FC-BabaiSeressConjectures.babai_seress_conjecture

V2-000640 | Babai-Seress Conjectures on the Diameter of Finite Groups | babai_seress_conjecture_alternating

research open | proof annotation: False | 官方代码 | 原文线索

Babai-Seress Conjecture (Conjecture 1.5): There exists an absolute constant C such that the diameter of the alternating group A_n satisfies $$\operatorname{diam}(A_n) \leq n^C.$$ *Reference:* [L. Babai and Á. Seress, *On the diameter of permutation groups*, European Journal of Combinatorics 13 (1992), Conjecture 1.5](https://doi.org/10.1016/S0195-6698(05)80029-0)

```
theorem babai_seress_conjecture_alternating :
∃ C : ℕ, ∀ n : ℕ,
(groupDiam (alternatingGroup (Fin n)) : ℝ) ≤ (n : ℝ) ^ C
```

FC-BabaiSeressConjectures.babai_seress_conjecture_alternating

Diophantine m-tuples

MODULE CONTEXT: # Diophantine m-tuples A **Diophantine m-tuple** is a set of m distinct positive integers $\{a_1, \dots, a_m\}$ such that $a_i a_j + 1$ is a perfect square for every $i \neq j$. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Diophantine_quintuple) - [Du16] Dujella, Andrej, [*What is... a Diophantine m-tuple?*](https://www.ams.org/journals/notices/201607/rnoti-p772.pdf), Notices Amer. Math. Soc. 63 (2016), 772--774. - [Du] Dujella, Andrej, [*Diophantine quintuple conjecture*](https://web.math.pmf.unizg.hr/~duje/quint.html). - [HTZ19] He, Bo, Togbé, Alain and Ziegler, Volker, [*There is no Diophantine quintuple*](https://arxiv.org/abs/1610.04020), Trans. Amer. Math. Soc. 371 (2019), 6665--6709. - [BD69] Baker, Alan and Davenport, Harold, The equations $3x^2-2=y^2$ and $8x^2-7=z^2$. Quart. J. Math. Oxford Ser. (2) 20 (1969), 129--137. - [Gi99] Gibbs, Philip, [*Some rational Diophantine sextuples*](https://arxiv.org/abs/math/9902081), (1999).

V2-000641 | Diophantine m-tuples | hasUniqueExtension_of_1_3_8

research solved | proof annotation: False | 官方代码 | 原文线索

`HasUniqueExtension` is known to hold for certain triples, including $\{1, 3, 8\}$: this is essentially the Baker-Davenport theorem [BD69], which states that 120 is the only integer d such that $\{1, 3, 8, d\}$ is a Diophantine tuple. Together with a finite check ruling out $d < 8$, this shows that no integral Diophantine tuple properly extends $\{1, 3, 8, 120\}$.

theorem hasUniqueExtension_of_1_3_8 : HasUniqueExtension 1 3 8

FC-DiophantineTuple.hasUniqueExtension_of_1_3_8

V2-000642 | Diophantine m-tuples | hasUniqueExtension_of_forall

research open | proof annotation: False | 官方代码 | 原文线索

The "strong Diophantine 5-tuple conjecture", so-called because it implies the Diophantine 5-tuple theorem (see `noIntegralDiophantineFiveTuple_of_hasUniqueExtensionOfForall`). [Du]

theorem hasUniqueExtension_of_forall : HasUniqueExtensionOfForall

FC-DiophantineTuple.hasUniqueExtension_of_forall

V2-000643 | Diophantine m-tuples | noIntegralDiophantineFiveTuple

research solved | proof annotation: False | 官方代码 | 原文线索

The Diophantine 5-tuple theorem: there does not exist an integral Diophantine 5-tuple. [HTZ19]

theorem noIntegralDiophantineFiveTuple : NoIntegralDiophantineFiveTuple

FC-DiophantineTuple.noIntegralDiophantineFiveTuple

V2-000644 | Diophantine m-tuples | rational_7_tuple

research open | proof annotation: False | 官方代码 | 原文线索

Does there exist a rational Diophantine 7-tuple? [Du16]

theorem rational_7_tuple :
∃ t, IsRationalDiophantineTuple t ∧ t.card = 7

FC-DiophantineTuple.rational_7_tuple

Some conjectures about ranks of elliptic curves over ℚ

MODULE CONTEXT: # Some conjectures about ranks of elliptic curves over ℚ *References:* - [PPVW2016] Jennifer Park, Bjorn Poonen, John Voight, and Melanie Matchett Wood. A heuristic for boundedness of ranks of elliptic curves, https://ems.press/journals/jems/articles/16228 - [BS2013] Manjul Bhargava and Arul Shankar. The average size of the 5-Selmer group of elliptic curves is 6, and the average rank is less than 1, https://arxiv.org/pdf/1312.7859 - [Goldfeld1979] Dorian Goldfeld. Conjectures on elliptic curves over quadratic fields, Number Theory Carbondale 1979, Lecture Notes in Math. 751, 108-118, https://doi.org/10.1007/BFb0062705 - [Smith2025] Alexander Smith. The Birch and Swinnerton-Dyer conjecture implies Goldfeld's conjecture, https://arxiv.org/abs/2503.17619 - [Wikipedia](https://en.wikipedia.org/wiki/Rank_of_an_elliptic_curve) - [ICARM](https://elliptic-rank.icarm.cloud/curve/273) - [Stoll] Michael Stoll. EllipticCurves, a Lean 4 formalization of the Mordell-Weil theorem and explicit 2-descent, https://github.com/MichaelStollBayreuth/EllipticCurves

V2-000645 | Some conjectures about ranks of elliptic curves over ℚ | _02062_le_density_rank_zero

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 5 of [BS2013]: when elliptic curves over ℚ are ordered by height, a density of at least 20.62% have rank 0.

theorem _02062_le_density_rank_zero : 0.2062 ≤ atTop.liminf
fun H ↦ ({E ∈ heightLE H | E.rank = 0}.ncard / (heightLE H).ncard : ℝ)

FC-EllipticCurveRank.RatEllipticCurve._02062_le_density_rank_zero

V2-000646 | Some conjectures about ranks of elliptic curves over ℚ | _08375_le_density_rank_zero_one

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 4 of [BS2013]: when elliptic curves over ℚ are ordered by height, a density of at least 83.75% have rank 0 or 1.

theorem _08375_le_density_rank_zero_one : 0.8375 ≤ atTop.liminf
fun H ↦ ({E ∈ heightLE H | E.rank = 0 ∨ E.rank = 1}.ncard / (heightLE H).ncard : ℝ)

FC-EllipticCurveRank.RatEllipticCurve._08375_le_density_rank_zero_one

V2-000647 | Some conjectures about ranks of elliptic curves over ℚ | avg_rank_lt_0885

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 3 of [BS2013]: when elliptic curves over ℚ are ordered by height, their average rank is < .885.

```
theorem avg_rank_lt_0885 :
atTop.limsup (fun H ↦ ((∑ᶠ E : heightLE H, E.1.rank) / (heightLE H).ncard : ℝ)) < 0.885
```

FC-EllipticCurveRank.RatEllipticCurve.avg_rank_lt_0885

V2-000648 | Some conjectures about ranks of elliptic curves over ℚ | exists_rank_ge_thirtyone

research solved | proof annotation: False | 官方代码 | 原文线索

Is there an elliptic curve over ℚ of rank at least 31? The answer is yes: such a curve was found by Claude, Levent Alpöge and Ava Howell in 2026. See https://elliptic-rank.icarm.cloud/curve/302 and `WeierstrassCurve.claudeAlpogeHowell31` below.

```
theorem exists_rank_ge_thirtyone : ∃ E : RatEllipticCurve, 31 ≤ E.rank
```

FC-EllipticCurveRank.RatEllipticCurve.exists_rank_ge_thirtyone

V2-000649 | Some conjectures about ranks of elliptic curves over ℚ | exists_rank_ge_thirtytwo

research open | proof annotation: False | 官方代码 | 原文线索

Is there an elliptic curve over ℚ of rank at least 32? The largest known rank of an elliptic curve over ℚ as of 2026 is at least 31 (and exactly 31 assuming the generalized Riemann hypothesis and Birch and Swinnerton-Dyer conjecture). See https://elliptic-rank.icarm.cloud/curve/302.

```
theorem exists_rank_ge_thirtytwo : ∃ E : RatEllipticCurve, 32 ≤ E.rank
```

FC-EllipticCurveRank.RatEllipticCurve.exists_rank_ge_thirtytwo

V2-000650 | Some conjectures about ranks of elliptic curves over ℚ | finite_twentyone_lt_finrank

research open | proof annotation: False | 官方代码 | 原文线索

From [PPVW2016], Section 8.2: "Our heuristic predicts (a) All but finitely many E ∈ $\mathcal{E}$ satisfy rk E(ℚ) ≤ 21". In other words, there are only finitely many elliptic curves over ℚ (up to isomorphism) with rank greater than 21. Notice that this contradicts the previous conjecture.

```
theorem finite_twentyone_lt_finrank : {E : RatEllipticCurve | 21 < E.rank}.Finite
```

FC-EllipticCurveRank.RatEllipticCurve.finite_twentyone_lt_finrank

V2-000651 | Some conjectures about ranks of elliptic curves over ℚ | goldfeld_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Goldfeld's conjecture ([Goldfeld1979], Conjecture B), in the form stated in the introduction of [Smith2025]: when the quadratic twists E^d of an elliptic curve E over $\mathbb{Q}$ are ordered by $|d|$, 50% of them have rank 0 and 50% have rank 1. See `goldfeld_conjecture.variants.two_le` for the remaining case $2 \leq r$. Goldfeld states the conjecture for the analytic rank of E^d, which the Birch and Swinnerton-Dyer conjecture predicts is equal to the Mordell-Weil rank used here. Corollary 1.2 of [Smith2025] proves that the Birch and Swinnerton-Dyer conjecture for the quadratic twist family of E implies Goldfeld's conjecture for E.

```
theorem goldfeld_conjecture (E : RatEllipticCurve) (r : ℕ) (hr : r = 0 ∨ r = 1) :
atTop.Tendsto
(fun H ↦ ({d ∈ twistIndexLE H | E.twistRank d = r}.ncard / (twistIndexLE H).ncard : ℝ))
(𝓝 (1 / 2))
```

FC-EllipticCurveRank.RatEllipticCurve.goldfeld_conjecture

V2-000652 | Some conjectures about ranks of elliptic curves over ℚ | half_rank_zero_and_half_rank_one

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture by Goldfeld and Katz-Sarnak: if elliptic curves over ℚ are ordered by their heights, then 50% of the curves have rank 0 and 50% have rank 1. See p. 28 of https://people.maths.bris.ac.uk/~matyd/BSD2011/bsd2011-Bhargava.pdf.

```
theorem half_rank_zero_and_half_rank_one (r : ℕ) (hr : r = 0 ∨ r = 1) :
atTop.Tendsto
(fun H ↦ ({E ∈ heightLE H | E.rank = r}.ncard / (heightLE H).ncard : ℝ)) (𝓝 (1 / 2))
```

FC-EllipticCurveRank.RatEllipticCurve.half_rank_zero_and_half_rank_one

V2-000653 | Some conjectures about ranks of elliptic curves over ℚ | rank_height_count_asymptotic

research open | proof annotation: False | 官方代码 | 原文线索

[PPVW2016] 8.2(b): for 1 ≤ r ≤ 20, the number of elliptic curves over ℚ with rank at least `r` and naïve height at most `H` is asymptotically `H ^ ((21 - r) / 24 + o(1))`. Note: $\mathcal{E}$_H in 8.2(b) should be $\mathcal{E}$_{≤H}, see the statement of Theorem 7.3.3. When `r = 1`, the exponent is `20 / 24 = 5 / 6`, which agrees with the exponent in `card_heightLE_div_pow_five_div_six_tensto` and is consistent with `half_rank_zero_and_half_rank_one`. The equality is asserted only for large `H`, matching the `o(1)` above: it cannot hold at small `H`, because `heightLE H` is empty for `H ≤ 3` (`naiveHeight E ≤ 3` forces `A = B = 0`, which both `Δ_ne_zero` and `reduced` exclude) and consists only of the two rank-zero curves `y² = x³ ± x` for `4 ≤ H ≤ 26`, while the right hand side is `Real.rpow` and hence strictly positive.

```
theorem rank_height_count_asymptotic (r : ℕ) (h₁ : 1 ≤ r) (h₂ : r ≤ 20) :
∃ f : ℕ → ℝ, atTop.Tendsto f (𝓝 0) ∧
∀ᶠ H : ℕ in atTop,
{E ∈ heightLE H | r ≤ E.rank}.ncard = (H : ℝ) ^ ((21 - r) / 24 + f H)
```

FC-EllipticCurveRank.RatEllipticCurve.rank_height_count_asymptotic

V2-000654 | Some conjectures about ranks of elliptic curves over ℚ | twentyone_le_rank_height_count_asymptotic

research open | proof annotation: False | 官方代码 | 原文线索

[PPVW2016] 8.2(c): the number of elliptic curves over ℚ with rank ≥ 21 and naïve height at most `H` is asymptotically at most `H ^ o(1)`.

```
theorem twentyone_le_rank_height_count_asymptotic :
∃ f : ℕ → ℝ, atTop.Tendsto f (𝓝 0) ∧
∀ H : ℕ, 1 < H → {E ∈ heightLE H | 21 ≤ E.rank}.ncard ≤ (H : ℝ) ^ f H
```

FC-EllipticCurveRank.RatEllipticCurve.twentyone_le_rank_height_count_asymptotic

V2-000655 | Some conjectures about ranks of elliptic curves over ℚ | unbounded_rank_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

From [PPVW2016], Section 3.1: "from the mid-1960s to the present, it seems that most experts conjectured unboundedness."

```
theorem unbounded_rank_conjecture (n : ℕ) : ∃ E : RatEllipticCurve, n ≤ E.rank
```

FC-EllipticCurveRank.RatEllipticCurve.unbounded_rank_conjecture

V2-000656 | Some conjectures about ranks of elliptic curves over ℚ | rank_claudeAlpogeHowell31

research open | proof annotation: False | 官方代码 | 原文线索

The rank of the Claude-Alpöge-Howell curve is exactly 31. It has rank exactly 31 assuming the generalized Riemann hypothesis and the Birch and Swinnerton-Dyer conjecture.

```
theorem rank_claudeAlpogeHowell31 : finrank ℤ claudeAlpogeHowell31.Point = 31
```

FC-EllipticCurveRank.WeierstrassCurve.rank_claudeAlpogeHowell31

V2-000657 | Some conjectures about ranks of elliptic curves over ℚ | rank_elkies28

research open | proof annotation: False | 官方代码 | 原文线索

The rank of the Elkies curve is exactly 28.

```
theorem rank_elkies28 : finrank ℤ elkies28.Point = 28
```

FC-EllipticCurveRank.WeierstrassCurve.rank_elkies28

V2-000658 | Some conjectures about ranks of elliptic curves over ℚ | rank_elkiesKlagsbrun29

research open | proof annotation: False | 官方代码 | 原文线索

The rank of the Elkies-Klagsbrun curve is exactly 29.

```
theorem rank_elkiesKlagsbrun29 : finrank ℤ elkiesKlagsbrun29.Point = 29
```

FC-EllipticCurveRank.WeierstrassCurve.rank_elkiesKlagsbrun29

V2-000659 | Some conjectures about ranks of elliptic curves over ℚ | rank_ranksunbounded30

research open | proof annotation: False | 官方代码 | 原文线索

The rank of the ranksunbounded curve is exactly 30.

```
theorem rank_ranksunbounded30 : finrank ℤ ranksunbounded30.Point = 30
```

FC-EllipticCurveRank.WeierstrassCurve.rank_ranksunbounded30

V2-000660 | Some conjectures about ranks of elliptic curves over ℚ | thirty_le_rank_ranksunbounded30

research solved | proof annotation: False | 官方代码 | 原文线索

The rank of the ranksunbounded curve is at least 30.

```
theorem thirty_le_rank_ranksunbounded30 : 30 ≤ finrank ℤ ranksunbounded30.Point
```

FC-EllipticCurveRank.WeierstrassCurve.thirty_le_rank_ranksunbounded30

V2-000661 | Some conjectures about ranks of elliptic curves over ℚ | thirtyone_le_rank_claudeAlpogeHowell31

research solved | proof annotation: False | 官方代码 | 原文线索

The rank of the Claude-Alpöge-Howell curve is at least 31.

```
theorem thirtyone_le_rank_claudeAlpogeHowell31 : 31 ≤ finrank ℤ claudeAlpogeHowell31.Point
```

FC-EllipticCurveRank.WeierstrassCurve.thirtyone_le_rank_claudeAlpogeHowell31

V2-000662 | Some conjectures about ranks of elliptic curves over ℚ | twentyeight_le_rank_elkies28

research solved | proof annotation: False | 官方代码 | 原文线索

The rank of the Elkies curve is at least 28.

```
theorem twentyeight_le_rank_elkies28 : 28 ≤ finrank ℤ elkies28.Point
```

FC-EllipticCurveRank.WeierstrassCurve.twentyeight_le_rank_elkies28

V2-000663 | Some conjectures about ranks of elliptic curves over ℚ | twentynine_le_rank_elkiesKlagsbrun29

research solved | proof annotation: False | 官方代码 | 原文线索

The rank of the Elkies-Klagsbrun curve is at least 29.

theorem twentynine_le_rank_elkiesKlagsbrun29 : 29 ≤ finrank ℤ elkiesKlagsbrun29.Point

FC-EllipticCurveRank.WeierstrassCurve.twentynine_le_rank_elkiesKlagsbrun29

V2-000664 | Some conjectures about ranks of elliptic curves over ℚ | mordell_weil

research solved | proof annotation: True | 官方代码 | 原文线索

The rank of an elliptic curve over a number field is always finite by the Mordell-Weil theorem. Consequently, the rank is always finite, so `finrank ℤ E(K) = 0` really means that the group of rational points is torsion, not that it is of infinite rank. The proof linked via `formal_proof` is Michael Stoll's formalization of the Mordell-Weil theorem [Stoll]

theorem mordell_weil {K} [Field K] [NumberField K] [DecidableEq K] (E : Affine K) [E.IsElliptic] : Module.Finite ℤ E.Point

FC-EllipticCurveRank.mordell_weil

Emirps

MODULE CONTEXT: # Emirps An emirp is a prime whose decimal reversal is a different prime. It is unknown whether there are infinitely many emirps. *References:* * [Wikipedia, Emirp](https://en.wikipedia.org/wiki/Emirp) * [OEIS A006567](https://oeis.org/A006567)

V2-000665 | Emirps | infinitude_question

research open | proof annotation: False | 官方代码 | 原文线索

[Wikipedia's article on emirps](https://en.wikipedia.org/wiki/Emirp) states: "It is not known whether there are infinitely many emirps."

theorem infinitude_question :
{p : ℕ | IsEmirp p}.Infinite

FC-Emirp.infinitude_question

Euclid Numbers conjecture

MODULE CONTEXT: # Euclid Numbers conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Euclid_number)

V2-000666 | Euclid Numbers conjecture | infinite_prime_euclid_numbers

research open | proof annotation: False | 官方代码 | 原文线索

It is not known whether there is an infinite number of prime Euclid numbers.

theorem infinite_prime_euclid_numbers : {n | (Euclid n).Prime}.Infinite

FC-EuclidNumbers.infinite_prime_euclid_numbers

Open questions regarding the existence of Euler bricks

MODULE CONTEXT: # Open questions regarding the existence of Euler bricks *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Euler_brick) - [stackexchange](https://math.stackexchange.com/questions/2264401/euler-bricks-and-the-4th-dimension) - [Sh12] Sharipov, Ruslan. Perfect cuboids and irreducible polynomials. https://arxiv.org/abs/1108.5348

V2-000667 | Open questions regarding the existence of Euler bricks | cuboidOne

research solved | proof annotation: True | 官方代码 | 原文线索

The first Cuboid conjecture The DeepMind prover agent has found a formal proof of this statement. An (independent) informal solution can be found here: *Reference:* [arxiv/2510.11768](https://arxiv.org/abs/2510.11768) **Irreducibility of the Cuboid Polynomial P_{a,u}(t) via a Rank-Zero Elliptic Curve** by *Valery Asiryan*

theorem cuboidOne : CuboidOne

FC-EulerBrick.cuboidOne

V2-000668 | Open questions regarding the existence of Euler bricks | cuboidThree

research open | proof annotation: False | 官方代码 | 原文线索

The third Cuboid conjecture

theorem cuboidThree : CuboidThree

FC-EulerBrick.cuboidThree

V2-000669 | Open questions regarding the existence of Euler bricks | cuboidTwo

research open | proof annotation: False | 官方代码 | 原文线索

The second Cuboid conjecture

theorem cuboidTwo : CuboidTwo

FC-EulerBrick.cuboidTwo

V2-000670 | Open questions regarding the existence of Euler bricks | cuboid_perfect_euler_brick

research solved | proof annotation: False | 官方代码 | 原文线索

In [Sh12], Ruslan notes that a perfect Euler brick does not exist if all three Cuboid conjectures hold.

theorem cuboid_perfect_euler_brick (h₁ : CuboidOne) (h₂ : CuboidTwo) (h₃ : CuboidThree) : ¬ ∃ a b c : ℕ+, IsPerfectCuboid a b c

FC-EulerBrick.cuboid_perfect_euler_brick

V2-000671 | Open questions regarding the existence of Euler bricks | four_dim_euler_brick_existence

research open | proof annotation: False | 官方代码 | 原文线索

Is there an Euler brick in 4-dimensional space?

```
theorem four_dim_euler_brick_existence :
∃ sides : Fin 4 → ℕ+, IsEulerHyperBrick 4 sides
```

FC-EulerBrick.four_dim_euler_brick_existence

V2-000672 | Open questions regarding the existence of Euler bricks | n_dim_euler_brick_existence

research open | proof annotation: False | 官方代码 | 原文线索

Is there an Euler brick in n-dimensional space for any $n > 3$?

```
theorem n_dim_euler_brick_existence :
∀ n > 3, ∃ sides : Fin n → ℕ+, IsEulerHyperBrick n sides
```

FC-EulerBrick.n_dim_euler_brick_existence

V2-000673 | Open questions regarding the existence of Euler bricks | perfect_euler_brick_existence

research open | proof annotation: False | 官方代码 | 原文线索

Is there a perfect Euler brick?

```
theorem perfect_euler_brick_existence :
∃ a b c : ℕ+, IsPerfectCuboid a b c
```

FC-EulerBrick.perfect_euler_brick_existence

Euler's sum of powers conjecture

MODULE CONTEXT: # Euler's sum of powers conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Euler's_sum_of_powers_conjecture)

V2-000674 | Euler's sum of powers conjecture | eulers_sum_of_powers_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Euler's sum of powers conjecture states that for integers $n > 1$ and $k > 1$, if the sum of n positive integers each raised to the k-th power equals another integer raised to the k-th power, then $n ≥ k$. The conjecture is known to be false for $k = 4$ and $k = 5$, but remains open for $k ≥ 6$.

```
theorem eulers_sum_of_powers_conjecture (n k b : ℕ) (hn : 1 < n) (hk : 5 < k) (a : Fin n → ℕ)
(ha : ∀ i, a i > 0) (hsum : ∑ i, (a i) ^ k = b ^ k) : k ≤ n
```

FC-EulerSumOfPowers.eulers_sum_of_powers_conjecture

Exponentials conjectures and theorems

MODULE CONTEXT: # Exponentials conjectures and theorems *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Four_exponentials_conjecture)

V2-000675 | Exponentials conjectures and theorems | four_exponentials_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Four exponentials conjecture Let x_0, x_1 and y_0, y_1 be $\mathbb Q$-linearly independent pairs of complex numbers, then some $e^{x_i y_j}$ is transcendental.

```
theorem four_exponentials_conjecture (x : Fin 2 → ℂ) (y : Fin 2 → ℂ)
(h1 : LinearIndependent ℚ x) (h2 : LinearIndependent ℚ y) :
∃ i j : Fin 2, Transcendental ℚ (exp (x i * y j))
```

FC-Exponentials.four_exponentials_conjecture

V2-000676 | Exponentials conjectures and theorems | two_pow_three_pow_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

The four exponential conjecture would imply that for any irrational number t, at least one of the numbers 2^t and 3^t is transcendental.

```
theorem two_pow_three_pow_transcendental (t : ℝ) (h : Irrational t) :
Transcendental ℚ (2 ^ t : ℝ) ∨ Transcendental ℚ (3 ^ t : ℝ)
```

FC-Exponentials.two_pow_three_pow_transcendental

Factorial primes

MODULE CONTEXT: # Factorial primes A factorial prime is a prime that is one more or one less than a factorial. It is conjectured that there are infinitely many factorial primes. *References:* - [Wikipedia, Factorial prime](https://en.wikipedia.org/wiki/Factorial_prime) - [OEIS A088054](https://oeis.org/A088054)

V2-000677 | Factorial primes | infinitely_many_factorial_primes

research open | proof annotation: False | 官方代码 | 原文线索

There are infinitely many factorial primes.

```
theorem infinitely_many_factorial_primes :
Set.Infinite {p : ℕ | IsFactorialPrime p}
```

FC-FactorialPrime.infinitely_many_factorial_primes

Falconer's distance set conjecture

MODULE CONTEXT: # Falconer's distance set conjecture If $d \ge 2$ and $E \subseteq \mathbb{R}^d$ is compact with $\dim_H E > \frac{d}{2}$, then the distance set $$\{ |x - y| \mid x, y \in E \}$$ has positive Lebesgue measure. The restriction $d \ge 2$ is needed; see `falconer_conjecture`. ## References * [K. Falconer, *On the Hausdorff dimensions of distance sets*](https://doi.org/10.1112/S0025579300010998) * [Wikipedia, *Falconer's conjecture*](https://en.wikipedia.org/wiki/Falconer%27s_conjecture) * [OAI26a] OpenAI, *The Falconer distance conjecture in all dimensions*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Falconer-distance-conjecture-in-all-dimensions-September-23-2026/paper.pdf

V2-000678 | Falconer's distance set conjecture | falconer_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Falconer's distance set conjecture in dimension $d \ge 2$. The hypothesis $2 \le d$ excludes a degenerate case: the conclusion is false for $d = 1$. The base-7 Cantor set $E = \{\sum_{n \ge 1} a_n 7^{-n} : a_n \in \{0, 1, 2\}\}$ is compact with $\dim_H E = \log 3 / \log 7 > 1/2$, but $E - E$ has digits in $\{-2, \dots, 2\}$, so after fixing n digits it is covered by 5^n intervals of length $\frac{2}{3} 7^{-n}$ and is therefore null; its distance set is the image of $E - E$ under $|\cdot|$ and is null too. This was proved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof.

lemma falconer_conjecture (d : ℕ) (h2d : 2 ≤ d) (E : Set <| ℝ^d) (hc : IsCompact E)
(hd : d < 2 * dimH E) :
0 < volume (image2 dist E E)

FC-Falconer.falconer_conjecture

V2-000679 | Falconer's distance set conjecture | falconer_conjecture_two

research solved | proof annotation: True | 官方代码 | 原文线索

Falconer's distance set conjecture, $d = 2$ case. This was proved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof.

lemma falconer_conjecture_two (E : Set <| ℝ²) (hc : IsCompact E) (hd : 2 < 2 * dimH E) :
0 < volume (image2 dist E E)

FC-Falconer.falconer_conjecture_two

Open questions about Fermat numbers

MODULE CONTEXT: # Open questions about Fermat numbers *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Fermat_number)

V2-000680 | Open questions about Fermat numbers | fermat_number_are_composite

research open | proof annotation: False | 官方代码 | 原文线索

Are Fermat numbers composite for all `n > 4`?

theorem fermat_number_are_composite : ∀ n > 4, ¬Prime n.fermatNumber

FC-Fermat.fermat_number_are_composite

V2-000681 | Open questions about Fermat numbers | infinite_fermat_composite

research open | proof annotation: False | 官方代码 | 原文线索

Are there infinitely many composite Fermat numbers?

theorem infinite_fermat_composite : Infinite {n : ℕ | ¬Prime n.fermatNumber}

FC-Fermat.infinite_fermat_composite

V2-000682 | Open questions about Fermat numbers | infinite_fermat_primes

research open | proof annotation: False | 官方代码 | 原文线索

Are there infinitely many Fermat primes?

theorem infinite_fermat_primes : Infinite {n : ℕ | Prime n.fermatNumber}

FC-Fermat.infinite_fermat_primes

Fibonacci Primes

MODULE CONTEXT: # Fibonacci Primes *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Fibonacci_prime)

V2-000683 | Fibonacci Primes | fib_primes_infinite

research open | proof annotation: False | 官方代码 | 原文线索

There are infinitely many Fibonacci primes, i.e., Fibonacci numbers that are prime It is also a barrier to defining a benchmark from this paper: https://arxiv.org/html/2505.13938v1 (see Figure 8).

theorem fib_primes_infinite : {n : ℕ | (∃ m : ℕ, m.fib = n) ∧ n.Prime}.Infinite

FC-FibonacciPrimes.fib_primes_infinite

Gap conjecture

MODULE CONTEXT: # Gap conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Gromov%27s_theorem_on_groups_of_polynomial_growth#The_gap_conjecture) - [On the Gap Conjecture concerning group growth](https://arxiv.org/pdf/1202.6044) by *Rostislav Grigorchuk*

V2-000684 | Gap conjecture | gap_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

If a finitely generated group has superpolynomial growth, then with respect to any finite generating set its growth function is at least $e^{\sqrt n}$ in Grigorchuk's preorder on growth functions, where the comparison is witnessed by linearly rescaling the radius.

```
theorem gap_conjecture :
∀ (G : Type) [Group G] (S : Set G), S.Finite → Subgroup.closure S = ⊤ →
HasSuperPolynomialGrowth G →
∃ C : ℕ, 0 < C ∧
∀ᶠ n : ℕ in atTop, Real.exp (Real.sqrt (n : ℝ)) ≤
(GrowthFunction S (C * n) : ℝ)
```

FC-GapConjecture.gap_conjecture

Goodman's conjecture on coefficients of p-valent functions

MODULE CONTEXT: # Goodman's conjecture on coefficients of p-valent functions A regular (analytic) function is **p-valent** on the unit disk if it assumes each value at most p times there, and some value exactly p times. For such a function $f(z) = \sum_{n \ge 1} b_n z^n$ (so $f(0) = 0$), **Goodman's conjecture** (1948) asserts that its coefficients satisfy $$|b_n| \le \sum_{k=1}^{p} \frac{2k (n+p)!}{(p-k)!\,(p+k)!\,(n-p-1)!\,(n^2-k^2)} |b_k| \qquad (n > p).$$ This generalises the classical coefficient bounds for univalent functions (the case $p = 1$, where the bound reads $|b_n| \le n\,|b_1|$, i.e. $|b_n| \le n$ under the classical normalisation $b_1 = 1$) to p-valent functions. It has been verified in several special cases, including p-valent typically-real functions. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Goodman%27s_conjecture) - Goodman, A. W., *On some determinants related to p-valent functions*, Trans. Amer. Math. Soc. 63 (1948), 175-192.

V2-000685 | Goodman's conjecture on coefficients of p-valent functions | goodman_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Goodman's conjecture. For every p-valent normalised function f on the unit disk and every $n > p$, the n-th coefficient is bounded by the Goodman bound: $$|b_n| \le \sum_{k=1}^{p} \frac{2k (n+p)!}{(p-k)!\,(p+k)!\,(n-p-1)!\,(n^2-k^2)} |b_k|.$$

```
theorem goodman_conjecture (f : ℂ → ℂ) (p n : ℕ) (hf : IsPValent f p)
(hf' : IsNormalized f) (hn : p < n) :
‖coeff f n‖ ≤ goodmanBound f p n
```

FC-GoodmanConjecture.goodman_conjecture

Gromov's theorem on groups of polynomial growth

MODULE CONTEXT: # Gromov's theorem on groups of polynomial growth *Reference:* - [Wikipedia](https://en.wikipedia.org/wiki/Gromov%27s_theorem_on_groups_of_polynomial_growth)

V2-000686 | Gromov's theorem on groups of polynomial growth | GromovPolynomialGrowthTheorem

research solved | proof annotation: False | 官方代码 | 原文线索

Gromov's Polynomial Growth Theorem : A finitely generated group has polynomial growth if and only if it is virtually nilpotent.

```
theorem GromovPolynomialGrowthTheorem [Group.FG G] :
HasPolynomialGrowth G ↔ Group.IsVirtuallyNilpotent G
```

FC-GromovPolynomialGrowth.GromovPolynomialGrowthTheorem

Hajós' conjecture (1961) - disproved

MODULE CONTEXT: # Hajós' conjecture (1961) - disproved *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Hadwiger_conjecture_(graph_theory)#Generalizations) * [Ha61] Hajós, G. (1961). "Über eine Konstruktion nicht n-färbbarer Graphen." *Wiss. Z. Martin-Luther-Univ. Halle-Wittenberg Math.-Natur. Reihe* 10, pp. 116--117. * [Di52] Dirac, G. A. (1952). "A property of 4-chromatic graphs and some remarks on critical graphs." *J. London Math. Soc.* 27, pp. 85--92. * [Ca79] Catlin, P. A. (1979). "Hajós' graph-coloring conjecture: variations and counterexamples." *J. Combin. Theory Ser. B* 26, pp. 268--274. * [EF81] Erdős, P. and Fajtlowicz, S. (1981). "On the conjecture of Hajós." *Combinatorica* 1, pp. 141--143. * [Th05] Thomassen, C. (2005). "Some remarks on Hajós' conjecture." *J. Combin. Theory Ser. B* 93, pp. 95--105.

V2-000687 | Hajós' conjecture (1961) - disproved | hajos_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Hajós' conjecture (1961) - disproved. Hajós conjectured that every graph with chromatic number at least k contains a subdivision of K_k (a strengthening of Hadwiger's conjecture). This is **false**: Catlin [Ca79] gave counterexamples for every $k \ge 7$, and Erdős and Fajtlowicz [EF81] showed that almost all graphs are counterexamples.

```
theorem hajos_conjecture : ¬
∀ {V : Type} [Fintype V] (G : SimpleGraph V) (k : ℕ),
(k : ℕ∞) ≤ G.chromaticNumber → ContainsSubdivision (completeGraph (Fin k)) G
```

FC-HajosConjecture.hajos_conjecture

First Hardy-Littlewood conjecture

MODULE CONTEXT: # First Hardy-Littlewood conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/First_Hardy%E2%80%93Littlewood_conjecture)

V2-000688 | First Hardy-Littlewood conjecture | first_hardy_littlewood_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Let $P = (m_1, \dots, m_k)$ be a tuple of distinct positive even integers. Let $\pi_P(n)$ denote the number of primes $p\leq n$ such that $(p, p + m_1, \dots, p + m_k)$ forms an admissible prime constellation. Let $w(q; m_1, \dots, m_k)$ denote the number of distinct residues of $0, m_1, \dots, m_k$ modulo q, and let $$ C_P = 2 ^ k\prod_{\substack{q\ \text{prime} \\ q\geq 3}} \frac{1 - \frac{w(q; m_1, \dots, m_k)}{q}}{\left(1 - \frac{1}{q}\right)^{k+1}}. $$ Then $$ \pi_P(n)\sim C_P\int_2^n\frac{dt}{\log^{k+1}t}. $$

theorem first_hardy_littlewood_conjecture {k : ℕ} (m : Fin k.succ → ℕ) :
FirstHardyLittlewoodConjectureFor m

FC-HardyLittlewood.first_hardy_littlewood_conjecture

V2-000689 | First Hardy-Littlewood conjecture | not_first_and_secondHardyLittlewoodConjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Richards [Ri74] showed that only one of the two Hardy-Littlewood conjectures can be true. [Ri74] Richards, Ian (1974). _On the Incompatibility of Two Conjectures Concerning Primes_. Bull. Amer. Math. Soc. 80: 419-438.

theorem not_first_and_secondHardyLittlewoodConjecture :
(∀ {k : ℕ} (m : Fin k.succ → ℕ), FirstHardyLittlewoodConjectureFor m) →
¬(∀ {x y : ℕ} (hx : 2 ≤ x) (hy : 2 ≤ y), SecondHardyLittlewoodConjectureFor x y)

FC-HardyLittlewood.not_first_and_secondHardyLittlewoodConjecture

The Hasse--Weil conjecture for elliptic curves

MODULE CONTEXT: # The Hasse--Weil conjecture for elliptic curves The L-series $L(E, s)$ of an elliptic curve E over a number field converges absolutely on $\operatorname{Re} s > 3/2$ ([Tate1966], p. 416; [Gross2011], Lecture 2, §1). The **Hasse--Weil conjecture** predicts that it has an analytic continuation to the whole complex plane and satisfies a functional equation ([Tate1966], p. 416; [Gross2011], Conjecture 2.7, stated there for the completed L-function). Only the continuation is stated here. Over $\mathbb{Q}$ it follows from the modularity theorem. [Wikipedia] states the conjecture for the Hasse--Weil zeta function and asks only for a meromorphic continuation, which is equivalent to a meromorphic continuation of $L(E, s)$. That weaker form is the one the Birch and Swinnerton-Dyer conjecture rests on, so it is recorded here as well. [Gross2011], Conjecture 2.10 takes it as a hypothesis, and only near $s = 1$. The formulation of the Birch and Swinnerton-Dyer conjecture in `FormalConjectures/Millennium/BSD.lean` instead asserts a global meromorphic continuation, so it implies `exists_hasMeromorphicContinuation` below. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Hasse%E2%80%93Weil_zeta_function#Hasse%E2%80%93Weil_conjecture) - [Tate1966] John Tate. "On the conjectures of Birch and Swinnerton-Dyer and a geometric analog." Seminaire Bourbaki, Vol. 9, Exp. No. 306 (1966), 415-440, [numdam](https://www.numdam.org/item/SB_1964-1966__9__415_0/) - [Gross2011] Benedict H. Gross. "Lectures on the conjecture of Birch and Swinnerton-Dyer." Arithmetic of L-functions, IAS/Park City Math. Ser. 18, AMS (2011), 169-209, [math.harvard.edu](https://people.math.harvard.edu/~gross/preprints/lectures-pcmi.pdf)

V2-000690 | The Hasse--Weil conjecture for elliptic curves | exists_hasAnalyticContinuation

research open | proof annotation: False | 官方代码 | 原文线索

The **Hasse--Weil conjecture** ([Tate1966], p. 416; [Gross2011], Conjecture 2.7): the L-series of an elliptic curve over a number field has an analytic continuation to $\mathbb{C}$.

theorem exists_hasAnalyticContinuation (E : WeierstrassCurve K) [E.IsElliptic] :
∃ L, HasAnalyticContinuation E L

FC-HasseWeil.exists_hasAnalyticContinuation

V2-000691 | The Hasse--Weil conjecture for elliptic curves | exists_hasAnalyticContinuation_rat

research solved | proof annotation: False | 官方代码 | 原文线索

The **Hasse--Weil conjecture** over $\mathbb{Q}$: the L-series of an elliptic curve over $\mathbb{Q}$ has an analytic continuation to $\mathbb{C}$. This follows from the modularity theorem, see [Wikipedia].

theorem exists_hasAnalyticContinuation_rat (E : WeierstrassCurve ℚ) [E.IsElliptic] :
∃ L, HasAnalyticContinuation E L

FC-HasseWeil.exists_hasAnalyticContinuation_rat

V2-000692 | The Hasse--Weil conjecture for elliptic curves | exists_hasMeromorphicContinuation

research open | proof annotation: False | 官方代码 | 原文线索

The L-series of an elliptic curve over a number field has a meromorphic continuation to $\mathbb{C}$. This is the **Hasse--Weil conjecture** in the form stated in [Wikipedia], and a weak form of [Gross2011], Conjecture 2.7. It is also what the Birch and Swinnerton-Dyer conjecture rests on, as the hypothesis of [Gross2011], Conjecture 2.10.

theorem exists_hasMeromorphicContinuation (E : WeierstrassCurve K) [E.IsElliptic] :
∃ L, HasMeromorphicContinuation E L

FC-HasseWeil.exists_hasMeromorphicContinuation

Polynomial-time computability of factoring

MODULE CONTEXT: # Polynomial-time computability of factoring This file formalizes the open problem of whether natural number / integer factorization can be computed in polynomial time on a deterministic classical Turing machine. More precisely, it formalizes the following statement: Can the prime factorization of a natural number be computed in polynomial time (on a deterministic classical Turing machine)? *References:* - [Wikipedia: List of unsolved problems in computer science](https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_computer_science) - [Wikipedia: Integer factorization](https://en.wikipedia.org/wiki/Integer_factorization)

V2-000693 | Polynomial-time computability of factoring | isPolyTime_primeFactorsList

research open | proof annotation: False | 官方代码 | 原文线索

The integer factorization problem: Can the prime factorization of a positive integer be computed in polynomial time? We state the problem by asking if `Nat.primeFactorsList` is polynomial-time computable (assuming typical encodings of ℕ and List ℕ into bitstrings). *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Integer_factorization)

theorem isPolyTime_primeFactorsList : IsPolyTime Nat.primeFactorsList

FC-PolyTime.isPolyTime_primeFactorsList

Open questions on irrationality of numbers

MODULE CONTEXT: # Open questions on irrationality of numbers *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Irrational_number#Open_questions) - [OAI26a] OpenAI, *Catalan's constant is irrational*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Catalans-constant-is-irrational-September-24-2026/paper.pdf

V2-000694 | Open questions on irrationality of numbers | algebraicIndependent_e_pi

research open | proof annotation: False | 官方代码 | 原文线索

Are e and π algebraically independent?

theorem algebraicIndependent_e_pi :
AlgebraicIndependent ℚ ![e, π]

FC-Irrational.algebraicIndependent_e_pi

V2-000695 | Open questions on irrationality of numbers | irrational_catalanConstant

research solved | proof annotation: True | 官方代码 | 原文线索

Is the Catalan constant $$G = \sum_{n=0}^∞ (-1)^n / (2n + 1)^2 \approx 0.91596$$ irrational? This was proved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof.

theorem irrational_catalanConstant :
Irrational catalanConstant

FC-Irrational.irrational_catalanConstant

V2-000696 | Open questions on irrationality of numbers | irrational_e_plus_pi

research open | proof annotation: False | 官方代码 | 原文线索

Is $e + \pi$ irrational?

theorem irrational_e_plus_pi :
Irrational (e + π)

FC-Irrational.irrational_e_plus_pi

V2-000697 | Open questions on irrationality of numbers | irrational_e_times_pi

research open | proof annotation: False | 官方代码 | 原文线索

Is $e \pi$ irrational?

theorem irrational_e_times_pi :
Irrational (e * π)

FC-Irrational.irrational_e_times_pi

V2-000698 | Open questions on irrationality of numbers | irrational_e_to_e

research open | proof annotation: False | 官方代码 | 原文线索

Is $e ^ e$ irrational?

theorem irrational_e_to_e :
Irrational (e ^ e)

FC-Irrational.irrational_e_to_e

V2-000699 | Open questions on irrationality of numbers | irrational_ln_pi

research open | proof annotation: False | 官方代码 | 原文线索

Is $\ln(\pi)$ irrational?

theorem irrational_ln_pi :
Irrational (log π)

FC-Irrational.irrational_ln_pi

V2-000700 | Open questions on irrationality of numbers | irrational_pi_to_e

research open | proof annotation: False | 官方代码 | 原文线索

Is $\pi ^ e$ irrational?

theorem irrational_pi_to_e :
Irrational (π ^ e)

FC-Irrational.irrational_pi_to_e

V2-000701 | Open questions on irrationality of numbers | irrational_pi_to_pi

research open | proof annotation: False | 官方代码 | 原文线索

Is $\pi ^ \pi$ irrational?

theorem irrational_pi_to_pi :
Irrational (π ^ π)

FC-Irrational.irrational_pi_to_pi

Juggler conjecture

MODULE CONTEXT: # Juggler conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Juggler_sequence)

V2-000702 | Juggler conjecture | juggler_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Now form a sequence beginning with any positive integer, where each subsequent term is obtained by applying the operation defined above to the previous term. The **Juggler Conjecture** states that for any positive integer n, there exists a natural number m such that the m-th term of the sequence is 1.

theorem juggler_conjecture (n : ℕ) (hn : n > 0) : ∃ m, jugglerStep^[m] n = 1

FC-JugglerConjecture.juggler_conjecture

Kakeya problem

MODULE CONTEXT: # Kakeya problem *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Kakeya_set)

V2-000703 | Kakeya problem | kakeya_2d

research solved | proof annotation: False | 官方代码 | 原文线索

The two-dimensional case, proved by Davies [Da71]. [Da71] Davies, R. O., _Some remarks on the Kakeya problem_. Math. Proc. Cambridge Philos. Soc. 69 (1971), no. 3, 417-421.

theorem kakeya_2d : KakeyaSetConjectureDim 2

FC-Kakeya.kakeya_2d

V2-000704 | Kakeya problem | kakeya_3d

research solved | proof annotation: False | 官方代码 | 原文线索

The three-dimensional case, proved by Wang, Zahl [WaZa25]. [WaZa25] Wang, H. and Zahl, J., _Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions_. arXiv preprint, arXiv:2502.17655, 2025.

theorem kakeya_3d : KakeyaSetConjectureDim 3

FC-Kakeya.kakeya_3d

V2-000705 | Kakeya problem | kakeya_finite

research solved | proof annotation: False | 官方代码 | 原文线索

The finite field Kakeya conjecture asserts that any Kakeya set in `𝔽_qⁿ` has size at least `c_n · qⁿ` for some constant `c_n` depending only on `n`. This was first proved by Dvir [Dv08]. The best known bound to date, due to Bukh and Chao [BuCh21], establishes that any Kakeya set in `𝔽_qⁿ` has size at least `qⁿ / (2 - 1/q)^(n - 1)`. The dimension is positive, as in `kakeya_set_conjecture`; for `n = 0` the empty set is a Kakeya set and the bound would read `1 ≤ 0`. [Dv08] Dvir, Z., _On the size of Kakeya sets in finite fields_. Journal of the American Mathematical Society 22 (2009), no. 4, 1093-1097. [BuCh21] Bukh, B. and Chao, T.-W., _Sharp density bounds on the finite field Kakeya problem_. Discrete Analysis 26 (2021), 9 pp.

theorem kakeya_finite {F : Type*} [Field F] [Fintype F] {n : ℕ} (hn : 0 < n)
(K : Finset (Fin n → F)) (hK : IsKakeyaFinite K) :
card F ^ n / (2 - 1 / card F : ℚ) ^ (n - 1) ≤ K.card

FC-Kakeya.kakeya_finite

V2-000706 | Kakeya problem | kakeya_set_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Kakeya set conjecture: Kakeya sets in $\mathbb{R}^n$ have Hausdorff dimension n.

theorem kakeya_set_conjecture (n : ℕ) (hn : n > 0) :
KakeyaSetConjectureDim n

FC-Kakeya.kakeya_set_conjecture

Komlós conjecture

MODULE CONTEXT: # Komlós conjecture The Komlós conjecture in discrepancy theory: there is a universal constant K such that for all n, m and all vectors $v_1, \dots, v_n \in \mathbb{R}^m$ with $\|v_i\|_2 \le 1$, there exist signs $\varepsilon_i \in \{-1, +1\}$ such that $$\left\|\sum_{i=1}^n \varepsilon_i v_i\right\|_\infty \le K.$$ For a long time the best known bound was due to Banaszczyk, who proved that one can always achieve $O(\sqrt{\log n})$. The Beck-Fiala theorem on the discrepancy of sparse set systems is a special case (up to scaling), and the conjecture implies the Beck-Fiala conjecture that set systems of degree t have discrepancy $O(\sqrt{t})$. The conjecture was proved in September 2026 by Guo, Fang and Lu with $K = 3\sqrt{2\pi}$. Karingula and Lovett gave an elementary proof with $K = 36$. A formalisation of their proof is recorded in the `formal_proof` attributes below. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Discrepancy_theory#Major_open_problems) - [W. Banaszczyk, *Balancing vectors and Gaussian measures of n-dimensional convex bodies*, Random Structures & Algorithms **12** (1998), 351-360](https://doi.org/10.1002/(SICI)1098-2418(199807)12:4%3C351::AID-RSA3%3E3.0.CO;2-S) - [J. Spencer, *Six standard deviations suffice*, Trans. Amer. Math. Soc. **289** (1985), 679-706](https://doi.org/10.1090/S0002-9947-1985-0784009-0) - [S. Guo, E. X. Fang and J. Lu, *Vector balancing via directional total variation*, arXiv:2609.11189](https://arxiv.org/abs/2609.11189) - [S. R. Karingula and S. Lovett, *An elementary proof of the Komlós conjecture*, arXiv:2609.20979](https://arxiv.org/abs/2609.20979)

V2-000707 | Komlós conjecture | komlos_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The Komlós conjecture There exists a universal constant $K > 0$ such that for all $n, m \in \mathbb{N}$ and all vectors $v_1, \dots, v_n \in \mathbb{R}^m$ with $\|v_i\|_2 \le 1$ (encoded here as $\sum_j v_{ij}^2 \le 1$), there exist signs $\varepsilon_i \in \{-1, +1\}$ such that $\left\|\sum_i \varepsilon_i v_i\right\|_\infty \le K$, i.e. $\left|\sum_i \varepsilon_i v_{ij}\right| \le K$ for every coordinate j. Proved by Guo, Fang and Lu with $K = 3\sqrt{2\pi}$ and by Karingula and Lovett with $K = 36$.

```
theorem komlos_conjecture :
∃ K : ℝ, 0 < K ∧ ∀ (n m : ℕ) (v : Fin n → Fin m → ℝ),
(∀ i, ∑ j, (v i j) ^ 2 ≤ 1) →
∃ ε : Fin n → ℝ, (∀ i, ε i = 1 ∨ ε i = -1) ∧
∀ j, |∑ i, ε i * v i j| ≤ K
```

FC-KomlosConjecture.komlos_conjecture

Leinster Groups

MODULE CONTEXT: # Leinster Groups A finite group is a Leinster group if the sum of the orders of all its normal subgroups equals twice the group's order. *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Leinster_group) * Leinster, Tom (2001). "Perfect numbers and groups". [arXiv:math/0104012](https://arxiv.org/abs/math/0104012) TODO: The following properties from the Wikipedia article can also be formalized: - There are no Leinster groups that are symmetric or alternating. - There is no Leinster group of order p^2q^2 where p, q are primes. - No finite semi-simple group is Leinster. - No p-group can be a Leinster group. - All abelian Leinster groups are cyclic with order equal to a perfect number.

V2-000708 | Leinster Groups | abelian_is_leinster_iff_cyclic_perfect

research solved | proof annotation: False | 官方代码 | 原文线索

An abelian group is a Leinster group if and only if it is cyclic with order equal to a perfect number. Reference: Leinster, Tom (2001). "Perfect numbers and groups". Theorem 2.1.

```
theorem abelian_is_leinster_iff_cyclic_perfect (G : Type*) [CommGroup G] [Fintype G] :
IsLeinster G ↔ IsCyclic G ∧ Nat.Perfect (Fintype.card G)
```

FC-LeinsterGroup.abelian_is_leinster_iff_cyclic_perfect

V2-000709 | Leinster Groups | dihedral_is_leinster_iff_odd_perfect

research solved | proof annotation: False | 官方代码 | 原文线索

The dihedral group `DihedralGroup n` (of order `2n`) is a Leinster group if and only if `n` is an odd perfect number. This gives a one-to-one correspondence between dihedral Leinster groups and odd perfect numbers. In particular, the existence of odd perfect numbers is equivalent to the existence of dihedral Leinster groups. Reference: Leinster, Tom (2001). "Perfect numbers and groups".

```
theorem dihedral_is_leinster_iff_odd_perfect (n : ℕ) [NeZero n] :
IsLeinster (DihedralGroup n) ↔ Nat.Perfect n ∧ Odd n
```

FC-LeinsterGroup.dihedral_is_leinster_iff_odd_perfect

V2-000710 | Leinster Groups | exists_nonabelian_leinster_group

research solved | proof annotation: False | 官方代码 | 原文线索

Non-abelian Leinster groups exist. For example, `S₃ × C₅` (order 30) and `A₅ × C₁₅₁₂₈` are Leinster groups. Reference: Leinster, Tom (2001). "Perfect numbers and groups".

```
theorem exists_nonabelian_leinster_group :
∃ G : Type, ∃ (_ : Group G) (_ : Fintype G),
IsLeinster G ∧ ¬ ∀ (a b : G), a * b = b * a
```

FC-LeinsterGroup.exists_nonabelian_leinster_group

Lemoine's conjectures

MODULE CONTEXT: # Lemoine's conjectures *References:* - [Wikipedia](https://en.wikipedia.org/wiki/%C3%89mile_Lemoine#Lemoine's_conjecture_and_extensions) - [Ki85] Kiltinen, J. and Young P. (1985). Goldbach, Lemoine, and a Know/Don't Know Problem.

V2-000711 | Lemoine's conjectures | lemoine_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For all odd integers $n ≥ 7$ there are prime numbers p,q such that $n = p+2q$.

```
theorem lemoine_conjecture (n : ℕ) (hn : 6 < n) (odd : Odd n) :
∃ (p q : ℕ), p.Prime ∧ q.Prime ∧ p + 2 * q = n
```

FC-Lemoine.lemoine_conjecture

V2-000712 | Lemoine's conjectures | lemoine_conjecture_extension

research open | proof annotation: False | 官方代码 | 原文线索

For all odd integers $n ≥ 9$ there are odd prime numbers p,q,r,s and natural numbers a,b such that $p+2q = n$, $2+pq = 2^a+r$, $2p+q = 2^b+s$

```
theorem lemoine_conjecture_extension (n : ℕ) (hn : 8 < n) (odd : Odd n) :
∃ (p q r s a b : ℕ), OddPrime p ∧ OddPrime q ∧ OddPrime r ∧ OddPrime s ∧
p + 2 * q = n ∧ 2 + p * q = 2 ^ a + r ∧ 2 * p + q = 2 ^ b + s
```

FC-Lemoine.lemoine_conjecture_extension

Littlewood conjectures

MODULE CONTEXT: # Littlewood conjectures *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Littlewood_conjecture) - [Bernard de Mathan and Olivier Teulié, *Problèmes diophantiens simultanés*](https://doi.org/10.1007/s00605-003-0199-y)

V2-000713 | Littlewood conjectures | littlewood_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For any two real numbers α and β, $$ \liminf_{n\to\infty} n\||n\alpha\||\||n\beta\|| = 0 $$ where $\||x\|| := \min(|x - \lfloor x \rfloor|, |x - \lceil x \rceil|)$ is the distance to the nearest integer.

```
theorem littlewood_conjecture (α β : ℝ) :
atTop.liminf (fun (n : ℕ) ↦ n * distToNearestInt (n * α) * distToNearestInt (n * β)) = 0
```

FC-LittlewoodConjecture.littlewood_conjecture

V2-000714 | Littlewood conjectures | padic_littlewood_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For real number α and prime p, $$ \liminf_{n \to\infty} n |n|_{p}\||n\alpha\|| = 0 $$ where $\||x\|| := \min(|x - \lfloor x \rfloor|, |x - \lceil x \rceil|)$ is the distance to the nearest integer, and $|x|_{p}$ is the p-adic norm.

```
theorem padic_littlewood_conjecture (α : ℝ) (p : ℕ) (hp : p.Prime) :
atTop.liminf (fun (n : ℕ) ↦ n * padicNorm p n * distToNearestInt (n * α)) = 0
```

FC-LittlewoodConjecture.padic_littlewood_conjecture

Local uniformization

MODULE CONTEXT: # Local uniformization Let F be a finitely generated field extension of a field k and let $\mathcal{O}$ be a valuation ring of F containing k. *Local uniformization* asks for an affine model of $F \mid k$ on which the centre of $\mathcal{O}$ is a regular point: a finitely generated k-subalgebra $A \subseteq \mathcal{O}$ with fraction field F such that $A_\mathfrak{p}$ is regular, where $\mathfrak{p} = \mathfrak{m}_\mathcal{O} \cap A$ is the centre. This is the local form of resolution of singularities, one valuation at a time. Zariski introduced it and proved it in characteristic zero, and deduced resolution of singularities in dimension at most three from it. In positive characteristic it is known in dimension at most three, and open from dimension four on. A place is *Abhyankar* when the rational rank of its value group and the transcendence degree of its residue field over k add up to the transcendence degree of $F \mid k$; an Abhyankar place whose residue field is separable over k is uniformized in any dimension [KK2005]. The form asked here is the weak one: some affine model of $F \mid k$ works. The proofs give the strong form, where the model may moreover be required to contain a prescribed finite subset of $\mathcal{O}$, and that is the form that patches into a resolution of singularities. The conclusion here asks that the centre be a regular point, not a smooth one. Over a perfect field the two agree; over an imperfect field regularity is the right condition, since already $k(a^{1/p}) \mid k$ for $a \in k \setminus k^p$ has no model that is smooth over k. For a field extension, `Algebra.EssFiniteType k F` says that F is finitely generated as a field over k, and `Algebra.trdeg k F` is the transcendence degree, the dimension of a model. The centre `ValuationSubring.centerOn` and the predicate `ValuationSubring.HasLocalUniformization` are defined in `FormalConjecturesForMathlib.RingTheory.Valuation.LocalUniformization`. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Local_uniformization) - [Zar1940] O. Zariski, [Local uniformization on algebraic varieties](https://doi.org/10.2307/1968864), Ann. of Math. 41 (1940), 852--896. - [Abh1956] S. Abhyankar, [Local uniformization on algebraic surfaces over ground fields of characteristic $p \neq 0$](https://doi.org/10.2307/1970014), Ann. of Math. 63 (1956), 491--526. - [Abh1966] S. Abhyankar, Resolution of singularities of embedded algebraic surfaces, Monographs in Pure and Applied Mathematics 24, Academic Press, 1966; birational resolution of threefolds over an algebraically closed field of characteristic $p > 5$. - [Cut2009] S. D. Cutkosky, [Resolution of singularities for 3-folds in positive characteristic](https://doi.org/10.1353/ajm.0.0036), Amer. J. Math. 131 (2009), 59--127; a simplification of [Abh1966], also over an algebraically closed field. - [CP2008] V. Cossart and

O. Piltant, [Resolution of singularities of threefolds in positive characteristic I](https://doi.org/10.1016/j.jalgebra.2008.03.032), J. Algebra 320 (2008), 1051--1082. - [CP2009] V. Cossart and O. Piltant, [Resolution of singularities of threefolds in positive characteristic II](https://doi.org/10.1016/j.jalgebra.2008.11.030), J. Algebra 321 (2009), 1836--1976; quasi-projective threefolds over a field k with $[k : k^p] < \infty$, in every positive characteristic. - [CP2019] V. Cossart and O. Piltant, [Resolution of singularities of arithmetical threefolds](https://doi.org/10.1016/j.jalgebra.2019.02.017), J. Algebra 529 (2019), 268--535. - [KK2005] H. Knaf and F.-V. Kuhlmann, [Abhyankar places admit local uniformization in any characteristic](https://doi.org/10.1016/j.ansens.2005.09.001), Ann. Sci. École Norm. Sup. 38 (2005), 833--846; the hypotheses are that the place is Abhyankar and that its residue field is separable over the ground field. - [KK2009] H. Knaf and F.-V. Kuhlmann, [Every place admits local uniformization in a finite extension of the function field](https://doi.org/10.1016/j.aim.2008.12.009), Adv. Math. 221 (2009), 428--453. - [Tem2013] M. Temkin, [Inseparable local uniformization](https://doi.org/10.1016/j.jalgebra.2012.09.023), J. Algebra 373 (2013), 65--119.

V2-000715 | Local uniformization | local_uniformization

research open | proof annotation: False | 官方代码 | 原文线索

Local uniformization in positive characteristic. Let k be a field of characteristic $p > 0$, let F be a finitely generated field extension of k, and let $\mathcal{O}$ be a valuation ring of F containing k. Then $\mathcal{O}$ admits local uniformization over k. This is open from dimension four on: it is known when the transcendence degree of $F \mid k$ is at most three, see `local_uniformization_of_trdeg_le_three`.

```
theorem local_uniformization (k F : Type*) [Field k] [Field F] [Algebra k F]
[Algebra.EssFiniteType k F] (p : ℕ) [Fact p.Prime] [CharP k p] (𝒪 : ValuationSubring F)
(hk : ∀ x : k, algebraMap k F x ∈ 𝒪) :
𝒪.HasLocalUniformization k
```

FC-LocalUniformization.local_uniformization

V2-000716 | Local uniformization | local_uniformization_after_finite_extension

research solved | proof annotation: False | 官方代码 | 原文线索

Local uniformization after a finite extension. Every valuation ring of a finitely generated field extension $F \mid k$ that contains k admits local uniformization over k after a finite extension E of F, in any characteristic. This is the main theorem of [KK2009]. Compare [Tem2013], where the extension of F is purely inseparable but the ground field is extended too.

```
theorem local_uniformization_after_finite_extension (k F : Type*) [Field k] [Field F]
[Algebra k F] [Algebra.EssFiniteType k F] (𝒪 : ValuationSubring F)
(hk : ∀ x : k, algebraMap k F x ∈ 𝒪) :
∃ (E : IntermediateField F (AlgebraicClosure F)) (_ : FiniteDimensional F E)
(𝒪' : ValuationSubring E),
𝒪'.comap (algebraMap F E) = 𝒪 ∧ 𝒪'.HasLocalUniformization k
```

FC-LocalUniformization.local_uniformization_after_finite_extension

V2-000717 | Local uniformization | local_uniformization_of_charZero

research solved | proof annotation: False | 官方代码 | 原文线索

Zariski's local uniformization theorem. Every valuation ring of a finitely generated field extension $F \mid k$ that contains a field k of characteristic zero admits local uniformization over k. This is the main theorem of [Zar1940].

```
theorem local_uniformization_of_charZero (k F : Type*) [Field k] [Field F] [Algebra k F]
[Algebra.EssFiniteType k F] [CharZero k] (𝒪 : ValuationSubring F)
(hk : ∀ x : k, algebraMap k F x ∈ 𝒪) :
𝒪.HasLocalUniformization k
```

FC-LocalUniformization.local_uniformization_of_charZero

V2-000718 | Local uniformization | local_uniformization_of_trdeg_le_three

research solved | proof annotation: False | 官方代码 | 原文线索

Local uniformization holds in transcendence degree at most three over any field. In characteristic zero this is [Zar1940]. In positive characteristic, surfaces are [Abh1956]; threefolds over an algebraically closed field of characteristic $p > 5$ are [Abh1966], simplified in [Cut2009]; and the remaining characteristics $2, 3, 5$ are [CP2008] and [CP2009], for quasi-projective threefolds over a field k with $[k : k^p] < \infty$. It follows in the generality stated here from [CP2019], Theorem 1.1, which resolves the singularities of every reduced separated Noetherian quasi-excellent scheme of dimension at most three in any characteristic: the centre of $\mathcal{O}$ on a resolution of a proper model of $F \mid k$ is a regular point.

```
theorem local_uniformization_of_trdeg_le_three (k F : Type*) [Field k] [Field F] [Algebra k F]
[Algebra.EssFiniteType k F] (𝒪 : ValuationSubring F)
(hk : ∀ x : k, algebraMap k F x ∈ 𝒪) (hF : Algebra.trdeg k F ≤ 3) :
𝒪.HasLocalUniformization k
```

FC-LocalUniformization.local_uniformization_of_trdeg_le_three

The Lovász-Plummer conjecture (proved 2011) and Sheehan's conjecture

MODULE CONTEXT: # The Lovász-Plummer conjecture (proved 2011) and Sheehan's conjecture *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Petersen%27s_theorem#Related_conjectures) * [LP86] Lovász, L. and Plummer, M. D. (1986). *Matching Theory.* North-Holland. * [EKKKN11] Esperet, L., Kardoš, F., King, A. D., Král', D. and Norine, S. (2011). "Exponentially many perfect matchings in cubic graphs." *Adv. Math.* 227, pp. 1646--1664. [arXiv:1012.2878](https://arxiv.org/abs/1012.2878) * [Sh77] Sheehan, J. (1977). "The multiplicity of Hamiltonian circuits in a graph." In *Recent Advances in Graph Theory*, Academia, Prague, pp. 477--480. * [Th98] Thomassen, C. (1998). "Independent dominating sets and a second Hamiltonian cycle in regular graphs." *J. Combin. Theory Ser. B* 72, pp. 104--109.

V2-000719 | The Lovász-Plummer conjecture (proved 2011) and Sheehan's conjecture | lovasz_plummer_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

The Lovász-Plummer conjecture (1970s), proved by Esperet, Kardoš, King, Král' and Norine (2011). Every bridgeless cubic graph on n vertices has exponentially many perfect matchings: there is a constant $c > 0$ such that the number of perfect matchings is at least 2^{cn}. [EKKKN11] prove this with $2^{n/3656}$.

```
theorem lovasz_plummer_conjecture :
∃ c : ℝ, 0 < c ∧ ∀ {V : Type} [Fintype V] [DecidableEq V]
(G : SimpleGraph V) [DecidableRel G.Adj],
(∀ v, G.degree v = 3) → G.IsBridgeless →
(2 : ℝ) ^ (c * Fintype.card V) ≤ perfectMatchingCount G
```

FC-LovaszPlummerConjecture.lovasz_plummer_conjecture

V2-000720 | The Lovász-Plummer conjecture (proved 2011) and Sheehan's conjecture | sheehan_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Sheehan's conjecture (1977). Every 4-regular graph with a Hamiltonian cycle has a second Hamiltonian cycle (one with a different edge set). Sheehan's conjecture would settle the last open case of the question, raised by Smith's theorem for cubic graphs, of which regular Hamiltonian graphs have a second Hamiltonian cycle: Thomassen [Th98] proved it for all r-regular graphs with $r \ge 300$.

```
theorem sheehan_conjecture :
∀ {V : Type} [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
(∀ v, G.degree v = 4) →
∀ (v : V) (c : G.Walk v v), IsHamiltonianCycle G c →
∃ (w : V) (c' : G.Walk w w), IsHamiltonianCycle G c' ∧
c'.edges.toFinset ≠ c.edges.toFinset
```

FC-LovaszPlummerConjecture.sheehan_conjecture

Conjectures about the Mandelbrot and Multibrot sets

MODULE CONTEXT: # Conjectures about the Mandelbrot and Multibrot sets This file adds three conjectures about the Mandelbrot and Multibrot sets: - the *MLC conjecture*, stating that these sets are locally connected - the *density of hyperbolicity* conjecture, stating that parameters with attracting cycles are dense in the Mandelbrot and Multibrot sets - the conjecture that the boundaries of these sets have zero area. The first two conjectures are related in that the former implies the latter. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Mandelbrot_set#Local_connectivity) - [arxiv/math/9902155](https://arxiv.org/abs/math/9902155) - [mathoverflow/37229](https://mathoverflow.net/questions/37229/)

V2-000721 | Conjectures about the Mandelbrot and Multibrot sets | MLC_general_exponent

research open | proof annotation: False | 官方代码 | 原文线索

A stronger version of the MLC conjecture, stating that all multibrots are locally connected. Note that we don't need to require `2 ≤ n` because the conjecture holds in the trivial cases `n = 0` and `n = 1` too.

```
theorem MLC_general_exponent (n : ℕ) : LocallyConnectedSpace (multibrotSet n)
```

FC-Mandelbrot.MLC_general_exponent

V2-000722 | Conjectures about the Mandelbrot and Multibrot sets | density_of_hyperbolicity

research open | proof annotation: False | 官方代码 | 原文线索

The density of hyperbolicity conjecture, stating that the set of all parameters `c` for which `fun z ↦ z ^ 2 + c` has an attracting cycle is dense in the Mandelbrot set.

```
theorem density_of_hyperbolicity :
mandelbrotSet ⊆ closure {c | ∃ m z, IsAttractingCycle (fun z ↦ z ^ 2 + c) m z}
```

FC-Mandelbrot.density_of_hyperbolicity

V2-000723 | Conjectures about the Mandelbrot and Multibrot sets | density_of_hyperbolicity_general_exponent

research open | proof annotation: False | 官方代码 | 原文线索

The density of hyperbolicity conjecture for Multibrot sets, stating that the set of all parameters `c` for which `fun z ↦ z ^ n + c` has an attracting cycle is dense in `multibrotSet n`. Note that we need to require `2 ≤ n` because the conjecture is trivially false for `n = 1`.

```
theorem density_of_hyperbolicity_general_exponent {n : ℕ} (hn : 2 ≤ n) :
multibrotSet n ⊆ closure {c | ∃ m z, IsAttractingCycle (fun z ↦ z ^ n + c) m z}
```

FC-Mandelbrot.density_of_hyperbolicity_general_exponent

V2-000724 | Conjectures about the Mandelbrot and Multibrot sets | volume_frontier_mandelbrotSet_eq_zero

research open | proof annotation: False | 官方代码 | 原文线索

The boundary of the Mandelbrot set is conjectured to have zero area.

```
theorem volume_frontier_mandelbrotSet_eq_zero : volume (frontier mandelbrotSet) = 0
```

FC-Mandelbrot.volume_frontier_mandelbrotSet_eq_zero

V2-000725 | Conjectures about the Mandelbrot and Multibrot sets | volume_frontier_multibrotSet_eq_zero

research open | proof annotation: False | 官方代码 | 原文线索

The boundary of any Multibrot set is conjectured to have zero area. Note that we don't need to exclude the trivial cases `n = 0` and `n = 1` because the conjecture holds for them.

theorem volume_frontier_multibrotSet_eq_zero {n : ℕ} : volume (frontier (multibrotSet n)) = 0

FC-Mandelbrot.volume_frontier_multibrotSet_eq_zero

Mean value problem

MODULE CONTEXT: # Mean value problem *Reference:* - [Wikipedia](https://en.wikipedia.org/wiki/Mean_value_problem) - [The fundamental theorem of algebra and complexity theory](https://www.ams.org/journals/bull/1981-04-01/S0273-0979-1981-14858-8/) by Steve Smale Given a complex polynomial p of degree $d ≥ 2$ and a complex number z there is a critical point c of p, such that $|p(z)-p(c)|/|z-c| ≤ K* |p'(z)|$ for $K=1$. The conjecture has been proven for: * `K = 4` [The fundamental theorem of algebra and complexity theory](https://www.ams.org/journals/bull/1981-04-01/S0273-0979-1981-14858-8/) by *Steve Smale* * `K = (d-1)/d` at a point z if the normalised polynomial $p(X + z) - p(z)$ has only real roots, or if all its nonzero roots have the same norm. [Critical points and values of complex polynomials](https://doi.org/10.1016/0885-064X(89)90019-8) by *David Tischler*

V2-000726 | Mean value problem | mean_value_problem_leq_4

research solved | proof annotation: False | 官方代码 | 原文线索

The following weaker version of the mean value problem has been proven. Given a complex polynomial p of degree $d ≥ 2$ and a complex number z there a critical point c of p, such that $|p(z)-p(c)|/|z-c| ≤ 4|p'(z)|$.

lemma mean_value_problem_leq_4 (p : Polynomial ℂ) (hp : 2 ≤ p.degree) (z : ℂ) (K : ℝ):
∃ c : ℂ, p.derivative.eval c = 0 ∧
‖p.eval z - p.eval c‖ / ‖z - c‖ ≤ 4 *‖p.derivative.eval z‖

FC-MeanValueProblem.mean_value_problem_leq_4

V2-000727 | Mean value problem | mean_value_problem_of_real_roots

research solved | proof annotation: False | 官方代码 | 原文线索

The following tighter bound depending on the degree d of the polynomial p has been shown by Tischler. Let z be a complex number such that the normalised polynomial $p(X + z) - p(z)$ has only real roots, i.e. every x with $p(x) = p(z)$ satisfies $\operatorname{Im} x = \operatorname{Im} z$. Then there is a critical point c of p such that $|p(z)-p(c)|/|z-c| \le (d-1)/d \cdot |p'(z)|$.

lemma mean_value_problem_of_real_roots (p : Polynomial ℂ) (hp : 2 ≤ p.natDegree) (z : ℂ)
(h : ∀ x : ℂ, p.eval x = p.eval z → x.im = z.im) (K : ℝ) :
∃ c : ℂ, p.derivative.eval c = 0 ∧
‖p.eval z - p.eval c‖ / ‖z - c‖ ≤ (p.natDegree - 1)/ p.natDegree * ‖p.derivative.eval z‖

FC-MeanValueProblem.mean_value_problem_of_real_roots

V2-000728 | Mean value problem | mean_value_problem_of_roots_same_norm

research solved | proof annotation: False | 官方代码 | 原文线索

The following tighter bound depending on the degree d of the polynomial p has been shown by Tischler. Let z be a complex number such that all nonzero roots of the normalised polynomial $p(X + z) - p(z)$ have the same norm, i.e. all $x \ne z$ with $p(x) = p(z)$ have the same distance to z. Then there is a critical point c of p such that $|p(z) - p(c)|/|z-c| \le (d-1)/d \cdot |p'(z)|$.

lemma mean_value_problem_of_roots_same_norm (p : Polynomial ℂ) (hp : 2 ≤ p.natDegree) (z : ℂ)
(h : ∀ x y : ℂ, p.eval x = p.eval z → p.eval y = p.eval z → x ≠ z → y ≠ z →
‖x - z‖ = ‖y - z‖) (K : ℝ) :
∃ c : ℂ, p.derivative.eval c = 0 ∧
‖p.eval z - p.eval c‖ / ‖z - c‖ ≤ (p.natDegree - 1)/ p.natDegree * ‖p.derivative.eval z‖

FC-MeanValueProblem.mean_value_problem_of_roots_same_norm

Conjectures about Mersenne primes

MODULE CONTEXT: # Conjectures about Mersenne primes *References:* - [Wikipedia: Mersenne conjectures](https://en.wikipedia.org/wiki/Mersenne_conjectures) - [Wikipedia: Catalan's Mersenne conjecture](https://en.wikipedia.org/wiki/Catalan%27s_Mersenne_conjecture) - [MathWorld: Catalan-Mersenne Number](https://mathworld.wolfram.com/Catalan-MersenneNumber.html)

V2-000729 | Conjectures about Mersenne primes | new_mersenne_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

For any odd natural number `p` if two of the following conditions hold, then all three must hold: 1. 2^p-1 is prime 2. $(2^p+1)/3$ is prime 3. Exists a number `k` such that $p = 2^k \\pm 1$ or $p = 4^k \\pm 3$

theorem new_mersenne_conjecture (p : ℕ) (hp : Odd p) :
NewMersenneConjectureStatement p

FC-Mersenne.new_mersenne_conjecture

Mills' Theorem

MODULE CONTEXT: # Mills' Theorem There exists a real $A > 1$ such that $\lfloor A^{3^n}\rfloor$ is prime for every positive integer n, where $\lfloor\cdot\rfloor$ denotes the floor function. The least such A is known as *Mills' constant*. It is irrational, and assuming the Riemann hypothesis it is approximately $1.3063778838\ldots$. *References:* - [Wikipedia, Mills' constant](https://en.wikipedia.org/wiki/Mills%27_constant) - [A prime-representing function](https://www.ams.org/journals/bull/1947-53-06/S0002-9904-1947-08849-2/) by *W. H. Mills*, Bull. Amer. Math. Soc. **53** (1947), 604. - [Mills' constant](https://mathworld.wolfram.com/MillsConstant.html) on Wolfram MathWorld. - [Mills' constant is irrational](https://doi.org/10.1112/mtk.70027) by *Kota Saito*, Mathematika **71** (2025), no. 3, e70027, [arXiv:2404.19461](https://arxiv.org/abs/2404.19461). - [Determining Mills' Constant ..](https://cs.uwaterloo.ca/journals/JIS/VOL8/Caldwell/caldwell78.pdf) by *Chris K. Caldwell and Yuanyou Cheng*, J. Integer Seq. **8** (2005), Article 05.4.1. - [OEIS A051021](https://oeis.org/A051021) (decimal expansion of Mills' constant)

V2-000730 | Mills' Theorem | exists'

research solved | proof annotation: False | 官方代码 | 原文线索

Mills' theorem (Mills, 1947). There is a real number $A > 1$ such that $\lfloor A^{3^n}\rfloor$ is prime.

```
theorem exists' : ∃ A > 1, IsMills A
```

FC-Mills.exists'

V2-000731 | Mills' Theorem | exists_least

research solved | proof annotation: False | 官方代码 | 原文线索

Mills' constant. There is a *least* Mills number.

```
theorem exists_least : ∃ A, IsMinMills A
```

FC-Mills.exists_least

V2-000732 | Mills' Theorem | irrational

research solved | proof annotation: False | 官方代码 | 原文线索

Mills' constant is irrational (Saito, 2024).

```
theorem irrational {A} (hA : IsMinMills A) :
Irrational A
```

FC-Mills.irrational

V2-000733 | Mills' Theorem | lower_bound_of_RH

research solved | proof annotation: False | 官方代码 | 原文线索

Mills' constant lower bound (Caldwell-Cheng, 2005): assuming the Riemann hypothesis, Mills' constant begins at $1.3063778838\ldots$.

```
theorem lower_bound_of_RH (hRH : RiemannHypothesis) {A} (hA : IsMinMills A) :
A ∈ Set.Ioo (1.3063778838 : ℝ) 1.3063778839
```

FC-Mills.lower_bound_of_RH

Modularity conjecture

MODULE CONTEXT: # Modularity conjecture The **Modularity conjecture** (also known as the Shimura-Taniyama-Weil conjecture) states that every rational elliptic curve is modular, meaning that it can be associated with a modular form. We state the `a_p` version of the conjecture, which relates the coefficients of the modular form to the number of points on the elliptic curve over finite fields. Since we don't have the conductor of the elliptic curve, our definition of `a_p(E)` differs from that in the literature at primes of bad reduction. For this reason, we state the conjecture with the assumption that `p ∤ N`, in order to give an equivalent statement. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Modularity_theorem) - [F. Diamond and J. Shurman, *A First Course in Modular Forms*](https://doi.org/10.1007/978-0-387-27226-9)

V2-000734 | Modularity conjecture | modularity_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The Modularity Theorem (formerly Shimura-Taniyama-Weil conjecture): every elliptic curve over $\mathbb{Q}$ is modular.

```
theorem modularity_conjecture (E : WeierstrassCurve ℚ) [E.IsElliptic] : modularityConjecture E
```

FC-ModularityConjecture.modularity_conjecture

Moser's Worm

MODULE CONTEXT: # Moser's Worm *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Moser%27s_worm_problem) - [Tao's optimization constant 13a](https://teorth.github.io/optimizationproblems/constants/13a.html)

V2-000735 | Moser's Worm | convex_mosers_worm_problem_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

0.232239 is a lower bound on the area of a convex set that covers all worms. *Reference:* Khandhawit, Tirasan; Pagonakis, Dimitrios; Sriswasdi, Sira (2013), "Lower Bound for Convex Hull Area and Universal Cover Problems", International Journal of Computational Geometry & Applications, 23 (3): 197-212, arXiv:1101.5638, doi:10.1142/S0218195913500076, MR 3158583, S2CID 207132316.

```
theorem convex_mosers_worm_problem_lower_bound :
0.232239 ≤ convexWormCoverConstant
```

FC-MoserWorm.convex_mosers_worm_problem_lower_bound

V2-000736 | Moser's Worm | convex_mosers_worm_problem_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

There is a convex set of area 0.270911861 that covers all worms. *Reference:* Wang, Wei (2006), "An improved upper bound for the worm problem", Acta Mathematica Sinica, 49 (4): 835-846, MR 2264090.

```
theorem convex_mosers_worm_problem_upper_bound :
∃ X ∈ WormCovers, Convex ℝ X ∧ volume X = 0.270911861
```

FC-MoserWorm.convex_mosers_worm_problem_upper_bound

V2-000737 | Moser's Worm | mosers_worm_problem_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

There is a set of area 0.260437 that covers all worms. *Reference:* Norwood, Rick; Poole, George (2003), "An improved upper bound for Leo Moser's worm problem", Discrete and Computational Geometry, 29 (3): 409-417, doi:10.1007/s00454-002-0774-3, MR 1961007.

```
theorem mosers_worm_problem_upper_bound :
∃ X ∈ WormCovers, volume X = 0.260437
```

FC-MoserWorm.mosers_worm_problem_upper_bound

Rational_variety

MODULE CONTEXT: # Rational_variety *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Rational_variety)

V2-000738 | Rational_variety | noether_problem

research solved | proof annotation: False | 官方代码 | 原文线索

The **Noether Problem**: let `L` be the field of rational functions in `n` indeterminates over `K`, and let `G` be a finite group permuting these indeterminates. Is the fixed field `L^G` a rational extension of `K`, i.e. does `L/K` have the Noether property? Solution: False.

```
theorem noether_problem : ¬ ∀ (K L ι : Type)
[Field K] [Field L] [Fintype ι] [Algebra K L] [IsRationalExtension K L ι],
HasNoetherProperty K L ι
```

FC-NoetherProblem.noether_problem

Normality of mathematical constants

MODULE CONTEXT: # Normality of mathematical constants Despite extensive empirical evidence-billions of digits have been computed for π, e, and $\sqrt{2}$, all showing near-uniform digit distribution-it is an open problem whether any of the classical constants π, e, $\sqrt{2}$, $\ln 2$, or φ is normal in any base. *References:* * [Wikipedia (Normal number)](https://en.wikipedia.org/wiki/Normal_number) * [Wikipedia (Pi)](https://en.wikipedia.org/wiki/Pi)

V2-000739 | Normality of mathematical constants | pi_normal_base_ten

research open | proof annotation: False | 官方代码 | 原文线索

π is normal in base 10.

```
theorem pi_normal_base_ten : IsNormalInBase 10 π
```

FC-NormalNumber.pi_normal_base_ten

Ohba's conjecture (2002; proved by Noel, Reed and Wu 2015)

MODULE CONTEXT: # Ohba's conjecture (2002; proved by Noel, Reed and Wu 2015) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/List_coloring#Ohba's_conjecture) * [Oh02] Ohba, K. (2002). "On chromatic-choosable graphs." *J. Graph Theory* 40, pp. 130--135. * [NRW15] Noel, J. A., Reed, B. A. and Wu, H. (2015). "A proof of a conjecture of Ohba." *J. Graph Theory* 79, pp. 86--102. [arXiv:1211.1999](https://arxiv.org/abs/1211.1999) * [ERT79] Erdős, P., Rubin, A. L. and Taylor, H. (1979). "Choosability in graphs." *Congr. Numer.* 26, pp. 125--157.

V2-000740 | Ohba's conjecture (2002; proved by Noel, Reed and Wu 2015) | ohba_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

Ohba's conjecture (2002), proved by Noel, Reed and Wu (2015). Every graph G with at most $2\chi(G) + 1$ vertices is *chromatic-choosable*: it is $\chi(G)$-choosable, so its choice number equals its chromatic number.

```
theorem ohba_conjecture :
∀ {V : Type} [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
(Fintype.card V : ℕ∞) ≤ 2 * G.chromaticNumber + 1 →
Choosable G G.chromaticNumber.toNat
```

FC-OhbaConjecture.ohba_conjecture

Oppermann's Conjecture

MODULE CONTEXT: # Oppermann's Conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Oppermann%27s_conjecture) - [Luan Alberto Ferreira, *Real exponential sums over primes and prime gaps*](https://arxiv.org/abs/2307.08725)

V2-000741 | Oppermann's Conjecture | oppermann_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Oppermann's Conjecture: For every integer $x \ge 2$, the following hold: - There exists a prime between $x(x-1)$ and x^2. - There exists a prime between x^2 and $x(x+1)$.

```
theorem oppermann_conjecture (x : ℕ) (hx : 2 ≤ x) :
(∃ p ∈ Ioo (x * (x - 1)) (x^2), p.Prime) ∧
(∃ p ∈ Ioo (x^2) (x * (x + 1)), p.Prime)
```

FC-Oppermann.oppermann_conjecture

Pebbling number conjecture

MODULE CONTEXT: # Pebbling number conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Graph_pebbling) - [Pebbling on Graph Products and Other Binary Graph Constructions](https://arxiv.org/abs/1801.07808)

V2-000742 | Pebbling number conjecture | pebbling_number_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The pebbling number conjecture: the pebbling number of a Cartesian product of connected graphs is at most equal to the product of the pebbling numbers of the factors. See [Asplund, Hurlbert, and Kenter](https://arxiv.org/abs/1801.07808).

```
theorem pebbling_number_conjecture {W : Type} [Fintype V] [Fintype W] [DecidableEq W]
(G : SimpleGraph V) (H : SimpleGraph W) (hG : G.Connected) (hH : H.Connected) :
PebblingNumber (G □ H) ≤ PebblingNumber G * PebblingNumber H
```

FC-PebblingNumberConjecture.pebbling_number_conjecture

Infinitude of Pell number primes

MODULE CONTEXT: # Infinitude of Pell number primes *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Pell_number#Primes_and_squares) - [A86383](https://oeis.org/A86383) The Pell numbers P_n are defined by $P_0 = 0$, $P_1 = 1$, $P_{n+2} = 2*P_{n+1} + P_n$. [OEIS A129](https://oeis.org/A129) The conjecture says that there are infinitely many prime Pell numbers.

V2-000743 | Infinitude of Pell number primes | infinite_pellNumber_primes

research open | proof annotation: False | 官方代码 | 原文线索

There are infinitely many prime Pell numbers

```
theorem infinite_pellNumber_primes : Infinite {n : ℕ | Prime (pellNumber n)}
```

FC-PellNumbers.infinite_pellNumber_primes

Perfect numbers

MODULE CONTEXT: # Perfect numbers A perfect number is a positive integer that equals the sum of its proper divisors (i.e., all its positive divisors excluding the number itself). For example, 6 is perfect because its proper divisors are 1, 2, and 3, and 1 + 2 + 3 = 6. Similarly, 28 is perfect because 1 + 2 + 4 + 7 + 14 = 28. All known perfect numbers are even. Several open problems about perfect numbers are formalised here: * Are there infinitely many perfect numbers? * Are there infinitely many even perfect numbers? * Do odd perfect numbers exist? *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Perfect_number) - [Wikipedia, Odd perfect numbers](https://en.wikipedia.org/wiki/Perfect_number#Odd_perfect_numbers)

V2-000744 | Perfect numbers | infinitely_many_even_perfect

research open | proof annotation: False | 官方代码 | 原文线索

Infinitely many even perfect numbers conjecture. Are there infinitely many even perfect numbers? This is equivalent to asking whether there are infinitely many Mersenne primes, since by the Euclid-Euler theorem an even number is perfect if and only if it has the form $2^{p-1}(2^p - 1)$ where $2^p - 1$ is a Mersenne prime. *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Perfect_number)

```
theorem infinitely_many_even_perfect :
{n : ℕ | Perfect n ∧ Even n}.Infinite
```

FC-PerfectNumbers.infinitely_many_even_perfect

The Petersen colouring conjecture (Jaeger 1988)

MODULE CONTEXT: # The Petersen colouring conjecture (Jaeger 1988) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Petersen_graph#Petersen_coloring_conjecture) * [Ja88] Jaeger, F. (1988). "Nowhere-zero flow problems." In *Selected Topics in Graph Theory 3*, Academic Press, pp. 71--95. * [Mk13] Mkrtchyan, V. V. (2013). "A remark on the Petersen coloring conjecture of Jaeger." *Australas. J. Combin.* 56, pp. 145--151. * [HS14] Hägglund, J. and Steffen, E. (2014). "Petersen-colorings and some families of snarks." *Ars Math. Contemp.* 7, pp. 161--173. * [Zh97] Zhang, C.-Q. (1997). *Integer flows and cycle covers of graphs.* Marcel Dekker.

V2-000745 | The Petersen colouring conjecture (Jaeger 1988) | petersen_coloring_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Petersen colouring conjecture (Jaeger 1988). Every bridgeless cubic graph has a Petersen colouring: an edge colouring by the edges of the Petersen graph such that the three edges at each vertex are mapped onto the three edges at some vertex of the Petersen graph. Jaeger showed that this conjecture implies both the Berge-Fulkerson conjecture and the cycle double cover conjecture.

```
theorem petersen_coloring_conjecture :
∀ {V : Type} [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
(∀ v, G.degree v = 3) → G.IsBridgeless →
∃ φ : Sym2 V → Sym2 PetersenVertex, IsPetersenColoring G φ
```

FC-PetersenColoringConjecture.petersen_coloring_conjecture

Asymptotic density of powerful numbers

MODULE CONTEXT: # Asymptotic density of powerful numbers Let $Q(x)$ denote the number of powerful integers up to x. Erdős and Szekeres [ES35] proved $$Q(x) = \frac{\zeta(3/2)}{\zeta(3)} x^{1/2} + O(x^{1/3}),$$ and Bateman and Grosswald [BG58] sharpened this to $$Q(x) = \frac{\zeta(3/2)}{\zeta(3)} x^{1/2} + \frac{\zeta(2/3)}{\zeta(2)} x^{1/3} + O(x^{1/6}).$$ Improving the exponent $1/6$ in the error term unconditionally remains open; conditional improvements are known under the Riemann Hypothesis. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Powerful_number) - [ES35] Erdős, P. and Szekeres, G., *Über die Anzahl der Abelschen Gruppen gegebener Ordnung und über ein verwandtes zahlentheoretisches Problem*, Acta Sci. Math. (Szeged) 7 (1935), 95-102. - [BG58] Bateman, P. T. and Grosswald, E., *On a theorem of Erdős and Szekeres*, Illinois J. Math. 2 (1958), 88-98.

V2-000746 | Asymptotic density of powerful numbers | asymptotic_bateman_grosswald

research solved | proof annotation: False | 官方代码 | 原文线索

Bateman and Grosswald [BG58] proved the sharper asymptotic $$Q(x) = \frac{\zeta(3/2)}{\zeta(3)} x^{1/2} + \frac{\zeta(2/3)}{\zeta(2)} x^{1/3} + O(x^{1/6}).$$

```
theorem asymptotic_bateman_grosswald :
(fun x : ℝ => (Q x : ℝ) - A * x ^ ((1 : ℝ) / 2) - B * x ^ ((1 : ℝ) / 3)) =O[atTop]
fun x => x ^ ((1 : ℝ) / 6)
```

FC-PowerfulNumbersDensity.asymptotic_bateman_grosswald

V2-000747 | Asymptotic density of powerful numbers | asymptotic_erdos_szekeres

research solved | proof annotation: False | 官方代码 | 原文线索

Erdős and Szekeres [ES35] proved that the number of powerful integers up to x satisfies $$Q(x) = \frac{\zeta(3/2)}{\zeta(3)} x^{1/2} + O(x^{1/3}).$$ In particular $Q(x) \sim \frac{\zeta(3/2)}{\zeta(3)} \sqrt{x}$.

```
theorem asymptotic_erdos_szekeres :
(fun x : ℝ => (Q x : ℝ) - A * x ^ ((1 : ℝ) / 2)) =O[atTop]
fun x => x ^ ((1 : ℝ) / 3)
```

FC-PowerfulNumbersDensity.asymptotic_erdos_szekeres

V2-000748 | Asymptotic density of powerful numbers | error_term_improvement

research open | proof annotation: False | 官方代码 | 原文线索

Can the exponent $1/6$ in the error term of the Bateman-Grosswald asymptotic be improved unconditionally? That is, is there $\delta > 0$ such that $$Q(x) = \frac{\zeta(3/2)}{\zeta(3)} x^{1/2} + \frac{\zeta(2/3)}{\zeta(2)} x^{1/3} + O(x^{1/6 - \delta})?$$ Improvements are known under the Riemann Hypothesis.

```
theorem error_term_improvement :
∃ δ > (0 : ℝ),
(fun x : ℝ => (Q x : ℝ) - A * x ^ ((1 : ℝ) / 2) - B * x ^ ((1 : ℝ) / 3)) =O[atTop]
fun x => x ^ ((1 : ℝ) / 6 - δ)
```

FC-PowerfulNumbersDensity.error_term_improvement

Prime Triplet Conjecture

MODULE CONTEXT: # Prime Triplet Conjecture *Reference:* [Prime Triplet Wikipedia Page](https://en.wikipedia.org/wiki/Prime_triplet#Conjecture_on_prime_triplets)

V2-000749 | Prime Triplet Conjecture | prime_triplets

research open | proof annotation: False | 官方代码 | 原文线索

Are there infinitely many tuples of three consecutive primes (p, q, r) such that $r - p = 6$?

```
theorem prime_triplets :
{p : ℕ | Prime p ∧ (Prime (p + 2) ∨ Prime (p + 4)) ∧ Prime (p + 6)}.Infinite
```

FC-PrimeTriplets.prime_triplets

Primes and perfect squares

MODULE CONTEXT: # Primes and perfect squares *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Landau%27s_problems#Near-square_primes)

V2-000750 | Primes and perfect squares | infinite_prime_sq_add_one

research open | proof annotation: False | 官方代码 | 原文线索

Are there infinitely many primes p such that $p - 1$ is a perfect square? In other words: Are there infinitely many primes of the form $n^2 + 1$?

```
theorem infinite_prime_sq_add_one :
{n : ℕ | Prime (n^2 + 1)}.Infinite
```

FC-PrimesAndPerfectSquares.infinite_prime_sq_add_one

Ramanujan τ-function

MODULE CONTEXT: # Ramanujan τ-function There are two conjectures related to the Ramanujan τ-function: - Ramanujan-Petersson conjecture: For every prime `p`, the absolute value of the Ramanujan τ-function at `p` is bounded by `2 * p^(11/2)`. - Lehmer's conjecture: The Ramanujan τ-function is never zero for any positive integer `n`. *References:* - [Ramanujan-Petersson conjecture](https://en.wikipedia.org/wiki/Ramanujan%E2%80%93Petersson_conjecture) - [Lehmer's conjecture](https://en.wikipedia.org/wiki/Ramanujan_tau_function#Conjectures_on_the_tau_function)

V2-000751 | Ramanujan τ-function | lehmer_ramanujan_tau

research open | proof annotation: False | 官方代码 | 原文线索

Lehmer's conjecture: $\tau(n) \ne 0$ for all $n > 0$.

```
theorem lehmer_ramanujan_tau : ∀ n > 0, τ n ≠ 0
```

FC-RamanujanTau.lehmer_ramanujan_tau

V2-000752 | Ramanujan τ-function | ramanujan_petersson

research solved | proof annotation: False | 官方代码 | 原文线索

The Ramanujan-Petersson conjecture: $|\tau(p)| \le 2 p^{11/2}$ for primes p.

theorem ramanujan_petersson : ∀ p : ℕ, Prime p → abs (τ p) ≤ 2 * (p : ℝ) ^ ((11 : ℝ) / 2)

FC-RamanujanTau.ramanujan_petersson

Ramsey numbers

MODULE CONTEXT: # Ramsey numbers The (graph) Ramsey number $R(k,\ell)$ is the least natural number n such that every simple graph on n vertices contains either a clique of size k or an independent set of size ℓ (equivalently, the complement graph contains a clique of size ℓ). We formalize the classical open problem of determining $R(5,5)$, together with the currently best known bounds $43 \le R(5,5) \le 46$. Note: the diagonal Ramsey number $R(n,n)$ can also be formulated in terms of 2-colorings of 2-subsets, as `Combinatorics.hypergraphRamsey 2 n` (see `FormalConjecturesForMathlib/Combinatorics/Ramsey.lean`). *References:* - [Wikipedia: Ramsey number](https://en.wikipedia.org/wiki/Ramsey_number) - [Rad] S. P. Radziszowski, *Small Ramsey Numbers*, Electronic Journal of Combinatorics, Dynamic Survey DS1. (Updated periodically.) https://www.combinatorics.org/ojs/index.php/eljc/article/view/DS1 - [Exoo89] G. Exoo, *A lower bound for* $R(5,5)$, Journal of Graph Theory 13 (1989), 97-98. DOI: 10.1002/jgt.3190130113 - [AM24] V. Angeltveit and B. McKay, *$R(5,5) \le 46$*, arXiv:2409.15709 (2024). - [OEIS A212954](https://oeis.org/A212954) - [MathWorld: Ramsey Number](https://mathworld.wolfram.com/RamseyNumber.html)

V2-000753 | Ramsey numbers | ramsey_number_five_five

research open | proof annotation: False | 官方代码 | 原文线索

The open problem: determine the Ramsey number $R(5,5)$. It is known that $43 \le R(5,5) \le 46$.

theorem ramsey_number_five_five :
R(5, 5) = ramsey_number_five_five.answer

FC-RamseyNumbers.ramsey_number_five_five

V2-000754 | Ramsey numbers | ramsey_number_five_five_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Lower bound $43 \le R(5,5)$, equivalently: there exists a graph on 42 vertices with no 5-clique and no independent set of size 5.

theorem ramsey_number_five_five_lower_bound :
∃ G : SimpleGraph (Fin 42), G.CliqueFree 5 ∧ (Gᶜ).CliqueFree 5

FC-RamseyNumbers.ramsey_number_five_five_lower_bound

V2-000755 | Ramsey numbers | ramsey_number_five_five_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Upper bound $R(5,5) \le 46$, i.e. every graph on 46 vertices contains a 5-clique or an independent set of size 5.

theorem ramsey_number_five_five_upper_bound :
IsGraphRamsey 46 5 5

FC-RamseyNumbers.ramsey_number_five_five_upper_bound

V2-000756 | Ramsey numbers | ramsey_number_four_five

research solved | proof annotation: False | 官方代码 | 原文线索

$R(4,5) = 25$ (McKay-Radziszowski).

theorem ramsey_number_four_five : R(4, 5) = 25

FC-RamseyNumbers.ramsey_number_four_five

V2-000757 | Ramsey numbers | ramsey_number_six_six

research open | proof annotation: False | 官方代码 | 原文线索

The diagonal Ramsey number $R(6,6)$ is unknown. The best known bounds recorded in [Rad] are $102 \le R(6,6) \le 165$.

theorem ramsey_number_six_six : R(6, 6) = ramsey_number_six_six.answer

FC-RamseyNumbers.ramsey_number_six_six

V2-000758 | Ramsey numbers | ramsey_number_six_six_lower_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Lower bound $102 \le R(6,6)$, equivalently: there exists a graph on 101 vertices with no 6-clique and no independent set of size 6.

theorem ramsey_number_six_six_lower_bound :
∃ G : SimpleGraph (Fin 101), G.CliqueFree 6 ∧ (Gᶜ).CliqueFree 6

FC-RamseyNumbers.ramsey_number_six_six_lower_bound

V2-000759 | Ramsey numbers | ramsey_number_six_six_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

Upper bound $R(6,6) \le 165$, i.e. every graph on 165 vertices contains a 6-clique or an independent set of size 6.

theorem ramsey_number_six_six_upper_bound :
IsGraphRamsey 165 6 6

FC-RamseyNumbers.ramsey_number_six_six_upper_bound

V2-000760 | Ramsey numbers | ramsey_number_three_eight

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,8) = 28$.

theorem ramsey_number_three_eight : R(3, 8) = 28

FC-RamseyNumbers.ramsey_number_three_eight

V2-000761 | Ramsey numbers | ramsey_number_three_five

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,5) = 14$.

theorem ramsey_number_three_five : R(3, 5) = 14

FC-RamseyNumbers.ramsey_number_three_five

V2-000762 | Ramsey numbers | ramsey_number_three_four

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,4) = 9$.

theorem ramsey_number_three_four : R(3, 4) = 9

FC-RamseyNumbers.ramsey_number_three_four

V2-000763 | Ramsey numbers | ramsey_number_three_nine

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,9) = 36$ (Grinstead-Roberts).

theorem ramsey_number_three_nine : R(3, 9) = 36

FC-RamseyNumbers.ramsey_number_three_nine

V2-000764 | Ramsey numbers | ramsey_number_three_seven

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,7) = 23$.

theorem ramsey_number_three_seven : R(3, 7) = 23

FC-RamseyNumbers.ramsey_number_three_seven

V2-000765 | Ramsey numbers | ramsey_number_three_six

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,6) = 18$.

theorem ramsey_number_three_six : R(3, 6) = 18

FC-RamseyNumbers.ramsey_number_three_six

V2-000766 | Ramsey numbers | ramsey_number_three_three

research solved | proof annotation: False | 官方代码 | 原文线索

$R(3,3) = 6$.

theorem ramsey_number_three_three : R(3, 3) = 6

FC-RamseyNumbers.ramsey_number_three_three

Rational distance problem

MODULE CONTEXT: # Rational distance problem *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Unit_square#Rational_distance_problem) - [mathoverflow/418260](https://mathoverflow.net/questions/418260/) asked by user [Yuan Yang](https://mathoverflow.net/users/177957/yuan-yang) - D19 in [Unsolved Problems in Number Theory](https://doi.org/10.1007/978-0-387-26677-0) by *Richard K. Guy*

V2-000767 | Rational distance problem | rational_distance_problem

research open | proof annotation: False | 官方代码 | 原文线索

Does there exist a point in the plane at rational distance from all four vertices of the unit square?

theorem rational_distance_problem :
∃ P : ℝ² , ∀ i, ¬ Irrational (dist P (UnitSquareCorners i))

FC-RationalDistanceProblem.rational_distance_problem

Infinite Regular Primes

MODULE CONTEXT: # Infinite Regular Primes We define the notion of regular primes, which are prime numbers that are coprime with the cardinality of the class group of the `p`-th cyclotomic field. We also state that there are infinitely many regular primes. *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Regular_prime)

V2-000768 | Infinite Regular Primes | infinitude_of_irregularprimes

research solved | proof annotation: False | 官方代码 | 原文线索

The set of irregular primes is infinite.

theorem infinitude_of_irregularprimes : irregularPrimes.Infinite

FC-RegularPrimes.infinitude_of_irregularprimes

V2-000769 | Infinite Regular Primes | regularprime_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture: The set of regular primes is infinite.

theorem regularprime_conjecture : RegularPrimeConjecture

FC-RegularPrimes.regularprime_conjecture

Decimal Repunit Primes

MODULE CONTEXT: # Decimal Repunit Primes A decimal repunit is a natural number whose decimal representation consists entirely of ones. It is conjectured that infinitely many decimal repunits are prime. *References:* - [Wikipedia: Repunit](https://en.wikipedia.org/wiki/Repunit#Decimal_repunit_primes) - [OEIS A002275](https://oeis.org/A002275) - [OEIS A004022](https://oeis.org/A004022) - [OEIS A004023](https://oeis.org/A004023)

V2-000770 | Decimal Repunit Primes | infinitely_many_repunit_primes

research open | proof annotation: False | 官方代码 | 原文线索

[Wikipedia's repunit article](https://en.wikipedia.org/wiki/Repunit#Decimal_repunit_primes) states: "It has been conjectured that there are infinitely many repunit primes."

```
theorem infinitely_many_repunit_primes :
{n : ℕ | (repunit10 n).Prime}.Infinite
```

FC-RepunitPrime.infinitely_many_repunit_primes

Resolution of singularities

MODULE CONTEXT: # Resolution of singularities A variety X over a field k admits a resolution of singularities if there is a smooth k-variety Y and a proper birational morphism $Y \to X$. Hironaka proved that every variety over a field of characteristic zero admits one. In positive characteristic this is known over a perfect field in dimension at most three, and is open from dimension four on. Perfectness of k cannot be dropped from the statement in this form: over an imperfect field it fails already in dimension zero, see `exists_not_hasResolution_of_not_perfectField` below. Over an arbitrary field one asks instead that Y be regular. *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Resolution_of_singularities) * [Hir1964] H. Hironaka, Resolution of singularities of an algebraic variety over a field of characteristic zero, [I](https://doi.org/10.2307/1970486) and [II](https://doi.org/10.2307/1970547), Ann. of Math. 79 (1964), 109--203 and 205--326. * [Kol2007] J. Kollár, [Resolution of singularities -- Seattle lecture](https://arxiv.org/abs/math/0508332), Theorem 36. * [Hau2010] H. Hauser, [On the problem of resolution of singularities in positive characteristic (or: a proof we are still waiting for)](https://doi.org/10.1090/S0273-0979-09-01274-9), Bull. Amer. Math. Soc. 47 (2010), 1--30. * [Lip1978] J. Lipman, [Desingularization of two-dimensional schemes](https://doi.org/10.2307/1971141), Ann. of Math. 107 (1978), 151--207. * [CP2008] V. Cossart and O. Piltant, [Resolution of singularities of threefolds in positive characteristic I](https://doi.org/10.1016/j.jalgebra.2008.03.032), J. Algebra 320 (2008), 1051--1082. * [CP2009] V. Cossart and O. Piltant, [Resolution of singularities of threefolds in positive characteristic II](https://doi.org/10.1016/j.jalgebra.2008.11.030), J. Algebra 321 (2009), 1836--1976. * [CP2019] V. Cossart and O. Piltant, [Resolution of singularities of arithmetical threefolds](https://doi.org/10.1016/j.jalgebra.2019.02.017), J. Algebra 529 (2019), 268--535.

V2-000771 | Resolution of singularities | resolution_of_singularities

research open | proof annotation: False | 官方代码 | 原文线索

Resolution of singularities in positive characteristic. Let k be a perfect field of characteristic $p > 0$ and let X be an integral scheme that is separated and of finite type over k. Then there is an integral scheme Y that is smooth over k together with a proper birational morphism $Y \to X$. This is open from dimension four on; see [Hau2010]. Perfectness of k is needed for the conclusion as stated; see `exists_not_hasResolution_of_not_perfectField`.

```
theorem resolution_of_singularities (k : Type u) [Field k] [PerfectField k] (p : ℕ)
[Fact p.Prime] [CharP k p] {X : Scheme.{u}} (sX : X ⟶ Spec (.of k)) [IsIntegral X]
[LocallyOfFiniteType sX] [QuasiCompact sX] [IsSeparated sX] :
Scheme.HasResolution sX
```

FC-AlgebraicGeometry.resolution_of_singularities

V2-000772 | Resolution of singularities | resolution_of_singularities_of_charZero

research solved | proof annotation: False | 官方代码 | 原文线索

Hironaka's theorem. Every integral scheme that is separated and of finite type over a field of characteristic zero admits a resolution of singularities. This is the main theorem of [Hir1964]; see [Kol2007], Theorem 36, for a statement in the generality used here.

```
theorem resolution_of_singularities_of_charZero (k : Type u) [Field k] [CharZero k]
{X : Scheme.{u}} (sX : X ⟶ Spec (.of k)) [IsIntegral X] [LocallyOfFiniteType sX]
[QuasiCompact sX] [IsSeparated sX] :
Scheme.HasResolution sX
```

FC-AlgebraicGeometry.resolution_of_singularities_of_charZero

V2-000773 | Resolution of singularities | resolution_of_singularities_of_topologicalKrullDim_le_three

research solved | proof annotation: False | 官方代码 | 原文线索

Resolution of singularities holds in dimension at most three over any perfect field. In characteristic zero this is [Hir1964]. In positive characteristic, dimension at most two is [Lip1978] and dimension three is due to Cossart and Piltant: [CP2008] and [CP2009] prove it for quasi-projective varieties over a field that is differentially finite over a perfect subfield, and [CP2019], Theorem 1.1, removes both restrictions, resolving every reduced separated Noetherian quasi-excellent scheme of dimension at most three in any characteristic. That theorem gives an everywhere regular source, which over a perfect field is the same as a smooth one. The problem is open in positive characteristic from dimension four on.

```
theorem resolution_of_singularities_of_topologicalKrullDim_le_three (k : Type u) [Field k]
[PerfectField k] {X : Scheme.{u}} (sX : X ⟶ Spec (.of k)) [IsIntegral X]
[LocallyOfFiniteType sX] [QuasiCompact sX] [IsSeparated sX]
(hX : topologicalKrullDim X ≤ 3) :
Scheme.HasResolution sX
```

FC-AlgebraicGeometry.resolution_of_singularities_of_topologicalKrullDim_le_three

Particular values of the Riemann zeta function

MODULE CONTEXT: # Particular values of the Riemann zeta function *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Particular_values_of_the_Riemann_zeta_function)

V2-000774 | Particular values of the Riemann zeta function | exists_irrational_of_five_seven_nine_eleven

research solved | proof annotation: False | 官方代码 | 原文线索

At least one of $\zeta(5), \zeta(7), \zeta(9)$ or $\zeta(11)$ is irrational. [Zu01] W. Zudilin (2001). _One of the numbers ζ(5), ζ(7), ζ(9), ζ(11) is irrational_. Russ. Math. Surv. 56 (4): 774-776.

theorem exists_irrational_of_five_seven_nine_eleven :
{5, 7, 9, 11} ∩ { a | ∃ x, Irrational x ∧ riemannZeta a = x} |>.Nonempty

FC-RiemannZetaValues.exists_irrational_of_five_seven_nine_eleven

V2-000775 | Particular values of the Riemann zeta function | infinite_irrational_at_odd

research solved | proof annotation: False | 官方代码 | 原文线索

There are infinitely many $\zeta(2n + 1)$, $n \in \mathbb{N}$, that are irrational. [Ri00] Rivoal, T. (2000). _La fonction zeta de Riemann prend une infinité de valeurs irrationnelles aux entiers impairs_. Comptes Rendus de l'Académie des Sciences, Série I. 331 (4): 267-270.

theorem infinite_irrational_at_odd :
{ n : ℕ | ∃ x, Irrational x ∧ riemannZeta (2 * n + 1) = x }.Infinite

FC-RiemannZetaValues.infinite_irrational_at_odd

V2-000776 | Particular values of the Riemann zeta function | irrational_eleven

research open | proof annotation: False | 官方代码 | 原文线索

$\zeta(11)$ is irrational.

theorem irrational_eleven : ∃ x, Irrational x ∧ riemannZeta 11 = x

FC-RiemannZetaValues.irrational_eleven

V2-000777 | Particular values of the Riemann zeta function | irrational_five

research solved | proof annotation: True | 官方代码 | 原文线索

$\zeta(5)$ is irrational. [Fa26] A. Fauzan (2026). [_$\zeta(5)$ is irrational_](https://zenodo.org/records/22826419). Preprint, 17 September 2026.

theorem irrational_five : ∃ x, Irrational x ∧ riemannZeta 5 = x

FC-RiemannZetaValues.irrational_five

V2-000778 | Particular values of the Riemann zeta function | irrational_nine

research open | proof annotation: False | 官方代码 | 原文线索

$\zeta(9)$ is irrational.

theorem irrational_nine : ∃ x, Irrational x ∧ riemannZeta 9 = x

FC-RiemannZetaValues.irrational_nine

V2-000779 | Particular values of the Riemann zeta function | irrational_odd

research open | proof annotation: False | 官方代码 | 原文线索

$\zeta(2n + 1)$ is irrational for any $n\in\mathbb{N}^{+}$.

theorem irrational_odd (n : ℕ) (hn : 0 < n) :
∃ x, Irrational x ∧ riemannZeta (2 * n + 1) = x

FC-RiemannZetaValues.irrational_odd

V2-000780 | Particular values of the Riemann zeta function | irrational_seven

research open | proof annotation: False | 官方代码 | 原文线索

$\zeta(7)$ is irrational.

theorem irrational_seven : ∃ x, Irrational x ∧ riemannZeta 7 = x

FC-RiemannZetaValues.irrational_seven

V2-000781 | Particular values of the Riemann zeta function | irrational_three

research solved | proof annotation: False | 官方代码 | 原文线索

$\zeta(3)$ is irrational. [Ap79] Apéry, R. (1979). _Irrationalité de ζ(2) et ζ(3)_. Astérisque. 61: 11-13.

theorem irrational_three : ∃ x, Irrational x ∧ riemannZeta 3 = x

FC-RiemannZetaValues.irrational_three

Rudin's conjecture on squares in arithmetic progressions

MODULE CONTEXT: # Rudin's conjecture on squares in arithmetic progressions *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Rudin%27s_conjecture) - [Ru60] Rudin, W., *Trigonometric series with gaps*, J. Math. Mech. 9 (1960), 203-227. - González-Jiménez, E. and Xarles, X., *On a conjecture of Rudin on squares in arithmetic progressions*, LMS J. Comput. Math. 17 (2014), 58-76.

V2-000782 | Rudin's conjecture on squares in arithmetic progressions | rudins_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Rudin's conjecture. The maximal number of squares among the first N terms of a non-trivial arithmetic progression grows at most like $\sqrt{N}$: $$Q(N) = O(\sqrt{N}).$$

theorem rudins_conjecture :
(fun N : ℕ => (Qmax N : ℝ)) =O[atTop] fun N : ℕ => Real.sqrt N

FC-RudinsConjecture.rudins_conjecture

V2-000783 | Rudin's conjecture on squares in arithmetic progressions | rudins_conjecture_unique

research open | proof annotation: False | 官方代码 | 原文线索

The strongest form of Rudin's conjecture also asserts *uniqueness*: for $N \ge 6$, any non-trivial arithmetic progression attaining the maximum $Q(N)$ has common difference 24. (Its initial term is then forced by $\gcd(24, a) = 1$; the progression $24n + 1$ is the canonical representative.)

theorem rudins_conjecture_unique (N : ℕ) (hN : 6 ≤ N) (q a : ℕ)
(hqa : IsNontrivial q a) (hmax : Q N q a = Qmax N) : q = 24

FC-RudinsConjecture.rudins_conjecture_unique

Sato-Tate conjecture

MODULE CONTEXT: # Sato-Tate conjecture The **Sato-Tate conjecture** describes the distribution of the normalized Frobenius traces $a_p(E)/(2\sqrt{p})$ of a non-CM elliptic curve E over $\mathbb{Q}$, as p ranges over the primes of good reduction: they equidistribute in $[-1,1]$ with respect to the **Sato-Tate measure** $$ \frac{2}{\pi}\sqrt{1-x^2}\,dx. $$ Originally conjectured independently by Mikio Sato and John Tate around 1960, it is now a theorem: the case of elliptic curves over totally real fields with nonintegral j-invariant was established through the work of Clozel, Harris, and Taylor [CHT08], Taylor [Tay08], and Harris, Shepherd-Barron, and Taylor [HST10]. The remaining non-CM case was settled by Barnet-Lamb, Geraghty, Harris, and Taylor [BGHT11]. In particular, it holds unconditionally for every non-CM elliptic curve over $\mathbb{Q}$. We use `ModularityConjecture.WeierstrassCurve.ap`, which agrees with the Frobenius trace at every prime where the supplied Weierstrass equation has integral coefficients and nonsingular reduction. Only finitely many primes are exceptional. Including these primes does not change the limiting density, so the formal statement averages over all primes below N. The non-CM hypothesis is expressed using the classification of rational CM j-invariants. Complex multiplication here means complex multiplication over $\overline{\mathbb{Q}}$. ## References - [CHT08] L. Clozel, M. Harris, R. Taylor, *Automorphy for some l-adic lifts of automorphic mod l Galois representations*, Publications Mathématiques de l'IHÉS 108 (2008), 1-181. https://doi.org/10.1007/s10240-008-0016-1 - [Tay08] R. Taylor, *Automorphy for some l-adic lifts of automorphic mod l Galois representations. II*, Publications Mathématiques de l'IHÉS 108 (2008), 183-239. https://doi.org/10.1007/s10240-008-0015-2 - [HST10] M. Harris, N. Shepherd-Barron, R. Taylor, *A family of Calabi-Yau varieties and potential automorphy*, Annals of Mathematics 171 (2010), no. 2, 779-813. https://doi.org/10.4007/annals.2010.171.779 - [BGHT11] T. Barnet-Lamb, D. Geraghty, M. Harris, R. Taylor, *A family of Calabi-Yau varieties and potential automorphy II*, Publications of the Research Institute for Mathematical Sciences 47 (2011), no. 1, 29-98. https://doi.org/10.2977/PRIMS/31 - [Wikipedia](https://en.wikipedia.org/wiki/Sato%E2%80%93Tate_conjecture)

V2-000784 | Sato-Tate conjecture | satoTate_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The Sato-Tate conjecture: for a non-CM elliptic curve E over $\mathbb{Q}$ and $-1 \le a \le b \le 1$, the proportion of primes $p < N$ whose normalized coefficient belongs to $[a,b]$ tends to $$ \frac{2}{\pi}\int_a^b \sqrt{1-x^2}\,dx $$ as $N \to \infty$. The finitely many primes where the supplied equation fails to have integral coefficients and nonsingular reduction do not affect this limit. Established through the work of Clozel-Harris-Taylor [CHT08], Taylor [Tay08], Harris-Shepherd-Barron-Taylor [HST10], and Barnet-Lamb-Geraghty-Harris-Taylor [BGHT11].

```
theorem satoTate_conjecture
(E : WeierstrassCurve ℚ) [E.IsElliptic] (hCM : ¬ HasCM E)
(a b : ℝ) (ha : -1 ≤ a) (hab : a ≤ b) (hb : b ≤ 1) :
Filter.Tendsto
(fun N : ℕ ↦
(primeCountInInterval E a b N : ℝ) /
((Nat.primesBelow N).card : ℝ))
Filter.atTop
(nhds (satoTateMeasure a b))
```

FC-SatoTateConjecture.satoTate_conjecture

Hypothesis H

MODULE CONTEXT: # Hypothesis H *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Schinzel%27s_hypothesis_H)

V2-000785 | Hypothesis H | schinzel_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Schinzel conjecture (H hypothesis) If a finite set of polynomials f_i satisfies both Schinzel and Bunyakovsky conditions, there exist infinitely many natural numbers n such that $f_i(n)$ are primes for all i.

```
theorem schinzel_conjecture (fs : Finset ℤ[X]) (hfs : ∀ f ∈ fs, BunyakovskyCondition f)
(hfs' : SchinzelCondition fs) : Infinite {n : ℕ | ∀ f ∈ fs, (f.eval (n : ℤ)).natAbs.Prime}
```

FC-Schinzel.schinzel_conjecture

Selfridge's conjectures

MODULE CONTEXT: # Selfridge's conjectures *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/John_Selfridge#Selfridge's_conjecture_about_primality_testing)

V2-000786 | Selfridge's conjectures | selfridge_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

PSW conjecture (Selfridge's test) Let p be an odd number, with $p \equiv \pm 2 \pmod{5}$, $2^{p-1} \equiv 1 \pmod{p}$ and $F_{p+1} \equiv 0 \pmod{p}$, then p is a prime number.

```
theorem selfridge_conjecture (p : ℕ) (hp : IsSelfridge p) : p.Prime
```

FC-Selfridge.selfridge_conjecture

V2-000787 | Selfridge's conjectures | selfridge_seq_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Selfridge conjectured that the number of prime factors of the `n`-th Fermat number does not grow monotonically in n.

```
theorem selfridge_seq_conjecture : ¬ Monotone fermatFactors
```

FC-Selfridge.selfridge_seq_conjecture

Sendov's conjecture

MODULE CONTEXT: # Sendov's conjecture *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Sendov%27s_conjecture) - Lech Mazur, *A Computer-Assisted Proof of Sendov's Conjecture*, [ProofAtlas, August 5, 2026](https://www.proofatlas.ai/papers/sendov-conjecture/SENDOV_CONJECTURE_PROOF_AUGUST_5_2026.pdf). - Terence Tao, [*A digestion of the proof of Sendov's conjecture*](https://terrytao.wordpress.com/2026/08/12/a-digestion-of-the-proof-of-sendovs-conjecture/). - [Tao's Lean formalization of the digested proof](https://github.com/teorth/sendov). Tags: Sendov Conjecture, Ilieff's Conjecture.

V2-000788 | Sendov's conjecture | sendov_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Sendov's conjecture states that for a polynomial $$f(z)=(z-r_{1})\cdots (z-r_{n}),\qquad (n\geq 2)$$ with all roots $r_1, ..., r_n$ inside the closed unit disk $|z| ≤ 1$, each of the n roots is at a distance no more than 1 from at least one critical point.

```
theorem sendov_conjecture (n : ℕ) (hn : 2 ≤ n) : n.SatisfiesSendovConjecture
```

FC-Sendov.sendov_conjecture

Serre's multiplicity conjectures

MODULE CONTEXT: # Serre's multiplicity conjectures Let R be a regular local ring and let M, N be finitely generated R-modules such that $M \otimes_R N$ has finite length. Serre defined the intersection multiplicity $$\chi(M, N) = \sum_{i \ge 0} (-1)^i \ell_R(\operatorname{Tor}_i^R(M, N)).$$ He proved the dimension inequality $\dim M + \dim N \le \dim R$ for every regular local ring, and conjectured the following three properties: - **Non-negativity:** $\chi(M, N) \ge 0$. - **Vanishing:** if $\dim M + \dim N < \dim R$, then $\chi(M, N) = 0$. - **Positivity:** if $\dim M + \dim N = \dim R$, then $\chi(M, N) > 0$. Serre proved all three when R is of equal characteristic, or of mixed characteristic and unramified. Vanishing was proved in general by Roberts and, independently, by Gillet and Soulé. Non-negativity was proved in general by Gabber. Positivity is open in general. For Serre's two special cases, this file states only positivity, since non-negativity and vanishing are now known in general. The intersection multiplicity is `Module.intersectionMultiplicity`, defined in `FormalConjecturesForMathlib.RingTheory.IntersectionMultiplicity`. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Serre%27s_multiplicity_conjectures) - [Se00] J.-P. Serre, *Local algebra*, Springer Monographs in Mathematics, Springer, 2000, Chapter V, Part B. - [Ro85] P. Roberts, *The vanishing of intersection multiplicities of perfect complexes*, Bull. Amer. Math. Soc. 13 (1985), 127-130. - [GS87] H. Gillet, C. Soulé, *Intersection theory using Adams operations*, Invent. Math. 90 (1987), 243-277. - [Ro98] P. Roberts, *Recent developments on Serre's multiplicity conjectures: Gabber's proof of the nonnegativity conjecture*, Enseign. Math. 44 (1998), 305-324. - [Sk19] C. Skalit, *Positivity of intersection multiplicity over a two-dimensional base*, J. Pure Appl. Algebra 223 (2019), 1801-1816. [arXiv:1510.05146](https://arxiv.org/abs/1510.05146)

V2-000789 | Serre's multiplicity conjectures | intersectionMultiplicity_eq_zero_of_lt

research solved | proof annotation: False | 官方代码 | 原文线索

Vanishing. Let R be a regular local ring and let M, N be finitely generated R-modules such that $M \otimes_R N$ has finite length. If $\dim M + \dim N < \dim R$, then $\chi(M, N) = 0$. Conjectured by Serre and proved by Roberts [Ro85] and, independently, by Gillet and Soulé [GS87].

```
theorem intersectionMultiplicity_eq_zero_of_lt (h : IsFiniteLength R (M ⊗[R] N))
(hdim : Module.supportDim R M + Module.supportDim R N < ringKrullDim R) :
intersectionMultiplicity R M N = 0
```

FC-SerreMultiplicityConjectures.intersectionMultiplicity_eq_zero_of_lt

V2-000790 | Serre's multiplicity conjectures | intersectionMultiplicity_nonneg

research solved | proof annotation: False | 官方代码 | 原文线索

Non-negativity. Let R be a regular local ring and let M, N be finitely generated R-modules such that $M \otimes_R N$ has finite length. Then $\chi(M, N) \ge 0$. Conjectured by Serre and proved by Gabber in 1995 [Ro98].

```
theorem intersectionMultiplicity_nonneg (h : IsFiniteLength R (M ⊗[R] N)) :
0 ≤ intersectionMultiplicity R M N
```

FC-SerreMultiplicityConjectures.intersectionMultiplicity_nonneg

V2-000791 | Serre's multiplicity conjectures | intersectionMultiplicity_pos

research open | proof annotation: False | 官方代码 | 原文线索

Positivity conjecture. Let R be a regular local ring and let M, N be finitely generated R-modules such that $M \otimes_R N$ has finite length. If $\dim M + \dim N = \dim R$, then $\chi(M, N) > 0$. The hypothesis on dimensions forces M and N to be nonzero, since the dimension of the zero module is $\bot$.

```
theorem intersectionMultiplicity_pos (h : IsFiniteLength R (M ⊗[R] N))
(hdim : Module.supportDim R M + Module.supportDim R N = ringKrullDim R) :
0 < intersectionMultiplicity R M N
```

FC-SerreMultiplicityConjectures.intersectionMultiplicity_pos

V2-000792 | Serre's multiplicity conjectures | intersectionMultiplicity_pos_of_equalCharacteristic

research solved | proof annotation: False | 官方代码 | 原文线索

Serre proved the positivity conjecture when R is of equal characteristic, that is, when R contains a field [Se00].

```
theorem intersectionMultiplicity_pos_of_equalCharacteristic (k : Type*) [Field k] [Algebra k R]
(h : IsFiniteLength R (M ⊗[R] N))
(hdim : Module.supportDim R M + Module.supportDim R N = ringKrullDim R) :
0 < intersectionMultiplicity R M N
```

FC-SerreMultiplicityConjectures.intersectionMultiplicity_pos_of_equalCharacteristic

V2-000793 | Serre's multiplicity conjectures | intersectionMultiplicity_pos_of_unramified

research solved | proof annotation: False | 官方代码 | 原文线索

Serre proved the positivity conjecture when R is of mixed characteristic and unramified, that is, when the characteristic p of the residue field is not in the square of the maximal ideal of R [Se00]. The hypothesis $p \notin \mathfrak m^2$ excludes $p = 0$ and the equal characteristic case, where $p = 0$ in R.

```
theorem intersectionMultiplicity_pos_of_unramified (p : ℕ) [CharP (ResidueField R) p]
(hp : (p : R) ∉ maximalIdeal R ^ 2) (h : IsFiniteLength R (M ⊗[R] N))
(hdim : Module.supportDim R M + Module.supportDim R N = ringKrullDim R) :
0 < intersectionMultiplicity R M N
```

FC-SerreMultiplicityConjectures.intersectionMultiplicity_pos_of_unramified

V2-000794 | Serre's multiplicity conjectures | supportDim_add_supportDim_le_ringKrullDim

research solved | proof annotation: False | 官方代码 | 原文线索

Dimension inequality. Let R be a regular local ring and let M, N be finitely generated R-modules such that $M \otimes_R N$ has finite length. Then $\dim M + \dim N \le \dim R$. Proved by Serre for every regular local ring [Se00].

```
theorem supportDim_add_supportDim_le_ringKrullDim (h : IsFiniteLength R (M ⊗[R] N)) :
Module.supportDim R M + Module.supportDim R N ≤ ringKrullDim R
```

FC-SerreMultiplicityConjectures.supportDim_add_supportDim_le_ringKrullDim

Sidorenko's conjecture (1993)

MODULE CONTEXT: # Sidorenko's conjecture (1993) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Sidorenko%27s_conjecture) * [Si93] Sidorenko, A. (1993). "A correlation inequality for bipartite graphs." *Graphs Combin.* 9, pp. 201--204. * [CoFo10] Conlon, D. and Fox, J. (2010). "Bounds for graph regularity and removal lemmas." *Geom. Funct. Anal.* 22, pp. 1191--1256. * [KLL18] Kim, J.H., Lee, C., Lee, J. (2018). "Two approaches to Sidorenko's conjecture." *Trans. Amer. Math. Soc.* 370, pp. 8515--8552. * [ArXiv2605] [arXiv:2605.14138](https://arxiv.org/abs/2605.14138) * [BR65] Blakley, G. R. and Roy, P. (1965). "A Hölder type inequality for symmetric matrices with nonnegative entries." *Proc. Amer. Math. Soc.* 16, pp. 1244--1245. * [OAI26a] OpenAI, *A counterexample to Sidorenko's conjecture*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-counterexample-to-Sidorenkos-conjecture-September-23-2026/paper.pdf

V2-000795 | Sidorenko's conjecture (1993) | sidorenko_K2

research solved | proof annotation: False | 官方代码 | 原文线索

Case `H = K_2` (single edge): Sidorenko's inequality holds trivially with equality. When `H` is `K_2` (the single-edge graph on 2 vertices), `e(H) = 1`, so the RHS of Sidorenko's inequality is just `t(K_2, G)^1 = t(K_2, G) = t(H, G)`, which equals the LHS. Hence the inequality holds as equality. The proof records that `(completeGraph (Fin 2)).edgeFinset.card = 1` and then reduces the claim to `t(K_2, G) ≤ t(K_2, G)`, which is `le_refl`.

```
theorem sidorenko_K2 {W : Type} [Fintype W] [DecidableEq W]
(G : SimpleGraph W) [DecidableRel G.Adj] :
homDensity (completeGraph (Fin 2)) G ^
((completeGraph (Fin 2)).edgeFinset.card) ≤
homDensity (completeGraph (Fin 2)) G
```

FC-SidorenkoConjecture.sidorenko_K2

V2-000796 | Sidorenko's conjecture (1993) | sidorenko_K22

research solved | proof annotation: False | 官方代码 | 原文线索

Case `H = K_{2,2}` (four-cycle, also called `C_4`): Sidorenko's conjecture holds, by Cauchy-Schwarz. The textbook statement at `H = K_{2,2}` is `t(K_2, G)^{e(K_{2,2})} = t(K_2, G)^4 ≤ t(K_{2,2}, G)`. **Proof sketch.** Write `d(a) := G.degree a`. Then - `homCount(K_2, G) = ∑_a d(a) = 2·|E(G)|` (handshaking). - `homCount(K_{2,2}, G) = ∑_{b₀, b₁} |N(b₀) ∩ N(b₁)|²` (product structure of bipartite homomorphism). - `∑_{b₀, b₁} |N(b₀) ∩ N(b₁)| = ∑_a d(a)²` (swap sums). - Cauchy-Schwarz #1: `(∑_{b₀, b₁} |N(b₀) ∩ N(b₁)|)² ≤ |W|² · ∑_{b₀, b₁} |N(b₀) ∩ N(b₁)|²`. - Cauchy-Schwarz #2: `(∑_a d(a))² ≤ |W| · ∑_a d(a)²`. - Chain: `(∑_a d(a))⁴ ≤ |W|⁴ · homCount(K_{2,2}, G)`. - Divide by `|W|^8` to get `t(K_2, G)^4 ≤ t(K_{2,2}, G)`. The proof uses `Finset.sum_mul_sq_le_sq_mul_sq` (discrete Cauchy-Schwarz) from `Mathlib.Algebra.Order.BigOperators.Ring.Finset`. **Status (2026-04-22):** main theorem closed sorry-free. See [Si93].

```
theorem sidorenko_K22 {W : Type} [Fintype W] [DecidableEq W]
(G : SimpleGraph W) [DecidableRel G.Adj] :
homDensity (completeGraph (Fin 2)) G ^
((completeBipartiteGraph (Fin 2) (Fin 2)).edgeFinset.card) ≤
homDensity (completeBipartiteGraph (Fin 2) (Fin 2)) G
```

FC-SidorenkoConjecture.sidorenko_K22

V2-000797 | Sidorenko's conjecture (1993) | sidorenko_completeBipartiteGraph_graphon

research solved | proof annotation: False | 官方代码 | 原文线索

Case: `H = K_{a,b}` is a complete bipartite graph (Sidorenko 1993, graphon version). Every complete bipartite graph $K_{a,b}$ satisfies Sidorenko's inequality for all graphons on $[0, 1]$.

```
theorem sidorenko_completeBipartiteGraph_graphon {A B : Type*} [Fintype A] [Fintype B]
[DecidableEq A] [DecidableEq B] (W : LimitObjects.Graphon) :
(graphonEdgeDensity W) ^ (completeBipartiteGraph A B).edgeFinset.card ≤
graphonHomDensity (completeBipartiteGraph A B) W
```

FC-SidorenkoConjecture.sidorenko_completeBipartiteGraph_graphon

V2-000798 | Sidorenko's conjecture (1993) | sidorenko_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Sidorenko's conjecture (1993). For every finite bipartite simple graph H and every finite simple graph G: $t(H, G) \ge t(K_2, G)^{e(H)}$, where K_2 denotes the single-edge graph on 2 vertices (i.e. `completeGraph (Fin 2)`). This is false. It was disproved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof, which gives a bipartite H with 35 vertices and 66 edges and a finite host G with $t(H, G) < t(K_2, G)^{66}$.

```
theorem sidorenko_conjecture :
¬ ∀ {V W : Type} [Fintype V] [Fintype W] [DecidableEq V] [DecidableEq W] [Nonempty W]
(H : SimpleGraph V) (G : SimpleGraph W)
[DecidableRel H.Adj] [DecidableRel G.Adj],
H.IsBipartite →
homDensity (completeGraph (Fin 2)) G ^ H.edgeFinset.card ≤ homDensity H G
```

FC-SidorenkoConjecture.sidorenko_conjecture

V2-000799 | Sidorenko's conjecture (1993) | sidorenko_conjecture_graphon

research solved | proof annotation: False | 官方代码 | 原文线索

Sidorenko's conjecture for graphons (1993). For every finite bipartite simple graph H and every graphon W on $[0, 1]$ with Lebesgue measure: $t(H, W) \ge t(K_2, W)^{e(H)}$, where $t(K_2, W) = p(W)$ is the edge density of W, and $t(H, W)$ is the graphon homomorphism density of H in W. This is false. The finite counterexample of [OAI26a] (see `sidorenko_conjecture`) also refutes the graphon form: the step graphon of the finite host graph has the same homomorphism densities and the same edge density as the host. This reduction is not part of the linked Lean proof of the finite form.

```
theorem sidorenko_conjecture_graphon :
¬ ∀ {V : Type*} [Fintype V] [DecidableEq V] (H : SimpleGraph V) [DecidableRel H.Adj],
H.IsBipartite →
∀ (W : Graphon),
(graphonEdgeDensity W) ^ H.edgeFinset.card ≤ graphonHomDensity H W
```

FC-SidorenkoConjecture.sidorenko_conjecture_graphon

V2-000800 | Sidorenko's conjecture (1993) | sidorenko_even_cycle_graphon

research solved | proof annotation: False | 官方代码 | 原文线索

Case: `H = C_{2k}` is an even cycle (Sidorenko 1993, graphon version). Every even cycle C_{2k} satisfies Sidorenko's inequality for all graphons on $[0, 1]$.

```
theorem sidorenko_even_cycle_graphon (k : ℕ) (hk : 1 ≤ k)
(W : LimitObjects.Graphon) :
(graphonEdgeDensity W) ^ (cycleGraph (2 * k)).edgeFinset.card ≤
graphonHomDensity (cycleGraph (2 * k)) W
```

FC-SidorenkoConjecture.sidorenko_even_cycle_graphon

V2-000801 | Sidorenko's conjecture (1993) | sidorenko_tree

research solved | proof annotation: False | 官方代码 | 原文线索

Case: `H` is a tree (Sidorenko 1993). If `H` is a finite tree then Sidorenko's inequality holds. **Proof idea (Sidorenko 1993):** by induction on the tree, applying Jensen / convexity. The number of homomorphisms from a tree `T` with `v` vertices and `v - 1` edges into `G` factors nicely in the degree sequence of `G`, and AM-GM / convexity gives the required lower bound. **Current status (2026-04-22, partial):** - The subsingleton base case (`|V(T)| ≤ 1`) is dispatched via `sidorenko_tree_subsingleton`. - The main inductive step (leaf extraction + AM-GM on the degree sequence) is deferred; it requires a walk-parametrised homomorphism count for trees and a discrete Jensen inequality. See [Si93]. The full proof is left as `sorry`; the subsingleton case is closed.

```
theorem sidorenko_tree {V W : Type} [Fintype V] [Fintype W]
[DecidableEq V] [DecidableEq W] [Nonempty W]
(H : SimpleGraph V) (G : SimpleGraph W)
[DecidableRel H.Adj] [DecidableRel G.Adj]
(hTree : H.IsTree) :
homDensity (completeGraph (Fin 2)) G ^ (H.edgeFinset.card) ≤ homDensity H G
```

FC-SidorenkoConjecture.sidorenko_tree

V2-000802 | Sidorenko's conjecture (1993) | sidorenko_tree_graphon

research solved | proof annotation: False | 官方代码 | 原文线索

Case: `H` is a tree (Sidorenko 1993, graphon version). If `H` is a finite tree then Sidorenko's inequality holds for all graphons on $[0, 1]$.

```
theorem sidorenko_tree_graphon {V : Type*} [Fintype V] [DecidableEq V]
(H : SimpleGraph V) [DecidableRel H.Adj] (hTree : H.IsTree)
(W : LimitObjects.Graphon) :
(graphonEdgeDensity W) ^ H.edgeFinset.card ≤ graphonHomDensity H W
```

FC-SidorenkoConjecture.sidorenko_tree_graphon

V2-000803 | Sidorenko's conjecture (1993) | sidorenko_tree_subsingleton

research solved | proof annotation: False | 官方代码 | 原文线索

Base case of Sidorenko for trees. When the tree `H` has at most one vertex, both sides of Sidorenko's inequality evaluate to `1` (there are no edges, and every function is a homomorphism). Thus the inequality `1 ≤ 1` holds trivially. This covers the `|V(T)| = 1` base case of the induction on tree size.

```
theorem sidorenko_tree_subsingleton {V W : Type} [Fintype V] [Fintype W]
[Subsingleton V] [DecidableEq V] [DecidableEq W] [Nonempty W]
(H : SimpleGraph V) (G : SimpleGraph W)
[DecidableRel H.Adj] [DecidableRel G.Adj] :
homDensity (completeGraph (Fin 2)) G ^ (H.edgeFinset.card) ≤ homDensity H G
```

FC-SidorenkoConjecture.sidorenko_tree_subsingleton

V2-000804 | Sidorenko's conjecture (1993) | tournament_anti_sidorenko_single_even_degree_tree

research solved | proof annotation: False | 官方代码 | 原文线索

TAS Trees Conjecture: Trees with a single even-degree vertex. Proven case of TAS Trees Conjecture: any tree containing exactly one vertex of even degree possesses an orientation satisfying the Tournament Anti-Sidorenko inequality $t_{\vec{T}}(G) \le 2^{-e(T)}$.

```
theorem tournament_anti_sidorenko_single_even_degree_tree {V : Type*} [Fintype V]
[DecidableEq V]
(T : SimpleGraph V) [DecidableRel T.Adj] (hTree : T.IsTree)
(hEven : (Finset.univ.filter (fun v => Even (T.degree v))).card = 1) :
∃ (D : Digraph V),
D.IsOrientation T ∧
∀ {W : Type*} [Fintype W] [DecidableEq W] [Nonempty W]
(G : Digraph W) [DecidableRel G.Adj],
G.IsTournament →
Digraph.homDensity D G ≤ (1 / 2 : ℝ) ^ T.edgeFinset.card
```

FC-SidorenkoConjecture.tournament_anti_sidorenko_single_even_degree_tree

V2-000805 | Sidorenko's conjecture (1993) | tournament_anti_sidorenko_spider234

research solved | proof annotation: False | 官方代码 | 原文线索

TAS Trees Conjecture: The $(2,3,4)$-spider tree (Solved case). The $(2,3,4)$-spider tree satisfies the Tournament Anti-Sidorenko Trees Conjecture. *Reference:* * [ArXiv 2605.14138](https://arxiv.org/abs/2605.14138)

```
theorem tournament_anti_sidorenko_spider234 {V : Type*} [Fintype V] [DecidableEq V]
(T : SimpleGraph V) [DecidableRel T.Adj] (hSpider : IsSpider234 T) :
∃ (D : Digraph V),
D.IsOrientation T ∧
∀ {W : Type*} [Fintype W] [DecidableEq W] [Nonempty W]
(G : Digraph W) [DecidableRel G.Adj],
G.IsTournament →
Digraph.homDensity D G ≤ (1 / 2 : ℝ) ^ T.edgeFinset.card
```

FC-SidorenkoConjecture.tournament_anti_sidorenko_spider234

V2-000806 | Sidorenko's conjecture (1993) | tournament_anti_sidorenko_trees_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Tournament Anti-Sidorenko (TAS) Trees Conjecture. For every finite undirected tree T, there exists an orientation $\vec{T}$ of its edges such that for any finite tournament G, the homomorphism density satisfies: $$ t_{\vec{T}}(G) \le 2^{-e(T)} $$ where $e(T)$ is the total number of edges in T.

```
theorem tournament_anti_sidorenko_trees_conjecture :
∀ {V : Type*} [Fintype V] [DecidableEq V] (T : SimpleGraph V) [DecidableRel T.Adj],
T.IsTree →
∃ (D : Digraph V),
D.IsOrientation T ∧
∀ {W : Type*} [Fintype W] [DecidableEq W] [Nonempty W]
(G : Digraph W) [DecidableRel G.Adj],
G.IsTournament →
Digraph.homDensity D G ≤ (1 / 2 : ℝ) ^ T.edgeFinset.card
```

FC-SidorenkoConjecture.tournament_anti_sidorenko_trees_conjecture

V2-000807 | Sidorenko's conjecture (1993) | tournament_anti_sidorenko_trees_conjecture_tournamenton

research open | proof annotation: False | 官方代码 | 原文线索

Tournament Anti-Sidorenko (TAS) Trees Conjecture (Tournamenton limit version). For every finite undirected tree T, there exists an orientation $\vec{T}$ of its edges such that for every tournamenton $W : [0, 1]^2 \to [0, 1]$, the homomorphism density satisfies: $$ t_{\vec{T}}(W) \le 2^{-e(T)} $$ where $e(T)$ is the total number of edges in T.

```
theorem tournament_anti_sidorenko_trees_conjecture_tournamenton :
∀ {V : Type*} [Fintype V] [DecidableEq V] (T : SimpleGraph V) [DecidableRel T.Adj],
T.IsTree →
∃ (D : Digraph V),
D.IsOrientation T ∧
∀ (W : LimitObjects.Tournamenton),
tournamentonHomDensity D W ≤ (1 / 2 : ℝ) ^ T.edgeFinset.card
```

FC-SidorenkoConjecture.tournament_anti_sidorenko_trees_conjecture_tournamenton

Sierpiński number

MODULE CONTEXT: # Sierpiński number *References:* - [Wikipedia, Sierpiński number](https://en.wikipedia.org/wiki/Sierpi%C5%84ski_number) - [Si60] Sierpiński, W., Elementary Theory of Numbers. Państwowe Wydawnictwo Naukowe, Warsaw (1960). A positive odd integer k is a *Sierpiński number* if $k \cdot 2^n + 1$ is composite for all natural numbers n. In 1960, Sierpiński proved that there are infinitely many such numbers. John Selfridge proved in 1962 that 78557 is a Sierpiński number. It is conjectured to be the smallest. ## Sierpiński problem The *Sierpiński problem* asks: is 78557 the smallest Sierpiński number? ## Prime Sierpiński problem The *prime Sierpiński problem* asks: is 271129 the smallest *prime* Sierpiński number? ## Extended Sierpiński problem The *extended Sierpiński problem* asks: is 271129 the second-smallest Sierpiński number?

V2-000808 | Sierpiński number | extended_sierpinski_problem

research open | proof annotation: False | 官方代码 | 原文线索

The extended Sierpiński problem. Is 271129 the second-smallest Sierpiński number? Even if 78557 is confirmed as the smallest Sierpiński number, there could exist a composite Sierpiński number k with $78557 < k < 271129$. We formalize "second-smallest" as: the least Sierpiński number k such that there exists exactly one Sierpiński number below it.

```
theorem extended_sierpinski_problem :
IsLeast {k | k.IsSierpinskiNumber ∧
∃ k', k'.IsSierpinskiNumber ∧ k' < k} 271129
```

FC-SierpinskiNumber.extended_sierpinski_problem

V2-000809 | Sierpiński number | prime_sierpinski_problem

research open | proof annotation: False | 官方代码 | 原文线索

The prime Sierpiński problem. Is 271129 the smallest prime Sierpiński number? In 1976, Nathan Mendelsohn determined that the second provable Sierpiński number is the prime $k = 271129$.

```
theorem prime_sierpinski_problem :
IsLeast {k | k.IsSierpinskiNumber ∧ k.Prime} 271129
```

FC-SierpinskiNumber.prime_sierpinski_problem

V2-000810 | Sierpiński number | selfridge_78557

research solved | proof annotation: False | 官方代码 | 原文线索

Selfridge proved in 1962 that 78557 is a Sierpiński number by showing that all numbers of the form $78557 \cdot 2^n + 1$ have a factor in the covering set $\{3, 5, 7, 13, 19, 37, 73\}$.

```
theorem selfridge_78557 : Nat.IsSierpinskiNumber 78557
```

FC-SierpinskiNumber.selfridge_78557

V2-000811 | Sierpiński number | selfridge_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The Sierpiński problem (Selfridge's conjecture). Is 78557 the smallest Sierpiński number? Selfridge conjectured that 78557 is the smallest Sierpiński number. He proved in 1962 that 78557 is indeed a Sierpiński number by showing that all numbers of the form $78557 \cdot 2^n + 1$ have a factor in the covering set $\{3, 5, 7, 13, 19, 37, 73\}$.

```
theorem selfridge_conjecture :
IsLeast {k | k.IsSierpinskiNumber} 78557
```

FC-SierpinskiNumber.selfridge_conjecture

Six standard deviations suffice (Spencer's theorem)

MODULE CONTEXT: # Six standard deviations suffice (Spencer's theorem) Spencer's theorem in discrepancy theory: for every n and every family of n subsets $S_1, \dots, S_n$ of $\{1, \dots, n\}$, there is a colouring $\chi : \{1, \dots, n\} \to \{-1, +1\}$ such that $$\left|\sum_{j \in S_i} \chi(j)\right| \le 6\sqrt{n}$$ for every i. A uniformly random colouring only achieves the bound $O(\sqrt{n \log n})$, so Spencer's theorem ("six standard deviations suffice") is a genuine improvement over the basic probabilistic method; its proof uses a partial-colouring argument based on the entropy method, and the bound $\Theta(\sqrt{n})$ is tight up to the value of the constant. Spencer's original argument is non-constructive; polynomial-time algorithms achieving $O(\sqrt{n})$ were later found by Bansal (2010) and Lovett-Meka (2012). The Komlós conjecture (the subject of a companion file in this directory) would strengthen this theorem: applied to the incidence vectors of the sets, scaled by $1/\sqrt{n}$, it would give a discrepancy bound $O(\sqrt{n})$ for any number of sets, not just n of them. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Discrepancy_of_hypergraphs#General_hypergraphs) - [J. Spencer, *Six standard deviations suffice*, Trans. Amer. Math. Soc. **289** (1985), 679-706](https://doi.org/10.1090/S0002-9947-1985-0784009-0) - [N. Bansal, *Constructive algorithms for discrepancy minimization*, FOCS 2010, 3-10](https://doi.org/10.1109/FOCS.2010.7) - [S. Lovett and R. Meka, *Constructive discrepancy minimization by walking on the edges*, SIAM J. Comput. **44** (2015), 1573-1582](https://doi.org/10.1137/130929400)

V2-000812 | Six standard deviations suffice (Spencer's theorem) | six_standard_deviations

research solved | proof annotation: False | 官方代码 | 原文线索

Six standard deviations suffice (Spencer, 1985) For every n and every family of n subsets $S_1, \dots, S_n$ of $\{1, \dots, n\}$, there is a colouring $\chi : \{1, \dots, n\} \to \{-1, +1\}$ such that $\left|\sum_{j \in S_i} \chi(j)\right| \le 6\sqrt{n}$ for every i.

```
theorem six_standard_deviations :
∀ (n : ℕ) (S : Fin n → Finset (Fin n)),
∃ χ : Fin n → ℝ, (∀ j, χ j = 1 ∨ χ j = -1) ∧
∀ i, |∑ j ∈ S i, χ j| ≤ 6 * Real.sqrt n
```

FC-SixStandardDeviations.six_standard_deviations

The small Cohen-Macaulay modules conjecture

MODULE CONTEXT: # The small Cohen-Macaulay modules conjecture Let $(R, \mathfrak m)$ be a Noetherian local ring of Krull dimension d. A *system of parameters* of R is a sequence $x_1, \dots, x_d$ of d elements of R with $\sqrt{(x_1, \dots, x_d)} = \mathfrak m$, equivalently with $R / (x_1, \dots, x_d)$ Artinian. A *small Cohen-Macaulay module*, also called a maximal Cohen-Macaulay module, is a finitely generated R-module $M \ne 0$ such that some system of parameters of R is a regular sequence on M. A *balanced big Cohen-Macaulay module* is a module W, not necessarily finitely generated, with $\mathfrak m W \ne W$ and on which every system of parameters is a regular sequence. For finitely generated modules the two notions agree. **Small Cohen-Macaulay modules conjecture.** If R is complete, then R has a small Cohen-Macaulay module. Hochster conjectured this for complete local domains in the early 1970s. In the 2000s he conjectured the opposite, that there are complete local domains with no small Cohen-Macaulay module [Ho17]. Small Cohen-Macaulay modules are known to exist when $\dim R \le 2$, and when R is $\mathbb N$-graded over a perfect field of characteristic p with an isolated non-Cohen-Macaulay point at the origin, a case first observed by Hartshorne and rediscovered by Peskine and Szpiro [Ho75b]. In dimension at least three the conjecture is open in every characteristic, although classes of examples keep being found there, such as three-dimensional F-pure complete local $\mathbb F_p$-algebras, due to Schoutens, and three-dimensional completions of $\mathbb N$-graded rings over a field of characteristic p, due to Hochster; both are listed, with references, in [ST23]. The analogous statement for algebras is false: Bhatt constructed complete local normal domains of characteristic p admitting no module-finite extension ring that is Cohen-Macaulay [Bh14]. Balanced big Cohen-Macaulay modules, by contrast, are now known to exist over every Noetherian local ring, by Hochster in equal characteristic and by André in mixed characteristic. Systems of parameters and Cohen-Macaulay modules are defined in `FormalConjecturesForMathlib.RingTheory.SystemOfParameters` and `FormalConjecturesForMathlib.RingTheory.CohenMacaulayModule`. *References:* - [Wikipedia, Homological conjectures in commutative algebra](https://en.wikipedia.org/wiki/Homological_conjectures_in_commutative_algebra), conjectures 8 and 14. - [Ho75a] M. Hochster, *Topics in the homological theory of modules over commutative rings*, C.B.M.S. Regional Conf. Ser. in Math. 24, Amer. Math. Soc., 1975. - [Ho75b] M. Hochster, *Big Cohen-Macaulay modules and algebras and embeddability in rings of Witt vectors*, Queen's Papers in Pure and Applied Math. 42, 1975, 106-195. - [Ho17] M. Hochster, *Homological conjectures and lim Cohen-Macaulay sequences*, in Homological and Computational Methods in Commutative Algebra, Springer INdAM Ser. 20, Springer, 2017. [PDF](https://sites.lsa.umich.edu/hochster/wp-content/uploads/sites/1337/2024/08/DSlim2.pdf), Conjectures 2.1 and 2.2. - [An18] Y. André, *La conjecture du facteur direct*, Publ. Math. IHÉS 127 (2018), 71-93. [arXiv:1609.00345](https://arxiv.org/abs/1609.00345). This is the paper [Ho17] credits with the existence of big Cohen-Macaulay algebras; it rests on the perfectoid Abhyankar lemma of *Le lemme d'Abhyankar perfectoïde*, Publ. Math. IHÉS 127 (2018), 1-70. - [Stacks, Tag 00N6](https://stacks.math.columbia.edu/tag/00N6), on regular sequences in a Cohen-Macaulay module. - [ST23] K. Shimomoto, E. Tavanfar, *Remarks on the Small Cohen-Macaulay conjecture and new instances of maximal Cohen-Macaulay modules*, J. Algebra 634 (2023), 667-697. [arXiv:2203.10368](https://arxiv.org/abs/2203.10368) - [Bh14] B. Bhatt, *On the non-existence of small Cohen-Macaulay algebras*, J. Algebra 411 (2014), 1-11. [arXiv:1207.5413](https://arxiv.org/abs/1207.5413)

V2-000813 | The small Cohen-Macaulay modules conjecture | exists_isBalancedBigCohenMacaulay

research solved | proof annotation: False | 官方代码 | 原文线索

Existence of balanced big Cohen-Macaulay modules. Every Noetherian local ring has a balanced big Cohen-Macaulay module: conjecture 8 of the Wikipedia list, and Theorem CMM of [Ho17]. Proved by Hochster in equal characteristic [Ho75a], and, using perfectoid methods, by André in mixed characteristic [An18].

```
theorem exists_isBalancedBigCohenMacaulay :
∃ (W : Type u) (_ : AddCommGroup W) (_ : Module R W), IsBalancedBigCohenMacaulay R W
```

FC-SmallCohenMacaulayModules.exists_isBalancedBigCohenMacaulay

V2-000814 | The small Cohen-Macaulay modules conjecture | exists_isSmallCohenMacaulay

research open | proof annotation: False | 官方代码 | 原文线索

The small Cohen-Macaulay modules conjecture. If R is a complete Noetherian local ring, then there is a finitely generated R-module $M \ne 0$ such that some system of parameters of R is a regular sequence on M. Hochster stated the conjecture for complete local domains [Ho17, Conjecture 2.1]. The two forms are equivalent: for a minimal prime P of R with $\dim R/P = \dim R$, a small Cohen-Macaulay module over the complete local domain R/P is one over R. In the 2000s Hochster conjectured the opposite, that some complete local domain has no small Cohen-Macaulay module [Ho17, Conjecture 2.2], so this statement may well be false.

```
theorem exists_isSmallCohenMacaulay [IsAdicComplete (maximalIdeal R) R] :
∃ (M : Type u) (_ : AddCommGroup M) (_ : Module R M), IsSmallCohenMacaulay R M
```

FC-SmallCohenMacaulayModules.exists_isSmallCohenMacaulay

Snake in the box

MODULE CONTEXT: # Snake in the box *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Snake-in-the-box) - [Hypercube](https://en.wikipedia.org/wiki/Hypercube_graph) - [xkcd](https://xkcd.com/3125/) - [DK67] L. Danzer and V. Klee, *Lengths of snakes in boxes*, Journal of Combinatorial Theory 2 (1967), 258-265. - [Ze97] G. Zémor, *An upper bound on the size of the snake-in-the-box*, Combinatorica 17 (1997), 287-298.

V2-000815 | Snake in the box | coil_upper_bound

research solved | proof annotation: False | 官方代码 | 原文线索

For $n \geq 2$, an upper bound on the maximal length of the longest coil in a box is given by $$ 1 + 2^{n-1}\frac{6n}{6n + \frac{1}{6\sqrt{6}}n^{\frac 1 2} - 7} $$ (see [Ze97]). The case $n = 1$ is excluded since the right-hand side is negative there.

```
theorem coil_upper_bound (n : ℕ) (hn : 2 ≤ n) : LongestCoilInTheBox n
≤ (1 : ℝ) + 2 ^ (n - 1) * (6 * n) / (6 * n + (1 / (6 * √6) * √n) - 7)
```

FC-SnakeInBox.coil_upper_bound

Solitary Numbers

MODULE CONTEXT: # Solitary Numbers *References:* - [Solitary number (Wikipedia)](https://en.wikipedia.org/wiki/Solitary_number) - [Solitary number large clubs (Wikipedia)](https://en.wikipedia.org/wiki/Solitary_number#Large_clubs)

V2-000816 | Solitary Numbers | infinite_club_exists

research open | proof annotation: False | 官方代码 | 原文线索

Existence of an infinite club. A *club* is an abundancy equivalence class, i.e. the set of all positive integers friendly with a given n. It is unknown whether any club is infinite.

```
theorem infinite_club_exists :
∃ n, 0 < n ∧ {m : ℕ | Friendly m n}.Infinite
```

FC-SolitaryNumber.infinite_club_exists

Sparse Ruler

MODULE CONTEXT: # Sparse Ruler A sparse ruler of length L is a sequence of marks $0 = a_1 < a_2 < \dots < a_m = L$. A distance $k \in \mathbb{N}$ can be measured if there are $i, j \in \{1, \dots, m\}$, such that $k = a_j - a_i$. One question concerns the structure of *optimal* rulers. Wichmann [Wi63] gave a parametric family of sparse rulers and speculated that every sufficiently large optimal ruler is of his type. The Wikipedia article records that no optimal ruler of length $1, 13, 17, 23$ or 58 is a Wichmann ruler, and that every other optimal length up to 213 is attained by one; non-Wichmann optimal rulers also occur alongside Wichmann ones at lengths $9, 29, 50$ and 68. The asymptotic growth of the minimal number of marks of an optimal ruler of length L - i.e. the limit of $l(n)^2 / n$, conjectured to lie in $[2.434\ldots, 3]$ - is the subject of `FormalConjectures.ErdosProblems.«170»` (there phrased via $F(N)/\sqrt{N}$), and is not restated here. *References:* - [Wi63] Wichmann, B. "A note on restricted difference bases." Journal of the London Mathematical Society 38 (1963): 465-466. - [Wikipedia](https://en.wikipedia.org/wiki/Sparse_ruler)

V2-000817 | Sparse Ruler | wichmann_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Wichmann's conjecture on optimal rulers. Every optimal ruler of sufficiently large length is a Wichmann ruler $W(r, s)$ (up to reflection, i.e. reversing the segment list). Posed by Wichmann [Wi63]. The Wikipedia article records that no optimal ruler of length $1, 13, 17, 23$ or 58 is a Wichmann ruler, and that every other optimal length up to 213 is attained by one; non-Wichmann optimal rulers also occur alongside Wichmann ones at lengths $9, 29, 50$ and 68.

```
theorem wichmann_conjecture :
∃ N : ℕ, ∀ g : List ℕ, IsOptimal g → N < g.sum →
∃ r s : ℕ, g = wichmannGaps r s ∨ g = (wichmannGaps r s).reverse
```

FC-SparseRuler.wichmann_conjecture

Packing

MODULE CONTEXT: # Packing This file contains a number of open problems related to the minimal size of a square (or circle) that can contain a given number of unit squares (or circles). In each case, we provide a known upper bound, and ask for the least such size. *References:* - [Wikipedia on packing of squares](https://en.wikipedia.org/wiki/Square_packing) - [Wikipedia on packing of circles in a circle](https://en.wikipedia.org/wiki/Circle_packing_in_a_circle) - [Wikipedia on packing of circles in a square](https://en.wikipedia.org/wiki/Circle_packing_in_a_square) - Friedman, Erich (2009), "Packing unit squares in squares: a survey and new results", Electronic Journal of Combinatorics, 1000, Dynamic Survey 7 - Pirl, U. (1969), ["Der Mindestabstand von n in der Einheitskreisscheibe gelegenen Punkten"](https://doi.org/10.1002/mana.19690400110), Mathematische Nachrichten, 40: 111-124 - A website with visualizations of packings: [link](https://erich-friedman.github.io/packing/) - [Erich Friedman, Squares in Circles](https://erich-friedman.github.io/packing/squincir/) - [The-Anh Vu-Le, Lean 4 proof of Squares in Circles](https://github.com/vltanh/lean4-squares-in-circles) - [Kenta Kitamura, Lean proof of the exact three-square minimum for Formal Conjectures](https://github.com/KitaKen1/three-squares-in-a-circle-lean/blob/87bd2b6915933cf608a3cae6263c2500fe81f7ec/lean/ThreeSquares/Compressed/Main.lean#L216-L219)

V2-000818 | Packing | least_eleven_square_packing_in_square

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest square that can contain 11 unit squares? Reference: [Wikipedia](https://en.wikipedia.org/wiki/Square_packing#In_a_square)

theorem least_eleven_square_packing_in_square :
IsLeast {x : ℝ | Nonempty (Packing 11 UnitSquare (Square x))}
least_eleven_square_packing_in_square.answer

FC-SquarePacking.least_eleven_square_packing_in_square

V2-000819 | Packing | least_fifteen_circle_packing_in_circle

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest circle that can contain 15 unit circles? Reference: Graham RL, Lubachevsky BD, Nurmela KJ, Ostergard PRJ. Dense packings of congruent circles in a circle. Discrete Math 1998;181:139-154. [Pirl (1969)](https://doi.org/10.1002/mana.19690400110) conjectured this configuration to be optimal.

theorem least_fifteen_circle_packing_in_circle :
IsLeast {r : ℝ≥0 | Nonempty (Packing 15 UnitCircle (Circle r))}
least_fifteen_circle_packing_in_circle.answer

FC-SquarePacking.least_fifteen_circle_packing_in_circle

V2-000820 | Packing | least_seventeen_square_packing_in_square

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest square that can contain 17 unit squares? Reference: [Wikipedia](https://en.wikipedia.org/wiki/Square_packing#In_a_square)

theorem least_seventeen_square_packing_in_square :
IsLeast {x : ℝ | Nonempty (Packing 17 UnitSquare (Square x))}
least_seventeen_square_packing_in_square.answer

FC-SquarePacking.least_seventeen_square_packing_in_square

V2-000821 | Packing | least_three_square_packing_in_circle

research solved | proof annotation: True | 官方代码 | 原文线索

What is the smallest circle that can contain 3 unit squares? The least radius is $5\sqrt{17}/16$. The-Anh Vu-Le proved optimality in September 2026. Kenta Kitamura gave a separate Lean proof of this exact `IsLeast` statement using the `SquarePacking` definitions in this file. References: [Wikipedia](https://en.wikipedia.org/wiki/Square_packing#In_a_circle), [Friedman's table](https://erich-friedman.github.io/packing/squincir/), [Vu-Le's Lean proof](https://github.com/vltanh/lean4-squares-in-circles), and [Kitamura's FC-target Lean proof](https://github.com/KitaKen1/three-squares-in-a-circle-lean/blob/87bd2b6915933cf608a3cae6263c2500fe81f7ec/lean/ThreeSquares/Compressed/Main.lean#L216-L219).

theorem least_three_square_packing_in_circle :
IsLeast {r : ℝ≥0 | Nonempty (Packing 3 UnitSquare (Circle r))}
least_three_square_packing_in_circle.answer

FC-SquarePacking.least_three_square_packing_in_circle

V2-000822 | Packing | least_twenty_one_circle_packing_in_square

research open | proof annotation: False | 官方代码 | 原文线索

What is the smallest square that can contain 21 unit circles?

theorem least_twenty_one_circle_packing_in_square :
IsLeast {x : ℝ | Nonempty (Packing 21 UnitCircle (Square x))}
least_twenty_one_circle_packing_in_square.answer

FC-SquarePacking.least_twenty_one_circle_packing_in_square

Polynomial-time computability of comparisons of sums of square roots

MODULE CONTEXT: # Polynomial-time computability of comparisons of sums of square roots This file formalizes the "Square-root sum problem" which asks if one can decide in polynomial time on a Turing machine whether the sum of the square roots of a list of naturals is at most a cutoff (or alternatively, at most another such sum). This problem is relevant to questions in computational geometry (for example, in determining which of two polygonal paths in Euclidean space is longer). Wikipedia states the problem for positive integers, an integer threshold, and lists of equal length. The versions here allow zeros, a natural-number threshold, and lists of any length. This slightly more general formulation can be reduced to the fixed length form (by removing zeros and introducing copies of $\sqrt{4}$ to the longer list and $\sqrt{1} + \sqrt{1}$ to the shorter one). *References:* - [Wikipedia: List of unsolved problems in computer science](https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_computer_science) - [Wikipedia: Square-root sum problem](https://en.wikipedia.org/wiki/Square-root_sum_problem)

V2-000823 | Polynomial-time computability of comparisons of sums of square roots | isPolyTime_squareRootSum

research open | proof annotation: False | 官方代码 | 原文线索

Is the square-root sum problem polynomial time (single sum version)? *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Square-root_sum_problem)

theorem isPolyTime_squareRootSum :
IsPolyTime (Function.uncurry squareRootSum)

FC-PolyTime.isPolyTime_squareRootSum

V2-000824 | Polynomial-time computability of comparisons of sums of square roots | isPolyTime_squareRootSumTwoSided

research open | proof annotation: False | 官方代码 | 原文线索

Is the square-root sum problem polynomial time (two sum version)? *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Square-root_sum_problem)

theorem isPolyTime_squareRootSumTwoSided :
IsPolyTime (Function.uncurry squareRootSumTwoSided)

FC-PolyTime.isPolyTime_squareRootSumTwoSided

Steiner Systems

MODULE CONTEXT: # Steiner Systems A Steiner system $S(t, k, n)$ is a collection of k-element subsets (called blocks) of an n-element set such that every t-element subset is contained in exactly one block. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Steiner_system) - [Large Steiner Systems](https://epoch.ai/frontiermath/open-problems/large-steiner-systems) by Kunal Marwaha

V2-000825 | Steiner Systems | infinitely_many_steiner_t4

research solved | proof annotation: False | 官方代码 | 原文线索

There are infinitely many Steiner systems with $t = 4$. Keevash (2014) proved that for any fixed t and k, a Steiner system $S(t, k, n)$ exists for all sufficiently large n satisfying the necessary divisibility conditions. Since there are infinitely many such admissible n, this implies infinitely many $S(4, k, n)$ systems exist (for any fixed $k > 4$). The proof is nonconstructive. Explicit examples include $S(4, 5, 11)$ (the unique system, related to the Mathieu group M_{11}) and $S(4, 7, 23)$ (related to the Mathieu group M_{23}). Only nontrivial systems with $4 < k < n$ are counted: for every n the single block $\{1, \dots, n\}$ is an $S(4, n, n)$, and all 4-subsets of an n-set form an $S(4, 4, n)$.

theorem infinitely_many_steiner_t4 :
{s : Σ k n : ℕ, S(4, k, n) | 4 < s.1 ∧ s.1 < s.2.1}.Infinite

FC-SteinerSystems.infinitely_many_steiner_t4

V2-000826 | Steiner Systems | infinitely_many_steiner_t5

research solved | proof annotation: False | 官方代码 | 原文线索

There are infinitely many Steiner systems with $t = 5$. Keevash (2014) proved that for any fixed t and k, a Steiner system $S(t, k, n)$ exists for all sufficiently large n satisfying the necessary divisibility conditions. This settles the long-standing open problem of whether infinitely many $S(5, k, n)$ systems exist. The proof is nonconstructive. Only two explicit examples are known: $S(5, 6, 12)$ and $S(5, 8, 24)$, both Witt designs related to the Mathieu groups M_{12} and M_{24} respectively. Only nontrivial systems with $5 < k < n$ are counted, for the same reason as in `infinitely_many_steiner_t4`. No Steiner system with $t \geq 6$ has been explicitly constructed, though Keevash's result guarantees their existence nonconstructively as well.

theorem infinitely_many_steiner_t5 :
{s : Σ k n : ℕ, S(5, k, n) | 5 < s.1 ∧ s.1 < s.2.1}.Infinite

FC-SteinerSystems.infinitely_many_steiner_t5

V2-000827 | Steiner Systems | large_steiner_systems

research open | proof annotation: False | 官方代码 | 原文线索

Construct an $S(t, k, n)$-Steiner system with $n > k > t > 5$, $t < 10$, and $n < 200$. No example of a Steiner system with $t > 5$ is known, despite a 2014 existence theorem by Keevash showing that such systems must exist for sufficiently large n. *Reference:* [Large Steiner Systems](https://epoch.ai/frontiermath/open-problems/large-steiner-systems)

theorem large_steiner_systems : Nonempty LargeSteinerSystemWitness

FC-SteinerSystems.large_steiner_systems

V2-000828 | Steiner Systems | steiner_system_5_6_12

research solved | proof annotation: False | 官方代码 | 原文线索

Existence of $S(5, 6, 12)$: The small Witt design. There exists a unique Steiner system $S(5, 6, 12)$, known as the small Witt design. It was constructed by Witt (1938) and is closely related to the Mathieu group M_{12}. This is one of only two known Steiner systems with $t = 5$.

```
theorem steiner_system_5_6_12 : Nonempty S(5, 6, 12) :=
⟨{{0, 1, 2, 3, 4, 9},
{0, 1, 2, 3, 5, 11},
{0, 1, 2, 3, 6, 7},
{0, 1, 2, 3, 8, 10},
{0, 1, 2, 4, 5, 7},
{0, 1, 2, 4, 6, 8},
{0, 1, 2, 4, 10, 11},
{0, 1, 2, 5, 6, 10},
{0, 1, 2, 5, 8, 9},
{0, 1, 2, 6, 9, 11},
{0, 1, 2, 7, 8, 11},
{0, 1, 2, 7, 9, 10},
{0, 1, 3, 4, 5, 8},
{0, 1, 3, 4, 6, 10},
{0, 1, 3, 4, 7, 11},
{0, 1, 3, 5, 6, 9},
{0, 1, 3, 5, 7, 10},
{0, 1, 3, 6, 8, 11},
{0, 1, 3, 7, 8, 9},
{0, 1, 3, 9, 10, 11},
{0, 1, 4, 5, 6, 11},
{0, 1, 4, 5, 9, 10},
{0, 1, 4, 6, 7, 9},
{0, 1, 4, 7, 8, 10},
{0, 1, 4, 8, 9, 11},
{0, 1, 5, 6, 7, 8},
{0, 1, 5, 7, 9, 11},
{0, 1, 5, 8, 10, 11},
{0, 1, 6, 7, 10, 11},
{0, 1, 6, 8, 9, 10},
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{0, 2, 3, 4, 7, 10},
{0, 2, 3, 4, 8, 11},
{0, 2, 3, 5, 7, 8},
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{0, 2, 4, 6, 9, 10},
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{0, 5, 6, 9, 10, 11},
{0, 5, 7, 8, 9, 10},
{0, 6, 7, 8, 9, 11},
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{1, 2, 3, 4, 6, 11},
{1, 2, 3, 4, 7, 8},
{1, 2, 3, 5, 6, 8},
{1, 2, 3, 5, 7, 9},
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{1, 2, 3, 8, 9, 11},
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{1, 2, 4, 5, 8, 11},
{1, 2, 4, 6, 7, 10},
{1, 2, 4, 7, 9, 11},
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{1, 2, 5, 7, 8, 10},
{1, 2, 5, 9, 10, 11},
{1, 2, 6, 7, 8, 9},
{1, 2, 6, 8, 10, 11},
{1, 3, 4, 5, 6, 7},
```

{1, 3, 4, 5, 9, 11},
{1, 3, 4, 6, 8, 9},
{1, 3, 4, 7, 9, 10},
{1, 3, 4, 8, 10, 11},
{1, 3, 5, 6, 10, 11},
{1, 3, 5, 7, 8, 11},
{1, 3, 5, 8, 9, 10},
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{1, 4, 6, 9, 10, 11},
{1, 5, 6, 7, 9, 10},
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{2, 4, 5, 6, 7, 8},
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{2, 4, 6, 8, 9, 11},
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{2, 5, 7, 8, 9, 11},
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{3, 4, 5, 6, 9, 10},
{3, 4, 5, 7, 8, 10},
{3, 4, 6, 7, 10, 11},
{3, 4, 7, 8, 9, 11},
{3, 5, 6, 7, 8, 9},
{3, 5, 7, 9, 10, 11},
{3, 6, 8, 9, 10, 11},
{4, 5, 6, 7, 9, 11},
{4, 5, 8, 9, 10, 11},
{4, 6, 7, 8, 9, 10},
{5, 6, 7, 8, 10, 11}},
by native_decide, by native_decide)⟩

/--
Existence of $S(5, 8, 24)$: The large Witt design.

There exists a unique Steiner system $S(5, 8, 24)$, known as the large Witt design.
It was constructed by Witt (1938) and is closely related to the Mathieu group M_{24}.
This is one of only two known Steiner systems with $t = 5$.
-/
@[category research solved, AMS 5]
theorem steiner_system_5_8_24 : Nonempty S(5, 8, 24)

FC-SteinerSystems.steiner_system_5_6_12

V2-000829 | Steiner Systems | steiner_system_5_8_24

research solved | proof annotation: False | 官方代码 | 原文线索

Existence of $S(5, 8, 24)$: The large Witt design. There exists a unique Steiner system $S(5, 8, 24)$, known as the large Witt design. It was constructed by Witt (1938) and is closely related to the Mathieu group M_{24}. This is one of only two known Steiner systems with $t = 5$.

theorem steiner_system_5_8_24 : Nonempty S(5, 8, 24)

FC-SteinerSystems.steiner_system_5_8_24

Sum of three cubes

MODULE CONTEXT: # Sum of three cubes An integer `n : ℤ` can be written as a sum of three cubes (of integers) if and only if `n` is not `4` or `5` mod `9`. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Sums_of_three_cubes) - [mathoverflow/100324](https://mathoverflow.net/a/100324) asked by user [*David Feldman*](https://mathoverflow.net/users/10909/david-feldman)

V2-000830 | Sum of three cubes | isSumOfThreeCubesRat_any

research solved | proof annotation: False | 官方代码 | 原文线索

Any rational number is a sum of three rational cubes. First proved by Ryley in 1825, which can be found in [Ri1930]. The below parametrization is brought from the MSE answer [MSE]. [Ri1930] Richmond, H. W. "On Rational Solutions of $x^3 + y^3 + z^3 = R$." Proceedings of the Edinburgh Mathematical Society 2.2 (1930): 92-100. [MSE] Kieren MacMillan, Proving that any rational number can be represented as the sum of the cubes of three rational numbers, https://math.stackexchange.com/q/4480969

theorem isSumOfThreeCubesRat_any (r : ℚ) : IsSumOfThreeCubes r

FC-SumOfThreeCubes.isSumOfThreeCubesRat_any

V2-000831 | Sum of three cubes | isSumOfThreeCubes_iff_mod_9

research open | proof annotation: False | 官方代码 | 原文线索

An integer `n : ℤ` can be written as a sum of three cubes (of integers) if and only if `n` is not `4` or `5` mod `9`.

```
theorem isSumOfThreeCubes_iff_mod_9 :
∀ n : ℤ, IsSumOfThreeCubes n ↔ ¬(n ≡ 4 [ZMOD 9] ∨ n ≡ 5 [ZMOD 9])
```

FC-SumOfThreeCubes.isSumOfThreeCubes_iff_mod_9

Sumner's universal tournament conjecture (1971)

MODULE CONTEXT: # Sumner's universal tournament conjecture (1971) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Sumner%27s_conjecture) * [HT91] Häggkvist, R. and Thomason, A. (1991). "Trees in tournaments." *Combinatorica* 11, pp. 123--130. * [ES04] El Sahili, A. (2004). "Trees in tournaments." *J. Combin. Theory Ser. B* 92, pp. 183--187. * [KMO11] Kühn, D., Mycroft, R. and Osthus, D. (2011). "A proof of Sumner's universal tournament conjecture for large tournaments." *Proc. London Math. Soc.* 102, pp. 731--766. [arXiv:1010.4430](https://arxiv.org/abs/1010.4430)

V2-000832 | Sumner's universal tournament conjecture (1971) | sumner_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Sumner's universal tournament conjecture (1971). Every tournament on $2n - 2$ vertices contains every oriented tree on $n \ge 2$ vertices as a subgraph.

```
theorem sumner_conjecture :
∀ (n : ℕ), 2 ≤ n →
∀ {W : Type} [Fintype W] [DecidableEq W] (D : Digraph W) [DecidableRel D.Adj],
Fintype.card W = n → IsOrientedTree D →
∀ {V : Type} [Fintype V] [DecidableEq V] (G : Digraph V) [DecidableRel G.Adj],
Fintype.card V = 2 * n - 2 → G.IsTournament → Embeds D G
```

FC-SumnerUniversalTournamentConjecture.sumner_conjecture

Gottschalk's surjunctivity conjecture

MODULE CONTEXT: # Gottschalk's surjunctivity conjecture A group G is *surjunctive* if every injective, continuous, G-equivariant map $A^G \to A^G$ (for any finite alphabet A) is surjective. Here equivariance is with respect to the left shift action of G on A^G, defined by $(g \cdot x)(h) = x(g^{-1} h)$, and continuity is with respect to the product topology on A^G (where A carries the discrete topology). Gottschalk's conjecture (1973) states that every group is surjunctive. It is false: an internal OpenAI model gave counterexamples in September 2026 [OAI26a], [OAI26b]. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Surjunctive_group) - Gottschalk, W. H. (1973), "Some general dynamical notions" - [OAI26a] OpenAI, *A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Characteristic-Two-September-23-2026/paper.pdf - [OAI26b] OpenAI, *A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Odd-Characteristic-September-26-2026/paper.pdf

V2-000833 | Gottschalk's surjunctivity conjecture | counter_gottschalk_surjunctivity

research solved | proof annotation: True | 官方代码 | 原文线索

Gottschalk's surjunctivity conjecture is false: there is a group that is not surjunctive. There are a finite field K, a group G and $a, b \in K[G]$ with $ab = 1$ and $ba \neq 1$. The linear cellular automaton $x \mapsto (g \mapsto \sum_u b_u x(gu))$ on K^G is then injective but not surjective. In [OAI26a], K has characteristic two and G is finitely presented. In [OAI26b], K has odd characteristic and G is finitely generated. This was proved by an internal OpenAI model in September 2026. The linked Lean proof is for [OAI26b]. It shows that the map above is injective and not surjective. That the map is continuous and shift-equivariant, as `IsSurjunctive` requires, is not part of the linked proof.

```
theorem counter_gottschalk_surjunctivity :
∃ (G : Type) (_ : Group G), ¬ IsSurjunctive G
```

FC-GottschalkSurjunctivity.counter_gottschalk_surjunctivity

V2-000834 | Gottschalk's surjunctivity conjecture | gottschalk_surjunctivity_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Gottschalk's surjunctivity conjecture (1973): every group is surjunctive. That is, for every group `G` and every finite alphabet `A`, every injective cellular automaton on `A^G` is surjective. This is false. It was disproved by an internal OpenAI model in September 2026 [OAI26a], [OAI26b]; see `counter_gottschalk_surjunctivity` for the counterexample and for what the linked Lean proof covers.

```
theorem gottschalk_surjunctivity_conjecture :
¬ ∀ (G : Type) [Group G], IsSurjunctive G
```

FC-GottschalkSurjunctivity.gottschalk_surjunctivity_conjecture

Taxicab numbers

MODULE CONTEXT: # Taxicab numbers A *taxicab number* for natural numbers k, m, n is the smallest number x that can be expressed as a sum of m positive k-th powers in at least n distinct ways. The most famous taxicab number is $ 1729 = 1³ + 12³ = 9³ + 10³, $ also known as the Hardy-Ramanujan number. However, a taxicab number is not known for $k=5$, $m=2$, and any $n ≥ 2$: No positive integer is known that can be written as the sum of two 5th powers in more than one way, and it is not known whether such a number exists. In particular, it is not known whether there exists a taxicab number for $k=5$, $m=2$, and $n=2$. *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Taxicab_number) - [Generalized taxicab number](https://en.wikipedia.org/wiki/Generalized_taxicab_number) - [OEIS taxicab cubes](https://oeis.org/A001235) - [OEIS taxicab 4th powers](https://oeis.org/A018786) - [OEIS taxicab conjecture](https://oeis.org/A088703)

V2-000835 | Taxicab numbers | taxicab_for_5_2_2

research open | proof annotation: False | 官方代码 | 原文线索

Taxicab number for $k=5$, $m=2$, and $n=2$ is not known. Whether such a number exists is also not known.

theorem taxicab_for_5_2_2 : ∃ x : ℕ, IsTaxicabFor 5 2 2 x

FC-Taxicab.taxicab_for_5_2_2

V2-000836 | Taxicab numbers | taxicab_for_5_2_n

research open | proof annotation: False | 官方代码 | 原文线索

Taxicab number for $k=5$ and $m=2$ is not-known for any $n ≥ 2$. Whether such a number exists is also not known.

theorem taxicab_for_5_2_n : ∃ n : ℕ, n ≥ 2 ∧ (∃ x : ℕ, IsTaxicabFor 5 2 n x)

FC-Taxicab.taxicab_for_5_2_n

The total colouring conjecture (Behzad 1965, Vizing 1968)

MODULE CONTEXT: # The total colouring conjecture (Behzad 1965, Vizing 1968) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Total_coloring) * [Be65] Behzad, M. (1965). *Graphs and their chromatic numbers.* Ph.D. thesis, Michigan State University. * [Vi68] Vizing, V. G. (1968). "Some unsolved problems in graph theory." *Uspekhi Mat. Nauk* 23, pp. 117--134. * [Ko96] Kostochka, A. V. (1996). "The total chromatic number of any multigraph with maximum degree five is at most seven." *Discrete Math.* 162, pp. 199--214. * [MR98] Molloy, M. and Reed, B. (1998). "A bound on the total chromatic number." *Combinatorica* 18, pp. 241--280. * [Yap96] Yap, H. P. (1996). *Total colourings of graphs.* Lecture Notes in Mathematics 1623, Springer.

V2-000837 | The total colouring conjecture (Behzad 1965, Vizing 1968) | total_coloring_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The total colouring conjecture (Behzad 1965, Vizing 1968). Every finite simple graph G has a total colouring with at most $\Delta(G) + 2$ colours, i.e. $\chi''(G) \le \Delta(G) + 2$, where $\Delta(G)$ is the maximum degree.

theorem total_coloring_conjecture :
∀ {V : Type} [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
G.totalChromaticNumber ≤ G.maxDegree + 2

FC-TotalColoringConjecture.total_coloring_conjecture

Open questions on transcendence of numbers

MODULE CONTEXT: # Open questions on transcendence of numbers *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/Transcendental_number)

V2-000838 | Open questions on transcendence of numbers | exp_add_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$e + \pi$ is transcendental.

theorem exp_add_pi_transcendental : Transcendental ℚ (exp 1 + π)

FC-Transcendental.exp_add_pi_transcendental

V2-000839 | Open questions on transcendence of numbers | exp_exp_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

e^e is transcendental.

theorem exp_exp_transcendental : Transcendental ℚ (exp (exp 1))

FC-Transcendental.exp_exp_transcendental

V2-000840 | Open questions on transcendence of numbers | exp_mul_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$e\pi$ is transcendental.

theorem exp_mul_pi_transcendental : Transcendental ℚ (exp 1 * π)

FC-Transcendental.exp_mul_pi_transcendental

V2-000841 | Open questions on transcendence of numbers | exp_pow_pi_sq_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

e^{π^2} is transcendental.

theorem exp_pow_pi_sq_transcendental : Transcendental ℚ (exp (π ^ 2))

FC-Transcendental.exp_pow_pi_sq_transcendental

V2-000842 | Open questions on transcendence of numbers | pi_pow_exp_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

π^e is transcendental.

theorem pi_pow_exp_transcendental : Transcendental ℚ (π ^ (exp 1))

FC-Transcendental.pi_pow_exp_transcendental

V2-000843 | Open questions on transcendence of numbers | pi_pow_pi_pow_pi_pow_pi_not_integer

research open | proof annotation: False | 官方代码 | 原文线索

$\pi^{\pi^{\pi^\pi}}$ is not an integer. This would follow from $\pi^{\pi^{\pi^\pi}}$ being transcendental, but this formulation is of interest in its own right, as it could in principle be proven by direct computation. *Reference:* [YouTube](https://www.youtube.com/watch?v=BdHFLfv-ThQ)

theorem pi_pow_pi_pow_pi_pow_pi_not_integer : ¬ ∃ (n : ℤ), π ^ π ^ π ^ π = n

FC-Transcendental.pi_pow_pi_pow_pi_pow_pi_not_integer

V2-000844 | Open questions on transcendence of numbers | pi_pow_pi_pow_pi_pow_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\pi^{\pi^{\pi^\pi}}$ is transcendental.

theorem pi_pow_pi_pow_pi_pow_pi_transcendental : Transcendental ℚ (π ^ (π ^ (π ^ π)))

FC-Transcendental.pi_pow_pi_pow_pi_pow_pi_transcendental

V2-000845 | Open questions on transcendence of numbers | pi_pow_pi_pow_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\pi^{\pi^{\pi}}$ is transcendental.

theorem pi_pow_pi_pow_pi_transcendental : Transcendental ℚ (π ^ (π ^ π))

FC-Transcendental.pi_pow_pi_pow_pi_transcendental

V2-000846 | Open questions on transcendence of numbers | pi_pow_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

π^{π} is transcendental.

theorem pi_pow_pi_transcendental : Transcendental ℚ (π ^ π)

FC-Transcendental.pi_pow_pi_transcendental

V2-000847 | Open questions on transcendence of numbers | pi_pow_sqrt_two_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\pi^{\sqrt{2}}$ is transcendental.

theorem pi_pow_sqrt_two_transcendental : Transcendental ℚ (π ^ √2)

FC-Transcendental.pi_pow_sqrt_two_transcendental

V2-000848 | Open questions on transcendence of numbers | rlog_pi_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\log(\pi)$ is transcendental.

theorem rlog_pi_transcendental : Transcendental ℚ (log π)

FC-Transcendental.rlog_pi_transcendental

V2-000849 | Open questions on transcendence of numbers | rlog_rlog_two_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\log(\log(2))$ is transcendental.

theorem rlog_rlog_two_transcendental : Transcendental ℚ ((2 : ℝ).log.log)

FC-Transcendental.rlog_rlog_two_transcendental

V2-000850 | Open questions on transcendence of numbers | sin_exp_transcendental

research open | proof annotation: False | 官方代码 | 原文线索

$\sin(e)$ is transcendental.

theorem sin_exp_transcendental : Transcendental ℚ (Real.sin (exp 1))

FC-Transcendental.sin_exp_transcendental

V2-000851 | Open questions on transcendence of numbers | transcendental_catalanConstant

research open | proof annotation: False | 官方代码 | 原文线索

The Catalan constant G is transcendental.

theorem transcendental_catalanConstant : Transcendental ℚ catalanConstant

FC-Transcendental.transcendental_catalanConstant

V2-000852 | Open questions on transcendence of numbers | transcendental_eulerMascheroniConstant_or_gompertzConstant

research solved | proof annotation: False | 官方代码 | 原文线索

At least one of the Euler-Mascheroni constant γ and the Gompertz constant δ is transcendental. [Ri12] Rivoal, T. (2012). On the arithmetic nature of the values of the gamma function, Euler's constant, and Gompertz's constant. Michigan Mathematical Journal, 61(2), 239-254.

theorem transcendental_eulerMascheroniConstant_or_gompertzConstant :
Transcendental ℚ eulerMascheroniConstant ∨ Transcendental ℚ gompertzConstant

FC-Transcendental.transcendental_eulerMascheroniConstant_or_gompertzConstant

V2-000853 | Open questions on transcendence of numbers | transcendental_gamma_one_div

research open | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/n)$ for `n ≥ 2` is transcendental.

theorem transcendental_gamma_one_div (n : ℕ) (hn : 2 ≤ n) : Transcendental ℚ (1 / n : ℝ).Gamma

FC-Transcendental.transcendental_gamma_one_div

V2-000854 | Open questions on transcendence of numbers | transcendental_gamma_one_div_four

research solved | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/4)$ is transcendental. [Ch84] Chudnovsky, G. (1984). Contributions to the theory of transcendental numbers.

theorem transcendental_gamma_one_div_four : Transcendental ℚ (1 / 4 : ℝ).Gamma

FC-Transcendental.transcendental_gamma_one_div_four

V2-000855 | Open questions on transcendence of numbers | transcendental_gamma_one_div_six

research solved | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/6)$ is transcendental. [Ch84] Chudnovsky, G. (1984). Contributions to the theory of transcendental numbers.

theorem transcendental_gamma_one_div_six : Transcendental ℚ (1 / 6 : ℝ).Gamma

FC-Transcendental.transcendental_gamma_one_div_six

V2-000856 | Open questions on transcendence of numbers | transcendental_gamma_one_div_three

research solved | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/3)$ is transcendental. [Ch84] Chudnovsky, G. (1984). Contributions to the theory of transcendental numbers.

theorem transcendental_gamma_one_div_three : Transcendental ℚ (1 / 3 : ℝ).Gamma

FC-Transcendental.transcendental_gamma_one_div_three

V2-000857 | Open questions on transcendence of numbers | transcendental_gamma_one_div_two

research solved | proof annotation: False | 官方代码 | 原文线索

$\Gamma(1/2)$ is transcendental. [Ch84] Chudnovsky, G. (1984). Contributions to the theory of transcendental numbers.

theorem transcendental_gamma_one_div_two : Transcendental ℚ (1 / 2 : ℝ).Gamma

FC-Transcendental.transcendental_gamma_one_div_two

V2-000858 | Open questions on transcendence of numbers | transcendental_gompertzConstant

research open | proof annotation: False | 官方代码 | 原文线索

The Gompertz constant δ is transcendental.

theorem transcendental_gompertzConstant : Transcendental ℚ gompertzConstant

FC-Transcendental.transcendental_gompertzConstant

Vizing's conjecture (1968)

MODULE CONTEXT: # Vizing's conjecture (1968) *References:* * [Wikipedia](https://en.wikipedia.org/wiki/Vizing%27s_conjecture) * [Vi68] Vizing, V. G. (1968). "Some unsolved problems in graph theory." *Uspekhi Mat. Nauk* 23, pp. 117--134. * [ClSu00] Clark, W. E. and Suen, S. (2000). "An inequality related to Vizing's conjecture." *Electron. J. Combin.* 7, N4. * [SuTa12] Suen, S. and Tarr, J. (2012). "An improved inequality related to Vizing's conjecture." *Electron. J. Combin.* 19, P8. * [BDGHHKR12] Brešar, B., Dorbec, P., Goddard, W., Hartnell, B. L., Henning, M. A., Klavžar, S. and Rall, D. F. (2012). "Vizing's conjecture: a survey and recent results." *J. Graph Theory* 69, pp. 46--76.

V2-000859 | Vizing's conjecture (1968) | vizing_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Vizing's conjecture (1968). For all finite simple graphs G and H, the domination number of the Cartesian (box) product satisfies $\gamma(G \,\square\, H) \ge \gamma(G)\,\gamma(H)$.

theorem vizing_conjecture :
∀ {α β : Type} [Fintype α] [Fintype β] [DecidableEq α] [DecidableEq β]
(G : SimpleGraph α) (H : SimpleGraph β),
G.dominationNumber * H.dominationNumber ≤ (G □ H).dominationNumber

FC-VizingConjecture.vizing_conjecture

Infinitude of Wall-Sun-Sun primes

MODULE CONTEXT: # Infinitude of Wall-Sun-Sun primes *References:* - [Wikipedia](https://en.wikipedia.org/wiki/Wall%E2%80%93Sun%E2%80%93Sun_prime) - [EJ10] A.-S. Elsenhans and J. Jahnel, *The Fibonacci sequence modulo p^2 - An investigation by computer for $p < 10^{14}$*, [arXiv:1006.0824](https://arxiv.org/abs/1006.0824)

V2-000860 | Infinitude of Wall-Sun-Sun primes | exists_isWallSunSunPrime

research open | proof annotation: False | 官方代码 | 原文线索

A prime p is a Wall-Sun-Sun prime if and only if $L_p \equiv 1 \pmod{p^2}$, where L_p is the p-th Lucas number. It is conjectured that there is at least one Wall-Sun-Sun prime.

theorem exists_isWallSunSunPrime : ∃ p, IsWallSunSunPrime p

FC-WallSunSun.exists_isWallSunSunPrime

V2-000861 | Infinitude of Wall-Sun-Sun primes | infinite_isWallSunSunPrime

research open | proof annotation: False | 官方代码 | 原文线索

A prime p is a Wall-Sun-Sun prime if and only if $L_p \equiv 1 \pmod{p^2}$, where L_p is the p-th Lucas number. It is conjectured that there are infinitely many Wall-Sun-Sun primes.

theorem infinite_isWallSunSunPrime : {p : ℕ | IsWallSunSunPrime p}.Infinite

FC-WallSunSun.infinite_isWallSunSunPrime

V2-000862 | Infinitude of Wall-Sun-Sun primes | infinite_isWallSunSunPrime_of_disc_eq

research open | proof annotation: False | 官方代码 | 原文线索

Let K be a real quadratic field of discriminant D and let ε be a fundamental unit of K. Following [EJ10, Remark 2.2.8], an odd prime $p \nmid D$ is a Wall-Sun-Sun prime for K if, in $\mathcal{O}_K$, $\varepsilon^{p-1} \equiv 1 \pmod{p^2}$ when $\left(\tfrac{D}{p}\right) = 1$, and $\varepsilon^{2p+2} \equiv 1 \pmod{p^2}$ when $\left(\tfrac{D}{p}\right) = -1$. Both exponents are even, so the condition does not depend on the choice of ε, and it is equivalent to asking the same congruence for every unit of K. For $K = \mathbb{Q}(\sqrt{5})$ and $\varepsilon = \frac{1 + \sqrt{5}}{2}$ these are the classical Wall-Sun-Sun primes other than 2 and 5 [EJ10, Proposition 2.2.6]. It is conjectured that for every fundamental discriminant $D \neq 1$ there are infinitely many Wall-Sun-Sun primes with discriminant D (Wikipedia; [EJ10, §4.1] gives the heuristic for $\mathbb{Q}(\sqrt{5})$). It is stated here for $D > 0$ only. Wikipedia's sentence also covers $D < 0$, but [EJ10] gives the definition above only for real quadratic fields, and its literal extension to imaginary quadratic fields is degenerate: there every unit is a root of unity of order dividing 4 or 6, and that order divides the relevant exponent $p - 1$ or $2p + 2$ for every odd prime $p \nmid D$.

theorem infinite_isWallSunSunPrime_of_disc_eq {K : Type*} [Field K] [NumberField K] [IsQuadraticExtension ℚ K] [IsTotallyReal K] {D : ℤ} (hD : discr K = D) : {p : ℕ | p.Prime ∧ Odd p ∧ ¬ (p : ℤ) | D ∧ ∀ ε : (𝓞 K)ˣ, (p : 𝓞 K) ^ 2 | (ε : 𝓞 K) ^ (if J(D | p) = 1 then p - 1 else 2 * p + 2) - 1}.Infinite

FC-WallSunSun.infinite_isWallSunSunPrime_of_disc_eq

The $\frac 1 3$-$\frac 2 3$ conjecture

MODULE CONTEXT: # The $\frac 1 3$-$\frac 2 3$ conjecture *Reference:* [Wikipedia](https://en.wikipedia.org/wiki/1/3%E2%80%932/3_conjecture)

V2-000863 | The $\frac 1 3$-$\frac 2 3$ conjecture | conjecture_1_3_to_2_3

research open | proof annotation: False | 官方代码 | 原文线索

Does every finite partially ordered set that is not totally ordered contain two elements x and y such that the probability that x appears before y in a random linear extension is between $\frac 1 3$ and $\frac 2 3$? The set of all total order extensions is represented as order preserving bijections P of $1, ..., n$.

theorem conjecture_1_3_to_2_3 : ∀ (P : Type) [Finite P] [PartialOrder P] (not_total : ¬ Std.Total (α := P) (· ≤ ·)) (total_ext : Set <| OrderHom P ℕ) (total_ext_def : ∀ σ, σ ∈ total_ext ↔ Set.range σ = Set.Icc 1 (Nat.card P)), ∃ x y : P, ({σ ∈ total_ext | σ x < σ y}.ncard / total_ext.ncard : ℚ) ∈ Set.Icc (1/3) (2/3)

FC-Conjecture_1_3_to_2_3.conjecture_1_3_to_2_3

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Written on the Wall II - Conjecture 1

MODULE CONTEXT: # Written on the Wall II - Conjecture 1 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000864 | Written on the Wall II - Conjecture 1 | conjecture1

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 1](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G` the maximum number of leaves of a spanning tree satisfies `Ls(G) ≥ n(G) + 1 - 2·m(G)` where `n(G)` counts vertices and `m(G)` is the size of a maximum matching. A formal proof reduces to a spanning tree `T`. If `I` is the set of non-leaves of `T`, Hall's theorem applied to a bipartition of `T` gives a matching `M` with `|I| + 1 ≤ 2 * |M|`. Since `|V|` is the sum of the numbers of leaves and non-leaves, and every matching and leaf count constructed in `T` is admissible for the corresponding supremum in `G`, the stated inequality follows.

```
theorem conjecture1 {α : Type*} [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (h_conn : G.Connected) :
(Fintype.card α : ℝ) + 1 - 2 * matchingNumber G ≤ (Ls G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture1.conjecture1

Written on the Wall II - Conjecture 100

MODULE CONTEXT: # Written on the Wall II - Conjecture 100 **Verbatim statement (WOWII #100, status O):** > If G is a simple connected graph, then α(G) ≤ CEIL[(maximum of λ(v) + 0.5*length(G̅))/2] **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj100 The WOWII HTML uses `length(G̅)` (the bar denotes graph complement); the extracted JSON in our private repo previously dropped the overline. The formal statement below uses the Euclidean norm of the degree sequence of `Gᶜ`. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definition of graph length The WOWII definitions popup defines `length(H)` as the square root of the sum of the squares of the vertex degrees. This is `degreeL2Norm H` in Lean. Combined with the overline above, the inequality reads: `α(G) ≤ ⌈(max_v l(v) + 0.5 · degreeL2Norm(Gᶜ)) / 2⌉` where `l(v) = indepNeighbors G v`.

V2-000865 | Written on the Wall II - Conjecture 100 | conjecture100

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 100](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj100) (status O): For a simple connected graph `G`, `α(G) ≤ ⌈(max_v l(v) + 0.5 · degreeL2Norm(Gᶜ)) / 2⌉` where `α(G) = G.indepNum` is the independence number, `max_v l(v)` is the maximum over all vertices of the independence number of the neighbourhood (in `G`), and `degreeL2Norm(Gᶜ)` is the square root of the sum of the squares of the degrees in the complement `Gᶜ`. **Proof outline.** Choose a maximum independent set `S`, write `A = |S|`, let `T` be its complement, and let `L` be the maximum neighbourhood independence number. Connectedness gives at least one edge from every vertex of `S` into `T`, while each vertex of `T` has at most `L` neighbours in `S`; double counting therefore gives `A ≤ |T|L`. In the complement graph, `S` is a clique. The missing cross-edges force explicit lower bounds on the complement degrees of vertices in both `S` and `T`. Summing their squares gives a lower bound for `degreeL2Norm(Gᶜ)²`, and the resulting quadratic inequality rearranges to the stated ceiling bound. **Provenance.** Solved by Dominic Dabish. ProofOrchestrator, using OpenAI GPT-5.6 Thinking, assisted with the mathematical argument and Lean formalization; all formal claims were checked by the pinned Lean compiler.

```
theorem conjecture100 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let maxL := (Finset.univ.image (indepNeighborsCard G)).max' (by simp)
(G.indepNum : ℝ) ≤ ⌈((maxL : ℝ) + (1 / 2) * (degreeL2Norm Gᶜ : ℝ)) / 2⌉
```

FC-WrittenOnTheWallIII.GraphConjecture100.conjecture100

Written on the Wall II - Conjecture 101

MODULE CONTEXT: # Written on the Wall II - Conjecture 101 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [LM11] V. E. Levit and E. Mandrescu, [A Set and Collection Lemma](https://arxiv.org/abs/1101.4564), 2011, Corollaries 2.3-2.4. - [Li26] Wentao Li, [Lean proof of WOWII 101](https://github.com/VictorLiwentao/lean-graph-proofs/blob/9ce66be23570e0a9c8a6ae3f19f52e642878a8a9/proofs/WOWII101/PROOF.md), 2026. ## Definitions The **α-core** of a graph G, written `alphaCore G`, is the set of vertices v such that removing v strictly decreases the independence number: $$\mathrm{alphaCore}(G) = \{v \mid \alpha(G - v) < \alpha(G)\}$$ where $G - v$ is the subgraph of G induced on $V(G) \setminus \{v\}$. These vertices are also called "critical vertices for independence."

V2-000866 | Written on the Wall II - Conjecture 101 | conjecture101

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 101](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\alpha(G) \le \lfloor (n + |\mathrm{alphaCore}(G)|) / 2 \rfloor$ where $\alpha(G) = G.\mathrm{indepNum}$ is the independence number, n is the number of vertices, and $\mathrm{alphaCore}(G)$ is the set of vertices whose removal decreases the independence number. This is Hajnal's inequality in its independent-set form; see [LM11]. Lean formalization by Wentao Li, with AI assistance; see [Li26].

theorem conjecture101 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
G.indepNum ≤ (Fintype.card α + (alphaCore G).card) / 2

FC-WrittenOnTheWallIII.GraphConjecture101.conjecture101

Written on the Wall II - Conjecture 103

MODULE CONTEXT: # Written on the Wall II - Conjecture 103 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000867 | Written on the Wall II - Conjecture 103 | conjecture103

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 103](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\alpha(G) \le \lfloor b(G) - \ln(\mathrm{ecc_avg}(G)) \rfloor$ where $\alpha(G) = G.\mathrm{indepNum}$ is the independence number, $b(G)$ is the largest induced bipartite subgraph size, and $\mathrm{ecc_avg}(G) = G.\mathrm{averageEccentricity}$ is the average eccentricity of G. Uses `Real.log` (natural logarithm). This conjecture is false. The graph `wowii103Counterexample` has independence number 9, largest induced bipartite subgraph size 10, and average eccentricity $30/11$. Since $1 < \ln(30/11) < 2$, the proposed upper bound is 8.

theorem conjecture103 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) (_h : G.Connected),
(G.indepNum : ℝ) ≤ ⌊b G - Real.log (averageEccentricity G)⌋

FC-WrittenOnTheWallIII.GraphConjecture103.conjecture103

Written on the Wall II - Conjecture 109

MODULE CONTEXT: # Written on the Wall II - Conjecture 109 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000868 | Written on the Wall II - Conjecture 109 | conjecture109

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 109](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, the independence number $\alpha(G)$ was conjectured to satisfy $\alpha(G) \le \lfloor (\mathrm{residue}(G) + 2 \cdot b(G)) / 3 \rfloor$, where $\mathrm{residue}(G)$ is the Havel--Hakimi residue and $b(G)$ is the size of a largest induced bipartite subgraph. This is false. A connected graph on 21 vertices has an independent set of size 15, residue 8, and no induced bipartite subgraph with more than 18 vertices, so the conjectured right-hand side is at most 14. A smaller counterexample is the connected 13-vertex graph $\overline{K_7 \vee (K_3 \sqcup K_3)}$. It has independence number 7, residue 2, and largest induced bipartite subgraph size 9, so its conjectured right-hand side is 6.

theorem conjecture109 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_h : G.Connected),
(G.indepNum : ℝ) ≤ ⌊((residue G : ℝ) + 2 * b G) / 3⌋

FC-WrittenOnTheWallIII.GraphConjecture109.conjecture109

Written on the Wall II - Conjecture 13

MODULE CONTEXT: # Written on the Wall II - Conjecture 13 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000869 | Written on the Wall II - Conjecture 13 | conjecture13

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 13](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, the size `b(G)` of a largest induced bipartite subgraph satisfies `b(G) ≥ diam(G) + max_{v ∈ V} l(v) - 1`, where `diam(G)` is the diameter of `G` and `l(v) = indepNeighborsCard G v` is the independence number of the neighbourhood of `v`.

theorem conjecture13 (G : SimpleGraph α) (h : G.Connected) :
letI maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
(G.diam : ℝ) + (maxL : ℝ) - 1 ≤ b G

FC-WrittenOnTheWallIII.GraphConjecture13.conjecture13

Written on the Wall II - Conjecture 133

MODULE CONTEXT: # Written on the Wall II - Conjecture 133 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000870 | Written on the Wall II - Conjecture 133 | conjecture133

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 133](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/): For a simple connected graph G, $\operatorname{path}(G) \ge \operatorname{rad}(G) + (\mathrm{avg}_v\, I(v))^{cC_4(G)}$, where $\operatorname{path}(G)$ is the path number of the graph (number of vertices of a largest induced path), $\operatorname{rad}(G)$ is the radius (minimum eccentricity, as a natural number), $\mathrm{avg}_v\, I(v) = I(G)$ is the average independence number of vertex neighbourhoods, and $cC_4(G)$ is the C_4-free characteristic function (1 if G is C_4-free, not necessarily induced, and 0 otherwise).

```
theorem conjecture133 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let rad := G.radius.toNat
let hasC4 := ∃ a b c d : α, a ≠ b ∧ a ≠ c ∧ a ≠ d ∧ b ≠ c ∧ b ≠ d ∧ c ≠ d ∧
G.Adj a b ∧ G.Adj b c ∧ G.Adj c d ∧ G.Adj d a
let cC4 : ℕ := if hasC4 then 0 else 1
(rad : ℝ) + I G ^ cC4 ≤ (path G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture133.conjecture133

Written on the Wall II - Conjecture 141

MODULE CONTEXT: # Written on the Wall II - Conjecture 141 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000871 | Written on the Wall II - Conjecture 141 | conjecture141

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 141](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\mathrm{tree}(G) \ge (1/2) \cdot \mathrm{girth}(G) - 1 + \max_v I(v)$ where $\mathrm{tree}(G)$ is the number of vertices of a largest induced tree subgraph, $\mathrm{girth}(G)$ is the length of the shortest cycle (0 if acyclic), and $I(v)$ (`indepNeighborsCard G v`) is the independence number of the neighbourhood of v.

```
theorem conjecture141 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
(G.girth : ℝ) / 2 - 1 + ((Finset.univ.sup (indepNeighborsCard G) : ℕ) : ℝ) ≤
(largestInducedTreeSize G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture141.conjecture141

Written on the Wall II - Conjecture 142

MODULE CONTEXT: # Written on the Wall II - Conjecture 142 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000872 | Written on the Wall II - Conjecture 142 | conjecture142

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 142](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/): For a simple connected graph G, $\mathrm{tree}(G) \ge (2/3) \cdot \mathrm{girth}(G) + \mathrm{ecc}(B)$ where $\mathrm{tree}(G)$ is the largest induced tree size, $\mathrm{girth}(G)$ is the length of the shortest cycle (0 if acyclic), B is the set of boundary vertices (those of maximum eccentricity), and $\mathrm{ecc}(B)$ is the eccentricity of the set B. **Proof sketch.** The proof first strengthens the real-valued target to the integral bound $$ \mathrm{ecc}(B) + \mathrm{girth}(G) - \lfloor \mathrm{girth}(G) / 3 \rfloor \leq \mathrm{tree}(G). $$ For a cyclic graph, start with a shortest cycle K. It is chordless, so removing one vertex leaves an induced path on $\mathrm{girth}(G)-1$ vertices. Shortest paths from selected vertices to K are called *descents*. Pairwise noninteracting descents attach to the retained cycle path as disjoint branches. If two descents interact, the proof cuts at the first interaction encountered from the outer endpoint toward the cycle and splices the paths there. Minimality of that interaction makes the retained prefix disjoint and nonadjacent to the first descent; geodesicity excludes chords, while girth at least five rules out extra cross-edges and cycle attachments. Choose a vertex realizing $\mathrm{ecc}(B)$ and two endpoints of a diametral geodesic. Either two of their descents interact, directly giving a sufficiently long spliced tree, or all three are separate, in which case a three-point distance inequality forces their total length to be large enough. This proves the integral bound in the main range. Girths three and four, small $\mathrm{ecc}(B)$, and the two remaining congruence cases are handled by separate geodesic/cycle certificates with one or two descents; the acyclic case follows from a diametral path. Finally, $$ \mathrm{girth}(G)-\lfloor\mathrm{girth}(G)/3\rfloor = \lceil 2\,\mathrm{girth}(G)/3\rceil \geq 2\,\mathrm{girth}(G)/3, $$ which gives the stated real-valued inequality. The argument was developed with assistance from ChatGPT Pro. The linked Lean 4 formalization was produced with assistance from OpenAI Codex and checked by Lean.

```
theorem conjecture142 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let B : Set α := (maxEccentricityVertices G : Set α)
(2 : ℝ) / 3 * (G.girth : ℝ) + (eccSet G B : ℝ) ≤ (largestInducedTreeSize G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture142.conjecture142

Written on the Wall II - Conjecture 143

MODULE CONTEXT: # Written on the Wall II - Conjecture 143 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000873 | Written on the Wall II - Conjecture 143 | conjecture143

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 143](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj143): For a simple connected graph G, $\mathrm{tree}(G) \ge (\mathrm{girth}(G) + 1) / \sigma(G)$, where $\mathrm{tree}(G)$ is the largest induced tree size, $\mathrm{girth}(G)$ is the length of the shortest cycle, and $\sigma(G) = G.\mathrm{secondSmallestDegree}$ is the **second-smallest degree** of G's degree sequence (per WOWII defEntry 65). We state the inequality in denominator-free form to avoid the $\sigma = 0$ corner case ($n \le 1$). The proof splits into the acyclic case and two cyclic cases. For an acyclic graph the girth is zero. If $\sigma \ge 2$, deleting two consecutive edges from a shortest cycle leaves an induced tree on $\mathrm{girth}(G)-1$ vertices, and the factor $\sigma \ge 2$ yields the required inequality. If $\sigma = 1$, there are two degree-one vertices. A maximal induced tree containing them must have an external vertex with two attachments; those attachments create a cycle whose length forces the tree to contain at least $\mathrm{girth}(G)+1$ vertices.

```
theorem conjecture143 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected)
(hσ : 0 < secondSmallestDegree G) :
(G.girth : ℝ) + 1 ≤ (largestInducedTreeSize G : ℝ) * (secondSmallestDegree G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture143.conjecture143

Written on the Wall II - Conjecture 144

MODULE CONTEXT: # Written on the Wall II - Conjecture 144 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000874 | Written on the Wall II - Conjecture 144 | conjecture144

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 144](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\mathrm{tree}(G) \ge \mathrm{girth}(G) - 1 + \mathrm{ecc}(\mathrm{Centers})$ where $\mathrm{tree}(G)$ is the largest induced tree size, $\mathrm{girth}(G)$ is the length of the shortest cycle (0 if acyclic), $\mathrm{Centers} = G.\mathrm{center}$ is the set of vertices with minimum eccentricity (the center of G), and $\mathrm{ecc}(\mathrm{Centers})$ is the eccentricity of the center set - the maximum distance from any non-center vertex to the nearest center vertex. **Proof sketch.** Let $g = \mathrm{girth}(G)$ and $e = \mathrm{ecc}(G, \mathrm{center}(G))$. The acyclic and $e = 0$ cases are immediate. If $g \le e + 2$, the formalized Bacsó--Tuza induced-path argument, with an elementary $e = 1$ case, produces an induced tree on at least $2e + 1$ vertices. Since $g - 1 + e \le 2e + 1$, this proves the result. Suppose instead that $e + 3 \le g$, and fix a shortest cycle C. Removing one vertex of C leaves an induced tree on $g - 1$ vertices. Analyze the connected components outside C. If one has at least e vertices, it supplies an attached rooted induced tree of order e. Otherwise, let D be the sum of the components' attachment depths. If $D \ge e$, select pairwise edge-separated rooted branches of total order e and attach them to C, deleting a suitable cycle vertex. If $D < e$, shortest-cycle contact restrictions bound corresponding exclusion arcs. Their complement consists of central cycle vertices and D-dominates the graph, forcing $e \le D$, a contradiction. Finite graph search was used only during discovery to test candidate structural lemmas and reject false proof templates; no finite-search result is used in the universal proof.

```
theorem conjecture144 (G : SimpleGraph α) (h : G.Connected) :
(G.girth : ℝ) - 1 + (ecc G G.center : ℝ) ≤ (largestInducedTreeSize G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture144.conjecture144

Written on the Wall II - Conjecture 145

MODULE CONTEXT: # Written on the Wall II - Conjecture 145 The WOWII HTML uses $\lambda_{\min}(\overline{G})$ (the bar denotes graph complement). The formal statement below uses the local-independence minimum of G^c. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definitions The **local independence minimum** $\mathrm{lMin}(G)$ is: $$\mathrm{lMin}(G) = \min_{v \in V(G)} l(v)$$ where $l(v) = \mathrm{indepNeighborsCard}(G, v)$ is the independence number of the neighbourhood of v. This is the minimum over all vertices of the local independence number. The **boundary vertices** $B(G)$ of a connected graph are the vertices v such that the eccentricity of v equals the diameter of G. The **eccentricity of a set** $\mathrm{ecc}(S) = \max_{u \notin S} \min_{w \in S} \mathrm{dist}(u, w)$. In the conjecture below, $\mathrm{ecc}(B)$ is the eccentricity of the boundary set. **Conjecture 145:** $\mathrm{tree}(G) \ge 2 \cdot \mathrm{ecc}(B) / \lambda_{\min}(\overline{G})$ where $\mathrm{tree}(G)$ is `largestInducedTreeSize G`, $\mathrm{ecc}(B)$ is the eccentricity of the boundary vertices, and $\lambda_{\min}(\overline{G})$ is the local independence minimum of the complement $\overline{G}$.

V2-000875 | Written on the Wall II - Conjecture 145 | conjecture145

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 145](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\mathrm{tree}(G) \ge 2 \cdot \mathrm{ecc}(B) / \lambda_{\min}(\overline{G})$ where $\mathrm{tree}(G)$ is the number of vertices in a largest induced subtree, $\mathrm{ecc}(B)$ is the eccentricity of the boundary vertices (`eccSet` and `boundaryVertices`), and $\lambda_{\min}(\overline{G})$ is the minimum local independence number of the complement graph. We state the inequality in the form $\mathrm{tree}(G) \cdot \mathrm{lMin}(\overline{G}) \ge 2 \cdot \mathrm{ecc}(B)$ to avoid division. ## Provenance Solved by Dominic Dabish. ProofOrchestrator, using OpenAI GPT-5.6 Thinking, assisted with the mathematical argument and Lean formalization; all formal claims were checked by the pinned Lean compiler.

```
theorem conjecture145 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected)
(hlMin : 0 < localIndependenceMin Gᶜ) :
2 * eccSet G (maxEccentricityVertices G : Set α) ≤
largestInducedTreeSize G * localIndependenceMin Gᶜ
```

FC-WrittenOnTheWallII.GraphConjecture145.conjecture145

Written on the Wall II - Conjecture 146

MODULE CONTEXT: # Written on the Wall II - Conjecture 146 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definitions The **square** of a graph G, denoted G^2, is the graph on the same vertex set where two distinct vertices are adjacent if and only if their distance in G is at most 2. The **radius of G^2** is the minimum eccentricity of any vertex in G^2, i.e., $$\mathrm{rad}(G^2) = \min_{v \in V} \max_{u \in V} \mathrm{dist}_{G^2}(u, v).$$

V2-000876 | Written on the Wall II - Conjecture 146 | conjecture146

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 146](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\mathrm{tree}(G) \ge 2 \cdot \mathrm{ecc}(B) / \mathrm{rad}(G^2)$ where $\mathrm{tree}(G)$ is the number of vertices in a largest induced subtree, $\mathrm{ecc}(B)$ is the eccentricity of the boundary vertices of G (`eccSet` and `maxEccentricityVertices`), and $\mathrm{rad}(G^2)$ is the radius of the square graph of G. We state the inequality in the form $\mathrm{tree}(G) \cdot \mathrm{rad}(G^2) \ge 2 \cdot \mathrm{ecc}(B)$ to avoid division. ## Informal proof Write t for the largest induced-tree order, r and d for the radius and diameter of G, and p for the eccentricity of its peripheral set. First, the distance in the graph square is $\operatorname{dist}_{G^2}(u,v)=\lceil\operatorname{dist}_G(u,v)/2\rceil$, so $\operatorname{rad}(G^2)=\lceil r/2\rceil$. A diametral geodesic is an induced path, giving $t\ge d+1$, while $p\le d-1$. These bounds settle every case except $r=2$, $d=4$, and $p=3$. In that remaining configuration, choose a centre and shortest paths to two diametral vertices and to a vertex at distance three from the peripheral set. A finite case analysis on the two possible cross-arm edges (and then the possible chords) always produces an induced tree on at least six vertices. Hence $t\,\operatorname{rad}(G^2) \ge 2p$ in the exceptional case as well.

```
theorem conjecture146 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected)
(hrad : 0 < graphSquareRadius G) :
2 * eccSet G (maxEccentricityVertices G : Set α) ≤
largestInducedTreeSize G * graphSquareRadius G
```

FC-WrittenOnTheWallII.GraphConjecture146.conjecture146

Written on the Wall II - Conjecture 16

MODULE CONTEXT: # Written on the Wall II - Conjecture 16 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000877 | Written on the Wall II - Conjecture 16 | conjecture16

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 16](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, the size `b(G)` of a largest induced bipartite subgraph satisfies `b(G) ≥ 2 * (rad(G) - 1) + max_{v ∈ V} l(v)`, where `rad(G)` is the radius of `G` (the minimum eccentricity) and `l(v) = indepNeighborsCard G v` is the independence number of the neighbourhood of `v`.

```
theorem conjecture16 (G : SimpleGraph α) (h : G.Connected) :
letI maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
letI r := G.radius.toNat
2 * ((r : ℝ) - 1) + (maxL : ℝ) ≤ b G
```

FC-WrittenOnTheWallII.GraphConjecture16.conjecture16

Written on the Wall II - Conjecture 17

MODULE CONTEXT: # Written on the Wall II - Conjecture 17 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000878 | Written on the Wall II - Conjecture 17 | conjecture17

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 17](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, the size `b(G)` of a largest induced bipartite subgraph satisfies `b(G) ≥ α(G) + ⌈diam(G) / 3⌉`, where `α(G)` is the independence number of `G` and `diam(G)` is the diameter of `G`.

```
theorem conjecture17 (G : SimpleGraph α) (h : G.Connected) :
(G.indepNum : ℝ) + ⌈(G.diam : ℝ) / 3⌉ ≤ b G
```

FC-WrittenOnTheWallII.GraphConjecture17.conjecture17

Written on the Wall II - Conjecture 172

MODULE CONTEXT: # Written on the Wall II - Conjecture 172 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) The conjecture is false for the graph `D₉`, formed from two triangles by joining distinguished vertices with a path of nine edges. It has at most four leaves in a spanning tree. Its peripheral vertices have degree two, while the two maximum-degree vertices of its square are at distance seven in the original graph, so the claimed inequality would give `4 ≥ 8`.

V2-000879 | Written on the Wall II - Conjecture 172 | conjecture172

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII Conjecture 172 asked whether every nontrivial finite connected simple graph `G` satisfies `Ls G ≥ -1 + Δ(B(G)) + dist_min(G, M(G²))`. The answer is no, witnessed by `D₉`.

```
theorem conjecture172 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] [Nontrivial V]
(G : SimpleGraph V) [DecidableRel G.Adj], G.Connected →
Ls G ≥ (-1 : ℝ) + peripheryMaxDegree G +
distMin G (squareMaximumDegreeVertices G)
```

FC-WrittenOnTheWallII.GraphConjecture172.conjecture172

Written on the Wall II - Conjecture 176

MODULE CONTEXT: # Written on the Wall II - Conjecture 176 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) The conjecture is false for the graph `D₇`, formed from two triangles by joining distinguished vertices with a path of seven edges. It has at most four leaves in a spanning tree and largest induced bipartite subgraph order ten. The two maximum-degree vertices of its square are at distance five in the original graph, so the claimed inequality would give `14 ≥ 17`.

V2-000880 | Written on the Wall II - Conjecture 176 | conjecture176

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII Conjecture 176 asked whether every nontrivial finite connected simple graph `G` satisfies `Ls(G) + b(G) ≥ n(G) + dist_min(G, M(G²))`. The answer is no, witnessed by `D₇`.

```
theorem conjecture176 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] [Nontrivial V]
(G : SimpleGraph V) [DecidableRel G.Adj], G.Connected →
Ls G + b G ≥ (Fintype.card V : ℝ) +
distMin G (squareMaximumDegreeVertices G)
```

FC-WrittenOnTheWallII.GraphConjecture176.conjecture176

Written on the Wall II - Conjecture 18

MODULE CONTEXT: # Written on the Wall II - Conjecture 18 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000881 | Written on the Wall II - Conjecture 18 | conjecture18

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 18](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, the size $b(G)$ of a largest induced bipartite subgraph satisfies $b(G) \ge \alpha(G) + \lceil \sqrt{\mathrm{dist}_{\max}(M)} \rceil$, where $\alpha(G)$ is the independence number, M is the set of maximum-degree vertices, and $\mathrm{dist}_{\max}(M) = \max\{\mathrm{dist}_G(u,v) \mid u, v \in M\}$ is the maximum distance between two maximum-degree vertices (DeLaVina's `dist_max(M)`). Proven by Benny John (Feb. 2006), generalizing Schindl's proof of conjecture 17.

```
theorem conjecture18 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let M : Set α := {v | G.degree v = G.maxDegree}
(G.indepNum : ℝ) + ⌈Real.sqrt (distMaxSet G M : ℝ)⌉ ≤ b G
```

FC-WrittenOnTheWallII.GraphConjecture18.conjecture18

Written on the Wall II - Conjecture 181

MODULE CONTEXT: # Written on the Wall II - Conjecture 181 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) The expression `deg_avg(B(G²))` is formalized by measuring degrees in `G²`, the graph whose maximum-eccentricity vertices induce `B(G²)`. The conjecture is false for the triangular graph `T(7) = L(K₇)`: it has at most 16 leaves in a spanning tree, largest induced bipartite subgraph order 6, and independence number 3. Its square is `K₂₁`, whose vertices all have degree 20, so the claimed inequality would give `22 ≥ 23`.

V2-000882 | Written on the Wall II - Conjecture 181 | conjecture181

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII Conjecture 181 asked whether every nontrivial finite connected simple graph `G` satisfies `Ls G + b G ≥ G.indepNum + deg_avg(B(G²))`. The answer is no, witnessed by `T(7) = L(K₇)`.

```
theorem conjecture181 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] [Nontrivial V]
(G : SimpleGraph V) [DecidableRel G.Adj], G.Connected →
Ls G + b G ≥ G.indepNum + squarePeripheryAverage G
```

FC-WrittenOnTheWallIII.GraphConjecture181.conjecture181

Written on the Wall II - Conjecture 189

MODULE CONTEXT: # Written on the Wall II - Conjecture 189 **Verbatim statement (WOWII #189, status O):** > If G is a simple connected graph with n > 1 such that maximum { dist_even(v) : dist_even(v) is the number of vertices at even distance from v } ≤ 1 + σ(G), then G has a Hamiltonian path. **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj189 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000883 | Written on the Wall II - Conjecture 189 | conjecture189

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 189](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj189) (status O): For a simple connected graph `G`, if `max_v distEven(v) ≤ 1 + σ(G)`, then `G` has a Hamiltonian path. Here `distEven(v)` is the number of vertices at even distance from `v`, and `σ(G)` is the **second-smallest degree** of `G`'s degree sequence (per WOWII defEntry 65).

```
theorem conjecture189 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected)
(hσ : (Finset.univ.image (distEven G)).max' (by simp) ≤ 1 + secondSmallestDegree G) :
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallIII.GraphConjecture189.conjecture189

Written on the Wall II - Conjecture 19

MODULE CONTEXT: # Written on the Wall II - Conjecture 19 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000884 | Written on the Wall II - Conjecture 19 | conjecture19

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 19](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) If `G` is connected then the size `b(G)` of a largest induced bipartite subgraph satisfies `b(G) ≥ FLOOR((∑ ecc(v))/(|V|) + sSup (range (l G)))`, where `ecc(v)` denotes eccentricity and `l(G)` is the independence number of neighbourhoods.

```
theorem conjecture19 (G : SimpleGraph α) [Nontrivial α] (h_conn : G.Connected) :
⌊(∑ v ∈ Finset.univ, ((G.eccent v).toNat : ℝ)) / (Fintype.card α : ℝ) +
sSup (Set.range (indepNeighbors G))⌋ ≤ b G
```

FC-WrittenOnTheWallIII.GraphConjecture19.conjecture19

Written on the Wall II - Conjecture 194

MODULE CONTEXT: # Written on the Wall II - Conjecture 194 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample The conjecture is false. Take a clique on vertices `0, ..., 10`, four additional vertices `11, ..., 14` adjacent to every clique vertex, and three leaves attached at `11`, `12`, and `14`. This connected graph has independence number `4`, while the sum of the independence numbers of its vertex neighbourhoods is `54`, so their average is `3`. Thus it satisfies the conjecture's hypothesis with equality. However, its three leaves would all have to be endpoints of a Hamiltonian path, which is impossible.

V2-000885 | Written on the Wall II - Conjecture 194 | conjecture194

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 194](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, if `α(G) ≤ 1 + l_avg(G)`, then `G` has a Hamiltonian path. Here `α(G) = G.indepNum` is the independence number, and `l_avg(G) = averageIndepNeighbors G` is the average over all vertices of the independence number of the neighbourhood. A Hamiltonian path is a walk visiting every vertex exactly once. The answer is no, as witnessed by the 18-vertex graph described above. Counterexample (Graph6): `Q~~~~~~~~~~~~~}~}^~??G??_??_`

```
theorem conjecture194 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) (_h : G.Connected),
(G.indepNum : ℝ) ≤ 1 + averageIndepNeighbors G →
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallIII.GraphConjecture194.conjecture194

Written on the Wall II - Conjecture 198a

MODULE CONTEXT: # Written on the Wall II - Conjecture 198a *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000886 | Written on the Wall II - Conjecture 198a | conjecture198a

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 198a](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, if `b(G) ≤ 2 + ecc_avg(G)`, then `G` has a Hamiltonian path. Here `b(G)` is the number of vertices in a largest induced bipartite subgraph, and `ecc_avg(G)` is the average eccentricity of `G`. A Hamiltonian path is a walk visiting every vertex exactly once. Proofs and formalizations include: * [Ben Cohen, *Average Eccentricity, Induced Bipartite Subgraphs, and Hamiltonian Paths*](https://zenodo.org/records/21481944) (July 21, 2026); [Lean 4 formalization](https://github.com/bzcohen588/written-on-the-wall-198a/blob/5481927df49c374b06fa0931b5aa24f9296faa04/Conjecture198a.lean), publicly posted and verified September 16, 2026. * [Luke Kabbash, *A Hamiltonian-Path Criterion from Bipartite Number and Average Eccentricity*](https://github.com/lukekabbash/wowii-198a-lean/blob/49f080bbaee83821e9e9744eef14c5acaae2a101/paper/main.pdf); Lean 4 formalization publicly posted and passed its recorded verification July 23, 2026.

```
theorem conjecture198a (G : SimpleGraph α) (h : G.Connected)
(hb : b G ≤ 2 + averageEccentricity G) :
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallII.GraphConjecture198a.conjecture198a

Written on the Wall II - Conjecture 199

MODULE CONTEXT: # Written on the Wall II - Conjecture 199 **Verbatim statement (WOWII #199, status O):** > If G is a simple connected graph with n > 1 such that tree(G) - 2 ≤ κ(G), then G has a Hamiltonian path. **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj199 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000887 | Written on the Wall II - Conjecture 199 | conjecture199

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 199](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, if `tree(G) - 2 ≤ κ(G)`, then `G` has a Hamiltonian path. The nontrivial vertex type encodes the source assumption $n > 1$. Here `tree(G)` is the number of vertices in a largest induced tree and `κ(G) = vertexConnectivity G` is the vertex connectivity of `G`.

```
theorem conjecture199 (G : SimpleGraph α) (h : G.Connected)
(hκ : largestInducedTreeSize G - 2 ≤ vertexConnectivity G) :
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallII.GraphConjecture199.conjecture199

Written on the Wall II - Conjecture 2

MODULE CONTEXT: # Written on the Wall II - Conjecture 2 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [arxiv/2605.22763](https://arxiv.org/abs/2605.22763) *Advancing Mathematics Research with AI-Driven Formal Proof Search* by George Tsoukalas et al.

V2-000888 | Written on the Wall II - Conjecture 2 | conjecture2

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 2](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $Ls(G) \ge 2 \cdot (I(G) - 1)$ where $I(G)$ is the average independence number of the neighbourhoods of the vertices of G. A formal proof has been found with the methods described in [arxiv/2605.22763](https://arxiv.org/abs/2605.22763), where an informal proof is also provided. Another formal proof combines a spanning-tree leaf bound from connected domination with an ordered-pair double-counting argument for adjacent neighbourhoods. A third formal proof starts from a triangle-free spanning subgraph of G with the maximum number of edges. Maximality bounds the neighbourhood independence number of each vertex by its degree in that subgraph; an edge whose endpoint degrees sum to at least $2 \cdot I(G)$ then carries a double star, which is acyclic and extends to a spanning tree with at least $2 \cdot (I(G) - 1)$ leaves.

```
theorem conjecture2 (G : SimpleGraph α) (h : G.Connected) :
2 * (averageIndepNeighbors G - 1) ≤ Ls G
```

FC-WrittenOnTheWallII.GraphConjecture2.conjecture2

Written on the Wall II - Conjecture 20

MODULE CONTEXT: # Written on the Wall II - Conjecture 20 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [AKS87] N. Alon, J. Kahn and P. D. Seymour, [Large Induced Degenerate Subgraphs](https://doi.org/10.1007/BF01788542), Graphs and Combinatorics 3 (1987), 203-211. - [Li26] Wentao Li, [Lean proof of WOWII 20](https://github.com/VictorLiwentao/lean-graph-proofs/blob/9ce66be23570e0a9c8a6ae3f19f52e642878a8a9/proofs/WOWII20/PROOF.md), 2026.

V2-000889 | Written on the Wall II - Conjecture 20 | conjecture20

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 20](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, `b(G) ≥ n(G) / ⌊deg_avg(G)⌋`, where `b(G)` is the size of a largest induced bipartite subgraph, `n(G)` is the number of vertices, and `deg_avg(G) = (∑ v, deg(v)) / n(G)` is the average degree. Follows from the weighted induced-subgraph bound of Alon, Kahn and Seymour [AKS87]. Lean formalization by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture20 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let deg_avg : ℝ := (∑ v : α, (G.degree v : ℝ)) / (Fintype.card α : ℝ)
(Fintype.card α : ℝ) / (⌊deg_avg⌋ : ℝ) ≤ (b G : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture20.conjecture20

Written on the Wall II - Conjecture 200

MODULE CONTEXT: # Written on the Wall II - Conjecture 200 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample For every integer $q \geq 5$, let $C = \{a,b\} \cup D$ have q vertices and induce K_q minus the edge ab. Add two nonadjacent vertices x,y, each complete to C; a vertex z adjacent exactly to a,b; and, for every $d \in D$, a pendant vertex p_d adjacent exactly to d. The local-neighbourhood independence numbers are 2 at x,y,z, 1 at each p_d, and 3 at every vertex of C. Their sum is $4q+4$, while the graph has $2q+1$ vertices, so $$\left\lceil 1 + l_{\mathrm{avg}}(G)\right\rceil = 4.$$ The vertices $\{a,x,y,z\}$ induce a claw, and a case split on the number of vertices from C shows that no induced tree has more than four vertices. Thus $\operatorname{tree}(G)=4$. Finally, the graph has $q-2 \geq 3$ pendant vertices, but a Hamiltonian path has only two endpoints. The smallest member of this family has 11 vertices and graph6 encoding `J??FFBRq}N_`. This is a smallest counterexample overall: an exhaustive search over all 11,989,760 connected graphs on 4 ≤ n ≤ 10 vertices (via nauty's `geng`; graphs on at most 3 vertices are trivially traceable) found no graph satisfying the premise without a Hamiltonian path.

V2-000890 | Written on the Wall II - Conjecture 200 | conjecture200

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 200](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, if `tree(G) = ⌈1 + l_avg(G)⌉`, then `G` has a Hamiltonian path. Here `tree(G)` is the number of vertices of a largest induced tree subgraph, and `l_avg(G) = averageIndepNeighbors G` is the average over all vertices of the independence number of the neighbourhood. A Hamiltonian path is a walk visiting every vertex exactly once. This conjecture is false. The counterexample family in the module docstring satisfies the equality hypothesis and has no Hamiltonian path.

```
theorem conjecture200 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) (_h : G.Connected),
(largestInducedTreeSize G : ℝ) = ⌈1 + averageIndepNeighbors G⌉ →
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallII.GraphConjecture200.conjecture200

Written on the Wall II - Conjecture 217

MODULE CONTEXT: # Written on the Wall II - Conjecture 217 Per the WOWII definitions popup linked from this conjecture: - $L(G)$ is the **maximum number of leaves of a spanning tree** of G - i.e. `Ls G` in our invariant library. - $\chi_{\mathrm{residue}=2}(G)$ is the **characteristic function** for the predicate $\mathrm{residue}\, G = 2$, i.e. 1 when $\mathrm{residue}\, G = 2$ and 0 otherwise. It is not a connected-component count of any 2-core. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000891 | Written on the Wall II - Conjecture 217 | conjecture217

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 217](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj217): If G is a finite simple connected graph on $n > 1$ vertices and $L_s(G) \le 4 \cdot \chi_{\mathrm{residue}=2}(G) + 2$, then G has a Hamiltonian path. Here $L_s(G)$ is the maximum number of leaves over all spanning trees and $\chi_{\mathrm{residue}=2}(G)$ is the indicator of $\mathrm{residue}(G) = 2$.

```
theorem conjecture217 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected)
(hL : Ls G ≤ 4 * (residueEqTwoIndicator G : ℝ) + 2) :
∃ a b : α, ∃ p : G.Walk a b, p.IsHamiltonian
```

FC-WrittenOnTheWallII.GraphConjecture217.conjecture217

Written on the Wall II - Conjecture 23

MODULE CONTEXT: # Written on the Wall II - Conjecture 23 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000892 | Written on the Wall II - Conjecture 23 | conjecture23

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 23](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, `b(G) ≥ ⌊α(G) + dist_avg(M) / 2⌋`, where `b(G)` is the size of a largest induced bipartite subgraph, `α(G)` is the independence number, `M` is the set of maximum-degree vertices, and `dist_avg(M)` is the average distance between distinct vertices of `M`. This conjecture is false; there is a counterexample with `b(G) = 19`, `α(G) = 15`, and `dist_avg(M) = 10`.

```
theorem conjecture23 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected),
let M : Set α := {v | G.degree v = G.maxDegree}
⌊(G.indepNum : ℝ) + distAvgSet G M / 2⌋ ≤ (b G : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture23.conjecture23

Written on the Wall II - Conjecture 24

MODULE CONTEXT: # Written on the Wall II - Conjecture 24 **Verbatim statement (WOWII #24, status F):** > If G is a simple connected graph, then b(G) ≥ λ(G) + CEIL[minimum of dist_even(v)/3] **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj24 The WOWII page records this conjecture as disproved (status F). *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000893 | Written on the Wall II - Conjecture 24 | conjecture24

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 24](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj24) (status F, disproved): For every finite simple connected graph `G`, `b(G) ≥ λ(G) + ⌈(min_v dist_even(v)) / 3⌉` where `b(G)` is the size of a largest induced bipartite subgraph, `λ(G) := max_v l(v)` is the maximum over all vertices of the independence number of the open neighbourhood of `v`, and `dist_even(v)` is the number of vertices at even distance from `v`.

```
theorem conjecture24 :
¬ ∀ {α : Type*} [Fintype α] [DecidableEq α] [Nontrivial α],
∀ (G : SimpleGraph α) (_ : G.Connected),
let maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
let minDistEven := (Finset.univ.image (distEven G)).min' (by simp)
(maxL : ℝ) + ⌈(minDistEven : ℝ) / 3⌉ ≤ b G
```

FC-WrittenOnTheWallIII.GraphConjecture24.conjecture24

Written on the Wall II - Conjecture 25

MODULE CONTEXT: # Written on the Wall II - Conjecture 25 **Verbatim statement (WOWII #25, status F):** > If G is a simple connected graph, then b(G) ≥ 2 CEIL[(1 + minimum of dist_even(v))/3] **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj25 The WOWII page records this conjecture as disproved (status F), with the same counterexample as Conjecture 24. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000894 | Written on the Wall II - Conjecture 25 | conjecture25

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 25](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj25) (status F, disproved): For every finite simple connected graph G, $b(G) ≥ 2⌈(1 + \min_v \operatorname{dist}_{\mathrm{even}}(v))/3⌉$, where $b(G)$ is the size of a largest induced bipartite subgraph and $\operatorname{dist}_{\mathrm{even}}(v)$ counts the vertices at even distance from v. The nontrivial vertex type excludes graphs with fewer than two vertices. The source gives the same counterexample as for Conjecture 24.

```
theorem conjecture25 :
¬ ∀ {α : Type*} [Fintype α] [DecidableEq α] [Nontrivial α],
∀ (G : SimpleGraph α) (_ : G.Connected),
let minDistEven := (Finset.univ.image (distEven G)).min' (by simp)
2 * ⌈(1 + (minDistEven : ℝ)) / 3⌉ ≤ (b G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture25.conjecture25

Written on the Wall II - Conjecture 291

MODULE CONTEXT: # Written on the Wall II - Conjecture 291 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definitions For a vertex v in a graph G, **$T(v)$** is the number of triangles (3-cliques) incident to v, i.e., the number of 3-element cliques in G that contain v. The **triangle-frequency of the minimum** is the number of vertices that achieve the minimum value of $T(v)$. **k** is the **first step in the Havel-Hakimi process at which a zero appears**. Concretely, starting from the descending degree sequence s_0 of G, we set $s_{i+1} = \mathrm{havelHakimiStep}\, s_i$ and let k be the least $i \ge 0$ such that s_i contains a zero entry (or, vacuously, has been emptied entirely). Since each step is sorted descending and never increases entries, this is equivalent to the last (smallest) entry of s_i being 0, or s_i being $[]$. This is **strictly weaker** than $n - \mathrm{residue}(G)$: $n - \mathrm{residue}(G)$ is the *total* number of reduction steps until *every* entry is zero, whereas k only requires that *some* entry has hit zero - and a 0 typically appears well before the all-zero state is reached. **Conjecture 291:** For a simple connected graph G with $n > 2$, $\gamma_t(G) \le k + \mathrm{frequency}(t_{\min}(v))$ where $\gamma_t(G)$ is the total domination number, k is the Havel-Hakimi zero step, and $\mathrm{frequency}(t_{\min}(v))$ is the number of vertices achieving the minimum triangle count. The conjecture is **false**: on July 23, 2026, Zyad Tamimi sent a 12-vertex counterexample with $\gamma_t(G) = 4$, $\mathrm{frequency}(t_{\min}(v)) = 1$ and $k = 2$, and the source now lists Conjecture 291 as refuted.

V2-000895 | Written on the Wall II - Conjecture 291 | conjecture291

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 291](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G with $n > 2$, $\gamma_t(G) \le k + \mathrm{frequency}(t_{\min}(v))$ where: - $\gamma_t(G)$ is the total domination number, - k is the first step in which a zero appears in the Havel-Hakimi process, - $\mathrm{frequency}(t_{\min}(v))$ is the number of vertices achieving the minimum triangle count. This is false: the source records a 12-vertex counterexample by Zyad Tamimi (July 23, 2026) with $\gamma_t(G) = 4 > 2 + 1 = k + \mathrm{frequency}(t_{\min}(v))$. The hypothesis $n > 2$ is stated in the source's full list of conjectures (as for the neighbouring Conjectures 290, 292 and 293) but omitted on its list of open conjectures; the counterexample has 12 vertices, so it refutes both readings.

theorem conjecture291 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_h : G.Connected) (_hn : 2 < Fintype.card α),
G.totalDominationNumber ≤ havelHakimiZeroStep G + freqMinTriangles G

FC-WrittenOnTheWallIII.GraphConjecture291.conjecture291

Written on the Wall II - Conjecture 3

MODULE CONTEXT: # Written on the Wall II - Conjecture 3 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000896 | Written on the Wall II - Conjecture 3 | conjecture3

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 3](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a connected simple graph `G`, the number of leaves in a maximum spanning tree satisfies `Ls(G) ≥ G.indepDominationNumber * MaxTemp(G)`, where `G.indepDominationNumber` is the independent domination number and `MaxTemp(G)` is `max_v deg(v)/(n(G) - deg(v))`.

theorem conjecture3 {G : SimpleGraph α} [DecidableRel G.Adj] [Nontrivial α] (h_conn :
G.Connected) :
G.indepDominationNumber * MaxTemp G ≤ Ls G

FC-WrittenOnTheWallIII.GraphConjecture3.conjecture3

Written on the Wall II - Conjecture 309

MODULE CONTEXT: # Written on the Wall II - Conjecture 309 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [J. J. Gebendorfer, An Infinite Family of Counterexamples to Written on the Wall II, Conjecture 309](https://doi.org/10.5281/zenodo.21553295) The conjecture is false for the clique blow-ups $C_5[K_k]$ for every $k \geq 3$.

V2-000897 | Written on the Wall II - Conjecture 309 | conjecture309

research solved | proof annotation: True | 官方代码 | 原文线索

Does every finite simple connected graph G of order greater than two satisfy $$ \gamma_t(G) \leq \frac{1}{2}\left(\max_v(\operatorname{distEven}(v)-\operatorname{evenHorizontal}(v))+ \min_{e\in E(\overline G)}|N_{\overline G}(e)|\right)? $$ Gebendorfer disproved the statement with the family $C_5[K_k]$, $k \geq 3$.

theorem conjecture309 : ¬ conjecture309Statement

FC-WrittenOnTheWallIII.GraphConjecture309.conjecture309

Written on the Wall II - Conjecture 31

MODULE CONTEXT: # Written on the Wall II - Conjecture 31 The WOWII page records this as **Chung's theorem**. This radius bound is proved in Section 2 of P. Erdős, M. Saks, and V. T. Sós, *Maximum Induced Trees in Graphs*, J. Combin. Theory Ser. B **41** (1986), 61-79; the authors credit Fan Chung for the proof. We state it here as a theorem; the formal proof is left as `sorry` pending a Lean port of the argument. Here $\mathrm{path}(G)$ is the maximum number of vertices in an induced path, and $\mathrm{rad}(G)$ is the graph radius. *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [P. Erdős, M. Saks, V. T. Sós, Maximum Induced Trees in Graphs](https://doi.org/10.1016/0095-8956(86)90028-6)

V2-000898 | Written on the Wall II - Conjecture 31 | conjecture31

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 31](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj31) (Chung): For every simple connected graph G, $\mathrm{path}(G) \ge 2 \cdot \mathrm{rad}(G) - 1$, where $\mathrm{path}(G)$ is the maximum number of vertices in an induced path and $\mathrm{rad}(G)$ is the graph radius.

theorem conjecture31 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
2 * (G.radius.toNat : ℤ) - 1 ≤ (path G : ℤ)

FC-WrittenOnTheWallIII.GraphConjecture31.conjecture31

Written on the Wall II - Conjecture 314

MODULE CONTEXT: # Written on the Wall II - Conjecture 314 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000899 | Written on the Wall II - Conjecture 314 | conjecture314

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 314](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj314): For every finite simple connected graph G with $n > 1$ vertices, if G is triangle-free and $\mathrm{path}(G) \le 4$, then G is well totally dominated. Here $\mathrm{path}(G) = \mathrm{largestInducedPathSize}\, G$ is the **size of a largest induced path** in G, defined locally above. **Disambiguation.** Earlier revisions of this file used the `SimpleGraph.path` invariant, but that is the *floor of the average distance*, not the size of a largest induced path - a different quantity that makes Conjecture 314 vacuous in many cases. **Proof summary.** In a connected, K_{3}-free, induced-P_{5}-free graph G: 1. No minimal total dominating set has fewer than two vertices. 2. No minimal total dominating set has more than three vertices. 3. Minimal total dominating sets of sizes two and three cannot coexist. Thus, each minimal total dominating set in G has the same size. The Lean proof was generated by OpenAI GPT-5.6 Sol.

```
theorem conjecture314 [Nontrivial α] (G : SimpleGraph α) [DecidableRel G.Adj]
(hG : G.Connected)
(hTriFree : ∀ a b c : α, G.Adj a b → G.Adj b c → G.Adj c a → False)
(hPath : largestInducedPathSize G ≤ 4) :
IsWellTotallyDominated G
```

FC-WrittenOnTheWallIII.GraphConjecture314.conjecture314

Written on the Wall II - Conjecture 315

MODULE CONTEXT: # Written on the Wall II - Conjecture 315 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definitions The notions `IsTotalDominatingSet`, `IsMinimalTotalDominatingSet`, `IsWellTotallyDominated`, and `pendantVertices` live in `FormalConjecturesForMathlib.Combinatorics.SimpleGraph.GraphConjectures.WellTotallyDominated`.

V2-000900 | Written on the Wall II - Conjecture 315 | conjecture315

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 315](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) Let G be a simple connected graph and let P denote the set of pendant vertices (vertices of degree 1). If $\alpha(G) = |P|$, then G is well totally dominated. A formal proof of this conjectures has been obtained by Goran Žužić and Moritz Firsching using an experimental pipeline.

```
theorem conjecture315 (G : SimpleGraph α) [DecidableRel G.Adj] (hG : G.Connected)
(h : G.indepNum = (pendantVertices G).card) :
IsWellTotallyDominated G
```

FC-WrittenOnTheWallIII.GraphConjecture315.conjecture315

Written on the Wall II - Conjecture 316

MODULE CONTEXT: # Written on the Wall II - Conjecture 316 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000901 | Written on the Wall II - Conjecture 316 | conjecture316

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 316](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) Let `G` be a simple connected graph and let `P` denote the set of pendant vertices (vertices of degree 1). If `|P| ≥ deg_avg(Gᶜ)`, then `G` is well totally dominated, where `deg_avg(Gᶜ)` is the average degree of the complement of `G`. **Proof sketch.** In the trivial cases (`P = ∅`, or at most `2` vertices) `G` is complete, and complete graphs are well totally dominated. Otherwise the set `C` of non-pendant vertices satisfies `|C| ≤ 3` and is a clique of `G`, and a case split on the set `Q ⊆ C` of neighbours of pendant vertices shows that `G` is well totally dominated.

```
theorem conjecture316 (G : SimpleGraph α) [DecidableRel G.Adj] (hG : G.Connected)
(h : (averageDegree Gᶜ : ℚ) ≤ (pendantVertices G).card) :
IsWellTotallyDominated G
```

FC-WrittenOnTheWallIII.GraphConjecture316.conjecture316

Written on the Wall II - Conjecture 32

MODULE CONTEXT: # Written on the Wall II - Conjecture 32 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000902 | Written on the Wall II - Conjecture 32 | conjecture32

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 32](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\operatorname{path}(G) \ge \operatorname{dist}_{\operatorname{avg}}(A) + 0.5 \cdot \operatorname{ecc}_{\operatorname{avg}}(M)$, where $\operatorname{path}(G)$ is the number of vertices of a largest induced path of G, A is the set of minimum-degree vertices, M is the set of maximum-degree vertices, $\operatorname{dist}_{\operatorname{avg}}(A)$ is the average of all nonzero distances $\operatorname{dist}_G(u, v)$ with $u, v \in A$ (i.e. the average over ordered pairs of distinct vertices of A; it is taken to be 0 when A has fewer than two vertices), and $\operatorname{ecc}_{\operatorname{avg}}(M)$ is the average eccentricity of the vertices in M. The conjecture is false, the authors present a counterexample: "The path on 5 vertices is a counterexample, path = 5, distavg(A) = 4 and the average of eccentricity of maximum degree vertices is 8/3." Independent Lean formalization by Wentao Li, with AI assistance. An earlier Lean proof is recorded in the Epoch review archived in [tadamcz/fc-review-results](https://github.com/tadamcz/fc-review-results/blob/971290dd3e0e4dce063f98feaa8ecb9e51acd5ec/data/782312817c/files/WrittenOnTheWallIII/GraphConjecture32.json).

```
theorem conjecture32 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected),
let A : Finset α := Finset.univ.filter (fun v => G.degree v = G.minDegree)
let M : Finset α := Finset.univ.filter (fun v => G.degree v = G.maxDegree)
let distavgA : ℝ := (∑ p ∈ A.offDiag, (G.dist p.1 p.2 : ℝ)) / (A.offDiag.card : ℝ)
let eccavg (S : Finset α) : ℝ := (∑ v ∈ S, (G.eccent v).toNat) / (S.card : ℝ)
distavgA + (1 / 2 : ℝ) * eccavg M ≤ (path G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture32.conjecture32

Written on the Wall II - Conjecture 322

MODULE CONTEXT: # Written on the Wall II - Conjecture 322 *Reference:* [E. DeLaViña, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.uhd.edu/faculty/delavinae/research/wowII/) The source applies the local-independence hypothesis to the complement graph. Its conclusion also follows from the characterization of minimal total dominating sets of complete multipartite graphs in [M. Subramanian and A. Selvakumar, *Total Domination and Minimal Total Domination Polynomial of H-Join Graphs*](https://doi.org/10.2298/FIL2501267S).

V2-000903 | Written on the Wall II - Conjecture 322 | conjecture322

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 322](http://cms.uhd.edu/faculty/delavinae/research/wowII/open.html) Let `G` be a simple connected graph on `n ≥ 5` vertices. If the maximum over all vertices `v` of `l(v)` in the complement graph `Gᶜ` - the independence number of the neighborhood `N(v)` of `v` - is at most 1, then `G` is well totally dominated. Here `l(v) = α(Gᶜ[N(v)])` is the independence number of the subgraph induced by the open neighborhood of `v` in `Gᶜ`. This proof was provided by Samuel Schlesinger.

```
theorem conjecture322 (G : SimpleGraph α) [DecidableRel G.Adj] (_hG : G.Connected)
(hn : 5 ≤ Fintype.card α)
(h : ∀ v : α, indepNeighborsCard Gᶜ v ≤ 1) :
IsWellTotallyDominated G
```

FC-WrittenOnTheWallIII.GraphConjecture322.conjecture322

Written on the Wall II - Conjecture 327

MODULE CONTEXT: # Written on the Wall II - Conjecture 327 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000904 | Written on the Wall II - Conjecture 327 | conjecture327

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 327](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) Let `G` be a simple connected graph. If `3 · γ(G) = γ_i(G)`, then `G` is well totally dominated, where `γ(G)` is the domination number of `G` and `γ_i(G)` is the independent domination number of `G`. **Proof Sketch:** The conjecture states that if $3\gamma(G) = i(G)$ for a connected graph G, then G is well totally dominated. However, this conjecture is **FALSE**. **Counterexample:** Consider a graph G with 12 vertices: $u, v, a_0, a_1, a_2, a_3, a_4, b_0, b_1, b_2, b_3, b_4$. The edges are: - (u, v) - (u, a_i) for all $i \in \{0, 1, 2, 3, 4\}$ - (v, b_i) for all $i \in \{0, 1, 2, 3, 4\}$ - $(a_0, b_3), (a_1, b_3), (a_2, b_0), (a_3, b_0), (a_4, b_3), (a_4, b_4)$ Properties of G: 1. **Connected**: Yes, path exists between any two vertices through u and v. 2. **Domination Number $\gamma(G)$**: The set $\{u, v\}$ dominates all vertices. Since there is no universal vertex, $\gamma(G) = 2$. 3. **Independent Domination Number $i(G)$**: The minimum independent dominating set has size 6 (e.g., $\{u, b_0, b_1, b_2, b_3, b_4\}$). Thus $i(G) = 6$. 4. **Condition**: $3 \gamma(G) = 3 \times 2 = 6 = i(G)$. The condition holds. 5. **Well Totally Dominated**: A graph is well totally dominated if all minimal total dominating sets have the same size. - $\{u, v\}$ is a minimal total dominating set of size 2. - $\{v, b_0, b_3\}$ is a minimal total dominating set of size 3. Since $2 \neq 3$, G is NOT well totally dominated. The counterexample has been found by Moritz Firsching and Goran Žužić using an experimental pipeline.

```
theorem conjecture327 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj] (_hG :
G.Connected)
(_h : 3 * G.dominationNumber = G.indepDominationNumber),
IsWellTotallyDominated G
```

FC-WrittenOnTheWallIII.GraphConjecture327.conjecture327

Written on the Wall II - Conjecture 33

MODULE CONTEXT: # Written on the Wall II - Conjecture 33 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000905 | Written on the Wall II - Conjecture 33 | conjecture33

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 33](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\operatorname{path}(G) \ge \lceil 2 \operatorname{dist}_{\operatorname{avg}}(M, V) \rceil$, where $\operatorname{path}(G)$ is the number of vertices of a largest induced path of G, M is the set of maximum-degree vertices, and $\operatorname{dist}_{\operatorname{avg}}(M, V)$ is the average of all nonzero distances $\operatorname{dist}_G(m, v)$ with $m \in M$ and $v \in V$. The conjecture is false: the source records an October 2005 counterexample with $\operatorname{path}(G) = 7$ and $\operatorname{dist}_{\operatorname{avg}}(M, V) = 3.56$.

```
theorem conjecture33 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected),
let M : Set α := {v | G.degree v = G.maxDegree}
let distAvg : ℝ :=
open scoped Classical in
let pairs := (M.toFinset ×ˢ Finset.univ).filter (fun p => G.dist p.1 p.2 ≠ 0)
(∑ p ∈ pairs, (G.dist p.1 p.2 : ℝ)) / pairs.card
Int.ceil (2 * distAvg) ≤ (path G : ℤ)
```

FC-WrittenOnTheWallII.GraphConjecture33.conjecture33

Written on the Wall II - Conjecture 34

MODULE CONTEXT: # Written on the Wall II - Conjecture 34 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000906 | Written on the Wall II - Conjecture 34 | conjecture34

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 34](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $\operatorname{path}(G) \ge \lceil \operatorname{dist}_{\operatorname{avg}}(C, V) + \operatorname{dist}_{\operatorname{avg}}(M, V) \rceil$, where $\operatorname{path}(G)$ is the number of vertices of a largest induced path of G, C is the set of center vertices (those with minimum eccentricity), M is the set of maximum-degree vertices, and $\operatorname{dist}_{\operatorname{avg}}(S, V)$ is the average of all nonzero distances $\operatorname{dist}_G(s, v)$ with $s \in S$ and $v \in V$. The conjecture is false. The counterexample below was documented by [Salvatore Mercuri](https://github.com/google-deepmind/formal-conjectures/pull/5733). Let G be the tree on 39 vertices formed by a vertex with three pendant leaves, joined by a path of five edges to the root of a perfect binary tree of depth four. Then $\operatorname{path}(G) = 11$, while C and M are singletons whose distance sums are 154 and 266, so the bound is $\lceil (154 + 266) / 38 \rceil = 12$. Lean formalization by Wentao Li, with AI assistance.

```
theorem conjecture34 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected),
let C : Set α := center G
let M : Set α := {v | G.degree v = G.maxDegree}
let distAvg (S : Set α) : ℝ :=
open scoped Classical in
let pairs := (S.toFinset ×ˢ Finset.univ).filter (fun p => G.dist p.1 p.2 ≠ 0)
(∑ p ∈ pairs, (G.dist p.1 p.2 : ℝ)) / pairs.card
Int.ceil (distAvg C + distAvg M) ≤ (path G : ℤ)
```

FC-WrittenOnTheWallII.GraphConjecture34.conjecture34

Written on the Wall II - Conjecture 36

MODULE CONTEXT: # Written on the Wall II - Conjecture 36 The conjecture is false. The counterexample below has a subdivided triangle on vertices `0`, `1`, `9`; vertices `7`, `8` adjacent to all three triangle vertices; and leaves `3`, `2` attached to `7`, `8`, respectively. It has radius three, one diametrical pair, and largest induced-path order five. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000907 | Written on the Wall II - Conjecture 36 | conjecture36

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 36](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj36): For every finite simple connected graph G, $\operatorname{path}(G) \ge 2 \cdot \operatorname{rad}(G) / \operatorname{dp}(G)$, where $\operatorname{path}(G)$ is the number of vertices in a largest induced path, $\operatorname{rad}(G)$ is the radius of G, and $\operatorname{dp}(G)$ is the number of *diametrical pairs* of G - that is, the number of pairs of vertices at distance $\operatorname{diam}(G)$. Disproved by Waller in Oct 2003 (counterexample: path number 5, radius 3, dp 1).

```
theorem conjecture36 :
¬ ∀ {α : Type} [Fintype α] [DecidableEq α] [Nontrivial α],
∀ (G : SimpleGraph α) [DecidableRel G.Adj] (_ : G.Connected) (_ : 0 < dp G),
(2 * G.radius.toNat : ℝ) / (dp G : ℝ) ≤ (path G : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture36.conjecture36

Written on the Wall II - Conjecture 4

MODULE CONTEXT: # Written on the Wall II - Conjecture 4 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000908 | Written on the Wall II - Conjecture 4 | conjecture4

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 4](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) If `G` is a connected graph then the maximum number of leaves over all spanning trees satisfies `Ls(G) ≥ NG(G) - 1` where `NG(G)` is the minimal neighbourhood size of a non-edge of `G`.

```
theorem conjecture4 (G : SimpleGraph α) [DecidableRel G.Adj] [Nontrivial α] (h_conn :
G.Connected) :
NG G - 1 ≤ Ls G
```

FC-WrittenOnTheWallII.GraphConjecture4.conjecture4

Written on the Wall II - Conjecture 40

MODULE CONTEXT: # Written on the Wall II - Conjecture 40 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000909 | Written on the Wall II - Conjecture 40 | conjecture40

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 40](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a nontrivial connected graph `G` the size `f(G)` of a largest induced forest satisfies `f(G) ≥ ceil((p(G) + b(G) + 1)/2)` where `p(G)` is the path cover number and `b(G)` is the largest induced bipartite subgraph size.

```
theorem conjecture40 (h_conn : G.Connected) (h_nontrivial : 1 < Fintype.card α) :
⌈(((pathCoverNumber G : ℝ) + b G + 1) / 2)⌉ ≤ G.largestInducedForestSize
```

FC-WrittenOnTheWallIII.GraphConjecture40.conjecture40

Written on the Wall II - Conjecture 430a

MODULE CONTEXT: # Written on the Wall II - Conjecture 430a *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) The conjecture is false for the nonuniform clique blow-up of `P₇` with blob orders `(1,4,12,19,12,4,1)`. Its independent domination number is three, its center-neighborhood independence number is two, and its Caro--Wei sum is less than two. The claimed inequality therefore becomes `3 ≤ 2`.

V2-000910 | Written on the Wall II - Conjecture 430a | conjecture430a

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII Conjecture 430a asked whether every connected graph `G` of order greater than three satisfies `i(G) ≤ α(G[N(C)]) + 2 floor(CW(G)-1)`. The answer is no, witnessed by a nonuniform `P₇` clique blow-up.

```
theorem conjecture430a :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] [Nonempty V]
(G : SimpleGraph V) [DecidableRel G.Adj],
G.Connected → 3 < Fintype.card V →
(G.indepDominationNumber : ℤ) ≤
((G.induce
(setNeighborhood G (centerFinset G) : Set V)).indepNum : ℤ) +
2 * ⌊caroWei G - 1⌋
```

FC-WrittenOnTheWallIII.GraphConjecture430a.conjecture430a

Written on the Wall II - Conjecture 438b

MODULE CONTEXT: # Written on the Wall II - Conjecture 438b *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) WOWII 438b asks whether `alpha₂(G) ≤ alpha(G) + alpha(G[V \ H₂]) + |E(G[H₂])|`, where `H₂` is the set of vertices of degree at most two. The conjecture is true. In fact, the same inequality holds with an arbitrary vertex subset in place of `H₂`.

V2-000911 | Written on the Wall II - Conjecture 438b | conjecture438b

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII 438b states that every connected graph of order greater than three satisfies `alpha₂(G) ≤ alpha(G) + alpha(G[V \ H₂]) + |E(G[H₂])|`.

```
theorem conjecture438b :
∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V)
[DecidableRel G.Adj], G.Connected → 3 < Fintype.card V →
alphaTwo G ≤ G.indepNum +
indepNumOn G (Finset.univ \ lowDegreeLayer G) +
(internalEdges G (lowDegreeLayer G)).card
```

FC-WrittenOnTheWallIII.GraphConjecture438b.conjecture438b

Written on the Wall II - Conjecture 5

MODULE CONTEXT: # Written on the Wall II - Conjecture 5 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000912 | Written on the Wall II - Conjecture 5 | conjecture5

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 5](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, `Ls(G)` is bounded below by the maximal size of a sphere of radius `radius(G)` around the centres of `G`.

```
theorem conjecture5 (G : SimpleGraph V) (h_conn : G.Connected) :
letI centers := { v : V | G.eccent v = G.radius }
letI r_nat := G.radius.toNat
letI sphere_verts (v : V) : Set V := { w | G.dist v w = r_nat }
letI sphere_size (v : V) : ℝ := ↑(Finset.univ.filter (fun w => w ∈ sphere_verts v)).card
letI max_sphere_size := sSup (sphere_size '' centers)
max_sphere_size ≤ Ls G
```

FC-WrittenOnTheWallIII.GraphConjecture5.conjecture5

Written on the Wall II - Conjecture 58

MODULE CONTEXT: # Written on the Wall II - Conjecture 58 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample The conjecture is false. A counterexample is given by taking $K_{3,3}$ (with bipartition $\{0,1,2\}$, $\{3,4,5\}$) and K_{73} (on vertices $\{6,\ldots,78\}$), then adding edges between vertex 0 and every vertex of K_{73}. This graph G has $n = 79$ vertices and satisfies: - $b(G) \ge 7$: the set $\{0,1,2,3,4,5,6\}$ induces a bipartite subgraph - $f(G) \le 6$: the largest induced forest has at most 6 vertices - $l_{\mathrm{avg}}(G) = 92/79$ - $\lceil b(G) / l_{\mathrm{avg}}(G) \rceil \ge 7 > 6 \ge f(G)$

V2-000913 | Written on the Wall II - Conjecture 58 | conjecture58

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 58](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a connected graph `G`, the size `f(G)` of a largest induced forest satisfies `f(G) ≥ ceil( b(G) / average l(v) )` where `b(G)` is the largest induced bipartite subgraph and `l(v)` is the independence number of `G.neighborSet v`. This conjecture is false. A counterexample is the graph described in the module docstring above: a $K_{3,3}$ joined to a K_{73} via vertex 0, giving $\lceil b/l_{\mathrm{avg}} \rceil \ge 7 > 6 \ge f(G)$. The counterexample has been found by Moritz Firsching and Goran Žužić using an experimental pipeline.

```
theorem conjecture58 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_hG : G.Connected),
Nat.ceil (G.b / G.l_avg) ≤ G.largestInducedForestSize
```

FC-WrittenOnTheWallIII.GraphConjecture58.conjecture58

Written on the Wall II - Conjecture 59

MODULE CONTEXT: # Written on the Wall II - Conjecture 59 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000914 | Written on the Wall II - Conjecture 59 | conjecture59

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 59](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, the size $f(G)$ of a largest induced forest satisfies $f(G) \ge \lceil \sqrt{\mathrm{residue}(G) \cdot b(G)} \rceil$, where $\mathrm{residue}(G)$ is the Havel-Hakimi residue (the number of zeros remaining after applying the Havel-Hakimi algorithm to the degree sequence until termination) and $b(G)$ is the size of a largest induced bipartite subgraph. See: Favaron, Mahéo, Saclé (1991) for the residue; DeLaVina's Graffiti.pc for the conjecture. This conjecture is false. There is a connected counterexample on 123 vertices with `residue G = 101`, `b G = 122`, and `G.largestInducedForestSize = 111`. Indeed, `101 * 122 = 12322 = 111 ^ 2 + 1`, so the conjectured lower bound is 112. There is also an 18-vertex counterexample. Take a bipartite chain graph on vertices `0, ..., 9`, with parts `{0, ..., 4}` and `{5, ..., 9}`, a hub `10` adjacent to every core vertex, and leaves `11, ..., 17` attached at the hub. It has residue 10, `b = 17`, and largest induced forest 13. `⌈√170⌉ = 14`, so it falsifies the bound.

```
theorem conjecture59 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_ : G.Connected),
[Real.sqrt ((residue G : ℝ) * b G)] ≤ (G.largestInducedForestSize : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture59.conjecture59

Written on the Wall II - Conjecture 6

MODULE CONTEXT: # Written on the Wall II - Conjecture 6 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000915 | Written on the Wall II - Conjecture 6 | conjecture6

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 6](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a connected graph `G` we have `Ls(G) ≥ 1 + n(G) - m(G) - a(G)` where `a(G)` is defined via independent sets.

```
theorem conjecture6 (G : SimpleGraph α) [DecidableRel G.Adj] (h_conn : G.Connected) :
1 + (Fintype.card α : ℝ) - matchingNumber G - α(G) ≤ Ls G
```

FC-WrittenOnTheWallIII.GraphConjecture6.conjecture6

Written on the Wall II - Conjecture 61

MODULE CONTEXT: # Written on the Wall II - Conjecture 61 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000916 | Written on the Wall II - Conjecture 61 | conjecture61

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 61](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, the size $f(G)$ of a largest induced forest satisfies $f(G) \ge \mathrm{residue}(G) + \lceil \mathrm{diam}(G) / 3 \rceil$, where $\mathrm{residue}(G)$ is the Havel-Hakimi residue and $\mathrm{diam}(G)$ is the diameter of G. See: Favaron, Mahéo, Saclé (1991) for the residue; DeLaVina's Graffiti.pc for the conjecture.

```
theorem conjecture61 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
(residue G : ℝ) + ⌈(G.diam : ℝ) / 3⌉ ≤ (G.largestInducedForestSize : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture61.conjecture61

Written on the Wall II - Conjecture 63

MODULE CONTEXT: # Written on the Wall II - Conjecture 63 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample The conjecture is false for the lexicographic product $C_5[K_4]$. This graph has minimum even-distance count 9, while its largest induced forest and largest induced bipartite subgraph both have order 4. The conjectured lower bound is therefore $\lceil(9 + 4 + 1)/3\rceil = 5 > 4$.

V2-000917 | Written on the Wall II - Conjecture 63 | conjecture63

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 63](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) asked whether every simple connected graph G satisfies $f(G) \geq \lceil(\min_v \operatorname{distEven}(v) + b(G) + 1)/3\rceil$. The answer is no, as witnessed by $C_5[K_4]$.

```
theorem conjecture63 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) (_h : G.Connected),
let minDistEven := (Finset.univ.image (G.distEven ·)).min' (by simp)
⌈((minDistEven : ℝ) + G.b + 1) / 3⌉ ≤ (G.largestInducedForestSize : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture63.conjecture63

Written on the Wall II - Conjecture 64

MODULE CONTEXT: # Written on the Wall II - Conjecture 64 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [J. J. Gebendorfer, Clique Blow-ups of the 5-Cycle and Written on the Wall II Conjecture 64](https://doi.org/10.5281/zenodo.21595503) Gebendorfer recorded the disproof on 26 July 2026, crediting Kuberwastaken's earlier $C_5[K_4]$ certificate for the graph and its induced-forest computation. The linked Lean certificate formalizes this 20-vertex counterexample.

V2-000918 | Written on the Wall II - Conjecture 64 | conjecture64

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII Conjecture 64 proposed that every finite simple connected graph G satisfies $$ f(G) \geq \left\lceil\sqrt{\alpha(G)\bigl(1+(|V(G)|\bmod\Delta(G))\bigr)}\right\rceil, $$ where $f(G)$ is the maximum order of an induced forest. For $C_5[K_4]$, the forest number is 4, the independence number is 2, and the proposed lower bound is $\lceil\sqrt{2(1+9)}\rceil=5$. The remainder and product in the radicand are computed in $\mathbb{N}$ before coercion to $\mathbb{R}$. Nontriviality excludes the one-vertex graph, whose maximum degree is zero.

```
theorem conjecture64 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_h : G.Connected),
⌈Real.sqrt ((G.indepNum * (1 + Fintype.card α % G.maxDegree) : ℕ) : ℝ)⌉ ≤
(G.largestInducedForestSize : ℝ)
```

FC-WrittenOnTheWallII.GraphConjecture64.conjecture64

Written on the Wall II - Conjecture 65

MODULE CONTEXT: # Written on the Wall II - Conjecture 65 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample The conjecture is false. Start with the path $v_0-v_1-\cdots-v_{12}$ and attach one triangle at v_1 and another at v_{11}. The resulting graph has 17 vertices. Its Graph_6 string is `PhCGGC@?G?_@?@O?G?G?G?@C`. Its only minimum-degree vertices are v_0 and v_{12}, at distance 12, while its only maximum-degree vertices are v_1 and v_{11}, at distance 10. Thus the conjectured lower bound is $12 + \lceil 10/3 \rceil = 16$. Every induced forest must omit at least one vertex from each of the two vertex-disjoint triangles, so it has at most 15 vertices. Conversely, deleting one non-path vertex from each triangle leaves a tree on 15 vertices.

V2-000919 | Written on the Wall II - Conjecture 65 | conjecture65

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 65](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/): Does every simple connected graph G satisfy $f(G) \ge \operatorname{dist_min}(A) + \lceil \operatorname{dist_min}(M) / 3 \rceil$, where A is the set of minimum-degree vertices, M is the set of maximum-degree vertices, and $\operatorname{dist_min}(S)$ is the minimum distance between two distinct vertices of S (see `distMin`). The answer is no. `Counterexample.graph` has a conjectured lower bound of 16, but every induced forest in it has at most 15 vertices.

```
theorem conjecture65 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) [DecidableRel G.Adj] (_hG : G.Connected),
let A : Set α := {v | G.degree v = G.minDegree}
let M : Set α := {v | G.degree v = G.maxDegree}
(distMin G A : ℝ) + ⌈(distMin G M : ℝ) / 3⌉ ≤
(G.largestInducedForestSize : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture65.conjecture65

Written on the Wall II - Conjecture 7

MODULE CONTEXT: # Written on the Wall II - Conjecture 7 *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [DFW03] E. DeLaVina, S. Fajtlowicz and B. Waller, [On Some Conjectures of Griggs and Graffiti](https://www.uhd.edu/documents/academics/sciences/griggsngraffiti.pdf), 2002, revised 2003. - [Li26] Wentao Li, [Lean proof of WOWII 7](https://github.com/VictorLiwentao/lean-graph-proofs/blob/9ce66be23570e0a9c8a6ae3f19f52e642878a8a9/proofs/WOWII7/PROOF.md), 2026.

V2-000920 | Written on the Wall II - Conjecture 7 | conjecture7

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 7](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $L_s(G) \ge \max_v \lambda(v) - 1 + n - 2 \alpha(G)$, where $L_s(G)$ is the maximum number of leaves over all spanning trees of G, $n = |V(G)|$, $\alpha(G) = G.\mathrm{indepNum}$ is the independence number, and $\lambda(v) = \mathrm{indepNeighborsCard}\, G\, v$ is the independence number of the neighbourhood of v. Proved by DeLaVina, Fajtlowicz, Waller (2002); see [DFW03]. Lean formalization by Wentao Li, with AI assistance; see [Li26].

```
theorem conjecture7 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
((maxL : ℤ) - 1 + (Fintype.card α : ℤ) - 2 * (G.indepNum : ℤ) : ℝ) ≤ Ls G
```

FC-WrittenOnTheWallIII.GraphConjecture7.conjecture7

Written on the Wall II - Conjecture 72

MODULE CONTEXT: # Written on the Wall II - Conjecture 72 **Verbatim statement (WOWII #72, status O):** > If G is a simple connected graph, then tree(G) ≥ CEIL[average of ecc(v) + maximum of λ(v) /3] **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj72 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000921 | Written on the Wall II - Conjecture 72 | conjecture72

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 72](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G`, $\operatorname{tree}(G) \ge \lceil \operatorname{ecc}_{\mathrm{avg}}(G) + \max_v \lambda(v)/3 \rceil$ where `tree(G)` is the number of vertices in a largest induced tree, `ecc_avg(G)` is the average eccentricity of `G`, and `max_v l(v) = max_v indepNeighborsCard G v` is the maximum over all vertices of the independence number of the neighbourhood.

```
theorem conjecture72 (G : SimpleGraph α) (h : G.Connected) :
let maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
⌈averageEccentricity G + (maxL : ℝ) / 3⌉ ≤ (largestInducedTreeSize G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture72.conjecture72

Written on the Wall II - Conjecture 84

MODULE CONTEXT: # Written on the Wall II - Conjecture 84 **Verbatim statement (WOWII #84, status O):** > If G is a simple connected graph, then tree(G) ≥ 2 rad(G) / δ(G) **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj84 Here `δ(G)` is the **minimum degree** of `G` (lowercase delta in the WOWII HTML, rendered as `<font face="Symbol">d</font>`), not the average distance. *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000922 | Written on the Wall II - Conjecture 84 | conjecture84

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 84](http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj84) (status O): For a simple connected graph `G` with positive minimum degree, `tree(G) ≥ 2 · rad(G) / δ(G)`, where `tree(G)` is the number of vertices in a largest induced subtree, `rad(G)` is the radius, and `δ(G) = G.minDegree` is the minimum degree. The hypothesis `0 < G.minDegree` follows from connectedness on a nontrivial vertex set, but is stated explicitly so the theorem reads as a real-valued inequality without `x / 0` corner cases.

```
theorem conjecture84 (G : SimpleGraph α) [DecidableRel G.Adj]
(h : G.Connected) (hδ : 0 < G.minDegree) :
2 * (G.radius.toNat : ℝ) / (G.minDegree : ℝ) ≤ (largestInducedTreeSize G : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture84.conjecture84

Written on the Wall II - Conjecture 85

MODULE CONTEXT: # Written on the Wall II - Conjecture 85 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Counterexample The conjecture is false for the lexicographic product $C_5[K_4]$. This graph has minimum even-distance count 9 and largest induced-tree order 4. The conjectured lower bound is therefore $\lceil\sqrt{1 + 2 \cdot 9}\rceil = \lceil\sqrt{19}\rceil = 5 > 4$.

V2-000923 | Written on the Wall II - Conjecture 85 | conjecture85

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 85](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) asked whether every simple connected graph G satisfies $\operatorname{tree}(G) \geq \lceil\sqrt{1 + 2\min_v \operatorname{distEven}(v)}\rceil$. The answer is no, as witnessed by $C_5[K_4]$.

```
theorem conjecture85 :
¬ ∀ (α : Type) [Fintype α] [DecidableEq α] [Nontrivial α]
(G : SimpleGraph α) (_h : G.Connected),
let minDistEven := (Finset.univ.image (G.distEven ·)).min' (by simp)
⌈Real.sqrt (1 + 2 * (minDistEven : ℝ))⌉ ≤
(G.largestInducedTreeSize : ℝ)
```

FC-WrittenOnTheWallIII.GraphConjecture85.conjecture85

Written on the Wall II - Conjecture 91

MODULE CONTEXT: # Written on the Wall II - Conjecture 91 **Verbatim statement (WOWII #91, status O):** > If G is a simple connected graph, then b(G) ≤ 1 + f(G) * (CEIL[average of λ(v)])/2 **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj91 This conjecture is **false**: Max Talwar found an 11-vertex counterexample (graph6 `JBza_CB?wF_`): take `K₃,₃` minus an edge together with a `K₅`, joined by two edges into one `K₅`-vertex. There the average neighbourhood independence number is exactly `2`, the largest induced forest has `6` vertices, and the largest induced bipartite subgraph has `8`, so the conjectured bound reads `8 ≤ 1 + 6·⌈2⌉/2 = 7`. *References:* - [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) - [Counterexample certificate for WOWII 91 (Max Talwar)](https://github.com/maxtalwar/wowii91-counterexample/releases/tag/v1.0.0)

V2-000924 | Written on the Wall II - Conjecture 91 | conjecture91

research solved | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 91](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) (disproved, 2026): For a simple connected graph `G`, `b(G) ≤ 1 + f(G) · ⌈avg_v I(v)⌉ / 2` where `b(G)` is the largest induced bipartite subgraph size, `f(G) = largestInducedForestSize G` is the largest induced forest size, and `avg_v I(v) = I G` is the average independence number of the neighbourhoods. Disproved by an 11-vertex counterexample (Max Talwar, 2026; graph6 `JBza_CB?wF_`) with `b(G) = 8`, `f(G) = 6` and `I G = 2`, violating `8 ≤ 1 + 6·⌈2⌉/2 = 7`.

```
theorem conjecture91 :
¬ ∀ {α : Type*} [Fintype α] [DecidableEq α] [Nontrivial α],
∀ (G : SimpleGraph α) (_ : G.Connected),
b G ≤ 1 + (G.largestInducedForestSize : ℝ) * ⌈I G⌉ / 2
```

FC-WrittenOnTheWallIII.GraphConjecture91.conjecture91

Written on the Wall II - Conjecture 96

MODULE CONTEXT: # Written on the Wall II - Conjecture 96 **Verbatim statement (WOWII #96, status O):** > If G is a simple connected graph, then α(G) ≤ 1 + minimum dist_even(v)*(maximum of λ(v) -1) **Source:** http://cms.uhd.edu/faculty/delavinae/research/wowII/all.html#conj96 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/)

V2-000925 | Written on the Wall II - Conjecture 96 | conjecture96

research open | proof annotation: False | 官方代码 | 原文线索

WOWII [Conjecture 96](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph `G` on at least two vertices, `α(G) ≤ 1 + min_v distEven(v) · (max_v I(v) - 1)` where `α(G) = G.indepNum` is the independence number, `distEven(v)` is the number of vertices at even distance from `v`, and `max_v I(v)` is the maximum independence number of a vertex neighbourhood. We record the nontrivial-graph version: the literal source inequality fails on K_1, where the independence number and even-distance count are both one, but the maximum local independence number is zero, giving $1 ≤ 0$.

```
theorem conjecture96 (G : SimpleGraph α) (h : G.Connected) :
let minDistEven := (Finset.univ.image (distEven G)).min' (by simp)
let maxL := (Finset.univ.image (fun v => indepNeighborsCard G v)).max' (by simp)
(G.indepNum : ℝ) ≤ 1 + (minDistEven : ℝ) * ((maxL : ℝ) - 1)
```

FC-WrittenOnTheWallIII.GraphConjecture96.conjecture96

Written on the Wall II - Conjecture 160

MODULE CONTEXT: # Written on the Wall II - Conjecture 160 *Reference:* [E. DeLaVina, Written on the Wall II, Conjectures of Graffiti.pc](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) ## Definitions For a vertex v in G, **$T(v)$** is the number of triangles incident to v: $$T(v) = |\{\{u, w\} \subseteq N(v) \mid u \sim w\}|$$ i.e., the number of pairs of neighbors of v that are themselves adjacent. The invariant `maxTrianglesAtVertex G` is the maximum of $T(v)$ over all vertices. Conjecture 160 uses both $\max_v T(v)$ and $\chi_{C_4}(G)$, the C_4-free characteristic function: it is `1` when G contains no cycle of length four and `0` otherwise. The cycle need not be induced. These invariants lower-bound the WOWII invariant $L_s(G)$, the maximum number of leaves over all spanning trees of G (exposed as `SimpleGraph.Ls G : ℝ`). The earlier formalization used the number of induced four-cycles. The historical conjecture instead uses this binary C_4-free indicator. **Provenance.** Statement corrected by Dominic Dabish.

V2-000926 | Written on the Wall II - Conjecture 160 | conjecture160

research solved | proof annotation: True | 官方代码 | 原文线索

WOWII [Conjecture 160](http://cms.dt.uh.edu/faculty/delavinae/research/wowII/) For a simple connected graph G, $L_s(G) \ge \max_v l(v) + \max_v T(v) \cdot \chi_{C_4}(G)$ where: - $L_s(G) = \mathrm{SimpleGraph.Ls}\, G$ is the maximum number of leaves over all spanning trees of G, - $\max_v l(v)$ is the maximum local independence number over vertices, - $\max_v T(v)$ is the maximum number of triangles incident to any vertex, - $\chi_{C_4}(G)$ is `1` if G has no cycle of length four and `0` otherwise. A formal proof uses the C_4-free neighborhood identity $d(v) = \lambda(v) + T(v)$ with a star/geodesic case analysis, routed through the connected-seed spanning-tree bound developed for Conjecture 2. An independent second proof, via a closest maximizer pair joined by a shortest path, is linked from the pull request that recorded the solution.

```
theorem conjecture160 (G : SimpleGraph α) [DecidableRel G.Adj] (h : G.Connected) :
let maxL := (Finset.univ.image (indepNeighborsCard G)).max' (by simp)
let maxT := maxTrianglesAtVertex G
let cC4 : ℕ := if ∃ v : α, ∃ c : G.Walk v v, c.IsCycle ∧ c.length = 4 then 0 else 1
(maxL : ℝ) + (maxT : ℝ) * (cC4 : ℝ) ≤ Ls G
```

FC-WrittenOnTheWallII.GraphConjecture160.conjecture160

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Generic and maximal rank of 3-tensors

MODULE CONTEXT: # Generic and maximal rank of 3-tensors The rank of a tensor $T \in \mathbb{C}^{m_1} \otimes \mathbb{C}^{m_2} \otimes \mathbb{C}^{m_3}$ is the least r such that T is a sum of r decomposable tensors $a \otimes b \otimes c$. Mathlib records such a tensor as a `Holor ℂ [m₁, m₂, m₃]` and its rank as `Holor.cprank`. Two basic invariants of the format (m_1, m_2, m_3) are the *generic rank* $\operatorname{grank}(m_1, m_2, m_3)$, the rank of a generic tensor, and the *maximal rank* $\operatorname{mrank}(m_1, m_2, m_3)$, the largest rank attained by a tensor of that format. Counting parameters gives the lower bound $\operatorname{grank}(m_1, m_2, m_3) \geq r_0(m_1, m_2, m_3) := \lceil m_1m_2m_3 / (m_1 + m_2 + m_3 - 2) \rceil$. Once m_3 exceeds $(m_1-1)(m_2-1)$ the generic rank is known to be $\min(m_3, m_1m_2)$, so the interesting case is the *critical range* $3 \leq m_1 \leq m_2 \leq m_3 \leq (m_1-1)(m_2-1)$ [Fri12, Conjecture 5.1; BFZ24, (4.14)]. There the bound r_0 is attained in every case that has been settled, except for the formats $(3, 2p+1, 2p+1)$, where the generic rank is $r_0 + 1$. Friedland's conjecture, stated here as `isGenericRank_of_le_critical`, is that these are the only exceptions. The maximal rank is much less well understood. It is known for the formats $2 \times m \times n$ and for a range of small formats, but no formula is known even for (n, n, n), and $\operatorname{mrank}(3, 3, 5)$ is undetermined. Tensor rank is invariant under permuting the three factors, but `Holor ℂ [m₁, m₂, m₃]` and its permutations are different types and Mathlib has no transport lemma between them. Each statement below therefore writes the format in the order its own source uses. *References:* * [Fri12] S. Friedland, *On the generic and typical ranks of 3-tensors*, Linear Algebra Appl. 436 (2012), 478-497, https://doi.org/10.1016/j.laa.2011.05.008 ([arxiv/0805.3777](https://arxiv.org/abs/0805.3777)). Conjecture 5.1, and the survey of known values in Section 5. * [BFZ24] W. Bruzda, S. Friedland, K. Życzkowski, *Rank of a tensor and quantum entanglement*, Linear Multilinear Algebra 72 (2024), 1796-1859, https://doi.org/10.1080/03081087.2023.2211717 ([arxiv/1912.06854v4](https://arxiv.org/pdf/1912.06854v4)). A survey; the numbering used below is that of the arXiv v4 PDF. Conjecture 4.12 restates [Fri12, Conjecture 5.1], and Sections 4.5 and 4.6 collect the current status. * [CGG02] M. V. Catalisano, A. V. Geramita, A. Gimigliano, *Ranks of tensors, secant varieties of Segre varieties and fat points*, Linear Algebra Appl. 355 (2002), 263-285, https://doi.org/10.1016/S0024-3795(02)00352-X. Used for `isGenericRank_of_gt_critical`. * [Str83] V. Strassen, *Rank and optimal computation of generic tensors*, Linear Algebra Appl. 52/53 (1983), 645-685, https://doi.org/10.1016/0024-3795(83)80041-X. Used for `isGenericRank_three_even` and `isGenericRank_three_odd`. * [AOP09] H. Abo, G. Ottaviani, C. Peterson, *Induction for secant varieties of Segre varieties*, Trans. Amer. Math. Soc. 361 (2009), 767-792, https://doi.org/10.1090/S0002-9947-08-04725-9. Used for `isGenericRank_four`. * [Lic85] T. Lickteig, *Typical tensorial rank*, Linear Algebra Appl. 69 (1985), 95-120, https://doi.org/10.1016/0024-3795(85)90070-9. Used for `isGenericRank_cube`. * [AS79] M. D. Atkinson, N. M. Stephens, *On the maximal multiplicative complexity of a family of bilinear forms*, Linear Algebra Appl. 27 (1979), 1-8, https://doi.org/10.1016/0024-3795(79)90026-0. Used for `isMaxRank_two`, `cprank_le_of_three_slices`, `cprank_le_of_square_slices`, `isMaxRank_of_le_min` and `isMaxRank_three_three_five_bounds`. * [JaJa79] J. JáJá, *Optimal evaluation of pairs of bilinear forms*, SIAM J. Comput. 8 (1979), 443-462, https://doi.org/10.1137/0208037, and J. B. Kruskal, *Rank, decomposition, and uniqueness for 3-way and N-way arrays*, in Multiway Data Analysis, North-Holland (1989), 7-18. Cited by [Fri12] for `isMaxRank_two`. * [SMS10] T. Sumi, M. Miyazaki, T. Sakata, *About the maximal rank of 3-tensors over the real and the complex number field*, Ann. Inst. Statist. Math. 62 (2010), 807-822, https://doi.org/10.1007/s10463-010-0294-5 ([arxiv/0806.4048](https://arxiv.org/abs/0806.4048)). Theorem 4.5 proves the first of the two bounds that [AS79] states without proof; Proposition 4.9(1) is the special case used for `isMaxRank_three`. * [Ki26] K. Kitamura, *A Lean proof that the maximal rank of complex $3 \times 3 \times 5$ tensors is 6*, GitHub repository (2026), https://github.com/KitaKen1/mrank-3-3-5-lean. Used for `isMaxRank_three_three_five`. * [BT15] G. Blekherman, Z. Teitler, *On maximum, typical and generic ranks*, Math. Ann. 362 (2015), 1021-1031, https://doi.org/10.1007/s00208-014-1150-3 ([arxiv/1402.2371](https://arxiv.org/abs/1402.2371)), Theorem 1. Used for `isMaxRank_le_two_mul_isGenericRank`.

V2-000927 | Generic and maximal rank of 3-tensors | cprank_le_of_square_slices

research solved | proof annotation: False | 官方代码 | 原文线索

Every complex $m \times n \times n$ tensor has rank at most $\lfloor (m+1)n/2 \rfloor$ for $m,n \geq 3$ [BFZ24, (4.22)].

```
theorem cprank_le_of_square_slices {m n : ℕ} (hm : 3 ≤ m) (hn : 3 ≤ n)
(T : Holor ℂ [m, n, n]) : T.cprank ≤ (m + 1) * n / 2
```

FC-Arxiv.«0805.3777».cprank_le_of_square_slices

V2-000928 | Generic and maximal rank of 3-tensors | cprank_le_of_three_slices

research solved | proof annotation: False | 官方代码 | 原文线索

Every $n \times n \times 3$ tensor has rank at most $2n - 1$ [SMS10, Theorem 4.5].

```
theorem cprank_le_of_three_slices {n : ℕ} (hn : 1 ≤ n) (T : Holor ℂ [n, n, 3]) :
T.cprank ≤ 2 * n - 1
```

FC-Arxiv.«0805.3777».cprank_le_of_three_slices

V2-000929 | Generic and maximal rank of 3-tensors | isGenericRank_cube

research solved | proof annotation: False | 官方代码 | 原文线索

For $n \neq 3$ the generic rank of an $n \times n \times n$ tensor is $\lceil n^3 / (3n - 2) \rceil$ [Fri12, (5.8)].

```
theorem isGenericRank_cube {n : ℕ} (hn : 1 ≤ n) (hn3 : n ≠ 3) :
IsGenericRank [n, n, n] (n ^ 3 [/] (3 * n - 2))
```

FC-Arxiv.«0805.3777».isGenericRank_cube

V2-000930 | Generic and maximal rank of 3-tensors | isGenericRank_four

research solved | proof annotation: False | 官方代码 | 原文线索

The generic rank of a $4 \times m \times m$ tensor is $\lceil 4m^2 / (2m + 2) \rceil$ [Fri12, (5.7)]. This is the case $(4, m, m)$ of `isGenericRank_of_le_critical`, settled in [AOP09]; see [BFZ24, Section 4.5].

```
theorem isGenericRank_four {m : ℕ} (hm : 3 ≤ m) :
IsGenericRank [4, m, m] (4 * m ^ 2 ⌈/⌉ (2 * m + 2))
```

FC-Arxiv.«0805.3777».isGenericRank_four

V2-000931 | Generic and maximal rank of 3-tensors | isGenericRank_of_gt_critical

research solved | proof annotation: False | 官方代码 | 原文线索

Above the critical range, a generic tensor of format (m_1, m_2, m_3) has rank $\min(m_3, m_1m_2)$ [Fri12, (5.1)].

```
theorem isGenericRank_of_gt_critical {m₁ m₂ m₃ : ℕ} (h₁ : 1 ≤ m₁) (h₁₂ : m₁ ≤ m₂) (h₂₃ : m₂ ≤ m₃)
(h : (m₁ - 1) * (m₂ - 1) < m₃) :
IsGenericRank [m₁, m₂, m₃] (min m₃ (m₁ * m₂))
```

FC-Arxiv.«0805.3777».isGenericRank_of_gt_critical

V2-000932 | Generic and maximal rank of 3-tensors | isGenericRank_of_le_critical

research open | proof annotation: False | 官方代码 | 原文线索

Friedland's conjecture. In the critical range $m_3 \leq (m_1 - 1)(m_2 - 1)$, and away from the formats $(3, 2p+1, 2p+1)$, the generic rank of a tensor of format (m_1, m_2, m_3) is the value $\lceil m_1m_2m_3 / (m_1 + m_2 + m_3 - 2) \rceil$ predicted by a dimension count [Fri12, Conjecture 5.1]. It is known for $m_1 \leq 4$ and for (n, n, n), and has been verified numerically throughout the critical range for $m_3 \leq 20$ [BFZ24, Section 4.5].

```
theorem isGenericRank_of_le_critical {m₁ m₂ m₃ : ℕ} (h₁ : 3 ≤ m₁) (h₁₂ : m₁ ≤ m₂) (h₂₃ : m₂ ≤ m₃)
(h : m₃ ≤ (m₁ - 1) * (m₂ - 1)) (hexc : ∀ p : ℕ, (m₁, m₂, m₃) ≠ (3, 2 * p + 1, 2 * p + 1)) :
IsGenericRank [m₁, m₂, m₃] (m₁ * m₂ * m₃ ⌈/⌉ (m₁ + m₂ + m₃ - 2))
```

FC-Arxiv.«0805.3777».isGenericRank_of_le_critical

V2-000933 | Generic and maximal rank of 3-tensors | isGenericRank_three_even

research solved | proof annotation: False | 官方代码 | 原文线索

The generic rank of a $3 \times 2p \times 2p$ tensor is $\lceil 12p^2 / (4p + 1) \rceil$, the value r_0 predicted by a dimension count [Fri12, (5.3)]. The hypothesis $2 \leq p$ is the ordering $3 \leq 2p$ of the format.

```
theorem isGenericRank_three_even {p : ℕ} (hp : 2 ≤ p) :
IsGenericRank [3, 2 * p, 2 * p] (12 * p ^ 2 ⌈/⌉ (4 * p + 1))
```

FC-Arxiv.«0805.3777».isGenericRank_three_even

V2-000934 | Generic and maximal rank of 3-tensors | isGenericRank_three_odd

research solved | proof annotation: False | 官方代码 | 原文线索

The generic rank of a $3 \times (2p+1) \times (2p+1)$ tensor is $\lceil 3(2p+1)^2 / (4p + 3) \rceil + 1$, one more than the value r_0 predicted by a dimension count [Fri12, (5.4)]. These are the only formats in the critical range known to exceed r_0. The case $p = 1$ says that the generic rank of a $3 \times 3 \times 3$ tensor is `5`, not `4`.

```
theorem isGenericRank_three_odd {p : ℕ} (hp : 1 ≤ p) :
IsGenericRank [3, 2 * p + 1, 2 * p + 1] (3 * (2 * p + 1) ^ 2 ⌈/⌉ (4 * p + 3) + 1)
```

FC-Arxiv.«0805.3777».isGenericRank_three_odd

V2-000935 | Generic and maximal rank of 3-tensors | isMaxRank_cube

research open | proof annotation: False | 官方代码 | 原文线索

Open problem. Determine the maximal rank of a complex $n \times n \times n$ tensor. No formula is known: for $n = 1, 2, 3$ the values are `1`, `3` and `5`, while for $4 \leq n \leq 7$ only the bounds in `isMaxRank_cube_bounds`, such as $7 \leq \operatorname{mrank}(4,4,4) \leq 10$, are available [BFZ24, (4.25)-(4.28)].

```
theorem isMaxRank_cube {n : ℕ} (hn : 1 ≤ n) :
IsMaxRank [n, n, n] (isMaxRank_cube.answer n)
```

FC-Arxiv.«0805.3777».isMaxRank_cube

V2-000936 | Generic and maximal rank of 3-tensors | isMaxRank_cube_bounds

research solved | proof annotation: False | 官方代码 | 原文线索

For $n \geq 4$, the maximal rank R of a complex cubic tensor satisfies $\lceil n^3/(3n-2) \rceil \leq R \leq n(n+1)/2$ [BFZ24, (4.22), (4.25)-(4.28)]. The lower bound is the generic rank; the upper bound is the square-slice estimate.

```
theorem isMaxRank_cube_bounds {n R : ℕ} (hn : 4 ≤ n) (hR : IsMaxRank [n, n, n] R) :
n ^ 3 ⌈/⌉ (3 * n - 2) ≤ R ∧ R ≤ n * (n + 1) / 2
```

FC-Arxiv.«0805.3777».isMaxRank_cube_bounds

V2-000937 | Generic and maximal rank of 3-tensors | isMaxRank_le_two_mul_isGenericRank

research solved | proof annotation: False | 官方代码 | 原文线索

The maximal rank is at most twice the generic rank [BT15, Theorem 1].

```
theorem isMaxRank_le_two_mul_isGenericRank {ds : List ℕ} {r R : ℕ} (hr : IsGenericRank ds r)
(hR : IsMaxRank ds R) : R ≤ 2 * r
```

FC-Arxiv.«0805.3777».isMaxRank_le_two_mul_isGenericRank

V2-000938 | Generic and maximal rank of 3-tensors | isMaxRank_of_le_min

research solved | proof annotation: False | 官方代码 | 原文线索

For $u \leq \min(4, m)$ the maximal rank of an $m \times n \times (mn - u)$ tensor is $mn - \lceil u/2 \rceil$ [BFZ24, (4.24)].

theorem isMaxRank_of_le_min {m n u : ℕ} (hm : 3 ≤ m) (hmn : m ≤ n) (hu : u ≤ min 4 m) : IsMaxRank [m, n, m * n - u] (m * n - u ⌈/⌉ 2)

FC-Arxiv.«0805.3777».isMaxRank_of_le_min

V2-000939 | Generic and maximal rank of 3-tensors | isMaxRank_three

research solved | proof annotation: False | 官方代码 | 原文线索

The maximal rank of a $3 \times 3 \times 3$ tensor is `5`. The upper bound is [SMS10, Proposition 4.9(1)], a special case of `cprank_le_of_three_slices`; the lower bound is the generic rank `isGenericRank_three_odd` at $p = 1$.

theorem isMaxRank_three : IsMaxRank [3, 3, 3] 5

FC-Arxiv.«0805.3777».isMaxRank_three

V2-000940 | Generic and maximal rank of 3-tensors | isMaxRank_three_three_five

research solved | proof annotation: True | 官方代码 | 原文线索

Determine the maximal rank of a $3 \times 3 \times 5$ tensor. It was the one undetermined entry of the table of $\operatorname{mrank}(3, 3, p)$ for $p \leq 9$ [BFZ24, (4.20)]. The answer is `6`. A Lean proof is given in [Ki26].

theorem isMaxRank_three_three_five : IsMaxRank [3, 3, 5] isMaxRank_three_three_five.answer

FC-Arxiv.«0805.3777».isMaxRank_three_three_five

V2-000941 | Generic and maximal rank of 3-tensors | isMaxRank_three_three_five_bounds

research solved | proof annotation: False | 官方代码 | 原文线索

The maximal rank of a $3 \times 3 \times 5$ tensor is `6` or `7` [BFZ24, (4.20)].

theorem isMaxRank_three_three_five_bounds {R : ℕ} (hR : IsMaxRank [3, 3, 5] R) : 6 ≤ R ∧ R ≤ 7

FC-Arxiv.«0805.3777».isMaxRank_three_three_five_bounds

V2-000942 | Generic and maximal rank of 3-tensors | isMaxRank_two

research solved | proof annotation: False | 官方代码 | 原文线索

The maximal rank of a $2 \times m \times n$ tensor is $m + \min(m, \lfloor n/2 \rfloor)$ [Fri12, (5.13)]; equivalently [BFZ24, (4.9)].

theorem isMaxRank_two {m n : ℕ} (hm : 2 ≤ m) (hmn : m ≤ n) : IsMaxRank [2, m, n] (m + min m (n / 2))

FC-Arxiv.«0805.3777».isMaxRank_two

The Curling Number Conjecture

MODULE CONTEXT: # The Curling Number Conjecture *Reference:* [arxiv/0912.2382](https://arxiv.org/abs/0912.2382) **The Curling Number Conjecture** by *Benjamin Chaffin and N. J. A. Sloane*

V2-000943 | The Curling Number Conjecture | curling_number_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Starting from any nonempty finite integer sequence, repeatedly appending its curling number eventually gives a sequence with curling number 1 [Chaffin and Sloane, Conjecture 1].

theorem curling_number_conjecture (S₀ : List ℤ) (h : S₀ ≠ []) : ∃ m, k (S S₀ m) = 1

FC-Arxiv.«0912.2382».curling_number_conjecture

The first Atiyah--Sutcliffe conjecture

MODULE CONTEXT: # The first Atiyah--Sutcliffe conjecture Atiyah and Sutcliffe associate a homogeneous binary polynomial to each point in a configuration of distinct points in Euclidean three-space. Their first conjecture says that these polynomials are always linearly independent. *References:* - M. F. Atiyah and P. M. Sutcliffe, [The Geometry of Point Particles](https://doi.org/10.1098/rspa.2001.0913) - Marcin Mazur and Bogdan V. Petrenko, [On the conjectures of Atiyah and Sutcliffe](https://arxiv.org/abs/1102.4662)

V2-000944 | The first Atiyah--Sutcliffe conjecture | conjecture_one

research open | proof annotation: False | 官方代码 | 原文线索

[Atiyah-Sutcliffe Conjecture 1](https://doi.org/10.1098/rspa.2001.0913), stated as Conjecture 1.1 in [Mazur-Petrenko](https://arxiv.org/abs/1102.4662): the configuration polynomials are linearly independent.

theorem conjecture_one {n : ℕ} (x : Fin n → Point) (hx : Function.Injective x) : LinearIndependent ℂ (pointPolynomial x)

FC-AtiyahSutcliffe.conjecture_one

Cunningham chains - Jones's conjecture

MODULE CONTEXT: # Cunningham chains - Jones's conjecture A Cunningham chain is a sequence of primes satisfying either $p_{i+1}=2p_i+1$ (first kind) or $p_{i+1}=2p_i-1$ (second kind). It is conjectured that there are infinitely many chains of every positive exact length, of both kinds. A chain has **exact length k** when its first k terms are prime and the $(k+1)$-th generated term is composite. Lenny Jones conjectures that for every positive integer k, infinitely many primes start a chain of exact length k, for each of the two kinds. *References:* - Lenny Jones, [Polynomial Cunningham Chains](https://arxiv.org/abs/1104.1579) - [OEIS A181697](https://oeis.org/A181697), first-kind chain lengths - [OEIS A181715](https://oeis.org/A181715), second-kind chain lengths

V2-000945 | Cunningham chains - Jones's conjecture | infinitely_many_firstKind_chains

research open | proof annotation: False | 官方代码 | 原文线索

Jones's conjecture (first kind): for every positive integer k, there are infinitely many primes p that start a first-kind Cunningham chain of exactly length k.

```
theorem infinitely_many_firstKind_chains (k : ℕ) (hk : 0 < k) :
Set.Infinite {p : ℕ | IsFirstKindChainOfLength p k}
```

FC-CunninghamChain.infinitely_many_firstKind_chains

V2-000946 | Cunningham chains - Jones's conjecture | infinitely_many_secondKind_chains

research open | proof annotation: False | 官方代码 | 原文线索

Jones's conjecture (second kind): for every positive integer k, there are infinitely many primes p that start a second-kind Cunningham chain of exactly length k.

```
theorem infinitely_many_secondKind_chains (k : ℕ) (hk : 0 < k) :
Set.Infinite {p : ℕ | IsSecondKindChainOfLength p k}
```

FC-CunninghamChain.infinitely_many_secondKind_chains

Conway's subprime Fibonacci sequences

MODULE CONTEXT: # Conway's subprime Fibonacci sequences A subprime Fibonacci sequence starts from two positive integers. Each new term is the sum of the previous two terms if that sum is prime; if the sum is composite, it is divided once by its least prime factor. This file formalizes the no-infinite-chains component of the conjecture: every such sequence eventually enters a cycle. It does not assert that the seven known loops are complete or unique. *References:* - R. K. Guy, T. Khovanova, and J. Salazar, ["Conway's subprime Fibonacci sequences"](https://arxiv.org/abs/1207.5099), *Mathematics Magazine* **87** (2014), no. 5, 323-337. - [A214897](https://oeis.org/A214897), Conway's subprime Fibonacci sequence: cycle lengths.

V2-000947 | Conway's subprime Fibonacci sequences | conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Every subprime Fibonacci sequence starting from two positive integers eventually enters a cycle. This is the "no infinite chains" conjecture recorded in [A214897](https://oeis.org/A214897), associated with [Guy-Khovanova-Salazar](https://arxiv.org/abs/1207.5099).

```
theorem conjecture (x y : ℕ) (hx : 0 < x) (hy : 0 < y) : EventuallyPeriodic (x, y)
```

FC-Arxiv.«1207.5099».conjecture

Boxdot Conjecture

MODULE CONTEXT: # Boxdot Conjecture The Boxdot Conjecture was originally formulated by French and Humberstone and has been studied in several works. In particular, see: *References:* - [arxiv/1308.0994](https://arxiv.org/abs/1308.0994) **Cluster Expansion and the Boxdot Conjecture** by *Emil Jeřábek* - [The Boxdot Conjecture and the Generalized McKinsey Axiom](https://ojs.victoria.ac.nz/ajl/article/view/4891) by *Christopher Steinsvold*, Australasian Journal of Logic

V2-000948 | Boxdot Conjecture | BoxdotConjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Boxdot Conjecture: every normal modal logic that faithfully interprets KT by the boxdot translation is included in KT.

```
theorem BoxdotConjecture (L : NormalModalLogic) (H : ∀ φ, L ⊢ ■ φ ↔ KT ⊢ φ) : L.thms ⊆ KT.thms
```

FC-Arxiv.«1308.0994».BoxdotConjecture

Unique Crystal Components

MODULE CONTEXT: # Unique Crystal Components *Reference:* [arxiv/1601.03081](https://arxiv.org/abs/1601.03081) **The Biharmonic mean** by *Marco Abrate, Stefano Barbero, Umberto Cerruti, Nadir Murru*

V2-000949 | Unique Crystal Components | crystals_components_unique

research open | proof annotation: False | 官方代码 | 原文线索

If $n = ab$ is a crystal, then there are no other pairs of positive integers $c, d > 1$, different from the couple a, b, such that $n = cd$ and $B(c, d) ∈ ℕ$, i.e., the components of the crystals are unique.

```
theorem crystals_components_unique (n a b c d : ℕ)
(hab : IsCrystalWithComponents n a b) (hcd : IsCrystalWithComponents n c d) :
({a, b} : Finset ℕ) = {c, d}
```

FC-Arxiv.«1601.03081».crystals_components_unique

The length of an s-increasing sequence of r-tuples

MODULE CONTEXT: # The length of an s-increasing sequence of r-tuples This file contains the formalisation of [GoLo21] up to and including Conjecture 1.8. *References:* - [arxiv/1609.08688](https://arxiv.org/abs/1609.08688) **The length of an s-increasing sequence of r-tuples** by *W. T. Gowers, J. Long* - [GoLo21](https://www.cambridge.org/core/journals/combinatorics-probability-and-computing/article/abs/length-of-an-sincreasing-sequence-of-rtuples/7301418D47DB1ECD6BE71C20E8A98D0A) **The length of an s-increasing sequence of r-tuples** by *W. T. Gowers, J. Long*, Combinatorics, Probability and Computing (2021), 686-721

V2-000950 | The length of an s-increasing sequence of r-tuples | maximalLength_ge_of_isSquare

research solved | proof annotation: False | 官方代码 | 原文线索

Moreover, whenever n is a perfect square we have $F(n) \geq n^{3/2}$.

theorem maximalLength_ge_of_isSquare {n : ℕ} (h : IsSquare n) :
n.sqrt ^ 3 ≤ F n

FC-Arxiv.«1609.08688».maximalLength_ge_of_isSquare

V2-000951 | The length of an s-increasing sequence of r-tuples | maximalLength_le

research solved | proof annotation: False | 官方代码 | 原文线索

For all n we have $F(n) \leq n^2$. This is the upper bound in [GoLo21, Proposition 1.4], proved by applying the pigeonhole principle to the first two coordinates.

theorem maximalLength_le (n : ℕ) : F n ≤ n ^ 2

FC-Arxiv.«1609.08688».maximalLength_le

V2-000952 | The length of an s-increasing sequence of r-tuples | maximalLength_le_isBigO

research solved | proof annotation: False | 官方代码 | 原文线索

$F(n) \leq n^2 / \exp(\Omega(\log^*(n)))$, i.e. there is a constant $c > 0$ such that $F(n) \leq n^2 / \exp(c \log^*(n))$ for all sufficiently large n.

theorem maximalLength_le_isBigO : ∃ c > (0 : ℝ), ∀ᶠ n : ℕ in atTop,
(F n : ℝ) ≤ (n : ℝ) ^ 2 / Real.exp (c * (Real.iteratedLog n : ℝ))

FC-Arxiv.«1609.08688».maximalLength_le_isBigO

V2-000953 | The length of an s-increasing sequence of r-tuples | maximalLength_le_strong

research open | proof annotation: False | 官方代码 | 原文线索

$F(n) \leq n^{3/2}$.

theorem maximalLength_le_strong (n : ℕ) : F n ≤ Real.sqrt n ^ 3

FC-Arxiv.«1609.08688».maximalLength_le_strong

V2-000954 | The length of an s-increasing sequence of r-tuples | maximalLength_pow

research solved | proof annotation: False | 官方代码 | 原文线索

Suppose that for some n we have $F(n) = n ^ {\alpha}$. Then there are arbitrarily large m such that $F(m) \geq m^{\alpha}$.

theorem maximalLength_pow {n : ℕ} {e : ℝ} (hn : 1 < n) (h : F n = (n : ℝ) ^ e) :
∃ᶠ m : ℕ in Filter.atTop, (m : ℝ) ^ e ≤ F m

FC-Arxiv.«1609.08688».maximalLength_pow

V2-000955 | The length of an s-increasing sequence of r-tuples | maximalLength_pow_eventually

research solved | proof annotation: False | 官方代码 | 原文线索

Moreover, for every $\beta < \alpha$ and all sufficiently large m we have $F(m) \geq m^{\beta}$.

theorem maximalLength_pow_eventually {n : ℕ} {e : ℝ} (hn : 1 < n) (h : F n = (n : ℝ) ^ e)
{b : ℝ} (hb : b < e) :
∀ᶠ m : ℕ in Filter.atTop, (m : ℝ) ^ b ≤ F m

FC-Arxiv.«1609.08688».maximalLength_pow_eventually

Ringel's Conjecture and Kotzig's Conjecture for large n

MODULE CONTEXT: # Ringel's Conjecture and Kotzig's Conjecture for large n *Reference:* [arxiv/2001.02665](https://arxiv.org/abs/2001.02665) **A proof of Ringel's Conjecture** by *Richard Montgomery, Alexey Pokrovskiy, Benny Sudakov* The original conjecture of Ringel (1963), which remains open, is stated in `Paper/RingelConjecture.lean`. The referenced paper proves it for all sufficiently large n. Montgomery-Pokrovskiy-Sudakov actually prove the strictly stronger statement conjectured by Kotzig: the decomposition can always be realized by cyclic shifts of a single copy of T. Kotzig's conjecture (all n) remains open; see `Paper/KotzigConjecture.lean`.

V2-000956 | Ringel's Conjecture and Kotzig's Conjecture for large n | kotzig_conjecture_large

research solved | proof annotation: False | 官方代码 | 原文线索

For all sufficiently large n, the complete graph K_{2n+1} decomposes into $2n+1$ edge-disjoint copies of any tree T with n edges via cyclic shifts of a single embedding. The $2n+1$ copies are $f_0, f_1, \dots, f_{2n}$ where $f_i(v) = f_0(v) + i$ for all vertices v - that is, each copy is obtained by adding $i \pmod{2n+1}$ to every vertex of the base copy. This is strictly stronger than `ringel_conjecture_large`. Kotzig conjectured this holds for all n; see `Paper/KotzigConjecture.lean`. Montgomery, Pokrovskiy, and Sudakov prove it for all sufficiently large n.

```
theorem kotzig_conjecture_large :
∀ᶠ (n : ℕ) in Filter.atTop, ∀ {V : Type} [Finite V] (T : SimpleGraph V),
T.IsTree → T.edgeSet.ncard = n →
∃ f : Fin (2 * n + 1) → (V ↪ Fin (2 * n + 1)),
(∀ i v, f i v = f 0 v + i) ∧
Pairwise (fun i j => Disjoint (T.map (f i)).edgeSet (T.map (f j)).edgeSet) ∧
⨆ i, T.map (f i) = (⊤ : SimpleGraph (Fin (2 * n + 1)))
```

FC-Arxiv.«2001.02665».kotzig_conjecture_large

V2-000957 | Ringel's Conjecture and Kotzig's Conjecture for large n | ringel_conjecture_large

research solved | proof annotation: False | 官方代码 | 原文线索

For all sufficiently large n, the complete graph K_{2n+1} decomposes into $2n+1$ edge-disjoint copies of any tree T with n edges. A "copy" of T is the image $T.\text{map}(f_i)$ of T under a vertex embedding $f_i : V \hookrightarrow \text{Fin}(2n+1)$; the copies are pairwise edge-disjoint and together cover every edge of K_{2n+1}. This follows from `kotzig_conjecture_large`; see `Paper/RingelConjecture.lean` for the open form.

```
theorem ringel_conjecture_large :
∀ᶠ (n : ℕ) in Filter.atTop, ∀ {V : Type} [Finite V] (T : SimpleGraph V),
T.IsTree → T.edgeSet.ncard = n →
∃ f : Fin (2 * n + 1) → (V ↪ Fin (2 * n + 1)),
Pairwise (fun i j => Disjoint (T.map (f i)).edgeSet (T.map (f j)).edgeSet) ∧
⨆ i, T.map (f i) = (⊤ : SimpleGraph (Fin (2 * n + 1)))
```

FC-Arxiv.«2001.02665».ringel_conjecture_large

Barker sequences

MODULE CONTEXT: # Barker sequences A Barker sequence is a finite sequence of ± 1 values whose nontrivial aperiodic autocorrelations all have magnitude at most one. The Barker conjecture says that no such sequence has length greater than 13. *References:* * J. Willms, [A note on Barker sequences of even length](https://arxiv.org/abs/2104.00502) * [Barker code](https://en.wikipedia.org/wiki/Barker_code) * [OEIS A091704](https://oeis.org/A091704)

V2-000958 | Barker sequences | barker_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Every Barker sequence has length at most 13.

```
theorem barker_conjecture (a : List ℤ) (ha : IsBarkerSequence a) : a.length ≤ 13
```

FC-Arxiv.«2104.00502».barker_conjecture

Independent Domination of Regular Graphs, Conjecture 1.6

MODULE CONTEXT: # Independent Domination of Regular Graphs, Conjecture 1.6 *Reference:* [arxiv/2107.00295](https://arxiv.org/abs/2107.00295) **On independent domination of regular graphs** by *Eun-Kyung Cho, Ilkyoo Choi, Boram Park* The conjecture was subsequently proved by Cho, Kim, Kim and Oum, as Corollary 1.3 of: *Reference:* [arxiv/2202.09594](https://arxiv.org/abs/2202.09594) **Independent domination of graphs with bounded maximum degree** by *Eun-Kyung Cho, Jinha Kim, Minki Kim, Sang-il Oum*, J. Combin. Theory Ser. B 158 (2023), 341-352. Corollary 1.3 states that every graph G with maximum degree at most Δ and no isolated vertices has an independent dominating set of size at most $\left(1 - \frac{\Delta}{\lfloor (\Delta+2)^2/4 \rfloor}\right)\lvert V(G)\rvert$. Expanding the floor for even and odd Δ gives exactly the two statements below.

V2-000959 | Independent Domination of Regular Graphs, Conjecture 1.6 | independentDominationEven

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.6 (Even case). For a nonempty isolate-free graph G on n vertices, if D is even, then $(D + 2)^2 \cdot i(G) \leq (D^2 + 4) \cdot n$. Proved by Cho, Kim, Kim and Oum ([arxiv/2202.09594](https://arxiv.org/abs/2202.09594), Corollary 1.3).

```
theorem independentDominationEven (hIso : 0 < G.minDegree) (hEven : Even G.maxDegree) :
let D := G.maxDegree
let i := G.indepDominationNumber
let n := Fintype.card V
(D + 2)^2 * i ≤ (D^2 + 4) * n
```

FC-Arxiv.«2107.00295».independentDominationEven

V2-000960 | Independent Domination of Regular Graphs, Conjecture 1.6 | independentDominationOdd

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.6 (Odd case). For a nonempty isolate-free graph G on n vertices, if D is odd, then $(D + 1)(D + 3) \cdot i(G) \leq (D^2 + 3) \cdot n$. Proved by Cho, Kim, Kim and Oum ([arxiv/2202.09594](https://arxiv.org/abs/2202.09594), Corollary 1.3).

```
theorem independentDominationOdd (hIso : 0 < G.minDegree) (hOdd : Odd G.maxDegree) :
let D := G.maxDegree
let i := G.indepDominationNumber
let n := Fintype.card V
(D + 1) * (D + 3) * i ≤ (D^2 + 3) * n
```

FC-Arxiv.«2107.00295».independentDominationOdd

Digit 2 in base 3 representation of 2^n

MODULE CONTEXT: # Digit 2 in base 3 representation of 2^n *References:* - [Some Unconventional Problems in Number Theory](https://doi.org/10.2307/2689842) by *Paul Erdös*, Mathematics Magazine 52, no. 2, p. 67, 1979 - [arxiv/2107.12475](https://arxiv.org/abs/2107.12475) **Hardness of busy beaver value BB(15)** by *Tristan Stérin, Damien Woods* - [Hardness of Busy Beaver Value BB(15)](https://doi.org/10.1007/978-3-031-72621-7_9) by *Tristan Stérin, Damien Woods*, Reachability Problems, Lecture Notes in Computer Science 15050, Springer, Cham (2024)

V2-000961 | Digit 2 in base 3 representation of 2^n | CollatzLike

research open | proof annotation: False | 官方代码 | 原文线索

For $n > 8$, 2^n is not the the sum of distinct powers of 3. Expressed here in terms of the base 3 digits of n. This conjecture is equivalent to the halting of a 15-state 2-symbol Turing Machine. TODO(lezeau): Formalize the Turing Machine version of this problem. Source: *Hardness of Busy Beaver Value BB(15)*: https://link.springer.com/chapter/10.1007/978-3-031-72621-7_9 This is also https://arxiv.org/abs/2107.12475.

```
theorem CollatzLike (n : ℕ) (hn : 8 < n) : 2 ∈ Nat.digits 3 (2^n)
```

FC-Arxiv.«2107.12475».CollatzLike

Zariski Cancellation

MODULE CONTEXT: # Zariski Cancellation *References:* - [arxiv/2208.14736](https://arxiv.org/abs/2208.14736) **The Zariski Cancellation Problem and related problems in Affine Algebraic Geometry** by *Neena Gupta* - [OAI26a] OpenAI, *An explicit failure of complex affine-space cancellation*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/An-explicit-failure-of-complex-affine-space-cancellation-September-23-2026/paper.pdf

V2-000962 | Zariski Cancellation | zariski_cancellation_problem

research solved | proof annotation: True | 官方代码 | 原文线索

The **Zariski Cancellation Problem**: is every polynomial ring over a field `k` of characteristic `0` cancellative? The answer is no: the polynomial ring in four variables over $\mathbb{C}$ is not cancellative. This was disproved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof and `zariski_cancellation_problem.variants.false_complex_four`.

```
theorem zariski_cancellation_problem :
¬ ∀ {k : Type} [Field k] [CharZero k] {ι : Type} [Fintype ι],
IsCancellative k (MvPolynomial ι k)
```

FC-Arxiv.«2208.14736».zariski_cancellation_problem

Spectral sets and weak tiling

MODULE CONTEXT: # Spectral sets and weak tiling This file formalizes Problems 7.1 and 7.2 from Kolountzakis, Lev, and Matolcsi. *References:* - [KLM2023] Mihail N. Kolountzakis, Nir Lev, and Máté Matolcsi, [Spectral sets and weak tiling](https://arxiv.org/abs/2209.04540). - [GL16] Rachel Greenfeld and Nir Lev, Spectrality and tiling by cylindric domains, *Journal of Functional Analysis* 271 (2016), 2808-2821. - [GL20] Rachel Greenfeld and Nir Lev, Spectrality of product domains and Fuglede's conjecture for convex polytopes, *Journal d'Analyse Mathématique* 140 (2020), 409-441. - [Ki26] Kenta Kitamura, A Lean proof of Problem 7.2 of Kolountzakis-Lev-Matolcsi in dimension three, [GitHub repository](https://github.com/KitaKen1/klm-7-2-spectral-product-3d) (2026).

V2-000963 | Spectral sets and weak tiling | exists_nowhereDense_spectralSet

research open | proof annotation: False | 官方代码 | 原文线索

[KLM2023, Problem 7.1] asks whether a bounded, measurable, nowhere dense subset $\Omega \subset \mathbb{R}^d$ of positive measure can be spectral. The answer is known to be negative for $d = 1$, so the dimension is restricted to $d \ge 2$, where the problem is open.

```
theorem exists_nowhereDense_spectralSet :
∃ d : ℕ, 2 ≤ d ∧
∃ Ω : Set (Fin d → ℝ), Bornology.IsBounded Ω ∧ MeasurableSet Ω ∧
IsNowhereDense Ω ∧ 0 < volume Ω ∧ isSpectral Ω
```

FC-NowhereDenseSpectralSet.exists_nowhereDense_spectralSet

V2-000964 | Spectral sets and weak tiling | isSpectral_right_of_product_of_convexBody

research open | proof annotation: False | 官方代码 | 原文线索

[KLM2023, Problem 7.2] For every positive dimension n, a convex body A and a bounded, measurable set B, must spectrality of $A \times B$ imply spectrality of B?

```
theorem isSpectral_right_of_product_of_convexBody :
∀ (n m : ℕ), 0 < n → 0 < m →
spectralProductImpliesRightSpectral n m
```

FC-SpectralSetProduct.isSpectral_right_of_product_of_convexBody

V2-000965 | Spectral sets and weak tiling | isSpectral_right_of_product_one_dimensional

research solved | proof annotation: False | 官方代码 | 原文线索

[KLM2023, Problem 7.2; GL16] For a one-dimensional convex body A and a bounded, measurable set B, if $A \times B$ is spectral, then B is spectral.

```
theorem isSpectral_right_of_product_one_dimensional :
∀ (m : ℕ), 0 < m → spectralProductImpliesRightSpectral 1 m
```

FC-SpectralSetProduct.isSpectral_right_of_product_one_dimensional

V2-000966 | Spectral sets and weak tiling | isSpectral_right_of_product_three_dimensional

research solved | proof annotation: True | 官方代码 | 原文线索

[KLM2023, Problem 7.2] For a three-dimensional convex body A and a bounded, measurable set B, must spectrality of $A \times B$ imply spectrality of B? The answer is yes. For the Lean proof, see [Ki26].

```
theorem isSpectral_right_of_product_three_dimensional :
∀ (m : ℕ), 0 < m → spectralProductImpliesRightSpectral 3 m
```

FC-SpectralSetProduct.isSpectral_right_of_product_three_dimensional

V2-000967 | Spectral sets and weak tiling | isSpectral_right_of_product_two_dimensional

research solved | proof annotation: False | 官方代码 | 原文线索

[KLM2023, Problem 7.2; GL20] For a two-dimensional convex body A and a bounded, measurable set B, if $A \times B$ is spectral, then B is spectral.

```
theorem isSpectral_right_of_product_two_dimensional :
∀ (m : ℕ), 0 < m → spectralProductImpliesRightSpectral 2 m
```

FC-SpectralSetProduct.isSpectral_right_of_product_two_dimensional

Furstenberg's `times p, times q` conjectures

MODULE CONTEXT: # Furstenberg's `times p, times q` conjectures *Reference:* [arxiv/2303.01089](https://arxiv.org/abs/2303.01089) **Around Furstenberg's times p, times q conjecture: times p-invariant measures with some large Fourier coefficients** by *Catalin Badea, Sophie Grivaux*

V2-000968 | Furstenberg's `times p, times q` conjectures | conjecture_1_3

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.3 (the $\times p, \times q$ conjecture): the only atomless Borel probability measure on $\mathbb{T}$ which is both T_p- and T_q-invariant is the Lebesgue measure.

```
theorem conjecture_1_3 {p q : ℕ} (hp : 2 <= p) (hq : 2 <= q) (hpq : MultiplicativelyIndependent
p q)
{μ : Measure 𝕋} [IsProbabilityMeasure μ] [MeasureTheory.IsAtomLess μ]
(hmup : MeasurePreserving (Tn p) μ μ)
(hmuq : MeasurePreserving (Tn q) μ μ) :
μ = volume
```

FC-Arxiv.id2303_01089.conjecture_1_3

V2-000969 | Furstenberg's `times p, times q` conjectures | conjecture_1_4

research solved | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.4: if μ is an atomless T_p-invariant Borel probability measure on $\mathbb{T}$, then $T_{q^n}\mu$ converges weak-star to Lebesgue measure. This paper disproves the conjecture.

```
theorem conjecture_1_4 :
¬ ∀ p q : ℕ, 2 <= p → 2 <= q → MultiplicativelyIndependent p q →
∀ μ : ProbabilityMeasure 𝕋,
MeasureTheory.IsAtomLess μ.1 → MeasurePreserving (Tn p) μ μ →
Tendsto (fun n : ℕ => μ.map (Tn_continuous (q ^ n)).aemeasurable) atTop
(𝓝 UnitAddCircle.ProbabilityMeasure)
```

FC-Arxiv.id2303_01089.conjecture_1_4

The circulant Hadamard conjecture

MODULE CONTEXT: # The circulant Hadamard conjecture A circulant matrix is generated by cyclically shifting one row. The circulant Hadamard conjecture, also attributed to Ryser, says that no circulant Hadamard matrix has order greater than 4. *References:* * S. Steinerberger, [A note on approximate Hadamard matrices](https://arxiv.org/abs/2402.13202) * [OAI26a] OpenAI, *The circulant Hadamard conjecture*. OpenAI Math Release preprint (2026). https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-circulant-Hadamard-conjecture-September-23-2026/paper.pdf

V2-000970 | The circulant Hadamard conjecture | circulant_hadamard_conjecture

research solved | proof annotation: True | 官方代码 | 原文线索

Every circulant Hadamard matrix has order at most four. This was proved by an internal OpenAI model in September 2026 [OAI26a]; see the linked Lean proof, which shows that a real circulant Hadamard matrix of positive order n exists exactly when $n = 1$ or $n = 4$.

```
theorem circulant_hadamard_conjecture {n : ℕ} (v : Fin n → ℝ)
(hv : Hadamard.IsHadamard' (Matrix.circulant v)) : n ≤ 4
```

FC-CirculantHadamard.circulant_hadamard_conjecture

An Arithmetic Sum Associated with the Classical Theta Function

MODULE CONTEXT: # An Arithmetic Sum Associated with the Classical Theta Function *Reference:* [arxiv/2501.03234](https://arxiv.org/abs/2501.03234) **An Arithmetic Sum Associated with the Classical Theta Function** by *Bruce C. Berndt, Raghavendra N. Bhat, Jeffrey L. Meyer, Likun Xie, Alexandru Zaharescu*

V2-000971 | An Arithmetic Sum Associated with the Classical Theta Function | conjecture_1_1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.1: For any odd prime k, the sum associated with the classical theta function $θ_3$, $S(k)$ is positive.

```
theorem conjecture_1_1 (k : ℕ) (hprim : k.Prime) (hodd : Odd k) : 0 < S k
```

FC-Arxiv.«2501.03234».conjecture_1_1

V2-000972 | An Arithmetic Sum Associated with the Classical Theta Function | conjecture_4_1

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4.1: For any prime k larger than 5, $S(k) > k$.

```
theorem conjecture_4_1 (k : ℕ) (hprim : k.Prime) (hodd : Odd k) (hgt : k > 5) : k < S k
```

FC-Arxiv.«2501.03234».conjecture_4_1

V2-000973 | An Arithmetic Sum Associated with the Classical Theta Function | conjecture_4_2

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4.2: For any prime k larger than 233, $S(k) > 2k$.

```
theorem conjecture_4_2 (k : ℕ) (hprim : k.Prime) (hodd : Odd k) (hgt : k > 233) : 2 * k < S k
```

FC-Arxiv.«2501.03234».conjecture_4_2

V2-000974 | An Arithmetic Sum Associated with the Classical Theta Function | conjecture_4_3

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4.3: For any prime k larger than 3119, $S(k) > 3k$.

```
theorem conjecture_4_3 (k : ℕ) (hprim : k.Prime) (hodd : Odd k) (hgt : k > 3119) : 3 * k < S k
```

FC-Arxiv.«2501.03234».conjecture_4_3

V2-000975 | An Arithmetic Sum Associated with the Classical Theta Function | conjecture_4_4

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4.4: Given a natural number $n ∈ ℕ$, for all large enough odd prime k (depending on n), $nk < S(k)$.

```
theorem conjecture_4_4 (n : ℕ) : ∀ᶠ (k : ℕ) in Filter.atTop, k.Prime → Odd k → n * k < S k
```

FC-Arxiv.«2501.03234».conjecture_4_4

A conjecture by Margulis on matrix groups

MODULE CONTEXT: # A conjecture by Margulis on matrix groups *Reference:* [arxiv/2504.17644v3](https://arxiv.org/abs/2504.17644v3) **Bounded diagonal orbits in homogeneous spaces over function fields** by *Qianlin Huang, Ronggang Shi*

V2-000976 | A conjecture by Margulis on matrix groups | conjecture_1_1

research open | proof annotation: False | 官方代码 | 原文线索

Let `D` be the diagonal group of `SL_n(ℝ)` where n ≥ 3. Then any relatively compact `D`-orbit in `SL_n(ℝ) / SL_n(ℤ)` is closed.

```
theorem conjecture_1_1 {n : ℕ} (hn : 3 ≤ n)
(g : SL(n, ℝ) [U+29F8] Subgroup.map (map (Int.castRingHom ℝ)) ⊤)
(hg : IsCompact <| closure (MulAction.orbit (diagonalSubgroup (Fin n) ℝ) g)) :
IsClosed <| MulAction.orbit (diagonalSubgroup (Fin n) ℝ) g
```

FC-Margulis.conjecture_1_1

V2-000977 | A conjecture by Margulis on matrix groups | huang_shi_theorem_1_2

research solved | proof annotation: False | 官方代码 | 原文线索

Huang-Shi, Theorem 1.2 Let `F` be a finite field of characteristic `p ∈ {3, 5, 7, 11}`, and set `K = F((t⁻¹))`, `A = F[t]`. Here `K` is modelled by the Laurent series field `F[U+2E28]X[U+2E29]` with `X = t⁻¹`, so that `A` is embedded via `t ↦ X⁻¹`. Let * `D` be the diagonal subgroup of `SL₄(K)`, * `Γ = SL₄(A)` the lattice subgroup embedded into `SL₄(K)` via the natural inclusion `A →+* K`. Then there exists `z : SL₄(K)/Γ` such that the `D`-orbit of `z` has compact closure but is not closed.

```
theorem huang_shi_theorem_1_2
(hchar : ringChar F ∈ ({3, 5, 7, 11} : Finset ℕ)) :
∃ z : SL(4, F[U+2E28]X[U+2E29]) [U+29F8] ( Matrix.SpecialLinearGroup.map
(polyToLaurent F)).range,
IsCompact (closure (MulAction.orbit (diagonalSubgroup (Fin 4) (F[U+2E28]X[U+2E29])) z)) ∧
¬ IsClosed (MulAction.orbit (diagonalSubgroup (Fin 4) (F[U+2E28]X[U+2E29])) z)
```

FC-Margulis.huang_shi_theorem_1_2

TxGraffiti Conjecture 1: independence, annihilation, and residue

MODULE CONTEXT: # TxGraffiti Conjecture 1: independence, annihilation, and residue *Reference:* [arxiv/2507.17780](https://arxiv.org/abs/2507.17780) **In Reverie Together: Ten Years of Mathematical Discovery with a Machine Collaborator** by *Randy Davila, Boris Brimkov, Ryan Pepper* Conjecture 1, the oldest of the four TxGraffiti conjectures in the cited collection (open since 2016), asserts that every nontrivial connected graph G with maximum degree $\Delta \ge 2$ satisfies $$\alpha(G) \ge \frac{a(G) + R(G)}{\Delta(G)},$$ where α is the independence number, a the annihilation number, R the Havel-Hakimi residue, and Δ the maximum degree. This conjecture is **true**, proved in [arXiv:2606.29553](https://arxiv.org/abs/2606.29553) (C. Gupta).

V2-000978 | TxGraffiti Conjecture 1: independence, annihilation, and residue | tx_graffiti_conjecture_1

research solved | proof annotation: False | 官方代码 | 原文线索

TxGraffiti [Conjecture 1](https://arxiv.org/abs/2507.17780): for every nontrivial connected graph G with $\Delta(G) \ge 2$, $$\alpha(G) \ge \frac{a(G) + R(G)}{\Delta(G)}.$$ This conjecture is **true**. **Proof sketch.** Substitute the Caro-Wei lower bound $\alpha \ge W$ for α and reduce to $a \le (\Delta - 1) \alpha$. The case $\Delta = 2$ is trivial ($a = \alpha$). For $\Delta \ge 4$ an AM-HM argument with $m \le (n - a)\Delta$ yields a quadratic whose discriminant is negative. For $\Delta = 3$ the annihilation number has a closed form in the degree counts (n_1, n_2, n_3) and $a \le 2W$ is verified in three regimes. See [arXiv:2606.29553](https://arxiv.org/abs/2606.29553).

```
theorem tx_graffiti_conjecture_1 (V : Type) [Fintype V] [DecidableEq V]
(G : SimpleGraph V) [DecidableRel G.Adj] (_hConn : G.Connected)
(_hDeg : 2 ≤ G.maxDegree) :
(G.annihilationNumber + residue G : ℝ) / G.maxDegree ≤ G.indepNum
```

FC-Arxiv.«2507.17780».tx_graffiti_conjecture_1

TxGraffiti Conjecture 2: zero forcing versus independence (cubic graphs)

MODULE CONTEXT: # TxGraffiti Conjecture 2: zero forcing versus independence (cubic graphs) *Reference:* [arxiv/2507.17780](https://arxiv.org/abs/2507.17780) **In Reverie Together: Ten Years of Mathematical Discovery with a Machine Collaborator** by *Randy Davila, Boris Brimkov, Ryan Pepper* Conjecture 2 (open since 2017) asserts that for every connected graph G with maximum degree at most 3 and $G \ne K_4$, $$Z(G) \le \alpha(G) + 1,$$ where $Z(G)$ is the zero forcing number and $\alpha(G)$ the independence number.

V2-000979 | TxGraffiti Conjecture 2: zero forcing versus independence (cubic graphs) | tx_graffiti_conjecture_2

research solved | proof annotation: False | 官方代码 | 原文线索

TxGraffiti [Conjecture 2](https://arxiv.org/abs/2507.17780): for every connected graph G with $\Delta(G) \le 3$ and $G \ne K_4$, $$Z(G) \le \alpha(G) + 1.$$ This conjecture is **false**. Fischer gives a connected 24-vertex counterexample with maximum degree 3; see [arXiv:2607.23664](https://arxiv.org/abs/2607.23664).

```
theorem tx_graffiti_conjecture_2 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V)
[DecidableRel G.Adj] (_hConn : G.Connected)
(_hDeg : G.maxDegree ≤ 3)
(_hNotK4 : G = ⊤ → Fintype.card V ≠ 4),
zeroForcingNumber G ≤ G.indepNum + 1
```

FC-Arxiv.«2507.17780».tx_graffiti_conjecture_2

TxGraffiti Conjecture 3: independent domination versus saturation (regular)

MODULE CONTEXT: # TxGraffiti Conjecture 3: independent domination versus saturation (regular) *Reference:* [arxiv/2507.17780](https://arxiv.org/abs/2507.17780) **In Reverie Together: Ten Years of Mathematical Discovery with a Machine Collaborator** by *Randy Davila, Boris Brimkov, Ryan Pepper* Conjecture 3 (open since 2020) asserts that for every r-regular graph G (every vertex has degree r), $$i(G) \le \mu^*(G),$$ where $i(G)$ is the independent domination number and $\mu^*(G)$ the saturation number (minimum size of a maximal matching).

V2-000980 | TxGraffiti Conjecture 3: independent domination versus saturation (regular) | tx_graffiti_conjecture_3

research open | proof annotation: False | 官方代码 | 原文线索

TxGraffiti [Conjecture 3](https://arxiv.org/abs/2507.17780): for every r-regular graph G ($r \ge 1$), $$i(G) \le \mu^*(G).$$ This conjecture is **open**.

```
theorem tx_graffiti_conjecture_3 (V : Type) [Fintype V] [DecidableEq V]
(G : SimpleGraph V) [DecidableRel G.Adj]
(r : ℕ) (_hReg : ∀ v, G.degree v = r) (_hr : 1 ≤ r) :
G.indepDominationNumber ≤ G.saturationNumber
```

FC-Arxiv.«2507.17780».tx_graffiti_conjecture_3

TxGraffiti Conjecture 4: the saturation number versus the harmonic index

MODULE CONTEXT: # TxGraffiti Conjecture 4: the saturation number versus the harmonic index *Reference:* [arxiv/2507.17780](https://arxiv.org/abs/2507.17780) **In Reverie Together: Ten Years of Mathematical Discovery with a Machine Collaborator** by *Randy Davila, Boris Brimkov, Ryan Pepper* TxGraffiti is an automated conjecturing program, a successor of Fajtlowicz's *Graffiti* and DeLaViña's *Graffiti.pc*. Conjecture 4 in the cited collection asserts a bound between two graph invariants: * the *saturation number* $\mu^*(G)$ (`SimpleGraph.saturationNumber`), the minimum number of edges in a maximal matching of G; * the *harmonic index* $H(G)$ (`SimpleGraph.harmonicIndex`), $\sum_{uv \in E(G)} 2 / (\deg u + \deg v)$. The conjecture is **false**. It was first refuted by Bıyıkoğlu, who exhibited a family of counterexamples - a disjoint union of edges joined to an independent set - with unbounded ratio μ^*/H. That the smallest counterexample has order 9, namely the friendship graph F_4 formalised below, and that the windmill family has an exact limiting ratio, are established in Gupta. *References:* - [A Note on the TxGraffiti Conjecture about Harmonic Index and Minimum Maximal Matching Number](https://doi.org/10.46793/match.96-3.28425), Türker Bıyıkoğlu, *MATCH Commun. Math. Comput. Chem.* **96** (2026), no. 3, 1097-1099 - the first refutation. - [The saturation number is not bounded by the harmonic index](https://arxiv.org/abs/2606.15761), C. Gupta - the order-9 minimality of F_4 (formalised below) and the exact windmill limit.

V2-000981 | TxGraffiti Conjecture 4: the saturation number versus the harmonic index | tx_graffiti_conjecture_4

research solved | proof annotation: False | 官方代码 | 原文线索

TxGraffiti [Conjecture 4](https://arxiv.org/abs/2507.17780): for every nontrivial connected graph G, the saturation number is at most the harmonic index, $\mu^*(G) \le H(G)$. This conjecture is **FALSE**. **Counterexample.** The friendship graph F_4 (`friendshipF4`) is connected and nontrivial, with $\mu^*(F_4) = 4$ and $H(F_4) = 18/5$. Indeed the four rim edges $\{1,2\}, \{3,4\}, \{5,6\}, \{7,8\}$ form a maximal matching of size 4, and no maximal matching is smaller, so $\mu^*(F_4) = 4$; the eight spokes contribute $2/(8+2) = 1/5$ each and the four rim edges contribute $2/(2+2) = 1/2$ each, giving $H(F_4) = 8 \cdot 1/5 + 4 \cdot 1/2 = 18/5$. Since $4 > 18/5$, the bound fails. More generally the friendship graphs F_k satisfy $\mu^*(F_k) = k$ and $H(F_k) = 2k/(k+1) + k/2$, so the conjecture fails for every $k \ge 4$. An exhaustive search confirms that F_4 is the smallest counterexample (order 9). The unbounded separation was first shown by Bıyıkoğlu, whose family of edges joined to an independent set makes μ^*/H arbitrarily large; the windmill generalisation and its exact limit appear in [arXiv:2606.15761](https://arxiv.org/abs/2606.15761).

```
theorem tx_graffiti_conjecture_4 :
¬ ∀ (V : Type) [Fintype V] [DecidableEq V] [Nontrivial V] (G : SimpleGraph V)
[DecidableRel G.Adj] (_hConn : G.Connected),
(G.saturationNumber : ℚ) ≤ G.harmonicIndex
```

FC-Arxiv.«2507.17780».tx_graffiti_conjecture_4

First Proof, Theorem 4

MODULE CONTEXT: # First Proof, Theorem 4 *Reference:* [arxiv/2602.05192v2](https://arxiv.org/abs/2602.05192v2) **First Proof** by *Mohammed Abouzaid, Andrew J. Blumberg, Martin Hairer, Joe Kileel, Tamara G. Kolda, Paul D. Nelson, Daniel Spielman, Nikhil Srivastava, Rachel Ward, Shmuel Weinberger, Lauren Williams*

V2-000982 | First Proof, Theorem 4 | four

research solved | proof annotation: True | 官方代码 | 原文线索

Is it true that if $p(x)$ and $q(x)$ are monic real-rooted polynomials of degree n, then $$\frac{1}{\Phi_n(p\boxplus_n q)} \ge \frac{1}{\Phi_n(p)}+\frac{1}{\Phi_n(q)}?$$ [arxiv/2602.05192v2](https://arxiv.org/abs/2602.05192v2) contains a proof.

```
theorem four : ∀ (p q : ℝ[X]) (n : ℕ), FourProp p q n
```

FC-Arxiv.«2602.05192».four

V2-000983 | First Proof, Theorem 4 | four_2

research solved | proof annotation: False | 官方代码 | 原文线索

Is it true that if $p(x)$ and $q(x)$ are monic real-rooted polynomials of degree 2, then $$\frac{1}{\Phi_2(p\boxplus_n q)} \ge \frac{1}{\Phi_2(p)}+\frac{1}{\Phi_2(q)}?$$

```
theorem four_2 : ∀ (p q : ℝ[X]), FourProp p q 2
```

FC-Arxiv.«2602.05192».four_2

V2-000984 | First Proof, Theorem 4 | four_3

research solved | proof annotation: False | 官方代码 | 原文线索

Is it true that if $p(x)$ and $q(x)$ are monic real-rooted polynomials of degree 3, then $$\frac{1}{\Phi_3(p\boxplus_n q)} \ge \frac{1}{\Phi_3(p)}+\frac{1}{\Phi_3(q)}?$$

```
theorem four_3 : ∀ (p q : ℝ[X]), FourProp p q 3
```

FC-Arxiv.«2602.05192».four_3

First Proof, Theorem 6

MODULE CONTEXT: # First Proof, Theorem 6 *Reference:* [arxiv/2602.05192v2](https://arxiv.org/abs/2602.05192v2) **First Proof** by *Mohammed Abouzaid, Andrew J. Blumberg, Martin Hairer, Joe Kileel, Tamara G. Kolda, Paul D. Nelson, Daniel Spielman, Nikhil Srivastava, Rachel Ward, Shmuel Weinberger, Lauren Williams*

V2-000985 | First Proof, Theorem 6 | epsilon_light_subset_exists

research solved | proof annotation: True | 官方代码 | 原文线索

Does there exist a constant $c > 0$ so that for every graph G and every ϵ between 0 and 1, V contains an ϵ-light subset S of size at least $c \epsilon |V|$?

```
theorem epsilon_light_subset_exists :
∃ (c : ℝ), c > 0 ∧ ∀ (n : ℕ) (G : SimpleGraph (Fin n)) (ε : ℝ),
0 < ε → ε < 1 →
∃ (S : Finset (Fin n)), IsEpsilonLight G ε S ∧ (S.card : ℝ) ≥ c * ε * n
```

FC-Arxiv.«2602.05192».epsilon_light_subset_exists

Group structure via subgroup counts

MODULE CONTEXT: # Group structure via subgroup counts *Reference:* [arXiv:2604.08040v1](https://arxiv.org/abs/2604.08040v1) **Group Structure via Subgroup Counts** by *Angsuman Das, Hiranya Kishore Dey, Khyati Sharma* (2026) For a finite group G, let $\mathrm{cyc}(G)$ denote the number of cyclic subgroups of G, and let $t = \pi(G)$ denote the number of distinct prime divisors of $|G|$. The paper establishes several structural results of the form "if $\mathrm{cyc}(G)$ or $\mathrm{sub}(G)$ is small relative to 2^t, then G has a strong structural property": * $\mathrm{cyc}(G) < 5 \cdot 2^{t-2} \implies G$ is nilpotent (Theorem 3.1) * $\mathrm{cyc}(G) < 2^{t+1} \implies G$ is supersolvable (Theorem 4.2) * $\mathrm{sub}(G) < 59 \cdot 2^{t-3} \implies G$ is solvable (Theorem 5.3) **Conjecture 5.5** proposes the analogue for solvability via cyclic subgroup count: if $\mathrm{cyc}(G) < 2^{t+2}$, then G is solvable. * **In-Paper Location:** Conjecture 5.5, Section 5 "Solvability of a group from $\mathrm{sub}(G)$" ([PDF page 15](https://arxiv.org/pdf/2604.08040v1#page=15)) * **OpenConjecture ID:** 1512

V2-000986 | Group structure via subgroup counts | nilpotent_of_cyc_lt

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 3.1. Below $5 \cdot 2^{t-2}$ cyclic subgroups, the group is nilpotent.

theorem nilpotent_of_cyc_lt (h : cyc G < 5 * 2 ^ (numPrimeFactors G - 2)) :
Group.IsNilpotent G

FC-Arxiv.«2604.08040».nilpotent_of_cyc_lt

V2-000987 | Group structure via subgroup counts | solvable_of_cyc_lt

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 5.5 (Das, Dey, Sharma 2026): If a finite group `G` satisfies $\mathrm{cyc}(G) < 2^{t+2}$, where $t = \pi(G)$ is the number of distinct prime divisors of $|G|$, then `G` is solvable. This is Theorem 7.1 of the second version of the paper, [arXiv:2604.08040v2](https://arxiv.org/abs/2604.08040v2) (Das, Dey, Galindo, Sharma), proved there using the classification of finite simple groups. The Lean proof here by Kenta Kitamura instead uses only the Feit-Thompson theorem and the Gorenstein-Walter classification of groups with dihedral Sylow 2-subgroups, together with counting arguments on Sylow 2-subgroups and families of cyclic subgroups.

theorem solvable_of_cyc_lt :
∀ (G : Type) [Group G] [Fintype G],
cyc G < 2 ^ (numPrimeFactors G + 2) → Group.IsSolvable G

FC-Arxiv.«2604.08040».solvable_of_cyc_lt

V2-000988 | Group structure via subgroup counts | solvable_of_cyc_lt_two_pow_succ

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 4.2. Below 2^{t+1} cyclic subgroups, the group is supersolvable. Stated here as solvability, which supersolvability implies, since Mathlib does not currently have a standalone `IsSupersolvable` class.

theorem solvable_of_cyc_lt_two_pow_succ (h : cyc G < 2 ^ (numPrimeFactors G + 1)) :
Group.IsSolvable G

FC-Arxiv.«2604.08040».solvable_of_cyc_lt_two_pow_succ

Dean's conjecture on cycles of length divisible by `k`

MODULE CONTEXT: # Dean's conjecture on cycles of length divisible by `k` *References:* - [arxiv/2605.02731](https://arxiv.org/abs/2605.02731) **Existence of cycles of length divisible by 3 or 4** by *Ilkyoo Choi, Hojin Chu, Ringi Kim, Boram Park*, where this is Conjecture 1.1. - [De88] Dean, N., Open problem, in Cycles and Rays. (1988). - [DeLeSa93] Dean, N. and Lesniak, L. and Saito, A., Cycles of length 0 modulo 4 in graphs. Discrete Math. (1993), 133--139. - [ChSa94] Chen, G. and Saito, A., Graphs with a cycle of length divisible by three. J. Combin. Theory Ser. B (1994), 277--292. - [LuMaZh26] Luo, Yufan and Ma, Jie and Zhao, Ziyuan, Dean's conjecture and cycles modulo k. [arXiv:2601.13552](https://arxiv.org/abs/2601.13552) (2026).

V2-000989 | Dean's conjecture on cycles of length divisible by `k` | dean_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1.1 (Dean, 1988). For every integer $k \geq 3$, every finite simple graph with minimum degree at least k contains a cycle whose length is divisible by k. A cycle has length at least `3`, so the divisor is never `0` and the statement is not satisfied for a trivial reason. `SimpleGraph.minDegree` is `0` on a graph with no vertices and on a graph with no edges, so the hypothesis excludes both.

theorem dean_conjecture :
∀ (k : ℕ), 3 ≤ k → ∀ (V : Type) [Fintype V] [DecidableEq V]
(G : SimpleGraph V) [DecidableRel G.Adj], k ≤ G.minDegree →
∃ m ∈ G.cycleLengths, k | m

FC-Arxiv.«2605.02731».dean_conjecture

Fernandes' conjecture on the 2-generation of even direct product permutation groups

MODULE CONTEXT: # Fernandes' conjecture on the 2-generation of even direct product permutation groups *Reference:* [arxiv/2605.12342](https://arxiv.org/abs/2605.12342) **Groups of permutations that are even on maximal proper subsets, and related monoids** *Reference:* - [arxiv/2605.12342](https://arxiv.org/abs/2605.12342) **Groups of permutations that are even on maximal proper subsets, and related monoids** by *Vítor H. Fernandes* - [K26](https://github.com/KitaKen1/fernandes-conjecture-1-lean) For positive integers $m, n \ge 2$, let $\mathrm{S}_m \times \mathrm{S}_n$ be the direct product of symmetric groups on $[m] = \{1, \dots, m\}$ and $[n'] = \{1', \dots, n'\}$. Define $$ \Gamma_{m \oplus n} = \{(\sigma_1, \sigma_2) \in \mathrm{S}_m \times \mathrm{S}_n : \mathrm{sgn}(\sigma_1) = \mathrm{sgn}(\sigma_2)\} $$ to be the index-2 subgroup of pairs of permutations with equal parity, i.e., those whose product action on $[m] \cup [n']$ is an even permutation. It is known that $\mathrm{rank}(\Gamma_{2 \oplus 2}) = 1$, $\mathrm{rank}(\Gamma_{3 \oplus 3}) = 3$, $\mathrm{rank}(\Gamma_{4 \oplus 3}) = 3$, and $\mathrm{rank}(\Gamma_{4 \oplus 4}) = 3$. **Conjecture 1 (Fernandes, 2026):** For all integers $m \ge n \ge 2$ such that $(m, n) \notin \{(2,2), (3,3), (4,3), (4,4)\}$, the group $\Gamma_{m \oplus n}$ has rank 2 (i.e., is 2-generated).

V2-000990 | Fernandes' conjecture on the 2-generation of even direct product permutation groups | conjecture_1

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 1 (Fernandes, 2026): Let $m \ge n \ge 2$ be integers with $(m, n) \notin \{(2,2), (3,3), (4,3), (4,4)\}$. Then the group $$ \Gamma_{m \oplus n} = \{(\sigma_1, \sigma_2) \in \mathrm{S}_m \times \mathrm{S}_n : \mathrm{sgn}(\sigma_1) = \mathrm{sgn}(\sigma_2)\} $$ has rank 2, i.e., minimal generating set of size 2. Note: Fernandes states the conjecture for groups of exact rank 2, which is why $(2,2)$ is in the exception list: $\Gamma_{2 \oplus 2} \cong C_2$ has rank 1. The formalised conclusion `∃ g₁ g₂, closure {g₁, g₂} = ⊤` encodes 2-generation (at most 2 generators), which $\Gamma_{2 \oplus 2}$ also satisfies. The other three exceptions $(3,3), (4,3), (4,4)$ have rank 3 and are genuinely not 2-generated. The proof was developed and formalized by Kenta Kitamura, see [K26] for more details.

```
theorem conjecture_1 {m n : ℕ} (hm2 : 2 ≤ m) (hn2 : 2 ≤ n) (hmn : n ≤ m)
(h_except : (m, n) ∉ ({(2, 2), (3, 3), (4, 3), (4, 4)} : Set (ℕ × ℕ))) :
∃ g₁ g₂ : gammaSubgroup m n, Subgroup.closure {g₁, g₂} = ⊤
```

FC-Arxiv.«2605.12342».conjecture_1

Convexity of Fisher information along the heat flow

MODULE CONTEXT: # Convexity of Fisher information along the heat flow Can the functional $\mathcal{D}[f]$ be negative for an admissible density f? Equivalently, is the Fisher information of the Gaussian heat flow always convex in time? This is Problem (iii) of [ZFGW26, Section 7.3]; it is known in dimensions $d \le 4$ and left open for $d \ge 5$ [ZFGW26, Section 7.2]. References: - [ZFGW26] J. Zou, L. Fan, J. Gao and J. Wang, *A Hexagonal Counterexample to Log-Convexity of Fisher Information Along the Heat Flow*, arXiv:2605.18081 (2026). https://arxiv.org/abs/2605.18081 - [Ki26] K. Kitamura, *Fisher information along the heat flow is not convex*, Lean 4 formalization (2026), [GitHub repository](https://github.com/KitaKen1/fisher-information-nonconvexity/tree/a6ff274a04e5b6d813b6f06dc62d3d68a21ae31a).

V2-000991 | Convexity of Fisher information along the heat flow | fisherInformationConvexity

research solved | proof annotation: True | 官方代码 | 原文线索

Is the Fisher information convex along the heat flow for every admissible density [ZFGW26, Section 7.3, Problem (iii)]? The answer is no [Ki26].

```
theorem fisherInformationConvexity :
¬ ∀ (d : ℕ) (f : EuclideanSpace ℝ (Fin d) → ℝ), IsAdmissible f →
ConvexOn ℝ (Set.Ioi 0) (fun t => fisherInformation (heatFlow f t))
```

FC-Arxiv.«2605.18081».fisherInformationConvexity

Bondy's conjecture on longest cycles in highly connected graphs

MODULE CONTEXT: # Bondy's conjecture on longest cycles in highly connected graphs *References:* - [arxiv/2606.03696](https://arxiv.org/abs/2606.03696) **Longest cycles and Dirac-type results in highly connected graphs** by *Jie Ma, Bo Ning, Ziyuan Zhao*, where this is Conjecture 1. - [Bo80] Bondy, J. A., Longest paths and cycles in graphs of high degree. (1980). Take a `k`-connected graph with a large minimum degree. Bondy says that the graph outside any longest cycle holds no long path. The case `k = 1` is Dirac's theorem and the case `k = 2` is the theorem of Nash-Williams. The case `k = 3` is proved. The cases `k ≥ 4` are open.

V2-000992 | Bondy's conjecture on longest cycles in highly connected graphs | bondy_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1 (Bondy, 1980). Let $k \geq 1$ and let G be a k-connected graph on n vertices. If $\delta(G) \geq \frac{n + k(k-1)}{k+1}$, then for every longest cycle C of G, every path in $G - V(C)$ has at most $k-1$ vertices. The bound on the number of vertices of a path is written as `+ 1 ≤ k` rather than `≤ k - 1`, because subtraction on `ℕ` is truncated. The two forms agree for `k ≥ 1`. A longest cycle is a cycle whose length is the circumference of `G`.

```
theorem bondy_conjecture :
∀ (k : ℕ), 1 ≤ k → ∀ (V : Type) [Fintype V] [DecidableEq V]
(G : SimpleGraph V) [DecidableRel G.Adj], IsKConnected G k →
((Fintype.card V : ℝ) + k * (k - 1)) / (k + 1) ≤ G.minDegree →
∀ (a : V) (C : G.Walk a a), C.IsCycle → C.length = G.circumference →
∀ (u v : offWalk C) (P : (G.induce (offWalk C)).Walk u v), P.IsPath →
P.support.length + 1 ≤ k
```

FC-Arxiv.«2606.03696».bondy_conjecture

Planar L_p-Rogers-Shephard, the equality case

MODULE CONTEXT: # Planar L_p-Rogers-Shephard, the equality case *Reference:* [arxiv/2607.03582](https://arxiv.org/abs/2607.03582) **L_p-Rogers-Shephard type inequalities for L_p-zonoids and symmetric bodies** by *Matthieu Fradelizi, Auttawich Manui, Mark Meyer, Cheikh Saliou Ndiaye* *Formal proof of Conjecture 5:* Kenta Kitamura, [Lean proofs for the symmetric L_p Rogers-Shephard conjectures](https://github.com/KitaKen1/lp-rogers-shephard-conjectures-lean/tree/c646c668c0568b8533bf5534abcf02aab4d2df72). Corollary 29 bounds $|K \oplus_p -K|$ against $|K|$ for planar convex bodies with a centre of symmetry containing the origin, and notes that parallelograms with a vertex at the origin attain it. Conjecture 5 asks whether they are the only bodies that do.

V2-000993 | Planar L_p-Rogers-Shephard, the equality case | isParallelogramAtOrigin_of_volume_lpSum_eq

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 5 (Fradelizi-Manui-Meyer-Ndiaye, 2026). Among planar convex bodies with a centre of symmetry containing the origin, for $p > 1$, equality in Corollary 29 holds only for parallelograms with a vertex at the origin.

```
theorem isParallelogramAtOrigin_of_volume_lpSum_eq :
∀ (K : Set ℝ²), Convex ℝ K → IsCompact K → (interior K).Nonempty →
HasCentreOfSymmetry K → (0 : ℝ²) ∈ K → ∀ p q : ℝ, 1 < p → 1 / p + 1 / q = 1 →
volume (lpSum p K (-K)) = ENNReal.ofReal (rsConstant q) * volume K →
IsParallelogramAtOrigin K
```

FC-Arxiv.«2607.03582».isParallelogramAtOrigin_of_volume_lpSum_eq

V2-000994 | Planar L_p-Rogers-Shephard, the equality case | volume_lpSum_eq_of_isParallelogramAtOrigin

research solved | proof annotation: False | 官方代码 | 原文线索

Parallelograms with a vertex at the origin attain the bound, which is the sharpness half of Corollary 29.

```
theorem volume_lpSum_eq_of_isParallelogramAtOrigin (K : Set ℝ²)
(hpar : IsParallelogramAtOrigin K) (p q : ℝ) (hp : 1 < p) (hq : 1 / p + 1 / q = 1) :
volume (lpSum p K (-K)) = ENNReal.ofReal (rsConstant q) * volume K
```

FC-Arxiv.«2607.03582».volume_lpSum_eq_of_isParallelogramAtOrigin

V2-000995 | Planar L_p-Rogers-Shephard, the equality case | volume_lpSum_le

research solved | proof annotation: False | 官方代码 | 原文线索

Corollary 29 (Fradelizi-Manui-Meyer-Ndiaye, 2026). For a planar convex body K with a centre of symmetry containing the origin and $p > 1$, $$|K \oplus_p -K| \leq \left(\frac{2\Gamma(1+1/q)^2}{\Gamma(1+2/q)} + 2\right)|K|,$$ where q is the Hölder conjugate of p.

```
theorem volume_lpSum_le (K : Set ℝ²) (hK : Convex ℝ K) (hKc : IsCompact K)
(hKi : (interior K).Nonempty) (hsym : HasCentreOfSymmetry K) (h0 : (0 : ℝ²) ∈ K)
(p q : ℝ) (hp : 1 < p) (hq : 1 / p + 1 / q = 1) :
volume (lpSum p K (-K)) ≤ ENNReal.ofReal (rsConstant q) * volume K
```

FC-Arxiv.«2607.03582».volume_lpSum_le

The microscopic weighting on a metric space

MODULE CONTEXT: # The microscopic weighting on a metric space *Reference:* [arxiv/2607.05349](https://arxiv.org/abs/2607.05349) **The microscopic weighting on a metric space** by *Emily Roff, Simon Willerton*

V2-000996 | The microscopic weighting on a metric space | hasFiniteConcentration_of_hasMicroscopicWeighting

research solved | proof annotation: False | 官方代码 | 原文线索

One direction is known: a microscopic weighting implies finite concentration (Theorem 3.1(3)). Together with the conjecture above this leaves the converse, that finite concentration is enough.

```
theorem hasFiniteConcentration_of_hasMicroscopicWeighting
(h : HasMicroscopicWeighting X) : HasFiniteConcentration (distanceMatrix X)
```

FC-Arxiv.«2607.05349».hasFiniteConcentration_of_hasMicroscopicWeighting

V2-000997 | The microscopic weighting on a metric space | hasMicroscopicWeighting_iff_of_isUnit

research solved | proof annotation: False | 官方代码 | 原文线索

Theorem 3.8: the conjecture is known when the distance matrix is invertible, and the microscopic weighting is then the unique gauging.

```
theorem hasMicroscopicWeighting_iff_of_isUnit (h : IsUnit (distanceMatrix X).det) :
HasMicroscopicWeighting X ↔ HasFiniteConcentration (distanceMatrix X)
```

FC-Arxiv.«2607.05349».hasMicroscopicWeighting_iff_of_isUnit

V2-000998 | The microscopic weighting on a metric space | microscopic_weighting_iff_finite_concentration

research solved | proof annotation: True | 官方代码 | 原文线索

Conjecture 3.3 (Roff-Willerton, 2026). A finite metric space admits a microscopic weighting if and only if its distance matrix has finite concentration. `Nonempty` is needed and not just tidiness. On the empty space every gauging condition fails, since `∑ i, g i` is `0` rather than `1`, while `X → ℝ` is a subsingleton so the weighting converges trivially. The equivalence would be false there for reasons that have nothing to do with the question. The answer is false as proved by Kenta Kitamura assisted by ChatGPT 5.6 sol. The proof proceeds by constructing an explicit counterexample: a finite metric space on 10 points that has finite concentration but does not admit a microscopic weighting. The metric space is defined by the following 10×10 distance matrix A: $$ \begin{pmatrix} 0 & 116 & 236 & 231 & 260 & 124 & 64 & 290 & 266 & 64 \cr 116 & 0 & 312 & 268 & 296 & 112 & 64 & 280 & 296 & 64 \cr 236 & 312 & 0 & 68 & 40 & 236 & 288 & 68 & 36 & 288 \cr 231 & 268 & 68 & 0 & 34 & 237 & 288 & 72 & 40 & 288 \cr 260 & 296 & 40 & 34 & 0 & 264 & 320 & 40 & 68 & 320 \cr 124 & 112 & 236 & 237 & 264 & 0 & 64 & 280 & 264 & 64 \cr 64 & 64 & 288 & 288 & 320 & 64 & 0 & 312 & 320 & 120 \cr 290 & 280 & 68 & 72 & 40 & 280 & 312 & 0 & 40 & 312 \cr 266 & 296 & 36 & 40 & 68 & 264 & 320 & 40 & 0 & 320 \cr 64 & 64 & 288 & 288 & 320 & 64 & 120 & 312 & 320 & 0 \end{pmatrix} $$ The space has finite concentration, demonstrated by the explicit gauging g: $$ g = \frac{1}{24842973905} \begin{pmatrix} 3672468740 \cr 6389133731 \cr 9124217512 \cr 5612262448 \cr -6875621136 \cr 2754831248 \cr 0 \cr 10758780188 \cr -6593098826 \cr 0 \end{pmatrix} $$ which gives $A g = \frac{4111107017312}{24842973905} \mathbf{1}$. To prove that it does not admit a microscopic weighting, we provide a vector $v \in \ker A$: $$ v = \begin{pmatrix} -2 \cr -1 \cr 3 \cr 4 \cr -4 \cr -2 \cr 2 \cr 2 \cr -4 \cr 2 \end{pmatrix} $$ along with a row-space certificate q for the entrywise square matrix $B = A^{\circ 2}$ satisfying $q A = v^\top B$: $$ q^\top = \frac{1}{128472094291} \begin{pmatrix} 43681853675722 \cr -53873248293642 \cr -66890627544007 \cr -81187181670120 \cr 24308499983196 \cr 40819375894674 \cr 0 \cr 54690260644468 \cr 35226831040652 \cr 0 \end{pmatrix} $$ This certificate shows that the obstruction $v^\top B$ annihilates $\ker A$, making it impossible to construct a convergent weighting.

theorem microscopic_weighting_iff_finite_concentration :
¬ ∀ (X : Type) [Fintype X] [DecidableEq X] [Nonempty X] [MetricSpace X],
HasMicroscopicWeighting X ↔ HasFiniteConcentration (distanceMatrix X)

FC-Arxiv.«2607.05349».microscopic_weighting_iff_finite_concentration

Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$

MODULE CONTEXT: # Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$ *References:* - [arxiv/2607.05739](https://arxiv.org/abs/2607.05739) **Integer values of $\tan(\arctan 1+\arctan 2+\cdots+\arctan n)$ are rare** by *Ken Ono* - [AMM08] T. Amdeberhan, L. A. Medina, and V. H. Moll, *Arithmetical properties of a sequence arising from an arctangent sum*, J. Number Theory 128 (2008), no. 6, 1807-1846. - [TanArctan](https://github.com/AxiomMath/TanArctan), a Lean formalisation of the three results of [Ono26], MIT licensed. Its `P`, `A`, `B` and `x` are the definitions used here.

V2-000999 | Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$ | exceptional_ncard_le

research solved | proof annotation: True | 官方代码 | 原文线索

The exceptional indices are sparse: $\#(E \cap [1,N]) = O(\log N)$. This is the sense in which integer values are rare, and it is what [Ono26] proves.

theorem exceptional_ncard_le : ∃ (C : ℝ) (N₀ : ℕ), 0 < C ∧ ∀ N : ℕ, N₀ ≤ N →
((exceptional ∩ Set.Icc 1 N).ncard : ℝ) ≤ C * Real.log N

FC-Arxiv.«2607.05739».exceptional_ncard_le

V2-001000 | Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$ | exists_near_half_pi

research solved | proof annotation: True | 官方代码 | 原文线索

Every exceptional index sits close to a multiple of $\pi/2$ in angle.

theorem exists_near_half_pi {n : ℕ} (hn : n ∈ exceptional) :
∃ j : ℤ, |angleSum n - (j : ℝ) * (Real.pi / 2)| < 2 / (n : ℝ)

FC-Arxiv.«2607.05739».exists_near_half_pi

V2-001001 | Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$ | kernel_dvd_of_eq_intCast

research solved | proof annotation: True | 官方代码 | 原文线索

An integer value is divisible in a way that forces it to be large: if $x_n = m$ then $K_n \mid 1 + m^2$, and $|m| \geq \sqrt{K_n - 1}$ once $K_n > 1$.

theorem kernel_dvd_of_eq_intCast {n : ℕ} (hn : 1 ≤ n) (hA : A n ≠ 0) {m : ℤ}
(hx : x n = (m : ℚ)) :
((kernel n : ℤ) | (1 + m ^ 2)) ∧
(1 < kernel n → Real.sqrt ((kernel n : ℝ) - 1) ≤ |(m : ℝ)|)

FC-Arxiv.«2607.05739».kernel_dvd_of_eq_intCast

V2-001002 | Integer values of $\tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$ | tan_arctan_sum_not_integer

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture (Amdeberhan-Medina-Moll, 2008). For every integer $n \geq 5$, the value $$x_n = \tan(\arctan 1 + \arctan 2 + \cdots + \arctan n)$$ is not an integer.

theorem tan_arctan_sum_not_integer :
∀ n : ℕ, 5 ≤ n → ¬ IsIntegerValue n

FC-Arxiv.«2607.05739».tan_arctan_sum_not_integer

The Alon-Tarsi short cycle cover conjecture

MODULE CONTEXT: # The Alon-Tarsi short cycle cover conjecture *References:* - [AlTa85] Alon, N. and Tarsi, M., Covering multigraphs by simple circuits. SIAM J. Algebraic Discrete Methods (1985), 345--350. - [arxiv/2607.06396](https://arxiv.org/abs/2607.06396) **Some new results on Sylvester colorings of cubic graphs** by *Luca Ferrarini, Vahan Mkrtchyan*, where this is Conjecture 4. This file states the finite simple-graph specialization of the source, which allows parallel edges. Every bridgeless simple graph has a list of cycles covering every edge whose lengths sum to at most $\frac{7}{5}|E|$.

V2-001003 | The Alon-Tarsi short cycle cover conjecture | alon_tarsi_short_cycle_cover

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 4 (Alon-Tarsi, 1985), simple-graph specialization. Every bridgeless simple graph has a list of cycles covering every edge, with $\sum_{C} |E(C)| \leq \frac{7}{5}|E(G)|$.

```
theorem alon_tarsi_short_cycle_cover :
∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V)
[DecidableRel G.Adj], G.IsBridgeless →
∃ C : Multiset (Cycle G), IsCycleCover G C ∧
(totalLength C : ℚ) ≤ 7 / 5 * #G.edgeFinset
```

FC-Arxiv.«2607.06396».alon_tarsi_short_cycle_cover

Minimum modulus for the unique multiset-sum problem

MODULE CONTEXT: # Minimum modulus for the unique multiset-sum problem *References:* - [arxiv/2607.08366](https://arxiv.org/abs/2607.08366) **Minimum modulus for the unique multiset-sum problem** by *José A. R. Fonollosa* - [jarfo/min-modulus](https://github.com/jarfo/min-modulus), the author's Lean development of the paper's Main Theorem. Section 7 of the paper describes it. The paper's Main Theorem fixes the super-increasing set $\{2^k - 1\}$ and pins the least modulus at which *it* is valid. Conjecture 1 says no other set of n residues does better, and is open.

V2-001004 | Minimum modulus for the unique multiset-sum problem | exists_isValidMod_minModulus

research solved | proof annotation: True | 官方代码 | 原文线索

Theorem A (Fonollosa, 2026). `minModulus n` admits a valid set of `n` residues, the super-increasing set $\{2^k - 1 : 0 \leq k \leq n - 1\}$. This bounds the least valid modulus from above; that no smaller modulus works is the open half, stated in `min_modulus`.

```
theorem exists_isValidMod_minModulus (n : ℕ) (hn : 2 ≤ n) :
∃ A : Finset (ZMod (minModulus n)), #A = n ∧ IsValidMod A
```

FC-Arxiv.«2607.08366».exists_isValidMod_minModulus

V2-001005 | Minimum modulus for the unique multiset-sum problem | min_modulus

research open | proof annotation: False | 官方代码 | 原文线索

Conjecture 1 (Fonollosa, 2026). For every $n \geq 2$ and every $N < 2^n - 2^{\lfloor \log_2 n\rfloor}$, no set of n residues mod N is valid. Equivalently the super-increasing set $\{2^k - 1 : 0 \leq k \leq n-1\}$ attains the least valid modulus, which is `minModulus n`. `0 < N` excludes `N = 0`, where `ZMod 0` is `ℤ` rather than a finite modulus and `{1, 2}` is valid, which would make the statement false for a reason unrelated to the question.

```
theorem min_modulus :
∀ n N : ℕ, 2 ≤ n → 0 < N → N < minModulus n →
∀ A : Finset (ZMod N), #A = n → ¬ IsValidMod A
```

FC-Arxiv.«2607.08366».min_modulus

The Balister-Győri-Schelp conjecture

MODULE CONTEXT: # The Balister-Győri-Schelp conjecture *References:* - [Perfect Matchings with Prescribed Differences Beyond Hall: The Two-Hole Problem](https://arxiv.org/abs/2607.08630) by *Aryeh Lev Zabokritskiy (Yohananov)* (2026) - [Coloring vertices and edges of a graph by nonempty subsets of a set](https://doi.org/10.1016/j.ejc.2010.11.008) by *P. N. Balister, E. Győri, and R. H. Schelp*, European Journal of Combinatorics 32(4), 533-537 (2011)

V2-001006 | The Balister-Győri-Schelp conjecture | balister_gyori_schelp_conjecture

research open | proof annotation: False | 官方代码 | 原文线索

The **Balister-Győri-Schelp conjecture** ([Conjecture 1](https://arxiv.org/pdf/2607.08630#page=6)): for $s \ge 2$, every zero-sum list of 2^{s-1} nonzero vectors in $\mathbb{F}_2^s$ is the prescribed-difference profile of a perfect matching of $\mathbb{F}_2^s$. The equivalence e indexes the two endpoints of each pair and ensures that the pairs are disjoint and exhaust the ambient vector space.

```
theorem balister_gyori_schelp_conjecture (s : ℕ) (hs : 2 ≤ s)
(v : Fin (2 ^ (s - 1)) → 𝔽₂ s)
(hv_nonzero : ∀ i, v i ≠ 0)
(hv_sum : ∑ i, v i = 0) :
∃ e : Fin (2 ^ (s - 1)) × Fin 2 ≃ 𝔽₂ s,
∀ i, e (i, 0) + e (i, 1) = v i
```

FC-Arxiv.«2607.08630».balister_gyori_schelp_conjecture

Banach-Mazur Rotation Problem

MODULE CONTEXT: # Banach-Mazur Rotation Problem *References:* - [arxiv/math.0110202](https://arxiv.org/abs/math/0110202) **A note on Banach--Mazur problem** by *Beata Randrianantoanina* - [mathoverflow/41211](https://mathoverflow.net/questions/41211/easy-proof-of-the-fact-that-isotropic-spaces-are-euclidean) **Easy proof of the fact that isotropic spaces are Euclidean**

V2-001007 | Banach-Mazur Rotation Problem | banach_mazur_rotation_problem

research open | proof annotation: False | 官方代码 | 原文线索

The Banach--Mazur rotation problem asks whether every separable Banach space whose group of linear isometric equivalences acts transitively on the unit sphere is linearly isometric to a Hilbert space.

theorem banach_mazur_rotation_problem :
∀ (E : Type*) [NormedAddCommGroup E] [NormedSpace ℝ E] [CompleteSpace E]
[SeparableSpace E]
[IsPretransitive (E ≃ᵢ[ℝ] E) (sphere (0 : E) 1)], ∃ (H : Type*) (_ : NormedAddCommGroup H)
(_ : InnerProductSpace ℝ H),
Nonempty (E ≃ᵢ[ℝ] H)

FC-Arxiv.«math.0110202».banach_mazur_rotation_problem

Finite generation of the cohomology of finite-dimensional Hopf algebras

MODULE CONTEXT: # Finite generation of the cohomology of finite-dimensional Hopf algebras Let A be a finite-dimensional Hopf algebra over a field k, and view k as an A-module through the counit. The cohomology ring $$H^*(A, k) = \mathrm{Ext}^*_A(k, k) = \bigoplus_{n \ge 0} \mathrm{Ext}^n_A(k, k)$$ is a graded-commutative k-algebra under the Yoneda product. The finite generation conjecture asserts that it is a finitely generated k-algebra, and that $\mathrm{Ext}^*_A(k, M)$ is a finitely generated module over it for every finite-dimensional A-module M. Etingof and Ostrik state it for every finite tensor category; the finite-dimensional modules over a finite-dimensional Hopf algebra form the main class of examples, and the conjecture is open already in this case. The conjecture is not due to any one author. The question for a finite-dimensional Hopf algebra was asked by a number of mathematicians and had circulated since at least the 1990s: Friedlander and Suslin record it while expressly declining to answer it ("We do not know whether it is reasonable to expect finite generation of the cohomology of an arbitrary finite dimensional Hopf algebra"), and Friedlander and Negron later describe the conjecture as having "existed as a question at least since the 90's" and as "recently stated explicitly in the work of Etingof and Ostrik". What Etingof and Ostrik contribute is that explicit statement, together with its extension from Hopf algebras to every finite tensor category; it is their Conjecture 2.18 that is formalised here. The conjecture holds when A is cocommutative, that is, when A is the group algebra of a finite group scheme (Friedlander-Suslin). For the group algebra $k[G]$ of a finite group G it is the theorem of Evens and Venkov, proved for a p-group by Golod. Etingof and Ostrik note that it also holds when A is commutative. Etingof and Ostrik work throughout the section containing the conjecture under the standing assumption that k is algebraically closed, whereas the statements below are for an arbitrary field. The generalisation is valid. For a field extension K/k one has $\mathrm{Ext}^*_{A \otimes_k K}(K, K) \cong \mathrm{Ext}^*_A(k, k) \otimes_k K$, because k admits a resolution by finitely generated free A-modules and K is flat over k; and finite generation descends, because the finitely many elements of $R = \mathrm{Ext}^*_A(k, k)$ occurring in a finite set of K-algebra generators of $R \otimes_k K$ generate a k-subalgebra R' with $R' \otimes_k K = R \otimes_k K$, whence $R' = R$ by faithful flatness. The same argument applies to the module statement. The theorems of Friedlander-Suslin and of Evens-Venkov are stated below over an arbitrary field because that is the generality in which they are proved. Ext is taken in the category of all A-modules rather than in the finite tensor category of finite-dimensional ones. For finite-dimensional A the two agree on finite-dimensional modules, as the module docstring of `FormalConjecturesForMathlib.RingTheory.Bialgebra.Cohomology` explains. *References:** * [arXiv:math/0301027](https://arxiv.org/abs/math/0301027) P. Etingof, V. Ostrik, *Finite tensor categories*, Mosc. Math. J. 4 (2004), no. 3, 627-654; Conjecture 2.18. * [Friedlander-Suslin](https://doi.org/10.1007/s002220050119) E. M. Friedlander, A. Suslin, *Cohomology of finite group schemes over a field*, Invent. Math. 127 (1997), 209-270. * [Evens](https://doi.org/10.1090/S0002-9947-1961-0137742-1) L. Evens, *The cohomology ring of a finite group*, Trans. Amer. Math. Soc. 101 (1961), 224-239. * B. B. Venkov, *Cohomology algebras for some classifying spaces*, Dokl. Akad. Nauk SSSR 127 (1959), 943-944. * E. S. Golod, *The cohomology ring of a finite p-group*, Dokl. Akad. Nauk SSSR 125 (1959), 703-706. * [Friedlander-Negron](https://doi.org/10.2140/ant.2018.12.1281) E. M. Friedlander, C. Negron, *Cohomology for Drinfeld doubles of some infinitesimal group schemes*, Algebra Number Theory 12 (2018), 1281-1309; the conjecture and its history, p. 1281. * [Oberwolfach](https://doi.org/10.4171/OWR/2019/11) H. Krause, S. Witherspoon, J. J. Zhang (organisers), *Mini-Workshop: Cohomology of Hopf Algebras and Tensor Categories*, Oberwolfach Rep. (2019), 663-693; "asked by a number of mathematicians", p. 664.

V2-001008 | Finite generation of the cohomology of finite-dimensional Hopf algebras | cohomologyRing_finiteType

research open | proof annotation: False | 官方代码 | 原文线索

Finite generation conjecture (Etingof-Ostrik, Conjecture 2.18, algebra part). For every finite-dimensional Hopf algebra A over a field k, the cohomology ring $H^*(A, k) = \mathrm{Ext}^*_A(k, k)$ is a finitely generated k-algebra.

theorem cohomologyRing_finiteType : Algebra.FiniteType k (cohomologyRing k A)

FC-HopfAlgebraCohomology.cohomologyRing_finiteType

V2-001009 | Finite generation of the cohomology of finite-dimensional Hopf algebras | cohomologyRing_finiteType_monoidAlgebra

research solved | proof annotation: False | 官方代码 | 原文线索

Evens-Venkov. For a finite group G and a field k, the cohomology ring $H^*(G, k) = \mathrm{Ext}^*_{k[G]}(k, k)$ of the group algebra $k[G]$ is a finitely generated k-algebra.

theorem cohomologyRing_finiteType_monoidAlgebra (G : Type u) [Group G] [Finite G] :
Algebra.FiniteType k (cohomologyRing k (MonoidAlgebra k G))

FC-HopfAlgebraCohomology.cohomologyRing_finiteType_monoidAlgebra

V2-001010 | Finite generation of the cohomology of finite-dimensional Hopf algebras | cohomologyRing_finiteType_of_isCocomm

research solved | proof annotation: False | 官方代码 | 原文线索

Friedlander-Suslin. If A is a finite-dimensional cocommutative Hopf algebra over a field k, that is, the group algebra of a finite group scheme over k, then $H^*(A, k)$ is a finitely generated k-algebra.

theorem cohomologyRing_finiteType_of_isCocomm [Coalgebra.IsCocomm k A] : Algebra.FiniteType k (cohomologyRing k A)

FC-HopfAlgebraCohomology.cohomologyRing_finiteType_of_isCocomm

V2-001011 | Finite generation of the cohomology of finite-dimensional Hopf algebras | cohomology_finite

research open | proof annotation: False | 官方代码 | 原文线索

Finite generation conjecture (Etingof-Ostrik, Conjecture 2.18, module part). For every finite-dimensional Hopf algebra A over a field k and every finite-dimensional A-module M, the cohomology $H^*(A, M) = \mathrm{Ext}^*_A(k, M)$ is a finitely generated module over $H^*(A, k)$.

theorem cohomology_finite (M : ModuleCat.{u} A) [FiniteDimensional k M] : Module.Finite (cohomologyRing k A) (cohomology k A M)

FC-HopfAlgebraCohomology.cohomology_finite

V2-001012 | Finite generation of the cohomology of finite-dimensional Hopf algebras | cohomology_finite_of_isCocomm

research solved | proof annotation: False | 官方代码 | 原文线索

Friedlander-Suslin. If A is a finite-dimensional cocommutative Hopf algebra over a field k and M is a finite-dimensional A-module, then $H^*(A, M)$ is a finitely generated module over $H^*(A, k)$.

theorem cohomology_finite_of_isCocomm [Coalgebra.IsCocomm k A] (M : ModuleCat.{u} A) [FiniteDimensional k M] : Module.Finite (cohomologyRing k A) (cohomology k A M)

FC-HopfAlgebraCohomology.cohomology_finite_of_isCocomm

The Dixmier Conjecture

MODULE CONTEXT: # The Dixmier Conjecture *References:* - [Dix68] [Sur les algèbres de Weyl](http://www.numdam.org/item/BSMF_1968__96__209_0/) by *Jacques Dixmier*, Bull. Soc. Math. France 96 (1968), 209-242. The conjecture is Problem 1 (for the first Weyl algebra). - [BCW82] [The Jacobian conjecture: reduction of degree and formal expansion of the inverse](https://doi.org/10.1090/S0273-0979-1982-15032-7) by *Hyman Bass, Edwin H. Connell, David Wright*, Bull. Amer. Math. Soc. 7 (1982), 287-330. Proves $DC_n \Longrightarrow JC_n$. - [AvdE07] [arxiv/math.0608009](https://arxiv.org/abs/math/0608009) **On the equivalence of the Jacobian, Dixmier and Poisson Conjectures in any characteristic** by *Kossivi Adjamagbo, Arno van den Essen*. Published as *A proof of the equivalence of the Dixmier, Jacobian and Poisson conjectures*, Acta Math. Vietnam. 32 (2007), 205-214. - [Tsu05] [Endomorphisms of Weyl algebra and p-curvatures](https://doi.org/10.18910/7472) by *Yoshifumi Tsuchimoto*, Osaka J. Math. 42 (2005), 435-452. Proves the stable equivalence $JC \Longleftrightarrow DC$ independently. - [BKK07] [The Jacobian Conjecture is stably equivalent to the Dixmier Conjecture](https://doi.org/10.17323/1609-4514-2007-7-2-209-218) by *Alexei Belov-Kanel, Maxim Kontsevich*, Moscow Math. J. 7 (2007), 209-218. Independent proof of the stable equivalence. - [Bav20] [The Jacobian Conjecture implies the Dixmier Problem](https://arxiv.org/abs/math/0512250) by *Vladimir Bavula*; published in Springer Proceedings in Mathematics & Statistics 317 (2020). A third proof. - [Alp26] [Counterexample to the Jacobian conjecture](https://x.com/__alpoge__/status/2079028340955197566) by *Levent Alpöge* (2026), disproving the Jacobian conjecture in dimension 3 - [Lon26] [An Explicit Counterexample to the Rank-Two Poisson Conjecture](https://arxiv.org/abs/2608.23777) by *Christopher D. Long* (2026), disproving PC_2 The Dixmier Conjecture DC_n asserts that, over a field `K` of characteristic zero, every endomorphism of the `n`-th Weyl algebra $A_n(K)$ is an automorphism. For $n = 1$ this is Problem 1 of [Dix68] and remains open. By [BCW82], $DC_n \Longrightarrow JC_n$ (the Jacobian conjecture in dimension `n`) over a fixed field `K`. By [AvdE07], Theorem 7, the chain $$JC_{2n} \Longrightarrow PC_n \Longrightarrow DC_n \Longrightarrow JC_n$$ holds when each conjecture is quantified over all fields of characteristic zero ([AvdE07], Notations 5 and 7). Since the Jacobian conjecture fails in every dimension $n ≥ 3$ [Alp26], the contrapositive of [BCW82] gives that DC_n is **false for all $n ≥ 3$**. DC_1 and DC_2 remain open. The chain only propagates falsity upwards, so the failure of PC_2 [Lon26] does not settle DC_2. The Weyl algebra $A_n(K)$ is `WeylAlgebra K (Fin n)`, with the $2n$ generators indexed by `Fin n ⊕ Fin n` as in `FormalConjecturesForMathlib/Algebra/WeylAlgebra.lean`; the Poisson side of the chain is stated in `PoissonConjecture.lean` in this directory.

V2-001013 | The Dixmier Conjecture | dixmier_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The **Dixmier Conjecture** (generalised to all dimensions; Problem 1 of [Dix68] for $n = 1$): for every `n`, every endomorphism of the `n`-th Weyl algebra over a field of characteristic zero is an automorphism. This is **false**: the Jacobian conjecture fails in dimension 3 [Alp26], and $DC_n \Longrightarrow JC_n$ [BCW82], so DC_3 fails.

theorem dixmier_conjecture : ¬ ∀ n, DixmierConjectureFor K n

FC-Arxiv.«math.0608009».dixmier_conjecture

The Poisson Conjecture

MODULE CONTEXT: # The Poisson Conjecture *References:* - [AvdE07] [arxiv/math.0608009](https://arxiv.org/abs/math/0608009) **On the equivalence of the Jacobian, Dixmier and Poisson Conjectures in any characteristic** by *Kossivi Adjamagbo, Arno van den Essen*. Published as *A proof of the equivalence of the Dixmier, Jacobian and Poisson conjectures*, Acta Math. Vietnam. 32 (2007), 205-214. - [BCW82] [The Jacobian conjecture: reduction of degree and formal expansion of the inverse](https://doi.org/10.1090/S0273-0979-1982-15032-7) by *Hyman Bass, Edwin H. Connell, David Wright*, Bull. Amer. Math. Soc. 7 (1982), 287-330. - [Alp26] [Counterexample to the Jacobian conjecture](https://x.com/__alpoge__/status/2079028340955197566) by *Levent Alpöge* (2026), disproving the Jacobian conjecture in dimension 3 - [Lon26] [An Explicit Counterexample to the Rank-Two Poisson Conjecture](https://arxiv.org/abs/2608.23777) by *Christopher D. Long* (2026), disproving PC_2 The Poisson Conjecture PC_n ([AvdE07], Notations 5) asserts that, over a field of characteristic zero, every endomorphism of the `n`-th canonical Poisson algebra $P_n(K)$ - the polynomial algebra $K[X_1, \dots, X_{2n}]$ equipped with the canonical Poisson bracket - is an automorphism. By [AvdE07], Theorem 7 (the "United Conjectures Theorem"), for every `n` there is a chain of implications $$JC_{2n} \Longrightarrow PC_n \Longrightarrow DC_n \Longrightarrow JC_n,$$ where JC_n is the Jacobian Conjecture in dimension `n` and DC_n the Dixmier Conjecture for the `n`-th Weyl algebra (the implication $DC_n \Longrightarrow JC_n$ being classical [BCW82]). Since the Jacobian conjecture was disproved in dimension 3 [Alp26] - and hence in every dimension $n \geq 3$ by padding with identity coordinates - PC_n is false for all $n \geq 3$: composing the counterexample's failure of JC_3 through $PC_3 \Longrightarrow DC_3 \Longrightarrow JC_3$ refutes PC_3, and directly, the cotangent (symplectic) lift of the counterexample map ([AvdE07], Theorem 1 in reverse) is a non-invertible Poisson endomorphism of $P_3(K)$. PC_2 is false as well, by the explicit rank-two counterexample of [Lon26]. Only $n = 1$ remains open. Indexing note: we index the $2n$ variables of $P_n(K)$ by `Fin n ⊕ Fin n`, with `Sum.inl i` playing the role of X_i and `Sum.inr i` the role of X_{i+n} of [AvdE07], so that the canonical bracket reads $\{X_i, X_{i+n}\} = 1$; see `MvPolynomial.poissonBracket`.

V2-001014 | The Poisson Conjecture | poisson_conjecture

research solved | proof annotation: False | 官方代码 | 原文线索

The **Poisson Conjecture** ([AvdE07], the characteristic zero case): for every `n`, every endomorphism of the `n`-th canonical Poisson algebra over a field `K` of characteristic zero is an automorphism. This is **false**: the Jacobian conjecture fails in dimension 3 [Alp26], hence so does PC_3 (via [AvdE07], Theorem 7, or directly via the symplectic lift of the counterexample map).

```
theorem poisson_conjecture : ¬ ∀ n, PoissonConjectureFor K n
```

FC-Arxiv.«math.0608009».poisson_conjecture

Algebraic consequences of the Farrell-Jones conjecture

MODULE CONTEXT: # Algebraic consequences of the Farrell-Jones conjecture The assembly map in the Farrell-Jones conjecture is not yet available in Mathlib. This file states two of its algebraic consequences using projective modules and matrices over group rings: the vanishing of the reduced projective class group and the Bass conjecture for commutative integral domains. *Reference:* - [BLR08] [arxiv/math.0703548](https://arxiv.org/abs/math/0703548) **On the Farrell-Jones Conjecture and its applications** by *Arthur Bartels, Wolfgang Lück, Holger Reich*, J. Topol. 1 (2008), 57-86.

V2-001015 | Algebraic consequences of the Farrell-Jones conjecture | bass_conjecture_integral_domain

research open | proof annotation: False | 官方代码 | 原文线索

Bass conjecture for commutative integral domains, in idempotent-matrix form. Let `A` be an idempotent matrix over `R[G]`. If the order of `g` is not invertible in `R`, then the Hattori-Stallings trace of the projective module represented by `A` vanishes at the conjugacy class of `g`. This includes every infinite-order `g`, whose `orderOf` is zero. This is Conjecture 0.6 of [BLR08]. By Theorems 0.5 (ii) and 0.7 of [BLR08] it follows from the Farrell-Jones conjecture with coefficients in every field of prime characteristic.

```
theorem bass_conjecture_integral_domain {R : Type*} [CommRing R] [IsDomain R] {G : Type*}
[Group G] {n : Type*} [Fintype n] (A : Matrix n n (MonoidAlgebra R G))
(hA : A * A = A) (g : G) (hg : ¬IsUnit (orderOf g : R)) :
matrixHattoriStallingsTraceAt A g = 0
```

FC-Arxiv.«math.0703548».bass_conjecture_integral_domain

V2-001016 | Algebraic consequences of the Farrell-Jones conjecture | projective_module_isStablyFree

research open | proof annotation: False | 官方代码 | 原文线索

Vanishing of the reduced projective class group for integral group rings. If `G` is torsion-free, that is, if its only element of finite order is `1`, then every finitely generated projective module over `ℤ[G]` is stably free. This is the stable-freeness form of the vanishing of $\widetilde K_0(\mathbb{Z}[G])$, which is the conclusion of Theorem 0.2 (ii) of [BLR08] for the regular ring $\mathbb{Z}$. That theorem assumes the K-theoretic Farrell-Jones conjecture for `G` with coefficients in `ℤ`.

```
theorem projective_module_isStablyFree {G : Type*} [Group G] (hG : ∀ g : G, IsOfFinOrder g → g = 1)
(M : Type*) [AddCommGroup M] [Module (MonoidAlgebra ℤ G) M]
[Module.Finite (MonoidAlgebra ℤ G) M] [Module.Projective (MonoidAlgebra ℤ G) M] :
Module.IsStablyFree (MonoidAlgebra ℤ G) M
```

FC-Arxiv.«math.0703548».projective_module_isStablyFree

B MathDB 公开目录摘录：79,521 条

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B01 数论 / Number Theory | 6,747 条

V2-001017 | MATHDB-315487

Inclusions among two-dimensional p-bounded crystalline loci

摘录 / EXCERPT: How can the simple inclusions ${\mathcal Z}(\underline r)\subseteq{\mathcal Z}(\underline r')$ be classified as $\underline r$ ranges over p-bounded Hodge types, and can those in...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001018 | MATHDB-315509

Divisibility set for a generalized Euler totient

摘录 / EXCERPT: Define $\varphi_k(n)=\sum_{1\leq a\leq n,(a,n)=1}a^k$ and $\mathcal D_s=\{k\geq s:\varphi_s(n)\mid\varphi_k(n)\text{ for every }n\}$. Is $\mathcal D_1=\{1,3,15\}$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001019 | MATHDB-315530

Almost all primes are partially regular

摘录 / EXCERPT: For how large an initial range of even indices $2k$ can one prove that almost every odd prime p does not divide the numerator of B_{2k}, and hence that the corresponding even c...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001020 | MATHDB-315535

Ramanujan’s tau function misses almost all primes

摘录 / EXCERPT: The cited paper does not formulate this as a conjecture. Its exact conditional theorem is: Assuming the abc Conjecture, as $X\to\infty$, ... where ... Consequently, ... so Ramanujan's...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001021 | MATHDB-315617

Sun Conjecture 4.6(ii) on trigonometric permanents

摘录 / EXCERPT: For an odd prime p, do Sun’s normalized trigonometric permanents satisfy $s_p<0\iff p\equiv5\pmod{12}$ and $s'_p<0\iff p\equiv7\pmod8$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001022 | MATHDB-315622

Strong-majority 4-edge-coloring with at most two degree-3 vertices

摘录 / EXCERPT: Conjecture. Every admissible graph G satisfies $\operatorname{Maj}^{\prime}(G)\leq 4$, where $\operatorname{Maj}^{\prime}(G)$ is the least number of colors in a strong majority e...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-001023 | MATHDB-315623

OEIS A211420 fixed-divisor conjecture

摘录 / EXCERPT: For $a_n=\dfrac{(8n)!\,n!}{(4n)!(3n)!(2n)!}$, and fixed $k\in\{1,2,3\}$ and $r\geq1$, does there exist a constant $C(k,r)$ such that ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001024 | MATHDB-315689

Carlson’s associated-prime depth conjecture

摘录 / EXCERPT: Is the depth of every finite-group cohomology ring realized by the dimension of one of its associated primes?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001025 | MATHDB-315716

Au’s gamma-product conjecture for Witten zeta functions

摘录 / EXCERPT: Conjecture 1.3, Section 1.4: For any root system Φ, let $\xi_\Phi(s):=K_\Phi^{-s}\zeta_\Phi(s)$ be the normalized Witten zeta function. The residue of $\xi_\Phi(s)$ at its abs...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001026 | MATHDB-315726

Non-covering congruence systems over $\mathbb F_q[x]$

摘录 / EXCERPT: What is the asymptotic size of the largest minimum modulus degree in a distinct-modulus congruence system over $\mathbb F_q[x]$ that does not cover the whole polynomial ring?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001027 | MATHDB-315754

Superlinear complexity of the $(3/2)^n$ steering word

摘录 / EXCERPT: For $n\geq 0$, let $m_n=\operatorname{round}((3/2)^n)$, where rounding places half-integers in the half-open convention $[-1/2,1/2)$, and write $(3/2)^n=m_n+\varepsilon_n$ with...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-001028 | MATHDB-315755

Aperiodicity and subword complexity in the binary expansion of powers of three

摘录 / EXCERPT: Theorem A. For every fixed period $p\geq 1$, there is a constant $C_{p}>0$ such that the number $B_{p}(3^{m})$ of positions at which the binary expansion of 3^{m} fails period-...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001029 | MATHDB-315766

Binary digits of the Erdős-Borwein constant

摘录 / EXCERPT: For the Erdős-Borwein constant ... does the block 11 occur infinitely often in the base-2 expansion of E?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001030 | MATHDB-315769

Explicit presentation of the 2-adic absolute Galois group

摘录 / EXCERPT: Give an explicit profinite presentation of the absolute Galois group $\operatorname{Gal}(\overline{\mathbb{Q}}_2/\mathbb{Q}_2)$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001031 | MATHDB-315771

Two small Diophantine equations

摘录 / EXCERPT: Prove that each of the following nine equations has infinitely many solutions $(x,y,z)\in\mathbb{Z}^3$: 1. $z^2+y^2z+x^3-2=0$; 2. $z^2+y^2z+x^3+x+1=0$; 3. $z^2+y^2z+x^3-x-1=0$; 4....

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-001032 | MATHDB-315881

Serre's conjecture II

摘录 / EXCERPT: Let F be a perfect field, let F_s be a separable closure of F, and let $\Gamma = \mathrm{Gal}(F_s/F)$ be its absolute Galois group. For a prime p, let $\mathrm{cd}_p(F)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-001033 | MATHDB-315893

Dade's conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a p-block of G of positive defect (that is, the defect groups of B are nontrivial). A p-subgroup $P \le G$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-001034 | MATHDB-316050

Grimm's conjecture

摘录 / EXCERPT: For positive integers n and k, if the k consecutive integers $n+1, n+2, \dots, n+k$ are all composite, then there exists a sequence of distinct prime numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-001035 | MATHDB-316060

Gilbreath's conjecture

摘录 / EXCERPT: Let $(p_n)_{n\ge 1}=(2,3,5,7,11,13,\dots)$ be the increasing sequence of all prime numbers, and define an array of nonnegative integers d_n^k for $n\ge 1$, $k\ge 1$ by ... Then...

conjecture | open | MathDB 页面 | 原文线索

V2-001036 | MATHDB-316064

Minimum overlap problem

摘录 / EXCERPT: For a positive integer n, consider all ways of splitting $\{1,2,\dots,2n\}$ into two complementary sets $A=\{a_1,\dots,a_n\}$ and $B=\{b_1,\dots,b_n\}$, i.e....

open_problem | open | MathDB 页面 | 原文线索

V2-001037 | MATHDB-316065

Pollock's conjectures

摘录 / EXCERPT: For integers $n\ge 1$ define the tetrahedral numbers ... the octahedral numbers ... and the cubes $C_n=n^{3}$. For a set S of positive integers and an integer $k\ge 1$, say that a po...

conjecture | open | MathDB 页面 | 原文线索

V2-001038 | MATHDB-316066

Skolem problem

摘录 / EXCERPT: For an integer $k \ge 1$, integer coefficients $a_1,\dots,a_k$, and integer initial values $u_0,\dots,u_{k-1}$, let $(u_n)_{n \ge 0}$ be the sequence determined by ... and let ... be i...

open_problem | open | MathDB 页面 | 原文线索

V2-001039 | MATHDB-316067

Waring's problem

摘录 / EXCERPT: For an integer $k \ge 1$ and an integer $s \ge 1$, call a positive integer n s-representable in k-th powers if there exist non-negative integers $x_1,\dots,x_s$ with ... Let...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-001040 | MATHDB-316068

Szemerédi's theorem

摘录 / EXCERPT: For a subset $A \subseteq \mathbb{N}$, define its upper density by ... and say that A has positive upper density if this quantity is > 0. For an integer $k \geq 1$, an arithmetic...

open_problem | open | MathDB 页面 | 原文线索

V2-001041 | MATHDB-316070

Fontaine-Mazur conjecture

摘录 / EXCERPT: Fix a prime p and an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and let $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}\,|\,\mathbb{Q})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001042 | MATHDB-316071

Gan-Gross-Prasad conjecture

摘录 / EXCERPT: Let E/k be a quadratic extension of local fields with k not of characteristic 2, and let $\varepsilon \in \{+1,-1\}$. Let V be a finite-dimensional E-vector space equippe...

conjecture | open | MathDB 页面 | 原文线索

V2-001043 | MATHDB-316072

Greenberg's conjectures

摘录 / EXCERPT: Let ℓ be a prime number and let k be a finite extension of $\mathbb{Q}$. Inside the field $k(\mu_{\ell^{\infty}})$ generated by all ℓ-power roots of unity there is a un...

conjecture | open | MathDB 页面 | 原文线索

V2-001044 | MATHDB-316074

Hilbert's 9th problem

摘录 / EXCERPT: Let K be an algebraic number field with adele ring $\mathbf{A}_K$, and let L/K be a finite Galois extension with Galois group $G=\mathrm{Gal}(L/K)$. Let ... be an irreducible com...

open_problem | open | MathDB 页面 | 原文线索

V2-001045 | MATHDB-316075

Hilbert's 11th problem

摘录 / EXCERPT: Let K be an algebraic number field, i.e. a finite extension of $\mathbb{Q}$, and let $\mathcal{O}_K$ denote its ring of integers. For $n \ge 1$, consider the quadratic forms ... wi...

open_problem | open | MathDB 页面 | 原文线索

V2-001046 | MATHDB-316078

Leopoldt's conjecture

摘录 / EXCERPT: Let K be a number field with r_1 real embeddings and r_2 pairs of complex embeddings, let $\mathcal{O}_K$ be its ring of integers, and let $E = \mathcal{O}_K^{\times}$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-001047 | MATHDB-316079

Stark conjectures

摘录 / EXCERPT: Let K/k be a finite Galois extension of number fields with Galois group G, and let S be a finite set of places of k containing all archimedean places and all places ramifie...

conjecture | open | MathDB 页面 | 原文线索

V2-001048 | MATHDB-316080

Hardy-Littlewood zeta function conjectures

摘录 / EXCERPT: In mathematics, the Hardy-Littlewood zeta function conjectures, named after Godfrey Harold Hardy and John Edensor Littlewood, are two conjectures concerning the distances between z...

conjecture | open | MathDB 页面 | 原文线索

V2-001049 | MATHDB-316083

density hypothesis

摘录 / EXCERPT: For $n\in\mathbb{Z}_{\geq 1}$ let $\Lambda(n)=\log p$ if $n=p^{k}$ for some prime p and integer $k\geq 1$, and $\Lambda(n)=0$ otherwise, and let φ denote the Euler totien...

open_problem | open | MathDB 页面 | 原文线索

V2-001050 | MATHDB-316088

Ramanujan-Petersson conjecture

摘录 / EXCERPT: (1) Coefficients of holomorphic cusp forms. Let $k\geq 1$ be an integer and let $\Gamma\subseteq SL(2,\mathbb{Z})$ be a congruence subgroup containing...

conjecture | open | MathDB 页面 | 原文线索

V2-001051 | MATHDB-316089

Selberg's 1/4 conjecture

摘录 / EXCERPT: Let $\mathbb{H}=\{z=x+iy\in\mathbb{C}: y>0\}$ be the upper half-plane, equipped with the hyperbolic Laplace-Beltrami operator ... and let $\operatorname{SL}_2(\mathbb{Z})$ act on...

conjecture | open | MathDB 页面 | 原文线索

V2-001052 | MATHDB-316104

Brocard's problem

摘录 / EXCERPT: For a positive integer n let $n! = \prod_{k=1}^{n} k$ denote its factorial. Consider the Diophantine equation ... in positive integers n and m. The only pairs (n,m) of positi...

open_problem | open | MathDB 页面 | 原文线索

V2-001053 | MATHDB-316105

Congruent number problem

摘录 / EXCERPT: Call a positive rational number n congruent if there exist $a,b,c\in\mathbb{Q}_{>0}$ with ... that is, if n is the area of a right triangle all of whose sides are rational. For a...

open_problem | open | MathDB 页面 | 原文线索

V2-001054 | MATHDB-316106

Erdős-Moser equation

摘录 / EXCERPT: For integers $m \ge 2$ and $k \ge 1$ put ... The only pair of integers (m,k) with $m \ge 2$, $k \ge 1$ satisfying ... is $(m,k) = (3,1)$, i.e. $1^{1} + 2^{1} = 3^{1}$. Equivalently,...

open_problem | open | MathDB 页面 | 原文线索

V2-001055 | MATHDB-316110

uniqueness conjecture for Markov numbers

摘录 / EXCERPT: Call a triple (x,y,z) of positive integers a Markov triple if ... and call it normalized if $x\le y\le z$. A Markov number is a positive integer occurring as a coordinate of some M...

conjecture | open | MathDB 页面 | 原文线索

V2-001056 | MATHDB-316111

Pillai's conjecture

摘录 / EXCERPT: Call an integer p a perfect power if $p = x^n$ for some integers $x \ge 1$ and $n \ge 2$, and let $p_1 < p_2 < p_3 < \cdots$ be the increasing enumeration of the perfect powers...

conjecture | open | MathDB 页面 | 原文线索

V2-001057 | MATHDB-316112

Sums of three cubes

摘录 / EXCERPT: For every integer x one has $x^{3}\equiv 0,\,1,$ or $-1 \pmod 9$, so no sum of three integer cubes is congruent to 4 or 5 modulo 9. For $n\in\mathbb{Z}$ put ... where the var...

open_problem | open | MathDB 页面 | 原文线索

V2-001058 | MATHDB-316113

Sum of four cubes problem

摘录 / EXCERPT: For every $n \in \mathbb{Z}$ there exist $x, y, z, w \in \mathbb{Z}$ (not required to be nonnegative) such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-001059 | MATHDB-316117

Bunyakovsky conjecture

摘录 / EXCERPT: Let $f(x)=c_0+c_1x+\cdots+c_dx^d \in \mathbb{Z}[x]$ be a polynomial of degree $d\ge 1$ satisfying: 1. $c_d>0$; 2. f is irreducible over $\mathbb{Q}$; 3. $\gcd\{f(n)\}_{n\ge 1}=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-001060 | MATHDB-316119

Dickson's conjecture

摘录 / EXCERPT: Let $k \ge 1$ and let $a_1, \dots, a_k$ and $b_1, \dots, b_k$ be integers with $b_i \ge 1$ for every i, and consider the k linear forms ... Suppose that no prime divides all the...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-001061 | MATHDB-316123

Gaussian moat

摘录 / EXCERPT: Let $\mathbb{Z}[i]=\{a+bi : a,b\in\mathbb{Z}\}$ be the ring of Gaussian integers, with norm $N(a+bi)=a^{2}+b^{2}$ and Euclidean absolute value $|a+bi|=\sqrt{a^{2}+b^{2}}$. Call...

open_problem | open | MathDB 页面 | 原文线索

V2-001062 | MATHDB-316125

Landau's problems

摘录 / EXCERPT: Let $\mathbb{P} = \{2,3,5,7,\dots\}$ denote the set of prime numbers. Consider the following four propositions. (1) For every even integer $n > 2$ there exist $p, q \in \mathbb{P}$...

open_problem | open | MathDB 页面 | 原文线索

V2-001063 | MATHDB-316129

Linnik's theorem

摘录 / EXCERPT: Linnik's theorem in analytic number theory answers a natural question after Dirichlet's theorem on arithmetic progressions. It asserts that there exist positive c and L such that,...

open_problem | open | MathDB 页面 | 原文线索

V2-001064 | MATHDB-316131

Polignac's conjecture

摘录 / EXCERPT: Let $p_1 < p_2 < p_3 < \cdots$ denote the increasing sequence of all prime numbers, so that p_k and p_{k+1} are consecutive primes for every $k \ge 1$. For an even positive int...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-001065 | MATHDB-316133

Selfridge's conjecture

摘录 / EXCERPT: Call an odd natural number k a Sierpi'nski number if every member of the set ... is composite, i.e. $k\cdot 2^{n}+1$ is composite for all positive integers n. Then 78557 is a S...

conjecture | open | MathDB 页面 | 原文线索

V2-001066 | MATHDB-316134

The converse as a conjecture

摘录 / EXCERPT: For an integer $n\ge 2$, let ... denote the congruence between the binomial coefficient $\binom{2n-1}{n-1}$ and 1 modulo n^{k}, where k is a positive integer. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-001067 | MATHDB-316135

Wall-Sun-Sun prime

摘录 / EXCERPT: Let $(F_n)_{n\ge 0}$ be the Fibonacci sequence, defined by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \ge 1$. For an integer $m \ge 1$, the sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-001068 | MATHDB-316136

Siegel's conjecture

摘录 / EXCERPT: For an odd prime p let ζ_p denote a primitive pth root of unity, let h_p denote the class number of the cyclotomic field $\mathbb{Q}(\zeta_p)$, that is, the order of th...

conjecture | open | MathDB 页面 | 原文线索

V2-001069 | MATHDB-316155

Carlitz-Wan conjecture

摘录 / EXCERPT: Let q be a prime power, let $\mathbb{F}_q$ be the field with q elements, and let $\overline{\mathbb{F}_q}$ denote an algebraic closure of $\mathbb{F}_q$. For...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001070 | MATHDB-316165

Kemnitz's conjecture

摘录 / EXCERPT: In additive number theory, Kemnitz's conjecture states that every set of integer lattice points in the plane has a large subset whose centroid is also a lattice point. It was prove...

conjecture | open | MathDB 页面 | 原文线索

V2-001071 | MATHDB-316205

Duffin-Schaeffer theorem

摘录 / EXCERPT: The Koukoulopoulos-Maynard theorem, historically known as the Duffin-Schaeffer conjecture, is a theorem in mathematics, specifically Diophantine approximation. It was proposed as a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001072 | MATHDB-316206

Main conjecture in Vinogradov's mean-value theorem

摘录 / EXCERPT: In mathematics, Vinogradov's mean value theorem is an estimate for the number of equal sums of powers. It is an important inequality in analytic number theory, named for I. M. Vino...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001073 | MATHDB-316207

Goldbach's weak conjecture

摘录 / EXCERPT: Call an integer p prime if $p>1$ and its only positive divisors are 1 and p; the primes in a representation are not required to be distinct. For every odd integer $n>5$ there...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001074 | MATHDB-316209

Serre's modularity conjecture

摘录 / EXCERPT: Fix a prime ℓ and let $G_{\mathbb{Q}}=\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be the absolute Galois group of $\mathbb{Q}$. Let $F=\mathbb{F}_{\ell^{r}}$ be a finite f...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001075 | MATHDB-316210

Green-Tao theorem

摘录 / EXCERPT: In number theory, the Green-Tao theorem, proven by Ben Green and Terence Tao in 2004, states that the sequence of prime numbers contains arbitrarily long arithmetic progressions. I...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001076 | MATHDB-316212

Erdős-Graham problem

摘录 / EXCERPT: In combinatorial number theory, the Erdős-Graham problem is the problem of proving that, if the set $\{2,3,4,\dots \}$ of integers greater than one is partitioned into finitely man...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001077 | MATHDB-316214

Fermat's Last Theorem

摘录 / EXCERPT: In number theory, Fermat's Last Theorem (sometimes called Fermat's conjecture, especially in older texts) states that there are no positive integers a,b,c,n with $n>2$ such that...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001078 | MATHDB-316225

Geometrization conjecture

摘录 / EXCERPT: A 3-manifold is closed if it is compact and has empty boundary. A closed 3-manifold M is prime if whenever M is homeomorphic to a connected sum $M_1 \# M_2$, one of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001079 | MATHDB-316230

Bloch-Kato conjecture

摘录 / EXCERPT: Let k be a field, let ℓ be a prime number invertible in k, and fix a separable closure k_{sep} of k with absolute Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-001080 | MATHDB-316235

n! conjecture

摘录 / EXCERPT: For an integer m with $m\neq 0$, let $\operatorname{rad}(m)$ denote the product of the distinct prime factors of m (with $\operatorname{rad}(\pm 1)=1$). For integers...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-001081 | MATHDB-316237

Modularity theorem

摘录 / EXCERPT: In number theory, the modularity theorem states that elliptic curves over the field of rational numbers are related to modular forms in a particular way. Andrew Wiles and Richard T...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-001082 | MATHDB-316239

Jacquet-Langlands correspondence for symplectic similitude groups

摘录 / EXCERPT: Let $S_k(N)$ be the space of Hilbert-Siegel cusp forms of weight k and level N on the split symplectic similitude group, let $S_k^B(N)$ be the corresponding space of algebraic...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-001083 | MATHDB-316255

Infinitely many Abby-Normal numbers

摘录 / EXCERPT: For $n\geq2$, define the scaled total stopping time by ... With $a_0=2$, call a_k the k-th Abby-Normal number if ... The Abby-Normal-number conjecture. There exist infinitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-001084 | MATHDB-316271

Growth-constant bound for height-one-growth prime ideals

摘录 / EXCERPT: Growth-constant bound conjecture. There exists a function $F:(0,\infty)\rightarrow(0,\infty)$ such that A has at most $F({\rm GC}(A))$ prime ideals P satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-001085 | MATHDB-316277

Conjecture on sums of rational approximations with prime denominators

摘录 / EXCERPT: Sums of rational approximations conjecture. There are integers $a_1,\ldots,a_n$ and distinct primes $q_1,\ldots,q_n$ with $1\leq q_i\leq N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001086 | MATHDB-316278

Conjecture on rational approximations with prescribed numbers of prime divisors

摘录 / EXCERPT: Prescribed-prime-divisors conjecture. There is a rational number A/Q such that $Q\leq X^{1+\epsilon}$, $\omega(Q)=n$, and

conjecture | open | MathDB 页面 | 原文线索

V2-001087 | MATHDB-316286

Hesselholt-Rains conjecture on torsion in preprojective algebras

摘录 / EXCERPT: Hesselholt-Rains conjecture. For every prime $p\geq 2$, the classes $r^{(p^\ell)}$ are nonzero and generate the p-torsion of Π_{cyc}. Moreover, Π_{cyc}...

conjecture | open | MathDB 页面 | 原文线索

V2-001088 | MATHDB-316294

The strong converse to the Gorenstein homomorphism criterion

摘录 / EXCERPT: Let K be a field, let S be a commutative ring, and let $\sigma\colon K\to S$ be a homomorphism. Write $S^{\mathsf e}=S\otimes_K S$, and let ${\mathfrak q}$ be a minimal prime i...

conjecture | open | MathDB 页面 | 原文线索

V2-001089 | MATHDB-316298

The limiting slope conjecture for degree covers of elliptic curves in genus three

摘录 / EXCERPT: Let d be prime, let σ be a partition of the branch-cycle type $(5^{1}1^{d-5})$, and let $Y_{3,d,\sigma}$ denote the corresponding one-parameter family of degree-d covers...

conjecture | open | MathDB 页面 | 原文线索

V2-001090 | MATHDB-316321

Buzzard-Diamond-Jarvis modularity conjecture for totally real fields

摘录 / EXCERPT: Fix a prime p and a totally real number field F in which p is unramified. Let $\mathcal{O}_F$ be the ring of integers of F, let G_F be its absolute Galois group, and fix...

conjecture | open | MathDB 页面 | 原文线索

V2-001091 | MATHDB-316332

Kimmerle's prime graph conjecture

摘录 / EXCERPT: Let G be a finite group. Write $\#(G)$ for the set of primes dividing the order of G. The Gruenberg-Kegel graph, or prime graph, of G is the graph $\pi(G)$ whose vertices are...

conjecture | open | MathDB 页面 | 原文线索

V2-001092 | MATHDB-316341

Coates-Fukaya-Kato-Sujatha-Venjakob finiteness conjecture

摘录 / EXCERPT: Let K be a number field, let p be a prime, let E be an elliptic curve over K, and let L/K be a Galois extension with $G=\operatorname{Gal}(L/K)$ containing a closed norma...

conjecture | open | MathDB 页面 | 原文线索

V2-001093 | MATHDB-316352

The primitive-prime enumeration formula for quantum matrices with three rows

摘录 / EXCERPT: Primitive-prime enumeration conjecture. The number of primitive $\mathcal{H}$-primes in ${\mathcal O}_q(M_{3,n})$ is

conjecture | open | MathDB 页面 | 原文线索

V2-001094 | MATHDB-316359

Gouvêa-Mazur conjecture on slopes of overconvergent modular forms

摘录 / EXCERPT: Gouvêa-Mazur conjecture. Let k, l be integers, n be a positive integer, and $\alpha<n$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-001095 | MATHDB-316360

Conjecture on the distribution of slopes of classical p-oldforms

摘录 / EXCERPT: Slope-distribution conjecture. The probability that an element of R_k chosen with uniform distribution lies in the interval

conjecture | open | MathDB 页面 | 原文线索

V2-001096 | MATHDB-316363

The noncommutative inverse shuffle-product conjecture for rational series

摘录 / EXCERPT: Let p be a prime, let $\mathbb K\subset\overline{\mathbb F_p}$ be a field, and let $\mathfrak m$ be the ideal of nonconstant formal power series in the noncommutative series alge...

conjecture | open | MathDB 页面 | 原文线索

V2-001097 | MATHDB-316397

Rubin's conjecture on Euler-system elements and Perrin-Riou p-adic L-functions

摘录 / EXCERPT: Let $r\in\mathbb{Z}^+$ be prime to p, let T be the representation specified in the surrounding setup, and let A denote the associated module. Let $\mathbb{Q}(\mu_r)^+$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-001098 | MATHDB-316406

Monotonicity of the minimum odious-prime excess

摘录 / EXCERPT: Monotonicity conjecture. For all $n\in\mathbb{N}$, $n\neq 5,6$, the sequence m_n increases monotonically.

conjecture | open | MathDB 页面 | 原文线索

V2-001099 | MATHDB-316410

Uniform-distribution conjecture for small constellations in the Eratosthenes sieve

摘录 / EXCERPT: Let ${\mathcal G}(p)$ be the cycle of gaps in the p-sieve, and let ${\mathcal G}(P)$ be the corresponding cycle for a larger prime P. Under the recursion relating these cycles,...

conjecture | open | MathDB 页面 | 原文线索

V2-001100 | MATHDB-316411

Infinitude conjecture for sufficiently small sieve constellations

摘录 / EXCERPT: Let p_k be the kth prime, and let ${\mathcal G}(p_k)$ denote the cycle of gaps in the p_k-sieve. Consider constellations in ${\mathcal G}(p_k)$ whose total gap sum is less th...

conjecture | open | MathDB 页面 | 原文线索

V2-001101 | MATHDB-316444

Width-two conjecture for prime-indexed generalised Fabrykowski-Gupta groups

摘录 / EXCERPT: Let p be an odd prime, and let Γ_p be the corresponding generalised Fabrykowski-Gupta group. The width-two conjecture. Then Γ_p is a group of width 2. The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-001102 | MATHDB-316451

The G2 maximal cross-correlation conjecture for non-Germain prime fields

摘录 / EXCERPT: G2 cross-correlation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001103 | MATHDB-316452

The W1-G2 maximal cross-correlation conjecture in non-Germain prime fields

摘录 / EXCERPT: W1-G2 cross-correlation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001104 | MATHDB-316453

Conjectural Eisenstein subspaces in degree-five cohomology of \(\Gamma_0(p)\)

摘录 / EXCERPT: Fix a prime p. Let $\Gamma_0(p)\subset SL(4,\mathbb Z)$ be the congruence subgroup in the paper, and let d_f and d_η denote the degrees of the fields generated by the Heck...

conjecture | open | MathDB 页面 | 原文线索

V2-001105 | MATHDB-316458

Selmer-corank conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, and let $r_{\rm an}(E/\mathbb{Q})$ be the order of vanishing of $L(E/\mathbb{Q},s)$ at $s=1$. For a prime p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-001106 | MATHDB-316476

Conjecture on twists of symmetric-power L-values

摘录 / EXCERPT: Let $\varphi\in S_k(N,\omega)$ be a primitive form, let η be a primitive Dirichlet character, and let $L_f(s,\operatorname{Sym}^n\varphi,\eta)$ denote the finite L-function...

conjecture | open | MathDB 页面 | 原文线索

V2-001107 | MATHDB-316481

The universal standard-relations conjecture for MPVs

摘录 / EXCERPT: Universal standard-relations conjecture. If $N=1$ or 2, the map $\varphi_{\mathbb{R}}$ is injective, so the algebra of MPVs is isomorphic to R_{RDS}. If $N=p$ is prime with...

conjecture | open | MathDB 页面 | 原文线索

V2-001108 | MATHDB-316483

The prime-level weight-three dimension conjecture

摘录 / EXCERPT: Prime-level weight-three conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001109 | MATHDB-316485

Density-one lower bound conjecture for dynamical periods modulo primes

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a number field, let $\varphi:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\geq 2$, and let $P\in\mathbb{P}^N(K)$ be a point with infinite φ-...

conjecture | open | MathDB 页面 | 原文线索

V2-001110 | MATHDB-316496

Metric-diophantine genericity conjecture for Lie groups

摘录 / EXCERPT: Let G be the Lie-group setting considered in the paper, and let $(g_1,\ldots,g_k)\in G^{(k)}$ be generic in the measure sense. Metric-diophantine genericity conjecture. The tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-001111 | MATHDB-316500

Heath-Brown's rational point density conjecture for hypersurfaces

摘录 / EXCERPT: Let $F\in\mathbb{Z}[x_1,\ldots,x_n]$ be an absolutely irreducible form of degree $d\geqslant 2$, and let $X_F\subset\mathbb{P}^{n-1}$ be the hypersurface defined by F. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001112 | MATHDB-316508

The main conjecture on 2-adic valuations of Stirling numbers

摘录 / EXCERPT: Main conjecture. The first index for which there is a constant class is $m_{0}-1$: every class is part of the j-level for $1\leq j\leq m_{0}-2$. For every $m\geq m_{0}$, the m-...

conjecture | open | MathDB 页面 | 原文线索

V2-001113 | MATHDB-316516

Iosevich-Rudnev finite-field Erdős-Falconer conjecture

摘录 / EXCERPT: Let $E\subset {\mathbb F}_q^d$, where q is a power of an odd prime, and define the finite-field distance set ... Iosevich-Rudnev finite-field Erdős-Falconer conjecture. If, for som...

conjecture | open | MathDB 页面 | 原文线索

V2-001114 | MATHDB-316538

Leopoldt's conjecture for rational tori

摘录 / EXCERPT: Let $\mathbf{T}$ be a torus over $\mathbb{Q}$. For a compact open subgroup $K_f=K^pK_p$ of $\mathbf{T}(\mathbb{A}_f)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-001115 | MATHDB-316565

Van Hamme's p-adic analogue of Ramanujan's formula

摘录 / EXCERPT: Let p be an odd prime. The p-adic Gamma function Gamma_p is denoted by Γ_p. Van Hamme's conjecture. ... This is a p-adic analogue of Ramanujan's formula...

conjecture | open | MathDB 页面 | 原文线索

V2-001116 | MATHDB-316571

The very-large diophantine subset conjecture over the complex rational function field

摘录 / EXCERPT: Let $D subseteq \mathbb{C}(t)$ be a diophantine subset containing a Zariski open subset of $\mathbb{C}[t]$; for example, suppose this holds in the space of polynomials of degree at...

conjecture | open | MathDB 页面 | 原文线索

V2-001117 | MATHDB-316597

Castaño-Bernard's Heegner index conjecture for prime-conductor elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve of rank one over $\mathbb{Q}$ and prime conductor N_E. Let I_E be the index of the subgroup of $E(\mathbb{Q})$ generated by the Heegner points, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001118 | MATHDB-316611

Sondow-Schalm conjecture on Taylor partial sums converging to e

摘录 / EXCERPT: Let $e=\sum_{r=0}^{\infty}1/r!$, and write its nth Taylor partial sum as $A_n/n!$. Let p_k/q_k be the convergents of the simple continued fraction expansion of e. Sondow-Scha...

conjecture | open | MathDB 页面 | 原文线索

V2-001119 | MATHDB-316612

Zeros Conjecture for the 2-factors of $A(n)$ and $Q(n)$

摘录 / EXCERPT: For an integer x and prime p, let $[x]_p$ denote the largest power of p dividing x, with $[0]_p=\infty$. Let $A(n)$ be the factorial-numerator sequence from the Taylor part...

conjecture | open | MathDB 页面 | 原文线索

V2-001120 | MATHDB-316613

Strengthened 2-adic Zeros Conjecture for the continued-fraction denominators of e

摘录 / EXCERPT: Let $\mathbb{Z}_2$ be the 2-adic integers, let $\widetilde{Q}_i:\mathbb{Z}_2\to\mathbb{Z}_2$ be the continuous extensions of $Q_i(n)=Q(3n+i)$, and let c_k be the unique zero of...

conjecture | open | MathDB 页面 | 原文线索

V2-001121 | MATHDB-316631

Holomorphic lifting conjecture from $GSp_{2m}\times GSp_{2m}$ to GSp_{4m}

摘录 / EXCERPT: Let f and g be Siegel modular forms of genus $2m$, with weights $k>2m$ and $l=k-2m$. Let $(\alpha_0,\alpha_1,\ldots,\alpha_{2m})$ and $(\beta_0,\beta_1,\ldots,\beta_{2m})$ be s...

conjecture | open | MathDB 页面 | 原文线索

V2-001122 | MATHDB-316642

Twisted 2ℓ-th moment conjecture for the Riemann zeta function

摘录 / EXCERPT: Let ℓ be a positive integer, let $\alpha_1,\dots,\alpha_\ell,\beta_1,\dots,\beta_\ell$ be complex shifts, and let ... where w is a nice function supported in $[T/2,4T]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001123 | MATHDB-316648

Asymptotic periodicity conjecture for finite alternating Hilbert operators

摘录 / EXCERPT: Let $A_N=\left(\frac{1}{m-n}\right)_{m,n=1,\dots,N}$, with zero on the diagonal, and let $\operatorname{Spect}(A_N)$ denote its spectrum. For each positive integer k, write...

conjecture | open | MathDB 页面 | 原文线索

V2-001124 | MATHDB-316655

Almost-divisibility conjecture for integers in a power interval

摘录 / EXCERPT: Almost-divisibility conjecture. For every $\epsilon>0$, one can take $L=X^\epsilon$ provided X is sufficiently large in terms of ϵ.

conjecture | open | MathDB 页面 | 原文线索

V2-001125 | MATHDB-316681

The manifold g-conjecture for h-prime vectors

摘录 / EXCERPT: Let Δ be a homology manifold, let h'_i denote its h-prime vector, and let β_{i-1} denote its $(i-1)$st Betti number. Manifold g-conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-001126 | MATHDB-316735

Transience conjecture for the natural prime-generating recurrence

摘录 / EXCERPT: Let $a(n)$ be the sequence defined by the recurrence in the paper, and let $n_1\geq 1$ satisfy $a(n_1)\geq 1$. Transience conjecture. There exists an N such that $a(n)-a(n-1)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-001127 | MATHDB-316758

The abelianization conjecture for the external fundamental group of the rational field

摘录 / EXCERPT: Let ${}^{\circ}{\rm I}\!\Gamma_{\mathbb Q}$ be the external fundamental group associated with the rational field, let $\hat{{\rm I}\!\Gamma}_{\mathbb Q}$ be its standardization quo...

conjecture | open | MathDB 页面 | 原文线索

V2-001128 | MATHDB-316762

Ribet-Ihara-type conjecture for cohomology of Shimura curves

摘录 / EXCERPT: Let S be a finite set of primes not dividing $M\ell$. Fix $f\in S_2(\Gamma_0(N\Delta'\ell^2S),\psi)$ that is supercuspidal of type τ at ℓ and special at primes...

conjecture | open | MathDB 页面 | 原文线索

V2-001129 | MATHDB-316763

Ihara's lemma conjecture for Shimura-curve cohomology

摘录 / EXCERPT: Let q be a prime such that $q\nmid N\Delta'\ell^2$. Let ${\bf T}_0^{\widehat\psi}(N)$ act on $H^1({\bf X}_1(N),\mathcal O)^{\widehat\psi}$, and fix a maximal non-Eisenstein ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-001130 | MATHDB-316777

Monodromy-weight conjecture

摘录 / EXCERPT: Let $\Ubar$ be the geometric generic fibre in a proper flat family over a local base, and let $W_\bullet$ denote the weight filtration on $H^j(\Ubar,\Ql)$. Let ... denote the logarit...

conjecture | open | MathDB 页面 | 原文线索

V2-001131 | MATHDB-316782

The standard-Koszul description conjecture for symmetric-group blocks

摘录 / EXCERPT: Let p be prime, let Σ denote symmetric groups, and let $B_{\Sigma,w}$ be the set of blocks of symmetric groups of weight w. Let $\Phi_{B_{\Sigma,w}}$ denote the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-001132 | MATHDB-316811

The infinitude of primes of weight 3

摘录 / EXCERPT: Let p_n denote the nth prime, and let its weight be the weight in the paper's decomposition of primes. The smallest member of each twin-prime pair other than 3 has weight 3...

conjecture | open | MathDB 页面 | 原文线索

V2-001133 | MATHDB-316812

The infinitude of primes of every odd weight at least 3

摘录 / EXCERPT: Let p_n denote the nth prime, and let its weight be the weight in the paper's decomposition of primes. Odd-weight prime conjecture. For every integer $k\ge 3$ that is not a mul...

conjecture | open | MathDB 页面 | 原文线索

V2-001134 | MATHDB-316813

Prime weight primality for primes classified by level

摘录 / EXCERPT: Let p_n denote the nth prime, and let l_n and its weight be the quantities in the paper's decomposition of primes. A prime is classified by level when its decomposition has p...

conjecture | open | MathDB 页面 | 原文线索

V2-001135 | MATHDB-316814

The infinitude of primes of level $(1;i)$

摘录 / EXCERPT: Let p_n denote the nth prime, and let level $(1;i)$ be the level notation used in the paper's decomposition. Level-$(1;i)$ infinitude conjecture. For every integer $i\ge 1$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-001136 | MATHDB-316815

The converse jump divisibility conjecture

摘录 / EXCERPT: Let p_n denote the nth prime, let g_n be its jump, and let l_n be the lower component in the paper's prime decomposition. Converse jump divisibility conjecture. If l_n is...

conjecture | open | MathDB 页面 | 原文线索

V2-001137 | MATHDB-316816

The rarefaction of level-classified primes

摘录 / EXCERPT: Let the primes be ordered as $p_1<p_2<\cdots$, and consider the subset of primes classified by level in the paper's decomposition. Rarefaction conjecture. The primes classified by...

conjecture | open | MathDB 页面 | 原文线索

V2-001138 | MATHDB-316863

Meromorphicity conjecture for global moment factors

摘录 / EXCERPT: Let $f:X\mapsto Y/\mathbb{Z}[1/N]$ be a map, and let $R_d(f\otimes\mathbb{F}_p,T)$ denote the non-trivial factor of the moment zeta function described in the source. Meromorphicity...

open_problem | open | MathDB 页面 | 原文线索

V2-001139 | MATHDB-316865

Eventual ordinarity conjecture for integral polytopes

摘录 / EXCERPT: Let $\triangle$ be an integral polytope, and call a prime p ordinary for $\triangle$ when $GNP(\triangle,p)=HP(\triangle)$. Eventual ordinarity conjecture. For any $\triangle$,...

conjecture | refuted | MathDB 页面 | 原文线索

V2-001140 | MATHDB-316870

Uniform reduction conjecture for test ideals and multiplier ideals

摘录 / EXCERPT: Let R be a domain smooth over ${\mathbf Z}$ with $R\otimes_{\mathbf Z}{\mathbf Q}\neq 0$, let $f\in R$ be nonzero, and for each prime p let...

conjecture | open | MathDB 页面 | 原文线索

V2-001141 | MATHDB-316877

Fouvry-Kowalski-Shparlinski conjecture on the maximal modular inversion gap

摘录 / EXCERPT: Let $M(n)$ be the maximum of $|a-b|$ over integers a,b with $1\leq a,b\leq n-1$ and $ab\equiv 1\pmod n$. Let $\delta=1-(1+\log\log 2)/\log 2$. The function g is any function te...

conjecture | open | MathDB 页面 | 原文线索

V2-001142 | MATHDB-316878

Conjecture on the divergence of the number of convex-hull vertices

摘录 / EXCERPT: Let $v(n)$ denote the number of vertices of the convex hull associated with the graph of modular inversions modulo n, and let τ denote the divisor-counting function. Diverge...

conjecture | open | MathDB 页面 | 原文线索

V2-001143 | MATHDB-316879

Congruence conjecture for the number of modular-inversion hull vertices

摘录 / EXCERPT: Let $v(n)$ be the number of vertices of the convex hull associated with the graph of modular inversions modulo n, and let τ be the divisor-counting function. Congruence conj...

conjecture | open | MathDB 页面 | 原文线索

V2-001144 | MATHDB-316896

The p-primary structure conjecture for cuspidal divisor class groups of $X_1(p^n)$

摘录 / EXCERPT: Let p be a prime and n a positive integer. For the cuspidal divisor class group $\mathscr C_1^\infty(p^n)$, let $[p]$ denote multiplication by p: ... Define its p-rank to be...

conjecture | open | MathDB 页面 | 原文线索

V2-001145 | MATHDB-316906

Smoktunowicz-Bell conjecture on prime monomial algebras

摘录 / EXCERPT: Smoktunowicz-Bell conjecture. Every prime monomial algebra A is either PI, primitive, or has nonzero Jacobson radical.

conjecture | open | MathDB 页面 | 原文线索

V2-001146 | MATHDB-316916

Harbater's generator-rank conjecture for arithmetic fundamental groups

摘录 / EXCERPT: In the arithmetic case, let $U_n=\operatorname{Spec}(\mathbb{Z}[\frac{1}{n}])$ for a positive square-free integer n, and let $\pi^t_A(U_n)$ denote the groups occurring as Galois...

conjecture | open | MathDB 页面 | 原文线索

V2-001147 | MATHDB-316917

Euler sums in terms of generalized polylogarithm values

摘录 / EXCERPT: Let w be a positive integer. An alternating multiple zeta value is a value of the alternating multiple zeta function of weight w. For a vector $\vec{s}$, write $w(\vec{s})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-001148 | MATHDB-316922

Alternating multiple zeta values with 1-2 indices basis conjecture

摘录 / EXCERPT: Let w be a positive integer, and let the $\mathbb{Q}$-linear space be generated by values of the alternating multiple zeta function $\zeta(\vec{s};\vec{\sigma})$ of weight w. W...

conjecture | open | MathDB 页面 | 原文线索

V2-001149 | MATHDB-316936

The uniform primality conjecture for elliptic divisibility sequences

摘录 / EXCERPT: Let $B=(B_n)$ be an elliptic divisibility sequence generated by a rational point on an elliptic curve in minimal form. A prime term is a term B_n/B_1 that is prime. The uniform p...

conjecture | open | MathDB 页面 | 原文线索

V2-001150 | MATHDB-316981

Balazard's conjecture on reciprocals of decomposition primes in abelian extensions

摘录 / EXCERPT: Let K be a global field, and let L/K be an abelian extension of degree n. Write $\operatorname{Dec}(L/K)$ for the set of places of K that split completely in L. Balazard'...

conjecture | open | MathDB 页面 | 原文线索

V2-001151 | MATHDB-316990

McCallum-Sharifi conjecture on cyclotomic cup products

摘录 / EXCERPT: Let p be an odd prime, let $F={\bf Q}(\mu_p)$, and let A_F be the p-part of the class group of F. For an irregular pair (p,k), where k is positive and even and p divi...

conjecture | open | MathDB 页面 | 原文线索

V2-001152 | MATHDB-316991

Sharifi's cup-product and L-value conjecture

摘录 / EXCERPT: Let (p,k) be an irregular pair with k even and $k\geq 12$. Let $F={\bf Q}(\mu_p)$, let $A_F^{(1-k)}$ be the corresponding eigenspace of the p-part of the class group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001153 | MATHDB-317013

Polynomiality conjecture for nilpotent conjugacy classes in parabolic subgroups

摘录 / EXCERPT: Let q be a prime power and let $\mathbf l=(l_1,\ldots,l_s)$ be a tuple of nonnegative integers. For $m=l_1+\cdots+l_s$, let $P^{\mathbf l}(q)$ be the parabolic subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-001154 | MATHDB-317054

The cluster-variance conjecture for codes of T

摘录 / EXCERPT: Let T and F be the transformations associated with codes, and let ${\mathbf c}_1,{\mathbf c}_2$ be codes of T of equal length. For a code ${\mathbf c}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-001155 | MATHDB-317055

Non-Eisenstein Ihara conjecture for Shimura curves

摘录 / EXCERPT: Let q be a prime number such that $q\not|N\Delta\ell$. Fix a maximal non-Eisenstein ideal $\mathfrak m$ of the Hecke algebra ${\bf T}^{\widehat\psi}(N)$ acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-001156 | MATHDB-317056

Centrality conjecture for the Sigma-congruence kernel

摘录 / EXCERPT: Let $\Sigma=\{q,\infty\}$, where q is a prime as in the preceding construction, and let $B^{(1)}$ be the norm-one subgroup of the quaternion algebra B. Write $C(\Sigma)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-001157 | MATHDB-317070

Integrality conjecture for eigenvalues of unitary finite-Euclidean graphs

摘录 / EXCERPT: Let n and d be positive integers, and let $T_n^{(d)}$ denote the unitary finite-Euclidean graph. Integrality conjecture. All eigenvalues of $T_n^{(d)}$ are integers. The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-001158 | MATHDB-317096

Noncommutative p-adic zeta-function existence conjecture

摘录 / EXCERPT: Let F_∞/F be the extension considered in the paper, let G be its Galois group, and let $\Lambda(G)_S$ be the localized Iwasawa algebra. Let κ_F be the cyclotomic ch...

conjecture | open | MathDB 页面 | 原文线索

V2-001159 | MATHDB-317104

Hardy-sequence patterns in the primes conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ denote the Hardy field under consideration, let $[x]$ be the integer part of x, and let “the growth assumptions of Theorem A (or Conjecture A)” refer to the hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-001160 | MATHDB-317117

The abelianized anabelian Torelli conjecture for curves

摘录 / EXCERPT: Let $k=\overline{\mathbb F}_p$ and let C be a curve over k of genus $\mathsf g(C)\geq 2$. Let ${\mathcal G}^a_{(p)}=\prod_{\ell\neq p}{\mathcal G}^a_\ell$ be the prime-to-p a...

conjecture | open | MathDB 页面 | 原文线索

V2-001161 | MATHDB-317129

Speculative zeta-function connection for quantized vortex dynamics

摘录 / EXCERPT: Speculative zeta-function connection. A connection may exist between the zeros of Riemann's zeta function and the quantization of a $3/2$ degrees-of-freedom vortex monopole problem...

open_problem | open | MathDB 页面 | 原文线索

V2-001162 | MATHDB-317131

Meyer's rule of thumb for levels of modular Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a modular Calabi-Yau threefold and let $p \geq 5$ be a prime. Let f be the twist of minimal level N of the weight four newform associated with X. Meyer's rule of t...

conjecture | open | MathDB 页面 | 原文线索

V2-001163 | MATHDB-317138

The S-torsion conjecture for non-commutative Iwasawa modules

摘录 / EXCERPT: Let $p\neq 2$ be a prime, let F be a totally real number field, and let F^{∞}/F be a totally real Galois extension of infinite degree containing the cyclotomic...

conjecture | open | MathDB 页面 | 原文线索

V2-001164 | MATHDB-317151

The strong Siegel-Hall height conjecture for elliptic curves

摘录 / EXCERPT: Strong Siegel-Hall conjecture. For any rational point D on E, if $L(Q)\leq L$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001165 | MATHDB-317152

The geometric Bogomolov conjecture for nonconstant curves

摘录 / EXCERPT: Geometric Bogomolov conjecture. If C is not a constant curve, then

conjecture | open | MathDB 页面 | 原文线索

V2-001166 | MATHDB-317153

The strong effectivity conjecture for the geometric Bogomolov bound

摘录 / EXCERPT: Strong effectivity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-001167 | MATHDB-317154

Zhang's metric-graph lower-bound conjecture

摘录 / EXCERPT: Zhang's conjecture. There exists a positive constant $c(g)$ such that for every pm-graph $\overline{\Gamma}$ of genus g,

conjecture | open | MathDB 页面 | 原文线索

V2-001168 | MATHDB-317155

The conjectural constant in the metric-graph bound

摘录 / EXCERPT: Conjecture on the optimal constant. One may take

conjecture | open | MathDB 页面 | 原文线索

V2-001169 | MATHDB-317156

Strict endoscopic conjecture for Siegel modular forms of level 2

摘录 / EXCERPT: Let $\mathbb V_{l,m}$ be the local system on $\mathcal A_2[2]$ indexed by (l,m), and write ... Let $S[\Gamma_0(4),k']$ denote the motive-valued contribution associated with ellipti...

conjecture | open | MathDB 页面 | 原文线索

V2-001170 | MATHDB-317157

Yoshida-type lifting conjecture for level 2 and level 4 elliptic forms

摘录 / EXCERPT: Let $S_{j,k}(\Gamma_2[2])$ be the space of Siegel cusp forms of weight (j,k) on the principal level-2 subgroup, and let w_2 act on newforms of level 2. For new eigenforms ……

conjecture | open | MathDB 页面 | 原文线索

V2-001171 | MATHDB-317158

Expanded endoscopic conjecture for equal weights at level 2

摘录 / EXCERPT: Let $\mathbb V_{m,m}$ be the equal-weight local system on $\mathcal A_2[2]$, set $k=2m+4$, and write $\tau_{N,k}=\dim S_k(\Gamma_0(N))^{\rm new}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001172 | MATHDB-317175

Primes in irrational Beatty sequences

摘录 / EXCERPT: Let $\alpha>0$ be an irrational number and let β be a real number. The expression $\lfloor{\alpha p+\beta}$ denotes the largest integer not exceeding $\alpha p+\beta$. Primes...

conjecture | open | MathDB 页面 | 原文线索

V2-001173 | MATHDB-317186

The refined higher Brumer conjecture using Snaith's fractional Galois ideals

摘录 / EXCERPT: Let K/k be an abelian extension of number fields with Galois group G, and let S contain the places ramified in K/k. For each odd prime ℓ, let $\mathcal{J}^r(K/k)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-001174 | MATHDB-317234

Schur-type arithmetic progression conjecture for primes

摘录 / EXCERPT: Schur-type conjecture. For every $l\geq 3$, there exist monochromatic primes $p_0,p_1,p_2,\ldots,p_l$ such that $p_1,\ldots,p_l$ form an arithmetic progression with common differen...

conjecture | open | MathDB 页面 | 原文线索

V2-001175 | MATHDB-317235

Freiman's $(3k-4)$ conjecture for subsets of cyclic groups of prime order

摘录 / EXCERPT: Let p be prime and let S be a subset of the finite group $\mathbb Z/p\mathbb Z$. For a set S, write $2S=\{s_1+s_2:s_1,s_2\in S\}$. An arithmetic progression in...

conjecture | open | MathDB 页面 | 原文线索

V2-001176 | MATHDB-317252

The maximality conjecture for circle constructions

摘录 / EXCERPT: Let R have only prime factors p satisfying $p\equiv 1\pmod 4$. Consider the two plane integral point sets supplied by the circle constructions: $\operatorname{circle}(R)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-001177 | MATHDB-317253

The maximality conjecture for the $t=8$ circle construction

摘录 / EXCERPT: Let R have only prime factors p satisfying $p\equiv 1\pmod 4$, and let $\operatorname{circle}(R,8)$ be the integral point set obtained from the scaling construction by taking a...

conjecture | open | MathDB 页面 | 原文线索

V2-001178 | MATHDB-317256

Completeness of maximum integral point sets over prime-power residue rings

摘录 / EXCERPT: Let p be a prime and r a positive integer. An integral point set in $\mathbb{Z}_{p^r}^2$ is a set of points whose pairwise squared distances are quadratic residues in...

conjecture | open | MathDB 页面 | 原文线索

V2-001179 | MATHDB-317258

Unbounded integral point sets over prime residue rings

摘录 / EXCERPT: For a prime p, let $\dot{\mathcal{I}}(\mathbb{Z}_p,2)$ denote the maximum cardinality of an integral point set in the plane over $\mathbb{Z}_p$ in general position. Unboundedness...

conjecture | open | MathDB 页面 | 原文线索

V2-001180 | MATHDB-317272

Good-prime torsion conjecture for standard perverse sheaves

摘录 / EXCERPT: Let $G^\vee$ be the Langlands dual reductive group, let λ be a dominant coweight, and let ℓ be a prime that is good for $G^\vee$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-001181 | MATHDB-317280

The vanishing conjecture for localized second cohomology

摘录 / EXCERPT: Let $\Omega=\Lambda(\mathcal{G}_{\mathfrak f})$, and let $\mathfrak q$ be a height-one prime ideal of Ω containing p. Vanishing conjecture. ... This hypothesis is introduced...

conjecture | open | MathDB 页面 | 原文线索

V2-001182 | MATHDB-317293

Finite-field list conjecture for matroid representability

摘录 / EXCERPT: Let k be a prime power. The claim concerns matroids and representability over fields and finite fields $\operatorname{GF}(q)$. Finite-field list conjecture. There exists a number...

conjecture | open | MathDB 页面 | 原文线索

V2-001183 | MATHDB-317309

CFKRS-DGH conjecture for the first moment of quadratic Dirichlet L-functions

摘录 / EXCERPT: CFKRS-DGH conjecture. The first moment satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001184 | MATHDB-317310

CFKRS-DGH twisted moment conjecture for quadratic Dirichlet L-functions

摘录 / EXCERPT: CFKRS-DGH twisted moment conjecture. Uniformly for $|\operatorname{Re}(\alpha)|\ll(\log X)^{-1}$ and $|\operatorname{Im}(\alpha)|\ll(\log X)^2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-001185 | MATHDB-317361

The ABC conjecture with type ψ for number fields

摘录 / EXCERPT: ABC conjecture with type ψ. There exists a function ψ with $\psi(h)=o(h)$ such that, for every such point P,

conjecture | open | MathDB 页面 | 原文线索

V2-001186 | MATHDB-317373

Fujii-Zaharescu interval-distribution conjecture for fractional parts of zeta zeros

摘录 / EXCERPT: Let $\alpha>0$ be fixed, let $\mathbb T=\mathbb R/\mathbb Z$, and let $N(T)$ count the positive ordinates $0<\gamma\le T$ of zeros of $\zeta(s)$. Define g_α by the formula i...

conjecture | open | MathDB 页面 | 原文线索

V2-001187 | MATHDB-317374

Discrepancy asymptotic for fractional parts of zeta zeros

摘录 / EXCERPT: Let $\alpha>0$ be fixed and let $D_\alpha(T)$ be the discrepancy of the sequence $\{\alpha\gamma\}$ for $0<\gamma\le T$, with the supremum taken over intervals...

conjecture | open | MathDB 页面 | 原文线索

V2-001188 | MATHDB-317375

Uniform cancellation conjecture for exponential sums over zeta zeros

摘录 / EXCERPT: Let $A>1$ be fixed. For positive ordinates γ of zeros of $\zeta(s)$, uniform exponential-sum conjecture. Uniformly for ... we have ... This estimate would imply the interval-dis...

conjecture | open | MathDB 页面 | 原文线索

V2-001189 | MATHDB-317376

Prime number theorem conjecture in short intervals at square-root scale

摘录 / EXCERPT: Let $\psi(x)=\sum_{n\le x}\Lambda(n)$ be Chebyshev's second function. Short-interval prime-distribution conjecture. For every $\varepsilon>0$, when x is large and...

conjecture | open | MathDB 页面 | 原文线索

V2-001190 | MATHDB-317377

Fujii-Zaharescu interval-distribution conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let F be a Dirichlet L-function, let $N_F(T)$ count its positive zero ordinates $0<\gamma\le T$, and let $g_{F,\alpha}$ be the associated limiting density on...

conjecture | open | MathDB 页面 | 原文线索

V2-001191 | MATHDB-317378

Discrepancy asymptotic for zeros of Dirichlet L-functions

摘录 / EXCERPT: Let F be a Dirichlet L-function and let $D_{F,\alpha}(T)$ denote the discrepancy of the fractional parts $\{\alpha\gamma\}$ for its positive zero ordinates, with the supremum t...

conjecture | open | MathDB 页面 | 原文线索

V2-001192 | MATHDB-317379

Uniform exponential-sum conjecture for zeros of general L-functions

摘录 / EXCERPT: Let F belong to the class $\mathcal S$, let θ_F be the parameter appearing in the paper, and let γ range over positive ordinates of zeros of F. Fix $A>1$. General...

conjecture | open | MathDB 页面 | 原文线索

V2-001193 | MATHDB-317380

Murty-Perelli pair-correlation conjecture for the Selberg class

摘录 / EXCERPT: Murty-Perelli pair-correlation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001194 | MATHDB-317381

Smooth-variation conjecture for pair correlation of Selberg-class L-functions

摘录 / EXCERPT: Let $F\in\mathcal S$ and let $\mathcal D_F(x,T)$ be the normalized pair-correlation function defined above. Smooth-variation conjecture. For every $A>0$, there exists $c>0$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-001195 | MATHDB-317429

Conjecture on mean values of arithmetic subsequences of Hardy-Littlewood constants

摘录 / EXCERPT: Let C_{2r} denote the Hardy-Littlewood constant associated with the prime pair $(p,p+2r)$. An arithmetic subsequence of the index sequence $\{2r\}$ means a subsequence obtained b...

conjecture | open | MathDB 页面 | 原文线索

V2-001196 | MATHDB-317485

The dimension conjecture for generalized twisted characters

摘录 / EXCERPT: Generalized twisted-character dimension conjecture. The vector space of suitably defined generalized σ-twisted characters is $3m+1$-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-001197 | MATHDB-317516

Finiteness conjecture for the sets J of multiple harmonic sums

摘录 / EXCERPT: Finiteness conjecture. The set $J({\vec{s}}|p)$ is finite for every prime p.

conjecture | open | MathDB 页面 | 原文线索

V2-001198 | MATHDB-317517

Conjecture on 2-divisible sets of multiple harmonic sums

摘录 / EXCERPT: Conjecture on 2-divisible sets. For every positive integer d and every ${\vec{s}}\in{\mathbb N}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-001199 | MATHDB-317518

Conjectural structure of the reserved sets RJ for multiple harmonic sums

摘录 / EXCERPT: Reserved-set structure conjecture. If one of the following holds: (i) ${\vec{s}}=1$, (ii) ${\vec{s}}=(1,2,1)$, (iii) ${\vec{s}}=(2r-1,1)$ for some $r\geq1$, or (iv)...

conjecture | open | MathDB 页面 | 原文线索

V2-001200 | MATHDB-317519

Reserved density conjecture for multiple harmonic sums

摘录 / EXCERPT: Reserved density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001201 | MATHDB-317528

The asymptotic formula for the postage stamp function

摘录 / EXCERPT: Let $E(h,k)$ be the function studied in the postage stamp problem, and let ... For each fixed k, let $\underline{R}(k)\in(1/e,1)$ be the postage-stamp constant appearing in Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-001202 | MATHDB-317530

Finite intersection conjecture for special cycles and CM cycles

摘录 / EXCERPT: Let L be the relevant lattice, let U be the negative-definite subspace defining the CM cycle, and let P and N be the positive- and negative-definite lattices arising from a...

conjecture | open | MathDB 页面 | 原文线索

V2-001203 | MATHDB-317531

General Faltings height formula for arithmetic divisors and CM cycles

摘录 / EXCERPT: Let $f\in H_{1-n/2, \bar{\rho}_L}$, let $c^+(0,0)$ be its constant term, let $\hat{\mathcal{Z}}(f)$ be the associated arithmetic divisor, and let $\kappa(0,0)$ be the constant coef...

conjecture | open | MathDB 页面 | 原文线索

V2-001204 | MATHDB-317541

Katz's Lie-algebraic p-curvature conjecture

摘录 / EXCERPT: Katz's Lie-algebraic p-curvature conjecture. With this notation, $\mathfrak g$ is the smallest algebraic Lie subalgebra of $\operatorname{End}_{\mathcal O_X}V\otimes K(X)$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-001205 | MATHDB-317555

Yau-Zaslow product formula for genus-zero K3 BPS counts

摘录 / EXCERPT: Let $r_{0,h}$ denote the genus-zero BPS count associated to an effective class β on a K3 surface with norm ... Under the norm-dependence conjecture, this count is independent o...

conjecture | open | MathDB 页面 | 原文线索

V2-001206 | MATHDB-317556

The 2-adic valuation conjecture for Stirling-number-related quantities at powers of two

摘录 / EXCERPT: Power-of-two valuation conjecture. If $n=2^t$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001207 | MATHDB-317579

Subpolynomial refinement of the matrix-norm bound for the Mertens function

摘录 / EXCERPT: Let $\mu(k)$ be the Möbius function and let ... be the Mertens function. Let $\mathcal{M}=\left(\mathcal{M}_n\right)_{n\in\mathbb{N}^*}$ be the sequence of symmetric square matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-001208 | MATHDB-317593

Spacing of unit fractions

摘录 / EXCERPT: Let $r\geqslant 1$ be an integer, and let X be a positive parameter. Spacing of unit fractions. The number of $x_1,\dots,x_{2r}\in[X,2X)$ satisfying ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-001209 | MATHDB-317599

Maulik-Pandharipande quasi-modularity conjecture for reduced K3 descendents

摘录 / EXCERPT: Let S be an elliptically fibered $K3$ surface with section, with section and fiber classes $\mathbf{s},\mathbf{f}\in H^2(S,\mathbb{Z})$ spanning $\operatorname{Pic}(S)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001210 | MATHDB-317622

Moore-Witten type conjecture for generalized Donaldson invariants with massless monopoles

摘录 / EXCERPT: For $N_f=2,3$, let $\pmb{\mathrm{D}}^{N_f}_{m,2n}$ denote the generating function for the generalized $\operatorname{SO}(3)$-Donaldson invariants of $\mathbb{C}\mathrm{P}^2$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-001211 | MATHDB-317623

Cuspidal u-plane formula for generalized Donaldson invariants

摘录 / EXCERPT: Let $c=\operatorname{H}$, let $N_f=0,2,3$, and let m,n,k satisfy ... Write $\pmb{\mathrm{D}}^{N_f}_{m,2n}$ for the Donaldson invariant and $\pmb{\Phi}_{k,m,2n}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001212 | MATHDB-317630

The square-mean conjecture for cusp-form coefficients at Piatetski-Shapiro primes

摘录 / EXCERPT: Let f be a cusp form for the full modular group, let $1<c<8/7$, and let $\mathbbm{P}$ denote the set of primes. Write $\lambda_f(n)$ for the normalized Fourier coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-001213 | MATHDB-317635

Faithful invariant ideals are controlled by fixed points

摘录 / EXCERPT: Fixed-point control conjecture. If I is a Γ-invariant faithful prime ideal of kG, then I is controlled by the subgroup of Γ-fixed points G^Γ:

conjecture | open | MathDB 页面 | 原文线索

V2-001214 | MATHDB-317647

The DKSS conjecture on distinct products in finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group with $|G|>1$, and let $p(G)$ be the smallest prime divisor of $|G|$. Let $k<p(G)$ be a positive integer. Assume that $A=\{a_1,\ldots,a_k\}$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-001215 | MATHDB-317655

Lang's conjecture relating entire curves and integral sets

摘录 / EXCERPT: Lang's conjecture. There exists a non-constant holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-001216 | MATHDB-317670

Barrett and Jarvis's conjecture on the small eigenvalues of C_n

摘录 / EXCERPT: Let C_n be the matrix discussed above, with two large real eigenvalues and $\lfloor \log_2 n\rfloor-1$ remaining small nontrivial eigenvalues. For a small nontrivial eigenvalue,...

conjecture | open | MathDB 页面 | 原文线索

V2-001217 | MATHDB-317677

Koblitz's conjecture for prime orders of elliptic-curve reductions

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ of conductor N_E. For each prime p of good reduction, let $E(\mathbb{F}_p)$ denote the group of $\mathbb{F}_p$-points of the redu...

conjecture | open | MathDB 页面 | 原文线索

V2-001218 | MATHDB-317699

The relative Burnside-kernel generation conjecture

摘录 / EXCERPT: Let p be a prime, let G be a finite p-group, and let $H\cong \mathbb Z_p$. Write $\tilde G=G\times H$, let $N(G,H)$ denote the relative Burnside kernel, and for a subquotient...

conjecture | open | MathDB 页面 | 原文线索

V2-001219 | MATHDB-317701

Three-periodicity conjecture for lower central quotients of the groups Γ_T

摘录 / EXCERPT: Let $\Gamma=\Gamma_{\mathcal T}$ be one of the groups introduced in Section 4 of the cited source, associated to a prime power $q=p^3$ with $p\neq 3$. Let $\gamma_i(\Gamma)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-001220 | MATHDB-317766

Goldbach's function-field conjecture for polynomials over finite fields

摘录 / EXCERPT: Let ${\mathbf{F}_{q}}$ be a finite field of characteristic p and cardinality q, and let F be a monic polynomial in ${\mathbf{F}_{q}}[t]$ of degree n at least 2 which is e...

conjecture | open | MathDB 页面 | 原文线索

V2-001221 | MATHDB-317770

Cartan group conjecture for graded invariant factors

摘录 / EXCERPT: Let $ell\geq1$ and $d\geq0$, and let $X_{ell,d}$ be the matrix arising in the computation of Cartan invariants. Write the prime decomposition as $ell=\prod_i p_i^{r_i}$ and define...

conjecture | open | MathDB 页面 | 原文线索

V2-001222 | MATHDB-317812

Conjecture on infinitely many rational points on quintic hypersurfaces

摘录 / EXCERPT: Let $f(x)=x^5+ax^3+cx$, where $a,c\in\mathbb{Z}$ are not both zero, and let $\mathcal{V}_f$ be the hypersurface associated with the equation studied in the paper. Let T_f denote...

conjecture | open | MathDB 页面 | 原文线索

V2-001223 | MATHDB-317825

Sharpness of standard and non-standard relations for multiple polylogarithm values

摘录 / EXCERPT: Let N be a level. Call N standard if either $N=1,2,3$ or $N=p^n$ for a prime $p\ge 5$ and an integer $n\ge 1$; otherwise call it non-standard. Let $d(w,N)$ denote the dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-001224 | MATHDB-317843

Crystalline-lift criterion for weights of mod-p representations

摘录 / EXCERPT: Let $n\geq1$, let p be a prime, and let $\overline\rho:G_{\mathbb Q}\to\operatorname{GL}_n(\overline{\mathbb F}_p)$ be continuous, odd, and irreducible. Let $F(a_1,\ldots,a_n)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-001225 | MATHDB-317863

Gal's arithmetic positivity conjecture for Kerov polynomials

摘录 / EXCERPT: Let p be an odd prime, and let Σ_{p+1} be the Kerov character polynomial on a cycle of length $p+1$, with R_i denoting the free cumulants. Gal's conjecture. The expressi...

conjecture | open | MathDB 页面 | 原文线索

V2-001226 | MATHDB-317876

Margulis's orbit-closure conjecture for the three-dimensional space X_3

摘录 / EXCERPT: Special case of Margulis's conjecture. For $x\in X_3$, one of the following three options occurs: Ax is dense; Ax is closed; or Ax is contained in a closed orbit Hx of an i...

conjecture | open | MathDB 页面 | 原文线索

V2-001227 | MATHDB-317882

Conjectural slope formula for the U_p operator at genus-one primes

摘录 / EXCERPT: Let $p\in\{11,17,19\}$. For each such prime, let U_p denote the operator acting on overconvergent p-adic modular forms of weight 0, and let $s_k(j)$ be the functions presente...

conjecture | open | MathDB 页面 | 原文线索

V2-001228 | MATHDB-317891

The generalized Collatz conjecture for the $m=b-1$ family

摘录 / EXCERPT: Let b and m be the parameters defining the generalized Collatz map, and let (S_k) be the sequence obtained by iterating this map from an arbitrary positive integer. Generaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-001229 | MATHDB-317908

Lei Zhou's ternary Goldbach-type conjecture

摘录 / EXCERPT: Let p_1 and p_2 be odd primes. Lei Zhou's conjecture. Every odd integer $n\geq9$ can be represented in the form ... This is presented as a strengthening of the ternary Goldbach p...

conjecture | open | MathDB 页面 | 原文线索

V2-001230 | MATHDB-317909

Bounded-representation subsequence conjecture for the binary Goldbach-type problem

摘录 / EXCERPT: Let C_1 and C_2 be arbitrary positive numbers, and let T be a subsequence of the primes. Bounded-representation conjecture. There is no such T for which every odd integer...

conjecture | open | MathDB 页面 | 原文线索

V2-001231 | MATHDB-317910

Prime infinitude conjecture for the shifted Moser-de Bruijn sequences

摘录 / EXCERPT: For an odd integer c, define ... where $\{m_2(n)\}_{n\geq0}$ is the Moser-de Bruijn sequence characterized by unique representations $n=m_2(k)+2m_2(l)$. Prime-infinitude conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-001232 | MATHDB-317911

Pomerance's $e^{-\gamma}J_0(x)$ threshold conjecture

摘录 / EXCERPT: Let $a_1,a_2,\ldots$ be independent uniform random integers in $\{1,\ldots,x\}$, and let T be the least t such that some subsequence of $a_1,\ldots,a_t$ has product equal to a...

conjecture | open | MathDB 页面 | 原文线索

V2-001233 | MATHDB-317933

The extra-divergence Duffin-Schaeffer conjecture

摘录 / EXCERPT: Extra-divergence Duffin-Schaeffer conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-001234 | MATHDB-317961

The conditional-convergence conjecture for coefficients of L-functions

摘录 / EXCERPT: Conditional-convergence conjecture. For an entire L-function,

conjecture | open | MathDB 页面 | 原文线索

V2-001235 | MATHDB-317962

The q-denominators conjecture for q-analogues of Dirichlet's beta function

摘录 / EXCERPT: The q-denominators conjecture. There exist real numbers β and γ, depending only on A and r, such that

conjecture | open | MathDB 页面 | 原文线索

V2-001236 | MATHDB-317964

Gross's conjecture on non-solvable Galois extensions ramified at a single prime

摘录 / EXCERPT: Let p be a prime number. A finite Galois extension K of $\mathbb{Q}$ is ramified at p only if it is unramified at every prime other than p. Gross's conjecture. For every pr...

conjecture | open | MathDB 页面 | 原文线索

V2-001237 | MATHDB-317969

Conjecture on the canonical equivalence for coherent overconvergent F-isocrystals

摘录 / EXCERPT: Canonical equivalence conjecture. The canonical functor

conjecture | open | MathDB 页面 | 原文线索

V2-001238 | MATHDB-317970

Conjecture on the equivalence of overconvergent and surcoherent F-isocrystals

摘录 / EXCERPT: Equivalence conjecture. There is a canonical equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-001239 | MATHDB-318002

Higher-moment conjecture for products of two residues in short intervals

摘录 / EXCERPT: Let q be a modulus, let c be coprime to q, and let L be a positive integer. For each a_1,a_2, consider the number of pairs (q_1,q_2) with $a_i<q_i\leq a_i+L$, with the...

conjecture | open | MathDB 页面 | 原文线索

V2-001240 | MATHDB-318035

Lang's lower-bound conjecture for canonical heights on elliptic curves

摘录 / EXCERPT: Lang's conjecture. There is a positive constant $c(k)$, depending only on k, such that

conjecture | open | MathDB 页面 | 原文线索

V2-001241 | MATHDB-318063

The two-thirds conjecture for ternary cyclotomic coefficients

摘录 / EXCERPT: Two-thirds conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001242 | MATHDB-318100

The higher Apéry-like square-sum supercongruence for \(\tilde{J}_2\)

摘录 / EXCERPT: Square-sum supercongruence. For any odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-001243 | MATHDB-318101

Kimoto-Osburn's congruence conjecture for \(\tilde{J}_2\)

摘录 / EXCERPT: Kimoto-Osburn conjecture. For any odd prime p and any $m,r\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-001244 | MATHDB-318102

The ASD-type congruence conjecture for \(\tilde{J}_2\)

摘录 / EXCERPT: ASD-type congruence conjecture. For any odd prime p and $m,r\in\mathbb{Z}_{>0}$ with m odd,

conjecture | open | MathDB 页面 | 原文线索

V2-001245 | MATHDB-318103

The higher Apéry-like sum congruence for \(\tilde{J}_{2k}\)

摘录 / EXCERPT: Higher Apéry-like sum conjecture. For any odd prime p and any integer $k\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-001246 | MATHDB-318106

Local lattice-count identity for the Jacquet-Rallis fundamental lemma

摘录 / EXCERPT: Let (a,b) be a strongly regular semisimple collection of invariants with $a_i,b_i\in\mathcal{O}_E^{\sigma=(-1)^i}$, allowing arbitrary $b_0\in\mathcal{O}_F$. Let $\Delta_{a,b}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-001247 | MATHDB-318141

Average-time complexity conjecture for perfect sorting by reversals

摘录 / EXCERPT: Let n be the size of a random permutation, and let $T_{n,p}$ denote the number of strong interval trees with n leaves and p prime vertices. Define ... where T_n is the total...

conjecture | open | MathDB 页面 | 原文线索

V2-001248 | MATHDB-318146

Sun's prime-Fibonacci-Fibonacci representation conjecture

摘录 / EXCERPT: Let $F_0=0$, $F_1=1$, and $F_{i+1}=F_i+F_{i-1}$ for $i=1,2,3,\ldots$ be the Fibonacci sequence. A Fibonacci number is positive when its index is positive and its value is positive....

conjecture | open | MathDB 页面 | 原文线索

V2-001249 | MATHDB-318147

Sun's prime-Pell-number representation conjecture

摘录 / EXCERPT: Let $P_0=0$, $P_1=1$, and $P_{n+1}=2P_n+P_{n-1}$ for $n=1,2,3,\ldots$ be the Pell sequence. Sun's prime-Pell-number conjecture. Every integer $n>5$ can be written as the sum of an...

conjecture | open | MathDB 页面 | 原文线索

V2-001250 | MATHDB-318148

Hou and Zeng's prime-Fibonacci-Catalan representation conjecture

摘录 / EXCERPT: Let F_n denote the Fibonacci numbers, and let ... be the nth Catalan number. A Fibonacci number is positive when it is positive as an integer. Hou and Zeng's prime-Fibonacci-Cata...

conjecture | open | MathDB 页面 | 原文线索

V2-001251 | MATHDB-318149

Sun's prime-Fibonacci-power representation conjectures

摘录 / EXCERPT: Let F_n denote the Fibonacci sequence. A Fibonacci number is positive when its value is positive, and an odd Fibonacci number is one not divisible by 2. Sun's prime-Fibonacci-p...

conjecture | open | MathDB 页面 | 原文线索

V2-001252 | MATHDB-318150

Sun's Lucas and mixed Fibonacci-Lucas representation conjectures

摘录 / EXCERPT: Let F_n and L_n denote the Fibonacci and Lucas sequences, respectively. A sequence term is positive when its value is positive, and a term is odd when it is not divisible by...

conjecture | open | MathDB 页面 | 原文线索

V2-001253 | MATHDB-318165

Bogomolov's conjecture for non-isotrivial fibrations

摘录 / EXCERPT: Let X be a smooth projective surface over a field k, let Y be a smooth projective curve over k, and let $f:X\to Y$ be a semistable fibration whose smooth generic fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-001254 | MATHDB-318167

Zhang's first conjecture on the phi invariant of polarized metrized graphs

摘录 / EXCERPT: Let $f:X\to Y$ be as above, let $CV(f)$ denote the set of singular fibers, and for $y\in CV(f)$ let Γ_y be the polarized metrized graph of the fiber. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-001255 | MATHDB-318168

Zhang's second conjecture on the lambda invariant of polarized metrized graphs

摘录 / EXCERPT: Let $f:X\to Y$ be a semistable fibration, let $CV(f)$ be the set of singular fibers, and let Γ_y be the polarized metrized graph of the fiber over $y\in CV(f)$. For each nod...

conjecture | open | MathDB 页面 | 原文线索

V2-001256 | MATHDB-318169

Zhang's unified conjecture for polarized metrized graphs

摘录 / EXCERPT: Let Γ be an irreducible polarized metrized graph of genus $\bar{g}\geq2$, with total length $\ell(\Gamma)$, invariants $a(\Gamma)$ and $\epsilon(\Gamma)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001257 | MATHDB-318170

Grothendieck-Gillet-Soulé height-positivity conjecture for the Gross-Schoen cycle

摘录 / EXCERPT: Let k be a number field or a function field of positive characteristic, and let Δ_ξ be the canonical Gross-Schoen cycle associated with the relevant curve or fibration....

conjecture | open | MathDB 页面 | 原文线索

V2-001258 | MATHDB-318174

The nonadequacy conjecture for the family $K(9_{42},8_{20},n)$

摘录 / EXCERPT: For each natural number n, let $K(9_{42},8_{20},n)$ denote the link family obtained by the prime tangle construction described in the paper. A diagram is adequate when it satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-001259 | MATHDB-318192

The field-of-definition conjecture for differentially extremal forms

摘录 / EXCERPT: Let f be a differentially extremal element of the algebra $\widetilde{M}$ of Drinfeld quasi-modular forms. A form is defined over a field $L\subset C$ if $f\in L[E,g,h]$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-001260 | MATHDB-318244

The bu Whitehead conjecture

摘录 / EXCERPT: bu Whitehead conjecture. The complex is exact at the prime p.

conjecture | open | MathDB 页面 | 原文线索

V2-001261 | MATHDB-318245

The reduced bu Whitehead conjecture

摘录 / EXCERPT: Reduced bu Whitehead conjecture. The complex is exact at the prime p.

conjecture | open | MathDB 页面 | 原文线索

V2-001262 | MATHDB-318246

Delooping conjecture for the reduced bu Whitehead complex

摘录 / EXCERPT: Work at a prime p, and consider the connecting maps in the Taylor tower evaluated at ${\mathbb C}^0$: ... A contracting homotopy for the complex of spectra ${\overline{M}}(k)$ woul...

conjecture | open | MathDB 页面 | 原文线索

V2-001263 | MATHDB-318260

Distinctness conjecture for Chebyshev knot parametrizations

摘录 / EXCERPT: Let T_n denote the degree-n Chebyshev polynomial of the first kind. Let i,j be relatively prime positive integers, and let k,ℓ belong to the semigroup denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-001264 | MATHDB-318274

Noncyclotomic growth conjecture for solutions of q-difference equations

摘录 / EXCERPT: Let K be the coefficient field, let ${\mathcal P}_f$ be its finite places, and let ${\mathcal C}$ be the cyclotomic places. Suppose that ... is a solution of a q-difference equat...

conjecture | open | MathDB 页面 | 原文线索

V2-001265 | MATHDB-318277

The 2-adic shift conjecture for involution numbers

摘录 / EXCERPT: 2-adic shift conjecture. There is a 2-adic integer ρ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-001266 | MATHDB-318278

Average Dirichlet-series conjecture for quadratic prime pairs

摘录 / EXCERPT: Average quadratic prime-pair Dirichlet-series conjecture. For $\sigma>1/4$ and $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-001267 | MATHDB-318286

VIGRE conjecture on the complexity of Specht modules for hook partitions

摘录 / EXCERPT: Let p be a prime and let μ be a partition. Write S^μ for the Specht module corresponding to μ, and let the p-weight of μ be the usual number of removable p-ho...

conjecture | open | MathDB 页面 | 原文线索

V2-001268 | MATHDB-318290

The growth conjecture for the impulse response of the zeta-function system

摘录 / EXCERPT: Let ... be the transfer function, and let $g(t)=\mathcal{L}^{-1}\{G(s)\}(t)$ be the corresponding impulse response. Growth conjecture. For some $k\in\{0,1,2\}$, the impulse response...

conjecture | open | MathDB 页面 | 原文线索

V2-001269 | MATHDB-318312

Relative monodromy conjecture for principal monodromy

摘录 / EXCERPT: Let F be a number field, let K be a completion of F, and let $Z_{\Phi}(s,\chi,f_l,V_{l-1},K)$ be the local zeta function associated with the complex non-degenerate polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-001270 | MATHDB-318313

The direct-sum conjecture for special values of Witten multiple zeta values attached to \(\mathfrak{sl}_4\)

摘录 / EXCERPT: Let $\MZV(w,\le l)$ be the $\Q$-vector space generated by multiple zeta values of weight w and depth at most l, and let $\gz_\slfour$ denote the Witten multiple zeta function a...

conjecture | open | MathDB 页面 | 原文线索

V2-001271 | MATHDB-318319

The generic nonvanishing conjecture for definite Heegner-family L-values

摘录 / EXCERPT: Let $f_\mathfrak p^\dagger$ be the twisted specialization at an arithmetic prime $\mathfrak p$ of weight $k_\mathfrak p\ge2$, let χ be a fixed character of $\widetilde G_c$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-001272 | MATHDB-318333

Carlsson's conjecture on completed K-theory of semilinear representations

摘录 / EXCERPT: Let F) be a field, let k be an algebraically closed subfield of $\overline{F}$, and let G_F be the absolute Galois group of F. For a prime ℓ different from the charact...

conjecture | open | MathDB 页面 | 原文线索

V2-001273 | MATHDB-318336

Graham's zero-sum subsequence conjecture

摘录 / EXCERPT: Let p be a prime, let C_p be the cyclic group of order p, and let S be a sequence over C_p of length p. A zero-sum subsequence is a subsequence whose terms sum to 0 i...

conjecture | open | MathDB 页面 | 原文线索

V2-001274 | MATHDB-318337

Paramodular eigenform contribution to cuspidal cohomology of \SL_4(\mathbb Z)

摘录 / EXCERPT: Paramodular cohomology conjecture. The cuspidal cohomology $H^5_{\mathrm{cusp}}(\Gamma_0(N);\mathbb C)$ contains a $2d_h$-dimensional subspace spanned by Hecke eigenclasses. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-001275 | MATHDB-318346

Paulin's geometric level raising conjecture for the eigencurve

摘录 / EXCERPT: Fix distinct primes p and l, and an integer N coprime to pl. Let $\mathcal{E}$ be the cuspidal eigencurve of tame level $\Gamma_0(Nl)$, parametrising overconvergent cuspida...

conjecture | open | MathDB 页面 | 原文线索

V2-001276 | MATHDB-318354

Guo-Jouhet-Zeng gcd conjecture for alternating binomial sums

摘录 / EXCERPT: For positive integers m and n, define ... Here $\gcd(a_1,a_2,\ldots)$ denotes the greatest common divisor of the indicated integers. Guo-Jouhet-Zeng's gcd conjecture. ... Calkin's...

conjecture | open | MathDB 页面 | 原文线索

V2-001277 | MATHDB-318355

Guo-Jouhet-Zeng divisibility conjectures for alternating binomial products

摘录 / EXCERPT: Guo-Jouhet-Zeng's divisibility conjecture. For all positive integers r,s,t,n,

conjecture | open | MathDB 页面 | 原文线索

V2-001278 | MATHDB-318368

The stochastic-process conjecture for the zeros of the Riemann zeta function

摘录 / EXCERPT: Let the zeros of the Riemann zeta function be the complex numbers ρ satisfying $\zeta(\rho)=0$. A single-process conjecture. The zeros of the ζ function are derived from...

conjecture | open | MathDB 页面 | 原文线索

V2-001279 | MATHDB-318392

Palmieri's nilpotence conjecture for the power operation on Ext

摘录 / EXCERPT: Let p be a prime, let P_* be the polynomial part of the dual Steenrod algebra, and let P be its dual. Consider $\operatorname{Ext}_P^{*,*}(\mathbb{F}_p,\mathbb{F}_p)$ and the...

conjecture | open | MathDB 页面 | 原文线索

V2-001280 | MATHDB-318393

The one-step nilpotence conjecture for Palmieri's power operation

摘录 / EXCERPT: Let p be a prime and let $\mathcal{P}^0$ be the endomorphism of $\operatorname{Ext}_P^{*,*}(\mathbb{F}_p,\mathbb{F}_p)$ induced by the pth-power map on the polynomial part of t...

conjecture | open | MathDB 页面 | 原文线索

V2-001281 | MATHDB-318412

The linear bound conjecture for prime-chain counts

摘录 / EXCERPT: Let $N(x;p)$ denote the number of prime chains beginning at the prime p whose final-to-initial ratio is at most x. Prime-chain counting conjecture. One has ... The paper notes th...

conjecture | open | MathDB 页面 | 原文线索

V2-001282 | MATHDB-318413

The refined distribution conjecture for Pratt-tree height

摘录 / EXCERPT: For a prime p, let $H(p)$ be the height of its Pratt tree, let $\log_2 p=\log\log p$ and $\log_3 p=\log\log\log p$, and write $\pi(x)$ for the number of primes at most x. Refin...

conjecture | open | MathDB 页面 | 原文线索

V2-001283 | MATHDB-318414

The prime-chain tuples conjecture

摘录 / EXCERPT: A prime chain is a sequence of primes $(p_1,\ldots,p_k)$ in which each successive prime has the form $p_{j+1}=mp_j+1$ for a common positive integer m. Prime-chain tuples conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-001284 | MATHDB-318415

The convergence conjecture for the branching-random-walk maximum

摘录 / EXCERPT: Let B_n be the maximum at generation n in the branching random walk considered in the paper, and let b_n be its deterministic centering term. Branching-random-walk convergenc...

conjecture | open | MathDB 页面 | 原文线索

V2-001285 | MATHDB-318425

Uniform multiplication-table congruence bound

摘录 / EXCERPT: Let $\delta\in(0,1)$ be fixed, let q and c be integers with $(c,q)=1$, and let N be positive. Uniform multiplication-table congruence conjecture. One has ... uniformly for...

conjecture | open | MathDB 页面 | 原文线索

V2-001286 | MATHDB-318445

Unique prime decomposition conjecture for knotted graphs

摘录 / EXCERPT: A knotted graph is a compact graph embedded in a 3-manifold, considered up to homeomorphism, and a connected sum is the usual operation obtained by joining knotted graphs along s...

conjecture | open | MathDB 页面 | 原文线索

V2-001287 | MATHDB-318446

Unique prime decomposition conjecture for 3-dimensional orbifolds

摘录 / EXCERPT: A 3-dimensional orbifold is a pair (M,G_φ), where M is a 3-manifold and G_φ is a colored graph with vertices of valence at most 3, univalent vertices on...

conjecture | open | MathDB 页面 | 原文线索

V2-001288 | MATHDB-318470

Conjecture on tamely ramified adequate extensions

摘录 / EXCERPT: Conjecture on tamely ramified adequate extensions. There is a K-adequate Galois G-extension that is tamely ramified if and only if, for every prime p dividing $|G|$, a p-Sy...

conjecture | open | MathDB 页面 | 原文线索

V2-001289 | MATHDB-318471

The plus/minus Selmer characteristic-ideal conjecture

摘录 / EXCERPT: Let f be a modular form in the setting above, let θ be a character of the torsion subgroup of the cyclotomic Galois group, and let $\operatorname{Col}^{\pm,\theta}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-001290 | MATHDB-318487

Finite-translate perfect-squares conjecture

摘录 / EXCERPT: Let $\textsc{Squares}$ denote the set of perfect squares in $\mathbb Z$. Finite-translate perfect-squares conjecture. There exists a positive integer k such that there are no set...

open_problem | open | MathDB 页面 | 原文线索

V2-001291 | MATHDB-318489

Cohomological proficiency conjecture for universal covers

摘录 / EXCERPT: Let S be a finite simple group, let p be a prime, let $\tilde{S}$ be its universal cover, and let M be a nontrivial irreducible $\mathbb F_p\tilde{S}$-module. Cohomological p...

conjecture | open | MathDB 页面 | 原文线索

V2-001292 | MATHDB-318490

Second-cohomology bound for finite quasisimple groups

摘录 / EXCERPT: Let S be a finite quasisimple group, let p be a prime, let $\tilde{S}$ be the universal cover appearing in the source, and let M be an $\mathbb F_pS$-module. Second-cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-001293 | MATHDB-318531

The HS-L conjecture for Hilbert-Speiser and Leopoldt fields of type G

摘录 / EXCERPT: HS-L(K,G) conjecture. K is a Hilbert-Speiser field of type G if and only if K is a Leopoldt field of type G. This generalizes the equivalence of the unrestricted Hilbert-...

conjecture | open | MathDB 页面 | 原文线索

V2-001294 | MATHDB-318532

The modified HS-L conjecture for Hilbert-Speiser fields of type G

摘录 / EXCERPT: Let K be a number field and let G be a finite abelian group. For a G-Galois extension L/K, let $\mathcal{A}_{L/K}$ be the associated order and let ... be the adjusted ring of...

conjecture | open | MathDB 页面 | 原文线索

V2-001295 | MATHDB-318540

The genus-one Hasse principle violation conjecture

摘录 / EXCERPT: Genus-one Hasse principle violation conjecture. For every global field k, there exists a genus one curve $C_{/k}$ which violates the Hasse Principle. Equivalently, there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-001296 | MATHDB-318541

The abundance conjecture for torsion in Shafarevich-Tate groups

摘录 / EXCERPT: Abundance conjecture for Shafarevich-Tate torsion. There exists an elliptic curve $E_{/k_0}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-001297 | MATHDB-318542

The persistent genus-one Hasse principle violation conjecture over global field extensions

摘录 / EXCERPT: Persistent genus-one Hasse principle violation conjecture. For every global field k, there is a genus one curve C defined over k_0 such that C violates the Hasse Principle...

conjecture | open | MathDB 页面 | 原文线索

V2-001298 | MATHDB-318550

Rational generating-function conjecture for maximum independent sets of de Bruijn graphs

摘录 / EXCERPT: Rational generating-function conjecture. If D is a fixed odd prime number, then this exponential generating function is the ratio of two polynomials, each of degree $D-1$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001299 | MATHDB-318567

The even-prime cohomology conjecture for $(2l,2,1^q)$

摘录 / EXCERPT: Let $l\in\mathbb Z_{>0}$ and $q\in\mathbb Z_{\geq 0}$, and let $\mathcal C_2^2$ be the second cohomology graph at $p=2$. Set $\lambda=(2l,2,1^q)$. The even-prime cohomology conject...

conjecture | open | MathDB 页面 | 原文线索

V2-001300 | MATHDB-318570

The solvable base-change and prescribed type conjecture for Hilbert eigenforms

摘录 / EXCERPT: Let F be a totally real field, let f be a cuspidal parallel weight two Hilbert eigenform with coefficients in $\overline{\mathbb{Q}}_p$, unramified outside a finite set S, le...

conjecture | open | MathDB 页面 | 原文线索

V2-001301 | MATHDB-318578

The improved bound conjecture for primes between consecutive squares

摘录 / EXCERPT: For a positive integer n, let $leg(n)$ denote the number of primes between n^{2} and $(n+1)^{2}$. The improved bound conjecture for $leg(n)$. ... The source proposes this as a ti...

conjecture | open | MathDB 页面 | 原文线索

V2-001302 | MATHDB-318579

The generalized Bertrand conjecture

摘录 / EXCERPT: The generalized Bertrand conjecture. If $n\geq a$, then there are at least $k-1$ primes between n and kn. This conjecture generalizes Bertrand's postulate, which is the case...

conjecture | open | MathDB 页面 | 原文线索

V2-001303 | MATHDB-318580

The upper-bound conjecture for primes between n and kn

摘录 / EXCERPT: Let k be a positive integer, and let n be a positive integer. The upper-bound conjecture for primes between n and kn. The number of primes between n and kn is at most ……

conjecture | open | MathDB 页面 | 原文线索

V2-001304 | MATHDB-318584

Uniqueness of the base-3 characterization of repunit primes in Cantor sets

摘录 / EXCERPT: Uniqueness conjecture. It is possible to characterize base-N repunit primes as precisely the primes whose reciprocals belong to $\mathcal{C}_N$ only when $N=3$; for every...

conjecture | open | MathDB 页面 | 原文线索

V2-001305 | MATHDB-318609

Non-finite generation of Frobenius maps for the determinantal ring

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of prime characteristic p, let $R=\mathbb{K}[\![x,y,z,u,v,w]\!]$, and let I be the ideal generated by the 2×2 minors of the matrix ... Set...

conjecture | open | MathDB 页面 | 原文线索

V2-001306 | MATHDB-318624

Navarro conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group, let p be a prime dividing $|G|$, let D be a Sylow p-subgroup of G, and let $N_G(D)$ be the normalizer of D in G. Let $\operatorname{Irr}(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001307 | MATHDB-318625

Degree and leading-term conjecture for normal zeta functions of nilpotent Lie rings

摘录 / EXCERPT: Let L be a class-c-nilpotent Lie ring of rank n, with upper central series $(Z_i(L))_{i=0}^c$. Set $n_i:=\operatorname{rk}(L/Z_i(L))$, so that $n_0=n=\operatorname{rk}(L)$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-001308 | MATHDB-318626

Paajanen's abscissa formula for normal zeta functions of class-two Lie rings

摘录 / EXCERPT: Let L be a class-2-nilpotent Lie ring with $\operatorname{rk}(L/L')=d$, $\operatorname{rk}(Z(L))=m$, and $\operatorname{rk}(L/Z(L))=r$. Let $\alpha^{\triangleleft}$ be the absc...

conjecture | open | MathDB 页面 | 原文线索

V2-001309 | MATHDB-318628

The differential nonvanishing conjecture for simple Jacobians

摘录 / EXCERPT: Let $C/{\rm \a0Q}$ be a curve of genus $g \ge 2$ whose Jacobian J is simple over ${\rm \a0Q}$ and whose MordellWeil rank r satisfies $r<g$. For a prime p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-001310 | MATHDB-318629

The deep-information differential nonvanishing conjecture

摘录 / EXCERPT: Let $C/{\rm \a0Q}$ be a curve of genus $g \ge 2$ whose Jacobian J is simple and whose MordellWeil rank r satisfies $r<g$. For a prime p, let $\Omega(C_{{\rm \a0Q}_p})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-001311 | MATHDB-318630

The Topological Monodromy Conjecture for hypersurfaces

摘录 / EXCERPT: Let $f\in\mathbf{C}[x_1,\ldots,x_n]$ be a non-constant polynomial with $f(0)=0$, let $\mu:Y\to\mathbf{C}^n$ be a log resolution of $f^{-1}(0)$, and write ... for its local topologica...

conjecture | open | MathDB 页面 | 原文线索

V2-001312 | MATHDB-318672

Baker-Schmidt conjecture for the Hausdorff dimension of polynomial approximation sets

摘录 / EXCERPT: Let $n\geqslant 2$ be an integer, let $0<\mu<\frac{n+1}3$, and let $w>n-2\mu$. Define ... where $H(P)$ is the absolute height of P, and let $\dim$ denote Hausdorff dimension. Baker...

conjecture | open | MathDB 页面 | 原文线索

V2-001313 | MATHDB-318674

Averaged Sato-Tate conjecture for weight-one newforms

摘录 / EXCERPT: Averaged Sato-Tate conjecture. The measures μ_q converge weakly as $q\to\infty$ through primes congruent to 3 modulo 4; equivalently, for every continuous function...

conjecture | open | MathDB 页面 | 原文线索

V2-001314 | MATHDB-318689

Metric continued-fraction denominator conjecture

摘录 / EXCERPT: Metric continued-fraction denominator conjecture. The set $\mathcal{D}(\alpha)\cap\mathcal{Q}$ is infinite for almost all irrational points α in $\mathbb{R}/\mathbb{Z}$ if a...

conjecture | open | MathDB 页面 | 原文线索

V2-001315 | MATHDB-318697

p-adic Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let p be a prime and let $\psi:\mathbb{N}\rightarrow\mathbb{R}$ be non-negative. For each $n\in\mathbb{N}$, let $\mathcal{E}_n\subseteq\mathbb{Z}_p$ be the union of the p-adic...

conjecture | open | MathDB 页面 | 原文线索

V2-001316 | MATHDB-318714

Periodicity and Thue-Morse structure of the sequences associated with B_a and C_a

摘录 / EXCERPT: Periodicity conjecture. The sequence $\{\gamma_a(n)\}$ is periodic. If $2^m\mathrel{\|}a$, then its minimal period has 2^{m+1} terms. If a is evil, this period consists of the...

conjecture | open | MathDB 页面 | 原文线索

V2-001317 | MATHDB-318715

Conjecture on convergents to the Thue-Morse constant

摘录 / EXCERPT: Conjecture on convergents. For every $n\geq1$, the fraction $\tau_n=s_n/F_n$ is a convergent in the continued fraction expansion of τ.

conjecture | open | MathDB 页面 | 原文线索

V2-001318 | MATHDB-318725

Quadratic irrationality conjecture for the parameter tau

摘录 / EXCERPT: Let z and c be the parameters occurring in the paper's hypergeometric expansion, and let τ be the associated parameter. Quadratic irrationality conjecture. If z and c...

conjecture | open | MathDB 页面 | 原文线索

V2-001319 | MATHDB-318726

Algebraicity conjecture from rational k and j

摘录 / EXCERPT: Let k and j be the associated quantities in the paper's hypergeometric expansion, and let z, a, b, and c be its parameters. Algebraicity conjecture. If k and j are...

conjecture | open | MathDB 页面 | 原文线索

V2-001320 | MATHDB-318736

Pan's conjecture on distinct prime factors of shifted power expressions

摘录 / EXCERPT: Let $comega(m)$ denote the number of distinct prime factors of m. Fix a large $K>0$ and a small $cepsilon>0$. For sufficiently large x, consider integers $n":[x,(1+K^{-1})x]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001321 | MATHDB-318780

The bounded negativity conjecture for blow-ups of the plane at generic points

摘录 / EXCERPT: Bounded negativity conjecture for generic blow-ups. Then $C^2\ge -1$ for every prime divisor C, with equality if and only if C is a smooth rational curve with $K_X\cdot C=-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-001322 | MATHDB-318813

Borel's conjecture on binary string complexity of algebraic numbers

摘录 / EXCERPT: Let $r\in(0,1)$ be an irrational algebraic real, and let $p_r(m)$ denote the number of distinct binary words of length m occurring in the binary expansion of r. Borel's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-001323 | MATHDB-318828

The generalized Kummer conjecture for cyclotomic fields

摘录 / EXCERPT: Generalized Kummer's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001324 | MATHDB-318848

The Iwasawa main conjecture for totally real fields

摘录 / EXCERPT: Let p, F, and F_{∞}/F be as above. The p-adic zeta function $\xi_{F_{\infty}/F}$ is the uniquely existing element satisfying the interpolation formula, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001325 | MATHDB-318883

Prime-depleted Szpiro conjecture for elliptic curves

摘录 / EXCERPT: Prime-depleted Szpiro conjecture. There exist an integer $J\ge0$ and a constant C_3, depending only on K, such that for all elliptic curves E/K,

conjecture | open | MathDB 页面 | 原文线索

V2-001326 | MATHDB-318884

Prime-depleted ABC conjecture

摘录 / EXCERPT: Prime-depleted ABC-conjecture. There exist an integer $J\ge0$ and a constant C_5 such that

conjecture | open | MathDB 页面 | 原文线索

V2-001327 | MATHDB-318888

Quantitative refinement of Pillai's conjecture

摘录 / EXCERPT: Let $\varepsilon>0$, and let a,b,x,y be positive integers with $x,y\ge 2$ and $a^x\ne b^y$. Quantitative Pillai conjecture. There exists a constant $\kappa(\varepsilon)>0$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-001328 | MATHDB-318889

Hall's conjecture on the difference between a cube and a square

摘录 / EXCERPT: Let x,y be positive integers with $x^3\ne y^2$. Hall's conjecture. There exists an absolute constant $C>0$ such that ... This is a strengthening of the corresponding ε-...

conjecture | open | MathDB 页面 | 原文线索

V2-001329 | MATHDB-318890

Weak Hall's conjecture

摘录 / EXCERPT: Let x,y be positive integers with $x^3\ne y^2$. Weak Hall's conjecture. There exists an absolute constant $\kappa>0$ such that ... This weaker form replaces Hall's exponent $1/6$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-001330 | MATHDB-318891

Ideal Waring's conjecture

摘录 / EXCERPT: Ideal Waring conjecture. For every $k\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-001331 | MATHDB-318892

Conjecture on the occurrence of every digit in algebraic irrational expansions

摘录 / EXCERPT: Let x be a real irrational algebraic number, let $g\geqslant3$ be a positive integer, and let a be an integer satisfying $0\leqslant a\leqslant g-1$. The g-ary expansion of...

conjecture | open | MathDB 页面 | 原文线索

V2-001332 | MATHDB-318893

Borel's normality conjecture for algebraic irrational numbers

摘录 / EXCERPT: Let x be a real irrational algebraic number and let $g\geqslant2$ be a positive integer. A real number is normal in base g if it is simply normal in base g^m for every...

conjecture | open | MathDB 页面 | 原文线索

V2-001333 | MATHDB-318901

Rudnick-Sarnak-Heath-Brown conjecture on pair correlation of quadratic fractional parts

摘录 / EXCERPT: Rudnick-Sarnak-Heath-Brown conjecture. Assume β is Diophantine. Then the pair correlation for the fractional parts of $n^2\beta$ is Poissonian.

conjecture | open | MathDB 页面 | 原文线索

V2-001334 | MATHDB-318902

Conjecture on averaged divisor sums in arithmetic progressions

摘录 / EXCERPT: Averaged divisor-sum conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-001335 | MATHDB-318903

Heath-Brown's upper-bound conjecture for truncated divisor sums

摘录 / EXCERPT: Heath-Brown's upper-bound conjecture. Uniformly for $(u,q)=1$ and $q\text{?}N^{2-\text{?}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-001336 | MATHDB-318935

Analytic continuation of Dirichlet L series via nonstandard limits

摘录 / EXCERPT: Let $a_1,a_2,a_3,\ldots$ be a sequence of complex numbers, and let $\{b_n\}$ be its partial sum sequence, assumed to be bounded. Choose a hyperreal number system and define...

conjecture | open | MathDB 页面 | 原文线索

V2-001337 | MATHDB-318972

The regular-polytope conjecture for the index-two extensions of L_2(q)

摘录 / EXCERPT: Let q be an odd square prime power. Let $L_2(q)$ be the simple projective special linear group, and let b and c be the outer automorphisms described above, with...

conjecture | open | MathDB 页面 | 原文线索

V2-001338 | MATHDB-318985

Prime-order index conjecture with a repetition-length bound

摘录 / EXCERPT: Let Z_p be the cyclic group of p elements, let S be a sequence of p elements in Z_p, and let $h(S)$ denote the maximal repetition value of S. A subsequence has index...

conjecture | open | MathDB 页面 | 原文线索

V2-001339 | MATHDB-319010

A 12l upper bound for the height of the classical modular polynomial

摘录 / EXCERPT: Height upper-bound conjecture. For every prime $l>30$,

conjecture | open | MathDB 页面 | 原文线索

V2-001340 | MATHDB-319011

The 11.8 lower bound for the asymptotic height of classical modular polynomials

摘录 / EXCERPT: Asymptotic height lower-bound conjecture. For prime l,

conjecture | open | MathDB 页面 | 原文线索

V2-001341 | MATHDB-319014

Darmon's refined class number formula for real quadratic fields

摘录 / EXCERPT: Suppose F is a real quadratic field and $K_n=F(\boldsymbol{\mu}_n)$, where n is prime to the conductor of $F/\mathbf{Q}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001342 | MATHDB-319042

Mean-square comparison conjecture for prime progressions

摘录 / EXCERPT: Let $E(x;q,a)$ be the discrepancy ... where $\pi_{q,a}(x)$ counts primes at most x congruent to a modulo the prime q. Mean-square comparison conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-001343 | MATHDB-319053

Zywina's congruence-obstruction criterion for prime orders of elliptic curves

摘录 / EXCERPT: Let K be a number field, let E be an elliptic curve over K, and let S_E be its set of primes of bad reduction. For a positive integer t, and for each prime ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-001344 | MATHDB-319054

Goncharov's regulator conjecture for higher arithmetic Chow groups

摘录 / EXCERPT: Let X be an equidimensional compact complex algebraic manifold. Write $K_n(X)_{\operatorname{\mathbb{Q}}}$ for rational K-theory, $CH^{p}_{s}(X,n)_{\operatorname{\mathbb{Q}}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-001345 | MATHDB-319070

The precise symmetry conjecture for R-primes and L-primes

摘录 / EXCERPT: Let $\mathbf{R}$ and $\mathbf{L}$ be the sequences of $\mathbf{R}$-primes and $\mathbf{L}$-primes, respectively, and let $\mathbf{R}_n$ and $\mathbf{L}_n$ denote their n-th terms...

conjecture | open | MathDB 页面 | 原文线索

V2-001346 | MATHDB-319078

Kösters' shifted moments conjecture for the Riemann zeta function

摘录 / EXCERPT: Let k be a positive integer, let T tend to infinity, and let α_1,α_2 be real shifts. Write $M_{(k,k)}(T, (\alpha_1, \alpha_2))$ for the shifted moment measuring the...

conjecture | open | MathDB 页面 | 原文线索

V2-001347 | MATHDB-319093

The 3x+1 maximum excursion constant conjecture

摘录 / EXCERPT: 3x+1 maximum excursion constant conjecture. The maximum excursion constant ρ is finite and

conjecture | open | MathDB 页面 | 原文线索

V2-001348 | MATHDB-319134

Sun-Tauraso q-congruences for central q-binomial coefficients

摘录 / EXCERPT: Let p be a prime, $a\geq1$, let $\Phi_{p^a}(q)$ be the p^ath cyclotomic polynomial, and let $\left(\frac{p^a}{3}\right)$ denote the Legendre symbol. Sun-Tauraso q-conjecture. O...

conjecture | open | MathDB 页面 | 原文线索

V2-001349 | MATHDB-319135

Two divisibility conjectures for weighted central binomial sums

摘录 / EXCERPT: Let m be a positive integer. The paired divisibility conjecture. The congruences ... and ... hold. The source notes that these follow from the preceding prime-factor divisibility con...

conjecture | open | MathDB 页面 | 原文线索

V2-001350 | MATHDB-319136

Selected higher-power congruences for weighted central binomial sums

摘录 / EXCERPT: Let a be a positive integer. The selected congruences conjecture. The following congruences hold: ... ... and ... They are part of the open problems proposed after the preceding divisi...

conjecture | open | MathDB 页面 | 原文线索

V2-001351 | MATHDB-319137

Binomial-sum primality criterion for numbers of the form 4m plus or minus 1

摘录 / EXCERPT: Let m be a positive integer. The binomial-sum primality conjecture. If $m\neq30$ and ... then $4m-1$ is prime. If ... then $4m+1$ is prime. The authors report checking $m\leq1500$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-001352 | MATHDB-319138

5-adic congruences for products of central binomial coefficients

摘录 / EXCERPT: Let a and n be positive integers. The 5-adic product congruence conjecture. One has ... and ... The source describes this as a separate and particularly challenging open problem.

conjecture | open | MathDB 页面 | 原文线索

V2-001353 | MATHDB-319140

A local-parameter bound toward the generalized Ramanujan conjecture

摘录 / EXCERPT: Let E be a number field, let p be a rational prime that is unramified and does not split completely in E, and let $\nu\mid p$. Let f_p denote the relevant residue-degree qu...

conjecture | open | MathDB 页面 | 原文线索

V2-001354 | MATHDB-319219

Gun-Murty-Rath transcendence conjecture for shifted reciprocal-power series

摘录 / EXCERPT: Gun-Murty-Rath conjecture. The number

conjecture | open | MathDB 页面 | 原文线索

V2-001355 | MATHDB-319278

The constant-function mimicry conjecture for weighted multiplicative sums

摘录 / EXCERPT: Constant-function mimicry conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-001356 | MATHDB-319286

Chan-Cooper-Sica coefficient congruence conjecture for modular forms

摘录 / EXCERPT: Chan-Cooper-Sica coefficient congruence conjecture. If p is a prime with $\left(\frac{p}{23}\right)=1$ and $n\geq 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001357 | MATHDB-319327

A lower-bound conjecture for the coefficient C_0

摘录 / EXCERPT: Lower-bound conjecture for $\mathbf C_0$. For a random choice of the coefficients, one has

conjecture | open | MathDB 页面 | 原文线索

V2-001358 | MATHDB-319332

The non-CM aliquot-cycle counting conjecture for elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication. For an integer $\ell\ge1$, let $\mathcal Q_{E,\ell}(X)$ count normalized aliquot cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-001359 | MATHDB-319335

The Type-1 density conjecture for the CM j=0 family

摘录 / EXCERPT: Let $k\in\mathbb Z$ satisfy ${\operatorname{gcd}}(6,k)=1$. Define $\mathcal N_k(X)$ and $\mathcal N_k^{[1]}(X)$ as the counting functions for the prime-order reductions and Type...

conjecture | open | MathDB 页面 | 原文线索

V2-001360 | MATHDB-319336

The explicit Type-1 density conjecture for prime k

摘录 / EXCERPT: Let $k\ge5$ be a rational prime. Let $\mathcal N_k(X)$ and $\mathcal N_k^{[1]}(X)$ count the prime-order and Type 1 primes for $y^2=x^3+k$. The explicit Type-1 density conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-001361 | MATHDB-319376

Shevelev's parametrized twin-prime construction conjecture

摘录 / EXCERPT: Let $m\geq4$, set $A_m=\{1,\ldots,m-3\}$, and define ... with, for $n\geq2$, ... Let $n^*=n^*(m)$ be the point of the last nontrivial increment of $\{\tilde{c}^{(m)}(n)\}$ on A_m, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-001362 | MATHDB-319407

The prime-modulus minimum-color conjecture for links

摘录 / EXCERPT: Prime-modulus minimum-color conjecture. If $\langle r,\det L\rangle=\langle r,\det L'\rangle$, then $mincol_rL=mincol_rL'$. Moreover, m_p is a non-decreasing function of the prim...

conjecture | open | MathDB 页面 | 原文线索

V2-001363 | MATHDB-319411

The prime-curvature counting conjecture for Apollonian circle packings

摘录 / EXCERPT: Prime-curvature counting conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-001364 | MATHDB-319412

The kissing-prime counting conjecture for Apollonian circle packings

摘录 / EXCERPT: Kissing-prime counting conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-001365 | MATHDB-319419

Dahlquist-Essouabri conjecture on Euler products of polynomial factors

摘录 / EXCERPT: Let $W(x,y)=\sum_{n,m}a_{n,m}x^ny^m$ be an integral polynomial with $W(x,0)=1$, and define ... If $\beta=\max\{n/m:a_{n,m}\neq 0\}$, then the Euler-product continuation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-001366 | MATHDB-319420

Fujii's conjecture on additive relations among zeta zeros

摘录 / EXCERPT: Let Γ be the set of imaginary parts of the non-trivial zeros of the Riemann zeta-function. Fujii's conjecture. If $\gamma_1,\gamma_2,\gamma_3,\gamma_4\in\Gamma$ satisfy ... th...

conjecture | open | MathDB 页面 | 原文线索

V2-001367 | MATHDB-319441

The major-arcs conjecture for exceptional sets of additive problems

摘录 / EXCERPT: Let ${\mathcal A},{\mathcal B}\subseteq {\mathbb N}$. Write $\overline{{\mathcal A}+{\mathcal B}}$ for the exceptional set associated with the sumset ${\mathcal A}+{\mathcal B}$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-001368 | MATHDB-319445

Maximal complexity conjecture for classical constants

摘录 / EXCERPT: Maximal complexity conjecture. The complexity of α satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001369 | MATHDB-319480

The mixed Schmidt conjecture

摘录 / EXCERPT: The mixed Schmidt conjecture. For any (i_1,j_1) and (i_2,j_2) satisfying $0<i_r,j_r<1$ and $i_r+j_r=1$ for $r=1,2$,

conjecture | open | MathDB 页面 | 原文线索

V2-001370 | MATHDB-319481

The genuine mixed Schmidt conjecture for three weights

摘录 / EXCERPT: Let $\mathcal{D}$ be as above. For triples satisfying ... let $\mathbf{Bad}_{\mathcal{D}}(i,j,k)$ be the set of $(x,y)\in\mathbb{R}^2$ for which there exists $c(x,y)>0$ such that ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-001371 | MATHDB-319482

The countable mixed Schmidt dimension conjecture

摘录 / EXCERPT: Let (i_t,j_t,k_t) be a countable collection of triples of real numbers satisfying the source's condition $0<i_t,j_t<1$ and $i_t+j_t=1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001372 | MATHDB-319494

Droms-Mellinger-Meyer dimension conjecture for projective-plane LDPC codes

摘录 / EXCERPT: Let q be an odd prime power, and let ${\mathbf B}$ and ${\mathbf B}_0$ be the incidence matrices of internal points versus secant lines and external points versus passant lines,...

conjecture | open | MathDB 页面 | 原文线索

V2-001373 | MATHDB-319506

Conjecture on the asymptotic periodicity of zeros of zeta derivatives

摘录 / EXCERPT: Asymptotic periodicity conjecture. For all $\varepsilon>0$ and natural numbers M and $N\geq 2$ there is $K\in\mathbb{N}$ such that for all $k>K$ there exists $s\in\mathbb{C}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001374 | MATHDB-319507

Conjecture on rightward correspondence of zeros of successive zeta derivatives

摘录 / EXCERPT: Rightward correspondence conjecture. For $k\in\mathbb{N}$ there is a one-to-one correspondence between the zeros of $\zeta^{(k)}(s)$ and $\zeta^{(k+1)}(s)$, where the zero of...

conjecture | open | MathDB 页面 | 原文线索

V2-001375 | MATHDB-319514

Nonexistence conjecture for the equation $5x^2+q^{2n}=y^5$

摘录 / EXCERPT: Nonexistence conjecture. If the prime q is not of the form

conjecture | open | MathDB 页面 | 原文线索

V2-001376 | MATHDB-319531

Conjecture on derivative-zero density and small zeta-zero gaps

摘录 / EXCERPT: Derivative-zero density conjecture. If $m'(\nu)\gg\nu^\alpha$ for some $\alpha<2$, then $m(\nu)>0$ for all $\nu>0$. This is proposed as a refinement of Soundararajan's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-001377 | MATHDB-319573

Galois representations for Hecke eigenclasses in four homology theories

摘录 / EXCERPT: Let p be a prime, let $\overline{\mathbb F}_p$ be an algebraic closure of $\mathbb F_p$, and let M be an admissible S_N-module: a finite-dimensional $\mathbb F_p$-vector spac...

conjecture | open | MathDB 页面 | 原文线索

V2-001378 | MATHDB-319574

Linear-degree conjecture for Tverberg graphs

摘录 / EXCERPT: Let $q\geq 2$ be a prime power, $d\geq 1$, and let G be a graph on $(d+1)(q-1)+1$ vertices. A graph is a (d,q)--Tverberg graph if it has the corresponding Tverberg property for...

conjecture | open | MathDB 页面 | 原文线索

V2-001379 | MATHDB-319579

Artin's conjecture on zeros of p-adic forms

摘录 / EXCERPT: Artin's conjecture. If $n>d^2$, then the equation

conjecture | open | MathDB 页面 | 原文线索

V2-001380 | MATHDB-319587

Conjecture on unequal digits in the base-b expansion of $Z_b(b^k)$

摘录 / EXCERPT: Digit-discrepancy conjecture. The number of unequal digits between $Z_b(b^k)$ and the first k digits of the base-b expansion of $\vartheta(b)$ is smaller than or equal to the n...

conjecture | open | MathDB 页面 | 原文线索

V2-001381 | MATHDB-319609

Egami-Matsumoto effective additive independence conjecture for zeta zero ordinates

摘录 / EXCERPT: Egami-Matsumoto's effective independence conjecture. There exists some $\alpha<\frac{\pi}{2}$ such that either $\{\gamma_1,\gamma_2\}=\{\gamma_3,\gamma_4\}$, or

conjecture | open | MathDB 页面 | 原文线索

V2-001382 | MATHDB-319614

The conjecture that the maximum Frobenius discrepancy for three generators is 14

摘录 / EXCERPT: Frobenius discrepancy conjecture for three generators.

conjecture | open | MathDB 页面 | 原文线索

V2-001383 | MATHDB-319631

The p-adic monodromy weight conjecture

摘录 / EXCERPT: Let K be a non-archimedean local field, let X/K be smooth and projective, and let D^n be the finite-dimensional filtered (N,φ)-module associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-001384 | MATHDB-319632

The global H^2 vanishing conjecture for divisible geometric cohomology

摘录 / EXCERPT: Let K be a global field, let X/K be smooth and projective of dimension d, and let $DivA$ denote the maximal divisible submodule of a torsion Galois module A. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-001385 | MATHDB-319651

Artin's conjecture that the maximal abelian extension of the rationals is C_1

摘录 / EXCERPT: Let $\mathbf{Q}^{\mathrm{ab}}$ denote the maximal abelian extension of the rational numbers. Artin's conjecture. The field $\mathbf{Q}^{\mathrm{ab}}$ is (C_1). The source present...

conjecture | open | MathDB 页面 | 原文线索

V2-001386 | MATHDB-319655

Kervaire-Murthy adelic class-group conjecture

摘录 / EXCERPT: Let F_n be the cyclotomic field in question, let ${\mathbb A}(F_n)$ be its ring of adeles, and let $K_{p^{n+1}-1}$ be the subgroup ... of $GL(1,{\mathbb A}(F_n))$, where the produc...

conjecture | open | MathDB 页面 | 原文线索

V2-001387 | MATHDB-319673

Twisted inhomogeneous badly approximable sets have full Hausdorff dimension

摘录 / EXCERPT: Let i and j be real numbers satisfying the paper's condition on the weights, and let $\mathbf{x}\in[0,1]^2$ be an irrational vector. Write $\textup{\bf Bad}^{\mathbf{x}}(i,j)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001388 | MATHDB-319684

Goldbach-type conjecture for 43_{x^2+y^2}-primes

摘录 / EXCERPT: Let the $\oplus_{x^2+y^2}^{\mathbb{N}}$-primes be the positive integers that are not representable as x^2+y^2 with x,y positive integers; the source lists some of them as...

conjecture | open | MathDB 页面 | 原文线索

V2-001389 | MATHDB-319741

Cohen-Kamiyama homology conjecture for centers-of-mass configuration spaces

摘录 / EXCERPT: Let p be an odd prime and let n be a natural number. For $ atural$ numbers n and $ell<n$, let $M_{\ell}(\mathbb{C},n)$ be the Cohen-Kamiyama subspace of the ordered configura...

conjecture | open | MathDB 页面 | 原文线索

V2-001390 | MATHDB-319774

Chida's reciprocity conjecture for u_C

摘录 / EXCERPT: Let $[(r,\tau)]\in(\mathbb{Z}/f\mathbb{Z}\times H^{\mathcal{O}}(\mathfrak{N}))/\sim_f$ with τ reduced, set $\mathcal{O}=\mathcal{O}_{\tau}$, $\mathfrak{f}=f\mathcal{O}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001391 | MATHDB-319775

Dasgupta's algebraicity and reciprocity conjecture

摘录 / EXCERPT: Let $\mathfrak{b}$ be an invertible $\mathcal{O}$-ideal that is $\mathfrak{f}$-integral, let $\mathfrak{n}$ be an invertible $\mathcal{O}$-ideal satisfying ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001392 | MATHDB-319785

The optimal exponential bound conjecture for the arithmetic function f

摘录 / EXCERPT: Let f be the arithmetic function defined earlier in the paper. Optimal exponential bound conjecture. For every positive integer n, ... This conjecture would improve the preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-001393 | MATHDB-319810

Equality of the Darmon and Fontaine-Mazur L-invariants

摘录 / EXCERPT: Let V_p be the p-adic representation attached to the space of p-new modular forms, let $\mathbf{D}^{FM}=D_{st}(V_p)$ be its associated two-dimensional monodromy module, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-001394 | MATHDB-319811

Rationality conjecture for Darmon cohomology classes

摘录 / EXCERPT: Let K be the real quadratic field, let $H_{\mathcal{O}}/K$ be the narrow ring class field attached to an order $\mathcal{O}$, let V_p be the relevant p-adic representation, a...

conjecture | open | MathDB 页面 | 原文线索

V2-001395 | MATHDB-319812

Non-triviality conjecture for Darmon cohomology classes

摘录 / EXCERPT: Let $f\in S_k(\Gamma_0(pN^+))^{p\relax\protect\ifmmode\expandafter\text@\else\expandafter\text\fi{-new}}$ be a p-new eigenform, let $L_{f,\mathfrak{p}}$ be the completion of its...

conjecture | open | MathDB 页面 | 原文线索

V2-001396 | MATHDB-319813

Gross-Zagier non-vanishing conjecture for Darmon cycles

摘录 / EXCERPT: Let f be a p-new eigenform, let $\mathfrak{p}$ be a prime above p, let $V_{\mathfrak{p}}(f)$ be the corresponding p-adic representation, let ${\Greekmath 011F}$ be a charac...

conjecture | open | MathDB 页面 | 原文线索

V2-001397 | MATHDB-319816

The property $(\mathcal P)$ conjecture for the digits of Pi and the Euler constant

摘录 / EXCERPT: Let $m\ge 2$ be an integer, and consider the digits in basis m of the fractional part of either Pi or the Euler constant. Denote by $(\mathcal P)$ the large-deviation property de...

conjecture | open | MathDB 页面 | 原文线索

V2-001398 | MATHDB-319817

The dimension formulas for the codes C(1,q) and C(2,q) over odd prime powers

摘录 / EXCERPT: Let q be an odd prime power, and let $C(1,q)$ and $C(2,q)$ be the codes defined from the incidence structures in the paper. Their dimensions are the integers $\dim C(1,q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001399 | MATHDB-319854

The multiplicity estimate conjecture for Drinfeld quasi-modular forms

摘录 / EXCERPT: Let $q=p^e$ be a prime power, let $A=\mathbb{F}_q[\theta]$, and let f be a non-zero Drinfeld quasi-modular form of weight w and depth $l\geq 1$. Write $\nu_{\infty}(f)$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-001400 | MATHDB-319855

The polynomial-generation conjecture for almost A-quasi-modular forms

摘录 / EXCERPT: Let $\mathbb{T}_{>0}$ be the algebra of power series with positive convergence radius, and let g, h, and $\boldsymbol{s}_2$ be the almost A-quasi-modular forms used in the so...

conjecture | open | MathDB 页面 | 原文线索

V2-001401 | MATHDB-319859

The generalized Hall conjecture for repulsion of rational points

摘录 / EXCERPT: The generalized Hall conjecture. The algebraic points repel the rational points with length below k (the Siegel part). Moreover, for each fixed algebraic point Q, there is a co...

conjecture | open | MathDB 页面 | 原文线索

V2-001402 | MATHDB-319872

The graded Cartan elementary-divisor conjecture for Hecke algebras

摘录 / EXCERPT: Let p be a prime. For a partition $\boldsymbol{\mu}=(\mu^{(1)},\dots,\mu^{(p-1)})\in M_{p-1}(d)$, let $w_H(\mu^{(p-1)})$ be the corresponding weight, and let the block of p-wei...

conjecture | open | MathDB 页面 | 原文线索

V2-001403 | MATHDB-319881

Conjectural mean-size asymptotic for primitive odd class-group components

摘录 / EXCERPT: Mean-size conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001404 | MATHDB-319882

Dąbrowski's conjecture on smooth numbers of the form x^2-1

摘录 / EXCERPT: Dąbrowski's conjecture. The equation has exactly 28 solutions $(x;\alpha_1,\ldots,\alpha_k)$ in positive integers, namely the solutions listed in the statement: $(3;3)$;...

conjecture | open | MathDB 页面 | 原文线索

V2-001405 | MATHDB-319920

Refined Koblitz conjecture for elliptic curves with rational torsion

摘录 / EXCERPT: Elliptic-curve refinement. For $X\gg 0$, the number of primes $p\leq X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001406 | MATHDB-319924

The Ozsváth-Szabó realization conjecture for lens space surgery data

摘录 / EXCERPT: Let (p,k) be a coprime integer pair. Let $d434_{k,p}(t)$ denote the polynomial associated with the pair (p,k) in the source. Suppose that $d434_{k,p}(t)$ satisfies the alternat...

conjecture | open | MathDB 页面 | 原文线索

V2-001407 | MATHDB-319935

Hooley's Hypothesis K^* for diagonal Waring representations

摘录 / EXCERPT: Let $r_{k,k}(l)$ denote the number of representations of a positive integer l as a sum of k positive kth powers. Hooley's Hypothesis K^*. For $k=2$ and every $\epsilon>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001408 | MATHDB-319940

Sun's base- expansion conjectures

摘录 / EXCERPT: Sun's conjecture. (I) There are at least k non-zero digits in the base-m expansion of

conjecture | open | MathDB 页面 | 原文线索

V2-001409 | MATHDB-319958

Long-Maclachlan-Reid conjectures on rational homology 3-spheres

摘录 / EXCERPT: Long-Maclachlan-Reid conjectures. 1. There exist infinitely many pairs of prime ideals $\mathfrak{p}_1,\mathfrak{p}_2$ in $mathbb Z[i]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001410 | MATHDB-320014

The finitary partial-field excluded-minor conjecture

摘录 / EXCERPT: A partial field $\mathbb{P}$ is finitary if it admits a homomorphism to $\operatorname{GF}(q)$ for some prime power q. Let $\operatorname{\mathcal{M}}(\mathbb{P})$ denote the cla...

conjecture | open | MathDB 页面 | 原文线索

V2-001411 | MATHDB-320020

The torsion conjecture for modular-form Selmer groups

摘录 / EXCERPT: Let f be the modular form in the source, let $p\geq5$ be a good ordinary prime, let $F_\infty/\mathbb{Q}$ be the extension determined by the image of the associated representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-001412 | MATHDB-320022

The Paramodular Conjecture for rational abelian surfaces

摘录 / EXCERPT: Let p be a prime. A paramodular form here is a Hecke eigenform $F\in S^2(\Gamma^{\text{para}}[p])$ with rational eigenvalues; exclude Gritsenko lifts. Let $\mathcal{A}$ be an iso...

conjecture | open | MathDB 页面 | 原文线索

V2-001413 | MATHDB-320023

Paramodular Böcherer conjecture

摘录 / EXCERPT: Let $F\in S^k(\Gamma^{\text{para}}[p])^+$, where $S^k(\Gamma^{\text{para}}[p])^+$ is the plus space of paramodular cusp forms, and for a fundamental discriminant $D<0$ define ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-001414 | MATHDB-320042

Colliot-Thélène's approximation conjecture for zero-cycles

摘录 / EXCERPT: Colliot-Thélène's approximation conjecture. If $\{z_v\}\in\prod_{v\in\Omega_k}Z_0(X_v)$ is a degree-δ family orthogonal to $Br(X)$, then for every positive integer m there...

conjecture | open | MathDB 页面 | 原文线索

V2-001415 | MATHDB-320060

Brumer-Rosen conjecture on primary components of Brauer groups

摘录 / EXCERPT: Let F be a field and let $(\operatorname{Br}F)_p$ denote the p-primary component of its Brauer group. Brumer-Rosen conjecture. For each prime p, one of the following holds: 1...

conjecture | open | MathDB 页面 | 原文线索

V2-001416 | MATHDB-320081

The conjecture on invariants of a vector and covector for the general linear group

摘录 / EXCERPT: Let q be a prime power, let V be the natural n-dimensional representation of $\operatorname{GL}_n(\mathbb{F}_q)$ over $\mathbb{F}_q$, and let V^* be its dual. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-001417 | MATHDB-320104

Extremal Mahler measure conjecture for algebraic numbers

摘录 / EXCERPT: Extremal Mahler measure conjecture. For each $1\leq p\leq\infty$, there exists a constant c_p such that

conjecture | open | MathDB 页面 | 原文线索

V2-001418 | MATHDB-320133

Determinism conjecture for the generalized Cullen number primality algorithm

摘录 / EXCERPT: Determinism conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-001419 | MATHDB-320146

Three-dimensional multi-dimensional conjecture for integral differences of an \mathcal{N}-set

摘录 / EXCERPT: Let $A\subseteq\mathbb{R}^3$ be an $\mathcal{N}$-set, and let p_1,p_2,p_3 be three two-dimensional linear subspaces of $\mathbb{R}^3$. Multi-dimensional conjecture. One has ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-001420 | MATHDB-320149

The conjecture on concentration points counted by $I_2(M;K,L)$

摘录 / EXCERPT: Let p be prime, and let $I_2(M;K,L)$ denote the quantity defined in the paper that counts the relevant concentration points for the parameters M, K, and L. The I_2 concen...

conjecture | open | MathDB 页面 | 原文线索

V2-001421 | MATHDB-320150

The conjecture on triple products of short intervals modulo p

摘录 / EXCERPT: Let p be prime, and let $\mathcal{I}_1$, $\mathcal{I}_2$, and $\mathcal{I}_3$ be intervals in $\mathbb{F}_p^*$ satisfying ... for each i. The triple-product growth conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-001422 | MATHDB-320171

The generation conjecture for the cohomology algebra of \(\mathfrak{sl}_2(\mathbb{F}_p)\)

摘录 / EXCERPT: Let p be an odd prime, let $\mathfrak{sl}_2(\mathbb{F}_p)$ be the Lie algebra of traceless 2×2 matrices over $\mathbb{F}_p$, and let $Poly(ad^*)$ denote the polynomial al...

conjecture | open | MathDB 页面 | 原文线索

V2-001423 | MATHDB-320187

A supercongruence for a truncated hypergeometric series with reciprocal-square weights

摘录 / EXCERPT: Let $p>3$ be a prime and let $x=r/m$, where $0<r<m$ are integers with m prime to p. Write $Q_p(x)$ for the quantity defined in the preceding results of the source. The reciproc...

conjecture | open | MathDB 页面 | 原文线索

V2-001424 | MATHDB-320204

Kohnen-Mason conjecture on integral generalized modular forms

摘录 / EXCERPT: Kohnen-Mason conjecture. Under these conditions, f is a classical modular form.

conjecture | open | MathDB 页面 | 原文线索

V2-001425 | MATHDB-320208

Prime-order generation conjecture for finite nonabelian simple groups

摘录 / EXCERPT: Let G be a finite nonabelian simple group. Define ... to be the set of element orders of G. Prime-order generation conjecture. There exist two distinct primes...

conjecture | open | MathDB 页面 | 原文线索

V2-001426 | MATHDB-320211

The Koszulity conjecture for Milnor K-theory

摘录 / EXCERPT: Let K be a field and let l be prime such that K contains a primitive l-th root of unity. A positively graded algebra $A=k\oplus A_1\oplus A_2\oplus\dotsb$ is Koszul when...

conjecture | open | MathDB 页面 | 原文线索

V2-001427 | MATHDB-320214

Conjecture on the form of all 2-hyperperfect numbers

摘录 / EXCERPT: Conjecture on 2-hyperperfect numbers. Every 2-hyperperfect number has the form

conjecture | open | MathDB 页面 | 原文线索

V2-001428 | MATHDB-320215

Conjecture on super-hyperperfect numbers of 3-power type

摘录 / EXCERPT: Conjecture on the form of all solutions. All solutions for this generalization are 3^{p-1}-like numbers, where p and $(3^p-1)/2$ are prime.

conjecture | open | MathDB 页面 | 原文线索

V2-001429 | MATHDB-320225

Mazur's topology of rational points conjecture

摘录 / EXCERPT: Mazur's conjecture. The topological closure of $V(\mathbb{Q})$ in $V(\mathbb{R})$ possesses at most a finite number of connected components.

conjecture | open | MathDB 页面 | 原文线索

V2-001430 | MATHDB-320229

The half-power sign conjecture for Liouville sums

摘录 / EXCERPT: Half-power sign conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001431 | MATHDB-320267

Bruinier-Yang conjecture on arithmetic intersections of Hirzebruch-Zagier divisors

摘录 / EXCERPT: Bruinier-Yang conjecture. The arithmetic intersection satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001432 | MATHDB-320269

Vojta's height conjecture for families of higher-genus curves

摘录 / EXCERPT: Let ${\mathcal C}\to B$ be a family of higher-genus curves, let $b\in B({\mathbb Q})$, and let ${\mathcal C}_b$ be a smooth fiber. Let H_B be a non-logarithmic height on B, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001433 | MATHDB-320272

Poonen's Mordell-Weil sieve heuristic

摘录 / EXCERPT: Let C be a curve over ${\mathbb Q}$ with Jacobian J, let S be a finite set of primes of good reduction, and let $n\geq1$. Consider the maps ... and ... from the Mordell-Weil siev...

conjecture | open | MathDB 页面 | 原文线索

V2-001434 | MATHDB-320277

Hardy-Littlewood Goldbach representation conjecture

摘录 / EXCERPT: Let $R(n)$ denote the number of ordered pairs of odd primes (p,q) satisfying $p+q=n$, and let ... be the twin-prime constant. Hardy-Littlewood's conjecture. As n tends to infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-001435 | MATHDB-320295

A mod-n^3 congruence for alternating sums involving central Delannoy numbers

摘录 / EXCERPT: Central Delannoy congruence. If n is a power of 2, then

conjecture | open | MathDB 页面 | 原文线索

V2-001436 | MATHDB-320296

A generalized binomial divisibility conjecture with alternating weights

摘录 / EXCERPT: Let $m\in\mathbb{Z}^+$, let $a_1,\ldots,a_m\in\mathbb{Z}$, let $b_1,\ldots,b_m,r\in\mathbb{N}$, let $n\in\mathbb{Z}^+$, and let $\varepsilon\in\{\pm1\}$. Define the binomial coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-001437 | MATHDB-320297

A prime-power q-binomial congruence

摘录 / EXCERPT: Let p be a prime, let n be a power of p, and define the q-integer by ... For Gaussian binomial coefficients, write ${a\brack b}_q$. Prime-power q-binomial congruence. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-001438 | MATHDB-320298

The (3,3) conjecture for p-tower groups over quadratic imaginary fields

摘录 / EXCERPT: Let p be an odd prime, let K be a quadratic imaginary number field, and let G be its p-tower group. For a finite 2-generated p-tower group, define its Zassenhaus type t...

conjecture | open | MathDB 页面 | 原文线索

V2-001439 | MATHDB-320303

Infinitude conjecture for the first binomial gcd condition

摘录 / EXCERPT: First binomial gcd infinitude conjecture. There are infinitely many positive integers n such that

conjecture | open | MathDB 页面 | 原文线索

V2-001440 | MATHDB-320304

Infinitude conjecture for the second binomial gcd condition

摘录 / EXCERPT: Second binomial gcd infinitude conjecture. There are infinitely many positive integers n such that

conjecture | open | MathDB 页面 | 原文线索

V2-001441 | MATHDB-320307

The hypergeometric-zeta correspondence for the curves \(\mathcal{C}_{z}^{(m,s)}\)

摘录 / EXCERPT: Let l and q be odd primes such that $q\equiv 1 \pmod{l}$, and let $z\in\mathbb{F}_{q}$ with $z\neq 0,1$. For integers $1\leq m,s<l$ satisfying $m+s=l$, consider the smooth proj...

conjecture | open | MathDB 页面 | 原文线索

V2-001442 | MATHDB-320323

Ikeda's period conjecture for the Duke-Imamoglu-Ikeda lift

摘录 / EXCERPT: Let n be even, let f be a primitive form in ${\textfrak S}_{2k-n}(\varGamma^{(1)})$, and let h be the Hecke eigenform in ${\textfrak S}_{k-n/2+1/2}^{+}(\varGamma_0(4))$ corre...

conjecture | open | MathDB 页面 | 原文线索

V2-001443 | MATHDB-320345

Gaussian localization conjecture for orthogonal modular categories

摘录 / EXCERPT: Let $p\geq 3$ be an odd prime. Let X be either the simple spin object X_{ε} in the unitary modular tensor category $(SO(p),2)$ or the simple spin object...

conjecture | open | MathDB 页面 | 原文线索

V2-001444 | MATHDB-320346

Bhargava-Kane conjecture on Selmer groups of elliptic curves

摘录 / EXCERPT: Fix a global field k and a prime p. For an elliptic curve E/k, let $\operatorname{Sel}_p E$ denote its p-Selmer group, and define the probability by ordering elliptic curve...

conjecture | open | MathDB 页面 | 原文线索

V2-001445 | MATHDB-320347

Selmer-group conjecture for squarefree integers

摘录 / EXCERPT: Fix a global field k, a squarefree positive integer n, and let $\omega(n)$ be the number of prime factors of n. For each prime $p\mid n$, choose $d_p\in{\mathbb Z}_{\ge0}$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-001446 | MATHDB-320374

The induction-restriction correspondence conjecture for abelian Sylow subgroups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be an abelian Sylow p-subgroup of G. A signed bijection...

conjecture | open | MathDB 页面 | 原文线索

V2-001447 | MATHDB-320375

The weak induction-restriction correspondence conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. The properties (pRes-Syl), (pInd-Syl), and (WIRC-Syl) are defined using the subgroup of c...

conjecture | open | MathDB 页面 | 原文线索

V2-001448 | MATHDB-320376

The generalized weak induction-restriction correspondence conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let P be a p-subgroup of G, and set $H=N_G(P)$. The source defines $\operatorname{Irr}_0(G,P)$ as the irreducible characters of...

conjecture | open | MathDB 页面 | 原文线索

V2-001449 | MATHDB-320400

Greenberg's conjecture for arbitrary Galois extensions

摘录 / EXCERPT: Let bK/Q be an arbitrary Galois extension containing the p-th roots of unity, and let H/K_{∞} be the maximal unramified p-abelian extension of the cyclotomic Z_p-exte...

conjecture | open | MathDB 页面 | 原文线索

V2-001450 | MATHDB-320404

Conjecture on breaking the square-root barrier for deterministic prime decisions

摘录 / EXCERPT: Let N be large, and let $[a,b]$ be an interval contained in $[N,2N]$ whose length is at most $N^{1/2+c}$ for some absolute constant $c>0$. Square-root barrier conjecture. There e...

conjecture | open | MathDB 页面 | 原文线索

V2-001451 | MATHDB-320444

Bounded-offset conjecture for digit sums of multiples

摘录 / EXCERPT: Let b,r be positive integers with $\gcd(b-1,r)=1$, and let c be any integer. Write $s_b(n)$ for the sum of the base-b digits of n. Bounded-offset conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-001452 | MATHDB-320446

The odd-prime-isogeny local root-number formula

摘录 / EXCERPT: Let $\mathcal K$ be a local field of characteristic zero, and let $E/\mathcal K$ be an elliptic curve with a p-isogeny $\phi:E\to E'$ over $\mathcal K$. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-001453 | MATHDB-320451

The divisibility conjecture for maximal heights of pq^b

摘录 / EXCERPT: Let p be a fixed odd prime and let b be a fixed positive integer. As q ranges over primes satisfying $p<q$, consider the finite list of values $B(pq^b)$. The divisibility con...

conjecture | open | MathDB 页面 | 原文线索

V2-001454 | MATHDB-320452

The parity-product conjecture for maximal heights of square-free divisors

摘录 / EXCERPT: Let $n=p_1\cdots p_t$ be square-free, and let $\omega(d)$ denote the number of primes dividing d. The parity-product conjecture. $B(n)$ is attained by at least one of the two pol...

conjecture | open | MathDB 页面 | 原文线索

V2-001455 | MATHDB-320453

The divisibility and lower-bound conjecture for maximal heights of pqr^b

摘录 / EXCERPT: Let $p<q<r$ be primes and let $b>1$. The divisibility and lower-bound conjecture. For $n=pqr^b$, the value $B(n)$ is divisible by p and satisfies ... This conjecture is motivated b...

conjecture | open | MathDB 页面 | 原文线索

V2-001456 | MATHDB-320491

Sierpiński's conjecture on the number of solutions to Euler's totient equation

摘录 / EXCERPT: Let ϕ denote Euler's totient function. For an integer $s\geq 2$, consider the equation $\phi(x)=m$ in positive integers x. Sierpiński's conjecture. For any integer $s\geq 2$...

conjecture | open | MathDB 页面 | 原文线索

V2-001457 | MATHDB-320544

Reduced Khovanov homology detects the unknot among prime knots

摘录 / EXCERPT: Let U denote the unknot, and let K be a prime knot. Suppose that ${\widehat H}_{red}(K)$ and ${\widehat H}_{red}(U)$ are isomorphic as $delta$-graded vector spaces over...

conjecture | open | MathDB 页面 | 原文线索

V2-001458 | MATHDB-320589

NP-hardness conjecture for p-adic feasibility of sparse polynomials

摘录 / EXCERPT: For a prime p, let ${\mathcal F}^*_{n,n+1}$ denote the class of $(n+1)$-nomials in n variables, and let ${\text{{\tt FEAS}}}_{\mathbb{Q}_p}$ be the feasibility problem for rati...

conjecture | open | MathDB 页面 | 原文线索

V2-001459 | MATHDB-320596

Banks-Ford-Luca conjecture on repeated values of Carmichael's function

摘录 / EXCERPT: Let $\lambda(n)$ be the Carmichael function, the largest order of any number modulo n. Banks-Ford-Luca conjecture. For every positive integer n, there is an integer $m\ne n$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-001460 | MATHDB-320597

Conjecture on prime powers and the Pratt-tree function

摘录 / EXCERPT: Let p,q,r,s denote primes. For a prime q, construct a tree $T(q)$ rooted at q by linking each prime power p^e exactly dividing $q-1$ to q, and recursively linking each...

conjecture | open | MathDB 页面 | 原文线索

V2-001461 | MATHDB-320605

Kessar-Schaps conjecture on derived equivalence classes of spin blocks

摘录 / EXCERPT: Let G be a finite group and let k be a splitting field of characteristic p, where p is an odd prime. Let $\tilde S_n$ and $\tilde A_n$ be the covering groups of the symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-001462 | MATHDB-320608

Gross's nonsolvable extensions conjecture

摘录 / EXCERPT: Let p be a prime. Gross's conjecture. For every prime p, there exists a nonsolvable Galois number field K ramified only at p. This conjecture concerns the existence of numb...

conjecture | open | MathDB 页面 | 原文线索

V2-001463 | MATHDB-320631

Kowalski-Michel-VanderKam mass equidistribution conjecture in the level aspect

摘录 / EXCERPT: Let f be a classical holomorphic newform of fixed weight k and level q, with mass $d\nu_f=|f(z)|^2y^{k-2}\,dx\,dy$ on $Y_0(q)=\Gamma_0(q)\backslash\mathbb H$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001464 | MATHDB-320644

The supercongruence conjecture modulo p^3

摘录 / EXCERPT: Supercongruence conjecture modulo p^3. Under these conditions,

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-001465 | MATHDB-320656

Kellner's equivalent conjecture for power sums

摘录 / EXCERPT: Kellner's equivalent conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-001466 | MATHDB-320657

The even-exponent generalized Erdős-Moser conjecture

摘录 / EXCERPT: Let ... where $S_k(m)=1^k+2^k+\cdots+(m-1)^k$. Even-exponent generalized Erdős-Moser conjecture. The set $\mathcal A$ is empty. This is the remaining case identified in the source af...

conjecture | open | MathDB 页面 | 原文线索

V2-001467 | MATHDB-320658

Parity conjecture for Selmer ranks of elliptic curves

摘录 / EXCERPT: Selmer-rank parity conjecture. Let ℓ be a prime. We have

conjecture | open | MathDB 页面 | 原文线索

V2-001468 | MATHDB-320707

Weng's Riemann hypothesis for Weng's zeta functions

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{Q}$ and P a maximal parabolic subgroup, and let $\xi^{G/P}_{\mathbb{Q}}(s)$ denote the normalized Weng zeta function associa...

conjecture | open | MathDB 页面 | 原文线索

V2-001469 | MATHDB-320718

Strong independence conjecture for systems of linear forms

摘录 / EXCERPT: Fix a prime p) and $d\ge 1$. Let $\mathcal{L}=\{L_1,\ldots,L_m\}$ be a system of linear forms over $\mathbb{F}_p$ such that L_1^{d+1} is not linearly dependent on...

conjecture | open | MathDB 页面 | 原文线索

V2-001470 | MATHDB-320738

Removable-poles conjecture for generalized Sierpinski carpets

摘录 / EXCERPT: Let a generalized Sierpinski carpet have self-similar, possibly graph-directed, structures yielding the same Laplacian operator. Let d_k denote the relevant dimensions, with...

conjecture | open | MathDB 页面 | 原文线索

V2-001471 | MATHDB-320741

Pappas-Rapoport spin-condition conjecture for symplectic local models

摘录 / EXCERPT: Pappas-Rapoport spin-condition conjecture. For any periodic self-dual lattice chain ${\mathcal L}$,

conjecture | open | MathDB 页面 | 原文线索

V2-001472 | MATHDB-320742

Görtz's flatness conjecture for ramified symplectic local models

摘录 / EXCERPT: Görtz's flatness conjecture. For any self-dual periodic ${\mathcal O}_F$-lattice chain ${\mathcal L}$, $M^{\mathrm{naive}}_{G,\{\mu\},{\mathcal L}}$ is flat over

conjecture | open | MathDB 页面 | 原文线索

V2-001473 | MATHDB-320747

Batyrev-Manin conjecture for K3 surfaces

摘录 / EXCERPT: Batyrev-Manin conjecture for K3 surfaces. For every real number $\epsilon>0$ and every ample divisor L on X, there exists a nonempty Zariski open subset $U(\epsilon)\subset X$...

conjecture | open | MathDB 页面 | 原文线索

V2-001474 | MATHDB-320748

Manin's rational curve conjecture

摘录 / EXCERPT: Manin's rational curve conjecture. If there exist positive real constants δ and c such that

conjecture | open | MathDB 页面 | 原文线索

V2-001475 | MATHDB-320757

Stephan's third ratio conjecture for the Pascal triangle sequence

摘录 / EXCERPT: Stephan's third ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001476 | MATHDB-320762

Mersenne-prime conjecture for odd redundant divisors

摘录 / EXCERPT: A positive integer n is near-perfect if it is the sum of all its proper divisors except one proper divisor, called its redundant divisor. The number n is even when it is divisi...

conjecture | open | MathDB 页面 | 原文线索

V2-001477 | MATHDB-320771

The representation zeta function of the first principal congruence subgroup of \operatorname{SL}_2(\mathbb{Z}_2)

摘录 / EXCERPT: The representation zeta function conjecture. The zeta function is

conjecture | open | MathDB 页面 | 原文线索

V2-001478 | MATHDB-320774

Modified Vojta conjecture with a dimension-dependent discriminant coefficient

摘录 / EXCERPT: Modified Vojta conjecture. There exists a proper Zariski-closed subset

conjecture | open | MathDB 页面 | 原文线索

V2-001479 | MATHDB-320793

Multiplicative Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $n\geq 1$ and let $\psi:\mathbb{N}\to\mathbb{R}^{\geq 0}$ be a non-negative function. Define ... Here $\varphi(q)$ denotes Euler's phi function. Multiplicative Duffin-Schaeffer c...

conjecture | open | MathDB 页面 | 原文线索

V2-001480 | MATHDB-320819

Legendre-symbol triangulation polynomial congruence

摘录 / EXCERPT: Let p be an odd prime, and let E_p^+ and E_p^- be the convex $(p+1)$-near-edges associated respectively to the sequences of Legendre symbols and their negatives. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001481 | MATHDB-320821

The asymptotic independence conjecture for the counting functions of L_k

摘录 / EXCERPT: Asymptotic independence conjecture. For every $k>1$, the asymptotic behavior of C_k does not depend on k and, in particular,

conjecture | open | MathDB 页面 | 原文线索

V2-001482 | MATHDB-320822

The Carmichael-number infinitude conjecture for successive L_k

摘录 / EXCERPT: For each natural number k, let L_k denote the set of k-Lehmer numbers, and let $L_{k+1}\setminus L_k$ be the corresponding difference of successive sets. A Carmichael-number...

conjecture | open | MathDB 页面 | 原文线索

V2-001483 | MATHDB-320831

Irrationality conjecture for solutions of tan nx = n tan x

摘录 / EXCERPT: Irrationality conjecture. Then x is x/π is irrational. The claim is posed in connection with the arithmetic nature of the solutions to the tangent equation; the source provid...

conjecture | open | MathDB 页面 | 原文线索

V2-001484 | MATHDB-320833

Conjecture on the typical height of Pratt trees

摘录 / EXCERPT: For a prime p, let $H(p)$ be the height of its Pratt tree: the tree with root p, whose children are the prime factors of $p-1$, constructed recursively, stopping at $p=2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001485 | MATHDB-320843

Odoni-Stoll conjecture on stable reduction of x^2+1

摘录 / EXCERPT: Let $\overline{f}$ denote the reduction of a polynomial f modulo a prime. Odoni-Stoll conjecture. For ... $\overline{f}$ is stable for $p=3$ and for no other primes. Odoni observed...

conjecture | open | MathDB 页面 | 原文线索

V2-001486 | MATHDB-320849

The unbounded denominators conjecture for meromorphic modular forms

摘录 / EXCERPT: Unbounded denominators conjecture. The Fourier coefficients of f have bounded denominators if and only if f is a cusp form for a congruence subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-001487 | MATHDB-320858

Conjecture on the denominator ratios of the iterated integral of $\ln(1+x^2)$

摘录 / EXCERPT: Let $A_{n,2}(x)$ be the rational polynomial part of the nth iterated integral in the $\ln(1+x^2)$ case, let $\alpha_{n,2}$ be its denominator, and define ... Here p denotes a pri...

conjecture | open | MathDB 页面 | 原文线索

V2-001488 | MATHDB-320859

Conjecture on the denominators of the iterated integral of $\ln(1+x^2)$

摘录 / EXCERPT: Let $A_{n,2}(x)$ be the rational polynomial part of the nth iterated integral in the $\ln(1+x^2)$ case, and let $\alpha_{n,2}$ be its denominator. Let $m\ge1$ be an integer. Deno...

conjecture | open | MathDB 页面 | 原文线索

V2-001489 | MATHDB-320860

Conjecture on the auxiliary denominators for the iterated integral of $\ln(1+x^2)$

摘录 / EXCERPT: Let $G_n(x)$ be the auxiliary polynomial introduced by expanding the expression for $A_{n,2}(x)$, and let γ_n be its denominator. Auxiliary-denominator conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001490 | MATHDB-320866

Bateman-Pomerance-Vaughan conjecture on cyclotomic polynomial height

摘录 / EXCERPT: Let n be a product of k distinct primes, let Φ_n be the cyclotomic polynomial of order k, and let A_n denote its height. In the known bound ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-001491 | MATHDB-320867

Bateman-Pomerance-Vaughan lower-bound conjecture for cyclotomic polynomial heights

摘录 / EXCERPT: For each positive integer k, let A_n denote the height of the cyclotomic polynomial Φ_n, and call Φ_n a cyclotomic polynomial of order k when n has k distinct p...

conjecture | open | MathDB 页面 | 原文线索

V2-001492 | MATHDB-320901

Uniform shifted-block bound for pairwise-coprime-free sets

摘录 / EXCERPT: Shifted-block bound conjecture. For every such k,l,a,

conjecture | open | MathDB 页面 | 原文线索

V2-001493 | MATHDB-320902

Eventual uniqueness of the Erdős extremal set

摘录 / EXCERPT: Eventual uniqueness conjecture. For every integer $k\ge3$, there exists an integer n_k such that, for all $n\ge n_k$, if

conjecture | open | MathDB 页面 | 原文线索

V2-001494 | MATHDB-320903

Coprime-endpoint equality implies canonical offsets

摘录 / EXCERPT: Coprime-endpoint bound conjecture. For every such k,l,a,

conjecture | open | MathDB 页面 | 原文线索

V2-001495 | MATHDB-320904

Endpoint equality for shifted pairwise-coprime-free blocks

摘录 / EXCERPT: Endpoint equality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-001496 | MATHDB-320910

Upper-bound conjecture for Pomerance's constant

摘录 / EXCERPT: Pomerance's upper-bound conjecture. Pomerance's constant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001497 | MATHDB-320921

Rubinstein-Sarnak conjecture on unbiased prime number races

摘录 / EXCERPT: Rubinstein-Sarnak conjecture. When $r\geq 3$, the race $\{q;a_1,\dots,a_r\}$ is unbiased if and only if $r=3$ and the residue classes a_1,a_2,a_3 satisfy assumption (1.3).

conjecture | open | MathDB 页面 | 原文线索

V2-001498 | MATHDB-320938

Period conjecture for multiple Dedekind zeta values

摘录 / EXCERPT: Period conjecture. These multiple Dedekind zeta values are periods. This conjecture proposes that multiple Dedekind zeta values belong to the class of periods, connecting their ana...

conjecture | open | MathDB 页面 | 原文线索

V2-001499 | MATHDB-320967

The cubic binary-form perfect-power conjecture

摘录 / EXCERPT: Let F be a separable homogeneous cubic binary form with integer coefficients, let d be a fixed integer with $d\geq 1$, and let l be a prime number. Cubic binary-form perfect-...

conjecture | open | MathDB 页面 | 原文线索

V2-001500 | MATHDB-320988

Cassels-Swinnerton-Dyer conjecture on bounded diagonal orbits

摘录 / EXCERPT: Let $X_d=\operatorname{SL}_d(\mathbb{R})/\operatorname{SL}_d(\mathbb{Z})$ for $d\geq 2$, and let A be the subgroup of diagonal matrices with positive diagonal entries and determi...

conjecture | open | MathDB 页面 | 原文线索

V2-001501 | MATHDB-320994

Harbourne's bounded cohomology conjecture

摘录 / EXCERPT: Let X be a smooth projective surface that is either rational or defined in characteristic zero. The conjecture asserts that there exists a constant c_X such that every prime di...

conjecture | open | MathDB 页面 | 原文线索

V2-001502 | MATHDB-320999

The positive-radius conjecture for the symplectic zeta function

摘录 / EXCERPT: Let M be a compact symplectic manifold and let $\phi:M\rightarrow M$ be a symplectomorphism such that the Floer homology groups $HF_*(\phi^n)$ are well defined for every...

conjecture | open | MathDB 页面 | 原文线索

V2-001503 | MATHDB-321014

Galbraith-McKee conjecture for prime orders of elliptic curves

摘录 / EXCERPT: For a prime $p>3$, let $P_1(p)$ be the probability that a uniformly randomly chosen integer in the Hasse interval $[p+1-2\sqrt{p},p+1+2\sqrt{p}]$ is prime. Let $P_2(p)$ be the prob...

conjecture | open | MathDB 页面 | 原文线索

V2-001504 | MATHDB-321015

Rational-Weierstrass genus-2 Jacobian prime-order conjecture

摘录 / EXCERPT: For a prime $p>2$, let $P_1(p)$ be the probability that a uniformly randomly chosen integer in the Hasse-Weil interval $[(\sqrt p-1)^4,(\sqrt p+1)^4]$ is prime, and let $P_2(p)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-001505 | MATHDB-321053

Ziegler's conjecture on power bases of units in pure number-field orders

摘录 / EXCERPT: Let $d\geq 2$ be an integer and let $m\in\mathbb{Z}\setminus\{0\}$ be such that $\sqrt[d]{m}$ has algebraic degree d. The order $\mathbb{Z}[\sqrt[d]{m}]$ is the subring generated...

conjecture | open | MathDB 页面 | 原文线索

V2-001506 | MATHDB-321068

The universality conjecture for $T_{2n-1}(p^r)$

摘录 / EXCERPT: Let p be a prime, let n,r be positive integers, and let $P^{2n}(p^r)\xrightarrow{i}T_{2n-1}(p^r)$ be the canonical map. A space Z is strongly $(2n,r)$-subexponential if ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-001507 | MATHDB-321099

The binary-representation conjecture for fractional parts of binomial coefficients

摘录 / EXCERPT: Let $p\neq 2$ be a prime, let $m=2p$, and let $m_l\dots m_1m_0$ be the standard base-2 representation of m. Let $n\geq 1$. Write n_1 for the leading digit in the base...

conjecture | open | MathDB 页面 | 原文线索

V2-001508 | MATHDB-321120

Götsche's generating-function conjecture for virtual curve numbers

摘录 / EXCERPT: Let $(S,\mathscr{L})$ be a fixed polarized smooth projective surface, with Chern numbers $\partial=\mathscr{L}^2$, $k=\mathscr{L}\mathscr{K}_S$, $s=\mathscr{K}_S^2$, and $x=c_2(S)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001509 | MATHDB-321153

Craven's algebraicity-periodicity conjecture for modules of elementary abelian groups

摘录 / EXCERPT: Let p be an odd prime, let k be a field of characteristic p, and let $G=C_p\times C_p$. An absolutely indecomposable kG-module M is of dimension a multiple of p when...

conjecture | open | MathDB 页面 | 原文线索

V2-001510 | MATHDB-321169

Periodicity conjecture for integer sequences with bounded second-order recurrence errors

摘录 / EXCERPT: Let λ be a real number with $-2<\lambda<2$, and let (a_n) be an integer sequence satisfying ... for every relevant index n. Periodicity conjecture. The sequence (a_n) i...

conjecture | open | MathDB 页面 | 原文线索

V2-001511 | MATHDB-321191

The jumping champion conjecture

摘录 / EXCERPT: Jumping champion conjecture. The jumping champions greater than 1 are 4 and the primorials $p_1^{\sharp},p_2^{\sharp},p_3^{\sharp},\ldots$, namely $2,6,30,210,2310,\ldots$. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-001512 | MATHDB-321192

The weak jumping champion conjecture

摘录 / EXCERPT: For each x, let a jumping champion be a gap between consecutive primes up to x occurring with maximal frequency. A prime p is said to divide a jumping champion when it divide...

conjecture | open | MathDB 页面 | 原文线索

V2-001513 | MATHDB-321193

The relative-density conjecture for primitive smooth solutions

摘录 / EXCERPT: Relative-density conjecture for primitive solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-001514 | MATHDB-321201

The prime-index irreducibility conjecture for Stern polynomials

摘录 / EXCERPT: Let $B_p(t)$ be the Stern polynomial indexed by a prime number p. The prime-index irreducibility conjecture. For every prime number p, the polynomial $B_p(t)$ is irreducible. T...

conjecture | open | MathDB 页面 | 原文线索

V2-001515 | MATHDB-321202

The irreducibility conjecture for Stern polynomials with prescribed prime divisors

摘录 / EXCERPT: Let $B_n(t)$ denote the Stern polynomial indexed by the positive integer n. For a positive integer n, say that it has exactly k prime divisors, with the source's convention f...

conjecture | open | MathDB 页面 | 原文线索

V2-001516 | MATHDB-321210

A conjecture on the least prime in powers-of-two arithmetic progressions

摘录 / EXCERPT: Let $p(r,a)$ denote the least prime in the arithmetic progression a modulo r. Here r is a power of 2 and $a=1$. Prime progression conjecture. There exists a constant $c>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-001517 | MATHDB-321227

The exceptional-locus parameterization conjecture for almost large modules

摘录 / EXCERPT: Let A be a finitely generated basic algebra, module-finite over its prime center Z, and let d be the dimension vector of a large A-module. Assume that A is homologically...

conjecture | open | MathDB 页面 | 原文线索

V2-001518 | MATHDB-321269

Weakly strong Neuwirth conjecture for prime non-torus knots

摘录 / EXCERPT: Let K be a prime non-torus knot, let F be a non-orientable spanning surface for K, and let $E(K)$ be the exterior of K. The surface $F\cap E(K)$ is required to be geometric...

conjecture | open | MathDB 页面 | 原文线索

V2-001519 | MATHDB-321270

Strong even boundary slope conjecture for prime non-torus knots

摘录 / EXCERPT: Let K be a prime non-torus knot and let $E(K)$ be its exterior. A properly embedded orientable incompressible and boundary incompressible surface in $E(K)$ may have a boundary sl...

conjecture | open | MathDB 页面 | 原文线索

V2-001520 | MATHDB-321286

Kuijlaars and Saff's conjecture for the planar Riesz-energy constant

摘录 / EXCERPT: Let $C_{s,d}$ be the asymptotic Riesz-energy constant defined, for $d\geq 2$, by ... Here $\mathcal{E}_s([0,1]^d;N)$ denotes the optimal discrete Riesz s-energy of N points in th...

conjecture | open | MathDB 页面 | 原文线索

V2-001521 | MATHDB-321327

576-periodicity conjecture for free fermion conformal nets

摘录 / EXCERPT: Let $\mathit{Fer}(n)$ denote the conformal net of n free fermions, and let CN denote the 3-category of conformal nets, with its notion of CN-equivalence. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-001522 | MATHDB-321339

The order conjecture for strong Fermat solutions at p equals 3

摘录 / EXCERPT: Order conjecture. There exists such a solution (u,v) for which the order of v/u modulo q is at least m.

conjecture | open | MathDB 页面 | 原文线索

V2-001523 | MATHDB-321341

The finite-field matching conjecture for equal-sum collisions

摘录 / EXCERPT: Let p be a prime, let $k\leq p$, and let $a_1,\dots,a_k\in\mathbb F_p$ and $b_1,\dots,b_k\in\mathbb F_p$ be arbitrary elements. Finite-field matching conjecture. There is a permu...

conjecture | open | MathDB 页面 | 原文线索

V2-001524 | MATHDB-321354

Conjectural archimedean estimates for multiplicative character sums

摘录 / EXCERPT: Let $L(t)$ be the L-function associated to the sums (1.2), under the hypotheses (1.1) and (2.1), with all characters χ_i nontrivial as in (2.5). Archimedean estimate...

conjecture | open | MathDB 页面 | 原文线索

V2-001525 | MATHDB-321401

Nikolaev's Q-rank conjecture for CM elliptic curves

摘录 / EXCERPT: Let p be a prime with $p=3\pmod 4$, let ${\cal E}(p)={\cal E}_{CM}^{(-p,1)}$ be the associated $\mathbb Q$-curve, and let h_K be the class number of the relevant imaginary quad...

conjecture | open | MathDB 页面 | 原文线索

V2-001526 | MATHDB-321434

Cohomological finiteness conjecture for signed formal groups

摘录 / EXCERPT: Let K_∞ be the local extension occurring above a prime of L over p, let $\mathcal H=\operatorname{Gal}(K_\infty/\mathbb Q_{p,\infty})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001527 | MATHDB-321443

The restricted-radical ABC solution-count conjecture

摘录 / EXCERPT: Restricted-radical ABC solution-count conjecture. If $a+b+c>1$, then for sufficiently large N the number of such solutions lies between

conjecture | open | MathDB 页面 | 原文线索

V2-001528 | MATHDB-321446

The limiting-distribution conjecture for totient multiplicities

摘录 / EXCERPT: Let ϕ be Euler's totient function. For $x\geqslant 1$, let $V(x)$ be the number of totients at most x, and for an integer $k\geqslant 1$, let $V_k(x)$ be the number of totie...

conjecture | open | MathDB 页面 | 原文线索

V2-001529 | MATHDB-321452

Gross-Kohnen-Zagier-type conjecture for Darmon's points

摘录 / EXCERPT: Assume that $E(F)$ has rank 1, choose a generator P_0 of $E(F)$ modulo torsion, and for each admissible totally positive $t\in O_F$ let $P_t\in E(F)$ be the trace of the Darmon...

conjecture | open | MathDB 页面 | 原文线索

V2-001530 | MATHDB-321475

Nahm's conjecture on Bloch-group torsion and modularity

摘录 / EXCERPT: Let A be a positive definite symmetric $r\times r$ matrix with rational coefficients. Let $(Q_1,\dots,Q_r)$ range over the solutions in number fields of the algebraic system ... ma...

conjecture | open | MathDB 页面 | 原文线索

V2-001531 | MATHDB-321480

The faithful linear representation conjecture for p-compact groups

摘录 / EXCERPT: Let p be a prime and let $B\Gamma$ be the classifying space of a p-compact group. A faithful linear representation means a map from $B\Gamma$ to $BSU(n)$, for some n, whose h...

conjecture | open | MathDB 页面 | 原文线索

V2-001532 | MATHDB-321482

The distinct odd-prime refinement of Goldbach's conjecture

摘录 / EXCERPT: Let $Prime(q)$ be the quota complex formed from odd primes whose total sum is below q, and let $H_1(Prime(q),\mathbb{Q})$ denote its first homology group. The distinct odd-prime...

conjecture | open | MathDB 页面 | 原文线索

V2-001533 | MATHDB-321483

The three-distinct-odd-prime refinement of the weak Goldbach conjecture

摘录 / EXCERPT: Let $Prime(q)$ be the quota complex formed from odd primes whose total sum is below q, and let $H_2(Prime(q),\mathbb{Q})$ denote its second homology group. The three-distinct-odd...

conjecture | open | MathDB 页面 | 原文线索

V2-001534 | MATHDB-321484

The asymptotic counting conjecture for sums of distinct odd primes

摘录 / EXCERPT: The odd-prime sum-counting conjecture. For these constants,

conjecture | open | MathDB 页面 | 原文线索

V2-001535 | MATHDB-321485

The asymptotic counting conjecture for sums of distinct squares and cubes

摘录 / EXCERPT: The square-and-cube sum-counting conjecture. The asymptotics are

conjecture | open | MathDB 页面 | 原文线索

V2-001536 | MATHDB-321524

The uniform prime-order conjugacy-class generation conjecture for finite simple groups of Lie type

摘录 / EXCERPT: Prime-order generation conjecture. With finitely many exceptions, there exist conjugacy classes C and D of G consisting of elements of prime order such that

conjecture | open | MathDB 页面 | 原文线索

V2-001537 | MATHDB-321525

Definability of primality for radical differential ideals

摘录 / EXCERPT: Definability of primality. For every d there exists $r=r(d,n,m)$ such that, for every such k and S, the following are equivalent: (i) $\{S\}$ is prime; (ii) $\{S\}$ is proper...

conjecture | open | MathDB 页面 | 原文线索

V2-001538 | MATHDB-321533

The inverse sieve conjecture in dimension one

摘录 / EXCERPT: Let $[N]=\{1,\ldots,N\}$. Suppose $S\subseteq[N]$ is a set of integers with $|S|\ge N^{\varepsilon}$, occupying fewer than αp residue classes modulo every prime p, where...

conjecture | open | MathDB 页面 | 原文线索

V2-001539 | MATHDB-321544

Kenyon-Smillie conjecture on lattice-property rational triangles

摘录 / EXCERPT: Kenyon-Smillie conjecture. One of the following conditions holds: $n\leq 78$, $s+t=n$, $s+2t=n$, $2s+t=n$, or n is even and $|t-s|=\frac{n}{2}$. This arithmetic conjecture would...

conjecture | open | MathDB 页面 | 原文线索

V2-001540 | MATHDB-321562

Longest-run conjecture for Ramanujan and non-Ramanujan primes

摘录 / EXCERPT: Longest-run conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001541 | MATHDB-321563

Twin Ramanujan-prime ratio bounds

摘录 / EXCERPT: Twin Ramanujan-prime ratio conjecture. For all $x\ge10^5$,

conjecture | open | MathDB 页面 | 原文线索

V2-001542 | MATHDB-321564

Asymptotic equipartition of twin Ramanujan-prime pairs

摘录 / EXCERPT: Twin Ramanujan-prime asymptotic conjecture. If $\pi_2(x)\to\infty$ as $x\to\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001543 | MATHDB-321569

Schinzel's conjecture on almost-prime values in arithmetic progressions

摘录 / EXCERPT: Let $\mathcal{O}=b+q\mathbb{Z}$ be an arithmetic progression, and let $f(x)\in\mathbb{Q}[x]$ be integral on $\mathcal{O}$. The pair $(\mathcal{O},f)$ is primitive if for every inte...

conjecture | open | MathDB 页面 | 原文线索

V2-001544 | MATHDB-321570

Hardy-Littlewood prime-vector conjecture

摘录 / EXCERPT: Let Λ be a subgroup of $\mathbb{Z}^n$, let $b\in\mathbb{Z}^n$, and let $\mathcal{O}=b+\Lambda$. Assume each coordinate function x_j is nonconstant on Λ, and defin...

conjecture | open | MathDB 页面 | 原文线索

V2-001545 | MATHDB-321615

Asymptotic sharpness of the characteristic bound for Heisenberg-group modules

摘录 / EXCERPT: Asymptotic sharpness conjecture. There exists a prime q such that, for every prime $p \geq q$, there exists a $\frac{p+1}{2}$-dimensional module for H_1 over a field of charact...

conjecture | open | MathDB 页面 | 原文线索

V2-001546 | MATHDB-321652

Discrete fractional integral conjecture for $I_{k,\lambda}$

摘录 / EXCERPT: Discrete fractional integral conjecture. For $0<\lambda<1$ and $1\leq p,q\leq\infty$, $I_{k,\lambda}$ is a bounded operator from $\ell^p(\mathbb{Z})$ to $\ell^q(\mathbb{Z})$ if and...

conjecture | open | MathDB 页面 | 原文线索

V2-001547 | MATHDB-321663

Kummer-Vandiver conjecture on cyclotomic class-group eigenspaces

摘录 / EXCERPT: Let l be a prime, let $A=Cl(\mathbb Z[\mu_l])_l$, and let $A^{[i]}$ be the eigenspace corresponding to the i-th power of the Teichmüller character...

conjecture | open | MathDB 页面 | 原文线索

V2-001548 | MATHDB-321664

Mahler's rational-counting conjecture for the standard Cantor set

摘录 / EXCERPT: Let C be the standard Cantor set, and let ... Here $\gcd(p,q)=1$ means that the rational is written in reduced form. Mahler's rational-counting conjecture. For every $\epsilon>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001549 | MATHDB-321665

A subquadratic rational-counting conjecture for the standard Cantor set

摘录 / EXCERPT: Subquadratic rational-counting conjecture. There exists a constant $\gamma<2$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001550 | MATHDB-321680

Infinite generation conjecture for the Schur multiplier of Fabrykowski-Gupta groups

摘录 / EXCERPT: Let Γ_d be a Fabrykowski-Gupta group with d prime, and let $M(G)$ denote a Schur multiplier. Fabrykowski-Gupta conjecture. The Schur multiplier of Γ_d, modulo the D...

conjecture | open | MathDB 页面 | 原文线索

V2-001551 | MATHDB-321799

Kaneko-Ohno conjecture on duality differences of multiple zeta-star values

摘录 / EXCERPT: For positive integers k,n,s with $k\geq n+s$ and $n\geq s\geq 1$, let $X_0^{\star}(k,n,s)$ denote the sum of all multiple zeta-star values of weight k, depth n, and height...

conjecture | open | MathDB 页面 | 原文线索

V2-001552 | MATHDB-321806

The global Serre weight conjecture for unitary Galois representations

摘录 / EXCERPT: Fix an imaginary CM field F with maximal totally real subfield F^+ such that every prime of F^+ above p has residue field ${\mathbb{F}}_p$ and splits in F. Let S be the...

conjecture | open | MathDB 页面 | 原文线索

V2-001553 | MATHDB-321851

The asymptotic count of dismal primes

摘录 / EXCERPT: Let $\pi_b(k)$ denote the number of base b dismal primes with k digits. Prime-counting conjecture. As k tends to infinity, ... The conjecture predicts that asymptotically almos...

conjecture | open | MathDB 页面 | 原文线索

V2-001554 | MATHDB-321852

The maximum number of dismal divisors in base b

摘录 / EXCERPT: Let $d_b(n)$ denote the number of dismal divisors of n in base b, and let n range over the k-digit base-b numbers. Maximum-divisor conjecture. For every base $b\geq 3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001555 | MATHDB-321853

The maximum number of dismal divisors in base 2

摘录 / EXCERPT: Let $d_2(n)$ denote the number of dismal divisors of the k-digit binary number n. Binary maximum-divisor conjecture. Among all k-digit numbers, the maximal value of $d_2(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001556 | MATHDB-321854

The two largest dismal-divisor counts among odd binary numbers

摘录 / EXCERPT: Let $d_2(n)$ denote the number of dismal divisors of n, and let n range over the odd k-digit binary numbers. Odd-binary divisor-count conjecture. The unique maximum of...

conjecture | open | MathDB 页面 | 原文线索

V2-001557 | MATHDB-321855

The strict inequality between dismal divisor counts and their sum

摘录 / EXCERPT: Let $d_{10}(n)$ denote the number of dismal divisors of n in base 10, and let $\sigma_{10}(n)$ denote the dismal sum of those divisors. Dismal divisor-sum inequality. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-001558 | MATHDB-321873

Tutte and Kauffman polynomials detect primeness of knots and links

摘录 / EXCERPT: Primeness-detection conjecture. The Tutte polynomial and the Kauffman two-variable polynomial detect primeness: they are not factorizable for prime KLs.

conjecture | open | MathDB 页面 | 原文线索

V2-001559 | MATHDB-321958

Quantitative density conjecture for values of the Riemann zeta function

摘录 / EXCERPT: Let m denote Lebesgue measure on the complex plane, and let $\log\zeta(1/2+iu)$ use the normalization specified in the source. For a bounded Jordan-measurable set...

conjecture | open | MathDB 页面 | 原文线索

V2-001560 | MATHDB-321963

Realization conjecture for arithmetic invariants of hyperbolic 3-manifolds

摘录 / EXCERPT: Let k be a non-real concrete number field, and let A be a quaternion algebra over k. For a hyperbolic manifold M, write k and A for its invariant trace field and invari...

conjecture | open | MathDB 页面 | 原文线索

V2-001561 | MATHDB-321964

Rigidity conjecture for the homology of the discrete complex projective linear group

摘录 / EXCERPT: Let $\operatorname{PSL}(2,\mathbb C)^\delta$ denote $\operatorname{PSL}(2,\mathbb C)$ with the discrete topology, and let K range over all concrete number fields. The groups...

conjecture | open | MathDB 页面 | 原文线索

V2-001562 | MATHDB-321971

The field-theoretic model conjecture for periodic topological modular forms

摘录 / EXCERPT: The paper considers $2|1$-dimensional Euclidean field theories, whose partition functions are modular functions, and writes $2|1\text{-}\operatorname{EFT}^n_{loc}[X]$ for the degre...

conjecture | open | MathDB 页面 | 原文线索

V2-001563 | MATHDB-321984

The bounded exponent conjecture for fixed points of coprime operator groups

摘录 / EXCERPT: Bounded exponent conjecture. If $\gamma_{r-1}(C_G(a))$ has exponent dividing m for every $a\in A^\#$, then $\gamma_{r-1}(G)$ has $\{m,q,r\}$-bounded exponent. Moreover, if d is...

conjecture | open | MathDB 页面 | 原文线索

V2-001564 | MATHDB-321987

Asymptotic norm conjecture for restricted Saito-Kurokawa lifts

摘录 / EXCERPT: Asymptotic norm conjecture. As k tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-001565 | MATHDB-321992

The Sinnott-Kurihara integrality conjecture

摘录 / EXCERPT: Let L/K be a Galois extension with Galois group G. Assume that S contains all infinite and ramified places of K, that S is disjoint from a finite set T of places, and t...

conjecture | open | MathDB 页面 | 原文线索

V2-001566 | MATHDB-321993

A negative-integer integrality conjecture for equivariant L-values

摘录 / EXCERPT: Let L/K be a Galois extension of number fields with Galois group G, let $r<0$, and let $\mu_{1-r}(L):=(\mathbb{Q}/\mathbb{Z})(1-r)^{G_L}$, where G_L is the absolute Galois gr...

open_problem | open | MathDB 页面 | 原文线索

V2-001567 | MATHDB-321994

The non-abelian Brumer conjecture

摘录 / EXCERPT: Let L/K be a Galois CM-extension with Galois group G. Let S be a finite set of places of K containing all ramified and infinite places, let $\mathcal A_S$ be the module def...

conjecture | open | MathDB 页面 | 原文线索

V2-001568 | MATHDB-321995

The non-abelian Brumer-Stark conjecture

摘录 / EXCERPT: Let L/K be a Galois CM-extension with Galois group G. Let S be a finite set of places of K containing all ramified and infinite places. For $\alpha\in L^\times$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-001569 | MATHDB-322017

Mossinghoff-Trudgian sign-change conjecture for weighted Liouville sums

摘录 / EXCERPT: Mossinghoff-Trudgian conjecture. For $0<\alpha<1/2$, the weighted sum $L_{\alpha}(x)$ changes sign infinitely often. The source notes that this has been proved unconditionally for...

conjecture | open | MathDB 页面 | 原文线索

V2-001570 | MATHDB-322018

Refined conjecture for the half-weighted Liouville sum

摘录 / EXCERPT: Refined half-weight conjecture. As x tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-001571 | MATHDB-322030

Equivalent harmonic-sum congruence for Sun's Apéry-type conjecture

摘录 / EXCERPT: Let $p>3$ be a prime, and define ... with B_{p-5} the Bernoulli number of index $p-5$. Equivalent form of Sun's conjecture. ... The paper states that this congruence is equivalent, v...

conjecture | open | MathDB 页面 | 原文线索

V2-001572 | MATHDB-322044

The normal-closure inequality for torsion-free groups and p-groups

摘录 / EXCERPT: Let G be a torsion free discrete group, or a p-group for a prime p, and let $g_1,\dots,g_n$ be arbitrary elements of G. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-001573 | MATHDB-322058

Hall-Lang conjecture on integral points of elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve in minimal Weierstrass form, let $P \in E(\mathbb{Q})$ be an integral point of infinite order, let $\widehat{h}(P)$ denote its canonical hei...

conjecture | open | MathDB 页面 | 原文线索

V2-001574 | MATHDB-322068

The stronger uniform Hardy-Littlewood conjecture for two tuple sizes

摘录 / EXCERPT: Stronger uniform Hardy-Littlewood conjecture. For $n=k+1,k+2$, if $\mathfrak{S}(\{0\}\cup\mathcal{D}_n)\neq0$, then, as $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-001575 | MATHDB-322072

Tyszka's bound conjecture for solvable linear systems

摘录 / EXCERPT: Tyszka's conjecture. The system has a rational solution satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-001576 | MATHDB-322084

Kawanaka's conjecture on generalized Gelfand-Graev representations

摘录 / EXCERPT: Let $\bG$ be a connected reductive algebraic group, let p be a good prime for $\bG$, and write G for the associated finite group of fixed points. For each unipotent element u...

conjecture | open | MathDB 页面 | 原文线索

V2-001577 | MATHDB-322097

Selberg central limit theorem conjecture for small deviations of the zeta function

摘录 / EXCERPT: Small-deviation conjecture. If $\Delta=o(\sqrt{\log\log T})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001578 | MATHDB-322111

Canonical p-adic L-function basis conjecture

摘录 / EXCERPT: Let V be the representation attached to a weight-2 modular form over an imaginary quadratic field K, and let K_∞/K have Galois group G. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001579 | MATHDB-322126

Facet-count conjecture for three-orbit cyclic permutation polytopes

摘录 / EXCERPT: Let $a,b,c\geq 2$ be pairwise coprime integers, and let $P(a,b,c)$ be the permutation polytope associated with three disjoint cycles of lengths ab, ac, and bc. Facet-count co...

conjecture | open | MathDB 页面 | 原文线索

V2-001580 | MATHDB-322145

Gras's strong Fermat's last theorem conjecture

摘录 / EXCERPT: Let p be an odd prime, set $K=\text{\mathbb Q}(\zeta)$ and $\mathfrak p=(\zeta-1)\text{\mathbb Z}[\zeta]$. For coprime integers u,v, let $\delta\geq 0$ and let $\mathfrak w_1$...

conjecture | open | MathDB 页面 | 原文线索

V2-001581 | MATHDB-322146

Cyclotomic Kummer splitting criterion implying SFLT2

摘录 / EXCERPT: Let $p>3$ be prime and let $(p,u,v)\in\mathbb Z^3$ with u and v coprime and $p\mid v$. For a p-principal prime q with $q\nmid puv$, let n be the order of v/u modulo q...

conjecture | open | MathDB 页面 | 原文线索

V2-001582 | MATHDB-322163

Linear independence of Riemann zeta zero ordinates

摘录 / EXCERPT: Let $\zeta(s)$ be the Riemann zeta function, and consider the positive imaginary parts of its nontrivial zeros. The zeta linear-independence conjecture. These imaginary parts are l...

conjecture | open | MathDB 页面 | 原文线索

V2-001583 | MATHDB-322222

Nahm's conjecture on modular triples and Bloch-group torsion

摘录 / EXCERPT: Let $A=(a_{ij})$ be a positive definite symmetric $r\times r$ matrix with rational entries. Consider the system ... For a solution $\mathbf{x}=(x_1,\ldots,x_r)$ in a number field F...

conjecture | open | MathDB 页面 | 原文线索

V2-001584 | MATHDB-322225

The chromatic unit-map conjecture for Johnson-Wilson spectra

摘录 / EXCERPT: Let $BP\langle n\rangle$ denote the Johnson-Wilson spectrum at the prime 2, with coefficient ring $\mathbb{Z}_{(2)}[v_1,\dots,v_n]$. Assume that $BP\langle n\rangle$ admits the s...

conjecture | open | MathDB 页面 | 原文线索

V2-001585 | MATHDB-322288

Midpoint approximation for jumps of the Chebyshev psi function

摘录 / EXCERPT: Midpoint jump conjecture. The jump at the prime p_n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001586 | MATHDB-322289

Midpoint approximation for jumps at prime powers

摘录 / EXCERPT: Prime-power midpoint conjecture. The jump satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001587 | MATHDB-322290

Normalized Chebyshev criterion for generalized Chebyshev primes

摘录 / EXCERPT: Normalized Chebyshev-prime criterion. A Chebyshev prime of index l is a prime p_m satisfying the displayed inequality. This criterion is the formulation used to relate generali...

conjecture | open | MathDB 页面 | 原文线索

V2-001588 | MATHDB-322291

Negative-jump conjecture for the truncated psi-based prime-counting function

摘录 / EXCERPT: Negative-jump conjecture. Negative jumps of $\eta_N(x)$ occur at every point $x+1\in\mathcal P$, where $\mathcal P$ is the set of primes, and

conjecture | open | MathDB 页面 | 原文线索

V2-001589 | MATHDB-322363

Kuhn's realization conjecture for finite generation over the Steenrod algebra

摘录 / EXCERPT: Fix a finite prime field $\mathbb{F}_p$ and let $\mathcal{A}_p$ denote the Steenrod algebra. For a topological space X, write $H^*(X)$ for its cohomology with coefficients in...

conjecture | open | MathDB 页面 | 原文线索

V2-001590 | MATHDB-322392

Mean-square bound for shifted products of non-holomorphic cusp-form coefficients

摘录 / EXCERPT: Mean-square shifted-product conjecture. The displayed estimate should hold.

conjecture | open | MathDB 页面 | 原文线索

V2-001591 | MATHDB-322409

The conjecture on sums of distinct primes beyond the consecutive-prime bound

摘录 / EXCERPT: Let $s\ge 3$ be an integer, and let $p_1=2,p_2=3,\ldots$ denote the consecutive primes. Distinct-prime sum conjecture. Every integer $l>p_2+p_3+\cdots+p_{s+2}$ is the sum of exactl...

conjecture | open | MathDB 页面 | 原文线索

V2-001592 | MATHDB-322414

Rademacher's infinite partial fraction conjecture for partition generating functions

摘录 / EXCERPT: Let $p_N(n)$ denote the number of partitions of n into at most N parts, and write ... with partial-fraction coefficients $C_{h,k,l}(N)$ defined by ... Here $s(h,k)$ is the Dedekind...

conjecture | open | MathDB 页面 | 原文线索

V2-001593 | MATHDB-322415

Oscillation conjecture for the first Rademacher coefficient sequence

摘录 / EXCERPT: Let $C_{0,1,1}(N)$ be the Rademacher coefficient associated with the pole at $x=1$ in the partial-fraction decomposition of the generating function for partitions into at most N...

conjecture | open | MathDB 页面 | 原文线索

V2-001594 | MATHDB-322440

Bounded average simplification-length conjecture for 3-manifold triangulations

摘录 / EXCERPT: Let σ_n be the average simplification path length for 3-sphere triangulations at level n, and let π_n be the corresponding average for closed prime orientable 3-manifo...

conjecture | open | MathDB 页面 | 原文线索

V2-001595 | MATHDB-322445

The two-variable main conjecture of Iwasawa theory for elliptic curves

摘录 / EXCERPT: Fix a prime p. Let $G=\operatorname{Gal}(k_{\infty}/k)$, where k_{∞} is the compositum of all $\mathbb{Z}_p$-extensions of an imaginary quadratic field k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001596 | MATHDB-322450

Non-negativity conjecture for cylindric Kostka polynomials

摘录 / EXCERPT: Let $P_{{\Greekmath 0115} /d/{\Greekmath 0116} }^{\prime }(x_{1},\ldots,x_{n-1};t)$ be expanded in Schur functions indexed by partitions in $\mathcal{P}_{n,k}^{+}$: ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-001597 | MATHDB-322454

The conjecture on the spectrum of growth rate functions for exponentially dense matroid classes

摘录 / EXCERPT: Let q be a prime power, and let $\mathcal{M}$ be a base-q exponentially dense minor-closed class of matroids, meaning that its growth rate is eventually of order q^n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001598 | MATHDB-322455

The structural conjecture for extremal matroids in exponentially dense classes

摘录 / EXCERPT: Let q be a prime power, and let $\mathcal{M}$ be a base-q exponentially dense minor-closed class of matroids, meaning that its growth rate is eventually of order q^n. For a n...

conjecture | open | MathDB 页面 | 原文线索

V2-001599 | MATHDB-322462

The prime-ideal conjecture for pairs of power sums in four variables

摘录 / EXCERPT: Let $S=\mathbb{C}[x_1,x_2,x_3,x_4]$. The prime-ideal conjecture for pairs of power sums. The following assertions hold: (1) if a is prime, $a\ge5$, and $b=a+m+6d$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-001600 | MATHDB-322489

Kaplansky's conjecture for the three remaining ternary forms

摘录 / EXCERPT: Kaplansky's conjecture. Each of these ternary quadratic forms represents every positive odd integer; that is, for every positive odd integer n, the equation $Q(x,y,z)=n$ is solva...

conjecture | open | MathDB 页面 | 原文线索

V2-001601 | MATHDB-322508

Conjecture that tame Steinitz classes form a subgroup

摘录 / EXCERPT: Tame Steinitz class subgroup conjecture. For any number field k and any finite group G, $\operatorname{R}_t(k,G)$ is a subgroup of $\operatorname{Cl}(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-001602 | MATHDB-322515

Langlands program for p-adic coefficients

摘录 / EXCERPT: Let p be a prime, let k be a finite field with $q=p^s$ elements, and let X be a smooth proper geometrically connected curve over $\operatorname{Spec}(k)$ with function field...

conjecture | open | MathDB 页面 | 原文线索

V2-001603 | MATHDB-322532

Wilf's conjecture for numerical semigroups

摘录 / EXCERPT: Let Λ be a numerical semigroup, meaning a cofinite subsemigroup of the non-negative integers. Let $e(\Lambda)$ be the size of its minimal generating set, let...

conjecture | open | MathDB 页面 | 原文线索

V2-001604 | MATHDB-322546

Uniform Hardy-Littlewood prime k-tuple asymptotic in the logarithmic range

摘录 / EXCERPT: Uniform logarithmic-range prime k-tuple conjecture. For a positive constant λ, if $h\sim\lambda\log x$ as $x\to\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001605 | MATHDB-322553

Conjecture on the number of Beauville p-groups of order p^5

摘录 / EXCERPT: Let p be a prime, let $g(p)$ denote the number of Beauville p-groups of order p^{5}, and let H_3 and H_4 be the groups referred to in the preceding construction. The Beau...

conjecture | open | MathDB 页面 | 原文线索

V2-001606 | MATHDB-322564

Conjecture on the 1-level density beyond finite support

摘录 / EXCERPT: Let f be the test function in the average 1-level density, and let σ denote the finite support parameter. Theorem 3/2(3) gives an expansion involving the terms of order...

conjecture | open | MathDB 页面 | 原文线索

V2-001607 | MATHDB-322569

Four-degree complete-intersection gcd conjecture

摘录 / EXCERPT: Four-degree gcd conjecture. The integer G_n has the form 2^a3^b for some nonnegative integers a,b with $a+b>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-001608 | MATHDB-322598

Asymptotic density conjecture for p-practical numbers

摘录 / EXCERPT: Asymptotic density conjecture. For each prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-001609 | MATHDB-322600

The square-root conjecture for incomplete Kloosterman sums

摘录 / EXCERPT: Let c be a positive integer, let a and b be integers, and let $\mathcal{I}$ be an interval with $\#\mathcal{I}<c$. Define ... where $x\overline{x}\equiv1\pmod c$ and the asteri...

conjecture | open | MathDB 页面 | 原文线索

V2-001610 | MATHDB-322601

Hooley's R^*-conjecture for incomplete Kloosterman sums

摘录 / EXCERPT: Let c be a positive integer, let b be an integer, and let $\mathcal{I}$ be an interval satisfying $c^{1/4}<\#\mathcal{I}<c$. Define ... where $x\overline{x}\equiv1\pmod c$ and th...

conjecture | open | MathDB 页面 | 原文线索

V2-001611 | MATHDB-322624

Conjecture on the second moment of prime gaps in arithmetic progressions

摘录 / EXCERPT: Fix integers a and q with $(a,q)=1$, and let $p_i^{a,q}$ denote the i^{th} prime congruent to a modulo q. The arithmetic-progression prime-gap conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-001612 | MATHDB-322688

Faber-van der Geer Hecke-trace conjecture for vector-valued Siegel cusp forms

摘录 / EXCERPT: Let $a\geq b\geq 0$ with $(a,b)\neq(0,0)$, let $V_{a,b}$ be the corresponding local system on $\A{2}$, let $T(p)$ be the Hecke operator at p, and let $S_{a-b,b+3}$ be the space o...

conjecture | open | MathDB 页面 | 原文线索

V2-001613 | MATHDB-322698

Beukers-Stewart inequality

摘录 / EXCERPT: Let x, y, m, and n be the integers occurring in the relevant exponential Diophantine equation. Beukers-Stewart conjecture. The asserted inequality is ... The supplied excerpt...

conjecture | open | MathDB 页面 | 原文线索

V2-001614 | MATHDB-322722

Howard's nonvanishing criterion for Hilbert modular forms

摘录 / EXCERPT: Let ${\bf{f}}$ be the Hilbert modular eigenform and K/F the totally imaginary quadratic extension introduced in the paper. Let $\Lambda=\mathcal{O}[[G_{\mathfrak{p}^{\infty}}]]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-001615 | MATHDB-322723

Nonvanishing conjecture for Rankin-Selberg central values

摘录 / EXCERPT: Let ${\bf{f}} \in \mathcal{S}_2(\mathfrak{N})$ be a cuspidal Hilbert modular eigenform, and let K/F be a totally imaginary quadratic extension such that the root number of...

conjecture | open | MathDB 页面 | 原文线索

V2-001616 | MATHDB-322739

The odd-prime composition-factor and triple conjecture

摘录 / EXCERPT: Odd-prime composition-factor and triple conjecture. The following statements are equivalent: (i) G contains a composition factor whose order is divisible by pq; and (ii) G co...

conjecture | open | MathDB 页面 | 原文线索

V2-001617 | MATHDB-322745

Injectivity and differentiability conjecture for the quasi-periodic renormalization operator

摘录 / EXCERPT: Let $\Omega=\Omega_{\gamma,\tau}$ be the set of Diophantine rotation numbers, let $\mathcal{B}$ be the Banach space of maps, and let $\mathcal{D}(\mathcal{T})$ be the domain of the...

conjecture | open | MathDB 页面 | 原文线索

V2-001618 | MATHDB-322746

Asymptotic equivalence and nondegeneracy conjecture for reducibility-loss directions

摘录 / EXCERPT: Let $\{c(\alpha,\varepsilon)\}_{(\alpha,\varepsilon)\in A}$ be a family satisfying H1 and H2, let ω_0 be Diophantine, and let $\tilde v^{(n)}_{n-1}$ and v_{n-1} be the vec...

conjecture | open | MathDB 页面 | 原文线索

V2-001619 | MATHDB-322749

Berndt-Kim eventual sign conjecture for partial theta coefficients

摘录 / EXCERPT: Berndt-Kim conjecture. For sufficiently large n, the coefficients a_n have the same sign.

conjecture | open | MathDB 页面 | 原文线索

V2-001620 | MATHDB-322754

Moree-Lenstra-Murata conjecture on near-primitive root densities for square-free integers

摘录 / EXCERPT: Let $g>1$ be a square-free integer and $t\ge1$ an integer. Let $N_{g,t}$ be the set of primes p with $\nu_p(g)=0$ and residual index $r_g(p)=t$, and let $\delta(g,t)$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-001621 | MATHDB-322755

Moree's conjecture on near-primitive root densities for rational numbers

摘录 / EXCERPT: Let $g\in\mathbb Q\setminus\{-1,0,1\}$, let $t\ge1$, and let $N_{g,t}$ be the set of primes p with $\nu_p(g)=0$ and residual index $r_g(p)=t$. Denote its natural density, when it...

conjecture | open | MathDB 页面 | 原文线索

V2-001622 | MATHDB-322768

The impossibility of exactly simplifying the floor quotient below the square root

摘录 / EXCERPT: Let p be a prime number and let a be an integer satisfying $1<a<p$. Consider the quotient $\left\lfloor \frac{p}{a}\right\rfloor$ for $a\leq\left\lfloor\sqrt{p}\right\rfloor$....

conjecture | open | MathDB 页面 | 原文线索

V2-001623 | MATHDB-322773

Shumyatsky's centralizer conjecture for coprime automorphisms

摘录 / EXCERPT: Centralizer conjecture. If $\gamma_{k-2}(C_G(a))$ is nilpotent of class at most c for any $a\in A^{\#}$, then $\gamma_{k-2}(G)$ is nilpotent and has $\{c,k,p\}$-bounded nilpotenc...

conjecture | open | MathDB 页面 | 原文线索

V2-001624 | MATHDB-322788

The small-characteristic Hilbert-series conjecture for quotient singularities

摘录 / EXCERPT: Let V be a symplectic vector space over a field $\boldsymbol{F}$ of characteristic p, let G be a finite group of order coprime to p acting faithfully on V, and let A be...

conjecture | open | MathDB 页面 | 原文线索

V2-001625 | MATHDB-322793

Affine subspace hereditary properties are testable

摘录 / EXCERPT: Let p be a prime, and let an affine-invariant property $\mathcal{P}$ assign to functions $f:\mathbb{F}_p^n\to [R]$ a property that is preserved under affine transformations. The...

conjecture | open | MathDB 页面 | 原文线索

V2-001626 | MATHDB-322805

Congruence conjecture for signed power sums of Franel numbers

摘录 / EXCERPT: For nonnegative integers n and r, define ... In particular, $f_n^{(3)}=f_n$ are the Franel numbers. The signed Franel congruence conjecture. Let $n\geq 1$ and $r\geq 0$ be two in...

conjecture | open | MathDB 页面 | 原文线索

V2-001627 | MATHDB-322806

Multinomial power-sum congruence conjecture

摘录 / EXCERPT: For integers $m,n\geq 1$ and $r\geq 0$, define the multinomial power sums ... where $k_1,\ldots,k_m\geq 0$ and $k_1+\cdots+k_m=n$. The multinomial congruence conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001628 | MATHDB-322815

Lei's non-commutative signed Selmer-category conjecture

摘录 / EXCERPT: Let $E/\boldsymbol{Q}$ be an elliptic curve with good supersingular reduction at an odd prime p, let $\mathcal{F}$ be the $\boldsymbol{Z}_p$-cyclotomic extension of a finite Galo...

conjecture | open | MathDB 页面 | 原文线索

V2-001629 | MATHDB-322833

Perverse-filtration and Cherednik-filtration conjecture for compactified Jacobians

摘录 / EXCERPT: For coprime positive integers k,n, let $L_{k/n}$ be the finite-dimensional simple representation of the rational Cherednik algebra with parameter k/n, let $eL_{k/n}$ be its sph...

conjecture | open | MathDB 页面 | 原文线索

V2-001630 | MATHDB-322844

Nonexistence of prime nonsplit two-component links with full symmetry

摘录 / EXCERPT: Let L be a prime, nonsplit two-component link, and let $\Sigma(L)$ denote its intrinsic symmetry group. The link has full symmetry when $\Sigma(L)=\Gamma_2$, where Γ_2 is...

conjecture | open | MathDB 页面 | 原文线索

V2-001631 | MATHDB-322906

Mason's bounded-denominator conjecture for vector-valued modular forms

摘录 / EXCERPT: Let $\Gamma=SL_2(\mathbb{Z})$, let ρ be a representation of Γ, and let F be a vector-valued modular form for ρ. Say that the components of F have Fourier expan...

conjecture | open | MathDB 页面 | 原文线索

V2-001632 | MATHDB-322921

Equidistribution from large image for word maps on Chevalley groups

摘录 / EXCERPT: For a fixed prime p, let G_q be a family of Chevalley groups of fixed Lie type over $\mathbb{F}_q$, where $q=p^n$ varies. For a fixed word $w\in F_d$, with $d\ge 2$, let ... be t...

conjecture | open | MathDB 页面 | 原文线索

V2-001633 | MATHDB-322969

A zero-free-boundary estimate for Dirichlet L-functions

摘录 / EXCERPT: Let χ be a Dirichlet character modulo M, let t be real, and let $\delta>0$ be sufficiently small. The stated L-function conjecture. There exists a real number $\eta>0$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-001634 | MATHDB-322974

Negativity conjecture for prime divisors on blow-ups of the plane

摘录 / EXCERPT: Let X_n be the projective plane blown up at n general points, and let C be a prime divisor on X_n with self-intersection C^2 and arithmetic genus g. Negativity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-001635 | MATHDB-322980

The counting miracle for elliptic-curve non-abelian zeta invariants

摘录 / EXCERPT: Let $E/\mathbb F_q$ be an elliptic curve, and let $\alpha_{E,n}(0)$ and $\beta_{E,n}(0)$ denote the corresponding invariants of the rank-n pure non-abelian zeta function. Countin...

conjecture | open | MathDB 页面 | 原文线索

V2-001636 | MATHDB-322982

The coefficient form of the Riemann hypothesis for pure non-abelian zeta functions

摘录 / EXCERPT: Let $E/\mathbb F_q$ be an elliptic curve, let r be a rank, and set ... Here $\alpha_{E,r}(0)$ and $\beta_{E,r}(0)$ are the coefficients defining the numerator of the rank-r pure...

conjecture | open | MathDB 页面 | 原文线索

V2-001637 | MATHDB-322992

The p-integrality conjecture for the Narayana-related sequence

摘录 / EXCERPT: The p-integrality conjecture. For every $\mu\in\mathbb{N}$, there exist an initial condition $a_1(\mu)$ and a prime p such that the sequence $a_n(\mu)$ is p-integral.

conjecture | open | MathDB 页面 | 原文线索

V2-001638 | MATHDB-323016

Uniformity and the Riemann hypothesis for higher-rank zeta functions

摘录 / EXCERPT: Let F be a number field, let r be a positive integer, and let X be the curve over which the function field zeta function is defined. Write $\widehat\zeta_{F,r}(s)$ for the co...

conjecture | open | MathDB 页面 | 原文线索

V2-001639 | MATHDB-323018

Parabolic reduction, stability and the mass formula

摘录 / EXCERPT: Let G be a split reductive group with maximal parabolic subgroup P, and let c_P, ρ, Δ, w_J, and J_P have the meanings used in the source. For a number field...

conjecture | open | MathDB 页面 | 原文线索

V2-001640 | MATHDB-323026

Haselgrove condition for the modulus k

摘录 / EXCERPT: Let k be a modulus, let χ range over Dirichlet characters modulo k, and write $s=\sigma+it$. Haselgrove condition. There is a function $Z(k)$ with $0<Z(k)\leq 1$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-001641 | MATHDB-323027

Bentz's zero-free and nonvanishing conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let $s=\sigma+it$ be the complex variable for a Dirichlet L-function. Bentz's conjecture. The domain ... is zero free, and there is no zero at $s=\frac12$ for a Dirichlet L-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-001642 | MATHDB-323028

A zero-location inequality for Dirichlet L-functions

摘录 / EXCERPT: Let $\varrho:=\beta+i\gamma$ range over the zeros under consideration of Dirichlet L-functions. Zero-location conjecture. All zeros $\varrho:=\beta+i\gamma$ satisfy ... The source...

conjecture | open | MathDB 页面 | 原文线索

V2-001643 | MATHDB-323029

Kaczorowski's positive-density conjecture for prime-number races

摘录 / EXCERPT: Let k be a modulus, let $\varphi(k)$ denote Euler's totient, and let $a_1,\ldots,a_{\varphi(k)}$ be a permutation of the reduced residue classes modulo k. Write $\pi(m,k,a)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-001644 | MATHDB-323030

Rubinstein-Sarnak linear independence hypothesis for Dirichlet L-functions

摘录 / EXCERPT: Fix a modulus q, and consider the zeros of Dirichlet L-functions attached to primitive characters modulo q. For a zero $\varrho=\beta+i\gamma$, call γ its imaginary pa...

conjecture | open | MathDB 页面 | 原文线索

V2-001645 | MATHDB-323031

Lamzouri's bias conjecture for races of fixed distinct integers

摘录 / EXCERPT: Let $r\geq 3$ and let $a_1,\ldots,a_r$ be distinct nonzero integers. Define $\mathcal{Q}_{a_1,\ldots,a_r}$ to be the set of positive integers q such that the a_i are distinct m...

conjecture | open | MathDB 页面 | 原文线索

V2-001646 | MATHDB-323039

Mean-square shifted convolution conjecture for non-holomorphic cusp-form coefficients

摘录 / EXCERPT: Let $t(n)$ denote the Fourier coefficients of the non-holomorphic cusp form considered in the paper, and let $m\sim M$ mean $M<m\leq 2M$. Let $N\geq 1$, $1\leq L\leq N$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001647 | MATHDB-323119

The epimorphism conjecture for Weyl group invariants

摘录 / EXCERPT: Let p be an odd prime and let G be a simply-connected, simple, compact connected Lie group. Let T be a maximal torus of G, let W be the Weyl group, and let η^* be th...

conjecture | open | MathDB 页面 | 原文线索

V2-001648 | MATHDB-323121

Modularity conjecture for elliptic curves over $\mathbb{Q}(\sqrt{5})$

摘录 / EXCERPT: Let $F=\mathbb{Q}(\sqrt{5})$. An elliptic curve over F has an associated L-function, and a cuspidal Hilbert modular newform over F has a weight $(2,2)$ and Hecke eigenvalues;...

conjecture | open | MathDB 页面 | 原文线索

V2-001649 | MATHDB-323125

The projective-cover conjecture for logarithmic modules of the W_{3,2}-algebra

摘录 / EXCERPT: Let $\mathcal{W}_{3,2}$ be the logarithmic vertex operator algebra associated with the coprime pair $(3,2)$, and let $\overline{V(3,2)}$ and $\overline{MV(3,2)}$ be the logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-001650 | MATHDB-323126

The nilpotent-rank conjecture for finitely generated W_{p,p'}-modules

摘录 / EXCERPT: Let $p,p'\in\mathbb{Z}_{>0}$ with $p,p'\geq 2$ and p and p' relatively prime. Let $\mathcal{W}_{p,p'}$ be the vertex operator algebra defined as the intersection of the kernels...

conjecture | open | MathDB 页面 | 原文线索

V2-001651 | MATHDB-323127

The projective-cover conjecture for the constructed W_{p,p'}-modules

摘录 / EXCERPT: Let $p,p'\in\mathbb{Z}_{>0}$ with $p,p'\geq 2$ and p and p' relatively prime. Let $\mathcal{W}_{p,p'}$ be the logarithmic vertex operator algebra, and let $\overline{V(p,p')}$...

conjecture | open | MathDB 页面 | 原文线索

V2-001652 | MATHDB-323152

Guillera-Almkvist conjecture on rationality and algebraicity in Ramanujan-like series

摘录 / EXCERPT: Guillera-Almkvist conjecture. The quantities r, α_0, and τ_0^2 are rational if and only if z_0, c_0, b_0, and a_0 are algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-001653 | MATHDB-323154

Conjecture on modular identities relating two parameter values

摘录 / EXCERPT: Modular-identity conjecture. If one of the relations

conjecture | open | MathDB 页面 | 原文线索

V2-001654 | MATHDB-323173

Congruence conjecture for normalized critical p-adic L-values

摘录 / EXCERPT: Let f be the modular form under consideration, let ρ and σ be the Artin representations described above, and let ${\mathcal L}_p^{\rm can}(f,\rho,n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001655 | MATHDB-323176

Minimal-start conjecture for geometric-arithmetic progressions

摘录 / EXCERPT: Minimal-start conjecture. The minimal starting prime p_1 in a GAP-k is the smallest prime satisfying $p_1\geq k$.

conjecture | open | MathDB 页面 | 原文线索

V2-001656 | MATHDB-323177

Minimal-start-and-ratio conjecture for geometric-arithmetic progressions

摘录 / EXCERPT: Minimal-start-and-ratio conjecture. The minimal starting prime p_1 and minimal ratio r in a GAP-k are the smallest prime satisfying $p\geq k$, with $p_1=r$.

conjecture | open | MathDB 页面 | 原文线索

V2-001657 | MATHDB-323178

GAP-free tail conjecture for minimal geometric-arithmetic progressions

摘录 / EXCERPT: GAP-free conjecture. Whenever the first sequence has the minimal GAP-k, the remaining infinite sequence does not contain any GAPs of orders at least 3.

conjecture | open | MathDB 页面 | 原文线索

V2-001658 | MATHDB-323184

The dilated sumset conjecture in finite prime cyclic groups

摘录 / EXCERPT: Let k be a positive integer, let $lambda_1,...,lambda_k \in \mathbb{Z}$, and let $epsilon>0$. A sum of dilates of a set $A\subset \mathbb{Z}_p$ is the set...

conjecture | open | MathDB 页面 | 原文线索

V2-001659 | MATHDB-323198

The tau-conjecture on integer roots of straight-line polynomials

摘录 / EXCERPT: For $f\in\mathbb{Z}[T]$, define $\tau(f)$ as the least k for which f occurs in a sequence beginning with $f_{-1}(T)=1$ and $f_0(T)=T$, with each later polynomial obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-001660 | MATHDB-323208

Adams-type periodicity conjecture for mod-p cohomology

摘录 / EXCERPT: Let p be a prime, let M be a topological space, and let $x\in H^k(M;\mathbb{Z}_p)$ be nonzero. Say that x induces periodicity up to degree r when multiplication by x give...

conjecture | open | MathDB 页面 | 原文线索

V2-001661 | MATHDB-323209

Elliptic Lang-Waldschmidt conjecture

摘录 / EXCERPT: Let $n\geq1$ be an integer. Let K be a number field of degree d over $\mathbf{Q}$, and let E/K be given by $y^2=x^3+Ax+B$. Set $h_E=\max\{1,h(1:A:B)\}$. For rational integers...

conjecture | open | MathDB 页面 | 原文线索

V2-001662 | MATHDB-323216

The periodicity conjecture for the last non-zero digit sequence

摘录 / EXCERPT: Let $b\geq 2$ be an integer, and let $\mathfrak{S}_b(n)$ denote the last non-zero digit of n^n in base b. Periodicity conjecture. The sequence $\mathfrak{S}_b(n)$ is eventually...

conjecture | open | MathDB 页面 | 原文线索

V2-001663 | MATHDB-323235

The normal-order conjecture for iterated Carmichael lambda values

摘录 / EXCERPT: Let $\lambda(n)$ be the Carmichael lambda function, defined as the exponent of $(\mathbb{Z}/n\mathbb{Z})^{\times}$, and define its iterates by ... Let $L(n)$ be the smallest non-nega...

conjecture | open | MathDB 页面 | 原文线索

V2-001664 | MATHDB-323257

The general odd-prime torsion conjecture for matching complexes

摘录 / EXCERPT: Let $p=2q-1$ be an odd prime, and let $n=2k+1+3r$ and $d=k-1+r$. Then the relevant reduced homology group is $\tilde{H}_{k-1+r}(\mathsf{M}_{2k+1+3r};\mathbb{Z})$. The general odd-p...

conjecture | open | MathDB 页面 | 原文线索

V2-001665 | MATHDB-323270

The arithmetic fundamental lemma for regular semisimple elements

摘录 / EXCERPT: The arithmetic fundamental lemma. For $\gamma\in S_n(F)_{\rm rs}$,

conjecture | open | MathDB 页面 | 原文线索

V2-001666 | MATHDB-323271

The arithmetic intersection conjecture for arbitrary fundamental matrices

摘录 / EXCERPT: The annihilation conjecture. The element π^{r_1} annihilates the structure sheaf of

conjecture | open | MathDB 页面 | 原文线索

V2-001667 | MATHDB-323272

The discrete arithmetic fundamental lemma for intersection multiplicities

摘录 / EXCERPT: The discrete arithmetic fundamental lemma. If

conjecture | open | MathDB 页面 | 原文线索

V2-001668 | MATHDB-323273

A quadratic-twist functional-equation conjecture for elliptic curves

摘录 / EXCERPT: Fix a prime $p\geq 5$. Let E and E' be elliptic curves over $\mathbb{Q}$, and suppose that E' is a quadratic twist of E and that they have ordinary reduction of the same ty...

conjecture | open | MathDB 页面 | 原文线索

V2-001669 | MATHDB-323278

Noguchi's conjecture on the zeta function and Euler characteristic of finite categories

摘录 / EXCERPT: Noguchi's conjecture. The zeta function is a finite product of the form

conjecture | open | MathDB 页面 | 原文线索

V2-001670 | MATHDB-323296

The finite integrality conjecture for Schmidt-type coefficients

摘录 / EXCERPT: Finite integrality conjecture. For any $s>1$ and $n\geqslant 0$, there is an integer $r>s$ such that all the numbers $c_k^{(r,s)}$ for $0\leqslant k\leqslant n$ are integers. This...

conjecture | open | MathDB 页面 | 原文线索

V2-001671 | MATHDB-323297

The nonintegrality conjecture for Schmidt-type coefficients

摘录 / EXCERPT: Nonintegrality conjecture. For any $r>s>1$, there is a positive integer n such that $c_n^{(r,s)}$ is not an integer. Together with the finite integrality conjecture, this propose...

conjecture | open | MathDB 页面 | 原文线索

V2-001672 | MATHDB-323314

Karaś's multidegree conjecture for tame automorphisms

摘录 / EXCERPT: Let k be a field of characteristic 0, and let $3\leq p_1\leq d_2\leq d_3$ be integers with p_1 a prime number. Write $\mathrm{mdeg} F$ for the multidegree of a polynomial map...

conjecture | open | MathDB 页面 | 原文线索

V2-001673 | MATHDB-323331

The bijection conjecture between large odd perfect numbers and rational abundancy pairs

摘录 / EXCERPT: Abundancy-pair bijection conjecture. Each such N corresponds to exactly one ordered pair of rational numbers in this region, and conversely each such pair corresponds to an N....

conjecture | open | MathDB 页面 | 原文线索

V2-001674 | MATHDB-323339

Masser's polynomial bound conjecture for integral equivalence of quadratic forms

摘录 / EXCERPT: Let F and G be nonsingular integral quadratic forms in $N\geq 3$ variables. Suppose they are integrally equivalent, meaning that there is a matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-001675 | MATHDB-323358

The umbral moonshine module conjecture

摘录 / EXCERPT: Umbral moonshine module conjecture. For every $\ell\in\Lambda$, such a $G^{(\ell)}$-module exists and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001676 | MATHDB-323359

The umbral moonshine Rademacher-sum conjecture

摘录 / EXCERPT: Let $\Lambda=\{2,3,4,5,7,13\}$, let $G^{(\ell)}$ be the corresponding umbral group, and let $H^{(\ell)}_g$ be its vector-valued mock modular McKay-Thompson series. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001677 | MATHDB-323360

The umbral moonshine shadow conjecture

摘录 / EXCERPT: Let $\Lambda=\{2,3,4,5,7,13\}$, let $g\in G^{(\ell)}$, and let $H^{(\ell)}_g=(H^{(\ell)}_{g,r})$ be the vector-valued mock modular form obtained from the graded supertrace function...

conjecture | open | MathDB 页面 | 原文线索

V2-001678 | MATHDB-323361

The umbral moonshine multiplier-system conjecture

摘录 / EXCERPT: Let $\Lambda=\{2,3,4,5,7,13\}$, let $G^{(\ell)}$ be the corresponding umbral group, and let $H^{(\ell)}_g$ be its vector-valued mock modular McKay-Thompson series. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-001679 | MATHDB-323363

The discriminant dual-pair conjecture for umbral modules

摘录 / EXCERPT: Let $H^{(\ell)}$ be an umbral vector-valued mock modular form, let D be a discriminant of $H^{(\ell)}$, and let $K^{(\ell)}_{r,-D/4\ell}$ be the corresponding representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-001680 | MATHDB-323402

Matthews' generalized primitive-root conjecture

摘录 / EXCERPT: Matthews' generalized conjecture. The displayed asymptotic holds for the number of primes $\le x$. The source introduces this as a generalized version of Artin's conjecture and use...

conjecture | open | MathDB 页面 | 原文线索

V2-001681 | MATHDB-323409

Andrews' conjecture on partitions with no short sequences

摘录 / EXCERPT: Let $0<s<1$ and let $C_1,C_2,\ldots$ be independent events with probabilities ... Let A_k be the event ... so that no sequence of k consecutive events C_i fails to occur. Andrews...

conjecture | open | MathDB 页面 | 原文线索

V2-001682 | MATHDB-323418

The indicator-function residue conjecture for prime tower factorizations

摘录 / EXCERPT: Let q be the parameter defining the set $M^c(q)$, let $d(q)$ denote the density associated with this set, and let $I(n)$ be the indicator function of the event $n\in M^c(q)$. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-001683 | MATHDB-323419

The no-finite-relatively-prime-conditions conjecture for Frobenius meanders

摘录 / EXCERPT: A Frobenius meander is a meander whose associated seaweed Lie algebra has index zero; its block structure is the ordered list of block sizes. A relatively prime condition is a crit...

conjecture | open | MathDB 页面 | 原文线索

V2-001684 | MATHDB-323426

Rational-slope q,t-Catalan symmetry conjecture

摘录 / EXCERPT: Let m and n be coprime positive integers, let $Y_{m,n}$ be the set of lattice paths below the diagonal in an m by n rectangle, and define ... where $h_{+}^{\frac{n}{m}}(D)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-001685 | MATHDB-323444

The cuspidal cohomology conjecture for prime ideals in imaginary quadratic fields

摘录 / EXCERPT: Let K be an imaginary quadratic field, let ${\mathcal O}_K$ be its ring of integers, and let $\Gamma_0({\mathfrak p})$ denote the corresponding congruence subgroup for a prime id...

conjecture | open | MathDB 页面 | 原文线索

V2-001686 | MATHDB-323445

The modularity conjecture for elliptic curves and Bianchi newforms

摘录 / EXCERPT: Let K be an imaginary quadratic field. A Bianchi newform is a weight-2 automorphic form over K at level $\mathcal{K}_0(\mathfrak{n})$, and an elliptic curve over K has cond...

conjecture | open | MathDB 页面 | 原文线索

V2-001687 | MATHDB-323446

The Bergeron-Venkatesh torsion growth conjecture for Bianchi groups

摘录 / EXCERPT: Let $\Gamma_0({\mathfrak p})$ be a congruence subgroup inside a fixed Bianchi group, where ${\mathfrak p}$ is a prime ideal of residue degree one. Write $N{\mathfrak p}$ for its no...

conjecture | open | MathDB 页面 | 原文线索

V2-001688 | MATHDB-323465

Conjecture on the merit factor of Gordon-Mills-Welch sequences

摘录 / EXCERPT: Let $F=\mathbb{F}_{2^d}$ and let K be a subfield of size 2^k, where k divides d. Let θ be primitive in F, let ℓ be coprime to 2^k-1, and let the Gordon-Mill...

conjecture | open | MathDB 页面 | 原文线索

V2-001689 | MATHDB-323466

Conjecture on the merit factor of Sidelnikov sequences

摘录 / EXCERPT: Let q be an odd prime power, let θ be primitive in $\mathbb{F}_q$, and let $Z_{n,\theta}$ be the Sidelnikov sequence of length $n=q-1$ with respect to θ. Let F de...

conjecture | open | MathDB 页面 | 原文线索

V2-001690 | MATHDB-323500

The all-order topological-recursion conjecture for prime hyperbolic knot complements

摘录 / EXCERPT: Let $\mathfrak{M}$ be the complement of a prime hyperbolic knot, with a choice of determination as in the generalized volume conjecture. Let $J_N(\mathfrak{K},q)$ be the colored Jo...

conjecture | open | MathDB 页面 | 原文线索

V2-001691 | MATHDB-323503

Discrete restriction conjecture for the higher-dimensional sphere

摘录 / EXCERPT: Let $ncge 2$ and $clambdacge 1$ be integers, set $N=[clambda^{1/2}]+1$, and define ... For $1cle p,qclecinfty$, let ... where $e(z)=e^{2cpi i z}$. Discrete restriction conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-001692 | MATHDB-323511

Ash-Doud conjecture on Galois representations and eigenclasses in the homology of \(\operatorname{GL}(3,\mathbb{Z})\)

摘录 / EXCERPT: Let p be a prime and let $\bar{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$. Let $\rho:G_{\mathbb{Q}}\to\operatorname{GL}(3,\bar{\mathbb{F}}_p)$ be a Galois represent...

conjecture | open | MathDB 页面 | 原文线索

V2-001693 | MATHDB-323514

Leopoldt's conjecture for number fields

摘录 / EXCERPT: Let E be a number field. The units $\mathcal{O}_E^{\times}$ embed diagonally into the direct product of the completions $E_v^{\times}$ at the primes v of E lying over p. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-001694 | MATHDB-323520

Conjecture on the asymptotic Noether number for non-cyclic finite groups

摘录 / EXCERPT: Let $\mathcal{C}_q$ denote the set of isomorphism classes of non-cyclic finite groups whose order is not divisible by $\operatorname{char}(\mathbb{F})$ and whose smallest prime div...

conjecture | open | MathDB 页面 | 原文线索

V2-001695 | MATHDB-323538

The global bound conjecture for the oscillatory integral I(b,x)

摘录 / EXCERPT: Let $I(b,x)$ be the function studied in the paper, with parameters $b>1$ and $x>0$. Global bound conjecture. For all $b>1$ and $x>0$, one has ... This conjecture is consistent with t...

conjecture | open | MathDB 页面 | 原文线索

V2-001696 | MATHDB-323546

The prime k-tuples conjecture for admissible linear functions

摘录 / EXCERPT: Prime k-tuples conjecture. Given an admissible set of integer linear functions $L_i(x)$ for $i\in\{1,\dots,k\}$, there are infinitely many integers n for which all the $L_i(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001697 | MATHDB-323581

Böcherer's paramodular conjecture for prime-level eigenforms

摘录 / EXCERPT: Let p) be prime and let $F\in S^k(\Gamma^{\mathrm{para}}[p])^+$ be a paramodular Hecke eigenform of even weight k. For each fundamental discriminant $D<0$, let $A_F(D)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-001698 | MATHDB-323582

Twisted Böcherer conjecture for squarefree-level paramodular forms

摘录 / EXCERPT: Twisted Böcherer conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001699 | MATHDB-323583

Parametric twisted Böcherer conjecture for paramodular forms

摘录 / EXCERPT: Let N be squarefree, and let $F\in S^k(\Gamma^{\mathrm{para}}[N])$ be a Hecke eigenform that is not a Gritsenko lift. For fundamental discriminants ℓ and D satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-001700 | MATHDB-323585

The bounded prime-gap conjecture

摘录 / EXCERPT: Let p_n denote the nth prime. Bounded prime-gap conjecture. ... This asserts that the gaps between consecutive primes are bounded infinitely often. It is presented as equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-001701 | MATHDB-323586

Greenberg's local non-semisimplicity conjecture for ordinary modular eigenforms

摘录 / EXCERPT: Let f be a modular eigenform that is ordinary at a prime p, and let ρ_f be its associated Galois representation; write G_p for a decomposition group at p. Greenberg's...

conjecture | open | MathDB 页面 | 原文线索

V2-001702 | MATHDB-323590

Nagell-Ljunggren conjecture

摘录 / EXCERPT: Nagell-Ljunggren conjecture. The only solutions in integers $x,y>1$, $n>2$, and $q\geq 2$ are

conjecture | open | MathDB 页面 | 原文线索

V2-001703 | MATHDB-323614

The rational-pair correspondence conjecture for large odd perfect numbers

摘录 / EXCERPT: Rational-pair correspondence conjecture. For each such N with $N>10^{300}$, there corresponds exactly one ordered pair lying in the region

conjecture | open | MathDB 页面 | 原文线索

V2-001704 | MATHDB-323620

Necessity of the zeta-function derivative condition for the Riemann hypothesis

摘录 / EXCERPT: Necessary-condition conjecture. Condition (A) is necessary for the validity of the Riemann hypothesis.

conjecture | open | MathDB 页面 | 原文线索

V2-001705 | MATHDB-323624

Hirasawa-Murasugi conjecture on dihedral twisted Alexander polynomials

摘录 / EXCERPT: Let K be a knot and let ... be a non-abelian linear representation, where D_ℓ is the dihedral group of order 2ℓ and ℓ is an odd prime. Hirasawa-Murasugi conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-001706 | MATHDB-323677

The periodicity conjecture for discretized rotations

摘录 / EXCERPT: Periodic-orbit conjecture. The sequence (a_n) is periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-001707 | MATHDB-323682

Uniqueness conjecture for lattice cross tilings of prime order

摘录 / EXCERPT: Cross-tiling uniqueness conjecture. If $2n+1$ is prime, then there exists, up to congruence, only one Z-tiling of R^n by crosses.

conjecture | open | MathDB 页面 | 原文线索

V2-001708 | MATHDB-323689

Parity conjecture for the Rado number of $x+qy=q^2z$

摘录 / EXCERPT: Let $q\in\mathbb{N}$, and let $R(x+qy=q^2z)$ denote the 2-color Rado number of the equation $x+qy=q^2z$. Parity conjecture. ... The claim is presented as an empirical conjecture base...

conjecture | open | MathDB 页面 | 原文线索

V2-001709 | MATHDB-323690

Four-variable Rado number conjecture for $x+a(x+y)=2aw$

摘录 / EXCERPT: Let a be a positive integer, and let $R(x+a(x+y)=2aw)$ denote the 2-color Rado number of the four-variable equation $x+a(x+y)=2aw$. Four-variable Rado number conjecture. ... The co...

conjecture | open | MathDB 页面 | 原文线索

V2-001710 | MATHDB-323711

Conjectural local-factor formula for singular moduli with relatively prime conductors

摘录 / EXCERPT: Let d_1 and d_2 be quadratic imaginary discriminants with relatively prime conductors. Write f for the product of the two conductors, and for each prime p let...

conjecture | open | MathDB 页面 | 原文线索

V2-001711 | MATHDB-323727

Factorisation of the supersymmetry singlet norm

摘录 / EXCERPT: For twist angle $\phi=\pi$, write the supersymmetry singlet as $|\Psi\rangle=\sum_\mu\psi_\mu|\mu\rangle$, with the components chosen as coprime polynomials in x,y. The norm-fact...

conjecture | open | MathDB 页面 | 原文线索

V2-001712 | MATHDB-323728

Integral polynomial degree for singlet components

摘录 / EXCERPT: In the trigonometric case $y=0$, choose the singlet components to be coprime polynomials in x and normalise their leading terms as specified in the source. Let N be written as...

conjecture | open | MathDB 页面 | 原文线索

V2-001713 | MATHDB-323750

Nonvanishing conjecture for Galois averages of Rankin-Selberg L-values

摘录 / EXCERPT: Let K be the number field in the setup, let p be the fixed prime, and let $\mathcal{W}=\rho\chi\circ{\bf{N}}$ be a finite-order Hecke character of K, unramified outside p a...

conjecture | open | MathDB 页面 | 原文线索

V2-001714 | MATHDB-323787

The generalised Hardy-Littlewood conjecture for affine-linear systems

摘录 / EXCERPT: Let N, d, t, and L be positive integers, and let ... be a system of affine-linear forms, none constant and no two rational multiples of one another. Write each form as...

conjecture | open | MathDB 页面 | 原文线索

V2-001715 | MATHDB-323791

Asymptotic with second-order error for absolute Fourier coefficients

摘录 / EXCERPT: Let a_n be the Fourier coefficients of a modular form, and let p range over primes. Absolute-coefficient asymptotic. There is a constant B, depending on the form, such that .....

conjecture | open | MathDB 页面 | 原文线索

V2-001716 | MATHDB-323808

Ramachandra's lower-bound conjecture for Dirichlet polynomials

摘录 / EXCERPT: Let $N\geq 2$ be an integer and let a_n be complex numbers for $2\leq n\leq N$. Ramachandra's conjecture. For each $\delta>0$, there exists an $H>0$ such that ... for all integers...

conjecture | open | MathDB 页面 | 原文线索

V2-001717 | MATHDB-323814

The exact integer complexity conjecture for powers of 2 and 3

摘录 / EXCERPT: Let $\|n\|$ denote the least number of ones needed to write the positive integer n using addition, multiplication, and parentheses. For integers $a,k\geq 0$ with $a+k\geq 1$, con...

conjecture | open | MathDB 页面 | 原文线索

V2-001718 | MATHDB-323818

The conjecture that there are no non-trivial cycles of two runs in subprime Fibonacci sequences

摘录 / EXCERPT: Two-run cycle conjecture. There are no non-trivial cycles of two runs, that is, no non-trivial cycles containing two even terms.

conjecture | open | MathDB 页面 | 原文线索

V2-001719 | MATHDB-323821

Mathieu moonshine conjecture for Fourier coefficients of a mock modular form

摘录 / EXCERPT: Let $H(\tau)$ be a mock modular form with Fourier coefficients A_n, and let M_{24} be the simple sporadic Mathieu group. Mathieu moonshine conjecture. The Fourier coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-001720 | MATHDB-323831

Flatness classification conjecture for ternary cyclotomic polynomials

摘录 / EXCERPT: Let $p<q<r$ be odd primes, and let $A(pqr)$ denote the height of $\Phi_{pqr}(x)$. Flatness classification conjecture. If $A(pqr)=1$, then either ... or ... Numerical data suggests that...

conjecture | open | MathDB 页面 | 原文线索

V2-001721 | MATHDB-323832

Broadhurst's third-category conjecture for flat ternary cyclotomic polynomials

摘录 / EXCERPT: Let $p<q<r$ be odd primes, let $A(pqr)$ denote the height of $\Phi_{pqr}(x)$, and let w be the smallest positive integer such that ... Broadhurst's third-category conjecture. Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-001722 | MATHDB-323833

Flatness classification conjecture for pseudocyclotomic ternary polynomials

摘录 / EXCERPT: Let $p<q<r$ be pairwise relatively prime positive integers, and let $\widetilde\Phi_{p,q,r}(x)$ be the corresponding pseudocyclotomic polynomial. Say that it is flat when all its c...

conjecture | open | MathDB 页面 | 原文线索

V2-001723 | MATHDB-323834

Broadhurst's third-category conjecture for pseudocyclotomic ternary polynomials

摘录 / EXCERPT: Let $p<q<r$ be relatively prime positive integers, let $\widetilde\Phi_{p,q,r}(x)$ be flat, and let w be the smallest positive integer such that ... Pseudocyclotomic third-category...

conjecture | open | MathDB 页面 | 原文线索

V2-001724 | MATHDB-323835

Classification conjecture for flat quaternary cyclotomic polynomials

摘录 / EXCERPT: Let $p<q<r<s$ be odd primes, and let $A(pqrs)$ denote the height of the quaternary cyclotomic polynomial $\Phi_{pqrs}(x)$. Quaternary flatness classification conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-001725 | MATHDB-323836

Height-two conjecture for quaternary cyclotomic polynomials in the ±1 residue case

摘录 / EXCERPT: Let $p<q<r<s$ be primes, and let $A(pqrs)$ denote the height of $\Phi_{pqrs}(x)$. Quaternary height-two conjecture. If ... but $q\not\equiv-1\pmod p$, then ... The conjecture is sugges...

conjecture | open | MathDB 页面 | 原文线索

V2-001726 | MATHDB-323837

Non-flatness conjecture for quinary cyclotomic polynomials

摘录 / EXCERPT: Let $2<p<q<r<s<t$ be primes, and let $\Phi_{pqrst}(x)$ be a quinary cyclotomic polynomial. Say that it is flat when all its coefficients have absolute value at most 1. Quinary no...

conjecture | open | MathDB 页面 | 原文线索

V2-001727 | MATHDB-323838

Height persistence conjecture under multiplication by a prime

摘录 / EXCERPT: Let n be a positive integer, let p be a prime, and let $A(n)$ denote the height of the cyclotomic polynomial $\Phi_n(x)$. Height persistence conjecture. If ... then ... The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-001728 | MATHDB-323858

The equality of scaling, geometric and symbolic multifractal spectra

摘录 / EXCERPT: Spectrum equality conjecture. For a self-similar measure μ and all $t\in[t_{\min},t_{\max}]$,

conjecture | open | MathDB 页面 | 原文线索

V2-001729 | MATHDB-323871

Böcherer's Gross-Prasad type non-vanishing conjecture for Siegel cusp forms

摘录 / EXCERPT: Let $F \in S_k^{(2)}(N)$ be a Hecke eigenform, let K be an imaginary quadratic field, and let Λ be an ideal class group character of K. Suppose that the quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-001730 | MATHDB-323880

Deligne's conjecture on Frobenius invariants of lisse sheaves with fixed local monodromy

摘录 / EXCERPT: Let C be a smooth curve over a finite field $\mathbb F_q$, let $\bar C$ be its smooth compactification, and let M be the set of irreducible lisse $\bar{\mathbb Q}_\ell$-sheaves...

conjecture | open | MathDB 页面 | 原文线索

V2-001731 | MATHDB-323889

Extension of syntomic regulator formulae from integral models to curves

摘录 / EXCERPT: Let F be the function field associated with the curve C, and let $\mathscr{C}$ denote the corresponding curve in place of the integral model $\mathcal{O}$. The preceding theore...

conjecture | open | MathDB 页面 | 原文线索

V2-001732 | MATHDB-323898

Finiteness conjecture for anisotropic Euclidean quadratic forms

摘录 / EXCERPT: Let R be a Hasse domain, and let q be an anisotropic Euclidean quadratic form over R. Finiteness conjecture. There are only finitely many isomorphism classes of such forms...

conjecture | open | MathDB 页面 | 原文线索

V2-001733 | MATHDB-323899

Class-number-one conjecture for anisotropic Euclidean quadratic forms

摘录 / EXCERPT: Let R be a Hasse domain, and let q be an anisotropic Euclidean quadratic form over R. Class-number-one conjecture. The form q has class number one. The conjecture would con...

conjecture | open | MathDB 页面 | 原文线索

V2-001734 | MATHDB-323912

Boundedness conjecture for discretised rotations

摘录 / EXCERPT: Let ... where λ is real and $|\lambda|<2$. The map is the discretisation of the linear rotation-like map $(x,y)\mapsto(\lambda x-y,x)$, with $\lambda=2\cos(2\pi\nu)$. Bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-001735 | MATHDB-323914

The Iwasawa main conjecture for the symmetric-power representation V_m

摘录 / EXCERPT: Main conjecture for V_m. For all $\mathfrak{s}\in\mathfrak{S}$ and $a\in\mathbb Z/(p-1)\mathbb Z$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-001736 | MATHDB-323930

Conjecture on long-period rational points in the Cantor set

摘录 / EXCERPT: Long-period conjecture. For every $K>2/\log(3/2)$, one has $S_n^{(K)}=S_n$ for all sufficiently large n; in particular,

conjecture | open | MathDB 页面 | 原文线索

V2-001737 | MATHDB-323933

Hooley's Riemann hypothesis for cubic six-variable Hasse-Weil L-functions

摘录 / EXCERPT: HRH. The function $\xi({\mathbf m};s)$ has a meromorphic continuation to $\mathbb C$ of finite order, with only possible poles at $s=\frac{3}{2}$ and $s=\frac{5}{2}$; there is a si...

conjecture | open | MathDB 页面 | 原文线索

V2-001738 | MATHDB-323947

Conjecture on the infinitude of parabolic primes

摘录 / EXCERPT: The source refers to parabolic primes p and to n-ghost right-triangles, but the supplied context does not define these terms. Conjecture on the infinitude of parabolic primes....

conjecture | open | MathDB 页面 | 原文线索

V2-001739 | MATHDB-323950

Gauss's conjectures on average class numbers of binary quadratic forms

摘录 / EXCERPT: Let $D=4(b^{2}-ac)$ be a discriminant of a binary quadratic form, and let h_D denote the number of equivalence classes of such forms with discriminant D. For positive discrimin...

conjecture | open | MathDB 页面 | 原文线索

V2-001740 | MATHDB-323954

Multiplicity one for Siegel cusp forms of degree 2 and full level

摘录 / EXCERPT: Let $S_k(\Gamma_2)$ be the space of holomorphic Siegel cusp forms of weight k for $\Gamma_2=\operatorname{Sp}_4(\mathbb{Z})$. Let $f_1,f_2\in S_k(\Gamma_2)$ be Hecke eigenforms,...

conjecture | open | MathDB 页面 | 原文线索

V2-001741 | MATHDB-323955

Weak generalized Böcherer conjecture for class-group twists

摘录 / EXCERPT: Let $S_k(\Gamma_2)$ be the space of holomorphic Siegel cusp forms of weight k for $\Gamma_2=\operatorname{Sp}_4(\mathbb{Z})$. For an imaginary quadratic field...

conjecture | open | MathDB 页面 | 原文线索

V2-001742 | MATHDB-323956

The conjecture for desirable triples of the form q^a-2

摘录 / EXCERPT: Let p be prime, $q=p^s$, and let $g_{n,q}$ be defined by ... Call $(n,e;q)$ desirable if $g_{n,q}$ is a permutation polynomial of $\Bbb F_{q^e}$. The conjecture for desirable tripl...

conjecture | open | MathDB 页面 | 原文线索

V2-001743 | MATHDB-323959

The abc-conjecture for number fields

摘录 / EXCERPT: The abc-conjecture for number fields. For any $\epsilon>0$, there exists a constant $C_{K,\epsilon}>0$ such that for all $a,b,c\in K^*$ satisfying $a+b=c$,

conjecture | open | MathDB 页面 | 原文线索

V2-001744 | MATHDB-323981

The Gray-Theriault conjecture for the homotopy type of BW_n

摘录 / EXCERPT: Let p be a prime, let n be a positive integer, and let r be a positive integer. Set $T_{2n}=S^{2n+1}\{p^r\}$, and let BW_n be the space occurring in the EHP sequence for th...

conjecture | open | MathDB 页面 | 原文线索

V2-001745 | MATHDB-323989

The nonvanishing conjecture for quaternionic theta lifts of Hilbert newforms

摘录 / EXCERPT: Let F be a totally real number field, let $\boldsymbol{c}$ be an odd square-free level, let $g\in S_{\mathbf{2}}(\boldsymbol{c})$ be a newform, and let v_g be a Hecke eigenvect...

conjecture | open | MathDB 页面 | 原文线索

V2-001746 | MATHDB-323994

Strong ideological generation of the real numbers by $cmathfrak{W}_{cinfty}$

摘录 / EXCERPT: Let $\text{\usefont{U}{euf}{m}{n}W}_{\infty}$ denote the class of real numbers with the strong ideological generation property. Strong ideological generation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-001747 | MATHDB-323995

Stable ideology-classification conjecture

摘录 / EXCERPT: Let $\widetilde{\mathcal C\ell}_{\infty}(\mathbb K)$ be the direct limit of the quotient sets $\mathbb K^n/{\rm PGL}_2(\mathcal O)^n$, and let $\mathcal C\ell_{\infty}(\mathbb K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001748 | MATHDB-324008

Noe-Post conjecture on intersections of generalized Fibonacci sequences

摘录 / EXCERPT: Noe-Post conjecture. The Diophantine equation

conjecture | open | MathDB 页面 | 原文线索

V2-001749 | MATHDB-324057

Nonvanishing signature conjecture for quantum representations

摘录 / EXCERPT: Let p be a prime with $p\geq 5$, let ζ be a primitive $2p$-th root of unity, and define ... for odd p, and $A_p=\pm\exp\left(\frac{\pi i}{p}\right)$ for even p. Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-001750 | MATHDB-324069

Uniqueness conjecture for complete weight enumerators from theta functions

摘录 / EXCERPT: Let p be a prime, let $\ell>0$ be square-free with $\ell\equiv 3\pmod 4$ and $p\nmid\ell$, and let $K=\mathbb{Q}(\sqrt{-\ell})$ with ring of integers $\mathcal O_K$. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-001751 | MATHDB-324071

Kim's finite-level cohomological globalization conjecture for integral points

摘录 / EXCERPT: Let $\mathcal{X}$ be a hyperbolic curve over the relevant arithmetic base, let p be a prime, and let $\mathcal{X}(\mathbb{Z})\subset\mathcal{X}(\mathbb{Z}_p)$ denote the global a...

conjecture | open | MathDB 页面 | 原文线索

V2-001752 | MATHDB-324095

Cohen-Macaulay cohomology conjecture for simple algebraic groups

摘录 / EXCERPT: Let G be a simple algebraic group defined over an algebraically closed field k of characteristic p, where p is a good prime for G. Let B be a Borel subgroup of G, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001753 | MATHDB-324158

Special uniformity conjecture for zeta functions of Riemann surfaces

摘录 / EXCERPT: Let M be a compact Riemann surface, let $\widehat\zeta_{M,r}(s)$ denote the rank-r zeta function of M, and let $\widehat\zeta_M^{SL(n)/P_{n-1,1}}$ denote the zeta function wi...

conjecture | open | MathDB 页面 | 原文线索

V2-001754 | MATHDB-324159

Tamagawa number conjecture for SL_n over a function field

摘录 / EXCERPT: Let $F=\mathbb C(X)$ be the function field associated with the Riemann surface M, let $\mathbb A$ be its adelic ring, and let $\mathbb K$ be the specified maximal compact subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-001755 | MATHDB-324160

General uniformity conjecture for zeta functions on Riemann surfaces

摘录 / EXCERPT: Let M be a connected compact Riemann surface. For a reductive group G, let $m_{M;G}(\nu_G')$ and $m^{\mathrm{ss}}_{M;G}(\nu_G')$ denote the total and semistable masses at slope...

conjecture | open | MathDB 页面 | 原文线索

V2-001756 | MATHDB-324222

The Euler-product characterization of GRH for nonprincipal Dirichlet L-functions

摘录 / EXCERPT: Euler-product characterization of GRH. If $\chi\ne\1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001757 | MATHDB-324236

Conjecture that radically weak Carmichael numbers outnumber Carmichael numbers

摘录 / EXCERPT: Let $K(x)$ be the number of integers up to x satisfying the paper's radically weakened Carmichael condition, and let $C(x)$ be the number of Carmichael numbers up to x. Growth-...

conjecture | open | MathDB 页面 | 原文线索

V2-001758 | MATHDB-324243

Lang-Silverman conjecture for Drinfeld modules

摘录 / EXCERPT: Let K be the global function field in the paper's setting. For every finite extension L/K, every integer $r\geq 1$, every Drinfeld module ϕ/L of rank r, and every non-to...

conjecture | open | MathDB 页面 | 原文线索

V2-001759 | MATHDB-324275

The bounded-degree hyperplane inequality in the projective plane

摘录 / EXCERPT: Let S be a finite set of places of a number field k. Let $L_1,\ldots,L_q$ be lines over k in general position in $\mathbb{P}^2$, and set ... Let δ be a positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-001760 | MATHDB-324291

Squarefree-value conjecture for elliptic-curve sequences

摘录 / EXCERPT: Squarefree-value conjecture. As $X\to\infty$,

conjecture | solved | MathDB 页面 | 原文线索

V2-001761 | MATHDB-324295

Artin's conjecture in arithmetic progressions

摘录 / EXCERPT: Let $a,b,r\in\mathbb{Z}$ with $\gcd(a,b)=1$, and suppose that $r\neq -1$ and that r is not a perfect mth power for any $m>1$. An integer r is a primitive root modulo a prime...

conjecture | open | MathDB 页面 | 原文线索

V2-001762 | MATHDB-324333

The braid-index conjecture for torsion in Khovanov homology

摘录 / EXCERPT: Let a closed n-braid be the link obtained by taking the braid closure of an n-strand braid, and let p be prime. Braid-index conjecture. (1) Khovanov homology of a closed 3-...

conjecture | open | MathDB 页面 | 原文线索

V2-001763 | MATHDB-324342

Conference-matrix MDS code conjecture

摘录 / EXCERPT: Let p be a prime, and let ${\mathbb C}$ be a code of length $p+1$ constructed from a generator matrix that is a conference matrix. Conference-matrix MDS conjecture. The code...

conjecture | open | MathDB 页面 | 原文线索

V2-001764 | MATHDB-324371

Mochizuki's uniform abc conjecture for curves over number fields

摘录 / EXCERPT: Mochizuki's uniform abc conjecture. If $\epsilon>0$, then there exists a constant $c_{\epsilon,d,\mathcal{X},\mathcal{D}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001765 | MATHDB-324385

The double-exponential generalized Skewes-number conjecture

摘录 / EXCERPT: Double-exponential generalized Skewes-number conjecture. As $\rho(q)/\log q'$ tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-001766 | MATHDB-324397

Reversal-multiplication polynomiality conjecture

摘录 / EXCERPT: Let A be a positive integer and let A^{*} denote its reversal. A pair (A,A^{*}) is polynomial if the multiplication $A \times A^{*}$ can be performed without carry, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001767 | MATHDB-324400

Conjecture on maximal affine Fourier Hadamard families in dimensions twice a prime

摘录 / EXCERPT: Maximal-family conjecture. The maximal affine family of complex Hadamard matrices stemming from the Fourier matrix F_d in dimensions $d=2p$, where p is prime, is given by Eqs....

conjecture | open | MathDB 页面 | 原文线索

V2-001768 | MATHDB-324404

The local factor formula for the arithmetic intersection number of quadratic imaginary orders

摘录 / EXCERPT: Let d_1 and d_2 be discriminants of quadratic imaginary orders, and fix an integer t. Let ... Assume that the conductors of d_1 and d_2, together with m, have no simultan...

conjecture | open | MathDB 页面 | 原文线索

V2-001769 | MATHDB-324410

Generalised non-abelian Coates-Sinnott conjecture

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with Galois group G, let $r<0$, and let S contain all archimedean places and all places ramified in L/K. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001770 | MATHDB-324420

The effective geometric Bogomolov conjecture for curves

摘录 / EXCERPT: Let C be a smooth projective curve over $\overline{K}$ of genus at least 2, let J_C be its Jacobian, let D be a degree-one divisor, let $j_D:C\to J_C$ be the associated emb...

conjecture | open | MathDB 页面 | 原文线索

V2-001771 | MATHDB-324465

Prime-dimension existence conjecture for binary Steiner structures

摘录 / EXCERPT: Let n be a prime satisfying ... A coset complete 3-dimensional subspace is a 3-dimensional subspace of $\mathbb{F}_2^n$ used as a generator for the construction described in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-001772 | MATHDB-324490

The Mathieu moonshine representation-pair conjecture

摘录 / EXCERPT: Let $K=\bigoplus_{n=-1}^{\infty}K_n q^{n/8}$ be the virtual graded M_{24}-module determined by the McKay-Thompson series, with K_n the representation whose character is given b...

conjecture | open | MathDB 页面 | 原文线索

V2-001773 | MATHDB-324507

Nonexistence conjecture for strengthened power-sum Bernoulli congruences

摘录 / EXCERPT: Let $S_m(n)=\sum_{j=1}^{n-1}j^m$ and let B_m denote the mth Bernoulli number. The strengthened congruences conjecture. For every integer $n\ge 2$, neither of the congruences .....

conjecture | open | MathDB 页面 | 原文线索

V2-001774 | MATHDB-324508

Nonexistence conjecture for a fourth-power Wolstenholme congruence

摘录 / EXCERPT: Let $S_m(n)=\sum_{j=1}^{n-1}j^m$. Fourth-power Wolstenholme congruence conjecture. For no integer $n\ge 2$ does the congruence ... hold. The source presents this as a computationally...

conjecture | open | MathDB 页面 | 原文线索

V2-001775 | MATHDB-324509

Irregular-prime characterization for a half-power-sum congruence

摘录 / EXCERPT: Let $S_m(n)=\sum_{j=1}^{n-1}j^m$, and let B_m denote the mth Bernoulli number. An irregular prime is a prime dividing the numerator of some Bernoulli number B_{2r} with...

conjecture | open | MathDB 页面 | 原文线索

V2-001776 | MATHDB-324510

Nonexistence conjecture for the half-power-sum congruence in the 1 modulo 4 case

摘录 / EXCERPT: Let $S_m(n)=\sum_{j=1}^{n-1}j^m$. Half-power-sum nonexistence conjecture. For no odd positive integer $n\ge 5$ with $n\equiv 1\pmod{4}$ does ... hold. The source proposes this after...

conjecture | open | MathDB 页面 | 原文线索

V2-001777 | MATHDB-324511

Even-modulus power-sum binomial congruence

摘录 / EXCERPT: Let $S_m(n)=\sum_{j=1}^{n-1}j^m$, and let k be a positive integer. Even-modulus power-sum conjecture. For every $k\ge 1$ and every even positive integer n, the congruence ... hol...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001778 | MATHDB-324519

Finiteness of splitting classes for hyperbolic profinite extensions

摘录 / EXCERPT: Finiteness conjecture for splitting classes. There are only finitely many H-conjugacy classes of splittings of this short exact sequence. The source says that this conjecture wou...

conjecture | open | MathDB 页面 | 原文线索

V2-001779 | MATHDB-324521

Finiteness of integral points under pro-p Galois hyperbolicity

摘录 / EXCERPT: Let k be a finitely generated field and let X be a hyperbolic algebraic curve defined over k. Let S be a finite set of primes, assume that $Gal(\bar{k}/k)(p)$ is hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-001780 | MATHDB-324522

Pro-p detection of profinite hyperbolicity

摘录 / EXCERPT: Let H be a finitely generated profinite group, and for each prime p let $H(p)$ denote its relevant pro-p quotient. Pro-p detection conjecture. If $H(p)$ is hyperbolic profi...

conjecture | open | MathDB 页面 | 原文线索

V2-001781 | MATHDB-324523

Uniform boundedness of rational points on hyperbolic curves

摘录 / EXCERPT: Let X be a hyperbolic curve defined over a number field k, and let g be its genus. Uniform rational-point bound conjecture. The number of rational points of X is bounded ab...

conjecture | open | MathDB 页面 | 原文线索

V2-001782 | MATHDB-324524

Generalized Lang-type conjecture for arithmetic fractals

摘录 / EXCERPT: Arithmetic-fractal intersection conjecture. For each such component B_i, either B_i is a point, or $B_i(\bar K)\cap F$ is an arithmetic fractal with respect to some induced end...

conjecture | open | MathDB 页面 | 原文线索

V2-001783 | MATHDB-324526

Fractal dimension conjecture for self-similar subsets of rings of integers

摘录 / EXCERPT: Fractal-dimension conjecture. The above notion of fractal dimension for self-similar subsets of O_K is well-defined and well-behaved with respect to inclusion: fractal dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-001784 | MATHDB-324584

Bonnafé-Kemper's Gorenstein conjecture for the modular invariant ring

摘录 / EXCERPT: Let $K=F_q$ be a finite field with $q=p^r$ elements, where p is prime and $r\geq 1$. Let V be the natural n-dimensional representation of $GL_n(F_q)$, and let $V^{\ast}$ be i...

conjecture | open | MathDB 页面 | 原文线索

V2-001785 | MATHDB-324614

The asymptotic conjecture for elliptic aliquot sequences

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ without complex multiplication, and let $L\geq 2$ be fixed. Let $\pi_{E,L}^{\mathrm{ali\text{-}sequence}}(x)$ count aliquot sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-001786 | MATHDB-324617

Berry-Tabor conjecture for Laplacian spacings on Diophantine irrational flat tori

摘录 / EXCERPT: Let ${\mathcal L}_0$ be a Diophantine irrational lattice. Write n_j for the distinct Laplacian eigenvalues, let $\delta_j=n_j-n_{j-1}>0$, let ... and define the mean-normalized spa...

conjecture | open | MathDB 页面 | 原文线索

V2-001787 | MATHDB-324672

Uniform boundedness for rational polynomial values in multiplicative groups

摘录 / EXCERPT: Let $g(x)$ be a polynomial over the rationals of degree d, and let G be a complex multiplicative group of rank r. Uniform boundedness conjecture. There is a function $b(d,r)$...

conjecture | open | MathDB 页面 | 原文线索

V2-001788 | MATHDB-324680

Power-law growth conjecture for Pratt-tree avoiding prime sets

摘录 / EXCERPT: Growth conjecture for $\mathcal{P}_r$. For each r, there is a number $\delta_r>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001789 | MATHDB-324702

Conjecture on the logarithmic order of the restriction norm of a Maass form

摘录 / EXCERPT: Restriction-norm conjecture. Based on the conjectures of Conrey, Farmer, Keating, Rubinstein, and Snaith, one conjectures

conjecture | open | MathDB 页面 | 原文线索

V2-001790 | MATHDB-324715

Conjectured unification of short-interval Goldbach and twin-prime results

摘录 / EXCERPT: Conjectured unification. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-001791 | MATHDB-324724

Irreducible-module and torus one-point-function conjecture for A_m orbifolds

摘录 / EXCERPT: Let $p\geq 2$, let m be the orbifold parameter, and let $\mathcal{W}(p)^{A_m}$ be the corresponding A_m-orbifold of the triplet vertex algebra. Am orbifold representation and o...

conjecture | open | MathDB 页面 | 原文线索

V2-001792 | MATHDB-324731

Infinitely many Carmichael numbers in reduced residue classes

摘录 / EXCERPT: Let M be a positive integer, and let a be an integer such that $(a,M)=1$. A Carmichael numbers in arithmetic progressions conjecture. There are infinitely many Carmichael numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-001793 | MATHDB-324732

Pollio-Rapinchuk's multinorm principle conjecture

摘录 / EXCERPT: Pollio-Rapinchuk's conjecture. If every extension P of k contained in $L_1\cap L_2$ satisfies the norm principle, then the pair L_1,L_2 satisfies the multinorm principle. It...

conjecture | open | MathDB 页面 | 原文线索

V2-001794 | MATHDB-324745

Bruckman-Anderson conjecture on the density of Fibonacci entry-point divisibility

摘录 / EXCERPT: Bruckman-Anderson conjecture. If $m=q^e$ is a prime power with $e\geq 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001795 | MATHDB-324766

Coprime minimal-model W-algebra conjecture for the two-screening kernel

摘录 / EXCERPT: Let p,p' be positive integers with $(p,p')=1$, let $L=\sqrt{pp'}\mathbb{Z}\alpha_1$ for $Q=A_1$, and define the two-screening vertex algebra ... where $\mathcal{Q}=e^{\alpha/p'}_0$...

conjecture | open | MathDB 页面 | 原文线索

V2-001796 | MATHDB-324780

General minimal-model reducibility conjecture for tensor-product Virasoro modules

摘录 / EXCERPT: Let $c=c_{p,q} eq 0$ and let $L(c,h)$ be a minimal model. Let $V_{\alpha,\beta}^{\prime}$ be an irreducible module from the intermediate series, and let an admissible triple mean a...

conjecture | open | MathDB 页面 | 原文线索

V2-001797 | MATHDB-324787

Nonexistence of small positive solutions to the castling equation

摘录 / EXCERPT: Nonexistence conjecture. There exists no positive integer solution $(k_1,\ldots,k_j)$ of this equation such that $k_i\leq b1-1$ for every i.

conjecture | open | MathDB 页面 | 原文线索

V2-001798 | MATHDB-324788

Brignall's prime bound conjecture for graphs

摘录 / EXCERPT: Let G be a graph with $|V(G)|\geq 2$. The prime bound $p(G)$ is the smallest integer such that G admits a prime $p(G)$-extension; here $\alpha(G)$ denotes the stability number...

conjecture | open | MathDB 页面 | 原文线索

V2-001799 | MATHDB-324789

Brumer's conjecture for equivariant partial zeta values

摘录 / EXCERPT: Brumer's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001800 | MATHDB-324790

Brumer-Stark conjecture for Abelian extensions

摘录 / EXCERPT: Brumer-Stark conjecture. There exists $\varepsilon_{\mathfrak{a},K/k,S}\in K^{\times}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001801 | MATHDB-324798

Arithmetic quantum chaos conjecture for eigenvalue distributions

摘录 / EXCERPT: Arithmetic quantum chaos conjecture. The distribution of the discrete eigenvalues approaches a Poisson distribution.

conjecture | open | MathDB 页面 | 原文线索

V2-001802 | MATHDB-324814

Lie triple isomorphism decomposition conjecture for nest algebras

摘录 / EXCERPT: Let $\mathcal{N}$ and $\mathcal{N}^\prime$ be nests on a Banach space $\mathbf{X}$, and let $\mathcal{A}\!\operatorname{lg}(\mathcal{N})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001803 | MATHDB-324832

Soundararajan's conjecture on small zeta-zero gaps and zeros of the derivative

摘录 / EXCERPT: Let $\rho=\beta+i\u03b3$ and $\rho'=\beta'+i\u03b3'$ denote non-trivial zeros of $\zeta(s)$ and $\zeta'(s)$, respectively, and assume the Riemann Hypothesis. Let γ^+ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-001804 | MATHDB-324833

Farmer-Ki conjecture on derivative-zero distributions and small zeta-zero gaps

摘录 / EXCERPT: Assume the Riemann Hypothesis. Let $w(x)$ be the indicator function of the unit interval, and define ... ... where γ and γ' are ordinates of zeros of $\zeta(s)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001805 | MATHDB-324853

Winning conjecture for badly approximable points on non-degenerate curves

摘录 / EXCERPT: For an n-tuple of non-negative real numbers $\mathbf{i}=(i_1,\ldots,i_n)$ satisfying ... let $\mathbf{Bad}(\mathbf{i})$ be the corresponding weighted badly approximable set in...

conjecture | open | MathDB 页面 | 原文线索

V2-001806 | MATHDB-324871

The monodromy conjecture for meromorphic germs

摘录 / EXCERPT: Let $f=\frac{P}{Q}$ be a germ of a meromorphic function on $({\mathbb C}^{n+1},0)$, and let $Z_{\operatorname{top},f}$ denote its topological zeta function. Let h_f be the monodr...

conjecture | open | MathDB 页面 | 原文线索

V2-001807 | MATHDB-324901

Linear bound for detecting failure of optimal domination

摘录 / EXCERPT: Let p be the prime defining the quaternion algebra B_p, and let $\mathcal{O},\mathcal{O}'\subset B_p$ be maximal orders of different types. For an integer $b>0$, say that...

conjecture | open | MathDB 页面 | 原文线索

V2-001808 | MATHDB-324911

The generalized binomial coefficient residue conjecture

摘录 / EXCERPT: Residue conjecture. For every $r\in\{0,1,\ldots,p-1\}$, there are infinitely many positive integers n such that

conjecture | open | MathDB 页面 | 原文线索

V2-001809 | MATHDB-324913

The equidistribution conjecture for Selmer elements on hyperelliptic curves

摘录 / EXCERPT: Fix $g\geq 1$ and a prime p. Let U be a trivializing congruence class, let $C=\mathscr{C}_m$ vary over curves with $m\in\mathcal{F}_g\cap U$, and let $J=\mathscr{J}_m$ be the J...

conjecture | open | MathDB 页面 | 原文线索

V2-001810 | MATHDB-324914

The uniqueness conjecture for generalized Wolstenholme congruences

摘录 / EXCERPT: Uniqueness conjecture. Then $b_j=b_{j,n}(k)$ for $j=0,1,\ldots,n$.

conjecture | open | MathDB 页面 | 原文线索

V2-001811 | MATHDB-324915

The strong uniqueness conjecture for generalized Wolstenholme congruences

摘录 / EXCERPT: Strong uniqueness conjecture. Then there are constants $c_0,c_1,\ldots\in\mathbb{Q}$ such that, if $b_0,b_1,\ldots$ are defined by the generalized Wolstenholme coefficient construc...

conjecture | open | MathDB 页面 | 原文线索

V2-001812 | MATHDB-324927

Granville's conjecture on primes in short intervals

摘录 / EXCERPT: Granville's conjecture. If $\Phi(x)>x^\epsilon$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001813 | MATHDB-324928

Conjecture on primes in arithmetic progressions with growing moduli

摘录 / EXCERPT: Conjecture on growing moduli. For every $\delta>0$, this asymptotic holds in the range

conjecture | open | MathDB 页面 | 原文线索

V2-001814 | MATHDB-324929

The subsingular-vector factor conjecture for twisted Heisenberg-Virasoro modules

摘录 / EXCERPT: Let $V(c,\frac{1-r}{2},0)$ be the module in question, let $J^{\prime}(c,\frac{1-r}{2},0)$ denote the specified submodule, and let u be a subsingular vector in...

conjecture | open | MathDB 页面 | 原文线索

V2-001815 | MATHDB-324930

The reducibility criterion for modules over the twisted Heisenberg-Virasoro algebra

摘录 / EXCERPT: Let $p\in\mathbb{N}$, and let c,h,h_W,r be the parameters of the module $L^{\prime}(c,h,h_W)$. Assume ... Reducibility criterion. The module $L^{\prime}(c,h,h_W)$ is reducible if a...

conjecture | open | MathDB 页面 | 原文线索

V2-001816 | MATHDB-324932

Wills's version of the Lonely Runner Conjecture

摘录 / EXCERPT: Wills's conjecture. If $1\le m_1<m_2<\ldots<m_k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001817 | MATHDB-324948

Linnik's microsquares conjecture for sums of three squares

摘录 / EXCERPT: Linnik's microsquares conjecture. Whenever n is sufficiently large in terms of ε, this equation possesses a solution in which

conjecture | open | MathDB 页面 | 原文线索

V2-001818 | MATHDB-324977

Erdős's conjecture on moments of gaps between reduced residues

摘录 / EXCERPT: Let q be a natural number, let $P=\phi(q)/q$, and let $1=a_1<a_2<\cdots$ be the integers coprime to q. Define ... Erdős's conjecture. The gaps between consecutive reduced residue...

conjecture | open | MathDB 页面 | 原文线索

V2-001819 | MATHDB-324978

The prime-tuple gap moment conjecture for admissible sets

摘录 / EXCERPT: Let $\mathcal{D}=\{h_1,h_2,\ldots,h_s\}$ be a fixed admissible set of integers, meaning that if $\nu_p(\mathcal{D})$ is the number of distinct elements of $\mathcal{D}$ modulo p,...

conjecture | open | MathDB 页面 | 原文线索

V2-001820 | MATHDB-325053

Conjecture on moments of the hybrid zero factor derivative

摘录 / EXCERPT: Assume the Riemann Hypothesis. Let ρ range over the non-trivial zeros of $\zeta(s)$, let $N(T)$ count those zeros with $0<\operatorname{Im}(\rho)\leq T$, and let $Z_X(s)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-001821 | MATHDB-325054

Hybrid Euler-Hadamard independence conjecture for ζ′ moments

摘录 / EXCERPT: Let ρ range over the non-trivial zeros of $\zeta(s)$, let $N(T)$ count zeros with $0<\operatorname{Im}(\rho)\leq T$, and let $P_X(s)$ and $Z_X(s)$ denote the Euler and zero fa...

conjecture | open | MathDB 页面 | 原文线索

V2-001822 | MATHDB-325065

The prime-index inequality conjecture

摘录 / EXCERPT: Let p_n denote the n-th prime. Prime-index inequality conjecture. For all integers $n\geq 2$ and all integers k satisfying $1\leq k\leq n-1$, ... The inequality would imply tha...

conjecture | open | MathDB 页面 | 原文线索

V2-001823 | MATHDB-325066

The Apéry-sequence Brun criterion conjecture

摘录 / EXCERPT: Apéry-sequence Brun criterion conjecture. There is an unbounded subsequence of positive integers $n_k\to\infty$ such that $\delta_{n_k}<0$. The sequence y_n/x_n is increasing and...

conjecture | open | MathDB 页面 | 原文线索

V2-001824 | MATHDB-325067

The Apéry-sequence consecutive negativity conjecture

摘录 / EXCERPT: Apéry-sequence consecutive negativity conjecture. For every integer $N\geq2$, there is a positive integer n such that

conjecture | open | MathDB 页面 | 原文线索

V2-001825 | MATHDB-325113

Vanishing conjecture for the arithmetic theta lift

摘录 / EXCERPT: Vanishing conjecture. If $\xi(f)=0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-001826 | MATHDB-325114

Modularity conjecture for the arithmetic theta series

摘录 / EXCERPT: Modularity conjecture. The formal generating series is the q-expansion of a function

conjecture | open | MathDB 页面 | 原文线索

V2-001827 | MATHDB-325139

The irrationality conjecture for Euler's constant

摘录 / EXCERPT: Euler's constant irrationality conjecture. Euler's constant is irrational. This is a long-standing open problem concerning the arithmetic nature of Euler's constant.

conjecture | open | MathDB 页面 | 原文线索

V2-001828 | MATHDB-325140

The Granville-Soundararajan maximal-order conjecture for the Riemann zeta function

摘录 / EXCERPT: Let $\zeta(s)$ denote the Riemann zeta function, let γ be Euler's constant, and let $T\geq10$. Granville-Soundararajan conjecture. There is a constant C_1 such that ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-001829 | MATHDB-325141

Montgomery-Vaughan maximal-order conjectures for quadratic Dirichlet L-functions

摘录 / EXCERPT: Let χ_{-d} be the real primitive quadratic character associated with the imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$, and let $L(s,\chi_{-d})$ be its Dirichlet L-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-001830 | MATHDB-325153

Divisibility conjecture for the components of odd perfect numbers

摘录 / EXCERPT: Let $N=q^k n^2$ be an odd perfect number in Eulerian form: q is prime, $q\equiv k\equiv 1\pmod 4$, and $\gcd(q,n)=1$. Let $\sigma(m)$ denote the sum of the positive divisors of...

conjecture | open | MathDB 页面 | 原文线索

V2-001831 | MATHDB-325154

Greenberg's conjecture for cyclotomic towers

摘录 / EXCERPT: Let k be a number field, let p be an odd prime, let k_∞/k be the cyclotomic $\mathbb{Z}_p$-extension, and let $\Gamma=\operatorname{Gal}(k_\infty/k)$. For each layer...

conjecture | open | MathDB 页面 | 原文线索

V2-001832 | MATHDB-325155

The bounded Freiman ring-isomorphism conjecture over finite fields

摘录 / EXCERPT: Let $k\geq 3$ be an integer. A bijection preserving the vanishing of every k-bounded integer polynomial is called an F_k-ring-isomorphism. For a prime p, write $\mathbb{F}_p$...

conjecture | open | MathDB 页面 | 原文线索

V2-001833 | MATHDB-325161

The two-fold sumset threshold conjecture for multiplicative subgroups

摘录 / EXCERPT: Let p be a prime and let A be a multiplicative subgroup of $\mathbb Z_p^*$. For a positive integer k, write $kA=\{x_1+\dots+x_k:x_i\in A,1\leq i\leq k\}$. Two-fold sumset thr...

conjecture | open | MathDB 页面 | 原文线索

V2-001834 | MATHDB-325195

Beilinson-Bloch conjecture for modular-form motives

摘录 / EXCERPT: Let E be a number field, let f be the modular form under consideration, let $L(f\otimes E,s)$ be its complex L-function over E, and let $X_\mathfrak{p}(E)$ be the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-001835 | MATHDB-325205

Goresky-Kottwitz-Macpherson purity conjecture for affine Springer fibers

摘录 / EXCERPT: Let $k=\mathbf{F}_{q}$, let G be a connected reductive group split over k, and let $\gamma\in \mathfrak{t}(\mathcal{O})$ be regular, with affine Springer fiber ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-001836 | MATHDB-325217

The Zarankiewicz number conjecture for orders three below a projective-plane order

摘录 / EXCERPT: Let $z(n)$ denote the maximum number of edges in a bipartite graph with partite sets of order n that contains no cycle of length four. For a prime power k, consider the order...

conjecture | open | MathDB 页面 | 原文线索

V2-001837 | MATHDB-325229

The full moment conjecture for quadratic Dedekind zeta functions

摘录 / EXCERPT: Let $\mathbb{K}$ be a quadratic extension, let χ be the associated real Dirichlet character, and let $a(k)$ be the arithmetic factor defined by the Euler product in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-001838 | MATHDB-325230

The non-primitive Selberg-class moment conjecture

摘录 / EXCERPT: The non-primitive Selberg-class moment conjecture. For $k>-1/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-001839 | MATHDB-325233

Atkin's 13-adic congruences for normalized coefficients of the j-function

摘录 / EXCERPT: Atkin's conjecture. For any prime $p\neq13$ such that $c_{p^m}(j)\neq0$ for all $m\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-001840 | MATHDB-325234

A 5-adic congruence for a weight-1 noncongruence modular form

摘录 / EXCERPT: The 5-adic congruence conjecture. For $p=5$, $m\geq1$, and odd $n\geq1$, one has $c_{5^m}(f)\neq0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-001841 | MATHDB-325258

Joshi-Petrov conjecture on nonsmooth Hecke algebras of Drinfeld modular forms

摘录 / EXCERPT: Joshi-Petrov conjecture. Given q, there exist $k\gg 0$ and m, with $0\leq m<q$ and $k\equiv 2m\pmod{q-1}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-001842 | MATHDB-325262

Chowla's conjecture on Möbius correlations

摘录 / EXCERPT: Let $r\geq 1$, let $n_1,\ldots,n_r\in\mathbb{N}$, and let $k_1,\ldots,k_r\in\{1,2\}$, with the k_i not all even. Then ... as $N\to\infty$. Chowla's conjecture. The displayed asympt...

conjecture | open | MathDB 页面 | 原文线索

V2-001843 | MATHDB-325266

Fyodorov and Keating's short-interval maximum conjecture for the Riemann zeta function

摘录 / EXCERPT: Fyodorov and Keating's conjecture. Let $\epsilon>0$. For T_1 sufficiently large depending on ϵ, for all $T_1\leq T\leq 2T_1$, outside a set of “bad” values of T of mea...

conjecture | open | MathDB 页面 | 原文线索

V2-001844 | MATHDB-325267

Typical short-interval bounds for the maximum of the Riemann zeta function

摘录 / EXCERPT: Typical short-interval maximum conjecture. Let $\epsilon>0$. For T_1 sufficiently large depending on ϵ, for all $T_1\leq T\leq 2T_1$, outside a set of “bad” values of...

conjecture | open | MathDB 页面 | 原文线索

V2-001845 | MATHDB-325268

Determinant conjecture for the quintic multisection matrices

摘录 / EXCERPT: Let $\mathcal{A}_{d}$ be the matrices defined in Corollary. Determinant conjecture. The determinant of $\mathcal{A}_{d}$ is equal to $\pm 5^{f(d)}$ for some $f(d)\in\mathbb{N}$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-001846 | MATHDB-325269

Higher-power congruence conjecture for partition coefficients

摘录 / EXCERPT: Let $p_r(n)$ denote the coefficients in the partition-generating series used in the paper. For a positive integer λ, let μ_{λ} and δ_{λ} satisfy ... Hi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-001847 | MATHDB-325270

The determinant conjecture for cubic Eisenstein multisection matrices

摘录 / EXCERPT: Determinant conjecture. The exponent $w(n)$ exists for every $n\in\mathbb{N}$, and for even indices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001848 | MATHDB-325333

The superpolynomial lattice-path formula

摘录 / EXCERPT: Let m and n be coprime positive integers, let $\mathcal{P}_{n,m}(u,q,t)$ denote the superpolynomial, and let D range over lattice paths below the main diagonal in an...

conjecture | open | MathDB 页面 | 原文线索

V2-001849 | MATHDB-325343

Brylinski's positive-energy representation conjecture for elliptic cohomology

摘录 / EXCERPT: Let m be an integer, and let $P_m(d)$ be the free abelian group generated by the isomorphism classes of irreducible positive-energy representations of the universal central exten...

conjecture | open | MathDB 页面 | 原文线索

V2-001850 | MATHDB-325346

Lalín-Rodrigue-Rogers unimodularity conjecture for Bernoulli-Euler polynomials

摘录 / EXCERPT: Let B_m denote the Bernoulli numbers and E_m the Euler numbers. For an even positive integer k, define ... Call a polynomial unimodular if all of its zeros lie on the unit circ...

conjecture | open | MathDB 页面 | 原文线索

V2-001851 | MATHDB-325351

The Relatively Prime Conjecture for factorial polynomials

摘录 / EXCERPT: Relatively Prime Conjecture. For all such ${\bf a},{\bf b}$ and all $n\in\mathbb N$, the polynomials $P_{{\bf a},{\bf b},n}(X)$ and $P_{{\bf a},{\bf b},n+1}(X)$ have no common zero...

conjecture | open | MathDB 页面 | 原文线索

V2-001852 | MATHDB-325369

The branch-count conjecture for total periodic diagrams of perturbed monomial systems

摘录 / EXCERPT: Let $n\geq 2$ be an integer, let p range over primes, and let $h_c(x)=x^n+c\bmod{p}$. The total periodic diagram $\operatorname{TPD}(h_c,m)$ is the diagram for the first m prim...

conjecture | open | MathDB 页面 | 原文线索

V2-001853 | MATHDB-325372

Independence of recursion coefficients for quadratic polynomials over finite fields

摘录 / EXCERPT: Let p be a prime and let c be an element of the finite field with $c\neq\frac14\pmod{5}$. Consider the recursion for the sequence $a_{1+x+x^2,p}(n)$ associated with the polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-001854 | MATHDB-325373

Mahler-type functional equation for generating functions of polynomial powers

摘录 / EXCERPT: Let $f(x)$ be a polynomial over a finite field, let p be the relevant prime, and let $g_{f(x),p}(z)$ denote the generating function associated with the coefficient sequence for p...

conjecture | open | MathDB 页面 | 原文线索

V2-001855 | MATHDB-325375

Asymptotic maximal eigenvalue for polynomial degree

摘录 / EXCERPT: For a prime p, let $\lambda(f)$ be the growth rate associated with a polynomial $f(x)$, and let Λ_k be the maximal value of $\lambda(f)$ among polynomials satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-001856 | MATHDB-325397

The completion invariance conjecture for the c_2-invariant

摘录 / EXCERPT: Let G_1 and G_2 be primitive-divergent graphs with isomorphic completions, and let $c_2(G_i)_q\in\mathbb{Z}/q\mathbb{Z}$ denote their c_2-invariants at a prime power q. Com...

conjecture | open | MathDB 页面 | 原文线索

V2-001857 | MATHDB-325399

The no-weight-two conjecture for modular completed primitive graphs

摘录 / EXCERPT: Let Γ be a completed primitive graph and let f be a modular form such that Γ is modular with respect to f. The no-weight-two conjecture. The weight of f is at l...

conjecture | open | MathDB 页面 | 原文线索

V2-001858 | MATHDB-325415

Conjecture on the index of Chebyshev radical extensions

摘录 / EXCERPT: Let θ be a root of θ's defining polynomial ... . Index conjecture. If $p eq\ell$ is an odd prime for which $t\equiv\pm2\pmod{p^2}$, then ... This conjectural formula is i...

conjecture | open | MathDB 页面 | 原文线索

V2-001859 | MATHDB-325416

Newton-polygon lift conjecture for Chebyshev radical extensions

摘录 / EXCERPT: Let p be an odd prime, let Φ be the defining polynomial in the paper, and let ϕ_i be an irreducible factor of its reduction modulo p. A lift $\hat\phi_i$ is a monic p...

conjecture | open | MathDB 页面 | 原文线索

V2-001860 | MATHDB-325421

Strong Fermat's Last Theorem conjecture

摘录 / EXCERPT: Let $p>3$ be prime, let $K=\mathbb Q(\zeta)$ be the p-th cyclotomic field, and let $u,v\in\mathbb Z\setminus\{0\}$ be coprime. The equation ... according as $u+v\not\equiv0\pmod p$...

conjecture | open | MathDB 页面 | 原文线索

V2-001861 | MATHDB-325422

SFLT2 criterion via a nonsplit p-principal prime

摘录 / EXCERPT: Let $p>3$ be prime. For a triple $(p,u,v)\in\mathbb Z^3$ with u and v coprime and $p\mid v$, let q be a p-principal prime with $puv\not\equiv0\pmod q$, let n be the order...

conjecture | open | MathDB 页面 | 原文线索

V2-001862 | MATHDB-325423

Reduced-form conjecture for SFLT2 solutions

摘录 / EXCERPT: Let p be an odd prime, let $K=\mathbb Q(\zeta)$ be the p-th cyclotomic field, and consider a solution of the SFLT2 equation. Reduced-form conjecture. If the SFLT2 conjecture fa...

conjecture | open | MathDB 页面 | 原文线索

V2-001863 | MATHDB-325451

Chen-Fang's additive complement conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a set of nonnegative integers, and write ... Let $\mathcal{B}$ be an additive complement of $\mathcal{A}$, meaning that every sufficiently large nonnegative inte...

conjecture | open | MathDB 页面 | 原文线索

V2-001864 | MATHDB-325455

Random-matrix conjecture for the squarefree-number detection exponent

摘录 / EXCERPT: Random-matrix conjecture. The limiting exponent is

conjecture | open | MathDB 页面 | 原文线索

V2-001865 | MATHDB-325468

Conjecture on the cycles of Fermat numbers and their factors

摘录 / EXCERPT: Let $F_n=2^{2^n}+1$ be the nth Fermat number. For a positive integer q, let G_q/R_q denote the cycle decomposition of the multiplication-by-2 map modulo q, and call each...

conjecture | open | MathDB 页面 | 原文线索

V2-001866 | MATHDB-325469

Conjecture on the cycle counts of Mersenne numbers and their factors

摘录 / EXCERPT: Let $M_p=2^p-1$ be the Mersenne number associated with a prime number p. For a positive integer q, let G_q/R_q denote the cycle decomposition of the multiplication-by-2 map...

conjecture | open | MathDB 页面 | 原文线索

V2-001867 | MATHDB-325492

Hurwitz's linearity conjecture for class-number sums modulo 5

摘录 / EXCERPT: Hurwitz's conjecture. For every $a,L\in\mathbb{Z}$, there exist constants $c_1,c_2\in\mathbb{Q}$, given explicitly by the following cases, such that for every prime...

conjecture | open | MathDB 页面 | 原文线索

V2-001868 | MATHDB-325493

Hurwitz's partial linearity conjecture for class-number sums modulo 7

摘录 / EXCERPT: Hurwitz's conjecture. If $L=3,5,6$ and $a\in\mathbb{Z}$, or if $(a,L)\equiv(\pm1,1)\pmod{7}$, then there exist constants $c_1,c_2\in\mathbb{Q}$ such that for every prime...

conjecture | open | MathDB 页面 | 原文线索

V2-001869 | MATHDB-325494

Hurwitz's explicit class-number-sum conjecture modulo 7

摘录 / EXCERPT: Hurwitz's conjecture. For a prime ℓ and $a\in\mathbb{Z}$,

conjecture | open | MathDB 页面 | 原文线索

V2-001870 | MATHDB-325505

Binary-valued conjecture for the quantum invariants $d_l(\alpha_{m,n})$

摘录 / EXCERPT: Let l be a prime and let m,n satisfy ... For each such pair, let $d_l(\alpha_{m,n})$ be the quantity defined in the paper. Binary-valued conjecture. ... This conjecture predicts th...

conjecture | open | MathDB 页面 | 原文线索

V2-001871 | MATHDB-325510

Conjectures on free infinite divisibility of beta, beta prime, and gamma distributions

摘录 / EXCERPT: Let $\bm{\beta}_{p,q}$ and $\bm{\beta}'_{p,q}$ denote the beta and beta prime distributions, respectively, and let $\bm{\gamma}_p$ denote the gamma distribution, with parameters in...

conjecture | open | MathDB 页面 | 原文线索

V2-001872 | MATHDB-325530

Unique prime factorization conjecture for non-thin hypergraphs

摘录 / EXCERPT: Let H be a connected, simple, non-thin hypergraph. Write ${\weithutzl{\boxtimes}}\,$ and ${\weitdacherl{\boxtimes}}\,$ for the normal and strong hypergraph products, respectively...

conjecture | open | MathDB 页面 | 原文线索

V2-001873 | MATHDB-325533

Density-one conjecture for weak Carmichael prime factors

摘录 / EXCERPT: For integers e,f with $1\le e\le f$ and $e+f\ge 3$, let ${\mathcal Q}_w(e,f)$ be the set of primes occurring in a weak Carmichael number of the form p^eq^f or q^ep^f for some...

conjecture | open | MathDB 页面 | 原文线索

V2-001874 | MATHDB-325534

Infinite weak Carmichael families with a fixed prime

摘录 / EXCERPT: For integers e,f with $1\le e\le f$ and $e+f\ge 3$, and an odd prime q, let ${\mathcal Q}_w(q;e,f)$ be the set of primes p such that p^eq^f or q^ep^f is a weak Carmichael...

conjecture | open | MathDB 页面 | 原文线索

V2-001875 | MATHDB-325535

Generalized infinitude conjecture for weak Carmichael numbers

摘录 / EXCERPT: Let $s\ge 2$ be an integer, and let $(e_1,e_2,\ldots,e_s)$ be a fixed tuple with $e_1\ge e_2\ge\cdots\ge e_s\ge 1$ and $\sum_{i=1}^s e_i\ge 3$. Generalized weak Carmichael infinitu...

conjecture | open | MathDB 页面 | 原文线索

V2-001876 | MATHDB-325537

Twin Carmichael numbers conjecture

摘录 / EXCERPT: Two Carmichael numbers are twin Carmichael numbers if no weak Carmichael number lies strictly between them. Twin Carmichael numbers conjecture. There are infinitely many pairs of t...

conjecture | open | MathDB 页面 | 原文线索

V2-001877 | MATHDB-325538

Smallest weak Carmichael numbers with a fixed number of prime factors

摘录 / EXCERPT: For a composite integer, call the number of distinct prime divisors its number of prime factors. Smallest weak Carmichael number conjecture. For every integer $k\ge 3$, the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-001878 | MATHDB-325539

No odd prime power is super Carmichael

摘录 / EXCERPT: A super Carmichael number is a weak Carmichael number satisfying ... No odd prime-power super Carmichael conjecture. For every odd prime p and every $f\ge 2$, the prime power p^f...

conjecture | open | MathDB 页面 | 原文线索

V2-001879 | MATHDB-325540

Infinitely many super Carmichael numbers

摘录 / EXCERPT: A super Carmichael number is a weak Carmichael number satisfying ... Super Carmichael infinitude conjecture. There are infinitely many super Carmichael numbers. The source motivates...

conjecture | open | MathDB 页面 | 原文线索

V2-001880 | MATHDB-325542

The odd-denominator initial-segment conjecture for hypergeometric Bernoulli numbers

摘录 / EXCERPT: Let b be a positive integer, let $B_n^{(b)}$ denote the hypergeometric Bernoulli numbers, and define ... Let $\nu_2(b)$ be the exponent of 2 in b, and let $\alpha(b)$ be the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-001881 | MATHDB-325543

The prime-bound conjecture for denominators of hypergeometric Bernoulli numbers

摘录 / EXCERPT: Let b and n be nonnegative integers, and let $B_n^{(b)}$ denote the hypergeometric Bernoulli numbers. Prime-bound conjecture. Every prime p dividing the denominator of...

conjecture | open | MathDB 页面 | 原文线索

V2-001882 | MATHDB-325560

Continuity conjecture for the dimension of Diophantine approximation sets on smooth curves

摘录 / EXCERPT: For a smooth function $f\colon\mathbb{R}\to\mathbb{R}$, let $W_{\tau}(f)$ be the set of points defined by the simultaneous Diophantine approximation conditions used above, with inf...

conjecture | open | MathDB 页面 | 原文线索

V2-001883 | MATHDB-325589

Lang's hyperbolicity and Mordellicity conjectures

摘录 / EXCERPT: Lang's conjectures. If X is hyperbolic, then it is mordellic. Moreover, $X\setminus\operatorname{Exc}(X)$ is mordellic.

conjecture | open | MathDB 页面 | 原文线索

V2-001884 | MATHDB-325609

The classification conjecture for tuples with the Khintchine property

摘录 / EXCERPT: Let p be a prime, let $k<p$, and let $c_0,\ldots,c_k$ be distinct elements of $\mathbb{F}_p$. A tuple has the Khintchine property when it satisfies the recurrence property define...

conjecture | open | MathDB 页面 | 原文线索

V2-001885 | MATHDB-325618

Cohen's modularity conjecture for class-number generating series

摘录 / EXCERPT: Let ... where $q=e^{2\pi i\tau}$ with $\operatorname{Im}(\tau)>0$, $H(n)$ is the Hurwitz class number, and $\lambda_{2k+1}(n)$ denotes the coefficients occurring in Cohen's formal po...

conjecture | open | MathDB 页面 | 原文线索

V2-001886 | MATHDB-325621

The Diophantine-dimension conjecture for effective index

摘录 / EXCERPT: Effective-index conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-001887 | MATHDB-325623

Lehmer's conjecture for the Ramanujan tau coefficients

摘录 / EXCERPT: Let $q=e^{2\pi i z}$, and let ... be a newform, where $k\in\{12,16,18,20,22,26\}$. Lehmer's conjecture. For every such k and every positive integer n, ... For $k=12$, this is Lehme...

conjecture | open | MathDB 页面 | 原文线索

V2-001888 | MATHDB-325656

The DHL$(k,2)$ conjecture on two primes in admissible tuples

摘录 / EXCERPT: Let $\mathcal H=\{h_i\}_{i=1}^k$ be an admissible k-tuple, meaning that its components avoid covering every residue class modulo any prime. The DHL$(k,2)$ conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-001889 | MATHDB-325657

Average first negative value conjecture for Rankin-Selberg Fourier coefficients

摘录 / EXCERPT: Average first negative value conjecture. As $x\to\infty$, one expects

conjecture | open | MathDB 页面 | 原文线索

V2-001890 | MATHDB-325659

Non-triadic-base approximation conjecture on the Cantor set

摘录 / EXCERPT: Let K be the middle-third Cantor set, let $\kappa=\operatorname{dim_{H}}K$, let b be an integer that is not a power of three, and let $v_b(\xi)$ be the approximation exponent a...

conjecture | open | MathDB 页面 | 原文线索

V2-001891 | MATHDB-325660

Baker-Schmidt-type algebraic approximation conjecture on the Cantor set

摘录 / EXCERPT: Let K be the middle-third Cantor set, let $\kappa=\operatorname{dim_{H}}K$, and let $w_n^{\ast}(\xi)$ be the exponent governing approximation of ξ by algebraic numbers of deg...

conjecture | open | MathDB 页面 | 原文线索

V2-001892 | MATHDB-325673

Average canonical height conjecture for higher-dimensional elliptic-curve families

摘录 / EXCERPT: Average canonical height conjecture. For any $P\in E(K)$,

conjecture | open | MathDB 页面 | 原文线索

V2-001893 | MATHDB-325695

Large-rank conjecture for rank-k flats in matroids with no uniform minor

摘录 / EXCERPT: Let $\ell\ge 2$ be an integer, let M be a rank-r matroid, and let $U_{2,\ell+2}$ denote the rank-2 uniform matroid on $\ell+2$ elements. Let q be the largest prime power su...

conjecture | open | MathDB 页面 | 原文线索

V2-001894 | MATHDB-325696

The Gonzales-Acuña-Short cabling conjecture for knots in the 3-sphere

摘录 / EXCERPT: Gonzales-Acuña-Short's cabling conjecture. If a knot in S^3 admits a surgery to a non-prime 3-manifold, then the knot is either a torus knot or a cabled knot.

conjecture | open | MathDB 页面 | 原文线索

V2-001895 | MATHDB-325698

The classification conjecture for hyperbolic doubly primitive knots in non-prime manifolds

摘录 / EXCERPT: Doubly primitive knot classification conjecture. If K is hyperbolic, then

conjecture | open | MathDB 页面 | 原文线索

V2-001896 | MATHDB-325701

Smooth numbers in short intervals conjecture

摘录 / EXCERPT: For real x,y, let $\Psi(x,y)$ denote the number of positive integers at most x all of whose prime factors are at most y. Such integers are called y-smooth numbers. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001897 | MATHDB-325704

Conjecture on the extremal divisor count for almost all parameter pairs

摘录 / EXCERPT: Divisor-count conjecture. For most values of k and n, specifically for all but finitely many values of k and n,

conjecture | open | MathDB 页面 | 原文线索

V2-001898 | MATHDB-325718

Young's shifted convolution conjecture for holomorphic cusp forms

摘录 / EXCERPT: Let $U(z)=y^{k/2}f(z)$, where f is a holomorphic Hecke cusp form of weight k, with Fourier coefficients $\lambda(n)$ and first coefficient $\rho(1)$. For fixed $0<a\le y\le b$...

conjecture | open | MathDB 页面 | 原文线索

V2-001899 | MATHDB-325725

Ash's mod-p Langlands conjecture for cohomological Hecke eigenvalues

摘录 / EXCERPT: Let X_K be a locally symmetric space and let $H^i(X_K,\mathbb{F}_p)$ carry the action of Hecke operators. A system of Hecke eigenvalues in this cohomology is a compatible collect...

conjecture | open | MathDB 页面 | 原文线索

V2-001900 | MATHDB-325737

Twisted Kazhdan-Lusztig monotonicity property B-prime

摘录 / EXCERPT: Twisted property B-prime. The polynomials $P^\pm_{y,w}$ are decreasing for fixed w: whenever $y,z,w\in\mathbf I_*$ and $y\leq z$, both differences $P^+_{y,w}-P^+_{z,w}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-001901 | MATHDB-325738

Twisted Kazhdan-Lusztig structure-constant unimodality property D-prime

摘录 / EXCERPT: Let $(W,S,*)$ be a triple consisting of a Coxeter system (W,S) and an S-preserving involution $*\in\operatorname{Aut}(W)$, with corresponding twisted involutions $\mathbf I_*$....

conjecture | open | MathDB 页面 | 原文线索

V2-001902 | MATHDB-325739

Twisted Kazhdan-Lusztig positivity package

摘录 / EXCERPT: Let $(W,S,*)$ be a triple in which (W,S) is a Coxeter system and $*\in\operatorname{Aut}(W)$ is an S-preserving involution. Let Properties A-prime, B-prime, C-prime, and D-prim...

conjecture | open | MathDB 页面 | 原文线索

V2-001903 | MATHDB-325772

Exceptional-zero conjecture for the symmetric-square p-adic L-function

摘录 / EXCERPT: Let $\mathbf{f}$ be the Hilbert modular form in the paper, let F be its totally real coefficient field, and let m be the parameter for which the trivial zero occurs at $s=m+1$....

conjecture | open | MathDB 页面 | 原文线索

V2-001904 | MATHDB-325779

Transcendence conjecture for Artin-Mazur zeta functions of separable rational maps

摘录 / EXCERPT: Let p be a prime, let $f\in \overline{\mathbb{F}}_p(x)$ be a separable rational map, and let $\zeta(f,\mathbb{P}^1(\overline{\mathbb{F}}_p);t)$ denote its Artin-Mazur zeta functi...

conjecture | open | MathDB 页面 | 原文线索

V2-001905 | MATHDB-325789

Finiteness conjecture for motives of bounded height

摘录 / EXCERPT: Let K be a number field and let $c>0$. Fix a type $\Phi=(w,(h^r)_{r\in\mathbb{Z}})$ of motives, with $h^r=\dim \operatorname{gr}_{dR}^r$, and choose a,b such that $h^r=0$ unles...

conjecture | open | MathDB 页面 | 原文线索

V2-001906 | MATHDB-325810

Algebraic principal-part conjecture for harmonic weak Maass forms

摘录 / EXCERPT: Let $m>0$ and let f be a harmonic weak Maass form of weight $\frac{1}{2}-m$ for the representation ρ_L. Assume that $\xi_{\frac{1}{2}-m}f$ is holomorphic also at ∞, a...

conjecture | open | MathDB 页面 | 原文线索

V2-001907 | MATHDB-325815

The non-exceptionality conjecture for 11 over quadratic fields

摘录 / EXCERPT: Let K be a quadratic field. An exceptional prime for K is a prime l for which there exists an elliptic curve over K whose mod-l Galois representation has exceptional imag...

conjecture | open | MathDB 页面 | 原文线索

V2-001908 | MATHDB-325823

The 1089 graph divisibility conjecture for reverse multiples

摘录 / EXCERPT: 1089 graph conjecture. The (g,k) Young graph is isomorphic to the 1089 graph if and only if $k+1$ divides g.

conjecture | open | MathDB 页面 | 原文线索

V2-001909 | MATHDB-325824

The 1089 graph characterization by endpoint digits

摘录 / EXCERPT: 1089 graph endpoint conjecture. If there is a (g,k)-reverse multiple with $a_0+a_{n-1}=g$, then the Young graph is the 1089 graph. Conversely, if the graph is the 1089 graph, the...

conjecture | open | MathDB 页面 | 原文线索

V2-001910 | MATHDB-325825

The diagonal-label characterization of complete Young graphs

摘录 / EXCERPT: Complete Young graph conjecture. A Young graph is a complete graph if and only if it has at least one nonzero diagonal-labelled node $[r,r]$.

conjecture | open | MathDB 页面 | 原文线索

V2-001911 | MATHDB-325848

The zero-digit conjecture for products in base q

摘录 / EXCERPT: Fix an integer base $q>1$. A product of k digits greater than 1 for base q means a product of k elements of $\{2,\ldots,q-1\}$. Zero-digit conjecture for products in base...

conjecture | open | MathDB 页面 | 原文线索

V2-001912 | MATHDB-325849

The equidistribution conjecture for digits of powers in base q

摘录 / EXCERPT: Let p be a prime, let q be a base that is not a power of p, and write the base-q expansion of p^n as having $k_n=\lceil n\log_q p\rceil$ digits. Let $d_{j,n}$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-001913 | MATHDB-325870

Uniform Mordell-Lang conjecture for curves

摘录 / EXCERPT: Let $g\ge 2$ and $r\ge 0$. For a curve C over $\mathbb{C}$ of genus g, let J be its Jacobian and let $\iota\colon C\to J$ be an embedding. Let $\Gamma\subset J(\mathbb{C})$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-001914 | MATHDB-325881

Conjecture on regular prime graphs of finite groups

摘录 / EXCERPT: Let G be a finite group, let $\rho(G)$ denote the set of primes dividing character degrees of G, and let $\Delta(G)$ be the prime graph of G, whose vertices are the elements...

conjecture | open | MathDB 页面 | 原文线索

V2-001915 | MATHDB-325884

Greenberg's rationality conjecture for the point P_ψ

摘录 / EXCERPT: Greenberg's rationality conjecture. The local point P_ψ is a global point, and more precisely

conjecture | open | MathDB 页面 | 原文线索

V2-001916 | MATHDB-325913

Bogomolov conjecture for powers of the multiplicative group

摘录 / EXCERPT: Bogomolov conjecture. If, for each $\epsilon>0$, the set $V(\overline{\mathbb Q})\cap S_{\epsilon}$ is Zariski dense in V, then V is a torsion translate of an algebraic subgrou...

conjecture | solved | MathDB 页面 | 原文线索

V2-001917 | MATHDB-325923

Hirschhorn-Sellers congruence for overpartitions

摘录 / EXCERPT: Let an overpartition of a nonnegative integer be a partition in which the first occurrence of each part may be overlined, and let $\bar{p}(n)$ denote the number of overpartitions o...

conjecture | open | MathDB 页面 | 原文线索

V2-001918 | MATHDB-325964

Theta-number conjecture for 1-intersecting invertible matrices

摘录 / EXCERPT: Let q be a prime power, let $\Gamma=\operatorname{GL}(n,\mathbb{F}_q)$, and let ... Here $\alpha(G)$ denotes the independence number and $\vartheta(G)$ the Lovász theta-number. The...

conjecture | open | MathDB 页面 | 原文线索

V2-001919 | MATHDB-325965

Erdős-Ko-Rado conjecture for k-intersecting invertible matrices

摘录 / EXCERPT: Let q be a prime power, let $k\in\mathbb{N}$, and let $\Gamma=\operatorname{GL}(n,\mathbb{F}_q)$. Define ... The graph's independent sets are the k-intersecting families of inver...

conjecture | open | MathDB 页面 | 原文线索

V2-001920 | MATHDB-325966

A regularised cosine sum over the ordinates of zeta zeros

摘录 / EXCERPT: Let $\gamma_1,\gamma_2,\ldots$ denote the positive imaginary parts of the non-trivial zeros of the Riemann zeta function, counted in the ordering used by the source, and let $t>0$....

conjecture | open | MathDB 页面 | 原文线索

V2-001921 | MATHDB-325970

The umbral McKay-Thompson shadow conjecture

摘录 / EXCERPT: Let X be a Niemeier root system with Coxeter number m, let G^X be its associated group, and let S^X_g be the attached cusp form for $g\in G^X$. For $g\in G^X$, let n_g be...

conjecture | open | MathDB 页面 | 原文线索

V2-001922 | MATHDB-326008

The mod-ell Galois representation conjecture for real analytic Siegel cusp forms

摘录 / EXCERPT: Let $F\in S_{(2,1)}(\Gamma(N),-\frac{5}{12},0)$ be a Hecke eigenform with eigenvalues $\lambda(p^i)$ for $T(p^i)$, and suppose $F|[S_{p,1}]=\chi_1(p)F$ and $F|[S_{p,p}]=\chi_2(p)F$...

conjecture | open | MathDB 页面 | 原文线索

V2-001923 | MATHDB-326045

The average-size conjecture for coprime core partitions

摘录 / EXCERPT: Let $a,b\in\mathbb{N}$ be coprime. An (a,b)-core is a partition with no hook length divisible by a or b; a self-conjugate (a,b)-core is one equal to its conjugate. The aver...

conjecture | open | MathDB 页面 | 原文线索

V2-001924 | MATHDB-326046

The skew-length generating-function conjecture for core partitions

摘录 / EXCERPT: Let $a<b$ be coprime. For an (a,b)-core partition λ, let $\ell(\lambda)$ be its number of nonzero rows. Define its skew length $s\ell(\lambda)$ to be the number of boxes...

conjecture | open | MathDB 页面 | 原文线索

V2-001925 | MATHDB-326057

Asymptotic formula for Carmichael numbers with three prime factors

摘录 / EXCERPT: Let $C(x)$ denote the number of Carmichael numbers $n\leq x$ with exactly three prime factors. Three-factor Carmichael number conjecture. As $x\to\infty$, one has ... where C is an...

conjecture | open | MathDB 页面 | 原文线索

V2-001926 | MATHDB-326137

Krause-Zahlten's cross number conjecture for finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group. Write ... with each summand of prime-power order, and define ... Let $\mathsf{K}^{*}(G)=\mathsf{k}^{*}(G)+1/\operatorname{exp}(G)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-001927 | MATHDB-326139

The Narkiewicz constant conjecture for elementary abelian p-groups

摘录 / EXCERPT: Let p be prime, let n be a positive integer, and let $G=C_p^n$ be the elementary abelian p-group. The elementary abelian p-group conjecture. If $G=C_p^n$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-001928 | MATHDB-326140

The unique-subsequence-sum conjecture for zero-sum free sequences

摘录 / EXCERPT: Let p be prime, let k be a positive integer, and let $G=C_p^k$. A sequence over G is zero-sum free if none of its nonempty subsequences has sum zero. The short-subsequence co...

conjecture | open | MathDB 页面 | 原文线索

V2-001929 | MATHDB-326141

The cyclic-subgroup strengthening for zero-sum free sequences

摘录 / EXCERPT: Let p be prime, let k be a positive integer, and let $G=C_p^k$. For a cyclic subgroup H of G, let T_H denote the subsequence of T consisting of terms in H, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001930 | MATHDB-326172

Non-commutative Hilbert modular symbol 1-cocycle conjecture

摘录 / EXCERPT: Let $c^1_{p_1,p_2;p_3}$ denote the non-commutative Hilbert modular symbol associated with points p_1,p_2,p_3. Non-commutative Hilbert modular symbol 1-cocycle conjecture. The sym...

conjecture | open | MathDB 页面 | 原文线索

V2-001931 | MATHDB-326173

Homology conjecture for non-commutative Hilbert modular 2-cocycles

摘录 / EXCERPT: Let $c^2_p(\beta,\gamma)$ and $c^2_q(\beta,\gamma)$ be the conjectural non-commutative Hilbert modular 2-cocycles associated with points p and q. Let b_{pq}^1 be the interv...

conjecture | open | MathDB 页面 | 原文线索

V2-001932 | MATHDB-326183

Cubic residue asymptotic ordered by the first successive minimum

摘录 / EXCERPT: Let E range over totally real cubic number fields, and let $m_1(E)$ denote the first successive minimum of the positive definite quadratic form Q_E on $\mathcal{O}_E/\mathbb Z$...

conjecture | open | MathDB 页面 | 原文线索

V2-001933 | MATHDB-326196

Main exponential-sum conjecture for powers of 3 modulo powers of 2

摘录 / EXCERPT: Main exponential-sum conjecture. There are constants $c<\infty$ and $\delta>0$ such that for all n and all nonzero $m\in\mathbb Z/2^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-001934 | MATHDB-326197

Branch-jumping conjecture for base 3/2 expansions

摘录 / EXCERPT: Branch-jumping conjecture. There is a constant $c>0$ such that, whenever this condition holds,

conjecture | open | MathDB 页面 | 原文线索

V2-001935 | MATHDB-326214

Best-possible bounds for the abundancy index of almost perfect numbers

摘录 / EXCERPT: Best-possible bounds conjecture. The bounds in Theorem 1 are best-possible.

open_problem | open | MathDB 页面 | 原文线索

V2-001936 | MATHDB-326229

Monotonicity conjecture for the sequences t_n and T_n

摘录 / EXCERPT: Let $(t_n)_{n=1}^{\infty}$ and $(T_n)_{n=1}^{\infty}$ be the sequences whose values are listed in the table preceding the conjecture. Monotonicity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-001937 | MATHDB-326234

The adelic root bound hypothesis for sums of products of sparse polynomials

摘录 / EXCERPT: For $k,m,t\in\mathbb{N}$, let $\mathrm{SPS}(k,m,t)$ denote the class of polynomials represented as sums of k products of m integer polynomials, each having at most t nonzero...

open_problem | open | MathDB 页面 | 原文线索

V2-001938 | MATHDB-326237

Ostmann's conjecture on additive irreducibility of the primes

摘录 / EXCERPT: Let ${\cal S}$ be a set of positive integers. It is asymptotically additively irreducible if there do not exist sets ${\cal A}$ and ${\cal B}$, each containing at least two element...

conjecture | open | MathDB 页面 | 原文线索

V2-001939 | MATHDB-326238

Sárközy's conjecture on additive irreducibility of smooth numbers

摘录 / EXCERPT: Fix $0<\epsilon<1$. An integer is y-smooth if all of its prime factors are at most y, and let $\mathcal{S}_{n^{\epsilon}}$ denote the set of integers n with no prime factor g...

conjecture | open | MathDB 页面 | 原文线索

V2-001940 | MATHDB-326254

Asymptotic-extension conjecture for weighted congruences

摘录 / EXCERPT: Asymptotic-extension conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-001941 | MATHDB-326328

The tropical prime form conjecture

摘录 / EXCERPT: Tropical prime form conjecture. At any loop order, the tropical prime form equals the scalar s of Dai, namely the graph distance $d_\gamma(X,Y)$ between X and Y along a path...

conjecture | open | MathDB 页面 | 原文线索

V2-001942 | MATHDB-326332

Squarefree-level conjecture for 1⊕χ_l^{k-1}

摘录 / EXCERPT: Let k be an even integer with $k\geq4$, let $l>k+1$, and let N be squarefree. Consider the representation $\mathbf{1}\oplus\chi_l^{k-1}$, where χ_l is the mod l cyclotom...

conjecture | open | MathDB 页面 | 原文线索

V2-001943 | MATHDB-326333

Sufficiency of level raising for 1⊕χ_l^{k-1}

摘录 / EXCERPT: Level-raising sufficiency conjecture. If this condition holds at p, then ρ arises from a weight-k newform in $\mathcal{S}_k(\Gamma_0(Np))$. This is the precise conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-001944 | MATHDB-326344

The equivariant epsilon conjecture for one-dimensional Lubin-Tate groups

摘录 / EXCERPT: Let $L/\mathbb{Q}_p$ be the extension, let T be the p-adic Tate module of the formal group $\mathcal{F}$, and let $\tilde{y}$ be the component of the equivariant epsilon isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-001945 | MATHDB-326356

Gerstenhaber-Giaquinto conjecture on boundary triangular r-matrices

摘录 / EXCERPT: Let m and n be relatively prime positive integers with $m<n$. Let $\mathfrak{p}_{m,n}\subseteq\mathfrak{sl}_n$ be the maximal parabolic subalgebra obtained by deleting the m-...

conjecture | open | MathDB 页面 | 原文线索

V2-001946 | MATHDB-326361

The Mersennian preon charge model

摘录 / EXCERPT: Mersennian preon charge model. Preons of order f_{n+1} are either up-type or down-type, with charges transforming according to

conjecture | open | MathDB 页面 | 原文线索

V2-001947 | MATHDB-326362

The hypersphere divisibility conjecture for preon constituents

摘录 / EXCERPT: Let a preon be a constituent of a superordinate preon, and let the constituent structure use a number of hyperspheres. Let the kissing number of a space be the maximum number of no...

conjecture | open | MathDB 页面 | 原文线索

V2-001948 | MATHDB-326378

Indexed strong Goldbach representation conjecture

摘录 / EXCERPT: Indexed strong Goldbach representation conjecture. Let $t_n^\alpha\in\mathbb{N}^*$ be a positive even integer. Then there esist two prime numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-001949 | MATHDB-326414

Helleseth's zero-value conjecture for decimated m-sequence cross correlation

摘录 / EXCERPT: Let p be a prime, let n be a positive integer, and let d be a decimation parameter. Define $\text{S}_d(z)=\text{C}_d(z)+1$, where $z\in\operatorname{GF}(p^n)^*$. Helleseth's...

conjecture | open | MathDB 页面 | 原文线索

V2-001950 | MATHDB-326431

Cohomology stabilization conjecture for coclass families

摘录 / EXCERPT: Let p be a prime, let k be a field of characteristic p, and let $(G_i \mid i \geq 0)$ be a coclass family of finite p-groups. Cohomology stabilization conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-001951 | MATHDB-326433

The generalized Fermat equation conjecture for prime exponents and prime-power gcd

摘录 / EXCERPT: Let $A,B,C,D,n\in\mathbb{Z}_{+}$ with $n\geq3$, and suppose ... Assume that $\gcd(A,B,C)=p^k$, where p is an odd prime and $k\in\mathbb{Z}_{+}$. Generalized Fermat equation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-001952 | MATHDB-326434

The prime-power gcd conjecture for the generalized Fermat equation

摘录 / EXCERPT: Let $A,B,C,D,n\in\mathbb{Z}_{+}$ with $n\geq3$, and suppose ... Assume that $n>3$ is prime, $n\equiv r\pmod3$ with $1\leq r\leq2$, and $\gcd(A,B,C)=p^k$ for a prime p and...

conjecture | open | MathDB 页面 | 原文线索

V2-001953 | MATHDB-326435

The odd-prime gcd conjecture for the generalized Fermat equation

摘录 / EXCERPT: Let $A,B,C,D,n\in\mathbb{Z}_{+}$ with $n\geq3$, and suppose ... Assume that n is an odd prime and that $\gcd(A,B,C)=p$ for a prime p. Odd-prime gcd conjecture. Then the system ha...

conjecture | open | MathDB 页面 | 原文线索

V2-001954 | MATHDB-326451

The consecutive-integers prime conjecture between consecutive squares

摘录 / EXCERPT: Consecutive-integers prime conjecture. At least one of $a_1,\ldots,a_n$ is prime.

conjecture | open | MathDB 页面 | 原文线索

V2-001955 | MATHDB-326478

The Hardy-Littlewood and Bateman-Horn prime-tuples conjecture

摘录 / EXCERPT: Let $f_1(x),\ldots,f_r(x)\in\mathbb{Z}[x]$ and let $f(x)=f_1(x)\cdots f_r(x)$. Assume that the f_i are distinct and irreducible in $\mathbb{Z}[x]$, have positive leading coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-001956 | MATHDB-326479

The type I exceptional-prime asymptotic conjecture

摘录 / EXCERPT: For $c\in\{-5,-3,-1,1,3,5\}$, let $f_c(k)=k^2+5k+c$, and let $\pi_E^{(\mathrm{I})}(c;x)$ count integers $k\le x$ for which $f_c(k)$ is exceptional of type (I). Put...

conjecture | open | MathDB 页面 | 原文线索

V2-001957 | MATHDB-326480

The type II exceptional-prime asymptotic conjecture

摘录 / EXCERPT: For $c\in\{-5,-3,-1,1,3,5\}$, let $\pi_E^{(\mathrm{II})}(c;x)$ count integers $k\le x$ for which the associated value is exceptional of type (II). Type II exceptional-pr...

conjecture | open | MathDB 页面 | 原文线索

V2-001958 | MATHDB-326486

Rankin's set conjecture for the maximal density of geometric-progression-free sets

摘录 / EXCERPT: Rankin's set conjecture. Rankin's set G^*_3 has the largest possible density among geometric-progression-free sets which have a density, so

conjecture | open | MathDB 页面 | 原文线索

V2-001959 | MATHDB-326495

The half-factorial residue distribution conjecture for primes congruent to 3 modulo 4

摘录 / EXCERPT: Half-factorial residue distribution conjecture. This proportion approaches $\frac{1}{2}$ as $p \to \infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-001960 | MATHDB-326496

The Wilson congruence zero-solution proportion conjecture

摘录 / EXCERPT: Wilson congruence zero-solution proportion conjecture. The proportion of primes for which $$ has no nontrivial solutions is

conjecture | open | MathDB 页面 | 原文线索

V2-001961 | MATHDB-326497

The Wilson congruence solution-count distribution conjecture

摘录 / EXCERPT: Wilson congruence solution-count distribution conjecture. The proportion of primes p for which $$ has exactly N nontrivial solutions is

conjecture | open | MathDB 页面 | 原文线索

V2-001962 | MATHDB-326509

Stopple's weak Newman conjecture for quadratic Dirichlet L-functions

摘录 / EXCERPT: For every fundamental discriminant $D\in\mathbb Z$, let Λ_D be the associated De Bruijn-Newman constant. Weak Newman conjecture. ... where the supremum is taken over all fund...

conjecture | open | MathDB 页面 | 原文线索

V2-001963 | MATHDB-326510

Newman's conjecture for reductions of a fixed polynomial

摘录 / EXCERPT: Fix a square-free polynomial $\mathcal D\in\mathbb Z[T]$. For each prime p, let $D_p\in\mathbb F_p[T]$ be the reduction of $\mathcal D$ modulo p, and let (p,D_p) range over g...

conjecture | open | MathDB 页面 | 原文线索

V2-001964 | MATHDB-326548

Weng's special uniformity conjecture for non-abelian zeta functions

摘录 / EXCERPT: Let X be a curve over $\bF_q$. For rank r, write $\what\zeta_{X,r}(s)=\what Z_{X,r}(q^{-rs})$ for the non-abelian zeta function of rank-r semistable bundles, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-001965 | MATHDB-326552

Uniform distribution conjecture for prime values of a polynomial

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ produce infinitely many primes. A congruence class modulo a positive integer m is m-allowable for $f(x)$ if $(f(r),m)=1$ for every integer r in tha...

conjecture | open | MathDB 页面 | 原文线索

V2-001966 | MATHDB-326553

Generalized Artin density conjecture for polynomial prime values

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ produce infinitely many primes. For an integer g, define ... Here an Artin prime for g is a prime for which g is a primitive root modulo that prime....

conjecture | open | MathDB 页面 | 原文线索

V2-001967 | MATHDB-326554

Moree's density formula for polynomial Artin primes

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ produce infinitely many primes, and let g be a square-free integer such that all but finitely many primes produced by $f(x)$ remain inert in...

conjecture | open | MathDB 页面 | 原文线索

V2-001968 | MATHDB-326559

The squarefreeness conjecture for the polynomials $F_{m,n}$

摘录 / EXCERPT: Squarefreeness conjecture. The polynomial $F_{m,n}$ has no multiple roots.

conjecture | open | MathDB 页面 | 原文线索

V2-001969 | MATHDB-326560

The multiple-factor conjecture for quadrinomials

摘录 / EXCERPT: Let $a\in\mathbb{Q}^{*}$ and let ... where $n>m>k$. Multiple-factor conjecture. If $f_{n,m,k}(a,x)$ has multiple factors, then $a=1$ and ... for some $d\geq 1$ and odd integers...

conjecture | open | MathDB 页面 | 原文线索

V2-001970 | MATHDB-326572

Conjecture on the asymptotic growth of the excess normalized Gauss sums

摘录 / EXCERPT: Asymptotic-growth conjecture. There is a constant $c\in(1,3)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001971 | MATHDB-326578

The fast general linear group conjecture

摘录 / EXCERPT: Let q be a prime number. The group $\mathrm{GL}(n,q)$ acts naturally on $\mathrm{GF}(q)^n$, and is regarded as a subgroup of $\mathrm{Sym}(\mathrm{GF}(q)^n)$. Fast general linear...

conjecture | open | MathDB 页面 | 原文线索

V2-001972 | MATHDB-326590

Benois's trivial-zero formula for arithmetic L-invariants

摘录 / EXCERPT: Let p be an odd prime and let $\rho:G_{\mathbb{Q}}\to \operatorname{GL}(n,\overline{\mathbb{Q}}_p)$ be a continuous Galois representation unramified at all but finitely many prim...

conjecture | open | MathDB 页面 | 原文线索

V2-001973 | MATHDB-326606

The sharp symmetry-rank conjecture for cyclic fundamental groups

摘录 / EXCERPT: Let M^{2m-1} be a closed Riemannian manifold with positive sectional curvature. Let q denote the smallest prime divisor of m, and let an r-torus act effectively and isometr...

conjecture | open | MathDB 页面 | 原文线索

V2-001974 | MATHDB-326609

Conjecture on the periodic discriminant equality for real cubic fields

摘录 / EXCERPT: Let K be a real cubic field and let $r=5/2$. Write Δ_K for the discriminant of K and $\Delta_{K,r}^{\mathcal{P}er}$ for the associated periodic discriminant quantity. Peri...

conjecture | open | MathDB 页面 | 原文线索

V2-001975 | MATHDB-326658

Sárközy-Stewart conjecture on the greatest prime factor of products plus one

摘录 / EXCERPT: Sárközy-Stewart conjecture. For every ϵ satisfying $0<\epsilon<1$, there exist $N(\epsilon)$ and $C(\epsilon)>0$ such that, for every integer $N\geq N(\epsilon)$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-001976 | MATHDB-326694

Agoh's Bernoulli-number characterization of primes

摘录 / EXCERPT: Agoh's conjecture. The integer n is prime if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-001977 | MATHDB-326695

The conjectured minimum of the prime-indexed density constants

摘录 / EXCERPT: Density-minimum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-001978 | MATHDB-326697

Genericity conjecture for the generalized Liouville sequence

摘录 / EXCERPT: Assume $b_k=a_k^2$, with the integers a_k pairwise relatively prime, and define ... Let Σ denote the parameter from the preceding construction, and let β be the probab...

conjecture | open | MathDB 页面 | 原文线索

V2-001979 | MATHDB-326703

The odd-order-free Fitting-height bound for coprime factorisations

摘录 / EXCERPT: Let $G=AB$ be a finite soluble group factorised by its proper subgroups A and B with $\gcd(|A|,|B|)=1$. Here $h(X)$ denotes the Fitting height of a soluble group X, and...

conjecture | open | MathDB 页面 | 原文线索

V2-001980 | MATHDB-326710

Holomorphic extension and zero localization conjecture for the semiclassical zeta function

摘录 / EXCERPT: Holomorphic extension and zero localization conjecture. The semiclassical zeta function $Z_{sc}(s)$ for contact Anosov flows has a holomorphic extension to the whole complex plane...

conjecture | open | MathDB 页面 | 原文线索

V2-001981 | MATHDB-326717

The comparison conjecture for completed algebraic K-groups

摘录 / EXCERPT: Let F be the number field and let S be the set of places consisting of the archimedean places and the places above p, with $\mathscr{O}$ the ring of S-integers and...

conjecture | open | MathDB 页面 | 原文线索

V2-001982 | MATHDB-326721

Finiteness conjecture for circulant complex Hadamard matrices

摘录 / EXCERPT: Let $C_n^{\operatorname{circ}}(\infty)$ denote the set of circulant complex Hadamard matrices of order n, with no fixed bound on the orders of their root-of-unity entries, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-001983 | MATHDB-326744

Unramified zeta integral continuation and functional equation conjecture

摘录 / EXCERPT: Unramified zeta integral conjecture. The zeta integral meromorphically extends to the complex plane and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-001984 | MATHDB-326745

Mean-periodicity conjecture for arithmetic-surface zeta functions

摘录 / EXCERPT: Let $\mathcal{S}\to\operatorname{Spec}(\mathcal{O}_k)$ be a proper, regular model of a smooth projective curve over a number field k. For a finite set of finite Galois extensions...

conjecture | open | MathDB 页面 | 原文线索

V2-001985 | MATHDB-326746

Marshall Hall's conjecture for integral points on Mordell curves

摘录 / EXCERPT: Hall's conjecture. For every $\epsilon>0$, there exists a positive constant C_ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-001986 | MATHDB-326766

The Gauss sphere problem conjecture

摘录 / EXCERPT: Gauss sphere problem conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-001987 | MATHDB-326802

Fuchs-Sanden conjecture for prime and twin-prime circles

摘录 / EXCERPT: Let $\mathcal P$ be a bounded primitive integral Apollonian packing. Define ... Here $N_{\mathcal P}(T)$ denotes the relevant circle-counting function. Fuchs-Sanden conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-001988 | MATHDB-326813

Asymptotic uniformity conjecture for constellations under the gap-cycle recursion

摘录 / EXCERPT: Let ${\mathcal G}(p)$ denote the cycle of gaps at a sieve stage, let s be a constellation in ${\mathcal G}(p)$, and let $\sigma(s)$ be the sum of its gaps. Assume $\sigma(s)<2p$....

conjecture | open | MathDB 页面 | 原文线索

V2-001989 | MATHDB-326814

Cooper's one-term supercongruences for the sporadic sequences s_7 and s_{18}

摘录 / EXCERPT: Let ... and ... with $s_{18}(0)=1$. Cooper's conjecture. For every prime $p\geqslant3$, ... and, for every prime p, ... These congruences are sporadic-sequence analogues of known congrue...

conjecture | open | MathDB 页面 | 原文线索

V2-001990 | MATHDB-326819

Galois structure conjecture for torsion cohomology

摘录 / EXCERPT: Galois structure conjecture. The graded $\mathbb{Z}[\sigma]$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-001991 | MATHDB-326824

The vanishing n-fold Massey product conjecture for absolute Galois groups

摘录 / EXCERPT: Let p be a prime number and $n\geq 3$ an integer. Let F be a field, which contains a primitive p-th root of unity if $\operatorname{char}(F)\not=p$. Let G_F denote the abso...

conjecture | open | MathDB 页面 | 原文线索

V2-001992 | MATHDB-326838

Ryan-Pomerance conjecture on divisibility of maximal divisor heights

摘录 / EXCERPT: Let p be a fixed odd prime and let b be a fixed positive integer. For a positive integer n, define ... where $H(f)$ is the largest absolute value of a coefficient of f. Ryan-...

conjecture | open | MathDB 页面 | 原文线索

V2-001993 | MATHDB-326839

Ryan-Pomerance conjecture on maximal heights for pq^b

摘录 / EXCERPT: Let $p<q$ be odd primes, and define ... where $H(f)$ is the largest absolute value of a coefficient of f. Ryan-Pomerance height conjecture. For every positive integer b, ... and fo...

conjecture | open | MathDB 页面 | 原文线索

V2-001994 | MATHDB-326860

Adelic commutator openness conjecture

摘录 / EXCERPT: Let X be as above, let H be the Mumford-Tate group, and let H^{sc} be the simply connected cover of its derived group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-001995 | MATHDB-326864

Stark's conjecture over the integers for higher-rank Rubin-Stark elements

摘录 / EXCERPT: Stark's conjecture over $\mathbb{Z}$. There is a unique element

conjecture | open | MathDB 页面 | 原文线索

V2-001996 | MATHDB-326865

Twisted-trace conjecture for Stark elements in abelian towers

摘录 / EXCERPT: Twisted-trace conjecture. The following two assertions hold: 1....

conjecture | open | MathDB 页面 | 原文线索

V2-001997 | MATHDB-326872

Armstrong's average-size conjecture for simultaneous core partitions

摘录 / EXCERPT: Let s and t be coprime positive integers. An (s,t)-core is a partition that is simultaneously an s-core and a t-core; its size is the sum of its parts. Armstrong's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-001998 | MATHDB-326884

The local-global conjecture for Apollonian 3-circle packings

摘录 / EXCERPT: Local-global conjecture. Every sufficiently large admissible integer from $\mathcal{P}$ is a curvature. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-001999 | MATHDB-326886

Conjecture for the characteristic-prime case of CPP exponents

摘录 / EXCERPT: Characteristic-prime CPP conjecture. The polynomial $a^{-1}x^d$ is a complete permutation polynomial over $\mathbb{F}_{p^{rk}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002000 | MATHDB-326890

Classification of depth-two zeta-like primitive tuples up to weight q^2

摘录 / EXCERPT: Let q be a prime power, and consider primitive tuples of depth 2 and weight at most q^2. Depth-two classification conjecture. The zeta-like primitive tuples are exactly ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-002001 | MATHDB-326924

Triviality conjecture for the Ekedahl invariants of the p-discrete Heisenberg group

摘录 / EXCERPT: Let p be a prime and let H_p denote the p-discrete Heisenberg group. Its Ekedahl invariants are the invariants $\operatorname{e}_i(H_p)$ associated with the classifying space...

conjecture | open | MathDB 页面 | 原文线索

V2-002002 | MATHDB-326934

The local-global equality for the non-abelian reciprocity filtration

摘录 / EXCERPT: Local-global equality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-002003 | MATHDB-326935

Strong Kurepa's hypothesis for Kurepa's determinants

摘录 / EXCERPT: For each integer $n\ge 7$, let K_n be the Kurepa determinant of order $n-4$. The strong Kurepa's hypothesis. For each integer $n\ge 7$, the Kurepa's determinant K_n is not divi...

conjecture | open | MathDB 页面 | 原文线索

V2-002004 | MATHDB-326936

Infinitude conjecture for Bell-number congruences

摘录 / EXCERPT: Let B_m denote the mth Bell number. Bell-number congruence conjecture. There are infinitely many positive integers n such that ... The paper reports computational examples of s...

conjecture | open | MathDB 页面 | 原文线索

V2-002005 | MATHDB-326946

Effective boundedness conjecture for parametrized Thue inequalities

摘录 / EXCERPT: Let α be an algebraic number of degree $d\geq 3$ over $\mathbf{Q}$, let $K=\mathbf{Q}(\alpha)$, and let $\mathbf{Z}_K^\times$ be the group of units of K. For a unit...

conjecture | open | MathDB 页面 | 原文线索

V2-002006 | MATHDB-326948

The c_2-invariant conjecture for primitive log-divergent graphs

摘录 / EXCERPT: Let G_1 and G_2 be primitive log-divergent graphs with graph periods I_{G_1} and I_{G_2}. For each finite field size q, define the c_2-invariant by ... where X_G is the...

conjecture | open | MathDB 页面 | 原文线索

V2-002007 | MATHDB-326953

The average-size conjecture for n-Selmer groups of elliptic curves

摘录 / EXCERPT: Let n be any positive integer. For an elliptic curve E over ${\mathbb Q}$, let $S_n(E)$ denote its n-Selmer group, and order all elliptic curves by height. Average-size conje...

conjecture | open | MathDB 页面 | 原文线索

V2-002008 | MATHDB-326966

The union-measure form of the Duffin-Schaeffer conjecture

摘录 / EXCERPT: For a non-negative function $\psi:\mathbb{N}\to\mathbb{R}$, let ${\mathcal E}_n(\psi)$ and $W(\psi)$ be as above, and define ... Let λ denote Lebesgue measure on...

conjecture | open | MathDB 页面 | 原文线索

V2-002009 | MATHDB-326967

Beresnevich-Velani's Hausdorff-measure Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $\psi:\mathbb{N}\to\mathbb{R}$ be non-negative, let $W(\psi)$ be the associated limsup set of reduced rational approximation intervals, and let $f:[0,\infty)\to[0,\infty)$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-002010 | MATHDB-326968

The union-measure conjecture for formal Laurent series

摘录 / EXCERPT: Let $\mathbb{F}$ be a finite field, let $\mathbb{L}\subset\mathbb{F}((X^{-1}))$ be the negative-degree series, and let ν_d be the product Haar measure on $\mathbb{L}^d$. For no...

conjecture | open | MathDB 页面 | 原文线索

V2-002011 | MATHDB-326969

The finite-block Duffin-Schaeffer lower-bound conjecture

摘录 / EXCERPT: Let $S_h(\psi)$ and $B_h(\psi)$ denote the quantities defined in the paper for the finite block indexed by h, with $\psi:\mathbb{N}\to\mathbb{R}$ non-negative. Finite-block lower...

conjecture | open | MathDB 页面 | 原文线索

V2-002012 | MATHDB-326970

The scaled finite-block Duffin-Schaeffer lower-bound conjecture

摘录 / EXCERPT: Let $S_h(\psi)$ and $B_h(\psi)$ denote the paper's finite-block quantities for $h\in\mathbb{N}$ and non-negative $\psi:\mathbb{N}\to\mathbb{R}$. Scaled finite-block conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-002013 | MATHDB-326971

The positive-measure large-sum conjecture for rational approximation

摘录 / EXCERPT: Let $d\in\mathbb{N}$, let $\gamma\in(0,1)$, let $\psi:\mathbb{N}\to\mathbb{R}$ be non-negative, and let ${\mathcal E}_n(\psi)$ be the real reduced rational approximation sets. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-002014 | MATHDB-326973

The functional-equation conjecture for normal zeta functions of Heisenberg groups

摘录 / EXCERPT: Let K be a number field, let g be the number of primes of K above a rational prime, and let $(\mathbf{e},\mathbf{f})\in\mathbb{N}^g\times\mathbb{N}^g$ satisfy ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-002015 | MATHDB-327002

Askey-Gasper rational-function coefficient supercongruence

摘录 / EXCERPT: Let ... and write its Taylor expansion as ... This is the Askey-Gasper rational function. Askey-Gasper coefficient supercongruence. For every prime $p\geqslant 5$ and integer...

conjecture | open | MathDB 页面 | 原文线索

V2-002016 | MATHDB-327005

Amdeberhan-Moll divisibility conjecture for binomial coefficients

摘录 / EXCERPT: Let a,b and n be positive integers with $a>b$. For integers x and y, write $x\mid y$ when x divides y. Amdeberhan-Moll's divisibility conjecture. ... The conjecture conce...

conjecture | open | MathDB 页面 | 原文线索

V2-002017 | MATHDB-327014

Minimal Markov basis conjecture for contingency-table margin equations

摘录 / EXCERPT: Let C denote the set of small conditionals, let $J-1$ be the dimension of the underlying lattice, and consider the coefficient matrix of the Diophantine equation governing the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-002018 | MATHDB-327018

The Schenker sums 5-adic lifting conjecture

摘录 / EXCERPT: Let $a_n=\sum_{j=0}^{n}\frac{n!}{j!}n^j$ for $n\in\mathbb{N}_+$, and let v_5 denote the 5-adic valuation. The 5-adic lifting conjecture. Assume that n_k is the unique natural n...

conjecture | open | MathDB 页面 | 原文线索

V2-002019 | MATHDB-327019

The Schenker prime valuation-tree conjecture

摘录 / EXCERPT: Let $a_n=\sum_{j=0}^{n}\frac{n!}{j!}n^j$ for $n\in\mathbb{N}_+$, and call an odd prime p a Schenker prime when $p\mid a_n$ for some positive integer n with $p\nmid n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002020 | MATHDB-327097

The nonexistence conjecture for Diophantine quintuples

摘录 / EXCERPT: A Diophantine quintuple is a set of five distinct positive integers such that the product of any two distinct elements, increased by 1, is a perfect square. Nonexistence conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-002021 | MATHDB-327101

Modularity conjecture for geometric Galois representations

摘录 / EXCERPT: Let F be a number field, let ℓ be a prime, and let $\rho:\operatorname{Gal}(\overline{F}/F)\to\operatorname{GL}_n(\overline{\mathbb{Q}}_\ell)$ be a Galois representation tha...

conjecture | open | MathDB 页面 | 原文线索

V2-002022 | MATHDB-327111

Rodriguez-Villegas-type q-supercongruences modulo the square of a q-integer

摘录 / EXCERPT: Let $p\geqslant 5$ be a prime. Define the q-shifted factorial by $(a;q)_0=1$ and ... for $n\geq 1$, and let $[p]=1+q+\cdots+q^{p-1}$. Let $(\frac{\cdot}{p})$ denote the Legendre sy...

conjecture | open | MathDB 页面 | 原文线索

V2-002023 | MATHDB-327112

The strengthened q-congruences for the second, third and fourth Rodriguez-Villegas cases

摘录 / EXCERPT: Let $p\geqslant 5$ be a prime, and let $[p]=1+q+\cdots+q^{p-1}$. Strengthened q-congruences. The three congruences in Corollary ${\rm}$ hold modulo $[p]^2$; explicitly, ... This is d...

conjecture | open | MathDB 页面 | 原文线索

V2-002024 | MATHDB-327113

The exponent recurrence conjecture for q-supercongruences

摘录 / EXCERPT: Let p be an odd prime, and let m and r be positive integers with $p\nmid m$ and $m\nmid r$. Let $\langle -r/m\rangle_p$ denote the least nonnegative residue of $-r/m$ modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-002025 | MATHDB-327114

The explicit exponent formula for q-supercongruences

摘录 / EXCERPT: Let p be an odd prime, and let m and r be positive integers with $r<m$ and $p\equiv\pm1\pmod m$. Let $f_{p,m,r}$ be the unique integer from the exponent recurrence conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-002026 | MATHDB-327115

The symmetry conjecture for generalized q-Legendre polynomials

摘录 / EXCERPT: Let p be an odd prime, and let m and r be positive integers with $p\nmid m$. Define ... Let $\langle-r/m\rangle_p$ denote the least nonnegative residue of $-r/m$ modulo p, an...

conjecture | open | MathDB 页面 | 原文线索

V2-002027 | MATHDB-327123

Lagarias-Soundararajan's strong XYZ conjecture for ABC solutions

摘录 / EXCERPT: XYZ conjecture (strong form). For every $\epsilon>0$, both of the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-002028 | MATHDB-327124

A weak XYZ conjecture for ABC solutions

摘录 / EXCERPT: Weak XYZ conjecture. For every $\delta>0$, there are only finitely many solutions $(A,B,C)\in\frak X$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-002029 | MATHDB-327146

The rational local p-th power conjecture

摘录 / EXCERPT: Let $a\in\mathbb{Q}^{\times}$. For a prime p, write $\mathbb{Q}_p^{\times p}$ for the subgroup of p-th powers in $\mathbb{Q}_p^{\times}$. The rational local p-th power conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-002030 | MATHDB-327147

The number-field local p-th power conjecture

摘录 / EXCERPT: Let K be a number field and let $\eta\in K^{\times}$. For each rational prime p, let K_v be the completion at a place v above p, and write $\prod_{v\mid p}K_v^{\times p}$...

conjecture | open | MathDB 页面 | 原文线索

V2-002031 | MATHDB-327149

The finiteness conjecture for the p-ramification invariants

摘录 / EXCERPT: Let K be a number field, and for each prime p let ${\mathcal T}_p={\rm Gal}(H^{p\rm ra}_{\!(p)}/\widehat K_{\!(p)})$, where $H^{p\rm ra}_{\!(p)}$ is the maximal Abelian pro-p...

conjecture | open | MathDB 页面 | 原文线索

V2-002032 | MATHDB-327158

Mitra-Paul-Sarkar's conjecture on primes in multiplicative intervals

摘录 / EXCERPT: Let $m,n\in\mathds{N}$, let $\pi(x)$ denote the prime-counting function, and let $\lceil\cdot\rceil$ denote the ceiling function. Mitra-Paul-Sarkar's conjecture. If ... then ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-002033 | MATHDB-327174

Equality of lower and upper accumulation exponents for Pisot numbers

摘录 / EXCERPT: Let ζ be a Pisot number and let α be real. The quantities $\underline{\sigma}(\alpha,\zeta)$ and $\overline{\sigma}(\alpha,\zeta)$ denote the lower and upper rates of...

conjecture | open | MathDB 页面 | 原文线索

V2-002034 | MATHDB-327175

Uniform upper bound for accumulation exponents outside exceptional algebraic pairs

摘录 / EXCERPT: Let α be a nonzero real algebraic number and let ζ be real. Exclude pairs of the form ... where $L,M,A,B,g>0$ are integers. The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-002035 | MATHDB-327185

The index conjecture for length-four minimal zero-sum sequences

摘录 / EXCERPT: Let G be a finite cyclic group of order $|G|$, and let S be a sequence over G. Write ... where g generates G, and define ... A sequence is minimal zero-sum if it has sum zero...

conjecture | solved | MathDB 页面 | 原文线索

V2-002036 | MATHDB-327198

Weight-six generation conjecture for integral modular forms on \(\Gamma_0(p)\)

摘录 / EXCERPT: Let p be the prime under consideration, let $M(\Gamma_0(p),\mathbb{Z})$ be the graded algebra of integral modular forms on $\Gamma_0(p)$, and let ... Here $\widetilde f$ denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-002037 | MATHDB-327199

Generator-weight conjecture for integral modular forms on \(\Gamma_0(p)\)

摘录 / EXCERPT: Let $M(\Gamma_0(p),\mathbb{Z})$ be the graded algebra of integral modular forms on $\Gamma_0(p)$, and let $T\in M_{p-1}(\Gamma_0(p),\mathbb{Z})$ be the T-form. Generator-weight c…

conjecture | open | MathDB 页面 | 原文线索

V2-002038 | MATHDB-327235

ℓ-adic Schanuel conjecture

摘录 / EXCERPT: ℓ-adic Schanuel conjecture. The numbers $\log x_1,\ldots,\log x_n$ are algebraically independent over $\mathbb Q$.

conjecture | open | MathDB 页面 | 原文线索

V2-002039 | MATHDB-327237

The ℓ-adic regulator conjecture

摘录 / EXCERPT: The ℓ-adic regulator conjecture. Define

conjecture | open | MathDB 页面 | 原文线索

V2-002040 | MATHDB-327238

The feeble ℓ-adic regulator conjecture

摘录 / EXCERPT: Feeble ℓ-adic regulator conjecture. There exists an index n_0, depending only on K and ℓ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002041 | MATHDB-327239

A cyclic-generator conjecture for principal ℓ-adic units

摘录 / EXCERPT: Cyclic-generator conjecture. There exists $\omega\in U_{S,1}(K)$ such that $\tau(\omega)=\omega$ and the set

conjecture | open | MathDB 页面 | 原文线索

V2-002042 | MATHDB-327274

The Iwasawa distinguished-polynomial square-freeness and order conjecture

摘录 / EXCERPT: Let k be a totally real number field, let p be a prime, let E be a finite extension of $\mathbb Q_p$, and let $\chi:G_k\to E$ be an even Artin character realizable over E….

conjecture | open | MathDB 页面 | 原文线索

V2-002043 | MATHDB-327297

Congruent skein relations for reformulated colored HOMFLY-PT invariants

摘录 / EXCERPT: Let $\mathcal{L}$ be a link and let p be a prime number. Write $\check{\mathcal{Z}}_p(\mathcal{L};q,t)$ for the reformulated colored HOMFLY-PT invariant associated with the p-r…

conjecture | open | MathDB 页面 | 原文线索

V2-002044 | MATHDB-327315

The Poisson avoidance conjecture for a system of linear forms

摘录 / EXCERPT: Let q be a prime, let $\mathbb F_q$ be the finite field with q elements, let $d=\pi(q-1)$, and let L_n be the linear forms from the paper on $\mathbb F_q^d$. For…

conjecture | open | MathDB 页面 | 原文线索

V2-002045 | MATHDB-327321

Cohomological comparison conjecture for loop spaces of spin groups

摘录 / EXCERPT: Let p be an odd prime, let q be a power of p, and let $\mathbb{F}_q$ be the finite field with q elements. For $n\geq 3$, let $\mathrm{Spin}(n)$ be the spin group, let…

conjecture | open | MathDB 页面 | 原文线索

V2-002046 | MATHDB-327333

Density conjecture for squarefree trinomial discriminant values

摘录 / EXCERPT: Let S be the set of positive integers n for which $n^n+(-1)^n(n-1)^{n-1}$ is squarefree. Squarefree-value density conjecture. The set S has density … correct to that many dec…

conjecture | open | MathDB 页面 | 原文线索

V2-002047 | MATHDB-327334

Poisson distribution conjecture for six-packs of consecutive pth powers

摘录 / EXCERPT: For each prime p, let f_p be the polynomial defined in the paper, and let $\rho(m)$ be the relative density of primes p for which f_p has exactly m roots; equivalently, t…

conjecture | open | MathDB 页面 | 原文线索

V2-002048 | MATHDB-327335

Average-size conjecture for sporadic square divisibility classes

摘录 / EXCERPT: For each prime p, let ${\mathcal C}_p$ be the set defined in the paper that records the relevant sporadic roots or residue classes. Average-size conjecture for ${\mathcal C}_p$….

conjecture | open | MathDB 页面 | 原文线索

V2-002049 | MATHDB-327339

Iwasawa's cyclicity conjecture for eigenspaces of the class group

摘录 / EXCERPT: Let l be an odd prime, let A be the l-torsion part of the class group of $\mathbb Z[\mu_l]$, and let ω be the Teichmüller character of $G(\mathbb Q(\mu_l)/\mathbb Q)$….

conjecture | open | MathDB 页面 | 原文线索

V2-002050 | MATHDB-327365

The divisibility conjecture for truncated northeast triangular posets

摘录 / EXCERPT: Let $\operatorname*{NEtri}(p)$ be the northeast triangular poset, let $s\in\mathbb{N}$, and define the subposet … Let $R_{\operatorname*{NEtri}\nolimits^{\prime}(p)}$ denote birati…

conjecture | open | MathDB 页面 | 原文线索

V2-002051 | MATHDB-327366

The modular Rayuela conjecture

摘录 / EXCERPT: Let F be a totally real number field, and let ell_0 and ell_∞ be prime numbers. Let … be two absolutely irreducible odd Galois representations attached to Hilbert modula…

conjecture | open | MathDB 页面 | 原文线索

V2-002052 | MATHDB-327369

Friedl-Suciu's conjecture on nonprime quasiprojective 3-manifold groups

摘录 / EXCERPT: Friedl-Suciu's conjecture. If $\pi_1(N)$ is a quasiprojective group and N is not prime, then N is the connected sum of spherical 3-manifolds and manifolds which are either di...

conjecture | open | MathDB 页面 | 原文线索

V2-002053 | MATHDB-327374

Local Density Conjecture for Salem numbers

摘录 / EXCERPT: Let ${\rm T}$ be the set of Salem numbers, let $c>1$, and let A_q denote the classes of Salem numbers used in the source. Local Density Conjecture. For every $c>1$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-002054 | MATHDB-327379

Weak Serre conjecture over totally real fields

摘录 / EXCERPT: Weak Serre conjecture. If $\overline{\rho}$ is continuous, irreducible, and totally odd, then it is modular.

conjecture | open | MathDB 页面 | 原文线索

V2-002055 | MATHDB-327380

Strong Serre conjecture on modularity and Serre weights

摘录 / EXCERPT: Strong Serre conjecture. If $\overline{\rho}$ is continuous, irreducible, and totally odd, then it is modular. Moreover, $\overline{\rho}$ is modular of weight σ if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-002056 | MATHDB-327418

Good-characteristic parity conjecture for standard sheaves

摘录 / EXCERPT: Good-characteristic parity conjecture. If p is a good prime for ${\mathbf G}$, then the standard sheaves with coefficients in a field of characteristic p are $*$-parity.

conjecture | open | MathDB 页面 | 原文线索

V2-002057 | MATHDB-327419

Multiplicative independence conjecture for shifted roots of unity

摘录 / EXCERPT: Multiplicative independence conjecture. The algebraic integers

conjecture | open | MathDB 页面 | 原文线索

V2-002058 | MATHDB-327460

Meromorphicity of the $(2,2)$ theta lift for non-isotropic lattices

摘录 / EXCERPT: Non-isotropic $(2,2)$ meromorphicity conjecture. The image of $\frac{i^{m}}{2}\Phi_{L,m,m,0}$ under $\frac{1}{-4\pi^{2}}R_{m}^{(2)}$, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-002059 | MATHDB-327512

Existence conjecture for mod ℓ Galois representations attached to cuspidal representations of GL_n

摘录 / EXCERPT: Let π be a cuspidal representation of $GL_n/\mathbb Q$ as above, let N be its conductor, and let ℓ be an odd prime coprime to N. For each finite place λ of the...

conjecture | open | MathDB 页面 | 原文线索

V2-002060 | MATHDB-327521

Szalay-Ziegler conjecture on S-Diophantine quadruples

摘录 / EXCERPT: Szalay-Ziegler conjecture. No $\{p,q\}$-Diophantine quadruple exists.

conjecture | open | MathDB 页面 | 原文线索

V2-002061 | MATHDB-327562

Sprindžuk's extremality conjecture for non-degenerate manifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be a submanifold of $\mathbb{R}^n$. It is non-degenerate when it does not lie locally in any proper affine subspace, and it is extremal when almost every point of...

conjecture | open | MathDB 页面 | 原文线索

V2-002062 | MATHDB-327563

A. Baker's conjecture for approximation on the Veronese curve

摘录 / EXCERPT: Let $\psi:\mathbb{N}\to\mathbb{R}^+$ be decreasing, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002063 | MATHDB-327564

The generalized Baker-Schmidt problem for Hausdorff dimension

摘录 / EXCERPT: Generalized Baker-Schmidt problem for Hausdorff dimension. If $\mathcal{M}$ is non-degenerate everywhere except possibly on a set of Hausdorff dimension at most D, then

open_problem | open | MathDB 页面 | 原文线索

V2-002064 | MATHDB-327565

The generalized Baker-Schmidt problem for Hausdorff measure

摘录 / EXCERPT: Generalized Baker-Schmidt problem for Hausdorff measure.

conjecture | open | MathDB 页面 | 原文线索

V2-002065 | MATHDB-327623

Vinogradov's main conjecture for the mean value $J_{r,d}(X)$

摘录 / EXCERPT: Let $J_{r,d}(X)$ denote the number of solutions in integers $1\leq x_1,\ldots,x_{2r}\leq X$ to ... where $r,d\geq 1$, and set $D=d(d+1)/2$. Vinogradov's main conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002066 | MATHDB-327625

The unbounded-drop conjecture for integer complexity

摘录 / EXCERPT: For each positive integer n, let $\Vert n\Vert$ be the least number of 1's needed to represent n using addition and multiplication. The quantity $\Vert n-1\Vert-\Vert n\Vert$...

conjecture | open | MathDB 页面 | 原文线索

V2-002067 | MATHDB-327630

Cohen-Martinet conjecture for relative class groups

摘录 / EXCERPT: Let $\Sigma=(H,K_0,\sigma)$ be a situation satisfying the stated character-theoretic assumptions, let ${\mathcal{O}}={\mathcal{O}}(\Sigma)$, and let ${\mathfrak{p}}$ be a prime ide...

conjecture | open | MathDB 页面 | 原文线索

V2-002068 | MATHDB-327631

Cohen-Martinet distribution conjecture with roots of unity

摘录 / EXCERPT: Let p be a prime, let $\Sigma=(H,K_0,\sigma)$ be a situation with $\gcd(p,|H|)=1$ and ${\mathcal{O}}(\Sigma)={\mathbb{Z}}$, and suppose that K_0 contains the p^mth but not th...

conjecture | open | MathDB 页面 | 原文线索

V2-002069 | MATHDB-327672

Conjecture on Polignac numbers in non-trivial arithmetic progressions

摘录 / EXCERPT: Arithmetic-progression conjecture for Polignac numbers. Every non-trivial arithmetic progression contains infinitely many Polignac numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-002070 | MATHDB-327673

Conjecture that Polignac numbers contain arbitrarily long arithmetic progressions

摘录 / EXCERPT: Arithmetic-progression conjecture for Polignac numbers. The Polignac numbers contain infinitely long arithmetic progressions.

conjecture | open | MathDB 页面 | 原文线索

V2-002071 | MATHDB-327693

Stability conjecture for distinguishing newforms at squarefree level

摘录 / EXCERPT: Let $n_0(N,k)$ be the smallest nonnegative integer such that, for any newforms $f,g\in S_k(\Gamma_0(N))$, equality $a_n(f)=a_n(g)$ for all $n\leq n_0(N,k)$ implies $f=g$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002072 | MATHDB-327699

Gorsky-Negut rational Shuffle Conjecture for Hikita polynomials

摘录 / EXCERPT: Gorsky-Negut rational Shuffle Conjecture. For all coprime pairs of positive integers (m,n),

conjecture | open | MathDB 页面 | 原文线索

V2-002073 | MATHDB-327700

Non-coprime rational parking-function Shuffle Conjecture

摘录 / EXCERPT: Non-coprime rational Shuffle Conjecture. For all coprime pairs of positive integers (m,n) and any $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002074 | MATHDB-327703

Kimoto-Wakayama's supercongruence for Apéry-like numbers

摘录 / EXCERPT: Kimoto-Wakayama's conjecture. For every prime $p\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-002075 | MATHDB-327716

Brumer-Kramer paramodularity conjecture for abelian surfaces

摘录 / EXCERPT: Let B be an abelian surface defined over $\mathbb{Q}$ of conductor N such that $\operatorname{End}_{\mathbb{Q}}(B)=\mathbb{Z}$. A Brumer-Kramer paramodularity conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-002076 | MATHDB-327724

Inverse binary Goldbach conjecture for dense subsets

摘录 / EXCERPT: Inverse binary Goldbach conjecture. For N sufficiently large depending on δ, the sumset A_1+A_2 contains a composite number.

conjecture | open | MathDB 页面 | 原文线索

V2-002077 | MATHDB-327725

Sharpness conjecture for the larger sieve in the small-density regime

摘录 / EXCERPT: Larger sieve sharpness conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-002078 | MATHDB-327763

Guo's positive-base Franel-number congruence family

摘录 / EXCERPT: Let the Franel numbers be $f_n=\displaystyle\sum_{j=0}^n {n\choose j}^3$. Guo's conjecture. For every positive integer $n>1$ and every ... ... The authors report verification for n u...

conjecture | open | MathDB 页面 | 原文线索

V2-002079 | MATHDB-327772

Artin's conjecture for the prime base 2

摘录 / EXCERPT: Artin's conjecture. There are infinitely many primes p such that 2 is a generator of the multiplicative group $\mathbb{Z}_p^\times$.

conjecture | open | MathDB 页面 | 原文线索

V2-002080 | MATHDB-327814

Generalized Chowla's conjecture for weighted Möbius correlations

摘录 / EXCERPT: Generalized Chowla's conjecture. For fixed $0\le\kappa_1<\kappa_2$ and $y=\exp((\log x)^\delta)$ with fixed $0<\delta<1$,

conjecture | open | MathDB 页面 | 原文线索

V2-002081 | MATHDB-327852

Harari-Voloch conjecture for integral points on rational hyperbolic curves

摘录 / EXCERPT: Harari-Voloch conjecture. The diagonal map

conjecture | open | MathDB 页面 | 原文线索

V2-002082 | MATHDB-327949

Middle-level node-width conjecture for divisor Hasse diagrams

摘录 / EXCERPT: Let $G^H(n)$ be the Hasse diagram associated with the divisors of n, let $V_l(n)$ denote its vertices at level l, and let $W_v(n)$ denote the node width. Node width at middle l...

conjecture | open | MathDB 页面 | 原文线索

V2-002083 | MATHDB-327950

Coincidence of node and arc widths in divisor Hasse diagrams

摘录 / EXCERPT: Let $G^H(n)$ be the Hasse diagram associated with the divisors of n. For a level l, let $V_l(n)$ and $E^H_l(n)$ denote its vertices and arcs, respectively, and let $W_v(n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002084 | MATHDB-327979

The degree conjecture for topological zeta functions of algebras

摘录 / EXCERPT: Let $\cA$ be a non-associative $\fo$-algebra that is free of rank d as an $\fo$-module, where $\fo$ is the ring of integers in a number field k. Degree conjecture. ... This is pr...

conjecture | open | MathDB 页面 | 原文线索

V2-002085 | MATHDB-327980

The zero-pole conjecture for topological zeta functions of algebras

摘录 / EXCERPT: Let $\cA$ be a non-associative $\fo$-algebra that is free of rank d as an $\fo$-module, where $\fo$ is the ring of integers in a number field k. Zero-pole conjecture. The topol...

conjecture | open | MathDB 页面 | 原文线索

V2-002086 | MATHDB-327981

The root-location conjecture for topological zeta functions of algebras

摘录 / EXCERPT: Let $\cA$ be a non-associative $\fo$-algebra that is free of rank d as an $\fo$-module, where $\fo$ is the ring of integers in a number field k. Root-location conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-002087 | MATHDB-327982

The topological residue conjecture for nilpotent algebras

摘录 / EXCERPT: Let $\cA$ be a Lie algebra or a non-unital associative $\fo$-algebra, free of rank d as an $\fo$-module, where $\fo$ is the ring of integers in a number field k, and suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-002088 | MATHDB-327983

The p-adic residue conjecture for nilpotent algebras

摘录 / EXCERPT: Let $\cA$ be a nilpotent $\fo$-algebra that is free of rank d as an $\fo$-module, where $\fo$ is the ring of integers in a number field k. Let $K\supset k$ be a p-adic field...

conjecture | open | MathDB 页面 | 原文线索

V2-002089 | MATHDB-327988

Restricted-discriminant Hilbert modular twist vanishing conjecture

摘录 / EXCERPT: Restricted-discriminant twist vanishing conjecture. There exist $b_{f,\mathbb Z},C_{f,\mathbb Z}\geq 0$ depending on f such that, as $X\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002090 | MATHDB-327989

Nesterenko's transcendence conjecture for modular values

摘录 / EXCERPT: Let ${\mathfrak H}$ be the upper half-plane, let $\alpha\in{\mathfrak H}$, and let E_2, E_4, and E_6 denote the corresponding Eisenstein series. A point α in...

conjecture | open | MathDB 页面 | 原文线索

V2-002091 | MATHDB-327990

Malle's relative class-group distribution conjecture with roots of unity

摘录 / EXCERPT: Let ℓ be a prime, let $\mathcal{G}$ denote the set of finite abelian ℓ-groups, and suppose that $A\in\mathcal{G}$ has ℓ-rank r. Let K_0 be a number field contain...

conjecture | open | MathDB 页面 | 原文线索

V2-002092 | MATHDB-328001

Equivalent minimal-prime finiteness conjecture for difference algebras

摘录 / EXCERPT: Let k be a σ-field and R a finitely σ-generated k-σ-algebra. A prime ... of R is minimal if every prime ... with $\subset$ satisfies $=$. Minimal-prime fin...

conjecture | open | MathDB 页面 | 原文线索

V2-002093 | MATHDB-328012

Conjecture on exponential sums associated with Mordell curves

摘录 / EXCERPT: Exponential-sum conjecture. If D is not a square, then, uniformly for $D\ll Y^{1+\varepsilon}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002094 | MATHDB-328023

The Beauville p^5-group counting conjecture

摘录 / EXCERPT: Let p be a prime. A Beauville p-group is a Beauville group whose order is a power of p. The Beauville p^5-group counting conjecture. For every $p\geq 5$, the number of Beau...

conjecture | open | MathDB 页面 | 原文线索

V2-002095 | MATHDB-328037

Lang-Silverman conjecture for elliptic curves

摘录 / EXCERPT: Let K be a number field, let E/K be an elliptic curve, and let $P\in E(K)$. Let $L=3(O)$, where O is the origin of E, and write $\widehat{h}(P)=\widehat{h}_{E,L}(P)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-002096 | MATHDB-328042

The modal multiplicity conjecture for sieve patterns

摘录 / EXCERPT: Let p_n denote the nth prime, and let $\mathcal{P}_{p_n}$ denote the corresponding sieve pattern. Modal multiplicity conjecture. The number 6 occurs most frequently among the...

conjecture | open | MathDB 页面 | 原文线索

V2-002097 | MATHDB-328043

The characteristic-function interval inequality for primes

摘录 / EXCERPT: Let $\pi(x)$ be the prime-counting function. For a positive integer c, let $I_k=[(k-1)c,kc)$ and define $\chi_{I_k}=1$ if I_k contains one or more primes, and $\chi_{I_k}=0$ ot...

conjecture | open | MathDB 页面 | 原文线索

V2-002098 | MATHDB-328079

The local-global principle for power maps on the rational numbers

摘录 / EXCERPT: Let $f:\mathbb{Q}\longrightarrow\mathbb{Q}$. For each prime p, define ... where $\mathbb{Z}_{(p)}^\times$ is the group of elements of $\mathbb{Q}$ whose numerator and denominator a...

conjecture | open | MathDB 页面 | 原文线索

V2-002099 | MATHDB-328080

Subbarao-Fabrykowski conjecture on quasi-multiplicative functions

摘录 / EXCERPT: Subbarao-Fabrykowski conjecture. If T_f is infinite, then either $f(n)=0$ for every $n\in\mathbb{N}$ or f is a global power map, meaning $f(n)=n^k$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-002100 | MATHDB-328084

Uniform boundedness of S-units in rational-function images

摘录 / EXCERPT: Let K be a number field, let S be a finite set of places of K containing all Archimedean places, and let $\operatorname{{\mathfrak o}}^*_S$ denote the group of S-units. For...

conjecture | open | MathDB 页面 | 原文线索

V2-002101 | MATHDB-328091

Erdős-Rosenfeld conjecture on divisors near the square root

摘录 / EXCERPT: Erdős-Rosenfeld conjecture. There is an absolute constant K such that, for every c, the number of such divisors is at most K whenever $n>n_0(c)$.

conjecture | open | MathDB 页面 | 原文线索

V2-002102 | MATHDB-328092

Ruzsa's conjecture on divisors in shrinking intervals

摘录 / EXCERPT: Ruzsa's conjecture. There is a constant K_ϵ such that, for every positive integer n, the number of such divisors is at most K_ϵ.

conjecture | open | MathDB 页面 | 原文线索

V2-002103 | MATHDB-328093

Conjecture on lattice points of circles in short horizontal intervals

摘录 / EXCERPT: Circle lattice-point conjecture. There is a constant C_α such that, for every N, the number of these lattice points is at most C_α.

conjecture | open | MathDB 页面 | 原文线索

V2-002104 | MATHDB-328094

Unimodularity conjecture for generalized Ramanujan polynomials

摘录 / EXCERPT: Let k be an integer for which the polynomial $R_k(X;\chi,\psi)$ is defined, and let χ and ψ be nonprincipal real Dirichlet characters. Unimodularity conjecture. The pol...

conjecture | open | MathDB 页面 | 原文线索

V2-002105 | MATHDB-328095

Unimodularity conjecture for diagonal generalized Ramanujan polynomials

摘录 / EXCERPT: Let k be an integer for which the polynomial $S_k(X;\chi,\chi)$ is defined, and let χ be a nonprincipal real Dirichlet character. Unimodularity conjecture. All nonzero roots...

conjecture | open | MathDB 页面 | 原文线索

V2-002106 | MATHDB-328114

Divisibility conjecture for Taylor coefficients at CM points

摘录 / EXCERPT: Let f be a holomorphic modular form of weight k with integer Fourier coefficients, and let τ be a CM point in $\mathbb{Q}(\sqrt{-d})$. Let p be a prime satisfying ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-002107 | MATHDB-328118

The global equivariant epsilon constant conjecture

摘录 / EXCERPT: Let F/E be a Galois extension with Galois group $Gamma$, and let $T\Omega^\mathrm{loc}(F/E,1)$ be the locally defined element of $K_0(\mathbb Z[\Gamma],\mathbb R)$. Global equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-002108 | MATHDB-328130

Cyclic sieving conjecture for affine bicolor dichotomy patterns

摘录 / EXCERPT: Cyclic sieving conjecture. If k is equal to 1, 2, or a power of an odd prime, then $|Q_i(-1)|$ counts the number of self-complementary dichotomies with automorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-002109 | MATHDB-328138

Legendre-symbol construction conjecture for optimal restricted isometry matrices

摘录 / EXCERPT: Legendre-symbol RIP conjecture. There exists a universal constant C such that for every $\delta>0$, there exist $N_0>0$ and $P_0(N)=O(2^{\operatorname{poly}(N)})$ such that, when...

conjecture | open | MathDB 页面 | 原文线索

V2-002110 | MATHDB-328144

Positive and negative approximability conjectures for random regular graphs

摘录 / EXCERPT: For even $d\ge 4$, let $\mathcal{G}_{n,d}$ be the random model of d-regular graphs. For a contour $C^+_{\epsilon,\delta}$ and formal coefficient functions $\mathcal{P}_i(u)$, def...

conjecture | open | MathDB 页面 | 原文线索

V2-002111 | MATHDB-328150

Darmon's conjecture on cyclotomic units and class numbers

摘录 / EXCERPT: Let F be the field and n the integer in the source, write $n_+=\prod_{i=1}^{\nu_+}\ell_i$, let ν_- be the complementary prime-count parameter, let h_n be the n-class nu...

conjecture | open | MathDB 页面 | 原文线索

V2-002112 | MATHDB-328153

Conjecture on a prime among every n consecutive integers between consecutive squares

摘录 / EXCERPT: The consecutive-integers prime conjecture. Among the n integers $a_1,a_2,\ldots,a_n$, there is at least one prime number.

conjecture | open | MathDB 页面 | 原文线索

V2-002113 | MATHDB-328158

Sano-Mazur-Rubin higher-norm conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let $K/L/k$ be a tower of finite extensions of global fields with K/k abelian, let S and T satisfy the paper's standing conditions, and let $V\subset V'\subset S$ be proper s...

conjecture | open | MathDB 页面 | 原文线索

V2-002114 | MATHDB-328178

The prime linear-representation conjecture

摘录 / EXCERPT: Let a and b be positive integers with $(a,b)=1$. The prime linear-representation conjecture. When $n\geq(a-1)(b-1)$, there always exists a pair of primes (x,y) such that ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-002115 | MATHDB-328179

The prime-plus-proper-figurate-prime conjecture

摘录 / EXCERPT: A proper figurate prime is a figurate prime that is not a prime. The prime-plus-proper-figurate-prime conjecture. Every positive integer $n>5$ can be expressed as the sum of a prim...

conjecture | open | MathDB 页面 | 原文线索

V2-002116 | MATHDB-328180

The cube-full Catalan-type equation conjecture

摘录 / EXCERPT: For positive integers A and B, call AB cube-full when every prime divisor of AB occurs to exponent at least 3. The cube-full Catalan-type equation conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002117 | MATHDB-328181

The prime-gap cube-full equation conjecture

摘录 / EXCERPT: For positive integers A and B, call AB cube-full when every prime divisor of AB occurs to exponent at least 3. The prime-gap cube-full equation conjecture. There are infi...

conjecture | open | MathDB 页面 | 原文线索

V2-002118 | MATHDB-328195

Average prime k-tuple counting conjecture

摘录 / EXCERPT: Let $\mathcal{H}_k$ be an admissible k-tuple and let $\overline{J_{(k)}(\mathsf{x})}$ denote the average k-tuple analog of Riemann's counting function. Write μ for the Möbi...

conjecture | open | MathDB 页面 | 原文线索

V2-002119 | MATHDB-328196

Prime double Chebyshev-function conjecture

摘录 / EXCERPT: Let $\overline{\psi_{(2)}(\mathsf{x})}$ be the average prime double Chebyshev function defined from the singular series and the corresponding logarithmic integral, and let μ de...

conjecture | open | MathDB 页面 | 原文线索

V2-002120 | MATHDB-328197

Euler-series representation conjecture for the k-tuple zeta function

摘录 / EXCERPT: Let $\mathfrak{N}_+^k$ be the pair-wise coprime k-lattice, let $\mathfrak{r}_k$ be a ray in this lattice, and let $n_{(k)}$ be the geometric mean of a point...

conjecture | open | MathDB 页面 | 原文线索

V2-002121 | MATHDB-328209

The conjecture on cyclic branched covers of prime satellite knots

摘录 / EXCERPT: Satellite branched-cover conjecture. For every prime satellite knot K, the manifold $cSigma_n(K)$ is excellent for all sufficiently large integers n:

conjecture | open | MathDB 页面 | 原文线索

V2-002122 | MATHDB-328214

Sun's characterization conjecture for primitive covering numbers

摘录 / EXCERPT: Let L be a primitive covering number, meaning a positive integer that is a covering number but has no proper divisor that is a covering number. Write its prime factorization as .....

conjecture | open | MathDB 页面 | 原文线索

V2-002123 | MATHDB-328221

Friedlander-Granville-Montgomery conjecture on primes in arithmetic progressions

摘录 / EXCERPT: Let x be real, let $q<x$ be a positive integer, let a satisfy $(a,q)=1$, and let $\pi(x;q,a)$ count primes $p\leq x$ with $p\equiv a\pmod q$. Let $\varphi(q)$ denote Euler's to...

conjecture | open | MathDB 页面 | 原文线索

V2-002124 | MATHDB-328225

Rohrlich's conjecture on algebraic products of gamma values

摘录 / EXCERPT: Let $a_1, a_2, \ldots, a_r$ be rational numbers that are not in $\{0, -1, -2, \ldots\}$. Let D be a common denominator of the a_i's. Rohrlich's conjecture. The product ... is an...

conjecture | open | MathDB 页面 | 原文线索

V2-002125 | MATHDB-328252

Lang's uniformity conjecture for rational points on curves

摘录 / EXCERPT: Soient $d\geq 2$ et $D\geq 1$ deux entiers naturels. Pour une courbe X/K de genre d, notons J sa variété jacobienne et $\operatorname{rang}(J(K))$ le rang du groupe abélien...

conjecture | open | MathDB 页面 | 原文线索

V2-002126 | MATHDB-328253

Buzzard-Kilford boundary eigencurve conjecture for ramified nebentypus slopes

摘录 / EXCERPT: Let p be an odd prime, let N be coprime to p, and let ψ_0 be a character of $(\mathbb{Z}/p\mathbb{Z})^\times$ with $\psi_0(-1)=-1$. For $m\geq 2$, let ψ be a charac...

conjecture | open | MathDB 页面 | 原文线索

V2-002127 | MATHDB-328277

The imprimitive Iwasawa-Greenberg main conjecture

摘录 / EXCERPT: Let f, L, $\mathcal O$, $\Lambda_{\mathcal O}$, and the associated Selmer-theoretic objects be as in the preceding sections. For a finite set of primes Σ containing p,...

conjecture | open | MathDB 页面 | 原文线索

V2-002128 | MATHDB-328308

Existence of a nonzero Kurihara-Kolyvagin derivative

摘录 / EXCERPT: Let $E/{\bf Q}$ be the elliptic curve under consideration, let p be the fixed prime, and let $N\geq 1$. Let ${\cal P}_{1}^{(N)}$ be the set of primes ℓ for which...

conjecture | open | MathDB 页面 | 原文线索

V2-002129 | MATHDB-328309

The delta-minimal computation conjecture for Selmer groups

摘录 / EXCERPT: Let $E/{\bf Q}$ be the elliptic curve under consideration and let p be the fixed prime. For $m\in{\cal N}_{1}^{(1)}$, regard $\tilde{\delta}_{m}$ as an element of ${\bf F}_{p}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-002130 | MATHDB-328311

Zhao's Bernoulli-number nondivisibility conjecture

摘录 / EXCERPT: Let B_m denote the mth Bernoulli number, and let $h\in\Q[x_3,x_5,\ldots,x_{2n+1}]$ be a nonzero homogeneous polynomial with $\deg(x_{2k+1})=2k+1$. Zhao's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002131 | MATHDB-328313

Monotonicity conjecture for the proportion of dominant integer polynomials

摘录 / EXCERPT: Monotonicity conjecture. For every integer $n\ge 2$, $P_n(H)$ is an increasing function of H. If true, this would imply that $\lim_{H\to\infty}P_n(H)$ exists for $n\ge 3$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-002132 | MATHDB-328324

Asymptotic limit conjecture for Beiter constants

摘录 / EXCERPT: Asymptotic limit conjecture for Beiter constants. There exists a limit

conjecture | open | MathDB 页面 | 原文线索

V2-002133 | MATHDB-328337

Mishou's joint strong universality conjecture for Lerch zeta-functions

摘录 / EXCERPT: Let α be a transcendental real number with $0<\alpha<1$, and let $\lambda_1,\ldots,\lambda_r$ be distinct real numbers satisfying $0\leq\lambda_j<1$. Mishou's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-002134 | MATHDB-328355

Modified maximum subset-divisor conjecture

摘录 / EXCERPT: Let A be a finite set of n positive integers, let $d_k(A)$ be the number of k-subsets B of A satisfying $\sum B\mid\sum A$, and let $d(k,n)$ be the maximum of $d_k(A)$ ov...

conjecture | open | MathDB 页面 | 原文线索

V2-002135 | MATHDB-328358

Squarefree-value residue class conjecture

摘录 / EXCERPT: Squarefree-value residue class conjecture. For all but finitely many primes p, the set $\mathcal S(f)$ contains infinitely many elements from every nonzero residue class modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-002136 | MATHDB-328359

Quadratic-twist parity conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbf Q$ with conductor N_E, and let d be a squarefree integer coprime to $2\cdot N_E$. Write E_d for the quadratic twist of E by d,...

conjecture | open | MathDB 页面 | 原文线索

V2-002137 | MATHDB-328364

Armstrong's average-size conjecture for (a,b)-core partitions

摘录 / EXCERPT: Let a and b be positive integers, and let an (a,b)-core partition be a partition with no hook length divisible by either a or b. For a partition λ, write...

conjecture | open | MathDB 页面 | 原文线索

V2-002138 | MATHDB-328405

The logarithm-rooted-tree degree asymptotic

摘录 / EXCERPT: Logarithm-rooted-tree degree conjecture. The degree of the rooted tree associated with a prime p is asymptotic to $\ln(p)$. This is proposed as a natural measure on the linearize...

conjecture | open | MathDB 页面 | 原文线索

V2-002139 | MATHDB-328410

Okounkov's derivation conjecture for q-multiple zeta values

摘录 / EXCERPT: Let $\operatorname{\mathsf{qMZV}}$ denote the algebra of q-multiple zeta values, containing the ring of quasi-modular forms $\widetilde{M}_{\mathbb{Q}}=\mathbb{Q}[G_2,G_4,G_6]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-002140 | MATHDB-328418

Thakur's conjecture on degrees of Wieferich primes

摘录 / EXCERPT: Let $A=\mathbb{F}_q[T]$ have characteristic p, and let a Wieferich prime in A mean a prime satisfying the Wieferich condition discussed in the paper. Thakur's conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-002141 | MATHDB-328431

Weak Lang-Vojta conjecture for integral points on log-general-type surfaces

摘录 / EXCERPT: Let X be a quasi-projective surface of log-general type, defined over a number field K, and let $\mathcal O_S$ be the ring of S-integers for a finite set of places of K con...

conjecture | open | MathDB 页面 | 原文线索

V2-002142 | MATHDB-328434

The extremal-ray conjecture for cyclic divisors on $\overline{M}_{0,n}$

摘录 / EXCERPT: Let $m\geq 5$ be prime, let $m\mid n$, and let $d_1=\cdots=d_n$. For the associated $\mathbb{Z}_m$-tuple, let $\mathfrak{D}(\mathbb{Z}_m,E_m;\vec d)$ be the divisor defined in equa...

conjecture | open | MathDB 页面 | 原文线索

V2-002143 | MATHDB-328447

Prime-graph equality conjecture for intransitive subgroups of alternating groups

摘录 / EXCERPT: Let $G=A_n$ and $H=(S_k\times S_{n-k})\cap G$, where $1<k<n$ and $p\leqslant k$ for every prime number $p\leqslant n$. Prime-graph equality conjecture. If $n\geqslant 12$, then ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-002144 | MATHDB-328448

Distinct-prime Goldbach variation below the largest prime

摘录 / EXCERPT: Let $n\geqslant 12$ be an even integer, and let p be the largest prime less than n. Distinct-prime Goldbach variation. There exist distinct primes r and s such that ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-002145 | MATHDB-328468

The HWM convergent calculation conjecture for Champernowne's constant

摘录 / EXCERPT: HWM convergent calculation conjecture. The HWM convergent calculates C_b correctly until the integer $T_b(X+1)-2$ is reached, and calculates that integer as $T_b(X+1)-1$ instead.

conjecture | open | MathDB 页面 | 原文线索

V2-002146 | MATHDB-328469

The NCD formula conjecture for HWM convergents of Champernowne's constant

摘录 / EXCERPT: NCD formula conjecture. The NCD of the HWM convergent is

conjecture | open | MathDB 页面 | 原文线索

V2-002147 | MATHDB-328470

The even HWM coefficient-number conjecture

摘录 / EXCERPT: Even HWM coefficient-number conjecture. The coefficient numbers of the HWM coefficients are even.

conjecture | open | MathDB 页面 | 原文线索

V2-002148 | MATHDB-328471

The HWM convergent error formula conjecture

摘录 / EXCERPT: HWM convergent error formula conjecture. The error of the HWM convergent is

conjecture | open | MathDB 页面 | 原文线索

V2-002149 | MATHDB-328472

The HWM coefficient-length formula conjecture

摘录 / EXCERPT: HWM coefficient-length conjecture. Its length is

conjecture | open | MathDB 页面 | 原文线索

V2-002150 | MATHDB-328477

The square-root interval conjecture for the prime number theorem

摘录 / EXCERPT: Let p_k denote the kth prime, let $s_k:=\{p_k^2,\dots,p_{k+1}^2-1\}$, and let π_k be the number of primes in s_k. For x and an interval length $y=y(x)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-002151 | MATHDB-328482

A generalized q-supercongruence with shifted summation

摘录 / EXCERPT: Let p be an odd prime, let m,r be positive integers with $p\nmid m$, and let $s\leqslant p-1$ be nonnegative. Write $\langle a\rangle_p$ for the least nonnegative residue of...

conjecture | open | MathDB 页面 | 原文线索

V2-002152 | MATHDB-328507

Abhyankar's affine arithmetical conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let A be a finite group. Abhyankar's affine arithmetical conjecture. There is a surjection...

conjecture | open | MathDB 页面 | 原文线索

V2-002153 | MATHDB-328508

Abhyankar's total arithmetical conjecture

摘录 / EXCERPT: Let $\pi_A(\mathbb A^1_{\mathbb F_p,\infty})$ denote the set of finite groups that occur as Galois groups of finite Galois extensions of the rational function field...

conjecture | open | MathDB 页面 | 原文线索

V2-002154 | MATHDB-328531

Conjecture on the dependence of the order-growth exponent on degree

摘录 / EXCERPT: Let $K/{\mathbb Q}$ be a number field of degree d. Let $N_d(B)$ denote the counting function from Theorem, and let α_K be the associated exponent. Degree-dependence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-002155 | MATHDB-328577

Finiteness conjecture for strong eigenforms at bounded weight

摘录 / EXCERPT: Fixed-weight finiteness conjecture. The statement $\mathbf{Fin}_m$ holds for every m.

conjecture | open | MathDB 页面 | 原文线索

V2-002156 | MATHDB-328578

Finiteness conjecture for reductions of Galois-representation characters

摘录 / EXCERPT: Galois-character finiteness conjecture. The statement $\mathbf{Rep}_m$ holds for every m.

conjecture | open | MathDB 页面 | 原文线索

V2-002157 | MATHDB-328588

Conjecture on the convergence of the reciprocal extremal-prime series

摘录 / EXCERPT: Let $\mathbb E=(e_k)_{k\geq 1}$ be the sequence of extremal prime numbers, listed in increasing order. Reciprocal-series conjecture. The series ... is convergent. Numerical data give...

conjecture | open | MathDB 页面 | 原文线索

V2-002158 | MATHDB-328589

Conjecture on the positive growth exponent of the extremal-prime counting function

摘录 / EXCERPT: Positive-exponent conjecture. There exists an infimum

conjecture | open | MathDB 页面 | 原文线索

V2-002159 | MATHDB-328590

Conjecture on consecutive extremal-prime ratios

摘录 / EXCERPT: Let $\mathbb E=(e_k)_{k\geq 1}$ be the sequence of extremal prime numbers, listed in increasing order. Consecutive-ratio conjecture. In this notation, ... The conjecture is supported...

conjecture | open | MathDB 页面 | 原文线索

V2-002160 | MATHDB-328592

Rational independence of the eigenvalues for odd polygons

摘录 / EXCERPT: Let n be odd, write $n=2m+1$, and let $bbda_0=0,bbda_1,bbda_{n-1}$ be the eigenvalues of the circulant matrix governing the linearized system, with $bbda_j=-bbda_{n-j}$ and, for...

conjecture | open | MathDB 页面 | 原文线索

V2-002161 | MATHDB-328598

Residue criterion for the primitive height zeta function

摘录 / EXCERPT: In the same quotient setting, let $\fie GK$ denote the set of isomorphism classes of large G-fields over K, and let $\delta_{F,m}$ be the residue invariant associated with F...

conjecture | open | MathDB 页面 | 原文线索

V2-002162 | MATHDB-328607

Truncated hook-length and restricted colored-partition linear-term conjecture

摘录 / EXCERPT: Truncated hook-length conjecture. For every n and fixed m, the linear terms in b of $H_{n,m}(b)$ and the coefficient of q^n in $C_{1-b,m}^{\prime}(q)$ are equal. The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-002163 | MATHDB-328626

Unit-circle roots conjecture for the Goldbach polynomials

摘录 / EXCERPT: For each positive integer N, let F_N be the polynomial defined from the indicator function of the odd primes by ... Let φ denote Euler's totient function. Unit-circle roo...

conjecture | open | MathDB 页面 | 原文线索

V2-002164 | MATHDB-328647

Optimality of the inhomogeneous rational-approximation constant

摘录 / EXCERPT: Let a,b,r,s be integers as in the (a,b,r,s)-approximation setting, and let θ denote the constant on the right-hand side of the relevant approximation inequality. The con...

conjecture | open | MathDB 页面 | 原文线索

V2-002165 | MATHDB-328648

Full dimension conjecture for uniformly approximable non-Bad numbers

摘录 / EXCERPT: Let a,b,r,s be the fixed parameters of the (a,b,r,s)-approximation setting, and let Bad denote the set of badly approximable real numbers. Consider the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-002166 | MATHDB-328663

Bhargava's ramification-density conjecture for S_n-number fields

摘录 / EXCERPT: Bhargava's ramification-density conjecture. For every prime p and positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-002167 | MATHDB-328664

Bhargava's Chebotarev splitting conjecture for S_n-number fields

摘录 / EXCERPT: Bhargava's Chebotarev splitting conjecture. For every prime p, positive integer n, and $\mu\in T_n^{\ast}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002168 | MATHDB-328690

The asymptotic existence conjecture for modular Hadamard matrices

摘录 / EXCERPT: Let p be an odd prime. An m-modular Hadamard matrix of size n, denoted $MH(n,m)$, is an $n\times n$ matrix with entries ± 1 satisfying $HH^\top\equiv nI\pmod m$. A quadra...

conjecture | open | MathDB 页面 | 原文线索

V2-002169 | MATHDB-328701

Jang-Lee prime-knot conjecture for Whitehead doubles

摘录 / EXCERPT: Let K be a prime knot, let $c(K)$ denote its crossing number, and let $W(K)$ be its Whitehead double. Jang-Lee's conjecture. The crossing number of any prime knot coincides with...

conjecture | open | MathDB 页面 | 原文线索

V2-002170 | MATHDB-328727

Davenport constant equality for multiplicative semigroups of polynomial quotient rings

摘录 / EXCERPT: Let $p>2$ be a prime, let $f(x)\in \mathbb{F}_p[x]$ be a non-constant polynomial, and let $\mathcal{S}_{f(x)}^p$ denote the multiplicative semigroup of the quotient ring...

conjecture | open | MathDB 页面 | 原文线索

V2-002171 | MATHDB-328731

The spectrum conjecture for the first approximation constant of powers

摘录 / EXCERPT: Let k be a positive integer. For a real number $zeta$ not algebraic of degree at most k, let $lambda_{k,1}(zeta)$ denote its first approximation constant for the vector...

conjecture | open | MathDB 页面 | 原文线索

V2-002172 | MATHDB-328732

Weak spectrum conjecture for higher approximation constants

摘录 / EXCERPT: Let k be a positive integer and $1\leq j\leq k+1$. For a real number ζ not algebraic of degree at most k, let $\lambda_{k,j}(\zeta)$ denote the j-th approximation const...

conjecture | open | MathDB 页面 | 原文线索

V2-002173 | MATHDB-328733

Strong spectrum conjecture for approximation constants in the Cantor set

摘录 / EXCERPT: Let k be a positive integer and $1\leq j\leq k+1$. For an irrational number ζ in the Cantor middle-third set, let $\lambda_{k,j}(\zeta)$ denote the j-th approximation con...

conjecture | open | MathDB 页面 | 原文线索

V2-002174 | MATHDB-328735

Strong conjecture on the simultaneous Diophantine approximation exponent of Cantor-type numbers

摘录 / EXCERPT: Let $k\geq 2$ and $b\geq 2$ be integers, and let $\rho>0$. Let $(a_n)_{n\geq 1}$ be a strictly increasing sequence of positive integers satisfying the property in ... . The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-002175 | MATHDB-328740

Knuth-Stolarsky conjecture on binary weight and addition-subtraction length

摘录 / EXCERPT: For a positive integer n, let $\nu_2(n)$ denote the number of 1's in its binary expansion, and let $s(n)$ denote the addition-subtraction chain length used in the paper. KnuthS...

conjecture | open | MathDB 页面 | 原文线索

V2-002176 | MATHDB-328741

Conjecture on exact lower-bound functions for addition-chain defects

摘录 / EXCERPT: Let $f^\ell(k)$ and $f^\ell_{\mathrm{st}}(k)$ be the lower-bound functions for ordinary and stable addition-chain defects, and let $f^*(k)$ and $f^*_{\mathrm{st}}(k)$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-002177 | MATHDB-328761

Ruzsa's conjecture on nondegenerate induced cycles in S-unit graphs

摘录 / EXCERPT: Let S be a finite nonempty set of primes. Let $\mathcal{G}$ be the graph with vertex set $\mathbb{Q}$ in which two rationals are adjacent when their difference is an S-unit. A...

conjecture | open | MathDB 页面 | 原文线索

V2-002178 | MATHDB-328762

The conjecture on nondegenerate induced cycles for sets with at least two primes

摘录 / EXCERPT: Let S be a finite set of primes with $\#S\geq2$, and let $\mathcal{G}$ be its S-unit graph. A cycle is nondegenerate if its edge differences have no proper nonempty zero subsum...

conjecture | open | MathDB 页面 | 原文线索

V2-002179 | MATHDB-328765

Dobbertin-Helleseth-Kumar-Martinsen conjecture on a three-valued family of Weil sums

摘录 / EXCERPT: Let F be a finite field of order $q=3^n$, where n is odd and $n>1$, and let $d=3^r+2$ satisfy ... For $a\in F$, define the Weil sum ... where $\psi\colon F\to\mathbb C$ is the cano...

conjecture | open | MathDB 页面 | 原文线索

V2-002180 | MATHDB-328783

de Mathan-Teullié's Mixed Littlewood conjecture

摘录 / EXCERPT: de Mathan-Teullié's Mixed Littlewood conjecture. For every real number α and every pseudo-absolute sequence $\mathcal{D}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002181 | MATHDB-328801

Rhoades' finiteness conjecture for modular-form corrections of a mock theta function

摘录 / EXCERPT: For fixed integers a,b,A,B, let $f_{a,b,A,B,h,k}$ be the modular forms used to cancel the singularities of $g_2(\zeta_b^a q^A;q^B)$ as h/k ranges over $\mathbb{Q}$. Rhoades' fi...

conjecture | open | MathDB 页面 | 原文线索

V2-002182 | MATHDB-328834

The invariance conjecture for the degree of topological zeta functions

摘录 / EXCERPT: Invariance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-002183 | MATHDB-328842

The prime power conjecture for linear perfect codes in A_n

摘录 / EXCERPT: Prime power conjecture. There exists a linear 1-perfect code in (A_n,d), equivalently in $(\mathbb{Z}^n,d_a)$, if and only if n is a prime power.

conjecture | open | MathDB 页面 | 原文线索

V2-002184 | MATHDB-328860

Bergström-Faber-van der Geer trace conjecture for genus-three Siegel forms

摘录 / EXCERPT: Let $\lambda=(a,b,c)$ with $a\geq b\geq c\geq0$. Let ${\mathbb V}_{\lambda}$ be the corresponding local system on ${\mathcal A}_3$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002185 | MATHDB-328878

The algebraic-curve conjecture for functions over finite fields

摘录 / EXCERPT: Let p be prime, let $f:{\rm GF}(p)\to{\rm GF}(p)$, and let $I(f)$ denote the minimum defined by ... Let ϵ and t be as in the stated hypothesis. The algebraic-curve conje...

conjecture | open | MathDB 页面 | 原文线索

V2-002186 | MATHDB-328879

The cylinder conjecture in affine 3-space

摘录 / EXCERPT: Let p be prime, let U be a set of p^2 points in the affine space ${\rm AG}(3,p)$, and let N be the set of directions not determined by pairs of points of U. The cylinder...

conjecture | open | MathDB 页面 | 原文线索

V2-002187 | MATHDB-328880

The strong cylinder conjecture in affine 3-space

摘录 / EXCERPT: Let p be prime and let U be a set of p^2 points in the affine space ${\rm AG}(3,p)$. Strong cylinder conjecture. If every plane meets U in a number of points divisible by...

conjecture | open | MathDB 页面 | 原文线索

V2-002188 | MATHDB-328881

Hida's dimension conjecture for ordinary completed cohomology

摘录 / EXCERPT: Let F be the number field, p the fixed prime, U a product subgroup as above, S a finite set of finite places containing those dividing p, and $Lambda$ the ordinary weight...

conjecture | open | MathDB 页面 | 原文线索

V2-002189 | MATHDB-328884

Leopoldt's abelian conjecture for totally real fields

摘录 / EXCERPT: Let K be a totally real field and let p be a prime. Let Δ be the Galois group of the maximal abelian pro-p extension of K unramified outside p. Leopoldt's abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-002190 | MATHDB-328885

Non-abelian Leopoldt conjecture for ordinary completed cohomology

摘录 / EXCERPT: Let F be a number field, p a prime, n a positive integer, $U\subset\operatorname{GL}_n({\mathbb A}_F^\infty)$ a good subgroup, and...

conjecture | open | MathDB 页面 | 原文线索

V2-002191 | MATHDB-328889

The inverse sieve conjecture in rough form

摘录 / EXCERPT: Let $X\subset [N]$ be a subset. Suppose that, for every prime p, the reduction $X\pmod p$ occupies at most $0.9p$ residue classes. Inverse sieve conjecture, rough form. Either...

conjecture | open | MathDB 页面 | 原文线索

V2-002192 | MATHDB-328891

The inverse large sieve conjecture

摘录 / EXCERPT: Let $X\subset\mathbb{Z}$ be a subset and let $\epsilon>0$ be real. Assume that, for every prime $p\geq\epsilon^{-1}$, the reduction $X\pmod p$ occupies at most $0.99p$ residue clas...

conjecture | open | MathDB 页面 | 原文线索

V2-002193 | MATHDB-328892

The finitary inverse large sieve conjecture

摘录 / EXCERPT: Let $X\subset [N]$ be a subset and let $\epsilon>0$ be real. Assume that, for every prime $p\geq\epsilon^{-1}$, the reduction $X\pmod p$ occupies at most $0.99p$ residue classes. F...

conjecture | open | MathDB 页面 | 原文线索

V2-002194 | MATHDB-328903

The square-free values density conjecture for polynomials at primes

摘录 / EXCERPT: Square-free values density conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002195 | MATHDB-328920

The four-solution conjecture for a parametrized exponential Diophantine equation

摘录 / EXCERPT: The four-solution conjecture. This equation has exactly four solutions in integers (x,n), with

conjecture | open | MathDB 页面 | 原文线索

V2-002196 | MATHDB-328921

The second four-solution conjecture for a parametrized exponential Diophantine equation

摘录 / EXCERPT: The second four-solution conjecture. This equation has exactly four solutions in integers (x,n), with

conjecture | open | MathDB 页面 | 原文线索

V2-002197 | MATHDB-328922

The three-solution conjecture for $x^2+k^n=B$

摘录 / EXCERPT: Three-solution conjecture. For any such k and B, the equation has at most 3 solutions in non-negative integers (x,n). The conjecture is motivated by numerical experiments f...

conjecture | open | MathDB 页面 | 原文线索

V2-002198 | MATHDB-328923

Pasol-Stevens conjecture on the evil Eisenstein eigensymbol

摘录 / EXCERPT: Let p and ℓ be primes, let ... where ... Let $\Phi\in\operatorname{Symb}^+_{\Gamma_0(p\ell)}({\mathcal D}_0)$ be an overconvergent eigensymbol with the same Hecke eigenvalues as...

conjecture | open | MathDB 页面 | 原文线索

V2-002199 | MATHDB-328940

Arbitrarily long consecutive sequences of attracted numbers for augmented generalized happy functions

摘录 / EXCERPT: Consecutive-sequence conjecture. For each $u\in U_{[c,b]}$, there exist arbitrarily long finite sequences of consecutive u-attracted numbers for $S_{[c,b]}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002200 | MATHDB-328957

Bloch-Kato's p-part of the Tamagawa number conjecture

摘录 / EXCERPT: Tamagawa number conjecture. For every finite set of primes $S\supset\{p\}$, there exists a one-dimensional $\mathbb Q$-vector space $\Delta_S(M/\mathbb Q)$, canonical period isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-002201 | MATHDB-328959

Microscopic zeta-zero process conjecture for the sine-kernel limit

摘录 / EXCERPT: Microscopic zeta-zero conjecture. Assuming the Riemann hypothesis, for every real-valued function f satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-002202 | MATHDB-328960

Microscopic zeta-function process convergence to the limiting characteristic function

摘录 / EXCERPT: Microscopic zeta-function convergence conjecture. In law, uniformly for z on every compact subset of $\mathbb C$,

conjecture | open | MathDB 页面 | 原文线索

V2-002203 | MATHDB-328961

Microscopic zeta logarithmic-derivative moment conjecture

摘录 / EXCERPT: Microscopic logarithmic-derivative moment conjecture. The following two asymptotic identities should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-002204 | MATHDB-328967

The zeta-function conjecture for almost all spectrally undetermined graphs

摘录 / EXCERPT: Zeta-function conjecture. Almost all graphs that are not DS are DZ. More precisely, if $\mathcal H_n$ is the set of non-DS graphs of order n and $\mathcal H'_n$ is the set of non...

conjecture | open | MathDB 页面 | 原文线索

V2-002205 | MATHDB-328987

Carries modulo n minus one conjecture for symmetric palintiples

摘录 / EXCERPT: Let $p=(d_k,d_{k-1},\ldots,d_0)_b$ be an (n,b)-palintiple with carries $c_k,c_{k-1},\ldots,c_0$. A palintiple is symmetric when its digits satisfy the symmetry condition defined...

conjecture | open | MathDB 页面 | 原文线索

V2-002206 | MATHDB-329000

Buhler-Pomerance-Robertson stabilization conjecture for real cyclotomic class numbers

摘录 / EXCERPT: For a prime p and a positive integer n, let ... be the real cyclotomic field of conductor p^n, and write $h(K)$ for its class number.
Buhler-Pomerance-Robertson conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-002207 | MATHDB-329001

Main conjecture on class numbers in cyclotomic $\mathbb{Z}_p$-extensions

摘录 / EXCERPT: For a prime p and a positive integer n, let $\mathbb{B}_{p,n}$ be the nth layer of the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$. For odd p it is the unique real...

conjecture | open | MathDB 页面 | 原文线索

V2-002208 | MATHDB-329003

Class-number-one conjecture for the first layers of cyclotomic $\mathbb{Z}_p$-extensions

摘录 / EXCERPT: For every prime p, let $\mathbb{B}_{p,1}$ be the first layer of the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$. First-layer class-number conjecture. For every prime p,...

conjecture | open | MathDB 页面 | 原文线索

V2-002209 | MATHDB-329016

Game distinguishing number conjecture for prime cycles

摘录 / EXCERPT: Let C_p be a cycle of prime order p, and let $D_\mathcal{G}(C_p)$ denote its Gentle-game distinguishing number. Prime-cycle conjecture. If $p\geq 11$, then ... The conjecture is...

conjecture | open | MathDB 页面 | 原文线索

V2-002210 | MATHDB-329025

Maximal oscillation conjecture for the Mertens product

摘录 / EXCERPT: Maximal oscillation conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002211 | MATHDB-329037

Divisor approximation conjecture for minimal Mahler measure

摘录 / EXCERPT: Let d be a square-free positive integer. The notation $o(\sqrt d)$ means a quantity whose ratio to $\sqrt d$ tends to zero as d tends to infinity, and $c=(1+o(1))\sqrt d$ means...

conjecture | open | MathDB 页面 | 原文线索

V2-002212 | MATHDB-329038

Conjectured global supremum for normalized minimal Mahler measure

摘录 / EXCERPT: Let $L(d)$ and D be as above, with d ranging over the relevant square-free positive integers. Global supremum conjecture. ... The paper says that computations suggest the largest...

conjecture | open | MathDB 页面 | 原文线索

V2-002213 | MATHDB-329039

Conjectured residue-class suprema for minimal Mahler measure

摘录 / EXCERPT: Let $L(d)$ and D be as above, and restrict d to the indicated congruence classes. Residue-class supremum conjecture. ... ... These are computationally suggested refinements after s...

conjecture | open | MathDB 页面 | 原文线索

V2-002214 | MATHDB-329040

Quadratic-residue conjecture for squarefree integers

摘录 / EXCERPT: Let d range over square-free positive integers, and let $\left(\frac{d}{p}\right)$ denote the Legendre symbol for an odd prime p. Quadratic-residue conjecture. All but finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-002215 | MATHDB-329041

Conjectured residue-class suprema modulo 8 and 3

摘录 / EXCERPT: Let $L(d)$ and D be as above, and restrict d to the indicated congruence classes. Residue-class supremum conjecture. ... ... These values are presented as computational improvement...

conjecture | open | MathDB 页面 | 原文线索

V2-002216 | MATHDB-329042

Uniformity conjecture for local representation zeta functions of type A groups

摘录 / EXCERPT: Let $n\in\mathbb{N}_{\geq 2}$. For a divisor $\iota(v)=\gcd(q_v-\varepsilon(v),n)$ of n, where $\varepsilon(v)\in\{1,-1\}$ records whether the relevant quadratic extension splits...

conjecture | open | MathDB 页面 | 原文线索

V2-002217 | MATHDB-329043

Ennola duality conjecture for character degrees of general linear and unitary congruence quotients

摘录 / EXCERPT: Let $n\in\mathbb{N}$. For $\varepsilon\in\{-1,1\}$, let $\mathfrak{o}$ be a compact discrete valuation ring with residue cardinality q, let G be one of the...

conjecture | open | MathDB 页面 | 原文线索

V2-002218 | MATHDB-329049

The local-global cohomological conjecture for relative K-groups

摘录 / EXCERPT: Let F be a number field and let G be a finite group on which Ω_F acts trivially. Define ... and set $K\mathcal R(O_FG)=\operatorname{Im}(\Psi)$. For each finite place v...

conjecture | open | MathDB 页面 | 原文线索

V2-002219 | MATHDB-329070

The classification conjecture for finite p-groups with transitive fusion systems

摘录 / EXCERPT: Let p be a prime, let P be a finite p-group, and let ${\mathcal F}$ be a saturated fusion system on P. The fusion system ${\mathcal F}$ is transitive if any two nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-002220 | MATHDB-329077

Jones's refined conjecture on amicable pairs for non-CM elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication, and let $\pi_{E,2}(X)$ denote the number of amicable pairs of E up to X. Jones's refined conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-002221 | MATHDB-329083

Conjecture on four consecutive primitive elements in finite fields

摘录 / EXCERPT: Let q be a prime power and let $\mathbb{F}_q$ denote the finite field with q elements. A field has consecutive primitive elements if some sequence of consecutive elements of it...

conjecture | open | MathDB 页面 | 原文线索

V2-002222 | MATHDB-329084

Conjecture on five consecutive primitive elements in finite fields

摘录 / EXCERPT: Let q be a prime power and let $\mathbb{F}_q$ denote the finite field with q elements. A sequence of consecutive primitive elements is a sequence of consecutive elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-002223 | MATHDB-329085

Conjecture on six consecutive primitive elements in finite fields

摘录 / EXCERPT: Let q be a prime power and let $\mathbb{F}_q$ denote the finite field with q elements. Consecutive primitive elements are consecutive field elements that are all generators of...

conjecture | open | MathDB 页面 | 原文线索

V2-002224 | MATHDB-329088

Chung's Paley discrepancy conjecture

摘录 / EXCERPT: For primes $p\equiv1\bmod4$, let χ be the Legendre symbol modulo p. Define $\operatorname{PaleyDiscrepancy}[\alpha,\beta]$ to mean that, for every sufficiently large p, ...

conjecture | open | MathDB 页面 | 原文线索

V2-002225 | MATHDB-329097

Krause-Zahlten conjecture on the cross number of finite abelian groups

摘录 / EXCERPT: Krause-Zahlten conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-002226 | MATHDB-329098

Structural conjecture for extremal zero-sum sequences over finite abelian groups

摘录 / EXCERPT: Structural conjecture. Each minimal zero-sum sequence U over G with $\mathsf{k}(U)=\mathsf{K}(G)$ has the form

conjecture | open | MathDB 页面 | 原文线索

V2-002227 | MATHDB-329113

The primitive-number square-free and order conjecture

摘录 / EXCERPT: Primitive-number conjecture. The number m is square-free, and there exists an index i such that

conjecture | open | MathDB 页面 | 原文线索

V2-002228 | MATHDB-329114

The coprime-order criterion for completeness

摘录 / EXCERPT: Coprime-order completeness conjecture. Then m is complete.

conjecture | open | MathDB 页面 | 原文线索

V2-002229 | MATHDB-329119

Type II distribution conjecture

摘录 / EXCERPT: Let $0<\varpi<1/4$, and let $\delta>0$ be a sufficiently small fixed quantity depending on ϑ. Let x tend to infinity, let P be a square-free number all of whose prime...

conjecture | open | MathDB 页面 | 原文线索

V2-002230 | MATHDB-329120

Least nontrivial value conjecture for primitive Dirichlet characters

摘录 / EXCERPT: Let χ be a Dirichlet character, and let n_χ be the first natural number such that $\chi(n_\chi)\neq1$. Character least nontrivial value conjecture. For every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-002231 | MATHDB-329130

Functional-equation formulation of the Riemann hypothesis for graph spectral zeta functions

摘录 / EXCERPT: Functional-equation conjecture. For every $s\in\mathbb{C}$ with $0<\operatorname{Re}(s)<1$,

conjecture | open | MathDB 页面 | 原文线索

V2-002232 | MATHDB-329140

Restricted inverse Galois problem over function fields

摘录 / EXCERPT: Let p be a prime and let G be a nontrivial finite group. Write G^{ab} for the abelianization of G, let $p(G)$ be the normal subgroup generated by the elements of p-power...

conjecture | open | MathDB 页面 | 原文线索

V2-002233 | MATHDB-329175

Finiteness conjecture for irreducible non-weakly-exceptional quotient singularities

摘录 / EXCERPT: Finiteness conjecture. For any prime p, there are only finitely many finite groups $G\subset\operatorname{SL}_{p}(\mathbb{C})$, up to conjugation, such that G acts irreducibly...

conjecture | open | MathDB 页面 | 原文线索

V2-002234 | MATHDB-329176

Generalized Borchardt mean identity for transformed theta invariants

摘录 / EXCERPT: Generalized Borchardt mean identity. With the notation above, the displayed identity holds for all such τ and γ. This extends the preceding Borchardt mean formula from...

conjecture | open | MathDB 页面 | 原文线索

V2-002235 | MATHDB-329177

Denominator conjecture for modular polynomials in the b'-invariants

摘录 / EXCERPT: Denominator conjecture. The polynomial L'_p is the denominator of every coefficient of the three modular polynomials in the functions b'_1,b'_2,b'_3.

conjecture | open | MathDB 页面 | 原文线索

V2-002236 | MATHDB-329187

Lalín-Rodrigue-Rogers rationality conjecture for secant zeta values

摘录 / EXCERPT: Let d and k be positive integers such that d is not a square. The secant zeta function is ... for sufficiently large real part of s. Lalín-Rodrigue-Rogers conjecture. ... The c...

conjecture | open | MathDB 页面 | 原文线索

V2-002237 | MATHDB-329191

Andrews-Garvan-Straub conjecture on linear congruences for the function $\xi(n)$

摘录 / EXCERPT: Andrews-Garvan-Straub conjecture. There exists a $B\in\mathbb N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002238 | MATHDB-329212

Newman's conjecture for function-field L-functions

摘录 / EXCERPT: Let $\mathcal F$ be a family of L-functions over a function field $\mathbb F_q[x]$. For each parameter $D\in\mathcal F$, let Λ_D denote the associated De Bruijn-Newman co...

conjecture | open | MathDB 页面 | 原文线索

V2-002239 | MATHDB-329213

Andrade-Chang-Miller conjecture for fixed odd-prime-power fields

摘录 / EXCERPT: Let q be a power of an odd prime. For every good pair (q,D), let Λ_D be the associated De Bruijn-Newman constant. Andrade-Chang-Miller conjecture. One has ... This is a p...

conjecture | open | MathDB 页面 | 原文线索

V2-002240 | MATHDB-329215

Andrade-Chang-Miller conjecture for reductions of a fixed square-free polynomial

摘录 / EXCERPT: Fix a square-free polynomial $D\in\mathbb Z[T]$. For each prime p, let $D_p\in\mathbb F_p[T]$ be the polynomial obtained by reducing D modulo p, and let Λ_{D_p} be th...

conjecture | open | MathDB 页面 | 原文线索

V2-002241 | MATHDB-329216

The folklore conjecture on quasi-modularity for Calabi-Yau 1-folds

摘录 / EXCERPT: Calabi-Yau 1-fold quasi-modularity conjecture. The ancestor Gromov-Witten correlation functions defined above are quasi-modular forms. This is the one-dimensional formulation of th...

conjecture | open | MathDB 页面 | 原文线索

V2-002242 | MATHDB-329219

The radius of convergence of the cotangent-sum moment series

摘录 / EXCERPT: The radius-of-convergence conjecture. The radius of convergence of this series is π^2. The preceding results establish convergence for a positive range of x and divergence fo...

conjecture | open | MathDB 页面 | 原文线索

V2-002243 | MATHDB-329221

Uniform block-product boundedness assumption for the Fibonacci decomposition

摘录 / EXCERPT: Let ω be the golden rotation number, let F_s denote the Fibonacci numbers, and write an integer $k\ge1$ in Fibonacci form $k=\sum_{s=1}^{m}b_sF_s$, with $b_s\in\{0,1\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-002244 | MATHDB-329225

Friedlander-Granville conjecture on primes in arithmetic progressions

摘录 / EXCERPT: Friedlander-Granville conjecture. For every fixed $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-002245 | MATHDB-329226

Squarefree-number error conjecture in arithmetic progressions

摘录 / EXCERPT: Squarefree error conjecture. For every fixed $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-002246 | MATHDB-329243

The exceptional-prime conjecture for type I(a) solutions

摘录 / EXCERPT: Exceptional-prime conjecture. The only prime p for which there is no type I(a) solution is $p=193$.

conjecture | open | MathDB 页面 | 原文线索

V2-002247 | MATHDB-329244

Existence conjecture for type I(b) solutions

摘录 / EXCERPT: Type I(b) existence conjecture. Every prime p has a type I(b) solution.

conjecture | open | MathDB 页面 | 原文线索

V2-002248 | MATHDB-329246

The Sylow vertex conjecture for hook Specht modules

摘录 / EXCERPT: Let p be an odd prime, let k be a positive integer, and let r be an integer with $0\leq r\leq kp-1$. Write $S^{(kp-r,1^r)}$ for the hook Specht module for the symmetric group...

conjecture | open | MathDB 页面 | 原文线索

V2-002249 | MATHDB-329256

Gross's higher-order vanishing conjecture for partial zeta values

摘录 / EXCERPT: Let $L\supseteq K\supseteq F$ be a tower of finite abelian extensions, let $G=\operatorname{Gal}(K/F)$, let $\widetilde{G}=\operatorname{Gal}(L/F)$, let $H=\operatorname{Gal}(L/K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-002250 | MATHDB-329258

Cyclotomic root conjecture for the inversion polynomial

摘录 / EXCERPT: Let $b=ck$, and let ζ_m denote a primitive mth root of unity. The inversion polynomial $f_b(x)$ is defined by its roots indexed by the inversion numbers $\inv(a,b)$. Cyclot...

conjecture | open | MathDB 页面 | 原文线索

V2-002251 | MATHDB-329259

Conjecture on the six smallest inversion numbers

摘录 / EXCERPT: For a fixed positive integer b, let $\operatorname{inv}(a,b)$ denote the inversion number and let $I_n(b)$ be its nth smallest value, where $1\leq n\leq6$. Six-smallest-values...

conjecture | open | MathDB 页面 | 原文线索

V2-002252 | MATHDB-329270

Conjectural square-root cancellation for twisted Kloosterman correlation sums

摘录 / EXCERPT: Let q be prime, let a be an integer coprime with q, let $\boldsymbol{b}$ lie in the generic set ... and let $[?]$ be elements of $[?]$. For the correlation sum $[?]$ formed usi...

conjecture | open | MathDB 页面 | 原文线索

V2-002253 | MATHDB-329271

Supersingular Heegner-point corank conjecture for strong Weil curves

摘录 / EXCERPT: Let E be an elliptic curve over ${\mathbb{Q}}$, let K be an imaginary quadratic field, let p be a prime, and let K_{∞}/K be the anticyclotomic ${\mathbb{Z}_p}$-extensi...

conjecture | open | MathDB 页面 | 原文线索

V2-002254 | MATHDB-329289

Differentiability conjecture for the arithmetic Fourier series F_k and G_k

摘录 / EXCERPT: Let F_k and G_k be the arithmetic Fourier series considered in the paper, and let $q_n(x)$ denote the denominators of the continued-fraction convergents of x. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002255 | MATHDB-329294

Polylogarithmic root bound for sparse trinomials over finite fields

摘录 / EXCERPT: Polylogarithmic root-bound conjecture. There is an absolute constant $c\geq 1$ such that this trinomial has no more than $(\log p)^c$ roots in $\mathbb{F}_p$.

conjecture | open | MathDB 页面 | 原文线索

V2-002256 | MATHDB-329297

Conway--Norton monstrous moonshine conjecture

摘录 / EXCERPT: Let $\mathbb{M}$ be the monster, and for each $g\in\mathbb{M}$ let T_g be its monstrous McKay--Thompson series. A normalized principal modulus is a generator of the function fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-002257 | MATHDB-329312

The sharp lower-bound conjecture for heights on GIT quotients

摘录 / EXCERPT: Sharp lower-bound conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-002258 | MATHDB-329355

Kloosterman-fraction trilinear estimate conjecture

摘录 / EXCERPT: Let ... where $M\leq m<2M$, $N\leq n<2N$, $A\leq a<2A$, and $\|\cdot\|$ and $\|\cdot\|_\infty$ denote the L_2 and L_∞ norms. The conjecture concerns the range...

conjecture | open | MathDB 页面 | 原文线索

V2-002259 | MATHDB-329372

The conjectural formula for the geography constant of surfaces of general type in characteristic p

摘录 / EXCERPT: Let p be a prime, and let κ_p be the infimum of χ/c_1^2 over minimal algebraic surfaces of general type defined over an algebraically closed field of characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-002260 | MATHDB-329396

The exact signed sumset minimum conjecture for rank-two elementary abelian groups

摘录 / EXCERPT: Let p be an odd prime and write $m=cp+v$, where $0\leq c\leq p-1$ and $1\leq v\leq p$. The quantities $\rho(\mathbb Z_p^2,m,2)$, $\rho_{\pm}(\mathbb Z_p^2,m,2)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-002261 | MATHDB-329425

Conjecture on the largest prime divisor of rank-proportion denominators

摘录 / EXCERPT: Let c_k be the limiting proportion of vertices of rank k, and let $denom(c_k)$ denote its denominator when c_k is written in lowest terms. Denominator prime-divisor bound con...

conjecture | open | MathDB 页面 | 原文线索

V2-002262 | MATHDB-329432

The exponent-one conjecture for 2-powerfully perfect numbers in the Gaussian and Eisenstein-type rings

摘录 / EXCERPT: Let k be the exponent appearing in Theorems 2.2 and 2.4 for the classification of 2-powerfully perfect numbers in the relevant quadratic rings. Exponent-one conjecture. The val...

conjecture | open | MathDB 页面 | 原文线索

V2-002263 | MATHDB-329433

Cubic prime-divisor conjecture for sparsely Schemmel totient numbers

摘录 / EXCERPT: Let r be a positive integer, let $n>1$ be an integer with $n\in F_r$, and let $P_1(n)$ denote the smallest prime divisor of n. Cubic prime-divisor conjecture. ... This generalize...

conjecture | open | MathDB 页面 | 原文线索

V2-002264 | MATHDB-329480

The exact balancing-set conjecture for vectors with pth roots of unity

摘录 / EXCERPT: Exact balancing-set conjecture. For every integer n and every prime factor p of n,

conjecture | open | MathDB 页面 | 原文线索

V2-002265 | MATHDB-329482

Existence of fixed multipliers for odd-power binomial-sum congruences

摘录 / EXCERPT: Multiplier conjecture. There exist integers a_{2r-1} and b_r, independent of n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002266 | MATHDB-329483

Small-index multipliers for binomial-sum congruences

摘录 / EXCERPT: Small-index multiplier conjecture. Set

conjecture | open | MathDB 页面 | 原文线索

V2-002267 | MATHDB-329487

The unique non-trivial solution conjecture for the equation $X^n-1=BZ^n$

摘录 / EXCERPT: Let $B,n\in\mathbb{N}_{>1}$ satisfy ... Consider the equation ... where solutions with $Z\in\{-1,0,1\}$ are called trivial. Unique-solution conjecture. Under these conditions, the equa...

conjecture | open | MathDB 页面 | 原文线索

V2-002268 | MATHDB-329510

The Tate conjecture for K3 surfaces

摘录 / EXCERPT: The Tate conjecture. This map should be an isomorphism. It would calculate the Picard group, and hence the Picard number, of a K3 surface from its ell-adic cohomology. The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-002269 | MATHDB-329515

Maximal Hausdorff dimension conjecture for badly approximable complex matrices

摘录 / EXCERPT: Let m,n be positive integers, let $\textbf{Bad}_{\mathbb{C}}(m,n)\subset\mathbb{C}^{mn}$ denote the set of badly approximable points, and let $K\subset\mathbb{C}^{mn}$ be any com...

conjecture | open | MathDB 页面 | 原文线索

V2-002270 | MATHDB-329540

Guo's Delannoy polynomial valuation conjecture

摘录 / EXCERPT: Let $D_k(x)$ be the Delannoy polynomial, defined by ... For a prime p, let ν_p denote the p-adic valuation. Let n be a positive integer and let x be a p-adic integer. G...

conjecture | open | MathDB 页面 | 原文线索

V2-002271 | MATHDB-329541

The mod-p^3 refinement for alternating fourth powers of Delannoy polynomials

摘录 / EXCERPT: Let $D_k(x)$ be the Delannoy polynomial, defined by ... Let x be an integer and let p be an odd prime. The mod-p^3 refinement. ... The paper proves the corresponding congruence m...

conjecture | open | MathDB 页面 | 原文线索

V2-002272 | MATHDB-329549

Almost-square conjecture for intervals around almost all integers

摘录 / EXCERPT: For $0\leq \theta\leq 1/2$ and $C>0$, call an integer n a (θ,C)-almost square if $n=ab$ for integers $a,b\in[n^{1/2}-Cn^\theta,n^{1/2}+Cn^\theta]$. Almost-square conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-002273 | MATHDB-329551

The Vojta-Hall-Lang height bound for integral points on hyperelliptic curves

摘录 / EXCERPT: Vojta-Hall-Lang conjecture. There exist constants $C_1=C_1(d)$ and $C_2=C_2(d)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002274 | MATHDB-329577

Divisibility conjecture for shifted sums involving $R_{k,r}$

摘录 / EXCERPT: Divisibility conjecture for shifted sums. There exists an integer a_r, independent of n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002275 | MATHDB-329584

The balanced orbit's product-majorization conjecture

摘录 / EXCERPT: Let $1\leq p<q$ be coprime integers. For a word $w\in\mathbb{W}_{p,q}$, let $\mathcal{I}(w)=(w_1,\ldots,w_q)$ be its base-2 orbit and define the partial products by ... The relation...

conjecture | open | MathDB 页面 | 原文线索

V2-002276 | MATHDB-329593

Extreme class number upper-bound conjecture for real quadratic fields

摘录 / EXCERPT: Let d be a positive fundamental discriminant, and let $h(d)$ denote the class number of the real quadratic field $\mathbb{Q}(\sqrt{d})$. Extreme class number upper-bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-002277 | MATHDB-329618

Injectivity conjecture for modular-form classes in rank-two assembly

摘录 / EXCERPT: Let $\mathcal{X}_{2m,0}$ be the space appearing in the modular-form construction, and let ... be the map obtained by gluing two rank-2 graphs along all their leaves. Injectivity co...

conjecture | open | MathDB 页面 | 原文线索

V2-002278 | MATHDB-329619

Injectivity conjecture for odd-dimensional modular-form classes

摘录 / EXCERPT: Odd-dimensional injectivity conjecture. This map is injective for all k. The target lies in codimension 2 because the virtual cohomological dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-002279 | MATHDB-329634

The cohomological modularity conjecture for new Hecke eigenclasses

摘录 / EXCERPT: Let K be a number field with n real places and s pairs of complex places. Let $\mathfrak D$ be the discriminant ideal of an inner form and let $\mathfrak m$ be a level ideal....

conjecture | open | MathDB 页面 | 原文线索

V2-002280 | MATHDB-329635

The cohomological p-adic Tate lattice conjecture

摘录 / EXCERPT: Let K be a number field, let $\mathfrak p$ be a prime of K, and let f be a new rational Hecke eigenclass at level $\mathfrak p\mathfrak n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002281 | MATHDB-329644

Finite generation conjecture for the module \(\mathscr{M}_{\zeta}\)

摘录 / EXCERPT: Finite generation conjecture. For every $c13\zeta\in S$, the set $c13\mathscr{M}_{\zeta}$ is finitely generated as a $c13\mathbb{Z}[\zeta,\zeta^{-1}]$-module. More generally,

conjecture | open | MathDB 页面 | 原文线索

V2-002282 | MATHDB-329663

A companion truncated hypergeometric congruence for the p-adic Gamma function

摘录 / EXCERPT: Let p be a prime with $p\equiv1\pmod4$, let $(a)_k$ denote the Pochhammer symbol, and let Γ_p denote the p-adic Gamma function. The companion congruence. The following c...

conjecture | open | MathDB 页面 | 原文线索

V2-002283 | MATHDB-329664

A level 5 modular-form congruence for truncated hypergeometric series

摘录 / EXCERPT: Let $n>1$ be an integer and let p be a prime with $p\equiv1\pmod n$. Let $E_1(z)$ and $E_2(z)$ be the two explicit level 5 weight 3 Eisenstein series with coefficients in...

conjecture | open | MathDB 页面 | 原文线索

V2-002284 | MATHDB-329671

Polynomial ideal conjecture for pairwise coprime linear forms

摘录 / EXCERPT: Let $f_1,\ldots,f_m\in\mathbb C[x,y]$ be linear forms such that $\gcd(f_i,f_j)=1$ for all $i\neq j$. For each pair $i<j$, let $\langle f_i+z,f_j+z\rangle$ denote the ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-002285 | MATHDB-329686

Arkin-Hoggatt-Straus conjecture on Diophantine quadruples

摘录 / EXCERPT: Arkin-Hoggatt-Straus conjecture. If $\{a,b,c,d\}$ is a quadruple, then $d=d_{+}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002286 | MATHDB-329708

Modularity conjecture for elliptic Calabi-Yau expansion coefficients

摘录 / EXCERPT: Modularity conjecture. The coefficient of Z^n in each expansion is a modular form. Its modular weight is $18n$ for the expansions of $z_1(1-432z_1)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-002287 | MATHDB-329727

The unified Mordell-Lang conjecture for uniformizable T-modules

摘录 / EXCERPT: Let $\mathcal{A}=(\mathbb{G}_{a}^{m},\Phi)$ be a uniformizable T-module such that there exists $i\in\mathbb{N}\setminus\{0\}$ for which the leading coefficient matrix of...

conjecture | open | MathDB 页面 | 原文线索

V2-002288 | MATHDB-329728

HAW conjecture on bounded points in the expanding subgroup

摘录 / EXCERPT: HAW conjecture on H. The set

conjecture | open | MathDB 页面 | 原文线索

V2-002289 | MATHDB-329761

Buratti's prime-cyclic multiset sequencing conjecture

摘录 / EXCERPT: Let p be a prime and let S be a multiset containing $p-1$ nonzero elements of ${\mathbb Z}_p$. Buratti's conjecture. There exists an arrangement of the elements of S such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-002290 | MATHDB-329784

The graded associated Lannes-Zarati conjecture

摘录 / EXCERPT: Let p be an odd prime, let A be the mod p Steenrod algebra, and let φ_s denote the mod p Lannes-Zarati homomorphism in Adams filtration s, with positive stem t....

conjecture | open | MathDB 页面 | 原文线索

V2-002291 | MATHDB-329805

Necessary conditions for divisibility of two-part-size partition counts

摘录 / EXCERPT: Let $\nu_2(m)$ denote the relevant counting function for partitions into two sizes of part, and let A and B be integers such that $An+B$ is the progression under consideration....

conjecture | open | MathDB 页面 | 原文线索

V2-002292 | MATHDB-329806

The converse relation between parity of ν_3 and divisibility of ν_2

摘录 / EXCERPT: Let $\nu_2(m)$ and $\nu_3(m)$ denote the relevant counting functions for partitions into two and three sizes of part, respectively, and let $An+B$ be an arithmetic progression. Par...

conjecture | open | MathDB 页面 | 原文线索

V2-002293 | MATHDB-329809

The asymptotic AECC intersection bound for DNA sequence profiles

摘录 / EXCERPT: Let q, ℓ, and d be fixed. Choose a matrix ${\bf H}$ and a prime p such that $\mathcal C({\bf H},{\bf 0})$ is an $(N,d+1)$-AECC containing ${\bf 1}$. Let $c(q,\ell)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002294 | MATHDB-329810

Cohen-Lenstra density conjecture for divisibility of imaginary quadratic class numbers by powers of 2

摘录 / EXCERPT: Let $S(n)=\{p\text{ prime}:n\mid h(-p)\}$, where $h(-p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$. For $X\geq 2$, define ... The natural density $\rho(S)$ is the limit of...

conjecture | open | MathDB 页面 | 原文线索

V2-002295 | MATHDB-329829

The predicted equality between ray class group exponents and unit quotient orders

摘录 / EXCERPT: Let $\mathfrak{a}\subseteq\mathfrak{o}$ be a G-stable ideal. Write $E_{\mathfrak{a}}$ for the relevant ray-unit group,...

conjecture | open | MathDB 页面 | 原文线索

V2-002296 | MATHDB-329837

A Gaussian binomial coefficient identity

摘录 / EXCERPT: Let u and i be nonnegative integers and let q be a prime power. Write $\binom{n}{r}_{q}$ for the Gaussian binomial coefficient, namely the number of r-dimensional subspaces...

conjecture | open | MathDB 页面 | 原文线索

V2-002297 | MATHDB-329844

Conjecture on the moments of greatest common divisors of Gaussian ideals

摘录 / EXCERPT: Let $\mathfrak{n}$ and $\mathfrak{m}$ be nonzero ideals chosen independently and uniformly at random from the set of ideals in $\mathbb{Z}[i]$ with norm at most x. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-002298 | MATHDB-329859

Gouvêa's limiting-measure conjecture for slopes of p-oldforms

摘录 / EXCERPT: Gouvêa's limiting-measure conjecture. As k tends to infinity, these slopes converge to the measure that is uniform on

conjecture | open | MathDB 页面 | 原文线索

V2-002299 | MATHDB-329860

Irreducibility conjecture for reductions of crystalline representations

摘录 / EXCERPT: Irreducibility conjecture. If $v(a_p)\notin{\mathbb{Z}}$, then $\overline{V}_{k,a_p}$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-002300 | MATHDB-329861

Irreducibility conjecture for reductions at the boundary of weight space

摘录 / EXCERPT: Boundary irreducibility conjecture. If $v_\chi(\alpha)\notin{\mathbb{Z}}$, then $\overline{V}_{k,\alpha,\chi}$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-002301 | MATHDB-329862

Local constancy conjecture for crystalline lifts

摘录 / EXCERPT: Local constancy conjecture. There is a crystalline lift of $\bar r$ with Hodge-Tate weights $0,k-1$ and slope n if and only if there is a crystalline lift of $\bar r$ with Hodge-...

conjecture | open | MathDB 页面 | 原文线索

V2-002302 | MATHDB-329865

The arithmetic fundamental lemma, Lie algebra version

摘录 / EXCERPT: Let F_0 be the base p-adic field, let $\mathfrak{s}(F_0)_{\mathrm{rs}}$ denote the regular semisimple elements of the symmetric Lie algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002303 | MATHDB-329898

Optimality conjecture for trivariate Lissajous curve frequency triples

摘录 / EXCERPT: Optimality conjecture. There exists an integer triple (i,j,k), not all zero, with $i+j+k\leq 2n$, such that at least one of

conjecture | open | MathDB 页面 | 原文线索

V2-002304 | MATHDB-329904

A part of Stark's conjecture for totally real abelian extensions

摘录 / EXCERPT: A part of Stark's conjecture. For every $\sigma\in G$,

conjecture | open | MathDB 页面 | 原文线索

V2-002305 | MATHDB-329914

Generalized Iwasawa main conjecture for zeta isomorphisms

摘录 / EXCERPT: Generalized Iwasawa main conjecture. For every pair (R,T), there exists such a zeta isomorphism, compatible with arbitrary base change and exact sequences; when $(R,T)=(L,V)$ com...

conjecture | open | MathDB 页面 | 原文线索

V2-002306 | MATHDB-329915

Local epsilon-isomorphism conjecture

摘录 / EXCERPT: Fix a prime l. For a coefficient ring R, an R-representation T of $G_{\mathbb{Q}_l}$, and a $\mathbb{Z}_l$-basis ζ of...

conjecture | open | MathDB 页面 | 原文线索

V2-002307 | MATHDB-329919

Green's inverse conjecture for the large sieve

摘录 / EXCERPT: Green's inverse conjecture for the large sieve. Then $|A|\ll X^\epsilon$ unless A is contained, apart from a set of size X^ϵ, in the set of values of a quadratic polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-002308 | MATHDB-329920

Inverse conjecture for the large sieve with additively structured sieving sets

摘录 / EXCERPT: Refined inverse conjecture for the large sieve. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-002309 | MATHDB-329922

Puncture-code distance conjecture for cyclic quantum MDS codes

摘录 / EXCERPT: Let q be a prime power and let $C^*=[q^2+1,q^2+1-d,d]_{q^2}$ be an F_{q^2}-linear (consta)cyclic MDS code. Let $P(C)$ denote its corresponding puncture code. Puncture-code dist...

conjecture | open | MathDB 页面 | 原文线索

V2-002310 | MATHDB-329935

The local ratios conjecture for real translations

摘录 / EXCERPT: Local ratios conjecture. For these fixed parameters,

conjecture | open | MathDB 页面 | 原文线索

V2-002311 | MATHDB-329938

The eventual quasi-polynomial conjecture for the parametric Frobenius problem

摘录 / EXCERPT: Let $a_i:bbZ_+crightarrowbbZ_+$, $1cle icle n$, be eventually positive functions in $cmathrm{EQP}$, where $F(a_1,cdots,a_n)$ denotes the largest element of the group...

conjecture | open | MathDB 页面 | 原文线索

V2-002312 | MATHDB-329945

Vinogradov's generalized shifted divisor conjecture

摘录 / EXCERPT: Let k be a positive integer, let $d_k(n)$ be the k-fold divisor function, ... and let f be a smooth function on $\mathbb{R}^{+}\times\mathbb{R}^{+}$. Vinogradov's generalized s...

conjecture | open | MathDB 页面 | 原文线索

V2-002313 | MATHDB-329956

Lubin's conjecture on stable power series and Lubin-Tate formal groups

摘录 / EXCERPT: Lubin's conjecture. Then g is an endomorphism of some Lubin-Tate formal group $F(x,y)$ over ${\mathcal O}_K$, and

conjecture | open | MathDB 页面 | 原文线索

V2-002314 | MATHDB-329958

Chow's stability conjecture for holomorphic newforms

摘录 / EXCERPT: Let q be a positive squarefree integer, let k be a positive even integer, and let $n_0(q,k)$ be the least integer such that equality of Hecke eigenvalues at all primes...

conjecture | open | MathDB 页面 | 原文线索

V2-002315 | MATHDB-329976

Primitive polynomial and integer solutions conjecture for quadratic forms

摘录 / EXCERPT: Let $F\in\mathbb{Z}[x,y,z]$ be a quadratic form, and consider the Diophantine equation ... where G is the polynomial defined in the source by equation. A solution (x,y,z) is prim...

conjecture | open | MathDB 页面 | 原文线索

V2-002316 | MATHDB-329977

Primitive quintic diagonal equation conjecture

摘录 / EXCERPT: Let $a,b,c\in\mathbb{Z}\setminus\{0\}$ and $d\in\mathbb{Z}$. Consider the Diophantine equation ... The integers X_1,X_2,X_3,X_4 are co-prime when their greatest common divisor is...

conjecture | open | MathDB 页面 | 原文线索

V2-002317 | MATHDB-329994

Cohen-Lenstra conjecture on ranks of imaginary quadratic class groups

摘录 / EXCERPT: Let l be an odd prime, let $\Delta<0$ be a negative discriminant, and let Cl_Δ denote the class group of the corresponding imaginary quadratic field. Define ... The l-rank...

conjecture | open | MathDB 页面 | 原文线索

V2-002318 | MATHDB-330000

Montgomery's conjectures on the maximal order of the Farey fraction remainder

摘录 / EXCERPT: Montgomery's conjectures. As $x\to\infty$, the remainder satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002319 | MATHDB-330041

Cheng-Duncan-Harvey umbral moonshine module conjecture

摘录 / EXCERPT: Umbral moonshine module conjecture. There is a naturally defined module $\check{K}^X$ such that, for $g\in G^X$ and $r\in I^X$,

conjecture | open | MathDB 页面 | 原文线索

V2-002320 | MATHDB-330050

The equality criterion for the left-hand Hausdorff-dimension bound

摘录 / EXCERPT: Let $k\geq 1$ be an integer and let $\mathscr{C}$ be a curve as in the paper. Write $\mathscr{H}^{k}_{\lambda}$ for the set of points satisfying the relevant Diophantine approximat...

conjecture | open | MathDB 页面 | 原文线索

V2-002321 | MATHDB-330072

Chen-Yang integrality conjecture for Turaev-Viro invariants of torus links

摘录 / EXCERPT: Let $T_{(m,n)}$ be the (m,n)-torus link in S^3, with $m<n$. Let $TV_r(S^3\setminus T_{(m,n)})$ be its Turaev-Viro invariant. Chen-Yang integrality conjecture. If r is coprime...

conjecture | open | MathDB 页面 | 原文线索

V2-002322 | MATHDB-330084

Selberg's orthonormality conjecture for the Selberg class

摘录 / EXCERPT: Let ${\mathcal S}$ be the Selberg class of functions $L(s)=\sum_{n=1}^{\infty}a_L(n)n^{-s}$ satisfying the Ramanujan hypothesis, analytic continuation, a functional equation, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-002323 | MATHDB-330091

The square-root abundancy conjecture for spoof odd perfect numbers

摘录 / EXCERPT: Let $n=km>1$ be a spoof odd perfect number. If $\gcd(m,k)=1$, then $\sqrt{k}$ is an integer and the sum-of-divisors function σ is evaluated at this factor. Square-root abund...

conjecture | open | MathDB 页面 | 原文线索

V2-002324 | MATHDB-330109

Boundedness conjecture for the next odd Collatz value

摘录 / EXCERPT: Boundedness conjecture. The next odd number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002325 | MATHDB-330110

Fundamental-set conjecture for odd integers in the Collatz system

摘录 / EXCERPT: Let d_i denote the values used in the proposed Collatz-based number system, and let the odd integers be the odd members of the system. Fundamental-set conjecture. The values...

conjecture | open | MathDB 页面 | 原文线索

V2-002326 | MATHDB-330111

18-formula conjecture for Collatz stopping times

摘录 / EXCERPT: Consider the proposed fundamental covering system of odd integers for the Collatz map, and let a total stopping time function assign to each integer the number of iterations needed...

conjecture | open | MathDB 页面 | 原文线索

V2-002327 | MATHDB-330150

Coates-Sujatha fine Selmer conjecture over cyclotomic extensions

摘录 / EXCERPT: Let l be an odd prime, let F be a number field, and let $\mathcal{E}$ be an elliptic curve over F. For an extension L/F in the maximal extension unramified outside the pres...

conjecture | open | MathDB 页面 | 原文线索

V2-002328 | MATHDB-330151

Anticyclotomic fine Selmer cofiniteness conjecture

摘录 / EXCERPT: Let l be an odd prime, let F be an imaginary quadratic field, and let $\mathcal{E}$ be an elliptic curve over F. Let F^{anti} be the anticyclotomic $\mathbb{Z}_l$-...

conjecture | open | MathDB 页面 | 原文线索

V2-002329 | MATHDB-330152

Local Heegner-point infinitude conjecture

摘录 / EXCERPT: Let K be an imaginary quadratic field, let $p\geq 5$ be a prime, let E be an elliptic curve over $\mathbb{Q}$ with modular parametrization π, and let K_{∞}/K be the...

conjecture | open | MathDB 页面 | 原文线索

V2-002330 | MATHDB-330167

The semisimple proper-intersection criterion

摘录 / EXCERPT: Let $G_{W_1}(F_0)$ act on the relevant moduli spaces, let Δ be the diagonal correspondence, and let Δ_g be its translate by g. A semisimple element is one whose orb...

conjecture | open | MathDB 页面 | 原文线索

V2-002331 | MATHDB-330176

Gao-Thangadurai-Kubertin conjecture for finite abelian p-groups

摘录 / EXCERPT: Let p be a prime, let G be a finite abelian p-group, and write $q=\exp(G)$. For a positive integer k, let $s_{kq}(G)$ be the smallest length such that every sequence over...

conjecture | open | MathDB 页面 | 原文线索

V2-002332 | MATHDB-330191

Buzzard-Gee irreducibility conjecture with explicit exceptional conditions

摘录 / EXCERPT: Let r be a positive integer written as $r=t(p-1)+s$ with $t\geqslant 0$ and $s\in\{1,\ldots,p-1\}$, let $a\in\overline{\mathbb Q}_p$ with $v=v(a)\geqslant 1$, and define the risi...

conjecture | open | MathDB 页面 | 原文线索

V2-002333 | MATHDB-330197

The Beatty-sequence logarithm bound by $\pi^2/6$

摘录 / EXCERPT: Let $a>1$ and $b>1$ be irrational numbers. The Beatty-sequence logarithm bound. ... This conjecture is motivated by estimating the corresponding series through the quantities...

conjecture | open | MathDB 页面 | 原文线索

V2-002334 | MATHDB-330198

The Beatty-sequence logarithm bound by 1

摘录 / EXCERPT: Let $a>1$ and $b>1$ be irrational numbers. The Beatty-sequence logarithm bound by 1. ... Numerical data in the source suggest this stronger bound than the preceding conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-002335 | MATHDB-330207

Wu's divisibility-by-3 conjecture for symmetric concatenations

摘录 / EXCERPT: Wu's divisibility-by-3 conjecture. If n is not a multiple of 11 with an even number of digits and $a(n)=0$, then n is a multiple of 3.

conjecture | open | MathDB 页面 | 原文线索

V2-002336 | MATHDB-330208

The OEIS A256480 conjecture for prepended-digit primes

摘录 / EXCERPT: OEIS A256480 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-002337 | MATHDB-330209

Wu's divisibility-by-3 conjecture for appended-digit primes

摘录 / EXCERPT: Wu's divisibility-by-3 conjecture. If $c(n)=0$, then n is divisible by 3.

conjecture | open | MathDB 页面 | 原文线索

V2-002338 | MATHDB-330224

Ciucu's prime-factor conjecture for trimmed Aztec rectangles

摘录 / EXCERPT: Let m,n,h_1,h_2 be the parameters defining the trimmed Aztec rectangles $TA_{m,n}^{h_1,h_2}$ and $TB_{m,n}^{h_1,h_2}$: $TA_{m,n}^{h_1,h_2}$ is obtained by trimming a...

conjecture | open | MathDB 页面 | 原文线索

V2-002339 | MATHDB-330247

Beilinson's conjecture for integral tame K_2 of curves

摘录 / EXCERPT: Let C be a smooth, projective, geometrically irreducible curve of genus g defined over a number field K with ring of integers $\mathcal{O}_K$. Let $K_2(C;\mathcal{O}_K)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-002340 | MATHDB-330255

Togbé's sharp solution-count conjecture for y^2=px(Ax^2+2)

摘录 / EXCERPT: Togbé's conjecture. If (A,p) is congruent modulo 8 to one of $(1,1)$, $(1,5)$, $(1,7)$, $(3,1)$, $(3,3)$, $(3,7)$, $(5,1)$, $(5,5)$, $(5,7)$, $(7,3)$, or $(7,5)$, then the equa...

conjecture | open | MathDB 页面 | 原文线索

V2-002341 | MATHDB-330256

The rational Catalan area-codinv conjecture for coprime Dyck paths

摘录 / EXCERPT: Let m and n be coprime positive integers. An (m,n)-Dyck path is a lattice path from $(0,0)$ to (m,n) staying on the prescribed side of the main diagonal; let...

conjecture | open | MathDB 页面 | 原文线索

V2-002342 | MATHDB-330257

The rational Catalan length-skew-length conjecture for core partitions

摘录 / EXCERPT: Let m and n be coprime positive integers. An (m,n)-core is a partition that is simultaneously an m-core and an n-core; denote the set of such cores by $\mathcal P_{m,n}$....

conjecture | open | MathDB 页面 | 原文线索

V2-002343 | MATHDB-330269

The bounded completeness conjecture for Collatz sequences

摘录 / EXCERPT: Bounded completeness conjecture. For all $m_k>1$, there exists a constant $C_{\max}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002344 | MATHDB-330291

The Schneider conjecture on higher K-theory of number fields

摘录 / EXCERPT: Schneider's conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-002345 | MATHDB-330294

Ramanujan's conjecture for half-integral weight modular forms

摘录 / EXCERPT: Let $f(z) = \sum^{\infty}_{n=1} a_f(n) n^{\frac{k-1}{2}} q^n$ be a half-integral weight modular form of weight $k=\ell+\frac{1}{2}$ on $\Gamma_0(4N)$, where $k \in \mathbb{N}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-002346 | MATHDB-330295

Siegel-type lower bound for nonzero quadratic twists

摘录 / EXCERPT: Let g be the classical modular form associated with the half-integral weight modular form f via Shimura's correspondence, and let χ_p denote the quadratic character attach...

conjecture | open | MathDB 页面 | 原文线索

V2-002347 | MATHDB-330296

Quattrini's Fourier coefficient formula for Cohen-Eisenstein series

摘录 / EXCERPT: Let B be a definite quaternion algebra ramified at exactly one finite prime p, and let $N=pM$ with $p \mid M$ be square-free. Write ... For $D\in\mathbb{N}$, let $-D$ be a fundame...

conjecture | open | MathDB 页面 | 原文线索

V2-002348 | MATHDB-330315

Cloitre's prime-or-one conjecture for the least-common-multiple sequence

摘录 / EXCERPT: Define ... and, for $n\geq2$, define $b_n=a_n/a_{n-1}-1$. The sequence (b_n) consists of positive integers. Cloitre's conjecture. For any $n\geq2$, b_n is either 1 or a prime n...

conjecture | open | MathDB 页面 | 原文线索

V2-002349 | MATHDB-330316

Cloitre's conjecture on the primes occurring in the least-common-multiple sequence

摘录 / EXCERPT: Let a_n and b_n be defined by ... and $b_n=a_n/a_{n-1}-1$ for $n\geq2$. Cloitre's prime-occurrence conjecture. The sequence (b_n) contains every prime number other than 3, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-002350 | MATHDB-330317

The prime progression-count conjecture modulo a prime

摘录 / EXCERPT: For real x, a positive integer modulus q, and an integer residue r, let $\pi(x;q,r)$ denote the number of primes less than x that are congruent to r modulo q. Prime pro...

conjecture | open | MathDB 页面 | 原文线索

V2-002351 | MATHDB-330341

Armstrong's rank and co-skew-length symmetry conjecture for coprime cores

摘录 / EXCERPT: Armstrong's rank and co-skew-length symmetry conjecture. The following q,t-polynomials are equal:

conjecture | open | MathDB 页面 | 原文线索

V2-002352 | MATHDB-330342

Uniqueness of the eta partner for an image of the zeta map

摘录 / EXCERPT: Let a and b be relatively prime positive integers. Let $\mathfrak{D}_{a,b}$ be the set of (a,b)-Dyck paths, and let ... be the image of the pair (ζ,η). Uniqueness conj...

conjecture | open | MathDB 页面 | 原文线索

V2-002353 | MATHDB-330355

Erdős-Pólya-Turán conjecture for linear forms in consecutive prime gaps

摘录 / EXCERPT: Erdős-Pólya-Turán conjecture. The expression $\sum_{i=1}^{\ell}\alpha_i d_{n+i}$ changes sign infinitely often as n runs through all integers if and only if the nonzero elements...

conjecture | open | MathDB 页面 | 原文线索

V2-002354 | MATHDB-330374

Erdős-Fuchs conjecture on additive representation functions

摘录 / EXCERPT: Let $\mathcal{A} = \{a_{1}, a_{2},\ldots\}$ be a set of nonnegative integers satisfying $a_n \leq cn^2$ for all n, where $c>0$ is a real constant. Let $R_A(n)$ denote the number...

conjecture | open | MathDB 页面 | 原文线索

V2-002355 | MATHDB-330376

The Topological Monodromy Conjecture and its strong form

摘录 / EXCERPT: Topological Monodromy Conjecture. Any pole of $Z_f(s)$ yields under exponentiation an eigenvalue of the monodromy operator at some $p\in\operatorname{Var}(f)$. The strong form asse...

conjecture | open | MathDB 页面 | 原文线索

V2-002356 | MATHDB-330385

The prime-torsion conjecture for poset Lie algebra homology

摘录 / EXCERPT: Let $\preceq$ be the poset defining the poset Lie algebra $\frak{gl}_n^\preceq$, and let $H_\ast(\frak{gl}_n^\preceq;\mathbb{Z})$ denote its homology with integer coefficients. For...

conjecture | open | MathDB 页面 | 原文线索

V2-002357 | MATHDB-330399

Type-D0 constituent conjecture for Thompson moonshine modules

摘录 / EXCERPT: Let ${\rm Th}$ be the Thompson sporadic group, let ${\cal F}_3$ be the relevant weight one-half weakly holomorphic modular form, and let W_m be the representation attached to the...

conjecture | open | MathDB 页面 | 原文线索

V2-002358 | MATHDB-330408

Exact valuation conjecture for Somos-4 and Somos-5 sequences

摘录 / EXCERPT: Exact valuation conjecture. For the relevant integers m and l, one has

conjecture | open | MathDB 页面 | 原文线索

V2-002359 | MATHDB-330418

The sweep map bijectivity conjecture for rational Dyck paths

摘录 / EXCERPT: Given coprime positive integers m and n, let $\mathcal D_{m,n}$ be the set of rational (m,n)-Dyck paths, and let Φ be the sweep map that sorts the steps of a path accord...

conjecture | open | MathDB 页面 | 原文线索

V2-002360 | MATHDB-330434

Moment conjecture for Steinhaus random multiplicative functions

摘录 / EXCERPT: Random-multiplicative-function moment conjecture. As $N\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002361 | MATHDB-330444

Kato's finiteness conjecture for integral motives

摘录 / EXCERPT: Fix nonnegative integers $h^{r,w-r}$ with $h^{r,w-r}=0$ for $r<0$ and $r>w$. Let F be a number field, let w be an integer, and let $h(M)$ denote the height of a pure...

conjecture | open | MathDB 页面 | 原文线索

V2-002362 | MATHDB-330447

The Shafarevich finiteness conjecture for motives

摘录 / EXCERPT: Fix nonnegative integers $h^{r,w-r}$ as above and a set S of prime numbers. Let F be the underlying number field, let w be the weight, and let M be a pure $\mathbb Z$-motiv...

conjecture | open | MathDB 页面 | 原文线索

V2-002363 | MATHDB-330449

Oort's lifting conjecture for local cyclic extensions

摘录 / EXCERPT: Let S be the fixed field of $\ker(\pi)$, identified with $k((\overline{u}))$ for a parameter $\overline{u}$. The action of G_F defines a continuous k-linear action τ of...

conjecture | open | MathDB 页面 | 原文线索

V2-002364 | MATHDB-330452

Shafarevich conjecture for complete intersections

摘录 / EXCERPT: Shafarevich conjecture for complete intersections. The set of K-linear isomorphism classes of smooth complete intersections of type T over K with good reduction outside S i...

conjecture | open | MathDB 页面 | 原文线索

V2-002365 | MATHDB-330453

The exact equality between Smarandache primitive numbers and maximal Smarandache deficiency

摘录 / EXCERPT: The exact-equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-002366 | MATHDB-330465

Graham-Lagarias-Mallows-Wilks-Yan local-global conjecture for Apollonian circle packings

摘录 / EXCERPT: Graham-Lagarias-Mallows-Wilks-Yan conjecture. All sufficiently large integers with residues in S occur as curvatures in $\mathcal P$.

conjecture | open | MathDB 页面 | 原文线索

V2-002367 | MATHDB-330514

The nonperiodicity conjecture for pure cubic Jacobi-Perron expansions

摘录 / EXCERPT: Let $x\geq 2$ be a positive integer and set ... The nonperiodicity conjecture. The Jacobi-Perron algorithm expansion of $(m^{1/3},m^{2/3})$ is not periodic. This conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-002368 | MATHDB-330532

The weakly Gorenstein conjecture for Eisenstein Hecke algebras

摘录 / EXCERPT: Let $\mathfrak{H}$ and $\mathfrak{h}$ be the Eisenstein components of the Hida Hecke algebras for modular forms and cusp forms. For every relevant height-one prime $\mathfrak{p}$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-002369 | MATHDB-330572

Erdős's conjecture on primes of prescribed multiplicative order

摘录 / EXCERPT: For an integer a and a positive integer r, define ... where $\operatorname{ord}_p(a)$ is the multiplicative order of a modulo the prime p. Erdős's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002370 | MATHDB-330579

Goodearl's quantum-Poisson spectrum correspondence conjecture

摘录 / EXCERPT: Goodearl's quantum-Poisson correspondence conjecture. The algebra A should belong to a flat family of k-algebras with semiclassical limit $\mathcal{O}(V)$, and there should be...

conjecture | open | MathDB 页面 | 原文线索

V2-002371 | MATHDB-330580

Brown-Goodearl conjecture on the maps between quantum strata

摘录 / EXCERPT: Brown-Goodearl conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-002372 | MATHDB-330588

Rank-two unit Thue inequality conjecture

摘录 / EXCERPT: Rank-two unit conjecture. There exists a positive absolute constant κ such that, whenever n,s,t,x,y,m are integers satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-002373 | MATHDB-330632

Prime-power Olson degree conjecture

摘录 / EXCERPT: An integer d is an Olson degree if every torsion subgroup of a CM elliptic curve defined over a number field of degree d is determined by the curve's CM data as in the paper. P...

conjecture | open | MathDB 页面 | 原文线索

V2-002374 | MATHDB-330645

The multiplicity conjecture for Whitehead groups of $C_{p^n}\times C_{p^n}$

摘录 / EXCERPT: Let p be an odd prime and let i be a positive integer. For $n\geq 2i$, define ... where $E(i)$ is 1 if i is even and 0 if it is odd. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002375 | MATHDB-330654

Rahm-Wendt's refinement of Quillen's conjecture

摘录 / EXCERPT: Let K be a number field. Fix a prime ℓ such that $\zeta_\ell\in K$, and an integer $n<\ell$. Let S be a set of places containing the infinite places and the places lying o...

conjecture | open | MathDB 页面 | 原文线索

V2-002376 | MATHDB-330719

The log abundance conjecture for log canonical pairs

摘录 / EXCERPT: Let (X,Δ) be a log pair, with X normal and Δ an effective formal $\mathbb{Q}$-linear combination of prime divisors. The pair is log canonical if all discrepancies a...

conjecture | open | MathDB 页面 | 原文线索

V2-002377 | MATHDB-330728

Wan-Xiao-Zhang slope conjecture for overconvergent p-adic modular forms

摘录 / EXCERPT: Let p be a prime, let $\mathcal{W}$ be the p-adic weight space, and let $\mathcal{W}_\eta$ be one of its components. For a weight $\kappa\in\mathcal{W}_\eta$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-002378 | MATHDB-330739

Solitary powers conjecture in quadratic integer rings

摘录 / EXCERPT: Solitary powers conjecture. The element p^k is n-powerfully solitary in $\mathcal O_{\mathbb{Q}(\sqrt{d})}$ for every positive integer n. More strongly,...

conjecture | open | MathDB 页面 | 原文线索

V2-002379 | MATHDB-330740

The logarithmic lower-bound conjecture for the Schemmel iteration function R_2

摘录 / EXCERPT: Let R_2 be the function introduced in the paper, and let x be an odd integer greater than 3. The logarithmic lower-bound conjecture. ... The conjecture proposes a lower bound f...

conjecture | open | MathDB 页面 | 原文线索

V2-002380 | MATHDB-330800

Vojta-Lang conjecture for pairs openly of log general type

摘录 / EXCERPT: Let K be a number field and let S be a finite set of places of K. Let (X,D) be a pair openly of log general type defined over K. Vojta-Lang conjecture. The set of S-int...

conjecture | open | MathDB 页面 | 原文线索

V2-002381 | MATHDB-330814

The p-component Rubin-Stark conjecture

摘录 / EXCERPT: The p-component Rubin-Stark conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-002382 | MATHDB-330818

Local explicit Iwasawa main conjecture for Rubin-Stark elements

摘录 / EXCERPT: Local explicit Iwasawa main conjecture. For every height-one prime ideal $\mathfrak{p}$ of Λ,

conjecture | open | MathDB 页面 | 原文线索

V2-002383 | MATHDB-330820

Odlyzko-Stanley conjecture on the growth of p-Stanley sequences

摘录 / EXCERPT: Let p be an odd prime. A p-Stanley sequence $S_p(A)=a_1,a_2,\ldots$ is generated greedily from a p-free set A of nonnegative integers, by adjoining at each step the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-002384 | MATHDB-330836

The Almkvist-Zudilin supercongruence

摘录 / EXCERPT: Let $\mathbb{N}^+$ denote the positive integers, let $a_i(n)$ be the sequence defined by ... and let $p\geq 5$ be prime. The Almkvist-Zudilin supercongruence. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002385 | MATHDB-330837

The higher Almkvist-Zudilin supercongruence

摘录 / EXCERPT: Let $\mathbb{N}^+$ denote the positive integers, and for integers $i\geq 0$ and $n\geq 1$ define ... Let p be a prime satisfying $p>2i$. The higher Almkvist-Zudilin supercongruence...

conjecture | open | MathDB 页面 | 原文线索

V2-002386 | MATHDB-330850

Farhi's three-term representation conjecture for quadratic-floor sequences

摘录 / EXCERPT: Farhi's conjecture. Every natural number can be represented as the sum of three terms of this sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-002387 | MATHDB-330861

Bloch-Kato conjecture for the Asai L-function

摘录 / EXCERPT: Let $K/\mathbf{Q}$ be the quadratic extension, let π be the automorphic representation and let $q>2k$ be a prime at which neither π nor $K/\mathbf{Q}$ is ramified. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002388 | MATHDB-330904

Xia's divisibility conjecture for 3-core partition triples

摘录 / EXCERPT: Let $A_{3}(n)$ denote the number of partition triples of n whose components are 3-cores. For a positive integer j and a prime p, consider the values of $A_{3}(n)$ in the arit...

conjecture | open | MathDB 页面 | 原文线索

V2-002389 | MATHDB-330907

The Browkin-Brzeziński conjecture for zero-sum integers

摘录 / EXCERPT: Let $n>2$ be an integer and let $\epsilon>0$. For integers $a_1,\dots,a_n$ satisfying ... with no proper subset having zero sum and with $\gcd(a_1,\dots,a_n)=1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002390 | MATHDB-330933

Algebraic independence conjecture for finite Bernoulli numbers and Fermat quotients

摘录 / EXCERPT: For $k\ge0$, let ... be the $\calA_1$-Bernoulli numbers, and for $k\ge2$ let ... be the kth $\calA_1$-Fermat quotient. Set $\gb_1=1$. Suppose $n_1,\dots,n_r\in\N$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002391 | MATHDB-330937

Superbity-two dimension conjecture for finite multiple zeta values

摘录 / EXCERPT: For each positive integer w, let $\FMZ_{w,2}$ be the $\Q$-vector space generated by finite multiple zeta values of weight w and superbity two. Define $d_{0,2}=1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002392 | MATHDB-330967

Finiteness conjecture for maximal Cohen-Macaulay elements in the class group

摘录 / EXCERPT: Let R be a complete local Cohen-Macaulay ring satisfying Serre's condition (R_2), meaning that $R_{\mathfrak p}$ is regular for every prime $\mathfrak p$ of height at most two....

conjecture | open | MathDB 页面 | 原文线索

V2-002393 | MATHDB-331032

The indecomposability conjecture for flag Gorenstein* complexes

摘录 / EXCERPT: Indecomposability conjecture. This graded ring is indecomposable, and consequently $\mathcal{Z}_{K}$ is a prime manifold. The analogous assertion is proved in the paper for flag...

conjecture | open | MathDB 页面 | 原文线索

V2-002394 | MATHDB-331037

The Apéry-number prime divisibility proportion conjecture

摘录 / EXCERPT: Let $A(n)$ denote the Apéry numbers, and let a prime divide the sequence if it divides at least one term $A(n)$. The Apéry-number proportion conjecture. The proportion of primes th...

conjecture | open | MathDB 页面 | 原文线索

V2-002395 | MATHDB-331038

The prime divisibility proportion conjecture for sporadic Apéry-like sequences

摘录 / EXCERPT: Let $C(n)$ range over the Apéry-like sequences listed in Tables 2 and 3 of the source, including (η) and the sequences (δ), (α), (ϵ), (ζ), and...

conjecture | open | MathDB 页面 | 原文线索

V2-002396 | MATHDB-331063

Bouniakowski's conjecture on prime values of polynomials

摘录 / EXCERPT: Let $f(x)\in\mathbb Z[x]$ be a non-constant irreducible polynomial such that $f(x)$ is not identically zero modulo any prime p. Bouniakowski's conjecture. The value $f(n)$ is pri...

conjecture | open | MathDB 页面 | 原文线索

V2-002397 | MATHDB-331072

Conjecture on infimum sets for prime approximations

摘录 / EXCERPT: The infimum-set conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-002398 | MATHDB-331075

Chen-Guo-Wang's log-concavity conjecture for the zeta-Gamma function

摘录 / EXCERPT: For $x>6$, define ... where ... is the Riemann zeta function and $\Gamma(x)$ is the Gamma function. Chen-Guo-Wang's conjecture. The function $\theta(x)$ is log-concave on $(6,\infty)$....

conjecture | open | MathDB 页面 | 原文线索

V2-002399 | MATHDB-331077

The derangement analogue of Kurepa's conjecture

摘录 / EXCERPT: Derangement factorial-sum conjecture. For every prime number $p>11$,

conjecture | open | MathDB 页面 | 原文线索

V2-002400 | MATHDB-331095

The prime-square subgroup conjecture for Helly numbers

摘录 / EXCERPT: Let $h(S)$ denote the Helly number of a subset $S\subseteq\mathbb{R}^d$, and let $\mathbb{P}$ be the set of prime numbers. Prime-square Helly conjecture. ... This conjecture predicts...

conjecture | open | MathDB 页面 | 原文线索

V2-002401 | MATHDB-331103

The minimal-polynomial degree conjecture for recursive sequences modulo prime powers

摘录 / EXCERPT: Let p be prime and let p^γ, with $\gamma\ge1$, be a modulus. A minimal polynomial is a polynomial of least degree giving the relevant recursive-sequence relation modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-002402 | MATHDB-331117

Infinitely many prime triples from the cubic parametrization

摘录 / EXCERPT: Let $\mathcal{P}_3$ denote the set of prime triples arising from the cubic parametrization, and let $m(P)$ denote the associated quantity defined for $P\in\mathcal{P}_3$. Infinitud...

conjecture | open | MathDB 页面 | 原文线索

V2-002403 | MATHDB-331118

Infinitely many loops in the Euclid-Mullin graph

摘录 / EXCERPT: Let G_1 be the Euclid-Mullin graph associated with the initial node 1, whose edges and loops are defined by the graph construction in the paper. Infinitely many loops conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-002404 | MATHDB-331126

Analytic properties of representation zeta functions of Heisenberg groups

摘录 / EXCERPT: Let R be a ring which is torsion-free and finitely generated over $\mathbb{Z}$, and let $\mathbf{H}(R)$ be the corresponding Heisenberg group. Write $\alpha(\mathbf{H})$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-002405 | MATHDB-331134

Prime Coxeter number conjecture for dominant numbers of dual fundamental series

摘录 / EXCERPT: Let $\mathfrak{g}$ be the simple Lie algebra considered in the topological ODE construction, with Coxeter number h, rank n, and dual fundamental series $\phi_{a;a}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-002406 | MATHDB-331140

Strong local-global conjecture for periodic points of quadratic polynomials

摘录 / EXCERPT: Let $f\in\mathbf Q[x]$ be a quadratic polynomial, and let $n>3$ be an integer. For a prime p, write $\mathbf Q_p$ for the field of p-adic numbers. A point has period n for...

conjecture | open | MathDB 页面 | 原文线索

V2-002407 | MATHDB-331146

Finiteness conjecture for zero terms in the smallest-representation sequence

摘录 / EXCERPT: Finiteness conjecture. The sequence $s(n)$ contains only finitely many zero terms. This asserts that p_n+k-3 is represented by F_k for some $k\geq3$ for all but finitely many p...

conjecture | open | MathDB 页面 | 原文线索

V2-002408 | MATHDB-331159

Dynamically defined covering-systems conjecture

摘录 / EXCERPT: Dynamically defined covering-systems conjecture. These choices can satisfy both $\max_i\|n_my_i-w_i\|\leq\varepsilon/2$ for every $m=1,\dots,M$, and

conjecture | open | MathDB 页面 | 原文线索

V2-002409 | MATHDB-331188

Weak-order maximality conjecture for affine Weyl group cores

摘录 / EXCERPT: Let $\widetilde{W}$ be a simply-laced affine Weyl group with Coxeter number h, let b be coprime to h, and let $\mathcal{S}_\Phi(b)$ be the relevant Sommers region. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002410 | MATHDB-331204

Best-constant Stolarsky-mean approximation conjecture for the elliptic integral

摘录 / EXCERPT: Let $r^{\prime}=\sqrt{1-r^{2}}$. For $p\neq 0$, let $S_{p,q}(a,b)$ denote the Stolarsky mean, and let $p_{0}\approx 1.763135$ be the unique solution on $(0,9/4]$ of ... that is,...

conjecture | open | MathDB 页面 | 原文线索

V2-002411 | MATHDB-331210

Ledrappier-Young conjecture for quasi-decay of smooth dynamical measures

摘录 / EXCERPT: Let (X,T,μ) be a smooth ergodic measure-preserving dynamical system of dimension d. Let $\lambda_1<\ldots<\lambda_k$ be its Lyapunov exponents, and let $d_1,\ldots,d_k$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-002412 | MATHDB-331218

Conjecture B on the averaged binary Goldbach representation count

摘录 / EXCERPT: Let $\mathbb{P}$ be the set of primes, and let $r_2(n;\mathbb{P})$ denote the number of representations of n as a sum of two primes; let φ be Euler's totient function. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-002413 | MATHDB-331223

The backward orbit conjecture for the powering map over global fields

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{F}_p(t)$, let $\phi:\mathbb{P}^1\to\mathbb{P}^1$ be a rational map of degree $d\geq 2$ defined over K, and let $\overline{K}$ be the sep...

conjecture | open | MathDB 页面 | 原文线索

V2-002414 | MATHDB-331262

Kumar Murty's conjecture on rational Hecke eigenvalues

摘录 / EXCERPT: Let f be a weight 2 normalized cuspidal Hecke eigenform of level $\Gamma_1(N)$ with quadratic coefficient field and without inner twists. Write $a_p(f)$ for its p-th Hecke co...

conjecture | open | MathDB 页面 | 原文线索

V2-002415 | MATHDB-331270

Zywina's surjectivity conjecture for mod-ell representations of elliptic curves

摘录 / EXCERPT: Let E be a non-CM elliptic curve over $\mathbb{Q}$, let j_E be its j-invariant, and let ... For each prime ℓ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002416 | MATHDB-331271

Inversion of adjunction for arithmetic log pairs

摘录 / EXCERPT: Let $C=\operatorname{Spec}(\mathcal{O}_{K})$ be the spectrum of the ring of integers of a number field, let $\pi\colon\mathcal{X}\to C$ be as in the preceding notation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002417 | MATHDB-331276

Arakelov K-moduli conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ denote the proposed moduli object for globally Arakelov K-semistable polarized models, and let M be its coarse moduli space. Arakelov K-moduli conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-002418 | MATHDB-331291

Herman's conjecture on KAM stability of Diophantine elliptic equilibria

摘录 / EXCERPT: Let H be a real-analytic Hamiltonian with an elliptic equilibrium at the origin, and let $\alpha\in\mathbb R^n$ be the frequency vector of the linear part of the flow. Recall tha...

conjecture | open | MathDB 页面 | 原文线索

V2-002419 | MATHDB-331298

The discrete Lambert sum conjecture for prime-power moduli

摘录 / EXCERPT: Let p be an odd prime, $m_p=\operatorname{ord}_p(g)$, and $m_{p^e}=\operatorname{ord}_{p^e}(g)$. For fixed g and c with $p\nmid g$ and $p\nmid c$, define ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-002420 | MATHDB-331306

Conjecture on reciprocal Carmichael-function sums over integers with prescribed prime signatures

摘录 / EXCERPT: Let B, σ, t, m, and ν be as above, and let $\mathcal{K}(t,\sigma)$ be the set of integers with exactly t distinct prime divisors, each having signature matching...

conjecture | open | MathDB 页面 | 原文线索

V2-002421 | MATHDB-331355

The gcd conjecture for cyclotomic factors of a divisibility sequence

摘录 / EXCERPT: Gcd conjecture. If $(b_m,b_n)>1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-002422 | MATHDB-331356

Square-freeness conjecture for the factors of a divisibility sequence

摘录 / EXCERPT: Square-freeness conjecture. If n is square-free, then b_n is square-free.

conjecture | open | MathDB 页面 | 原文线索

V2-002423 | MATHDB-331357

Primitive divisor conjecture with a finite exceptional divisor bound

摘录 / EXCERPT: Primitive divisor conjecture. The term a_n has a primitive prime divisor for all but finitely many n. Moreover, there exists a positive integer M such that, whenever a_n do...

conjecture | open | MathDB 页面 | 原文线索

V2-002424 | MATHDB-331368

The square-root cancellation conjecture for prime values of non-principal Dirichlet characters

摘录 / EXCERPT: Square-root cancellation conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002425 | MATHDB-331369

The square-root cancellation conjecture for Fourier coefficients of cusp forms

摘录 / EXCERPT: Square-root cancellation conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002426 | MATHDB-331383

The conjecture on representations by two quadratic constraints

摘录 / EXCERPT: Representation-count conjecture. The number of such integer solutions is $o((|m|+|n|)^{\epsilon})$ for every $\epsilon>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-002427 | MATHDB-331407

Reciprocity law I for pro-semisimple fundamental groups

摘录 / EXCERPT: Assume that $\dim X=1$. Let P be the weight group of $\Delta_{\hat\Pi}$, let $Q\subset P$ be the subgroup generated by the roots, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002428 | MATHDB-331408

Reciprocity law II for Brauer obstructions

摘录 / EXCERPT: Assume that $\dim X=1$. Let $\rho\in\operatorname{Irr}(\hat\Pi)$, let $E\subset\overline{\mathbb{Q}}$ be the finite field generated by the coefficients of the Frobenius characteris...

conjecture | open | MathDB 页面 | 原文线索

V2-002429 | MATHDB-331409

The zero-congruence conjecture for Chocolate Numbers

摘录 / EXCERPT: Let (B_n) be the sequence of Chocolate Numbers, and let p be a prime. Consider the sequence modulo p. Zero-congruence conjecture. The sequence B_n considered modulo prime...

conjecture | open | MathDB 页面 | 原文线索

V2-002430 | MATHDB-331410

Period-divisibility conjecture for Chocolate Numbers modulo primes

摘录 / EXCERPT: Let (B_n) be the sequence of Chocolate Numbers, and let p be a prime such that $p\ne2$, $p\ne5$, and $p\not\equiv\pm1\pmod{5}$. Period-divisibility conjecture. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-002431 | MATHDB-331469

Cooper-Lam representation formula conjecture for diagonal ternary quadratic forms

摘录 / EXCERPT: Cooper-Lam's conjecture. The number of representations is of the form

conjecture | solved | MathDB 页面 | 原文线索

V2-002432 | MATHDB-331487

Finiteness conjecture for permutation binomials over quadratic finite fields

摘录 / EXCERPT: Let q be a prime power, let $\Bbb F_{q^2}$ be the finite field with q^2 elements, and let $r>2$ be a fixed prime. For $a\in\Bbb F_{q^2}^*$, define ... A polynomial is a permutati...

conjecture | open | MathDB 页面 | 原文线索

V2-002433 | MATHDB-331529

The absence of divergent trajectories in the $3x+1$ problem

摘录 / EXCERPT: Absence of divergent trajectory. For every positive integer n, the trajectory of n is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-002434 | MATHDB-331530

The absence of non-trivial cycles in the $3x+1$ problem

摘录 / EXCERPT: Absence of non-trivial cycle. There do not exist integers $n>2$ and $j>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002435 | MATHDB-331592

Soundararajan's conjecture on zeros of the zeta-function derivative and gaps between zeta zeros

摘录 / EXCERPT: Soundararajan's conjecture. The following two statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-002436 | MATHDB-331595

Nonexistence conjecture for elementary abelian class groups

摘录 / EXCERPT: For an odd prime p, an elementary abelian p-group is a group isomorphic to $(\mathbb{Z}/p\mathbb{Z})^r$. Nonexistence conjecture for elementary abelian class groups. No element...

conjecture | open | MathDB 页面 | 原文线索

V2-002437 | MATHDB-331598

Nonemptiness conjecture for arithmetic triangle triples

摘录 / EXCERPT: For each integer $r\geq 1$, let $\mathcal{T}(r)$ denote the set of r-arithmetic triples. Nonemptiness conjecture. The set $\mathcal{T}(r)$ is nonempty for all $r\geq 1$. The prec...

conjecture | open | MathDB 页面 | 原文线索

V2-002438 | MATHDB-331613

Schinzel-Tijdeman conjecture for the equation $y^2z^3=P(x)$

摘录 / EXCERPT: Let P be a polynomial with rational coefficients. Schinzel-Tijdeman conjecture. If P has at least three simple zeros, then the equation ... has only finitely many solutions in in...

conjecture | open | MathDB 页面 | 原文线索

V2-002439 | MATHDB-331616

Sander's conjecture on rational points of Erdos-Selfridge superelliptic curves

摘录 / EXCERPT: For integers $k\geq 2$ and $\ell\geq 2$, consider the superelliptic curve ... The rational points include the trivial points with $y=0$ and the families ... ... for integers a,b and po...

conjecture | open | MathDB 页面 | 原文线索

V2-002440 | MATHDB-331623

Pollack's lifting conjecture for cusp-form relations

摘录 / EXCERPT: Pollack's lifting conjecture. There exists a linear combination T of brackets of ε_{2i}, containing at least three ε_{2i} with $i>1$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002441 | MATHDB-331624

The stronger Diophantine quadruple conjecture

摘录 / EXCERPT: Stronger Diophantine quadruple conjecture. If $\{a,b,c,d\}$ is a Diophantine quadruple and $d>\max\{a,b,c\}$, then $d=d_+$.

conjecture | open | MathDB 页面 | 原文线索

V2-002442 | MATHDB-331637

Spherical or planar shape conjecture for elastic prime knots

摘录 / EXCERPT: An elastic knot is a knot representative minimizing the bending energy within its knot class, and a prime knot is a knot that cannot be expressed as a nontrivial connected sum. A c...

conjecture | open | MathDB 页面 | 原文线索

V2-002443 | MATHDB-331640

The zeta-invariant inequalities for smooth real functions on the circle

摘录 / EXCERPT: Zeta-invariant inequalities. Inequalities (3.1) hold for every real function $a\in C^\infty({\mathbb S})$.

conjecture | open | MathDB 页面 | 原文线索

V2-002444 | MATHDB-331647

Conjecture on descents in the sequence of rational-square counts

摘录 / EXCERPT: Descent conjecture. For every integer $k>0$, there exists some s such that

conjecture | open | MathDB 页面 | 原文线索

V2-002445 | MATHDB-331671

Zhao's non-divisibility conjecture for gcd-closed sets

摘录 / EXCERPT: Zhao's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-002446 | MATHDB-331672

Non-divisibility conjecture for gcd-closed sets failing condition M

摘录 / EXCERPT: Non-divisibility conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-002447 | MATHDB-331682

Short geodesic conjecture for arithmetic hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a closed, orientable, arithmetic, hyperbolic 3-manifold. The Short Geodesic Conjecture. There is a constant $C>0$, independent of M, such that the length of the shor...

conjecture | open | MathDB 页面 | 原文线索

V2-002448 | MATHDB-331691

Ramakrishnan's refined strong multiplicity-one conjecture

摘录 / EXCERPT: Let F be a number field, let S be a set of places of F, and let π and π' be cuspidal automorphic representations of $\operatorname{GL}_n(\mathbb A_F)$. Suppose their...

conjecture | open | MathDB 页面 | 原文线索

V2-002449 | MATHDB-331701

The Masser-Oesterlé-Szpiro abc conjecture for number fields

摘录 / EXCERPT: Let α be the quadratic algebraic integer used to define the ring $\mathbb{Z}[\alpha]$, and let $(a,b,c)\in\mathbb{Z}[\alpha]^3$ be written in the lowest term, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-002450 | MATHDB-331702

Log K3 integral-point counting conjecture

摘录 / EXCERPT: Integral-point counting conjecture. There exists a Zariski open subset $\mathcal{U} \subseteq \mathcal{X}$ and a constant b such that

conjecture | open | MathDB 页面 | 原文线索

V2-002451 | MATHDB-331714

The order growth conjecture for algebraic numbers modulo primes

摘录 / EXCERPT: The order growth conjecture. For all but finitely many primes p that are not totally split in $K/\mathbb{Q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002452 | MATHDB-331715

The quadratic-case divisor-sum conjecture

摘录 / EXCERPT: Quadratic-case divisor-sum conjecture. As $p\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002453 | MATHDB-331720

Chevyrev-Galbraith's optimal domination conjecture for maximal quaternion orders

摘录 / EXCERPT: Let p be a prime and let B_p be the quaternion algebra over $\mathbb{Q}$ ramified precisely at p and ∞. Let $\mathcal{O}$ and $\mathcal{O}'$ be maximal orders of B_p...

conjecture | open | MathDB 页面 | 原文线索

V2-002454 | MATHDB-331744

The conjecture for the number of left braces of order $4p$

摘录 / EXCERPT: Let $b(n)$ denote the number of non-isomorphic left braces of size n. Let $p>3$ be a prime. The order-$4p$ counting conjecture. ... The statement is presented as a conjecture sugge...

conjecture | open | MathDB 页面 | 原文线索

V2-002455 | MATHDB-331745

The conjecture for the number of left braces of order $9p$

摘录 / EXCERPT: Let $b(n)$ denote the number of non-isomorphic left braces of size n, and let $p>3$ be a prime. The order-$9p$ counting conjecture. ... The conjecture is suggested by the table of...

conjecture | open | MathDB 页面 | 原文线索

V2-002456 | MATHDB-331746

The conjecture for the number of left braces of order p^2q

摘录 / EXCERPT: Let $b(n)$ denote the number of non-isomorphic left braces of size n. Let p,q be prime numbers such that $p<q$ and $q\not\equiv1\bmod p$. The order-p^2q counting conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-002457 | MATHDB-331776

The Mathieu-moonshine twisted black-hole counting conjecture

摘录 / EXCERPT: Twisted black-hole counting conjecture. For every $g\in M_{24}$ and $h\in C_{M_{24}}(g)$,

conjecture | open | MathDB 页面 | 原文线索

V2-002458 | MATHDB-331779

The non-overlapping solitons conjecture

摘录 / EXCERPT: Let p be a prime and let $g_1,g_2\geq 2$. For each g, let $\mathfrak S(p,g)$ be the connected collection of cells that starts at p^g and contains only powers of p larger th...

conjecture | open | MathDB 页面 | 原文线索

V2-002459 | MATHDB-331780

The multiplicative Gilbreath conjecture for the left edge

摘录 / EXCERPT: Let $T_{\mathbb N^*}$ be the infinite triangle generated by iterating the Z-rule on the positive integers, and call its first column the left edge. A number is square-free when n...

conjecture | open | MathDB 页面 | 原文线索

V2-002460 | MATHDB-331781

The formula for $W_{\mathbb N^*}(2^g+1)$

摘录 / EXCERPT: Let $W_{\mathbb N^*}(m)$ denote the m-th entry of the west edge of the triangle generated from the positive integers by the Z-process. For each integer $g\geq 0$, let D_g be...

conjecture | open | MathDB 页面 | 原文线索

V2-002461 | MATHDB-331787

The connected-component decomposition conjecture for \fe(p,2) and \fe'(p,2)

摘录 / EXCERPT: Let $p\ge 5$ be prime. Write $\Omega(p,2)$ and $\Omega'(p,2)$ as disjoint unions of connected graph components, where K_n denotes the complete graph on n vertices, P_n denote...

conjecture | open | MathDB 页面 | 原文线索

V2-002462 | MATHDB-331814

Ravenel's Local-Global Conjecture for formal module Ext groups

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a finite field extension with ring of integers A. For a prime ideal $\mathfrak{p}$ of A, write $A_{\mathfrak{p}}$ for the localization of A at...

conjecture | open | MathDB 页面 | 原文线索

V2-002463 | MATHDB-331815

Ravenel's Global Conjecture for formal module Ext groups, precise form

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a finite Galois extension with ring of integers A. An ideal I of A is n-congruing if, for every $a\in A$, there exists $N\in\mathbb{N}$ such that...

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Ravenel's Global Conjecture for formal module Ext groups, stronger precise form

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a finite Galois extension with ring of integers A. Let J_n denote the minimal n-congruing ideal of A. Ravenel's Global Conjecture, stronger precise fo...

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V2-002465 | MATHDB-331828

Topological invariance and existence conjecture for level-k series

摘录 / EXCERPT: Let K be a knot, let ζ be a root of unity, and let γ be a non-degenerate Neumann-Zagier datum arising from a triangulation of the knot complement. Topological invari...

conjecture | open | MathDB 页面 | 原文线索

V2-002466 | MATHDB-331829

Rank conjecture for X-base Fibonacci-Wieferich primes

摘录 / EXCERPT: Let $X=\{a_1,\ldots,a_m\}$ be a finite set of algebraic numbers, and let K be a number field containing all a_i. An X-base Fibonacci-Wieferich prime is a prime ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-002467 | MATHDB-331830

The rational abc-conjecture

摘录 / EXCERPT: The rational abc-conjecture. For every $\varepsilon>0$, there is a constant $C_\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002468 | MATHDB-331834

Generalized Collatz eventual cyclicity conjecture

摘录 / EXCERPT: Let $d>1$ be an integer, let $m=(m_0,\dots,m_{d-1})$ and $r=(r_0,\dots,r_{d-1})$ be d-tuples of integers with $m_i\neq 0$ and $m_i i+r_i\equiv 0\pmod d$ for every i, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-002469 | MATHDB-331843

Eichler-Shimura-Manin correspondence for newforms of levels 2 and 3

摘录 / EXCERPT: Eichler-Shimura-Manin correspondence. There are isomorphisms defined over $\mathbb C$

conjecture | open | MathDB 页面 | 原文线索

V2-002470 | MATHDB-331844

Generating-series conjecture for newform dimensions at levels 2 and 3

摘录 / EXCERPT: For $N=2,3$, let $\mathcal S_k^{\mathrm{new}}(\Gamma_0(N))^{\pm}$ denote the subspaces of newforms of weight k and level $\Gamma_0(N)$ on which the Atkin-Lehner involution acts b...

conjecture | open | MathDB 页面 | 原文线索

V2-002471 | MATHDB-331847

Non-vanishing conjecture for Bernoulli numbers modulo primes

摘录 / EXCERPT: Non-vanishing conjecture. There exist infinitely many primes p such that

conjecture | open | MathDB 页面 | 原文线索

V2-002472 | MATHDB-331856

Harvey Friedman's conjecture on finite-solution Diophantine equations

摘录 / EXCERPT: Consider the collection of Diophantine equations over the rationals that have only finitely many rational solutions. Harvey Friedman's conjecture. This collection is not recursivel...

conjecture | open | MathDB 页面 | 原文线索

V2-002473 | MATHDB-331857

Noncomputability conjecture for finite-solution Diophantine equations

摘录 / EXCERPT: Consider the collection of Diophantine equations over the rationals that have only finitely many rational solutions. Noncomputability conjecture. This collection is not computable....

conjecture | open | MathDB 页面 | 原文线索

V2-002474 | MATHDB-331858

Computable bound on the number of rational solutions of a Diophantine equation

摘录 / EXCERPT: Let $D(x_1,\ldots,x_p)=0$ be a Diophantine equation, and suppose its set of rational solutions is finite. Computable solution-count bound conjecture. There is an algorithm that tak...

conjecture | open | MathDB 页面 | 原文线索

V2-002475 | MATHDB-331861

Conjecture on reciprocal fractional-part sums for algebraic irrationals

摘录 / EXCERPT: Algebraic-irrational conjecture. For every algebraic $\alpha\in\mathbb{R}\setminus\mathbb{Q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-002476 | MATHDB-331862

Inhomogeneous Gallagher conjecture

摘录 / EXCERPT: Inhomogeneous Gallagher conjecture. If the sum specified in the source is divergent, then for almost all $(\alpha,\beta)\in\mathbb{R}^2$, the inequality holds for infinitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-002477 | MATHDB-331866

Prime-divisor conjecture for periodic points of cyclic SDS maps

摘录 / EXCERPT: Prime-divisor conjecture. Every prime divisor of $\lvert\operatorname{Per}_1(F_{m,n}^r)\rvert$ divides m, and every prime divisor of $\lvert\operatorname{Per}_1(G_{m,n}^r)\rvert$...

conjecture | open | MathDB 页面 | 原文线索

V2-002478 | MATHDB-331896

Rodriguez Villegas's Mahler measure conjecture for a four-variable linear polynomial

摘录 / EXCERPT: For a nonzero Laurent polynomial P, let its logarithmic Mahler measure be ... Let f be the modular form appearing in the surrounding discussion, and consider the four-variable po...

conjecture | open | MathDB 页面 | 原文线索

V2-002479 | MATHDB-331902

Sharpness conjecture for local-point bounds on fibrations over projective space

摘录 / EXCERPT: Let k be a number field and let $\pi:X\to\mathbb{P}_k^n$ be as in Theorem. Assume that at least one fibre of π is everywhere locally solvable and that the fibre of π over...

conjecture | open | MathDB 页面 | 原文线索

V2-002480 | MATHDB-331924

Generalized Schur-Siegel-Smyth lower-bound conjecture for coefficient profiles

摘录 / EXCERPT: Generalized Schur-Siegel-Smyth conjecture. Fix $\epsilon>0$. Then there exist only finitely many $f(x)\in\mathcal{F}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002481 | MATHDB-331938

Goodson's point-counting conjecture for Dwork hypersurfaces

摘录 / EXCERPT: Let d be an odd prime, let p be a prime such that $p\not\equiv 1 \pmod{d}$, and let ... be the Dwork hypersurface over $\mathbb{F}_p$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-002482 | MATHDB-331995

Hochster-Huneke's (LC) conjecture

摘录 / EXCERPT: Throughout, let $R:=\bigoplus_{n\geq 0}R_n$ be a standard graded algebra over a field of prime characteristic $p>0$, let $\mathfrak{m}:=\bigoplus_{n>0}R_n$ be its irrelevant ideal,...

conjecture | open | MathDB 页面 | 原文线索

V2-002483 | MATHDB-332009

Restricted sumset formula for the small-set regime

摘录 / EXCERPT: Let G be a finite abelian group of order n, let p be the smallest prime divisor of n, and let $\rho\hat{\;}(G,m,h)$ denote the minimum restricted h-fold sumset size amon...

conjecture | open | MathDB 页面 | 原文线索

V2-002484 | MATHDB-332010

Inverse restricted-sumset conjecture above the small-set threshold

摘录 / EXCERPT: Let G be a finite abelian group of order n, let p be the smallest prime divisor of n, and let A be an m-subset of G. Inverse restricted-sumset conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-002485 | MATHDB-332011

Bajnok and Matzke's exceptional signed-sumset conjecture for elementary abelian groups

摘录 / EXCERPT: Let p be an odd prime and let $\mathbb{Z}_p^r$ be an elementary abelian group. Define $\delta=0$ if $p-1$ is divisible by h and $\delta=1$ otherwise. Let k be the largest int...

conjecture | open | MathDB 页面 | 原文线索

V2-002486 | MATHDB-332041

Conjecture on unit-modulus conjugates and representations of zero

摘录 / EXCERPT: Unit-modulus conjugate conjecture. If β has a conjugate of modulus 1, then $W_{\beta,\lceil \beta\rceil-1}$ is not recognizable by a finite automaton.

conjecture | open | MathDB 页面 | 原文线索

V2-002487 | MATHDB-332052

The tensor-product conjecture for torsion subalgebras of integral-dynamics Kirchberg algebras

摘录 / EXCERPT: Let $S\subset\mathbb{N}^{\times}\setminus\{1\}$ be a family of relatively prime numbers with $\lvert S\rvert\geq 2$. Let $\mathcal{A}_S$ be the torsion subalgebra associated to S...

conjecture | open | MathDB 页面 | 原文线索

V2-002488 | MATHDB-332061

Caporaso-Harris-Mazur uniformity conjecture for rational points on curves

摘录 / EXCERPT: Let g be a nonnegative integer and let K be a number field. For a smooth projective curve C of genus $g\thinspace\geq\thinspace2$ defined over K, write $C(K)$ for its set o...

conjecture | open | MathDB 页面 | 原文线索

V2-002489 | MATHDB-332062

Rank-bounded uniformity conjecture for rational points on curves

摘录 / EXCERPT: Let g and m be nonnegative integers and let K be a number field. For a smooth projective curve C of genus $g\geq2$ defined over K, suppose that the Mordell-Weil rank of i...

open_problem | open | MathDB 页面 | 原文线索

V2-002490 | MATHDB-332068

The CK(n,m)-domain existence conjecture for $m=2$ and $m=3$

摘录 / EXCERPT: Let n be a positive integer. A primefree CK$(n,m;0)$-domain is a primefree atomic domain of the indicated CK type with parameters n,m and auxiliary parameter 0. The CK(n,m)...

conjecture | open | MathDB 页面 | 原文线索

V2-002491 | MATHDB-332126

Finiteness conjecture for Fano threefolds with good reduction

摘录 / EXCERPT: Finiteness conjecture. The set of K-isomorphism classes of Fano threefolds over K of Picard number 1 with good reduction outside of S is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-002492 | MATHDB-332127

Böcherer's conjecture for degree-two Siegel cusp forms

摘录 / EXCERPT: Böcherer's conjecture. There exists a constant c_f depending only on f such that, for every such K,

conjecture | open | MathDB 页面 | 原文线索

V2-002493 | MATHDB-332128

Refined Böcherer conjecture for class-group characters

摘录 / EXCERPT: Refined Böcherer conjecture. For every such K and Λ,

conjecture | open | MathDB 页面 | 原文线索

V2-002494 | MATHDB-332145

Cohen-Lenstra conjectural density for 8-rank in the case $d=-4$

摘录 / EXCERPT: Let $d=-4$. For primes p,q with $pq\to\text{?}$, write $\mathrm{Cl}(dpq)$ for the relevant narrow class group, and let rk_4 and rk_8 denote its 4-rank and 8...

conjecture | open | MathDB 页面 | 原文线索

V2-002495 | MATHDB-332152

Distinct prime-order class-size conjecture for finite simple groups

摘录 / EXCERPT: Distinct prime-order class-size conjecture. For any two prime divisors p and q of the order of G,

conjecture | open | MathDB 页面 | 原文线索

V2-002496 | MATHDB-332187

Nonvanishing conjecture for signed p-adic L-functions

摘录 / EXCERPT: Let f be a modular form as above, let p be a good supersingular prime, and let ω^i denote the relevant character component. The signed and completed signed p-adic L-...

conjecture | open | MathDB 页面 | 原文线索

V2-002497 | MATHDB-332203

The conjecture on linear independence of successive best approximations

摘录 / EXCERPT: Let $n\geq1$ and let ζ be a real transcendental number. A best approximation for (n,ζ) is a polynomial in $\mathbb{Z}[X]$ of degree at most n for which no polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-002498 | MATHDB-332211

Eisenbud-Mazur conjecture on symbolic squares of prime ideals

摘录 / EXCERPT: Let $P\subset\mathbb C[[x_1,\ldots,x_N]]$ be a prime ideal in the ring of formal power series, and let $M=\langle x_1,\ldots,x_N\rangle$. Eisenbud-Mazur conjecture. The second symb...

conjecture | open | MathDB 页面 | 原文线索

V2-002499 | MATHDB-332234

Folklore conjecture on algebraic badly approximable numbers

摘录 / EXCERPT: Folklore conjecture. The only algebraic irrationals that belong to $\mathbf{Bad}$ are the quadratic irrationals.

conjecture | open | MathDB 页面 | 原文线索

V2-002500 | MATHDB-332235

Algebraic continued-fraction partial-quotient conjecture

摘录 / EXCERPT: Algebraic partial-quotient conjecture. For every such α,

conjecture | open | MathDB 页面 | 原文线索

V2-002501 | MATHDB-332236

Khintchine's theorem on fibers

摘录 / EXCERPT: Fiber Khintchine conjecture. If $\psi\colon\mathbb{N}\to\mathbb{R}^+$ is monotonic and $\alpha\in\mathbb{R}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-002502 | MATHDB-332252

The product-dimension conjecture for one-dimensional Diophantine approximation sets

摘录 / EXCERPT: Let $s\geq 1$ be an integer, and let $\eta_{1},\ldots,\eta_{s}$ be real parameters satisfying $\eta_i\geq 1$ for every i. For each i, let $\mathscr{H}^{1}_{\eta_i}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-002503 | MATHDB-332253

Conjecture on the spectrum of the Veronese-curve exponent

摘录 / EXCERPT: Let $k\geq 1$ be an integer. For a real number ζ, write ... and consider the exponent χ_k restricted to points on the Veronese curve. Spectrum conjecture. The spectrum of...

conjecture | open | MathDB 页面 | 原文线索

V2-002504 | MATHDB-332254

The twin-prime distribution conjecture in arithmetic progressions

摘录 / EXCERPT: Let a,q be positive integers such that $(a(a+2),q)=1$. Let Π_2 denote the twin-prime constant, let $\varphi_2(q)$ denote the corresponding arithmetic-progression factor, and...

conjecture | open | MathDB 页面 | 原文线索

V2-002505 | MATHDB-332281

Conjecture on arithmetic realizations with arbitrarily large mu-invariant

摘录 / EXCERPT: Let p be a prime and let Γ be a uniform pro-p group. Suppose that Γ has a non-trivial fixed-point-free automorphism σ of order m coprime to p; in parti...

conjecture | open | MathDB 页面 | 原文线索

V2-002506 | MATHDB-332282

Conjecture on cyclic extensions with p-rational total field

摘录 / EXCERPT: Let p be a prime and let $m\geq 1$ be an integer coprime to p. A number field is p-rational when it has the standard p-rationality property used in the paper. Cyclic p-rati...

conjecture | open | MathDB 页面 | 原文线索

V2-002507 | MATHDB-332283

Cohen-Lenstra-inspired conjecture on p-rational cyclic extensions

摘录 / EXCERPT: Fix a prime $p\geq 5$ and an integer $m>1$ not divisible by p. Let K_0 be a p-rational quadratic imaginary field, and let $\mathcal F_m(K_0)$ be the family of cyclic extensio...

conjecture | open | MathDB 页面 | 原文线索

V2-002508 | MATHDB-332305

Consecutive Piatetski-Shapiro primes conjecture

摘录 / EXCERPT: Let k_0 be a positive integer and let $c>1$ be non-integral. The Piatetski-Shapiro sequence is given by the integer parts $[n^c]$ for positive integers n. Consecutive Piatetski...

conjecture | open | MathDB 页面 | 原文线索

V2-002509 | MATHDB-332310

Sun's permutation conjecture for nonvanishing products modulo n

摘录 / EXCERPT: Sun's conjecture. There exists a permutation σ of $\{1,\ldots,n-1\}$ such that $n\nmid s m_{\sigma(s)}$ for every $s=1,\ldots,n-1$. Theorem 1.1 in the paper confirms this co...

conjecture | open | MathDB 页面 | 原文线索

V2-002510 | MATHDB-332349

Torsion-homology conjecture for strongly isospectral hyperbolic 3-manifolds

摘录 / EXCERPT: Torsion-homology conjecture. For every prime number p, there exist strongly isospectral closed hyperbolic 3-manifolds M_1 and M_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-002511 | MATHDB-332373

Factorial decay conjecture for roots of sparse polynomials

摘录 / EXCERPT: Factorial decay conjecture. There exists a constant $\gamma>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002512 | MATHDB-332384

Gross's leading term conjecture

摘录 / EXCERPT: Let $(H/F,S,T,(v_i)_i,(w_i)_i)$ be as above, let K/F be a finite abelian extension unramified outside S with $H\subset K$ and torsion-free $\mathcal{O}_{K,S,T}^{\times}$, and p...

conjecture | open | MathDB 页面 | 原文线索

V2-002513 | MATHDB-332385

Order-of-vanishing conjecture for Gross's conjecture

摘录 / EXCERPT: With K/F, G, V, and $D'=\prod_{v\in V}I_{G_v,G}\subset\mathbb{Z}[G]$ as above, let $\Theta_{K,S,T}$ be the Stickelberger element. Order-of-vanishing conjecture. ... This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-002514 | MATHDB-332386

Shintani construction conjecture for Rubin-Stark elements

摘录 / EXCERPT: Assume that (S,T,V) satisfies (C1), let $\hat{\Theta}_{S,T,V,J}\in D/I_{F^{\times}}D$ be the Shintani-datum element, and let $\hat{R}_{H/F,S,T,J}$ be the corresponding regulator....

conjecture | open | MathDB 页面 | 原文线索

V2-002515 | MATHDB-332387

Inverse-limit Shintani-Rubin-Stark conjecture

摘录 / EXCERPT: Assume that (S,T,V) satisfies (C1), and let $\hat{\Theta}_{S,T,V}$, p_H, and $\hat{R}_{H/F,S,T}$ be the inverse-limit Shintani element, projection, and regulator defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-002516 | MATHDB-332388

Localization conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let $\epsilon_{H,S,T,V}$ be the Rubin-Stark element and let $\epsilon_{H,S,T,V}^{\mathrm{an}}$ be the analytic Rubin-Stark element defined in the inverse-limit localized module. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-002517 | MATHDB-332409

Conjecture on representations by three signed triangular-number forms

摘录 / EXCERPT: Let $\mathbb{N}_0$ denote the set of nonnegative integers. Consider the two forms ... and ... where $a,b,c\in\mathbb{N}_0$. Three-form representation conjecture. Every natural number o...

conjecture | open | MathDB 页面 | 原文线索

V2-002518 | MATHDB-332426

Existence conjecture for irreducible polynomials with prescribed finite-field coefficients

摘录 / EXCERPT: Let q be a prime power, and let k,l be positive integers. Write $c\mathcal{F}_k(q)$ for the set of elements of degree k over $c\mathbb{F}_q$. Existence conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-002519 | MATHDB-332429

Finiteness conjecture for power-sum congruence solutions

摘录 / EXCERPT: Finiteness conjecture. For every $m\in\mathbb{N}$, the set of solutions to the congruence

conjecture | open | MathDB 页面 | 原文线索

V2-002520 | MATHDB-332430

The least square-free primitive root bound beyond 409

摘录 / EXCERPT: Let $g^{\square}(p)$ denote the least square-free primitive root modulo a prime p. Least square-free primitive root conjecture. For all $p>409$ we have ... The preceding explicit b...

conjecture | open | MathDB 页面 | 原文线索

V2-002521 | MATHDB-332437

The quasi-irreducibility criterion for holomorphic eta quotients

摘录 / EXCERPT: Let f be a holomorphic eta quotient of level N. It is quasi-irreducible if it is not factorizable on $\Gamma_0(N)$, whereas it is irreducible if its only factors are 1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-002522 | MATHDB-332438

The rescaling irreducibility conjecture for holomorphic eta quotients

摘录 / EXCERPT: Let f be an irreducible holomorphic eta quotient, and for a positive integer ν let its rescaling be $f_\nu(z)=f(\nu z)$. Rescaling irreducibility conjecture. The rescalings o...

conjecture | open | MathDB 页面 | 原文线索

V2-002523 | MATHDB-332439

The Atkin-Lehner irreducibility conjecture for holomorphic eta quotients

摘录 / EXCERPT: Let f be an irreducible holomorphic eta quotient, and let an Atkin-Lehner involution act on eta quotients in the usual way. Atkin-Lehner irreducibility conjecture. The images of...

conjecture | open | MathDB 页面 | 原文线索

V2-002524 | MATHDB-332440

The extract criterion for irreducibility of holomorphic eta quotients

摘录 / EXCERPT: Call an eta quotient primitive if it is not a rescaling of another eta quotient by a positive integer. For an eta quotient f, its extract is the primitive eta quotient f_0 of w...

conjecture | open | MathDB 页面 | 原文线索

V2-002525 | MATHDB-332525

Skalba's density conjecture for primes dividing binary sums

摘录 / EXCERPT: Let T_k denote the set of primes p dividing an integer of the form $2^{a_1}+\dots+2^{a_k}+1$, where $a_1,\ldots,a_k\in\mathbb{N}$, and let $T_k(X)$ be the number of such primes...

conjecture | open | MathDB 页面 | 原文线索

V2-002526 | MATHDB-332549

Andrade-Miller-Pratt-Trinh conjecture on infinite non-trivial F-sets

摘录 / EXCERPT: Throughout, let q be a prime power, let I_q be the set of all monic irreducible polynomials in $\mathbb{F}_q[x]$, and call a subset A of I_q an F-set if, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-002527 | MATHDB-332554

Degree and limiting-value conjecture for submodule zeta functions of nilpotent endomorphism algebras

摘录 / EXCERPT: Degree and limiting-value conjecture. If $({\mathcal{L}},{\mathcal{E}})$ is almost uniform, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-002528 | MATHDB-332559

Minimalist non-vanishing conjecture for anticyclotomic twists of elliptic modular forms

摘录 / EXCERPT: Let f be an eigenform that is non-exceptional in the sense of Cornut and Vatsal. Let K be an imaginary quadratic field, let p be a prime, and let χ range over ring class...

conjecture | open | MathDB 页面 | 原文线索

V2-002529 | MATHDB-332592

The form conjecture for odd norm-perfect Eisenstein integers

摘录 / EXCERPT: Odd norm-perfect form conjecture. The integer α must have the form

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V2-002530 | MATHDB-332599

Horak's uniqueness conjecture for prime-order crosses

摘录 / EXCERPT: The n-cross is the cluster in $\mathbb{R}^{n}$ consisting of $2n+1$ unit cubes: a central cube and its reflections in all faces; its centers form...

conjecture | open | MathDB 页面 | 原文线索

V2-002531 | MATHDB-332600

The prime-size tile uniqueness and lattice-tiling conjecture

摘录 / EXCERPT: Let q be prime and let ... where $\{\mathbf{v}_{1},\dots,\mathbf{v}_{q-1}\}$ generates $\mathbb{Z}^{n}$. Suppose that V tiles $\mathbb{Z}^{n}$ by translates. Prime-size tile conj...

conjecture | open | MathDB 页面 | 原文线索

V2-002532 | MATHDB-332606

Conjecture on avoiding primes for elementary symmetric Boolean functions

摘录 / EXCERPT: Let $\sigma_{n,k}$ be the elementary symmetric Boolean function of degree k in n variables, and let $S(\sigma_{n,k})$ denote its exponential sum. Suppose that k is a natural...

conjecture | open | MathDB 页面 | 原文线索

V2-002533 | MATHDB-332621

Greenberg-Benois trivial zeros conjecture for p-adic L-functions

摘录 / EXCERPT: Let M be an irreducible motive, pure of weight 0 over $\mathbb{Q}$, such that $s=0$ is a critical integer in the sense of Deligne and $L(M,0)\neq 0$. Let p be a prime, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002534 | MATHDB-332635

Boundary-zero conjecture for products of Eisenstein series

摘录 / EXCERPT: Boundary-zero conjecture. All the zeros of $\Delta_{k,\ell}$ lying in the standard fundamental domain are on the boundary, $|z|=1$ or $x=\pm1/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-002535 | MATHDB-332636

Arc-zero conjecture for \(\Delta_{k,4}\), \(\Delta_{k,6}\), and \(\Delta_{k,8}\)

摘录 / EXCERPT: Arc-zero conjecture. For $k\geq14$, all of the zeros of $\Delta_{k,4}$, $\Delta_{k,6}$, and $\Delta_{k,8}$ in the fundamental domain lie on the arc $|z|=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-002536 | MATHDB-332645

The growth-ratio conjecture for van der Waerden numbers in the number of colours

摘录 / EXCERPT: Let $W(r,k)$ denote the van der Waerden number, and write $W(r,k) = (r)^{n_r}$ and $W(r+1,k)=(r+1)^{n_{r+1}}$ for the corresponding exponents. Growth-ratio conjecture. As r grows...

conjecture | open | MathDB 页面 | 原文线索

V2-002537 | MATHDB-332646

The quadratic growth-ratio conjecture for van der Waerden numbers in the progression length

摘录 / EXCERPT: Let $W(r,k)$ denote the van der Waerden number. Quadratic growth-ratio conjecture. As k grows large with $k>r$, ... This is the simplified form obtained in the source under the add...

conjecture | open | MathDB 页面 | 原文线索

V2-002538 | MATHDB-332647

Integrality and nonnegativity of the hypergeometric coefficients $X^{(d)}_{m,h}(n;\mu)$

摘录 / EXCERPT: The quantities $X^{(d)}_{m,h}(n;\mu)$ are defined by the explicit formula preceding these observations, for the parameters and partition μ used there. Integrality conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-002539 | MATHDB-332678

Rationality conjecture for Diophantine exponents of nilpotent Lie groups

摘录 / EXCERPT: Let G be a connected nilpotent real Lie group endowed with a left-invariant geodesic metric, and let $\widehat{\beta}_k$ denote its Diophantine exponent on k letters. Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-002540 | MATHDB-332779

Conrey-Keating shifted-divisor correlation conjecture

摘录 / EXCERPT: Let A and B be sets of complex shifts, with $|b1|,|b2|<\epsilon$ for all $\alpha\in A$ and $\beta\in B$, where $\epsilon>0$ is fixed and small. Define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002541 | MATHDB-332800

Rank-one conjecture for vectorial modular forms associated to tensor-product representations

摘录 / EXCERPT: We consider $\mathbb F_q$-algebra representations $\sigma_1,\ldots,\sigma_s:A\rightarrow\operatorname{Mat}_{d\times d}(\boldsymbol{K})$ and set $\sigma=\sigma_1\cdots\sigma_s$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002542 | MATHDB-332812

The conjecture that 8 is not in A_4

摘录 / EXCERPT: Let A_4 be the set of positive integers represented by sums of four cyclic ratios. The conjecture on A_4. There exists a positive integer $k \geq 4$ such that $k \notin A_4$. I...

conjecture | open | MathDB 页面 | 原文线索

V2-002543 | MATHDB-332819

Noguchi's lower-bound conjecture for α_n

摘录 / EXCERPT: For each relevant integer n, let d_n denote the dimension parameter associated with the poset P_n, and let α_n be defined by ... Here $\gg$ denotes a lower bound up to a...

conjecture | open | MathDB 页面 | 原文线索

V2-002544 | MATHDB-332834

The strong Neuwirth conjecture for prime knots

摘录 / EXCERPT: Strong Neuwirth conjecture. There exists an essential non-orientable spanning surface for K.

conjecture | open | MathDB 页面 | 原文线索

V2-002545 | MATHDB-332850

Cilleruelo-Granville conjecture on lattice points in short arcs

摘录 / EXCERPT: Cilleruelo-Granville conjecture. For every $\delta>0$, there exists a constant C_{δ} such that every arc of length $R^{1-\delta}$ contains at most C_{δ} lattice point...

conjecture | open | MathDB 页面 | 原文线索

V2-002546 | MATHDB-332875

The general abc conjecture for finitely many points on the projective line

摘录 / EXCERPT: Let k be a number field and let S be a finite set of places of k containing the archimedean places. Let $\alpha_1,\ldots,\alpha_q\in\mathbb{P}^1(k)$ be distinct points, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002547 | MATHDB-332876

Maeda's conjecture for cuspidal eigenforms on \(\operatorname{SL}_2(\mathbf Z)\)

摘录 / EXCERPT: We specialize to $M=1$, $K=\mathbf Q$, and trivial character. Let $P_k(\operatorname{SL}_2(\mathbf Z))$ denote the set of normalized cuspidal Hecke eigenforms of weight k for...

conjecture | open | MathDB 页面 | 原文线索

V2-002548 | MATHDB-332922

Polynomiality conjecture for medial quasigroups of prime-power order

摘录 / EXCERPT: Let k be any natural number, let p be a prime, and write $mq(\mathbb Z_p^k)$ for the number of medial quasigroup structures on the elementary abelian group $\mathbb Z_p^k$, whi...

conjecture | open | MathDB 页面 | 原文线索

V2-002549 | MATHDB-332927

Boston-Bush-Hajir conjecture for nonabelian class tower groups

摘录 / EXCERPT: Fix an odd prime p. Let D_X be the set of imaginary quadratic fields of absolute discriminant less than X. For each such field K, let G_K be the Galois group of its maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-002550 | MATHDB-332930

Nicholson's conjecture on ratios of consecutive primes

摘录 / EXCERPT: Let p_n be the nth prime and let $\log$ denote the natural logarithm. Nicholson's conjecture. For every integer $n \geq 5$, ... This is one of several conjectures related to Firo...

conjecture | open | MathDB 页面 | 原文线索

V2-002551 | MATHDB-332931

Farhadian's conjecture on powers of consecutive-prime ratios

摘录 / EXCERPT: Let p_n be the nth prime, let $[x]$ denote the integer-part expression used in the source, and let $n \geq 5$. Farhadian's conjecture. For every $n \geq 5$, ... This is the third...

conjecture | open | MathDB 页面 | 原文线索

V2-002552 | MATHDB-332953

The bounded-degree torsion sparsity conjecture for Jacobian families

摘录 / EXCERPT: Jacobian torsion sparsity conjecture. The set ${\mathbf{T}}(d)$ is not Zariski dense in S. By the Cadoret-Tamagawa result cited in the paper, restricting the bounded-degree torsi...

conjecture | open | MathDB 页面 | 原文线索

V2-002553 | MATHDB-332961

Point-count conjecture for the double octic O_5

摘录 / EXCERPT: Let O_5 be the double cover of P^3 branched over the octic displayed in the source. For each prime $p>2$, let a_p denote the eigenvalue for T_p of the newform of weight 4...

conjecture | open | MathDB 页面 | 原文线索

V2-002554 | MATHDB-332962

Point-count conjecture for the double octic O_6

摘录 / EXCERPT: Let O_6 be the double octic defined by the equation displayed in the source, and define ... For each prime $p>3$, let a_p be the eigenvalue of T_p for the newform of weight 4...

conjecture | open | MathDB 页面 | 原文线索

V2-002555 | MATHDB-332964

Algorithmic Mordell conjecture

摘录 / EXCERPT: Let K be a number field, let X be a smooth projective geometrically connected curve over K of genus at least two, and let L be a number field over K. Algorithmic Mordell...

conjecture | open | MathDB 页面 | 原文线索

V2-002556 | MATHDB-332970

The conjecture on infinitely many trivially squared consecutive prime pairs

摘录 / EXCERPT: Trivially squared prime-pair conjecture. There are infinitely many consecutive primes p and q ($p>q$) which are trivially squared.

conjecture | open | MathDB 页面 | 原文线索

V2-002557 | MATHDB-332971

The conjecture on infinitely many nontrivially squared consecutive prime pairs

摘录 / EXCERPT: Nontrivially squared prime-pair conjecture. There are infinitely many consecutive primes p and q ($p>q$) which are nontrivially squared.

conjecture | open | MathDB 页面 | 原文线索

V2-002558 | MATHDB-332972

The finiteness conjecture for square differences of consecutive prime powers

摘录 / EXCERPT: Finiteness conjecture for consecutive primes. For any pair of consecutive primes p and q ($p>q$), there are only a finite number of pairs of natural numbers (x,y) such that...

conjecture | open | MathDB 页面 | 原文线索

V2-002559 | MATHDB-332979

Correlation conjecture for shifted values of $I_{\alpha,\gamma}(n)$

摘录 / EXCERPT: Let $\alpha,\beta,\gamma,\delta$ be complex parameters, let h be a positive integer, and define $I_{\alpha,\gamma}(n)=\sum_{de=n}\mu(e)d^{-\alpha}e^{-\gamma}$. Let $r_q(h)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002560 | MATHDB-333011

The prime k-tuple conjecture

摘录 / EXCERPT: Prime k-tuple conjecture. For every admissible set, there are infinitely many integers n such that

conjecture | open | MathDB 页面 | 原文线索

V2-002561 | MATHDB-333026

Prime-set determination conjecture for complete bipartite graphs

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with two partite sets of n vertices. A minimal coprime labeling assigns distinct labels from $\{1,2,\ldots,k\}$ to its vertices, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-002562 | MATHDB-333099

Kaneko-Yamamoto conjecture on integral-series relations for multiple zeta values

摘录 / EXCERPT: Kaneko-Yamamoto conjecture. Any linear dependency of multiple zeta values over $\mathbb{Q}$ can be deduced from for suitable indices $\boldsymbol{k}$ and $\boldsymbol{l}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002563 | MATHDB-333102

Nathanson's conjecture on tiling intervals with two repeated gap lengths

摘录 / EXCERPT: Let k and l be positive integers, and let $p_1,\ldots,p_k,q_1,\ldots,q_l$ be positive integer gap lengths. A set of $k+l+1$ integers has gap sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-002564 | MATHDB-333120

Projective-generator pseudo-trace isomorphism conjecture for a vertex operator algebra

摘录 / EXCERPT: Let $\operatorname{Rep}(\mathcal{V})$ be the representation category of a vertex operator algebra $\mathcal{V}$, let G be a projective generator of this category, let E be its...

conjecture | open | MathDB 页面 | 原文线索

V2-002565 | MATHDB-333125

The logarithmically averaged Chowla conjecture

摘录 / EXCERPT: Let $k\geq 1$, let $a_1,\dots,a_k$ be natural numbers, and let $b_1,\dots,b_k$ be distinct nonnegative integers satisfying ... for $1\leq i<j\leq k$. Let λ denote the Liouvil...

conjecture | open | MathDB 页面 | 原文线索

V2-002566 | MATHDB-333131

Horizontal-line finiteness conjecture for the Rossby wavenumber set

摘录 / EXCERPT: Let Λ be the wavenumber set, whose elements are pairs (a,b). For positive a,b, consider resonant Rossby waves with fixed meridional wavenumber b. Horizontal-line fini...

conjecture | open | MathDB 页面 | 原文线索

V2-002567 | MATHDB-333132

Infinitude conjecture for vertical lines in the primitive Rossby wavevector set

摘录 / EXCERPT: Let Λ' be the primitive resonant wavevector set, consisting of integer pairs (a,b). Fix a nonzero integer a and consider the integer values b for which (a,b) is pri...

conjecture | open | MathDB 页面 | 原文线索

V2-002568 | MATHDB-333135

Asymptotic conjecture for short weighted averages of prime-pair counts

摘录 / EXCERPT: Short-average prime-pair conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002569 | MATHDB-333151

Igusa cusp-form formula for K3 times an elliptic curve

摘录 / EXCERPT: Let $X=S\times E$, where S is a nonsingular projective K3 surface and E is a nonsingular elliptic curve. For a primitive curve class $\beta_h\in H_2(S,\mathbb{Z})$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-002570 | MATHDB-333157

Bloch-Kato Tamagawa number conjecture for modular forms

摘录 / EXCERPT: Let f be a newform of even weight k, let $0\leq j\leq k-2$, and assume $L(f,j+1)\neq0$. Let $\Omega^{(-1)^{j+1}}$ denote the relevant period, u_{j+1} the associated unit fact...

conjecture | open | MathDB 页面 | 原文线索

V2-002571 | MATHDB-333172

Z.-W. Sun's supercongruences for truncated binomial hypergeometric sums

摘录 / EXCERPT: Let $p\ge 5$ be a prime and let n be a positive integer. Let E_m denote the m-th Euler number, and let $E_m(x)$ and $B_m(x)$ denote the Euler and Bernoulli polynomials of deg...

conjecture | open | MathDB 页面 | 原文线索

V2-002572 | MATHDB-333193

Conjecture on logarithmic lower bounds for p-adic valuations of elementary symmetric harmonic sums

摘录 / EXCERPT: For integers $n\geq k\geq1$, define ... and let ν_p denote the p-adic valuation. The logarithmic lower-bound conjecture. For any prime number p and any integer $k\geq1$, ther...

conjecture | open | MathDB 页面 | 原文线索

V2-002573 | MATHDB-333242

Xiong's stronger positivity conjecture for prime-indexed overpartition series

摘录 / EXCERPT: Let $h(q)$ be the q-series ... let $(q)_\infty:=\prod_{n=1}^{\infty}(1-q^n)$, and let $p_1>p_2\geq 2$ be primes. Xiong's stronger positivity conjecture. The function ... has positive...

conjecture | open | MathDB 页面 | 原文线索

V2-002574 | MATHDB-333250

Conjecture on the unbounded and recurrent values of $h(n)$

摘录 / EXCERPT: Conjecture on the values of $h(n)$. The function $h(n)$ attains each positive integer infinitely many times. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-002575 | MATHDB-333251

Conjecture on the zero coefficients of the half Lerch sum

摘录 / EXCERPT: Conjecture on zero coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-002576 | MATHDB-333252

Conjecture on the coefficients equal to two of the half Lerch sum

摘录 / EXCERPT: Conjecture on coefficients equal to two.

conjecture | open | MathDB 页面 | 原文线索

V2-002577 | MATHDB-333253

Conjecture on the coefficients equal to one of the half Lerch sum

摘录 / EXCERPT: Conjecture on coefficients equal to one.

conjecture | open | MathDB 页面 | 原文线索

V2-002578 | MATHDB-333259

The field-of-rationality lower-bound conjecture for positive-depth supercuspidals

摘录 / EXCERPT: Let G/F be a connected reductive group, and let $|W|$ be the cardinality of the Weyl group of a maximal torus in $G\times\overline{F}$. Let $\mathfrak{p}$ be a prime such that...

conjecture | open | MathDB 页面 | 原文线索

V2-002579 | MATHDB-333271

The semisimple-quotient conjecture for submodule zeta functions

摘录 / EXCERPT: Semisimple-quotient conjecture. For almost all $v\in\mathcal V_k$,

conjecture | open | MathDB 页面 | 原文线索

V2-002580 | MATHDB-333302

Coordinate independence of the q-de Rham complex

摘录 / EXCERPT: Let R be a smooth $\mathbb Z$-algebra, and let $\square:\mathbb Z[T_1,\ldots,T_d]\to R$ be an étale framing. The framed complex...

conjecture | open | MathDB 页面 | 原文线索

V2-002581 | MATHDB-333303

The Frobenius operator on q-de Rham complexes

摘录 / EXCERPT: Let R be a smooth $\mathbb Z$-algebra with p-adic completion $\widehat R$. The Frobenius conjecture. The p-adic completion $\widehat{q\!\operatorname-\!\Omega_R}$ admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-002582 | MATHDB-333304

Coordinate independence of q-connections

摘录 / EXCERPT: Let R be a smooth $\mathbb Z$-algebra, and let a framing $\square$ define the category of finite projective $R[[q-1]]$-modules equipped with flat q-connections with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-002583 | MATHDB-333305

Homocyclic Sylow subgroup conjecture for irreducible p-constant characters

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let ρ be an irreducible p-constant character of G. Write c_ρ for its constant value on the p-singular elements. H...

conjecture | open | MathDB 页面 | 原文线索

V2-002584 | MATHDB-333306

Perfect-group conjecture for irreducible p-constant characters

摘录 / EXCERPT: Let G be a finite perfect group, let p be a prime, and let χ be an irreducible p-constant character with constant value c_χ on the p-singular elements. Perfect-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-002585 | MATHDB-333327

The Diophantine inheritance meta-conjecture for nondegenerate smooth submanifolds

摘录 / EXCERPT: Let a Diophantine property be a property of vectors in an ambient space such as $\mathbb{R}^{n}$, and let $\mathcal{M}$ be a nondegenerate smooth submanifold of that space. Diophan...

conjecture | open | MathDB 页面 | 原文线索

V2-002586 | MATHDB-333328

Hurwitz zeta universality conjecture for algebraic irrational parameters

摘录 / EXCERPT: Let α be a parameter with $0<\alpha<1/2$ or $1/2<\alpha<1$, and let Theorem 1 denote the universality statement for the Hurwitz zeta-function in the preceding theorem. Hurwi...

conjecture | open | MathDB 页面 | 原文线索

V2-002587 | MATHDB-333329

Dubickas conjecture on multiplicative dependence of shifted algebraic numbers

摘录 / EXCERPT: For an algebraic number α, consider the set ${\mathbb N}+\alpha=\{n+\alpha:n\in{\mathbb N}\}$. A set of nonzero algebraic numbers is multiplicatively independent if no nontr...

conjecture | open | MathDB 页面 | 原文线索

V2-002588 | MATHDB-333330

Smooth-values conjecture for irreducible polynomials

摘录 / EXCERPT: Let $P\in{\mathbb Z}[x]$ be an irreducible polynomial of degree d, and let $\Psi_P(x,y)$ be the number of integers in the set $\{P(n):1\leq n<x\}$ whose prime factors are all at...

conjecture | open | MathDB 页面 | 原文线索

V2-002589 | MATHDB-333331

Weak smooth-values conjecture for polynomial sequences

摘录 / EXCERPT: For a polynomial $P\in{\mathbb Z}[x]$, let $\Psi_P(x,y)$ count the integers among $\{P(n):1\leq n<x\}$ whose prime factors are all at most y. Weak smooth-values conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-002590 | MATHDB-333332

Logarithmic-dimension conjecture for shifted algebraic numbers

摘录 / EXCERPT: Let α be an algebraic number of degree $d\geq2$. Consider the rational vector space spanned by the logarithms of its integer shifts,...

conjecture | open | MathDB 页面 | 原文线索

V2-002591 | MATHDB-333353

Papadima-Suciu conjecture on prime-power monodromy orders

摘录 / EXCERPT: Let ${\mathcal A}$ be a real arrangement, let F be its Milnor fiber, and let monodromy mean the action induced on the homology of F. A nontrivial monodromy has prime-power orde...

conjecture | open | MathDB 页面 | 原文线索

V2-002592 | MATHDB-333354

Asymptotic equality of mod-ℓ and ℓ-adic rank-2 representation counts

摘录 / EXCERPT: Let X be a smooth projective curve over ${\mathbb F}_q$, let ℓ be a prime different from the characteristic of ${\mathbb F}_q$, and let $T(X,2,q^n,\ell)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002593 | MATHDB-333355

Conjecture on the complexity of products of powers of 2 and 3

摘录 / EXCERPT: Let $\|n\|$ denote the smallest number of ones needed to write the positive integer n using addition and multiplication. For integers $k,\ell\geq 0$, with k and ℓ not both...

conjecture | open | MathDB 页面 | 原文线索

V2-002594 | MATHDB-333366

The divergence conjecture for Mahler measures outside fixed closed sets

摘录 / EXCERPT: Let $\mathcal L$ be Boyd's set of Mahler measures $m(F)$, where F ranges over integer polynomials in any number of variables. For each $F\in\mathcal L$, let $\mathcal M(F)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-002595 | MATHDB-333373

Reduced zeta-function conjecture for graded ideals in free nilpotent Lie rings

摘录 / EXCERPT: Assume the uniformity conjecture, and define the reduced graded ideal zeta function by ... Let $r=\sum_{i=1}^cW_d(i)$ and $[N]_0=\{0,\ldots,N\}$. Reduced zeta-function conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-002596 | MATHDB-333374

Degree conjecture for topological graded ideal zeta functions

摘录 / EXCERPT: Let $c\in\mathbb{N}$ and $d\in\mathbb{N}_{\geq2}$, let $\mathfrak{f}_{c,d}$ be the free nilpotent Lie ring of class c on d generators, and put...

conjecture | open | MathDB 页面 | 原文线索

V2-002597 | MATHDB-333375

Topological graded ideal zeta function at infinity conjecture

摘录 / EXCERPT: Assume the uniformity conjecture, so that the reduced graded ideal zeta function $\zeta_{\mathfrak{f}_{c,d},\operatorname{red}}^{\triangleleft_{\operatorname{gr}}}(Y)$ is defined....

conjecture | open | MathDB 页面 | 原文线索

V2-002598 | MATHDB-333376

Topological graded ideal zeta function at zero conjecture

摘录 / EXCERPT: Let $c\in\mathbb{N}$ and $d\in\mathbb{N}_{\geq2}$, let $W=W_d$ be the Witt function, and let $\mathfrak{f}_{c,d}$ be the free nilpotent Lie ring of class c on d generators. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-002599 | MATHDB-333422

Cohen's refined Zaremba multiplicity conjecture

摘录 / EXCERPT: Cohen's refined Zaremba conjecture. The multiplicity satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002600 | MATHDB-333504

Uniform boundedness conjecture for Brauer groups of lattice-polarized K3 surfaces

摘录 / EXCERPT: Fix a number field k and a lattice Λ with a primitive embedding ... Let X be a K3 surface over k such that $\operatorname{NS}({\overline{X}})\cong\Lambda$. Uniform bound...

conjecture | open | MathDB 页面 | 原文线索

V2-002601 | MATHDB-333522

Jaeger-Linial-Payan-Tarzi additive bases conjecture for finite vector spaces

摘录 / EXCERPT: Jaeger-Linial-Payan-Tarzi conjecture. For every prime p, there exists a constant k_p such that the union, with repetitions, of any k_p bases for $\mathbb{Z}_p^n$ forms an add...

conjecture | open | MathDB 页面 | 原文线索

V2-002602 | MATHDB-333527

The Weak Diversity Conjecture for number fields in parametric families

摘录 / EXCERPT: Let X be a curve over $\mathbb{Q}$ and let $\min \mathbb{Q}(X)$ be a non-constant $\mathbb{Q}$-rational function of degree at least 2. For the points $P_1,\ldots,P_N$ in the abo...

conjecture | open | MathDB 页面 | 原文线索

V2-002603 | MATHDB-333536

Anti-symplectic Frey-Mazur conjecture

摘录 / EXCERPT: Let E and E' be elliptic curves over $\mathbb{Q}$, let p be a prime, and let $E[p]$ and $E'[p]$ denote their p-torsion modules with Galois action by $G_{\mathbb{Q}}$. An is...

conjecture | open | MathDB 页面 | 原文线索

V2-002604 | MATHDB-333543

Strong Approximation Conjecture for the Markoff surface

摘录 / EXCERPT: Strong Approximation Conjecture. For any prime p, $\mathbb X(\mathbb Z/p\mathbb Z)$ consists of exactly two Γ-orbits, namely

conjecture | open | MathDB 页面 | 原文线索

V2-002605 | MATHDB-333564

Rationality criterion for admissible 2-adic series

摘录 / EXCERPT: Rationality criterion. The power series $G_\psi(T)$ is rational if and only if $G_\psi(\frac{1}{3})$ is a rational element of ${\bf Q}_2$. This is presented as a reformulation of t...

conjecture | open | MathDB 页面 | 原文线索

V2-002606 | MATHDB-333578

The paramodular conjecture for abelian surfaces

摘录 / EXCERPT: Let $\mathcal{A}/\mathbb{Q}$ be an abelian surface of conductor N with $\operatorname{End}_{\mathbb{Q}}\mathcal{A}=\mathbb{Z}$. Let F be a weight 2 cuspidal paramodular newfo...

conjecture | open | MathDB 页面 | 原文线索

V2-002607 | MATHDB-333579

Quantum modular invariant conjecture for real quadratic number fields

摘录 / EXCERPT: Let $\uptheta\in\mathbb R-\mathbb Q$ be a fundamental quadratic unit and let $K=\mathbb Q(\uptheta)$, with fundamental discriminant D. The multivalued quantum modular invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-002608 | MATHDB-333580

Harvey-Rayhaun's Thompson Moonshine conjecture

摘录 / EXCERPT: Harvey-Rayhaun's Thompson Moonshine conjecture. There exists such a graded Th-supermodule W for which, for every $g\in Th$, the series $\mathcal{T}_{[g]}(\tau)$ is a specifical...

conjecture | open | MathDB 页面 | 原文线索

V2-002609 | MATHDB-333586

Rudin-Ruzsa conjecture for sums of perfect squares

摘录 / EXCERPT: Rudin-Ruzsa conjecture. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-002610 | MATHDB-333598

Density and interior conjectures for shapes of unit lattices

摘录 / EXCERPT: Shapes of unit lattices conjectures. The closure $\overline{\Omega}$ in the modular surface is non-compact, the closure $\overline{\Omega}$ in the modular surface has non-empty int...

conjecture | open | MathDB 页面 | 原文线索

V2-002611 | MATHDB-333608

The modified ghost conjecture for 2-adic slopes

摘录 / EXCERPT: Modified ghost conjecture. For each $\kappa\in\mathcal W$,

conjecture | open | MathDB 页面 | 原文线索

V2-002612 | MATHDB-333627

Extension of Fermat's last theorem to real algebraic integer exponents

摘录 / EXCERPT: Let n be a real algebraic integer with $n>2$, and let x,y,z be positive integers. Extension of Fermat's last theorem. The equation ... has no solution in positive integers. This...

conjecture | open | MathDB 页面 | 原文线索

V2-002613 | MATHDB-333665

Asai p-adic L-function interpolation conjecture

摘录 / EXCERPT: Asai p-adic L-function conjecture. Both $L_{\mathfrak{p},\mathrm{Asai}}(\mathcal{F})$ and $L_{\mathfrak{p},\mathrm{Asai}}^{\operatorname{imp}}(\mathcal{F})$ should be analytic...

conjecture | open | MathDB 页面 | 原文线索

V2-002614 | MATHDB-333740

Infinitude of full-cycle primes for odd-degree maps

摘录 / EXCERPT: Let $\phi(x)=x^d+c\in\mathbb{Z}[x]$ with $d\geq 2$, and let $S_{d,c}$ be the set of primes p such that the dynamical system ... consists of a single p-cycle. Infinitude conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-002615 | MATHDB-333761

Bouc-Zimmermann conjecture on centres of principal blocks of $PSU(3,p^r)$

摘录 / EXCERPT: Let p be a prime, let r be a positive integer, let $G=PSU(3,p^r)$, let P be a Sylow p-subgroup of G, let k be the algebraically closed field of characteristic p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002616 | MATHDB-333791

The bounded-distance characterization of mock characters

摘录 / EXCERPT: Bounded-distance conjecture. For any mock character κ, there exists a Dirichlet character χ such that $\mathbb D(\kappa,\chi;y)$ is bounded. Conversely, if...

conjecture | open | MathDB 页面 | 原文线索

V2-002617 | MATHDB-333794

The Covering Conjecture for standard t-ary expansions

摘录 / EXCERPT: Let t be an integer with $t\geq 2$, and let n and d be positive integers. For an element of ${\mathbb Z}/(t^n-1){\mathbb Z}$, write its standard t-ary expansion with digits...

conjecture | open | MathDB 页面 | 原文线索

V2-002618 | MATHDB-333819

Schmid-Zhuang's equality conjecture for zero-sum invariants of large-exponent p-groups

摘录 / EXCERPT: Let p be an odd prime and let G be a finite abelian p-group with ... Here ${\mathsf{D}}(G)$ is the Davenport constant, ${\mathsf{s}}(G)$ is the Erdős-Ginzburg-Ziv constant, and...

conjecture | open | MathDB 页面 | 原文线索

V2-002619 | MATHDB-333831

The conjecture that the positive Pillai equation has at most two primitive solutions

摘录 / EXCERPT: For a positive integer d, consider the equation ... with integer parameters $g\geq 2$ and $1\leq x<y$. A solution is primitive when the relevant parameters satisfy the coprimality...

conjecture | open | MathDB 页面 | 原文线索

V2-002620 | MATHDB-333844

The coefficient-bound conjecture for the critical-slope regulator form

摘录 / EXCERPT: Let b_n be the coefficient of the n-th basis vector in the Kolberg-basis expansion of $h=\iota^*(\Theta_1^{-1}\mathcal{F}^{[\mathfrak{p}_1]})$. Coefficient-bound conjecture. On...

conjecture | open | MathDB 页面 | 原文线索

V2-002621 | MATHDB-333851

The minor nonvanishing conjecture for Parsell-Vinogradov matrices

摘录 / EXCERPT: Minor nonvanishing conjecture. For each $d\ge2$ and $k\ge2$, each $1\le l\le k-1$, and each linear subspace $V\subset\mathbb{R}^n$ with dimension $\dim(V)$, the matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-002622 | MATHDB-333913

Conjecture on Legendre-symbol control of Gaussian-prime walk equivalence

摘录 / EXCERPT: Let $a+bi$ be a Gaussian prime and let $\frac{q}{p}$ be the Legendre symbol appearing in the comparison of the two Gaussian-prime ratios $\frac{a+bi}{\beta+\beta i}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002623 | MATHDB-333924

Fyodorov-Keating multiplicative-chaos conjecture for zeta-function densities

摘录 / EXCERPT: Let $T\to\infty$, let $\beta\in(0,2)$, and consider the random densities on $[0,1]$ given by ... Multiplicative-chaos conjecture. These random densities should converge in distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-002624 | MATHDB-333925

Critical multiplicative-chaos conjecture for zeta-function densities

摘录 / EXCERPT: Let $\beta_c=2$ and consider the random densities from the preceding conjecture, with the critical exponent $\beta=\beta_c$. Critical multiplicative-chaos conjecture. The preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-002625 | MATHDB-333958

The strong union-of-lines conjecture

摘录 / EXCERPT: Let $\epsilon>0$ be any constant and let q be a sufficiently large prime power. Let L be a set of lines in $\mathbb{F}_q^3$ satisfying ... and such that no plane contains more th...

conjecture | open | MathDB 页面 | 原文线索

V2-002626 | MATHDB-333959

The quadratic-prime conjecture for n^2+1

摘录 / EXCERPT: Quadratic-prime conjecture. There are infinitely many primes of the form $p=n^2+1$, and

conjecture | open | MathDB 页面 | 原文线索

V2-002627 | MATHDB-333960

The quartic-prime conjecture for n^4+1

摘录 / EXCERPT: Quartic-prime conjecture. The weighted counting function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002628 | MATHDB-333969

Simplified additive divisor conjecture for shifted divisor correlations

摘录 / EXCERPT: Let k and $cell$ be integers with $k,cell\ge 2$, let $\varepsilon>0$, and define ... where τ_k is the k-th divisor function. Additive divisor conjecture, simplified version....

conjecture | open | MathDB 页面 | 原文线索

V2-002629 | MATHDB-333970

Additive divisor conjecture with a power-saving error term

摘录 / EXCERPT: Let $k,\ell>1$ be natural numbers, let x be large, and let h satisfy $1\le h\le x^{1-\varepsilon}$. Let $D_{k,\ell}(x,h)=\sum_{n\le x}\tau_k(n)\tau_\ell(n+h)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002630 | MATHDB-333985

The first non-trivial primitive number for repunit parameters

摘录 / EXCERPT: Let m be the number whose base-g expansion consists of g digits equal to 1, and let $g=p+1$ where p is a prime number. Primitive-number conjecture. Then m is the first...

conjecture | open | MathDB 页面 | 原文线索

V2-002631 | MATHDB-333987

The strong Lonely Runner Conjecture

摘录 / EXCERPT: Let m be a positive integer, let d be a parameter, and let $v_1,\dotsc,v_m\in\mathbb{R}_{>0}$ satisfy $\dim_{\mathbb{Q}}(v_1,\dotsc,v_m)\geq d$. The strong Lonely Runner Conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-002632 | MATHDB-334012

Short-interval variance conjecture for sums of the Möbius function times the number-of-prime-factors function

摘录 / EXCERPT: Let $\mu(n)$ denote the Möbius function and let $\omega(n)$ denote the number of distinct prime factors of n. For $X\to\infty$, let $H=X^\delta$ with $\delta\in(0,1)$. Short-inte...

conjecture | open | MathDB 页面 | 原文线索

V2-002633 | MATHDB-334022

The Vojta conjecture for integral points on hyperelliptic curves

摘录 / EXCERPT: For each integer $d\ge 3$, let $f\in\mathbb{Z}[x]$ have degree d and nonzero discriminant, and let $h(f)$ be the maximum of the logarithmic heights of the coefficients of f. Vo...

conjecture | open | MathDB 页面 | 原文线索

V2-002634 | MATHDB-334023

Chowla's polynomial conjecture for the Liouville function

摘录 / EXCERPT: Let λ be the Liouville function. Let $f(x)\in\mathbb Z[x]$ be a polynomial that is not of the form $cg(x)^2$, where $c\in\mathbb Z$ and $g(x)\in\mathbb Z[x]$. Chowla's poly...

conjecture | open | MathDB 页面 | 原文线索

V2-002635 | MATHDB-334032

Fried's analytic torsion conjecture for negatively curved manifolds

摘录 / EXCERPT: Let (Σ,g) be a negatively curved oriented compact Riemannian manifold of dimension $n>2$, and let α be an acyclic unitary representation of $\pi_1(\Sigma)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002636 | MATHDB-334047

Lower bound conjecture for generalised divisor sums

摘录 / EXCERPT: Let K be a number field, fix an ideal class representative $ u\in\mathcal{I}_K$, let $f\in\mathcal{Z}_K$, and let $\mathfrak{F}$ be a system of binary forms as in the paper. For...

conjecture | open | MathDB 页面 | 原文线索

V2-002637 | MATHDB-334053

The arithmetic lacunary spherical maximal weak-type $(1,1)$ conjecture

摘录 / EXCERPT: Let $d\geq5$, let $\mathcal{R}$ be a lacunary subsequence of $\mathcal{R}_{full}$, and let $A_{\mathcal{R}}^*$ be the associated discrete spherical maximal operator. Weak-type...

conjecture | open | MathDB 页面 | 原文线索

V2-002638 | MATHDB-334056

The conjecture on odd exponents for sums of two squares

摘录 / EXCERPT: Let a be a positive integer that is not a perfect power, meaning $a\ne c^k$ for every positive integer c and every $k>1$. Let m be the smallest positive integer such that...

conjecture | open | MathDB 页面 | 原文线索

V2-002639 | MATHDB-334060

The monotonicity-free approximation regularity conjecture for conformal iterated function systems

摘录 / EXCERPT: Let Φ be an iterated function system with attractor X, and let θ range over the class of functions used in the definition of an approximation regular pair. Theorem 1....

conjecture | open | MathDB 页面 | 原文线索

V2-002640 | MATHDB-334105

A uniform bound conjecture for primes of bad reduction of CM curves

摘录 / EXCERPT: Uniform bad-reduction bound conjecture. There is a constant $e\in\mathbb{R}_{\geq 0}$, independent of the curve and the prime, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002641 | MATHDB-334134

Akiyama-Pethő integrality conjecture for contractive-polynomial volumes

摘录 / EXCERPT: Let $\mathcal{E}_d^{(s)}$ be the set of coefficient vectors of monic degree-d real polynomials whose roots include exactly s pairs of nonreal conjugates, and let ... Here d is...

conjecture | open | MathDB 页面 | 原文线索

V2-002642 | MATHDB-334161

The lower-bound conjecture for the normalized arithmetic and geometric means of primes

摘录 / EXCERPT: For each positive integer n, let p_n be the nth prime, let A_n and G_n denote the arithmetic and geometric means of the first n prime numbers, and let $D(n)$ be the qua...

conjecture | open | MathDB 页面 | 原文线索

V2-002643 | MATHDB-334171

The positive finite-rate conjecture for the distributional asymptotics of $(\log_b n)$ modulo 1

摘录 / EXCERPT: Let b be the base, let ν_N denote the empirical distribution associated with the fractional parts of $(\log_b n)$ up to N, let E_b be the relevant exponential distributio...

conjecture | open | MathDB 页面 | 原文线索

V2-002644 | MATHDB-334173

Quillen's freeness conjecture for cohomology of arithmetic groups

摘录 / EXCERPT: Let $cell$ be a prime number. Let K be a number field with $\czeta_\ell\in K$, and let S be a finite set of places containing the infinite places and the places over ell. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002645 | MATHDB-334174

The function-field version of Quillen's conjecture

摘录 / EXCERPT: Let K) be a function field of a smooth projective curve $C/\mathbb{F}_q$, let S be a nonempty finite set of places of K, and let ℓ be a prime with $\ell\mid q-1$. Let...

conjecture | refuted | MathDB 页面 | 原文线索

V2-002646 | MATHDB-334175

The filtered Quillen conjecture

摘录 / EXCERPT: Let K be a global field, let ℓ be a prime different from the characteristic of K with $\zeta_\ell\in K$, and let S be a finite set of places containing the places dividi...

conjecture | open | MathDB 页面 | 原文线索

V2-002647 | MATHDB-334181

Barban-Davenport-Halberstam conjecture for primes in short intervals and arithmetic progressions

摘录 / EXCERPT: Barban-Davenport-Halberstam conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-002648 | MATHDB-334206

Existence conjecture for odd-dimension linear MDS b-symbol codes

摘录 / EXCERPT: Let q be a prime power, let $b\ge 2$, and let n satisfy ... A linear MDS $(n,2b+1)_q$ b-symbol code should exist. This conjecture would provide a general inductive construction...

conjecture | open | MathDB 页面 | 原文线索

V2-002649 | MATHDB-334207

Existence conjecture for even-dimension linear MDS b-symbol codes

摘录 / EXCERPT: Let q be a prime power, let $b\ge 2$, and let $n\ge 2b$. A linear MDS $(n,2b)_q$ b-symbol code should exist. This conjecture follows the constructions and conclusions discussed...

conjecture | open | MathDB 页面 | 原文线索

V2-002650 | MATHDB-334277

Shevelev-Moses conjecture on merging prime-preserving sequences

摘录 / EXCERPT: Let H_a be the lexicographically first strictly increasing sequence beginning with $H_a(2)=a$ such that $H_a(n)$ is prime if and only if n is prime. For primes $a>b\geq 3$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-002651 | MATHDB-334298

The regularity conjecture for Diophantine quadruples

摘录 / EXCERPT: The regularity conjecture for Diophantine quadruples. If $d>\max\{a,b,c\}$, then $d=d_+$. All known Diophantine quadruples were regular when this conjecture was formulated, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-002652 | MATHDB-334306

Anavi-Pollack-Pomerance conjecture on sporadic divisor-sum congruence solutions

摘录 / EXCERPT: Anavi-Pollack-Pomerance conjecture. Uniformly for integers k satisfying $|k|\leq x/2$, the number of sporadic solutions $n\leq x$ is at most $(\log x)^{O(1)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002653 | MATHDB-334333

CFKRS shifted moment conjecture

摘录 / EXCERPT: Let $T>3$, let $\mathcal I=\{a_1,\ldots,a_k\}$ and $\mathcal J=\{b_1,\ldots,b_k\}$ with $|a_i|,|b_j|\ll(\log T)^{-1}$. Let Φ_j and Ψ_j denote the j-element subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-002654 | MATHDB-334344

Characterization of rational vertex operator algebras by modular q-characters

摘录 / EXCERPT: Modularity characterization conjecture. The following statements hold: (1) U is rational if and only if the q-character of each irreducible U-module is a modular function on...

conjecture | open | MathDB 页面 | 原文线索

V2-002655 | MATHDB-334345

Converse to Zhu's modularity theorem for vertex operator algebras

摘录 / EXCERPT: Converse-to-Zhu conjecture. The vertex operator algebra U is rational if and only if the space spanned by

conjecture | open | MathDB 页面 | 原文线索

V2-002656 | MATHDB-334346

Mohanty's conjecture on bounded y-arithmetic progressions on Mordell curves

摘录 / EXCERPT: Let $b\in\mathbb{Q}^*$, and consider the Mordell elliptic curve ... A y-arithmetic progression is a sequence of points whose y-coordinates form an arithmetic progression. Mohanty...

conjecture | open | MathDB 页面 | 原文线索

V2-002657 | MATHDB-334352

The coefficient-shape conjecture for $P_k(t,x)$

摘录 / EXCERPT: Write ... Thus $p_{k,j}(t)$ denotes the coefficient functions in this expansion. Coefficient-shape conjecture. The coefficients satisfy ... where ... with rational positive nonzero numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-002658 | MATHDB-334386

Prescribed-coefficients point-count formula for irreducible polynomials over finite fields

摘录 / EXCERPT: Prescribed-coefficients enumeration conjecture. There exist $\omega_1,\ldots,\omega_N\in\overline{\mathbb{Z}}$, all of norm $\sqrt q$, $\upsilon_1,\ldots,\upsilon_N\in\mathbb{Z}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-002659 | MATHDB-334388

Variance conjecture for divisor sums in arithmetic progressions

摘录 / EXCERPT: Variance conjecture for divisor sums in arithmetic progressions. If $X,q\to\infty$ in such a way that $\frac{\log X}{\log q}\to c\in(0,k)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-002660 | MATHDB-334407

The average-of-roots-of-unity integrality conjecture for functions on cyclic groups

摘录 / EXCERPT: Let n be a positive integer, let $f:\mathbb{Z}_n\longrightarrow\mathbb{Z}_n$ be a function, and let $\mu_f^{a,b}$ denote the average associated with f for $a,b\in\mathbb{Z}_n$....

conjecture | open | MathDB 页面 | 原文线索

V2-002661 | MATHDB-334415

The large-prime-square-factor form of the squarefree values conjecture

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x_1,\ldots,x_h]$ be primitive and squarefree, and let $\mathbb{Z}^h(X)=\{{\rm v}\in\mathbb{Z}^h:|{\rm v}|\leq X\}$. Squarefree conjecture, alternative version....

conjecture | open | MathDB 页面 | 原文线索

V2-002662 | MATHDB-334422

The conjecture on monotonicity of the zeta-function expression $F(s)$

摘录 / EXCERPT: Monotonicity conjecture. The function $F(s)$ is strictly increasing on $(1,\infty)$. In particular, $s=2$ is the unique zero of $F(s)$.

conjecture | open | MathDB 页面 | 原文线索

V2-002663 | MATHDB-334459

The locally modular homomorphism count conjecture for squarefree conductors

摘录 / EXCERPT: Let p be a prime, and let U be a “large" subset of the squarefree integers. Define ... There are constants $C_1,C_2>0$ such that ... In particular, if...

conjecture | open | MathDB 页面 | 原文线索

V2-002664 | MATHDB-334460

The locally modular representation count conjecture with a Frobenius trace condition

摘录 / EXCERPT: Let p be a prime, let U be a “large" subset of the odd squarefree integers, and let $h_1,\ldots,h_5\in\mathbb F_p$ be the reductions of the five integers of absolute value at m...

conjecture | open | MathDB 页面 | 原文线索

V2-002665 | MATHDB-334470

Random permutation model for Hecke operators modulo prime level

摘录 / EXCERPT: Let p be a prime and write $D_p^\pm=\dim_{\mathbb{Q}}S_2(\Gamma_0(p);\mathbb{Q})^\pm$. Fix a prime ℓ. Let σ_+ and σ_- be independently chosen uniformly random...

conjecture | open | MathDB 页面 | 原文线索

V2-002666 | MATHDB-334471

Dickman distribution conjecture for large Hecke factors

摘录 / EXCERPT: For primes $p\ne\ell$, let $D_p=\dim_{\mathbb{Q}}S_2(\Gamma_0(p);\mathbb{Q})$, let $D_p^\pm=\dim_{\mathbb{Q}}S_2(\Gamma_0(p);\mathbb{Q})^\pm$, and define ... where the tilded polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-002667 | MATHDB-334476

The cyclotomic Iwasawa main conjecture for supersingular elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve, let p be an odd supersingular prime for E, and let $L^\sharp(X),L^\flat(X)\in\mathbb{Z}_p[[X]]$ be the two signed p-adic L-function...

conjecture | open | MathDB 页面 | 原文线索

V2-002668 | MATHDB-334483

Upper-bound conjecture for Schur-function vanishing probabilities

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\ldots,\lambda_k)$ be a partition with k parts, let q be a prime power, and let $P(s_\lambda\mapsto0)$ denote the probability that s_λ evaluates...

conjecture | open | MathDB 页面 | 原文线索

V2-002669 | MATHDB-334485

Vanishing probability conjecture for 2-staircases

摘录 / EXCERPT: Let q be a prime power and let $\lambda=(2n,2n-2,\ldots,4,2)$ be the 2-staircase partition. Let $P(s_\lambda\mapsto0)$ denote the probability that its Schur function evaluates to...

conjecture | open | MathDB 页面 | 原文线索

V2-002670 | MATHDB-334498

Asymptotic conjecture for periodic points of quadratic maps modulo primes

摘录 / EXCERPT: Periodic-point asymptotic conjecture. As N tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-002671 | MATHDB-334517

Conjecture B on the number of Brauer constituents of a character

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $\chi\in\operatorname{Irr}(G)$. Let χ^0 be the restriction of χ to the elements of G of order prime to p, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-002672 | MATHDB-334528

The conjectural dimension and relations of p-adic multiple zeta values

摘录 / EXCERPT: For each $n\in\mathbb{N}$, prime p, and $\alpha\in(\mathbb{Z}\cup\{\pm\infty\})\setminus\{0\}$, let $\mathcal{Z}^{(p,\alpha)}_n$ be the $\mathbb{Q}$-vector space generated by the...

conjecture | open | MathDB 页面 | 原文线索

V2-002673 | MATHDB-334530

Generation of relations among multiple harmonic values by polylogarithmic expansions

摘录 / EXCERPT: Consider convergent sums of multiple harmonic values that are both p-adically and weight-adically convergent. Generation conjecture for multiple harmonic-value relations. Every r...

conjecture | open | MathDB 页面 | 原文线索

V2-002674 | MATHDB-334531

Kontsevich-Zagier-type conjecture for multiple harmonic values

摘录 / EXCERPT: Consider the domains and summands described in the source: domains are formed from finite disjoint unions and finite products of sets defined by chains of equalities and inequaliti...

conjecture | open | MathDB 页面 | 原文线索

V2-002675 | MATHDB-334569

The degree conjecture for subgroup-counting polynomials of finite abelian p-groups

摘录 / EXCERPT: Let p be prime and let $N(p^a,p^b,p^c,p^d)$ denote the polynomial counting subgroups of the finite abelian group...

conjecture | open | MathDB 页面 | 原文线索

V2-002676 | MATHDB-334570

The degree and leading-coefficient conjecture for equal-exponent subgroup-counting polynomials

摘录 / EXCERPT: Let p be prime, let $m\ge 1$, and let $N(p^m,p^m,p^m,p^m)$ denote the polynomial counting subgroups of $(\mathbb{Z}_{p^m})^4$. Equal-exponent degree conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002677 | MATHDB-334576

The $p=2$ ghost conjecture for modular-form slopes

摘录 / EXCERPT: The $p=2$ ghost conjecture. If 2 is $\Gamma_0(N)$-regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-002678 | MATHDB-334586

The character finiteness conjecture for admissible genera

摘录 / EXCERPT: Character finiteness conjecture. For each admissible genus $(\mathcal{C},c)$, there are finitely many character vectors of VOA realizations. This is a weaker form of genus finitene...

conjecture | open | MathDB 页面 | 原文线索

V2-002679 | MATHDB-334621

The enumeration of right Bol loops of order $3p$ up to isotopism

摘录 / EXCERPT: Let $p>3$ be a prime, and consider right Bol loops of order $3p$, with two such loops identified when they are isotopic. Enumeration claim. The number of right Bol loops of order...

conjecture | open | MathDB 页面 | 原文线索

V2-002680 | MATHDB-334655

Finite-level and nonexistence conjecture for rational newforms

摘录 / EXCERPT: For each even weight k and level N, let $Q_k(N)$ be the set of quadratic-twist classes of classical newforms with rational coefficients and no complex multiplication, represent...

conjecture | open | MathDB 页面 | 原文线索

V2-002681 | MATHDB-334679

Conjectural supercongruences for truncated hypergeometric series and the p-adic Gamma function

摘录 / EXCERPT: Let $p\ge 5$ be a prime. For a rational number x and a positive integer n, write ... for the truncated hypergeometric series, where $(a)_k$ is the rising factorial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002682 | MATHDB-334680

Parametric supercongruence for a truncated hypergeometric series

摘录 / EXCERPT: Let $p\ge 5$ be a prime. For a p-adic integer a, let $\langle a\rangle_p$ denote its least nonnegative residue modulo p, and let Γ_p denote the p-adic Gamma function...

conjecture | open | MathDB 页面 | 原文线索

V2-002683 | MATHDB-334681

The higher-order exponential Diophantine finiteness conjecture

摘录 / EXCERPT: Let α, β and γ be positive real algebraic numbers. For positive integers k, m, and n greater than 1, consider the equation ... where α, β, an...

conjecture | open | MathDB 页面 | 原文线索

V2-002684 | MATHDB-334684

Sarnak-Strömbergsson conjecture on cubic lattice optimality

摘录 / EXCERPT: Let ζ_L and θ_L denote the Epstein zeta function and theta function of a Bravais lattice L, and let D_3 and D_3^* denote the FCC and BCC lattices, respectively,...

conjecture | open | MathDB 页面 | 原文线索

V2-002685 | MATHDB-334686

The folklore conjecture on simultaneous rank-zero twists and Tate-Shafarevich orders

摘录 / EXCERPT: For positive integers m and k, consider elliptic curves over the rationals and, for a prime p, their p-twists. Folklore conjecture. For any positive integers m and k, t...

conjecture | open | MathDB 页面 | 原文线索

V2-002686 | MATHDB-334688

Termwise divisibility conjecture for the symmetric-function expansion of g(m,n)

摘录 / EXCERPT: Let p be prime, let $e<f$ be positive integers, and set $m=p^e$, $n=p^f$ as above. With $n=km$, consider the symmetric-function expression ... where the sum runs over partitions fi...

conjecture | open | MathDB 页面 | 原文线索

V2-002687 | MATHDB-334689

Kostka-matrix termwise divisibility conjecture

摘录 / EXCERPT: Let p be prime, let $e<f$ be positive integers, and set $m=p^e$, $n=p^f$, and $k=n/m=p^{f-e}$. Let M^{se} and M^{es} denote the change-of-basis matrices between the Schur and...

conjecture | open | MathDB 页面 | 原文线索

V2-002688 | MATHDB-334716

The CFZ-CS ratios conjecture for averages of zeta-function ratios

摘录 / EXCERPT: The CFZ-CS ratios conjecture. Suppose the sets A,B,C,D are as above and the imaginary parts of all parameters are $O(T^{1-\xi})$ for some $\xi>0$. Then, for some $\eta>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-002689 | MATHDB-334750

Beresnevich-Haynes-Velani inhomogeneous Gallagher conjecture

摘录 / EXCERPT: Let $\psi:\mathbb N\to[0,\infty)$ be an approximating function, let $\bftheta\in\mathbb R^d$, and let $\WW^d_{\psi,\bftheta}\subset\mathbb R^d$ denote the set of points $\xx$ for w...

conjecture | open | MathDB 页面 | 原文线索

V2-002690 | MATHDB-334751

Higher-dimensional inhomogeneous multiplicative Gallagher conjecture

摘录 / EXCERPT: Let $m,d\geq1$ be integers, let $\Psi:\mathbb Z^m\to(0,\infty)$, and fix $\bftheta\in\I^d$. For $X=(\mathbf x^{(1)},\ldots,\mathbf x^{(d)})\in\I^{md}$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-002691 | MATHDB-334752

Higher-dimensional inhomogeneous Hausdorff-measure divergence conjecture

摘录 / EXCERPT: Higher-dimensional Hausdorff-measure divergence conjecture. If $\Psi(\mathbf q)=\psi(|\mathbf q|)$ for a monotonically decreasing $\psi:\mathbb N\to(0,\infty)$, this series diverge...

conjecture | open | MathDB 页面 | 原文线索

V2-002692 | MATHDB-334753

Doubly metric higher-dimensional Hausdorff-measure conjecture

摘录 / EXCERPT: Let $m,d\geq1$, let $\Psi:\mathbb Z^m\to(0,\infty)$ be arbitrary, and let f be a dimension function such that the source's series ... diverges and $x\mapsto x^{-md+1}f(x)$ is monot...

conjecture | open | MathDB 页面 | 原文线索

V2-002693 | MATHDB-334758

Convergence conjecture for nonresonant billiard normal forms

摘录 / EXCERPT: Convergence conjecture. If the eigenvalues additionally satisfy good Diophantine properties, then the formal Taylor series defining the function f is locally convergent. The form...

conjecture | open | MathDB 页面 | 原文线索

V2-002694 | MATHDB-334760

The order conjecture for the generalized Lagrangian germ

摘录 / EXCERPT: For coprime positive integers m,n, let $W_{m,n}$ be the geometric model obtained by attaching a disk to the one-component link of the singularity $x^m=y^n$, let $Z_{m,n}$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-002695 | MATHDB-334771

Billerey-Menares refinement of Ramanujan's congruence for square-free levels

摘录 / EXCERPT: Let $N=MN'$ be square-free and let $k\geq 4$. Let $g\in M_k(M)$ be a newform of level M, with Hecke eigenvalues λ_p for $p\mid N'$. Let $\varepsilon:D(N')\to\{\pm1\}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-002696 | MATHDB-334772

Monotonicity of the number of functions determined by additive relations

摘录 / EXCERPT: Let E be a set and S a set of arithmetic functions. For positive integers k and ell, let $n(E,S,k)$ denote the number of functions in S determined by the relation ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-002697 | MATHDB-334773

Existence of additive uniqueness sets with prescribed minimal number of summands

摘录 / EXCERPT: Let n be an integer satisfying $n\le 2$. Existence conjecture for additive uniqueness sets. There exist $E\subseteq\mathbb{N}$ and a set S of arithmetic functions such that E...

conjecture | open | MathDB 页面 | 原文线索

V2-002698 | MATHDB-334792

Holomorphy and Siegel modularity conjecture for normalized genus-two partition functions

摘录 / EXCERPT: Let V be a vertex operator algebra with central charge c, and let ... be its normalized genus-two partition function, where $Z_M^{(2)}$ is the genus-two partition function of the...

conjecture | open | MathDB 页面 | 原文线索

V2-002699 | MATHDB-334811

Uniform weak error conjecture for prime-pair counts

摘录 / EXCERPT: For $d>0$, define ... and write $G(x,d)=\widetilde{G}(x,d)+E(x,d)$. The uniform weak error conjecture for prime-pair counts. The variable-difference Hardy-Littlewood formula holds wi...

conjecture | open | MathDB 页面 | 原文线索

V2-002700 | MATHDB-334816

The square-free sieve conjecture

摘录 / EXCERPT: Let $P(x)\in\mathbb{Z}[x]$ be square-free. The square-free sieve conjecture. The set of integers n for which $P(n)$ is divisible by the square of a prime larger than $\sqrt n$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-002701 | MATHDB-334825

The two-thirds regularity conjecture for prime knots

摘录 / EXCERPT: Let $\operatorname{cr}(X)$ denote the crossing number of a knot X. A prime knot P is λ-regular if ... whenever P is a factor of a knot K. Two-thirds regularity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-002702 | MATHDB-334826

The strong property PT conjecture for prime knots

摘录 / EXCERPT: Let a knot diagram have strong property PT when it has the strong property PT defined in the paper. Strong property PT conjecture. Each prime knot has strong property PT. The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-002703 | MATHDB-334844

The conjecture on the residue of the nilpotent-orbit zeta function

摘录 / EXCERPT: The conjecture on the residue of $H_{\mathfrak{o}}$. The meromorphic function $H_{\mathfrak{o}}(s)$ has a simple pole at $s=0$ with residue $J_{\mathfrak{o}}(f)$.

conjecture | open | MathDB 页面 | 原文线索

V2-002704 | MATHDB-334859

Prism manifold realization conjecture for positive surgery parameters

摘录 / EXCERPT: Let $p>1$ and $q>0$ be relatively prime integers. Let $P(p,q)$ be the prism manifold associated with these parameters, let K be a knot in S^3, and let $\mathcal P^+$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-002705 | MATHDB-334871

MDS-extension conjecture for Reed-Solomon codes

摘录 / EXCERPT: Let q be a prime power and let $2\leq m\leq q-2$. An MDS extension by one digit of an m-dimensional Reed-Solomon code with evaluation set $GF(q)$ is an MDS code obtained by add...

conjecture | open | MathDB 页面 | 原文线索

V2-002706 | MATHDB-334872

RNC completeness conjecture for MDS extensions

摘录 / EXCERPT: Let q be a prime power and let m be an integer for which the stated MDS codes are defined. An $[q+1,m]_q$ Reed-Solomon code corresponds to a normal rational curve (RNC) in...

conjecture | open | MathDB 页面 | 原文线索

V2-002707 | MATHDB-334891

Guo-Lei-Zhao's weighted sum formula conjecture for multiple zeta values

摘录 / EXCERPT: Guo-Lei-Zhao's conjecture. For every positive integer $k\geqslant n$, there exist polynomials $e_{F,l}(x),c_{F,l}(x)\in\mathbb{Q}[x]$, depending only on l and F, such that

conjecture | open | MathDB 页面 | 原文线索

V2-002708 | MATHDB-334892

Fixed-class cohomology finiteness conjecture for p-groups

摘录 / EXCERPT: Let p be a prime, and let c and d be non-negative integers. Consider the mod-p cohomology ring $H^*(G;{\mathbb F}_p)$ of a d-generated finite p-group G of nilpotency...

conjecture | open | MathDB 页面 | 原文线索

V2-002709 | MATHDB-334893

Rank-bounded cohomology finiteness conjecture for p-groups

摘录 / EXCERPT: Let p be a prime, and let r be a non-negative integer. The rank of a finite p-group G is the least integer such that every subgroup of G can be generated by at most that...

conjecture | open | MathDB 页面 | 原文线索

V2-002710 | MATHDB-334908

Nonexistence conjecture for hemisystems of dual polar spaces \(\mathsf{DW}(5,q)\)

摘录 / EXCERPT: Let q be a prime power, and let a hemisystem of the dual polar space $\mathsf{DW}(5,q)$ mean the object discussed in the paper. Nonexistence conjecture. There are no hemisystems...

conjecture | open | MathDB 页面 | 原文线索

V2-002711 | MATHDB-334909

Existence conjecture for hemisystems of \(\mathsf{DQ}(6,q)\)

摘录 / EXCERPT: Let q be an odd prime power, and let a hemisystem of the dual polar space $\mathsf{DQ}(6,q)$ mean the object discussed in the paper. Existence conjecture. For each odd prime powe...

conjecture | open | MathDB 页面 | 原文线索

V2-002712 | MATHDB-334965

Erdős-Pomerance-Sarkőzy conjecture on the order of equal consecutive divisor values

摘录 / EXCERPT: Erdős-Pomerance-Sarkőzy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-002713 | MATHDB-334969

Maximal-weight conjecture for simple holomorphic eta quotients of odd prime-power level

摘录 / EXCERPT: For a prime p and an integer $n>3$, let $f_{p,n}$ be the eta quotient defined by ... Here η_m denotes $\eta(mz)$, and a simple holomorphic eta quotient is one that is both pri...

conjecture | open | MathDB 页面 | 原文线索

V2-002714 | MATHDB-334970

The inert-case uniformity conjecture for p-adic regulator matrices

摘录 / EXCERPT: Let $p>3$ be a prime. Let $\mathbb Q_p^{(3)}$ be the unique unramified cubic extension of $\mathbb Q_p$, with ring of integers $\mathbb Z_p^{(3)}$ and cyclic Galois group generated...

conjecture | open | MathDB 页面 | 原文线索

V2-002715 | MATHDB-334971

The inert-case valuation conjecture for p-adic regulator matrices

摘录 / EXCERPT: Let $p>3$ be a prime, and let M_p^{inert} be the measured space of matrices associated with the inert case. Define the random variable ... where $f=X^2+3Y^2$. Inert-case v...

conjecture | open | MathDB 页面 | 原文线索

V2-002716 | MATHDB-334995

The sums-of-two-squares k-tuple conjecture

摘录 / EXCERPT: Let $\mathbb{S}$ denote the set of integers expressible as a sum of two squares. For $k\geq 1$ and a set $\boldsymbol{h}=\{h_1,\ldots,h_k\}\subseteq\mathbb{Z}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-002717 | MATHDB-334996

The qualitative sums-of-two-squares k-tuple conjecture

摘录 / EXCERPT: Let $\mathbb{S}$ be the set of integers expressible as a sum of two squares. For $k\geq 1$ and a set $\boldsymbol{h}=\{h_1,\ldots,h_k\}\subseteq\mathbb{Z}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-002718 | MATHDB-335017

The lonely runner conjecture in terms of the gap of loneliness

摘录 / EXCERPT: For $t\in\mathbf{R}/\mathbf{Z}$, let $\|t\|_{\mathbf{R}/\mathbf{Z}}$ be the distance from any representative of t to the nearest integer. For an n-tuple of non-zero integers...

conjecture | open | MathDB 页面 | 原文线索

V2-002719 | MATHDB-335022

The Virtual Cosmetic Surgery Conjecture

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $\gamma=p/q$ and $\gamma'=p'/q'$ be surgery slopes in the homological framing, with p,q and p',q' relatively prime. Virtual Cosmetic Surgery C...

conjecture | open | MathDB 页面 | 原文线索

V2-002720 | MATHDB-335027

Positive-rank twist conjecture for the elliptic curves associated with the quartic equation

摘录 / EXCERPT: Positive-rank twist conjecture. There exists at least one rational number t such that the rank of $E(h)_t$ is positive.

conjecture | open | MathDB 页面 | 原文线索

V2-002721 | MATHDB-335028

Coverage conjecture for the rational values generated by positive-rank elliptic curves

摘录 / EXCERPT: Let ... be the elliptic curve associated with a rational parameter Z, and define ... For $Z\in\Omega$, let $H(Z)$ denote the set of rational values of h obtained from the rational...

conjecture | open | MathDB 页面 | 原文线索

V2-002722 | MATHDB-335029

Finiteness conjecture for p-ramification invariants

摘录 / EXCERPT: Let K be a number field satisfying the Leopoldt conjecture for every prime p, and let ${\mathcal T}_p$ be the torsion subgroup associated with the maximal Abelian p-ramified...

conjecture | open | MathDB 页面 | 原文线索

V2-002723 | MATHDB-335057

Duke-Li valuation conjecture for CM coefficients

摘录 / EXCERPT: Let $l>3$ be a prime with $l\equiv 3\pmod{4}$, let ${\mathsf{k}}=\mathbb{Q}(\sqrt{-l})$, and let p be a prime that is non-split in ${\mathsf{k}}$. Let $\mathfrak{P}_0\mid p$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-002724 | MATHDB-335072

Lara Rodríguez's full conjecture for double zeta values in function fields

摘录 / EXCERPT: Lara Rodríguez's full conjecture. For general q, we have

conjecture | open | MathDB 页面 | 原文线索

V2-002725 | MATHDB-335077

The coprime-width formula for the generalized descent difference polynomial

摘录 / EXCERPT: Let n and k be positive integers with $1\leq k<n$, and define ... Here $A_{n-1}(q)$ denotes the Eulerian polynomial of degree $n-1$. Coprime-width formula. If $\gcd(k,n)=1$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-002726 | MATHDB-335101

Double Goldbach conjecture

摘录 / EXCERPT: Double Goldbach conjecture. Every positive even integer n can be written as a sum of two primes in two different ways, with none of the primes being the largest prime less than...

conjecture | open | MathDB 页面 | 原文线索

V2-002727 | MATHDB-335119

Krattenthaler conjecture on prescribed binomial divisibility

摘录 / EXCERPT: Let m be a positive integer. Krattenthaler's conjecture. There exist positive integers a and b satisfying $am>b$ such that ... for every integer $n\geq1$. This conjecture conce...

conjecture | open | MathDB 页面 | 原文线索

V2-002728 | MATHDB-335138

Conjectural formula for the zero-section contribution to curve counts on a Weierstrass model

摘录 / EXCERPT: Let X be the elliptic Calabi-Yau threefold over $\mathbb P^2$ considered in the source, let ℓ be the class of a section over a line and let f be the fiber class. For...

conjecture | open | MathDB 页面 | 原文线索

V2-002729 | MATHDB-335145

The short-interval prime number conjecture

摘录 / EXCERPT: Let $<\varepsilon<1$ and let $\Phi(x)=x^{\varepsilon}$, where $\Phi(x)$ determines the intervals $I(x,\Phi)=[x-\Phi(x),x+\Phi(x)]$. Write $\pi(I(x,\Phi))$ for the number of prime i...

conjecture | open | MathDB 页面 | 原文线索

V2-002730 | MATHDB-335147

The prime-orbit decomposition conjecture for vanishing sums of roots of unity

摘录 / EXCERPT: Let μ_{2k+1} denote the set of $(2k+1)$-th roots of unity. For every prime factor p of $2k+1$ and $\beta_p\in\left[0:\frac{2k+1}{p}-1\right]$, let ... be the corresponding p-...

conjecture | open | MathDB 页面 | 原文线索

V2-002731 | MATHDB-335190

Generalized Borel conjecture for algebraically independent base-b expansions

摘录 / EXCERPT: Let $b\geq 2$ be an integer base, and let $(\alpha_1,\ldots,\alpha_n)$ be an n-tuple of real numbers. Writing their base-b expansions in aligned columns, call the tuple normal...

conjecture | open | MathDB 页面 | 原文线索

V2-002732 | MATHDB-335191

A gcd bound for values of two algebraic polynomials at powers of the base

摘录 / EXCERPT: Let $b\geq 2$ be the fixed integer base. Let $\epsilon>0$, let k,l be positive integers, and let ... ... be polynomials with real algebraic coefficients, where a_k and c_l are li...

conjecture | open | MathDB 页面 | 原文线索

V2-002733 | MATHDB-335192

A nonlinear gcd bound for products of two fractional-part approximations

摘录 / EXCERPT: The first nonlinear gcd conjecture. For sufficiently large m,

conjecture | open | MathDB 页面 | 原文线索

V2-002734 | MATHDB-335193

A nonlinear gcd bound for a nested algebraic approximation

摘录 / EXCERPT: The third nonlinear gcd conjecture. For sufficiently large m,

conjecture | open | MathDB 页面 | 原文线索

V2-002735 | MATHDB-335200

Rudnick-Sarnak conjecture on Poissonian gap statistics for polynomial sequences

摘录 / EXCERPT: Let $\varphi(z)=\alpha_q z^q+\alpha_{q-1}z^{q-1}+\cdots+\alpha_0$ be a polynomial of degree q, and let $P_N(\,\cdot\,;\varphi)$ denote the gap distribution of...

conjecture | open | MathDB 页面 | 原文线索

V2-002736 | MATHDB-335201

Heisenberg-group equidistribution conjecture for Diophantine parameters

摘录 / EXCERPT: Let $\mathbb{H}(\mathbb{R})$ be the Heisenberg group, let $\mathbb{H}(\mathbb{Z})$ be its integer subgroup, and let μ_H be the invariant probability measure on...

conjecture | open | MathDB 页面 | 原文线索

V2-002737 | MATHDB-335216

Weak Generalized Riemann Hypothesis

摘录 / EXCERPT: Weak Generalized Riemann Hypothesis. There exists a constant $\frac{1}{2}>\epsilon\geq 0$ such that all nontrivial roots of $L(s,\chi)$ lie in

conjecture | open | MathDB 页面 | 原文线索

V2-002738 | MATHDB-335219

A lower-bound conjecture for the 2-adic complexity of LSB sequences of p-ary m-sequences

摘录 / EXCERPT: Let p be any odd prime, let n be a positive integer, and set $N=p^n-1$. Let $\{s_t\}_{t=0}^{N-1}$ be the LSB sequence of a p-ary m-sequence of order n, and let C_p deno...

conjecture | open | MathDB 页面 | 原文线索

V2-002739 | MATHDB-335284

Finite-field exceptional-set dimension conjecture for planar projections

摘录 / EXCERPT: Let p be prime, let s,t satisfy ... and let $E\subset \mathbb{F}_{p}^{2}$ satisfy ... For $W\in G(2,1)$, write $\pi^{W}(E)$ for the projection of E onto the quotient by W. Fini...

conjecture | open | MathDB 页面 | 原文线索

V2-002740 | MATHDB-335304

Symmetry conjecture for 4-digit permutiples

摘录 / EXCERPT: Let $(\frac{a_0}{a_1},\frac{a_2}{a_3})$ be a 4-digit (σ,k)-permutiple, where σ is a permutation of the digit positions and k is the multiplication factor. A permuti...

conjecture | open | MathDB 页面 | 原文线索

V2-002741 | MATHDB-335343

Erdős-Heilbronn conjecture

摘录 / EXCERPT: Let p be a prime and let A,B be non-empty subsets of $\mathbb{F}_p$. For subsets $A,B\subset\mathbb{F}_p$, write $A+B=\{a+b\mid a\in A,\ b\in B\}$ and define the restricted sum...

conjecture | open | MathDB 页面 | 原文线索

V2-002742 | MATHDB-335344

Exceptional-set conjecture for restricted subsums

摘录 / EXCERPT: Let p be a prime number, let $A\subset\mathbb{F}_p$ satisfy $A\cap(-A)=\emptyset$, and let $\Sigma_\alpha^\beta(A)$ denote the restricted subsum set used in the source, with para...

conjecture | open | MathDB 页面 | 原文线索

V2-002743 | MATHDB-335353

Possible tile numbers for space-efficient 7-mosaics with no empty rows or columns

摘录 / EXCERPT: Let K be a prime knot and let a space-efficient 7-mosaic of K be a 7-mosaic whose tile number is minimal among mosaics of K. Suppose that every column or every row of the mos...

conjecture | open | MathDB 页面 | 原文线索

V2-002744 | MATHDB-335354

Arithmetic conjectures for the normalized denominators of the Fabius function

摘录 / EXCERPT: Conjectured arithmetic properties. The following statements appear to be true: $A_{2n}=A_{2n+1}$ for $n\geq 1$; $2(2n-1)!$ divides K_n; and H_n is an odd integer.

conjecture | open | MathDB 页面 | 原文线索

V2-002745 | MATHDB-335395

The conjecture that the Markoff block action is alternating or symmetric

摘录 / EXCERPT: Let $\mathbb{X}$ be the affine surface defined by ... For a prime p, let $X^{*}(p)$ be the set of nonzero solutions in $\mathbb{Z}/p\mathbb{Z}$, and partition it into blocks obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-002746 | MATHDB-335406

Conjecture on algebraic independence of derivatives at quadratic points

摘录 / EXCERPT: Let $\mathcal{H}$ be the upper half-plane, and let $\tau_1,\dots,\tau_n\in\mathcal{H}$ be quadratic points lying in distinct $\operatorname{GL}_2^+(\mathbb{Q})$-orbits, with none l...

conjecture | open | MathDB 页面 | 原文线索

V2-002747 | MATHDB-335485

Unboundedness of the coefficient sequence $t_m(n)$

摘录 / EXCERPT: Let $t_m(n)$ denote the coefficient sequence associated with the power series in the paper, and let $m\in\mathbb{N}_{\geq 2}$. Unboundedness conjecture. ... The preceding estimate gi...

conjecture | open | MathDB 页面 | 原文线索

V2-002748 | MATHDB-335486

Nonvanishing conjecture for $t_m(n)$ when $m\geq4$

摘录 / EXCERPT: Let $t_m(n)$ be the coefficient sequence considered in the paper, with $m\in\mathbb{N}_{\geq4}$ and n a positive integer. Nonvanishing conjecture. The equation ... has no solution...

conjecture | open | MathDB 页面 | 原文线索

V2-002749 | MATHDB-335492

Erdős-Komornik conjecture on non-Pisot beta-expansions

摘录 / EXCERPT: Let $1<\beta<\dfrac{1+\sqrt{5}}{2}$, and let $L^{1}(\beta)$ denote the Lebesgue measure of the set of points in $[0,(\beta-1)^{-1}]$ that have a universal expansion in base β...

conjecture | open | MathDB 页面 | 原文线索

V2-002750 | MATHDB-335500

Rational Catalan dimension conjecture for small quantum group centers

摘录 / EXCERPT: Let $n\geq 2$, let l be a prime number greater than $n+1$, and let q be a primitive lth root of unity. Let ${\mathsf u}_q(\mathfrak{sl}_n)$ be the small quantum group and...

conjecture | open | MathDB 页面 | 原文线索

V2-002751 | MATHDB-335501

Trivial-representation conjecture for centers of type A small quantum groups

摘录 / EXCERPT: Let $n\geq 2$, let l be a prime number greater than n, and let q be a primitive lth root of unity. Let ${\mathsf u}_q(\mathfrak{sl}_n)$ be the small quantum group and...

conjecture | open | MathDB 页面 | 原文线索

V2-002752 | MATHDB-335503

Recursive width conjecture for torus knots

摘录 / EXCERPT: Let $T_{p,q}$ denote the torus knot associated to positive coprime integers p and q, and let $\widehat{HFK}(T_{p,q})$ be its knot Floer homology. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-002753 | MATHDB-335519

Rational Dyck path generating-series conjecture for torus links

摘录 / EXCERPT: Let N and M be positive integers, write $d=\frac{\gcd(N,M)}{}$, $N=dn$, and $M=dm$ with n and m coprime, and define ... Let $P_{n,m}$ be the operator from the coprime case, a...

conjecture | open | MathDB 页面 | 原文线索

V2-002754 | MATHDB-335520

Cramer-type conjecture for values of binary quadratic forms representing sums of two squares

摘录 / EXCERPT: Let $Q(x,y):=Ax^2+2Bxy+Cy^2+Dx+Ey+F$ be a binary quadratic form with integral coefficients satisfying ... and with discriminant ... for a prime $q=4k+1$. Suppose that $Q(x,y)$ is posit...

conjecture | open | MathDB 页面 | 原文线索

V2-002755 | MATHDB-335531

Euler's refinement of Lagrange's four-square theorem

摘录 / EXCERPT: Euler's refinement. Every odd positive integer n has a representation as four integer squares satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-002756 | MATHDB-335541

Schmidt's conjecture on intersecting weighted badly approximable sets

摘录 / EXCERPT: Let an n-dimensional weight be a vector $\boldsymbol{\mathbf{r}}=(r_1,\dots,r_n)$ with $r_i\geq 0$ and $r_1+\cdots+r_n=1$. A vector $\mathbf{x}\in\mathbb{R}^n$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-002757 | MATHDB-335551

Rademacher-sum conjecture for umbral McKay-Thompson series

摘录 / EXCERPT: Let X be a Niemeier root system and let $g\in G^X$. Let $\check{H}^X_g$ denote the vector formed by the components of H^X_g indexed by $0<r<m$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002758 | MATHDB-335557

Chida-Mok-Park's Jacquet-Langlands conjecture for Hilbert modular L-invariants

摘录 / EXCERPT: Fix a prime number p. Let F be a totally real field, set $g=[F:{\mathbb Q}]$, and let ${\mathfrak p}$ be a prime ideal of F above p. Let f_∞ be a Hilbert eigen newfo...

conjecture | open | MathDB 页面 | 原文线索

V2-002759 | MATHDB-335593

The primitive-divisor conjecture for the sequence 2^d-3

摘录 / EXCERPT: For each positive integer i, consider the integer 2^i-3. A primitive prime divisor of 2^d-3 is a prime divisor that divides none of the earlier terms 2^i-3 with $i<d$. Prim...

conjecture | open | MathDB 页面 | 原文线索

V2-002760 | MATHDB-335595

Mubayi-Williford's conjecture on triangle-free induced subgraphs of Erdős-Rényi polarity graphs

摘录 / EXCERPT: Let q be a power of an odd prime, and let ER_q be the polarity graph associated with the orthogonal polarity of $PG(2,q)$. An absolute point is a vertex corresponding to a poin...

conjecture | open | MathDB 页面 | 原文线索

V2-002761 | MATHDB-335612

The p-adic Stark conjecture at $s=1$

摘录 / EXCERPT: Let E/F be a finite Galois extension of totally real number fields, with $G=\operatorname{Gal}(E/F)$. Let p be a prime, let Σ be a finite set of places of F containing...

conjecture | open | MathDB 页面 | 原文线索

V2-002762 | MATHDB-335619

Inhomogeneous multiplicative approximation conjecture for a non-Liouville number

摘录 / EXCERPT: Inhomogeneous multiplicative approximation conjecture. For almost all ${\beta}\in\mathbb R$, there exist infinitely many $n\in\mathbb N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002763 | MATHDB-335620

Duffin--Schaeffer conjecture for coprime rational approximations

摘录 / EXCERPT: Let $\Phi:\mathbb N\to\mathbb R_{\geqslant 0}$ satisfy the hypothesis denoted by in the source. Duffin--Schaeffer conjecture. For almost all ${\beta}\in\mathbb R$, the inequality ……

conjecture | open | MathDB 页面 | 原文线索

V2-002764 | MATHDB-335628

Generalized nonintegrality conjecture for multiple reciprocal sums of polynomials

摘录 / EXCERPT: Let $f(x)$ be a nonzero polynomial of integer coefficients. Let $Z_f:=\{x\in\mathbb{Z}:f(x)=0\}$, let $\{a_k\}_{k=1}^{\infty}:=\mathbb{Z}^+\setminus Z_f$ be arranged in increasing...

conjecture | open | MathDB 页面 | 原文线索

V2-002765 | MATHDB-335672

Zeta cycles and spiral structure of backward orbits of Riemann zeros

摘录 / EXCERPT: Let $zeta$ denote the Riemann zeta function, let rho_n be the nth nontrivial Riemann zero, and let A_phi and A_infty denote the corresponding immediate basins. A branch...

conjecture | open | MathDB 页面 | 原文线索

V2-002766 | MATHDB-335673

Quantitative spiral fit for backward branches of nontrivial zeta zeros

摘录 / EXCERPT: Let rho_n be the nth nontrivial Riemann zero, let psi_n be the associated zeta fixed point for $L=1$, and let B_{rho_n,psi_n} be the corresponding backward branch. Let c...

conjecture | open | MathDB 页面 | 原文线索

V2-002767 | MATHDB-335688

Wild-kernel annihilation conjecture

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with Galois group G, let $r>1$ be an integer, let $p\ne2$ be a prime, and let S contain...

conjecture | open | MathDB 页面 | 原文线索

V2-002768 | MATHDB-335693

Hooley's conjecture on the variance of primes in arithmetic progressions

摘录 / EXCERPT: For a fixed modulus c, define ... where $\Lambda(n)$ is the von Mangoldt function and $\phi(c)$ is Euler's totient function. Hooley's conjecture. ... Hooley did not specify the size...

conjecture | open | MathDB 页面 | 原文线索

V2-002769 | MATHDB-335717

The p-adic Stark-unit conjecture for Gross's p-adic regulator

摘录 / EXCERPT: Gross's p-adic conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-002770 | MATHDB-335718

The p-adic Yoshida-invariant conjecture for CM fields

摘录 / EXCERPT: Let K be a CM-field abelian over a totally real field F, with conductor $\mathfrak f_{K/F}$, and assume that $\mathfrak p_\iota$ splits completely in K/F. Let χ' be an o...

conjecture | open | MathDB 页面 | 原文线索

V2-002771 | MATHDB-335744

Functoriality of automorphic cL-invariants under base change

摘录 / EXCERPT: Let E/F be the finite extension and let $S_p(cpi)$ denote the relevant set of primes above p. For a prime $\tilde{\mathfrak{p}}\in S_p(\pi)$, let $e_{\tilde{\mathfrak{p}}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002772 | MATHDB-335778

Generation of $\Gamma_1(N)$ by fixed-upper-left-entry matrices

摘录 / EXCERPT: Generation conjecture. There exists $q_0=q_0(N)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002773 | MATHDB-335784

Hensley's conjecture on denominators for finite alphabets

摘录 / EXCERPT: Let $\mathcal A$ be a finite alphabet of positive integers, let $\mathfrak D_{\mathcal A}$ be the set of denominators of reduced rationals whose continued-fraction partial quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-002774 | MATHDB-335785

McMullen's conjecture on uniformly bounded partial quotients in quadratic fields

摘录 / EXCERPT: For a positive squarefree integer d, let $\mathbb{Q}(\sqrt d)$ be the associated real quadratic field, let $\mathfrak C_m$ denote the set of numbers with uniformly bounded partia...

conjecture | open | MathDB 页面 | 原文线索

V2-002775 | MATHDB-335787

Bourgain-Kontorovich's revised Hensley conjecture with admissibility

摘录 / EXCERPT: Let $\mathcal A$ be a finite alphabet, let $\mathfrak D_{\mathcal A}$ be its denominator set, let $\mathfrak U_{\mathcal A}$ be the set of admissible integers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002776 | MATHDB-335791

The powerful-number prime-gap conjecture

摘录 / EXCERPT: A positive integer is powerful if every prime divisor of it occurs with exponent at least two. Powerful-number prime-gap conjecture. If $a>1$ is powerful, then there is a prime p...

conjecture | open | MathDB 页面 | 原文线索

V2-002777 | MATHDB-335813

The rigidity conjecture for Hausdorff dimensions under inhomogeneous shifts

摘录 / EXCERPT: Let $B_n\subseteq\{0,1,\dots,n-1\}$ for each integer n. For an approximation function f and an inhomogeneous shift θ, define ... Write $\mathbf{0}$ for the zero shift. The...

conjecture | open | MathDB 页面 | 原文线索

V2-002778 | MATHDB-335814

The dimension bounds conjecture for restricted numerator systems

摘录 / EXCERPT: Let $B_n\subseteq\{0,1,\dots,n-1\}$ for each integer n. For an approximation function f and an inhomogeneous shift θ, define $W_B(f,\theta)$ as the set of $x\in[0,1]$ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-002779 | MATHDB-335815

The cancellation conjecture for Ramanujan-sum correlations

摘录 / EXCERPT: Let f be an approximation function, let θ be an inhomogeneous shift, and let ϕ and d denote Euler's totient and divisor-counting functions, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002780 | MATHDB-335821

Subrepresentation conjecture for fine Selmer finiteness

摘录 / EXCERPT: Let K be a number field, let p be a prime, and let M be a representation of G_K on a finite-dimensional vector space over $\mathbb{F}_p$. For such a represen...

conjecture | open | MathDB 页面 | 原文线索

V2-002781 | MATHDB-335822

Cohomological subrepresentation injectivity conjecture

摘录 / EXCERPT: Let K be a number field, let p be a prime, let M be a representation of G_K on a finite-dimensional vector space over $\mathbb{F}_p$, and let S contain the...

conjecture | open | MathDB 页面 | 原文线索

V2-002782 | MATHDB-335823

Fouvry's conjectural short exponential-sum estimate

摘录 / EXCERPT: Fouvry's conjectural estimate. There exists an absolute constant $\vartheta\in[\frac12,1)$ such that, uniformly under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-002783 | MATHDB-335825

The weighted vector-composition prime criterion

摘录 / EXCERPT: Let $N\geq 1$, let $\mathbb{N}$ denote the nonnegative integers, and let $\mathbf{1}=(1,\ldots,1)\in\mathbb{N}^N$. For $\mathbf{s}=(s_1,\ldots,s_N)\in\mathbb{N}^N$, define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002784 | MATHDB-335828

Conjecture on extending the rational rotation-number height bound

摘录 / EXCERPT: Let $\lambda=a/b$ and $\delta=r/s$ be rational numbers with $0<\lambda,\delta<1$, and let $\gamma>1$. Theorem 2 concerns the rational rotation number $\rho_{\lambda,\delta}=p/q$ un...

conjecture | open | MathDB 页面 | 原文线索

V2-002785 | MATHDB-335836

Continuity and computability conjecture for dimensions of nontrivially singular matrices

摘录 / EXCERPT: For m,n positive integers, let $\operatorname{Sing}_{m,n}^*(\omega)$ denote the set of $m\times n$ matrices with Diophantine exponent at least ω that are not trivially sin...

conjecture | open | MathDB 页面 | 原文线索

V2-002786 | MATHDB-335837

The period-index conjecture over fields of finite transcendence degree

摘录 / EXCERPT: The period-index conjecture. For every $\alpha\in\operatorname{Br}(K)$,

conjecture | open | MathDB 页面 | 原文线索

V2-002787 | MATHDB-335876

Euler's totient formulation of the binary Goldbach conjecture

摘录 / EXCERPT: Euler's totient formulation of the binary Goldbach conjecture. The system has an integral solution for every natural number n distinct from unity. This is presented as an alterna...

conjecture | open | MathDB 页面 | 原文线索

V2-002788 | MATHDB-335877

Euler's totient formulation of the ternary Goldbach theorem with a congruence

摘录 / EXCERPT: Euler's totient formulation of the ternary Goldbach theorem with a congruence. The system has a solution (x,y) in non-negative integers for every odd integer n greater than fiv...

conjecture | open | MathDB 页面 | 原文线索

V2-002789 | MATHDB-335904

The Linear Independence Conjecture for zeta zeros

摘录 / EXCERPT: Assume that the Riemann zeta-function $\zeta(s)$ satisfies the Riemann Hypothesis. Let the positive ordinates of the non-trivial zeros of $\zeta(s)$ be counted with their distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-002790 | MATHDB-335905

The conjectured lower bound for Riesz means of the Möbius function

摘录 / EXCERPT: Let $M_\tau(x)$ be the Riesz mean ... where μ is the Möbius function. Let ρ be any non-trivial zero of the Riemann zeta-function, with multiplicity $m(\rho)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002791 | MATHDB-335913

The three-cubes conjecture for the sequence P_n

摘录 / EXCERPT: For $n\,in\mathbb{N}_{+}$, define ... The Diophantine equation under consideration is ... where x,y,z are integers. The three-cubes conjecture for P_n. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002792 | MATHDB-335938

Mazur-Tate-Teitelbaum exceptional zero conjecture

摘录 / EXCERPT: Let f be a newform of even weight $k\geq 2$ for $\Gamma_0(pN)$, where p is prime and $p\nmid N$, and let χ be a Dirichlet character of conductor prime to pN with...

conjecture | open | MathDB 页面 | 原文线索

V2-002793 | MATHDB-335959

Conjecture on perfect powers in alternating sums of consecutive cubes

摘录 / EXCERPT: Let $r=2d$ with $1\leq d\leq 50$, let $p\geq 5$ be prime, and consider the equation ... Here d,x,z are integers. Alternating-cube perfect-power conjecture. The possible integer sol...

conjecture | open | MathDB 页面 | 原文线索

V2-002794 | MATHDB-335977

The local-global realizability conjecture for inertia candidates in \(\operatorname{GL}_2(\mathbb{F}_p)\)

摘录 / EXCERPT: Let $p>2$ be a prime, let $G=\operatorname{GL}_2(\mathbb{F}_p)$, and let (G,I,p) denote the realizability problem with inertia subgroup I. An inertia candidate is a subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-002795 | MATHDB-335987

Artin-group conjecture for complements of hypocycloids

摘录 / EXCERPT: A hypocycloid is determined by coprime integers $0<\ell<k$; set $N:=k+\ell$, and let $\mathcal{C}_{k,\ell}$ denote its complexified plane algebraic curve. The Artin-group conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-002796 | MATHDB-336007

Modularity conjecture for rigid Calabi-Yau threefolds

摘录 / EXCERPT: Modularity conjecture. Any rigid Calabi-Yau threefold X defined over $\mathbb{Q}$ is modular: its L-series $L(X,s)$ coincides up to finitely many Euler factors with the Mellin...

conjecture | open | MathDB 页面 | 原文线索

V2-002797 | MATHDB-336020

Conjecture on arbitrarily long sequences of consecutive happy numbers in even negative bases

摘录 / EXCERPT: Conjecture. For every even integer $b\leq -12$, there exist arbitrarily long finite sequences of consecutive b-happy numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-002798 | MATHDB-336023

The affine almost-invariance lower-bound conjecture

摘录 / EXCERPT: Let $\varepsilon>0$, let p be a prime, and let $A\subset\mathbb{F}_p\setminus\{0\}$. A nonempty set $X\subset\mathbb{F}_p$ satisfies the required affine almost-invariance conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-002799 | MATHDB-336042

Supercongruence for leading coefficients of convergent cellular integrals

摘录 / EXCERPT: For each integer $N\geq 5$ and each convergent σ_N, let $A_{\sigma_N}(n)$ denote the leading coefficients associated with σ_N. The leading-coefficient supercongruence...

conjecture | open | MathDB 页面 | 原文线索

V2-002800 | MATHDB-336046

Density conjecture for sets without k+1 pairwise coprime integers

摘录 / EXCERPT: Let $\mathbb{Z}^{+}$ be the positive integers, let p_i be the ith prime, and let C_k be the family of subsets of $\mathbb{Z}^{+}$ containing no $k+1$ pairwise coprime integer...

conjecture | open | MathDB 页面 | 原文线索

V2-002801 | MATHDB-336047

Quantitative density conjecture for sets without k+1 pairwise coprime integers

摘录 / EXCERPT: Let p_i denote the ith prime and set $P_k=p_1p_2\cdots p_k$. Let C_k be the family of subsets of the positive integers containing no $k+1$ pairwise coprime integers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002802 | MATHDB-336048

Polylogarithmic gap conjecture for sets outside E_k

摘录 / EXCERPT: Let p_i denote the ith prime. Let $C_k(n)$ be the subsets of $\{1,\ldots,n\}$ containing no $k+1$ pairwise coprime integers, and let $E_k(n)$ be the integers at most n divisi...

conjecture | open | MathDB 页面 | 原文线索

V2-002803 | MATHDB-336055

Soydan-Németh-Szalay conjecture on weighted power sums of Fibonacci numbers

摘录 / EXCERPT: Soydan-Németh-Szalay conjecture. Apart from these trivial solutions, the only solutions are

conjecture | open | MathDB 页面 | 原文线索

V2-002804 | MATHDB-336119

Smooth Gaussian-prime variance conjecture

摘录 / EXCERPT: Smooth Gaussian-prime variance conjecture. The variance satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002805 | MATHDB-336120

Gaussian-prime zero-statistics form-factor conjecture

摘录 / EXCERPT: Gaussian-prime zero-statistics form-factor conjecture. As $K\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002806 | MATHDB-336121

Conjecture on the structure of the sets of bounded-coefficient $(q;q)_m$ polynomials

摘录 / EXCERPT: Let h be a positive integer, and let S_h be the set of all nonnegative integers m such that every coefficient of $(q;q)_m$ lies between $-h$ and h, with at least one coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-002807 | MATHDB-336165

Unique extension conjecture for Diophantine quadruples

摘录 / EXCERPT: Unique extension conjecture. Then $d=d_{+}$.

conjecture | open | MathDB 页面 | 原文线索

V2-002808 | MATHDB-336166

Dujella's conjecture on Diophantine quadruples

摘录 / EXCERPT: Dujella's conjecture. For every $n\in S$, there does not exist a $D(n)$-quadruple of natural numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-002809 | MATHDB-336168

The folklore conjecture on Tamagawa factors of elliptic curves

摘录 / EXCERPT: For an elliptic curve E over $\mathbb{Q}$, let N_E be its conductor and let $\operatorname{Tam}(E)=\prod_p\operatorname{Tam}_p(E)$ be its global Tamagawa factor. Tamagawa-facto...

conjecture | open | MathDB 页面 | 原文线索

V2-002810 | MATHDB-336176

Gross's conjecture on the leading term of the p-adic L-function

摘录 / EXCERPT: Let F be a number field, let H/F be a finite cyclic CM extension, let p be a prime, and let R, R_0, and R_1 be the subsets of primes of F above p that split, ramify...

conjecture | open | MathDB 页面 | 原文线索

V2-002811 | MATHDB-336186

Stark's conjecture over the rationals for Artin L-values

摘录 / EXCERPT: Stark's conjecture over the rationals. For every $\chi\in\widehat{G}$, one has $A(\chi,\mathcal{A})\in\overline{\mathbb{Q}}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-002812 | MATHDB-336234

F-regularity conjecture for partial commutator ideals

摘录 / EXCERPT: F-regularity conjecture for partial commutator ideals. The quotient $R/\mathcal{I}_Z$ is F-regular. In particular, $\mathcal{I}_Z$ is a prime ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-002813 | MATHDB-336241

Tate's alternating Artin-Tate pairing conjecture

摘录 / EXCERPT: Let X be a smooth, projective, geometrically connected surface over $\mathbf{F}_q$, where $\operatorname{char} \mathbf{F}_q=p>2$, and let $\ell\neq p$ be a prime. The Artin-Tate...

conjecture | open | MathDB 页面 | 原文线索

V2-002814 | MATHDB-336244

Tightened Goldbach conjecture

摘录 / EXCERPT: Tightened Goldbach conjecture. For all $n\geq 6$, there exist primes $q_1,q_2\in\mathbb{P}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002815 | MATHDB-336245

Upper-bound conjecture for the primorial paired Jacobsthal function

摘录 / EXCERPT: Let p_n denote the nth prime, and let $h_2(n)$ denote the primorial paired Jacobsthal function, defined as the smallest length of a paired progression containing a pair coprime...

conjecture | open | MathDB 页面 | 原文线索

V2-002816 | MATHDB-336258

Truncation conjecture for prime Fourier matrices

摘录 / EXCERPT: Let $F_{S,p}$ denote the partial Fourier matrix associated to a subset $S\subset\mathbb Z_p$, where p is prime. Truncation conjecture. There exists a constant $\varepsilon>0$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-002817 | MATHDB-336264

Isolated master Hadamard conjecture

摘录 / EXCERPT: A square master Hadamard matrix is a master Hadamard matrix of square size, and F_p denotes the Fourier matrix of prime order p. A matrix is isolated when it has no nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-002818 | MATHDB-336265

Arithmetic isolation conjecture for Gauss-symbol Hadamard constructions

摘录 / EXCERPT: Let $q\geq3$ be prime, let $K\in M_{M\times N}(\mathbb C)$ be an isolated truncation of F_p with p prime, and let S,T be the sets used in Theorem 4.4. That theorem constructs...

conjecture | open | MathDB 页面 | 原文线索

V2-002819 | MATHDB-336267

The infinitude conjecture for prime power Giuga numbers

摘录 / EXCERPT: A prime power Giuga number is a Giuga number whose prime factors occur with arbitrary positive prime-power exponents. Infinitude conjecture. There are infinitely many prime power G...

conjecture | open | MathDB 页面 | 原文线索

V2-002820 | MATHDB-336268

The extended Fermat prime infinitude conjecture

摘录 / EXCERPT: Extended Fermat prime infinitude conjecture. There exists an extended Fermat prime p such that $(p-1)p^k+1$ is an extended Fermat prime for infinitely many values of k. The pap...

conjecture | open | MathDB 页面 | 原文线索

V2-002821 | MATHDB-336269

The existence conjecture for level-ℓ extended Fermat primes

摘录 / EXCERPT: Level-ℓ existence conjecture. Given any positive integer ℓ, there exists a level-ℓ extended Fermat prime. This asserts the existence of extended Fermat primes at eve...

conjecture | open | MathDB 页面 | 原文线索

V2-002822 | MATHDB-336306

Fixed-degree class-group l-rank conjecture

摘录 / EXCERPT: Fix an integer $d\ge 2$ and a prime l. For a number field k of degree d, let $\mathcal{D}_k$ be its discriminant and let $\mathrm{rk}_l(\mathrm{Cl}(k))$ be the dimension of t...

conjecture | open | MathDB 页面 | 原文线索

V2-002823 | MATHDB-336307

Weak fixed-degree class-group l-rank conjecture

摘录 / EXCERPT: Fix an integer $d\ge 2$ and a prime l, and let k be a number field of degree d. Write $\mathcal{D}_k$ for its discriminant and $\mathrm{rk}_l(\mathrm{Cl}(k))$ for the l-ran...

conjecture | open | MathDB 页面 | 原文线索

V2-002824 | MATHDB-336324

Conjecture on polylogarithmically many solutions to σ(n)=kn+a

摘录 / EXCERPT: Let $\sigma(n)$ denote the sum-of-divisors function. For integers $k\geq 2$, $x\geq 3$, and $a\in\mathbb{Z}$ satisfying ... consider the solutions $n\leq x$ of $\sigma(n)=kn+a$. Poly...

conjecture | open | MathDB 页面 | 原文线索

V2-002825 | MATHDB-336327

Gross-Stark conjecture

摘录 / EXCERPT: Assume, in addition to the Stark setup, that F is totally real, K is a CM-field, v is a finite place above a rational prime p, and S contains all places of F above p....

conjecture | open | MathDB 页面 | 原文线索

V2-002826 | MATHDB-336337

The derived Hecke action formula for Taylor--Wiles primes

摘录 / EXCERPT: Let E be the coefficient field, let $\mathcal{O}$ be its ring of integers, and let $\mathfrak{p}$ be a prime satisfying the conditions of the paper. Let $n\geqslant1$, let q be...

conjecture | open | MathDB 页面 | 原文线索

V2-002827 | MATHDB-336338

The Merel-unit and cubic-field unit conjecture

摘录 / EXCERPT: Let K be a cubic extension with negative discriminant $-D$, let L be its sextic Galois closure, and let g be the associated weight one form of level D. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002828 | MATHDB-336339

Kowalski's conjecture on the average of the first elliptic-curve divisor

摘录 / EXCERPT: Kowalski's conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002829 | MATHDB-336346

The semi-cross lattice tiling conjecture

摘录 / EXCERPT: Semi-cross lattice tiling conjecture. Every tiling of $\mathbb{Z}^{p-1}$ by V_{p-1} is lattice. The paper states that this special case is equivalent to the prime-tile lattice ti...

conjecture | open | MathDB 页面 | 原文线索

V2-002830 | MATHDB-336351

The prime-order perfect Lee code linearity conjecture

摘录 / EXCERPT: Let $PL(n,e,q)$ denote a perfect e-error-correcting Lee code over $\mathbb{Z}_q^n$. Assume that $2n+1$ is prime. Prime-order perfect Lee code conjecture. Every $PL(n,1,q)$-code i...

conjecture | open | MathDB 页面 | 原文线索

V2-002831 | MATHDB-336358

The holomorphy and shadow conjecture for weight $1/2$ Eisenstein series

摘录 / EXCERPT: Holomorphy and shadow conjecture. The value $E_{1/2}^*(\tau,0)$ is a holomorphic modular form of weight $1/2$; moreover, it is a linear combination of the shadows of the mock Eisen...

conjecture | open | MathDB 页面 | 原文线索

V2-002832 | MATHDB-336363

Conjecture on the variance of the normalized Euler totient function in intervals

摘录 / EXCERPT: Let $R_0(x):=\sum_{n\leq x}\frac{\varphi(n)}{n}-\frac{x}{\zeta(2)}$ and, for an interval of size H, let ... Let $H=\Theta(x^\delta)$ for some fixed $0<\delta\leq 1$. Variance conje...

conjecture | open | MathDB 页面 | 原文线索

V2-002833 | MATHDB-336370

The asymptotic eigenvalue conjecture for the generalized Gauss-Kuzmin-Wirsing operator

摘录 / EXCERPT: Let $d4d8_p$ be the operator on functions defined by ... For every positive integer p, let $d4d8_p$ denote the n-th eigenvalue of the generalized Gauss-Kuzmin-Wirsing operator an...

conjecture | open | MathDB 页面 | 原文线索

V2-002834 | MATHDB-336371

Existence of connected quadratic functional graphs modulo every prime

摘录 / EXCERPT: Let p be prime, and let I_p denote the number of parameters $a\in\mathbb F_p$ for which the functional graph ${\mathcal G}_a$ of $f_a(X)=X^2+a$ is connected. The existence conj...

conjecture | open | MathDB 页面 | 原文线索

V2-002835 | MATHDB-336372

The intersection bound for quadratic graph parameter sets

摘录 / EXCERPT: Let $p\geq3$ be prime. Let ${\mathscr A}_p$ and ${\mathscr B}_p^*$ be the parameter sets defined in the paper, with ${\mathscr A}_p$ denoting the relevant set of parameters and...

conjecture | open | MathDB 页面 | 原文线索

V2-002836 | MATHDB-336374

Miller's Cohen-Lenstra-Martinet density conjecture for cyclic prime-degree fields

摘录 / EXCERPT: Let K be a cyclic extension of $\mathbb{Q}$ of odd prime degree q, let p be a prime not dividing q, and let h_K denote the class number of K. Let ω be the multip...

conjecture | open | MathDB 页面 | 原文线索

V2-002837 | MATHDB-336376

Hofman-Zhang regulator divisibility conjecture for cyclic cubic fields

摘录 / EXCERPT: Let $p>3$ be a prime, let K range over cyclic cubic fields, and let $R_{K,p}$ denote the normalized p-adic regulator. Hofman-Zhang's conjecture. ... The source says that this pre...

conjecture | open | MathDB 页面 | 原文线索

V2-002838 | MATHDB-336377

Cohen-Lenstra-Martinet density heuristics for elementary abelian 2- and 3-extensions

摘录 / EXCERPT: Let t be a positive integer, let p be an odd prime in the degree-2 case, and let $p\geq 5$ be prime in the degree-3 case. Write $(p)_\infty=\prod_{k\geq1}(1-p^{-k})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-002839 | MATHDB-336399

Terai-Jeśmanowicz' conjecture for coprime bases

摘录 / EXCERPT: Let $a,b,c,p,q,r\in\mathbb{N}$ satisfy ... with $a,b,c,p,q,r\geq 2$ and $\gcd(a,b)=1$. Terai-Jeśmanowicz' conjecture. The equation ... has only the solution $(x,y,z)=(p,q,r)$ with...

conjecture | refuted | MathDB 页面 | 原文线索

V2-002840 | MATHDB-336400

Le's uniqueness conjecture for positive solutions of exponential equations

摘录 / EXCERPT: Let a,b,c be positive integers. Le's conjecture. The equation ... has at most one solution (x,y,z) satisfying $\min(x,y,z)>1$. This conjecture is proposed after counterexamples t...

conjecture | open | MathDB 页面 | 原文线索

V2-002841 | MATHDB-336407

Schinzel's Strong Diversity conjecture for branched covers

摘录 / EXCERPT: Let $X\rightarrow \mathbb{A}^1_{\mathbb{Q}}$ be a geometrically irreducible branched cover of curves over $\mathbb{Q}$, such that not all of its branch points are $\mathbb{Q}$-rati...

conjecture | open | MathDB 页面 | 原文线索

V2-002842 | MATHDB-336408

Bilu-Luca's Weak Diversity conjecture for branched covers

摘录 / EXCERPT: Let K be a number field, and let $X\rightarrow \mathbb{A}^1_K$ be a non-trivial geometrically irreducible branched cover of curves over K. Weak Diversity conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-002843 | MATHDB-336409

Uniform Faltings' theorem conjecture for curves over bounded-degree number fields

摘录 / EXCERPT: Let $g\geq 2$ and d be natural numbers. Uniform Faltings' theorem conjecture. There exists a constant $B_{d,g}$ such that for every number field L of degree d over...

conjecture | open | MathDB 页面 | 原文线索

V2-002844 | MATHDB-336438

Distribution conjecture for Hasse failures on Markoff-type cubic surfaces

摘录 / EXCERPT: Distribution conjecture. The number of Hasse failures for $0\leq k\leq K$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002845 | MATHDB-336442

Quadratic inverse conjecture for the large sieve

摘录 / EXCERPT: Let $A\subset [1,N]$ be a set of integers. For each prime p, write A_p for the set of residue classes modulo p represented by elements of A. The notation $X\ll Y$ means tha...

conjecture | open | MathDB 页面 | 原文线索

V2-002846 | MATHDB-336452

Polylogarithmic gap conjecture for sums of two squares

摘录 / EXCERPT: Gap conjecture for sums of two squares. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-002847 | MATHDB-336469

The stronger polynomial selection conjecture for small-characteristic DLP

摘录 / EXCERPT: Let $\mathbb F_q$ be a finite field of odd characteristic. The polynomials $h_1,h_2\in\mathbb F_{q^d}[X]$ are required to be coprime and have degree at most 2. Stronger polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-002848 | MATHDB-336472

Distinct cubic fields conjecture for the initial point sets $I_{0,c}$

摘录 / EXCERPT: Let $I_{0,c}$ be the initial point set associated with the positive integer c, whose elements are cubic algebraic integers. Distinct cubic fields conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002849 | MATHDB-336480

The new density conjecture for symmetric pairs of roots

摘录 / EXCERPT: Let $f(x)$ be a polynomial of degree n. For a prime p in $Spl(f)$, number the roots of $f(x)$ modulo p as ... For a subset $S\subseteq\{1,2,\dots,n\}$, write $Pr(f,S)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-002850 | MATHDB-336490

The unique-zero conjecture for the truncated Riemann kernel

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geqslant 2$. Let $\Phi_n(t)$ be the truncated kernel considered above, let $\omega_n=(1/2)\log n$, and let τ_n denote its smallest positive zero....

conjecture | open | MathDB 页面 | 原文线索

V2-002851 | MATHDB-336500

Kellner's large-prime-factor conjecture for Bernoulli-polynomial denominators

摘录 / EXCERPT: For a positive integer n, let $s_p(n)$ denote the sum of the base-p digits of n, and define ... Let $P(m)$ be the largest prime factor of m. Kellner's conjecture. For $n>192$...

conjecture | open | MathDB 页面 | 原文线索

V2-002852 | MATHDB-336501

Kellner's asymptotic conjecture for large prime factors of Bernoulli-polynomial denominators

摘录 / EXCERPT: For a positive integer n, define ... where $s_p(n)$ is the sum of the base-p digits of n. Let $\omega(m)$ denote the number of distinct prime factors of m. Kellner's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-002853 | MATHDB-336515

The Main Conjecture for Vinogradov mean values

摘录 / EXCERPT: The Main Conjecture. For all integers $s,k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-002854 | MATHDB-336527

Hellus's associated-prime conjecture for Matlis duals of local cohomology

摘录 / EXCERPT: Hellus's conjecture. If $\operatorname{H}^i_I(R)\neq 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-002855 | MATHDB-336540

Ohno's conjecture on zeta functions of pairs of ternary quadratic forms

摘录 / EXCERPT: Let L and $\hat{L}$ be the two lattices in the prehomogeneous vector space whose zeta functions are $\xi_i(L,s)$ and $\xi_i(\hat{L},s)$ for $i=1,2,3$, as defined in the paper. Oh...

conjecture | open | MathDB 页面 | 原文线索

V2-002856 | MATHDB-336560

The Ramanujan nearest-integer conjecture for the prime-counting function

摘录 / EXCERPT: Let $\mathcal{R}\subset \mathbb{R}_+$ be the Ramanujan set of values $x\in \mathbb{R}_+$ for which $\pi(x)$ is the nearest integer to ... where $\pi(x)$ is the prime-counting functio...

conjecture | open | MathDB 页面 | 原文线索

V2-002857 | MATHDB-336563

The rank-order conjecture for Mazur's Eisenstein ideal

摘录 / EXCERPT: Rank-order conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-002858 | MATHDB-336564

Non-vanishing conjecture for adjoint p-adic cyclotomic multiple zeta values

摘录 / EXCERPT: Non-vanishing conjecture. For every such choice of $(n_i)_d$ and $(\xi_i)_d$, the sequence $\zeta_{p,\alpha}^{\operatorname{Ad}}\big((n_i)_d;(\xi_i)_d\big)$ is non-zero. More preci...

conjecture | open | MathDB 页面 | 原文线索

V2-002859 | MATHDB-336572

The prime-power conjecture for finite projective planes

摘录 / EXCERPT: Prime-power conjecture. If a finite projective plane of order n exists, then n is a power of some prime p.

conjecture | open | MathDB 页面 | 原文线索

V2-002860 | MATHDB-336601

Guralnick's positive-characteristic genus conjecture

摘录 / EXCERPT: Let $p>0$ be a prime, let S be a nonabelian simple group, and say that S has genus g in characteristic p if it is a composition factor of the monodromy group of a finite se...

conjecture | open | MathDB 页面 | 原文线索

V2-002861 | MATHDB-336606

The ideal-class reformulation of Gompf's conjecture

摘录 / EXCERPT: Let Θ_n be a root of $f_n(x)=x^3-nx^2+(n-1)x-1$, and define ... Let $\sim$ be the equivalence relation on $\mathcal{C}\mathcal{S}$ generated by $\sim_S$ and $\sim_G$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-002862 | MATHDB-336633

Conjecture for minimum coprime numbers of joins of paths

摘录 / EXCERPT: Let P_m and P_n be paths on $m+1$ and $n+1$ vertices, respectively, and let P_m+P_n denote their graph join. The minimum coprime number $\mathfrak{pr}(G)$ is the least M fo...

conjecture | open | MathDB 页面 | 原文线索

V2-002863 | MATHDB-336639

Bourgain-Gamburd-Sarnak conjecture on almost-prime values on algebraic groups

摘录 / EXCERPT: Let $G\subset GL_n(\mathbb{R})$ be the group of real points of an algebraically connected, algebraically simply connected, absolutely almost simple linear algebraic group defined o...

conjecture | open | MathDB 页面 | 原文线索

V2-002864 | MATHDB-336641

Deligne-Beilinson-Scholl conjecture on derivatives of modular L-functions

摘录 / EXCERPT: Let f be a weight k Hecke eigencuspform for $\operatorname{SL}_2(\mathbb{Z})$, let $L_f(s)$ be its L-function, and let m be an integer. Let $\mathcal P$ denote the ring of...

conjecture | open | MathDB 页面 | 原文线索

V2-002865 | MATHDB-336645

Conjecture on the number of generalized degenerate monic integer polynomials

摘录 / EXCERPT: Let $M_n(H)$ be the set of generalized degenerate monic integer polynomials of degree n and height at most H. Growth conjecture. For every integer $n\ge 2$, ... The preceding res...

conjecture | open | MathDB 页面 | 原文线索

V2-002866 | MATHDB-336646

Arbitrarily large exponent conjecture for the three propositions

摘录 / EXCERPT: Let the three propositions above be the results concerning the sets and quantities introduced in this section, each currently stated for a restricted range of $calpha$. Exponent co...

conjecture | open | MathDB 页面 | 原文线索

V2-002867 | MATHDB-336655

Vojta's bounded-degree proximity conjecture

摘录 / EXCERPT: Let k be a number field, let $\overline{k}$ be its algebraic closure, and let $S\subset \mathcal P_k$ be a finite set containing the infinite places $\mathcal P_k^\infty$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002868 | MATHDB-336667

Sylvester's conjecture on sums of two rational cubes

摘录 / EXCERPT: Let n be a positive cubefree integer, and let E_n be the elliptic curve over $ Q$ defined projectively by ... For a prime p, Sylvester's conjecture. If ... then ... This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-002869 | MATHDB-336671

Buniakowski's conjecture for quadratic polynomials

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ be a quadratic irreducible polynomial such that the sequence $\\{f(n)\\}_{n\geq 1}$ does not have a common factor. Buniakowski's conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-002870 | MATHDB-336674

Mean-value conjecture for well-conditioned polynomial tuples

摘录 / EXCERPT: Let X be sufficiently large, and let ${\mathbf f}=(f_1,\ldots,f_t)$ be a well-conditioned t-tuple of polynomials with integral coefficients, where $\deg(f_j)=k_j$. Define ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-002871 | MATHDB-336679

Lang's conjecture on logarithmically badly approximable algebraic numbers

摘录 / EXCERPT: Let $\alpha\in\overline{\mathbb{Q}}$. A real number is $\frac{1}{q^2\cdot\log^A q}$-badly approximable if there is a positive constant, depending on the number, such that its ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-002872 | MATHDB-336684

de Weger's conjecture on binomial collisions

摘录 / EXCERPT: de Weger's conjecture. There are no other collisions than those given above.

conjecture | open | MathDB 页面 | 原文线索

V2-002873 | MATHDB-336685

The conjecture on binomial near collisions with difference one

摘录 / EXCERPT: Difference-one near-collision conjecture. There are no other near collisions with difference 1 than those given above.

conjecture | open | MathDB 页面 | 原文线索

V2-002874 | MATHDB-336686

Finiteness conjecture for binomial near collisions of fixed difference

摘录 / EXCERPT: Fixed-difference finiteness conjecture. Given a fixed difference d, the number of near collisions with difference d is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-002875 | MATHDB-336700

Zeta-value conjecture for renewal sequence point probabilities

摘录 / EXCERPT: Let $C_k^{1,1}$ be the renewal-sequence random variable whose point probabilities are under consideration, and let $\zeta(j)$ denote the Riemann zeta function. For each $k\geq 1$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-002876 | MATHDB-336702

Equidistribution conjecture for highly non-pretentious multiplicative functions

摘录 / EXCERPT: Let $f:\mathbb{N}\to\mathbb{T}$ be a highly non-pretentious multiplicative function, meaning that it does not closely correlate with functions of the form $n^{it}\chi(n)$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-002877 | MATHDB-336709

Complementary Cusick conjecture on strict Hamming-weight increase

摘录 / EXCERPT: Complementary Cusick conjecture. For every nonnegative integer t,

conjecture | open | MathDB 页面 | 原文线索

V2-002878 | MATHDB-336717

Weak positivity conjecture for q-hit numbers

摘录 / EXCERPT: Let $B\subseteq[m]\times[n]$ be a board, let r be a rank, and let q be a prime power. The q-hit number $H_r(B,q)$ is defined for this board and rank. Weak positivity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-002879 | MATHDB-336751

The alternating-group quotient conjecture for triangle groups

摘录 / EXCERPT: Let $G=\Delta(p,q,r)$ be the triangle group with presentation ... Assume that p is prime and $p\leq q\leq r$. Write C_p for the cyclic group of order p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002880 | MATHDB-336752

The wreath-product quotient conjecture for triangle groups

摘录 / EXCERPT: Let $G=\Delta(p,q,r)$ be the triangle group with presentation ... Assume that p is prime and $p\leq q\leq r$, and choose an integer m such that the alternating group ${\cal A}_m$...

conjecture | open | MathDB 页面 | 原文线索

V2-002881 | MATHDB-336753

The Manhattan-circle mean conjecture for pi

摘录 / EXCERPT: Manhattan-circle mean conjecture. The arithmetic mean converges to π in the asymptotic limit:

conjecture | open | MathDB 页面 | 原文线索

V2-002882 | MATHDB-336761

Conjecture on irreducible factors of rational-function iterates

摘录 / EXCERPT: Let q be a prime power and let $n\geq 1$. Write ... for the least power of q that is at least n, and set ... Let T be the set of pairs $(g,h)\in\mathbb{F}_q[X]^2$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-002883 | MATHDB-336810

Quillen's freeness conjecture for special linear groups over rings of integers

摘录 / EXCERPT: Let ℓ be a prime number. Let K be a number field with $\zeta_\ell\in K$, and let S be a finite set of places containing the infinite places and the places over ℓ. The...

conjecture | open | MathDB 页面 | 原文线索

V2-002884 | MATHDB-336816

Random Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $f:\mathbb{N}\to\mathbb{N}_0$ satisfy $0\leqslant f(n)\leqslant n$, let Ω_f be the associated random space of numerator sets, and let $W^P(\Psi)$ denote the set of numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-002885 | MATHDB-336817

Weak Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $\Psi:\mathbb{N}\to\mathbb{R}_{\geqslant 0}$ and define $W(\Psi)$ as the set of numbers in $[0,1]$ approximable by a/n within $\Psi(n)$ for infinitely many $n\in\mathbb{N}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-002886 | MATHDB-336825

Prime-divisibility conjecture for symmetric-group character tables

摘录 / EXCERPT: Let S_n be the symmetric group, and let its character table have entries given by the values of irreducible characters on conjugacy classes. For a fixed prime number p, choose...

conjecture | open | MathDB 页面 | 原文线索

V2-002887 | MATHDB-336835

Conjecture on the explicit slope-and-$\mathscr L$-invariant list formula

摘录 / EXCERPT: Let p be a prime and $N=1$. For a p-new eigenform $f\in S_k(\Gamma)$ of slope h, let $\mathscr L_f$ be its $\mathscr L$-invariant and define ... Here $\widetilde m_p(h)$ is a n...

conjecture | open | MathDB 页面 | 原文线索

V2-002888 | MATHDB-336843

Density of conditioned directional lattices near the projective triangle

摘录 / EXCERPT: Let L and S be as in Corollary, and let ... Let $\mathfrak{D}_S(L)$ denote the corresponding set of conditioned directional lattices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002889 | MATHDB-336858

Sylvester's cube-sum conjecture for primes

摘录 / EXCERPT: Let p be a prime number. A nonzero rational number is a cube sum if it can be written as a^3+b^3 with $a,b\in {\mathbb Q}^\times$. Sylvester's conjecture. Every prime number...

conjecture | open | MathDB 页面 | 原文线索

V2-002890 | MATHDB-336874

Sarnak's conjecture for zero-entropy flows

摘录 / EXCERPT: Sarnak's conjecture. The Möbius function is linearly disjoint from, equivalently orthogonal to, all flows with zero topological entropy.

conjecture | open | MathDB 页面 | 原文线索

V2-002891 | MATHDB-336903

A characteristic-prime criterion for complete permutation monomials

摘录 / EXCERPT: Let p be an odd prime, let $n+1=p$, and set ... Let $a\in\mathbb{F}_{p^{nk}}^*$ satisfy $a^{p^k-1}=-1$. Complete permutation monomial conjecture. The monomial $a^{-1}x^d$ is a comp...

conjecture | solved | MathDB 页面 | 原文线索

V2-002892 | MATHDB-336911

Broadhurst-Kreimer dimension conjecture for multiple zeta values

摘录 / EXCERPT: Broadhurst-Kreimer's conjecture. The generating series of the dimensions of the weight- and depth-graded parts is

conjecture | open | MathDB 页面 | 原文线索

V2-002893 | MATHDB-336912

The decomposition conjecture for q-analogues of multiple zeta values

摘录 / EXCERPT: Let $\mathcal Z_q$ be the q-series algebra in the source, let $\widetilde{M}_{\mathbb Q}(\operatorname{SL}_2(\mathbb Z))$ be the indicated quasi-modular-form subalgebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002894 | MATHDB-336916

The Hardy-Littlewood conjecture for prime tuples on curves over finite fields

摘录 / EXCERPT: Let C be a smooth projective geometrically irreducible curve over $\mathbb F_q$, and let $E=m_1\mathfrak p_1+\cdots+m_s\mathfrak p_s$ be an effective divisor on C. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002895 | MATHDB-336943

The coefficient congruence conjecture for Böttcher coordinates of a perturbed monomial

摘录 / EXCERPT: Let p be a prime and let ... Write the Böttcher coordinate of φ as ... where the coefficients a_k are integers. Coefficient congruence conjecture. The following congruences...

conjecture | open | MathDB 页面 | 原文线索

V2-002896 | MATHDB-336944

The asymptotic valuation and radius conjecture for perturbed monomial Böttcher coordinates

摘录 / EXCERPT: Let p be a prime and let $r\ge0$. Define ... and let ... be the associated Böttcher coordinate. Asymptotic valuation and radius conjecture. For $k\equiv0\pmod p$, ... Moreover, the rad...

conjecture | open | MathDB 页面 | 原文线索

V2-002897 | MATHDB-336955

Igusa's monodromy conjecture for local zeta functions

摘录 / EXCERPT: Let f be a non-constant polynomial in $\mathbb{Z}[x_1,\ldots,x_n]$, for some $n\geq 1$, with $f(\mathbf{0})=0$. For every prime number p, let $Z_{f,p}(s)$ be Igusa's p-adic z...

conjecture | open | MathDB 页面 | 原文线索

V2-002898 | MATHDB-336973

Uniform Lindelöf-strength error bound conjecture for relatively r-prime lattice points

摘录 / EXCERPT: Uniform error bound conjecture. For every $\varepsilon>0$, as $xD^2\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-002899 | MATHDB-336980

Laurent irreducibility conjecture for toric-digraph R-systems

摘录 / EXCERPT: Toric-digraph Laurent irreducibility conjecture. The rational functions $Y_v(t)$ are pairwise coprime irreducible polynomials in $\mathbf{x}_V$; more precisely, they are irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-002900 | MATHDB-337005

Oberdieck's quasi-modularity conjecture for primitive abelian-surface Gromov-Witten classes

摘录 / EXCERPT: Let E_1 and E_2 be nonsingular elliptic curves, let $\mathsf{A}=E_1\times E_2$ be fibered over E_1, and let ${\mathcal A}_{g,n}(\gamma_1,\ldots,\gamma_n)$ denote the reduced...

conjecture | open | MathDB 页面 | 原文线索

V2-002901 | MATHDB-337011

Yoshida-lift generation conjecture for vector-valued Siegel modular forms

摘录 / EXCERPT: Let $S_{j,2}(Gamma_2[2])$ denote the space of vector-valued Siegel modular forms of degree 2, weight $(j,2)$, and level $Gamma_2[2]$. A Yoshida-lift generation conjecture. The sp...

conjecture | open | MathDB 页面 | 原文线索

V2-002902 | MATHDB-337012

Baker's lens space cabling conjecture

摘录 / EXCERPT: Let K be a knot in a lens space L and let surgery on K produce a non-prime 3-manifold Y. A knot K is hyperbolic if its exterior admits a complete finite-volume hyperbol...

conjecture | open | MathDB 页面 | 原文线索

V2-002903 | MATHDB-337013

Modularity conjecture for admissible affine Lie algebra modules

摘录 / EXCERPT: Let $\mathfrak g$ be a simple Lie algebra, with dual Coxeter number $h^\vee$ and lacety d. Let $\ell=-h^\vee+\frac{a}{b}$ be an admissible level, where a,b are coprime positive...

conjecture | open | MathDB 页面 | 原文线索

V2-002904 | MATHDB-337026

Cluckers-Veys conjecture on exponential sums and log-canonical thresholds

摘录 / EXCERPT: Let $n 1$ be a natural number and let $f\in\mathbb{Z}[x_1,\ldots,x_n]$ be a non-constant polynomial with $f(0,\ldots,0)=0$. For a prime p, an integer $m\geq 1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-002905 | MATHDB-337059

Pair Sato-Tate equidistribution conjecture for non-CM primitive forms

摘录 / EXCERPT: For $i=1,2$, let $g_i=\sum_{n=1}^{\infty}b_i(n)q^n$ be primitive eigenforms of weight $2k_i$ and level $2N_i$, and define ... Let g_1,g_2 be distinct non-CM primitive forms of weig...

conjecture | open | MathDB 页面 | 原文线索

V2-002906 | MATHDB-337071

Conjecture on the asymptotic formula for the nth prime

摘录 / EXCERPT: For each positive integer $n>2$, define ... Here p_n denotes the nth prime number. Asymptotic formula for the nth prime. There exist some $\epsilon\in[0,1]$ and $\delta\in[-2,2]$...

conjecture | open | MathDB 页面 | 原文线索

V2-002907 | MATHDB-337072

Asymptotic binary Goldbach and Lemoine conjecture

摘录 / EXCERPT: Let n be an even positive integer. An odd prime is a prime number other than 2. Asymptotic binary Goldbach and Lemoine conjecture. For every sufficiently large even number n,...

conjecture | open | MathDB 页面 | 原文线索

V2-002908 | MATHDB-337087

The Hilbert class field tower conjecture for curves over finite fields

摘录 / EXCERPT: Let X and Y be smooth projective curves of genus at least two over a finite field ${\mathbb F}_q$. Let $H_0(X):=X$, let $H_1(X):=H(X)$, and recursively let...

conjecture | open | MathDB 页面 | 原文线索

V2-002909 | MATHDB-337088

Mazur-Greenberg's Iwasawa main conjecture for modular forms

摘录 / EXCERPT: Let f be a modular form with $a_p(f)$ a λ-adic unit satisfying Condition (NA), let $A_f(1)$ be the associated Tate-twisted representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002910 | MATHDB-337089

Kobayashi's signed Iwasawa main conjecture for supersingular modular forms

摘录 / EXCERPT: Suppose that $a_p(f)=0$ and $\psi=\mathbf{1}$. Let $\mathrm{Sel}^{\pm}(\mathbb{Q}_\infty,A_f(1))^\vee$ be the Pontryagin duals of the signed Selmer groups, let...

conjecture | open | MathDB 页面 | 原文线索

V2-002911 | MATHDB-337096

The $k_{\max}$-conjecture for integer roots of Moser polynomials

摘录 / EXCERPT: Let $F_{s,k}(n)$ be the Moser polynomial, and let $k_{\max}(s)$ denote the largest positive integer k for which $F_{s,k}$ has an integer root. $k_{\max}$-conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-002912 | MATHDB-337097

Magical Triplet Conjecture for s-equivalent multisets

摘录 / EXCERPT: Let $s>2$ and $n>2s$, and let an n-multiset be a multiset with n elements. Two n-multisets are s-equivalent when they have the same multiset of s-sums. Magical Triplet Co...

conjecture | open | MathDB 页面 | 原文线索

V2-002913 | MATHDB-337112

Brassil-Reichstein's conjecture on a system of cubic equations

摘录 / EXCERPT: Let $a>b\geq 0$ be integers and let $[Z_1:Z_2:Z_3]\in \mathbb{P}^2(\mathbb{Q})$. Brassil-Reichstein's conjecture. The system ... ... has no solution $(a,b,[Z_1:Z_2:Z_3])$. This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-002914 | MATHDB-337131

Bounded gaps between powers of consecutive primes

摘录 / EXCERPT: Bounded-power-gap conjecture. For every $0\leq\alpha<1$, there exists a constant $C(\alpha)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002915 | MATHDB-337132

Uniform boundedness of weighted gaps between powers of consecutive primes

摘录 / EXCERPT: Weighted power-gap conjecture. There exists a constant $C(\alpha,\beta)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002916 | MATHDB-337135

Ayad-Kihel conjecture on determining prime splitting from fixed-divisor data

摘录 / EXCERPT: Let $\mathbb{K}$ be a Galois extension of $\mathbb{Q}$ of degree n, and let p be a prime such that $p\mid\mathfrak{J}(\mathbb{K}\mid\mathbb{Q})$. Let $\rho(p)$ denote the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-002917 | MATHDB-337139

The trivial zero conjecture for geometric p-adic representations

摘录 / EXCERPT: Let V be a geometric p-adic representation of $\mathrm{G}_{\mathbb{Q}}$ that is semistable at p and critical in the sense of Deligne. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-002918 | MATHDB-337141

Fraenkel's conjecture on the moduli of disjoint covering systems

摘录 / EXCERPT: Fraenkel's conjecture. If the system has no multiplicity, then $\alpha_i=p/q_i$, where

conjecture | open | MathDB 页面 | 原文线索

V2-002919 | MATHDB-337148

Prime-divisor conjecture for primitive-sequence sums

摘录 / EXCERPT: For a positive integer n, let $\Omega(n)$ be the number of prime factors counted with multiplicity. For a sequence $\mathscr{A}$, let ... Let $\mathscr{A}$ be a primitive sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-002920 | MATHDB-337149

Cardinality-bounded prime-sum conjecture for primitive sequences

摘录 / EXCERPT: Let $\mathscr{P}={(p_n)}_{n\geq 1}$ be the sequence of primes, and let $\operatorname{Card}\mathscr{A}$ denote the cardinality of a sequence $\mathscr{A}$. Let $\mathscr{A}$ be a p...

conjecture | open | MathDB 页面 | 原文线索

V2-002921 | MATHDB-337150

Homogeneous-sequence maximizer conjecture

摘录 / EXCERPT: For a natural number k, define ... where $\Omega(n)$ counts prime factors with multiplicity. In particular, $\mathscr{P}_1$ is the sequence of primes. Homogeneous-sequence maximize...

conjecture | open | MathDB 页面 | 原文线索

V2-002922 | MATHDB-337168

The nonsquare supersingular lambda-invariant conjecture

摘录 / EXCERPT: Assume N is a prime with $N\equiv 1\pmod 4$, let L be the set of supersingular lambda invariants in $\mathbf{F}_{N^2}$, and let H' denote the derivative of the polynomial H...

conjecture | open | MathDB 页面 | 原文线索

V2-002923 | MATHDB-337193

Dedekind's conjecture on zeta-function quotients

摘录 / EXCERPT: Let L/k be a finite algebraic extension of number fields, and let $\zeta_L(s)$ and $\zeta_k(s)$ denote their Dedekind zeta functions. Dedekind's conjecture. The quotient ... is ent...

conjecture | open | MathDB 页面 | 原文线索

V2-002924 | MATHDB-337238

Finiteness conjecture for eigenforms modulo prime powers

摘录 / EXCERPT: Let p be a prime, let $N\geq 1$, and let $S(N,\overline{\mathbb{Z}}_p)$ be the module spanned by modular forms of level N and all weights. For eigenforms f,g, write...

conjecture | open | MathDB 页面 | 原文线索

V2-002925 | MATHDB-337239

Conjecture on the mod-9 value of the second Hecke eigenvalue

摘录 / EXCERPT: Let f be an eigenform of level 1 and weight $w+2$, with w an integer. Mod-9 second-eigenvalue conjecture. The eigenvalue $a_2(f)$ satisfies ... This conjecture would constrain...

conjecture | open | MathDB 页面 | 原文线索

V2-002926 | MATHDB-337275

Akbary-Park conjecture on fixed traces for a pair of elliptic curves

摘录 / EXCERPT: Akbary-Park fixed-traces conjecture. There exists a constant $C(E_1,E_2,t_1,t_2)\geq0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-002927 | MATHDB-337296

The complementary exceptional propus-parameter conjecture

摘录 / EXCERPT: Complementary exceptional propus-parameter conjecture. In positive integers, this equation has a unique solution (a,b) satisfying $1<a\le(s-1)/2$; moreover, $a-b$ is either a squ...

conjecture | open | MathDB 页面 | 原文线索

V2-002928 | MATHDB-337321

The powers-of-singular-moduli linear equation conjecture

摘录 / EXCERPT: Powers-of-singular-moduli linear equation conjecture. Then one of the following holds: $A+B=C=0$, $x=y$, and $m=n$; $x,y\in\mathbb Q$; or $x\ne y$ and $\mathbb Q(x)=\mathbb Q(y)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-002929 | MATHDB-337335

Density conjecture for sums of three binary squares

摘录 / EXCERPT: Density conjecture. The lower asymptotic density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-002930 | MATHDB-337336

Four positive binary squares conjecture

摘录 / EXCERPT: Four positive binary squares conjecture. Every natural number $N>1772$ is the sum of exactly four positive binary squares. The 112 exceptions are

conjecture | open | MathDB 页面 | 原文线索

V2-002931 | MATHDB-337342

The negative-curve conjecture for blow-ups of the plane

摘录 / EXCERPT: Let C be a prime divisor on ${\rm Bl}_{s}(\mathbb{P}^{2})$. Negative-curve conjecture. If ... then C is a (-1)-curve. This conjecture describes the expected negative curves on...

conjecture | open | MathDB 页面 | 原文线索

V2-002932 | MATHDB-337357

Asymptotic conjecture for the least smooth square-free representative

摘录 / EXCERPT: Asymptotic conjecture for $M(p)$. As p tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-002933 | MATHDB-337377

Conjecture on elliptic Korselt numbers of Type I and anomalous primes

摘录 / EXCERPT: Let $M\geq 7$. Choose distinct primes p,q uniformly at random subject to $5\leq p,q\leq M$, and set $N=pq$. Choose an elliptic curve $E(\mathbb{Z}/N\mathbb{Z})$ uniformly at rand...

conjecture | open | MathDB 页面 | 原文线索

V2-002934 | MATHDB-337387

Cohen-Lenstra conjecture for the largest torsion group in random complexes

摘录 / EXCERPT: Let $LT(n)$ be the largest torsion group found in the first homology during the torsion burst of the Linial-Meshulam process. For a fixed prime q, let the Sylow q-subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-002935 | MATHDB-337388

Cohen-Lenstra conjecture for torsion in random 2-trees

摘录 / EXCERPT: Let $\mathcal{T}^2(n)$ be the set of 2-trees on n vertices, and let X be drawn uniformly from $\mathcal{T}^2(n)$. For a fixed prime q, let the Sylow q-subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-002936 | MATHDB-337393

Bloch's conductor formula conjecture

摘录 / EXCERPT: Let $S=\operatorname{Spec} A$ be a discrete valuation ring with perfect residue field k and fraction field K. Let $p:X\to S$ be proper and flat of finite type and relative dime...

conjecture | open | MathDB 页面 | 原文线索

V2-002937 | MATHDB-337421

The arithmetic Gan-Gross-Prasad conjecture for fixed level

摘录 / EXCERPT: Arithmetic Gan-Gross-Prasad conjecture. Conditions (a) $\dim\operatorname{Hom}_{{\mathscr H}_K}(\pi_f^K,{\mathcal Z}_{K,0})=1$ and (b) $\operatorname{ord}_{s=1/2}L(s,\pi,R)=1$, tog...

conjecture | open | MathDB 页面 | 原文线索

V2-002938 | MATHDB-337422

The global arithmetic intersection formula at trivial level

摘录 / EXCERPT: Global arithmetic intersection conjecture at trivial level. One has

conjecture | open | MathDB 页面 | 原文线索

V2-002939 | MATHDB-337423

The global arithmetic intersection formula at nontrivial level

摘录 / EXCERPT: Global arithmetic intersection conjecture at nontrivial level. One has

conjecture | open | MathDB 页面 | 原文线索

V2-002940 | MATHDB-337424

The semi-global arithmetic intersection formula

摘录 / EXCERPT: Semi-global arithmetic intersection conjecture. If v_0 is non-archimedean of hyperspecial type and $f'_{v_0}=\mathbf 1_{G'(O_{F_0,v_0})}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-002941 | MATHDB-337428

The odd-prime-power torus packing conjecture

摘录 / EXCERPT: Odd-prime-power torus packing conjecture. There exists n_0 such that, for all $n \ge n_0$, C_k^n admits a perfect induced H-packing.

conjecture | open | MathDB 页面 | 原文线索

V2-002942 | MATHDB-337458

Corrected Sister Beiter conjecture for ternary cyclotomic coefficients

摘录 / EXCERPT: Let $p<q<r$ be primes, and let $a_{pqr}(k)$ denote the coefficient of x^k in the ternary cyclotomic polynomial $\Phi_{pqr}(x)$. Corrected Sister Beiter conjecture. For every k,...

conjecture | open | MathDB 页面 | 原文线索

V2-002943 | MATHDB-337473

Unramified Fontaine-Mazur conjecture via uniform quotients

摘录 / EXCERPT: Let K be a number field and p a prime. Let $K^{ur}(p)$ be the maximal unramified pro-p extension of K, and set ... A quotient is uniform if it is a uniform pro-p group. Unr...

conjecture | open | MathDB 页面 | 原文线索

V2-002944 | MATHDB-337484

Smallest classical weight conjecture for boundary slope seeds

摘录 / EXCERPT: Let $V\in\{U_p,U_{\mathfrak p}\}$, let $\kappa_{\min,\chi\tau^t}$ be the smallest classical weight associated with the character $\chi\tau^t$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002945 | MATHDB-337488

Provisional Bessel K decomposition conjecture for the Riemann zeta function

摘录 / EXCERPT: Let $r\in\{1,2,3\}$, let $\zeta^{*}(s,r)$ be the completed zeta function defined in the paper, and let $a_r(n)$ and $b_r(n)$ be the coefficient sequences introduced by the theta-fu...

conjecture | open | MathDB 页面 | 原文线索

V2-002946 | MATHDB-337497

Brumer-Kramer Paramodular Conjecture

摘录 / EXCERPT: Let A range over isogeny classes of abelian surfaces over $\mathbf{Q}$ of conductor N with $\operatorname{End}_{\mathbf{Q}} A=\mathbf{Z}$. Let $K(N)$ be the paramodular group ……

conjecture | open | MathDB 页面 | 原文线索

V2-002947 | MATHDB-337529

Odlyzko-Rubinstein-Wolf conjecture on the divisibility of jumping champions

摘录 / EXCERPT: Odlyzko-Rubinstein-Wolf conjecture. The jumping champions tend to infinity. Furthermore, any fixed prime p divides all sufficiently large jumping champions.

conjecture | open | MathDB 页面 | 原文线索

V2-002948 | MATHDB-337556

Divisibility refinement for p-solvable groups

摘录 / EXCERPT: Let p be a prime, let G be a p-solvable finite group, and let P be a Sylow p-subgroup of G. Let $\operatorname{N}_{G}(P)$ be the normalizer of P, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002949 | MATHDB-337557

Divisibility refinement for symmetric groups of prime degree

摘录 / EXCERPT: Let p be a prime, let S_p be the symmetric group on p elements, and let P be a Sylow p-subgroup of S_p. Let $\operatorname{N}_{S_p}(P)$ be its normalizer, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002950 | MATHDB-337558

Divisibility refinement for symmetric groups of prime-power degree

摘录 / EXCERPT: Let p be a prime, let $a\in\mathbb{N}$, let S_{p^a} be the symmetric group on p^a elements, and let P be a Sylow p-subgroup of S_{p^a}. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-002951 | MATHDB-337575

The Main Conjecture on rational points near curved submanifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be a compact m-dimensional submanifold of $\mathbb{R}^n$ with codimension $k=n-m$ and proper curvature conditions. Let $N_{\mathcal{M}}(Q,\delta)$ count rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-002952 | MATHDB-337576

The generalized Baker-Schmidt conjecture for simultaneous approximation on nondegenerate manifolds

摘录 / EXCERPT: Let $\psi:\mathbb{N}\to\mathbb{R}^+$ be a decreasing approximation function, and define ... Here $\|\cdot\|$ denotes distance to the nearest integer. Let k be the codimension of an...

conjecture | open | MathDB 页面 | 原文线索

V2-002953 | MATHDB-337578

The Height Uniformity Conjecture for hyperelliptic curves

摘录 / EXCERPT: Let K be a number field. For each integer $d\geq 5$, consider polynomials $F\in K[x]$ of degree d with nonzero discriminant, and points $(x,y)\in K^2$ satisfying $y^2=F(x)$. He...

conjecture | open | MathDB 页面 | 原文线索

V2-002954 | MATHDB-337580

Finite-monodromy classification for the families with p=2

摘录 / EXCERPT: Let $p=2$, and let ${\mathcal F}_d$ denote the family considered in the paper. The preceding corollaries establish finite monodromy when $d=2^a+1$ for some $a\geq 1$, and when...

conjecture | open | MathDB 页面 | 原文线索

V2-002955 | MATHDB-337593

Rodriguez-Villegas conjectures for logarithmic Mahler measures

摘录 / EXCERPT: For a Laurent polynomial $P\in\mathbb C[x_1^{\pm1},\ldots,x_n^{\pm1}]$, let $m(P)$ denote its logarithmic Mahler measure. Let $f_{3,15}$ and $f_{4,6}$ be the modular forms defined...

conjecture | open | MathDB 页面 | 原文线索

V2-002956 | MATHDB-337605

Mennicke's modularity conjecture for elliptic curves over imaginary quadratic fields

摘录 / EXCERPT: Mennicke's modularity conjecture. Every such elliptic curve E is modular.

conjecture | open | MathDB 页面 | 原文线索

V2-002957 | MATHDB-337606

Ribet-style level-lowering conjecture for Bianchi newforms

摘录 / EXCERPT: Ribet-style level-lowering conjecture. Under these hypotheses, such an automorphic form G exists at level with the prime factor $\operatorname{id}{p}$ removed.

conjecture | open | MathDB 页面 | 原文线索

V2-002958 | MATHDB-337608

Uniqueness conjecture for Markov-number sum-of-squares representations

摘录 / EXCERPT: Let $m>2$ be a Markov number, meaning an integer belonging to a triple of positive integers satisfying ... For coprime positive integers $a<b$, the palindromification of the snake gr...

conjecture | open | MathDB 页面 | 原文线索

V2-002959 | MATHDB-337609

Uniqueness conjecture for Markov-number continued fractions

摘录 / EXCERPT: Let $m>2$ be a Markov number, meaning an integer belonging to a triple of positive integers satisfying ... For coprime positive integers $a<b$, write $b/a=[a_1,a_2,\ldots,a_n]$ as a...

conjecture | open | MathDB 页面 | 原文线索

V2-002960 | MATHDB-337614

Katok-Herman conjecture on cohomological rigidity

摘录 / EXCERPT: Katok-Herman conjecture. The only examples, up to smooth diffeomorphism and conjugacy, of $CR(M)$ diffeomorphisms are Diophantine translations in tori.

conjecture | open | MathDB 页面 | 原文线索

V2-002961 | MATHDB-337616

The limiting constant for the ell_infty number-trail length

摘录 / EXCERPT: Let $L_infty(N)$ denote the total length of the number trail through the prime grid from 1 to N, measured in the ell_infty metric. The limiting-constant conjecture. The limit...

conjecture | open | MathDB 页面 | 原文线索

V2-002962 | MATHDB-337617

Modified prime number theorem for the number trail

摘录 / EXCERPT: Modified prime number theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-002963 | MATHDB-337619

Conjecture on the growth of the special-zero constants

摘录 / EXCERPT: Let Δ_n be the simplicial complexes considered in the paper, let $d_n=\dim \Delta_n$, and write ... for constants α_n, as allowed by the preceding theorem. Growth conje...

conjecture | open | MathDB 页面 | 原文线索

V2-002964 | MATHDB-337631

The nonexistence conjecture for new points on curves of fixed genus

摘录 / EXCERPT: Nonexistence conjecture. Given any $g\geq 2$, there exist a field K and a separable extension L/K such that no smooth projective geometrically connected curve X/K of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-002965 | MATHDB-337647

The conjecture on infinitely many Zubrilina numbers

摘录 / EXCERPT: Zubrilina-number conjecture. There are infinitely many Zubrilina numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-002966 | MATHDB-337674

Base change equivalence for Hilbert newforms

摘录 / EXCERPT: Let F/E be a Galois extension with group $G=\operatorname{Gal}(F/E)$, and let f be a Hilbert newform of G-invariant level $\mathfrak{N}$, weight k, and character χ. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-002967 | MATHDB-337695

Rank-preserving elliptic curves conjecture for prime cyclic extensions

摘录 / EXCERPT: The rank-preserving elliptic curves conjecture. There exists such a d for which $(0,d)$ has infinite order and generates $E(L)\otimes\mathbb Q$. This would provide a rank-preserv...

conjecture | open | MathDB 页面 | 原文线索

V2-002968 | MATHDB-337696

Rank-preserving elliptic curves conjecture for arbitrary number-field extensions

摘录 / EXCERPT: Let $K\subset L$ be number fields. Let $d\in\mathcal O_K$ be nonzero and satisfy the hypotheses of Theorem--in particular, the hypotheses referred to as -- in the source. For the c...

conjecture | open | MathDB 页面 | 原文线索

V2-002969 | MATHDB-337701

Joshi's elliptic progression conjecture

摘录 / EXCERPT: Let $E/{\mathbb Q}$ be an elliptic curve. For a prime p of good reduction, let $N_p(E)$ denote the number of points on the reduction of E modulo p, and let $G_1(E,n)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-002970 | MATHDB-337708

Tao's logarithmic Chowla conjecture

摘录 / EXCERPT: Let $\bm\mu$ denote the Möbius function. For integers $r\geq 0$, shifts $1\leq a_1<\dots<a_r$, and exponents $i_s\in\{1,2\}$ that are not all equal to 2, consider the logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-002971 | MATHDB-337720

Deligne's conjecture for coefficient objects

摘录 / EXCERPT: Deligne's conjecture. The following statements hold: $\mathcal{E}_0$ is pure of weight 0; it is E-rational for some number field $E\subseteq\overline{\mathbb{Q}}_{\ell}$; it is...

conjecture | open | MathDB 页面 | 原文线索

V2-002972 | MATHDB-337721

The mixedness conjecture for coefficient objects

摘录 / EXCERPT: Mixedness conjecture. Every $\overline{\mathbb{Q}}_{\ell}$-coefficient object on X_0 is ι-mixed.

conjecture | open | MathDB 页面 | 原文线索

V2-002973 | MATHDB-337726

The asymptotic size conjecture for the generating set of G(r,p)

摘录 / EXCERPT: Generating-set size conjecture. The generating set Σ has at most $2r$ distinct elements for each prime p, and

conjecture | open | MathDB 页面 | 原文线索

V2-002974 | MATHDB-337747

Ihara's conjecture on Euler-Kronecker constants of prime cyclotomic fields

摘录 / EXCERPT: Let $q\ge 3$ be a prime, and let γ_q denote the Euler-Kronecker constant of $\mathbb Q(\zeta_q)$. Ihara's conjecture. First, $\gamma_q>0$. Second, for every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-002975 | MATHDB-337753

Khintchine's conjecture for non-degenerate manifolds

摘录 / EXCERPT: Let $\mathcal{M}\subset\mathbb{R}^n$ be a d-dimensional non-degenerate submanifold, let μ_d be the normalised d-dimensional Lebesgue measure induced on $\mathcal{M}$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-002976 | MATHDB-337754

The Kurzweil-type characterization of non-singular vectors

摘录 / EXCERPT: Let $\boldsymbol{i}=(i_1,\ldots,i_n)$ be a weight vector, let $\mathcal{S}_n(\boldsymbol{i})$ denote the set of $\boldsymbol{i}$-singular vectors, and for each $\varepsilon>0$ defi...

conjecture | open | MathDB 页面 | 原文线索

V2-002977 | MATHDB-337811

Random maximal-isotropic model for p^e-Selmer groups

摘录 / EXCERPT: Fix a prime p and an integer $e\geq1$. Let Q be the quadratic form in the source and set ... A maximal isotropic direct summand is a maximal isotropic subgroup that is also a dir...

conjecture | open | MathDB 页面 | 原文线索

V2-002978 | MATHDB-337813

Bhargava-Kane-Lenstra-Poonen-Rains conjecture for ranks and Sha

摘录 / EXCERPT: Fix a prime p and let $V_n=(\mathbb{Z}_p^{2n},Q)$. Choose maximal isotropic direct summands Z,W at random as in the inverse-limit measure, and define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-002979 | MATHDB-337842

Conway-Norton Moonshine conjecture

摘录 / EXCERPT: Let $V=\bigoplus_{n=0}^\fty V_n$ be a graded infinite-dimensional $\mathbb C$-vector space with $\dim V_n<\infty$ and a natural algebraic structure whose automorphism group is the...

conjecture | open | MathDB 页面 | 原文线索

V2-002980 | MATHDB-337854

The regularity-index conjecture for ideals with linear associated primes

摘录 / EXCERPT: Let S be the standard graded polynomial ring in the paper, and let $I\subset S$ be an unmixed radical graded ideal. Assume that every associated prime of I is generated by line...

conjecture | open | MathDB 页面 | 原文线索

V2-002981 | MATHDB-337865

V. Arnold's face-structure conjecture for algebraic Klein sails

摘录 / EXCERPT: V. Arnold's conjecture. The torus decomposition of every algebraic sail in the sense of Klein contains at least one face that is integer congruent to a triangle, at the unit intege...

conjecture | open | MathDB 页面 | 原文线索

V2-002982 | MATHDB-337887

Refined Sato-Tate conjecture for genus-2 abelian surfaces

摘录 / EXCERPT: Let A be an abelian surface over a number field k, and let $\operatorname{ST}_{A}\subset\operatorname{USp}(4)$ be its Sato-Tate group. For each prime $\mathfrak{p}$ of good red...

conjecture | open | MathDB 页面 | 原文线索

V2-002983 | MATHDB-337892

Slope stability conjecture for ordinary algebraic-geometric \p-adic towers

摘录 / EXCERPT: Slope stability conjecture. The following hold: (1) the q-slopes are equidistributed in $[0,1]$ as $n\to\infty$; (2) for each fixed rational $\alpha\in[0,1]$, there is a polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-002984 | MATHDB-337906

Miyawaki's conjecture for degree-three Siegel modular forms

摘录 / EXCERPT: Let k be such that the indicated spaces are defined, and let $f \in S_{2k-4}(\operatorname{SL}_2(\mathbb{Z}))$ and $g \in S_k(\operatorname{SL}_2(\mathbb{Z}))$ be normalized Heck...

conjecture | open | MathDB 页面 | 原文线索

V2-002985 | MATHDB-337907

Ikeda's non-vanishing conjecture for Miyawaki liftings

摘录 / EXCERPT: Let k be positive and even, let $f \in S_{2k-2(n+r)}(\operatorname{SL}_2(\mathbb{Z}))$ be a normalized Hecke eigenform, let $F^{(2n+2r)}$ be its Ikeda lift, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-002986 | MATHDB-337916

Diamond-Sasaki geometric Serre weight conjecture

摘录 / EXCERPT: Let F be a totally real field in which p is unramified, let Σ be the set of embeddings $F\to L$, and let G_F denote the absolute Galois group of F. For a weight...

conjecture | open | MathDB 页面 | 原文线索

V2-002987 | MATHDB-337931

The cauliflower convergence conjecture for Euclid polynomial series

摘录 / EXCERPT: Let $E_k(\lambda)$ denote the Euclid polynomials, and consider the series ... The “cauliflower” is the region depicted in Figure. Cauliflower convergence conjecture. There is converg...

conjecture | open | MathDB 页面 | 原文线索

V2-002988 | MATHDB-337933

Benford error conjecture for Mersenne numbers

摘录 / EXCERPT: Let $M_n=2^{p_n}-1$, where p_n is the n-th prime, let $D(M_n)$ denote the leading digit of M_n, and define the Benford error by ... Here $P(d)$ is the Benford probability of th...

conjecture | open | MathDB 页面 | 原文线索

V2-002989 | MATHDB-337934

Local Benford distribution conjecture for Mersenne numbers

摘录 / EXCERPT: Let $M_n=2^{p_n}-1$, where p_n is the n-th prime, and consider the leading digits of consecutive terms $M_n,M_{n+1},\ldots,M_{n+k-1}$. A digit is Benford-distributed when its p...

conjecture | open | MathDB 页面 | 原文线索

V2-002990 | MATHDB-337935

Benford waiting-times conjecture for Mersenne numbers

摘录 / EXCERPT: Let $M_n=2^{p_n}-1$, where p_n is the n-th prime, and let $P(d)$ denote the Benford probability of leading digit d. The waiting time between successive occurrences of a leadi...

conjecture | open | MathDB 页面 | 原文线索

V2-002991 | MATHDB-337986

The third and fourth largest coset leader conjecture for odd-characteristic BCH codes

摘录 / EXCERPT: Let q be an odd prime power and let $n=q^m+1$, with the quantities δ_3 and δ_4 defined as in the preceding cases for even m. Third and fourth largest coset leader...

conjecture | open | MathDB 页面 | 原文线索

V2-002992 | MATHDB-338047

Congruences for the fourth odd-rank moment

摘录 / EXCERPT: Let $\eta_{4}(n)$ denote the fourth symmetrized moment of the odd rank. Conjecture. For every integer $n\geq 0$, the congruences ... and ... hold. These congruences are presented as nu...

open_problem | open | MathDB 页面 | 原文线索

V2-002993 | MATHDB-338056

Anderson-Mullen conjecture on primitive and 2-primitive normal elements

摘录 / EXCERPT: Let $p\ge 5$ be a prime and let $n\ge 3$. For $\alpha\in\mathbb{F}_{p^n}$, call α k-normal over $\mathbb{F}_p$ if the $\mathbb{F}_p$-vector space spanned by ... has dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-002994 | MATHDB-338094

Wada's conjecture on class groups of imaginary quadratic fields

摘录 / EXCERPT: Let $K=\mathbb{Q}(\mathbb{sqrt{-D}})$ be an imaginary quadratic field, where $D>1$ is square-free, and write its class group as a direct product of cyclic groups. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-002995 | MATHDB-338095

Infinitely many imaginary quadratic fields with class number divisible by a prescribed odd integer

摘录 / EXCERPT: The authors' conjecture. For each such p, q, n, and m, there are infinitely many imaginary quadratic fields of this form whose class number is divisible by n. This extend...

conjecture | open | MathDB 页面 | 原文线索

V2-002996 | MATHDB-338100

Girstmair's numerator restriction conjecture for normalized Dedekind sums

摘录 / EXCERPT: Let $S(a,b)$ be the normalized Dedekind sum, and write a value in lowest terms as $S(a,b)=k/q$, where $q\geq 2$ and $\gcd(k,q)=1$. Girstmair's necessary conditions are that...

conjecture | open | MathDB 页面 | 原文线索

V2-002997 | MATHDB-338101

The range conjecture for the Dedekind-sum auxiliary function λ

摘录 / EXCERPT: For an integer $q\geq 2$, let Λ_q be the set of residue classes produced by the auxiliary function $\lambda(a,t,t^{*})$, and define ... The range conjecture. For every intege...

conjecture | open | MathDB 页面 | 原文线索

V2-002998 | MATHDB-338122

The conjectural bounded-distance estimate beyond the square-spacing threshold

摘录 / EXCERPT: Let $a\ge0$, $d\ge1$, and $N\ge1$ be integers, and define ... Assume that $a\le\left(\frac{Nd-1}{2}\right)^2$ and $N>d+2\sqrt a$. The conjectural bounded-distance estimate. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-002999 | MATHDB-338123

The conjectural bound for twisted incomplete Salié sums

摘录 / EXCERPT: Let a and q be integers with $q\ge2$, $(a,q)=1$, and q not a perfect square. Let $H,K\ge1$ and let λ,μ be real numbers. Write $e(t)=e^{2\pi i t}$, let $\overline h$...

conjecture | open | MathDB 页面 | 原文线索

V2-003000 | MATHDB-338138

The conjecture that all trees are neighborhood-prime

摘录 / EXCERPT: Neighborhood-prime tree conjecture. Every tree is neighborhood-prime.

conjecture | open | MathDB 页面 | 原文线索

V2-003001 | MATHDB-338173

Cluckers' quasi-homogeneous generalization of Igusa's conjecture

摘录 / EXCERPT: Cluckers' generalization of Igusa's conjecture. There exists a constant $c\in\mathbf{R}_{>0}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003002 | MATHDB-338174

Cluckers-Veys conjecture on exponential sums and local log canonical thresholds

摘录 / EXCERPT: Cluckers-Veys conjecture. There exists a constant $c\in\mathbf{R}_{>0}$ such that the bound

conjecture | open | MathDB 页面 | 原文线索

V2-003003 | MATHDB-338178

Wan's genus-stability conjecture for p-adic Lie towers

摘录 / EXCERPT: Wan's genus-stability conjecture. If ρ is algebraic-geometric and ordinary, then the sequence (g_i) is given by a polynomial in p^i for $i\gg 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-003004 | MATHDB-338179

Multivariable Bateman-Horn conjecture

摘录 / EXCERPT: Multivariable Bateman-Horn conjecture. As $P\to+\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003005 | MATHDB-338182

Solution-count conjecture for Vinogradov systems modulo \(N\)

摘录 / EXCERPT: Let $\gamma(t)=(t,t^2,\ldots,t^n)$ and define $\Phi(\vec t)=\sum_{i=1}^n\gamma(t_i)$. For $\vec t\in[\mathbb Z/N\mathbb Z]^n$, let $\mathbf N(\vec t;N)$ be the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-003006 | MATHDB-338188

Conjecture on distinct determinants of matrices over prime fields

摘录 / EXCERPT: Let p be a prime, let $A\subset \mathbb{F}_p$, and let $f_d(A)$ denote the number of distinct determinants determined by $d\times d$ matrices with entries in A. Determinant-val...

conjecture | open | MathDB 页面 | 原文线索

V2-003007 | MATHDB-338189

Conjecture on distinct permanents of matrices over prime fields

摘录 / EXCERPT: Let p be a prime, let $A\subset \mathbb{F}_p$, and let $g_d(A)$ denote the number of distinct permanents determined by $d\times d$ matrices with entries in A. Permanent-value c...

conjecture | open | MathDB 页面 | 原文线索

V2-003008 | MATHDB-338201

Limsup-one conjecture for p-adic torsion growth

摘录 / EXCERPT: Let ${\mathcal K}_{\rm real}$ be the set of all totally real number fields, let D_K be the discriminant of K, and fix a prime $p\geq 2$. For each K, let ${\mathcal T}_K$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-003009 | MATHDB-338211

Bloch-Kato conjecture for Hilbert modular forms

摘录 / EXCERPT: Let g be a normalised cuspidal Hilbert modular newform of weight $(k_1,k_2,\ldots,k_d,w=2)$ and level $\Gamma_0(\mathfrak{n})$ over a totally real number field F, with each...

conjecture | open | MathDB 页面 | 原文线索

V2-003010 | MATHDB-338212

Siar-Keskin conjecture for the Pell and Pell Lucas families

摘录 / EXCERPT: Siar-Keskin's Pell-family conjecture. For every odd integer $k>3$, the equation has only the solution

conjecture | open | MathDB 页面 | 原文线索

V2-003011 | MATHDB-338213

Siar-Keskin bounded-exponent conjecture for $(a^n-2)(b^n-2)=x^2$

摘录 / EXCERPT: Siar-Keskin's bounded-exponent conjecture. If this equation has a solution n,x, then

conjecture | open | MathDB 页面 | 原文线索

V2-003012 | MATHDB-338219

The rational Dyck-path derived-equivalence conjecture

摘录 / EXCERPT: Let a and b be two co-prime integers. Let $\mathcal{L}_{a,b}$ be the poset of lattice paths in the rectangle $a\times b$, and let $\mathrm{Dyck}_{a,b}$ be the poset of (a,b)-...

conjecture | open | MathDB 页面 | 原文线索

V2-003013 | MATHDB-338228

Broadhurst-Mellit conjecture relating a 7-Bessel moment to a zeta function

摘录 / EXCERPT: Broadhurst-Mellit conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-003014 | MATHDB-338231

Darmon's rationality conjecture with Shimura reciprocity for Darmon points

摘录 / EXCERPT: Fix an integer c prime to $D\cdot N$, let $\mathcal{O}_c$ be the order of the real quadratic field F of conductor c, and let $Q\in\mathcal{F}_{Dc^2}$ represent a class of qua...

conjecture | open | MathDB 页面 | 原文线索

V2-003015 | MATHDB-338239

Vinogradov's conjecture on the least primitive root

摘录 / EXCERPT: Vinogradov's conjecture. For every $\epsilon>0$, for all sufficiently large primes p, one has

conjecture | open | MathDB 页面 | 原文线索

V2-003016 | MATHDB-338261

Singhi's conjecture on prime projective planes

摘录 / EXCERPT: A projective plane σ is a subplane of a projective plane π if σ is a subsystem of π. A projective plane is prime if it has no proper projective subplane. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003017 | MATHDB-338262

The minimum-weight dual-code characterization of the Desarguesian plane

摘录 / EXCERPT: Let π be a projective plane of prime order p, and let $C_p^\perp(\pi)$ denote the dual of its p-ary code. For a codeword, its Hamming weight is the number of nonzero coordi...

conjecture | open | MathDB 页面 | 原文线索

V2-003018 | MATHDB-338265

The fixed denominator conjecture for Markov numbers

摘录 / EXCERPT: Let $m_{p/q}$ denote the Markov number indexed by a reduced rational number p/q with $0<p<q$. Fixed denominator conjecture. For positive integers p,q,i such that $p+i<q$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003019 | MATHDB-338267

The categorical blob versus Soergel conjecture

摘录 / EXCERPT: Let p be a prime, let $l\geq 2$, and fix the data (e,l,n,κ) of a generalized blob algebra. For a regular $\boldsymbol{\lambda}\in P_1^l(n)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003020 | MATHDB-338276

Prime correlation conjecture for Kloosterman sums and cusp-form eigenvalues

摘录 / EXCERPT: Let f be a fixed primitive cusp form, holomorphic or Maass, and let $\lambda_f(p)$ denote its Hecke eigenvalue at a prime p. Write $\mathcal{L}=\log X$. Prime correlation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-003021 | MATHDB-338304

Conjecture on the Goldbach explicit-formula remainder under RH

摘录 / EXCERPT: Let N be a positive integer, let ρ_1 and ρ_2 range over the nontrivial zeros of the Riemann zeta function, and let $B_x(a,b)$ denote the incomplete beta function. The n...

conjecture | open | MathDB 页面 | 原文线索

V2-003022 | MATHDB-338350

Refinements of four Ramanujan-type supercongruences

摘录 / EXCERPT: The four supercongruence refinements. The following four congruences are proposed:

conjecture | open | MathDB 页面 | 原文线索

V2-003023 | MATHDB-338356

Infinitely many primes in the Chebyshev-related sequence $s_k(n)$

摘录 / EXCERPT: Let $(s_k(n))_{k\geq 0}$ be the sequence associated with the positive integer parameter n, and let $\mathcal{T}_p(j)$ denote the dilated Chebyshev polynomial of the first kind ev...

conjecture | open | MathDB 页面 | 原文线索

V2-003024 | MATHDB-338357

Asymptotic distribution of prime terms in $s_k(n)$

摘录 / EXCERPT: Let $(s_k(n))_{k\geq 0}$ be the sequence associated with the positive integer n. For a fixed n, define $(k_N)_{N\geq 1}$ by requiring that $s_{k_N}(n)$ is the Nth prime term...

conjecture | open | MathDB 页面 | 原文线索

V2-003025 | MATHDB-338358

Infinitely many primes in the companion sequence $r_k(n)$

摘录 / EXCERPT: Let $(r_k(n))_{k\geq 0}$ be the companion sequence associated with the positive integer parameter n, and let $\mathcal{T}_p(j)$ denote the relevant dilated Chebyshev value. Prime...

conjecture | open | MathDB 页面 | 原文线索

V2-003026 | MATHDB-338364

Jones' conjecture on binomial congruences

摘录 / EXCERPT: For $n\in\mathbb{N}$, let ... A prime p satisfying $w_p\equiv 1\pmod{p^3}$ is considered in the context of Wolstenholme's theorem. Jones' conjecture. ... This asserts that the known...

conjecture | open | MathDB 页面 | 原文线索

V2-003027 | MATHDB-338365

Hussein's conjecture on large prime factors of binomial congruences

摘录 / EXCERPT: For $n\in\mathbb{N}$, let ... Consider pairs of distinct primes p and q for which the congruence $w_p\equiv 1\pmod{q^2}$ holds. Hussein's conjecture. For all but at most finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-003028 | MATHDB-338377

Williamson's prime-factor conjecture implying Goldbach's conjecture

摘录 / EXCERPT: Let $p_{1},\ldots,p_{k}$ denote the first k odd primes in succession, and let n be an even integer such that $n>p_{k}$. Williamson's prime-factor conjecture. If, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-003029 | MATHDB-338412

Somekawa's generalized Galois symbol injectivity conjecture

摘录 / EXCERPT: Somekawa's conjecture. The generalized Galois symbol s_n is always injective.

conjecture | open | MathDB 页面 | 原文线索

V2-003030 | MATHDB-338420

The injectivity conjecture for Keller maps over the integers

摘录 / EXCERPT: Injectivity conjecture. Then F is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-003031 | MATHDB-338433

The unique coherent extension conjecture for balanced dessins

摘录 / EXCERPT: Unique coherent extension conjecture. This partial choice of factors has a unique coherent extension to $\tilde{\mathcal{A}}$, meaning that the ideal generated by $Z_\psi(\psi)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003032 | MATHDB-338449

The finite-set completion conjecture

摘录 / EXCERPT: Let $\mathcal{T}$ be a finite subset of the positive integers $\mathbb{N}$. Finite-set completion conjecture. The set $\mathcal{T}$ can be completed in $\mathbb{N}$, meaning that t...

conjecture | open | MathDB 页面 | 原文线索

V2-003033 | MATHDB-338450

The complete-subset counting upper-bound conjecture

摘录 / EXCERPT: Let $\mathcal{H}:=\{1,2,\ldots,N\}$, and let $\mathcal{C}(N)$ be the number of complete subsets of $\mathcal{H}$. Complete-subset counting conjecture. ... The paper studies the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-003034 | MATHDB-338472

The bounded-coefficient Dirichlet-series pole conjecture

摘录 / EXCERPT: Bounded-coefficient Dirichlet-series conjecture. Every function in this class has no poles to the right of the critical line $\operatorname{Re}(z)=1/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-003035 | MATHDB-338473

The critical-line conjecture for generalized zeta derivatives

摘录 / EXCERPT: Generalized zeta-derivative critical-line conjecture. The function ζ_f' has nontrivial zeroes only on the critical line $\operatorname{Re}(z)=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-003036 | MATHDB-338474

The turbulent Riemann hypothesis for generalized L-functions

摘录 / EXCERPT: Turbulent Riemann hypothesis. The analytic extension of $\zeta_{(f,g)}$ to the complex numbers has nontrivial zeroes only on the critical line

conjecture | open | MathDB 页面 | 原文线索

V2-003037 | MATHDB-338475

Weak pair correlation conjecture for Hilbert characters

摘录 / EXCERPT: Let $D<0$ be a fundamental discriminant, let $H(D)$ be the ideal class group of $\mathbb{Q}(\sqrt{D})$, and let $\hat{H}(D)$ be the family of unramified Hecke characters associated...

conjecture | open | MathDB 页面 | 原文线索

V2-003038 | MATHDB-338494

Infinitude conjecture for primitive weird numbers

摘录 / EXCERPT: A primitive weird number (PWN) is a weird number that is not a multiple of any smaller weird number. Infinitude conjecture. There exist infinitely many PWN. Despite the positive as...

conjecture | open | MathDB 页面 | 原文线索

V2-003039 | MATHDB-338519

Mixed involution and odd-prime count conjecture

摘录 / EXCERPT: Mixed counting conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-003040 | MATHDB-338521

Cohomological freeness conjecture for ordinary modular Hecke cohomology

摘录 / EXCERPT: Let H^d be the cohomological group associated with the ordinary Hecke algebra $\mathbf{T}^\circ_{L^+}$, where d is the relevant cohomological degree. Cohomological freeness con...

conjecture | open | MathDB 页面 | 原文线索

V2-003041 | MATHDB-338561

Uniform upper-bound conjecture for partial cubic exponential sums

摘录 / EXCERPT: Let $p\geq 3$ be prime, let $a\in\mathbb{F}_p^{\times}$, and let x satisfy $1<x\leq p$. Uniform partial-sum conjecture. There exists a positive constant C_0 such that ... This wo...

conjecture | open | MathDB 页面 | 原文线索

V2-003042 | MATHDB-338577

Benkoski-Erdős conjecture on infinitely many primitive weird numbers

摘录 / EXCERPT: Let a primitive weird number be a weird number that is not a multiple of any other weird number. Benkoski-Erdős conjecture. There exist infinitely many primitive weird numbers. The...

conjecture | open | MathDB 页面 | 原文线索

V2-003043 | MATHDB-338578

Conjecture on primitive weird numbers with arbitrarily many distinct prime factors

摘录 / EXCERPT: Let $k\geq 3$ be an integer. Conjecture. There exists a primitive weird number having at least k distinct prime factors. This would show that primitive weird numbers occur with a...

conjecture | open | MathDB 页面 | 原文线索

V2-003044 | MATHDB-338612

Uniform-distribution conjecture for the random variable X_l(m,N)

摘录 / EXCERPT: Let $N\ge 3$, and let l and m be positive integers such that $1\le l\le N$, $2\le m\le N-1$, and both l and m are relatively prime to N. Let $X_l(m,N)$ be the random vari...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-003045 | MATHDB-338638

Generalized Minkowski bound for prime ideals in ideal class groups

摘录 / EXCERPT: Let K) be a number field of bounded degree n over $\mathbb{Q}$, with discriminant D_K and class number h_K. An ideal class is said to contain a prime ideal if some prime id...

conjecture | open | MathDB 页面 | 原文线索

V2-003046 | MATHDB-338664

(LC) conjecture on Frobenius powers of ideals

摘录 / EXCERPT: Let $(R,\mathfrak{m},k)$ be a local or graded ring of prime characteristic $p>0$, and let $I\subseteq R$ be an ideal, homogeneous in the graded case. Write $I^{[p^e]}$ for the idea...

conjecture | open | MathDB 页面 | 原文线索

V2-003047 | MATHDB-338668

The density conjecture for prime divisors absent from the r-derangement numbers

摘录 / EXCERPT: Density conjecture. The set $\mathcal{A}_r$ is infinite. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-003048 | MATHDB-338669

Amdeberhan-Manna-Moll conjecture on 2-adic valuations of Stirling numbers

摘录 / EXCERPT: Amdeberhan-Manna-Moll conjecture. Fix a number $k\in\mathbb{N}_+$. Then there exist $m_0(k)\in\mathbb{N}_+$ and $\mu(k)\in\mathbb{N}$ such that for any integer $m\geq m_0(k)$ there...

conjecture | solved | MathDB 页面 | 原文线索

V2-003049 | MATHDB-338670

Berrizbeitia-Medina-Moll-Moll-Noble conjecture on p-adic valuations of Stirling numbers

摘录 / EXCERPT: Berrizbeitia-Medina-Moll-Moll-Noble conjecture. Fix a positive integer k and a prime number p such that $k<p$. Then there exist $m_0(k)\in\mathbb{N}_+$ and...

conjecture | solved | MathDB 页面 | 原文线索

V2-003050 | MATHDB-338677

The Bézout tree conjecture for Pythagorean pairs

摘录 / EXCERPT: Consider the trinary trees generated by $(2,1)$ and $(3,1)$. For a relatively prime pair (m,n), let (u,v) be the pair in a Bézout tree corresponding to (m,n), and let (U,V)...

conjecture | open | MathDB 页面 | 原文线索

V2-003051 | MATHDB-338693

The folklore conjecture on unbounded class-group ranks

摘录 / EXCERPT: Let $d>1$ and $m>1$ be integers. For a number field k, let $\operatorname{Cl}(k)$ denote its ideal class group, let $[k:\mathbb{Q}]$ denote its degree over $\mathbb{Q}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003052 | MATHDB-338719

D-transcendence conjecture for prime-related generating functions

摘录 / EXCERPT: Prime-series D-transcendence conjecture. The generating functions

conjecture | open | MathDB 页面 | 原文线索

V2-003053 | MATHDB-338720

D-transcendence conjecture for the cubic theta series

摘录 / EXCERPT: Cubic theta-series conjecture. The series $F(t)$ is D-transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-003054 | MATHDB-338737

Fundamental-solution conjecture for Helmholtz operators on spheres

摘录 / EXCERPT: Fundamental-solution conjecture. The candidate function $\mathsf S_{R,\beta}^{d,-}(\mathbf x,\mathbf x^\prime)$ is a fundamental solution for the Helmholtz operator...

conjecture | open | MathDB 页面 | 原文线索

V2-003055 | MATHDB-338746

Sparse large sieve conjecture for function fields

摘录 / EXCERPT: Let $\boldsymbol{F}_q[t]$ be the polynomial ring over the finite field with q elements. Let $\boldsymbol{S}$ be a set of nonzero monic polynomials in $\boldsymbol{F}_q[t]$ of deg...

conjecture | open | MathDB 页面 | 原文线索

V2-003056 | MATHDB-338750

The equivalent linear arithmetic fundamental lemma identity

摘录 / EXCERPT: Let K/F be an unramified quadratic extension with odd residue characteristic, let (φ,τ) be a pair of isomorphisms, and let f be a spherical Hecke function. For...

conjecture | open | MathDB 页面 | 原文线索

V2-003057 | MATHDB-338773

The prime-divisor positivity implication for Goldbach convolution sums

摘录 / EXCERPT: Let N be an integer, let p range over the primes dividing N, and let Λ_0 denote the function used in the source. Prime-divisor positivity conjecture. If ... then ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-003058 | MATHDB-338778

Erez's conjecture on the square root of the inverse different

摘录 / EXCERPT: Let F_{π}/F be a finite Galois-algebra extension of number fields with group G. Write $\mathfrak{D}(\pi)$ for its different, and suppose that the square root $A(\pi)$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-003059 | MATHDB-338796

The conjecture on complex dimensions of Cartesian products

摘录 / EXCERPT: Complex dimensions of Cartesian products. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-003060 | MATHDB-338827

Malle-Navarro inequalities for character and module counts in blocks

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a Brauer p-block of G with defect group D. Write $D'=[D,D]$ for the derived subgroup of D. Let $k(B)$ be the n...

conjecture | open | MathDB 页面 | 原文线索

V2-003061 | MATHDB-338839

Almost-prime conjecture for degenerate horizon cross-sections

摘录 / EXCERPT: Almost-prime conjecture. Each such $\mathcal{H}$ is almost prime: if it is expressed as a connected sum of two manifolds, then at least one of M_1 and M_2 is simply connected.

conjecture | open | MathDB 页面 | 原文线索

V2-003062 | MATHDB-338865

Higher-dimensional grid graph conjecture for neighborhood-prime labelings

摘录 / EXCERPT: Higher-dimensional grid conjecture. The graph $P_{n_1}\times P_{n_2}\times\cdots\times P_{n_t}$ is neighborhood-prime.

conjecture | open | MathDB 页面 | 原文线索

V2-003063 | MATHDB-338866

Hamiltonian edge-density conjecture for neighborhood-prime graphs

摘录 / EXCERPT: Hamiltonian edge-density conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-003064 | MATHDB-338867

Neighborhood-prime conjecture for unions of cycles with one arbitrary component

摘录 / EXCERPT: Cycle-union conjecture. The union

conjecture | open | MathDB 页面 | 原文线索

V2-003065 | MATHDB-338868

Cloys and Fox's conjecture on neighborhood-prime lobsters

摘录 / EXCERPT: Cloys and Fox's lobster conjecture. Every lobster is neighborhood-prime.

conjecture | open | MathDB 页面 | 原文线索

V2-003066 | MATHDB-338905

Uniform square-root conjecture for the Piltz divisor error term over number fields

摘录 / EXCERPT: Uniform square-root conjecture. If K runs through number fields with $D_K<x$, then

conjecture | open | MathDB 页面 | 原文线索

V2-003067 | MATHDB-338921

Conjecture on integer ratios of consecutive odd power sums

摘录 / EXCERPT: Conjecture on odd power sums.

conjecture | open | MathDB 页面 | 原文线索

V2-003068 | MATHDB-338922

Conjecture equating integer ratios for ordinary and odd power sums

摘录 / EXCERPT: Equality conjecture for power-sum ratios.

conjecture | open | MathDB 页面 | 原文线索

V2-003069 | MATHDB-338944

Lattice optimality conjecture for Riesz energy constants in dimensions 2, 4, 8, and 24

摘录 / EXCERPT: Let $C_{s,d}$ be the asymptotic Riesz s-energy constant for d-rectifiable sets, and let ... be the Epstein zeta function of a lattice $\Lambda\subset\mathbb{R}^d$ with covolume...

conjecture | open | MathDB 页面 | 原文线索

V2-003070 | MATHDB-338954

Conjectural discrete moment bound for Weyl sums

摘录 / EXCERPT: Let $k\geq 3$, set ... and suppose that $|\alpha_{k}-a/q|<q^{-2}$ for coprime integers $(a,q)=1$. Let $S_{az}(\boldsymbol{\alpha})$ denote the Weyl sum used in the paper. Conjectural...

conjecture | open | MathDB 页面 | 原文线索

V2-003071 | MATHDB-338986

Isaacs--Navarro refinement of McKay's conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Isaacs--Navarro refinement. In the situation of McKay's conjecture, there exists a bijection .....

conjecture | open | MathDB 页面 | 原文线索

V2-003072 | MATHDB-338987

Brauer's height-zero conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a p-block with defect group D. The characters in ... are the characters of height zero. Brauer's height-zero conject...

conjecture | open | MathDB 页面 | 原文线索

V2-003073 | MATHDB-338989

Natural basic sets for blocks of quasi-simple groups of Lie type

摘录 / EXCERPT: Let G be a quasi-simple group of Lie type and let ℓ be a prime. An ℓ-block B of G has a basic set if a linearly independent subset $X\subseteq\operatorname{Irr}(B)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003074 | MATHDB-339008

Existence of bounded-degree Grassmann eposets

摘录 / EXCERPT: Let q be a prime power, let $d\in\mathbb{N}$, and let $0<\gamma<1$. A d-dimensional measured poset X is sparse when $|X(0)|=n$ and $|X(d)|=O(n)$, where the implied constant m...

conjecture | open | MathDB 页面 | 原文线索

V2-003075 | MATHDB-339022

Structure conjecture for prime Hopf algebras of GK-dimension one

摘录 / EXCERPT: Let H be a prime Hopf algebra of GK-dimension one. An exact sequence of Hopf algebras is a sequence consisting of Hopf algebra morphisms with the usual exactness property. Here,...

conjecture | open | MathDB 页面 | 原文线索

V2-003076 | MATHDB-339032

Apagodu's congruences for sums of super Catalan numbers

摘录 / EXCERPT: Let $S(m,n)$ denote the super Catalan numbers ... For an odd prime p, let $\left(\frac{p}{3}\right)$ denote the Legendre symbol. Apagodu's conjecture. For any odd prime p, ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-003077 | MATHDB-339034

Kuca's regularity classification conjecture for Ulam and modified Ulam sequences

摘录 / EXCERPT: Let $a<b$ be relatively prime positive integers. Let $\mathcal{U}(a,b)$ denote the Ulam sequence generated by a,b, let $\mathcal{V}(a,b)$ denote the corresponding modified Ulam s...

conjecture | open | MathDB 页面 | 原文线索

V2-003078 | MATHDB-339035

Kuca's even-term conjecture for modified Ulam sequences with fixed even generator

摘录 / EXCERPT: Let a,b be relatively prime, with a even and $b>2a$ odd. The modified Ulam sequence $\mathcal{V}(a,b)$ has explicitly listed even terms in the cases $a\in\{4,6,8,10,12,14,16\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-003079 | MATHDB-339036

Finch's even-term conjecture for two-generator Ulam sequences

摘录 / EXCERPT: Let a,b be relatively prime positive integers, with one of a,b even. For an Ulam sequence $\mathcal{U}(a,b)$, Finch's even-term conjecture. If a is even, then...

conjecture | open | MathDB 页面 | 原文线索

V2-003080 | MATHDB-339045

Conjecture on the extreme values of quadratic Dirichlet L-functions over function fields

摘录 / EXCERPT: Extreme-value conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003081 | MATHDB-339065

Tauraso's q-supercongruence conjecture for central q-binomial coefficients

摘录 / EXCERPT: Tauraso's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003082 | MATHDB-339102

Equality of automorphic L-invariants across cohomology degrees

摘录 / EXCERPT: Let F be a number field, let $b\mathfrak{p}d$ be a fixed prime above p, let π be the automorphic representation and let i and s be the cohomological indices occurring i...

conjecture | open | MathDB 页面 | 原文线索

V2-003083 | MATHDB-339134

Existence conjecture for double blocking sets with two long secants

摘录 / EXCERPT: Let q be a prime power satisfying $q\geq 13$ and $q\not\equiv 2\pmod 3$. A double blocking set in $\operatorname{PG}(2,q)$ is a set meeting every line in at least two points; a...

conjecture | open | MathDB 页面 | 原文线索

V2-003084 | MATHDB-339144

Finiteness conjecture for Type (i) and Type (iia) power mappings

摘录 / EXCERPT: Let n be a positive integer, let p be a prime with $p>2n$, and partition the reduced residues modulo p into residue classes I_i modulo n. Let $f(x)=Ax^k\pmod p$ be a powe...

conjecture | open | MathDB 页面 | 原文线索

V2-003085 | MATHDB-339145

The rationality conjecture for Morita Frobenius numbers of blocks

摘录 / EXCERPT: Let ℓ be a prime number, let $k\cong\overline{\mathbb{F}}_\ell$ be the residue field of an absolutely unramified ℓ-modular system, and let D be a finite ℓ-group. T...

conjecture | open | MathDB 页面 | 原文线索

V2-003086 | MATHDB-339151

Conjecture on birational models of modular curves from primitive cusp forms

摘录 / EXCERPT: The birational-model conjecture. The map defined in the paper is a birational equivalence, and the curve $\mathcal C(f,g,h)$ has degree

conjecture | open | MathDB 页面 | 原文线索

V2-003087 | MATHDB-339160

Stronger supercongruences for selected hypergeometric parameters

摘录 / EXCERPT: Fix an integer $d\ge2$, let p be an odd prime, and use the truncated hypergeometric series $F_p(\epsilon)$ and unit-root limit $f_p(\epsilon)$ defined above, with $\epsilon=\pm1$...

conjecture | open | MathDB 页面 | 原文线索

V2-003088 | MATHDB-339161

Modular-form coefficient conjectures for negative evaluation of hypergeometric series

摘录 / EXCERPT: Negative-value modular-form conjectures. Assuming that $F_p(-1)$ is a p-adic unit, for $d=3$ one expects

conjecture | open | MathDB 页面 | 原文线索

V2-003089 | MATHDB-339162

Higher-level supercongruence for truncated hypergeometric series

摘录 / EXCERPT: Fix an integer $d\ge2$, let p be an odd prime, and define ... and ... Let $\epsilon=\pm1$ and suppose that $F_p(\epsilon)$ is a p-adic unit. Higher-level supercongruence. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-003090 | MATHDB-339200

Fixed-dimensional k-Frobenius Problem hardness conjecture

摘录 / EXCERPT: Let k-Frobenius Problem denote the problem defined earlier in the source. The fixed-dimensional k-Frobenius Problem hardness conjecture. The k-Frobenius Problem is...

conjecture | open | MathDB 页面 | 原文线索

V2-003091 | MATHDB-339238

Linear independence conjecture for Dedekind zeta-functions

摘录 / EXCERPT: Let the nontrivial zeros of a Dedekind zeta-function be written as $\rho=\beta+i\gamma$, and consider their positive imaginary parts $\gamma>0$. Linear independence conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-003092 | MATHDB-339239

Asymptotic reciprocal-derivative conjecture for Dirichlet and Dedekind zeta-functions

摘录 / EXCERPT: Let χ be a primitive Dirichlet character, and let K be an Abelian number field. Write $\rho=\beta+i\gamma$ for nontrivial zeros of the relevant Dirichlet L-function...

conjecture | open | MathDB 页面 | 原文线索

V2-003093 | MATHDB-339240

Ailon-Rudnick conjecture on coprime values of exponential recurrences

摘录 / EXCERPT: Ailon-Rudnick conjecture. The set $\mathcal{A}_{F,G}$ is infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-003094 | MATHDB-339244

Irreducibility conjecture for the non-cyclotomic factor of f_k

摘录 / EXCERPT: Let ... be the polynomial considered above, and let Φ_n denote the nth cyclotomic polynomial. Write $(f_k,\Phi_6\Phi_{10}\Phi_{12})$ for the greatest common divisor of f_k a...

conjecture | open | MathDB 页面 | 原文线索

V2-003095 | MATHDB-339321

Non-factorability conjecture for the character on smooth polynomial congruences

摘录 / EXCERPT: Let n, m, and α and the notation above be given. Let $\mathcal{P}_S$ be the set of smooth elements of the multiplicative semigroup $\mathcal{P}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003096 | MATHDB-339323

Conjecture on infinitely many non-covering numbers

摘录 / EXCERPT: Let $\mathscr{E}$ be the set of all integers that are covering numbers, meaning integers of the form $\sigma(G)$ for some group G. Non-covering-number infinitude conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-003097 | MATHDB-339348

Lagarias-Rains-Sloane asymptotic conjecture for the EKG sequence

摘录 / EXCERPT: Lagarias-Rains-Sloane conjecture. In the EKG sequence, if $a(n)$ is neither a prime nor three times a prime, then

conjecture | open | MathDB 页面 | 原文线索

V2-003098 | MATHDB-339353

Byszewski-Ulas digit-sum summation conjecture for powers of binary sums

摘录 / EXCERPT: Let $r\ge 1$, let $N_1,\ldots,N_r$ be positive integers, and let $x,y_1,\ldots,y_r$ be variables. For each nonnegative integer n, write $s_2(n)$ for the sum of the binary digits...

conjecture | open | MathDB 页面 | 原文线索

V2-003099 | MATHDB-339354

Byszewski-Ulas digit-sum factorization conjecture

摘录 / EXCERPT: Let $r\ge 1$, let $N_1,\ldots,N_r$ be positive integers, and let $x_1,\ldots,x_r,y_1,\ldots,y_r$ be variables. For each nonnegative integer n, write $s_2(n)$ for the sum of the b...

conjecture | open | MathDB 页面 | 原文线索

V2-003100 | MATHDB-339367

Pillai's conjecture for alternating sums of consecutive primes

摘录 / EXCERPT: Let p_i) denote the ith prime. Pillai's conjecture. As $k\to\infty$, the absolute value of the alternating sum of the first k primes satisfies ... This conjecture is attributed...

conjecture | open | MathDB 页面 | 原文线索

V2-003101 | MATHDB-339368

Lower asymptotic bound for primes in the alternating-sum sequence

摘录 / EXCERPT: Let (A_n) be defined by $A_n=\sum_{i=1}^{2n}(-1)^ip_i$, and let $r_k=A_m$ be the kth prime occurring in this sequence. Alternating-sum prime lower-bound conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-003102 | MATHDB-339369

Restricted Prime Number Theorem for shifted-index alternating sums

摘录 / EXCERPT: Shifted-index restricted-prime conjecture. For every fixed positive integer k, the sequence $(A_n^{(k)})$ satisfies the Restricted Prime Number Theorem. The conjecture is present...

conjecture | open | MathDB 页面 | 原文线索

V2-003103 | MATHDB-339370

Dirichlet's theorem for the alternating-sum sequences

摘录 / EXCERPT: Let a and d be relatively prime positive integers. Dirichlet-type conjecture for alternating sums. Each of the sequences (A_n) and (T_n) contains infinitely many primes of...

conjecture | open | MathDB 页面 | 原文线索

V2-003104 | MATHDB-339371

Bounds for the deviation of alternating sums from the kth prime

摘录 / EXCERPT: Let p_n be the nth prime and define $A_n=\sum_{i=1}^{2n}(-1)^ip_i$. Deviation-bound conjecture. For every $n\ge50$, ... and for every $n\ge10^8+5\cdot10^7$, ... These two numerical...

conjecture | open | MathDB 页面 | 原文线索

V2-003105 | MATHDB-339375

Natural-density conjecture for inert degree-n number fields

摘录 / EXCERPT: For $d\in\mathbb{N}^*$ and $n\in\mathbb{N}^*$, let $N_n(d)$ be the number of degree-n fields over $\mathbb{Q}$ whose discriminants have absolute value at most d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003106 | MATHDB-339389

Combined Freiman covering conjecture in a prime cyclic group

摘录 / EXCERPT: Let $\mathcal{A}\subseteq\mathbb{Z}_p$ be a set. Combined Freiman covering conjecture. If either ... or ... then $\mathcal{A}$ can be covered by an arithmetic progression of length at...

conjecture | open | MathDB 页面 | 原文线索

V2-003107 | MATHDB-339392

The uniqueness conjecture for Leopoldt fields

摘录 / EXCERPT: Let K be a number field. Call K a Leopoldt field if every tame abelian extension L/K has a weak normal integral basis, meaning that, for the maximal order $\mathcal{M}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-003108 | MATHDB-339421

Refined Goldbach conjecture modulo 4

摘录 / EXCERPT: Let p and q be primes. The mod-4 refined Goldbach conjecture. The following assertions hold: 1. Every even $n>4$ has a representation $n=p+q$ with $p\equiv3\pmod4$. 2. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-003109 | MATHDB-339422

Refined Goldbach conjectures for selected moduli

摘录 / EXCERPT: Let p and q be primes. The selected-moduli refined Goldbach conjecture. The following assertions hold: 1. Every even $n\equiv0\pmod3$ except $n=6$ has a representation with...

conjecture | open | MathDB 页面 | 原文线索

V2-003110 | MATHDB-339423

Exceptional-set growth conjecture for Goldbach representations in progressions

摘录 / EXCERPT: Exceptional-set growth conjecture. As $m\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003111 | MATHDB-339424

Existence of infinitely many empty exceptional sets

摘录 / EXCERPT: Let $E_{a,b,m}$ be the exceptional set of positive even integers in the residue class $a+b\pmod m$ that do not have a representation $n=p+q$ with primes $p\equiv a\pmod m$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003112 | MATHDB-339425

Ternary Goldbach conjecture with primes in one progression modulo 3

摘录 / EXCERPT: Let p,q,r be primes. The ternary progression Goldbach conjecture. Every odd integer $n>5$ can be written as ... with ... This is stated as a ternary analogue of one of the explicit m...

conjecture | open | MathDB 页面 | 原文线索

V2-003113 | MATHDB-339442

Polynomial bound for solutions in almost subgroups

摘录 / EXCERPT: Polynomial almost-subgroup conjecture. There is a constant $c(k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003114 | MATHDB-339452

Giannelli's conjecture on linear constituents of restricted characters

摘录 / EXCERPT: Let p be a prime, let G be a finite group, and let P be a Sylow p-subgroup of G. Write $\operatorname{Irr}(G)$ for the set of ordinary irreducible characters of G, and...

conjecture | open | MathDB 页面 | 原文线索

V2-003115 | MATHDB-339453

The global equality between Galois-Gauss and twisted unramified classes

摘录 / EXCERPT: Let L/K be a weakly ramified odd-degree Galois extension of number fields. Let $\mathfrak{a}_{L/K}$ and $\mathfrak{c}_{L/K}$ be the global elements defined in the source. Global...

conjecture | open | MathDB 页面 | 原文线索

V2-003116 | MATHDB-339462

Deines-Fuselier-Long-Swisher-Tu supercongruence for truncated hypergeometric series

摘录 / EXCERPT: Let $n\geq 3$ be an integer and let p be a prime such that $p\equiv 1\pmod{n}$. For parameters $x_1,\ldots,x_n$, $y_1,\ldots,y_{n-1}$, define the truncated hypergeometric series...

conjecture | open | MathDB 页面 | 原文线索

V2-003117 | MATHDB-339499

Mu-invariant conjecture for Eisenstein Hida families

摘录 / EXCERPT: Let (p,j) be an irregular pair satisfying the condition $(r1)$, let f_k be a classical specialization with $k\equiv j\pmod{p-1}$, and let $\mu(f_k,\omega^a)$ denote its μ-i...

conjecture | open | MathDB 页面 | 原文线索

V2-003118 | MATHDB-339500

Main conjecture for Selmer groups and p-adic L-functions

摘录 / EXCERPT: Let f be a classical modular form, let $A_{f,j}$ be the associated Galois module in the ω^j-twist, and let $\mathbb Q_\infty$ be the cyclotomic $\mathbb Z_p$-extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-003119 | MATHDB-339501

Greenberg's algebraic mu-invariant conjecture for modular forms

摘录 / EXCERPT: Let f be a classical form in the setting of the preceding section, and set $A_f=V_f/T_{\operatorname{eis}(f)}$, where $T_{\operatorname{eis}(f)}$ is the chosen Eisenstein lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-003120 | MATHDB-339517

Hamada-Sachar conjecture on the p-rank of projective planes

摘录 / EXCERPT: Let Π be a projective plane of order p^s, where p is prime, and let its incidence matrix have rank over $\mathbb{F}_p$ called its p-rank. Hamada-Sachar conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-003121 | MATHDB-339524

Multiplicity conjecture for mod-2 dihedral cuspforms of prime conductor

摘录 / EXCERPT: Let N be an odd prime and let $\mathfrak{m}$ be a maximal ideal of $\mathbb{T}_2(N)$. For a quadratic field K, say that $\mathfrak{m}$ is K-dihedral when the associated mod-2...

conjecture | open | MathDB 页面 | 原文线索

V2-003122 | MATHDB-339527

Primitive-representation local converse conjecture over finite fields

摘录 / EXCERPT: Let r be a prime divisor of n. For a regular character χ of $\mathbb F_{q^n}^{\times}$, let π_χ be the associated primitive cuspidal representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-003123 | MATHDB-339529

Gauss-sum local converse conjecture in prime characteristic

摘录 / EXCERPT: Let n be prime and let ω be a character of $\mathbb F_{p^n}^{\times}$; write $S(\chi)$ for the corresponding Gauss sum. Let $\hat k=k(p^n-1)/(p-1)$. Prime-characteristic G...

conjecture | open | MathDB 页面 | 原文线索

V2-003124 | MATHDB-339530

Kloosterman-sheaf rigidity conjecture for prime rank

摘录 / EXCERPT: Let n be prime and let $\psi:\mathbb F_q\to\overline{\mathbb Q}_\ell^{\times}$ be nontrivial. Let $\Gamma=\varinjlim\widehat{\mathbb F_{q^n}^{\times}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003125 | MATHDB-339558

Prime-degree extremal polynomial conjecture in the 5 modulo 6 case

摘录 / EXCERPT: Let $d\geq 17$ be prime with $d\equiv 5\pmod 6$. Define ... Let an algebraic integer of degree d that is not a root of unity be extremal if it has the minimum possible house among...

conjecture | open | MathDB 页面 | 原文线索

V2-003126 | MATHDB-339637

Ash's mod-p Galois representation conjecture for Hecke eigensystems

摘录 / EXCERPT: Let S be a finite set of prime numbers. If $\varphi:\mathscr{H}^S_{{\mathbf{GL}}_n}\longrightarrow\mathbb{F}_p$ is a character occurring in...

conjecture | open | MathDB 页面 | 原文线索

V2-003127 | MATHDB-339651

Farhi's almost-universality conjecture for equal floor-square sums

摘录 / EXCERPT: Farhi's conjecture. For each integer $m\geq 3$, every natural number n can be represented as

conjecture | open | MathDB 页面 | 原文线索

V2-003128 | MATHDB-339696

Leopoldt's conjecture on the rank of global units

摘录 / EXCERPT: Leopoldt's conjecture. The $\mathbb{Z}$-rank of $\mathcal{O}_M^{\times}$ equals the $\mathbb{Z}_p$-rank of the topological closure of $D(U)$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-003129 | MATHDB-339712

Constructive Denef--Loeser conjecture

摘录 / EXCERPT: Constructive Denef--Loeser conjecture. For every embedded toric resolution π of a singularity f and every pole q of the topological zeta-function of f, we have

conjecture | open | MathDB 页面 | 原文线索

V2-003130 | MATHDB-339716

Umbral moonshine conjecture for the modules K^X

摘录 / EXCERPT: For each of the 23 umbral cases, let G^X be the corresponding finite group, let m be its associated positive integer, and let $I^X\subset\{1,2,\dots,m-1\}$ be the specified sub...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-003131 | MATHDB-339760

Mazur-Rubin-Stein conjecture on shifted modular-symbol convolutions

摘录 / EXCERPT: Mazur-Rubin-Stein conjecture. For each $x\in[0,1]$, the limits satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-003132 | MATHDB-339762

Oura's conjecture on analogies between Eisenstein series and Eisenstein polynomials

摘录 / EXCERPT: Oura's conjecture. The analogous properties hold for the transforms of the normalized Eisenstein polynomials, namely:

conjecture | open | MathDB 页面 | 原文线索

V2-003133 | MATHDB-339765

Two-prime S-Diophantine quadruple conjecture

摘录 / EXCERPT: Let $p<q$ be primes and let $S=\{p,q\}$. An S-Diophantine quadruple is a quadruple of distinct positive integers such that every pairwise product plus one has all prime divisors...

conjecture | open | MathDB 页面 | 原文线索

V2-003134 | MATHDB-339766

Wieferich-free two-prime S-Diophantine quadruple conjecture

摘录 / EXCERPT: Let $p<q$ be primes. A pair (p,q) is called a Wieferich pair when it satisfies the paper's Wieferich condition. Wieferich-free two-prime S-Diophantine quadruple conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-003135 | MATHDB-339767

The {5,q}-Diophantine quadruple conjecture

摘录 / EXCERPT: Let q be a prime. The $\{5,q\}$-Diophantine quadruple conjecture. If $q\not\equiv1\pmod4$, then no $\{5,q\}$-Diophantine quadruple exists. The authors say that this should be pro...

conjecture | open | MathDB 页面 | 原文线索

V2-003136 | MATHDB-339793

A modulo p^5 extension of van Hamme's (A.2) supercongruence

摘录 / EXCERPT: Let $p\geq 5$ be a prime satisfying $p\equiv 3\pmod{4}$, and let Γ_p denote the p-adic Gamma function. A modulo p^5 extension of van Hamme's (A.2) supercongruence. ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-003137 | MATHDB-339844

Van der Waerden's conjecture on exceptional Galois groups of polynomials

摘录 / EXCERPT: Van der Waerden's conjecture. For $n\geqslant 3$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-003138 | MATHDB-339865

Non-vanishing conjecture for the Rankin convolution of an Ikeda lift

摘录 / EXCERPT: Non-vanishing conjecture. The Rankin convolution $R(k+n+\frac{r-1}{2},H,\bar g)$ is non-vanishing. The series is known to converge absolutely when $2n>r+2$, but its non-vanishing i...

conjecture | open | MathDB 页面 | 原文线索

V2-003139 | MATHDB-339878

The modular order conjecture for monomials in Eisenstein series

摘录 / EXCERPT: Let E_4 and E_6 be the Eisenstein series, and for a modular form ϕ let $\mathrm{mord}(\phi)$ denote the least nonnegative integer n such that some element of...

conjecture | open | MathDB 页面 | 原文线索

V2-003140 | MATHDB-339910

The paper's sieve bound for integers with large least prime divisor

摘录 / EXCERPT: Let a,b be positive integers with $\gcd(a,b)=1$, let x and y be positive real parameters, and let $p(n)$ denote the smallest prime divisor of the integer n. Sieve bound con...

conjecture | open | MathDB 页面 | 原文线索

V2-003141 | MATHDB-339912

Aka-Einsiedler-Shapira joint equidistribution conjecture

摘录 / EXCERPT: Joint equidistribution conjecture. As $D\to\infty$ along $\mathbb{D}(d)$, the measures μ_D^{Grid} converge weak-$*$ to normalized Haar measure on

conjecture | open | MathDB 页面 | 原文线索

V2-003142 | MATHDB-339913

Aka-Einsiedler-Shapira weaker joint equidistribution conjecture in dimension three

摘录 / EXCERPT: Weaker joint equidistribution conjecture. As $D\to\infty$ through $\mathbb{D}(3)$, the measures μ_D^{Lat} converge weak-$*$ to normalized Haar measure on

conjecture | open | MathDB 页面 | 原文线索

V2-003143 | MATHDB-339921

Nonvanishing harmonic sum conjecture for odd primes

摘录 / EXCERPT: Nonvanishing harmonic sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003144 | MATHDB-339943

The $p=2$ conjecture on symmetries of $\mathcal{L}$-invariant slopes

摘录 / EXCERPT: Given a positive even integer k, let $\nu_\mathcal{L}^{\varepsilon}(k,p,N)$ be the finite sequence of slopes of the p-adic $\mathcal{L}$-invariants attached to forms in...

conjecture | open | MathDB 页面 | 原文线索

V2-003145 | MATHDB-339979

The jumping champions conjecture for prime gaps

摘录 / EXCERPT: For $t>0$, let the jumping champion be the integer g occurring most frequently as a gap between two successive primes less than or equal to t. Jumping champions conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-003146 | MATHDB-339986

Conjectured totient formula for $C_{2,2}$

摘录 / EXCERPT: Totient-formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003147 | MATHDB-339992

Prime-partition equivalence conjecture for consumption and dominance orderings

摘录 / EXCERPT: Let a partition of n be ordered by the dominance ordering, and let the consumption ordering be the ordering introduced in the paper for distinguishing partitions. The prime-parti...

conjecture | open | MathDB 页面 | 原文线索

V2-003148 | MATHDB-339993

Nonprime partition nonequivalence conjecture for consumption and dominance orderings

摘录 / EXCERPT: Let the consumption ordering and dominance ordering be the two orderings on partitions of n introduced in the paper. The nonprime partition nonequivalence conjecture. If n is n...

conjecture | open | MathDB 页面 | 原文线索

V2-003149 | MATHDB-340014

The weak acyclic matching conjecture for cyclic groups of prime order

摘录 / EXCERPT: Weak acyclic matching conjecture. There exists a prime p such that $\mathbb{Z}/p\mathbb{Z}$ does not satisfy the weak acyclic matching property.

conjecture | open | MathDB 页面 | 原文线索

V2-003150 | MATHDB-340016

Prime-degree linear acyclic matching conjecture for field extensions

摘录 / EXCERPT: Prime-degree linear acyclic matching conjecture. There are infinitely many primes p for which there exists a field extension $K\subset L$ with $[L:K]=p$ such that $K\subset L$ do...

conjecture | open | MathDB 页面 | 原文线索

V2-003151 | MATHDB-340025

The Hecke-commensurator conjecture for triangle groups

摘录 / EXCERPT: Let Γ be a discrete triangle group of type (m_1,m_2,∞), and let $G={\rm GL}(2,\mathbb{C})$. Define ... Let $Z(G)\cong \mathbb{C}^*$ be the group of scalar matrices. The...

conjecture | open | MathDB 页面 | 原文线索

V2-003152 | MATHDB-340027

Conjecture on the mean square of the zeta-zero ordinate sum

摘录 / EXCERPT: Let $G(s)$ be the function considered above, and let T tend to infinity. Mean-square conjecture. ... The preceding theorem gives the unconditional bounds...

conjecture | open | MathDB 页面 | 原文线索

V2-003153 | MATHDB-340028

The principal capitulation conjecture for unramified abelian extensions

摘录 / EXCERPT: Principal capitulation conjecture. The order of the capitulation kernel is a multiple of the degree of the extension:

conjecture | open | MathDB 页面 | 原文线索

V2-003154 | MATHDB-340044

Oscillation conjecture for the Hardy-Littlewood prime-counting inequality

摘录 / EXCERPT: Let $x\geq 2$ be large, let $r>0$ be real, and set $y=\log^r x$. With $\pi(t)=\#\{p\leq t:p\text{ is prime}\}$, the oscillation conjecture. Both inequalities ... and ... occur infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-003155 | MATHDB-340045

LI-conditioned extremal-order conjecture for the Chebyshev psi function

摘录 / EXCERPT: Let $\psi(x)=\sum_{p^k\leq x}\log p^k$ be the weighted prime-counting function. Assume the Riemann hypothesis, and let the nontrivial zeros of the zeta function be written as...

conjecture | open | MathDB 页面 | 原文线索

V2-003156 | MATHDB-340067

Toral Affine Sieve Conjecture for hyperbolic toral orbits

摘录 / EXCERPT: Let $\g\in\operatorname{GL}_2(\mathbb{Q})$ be a hyperbolic matrix, that is, one having two distinct real eigenvalues; equivalently ... Let $\G=\langle\g\rangle^+:=\{\g^n:n\ge0\}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-003157 | MATHDB-340069

Refined modularity conjecture for Vafa-Witten generating functions

摘录 / EXCERPT: Let S be a smooth projective surface satisfying $H_1(S,\mathbb Z)=0$ and $p_g>0$. Let H be a polarization, let $r=1$ or a prime, and let $c_1\in H^2(S,\mathbb Z)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003158 | MATHDB-340071

Refined K3 Vafa-Witten formula for prime rank

摘录 / EXCERPT: Let S be a K3 surface with polarization H, first Chern class c_1, and prime rank r. Let $\mathsf Z_{K3,H,r,c_1}^{\mathrm{inst}}(q,y)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003159 | MATHDB-340084

Everest-Miller-Stephens primality conjecture for elliptic divisibility sequences

摘录 / EXCERPT: Let $D=(D_n)_{n\in\mathbb{N}}$ be an elliptic divisibility sequence. Primality conjecture. The sequence D contains only finitely many prime terms. Everest, Miller, and Stephens p...

conjecture | open | MathDB 页面 | 原文线索

V2-003160 | MATHDB-340092

Broadhurst-Zudilin conjecture on the divisibility of coefficients of a meromorphic modular form

摘录 / EXCERPT: Let ... where $q_z=e^{2\pi i z}$ and ... The function ϕ is a meromorphic modular form of weight 4 on $\Gamma_0(8)$ with double poles at $z_\pm=(\pm1+i)/4$. Broadhurst-Zudilin's...

conjecture | open | MathDB 页面 | 原文线索

V2-003161 | MATHDB-340101

Uniformity Conjecture for mod-\ell Galois representations of elliptic curves

摘录 / EXCERPT: Let k be a number field, and let E/k be a non-CM elliptic curve. For each prime ℓ, consider the mod-ℓ Galois representation of E. Uniformity Conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-003162 | MATHDB-340102

Strong Uniformity Conjecture for elliptic-curve Galois representations

摘录 / EXCERPT: Fix a positive integer d. For every degree d number field k and every non-CM elliptic curve E/k, consider the mod-ℓ Galois representation of E for each prime ℓ....

conjecture | open | MathDB 页面 | 原文线索

V2-003163 | MATHDB-340105

Serre's elliptic-curve or fake-elliptic-curve conjecture over number fields

摘录 / EXCERPT: Let K be a number field and let $\mathfrak f$ be a non-trivial, new, weight 2 complex eigenform of level $\mathcal N$ with rational integer Hecke eigenvalues. For a prime ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-003164 | MATHDB-340107

Submultiplicativity conjecture for the primitive-set ratio function

摘录 / EXCERPT: For $k>1$, let $Q(k)$ be the number of primitive subsets of $\{1,2,\ldots,k\}$ and define ... Thus $g(k)$ is the ratio of primitive subsets of $\{1,2,\ldots,k\}$ that contain k...

conjecture | open | MathDB 页面 | 原文线索

V2-003165 | MATHDB-340149

Miyazaki's shuffle conjecture for Jeśmanowicz equations

摘录 / EXCERPT: Miyazaki's shuffle conjecture. If $f=g+1$, the only solution is $(x,y,z)=(1,1,2)$; otherwise, there are no solutions. The source describes this as unsolved, with verification in se...

conjecture | open | MathDB 页面 | 原文线索

V2-003166 | MATHDB-340150

Miyazaki's shuffle conjecture for Terai-Jeśmanowicz equations

摘录 / EXCERPT: Miyazaki's shuffle conjecture. If $q=r=2$ and $b+1=c$, the only solution is $(X,Y,Z)=(1,1,p)$; otherwise, there are no solutions. The source presents this as a final open conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-003167 | MATHDB-340176

The infinitude conjecture for Zubrilina numbers

摘录 / EXCERPT: A Zubrilina number is a number arising as a value of the divisor function in the sense defined in the paper. Infinitude conjecture. There are infinitely many Zubrilina numbers. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-003168 | MATHDB-340196

The general kernel formula for higher-depth mock modular forms

摘录 / EXCERPT: Let $\mathcal{G}$ be the instanton generating function, expanded in terms of the completed generating functions $\widehat h_{p_i,\mu_i}$ and theta series...

conjecture | open | MathDB 页面 | 原文线索

V2-003169 | MATHDB-340213

The LCM-closed sets conjecture

摘录 / EXCERPT: Let $\mathcal{N}\subset\mathbb{N}^+$ be a finite LCM-closed set, meaning that $\operatorname{lcm}(m,n)\in\mathcal{N}$ whenever $m,n\in\mathcal{N}$, and suppose that $\mathcal{N}$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-003170 | MATHDB-340214

The dual GCD-closed sets conjecture

摘录 / EXCERPT: Let $\mathcal{N}\subset\mathbb{N}^+$ be a finite GCD-closed set, meaning that $\operatorname{gcd}(m,n)\in\mathcal{N}$ whenever $m,n\in\mathcal{N}$, and suppose that $\mathcal{N}$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-003171 | MATHDB-340220

Constrained random model conjecture for the prime-counting function

摘录 / EXCERPT: Let ${\operatorname{RM}}_c$ denote the constrained ${\operatorname{RM}}$ described in the source, and let $\pi_{\operatorname{RM}_c}(x)$ be its prime-counting function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003172 | MATHDB-340221

Equivalent Riemann hypothesis conjecture for the constrained random model

摘录 / EXCERPT: Let RM be the random model, with a realisation determined by the shifts a and factors $\rho_k(n+a)$, and let $\pi_{\operatorname{RM}}(x)$ be its counting functio...

conjecture | open | MathDB 页面 | 原文线索

V2-003173 | MATHDB-340248

The lower box-dimension conjecture for multi-rotation orbits

摘录 / EXCERPT: Let $k\geq 2$ be an integer and let $\Lambda=\{\alpha_1,\dots,\alpha_k\}$ be a set of k rationally independent numbers. Let $\omega\in\Lambda^\mathbb{N}$ be an arbitrary sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-003174 | MATHDB-340295

Generalised uniqueness conjecture for generalised Markov triple-graphs

摘录 / EXCERPT: Let $T_{\mu,\nu}$ be a generalised Markov triple-graph, and let a generalised Markov number be a number associated with this graph. A maximal (middle) element is the middle entry o...

conjecture | open | MathDB 页面 | 原文线索

V2-003175 | MATHDB-340311

Free-rank conjecture for products of spheres

摘录 / EXCERPT: Let p be a prime, let $n_1,\ldots,n_k$ be positive integers, and let $(\mathbb{Z}_p)^r$ act freely on the product $S^{n_1}\times\cdots\times S^{n_k}$. Free-rank conjecture for pr...

conjecture | open | MathDB 页面 | 原文线索

V2-003176 | MATHDB-340313

Heun polynomial representations for supersingular polynomials at levels 5 and 7

摘录 / EXCERPT: Heun polynomial representation conjecture. For every prime $p\geq7$,

conjecture | open | MathDB 页面 | 原文线索

V2-003177 | MATHDB-340314

Harada-Norton and Held group characterization by linear supersingular factors

摘录 / EXCERPT: Sporadic-group linear-factor conjecture. For every prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-003178 | MATHDB-340315

Apéry-like coefficient formulas for squares of level-5 and level-7 supersingular polynomials

摘录 / EXCERPT: Apéry-like supersingular-polynomial conjecture. For every prime $p\geq7$,

conjecture | open | MathDB 页面 | 原文线索

V2-003179 | MATHDB-340317

Class-number formula for the number of linear factors at general levels

摘录 / EXCERPT: General-level linear-factor formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003180 | MATHDB-340318

General-level degree formula for supersingular polynomials

摘录 / EXCERPT: General-level degree formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003181 | MATHDB-340319

Degree and linear-factor conjecture for class 3C supersingular polynomials

摘录 / EXCERPT: Class 3C supersingular-polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003182 | MATHDB-340320

Thompson group characterization for class 3C supersingular polynomials

摘录 / EXCERPT: Thompson-group conjecture. For every prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-003183 | MATHDB-340335

Germain prime conjecture

摘录 / EXCERPT: Let $x\geq 1$ be large, and let Λ be the von Mangoldt function. Germain prime conjecture. ... where ... This predicts a quantitative distribution for Sophie Germain primes, nam...

conjecture | open | MathDB 页面 | 原文线索

V2-003184 | MATHDB-340338

Converse conjecture for p-adic annihilation of monstrous moonshine Hauptmoduln

摘录 / EXCERPT: Let p be a prime and let f be a modular function. Say that f is p-adically annihilated if the sequence $f\mid U_p^n$ uniformly converges to 0 in the p-adic limit as...

conjecture | open | MathDB 页面 | 原文线索

V2-003185 | MATHDB-340341

The cyclotomic difference-set power-of-two conjecture

摘录 / EXCERPT: Let $q=p^{\ell}$ be an odd prime power, let $N\geq 2$ be an even divisor of $q-1$, and let $C_0^{(N,q)}$ be the subgroup of $\mathbb{F}_q^*$ of index N. A subset...

conjecture | open | MathDB 页面 | 原文线索

V2-003186 | MATHDB-340343

The Schmidt-White conjecture on cyclotomic strongly regular graphs

摘录 / EXCERPT: Let $q=p^{\ell}$ be a prime power, let m be a positive integer, and let N divide $(q^m-1)/(q-1)$. Let $C_0^{(N,q^m)}$ be the subgroup of $\mathbb{F}_{q^m}^*$ of index N, and...

conjecture | open | MathDB 页面 | 原文线索

V2-003187 | MATHDB-340349

Kellner's conjecture on integer ratios of consecutive power sums

摘录 / EXCERPT: Let $S_k(n)=\sum_{j=1}^{n-1}j^k$ for positive integers k and n. Kellner's conjecture. Let $k\ge 1$ and $n\ge 3$ be integers. Then the ratio ... is an integer if and only if $n=3$...

conjecture | open | MathDB 页面 | 原文线索

V2-003188 | MATHDB-340350

Extension of the logarithmic linear independence lemma to algebraic irrational parameters

摘录 / EXCERPT: Extension conjecture. Lemma 3 holds also for algebraic irrational numbers $alpha>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-003189 | MATHDB-340375

Conjecture on multiple roots of sums of products of consecutive integers

摘录 / EXCERPT: Multiple-root conjecture. The polynomial $g_T(x)$ has multiple roots if and only if one of the following holds: (1) $T=(n-4)$ for $n\geq4$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-003190 | MATHDB-340376

Conjectured rational points on genus-two curves associated with products of consecutive integers

摘录 / EXCERPT: Let $T\in A_5$, and let C_T be the corresponding genus-two curve, with $r(T)$ denoting the rank quantity used in the source. Rational-points conjecture. For $r(T)>1$, the values...

conjecture | open | MathDB 页面 | 原文线索

V2-003191 | MATHDB-340384

The conjecture on valuations of colored p-ary partitions for exceptional exponents

摘录 / EXCERPT: Let $A_{p,k}(n)$ denote the number of k-colored partitions of n into powers of the prime p, and let $ u_p$ be the p-adic valuation. Write $\mathbb{P}_{\geq 3}$ for the prim...

conjecture | open | MathDB 页面 | 原文线索

V2-003192 | MATHDB-340469

The modularity conjecture for Gromov-Witten potentials of elliptic fibrations

摘录 / EXCERPT: The modularity conjecture. The relative potential has the form

conjecture | open | MathDB 页面 | 原文线索

V2-003193 | MATHDB-340473

Farey-interval shape conjecture for Mullineux wall-crossing of the sign representation

摘录 / EXCERPT: Let n be a prime number, let sgn_n be the partition of n corresponding to the sign representation of S_n, and let r be a term in the n-th Farey sequence....

conjecture | open | MathDB 页面 | 原文线索

V2-003194 | MATHDB-340477

Prime factorization conjecture for factorials plus one

摘录 / EXCERPT: Prime factorization conjecture for factorials plus one. For every $n\in\mathbb N\setminus S$, the number $n!+1$ is square-free.

conjecture | open | MathDB 页面 | 原文线索

V2-003195 | MATHDB-340490

Brauer's Cartan invariant conjecture for blocks with prescribed defect group

摘录 / EXCERPT: Let p be a prime, let $k=\overline{\mathbb F}_p$, and let P be a finite p-group. For a finite group G, let B be a block of the group algebra kG with defect groups isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-003196 | MATHDB-340491

The strong Frobenius number conjecture for blocks

摘录 / EXCERPT: Let p be a prime, let $(K,\mathcal{O},k)$ be a p-modular system, and let B be a block of $\mathcal{O}G$ for a finite group G. The strong $\mathcal{O}$-Frobenius number...

conjecture | open | MathDB 页面 | 原文线索

V2-003197 | MATHDB-340495

The local density conjecture for G-irregular primes

摘录 / EXCERPT: Given coprime positive integers a and d, let ${\mathcal P}_G(d,a)$ be the set of G-irregular primes congruent to a modulo d, and let $\pi(x;d,a)$ count primes $p\le x$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-003198 | MATHDB-340496

The positive-density conjecture for G-regular primes in primitive residue classes

摘录 / EXCERPT: Let a and d be coprime positive integers, and consider primes in the primitive residue class $a\pmod d$. The G-regular-prime density conjecture. The subset of G-regular primes...

conjecture | open | MathDB 页面 | 原文线索

V2-003199 | MATHDB-340544

Extension of the Rabinowitz-Bruckman theorem to higher-order recurrences

摘录 / EXCERPT: Higher-order extension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-003200 | MATHDB-340546

Short character-sum conjecture

摘录 / EXCERPT: Let $n\geq 3$, let $\epsilon>0$, let M be an integer, let $Q\geq 3$ be an integer, let $N\leq Q^{1/n}$ be an integer, and let χ be a real non-principal character of modulus...

conjecture | open | MathDB 页面 | 原文线索

V2-003201 | MATHDB-340552

Allen's Ramanujan conjecture for the graphs $X_p(\delta,a,c)$

摘录 / EXCERPT: Let $p\geq 5$ be an odd prime, let $\delta,a,c\in\mathbb{F}_p^{\times}$, and let $X_p(\delta,a,c)$ be Allen's Cayley graph on the specified subgroup of $\operatorname{GL}(3,p)$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-003202 | MATHDB-340590

Refined K3 Vafa-Witten generating-series conjecture

摘录 / EXCERPT: Let S be a K3 surface, let r be a positive integer, and let $\mathsf{VW}_{r,n}(t)$ denote the refined Vafa-Witten invariant for charge $(r,0,n)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003203 | MATHDB-340593

The Hlawka zeta isometry conjecture for nonconstant radial functions

摘录 / EXCERPT: Let $r\bigl(\theta\bigr)$ be a positive continuous function on the circle, viewed as a radial function, and suppose that infinitely many Fourier coefficients $\hat{r}(n)$ are nonze...

conjecture | open | MathDB 页面 | 原文线索

V2-003204 | MATHDB-340594

The Hlawka zeta orbit conjecture for differentiable radial functions

摘录 / EXCERPT: Let r and $r^{\ast}$ be positive continuously differentiable functions on the circle, and let $Z_r(s)$ denote the Hlawka zeta function associated with r. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-003205 | MATHDB-340605

Donovan-Wemyss classification conjecture for simple formal flopping contractions

摘录 / EXCERPT: Let $({\widehat{Y}},{\widehat{f}},R)$ and $({\widehat{Y}}^{\prime},{\widehat{f}}^{\prime},R^{\prime})$ be two three-dimensional simple formal flopping contractions, with associated...

conjecture | open | MathDB 页面 | 原文线索

V2-003206 | MATHDB-340629

Broué-Rickard splendid equivalence conjecture

摘录 / EXCERPT: Let G be a finite group and let ℓ be a prime. Let B be an ℓ-block of G with abelian defect group D, and let b be the corresponding ℓ-block of $N_G(D)$ unde...

conjecture | open | MathDB 页面 | 原文线索

V2-003207 | MATHDB-340632

Geometric Broué's abelian defect conjecture

摘录 / EXCERPT: Let ℓ be a prime and let $G=\mathbb{G}(q)$ be a finite group of Lie type, where $p\ne\ell$ and q is a power of p. Let $D\in\operatorname{Syl}_{\ell}(G)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003208 | MATHDB-340638

Tsaknias's generalized Maeda conjecture for square-free levels

摘录 / EXCERPT: Let $k\geq 2$ be even and sufficiently large, and let N be square-free. Write $d_{k,N}$ for the dimension of the largest simple Hecke submodule of $S_k(\Gamma_0(N))$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003209 | MATHDB-340662

Kohl's primitive-root permutation sign conjecture

摘录 / EXCERPT: Let p be an odd prime, let g be a primitive root modulo p, and define σ_g on $\mathbb Z/p\mathbb Z=\{0,1,\ldots,p-1\}$ by ... for $b\in\{1,\ldots,p-1\}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-003210 | MATHDB-340663

The Lyapunov condition for generalized Zeckendorf summand sets

摘录 / EXCERPT: Let A_n be the summand set ... The Lyapunov condition is the condition used for the relevant triangular array in the preceding central limit theorem argument. Lyapunov condition co...

conjecture | open | MathDB 页面 | 原文线索

V2-003211 | MATHDB-340670

Instability conjecture for finite-field polynomials of degree divisible by the characteristic

摘录 / EXCERPT: Let p be a prime, let ${\mathbb F}_p$ be the finite field with p elements, and let $f(x)\in{\mathbb F}_p[x]$ be a polynomial whose degree is divisible by p. A polynomial over...

conjecture | open | MathDB 页面 | 原文线索

V2-003212 | MATHDB-340683

Zero residues and bounded runs in generalized Thue-Morse sequences

摘录 / EXCERPT: Zero-residue conjecture. For every prime $p>3$, every $n\in\mathbb{N}$, and every such i, one has

conjecture | open | MathDB 页面 | 原文线索

V2-003213 | MATHDB-340692

Inhomogeneous Duffin-Schaeffer theorem

摘录 / EXCERPT: Let $\delta\in\mathbb R$, and let $\Phi:\mathbb N\to\mathbb R_{\geqslant 0}$ satisfy the two conditions denoted by ... in the source. Inhomogeneous Duffin-Schaeffer theorem. Then for...

conjecture | open | MathDB 页面 | 原文线索

V2-003214 | MATHDB-340693

Inhomogeneous multiplicative Khintchine-type conjecture

摘录 / EXCERPT: Inhomogeneous multiplicative Khintchine-type conjecture. Then for almost all $\alpha_k\in\mathbb R$, there exist infinitely many $(n_1,\ldots,n_k)\in\mathbb Z^k$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003215 | MATHDB-340701

Lozano-Robledo's polynomial torsion-growth conjecture

摘录 / EXCERPT: Let E be an elliptic curve over a number field K of degree $d=[K:\mathbb{Q}]$. Suppose that E has a point of exact order p^n over K, where p is prime and $n\geq 1$. Loz...

conjecture | open | MathDB 页面 | 原文线索

V2-003216 | MATHDB-340766

Lemmermeyer's conjecture on 3-class groups of pure cubic fields

摘录 / EXCERPT: Let p be a prime number such that $p\equiv 1\, (\mathrm{mod}\,3)$, let $L=\mathbb{Q}(\sqrt[3]{p})$ be the corresponding pure cubic field, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003217 | MATHDB-340777

The sixth-power norm conjecture for differences of lambda invariants at CM points

摘录 / EXCERPT: Let d_1 and d_2 be the discriminants under consideration, with $d_1 e d_2$ and $d_1 e 0$, $d_2 e 0$. Assume $d_1 equiv d_2 equiv 5\pmod 8$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-003218 | MATHDB-340798

Divisor-count conjecture for classes of upper k-gap balancing numbers

摘录 / EXCERPT: Let $k\geq 0$ be an integer. An upper k-gap balancing number is a balancing number associated with a gap of k, and solutions are grouped into classes as in the paper. Divisor-c...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-003219 | MATHDB-340811

Generalized Hardy-Littlewood conjecture for affine prime patterns

摘录 / EXCERPT: Fix $R\in\mathbb{N}$ and $C>0$. Let $a_i,b_i\in\mathbb{Z}$ satisfy $(a_i,b_i)=1$ and $a_i>0$ for $1\leq i\leq R$. Define the Hardy-Littlewood singular series ... where $\rho(p)$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-003220 | MATHDB-340818

Kátai's conjecture on multiplicative pairs

摘录 / EXCERPT: Let $\mathbb{T}$ denote the unit circle, and let $f,g:\mathbb{N}\to\mathbb{T}$ be completely multiplicative functions. Suppose that the sequence $\{(f(n),g(n+1))\}_n$ is not dense...

conjecture | open | MathDB 页面 | 原文线索

V2-003221 | MATHDB-340847

Conjecture on the zeros of the half-integral weight Eisenstein series on $\Gamma_0(4)$

摘录 / EXCERPT: Zero-location conjecture. All but one of the zeroes of $E_\infty(z)$ in F_∞ lie on the circle

conjecture | open | MathDB 页面 | 原文线索

V2-003222 | MATHDB-340880

The squarefree conjecture on arithmetic progressions

摘录 / EXCERPT: Let $f\in\mathbb{Z}[T]$ be primitive and squarefree, let $\mathcal{A}=a+N\mathbb{Z}\subseteq\mathbb{Z}$ be an arithmetic progression, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003223 | MATHDB-340892

Conjecture on the relative strength of Ochem-Rao restrictions

摘录 / EXCERPT: Let E be the set of numbers of Euler's form for an odd perfect number, and for real numbers α and β let $OR_{\alpha,\beta}$ denote the property...

conjecture | open | MathDB 页面 | 原文线索

V2-003224 | MATHDB-340893

Existence of an obstruction-free bound for odd perfect numbers

摘录 / EXCERPT: An n-obstruction is a set of primes greater than 3, a_i,b_i,c_i for $1\leq i\leq n$, together with an odd prime p, such that, for every i,...

conjecture | open | MathDB 页面 | 原文线索

V2-003225 | MATHDB-340897

Prime-index density-zero conjecture for reducible cyclotomic compositions

摘录 / EXCERPT: Let $\bar D(t)$ count good pairs (m,n) with m,n both prime and $m\leq t$, $n\leq t$, where goodness means that $\Phi_m(\Phi_n(x))$ factors over the integers. Let $\pi(t)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-003226 | MATHDB-340898

Generalized prime-index density-zero conjecture for reducible cyclotomic compositions

摘录 / EXCERPT: Let $f(t)$ and $g(t)$ be strictly increasing functions tending to infinity as t tends to infinity. Let $\bar D_{f,g}(t)$ count good pairs (m,n) with m,n prime, $m\leq f(t)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003227 | MATHDB-340924

Stolz-Teichner simplified conjecture for holomorphic supersymmetric VOAs

摘录 / EXCERPT: Let V be a holomorphic vertex operator algebra of central charge c equipped with $=1$ supersymmetry. Let $Z_{RR}(V)\in\mathbb Z$ denote its supersymmetric index, and let η...

conjecture | open | MathDB 页面 | 原文线索

V2-003228 | MATHDB-340925

Divisibility conjecture for the index of holomorphic N=1 SCFTs

摘录 / EXCERPT: Let V be a holomorphic $\mathcal N=1$ superconformal field theory of central charge $c=12k$, and let $Z_{RR}(V)$ be its supersymmetric index. Index divisibility conjecture. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-003229 | MATHDB-340926

Sen's conjecture on L^2 harmonic forms on strongly centred monopole spaces

摘录 / EXCERPT: Sen's conjecture. If k and ℓ are coprime, then

conjecture | open | MathDB 页面 | 原文线索

V2-003230 | MATHDB-340959

The Bloch-Kato Tamagawa number conjecture for the standard representation of a Siegel modular form

摘录 / EXCERPT: Let $\rho^\theta_\blambda$ be the p-adic realization of the motive associated with the Siegel modular form, let $\rho^\theta_\blambda atural\theta$ denote its twist by θ,...

conjecture | open | MathDB 页面 | 原文线索

V2-003231 | MATHDB-341000

Rogers-Zudilin type conjecture for the $1/6,5/6$ elliptic family

摘录 / EXCERPT: Let $a\in\mathbb Q\setminus\{0,1\}$ and let X_a be the elliptic curve over $\mathbb Q$ defined by ... Suppose that ... is integral. Let $p>3$ be a prime such that X_a has good ordi...

conjecture | open | MathDB 页面 | 原文线索

V2-003232 | MATHDB-341040

Gaussian Primes Conjecture for weighted coordinate sums

摘录 / EXCERPT: Gaussian Primes Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-003233 | MATHDB-341054

Finiteness conjecture for Fermat curves with coefficients

摘录 / EXCERPT: Finiteness conjecture for Fermat curves with coefficients. The set $AF_{a,b,c}$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-003234 | MATHDB-341057

Primitive Donaldson-Thomas partition function conjecture for CHL models

摘录 / EXCERPT: Let X be an elliptic CHL model of order N, and let $\mathsf{Z}^X(q,t,p)$ be its primitive Donaldson-Thomas partition function. Write ... and let $\widetilde{\Phi}_N(Z)$ be the Bo...

conjecture | open | MathDB 页面 | 原文线索

V2-003235 | MATHDB-341058

Untwisted Donaldson-Thomas series conjecture for order-two CHL models

摘录 / EXCERPT: Let X be an order-two CHL model, and let $\mathsf{Z}^{\mathrm{untw}}(q,t,p)$ denote the primitive Donaldson-Thomas partition function formed from untwisted curve classes. Let Z...

conjecture | open | MathDB 页面 | 原文线索

V2-003236 | MATHDB-341074

The sum-of-two-squares conjecture for the sets $A_{Q,r}$

摘录 / EXCERPT: Let $\mathbf t=(t_0,\ldots,t_{d-2})$ and ... where $N,b_0,\ldots,b_{d-2}$ are integers, $N\equiv\sum_{i=0}^{d-2}b_i^2\pmod q$, and $\gcd(N,q)=1$. For $r>0$, let ... The sum-of-two-squa...

conjecture | open | MathDB 页面 | 原文线索

V2-003237 | MATHDB-341075

The optimal upper-bound conjecture for the diophantine exponent

摘录 / EXCERPT: Let $\mathbf a\in S^{d-2}(\mathbb Z/q\mathbb Z)\subset S^d(\mathbb Z/q\mathbb Z)$, let $p\leq\log(q)$ be prime, and let $d\geq3$. Let $w_p(\mathbf a)$ be the diophantine exponent a...

conjecture | open | MathDB 页面 | 原文线索

V2-003238 | MATHDB-341092

Linear independence of eta-quotients from the two construction theorems

摘录 / EXCERPT: Let the eta-quotients obtained from Theorems and be the eta-quotients constructed in those results. Linear independence conjecture. All eta-quotients gained from Theorems and are l...

conjecture | open | MathDB 页面 | 原文线索

V2-003239 | MATHDB-341101

Non-vanishing conjecture for quadratic twists of the Gross curve

摘录 / EXCERPT: Let q be a prime with $q \equiv 3 \ \mathrm{mod}\ 8$, let $K=\mathbb{Q}(\sqrt{-q})$, let H be the Hilbert class field of K, and let A be the Gross elliptic curve over H....

conjecture | open | MathDB 页面 | 原文线索

V2-003240 | MATHDB-341109

Finite restriction conjecture for weighted four-square representations

摘录 / EXCERPT: Let $a,b,c,d\in\mathbb{N}$ satisfy $a^2+b^2+c^2+d^2\ne0$. Finite restriction conjecture. There exists a finite set $\mathcal{S}\subseteq\mathbb{N}$ such that every sufficiently lar...

conjecture | open | MathDB 页面 | 原文线索

V2-003241 | MATHDB-341118

Cyclotomic factor conjecture for necklace polynomials

摘录 / EXCERPT: Cyclotomic factor conjecture. All cyclotomic factors of $M_d(x)$ may be accounted for by these divisibilities; equivalently, if $cPhi_m(x)$ divides $M_d(x)$ for some $m,d\geq 1$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-003242 | MATHDB-341125

The branched-cover L-space conjecture for prime strongly quasipositive links

摘录 / EXCERPT: Let L be a link in S^3, and for each integer $n\geq 2$ let $\Sigma_n(L)$ denote the canonical n-fold cyclic cover branched over L. A link is simply laced arborescent if it...

conjecture | open | MathDB 页面 | 原文线索

V2-003243 | MATHDB-341126

The low BKL-exponent obstruction conjecture for L-space branched covers

摘录 / EXCERPT: Let L be a prime, fibred, strongly quasipositive link. Write $k(L)$ for its BKL-exponent, defined as the maximum integer k such that L is the closure of δ_n^kP for som...

conjecture | open | MathDB 页面 | 原文线索

V2-003244 | MATHDB-341134

Universal Koszulity conjecture for Galois cohomology

摘录 / EXCERPT: Let p be prime, let F be a field containing a primitive pth root of unity, and assume that $G_F(p)$ is finitely generated. Write $H^\bullet(G_F(p),\mathbb{F}_p)$ for the cont...

conjecture | open | MathDB 页面 | 原文线索

V2-003245 | MATHDB-341153

Shumyatsky's nilpotency conjecture for multilinear commutator values

摘录 / EXCERPT: Let w be a multilinear commutator word and let G be a finite group in which $|ab|=|a||b|$ whenever the w-values a,b have coprime orders. Shumyatsky's conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-003246 | MATHDB-341168

Short sums conjecture for Kloosterman sums

摘录 / EXCERPT: Let p be prime, let $a\in\mathbb{F}_p^\times$, let N and H be integers, and write $\overline{x}$ for the multiplicative inverse of x modulo p. Short sums conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-003247 | MATHDB-341179

The arithmetic-progression conjecture for digit ones in powers of 3

摘录 / EXCERPT: For each integer $k\geq 0$, write the binary expansion of 3^k as a sequence of digits in $\{0,1\}$, and let the positions of its digit 1 be regarded as a subset of the nonnegat...

conjecture | open | MathDB 页面 | 原文线索

V2-003248 | MATHDB-341180

The weak digit-one density conjecture for initial base-q digits

摘录 / EXCERPT: Let p,q be two distinct prime numbers. For each k and each admissible integer m, let $d_{k,m}$ denote the number of digit ones among the first m digits of the q-ary expan...

conjecture | open | MathDB 页面 | 原文线索

V2-003249 | MATHDB-341184

The logarithmic exceptional-set conjecture for powers of 3

摘录 / EXCERPT: Logarithmic exceptional-set conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-003250 | MATHDB-341243

The powers-of-three complexity conjecture

摘录 / EXCERPT: Powers-of-three complexity conjecture. For every $k\in\mathbb N$,

conjecture | open | MathDB 页面 | 原文线索

V2-003251 | MATHDB-341244

The sequence-growth conjecture for optimal presentations

摘录 / EXCERPT: Sequence-growth conjecture. For all $k\in\mathbb N$,

conjecture | open | MathDB 页面 | 原文线索

V2-003252 | MATHDB-341245

The Ford-type conjecture on gaps between integers with restricted factors

摘录 / EXCERPT: Restricted-factor gap conjecture. The length of (a_i) is at most of order $(\log N)^\delta$ for some fixed δ.

conjecture | open | MathDB 页面 | 原文线索

V2-003253 | MATHDB-341246

The non-Diophantineness conjecture for the integers over the rationals

摘录 / EXCERPT: Let $\mathbb{Z}$ denote the ring of integers and $\mathbb{Q}$ the field of rational numbers. A subset of $\mathbb{Q}$ is Diophantine if it is defined by the existence of solutions...

conjecture | open | MathDB 页面 | 原文线索

V2-003254 | MATHDB-341249

Martin Davis's conjecture on simple sets and Hilbert's tenth problem over the rationals

摘录 / EXCERPT: Let $S\subseteq\mathbb{N}$ be a simple set, meaning that $\mathbb{N}\setminus S$ is infinite and contains no infinite recursively enumerable subset. Let $U\subseteq\mathbb{Q}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-003255 | MATHDB-341251

Local higher-order Fourier uniformity conjecture for the Liouville function

摘录 / EXCERPT: Let $s \geq 0$. Let $G \backslash \Gamma$ be an s-step nilmanifold. Let $F:G \backslash \Gamma \rightarrow \mathbb{C}$ be Lipschitz continuous and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003256 | MATHDB-341253

Littlewood's maximum-order conjecture for the Riemann zeta function on the 1-line

摘录 / EXCERPT: Let $\zeta(s)$ denote the Riemann zeta function, let $T\geq 1$, and write γ for the Euler-Mascheroni constant. Littlewood's maximum-order conjecture. ... This conjecture predi...

conjecture | open | MathDB 页面 | 原文线索

V2-003257 | MATHDB-341254

The conjecture that the maximal order of S(t) is one half

摘录 / EXCERPT: The maximal-order conjecture for $S(t)$. Conditionally on the Riemann hypothesis,

conjecture | open | MathDB 页面 | 原文线索

V2-003258 | MATHDB-341269

Prime-monomial quiver Grassmannian coefficient conjecture

摘录 / EXCERPT: Let Q be a quiver and let $v_1,\ldots,v_n$ be a sequence of mutations. For each vertex i, let $\phi_{n,i}$ be the representation on the base quiver corresponding to the last mu...

conjecture | open | MathDB 页面 | 原文线索

V2-003259 | MATHDB-341277

Conjectures on Hecke operators, diagonalizability, and oldforms for Drinfeld cusp forms

摘录 / EXCERPT: The oldform and Hecke conjectures. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-003260 | MATHDB-341278

Goss-Gekeler slope multiplicity congruence conjecture

摘录 / EXCERPT: Slope multiplicity conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-003261 | MATHDB-341294

Strong Andrica conjecture for consecutive primes

摘录 / EXCERPT: Let p_n denote the nth prime and let $g_n:=p_{n+1}-p_n$. Strong Andrica conjecture. Except for $p_n in\{3,7,13,23,31,113\}$, equivalently $n\in\{2,4,6,9,11,30\}$, one has ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-003262 | MATHDB-341295

Strong Legendre conjecture on primes between consecutive squares

摘录 / EXCERPT: Let m be an integer. Strong Legendre conjecture. The interval ... contains at least two primes. This is stronger than standard Legendre's conjecture and is presented as a consequen...

conjecture | open | MathDB 页面 | 原文线索

V2-003263 | MATHDB-341296

Strong Brocard conjecture on primes between consecutive prime squares

摘录 / EXCERPT: Let p_n be the nth prime and define $g_n:=p_{n+1}-p_n$. Strong Brocard conjecture. The interval ... contains at least $2g_n$ primes. The conjecture is described as stronger than...

conjecture | open | MathDB 页面 | 原文线索

V2-003264 | MATHDB-341297

Weakened Andrica conjecture with constant 2

摘录 / EXCERPT: Let p_n be the nth prime and define $g_n:=p_{n+1}-p_n$. Weakened Andrica conjecture. For every integer $n\geq1$, ... The source also discusses the more general proposed bound wit...

conjecture | open | MathDB 页面 | 原文线索

V2-003265 | MATHDB-341298

Square root conjecture for consecutive prime gaps

摘录 / EXCERPT: Let p_n be the nth prime and define $g_n:=p_{n+1}-p_n$. Square root conjecture. Except for $p_n\in\{3,7,13,23,31,113\}$, equivalently $n\in\{2,4,6,9,11,30\}$, one has ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-003266 | MATHDB-341303

Beraldo-Kanel-Belov-Kontsevich conjecture on the image over the infinite-prime base

摘录 / EXCERPT: Let R be a commutative ring, let R_{∞} be the reduction modulo the infinite prime, let $i:R\rightarrow R_{\infty}$ be the tautological inclusion, and let ... be the group ho...

conjecture | open | MathDB 页面 | 原文线索

V2-003267 | MATHDB-341304

Infinite-prime independence conjecture for the Weyl-to-Poisson homomorphism

摘录 / EXCERPT: Let Φ be the homomorphism from automorphisms of the Weyl algebra to polynomial symplectomorphisms constructed by reduction modulo an infinite prime, restriction to centers, in...

conjecture | open | MathDB 页面 | 原文线索

V2-003268 | MATHDB-341312

Simplicity of the roots generated by the Newton sequence

摘录 / EXCERPT: Let $Y_{n,m}(t)$ be the sequence of functions used to generate the approximations, and define ... The roots generated by this sequence are the zeros y_n of the corresponding functi...

conjecture | open | MathDB 页面 | 原文线索

V2-003269 | MATHDB-341313

Contraction conjecture for the Lipschitz constants

摘录 / EXCERPT: For each n, let y_n be the sequence-generated root and define ... This is the Lipschitz constant in the referenced formula. Contraction conjecture. It is always possible to choos...

conjecture | open | MathDB 页面 | 原文线索

V2-003270 | MATHDB-341322

Finiteness conjecture for integral overconvergent de Rham-Witt cohomology of affine curves

摘录 / EXCERPT: Let X be an affine smooth curve over k, and let W denote the ring of Witt vectors of k. Write X_1 for the affine curve obtained by removing all but one of the points at i...

conjecture | open | MathDB 页面 | 原文线索

V2-003271 | MATHDB-341334

Base-change conjecture for the top-down local Langlands correspondence

摘录 / EXCERPT: Let $R^_{\overline{\rho}}$ be the universal deformation ring of the residual representation and let $\rho^\square$ be the universal lift. For a prime ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-003272 | MATHDB-341341

The TMF splitting conjecture for modular curves of level \a0n

摘录 / EXCERPT: TMF splitting conjecture. The $\operatorname{TMF}_{(3)}$-module $\operatorname{TMF}_0(n)_{(3)}$ decomposes into possibly shifted copies of $\operatorname{TMF}_1(2)_{(3)}$ and of...

conjecture | open | MathDB 页面 | 原文线索

V2-003273 | MATHDB-341383

The three-term inequality for weights of free bases of vector-valued modular forms

摘录 / EXCERPT: Let ρ be an irreducible representation, and let m_k denote the multiplicity of forms of even weight k among a free basis for $M(\rho)$ over the ring of scalar-valued modul...

conjecture | open | MathDB 页面 | 原文线索

V2-003274 | MATHDB-341397

Conjecture on the ninth primary pseudoperfect number

摘录 / EXCERPT: A primary pseudoperfect number is a positive integer K with the defining property that, for every prime divisor p of K, one has $1/K+1/p$ equal to an integer. Let K_r denot...

conjecture | open | MathDB 页面 | 原文线索

V2-003275 | MATHDB-341398

Congruence progression conjecture for primary pseudoperfect numbers

摘录 / EXCERPT: Let K_r denote a primary pseudoperfect number with r prime factors. The claim concerns those K_r divisible by 6. Primary pseudoperfect congruence conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-003276 | MATHDB-341403

BDHPS conjecture for the Euler totient partial sum

摘录 / EXCERPT: Let ... S{varphi}(x):=sum{nleq x}varphileft(leftlfloor x/n rightrfloorright). ... S{varphi}(x)=dfrac{xlog(x)}{zeta(2)}(1+o(1)). ... denotes the Riemann zeta function. The conjecture pred...

conjecture | open | MathDB 页面 | 原文线索

V2-003277 | MATHDB-341404

Numerical $_6F_5(1)$ supercongruences for weight-4 modular forms

摘录 / EXCERPT: Let $p\ge7$ be prime, let $\left(\frac{\cdot}{p}\right)$ be the Legendre symbol, and let $a_p(f)$ denote the p-th Fourier coefficient of the indicated modular form. The weight-4...

conjecture | open | MathDB 页面 | 原文线索

V2-003278 | MATHDB-341416

Pairwise coprimality conjecture for prime-indexed Lucas sequence terms

摘录 / EXCERPT: Pairwise coprimality conjecture. For any two different primes p and q, the terms u_p and u_q are relatively prime:

conjecture | open | MathDB 页面 | 原文线索

V2-003279 | MATHDB-341424

Ali-Fošner's generalized (m,n)-Jordan derivation conjecture

摘录 / EXCERPT: Let $m,n\geq 1$ be fixed integers, let R be a semiprime ring satisfying suitable torsion restrictions, and let $F:R\longrightarrow R$ be a nonzero generalized (m,n)-Jordan deri...

conjecture | open | MathDB 页面 | 原文线索

V2-003280 | MATHDB-341425

Fošner's generalized (m,n)-Jordan centralizer conjecture

摘录 / EXCERPT: Let $m,n\geq 1$ be fixed integers, let R be a semiprime ring satisfying suitable torsion restrictions, and let $T:R\longrightarrow R$ be a generalized (m,n)-Jordan centralizer....

conjecture | open | MathDB 页面 | 原文线索

V2-003281 | MATHDB-341432

Gross-Kohnen-Zagier algebraicity conjecture for higher Green's functions

摘录 / EXCERPT: Let $k\geq 2$, let $f\in M^!_{2-2k}$ be a weakly holomorphic modular form with rational Fourier coefficients, and let $G_{k,f}$ be the higher Green's function associated to f. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-003282 | MATHDB-341436

Generalized nonintegrality conjecture for reciprocal power sums

摘录 / EXCERPT: Generalized nonintegrality conjecture. There is a positive integer N such that for every integer $n\geq N$ and every integer k with $1\leq k\leq n$, neither...

conjecture | open | MathDB 页面 | 原文线索

V2-003283 | MATHDB-341440

Van Hamme's supercongruence (A.2) for a truncated hypergeometric series

摘录 / EXCERPT: Let p be an odd prime. For parameters $a_1,,a_6$ and $b_1,,b_5$, write the truncated hypergeometric series as ... where $(a)_0=1$ and $(a)_k=a(a+1)\cdots(a+k-1)$ for $k\geq 1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003284 | MATHDB-341442

Tolue's prime-degree non-centralizer graph conjecture

摘录 / EXCERPT: Let G be a finite group, let $Z(G)$ be its center, and let Υ_G be the simple graph whose vertices are the elements of G, with distinct vertices x and y adjacent wh...

conjecture | open | MathDB 页面 | 原文线索

V2-003285 | MATHDB-341455

General-type conjecture for portrait moduli spaces

摘录 / EXCERPT: Let $N\ge1$ and $d\ge2$. An unweighted preperiodic portrait is a pair ${\mathcal P}=({\mathcal V},\Phi)$, and ${\mathcal M}_d^N[{\mathcal P}]$ denotes its moduli space;...

conjecture | open | MathDB 页面 | 原文线索

V2-003286 | MATHDB-341461

Erdős-Odlyzko-Sárközy conjecture on products of two small primes

摘录 / EXCERPT: Let k be a positive integer and let a be an integer coprime to k. The primes p_1 and p_2 are required to satisfy $p_1,p_2\leq k$. Erdős-Odlyzko-Sárközy conjecture. For al...

conjecture | open | MathDB 页面 | 原文线索

V2-003287 | MATHDB-341462

Liu's mod-p^5 truncated hypergeometric supercongruence

摘录 / EXCERPT: Let $p\geq5$ be a prime with $p\equiv3\pmod{4}$, and let Γ_p denote the p-adic Gamma function. Liu's conjecture. ... The paper's abstract identifies this as a congruence con...

conjecture | open | MathDB 页面 | 原文线索

V2-003288 | MATHDB-341501

Ill-distribution Conjecture A for definable sets

摘录 / EXCERPT: Let K be a number field and let $\widetilde{\mathbb{R}}=(\mathbb{R},<,+,\cdot,f_{1},\ldots,f_{r})$ be a model-complete expansion of the real field by a Pfaffian chain. For...

conjecture | open | MathDB 页面 | 原文线索

V2-003289 | MATHDB-341503

Uniform ill-distribution conjecture for definable intersections

摘录 / EXCERPT: Let K be a number field and let $\widetilde{\mathbb{R}}=(\mathbb{R},<,+,\cdot,f_{1},\ldots,f_{r})$ be a model-complete expansion of the real field by a Pfaffian chain. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003290 | MATHDB-341507

Large prime divisor conjecture for cyclotomic-type quotients

摘录 / EXCERPT: Large prime divisor conjecture. There exists a prime divisor p of this integer such that $p>z$. The paper presents this as an observation suggested by computations and proposes i...

conjecture | open | MathDB 页面 | 原文线索

V2-003291 | MATHDB-341553

The random linear-code list-decoding conjecture

摘录 / EXCERPT: Let q be a prime power, let $0<p<1-1/q$, and let $H_q(p)$ denote the q-ary entropy. A random linear code of rate $1-H_q(p)-\delta$ is chosen uniformly among linear codes of tha...

conjecture | open | MathDB 页面 | 原文线索

V2-003292 | MATHDB-341555

Characterization of Jacobian linear type for hypersurfaces with one-dimensional singularities

摘录 / EXCERPT: Let $X=V(f)\subseteq \mathbb{A}_k^n$ be an affine hypersurface with one-dimensional singularities. For a minimal prime $\mathfrak q$ of the Jacobian ideal $I(f)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003293 | MATHDB-341567

Infinite sign-change conjecture for the orthorecursive coefficients

摘录 / EXCERPT: Sign-change conjecture. The sequence $(c_n)_{n\geq0}$ changes sign infinitely often.

conjecture | open | MathDB 页面 | 原文线索

V2-003294 | MATHDB-341568

Logarithmic-phase asymptotic conjecture for the orthorecursive coefficients

摘录 / EXCERPT: Logarithmic-phase asymptotic conjecture. For all $n>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-003295 | MATHDB-341582

The transcendental-coefficient conjecture for linear forms in primes

摘录 / EXCERPT: Let L, $\mathbf{v}$, F, and G be as in Theorem 1, without assuming that L has algebraic coefficients. Let $W=\prod_{p\leq w}p$. Transcendental-case conjecture. There should...

conjecture | open | MathDB 页面 | 原文线索

V2-003296 | MATHDB-341583

The pseudorandomness conjecture for nondegenerate linear maps

摘录 / EXCERPT: Let m,d be natural numbers with $d\geqslant m+2$, and let $L:\mathbb{R}^d\to\mathbb{R}^m$ be a surjective linear map. Let $V^*_{\operatorname{degen}}(m,d)$ denote the degenerate...

conjecture | open | MathDB 页面 | 原文线索

V2-003297 | MATHDB-341604

The relation-number formulation of the Main Conjecture

摘录 / EXCERPT: Relation-number conjecture. Every rational number in $(-4,4)$ is an ℓ-step relation number for some $\ell\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-003298 | MATHDB-341613

The quantitative Schmidt conjecture for dimensions of k-singular matrices

摘录 / EXCERPT: For integers m,n with $(m,n)\neq(1,1)$, let $f_{m,n}(k)=mn-\frac{k(m+n-k)mn}{(m+n)^2}-\left\{\frac{km}{m+n}\right\}\left\{\frac{kn}{m+n}\right\}$, where $\{x\}$ denotes the fract...

conjecture | open | MathDB 页面 | 原文线索

V2-003299 | MATHDB-341616

The conjecture on birational superrigidity of quasi-smooth prime Fano 3-folds

摘录 / EXCERPT: Let X be a general quasi-smooth prime Fano 3-fold of codimension at least 2. A birational superrigidity conjecture. X is birationally superrigid if and only if X is singu...

conjecture | open | MathDB 页面 | 原文线索

V2-003300 | MATHDB-341626

Ratios-Conjecture prediction for the variance of Gaussian primes

摘录 / EXCERPT: Let $K=X^\lambda$ with $0<\lambda<1$, and let C_f, C_Φ, C'_Φ, $\widetilde\Phi$, $C_{\Phi,\zeta}$, $C_{\Phi,L}$, and A'_Φ be the constants defined in the paper. Rat...

conjecture | open | MathDB 页面 | 原文线索

V2-003301 | MATHDB-341627

Mixed logarithmic-derivative moments conjecture for the Riemann zeta-function

摘录 / EXCERPT: Let $Z(t)$ be Hardy's function, so that $|Z(t)|=|\zeta(1/2+it)|$, and let $\sigma_{I\!I\!I'}$ be the function defined by the Painlevé equation referenced in the source. For non-neg...

conjecture | open | MathDB 页面 | 原文线索

V2-003302 | MATHDB-341628

Near-critical logarithmic-derivative moment conjecture for the Riemann zeta-function

摘录 / EXCERPT: Let K be a positive integer, let $a>0$, and consider the logarithmic derivative of the Riemann zeta-function at $1/2+a/\log T+it$. Near-critical logarithmic-derivative moment con...

conjecture | open | MathDB 页面 | 原文线索

V2-003303 | MATHDB-341643

Parametric Khintchine-like behaviour conjecture for iterated function systems

摘录 / EXCERPT: Let U be an open subset of $\mathbb{R}^k$, and for each $u\in U$ let $\Phi_u=\{\phi_{i,u}(x)=A_i(u)x+t_i(u)\}_{i=1}^l$ be an iterated function system with attractor X_u. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003304 | MATHDB-341646

Weak even Goldbach conjecture for sums of consecutive progression terms

摘录 / EXCERPT: Weak even Goldbach conjecture. For each even positive integer $a>2$, there exist a positive integer n and odd primes p and q such that

conjecture | open | MathDB 页面 | 原文线索

V2-003305 | MATHDB-341647

Weak odd Goldbach conjecture for sums of consecutive progression terms

摘录 / EXCERPT: Weak odd Goldbach conjecture. For each odd positive integer $a>2$, there exist a positive integer n and odd primes p and q such that

conjecture | open | MathDB 页面 | 原文线索

V2-003306 | MATHDB-341648

Unified weak Goldbach conjecture for prime-started arithmetic progressions

摘录 / EXCERPT: Let $A(p,q)$ be the arithmetic progression whose first two terms are odd primes p and q, and let n be a positive integer. Unified weak Goldbach conjecture. For each positive...

conjecture | open | MathDB 页面 | 原文线索

V2-003307 | MATHDB-341676

The vanishing conjecture for Morava stabilizer cohomology

摘录 / EXCERPT: Let p be a prime, let n be a height, let $\mathbb{G}_n$ be the Morava stabilizer group, let $\mathbb{W}_n$ be the Witt-vector module, and let $(E_n)_0$ be the degree-zero coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-003308 | MATHDB-341677

The large-prime exterior-algebra conjecture for Morava stabilizer cohomology

摘录 / EXCERPT: Let $\mathbb{G}_n$ be the Morava stabilizer group, let $\mathbb{W}_n$ be the Witt-vector module, and let $x_1,\ldots,x_n$ be generators with cohomological degrees $\deg(x_i)=2i-1$....

conjecture | open | MathDB 页面 | 原文线索

V2-003309 | MATHDB-341752

Marton's Polynomial Freiman-Ruzsa conjecture over finite fields

摘录 / EXCERPT: Let p be a prime, let $A\subseteq\mathbb{F}_p^n$ satisfy $|A+A|\leqslant K|A|$, and let $S\subseteq\mathbb{F}_p^N$ be a set. Equivalently, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003310 | MATHDB-341753

Strong Polynomial Freiman-Ruzsa conjecture for approximate homomorphisms

摘录 / EXCERPT: Let p be a prime, and let $\phi:\mathbb{F}_p^n\to\mathbb{F}_p^N$ be a function such that $\phi(x+y)-\phi(x)-\phi(y)$ takes values in a set S. Let K be the doubling parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-003311 | MATHDB-341790

Primality conjecture for almost every 3-IET

摘录 / EXCERPT: 3-IET primality conjecture. Almost every 3-IET is prime.

conjecture | open | MathDB 页面 | 原文线索

V2-003312 | MATHDB-341791

Primality conjecture for first return maps of odometers

摘录 / EXCERPT: Odometer return-map primality conjecture. For almost every $x\in[0,1]$, the first return map of T to $[0,x]$ is prime.

conjecture | open | MathDB 页面 | 原文线索

V2-003313 | MATHDB-341805

The Omega-algebraicity conjecture for modules of elementary abelian rank two

摘录 / EXCERPT: Let p be a prime, let $G=\mathbb{Z}/p\times\mathbb{Z}/p$, and let M be an absolutely indecomposable kG-module, meaning that it remains indecomposable after extending the fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-003314 | MATHDB-341806

The prime-set level conjecture

摘录 / EXCERPT: Prime-set level conjecture. For every integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-003315 | MATHDB-341811

Calegari-Mazur conjecture on infinitesimally classical deformations

摘录 / EXCERPT: Let F be an imaginary quadratic field and let p be a split prime. Suppose that $\rho:\operatorname{Gal}(\overline{F}/F)\to\operatorname{GL}_2(E)$ is continuous, irreducible, ne...

conjecture | open | MathDB 页面 | 原文线索

V2-003316 | MATHDB-341839

Gonek's conjecture for the Mertens function

摘录 / EXCERPT: Let $\mu(n)$ be the Möbius function and define the Mertens function by ... Let $\overline{\varliminf}$ denote the upper limit of the lower limit. Gonek's conjecture. There exists a n...

conjecture | open | MathDB 页面 | 原文线索

V2-003317 | MATHDB-341849

Asymptotic symmetry conjecture for coefficients of cyclotomic polynomials

摘录 / EXCERPT: Let A be the set of points (c,n) such that c is a nontrivial coefficient of Φ_n, and let A_k consist of those points in A with $n\leq k$. For each k, write...

conjecture | open | MathDB 页面 | 原文线索

V2-003318 | MATHDB-341856

The mod-9 partition conjectures

摘录 / EXCERPT: Let $n=\lambda_1+\lambda_2+\cdots+\lambda_k$ be a partition written in weakly increasing order. Say that it satisfies Condition(i) if ... and, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-003319 | MATHDB-341857

The mod-9 q-series conjectures

摘录 / EXCERPT: For q a formal variable and $(a_1,\ldots,a_r;q^9)_\infty=\prod_{m\geq0}\prod_{j=1}^r(1-a_jq^{9m})$, the five partition conjectures are equivalently expressed by the following q...

conjecture | open | MathDB 页面 | 原文线索

V2-003320 | MATHDB-341865

Multiplicity stability conjecture for $VIC(\mathbb{Z}/p^k\mathbb{Z})$-modules

摘录 / EXCERPT: For a prime p and a positive integer k, consider the irreducible complex representations of $GL_n(\mathbb{Z}/p^k\mathbb{Z})$ for all n, together with a compatible collection...

conjecture | open | MathDB 页面 | 原文线索

V2-003321 | MATHDB-341866

Logarithmic representation-growth conjecture for $SL_n(\mathbb{Z}_p)$

摘录 / EXCERPT: Let $C>0$ be a constant. For each n and prime p, consider the irreducible smooth representations of $SL_n(\mathbb{Z}_p)$, with dimension measured as the vector-space dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-003322 | MATHDB-341868

Local Pre-Malle conjecture for reduced discriminants

摘录 / EXCERPT: Local Pre-Malle conjecture. There is a finite set of primes S_0 depending on G such that, for every prime $p\notin S_0$ and every integer $k\not\equiv 0\pmod p$, some G-exten...

conjecture | open | MathDB 页面 | 原文线索

V2-003323 | MATHDB-341894

General extremal-function conjecture for matroid classes over prime powers

摘录 / EXCERPT: For a prime power q, let $\mathcal{AC}_q$, $\mathcal{SL}_q$, and $\mathcal{AF}_q$ be the corresponding classes of matroids, with extremal functions $h_{\mathcal{AC}_q}(r)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003324 | MATHDB-341901

Rigidity conjecture for random $SL_2(\mathbb{F}_p)$-circulant matrices

摘录 / EXCERPT: Let p be a prime, let $G=SL_2(\mathbb{F}_p)$, and let $f:G\rightarrow\{0,1\}$ be chosen uniformly at random. Write $M_G(f)$ for the corresponding G-circulant matrix. Rigidity c...

conjecture | open | MathDB 页面 | 原文线索

V2-003325 | MATHDB-341945

Generalized measurable Sarnak's Conjecture

摘录 / EXCERPT: Let $\boldsymbol{K}=(K,\mathcal{O}_{K},D,\mathcal{B}=\{b_{1},\dots,b_{D}\})$ be an integral tuple and $\chi\in\mathcal{M}^{a}_{K}$. Let $(X,\mu,T_{1},\dots,T_{d})$ be a measure-pre...

conjecture | open | MathDB 页面 | 原文线索

V2-003326 | MATHDB-341972

Infinitude conjecture for primary Carmichael numbers

摘录 / EXCERPT: Infinitude conjecture for primary Carmichael numbers. The following claims are true: the set $4\mathcal{C}'4$ is infinite, and the set $4\mathcal{C}\setminus\mathcal{C}'4$ is infin...

conjecture | open | MathDB 页面 | 原文线索

V2-003327 | MATHDB-341973

Growth conjecture for primary Carmichael numbers

摘录 / EXCERPT: Growth conjecture for primary Carmichael numbers. For sufficiently small $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-003328 | MATHDB-341974

Infinitude conjecture for the sets of digit-sum classes

摘录 / EXCERPT: Infinitude conjecture for the sets of digit-sum classes. For each $d\geq2$, the sets $\mathcal{S}_d$ and $\mathcal{S}\setminus\mathcal{S}_d$ are infinite. Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-003329 | MATHDB-341977

Positive characteristic form of the Langlands correspondence

摘录 / EXCERPT: Let G be a connected reductive group over $\mathbb{Q}$, split over a number field F, and let K be a level of G. Fix a prime p, let f be a mod p Hecke eigenform of lev...

conjecture | open | MathDB 页面 | 原文线索

V2-003330 | MATHDB-341982

The conjecture that almost all three-factor Carmichael numbers are primary

摘录 / EXCERPT: For $x>0$, let $C_3(x)$ denote the number of three-factor Carmichael numbers at most x, and let $C'_3(x)$ denote the number of primary three-factor Carmichael numbers at most x...

conjecture | open | MathDB 页面 | 原文线索

V2-003331 | MATHDB-342023

The Conrey-Gonek-Ivić additive divisor sum conjecture

摘录 / EXCERPT: Let $d_k(n)$ denote the k-fold divisor function, and define the additive divisor correlation $D_{h,k,l}(x)$ as in the paper. For fixed $k,l\in\mathbb{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003332 | MATHDB-342026

The -adic Tate conjecture for divisors

摘录 / EXCERPT: The ℓ-adic Tate conjecture for divisors. This map should be surjective. The conjecture predicts that divisor classes account for all cohomology classes fixed by the absolute G...

conjecture | open | MathDB 页面 | 原文线索

V2-003333 | MATHDB-342028

Concentration conjecture for coprime nonreal cyclotomic coincidence roots

摘录 / EXCERPT: Let $S_{\mathbb{C}}$ denote the set of all nonreal complex numbers z such that $\Phi_m(z)=\Phi_n(z)$ for some distinct coprime positive integers m and n. Concentration conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003334 | MATHDB-342057

Glasby's cyclotomic ordering conjecture for the positive integers

摘录 / EXCERPT: Let $\mathcal{P}$ be the set of positive integers, and let $\Phi_d(t)$ be the dth cyclotomic polynomial. Define a relation $\preceq$ on $\mathcal{P}$ by ... and define $m\prec n$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-003335 | MATHDB-342058

Glasby's adjacency conjecture for powers of 3 in the cyclotomic ordering

摘录 / EXCERPT: Let $\mathcal{P}$ be the positive integers equipped with the relation $\prec$ defined by $m\prec n$ if $\Phi_m(q)\leqslant\Phi_n(q)$ for every integer $q\geqslant 2$ and $m\ne n$....

conjecture | open | MathDB 页面 | 原文线索

V2-003336 | MATHDB-342115

The conjectural bound for the spectral exponential sum over the Picard manifold

摘录 / EXCERPT: Let $\Gamma=\mathrm{PSL}_{2}(\mathcal{O})$, let $\lambda_j=1+t_j^2$ be the eigenvalues of the hyperbolic Laplacian on $\Gamma\backslash\mathfrak{h}^{3}$, and let $S(T,X)$ be the sp...

conjecture | open | MathDB 页面 | 原文线索

V2-003337 | MATHDB-342118

Relative Kato homology conjecture for semilocal global models

摘录 / EXCERPT: Let K be a global field, let $\mathcal{O}_S$ be a regular semilocal subring of K, and set $\mathcal{S}=\operatorname{Spec}(\mathcal{O}_S)$. Let $\mathcal{X}$ be a regular schem...

conjecture | open | MathDB 页面 | 原文线索

V2-003338 | MATHDB-342157

Inhomogeneous extremality dichotomy for analytic manifolds

摘录 / EXCERPT: Let $\mathcal{M} \subset \mathbb{R}^n$ be a connected analytic manifold. For $\mathbf{x}\in\mathcal{M}$ and $\theta\in\mathbb{R}$, let $\omega(\mathbf{x},\theta)$ denote the inhomo...

conjecture | open | MathDB 页面 | 原文线索

V2-003339 | MATHDB-342222

The Shapiro-class formula for the prime-counting function

摘录 / EXCERPT: Shapiro-class formula for $\pi(n)$. Then $G(n)\sim\pi(n)$ for a constant b, where

conjecture | open | MathDB 页面 | 原文线索

V2-003340 | MATHDB-342223

The Shapiro-class formula for the nth prime

摘录 / EXCERPT: Let p_n denote the nth prime, and let $S(n)$ denote the Shapiro-class function. Let $B=\gamma/\log 2$, where γ is the Euler-Mascheroni constant. Then Shapiro-class formu...

conjecture | open | MathDB 页面 | 原文线索

V2-003341 | MATHDB-342258

The bounded-error conjecture for the minimum degree of linearly dependent powers

摘录 / EXCERPT: Bounded-error conjecture. There is a small constant M such that, for all k and d,

conjecture | open | MathDB 页面 | 原文线索

V2-003342 | MATHDB-342260

Jacobsthal's conjecture on the maximum and primorial Jacobsthal functions

摘录 / EXCERPT: Jacobsthal's conjecture. For all $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-003343 | MATHDB-342261

Conjecture that the maximum Jacobsthal function eventually exceeds the primorial function

摘录 / EXCERPT: Eventual strict inequality conjecture. For all $k\ge 33$,

conjecture | open | MathDB 页面 | 原文线索

V2-003344 | MATHDB-342262

Conjecture on the doubling bound for the maximum Jacobsthal function

摘录 / EXCERPT: Doubling-bound conjecture. For all $k\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-003345 | MATHDB-342271

Divisibility conjecture for an elliptic Dedekind sum

摘录 / EXCERPT: Let a be an even integer and b an odd integer, and let $s_{\sqrt{-1}}(a;b)$ and $Q(a;b)$ denote the elliptic Dedekind sum and the associated quantity used in the paper. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-003346 | MATHDB-342288

Deligne-Ihara conjecture for the pro--version Galois Lie algebra

摘录 / EXCERPT: Let $X=\mathbb P^1\backslash\{0,1,\infty\}$, let ℓ be a prime, and let $\mathfrak{g}^{\ell}=\bigoplus_{m\geq 1}\operatorname{gr}^m_{I_{\ell}}G_{\mathbb Q}$ be the graded Lie a...

conjecture | open | MathDB 页面 | 原文线索

V2-003347 | MATHDB-342289

Conjecture on Eisenstein multiple modular values and multiple zeta values

摘录 / EXCERPT: Let $a_1,\ldots,a_r\geq 1$, set $a=a_1+\cdots+a_r$, and let $\mathbb G_{2k}$ denote the normalized level-one Eisenstein series of weight $2k$. Let Λ denote the multiple mod...

conjecture | open | MathDB 页面 | 原文线索

V2-003348 | MATHDB-342302

A supercongruence for a truncated hypergeometric series with parameter r

摘录 / EXCERPT: Let r be a positive integer and let p be a prime with $p>2r+1$. The supercongruence. ... This is one of three proposed strengthenings related to the paper's earlier supercongruen...

conjecture | open | MathDB 页面 | 原文线索

V2-003349 | MATHDB-342303

A higher-power supercongruence for a truncated hypergeometric series

摘录 / EXCERPT: Let $p>3$ be a prime, and let r be a positive integer. The supercongruence. First, ... More generally, ... This conjecture concerns supercongruences modulo high prime powers and gene...

conjecture | open | MathDB 页面 | 原文线索

V2-003350 | MATHDB-342305

A refined q-supercongruence for the positive-parameter case

摘录 / EXCERPT: Let p be a prime. Define the q-integer $[p]=(1-q^p)/(1-q)$ and let $(a;q)_k$ denote the q-shifted factorial. The q-supercongruence. ... The paper explicitly describes this as c...

conjecture | open | MathDB 页面 | 原文线索

V2-003351 | MATHDB-342306

A refined q-supercongruence for the negative-parameter case

摘录 / EXCERPT: Let p be a prime. Define the q-integer $[p]=(1-q^p)/(1-q)$ and let $(a;q)_k$ denote the q-shifted factorial. The q-supercongruence. ... This is the complete q-analogue mentio...

conjecture | open | MathDB 页面 | 原文线索

V2-003352 | MATHDB-342308

Firoozbakht's conjecture on successive prime roots

摘录 / EXCERPT: Let p_n denote the nth prime, and let $n\geq1$. Firoozbakht's conjecture. ... This conjecture asserts a strong monotonicity property for the distribution of the primes. It has be...

conjecture | open | MathDB 页面 | 原文线索

V2-003353 | MATHDB-342309

The combined Firoozbakht-Nicholson-Farhadian conjectures for prime gaps

摘录 / EXCERPT: Let p_n be the nth prime and define the successive prime gap by $g_n:=p_{n+1}-p_n$. Firoozbakht-Nicholson-Farhadian conjectures. For the indicated ranges of n, the following...

conjecture | open | MathDB 页面 | 原文线索

V2-003354 | MATHDB-342311

The folklore conjecture on Hardy pairs

摘录 / EXCERPT: Let ... be the Hardy set, where $\|x\|$ denotes the distance from x to the nearest integer. The folklore conjecture. If $(\lambda, \alpha) \in H$, then α is a Pisot-Vijayara...

conjecture | open | MathDB 页面 | 原文线索

V2-003355 | MATHDB-342312

Malle's degree-matching conjecture for cuspidal characters of finite Chevalley groups

摘录 / EXCERPT: Let $q=p^f$ be a power of the prime p, let G be a finite Chevalley group defined over $\mathbb F_q$ with p a bad prime for G, and let U be a Sylow p-subgroup of G. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003356 | MATHDB-342362

The p-adic root conjecture for zeros of $H_2(n)$

摘录 / EXCERPT: Let $H_2(n)$ be the sequence considered in the paper, let p be an odd prime, and let $ u_p$ denote the p-adic valuation. Assume that $p\mid H_2(n)$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-003357 | MATHDB-342363

The conjecture on prime divisors omitted by $H_d(n)$

摘录 / EXCERPT: Prime-divisor density conjecture. For every $d\in\mathbb{N}_{\geq 2}$, the set $\mathbb{P}\setminus\mathbb{P}_d$ is infinite and has asymptotic density $1/e$ in $\mathbb{P}$.

conjecture | open | MathDB 页面 | 原文线索

V2-003358 | MATHDB-342367

The conjectures on valuations for non-prime-power colored partition functions

摘录 / EXCERPT: Let $S_{k,m}(n)$ be the nth coefficient in the power-series expansion defining the m-colored k-ary partition function, and let ... Non-prime-power valuation conjectures. If k...

conjecture | open | MathDB 页面 | 原文线索

V2-003359 | MATHDB-342368

Conjecture on period lengths of square roots of primes

摘录 / EXCERPT: Period-length distribution conjecture. For each $i\in\mathbb{N}_{+}$, the limits

conjecture | open | MathDB 页面 | 原文线索

V2-003360 | MATHDB-342388

Bell-Bruin-Coons conjecture on multiplicative automatic functions

摘录 / EXCERPT: Bell-Bruin-Coons conjecture. There exists an eventually periodic function $g:\mathbb{N}\to\mathbb{C}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003361 | MATHDB-342403

Mock Jacobi property of the generating function for planar hexagons

摘录 / EXCERPT: Let $f_5(\boldsymbol{z};\tau)$ be the generating function defined by ... where the parameters α_k, for $k\in\{1,2,3,4\}$, satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-003362 | MATHDB-342419

Kantor's conjecture on the abundance of additive MRD codes

摘录 / EXCERPT: Let p be prime and let n be a positive integer. Consider additive MRD codes in $M_{n\times n}(\mathbb{F}_p)$ with minimum distance n, where equivalence is taken under the usu...

conjecture | open | MathDB 页面 | 原文线索

V2-003363 | MATHDB-342421

Böckle-Juschka irreducible-component conjecture for framed deformation spaces

摘录 / EXCERPT: Let p be a prime, let n be a positive integer, let K be a finite extension of cQ_p, and let $\overline\rho:G_K\to\operatorname{GL}_n(k_L)$ be a continuous representation. L...

conjecture | open | MathDB 页面 | 原文线索

V2-003364 | MATHDB-342422

Density conjecture for regular crystalline points in framed deformation spaces

摘录 / EXCERPT: Let p be a prime, let n be a positive integer, let K be a finite extension of $\mathbf{Q}_p$, and let $\overline\rho:G_K\to\operatorname{GL}_n(k_L)$ be a continuous represent...

conjecture | open | MathDB 页面 | 原文线索

V2-003365 | MATHDB-342427

Scholz's conjecture on distinguished dihedral cubic fields

摘录 / EXCERPT: Scholz's conjecture. There should exist such fields L for which either $f=1$, $\varrho_3(K)=2$, the complete 3-elementary class group of K capitulates in N, and $U_N=U_0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003366 | MATHDB-342428

Mayer's refinement of Scholz's conjecture for ramified dihedral cubic fields

摘录 / EXCERPT: Mayer's conjecture. Non-Galois totally real cubic fields with these properties should exist in each of the following situations: type α_1 with...

conjecture | open | MathDB 页面 | 原文线索

V2-003367 | MATHDB-342436

Massey vanishing conjecture for Galois cohomology

摘录 / EXCERPT: Let k be a field, let p be a prime, and let $n\geq 3$. For cohomology classes ... consider the n-fold Massey product...

conjecture | open | MathDB 页面 | 原文线索

V2-003368 | MATHDB-342437

Finiteness conjecture for Diophantine quadruples with nonsquare product parameter

摘录 / EXCERPT: For a nonzero integer n, call a set of positive integers a $D(n)$-quadruple if the product of any two distinct elements plus n is a perfect square. Finiteness conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-003369 | MATHDB-342446

The large-prime surjectivity conjecture for non-CM elliptic curves

摘录 / EXCERPT: Large-prime surjectivity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-003370 | MATHDB-342450

Mazur's uniform boundedness conjecture for rational points on curves

摘录 / EXCERPT: Let $g\ge 2$ be an integer and let k be a number field. For a smooth projective curve C of genus g defined over k, write $\operatorname{Jac}(C)$ for its Jacobian and...

conjecture | open | MathDB 页面 | 原文线索

V2-003371 | MATHDB-342451

Cancellation conjecture for finitely generated prime algebras

摘录 / EXCERPT: Cancellation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003372 | MATHDB-342470

Serre's cyclicity conjecture for elliptic curves

摘录 / EXCERPT: Let $E/{\mathbb Q}$ be an elliptic curve, and for each prime p of good reduction let $\widetilde{E}({\mathbb F}_p)$ denote the group of rational points of its reduction modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-003373 | MATHDB-342471

Cyclicity conjecture for elliptic curves in arithmetic progressions

摘录 / EXCERPT: Let E be an elliptic curve over ${\mathbb Q}$, let f and a be relatively prime positive integers, and let $K_d={\mathbb Q}(E[d])$ be the d-division field. For a primitive...

conjecture | open | MathDB 页面 | 原文线索

V2-003374 | MATHDB-342479

Cusick's simplified conjecture for paired binary shifts

摘录 / EXCERPT: Cusick's simplified conjecture. For all $t\geq 0$, one has $c_t+c_{t'}>1$; equivalently, at least one of t or t' satisfies Cusick's conjecture. This is presented as an easier s...

conjecture | open | MathDB 页面 | 原文线索

V2-003375 | MATHDB-342481

Nonvanishing conjecture for Gross-curve twists when q is 3 modulo 8

摘录 / EXCERPT: Nonvanishing conjecture. For every prime q satisfying $q\equiv3\pmod 8$,

conjecture | open | MathDB 页面 | 原文线索

V2-003376 | MATHDB-342485

Valuation conjecture for norms of singular moduli of discriminant $-3f^2$

摘录 / EXCERPT: Let p be a prime number, let $v_p:4^*\to\mathbb{Z}$ be the usual p-adic valuation, and let $D=-3f^2$ be a discriminant. Let j_D be any singular modulus relative to D, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-003377 | MATHDB-342489

Romik's conjecture on the periodicity of theta-function Taylor coefficients

摘录 / EXCERPT: Romik's conjecture. Let p be an odd prime. If $p\equiv3\pmod 4$, then $d(n)\equiv0\pmod p$ for sufficiently large n; if $p\equiv1\pmod 4$, then the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-003378 | MATHDB-342501

Ostrik's conjecture on the cohomology of cyclic number fields

摘录 / EXCERPT: Let $\mathbb{K}$ be a cyclic extension of $\mathbb{Q}$ of degree $n\in\mathbb{Z}_{\geq 2}$. Let G be the Galois group of $\mathbb{K}$ over $\mathbb{Q}$ and let U be the unit gr...

conjecture | open | MathDB 页面 | 原文线索

V2-003379 | MATHDB-342502

The discreteness conjecture for ordered totally real d-numbers

摘录 / EXCERPT: Discreteness conjecture. This subset is not discrete in $\mathbb{R}$. The claim predicts accumulation among totally ordered algebraic integers satisfying the d-number condition;...

conjecture | open | MathDB 页面 | 原文线索

V2-003380 | MATHDB-342504

The isomorphism criterion for monomial digraphs

摘录 / EXCERPT: Let q be a prime power, and let m_1,n_1,m_2,n_2 be integers from $\{1,2,\ldots,q-1\}$. For a monomial digraph $D(q;m,n)$, write $D_1=D(q;m_1,n_1)$ and $D_2=D(q;m_2,n_2)$. The m...

conjecture | open | MathDB 页面 | 原文线索

V2-003381 | MATHDB-342523

Oberdieck-Pixton conjecture on Gromov-Witten potentials of elliptic fibrations

摘录 / EXCERPT: Oberdieck-Pixton conjecture. For $\gamma\in H_2(B,\mathbb Z)$,

conjecture | open | MathDB 页面 | 原文线索

V2-003382 | MATHDB-342524

Quasimodularity conjecture for quasimap potentials of elliptic fibrations

摘录 / EXCERPT: Quasimap elliptic-fibration conjecture. For $\gamma\in H_2(B,\mathbb Z)$,

conjecture | open | MathDB 页面 | 原文线索

V2-003383 | MATHDB-342554

The unbounded multiplicity conjecture for differences of tetrahedral numbers

摘录 / EXCERPT: Unbounded multiplicity conjecture. For each $N\in\mathbb{N}$ there is $d_N\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003384 | MATHDB-342557

Conjecture on cocyclic Hadamard matrices of order 4p

摘录 / EXCERPT: Let p be an odd prime, and let W,X,Y,Z be circulant blocks defining cocyclic Hadamard matrices of order $4p$. Write τ for the relevant nontrivial action and ι for th...

conjecture | open | MathDB 页面 | 原文线索

V2-003385 | MATHDB-342558

Fuglede's conjecture for elementary abelian groups

摘录 / EXCERPT: Let p be a prime and $d\geq 1$. A set $E\subseteq (\mathbb Z/p\mathbb Z)^d$ is tiling if there exists $T\subseteq (\mathbb Z/p\mathbb Z)^d$ such that the translates...

conjecture | open | MathDB 页面 | 原文线索

V2-003386 | MATHDB-342571

The Mersenne-primality conjecture for the binary happy-number sequence

摘录 / EXCERPT: Let $\gamma_2(\eta)$ denote the sequence introduced in the paper, and let a Mersenne prime be a prime number of the form 2^p-1, where p is prime. Mersenne-primality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003387 | MATHDB-342612

The folklore conjecture on integers with binary digits in bases 3, 4 and 5

摘录 / EXCERPT: For an integer, say that its base-b expansion contains solely the digits 0 and 1 when no other digit occurs. Folklore base-expansion conjecture. The only integers whose base-...

conjecture | open | MathDB 页面 | 原文线索

V2-003388 | MATHDB-342613

Pomerance's conjecture on the count of central binomial coefficients coprime to 105

摘录 / EXCERPT: Let $G(N)$ be the number of positive integers $n \leq N$ such that the central binomial coefficient $\binom{2n}{n}$ is coprime to 105. Pomerance's conjecture. For all sufficientl...

conjecture | open | MathDB 页面 | 原文线索

V2-003389 | MATHDB-342625

Euclidean distortion conjecture for special linear groups over finite fields

摘录 / EXCERPT: Special linear group Euclidean distortion conjecture. For every $k\in\mathbb{N}$ and every prime power $\mathfrak{q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-003390 | MATHDB-342626

Low-dimensional embedding conjecture for special linear groups

摘录 / EXCERPT: Low-dimensional special linear group conjecture. For every prime power $\mathfrak{q}$ there exist $D=D(\mathfrak{q})\geqslant1$ and positive constants $c=c(\mathfrak{q})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003391 | MATHDB-342649

Unboundedness conjecture for the prime-index partial sums

摘录 / EXCERPT: Unboundedness conjecture. For every $N\geq N_0$, where $N_0>0$ is sufficiently large, the supremum of these partial sums satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-003392 | MATHDB-342711

Near-miss conjecture for reciprocal powers and long arithmetic progressions

摘录 / EXCERPT: Let $A_t = \{ n^{-t}: n \in \mathbb{N}\}$ for a fixed integer $t\geq 2$ and let $\alpha \in (0,1)$. An arithmetic progression has gap length Δ if the difference between cons...

conjecture | open | MathDB 页面 | 原文线索

V2-003393 | MATHDB-342712

Positivity conjecture for the approximation exponents f_t

摘录 / EXCERPT: For integers $t\geq 3$, define ... The numerical evidence suggests that $f_3=0$ and $f_t>0$ for $t\geq 4$. Positivity conjecture. For integers $t\geq 4$, $f_t>0$. On the other hand,...

conjecture | open | MathDB 页面 | 原文线索

V2-003394 | MATHDB-342737

Iosevich's sum-product conjecture over prime fields

摘录 / EXCERPT: Let $A\subset\mathbb{F}_p$, and let $\epsilon>0$. Iosevich's conjecture. If ... then ... The conjecture concerns simultaneous additive-multiplicative expansion in prime fields and is s...

conjecture | open | MathDB 页面 | 原文线索

V2-003395 | MATHDB-342770

Prime-ancestor exhaustion conjecture for c_2-invariants

摘录 / EXCERPT: A prime ancestor is a graph with at least five vertices that is 4-regular, internally 6-connected, has vertex connectivity 4, and has no edge shared by exactly two triangles. A pri...

conjecture | open | MathDB 页面 | 原文线索

V2-003396 | MATHDB-342771

The K_5 conjecture for ϕ^4 prime ancestors

摘录 / EXCERPT: A ϕ^4 prime ancestor is a prime ancestor arising in ϕ^4 theory, and $c_2^{(q)}(G)$ is its c_2-invariant at the prime power q. The K_5 conjecture. The only ϕ^4...

conjecture | open | MathDB 页面 | 原文线索

V2-003397 | MATHDB-342772

Lower bound on modular weights of primitive ϕ^4 graphs

摘录 / EXCERPT: Let G be a primitive ϕ^4 graph, and suppose its c_2-invariant is modular with respect to a modular form f of weight w. Weight lower-bound conjecture. Then ... Computatio...

conjecture | open | MathDB 页面 | 原文线索

V2-003398 | MATHDB-342776

Low-dimensional modularity conjecture for graph c_2-invariants

摘录 / EXCERPT: A c_2-invariant has a dimension determined by the minimal model, and it is called modular when its sequence is associated with a modular form. Graphs are classified as ϕ^4 o...

conjecture | open | MathDB 页面 | 原文线索

V2-003399 | MATHDB-342790

The EZ lattice-generation conjecture

摘录 / EXCERPT: Suppose $c_0=j/k$ with k and j relatively prime positive integers, and $2\leq k\leq n$. Let $\mathbf{C}[\mathfrak{h}]$ be the polynomial representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003400 | MATHDB-342792

The character-sum non-triviality conjecture

摘录 / EXCERPT: Character-sum conjecture. For every $\varepsilon>0$, once p is sufficiently large in terms of ε,

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V2-003401 | MATHDB-342793

Shahriari and Yu's conjecture for forbidden copies of Y_k and Y'_k

摘录 / EXCERPT: Let q be a prime power, and let $\mathrm{La}_q(n,\{Y_k,Y'_k\})$ denote the maximum size of a family of subspaces of an n-dimensional vector space over $\mathbb{F}_q$ containing...

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V2-003402 | MATHDB-342811

Conjectural one-level density formula for the family $\mathcal{F}(K)$

摘录 / EXCERPT: Let $\mathcal{F}(K)=\{L_k(s):1\le k\le K\}$, let $M=\log K$, let f be the test function, and let W_f, S_ζ, S_L, $S_{A'}$, and S_Γ be the integrals defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-003403 | MATHDB-342827

Bass conjecture on finite generation of Quillen K-theory groups

摘录 / EXCERPT: Let ${\cal X}$ be the curve over the ring of integers localized away from the prime l considered above, and let $K_m({\cal X})$ denote its Quillen K-theory groups for positive...

conjecture | open | MathDB 页面 | 原文线索

V2-003404 | MATHDB-342836

Injection dimension of weighted projective spaces with two coprime weights

摘录 / EXCERPT: Let $d,e\geq 2$ be relatively prime integers and let $r,s\geq 2$. Let $\mathbb{P}(\underbrace{d,\ldots,d}_{r},\underbrace{e,\ldots,e}_{s})$ denote the weighted projective space wit...

conjecture | open | MathDB 页面 | 原文线索

V2-003405 | MATHDB-342886

Coleman's conjecture on circular distributions

摘录 / EXCERPT: Let $\mathcal{F}^{\rm d}$ be the group of distributions, let $\mathcal{F}^{\rm c}$ be the R-submodule of strict distributions generated by the cyclotomic distribution Φ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-003406 | MATHDB-342900

Chernousov-Rapinchuk finiteness conjecture for forms of algebraic groups

摘录 / EXCERPT: Let $K=k(C)$ be the function field of a smooth affine geometrically integral curve over a field k, and let V be the set of discrete valuations associated with the closed points...

conjecture | open | MathDB 页面 | 原文线索

V2-003407 | MATHDB-342908

The main conjecture on basic Euler systems over number fields

摘录 / EXCERPT: Let K be a number field, let $\mathcal{A}^{\rm s}_K$ be the specified ideal in the relevant group ring, let ${\rm ES}_K$ be the module of Euler systems of rank r_K for...

conjecture | open | MathDB 页面 | 原文线索

V2-003408 | MATHDB-342920

Finiteness conjecture for universal diagonal ternary quadratic forms

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many primes p and integers ℓ with $1\leq\ell<p$ for which a diagonal ternary positive (p,ℓ)-universal quadratic form exis...

conjecture | open | MathDB 页面 | 原文线索

V2-003409 | MATHDB-342921

Exhaustive classification conjecture for universal diagonal ternary quadratic forms

摘录 / EXCERPT: Exhaustive classification conjecture. The complete list of diagonal ternary positive (p,ℓ)-universal quadratic forms is the one displayed in the source, namely the forms liste...

conjecture | open | MathDB 页面 | 原文线索

V2-003410 | MATHDB-342933

Cohen-Macaulayness and acyclicity of Gröbner-smoothable simplicial complexes

摘录 / EXCERPT: Let Δ be a simplicial complex on n vertices. It is Gröbner-smoothable over $\mathbb{K}$ if there exist a homogeneous prime ideal $I\subset\mathbb{K}[x_1,\ldots,x_n]$ defin...

conjecture | open | MathDB 页面 | 原文线索

V2-003411 | MATHDB-342937

The modified Bateman-Horn conjecture

摘录 / EXCERPT: Let $f_{1},f_{2},\cdots,f_{M}\in\mathbb{Z}[x]$ be distinct irreducible polynomials with positive leading coefficients. Define ... Put $f(n)=\prod_{i=1}^{M}f_{i}(n)$, suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-003412 | MATHDB-342949

Ibukiyama's transfer conjecture for paramodular Siegel modular forms

摘录 / EXCERPT: Let D) be the quaternion algebra over $\mathbb{Q}$ that is non-split over $\mathbb{R}$ and $\mathbb{Q}_p$ for a prime p and split at all other places. Let $G=GU_2(D)$ be the as...

conjecture | open | MathDB 页面 | 原文线索

V2-003413 | MATHDB-342958

The basic exact sequence conjecture for equivariant Tamagawa numbers

摘录 / EXCERPT: Let F be the number field, M the motive, and A the coefficient algebra under consideration. Write $M^{\ast}(1)$ for the dual Tate twist, $H^i(F,-)$ for global cohomology,...

conjecture | open | MathDB 页面 | 原文线索

V2-003414 | MATHDB-342971

The newness conjecture for the maximum scattered linear set \(\mathcal{L}\)

摘录 / EXCERPT: Let q be a prime power, and let $\mathcal{L}$ denote the linear set constructed in the paper in $\operatorname{PG}(1,q^6)$. A linear set is maximum scattered if it is scattered a...

conjecture | open | MathDB 页面 | 原文线索

V2-003415 | MATHDB-342975

Prime-cycle covering conjecture for finite digraph pp-constructability types

摘录 / EXCERPT: Let $\mathfrak{P}_\text{D}$ be the poset of primitive-positive constructability types of finite digraphs. For a prime number p, let $\mathbb{C}_p$ denote the directed cycle of le...

conjecture | open | MathDB 页面 | 原文线索

V2-003416 | MATHDB-342977

Infinitude conjecture for even strong, weak and very weak alpha numbers

摘录 / EXCERPT: For real x, let $N_e(x,n)$, $\bar{N}_e(x,n)$ and $\bar{\bar{N}}_e(x,n)$ count respectively the strong, weak and very weak even alpha numbers n not exceeding x. Even alpha-num...

conjecture | open | MathDB 页面 | 原文线索

V2-003417 | MATHDB-342981

Euclidean algorithm quotient-sum conjecture

摘录 / EXCERPT: For each integer $n\ge 2$, choose an integer d with $1\le d\le n-1$ and $\gcd(n,d)=1$, and let $s(n,d)$ denote the sum of the quotients obtained by running the Euclidean algorith...

conjecture | open | MathDB 页面 | 原文线索

V2-003418 | MATHDB-343000

The Asai recognition conjecture for special hypergeometric motives

摘录 / EXCERPT: Let $\boldsymbol\alpha$ be a set of parameters from Table 1 and let $\boldsymbol\beta=\{1,1,1,1,1\}$. For a hypergeometric motive $H(\boldsymbol\alpha,\boldsymbol\beta\mid z)$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-003419 | MATHDB-343006

Recursive good reduction conjecture for towers of Shimura curves

摘录 / EXCERPT: Let a tower of Shimura curves be given, and suppose it has good reduction at a prime. Recursive good reduction conjecture. Good reductions of towers of Shimura curves should be rec...

open_problem | open | MathDB 页面 | 原文线索

V2-003420 | MATHDB-343008

Prime-field descent conjecture for dense optimal families of curves

摘录 / EXCERPT: Let p be a prime number, let $2t\geq 4$ be even, and put $q=p^{2t}$. Let $A'_r(q)$ denote the relevant dense-family asymptotic invariant. Prime-field descent conjecture. ... Equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-003421 | MATHDB-343009

Dense odd-determinant conjecture for Hecke operators

摘录 / EXCERPT: Let $p>2$ be an odd prime. For each positive integer N, let $T_p(N)$ be the Hecke operator acting on the space of weight-2 cusp forms $S_2(\Gamma_0(N))$. Odd-determinant conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003422 | MATHDB-343021

Lehmer's conjecture on nonvanishing of the Ramanujan tau function

摘录 / EXCERPT: Let $\tau(n)$ be Ramanujan's tau function, the Fourier coefficients of the discriminant modular form Δ, and let p range over primes. Lehmer's conjecture. If $n\geq 1$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-003423 | MATHDB-343039

The divided beta family conjecture for explicit modular forms

摘录 / EXCERPT: Let $p\geq 5$ be a prime. For a positive integer i written as $i=rp^n$ with $(r,p)=1$, let $f_{i/j}$ denote the modular form associated to the divided beta family, for those valu...

conjecture | open | MathDB 页面 | 原文线索

V2-003424 | MATHDB-343057

Equality conjecture for linkage construction bounds

摘录 / EXCERPT: Let q be a prime power, and let $B_q(v_1,v_2,d;k)$ denote the maximum size of a constant-dimension code in a v_1-dimensional vector space whose codewords have dimension k, mi...

conjecture | open | MathDB 页面 | 原文线索

V2-003425 | MATHDB-343059

Uniqueness conjecture for odd deficient perfect numbers with four distinct prime factors

摘录 / EXCERPT: An odd deficient perfect number is an odd positive integer whose proper-divisor sum is less than the number itself; it has four distinct prime factors when its prime factorization...

conjecture | open | MathDB 页面 | 原文线索

V2-003426 | MATHDB-343060

Finiteness conjecture for odd deficient perfect numbers with a fixed number of prime factors

摘录 / EXCERPT: For a positive integer $k\geq 3$, consider odd deficient perfect numbers whose prime factorization has exactly k distinct prime factors. Finiteness conjecture. For any positive i...

conjecture | open | MathDB 页面 | 原文线索

V2-003427 | MATHDB-343069

Singular-weight Borcherds products conjecture

摘录 / EXCERPT: Let F be a Borcherds product of singular weight for an even lattice M of signature $(2,n)$ with $n\geq 3$. Singular-weight Borcherds products conjecture. There exists an even l...

conjecture | open | MathDB 页面 | 原文线索

V2-003428 | MATHDB-343071

The irrationality conjecture for the fifth zeta value and its reciprocal

摘录 / EXCERPT: The Riemann zeta function is denoted by ζ, and $\zeta(5)$ is its value at 5; $1/\zeta(5)$ denotes its reciprocal. Irrationality conjecture. The real numbers $\zeta(5)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003429 | MATHDB-343073

Gukov's conjecture on the depth of quantum modular forms from plumbing graphs

摘录 / EXCERPT: Gukov's conjecture. The function $\widehat{Z}_{\boldsymbol{a}}(q)$ is a depth r quantum modular form whose quantum set is a subset of $\mathbb Q$. Moreover, for any unimodular pl...

conjecture | open | MathDB 页面 | 原文线索

V2-003430 | MATHDB-343087

Spectrality of tiles in powers of p-groups

摘录 / EXCERPT: Let p be a prime, and let $n,d\geq 1$. Consider the finite abelian group $\mathbb{Z}_{p^n}^d$. The p-group tiling conjecture. Every tile in $\mathbb{Z}_{p^n}^d$ is a spectral set...

conjecture | open | MathDB 页面 | 原文线索

V2-003431 | MATHDB-343088

Galois-McKay conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. Set $H=N_G(P)$, the normalizer of P in G. Let $\mathcal H$ be the subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-003432 | MATHDB-343089

Tsfasman-Vlăduț generalized Brauer-Siegel conjecture

摘录 / EXCERPT: Let $\mathcal{K}=\{K_i\}_{i\in\mathbb{N}}$ be an asymptotically exact family of number fields. For each i, let h_{K_i} be the class number, R_{K_i} the regulator,...

conjecture | open | MathDB 页面 | 原文线索

V2-003433 | MATHDB-343092

Prime planar integral point sets containing any prescribed distance

摘录 / EXCERPT: Let $\mathfrak{M}(2,n)$ denote planar integral point sets with n points. An integral point set is prime when it cannot be obtained by scaling a smaller integral point set while p...

conjecture | open | MathDB 页面 | 原文线索

V2-003434 | MATHDB-343098

Exact unconnected-graph count conjecture for linear equational graphs

摘录 / EXCERPT: Let p be a prime, let $L(p)$ be defined by ... and let $U_1(p)$ denote the number of unconnected graphs in the linear family. Exact-count conjecture. For any prime $p>31$,...

conjecture | open | MathDB 页面 | 原文线索

V2-003435 | MATHDB-343099

Connectedness conjecture for cubic equational graphs over finite fields

摘录 / EXCERPT: Let p be an odd prime satisfying $3\nmid p-1$, let ${\mathbb F}_p$ be the field with p elements, and let ${\mathcal G}(\lambda,X^3+a)$ denote the associated cubic equational gr...

conjecture | open | MathDB 页面 | 原文线索

V2-003436 | MATHDB-343100

Exact unconnected-graph count conjecture for cubic equational graphs

摘录 / EXCERPT: Let p be a prime satisfying $3\nmid p-1$, let $L(p)$ be the piecewise function defined by ... and let $U_3(p)$ denote the number of unconnected graphs ${\mathcal G}(\lambda,X^3+a)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003437 | MATHDB-343101

Two-component conjecture for unconnected cubic equational graphs

摘录 / EXCERPT: Let p be a prime with $p\ge5$ satisfying $3\nmid p-1$, and let ${\mathcal G}(\lambda,X^3+a)$ be a cubic equational graph over ${\mathbb F}_p$, where $\lambda\in{\mathbb N}_p$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003438 | MATHDB-343102

The linear arithmetic fundamental lemma for GL$_{2h}$

摘录 / EXCERPT: Let F be a non-Archimedean local field with residue field of cardinality q, let K/F be a quadratic étale extension, and let D be a division algebra over F of invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-003439 | MATHDB-343105

Theta-block lift conjecture

摘录 / EXCERPT: Let $\Theta\in J_{k,t}$ be a theta block with trivial character and with order of vanishing 1 in q. Here $J_{k,t}$ denotes the space of Jacobi forms of weight k and index t...

conjecture | open | MathDB 页面 | 原文线索

V2-003440 | MATHDB-343115

Sondow's conjecture on the Smarandache function

摘录 / EXCERPT: For a positive integer n, let $S(n)$ be the smallest positive integer j such that n divides $j!$. Sondow's conjecture. The inequality ... holds for almost all positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-003441 | MATHDB-343128

Levin's arithmetic hyperbolicity conjecture for complements of ample divisors

摘录 / EXCERPT: Levin's conjecture. If $q\geq n+2$, then $X\setminus D$ is arithmetically quasi-hyperbolic; if $q\geq 2n+1$, then $X\setminus D$ is arithmetically hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-003442 | MATHDB-343150

Generalized Kaneko-Zagier homomorphism conjecture for colored multiple zeta values

摘录 / EXCERPT: Generalized Kaneko-Zagier conjecture. For each $\alpha\in(\mathbb{Z}/N\mathbb{Z})^\times$, the $\mathbb{Q}(\zeta_N)$-linear map

conjecture | open | MathDB 页面 | 原文线索

V2-003443 | MATHDB-343151

Helfgott-Venkatesh and Croot-Elsholtz inverse sieve conjecture

摘录 / EXCERPT: Let $S\subseteq \{0,\ldots,N\}$ satisfy $|S|\geq N^{\varepsilon}$ and occupy fewer than αp residue classes modulo every prime p, where $0<\alpha<1$. Helfgott-Venkatesh an...

conjecture | open | MathDB 页面 | 原文线索

V2-003444 | MATHDB-343152

The classification conjecture for three-dimensional translation-invariant Clifford QCAs

摘录 / EXCERPT: Let p be a prime, and let $\mathfrak C(D,p)$ denote the group of translation-invariant Clifford quantum cellular automata in spatial dimension D, modulo Clifford circuits and s...

conjecture | open | MathDB 页面 | 原文线索

V2-003445 | MATHDB-343161

Weyl-sum orbit density conjecture

摘录 / EXCERPT: For an integer $d\geqslant2$ and $\mathbf x\in\mathsf{T}_d$, let ... be the orbit of the Weyl sums. Weyl-sum orbit density conjecture. For almost all $\mathbf x\in\mathsf{T}_d$ in th...

conjecture | open | MathDB 页面 | 原文线索

V2-003446 | MATHDB-343162

Parabolic restriction dimension conjecture for large Weyl sums

摘录 / EXCERPT: For an integer $d\geqslant2$ and $\alpha\in(0,1)$, define ... Let $\mathcal X=\{(t,t^2):t\in[0,1]\}\subset\mathsf{T}_2$. Parabolic restriction dimension conjecture. ... This is a conje...

conjecture | open | MathDB 页面 | 原文线索

V2-003447 | MATHDB-343163

Parabolic natural-measure conjecture for large quadratic Weyl sums

摘录 / EXCERPT: Define ... Let $\mathcal X=\{(t,t^2):t\in[0,1]\}$ and let μ be the natural probability measure on $\mathcal X$. Parabolic natural-measure conjecture. ... This asserts that the larg...

conjecture | open | MathDB 页面 | 原文线索

V2-003448 | MATHDB-343170

SO(5) mass-formula congruence conjecture for Siegel modular forms

摘录 / EXCERPT: Let G be the relevant inner form of $\operatorname{SO}(5)$, let K be the maximal compact subgroup in the notation of the SO(5) mass formula, and let $S=S_1\sqcup S_2\sqcup S_3$...

conjecture | open | MathDB 页面 | 原文线索

V2-003449 | MATHDB-343171

Squarefree-level weight-four Eisenstein congruence conjecture

摘录 / EXCERPT: Let N_1 and N_2 be coprime squarefree positive integers, with $\omega(N_1)$ odd, and let E_4 be the weight-4 Eisenstein series on $\operatorname{SL}_2(\mathbb Z)$. Squarefr...

conjecture | open | MathDB 页面 | 原文线索

V2-003450 | MATHDB-343176

Mináč-Pál-Topaz-Wittenberg conjecture on universal Koszulity of maximal pro-p Galois groups

摘录 / EXCERPT: Let p be a prime, let $\mathbb{K}$ be a field containing a root of unity of order p, and let $\mathcal{G}_{\mathbb{K}}$ be its maximal pro-p Galois group. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-003451 | MATHDB-343181

Rationality conjecture for cycle integrals of twisted meromorphic modular forms

摘录 / EXCERPT: Rational cycle-integral conjecture. If this function has no poles on the cycle, then its cycle integrals are rational.

conjecture | open | MathDB 页面 | 原文线索

V2-003452 | MATHDB-343200

Topological multivariable strong monodromy conjecture

摘录 / EXCERPT: Topological multivariable strong monodromy conjecture. The polar locus of $Z_{F}^{\operatorname{top}}(S)$ is contained in $V(B_{F})$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-003453 | MATHDB-343210

The linear-programming improvement conjecture for Paley graph clique bounds

摘录 / EXCERPT: Let G_p be the Paley graph for a prime $p\equiv 1~(\operatorname{mod}4)$, let $\operatorname{LS}(p)$ denote the linear programming bound obtained in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003454 | MATHDB-343212

Ihara's lemma for Shimura curves with arbitrary weight

摘录 / EXCERPT: Let F be a totally real field, let D be a quaternion algebra over F ramified at exactly one infinite place, and let $\mathfrak{p}$ be a finite place at which D is unramifie...

conjecture | open | MathDB 页面 | 原文线索

V2-003455 | MATHDB-343213

Rado conditions conjecture for nonlinear Diophantine equations with linear diagonal polynomial

摘录 / EXCERPT: Let $P\in \mathbb{Z}[x_{1},\ldots,x_{n}]$ be a polynomial, and define its corresponding monovariate polynomial by ... Assume that $\widetilde{P}$ is a nonzero homogeneous linear poly...

conjecture | open | MathDB 页面 | 原文线索

V2-003456 | MATHDB-343214

Odd-characteristic quadratic-function design conjecture

摘录 / EXCERPT: Let p be an odd prime with $p\equiv3\pmod4$, let $m\geq3$ be odd, and set $q=p^m$. Let ℓ be a positive integer satisfying $\frac{m-2}{4}\leq\ell\leq m-1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003457 | MATHDB-343245

Jena-Genin-Mosca conjecture on Clifford Pauli partitioning

摘录 / EXCERPT: Let q be prime, let $\mathcal{S}\subseteq \pauli{q}[*]$ be a set of generalized Pauli operators, and let $\mathcal{C}$ denote the generalized Clifford group. A partition is a col...

conjecture | open | MathDB 页面 | 原文线索

V2-003458 | MATHDB-343246

Jena-Genin-Mosca conjecture on single-qudit Clifford partitioning

摘录 / EXCERPT: Let q be prime, let $\mathcal{S}\subseteq \pauli{q}[*]$ be a set of generalized Pauli operators, let $sq\mathcal{C}$ denote the gate set of single-qudit Clifford operators, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-003459 | MATHDB-343273

Conjecture on positivity of coefficients of the quasimodular forms f_w

摘录 / EXCERPT: Coefficient-positivity conjecture. Every coefficient in this expansion is positive:

conjecture | open | MathDB 页面 | 原文线索

V2-003460 | MATHDB-343274

The height bound conjecture for rational points on hyperelliptic curves

摘录 / EXCERPT: For an integer $d\geq 3$, let $f\in\mathbb{Q}[x]$ be a polynomial of degree d with nonzero discriminant, and let $h(f)$ denote the maximum of the logarithmic heights of its coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-003461 | MATHDB-343324

Parity conjecture for fixed points of the sequences A(p)

摘录 / EXCERPT: Parity conjecture. The fixed points of sequences $A(p)$, for p an odd prime, are odd.

conjecture | open | MathDB 页面 | 原文线索

V2-003462 | MATHDB-343342

Arithmetic persistence conjecture

摘录 / EXCERPT: Arithmetic Persistence Conjecture. The base change X_L is arithmetically hyperbolic over L.

conjecture | open | MathDB 页面 | 原文线索

V2-003463 | MATHDB-343366

Infinitely many squarefree second-species radicands yielding M0-fields

摘录 / EXCERPT: M0-field infinitude hypothesis. There exist infinitely many such radicands d for which L is an $\mathrm{M}0$-field.

conjecture | open | MathDB 页面 | 原文线索

V2-003464 | MATHDB-343374

Asplund-Fox extension conjecture for corona coprime numbers

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $\overline{K}_m$ be the edgeless graph on m vertices, and let $\mathfrak{pr}(G)$ denote the minimum coprime number of a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-003465 | MATHDB-343375

Asplund-Fox conjecture for minimum coprime numbers of joins of paths

摘录 / EXCERPT: Let P_m and P_n be paths on m and n vertices, respectively, and let $\mathfrak{pr}(G)$ denote the minimum coprime number of a graph G. Asplund-Fox's path-join conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003466 | MATHDB-343378

Prime-translate conjecture for ranks of polynomial-tuple expansions

摘录 / EXCERPT: Let $\mathcal{S}$ be a tuple of polynomials in $\mathbb{R}[x]$ whose integer coefficients all have the same parity, and let $\mathcal{R}(\mathcal{S})$ denote the rank of its expans...

conjecture | open | MathDB 页面 | 原文线索

V2-003467 | MATHDB-343392

Necessary condition for modular Hilbert-series changes of quasi-invariants

摘录 / EXCERPT: Let $Q_m(n)$ denote the space of quasi-invariant polynomials in n variables with parameter m, and let p be prime. Suppose the Hilbert series of $Q_m(n)$ over $\mathbb F_p$ di...

conjecture | open | MathDB 页面 | 原文线索

V2-003468 | MATHDB-343404

The refined $3x+1$ conjecture

摘录 / EXCERPT: Let $T_R:\mathbb{Z}\to\mathbb{Z}$ be defined by ... For $k\geq 0$, define $T_R^{0}(n)=n$ and $T_R^{k+1}(n)=T_R(T_R^{k}(n))$, and call $(T_R^{k}(n))_{k\geq 0}$ the T_R-trajectory of...

conjecture | open | MathDB 页面 | 原文线索

V2-003469 | MATHDB-343405

Sharpness of the divisor concentration bounds

摘录 / EXCERPT: Let $\zeta=\limsup_{k\to\infty}\frac{\log k}{\log(1/\beta_k)}$, where β_k is the equal-sums exponent for logarithmic random sets. For an integer n, let $\Delta(n)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-003470 | MATHDB-343406

Sharpness of the lower bound relating divisor packing to equal sums

摘录 / EXCERPT: For each $k\geqslant2$, let β_k be the equal-sums exponent for logarithmic random sets, and let α_k be the supremum of the real numbers α such that almost every...

conjecture | open | MathDB 页面 | 原文线索

V2-003471 | MATHDB-343436

The unbounded total-domination gap conjecture for direct products of complete graphs

摘录 / EXCERPT: Let X_n be the direct product of complete graphs associated with the positive integer n, let $[? g(n)$ denote the Jacobsthal function, and let $[? \gamma_t(X_n)$ denote the tot...

conjecture | open | MathDB 页面 | 原文线索

V2-003472 | MATHDB-343437

The Jacobsthal-function gap conjecture for products of primes

摘录 / EXCERPT: Let $g(m)$ denote the Jacobsthal function of a positive integer m. Jacobsthal-function gap conjecture. For all positive integers j, there exists some positive integer k along...

conjecture | open | MathDB 页面 | 原文线索

V2-003473 | MATHDB-343446

The generalized Brauer-Siegel conjecture for asymptotically exact families

摘录 / EXCERPT: Tsfasman-Vlăduț conjecture. For every asymptotically exact family $\mathcal{K}$,

conjecture | open | MathDB 页面 | 原文线索

V2-003474 | MATHDB-343467

Postnikov's periodicity conjecture for weighted Catalan numbers modulo powers of 3

摘录 / EXCERPT: Postnikov's periodicity conjecture. For $r\geq 3$, the sequence $\{L_n\pmod{3^r}\}$ is purely periodic with period $2\cdot 3^{r-3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-003475 | MATHDB-343468

Postnikov's 2-adic valuation conjecture for Morse link and Catalan numbers

摘录 / EXCERPT: Postnikov's 2-adic valuation conjecture. There exists such a 2-adic integer α satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-003476 | MATHDB-343469

The generalized 2-adic valuation conjecture for weighted Catalan numbers

摘录 / EXCERPT: Generalized 2-adic valuation conjecture. There exists a 2-adic integer α_k and a nonnegative integer c_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-003477 | MATHDB-343470

Postnikov's 5-adic valuation conjecture for Morse link numbers

摘录 / EXCERPT: Postnikov's 5-adic valuation conjecture. For $n\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-003478 | MATHDB-343471

The 3-adic valuation pattern conjecture for Morse link numbers

摘录 / EXCERPT: 3-adic valuation pattern conjecture. There exists a 3-adic integer α such that, for odd $n\geq 3$, the quantity $\xi_3(L_n-1)$ depends only on $\xi_3(n-\alpha)$ and the last...

conjecture | open | MathDB 页面 | 原文线索

V2-003479 | MATHDB-343475

Normal-distribution conjecture for Khovanov ranks of weaving knots

摘录 / EXCERPT: Fix an even integer n, and consider the weaving knot $W(n+1,m)$, where m is coprime to $n+1$. Along either of the two lines specified in Theorem 3, consider the normalized Khov...

conjecture | open | MathDB 页面 | 原文线索

V2-003480 | MATHDB-343483

Small quadratic non-residue conjecture

摘录 / EXCERPT: Small quadratic non-residue conjecture. There exists a prime $q<2\sqrt{p}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003481 | MATHDB-343496

Self-converse classification conjecture for entropic Mendelsohn triple systems

摘录 / EXCERPT: Self-converse classification conjecture. The system $\emph{Lin}(M,R)$ is self-converse if and only if no prime dividing the order of M is congruent to 1 modulo 3.

conjecture | open | MathDB 页面 | 原文线索

V2-003482 | MATHDB-343506

Containment conjecture for cyclic-prime shuffle groups

摘录 / EXCERPT: Let k be an odd prime, let $n>k$, and suppose that n is not a power of k. Cyclic-prime shuffle-group conjecture. Then ... This is presented as a computationally verified expect...

conjecture | open | MathDB 页面 | 原文线索

V2-003483 | MATHDB-343508

The higher-wise independence conjecture for projective norm graphs

摘录 / EXCERPT: Higher-wise independence conjecture. For any prime power q and integers $4\leq\ell<t$, all but $o(n^\ell)$ sets of ℓ vertices in $\operatorname{NG}(q,t)$ have...

conjecture | open | MathDB 页面 | 原文线索

V2-003484 | MATHDB-343510

Le Brun et al.'s lunar divisor maximum conjecture in bases at least three

摘录 / EXCERPT: Let $b\geq 3$ and let n range over the k-digit numbers in base b. Write $d_b(n)$ for the number of lunar divisors of n in base b, and write $111\ldots 1|_b$ for the base-...

conjecture | open | MathDB 页面 | 原文线索

V2-003485 | MATHDB-343511

Le Brun et al.'s binary lunar divisor maximum conjecture

摘录 / EXCERPT: Let $d_2(n)$ denote the number of binary lunar divisors of n, and let $111\ldots 10|_2$ denote the binary number with $k-1$ initial 1's followed by 0. Le Brun et al.'s conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003486 | MATHDB-343512

Applegate, Le Brun, and Sloane's lunar prime asymptotic conjecture

摘录 / EXCERPT: For each base b, let $\pi_b(k)$ denote the number of base-b lunar primes with k digits. Applegate, Le Brun, and Sloane's conjecture. As k tends to infinity, ... This predicts...

conjecture | open | MathDB 页面 | 原文线索

V2-003487 | MATHDB-343532

A diagonal valuation formula for Stirling numbers of the first kind

摘录 / EXCERPT: Let p be an odd prime. Let a,n,k be positive integers with $1\le a\le p-1$ and $2\le k\le a(p-1)p^{n-1}+1$. Define $\epsilon_k=0$ if k is even and $\epsilon_k=1$ if k is od...

conjecture | open | MathDB 页面 | 原文线索

V2-003488 | MATHDB-343539

The conjecture on odd-degree torsion in the cohomology of M

摘录 / EXCERPT: Let M and X be the spaces in the paper, let p be the prime under consideration, and let k be as above. Write $t_p^i(M)$ for the exponent of the p-primary torsion in...

conjecture | open | MathDB 页面 | 原文线索

V2-003489 | MATHDB-343541

Minimum coprime number conjecture for odd generalized Petersen graphs with k=3

摘录 / EXCERPT: Let $GP(n,3)$ be the generalized Petersen graph, with $n\geq 7$ odd, and let $\mathfrak{pr}(G)$ denote its minimum coprime number. The $k=3$ minimum-coprime-number conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-003490 | MATHDB-343542

Minimum coprime number conjecture for GP(3k,k)

摘录 / EXCERPT: Let $k\geq 2$, and let $GP(3k,k)$ be the generalized Petersen graph. Let $\mathfrak{pr}(G)$ denote its minimum coprime number. The $GP(3k,k)$ minimum-coprime-number conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-003491 | MATHDB-343543

Minimum coprime number conjecture for stacked odd-gon prism graphs

摘录 / EXCERPT: Let $k\geq 1$ and $n\geq 1$. The stacked $(2k+1)$-gon prism graph $Y_{2k+1,n}$ consists of n stacked copies of a $(2k+1)$-cycle prism. Let $\mathfrak{pr}(G)$ denote the minimum c...

conjecture | open | MathDB 页面 | 原文线索

V2-003492 | MATHDB-343566

Lemke Oliver-Soundararajan conjecture for consecutive prime residue patterns

摘录 / EXCERPT: Let $q\ge3$ be a positive integer, let a and b be reduced residue classes modulo q, and set $h\equiv b-a\pmod q$. Define ... Let $\mathfrak{S}_{q,0}$ denote the inclusion-exclu...

conjecture | open | MathDB 页面 | 原文线索

V2-003493 | MATHDB-343606

Largest-prime-gap asymptotic via the interval sieve

摘录 / EXCERPT: Let W_y be the minimum of $|[0,y]\cap\mathcal{S}_{(y/\log y)^{1/2}}|$ over all choices of residue classes modulo primes up to $(y/\log y)^{1/2}$, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003494 | MATHDB-343617

Oppermann's conjecture on primes in quadratic intervals

摘录 / EXCERPT: Let n be an integer with $n>1$. Oppermann's conjecture. There is at least one prime number between $n(n-1)$ and n^2, and at least another prime number between n^2 and...

conjecture | open | MathDB 页面 | 原文线索

V2-003495 | MATHDB-343618

Brocard's conjecture on primes between squares of consecutive primes

摘录 / EXCERPT: Let p and q be consecutive prime numbers other than the exceptional pair 2 and 3. Brocard's conjecture. There are at least four primes strictly between p^2 and q^2. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-003496 | MATHDB-343627

Squarefreeness conjecture for sigma-2,2 pairs

摘录 / EXCERPT: Let p and q be a $\sigma_{2,2}$ pair, meaning that they are distinct odd primes satisfying the paper's definition of a $\sigma_{2,2}$ pair. Squarefreeness conjecture. The integ...

conjecture | open | MathDB 页面 | 原文线索

V2-003497 | MATHDB-343647

Vatwani's Gaussian prime pair conjecture

摘录 / EXCERPT: Let m_1,m_2 be integers of the same parity. Vatwani's Gaussian prime pair conjecture. There are infinitely many pairs of rational primes (p_1,p_2) of the form ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-003498 | MATHDB-343648

The vanishing ratio conjecture for primes with fixed and diagonal indices

摘录 / EXCERPT: Let $p^{(k)}_n$ denote the k-th prime in the subsequence of primes having prime index, with n specifying the subsequence, and let $p^{(k)}_k$ be the corresponding diagonal term...

conjecture | open | MathDB 页面 | 原文线索

V2-003499 | MATHDB-343649

The compatible Galois representations conjecture for regular L-algebraic cuspidal representations

摘录 / EXCERPT: Let L) be a number field and let π be a cuspidal automorphic representation of $\operatorname{GL}_n(\mathbb A_L)$ that is L-algebraic and regular. For a prime p and an is...

conjecture | open | MathDB 页面 | 原文线索

V2-003500 | MATHDB-343671

Modularity conjecture for elliptic curves over quadratic imaginary fields

摘录 / EXCERPT: Let K be a quadratic imaginary field of class number one, let $p>3$ be a prime, and let E be an elliptic curve over K. Write G_K for the absolute Galois group of K, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-003501 | MATHDB-343685

The rational-point counting conjecture for Cantor sets

摘录 / EXCERPT: Let ${\mathcal{C}}$ be Cantor's middle-thirds set, and more generally let ${\mathcal{C}}$ be the set of numbers $x=\sum_{i=1}^{\infty}a_i b^{-i}$ with a_i in a proper subset...

conjecture | open | MathDB 页面 | 原文线索

V2-003502 | MATHDB-343686

Mahler's intrinsic approximation conjecture for the Cantor set

摘录 / EXCERPT: Let ${\mathcal{C}}$ be Cantor's middle-thirds set, equipped with its coin-tossing measure, which assigns equal probability to the digits 0 and 2 in the ternary expansion. A rat...

conjecture | open | MathDB 页面 | 原文线索

V2-003503 | MATHDB-343689

Equidistribution conjecture for visible points along irrational slopes

摘录 / EXCERPT: Let $0<\rho<1$ be an irrational number with irrationality measure 2. For each positive integer n, let $V(\lceil\rho n\rceil,n)$ denote the number of visible lattice points in t...

conjecture | open | MathDB 页面 | 原文线索

V2-003504 | MATHDB-343693

Romik's conjectures on congruences for Jacobi theta coefficients

摘录 / EXCERPT: Romik's conjectures. The following statements are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-003505 | MATHDB-343703

Lee-Szczarba conjecture on top-dimensional cohomology of congruence subgroups

摘录 / EXCERPT: Lee-Szczarba conjecture. For a prime p and $n\geq 2$, the induced map

conjecture | open | MathDB 页面 | 原文线索

V2-003506 | MATHDB-343705

Jacquet-Rallis fundamental lemma conjecture

摘录 / EXCERPT: Let F/F_0 be an unramified quadratic extension of p-adic fields for an odd prime p. Let S_n be the Jacquet-Rallis symmetric space, let $G_i=\mathrm{U}(V_i^\sharp)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003507 | MATHDB-343707

Ihara's lemma for generic irreducible submodules

摘录 / EXCERPT: Ihara's lemma. Then π is generic. This is the precise weak Ihara lemma attributed in the source to Clozel, Harris, and Taylor. The introduction explains that the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-003508 | MATHDB-343746

Prime-polynomial symmetry conjecture for weighted Dirichlet polynomials

摘录 / EXCERPT: Prime-polynomial symmetry conjecture. The positive and negative real parts, the real and imaginary parts, and the positive and negative imaginary parts have comparable maximal size...

conjecture | open | MathDB 页面 | 原文线索

V2-003509 | MATHDB-343747

Radziwiłł's Gaussian value-distribution conjecture for log-modulus of the Riemann zeta-function

摘录 / EXCERPT: Radziwiłł's Gaussian value-distribution conjecture. If $V=o(\sqrt{\log\log T})$ as $T\to+\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-003510 | MATHDB-343766

Normalized Vafa-Witten partition functions satisfy S-duality

摘录 / EXCERPT: Let S be a smooth projective surface, let $q=e^{2\pi i\tau}$, and let $\eta(q)=q^{1/24}\prod_{k\geq 1}(1-q^k)$ be the Dedekind eta function. Define the normalized partition funct...

conjecture | open | MathDB 页面 | 原文线索

V2-003511 | MATHDB-343815

Galois-orbit equidistribution conjecture for torsion points and Laurent polynomials

摘录 / EXCERPT: Let $d\geq 1$, let $P\in \overline{\mathbb{Q}}[X_1^{\pm 1},\ldots,X_d^{\pm 1}]\setminus\{0\}$, and let $m(P)$ denote the logarithmic Mahler measure of P. For...

conjecture | open | MathDB 页面 | 原文线索

V2-003512 | MATHDB-343824

The characterization of pairs occurring infinitely often among slowest walks

摘录 / EXCERPT: Characterization conjecture. There exist infinitely many n with $(\alpha,\beta)\in\mathbf{S}_T(n)$ if and only if $(\alpha,\beta)\in R_T'$. This concerns which relatively prime p...

conjecture | open | MathDB 页面 | 原文线索

V2-003513 | MATHDB-343825

A subsequence conjecture for pairs in R_T'

摘录 / EXCERPT: Let T be a set of relatively prime pairs, and let $(\alpha,\beta)\in R_T'$, where ... Let $g_t^{\alpha,\beta}$ denote the sequence associated with the (α,β)-walk, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-003514 | MATHDB-343834

Bimodule-summand conjecture for the Khovanov complex of L_p

摘录 / EXCERPT: For an odd prime p, let L_p be the two-component link obtained by closing the braid word $(\sigma_1\cdots\sigma_p)^{p+1}\sigma_1\cdots\sigma_{p-1}$. Let R_p be the coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-003515 | MATHDB-343836

The 4/3 diameter conjecture for Morgenstern Ramanujan graphs

摘录 / EXCERPT: Let q be a prime power, let $\mathbb{F}_q$ be the finite field with q elements, and let $g\in\mathbb{F}_q[t]$ satisfy $\gcd(g,t(t-1))=1$. Let $X^{q,g}$ be the associated $q+1$-...

conjecture | open | MathDB 页面 | 原文线索

V2-003516 | MATHDB-343848

The standard-prime ideal conjecture for solvable Iwasawa algebras

摘录 / EXCERPT: Standard-prime ideal conjecture. If P is a prime ideal of KG, then P is standard.

conjecture | open | MathDB 页面 | 原文线索

V2-003517 | MATHDB-343862

Montgomery's bound for Rudin-Shapiro polynomial intervals

摘录 / EXCERPT: Let $P_{[m,n)}(z)=\sum_{i=m}^{n-1}a_i z^i$ denote the polynomial formed from the Rudin-Shapiro coefficients, where $n\ge m\ge 0$, and let z lie on the unit circle, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-003518 | MATHDB-343877

Minimum minimizing index conjecture for doubled regular odd prime-gons

摘录 / EXCERPT: Let X_p be the doubled regular p-gon, where p is an odd prime. The minimizing index $\operatorname{minind}(X_p)$ is the smallest positive integer k such that X_p admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-003519 | MATHDB-343880

The Möbius-transform conjecture for l-behaved elements

摘录 / EXCERPT: Möbius-transform conjecture. For any given $u \in \mathbb{F}_p(t)\setminus\mathbb{F}_p$, there exist $\alpha_u,\beta_u,\gamma_u,\delta_u\in\mathbb{F}_p$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003520 | MATHDB-343910

The irreducibility conjecture for Q_n^{(k)} and V_n

摘录 / EXCERPT: Let $Q_n^{(k)}(x)$ and $V_n(x)$ be the polynomials defined in the paper, with integers $n\geq 1$ and $0\leq k\leq n-1$. Irreducibility conjecture. For all integers $n\geq 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003521 | MATHDB-343918

The global Langlands conjecture

摘录 / EXCERPT: Let F be a finite extension of $\mathbb Q$, let $\mathbb A_F$ be its ring of adeles, and fix a prime p together with an isomorphism $\overline{\mathbb Q}_p\cong\mathbb C$. Cons...

conjecture | open | MathDB 页面 | 原文线索

V2-003522 | MATHDB-343923

The birational section conjecture over finitely generated fields

摘录 / EXCERPT: Let k be a finitely generated field over $\operatorname{\mathbb{Q}}$, and let X be a smooth, projective, geometrically connected curve over k. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-003523 | MATHDB-343933

Cancellation conjecture for quadratic exponential sums

摘录 / EXCERPT: Let $q(l)$ be a quadratic polynomial, with the sum taken over integers l satisfying $q(l)<x$, and let x tend to infinity. Cancellation conjecture. ... This conjectures cancellati...

conjecture | open | MathDB 页面 | 原文线索

V2-003524 | MATHDB-343944

Conjecture on the optimal bound for primitive pairs of type $(1,1)$

摘录 / EXCERPT: Let $Q_{1,1}$ be the set of prime powers q for which every rational function in $R_{1,1}$ admits a primitive pair $(\alpha,f(\alpha))$ with $\alpha\in\mathbb{F}_q$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003525 | MATHDB-343954

Guillera-Zudilin-type supercongruence for a fifth-power hypergeometric sum

摘录 / EXCERPT: Let p be an odd prime, let r be a positive integer, and let $\delta\in\{1,2\}$. Guillera-Zudilin-type conjecture. ... The paper states that this stronger modulus cannot be proved...

conjecture | open | MathDB 页面 | 原文线索

V2-003526 | MATHDB-343978

Gibbs's concentration conjecture for the Ulam sequence U(1,2)

摘录 / EXCERPT: Let $U(1,2)$ be the Ulam sequence, and for a positive real number λ interpret reduction modulo λ by taking representatives in an interval of length λ. Gibbs...

conjecture | open | MathDB 页面 | 原文线索

V2-003527 | MATHDB-343979

Hinman's generalized period conjecture for Ulam sequences U(1,n)

摘录 / EXCERPT: For integers $1\leq a<b$, let $U(a,b)$ be the Ulam sequence whose later terms are the smallest integers representable as sums of two distinct prior terms in exactly one way. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-003528 | MATHDB-343980

The asymptotic period conjecture for coefficients of U(1,X)

摘录 / EXCERPT: Let $U(1,X)$ denote the polynomial-ring analogue of the Ulam sequence, and let a_i,b_i,c_i,d_i be its coefficients. For real X, reduction modulo $3X+\lambda'$ is taken using re...

conjecture | open | MathDB 页面 | 原文线索

V2-003529 | MATHDB-344002

The uniqueness conjecture for extreme subsets of prime cyclic groups

摘录 / EXCERPT: Prime-cyclic uniqueness conjecture. All extreme n-element subsets of $\mathbb Z_p$ are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-003530 | MATHDB-344004

The prime-cardinality frequency conjecture for extreme sets

摘录 / EXCERPT: Prime-cardinality frequency conjecture. Extreme sets with prime cardinality appear more frequently than sets whose cardinality is neither prime nor a perfect square.

conjecture | open | MathDB 页面 | 原文线索

V2-003531 | MATHDB-344007

Ternary cyclotomic height universality conjecture

摘录 / EXCERPT: Ternary height universality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003532 | MATHDB-344008

Optimal ternary height conjecture

摘录 / EXCERPT: Optimal ternary height conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003533 | MATHDB-344009

Prime-gap conjecture below the square-root scale

摘录 / EXCERPT: Let p_n denote the nth prime. Prime-gap conjecture below the square-root scale. For every $n\ge31$ (so that $p_n\ge127$), ... The paper states that this conjecture implies its cy...

conjecture | open | MathDB 页面 | 原文线索

V2-003534 | MATHDB-344010

Cyclotomic coefficient-set coverage conjecture

摘录 / EXCERPT: Let $\mathcal R$ be the set of natural numbers obtained by the paper's explicit prime-parametrized construction of ternary cyclotomic heights. Cyclotomic coefficient-set coverage c...

conjecture | open | MathDB 页面 | 原文线索

V2-003535 | MATHDB-344012

Optimal ternary extreme-coefficient coverage conjecture

摘录 / EXCERPT: Let $\mathcal R^{\pm}$ be the union of the two prime-parametrized sets defined by ... Let $\mathcal A^+_{\mathrm{opt}}$ and $\mathcal A^-_{\mathrm{opt}}$ denote the sets of maximum a...

conjecture | open | MathDB 页面 | 原文线索

V2-003536 | MATHDB-344016

Linnik's microsquare conjecture

摘录 / EXCERPT: Let $\varepsilon>0$. For each sufficiently large odd squarefree integer $n\not\equiv7\pmod{8}$, consider representations of n by three integer squares. Linnik's conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-003537 | MATHDB-344018

Abe-Tagami and Baker's linear independence conjecture for prime fibered strongly quasi-positive knots

摘录 / EXCERPT: Let $\mathcal{C}$ denote the smooth knot concordance group. A knot is strongly quasi-positive if it admits a strongly quasi-positive braid representative. Abe-Tagami and Baker's co...

conjecture | open | MathDB 页面 | 原文线索

V2-003538 | MATHDB-344038

Coprime summatory conjecture for the s-th Möbius function

摘录 / EXCERPT: Coprime s-free summatory conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-003539 | MATHDB-344043

Quaternion-ring analogue of Waring's problem

摘录 / EXCERPT: For positive integers a,b, let $LQ_{a,b}$ be the quaternion ring ... and let $LQ_{a,b}^k$ denote the additive group generated by all kth powers in $LQ_{a,b}$. Quaternion-ring War...

conjecture | open | MathDB 页面 | 原文线索

V2-003540 | MATHDB-344045

Hardy-Littlewood's asymptotic density conjecture for admissible prime tuples

摘录 / EXCERPT: Let $(a_1,a_2,\ldots,a_k)$ be a monotonically increasing sequence of positive even integers. Let p be prime, and let $P=(p,p+a_1,p+a_2,\ldots,p+a_k)$ be an admissible prime k-t...

conjecture | open | MathDB 页面 | 原文线索

V2-003541 | MATHDB-344046

Wolf's sign-change conjecture for the twin-prime counting error

摘录 / EXCERPT: Wolf's conjecture. The number of sign changes of $d_2(n)$ for $n\in(1,T)$ is $\frac{\sqrt T}{\log T}$. This numerical prediction concerns fluctuations around the Hardy-Littlewood e...

conjecture | open | MathDB 页面 | 原文线索

V2-003542 | MATHDB-344047

The Skewes-number conjecture for admissible prime tuples

摘录 / EXCERPT: Skewes-number conjecture. Every admissible prime k-tuple has a Skewes number. The conjecture is based on the computational results in the paper; no proof or disproof is supplied.

conjecture | open | MathDB 页面 | 原文线索

V2-003543 | MATHDB-344057

Sharifi's conjecture on the Eisenstein quotient and cyclotomic K-theory

摘录 / EXCERPT: Let N be the prime level and p a prime with $p\nmid N$. Let H_+ be the plus part of the modular-symbol homology group, let I be the Eisenstein ideal acting on H_+, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003544 | MATHDB-344134

Saad-Wolf commonness conjecture for even-variable linear equations

摘录 / EXCERPT: Let q be a prime power, let n be a positive integer, and let $k\ge 2$. Consider the linear equation over $\mathbb{F}_q^n$ ... The equation is common if, for every two-coloring of...

conjecture | open | MathDB 页面 | 原文线索

V2-003545 | MATHDB-344162

Guo-Zudilin q-analogue with full truncation

摘录 / EXCERPT: Let $(a;q)_k=\prod_{j=0}^{k-1}(1-aq^j)$ be the q-shifted factorial, let $\Phi_n(q)$ be the nth cyclotomic polynomial, and let $n>1$ be odd and $r\geqslant 1$. Guo-Zudilin's ful...

conjecture | open | MathDB 页面 | 原文线索

V2-003546 | MATHDB-344186

Binary Collatz conjecture

摘录 / EXCERPT: Binary Collatz conjecture. For any $x\in\mathbb{N}$, with $y=M(x)$, one has $y\in I$ and

conjecture | open | MathDB 页面 | 原文线索

V2-003547 | MATHDB-344187

Convergence conjecture for the exceptional set $\mathcal{C}$

摘录 / EXCERPT: Convergence conjecture for $\mathcal{C}$. There exists $k\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003548 | MATHDB-344225

Gonek's conjecture on zeros of the Hurwitz zeta function

摘录 / EXCERPT: Let a be a real number with $0<a<1$, and let $\zeta(s,a)$ denote the Hurwitz zeta function. For $T>0$, consider the segment from $1/2$ to $1/2+iT$ in the complex plane, and count...

conjecture | open | MathDB 页面 | 原文线索

V2-003549 | MATHDB-344252

Existence conjecture for divisible design Cayley digraphs with prime-power parameters

摘录 / EXCERPT: Let n be an odd prime power. A divisible design Cayley digraph is a Cayley digraph that forms a divisible design with parameters $(v,k,\lambda_1,\lambda_2,m,n)$, where the parame...

conjecture | open | MathDB 页面 | 原文线索

V2-003550 | MATHDB-344266

Conjecture on integral Fourier coefficients of extremal quasi-modular forms

摘录 / EXCERPT: Let $f_{l,w}$ be the normalised extremal quasi-modular form of weight w and depth at most l, and let ... be the set of weights for which its Fourier coefficients are integral. In...

conjecture | open | MathDB 页面 | 原文线索

V2-003551 | MATHDB-344286

Serre's weak conjecture over totally real fields

摘录 / EXCERPT: Let L be a totally real number field, let ℓ be an odd prime, and let ... be an irreducible and totally odd Galois representation, where totally odd means that...

conjecture | open | MathDB 页面 | 原文线索

V2-003552 | MATHDB-344291

Conjecture on the correspondence between Tate-algebra zeta values and periodic multiple sums

摘录 / EXCERPT: The correspondence conjecture. The correspondence $\zeta_A(\mathcal{C})\leftrightarrow\varphi_A(\mathcal{C})$ induces an isomorphism of $\mathbb{F}_p$-algebras

conjecture | open | MathDB 页面 | 原文线索

V2-003553 | MATHDB-344292

Conjectural algebra and modular correspondence for multiple Eisenstein series

摘录 / EXCERPT: Multiple Eisenstein correspondence conjecture. The following properties hold: (1)...

conjecture | open | MathDB 页面 | 原文线索

V2-003554 | MATHDB-344296

Finite generation conjecture for elliptic curves over real abelian extensions

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ and let $F/\mathbb{Q}$ be a real abelian extension containing only finitely many subfields of degree 2, 3, or 5 over...

conjecture | open | MathDB 页面 | 原文线索

V2-003555 | MATHDB-344298

Non-concentration conjecture for normalized theta-coefficients

摘录 / EXCERPT: Let E be a semistable elliptic curve over $\mathbb{Q}$ with conductor N, and for every real cyclic extension $F/\mathbb{Q}$ of degree $d\geq3$ and conductor m define the norm...

conjecture | open | MathDB 页面 | 原文线索

V2-003556 | MATHDB-344305

Arratia's divisibility conjecture for prime factors of a uniform random integer

摘录 / EXCERPT: Let $N(n)$ be uniformly distributed on $\{1,\ldots,n\}$, with prime factorization ... For each prime $p\le n$, let Z_p be an independent geometric random variable with parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-003557 | MATHDB-344311

Cilleruelo's LCM growth conjecture for irreducible polynomial sequences

摘录 / EXCERPT: Cilleruelo's conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003558 | MATHDB-344315

Finiteness conjecture for unit equations in finite-dimensional division algebras

摘录 / EXCERPT: Let A be a finite-dimensional division algebra over $\mathbb{Q}$, and let Γ_1,Γ_2 be finitely generated semigroups of the multiplicative group $A^\times$. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-003559 | MATHDB-344318

Cramér's weak conjecture on maximal prime gaps

摘录 / EXCERPT: Cramér's weak conjecture. The largest such gap satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-003560 | MATHDB-344326

The prime-generator index-order conjecture for Collatz processes

摘录 / EXCERPT: Let f be the Collatz function and let $\{f^{s}(b)\}_{s=1}^{\infty}$ be a full Collatz process. The prime-generator index-order conjecture. If the process converges, then ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-003561 | MATHDB-344327

The Sophie Germain prime distribution conjecture for backward Collatz translates

摘录 / EXCERPT: Let f be a Collatz function and let $\{f^{s}(b)\}_{s=1}^{\infty}$ be the corresponding Collatz process. A process is full in the sense used by the source, and define ... Let ρ...

conjecture | open | MathDB 页面 | 原文线索

V2-003562 | MATHDB-344328

The Möbius-nonvanishing conjecture for Collatz trajectories of primes

摘录 / EXCERPT: Let f be the Collatz function, let $p>2$ be prime with $p\equiv 3\pmod 4$, and let μ denote the Möbius function. The Möbius-nonvanishing conjecture. There exists an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-003563 | MATHDB-344329

The odd-prime occurrence conjecture for full Collatz processes

摘录 / EXCERPT: Let f be the Collatz function and let $\{f^{s}(b)\}_{s=1}^{\infty}$ be a full Collatz process. The odd-prime occurrence conjecture. There exists an odd prime p such that ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-003564 | MATHDB-344330

The prime-factor bound for generators of full Collatz processes

摘录 / EXCERPT: Let f be the Collatz function and let $\{f^{s}(a)\}_{s=1}^{\infty}$ be the corresponding Collatz process. Let $\Omega(a)$ denote the total number of prime factors of a, counted...

conjecture | open | MathDB 页面 | 原文线索

V2-003565 | MATHDB-344334

Littlewood-admissibility conjecture for pairs with two interior zeros

摘录 / EXCERPT: Let $n\geq 1$, and let (k,n) be a valid pair for Littlewood polynomials. A pair is Littlewood-admissible if some Littlewood polynomial of degree n has exactly k zeros in the...

conjecture | open | MathDB 页面 | 原文线索

V2-003566 | MATHDB-344345

The semi-local framework conjecture for the Weil inequality

摘录 / EXCERPT: Semi-local framework conjecture. The semi-local operator theoretic framework with $S:=\{\infty\}\cup \{p\mid p<q\}$ suffices to prove the Weil inequality for all test functions wit...

conjecture | open | MathDB 页面 | 原文线索

V2-003567 | MATHDB-344352

Trivial zero conjecture for the cohomological and Fontaine-Mazur $mathcal{L}$-invariants

摘录 / EXCERPT: Let f be the Bianchi cuspidal eigenform and $V_p(f)$ its attached p-adic Galois representation. For each embedding $\sigma:F_{\mathfrak p}\hookrightarrow L$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003568 | MATHDB-344353

Global rationality and Shimura reciprocity conjecture for Stark-Heegner cycles

摘录 / EXCERPT: Let $V_p(f)$ be the two-dimensional representation attached to a Bianchi modular form, assume that $V_p(f)|_{G_{F_{\mathfrak p}}}$ is semistable, and let $\mathfrak p$, K,...

conjecture | open | MathDB 页面 | 原文线索

V2-003569 | MATHDB-344358

Chowla-Walum conjecture for restricted Bernoulli sums

摘录 / EXCERPT: For $j\in\mathbb{Z}_{\geqslant 1}$ and $\alpha\geqslant0$, define the Chowla-Walum sum ... where B_j is the j-th Bernoulli function. Chowla-Walum conjecture. For all sufficientl...

conjecture | open | MathDB 页面 | 原文线索

V2-003570 | MATHDB-344359

Extended Chowla-Walum conjecture for restricted divisor sums

摘录 / EXCERPT: Let $a>1$, $\alpha\geqslant0$, and $j\in\mathbb{Z}_{\geqslant1}$. Define ... where B_j is the j-th Bernoulli function. Extended Chowla-Walum conjecture. For all $\varepsilon>0$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-003571 | MATHDB-344367

Cilleruelo's least-common-multiple conjecture for polynomial sequences

摘录 / EXCERPT: Let $f\in\mathbb Z[x]$ be irreducible over $\mathbb Q$ of degree $d\ge 2$, and define ... ignoring zero values and taking the least common multiple of an empty set to be 1. Cilleru...

conjecture | open | MathDB 页面 | 原文线索

V2-003572 | MATHDB-344368

The radical least-common-multiple conjecture for polynomial sequences

摘录 / EXCERPT: Let $f\in\mathbb Z[x]$ be irreducible over $\mathbb Q$ of degree $d\ge 2$, and define ... where $\operatorname{rad}(n)$ is the product of the distinct primes dividing n. Radical LC...

conjecture | open | MathDB 页面 | 原文线索

V2-003573 | MATHDB-344380

Uniqueness conjecture for general Markov numbers

摘录 / EXCERPT: Let a and b be positive integers with $a<b$. The graph of general Markov numbers is the graph $\mathcal{T}\big((a,a),(b,b)\big)$, whose vertices are triples of integers constru...

conjecture | refuted | MathDB 页面 | 原文线索

V2-003574 | MATHDB-344385

DFI equidistribution conjecture for polynomial roots modulo primes

摘录 / EXCERPT: DFI conjecture. As $|S(x)| \to \infty$, the probability that $\alpha \leq \frac{\nu}{p} < \beta$ approaches $\beta-\alpha$.

conjecture | open | MathDB 页面 | 原文线索

V2-003575 | MATHDB-344386

Extended DFI equidistribution conjecture over number fields

摘录 / EXCERPT: Extended DFI conjecture. The probability measures $\mu[L,a;x]$ converge weakly to Haar measure on ${\mathsf{T}}$ as $x \to \infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-003576 | MATHDB-344423

Montgomery-Dyson conjecture on zeta zeros and random unitary matrices

摘录 / EXCERPT: Let T be a height on the critical line for the Riemann zeta function, and let N be the size of a random unitary matrix. The notation $N\approx \log(T/(2\pi))$ relates the heigh...

conjecture | open | MathDB 页面 | 原文线索

V2-003577 | MATHDB-344424

Fujii's conjecture on Gram interval statistics

摘录 / EXCERPT: Let $U(N)$ be the group of $N\times N$ unitary matrices, and let $J\subset[-\pi,\pi)$ be an interval of length $2\pi/N$. Define $E_{U(N)}(k,J)$ to be the probability that a random...

conjecture | open | MathDB 页面 | 原文线索

V2-003578 | MATHDB-344425

Alternative SU(N) conjecture for Gram interval statistics

摘录 / EXCERPT: Let $E_{SU(N)}(k,\mathcal J)$ denote the probability that a random special unitary matrix has exactly k eigenvalues in the corresponding $SU(N)$ Gram interval $\mathcal J$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-003579 | MATHDB-344464

Birkar's bounded smallest weight conjecture for terminal weighted blowups

摘录 / EXCERPT: Let $n\in\mathbb{N}_+$ and let $a_1,a_2,\ldots,a_n$ be coprime positive integers with $a_1\leq a_2\leq\cdots\leq a_n$. Let $X_{\mathbf a}$ be the weighted blowup of $\mathbb{A}^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-003580 | MATHDB-344465

Birkar's bounded smallest weight conjecture for epsilon-log-canonical weighted blowups

摘录 / EXCERPT: Fix a positive integer n and a positive real number $\epsilon\in(0,1]$. Let $a_1,a_2,\ldots,a_n$ be coprime positive integers with $a_1\leq a_2\leq\cdots\leq a_n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003581 | MATHDB-344469

Tenner's iterative Glaisher bijection conjecture for noncoprime regular and distinct partitions

摘录 / EXCERPT: Tenner's conjecture. Iteration of the map $\phi_s\phi_t$ suffices to produce a bijection: for every s-regular, t-distinct partition λ, there exists an integer ℓ de...

conjecture | open | MathDB 页面 | 原文线索

V2-003582 | MATHDB-344471

Conjecture on 2-divisibility of class numbers of pure quartic fields

摘录 / EXCERPT: Let p be a prime and let K denote the pure quartic field considered in the paper, with class number h_K. Let $h(-2p)$ denote the class number of the imaginary quadratic field...

conjecture | open | MathDB 页面 | 原文线索

V2-003583 | MATHDB-344474

The enumeration conjecture for medial racks and quandles of prime order

摘录 / EXCERPT: Enumeration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003584 | MATHDB-344508

Linear recurrence conjecture for Mahler measure iterates of algebraic units

摘录 / EXCERPT: Let M denote Mahler's measure, and write $M^{(n)}$ for its n-fold iterate. For an algebraic unit α, consider the sequence of real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-003585 | MATHDB-344511

Birkar's bounded-weight conjecture for weighted blowups

摘录 / EXCERPT: Let $\mathbb A^d_{\mathbf n}$ be the weighted blowup of $\mathbb A^d$ at $0\in\mathbb A^d$ with coprime, meaning primitive, weights $\mathbf n=(n_1,\ldots,n_d)\in\mathbb N^d$. A va...

conjecture | open | MathDB 页面 | 原文线索

V2-003586 | MATHDB-344517

Motivated Mumford--Tate conjecture

摘录 / EXCERPT: Let k be a finitely generated subfield of $\mathbf C$, let ℓ be a prime, and let $M\in\operatorname{AM}(k)$. Let $r_{\mathrm B}(M)$ and $r_\ell(M)$ be its Betti and ℓ-a...

conjecture | open | MathDB 页面 | 原文线索

V2-003587 | MATHDB-344538

Quadratic-residue distribution conjecture for breaking the square-root barrier

摘录 / EXCERPT: Let p be prime, let $\chi(x)=\left(\frac{x}{p}\right)$ denote the Legendre symbol, and let $\{r_1,\ldots,r_k\}\subseteq\mathbb{Z}_p$. For distinct indices i,j, set $a=r_j-r_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-003588 | MATHDB-344542

Mercat's conjecture on periodic continued fractions with partial quotients 1 or 2

摘录 / EXCERPT: Mercat's conjecture. Every real quadratic field contains a periodic continued fraction with partial quotients equal to 1 or 2.

conjecture | open | MathDB 页面 | 原文线索

V2-003589 | MATHDB-344549

Primary decomposition conjecture for knot concordance

摘录 / EXCERPT: Primary decomposition conjecture. Let K and J be nonslice knots. If K and J have coprime Alexander polynomials, then K and J are not concordant.

conjecture | open | MathDB 页面 | 原文线索

V2-003590 | MATHDB-344583

Oddness conjecture for the rational Korselt witness set of pq

摘录 / EXCERPT: Let $N=pq$, where p and q are the prime factors under consideration, and let $\mathbb{Q}$-$\mathcal{KW}(N)$ denote the rational Korselt witness set associated with N. Oddness...

conjecture | open | MathDB 页面 | 原文线索

V2-003591 | MATHDB-344606

Referee's supercongruence for a truncated double sum with alternating powers of 8

摘录 / EXCERPT: The first supercongruence conjecture. For every odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-003592 | MATHDB-344609

Conjecture on intermediate values of Mahler measure

摘录 / EXCERPT: Let $M_{\inf}$ be the limit infimum of the Mahler measures ${\rm M}(\alpha)$ as $\lvert\alpha\rvert\to1^+$, and let Λ be the limiting constant introduced for the relevant f...

conjecture | open | MathDB 页面 | 原文线索

V2-003593 | MATHDB-344611

The majority nonnegativity conjecture for Legendre-symbol sums

摘录 / EXCERPT: Let p be an odd prime, let $0\leq\alpha\leq\frac{1}{2}$, and define ... where $\left(\frac{n}{p}\right)$ is the Legendre symbol. Majority nonnegativity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-003594 | MATHDB-344613

Cotorsion conjecture for signed Selmer groups over cyclotomic extensions

摘录 / EXCERPT: Let E be the elliptic curve and let F_∞ be the cyclotomic extension considered in the paper. For each supersingular prime w, choose $s_w\in\{\flat,\sharp\}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-003595 | MATHDB-344644

Baer-Suzuki conjecture for the π-radical

摘录 / EXCERPT: Let π be a proper subset of the set of all primes containing at least two elements, and let r be the minimal prime not in π. The invariant $\operatorname{BS}(\pi)$ is def...

conjecture | open | MathDB 页面 | 原文线索

V2-003596 | MATHDB-344645

Bound conjecture for β-subgroups in finite simple groups

摘录 / EXCERPT: Let r be a prime, let L be a nonabelian simple group whose order is divisible by r, and let $x\in\operatorname{Aut}(L)$ be an automorphism of prime order. Bound conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003597 | MATHDB-344648

Gouvêa's limiting-measure conjecture for supersingularities

摘录 / EXCERPT: Let p be a prime number, and let $k\geqslant 2$ and N be positive integers with N coprime to p. For weight k and level $\Gamma_0(Np)$ eigenforms, define the supersingula...

conjecture | open | MathDB 页面 | 原文线索

V2-003598 | MATHDB-344651

The uniqueness and ATLAS-ordering conjecture for supercharacter theories of cyclic groups of prime order

摘录 / EXCERPT: Uniqueness and ATLAS-ordering conjecture. For each divisor r of $p-1$, there exists exactly one supercharacter theory $(\mathcal{X},\mathcal{K})$ of Z_p such that every non-tri...

conjecture | open | MathDB 页面 | 原文线索

V2-003599 | MATHDB-344691

Quadratic successive-maxima conjecture for non-genus class numbers

摘录 / EXCERPT: Quadratic successive-maxima conjecture. As D increases from its minimum, the successive maxima of $C_{\varepsilon}(D)$ occur only for prime conductors $f_K=\ell$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-003600 | MATHDB-344707

Guo-Schlosser's supercongruence for primes congruent to 2 modulo 3

摘录 / EXCERPT: Let $(x)_0=1$ and $(x)_k=x(x+1)\cdots(x+k-1)$ denote the Pochhammer symbol. Guo-Schlosser's supercongruence. For every prime $p\equiv 2\pmod{3}$, one has ... This is a supercongruenc...

conjecture | open | MathDB 页面 | 原文线索

V2-003601 | MATHDB-344713

David's height-degree conjecture for elliptic curves

摘录 / EXCERPT: Soit E une courbe elliptique définie sur un corps de nombres K. Notons E_{tors} le groupe des points de torsion de E, et $K(E_{\mathrm{tors}},P)$ the field generat...

conjecture | open | MathDB 页面 | 原文线索

V2-003602 | MATHDB-344715

Equality conjecture for two truncated hypergeometric sums

摘录 / EXCERPT: Let p be an odd prime and let r be a positive integer. Equality conjecture. One should have ... The conjecture is proposed because both sides are predicted to have the same Berno...

conjecture | open | MathDB 页面 | 原文线索

V2-003603 | MATHDB-344726

El Bachraoui's double binomial sum supercongruences

摘录 / EXCERPT: El Bachraoui's conjecture. For every odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-003604 | MATHDB-344730

Splitting-field growth conjecture for zeta functions of \(\mathbb{Z}_p\)-towers

摘录 / EXCERPT: Let $P(C_n,s)$ be the numerator of the zeta function of the curve C_n. Let $\mathbb{Q}_n$ and $\mathbb{Q}_{p,n}$ be its splitting fields over $\mathbb{Q}$ and $\mathbb{Q}_p$, res...

conjecture | open | MathDB 页面 | 原文线索

V2-003605 | MATHDB-344757

The structured ring spectrum conjecture for $\Mell^\dR$ and $\TMF/\!\/\nu$

摘录 / EXCERPT: Structured ring spectrum conjecture. The global sections satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-003606 | MATHDB-344767

Buzzard-Calegari conjecture on the Newton polygon of T_2

摘录 / EXCERPT: Buzzard-Calegari conjecture. The polynomial $P_{BC}(X)$ has the same 2-adic Newton polygon as $P_{T_2}(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-003607 | MATHDB-344784

Symmetry conjecture for Littlewood polynomials and equipowerful bipartitions

摘录 / EXCERPT: Symmetry conjecture. If there is an LP of length n and exact order m, then there is one that is $(-1)^m$-symmetric. Alternatively, if there is an equipowerful bipartion of leng...

conjecture | open | MathDB 页面 | 原文线索

V2-003608 | MATHDB-344806

Binary logarithmic equation in prime numbers

摘录 / EXCERPT: Let N be a sufficiently large positive integer. Consider the logarithmic equation ... Here p_1 and p_2 are required to be prime numbers. Binary logarithmic equation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003609 | MATHDB-344823

The main inequalities conjecture for points of bounded degree

摘录 / EXCERPT: The main inequalities for points of bounded degree. There exists a constant $\mathrm{c}(r,d)>0$, depending only on r and d, such that, for every fixed $\delta>0$, if...

conjecture | open | MathDB 页面 | 原文线索

V2-003610 | MATHDB-344829

Wan's asymptotic p-adic Riemann Hypothesis

摘录 / EXCERPT: Let n and d be the parameters defining the generic Newton polygon ${\rm GNP}(n,d;p)$, the Hodge polygon ${\rm HP}(n,d)$, and a prime p. Wan's asymptotic p-adic Riemann Hypo...

conjecture | open | MathDB 页面 | 原文线索

V2-003611 | MATHDB-344830

The p-adic Riemann Hypothesis for exponential sums

摘录 / EXCERPT: Let n and d be the parameters defining the generic Newton polygon ${\rm GNP}(n,d;p)$ and the Frobenius polygon ${\rm FP}(n,d;p)$, and let p be a prime. The p-adic Riemann H...

conjecture | open | MathDB 页面 | 原文线索

V2-003612 | MATHDB-344833

Multiplicativity conjecture for commutativity degrees of coprime direct products

摘录 / EXCERPT: Multiplicativity conjecture. For any two such groups,

conjecture | open | MathDB 页面 | 原文线索

V2-003613 | MATHDB-344838

The cubic analogue of the Ankeny-Artin-Chowla-Mordell conjecture

摘录 / EXCERPT: Let $p\neq 3$ be a prime number, let $P=p$ or $2p$, and let ... be the fundamental unit of $\mathbb{Q}(P^{1/3})$, where $\alpha,\beta,\gamma\in\mathbb{Z}$. The cubic analogue of the...

conjecture | open | MathDB 页面 | 原文线索

V2-003614 | MATHDB-344859

Base-change compatibility conjecture for automorphic \a3-invariants

摘录 / EXCERPT: Let E/F be a solvable extension, let $\mathfrak{q}$ be a prime of E above $\mathfrak{p}$, and let π_E be the base-change lift of π to $PGL_{n,E}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003615 | MATHDB-344864

McKay-Navarro conjecture for principal blocks

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. Let B_0 be the principal p-block of G and b_0 the principal p-block of $N_G(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003616 | MATHDB-344871

Sutherland-Zywina conjecture on mod-p images of elliptic curves

摘录 / EXCERPT: Let $E/\mathbb Q$ be an elliptic curve, let p be a prime, and let $G\subseteq \operatorname{GL}(2,\mathbb Z/p\mathbb Z)$ be the image of the mod-p Galois representation...

conjecture | open | MathDB 页面 | 原文线索

V2-003617 | MATHDB-344872

Mizuno's modularity conjecture for partition q-series

摘录 / EXCERPT: Let $\alpha=(A_+,A_-,D)$ be a Cartan-like T-datum of finite type. Let S_α be the finite abelian group associated with α, let $\mathcal{Z}_{\alpha,\sigma}(q)$ be its p...

conjecture | open | MathDB 页面 | 原文线索

V2-003618 | MATHDB-344873

Mizuno's modularity conjecture for normalized partition q-series

摘录 / EXCERPT: Let $\alpha=(A_+,A_-,D)$ be a Cartan-like T-datum of finite type, let S_α be its associated finite abelian group, and let $\mathcal{Z}_{\alpha,\sigma}(q)$ be the partition...

conjecture | open | MathDB 页面 | 原文线索

V2-003619 | MATHDB-344879

The weak Loneliness Spectrum Conjecture for one very fast speed

摘录 / EXCERPT: Weak Loneliness Spectrum Conjecture. For every integer $n\geq4$, there exists a function $f_n:\mathbb{N}\to\mathbb{N}$ such that, whenever $v_n>f_n(v_{n-1})$, either

conjecture | open | MathDB 页面 | 原文线索

V2-003620 | MATHDB-344880

The short-progression exclusion for the Loneliness Spectrum

摘录 / EXCERPT: Short-progression exclusion conjecture. One has

open_problem | open | MathDB 页面 | 原文线索

V2-003621 | MATHDB-344889

Foil diagrams maximize trefoil descendants among minimal prime diagrams

摘录 / EXCERPT: Let n be odd with $n\geq 3$. A minimal prime free knot diagram is a minimal free knot diagram representing a prime knot, and a trefoil descendant is a resultant trefoil obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-003622 | MATHDB-344893

The 2n diagrams attain the lower bound on trefoils

摘录 / EXCERPT: A minimal prime algebraic free knot diagram is a minimal free knot diagram representing a prime knot and arising from an algebraic tangle. A free $2n$ knot is the family considered...

conjecture | open | MathDB 页面 | 原文线索

V2-003623 | MATHDB-344899

Thangadurai's conjectural refinement of Adams's theorem for Bernoulli numerators

摘录 / EXCERPT: Let $p>3$ be a prime, let n be a positive even integer, and write $p^l\mathrel{\|}n$ to mean that $p^l\mid n$ but $p^{l+1}\nmid n$. Assume that $p-1\nmid n$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-003624 | MATHDB-344903

On-average Maeda-type conjecture in the level aspect

摘录 / EXCERPT: Let $k\ge 2$ be even, let Sq_r be the set of squarefree positive integers with exactly r prime factors, and define ... For a newspace $S_k^{\mathrm{new}}(N)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003625 | MATHDB-344904

Roberts's finiteness conjecture for rational newforms

摘录 / EXCERPT: Fix an even weight $k\ge 6$. Let $\mathcal F_k$ be the set of newforms lying in some $S_k(N)$ with N squarefree, and call a newform rational when its Fourier coefficients are rat...

conjecture | open | MathDB 页面 | 原文线索

V2-003626 | MATHDB-344918

The horror conjecture on recursive axiomatization of curve-counting conditions

摘录 / EXCERPT: The horror conjecture. In the language of T^+, there are sentences $\tau_{g,p^k,n}$ depending recursively on the parameters such that, for every $g\geq 1$, every prime power...

conjecture | open | MathDB 页面 | 原文线索

V2-003627 | MATHDB-344933

The mock modularity conjecture for quantum invariants of orientation-reversed three-manifolds

摘录 / EXCERPT: Let M_3 be a three-manifold whose quantum invariants have the form ... where $c\in\mathbb{Q}$, $\widetilde g(\tau)$ is the Eichler integral of a weight-$\frac{3}{2}$ theta function...

conjecture | open | MathDB 页面 | 原文线索

V2-003628 | MATHDB-344935

Double Goldbach plus epsilon conjecture

摘录 / EXCERPT: Let n be a positive even integer. A pair of Goldbach representations consists of primes q_1,q_2,q_4,q_5 satisfying ... and ... There is also a prime q_3 with $q_5<q_3<n$. Double...

conjecture | open | MathDB 页面 | 原文线索

V2-003629 | MATHDB-344947

Conjecture on the maximum of Q for signature parameter r_2=1 in degree five

摘录 / EXCERPT: Let $Q(5,1,\{4\},(x,y,1,g))$ be the function obtained after reducing to $z=1$ and $g\neq\pm1$, and let (x,y,z,g) denote its parameters. Maximum-value conjecture. The maximum of...

conjecture | open | MathDB 页面 | 原文线索

V2-003630 | MATHDB-344948

Conjecture on the tabulated maximum values M(n,r_2)

摘录 / EXCERPT: For each degree n and signature parameter r_2, let $M(n,r_2)$ denote the maximum value of the corresponding function $Q(n,r_2,\{j_1,\ldots,j_{r_2}\},\cdot)$, with the conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-003631 | MATHDB-344949

Conjectured maximizing points for M(n,1)

摘录 / EXCERPT: Let $M(n,1)$ denote the maximum value of the regulator-related function for $r_2=1$, and let $C(1)$ be the corresponding threshold. Maximizer conjecture for $r_2=1$. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-003632 | MATHDB-344983

The odd-characteristic spanning conjecture for v-adic multiple zeta relations

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$ be a rational function field and let v be a finite monic prime. Write $W_v(K)$ for the vector space of v-adic multiple zeta values at integer tuples. Od...

conjecture | open | MathDB 页面 | 原文线索

V2-003633 | MATHDB-345025

The natural-hyperbolic-structure conjecture for prime links

摘录 / EXCERPT: Let L be an n-component prime link whose complement has only hyperbolic pieces. Let $\vec M=(M_1,\ldots,M_n)$ be a coloring, with limiting ratios...

conjecture | open | MathDB 页面 | 原文线索

V2-003634 | MATHDB-345048

The ramification-jump lifting conjecture for non-abelian cyclic p-Sylow groups

摘录 / EXCERPT: Let p be a prime, let $G e \mathbb{Z}/p^n\rtimes\mathbb{Z}/m$ be non-abelian with $p\nmid m$, and let $k[[z]]/k[[t]]$ be a local G-extension whose $\mathbb{Z}/p^n$-subextension...

conjecture | open | MathDB 页面 | 原文线索

V2-003635 | MATHDB-345077

Three odd Gaussian primes conjecture for the first quadrant

摘录 / EXCERPT: Let $A=\{z\in\mathbb{Z}[i]\mid\operatorname{Re}(z)>0\text{ and }\operatorname{Im}(z)>0\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003636 | MATHDB-345078

Bounded odd Gaussian-prime representation conjecture

摘录 / EXCERPT: Bounded odd Gaussian-prime representation conjecture. There are constants c_0 and k_0 such that, for every $z\in\Gamma_G$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-003637 | MATHDB-345080

Equidistribution of prime pairs among coset lattice congruence classes

摘录 / EXCERPT: Let z and q be fixed, let $\mathcal{N}_q(a,b)$ denote a coset lattice congruence class for prime pairs of gap z, and let $\pi_z(x;\mathcal{N}_q(a,b),q)$ count the pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-003638 | MATHDB-345081

Von Mangoldt reformulation of prime-pair equidistribution

摘录 / EXCERPT: Let z and q be fixed, let $\mathcal{N}_q(a,b)$ denote a coset lattice congruence class for prime pairs (p_i,p_{i+z}), and let $\Psi_z(x;\mathcal{N}_q(a,b),q)$ be the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-003639 | MATHDB-345126

The topological monodromy conjecture for complex plane curve singularities

摘录 / EXCERPT: Topological monodromy conjecture. If θ is a pole of $Z_f^{\mathrm{top}}(s)$, then $\exp(2\pi i\theta)$ is an eigenvalue of the monodromy $M_x^{(q)}$ for some $q\in\mathbb N$...

conjecture | open | MathDB 页面 | 原文线索

V2-003640 | MATHDB-345133

Quaternionic modular forms have asymptotically only trivial zeroes at fixed factor count

摘录 / EXCERPT: Let r be odd. For each $N\in\mathrm{Sq}_r$, choose a maximal order $\mathcal O_N$ in the definite quaternion algebra B_N of discriminant N. An eigenform in $S(\mathcal O_N)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003641 | MATHDB-345134

Quaternionic eigenform degree conjecture for nontrivial zeroes

摘录 / EXCERPT: Let $\varphi\in S(\mathcal O_N)$ be an eigenform, and let K_φ be the number field generated by its Hecke eigenvalues. Define the degree of φ to be...

conjecture | open | MathDB 页面 | 原文线索

V2-003642 | MATHDB-345135

Maeda-type Galois orbit conjecture for squarefree levels

摘录 / EXCERPT: Fix $k\in 2\mathbb Z_{>0}$ and $r\in\mathbb Z_{>0}$. Let Sq_r be the positive squarefree integers with exactly r prime factors, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003643 | MATHDB-345136

Degree-one dominance conjecture for analytic-rank-two newforms

摘录 / EXCERPT: Consider weight 2 newforms, ordered by level, and call a newform’s degree the degree of its rationality field over $\mathbb Q$. Degree-one analytic-rank conjecture. Among all wei...

conjecture | open | MathDB 页面 | 原文线索

V2-003644 | MATHDB-345137

Growth conjecture for nontrivial zeroes of quaternionic modular forms

摘录 / EXCERPT: For $N\in\mathrm{Sq}_{\mathrm{odd}}$, choose a maximal quaternionic order $\mathcal O_N$ of level N in B_N, and let a nontrivial zero be a zero of an eigenform in...

conjecture | open | MathDB 页面 | 原文线索

V2-003645 | MATHDB-345138

Degree-one dominance conjecture for nontrivial quaternionic zeroes

摘录 / EXCERPT: For $N\in\mathrm{Sq}_{\mathrm{odd}}$, choose a maximal order $\mathcal O_N$ in the definite quaternion algebra B_N. For an eigenform $\varphi\in S(\mathcal O_N)$, its degree is t...

conjecture | open | MathDB 页面 | 原文线索

V2-003646 | MATHDB-345162

The classification conjecture for extensions of the pair {k,k+1}

摘录 / EXCERPT: Classification conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-003647 | MATHDB-345174

The purely-prime ideal conjecture

摘录 / EXCERPT: Let A be a commutative ring. An ideal of A is purely-prime if it is prime and pure, and purely-maximal if it is maximal among the proper pure ideals. Purely-prime ideal conject...

conjecture | open | MathDB 页面 | 原文线索

V2-003648 | MATHDB-345175

Strong Bertrand conjecture for Gaussian lines

摘录 / EXCERPT: Let L be a primitive Gaussian line, and let α_n denote the indexed Gaussian integers on L. For $\beta=x+iy\in\mathbb{Z}[i]$, define ... Here N is the Gaussian norm. Stro...

conjecture | open | MathDB 页面 | 原文线索

V2-003649 | MATHDB-345176

Weak Bertrand conjecture for Gaussian lines

摘录 / EXCERPT: Let L be a primitive Gaussian line, and let α_n denote the indexed Gaussian integers on L. For $\beta=x+iy\in\mathbb{Z}[i]$, write $N(\beta)$ for its Gaussian norm. Weak...

conjecture | open | MathDB 页面 | 原文线索

V2-003650 | MATHDB-345198

Serre's conjecture on characteristic polynomials at bad primes

摘录 / EXCERPT: Serre's conjecture. For every non-archimedean prime v, the polynomial $P_v(\mathbf M_\ell,T)$ lies in $\mathbb Z[T]$ and is independent of ℓ, and it factors as

conjecture | open | MathDB 页面 | 原文线索

V2-003651 | MATHDB-345206

Craven's perversity-unitriangularity conjecture for unipotent blocks

摘录 / EXCERPT: Let $G={\mathbf{G}}^F$ for ${\mathbf{G}}$ connected reductive defined over ${\mathbb{F}}_q$. Let ℓ be a prime such that Sylow ℓ-subgroups of G are abelian, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003652 | MATHDB-345217

LI- conjecture on linear independence of Artin L-function zeros

摘录 / EXCERPT: Let $L/\mathbb Q$ be a number-field extension and let $L_0/\mathbb Q$ be its Galois closure. Write ... where the union is a multiset. LI- conjecture. The multiset...

conjecture | open | MathDB 页面 | 原文线索

V2-003653 | MATHDB-345222

Sørensen's conjecture on Hermitian surface intersections

摘录 / EXCERPT: Let q be a prime power, let $F\in\mathbb{F}_{q^2}[x_0,x_1,x_2,x_3]$ be homogeneous of degree d, and let $V_2\subset\mathbb{P}^3$ be a non-degenerate Hermitian surface defined o...

conjecture | open | MathDB 页面 | 原文线索

V2-003654 | MATHDB-345247

Guo and Zudilin's product congruence for prime powers

摘录 / EXCERPT: Let p be a prime with $p\equiv 3\;\allowbreak(\operatorname{mod}4)$, and let r be a positive integer. Guo and Zudilin's product conjecture. ... The conjecture is proposed by comp...

conjecture | open | MathDB 页面 | 原文线索

V2-003655 | MATHDB-345258

Guo's product congruence conjecture

摘录 / EXCERPT: Let p be a prime with $p e 2$ and $p eq 1\begin{pmatrix} \end{pmatrix}$ - more precisely, $p\equiv3\pmod4$ - and let r be a positive integer. Guo's conjecture. ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-003656 | MATHDB-345272

Power-saving sieve bound for quadratic Hecke sums

摘录 / EXCERPT: Let $(\varphi_k)_k$, $(Q_k)_k$, and $(f_k)_k$ satisfy the hypotheses of the quadratic Hecke-sum conjecture: the forms φ_k are non-dihedral, the Q_k are irreducible intege...

conjecture | open | MathDB 页面 | 原文线索

V2-003657 | MATHDB-345294

Goldston-Montgomery conjecture on the variance of primes in short intervals

摘录 / EXCERPT: Let $\Lambda(n)$ denote the von Mangoldt function. For $\delta\in(0,1)$, set $H=X^\delta$. Goldston-Montgomery conjecture. As $X\to\infty$, ... This predicts the variance of sums of...

conjecture | open | MathDB 页面 | 原文线索

V2-003658 | MATHDB-345295

Universal variance conjecture for primes in short intervals over number fields

摘录 / EXCERPT: Universal variance conjecture. For $\delta\in(0,1)$ and $H=X^\delta$,

conjecture | open | MathDB 页面 | 原文线索

V2-003659 | MATHDB-345298

Le's conjecture for $x^2-D=4p^n$

摘录 / EXCERPT: Let D be an odd positive integer, let p be an odd prime, and let $N(-D,4,p)$ denote the number of solutions $(x,n)\in\mathbb{N}^2$ of $x^2-D=4p^n$. Let cases (i) and (ii) b...

conjecture | open | MathDB 页面 | 原文线索

V2-003660 | MATHDB-345299

Terai's conjecture for primitive Pythagorean triples

摘录 / EXCERPT: Let (a,b,c) be a primitive Pythagorean triple with ... where $a,b,c\in\mathbb{N}$, $\gcd(a,b)=1$, and $a\equiv0\pmod{2}$. Terai's conjecture. The equation ... has only the solution...

conjecture | open | MathDB 页面 | 原文线索

V2-003661 | MATHDB-345300

The corrected Cao-Dong conjecture

摘录 / EXCERPT: Corrected Cao-Dong conjecture. The equation

conjecture | open | MathDB 页面 | 原文线索

V2-003662 | MATHDB-345301

Terai's conjecture with prescribed exponents

摘录 / EXCERPT: Let b,c,r,t be fixed positive integers satisfying ... Let $x,y,z\in\mathbb{N}$ satisfy $x^2+b^y=c^z$. Terai's prescribed-exponent conjecture. The only solution is ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-003663 | MATHDB-345302

Ma's second conjecture on a quadratic exponential equation

摘录 / EXCERPT: Let p be an odd prime, $a\in\mathbb{Z}$ with $a\geq0$, and $x,m,n\in\mathbb{N}$. Ma's second conjecture. The equation ... has only the solution $(p,x,a,m,n)=(5,49,3,2,1)$. The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-003664 | MATHDB-345303

The conjecture for $x^2-2=y^n$

摘录 / EXCERPT: Let $x,y,n\in\mathbb{N}$ satisfy $\gcd(x,y)=1$ and $n>2$. The conjecture for $x^2-2=y^n$. The equation ... has no solutions (x,y,n). The source presents this as an unresolved conje...

conjecture | open | MathDB 页面 | 原文线索

V2-003665 | MATHDB-345338

Fermat's conjecture over arithmetic function fields

摘录 / EXCERPT: Fermat's conjecture over an arithmetic function field. For every arithmetic function field K, there exists a positive integer N_0 depending on K such that F_N has Fermat's...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-003666 | MATHDB-345348

The characterization of vanishing Hecke-obstruction quantity for mod- \ell eigenforms

摘录 / EXCERPT: Vanishing characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003667 | MATHDB-345359

The Wall-Sun-Sun prime conjecture for Fibonacci numbers

摘录 / EXCERPT: Let F_m denote the mth Fibonacci number, and let $\left(\frac{p}{5}\right)$ be the Legendre symbol for a prime number p. Wall-Sun-Sun prime conjecture. There are no prime num...

conjecture | open | MathDB 页面 | 原文线索

V2-003668 | MATHDB-345373

The first negative eigenvalue conjecture for Siegel Hecke eigenforms

摘录 / EXCERPT: First negative eigenvalue conjecture. For every $\epsilon>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-003669 | MATHDB-345374

Heuristic for counting principally polarized abelian surfaces over the rationals

摘录 / EXCERPT: Let $\mathcal{N}_{2,\mathbb{Q}}(\mathcal{B})$ count principally polarized abelian surfaces $A=\operatorname{Jac}(X)$, where X is a stable genus 2 curve over $\mathbb{Z}$ satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-003670 | MATHDB-345375

Heuristic for counting hyperelliptic curves over the rationals

摘录 / EXCERPT: For $g\geq3$, let $\mathcal{Z}_{g,\mathbb{Q}}(\mathcal{B})$ count stable hyperelliptic genus g curves with a marked rational Weierstrass point over $\mathbb{Z}$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-003671 | MATHDB-345417

Lower-tail conjecture for first-occurrence prime gaps

摘录 / EXCERPT: Let $q>r\geq1$ be fixed coprime integers, let (P) be the progression $r,r+q,r+2q,\ldots$, and let $d_{q,r}(x)$ be the size of the first occurrence, closest to x, of a gap betwe...

conjecture | open | MathDB 页面 | 原文线索

V2-003672 | MATHDB-345418

Generalized Polignac conjecture for prime gaps in arithmetic progressions

摘录 / EXCERPT: Let $q>r\geq1$ be fixed coprime integers and consider the progression $(P):r,r+q,r+2q,\ldots$. For a gap size d, let $P_f(d;q,r)$ be the prime starting the first occurrence of th...

conjecture | open | MathDB 页面 | 原文线索

V2-003673 | MATHDB-345419

First-occurrence location conjecture for prime gaps in arithmetic progressions

摘录 / EXCERPT: Let $(P):r,r+q,r+2q,\ldots$ be a fixed arithmetic progression with q and r coprime, and let $P_f(d;q,r)$ denote the starting prime of the first occurrence of a gap d in (P)...

conjecture | open | MathDB 页面 | 原文线索

V2-003674 | MATHDB-345420

Sequential growth conjecture for maximal gaps

摘录 / EXCERPT: Let $(P):r,r+q,r+2q,\ldots$ be a fixed arithmetic progression with q and r coprime, and let $R(n;q,r)$ be the size of the n-th record (maximal) gap between primes in (P). S...

conjecture | open | MathDB 页面 | 原文线索

V2-003675 | MATHDB-345424

The unimodality conjecture for the function beta

摘录 / EXCERPT: Let $\beta(t)$ be the function defined in the paper, and let μ and α be the constants used there. Unimodality conjecture for β. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-003676 | MATHDB-345432

Stolarsky's conjecture on flimsy numbers

摘录 / EXCERPT: Let j be the number of 0's in the binary representation of a number, and let s be a binary string with length $|s|=j$ that begins with 1 and ends with 0. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-003677 | MATHDB-345456

Fried's conjecture on Ruelle zeta functions and analytic torsion

摘录 / EXCERPT: Let (M,E,X) be the geometric data associated with the unit-cotangent bundle $M=S^*_g\Sigma$ of a closed manifold Σ, where E is the flat vector bundle determined by a rep...

conjecture | open | MathDB 页面 | 原文线索

V2-003678 | MATHDB-345461

Surgery-limit invariants for \(\mathit{HP}\) and \(\mathit{HP}_{\#}\)

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let p be fixed, and let $S^3_{p/q}(K)$ denote p/q surgery, with q tending to infinity through integers relatively prime to p. For each degree...

conjecture | open | MathDB 页面 | 原文线索

V2-003679 | MATHDB-345474

The noncommutative Tate conjecture for saturated dg-categories

摘录 / EXCERPT: Let k be a perfect field, let l be a prime invertible in k, and let $\operatorname{\mathcal{A}}$ be a saturated dg-category over k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003680 | MATHDB-345487

The Atkin-Lehner equidistribution conjecture for level-2 modular forms

摘录 / EXCERPT: Atkin-Lehner equidistribution conjecture. For all $k\geq 2$, the space $S_k(\Gamma_0(2))$ decomposes under W_2 into Hecke-irreducible subspaces of dimensions $\lfloor d/2\rfloor$...

conjecture | open | MathDB 页面 | 原文线索

V2-003681 | MATHDB-345517

Bringmann--Mahlburg's conjecture on partitions into squares modulo three

摘录 / EXCERPT: Bringmann--Mahlburg's conjecture. The ordering is

conjecture | open | MathDB 页面 | 原文线索

V2-003682 | MATHDB-345554

Robin's inequality for products of the first distinct primes

摘录 / EXCERPT: Let $p_1,p_2,\ldots,p_m$ be the first m distinct primes, and let $n=\prod_{j=1}^{m}p_j>5040$. Robin's inequality is the assertion that $\sigma(n)/n<e^\gamma\log\log n$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-003683 | MATHDB-345555

Monotonicity of Robin's inequality under increasing prime exponents

摘录 / EXCERPT: Let $n= p_1^{k_1}p_2^{k_2}\dots p_m^{k_m}>5040$, where the p_i are distinct primes and the k_i are positive integers. For an index j, form the integer...

conjecture | open | MathDB 页面 | 原文线索

V2-003684 | MATHDB-345557

Missing-digit Cantor sets and Liouville numbers product-dimension conjecture

摘录 / EXCERPT: Let $b\geq 3$ and let $W\subseteq\{0,1,\ldots,b-1\}$. Define the missing-digit Cantor set ... and let $\mathscr{L}$ denote the set of Liouville numbers. For $n\geq 3$, the missing-di...

conjecture | open | MathDB 页面 | 原文线索

V2-003685 | MATHDB-345558

Mahler U-number product-dimension conjecture

摘录 / EXCERPT: For $m\geq 1$, let U_m be the set of real U_m-numbers in Mahler's classification, and for $n\geq 1$ let U_m^n denote their n-fold Cartesian product. The U_m product-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-003686 | MATHDB-345559

Irrationality-exponent sets product-dimension bounds conjecture

摘录 / EXCERPT: For $\lambda\in[2,\infty]$, let $\mathscr{W}_\lambda$ denote the set of real numbers with the corresponding irrationality exponent, and let $n\geq 1$. The irrationality-exponent pr...

conjecture | open | MathDB 页面 | 原文线索

V2-003687 | MATHDB-345560

Irrationality-exponent sumset complement conjecture

摘录 / EXCERPT: For $\lambda_0,\lambda_1\in[2,\infty]$, let $\mathscr{W}_{\lambda_j}$ be the corresponding sets of real numbers with prescribed irrationality exponent, and write A^c for the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-003688 | MATHDB-345590

The shifted moments conjecture for the Riemann zeta-function

摘录 / EXCERPT: Let $k\in\mathbb N$, let $Y=Y(T)$, and let R be a piecewise smooth positively oriented path encircling 0 and $-s$. Shifted moments conjecture. For $T\geq1$, ... This holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-003689 | MATHDB-345591

The divisor variance conjecture for short intervals

摘录 / EXCERPT: For $\alpha>0$ and $k\in\mathbb N$, define $\mathcal C_k(\alpha)$ as the normalized limiting variance of the k-fold divisor function in short intervals, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003690 | MATHDB-345592

The divisor variance conjecture in arithmetic progressions

摘录 / EXCERPT: Let q tend to infinity through primes, let $\mathcal D_k(\alpha)$ be the normalized variance of the k-fold divisor function in reduced residue classes, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003691 | MATHDB-345642

Conjecture on coprime splitter sets for purely singular perfect splittings

摘录 / EXCERPT: Coprime splitter conjecture. Let k_1, k_2 be integers with $0 \le k_1 \leq k_2$ and $k_1 + k_2\ge 4$. If there exists a purely singular perfect $B-k_1,k_2$ set with splitter set...

conjecture | open | MathDB 页面 | 原文线索

V2-003692 | MATHDB-345644

The ABC conjecture over number fields

摘录 / EXCERPT: Let K be a number field and let $a,b,c\in K$ satisfy $a+b+c=0$. For $\alpha\in\mathcal{O}_K$, define ... and let $H(a,b,c)$ be the multiplicative Weil height of the triple. ABC con...

conjecture | open | MathDB 页面 | 原文线索

V2-003693 | MATHDB-345716

Conjecture on the irrationality exponent of the first odd zeta constant

摘录 / EXCERPT: Irrationality-exponent conjecture. The irrationality exponent of the first odd zeta constant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-003694 | MATHDB-345718

Conjecture on fusion systems of Sylow subgroups of Lie type groups

摘录 / EXCERPT: Let p be a prime, and let $\mathcal{S}$ be the collection of pairs (G,S) such that G is a simple Lie type group defined in characteristic p, G is not isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-003695 | MATHDB-345751

Bosek's arithmetic graph chromatic conjecture

摘录 / EXCERPT: For a positive integer k, define the arithmetic proximity of positive integers a,b by $\max\{a/d,b/d\}$, where $d=\gcd(a,b)$. The arithmetic graph B_k has vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-003696 | MATHDB-345752

The conjecture that Rejmer's algorithm never stops

摘录 / EXCERPT: A golden seed is a finite sequence of signs satisfying the balance and multiplicativity conditions maintained by Rejmer's greedy construction. Rejmer's algorithm conjecture. Rejmer...

conjecture | open | MathDB 页面 | 原文线索

V2-003697 | MATHDB-345753

The positive-density balanceable-set conjecture

摘录 / EXCERPT: A set $B\subseteq\mathbb N$ is balanceable if there are a constant C and a 2-coloring of $\mathbb N$ such that every cascade whose length belongs to B is C-balanced. The po...

conjecture | open | MathDB 页面 | 原文线索

V2-003698 | MATHDB-345754

The divine coloring conjecture

摘录 / EXCERPT: Let f be a coloring of $\mathbb N$ by the two colors $\{-1,+1\}$. A finite set $A=\{a_1<\cdots<a_n\}$ is divinely colored if at least half of $a_2,\dots,a_n$ have a color differe...

conjecture | open | MathDB 页面 | 原文线索

V2-003699 | MATHDB-345755

The bounded multiplicative coloring conjecture

摘录 / EXCERPT: A multiplicative 2-coloring is a multiplicative function $f:\mathbb N\to\{-1,+1\}$. The bounded multiplicative coloring conjecture. There exists a constant C and a multiplicati...

conjecture | open | MathDB 页面 | 原文线索

V2-003700 | MATHDB-345814

The prime power conjecture for perfect difference sets

摘录 / EXCERPT: Let $D\subset \mathbb{Z}/m\mathbb{Z}$ be a perfect difference set if every nonzero $a\in\mathbb{Z}/m\mathbb{Z}$ can be written uniquely as the difference of two elements of D. A...

conjecture | open | MathDB 页面 | 原文线索

V2-003701 | MATHDB-345858

The elementary proof conjecture for the real Hardy-Ramanujan formula

摘录 / EXCERPT: Elementary proof conjecture. There exists a proof of Proposition 1 via elementary methods.

conjecture | open | MathDB 页面 | 原文线索

V2-003702 | MATHDB-345868

Badziahin-Velani conjecture for the logarithmic spectrum of multiplicatively badly approximable matrices

摘录 / EXCERPT: For $lambda eu0007\mathbb{R}$, let $\operatorname{Mad}^{\lambda}(2,1)$ denote the corresponding logarithmic approximation set in the multiplicative setting. Badziahin-Velani conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003703 | MATHDB-345917

Barlak-Omland-Stammeier conjecture for the K-theory of $\mathcal{Q}_S$

摘录 / EXCERPT: Let $S\subset\mathbb{N}^{\times}\setminus\{1\}$ be nonempty and consist of mutually coprime numbers. Let $\mathcal{Q}_S$ be the unital C-algebra defined by Barlak, Omland, and Stam...

conjecture | open | MathDB 页面 | 原文线索

V2-003704 | MATHDB-345929

Partition regularity of the exceptional linear-quadratic equation

摘录 / EXCERPT: Partition-regularity conjecture. For every choice of non-zero integers a,b,c, this equation is partition regular.

conjecture | open | MathDB 页面 | 原文线索

V2-003705 | MATHDB-345938

Eventual vanishing conjecture for rational approximants

摘录 / EXCERPT: Let L be the linear functional in the preceding generalization, with integer parameters $alpha$, $beta_i$, and $gamma_i$, and let P_n/Q_n be the associated rational approximant...

conjecture | open | MathDB 页面 | 原文线索

V2-003706 | MATHDB-345956

The asymptotic Fermat conjecture over number fields

摘录 / EXCERPT: Let K be a number field. A solution $(x,y,z)\in K^3$ of the Fermat equation ... is trivial if $xyz=0$ and non-trivial otherwise. The asymptotic Fermat conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-003707 | MATHDB-345993

Equidistribution of polynomial roots modulo primes with root-count normalization

摘录 / EXCERPT: Let $f\be\in\mathbf{Z}[X]$ be a monic irreducible polynomial of degree at least 2. Let $\Pi_f(x)$ denote the set of primes $p\leqslant x$ for which f has at least one root modu...

conjecture | open | MathDB 页面 | 原文线索

V2-003708 | MATHDB-345994

Hooley-measure equidistribution conjecture for polynomial roots modulo primes

摘录 / EXCERPT: Let $f\in\mathbf{Z}[X]$ be a monic irreducible polynomial of degree at least 2. For each prime p, let $\varrho_f(p)$ be the number of roots of f modulo p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003709 | MATHDB-346005

Furstenberg-type periodicity conjecture for even parameters

摘录 / EXCERPT: Let E denote the set of even integers, let $k\in E$, and let n be an integer. The periodicity conjecture. The n-orbit is periodic. The preceding analysis shows divergence for...

conjecture | open | MathDB 页面 | 原文线索

V2-003710 | MATHDB-346010

Chbili-Qazaqzeh Jones-coefficient conjecture for quasi-alternating links

摘录 / EXCERPT: Let L be a link whose Jones polynomial is written as ... where $m\geq 0$, $a_0\neq 0$, and $a_m\neq 0$. Chbili-Qazaqzeh conjecture. If L is a prime quasi-alternating link, other...

conjecture | open | MathDB 页面 | 原文线索

V2-003711 | MATHDB-346025

The local characterization conjecture for multiplier sets and cyclic direct factors

摘录 / EXCERPT: Let m, n, and k be positive integers with $m\geq 2$, and suppose ... For a prime p dividing $|G|$, write $\operatorname{ord}_p(m)$ for the multiplicative order of m modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-003712 | MATHDB-346070

Improved upper-bound conjecture for clique numbers of Paley graphs

摘录 / EXCERPT: Let p be a prime with $p\equiv 1\pmod{4}$, and let $q=p^{2s+1}$ for some positive integer s. Write $\omega(P_q)$ for the clique number of the Paley graph P_q. Improved Paley-...

conjecture | open | MathDB 页面 | 原文线索

V2-003713 | MATHDB-346073

A prime-representing quadratic-exponent conjecture following Mills

摘录 / EXCERPT: Prime-representing function conjecture. There is a constant A, possibly near 1.1966746500705764022, such that $f(n)$ is prime for every integer $n\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-003714 | MATHDB-346087

Montgomery's conjecture for primes in arithmetic progressions

摘录 / EXCERPT: Let $a<q$ be integers with $\gcd(a,q)=1$, and let $q\leq x$. Write $\psi(x,q,a)$ for the weighted prime-counting function in the progression $a\pmod q$, $\pi(x,q,a)$ for the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-003715 | MATHDB-346103

Supercritical discrete restriction conjecture for positive definite forms

摘录 / EXCERPT: Supercritical discrete restriction conjecture. For each $1\leq p\leq\infty$ and $\epsilon>0$, there exists a positive constant $C_{\mathcal{Q},p,\epsilon}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003716 | MATHDB-346105

Hooley's conjecture on small fundamental units in real quadratic fields

摘录 / EXCERPT: Hooley's conjecture. For every $\alpha>0$, there exists a constant $b(\alpha)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003717 | MATHDB-346140

A regulator-sum bound for Scholz's conjecture

摘录 / EXCERPT: Scholz's conjecture reduction. The following inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-003718 | MATHDB-346153

Furstenberg-type restricted-digit counting conjecture

摘录 / EXCERPT: Let $p_1,\dots,p_k$ be $k\geq 2$ different odd prime numbers. For each $i\in\\{1,\dots,k\\}$, set ... Define ... Here $N_{p_1,\dots,p_k}^{B_1,\dots,B_k}$ denotes the set of nonnegative...

conjecture | open | MathDB 页面 | 原文线索

V2-003719 | MATHDB-346156

Finiteness conjecture for intersections of restricted-digit sets

摘录 / EXCERPT: For an integer $b\geq 3$, let B_b be the set of positive integers whose base-b expansion contains only the digits 0 and 1. Let a,b be multiplicatively independent integer...

conjecture | open | MathDB 页面 | 原文线索

V2-003720 | MATHDB-346171

Higher-genus quasimodularity for d-elliptic loci

摘录 / EXCERPT: Higher-genus quasimodularity conjecture. The statement of this theorem holds for all $g\ge2$.

conjecture | open | MathDB 页面 | 原文线索

V2-003721 | MATHDB-346224

Second Borwein conjecture on coefficient positivity

摘录 / EXCERPT: Second Borwein conjecture. Each of $\alpha_n(q)$, $\beta_n(q)$ and $\gamma_n(q)$ has non-negative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-003722 | MATHDB-346240

Asymptotic prime count conjecture for two-generated numerical semigroups

摘录 / EXCERPT: Let $1\leq a<b$ be two relatively prime integers, let $S=\langle a,b\rangle$ be the numerical semigroup generated by a and b, and let π_S denote the number of primes in S...

conjecture | open | MathDB 页面 | 原文线索

V2-003723 | MATHDB-346244

The characteristic-polynomial conjecture for $y^2=x^p-1$

摘录 / EXCERPT: Let C_p be the genus g curve ... where $p=2g+1$ is prime. For the characteristic polynomial $P_{\gamma^d}(T)$ associated with the indicated component representative, the characte...

conjecture | open | MathDB 页面 | 原文线索

V2-003724 | MATHDB-346245

The characteristic-polynomial conjecture for $y^2=x^{2p}-1$

摘录 / EXCERPT: Let C_{2p} be the genus g curve ... where p is prime. For the characteristic polynomials $P_{d,j}(T)$ associated with the component representatives indexed by d and j, the...

conjecture | open | MathDB 页面 | 原文线索

V2-003725 | MATHDB-346246

Mixed signed Selmer torsion conjecture over multiple \mathbb{Z}_p-extensions

摘录 / EXCERPT: Let p be an odd prime, let E be an elliptic curve over a number field F, and let Σ^{ss} be the nonempty set of primes of F above p at which E has good supersingu...

conjecture | open | MathDB 页面 | 原文线索

V2-003726 | MATHDB-346250

Infinitely many prime divisors for non-geometric generalized derangement sequences

摘录 / EXCERPT: The non-geometric prime-divisor conjecture. If $\mathbf{a}\in\mathcal{R}$ is not a geometric progression, then the set $\mathcal{P}_{\mathbf a}$ is infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-003727 | MATHDB-346251

Rigidity of ultimately geometric generalized derangement sequences

摘录 / EXCERPT: Let $\mathbb{N}$ and $\mathbb{Z}$ denote the non-negative integers and integers, respectively. For $f,g,h\in\mathbb{Z}[X]$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003728 | MATHDB-346290

Hardy-Littlewood discrepancy conjecture for reciprocal quadratic surds

摘录 / EXCERPT: Hardy-Littlewood discrepancy conjecture. The function $C_{a_0}(n)+C_{1/a_0}(n)$ is bounded if and only if $a_0+1/a_0\in\mathbb{N}$. If it is bounded, it is identically zero. If it...

conjecture | open | MathDB 页面 | 原文线索

V2-003729 | MATHDB-346302

Conjecture on the number and modular invariance of critical points of the interaction functional

摘录 / EXCERPT: Critical-point conjecture. The function $\mathcal{J}(z;a,b)$ has either four or six critical points with respect to a,b, and the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-003730 | MATHDB-346327

Rank-one Bloch-Kato conjecture for the triple product motive

摘录 / EXCERPT: Let f_1,f_2,f_3 be the triple of modular forms in the paper, let $\mathrm{V}(\underline{\mathbf{f}})$ be the associated l-adic representation over $E_{\underline{\lambda}}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-003731 | MATHDB-346341

Delta-prime formula for the bosonic coinvariant characteristic

摘录 / EXCERPT: Let $\mathcal{E}_n=\sum_{\mu\vdash n}\mathcal{C}_\mu\otimes s_\mu(\bm{z})$ be the pure bosonic multivariate coinvariant characteristic, where...

conjecture | open | MathDB 页面 | 原文线索

V2-003732 | MATHDB-346351

Rudnick-Sarnak-Zaharescu Diophantine criterion for Poissonian statistics

摘录 / EXCERPT: For $\alpha\in[0,1]$, say that α is of type ω if there are only finitely many pairs (a,q) satisfying $\lvert\alpha-a/q\rvert<q^{-\omega}$. Let a/q range over the...

conjecture | open | MathDB 页面 | 原文线索

V2-003733 | MATHDB-346357

De Clercq-Florens's Smoothness Conjecture for profinite groups

摘录 / EXCERPT: Let p be a prime. A p-orientation of a profinite group G is a continuous homomorphism $\theta:G\to\mathbb{Z}_p^{\times}$. For the resulting oriented group (G,θ), let...

conjecture | open | MathDB 页面 | 原文线索

V2-003734 | MATHDB-346362

Positive-density conjecture for words avoiding forbidden signatures

摘录 / EXCERPT: Let $\pmb{\omega}=(\omega_0,\omega_1,\ldots,\omega_k)\notin\mathcal{F}$ be a finite-length word that contains no $\pmb{\tau}\in\mathcal{F}$ as a subsequence of consecutive symbols....

conjecture | open | MathDB 页面 | 原文线索

V2-003735 | MATHDB-346363

Modified Polignac's conjecture for number-trail prime gaps

摘录 / EXCERPT: Let $\mathcal{D}_k^1=L_{\infty}(p_{k+1})-L_{\infty}(p_k)$ be the prime gaps along the number trail, and let $\mathbb{N}$ denote the natural numbers. Modified Polignac's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003736 | MATHDB-346377

Primality conjecture for integers of the form $am+1$

摘录 / EXCERPT: Let $n=am+1$, where a and m are positive integers, and let p be the least prime divisor of m. Primality conjecture. If $a<p$ and $m\mid\phi(n)$, then n is prime. This con...

conjecture | open | MathDB 页面 | 原文线索

V2-003737 | MATHDB-346388

Quillen's conjecture on the poset of elementary abelian p-subgroups

摘录 / EXCERPT: Let G be a finite group and let p be a prime dividing $|G|$. Let ${\mathcal{A}}_p(G)$ be the poset of non-trivial elementary abelian p-subgroups of G, ordered by inclusion,...

conjecture | open | MathDB 页面 | 原文线索

V2-003738 | MATHDB-346409

Small-height zero abundance conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let $N(T,\chi)$ denote the number of zeros of the Dirichlet L-function associated with a character χ having imaginary part in the relevant height range, and let the conducto...

conjecture | open | MathDB 页面 | 原文线索

V2-003739 | MATHDB-346410

Logarithmic error bound conjecture for Dirichlet L-function zero counts

摘录 / EXCERPT: Let χ be a character with conductor $q>1$, and define ... Assume $T\ge5/7$. Logarithmic error bound conjecture. The zero-counting function satisfies ... This is a speculative expl...

conjecture | open | MathDB 页面 | 原文线索

V2-003740 | MATHDB-346423

The sign-identity conjecture for the theta-series kernel

摘录 / EXCERPT: Let v_0 be a null vector and let N_k,q_k be the charge data entering the pairings ... Assume that all $b^{\,0}_{k\ell}$ are nonzero. Let Φ_n be the function defined in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-003741 | MATHDB-346429

The uniform exponential-sum bound for polynomial phases

摘录 / EXCERPT: Uniform exponential-sum bound. Given f, n, s, and d as above and any $\varepsilon>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-003742 | MATHDB-346458

The Markoff graph connectivity conjecture modulo primes

摘录 / EXCERPT: Let $\mathcal M$ be the set of positive integer solutions to ... and let $\mathcal M_p$ be the set of non-zero solutions modulo a prime p. The functional graph is generated by the...

conjecture | open | MathDB 页面 | 原文线索

V2-003743 | MATHDB-346459

The bounded Möbius-action intersection conjecture

摘录 / EXCERPT: There exist constants $\varepsilon_0>0$ and A such that, for every prime p, every multiplicative subgroup $\mathcal G\subseteq\overline{\mathbb F}_p$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-003744 | MATHDB-346476

Tout-Dabboucy-Howalla conjecture on prime labelings of 2-regular graphs

摘录 / EXCERPT: Let $G=C_{n_1}\cup\cdots\cup C_{n_m}$ be a 2-regular graph, where each $n_i\ge 3$, and let a graph be prime when it admits a labeling by distinct integers from...

conjecture | open | MathDB 页面 | 原文线索

V2-003745 | MATHDB-346477

Schroeder's odd-start cycle labeling conjecture

摘录 / EXCERPT: For odd integers n and k with $n\ge 3$, consider the cycle C_n and the consecutive label set $\{k,k+1,\ldots,k+n-1\}$. A prime labeling assigns these labels bijectively to th...

conjecture | open | MathDB 页面 | 原文线索

V2-003746 | MATHDB-346492

Kaneko-Koike positivity and denominator conjecture for extremal quasimodular forms

摘录 / EXCERPT: Kaneko-Koike's conjecture. If $\ell\leq 4$ and $w>2$, then every Fourier coefficient a_n is positive. Moreover, no prime factor of the denominator of any such coefficient is grea...

conjecture | open | MathDB 页面 | 原文线索

V2-003747 | MATHDB-346502

Mordell's conjecture on Pell equation solutions for primes congruent to 3 modulo 4

摘录 / EXCERPT: Mordell's conjecture. The prime p does not divide y.

conjecture | open | MathDB 页面 | 原文线索

V2-003748 | MATHDB-346510

Mazur-Rubin's variance asymptotic conjecture for modular symbols

摘录 / EXCERPT: Let $f\in S_2(\Gamma_0(q))$, where q is a positive integer. For each positive integer c, define ... where $\langle r\rangle=\int_{i\infty}^r\operatorname{Re}(f(z)\,dz)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003749 | MATHDB-346511

Mazur-Rubin's central limit conjecture for modular symbols

摘录 / EXCERPT: Let $f\in S_2(\Gamma_0(q))$, let c tend to infinity through positive integers satisfying $(c,q)=d$, where d divides q, and let a range over $(\mathbb Z/c\mathbb Z)^*$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003750 | MATHDB-346548

Conjecture on the third term of lens surgery polynomials

摘录 / EXCERPT: Let p and k be relatively prime positive integers. Let $K_{p,k}$ be the lens space knot in the homology sphere $Y_{p,k}$ dual to a simple $(1,1)$-knot in a lens space, with par...

conjecture | open | MathDB 页面 | 原文线索

V2-003751 | MATHDB-346550

Mustață-Takagi-Watanabe conjecture on equality of log canonical and F-pure thresholds

摘录 / EXCERPT: Let $f\in\mathbb{Q}[x_1,\ldots,x_n]$, and for each prime number p let $f_p\in\mathbb{F}_p[x_1,\ldots,x_n]$ be the polynomial obtained by reducing each coefficient of f modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-003752 | MATHDB-346554

Velani's dyadic approximation conjecture for the middle-third Cantor set

摘录 / EXCERPT: Let $\psi:{\mathbb{N}}\to{\mathbb{R}}^+$ be monotonic, and let ... where $\mathcal A(2)=\{2^n:n=0,1,2,\ldots\}$ and μ denotes Lebesgue measure. Velani's conjecture. ... This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-003753 | MATHDB-346577

Conjecture C_η on short character sums

摘录 / EXCERPT: Let η be a real number satisfying $0<\eta\leq 1$. A real character is a Dirichlet character taking real values, and a character is non-principal when it is not the principal c...

conjecture | open | MathDB 页面 | 原文线索

V2-003754 | MATHDB-346588

Conjecture on the size of the minimum logarithmic derivative of Dirichlet L-functions

摘录 / EXCERPT: Conjecture on m_q. For all $\varepsilon>0$ and for all odd primes q,

conjecture | open | MathDB 页面 | 原文线索

V2-003755 | MATHDB-346600

Nonexistence conjecture for closed embedded totally geodesic hypersurfaces in minimal-volume cusped arithmetic hyperbolic manifolds

摘录 / EXCERPT: Nonexistence conjecture. If W^n contains a closed embedded orientable totally geodesic hypersurface, then

conjecture | open | MathDB 页面 | 原文线索

V2-003756 | MATHDB-346602

The character sum cancellation conjecture for rough integers

摘录 / EXCERPT: Let q be large and let χ be a non-principal character modulo q. For real numbers x and y satisfying ... and writing $P(n)$ for the largest prime factor of n and...

conjecture | open | MathDB 页面 | 原文线索

V2-003757 | MATHDB-346611

Exponent conjecture for finite p-groups

摘录 / EXCERPT: Let G be a finite p-group, where p is prime. Exponent conjecture for finite p-groups. ... This is attributed to the authors of and gives a bound for the exponent of the Schur...

conjecture | open | MathDB 页面 | 原文线索

V2-003758 | MATHDB-346613

Chinburg's $\Omega(3)$ conjecture for number-field extensions

摘录 / EXCERPT: Chinburg's $\Omega(3)$ conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003759 | MATHDB-346614

Chinburg's $\Omega(3)$ conjecture modulo the kernel group

摘录 / EXCERPT: Chinburg's $\Omega(3)$ conjecture modulo the kernel group.

conjecture | open | MathDB 页面 | 原文线索

V2-003760 | MATHDB-346624

Schmidt's intersection conjecture for weighted badly approximable vectors

摘录 / EXCERPT: Let $\mathcal{W}_2=\{\mathbf{w}=(w_1,w_2)\in\mathbb{R}^2:w_1,w_2\geq0,\ w_1+w_2=1\}$. For $\mathbf{w}\in\mathcal{W}_2$, let $\mathbf{Bad}(\mathbf{w})$ be the set of vectors...

conjecture | solved | MathDB 页面 | 原文线索

V2-003761 | MATHDB-346627

Primitive non-deficient number component conjecture

摘录 / EXCERPT: Primitive non-deficient number component conjecture. There is an i, with $1\leq i\leq k$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-003762 | MATHDB-346628

Eventual unbounded growth of the generalized prime-factor function

摘录 / EXCERPT: The generalized growth conjecture. For any $\alpha>1$ and $C>0$, for sufficiently large n one has

conjecture | open | MathDB 页面 | 原文线索

V2-003763 | MATHDB-346629

Factorial-plus-one bump conjecture for $A(n)$

摘录 / EXCERPT: The $A(n)$ bump conjecture. There are arbitrarily large n such that

conjecture | open | MathDB 页面 | 原文线索

V2-003764 | MATHDB-346630

Factorial-minus-one bump conjecture for $B(n)$

摘录 / EXCERPT: The $B(n)$ bump conjecture. There are arbitrarily large n such that

conjecture | open | MathDB 页面 | 原文线索

V2-003765 | MATHDB-346631

Factorial-plus-one bump conjecture for $C(n)$

摘录 / EXCERPT: The $C(n)$ bump conjecture. There are arbitrarily large n such that

conjecture | open | MathDB 页面 | 原文线索

V2-003766 | MATHDB-346632

Factorial-plus-one bump conjecture for $D(n)$

摘录 / EXCERPT: The $D(n)$ bump conjecture. There are arbitrarily large n such that

conjecture | open | MathDB 页面 | 原文线索

V2-003767 | MATHDB-346633

Limit conjecture for the generalized primorial-type function H

摘录 / EXCERPT: The factorial-power limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003768 | MATHDB-346641

Non-existence conjecture for almost m-ary NPSs with two consecutive zero-symbols

摘录 / EXCERPT: Non-existence conjecture. Let $m\geq 5$ be a prime, $\gamma_1,\gamma_2\in\mathbb Z$, and $n\in\mathbb Z^+$. There does not exist a non-trivial almost m-ary NPS of type...

conjecture | open | MathDB 页面 | 原文线索

V2-003769 | MATHDB-346643

Conjecture on analytic rank and Schmidt rank over finite fields

摘录 / EXCERPT: Let q be a prime power, let $\mathbb F_q$ be the finite field with q elements, and let P be a multilinear polynomial of degree d. Write $a_{\mathbb F_q}(P)$ for its analyti...

conjecture | open | MathDB 页面 | 原文线索

V2-003770 | MATHDB-346648

Thurston's geometric decomposition conjecture for compact prime 3-manifolds

摘录 / EXCERPT: Let M be a compact, orientable, prime 3-manifold. The eight Thurston geometries are the homogeneous geometries that admit compact quotient manifolds. Thurston's geometric decom...

conjecture | open | MathDB 页面 | 原文线索

V2-003771 | MATHDB-346662

Sun's refined 1-3-5 conjecture with a square or three-times-square coordinate

摘录 / EXCERPT: Sun's refined 1-3-5 conjecture. Any positive integer can be represented in this way with $x+3y+5z$ a square, and with at least one of the following properties: x is three times a...

conjecture | open | MathDB 页面 | 原文线索

V2-003772 | MATHDB-346663

The computational refinement of the 1-3-5 conjecture

摘录 / EXCERPT: The computational refinement of the 1-3-5 conjecture. Except for $m\in\{31,43,111,151,168,200,248,263,319,456,479,871,1752,1864,3544\}$, every such m has a representation for whi...

conjecture | open | MathDB 页面 | 原文线索

V2-003773 | MATHDB-346671

Prime-norm global-dimension conjecture for fusion categories

摘录 / EXCERPT: Prime-norm global-dimension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-003774 | MATHDB-346679

Conjecture for joint moments of Hardy's function and its derivative

摘录 / EXCERPT: Let ... be Hardy's function, and let $\mathsf{a}(s)$ be the arithmetic factor ... For $s>-\frac12$ and $0\leq h<s+\frac12$, the conjecture asserts ... as $T\to\infty$. This refines the z...

conjecture | open | MathDB 页面 | 原文线索

V2-003775 | MATHDB-346689

Rejmer's conjecture on positive values at infinitely many primes

摘录 / EXCERPT: Assuming Rejmer's algorithm runs indefinitely, let $R(n)$ be the resulting recursive binary sequence with values in $\{-1,+1\}$. Rejmer's prime-value conjecture. There exist infini...

conjecture | open | MathDB 页面 | 原文线索

V2-003776 | MATHDB-346697

Depth-two conjecture for the finite p-groups $G_r=C_p\ltimes T_0/T_r$

摘录 / EXCERPT: Let p be an odd prime and let $r\geq p-1$ be an integer. Let ... be the finite p-group defined in the source. Depth-two conjecture. The mod-p cohomology ring...

conjecture | open | MathDB 页面 | 原文线索

V2-003777 | MATHDB-346702

The primeness conjecture for the equivalence relation $\mathbb E_1$

摘录 / EXCERPT: Let $\mathbb E_1$ be the equivalence relation on sequences of reals defined by eventual agreement, and let a Borel equivalence relation be prime if it is Borel-reducible to no prod...

conjecture | open | MathDB 页面 | 原文线索

V2-003778 | MATHDB-346732

The Steinberg-homology conjecture for real quadratic fields

摘录 / EXCERPT: Let N be a level, let E be a real quadratic field, and let $\psi_{\Gamma,E}$ denote the map associated with a subgroup Γ as in the conjecture. Write $A_E(N)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003779 | MATHDB-346755

Ruzsa's analogue for alpha-primary pseudo-polynomials

摘录 / EXCERPT: The alpha-primary growth conjecture. If $f(n)=O(c^n)$ for some $c<e^\alpha$, then f is a polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-003780 | MATHDB-346756

Periodicity and exponential error conjecture for linear programming scaling dimensions

摘录 / EXCERPT: Scaling-dimension periodicity conjecture. The quantity $E(c)$ vanishes if and only if c is an integer congruent to 4 modulo 8. Moreover, there is a constant α such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-003781 | MATHDB-346757

Degeneracy bounds conjecture for the modular bootstrap

摘录 / EXCERPT: Degeneracy-ratio conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-003782 | MATHDB-346767

The SCFT spectrum conjecture for topological modular forms

摘录 / EXCERPT: Let $\SQFT^\bullet$ denote the spectrum whose spaces classify compact unitary $\mathcal N=(0,1)$ superconformal field theories, graded by anomaly. Let $\mathrm{TMF}^\bullet$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-003783 | MATHDB-346768

The equivariant topological modular forms refinement conjecture for Mathieu Moonshine

摘录 / EXCERPT: Let $\nu\in\mathrm{TMF}^{-3}(\mathrm{pt})$ be the class corresponding to $[\overline{\operatorname{Fer}(3)}]$, let $\{24\Delta\}\in\mathrm{TMF}^{-24}(\mathrm{pt})$ be the class cor...

conjecture | open | MathDB 页面 | 原文线索

V2-003784 | MATHDB-346770

The Tate-curve index conjecture for SQFTs with cylindrical ends

摘录 / EXCERPT: Let $\mathcal F$ be an $\mathcal N=(0,1)$ SQFT with cylindrical ends and boundary $\partial\mathcal F$, satisfying $Z_{RR}(\partial\mathcal F)=0$, and suppose $\partial\mathcal F$...

conjecture | open | MathDB 页面 | 原文线索

V2-003785 | MATHDB-346823

Oort's amalgamation conjecture for Newton polygons of curves

摘录 / EXCERPT: Let p be a prime and let C and C' be smooth proper irreducible curves of genera g and g', with Newton polygons P and P', respectively. A Newton polygon is determined...

conjecture | open | MathDB 页面 | 原文线索

V2-003786 | MATHDB-346847

Decidability conjecture for deficiency-one metabelian groups

摘录 / EXCERPT: Deficiency-one decidability conjecture. If $|A|-|R|=1$, then the Diophantine problem in G is decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-003787 | MATHDB-346862

The three Linnik primes Diophantine approximation conjecture

摘录 / EXCERPT: Three-Linnik-prime approximation conjecture. There exist infinitely many triples of primes p_1,p_2,p_3 such that

conjecture | open | MathDB 页面 | 原文线索

V2-003788 | MATHDB-346865

The Racah polynomial boundedness conjecture

摘录 / EXCERPT: Racah polynomial boundedness conjecture. For any integers k,n,T with $0\leq k,n\leq T-1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-003789 | MATHDB-346866

Cramér's conjecture on gaps between consecutive primes

摘录 / EXCERPT: Let p_n denote the n-th prime. Cramér's conjecture. There exist absolute constants $n_0,c>0$ such that if $n>n_0$, then ... This conjectural bound on maximal gaps between consecu...

conjecture | open | MathDB 页面 | 原文线索

V2-003790 | MATHDB-346874

Serre weight conjecture for GSp_4 over totally real fields

摘录 / EXCERPT: Let F be a totally real field, let p be a prime split completely in F, and let $\overline{\rho}:G_F\longrightarrow {\rm GSp}_4(\overline{\mathbb{F}}_p)$ be a mod p irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-003791 | MATHDB-346877

Conjectures on the recursively defined prime-counting sequence

摘录 / EXCERPT: The conjectures. (A) For every integer $k\ge0$, the set $\{n\ge1:a_n=k\}$ has a positive density δ_k, and

conjecture | open | MathDB 页面 | 原文线索

V2-003792 | MATHDB-346894

Gross-Hacking-Keel Frobenius structure conjecture

摘录 / EXCERPT: Let S be the dual intersection complex of D, let B be the cone over S, and let $B_0=B\setminus\{0\}$. The integer points $q\in B(\mathbb Z)$ index prime fundamental classes...

conjecture | open | MathDB 页面 | 原文线索

V2-003793 | MATHDB-346904

Asymptotic counting conjecture for powers of multiplicatively independent complex numbers

摘录 / EXCERPT: The multiplicative-independence counting conjecture. The numbers α and β are multiplicatively independent if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-003794 | MATHDB-346905

Asymptotic counting conjecture for differences of powers of π and e

摘录 / EXCERPT: The π-e counting conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003795 | MATHDB-346939

VafaWitten S-duality conjecture for SU(r) and SU(r)/Z_r

摘录 / EXCERPT: VafaWitten S-duality conjecture. The two partition functions are Fourier expansions of these meromorphic functions, which satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-003796 | MATHDB-346943

Conjectured Hausdorff-measure zero-one law for algebraic points near manifolds

摘录 / EXCERPT: Manifold Khintchine-type conjecture. The Hausdorff g-measure of $\dualappr_{n,\bm f}^*(\psi_0,\ldots,\psi_m)$ is zero if $S_{n,d}^g(\psi_0,\ldots,\psi_m)$ converges, and equals...

conjecture | open | MathDB 页面 | 原文线索

V2-003797 | MATHDB-346960

The fractional $3n+1$ conjecture for the $Delta$-sequence

摘录 / EXCERPT: The fractional $3n+1$ conjecture. Iterating Δ will eventually either converge to zero, or loop over sequences of real numbers whose successive integer parts are

conjecture | open | MathDB 页面 | 原文线索

V2-003798 | MATHDB-346987

The theta block conjecture for Borcherds products

摘录 / EXCERPT: Let the pure theta block Θ_f be a holomorphic Jacobi form of weight k and index m with vanishing order 1 in q, where $k,m\in\mathbb Z_{>0}$. Define the nearly holomo...

conjecture | open | MathDB 页面 | 原文线索

V2-003799 | MATHDB-346990

The conjectured strict recurrence inequality for Li coefficients

摘录 / EXCERPT: Let λ_n denote the Li coefficients of the Riemann zeta function. In view of the recurrence relation for these coefficients, strict recurrence inequality for Li coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-003800 | MATHDB-346996

Prime running function main-term conjecture

摘录 / EXCERPT: For integers $d\geq 2$ and a reduced residue $a\pmod d$, let ... where $\lfloor n\rfloor_{\mathcal P}$ is the largest prime at most n, with $\lfloor 1\rfloor_{\mathcal P}=0$. Prime...

conjecture | open | MathDB 页面 | 原文线索

V2-003801 | MATHDB-346997

Prime running bias conjecture

摘录 / EXCERPT: Let $\Phi(x;d,a)$ be the prime running function for integers $d\geq 2$ and residue classes a coprime to d. A bias constant is a constant $R(d;a)$ associated with the class a....

conjecture | open | MathDB 页面 | 原文线索

V2-003802 | MATHDB-346998

Existence of limiting rescaled bias constants for prime running functions

摘录 / EXCERPT: For integers $d\geq 2$ and residues a with $\gcd(a,d)=1$, define the rescaled bias function ... where $\Phi(x;d,a)$ is the prime running function. Existence of limiting rescaled bi...

conjecture | open | MathDB 页面 | 原文线索

V2-003803 | MATHDB-347003

F-regularity conjecture for the commutator ideals

摘录 / EXCERPT: Let K be a field of positive prime characteristic, and let R, $\mathcal{I}$, and J be the ring and ideals defined from the commutator matrix in the paper. F-regularity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003804 | MATHDB-347024

Bloch-Kato conjecture for quadratic Hilbert modular forms

摘录 / EXCERPT: Bloch-Kato conjecture. If $L(\pi,1+j)\ne 0$ for some integer j with $t_i\leqslant j\leqslant k_i+t_i-2$ for all i, then

conjecture | open | MathDB 页面 | 原文线索

V2-003805 | MATHDB-347027

Várilly-Alvarado's strong uniform boundedness conjecture for K3 Brauer groups

摘录 / EXCERPT: Fix a positive integer n and a primitive lattice embedding ... Let X be a K3 surface over a number field of degree n such that the geometric Néron-Severi lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-003806 | MATHDB-347031

Asymptotic existence conjecture for (a,b)-necklaces

摘录 / EXCERPT: Let a and b be positive coprime integers. An (a,b)-necklace is the combinatorial object considered in the paper, with length denoted by n. Asymptotic existence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003807 | MATHDB-347056

Generalized Fermat conjecture on primitive solutions

摘录 / EXCERPT: Let p,q and r be positive integers satisfying ... A solution (X,Y,Z) is called primitive here when $XYZ\ne0$ and $\gcd(X,Y)=1$. Generalized Fermat conjecture. All primitive sol...

conjecture | open | MathDB 页面 | 原文线索

V2-003808 | MATHDB-347061

Wirsing-Schmidt conjecture on approximation by algebraic numbers

摘录 / EXCERPT: Let $n\ge2$ and let ξ be a real number that is not algebraic of degree at most n. The height $H(\alpha)$ of an algebraic number is the height of its minimal defining polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-003809 | MATHDB-347063

Linear-growth conjecture for Split Smallest and Combine Smallest games

摘录 / EXCERPT: Linear-growth conjecture. For Split Smallest, the number of moves grows linearly with n, with the asymptotic constant appearing numerically to be the square of the golden mean. F...

conjecture | open | MathDB 页面 | 原文线索

V2-003810 | MATHDB-347064

Longest-game conjecture for the Split Smallest strategy

摘录 / EXCERPT: Longest-game conjecture. The longest game on any n is achieved by applying splitting moves whenever possible: first adding 1's, then splitting from smallest to largest, and fin...

conjecture | open | MathDB 页面 | 原文线索

V2-003811 | MATHDB-347073

The noncongruence of powers of two

摘录 / EXCERPT: Let n be a positive integer power of 2, so that $n\in 2^{\mathbb{N}}$. A positive integer is congruent if it is the area of a right triangle with rational side lengths. Noncong...

conjecture | open | MathDB 页面 | 原文线索

V2-003812 | MATHDB-347083

Hu-Li conjecture on the upper bound for $f(p)$

摘录 / EXCERPT: Let $p\ge 5$ be a prime, and let $f(p)$ be the smallest number such that every planar graph of girth p with no cycles of lengths from $p+1$ through $f(p)$ admits a homomorphism t...

conjecture | open | MathDB 页面 | 原文线索

V2-003813 | MATHDB-347100

The minimal-degree conjecture for triquadratic fields

摘录 / EXCERPT: Let $L = \mathbb{Q}(\sqrt{A}, \sqrt{B}, \sqrt{C})$ for distinct square-free non-zero integers A, B, and C, and let $M = \mathbb{Q}(\sqrt{A}, \sqrt{B})$. Here $\min\deg_L(v)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003814 | MATHDB-347119

The primorial Jacobsthal lower-bound conjecture for missing differences

摘录 / EXCERPT: Let $k\in\mathbb{N}$ with $k>1$. Write $h(k-1)$ for the primorial Jacobsthal function at $k-1$, and let $N_{min}(k)$ be the minimum even positive integer that does not occur as a d...

conjecture | open | MathDB 页面 | 原文线索

V2-003815 | MATHDB-347129

Banks-Martin conjecture for Erdős sums

摘录 / EXCERPT: Banks-Martin conjecture. The sequence of sums is strictly decreasing:

conjecture | open | MathDB 页面 | 原文线索

V2-003816 | MATHDB-347131

The monodromy conjecture for tuples of polynomials

摘录 / EXCERPT: Let $F=(f_1,\ldots,f_p)$ be a tuple of polynomials $f_i\in\mathbb{C}[x_1,\ldots,x_n]$. Let $Z_F^{top}(s_1,\ldots,s_p)$ be the topological zeta function of F, let...

conjecture | open | MathDB 页面 | 原文线索

V2-003817 | MATHDB-347132

The strong monodromy conjecture for tuples of polynomials

摘录 / EXCERPT: Let $F=(f_1,\ldots,f_p)$ be a tuple of polynomials $f_i\in\mathbb{C}[x_1,\ldots,x_n]$. Let $Z_F^{top}(s_1,\ldots,s_p)$ be the topological zeta function of F, with polar locus...

conjecture | open | MathDB 页面 | 原文线索

V2-003818 | MATHDB-347133

The smooth numbers conjecture in arithmetic progressions

摘录 / EXCERPT: Smooth numbers conjecture. If y and q are large with $q\leq y^A$, then, as $\frac{\log x}{\log y}\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003819 | MATHDB-347134

Conjecture on the asymptotic classes and equality of NFS minimizer series

摘录 / EXCERPT: Let a,b,d be the minimizers of Problem, and let $\mathbf A,\mathbf B\in\mathbb R[[X,Y]]$ be the series associated to a and b, respectively. Let $\mathcal C^{[1/3,2/3]}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003820 | MATHDB-347148

Universal odd-prime valuation bound for the sum of divisors

摘录 / EXCERPT: Universal odd-prime valuation conjecture. Every $n\in\mathbb{N}$ satisfies condition (1) or (2). Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-003821 | MATHDB-347163

The APT p-group bound for the Schur multiplier

摘录 / EXCERPT: Let G be a finite p-group, where p is prime. Write $\operatorname{exp}(G)$ for the exponent of G, and let $H_2(G,\mathbb{Z})$ be its second homology group with integer coef...

conjecture | open | MathDB 页面 | 原文线索

V2-003822 | MATHDB-347204

The weight part of Serre's conjecture in the generic tame setting

摘录 / EXCERPT: Let F^+ be the relevant totally real field, let $\overline r:G_{F^+}\rightarrow\mathcal{G}_n(\mathbb{F})$ be automorphic, and let $\overline r_p$ be its associated local L-homo...

conjecture | open | MathDB 页面 | 原文线索

V2-003823 | MATHDB-347217

Uniform symmetry-factor conjecture for pro-isomorphic zeta functions of graded Lie lattices

摘录 / EXCERPT: Let $\mathcal{L}$ be a graded Lie lattice. A grading is a decomposition of the underlying free $\mathbb{Z}$-module compatible with the Lie bracket, and its weight is the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-003824 | MATHDB-347232

Conjecture on recurrence divisibility for algebraic integers

摘录 / EXCERPT: Let $alpha \in \bar{\Z}$, let $K=\Q(\alpha)$, and let $d_k(\alpha)$ denote the largest positive integer such that ... Assume that either (a) $[\Q(\alpha^r):\Q]\geq 3$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-003825 | MATHDB-347244

The Furusho conjecture on algebraic relations between real and p-adic multiple zeta values

摘录 / EXCERPT: Let p) be a prime, and let $\zeta(\mathfrak{s})$ and $\zeta_p(\mathfrak{s})$ denote the real-valued and p-adic multiple zeta values associated with an admissible index...

conjecture | open | MathDB 页面 | 原文线索

V2-003826 | MATHDB-347281

The genus-count formula for the exterior of K13n586

摘录 / EXCERPT: Let B be the exterior of the knot $K13n586$, and let $\atil_B(n)=a_B(n+1)$ count connected essential surfaces of genus $n+1$. Let $\phi(n)$ denote Euler's totient function. Genus...

conjecture | open | MathDB 页面 | 原文线索

V2-003827 | MATHDB-347294

Robinson's solvability conjecture for finite groups with few Sylow subgroups

摘录 / EXCERPT: Let G be a finite group, and let $v_p(G)$ denote the number of Sylow p-subgroups of G. Robinson's solvability conjecture. If ... for each odd prime number p, then G is solv...

conjecture | open | MathDB 页面 | 原文线索

V2-003828 | MATHDB-347299

Twisted Leopoldt conjecture

摘录 / EXCERPT: Let F be a number field, let p be an odd prime, let S be the set of p-adic and infinite primes of F, and let $G_S(F)$ be the Galois group of the maximal algebraic extensi...

conjecture | open | MathDB 页面 | 原文线索

V2-003829 | MATHDB-347313

Shimizu-Hirakawa conjecture on coefficients of fundamental units

摘录 / EXCERPT: Let $p\neq 3$ be a prime number, let $P=p$ or $2p$, and let ... be the fundamental unit of $\mathbb{Q}(P^{1/3})$, where $\alpha,\beta,\gamma\in(1/3)\mathbb{Z}$. Shimizu-Hirakawa conj...

conjecture | open | MathDB 页面 | 原文线索

V2-003830 | MATHDB-347330

The weak converse of Zeckendorf's theorem for measurable Zeckendorf subsets

摘录 / EXCERPT: Weak converse of Zeckendorf's theorem. Given a finite list of positive integers ... -Zeckendorf subset of $$ that is not an open interval has measure zero.

conjecture | open | MathDB 页面 | 原文线索

V2-003831 | MATHDB-347331

Bourgain-Gamburd-Sarnak's Markoff transitivity conjecture

摘录 / EXCERPT: Bourgain-Gamburd-Sarnak's conjecture. For every prime ℓ, the action of Γ on $X^*(\ell)$ is transitive.

conjecture | open | MathDB 页面 | 原文线索

V2-003832 | MATHDB-347333

The vanishing conjecture for the anti-periodization at reciprocal square roots

摘录 / EXCERPT: Let f be the anti-periodization of the second zeta function considered in the paper, and let n be a positive integer. Vanishing conjecture. ... The claim addresses cases where th...

conjecture | open | MathDB 页面 | 原文线索

V2-003833 | MATHDB-347344

Equidistribution conjecture for bilinearly enhanced Picard groups of hyperelliptic curves

摘录 / EXCERPT: Fix an odd prime ℓ, a finite field $\mathbb F_q$ with $q\equiv 1\pmod{\ell^n}$ but $q\not\equiv1\pmod{\ell^{n+1}}$, and let $\mathbb cH_{g,\mathbb F_q}$ be the set of smooth p...

conjecture | open | MathDB 页面 | 原文线索

V2-003834 | MATHDB-347350

Conjecture on the rarity of block repdigits in Narayana's cows sequence

摘录 / EXCERPT: Let (N_n) be the Narayana sequence, and let an m-block repdigit be a positive integer formed by repeating the same decimal block of length m at least twice. Block-repdigit ra...

conjecture | open | MathDB 页面 | 原文线索

V2-003835 | MATHDB-347369

Brent's conjecture on polynomial interpolation of normalized Hecke-group invariants

摘录 / EXCERPT: Brent's interpolation conjecture. For every integer $n>-2$, there exists a polynomial $A_n(x)\in\mathbb{Q}[x]$ such that $\alpha_n(m)=A_n(m)$ for $m=3,4,\ldots$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-003836 | MATHDB-347370

Brent's nonvanishing conjecture for Fourier coefficients of Hecke-group invariants

摘录 / EXCERPT: Brent's nonvanishing conjecture. For each fixed integer $n\geq -1$ and every integer $m\geq 3$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-003837 | MATHDB-347371

Brent's root-location conjecture for the interpolation polynomials

摘录 / EXCERPT: Let $a_n(x)$ be the irreducible factor occurring in the conjectured factorization of $A_n(x)$. Brent's root-location conjecture. If $n>1$, $a_n(\rho)=0$, and $\rho\neq\pm2$, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-003838 | MATHDB-347372

Brent's polynomial interpolation conjecture for the normalized Hecke-group form $H_{4,m}$

摘录 / EXCERPT: Let $H_{4,m}$ have Fourier expansion ... where $q_m(z)=\exp(2\pi i z/\lambda_m)$ and $m\geq3$. Brent's $H_{4,m}$ interpolation conjecture. For every n, there is a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-003839 | MATHDB-347373

Brent's polynomial interpolation conjecture for the normalized Hecke-group form $H_{6,m}$

摘录 / EXCERPT: Let $H_{6,m}$ have Fourier expansion ... where $q_m(z)=\exp(2\pi i z/\lambda_m)$ and $m\geq3$. Brent's $H_{6,m}$ interpolation conjecture. For every n, there is a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-003840 | MATHDB-347374

Brent's nonvanishing conjecture for interpolated Hecke-group cusp-form coefficients

摘录 / EXCERPT: Brent's cusp-coefficient nonvanishing conjecture. There are corresponding polynomials $T_n\in\mathbb{Q}[x]$ satisfying $\tau_m(n)=T_n(m)$ for $m=3,4,\ldots$, together with the fact...

conjecture | open | MathDB 页面 | 原文线索

V2-003841 | MATHDB-347379

Unbounded torsion-prime conjecture for syzygies of Veronese embeddings

摘录 / EXCERPT: Let $r\geq 7$. For each d, consider the d-uple embedding of $\mathbb P^r$ and its Betti table. Unbounded torsion-prime conjecture. As $d\to\infty$, the number of primes ℓ...

conjecture | open | MathDB 页面 | 原文线索

V2-003842 | MATHDB-347412

Moretó's element-order-count characterization conjecture for finite simple groups

摘录 / EXCERPT: Let S be a finite simple group, and let p be the largest prime divisor of $|S|$. For a group H, let $H(p)$ denote the set of elements of order p; equivalently, the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-003843 | MATHDB-347448

The Mersenne divisor conjecture for sums of powers

摘录 / EXCERPT: Let M be a Mersenne prime and let $h\geq 2$. The polynomial sum-of-divisors function is denoted by σ. Mersenne divisor conjecture. The integer $\sigma(M^{2h})$ is divisibl...

conjecture | open | MathDB 页面 | 原文线索

V2-003844 | MATHDB-347449

The 2-Mersenne divisor conjecture

摘录 / EXCERPT: Let S be a 2-Mersenne prime and let $h\geq 2$. The polynomial sum-of-divisors function is denoted by σ. 2-Mersenne divisor conjecture. The integer $\sigma(S^{2h})$ is divi...

conjecture | open | MathDB 页面 | 原文线索

V2-003845 | MATHDB-347450

The odd-divisor classification conjecture for even nonsplitting perfect polynomials

摘录 / EXCERPT: Let A be an even nonsplitting perfect polynomial over $\mathbb{F}_2$, and consider its odd divisors. A prime is called Mersenne or 2-Mersenne according to the terminology of the...

conjecture | open | MathDB 页面 | 原文线索

V2-003846 | MATHDB-347451

Weak Mertens conjecture

摘录 / EXCERPT: Weak Mertens conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003847 | MATHDB-347454

Bounded-size conjecture for the set of minors of LT-superregular matrices

摘录 / EXCERPT: Let $\gamma\geq 2$ and let p be an odd prime. For a vector $(a_1,a_2,\dots,a_{\gamma-1})\in\mathbb{F}_p^{\gamma-1}$, let S_γ be the set associated with this vector in the...

conjecture | open | MathDB 页面 | 原文线索

V2-003848 | MATHDB-347487

Conjecture for iterated irreducibility over global function and number fields

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}$ or $\mathbb{F}_p(t)$, where p is prime. For integers $d\geq 2$, define $K\in\mathcal{N}_{d,n}^{\infty}$ to mean that K admits the...

conjecture | open | MathDB 页面 | 原文线索

V2-003849 | MATHDB-347490

The density asymptotic conjecture for prime-power map preimages

摘录 / EXCERPT: Let $\mathbf{P}$ be the prime-power map, let A denote the auxiliary map defined in the paper, and let $\mathbf{d}_x(S)$ denote the proportion of elements of S in...

conjecture | open | MathDB 页面 | 原文线索

V2-003850 | MATHDB-347494

Infinitude of admissible offsets in the Erdős-Graham equation

摘录 / EXCERPT: Infinitude conjecture. The set $\mathcal{U}$ is infinite; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-003851 | MATHDB-347497

The torsion and B-free conjectures for cyclotomic points on hyperelliptic Jacobians

摘录 / EXCERPT: Let ℓ be a prime, $k\geq 0$, and d a fixed dimension. A cyclotomic point means a rational point defined over a cyclotomic extension, and $D_{\ell^{k+1}}$ denotes the dihedra...

conjecture | open | MathDB 页面 | 原文线索

V2-003852 | MATHDB-347520

Atkin-Swinnerton-Dyer integrality conjecture for vector-valued modular functions

摘录 / EXCERPT: Let $\vec v(q)$ be a vector-valued modular function with components $v_i(q)$, and write the Fourier expansion of $v_i(q)$ with leading exponent $m_0^{(i)}$ and Fourier coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-003853 | MATHDB-347614

Ordinary Weight Conjecture for spetses

摘录 / EXCERPT: Let ${\mathbb{G}}$ be a simply connected ${\mathbb{Z}}_\ell$-spets, let $\ell>2$ be very good for ${\mathbb{G}}$, and let q be a power of a prime different from ℓ. Let B_0...

conjecture | open | MathDB 页面 | 原文线索

V2-003854 | MATHDB-347615

Ordinary Weight Conjecture for ell-compact groups

摘录 / EXCERPT: Let X be a simply connected ℓ-compact group with Weyl group (W,L), with $\ell>2$ very good, and let τ be an automorphism of X of finite order prime to ℓ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003855 | MATHDB-347647

Root-number bias conjecture for modular newforms of arbitrary level

摘录 / EXCERPT: Root-number bias conjecture. For any $M>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-003856 | MATHDB-347671

Belov-Kontsevich conjecture on Weyl algebra automorphisms and polynomial Poisson automorphisms

摘录 / EXCERPT: Let $A_{n,\mathbb{C}}$ be the Weyl algebra and let $P_{n,\mathbb{C}}$ be the commutative polynomial algebra equipped with its Poisson bracket. For an infinite prime $[p]$, let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-003857 | MATHDB-347674

Serre-Sutherland-Zywina surjectivity conjecture for mod-\ell representations

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a non-CM elliptic curve, and let $\rho_{E,\ell}$ denote its mod-ℓ Galois representation. Serre-Sutherland-Zywina conjecture. If $\ell>37$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-003858 | MATHDB-347685

Bateman-Horn conjecture over function fields

摘录 / EXCERPT: Function-field Bateman-Horn conjecture. As $X\to\infty$ through powers of q,

conjecture | open | MathDB 页面 | 原文线索

V2-003859 | MATHDB-347695

Fink's conjecture on ample numbers avoiding the first primes

摘录 / EXCERPT: Let $a(n)$ denote the recursive divisor function, and call a positive integer n ample when $a(n)>n$. Let $p_1,p_2,\ldots$ be the primes in increasing order. Fink's conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003860 | MATHDB-347702

Bounded complement approximation conjecture for delta invariants

摘录 / EXCERPT: Bounded complement approximation conjecture. There is a natural number N depending only on (X,Δ) and a sequence of prime divisors $\{E_i\}$ over X such that

conjecture | open | MathDB 页面 | 原文线索

V2-003861 | MATHDB-347707

Jakobson's variation of the prime k-tuples conjecture for sums of two squares

摘录 / EXCERPT: Let $\mathcal{H}^*=\{h_1,h_2,\ldots\}$ be an ordered set of integers. It is admissible if every finite truncation $\{h_1,\ldots,h_k\}$ is admissible, meaning that its reduction mod...

conjecture | open | MathDB 页面 | 原文线索

V2-003862 | MATHDB-347764

Dimension-generating-function conjecture for multiple t-values and multiple Morita values

摘录 / EXCERPT: Let MtV_w, MMV_w, and MMVe_w denote the weight-w spaces of multiple t-values, multiple Morita values, and multiple Morita values with the indicat...

conjecture | open | MathDB 页面 | 原文线索

V2-003863 | MATHDB-347775

The Diaconu-Goldfeld-Hoffstein secondary-term conjecture for quadratic Dirichlet moments

摘录 / EXCERPT: Diaconu-Goldfeld-Hoffstein conjecture. As $D\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003864 | MATHDB-347776

The absolute-convergence conjecture for the auxiliary Z-series

摘录 / EXCERPT: Absolute-convergence conjecture. The series

conjecture | open | MathDB 页面 | 原文线索

V2-003865 | MATHDB-347783

Mazur's main conjecture for modular forms

摘录 / EXCERPT: Mazur's main conjecture for modular forms.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-003866 | MATHDB-347789

Prime realization conjecture for exceptional harmonic-sum values

摘录 / EXCERPT: For a prime p, let condition (4.1) be the harmonic-sum congruence defined in the paper, and count the integers d with $d\leq p-1$ for which that condition is not satisfied. Pri...

conjecture | open | MathDB 页面 | 原文线索

V2-003867 | MATHDB-347831

Pieropan-Smeets-Tanimoto-Várilly-Alvarado counting conjecture for Campana points

摘录 / EXCERPT: Let K be a number field, let (X,D_{ϵ}) be a smooth klt Fano Campana orbifold over K, and let $(\mathcal{X},\mathcal{D}_{\epsilon})$ be a regular $\mathcal{O}_{K,S}$-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-003868 | MATHDB-347832

Uniform multidimensional equidistribution conjecture for prime products

摘录 / EXCERPT: Let $q>1$ be an integer, let $F=\{p_1,\dots,p_k\}$ be a finite set of primes that does not contain all the primes dividing q, and let a be a positive integer. Say that an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-003869 | MATHDB-347845

Shafarevich conjecture for projective hypersurfaces

摘录 / EXCERPT: Let $r\geq 2$ and $n\geq 1$. For a number field K and a finite set S of finite places of K, write $\mathcal{O}_{K,S}$ for the ring of S-integers. A smooth hypersurface of d...

conjecture | open | MathDB 页面 | 原文线索

V2-003870 | MATHDB-347880

Sarközy's local extrema conjecture for multiplicative functions

摘录 / EXCERPT: Let $f:\mathbb{N}\to\mathbb{N}$ be a multiplicative function such that, for every $k\geq 0$, there are at least two distinct primes p for which $f(p)\ne p^k$. Sarközy's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-003871 | MATHDB-347881

Elliott's conjecture on correlations of multiplicative functions

摘录 / EXCERPT: Elliott's conjecture. Then, as $X\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-003872 | MATHDB-347883

Connectedness conjecture for the lattice of superabundant numbers

摘录 / EXCERPT: Connectedness conjecture. Given a superabundant number n, either there is a prime p such that np is superabundant or there is a prime q such that n/q is superabundant; mo...

conjecture | open | MathDB 页面 | 原文线索

V2-003873 | MATHDB-347890

Krantz's approximation conjecture for dyadic and prime-power rationals

摘录 / EXCERPT: Krantz's conjecture. Does there exist such a prime p, integer β, and integer n such that

open_problem | open | MathDB 页面 | 原文线索

V2-003874 | MATHDB-347899

Anton's corrected vanishing conjecture for GL_2

摘录 / EXCERPT: Let p be a regular odd prime, let ζ_p be a primitive pth root of unity, set $R=\mathbb Z[\zeta_p,1/p]$, and let $k=\mathbb F_p$. Let $D_1(R)$ denote the diagonal subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-003875 | MATHDB-347918

Assiotis's mixed-moment asymptotics for the Riemann zeta function

摘录 / EXCERPT: Let $\mathcal{Z}(y)$ denote the Hardy-type real-valued normalization of the Riemann zeta function on the critical line. For h in the ranges specified below, consider logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-003876 | MATHDB-347921

The coprime forest conjecture for friends-and-strangers graphs

摘录 / EXCERPT: Coprime forest conjecture. If X is biconnected, then $\operatorname{FS}(X,Y)$ is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-003877 | MATHDB-347922

Infinitude conjecture for nontrivial relative torsion groups in cyclotomic fields

摘录 / EXCERPT: Let $N\geq 2$ and let p be a prime with $p\nmid N$. Let ψ_N be a character of $\mathbb{Q}(\mu_N)$ of order N, and let ${\mathfrak p}_N$ be a prime above p in...

conjecture | open | MathDB 页面 | 原文线索

V2-003878 | MATHDB-347953

Soundararajan's conjecture on smooth numbers in arithmetic progressions

摘录 / EXCERPT: Soundararajan's conjecture. For any fixed value of $A>0$, if q is sufficiently large, depending only on A, with $q\le y^A$ and $(a,q)=1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-003879 | MATHDB-347957

Beilinson's standard conjecture for homologically trivial cycles

摘录 / EXCERPT: Let X_K be the generic fiber of the arithmetic or geometric family, let ${\operatorname{Ch}}^i(X_K)^0$ denote the subgroup of homologically trivial codimension-i cycles, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-003880 | MATHDB-347969

Grand-orbit intersection conjecture for products of projective lines

摘录 / EXCERPT: Let $\mathcal{K}$ be the field under consideration. For an endomorphism g of $\mathbb{P}_1$ and $\alpha\in\mathbb{P}_1(\mathcal{K})$, define its grand orbit by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003881 | MATHDB-347979

The generalized principal-coefficient formula for Binet expansions

摘录 / EXCERPT: Generalized principal-coefficient conjecture. The coefficient a_1 satisfies

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-003882 | MATHDB-347980

The inverse relationship between Zeroing Algorithm runtime and $Q_0(r)$

摘录 / EXCERPT: Inverse-relationship conjecture. $Q_0(r)$ and the run-time have an inverse relationship.

conjecture | open | MathDB 页面 | 原文线索

V2-003883 | MATHDB-347992

Conjecture on the limiting logarithmic average of Fibonacci partitions

摘录 / EXCERPT: Limiting-average conjecture. There exists $B>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-003884 | MATHDB-347998

Generic contact Anosov flow conjecture for zero resonant states

摘录 / EXCERPT: Let M be a compact $(2n+1)$-dimensional manifold and let α be a contact 1-form on M whose corresponding flow is Anosov with orientable stable and unstable bundles. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003885 | MATHDB-348001

The no-gap conjecture for Khovanov homology of prime quasi-alternating links

摘录 / EXCERPT: No-gap conjecture. If L is a prime quasi-alternating link that is not a $(2,n)$-torus link, then the differential grading of $\mathcal{H}(L)$ has no gap.

conjecture | open | MathDB 页面 | 原文线索

V2-003886 | MATHDB-348019

Integrality and irreducibility conjecture for level-N class polynomials

摘录 / EXCERPT: Class-polynomial conjecture. The polynomial $H_{D,N,m}(X)$ belongs to $\mathbb{Z}[X]$ and is irreducible. This is proposed as an analogue of the preceding classical class-polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-003887 | MATHDB-348020

Modularity conjecture for the $K3$ hypersurfaces in the paper

摘录 / EXCERPT: Consider each $K3$ hypersurface listed above, after desingularization when necessary. $K3$ modularity conjecture. Each such $K3$ hypersurface is modular with the weight-3 newform...

conjecture | open | MathDB 页面 | 原文线索

V2-003888 | MATHDB-348021

The period-characterisation conjecture for geometric Eisenstein words

摘录 / EXCERPT: Let $V_{2n}^{\operatorname{dR}}$ be the SL_2-representation appearing above, let $\mathcal{O}(\mathcal{U}_E^{\operatorname{dR}})$ be the Hopf subalgebra generated by Eisenstein w...

conjecture | open | MathDB 页面 | 原文线索

V2-003889 | MATHDB-348048

Rodriguez Villegas's exterior-product conjecture for logarithmic embeddings

摘录 / EXCERPT: Let k be a number field, let S_{∞} be the set of its archimedean places with $|S_{\infty}|\geq 2$, and let $\Lambda_{S_{\infty}}(k)$ be the logarithmic embedding, of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-003890 | MATHDB-348059

Nonexistence conjecture for primitive graphs with prime distance sequence

摘录 / EXCERPT: A primitive graph is a graph whose automorphism group acts primitively on its vertices, and its distance sequence records the distinct finite and infinite distances from a vertex....

conjecture | open | MathDB 页面 | 原文线索

V2-003891 | MATHDB-348096

The Uniformity Conjecture over the rationals

摘录 / EXCERPT: Let $g\geq 2$ be an integer. A smooth curve is a curve defined over $\mathbb{Q}$, and its rational points are its points over $\mathbb{Q}$. Uniformity Conjecture over $\mathbb{Q}$....

conjecture | open | MathDB 页面 | 原文线索

V2-003892 | MATHDB-348097

Distribution conjecture for ${\mathcal T}_2$ in prime-parametrized subfamilies

摘录 / EXCERPT: Subfamily distribution conjecture. The following four limiting densities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-003893 | MATHDB-348121

Degree periodicity conjecture for polynomial maps

摘录 / EXCERPT: Let L/K be a finite extension of number fields and let $m\in\mathbb{N}$. Let ... be a polynomial map from L^m to L^m. For an element $a=(a_1,\ldots,a_m)\in L^m$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-003894 | MATHDB-348123

Harris's period rationality conjecture for weight-one Hilbert modular forms

摘录 / EXCERPT: Harris period conjecture. These combinations give rational bases of the corresponding coherent cohomology groups; in particular, $\omega_f^{\Sigma_\infty}/\det R_f$ is rational.

conjecture | open | MathDB 页面 | 原文线索

V2-003895 | MATHDB-348124

Stark's special-value conjecture for Artin representations

摘录 / EXCERPT: Stark conjecture. If ϱ does not contain the trivial representation, then

conjecture | open | MathDB 页面 | 原文线索

V2-003896 | MATHDB-348140

Bertolin's modular Schanuel conjecture for pairwise non-isogenous elliptic curves

摘录 / EXCERPT: Bertolin's modular Schanuel conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003897 | MATHDB-348141

Modular Schanuel conjecture for the modular j-function

摘录 / EXCERPT: Modular Schanuel conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003898 | MATHDB-348161

Going-up spectrum realization conjecture for simultaneous Diophantine exponents

摘录 / EXCERPT: Going-up spectrum realization conjecture. There is $\underline{\xi}\in\mathbb{R}^{\mathbb{N}}$ such that, for every $N\geq 1$, its first N coordinates $\underline{\xi}_N$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-003899 | MATHDB-348167

The exceptional weak Hardy--Littlewood conjecture

摘录 / EXCERPT: Let $g(n)$ denote the number of representations of an even integer n as a sum of two primes. Exceptional weak Hardy--Littlewood conjecture. Suppose that x is sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-003900 | MATHDB-348178

Completeness bound for positive linear recurrence sequences with coefficients $[1,1,0^k,N]$

摘录 / EXCERPT: Let (H_n) be the positive linear recurrence sequence generated by the coefficient list $[1,1,\underbrace{0,\dots,0}_{k},N]$. Let $f_1=1$ and $f_2=2$, with f_n denoting the Fibo...

conjecture | open | MathDB 页面 | 原文线索

V2-003901 | MATHDB-348179

Monotonicity conjecture for completeness in families of positive linear recurrence sequences

摘录 / EXCERPT: Monotonicity conjecture. If the sequence generated by $[\underbrace{1,\dots,1}_{g},\underbrace{0,\dots,0}_{k},N]$ is complete, then so is the sequence generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-003902 | MATHDB-348181

Restricted smallest-root conjecture for incomplete positive linear recurrence sequences

摘录 / EXCERPT: Restricted smallest-root conjecture. If the sequence generated by $[c_1,\ldots,c_L]$ is incomplete and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-003903 | MATHDB-348184

A Brun-Titchmarsh estimate on average for splitting primes in Kummer extensions

摘录 / EXCERPT: Let a and b be fixed coprime integers with $|a|,|b|>1$. For sufficiently small $0<\epsilon'<\epsilon$, let m be a positive integer and let m' be a prime such that...

conjecture | open | MathDB 页面 | 原文线索

V2-003904 | MATHDB-348190

The modular-product q-series identity of type B

摘录 / EXCERPT: Modular-product q-series conjecture. After setting $y_1=y_2=1$ in the charge-refined identity, the following identity holds:

conjecture | open | MathDB 页面 | 原文线索

V2-003905 | MATHDB-348207

The period dichotomy conjecture for concatenated decimal numbers

摘录 / EXCERPT: Let n be a natural number with $n e 10$, and let $r(n)$ denote the reversal of the decimal digits of n. Let $omega(n)$ be the smallest period and $omega_f(n)$ the constructivel...

conjecture | open | MathDB 页面 | 原文线索

V2-003906 | MATHDB-348208

The density-one conjecture for numbers never reaching a v-palindromic number

摘录 / EXCERPT: Density-one conjecture. The asymptotic density of T in S is 1. The paper proves that an explicitly described infinite family has $c(n)=\infty$, and asks for a simple way to d...

conjecture | open | MathDB 页面 | 原文线索

V2-003907 | MATHDB-348209

Coates-Sinnott conjecture for higher K-groups

摘录 / EXCERPT: Let L/K be a finite abelian extension of number fields and let $G=\operatorname{Gal}(L/K)$. Let r be a negative integer, and let S be a finite set of places of K containing...

conjecture | open | MathDB 页面 | 原文线索

V2-003908 | MATHDB-348226

Velani-Badziahin strengthening of Littlewood's conjecture

摘录 / EXCERPT: Let α,β be any two real numbers. Velani-Badziahin conjecture. There exist infinitely many positive integers n such that ... This is presented as a strengthening of Littl...

conjecture | open | MathDB 页面 | 原文线索

V2-003909 | MATHDB-348229

The $2L-1$ conjecture for completeness of positive linear recurrence sequences

摘录 / EXCERPT: Let $[c_1,\dots,c_L]$ define a positive linear recurrence sequence (PLRS) $\\{H_n\\}$, and let $B_{H,n}$ denote its nth Brown's gap. The $2L-1$ conjecture. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-003910 | MATHDB-348246

Weak Greenberg's generalized conjecture for maximal multiple $\p$-extensions

摘录 / EXCERPT: Let p be a prime number and k a number field. Let $\widetilde{k}$ be the maximal multiple $\mathbb{Z}_p$-extension field of k, let $L_{\widetilde{k}}$ be its maximal unramifi...

conjecture | open | MathDB 页面 | 原文线索

V2-003911 | MATHDB-348286

Quaternionic pairing conjecture for logarithms of algebraic numbers

摘录 / EXCERPT: Let F be a totally real number field of narrow class number 1, let $\mathfrak p$ be a prime of F, and let K_i/F be quadratic extensions in which $\mathfrak p$ is inert, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-003912 | MATHDB-348305

Descent conjecture for the Hodge-correlator map to the modular complex

摘录 / EXCERPT: Let E be one of the CM elliptic curves $E=\mathbb{C}/\mathbb{Z}[i]$ or $E=\mathbb{C}/\mathbb{Z}[\rho]$, let $\mathfrak{p}$ be the relevant prime ideal, let T_k be the coefficie...

conjecture | open | MathDB 页面 | 原文线索

V2-003913 | MATHDB-348309

Existence conjecture for nice Browkin continued fractions

摘录 / EXCERPT: Let p be an odd prime, let $\mathcal{Y}$ denote the set used in the paper for the allowed partial quotients, and let a nice BCF be a finite Browkin continued fraction satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-003914 | MATHDB-348310

Extension conjecture for nice Browkin continued fractions

摘录 / EXCERPT: Let p be an odd prime, let $\mathcal{Y}$ be the set of allowed partial quotients, and let $a_0,\ldots,a_{t-2}\in\mathcal{Y}$ satisfy $a_i>0$ and $|a_0|_\infty<\frac p4$. Extensio...

conjecture | open | MathDB 页面 | 原文线索

V2-003915 | MATHDB-348311

Infinitely many quadratic irrationals with every admissible even BCF period

摘录 / EXCERPT: Let p be a prime and let $t\geq1$, excluding the case $p=3$, $t=1$. A Browkin continued fraction (BCF) is a p-adic continued fraction expansion, and $\mathcal{Y}$ is the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-003916 | MATHDB-348323

Le's uniqueness conjecture for exponential Diophantine equations

摘录 / EXCERPT: Le's uniqueness conjecture. The equation has at most one solution in integers $x,y,z>1$. This conjecture concerns the uniqueness of solutions to a class of exponential Diophantine...

conjecture | open | MathDB 页面 | 原文线索

V2-003917 | MATHDB-348324

Jeśmanowicz' conjecture for primitive Pythagorean triples

摘录 / EXCERPT: Jeśmanowicz' conjecture. The equation has only the positive integer solution $(x,y,z)=(2,2,2)$. Jeśmanowicz' conjecture is a classical uniqueness problem for exponential Diophantin...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-003918 | MATHDB-348332

Conjecture on nonconstant values of multiplicative functions at polynomial arguments

摘录 / EXCERPT: Multiplicative functions at polynomials conjecture. If (P,g) is admissible and $v\in g(\mathbb{N})$, then there are infinitely many $n\in\mathbb{N}$ such that $g(P(n))\neq v$.

conjecture | open | MathDB 页面 | 原文线索

V2-003919 | MATHDB-348338

The projective-prime infinitude conjecture

摘录 / EXCERPT: Let $n\ge 2$ be an integer and let $q\ge 2$ be a prime power. Define the projective degree ... A projective prime is a prime value of m. The projective-prime infinitude conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003920 | MATHDB-348339

The uniqueness conjecture for common projective-group degrees

摘录 / EXCERPT: For integers $n,k\ge 2$ and prime powers $q,r\ge 2$, let ${\rm L}_n(q)$ and ${\rm L}_k(r)$ be projective special linear groups with natural degrees ... respectively. A natural degree...

conjecture | open | MathDB 页面 | 原文线索

V2-003921 | MATHDB-348340

The fixed-exponent projective-prime conjecture

摘录 / EXCERPT: Fix a prime integer $n\ge 3$, and let p range over prime numbers. Define ... The fixed-exponent projective-prime conjecture. For every fixed prime $n\ge 3$, there are infinitely ma...

conjecture | open | MathDB 页面 | 原文线索

V2-003922 | MATHDB-348341

The cubic-prime-power projective-prime conjecture

摘录 / EXCERPT: Let p be a prime, set $q=p^3$, and define ... The cubic-prime-power projective-prime conjecture. There are infinitely many values of $q=p^3$ such that $m=1+q+q^2$ is prime. The con...

conjecture | open | MathDB 页面 | 原文线索

V2-003923 | MATHDB-348350

Equality of power indices for divisor voting systems of prime multiples

摘录 / EXCERPT: Prime-multiple power-index conjecture. The divisors of the divisor voting systems of pn and mn have the same Shapley-Shubik power indices and the same Banzhaf power indices.

conjecture | open | MathDB 页面 | 原文线索

V2-003924 | MATHDB-348356

The non-Liouville inhomogeneous multiplicative approximation conjecture

摘录 / EXCERPT: Non-Liouville inhomogeneous multiplicative approximation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-003925 | MATHDB-348357

The one-dimensional Diophantine condition conjecture for inhomogeneous approximation

摘录 / EXCERPT: One-dimensional Diophantine condition conjecture. Suppose that

conjecture | open | MathDB 页面 | 原文线索

V2-003926 | MATHDB-348359

Dual inhomogeneous Gallagher problem in higher dimensions

摘录 / EXCERPT: Dual Gallagher problem. For almost all $\alpha_k\in\mathbb R$, there exist infinitely many $(n_1,\ldots,n_k)\in\mathbb Z^k$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-003927 | MATHDB-348362

The effective abc conjecture

摘录 / EXCERPT: Effective abc conjecture. There exists a function ψ such that, for every abc sum in K,

conjecture | open | MathDB 页面 | 原文线索

V2-003928 | MATHDB-348363

The uniform abc conjecture

摘录 / EXCERPT: Uniform abc conjecture. There exists a function ψ such that, for every such sum and every field of definition K,

conjecture | open | MathDB 页面 | 原文线索

V2-003929 | MATHDB-348364

The falsifiable abc conjecture

摘录 / EXCERPT: Falsifiable abc conjecture. For every such sum,

conjecture | open | MathDB 页面 | 原文线索

V2-003930 | MATHDB-348373

Main parity conjecture for eta-quotients

摘录 / EXCERPT: Main parity conjecture for eta-quotients. The following assertions hold: (i) for every F, δ_F exists and satisfies $\delta_F\le 1/2$; (ii) if $\delta_F=1/2$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-003931 | MATHDB-348374

Self-similarity conjecture for the 9-regular partition function

摘录 / EXCERPT: Self-similarity conjecture for b_9. Theorem ... hold for every prime $p\equiv\pm1\pmod 9$.

conjecture | open | MathDB 页面 | 原文线索

V2-003932 | MATHDB-348395

Unicity conjecture for the generalized Markov values m(c(n,d))

摘录 / EXCERPT: Let $c(n,d)$ be the path associated with (n,d) in the generalized Markov construction, and let $m(c(n,d))$ denote its associated value. Unicity conjecture. If ... then ... Thus, with...

conjecture | open | MathDB 页面 | 原文线索

V2-003933 | MATHDB-348426

The level Hurwitz class number identity for square indices

摘录 / EXCERPT: Let M satisfy $2\leq M\leq 10$ or $M\in\{12,13,16,18,25\}$, and let N be a positive integer coprime to M. Let $\delta_M(1,N)$ be the number defined in Theorem ... . Level Hurwi...

conjecture | open | MathDB 页面 | 原文线索

V2-003934 | MATHDB-348465

Indecomposability conjecture for multiple Dedekind symbols

摘录 / EXCERPT: Let Γ be the modular group, let $\mathcal{A}$ be the indexing set used to define the multiple modular form space $\mathcal{M}_{\mathcal{A}}(\Gamma)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003935 | MATHDB-348472

Kim-Lim-Lovejoy odd-balanced unimodal sequence congruence conjecture

摘录 / EXCERPT: Let $v(n)$ count odd-balanced unimodal sequences of size $2n+2$, and let $v(m,n)$ count those with rank m. Let $\ell\not\equiv-1\pmod8$ be prime and let k be a positive integer...

conjecture | refuted | MathDB 页面 | 原文线索

V2-003936 | MATHDB-348473

The Andrews spt-function mod 4 conjecture

摘录 / EXCERPT: Let $\operatorname{spt}(n)$ denote the number of occurrences of the smallest part among the partitions of n. Let $\ell>3$ be prime with $\ell\not\equiv23\pmod{24}$, and define .....

conjecture | refuted | MathDB 页面 | 原文线索

V2-003937 | MATHDB-348474

Kim-Lim-Lovejoy congruence conjecture for the mock theta function A(q)

摘录 / EXCERPT: Let $N_A(n)$ be defined by ... Let $p\not\equiv7\pmod8$ be an odd prime, let δ_p satisfy ... and let $k,n\in\mathbb Z$ with $\left(\frac{k}{p}\right)=1$. Kim-Lim-Lovejoy's...

conjecture | open | MathDB 页面 | 原文线索

V2-003938 | MATHDB-348483

Uniform boundedness conjecture for Waring's problem over diagonal forms

摘录 / EXCERPT: Let $k\in\mathbb{N}$. For each positive integer-valued function $C:\mathbb{N}\rightarrow\mathbb{N}$, let $P_C(k)=\sup_{l\geq 2}\{S_C(k,l)\}$, where $S_C(k,l)$ is the minimum number...

conjecture | open | MathDB 页面 | 原文线索

V2-003939 | MATHDB-348488

Nilpotent freeness conjecture for the May spectral sequence

摘录 / EXCERPT: Let HX denote the algebra given by the E_2 page of the May spectral sequence at the prime 2. An element is nilpotent if some positive power is zero. Nilpotent freeness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003940 | MATHDB-348496

Erdős's infinitude conjecture for Ruth-Aaron numbers

摘录 / EXCERPT: For a positive integer $n=\prod_{i=1}^k p_i^{a_i}$, define ... A Ruth-Aaron number is an integer n satisfying $f(n)=f(n+1)$. Erdős's infinitude conjecture. There are infinitely man...

conjecture | open | MathDB 页面 | 原文线索

V2-003941 | MATHDB-348497

The finite Ruth-Aaron triple conjecture

摘录 / EXCERPT: For a positive integer $n=\prod_{i=1}^k p_i^{a_i}$, define ... A Ruth-Aaron triple is a sequence $n,n+1,n+2$ of consecutive integers with equal values of f. Finite Ruth-Aaron tripl...

conjecture | open | MathDB 页面 | 原文线索

V2-003942 | MATHDB-348520

Artin prime-pair asymptotic conjecture

摘录 / EXCERPT: Fix an even integer d and an integer $g\in\mathbb{Z}\setminus\{-1\}$ satisfying the hypotheses in the source, and let $[d,g]$ be the Artin-admissibility condition. Let $\pi_d(x)$...

conjecture | open | MathDB 页面 | 原文线索

V2-003943 | MATHDB-348521

Artin prime-pair error-term conjecture

摘录 / EXCERPT: Let $E_{d,g}(x)$ be the error contribution defined by ... where $S_{k,l}(x;d,g)$ is the sum defined in the source, and let $[d,g]$ be Artin admissible. Artin prime-pair error-term co...

conjecture | open | MathDB 页面 | 原文线索

V2-003944 | MATHDB-348524

The rank 1 abelian Stark conjecture for differenced ray class zeta functions

摘录 / EXCERPT: Let K be a real quadratic field and let $\{\rho_1,\rho_2\}$ be its real embeddings. Let $\cc$ be an ideal of the maximal order $\OO_K$, let $S=\{\rho_2\}$, and let A be a ray i...

conjecture | open | MathDB 页面 | 原文线索

V2-003945 | MATHDB-348570

Conjecture for mixed moments of iterated zeta derivatives

摘录 / EXCERPT: Let U be uniformly distributed on $[0,1]$, let $m\in\mathbb{N}$, and let ${\boldsymbol h}=(h_1,\ldots,h_m)$ with $|{\boldsymbol h}|=h_1+\cdots+h_m$. Write $\zeta^{(j)}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-003946 | MATHDB-348571

Keating-Rodgers-Roditty-Gershon-Rudnick conjecture for divisor-function short intervals

摘录 / EXCERPT: Let U_n be uniformly distributed on $\{n+1,\ldots,2n\}$, let $d_k(N)$ be the k-th divisor function, and set $\Delta_k(U_n)=d_k(U_n)-\mathbb{E}(d_k(U_n))$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003947 | MATHDB-348573

Asymptotic polarity conjecture for weak Jacobi forms

摘录 / EXCERPT: For each positive integer m, let $J_{0,m}$ denote the space of weak Jacobi forms of weight 0 and index m, and define ... Let $P(m)$ be the integer such that $J^{P(m)}_{0,m}=0$...

conjecture | open | MathDB 页面 | 原文线索

V2-003948 | MATHDB-348584

Determinantal positivity conjecture for the Nyman-Báez-Duarte functions

摘录 / EXCERPT: For each integer $k\ge2$, let $f_k:(0,1]\to\mathbb{R}$ be defined by ... Equip $L^2(0,1)$ with inner product $\langle g,h\rangle:=\int_0^1g(x)h(x)\,dx$. Determinantal positivity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-003949 | MATHDB-348587

Paley graph conjecture

摘录 / EXCERPT: Paley graph conjecture. For each $0<\alpha\leq1$, there exist $\beta=\beta(\alpha)>0$ and $p(\alpha)>0$ such that $\mathcal{P}(\alpha,\beta)$ holds for every prime $p>p(\alpha)$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-003950 | MATHDB-348605

Arithmetic Simpson correspondence for periodic parabolic bundles

摘录 / EXCERPT: Let C be a smooth projective curve. The grading functor Gr sends periodic parabolic de Rham bundles over C to periodic parabolic Higgs bundles over C. Arithm...

conjecture | open | MathDB 页面 | 原文线索

V2-003951 | MATHDB-348621

Ternary Diophantine approximation conjecture for primes of the form x^2+y^2+1

摘录 / EXCERPT: Conjecture on ternary approximation. There exists $c_0>1$ such that, for any fixed $1<c<c_0$ and every sufficiently large positive number N, the inequality

conjecture | open | MathDB 页面 | 原文线索

V2-003952 | MATHDB-348680

The n-dimensional finite-field (U_k)-pattern conjecture

摘录 / EXCERPT: Fix a prime p and let $\pi_p:\mathbb{Z}\to\mathbb{F}_p$ be the canonical projection. For $U\subset\mathbb{Z}$, a U-pattern in $\mathbb{F}_p^n$ is a set of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-003953 | MATHDB-348685

The degree conjecture for the extended Selberg class

摘录 / EXCERPT: Let ${\mathfrak A}^{\#}_d$ denote the set of functions $F(s)$ in the extended Selberg class ${\mathfrak A}^{\#}$ whose degree is $d_F=d$, where d_F is the degree determined b...

conjecture | open | MathDB 页面 | 原文线索

V2-003954 | MATHDB-348687

Stauduhar's conjecture on the number of distinct factorial residues

摘录 / EXCERPT: Stauduhar's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-003955 | MATHDB-348691

The explicit ghost conjecture for U_5

摘录 / EXCERPT: For $a=0,\dots,3$, define the ghost series ... where ... with the exponents $m_n^{(a)}(l)$ prescribed by the source. Set ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003956 | MATHDB-348737

Maslova's conjecture on prime graphs of finite groups

摘录 / EXCERPT: Maslova's conjecture. If the prime graph of G does not contain 3-cocliques, then it is isomorphic to the Gruenberg-Kegel graph of some finite solvable group.

conjecture | open | MathDB 页面 | 原文线索

V2-003957 | MATHDB-348758

The logarithmic upper-bound conjecture for the least quadratic nonresidue

摘录 / EXCERPT: Let p be a prime, and let n_p denote the least positive quadratic nonresidue modulo p. Here $f(p)\ll g(p)$ means that there is a constant $C>0$ such that $f(p)\leq Cg(p)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-003958 | MATHDB-348769

Lian's quasimodularity conjecture for elliptic Hurwitz loci

摘录 / EXCERPT: Fix an integer $g\ge2$. Let $\overline{\mathcal{M}}_g$ be the moduli space of stable curves of genus g, and for each $d\ge1$ let $\phi_{g/1,d}$ be the morphism from the compactif...

conjecture | open | MathDB 页面 | 原文线索

V2-003959 | MATHDB-348798

Dold-condition conjecture for discriminant-scaled Lucas sequences along square indices

摘录 / EXCERPT: For integers P,Q, define the Lucas sequence $(U_n(P,Q))$ by ... The sequence is said to satisfy the Dold condition when its associated Dold congruences hold. Lucas-sequence conject...

conjecture | open | MathDB 页面 | 原文线索

V2-003960 | MATHDB-348822

Transcendence-degree conjecture for values of modular forms

摘录 / EXCERPT: Let $\tau\in\mathbf{H}$, where $\mathbf{H}$ is the upper half-plane, and let E_2,E_4,E_6 denote the Eisenstein series and j the modular invariant. Transcendence-degree conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-003961 | MATHDB-348825

The Batyrev-Tschinkel conjecture on the leading constant for point counts

摘录 / EXCERPT: Batyrev-Tschinkel conjecture. Then

conjecture | solved | MathDB 页面 | 原文线索

V2-003962 | MATHDB-348832

Conjectural tt* morphism for the LG/CY correspondence

摘录 / EXCERPT: Let $r^{\prime}:H^{\operatorname{LG}}\to H^{\operatorname{CY}}$ be the correspondence map, and let ... and ... be the Landau-Ginzburg and Calabi-Yau tt^* structures. Conjectural...

conjecture | open | MathDB 页面 | 原文线索

V2-003963 | MATHDB-348834

Infinitely many prime values of the j-function coefficients

摘录 / EXCERPT: Let c_n denote the coefficients of the j-function considered in the paper. Prime-value conjecture. There are infinitely many n such that c_n is prime. The computation finds...

conjecture | open | MathDB 页面 | 原文线索

V2-003964 | MATHDB-348876

Cohen-Sun-Vsemirnov's integral-square conjecture for S(1,p)

摘录 / EXCERPT: Cohen-Sun-Vsemirnov's conjecture. The number $S(1,p)/a$ is an integral square.

conjecture | open | MathDB 页面 | 原文线索

V2-003965 | MATHDB-348890

Infinitely many fully connected CoP pairs over the primes

摘录 / EXCERPT: Let $\mathcal{C}(n,\mathbb{P})$ denote a circle of partition (CoP) based on the set $\mathbb{P}$ of prime numbers. Prime-CoP connectivity conjecture. There are infinitely many pair...

conjecture | open | MathDB 页面 | 原文线索

V2-003966 | MATHDB-348906

Conjecture on simple inter-argument zeros of multiple zeta-functions

摘录 / EXCERPT: Let $r\geq 2$. For $2\leq k\leq r$, let $I_r(k)$ denote the number of inter-argument zeros (IAZs) of $\zeta_r(s)$ in the interval $(1/k,1/(k-1))$. Simple IAZ conjecture. All IAZs o...

conjecture | open | MathDB 页面 | 原文线索

V2-003967 | MATHDB-348909

Conjecture on simple real zeros of multiple zeta-functions

摘录 / EXCERPT: Let $r\geq 2$. For $2\leq k\leq r$, let $I_r(k)$ denote the number of inter-admissible zeros of $\zeta_r(s)$ in the interval $(1/k,1/(k-1))$, where an inter-admissible zero is a re...

conjecture | open | MathDB 页面 | 原文线索

V2-003968 | MATHDB-348910

Conjecture on inter-trivial zeros of multiple zeta-functions

摘录 / EXCERPT: Let $r\geq 1$, and let an inter-trivial zero (ITZ) of $\zeta_r(s)$ mean a real zero on the negative real line other than the known zeros at the negative even integers. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-003969 | MATHDB-348946

Odoni's conjecture on surjective arboreal Galois representations

摘录 / EXCERPT: Let K be a Hilbertian field of characteristic zero, and let $d>1$ be an integer. For a degree-d polynomial $f\in K[x]$, let $\mathrm{T}_\infty(d)$ be the rooted d-ary tree fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003970 | MATHDB-348955

Strong Goldbach conjecture

摘录 / EXCERPT: Strong Goldbach conjecture. Every even natural number greater than six can be written as the sum of two distinct odd primes.

conjecture | open | MathDB 页面 | 原文线索

V2-003971 | MATHDB-348958

Arithmetic nature conjectures for odd Bessel problems

摘录 / EXCERPT: Let k be an odd integer, and let G_{2k+2}^{BR} be a Gröbner basis generating the Broadhurst-Roberts ideal ... Let $\mathbf N_k$ be the matrix of on-shell Bessel moments....

conjecture | open | MathDB 页面 | 原文线索

V2-003972 | MATHDB-348975

Doubling-crossing conjecture for minimal link diagrams

摘录 / EXCERPT: Let $\mathcal{D}$ be an oriented link diagram. Let $\mathcal{D}^\prime$ be a diagram obtained from $\mathcal{D}$ by doubling a crossing, meaning replacing one crossing by the diagr...

conjecture | open | MathDB 页面 | 原文线索

V2-003973 | MATHDB-349002

Sun's infinitude conjecture for the determinants $[c,d]_p$

摘录 / EXCERPT: Sun's conjecture. If d is non-zero and $c^2-4d\ne0$, then $[c,d]_p=0$ for infinitely many primes p.

conjecture | open | MathDB 页面 | 原文线索

V2-003974 | MATHDB-349013

Subexponential upper-bound conjecture for representations by unit fractions

摘录 / EXCERPT: For fixed $k,m\in\mathbb{N}$, let $f_k(m,n)$ denote the number of representations ... with $(a_1,\ldots,a_k)\in\mathbb{N}^k$ in increasing order. Subexponential representation conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-003975 | MATHDB-349021

The four Linnik primes Piatetski-Shapiro inequality conjecture

摘录 / EXCERPT: Four Linnik primes conjecture. There exists $c_0>1$ such that, for any fixed $1<c<c_0$ and every sufficiently large positive number N, the inequality

open_problem | open | MathDB 页面 | 原文线索

V2-003976 | MATHDB-349042

Rosen's lifting conjecture for multiple harmonic sums

摘录 / EXCERPT: Let k be a positive integer and let p_1 be a prime greater than $k+1$. For each index $\boldsymbol{k}$, let $H_{p-1}(\boldsymbol{k})$ be the multiple harmonic sum, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003977 | MATHDB-349043

The refined Kaneko-Zagier conjecture for completed finite and symmetric multiple zeta values

摘录 / EXCERPT: Let $\boldsymbol{k}$ be an index, let $\operatorname{wt}(\boldsymbol{k})$ denote its weight, and let $\zeta_{\widehat{\mathcal{A}}}(\boldsymbol{k})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-003978 | MATHDB-349052

Bremner and Ulas's reducibility-type conjecture for trinomials

摘录 / EXCERPT: Bremner and Ulas's conjecture. The reducibility type $(1,1,1,n-3)$ does not occur for $f(x)$.

conjecture | open | MathDB 页面 | 原文线索

V2-003979 | MATHDB-349063

Main trace conjecture for genuine Picard modular forms

摘录 / EXCERPT: Let $\lambda\neq(0,0,0)$, let ν and μ be as in the paper, and write $\lambda=(a,b,i)$ so that $i\equiv_2 a+b$. Let $T(\nu)$ be the Hecke operator, F_ν the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-003980 | MATHDB-349064

Dimension conjecture for genuine Picard modular forms

摘录 / EXCERPT: Let $E_c(\lambda)$ and $E_{\rm extr}(\lambda)$ be the representations obtained from the compactly supported Euler characteristic and the extraneous contributions, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-003981 | MATHDB-349065

Modular-9 Hecke-trace congruence for genuine forms

摘录 / EXCERPT: Let $j,k,l\geq0$, let ν be a Hecke parameter, and let $\mu\vdash4$ be a partition indexing an $\mathfrak S_4$-isotypic component. The modular-9 trace congruence conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-003982 | MATHDB-349066

Conjectural normalized Euler characteristic formula for genus 2

摘录 / EXCERPT: Let $k>0$ and $l\geq0$ satisfy $k\equiv_6 l$. Let $\mathcal X^{(2)}_{\Gamma[\sqrt{-3}]}$ be the genus-2 moduli space considered in the paper, let $\mathbb W_{k,l}$ be the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-003983 | MATHDB-349067

Harder's conjecture for Picard modular forms

摘录 / EXCERPT: Let m and n be in the critical range for the algebraic Hecke character ψ_m, and define ... Suppose a prime $\ell>m$ divides the denominator of $Q(m,n)$. Harder's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-003984 | MATHDB-349099

The generalized L-function rigidity conjecture for pointed curves

摘录 / EXCERPT: Let $(C,[M])$ and $(C',[M'])$ be smooth irreducible projective pointed curves over a finite field ${\mathbb F}_q$. Let ${\mathfrak m},{\mathfrak m}'$ be moduli on C,C' disjoint f...

conjecture | open | MathDB 页面 | 原文线索

V2-003985 | MATHDB-349100

The generalized Jacobian reconstruction conjecture for pointed curves

摘录 / EXCERPT: Let $(C,[M])$ and $(C',[M'])$ be smooth, projective, irreducible pointed curves over ${\mathbb F}_q$, with moduli ${\mathfrak m},{\mathfrak m}'$ on C,C' respectively. Suppose the...

conjecture | open | MathDB 页面 | 原文线索

V2-003986 | MATHDB-349101

Willems' conjecture on Brauer character degrees

摘录 / EXCERPT: Let G be a finite group and let p be a prime. Write $|G|_{p'}$ for the prime-to-p part of the order of G, and let $\operatorname{IBr}(G)$ denote the irreducible p-Brauer...

conjecture | open | MathDB 页面 | 原文线索

V2-003987 | MATHDB-349109

Hanani's conjecture on additive complement counting functions

摘录 / EXCERPT: Hanani's conjecture. If A and B are infinite additive complements, then

conjecture | open | MathDB 页面 | 原文线索

V2-003988 | MATHDB-349133

Short-arc lattice-point conjecture for circles

摘录 / EXCERPT: Let $N\in\mathbb{N}$ and consider the circle centered at the origin with radius $\sqrt{N}$. An arc of length N^{η}, with $\eta<1/2$, is a short arc on this circle. Short-arc l...

conjecture | open | MathDB 页面 | 原文线索

V2-003989 | MATHDB-349155

Type-2 isomorphism conjecture for circulant graphs with coprime gcd parameters

摘录 / EXCERPT: Let $l,m,n,r otin\mathbb{N}$?

conjecture | open | MathDB 页面 | 原文线索

V2-003990 | MATHDB-349171

Kuznetsov's Fano threefold conjecture on Kuznetsov components

摘录 / EXCERPT: Let $\mathcal{MF}^i_d$ be the moduli spaces of prime Fano threefolds of index i and degree d. A correspondence conjecture. There is a correspondence...

conjecture | open | MathDB 页面 | 原文线索

V2-003991 | MATHDB-349179

Prime-order balanced shift-pattern coloring conjecture

摘录 / EXCERPT: Let $G_{m,n}$ be the grid in question, and let a shifted k-subgrid mean a k-subgrid satisfying the paper's shift-pattern condition. Prime-order balanced coloring conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-003992 | MATHDB-349185

Zauner's conjecture over finite fields

摘录 / EXCERPT: Let d be a positive integer, let q be a prime power, and let $\mathbb{F}_{q^2}$ be the quadratic extension of the finite field with q elements. A unitary geometry on...

conjecture | open | MathDB 页面 | 原文线索

V2-003993 | MATHDB-349219

The algebraicity conjecture for left-Diophantine numbers

摘录 / EXCERPT: A real number $x\in\mathbb{R}$ is left-Diophantine if it is the supremum of a Diophantine subset of $\mathbb{Q}$. Let $\mathscr{D}$ be the set of left-Diophantine numbers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-003994 | MATHDB-349220

The field conjecture for left-Diophantine numbers

摘录 / EXCERPT: Let $\mathscr{D}\subseteq\mathbb{R}$ be the set of left-Diophantine numbers. Field conjecture. The set $\mathscr{D}$ is a field. This is a weaker consequence of the algebraicity co...

conjecture | open | MathDB 页面 | 原文线索

V2-003995 | MATHDB-349221

The weak separation conjecture for left-listable and left-Diophantine numbers

摘录 / EXCERPT: A real number α is left-listable if $L(\alpha)=\{q\in\mathbb{Q}:q<\alpha\}$ is listable. Let $\mathscr{D}$ be the set of left-Diophantine numbers and let Λ be the se...

conjecture | open | MathDB 页面 | 原文线索

V2-003996 | MATHDB-349222

Unbounded positive existential rank for global fields

摘录 / EXCERPT: Let K be a global field and let $\mathscr{G}$ be a finite set of field generators. Consider K as a structure over $\mathscr{L}=\mathscr{L}_a\cup\mathscr{G}$. Global-field rank...

conjecture | open | MathDB 页面 | 原文线索

V2-003997 | MATHDB-349246

Trautmann's conjecture on maximum-size cyclic subspace codes

摘录 / EXCERPT: Let q be a prime power, let n and k be positive integers with $k\leq n/2$, and let $\mathcal{G}_q(n,k)$ denote the set of k-dimensional $\operatorname{\mathbb{F}}_q$-subspa...

conjecture | open | MathDB 页面 | 原文线索

V2-003998 | MATHDB-349255

Jesmanowicz's conjecture on exponential Diophantine equations

摘录 / EXCERPT: Let a,b,c be relatively prime positive integers satisfying ... A solution (x,y,z) in positive integers is nontrivial if it is not $(2,2,2)$. Jesmanowicz's conjecture. The equatio...

conjecture | open | MathDB 页面 | 原文线索

V2-003999 | MATHDB-349272

Odd-power symmetric sum conjecture

摘录 / EXCERPT: Let $m\in\mathbb{N}$ and let x be a positive integer. The coefficients ${\mathbf{A}}_{m,0}, {\mathbf{A}}_{m,1},\ldots,{\mathbf{A}}_{m,m}$ are constants independent of x and k...

conjecture | open | MathDB 页面 | 原文线索

V2-004000 | MATHDB-349311

The hypoenergeticity conjecture for group-annihilator graphs of cyclic prime-power groups

摘录 / EXCERPT: Let p be a prime and let α be a positive integer. Consider the group-annihilator graph realised by the group $\mathbb{Z}/p^{\alpha}\mathbb{Z}$. Hypoenergeticity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-004001 | MATHDB-349341

Erdős-Graham conjecture for equal-colored variables

摘录 / EXCERPT: An equation is partition regular if every finite coloring of $\mathbb{N}$ admits a solution whose variables all have the same color. Erdős-Graham conjecture. For every finite color...

conjecture | open | MathDB 页面 | 原文线索

V2-004002 | MATHDB-349345

Weak monotonicity conjecture for the coefficients of $R^{(k)}_1$

摘录 / EXCERPT: Weak monotonicity conjecture. The coefficients of $R^{(k)}_1$ are weakly increasing for all $k\geq 3$; that is,

conjecture | open | MathDB 页面 | 原文线索

V2-004003 | MATHDB-349360

The packing problem for square-free monomial ideals

摘录 / EXCERPT: Let I be a square-free monomial ideal. Its symbolic power is $I^{(m)}=P_1^m\cap\cdots\cap P_s^m$ when $I=P_1\cap\cdots\cap P_s$ is its irredundant prime decomposition. The ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-004004 | MATHDB-349372

The $AP(\theta)$ conjecture for primes in arithmetic progressions

摘录 / EXCERPT: For $0<\theta\leqslant1$, let $\psi(x;q,a)$ denote the weighted prime-counting function in the arithmetic progression $a\pmod q$, and let $\varphi(q)$ be Euler's totient function....

conjecture | open | MathDB 页面 | 原文线索

V2-004005 | MATHDB-349373

The conjecture on the variance of primes in short intervals

摘录 / EXCERPT: Let $\psi(x)=\sum_{n\leqslant x}\Lambda(n)$ be the Chebyshev function. Short-interval variance conjecture. For all $\varepsilon>0$, if $h\leqslant X^{1-\varepsilon}$, then ... as...

conjecture | open | MathDB 页面 | 原文线索

V2-004006 | MATHDB-349387

The mixed Littlewood conjecture

摘录 / EXCERPT: The mixed Littlewood conjecture. For every real number $\alpha\in\mathbb{R}$ and every pseudo-absolute sequence $\mathcal{D}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-004007 | MATHDB-349416

The GCP(1) conjecture for missing digits and self-similar measures

摘录 / EXCERPT: GCP(1) conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-004008 | MATHDB-349417

Velani's lacunary Khinchine convergence conjecture for the middle-third Cantor measure

摘录 / EXCERPT: Velani's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-004009 | MATHDB-349424

Dichotomy conjecture for root-finding in orders

摘录 / EXCERPT: Dichotomy conjecture. Π_f lies in $\mathcal{P}$ or in $\mathcal{NPC}$.

conjecture | open | MathDB 页面 | 原文线索

V2-004010 | MATHDB-349427

Evans's conjectural hypergeometric relation for weight-two newforms

摘录 / EXCERPT: Let $a_3(n)$ and $a_4(n)$ be the Fourier coefficients of newforms ... whose coefficient fields are $\mathbb{Q}\left(\sqrt{2}\right)$ and $\mathbb{Q}\left(\sqrt{3}\right)$, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-004011 | MATHDB-349428

Beilinson-Deligne conjecture for arithmetic surfaces at $s=1$

摘录 / EXCERPT: Let $S=\operatorname{Spec}\mathcal O_F$ for a number field F, and let X be an arithmetic surface over S, with special zeta value $\zeta^*(X,1)$ and generalized Euler characte...

conjecture | open | MathDB 页面 | 原文线索

V2-004012 | MATHDB-349456

Conjecture on the exact value of the exponent of approximation by rational subspaces

摘录 / EXCERPT: Exact-exponent conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004013 | MATHDB-349503

Maximality conjecture for subfield cliques in generalized Paley graphs

摘录 / EXCERPT: Let d be a positive integer greater than 1. Let $q \equiv 1 \pmod {2d}$ be a power of a prime p, and let r be the largest integer such that $d \mid \frac{q-1}{p^r-1}$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-004014 | MATHDB-349504

Strict clique-number conjecture for quartic Peisert graphs

摘录 / EXCERPT: Let q be a power of a prime p satisfying $p\equiv 3\pmod 4$ and $q>3$. Let $P_{q^4}^*$ be the Peisert graph of order q^4, and let $\omega(P_{q^4}^*)$ denote its clique number...

conjecture | open | MathDB 页面 | 原文线索

V2-004015 | MATHDB-349505

Toth's zero asymptotic-density conjecture for spoof odd perfect numbers

摘录 / EXCERPT: Let $\mathcal{S}$ be the set of odd positive integers n such that $2n/\sigma(n)-1=1/x$ for some positive integer x. Its asymptotic density is the limit, when it exists, of ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-004016 | MATHDB-349506

Toth's zero Schnirelmann-density conjecture for spoof odd perfect numbers

摘录 / EXCERPT: Let $\mathcal{S}$ be the set of odd positive integers n such that $2n/\sigma(n)-1=1/x$ for some positive integer x. For a set of nonnegative integers, its Schnirelmann density...

conjecture | open | MathDB 页面 | 原文线索

V2-004017 | MATHDB-349507

Golden Automaton conjecture on inductive Collatz convergence

摘录 / EXCERPT: Let n be a positive integer, and consider the Collatz dynamic on odd positive integers. The Golden Automaton is the finite proof system described in the source for establishing C...

conjecture | open | MathDB 页面 | 原文线索

V2-004018 | MATHDB-349516

Artin-Tate conjecture for smooth proper surfaces over finite fields

摘录 / EXCERPT: Let $k=\mathbb F_q$ be a finite field, let $T=\operatorname{Spec} k$, and let X be a smooth proper surface over T. Let $P_2(X,t)$ be the characteristic polynomial of Frobenius...

conjecture | open | MathDB 页面 | 原文线索

V2-004019 | MATHDB-349522

Author's even-weight formulas for quadratic Dirichlet L-functions

摘录 / EXCERPT: Author's conjecture. For every even integer $k\ge2$, there exist universal coefficients $c_{j,N}^k$ such that, for every positive fundamental discriminant D-which must additional...

conjecture | open | MathDB 页面 | 原文线索

V2-004020 | MATHDB-349554

Algebraicity characterization for rank-three lift geometries

摘录 / EXCERPT: Let Γ be a nontrivial finite group, and let p be a prime. Lift-geometry algebraicity conjecture. The matroid $\operatorname{LG}^+(3,\Gamma)$ is algebraic over...

conjecture | open | MathDB 页面 | 原文线索

V2-004021 | MATHDB-349555

Algebraicity characterization for rank-three Dowling geometries

摘录 / EXCERPT: Let Γ be a nontrivial finite group, and let p be a prime. Dowling-geometry algebraicity conjecture. The matroid $\operatorname{DG}(3,\Gamma)$ is algebraic over...

conjecture | open | MathDB 页面 | 原文线索

V2-004022 | MATHDB-349566

Positive-discriminant class group p-torsion average conjecture

摘录 / EXCERPT: Let p be a prime number such that $p \neq 3$. The quantity $\mu_p(\mathcal{F}^{+}(Y))$ denotes the average size of the p-torsion subgroup in the class groups of the positive-di...

conjecture | open | MathDB 页面 | 原文线索

V2-004023 | MATHDB-349571

Periodicity conjecture for the heuristic APD-algorithm on cubic conjugate vectors

摘录 / EXCERPT: Periodicity conjecture. The heuristic APD-algorithm is periodic for all triples of cubic conjugate vectors.

conjecture | open | MathDB 页面 | 原文线索

V2-004024 | MATHDB-349582

The first prime-based upper-bound conjecture for domination in Knödel graphs

摘录 / EXCERPT: Let KG_n be the Knödel graph on even n vertices, and let $\gamma(KG_n)$ denote its domination number. Let p be an odd prime with $p\leq\lceil\log n\rceil$, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-004025 | MATHDB-349583

The prime-power upper-bound conjecture for domination in Knödel graphs

摘录 / EXCERPT: Let KG_n be the Knödel graph on even n vertices, and let $\gamma(KG_n)$ denote its domination number. Let p be an odd prime and let $k\geq2$ be an integer. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-004026 | MATHDB-349592

The good reduction conjecture for K3 surfaces

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$ and let F be a finite extension of $W[1/p]$. Let X_F be a K3 surface over F such that...

conjecture | open | MathDB 页面 | 原文线索

V2-004027 | MATHDB-349599

The diagonal conjecture for primary pseudo-polynomials

摘录 / EXCERPT: Diagonal conjecture. If the generating series of a primary pseudo-polynomial is the diagonal of a rational fraction, then $(a_n)_{n\geq 0}$ is a polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-004028 | MATHDB-349616

Density-one vanishing conjecture for signed Selmer groups of rank-zero elliptic curves

摘录 / EXCERPT: Fix a prime p. Let $\mathcal{J}_p$ be the set of rank-zero elliptic curves with good reduction at p, and let $\mathcal{Z}_p$ be the subset for which ... For a set $\mathcal{S}$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-004029 | MATHDB-349631

Cori's lower-bound conjecture for the sequence F_n

摘录 / EXCERPT: Let $(F_n)_{n\geq 0}$ be the sequence defined by the matrix construction described in the paper. Cori's conjecture. ... for every $n>7$. This is one of the number-theoretic conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-004030 | MATHDB-349632

Cori's divergence conjecture for the sequence F_n

摘录 / EXCERPT: Let $(F_n)_{n\geq 0}$ be the sequence defined by the matrix construction described in the paper. Cori's divergence conjecture. ... This is the second conjecture posed by Cori and is...

conjecture | open | MathDB 页面 | 原文线索

V2-004031 | MATHDB-349633

Divergence conjecture for the sequence a_n

摘录 / EXCERPT: For each n, let a_n denote the number of elements in row n of the matrix F having track vector of the form $(a,(1)^p,b)$ with $a,b,p\geq 1$. Conjecture on a_n. ... more pre...

conjecture | open | MathDB 页面 | 原文线索

V2-004032 | MATHDB-349634

Breadth-three divergence conjecture for the matrix F

摘录 / EXCERPT: For each n, let c_n be the number of elements in row n of the matrix F with breadth 3. Breadth-three conjecture. ... The source describes this conjecture as being supported...

conjecture | open | MathDB 页面 | 原文线索

V2-004033 | MATHDB-349635

Conjecture on the extremal rows of elements created at each step

摘录 / EXCERPT: Consider the elements of the matrix F created at Step i, and let r_i be the maximal index of a row containing such an element. Extremal-row conjecture. The odd-indexed sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-004034 | MATHDB-349636

Context-free grammar for minimal recursive prime factorizations

摘录 / EXCERPT: Minimal-spelling grammar conjecture. The language underlying $\text{RPF}_{\mathbb{N}_{r_{\text{min}}}}$ is generated by this context-free grammar. The grammar is ambiguous, with tw...

conjecture | open | MathDB 页面 | 原文线索

V2-004035 | MATHDB-349670

Farjoun's finite p-group surjectivity conjecture for group localizations

摘录 / EXCERPT: Let L be a localization on the category of groups, and let G be a finite p-group, meaning a finite group whose order is a power of the prime p. Write $G\to LG$ for the loca...

conjecture | open | MathDB 页面 | 原文线索

V2-004036 | MATHDB-349673

A general partial-fraction relation for derivatives of an entire function

摘录 / EXCERPT: Let F be the function under consideration, with zeros a_k indexed by $k\geq 0$, let L be the derivative order, let m be a positive integer, and let ω_m denote an m...

conjecture | open | MathDB 页面 | 原文线索

V2-004037 | MATHDB-349675

The standardness conjecture for prime ideals in solvable uniform Iwasawa algebras

摘录 / EXCERPT: The standardness conjecture. Let G be a solvable, uniform pro-p group, and let P be a prime ideal in $\mathcal{O}G$. Then P is virtually standard, and moreover if $p\in P$...

conjecture | open | MathDB 页面 | 原文线索

V2-004038 | MATHDB-349709

Meagher-Spiga intersection-density conjecture for prime-product degrees

摘录 / EXCERPT: Intersection-density conjecture for prime-product degrees. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-004039 | MATHDB-349711

Asymptotic nonintegrality conjecture for the affine linear group intersection density

摘录 / EXCERPT: Affine intersection-density conjecture. For every $\varepsilon>0$, there exists a prime power q_0 such that for every prime power $q\geq q_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004040 | MATHDB-349712

The infinitude conjecture for equal values of P_3 and P_A

摘录 / EXCERPT: Infinitude conjecture for P_3 and P_A. There are infinitely many values of a such that the Diophantine equation

conjecture | open | MathDB 页面 | 原文线索

V2-004041 | MATHDB-349713

The equal-values conjecture for partition functions P_A and P_B

摘录 / EXCERPT: Equal-values conjecture for P_A and P_B. The Diophantine equation

conjecture | open | MathDB 页面 | 原文线索

V2-004042 | MATHDB-349714

The square-value conjecture for partition functions P_n

摘录 / EXCERPT: Square-value conjecture for P_n. The only positive integer solution is

conjecture | open | MathDB 页面 | 原文线索

V2-004043 | MATHDB-349715

The finiteness conjecture for equal values of P_m and P_n

摘录 / EXCERPT: Finiteness conjecture for P_m and P_n. This equation has only finitely many solutions in non-negative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-004044 | MATHDB-349728

Equivariant Fitting ideal conjecture for adjoint Hilbert modular L-values

摘录 / EXCERPT: Let F be the totally real field, let S be the finite set of places occurring in the construction, and let E/F be a totally real abelian extension ramified away from S or at...

conjecture | open | MathDB 页面 | 原文线索

V2-004045 | MATHDB-349729

Equivariant adjoint L-value congruence conjecture for Hilbert modular forms

摘录 / EXCERPT: Let F be the totally real field, let S be the set of places dividing the level and p, and let E/F be a totally real abelian extension ramified away from S or at p. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-004046 | MATHDB-349730

Finiteness conjecture for singular S-units

摘录 / EXCERPT: Finiteness conjecture for singular S-units. For every $s \in \mathbb{N}$, the number of singular moduli that are S-units for some set of rational primes S with $\#S=s$ is finit...

conjecture | open | MathDB 页面 | 原文线索

V2-004047 | MATHDB-349731

Uniformity conjecture for singular S-units

摘录 / EXCERPT: Uniformity conjecture for singular S-units. For every $s \in \mathbb{N}$, the set $\mathcal{J}_s$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-004048 | MATHDB-349746

The p-adic valuation conjecture for the repair factor of Stirling sequences

摘录 / EXCERPT: Let $S^{(2)}_k$ denote the relevant sequence and let ${\rm Fail}(S^{(2)}_k)$ be its repair factor. For a prime p, write $|\cdot|_p$ for the p-adic absolute value. The p-adic...

conjecture | open | MathDB 页面 | 原文线索

V2-004049 | MATHDB-349747

The large-prime valuation conjecture for the repair factor of Stirling sequences

摘录 / EXCERPT: Let $S^{(2)}_k$ denote the relevant sequence and let ${\rm Fail}(S^{(2)}_k)$ be its repair factor. For a prime p, write $|\cdot|_p$ for the p-adic absolute value. The large-pri...

conjecture | open | MathDB 页面 | 原文线索

V2-004050 | MATHDB-349748

The radical-growth conjecture for the repair factor of Stirling sequences

摘录 / EXCERPT: Let $S^{(2)}_k$ denote the relevant sequence, let ${\rm Fail}(S^{(2)}_k)$ be its repair factor, and let ${\rm Rad}(n)$ denote the radical of n, the product of its distinct prime...

conjecture | open | MathDB 页面 | 原文线索

V2-004051 | MATHDB-349749

The prime-power repair-factor conjecture for Stirling sequences

摘录 / EXCERPT: Let $S^{(2)}_k$ denote the relevant sequence and let ${\rm Fail}(S^{(2)}_k)$ be its repair factor. For a prime p and a positive integer j, the following assertions are conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004052 | MATHDB-349762

p-adic L-function conjecture at the parameter a=1

摘录 / EXCERPT: The $a=1$ p-adic L-function conjecture. There is a constant $C_1\in {\mathbb Q}^{\times}$ independent of p such that

conjecture | open | MathDB 页面 | 原文线索

V2-004053 | MATHDB-349767

Heuristic one-level density formula for logarithmic derivatives in cubic field families

摘录 / EXCERPT: The one-level density conjecture. The displayed asymptotic formula should hold, and the two sums on its right-hand side are absolutely convergent.

conjecture | open | MathDB 页面 | 原文线索

V2-004054 | MATHDB-349804

Conjecture on sharp bounds for diagonal equations with restricted solution sets

摘录 / EXCERPT: Let $d_1,\dots,d_s$ be the exponents of a diagonal equation, and suppose there is a bound for the number of its solutions with restricted coordinates, explicitly expressed in terms...

conjecture | open | MathDB 页面 | 原文线索

V2-004055 | MATHDB-349826

Orthosymplectic W-algebra rationality conjecture

摘录 / EXCERPT: Let $m\geq 1$ and set ... Assume that $p,q\in\mathbb{N}$ are coprime, that $p\geq 2m-1$ when q is odd, and that $p\geq 2m$ when q is even. Orthosymplectic W-algebra rationality c...

conjecture | open | MathDB 页面 | 原文线索

V2-004056 | MATHDB-349827

Principal orthosymplectic W-superalgebra rationality conjecture

摘录 / EXCERPT: Let $n\geq 1$ and set ... Assume that $p,q\in\mathbb N$ are coprime, that $p\geq 2n-1$ when q is odd, and that $p\geq 2n$ when q is even. Principal orthosymplectic W-superalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-004057 | MATHDB-349843

The Sato-Tate conjecture for Frobenius conjugacy classes of K3 surfaces

摘录 / EXCERPT: Let X be as above, let i be as above, and let S be the finite set of bad primes. For each good prime $p\in\mathbb P\setminus S$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004058 | MATHDB-349865

Unrefined arithmetic Gan-Gross-Prasad conjecture for U(n) × U(n)

摘录 / EXCERPT: Unrefined arithmetic Gan-Gross-Prasad conjecture. The following are equivalent: (a) $\operatorname{FJ}^{\natural}_\varepsilon\neq0$; (b)...

conjecture | open | MathDB 页面 | 原文线索

V2-004059 | MATHDB-349878

Conjecture on the oscillation of the cosine-sine divisor sum

摘录 / EXCERPT: Let … with the parameters and primed summation convention used in the paper. Cosine-sine oscillation conjecture. … This predicts unbounded oscillation at scale $x^{1/4}$. The motiv...

conjecture | open | MathDB 页面 | 原文线索

V2-004060 | MATHDB-349879

General oscillation conjecture for weighted multiple sine sums

摘录 / EXCERPT: General weighted sine-sum oscillation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004061 | MATHDB-349888

Asymptotic frequency conjecture for non-split prime-power conductors

摘录 / EXCERPT: Let $K=\mathbb{Q}(\sqrt{d})$ be a quadratic field with $d\equiv 2\pmod 3$ and 3-class rank $\varrho=0$. Consider 3-admissible non-split prime or prime-power conductors $f=q$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004062 | MATHDB-349894

The attractor conjecture for polarized Calabi-Yau threefolds

摘录 / EXCERPT: Attractor conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004063 | MATHDB-349925

The modular S-matrix conjecture for rational lisse vertex algebras

摘录 / EXCERPT: Let V be a rational, lisse, simple, self-dual conformal vertex algebra, and let $\{L_0,\ldots,L_r\}$ be its complete set of simple modules. Denote the conformal dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-004064 | MATHDB-349929

Breuil-Buzzard-Emerton irreducibility conjecture for reductions of crystalline representations

摘录 / EXCERPT: Let p be an odd prime, let $k\geqslant 2$ be an integer, and let $a\in\overline{\mathbb{Z}}_p$ satisfy $v_p(a)>0$. Let $V_{k,a}$ be the associated two-dimensional crystalline p...

conjecture | open | MathDB 页面 | 原文线索

V2-004065 | MATHDB-349956

The fundamental-period dichotomy for v-palindromes

摘录 / EXCERPT: Fundamental-period conjecture. Either

conjecture | open | MathDB 页面 | 原文线索

V2-004066 | MATHDB-349957

The type-independence conjecture for v-palindromes

摘录 / EXCERPT: Type-independence conjecture. The type of m with respect to n_1 is the same as the type of m with respect to n_2.

conjecture | open | MathDB 页面 | 原文线索

V2-004067 | MATHDB-349958

The Eisenstein p-adic L-function conjecture for the adjoint of a weight one modular form

摘录 / EXCERPT: Let g be a weight one modular form, let g^* be its dual, and let u_1 denote the unit attached to the associated Galois representation. Let $\kappa(g_{\alpha},g_{1/\alpha}^*)$...

conjecture | open | MathDB 页面 | 原文线索

V2-004068 | MATHDB-349979

Order of rank bias for rational elliptic curves

摘录 / EXCERPT: Order of rank bias. If $r\geq0$ and $\mathcal E_r$ is infinite, then there exists $\delta>0$ such that, for every $\epsilon>0$, if $\phi(N)\ll(\log N)^{\delta-\epsilon}$ then...

conjecture | open | MathDB 页面 | 原文线索

V2-004069 | MATHDB-349980

The asymptotic irregularity conjecture for orthomorphisms

摘录 / EXCERPT: Let q range over prime powers, and let an irregular orthomorphism be an orthomorphism whose translate $T_g\theta=\theta(x+g)-\theta(g)$ is non-cyclotomic for every...

conjecture | open | MathDB 页面 | 原文线索

V2-004070 | MATHDB-349984

Shintani's exact analytic formula for Brumer-Stark units

摘录 / EXCERPT: Let F be a totally real field, let $\mathfrak{n}$ be a nonzero ideal, let $F(\mathfrak{n})$ be its narrow ray class field, and let H be the maximal CM subfield in which...

conjecture | open | MathDB 页面 | 原文线索

V2-004071 | MATHDB-349994

Inhomogeneous Littlewood-type counting conjecture for inhomogeneously badly approximable numbers

摘录 / EXCERPT: Inhomogeneous Littlewood-type counting conjecture. There exists $G=G(\alpha,\gamma,\delta)\subseteq\operatorname{Bad}(\delta)$ with Hausdorff dimension

conjecture | open | MathDB 页面 | 原文线索

V2-004072 | MATHDB-349996

Second-moment negative-bias conjecture for elliptic-curve families

摘录 / EXCERPT: Second-moment negative-bias conjecture. The largest non-averaging lower-order term in the second-moment expansion of a family of elliptic curves is on average negative.

conjecture | open | MathDB 页面 | 原文线索

V2-004073 | MATHDB-350000

The two-element commutative p-base conjecture

摘录 / EXCERPT: The two-element commutative p-base conjecture. Every finite group has a commutative p-base of size 2 for every prime p.

conjecture | open | MathDB 页面 | 原文线索

V2-004074 | MATHDB-350014

Fermat prime finiteness conjecture

摘录 / EXCERPT: Fermat prime finiteness conjecture. There are only finitely many Fermat primes, that is, only finitely many indices n for which F_n is prime. The heuristic is based on summing...

conjecture | open | MathDB 页面 | 原文线索

V2-004075 | MATHDB-350015

Extended Goldbach conjecture with Hardy-Littlewood asymptotic

摘录 / EXCERPT: Let k be a positive integer and let $C_{2,k}$ be the prime-pair constant. Extended Goldbach conjecture. The number of ways of writing $2k$ as a sum of two primes is asymptotic to...

conjecture | open | MathDB 页面 | 原文线索

V2-004076 | MATHDB-350016

Prime arithmetic progression counting conjecture

摘录 / EXCERPT: Let k be the length of the progression, let d be its common difference, and let $A_{k,d}$ be the adjustment factor defined by the local prime-divisibility probabilities. Prime...

conjecture | open | MathDB 页面 | 原文线索

V2-004077 | MATHDB-350017

Grosswald's conjecture on prime arithmetic progressions

摘录 / EXCERPT: Grosswald's conjecture. The number of such progressions satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004078 | MATHDB-350019

Cunningham chain counting heuristic

摘录 / EXCERPT: Cunningham chain conjecture. The number of Cunningham chains of length k beginning with a prime $p\leq N$ is approximately

conjecture | open | MathDB 页面 | 原文线索

V2-004079 | MATHDB-350020

Prime values of n^2+1 conjecture

摘录 / EXCERPT: Let ... where $\left(\frac{-1}{p}\right)$ is the Legendre symbol. n^2+1 prime-value conjecture. The expected number of primes of the form n^2+1 less than or equal to N is ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-004080 | MATHDB-350021

Cullen and Woodall prime counting heuristic

摘录 / EXCERPT: Define the Cullen and Woodall sequences by $C(n)=n2^n+1$ and $W(n)=n2^n-1$. Cullen and Woodall prime conjecture. The expected number of Cullen primes and, separately, Woodall prime...

conjecture | open | MathDB 页面 | 原文线索

V2-004081 | MATHDB-350022

Primorial prime counting heuristic

摘录 / EXCERPT: For a prime p, let $p\#$ denote the primorial, the product of all primes at most p, and let γ be Euler's constant. Primorial prime conjecture. The expected number of pri...

conjecture | open | MathDB 页面 | 原文线索

V2-004082 | MATHDB-350023

Factorial prime counting heuristic

摘录 / EXCERPT: For an integer n, the factorial primes are primes of the forms $n!+1$ and $n!-1$, and let γ be Euler's constant. Factorial prime conjecture. The expected number of factori...

conjecture | open | MathDB 页面 | 原文线索

V2-004083 | MATHDB-350031

Generalized Carmichael counting-function symmetry conjecture

摘录 / EXCERPT: Generalized Carmichael symmetry conjecture. For all integers k,

conjecture | open | MathDB 页面 | 原文线索

V2-004084 | MATHDB-350043

Huang's asymptotic conjecture for rational points near curved manifolds

摘录 / EXCERPT: Huang's conjecture. The asymptotic above should hold in the stated range of δ.

conjecture | open | MathDB 页面 | 原文线索

V2-004085 | MATHDB-350052

Conjectural formula for minimal permutation representations of general linear groups

摘录 / EXCERPT: Let $q \geq 5$ be an odd prime power, let $g=\gcd(n,q-1)$, and write the prime factorization of g as $g=p_1^{r_1}\cdots p_s^{r_s}$. For a natural number m, write ... with the...

conjecture | open | MathDB 页面 | 原文线索

V2-004086 | MATHDB-350053

Large-field validity conjecture for minimal permutation representations of special linear groups

摘录 / EXCERPT: Let n be any natural number and let q be a prime power. Theorem is the theorem giving the minimal faithful permutation representation of ${\rm SL}_n(\mathbb{F}_q)$ under its st...

conjecture | open | MathDB 页面 | 原文线索

V2-004087 | MATHDB-350059

A p-adic supercongruence for a Ramanujan-type sum

摘录 / EXCERPT: Let $p\equiv 5\pmod{8}$ be a prime, and let $(a)_k=a(a+1)\cdots(a+k-1)$ denote the rising factorial. The p-adic supercongruence conjecture. ... This is presented as a stronger versio...

conjecture | open | MathDB 页面 | 原文线索

V2-004088 | MATHDB-350096

Conjecture on Diophantine exponents of aperiodic episturmian words

摘录 / EXCERPT: Diophantine-exponent conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004089 | MATHDB-350120

Granville's conjecture on integral points on quadratic twists of hyperelliptic curves

摘录 / EXCERPT: Let C) be a hyperelliptic curve given by the integral model $y^2=f(x)$, where $f\in\mathbb{Z}[x]$ has degree at least three, is separable, and hence defines a curve of genus at l...

conjecture | open | MathDB 页面 | 原文线索

V2-004090 | MATHDB-350121

Granville's asymptotic conjecture for integral points on elliptic-curve twists

摘录 / EXCERPT: Granville's elliptic-curve twist conjecture. As $D\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004091 | MATHDB-350128

Minimal Euler characteristic conjecture for compact 3-manifold groups

摘录 / EXCERPT: Let M be a compact 3-manifold with prime decomposition ... where the prime factors are as in (1.1) and categories (i)-(iv): the M_i are closed prime 3-manifolds with infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-004092 | MATHDB-350135

Smyth's conjecture on linear relations among Galois conjugates

摘录 / EXCERPT: Smyth's conjecture. There exist Galois conjugates $\gamma_1,\dots,\gamma_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004093 | MATHDB-350136

Smyth's generalized conjecture for Smyth tuples over number fields

摘录 / EXCERPT: Smyth's generalized conjecture. The tuple $(a_1,\dots,a_n)$ is a Smyth tuple if and only if it satisfies the strong absolute value criteria over K, or there exist roots of unity...

conjecture | open | MathDB 页面 | 原文线索

V2-004094 | MATHDB-350137

Smyth's strong-criteria conjecture over number fields

摘录 / EXCERPT: Smyth's strong-criteria conjecture. If $(a_1,\dots,a_n)$ satisfies the strong absolute value criteria, then $(a_1,\dots,a_n)$ is a Smyth tuple.

conjecture | open | MathDB 页面 | 原文线索

V2-004095 | MATHDB-350175

Mahler's conjecture on the ternary Cantor set

摘录 / EXCERPT: Let C denote the ternary Cantor set. Mahler's conjecture. Every number in C is either rational or transcendental. This conjecture concerns the arithmetic nature of points in th...

conjecture | open | MathDB 页面 | 原文线索

V2-004096 | MATHDB-350185

A Maeda-style conjecture for special eigenvalues of Drinfeld cusp forms

摘录 / EXCERPT: Let $A=\mathbb{F}_3[t]$, let $n\geq 2$, and set $k_n=1+3^n$. Let $\Sigma_n\subset(\mathbb{F}_3^\times)^n$ be the set of sign tuples with exactly i entries equal to -1 for some...

conjecture | open | MathDB 页面 | 原文线索

V2-004097 | MATHDB-350190

The Hurwitz zeta moment conjecture for rational parameters

摘录 / EXCERPT: Hurwitz zeta moment conjecture. For every fixed real number $k>0$, there is a constant $c_k(\alpha)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004098 | MATHDB-350229

Finiteness conjecture for vertex-transitive empty simplices

摘录 / EXCERPT: Let d be a positive integer such that $d+1$ is prime. Consider empty d-simplices modulo unimodular equivalence that admit a unimodular symmetry acting transitively on vertices,...

conjecture | open | MathDB 页面 | 原文线索

V2-004099 | MATHDB-350276

Montgomery's conjecture on Dirichlet polynomials

摘录 / EXCERPT: Let $N\geq 1$, $T>0$, and let B_T be any ball of radius T. For coefficients $\{b_n\}_{n=N+1}^{2N}\subset\mathbb{C}$, write ... for the associated Dirichlet polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004100 | MATHDB-350301

Darmon-Vonk conjecture on prime exponents of real quadratic singular moduli

摘录 / EXCERPT: Let D_1,D_2 be coprime fundamental discriminants, let τ_1,τ_2 be the corresponding real quadratic points, and let $p\leq 13$ be prime. Let $J_p(\tau_1,\tau_2)$ be the Dar...

conjecture | open | MathDB 页面 | 原文线索

V2-004101 | MATHDB-350315

Arithmetic Fundamental Lemma conjecture, semi-Lie algebra version

摘录 / EXCERPT: Let F_0 be the p-adic field with ring of integers O_{F_0} and residue-field cardinality q. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004102 | MATHDB-350341

The residue classification conjecture for Mersenne primes

摘录 / EXCERPT: Let ρ be a prime and let $M_\rho=2^\rho-1$. The Mersenne-prime residue conjecture. If M_ρ is a Mersenne prime, then exactly one of the following holds:...

conjecture | open | MathDB 页面 | 原文线索

V2-004103 | MATHDB-350342

The upper-bound conjecture for the Ova-Ova-Prime-Ova construction

摘录 / EXCERPT: Let the preceding Ova-Ova-Prime-Ova construction be given, and let N denote the number of primes it produces. The Ova-Ova-Prime-Ova counting conjecture. Although the preceding th...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-004104 | MATHDB-350343

The convergence conjecture for reciprocal prime-gap series

摘录 / EXCERPT: Let a series be formed from reciprocals associated with twin primes and with prime pairs separated by $2k$. The reciprocal-series conjecture. The series of reciprocals of twin prim...

conjecture | open | MathDB 页面 | 原文线索

V2-004105 | MATHDB-350344

The nonexistence conjecture for polynomial enumeration of divisor-count classes

摘录 / EXCERPT: For k with $0<k<n$, define $A_{2k}=\{n\in\mathbb{N}:\nu(n)=2k\}$, where $\nu(n)$ is the number of positive divisors of n. The divisor-class polynomial conjecture. No polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-004106 | MATHDB-350346

The intersection density conjecture for twice a prime

摘录 / EXCERPT: Let ${\cal I}_n$ be the set of intersection densities of transitive permutation groups of degree n, and let $I(n)$ be the maximum value in ${\cal I}_n$. The intersection density...

conjecture | solved | MathDB 页面 | 原文线索

V2-004107 | MATHDB-350360

Finite Gilbreath conjecture for prime circuits

摘录 / EXCERPT: Finite Gilbreath conjecture. For every such prime originator,

conjecture | open | MathDB 页面 | 原文线索

V2-004108 | MATHDB-350367

The DEDHP hardness conjecture in Sylow q-subquasigroups

摘录 / EXCERPT: Let $\mathbb{E}_{p^2}$ be an entropoid with $p=2q+1$ a safe prime, and let g_q generate its Sylow q-subquasigroup $\mathbb{E}^*_{q^2}$. The Sylow DEDHP hardness conjecture. No...

conjecture | open | MathDB 页面 | 原文线索

V2-004109 | MATHDB-350368

The computational discrete entropoid root hardness conjecture

摘录 / EXCERPT: Let $\mathbb{E}_{p^2}$ be an entropoid, with $p=2q+1$ a safe prime having λ bits, and let g generate $\mathbb{E}^*_{(p-1)^2}$. Let $\mathcal{A}$ be an algorithm for CDERP...

conjecture | open | MathDB 页面 | 原文线索

V2-004110 | MATHDB-350418

Radziwiłł's large-deviation conjecture for Selberg's central limit theorem

摘录 / EXCERPT: Let τ be uniformly distributed on $[T,2T]$, and set ... For $k>0$, consider the regime $V\sim k\sqrt{2\log\log T}$. Radziwiłł's conjecture. In this regime, ... This predicts that...

conjecture | open | MathDB 页面 | 原文线索

V2-004111 | MATHDB-350419

Refined Fyodorov-Hiary-Keating conjecture for zeta maxima in macroscopic intervals

摘录 / EXCERPT: Refined Fyodorov-Hiary-Keating conjecture. As a sequence of random variables, the maximum satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004112 | MATHDB-350445

The class-number distribution conjecture for prime-generated real quadratic fields

摘录 / EXCERPT: Let p_i) be the i-th prime, let $\mathfrak{P}=\{\mathbb{Q}(\sqrt{p})\mid p\text{ prime}\}$, and define ... Assume these probabilities exist for all positive odd integers q. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-004113 | MATHDB-350460

The weakened Goldbach upper-bound conjecture

摘录 / EXCERPT: Let Λ be the von Mangoldt function, let ... and let $\mathfrak{S}(n)$ be the singular series defined by ... for even n, with $C_2=\prod_{p>2}(1-(p-1)^{-2})$. The weakened Gol...

conjecture | open | MathDB 页面 | 原文线索

V2-004114 | MATHDB-350461

The weakened prime-pair upper-bound conjecture

摘录 / EXCERPT: For $k\geq0$, define ... where Λ is the von Mangoldt function and $\mathfrak{S}(k)$ is the singular series. The weakened prime-pair upper-bound conjecture. Given a fixed cons...

conjecture | open | MathDB 页面 | 原文线索

V2-004115 | MATHDB-350499

Incomplete Vinogradov mean-value conjecture for monomial exponential sums

摘录 / EXCERPT: Incomplete Vinogradov mean-value conjecture. For every $p\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-004116 | MATHDB-350502

Cilleruelo-Granville arc conjecture for lattice points on semicircles

摘录 / EXCERPT: For $N\in\mathbb{N}$, let $\mathbb{S}^1_+(N)$ be the upper semicircle of radius $\sqrt N$, and let $S_N=\mathbb{S}^1_+(N)\cap\mathbb{Z}^2$. Cilleruelo-Granville conjecture. For eve...

conjecture | open | MathDB 页面 | 原文线索

V2-004117 | MATHDB-350516

Extension of the projective-prime dessin conjecture

摘录 / EXCERPT: Projective-prime dessin conjecture. For each projective prime $p=1+q+\cdots+q^{n-1}>3$, there exists at least one dessin of type $(3,2,p)$ and degree p with monodromy group...

conjecture | open | MathDB 页面 | 原文线索

V2-004118 | MATHDB-350517

Conjecture on projective primes with prime-power bases

摘录 / EXCERPT: Conjecture on projective primes with prime-power bases. For each fixed prime $n\ge 3$ and each fixed power e of n, there are infinitely many projective primes of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-004119 | MATHDB-350526

The sporadic unit-equation conjecture for simplest cubic orders

摘录 / EXCERPT: Let K_a be the simplest cubic field considered in the paper, let ρ generate the order $\mathbb{Z}[\rho]$, and let $u_1,u_2\in\mathbb{Z}[\rho]^*$ and $n\in\mathbb{Z}$. Two so...

conjecture | open | MathDB 页面 | 原文线索

V2-004120 | MATHDB-350527

The maximal-order sporadic unit-equation conjecture

摘录 / EXCERPT: Let K_a be the simplest cubic field considered in the paper, let $\mathcal{O}_{K_a}$ be its maximal order, and let $u_1,u_2\in\mathcal{O}_{K_a}^*$ and $n\in\mathbb{Z}$ satisfy th...

conjecture | open | MathDB 页面 | 原文线索

V2-004121 | MATHDB-350540

Dual endomorphism conjecture for Anderson self-dual ring spectra

摘录 / EXCERPT: Let R be an $\mathbf{E}_1$-ring spectrum and write $A=\pi_0 R$. Suppose that there is a class $D\in \pi_{-d}R$ witnessing the Anderson self-duality of R. Let $F\colon R\to R$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-004122 | MATHDB-350545

Transformation formula between logarithmic and Dwork p-adic hypergeometric functions

摘录 / EXCERPT: Let $\sigma(t)=ct^p$ and $\widehat{\sigma}(t)=c^{-1}t^p$. Let $a\in\mathbb{Z}_p\backslash\mathbb{Z}_{\leq 0}$, and suppose that the rth Dwork prime satisfies $a^{(r)}=a$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-004123 | MATHDB-350547

Parity-corrected Dwork p-adic hypergeometric inversion conjecture

摘录 / EXCERPT: Parity-corrected Dwork inversion conjecture. For an odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-004124 | MATHDB-350558

Browning-Van Valckenborgh prediction for squareful triples

摘录 / EXCERPT: Browning-Van Valckenborgh prediction. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004125 | MATHDB-350560

Cassaigne et al.'s sign-change conjecture for the Liouville function

摘录 / EXCERPT: Cassaigne et al.'s conjecture. The values $\lambda(f(n))$ change sign infinitely often.

conjecture | open | MathDB 页面 | 原文线索

V2-004126 | MATHDB-350564

Greenberg's μ=0 conjecture for signed Selmer groups

摘录 / EXCERPT: Let g be a Hecke eigencuspform satisfying condition $(\star)$, let $\ddagger\in\{+,-\}$ be a choice of sign, and suppose that the residual representation $\bar{\rho}_g$ is irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-004127 | MATHDB-350570

Adelic conjecture for local zeta functions and string amplitudes

摘录 / EXCERPT: Fix N and let $Z_{\mathbb{R}}^{(N)}(\boldsymbol{s})$ and $Z_{\mathbb{Q}_p}^{(N)}(\boldsymbol{s})$ denote the archimedean and p-adic local zeta functions, respectively, regarded...

conjecture | open | MathDB 页面 | 原文线索

V2-004128 | MATHDB-350583

The p-adic Borel regulator injectivity conjecture

摘录 / EXCERPT: The p-adic Borel regulator injectivity conjecture. For every number field k, every rational prime p, and every odd dimension $2m-1\geq 3$, this map is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-004129 | MATHDB-350606

The negative bias conjecture for Gaussian-prime representations by sums of squares

摘录 / EXCERPT: Negative bias conjecture. There is a bias towards negative values in the distribution of D_1: for more than half of the $x\in[2,\infty)$ in logarithmic scale, more than half of t...

conjecture | open | MathDB 页面 | 原文线索

V2-004130 | MATHDB-350607

The complete positive bias conjecture for Gaussian-prime representations

摘录 / EXCERPT: Let ... Here p ranges over primes, and logarithmic scale is measured by logarithmic density, using the variable $y=\log x$; thus a property holds for almost all x in logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-004131 | MATHDB-350632

The subpolynomial rational-point counting conjecture for elliptic curves

摘录 / EXCERPT: Subpolynomial rational-point counting conjecture. For every $\epsilon>0$, there exists a constant $c'(\epsilon)$ depending only on ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004132 | MATHDB-350650

Corrected Lusztig-Williamson Billiards Conjecture

摘录 / EXCERPT: Let p be the prime fixed in the paper, and let $\widetilde{Z}$ be the multiset produced by the corrected billiards algorithm at $\ell=p$. Define the map...

conjecture | open | MathDB 页面 | 原文线索

V2-004133 | MATHDB-350659

Wilkie's conjecture on rational points in the exponential real field

摘录 / EXCERPT: Wilkie's conjecture. There exist constants $c=c(X)$ and $d=d(X)$ such that, for every $H\in\mathbb{N}$ with $H>e$,

conjecture | open | MathDB 页面 | 原文线索

V2-004134 | MATHDB-350665

Hardy-Littlewood conjecture for sums of two squares in arithmetic progressions

摘录 / EXCERPT: For $k\geq 1$, let $\mathcal{H}=\{h_1,\dots,h_k\}\subseteq\mathbb{Z}$, let $q\equiv 1\,(\mathrm{mod}\,4)$ be prime, let $a\in\mathbb{Z}$, and let $R_k(\mathcal{H};x,q,a)$ be the nor...

conjecture | open | MathDB 页面 | 原文线索

V2-004135 | MATHDB-350666

Refined Hardy-Littlewood conjecture in arithmetic progressions

摘录 / EXCERPT: For $k\geq1$, let $\mathcal{H}=\{h_1,\dots,h_k\}\subseteq\mathbb{Z}$, let $q\equiv1\,(\mathrm{mod}\,4)$ be prime, let $a\in\mathbb{Z}$, and let $R_k(\mathcal{H};x,q,a)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004136 | MATHDB-350667

Asymptotic conjecture for successive sums of two squares in arithmetic progressions

摘录 / EXCERPT: Fix a prime $q\equiv1\,(\mathrm{mod}\,4)$, an integer $r\geq2$, and $\mathbf{a}=(a_1,\dots,a_r)\in\mathbb{N}^r$. Let E_n be the increasing sequence of sums of two squares, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-004137 | MATHDB-350668

The $abcd$-conjecture

摘录 / EXCERPT: Let K be a number field or a one-dimensional function field of characteristic zero, and let $n\ge 3$. Let $[Z_1:\cdots:Z_n]$ be the standard homogeneous coordinates on...

conjecture | open | MathDB 页面 | 原文线索

V2-004138 | MATHDB-350681

Asymptotic conjecture for graph series of type A

摘录 / EXCERPT: For each $n\geq 1$, let $H_{A_m}(q)$ denote the graph series associated with the graph of type A_m, let $(q)_\infty=\prod_{k\geq 1}(1-q^k)$, and let $t\to0^+$. Asymptotic conject...

conjecture | open | MathDB 页面 | 原文线索

V2-004139 | MATHDB-350708

Conjecture on uniqueness of rational maps from a numerator of a 3-cycle

摘录 / EXCERPT: Let $A(m,n)$ and $B(m,n)$ be the integer forms parametrizing rational 3-cycles of maps $f_c(x)=x^2+c$, and let $(m,n)\in\mathbb{Z}^2$ satisfy $mn(m+n)\neq0$ and $\gcd(m,n)=1$. Nu...

conjecture | open | MathDB 页面 | 原文线索

V2-004140 | MATHDB-350709

Conjecture on the graph of numerators of rational 3-cycles

摘录 / EXCERPT: Let Γ be the graph whose vertices are the absolute values of numerators of rational points in rational 3-cycles of maps $f_c(x)=x^2+c$, with two vertices joined when they...

conjecture | open | MathDB 页面 | 原文线索

V2-004141 | MATHDB-350710

Conjecture on powers in the cubic number-field module

摘录 / EXCERPT: Let $K=\mathbb{Q}(\gamma)$ and $M=\mathbb{Z}[1,\gamma^2]$. For a pair (m,n) with $mn(m+n)\neq0$ and $\gcd(m,n)=1$, suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-004142 | MATHDB-350748

The lower bound for binary ones in triangular numbers

摘录 / EXCERPT: Binary-weight conjecture. If $n\in\mathbb{N}$ is such that $|(n)_2^1|\geq 6$, then $|(t_n)_2^1|\geq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-004143 | MATHDB-350749

The arithmetic-progression conjecture for very triangular numbers

摘录 / EXCERPT: Very triangular arithmetic-progression conjecture. For any positive integer k, there exists an arithmetic progression $AP(k)$ of length k such that $t_j\in VT$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-004144 | MATHDB-350769

Integrality Conjecture for Bloch-Kato elements

摘录 / EXCERPT: Let M/K be a rank-one motive with coefficient ring R, let T be its p-adic realization, and put $r=r(M/K,R)$. Assume that the $\mathcal{R}_p$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-004145 | MATHDB-350772

Büyükkboduk-Lei Perrin-Riou-Stark conjecture

摘录 / EXCERPT: Let K be a CM field, let $g=\tfrac12[K:\mathds{Q}]$, and fix a finite-prime-to-p character $\chi:G_K\to\overline{\mathds{Q}}_p^{\times}$ satisfying the conditions in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-004146 | MATHDB-350774

Bag, Li and Zhang's generalized quadratic Gauss-sum moment conjecture

摘录 / EXCERPT: Let p be a sufficiently large prime, let m be a positive integer, let n be an integer, and let χ range over Dirichlet characters modulo p. Set $e(y)=e^{2\pi iy}$. Bag,...

conjecture | open | MathDB 页面 | 原文线索

V2-004147 | MATHDB-350775

Bag, Li and Zhang's weighted generalized quadratic Gauss-sum moment conjecture

摘录 / EXCERPT: Let p be a sufficiently large prime, let m be a positive integer, and let n be an integer satisfying $(n,p)=1$. Let χ_0 be the principal character modulo p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004148 | MATHDB-350794

Homogeneous growth conjecture for unvisited points in the prime walk

摘录 / EXCERPT: Let the prime walk (PW) be the path constructed from the sequence of prime numbers, and let N denote the number of steps. Let the area's perimeter mean the boundary of the region...

conjecture | open | MathDB 页面 | 原文线索

V2-004149 | MATHDB-350819

Equidistribution conjecture over finitely generated fields

摘录 / EXCERPT: Let k be either $\mathbb{Z}$ or a field, and let F be a finitely generated field over k. Let v be a non-trivial valuation of F, whose restriction to k is trivial if k...

conjecture | open | MathDB 页面 | 原文线索

V2-004150 | MATHDB-350826

The dimension threshold conjecture for nondegenerate manifolds of dimension greater than one

摘录 / EXCERPT: Let $\tau_{n,d}$ denote the threshold governing the dimension of the set of simultaneously Diophantine-approximable points on nondegenerate d-dimensional manifolds in...

conjecture | open | MathDB 页面 | 原文线索

V2-004151 | MATHDB-350827

The conjecture for the dimension threshold of nondegenerate curves

摘录 / EXCERPT: For a nondegenerate curve in $\mathbb{R}^n$, let $\tau_{n,1}$ denote the threshold governing the dimension of the set of simultaneously Diophantine-approximable points on such curv...

conjecture | open | MathDB 页面 | 原文线索

V2-004152 | MATHDB-350843

Reflective Borcherds-product conjecture for free unitary modular-form algebras

摘录 / EXCERPT: Let d be the discriminant parameter of the imaginary quadratic field underlying the unitary group, let L be a Hermitian lattice of signature $(n,1)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004153 | MATHDB-350844

Conjecture that freeness descends to reflection overgroups

摘录 / EXCERPT: Let L be a Hermitian lattice, let $\mathop{\mathrm{U}}(L)$ denote its unitary group, and let $\Gamma<\mathop{\mathrm{U}}(L)$ be a finite-index subgroup. Let Γ_1 be a refle...

conjecture | open | MathDB 页面 | 原文线索

V2-004154 | MATHDB-350855

Generalized Iizuka conjecture for successive quadratic fields

摘录 / EXCERPT: Generalized Iizuka conjecture. For any odd integer $n\geq 3$ and any integer $m\geq 1$, there is an infinite family of $m+1$ successive imaginary or real quadratic fields of this f...

conjecture | open | MathDB 页面 | 原文线索

V2-004155 | MATHDB-350875

Mullin's conjecture on maximum cliques in Peisert graphs

摘录 / EXCERPT: Let $q\equiv 3\pmod 4$ be a prime power, and let $P^*_{q^2}$ denote the Peisert graph of order q^2. A maximum clique is a clique of largest possible size. Mullin's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-004156 | MATHDB-350885

Conjecture on the decreasing quotient sequence of factorial minimality indices

摘录 / EXCERPT: Let $n_1,n_2,n_3,\ldots$ be the sequence defined by $\sigma_{n_\ell}=\ell+1$ for all $\ell\geq 1$, where the values of σ_n arise from the paper's preceding construction. Quo...

conjecture | open | MathDB 页面 | 原文线索

V2-004157 | MATHDB-350890

Prime-distance characterization of circulant graphs

摘录 / EXCERPT: For integers n and k with $k\leq n/2$, let $\operatorname{Circ}(n,k)$ denote the circulant graph with these parameters. A graph is a prime distance graph if its vertices can be...

conjecture | open | MathDB 页面 | 原文线索

V2-004158 | MATHDB-350894

Primitive and prime spectrum equivalence for the Witt algebra and its positive part

摘录 / EXCERPT: Let W be the Witt algebra and $W_{\geq -1}$ its subalgebra spanned by the basis elements of index at least -1. Let $\operatorname{U}(W)$ and $\operatorname{U}(W_{\geq -1})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-004159 | MATHDB-350895

Prime ideals of the Witt enveloping algebra contain primitive ideals

摘录 / EXCERPT: Let W be the Witt algebra and let $\operatorname{U}(W)$ be its universal enveloping algebra. The Witt prime-to-primitive containment conjecture. Every proper prime ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-004160 | MATHDB-350896

Codimension-two prime ideals of the Witt enveloping algebra

摘录 / EXCERPT: Let W be the Witt algebra, let $\operatorname{U}(W)$ be its universal enveloping algebra, and let π_γ be the family of representations defined in the paper, with kernels...

conjecture | open | MathDB 页面 | 原文线索

V2-004161 | MATHDB-350917

Greenberg-Ochiai's specialized main conjecture at a height-one prime

摘录 / EXCERPT: Let $\mathbb{I}$ be a UFD and Gorenstein, assume $\chi|_{((\mathbb{Z}/p\mathbb{Z})^{\times})}\neq\mathbf{1},\omega$, $p\nmid\varphi(N)$, and that the Eisenstein ideal $\mathbb{J}$...

conjecture | open | MathDB 页面 | 原文线索

V2-004162 | MATHDB-350921

Positselski's Koszulity conjecture for maximal pro-p Galois cohomology

摘录 / EXCERPT: Let p be a prime and let $\mathbb{K}$ be a field containing a root of unity of order p. Write $G_{\mathbb{K}}(p)$ for the maximal pro-p Galois group of $\mathbb{K}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004163 | MATHDB-350922

Quadratic-dual and Koszulity conjectures for maximal pro-p Galois groups

摘录 / EXCERPT: Let p be a prime and let $\mathbb{K}$ be a field containing a root of unity of order p. For a pro-p group G, let $\mathfrak{L}(G)$ be the restricted Lie algebra induced by...

conjecture | open | MathDB 页面 | 原文线索

V2-004164 | MATHDB-350939

The integral restriction conjecture for characters of finite groups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let P be a Sylow p-subgroup of G, and let $\chi\in\operatorname{Irr}(G)$. For a subgroup $Q\le P$, write $\chi(1)_p$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-004165 | MATHDB-350940

Fully faithful Fourier-Mukai conjecture for moduli spaces of vector bundles

摘录 / EXCERPT: Let X be a connected smooth projective curve of genus $g\ge2$, let $r\ge2$ and let $d<r$ be coprime positive integers, let L be a line bundle of degree d on X, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004166 | MATHDB-350950

Sharpness conjecture for characteristic-polynomial congruence bounds of q-Seidel matrices

摘录 / EXCERPT: Sharpness conjecture. For $e \in \mathbb N$ with $e \geqslant 3$ and q a prime power, there exists $N \in \mathbb N$ such that for all $n \geqslant N$ we have equality in the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-004167 | MATHDB-350972

Conjecture on neural ring homomorphisms of prime-support circulant codes

摘录 / EXCERPT: Let $\mathcal{C}$ be a circulant code on n neurons with support p. Let $\operatorname{NRH}\{\mathcal{R}_{\mathcal{C}}\}$ denote the set of neural ring homomorphisms associated...

conjecture | open | MathDB 页面 | 原文线索

V2-004168 | MATHDB-350981

Conjecture on real branch singularities of the principal inverse zeta branch

摘录 / EXCERPT: Let $s_1=zeta^{-1}(w)$ be the principal branch of the inverse Riemann zeta function, and let j_1 denote the lowest zero described in the surrounding discussion. The strip...

conjecture | open | MathDB 页面 | 原文线索

V2-004169 | MATHDB-351022

Abhyankar's arithmetic conjecture for the affine line

摘录 / EXCERPT: Let p be a prime, let q be a power of p, and let G be a finite group. Write $p(G)$ for the subgroup generated by the p-Sylow subgroups of G; thus G is cyclic-by-quasi...

conjecture | open | MathDB 页面 | 原文线索

V2-004170 | MATHDB-351036

Integrality and closed form for a greatest-common-divisor exponential sum

摘录 / EXCERPT: Let n, a, and d be positive integers, and set ... Let $\varphi(n)$ denote Euler's totient function and let ζ_k be a primitive k-th root of unity. Arithmetic conjecture....

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-004171 | MATHDB-351045

Gall-Rahav conjecture on non-Mersenne divisors of cyclotomic polynomial values

摘录 / EXCERPT: Gall-Rahav conjecture. The polynomial $\sigma(M^{2h})$ is divisible by a non-Mersenne prime.

conjecture | open | MathDB 页面 | 原文线索

V2-004172 | MATHDB-351062

Completeness conjecture for rational exceptional j-invariants of prime-power level

摘录 / EXCERPT: Exceptional j-invariant completeness conjecture. Table lists all rational exceptional j-invariants of prime-power level.

conjecture | open | MathDB 页面 | 原文线索

V2-004173 | MATHDB-351081

The irrational-exponent conjecture for Fermat-type equations

摘录 / EXCERPT: Irrational-exponent conjecture. Then n is irrational. Equivalently, no such equation has a rational exponent n under these hypotheses. This statement is presented after the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-004174 | MATHDB-351086

The trinomial constructions of iso-dual polycyclic codes

摘录 / EXCERPT: Let q be a prime power and consider the indicated trinomial-associated polycyclic codes. Iso-dual construction conjecture. The following codes are isodual: - If q is a power of...

conjecture | open | MathDB 页面 | 原文线索

V2-004175 | MATHDB-351111

Algebraicity conjecture for higher Green functions at CM points

摘录 / EXCERPT: Let $Z(W_\mathbb{Q})$ be the CM cycle and let $\Phi^r_L(z,h,f)$ be the higher Green function in the setting of Theorem. Assume that f has integral Fourier coefficients and that t...

conjecture | open | MathDB 页面 | 原文线索

V2-004176 | MATHDB-351112

He-Shiue-Venkat decomposition conjecture for three-variable linear Diophantine equations

摘录 / EXCERPT: He-Shiue-Venkat conjecture. For any solution $(\hat{x},\hat{y},\hat{z})$ of $ax+by+cz=n$, there exist $s_i\in\hat{S_i}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004177 | MATHDB-351122

The three-prime-factor reduction conjecture for cyclic tilings

摘录 / EXCERPT: Three-prime-factor reduction conjecture. The answers to Questions q-slab (both versions) and q-obstruction are affirmative when M has at most 3 distinct prime factors.

conjecture | open | MathDB 页面 | 原文线索

V2-004178 | MATHDB-351129

Gordon-Lidman conjecture on cyclic branched covers of satellite knots

摘录 / EXCERPT: Let K be a prime satellite knot in S^3, and for each positive integer n let $Sigma_n(K)$ denote the n-fold cyclic branched cover of S^3 branched over K. A Gordon-Lidman...

conjecture | open | MathDB 页面 | 原文线索

V2-004179 | MATHDB-351142

Powerfree preservation conjecture for the sum-of-proper-divisors function

摘录 / EXCERPT: Powerfree preservation conjecture. On a set of integers n of asymptotic density 1,

conjecture | open | MathDB 页面 | 原文线索

V2-004180 | MATHDB-351159

The weak power-bound abc conjecture

摘录 / EXCERPT: Weak power-bound abc conjecture. There is an absolute constant M such that

conjecture | open | MathDB 页面 | 原文线索

V2-004181 | MATHDB-351196

Rudnick-Waxman symplectic variance conjecture for prime angles

摘录 / EXCERPT: Let $\mathfrak p=\langle\alpha\rangle$ be a prime ideal of $\mathbb Z[i]$, let $\theta_{\mathfrak p}=\arg\alpha$, and define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-004182 | MATHDB-351227

Multiplicity conjecture for Dedekind zeta zeros in non-Galois extensions

摘录 / EXCERPT: Let L/K be an extension of number fields, let M be the Galois closure of L/K, and write ... If the induced representation $\operatorname{Ind}_H^G(1_H)$ contains an irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-004183 | MATHDB-351229

Atkin-Serre lower-bound conjecture for Ramanujan's tau function

摘录 / EXCERPT: Atkin-Serre conjecture. For every $\varepsilon>0$, there exist constants $c(\varepsilon),d(\varepsilon)>0$ such that, for all primes $p>d(\varepsilon)$,

conjecture | open | MathDB 页面 | 原文线索

V2-004184 | MATHDB-351239

The group-ring lifting conjecture for the Alon-Jaeger-Tarsi identity

摘录 / EXCERPT: Let $p>3$ be prime, let M be a nonsingular $n\times n$ matrix over $\mathbb{F}_p$ with row vectors $a_1,\dots,a_n$, and let $e_1,\dots,e_n$ be the standard basis vectors of...

conjecture | open | MathDB 页面 | 原文线索

V2-004185 | MATHDB-351241

Five-prime Linnik representation conjecture for Diophantine inequalities

摘录 / EXCERPT: Five-prime Linnik representation conjecture. For every fixed $1<c<c_0$ and every sufficiently large positive number N, the displayed Diophantine inequality has a solution in prim...

conjecture | open | MathDB 页面 | 原文线索

V2-004186 | MATHDB-351253

Conjecture on second-largest denominators in Egyptian fraction representations of 1

摘录 / EXCERPT: An Egyptian fraction representation of 1 is a representation of the form ... with positive integers as denominators. An integer d is the second-largest denominator when it is the...

conjecture | open | MathDB 页面 | 原文线索

V2-004187 | MATHDB-351301

Exponential correlation decay for non-classical polynomial phases

摘录 / EXCERPT: Let p be a prime and let $k\geq p+2$ be an integer. For each n, consider functions $f\colon\mathbb F_p^n\to\mathbb C$ with $\|f\|_{\infty}\leq 1$ and Gowers norm...

conjecture | open | MathDB 页面 | 原文线索

V2-004188 | MATHDB-351362

The Hamming graph efficient domination divisibility conjecture

摘录 / EXCERPT: Let $q=p^b$ for a prime p, and let $H(q,d)$ denote the Hamming graph with alphabet size q and dimension d. An efficient $(1,k)$-dominating function is a function of the type...

conjecture | open | MathDB 页面 | 原文线索

V2-004189 | MATHDB-351398

Lifting-length conjecture for Markoff triples

摘录 / EXCERPT: Let G_p be the graph of Markoff triples modulo a prime p, and let a triple in G_p be lifted to a triple in the Markoff tree over $\mathbb Z$. The lifting-length conjecture st...

conjecture | open | MathDB 页面 | 原文线索

V2-004190 | MATHDB-351405

Generalized instanton surgery exact-triangle conjecture

摘录 / EXCERPT: Let Y be a closed three-manifold, let $K\subset Y$ be a knot, and let $Y_n(K)$ denote the result of n-surgery on K. Let $W_n^\prime(K)$ be the cobordism defined above, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-004191 | MATHDB-351408

The bounded Syracuse falling-time conjecture

摘录 / EXCERPT: Let syr be the Syracuse map on odd positive integers, and let $\operatorname{sft}(n)$ denote the Syracuse falling time of an odd integer $n\geq 3$, namely the leas...

conjecture | open | MathDB 页面 | 原文线索

V2-004192 | MATHDB-351409

The Syracuse falling-time conjecture for Mersenne-type integers

摘录 / EXCERPT: Let $\operatorname{sft}(n)$ denote the Syracuse falling time of an odd positive integer n. Mersenne Syracuse falling-time conjecture. ... for all $\ell\geq 4625$. The assertion agr...

conjecture | open | MathDB 页面 | 原文线索

V2-004193 | MATHDB-351410

The large-number falling-time conjecture

摘录 / EXCERPT: Let $\operatorname{ft}(n)$ denote the falling time associated with the accelerated $3x+1$ map. Large-number falling-time conjecture. ... for all $n\geq 2^{500}$. This is a strong qua...

conjecture | open | MathDB 页面 | 原文线索

V2-004194 | MATHDB-351411

The large-number Syracuse falling-time conjecture

摘录 / EXCERPT: Let $\operatorname{sft}(n)$ denote the Syracuse falling time of an odd positive integer n. Large-number Syracuse falling-time conjecture. ... for all odd $n\geq 2^{5000}$. This con...

conjecture | open | MathDB 页面 | 原文线索

V2-004195 | MATHDB-351432

The T-module normalizer conjecture for finite monomial groups

摘录 / EXCERPT: Let p be prime, let $D_{\{p\}'}$ denote the relevant diagonal module, and let T be a non-solvable transitive subgroup of $\mathrm{Sym}(p)$. Write $N_{\mathrm{Sym}(p)}(T)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-004196 | MATHDB-351433

The equal-extension-count conjecture for finite monomial groups

摘录 / EXCERPT: Let p be prime, let T be a non-solvable transitive subgroup of $\mathrm{Sym}(p)$, and let D be the diagonal subgroup appearing in the monomial group...

conjecture | open | MathDB 页面 | 原文线索

V2-004197 | MATHDB-351434

The uniseriality conjecture for indecomposable T-submodules

摘录 / EXCERPT: Let p be prime, let T be a non-solvable transitive subgroup of $\mathrm{Sym}(p)$, and let D be the diagonal module used in the monomial-group classification. A module is unis...

conjecture | open | MathDB 页面 | 原文线索

V2-004198 | MATHDB-351435

Conjecture on the non-monomial associated prime of $I_n(B)$

摘录 / EXCERPT: Let $I_n(B)$ be the ideal associated with the matrix B in the construction of the Rees algebra, and let $\mathcal{P}$ denote the non-linear part of the defining ideal of that Ree...

conjecture | open | MathDB 页面 | 原文线索

V2-004199 | MATHDB-351438

Tu's elliptic expansion conjecture for the modular j-function

摘录 / EXCERPT: Let $s(w)=2\pi\Omega^2S(w)$ be the rescaled inverse mirror-map coordinate, and let $g(t)$ and $h(t)$ be the power series associated with the elliptic quartic as defined above. Arou...

conjecture | open | MathDB 页面 | 原文线索

V2-004200 | MATHDB-351462

Schlage-Puchta's classification conjecture for triangular billiards

摘录 / EXCERPT: Schlage-Puchta's classification conjecture. Then at least one of the following holds: $n\leq 78$, $s+t=n$, $s+2t=n$, $2s+t=n$, or n is even and $|s-t|=\frac{n}{2}$. This classifi...

conjecture | open | MathDB 页面 | 原文线索

V2-004201 | MATHDB-351470

The prime-indexed Abelian limit conjecture

摘录 / EXCERPT: Let $f(n)$ be an arithmetic function, let p_n denote the nth prime, and let $\operatorname{idx}(p_n)=n$. Suppose that the Cesàro mean ... exists. Prime-indexed Abelian limit conj...

conjecture | open | MathDB 页面 | 原文线索

V2-004202 | MATHDB-351471

The Alladi-like supernormal identity conjecture

摘录 / EXCERPT: Let $f(n)$ be an arithmetic function, let $\operatorname{idx}(p_n)=n$, let $p_{\operatorname{min}}(n)$ be the prime associated with the least prime factor of n, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-004203 | MATHDB-351472

The short-partition model conjecture for prime gaps

摘录 / EXCERPT: Let p_n be the nth prime, let $\mathcal P$ denote the set of partitions, and let $\widehat{N}$ be the supernorm map. A partition has no 1's when none of its parts equals 1,...

conjecture | open | MathDB 页面 | 原文线索

V2-004204 | MATHDB-351498

Symmetry conjecture for the generalized alternating hyperharmonic coefficients

摘录 / EXCERPT: Let $r,m,j\in\mathbb N$, and let $f(r,s_{1},s_{2})$ be the index bound occurring in the coefficient recursion. For $0\leq m\leq f(r,s_{1},s_{2})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004205 | MATHDB-351499

Kronecker-delta sum conjecture for the generalized alternating hyperharmonic coefficients

摘录 / EXCERPT: Let $r,m\in\mathbb N$, let $f(r,s_{1},s_{2})$ be the index bound occurring in the coefficient recursion, and let δ_{mn} denote the Kronecker delta, with $\delta_{mm}=1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004206 | MATHDB-351521

The prime $4a+1$ well-defined equation conjecture

摘录 / EXCERPT: Let a be a natural number such that $w=4a+1$ is prime. The variables r,v,s are natural numbers, and divisibility is in the integers. Prime $4a+1$ well-defined equation conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004207 | MATHDB-351522

The least-common-multiple formulation of the Erdős-Straus conjecture

摘录 / EXCERPT: Let $a\in\mathbb{N}\setminus\{1\}$, and let $b,c,d\in\mathbb{N}$ with $c\mid b+d$. The modulus is $4\operatorname{lcm}(b,d)$. Least-common-multiple formulation. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-004208 | MATHDB-351523

Liu's Fundamental Lemma conjecture

摘录 / EXCERPT: Assume that F/F_0 is an unramified quadratic extension of p-adic fields for an odd prime p, and retain the groups $G(V_0)$ and $G(V_1)$, the symmetric space $S_n(F_0)$, and t...

conjecture | open | MathDB 页面 | 原文线索

V2-004209 | MATHDB-351551

Categorical intermediate-Jacobian conjecture for prime Fano threefolds

摘录 / EXCERPT: Let X and X' be prime Fano threefolds of index one or two, with intermediate Jacobians $J(X)$ and $J(X')$ and Kuznetsov components $\mathcal{K}u(X)$ and $\mathcal{K}u(X')$. Cat...

conjecture | open | MathDB 页面 | 原文线索

V2-004210 | MATHDB-351562

Linnik-prime representation conjecture for three-variable Piatetski-Shapiro equations

摘录 / EXCERPT: Let $c_0>1$. For fixed c with $1<c<c_0$ and every sufficiently large positive integer N, consider the equation ... where p_1,p_2,p_3 are prime numbers and each x_i,y_i is an...

conjecture | open | MathDB 页面 | 原文线索

V2-004211 | MATHDB-351582

Northcott conjecture for stacky curves with coarse space $\mathbb{P}^1$

摘录 / EXCERPT: For a stacky curve $\mathfrak{X}=\mathfrak{X}(\mathbb{P}^1:({\mathbf a},{\mathbf m}))$, let $\gamma(\mathfrak{X})$ be the infimum of the real numbers δ for which the modifie...

conjecture | open | MathDB 页面 | 原文线索

V2-004212 | MATHDB-351590

The spacing conjecture for the transformed zeta zeros

摘录 / EXCERPT: Let ρ_k be the nontrivial zeros of the Riemann zeta function with positive imaginary part, and set ... Let Λ be the associated entire function, and let its spacing const...

conjecture | open | MathDB 页面 | 原文线索

V2-004213 | MATHDB-351591

The strong spacing conjecture for transformed zeta zeros

摘录 / EXCERPT: Let $\lambda_j=\eta_j^2$ be the transformed nontrivial zeta zeros, ordered as in the source. For a parameter γ with ... consider consecutive transformed zeros that are not con...

conjecture | open | MathDB 页面 | 原文线索

V2-004214 | MATHDB-351609

Uniform distribution of residues along powers of degree-one ideals

摘录 / EXCERPT: Let L be a number field and let $\alpha\in L$ be irrational. Let $\mathfrak{a}$ be an unramified ideal of degree one that is prime to the denominator of the principal ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-004215 | MATHDB-351610

Bianchi modular generalization of Stevens' non-vanishing conjecture

摘录 / EXCERPT: Let F be the number field in the Bianchi modular setting, let p) be a prime satisfying the residual non-vanishing hypotheses, and let $\mathfrak{Y}$ be the set of finite order...

conjecture | open | MathDB 页面 | 原文线索

V2-004216 | MATHDB-351618

Henk-Weismantel conjecture on Hilbert bases of linear Diophantine equations

摘录 / EXCERPT: Henk-Weismantel conjecture. Every minimal solution lies in this convex hull:

conjecture | open | MathDB 页面 | 原文线索

V2-004217 | MATHDB-351632

Higher-girth edge-girth-regular graph families

摘录 / EXCERPT: Higher-girth existence conjecture. For $q\geq 3$ a prime power and $g\in\{8,12\}$, there exists a family of

conjecture | open | MathDB 页面 | 原文线索

V2-004218 | MATHDB-351644

The q-floor and q-ceiling determinant conjecture

摘录 / EXCERPT: Let a be an integer and let $n>1$ be odd. Write $[x]_q$ for the q-analogue used in the paper, and let $\left(\frac{\cdot}{n}\right)$ denote the relevant quadratic character. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-004219 | MATHDB-351645

The permanent congruence conjecture for absolute-difference matrices

摘录 / EXCERPT: For an odd prime p, let per denote the permanent. For a prime $p>3$ and an integer $a\not\equiv0\pmod p$, consider permutations $\tau\in S_p$ such that p divid...

conjecture | open | MathDB 页面 | 原文线索

V2-004220 | MATHDB-351647

The prime reciprocal-permutation sum conjecture

摘录 / EXCERPT: Let p be an odd prime. For $a\in\mathbb Z$, sum over permutations $\tau\in S_{p-1}$ for which p divides none of the integers $a+j\tau(j)$. If $p\nmid a$, the same restriction i...

conjecture | open | MathDB 页面 | 原文线索

V2-004221 | MATHDB-351677

Twisted derived Torelli conjecture for K3 surfaces in positive characteristic

摘录 / EXCERPT: Let S) and S' be K3 surfaces over a finite field k with $\operatorname{char}(k)=p>0$. A prime-to-$p$ derived isogeny is an equivalence of derived categories arising from a pr...

conjecture | open | MathDB 页面 | 原文线索

V2-004222 | MATHDB-351697

Krikorian's rigidity conjecture for Diophantine torus translations

摘录 / EXCERPT: Let T be a C^∞ diffeomorphism of the torus ${\mathbb T}^d$. Assume that T is topologically conjugate to the linear translation $T_\omega:{\bf x}\mapsto {\bf x}+\omega$ m...

conjecture | open | MathDB 页面 | 原文线索

V2-004223 | MATHDB-351698

Geometric rigidity conjecture for Diophantine step-two nilflows

摘录 / EXCERPT: A nilflow is a flow generated by a one-parameter subgroup on a nilmanifold; a flow is geometrically rigid when continuous conjugacies within the relevant smooth class are smooth. C...

conjecture | open | MathDB 页面 | 原文线索

V2-004224 | MATHDB-351700

Strict growth conjecture for prime knot counts by crossing number

摘录 / EXCERPT: Strict growth conjecture for prime knots. If $3<n<m$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004225 | MATHDB-351736

Conjectural asymptotic formula for the first moment of multiplicative chaos coefficients

摘录 / EXCERPT: Let $(A(N))_{N \geqslant 0}$ be defined by the formal power series associated with a sequence $(X(k))_{k \geqslant 1}$ of independent standard complex Gaussians. First-moment asymp...

conjecture | open | MathDB 页面 | 原文线索

V2-004226 | MATHDB-351737

Extension of the first-moment asymptotic conjecture to real Gaussian and Rademacher coefficients

摘录 / EXCERPT: Let $(A(N))_{N \geqslant 0}$ be defined by the same formal power series from a sequence $(X(k))_{k \geqslant 1}$. Real-Gaussian and Rademacher extension conjecture. The first-momen...

conjecture | open | MathDB 页面 | 原文线索

V2-004227 | MATHDB-351738

Conditional first-moment asymptotic conjecture for Rademacher coefficients

摘录 / EXCERPT: Let $(X(k))_{k \geqslant 1}$ be a sequence of independent random variables uniform on $\{\pm 1\}$, and define $(A(N))_{N \geqslant 0}$ by the paper's formal power series. For any f...

conjecture | open | MathDB 页面 | 原文线索

V2-004228 | MATHDB-351742

The square-root Omega conjecture for sums of multiplicative functions

摘录 / EXCERPT: Let f be a multiplicative function in the class of MO functions, and let x tend to infinity. Square-root Omega conjecture. We have ... The paper notes that its results do not p...

conjecture | open | MathDB 页面 | 原文线索

V2-004229 | MATHDB-351768

Vanishing conjecture for localized cohomology of arithmetic locally symmetric spaces

摘录 / EXCERPT: Let $X_\mathbf{G}^U$ be the locally symmetric space, let $\widetilde{V}_{\lambda}(\mathcal{O})$ be the associated local system, and let ${\mathfrak m}$ be the relevant maximal idea...

conjecture | open | MathDB 页面 | 原文线索

V2-004230 | MATHDB-351773

The finiteness conjecture for bases without interesting reversed sum-product pairs

摘录 / EXCERPT: Let $\beta\geq 2$ be a base. A pair (a,b) is a reversed sum-product pair in base β when the sum of a and b, written in base β, is the reversal of the base-β...

conjecture | open | MathDB 页面 | 原文线索

V2-004231 | MATHDB-351776

Equivariant Segal-Stolz-Teichner conjecture

摘录 / EXCERPT: Let G be a symmetry group, let ν be the gravitational anomaly, and let k be the 't Hooft anomaly, with $\nu\oplus k\in (I_\mathbb{Z}\Omega^\text{spin})^4(BG)$. For a simple...

conjecture | open | MathDB 页面 | 原文线索

V2-004232 | MATHDB-351777

Twisted Segal-Stolz-Teichner conjecture

摘录 / EXCERPT: Let X be a parameter space, and let $\nu\oplus k\in (I_\mathbb{Z}\Omega^\text{spin})^4(X)$ be the anomaly of a 2d spin $\mathcal{N}=(0,1)$ supersymmetric theory. Twisted Segal-St...

conjecture | open | MathDB 页面 | 原文线索

V2-004233 | MATHDB-351782

The cohomology computation for model solvable Lie superalgebras in arbitrary characteristic

摘录 / EXCERPT: Cohomology conjecture. The cohomology space $H^2(SL^{n,m};SL^{n,m})$ has the following dimensions and generators for every prime p: it is trivial when $m<p$ and $n<p+1$; one-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-004234 | MATHDB-351786

Conjecture on cusp-form moments in the cusp

摘录 / EXCERPT: Cusp-moment conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004235 | MATHDB-351795

Daemi-Scaduto's torus-knot Slice-Ribbon conjecture

摘录 / EXCERPT: Let $T_{p,q}$ be the (p,q)-torus knot in S^3, where p and q are positive coprime integers, and let K be a knot concordant to $T_{p,q}$. A ribbon concordance is a concorda...

conjecture | open | MathDB 页面 | 原文线索

V2-004236 | MATHDB-351798

Gaussian limit conjecture for random Bergman game lengths

摘录 / EXCERPT: Let ${}_0(n)$ denote the initial state of the Bergman Game, and let $\mathcal{L}_n$ be the distribution of the length of a random game played from ${}_0(n)$, where at each state an...

conjecture | open | MathDB 页面 | 原文线索

V2-004237 | MATHDB-351809

The curvature realization conjecture for the 24-cell Apollonian packing

摘录 / EXCERPT: Curvature realization conjecture. The set of curvatures of $\Omega(\mathcal{B}_{\mathcal{R}^4})$ is $\mathbb N$.

conjecture | open | MathDB 页面 | 原文线索

V2-004238 | MATHDB-351810

The consecutive-curvature sequence conjecture for the 24-cell Apollonian packing

摘录 / EXCERPT: Consecutive-curvature sequence conjecture. There is a sequence of consecutive tangent spheres

conjecture | open | MathDB 页面 | 原文线索

V2-004239 | MATHDB-351814

Irreducible characteristic polynomial conjecture for matrix Kloosterman sums

摘录 / EXCERPT: Let a be an $n\times n$ matrix over $\mathbb{F}_q$, with characteristic polynomial P_a irreducible over $\mathbb{F}_q$, and let $\alpha\in\mathbb{F}_{q^n}$ be an eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-004240 | MATHDB-351834

Zagier's conjecture for Dedekind zeta values

摘录 / EXCERPT: Soient F un corps de nombres et $n\geq 2$. Notons $d_n=r_1+r_2$ si n est impair et $d_n=r_2$ si n est pair, et soit...

conjecture | open | MathDB 页面 | 原文线索

V2-004241 | MATHDB-351842

Akiyama-Brunotte-Pethő-Steiner periodic-orbit conjecture for digital rotations

摘录 / EXCERPT: Akiyama-Brunotte-Pethő-Steiner conjecture. Every orbit of r_{φ} is periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-004242 | MATHDB-351863

The automorphism-group order conjecture for the graph R

摘录 / EXCERPT: Let $q=p^e$ be an odd prime power, and let R be the graph under discussion with automorphism group $Aut(R)$. Automorphism-group order conjecture. For all odd prime powers q, ……

conjecture | open | MathDB 页面 | 原文线索

V2-004243 | MATHDB-351867

Landau-Ramanujan comparison conjecture for gcd parameter 3

摘录 / EXCERPT: Let k be a positive integer and q a prime with $(k,q-1)=3$. Let $S_{k,q}(x)$ and $S'_{k,q}(x)$ denote the two sums considered in the paper, and compare their Landau and Ramanuj...

conjecture | open | MathDB 页面 | 原文线索

V2-004244 | MATHDB-351878

Fractional-sum counting conjecture for singular-series analysis

摘录 / EXCERPT: Let $\delta>0$ and $Q>1/\delta$. For fixed k, let S be the number of tuples … with $q_i\in[Q,2Q]$, a_i taken modulo q_i, satisfying $\left\|a_i/q_i\right\|\leq\delta$ for e...

open_problem | open | MathDB 页面 | 原文线索

V2-004245 | MATHDB-351881

The inhomogeneous Khintchine-Groshev conjecture in the critical cases

摘录 / EXCERPT: Let n,m be positive integers with $nm=2$, let $\psi:\mathbb{N}\to\mathbb{R}_{\geq 0}$, and let $\mathbf{y}\in\mathbb{R}^m$. Let $\mathcal A_{n,m}^{\mathbf{y}}(\psi)$ denote the s...

conjecture | open | MathDB 页面 | 原文线索

V2-004246 | MATHDB-351882

The general inhomogeneous Khintchine-Groshev conjecture for critical dimensions

摘录 / EXCERPT: Let n,m be positive integers with $nm=2$, and let $\Psi=(B_q)_{q\in\mathbb{N}}$ be any sequence of balls in $\mathbb{R}^m/\mathbb{Z}^m$. Let $\mathcal A_{n,m}(\Psi)$ denote the c...

conjecture | open | MathDB 页面 | 原文线索

V2-004247 | MATHDB-351889

The generalized Landau conjecture for quadratic polynomials

摘录 / EXCERPT: Generalized Landau conjecture. The polynomial f represents infinitely many primes.

conjecture | open | MathDB 页面 | 原文线索

V2-004248 | MATHDB-351902

Chinburg-Reid-Stover's surgery-prime conjecture for twist knots

摘录 / EXCERPT: Chinburg-Reid-Stover's conjecture. In the notation above,

conjecture | open | MathDB 页面 | 原文线索

V2-004249 | MATHDB-351914

EKR conjecture for PSL(2,q) on pairs of projective-line points

摘录 / EXCERPT: Let q be a prime power, and let $\operatorname{PSL}(2,q)$ act on the 2-subsets of the projective line $\operatorname{PG}(1,q)$. Recall that the group has the Erdős-Ko-Rado (EKR...

conjecture | open | MathDB 页面 | 原文线索

V2-004250 | MATHDB-351974

The Iwasawa-Greenberg main conjecture for non-ordinary modular forms

摘录 / EXCERPT: Let $\mathcal{K}$ be the imaginary quadratic field, let $\Gamma_\mathcal{K}$ be the relevant Galois group, and let $\mathcal{O}^{\mathrm{ur}}_L$ be the ring of integers in the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-004251 | MATHDB-351975

Visibility density conjecture for polynomially defined lattice points

摘录 / EXCERPT: Let $V(a_n,a_{n-1},\dots,a_1)$ denote the set of lattice points associated with the coefficient vector $(a_n,a_{n-1},\dots,a_1)$, and let dens denote its density....

conjecture | open | MathDB 页面 | 原文线索

V2-004252 | MATHDB-351976

A sufficient condition for full density of $V(1,1)$

摘录 / EXCERPT: Full-density criterion for $V(1,1)$. If

conjecture | open | MathDB 页面 | 原文线索

V2-004253 | MATHDB-351978

Romik's conjecture on the p-adic properties of the sequence d(n)

摘录 / EXCERPT: Let $d(n)$ be the integer sequence defined by ... where $\theta_3(z):=\sum_{n=-\infty}^{\infty}e^{\pi i n^2z}$. Romik's conjecture. Let p be an odd prime. Then: 1. If...

conjecture | open | MathDB 页面 | 原文线索

V2-004254 | MATHDB-351983

Vogan's quantization conjecture for nilpotent covers

摘录 / EXCERPT: Vogan's quantization conjecture. For each such cover $\widetilde{\mathcal{O}}$, there is a canonically defined completely prime primitive ideal $I_0(\widetilde{\mathcal{O}})$ in th...

conjecture | open | MathDB 页面 | 原文线索

V2-004255 | MATHDB-351986

Algebraic dependence conjecture for hyperelliptic constants

摘录 / EXCERPT: Let π_n denote the hyperelliptic constant associated with the index n. Let $n\in\mathbb{N}$ be composite and not a prime power, so that $n\neq p^m$ for $p\in\mathbb{P}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004256 | MATHDB-351995

Uniqueness conjecture for maximal entries in the associated solution tree

摘录 / EXCERPT: Let $\mathbb T$ be the tree constructed from positive integer solutions of the paper's Diophantine equation, and let (x,y,z) be a positive integer solution of that equation. The...

conjecture | open | MathDB 页面 | 原文线索

V2-004257 | MATHDB-352013

The SP Goldbach conjecture

摘录 / EXCERPT: SP Goldbach conjecture. Every sufficiently large natural number, whether even or odd, can be written as a sum of two SP numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-004258 | MATHDB-352014

The SP Goldbach conjecture for SP numbers

摘录 / EXCERPT: SP Goldbach conjecture for SP numbers. Every SP number greater than 27 can be written as a sum of two SP numbers in at least one way.

conjecture | open | MathDB 页面 | 原文线索

V2-004259 | MATHDB-352015

The SP-square interval conjecture

摘录 / EXCERPT: SP-square interval conjecture. There exists at least one SP number between consecutive square numbers greater than 500.

conjecture | open | MathDB 页面 | 原文线索

V2-004260 | MATHDB-352025

Harder's Euler-factor conjecture for Siegel modular forms

摘录 / EXCERPT: Let k and j be non-negative integers with j even and $k\geq 3$. Let $f=\sum a(n,f)q^n\in S_{2k+j-2}(SL_2(\mathbb Z))$ be a primitive form, and suppose that a sufficiently lar...

conjecture | open | MathDB 页面 | 原文线索

V2-004261 | MATHDB-352026

Minimal-weight conjecture for Weyl-invariant E8 weak Jacobi forms

摘录 / EXCERPT: Let t be an index, and let $W(E_8)$ denote the Weyl group of the E_8 root system. A weak Jacobi form is allowed to have Fourier terms with negative discriminant, and its weight...

conjecture | open | MathDB 页面 | 原文线索

V2-004262 | MATHDB-352028

Non-unimodular lattice conjecture for index-one weak Jacobi forms

摘录 / EXCERPT: Let L be an even positive definite lattice, meaning $(v,v)\in2\mathbb Z$ for $v\in L$, and assume that L is irreducible and not unimodular. Consider the free module of weak Jac...

conjecture | open | MathDB 页面 | 原文线索

V2-004263 | MATHDB-352037

Catlin's conjecture on rational approximations

摘录 / EXCERPT: Let $Delta_1,Delta_2,\dots\geqslant0$, define ... and let CA be the set of $x\in[0,1]$ for which $|x-a/q|<\Delta_q$ holds for infinitely many pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-004264 | MATHDB-352047

Prime rational-map common-measure conjecture

摘录 / EXCERPT: Prime common-measure conjecture. If Q and R have no common iteration, then either μ is discrete or μ is the measure of maximal entropy of every element of the semigroup...

conjecture | open | MathDB 页面 | 原文线索

V2-004265 | MATHDB-352063

The level-two finite multiple zeta value conjecture for finite Euler sums

摘录 / EXCERPT: Let $\mathcal{Z}_{\mathcal{A}}^{(2)}$ and $\mathcal{E}$ be the $\mathbb{Q}$-vector spaces of level-two finite multiple zeta values and finite Euler sums, respectively. For an index...

conjecture | open | MathDB 页面 | 原文线索

V2-004266 | MATHDB-352064

The odd-index basis conjecture for ordinary finite multiple zeta values

摘录 / EXCERPT: Let $\mathcal{Z}_{\mathcal{A}}^{(1)}$ be the $\mathbb{Q}$-vector space of ordinary finite multiple zeta values, and let $\zeta_{\mathcal{A}}^{(2)}(k_1,\ldots,k_r)$ denote a level-t...

conjecture | open | MathDB 页面 | 原文线索

V2-004267 | MATHDB-352065

The level-two weighted sum formula conjecture

摘录 / EXCERPT: For $r\geq 1$ and $0\leq a\leq r$, consider all indices $(k_1,\ldots,k_r)$ with $k_i\in\{1,2\}$ and exactly a entries equal to 2. Let $\#\{i\mid i\text{ odd},\ k_i=2\}$ count t...

conjecture | open | MathDB 页面 | 原文线索

V2-004268 | MATHDB-352085

The prime-value conjecture for the odd-perfect-number factor t

摘录 / EXCERPT: Prime-value conjecture. Then $t=2^r-1$ is prime. The claim is presented as a natural prediction in the paper's discussion of remaining cases; it is not resolved there and is intend...

conjecture | open | MathDB 页面 | 原文线索

V2-004269 | MATHDB-352110

Automaticity conjecture for the sequence of integer parts of consecutive-term ratios

摘录 / EXCERPT: Automaticity conjecture. The sequence $(a(n))_{n\in\mathbb{N}}$ is 2-automatic. The paper establishes that $a(n)\in\{0,1,2,3,4\}$, while the stronger automaticity assertion remai...

conjecture | open | MathDB 页面 | 原文线索

V2-004270 | MATHDB-352117

Density conjecture for primes in the rational 2-Selmer group of S_n-fields

摘录 / EXCERPT: Let p be a prime and let $n=2m$. Write $\operatorname{Sel}_2^{\mathbb{Q}}(F)$ for the rational part of the 2-Selmer group of an S_n-number field F, and let $c(n,p)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-004271 | MATHDB-352118

Independence conjecture for rational 2-Selmer prime conditions

摘录 / EXCERPT: Let $n=2m$, let S be a finite set of primes, let T be a finite set of primes (or an infinite set when $n>2$) disjoint from S, and let $X_{S,T}$ be the set of S_n-number fie...

conjecture | open | MathDB 页面 | 原文线索

V2-004272 | MATHDB-352157

The multiplicative persistence conjecture

摘录 / EXCERPT: Let $f(n)$ be the product of the decimal digits of n, let $Xi(n)$ be the smallest nonnegative integer k such that $f^k(n)\leq 9$, and let $\Delta(n)$ be the resulting one-digit...

conjecture | open | MathDB 页面 | 原文线索

V2-004273 | MATHDB-352158

The bounded-power-of-two conjecture for decimal digit-product fibers

摘录 / EXCERPT: For integers $n_2,\ldots,n_9$, define the set of decimal integers with prescribed prime-factor data by ... Here dec denotes the decimal-digit construction used in th...

conjecture | open | MathDB 页面 | 原文线索

V2-004274 | MATHDB-352159

Lillie's conjecture on primorial twin prime pairs

摘录 / EXCERPT: Lillie's conjecture. The total expected number of primorial twin prime pairs is approximately three.

conjecture | open | MathDB 页面 | 原文线索

V2-004275 | MATHDB-352160

The universal primorial form of the twin prime conjecture

摘录 / EXCERPT: Universal primorial conjecture. There are expected to be infinitely many numbers $x=Kp_n\#+g$ such that both x and $x+2$ are prime.

conjecture | open | MathDB 页面 | 原文线索

V2-004276 | MATHDB-352161

Caldwell and Gallot's conjecture on the expected number of primorial primes

摘录 / EXCERPT: Caldwell and Gallot's conjecture. The expected number of primorial primes of each of the forms $p_n\#\pm1$ less than or equal to $p_N\#\pm1$, respectively, are both approximately

conjecture | open | MathDB 页面 | 原文线索

V2-004277 | MATHDB-352163

Bounded complete weight vectors for hypersurface minimization

摘录 / EXCERPT: Let (T,w) be a weight system, where $T\in\operatorname{GL}(n+1,\mathbb{Z})$ and $w\in\mathbb{Z}_{\ge 0}^{n+1}$ is its weight vector. For a homogeneous polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-004278 | MATHDB-352184

The Omega conjecture for the prime numbers

摘录 / EXCERPT: Omega conjecture for the prime numbers. Based on the properties of Omega sequences, Omega spaces, and the Kernel of Omega space, there is an algorithm to determine the prime...

conjecture | open | MathDB 页面 | 原文线索

V2-004279 | MATHDB-352194

Xiao-Zhang's conjecture on unitary Friedberg-Jacquet periods

摘录 / EXCERPT: Let F be a CM field with maximal totally real subfield F^+, let p be an odd rational prime splitting completely in $F/\mathbb{Q}$, and assume $F^+\neq\mathbb{Q}$, F contain...

conjecture | open | MathDB 页面 | 原文线索

V2-004280 | MATHDB-352197

Cohomological P-invariant conjecture for Hilbert modular forms

摘录 / EXCERPT: Let Σ_{∞} be the set of archimedean places, let $S\subset\Sigma_{\infty}$, and let $r\in\{0,1\}^{S}$. Let π, τ, the cohomology classes $\mathbf{g}^{(\delta)}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004281 | MATHDB-352241

Infinite-family conjecture for non-Paley pseudo-Paley graphs

摘录 / EXCERPT: For a prime $p\geq 3$, consider subsets $I\subset\{0,1,\ldots,p\}$ with $0\in I$ and $|I|=(p+1)/2$, and the associated graph $PP(p^4,p+1,I)$. Let g be the primitive element used...

conjecture | open | MathDB 页面 | 原文线索

V2-004282 | MATHDB-352254

The salvaged residual-representation ghost conjecture

摘录 / EXCERPT: Let $p\geq 5$, and let $\overline{\rho}$ be a residual Galois representation that is regular in the sense specified in the paper. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-004283 | MATHDB-352267

Binary tangent inequality conjecture over primes

摘录 / EXCERPT: Binary tangent inequality conjecture. The assertion above holds.

conjecture | open | MathDB 页面 | 原文线索

V2-004284 | MATHDB-352268

Ternary tangent inequality conjecture over primes

摘录 / EXCERPT: Ternary tangent inequality conjecture. The assertion above holds.

conjecture | open | MathDB 页面 | 原文线索

V2-004285 | MATHDB-352282

The polynomial bound conjecture for intersecting subsets of the affine general linear group

摘录 / EXCERPT: Let q be a prime power, let $\AGL(2,q)$ act on the affine plane, and let $\mathcal{F}\subseteq\AGL(2,q)$ be intersecting, meaning that any two elements of $\mathcal{F}$ agree on...

conjecture | open | MathDB 页面 | 原文线索

V2-004286 | MATHDB-352304

The conjecture that squarefree numbers have arbitrarily small power-sized gaps

摘录 / EXCERPT: Squarefree-gap conjecture. For any $\epsilon>0$ and x sufficiently large in terms of ϵ, the interval $(x,x+x^\epsilon]$ always contains a squarefree number.

conjecture | open | MathDB 页面 | 原文线索

V2-004287 | MATHDB-352309

Standard Verlinde conjecture for Bershadsky-Polyakov minimal models

摘录 / EXCERPT: Let k be admissible-nondegenerate. For $\ell,\ell'\in\mathbb{Z}$, $[j],[j']\in\mathbb{R}/\mathbb{Z}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-004288 | MATHDB-352378

Leclere-Morier-Genoud-Ovsienko-Veselov conjecture on q-real radii of convergence

摘录 / EXCERPT: For a real number $x otin\frac{r}{s}$? Wait. Let $[x]_q$ denote the q-deformation of a real number $x e$? defined as the stabilized formal power series associated with rational a...

conjecture | open | MathDB 页面 | 原文线索

V2-004289 | MATHDB-352392

The loop equation conjecture for every element of S

摘录 / EXCERPT: Let S be the set of odd primes, and for $a,b\in S$ let $a\bullet b$ denote the smallest prime strictly greater than $|a-b|$. Loop equation conjecture. For every $x\in S$, the equ...

conjecture | open | MathDB 页面 | 原文线索

V2-004290 | MATHDB-352414

Strong Lang-Vojta conjecture for exceptional sets

摘录 / EXCERPT: Strong Lang-Vojta conjecture. The pair (X,D) is of log general type if and only if there exists a proper closed subset

conjecture | open | MathDB 页面 | 原文线索

V2-004291 | MATHDB-352421

Okounkov's structural conjecture for qMZVs

摘录 / EXCERPT: Okounkov's conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-004292 | MATHDB-352429

Local-global compatibility for automorphic Galois representations

摘录 / EXCERPT: Let F be a number field, let $\pi=\otimes'_v\pi_v$ be an algebraic cuspidal automorphic representation of $\operatorname{GL}_n(\mathbb{A}_F)$, let l be a rational prime, and fi...

conjecture | open | MathDB 页面 | 原文线索

V2-004293 | MATHDB-352434

The three-prime floor-function set formula

摘录 / EXCERPT: Let $G(x)$ denote the number of primes in the set $\left\{\left\lfloor x/n\right\rfloor:1\le n\le x\right\}$. Suppose that $x=p_1p_2p_3$, where p_1,p_2,p_3 are primes satisfying...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-004294 | MATHDB-352449

Short-interval conjecture for integers without small prime factors

摘录 / EXCERPT: Let X, y and H be parameters satisfying the conditions denoted by ... , and let $\alpha_y(n)$ be the indicator of integers without prime factors at most y, with mean density...

conjecture | open | MathDB 页面 | 原文线索

V2-004295 | MATHDB-352480

Geography of log adjunction

摘录 / EXCERPT: Let X/Z be a proper morphism, let η be the general point of Z, and let S_i be distinct prime b-divisors of X. Let $\mathfrak P$ be a polyhedron in the wlc geography of...

conjecture | open | MathDB 页面 | 原文线索

V2-004296 | MATHDB-352492

Bounded hyperplane order in Green-Sanders examples

摘录 / EXCERPT: For a prime p, let $\mathrm{GS}(n,p)$ denote the Green-Sanders example, and let an m-HOP$_2$ mean the paper's m-th order hyperplane order property. Green-Sanders HOP2 conject...

conjecture | open | MathDB 页面 | 原文线索

V2-004297 | MATHDB-352497

Asymptotic constant conjecture for the Steinhaus function

摘录 / EXCERPT: Let $s(d)$ denote the function measuring the extremal quantity studied in the paper, where d is the relevant dimension parameter. Asymptotic constant conjecture. The limit ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-004298 | MATHDB-352503

Density conjecture for Serre curves

摘录 / EXCERPT: Density conjecture. The density of primes p for which $\operatorname{gcd}(\#E(\mathbb{F}_p),p-1)=1$ is

conjecture | open | MathDB 页面 | 原文线索

V2-004299 | MATHDB-352512

The two-prime tangent equation conjecture

摘录 / EXCERPT: Two-prime tangent equation conjecture. There exists $c_0>1$ such that, for every fixed $1<c<c_0$, this equation has a solution in prime numbers p_1,p_2. The conjecture predicts s...

conjecture | open | MathDB 页面 | 原文线索

V2-004300 | MATHDB-352537

Eventual complexity formula for products of the form p(q3^j+1)

摘录 / EXCERPT: Let $\|n\|$ denote the complexity of a natural number, and let p,q be natural numbers. Product complexity conjecture. For each pair of natural numbers p and q, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-004301 | MATHDB-352540

Characterization of the a-sequence by complexity and the set A

摘录 / EXCERPT: Let $\|n\|$ denote the complexity of a natural number, and define ... Let $(a_\alpha)_{\alpha<\xi}$ be the transfinite sequence associated with numbers of complexity divisible by 3...

conjecture | open | MathDB 页面 | 原文线索

V2-004302 | MATHDB-352556

Beiter's conjecture on the height of ternary cyclotomic polynomials

摘录 / EXCERPT: For odd primes $p<q<r$, let $A(n)$ denote the height of the cyclotomic polynomial $4\Phi_n(X)$. Beiter's conjecture. ... for all odd primes $p<q<r$. The conjecture was disproved by L...

conjecture | open | MathDB 页面 | 原文线索

V2-004303 | MATHDB-352557

Gallot-Moree conjecture on the maximal ternary cyclotomic height

摘录 / EXCERPT: Let $A(n)$ denote the height of $4\Phi_n(X)$, and for an odd prime p define ... where q and r range over primes. Gallot-Moree conjecture. ... for every prime p. The survey repo...

conjecture | open | MathDB 页面 | 原文线索

V2-004304 | MATHDB-352581

Vinogradov's least quadratic nonresidue conjecture

摘录 / EXCERPT: Let p be a prime, and let $n(p)$ denote the least natural number that is not a quadratic residue modulo p. For any fixed $\epsilon>0$, the least quadratic nonresidue satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-004305 | MATHDB-352584

The full wreath-product Galois group conjecture for f_p

摘录 / EXCERPT: Let p be a prime, let f_p be the polynomial studied in the source, let g_p be its associated auxiliary polynomial, and write ... Let $\mathbb Q(f_p)$ denote the splitting field...

conjecture | open | MathDB 页面 | 原文线索

V2-004306 | MATHDB-352593

Waring conjecture for middle-$\frac{1}{\alpha}$ Cantor sets

摘录 / EXCERPT: Waring conjecture for C_α.

conjecture | open | MathDB 页面 | 原文线索

V2-004307 | MATHDB-352597

The alternating-group soluble-graph diameter conjecture

摘录 / EXCERPT: Let $p\geqslant 11$ be a prime with $p\equiv 3\pmod{4}$. For the alternating group A_p, let $\delta_{\mathcal{S}}(A_p)$ denote the diameter of its soluble graph. Alternating-grou...

conjecture | open | MathDB 页面 | 原文线索

V2-004308 | MATHDB-352604

The generalized Collatz finite-cycles conjecture

摘录 / EXCERPT: Let p,q and $r:\mathbb{Z}/p\mathbb{Z}\setminus\{0\}\to\mathbb{Z}$ be as in the paper's definition of the generalized Collatz maps S and C, and suppose that p and q are co...

conjecture | open | MathDB 页面 | 原文线索

V2-004309 | MATHDB-352606

Granville's least-prime primitive-root conjecture

摘录 / EXCERPT: Let q be a fixed integer with $q\ne 1$ and $q\ne n^2$, and let p be a prime. A number q is a primitive root modulo p when its residue class generates the multiplicative gro...

conjecture | open | MathDB 页面 | 原文线索

V2-004310 | MATHDB-352623

Conjecture on recursive uniformizers and eventual stability of truncated expansions

摘录 / EXCERPT: Recursive uniformizer and stability conjecture. For each prime $p\geq 3$ and integer $n\geq 1$, there exists a series of polynomials $\{\mathcal R_p^{m,n}(T)\}_{m\geq 2}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-004311 | MATHDB-352640

Local Irregularity Conjecture for the locally irregular chromatic index

摘录 / EXCERPT: Let $\mathfrak{T}^{\prime}$ be the family consisting of all odd-length paths, all odd-length cycles, and the graphs obtained from the triangle by the recursive construction describ...

conjecture | open | MathDB 页面 | 原文线索

V2-004312 | MATHDB-352641

Four-color local irregularity conjecture for connected graphs

摘录 / EXCERPT: Let $\mathfrak{T}^{\prime}$ be the family of odd-length paths, odd-length cycles, and the recursively constructed cactus graphs described above. For a colorable graph G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004313 | MATHDB-352654

The statistical valuation conjecture for Stirling numbers

摘录 / EXCERPT: Let c be a positive integer, let $S(n,k)$ denote a Stirling number of the second kind, let bd_2 be the 2-adic valuation, and let $c3_2(m)$ denote the sum of the binary digits...

conjecture | open | MathDB 页面 | 原文线索

V2-004314 | MATHDB-352655

The $a\equiv3\pmod 4$ Stirling valuation conjecture

摘录 / EXCERPT: Let $f(b,c)=\sigma_2(b)-\sigma_2(c)+\nu_2\left(\binom{2c-b-1}{c}\right)$, where $c3_2$ is the binary digit sum and bd_2 is the 2-adic valuation. Let a,b,c,h be positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-004315 | MATHDB-352662

Fine Selmer torsion conjecture for elliptic modular forms

摘录 / EXCERPT: Let f be a normalized cuspidal eigenform as in the source, let A_f be the associated discrete Galois representation, and let F_∞ be a $\mathbb{Z}_{p}$-extension of a num...

conjecture | open | MathDB 页面 | 原文线索

V2-004316 | MATHDB-352663

Pseudo-nullity conjecture for fine Selmer groups of elliptic modular forms

摘录 / EXCERPT: Let L_∞ be a $\mathbb{Z}_{p}^{2}$-extension of a number field F containing the cyclotomic $\mathbb{Z}_{p}$-extension F^{cyc}. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004317 | MATHDB-352666

Classification conjecture for Sidon polynomials over prime fields

摘录 / EXCERPT: Sidon-polynomial classification conjecture. If $P\in\mathbb{F}_p[x]$ is a Sidon polynomial, then either

conjecture | open | MathDB 页面 | 原文线索

V2-004318 | MATHDB-352667

Hardy-Littlewood-Elliott conjecture

摘录 / EXCERPT: Hardy-Littlewood-Elliott conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-004319 | MATHDB-352670

Arithmetic progressions of twin primes of every length

摘录 / EXCERPT: Twin-prime progression conjecture. For every natural number l with $l>2$, there exist arithmetic progressions of twin primes with l terms.

conjecture | open | MathDB 页面 | 原文线索

V2-004320 | MATHDB-352671

Asymptotic absence of type-II twin-prime bias

摘录 / EXCERPT: Type-II bias conjecture. For any two residue classes a_i and a_j,

conjecture | open | MathDB 页面 | 原文线索

V2-004321 | MATHDB-352672

Asymptotic absence of type-III prime and twin-prime bias

摘录 / EXCERPT: Type-III bias conjecture. As $x\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004322 | MATHDB-352689

The squarefree v-palindromes conjecture

摘录 / EXCERPT: Squarefree v-palindromes conjecture. There are infinitely many v-palindromes n in base b such that both n and $r_b(n)$ are squarefree.

conjecture | open | MathDB 页面 | 原文线索

V2-004323 | MATHDB-352690

The arbitrarily long gaps between v-palindromes conjecture

摘录 / EXCERPT: Arbitrarily long gaps conjecture. Consecutive positive integers each not a v-palindrome in base b can be arbitrarily long.

conjecture | open | MathDB 页面 | 原文线索

V2-004324 | MATHDB-352693

Diophantine conjecture on the square condition for $(k^2-n^2)^2+4k^4n^4$

摘录 / EXCERPT: Conjecture A. Every such integer solution satisfies either $n=\pm k$, or $n=0$, or $k=0$. The conjecture is motivated by the absence of nontrivial examples in a search with...

conjecture | open | MathDB 页面 | 原文线索

V2-004325 | MATHDB-352709

The squarefree-values conjecture for polynomials

摘录 / EXCERPT: Squarefree-values conjecture. Then $f(n)/B'$ is squarefree for infinitely many values of n.

conjecture | open | MathDB 页面 | 原文线索

V2-004326 | MATHDB-352713

Cluckers-Nguyen uniform bound conjecture for exponential sums

摘录 / EXCERPT: Let f be a polynomial in n variables, with s and d as in the preceding setup, and define ... Here $N\geq 1$ and $\psi:\mathbb Z/N\mathbb Z\to\mathbb C^\times$ is a group mono...

conjecture | open | MathDB 页面 | 原文线索

V2-004327 | MATHDB-352715

Strong monodromy conjecture for Igusa local zeta functions

摘录 / EXCERPT: Let L be a non-Archimedean local field over O_K, let $f\in\operatorname{O}_K[x_1,\ldots,x_n]$ be a nonconstant polynomial, let Φ be a Schwartz-Bruhat functi...

conjecture | open | MathDB 页面 | 原文线索

V2-004328 | MATHDB-352749

Navarro's Brauer character Glauberman correspondence conjecture

摘录 / EXCERPT: Navarro's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004329 | MATHDB-352755

Existence of small nontrivial μ-Sondow numbers

摘录 / EXCERPT: Small μ-Sondow number conjecture. For all $ \notin \{0,1,-1,2,-2,4,16\}$, $$

conjecture | open | MathDB 页面 | 原文线索

V2-004330 | MATHDB-352756

Hyperbolicity conjecture for prime twist-reduced 3-highly twisted link diagrams

摘录 / EXCERPT: Let L be a link in S^3 with a diagram that is prime, twist-reduced, and 3-highly twisted, and that has at least two twist regions. Hyperbolicity conjecture. The link L is h...

conjecture | open | MathDB 页面 | 原文线索

V2-004331 | MATHDB-352757

The Coates-Schmidt-Greenberg cotorsion conjecture

摘录 / EXCERPT: Let p be a prime, let $\mathbb{Q}_{\operatorname{cyc}}$ be the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$, and let Λ be its Iwasawa algebra. For $i=1,2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004332 | MATHDB-352758

Linearity conjecture for totally invariant prime divisors of projective space

摘录 / EXCERPT: Linearity conjecture. Every totally invariant prime divisor of $\mathbb{P}^{n}$ is linear.

conjecture | open | MathDB 页面 | 原文线索

V2-004333 | MATHDB-352774

Conjecture on cross-characteristic composition factors in isospectral groups

摘录 / EXCERPT: Cross-characteristic composition-factor conjecture. The group S cannot be a group of Lie type in characteristic coprime to q.

conjecture | open | MathDB 页面 | 原文线索

V2-004334 | MATHDB-352778

The prime-level dimension conjecture for spaces of cusp forms

摘录 / EXCERPT: Let $k\ge 4$ be an even integer and let $u>1$ be a real number. For a prime p, write $d_k(p)$ for the dimension of the space of weight-k cusp forms of prime level p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004335 | MATHDB-352793

A triple-sum Ramanujan-type supercongruence

摘录 / EXCERPT: Triple-sum Ramanujan-type congruence. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004336 | MATHDB-352796

Prefix-closure conjecture for square-digit solutions

摘录 / EXCERPT: Let $\langle a_1',\dots,a_l'\rangle$ be a solution of length $l\ge 2$, where a solution is a digit sequence satisfying the paper's square-digit condition. Prefix-closure conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-004337 | MATHDB-352797

Asymptotic formula for suffix-restricted squares in base 10

摘录 / EXCERPT: For a base $B\ge 2$ and a digit set $D\subseteq\{0,1,\dots,B-1\}$ with $|D|\ge 2$, let $\omega_B^D(n)$ denote the number of n-digit suffixes in base B from D that occur as th...

conjecture | open | MathDB 页面 | 原文线索

V2-004338 | MATHDB-352798

Asymptotic formula for leading-digit-restricted squares

摘录 / EXCERPT: Let $B\in\mathbb{N}_{\ge 2}$ and let $D\subseteq\{0,1,\dots,B-1\}$ satisfy $|D|\ge 2$. Let $\alpha_B^D(n)$ be the number of integers $B^{n-\frac12}\le x<B^{n+\frac12}$ such that th...

conjecture | open | MathDB 页面 | 原文线索

V2-004339 | MATHDB-352800

Three-variable squared-argument undecidability conjecture for exponential Diophantine equations

摘录 / EXCERPT: An exponential Diophantine equation over $\mathbb Q$ is an equation built using variables, rational constants, arithmetic operations, and exponentiation with nonnegative base and e...

conjecture | open | MathDB 页面 | 原文线索

V2-004340 | MATHDB-352808

The Schur multiplier exponent conjecture for finite p-groups

摘录 / EXCERPT: Let G be a finite p-group, where p is prime. The Schur multiplier exponent conjecture. ... This conjecture strengthens the known bound suggested by the counterexamples to Schur...

conjecture | open | MathDB 页面 | 原文线索

V2-004341 | MATHDB-352838

The main conjecture equating arithmetic and analytic invariants of p-class and p-ramified torsion groups

摘录 / EXCERPT: Let $\Phi=\Phi^+\cup\Phi^-$ be the set of abelian p-adic irreducible characters. For each $\varphi\in\Phi$, let $m^{\rm ar}_\varphi(\mathcal H^+)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004342 | MATHDB-352872

Kurihara's determinant conjecture for minus theta-kernel bases

摘录 / EXCERPT: Let H/F be a finite abelian CM-extension, let p be an odd prime, and put $r=\#S'_f-1$. Let $\operatorname{Ker}(\theta_V)_H$ be the theta-kernel module, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004343 | MATHDB-352888

Almost-flat fiducial conjecture for SICs in dimensions n^2+3

摘录 / EXCERPT: Let $d=n^2+3=p$ be a prime in the specified sequence of dimensions, and let $ obreak\hat{\Psi}=(x_0,x_1,\ldots,x_{d-1})^{\rm T}$ be an un-normalized SIC fiducial. Almost-flat fiduc...

conjecture | open | MathDB 页面 | 原文线索

V2-004344 | MATHDB-352889

Small-ray-class-field conjecture for SIC fiducials

摘录 / EXCERPT: Let $d=n^2+3=p$ be a prime in the specified sequence, let K be the real quadratic base field, and let $\hat{\Psi}$ be the un-normalized SIC fiducial with components x_j. The tw...

conjecture | open | MathDB 页面 | 原文线索

V2-004345 | MATHDB-352890

Stark phase unit conjecture for SIC fiducials

摘录 / EXCERPT: Let $d=n^2+3=p$ be a prime in the specified sequence, let K be the real quadratic field used in the construction, and let $e^{i\vartheta_j}$ be the phases in the almost-flat fidu...

conjecture | open | MathDB 页面 | 原文线索

V2-004346 | MATHDB-352891

Non-square Stark unit conjecture

摘录 / EXCERPT: Let $K^{\mathfrak p\mathfrak j}$ be the ray class field appearing in Theorem, let ϵ_σ be its Stark unit, and let ξ_ℓ and x_0 be the associated quantities in...

conjecture | open | MathDB 页面 | 原文线索

V2-004347 | MATHDB-352894

Merca's conjecture on divisibility of divisor sums in arithmetic progressions

摘录 / EXCERPT: For an integer $k\geq0$, let ... denote the number of positive divisors of n. Merca's conjecture. If ... for every $n\geq0$, then there is a sequence of prime numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-004348 | MATHDB-352895

The conjecture that Euler's totient has no nontrivial prime congruences

摘录 / EXCERPT: Let $p>2$ be prime, and let A' and B' be coprime integers. A Ramanujan-type congruence for Euler's totient function $cvarphi$ would have the form ... for every $n\geq0$. The toti...

conjecture | open | MathDB 页面 | 原文线索

V2-004349 | MATHDB-352903

A.P. Mani and R.J. Stones' conjecture for the Tutte polynomial of complete graphs

摘录 / EXCERPT: Let $T(K_n;a,b)$ denote the Tutte polynomial of the complete graph K_n, and let p be an odd prime and k a positive integer. Let $ u$ denote congruence modulo the indicated po...

conjecture | open | MathDB 页面 | 原文线索

V2-004350 | MATHDB-352957

Conjecture on common roots of Ramanujan-type polynomials and roots of unity

摘录 / EXCERPT: Common-root conjecture. If $2\mid k$, then the only common roots of $\mathfrak{R}_{2k+1,p}(z)$ and $(pz)^{2k}-1$ are $z=\pm i/p$; if $2\nmid k$, then the two polynomials have no co...

conjecture | open | MathDB 页面 | 原文线索

V2-004351 | MATHDB-352969

Precise inversion of adjunction conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $(X,\mathfrak{a})$ be a log pair, let D be a normal Cartier prime divisor, and let $x\in D$ be a closed point. Suppose that D is not contained in the cosupport of the...

conjecture | open | MathDB 页面 | 原文线索

V2-004352 | MATHDB-352980

The universal torsion conjecture for infinite-type Kac-Moody groups

摘录 / EXCERPT: Let $K(A)$ be a Kac-Moody group of infinite type, and let $G(A)$ denote the associated group. For a prime number p, consider the integral cohomology ring $H^*(BG(A);\mathbb Z)$....

conjecture | open | MathDB 页面 | 原文线索

V2-004353 | MATHDB-352986

Primality conjecture for invariant ideals in skew power series rings

摘录 / EXCERPT: Let R be a Noetherian ring, complete with respect to a positive filtration topology, and let (σ,δ) be a continuous skew derivation compatible with the filtration. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-004354 | MATHDB-352993

Relative Brauer relation generation conjecture for finite p-groups

摘录 / EXCERPT: Let p be a prime, let G be a finite p-group, and let $K(G,C_p)$ denote the module of relative (G,C_p)-Brauer relations. For a sub-quotient $P=K/N$ of $G\times C_p$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-004355 | MATHDB-352994

Bandini-Valentino's Atkin-Lehner conjecture for Drinfeld modular forms at level T

摘录 / EXCERPT: Let A be the coefficient ring and let T be the prime defining the level. For weights k,l, write $S_{k,l}(\mathrm{GL}_2(A))$ for the space of Drinfeld cusp forms for...

conjecture | open | MathDB 页面 | 原文线索

V2-004356 | MATHDB-352995

Classification conjecture for monomially irreducible integers

摘录 / EXCERPT: Classification conjecture. N is monomially irreducible if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-004357 | MATHDB-353002

TMF divisibility conjecture for rational vertex operator algebras

摘录 / EXCERPT: Let V be a rational vertex operator algebra with central charge $c=24n$, where n is an integer, and suppose that V has only one irreducible module. Let the character of V b...

conjecture | open | MathDB 页面 | 原文线索

V2-004358 | MATHDB-353010

Poissonian correlation conjecture for fractional parts of αn^θ

摘录 / EXCERPT: Let $m\ge 2$, let $\alpha>0$, and let $0<\theta<1/m$. Consider the sequence of fractional parts $(\alpha n^\theta\bmod 1)_{n>0}$. Poissonian correlation conjecture. For every such...

conjecture | open | MathDB 页面 | 原文线索

V2-004359 | MATHDB-353018

The prime-field cylinder conjecture

摘录 / EXCERPT: Prime-field cylinder conjecture. The cylinder conjecture is true for all parameter triples (v,r,p), where p is a prime.

conjecture | open | MathDB 页面 | 原文线索

V2-004360 | MATHDB-353021

Greenberg's algebraic-rank conjecture for Coleman families

摘录 / EXCERPT: Let $\operatorname{\boldsymbol f}_U$ be a Coleman family, let $r_{\mathrm{min}}(\operatorname{\boldsymbol f}_U)$ be its minimal admissible generic rank, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004361 | MATHDB-353033

A prime-power cubic binomial supercongruence for primes congruent to 3 modulo 4

摘录 / EXCERPT: Let $p>3$ be a prime with $p\equiv3\pmod4$, and let r be a positive integer. The supercongruence. ... This is described as a generalization of an earlier case distinction; the stat...

conjecture | open | MathDB 页面 | 原文线索

V2-004362 | MATHDB-353059

Deligne's algebraicity conjecture for twisted symmetric fifth L-functions

摘录 / EXCERPT: Let $f(\tau)=\sum_{n=1}^{\infty}a_f(n)q^n\in S_\kappa(N,\omega)$ be a normalized elliptic newform of weight $\kappa\geq 2$, and let χ be a Dirichlet character. For a critical...

conjecture | open | MathDB 页面 | 原文线索

V2-004363 | MATHDB-353073

David-Weston conjecture on local torsion primes

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication. A prime p is a local torsion prime when E has good reduction at p and $E(\mathbb{Q}_p)[p]\neq 0$. Davi...

conjecture | open | MathDB 页面 | 原文线索

V2-004364 | MATHDB-353074

Wuthrich's μ-λ conjecture for fine Selmer groups

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve, let p be a prime of good reduction, and let $\mathcal{R}_{p^\infty}(E/\mathbb{Q}_{\operatorname{cyc}})$ be the fine Selmer group over the...

conjecture | open | MathDB 页面 | 原文线索

V2-004365 | MATHDB-353075

Delaunay's proportion conjecture for p-divisible Tate-Shafarevich orders

摘录 / EXCERPT: Fix a prime p. For elliptic curves $E/\mathbb{Q}$ ordered by height, consider the proportion of curves for which $p\mid\#Ш(E/\mathbb{Q})$. Delaunay's proportion conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-004366 | MATHDB-353105

The generalized Kaprekar height conjecture

摘录 / EXCERPT: Fix a base $b\geq 2$. Write every $u\geq 1$ uniquely as a tower of powers of b ending in $0<\omega\leq 1$, and let $\operatorname{ht}(u)$ be the height of this tower. Let $K(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-004367 | MATHDB-353120

Bell-Miles-Ward's rationality-natural-boundary dichotomy for dynamical zeta functions

摘录 / EXCERPT: Bell-Miles-Ward's conjecture. The function $\tilde{\zeta}_{\theta}(z)$ is either rational or admits a natural boundary. This is a Pólya-Carlson-type dichotomy for a dynamical zeta...

conjecture | open | MathDB 页面 | 原文线索

V2-004368 | MATHDB-353146

Polynomial divisor-sum conjecture in short intervals

摘录 / EXCERPT: Polynomial divisor-sum conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004369 | MATHDB-353147

Polylogarithmic bounds conjecture for exceptional three-variable integers

摘录 / EXCERPT: Let ... where $R_3(n)$ counts positive integral solutions of $n=xyz+x+y+z$. Polylogarithmic exceptional-set conjecture. There exist positive constants $A<B$ such that ... The authors e...

conjecture | open | MathDB 页面 | 原文线索

V2-004370 | MATHDB-353148

Eventual representability conjecture for four-variable sum-product equations

摘录 / EXCERPT: Eventual representability conjecture. Every integer $n>45752$ has at least one such representation; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-004371 | MATHDB-353195

Baruah-Begum conjecture on congruences for the smallest parts function of restricted overpartitions

摘录 / EXCERPT: Let $\overline{\operatorname{spt}}_\omega(n)$ denote the smallest-parts function of restricted overpartitions. For integers $\ell\geq1$, $k\geq0$, and $n\geq0$, consider its values...

conjecture | open | MathDB 页面 | 原文线索

V2-004372 | MATHDB-353254

Average eigenvalue bound for Hecke operators on Hermitian modular forms

摘录 / EXCERPT: Let K be the field and level data used in the paper, let $M_0(K)$ be the cuspidal space, and let $HE(K)$ be its Hecke-eigenform basis. Write ... For a prime $\ell\ne p$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004373 | MATHDB-353255

Spectral bounds for superspecial abelian-surface isogeny graphs

摘录 / EXCERPT: Let $p\ge 5$ and let $\ell\ne p$ be prime. Let $\mathcal{G}^{SS}_{2}(\ell,p)$ be the superspecial isogeny graph, with normalized adjacency eigenvalues ... where $m=|SS_2(p)|$, and se...

conjecture | open | MathDB 页面 | 原文线索

V2-004374 | MATHDB-353293

Rémond's height conjecture for division groups

摘录 / EXCERPT: Rémond's conjecture. The following forms are expected to hold:

conjecture | open | MathDB 页面 | 原文线索

V2-004375 | MATHDB-353302

Square-free smooth representative conjecture for finite-field residue classes

摘录 / EXCERPT: Square-free smooth representative conjecture. Theorem holds for any prime power $q\geq 3$, and for $q=2$ as long as $r\geq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-004376 | MATHDB-353304

Sun and Wang's lower-bound conjecture for weights of $W(E_8)$-invariant weak Jacobi forms

摘录 / EXCERPT: Let m be an index and let k be the weight of a $W(E_8)$-invariant weak Jacobi form. Sun and Wang's lower-bound conjecture. Every non-zero $W(E_8)$-invariant weak Jacobi form of...

conjecture | open | MathDB 页面 | 原文线索

V2-004377 | MATHDB-353305

A hypergeometric supercongruence for even d

摘录 / EXCERPT: Let $(x)_k$ denote the rising factorial, with $(x)_0=1$. Let $d\geqslant 2$ be an even integer, and let $p\equiv d+1\pmod{2d}$ be a prime. The hypergeometric supercongruence. ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-004378 | MATHDB-353306

A Dwork-type q-supercongruence for n congruent to 1 modulo 4

摘录 / EXCERPT: Let $n\equiv 1\pmod{4}$ be a positive integer and let $r\geqslant 1$. Write $(a;q)_k$ for the q-shifted factorial, $[m]=(1-q^m)/(1-q)$, and let $\Phi_m(q)$ denote the mth cyclo...

conjecture | open | MathDB 页面 | 原文线索

V2-004379 | MATHDB-353328

Dynamical-unit conjecture for Misiurewicz polynomials

摘录 / EXCERPT: Let $d=p^e$, where p is a prime and $e\geq 1$. Let c_0 be a root of $G_{d,m,n}^{\zeta}$ for some $m\geq 2$, $n\geq 1$, and let $\zeta\neq 1$ be a d-th root of unity. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-004380 | MATHDB-353411

Full Hausdorff dimension of the intermediate Dirichlet-improvable set

摘录 / EXCERPT: Let $n\geqslant 2$. Define ... where $\mathbf{DI}_n$ is the set of Dirichlet improvable points in $\mathbb R^n$, ${\mathbf{Bad}}_n$ is the set of badly approximable points, and...

conjecture | open | MathDB 页面 | 原文线索

V2-004381 | MATHDB-353437

Conjecture on even-period Browkin II expansions of square roots

摘录 / EXCERPT: Let $h\in\mathbb{Z}$ be even. Consider the p-adic continued fraction expansion produced by the Browkin II algorithm. Even-period Browkin II conjecture. For every even...

conjecture | open | MathDB 页面 | 原文线索

V2-004382 | MATHDB-353453

Conjecture on the distribution of integers by number of arithmetic anti-derivatives

摘录 / EXCERPT: Let $D:\mathbb{Z}\to\mathbb{Z}$ be the arithmetic derivative, and call an integer y an integral anti-derivative of x when $D(y)=x$. Let $i(x)$ denote the number of anti-derivat...

conjecture | open | MathDB 页面 | 原文线索

V2-004383 | MATHDB-353455

The bounded-exception conjecture for Markoff surface orbits

摘录 / EXCERPT: Let $a,k\in\mathbb Z$, and let ... be the Markoff surface over $\mathbb Z$, with $\mathcal G_{\mathcal M}$ the group generated by the three involutions, double sign changes, and coor...

conjecture | open | MathDB 页面 | 原文线索

V2-004384 | MATHDB-353465

Smooth-number approximation conjecture for Dirichlet polynomials

摘录 / EXCERPT: Smooth-number approximation conjecture. There exists a constant $A>0$ such that, for any $1\le x\le T$ and $2T\le t\le 5T$, uniformly as $T\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004385 | MATHDB-353467

The F A-prime subgroup conjecture for Artin groups

摘录 / EXCERPT: A group G has property ${\rm F}\mathcal{A}'$ if every simplicial action of G on a tree without inversion is locally elliptic. A subgroup is non-trivial if it is not the trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-004386 | MATHDB-353468

The F C-prime subgroup conjecture for Artin groups

摘录 / EXCERPT: A group G has property ${\rm F}\mathcal{C}'$ if every cellular action of G on a finite-dimensional CAT(0) cube complex is locally elliptic. A subgroup is non-trivial if it is...

conjecture | open | MathDB 页面 | 原文线索

V2-004387 | MATHDB-353505

Poisson genericity of Fibonacci, Rudin-Shapiro, and Thue-Morse sequences

摘录 / EXCERPT: Let an infinite sequence over a finite alphabet be formed either by concatenating the Fibonacci numbers in any base, by taking the Rudin-Shapiro sequence along the squares, or by t...

conjecture | open | MathDB 页面 | 原文线索

V2-004388 | MATHDB-353525

Fagan's Tchoukaillon Solitaire conjecture

摘录 / EXCERPT: Fagan's Tchoukaillon Solitaire conjecture. There exist infinitely many Tchoukaillon solitaire boards that satisfy the Fagan property.

conjecture | open | MathDB 页面 | 原文线索

V2-004389 | MATHDB-353527

Erdős-Sárközy-Szemerédi conjecture on primitive sets

摘录 / EXCERPT: For $x\ge 1$, let $[x,\infty)$ denote the integers at least x, and let $f(A)=\sum_{a\in A}1/(a\log a)$ for a primitive set A. Erdős-Sárközy-Szemerédi conjecture. ... This questio...

conjecture | open | MathDB 页面 | 原文线索

V2-004390 | MATHDB-353528

Banks-Martin conjecture for odd-prime-supported primitive sets

摘录 / EXCERPT: Let $\Omega(n)$ be the number of prime factors of n, counted with multiplicity, and let $\mathbb{N}_k=\{n:\Omega(n)=k\}$. For a set of odd primes $\mathcal Q$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-004391 | MATHDB-353529

The Epstein-Riemann conjecture for the identity Epstein zeta function

摘录 / EXCERPT: Let $\alpha\in\mathbb{N}$, let $\zeta_Q(s)$ be the Epstein zeta function associated with a real-valued positive definite $\alpha\times\alpha$ matrix Q, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004392 | MATHDB-353530

The asymptotic functional-relation conjecture for the discrete torus zeta function

摘录 / EXCERPT: Let $H_n(s)$ be the function defined from the discrete-torus spectral zeta function by the asymptotic expansion ... where ... Here $\zeta(\Delta,s)$ is the spectral zeta function of th...

conjecture | open | MathDB 页面 | 原文线索

V2-004393 | MATHDB-353572

The stochastic zeta conjecture

摘录 / EXCERPT: Let $d4a6$ be uniformly distributed on $[0,1]$, let $T>0$, and let $d709_\infty$ be the random holomorphic function defined from a sine-kernel point process by ... with the product i...

conjecture | open | MathDB 页面 | 原文线索

V2-004394 | MATHDB-353578

The common-largest-elements conjecture for D(4)-triples

摘录 / EXCERPT: Common-largest-elements conjecture. Then $\{a_1,a_2,b,c\}$ is a $D(4)$-quadruple.

conjecture | open | MathDB 页面 | 原文线索

V2-004395 | MATHDB-353587

Long's hypergeometric supercongruence conjecture

摘录 / EXCERPT: Let (r_1,r_2,q) range over ... Set $\mathbf{\alpha}_{(r_1,r_2)}=\{r_1,1-r_1,r_2,1-r_2\}$ and $\mathbf{\beta}_q=\{1,1,q,2-q\}$. Let $\mathrm{HD}_{(r_1,r_2,q)}$ denote the hypergeome...

conjecture | open | MathDB 页面 | 原文线索

V2-004396 | MATHDB-353605

Finiteness conjecture for high-degree arithmetic points in strata

摘录 / EXCERPT: Let $\textnormal{S}_{\alpha}$ be a stratum of abelian differentials up to scaling. An arithmetic point has a degree, and points of degree at least 3 are those whose arithmetic de...

conjecture | open | MathDB 页面 | 原文线索

V2-004397 | MATHDB-353609

Conjecture for consecutive Piatetski-Shapiro primes

摘录 / EXCERPT: Consecutive-prime conjecture. For every fixed $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004398 | MATHDB-353632

Ghosh-Sarnak conjecture on the number of Hasse failures for Markoff surfaces

摘录 / EXCERPT: For each integer m, let $\mathcal{U}_m$ denote the corresponding integral Markoff surface, and let $\mathbb{A}_{\mathbb{Z}}$ denote the adelic space used to formulate the integra...

conjecture | refuted | MathDB 页面 | 原文线索

V2-004399 | MATHDB-353634

The $b7$-Leopoldt conjecture for $b7$

摘录 / EXCERPT: The $b7$-Leopoldt conjecture. The map $\log_{\Sigma,p}$ is injective. This is a $b7$-component form of Leopoldt's conjecture and is used to control the non-vanishing of the relevan...

conjecture | open | MathDB 页面 | 原文线索

V2-004400 | MATHDB-353647

The zero-density conjecture for equal GCDs involving $sigma(m^2)$

摘录 / EXCERPT: Zero-density conjecture. The density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004401 | MATHDB-353655

The traces-of-singular-moduli conjecture for Thompson McKay-Thompson series

摘录 / EXCERPT: Let $g\in \textsl{Th}$, with g not in the classes $\mathrm{9B}$, $\mathrm{18B}$ or $\mathrm{27ABC}$. Let $\breve{F}_{\mathrm{3C},g}(\tau)$ be the McKay-Thompson series, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004402 | MATHDB-353662

Bounded-solution conjecture for generalized Ramanujan-Nagell equations

摘录 / EXCERPT: Bounded-solution conjecture. Equation (5) has at most 5 positive integer solutions, where D is an integer-coefficient polynomial of n.

conjecture | open | MathDB 页面 | 原文线索

V2-004403 | MATHDB-353663

Finite-supremum conjecture for polynomial generalized Ramanujan-Nagell equations

摘录 / EXCERPT: Finite-supremum conjecture. The supremum of the number of solutions of this equation, as D ranges over integer-coefficient polynomials with $\deg(D)\leq k$, exists; denote it by...

conjecture | open | MathDB 页面 | 原文线索

V2-004404 | MATHDB-353666

The normality conjecture for bases with four zeros in their Pisano period

摘录 / EXCERPT: Normality conjecture for bases with four zeros. For every base β with $\omega(\beta)=4$ and every nonnegative integer y, the Fibonacci concatenation is normal in base...

conjecture | open | MathDB 页面 | 原文线索

V2-004405 | MATHDB-353667

The maximum conjecture for uniform-distribution thresholds

摘录 / EXCERPT: Maximum conjecture. If coprime bases β_1 and β_2 satisfy $\Upsilon(\beta_1)=u$ and $\Upsilon(\beta_2)=v$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004406 | MATHDB-353668

The absolute-normality conjecture for the Fibonacci concatenation

摘录 / EXCERPT: Absolute-normality conjecture. The Fibonacci concatenation is absolutely normal.

conjecture | open | MathDB 页面 | 原文线索

V2-004407 | MATHDB-353669

The simple-normality conjecture for the Fibonacci concatenation

摘录 / EXCERPT: Simple-normality conjecture. The Fibonacci concatenation is simply normal in every base.

conjecture | open | MathDB 页面 | 原文线索

V2-004408 | MATHDB-353686

Generalized μ=0 conjecture for fine Selmer groups

摘录 / EXCERPT: Let p be a prime number, let F be a number field, let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbb{Q}_p$, and let S be a finite set of primes of...

conjecture | open | MathDB 页面 | 原文线索

V2-004409 | MATHDB-353700

Classification conjecture for planar monomials up to graph equivalence

摘录 / EXCERPT: Let p be an odd prime and n a positive integer. A polynomial $f\in\mathbb{F}_{p^n}[X]$ is planar if, for every $a\in\mathbb{F}_{p^n}\setminus\{0\}$, the polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-004410 | MATHDB-353743

Polyhedrality conjecture for the modular cone

摘录 / EXCERPT: Let $\cone_k$ denote the modular cone associated with weight k. Polyhedrality conjecture. Suppose that $k\equiv 2$ modulo 4 and $k>4$. The cone $\cone_k$ is polyhedral. The pap...

conjecture | open | MathDB 页面 | 原文线索

V2-004411 | MATHDB-353744

Fourier-coefficient relation conjecture for Jacobi cusp forms

摘录 / EXCERPT: Let $k>4$ with $k\equiv 2$ modulo 4. For every reduced matrix $T\in\halfint^+_2$ with bottom-right entry m, let $S_m(n,T)$ be the set of integers r such that...

conjecture | open | MathDB 页面 | 原文线索

V2-004412 | MATHDB-353747

Faber-Grantham conjecture on curves with fixed gonality and many rational points

摘录 / EXCERPT: Faber-Grantham conjecture. For every fixed prime power q and integer $\gamma \ge 2$, for all sufficiently large genera g,

conjecture | open | MathDB 页面 | 原文线索

V2-004413 | MATHDB-353756

Conjecture on infinitely many nonterminating cases of the heuristic palindrome algorithm

摘录 / EXCERPT: For a natural number N, consider the heuristic algorithm used to search for palindromic numbers A and B satisfying $N=A/B$. Nontermination conjecture. There are infinitely ma...

conjecture | open | MathDB 页面 | 原文线索

V2-004414 | MATHDB-353757

Superpolynomial-size conjecture for smallest palindromic quotient representations

摘录 / EXCERPT: Let N be a natural number for which there exist palindromic numbers A and B with $N=A/B$, and measure the size of a solution by the length of the relevant base-2 representa...

conjecture | open | MathDB 页面 | 原文线索

V2-004415 | MATHDB-353760

Superpolynomial-size conjecture for smallest antipalindromic quotient representations

摘录 / EXCERPT: Let N be a natural number for which there exist antipalindromic numbers A and B with $N=A/B$, and measure the size of a solution by the length of the relevant base-2 repres...

conjecture | open | MathDB 页面 | 原文线索

V2-004416 | MATHDB-353769

The extended main conjecture in Vinogradov's mean value theorem

摘录 / EXCERPT: The extended main conjecture. Suppose that $k\in\mathbb N$ and ${\mathfrak B}\subseteq[0,1)^k$ is measurable. Whenever $s\in\mathbb R$ satisfies $s\geq\tfrac14k(k+1)+1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-004417 | MATHDB-353770

The extended main conjecture for large measurable sets in Vinogradov's mean value theorem

摘录 / EXCERPT: The large-set extension. Whenever $s>0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-004418 | MATHDB-353784

Few-differentials conjecture for the Adams spectral sequence

摘录 / EXCERPT: Let $\pi_n\mathbb{S}$ denote the nth stable stem and let rank_p denote its rank at the prime p. Few-differentials conjecture. The stable stems satisfy ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-004419 | MATHDB-353799

Non-sharpness of the Paley graph Shannon-capacity bounds

摘录 / EXCERPT: Paley Shannon-capacity conjecture. There exist constants $a_k,b_k>0$ such that, for every prime $p\equiv1\pmod{k}$,

conjecture | open | MathDB 页面 | 原文线索

V2-004420 | MATHDB-353815

The irrationality-measure conjecture for Catalan's constant

摘录 / EXCERPT: Let ... be Catalan's constant, and let $\mu(\beta(2))$ denote its irrationality measure. Irrationality-measure conjecture. The irrationality measure of Catalan's constant is ... This i...

conjecture | open | MathDB 页面 | 原文线索

V2-004421 | MATHDB-353817

Extended non-integrality conjecture for generalized binomial sums

摘录 / EXCERPT: Let $n,r\geq 1$, let ... and let $\mathcal{S}_{(r,n)}(\ell)$ denote the generalized binomial sum defined in the paper. Extended non-integrality conjecture. If $\ell\in\mathcal{L}$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-004422 | MATHDB-353834

The asymptotic conjecture for coprime permutations

摘录 / EXCERPT: Let $C(n)$ denote the number of permutations σ of $[n]$ such that $\gcd(j,\sigma(j))=1$ for every $j\in[n]$. After defining constants c_k for the corresponding restriction...

conjecture | open | MathDB 页面 | 原文线索

V2-004423 | MATHDB-353837

The twisted Iwasawa detection conjecture for non-fibered knots

摘录 / EXCERPT: Let K be a knot, let p be a fixed prime number, and let π be its knot group. For a representation $\rho:\pi\to \operatorname{GL}_N O_{\mathfrak{p}}$, let μ_1 denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-004424 | MATHDB-353849

Uniform upper-bound conjecture for matrices with a given characteristic polynomial

摘录 / EXCERPT: Let $R_n(H;f)$ denote the number of matrices $A\in\mathcal M_n(\mathbb Z;H)$ with characteristic polynomial $f\in\mathbb Z[X]$, where $\mathcal M_n(\mathbb Z;H)$ is the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-004425 | MATHDB-353850

Conjecture on the limsup of M(n)

摘录 / EXCERPT: Conjecture on the limsup of $M(n)$. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004426 | MATHDB-353851

Robertson-Myers conjecture for the Rado number of x+y+kz=ℓw

摘录 / EXCERPT: Let ℓ and k be positive integers satisfying ... For the 2-color Rado number $\operatorname{RR}(x+y+kz=\ell w)$, Robertson-Myers conjecture. for fixed ℓ, ... This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-004427 | MATHDB-353852

Modified Robertson-Myers conjecture for the Rado number of x+y+kz=ℓw

摘录 / EXCERPT: Let ℓ and k be positive integers satisfying ... For the 2-color Rado number $\operatorname{RR}(x+y+kz=\ell w)$, Modified Robertson-Myers conjecture. for fixed ℓ, ... The pa...

conjecture | open | MathDB 页面 | 原文线索

V2-004428 | MATHDB-353853

The characterization of rings with strong avoidance

摘录 / EXCERPT: Let R be a commutative ring. Say that R has strong avoidance if every pair of ideals of R whose union is R has one ideal equal to R. For each prime ideal $\mathfrak{p}$ o...

open_problem | open | MathDB 页面 | 原文线索

V2-004429 | MATHDB-353870

Finite-exception conjecture for the residual solutions of $s(n^2)=5$

摘录 / EXCERPT: Finite-exception conjecture. The residual finite set is

conjecture | open | MathDB 页面 | 原文线索

V2-004430 | MATHDB-353891

Kernel-dimension conjecture for degree-two theta series of rank-six lattices

摘录 / EXCERPT: Let G_p be the genus of lattices of rank 6 and discriminant p, and let k_p be the number of isomorphism classes in G_p of lattices with no automorphism of determinant...

conjecture | open | MathDB 页面 | 原文线索

V2-004431 | MATHDB-353892

Kernel-dimension conjecture for degree-two theta series of prime-discriminant rank-six genera

摘录 / EXCERPT: Let G_p be the genus of lattices of rank 6 and discriminant $D=p$. Kernel-dimension conjecture. The kernel of $\theta^{(2)}$ on G_p has dimension equal to the number of class...

conjecture | solved | MathDB 页面 | 原文线索

V2-004432 | MATHDB-353893

Siegel eigenform congruence from a symmetric-square L-value

摘录 / EXCERPT: Let $N\geq 1$ be square-free, let χ be a quadratic character mod N, let $j\geq 0$ and $k\geq 3$. Suppose $f\in S_{j+k}(\Gamma_0^{(1)}(N),\chi)$ is an eigenform with...

conjecture | solved | MathDB 页面 | 原文线索

V2-004433 | MATHDB-353918

The upper-bound conjecture for the general Smarandache sigma product

摘录 / EXCERPT: Upper-bound conjecture. For every $n\in\mathbb{N}\setminus\{10^k-1:k\in\mathbb{Z}^{+}\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-004434 | MATHDB-353919

The bounds conjecture for the ratio of the even and odd digit-product sums

摘录 / EXCERPT: Let $a(n)$ and $b(n)$ be the previously defined sums associated with the even and odd numbers, respectively. Bounds conjecture. For every $n\in\mathbb{N}$, ... and for every $n\geq5$...

conjecture | open | MathDB 页面 | 原文线索

V2-004435 | MATHDB-353928

Symmetry conjecture for toric periods

摘录 / EXCERPT: Symmetry conjecture. For $x>0$ and sufficiently small $\delta>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004436 | MATHDB-353950

Finiteness conjecture for admissible triangular modular curves of fixed genus

摘录 / EXCERPT: Let $\Delta=\Delta(a,b,c)$ be a triangle group, let A be the associated quaternion algebra, and let $\Gamma\leq\Gamma(a,b,c;\mathfrak{N})$ be an admissible subgroup defining a tr...

conjecture | open | MathDB 页面 | 原文线索

V2-004437 | MATHDB-353963

Krachun--Petrov conjecture on sums with an algebraic multiplier

摘录 / EXCERPT: Let λ be a real algebraic number. Write its minimal polynomial as $f(x)=\prod_{i=1}^d(a_i x+b_i)$ over the complex numbers, with coprime integer coefficients, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-004438 | MATHDB-353965

The conjectured optimal error term for sums of linear transformations

摘录 / EXCERPT: Let $\mathcal{L}_1,\ldots,\mathcal{L}_k\in\operatorname{Mat}_d(\mathbb{Z})$ be irreducible and coprime. The optimal error-term conjecture. There is a constant D such that, for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-004439 | MATHDB-353967

Jones's eventual stability conjecture

摘录 / EXCERPT: Let K be a number field, let $\phi\in K(x)$, and let $\alpha\in K$. The pair (ϕ,α) is eventually stable when, writing $\phi^n=p_n/q_n$ with coprime $p_n,q_n\in K[x]$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-004440 | MATHDB-353972

Constant-depth Frege oddtown conjecture for nontrivial prime moduli

摘录 / EXCERPT: Constant-depth Frege oddtown conjecture. For each $d\in\mathbb{N}$ and each prime $p\neq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-004441 | MATHDB-354003

The precise 2-adic valuation conjecture for Eisenstein-series coefficients

摘录 / EXCERPT: 2-adic valuation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004442 | MATHDB-354009

The derived Riemann hypothesis for curves over finite fields

摘录 / EXCERPT: ${\bf n}_m$-derived Riemann hypothesis. The function $\widehat\zeta^{\,({\bf n}_m)}_{X,\Bbb F_q}(s)$ satisfies the Riemann hypothesis: all its zeros lie on

conjecture | open | MathDB 页面 | 原文线索

V2-004443 | MATHDB-354011

Invertibility conjecture for a cyclotomic circulant matrix

摘录 / EXCERPT: Let $q=p^s$ be a prime power, let $Q=q^m$, and let N be a divisor of $Q-1$. Write $N_1=(Q-1)/N$ and let A be the circulant matrix defined in equation from the relevant Gaussian...

conjecture | open | MathDB 页面 | 原文线索

V2-004444 | MATHDB-354014

The nested-clique B-form code distance conjecture

摘录 / EXCERPT: Let t be an odd prime number. Consider the nested clique graph with respect to the permutations in beqrefbpermsd, and let $bmathcal Cd$ be the associated B-form code. Its lengt...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-004445 | MATHDB-354022

Equality of modified and primitive Kato zeta-element modules

摘录 / EXCERPT: Let T_f be a Galois-stable $\mathcal{O}_\pi$-lattice of V_f satisfying Assumption (Im0), and let $r\in\mathbb{Z}$. Assume that p does not divide N. For $i=0,\ldots,p-2$, le...

open_problem | open | MathDB 页面 | 原文线索

V2-004446 | MATHDB-354025

The quantitative Kurihara-number conjecture

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ of conductor N, let $p\geq5$ be a prime, and let $\widetilde{\boldsymbol{\delta}}$ be the collection of Kurihara numbers. Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-004447 | MATHDB-354026

The p-adic Birch--Swinnerton-Dyer conjecture for Kato's zeta elements

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve, let p be a prime, let T be its p-adic Tate module, let $\mathbb{Q}_\infty$ be the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-004448 | MATHDB-354052

Non-vanishing conjecture for principal specializations of monomial symmetric polynomials

摘录 / EXCERPT: Let n be a prime power and let k be a positive integer. Let Λ_n^k be the indexing set of partitions used in the paper, let $|\lambda|$ denote the size of λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004449 | MATHDB-354077

Quasi-modularity of symmetrized generalized multiple q-zeta values

摘录 / EXCERPT: Let Δ be a root system of ADE type, let Δ_+ be its set of positive roots, and let $r=|\Delta_+|$. For integers $k_\alpha\geq 1$, write ... for the associated generalize...

conjecture | open | MathDB 页面 | 原文线索

V2-004450 | MATHDB-354078

Quasi-modularity and MLDE conjecture for type-A generalized q-zeta values

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{sl}(n+1)$ with $n\geq 1$, let $k\geq 3$, and let $\mathcal{QM}=\mathbb{C}[E_2(\tau),E_4(\tau),E_6(\tau)]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004451 | MATHDB-354080

Symmetric quasi-modularity conjecture for type-A bi-brackets

摘录 / EXCERPT: Let $n\geq 1$, let $k_{[i,j]}\geq 1$ and $s_{[i,j]}\geq 0$ be integers for $1\leq i\leq j\leq n$, and let $\zeta_{\mathfrak{sl}(n+1),q}[\cdots]$ denote the associated type-A bi-bra...

conjecture | open | MathDB 页面 | 原文线索

V2-004452 | MATHDB-354091

Algebraicity conjecture for plectic Stark-Heegner points

摘录 / EXCERPT: Algebraicity conjecture. If $r_{\mathrm{alg}}(A/E)\geq r$, then there exists $w_{A,S}\in\bigwedge^r(A(E)_{\mathbb Q})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004453 | MATHDB-354092

Maximality conjecture for the $(\mathbb{F}_q,\alpha)$ and $(\alpha\mathbb{F}_q,1)$ constructions

摘录 / EXCERPT: Let q be an odd prime power, let $m\geq 2$ satisfy $m\mid (q+1)/2$, and consider the cliques in $\mathrm{GP}(q^2,m)$ produced by the $(\mathbb F_q,\alpha)$-construction and the...

conjecture | open | MathDB 页面 | 原文线索

V2-004454 | MATHDB-354093

Maximality conjecture for the Q_0 and αQ_0 constructions

摘录 / EXCERPT: Let q be an odd prime power and let $m\geq 2$ divide $q+1$. The Q_0-construction and the αQ_0-construction produce cliques in the generalised Paley graph...

conjecture | open | MathDB 页面 | 原文线索

V2-004455 | MATHDB-354094

The conjecture of $\mathbb{Q}$-independent spectrum for metric graphs

摘录 / EXCERPT: $\mathbb{Q}$-independent spectrum conjecture. For every graph Γ, possibly except for some pathological cases, there is a generic set $G\subset\mathbb{R}_{+}^{N}$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-004456 | MATHDB-354116

Modularity conjecture for q-holonomic modules of proper q-hypergeometric sums

摘录 / EXCERPT: Let a proper q-hypergeometric multi-dimensional sum be a proper q-hypergeometric sum in multiple variables, and let its associated q-holonomic module be the corresponding mod...

conjecture | open | MathDB 页面 | 原文线索

V2-004457 | MATHDB-354130

Merino's irreducibility conjecture for Chebyshev-Lissajous polynomials

摘录 / EXCERPT: Merino's irreducibility conjecture. This polynomial is irreducible over $\mathbb{R}$ if and only if m and n are coprime. The claim gives a precise arithmetic criterion for irre...

conjecture | open | MathDB 页面 | 原文线索

V2-004458 | MATHDB-354155

Adelic-sector conjecture for the zeta eigenvalues

摘录 / EXCERPT: Adelic-sector conjecture. The zeta eigenvalues are adelic in nature, corresponding to prime numbers and their adelic sectors, while the trivial zeros correspond to the infinite pri...

conjecture | open | MathDB 页面 | 原文线索

V2-004459 | MATHDB-354174

Uniformity conjecture for large-prime square divisors of discriminants

摘录 / EXCERPT: Uniformity conjecture. For every $M>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004460 | MATHDB-354198

The prime-detection conjecture for prime-to-prime principal-block characters

摘录 / EXCERPT: Prime-detection conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-004461 | MATHDB-354200

The principal-block intersection conjecture for different primes

摘录 / EXCERPT: Let G be a finite group and let p and q be different primes. For a prime r, write $B_r(G)$ for the principal r-block of G, and let $\operatorname{Irr}_{r'}(B_r(G))$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-004462 | MATHDB-354201

The distinct-primes principal-block set conjecture

摘录 / EXCERPT: Let G be a finite group, and let p and q be primes dividing the order of G. For a prime r, write $B_r(G)$ for the principal r-block of G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004463 | MATHDB-354202

The Sylow-commutation character-degree conjecture

摘录 / EXCERPT: Let G be a finite group, and let p and q be different primes. For a prime r, write $B_r(G)$ for the principal r-block of G, and let $\operatorname{Irr}_{r'}(B_r(G))$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-004464 | MATHDB-354215

Erdős's nonvanishing conjecture for periodic sign functions

摘录 / EXCERPT: Let N be a positive integer, and let f be an arithmetic function with period N such that $f(n)\in\{-1,1\}$ for $n=1,2,\dots,N-1$ and $f(n)=0$ whenever $n\equiv 0\pmod{N}$. Er...

conjecture | open | MathDB 页面 | 原文线索

V2-004465 | MATHDB-354219

Dynamical Lehmer's conjecture

摘录 / EXCERPT: Let $f\in\mathbb{Z}[z]$ be a fixed monic polynomial of degree $d\geq 2$, and let $\mathrm{m}_f(P)$ denote the dynamical Mahler measure of a single-variable polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-004466 | MATHDB-354262

Finiteness conjecture for the componentwise Albanese kernel

摘录 / EXCERPT: Let $X=C_1\times\cdots\times C_d$ be a product of smooth projective curves over a number field k, and let $F^2(X)$ be the kernel of the Albanese map on degree-zero zero-cycles. D...

conjecture | open | MathDB 页面 | 原文线索

V2-004467 | MATHDB-354271

Lattice-polygon knot-statistics asymptotic conjecture

摘录 / EXCERPT: For a knot type K, let $p_n(K)$ be the number of n-edge polygons of knot type K. Let 0_1 be the unknot, let f_K be the number of prime knot factors in the knot decomposit...

conjecture | solved | MathDB 页面 | 原文线索

V2-004468 | MATHDB-354297

Stability conjecture for exterior Fraser-Sargent surfaces

摘录 / EXCERPT: Let k and l be relatively prime integers with $k>l>0$, and let $EFS_{k,l}$ be the exterior Fraser-Sargent surface in $\mathbb E^4\setminus\mathring{\mathbb B}^4$ associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-004469 | MATHDB-354298

Lower-bound conjecture for the orientable covers of Fraser-Sargent surfaces

摘录 / EXCERPT: Let k and l be relatively prime integers with $k>l>0$, and let $\widetilde{IFS}_{k,l}$ denote the orientable cover of the Fraser-Sargent free-boundary minimal surface...

conjecture | open | MathDB 页面 | 原文线索

V2-004470 | MATHDB-354311

Jakobson-Naud's half-plane conjecture for zeros of the Selberg zeta function

摘录 / EXCERPT: Jakobson-Naud's conjecture. There are only finitely many zeros in the half-plane

conjecture | open | MathDB 页面 | 原文线索

V2-004471 | MATHDB-354337

The explicit Brumer-Stark unit conjecture via Shintani integrals

摘录 / EXCERPT: Let F be a totally real field, let H/F be a finite abelian CM extension, and let $\mathfrak p$ split completely in H. For a prime ideal $\mathfrak b$ prime to the prescribed...

conjecture | open | MathDB 页面 | 原文线索

V2-004472 | MATHDB-354338

The modified explicit Brumer-Stark unit conjecture

摘录 / EXCERPT: Let F_p^* be the relevant local multiplicative group and let $\widehat{F}_p^*=F_p^*\widehat\otimes\mathbf{Z}_p$. Define $v_p(\mathfrak b)^{\mathrm{an}}\in\widehat{F}_p^*$ from th...

conjecture | open | MathDB 页面 | 原文线索

V2-004473 | MATHDB-354347

Delaunay-Poonen-Rains conjecture for the density of elliptic curves with nontrivial Tate-Shafarevich group

摘录 / EXCERPT: Let p be an odd prime. Let $\mathfrak{T}_p$ denote the set of elliptic curves $E_{/\mathbb{Q}}$ such that the p-primary part of the Tate-Shafarevich group...

conjecture | open | MathDB 页面 | 原文线索

V2-004474 | MATHDB-354366

Reality conjecture for trees in prime graphs

摘录 / EXCERPT: Reality conjecture for trees. Every tree is real.

conjecture | open | MathDB 页面 | 原文线索

V2-004475 | MATHDB-354380

Divisibility conjecture for odd derivatives of cyclotomic polynomials

摘录 / EXCERPT: Divisibility conjecture. For any non-negative integer k, $\Phi_n^{(2k+1)}(1)/\Phi_n(1)$ is divisible by $\phi(n)-2k$ in the polynomial ring ...

conjecture | open | MathDB 页面 | 原文线索

V2-004476 | MATHDB-354385

Uniqueness conjecture for generalized Markov numbers

摘录 / EXCERPT: For each pair $(q,p)\in \mathbb Z_{>0}^2$ with $q\geq p$, let $m_{q,p}$ denote the generalized Markov number associated with (q,p). Uniqueness conjecture for generalized Markov nu...

conjecture | open | MathDB 页面 | 原文线索

V2-004477 | MATHDB-354401

Quantitative extension of Dickson's prime k-tuples conjecture

摘录 / EXCERPT: A finite set of integers $\{a_1,a_2,\ldots,a_k\}$ is admissible if, for every prime q, its residue classes do not cover all residue classes modulo q. Quantitative extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-004478 | MATHDB-354407

The prime-length non-dual-containing conjecture for polycyclic codes

摘录 / EXCERPT: Let C be a polycyclic code with generator polynomial $g(x)$ satisfying ... and suppose that n is prime. Prime-length non-dual-containing conjecture. The code C is not dual-cont...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-004479 | MATHDB-354425

Skolem's exponential local-global principle for simple rational LRBS

摘录 / EXCERPT: Let $\boldsymbol u=\langle u_n\rangle_{n=-\infty}^{\infty}$ be a simple rational LRBS taking values in $\mathbb{Z}[\frac{1}{b}]$ for some non-zero integer b. A sequence has no ze...

conjecture | open | MathDB 页面 | 原文线索

V2-004480 | MATHDB-354428

Conjecture on smooth approximations to oscillatory sums

摘录 / EXCERPT: Smooth approximation conjecture. There exists a constant $A>0$ such that, uniformly for $1\leqslant x\leqslant T$ and $2T\leqslant t\leqslant5T$,

conjecture | open | MathDB 页面 | 原文线索

V2-004481 | MATHDB-354429

Mixed Granville-Soundararajan conjecture for twisted character sums

摘录 / EXCERPT: Mixed conjecture. There exists a constant $A>0$ such that, uniformly in these ranges,

conjecture | open | MathDB 页面 | 原文线索

V2-004482 | MATHDB-354430

Conjecture on extreme values of derivatives of the zeta function

摘录 / EXCERPT: Derivative extreme-value conjecture. As $T\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004483 | MATHDB-354436

The prime-tuple hypothesis for simultaneous primality of linear forms

摘录 / EXCERPT: Prime-tuple hypothesis. If no congruence obstruction exists, then there are infinitely many values of x such that every $a_i x+b_i$ is prime.

conjecture | open | MathDB 页面 | 原文线索

V2-004484 | MATHDB-354442

Toroczkai's independent-edge conjecture for prime gap graphs

摘录 / EXCERPT: Let p_n denote the n-th prime number, with $p_0=1$. A prime gap graph on n vertices is a simple graph whose vertex degrees are $p_1-p_0,\dots,p_n-p_{n-1}$; two edges are ind...

conjecture | open | MathDB 页面 | 原文线索

V2-004485 | MATHDB-354462

The square-root bound conjecture for Dedekind sums at Eisenstein norms

摘录 / EXCERPT: Square-root bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004486 | MATHDB-354463

The order-sensitive bound conjecture for Dedekind sums modulo primes

摘录 / EXCERPT: Order-sensitive Dedekind-sum conjecture. There exists an absolute constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004487 | MATHDB-354492

Smooth-cutoff variance conjecture for the k-fold divisor function in arithmetic progressions

摘录 / EXCERPT: Fix $k\ge2$. Let $w(y)$ be a smooth function supported in $[1,2]$ such that ... and whose Mellin transform $\mathcal{M}[w]$ satisfies ... uniformly for $|\sigma|\le A$, for every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-004488 | MATHDB-354505

Infinitely many Ramanujan-type congruences for 3- and 5-regular partitions

摘录 / EXCERPT: Let $b_k(n)$ denote the number of k-regular partitions of n, where $k\in\{3,5\}$, and let m be a positive integer. A Ramanujan-type congruence is a congruence of the form ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-004489 | MATHDB-354579

The intersection-density conjecture for \operatorname{PSL}_2(q) on 3-sets

摘录 / EXCERPT: Let q be a prime power, and let $\operatorname{PSL}_2(q)$ act on the 3-sets of its natural degree-$q+1$ action. Intersection-density conjecture. If $q^2\equiv 1\pmod{5}$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-004490 | MATHDB-354580

The intersection-density conjecture for \operatorname{PGL}_2(q) on 2-sets

摘录 / EXCERPT: Let q be an odd prime power, and let $\operatorname{PGL}_2(q)$ act on the 2-sets of $\{1,\dots,q+1\}$. Intersection-density conjecture. The action has intersection density 1....

conjecture | open | MathDB 页面 | 原文线索

V2-004491 | MATHDB-354581

The maximum-density conjecture for \operatorname{PSL}_2(q) on the Kneser graph K(q+1,3)

摘录 / EXCERPT: Let q be a prime power with $q\equiv 3\pmod{4}$, and consider the Kneser graph $K(q+1,3)$ and its automorphism group. Maximum-density conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-004492 | MATHDB-354616

A uniform upper-bound conjecture for the smoothed pair correlation function

摘录 / EXCERPT: For $T\ge 3$, let $\mathcal{L}(T)$ and $\beta=\beta(T)$ be non-decreasing continuous functions satisfying ... Let $W=W(x)$ be increasing with $W(x)\gg(\log x)^A$ for a sufficiently l...

conjecture | open | MathDB 页面 | 原文线索

V2-004493 | MATHDB-354647

Caldwell's conjecture for the distribution of Cunningham chain lengths

摘录 / EXCERPT: Caldwell's conjecture. The number of primes $p\leq N$ with $l(p)\geq k$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004494 | MATHDB-354648

Lower-bound conjecture for the length of Cunningham chains

摘录 / EXCERPT: Let $l(p)$ be the length of the Cunningham chain generated by the prime p, and let $\mathbb P$ denote the set of primes. Cunningham-chain length conjecture. On the primes, ... Equi...

conjecture | open | MathDB 页面 | 原文线索

V2-004495 | MATHDB-354655

The Lang-Waldschmidt conjecture on linear forms in logarithms

摘录 / EXCERPT: Let $\epsilon>0$, let $a_1,\ldots,a_n$ be positive integers, and let $b_1,\ldots,b_n$ be non-zero integers such that $a_1^{b_1}\cdots a_n^{b_n}\ne 1$. Lang-Waldschmidt conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-004496 | MATHDB-354671

Extremal t-intersecting sets in finite general linear groups

摘录 / EXCERPT: Let q be a prime power and let $G_n=\operatorname{GL}(n,q)$. For positive integers t and sufficiently large n compared to t, let $Y\subseteq G_n$ be a t-intersecting set,...

conjecture | open | MathDB 页面 | 原文线索

V2-004497 | MATHDB-354672

Extremal t-space-intersecting sets in finite general linear groups

摘录 / EXCERPT: Let q be a prime power and let $G_n=\operatorname{GL}(n,q)$. A subset $Y\subseteq G_n$ is t-space-intersecting when every pair of elements of Y is t-space-intersecting. For...

conjecture | open | MathDB 页面 | 原文线索

V2-004498 | MATHDB-354673

Extremal cross-space-intersecting sets in finite general linear groups

摘录 / EXCERPT: Let q be a prime power and let $G_n=\operatorname{GL}(n,q)$. Let $Y,Z\subseteq G_n$ be t-space-cross-intersecting sets, meaning that every pair in $Y\times Z$ is t-space-inte...

conjecture | open | MathDB 页面 | 原文线索

V2-004499 | MATHDB-354677

Chen's Romanov-type lower-density conjecture

摘录 / EXCERPT: Let $\mathscr{A}$ and $\mathscr{B}$ be two sets of positive integers. For a set S of positive integers, write $S(x)=|S\cap[1,x]|$. Chen's Romanov-type conjecture. If there exists...

conjecture | refuted | MathDB 页面 | 原文线索

V2-004500 | MATHDB-354697

Rowland-Yassawi's conjecture on Motzkin numbers modulo prime squares

摘录 / EXCERPT: Rowland-Yassawi's conjecture. For each $p\in\{31,37,61\}$ and every $n\in\mathbb{Z}_{\geq 0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-004501 | MATHDB-354743

Logarithmic growth conjecture for the set T(α)

摘录 / EXCERPT: Let $T(\alpha)$ be the set defined in the paper, and write $T_{\leq x}(\alpha)$ for its elements up to x. Logarithmic growth conjecture. For suitable $\alpha\in(0,1)$, there exis...

conjecture | open | MathDB 页面 | 原文线索

V2-004502 | MATHDB-354744

Infinitude conjecture for four simultaneous perfect-square expressions

摘录 / EXCERPT: Infinitude conjecture. There exist infinitely many $(n_1,n_2,n_3)\in\mathbb N^3$ with $n_1,n_2,n_3\geq2$ such that all four displayed integers are perfect squares.

conjecture | open | MathDB 页面 | 原文线索

V2-004503 | MATHDB-354782

Ruzsa's short-interval divisor conjecture

摘录 / EXCERPT: For integers N and real $a\leqslant b$, let $\tau(N;a,b)$ denote the number of divisors of N in $[a,b]$. Ruzsa's conjecture. For every $\varepsilon>0$ there is...

conjecture | open | MathDB 页面 | 原文线索

V2-004504 | MATHDB-354826

The continuity conjecture for p-rationality levels

摘录 / EXCERPT: Let p be a prime, G a finite group, and let $\alpha\in\mathbb{Z}^{\geq 2}$. The p-rationality level of a character is the integer determined by the p-part of its conductor....

conjecture | open | MathDB 页面 | 原文线索

V2-004505 | MATHDB-354827

The p-local lower-bound conjecture for p'-degree characters

摘录 / EXCERPT: Let p be a prime, let G be a finite group whose order is divisible by p, and let $P\in\operatorname{Syl}_p(G)$. p-local lower-bound conjecture. The number of irreducible p'...

conjecture | open | MathDB 页面 | 原文线索

V2-004506 | MATHDB-354828

Isaacs-Navarro's conjecture on p-rationality levels

摘录 / EXCERPT: Let p be a prime, let G be a finite group, let $P\in\operatorname{Syl}_p(G)$, and let $\alpha\in\mathbb{Z}^{+}$. The p-rationality level of an irreducible character is determ...

conjecture | open | MathDB 页面 | 原文线索

V2-004507 | MATHDB-354829

Continuity of p-rationality under p'-index subgroups

摘录 / EXCERPT: Let p be a prime, let G be a finite group, let $\alpha\in\mathbb{Z}^{\geq 2}$, and let M be a subgroup of G of p'-index. Continuity under p'-index subgroups. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-004508 | MATHDB-354830

Navarro-Tiep's conjecture on achieving p-rationality levels at p-elements

摘录 / EXCERPT: Let p be a prime, let G be a finite group, and let χ be an irreducible p'-degree character of G. The p-rationality level of a character value is defined using the...

conjecture | open | MathDB 页面 | 原文线索

V2-004509 | MATHDB-354831

Continuity for invariant characters of almost simple groups

摘录 / EXCERPT: Let p be a prime, let S be a finite nonabelian simple group, and let $S\trianglelefteq X\leq\operatorname{Aut}(S)$, where X is almost simple and X/S is a p-group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004510 | MATHDB-354842

Bost's tensor-product minimal-height conjecture for rigid adelic spaces

摘录 / EXCERPT: Let E and F be rigid analytic spaces over a number field K, and let $H_{min}(E)$ denote the minimal height arising from the slope filtration of E. Bost's conjecture. The mi...

conjecture | open | MathDB 页面 | 原文线索

V2-004511 | MATHDB-354843

Isodual multiplicity-two tensor-product conjecture for rigid adelic spaces

摘录 / EXCERPT: Let (E,σ) be an isodual rigid adelic space over a number field K with automorphism group G. Assume that E and $E^{\vee}$ are isomorphic as $K[G]$-modules, and that ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-004512 | MATHDB-354865

Height comparison conjecture for nodal odd-dimensional hypersurfaces

摘录 / EXCERPT: Let X_0 be an odd-dimensional geometrically integral projective hypersurface of degree at least three over a number field, with a single nodal singularity. Let Q be the excepti...

conjecture | open | MathDB 页面 | 原文线索

V2-004513 | MATHDB-354869

The embedding conjecture for infinite prime graphs

摘录 / EXCERPT: A prime graph is a graph with no nontrivial module, and an infinite prime graph is a prime graph with infinitely many vertices. Let G_μ denote the graph associated with a 0-...

conjecture | open | MathDB 页面 | 原文线索

V2-004514 | MATHDB-354880

Blomer's sup-norm conjecture for Siegel Hecke eigenforms

摘录 / EXCERPT: Let S_k^n denote the space of Siegel cusp forms of degree n and weight k, and let $F\in S_k^n$ be an L^2-normalised Hecke eigenform. Write $\|F\|_\infty$ for its sup-norm....

conjecture | open | MathDB 页面 | 原文线索

V2-004515 | MATHDB-354881

The Bergman-kernel growth conjecture for Siegel cusp forms

摘录 / EXCERPT: Bergman-kernel growth conjecture. With this notation,

conjecture | open | MathDB 页面 | 原文线索

V2-004516 | MATHDB-354883

Pinsker reduction conjecture for polynomial ergodic averages along primes

摘录 / EXCERPT: Let $(X,\mathcal B,\mu,\mathbb Z^d)$ be a measure-preserving system generated by invertible transformations $T_1,\ldots,T_d$, and let $p_{i,j}(n)\in\mathbb Z[n]$ be the polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-004517 | MATHDB-354886

Shafarevich's finiteness conjecture for Néron-Severi lattices of K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface over a number field. Write $\operatorname{NS}(\overline{X})$ for its geometric Néron-Severi lattice. Shafarevich's conjecture. Only finitely many lattices c...

conjecture | open | MathDB 页面 | 原文线索

V2-004518 | MATHDB-354887

Folklore finiteness conjecture for bounded-degree isogenous K3 surfaces

摘录 / EXCERPT: Let N be an integer and let X be a K3 surface over $\overline{\mathbf{Q}}$. Consider the K3 surfaces isogenous to X that can be defined over extensions of degree at most N....

conjecture | open | MathDB 页面 | 原文线索

V2-004519 | MATHDB-354894

Kurokawa's Deep Riemann Hypothesis for Artin L-functions

摘录 / EXCERPT: Let K be a global field and let $\rho\colon\operatorname{Gal}(K^{\mathrm{sep}}/K)\to\operatorname{GL}_n(\mathbb C)$ be a nontrivial irreducible representation. For a prime v of...

conjecture | open | MathDB 页面 | 原文线索

V2-004520 | MATHDB-354905

Chandee's conjecture on shifted moments of the Riemann zeta function

摘录 / EXCERPT: Chandee's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004521 | MATHDB-354918

Existence of infinitely many Ramanujan-type congruences for 5-regular partitions

摘录 / EXCERPT: Let $b_5(n)$ denote the number of 5-regular partitions of n, and let m be a positive integer. A Ramanujan-type congruence is a congruence of the form $b_5(An+B)\equiv 0\pmod m$...

conjecture | open | MathDB 页面 | 原文线索

V2-004522 | MATHDB-354955

Serre's ordinary-reduction density conjecture

摘录 / EXCERPT: Serre's ordinary-density conjecture. There is a finite extension L/K such that a positive-density set of primes v of L gives X_L a good ordinary reduction at v.

conjecture | open | MathDB 页面 | 原文线索

V2-004523 | MATHDB-354968

Harris-Venkatesh conjecture for triple product L-functions

摘录 / EXCERPT: Let N and $p\geq 5$ be primes with $p\mid N-1$. Let $g\in S_1(\Gamma_1(d))$ be a weight-one cuspidal newform of conductor d with $\gcd(Np,d)=1$, let g^* be its dual, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004524 | MATHDB-354977

Iwaniec's conjecture for primes represented by x² + y² + 1

摘录 / EXCERPT: Iwaniec's conjecture. The limit exists and equals

conjecture | open | MathDB 页面 | 原文线索

V2-004525 | MATHDB-354978

The conjectural least-common-multiple asymptotic for x² + y² + 1

摘录 / EXCERPT: The conjecture for ψ_F. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-004526 | MATHDB-354994

Density-zero conjecture for p-divisible p-adic regulators

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ and let K be a number field. For each prime p, let $R_p(E/K)$ denote the p-adic regulator. Density-zero regulator conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-004527 | MATHDB-355013

Schnetz's prime reduction conjecture for c_2 invariants

摘录 / EXCERPT: Let G_1 and G_2 be graphs, and for a prime power $q=p^s$ let $c_2^{(q)}(G)$ denote their c_2 invariants. Schnetz's prime reduction conjecture. If ... for every prime p, then...

conjecture | open | MathDB 页面 | 原文线索

V2-004528 | MATHDB-355028

Carmichael's fixed-prime-factor infinitude conjecture

摘录 / EXCERPT: Carmichael's fixed-prime-factor infinitude conjecture. For any $R\in\mathbb N$ with $R\geq 3$, there exist infinitely many Carmichael numbers with R prime factors.

conjecture | open | MathDB 页面 | 原文线索

V2-004529 | MATHDB-355039

Prime-folding characterization of extremal tensors

摘录 / EXCERPT: Let $2\le n=\prod_{k=1}^{s_n}p_k$ be the prime factorization, with $2\le p_1\le p_2\le\dots\le p_{s_n}$. Consider a tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-004530 | MATHDB-355052

Vanishing conjecture for localized homology of arithmetic manifolds

摘录 / EXCERPT: Let F be a number field with r_1 real places and r_2 pairs of complex places. Let $K\subseteq\operatorname{PGL}_2(\mathbb{A}_F^{\infty})$ be a compact open subgroup, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004531 | MATHDB-355053

Ihara's lemma in the bottom homological degree

摘录 / EXCERPT: Let K and v be as in the paper's level-lowering setup, and let $\mathfrak{m}\subseteq\mathbb{T}^S(K)$ be a non-Eisenstein maximal ideal. Let ... be the level-lowering map. Ihara'...

conjecture | open | MathDB 页面 | 原文线索

V2-004532 | MATHDB-355065

Ramanujan-Hardy-Littlewood conjecture for the Möbius function

摘录 / EXCERPT: Let ρ range over the non-trivial zeros of the Riemann zeta function $\zeta(s)$. Let $b[?25l\alpha$ and β be positive real numbers satisfying ... Assume that the series ... c...

conjecture | open | MathDB 页面 | 原文线索

V2-004533 | MATHDB-355077

Buzzard-Diamond-Jarvis conjecture for quaternionic mod-p cohomology

摘录 / EXCERPT: Let F) be a totally real field and let ... be a continuous, irreducible and totally odd representation. Let D be a quaternion algebra over F that splits at exactly one infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-004534 | MATHDB-355078

Serre-type eigenvalue correspondence for definite quaternion algebras

摘录 / EXCERPT: Let D and $\bar D$ be the quaternion algebras introduced earlier, let Σ be the specified finite set of places, let $\mathbf{F}$ be the space of mod-p algebraic functions...

conjecture | open | MathDB 页面 | 原文线索

V2-004535 | MATHDB-355079

Weak Szpiro's conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ with minimal discriminant Δ_E and conductor N_E. Weak Szpiro's conjecture. There exist positive real numbers α and...

conjecture | open | MathDB 页面 | 原文线索

V2-004536 | MATHDB-355080

Strong Szpiro's conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ with minimal discriminant Δ_E and conductor N_E. Strong Szpiro's conjecture. For all $\epsilon>0$, there is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-004537 | MATHDB-355082

The abcd conjecture for coprime positive integer quadruples

摘录 / EXCERPT: Let (a,b,c,d) be a quadruple of coprime positive integers satisfying $a+b+c=d$. The abcd conjecture. For every positive real number ϵ, there exists a positive real numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-004538 | MATHDB-355083

The abcd conjecture in height form

摘录 / EXCERPT: Let K be a number field or a one-dimensional function field of characteristic 0, let $n\geq3$, let $[Z_1,\ldots,Z_n]$ be standard coordinates on $\mathbb{P}^{n-1}_K$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004539 | MATHDB-355085

Conjectural factor counts for the Hasse invariant of the Tate normal form E_7

摘录 / EXCERPT: Let $l\neq 2,3,7$ be prime. Let $\hat H_{7,l}(x)$ be the Hasse-invariant polynomial of the Tate normal form E_7 over $\mathbb F_l$, and let $h(-l)$ be the relevant class number....

conjecture | solved | MathDB 页面 | 原文线索

V2-004540 | MATHDB-355086

Conjectural quadratic-factor counts for the Hasse invariant of E_7

摘录 / EXCERPT: Let $l>7$ be prime with $l\equiv1,6\pmod7$. Let $\hat H_{7,l}(x)$ be the Hasse-invariant polynomial, and let N_2 count irreducible quadratic factors x^2+ax+b dividing it modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-004541 | MATHDB-355124

Gouvêa-Mazur conjecture on local constancy of slopes

摘录 / EXCERPT: Let p be a prime number, let N be a positive integer coprime to p, and normalize the p-adic valuation on $\overline{\mathbb{Q}}_p$ by $v_p(p)=1$. For $r\in\mathbb{Q}$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-004542 | MATHDB-355132

Congruence conjectures for permanents of shifted difference matrices

摘录 / EXCERPT: For a positive integer n, let $[j-k+\delta_{jk}]_{1\leqslant j,k\leqslant n}$ denote the $n\times n$ matrix whose (j,k) entry is $j-k+\delta_{jk}$, where δ_{jk} is the K...

conjecture | open | MathDB 页面 | 原文线索

V2-004543 | MATHDB-355133

Congruence conjecture for permanents of the absolute-difference matrix

摘录 / EXCERPT: Let p be an odd prime, let δ_{jk} be the Kronecker delta, and let $\operatorname{per}(M)$ denote the permanent of a matrix M. Absolute-difference permanent conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-004544 | MATHDB-355141

Hardy-Littlewood-Chowla conjecture for prime and Möbius correlations

摘录 / EXCERPT: Let $x\to\fty$. Let $a_0,a_1,\ldots,a_{k-1}$ be a fixed admissible k-tuple, and let $b_0<b_1<\cdots<b_{l-1}$ be a fixed subset of small integers. Hardy-Littlewood-Chowla conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004545 | MATHDB-355161

Stewart's uniform bound conjecture for primitive solutions of Thue equations

摘录 / EXCERPT: Let $F\in\mathbb Z[x,y]$ be a binary form with nonzero discriminant and degree at least three. For an integer m, consider the Thue equation ... A solution (x,y) is primitive when...

conjecture | open | MathDB 页面 | 原文线索

V2-004546 | MATHDB-355171

Scott-Styer's conjecture on solutions to a ternary purely exponential Diophantine equation

摘录 / EXCERPT: Scott-Styer's conjecture. If $a<b$, then $N(a,b,c)\leq 1$, apart from the thirteen explicitly listed exceptional cases and solutions in the source statement: the family...

conjecture | open | MathDB 页面 | 原文线索

V2-004547 | MATHDB-355172

Scott-Styer's prime-base conjecture on solutions to a ternary purely exponential Diophantine equation

摘录 / EXCERPT: Scott-Styer's prime-base conjecture. We have $N(a,b,c)\leq1$, apart from the six explicitly listed exceptional cases: $(2,3,5)$, $(2,3,11)$, $(2,5,3)$, $(2,7,3)$, $(3,5,2)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-004548 | MATHDB-355173

Conjecture on the number of solutions to $a^x+b^y=c^z$

摘录 / EXCERPT: One-solution conjecture. There is at most one solution (x,y,z), except when (a,b,c) or (b,a,c) is one of $(5,2,3)$, $(7,2,3)$, $(3,2,11)$, $(3,2,35)$, $(3,2,259)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004549 | MATHDB-355191

Uniqueness of the mountain-range diagram for non-loose torus knots

摘录 / EXCERPT: Let (p,q) be a pair of coprime integers specifying a torus knot, and consider an overtwisted contact structure that supports non-loose Legendrian (p,q)--torus knots. The mounta...

conjecture | open | MathDB 页面 | 原文线索

V2-004550 | MATHDB-355197

Eventual linearity of associated-prime counts for Inc-invariant monomial chains

摘录 / EXCERPT: Let ${\mathcal I}=(I_n)_{n\ge 1}$ be an Inc-invariant chain of monomial ideals. Associated-prime count conjecture. The number of associated primes of I_n is even...

conjecture | open | MathDB 页面 | 原文线索

V2-004551 | MATHDB-355201

Large-prime strong-counterexample conjecture for abelian surfaces

摘录 / EXCERPT: Let K be a number field, let ℓ be a prime, and let (A,ℓ) be a strong counterexample to the local-global principle for isogenies of prime degree between abelian surfaces...

conjecture | open | MathDB 页面 | 原文线索

V2-004552 | MATHDB-355204

The local ghost conjecture for primitive augmented modules

摘录 / EXCERPT: Let $\bar\rho=\begin{pmatrix}\omega_1^{a+b+1}&*\\0&\omega_1^b\end{pmatrix}:\operatorname{I}_{\mathbb{Q}_p}\to\operatorname{GL}_2(\mathbb{F})$ be a character with...

conjecture | open | MathDB 页面 | 原文线索

V2-004553 | MATHDB-355228

Maximal gcd conjecture for Lucas sequences

摘录 / EXCERPT: Let $(u_n)_{n\geqslant 0}$ be a Lucas sequence, and let $g_u(n)$ denote the gcd function associated with it. Assume that $\Delta_u\ne 1$. Then the maximal gcd conjecture asserts th...

conjecture | open | MathDB 页面 | 原文线索

V2-004554 | MATHDB-355238

Darmon-Harris-Venkatesh Stark-unit conjecture for the derived Hecke pairing

摘录 / EXCERPT: Let R be the ring of integers of L, with $6N$ inverted, and let g be the weight-one form with associated representation ρ_g. Define ... Let $\mathfrak{S}$ be the Shimura c...

conjecture | open | MathDB 页面 | 原文线索

V2-004555 | MATHDB-355247

Modular congruence criterion for non-unirational surfaces

摘录 / EXCERPT: Let Y be a surface satisfying the projective-resolution and congruence assumptions, and let $\widetilde{Y}$ be a fixed minimal smooth model. Suppose that for some...

conjecture | open | MathDB 页面 | 原文线索

V2-004556 | MATHDB-355260

Dénes conjecture on arithmetic progressions of like powers

摘录 / EXCERPT: Dénes conjecture. Then $xyz=0$ or $|x|=|y|=|z|$. The paper uses this assertion, together with related results on equal sums of like powers, to rule out reflecting numbers of type...

conjecture | open | MathDB 页面 | 原文线索

V2-004557 | MATHDB-355277

The plectic polylogarithm specialization conjecture for Lerch zeta values

摘录 / EXCERPT: Plectic polylogarithm specialization conjecture. The specialization satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004558 | MATHDB-355283

Quadratic-irrational periodicity and exponential-tail conjecture for Haros graphs

摘录 / EXCERPT: Let x be a quadratic irrational, and let G_x be its associated Haros graph with degree distribution $P(k,x)$. The continued-fraction expansion of x is eventually periodic. Qu...

conjecture | open | MathDB 页面 | 原文线索

V2-004559 | MATHDB-355284

Characterisation of quadratic irrationals by Haros-graph degree distributions

摘录 / EXCERPT: Let x be an irrational number and let G_x be its associated Haros graph with degree distribution $P(k,x)$. Call the properties in the preceding conjecture the combination of a...

conjecture | open | MathDB 页面 | 原文线索

V2-004560 | MATHDB-355288

The higher-rank torus-knot tail conjecture for $p<r$

摘录 / EXCERPT: Let $2\leq p<p'$ be coprime positive integers, let $2\leq r$ with $p<r$, and let $T(p,p')$ be the (p,p') torus knot. Let $L_r(n\Lambda_1)$ denote the irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-004561 | MATHDB-355338

Ramanujan conjecture for non-CAP Siegel cusp forms of degree 2

摘录 / EXCERPT: Let $F\in S_k(\Gamma^{(2)}_0(N))$ be a Hecke eigenform at good primes that is non-CAP, meaning that it is of general type or Yoshida type. For each prime p with $\gcd(p,N)=1$, le...

conjecture | open | MathDB 页面 | 原文线索

V2-004562 | MATHDB-355341

Random matrix distribution conjecture for Iwasawa lambda-invariants

摘录 / EXCERPT: Let p be an odd prime and let θ be a Dirichlet character with $p \mid \text{cond}(\theta)$. Write $\boldsymbol\chi=\theta\omega^i$ for a character of the first kind, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004563 | MATHDB-355342

Proportion conjecture for chi-regular primes

摘录 / EXCERPT: Let χ be a Dirichlet character, and let $\operatorname{ord}(\chi)$ denote its order. A prime p is χ-regular when the relevant even twists $\chi\omega^i$ have vanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-004564 | MATHDB-355343

Random matrix prediction for regular primes in totally real abelian extensions

摘录 / EXCERPT: Let $F/\mathbb{Q}$ be a totally real abelian extension of order m, and suppose ... For primes $p\gg 0$ satisfying $p\nmid m\,\operatorname{disc}_F$, write F-regular for the condi...

conjecture | open | MathDB 页面 | 原文线索

V2-004565 | MATHDB-355344

Asymptotic distribution conjecture for twisted Iwasawa lambda-invariants

摘录 / EXCERPT: Let χ be a fixed Dirichlet character, set $m=\operatorname{ord}(\chi)$, and let $\theta_{\leqslant X}^{(r)}(\chi)$ count primes $p\leqslant X$ for which the corrected χ-t...

conjecture | open | MathDB 页面 | 原文线索

V2-004566 | MATHDB-355345

McLeman's Zassenhaus type conjecture for imaginary quadratic fields

摘录 / EXCERPT: Let $p>3$ be a prime, and let K be an imaginary quadratic field whose class group has p-rank two. The Zassenhaus type is the pair (i,j) occurring in the Zassenhaus polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-004567 | MATHDB-355347

Distinctness conjecture for the ordering statistic of k-indivisible partitions

摘录 / EXCERPT: Let $k,t\geq 2$ be coprime integers, and let $1\leq r<s\leq t$. The quantity $\psi_{k,t}(r)$ measures the asymptotic bias associated with the congruence class r modulo t among...

conjecture | open | MathDB 页面 | 原文线索

V2-004568 | MATHDB-355348

Growth conjecture for the number of orderings of congruence-class biases

摘录 / EXCERPT: Let $\mathscr{O}(t)$ denote the number of possible orderings arising from the biases among congruence classes for k-indivisible partitions, and let $\varphi(t)$ be Euler's totien...

conjecture | open | MathDB 页面 | 原文线索

V2-004569 | MATHDB-355366

The conjectural square-root error term for the asymptotic of \(\log \sigma_n\)

摘录 / EXCERPT: Let σ_n be the quantity studied in the paper, and let n tend to infinity. Improving theorem. One would have ... This improves the preceding proved estimate with error term...

conjecture | open | MathDB 页面 | 原文线索

V2-004570 | MATHDB-355367

Two-sided prime-count conjecture for \(\mathscr A_n\)

摘录 / EXCERPT: For $n\in\mathbb N^*$, let ... so that $\operatorname{Card}\,\mathscr A_n$ counts the primes among the $\lfloor\sqrt n\rfloor$ displayed integers. Two-sided prime-count conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-004571 | MATHDB-355380

Short-interval upper-bound conjecture for k-full numbers

摘录 / EXCERPT: Let $k\ge 2$ be an integer, let $x\ge 1$ be real, and let $Q_k(t)$ denote the number of k-full numbers at most t. A number is k-full if every exponent in its prime factorizat...

conjecture | open | MathDB 页面 | 原文线索

V2-004572 | MATHDB-355385

Conjecture that the two layered-number sequences coincide

摘录 / EXCERPT: For each positive integer i, let g_i and k_i denote the two integer sequences defined in the paper, where g_i is the least integer admitting an i-layered partition and...

conjecture | open | MathDB 页面 | 原文线索

V2-004573 | MATHDB-355386

Conjecture on good k-abundant practical numbers

摘录 / EXCERPT: Let n be a practical number. A number is good k-abundant when it satisfies the paper's definition of that property, and a k-layered number is one admitting the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-004574 | MATHDB-355387

Conjecture that every odd abundant number is semiperfect

摘录 / EXCERPT: An integer n is abundant if $\sigma(n)>2n$, and it is semiperfect if it equals the sum of some of its proper divisors. Odd-abundant semiperfectness conjecture. Every odd abundant...

conjecture | open | MathDB 页面 | 原文线索

V2-004575 | MATHDB-355388

Clark's conjecture on even Zumkeller numbers

摘录 / EXCERPT: An integer is Zumkeller if its positive divisors can be partitioned into two sets with equal sum, and it is half-Zumkeller if it satisfies the paper's corresponding half-Zumkeller...

conjecture | open | MathDB 页面 | 原文线索

V2-004576 | MATHDB-355399

The converse to chromatic blueshift for finite abelian p-groups

摘录 / EXCERPT: Let p be a prime, let A be a finite abelian p-group, and let $\mathcal{F}$ be a proper family of subgroups of A. Let R be a commutative algebra regarded as a Borel-equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-004577 | MATHDB-355402

Denef-Lipshitz conjecture on Diophantine definability of the integers

摘录 / EXCERPT: Sea F un campo de números y sea O_F su anillo de enteros. Un subconjunto de O_F es diofantino si existe una ecuación polinómica con coeficientes en O_F cuyos valores de las...

conjecture | open | MathDB 页面 | 原文线索

V2-004578 | MATHDB-355410

Lenstra's successive-minima multiplicativity conjecture for tower types

摘录 / EXCERPT: Fix a tower type $\mathfrak{T}=(n_1,\dots,n_t)$, let n be its degree, and choose integers $0\leq i,j\leq i+j<n$. For an order $\mathcal{O}$ in the family...

conjecture | open | MathDB 页面 | 原文线索

V2-004579 | MATHDB-355420

Span conjecture for diagonal restrictions of Hilbert theta functions

摘录 / EXCERPT: Span conjecture. Equation $({\mathcal{M}}_F^\theta)^\Delta={\mathcal{M}}_\mathbb{Q}^{(4d/d_2)}$ holds for any totally real field F of degree d. This predicts that diagonal rest...

conjecture | open | MathDB 页面 | 原文线索

V2-004580 | MATHDB-355426

Common-factor conjecture for numerators of triangular-grid edge resistances

摘录 / EXCERPT: Let $L_s^{(i)}$, for $i\in\{1,2\}$, be the edge-resistance quantities defined in the paper, and let $\operatorname{num}(x)$ denote the numerator of the maximally reduced fractional...

conjecture | open | MathDB 页面 | 原文线索

V2-004581 | MATHDB-355440

Miyazaki's torsion conjecture for cables

摘录 / EXCERPT: Let K be a knot, let p and q be relatively prime integers with $|p|>1$, and let $K_{p,q}$ denote the (p,q)-cable of K. Let $\mathcal{C}$ denote the smooth knot concordanc...

conjecture | open | MathDB 页面 | 原文线索

V2-004582 | MATHDB-355450

Bergdall-Pollack slope-invariant conjecture for ghost series under direct sum

摘录 / EXCERPT: Let $p\geq 7$ be prime, let N be coprime to p, and let $\mathrm{K}_p=\operatorname{GL}_2(\mathbb{Z}_p)$. Let $\mathcal{O}_{\mathbb{C}_p}$ be the ring of integers of...

conjecture | open | MathDB 页面 | 原文线索

V2-004583 | MATHDB-355454

The asymptotic probability of being a 2-Sophie Germain prime

摘录 / EXCERPT: The 2-Sophie Germain prime probability conjecture. The probability that a given prime p is 2-SGP converges to 0.76 as $p\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-004584 | MATHDB-355455

The asymptotic probability of being an a-Sophie Germain prime

摘录 / EXCERPT: The a-Sophie Germain prime probability conjecture. The probability that a given prime p is a-SGP converges, as $p\to\infty$, to the common asymptotic value represented by

conjecture | open | MathDB 页面 | 原文线索

V2-004585 | MATHDB-355462

Ferdinand's girth conjecture for the bipartite graph D(k,q)

摘录 / EXCERPT: Let $D(k,q)$ be the bipartite graph originally proposed by Lazebnik and Ustimenko, where k is an integer not less than 2 and q is a prime power. The graph has girth at least...

conjecture | open | MathDB 页面 | 原文线索

V2-004586 | MATHDB-355472

A Legendre-symbol formula for $D_p(3,1)$

摘录 / EXCERPT: Let $D_p(c,d)$ denote the determinant associated with the quadratic form $i^2+cij+dj^2$ in the paper. Legendre-symbol formula. For any odd prime $p\equiv\pm2\pmod5$, ... The supplied...

conjecture | open | MathDB 页面 | 原文线索

V2-004587 | MATHDB-355473

Derangement reciprocal-product congruences

摘录 / EXCERPT: For a positive integer n, let $D(n)=\{\tau\in S_n:\tau(j)\ne j\text{ for all }j=1,\ldots,n\}$ be the set of derangements, and let $\operatorname{sign}(\tau)$ denote the sign of a...

conjecture | open | MathDB 页面 | 原文线索

V2-004588 | MATHDB-355474

Permutation sums with excluded fixed points

摘录 / EXCERPT: Let S_m be the symmetric group on $1,\ldots,m$. Excluded-fixed-point congruences. Let p be an odd prime. Then ... If $p\equiv3\pmod4$, then ... The supplied text gives no resolutio...

conjecture | open | MathDB 页面 | 原文线索

V2-004589 | MATHDB-355475

Signed quadratic quotient-product congruence

摘录 / EXCERPT: Let $S_{(p-1)/2}$ be the symmetric group on $1,\ldots,(p-1)/2$, and let $\operatorname{sign}(\tau)$ denote permutation sign. Signed quadratic quotient-product conjecture. Let $p>3$...

conjecture | open | MathDB 页面 | 原文线索

V2-004590 | MATHDB-355492

Conjecture of capitulation for totally real number fields

摘录 / EXCERPT: Let K be any totally real number field, let p be a prime, and let ${\mathcal H}_K$ be its p-class group of exponent p^e. For an integer $N\geq e$, consider primes ℓ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-004591 | MATHDB-355493

Distribution conjecture for primes represented by irreducible homogeneous polynomials

摘录 / EXCERPT: Distribution conjecture. The distribution of the slopes $t=b/a$ is F^{-1}.

conjecture | open | MathDB 页面 | 原文线索

V2-004592 | MATHDB-355494

Pair-prime distribution conjecture for imaginary quadratic norm forms

摘录 / EXCERPT: Imaginary quadratic pair-prime conjecture. The distribution is

conjecture | open | MathDB 页面 | 原文线索

V2-004593 | MATHDB-355495

Equidistribution conjecture for slopes of primes represented by homogeneous polynomials

摘录 / EXCERPT: Let $f(x,y)$ be a homogeneous polynomial of degree n, and define ... Let $C_f(t_1,t_2,M)$ count primes $p=f(a,b)<M$ satisfying $F^{-1}(t_1)<b/a<F^{-1}(t_2)$. For $t_1<t_2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004594 | MATHDB-355496

Continuous multiple Dedekind zeta conjecture for prime slopes

摘录 / EXCERPT: Continuous multiple Dedekind zeta conjecture. For fixed large M, as T varies, the distribution of the number of represented primes is given by

conjecture | open | MathDB 页面 | 原文线索

V2-004595 | MATHDB-355497

Class-number ratio conjecture for split primes in imaginary quadratic fields

摘录 / EXCERPT: Let $K=\mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field, and let C_1 be the cone whose elements have angle between $\pi/36$ and $2\pi/36$. Let $P_M(-d)$ count primes less t...

conjecture | open | MathDB 页面 | 原文线索

V2-004596 | MATHDB-355498

Squared class-number ratio conjecture for pairs of split primes in imaginary quadratic fields

摘录 / EXCERPT: Let $\mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field and let C_1 be the fixed cone. Let $PQ_M(-d)$ count pairs of primes (p,q) less than M that split completely, with...

conjecture | open | MathDB 页面 | 原文线索

V2-004597 | MATHDB-355499

Residue ratio conjecture for split primes in imaginary quadratic fields

摘录 / EXCERPT: For imaginary quadratic fields, let $P_M(-d)$ count primes less than M that split completely and whose factors lie in the cone C_1. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004598 | MATHDB-355500

Remainder-class independence conjecture for prime slopes

摘录 / EXCERPT: Let $f(x,y)$ be a homogeneous polynomial of degree n, let $F(t)=\int_0^t[\sqrt[n]{f(1,y)}]^{-2}\,dy$, and let $\mathfrak{a}$ be an ideal in $\mathbb{Q}[\xi]$, where ξ is a ro...

conjecture | open | MathDB 页面 | 原文线索

V2-004599 | MATHDB-355501

The graded L'vov-Kaplansky conjecture for matrix algebras of prime order

摘录 / EXCERPT: Let K be a field and let n be a prime number. Let $K\langle X\mid \mathbb{Z}_n\rangle$ denote the free $\mathbb{Z}_n$-graded algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004600 | MATHDB-355511

Wang-Sun formula for quasipolarities when the modulus is even

摘录 / EXCERPT: Let k be even, let ζ be a primitive $2k$-th root of unity, and let Q_k denote the set of quasipolarities. For each permutation $\pi\in Q_k$, consider the product over...

conjecture | open | MathDB 页面 | 原文线索

V2-004601 | MATHDB-355533

Cheng-Duncan conjecture on Mathieu moonshine Siegel modular forms

摘录 / EXCERPT: Cheng-Duncan conjecture. For each conjugacy class g of M_{24}, the product Φ_g defines a Siegel modular form with some multiplier on the congruence subgroup...

conjecture | solved | MathDB 页面 | 原文线索

V2-004602 | MATHDB-355545

Abramovich-Harris conjecture on low-degree maps from curves

摘录 / EXCERPT: Abramovich-Harris conjecture. Suppose

conjecture | refuted | MathDB 页面 | 原文线索

V2-004603 | MATHDB-355574

The recipe conjecture for averages of quadratic twists of Dirichlet polynomials

摘录 / EXCERPT: Let $\cal D$ be a family of fundamental discriminants in a fixed arithmetic progression, let $\cal M_A(D;\ell)$ denote the corresponding smoothed average of...

conjecture | open | MathDB 页面 | 原文线索

V2-004604 | MATHDB-355586

Congruence speed stabilization for tetration bases

摘录 / EXCERPT: Congruence speed stabilization conjecture. If $a \not\equiv \{0,3,7\} \pmod {10}$ and $b \geq \operatorname{len}(a)+2$, then $V(a,b)=V(a)$. This gives a sufficient tetration height...

conjecture | open | MathDB 页面 | 原文线索

V2-004605 | MATHDB-355587

Universal height-two stabilization conjecture for polymorphic tetration bases

摘录 / EXCERPT: Let $\smalloverbrace{\mathbb{A}}$ denote the set of polymorphic tetration bases, let $V(\smalloverbrace{a}^{*})$ be the limiting congruence speed of $\smalloverbrace{a}^{*}$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-004606 | MATHDB-355614

Random-model counting conjecture for rational points near nondegenerate manifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be a d-dimensional submanifold of $\mathbb{R}^n$ in Monge form, with codimension $m=n-d$, and let $N_{\mathcal{M}}(Q,\delta)$ count rational points of denominat...

conjecture | open | MathDB 页面 | 原文线索

V2-004607 | MATHDB-355636

Conjectural period-polynomial relations for double T- and double tilde-T-values

摘录 / EXCERPT: For $N=2$ or 4, even integers $k\ge4$, and integers j with $1\le j\le (k-2)/2$, define coefficients a_i and b_i by ... and ... Here $P_{N,k,j}^{(+)}$ and $P_{N,k,j}^{(-)}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-004608 | MATHDB-355650

Conjecture on the asymptotic error term for floor function sums over powers

摘录 / EXCERPT: Let $r\geq 2$ be an integer. Let f be given by the paper's definition in, with h satisfying $|h(d)|\ll d^\alpha$ for some $\alpha\in[0,\frac{1}{2r-1}]$, and let C_f be the co...

conjecture | open | MathDB 页面 | 原文线索

V2-004609 | MATHDB-355651

Exponent-pair approximation conjecture

摘录 / EXCERPT: Let $r\geq 1$ be real and let an exponent pair (k,ℓ) mean an exponent pair in the standard sense. Exponent-pair approximation conjecture. For every real $\varepsilon>0$, there...

conjecture | open | MathDB 页面 | 原文线索

V2-004610 | MATHDB-355684

Parker, Rushall and Hunt's form conjecture for odd norm-perfect Eisenstein integers

摘录 / EXCERPT: Parker, Rushall and Hunt's conjecture. Every odd norm-perfect Eisenstein integer has the form

conjecture | open | MathDB 页面 | 原文线索

V2-004611 | MATHDB-355685

Conjectural classification of even perfect and norm-perfect integers in higher cyclotomic fields

摘录 / EXCERPT: Classification conjecture for higher cyclotomic fields. The even perfect integers in $\mathbb{Z}[\zeta_p]$ are exactly

conjecture | open | MathDB 页面 | 原文线索

V2-004612 | MATHDB-355705

The Diophantine problem conjecture for star-shaped right-angled Baumslag-Solitar-Artin groups

摘录 / EXCERPT: Let G be a group, and let $B(\Gamma)$ be a right-angled Baumslag-Solitar-Artin group whose underlying undirected graph is a star graph. Suppose that G is virtually $B(\Gamma)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004613 | MATHDB-355711

Integrality Conjecture for vector-valued modular forms

摘录 / EXCERPT: Let $f_i(\tau)$ be a component of a vector-valued modular form with Fourier expansion ... where $n^{(i)}_0=p_i/N_i$ and $(p_i,N_i)=1$. Suppose that all coefficients $a_n^{(i)}$ are a...

conjecture | open | MathDB 页面 | 原文线索

V2-004614 | MATHDB-355712

Upper bound for categorifiable near-group fusion rings over cyclic prime groups

摘录 / EXCERPT: Let C_p be the cyclic group of prime order p, let $R(C_p,\ell)$ be the near-group fusion ring with group of invertible elements C_p and parameter ℓ, and call it categori...

conjecture | open | MathDB 页面 | 原文线索

V2-004615 | MATHDB-355729

Kudla-Rapoport conjecture for Krämer models at a ramified prime

摘录 / EXCERPT: Let F/F_0 be the ramified quadratic extension at the place under consideration, and let L be an integral Hermitian lattice of rank n with $\epsilon=\chi(L)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004616 | MATHDB-355730

Local Kudla-Rapoport conjecture for Krämer models via Whittaker functions

摘录 / EXCERPT: Fix a place v and let q be the residue-field cardinality. Let $\mathbb{V}_v$ be the space of special quasi-homomorphisms with $\chi_v(\mathbb{V}_v)=-\epsilon$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004617 | MATHDB-355733

Gross-Kohnen-Zagier conjecture on Green's functions at CM points

摘录 / EXCERPT: Let $\Gamma=SL_2(\mathbb Z)$, let $G_s(z_1,z_2)$ be the Green's function, and let ... For a weakly holomorphic modular form $f=\sum_m c_f(m)q^m$ of weight $-2j$, define ... For a discr...

conjecture | open | MathDB 页面 | 原文线索

V2-004618 | MATHDB-355739

Smoothed weighted one-level density conjecture for the Riemann zeta-function

摘录 / EXCERPT: Smoothed weighted one-level density conjecture. As $T\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004619 | MATHDB-355742

The polygonal stacked prism odd-prime labeling conjecture

摘录 / EXCERPT: Let $Y_{k,n}$ be the k-polygonal stacked prism with $k\geq 3$ sides and $n\geq 1$ polygons. Polygonal stacked prism conjecture. $Y_{k,n}$ is odd prime for all $k\geq 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004620 | MATHDB-355743

The all-trees odd-prime labeling conjecture

摘录 / EXCERPT: Let T be a tree. All-trees conjecture. Every tree T is odd prime. Paths, unions of paths, star graphs, and several additional classes are known to be odd prime, but the asserti...

conjecture | open | MathDB 页面 | 原文线索

V2-004621 | MATHDB-355744

The prime graph odd-prime labeling conjecture

摘录 / EXCERPT: Let G be a graph. A graph is prime if it admits a prime labeling, and odd prime if its vertices can be injectively labeled by $1,3,\ldots,2|V(G)|-1$ so that adjacent vertices rec...

conjecture | open | MathDB 页面 | 原文线索

V2-004622 | MATHDB-355762

Conjecture on joint moments of quadratic Dirichlet L-functions and their second derivatives

摘录 / EXCERPT: Joint-moment conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004623 | MATHDB-355788

Diophantine undecidability conjecture for subfields of the maximal abelian extension

摘录 / EXCERPT: Let $\mathbb{Q}^{\operatorname{ab}}$ denote the maximal abelian extension of $\mathbb{Q}$, and let $\mathcal{O}_{\mathbf L}$ denote the ring of integers of a subfield $\mathbf L$....

conjecture | open | MathDB 页面 | 原文线索

V2-004624 | MATHDB-355796

The linear arithmetic fundamental lemma for general Hecke functions

摘录 / EXCERPT: Linear arithmetic fundamental lemma. For every $f\in\mathcal{H}$, there is an equality up to sign,

conjecture | open | MathDB 页面 | 原文线索

V2-004625 | MATHDB-355817

The p-adic Hilbert-Smith conjecture

摘录 / EXCERPT: For a prime p, let $\widehat{\mathbb{Z}}_p$ denote the p-adic group. The p-adic Hilbert-Smith conjecture. For every prime p, there is no faithful action of...

conjecture | open | MathDB 页面 | 原文线索

V2-004626 | MATHDB-355834

Chen-Chen conjecture on large prime factors of shifted primes

摘录 / EXCERPT: Chen-Chen conjecture. For any integer $k\ge 1$ and any $c\in [\frac{1}{k+1},\frac{1}{k})$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-004627 | MATHDB-355838

Positivity of the rank for elliptic surface fibres

摘录 / EXCERPT: Positivity of the rank. There is $c>0$ such that for all sufficiently large x, $N(x)\ge c\cdot x$. This conjecture is motivated by work of Helfgott and, together with a parity st...

conjecture | open | MathDB 页面 | 原文线索

V2-004628 | MATHDB-355839

Averages of root numbers in elliptic-surface families

摘录 / EXCERPT: Averages of root numbers conjecture. The average root number in this family is zero. Helfgott gives strong support and proves the assertion for several elliptic surfaces under suit...

conjecture | open | MathDB 页面 | 原文线索

V2-004629 | MATHDB-355840

Equidistribution of root numbers in elliptic-surface families

摘录 / EXCERPT: Equidistribution of root numbers conjecture. Both S_{-1} and S_1 have natural density $1/2$ in $\mathbb{N}$. This is the equivalent sign-equidistribution formulation of the con...

conjecture | open | MathDB 页面 | 原文线索

V2-004630 | MATHDB-355885

Mináč-Tăn's vanishing conjecture for higher Massey products

摘录 / EXCERPT: Mináč-Tăn's conjecture. For every field F, every prime p, every $n\geq 3$, and all $\chi_1,\dots,\chi_n\in H^1(F,\mathbb Z/p\mathbb Z)$, if the Massey product

conjecture | open | MathDB 页面 | 原文线索

V2-004631 | MATHDB-355940

The universal non-Gorenstein $R=T$ conjecture for squarefree levels

摘录 / EXCERPT: Let p be a prime and let M be a squarefree level satisfying the conditions of Section. Let R_M and $\mathbb{T}_M$ denote the corresponding deformation ring and Hecke algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-004632 | MATHDB-355956

Yip's maximal subfield clique conjecture for generalized Paley graphs

摘录 / EXCERPT: Let $d>1$ be an integer. Let $q \equiv 1 \pmod {2d}$ be a power of a prime p, and let r be the largest integer such that $d \mid \frac{q-1}{p^r-1}$. The subfield...

conjecture | open | MathDB 页面 | 原文线索

V2-004633 | MATHDB-355958

Stoll's conjecture on weak approximation with Brauer-Manin obstruction for curves

摘录 / EXCERPT: Stoll's conjecture. The set $C(K)$ is dense in

conjecture | open | MathDB 页面 | 原文线索

V2-004634 | MATHDB-355978

Keith-Zanello's self-similarity conjecture for 19-regular partitions

摘录 / EXCERPT: Keith-Zanello's self-similarity conjecture. For a positive proportion of primes p, one has

conjecture | open | MathDB 页面 | 原文线索

V2-004635 | MATHDB-355979

Keith-Zanello's parity-density conjecture for t-regular partitions

摘录 / EXCERPT: Keith-Zanello's parity-density conjecture. The series f_t/f_1 has odd density $1/2$. This conjecture is paired in the source with a non-congruence assertion, and concerns the glo...

conjecture | open | MathDB 页面 | 原文线索

V2-004636 | MATHDB-355983

Fredi's girth conjecture for the algebraic bipartite graph D(k,q)

摘录 / EXCERPT: Let $D(k,q)$ be the algebraic bipartite graph proposed by Lazebnik and Ustimenko, where $k\geq 2$ and $q=p^m$ is a prime power. Fredi's conjecture. $D(k,q)$ has girth $k+5$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-004637 | MATHDB-355999

Discriminator conjecture for half-integer quadratic sequences with coefficient ratio divisible by 3

摘录 / EXCERPT: Let b and c be non-zero integers such that $3\nmid bc$. For a sequence ... let $D_{\operatorname{qr}}(n)$ denote its discriminator, the least modulus for which the first n sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-004638 | MATHDB-356009

The local p-adic monodromy conjecture

摘录 / EXCERPT: Let $\Bbbk$ be a number field, let f be a polynomial over $\Bbbk$, and let $Z_\mathfrak p(s)$ denote its local $\mathfrak p$-adic zeta function. For a pole α of...

conjecture | open | MathDB 页面 | 原文线索

V2-004639 | MATHDB-356010

The local topological monodromy conjecture

摘录 / EXCERPT: Let f be a polynomial with coefficients in a field of characteristic 0, and let $Z_\mathrm{top}(s)$ be its local topological zeta function. If α is a pole of...

conjecture | open | MathDB 页面 | 原文线索

V2-004640 | MATHDB-356032

Greenberg's Iwasawa main conjecture for modular forms

摘录 / EXCERPT: Let $F\in S_k(\Gamma_1(N))$ be a p-ordinary eigenform, let φ be a Dirichlet character, and let $\mu_{p,F,\varphi,j}$ denote the p-adic L-function attached to...

conjecture | open | MathDB 页面 | 原文线索

V2-004641 | MATHDB-356085

Six-primes conjecture for orders of ${\rm PSL}_2(p)$

摘录 / EXCERPT: Consider the four cases (a)-(d) for a prime p, in which the relevant factors of $p-1$ and $p+1$ are required to be prime; equivalently, each case gives a triple of linear polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-004642 | MATHDB-356093

The strong form of the p-adic Zauner conjecture

摘录 / EXCERPT: Let p be a prime, let $d\in\mathbb{N}$, and let $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ denote the inner product and norm on $\mathbb{Q}_p^d$. The vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-004643 | MATHDB-356095

Conjecture on parametric Euler sums in terms of Hurwitz zeta values

摘录 / EXCERPT: Parametric Euler-sum conjecture. These sums can be evaluated in terms of Hurwitz zeta values and digamma functions. This extends the preceding explicitly known cases, but the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-004644 | MATHDB-356104

GSS uniformity conjecture for free nilpotent Lie rings

摘录 / EXCERPT: Let $\mathfrak{f}_{c,d}$ denote the free class-c nilpotent Lie rings on d generators defined over $\mathbb{Z}$. For $c,d\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004645 | MATHDB-356105

Existence of a non-uniform graded ideal zeta function

摘录 / EXCERPT: For each $c\in\mathbb{N}$, let $\mathfrak{f}_{c,2}$ be the free class-c nilpotent Lie ring on two generators, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004646 | MATHDB-356123

Siegel's asymptotic conjecture for B-regular primes

摘录 / EXCERPT: Let $R_B(x)$ denote the number of B-regular primes up to x, where an odd prime is B-regular if it divides none of the numerators of $B_2,B_4,\ldots,B_{p-3}$. Siegel's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-004647 | MATHDB-356124

Asymptotic conjecture for \ell-Genocchi irregular primes in arithmetic progressions

摘录 / EXCERPT: Let ℓ be an odd prime, let $1\leq a<d$ with $(a,d)=1$, and let $\mathcal P_G(d,a)(x)$ count the primes $p\leq x$ satisfying $p\equiv a\pmod d$ that are ℓ-Genocchi irregul...

conjecture | open | MathDB 页面 | 原文线索

V2-004648 | MATHDB-356125

Positive-density conjecture for \ell-Genocchi regular primes in arithmetic progressions

摘录 / EXCERPT: Let ℓ be a prime and let $1\leq a<d$ with $(a,d)=1$. A prime is ℓ-Genocchi regular if it is not ℓ-Genocchi irregular. Positive-density conjecture. If ℓ is odd,...

conjecture | open | MathDB 页面 | 原文线索

V2-004649 | MATHDB-356126

B-regular refinement of divisor-counting estimates

摘录 / EXCERPT: Let $\mathcal P(g,t)$ and $\mathcal Q_{\alpha,\beta}$ be the prime sets defined earlier in the paper, and let $\delta(g,t)$ and $\rho_{\alpha,\beta}(a,d)$ be the corresponding cons...

conjecture | open | MathDB 页面 | 原文线索

V2-004650 | MATHDB-356127

Asymptotic equivalence between sequence-divisor sets and irregular-prime sets

摘录 / EXCERPT: Let $\mathcal Q_H(x)$, $\mathcal Q_{H^-}(x)$, and $\mathcal Q_{H^+}(x)$ count the prime divisors up to x of the H-, H^--, and H^+-sequences, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004651 | MATHDB-356158

Harpaz-Wittenberg conjecture on specializations of norm equations

摘录 / EXCERPT: Let $P_1,\ldots,P_n\in\mathbb{Q}[t]$ be pairwise distinct irreducible monic polynomials. Let $k_i=\mathbb{Q}[t]/(P_i(t))$ be the corresponding number fields, and let $a_i\in k_i$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-004652 | MATHDB-356162

Three-character covering conjecture for prime divisors of character degrees

摘录 / EXCERPT: Let G be a finite group, and let $\rho(G)$ be the set of primes dividing the degree of some irreducible character of G. Write $\operatorname{Irr}(G)$ for the set of irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-004653 | MATHDB-356184

The Weil-Petersson random prime geodesic conjecture

摘录 / EXCERPT: Weil-Petersson random prime geodesic conjecture. Theorem 1.1 still holds after replacing the exponent $3/4$ by $1/2$; equivalently, there is a universal constant $c>0$ such that fo...

conjecture | open | MathDB 页面 | 原文线索

V2-004654 | MATHDB-356186

Fuchs-Litman-Silverman-Tran orbit conjecture for Markoff-type K3 surfaces

摘录 / EXCERPT: Let q be an odd prime power, let $\mathcal{W}_k:x^2+y^2+z^2+x^2y^2z^2=kxyz$ be the Markoff-type tri-involutive K3 surface, and let Γ be the subgroup of automorphisms gener...

conjecture | open | MathDB 页面 | 原文线索

V2-004655 | MATHDB-356208

Goldfeld-Hinkle pole-free continuation conjecture for the cubic Pell equation L-function

摘录 / EXCERPT: Goldfeld-Hinkle's pole-free continuation conjecture. The function $L_d(s)$ has meromorphic continuation to $\textup{Re}(s)>\frac12$ with at most a simple pole at $s=\frac23$. In th...

conjecture | open | MathDB 页面 | 原文线索

V2-004656 | MATHDB-356225

The magic sign-smoothing conjecture for Dirichlet inverse functions

摘录 / EXCERPT: Magic sign-smoothing conjecture. If $f(n)$ has constant sign for all $n\geq 1$, then $M_1[f^{-1}]$ and $M_2[f^{-1}]$ are bounded, with eventual signs respectively satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-004657 | MATHDB-356231

The coprime-coefficients conjecture for restricted Frobenius representations

摘录 / EXCERPT: Let $s\ge 3$. Let a_k, where $k\in [s]$, be positive integers such that $(a_1,a_2,\ldots,a_s)=1$. Then $S_{s,s-2}^m(a_1,a_2,\ldots,a_s)$ denotes the set of positive integers sati...

conjecture | open | MathDB 页面 | 原文线索

V2-004658 | MATHDB-356235

The rank 0-or-1 conjecture for prime quadratic twists of an elliptic curve

摘录 / EXCERPT: Let E^d denote the quadratic twist of the elliptic curve E by d. Consider quadratic twists with $|d|$ prime. Rank 0-or-1 conjecture. 100% of quadratic twists E^d where...

conjecture | open | MathDB 页面 | 原文线索

V2-004659 | MATHDB-356236

The parity conjecture for prime quadratic twists

摘录 / EXCERPT: Let E^d denote the quadratic twist of the elliptic curve E by d, let $\operatorname{rank}(E^d)$ be its Mordell-Weil rank, and let $w(E^d)$ be the root number of E^d. For...

conjecture | open | MathDB 页面 | 原文线索

V2-004660 | MATHDB-356274

The petal-number bound for torus knots

摘录 / EXCERPT: Let r and s be relatively prime integers with $1 < r < s$, and let $T_{r,s}$ denote the corresponding torus knot. The petal-number bound for torus knots. ... The authors are led...

conjecture | open | MathDB 页面 | 原文线索

V2-004661 | MATHDB-356287

Uniqueness conjecture for Markov fractions

摘录 / EXCERPT: Let q be a Markov number, and call a rational number a Markov fraction when it is one of the Markov fractions defined in the paper; its denominator is the denominator in lowest t...

conjecture | open | MathDB 页面 | 原文线索

V2-004662 | MATHDB-356294

Erdős's conjecture on three consecutive powerful numbers

摘录 / EXCERPT: A positive integer is powerful if every prime divisor p satisfies $p^2\mid n$. For an integer N, the three consecutive integers are N, $N+1$, and $N+2$. Erdős's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-004663 | MATHDB-356331

A mod 32 class-number congruence for quadratic fields of discriminants $-8p$ and $8p$

摘录 / EXCERPT: Let p be a prime such that $p\equiv 3\pmod 4$. Let $h(-8p)$ and $h(8p)$ denote the class numbers associated with the discriminants $-8p$ and $8p$, respectively, and let Ψ be...

conjecture | open | MathDB 页面 | 原文线索

V2-004664 | MATHDB-356332

Infinite square-sequence Frobenius formula for large shifts

摘录 / EXCERPT: Large-shift infinite square-sequence conjecture. If $a>30$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004665 | MATHDB-356333

Euler's formula for the Waring number

摘录 / EXCERPT: Waring-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004666 | MATHDB-356343

The connected-sum conjecture for non-classical weakly prime virtual links

摘录 / EXCERPT: Let V_1 and V_2 be non-classical, weakly prime virtual links. A connected sum is formed by joining diagrams of two links along chosen arcs; for virtual links this operation can...

conjecture | open | MathDB 页面 | 原文线索

V2-004667 | MATHDB-356381

Terai's conjecture for the generalized Ramanujan equation

摘录 / EXCERPT: Terai's conjecture. The equation has only the solution

conjecture | open | MathDB 页面 | 原文线索

V2-004668 | MATHDB-356382

The even-power representation conjecture for a binary quadratic form

摘录 / EXCERPT: Let k be a positive integer such that $4\Vert k$ and $2k-1$ is a prime power. Write ... where a is square-free. The form $(a^2b^2,0,2k-1)$ is the binary quadratic form...

conjecture | open | MathDB 页面 | 原文线索

V2-004669 | MATHDB-356409

Klurman-Tenenbaum-Teräväinen conjecture on modified character partial sums

摘录 / EXCERPT: Klurman-Tenenbaum-Teräväinen conjecture. The partial sums satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-004670 | MATHDB-356428

Conjecture on simultaneous nonvanishing of real and imaginary amplitudes

摘录 / EXCERPT: Let x and t be integers satisfying ... Let $a(x,t)$ be the complex amplitude under consideration. Simultaneous nonvanishing conjecture. Then ... unless $(x,t)\in\{(-3,11),(5,11)\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-004671 | MATHDB-356430

Algebraic Fourier coefficients of harmonic Maass forms

摘录 / EXCERPT: Algebraicity conjecture. If all Fourier coefficients of the holomorphic part of $\tilde f$ are algebraic, then $\xi(\tilde f)$ is either a unary theta series with...

conjecture | open | MathDB 页面 | 原文线索

V2-004672 | MATHDB-356431

Várilly-Alvarado's uniform Brauer-group bound for K3 surfaces

摘录 / EXCERPT: Várilly-Alvarado's conjecture. The order

conjecture | open | MathDB 页面 | 原文线索

V2-004673 | MATHDB-356487

The prime-power wreath-product strategy conjecture for involution-generated groups

摘录 / EXCERPT: Let G be a finite group generated by involutions, and let H be a 2-group acting faithfully on the set of switches. The wreath product is denoted by $G\wr H$. Prime-power wrea...

conjecture | open | MathDB 页面 | 原文线索

V2-004674 | MATHDB-356489

The prime-power wreath-product surjective-strategy conjecture

摘录 / EXCERPT: Let G be a group generated by elements of order p^n, let H be a p-group acting faithfully on a set of switches Ω, and let $G\wr_\Omega H$ denote the corresponding wr...

conjecture | open | MathDB 页面 | 原文线索

V2-004675 | MATHDB-356497

Selberg's conjecture for arithmetic hyperbolic surfaces

摘录 / EXCERPT: An arithmetic hyperbolic surface is a hyperbolic surface arising from an arithmetic lattice, and let λ_1 denote the least positive eigenvalue of its Laplace-Beltrami operat...

conjecture | open | MathDB 页面 | 原文线索

V2-004676 | MATHDB-356529

Vinatier's freeness conjecture for the square root of the inverse different

摘录 / EXCERPT: Let L/K be a finite weakly ramified Galois extension of number fields, with $G=\operatorname{Gal}(L/K)$, ring of integers $\mathcal{O}_L$, and different $\mathcal{D}_{L/K}$. When...

conjecture | open | MathDB 页面 | 原文线索

V2-004677 | MATHDB-356530

Erez's class-group conjecture for the square root of the inverse different

摘录 / EXCERPT: Let L/K be a weakly ramified Galois extension of number fields for which the square root of the inverse different $\mathcal{A}_{L/K}$ exists, and set $G=\operatorname{Gal}(L/K)$....

conjecture | open | MathDB 页面 | 原文线索

V2-004678 | MATHDB-356531

Bley-Burns-Hahn's relative K-theory conjecture for weakly ramified extensions

摘录 / EXCERPT: Let L/K be a weakly ramified Galois extension of number fields of odd degree, with $G=\operatorname{Gal}(L/K)$. Let $\mathfrak{a}_{L/K}$ and $\mathfrak{c}_{L/K}$ be the canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-004679 | MATHDB-356532

Kollár's conjecture on Diophantine subsets of rational functions

摘录 / EXCERPT: Let $D\subseteq \mathbb{C}(z)$ be a Diophantine set. For each integer $n\geq 0$, write $\mathbb{C}[z]_n=\mathbb{C}^{n+1}$ for the polynomials of degree at most n. Kollár's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-004680 | MATHDB-356536

Vlasenko-Zwegers conjectural identities for the rank two Example 10 Nahm sums

摘录 / EXCERPT: Let q be a formal variable, and let $J_{a,b}$ and J_a denote the Jacobi-product notation used in the source. Vlasenko-Zwegers' conjecture. The following identities are conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004681 | MATHDB-356553

The odd-prime-power solubilizer order conjecture

摘录 / EXCERPT: Odd-prime-power solubilizer order conjecture. For every $x\in G$, $|\operatorname{Sol}_G(x)|\neq p^n$, where p is an odd prime and n is a natural number.

conjecture | open | MathDB 页面 | 原文线索

V2-004682 | MATHDB-356594

Super-base graph conjecture for the graphs $G_{n,k}$

摘录 / EXCERPT: Super-base graph conjecture. For every such n and k, if $G_{n,k}$ is a minimal prime graph, then $\overline{G_{n,k}}\setminus v$ is not a minimal prime graph for any vertex v...

conjecture | open | MathDB 页面 | 原文线索

V2-004683 | MATHDB-356615

The infinitude conjecture for Mersenne primes

摘录 / EXCERPT: A Mersenne prime is a prime number of the form ... with q prime. Mersenne-prime infinitude conjecture. There are infinitely many Mersenne primes. This conjecture is used to obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-004684 | MATHDB-356639

A lower-bound conjecture for nonzero differences of monomials

摘录 / EXCERPT: Monomial-difference lower-bound conjecture. If $D e 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004685 | MATHDB-356650

Monstrous Moonshine conjecture

摘录 / EXCERPT: Let $\mathbb{M}$ be the Monster finite simple group, and let $J(q)$ be the normalized Hauptmodul for $\operatorname{SL}(2,\mathbb{Z})$. Monstrous Moonshine conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-004686 | MATHDB-356661

Cassels-Swinnerton-Dyer's conjecture on positive norm minima and algebraic lattices

摘录 / EXCERPT: Cassels-Swinnerton-Dyer's conjecture. The condition $N(\Lambda)>0$ is equivalent to the existence of a non-degenerate diagonal operator D such that $D\Lambda$ is an algebraic lat...

conjecture | open | MathDB 页面 | 原文线索

V2-004687 | MATHDB-356669

Marques-Lengyel valuation conjecture for the Tribonacci sequence

摘录 / EXCERPT: Let p) be a prime number, and let $T(n)$ be the Tribonacci sequence defined by $T(0)=0$, $T(1)=T(2)=1$, and $T(n+3)=T(n+2)+T(n+1)+T(n)$. Write $ u_p(m)$ for the exponent of p i...

conjecture | open | MathDB 页面 | 原文线索

V2-004688 | MATHDB-356670

Rational-center valuation conjecture for the Tribonacci sequence

摘录 / EXCERPT: Let p be a prime number, and let $T(n)$ be the Tribonacci sequence defined by $T(0)=0$, $T(1)=T(2)=1$, and $T(n+3)=T(n+2)+T(n+1)+T(n)$. Write $\nu_p(m)$ for the exponent of p i...

conjecture | open | MathDB 页面 | 原文线索

V2-004689 | MATHDB-356671

Positive-density conjecture for exceptional Tribonacci primes

摘录 / EXCERPT: Let $\mathcal{ML}$ be the set of primes p for which the Marques-Lengyel conjecture holds, and let $\mathcal{NMLR}$ be the set of primes for which the rational-center variant fail...

conjecture | open | MathDB 页面 | 原文线索

V2-004690 | MATHDB-356693

Non-commutative Rubin-Stark conjecture

摘录 / EXCERPT: Let L/K, Π, and Π' be the fixed data, let $ Sigma$ be a subset of $ Sigma(L)$ with $ Sigma e Pi$, and let p be a prime such that $ O^\times_{L, Pi, Pi',p}$ is torsion-f...

conjecture | open | MathDB 页面 | 原文线索

V2-004691 | MATHDB-356696

The p-adic equivariant Tamagawa number conjecture for \mathbb{G}_m

摘录 / EXCERPT: Let L/K be the extension and $G={\rm Gal}(L/K)$, let $C=C_{L,S,T}$ be the perfect complex over $\mathbb{Z}[G]$, and for a prime p let C_p be its p-adic realization. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-004692 | MATHDB-356700

The Chowla conjecture for Kloosterman sums

摘录 / EXCERPT: Fix an integer $a\ne0$, positive integers $\nu_1,\ldots,\nu_s$, and integers $h_s>\cdots>h_1\geqslant0$. Let ${\mathcal K}_m^*(a)$ be the normalized Kloosterman-sum quantity and le...

conjecture | open | MathDB 页面 | 原文线索

V2-004693 | MATHDB-356738

Equality of the three Brumer-Stark unit formulas

摘录 / EXCERPT: Let $u_{\mathfrak p}$ be the Brumer-Stark element attached to the data $(S,T,H,\mathfrak P)$, and let $u_1,u_2,u_3\in F_{\mathfrak p}^*\otimes\mathbb Z[G]$ be the three formulas de...

conjecture | open | MathDB 页面 | 原文线索

V2-004694 | MATHDB-356739

Shintani-domain formula for the Brumer-Stark unit

摘录 / EXCERPT: Let e be the order of $\mathfrak p$ in $G_{\mathfrak f}$, and suppose that $\mathfrak p^e=(\pi)$ with π totally positive and $\pi\equiv1\pmod{\mathfrak f}$. Let $\mathcal D$...

conjecture | open | MathDB 页面 | 原文线索

V2-004695 | MATHDB-356744

Restricted Erdős-Heilbronn formula for abelian groups

摘录 / EXCERPT: Let G be an abelian group, and let p be the smallest prime divisor of the order of G. For positive integers m and h with $m\leq p$, write $\rho\hat{\;}(G,m,h)$ for the m...

conjecture | open | MathDB 页面 | 原文线索

V2-004696 | MATHDB-356745

Inverse restricted Erdős-Heilbronn conjecture below the prime threshold

摘录 / EXCERPT: Let G be an abelian group, let p be the smallest prime divisor of the order of G, and let A be an m-subset of G. Assume that m and h are positive integers satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-004697 | MATHDB-356747

Matzke's equality criterion for signed sumsets in elementary abelian groups

摘录 / EXCERPT: Let p be an odd prime, and let m,h,r be positive integers. Define $f_1=hm-h+1$. Let δ be 0 or 1 according as $p-1$ is divisible by h or not. Let k be the largest...

conjecture | open | MathDB 页面 | 原文线索

V2-004698 | MATHDB-356756

The prime-modulus conjecture for progression-free digit sets

摘录 / EXCERPT: Prime-modulus progression-free-set conjecture. The set $S(D,n)$ does not contain any arithmetic progression of length 4.

conjecture | open | MathDB 页面 | 原文线索

V2-004699 | MATHDB-356761

The 1439 extremal conjecture for integer complexity

摘录 / EXCERPT: The 1439 extremal conjecture. For all $n>1$, one has

conjecture | refuted | MathDB 页面 | 原文线索

V2-004700 | MATHDB-356763

The efficient-prime finite-control conjecture for integer complexity

摘录 / EXCERPT: The efficient-prime finite-control conjecture. For any $\alpha>\frac{2}{\log 2}$, there is a finite set of primes S_α such that, for any m, if v_p is sufficiently large...

conjecture | refuted | MathDB 页面 | 原文线索

V2-004701 | MATHDB-356790

Haensch-Kane spinor-genus decomposition conjecture for ternary lattice cosets

摘录 / EXCERPT: Let L be a quadratic lattice and let $\nu\in\mathbb{Q}L$. Define the proper spinor-genus theta series by ... where the sum is over representatives of the proper classes in the prop...

conjecture | open | MathDB 页面 | 原文线索

V2-004702 | MATHDB-356815

Strong Kolyvagin higher-weight non-vanishing conjecture

摘录 / EXCERPT: Let $\kappa^{\mathrm{st}}_{f,\infty}$ be the strict Kolyvagin set consisting of the classes $c_1(f,n)$ as n ranges over the Kolyvagin indices. Strong Kolyvagin conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-004703 | MATHDB-356839

Generic semisimplicity conjecture for time changes of transitive Anosov flows

摘录 / EXCERPT: Let X be a transitive Anosov vector field on a closed 3-manifold $\mathcal{M}$. Let Ω_0^1 denote the space of closed resonant 1-forms at zero, and let $\mathcal{L}_{fX}$...

conjecture | open | MathDB 页面 | 原文线索

V2-004704 | MATHDB-356840

The Density Hypothesis for the Riemann zeta-function and Dirichlet L-functions

摘录 / EXCERPT: Density Hypothesis. The implied constants being absolute, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-004705 | MATHDB-356891

The effective Lang-type approximation conjecture for cubic irrational numbers

摘录 / EXCERPT: Let $\xi\in\mathbb{R}$ be a real cubic irrational number, let $H(\xi)$ be the naive height of its minimal polynomial with coprime integer coefficients, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004706 | MATHDB-356894

A dyadic-denominator conjecture for lonely runner instances

摘录 / EXCERPT: Dyadic-denominator conjecture. For every $\mathbf{n}$ with $\gcd(\mathbf{n})=1$, there is always a suitable time of the form

conjecture | open | MathDB 页面 | 原文线索

V2-004707 | MATHDB-356911

Jaulent's weak cyclotomic conjecture for CM fields

摘录 / EXCERPT: Let K be a CM field, that is, a totally imaginary quadratic extension of a totally real subfield K^+. For a prime number ℓ, let ... be a partition of the places of K above...

conjecture | open | MathDB 页面 | 原文线索

V2-004708 | MATHDB-356923

Rarity of fixed inertia-type prime counts in number fields

摘录 / EXCERPT: Rarity conjecture. For every $r=0,1,2,\dots$, there exists an integer r' such that

conjecture | open | MathDB 页面 | 原文线索

V2-004709 | MATHDB-356926

Bhargava-Cremona-Fisher-Gajović density functional-equation conjecture

摘录 / EXCERPT: Let n be a positive integer and let σ be a splitting type of degree n. For a prime p, let $\rho(n,\sigma;p)$ be the Haar measure of the p-adic polynomials of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-004710 | MATHDB-356930

The double-series formula for the logarithmic quantity

摘录 / EXCERPT: Double-series formula. The quantity satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-004711 | MATHDB-356938

The higher-genus log-open correspondence for nef Looijenga pairs

摘录 / EXCERPT: Let (X,D) be a nef Looijenga pair admitting a deformation to a pair (X',D') with X' toric, D'_i prime toric divisors for $i<l$, and D'_l nef. Let (Y,L) be the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-004712 | MATHDB-356944

The Stein-degree boundedness conjecture for boundary components over characteristic-zero fields

摘录 / EXCERPT: Stein-degree boundedness conjecture over non-closed fields. The quantity

conjecture | open | MathDB 页面 | 原文线索

V2-004713 | MATHDB-356953

The abc-conjecture consequence for three-term monomial differences

摘录 / EXCERPT: The abc-conjecture consequence. If $D\neq 0$, then one has the lower bound

conjecture | open | MathDB 页面 | 原文线索

V2-004714 | MATHDB-356956

The Deligne-Milnor formula

摘录 / EXCERPT: Let S be a strictly henselian trait with closed point s, generic point η, and geometric generic point $\bar\eta$. Let $p:U\to S$ be flat, with U regular and purely of re...

conjecture | open | MathDB 页面 | 原文线索

V2-004715 | MATHDB-356964

Plectic comparison conjecture for partial Frobenius and filtrations

摘录 / EXCERPT: Let f be the Hilbert modular form and let $D_p(f)$ be its p-adic de Rham cohomology with partial Frobenius operators $\varphi_1,\dots,\varphi_d$ and partial filtrations...

conjecture | open | MathDB 页面 | 原文线索

V2-004716 | MATHDB-356967

The zig-zag conjecture for reductions of crystalline representations

摘录 / EXCERPT: Let p be an odd prime, let V_{k,a_p} be a two-dimensional crystalline representation of $\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$, and let $v_p(a_p)$ be a half...

conjecture | open | MathDB 页面 | 原文线索

V2-004717 | MATHDB-356975

Capitulation and stability conjecture for p-class groups in cyclotomic extensions

摘录 / EXCERPT: Let K be any real number field and let ${\mathcal H}_K$ be its p-class group, of exponent $p^{e(K)}$. For any prime number $\ell \equiv 1 \pmod {2p^N}$, with $N \geq e(K)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004718 | MATHDB-356980

Erdős-Graham-Granville-Selfridge conjecture on integral points on hyperelliptic curves

摘录 / EXCERPT: Erdős-Graham-Granville-Selfridge conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004719 | MATHDB-356985

Linowitz-Thompson conjecture for the average first negative Fourier coefficient

摘录 / EXCERPT: Let D_1,D_2 range over fundamental discriminants with $|D_1D_2|\to obreak\ter x$, and let $\eta(D_1,D_2)$ be the smallest prime p such that the sign of the corresponding Fourie...

conjecture | open | MathDB 页面 | 原文线索

V2-004720 | MATHDB-356999

The quadratic-orthomorphism cycle conjecture for primes

摘录 / EXCERPT: The cycle conjecture. For any prime $p\equiv 1\bmod 8$ or $p\equiv 3\bmod 8$, the permutation ϕ_p is a p-cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-004721 | MATHDB-357013

Berry's conjecture on the variance of zeta zeros

摘录 / EXCERPT: Let $S(t)$ be the argument function of the Riemann zeta function, and let γ_0 be Euler's constant. For $\delta>0$, consider the mean square of the increment...

conjecture | open | MathDB 页面 | 原文线索

V2-004722 | MATHDB-357014

Chan's extended strong pair correlation conjecture

摘录 / EXCERPT: Let $F_\Delta(\alpha)$ denote the shifted pair-correlation function and let $w(\Delta)=4/(4+\Delta^2)$. Chan's extended strong pair correlation conjecture. For $|\alpha|\geq1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004723 | MATHDB-357022

Komatsu-Morisawa-Okazaki trace conjecture for relative units

摘录 / EXCERPT: Let $\mathbb{B}_n$ be the maximal real subfield of $\mathbb{Q}(\zeta_{2^{n+2}})$, let E_n be its group of units, and let ${\rm N}_{n/n-1}:E_n\to E_{n-1}$ be the norm map. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-004724 | MATHDB-357023

Kashio-Yoshizaki refinement for traces of positive relative units

摘录 / EXCERPT: Let $\mathbb{B}_n$ be the maximal real subfield of $\mathbb{Q}(\zeta_{2^{n+2}})$. For the norm map ${\rm N}_{n/n-1}:E_n\to E_{n-1}$, set $RE_n^+=\ker({\rm N}_{n/n-1})$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-004725 | MATHDB-357024

Yoshizaki's generation conjecture for positive relative units

摘录 / EXCERPT: Let $\mathbb{B}_n$ be the maximal real subfield of $\mathbb{Q}(\zeta_{2^{n+2}})$, let $X_n=2\cos(2\pi/2^{n+2})$, and let $RE_n^+=\ker({\rm N}_{n/n-1}:E_n\to E_{n-1})$. Define ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-004726 | MATHDB-357056

Existence of tight irreducible affine vector space partitions from divisible dimensions

摘录 / EXCERPT: Let n be a positive integer, let q be a prime power, and let k be an integer with $1<k<n$ such that k divides n. An affine vector space partition is a partition of the po...

conjecture | open | MathDB 页面 | 原文线索

V2-004727 | MATHDB-357057

Extension conjecture for planes in the Klein quadric

摘录 / EXCERPT: Let q be a prime power, let $\mathcal{Q}=Q^+(5,q)$ be the Klein quadric, and fix a plane π in $\mathcal{Q}$. Let $\mathcal{P}$ be the set of planes in $\mathcal{Q}$ disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-004728 | MATHDB-357060

Polynomial formula conjecture for $g_{(3,2,2,1,1)}(p)$

摘录 / EXCERPT: Let p be a prime, and let $g_{(3,2,2,1,1)}(p)$ denote the quantity counting the relevant subring matrices associated with the composition $(3,2,2,1,1)$. Polynomial formula conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-004729 | MATHDB-357076

Möbius cancellation conjecture for ramified primes

摘录 / EXCERPT: Möbius cancellation conjecture. As $X\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-004730 | MATHDB-357085

Generalised Sylvester's conjecture for two prime factors

摘录 / EXCERPT: Generalised Sylvester's conjecture. If $n\not\equiv\pm1\pmod9$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004731 | MATHDB-357099

Milnor's linear independence conjecture for Hurwitz zeta values

摘录 / EXCERPT: Let p be a prime, let $k>1$ be an integer, and let the Hurwitz zeta function be ... for $0<x\le1$. Milnor's conjecture. The real numbers ... are linearly independent over...

conjecture | open | MathDB 页面 | 原文线索

V2-004732 | MATHDB-357116

Longer-than-polynomial lifespan conjecture for spherical water waves

摘录 / EXCERPT: Let ... denote the spherical water waves equation. Lifespan conjecture. If the Diophantine equation ... , the lifespan goes beyond $\varepsilon^{-(1+\alpha)}$ as $\varepsilon\to0$. Fin...

conjecture | open | MathDB 页面 | 原文线索

V2-004733 | MATHDB-357122

Harris--Venkatesh conjecture for derived Hecke operators on weight-1 forms

摘录 / EXCERPT: Harris--Venkatesh conjecture. There are an element $u\in\mathcal{U}(\operatorname{Ad}(\rho))$ and a positive integer m such that, for any primes $p,\ell\geq 5$ coprime to N,

conjecture | open | MathDB 页面 | 原文线索

V2-004734 | MATHDB-357126

Sawin-Wood conjecture for class groups of Gamma-extensions containing roots of unity

摘录 / EXCERPT: Let Γ be a finite group and let p be a prime with $p\nmid |\Gamma|$. Let ... Let K_0 be a number field with u infinite places containing the p^rth roots of unity but n...

conjecture | open | MathDB 页面 | 原文线索

V2-004735 | MATHDB-357133

Roth-type approximation conjecture for values of E-functions

摘录 / EXCERPT: Let f be an E-function and let $z_0\in\overline{\mathbb Q}$. For any $\varepsilon>0$, there exists $c>0$ such that for every $(p,q)\in\mathbb Z\times\mathbb N$ with $q\neq 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-004736 | MATHDB-357168

The Kauffman-Harary conjecture for prime-determinant alternating knot diagrams

摘录 / EXCERPT: Let D be a reduced alternating diagram of a knot K with determinant p, where p is prime. A Fox p-coloring is a coloring of the arcs of D by elements of $\mathbb Z_p$ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-004737 | MATHDB-357169

Goldbach-Intervals conjecture

摘录 / EXCERPT: Goldbach-Intervals conjecture. For every positive natural number n and every $m\in\mathcal{E}_{n+1}$, there are at least two prime numbers $p\in\mathcal{P}_n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004738 | MATHDB-357170

Goldbach-Twin conjecture

摘录 / EXCERPT: Goldbach-Twin conjecture. For any positive even natural number n, there are at least two prime numbers $p,q\in\mathcal{P}$ such that $m=p+q$ and p or q is a twin prime. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-004739 | MATHDB-357172

The asymptotic conjecture for the normalized sum involving J_n

摘录 / EXCERPT: Let n range over positive integers such that $p=4n-1$ is prime, and let J_n be the quantity defined earlier in the paper. Asymptotic conjecture for J_n. ... The paper presents...

conjecture | open | MathDB 页面 | 原文线索

V2-004740 | MATHDB-357173

The asymptotic floor-sum estimate for primes $p=4n-1$

摘录 / EXCERPT: Let n range over positive integers such that $p=4n-1$ is prime, and let M, R_m, and Q_m be the quantities defined earlier in the paper. Asymptotic floor-sum conjecture. ... M...

conjecture | open | MathDB 页面 | 原文线索

V2-004741 | MATHDB-357174

Pasten's conjecture on the Tamagawa product of elliptic curves

摘录 / EXCERPT: Let E) be an elliptic curve over $\mathbb{Q}$, with conductor $N=N(E)$ and Tamagawa product $\tau=\tau(E)$. Here $\ll$ denotes an absolute implied constant, and $\epsilon>0$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-004742 | MATHDB-357203

Penn-Calinescu-Wang modularity conjecture for tadpole Nahm sums

摘录 / EXCERPT: Penn-Calinescu-Wang modularity conjecture. The character $q^a\chi_0(1,\dots,1)$ is modular for some rational number a.

conjecture | open | MathDB 页面 | 原文线索

V2-004743 | MATHDB-357204

General modularity conjecture for tadpole Nahm sums

摘录 / EXCERPT: General tadpole modularity conjecture. For even $n=2k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-004744 | MATHDB-357207

De Loera-La Haye-Oliveros-Roldán-Pensado conjecture on the Helly number of the prime lattice

摘录 / EXCERPT: Let $\mathcal{P}$ be the set of prime numbers, and let $h(S)$ denote the Helly number of a set $S\subseteq\mathbb{R}^2$. De Loera-La Haye-Oliveros-Roldán-Pensado conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-004745 | MATHDB-357230

Conjecture on the induced forest number of Paley graphs of square order

摘录 / EXCERPT: Let $q>7$ be a prime power, and let $\mathcal{P}(q^2)$ denote the Paley graph on the finite field of order q^2. Let $\tau(G)$ denote the maximum order of an induced forest in a g...

conjecture | open | MathDB 页面 | 原文线索

V2-004746 | MATHDB-357242

Characterization of nontrivial common divisors of Pell-Lucas terms and indices

摘录 / EXCERPT: Let $(Q_n)_{n\geq 0}$ be the associated Pell sequence. For a prime p, let the rank of apparition (or entry point) $e_Q(p)$ be the smallest positive integer r such that p divi...

conjecture | open | MathDB 页面 | 原文线索

V2-004747 | MATHDB-357248

The K(1)-local Anderson duality conjecture for locally compact K-theory

摘录 / EXCERPT: Let K denote non-connective K-theory. Let p be an odd prime, let F be a number field, and let S be a finite set of finite places of F such that...

conjecture | open | MathDB 页面 | 原文线索

V2-004748 | MATHDB-357260

Roots-of-unity decomposition conjecture for generalized Vandermonde matrices

摘录 / EXCERPT: Let $X=\{x_0=1,x_1,\ldots,x_k\}$ be a set of pairwise distinct roots of unity, and let $B=\{b_0=0,b_1,\ldots,b_m\}$ be a set of coprime integer numbers. Let $M=(x_i^{b_j})$ be dege...

conjecture | open | MathDB 页面 | 原文线索

V2-004749 | MATHDB-357280

Coefficient relations for Berndt-type integrals

摘录 / EXCERPT: For every positive integer p, let i_p, j_p, g_p, and h_p be the numbers appearing in Theorems and. Coefficient relations. The following identities should hold: ... The conj...

conjecture | open | MathDB 页面 | 原文线索

V2-004750 | MATHDB-357300

Eventual stability conjecture for recurrence polynomials in the False Tate curve extension

摘录 / EXCERPT: For each integer $n\geq 1$, let $\left\{{\mathcal{R}}_p^{m,n}(T)\right\}_{m\geq 3}$ be the series of recurrence polynomials from the stated theorem. Eventual stability conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-004751 | MATHDB-357317

Congruence conjecture for prime-colored overpartitions

摘录 / EXCERPT: Let $\overline{p}_{-q}(n)$ denote the number of q-colored overpartitions of n, where q is prime. Congruence conjecture. For all $n\geq 0$ and primes q, one has ... The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-004752 | MATHDB-357331

Pro-l Galois-group isomorphism conjecture for the fields of Theorem 2

摘录 / EXCERPT: Let ℓ be a prime, and let K and L be any fields as in Theorem 2 of the source paper. Write $\mathcal{G}_{K}(\ell)$ for the relevant pro-ℓ Galois group. Pro-ℓ Gal...

conjecture | open | MathDB 页面 | 原文线索

V2-004753 | MATHDB-357332

The squarefree-level degree-count conjecture for weight 2 newforms

摘录 / EXCERPT: Let d be a positive integer, let X tend to infinity, and count weight 2 newforms of degree d and squarefree level $N\leq X$, up to the counting convention used in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-004754 | MATHDB-357344

Parity conjecture for 21-regular partitions at quadratic prime indices

摘录 / EXCERPT: Let $\mathcal{Q}$ be the set of primes specified in the paper, and let $b_{21}(n)$ denote the number of 21-regular partitions of n. Parity conjecture. If $p\in\mathcal{Q}$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-004755 | MATHDB-357345

Vanishing conjecture for the Bloch-Kato term h_{BK}

摘录 / EXCERPT: Bloch-Kato vanishing conjecture. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-004756 | MATHDB-357346

The v_2-local injectivity conjecture for the algebraic tmf resolution

摘录 / EXCERPT: Let tmf be the connective topological modular forms spectrum, let $\overline{\mathrm{tmf}}$ denote its cofiber in the unit decomposition, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004757 | MATHDB-357356

Coefficient-of-asymmetry reformulation of the Lonely Runner Conjecture

摘录 / EXCERPT: Let $\mathbf{n}\in\mathbb{Z}^d_{>0}$ be a velocity vector, meaning that its entries are distinct and relatively prime, and let ... An interior lattice point is a point...

conjecture | open | MathDB 页面 | 原文线索

V2-004758 | MATHDB-357360

Supercongruence conjecture for the sporadic Apéry-like sequences

摘录 / EXCERPT: Supercongruence conjecture. The displayed congruence holds for every one of the $6+6+3$ known sporadic Apéry-like sequences, with λ taking the specified value in each case.

conjecture | open | MathDB 页面 | 原文线索

V2-004759 | MATHDB-357367

Arithmetic mirror symmetry conjecture for reflective modular forms and hyperbolic lattices

摘录 / EXCERPT: Let M be an even lattice of signature $(l,2)$ with $l\geq 3$, and let c be a primitive norm-zero vector of M. Write ... A hyperbolic lattice is an even lattice of signature...

conjecture | open | MathDB 页面 | 原文线索

V2-004760 | MATHDB-357382

Betti-degree stability conjecture for symmetric-group cofixed spaces

摘录 / EXCERPT: Let p be a prime. For nonnegative integers k,m,n satisfying ... let $I^{\mathfrak{S}_m}\subset \mathbb{Z}_{(p)}[e_1,\ldots,e_m]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-004761 | MATHDB-357383

Modular Betti-shift stability conjecture for symmetric-group cofixed spaces

摘录 / EXCERPT: Let p be a prime. For each $r\geq 1$, define ... For an integer $i\geq 0$, let $A_i^R(M)$ be the multiset in which j occurs with multiplicity $\beta_{i,j}^R(M)$, for a graded mod...

conjecture | open | MathDB 页面 | 原文线索

V2-004762 | MATHDB-357397

The divisibility conjecture for morphisms of n-circular bayonet codes

摘录 / EXCERPT: Divisibility conjecture for n-circular bayonet codes. For any n-cbc X and any d prime to n, the set $\varphi_d(X)$ is an n-cbc.

conjecture | open | MathDB 页面 | 原文线索

V2-004763 | MATHDB-357433

The automorphism-group characterization for token graphs of connected Cartesian products

摘录 / EXCERPT: Let G) be a connected graph with $n\geq 5$ vertices and prime factor decomposition ... where $r>1$. Let ψ be the homomorphism from $\operatorname{Aut}(G)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-004764 | MATHDB-357437

The Halo conjecture for the $operatorname{GL}_2$ eigencurve

摘录 / EXCERPT: Let p be an odd prime, let $\mathcal{W}$ be the weight space for GL_2, and let $\mathcal{X}$ be an eigencurve with weight map...

conjecture | open | MathDB 页面 | 原文线索

V2-004765 | MATHDB-357452

Conjectural strengthened Hausdorff-dimension bounds for best-approximation sets

摘录 / EXCERPT: Conjectural refinement. These inequalities should hold for all $n\ge2$ and $w\in[n,\infty]$. This would improve the established lower bounds, whose corresponding numerators are 1...

conjecture | open | MathDB 页面 | 原文线索

V2-004766 | MATHDB-357475

The four-color conjecture for certain degenerate triangles in the max-norm plane

摘录 / EXCERPT: Let a,b be coprime positive integers with $a\not\equiv b\pmod{3}$, and let $T(a,b,a+b)$ be the degenerate triangle with side parameters a, b, and $a+b$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-004767 | MATHDB-357481

The good-pair conjecture for the Generalised Baker-Schmidt Problem

摘录 / EXCERPT: Let $l\geq 2$ and $n\geq 2l$. A pair (l,n) is called good when the conditions (I) and (II) defined earlier in the paper hold for the associated matrix construction. The induced m...

conjecture | open | MathDB 页面 | 原文线索

V2-004768 | MATHDB-357504

The Brumer-Stark conjecture for partial zeta values

摘录 / EXCERPT: Let F be totally real, let H/F be a finite abelian CM extension with Galois group G, and let $S=R\cup\{\mathfrak{p}\}$, where $\mathfrak{p}$ splits completely in H/F. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004769 | MATHDB-357505

Dasgupta-Spieß conjecture on the Brumer-Stark unit formula

摘录 / EXCERPT: Let F be totally real, let H/F be a finite abelian extension in which $\mathfrak{p}$ splits completely, and let u_2 denote the Brumer-Stark unit constructed from the second f...

conjecture | open | MathDB 页面 | 原文线索

V2-004770 | MATHDB-357520

Rodgers-Soundararajan variance conjecture for the divisor function in arithmetic progressions

摘录 / EXCERPT: Let k be the divisor-function parameter, let w be a smooth function supported in $[1,2]$ satisfying ... and whose Mellin transform ... satisfies rapid decay on every fixed vertical...

conjecture | open | MathDB 页面 | 原文线索

V2-004771 | MATHDB-357526

The Sato-Tate equidistribution conjecture

摘录 / EXCERPT: Under the algebraic Sato-Tate setup, let $\operatorname{ST}_K(V,\psi)$ be the associated compact Sato-Tate group. For a prime v of K, let s_v be the semisimple part of the no...

conjecture | open | MathDB 页面 | 原文线索

V2-004772 | MATHDB-357575

Existence of rigid symmetric monoidal categorifications of algebraic closures of finite fields

摘录 / EXCERPT: Let p be a prime. A rigid symmetric monoidal $b\mathbb{Z}d$-linear category is a rigid symmetric monoidal category enriched over $b\mathbb{Z}d$-modules; write $K_0(C)$ for its Gr...

conjecture | open | MathDB 页面 | 原文线索

V2-004773 | MATHDB-357590

Grazian-Monetta nilpotency conjecture for non-commuting graphs

摘录 / EXCERPT: Let p be a prime, and let $G=P\times A$ be a finite group where $P\in\operatorname{Syl}_p(G)$ is non-abelian and A is an abelian p'-group. Let H be a finite group whose non...

conjecture | open | MathDB 页面 | 原文线索

V2-004774 | MATHDB-357602

The logarithmic Burgess-like character-sum conjecture

摘录 / EXCERPT: Burgess-like character-sum conjecture. For a fixed $\epsilon>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-004775 | MATHDB-357603

The general Burgess-like character-sum conjecture

摘录 / EXCERPT: General Burgess-like character-sum conjecture. For a certain function $c:\mathbb{N}\rightarrow\mathbb{N}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-004776 | MATHDB-357615

The Gross--Kuz'min conjecture

摘录 / EXCERPT: Gross--Kuz'min conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004777 | MATHDB-357616

D--Kakde's Matrix Coefficient Conjecture

摘录 / EXCERPT: D--Kakde's Matrix Coefficient Conjecture. If $\det(M)=0$, then there exist nonzero vectors $w,v\in\mathbf Q^n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004778 | MATHDB-357631

The integrality conjecture for iterated harmonic numbers

摘录 / EXCERPT: For each integer $j\geq 1$, let $h_j(n)$ denote the jth iterated harmonic number, as defined in the paper. In particular, $h_j(1)=1$. Integrality conjecture. For each integer...

conjecture | open | MathDB 页面 | 原文线索

V2-004779 | MATHDB-357635

Positive-characteristic upper-bound conjecture for the Minkowski spectrum

摘录 / EXCERPT: Fix an integer $d\geq 2$ and a prime power q. Let $\left[\mathcal{L}_d\right]$ be the space of homothety classes of lattices, let $\tilde{\mathcal{K}}^d$ be the ambient vector sp...

conjecture | open | MathDB 页面 | 原文线索

V2-004780 | MATHDB-357637

Bezrukavnikov's invariant-comparison conjecture for combinatorial wall-crossing

摘录 / EXCERPT: Let n be a positive integer, let $\frac{r}{s}$ be a term of the n-th Farey sequence, and let $\tilde{M}_{r/s}$ and $\tilde{M}^{\prime}_{r/s}$ be the compositions of generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-004781 | MATHDB-357640

General irreducibility conjecture for integer polynomials with prime-power constant term

摘录 / EXCERPT: Irreducibility conjecture. If $p\nmid a_{n}a_{q}$, $q\nmid u$, and

conjecture | open | MathDB 页面 | 原文线索

V2-004782 | MATHDB-357656

The polynomial bound conjecture for real cyclotomic class numbers

摘录 / EXCERPT: Let m be a positive integer, let $\mathbb{Q}(\zeta_m+\zeta_m^{-1})$ be the maximal real subfield of the mth cyclotomic field, and let $h^+(m)$ denote its class number. Polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-004783 | MATHDB-357681

Strong inverse conjecture for the Gowers norm

摘录 / EXCERPT: Let p be prime, $k\geq1$, $G=\mathbb{F}_p^n$, and $\mathbb D=\{z\in\mathbb C:|z|\leq1\}$. Let $\varepsilon:\mathbb N\to\mathbb R^+$ be decreasing, and let $\mathrm{Poly}^k(G)$ be...

conjecture | solved | MathDB 页面 | 原文线索

V2-004784 | MATHDB-357731

The square-free values asymptotic conjecture for separable polynomials

摘录 / EXCERPT: Square-free values asymptotic conjecture. There is a positive constant

conjecture | open | MathDB 页面 | 原文线索

V2-004785 | MATHDB-357732

The cyclotomic prime-divisor sparsity conjecture

摘录 / EXCERPT: Cyclotomic prime-divisor sparsity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004786 | MATHDB-357755

The level one conjecture for the divisor function

摘录 / EXCERPT: Let $q\geq 1$ be a composite number, and let $W(x)$ be a smooth test weight supported on $[1,2]$. Its Mellin transform is ... Assume that, for every fixed $A>0$ and every positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-004787 | MATHDB-357771

Asymptotic parity conjecture for Fourier coefficients of the Hauptmodul j^*_{10}

摘录 / EXCERPT: Let A and B be positive integers such that ... Write $\mathcal{J}^*_{10}(m)$ for the m-th Fourier coefficient of $j^*_{10}(\tau)$. Asymptotic parity conjecture. There are such...

conjecture | open | MathDB 页面 | 原文线索

V2-004788 | MATHDB-357776

Finiteness conjecture for rational triangles fixed up to similarity by a weakly metric affine transformation

摘录 / EXCERPT: Finiteness conjecture. If $f\in V$ and $f\neq f_I$, there are only a finite number of dissimilar rational triangles $T\in X(T)$ such that $f(T)\sim T$.

conjecture | open | MathDB 页面 | 原文线索

V2-004789 | MATHDB-357815

Heap's conjecture on space-efficient knot-mosaic layouts

摘录 / EXCERPT: Heap's conjecture. Such space-efficient layouts produce the same set of prime knots.

conjecture | open | MathDB 页面 | 原文线索

V2-004790 | MATHDB-357898

Polylogarithmic independence number conjecture for Paley graphs

摘录 / EXCERPT: For a prime p, let G_p be the Paley graph, let $\overline G_p$ be its complement, and write $\alpha(G_p)=\omega(\overline G_p)$ for its independence number. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-004791 | MATHDB-357899

Minimum-eigenvalue conjecture for localized Paley graphs

摘录 / EXCERPT: Let G_p be the Paley graph of prime order p, let $a\geq1$, and let $\mathcal I_a(G_p)$ denote the independent sets of size a. For $I\in\mathcal I_a(G_p)$, let $A_{G_{p,I}}$ b...

conjecture | refuted | MathDB 页面 | 原文线索

V2-004792 | MATHDB-357900

Necklace character-sum conjecture for localized Paley graphs

摘录 / EXCERPT: Necklace character-sum conjecture. For every $k\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-004793 | MATHDB-357901

Improvement on the Hanson-Petridis bound via degree-2 localization

摘录 / EXCERPT: Let p be prime, let G_p be the Paley graph, let g be a multiplicative generator of $\mathbb F_p^\times$, and let ϑ denote the Lovász theta function. Improvement con...

conjecture | open | MathDB 页面 | 原文线索

V2-004794 | MATHDB-357902

Quartic-residue subgraph minimum-eigenvalue conjecture

摘录 / EXCERPT: Let p be a prime for which the quartic residues are defined, let F_p be the induced subgraph of the Paley graph G_p on the nonzero fourth powers modulo p, and let A_{F_p}...

conjecture | open | MathDB 页面 | 原文线索

V2-004795 | MATHDB-357919

Twin-prime occurrence in every interval between consecutive prime squares

摘录 / EXCERPT: Let p_n denote the nth prime. Twin-square-interval conjecture. For every integer $n\geq 3$, there exists at least one pair of twin primes lying in the interval ... This is a stro...

conjecture | open | MathDB 页面 | 原文线索

V2-004796 | MATHDB-357921

The generalised Goldbach conjecture for weighted prime sums

摘录 / EXCERPT: Generalised Goldbach conjecture. Under these conditions, there exist primes p and q such that

conjecture | open | MathDB 页面 | 原文线索

V2-004797 | MATHDB-357927

Aliev's generalized-hexagon conjecture for the critical determinant

摘录 / EXCERPT: For $n\geq3$, let $c(n)$ be the optimal constant defined by ... and let $\Delta(K)$ denote the critical determinant. Let $H_{1,\ldots,1}$ be the generalized hexagon in...

conjecture | open | MathDB 页面 | 原文线索

V2-004798 | MATHDB-357936

Conjecture on improving the Birkhoff-sum lower bound beyond $LB3$

摘录 / EXCERPT: Let $S_N\frac{1}{\left\Vert x\right\Vert }$ be the estimate in equation $\mathrm{EstLBSymm}$, and let $S_{b_nq_n}\frac{1}{\left\Vert x\right\Vert }$ be the current estimate, with l...

conjecture | open | MathDB 页面 | 原文线索

V2-004799 | MATHDB-357937

Conjecture on uniform linear growth of twisted Birkhoff sums

摘录 / EXCERPT: Let α be an irrational rotation parameter, let $\psi_M(x)=\psi(Mx)$, and define the twisted Birkhoff sum $S_N(\psi_M\phi)$ in the setting of lemma 84. Uniform twisted-sum co...

conjecture | open | MathDB 页面 | 原文线索

V2-004800 | MATHDB-357988

The asymptotic mixing-time conjecture for the S_k shuffle

摘录 / EXCERPT: Let N be the number of positions and let $k=o(N^{1/2})$. Denote by $t_{\mathrm{mix}}(\varepsilon)$ and $t_{\mathrm{mix}}^{\prime}(\varepsilon)$ the 4ε-mixing times o...

conjecture | open | MathDB 页面 | 原文线索

V2-004801 | MATHDB-358009

General quantum modularity conjecture for $b[?]$-invariants

摘录 / EXCERPT: General quantum modularity conjecture. The invariant $\widehat Z^G_{\vec{\underline{b}}}(M_3)$ is closely related to a quantum modular form of some kind.

conjecture | open | MathDB 页面 | 原文线索

V2-004802 | MATHDB-358010

Quantum modular depth conjecture for weakly definite plumbed three-manifolds

摘录 / EXCERPT: Quantum modular depth conjecture. The invariant $\widehat Z^G_{\vec{\underline{b}}}(M_3;\tau)$ is related to quantum modular forms of depth up to $r\times n$.

conjecture | open | MathDB 页面 | 原文线索

V2-004803 | MATHDB-358011

The continued-fraction hyperbola attack conjecture for RSA points

摘录 / EXCERPT: Continued-fraction hyperbola attack conjecture. For all $i\in\{1,2,3,4\}$, the associated coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-004804 | MATHDB-358027

The nonvanishing and sign conjecture for Hardy(k,ℓ,m,n)

摘录 / EXCERPT: Nonvanishing and sign conjecture. If $k+\ell+m+n$ is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-004805 | MATHDB-358032

The completion conjecture for the c_2-invariant of primitive ϕ^4 graphs

摘录 / EXCERPT: Let G_1 and G_2 be primitive ϕ^4 graphs, and let $c_2^{(q)}(G)$ denote the c_2-invariant modulo a prime power q. Let $\mathcal{P}(G)$ denote the Feynman period. Comple...

conjecture | open | MathDB 页面 | 原文线索

V2-004806 | MATHDB-358035

Hudson's conjecture on completely multiplicative functions of length three

摘录 / EXCERPT: Let $F_{2,3}$ denote the set of completely multiplicative functions $f:\mathbb{N}\to\{-1,1\}$ whose maximal run of consecutive $+1$ values has length 3. For primes...

conjecture | open | MathDB 页面 | 原文线索

V2-004807 | MATHDB-358037

Modified Elliott conjecture for non-pretentious multiplicative functions

摘录 / EXCERPT: Let $\mathbb{D}$ be the closed unit disc, let $\mathbb{D}(g_1,g_2;x)$ denote the pretentious distance, and let $\delta^+_{\log}$ denote upper logarithmic density. Let $k\ge1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-004808 | MATHDB-358039

Tao's asymptotic Elliott conjecture

摘录 / EXCERPT: Let $\mathbb{D}$ be the closed unit disc, and define the pretentious distance by ... Let $k\ge1$ and let $h_1,\ldots,h_k\in\mathbb{N}$ be fixed and distinct. For $x\ge A\ge2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004809 | MATHDB-358048

Manin's conjecture on rational hypersurfaces over C_1 fields

摘录 / EXCERPT: Let k be a C_1 field, and let V be a rational hypersurface over k. Manin's conjecture. The hypersurface V has a k-rational point. This is a weaker variant of Lang's con...

conjecture | open | MathDB 页面 | 原文线索

V2-004810 | MATHDB-358049

Conjecture on cancellation for prime algebras of low center or dimension three

摘录 / EXCERPT: Cancellation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004811 | MATHDB-358051

Undecidability conjecture for Diophantine problems over number fields and rings of algebraic integers

摘录 / EXCERPT: Let F be a number field, meaning a finite algebraic extension of $\mathbb{Q}$, and let $\mathcal{O}$ be a ring of algebraic integers. The Diophantine problem asks whether polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-004812 | MATHDB-358071

The hyperbola-based characterization of the factorization parameter

摘录 / EXCERPT: Hyperbola-based factoring conjecture. The set Γ has exactly three elements and satisfies

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-004813 | MATHDB-358094

Sparsity conjecture for sets satisfying the prime number theorem

摘录 / EXCERPT: For real x and $t>1$, let $S_{\left\lfloor x/n^t\right\rfloor,x}$ denote the set considered in the paper, with the function defining $S_{f,x}$ taken to be non-piecewise. Sparsity...

conjecture | open | MathDB 页面 | 原文线索

V2-004814 | MATHDB-358099

Sparse automatic set and zero-density automatic set intersection conjecture

摘录 / EXCERPT: Let k and ℓ be multiplicatively independent positive integers. A subset of $\mathbb{N}$ is sparse if its counting function is bounded by $O((\log x)^d)$ for some d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004815 | MATHDB-358100

Corrected Beiter conjecture for ternary cyclotomic heights

摘录 / EXCERPT: Corrected Beiter conjecture. Moree and Gallot proposed that

conjecture | open | MathDB 页面 | 原文线索

V2-004816 | MATHDB-358101

Finiteness near the corrected Beiter bound

摘录 / EXCERPT: Finiteness conjecture. For any real number r, there exist only finitely many primes p such that

conjecture | open | MathDB 页面 | 原文线索

V2-004817 | MATHDB-358103

Well-definedness conjecture for the minimal quotient function

摘录 / EXCERPT: Let ... For $a\in\mathcal{R}$, define ... A positive integer is fourth power-free if it is not divisible by the fourth power of any prime. Well-definedness conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-004818 | MATHDB-358128

Uniqueness of the optimal arithmetic distribution for the Schur-Siegel-Smyth problem

摘录 / EXCERPT: Uniqueness conjecture. There is a unique arithmetic probability distribution μ_{SSS} compactly supported in $\mathbb R^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004819 | MATHDB-358129

Zero-energy conjecture for optimal arithmetic distributions

摘录 / EXCERPT: Zero-energy conjecture. If μ is an arithmetic probability distribution compactly supported in $\mathbb R^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-004820 | MATHDB-358130

Support containment for measures associated with nested polynomial sets

摘录 / EXCERPT: Support-containment conjecture. If $A\subset B$, then

conjecture | open | MathDB 页面 | 原文线索

V2-004821 | MATHDB-358144

Conjecture on the roots of the Harada invariant for general linear groups

摘录 / EXCERPT: Root bound conjecture. For $n\geq 6$, the maximum real root of $v_q(h(\operatorname{GL}_n(k)))$ is less than 2.

conjecture | open | MathDB 页面 | 原文线索

V2-004822 | MATHDB-358161

Vanishing bounds for Romik's sequence $u(n)$ modulo odd prime powers

摘录 / EXCERPT: Let p be a prime with $p\equiv3\pmod 4$, let $e\ge1$, and let $(u(n))_{n\ge0}$ be the sequence defined in the paper. The first conjecture. (1) If $e\ge1$, then ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-004823 | MATHDB-358162

Vanishing bounds for Romik's sequence $v(n)$ modulo odd prime powers

摘录 / EXCERPT: Let p be a prime with $p\equiv3\pmod4$, let $e\ge1$, and let $(v(n))_{n\ge0}$ be the sequence defined in the paper. The second conjecture. If e is odd, then ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-004824 | MATHDB-358163

Vanishing bound for Romik's sequence $d(n)$ modulo primes $p\equiv3\pmod4$

摘录 / EXCERPT: Let p be a prime with $p\equiv3\pmod4$, let $e\ge1$, and let $(d(n))_{n\ge0}$ be the sequence defined in the paper. The third conjecture. ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-004825 | MATHDB-358164

Periodicity refinements for Romik's sequence $d(n)$ modulo powers of primes $p\equiv1\pmod4$

摘录 / EXCERPT: Let p be a prime with $p\equiv1\pmod4$, let $e\ge1$, and let $(d(n))_{n\ge1}$ be the sequence defined in the paper. The fourth conjecture. (1) The sequence $d(n)$ modulo p^e is...

conjecture | open | MathDB 页面 | 原文线索

V2-004826 | MATHDB-358165

Congruence behavior of the matrix $R(n,k)$

摘录 / EXCERPT: Let $R(n,k)$ be the matrix introduced in the paper, and let $R^{-1}(n,k)$ denote its inverse matrix. The fifth conjecture. (1) For fixed k and any prime power p^e with...

conjecture | open | MathDB 页面 | 原文线索

V2-004827 | MATHDB-358213

Anabelian characterization conjecture for the geometric fundamental group

摘录 / EXCERPT: Anabelian characterization conjecture. The subgroup Δ_X is the unique closed normal subgroup N of Π_X satisfying: (i) $\Pi_X/N\simeq\mathbb{Z}_p$; and (ii) for every o...

conjecture | open | MathDB 页面 | 原文线索

V2-004828 | MATHDB-358247

The McKay-Thompson moonshine conjecture

摘录 / EXCERPT: Let g be an element of the Monster group, and let $j_g(q)$ be its McKay-Thompson series. For a prime level N, let $\Gamma_0(N)$ be the congruence subgroup of the modular group...

conjecture | open | MathDB 页面 | 原文线索

V2-004829 | MATHDB-358255

The Main Conjecture on rational points near curved manifolds

摘录 / EXCERPT: Main Conjecture. The stated asymptotic holds under the proper curvature conditions above. This is presented as a basic open-ended conjecture motivated by the expected random distri...

conjecture | open | MathDB 页面 | 原文线索

V2-004830 | MATHDB-358256

The nondegenerate-manifold conjecture for rational-point asymptotics

摘录 / EXCERPT: Let $\mathscr{M}$ be an m-dimensional manifold in $\mathbb{R}^n$, with co-dimension $c=n-m$, and let $\mathrm{N}_{\mathscr{M}}(\delta,Q)$ be the counting function appearing in th...

conjecture | open | MathDB 页面 | 原文线索

V2-004831 | MATHDB-358261

Papadima-Suciu conjecture on prime-power roots of line-arrangement Alexander polynomials

摘录 / EXCERPT: Let C be a line arrangement that is not a union of concurrent lines, and let ${\alpha}$ be a root of its Alexander polynomial $\Delta^1_C(t)$. Suppose that the order of...

conjecture | open | MathDB 页面 | 原文线索

V2-004832 | MATHDB-358276

Conjecture on the spectral distribution of random Schreier graphs

摘录 / EXCERPT: Let q be a prime number, let S be a random subset of $\textit{GL}_k(\mathbb{Z}_q)$, and let G_k be the Schreier graph of $S \subset \textit{GL}_k(\mathbb{F}_q)$ acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-004833 | MATHDB-358277

Duality conjecture for modular Nahm sums

摘录 / EXCERPT: Duality conjecture. If the Nahm sum associated with (A,b,c,d) is modular, then the Nahm sum associated with (A^*,b^*,c^*,d^*) is also modular. The conjecture proposes a general...

conjecture | open | MathDB 页面 | 原文线索

V2-004834 | MATHDB-358278

Modular S-transformation conjecture for the rank-two Nahm sums

摘录 / EXCERPT: Modular transformation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-004835 | MATHDB-358279

Colliot-Thélène-Sansuc-Swinnerton-Dyer conjecture for intersections of two quadrics

摘录 / EXCERPT: Let k be a number field and let $X\subset \mathbb{P}^n_k$, with $n\geq 6$, be a non-conical geometrically integral intersection of two quadrics. Suppose that X has a smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-004836 | MATHDB-358302

Strengthened vanishing correlation conjecture for lattice points near circles

摘录 / EXCERPT: Strengthened vanishing correlation conjecture. These vectors converge in distribution, in the double limit $R\to\infty$ followed by $k\to\infty$, to

conjecture | open | MathDB 页面 | 原文线索

V2-004837 | MATHDB-358312

High-moment asymptotic for L-functions over small subgroups

摘录 / EXCERPT: High-moment asymptotic conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-004838 | MATHDB-358342

No Ties conjecture for residue-class biases in indivisible partitions

摘录 / EXCERPT: Let $k,t\geq 2$ be coprime, and let $\prec_{k,t}$ be the ordering on $\mathbb{Z}/t\mathbb{Z}$ defined by the eventual comparison of the quantities $D_k^\times(r,t;n)$: namely,...

conjecture | open | MathDB 页面 | 原文线索

V2-004839 | MATHDB-358343

Conjecture on extremal and minimal residue-class bias gaps

摘录 / EXCERPT: For coprime k,t, define ... ... and ... Here $\psi_{k,t}(r)$ is the bias parameter defined by the asymptotic formula for the residue-class counts. Extremal bias-gap conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-004840 | MATHDB-358349

The twisted E_8-equivariant lift conjecture for e_8

摘录 / EXCERPT: The twisted E_8-equivariant lift conjecture. The element e_8 lifts to

conjecture | open | MathDB 页面 | 原文线索

V2-004841 | MATHDB-358350

The VOSA-to-TMF class conjecture

摘录 / EXCERPT: The VOSA-to-TMF class conjecture. The VOSA should determine a class

conjecture | open | MathDB 页面 | 原文线索

V2-004842 | MATHDB-358351

The differential string trivialization conjecture for Witten Pfaffian lines

摘录 / EXCERPT: Let $(M_{8\ell+2},g_M^{\mathrm{spin}})$ be a closed differential spin manifold, and let g_M^{string} be a lift of its spin structure to a differential string structure....

conjecture | open | MathDB 页面 | 原文线索

V2-004843 | MATHDB-358352

The affine VOSA class conjecture

摘录 / EXCERPT: Let $\mathbb{V}$ be a unitary self-dual VOSA of central charge $c=n/2$ containing the affine Lie VOA $\mathfrak{g}_k$, and let G be the corresponding compact simply-connected gro...

conjecture | open | MathDB 页面 | 原文线索

V2-004844 | MATHDB-358353

The finite-group equivariant VOSA conjecture

摘录 / EXCERPT: Let $\mathbb{V}$ be a unitary self-dual VOSA of central charge $c=n/2$, equivariant under a finite group G with anomaly...

conjecture | open | MathDB 页面 | 原文线索

V2-004845 | MATHDB-358354

The kernel-generator conjecture for self-dual VOSAs

摘录 / EXCERPT: Let A_{-2c} be the kernel of ... For $c=32,31/2,15,14$, the kernel is respectively isomorphic to $\mathbb{Z}/3,\mathbb{Z}/2,\mathbb{Z}/2,\mathbb{Z}/2$. The kernel-generator conject...

conjecture | open | MathDB 页面 | 原文线索

V2-004846 | MATHDB-358383

Conjecture on coefficients of multivalued-group polynomials

摘录 / EXCERPT: Coefficient divisibility and nonvanishing conjecture. 1. If $n=p^m$ is a power of a prime p, then every coefficient of p_n except the coefficient of e_1^n is divisible by p...

conjecture | open | MathDB 页面 | 原文线索

V2-004847 | MATHDB-358386

Prime cyclotomic factor conjecture for unimodal generating functions

摘录 / EXCERPT: Let $\Phi^{\operatorname{uni}}$ denote the monoid of cyclotomic generating functions that are unimodal, and let $\Phi_p(q)$ denote the pth cyclotomic polynomial for a prime p....

conjecture | open | MathDB 页面 | 原文线索

V2-004848 | MATHDB-358419

Harris--Venkatesh plus Stark conjecture for weight-1 newforms

摘录 / EXCERPT: Let f be a newform of weight 1 and level N, with associated representation ρ_f. Let $\mathcal{U}(\operatorname{Ad}(\rho_f))$ be the equivariant unit module, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004849 | MATHDB-358429

The converse characterization of bases for asymptotically automatic sequences

摘录 / EXCERPT: Let V be a vector subspace of $\bigoplus_{i=1}^\infty \mathbb{Q}$. For a prime p, let ν_p denote the p-adic valuation, and let $p_1,p_2,\dots$ be the increasing enumerati...

conjecture | open | MathDB 页面 | 原文线索

V2-004850 | MATHDB-358480

The topological Tverberg sunflower conjecture

摘录 / EXCERPT: Let $d\geq 1$, let $r\geq 2$ be a prime power, and set $n=(r-1)(d+1)$. Let $N>n$, and let Δ_N be the N-dimensional simplex. Topological Tverberg sunflower conjecture. For...

open_problem | open | MathDB 页面 | 原文线索

V2-004851 | MATHDB-358530

Conjecture on primes with no minimal monomial solutions of size 2 modulo 4

摘录 / EXCERPT: For a prime p, consider the nonzero minimal monomial solutions associated with p, and let their size be the corresponding integer size. The prime classification conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-004852 | MATHDB-358532

Chapman's conjecture on a Legendre-symbol determinant

摘录 / EXCERPT: Let p) be an odd prime, let $varepsilon_p$ be the fundamental unit of $\mathbb{Q}(\sqrt{p})$, and let $h(p)$ be the class number of $\mathbb{Q}(\sqrt{p})$. Write ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-004853 | MATHDB-358557

Conjecture on affine Hermitian intersections with curves of Frobenius type

摘录 / EXCERPT: Let p be a prime, let $h\geq 2$, and set $q=p^{2h}$. In the affine plane $\mathrm{AG}(2,q^2)$, consider the affine Hermitian curve $\mathcal H(q^2)$ and the curves ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-004854 | MATHDB-358558

Relative stability and moduli conjecture for degenerating Fano categories

摘录 / EXCERPT: Let Y be a del Pezzo threefold, let $\cX/B$ be a family of prime Fano threefolds, and let $\bcA_{\cX}/B$ be the B-linear category constructed in Theorem 1.1. Let $o\in B$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-004855 | MATHDB-358561

GCD power-of-two conjecture for the coefficients of $c_m(c_1)$

摘录 / EXCERPT: GCD power-of-two conjecture. For every $m\geqslant2$,

conjecture | open | MathDB 页面 | 原文线索

V2-004856 | MATHDB-358585

Power-saving error term for q-orbits of closed geodesics

摘录 / EXCERPT: Power-saving error-term conjecture. The conclusion of the first theorem holds with a power-saving error term in terms of q.

open_problem | open | MathDB 页面 | 原文线索

V2-004857 | MATHDB-358588

Generalized Atkin polynomial congruence conjecture

摘录 / EXCERPT: Let $p\ge5$ be prime, and write ... where $m\in\mathbb{Z}_{\ge0}$ and $\delta,\varepsilon\in\{0,1\}$. Let $A_{u,a}^{(r)}(X)$ be the generalized Atkin polynomial associated with the n...

conjecture | open | MathDB 页面 | 原文线索

V2-004858 | MATHDB-358602

Functional equation conjecture for overlap type zeta functions of free class-2-nilpotent groups

摘录 / EXCERPT: Let $d\geq 2$, let d' denote the parameter used to define the overlap types, and let $w\in{\mathcal{D}}_{2d'}$. Let $\mathfrak{f}_{2,d}(\mathfrak{o})$ be the relevant free class-...

conjecture | open | MathDB 页面 | 原文线索

V2-004859 | MATHDB-358615

Condensation and densification conjecture for Vinogradov mean values

摘录 / EXCERPT: Condensation and densification conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004860 | MATHDB-358637

Uniqueness conjecture for the smallest element of a $D(4)$-quadruple

摘录 / EXCERPT: Smallest-element uniqueness conjecture. Then $\{a_2,b,c,d\}$ is not a $D(4)$-quadruple for any integer a_2 with $a_1\neq a_2<b$.

conjecture | open | MathDB 页面 | 原文线索

V2-004861 | MATHDB-358653

Commuting noninvertible formal power series conjecture

摘录 / EXCERPT: Let p be a prime, let $d\geq 1$, and write $X=(x_1,\ldots,x_d)$. For a power series $h\in\mathbb{Z}_p[[X]]^d$, let $\Lambda(h)$ denote its zero set. Commuting noninvertible forma...

conjecture | open | MathDB 页面 | 原文线索

V2-004862 | MATHDB-358654

Preperiodic points and zero sets of commuting formal power series

摘录 / EXCERPT: Let p be a prime, let $d\geq 1$, and write $X=(x_1,\ldots,x_d)$. Let $\operatorname{Preper}(u)$ denote the preperiodic points of u, and let $\Lambda(f)$ denote the zero set of...

conjecture | open | MathDB 页面 | 原文线索

V2-004863 | MATHDB-358659

Lang's conjecture on lower bounds for linear forms in logarithms

摘录 / EXCERPT: Let $\epsilon>0$. For rational positive numbers $a_1,\dots,a_n$ and integers $b_1,\dots,b_n$, define ... Assume that $b_1\log a_1+\cdots+b_n\log a_n\ne0$. Lang's conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-004864 | MATHDB-358661

An odd-parameter supercongruence for a terminating hypergeometric sum

摘录 / EXCERPT: Let $r\leq 1$ be an odd integer coprime with 5, and let p be an odd prime such that $p\equiv r\pmod{5}$ and $p\geq (5-3r)/2$. Then the proposed supercongruence. ... This is prese...

conjecture | open | MathDB 页面 | 原文线索

V2-004865 | MATHDB-358675

Non-simple telltale intersection-multiplicity conjecture

摘录 / EXCERPT: Let $R\to\mathbb P^1$ be a rational elliptic surface, let L determine a Severi curve with normalization $\widetilde{V}(L)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004866 | MATHDB-358690

Strengthened Lemoine's Conjecture

摘录 / EXCERPT: Let n be an odd integer with $n\geq 7$. Strengthened Lemoine's Conjecture. There exist primes p and q such that ... and ... This strengthening is introduced because it would supp...

conjecture | open | MathDB 页面 | 原文线索

V2-004867 | MATHDB-358697

Hyperbolic realization conjecture for knot symmetry types

摘录 / EXCERPT: Let G be a finite cyclic or dihedral group. A G-symmetry type of a prime knot is a symmetry type arising from an action of G on the knot. Hyperbolic realization conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-004868 | MATHDB-358699

Sutherland-Voloch anabelian conjecture for Hilbert class field towers of curves

摘录 / EXCERPT: Let C and D be smooth projective curves of equal genus at least 2 over a finite field ${\mathbb F}$. Embed C into its Jacobian and define its Hilbert class field cover...

conjecture | open | MathDB 页面 | 原文线索

V2-004869 | MATHDB-358712

Schäffer's conjecture on perfect powers of power sums

摘录 / EXCERPT: Schäffer's conjecture. If $(k,n)\notin S$, then the equation

conjecture | open | MathDB 页面 | 原文线索

V2-004870 | MATHDB-358773

The linear bound conjecture for reduced zero-sum-free sets in dimension two

摘录 / EXCERPT: Let p be prime, and let $h_p(2)$ denote the maximum size of a reduced weighted directed graph over a subgroup isomorphic to $\mathbb{Z}_p^2$ with no zero-weight directed cycle. L...

conjecture | open | MathDB 页面 | 原文线索

V2-004871 | MATHDB-358792

The visibility conjecture for prime ideals in tensor-category algebras

摘录 / EXCERPT: Let k be an algebraically closed field, let $\mathcal{C}$ be a tensor category of moderate growth, and let $A\in\mathsf{Alg}\mathcal{C}$ be an algebra object. A prime ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-004872 | MATHDB-358803

Cohomological p-newness conjecture for Eisenstein congruence classes

摘录 / EXCERPT: Cohomological p-newness conjecture. The class c' is p-new for every $p\in\mathcal{S}$, meaning that it does not come from the corresponding relaxed Selmer group with the condit...

conjecture | open | MathDB 页面 | 原文线索

V2-004873 | MATHDB-358804

Weight k-2 cohomological p-newness conjecture

摘录 / EXCERPT: Weight $k-2$ p-newness conjecture. The class $c^{”}$ is p-new for every $p\in\mathcal{S}$. This is motivated by the expectation that the general newform Eisenstein congruence con...

conjecture | open | MathDB 页面 | 原文线索

V2-004874 | MATHDB-358819

The Loneliness Spectrum Conjecture

摘录 / EXCERPT: Let $v_1,\ldots,v_n$ be positive integers, and let ... where $\lVert x\rVert$ denotes the absolute distance from x to the nearest integer. The Loneliness Spectrum Conjecture. One h...

conjecture | open | MathDB 页面 | 原文线索

V2-004875 | MATHDB-358820

The amended Loneliness Spectrum Conjecture

摘录 / EXCERPT: Let $v_1,\ldots,v_n$ be positive integers, and let ... where $\lVert x\rVert$ denotes the absolute distance from x to the nearest integer. The amended Loneliness Spectrum Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-004876 | MATHDB-358821

The restricted amended Loneliness Spectrum Conjecture for four speeds

摘录 / EXCERPT: Let v_1,v_2,v_3,v_4 be positive integers, and let ... where $\lVert x\rVert$ denotes the absolute distance from x to the nearest integer. The restricted amended Loneliness Spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-004877 | MATHDB-358843

The Eisenstein-family overconvergence-rate conjecture

摘录 / EXCERPT: Overconvergence-rate conjecture. For all $i,j\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004878 | MATHDB-358849

The quartic Gauss-sum $X^{3/4}$ bias conjecture over Gaussian primes

摘录 / EXCERPT: Let $e(x):=e^{2\pi i x}$ for $x\in\mathbb{R}$ and $\check{e}(z):=e(z+\overline{z})$ for $z\in\mathbb{C}$. Let $\mathbb{Q}(i)$ be the Gaussian quadratic field with ring of integers...

conjecture | open | MathDB 页面 | 原文线索

V2-004879 | MATHDB-358851

Weaker distance-tree embedding conjecture for finite vector spaces

摘录 / EXCERPT: Let q be a prime power, let d be a positive integer, and let $\Delta\geq 2$. A distance tree is a tree whose edges are assigned nonzero distances in $\mathbb{F}_q^d$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-004880 | MATHDB-358853

The finite-unit subring conjecture for algebraic numbers

摘录 / EXCERPT: Let $R\le \overline{\mathbb{Q}}$ be a subring with finitely many units. An order conjecture. There exists an integer $d\in\mathbb{Z}_{<0}$ such that R is an order in...

conjecture | open | MathDB 页面 | 原文线索

V2-004881 | MATHDB-358905

Conjecture on nice Browkin continued fractions of every length

摘录 / EXCERPT: Let p be a prime. A nice Browkin continued fraction is the special class of Browkin continued fractions introduced by Capuano, Murru, and Terracini. The nice-continued-fraction c...

conjecture | open | MathDB 页面 | 原文线索

V2-004882 | MATHDB-358906

The even-period conjecture for Browkin II continued fractions

摘录 / EXCERPT: Let p be a prime, and let $D\in\mathbb{Z}$ be not a perfect square. A Browkin II continued fraction is a continued fraction generated by Browkin's second p-adic continued-fract...

conjecture | open | MathDB 页面 | 原文线索

V2-004883 | MATHDB-358907

The finite-or-periodic conjecture for the p-adic Jacobi-Perron algorithm

摘录 / EXCERPT: Let $m\geq 1$, and let $1,\alpha_0^{(1)},\ldots,\alpha_0^{(m)}$ be p-adic numbers that are linearly dependent over $\mathbb{Q}$. The finite-or-periodic conjecture. The p-adic J...

conjecture | open | MathDB 页面 | 原文线索

V2-004884 | MATHDB-358916

Crowns are extremal for matroids omitting a complete-graphic minor

摘录 / EXCERPT: Let $ll in N$ with $ll ge 2$, and let q be the largest prime power with $q le ll$. For sufficiently large $t in N$, let $\falM$ be the class of matroids with no $U_{2,ll+2}$- or...

conjecture | open | MathDB 页面 | 原文线索

V2-004885 | MATHDB-358923

Infinitude conjecture for μ-hits

摘录 / EXCERPT: Infinitude conjecture for μ-hits. There are infinitely many μ-hits.

conjecture | open | MathDB 页面 | 原文线索

V2-004886 | MATHDB-358929

Sprindžuk's higher-dimensional Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $\psi:\mathbb{N}\to\mathbb{R}_{\geq 0}$. Define $\mathcal{A}_k”(\psi)\subset[0,1]^k$ to be the set of points x for which $\\|qx-\mathbf p\\|<\psi(q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-004887 | MATHDB-358930

Hausdorff measure Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $k\in\mathbb{N}$, let f be a dimension function such that $r^{-k}f(r)$ is monotonic, and let $\mathcal{A}_k”(\psi)\subset[0,1]^k$ be the set of simultaneously reduced ψ-...

conjecture | open | MathDB 页面 | 原文线索

V2-004888 | MATHDB-358933

A 7-adic congruence conjecture for 3-elongated plane partitions

摘录 / EXCERPT: Let $d_3(n)$ denote the 3-elongated plane partition diamond function. For integers $n,[?]$...

conjecture | open | MathDB 页面 | 原文线索

V2-004889 | MATHDB-358950

Smirnov's arithmetic defect conjecture for maps on the compactified spectrum of the integers

摘录 / EXCERPT: Let $q=a/b$ be a nonzero rational number, let $\varphi_q:\overline{\operatorname{Spec} \mathbb Z}^{\mathrm{Smi}}\to\overline{\operatorname{Spec} \mathbb Z}^{\mathrm{Smi}}$ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-004890 | MATHDB-358971

Primitive-point conjecture for curves over primitive number fields

摘录 / EXCERPT: Let C be a smooth projective geometrically irreducible curve over $\mathbb{Q}$ of genus $g\geq 2$, and let K be a primitive number field of degree $d\geq 2$, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-004891 | MATHDB-358984

Gonek-Graham-Lee conjecture for admissible sequences of positive integers

摘录 / EXCERPT: Let $\mathcal{N}$ be an admissible sequence of positive integers, and let $\mathrm{LH}(\mathcal{N})$ denote the Lindelöf hypothesis for that sequence. Gonek-Graham-Lee conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-004892 | MATHDB-358985

Gonek-Graham-Lee conjecture for generic admissible sequences

摘录 / EXCERPT: Let $\mathcal{N}$ be an admissible sequence of positive real numbers, and let $\mathrm{LH}(\mathcal{N})$ denote the Lindelöf hypothesis for that sequence. Gonek-Graham-Lee conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004893 | MATHDB-358991

The p-Hydra divergent-point correspondence conjecture

摘录 / EXCERPT: Let $H:\mathbb{Z}\rightarrow\mathbb{Z}$ be a p-Hydra map, where p is a finite prime, and suppose its numen $\chi_H:\mathbb{N}_0\rightarrow\mathbb{Q}$ extends to a function...

conjecture | open | MathDB 页面 | 原文线索

V2-004894 | MATHDB-358998

Nonexistence conjecture for prime v-palindromes

摘录 / EXCERPT: Let $v\colon\mathbb{N}\to\mathbb{Z}$ be the additive function defined by $v(p)=p$ for primes p and $v(p^\alpha)=p+\alpha$ for prime powers p^α with $\alpha\geq2$. For inte...

conjecture | open | MathDB 页面 | 原文线索

V2-004895 | MATHDB-359009

The finite-field t-adic Littlewood conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field. The finite-field t-adic Littlewood conjecture. The t-adic Littlewood conjecture is true over $\mathbb{F}_2$, but false over all other finite...

conjecture | open | MathDB 页面 | 原文线索

V2-004896 | MATHDB-359010

Hausdorff dimension of t-adic Littlewood counterexamples

摘录 / EXCERPT: Let t denote the polynomial used in the t-adic Littlewood conjecture, and let f be a growth function. Consider the set of counterexamples to t-LC with growth function f....

conjecture | open | MathDB 页面 | 原文线索

V2-004897 | MATHDB-359037

The real-moment limit conjecture for ideal class counting functions

摘录 / EXCERPT: Let $a(\beta,\beta')$ denote the quantities appearing in Theorem, and let $\beta>0$ be real. For β integral or half-integral, the corresponding β-moment has matching lo...

conjecture | open | MathDB 页面 | 原文线索

V2-004898 | MATHDB-359038

The real cuspidal moment conjecture

摘录 / EXCERPT: Let $a_{\mathrm{cusp}}(\sigma,n)$ be the cuspidal ideal-class counting function, let $\sigma\in N$ be an ideal class, and let $\beta>0$ be real. Theorem gives the order of growth o...

conjecture | open | MathDB 页面 | 原文线索

V2-004899 | MATHDB-359063

Lange-Trotter conjecture for non-CM elliptic curves

摘录 / EXCERPT: Let E be a nonsingular elliptic curve over the rational numbers $\mathbb{Q}$ without complex multiplication and conductor $N\geq 1$. Suppose that, for primes p of good reductio...

conjecture | open | MathDB 页面 | 原文线索

V2-004900 | MATHDB-359088

Finiteness conjecture for sporadic quadratic points on modular curves

摘录 / EXCERPT: Sporadic quadratic-point conjecture. Ranging over all $X_0(N)$ for $N\in{\mathbb Z}_{\ge 1}$, there are only finitely many sporadic quadratic points.

conjecture | open | MathDB 页面 | 原文线索

V2-004901 | MATHDB-359090

Ufnarovski-Åhlander conjecture on arithmetic-derivative iteration

摘录 / EXCERPT: Ufnarovski-Åhlander conjecture. For every $x\in\mathbb{N}$, exactly one of the following could happen: either $D^i(x)=0$ or p^p for some prime p for sufficiently large i, or

conjecture | open | MathDB 页面 | 原文线索

V2-004902 | MATHDB-359091

Ufnarovski-Åhlander conjecture on numbers with prescribed numbers of anti-derivatives

摘录 / EXCERPT: Ufnarovski-Åhlander conjecture. For every integer $n\geq 0$, there are infinitely many $x>0$ such that x has exactly n anti-derivatives.

conjecture | open | MathDB 页面 | 原文线索

V2-004903 | MATHDB-359113

The nonlinear Skolem-Mahler-Lech conjecture

摘录 / EXCERPT: Let $l\geq 1$, let $F\in\mathbb{C}[x_0,\dots,x_{l-1}]$, and let $\{A_n\}_{n\geq 0}$ be a recurrence sequence satisfying ... for all $n\geq 0$. The nonlinear Skolem-Mahler-Lech conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-004904 | MATHDB-359114

The Skolem-Mahler-Lech conjecture with dynamically varying coefficients

摘录 / EXCERPT: Let $l\geq 1$, let $g,a_i\in\mathbb{C}(x)$ for $i=0,\dots,l-1$, and let $\alpha\in\mathbb{C}$ satisfy $a_i(g^n(\alpha))\neq\infty$ for every $i=0,\dots,l-1$ and $n\geq 0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004905 | MATHDB-359125

Reversed Dickson first-kind permutation polynomial classification

摘录 / EXCERPT: Let $p>3$ be a prime, and let n satisfy $1leq nleq p^2-1$. Write $D_n(1,x)$ for the reversed Dickson polynomial of the first kind. A polynomial that permutes $\mathbb{F}_p$ is ca...

conjecture | open | MathDB 页面 | 原文线索

V2-004906 | MATHDB-359126

Complete permutation classification for reversed Dickson polynomials of the first kind

摘录 / EXCERPT: Let p be an odd prime, let $1\leq n\leq p^2-1$, and let $D_n(1,x)$ be the reversed Dickson polynomial of the first kind. A polynomial f is a complete permutation polynomial (CP...

conjecture | open | MathDB 页面 | 原文线索

V2-004907 | MATHDB-359127

Period conjecture for reversed Dickson second-kind values at one quarter for non-Mersenne primes

摘录 / EXCERPT: Let $p>3$ be a non-Mersenne prime, and let $E_n(1,x)$ denote the reversed Dickson polynomial of the second kind. Consider the sequence of values $E_n(1,\frac14)$ modulo p. Non-Me...

conjecture | open | MathDB 页面 | 原文线索

V2-004908 | MATHDB-359128

Period conjecture for reversed Dickson second-kind polynomials over finite rings

摘录 / EXCERPT: Let $p>3$ be a prime, let $n_1,n_2>1$, and let $E_n(1,x)$ denote the reversed Dickson polynomial of the second kind over $\mathbb{Z}_p$. Reversed Dickson second-kind period conject...

conjecture | open | MathDB 页面 | 原文线索

V2-004909 | MATHDB-359130

Reversed Dickson second-kind permutation classification for primes greater than seven

摘录 / EXCERPT: Let $p>7$ be a prime, and let $E_n(1,x)$ denote the reversed Dickson polynomial of the second kind. A permutation polynomial (PP) is a polynomial that permutes $\mathbb{F}_p$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-004910 | MATHDB-359131

Complete permutation classification for reversed Dickson second-kind polynomials

摘录 / EXCERPT: Let $p\geq 5$ be a prime, and let $E_n(1,x)$ denote the reversed Dickson polynomial of the second kind. A complete permutation polynomial (CPP) is a polynomial f such that both...

conjecture | open | MathDB 页面 | 原文线索

V2-004911 | MATHDB-359135

Serre's variant of Schinzel's Hypothesis (H)

摘录 / EXCERPT: Let k be a number field, let $f_i(t)$ be irreducible polynomials over k, and let S be a finite set of places of k containing all archimedean places, all finite places v w...

conjecture | open | MathDB 页面 | 原文线索

V2-004912 | MATHDB-359151

The exact failure criterion for the naive mod ℓ converse theorem for $\operatorname{GL}_2(\mathbb{F}_q)$

摘录 / EXCERPT: Let ℓ be a prime and let q be a prime power. The naive converse theorem for mod ℓ representations of $\operatorname{GL}_2(\mathbb{F}_q)$ asks whether the relevant mod...

conjecture | open | MathDB 页面 | 原文线索

V2-004913 | MATHDB-359167

Mixed zeta-ratio moment conjecture

摘录 / EXCERPT: Let $a,k,\beta\in(0,\infty)$ and let $\sigma\in(\frac{1}{2},1]$ be fixed. Mixed zeta-ratio moment conjecture. There is a constant ${\mathcal D}(\sigma,a,k,\beta)$, depending only o...

conjecture | open | MathDB 页面 | 原文线索

V2-004914 | MATHDB-359191

Legendre's conjecture on primes between consecutive squares

摘录 / EXCERPT: Legendre's conjecture. There always exist at least two prime numbers between n^2 and $(n+1)^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-004915 | MATHDB-359192

Sierpiński's conjecture on primes in rows of an n by n matrix

摘录 / EXCERPT: Sierpiński's conjecture. Every row of this matrix contains at least one prime number.

conjecture | open | MathDB 页面 | 原文线索

V2-004916 | MATHDB-359246

Modular equidistribution conjecture for prime-generated Proth-Gilbreath triangles

摘录 / EXCERPT: Let $\delta_k(r)$ be the rth element on the kth row of the Proth-Gilbreath triangle generated by the sequence of primes, where $r\ge 1$ is an integer. Modular equidistribution...

conjecture | open | MathDB 页面 | 原文线索

V2-004917 | MATHDB-359274

The EGPS conjecture on preimages of density-zero sets under the proper-divisor sum

摘录 / EXCERPT: EGPS conjecture. The preimage $s^{-1}(\mathcal{A})$ also has asymptotic density zero.

conjecture | open | MathDB 页面 | 原文线索

V2-004918 | MATHDB-359280

Hamilton-Berge connectivity conjecture for the factorisations \(\mathcal{F}_q\)

摘录 / EXCERPT: Let q be a prime power and let $\mathcal{F}_q$ denote the 1-factorisation considered above. A 1-factorisation is connected when the union of any two distinct 1-factors is connect...

conjecture | open | MathDB 页面 | 原文线索

V2-004919 | MATHDB-359293

Späth's Character Triple Conjecture

摘录 / EXCERPT: Let $G\unlhd A$ be finite groups, let p be a prime, and suppose that ${\mathbf{O}}_p(G)$, the largest normal p-subgroup of G, is contained in the centre of G. For a p-blo...

conjecture | open | MathDB 页面 | 原文线索

V2-004920 | MATHDB-359307

Primality conjecture for transient-number-one knots

摘录 / EXCERPT: Let K be a knot, and let $tr(K)$ denote its transient number. A knot is prime if it cannot be expressed as a nontrivial connected sum of knots. Primality conjecture for transient...

conjecture | open | MathDB 页面 | 原文线索

V2-004921 | MATHDB-359313

The norm-form conjecture for the resultant polynomial R_q

摘录 / EXCERPT: Let q be an odd prime, and let $R_q(y)$ be the resultant polynomial defined in the paper. Norm-form conjecture. There exist polynomials $h_1,h_2\in\mathbb Z[y]$ such that ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-004922 | MATHDB-359314

The irreducibility conjecture for Fekete polynomials at semiprime indices

摘录 / EXCERPT: Let p and q be distinct odd primes and set $n=pq$. Let f_n and g_n denote the reciprocal Fekete polynomial and its trace polynomial, respectively. Semiprime Fekete irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-004923 | MATHDB-359315

The product conjecture for fields with the Bogomolov property

摘录 / EXCERPT: Product conjecture. If E and F have the Bogomolov property, then the group product E^*F^* also has the Bogomolov property.

conjecture | open | MathDB 页面 | 原文线索

V2-004924 | MATHDB-359344

The conjecture on prime numbers that are palindromic in base b

摘录 / EXCERPT: Let $b\geq 2$ be an integer. A positive integer is b-palindromic if its sequence of base-b digits is a palindrome, and let $\mathscr{P}_b(x)$ denote the set of positive b-pal...

conjecture | open | MathDB 页面 | 原文线索

V2-004925 | MATHDB-359369

Bateman-Horn conjecture for pairs of linear forms

摘录 / EXCERPT: Let $f_1(t)=a_1t+b_1$ and $f_2(t)=a_2t+b_2$ be linear forms with integers a_1,a_2,b_1,b_2 and $a_1,a_2>0$. Set $f=f_1f_2$, and suppose that f is admissible, meaning that it doe...

conjecture | open | MathDB 页面 | 原文线索

V2-004926 | MATHDB-359386

Vafa-Witten S-duality conjecture for SU(r) and PSU(r) partition functions

摘录 / EXCERPT: Let (X,H) be a smooth polarized surface with $H_1(X,\mathbb{Z})=0$ and $h^{2,0}(X)>0$. Let r be prime and let $c_1\in H^2(X,\mathbb{Z})$ be algebraic. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004927 | MATHDB-359389

The abc conjecture of Oesterlé and Masser

摘录 / EXCERPT: The abc conjecture. For each positive real number ε there is a positive number $c(\varepsilon)$, which depends on ε only, such that for all pairwise coprime...

conjecture | open | MathDB 页面 | 原文线索

V2-004928 | MATHDB-359414

Gras's conjecture on the vanishing of tame ramification for large primes

摘录 / EXCERPT: Let K be a number field, let p be a prime, and let ${\mathcal T}_{S_p}$ denote the relevant tame ramification module for the maximal extension of K unramified outside the pla...

conjecture | open | MathDB 页面 | 原文线索

V2-004929 | MATHDB-359446

The conjecture on infinitely many odd-base anti-Niven arithmetic progressions

摘录 / EXCERPT: Anti-Niven progression conjecture. There exist infinitely many b-anti-Niven d-APs of length $p-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-004930 | MATHDB-359447

The conjecture on long Niven arithmetic progressions for even bases

摘录 / EXCERPT: Niven progression conjecture. There exist infinitely many pairs (b,d) for which there is a b-Niven d-AP of length

conjecture | open | MathDB 页面 | 原文线索

V2-004931 | MATHDB-359480

Coates-Sujatha's fine Selmer conjecture

摘录 / EXCERPT: Let E be an elliptic curve over a number field F, let p be a prime at which E has good reduction, and let F_∞ be the cyclotomic $\mathbb{Z}_p$-extension of F. Deno...

conjecture | open | MathDB 页面 | 原文线索

V2-004932 | MATHDB-359481

Conjecture on binary Kaprekar sequences with second digits

摘录 / EXCERPT: Binary Kaprekar sequence conjecture. (i) When the digit length is $2m$, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-004933 | MATHDB-359500

Quantitative Hardy-Littlewood prime tuples conjecture

摘录 / EXCERPT: Let $\mathcal{P}$ denote the set of primes, and let $1_{\mathcal{P}}$ be its indicator function. For a tuple of distinct integers ${\mathcal H}=\{h_1,\dots,h_k\}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004934 | MATHDB-359542

The Iwasawa-Greenberg main conjecture for the BDP p-adic L-function

摘录 / EXCERPT: Let K be an imaginary quadratic field in which the odd prime p splits as $(p)=\mathfrak p\overline{\mathfrak p}$, let Λ be the anticyclotomic Iwasawa algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004935 | MATHDB-359546

Prime-degree denseness conjecture for quotient sets of integral forms

摘录 / EXCERPT: Prime-degree denseness conjecture. There exists a positive integer m (in terms of q) such that for any form

conjecture | open | MathDB 页面 | 原文线索

V2-004936 | MATHDB-359555

The stationary descendent Gromov-Witten formula for primitive K3 classes

摘录 / EXCERPT: Let S be a K3 surface and let $\beta\in H_2(S,\mathbb Z)$ be primitive. For cohomology classes $\gamma_0,\gamma_1,\ldots\in H^*(S)$, define the partition function...

conjecture | open | MathDB 页面 | 原文线索

V2-004937 | MATHDB-359574

The characterization of Eisenstein-type orthogonal modular forms

摘录 / EXCERPT: The characterization of Eisenstein-type forms. The subspace of $M_{a,b}(\operatorname{SO}(\Lambda),\vartheta_d)$ spanned by forms of Eisenstein type is the span of constant functio...

conjecture | open | MathDB 页面 | 原文线索

V2-004938 | MATHDB-359591

Polynomial recurrence characterization for Lucas atoms at $t=0$

摘录 / EXCERPT: Let $s\in\mathbb Z$. The sequence $(P_n(s,0))_{n\geq 1}$ is called polynomially recurrent if it satisfies a polynomial recurrence in n. Polynomial-recurrence conjecture. The sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-004939 | MATHDB-359592

Conjecture on collapse and average logarithmic integer complexity

摘录 / EXCERPT: Collapse conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-004940 | MATHDB-359637

Yuan-Han conjecture on exceptional solutions of a ternary exponential equation

摘录 / EXCERPT: Let k be a positive integer, and let a,b be coprime positive integers with $\min\{a,b\}>1$. Consider the equation ... The solution $(x,y,z)=(1,1,1)$ is always present, and any so...

conjecture | open | MathDB 页面 | 原文线索

V2-004941 | MATHDB-359667

Watkins's prime-conductor classification conjecture

摘录 / EXCERPT: Let $N:=p$ be prime, and let m_E be the modular degree of an elliptic curve of conductor p. Watkins's prime-conductor classification conjecture. If ... then either $p=17$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-004942 | MATHDB-359673

The zero-residue bias conjecture for binary quadratic forms

摘录 / EXCERPT: Let D be any fundamental discriminant, let f be any reduced binary quadratic form of discriminant D, and let q be a prime such that $(q,2D)=1$. For a modulo q, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-004943 | MATHDB-359681

Full-rank conjecture for descendent matroids

摘录 / EXCERPT: Full-rank conjecture. The rank of $\mathcal{M}_k$ is always equal to $\operatorname{dim}(\operatorname{QM}_k)$. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-004944 | MATHDB-359682

Uniform-minor conjecture for positive descendent matroids

摘录 / EXCERPT: Uniform-minor conjecture. For every $k\geq 14$, there exists an integer $n>\operatorname{dim}(\operatorname{QM}_k)$ such that $\mathcal{M}_k\mid S^k$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-004945 | MATHDB-359688

Kaneko-Tsumura dimension and relation conjecture for double tilde-T values

摘录 / EXCERPT: Kaneko-Tsumura's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-004946 | MATHDB-359697

The quarter-exponent conjecture for Gauss's circle and Dirichlet's divisor problems

摘录 / EXCERPT: Quarter-exponent conjecture. In Gauss's Circle Problem and Dirichlet's Divisor Problem,

conjecture | open | MathDB 页面 | 原文线索

V2-004947 | MATHDB-359700

Prime Pair Error Conjecture

摘录 / EXCERPT: Let ... where $\psi_{2}(N,k)$ counts the von Mangoldt-weighted prime pairs of difference k, and $\mathfrak{S}(k)$ is the Hardy-Littlewood singular series. Prime Pair Error Conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-004948 | MATHDB-359701

Vaughan's L^1 conjecture for the exponential sum over primes

摘录 / EXCERPT: Let ... where Λ is the von Mangoldt function. Vaughan's L^1 conjecture. There exists a positive constant c_{6} satisfying $0<c_{6}\leq 1/\sqrt{2}$ such that ... This conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-004949 | MATHDB-359702

Prime Pair Error Lower Bound Conjecture

摘录 / EXCERPT: Let ... with $\psi_{2}(N,k)$ and $\mathfrak{S}(k)$ as above. Prime Pair Error Lower Bound Conjecture. ... The conjecture gives a lower bound of the expected order for the total prime-p...

conjecture | open | MathDB 页面 | 原文线索

V2-004950 | MATHDB-359703

Vaughan's L^1 lower bound conjecture for the exponential sum over primes

摘录 / EXCERPT: Let ... where Λ is the von Mangoldt function. Vaughan's L^1 lower bound conjecture. ... This is a lower-bound version of Vaughan's asymptotic L^1 conjecture and would give...

conjecture | open | MathDB 页面 | 原文线索

V2-004951 | MATHDB-359710

Reciprocal supernorm perimeter-statistic asymptotic conjecture

摘录 / EXCERPT: Let ${\rm Per}(n)$ be the perimeter ensemble of partitions indexed by n, and let $\widehat{W}_{\rm per}(n)$ denote its non-cumulative reciprocal supernorm statistic. Perimeter-en...

conjecture | open | MathDB 页面 | 原文线索

V2-004952 | MATHDB-359711

Reciprocal supernorm max-part-statistic asymptotic conjecture

摘录 / EXCERPT: Let ${\rm Max}(n)$ be the ensemble of partitions with largest part n, and let $\widehat{W}_{\rm max}(n)$ denote its non-cumulative reciprocal supernorm statistic. Max-part-ensemb...

conjecture | open | MathDB 页面 | 原文线索

V2-004953 | MATHDB-359712

Norm-reciprocal-supernorm relation for the size ensemble

摘录 / EXCERPT: Let $W_{\rm size}(n)$ be the size-ensemble norm statistic and let $\widehat{W}_{\rm size}(n)$ be the corresponding non-cumulative reciprocal supernorm statistic. Size-ensemble norm...

conjecture | open | MathDB 页面 | 原文线索

V2-004954 | MATHDB-359727

Conjectural Bernoulli recursion for coefficients of the Artin-Hasse exponential

摘录 / EXCERPT: Let p be an odd prime and let ... be the Artin-Hasse exponential series in $\mathbb F_p[[X]]$. For an integer n, let B_n denote the n-th Bernoulli number. The conjectural rec...

conjecture | open | MathDB 页面 | 原文线索

V2-004955 | MATHDB-359734

Ma-Chen-Wu asymptotic conjecture for truncated primes represented by the floor function

摘录 / EXCERPT: Let $0<\theta\leq 1$ be real, and let $\pi_{\theta}(x)$ denote the number of integers n with $1\leq n\leq x^{\theta}$ such that $\left\lfloor x/n\right\rfloor$ is prime. For a gi...

conjecture | open | MathDB 页面 | 原文线索

V2-004956 | MATHDB-359740

The prime-order even-list conjecture for perfect matchings

摘录 / EXCERPT: Let p be an odd prime, let K_{2p} be the complete graph on $2p$ circularly ordered vertices, and let L be a list of p positive integers less than p. A perfect matching...

conjecture | open | MathDB 页面 | 原文线索

V2-004957 | MATHDB-359756

The uniform Fourier-dimension conjecture for missing digits measures

摘录 / EXCERPT: Let $p>2$ be a prime number and let $\epsilon\in(0,1)$. Uniform Fourier-dimension conjecture. For every digit set D with ... we have ... where the $o(1)$ term is uniform across all s...

conjecture | open | MathDB 页面 | 原文线索

V2-004958 | MATHDB-359758

The rational-points conjecture for sparse missing digits on the unit circle

摘录 / EXCERPT: Let C be the unit circle ... For sufficiently large integers p, let $D\subset\{0,\dots,p-1\}^2$ satisfy $\#D\leq100$, and let $K_{p,D,I}$ denote the corresponding missing-digits...

open_problem | open | MathDB 页面 | 原文线索

V2-004959 | MATHDB-359782

Density-half conjecture for discrete stick fragmentation

摘录 / EXCERPT: Let $\mathfrak{S}\subseteq\mathbb{Z}_+$ be a stopping set such that the natural density exists, and write ... Assume that $r>0$. Density-half conjecture. The set of dead stick length...

conjecture | open | MathDB 页面 | 原文线索

V2-004960 | MATHDB-359834

The height-realisation conjecture for ternary cyclotomic polynomials

摘录 / EXCERPT: Height-realisation conjecture. For every odd prime p and every integer h with $1\le h\le M(p)$, there exist primes q and r such that

conjecture | open | MathDB 页面 | 原文线索

V2-004961 | MATHDB-359836

Obstruction realization conjecture for UHF algebra kernels

摘录 / EXCERPT: Let G be a countable discrete group with a reasonable finiteness condition for its cohomology, and let $\mathfrak{P}$ be a set of prime numbers. The UHF obstruction realization c...

conjecture | open | MathDB 页面 | 原文线索

V2-004962 | MATHDB-359845

Ellenberg-Jain-Venkatesh distribution conjecture for lambda-invariants

摘录 / EXCERPT: Fix a prime p. For imaginary quadratic fields K in which p is inert, let $\bb\b5_p(K)$ be the Iwasawa $\bb\b5$-invariant, and let r be a nonnegative integer. Ellenberg-Jain...

conjecture | open | MathDB 页面 | 原文线索

V2-004963 | MATHDB-359847

Generalized abc conjecture over number fields

摘录 / EXCERPT: Let K be a number field and let I be an ideal of $\mathcal{O}_K$. Define ... where $N(\mathfrak{p})=\#(\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)$. Generalized abc conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-004964 | MATHDB-359857

Proth-Gilbreath conjecture for the western edge of the prime difference triangle

摘录 / EXCERPT: Proth-Gilbreath conjecture. All the differences on the western edge of the triangle generated by the sequence of all primes are equal to 1.

conjecture | open | MathDB 页面 | 原文线索

V2-004965 | MATHDB-359881

Hardy-field prime recurrence conjecture for one function

摘录 / EXCERPT: Let $\mathcal{H}$ be the Hardy field under consideration, let $a\in\mathcal{H}$ have polynomial growth, and assume ... for every $c\in\mathbb{R}$ and $p(t)\in\mathbb{Z}[t]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-004966 | MATHDB-359882

Hardy-field prime recurrence conjecture for several functions

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $a_1,\ldots,a_k\in\mathcal{H}$ be functions of polynomial growth. Assume that every non-trivial linear combination a of $a_1,\ldots,a_k$ satisfies .....

conjecture | open | MathDB 页面 | 原文线索

V2-004967 | MATHDB-359895

The half-integral-weight Linnik-Selberg conjecture

摘录 / EXCERPT: Let $k=\frac12$ or $k=-\frac12$, and let u be a weight k multiplier system on $\Gamma_0(N)$. Let $S(m,n,c,\nu)$ be the corresponding half-integral-weight Kloosterman sum, let...

conjecture | open | MathDB 页面 | 原文线索

V2-004968 | MATHDB-359916

Existence of non-hyperplanar (q^2+q+1)-ovoids of the parabolic quadric

摘录 / EXCERPT: Let q be a prime power greater than 2, and let $\mathcal{Q}(8,q)$ be the parabolic quadric in projective 8-space. An (q^2+q+1)-ovoid is a set of points meeting every genera...

conjecture | open | MathDB 页面 | 原文线索

V2-004969 | MATHDB-359938

Conrey-Gonek conjecture for long divisor-coefficient Dirichlet polynomials

摘录 / EXCERPT: Let T be sufficiently large and let $K=T^{1+\eta}$ with $\eta\in(0,1)$. For $k\in\mathbb{N}$, define the k-th divisor function by ... Define ... ... where ... for $n\in\mathbb{Z}^+$ an...

conjecture | solved | MathDB 页面 | 原文线索

V2-004970 | MATHDB-359939

The k-ℓ additive divisor conjecture

摘录 / EXCERPT: k-ℓ additive divisor conjecture. There exists a triple $(\vartheta_{k,\ell},C_{k,\ell},\beta_{k,\ell})\in[1/2,1)\times[0,\infty)\times(0,1]$ such that, for every positive ab...

conjecture | open | MathDB 页面 | 原文线索

V2-004971 | MATHDB-359952

Categorical hyperkähler conjecture for quasi-BPS categories of K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $v=dv_0$ with v_0 primitive and $v_0,v_0=2g-2$, and let $n=\langle v,v\rangle/2+1$. For $w\in\mathbb{Z}$ coprime with v, consider the reduced quasi...

conjecture | open | MathDB 页面 | 原文线索

V2-004972 | MATHDB-359966

Mousavi's odd-prime solubilizer conjecture

摘录 / EXCERPT: Let G be a finite insoluble group, let $x\in G$, and write $\operatorname{Sol}_G(x)$ for the set of elements $y\in G$ such that $\langle x,y\rangle$ is soluble. Mousavi's odd-pri...

conjecture | open | MathDB 页面 | 原文线索

V2-004973 | MATHDB-359970

The prime-square conjecture for the unit-fraction Maker-Breaker game

摘录 / EXCERPT: Let k be a positive integer and let p be an odd prime such that $k+1=p^2$. Let $f(k,-1)$ denote the least board size for which Maker can force a solution to the unit-fraction e...

conjecture | open | MathDB 页面 | 原文线索

V2-004974 | MATHDB-359978

Ramírez Alfonsín-Skałba conjecture on primes represented by two generators

摘录 / EXCERPT: Let $1<c<d$ be relatively prime integers and set ... Let $\pi_{c,d}$ denote the number of primes not exceeding $g_{c,d}$ that can be written as $cx+dy$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-004975 | MATHDB-359980

Intersection density conjectures for transitive permutation groups

摘录 / EXCERPT: Intersection density conjectures. (i) If $|\Omega|$ is a prime power, then $\rho(G)=1$. (ii) If $|\Omega|=2p$, where p is an odd prime, then $1\leq\rho(G)\leq 2$. (iii) If...

conjecture | open | MathDB 页面 | 原文线索

V2-004976 | MATHDB-359987

Vojta's conjecture with multiplier ideal sheaves

摘录 / EXCERPT: Vojta's multiplier-ideal conjecture. For every real $\varepsilon>0$ and positive integer r, there is a proper Zariski-closed subset $Z\subsetneq\mathcal X$, depending only on...

conjecture | open | MathDB 页面 | 原文线索

V2-004977 | MATHDB-360004

Asymptotic conjecture for reversible binary prime pairs

摘录 / EXCERPT: Let ${\mathcal B}_n$ be the set of binary integers with n binary digits, let $\overleftarrow{a}$ denote the integer obtained by reversing the binary digits of a, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-004978 | MATHDB-360008

The weak Weyl-sum conjecture for prime-denominator incomplete sums

摘录 / EXCERPT: Weak Weyl-sum conjecture. There is a constant $C_{d,\varepsilon}>0$ such that, for every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-004979 | MATHDB-360021

Darmon-Dasgupta's real quadratic elliptic unit conjecture

摘录 / EXCERPT: Let p be a prime, let $\tau\in\mathcal{H}_p$ be a real quadratic point of discriminant D prime to p, and let H_τ be the narrow ring class field of the order defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-004980 | MATHDB-360036

p-adic cohomological Sarnak-Xue density hypothesis

摘录 / EXCERPT: Let G be a definite Gross inner form of SO_5 over a number field, let p be a prime of the number field, let $G_p=G(\mathbb{Q}_p)$, let $\Gamma_p(q)$ be the level-q p-arit...

conjecture | open | MathDB 页面 | 原文线索

V2-004981 | MATHDB-360044

Hiss-Schröer conjecture on bounded Fourier-Jacobi multiplicities

摘录 / EXCERPT: Let q be a power of a prime $p\neq 2$, let $\psi:\mathbb F_q\to\mathbb{\bar Q}_\ell^\times$ be a nontrivial additive character, and let $G_n=\mathrm U_n$ or Sp_{2n}. F...

conjecture | open | MathDB 页面 | 原文线索

V2-004982 | MATHDB-360048

Discrete Fourier restriction conjecture for the monomial curve

摘录 / EXCERPT: Let $d\geq 3$ be an integer, let $N\in\mathbb N$, let $a_1,\ldots,a_N\in\mathbb C$, and write $\mathbb T^2$ for the two-dimensional torus. For $p\geq 2(d+1)$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-004983 | MATHDB-360053

Residue distribution conjecture for the fundamental domain of Pascal's triangle modulo p

摘录 / EXCERPT: Residue distribution conjecture. As the prime modulus p tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-004984 | MATHDB-360054

Mean-value conjecture for character sums in Pascal's triangle modulo p

摘录 / EXCERPT: Mean-value conjecture. As p tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-004985 | MATHDB-360055

Infinitude conjecture for row-dominant characters

摘录 / EXCERPT: Infinitude conjecture. There are infinitely many row-dominant characters.

conjecture | open | MathDB 页面 | 原文线索

V2-004986 | MATHDB-360058

Holomorphic quantum unique ergodicity conjecture for Siegel cusp forms

摘录 / EXCERPT: Holomorphic QUE conjecture. Fix a bounded continuous function ϕ on $\operatorname{Sp}_{2n}(\mathbb Z)\backslash\mathbb H_n$. If $F\in S_k(\operatorname{Sp}_{2n}(\mathbb Z))$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-004987 | MATHDB-360061

Schinzel-Sierpiński hypothesis H

摘录 / EXCERPT: Let $m\in\mathbb{N}$ and let $f_1,\ldots,f_m\in\mathbb{Z}[x]$ be polynomials. Suppose that there is no prime number p such that ... for every $n\in\mathbb{N}$. Schinzel-Sierpiński...

conjecture | open | MathDB 页面 | 原文线索

V2-004988 | MATHDB-360084

Gaussian Goldbach conjecture with angular restrictions

摘录 / EXCERPT: Let n be an even Gaussian integer, let $N(n)$ denote its norm, and let $\arg(n)$ denote its argument. Gaussian Goldbach conjecture with angular restrictions. Every even Gaussian...

conjecture | open | MathDB 页面 | 原文线索

V2-004989 | MATHDB-360175

A bound on the number of distinct prime factors of Mersenne numbers

摘录 / EXCERPT: Let $p\ge 3$ be prime and set $N_p=2^p-1$. Write $\omega(N_p)$ for the number of distinct prime factors of N_p. The distinct-factor bound conjecture. There exists p_0 such that...

conjecture | open | MathDB 页面 | 原文线索

V2-004990 | MATHDB-360176

A squarefreeness conjecture for Mersenne numbers

摘录 / EXCERPT: Let $p\ge 3$ be prime and set $N_p=2^p-1$. The squarefreeness conjecture. There exists p_0 such that, if $p>p_0$, then N_p is squarefree. Squarefreeness would rule out repeated...

conjecture | open | MathDB 页面 | 原文线索

V2-004991 | MATHDB-360177

Existence of separating quadratic-residue tests for Mersenne factors

摘录 / EXCERPT: Let $p>p_0$ be prime, assume the distinct-factor bound conjecture and the uniform-distribution conjecture, and write ... where $q_1<\cdots<q_k$ are the prime factors and...

conjecture | open | MathDB 页面 | 原文线索

V2-004992 | MATHDB-360195

Finiteness and completeness conjecture for Bernoulli polynomial derivatives

摘录 / EXCERPT: The Bernoulli polynomials $\mathbf{B}_n(x)$ are defined by ... and let ... Finiteness and completeness conjecture. The set $\overline{\mathcal{S}}$ is finite and ... This conjecture asse...

conjecture | open | MathDB 页面 | 原文线索

V2-004993 | MATHDB-360196

Kellner's conjecture on the distribution of prime factors of $\mathbb{D}^+_n$

摘录 / EXCERPT: For a prime p, let $s_p(n)$ be the sum of the base-p digits of n, let $\omega(n)$ count the distinct prime divisors of n, and define ... Kellner's conjecture. The following s...

conjecture | open | MathDB 页面 | 原文线索

V2-004994 | MATHDB-360197

Conjecture on the radical characterization set for Bernoulli denominators

摘录 / EXCERPT: Define ... Radical characterization conjecture. One has ... The surrounding results show that $\overline{\mathcal{R}}$ is finite and constrain its elements; the conjecture remains unre...

conjecture | open | MathDB 页面 | 原文线索

V2-004995 | MATHDB-360207

Davenport's geometric-condition conjecture for cubic forms in ten variables

摘录 / EXCERPT: Davenport's geometric-condition conjecture. If C satisfies Davenport's Geometric Condition, then the asymptotic formula for the number of solutions to $C(\mathbf{x})=0$ holds wit...

conjecture | open | MathDB 页面 | 原文线索

V2-004996 | MATHDB-360208

Prime-modulus rank-count conjecture for cubic forms

摘录 / EXCERPT: Prime-modulus rank-count conjecture. For all n, uniformly in $1\leq H\leq R$, and for every relevant rank threshold r,

conjecture | open | MathDB 页面 | 原文线索

V2-004997 | MATHDB-360220

Koley and Reddy's irreducibility conjecture for integer polynomials

摘录 / EXCERPT: Let $f=a_0+a_1x+\cdots+a_nx^n\in\mathbb Z[x]$, where $a_0=\pm p^k$ for a prime p and positive integer k. Suppose that $a_1=a_2=\cdots=a_{q-1}=0$ for a prime $q\leq n$. Koley an...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-004998 | MATHDB-360221

Harrington's irreducibility conjecture for trinomial-like polynomials

摘录 / EXCERPT: Harrington's conjecture. The polynomial f is irreducible unless $f=x^2+4x+4$. The paper states that Zhang and Yuan provide an alternative proof using the Newton polygon technique...

conjecture | open | MathDB 页面 | 原文线索

V2-004999 | MATHDB-360242

The dimension conjecture for nondegenerate manifolds

摘录 / EXCERPT: Let $1\leq d<n$ be integers. Let $\widetilde{\mathcal M}_{n,d}$ be the class of submanifolds $\mathcal M\subset\mathbb R^n$ of dimension d that are nondegenerate at every point,...

conjecture | open | MathDB 页面 | 原文线索

V2-005000 | MATHDB-360245

Function-field analogue of Cilleruelo's least-common-multiple conjecture

摘录 / EXCERPT: Let $q=p^k$ be a prime power, and let $f\in\mathbb{F}_q[T][X]$ be a fixed irreducible polynomial with X-degree $d\geq 2$. Define ... Let $V_f:=\{g\in\mathbb{F}_q[T]:f(X+g)=f(X)\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-005001 | MATHDB-360290

Prime separation conjecture for Hecke eigenvalues of distinct Maass forms

摘录 / EXCERPT: Fix an integer $N\geq 2$. Let $\ell_1,\ell_2,\ldots,\ell_N$ be integers greater than one, and let $\eta_1,\eta_2,\ldots,\eta_N$ be distinct Maass forms, with η_j a Maass form...

conjecture | open | MathDB 页面 | 原文线索

V2-005002 | MATHDB-360309

Infinitude of regular primes

摘录 / EXCERPT: A prime p is regular if it does not divide any of the Bernoulli--Seki numbers $B_2,B_4,\ldots,B_{p-3}$. Infinitude conjecture for regular primes. There exist infinitely many regu...

conjecture | open | MathDB 页面 | 原文线索

V2-005003 | MATHDB-360310

Infinitude of non-Wieferich primes

摘录 / EXCERPT: A prime p is non-Wieferich if there is an integer N for which $N^{p-1}\not\equiv1\pmod{p^2}$. Infinitude conjecture for non-Wieferich primes. There exist infinitely many non-Wi...

conjecture | open | MathDB 页面 | 原文线索

V2-005004 | MATHDB-360311

Bernoulli--Seki nonvanishing consequence of regular-prime infinitude

摘录 / EXCERPT: Bernoulli--Seki nonvanishing consequence. Let $k\geq3$ be an odd integer. Then, there exist infinitely many primes p greater than k such that p does not divide B_{p-k}. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005005 | MATHDB-360318

Makowski's even perfect number power-sum conjecture

摘录 / EXCERPT: Let n be an even perfect number represented as $n=x^m+y^m$, where x and y are positive integers and m is an integer with $m\geq 2$. Makowski's conjecture. Then $m=3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005006 | MATHDB-360319

The explicit ABC conjecture with exponent $7/4$

摘录 / EXCERPT: Let A, B, and C be positive integers such that $A+B=C$ and $(A,B)=1$. Define $\operatorname{rad}(ABC)$ to be the product of the distinct prime divisors of ABC. Explicit ABC...

conjecture | open | MathDB 页面 | 原文线索

V2-005007 | MATHDB-360348

Distinctness of infinite local-product distributions

摘录 / EXCERPT: For each prime l, let Y'_{l,e_l} be the random variable associated with the local parameter e_l, and consider the infinite independent products $\prod_l Y'_{l,e_l}$. Distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-005008 | MATHDB-360373

Selberg's primitive-function and orthogonality conjectures

摘录 / EXCERPT: Let $\mathcal{S}$ be the Selberg class, let $F,F'\in\mathcal{S}$ be primitive functions, and let $a_F(n)$ and $a_{F'}(n)$ denote their Dirichlet coefficients. Selberg's primitive-f...

conjecture | open | MathDB 页面 | 原文线索

V2-005009 | MATHDB-360379

Almost-diameter conjecture for affine and projective actions

摘录 / EXCERPT: Almost-diameter conjecture. For every $\epsilon>0$, and for q large enough depending on ϵ, the following holds.

conjecture | open | MathDB 页面 | 原文线索

V2-005010 | MATHDB-360381

Betancort-Smirnov's F-threshold and Hilbert-Kunz conjecture

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a noetherian local ring of prime characteristic. Let $f_1,f_2,\ldots,f_r$ be part of a system of parameters, let J be an $\mathfrak m$-primary ideal, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005011 | MATHDB-360382

Corrected F-threshold and Hilbert-Kunz conjecture

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a noetherian local ring of prime characteristic. Let $f_1,f_2,\ldots,f_r$ be part of a system of parameters, let J be an ideal generated by a full system...

conjecture | open | MathDB 页面 | 原文线索

V2-005012 | MATHDB-360393

Gross-Zagier conjecture on the index of a Heegner point

摘录 / EXCERPT: Gross-Zagier conjecture. If P_L has infinite order in $E_L(L)$, then P_L generates a subgroup of finite index in $E_L(L)$, and this index equals

conjecture | open | MathDB 页面 | 原文线索

V2-005013 | MATHDB-360402

The number of nonsingular generalized Weierstrass elliptic curves over finite fields

摘录 / EXCERPT: Let q be a prime power and let a generalized Weierstrass elliptic curve over $\mathbb{F}_q$ be nonsingular when its discriminant is nonzero. The nonsingular generalized Weierstra...

conjecture | open | MathDB 页面 | 原文线索

V2-005014 | MATHDB-360403

The cubic-residue occurrence conjecture over prime-power rings

摘录 / EXCERPT: Let p be prime, let m be a positive integer, and write $\mathbb{Z}_{p^m}$ for the integers modulo p^m. Let CR denote a cubic residue and let $[i]$ denote the ith class in...

conjecture | open | MathDB 页面 | 原文线索

V2-005015 | MATHDB-360404

The upper-bound conjecture for reduced Weierstrass curves over prime-power rings

摘录 / EXCERPT: Let p be an odd prime and let m be a positive integer with $\gcd(p^m,6)=1$. Let $\Delta^i(p^m)$ denote the number of solutions to $\Delta \equiv i \pmod {p^m}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005016 | MATHDB-360416

Unbounded maximal elements in the Frobenius problem over real number fields

摘录 / EXCERPT: Unboundedness conjecture. For every positive integer $m\geqslant 1$, there exists some $\alpha\in\mathfrak{O}_K^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005017 | MATHDB-360424

The claim that the Collatz tree contains all reachable natural numbers

摘录 / EXCERPT: Collatz-tree reachability claim. The Collatz tree contains all the reachable natural numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-005018 | MATHDB-360425

The Hotel Collatz tree conjecture

摘录 / EXCERPT: Hotel Collatz conjecture. The Hotel Collatz graph is an infinite, connected, acyclic graph, hence a tree, with 1 as its root.

conjecture | open | MathDB 页面 | 原文线索

V2-005019 | MATHDB-360426

The Hotel Collatz partition conjecture

摘录 / EXCERPT: Hotel Collatz partition conjecture. The set $\overline{W}$ is a partition of N.

conjecture | open | MathDB 页面 | 原文线索

V2-005020 | MATHDB-360447

Odd Zumkeller-prime representation conjecture

摘录 / EXCERPT: Odd Zumkeller-prime representation conjecture. Every odd natural number greater than 8 and not equal to 21 is expressible as a sum of a Zumkeller number and a prime.

conjecture | open | MathDB 页面 | 原文线索

V2-005021 | MATHDB-360448

The Repulsion G-B-G conjecture for bad Gram points

摘录 / EXCERPT: Let g_n be a Gram point, and call it bad when it does not satisfy the usual Gram sign condition; call its consecutive neighbours g_{n-1} and g_{n+1} good when they do satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-005022 | MATHDB-360456

Unit-root subspace conjecture for canonical Mazur-Tate splittings

摘录 / EXCERPT: Let F be a number field, let v be a finite place above a prime p, and let J/F be a Jacobian with semistable ordinary reduction at v. Let Θ be a theta divisor and s...

conjecture | open | MathDB 页面 | 原文线索

V2-005023 | MATHDB-360471

Distinctness conjecture for the values of the divisor-sum ratio

摘录 / EXCERPT: Let $\sigma(x)$ denote the sum-of-divisors function for $x\in\mathbb{N}$. Distinctness conjecture. All values of ... for $x\in\mathbb{N}$ are pairwise distinct. No equal values are k...

conjecture | open | MathDB 页面 | 原文线索

V2-005024 | MATHDB-360472

Distinctness conjecture for coprime values of the divisor-sum ratio

摘录 / EXCERPT: Let $x,y\in\mathbb{N}$ be coprime and satisfy $x\ne y$. Distinctness conjecture. The values ... are distinct. Here σ is the sum-of-divisors function. Distinctness is known for...

conjecture | open | MathDB 页面 | 原文线索

V2-005025 | MATHDB-360481

Mean-value Lindelöf conjecture for real Dirichlet characters

摘录 / EXCERPT: Let $k\geqslant1$ be an integer, and let $Q,T\geqslant2$ be real numbers. Let $\mathcal{O}_2(Q)$ denote the family of real Dirichlet characters counted in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005026 | MATHDB-360487

The homological-dimension conjecture for Kuznetsov components of prime Fano threefolds

摘录 / EXCERPT: Let X be a cubic threefold, quartic double solid or Gushel--Mukai threefold, and let $\mathcal{K}u(X)$ be its Kuznetsov component. Let σ be a stability condition on...

conjecture | open | MathDB 页面 | 原文线索

V2-005027 | MATHDB-360488

Squarefree-value conjecture for polynomial values

摘录 / EXCERPT: Let $f(t)\in\mathbb{Z}[t]$ be a separable polynomial such that $f(t)\ne a(t)b(t)^2$ and $\operatorname{deg} f\geq 1$. Let μ denote the Möbius function and let $\rho_f(m)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-005028 | MATHDB-360489

Squarefree values of polynomials at primes

摘录 / EXCERPT: Let $f(t)\in\mathbb{Z}[t]$ be a separable polynomial of degree $\operatorname{deg} f\geq 1$. Let μ denote the Möbius function, let $\rho_f(m)$ denote the number of residue clas...

conjecture | open | MathDB 页面 | 原文线索

V2-005029 | MATHDB-360540

The conjectural double Fourier-Mellin transform formula over the complex numbers

摘录 / EXCERPT: Let p be an integer, let $\beta,\gamma,\nu\in\mathbb{C}$, and let $u,\varw\in\mathbb{C}$ satisfy ... For the Bessel function $\boldsymbol{B}_{\nu,p}$ and the associated function...

conjecture | open | MathDB 页面 | 原文线索

V2-005030 | MATHDB-360541

Conjecture on finitely many vanishing quadratic twists for higher-weight modular forms

摘录 / EXCERPT: Let f be a newform of level $\Gamma_0(N)$ and even weight k with $k\geq 6$, let π be the cuspidal automorphic representation of $\operatorname{PGL}_2(\mathbb{A})$ generated...

conjecture | open | MathDB 页面 | 原文线索

V2-005031 | MATHDB-360542

Conrey-Keating-Rubinstein-Snaith conjecture for prime quadratic twists

摘录 / EXCERPT: For a newform f and the cuspidal automorphic representation π of $\operatorname{PGL}_2(\mathbb{A})$ generated by f, let Δ_E be the discriminant of a quadratic field...

conjecture | open | MathDB 页面 | 原文线索

V2-005032 | MATHDB-360543

The possibility of non-square congruence subgroups for linear level polynomials

摘录 / EXCERPT: Non-square congruence subgroup conjecture. Both $\Gamma_1(\alpha T+\beta)$ and $\Gamma_0(\alpha T+\beta)$ may be non-square congruence subgroups for any choice of $\alpha\neq 0$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-005033 | MATHDB-360553

Finiteness conjecture for B-modular curves

摘录 / EXCERPT: Let F be a field, let B be a quaternion algebra over F, and let $g>1$. A curve C/F is B-modular if there is a Shimura curve X_K associated with a suitable compact open...

conjecture | open | MathDB 页面 | 原文线索

V2-005034 | MATHDB-360577

Finiteness conjecture for good quaternionic Shimura levels

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many open compact subgroups $K\subset B^\times(\mathbb A_{\mathrm f})$, respectively $K\subset PB^\times(\mathbb A_{\mathrm f})$, up...

conjecture | open | MathDB 页面 | 原文线索

V2-005035 | MATHDB-360580

Violet-prime embedding activities conjecture for the hypergraph Tutte polynomial

摘录 / EXCERPT: Let $\mathcal H$ be a hypergraph and let h be a hypertree. Define the order $<^{\mathrm{violet'}}_h$ from the tour of a violet Jaeger tree, and use this order to define embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-005036 | MATHDB-360584

The mod p Kurihara conjecture for elliptic curves

摘录 / EXCERPT: Fix a prime p and an elliptic curve $E/\mathbb{Q}$ of conductor N, with L-function coefficients $a_E(n)$, and assume that p is prime to the Manin constant c_E and that...

conjecture | open | MathDB 页面 | 原文线索

V2-005037 | MATHDB-360675

Frobenius's unicity conjecture for Markoff numbers

摘录 / EXCERPT: Let G_M be the graph whose vertices are the positive Markoff numbers, with edges arising from the Markoff transformations between coordinates of solutions to $x^2+y^2+z^2=3xyz$....

conjecture | open | MathDB 页面 | 原文线索

V2-005038 | MATHDB-360684

The Effective Linear Independence conjecture for Riemann zeta zeros

摘录 / EXCERPT: Effective Linear Independence conjecture. The above lower bound holds for all such T, ε, and integer coefficients. This is an effective quantitative strengthening of...

conjecture | open | MathDB 页面 | 原文线索

V2-005039 | MATHDB-360694

Neumann-Praeger bounded-degree conjecture for Kronecker classes

摘录 / EXCERPT: Let K/k be an extension of algebraic number fields of degree n, and let L/k be another extension that is Kronecker equivalent to K/k, meaning that K/k and L/k have the...

conjecture | open | MathDB 页面 | 原文线索

V2-005040 | MATHDB-360703

Uniqueness conjecture for a Stirling-number Diophantine equation

摘录 / EXCERPT: For integers n and m, consider the Diophantine equation ... The displayed Stirling-number identity shows that this equation arises from equating $\genfrac\{\}{0pt}{}{n}{n-3}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005041 | MATHDB-360704

Uniqueness conjecture for the equation referred to as diof2

摘录 / EXCERPT: Let n and m be the integer variables in the Diophantine equation denoted by ... (n,m)=(1,2),(3,4),text{ and }(5,16). ... is not included in the supplied context, so the precise mat...

conjecture | open | MathDB 页面 | 原文线索

V2-005042 | MATHDB-360711

The infinite sumset conjecture in the primes

摘录 / EXCERPT: Prime bounded finite-sums conjecture. For every $k\in\mathbb{N}$, there is an infinite set $B\subset\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005043 | MATHDB-360712

The ordered triple-sum conjecture in the primes

摘录 / EXCERPT: Ordered triple-prime sumset conjecture. There exist infinite sets $B,C,D\subset\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005044 | MATHDB-360713

The ordered two-set sumset conjecture in the primes

摘录 / EXCERPT: Ordered prime sumset conjecture. There exist such infinite sets B and C. The source presents this as an explicitly open ordered analogue of the prime bounded finite-sums conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-005045 | MATHDB-360717

The Dickson-Hardy-Littlewood prime-tuples conjecture

摘录 / EXCERPT: Dickson-Hardy-Littlewood conjecture. If $H\subset\mathbb{N}$ is finite and admissible, then there exist infinitely many $n\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005046 | MATHDB-360726

Fel'shtyn's rationality conjecture for Reidemeister zeta functions

摘录 / EXCERPT: Let Γ be a virtually polycyclic group and let $\varphi:\Gamma\to\Gamma$ be an endomorphism. When the Reidemeister zeta function is well-defined, it is given by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-005047 | MATHDB-360735

Complex-parameter extension of the Riemann Xi exponential-sum identity

摘录 / EXCERPT: Generalized Xi identity conjecture. For all $s_0\in\mathbb C$ and all $a\in\mathbb C$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005048 | MATHDB-360736

Generalized constraint identities for odd divisor functions

摘录 / EXCERPT: General constraint conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005049 | MATHDB-360737

Convergence of matrix inversion for divisor-function values

摘录 / EXCERPT: Matrix convergence conjecture. The first $M-1$ entries of

conjecture | open | MathDB 页面 | 原文线索

V2-005050 | MATHDB-360769

The algebraic-number sumset growth conjecture

摘录 / EXCERPT: The algebraic-number sumset growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005051 | MATHDB-360786

List-recoverability conjecture for Reed-Solomon codes over prime fields

摘录 / EXCERPT: Reed-Solomon list-recoverability conjecture. The number of polynomials $P(X) \in \mathbb{F}_p[X]$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-005052 | MATHDB-360787

Jutila-type gap conjecture for zeros of half-integral weight cusp-form L-functions

摘录 / EXCERPT: Jutila-type gap conjecture. For every $\epsilon>0$, there exists $T_0(\epsilon)$ such that, for all $T\geq T_0(\epsilon)$, $L(s,f)$ has a zero of the form

conjecture | open | MathDB 页面 | 原文线索

V2-005053 | MATHDB-360795

Super Strong Approximation for Markoff modulo-p graphs

摘录 / EXCERPT: For each prime p, let $\mathcal{G}_p$ be the corresponding Markoff modulo-p graph. Super Strong Approximation. The collection of Markoff modulo-p graphs $(\mathcal{G}_p)_p$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-005054 | MATHDB-360805

Conjecture on the compositeness of $1+3\cdot 2^{20n}$

摘录 / EXCERPT: Let n be a nonnegative integer, and interpret “$n=0$ or 1 mod 3” as $n\equiv 0$ or $1\pmod{3}$. Compositeness conjecture. The integer ... is not prime for every $n\equiv 0$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-005055 | MATHDB-360838

The rogue divisor conjecture for Cunningham chains

摘录 / EXCERPT: The rogue divisor conjecture. The least prime divisor of k is smaller than p:

conjecture | open | MathDB 页面 | 原文线索

V2-005056 | MATHDB-360839

The logarithmic Cunningham-chain bound beyond 5

摘录 / EXCERPT: Let $\mathbb{P}$ denote the primes, let $i\in\{1,2\}$, and let $C_i[p]$ be the Cunningham chain associated with p. The logarithmic Cunningham-chain bound. For every prime $p>5$,...

conjecture | open | MathDB 页面 | 原文线索

V2-005057 | MATHDB-360841

Ash's mod-ell conjecture on Galois representations from Hecke eigenclasses

摘录 / EXCERPT: Ash's conjecture. If $frak m\subset\mathbb T^S(K,\lambda)$ is maximal, then there exists a unique continuous semisimple representation

conjecture | open | MathDB 页面 | 原文线索

V2-005058 | MATHDB-360844

The GUE Hypothesis for the rescaled ordinates of zeta zeros

摘录 / EXCERPT: Let $\rho_n=\sigma_n+i\gamma_n$ be the nontrivial zeros of $\zeta(s)$, listed with multiplicity, and let $\mathbb{Z}_*^k$ denote the k-tuples of distinct indices. For a random or...

conjecture | open | MathDB 页面 | 原文线索

V2-005059 | MATHDB-360845

The conjecture for the variance of squarefull numbers in short intervals

摘录 / EXCERPT: Variance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005060 | MATHDB-360849

Goh-Wildberger factorization conjecture for zpread polynomials

摘录 / EXCERPT: Goh-Wildberger conjecture. There are polynomials $\Phi_d(x)\in\mathbb{Z}[x]$, $d\geq 1$, such that, for every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005061 | MATHDB-360864

Uniform polynomial Chowla conjecture over finite fields

摘录 / EXCERPT: Let q be a fixed power of an odd prime and let $c>0$ be fixed. For any polynomial $F\in\mathbb{F}_q[x][T]$ that is separable in the variable T, define the Liouville function by...

conjecture | open | MathDB 页面 | 原文线索

V2-005062 | MATHDB-360895

The infinitude of Fibonacci primes

摘录 / EXCERPT: The Fibonacci numbers are the sequence $(\phi_j)_{j\geq 0}$ defined by $\phi_0=0$, $\phi_1=1$, and $\phi_{j+1}=\phi_j+\phi_{j-1}$ for $j\geq 1$. Infinitude of Fibonacci primes. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005063 | MATHDB-360907

Directional maximum conjecture for logarithmic derivatives of Dirichlet L-functions

摘录 / EXCERPT: Let q tend to infinity through the primes. For a prime q, let χ_0 denote the principal character modulo q, and let $L(s,\chi)$ be the Dirichlet L-function associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-005064 | MATHDB-360908

Directional maximum conjecture for the logarithmic derivative of the Riemann zeta function

摘录 / EXCERPT: Fix $\epsilon\in(0,\frac12)$. For $T>0$, consider the interval $[T,2T]$, and let $\zeta(s)$ denote the Riemann zeta function. Directional maximum conjecture. As $T\to\infty$, ... uni...

conjecture | open | MathDB 页面 | 原文线索

V2-005065 | MATHDB-360909

Pointwise upper-bound conjecture for the logarithmic derivative of the Riemann zeta function

摘录 / EXCERPT: Fix $\sigma\in(\frac12,1)$, and let $\zeta(s)$ denote the Riemann zeta function. Pointwise upper-bound conjecture. For all sufficiently large t, ... The source lists this as a conj...

conjecture | open | MathDB 页面 | 原文线索

V2-005066 | MATHDB-360910

Pointwise upper-bound conjecture for logarithmic derivatives of Dirichlet L-functions

摘录 / EXCERPT: Fix $\sigma\in(\frac12,1)$. Let q be a sufficiently large integer, let χ be a primitive character modulo q, and let $L(s,\chi)$ be its Dirichlet L-function. Pointwise up...

conjecture | open | MathDB 页面 | 原文线索

V2-005067 | MATHDB-360911

Central-line maximal logarithmic derivative conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let q tend to infinity through the primes. For a prime q, let χ_0 be the principal character modulo q, and let $L(s,\chi)$ be the Dirichlet L-function associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-005068 | MATHDB-360922

The classification conjecture for unit reducible cyclotomic fields

摘录 / EXCERPT: Let $K_N=\mathbb{Q}(\zeta_N)$ be the cyclotomic field of conductor N, where ζ_N is a primitive Nth root of unity. A cyclotomic field is unit reducible if it has the unit-...

conjecture | open | MathDB 页面 | 原文线索

V2-005069 | MATHDB-360923

Uniform subpolynomial bound conjecture for integral plane curves

摘录 / EXCERPT: Let $f\in {\mathbb Z}[x,y]$ have degree d, and write f_d for its homogeneous part of highest degree. Say that f_d is 1-irreducible when it satisfies the source's notion of...

open_problem | open | MathDB 页面 | 原文线索

V2-005070 | MATHDB-360976

Lagarias's periodicity conjecture for the 2-adic Collatz map

摘录 / EXCERPT: Let $\mathbb{Z}_2$ be the ring of 2-adic integers, let $\overline{T}\colon\mathbb{Q}_2\to\mathbb{Q}_2$ be given by ... where at each iterate v is the residue modulo 2 of the cu...

conjecture | open | MathDB 页面 | 原文线索

V2-005071 | MATHDB-360988

The square-digit set conjecture for digit-restricted denominators

摘录 / EXCERPT: Square-digit set conjecture. The set $\mathfrak{D}_{b,2}$ is an $\mathcal{H}$-set. This extends the paper's investigation of digit-restricted denominators to squares of such denomi...

conjecture | open | MathDB 页面 | 原文线索

V2-005072 | MATHDB-360989

Exactness conjecture for quantum Drinfeld-Sokolov reduction functors

摘录 / EXCERPT: Let $p,q\geq 2$ be relatively prime. Let $\mathcal{F}_{p,q}: KL^{-2+p/q}(\mathfrak{sl}_2)\to\mathcal{O}_{c_{p,q}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005073 | MATHDB-361006

Rational spherical triangle measurement conjecture

摘录 / EXCERPT: Let a proper spherical triangle on the unit sphere have side lengths and area that are rational multiples of π, and let its measurement be the tuple (E,a,b,c), with...

conjecture | open | MathDB 页面 | 原文线索

V2-005074 | MATHDB-361007

Generalized L'Huilier equation classification conjecture

摘录 / EXCERPT: Generalized L'Huilier equation classification conjecture. The tuple (x_0,x_1,x_2,x_3,x_4) is a solution if and only if it is either in the infinite set Φ or in the set...

conjecture | open | MathDB 页面 | 原文线索

V2-005075 | MATHDB-361025

Galois Alperin weight conjecture

摘录 / EXCERPT: Fix a prime p and a finite group G. Let $\mathcal{H}=\operatorname{Gal}(\mathcal{K}/\mathbb{Q}_p)$ be the finite abelian group of Galois automorphisms associated with the p-m...

conjecture | open | MathDB 页面 | 原文线索

V2-005076 | MATHDB-361026

Galois Alperin weight conjecture for a finite group

摘录 / EXCERPT: Fix a prime p and a finite group G. Let $\mathcal{H}$ be the Galois automorphism group associated with the p-modular system in the source. Let $\mathcal{W}(G)$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-005077 | MATHDB-361044

Krashen's period-index conjecture from diophantine dimension

摘录 / EXCERPT: Let F be a field, let d be an integer, and let ℓ be the prime appearing in the cohomology group. The diophantine dimension $\operatorname{dd}(F)$ is the smallest integer...

conjecture | open | MathDB 页面 | 原文线索

V2-005078 | MATHDB-361051

Conjecture on the lower density of $\mathcal{W}_2$

摘录 / EXCERPT: Let $\mathcal{W}_2$ be the set of all positive integers a for which there is no integer $m>1$ such that $(a-2^k,m)>1$ for every positive integer k. The $\mathcal{W}_2$ density...

conjecture | open | MathDB 页面 | 原文线索

V2-005079 | MATHDB-361053

The formal multiple Eisenstein subalgebra conjecture

摘录 / EXCERPT: Let $\mathcal{E}^{\!f}$ be the subspace of $\operatorname{Fil}^{\mathrm{lwt}}_0\mathcal{G}^{\!f}$ spanned by $G^{\!f}(k_1,\dots,k_r)$ with $r\geq0$ and $k_1,\dots,k_r\geq2$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-005080 | MATHDB-361065

Greenberg's Iwasawa main conjecture for the adjoint of a modular form

摘录 / EXCERPT: Let f be the modular form under consideration, let $V=\operatorname{ad}(\rho_{f,v})$ be its 3-dimensional adjoint Galois representation, let T be a $G_{\mathbf{Q}}$-stable...

conjecture | open | MathDB 页面 | 原文线索

V2-005081 | MATHDB-361073

Classification conjecture for deep holes of extended MDS codes

摘录 / EXCERPT: Let q be a power of a prime, $(q-1)/2\le n-k$, and $k<n\le q$. Let ${\bf a}=(\alpha_1,\ldots,\alpha_n)$ and let ${\bf w}$ denote the multiplier vector from the associated MDS code...

conjecture | open | MathDB 页面 | 原文线索

V2-005082 | MATHDB-361074

ZWK conjecture on the covering radius of projective Reed-Solomon codes

摘录 / EXCERPT: Let q be a prime power and let $\operatorname{PRS}(k)$ denote the projective Reed-Solomon code of dimension k and length $q+1$. For $2\le k\le q-2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-005083 | MATHDB-361089

Clique-covering bound for relation graphs of shifted quadratic ideals

摘录 / EXCERPT: Let $d,k\in\mathbb{N}_{+}$, $s\in\mathbb{N}$, p be a prime, and let $M\colon\mathbb{F}_{p}^{d}\to\mathbb{F}_{p}$ be a non-degenerate quadratic form. Let G be the relation graph...

conjecture | open | MathDB 页面 | 原文线索

V2-005084 | MATHDB-361102

The two-term supercongruence conjecture for the original sporadic sequences

摘录 / EXCERPT: Two-term supercongruence conjecture. For all primes $p\geq 5$ and all integers $m,r\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005085 | MATHDB-361103

Uniqueness conjecture for generalized Markov triples

摘录 / EXCERPT: Let $k\in\mathbb Z_{\geq 0}$. A k-generalized Markov number is a number appearing in a k-generalized Markov triple. Uniqueness conjecture for generalized Markov triples. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-005086 | MATHDB-361115

The NCHO quasi-partition-function conjecture

摘录 / EXCERPT: NCHO quasi-partition-function conjecture. The following are expected: (1) $\widetilde{Z}_Q(t)=Z_Q(t)$; and (2), under (1), if

conjecture | open | MathDB 页面 | 原文线索

V2-005087 | MATHDB-361116

The quasi-partition-function conjecture for quantum systems

摘录 / EXCERPT: Let H be a Hamiltonian defining a quantum system, let $Z_H(t)$ be its partition function, and let $\widetilde{Z}_H(t)$ be its quasi-partition function as defined from the special...

conjecture | open | MathDB 页面 | 原文线索

V2-005088 | MATHDB-361131

Hildebrand's consecutive stable-set intersection conjecture

摘录 / EXCERPT: Hildebrand's conjecture. If $\mathcal{S}$ has positive lower density, then for every natural number k,

conjecture | open | MathDB 页面 | 原文线索

V2-005089 | MATHDB-361132

Hildebrand's nonemptiness conjecture for consecutive stable-set intersections

摘录 / EXCERPT: Hildebrand's nonemptiness conjecture. If $\mathcal{S}$ has positive lower density, then for every natural number k,

conjecture | open | MathDB 页面 | 原文线索

V2-005090 | MATHDB-361138

The lower uniform-exponent bound for the approximation discrepancy

摘录 / EXCERPT: Lower discrepancy conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-005091 | MATHDB-361152

Ramanujan-Petersson conjecture for weight-2 eigenforms over number fields

摘录 / EXCERPT: Let L be a number field. Let $\mathfrak{f}$ be a weight-2 complex eigenform over L of level $\mathfrak{N}$, and let $\mathfrak{p}\nmid\mathfrak{N}$ be a prime ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-005092 | MATHDB-361153

Non-degenerate odd Egyptian-fraction relation bound

摘录 / EXCERPT: Let $1\leq n\leq Q$ and let $k\geq 3$ be odd. A solution is a tuple $(a_1,q_1,\ldots,a_k,q_k)$ satisfying ... with $\gcd(a_i,q_i)=1$, $|a_i|\leq n$, and $q_i\leq Q$ for every i. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-005093 | MATHDB-361192

Monotonicity of (k,β)-Poissonian box correlations

摘录 / EXCERPT: Monotonicity conjecture. If x has (k,β_2)-Poissonian box correlations, then it also has (k,β_1)-Poissonian box correlations.

conjecture | open | MathDB 页面 | 原文线索

V2-005094 | MATHDB-361196

The congruence subgroup conjecture for $\Delta_{r/p}$

摘录 / EXCERPT: Let p be prime, let $r\in\mathbb{Z}$ satisfy $\gcd(r,p)=1$, and suppose that $r/p\in\mathbb{Q}\cap(0,4)$. Let ... Let $\overline{\Gamma}_1^{(p)}(r)$ denote the subgroup generated b...

conjecture | open | MathDB 页面 | 原文线索

V2-005095 | MATHDB-361198

The conjectures on relation numbers and congruence structure of $\Delta_{r/p}$

摘录 / EXCERPT: Let p be prime and $r\in\mathbb{Z}$ satisfy $\gcd(r,p)=1$, with r/p admissible as defined in the source. Let $\Delta_{r/p}$ be generated by the two parabolic matrices A and...

conjecture | open | MathDB 页面 | 原文线索

V2-005096 | MATHDB-361203

Bounded-height record-setter asymptotic conjecture for the function z

摘录 / EXCERPT: Bounded-height asymptotic conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-005097 | MATHDB-361204

Primary pseudoperfect strongly pseudoperfect uniqueness conjecture

摘录 / EXCERPT: A primary pseudoperfect number is a positive integer n satisfying ... where the sum is over the prime divisors p of n. A strongly pseudoperfect number is understood in the sens...

conjecture | open | MathDB 页面 | 原文线索

V2-005098 | MATHDB-361205

Weak primary pseudoperfect strongly pseudoperfect uniqueness conjecture

摘录 / EXCERPT: A weak primary pseudoperfect number is a positive integer n satisfying ... where the sum is over the prime divisors of n. A strongly pseudoperfect number is understood in the sen...

conjecture | open | MathDB 页面 | 原文线索

V2-005099 | MATHDB-361206

Finiteness conjecture for strongly pseudoperfect μ-Sondow numbers

摘录 / EXCERPT: For an integer μ, a μ-Sondow number is a positive integer n such that ... where the sum is over the prime divisors p of n. A strongly pseudoperfect number is understood...

conjecture | open | MathDB 页面 | 原文线索

V2-005100 | MATHDB-361209

The linear color function equals the DP color function for prime powers

摘录 / EXCERPT: Let k be a power of a prime. For a graph G, let $P_{\mathcal{L}_k}(G,k)$ be the minimum number of colorings over all $\mathcal{L}_k$-labelings, where $\mathcal{L}_k$ is the col...

conjecture | open | MathDB 页面 | 原文线索

V2-005101 | MATHDB-361213

Conjecture on odd-characteristic BCH codes with prescribed dual distance and locality

摘录 / EXCERPT: Let $q=p^s$, where p is an odd prime and $s\ge 2$, and set $h=\frac{p-1}{2}$. Let $\mathcal{C}_{(q,q+1,3,h)}$ denote the BCH code in question. BCH-code conjecture. The code...

conjecture | open | MathDB 页面 | 原文线索

V2-005102 | MATHDB-361214

Conjectured congruences for overpartition tuples with odd parts

摘录 / EXCERPT: Let $\overline{OPT}_k(n)$ denote the number of overpartition k-tuples with odd parts of size n. Let $i\geq 1$ and let r be odd, with $n\geq 0$ and $n,i\in\mathbb{Z}$. Conject...

conjecture | open | MathDB 页面 | 原文线索

V2-005103 | MATHDB-361215

Granville-Soundararajan conjecture on character sums and smooth numbers

摘录 / EXCERPT: Granville-Soundararajan conjecture. There exists a constant $A>0$ such that, for every non-principal character chi modulo q and every $1\leq x\leq q$, uniformly,

conjecture | open | MathDB 页面 | 原文线索

V2-005104 | MATHDB-361216

Conjecture on the maximal order of derivatives of the Riemann zeta function

摘录 / EXCERPT: Maximal-order conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005105 | MATHDB-361224

Modularity and topological invariance conjecture for completed numerator series

摘录 / EXCERPT: Let R be a plane curve germ over $\mathbb{C}$, and let $\widehat{\mathit{NZ}}_R(t)$ denote its completed numerator series. Modularity and topological invariance conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-005106 | MATHDB-361239

The quasi-modularity conjecture for Hodge products on an elliptic curve

摘录 / EXCERPT: Let E be an elliptic curve, let $\overline{M}_{g,n}$ be the moduli space of stable curves, and let $\mathcal C_g(\gamma_1,\ldots,\gamma_n)$ denote the class defined in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-005107 | MATHDB-361262

Paley graph conjecture for character sums over additive pairs

摘录 / EXCERPT: Let $epsilon>0$ be a real number. For subsets $A,B e\mathbb{F}_p$?

conjecture | open | MathDB 页面 | 原文线索

V2-005108 | MATHDB-361271

Two-summand criterion for Laurent series over semidomains

摘录 / EXCERPT: Let S be an additively reduced and additively atomic semidomain, and let $cmathcal{A}_{+}(S)$ denote the set of additive atoms of S and $S^{\times}$ its multiplicative unit gro...

conjecture | open | MathDB 页面 | 原文线索

V2-005109 | MATHDB-361279

Asymptotic formula for the totient sum over prime quotients

摘录 / EXCERPT: Asymptotic formula.

conjecture | open | MathDB 页面 | 原文线索

V2-005110 | MATHDB-361313

Cho's specialized intersection formula conjecture

摘录 / EXCERPT: Let $\{x_1,\dots,x_{n+h},y_1,\dots,y_{n-h}\}$ be a basis of $\mathbb{V}$ with moment matrix ... where T is an $n\times n$ matrix, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005111 | MATHDB-361326

The finite-place/closed-geodesic analogy for Dedekind zeta functions

摘录 / EXCERPT: Finite-place/closed-geodesic analogy. Finite places $\mathfrak{p}$ in the explicit formula for the Dedekind zeta function are closed geodesics $\gamma_{\mathfrak{p}}$ in the corres...

open_problem | open | MathDB 页面 | 原文线索

V2-005112 | MATHDB-361327

The exponent pair conjecture for monomial-phase exponential sums

摘录 / EXCERPT: Exponent pair conjecture. For every $\epsilon>0$, the pair $(\epsilon,\frac{1}{2}+\epsilon)$ is an exponent pair. This is a central conjecture in analytic number theory governing c...

conjecture | open | MathDB 页面 | 原文线索

V2-005113 | MATHDB-361331

Hardy-Littlewood asymptotic conjecture for BCC primes with $p\equiv1\pmod4$

摘录 / EXCERPT: BCC-prime asymptotic conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005114 | MATHDB-361337

The dimension conjecture for the losing set of lower digit frequencies

摘录 / EXCERPT: Let D^-_{c} denote the set of real numbers whose lower frequency of the digit 0 is greater than c, and let $D(D^-_{c})$ denote its set of Schmidt-game parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-005115 | MATHDB-361338

The winning criterion for the ternary badly approximable set

摘录 / EXCERPT: Let $alpha,beta$ be Schmidt-game parameters, and let $gamma_A$ and $gamma_B$ denote the auxiliary constants associated with Alice's and Bob's moves. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005116 | MATHDB-361339

The strong approximation conjecture for the Markoff graph

摘录 / EXCERPT: Let $\mathcal{M}$ be the set of Markoff triples over $\mathbb{Z}$, let $\mathcal{M}_p$ be the set of their reductions modulo a prime p, and let $\mathcal{G}^\times_p$ be the Mark...

conjecture | open | MathDB 页面 | 原文线索

V2-005117 | MATHDB-361355

Conjecture on squares in binary recurrence sequences

摘录 / EXCERPT: Let a, b, and d be positive integers with d nonsquare, let $\alpha=a+b^{2}\sqrt{d}$ have norm ... and let $\varepsilon=(t+u\sqrt{d})/2$ be a unit in...

conjecture | open | MathDB 页面 | 原文线索

V2-005118 | MATHDB-361356

Generalized conjecture on squares in the associated recurrence sequence

摘录 / EXCERPT: Let a, b, and d be positive integers with d nonsquare, let $\alpha=a+b^{2}\sqrt{d}$ have norm $N_{\alpha}=a^{2}-b^{4}d$, and let $\varepsilon=(t+u\sqrt{d})/2$ be a unit in...

conjecture | open | MathDB 页面 | 原文线索

V2-005119 | MATHDB-361407

Arithmetic-progression conjecture for Collatz stopping times

摘录 / EXCERPT: Let ${\mathcal{C}}$ be the classical Collatz map, and let the stopping time of an integer be the least iterate at which it reaches 1. For fixed integers $a,b\geq 0$, consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-005120 | MATHDB-361410

Parity congruence conjecture for finite Mordell--Tornheim sums

摘录 / EXCERPT: Let $r\in\N$, let p be a prime, and let $\bfs\in\N^d$ satisfy $p>|\bfs|+1$, where $|\bfs|$ is the weight and $Z_{p^r}(\bfs)$ is the corresponding finite Mordell--Tornheim sum. Fi...

conjecture | open | MathDB 页面 | 原文线索

V2-005121 | MATHDB-361440

The weak Hall-Lang conjecture for integral points on elliptic curves

摘录 / EXCERPT: Let E be the elliptic curve ... where $A,B\in\mathbb{Z}$ satisfy $4A^3+27B^2\neq 0$, and let $(x,y)\in E(\mathbb{Z})$ be an integral point. Weak Hall-Lang conjecture. There exist c...

conjecture | open | MathDB 页面 | 原文线索

V2-005122 | MATHDB-361441

The finite-codimension conjecture for prime ideals of Iwasawa algebras

摘录 / EXCERPT: Finite-codimension conjecture. If G is an almost simple uniform pro-p group, then every non-zero prime ideal of KG has finite codimension.

conjecture | open | MathDB 页面 | 原文线索

V2-005123 | MATHDB-361453

Irrationality conjecture for the roots associated with an infinite family of horospherical cones

摘录 / EXCERPT: For each integer $r\geq 5$, define ... The equation $Q(t,r)=0$ has a unique positive root. Irrationality conjecture. For all $r\geq 5$, this unique positive root is irrational. This...

conjecture | open | MathDB 页面 | 原文线索

V2-005124 | MATHDB-361454

Sah's radical conjecture for polynomial least common multiples

摘录 / EXCERPT: Sah's conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005125 | MATHDB-361469

Chapman's evil determinant conjecture

摘录 / EXCERPT: Chapman's evil determinant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005126 | MATHDB-361501

Intersective sets contain a difference of two primes

摘录 / EXCERPT: Intersective prime-difference conjecture. For every intersective set R,

conjecture | open | MathDB 页面 | 原文线索

V2-005127 | MATHDB-361529

Cinarcı-Keller conjecture on the minimum number of conjugacy classes

摘录 / EXCERPT: Let p be a prime. Let a and b be positive integers such that ... and such that $|a-b|$ is minimal. For a finite group G, let $k(G)$ denote the number of conjugacy classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-005128 | MATHDB-361538

Weakly consecutive sequence generation conjecture

摘录 / EXCERPT: A weakly consecutive sequence (WCS) is a permutation σ of $\{1,\ldots,k\}$ such that if an integer d divides $\sigma(i)$, then d also divides $\sigma(i\pm d)$ wherever t...

conjecture | open | MathDB 页面 | 原文线索

V2-005129 | MATHDB-361539

Endpoint-1 conjecture for weakly consecutive sequences

摘录 / EXCERPT: Let σ be a weakly consecutive sequence, meaning a permutation of $\{1,\ldots,k\}$ such that if d divides $\sigma(i)$, then d also divides $\sigma(i\pm d)$ wherever these...

conjecture | open | MathDB 页面 | 原文线索

V2-005130 | MATHDB-361540

Boundary-location conjecture for small entries in weakly consecutive sequences

摘录 / EXCERPT: Let σ be a weakly consecutive sequence, and let $m<k/2$. Boundary-location conjecture. Each m appears among either the first m or the last m terms of the sequence; equ...

conjecture | open | MathDB 页面 | 原文线索

V2-005131 | MATHDB-361541

Starting Sequence 4 dominance conjecture for weakly consecutive sequences

摘录 / EXCERPT: A weakly consecutive sequence (WCS) is a permutation of $\{1,\ldots,k\}$ satisfying the divisibility condition defining WCSs. The paper distinguishes four starting sequences and ru...

conjecture | open | MathDB 页面 | 原文线索

V2-005132 | MATHDB-361542

Density conjecture for minimal weakly consecutive sequence counts

摘录 / EXCERPT: Let $N(k)$ be the number of weakly consecutive sequences of length k. Minimal-count density conjecture. The value $N(k)=2$, which is minimal for $k>1$, occurs for infinitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-005133 | MATHDB-361602

Patterson's asymptotic conjecture for cubic exponential sums

摘录 / EXCERPT: Let p run through the primes with $p\equiv 1\pmod 3$, and define ... where $e(y)=e^{2\pi i y}$. Let ... where Γ is the classical gamma function. Patterson's conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-005134 | MATHDB-361603

Zhang's moment conjecture for quadratic Gauss sums weighted by L-functions

摘录 / EXCERPT: Let p be prime, let n be an integer, and for a Dirichlet character $\chi\bmod p$ define the generalized quadratic Gauss sum ... where $e(y)=e^{2\pi i y}$. Let χ_0 denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-005135 | MATHDB-361604

Bailey-Keating conjecture for moments of moments of the Riemann zeta function

摘录 / EXCERPT: Let $k,\gamma\in\mathbb{N}$ and $c>0$ be fixed, and define ... Let $a_{k,\gamma}$ be the arithmetic constant and $g_{k,\gamma}$ the geometric constant. Bailey-Keating conjecture. ... M...

conjecture | open | MathDB 页面 | 原文线索

V2-005136 | MATHDB-361612

Sun's quadratic-character conjecture for $D_p^{(3)}$

摘录 / EXCERPT: For positive integers m,n with n odd, define ... For primes $p\equiv1\pmod4$, this determinant is an integer square, so choose a square root $\sqrt{D_p^{(3)}}$. Sun's quadratic-c...

conjecture | open | MathDB 页面 | 原文线索

V2-005137 | MATHDB-361613

Finiteness conjecture for vanishing determinants $D_p^{(m)}$

摘录 / EXCERPT: For positive integers m,n with n odd, define ... For each positive odd integer m, let ... Finiteness conjecture. For every positive odd integer m, the set $E(m)$ is finite. In...

conjecture | open | MathDB 页面 | 原文线索

V2-005138 | MATHDB-361614

Finiteness of paths in the comma successor graph

摘录 / EXCERPT: For each base $b\geq 3$, define the successor graph G_s to have one node for every positive integer and a directed edge from i to j when j is the successor to i. A path h...

conjecture | open | MathDB 页面 | 原文线索

V2-005139 | MATHDB-361615

Ternary rule for comma-numbers

摘录 / EXCERPT: Write a positive integer n in base 3 as $n=d_1d_2\ldots d_m$, with digits $d_i\in\{0,1,2\}$, and let $cn(n)$ denote its comma-number. The notation d_md_1 denotes the two-digi...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005140 | MATHDB-361632

The quaternionic Maass Spezialschar Dirichlet-series conjecture for split \(\mathrm{SO}(8)\)

摘录 / EXCERPT: Let φ be a level-one cuspidal quaternionic modular form, let Π be the irreducible automorphic representation generated by φ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005141 | MATHDB-361636

Prime numerator Zaremba conjecture

摘录 / EXCERPT: Let $q>1$. Call a prime residue $p\in(\mathbb{Z}/q\mathbb{Z})^\times$ k-Zaremba for q if, in its continued-fraction expansion $p/q=[a_1,\ldots,a_n]$, one has $a_i\leq k$ for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-005142 | MATHDB-361651

The Bessenrodt-Ono type inequality for general A-partition functions

摘录 / EXCERPT: Bessenrodt-Ono type conjecture. For all sufficiently large values of a and b,

conjecture | open | MathDB 页面 | 原文线索

V2-005143 | MATHDB-361671

Gauss's class-number conjecture for imaginary quadratic fields

摘录 / EXCERPT: For an imaginary quadratic field K, let D_K and $h(K)$ denote its discriminant and class number, respectively. Gauss's class-number conjecture. Then ... This conjecture asserts t...

conjecture | open | MathDB 页面 | 原文线索

V2-005144 | MATHDB-361684

Prime-value conjecture for the periodic-continued-fraction polynomials

摘录 / EXCERPT: Let $f_n(x)$ be the polynomial sequence defined earlier in the paper, and let a be a positive integer. Prime-value conjecture. For every positive integer a, there are infinitel...

conjecture | open | MathDB 页面 | 原文线索

V2-005145 | MATHDB-361685

Baer-Suzuki conjecture for commutators of prime-order elements

摘录 / EXCERPT: Commutator Baer-Suzuki conjecture. If $[x,g]$ is either 1 or p-singular for every element $g\in G$, then $x\in O_p(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-005146 | MATHDB-361698

Three-rational-incidence saving conjecture

摘录 / EXCERPT: Three-rational-incidence saving conjecture. The number of such solutions is

conjecture | open | MathDB 页面 | 原文线索

V2-005147 | MATHDB-361710

The higher congruence characterization for finite simple groups of Lie type

摘录 / EXCERPT: Let G be a finite simple group of Lie type, let p be a prime different from the defining characteristic of G, and let m be a positive integer. For elements $g,g'\in G$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-005148 | MATHDB-361714

Conjectured evaluation of the Rogers-Ramanujan continued fraction

摘录 / EXCERPT: Conjectured evaluation. The evaluation

conjecture | open | MathDB 页面 | 原文线索

V2-005149 | MATHDB-361715

Transcendence conjecture for the Stirling-Ramanujan constants

摘录 / EXCERPT: Stirling-Ramanujan transcendence conjecture. The Stirling-Ramanujan constants are transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-005150 | MATHDB-361716

Transcendence conjecture for the Upsilon constants

摘录 / EXCERPT: Upsilon transcendence conjecture. Upsilon constants Υ_n are transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-005151 | MATHDB-361731

Refined Kim's conjecture for the thrice-punctured line

摘录 / EXCERPT: Let S be a finite set of primes, let $X=\mathbb P^1\smallsetminus\{0,1,\infty\}$, and let $p\notin S$ be a prime. For each depth $n\geq 1$, write $X(\mathbb Z_p)_{S,n}^{\min}$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-005152 | MATHDB-361732

Σ-refined Kim's conjecture for the thrice-punctured line

摘录 / EXCERPT: Let S be a finite set of primes, let $X=\mathbb P^1\smallsetminus\{0,1,\infty\}$, and let $p\notin S$ be a prime. A refinement condition is a tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-005153 | MATHDB-361749

The Scott-Styer conjecture on solutions to a ternary purely exponential Diophantine equation

摘录 / EXCERPT: Let a, b, and c be fixed coprime positive integers with >1 and let $N(a,b,c)$ denote the number of solutions in positive integers (x,y,z) to … Assume that a, b, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005154 | MATHDB-361807

Infinitude conjecture for cubic sums and fifth powers

摘录 / EXCERPT: Infinitude conjecture. For each choice of the signs, the equation

conjecture | open | MathDB 页面 | 原文线索

V2-005155 | MATHDB-361808

Polynomial solution conjecture for cubic sums and fourth or sixth powers

摘录 / EXCERPT: Polynomial solution conjecture. For each choice of the sign, there is a solution of

conjecture | open | MathDB 页面 | 原文线索

V2-005156 | MATHDB-361832

The weighted projective Serre bound conjecture

摘录 / EXCERPT: Let q be a prime power, let $a_0,\ots,a_n$ be positive weights, and let $e_q(d;a_0,a_1,\ots,a_n)$ be the maximum number of ${\mathbb F}_q$-rational points on a hypersurface of weig...

conjecture | open | MathDB 页面 | 原文线索

V2-005157 | MATHDB-361835

Growth conjecture for partial sums of the triangular-number Möbius function

摘录 / EXCERPT: Let $\mathcal T(i)=\frac12 i(i+1)$ be the i-th triangular number, let $i\leq_{\mathcal T}j$ mean that $\mathcal T(i)$ divides $\mathcal T(j)$, and let $\mu_{\mathcal T}$ be the M...

conjecture | open | MathDB 页面 | 原文线索

V2-005158 | MATHDB-361839

Frey's ABC conjecture for number fields

摘录 / EXCERPT: Let K be a number field with ring of integers $\mathcal{O}_K$ and discriminant Δ_K. For $\epsilon>0$ and a non-zero integral ideal $\mathfrak{a}$ of $\mathcal{O}_K$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-005159 | MATHDB-361841

Arthur's multiplicative formula conjecture for orbital integrals in \(\operatorname{GL}(2)\)

摘录 / EXCERPT: Let $K=\mathbb{Q}$, let γ be a contributing matrix, and let S_{γ} and D_{γ} be the positive integer and fundamental discriminant associated with γ as in...

conjecture | open | MathDB 页面 | 原文线索

V2-005160 | MATHDB-361847

The Brauer-Galois height zero conjecture for principal p-blocks

摘录 / EXCERPT: Let G be a finite group, let p be a prime number, and let $P\in\operatorname{Syl}_p(G)$. For the subgroup $\mathcal{J}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-005161 | MATHDB-361855

Distinct Frobenius-trace tuples for the shifted elliptic curves

摘录 / EXCERPT: Let p be prime and let $u\in\mathbb F_p$ be a parameter for which the 15 elliptic curves obtained from the five elliptic-curve factors and the shifts $u\mapsto u-22$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005162 | MATHDB-361873

The Ankeny-Artin-Chowla conjecture for real quadratic fields

摘录 / EXCERPT: Let $d\in\mathbb{N}_{\geq 2}$ be squarefree, let $K=\mathbb{Q}(\sqrt{d})$, and let $\mathcal{O}_K=\mathbb{Z}[\omega]$ be the ring of integers of K, where ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005163 | MATHDB-361915

Hikami's radial-limit conjecture for false theta functions

摘录 / EXCERPT: For $n\geq 0$, let $(x;q)_n=(1-x)(1-xq)\cdots(1-xq^{n-1})$ be the q-Pochhammer symbol, and let $\begin{bmatrix}n\\k\end{bmatrix}_q$ denote the q-binomial coefficient. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005164 | MATHDB-361916

The level of distribution one conjecture for shifted primes

摘录 / EXCERPT: Let ϑ denote the level of distribution of the shifted primes. Level of distribution one conjecture. The shifted primes have level of distribution ... A level of distributio...

conjecture | open | MathDB 页面 | 原文线索

V2-005165 | MATHDB-361926

Kudla's arithmetic modularity conjecture

摘录 / EXCERPT: Kudla's arithmetic modularity conjecture. The formal generating function $Z(g,\varphi)_K$ converges absolutely and defines an element in

conjecture | open | MathDB 页面 | 原文线索

V2-005166 | MATHDB-361950

Ihara's growth conjecture for Euler-Kronecker constants of cyclotomic fields

摘录 / EXCERPT: For m sufficiently large, let ${\mathbb Q}(\zeta_m)$ denote the m-th cyclotomic field and let $\gamma_{{\mathbb Q}(\zeta_m)}$ be its Euler-Kronecker constant. Ihara's growth co...

conjecture | open | MathDB 页面 | 原文线索

V2-005167 | MATHDB-361952

Conjecture on friable neighboring integers

摘录 / EXCERPT: For $u,v\geq 1$, let ... Here $P^+(m)$ denotes the largest prime factor of m, and ϱ is the Dickman function. Neighboring friable integers conjecture. For all $u,v\geq 1$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-005168 | MATHDB-361953

Elliott-Halberstam conjecture for friable integers

摘录 / EXCERPT: Let $\varepsilon\in(0,1)$, and let $v=\log x/\log z$ be fixed. Write $\Psi(x',z;a,q)$ for the count of z-friable integers up to x' in the reduced residue class $a\pmod q$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005169 | MATHDB-361957

Bloch-Kato conjecture for geometric irreducible representations

摘录 / EXCERPT: Let p be a prime, let cK be a p-adic field, and let V be a geometric irreducible representation of $\mathrm{Gal}(\overline{\mathbf Q}/\mathbf Q)$ over $\mathcal K$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005170 | MATHDB-361969

Kummer's asymptotic conjecture for the relative class number

摘录 / EXCERPT: Let q be a prime number, and let $h_1(q)$ denote the relative class number. Define ... The ratio $R(q)$ is called the Kummer ratio. Kummer's conjecture. As q tends to infinity, ...

conjecture | open | MathDB 页面 | 原文线索

V2-005171 | MATHDB-361981

Quandle-ring cocycle enhancement for prime knots up to 12 crossings

摘录 / EXCERPT: Let X be a quandle and let $\phi:X\times X\to A$ be a 2-cocycle. Let ... be the set of idempotents in the quandle ring $\mathbb{Z}_2[X]$, assume that Y is a quandle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005172 | MATHDB-361986

The conjecture that β(N,l) depends only on l

摘录 / EXCERPT: Let $N\geq 2$ and $1<l\mid N$. The homomorphism $\sigma_{N,l}$ has image $\operatorname{im}(\sigma_{N,l})=\beta(N,l)\mathbb{Z}$ for a unique positive integer $\beta(N,l)$. Dependen...

conjecture | open | MathDB 页面 | 原文线索

V2-005173 | MATHDB-361987

The conjectural formula for β(N)

摘录 / EXCERPT: Let $N\geq 2$ be an integer, and write $\beta(N)=\beta(N,N)$. Formula conjecture. 1. If $N\not\equiv 1,9\pmod{24}$, then $\beta(N)$ is periodic in N with period 24, with values...

conjecture | open | MathDB 页面 | 原文线索

V2-005174 | MATHDB-361988

The full-rank conjecture for the σ(N,l)-matrix

摘录 / EXCERPT: For $N\geq 2$, let t be the number of positive divisors of N, let $g_1,\ldots,g_r$ be the generators specified in the surrounding setup, and let $\sigma_{N,l}$ be the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-005175 | MATHDB-361992

The high-density $3k-4$ conjecture modulo a prime

摘录 / EXCERPT: The high-density $3k-4$ conjecture. There are arithmetic progressions $P_A,P_B,P_C\subseteq G$, all having a common difference, such that

conjecture | open | MathDB 页面 | 原文线索

V2-005176 | MATHDB-362088

Bilu et al.'s conjecture on p-adic valuations of third-order recurrence sequences

摘录 / EXCERPT: Let (x_n) be a third-order linear recurrence sequence, let p be a prime number, and let $ u_p$ denote the p-adic valuation. Bilu et al.'s conjecture. There exists a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-005177 | MATHDB-362092

Counting conjecture for rationals in missing-digit sets

摘录 / EXCERPT: Counting conjecture. For each $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005178 | MATHDB-362093

Localization accessibility of Theorem summod5

摘录 / EXCERPT: The localization method is applied to the functions and congruences described above, with $L_{\alpha}=L_{\alpha}^{(0)}+L_{\alpha}^{(1)}$ lying in $\mathcal{M}(\mathrm{X}_0(10))$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-005179 | MATHDB-362094

A combined integrality conjecture for generalized Frobenius partition functions

摘录 / EXCERPT: Let α and a be as in the preceding construction, let $\beta\in\{0,1\}$, let φ denote the relevant theta function, let q be the nome, let $W\tau$ denote the trans...

conjecture | open | MathDB 页面 | 原文线索

V2-005180 | MATHDB-362098

Gorsky-Oblomkov-Rasmussen filtration conjecture for rational Cherednik representations

摘录 / EXCERPT: Let m,n be coprime positive integers, let $\mathfrak{h}$ be the $(n-1)$-dimensional standard representation of S_n, and let $\mathrm{L}_{\frac{m}{n}}$ be the unique finite-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-005181 | MATHDB-362099

Oblomkov-Yun perverse and maximal-ideal filtration conjecture for compactified Jacobians

摘录 / EXCERPT: Let m,n be coprime positive integers, let $C_{\frac{m}{n}}$ be the closure of $y^m=x^n$ in $\mathbb{P}^2$, and let $J_{\frac{m}{n}}$ be its compactified Jacobian. Let P be the...

conjecture | open | MathDB 页面 | 原文线索

V2-005182 | MATHDB-362100

Gorsky-Oblomkov-Rasmussen algebraic and inductive filtration conjecture

摘录 / EXCERPT: Let m,n be coprime positive integers with $m>n$, and let $\mathrm{L}_{\frac{m}{n}}$ be the finite-dimensional irreducible representation of the rational Cherednik algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-005183 | MATHDB-362108

Weak Hardy-Littlewood conjecture for the Goldbach representation function

摘录 / EXCERPT: Let $R(n)$ denote the number of representations of an integer n as a sum of two primes. Weak Hardy-Littlewood conjecture. There exists an absolute constant $c_3>0$ such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-005184 | MATHDB-362109

Jia's strengthened Hardy-Littlewood conjecture for Goldbach representations

摘录 / EXCERPT: Let $R(n)$ denote the number of representations of an integer n as a sum of two primes, and let φ denote Euler's totient function. Jia's strengthened Hardy-Littlewood con...

conjecture | open | MathDB 页面 | 原文线索

V2-005185 | MATHDB-362110

Friedlander-Goldston-Iwaniec-Suriajaya conjecture for the weighted Goldbach function

摘录 / EXCERPT: Define the weighted Goldbach representation function ... and the singular series ... where C_2 is the twin prime constant. Friedlander-Goldston-Iwaniec-Suriajaya conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-005186 | MATHDB-362111

The distinct-prime-divisor conjecture for Mersenne-type numbers

摘录 / EXCERPT: Distinct-prime-divisor conjecture. If $\ell\geq 3$, then 2^n-1 is divisible by at least 2^ℓ distinct primes.

conjecture | open | MathDB 页面 | 原文线索

V2-005187 | MATHDB-362115

Amoroso's height conjecture for generators of symmetric Galois extensions

摘录 / EXCERPT: Let $\alpha\in\overline{\mathbb{Q}}$ generate a Galois extension of degree $d=n!$ over $\mathbb{Q}$ whose Galois group is isomorphic to the symmetric group $\mathfrak{S}_n$. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-005188 | MATHDB-362119

Odd-prime transchromatic shearing conjecture for equivariant spectral sequences

摘录 / EXCERPT: Odd-prime transchromatic shearing conjecture. There is a shearing isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-005189 | MATHDB-362137

Baker-Montgomery conjecture on zeros of Fekete polynomials

摘录 / EXCERPT: Let D be a fundamental discriminant, and let N_D denote the number of zeros of the Fekete polynomial F_D in $(0,1)$. Baker-Montgomery conjecture. For almost all fundamental d...

conjecture | open | MathDB 页面 | 原文线索

V2-005190 | MATHDB-362156

Quaternionic Lehmer conjecture

摘录 / EXCERPT: Let $P(x)\in\mathbb{Z}[x]$ be a monic irreducible polynomial, and let $m_{\mathbb{H}}(P)$ denote its quaternionic Mahler measure. Quaternionic Lehmer conjecture. There is a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-005191 | MATHDB-362159

Hulek-Verrill supercongruence conjecture

摘录 / EXCERPT: For the Hulek-Verrill family $1-tg(\boldsymbol{x})=0$, define ... and ... Let a_p be the coefficient of q^p in the weight 4 cusp form of level 14 beginning $q-2q^2+8q^3+\cdots$...

conjecture | open | MathDB 页面 | 原文线索

V2-005192 | MATHDB-362160

Level-54 attractor-point supercongruence conjecture

摘录 / EXCERPT: For every prime $p\ne3$, consider the truncated hypergeometric sum ... Let a_p be the coefficient of q^p in the weight 4 cusp form of level 54 beginning...

conjecture | open | MathDB 页面 | 原文线索

V2-005193 | MATHDB-362161

Conjecture on the Bogomolov property for Galois representations attached to modular forms

摘录 / EXCERPT: Let $k\geq 2$, let $f\in S_k(\Gamma_0(N))$ be a normalized eigenform, let ${\mathbb{K}}_f$ be the number field generated by its Hecke eigenvalues, and let $\mathcal{O}_f$ be its ri...

conjecture | open | MathDB 页面 | 原文线索

V2-005194 | MATHDB-362162

The multivariate topological monodromy conjecture

摘录 / EXCERPT: The multivariate monodromy conjecture. If F is such a tuple, then

conjecture | open | MathDB 页面 | 原文线索

V2-005195 | MATHDB-362182

Limit-point spectrum conjecture for maximal cliques in d-Paley graphs

摘录 / EXCERPT: Fix an integer $d\ge 2$. For each odd prime power $q\equiv1{\ifmmode\text{\rm\ (mod~d)}\else\discretionary{}{}{\text{ }}\rm(mod~d)\fi}$, let r_q be the cardinality of a maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-005196 | MATHDB-362193

The vector-space Szemerédi conjecture

摘录 / EXCERPT: Let q be a prime power, let $\mathbb{F}_{q}^{n}$ be the n-dimensional vector space over $\mathbb{F}_{q}$, and call a family of its subspaces of density at least δ when i...

conjecture | open | MathDB 页面 | 原文线索

V2-005197 | MATHDB-362203

Triple O'Nan conjecture for non-classical unitals

摘录 / EXCERPT: Let q be a prime power and let ${\mathcal U}$ be a non-classical unital in ${\textup{PG}}(2,q^2)$. A Triple O'Nan configuration is a set of six distinct lines pairwise intersecti...

conjecture | open | MathDB 页面 | 原文线索

V2-005198 | MATHDB-362214

The zero dual-entropy Möbius disjointness conjecture

摘录 / EXCERPT: Let $A\subseteq\mathbb{N}$, let 1_A be its indicator function, let $\AE^*(A)$ denote the dual entropy of A, and let μ be the Möbius function. Zero dual-entropy Möbius disjo...

conjecture | open | MathDB 页面 | 原文线索

V2-005199 | MATHDB-362231

Duke's integrality conjecture for higher Rademacher symbols

摘录 / EXCERPT: Let $\Gamma=\mathrm{SL}_2(\mathbb{Z})$. For each integer $k\in\mathbb{Z}_{\geq 2}$, let ... be Duke's higher Rademacher symbol. Duke's conjecture. For any integer...

conjecture | open | MathDB 页面 | 原文线索

V2-005200 | MATHDB-362242

Monotonicity conjecture for rational q-Catalan numbers

摘录 / EXCERPT: For coprime positive integers a and b, define the rational q-Catalan number ... Here $[n]_q=(1-q^n)/(1-q)$ and $\mathbb{N}[q]$ denotes polynomials in q with non-negative inte...

conjecture | open | MathDB 页面 | 原文线索

V2-005201 | MATHDB-362245

Standard-partition conjecture for q-Catalan germs

摘录 / EXCERPT: Let $\mathcal{L}=\mathbb{Z}^{a-1}$, let Box be the fundamental box, and for coprime a,c let c' be the largest integer less than c with $\gcd(a,c')=1$. A stan...

conjecture | open | MathDB 页面 | 原文线索

V2-005202 | MATHDB-362253

The additively twisted Linnik-Selberg conjecture over $\mathbb{Q}(i)$

摘录 / EXCERPT: Additively twisted Linnik-Selberg conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-005203 | MATHDB-362281

Positivity conjecture for the shadow Markoff tree

摘录 / EXCERPT: Positivity conjecture. Every triple (A,B,C) of positive integer solutions to the shadow Markoff equation can be obtained from an initial triple with $(\alpha:\beta:\gamma)$ a rat...

conjecture | open | MathDB 页面 | 原文线索

V2-005204 | MATHDB-362288

Solvabilizer number of PSL2 for prime powers congruent to one modulo four

摘录 / EXCERPT: PSL2 congruence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005205 | MATHDB-362300

Asymptotic equality of the v-function for maximal associated primes

摘录 / EXCERPT: Asymptotic maximal-prime conjecture. For all $\mathfrak{p}\in\operatorname{Max}^\infty(I)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-005206 | MATHDB-362301

Kogiso-Miyamoto-Ren-Wakui-Yanagawa arithmetic conjecture for prime denominators

摘录 / EXCERPT: Let p be an odd prime integer, and let a,b be positive integers coprime to p. The associated q-deformed rational-number polynomials are denoted by $\mathcal{S}_{a/p}(q)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-005207 | MATHDB-362308

Mori's congruence conjecture for the caliber of real quadratic fields

摘录 / EXCERPT: For prime numbers p and q with $p<q$ and $p\equiv q\equiv 1\pmod 4$, let x_p,y_p,x_q,y_q be the unique positive integers satisfying ... with $0<x_p<y_p$ and $0<x_q<y_q$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005208 | MATHDB-362316

Two-part criterion for odd-characteristic subgroup rigidity

摘录 / EXCERPT: Let F be a free group and let p be an odd prime. Let H be a finitely generated subgroup of F, and let $\widetilde H$ be its L^2-closure. The two-part odd-characteristic c...

conjecture | open | MathDB 页面 | 原文线索

V2-005209 | MATHDB-362332

Beresnevich-Haynes-Velani conjecture for coprime multiplicative approximation

摘录 / EXCERPT: Beresnevich-Haynes-Velani conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-005210 | MATHDB-362334

Cheng-Murray conjecture on deep holes of full-length Reed-Solomon codes

摘录 / EXCERPT: Let q be a prime power, let θ be a primitive element when relevant, and let ${\rm RS}_k(\mathbb{F}_q)$ denote the full-length Reed-Solomon code of dimension k. For a vec...

conjecture | open | MathDB 页面 | 原文线索

V2-005211 | MATHDB-362335

Completeness conjecture for deep holes of twisted Reed-Solomon codes

摘录 / EXCERPT: Let q be a prime power, let θ specify the twist defining the twisted Reed-Solomon code ${\rm TRS}_k(\mathbb{F}_q,\theta)$, and let $\mathcal{S}_{k,\theta}$ be the source-d...

conjecture | open | MathDB 页面 | 原文线索

V2-005212 | MATHDB-362383

The stronger density hypothesis for the Riemann zeta function

摘录 / EXCERPT: Stronger density hypothesis. For every $\varepsilon>0$, there exists $\delta=\delta(\varepsilon)\in(0,1)$ such that, for every $\nu\in[0,1/2-\varepsilon)$ and $T\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005213 | MATHDB-362387

Conjecture on sums of practical and polygonal numbers

摘录 / EXCERPT: Conjecture on practical and polygonal sums. If $s>3$, $s \not\equiv 0\pmod{12}$, and $s \not\equiv 4\pmod{12}$, then all sufficiently large natural numbers can be written as a sum...

conjecture | refuted | MathDB 页面 | 原文线索

V2-005214 | MATHDB-362388

Conjecture on sums of a practical number and two polygonal numbers

摘录 / EXCERPT: Conjecture on two polygonal summands. For $s\in\{4,5,6,7,8,10\}$, all natural numbers can be written as a sum of a practical number and two s-gonal numbers.

conjecture | refuted | MathDB 页面 | 原文线索

V2-005215 | MATHDB-362389

Erdős's strong prime power conjecture for projective planes

摘录 / EXCERPT: A finite projective plane has an integer n, called its order, such that each line contains $n+1$ points, each point lies on $n+1$ lines, and there are n^2+n+1 points and lines....

conjecture | open | MathDB 页面 | 原文线索

V2-005216 | MATHDB-362390

The weak prime power conjecture for perfect difference sets

摘录 / EXCERPT: Let n be a positive integer. A perfect difference set of order n is a subset K of the cyclic group $\mathds Z/(n^2+n+1)\mathds Z$ with $|K|=n+1$ such that every nonzero resid...

conjecture | open | MathDB 页面 | 原文线索

V2-005217 | MATHDB-362408

Log Manin's conjecture for klt Campana points

摘录 / EXCERPT: Let $(X, D_\epsilon = \sum_{\alpha \in \mathcal A} \epsilon_\alpha D_\alpha)$ be a klt Fano Campana orbifold over a number field F, let L be a big $\mathbb Q$-divisor on X, a...

conjecture | open | MathDB 页面 | 原文线索

V2-005218 | MATHDB-362409

Conjecture on simultaneous analytic images of Liouville matrices

摘录 / EXCERPT: Let $n\ge 2$ and let $I\subseteq\mathbb{R}$ be an open interval containing 0. Let $\mathscr{L}_{n,n}$ denote the set of Liouville matrices, and let $\textswab{f}_k$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005219 | MATHDB-362410

Growth conjecture for the original Treschev problem

摘录 / EXCERPT: Consider the original Treschev problem, with coefficients q_{2n} and a Diophantine rotation number. Here, $k>0$ is arbitrary and $C(k)$ denotes a constant depending on k. Tresc...

conjecture | open | MathDB 页面 | 原文线索

V2-005220 | MATHDB-362413

Nonsplit quantum unique ergodicity conjecture for Hilbert-Maaß forms

摘录 / EXCERPT: Nonsplit quantum unique ergodicity conjecture. For every fixed $H\in C_b(\Gamma\backslash\mathbb{H})$,

conjecture | open | MathDB 页面 | 原文线索

V2-005221 | MATHDB-362418

The false-mock conjecture for orientation-reversed three-manifolds

摘录 / EXCERPT: Let M_3 be a three-manifold such that ... where $\tau\in\mathbb H$, $c\in\mathbb Q$, $\widetilde\vartheta$ is the Eichler integral of a weight-$3/2$ unary theta function...

conjecture | open | MathDB 页面 | 原文线索

V2-005222 | MATHDB-362422

The exact Hausdorff distance conjecture for the Markov and Lagrange spectra

摘录 / EXCERPT: Exact Hausdorff distance conjecture. The Hausdorff distance between M and L is precisely δ_0.

conjecture | open | MathDB 页面 | 原文线索

V2-005223 | MATHDB-362431

Chimeral Ramanujan congruences for Klein-form quotients at all sufficiently large primes

摘录 / EXCERPT: Chimeral congruence conjecture. For primes $p\ge 11$, if $(a_0,\ldots,a_{(p-1)/2})$ lies in

conjecture | open | MathDB 页面 | 原文线索

V2-005224 | MATHDB-362432

Greenberg's algebraic rank conjecture for families

摘录 / EXCERPT: Let $\mathcal X$ be sufficiently small, let $W(f)$ be the global root number of the family, and let ${}^{\rm c}V_\mathcal X$ and ${}^{\rm c}V'_\mathcal X$ be the relevant critical...

conjecture | open | MathDB 页面 | 原文线索

V2-005225 | MATHDB-362444

Finiteness conjecture for nontrivial logarithmic class groups

摘录 / EXCERPT: Finiteness conjecture for logarithmic class groups. The set

conjecture | open | MathDB 页面 | 原文线索

V2-005226 | MATHDB-362445

Asymptotic formula conjecture for binary sums of Piatetski-Shapiro numbers

摘录 / EXCERPT: Asymptotic formula conjecture. For every $\alpha\in(1,2)\cup(2,3)$,

conjecture | open | MathDB 页面 | 原文线索

V2-005227 | MATHDB-362446

Asymptotic formula conjecture for arithmetic progressions of Piatetski-Shapiro numbers

摘录 / EXCERPT: Arithmetic-progression asymptotic conjecture. For every $\alpha\in(1,2)\cup(2,3)$,

conjecture | open | MathDB 页面 | 原文线索

V2-005228 | MATHDB-362470

Kurokawa-Mizumoto congruence for Klingen-Eisenstein series

摘录 / EXCERPT: Let k and ν be even integers with $k>4$. Let ... be a normalized Hecke eigenform, meaning that $a(1,f)=1$. Let $L(k-1,f,\mathrm{St})$ denote the relevant standard L-function,...

conjecture | open | MathDB 页面 | 原文线索

V2-005229 | MATHDB-362487

The arithmetic progression nonvanishing conjecture for Fermat quotients

摘录 / EXCERPT: For a prime $p>2$, write $q_2(p)=(2^{p-1}-1)/2$. Arithmetic progression nonvanishing conjecture. For every pair of positive integers $m>a>0$ with $\gcd(m,a)=1$, there are infinitel...

conjecture | open | MathDB 页面 | 原文线索

V2-005230 | MATHDB-362519

Supercongruence for the sporadic sequences B and F

摘录 / EXCERPT: Let $p\ge 5$ be a prime, and let $\{u_n\}_{n\ge 0}$ be one of the sequences $\mathbf{B}$ and $\mathbf{F}$, defined by ... or ... Here $\left(\frac{p}{3}\right)$ is the Legendre symbo...

conjecture | open | MathDB 页面 | 原文线索

V2-005231 | MATHDB-362541

Subpandigital and subpenholodigital square existence conjecture

摘录 / EXCERPT: Subpandigital and subpenholodigital square existence conjecture. Suppose $b>7$. A strict subpandigital square and a strict subpenholodigital square in base b exist if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-005232 | MATHDB-362542

Pandigital and penholodigital prime digit-sum conjecture

摘录 / EXCERPT: Pandigital and penholodigital prime digit-sum conjecture. For $b>3$, the smallest pandigital prime or penholodigital prime n satisfies $s_b(n)=b(b-1)/2+2$ if b is of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-005233 | MATHDB-362543

Subpandigital and subpenholodigital prime digit-sum conjecture

摘录 / EXCERPT: Subpandigital and subpenholodigital prime digit-sum conjecture. For $b>4$, the smallest subpandigital prime or smallest subpenholodigital prime n satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-005234 | MATHDB-362547

Conjecture on the integer part of non-perfect-power roots

摘录 / EXCERPT: Let $a,n\in\mathbb{Z}^+$ satisfy $a>2$, $\left\lfloor\log_2(a)\right\rfloor+1\geq n>1$, and suppose there is no $k\in\mathbb{Z}^+$ such that $k^n=a$. Integer-part root conjecture....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005235 | MATHDB-362553

Twisted fourth-moment hypothesis for Dirichlet L-functions

摘录 / EXCERPT: Twisted fourth-moment hypothesis. For every coefficient sequence $\boldsymbol{\alpha}$ and every $\varepsilon>0$, the displayed estimate holds. This conjecture asks for the average...

conjecture | open | MathDB 页面 | 原文线索

V2-005236 | MATHDB-362554

Hooley's Hypothesis R*

摘录 / EXCERPT: Let q be a positive integer, let $h\in\mathbf{Z}$, and let I be an interval satisfying ... For integers n coprime to q, write $\overline n$ for the multiplicative inverse of...

conjecture | open | MathDB 页面 | 原文线索

V2-005237 | MATHDB-362589

Kahle-Newman’s Cohen-Lenstra conjecture for determinantal hypertrees

摘录 / EXCERPT: Let T_n be a random 2-dimensional determinantal hypertree on the vertex set $[n]$, and let p be a prime. Write $\Gamma_{n,p}$ for the p-torsion subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-005238 | MATHDB-362591

Wagstaff's asymptotic conjecture for the least prime in an arithmetic progression

摘录 / EXCERPT: Let n be a positive integer, let $a<n$ satisfy $\operatorname{gcd}(a,n)=1$, and define ... ... where ϕ denotes Euler's totient function. Wagstaff's conjecture. As $n\to\infty$,...

conjecture | open | MathDB 页面 | 原文线索

V2-005239 | MATHDB-362592

Fiori's conjecture on large and small outliers for the least prime

摘录 / EXCERPT: Let $\epsilon>0$, and let $E_\epsilon(x,y)$ denote the counting function introduced in the paper for the relevant large outliers of $P(n)$; the supplied context does not reproduce...

conjecture | open | MathDB 页面 | 原文线索

V2-005240 | MATHDB-362593

Fiori's lower-bound conjecture for the least prime in an arithmetic progression

摘录 / EXCERPT: Fiori's lower-bound conjecture. For $0<\epsilon<1$, as $y\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005241 | MATHDB-362594

Fiori's explicit two-sided conjecture for the least prime in an arithmetic progression

摘录 / EXCERPT: Let n be a positive integer, let $a<n$ satisfy $\operatorname{gcd}(a,n)=1$, and define ... ... where ϕ denotes Euler's totient function. Fiori's explicit two-sided conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-005242 | MATHDB-362599

Mordell's Pellian equation conjecture

摘录 / EXCERPT: Let $d\in\mathbb{N}_{\geq 2}$ be squarefree, let $K=\mathbb{Q}(\sqrt{d})$, let $\mathcal{O}_K$ be the ring of algebraic integers of K, and let $\varepsilon>1$ be the fundamental...

conjecture | open | MathDB 页面 | 原文线索

V2-005243 | MATHDB-362601

The square power partition function inequality conjecture

摘录 / EXCERPT: Square power partition function inequality conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-005244 | MATHDB-362602

The initial log-concavity thresholds for k-th power partition functions

摘录 / EXCERPT: Initial log-concavity threshold conjecture. For $k\leq6$, the smallest such values are

conjecture | open | MathDB 页面 | 原文线索

V2-005245 | MATHDB-362624

EC-AAC for the families $E_{p,2}$ and $E_{p,4}$

摘录 / EXCERPT: Let p be an odd prime, and define ... Let $P=(X,Y)$ be a generator of one of these curves, and let k be a positive integer such that ... for integers u,v,d, with $p\mid d$. EC-AA...

conjecture | open | MathDB 页面 | 原文线索

V2-005246 | MATHDB-362627

Nonexistence of spanning thin codes for prime modulus

摘录 / EXCERPT: Let N be a prime with $N\geq 3$. A spanning simplex is a lattice simplex whose lattice points at height one generate the ambient lattice, and a non-degenerate thin code is a thin...

conjecture | open | MathDB 页面 | 原文线索

V2-005247 | MATHDB-362631

Magneticity characterized by vanishing residue-quotient cohomology class

摘录 / EXCERPT: Magneticity criterion. The form f is magnetic if and only if $C_f=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-005248 | MATHDB-362632

Decomposition of magnetic modular forms into strongly magnetic forms and derivatives

摘录 / EXCERPT: Decomposition conjecture. There is a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-005249 | MATHDB-362633

Algebraic CM-supported residues of magnetic modular forms

摘录 / EXCERPT: Residue conjecture. The residues of $\tau^m f(\tau)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-005250 | MATHDB-362634

Magneticity is preserved by the SL2 action

摘录 / EXCERPT: SL2-action conjecture. For every $\gamma\in\mathrm{SL}_2(\mathbb{Z})$, the action

conjecture | open | MathDB 页面 | 原文线索

V2-005251 | MATHDB-362635

Magneticity is preserved by Atkin-Lehner involutions

摘录 / EXCERPT: Atkin-Lehner invariance conjecture. The action of Atkin-Lehner involutions on $\mathcal{M}_k(\Gamma_1(N))$ restricts to the magnetic subspaces

conjecture | open | MathDB 页面 | 原文线索

V2-005252 | MATHDB-362651

The biMoore graph conjecture from perfect difference sets

摘录 / EXCERPT: Let s be a positive integer with $s-1$ a prime power, let S be a perfect difference set of size s, and let $G_m(S)$ be the graph constructed in Corollary. A bipartite graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-005253 | MATHDB-362653

Elliott's conjecture for correlations of non-pretentious completely multiplicative functions

摘录 / EXCERPT: Let $\mathcal{M}$ be the set of multiplicative functions $f:\mathbb{N}\to\mathbb{C}$ with $|f(n)|\leq 1$. For $f,g\in\mathcal{M}$, define ... A function is non-pretentious when it is...

conjecture | open | MathDB 页面 | 原文线索

V2-005254 | MATHDB-362670

Kronecker congruence conjecture for Ramanujan's continued fraction of order 10

摘录 / EXCERPT: Let $U(\tau)$ be the modular function considered above, and for an odd prime $p\neq 5$ let $U_p(X,Y)=0$ denote its modular equation of level p. Kronecker congruence conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-005255 | MATHDB-362679

The PIA conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $(X, \mathfrak{a})$ be a log pair over an algebraically closed field k of characteristic zero, and let D be a normal Cartier prime divisor on X. Let $x \in D$ be a closed...

conjecture | open | MathDB 页面 | 原文线索

V2-005256 | MATHDB-362723

Fixed points of the Jacobsthal sequence

摘录 / EXCERPT: Let the Jacobsthal sequence be defined by $J_0=0$, $J_1=1$, and $J_n=J_{n-1}+2J_{n-2}$, and let $m>1$ be an integer. A modulus m is a fixed point when its Pisano period equals...

conjecture | open | MathDB 页面 | 原文线索

V2-005257 | MATHDB-362725

Farmer-Gonek-Hughes conjecture on character sums

摘录 / EXCERPT: Farmer-Gonek-Hughes conjecture. As $q\to\infty$, there is a constant B such that

conjecture | open | MathDB 页面 | 原文线索

V2-005258 | MATHDB-362781

Rational quantum modularity conjecture for modular resurgent q-series

摘录 / EXCERPT: Let $f\colon\mathbb{Q}\to\mathbb{C}$ be a q-series with $q={\rm e}^{2\pi {\rm i}y}$. Suppose its asymptotic expansion $\tilde{f}(y)$ as $y\to0$ extends to $y\in\mathbb{C}$ and ha...

conjecture | open | MathDB 页面 | 原文线索

V2-005259 | MATHDB-362782

Asymptotically positive extensions and property (B)

摘录 / EXCERPT: Let $\mathcal{L}/\mathbb{Q}$ be an infinite extension. For a prime power q, let ψ_q denote the limiting normalized number of prime ideals of norm q in finite subextensions...

conjecture | open | MathDB 页面 | 原文线索

V2-005260 | MATHDB-362783

Relative Schinzel-Zassenhaus conjecture for generalized house

摘录 / EXCERPT: Fix a positive integer m. For an algebraic integer $\alpha\in\overline{\mathbb{Q}}$, define ... where $M_{\sigma(K)}(\alpha)$ is the relevant relative Mahler measure; for $m=1$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-005261 | MATHDB-362786

The P_5-free coloring conjecture for prime non-bipartite cores

摘录 / EXCERPT: Let H be a graph. A prime graph is one admitting no non-trivial decomposition with respect to the direct product, a core is a graph with no homomorphism onto a proper subgraph, a...

conjecture | open | MathDB 页面 | 原文线索

V2-005262 | MATHDB-362797

Dyadic-coefficient zero-one sparsity conjecture

摘录 / EXCERPT: Let $a_{r,\ell}$ be the coefficients in the dyadic expansions of $-3^{-\ell}$, let $[x_1,\ldots,x_k]$ be a Syracuse path with geometric measure $\mathbb G_{1-\mu^{-1}}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-005263 | MATHDB-362844

Conjecture on p^2-even eta-powers for many primes

摘录 / EXCERPT: For an odd integer $t\geq 1$, write ... For a prime p and $r\in\{0,\dots,p^2-1\}$, say that f_1^t is p^2-even with base r if ... for every $k\in\{1,\dots,p-1\}$. Prime-proporti...

conjecture | open | MathDB 页面 | 原文线索

V2-005264 | MATHDB-362845

Conjecture on infinitely many p^2-even eta-powers for each prime

摘录 / EXCERPT: For a positive integer t, write ... For a prime p and $r\in\{0,\dots,p^2-1\}$, say that f_1^t is p^2-even with base r if ... for every $k\in\{1,\dots,p-1\}$. Infinite-exponen...

conjecture | open | MathDB 页面 | 原文线索

V2-005265 | MATHDB-362846

Density conjecture for two eta-quotients in the $5t$ family

摘录 / EXCERPT: Let $t=2^\alpha t_0$, where t_0 is odd and $\alpha>0$. Assume ... The density of odd coefficients is meant in the same sense as in the paper. $5t$-family density conjecture. Both e...

conjecture | open | MathDB 页面 | 原文线索

V2-005266 | MATHDB-362847

The all-or-nothing density conjecture for eta-quotients

摘录 / EXCERPT: Let ... be an eta-quotient, shifted by a suitable power of q so that its powers are integral, and let δ_F be the odd density of its coefficients. All-or-nothing density conj...

conjecture | open | MathDB 页面 | 原文线索

V2-005267 | MATHDB-362867

The second Hardy-Littlewood conjecture for the prime-counting function

摘录 / EXCERPT: Let $\pi(t)$ denote the number of primes at most t, and let $x,y\geq 2$. Second Hardy-Littlewood conjecture. ... This is a classical subadditivity conjecture for the prime-counting...

conjecture | open | MathDB 页面 | 原文线索

V2-005268 | MATHDB-362868

Minculete's multiplicative inequality conjecture for the prime-counting function

摘录 / EXCERPT: Minculete's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005269 | MATHDB-362869

Young-type inequality conjecture for the prime-counting function

摘录 / EXCERPT: Let $\pi(t)$ denote the number of primes at most t, and let $p,q>1$ satisfy ... There exists a sufficiently large natural number n_0 such that, for all $x,y\geq n_0$, Young-type...

conjecture | open | MathDB 页面 | 原文线索

V2-005270 | MATHDB-362892

Levesque-Waldschmidt conjecture on doubly twisted simplest cubic Thue equations

摘录 / EXCERPT: Levesque-Waldschmidt conjecture. There exists a positive absolute constant κ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-005271 | MATHDB-362900

The constructive recognition complexity conjecture for special linear groups

摘录 / EXCERPT: Let $d\geq 4$, let q be a prime power, and let $\operatorname{SL}(d,q)$ be given by a generating set. Let ξ denote an upper bound on the number of field operations for comput...

conjecture | open | MathDB 页面 | 原文线索

V2-005272 | MATHDB-362904

Boston's generalization of the unramified Fontaine-Mazur conjecture

摘录 / EXCERPT: Boston's generalized unramified Fontaine-Mazur conjecture. Every continuous homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-005273 | MATHDB-362916

The positivity conjecture for character sums f_q

摘录 / EXCERPT: Let q be squarefree with $q\equiv 3 \pmod 8$, and let $f_q(x)$ denote the function defined earlier in the paper. Positivity conjecture. One has ... for $0\leqslant x\leqslant 1/2$....

conjecture | open | MathDB 页面 | 原文线索

V2-005274 | MATHDB-362917

The numerator nondivisibility conjecture for W(q,x)

摘录 / EXCERPT: Let q be a prime with $q\equiv 3 \pmod 8$, let x be rational with $0<x<1/2$, and define ... where $h(q)$ is the class number and χ_q is the relevant quadratic character. Num...

conjecture | open | MathDB 页面 | 原文线索

V2-005275 | MATHDB-362918

The conjectured RH inequality for primes congruent to 3 modulo 8

摘录 / EXCERPT: Let p and q be primes with $p\equiv q\equiv 3 \pmod 8$, satisfying $0<2a<p<q$, and let $\chi_q(p)=\left(\frac{p}{q}\right)$ be the Legendre symbol modulo q. Let ... holds for a...

conjecture | open | MathDB 页面 | 原文线索

V2-005276 | MATHDB-362922

The GL2 Main Conjecture over CM fields

摘录 / EXCERPT: Let $g\in S_2(\Gamma_0(\mathfrak n))$ be a Hilbert modular newform over a totally real field F, and let p be an odd prime unramified in F and good ordinary for g. Let M/F...

conjecture | open | MathDB 页面 | 原文线索

V2-005277 | MATHDB-362941

Birch and Swinnerton-Dyer conjecture for partial Euler products

摘录 / EXCERPT: Let $E/\mathbb Q$ be an elliptic curve with rank r. For each prime p, let $N_p=\#E_{\mathrm{ns}}(\mathbb F_p)$, where $E_{\mathrm{ns}}(\mathbb F_p)$ is the set of nonsingular...

conjecture | open | MathDB 页面 | 原文线索

V2-005278 | MATHDB-362943

Ramanujan-Petersson conjecture for unramified automorphic representations

摘录 / EXCERPT: Let $\pi=\bigotimes'\pi_v$ be an irreducible cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb A_\mathbb Q)$. At an unramified finite prime p, write the Satake parame...

conjecture | open | MathDB 页面 | 原文线索

V2-005279 | MATHDB-362975

The period-index conjecture for fields over algebraically closed fields

摘录 / EXCERPT: Period-index conjecture. For every $\alpha \in \operatorname{Br}(K)$,

conjecture | open | MathDB 页面 | 原文线索

V2-005280 | MATHDB-362979

Nonexistence conjecture for perfect codes over non-prime-power alphabets

摘录 / EXCERPT: Nonexistence conjecture. Let $q \ge 6$ be a non-prime power. Then, there are no perfect error correcting codes over Z_q.

conjecture | open | MathDB 页面 | 原文线索

V2-005281 | MATHDB-362992

Inhomogeneous Duffin-Schaeffer conjecture for systems of linear forms

摘录 / EXCERPT: Fix $m,n\in\mathbb N$. Let $\mathbf y\in\mathbb R^m$ and define, for $\mathbf q\in\mathbb Z^n$, ... Here $W_{n,m}^P(\psi)$ is the corresponding limsup set and m deno...

conjecture | open | MathDB 页面 | 原文线索

V2-005282 | MATHDB-362993

Weak inhomogeneous Duffin-Schaeffer conjecture for systems of linear forms

摘录 / EXCERPT: Fix $m,n\in\mathbb N$ and an inhomogeneous parameter $\mathbf y\in\mathbb R^m$. For every $\mathbf q\in\mathbb Z^n$, define ... Let $W_{n,m}^P(\psi)$ be the corresponding limsup set...

conjecture | open | MathDB 页面 | 原文线索

V2-005283 | MATHDB-362997

Harper's beyond square-root cancellation conjecture for character twists

摘录 / EXCERPT: Let $c\in\{\mu,\lambda\}$, where μ is the Möbius function and λ is the Liouville function. Let r be a prime modulus, let $\mathbb E_\chi$ denote the average over Diri...

conjecture | open | MathDB 页面 | 原文线索

V2-005284 | MATHDB-362999

Twisted Ratios conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let r be a prime, let χ_0 be the principal character modulo r, and let f be a completely multiplicative function with $f^*(n)=f(n)\mathbf 1_{\operatorname{gcd}(n,r)=1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-005285 | MATHDB-363006

The folklore local-to-global conjecture for cubic hypersurfaces

摘录 / EXCERPT: Folklore local-to-global conjecture. If $X(\mathbf{Q}_p)\neq\emptyset$ for every prime p, then

conjecture | open | MathDB 页面 | 原文线索

V2-005286 | MATHDB-363020

Bergeron-Venkatesh conjecture on torsion homology growth of arithmetic hyperbolic 3-manifolds

摘录 / EXCERPT: Suppose that ... is a tower of covers of congruence arithmetic hyperbolic 3-manifolds whose injectivity radius approaches infinity. Write $H_1(M_i,\mathbb{Z})_{tors}$ for the torsion...

conjecture | open | MathDB 页面 | 原文线索

V2-005287 | MATHDB-363034

The square-free v-palindrome conjecture in base ten

摘录 / EXCERPT: Let $v(n)$ denote the sum of the primes and exponents appearing in the prime factorization of n, and let $r_{10}(n)$ be the digit reversal of n in base ten. A v-palindrome in...

conjecture | open | MathDB 页面 | 原文线索

V2-005288 | MATHDB-363035

The uniqueness conjecture for a v-palindrome equation

摘录 / EXCERPT: Let $r(n)$ denote the digit reversal of n in base ten, and let $v(n)$ be the sum of the primes and exponents appearing in the prime factorization of n. Uniqueness conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-005289 | MATHDB-363048

Eventual non-omission of zero in ternary powers of 2

摘录 / EXCERPT: Eventual zero-digit conjecture. There is a positive integer k_0 such that for all $k\geq k_0$, the ternary expansion of 2^k cannot omit the digit 0. This is a weaker form of...

conjecture | open | MathDB 页面 | 原文线索

V2-005290 | MATHDB-363051

Blowup recolorability conjecture for prime graphs

摘录 / EXCERPT: Blowup recolorability conjecture. Every blowup of G is recolorable if and only if every induced subgraph of G is recolorable.

conjecture | open | MathDB 页面 | 原文线索

V2-005291 | MATHDB-363052

The completeness conjecture for MacMahon prime-detecting expressions

摘录 / EXCERPT: Let $M_a(n)$ be the partition functions defined by ... Let $\vec{P}(x):=(p_1(x),p_2(x),\dots,p_a(x))\in\mathbb{Z}[x]^a$ be a vector of relatively prime integer polynomials. For integ...

conjecture | open | MathDB 页面 | 原文线索

V2-005292 | MATHDB-363053

The Eisenstein-space conjecture for prime-detecting quasimodular forms

摘录 / EXCERPT: Let $\mathcal{E}$ be the quasimodular Eisenstein space, generated additively by the even-weight Eisenstein series and their derivatives. Let Ω be the space of prime-detectin...

conjecture | open | MathDB 页面 | 原文线索

V2-005293 | MATHDB-363055

Fyodorov-Hiary-Keating conjecture on zeta maxima over mesoscopic intervals

摘录 / EXCERPT: Let $\theta\in(-1,0]$, let T tend to infinity, and let τ be uniformly distributed on $[T,2T]$. Write $\mathcal{N}(0,1)$ for a standard Gaussian random variable, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005294 | MATHDB-363064

Mordell curve conjecture on integral points

摘录 / EXCERPT: Mordell curve conjecture. The number of integral points on this curve satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-005295 | MATHDB-363089

Paley graph extractor conjecture

摘录 / EXCERPT: Let p be an odd prime, let $n=\lceil\log_2 p\rceil$, and let $\operatorname{Ext}_p:\mathbb{F}_p\times\mathbb{F}_p\to\{0,1\}$ be the Paley graph extractor defined from the quadrat...

conjecture | open | MathDB 页面 | 原文线索

V2-005296 | MATHDB-363090

Paley ETF RIP-constant conjecture

摘录 / EXCERPT: Let $p\equiv 1\pmod{4}$ be prime, let Φ_p be the Paley equiangular tight frame, and let $\delta_K=\delta_K(\Phi_p)$ denote its RIP constant of order K. Paley ETF RIP-constan...

conjecture | open | MathDB 页面 | 原文线索

V2-005297 | MATHDB-363126

The higher-moment conjecture for the Hurwitz zeta function

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $0<\alpha\leqslant 1$ be irrational. Write $\zeta(s,\alpha)$ for the Hurwitz zeta function and let d denote the algebraic degree of an algebraic numb...

conjecture | open | MathDB 页面 | 原文线索

V2-005298 | MATHDB-363127

The Gaussian value-distribution conjecture for the Hurwitz zeta function

摘录 / EXCERPT: Let $S\subset\mathbb{C}$ be Borel, and let $\zeta(s,\alpha)$ denote the Hurwitz zeta function for a shift parameter $0<\alpha<1$. Gaussian value-distribution conjecture. For large...

conjecture | open | MathDB 页面 | 原文线索

V2-005299 | MATHDB-363145

Verstraëte's dichotomy conjecture for polynomial value sets

摘录 / EXCERPT: Verstraëte's dichotomy conjecture. For some constant $\rho=\rho(k,f)$ depending only on k and f, the maximal size of such a set A is either

conjecture | open | MathDB 页面 | 原文线索

V2-005300 | MATHDB-363150

Erdős and Graham's sparse admissible-set conjecture

摘录 / EXCERPT: Let $A\subseteq\mathbb{N}$ be admissible if there is no prime p such that A contains at least one element in every residue class modulo p. Erdős and Graham's conjecture. Ther...

conjecture | refuted | MathDB 页面 | 原文线索

V2-005301 | MATHDB-363154

Spira's Riemann-hypothesis conjecture for sections of the Hardy Z-function

摘录 / EXCERPT: Spira's conjecture. All the non-trivial zeros of $Z_{N(t)}(t)$ are real.

conjecture | open | MathDB 页面 | 原文线索

V2-005302 | MATHDB-363166

Equivariant quantum Kirwan--Steenrod compatibility conjecture

摘录 / EXCERPT: Let p be a prime, let $\Lambda_{\mathbb F_p}$ be the coefficient ring used in the paper, and let $V/\hspace{-0.1cm}/ K$ be the symplectic quotient. Let $Q\Sigma_b^K$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-005303 | MATHDB-363198

The $p(t)$-adic Littlewood Conjecture over finite fields

摘录 / EXCERPT: Let $\mathcal{F}_q$ be a finite field, let $\mathcal{F}_q((t^{-1}))$ be the field of formal Laurent series, and let $|\Theta(t)|=q^{\deg(\Theta(t))}$ be the norm. For an irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-005304 | MATHDB-363199

Adiceam-Nesharim-Lunnon's paperfolding counterexample conjecture

摘录 / EXCERPT: Let $\mathcal{Upsilon}(t)$ be the Laurent series associated with the Paperfolding sequence, and let t-LC denote the $p(t)$-adic Littlewood Conjecture with $p(t)=t$. Adiceam-Nes...

conjecture | open | MathDB 页面 | 原文线索

V2-005305 | MATHDB-363208

Weak generalized Ramanujan conjecture for L-function coefficients

摘录 / EXCERPT: Weak generalized Ramanujan conjecture. There exists $\eta>0$ such that, for every $n\geqslant 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005306 | MATHDB-363216

Kurihara's strong Mazur-Tate conjecture for Selmer-group Fitting ideals

摘录 / EXCERPT: Let E be an elliptic curve with good ordinary reduction at an odd prime p, let $\theta_n(f)$ be the Mazur-Tate element attached to the modular form f associated with E at...

conjecture | open | MathDB 页面 | 原文线索

V2-005307 | MATHDB-363224

Amoroso-David's obstruction-index conjecture for multiplicatively independent points

摘录 / EXCERPT: Amoroso-David's obstruction-index conjecture. There exists a real number $c(n)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005308 | MATHDB-363225

The adelic Lehmer conjecture for hypersurface heights

摘录 / EXCERPT: Adelic hypersurface Lehmer conjecture. There exists a positive constant $c(n)$, depending only on n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-005309 | MATHDB-363237

Conjecture on projective special linear and unitary Miyamoto groups over prime fields

摘录 / EXCERPT: Projective-group conjecture. There exist parameters $\alpha,\beta,\gamma,\psi\in F$ such that $T\simeq PSL(3,F)$, and a different set of parameters such that $T\simeq PSU(3,F)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-005310 | MATHDB-363276

The asymptotic dimension conjecture for Hilbert cusp forms

摘录 / EXCERPT: Let $n=n(m)$, let $\Gamma_m\subset\mathrm{GL}^+_2(\mathbb{R})^n$ be a sequence of groups commensurable with a Hilbert modular group, and suppose that whenever $m\ne m'$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005311 | MATHDB-363309

Asymptotic growth conjecture for the Laguerre inequalities of p_{24}(n)

摘录 / EXCERPT: Let $p_{24}(n)$ denote the sequence considered in the paper, and let $L_m(p_{24}(n))$ be its m-th Laguerre expression. For $m\in\mathbb{N}$, the asymptotic growth conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-005312 | MATHDB-363333

Stable martingale central limit theorem for multiplicative functions with prime variance parameter

摘录 / EXCERPT: Let $f\in\mathcal{M}$. Suppose that ... for some $\theta\in(0,1)$, and that there exists $c>0$ such that $|f(p^k)|^2=O(2^{k(1-c)})$ and $|f(p)|<1/c$ for all $k\geq 2$ and primes p....

conjecture | open | MathDB 页面 | 原文线索

V2-005313 | MATHDB-363367

Denef-Lipshitz conjecture on integral Diophantine extensions

摘录 / EXCERPT: Denef-Lipshitz conjecture. For every number field L, the extension

conjecture | open | MathDB 页面 | 原文线索

V2-005314 | MATHDB-363368

Existence conjecture for elliptic curves and imaginary quadratic fields at large primes

摘录 / EXCERPT: Large-prime existence conjecture. For all sufficiently large primes p, there exist an elliptic curve $E/\mathbb{Q}$ and an imaginary quadratic field K_0 such that the triple...

conjecture | open | MathDB 页面 | 原文线索

V2-005315 | MATHDB-363370

Montgomery-Vaughan conjecture on large values of quadratic Dirichlet L-functions

摘录 / EXCERPT: Let d range over fundamental discriminants with $|d|\leq X$. Write $\log_2 x=\log\log x$ and $\log_3 x=\log\log\log x$, and let θ and Θ be constants satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-005316 | MATHDB-363371

Coprime-universality conjecture for the ternary forms in Table 1

摘录 / EXCERPT: A ternary quadratic form is coprime-universal with respect to p if it represents every positive integer coprime to p. Coprime-universality conjecture. The ternary forms listed...

conjecture | open | MathDB 页面 | 原文线索

V2-005317 | MATHDB-363376

Congruence conjecture for coefficients of classical modular polynomials

摘录 / EXCERPT: Congruence conjecture. The following divisibilities hold: if $ell\ne 2$, then $a_{m,n}\equiv 0\pmod{2^{15c}}$; if $ell\ne 3$, then $a_{m,n}\equiv 0\pmod{3^{3c}}$, and if additional...

conjecture | open | MathDB 页面 | 原文线索

V2-005318 | MATHDB-363377

Additional 5-divisibility conjecture for modular-polynomial coefficients

摘录 / EXCERPT: Let ell be a prime with $0<m<ell$, and let $a_{ell,ell-m}$ be the coefficient of the ell-th classical modular polynomial $Phi_ell(X,Y)$. Suppose one of the following holds:...

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V2-005319 | MATHDB-363381

Maillet's conjecture on even numbers as differences of primes

摘录 / EXCERPT: Let $\mathbb{P}$ denote the set of primes. An even number is an element of $2\mathbb{N}$. Maillet's conjecture. Every even number is the difference of two primes. This is a foundat...

conjecture | open | MathDB 页面 | 原文线索

V2-005320 | MATHDB-363382

Kronecker's conjecture on infinitely many prime differences

摘录 / EXCERPT: Let $\mathbb{P}$ denote the set of primes. An even number is an element of $2\mathbb{N}$. Kronecker's conjecture. Every even number can be expressed in infinitely many ways as the...

conjecture | open | MathDB 页面 | 原文线索

V2-005321 | MATHDB-363405

Joint mass distribution conjecture for Hecke eigenforms

摘录 / EXCERPT: Let $\mathbb{H}=\{x+iy:x\in\mathbb{R},\ y>0\}$ be the upper half-plane, let $\Gamma=\operatorname{SL}_2(\mathbb{Z})$, and let H_k be a Hecke basis of holomorphic cusp forms on...

conjecture | open | MathDB 页面 | 原文线索

V2-005322 | MATHDB-363409

The Hasse-pair coverage conjecture

摘录 / EXCERPT: A pair of prime powers (q_1,q_2) is a Hasse pair when ... If both prime powers are odd, it is an odd Hasse pair. Hasse-pair coverage conjecture. Every prime power is a member of so...

conjecture | open | MathDB 页面 | 原文线索

V2-005323 | MATHDB-363410

Andrica's conjecture for consecutive prime powers

摘录 / EXCERPT: Let $q_1,q_2,\ldots$ be the consecutive prime powers. Andrica's conjecture for prime powers. The inequality ... holds for every n. This is a denser-sequence analogue of Andrica's c...

conjecture | open | MathDB 页面 | 原文线索

V2-005324 | MATHDB-363420

The monodromy conjecture for Bernstein-Sato polynomials

摘录 / EXCERPT: Monodromy conjecture. If $s_0\in\mathbb C$ is a pole of $Z_{f,\operatorname{top}}(s)$, then $e^{2\pi i s_0}$ is an eigenvalue of T_f and s_0 is a zero of $b_f(s)$.

conjecture | open | MathDB 页面 | 原文线索

V2-005325 | MATHDB-363435

Browkin-Brzeziński's n-terms abc conjecture

摘录 / EXCERPT: Let $n\ge3$. For integers $x_1,\ldots,x_n$, write $\operatorname{rad}(x_1x_2\cdots x_n)$ for the largest positive squarefree divisor of their product. The n-terms abc conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-005326 | MATHDB-363436

The strong n-terms abc conjecture

摘录 / EXCERPT: Let $n\ge3$. For non-zero integers $x_1,\ldots,x_n$, write $\operatorname{rad}(x_1x_2\cdots x_n)$ for the largest positive squarefree divisor of their product. The strong n-terms a...

conjecture | open | MathDB 页面 | 原文线索

V2-005327 | MATHDB-363443

The conjectured constant for the second moment of the shifted prime-divisor function

摘录 / EXCERPT: Second-moment constant conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-005328 | MATHDB-363456

Equality of the extremal function and the first-order comparison function

摘录 / EXCERPT: Let $U(z)$ and $d6d1_1(z)$ be the functions defined in the paper for $z\in(0,2)$. Equality conjecture. ... This is one of the paper's open questions concerning the relationship betwe...

conjecture | open | MathDB 页面 | 原文线索

V2-005329 | MATHDB-363463

Yuan's conjecture on simultaneous Pell equations with multiple solutions

摘录 / EXCERPT: Yuan's conjecture. If this system has at least two solutions in positive integers, then its coefficients are given by

conjecture | open | MathDB 页面 | 原文线索

V2-005330 | MATHDB-363477

Galois-invariant height-zero characterization of normal Sylow subgroups

摘录 / EXCERPT: Galois-invariant normal Sylow subgroup conjecture. Let G be a finite group and let q be a prime. Then G has a normal Sylow q-subgroup if and only if, for any $p\neq q$ that...

conjecture | open | MathDB 页面 | 原文线索

V2-005331 | MATHDB-363509

Strict Schnirelmann-density conjecture for visibility sets

摘录 / EXCERPT: Let $S\subseteq\mathbb{Z}^k$ be a finite set, and let $V(S)$ denote the set of lattice points mutually visible from every point of S. Write $D(V(S))$ for its asymptotic density a...

conjecture | open | MathDB 页面 | 原文线索

V2-005332 | MATHDB-363512

Noncommutative Tao uncertainty principle for prime dimensions

摘录 / EXCERPT: Noncommutative Tao uncertainty principle. For every $x\in\mathcal{A}^p\setminus\{0\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-005333 | MATHDB-363529

Density conjecture for the recursively defined sequence modulo m

摘录 / EXCERPT: Let (a_n) be the recursively defined sequence considered in the paper. For positive integers x and m, let $d_{x,m}$ denote the limiting density of indices n for which...

conjecture | open | MathDB 页面 | 原文线索

V2-005334 | MATHDB-363562

Sugawara's conjecture on ray class invariants for split 2

摘录 / EXCERPT: Let K) be an imaginary quadratic field with discriminant d_K, let $(2)=\mathfrak p_1\mathfrak p_2$, let $h(d_K)$ denote the class number, and let $\textsf{K}_1$ denote the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-005335 | MATHDB-363563

Sugawara's conjecture on automorphisms of a ray class field

摘录 / EXCERPT: Let d be an integer with quadratic discriminant $d_k=\prod_{p\mid d}p^*$, where $p^*=(-1)^{(p-1)/2}p$ for odd p, and let $2^*\in\{-4,8,-8\}$. Let $\textsf{K}_{\wp_5'}$ be the i...

conjecture | open | MathDB 页面 | 原文线索

V2-005336 | MATHDB-363564

Sugawara's conjecture on ramified prime-5 conjugates

摘录 / EXCERPT: Let K be an imaginary quadratic field with discriminant d_K, suppose $(5)=\wp_5^2$, let $\Sigma=\textsf{K}_1$, and let $\rho\in\Sigma$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005337 | MATHDB-363598

Stevenhagen's negative Pell equation density conjecture

摘录 / EXCERPT: Let D range over fundamental positive discriminants without prime factors congruent to 3 modulo 4. Define ... Stevenhagen's conjecture. The density of those D for which the n...

conjecture | open | MathDB 页面 | 原文线索

V2-005338 | MATHDB-363628

The classical-and-abelian composition-factor conjecture for the modular Plesken Lie algebra

摘录 / EXCERPT: Let G be a finite group, let p be a modular prime for G, and let $\mathcal{L}_p[\mathrm{G}]$ be the associated modular Plesken Lie algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005339 | MATHDB-363631

Level-zero quaternionic subgroup conjecture for the Burau representation modulo p

摘录 / EXCERPT: Let p be a prime satisfying $p\equiv1$ (mod 6). Let $\mathcal{Q}_{p}[0]$ be the quaternionic subgroup of level 0, let $\mathcal{K}_{p}$ be the relevant free group, let μ_p...

conjecture | open | MathDB 页面 | 原文线索

V2-005340 | MATHDB-363644

Matrix-form conjecture on solvability of the systems associated with A

摘录 / EXCERPT: Let $r=s+p^t(p-1)d$, with $p \mid d$, $s=b+c(p-1)$, and suppose that $2\leq b\leq p$, $1\leq c\leq p-1$, and $t\geq 2$. Let ... where the coefficients $\alpha(i,l)$ are those defined in th...

conjecture | open | MathDB 页面 | 原文线索

V2-005341 | MATHDB-363645

Monomial-form conjecture for the vectors $q(j)$

摘录 / EXCERPT: Let $r=s+p^t(p-1)d$, with $p \mid d$, $s=b+c(p-1)$, and suppose that $2\leq b\leq p$ and $1\leq c\leq p-1$. Fix a_p such that $1<\nu(a_p)<p-1$, and let $t>\nu(a_p)+c$. Suppose als...

conjecture | open | MathDB 页面 | 原文线索

V2-005342 | MATHDB-363703

The weak inhomogeneous Duffin-Schaeffer conjecture of Catlin

摘录 / EXCERPT: Weak inhomogeneous Duffin-Schaeffer conjecture. The condition denoted by ... , so the mathematical content and current status cannot be checked from the excerpt.

conjecture | open | MathDB 页面 | 原文线索

V2-005343 | MATHDB-363706

Weight-aspect Bergman-kernel size conjecture for Siegel cusp forms

摘录 / EXCERPT: Weight-aspect Bergman-kernel size conjecture. As $k\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005344 | MATHDB-363707

Level-aspect Bergman-kernel size conjecture for Siegel cusp forms

摘录 / EXCERPT: Let $n\geq 1$, let $\Gamma\subset\operatorname{Sp}_n(\mathbf Z)$ be a congruence subgroup of level N, and let $S^{(n)}_k(\Gamma)$ and $\mathbb B_{k,\Gamma}(Z)$ be as above. Level...

conjecture | open | MathDB 页面 | 原文线索

V2-005345 | MATHDB-363709

Level-aspect sup-norm conjecture for Siegel Hecke eigenforms

摘录 / EXCERPT: Level-aspect sup-norm conjecture. With F and Γ as above, as $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005346 | MATHDB-363710

Nonvanishing conjecture for Siegel Poincaré series of exponential type

摘录 / EXCERPT: Let $\Gamma_n=\operatorname{Sp}_n(\mathbf Z)$, let T be a positive-definite index for the Siegel Poincaré series P_{T,Γ_n}, and assume the usual parity condition for scala...

conjecture | open | MathDB 页面 | 原文线索

V2-005347 | MATHDB-363726

The arithmetic Mabuchi infimum conjecture for log Fano and log general type pairs

摘录 / EXCERPT: Let $\mathbb{F}$ be a number field and let $(X_{\mathbb{F}},D_{\mathbb{F}})$ be a log pair such that $\pm K_{(X_{\mathbb{F}},\Delta_{\mathbb{F}})}$ is ample. A polarized model...

conjecture | open | MathDB 页面 | 原文线索

V2-005348 | MATHDB-363760

The packing and covering density conjecture for symmetric four-element sets

摘录 / EXCERPT: Let $a<b$ be coprime positive integers, and let ... For a finite set $S\subset\mathbb{Z}$, write $d_p(S)$ for its packing density and $d_c(S)$ for its covering density. The packing a...

conjecture | open | MathDB 页面 | 原文线索

V2-005349 | MATHDB-363794

Huang's rational-point upper-bound conjecture for nondegenerate submanifolds

摘录 / EXCERPT: Let $\mathcal M$ be a bounded immersed submanifold of $\mathbb{R}^{M}$ with boundary, of dimension n and codimension R. Suppose that $\mathcal M$ is l-nondegenerate everywhere...

conjecture | open | MathDB 页面 | 原文线索

V2-005350 | MATHDB-363796

A dimension growth conjecture for nondegenerate manifolds

摘录 / EXCERPT: Let $\mathcal M$ be a bounded immersed submanifold of $\mathbb{R}^{M}$ with boundary, of dimension n and codimension R. Suppose that $\mathcal M$ is l-nondegenerate for...

conjecture | open | MathDB 页面 | 原文线索

V2-005351 | MATHDB-363817

Strong Ramanujan bound for traces of Hecke operators on Drinfeld cusp forms

摘录 / EXCERPT: Let $A=\mathbb{F}_q[T]$, let $\mathfrak p\triangleleft A$ be a maximal ideal of degree d, and let $n\geq 1$. For integers k,l, write $\mathbf{T}_{\mathfrak p}$ for the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-005352 | MATHDB-363818

The base-p digit-sum conjecture for planar monomial exponents

摘录 / EXCERPT: Let $q=p^n$, and write $s_p(m)$ for the sum of the base-p digits of the positive integer m. Let $e\star d$ denote the operation used in the source on the exponents modulo $q-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-005353 | MATHDB-363827

Weak Duffin-Schaeffer conjecture with a moving target

摘录 / EXCERPT: Weak Duffin-Schaeffer conjecture with a moving target. For every such ρ and $\mathbf y$,

conjecture | open | MathDB 页面 | 原文线索

V2-005354 | MATHDB-363866

Sun's finiteness conjecture for exceptional determinant primes

摘录 / EXCERPT: Let m be a positive odd integer, and define ... Here $S_{n,2}(d,p)$ is the determinant considered in the paper. Sun's finiteness conjecture. For every positive odd integer m, the...

conjecture | open | MathDB 页面 | 原文线索

V2-005355 | MATHDB-363868

Asymptotic representation-growth conjecture for Fuchsian groups

摘录 / EXCERPT: Representation-growth conjecture. (a) There exists a $2A$-periodic sequence $c_{q,1},c_{q,2},\ldots$ of positive numbers such that

conjecture | open | MathDB 页面 | 原文线索

V2-005356 | MATHDB-363910

The FKM conjecture on pairwise partition regularity of quadratic equations

摘录 / EXCERPT: FKM's pairwise partition-regularity conjecture. If at least one of ac, bc, or $(a+b)c$ is a square, then $a x^2+b y^2=c z^2$ is partition regular with respect to x,y.

conjecture | open | MathDB 页面 | 原文线索

V2-005357 | MATHDB-363926

Kummer's relative class number conjecture

摘录 / EXCERPT: Let q be a prime, let $h_1(q)$ be the relative class number of the cyclotomic field $\mathbb Q(\zeta_q)$ over its maximal real subfield, and define ... The Kummer conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-005358 | MATHDB-363927

Euler-Kronecker constant analogue of Kummer's conjecture

摘录 / EXCERPT: Let q be a prime, and let $\kappa(q)$ denote the Euler-Kronecker constant quantity defined in the paper for the maximal real cyclotomic subfield of $\mathbb Q(\zeta_q)$. The Eule...

conjecture | open | MathDB 页面 | 原文线索

V2-005359 | MATHDB-363928

Hardy-Littlewood lower-bound conjecture for admissible prime tuples

摘录 / EXCERPT: Let $\mathcal A=\{a_1,\ldots,a_s\}$ be an admissible set of distinct natural numbers, meaning that the associated linear forms satisfy the usual local non-obstruction condition. Ch...

conjecture | open | MathDB 页面 | 原文线索

V2-005360 | MATHDB-363946

The genus-zero covering subgroup conjecture

摘录 / EXCERPT: Let U_i be the genus-zero hyperbolic curves and let $\Pi_{U_i}^{(2,\ell)}$ be the groups used in the source for a prime ℓ. Let $(\overline{\Pi}_{U_i}^{2,\ell})^{[1]}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-005361 | MATHDB-363947

The p-complement structure conjecture for isolated p-regular classes

摘录 / EXCERPT: Let G be a p-separable group, for a given prime p. Suppose that $x^G,y^G\in\operatorname{Con}(G_{p'})$ are non-central classes such that any other class...

conjecture | open | MathDB 页面 | 原文线索

V2-005362 | MATHDB-363963

Clayton et al.'s nonvanishing conjecture for characteristic-polynomial coefficients

摘录 / EXCERPT: Clayton et al.'s nonvanishing conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-005363 | MATHDB-363969

Conjecture on vanishing and nonvanishing traces on Hecke newspaces

摘录 / EXCERPT: Let $m\geq 2$ be a fixed non-square integer, and let $T_m^{\operatorname{new}}(N,k)$ denote the restriction of the m-th Hecke operator to the new subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-005364 | MATHDB-364018

The Ratios Conjecture for quadratic twists of a modular form

摘录 / EXCERPT: Let f be a holomorphic cusp form with associated quadratic-twist family $\mathcal{F}_f^{+}(X)$, and let $R_f(\alpha,\gamma)$ denote the average of the ratio of shifted L-functi...

conjecture | open | MathDB 页面 | 原文线索

V2-005365 | MATHDB-364020

The conjecture on Waring numbers of ramified 2-adic rings

摘录 / EXCERPT: Let k be an even positive integer and let $e\geq 1$. Write $g_{2,e}(k)$ for the least number of k-th powers needed to represent every element of the ramified 2-adic ring of r...

conjecture | open | MathDB 页面 | 原文线索

V2-005366 | MATHDB-364034

The Fourier-coefficient growth criterion for cuspidality of Siegel modular forms

摘录 / EXCERPT: Let $c\in\mathbf R$, let $k_0\in\mathbf N$ be a scalar weight, and let $F\in M_k^n(\Gamma)$, where $\Gamma\subset\operatorname{Sp}_n(\mathbf Z)$ is a congruence subgroup. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-005367 | MATHDB-364035

The vector-valued generalization of the Fourier-coefficient cuspidality theorem

摘录 / EXCERPT: Let ρ be a representation and let $\widetilde{M}_n(k,\rho,\widetilde{\Gamma})$ denote the corresponding space of vector-valued Siegel modular forms. Vector-valued generalizati...

conjecture | open | MathDB 页面 | 原文线索

V2-005368 | MATHDB-364036

The positive-density refinement of the Fourier-coefficient cuspidality conjecture

摘录 / EXCERPT: Positive-density refinement. A positive answer to the Fourier-coefficient growth conjecture holds when the growth condition is imposed on a set S with $\delta_+(S)>0$. The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-005369 | MATHDB-364044

The heuristic for the oblong cubic surface

摘录 / EXCERPT: Oblong-surface conjecture. The main heuristic holds with

conjecture | open | MathDB 页面 | 原文线索

V2-005370 | MATHDB-364061

The distributional conjecture for normalized L-invariants

摘录 / EXCERPT: Distributional conjecture. For each sign $\pm$, the sets $\mathbf y_T^{\pm}(\overline{\rho})$ become equidistributed for Lebesgue measure on $[-1,0]$ as $T\to\infty$. This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-005371 | MATHDB-364096

Atkin-Serre conjecture for Fourier coefficients at composite indices

摘录 / EXCERPT: Generalized Atkin-Serre conjecture. Under these hypotheses, the stated lower bound holds for every nonzero Fourier coefficient $a_f(m)$. This extends the Atkin-Serre conjecture fro...

conjecture | open | MathDB 页面 | 原文线索

V2-005372 | MATHDB-364115

Cohen-Lenstra-Martinet average torsion conjecture for class groups of G-extensions

摘录 / EXCERPT: Let k be a number field, let G be a transitive permutation group, and let p be a prime such that $p\nmid \lvert G\rvert$. For a real number $X>1$, let $E_k(G,X)$ be the set o...

conjecture | open | MathDB 页面 | 原文线索

V2-005373 | MATHDB-364131

OEIS conjecture on moduli with one zero in the Pisano period

摘录 / EXCERPT: Let m be a positive integer. The Pisano period modulo m is one shortest period of the Fibonacci sequence modulo m, and its number of zeros is the order of m. OEIS conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-005374 | MATHDB-364132

Even-K extension of rank and order properties for K-Fibonacci sequences

摘录 / EXCERPT: Let K be even, and let $F_{K,n}$ be the K-Fibonacci sequence. For a modulus m, let $\pi_K(m)$ denote its Pisano period, let $\omega_K(m)$ denote the number of zeros in that p...

conjecture | refuted | MathDB 页面 | 原文线索

V2-005375 | MATHDB-364133

Multiplication table for orders of (a,b)-Fibonacci sequences with $b=-1$

摘录 / EXCERPT: Let (a,b)-Fibonacci sequences be defined by $\mathcal{U}_0=0$, $\mathcal{U}_1=1$, and $\mathcal{U}_n=a\mathcal{U}_{n-1}+b\mathcal{U}_{n-2}$. Let $\omega_{(a,-1)}(m)$ be the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-005376 | MATHDB-364134

Infinitely many moduli for every order in the infinite-range case

摘录 / EXCERPT: Let (a,b) be an (a,b)-Fibonacci sequence, and suppose it falls under case (vi) of the finite-orders conjecture, namely the case where the range of $\omega_{(a,b)}(m)$ is conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-005377 | MATHDB-364142

A gap conjecture for Sylow numbers

摘录 / EXCERPT: Let p be an odd prime, and let G be a finite group with subgroup H. For a finite group X, write $ u_p(X)$ for the number of Sylow p-subgroups of X. Gap conjecture for S...

conjecture | open | MathDB 页面 | 原文线索

V2-005378 | MATHDB-364147

Nonexistence of large-prime lens spaces as dual complexes of log Calabi-Yau fourfolds

摘录 / EXCERPT: Let p be a prime and let k be an integer. Consider a dlt log-Calabi-Yau 4-fold pair, meaning a dlt pair of dimension 4 whose log canonical divisor is numerically trivial, a...

conjecture | open | MathDB 页面 | 原文线索

V2-005379 | MATHDB-364150

Strong Chinburg's conjecture on Mahler measures and Dirichlet L-values

摘录 / EXCERPT: Let $m(P)$ denote the Mahler measure of a non-zero polynomial P, let $L'(\,\cdot\,,-n)$ denote the derivative of a Dirichlet L-function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005380 | MATHDB-364151

Generalized weak Chinburg's conjecture for primitive odd characters

摘录 / EXCERPT: Let $m(R)$ denote the Mahler measure of a non-zero rational function R, let $L'(\chi,-n)$ denote the derivative of the Dirichlet L-function associated with a character χ,...

conjecture | open | MathDB 页面 | 原文线索

V2-005381 | MATHDB-364152

Boyd's conjecture on the closedness of Mahler measures

摘录 / EXCERPT: Boyd's conjecture. The set $L^\sharp$ is closed with respect to the Euclidean topology.

conjecture | open | MathDB 页面 | 原文线索

V2-005382 | MATHDB-364182

Nesterenko's CM-point conjecture for modular values

摘录 / EXCERPT: Let $\tau\in\mathbb{H}$. The numbers ... are considered, where E_2, E_4, and E_6 denote the usual Eisenstein series. Nesterenko's conjecture. If at most three of these five num...

conjecture | open | MathDB 页面 | 原文线索

V2-005383 | MATHDB-364190

Factorization into two orthogonal quasi-prime matrices

摘录 / EXCERPT: Let n be an even natural number greater than 2. A quasi-prime matrix is a matrix that is the product of two quasi-invertible semi-prime matrices, where a semi-prime matrix is a...

conjecture | open | MathDB 页面 | 原文线索

V2-005384 | MATHDB-364191

Noncongruent quasi-prime matrices in prime dimensions

摘录 / EXCERPT: For matrices A and B of size $m\times n$, say that they are congruent if there are invertible square matrices U and V of orders m and n, respectively, such that $UA=BV$...

conjecture | open | MathDB 页面 | 原文线索

V2-005385 | MATHDB-364194

Non-triangular local resolution conjecture for repeated non-Desarguesian affine planes

摘录 / EXCERPT: Let q be a prime power, and let a BIBD consist of q copies of a non-Desarguesian affine plane of order q. A non-triangular local resolution system is a local resolution syste...

conjecture | open | MathDB 页面 | 原文线索

V2-005386 | MATHDB-364198

The Newton-polygon divisibility conjecture for Jacobian pairs

摘录 / EXCERPT: Let $\mathcal{R}=\mathbb{C}[x,y]$, and let $N^0(F)$ denote the Newton polygon of F with the origin as reference point. Let $a,b\in\mathbb{Z}_{>0}$ be relatively prime with...

conjecture | open | MathDB 页面 | 原文线索

V2-005387 | MATHDB-364268

Exponential path-length conjecture for comma-sequence graphs

摘录 / EXCERPT: For a base b, let G_b' be the finite graph constructed from comma-sequence danger intervals, and let P be a uniformly randomly chosen path in G_b'. Write $|P|$ for the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-005388 | MATHDB-364272

The Ratios Conjecture for local averages of three-cube sums

摘录 / EXCERPT: Fix a real $M\geqslant 0$. Let $\bm{a}\in \mathcal{O}^6$ and $d\in \mathcal{O}^+$ with $\lvert \bm{a}\rvert,\lvert d\rvert\leqslant \widehat M$. Let $\bm{b}\in K_\infty^6$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-005389 | MATHDB-364274

Browning-Wilsch conjecture for integral points on log K3 cubic surfaces

摘录 / EXCERPT: Let $U \subset \mathbb{A}^3$ be a cubic surface that is smooth and log K3 over $\mathbb{Q}$ and defined by a cubic polynomial $f \in \mathbb{Z}[x_1,x_2,x_3]$. Let $N_U^\circ(B)$ co...

conjecture | open | MathDB 页面 | 原文线索

V2-005390 | MATHDB-364284

Generalized Sun conjecture for the summatory function of $(-a)^{\Omega(n)}$

摘录 / EXCERPT: For real $a>0$, let $\Omega(n)$ denote the number of prime factors of n, counting multiplicity, and define ... For $a\geq 2$, the function has jumps of size $x^{\log a/\log 2}$ at...

conjecture | open | MathDB 页面 | 原文线索

V2-005391 | MATHDB-364285

Large-parameter conjecture for the normalized generalized summatory function

摘录 / EXCERPT: For real $a>0$, define ... Let $\widetilde{W}_a(x)$ and s_a be the normalized function and its limit superior defined in the source's preceding theorem. Large-parameter conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-005392 | MATHDB-364322

Gordon-Pomerance conjecture for integers of prescribed rank of apparition

摘录 / EXCERPT: Gordon-Pomerance conjecture. With the same hypotheses as Lemma Gordon-Pomerance, the displayed upper bound holds. The conjecture is used to obtain the conditional upper bound for t...

conjecture | open | MathDB 页面 | 原文线索

V2-005393 | MATHDB-364323

Pomerance's shifted-prime smoothness conjecture

摘录 / EXCERPT: For $X,Y\geq1$, let $P(n)$ be the greatest prime factor of n, and define ... Let $\Psi(X,Y)$ denote the count of positive integers at most X whose prime factors are all at most...

conjecture | open | MathDB 页面 | 原文线索

V2-005394 | MATHDB-364329

The odd-prime localization conjecture for symplectic annular Khovanov homology

摘录 / EXCERPT: Let L be a p-periodic link for an odd prime p, and let $\overline{L}$ be its quotient link, treated as an annular link with the axis of symmetry as the annular axis. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005395 | MATHDB-364357

Greenberg's μ=0 conjecture for elliptic curves over ℚ

摘录 / EXCERPT: Let $E/\mathbf{Q}$ be an elliptic curve, and let p be an odd prime of good ordinary reduction. The representation $E[p]$ is irreducible as a Galois module when it has no nontrivi...

conjecture | open | MathDB 页面 | 原文线索

V2-005396 | MATHDB-364363

Adelic hypergeometric homology freeness conjecture

摘录 / EXCERPT: For positive integers d and N, and $\lambda \in L(\mathbb{C})$, consider the smooth hypersurface $U_{N,\lambda}^{(d)} \subset \mathbb{A}^{d+1}$ defined by ... The group...

conjecture | open | MathDB 页面 | 原文线索

V2-005397 | MATHDB-364365

The Matrix Coefficient Conjecture for singular matrices of logarithms

摘录 / EXCERPT: Let $\mathscr{L}$ be the $\mathbf{Q}$-vector space of logarithms of algebraic numbers, and for a prime p let $\mathscr{L}_p$ be the corresponding space of p-adic logarithms. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-005398 | MATHDB-364379

Special-value conjecture for Ramanujan-Sato hypergeometric series and weight-3 modular forms

摘录 / EXCERPT: Special-value conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005399 | MATHDB-364408

Minimality conjecture for models of non-rational Hilbert modular surfaces

摘录 / EXCERPT: Let N be an integer coprime to 6, let r be as in the construction, and let $Z^\circ_{N,r}$ and $W_{N,r}^{\mathsf{small}}$ be the models defined in the nonsingular-model and g...

conjecture | open | MathDB 页面 | 原文线索

V2-005400 | MATHDB-364411

The coprime dice relabeling conjecture

摘录 / EXCERPT: Coprime dice relabeling conjecture. There are no ways to relabel a die of size r and a die of size s without changing the frequencies of their sums.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005401 | MATHDB-364413

The li_0-li_1 bounds conjecture for the prime-counting function

摘录 / EXCERPT: The li_0-li_1 bounds conjecture. For every real $x\ge e$,

conjecture | open | MathDB 页面 | 原文线索

V2-005402 | MATHDB-364414

The truncated asymptotic bounds conjecture for the prime-counting function

摘录 / EXCERPT: Truncated asymptotic bounds conjecture. For every real $x\ge e$,

conjecture | open | MathDB 页面 | 原文线索

V2-005403 | MATHDB-364420

The multiplicative Dream Theorem for non-degenerate manifolds

摘录 / EXCERPT: Let $\mathcal M$ be a non-degenerate submanifold of $\mathbb R^k$, and let $\psi:\mathbb N\to[0,1)$ be non-increasing. Let $W_k^\times(\psi)$ denote the set of points in $[0,1)^k$...

conjecture | open | MathDB 页面 | 原文线索

V2-005404 | MATHDB-364468

The critical-window conjecture for cokernels of random band matrices

摘录 / EXCERPT: Let p be a prime, let $(\nu_{\alpha})_{\alpha\in\mathbb{R}}$ be a proposed one-parameter family of distributions on the set of finite abelian p-groups, and let $n_1<n_2<\dots$...

conjecture | open | MathDB 页面 | 原文线索

V2-005405 | MATHDB-364485

Gao-Li conjecture on the Gao constant of finite non-cyclic groups

摘录 / EXCERPT: Let G be a finite group. Write $|G|$ for its order and let $\mathsf{E}(G)$ denote its Gao constant, namely the least integer such that every sequence over G of that length cont...

conjecture | open | MathDB 页面 | 原文线索

V2-005406 | MATHDB-364486

Dimitrov's conjecture on the ordered Davenport constant and Loewy length

摘录 / EXCERPT: Let p be a prime and let G be a finite p-group. The Loewy length $\mathsf{L}(G)$ is the nilpotency index of the Jacobson radical of the group algebra $\mathbb{F}_p[G]$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005407 | MATHDB-364499

The Jacobi-symbol conjecture for the integers c_p

摘录 / EXCERPT: Jacobi-symbol conjecture for c_p. For any odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-005408 | MATHDB-364517

Young's restricted quantum unique ergodicity conjecture for holomorphic cusp forms

摘录 / EXCERPT: Let $\psi\colon \mathbb{R}^+ \to \mathbb{R}$ be a smooth compactly supported function, and let $f(z)$ run over L^2-normalized holomorphic Hecke cusp forms of weight k. Young's...

conjecture | open | MathDB 页面 | 原文线索

V2-005409 | MATHDB-364520

Refined small-solution conjecture for generic ternary quadratic congruences

摘录 / EXCERPT: Let $q\in\mathbb{N}$ be odd and let $\alpha_1,\alpha_2,\alpha_3\in\mathbb{Z}$ satisfy $(\alpha_1\alpha_2\alpha_3,q)=1$. For $r\bmod q$ with $r\not\equiv0\pmod q$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-005410 | MATHDB-364528

Cheung's logarithmic growth conjecture for multidimensional best approximations

摘录 / EXCERPT: Let m be a positive integer and let $\theta\in\mathbb{R}^m$. For each best approximation $(p_l,q_l)\in\mathbb{Z}^m\times\mathbb{N}$ of θ, the only primitive vectors of the...

conjecture | open | MathDB 页面 | 原文线索

V2-005411 | MATHDB-364537

Sato's conjecture for $\kappa(qp)$

摘录 / EXCERPT: For distinct prime numbers p and q, let $H_{p,q}$ be the subgroup of $(\mathbb{Z}/p\mathbb{Z})^{\times}$ spanned by the images of q and -1 in $\mathbb{F}_{p}^{\times}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005412 | MATHDB-364549

Cooper's Lucas congruence conjecture for modular-form sequences

摘录 / EXCERPT: Cooper's conjecture. The sequences $\{T_{14,\pm}(n)\}$ satisfy the Lucas congruence modulo p if and only if $p=2$ or $p\equiv1,7\pmod{8}$, and the sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-005413 | MATHDB-364552

Zywina's refined Koblitz conjecture for elliptic curves

摘录 / EXCERPT: Zywina's refined Koblitz conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005414 | MATHDB-364557

Rational-coefficient representation conjecture for the modular anomaly function

摘录 / EXCERPT: Rational-coefficient representation conjecture. The coefficients $e_{\mathcal{T}}$ are rational numbers depending only on the topology of $\mathcal{T}$. This proposed representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-005415 | MATHDB-364583

The non-split Cartan conjecture for mod p representations of elliptic curves

摘录 / EXCERPT: Let p be a prime and let $E/\mathbb{Q}$ be a non-CM elliptic curve. Consider the mod p Galois representation of E, whose image is a subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-005416 | MATHDB-364584

Equidistribution conjecture for anomalous shadow lines in the receptacle

摘录 / EXCERPT: Let $E/\mathbf{Q}$ be an elliptic curve of analytic rank 2, let p be an odd prime of good ordinary anomalous reduction, and let K be an imaginary quadratic field satisfying t...

conjecture | open | MathDB 页面 | 原文线索

V2-005417 | MATHDB-364585

Independence of the natural line from the quadratic field

摘录 / EXCERPT: Let (E,p,K) be a triple in which E is an elliptic curve over $\mathbf{Q}$ of analytic rank 2, p is an odd prime of anomalous good ordinary reduction for E, and K is an...

conjecture | open | MathDB 页面 | 原文线索

V2-005418 | MATHDB-364596

Maximal divisibility-index conjecture for partial sums of Pell numbers

摘录 / EXCERPT: Let P_n) be the Pell numbers defined by $P_n=2P_{n-1}+P_{n-2}$ for $n>1$, with $P_0=0$ and $P_1=1$. Let S^P_n denote the sum of the first n Pell numbers, and let m^P_n be t...

conjecture | open | MathDB 页面 | 原文线索

V2-005419 | MATHDB-364627

Murmurations for Atkin-Lehner operators

摘录 / EXCERPT: Let $(\mathcal N,\mathcal Q)$ be an arithmetically compatible sequence of (N,Q)'s of Type I, II or III as above, and fix a weight k. Let $\mathcal F$ be the family of newforms...

conjecture | open | MathDB 页面 | 原文线索

V2-005420 | MATHDB-364628

Murmurations on Atkin-Lehner eigenspaces

摘录 / EXCERPT: Fix k,r and $0\le m<r$, and fix primes $p_1<\dots<p_m$. Let $\mathcal N$ be the set of squarefree levels $N=p_1\dots p_r$ with r prime factors, where the first m primes are f...

conjecture | open | MathDB 页面 | 原文线索

V2-005421 | MATHDB-364634

Polynomial height reduction conjecture for fixed degree and discriminant

摘录 / EXCERPT: Let $f\in\mathbb{Z}[X]$ be a polynomial, respectively a monic polynomial, of degree $n\geq 4$, respectively $n\geq 3$, with discriminant $D\ne 0$. Here $H(g)$ denotes the height of...

conjecture | open | MathDB 页面 | 原文线索

V2-005422 | MATHDB-364640

CFKRS moment conjecture for the random Euler-product model of function-field L-functions

摘录 / EXCERPT: Let r and $\tilde{r}$ be nonnegative integers, let $q>2$ be a prime power, let $s_1,\dots,s_{r+\tilde{r}}$ be complex numbers with real part $\frac{1}{2}$, and let A be a real...

conjecture | open | MathDB 页面 | 原文线索

V2-005423 | MATHDB-364642

Saikia-Sarma-Sellers conjecture for overpartition tuples with even length

摘录 / EXCERPT: Let $i\geq 1$, let $n\geq 0$, and let r be odd. The function $\overline{OPT}_{2^i r}(n)$ counts overpartition $2^i r$-tuples with odd parts.
Saikia-Sarma-Sellers' conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005424 | MATHDB-364658

Stevenhagen's conjecture for odd continued-fraction periods of products of two primes

摘录 / EXCERPT: Stevenhagen's conjecture. The following limit exists and equals $\frac{2}{3}$:

conjecture | open | MathDB 页面 | 原文线索

V2-005425 | MATHDB-364668

The density conjecture for scalar-valued P-adic modular forms

摘录 / EXCERPT: Density conjecture. The space

conjecture | open | MathDB 页面 | 原文线索

V2-005426 | MATHDB-364669

The P-ordinary classicality conjecture for scalar-valued cusp forms

摘录 / EXCERPT: Scalar P-ordinary classicality conjecture. The restriction

conjecture | open | MathDB 页面 | 原文线索

V2-005427 | MATHDB-364670

The independence-of-weight conjecture for the P-ordinary Hecke algebra

摘录 / EXCERPT: Independence-of-weight conjecture. If κ_1 and κ_2 are dominant characters such that $[\kappa_1]=[\kappa_2]$, then there is a canonical isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-005428 | MATHDB-364671

The finite-freeness and specialization conjecture for the big P-ordinary Hecke algebra

摘录 / EXCERPT: Big P-ordinary Hecke algebra conjecture. 1. For each tame character ϵ, $\mathbf{T}^{P\text{-ord}}_{K^p,[\kappa,\tau],R,\epsilon}$ is finite free over...

conjecture | open | MathDB 页面 | 原文线索

V2-005429 | MATHDB-364678

Partition rank versus analytic rank conjecture

摘录 / EXCERPT: Let $d\geq 2$, let q be a prime power, and let $T\in\mathbb{F}_q^{n_1}\otimes\cdots\otimes\mathbb{F}_q^{n_d}$, with $\operatorname{PR}_{\mathbb{F}_q}(T)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005430 | MATHDB-364679

Adiprasito-Schmidt stability conjecture for partition rank

摘录 / EXCERPT: Let $d\geq 2$, let q be a prime power, and let $T\in\mathbb{F}_q^{n_1}\otimes\cdots\otimes\mathbb{F}_q^{n_d}$. Write $\overline{\mathbb{F}}_q$ for an algebraic closure of...

conjecture | open | MathDB 页面 | 原文线索

V2-005431 | MATHDB-364680

Asymptotic direct sum conjecture for partition rank

摘录 / EXCERPT: Let $d\geq 2$, let q be a prime power, and let $T\in\mathbb{F}_q^{n_1}\otimes\cdots\otimes\mathbb{F}_q^{n_d}$. For $n\geq 1$, let $T^{\oplus n}$ denote the n-fold direct sum of...

conjecture | open | MathDB 页面 | 原文线索

V2-005432 | MATHDB-364712

Analytic continuation and pole conjecture for subform zeta-function terms

摘录 / EXCERPT: Let L be a $\mathbb{Z}$-lattice of rank $n\geq 2$ on a positive definite quadratic space V. Let $Z_{L,\pm 1}$ and $Z_{L,\pm 1}^{+}$ denote the series associated with the trivia...

conjecture | open | MathDB 页面 | 原文线索

V2-005433 | MATHDB-364713

Conlon-Lim's Brunn-Minkowski conjecture for discrete linear sumsets

摘录 / EXCERPT: Let $d,k4in\mathbb{N}$ and let $\mathcal{L}_1,\dots,\mathcal{L}_k\in\operatorname{Mat}_d(\mathbb{Z})$ be irreducible and coprime. Let $A\subseteq\mathbb{Z}^d$ be finite and non-emp...

conjecture | open | MathDB 页面 | 原文线索

V2-005434 | MATHDB-364718

The non-k-gonal pyramidal number formula for $k\geq 9$

摘录 / EXCERPT: Let $k\geq 9$, and let the n-th non-k-gonal pyramidal number be the n-th positive integer that is not a k-gonal pyramidal number. The preceding formula defines this sequenc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005435 | MATHDB-364724

The no-fragmentation criterion for plateaux in the expanded quantum infinite well

摘录 / EXCERPT: Let N be the parameter appearing in the expanded quantum infinite well, let $\Lambda\in\mathbb{Q}$, and assume that there is no fragmentation. A plateau is a plateau of the proba...

conjecture | open | MathDB 页面 | 原文线索

V2-005436 | MATHDB-364749

The toric section conjecture for hyperbolic curves over finitely generated fields

摘录 / EXCERPT: Toric section conjecture. This map is bijective. The conjecture is proposed as an intermediate step toward Grothendieck's section conjecture; the paper proves injectivity of the na...

conjecture | open | MathDB 页面 | 原文线索

V2-005437 | MATHDB-364761

McIver-Neumann conjecture on the enumeration of finite groups

摘录 / EXCERPT: Let $f(n)$ be the number of isomorphism classes of groups of order n, and let λ be the number of prime divisors of n, counted with multiplicities. McIver-Neumann conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-005438 | MATHDB-364773

Finiteness conjecture for modular curves with noncuspidal orbifold Q-subcanonical points

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many modular curves, up to isomorphism, over $\mathbb{C}$ that have orbifold $\mathbb{Q}$-subcanonical points outside cusps.

conjecture | open | MathDB 页面 | 原文线索

V2-005439 | MATHDB-364848

Dickson's conjecture for residue class patterns of consecutive primes

摘录 / EXCERPT: For $q,m \in \mathbb{N}$, let p_n denote the n-th prime and define ... for $\mathbf{a} \in \prod_{i=1}^m (\mathbb{Z}/q\mathbb{Z})^\times$. Dickson's conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-005440 | MATHDB-364861

Ramaekers's strong n-conjecture

摘录 / EXCERPT: Let $n\geq 3$. For an n-tuple $a=(a_1,\ldots,a_n)\in\mathbb{Z}^n$ with nonzero entries, write ... and for a sequence $A=\{a^{(1)},a^{(2)},\ldots\}$ of such tuples write...

conjecture | open | MathDB 页面 | 原文线索

V2-005441 | MATHDB-364898

Prime K-tuples conjecture

摘录 / EXCERPT: Let $\mathcal{L} = \{L_1,\ldots,L_K\}$ be a set of distinct linear forms $L_k(n) = a_kn+b_k$, where the coefficients a_k,b_k are positive integers. For a prime p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-005442 | MATHDB-364904

Modular rigidity conjecture for cuspidal representations of ${\rm GL}_n$

摘录 / EXCERPT: Let k be a global field, let ℓ be a prime different from the characteristic of k, and let $\iota:\mathbb{C}\to\overline{\mathbb{Q}}_\ell$ be a field isomorphism. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005443 | MATHDB-364907

The lowest-point algebraicity conjecture for the generalized Lagrange spectra

摘录 / EXCERPT: Lowest-point algebraicity conjecture. For all positive integers n, the lowest point $\min \mathcal{L}_n$ is an isolated point of $\mathcal{L}_n$ and is the irrational square root...

conjecture | open | MathDB 页面 | 原文线索

V2-005444 | MATHDB-364908

Strong Markoff uniqueness conjecture

摘录 / EXCERPT: Strong Markoff uniqueness conjecture. Each of the badly approximable irrationals ξ corresponding to a Markoff triple (x,y,z) has a different field of definition...

conjecture | open | MathDB 页面 | 原文线索

V2-005445 | MATHDB-364909

Maximum conjecture for the lowest generalized Lagrange-spectrum point

摘录 / EXCERPT: Maximum conjecture. The quantity $\min \mathcal{L}_p$ attains a maximum at $p=67$, and

conjecture | open | MathDB 页面 | 原文线索

V2-005446 | MATHDB-364910

Uniqueness and reversal conjecture for minimal prime-modulus approximations

摘录 / EXCERPT: Minimal-period uniqueness conjecture. For each prime p, the periodic part of the continued fraction of such a minimizing ξ is unique up to reversal. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-005447 | MATHDB-364948

Dual form of Littlewood's conjecture in d dimensions

摘录 / EXCERPT: Let $H(\mathbf m)$ denote the multiplicative height of $\mathbf m \in \mathbb Z^d$, and let ${\boldsymbol{\alpha}} \in \mathbb R^d$ with $d \geqslant 2$. Dual form of Littlewood's...

conjecture | open | MathDB 页面 | 原文线索

V2-005448 | MATHDB-364949

Littlewood's conjecture for products of linear forms

摘录 / EXCERPT: Let f be a product of k linear forms in $\mathbb R^k$, where $k \geqslant 3$, and assume that f is not proportional to a multiple of a polynomial with integer coefficients. L...

conjecture | open | MathDB 页面 | 原文线索

V2-005449 | MATHDB-364950

Special case of Margulis's conjecture

摘录 / EXCERPT: Let D be the set of diagonal matrices in $\mathrm{SL}_k(\mathbb R)$, where $k \geqslant 3$. Let $z \in \mathrm{SL}_k(\mathbb R)/\mathrm{SL}_k(\mathbb Z)$, and suppose that Dz h...

conjecture | open | MathDB 页面 | 原文线索

V2-005450 | MATHDB-364951

Smooth discrepancy unboundedness under a nonintegral product-of-forms condition

摘录 / EXCERPT: Let $\mathscr L_k$ denote the space of lattices of the form $A\mathbb Z^k$ with $A \in \mathrm{SL}_k(\mathbb R)$, and let $k \geqslant 3$. Let f be the product of the linear form...

conjecture | open | MathDB 页面 | 原文线索

V2-005451 | MATHDB-364955

Fried conjecture for twisted Ruelle zeta functions of Anosov flows

摘录 / EXCERPT: Let M be a closed 3-manifold, let φ be a smooth transitive Anosov flow on M, and let ρ be an acyclic unitary representation of $\pi_1(M)$. Define the twisted Rue...

conjecture | open | MathDB 页面 | 原文线索

V2-005452 | MATHDB-364969

Weak Premet-Topley conjecture on completely prime primitive ideals and minimal modules

摘录 / EXCERPT: Let ${\mathfrak g}_{\mathbb C}$ be a reductive Lie algebra over ${\mathbb C}$ with Chevalley basis ${\mathcal B}$, let $R={\mathbb Z}[1/q\mid q\text{ is a bad prime for }\Phi]$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-005453 | MATHDB-364986

Conjecture on the limiting variance of the normalized second moment

摘录 / EXCERPT: Let $\mathcal B_{2,\mathcal E}(p)$ denote the normalized second moment of a one-parameter family of elliptic curves, and consider its distribution as the prime p varies. Variance...

conjecture | open | MathDB 页面 | 原文线索

V2-005454 | MATHDB-365018

Spencer's Single Lonely Runner Conjecture

摘录 / EXCERPT: Let $V=\{v_1,\dots,v_n\}$ be an n-set of pairwise distinct real speeds, and let $\|x\|$ denote the distance from x to the nearest integer. Single Lonely Runner Conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005455 | MATHDB-365019

Fan-Sun amended Loneliness Spectrum Conjecture

摘录 / EXCERPT: For a positive-integer speed set V, let $\kappa(V)$ denote its gap of loneliness, and let $n=|V|$. Amended Loneliness Spectrum Conjecture. For every $n\in\mathbb{N}$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-005456 | MATHDB-365021

Bhardwaj-Narayanan-Venkataraman conjecture on lonely times

摘录 / EXCERPT: Let V be an n-set of positive integer speeds, let v_n denote the largest speed, and let $\|x\|$ be the distance from x to the nearest integer. Conjecture on lonely times. F...

conjecture | open | MathDB 页面 | 原文线索

V2-005457 | MATHDB-365022

Chow-Rimanic function-field Lonely Runner Conjecture

摘录 / EXCERPT: Function-field Lonely Runner Conjecture. For every prime power q and every finite set $F\subset\mathbb{F}_q[X]\setminus\{0\}$ of polynomials satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-005458 | MATHDB-365036

The prime and thickened prime component conjecture for Apollonian packings

摘录 / EXCERPT: Let ${\mathcal{P}^{\operatorname{full}}}$ be an ambient primitive integral Apollonian packing. A prime component ${\mathcal{P}^{\operatorname{pr}}}$ is a component consisting of th...

conjecture | open | MathDB 页面 | 原文线索

V2-005459 | MATHDB-365037

The growth conjecture for prime and thickened prime components

摘录 / EXCERPT: Let $C_{{\mathcal{P}^{\operatorname{pr}}}}(X)$ be the number of circles of curvature at most X in a prime component ${\mathcal{P}^{\operatorname{pr}}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005460 | MATHDB-365038

The prime-component counting conjecture for Apollonian packings

摘录 / EXCERPT: Let ${\mathcal{P}^{\operatorname{full}}}$ be a primitive integral Apollonian packing, and let $C_{{\mathcal{P}^{\operatorname{full}}}}(X)$ be the number of circles of curvature at...

conjecture | open | MathDB 页面 | 原文线索

V2-005461 | MATHDB-365058

Katz's intersection-cohomology Newton-Hodge inequality

摘录 / EXCERPT: Katz's intersection-cohomology Newton-Hodge inequality. There exists a finite set $S=S(X,n)$ of maximal ideals of $\mathscr{O}_F$ such that for every prime number ℓ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-005462 | MATHDB-365059

The intersection-cohomology ordinariness conjecture

摘录 / EXCERPT: Intersection-cohomology ordinariness conjecture. For every prime number ℓ, there exists an infinite set of maximal ideals $\mathfrak{p}$ of $\mathscr{O}_F$ with residue charac...

conjecture | open | MathDB 页面 | 原文线索

V2-005463 | MATHDB-365081

Arithmetic exponent pair hypothesis for smooth-conductor trace-function sums

摘录 / EXCERPT: Arithmetic exponent pair hypothesis. For every $\epsilon>0$, provided that $|I|<q\delta$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-005464 | MATHDB-365087

The full nonvanishing result for twisted elliptic transfer

摘录 / EXCERPT: Let E/F be the quadratic extension, let $ \mathfrak{n}_{\mathrm{ur}}$ denote the unramified part of the relevant conductor $ \mathfrak{n}$, and let ω be a character of...

conjecture | open | MathDB 页面 | 原文线索

V2-005465 | MATHDB-365092

PSTVA conjecture on integral points of bounded height on log Fano orbifolds

摘录 / EXCERPT: PSTVA conjecture. Suppose that L is nef and $(\mathcal{X},\mathcal{D})(\mathcal{O}_{K,S})$ is not thin. Then there exists a thin set...

conjecture | open | MathDB 页面 | 原文线索

V2-005466 | MATHDB-365100

Conjecture on the odd moments of singular series differences

摘录 / EXCERPT: Odd-moment conjecture. As h tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-005467 | MATHDB-365110

Bertrand-Rodriguez Villegas conjecture for exterior powers of units

摘录 / EXCERPT: Bertrand-Rodriguez Villegas conjecture. There exist absolute constants $c_0>0$ and $c_1>1$ such that, for every number field L, every $j\in\mathbb{Z}_{>0}$, and every nonzero

conjecture | open | MathDB 页面 | 原文线索

V2-005468 | MATHDB-365119

Soundararajan's maximal-order conjecture for the divisor problem

摘录 / EXCERPT: Let $\Delta(x)$ be the remainder term in the summatory divisor problem, ... where $d(n)$ is the divisor function, γ is the Euler-Mascheroni constant, and $\log_k$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-005469 | MATHDB-365125

Logarithmic growth conjecture for integral points on the cubic surface S_0

摘录 / EXCERPT: Let S_0 be the cubic surface introduced earlier in the paper, and consider its integral points with coordinates satisfying $1\leq x,y\leq N$. A growth conjecture asserts that the...

conjecture | open | MathDB 页面 | 原文线索

V2-005470 | MATHDB-365139

The exchange-pair conjecture for real prime modules

摘录 / EXCERPT: Let M and N be dominant monomials such that $L(M)$ and $L(N)$ are real prime modules, and let $\mathfrak{e}(\mathbf{g}_M,\mathbf{g}_N)$ be their generic symmetrized E-invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-005471 | MATHDB-365151

Unboundedness and surjectivity conjecture for corresponding-sequence lengths

摘录 / EXCERPT: Unboundedness and surjectivity conjecture. The function $T(n)$ is unbounded. Moreover, it is surjective onto the set of positive integers other than 2.

conjecture | open | MathDB 页面 | 原文线索

V2-005472 | MATHDB-365152

Characterization conjecture for equality in the bound on g(n)

摘录 / EXCERPT: Equality characterization conjecture. The upper bound is attained,

conjecture | open | MathDB 页面 | 原文线索

V2-005473 | MATHDB-365178

Ogg's cuspidal torsion conjecture for the Jacobian of X_0(p)

摘录 / EXCERPT: Let p be prime, let $J_0(p)$ be the Jacobian of $X_0(p)$, and let $\mathcal{C}(p)$ be its rational cuspidal divisor class group. Ogg's cuspidal torsion conjecture. The cyclic gro...

conjecture | open | MathDB 页面 | 原文线索

V2-005474 | MATHDB-365179

Ogg's conjecture on the Shimura subgroup of J_0(p)

摘录 / EXCERPT: Let p be prime. Let $\mathcal{S}(p)$ be the Shimura subgroup of $J_0(p)$, and let $\mathcal{M}(p)$ be the largest μ-type subgroup of $J_0(p)$. Ogg's Shimura-subgroup conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-005475 | MATHDB-365181

The Drinfeld analogue of Ogg's rational cyclic-subgroup conjecture

摘录 / EXCERPT: Drinfeld cyclic-subgroup conjecture. If $\mathfrak{p}$ is a prime of A with $\deg\mathfrak{p}\geq C$, then there is no F-rational A-submodule of $\phi[\mathfrak{p}]$ isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-005476 | MATHDB-365183

The Drinfeld analogue of Ogg's Shimura-subgroup conjecture

摘录 / EXCERPT: Let $\mathfrak{p}$ be a prime of A, let $J_0(\mathfrak{p})$ be the Jacobian of the Drinfeld modular curve $X_0(\mathfrak{p})$, let $\mathcal{S}(\mathfrak{p})$ be the Shimura subg...

conjecture | open | MathDB 页面 | 原文线索

V2-005477 | MATHDB-365190

Greenberg's analogue for S-ramified extensions of CM fields

摘录 / EXCERPT: Let p be an odd prime, let K be a CM field, and let K^+ be its maximal totally real subfield. Let S^+ be a set of primes of K^+ above p, each of which splits in K as...

conjecture | open | MathDB 页面 | 原文线索

V2-005478 | MATHDB-365199

Unimodality conjecture for totally nonnegative Grassmannian cardinalities

摘录 / EXCERPT: Let $q=p^r$, where p is an odd prime. For each n, consider the sequence ... where $\operatorname{Gr}_{k,n}^{\geq 0}(\mathbb{F}_q)$ denotes the totally nonnegative Grassmannian ov...

conjecture | open | MathDB 页面 | 原文线索

V2-005479 | MATHDB-365212

The Orbifold Mordell Conjecture for orbifold curves

摘录 / EXCERPT: Let (X,Δ) be a smooth proper orbifold curve over a number field K, where $\Delta=\sum_i(1-\frac{1}{m_i})D_i$, and call (X,Δ) Mordellic if, after every finite field...

conjecture | open | MathDB 页面 | 原文线索

V2-005480 | MATHDB-365233

The Hahn-Wilson conjecture on fp-spectra

摘录 / EXCERPT: Let p be a prime, and let a spectrum of fp-type n mean a bounded below, p-complete fp-spectrum whose associated thick subcategory of finite spectra has type at least $n+1$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-005481 | MATHDB-365257

Erdős's stronger least-common-multiple non-monotonicity conjecture

摘录 / EXCERPT: Erdős's stronger conjecture. There exist integers k and $0<x<y$ with $y>x+k$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005482 | MATHDB-365303

Log-concavity conjecture for marginals of the sine process

摘录 / EXCERPT: Consider a determinantal point process on the real line with correlation kernel ... Label its points by $\cdots<Z_{-1}<Z_0<Z_1<Z_2<\cdots$ so that $Z_{-1}<0\leq Z_0$. The sine-proces...

conjecture | open | MathDB 页面 | 原文线索

V2-005483 | MATHDB-365331

Generalized Lehmer conjecture for Hecke polynomial coefficients

摘录 / EXCERPT: Fix integers $m\geq 1$ and $r\geq 1$. Let $T_m(N,k)(x)$ be the Hecke polynomial, with its r-th coefficient understood in the source's indexing. Generalized Lehmer conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-005484 | MATHDB-365340

Florence's lifting conjecture for Borel representations

摘录 / EXCERPT: Let p be a prime number, let F be a field containing a primitive p^2-th root of unity, and let $n\geq 1$ be an integer. Let Γ_F denote the absolute Galois group of F...

conjecture | open | MathDB 页面 | 原文线索

V2-005485 | MATHDB-365368

Prime-dimensional stabilizer 3-design conjecture

摘录 / EXCERPT: Let d be prime, let $D=d^n$, and let $\mathrm{Stab}(n,d)$ denote the ensemble of stabilizer states. The source considers its shadow norm $\|\mathrm{Stab}(n,d)\|_{\mathrm{sh}}$. P...

conjecture | open | MathDB 页面 | 原文线索

V2-005486 | MATHDB-365401

Zhu et al.'s five-color conjecture for generalized Petersen graphs

摘录 / EXCERPT: Let n and k be integers with $n\geq 2k$, and let $GP(n,k)$ be the generalized Petersen graph with star chromatic index $\chi^\prime_s(GP(n,k))$, the smallest number of colors i...

conjecture | open | MathDB 页面 | 原文线索

V2-005487 | MATHDB-365402

Dvořák et al.'s six-color conjecture for subcubic graphs

摘录 / EXCERPT: Let G be a graph of maximum degree at most three, and let $\chi^\prime_s(G)$ denote its star chromatic index, the smallest number of colors in a proper edge coloring in which no...

conjecture | open | MathDB 页面 | 原文线索

V2-005488 | MATHDB-365429

Generalized Billerey-Menares conjecture for squarefree levels

摘录 / EXCERPT: Let $k\geq 4$ and let $\ell>k+1$ be prime. Let $N=p_1\cdots p_t$ be squarefree, with distinct prime factors p_i, and let ε be an Atkin-Lehner eigensystem for...

conjecture | open | MathDB 页面 | 原文线索

V2-005489 | MATHDB-365462

The exceptional-group count conjecture for groups of order p^6

摘录 / EXCERPT: Let $p\geq 5$ be prime, and let an exceptional group be a finite group G having a normal subgroup N such that $\mu(G/N)>\mu(G)$, where $\mu(G)$ is the least integer n for whi...

conjecture | open | MathDB 页面 | 原文线索

V2-005490 | MATHDB-365464

Kuperberg-Lalin symplectic variance conjecture for quadratic residues modulo primes

摘录 / EXCERPT: Kuperberg-Lalin's symplectic prime variance conjecture. For $x^{1/k+\varepsilon}\leq y$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-005491 | MATHDB-365472

The hyperbolic-genus inner-neighbor conjecture for holomorphic vertex operator algebras

摘录 / EXCERPT: Let V and V' be strong-rational, holomorphic vertex operator algebras, and fix a sufficiently large prime p. Hyperbolic-genus inner-neighbor conjecture. V and V' lie in t...

conjecture | open | MathDB 页面 | 原文线索

V2-005492 | MATHDB-365478

The balanced heavenly elliptic curve conjecture

摘录 / EXCERPT: Let ℓ be a prime, let K be a quadratic number field, and let E/K be an elliptic curve. An elliptic curve is heavenly at ℓ when its ℓ-power torsion field satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-005493 | MATHDB-365479

The finiteness conjecture for heavenly elliptic curves over quadratic fields

摘录 / EXCERPT: Let $\overline{\mathcal{H}}_{\geq 7}(2,1)$ denote the set defined in the paper, consisting of the relevant heavenly elliptic-curve data over quadratic fields with prime parameter a...

conjecture | open | MathDB 页面 | 原文线索

V2-005494 | MATHDB-365486

Gap-decay and extremal-frequency conjecture for the critical almost Mathieu operator

摘录 / EXCERPT: Let p and q be coprime integers with q odd. For the periodic problem $\Sigma(p/q)$, let $\gamma_k(p/q)$ denote the length of the gap with label k. Gap-decay and extremal-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-005495 | MATHDB-365487

Thouless-type spectral-measure inequality for the critical almost Mathieu operator

摘录 / EXCERPT: Let p and q be coprime integers with q odd, and let $\Sigma_{p/q}$ denote the periodic almost Mathieu spectrum. Let $C=0.915\ldots$ be Catalan's constant. Thouless-type spect...

conjecture | open | MathDB 页面 | 原文线索

V2-005496 | MATHDB-365515

Supersingular genus-six curves from double covers branched at two points

摘录 / EXCERPT: Let p be an odd prime. Let X' be a smooth curve of genus 3 over $\overline{\mathbf F}_p$, and let $\pi':Y'\to X'$ be a double cover branched at two points. By the Riemann-Hur...

conjecture | open | MathDB 页面 | 原文线索

V2-005497 | MATHDB-365535

The prime-graph complement characterization for strictly T-solvable groups

摘录 / EXCERPT: Prime-graph complement characterization. For every strictly T-solvable group G, there exists a set $Y\subseteq\bigcup_{p\in\pi(T)}\operatorname{IBr}_p(T)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005498 | MATHDB-365538

Louboutin's polynomial identity conjecture for S and R

摘录 / EXCERPT: For $d\geq 1$, define ... Let $a,b\in\mathbb{Z}$ be nonzero, put $c:=a+b\neq 0$, and define ... For $d\in\{a,b,c\}$, ... If a is even and b is odd, also define ... Then ... This identity...

conjecture | open | MathDB 页面 | 原文线索

V2-005499 | MATHDB-365539

Louboutin's degree conjecture for the auxiliary Laurent polynomial

摘录 / EXCERPT: Let $a,b\in\mathbb{Z}$ with $a\neq 0$ and $b\geq 1$, and let $m_{a,b}:=a^2+ab+b^2$. Using the polynomials and Laurent polynomials defined in Proposition 19, let $G_{a,b,m}(T)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005500 | MATHDB-365543

Asymptotic cryptarithmic uniqueness of the prime sequence

摘录 / EXCERPT: Let the sequence of primes be ordered by increasing value, and call a prime cryptarithmically unique when no other prime has the same multiset of decimal digits. A sequence is asym...

conjecture | open | MathDB 页面 | 原文线索

V2-005501 | MATHDB-365544

Asymptotic ratio of non-primonumerophobic and digit-pattern sequences

摘录 / EXCERPT: Let $A267013(d)$ denote the number of non-primonumerophobic digit patterns of length d, and let $A164864(d)$ denote the total number of available digit patterns of length d. As...

conjecture | open | MathDB 页面 | 原文线索

V2-005502 | MATHDB-365555

The Tijdeman-Zagier conjecture for products

摘录 / EXCERPT: Tijdeman-Zagier conjecture for products. The integer $x^n\pm y^m$ is not a product of degree d satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-005503 | MATHDB-365581

Gupta-Rai conjecture on permutation trinomials over finite fields

摘录 / EXCERPT: Gupta-Rai conjecture. The trinomial $f(X)$ is a permutation polynomial over $\mathbb F_{q^2}$ if and only if $\alpha=-1$ and $k=2$. This conjecture concerns the permutation behavio...

conjecture | open | MathDB 页面 | 原文线索

V2-005504 | MATHDB-365589

Selberg-type second-moment conjecture for prime coefficients

摘录 / EXCERPT: Let $\mathcal{L}\in\mathcal{S}$, and let $a(p)$ denote its prime coefficients. Let $\mathcal{C}_{\mathcal{L}}^{P_1}(x)$ and $\mathcal{C}_{\mathcal{L}}^{P_2}$ satisfy the stated gro...

conjecture | open | MathDB 页面 | 原文线索

V2-005505 | MATHDB-365590

Prime-coefficient first-moment conjecture for the Selberg class

摘录 / EXCERPT: Let $\mathcal{L}\in\mathcal{S}$, and let $a(p)$ denote its prime coefficients. Let $\widehat{\mathcal{C}_{\mathcal{L}}^{P_1}}(x)$ and $\widehat{\mathcal{C}_{\mathcal{L}}^{P_2}}$ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-005506 | MATHDB-365592

Eaton-Moretó conjecture on minimal positive heights of blocks

摘录 / EXCERPT: Let p be a prime, let G be a finite group, and let B be a p-block of G with defect group D. For a finite p-group D, define ... with value ∞ when D is abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-005507 | MATHDB-365605

Conjecture on the asymptotic density of the sets J_p

摘录 / EXCERPT: For each prime $p\geq 13$ satisfying $p\equiv 1\pmod{4}$, let J_p be the set defined in the paper's Theorem 3. The ratio $\#J_p/p$ is less than $1/2$. Asymptotic density conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-005508 | MATHDB-365606

The p-converse to the Gross-Zagier-Kolyvagin theorem

摘录 / EXCERPT: Let E be an elliptic curve defined over $\bQ$, let p be a prime, and let $r\in\bZ$. p-converse theorem conjecture. ... This is the conjectural converse to the implication from an...

conjecture | open | MathDB 页面 | 原文线索

V2-005509 | MATHDB-365607

The Heegner Point Main Conjecture

摘录 / EXCERPT: Let $E/\bQ$ be an elliptic curve, let $p>2$ be a prime of good ordinary reduction, and let K be an imaginary quadratic field satisfying the Heegner hypothesis. Let Λ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-005510 | MATHDB-365622

Degree and nilpotence-index conjectures for the modified Hecke operator at 19 modulo 5

摘录 / EXCERPT: Degree and nilpotence-index conjecture. The degree $D_{19}(k,5)$ is given by the displayed residue-class formulas in the source; moreover, $c_0=0$, $c_1=2$, and

conjecture | open | MathDB 页面 | 原文线索

V2-005511 | MATHDB-365623

Base-7 digit conjectures for nilpotency indices at 13 and 29 modulo 7

摘录 / EXCERPT: Base-7 digit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005512 | MATHDB-365624

Base-3 digit conjectures for nilpotency indices at 11 and 7 modulo 3

摘录 / EXCERPT: Base-3 digit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005513 | MATHDB-365625

Further overcubic partition triple congruences

摘录 / EXCERPT: Let $\overline{bt}(n)$ denote the overcubic partition triple function, and let $n,\alpha\geq 0$. The further overcubic partition triple congruence conjecture. ... ... These conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-005514 | MATHDB-365654

Higher-uniformity extremal construction conjecture for tight cycles

摘录 / EXCERPT: Higher-uniformity extremal construction conjecture. For all sufficiently long L relatively prime to r, the extremal $C_L^{(r)}$-hom-free constructions on sufficiently large num...

conjecture | open | MathDB 页面 | 原文线索

V2-005515 | MATHDB-365656

Erdős's squarefree-plus-power-of-two conjecture

摘录 / EXCERPT: Erdős's conjecture. There exist a squarefree integer m and an integer k such that

conjecture | open | MathDB 页面 | 原文线索

V2-005516 | MATHDB-365659

Conjecture on extremal subgroup counts for ${\rm PSL}_2(p)$

摘录 / EXCERPT: For a finite group G, let $i(G)$ be the number of isomorphism classes of non-identity proper subgroups of G, let $c(G)$ be the number of conjugacy classes of such subgroups, le...

conjecture | open | MathDB 页面 | 原文线索

V2-005517 | MATHDB-365673

Bergdall-Pollack equidistribution conjecture for slopes of $mathcal{L}$-invariants

摘录 / EXCERPT: Bergdall-Pollack equidistribution conjecture. As T tends to infinity, the set $Y_T\subseteq(-\infty,+\infty)$ is evenly distributed on $[0,1]$.

conjecture | open | MathDB 页面 | 原文线索

V2-005518 | MATHDB-365674

Gouvêa's distribution conjecture for slopes of modular forms

摘录 / EXCERPT: Gouvêa's distribution conjecture. The set X_T is equidistributed on $[0,1]$ as T tends to infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-005519 | MATHDB-365688

The multiplicative generation conjecture for generalized Motzkin numbers

摘录 / EXCERPT: Let $a,b\in\mathbb{Z}$, let p be a prime, and define … Assume that p does not divide any member of S_p. Multiplicative generation conjecture. The set S_p generates...

conjecture | open | MathDB 页面 | 原文线索

V2-005520 | MATHDB-365718

Bennett-Siksek conjecture on rational points on Erdős-Selfridge curves

摘录 / EXCERPT: Bennett-Siksek conjecture. The only rational points satisfy $y=0$, or are one of the following:

conjecture | open | MathDB 页面 | 原文线索

V2-005521 | MATHDB-365722

Conjecture on rank-two tuples on the curve modulo p

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x,y]$ and, for each prime p, let C_p be the curve of points modulo p defined by $f(x,y)\equiv 0\pmod p$. For a k-tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-005522 | MATHDB-365723

Uniform-distribution conjecture for the lifted curve points

摘录 / EXCERPT: Uniform-distribution conjecture. As $p\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005523 | MATHDB-365724

Exponential-sum conjecture for shifted polynomial values modulo p squared

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x,y]$, let C_p be the curve modulo p defined by $f(x,y)\equiv0\pmod p$, and write $e_{p^2}(u)=\exp(2\pi i u/p^2)$. Exponential-sum conjecture. There exists...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005524 | MATHDB-365734

Conjecture on algebraicity of nonmaximal-order Shintani--Faddeev RM values

摘录 / EXCERPT: Let $\beta\in\mathbb R$ satisfy $a\beta^2+b\beta+c=0$ with $a,b,c\in\mathbb Z$ and b^2-4ac nonsquare, and let $\r\in\mathbb Q^2$. Let $\shin^{\r}[\beta]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005525 | MATHDB-365745

The probabilistic independence conjecture for Goldbach events

摘录 / EXCERPT: Goldbach-event independence conjecture. The events A and A' satisfy

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005526 | MATHDB-365759

Cusick's conjecture on the sum-of-digits correlation measure

摘录 / EXCERPT: Cusick's conjecture. For all $t\geq 0$ we have

conjecture | open | MathDB 页面 | 原文线索

V2-005527 | MATHDB-365760

The block-appending conjecture for the sum-of-digits correlation measure

摘录 / EXCERPT: Let $t\geq 1$ be an odd integer, and let c_t denote the correlation quantity associated with the binary sum-of-digits difference for t. For each $K\geq 1$, consider the integer...

conjecture | open | MathDB 页面 | 原文线索

V2-005528 | MATHDB-365764

Allen-Ramírez conjecture for the inhomogeneous Khintchine-Groshev theorem

摘录 / EXCERPT: For positive integers n,m, a function $\psi: \mathbb{N} \to \mathbb{R}_{\geq 0}$, and an inhomogeneous parameter $\mathbf{y} \in \mathbb{R}^m$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-005529 | MATHDB-365769

Erdős's sum-product conjecture for positive integers

摘录 / EXCERPT: Erdős's sum-product conjecture. For every $\epsilon>0$ there is an n_0 such that, whenever $|A|\ge n_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005530 | MATHDB-365770

Solymosi-Volkmann sum-product inequality conjecture

摘录 / EXCERPT: Solymosi-Volkmann conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-005531 | MATHDB-365771

Sum-product exponent conjecture for positive integers

摘录 / EXCERPT: Sum-product exponent conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-005532 | MATHDB-365773

Small-representative conjecture for sum-product pairs

摘录 / EXCERPT: Small-representative conjecture. If $(i,j)\in\operatorname{SPP}(n)$, then there is a set $A\subseteq\mathbb N$ with

conjecture | open | MathDB 页面 | 原文线索

V2-005533 | MATHDB-365776

Aymone's conjecture for sums over k-free integers

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let $\mu^{(k)}$ denote the indicator function of the k-free integers, and let $h:\mathbb{N}\to\{-1,1\}$ be multiplicative. Aymone's conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-005534 | MATHDB-365803

Landesman's conjecture on resultant equation counts

摘录 / EXCERPT: Let $n\geqslant 2$ and let ... Landesman's conjecture. There exists $m\geqslant 0$ such that ... The conjecture predicts that, for every fixed degree $n\geqslant 2$, the quadratic poly...

conjecture | open | MathDB 页面 | 原文线索

V2-005535 | MATHDB-365810

Mizuno's duality conjecture for modular Nahm quadruples

摘录 / EXCERPT: Mizuno's duality conjecture. If (A,B,C,D) is a modular quadruple, then $(A^\star,B^\star,C^\star,D^\star)$ is also a modular quadruple.

conjecture | open | MathDB 页面 | 原文线索

V2-005536 | MATHDB-365819

Prime-product coefficient conjecture for Gröbner bases

摘录 / EXCERPT: Prime-product coefficient conjecture. Each coefficient occurring in the Gröbner basis of $I_{a_1,\ldots,a_{n+1}}$ is a possibly empty product of primes less than...

conjecture | open | MathDB 页面 | 原文线索

V2-005537 | MATHDB-365871

The Weil abscissa conjecture for procyclic groups

摘录 / EXCERPT: The Weil abscissa conjecture. For every set S of primes,

conjecture | open | MathDB 页面 | 原文线索

V2-005538 | MATHDB-365876

Katz's Sato-Tate conjecture for Kloosterman sums

摘录 / EXCERPT: Katz's Sato-Tate conjecture. For every $f\in\mathcal{C}([0,\pi])$,

conjecture | open | MathDB 页面 | 原文线索

V2-005539 | MATHDB-365894

Nonrepresentability of infinitely many multiples of four by a quadratic form

摘录 / EXCERPT: Nonrepresentability conjecture. If $(a,b)\ne(3,2)$, then infinitely many positive multiples of four are not represented by $Q_{a,b}$; that is, there are infinitely many positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-005540 | MATHDB-365895

Eventual three-term representations by generalized pentagonal numbers

摘录 / EXCERPT: Let $p_m(x)$ and $\bar p_m(x)$ denote the generalized polygonal-number functions used in the paper, and let $\mathbb N$ denote the nonnegative integers. Eventual representation con...

conjecture | open | MathDB 页面 | 原文线索

V2-005541 | MATHDB-365896

Exceptional sets for four ternary quadratic forms

摘录 / EXCERPT: Let $\mathbb N$ denote the nonnegative integers and let $x,y,z\in\mathbb Z$. Exceptional-set conjecture. The following representation sets are exactly: ... ... These claims give pr...

conjecture | open | MathDB 页面 | 原文线索

V2-005542 | MATHDB-365897

A four-term representation by pentagonal quadratic polynomials

摘录 / EXCERPT: Let $\mathbb N$ denote the nonnegative integers. Four-term pentagonal representation conjecture. Every integer $n>51$ can be written as ... with $w,x,y,z\in\mathbb N$. This is an exp...

conjecture | open | MathDB 页面 | 原文线索

V2-005543 | MATHDB-365898

Eventual universality for four signed pentagonal forms

摘录 / EXCERPT: Signed pentagonal universality conjecture. For each $r\in\{\pm1,\pm3\}$, every integer $n>N(r)$ can be written as

conjecture | open | MathDB 页面 | 原文线索

V2-005544 | MATHDB-365945

Gross's conjecture on quaternionic cusp-form lifts from elliptic modular forms

摘录 / EXCERPT: Let k be the weight parameter, let $f\in S_{2k}({\mathop{\mathrm{SL}}}_2(\mathbb Z))$ be a Hecke eigenform, and let F_f denote the associated quaternionic form. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005545 | MATHDB-365955

Vanishing-correlation conjecture for distinct irreducible binary quadratic forms

摘录 / EXCERPT: Vanishing-correlation conjecture. If $P_1,P_2\in\mathbb{Z}[m,n]$ are irreducible binary quadratic forms that are not multiples of each other, then they are good for vanishing of co...

conjecture | open | MathDB 页面 | 原文线索

V2-005546 | MATHDB-365959

Fractional-polynomial prime-pair correlation conjecture

摘录 / EXCERPT: Fractional-polynomial prime-pair conjecture. There exists $B=B(f,A)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005547 | MATHDB-365960

Frantzikinakis's prime progressions with fractional-power differences conjecture

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let c be a positive real number. Write $\mathbb{P}$ for the set of primes. Frantzikinakis's conjecture. For infinitely many pairs...

open_problem | open | MathDB 页面 | 原文线索

V2-005548 | MATHDB-365969

Xue-Zhu conjecture on determining level-one eigenforms by Fourier coefficients

摘录 / EXCERPT: Let $m\geq 2$ be fixed, and let f be a normalized Hecke eigenform of level one. Its m-th Fourier coefficient is the coefficient of q^m in its normalized Fourier expansion. Xu...

conjecture | open | MathDB 页面 | 原文线索

V2-005549 | MATHDB-365983

The strong Diophantine conjecture on bounded continued fractions

摘录 / EXCERPT: Let t,u be positive integers with $t<u$ and $\gcd(t,u)=1$. Consider continued-fraction expansions of the form ... where $b_1,\ldots,b_m\geq 1$. Diophantine conjecture. There is a u...

conjecture | open | MathDB 页面 | 原文线索

V2-005550 | MATHDB-365984

Zaremba's bounded-partial-quotient conjecture

摘录 / EXCERPT: Let $u\geq 1$ be an integer, and let $1\leq t<u$ be coprime to u. Write the ordinary continued-fraction expansion of t/u as $[a_1,\ldots,a_\ell]$. Zaremba's conjecture. There i...

conjecture | open | MathDB 页面 | 原文线索

V2-005551 | MATHDB-366005

The integer-complexity conjecture for powers of 2 and 3

摘录 / EXCERPT: Integer-complexity conjecture. For $a\geq 1$ and $b\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005552 | MATHDB-366006

The 2-complexity conjecture for powers of 6 and 10

摘录 / EXCERPT: 2-complexity conjecture. For $m\geq0$ and $r\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005553 | MATHDB-366016

Yıldırım's pair correlation conjecture for Dirichlet L-functions

摘录 / EXCERPT: Pair correlation for Dirichlet L-functions. Under GRH, as $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005554 | MATHDB-366023

Algebraicity conjecture for Heegner-cycle vanishing and central L-function derivatives

摘录 / EXCERPT: Let $G\in S_{2\kappa}^{new}(N)$ be a normalized newform, let F_G be the totally real field generated by its eigenvalues, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005555 | MATHDB-366039

Greenberg's class number conjecture for p-adic towers

摘录 / EXCERPT: Let p) be a fixed prime, let d be a positive integer, and let ... be a $\mathbb{Z}_p^d$-extension tower in which K_n/K corresponds to the subgroup $(p^n\mathbb{Z})^d$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005556 | MATHDB-366065

The weak p-adic digit distribution conjecture for algebraic irrational numbers

摘录 / EXCERPT: Let $\alpha\in\overline{\mathbf{Q}}\setminus\mathbf{Q}$, and let S be the set of rational primes that are totally split in $\mathbf{Q}(\alpha)$. Let r and s be integers, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005557 | MATHDB-366085

The weak Collatz conjecture on finite cycles

摘录 / EXCERPT: Let $C:\mathbb{Z}\to\mathbb{Z}$ be the Collatz map, and let $\mathbb{N}_{1}$ denote the positive integers. A periodic point is an integer lying on a finite cycle of C. Weak Colla...

conjecture | open | MathDB 页面 | 原文线索

V2-005558 | MATHDB-366086

The divergent trajectories conjecture for the Collatz map

摘录 / EXCERPT: Let $C:\mathbb{Z}\to\mathbb{Z}$ be the Collatz map and let $\mathbb{N}_{1}$ denote the positive integers. A trajectory is divergent when its iterates are unbounded. Divergent traje...

conjecture | open | MathDB 页面 | 原文线索

V2-005559 | MATHDB-366087

Möller's conjecture on generalized Collatz maps

摘录 / EXCERPT: Fix coprime integers $m,d>1$ with $m>d$, and define $T_{\mathrm{Möller}}:\mathbb{Z}\to\mathbb{Z}$ by the residue-class formula given in the source. Möller's conjecture. The map...

conjecture | open | MathDB 页面 | 原文线索

V2-005560 | MATHDB-366088

Matthews' conjectures for relatively prime generalized Collatz maps

摘录 / EXCERPT: Let $d\geq2$, let $m_{0},\ldots,m_{d-1}$ be nonzero integers, and choose $r_j\in\mathbb{Z}$ with $r_j\equiv jm_j\pmod d$. Define $T_{\mathrm{Matthews}}:\mathbb{Z}\to\mathbb{Z}$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-005561 | MATHDB-366089

The correspondence principle for divergent points

摘录 / EXCERPT: Let p be a prime, let $H:\mathbb{Z}\to\mathbb{Z}$ be a suitable p-adic Hydra map satisfying prerequisites such as those in the source's theorem, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005562 | MATHDB-366090

The divergence conjecture for the shortened qx+1 map

摘录 / EXCERPT: Let q be an odd prime, and let $\chi_q:\mathbb{Z}_2\to\mathbb{Z}$ denote the numen of the shortened $qx+1$ map. Divergence conjecture for $qx+1$. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-005563 | MATHDB-366109

The adjugate criterion for inner twisted derivations

摘录 / EXCERPT: Let p be an odd prime, let $K=\mathbb{Q}(\zeta)$ with $O_K=\mathbb{Z}[\zeta]$, and let A be the matrix defined by ... for $\beta=\sum_{i=0}^{p-2}b_i\zeta^i\in K$. Let D be a...

conjecture | open | MathDB 页面 | 原文线索

V2-005564 | MATHDB-366110

Sufficiency of the arithmetic conditions for multiplicative recurrence along linear-fractional patterns

摘录 / EXCERPT: Multiplicative recurrence conjecture. The set S is a set of measurable multiplicative recurrence if and only if $a=c$, and $b=d$ or $a\mid bd$.

conjecture | open | MathDB 页面 | 原文线索

V2-005565 | MATHDB-366111

Finite-dimensional criterion for multiplicative recurrence of completely multiplicative functions

摘录 / EXCERPT: Finite-dimensional multiplicative recurrence conjecture. The displayed conclusion holds.

conjecture | open | MathDB 页面 | 原文线索

V2-005566 | MATHDB-366112

Determinant conjecture for twisted derivations when the conductor is $2^r p$

摘录 / EXCERPT: Let $\mathbb{K}=\mathbb{Q}(\zeta)$ be the nth cyclotomic number field with $n=2^r p$, where $r\in\mathbb{N}$ and p is an odd rational prime, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005567 | MATHDB-366113

Innerness criterion for twisted derivations when the conductor is $2^r p$

摘录 / EXCERPT: Innerness conjecture. Under the hypotheses of the determinant conjecture, if $1\le e_1\le r-1$ and $e_2\ge1$, then D is inner if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-005568 | MATHDB-366114

Determinant conjecture for twisted derivations of conductor p^k

摘录 / EXCERPT: Let $\mathbb{K}=\mathbb{Q}(\zeta)$ be the p^kth cyclotomic number field, where $k\in\mathbb{N}\setminus\{1\}$ and p is a rational prime, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005569 | MATHDB-366115

Outer-derivation criterion for conductor p^k

摘录 / EXCERPT: Outer-derivation conjecture. Under the hypotheses of the determinant conjecture, D is outer if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-005570 | MATHDB-366156

Defect-normalizer conjecture for the p-rationality level of height-zero characters

摘录 / EXCERPT: Let p be a prime, G a finite group, and let B be a p-block of G with defect group D. Let u be the p-adic valuation, let $c(\rho)$ denote the conductor of the value...

conjecture | open | MathDB 页面 | 原文线索

V2-005571 | MATHDB-366157

Defect-normalizer refinement of the Alperin-McKay-Navarro conjecture

摘录 / EXCERPT: Let p be a prime and G a finite group. Let B be a p-block of G with defect group D, let b be its Brauer correspondent in $\mathbf{N}_G(D)$, and let $\mathcal{H}_B$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-005572 | MATHDB-366177

Topological invariance conjecture for quasihomogeneous singularities

摘录 / EXCERPT: Topological invariance conjecture. The function $Z_{f,\bm 0}^{\mathrm{top}}(s)$ depends only on $(\bm w,d)$, and is a topological invariant of the family $\mathbf{QH}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-005573 | MATHDB-366188

The irrationality conjecture for a ratio of hypergeometric values

摘录 / EXCERPT: Let ... be the hypergeometric value used in the paper. Irrationality conjecture. ... This mild irrationality assumption is introduced for class 8, where the relevant even twist is not...

conjecture | open | MathDB 页面 | 原文线索

V2-005574 | MATHDB-366206

Hamiltonian circle conjecture for two-ended Cayley graphs with cyclic prime-power commutator subgroup

摘录 / EXCERPT: Let $G=\langle S\rangle$ be a two-ended group, and suppose that its commutator subgroup satisfies ... The Hamiltonian circle conjecture. The Cayley graph ... has a hamiltonian circle....

conjecture | open | MathDB 页面 | 原文线索

V2-005575 | MATHDB-366214

Primitive pairs avoiding affine hyperplanes in finite fields

摘录 / EXCERPT: Let q be a prime power, let m be a positive integer, and identify $\mathbb{F}_{q^m}$ with an m-dimensional vector space over $\mathbb{F}_q$. Let $C=\{A_1,\ldots,A_m\}$ be a s...

conjecture | open | MathDB 页面 | 原文线索

V2-005576 | MATHDB-366215

Nonvanishing conjecture for principal submatrices of the Fourier matrix

摘录 / EXCERPT: Let n be a square-free positive integer, let ζ_n be an n-th root of unity, and consider the matrix $F_n=(\zeta_n^{ij})_{0\leq i,j\leq n-1}$. A principal submatrix is a ma...

conjecture | open | MathDB 页面 | 原文线索

V2-005577 | MATHDB-366224

The p-regular product conjecture for coprime conjugacy class sizes

摘录 / EXCERPT: Let G be a group, and let p be a prime. An element of G is p-regular if its order is not divisible by p, and the conjugacy class size of an element x is $|G:C_G(x)|$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005578 | MATHDB-366225

Sun's factorial valuation ratio conjecture for distinct primes

摘录 / EXCERPT: Let $p_1,\ldots,p_k$ be distinct primes, and let $v_p(n!)$ denote the exponent of p in the prime factorization of $n!$. Sun's conjecture. There are infinitely many positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-005579 | MATHDB-366260

Kurokawa-Ochiai vanishing conjecture for Witten zeta functions

摘录 / EXCERPT: Let G be a compact Hausdorff topological group, and define its Witten zeta function by ... where the sum runs over the finite-dimensional irreducible representations of G. Kuroka...

conjecture | open | MathDB 页面 | 原文线索

V2-005580 | MATHDB-366261

Optimality conjecture for vanishing orders of Witten zeta functions

摘录 / EXCERPT: Optimality conjecture. In the theorem, all stated inequalities for orders of vanishing are actually equalities. The theorem establishes the displayed lower bounds for compact simpl...

conjecture | open | MathDB 页面 | 原文线索

V2-005581 | MATHDB-366262

The abc conjecture for generalized Fermat equations with variable coefficients

摘录 / EXCERPT: Let a,b,c be nonzero integers, and let (x,y,z,p,q,r) be integers satisfying the generalized Fermat equation $eqref{eq:genfermat}$, with $gcd(x,y,z)=1$, $p,q,r>0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005582 | MATHDB-366263

The generalized Fermat conjecture with coefficients supported on a finite prime set

摘录 / EXCERPT: Let S be a finite set of primes. Consider integer solutions (a,b,c,x,y,z,p,q,r) of the generalized Fermat equation $eqref{eq:genfermat}$ satisfying $gcd(ax,by,cz)=1$, $p,q,r>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-005583 | MATHDB-366264

Littlewood's conjecture in k dimensions

摘录 / EXCERPT: Let $\mathbf{x}=(x_1,\ldots,x_k)\in\mathbb{R}^k$, where $k\geqslant2$. Littlewood's conjecture in k dimensions. ... This is the multiplicative form of simultaneous Diophantine appr...

conjecture | open | MathDB 页面 | 原文线索

V2-005584 | MATHDB-366265

The Khinchin and Gallagher conjecture for missing-digit measures

摘录 / EXCERPT: Let k be a positive integer, and let μ be a missing-digit measure on $\mathbb{R}^k$ that is not supported on any affine hyperplane. A probability measure is Khinchin if Khinc...

conjecture | open | MathDB 页面 | 原文线索

V2-005585 | MATHDB-366266

Broderick-Fishman-Reich conjecture for intrinsic approximation on missing-digit fractals

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let $\mathcal{K}$ be a proper missing-digit fractal in $[0,1]^k$ that is not contained in any proper affine subspace. Let μ be its Cantor-Lebesgue mea...

conjecture | open | MathDB 页面 | 原文线索

V2-005586 | MATHDB-366267

Mordell-Schinzel conjecture for cubic Diophantine equations

摘录 / EXCERPT: Mordell-Schinzel conjecture. The equation has infinitely many integer solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-005587 | MATHDB-366301

Infinite sign-change conjecture for quadratic-form prime biases

摘录 / EXCERPT: For a primitive positive definite binary quadratic form, let $F(N;8,1)$, $F(N;8,5)$, $F(N;12,1)$, and $F(N;12,7)$ denote the functions introduced in the paper for the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-005588 | MATHDB-366302

Negative-bias conjecture for the separated sum-of-two-squares counting functions

摘录 / EXCERPT: Negative-bias conjecture. There is a bias towards negative values in the distribution of the values of $D_1(x)$ and $D_2(x)$. The conjecture is based on numerical data for x up t...

conjecture | open | MathDB 页面 | 原文线索

V2-005589 | MATHDB-366304

Dense hypergraphs contain cycles with length coprime to the uniformity

摘录 / EXCERPT: Coprime-cycle conjecture. The hypergraph G contains a cycle whose length is coprime to k.

conjecture | open | MathDB 页面 | 原文线索

V2-005590 | MATHDB-366323

Conjectures on the frequency and sign distribution of extreme values

摘录 / EXCERPT: Conjectures on extreme values. One may conjecture that the number of extreme values up to height T is less than $C\log T$ for some constant C, and that the sets of indices n...

conjecture | open | MathDB 页面 | 原文线索

V2-005591 | MATHDB-366324

Prime-power covering-set index conjecture

摘录 / EXCERPT: For a finite group A and a subgroup $U\leqslant A$, let $\mathcal{P}(U)$ be the set of elements of U of prime-power order, and define ... the union of the A-conjugacy classes o...

conjecture | open | MathDB 页面 | 原文线索

V2-005592 | MATHDB-366325

Normal-subgroup prime-power covering-set conjecture

摘录 / EXCERPT: Let A be a finite group with normal subgroup G of index $|A:G|=n$, and let $U\leqslant G$ satisfy $P_A(U)=P_A(G)$, where $P_A(V)$ is the union of the A-conjugacy classes of e...

conjecture | open | MathDB 页面 | 原文线索

V2-005593 | MATHDB-366335

The conjecture on sporadic points of every degree on modular curves $X_0(n)$

摘录 / EXCERPT: For every positive integer d, consider modular curves $X_0(n)$ and points on them defined over number fields. A point is sporadic if its degree is below the relevant uniform thre...

conjecture | open | MathDB 页面 | 原文线索

V2-005594 | MATHDB-366338

Hung's principal-block continuity conjecture for p-rationality

摘录 / EXCERPT: Let G be a finite group and p a prime. Let $B_0(G)$ denote the principal p-block of G, and write $c(\theta)_p=p^r$ for the p-rationality level of a character θ. Hu...

conjecture | open | MathDB 页面 | 原文线索

V2-005595 | MATHDB-366339

The simple-group reduction conjecture for p-rational characters

摘录 / EXCERPT: Let p be a prime and let S be a finite non-abelian simple group whose order is divisible by p. Simple-group reduction conjecture. The following assertions hold. (a) Let X b...

conjecture | open | MathDB 页面 | 原文线索

V2-005596 | MATHDB-366340

Navarro's principal-block characterization of the continuity gap

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Define ... and let $Q\in\operatorname{Syl}_p(K)$. Call a character almost p-rational when it h...

conjecture | open | MathDB 页面 | 原文线索

V2-005597 | MATHDB-366341

The generalized congruence conjecture for T_2

摘录 / EXCERPT: The generalized T_2 congruence conjecture. For every integer $n\geq 0$ and every integer j with $1\leq j\leq p-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005598 | MATHDB-366344

The Z_p periodicity conjecture

摘录 / EXCERPT: Let X be a topological space and let p be an odd prime. A nonzero element $x\in H^k(X;\mathbb{Z}_p)$ induces periodicity if multiplication by x gives isomorphisms ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-005599 | MATHDB-366345

The Z_p periodicity conjecture in the truncated case

摘录 / EXCERPT: Let X^n be a space and let p be a prime. Say that $H^*(X;\mathbb{Z}_p)$ is k-periodic up to degree c when multiplication by a periodicity element induces successive isomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-005600 | MATHDB-366350

Büchi's problem on bounded sequences of squares

摘录 / EXCERPT: Let $x_1^2,\ldots,x_n^2$ be a sequence of increasing squares whose constant second difference is equal to 2. Büchi's problem. There exists a positive integer N such that every...

conjecture | open | MathDB 页面 | 原文线索

V2-005601 | MATHDB-366361

Eisenstein-series identities for the rank-two and rank-three Witten zeta functions

摘录 / EXCERPT: Let n be a positive integer, let $f_B(x)=(1+x)(1+2x)$ and $f_G(x)=(1+x)(1+2x)(1+3x)(2+3x)$, and let $E_{2k}(\tau)$ denote the Eisenstein series for τ in the upper half-plane...

conjecture | open | MathDB 页面 | 原文线索

V2-005602 | MATHDB-366362

Nonvanishing conjecture for the rank-three recurrence sequence

摘录 / EXCERPT: Let $\{g_n\}_{n\geq 0}$ be the sequence defined earlier in the paper through the rank-three Witten zeta function of type G. Nonvanishing conjecture for g_n. If...

conjecture | open | MathDB 页面 | 原文线索

V2-005603 | MATHDB-366380

Independence conjecture for almost-prime patterns

摘录 / EXCERPT: Let $\mathbb{P}_\ell$ denote the set of integers with the relevant almost-prime parameter ℓ, let $\1_{\mathbb{P}_\ell}$ be its indicator, and let $\overline{\pi}_\ell(N)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-005604 | MATHDB-366381

Joint-distribution independence conjecture for consecutive values of Ω

摘录 / EXCERPT: Let $\Omega(n)$ denote the total number of prime factors of n, counted with multiplicity, let $[N]=\{1,\ldots,N\}$, and for a finite set S write...

conjecture | open | MathDB 页面 | 原文线索

V2-005605 | MATHDB-366383

Mazur-Tate's multiplicative refined BSD conjecture

摘录 / EXCERPT: Let $E/\mathbb{Q}$ have split multiplicative reduction at a prime p, put $G_p=\left(\mathbb{Z}/p\mathbb{Z}\right)^*/\langle-1\rangle$, and let $\lambda(a,p)$ be the even modular...

conjecture | open | MathDB 页面 | 原文线索

V2-005606 | MATHDB-366423

The first Hardy-Littlewood conjecture for prime constellations

摘录 / EXCERPT: First Hardy-Littlewood conjecture. Unless P forms a complete residue class with respect to some prime, $\pi_P(x)$ is asymptotic to

conjecture | open | MathDB 页面 | 原文线索

V2-005607 | MATHDB-366427

Two-variable Iwasawa main conjectures for elliptic curves over imaginary quadratic fields

摘录 / EXCERPT: Let $E/{\mathbb Q}$ be an elliptic curve of conductor N, let $p>2$ be a prime at which E has good ordinary reduction, and let K be an imaginary quadratic field in which...

conjecture | open | MathDB 页面 | 原文线索

V2-005608 | MATHDB-366430

Hejhal's triple correlation conjecture for Riemann zeros

摘录 / EXCERPT: Let $\gamma,\gamma',\gamma”$ range over ordinates of nontrivial zeros of the Riemann zeta function, put $\widetilde\gamma=(\log T)/(2\pi)\,\gamma$, and let $\hat r$ be the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-005609 | MATHDB-366431

Strong twisted pair correlation conjecture

摘录 / EXCERPT: Strong twisted pair correlation conjecture. Fix $\varepsilon>0$. Uniformly for all prime powers $n\leq T^{1-\varepsilon}$,

conjecture | open | MathDB 页面 | 原文线索

V2-005610 | MATHDB-366432

Twisted pair correlation conjecture

摘录 / EXCERPT: Twisted pair correlation conjecture. Fix $\varepsilon>0$ and let $r\in\mathfrak C(\mathbb R)$. Uniformly for all prime powers $n=q^a\leq T^{1-\varepsilon}$,

conjecture | open | MathDB 页面 | 原文线索

V2-005611 | MATHDB-366433

Uniform Hardy-Littlewood conjecture for shifted prime correlations

摘录 / EXCERPT: Let Λ be the von Mangoldt function, define the singular series ... and for $h\ne0$ let $\mathfrak S(h)=0$ when h is odd, while for even h let ... Set...

conjecture | open | MathDB 页面 | 原文线索

V2-005612 | MATHDB-366474

Asymptotic bounds from Models 1, 2, and 2* for the prime-counting function

摘录 / EXCERPT: Model bounds conjecture. As $n\to\infty$, the estimates satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-005613 | MATHDB-366482

Infinitely many Catalan numbers with proportional 2- and 3-adic valuations

摘录 / EXCERPT: Let $C_n=\frac{1}{n+1}\binom{2n}{n}$ denote the nth Catalan number, and let $\nu_p(m)$ be the exponent of the prime p in the prime factorization of m. Assume that $a,b\geq 1$...

conjecture | open | MathDB 页面 | 原文线索

V2-005614 | MATHDB-366502

Conjecture on singularity of the Pell matrix at exceptional primes

摘录 / EXCERPT: Singularity conjecture. The matrix $B_p(n-1)$ is singular if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-005615 | MATHDB-366524

The set-system intersection bound modulo prime powers

摘录 / EXCERPT: Let p be a prime and let $q=p^\alpha$, with $\alpha\geq 1$. Let $k\geq 2$ be an integer, let $L\subseteq \{0,1,\ldots,q-1\}$ be a proper subset with size $s>0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005616 | MATHDB-366532

Stronger linear lower-bound conjecture for the theta-function arithmetic sum

摘录 / EXCERPT: Stronger linear lower-bound conjecture. For each prime $k>3119$,

conjecture | open | MathDB 页面 | 原文线索

V2-005617 | MATHDB-366533

Near-linear growth conjecture for the theta-function arithmetic sum

摘录 / EXCERPT: For positive integers h and k, define ... where $\lfloor x\rfloor$ is the greatest integer less than or equal to x, and define ... Let C_1 and C_2 be any fixed positive numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-005618 | MATHDB-366543

Refinement of the perfect-power repulsion conjecture for partition numbers

摘录 / EXCERPT: Refinement of the perfect-power repulsion conjecture. If $k>1$, then $M_k(0)=1$, and for every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005619 | MATHDB-366551

Twisted convolution conjecture for admissible tuples

摘录 / EXCERPT: Let $t=(d,r,Q)$ be an admissible tuple, and let $_t$ be its set of shifts, defined by the twisted convolution identity in the source. Twisted convolution conjecture. For every admi...

conjecture | open | MathDB 页面 | 原文线索

V2-005620 | MATHDB-366552

Minimalist real multiplication values conjecture

摘录 / EXCERPT: Let $\theta given$ be a real quadratic irrational satisfying ... where $a,b,c e? $ are integers and $Delta=b^2-4ac$ is nonsquare. Let $rho e? $ be in $Q^2\setminusZ^2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005621 | MATHDB-366553

Fundamental real multiplication values conjecture

摘录 / EXCERPT: Let $theta$ be real and satisfy ... with $a,b,c e? $ integers and fundamental discriminant $Delta=b^2-4ac$. Let $rho e? $ lie in $Q^2\setminusZ^2$, and let $A e? $ lie in...

conjecture | open | MathDB 页面 | 原文线索

V2-005622 | MATHDB-366554

Ray class field prediction for RM-value fields

摘录 / EXCERPT: Let $t=(d,r,Q)$ be an admissible tuple with associated real quadratic field K, let $d=d_{j,m}$, and let f be the conductor of Q. Let $s=(d,r,Q,G,g)$ be a fiducial datum. Deno...

conjecture | open | MathDB 页面 | 原文线索

V2-005623 | MATHDB-366565

Infinitude conjecture for deficient KL-primitive numbers

摘录 / EXCERPT: A positive integer n is deficient when its proper-divisor sum is less than n, and it is KL-primitive when ${\rm KL}(n)\geq 0$ while ${\rm KL}(d)<0$ for every proper divisor d...

conjecture | open | MathDB 页面 | 原文线索

V2-005624 | MATHDB-366566

Conjectural lower bound for the weighted divisor statistic $v(n)$

摘录 / EXCERPT: For a positive integer n, define ... Let p be the smallest prime factor of n. Lower-bound conjecture for $v(n)$. If ... then ... This would strengthen the proven bound...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005625 | MATHDB-366571

The directional p-adic Littlewood conjecture for vectors

摘录 / EXCERPT: Let $\overline{\alpha}\in\mathbb{R}^n$ be a vector and let p be a prime. Denote by $\langle\cdot\rangle:\mathbb{R}^n\to[0,1)^n$ the ℓ^∞ distance to the nearest integer...

conjecture | open | MathDB 页面 | 原文线索

V2-005626 | MATHDB-366574

Bouniakowsky's conjecture on prime values of irreducible polynomials

摘录 / EXCERPT: Bouniakowsky's conjecture. Then $f(n)$ is prime for infinitely many positive integer values of n. This is the motivating conjecture for studying eventually prime-free sequences,...

conjecture | open | MathDB 页面 | 原文线索

V2-005627 | MATHDB-366575

Conjecture on prime values of polynomial floor sequences

摘录 / EXCERPT: Prime-value conjecture for odd powers. (a) If $t\equiv 0\pmod 3$ and $d\neq 2,9$, then there are infinitely many primes of the form

conjecture | open | MathDB 页面 | 原文线索

V2-005628 | MATHDB-366637

Asymptotic associated-prime conjecture for homological shift ideals of edge ideals

摘录 / EXCERPT: Asymptotic associated-prime conjecture. For all $i\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005629 | MATHDB-366651

Adams's detection conjecture for mod-p cohomology of compact Lie groups

摘录 / EXCERPT: Adams's detection conjecture. For every compact connected Lie group G and every odd prime p, the map Φ is injective. This strengthens Quillen's result that the kernel cont...

conjecture | open | MathDB 页面 | 原文线索

V2-005630 | MATHDB-366660

Navarro-Tiep's character-value conjecture for finite quasi-simple groups

摘录 / EXCERPT: Let p be an odd prime, let G be a finite group, and let $\rho\in\operatorname{Irr}_{p'}(G)$ have conductor $p^a m$, where $a\in\mathbb{Z}_{\geq 0}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005631 | MATHDB-366673

Maistret-Shukla conjecture on twisted L-values and Galois modules of Tate-Shafarevich groups

摘录 / EXCERPT: Maistret-Shukla conjecture. There is an embedding $\iota:\mathbb{Q}(\zeta_d)\to\mathbb{C}$ such that, for every elliptic curve $E/\mathbb{Q}$ with conductor $\frak{f}_E$ coprime to...

conjecture | open | MathDB 页面 | 原文线索

V2-005632 | MATHDB-366685

The sparsity conjecture for equal partition counts

摘录 / EXCERPT: Sparsity conjecture. Fix integers d,e with $2<d<e$. If $p_d^n=p_e^m$ for some integers n,m, then $d=3$.

conjecture | open | MathDB 页面 | 原文线索

V2-005633 | MATHDB-366687

Conjecture on the least stationary primitive root modulo p

摘录 / EXCERPT: Let p be a prime, and let $g_s(p)$ denote the least stationary primitive root in $\mathbb{Z}/p\mathbb{Z}$. Let $\varepsilon>0$ be a small number. Least stationary primitive root...

conjecture | open | MathDB 页面 | 原文线索

V2-005634 | MATHDB-366701

Residue invariance of weight-one Kato homology under regular proper models

摘录 / EXCERPT: Residue invariance conjecture. If the model $\mathscr{X}$ is proper over $\mathcal{O}_F$ and regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-005635 | MATHDB-366706

Gonek's negative moment conjecture for the Riemann zeta-function

摘录 / EXCERPT: Let $k>0$ be fixed, and let T tend to infinity with $[?]$-the integrals below are taken along the line $[?]$. Gonek's conjecture. Uniformly for $1\leq\delta\leq\log T$, ... while u...

conjecture | open | MathDB 页面 | 原文线索

V2-005636 | MATHDB-366724

Higman's polynomiality conjecture for unitriangular groups

摘录 / EXCERPT: Let $n\in\mathbb{N}^{*}$, let q be a prime power, and let $U_n(\mathbb{F}_q)$ be the group of $n\times n$ upper triangular matrices over $\mathbb{F}_q$ with ones on the diagonal....

conjecture | open | MathDB 页面 | 原文线索

V2-005637 | MATHDB-366725

The conjecture that the even and odd vector-space limits differ

摘录 / EXCERPT: Let p be an odd prime and let g be a natural number. Define ... Theorem 1 establishes that these limits exist and are positive real numbers. Even-odd limit conjecture. The two li...

conjecture | open | MathDB 页面 | 原文线索

V2-005638 | MATHDB-366758

Relative Fontaine-Mazur conjecture for semisimple lisse sheaves

摘录 / EXCERPT: Let k be a finitely generated field, let $\overline{k}$ be a separable closure, let ℓ be different from the characteristic of k, let Λ be a finite extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-005639 | MATHDB-366772

Li-Sheng's Beauville-number conjecture

摘录 / EXCERPT: Let λ be a parameter, written in characteristic p as a lifting (λ_0,λ_1), and let $M_p(\lambda_0,\lambda_1)$ denote the matrix associated with the periodici...

conjecture | open | MathDB 页面 | 原文线索

V2-005640 | MATHDB-366778

Gross-Kudla conjecture on triple-product L-functions and Bloch-Beilinson heights

摘录 / EXCERPT: Gross-Kudla conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005641 | MATHDB-366803

Markov polynomial classification conjecture for Laurent-polynomial solutions

摘录 / EXCERPT: Let $X,Y,Z\in\mathbb Z[x^{\pm1},y^{\pm1},z^{\pm1}]$ be integer Laurent polynomials satisfying equation ... in terms of the Markov-polynomial construction; the source gives no resolut...

conjecture | open | MathDB 页面 | 原文线索

V2-005642 | MATHDB-366806

Factor 4 conjecture for Markov-polynomial coefficients

摘录 / EXCERPT: Let $\rho=a/b$ be rational, and let the critical triangle be the set of lattice points satisfying ... Consider the coefficients of the Markov polynomial associated with these lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-005643 | MATHDB-366816

Lemma nice criterion for non-intersecting integers

摘录 / EXCERPT: Lemma nice criterion. If p is the smallest prime divisor of n and n/p has fewer than p distinct prime divisors, then n is non-intersecting.

conjecture | open | MathDB 页面 | 原文线索

V2-005644 | MATHDB-366843

Montgomery's conjecture on large values of Dirichlet polynomials

摘录 / EXCERPT: For $1/2<\sigma\leq 1$ and $\tau\geq 0$, let $\mathrm{LV}(\sigma,\tau)$ denote the large-values exponent for Dirichlet polynomials in the source: it measures the exponent governing...

conjecture | open | MathDB 页面 | 原文线索

V2-005645 | MATHDB-366970

The alternative-measure conjecture for the polynomial $1+x_1+3+x_k$

摘录 / EXCERPT: Alternative-measure conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005646 | MATHDB-366971

The iterated-limit conjecture for alternative measures of sparse polynomials

摘录 / EXCERPT: Let $c(P)$ be the alternative measure of a polynomial, and let $k\geq 2$. Consider the integer-exponent family $1+x+x^{n_2}+x^{n_3}+\cdots+x^{n_k}$ and take the iterated limit with...

conjecture | open | MathDB 页面 | 原文线索

V2-005647 | MATHDB-366972

The limiting alternative-measure conjecture for $P_{m,1}(x)$

摘录 / EXCERPT: Limiting alternative-measure conjecture. The sequence has a limit, and

conjecture | open | MathDB 页面 | 原文线索

V2-005648 | MATHDB-366992

The q-rious positivity conjecture

摘录 / EXCERPT: Let $\boldsymbol{a}=(a_1,\dotsc,a_r)$ and $\boldsymbol{b}=(b_1,\dotsc,b_s)$ be tuples of positive integers satisfying ... For a positive integer n, write $[n]!=[1][2]\dotsm[n]$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-005649 | MATHDB-366993

The q-rious unimodality conjecture

摘录 / EXCERPT: Let $(\boldsymbol{a},\boldsymbol{b})$ be tuples of positive integers satisfying Landau's criterion, namely ... Let $D(\boldsymbol{a},\boldsymbol{b};q)$ be the corresponding q-facto...

conjecture | open | MathDB 页面 | 原文线索

V2-005650 | MATHDB-367015

Gonek-Ki conjecture for zeros of real and imaginary parts of the zeta function

摘录 / EXCERPT: Fix $\theta\in[0,2\pi)$, let $a=1/2-c/\log T$ with $0<c=o(\log T)$, and let $A>0$ be arbitrarily large but fixed. Let $f_a(w,\theta)$ be the function in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005651 | MATHDB-367028

Cossey's conjecture on lifts of irreducible Brauer characters

摘录 / EXCERPT: Let p be a prime, let G be a p-solvable group, and let $\varphi\in\operatorname{IBr}(G)$ be an irreducible Brauer character. A lift of φ is a character...

conjecture | open | MathDB 页面 | 原文线索

V2-005652 | MATHDB-367051

Quantum modularity conjecture for generating functions of real origamis

摘录 / EXCERPT: Quantum modularity conjecture. This generating function is a quantum modular form.

conjecture | open | MathDB 页面 | 原文线索

V2-005653 | MATHDB-367052

Quantum modularity conjecture for 2-shifted symmetric functions

摘录 / EXCERPT: Quantum modularity conjecture. The q-bracket $\langle f\rangle_q$ is a quantum modular form of weight k.

conjecture | open | MathDB 页面 | 原文线索

V2-005654 | MATHDB-367053

Jack-polynomial interpolation conjecture for real and complex origamis

摘录 / EXCERPT: Jack-polynomial interpolation conjecture. The generating series $\mathcal{R}_\mu(t,\mathbf{p};b)$ interpolates between the quasimodular forms given by the generating functions for...

open_problem | open | MathDB 页面 | 原文线索

V2-005655 | MATHDB-367060

The Batyrev-Manin conjecture for DM stacks in positive characteristic

摘录 / EXCERPT: Batyrev-Manin conjecture for DM stacks. There exists a thin subset $T\subset\mathcal{X}\langle F\rangle$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005656 | MATHDB-367061

The Batyrev-Manin conjecture for quasi-nice DM stacks

摘录 / EXCERPT: Batyrev-Manin conjecture for quasi-nice DM stacks. There exists a thin subset $T\subset\mathcal{X}\langle F\rangle$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005657 | MATHDB-367086

Tan-Zhou's Hausdorff-dimension conjecture for Lüroth approximation sets

摘录 / EXCERPT: Let $L(\psi)$ be the set of points defined by the Lüroth approximation condition associated with a function $\psi:\mathbb{N}\to(0,1]$, and let $\underline{\tau}_\psi$ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-005658 | MATHDB-367136

Schinzel's bounded orthogonal relation conjecture

摘录 / EXCERPT: Schinzel's conjecture. There exists a nonzero vector ${\bf b}\in\mathbb{Z}^n$ orthogonal to ${\bf a}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005659 | MATHDB-367146

The unimodular conjecture over the p-adic integers

摘录 / EXCERPT: Let p be a prime, let $n\geq 1$, and let ... satisfy $\det JF\in\mathbb{Z}_p^\times$. A vector in $\mathbb{Z}_p^n$ is unimodular when the ideal generated by its entries is...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005660 | MATHDB-367147

The global Azumaya algebra conjecture for automorphic Galois representations

摘录 / EXCERPT: Global Azumaya algebra conjecture. There exists a central simple algebra D over $\mathbb Q(\pi)$ such that, for every prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-005661 | MATHDB-367153

Fermat's divisibility conjecture for Mersenne numbers

摘录 / EXCERPT: Let p be an odd prime, and define the least occurrence exponent k to be the least positive integer such that p divides the Mersenne number 2^k-1, when such an integer exist...

conjecture | open | MathDB 页面 | 原文线索

V2-005662 | MATHDB-367160

Density-zero conjecture for prime divisors in the orbit of zero under x^2-1

摘录 / EXCERPT: Let $f(x)=x^2-1$, and for an integer $n\geqslant 2$ let $P(n)$ denote the set of prime numbers arising in the associated sequence defined in the paper. Density-zero conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005663 | MATHDB-367176

Malle-Navarro-Tiep conjecture on Brauer characters and normal Sylow subgroups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. Write $\operatorname{IBr}(G)$ for the set of p-Brauer characters of G. Malle-Navarro-...

conjecture | open | MathDB 页面 | 原文线索

V2-005664 | MATHDB-367178

Sylow-AWC conjecture

摘录 / EXCERPT: Fix a prime p. Let $G\unlhd A$ be finite groups, let P be a Sylow p-subgroup of G, and write $\operatorname{IBr}(H)$ for the set of p-Brauer characters of a group H. Sy...

conjecture | open | MathDB 页面 | 原文线索

V2-005665 | MATHDB-367180

Ruzsa's conjecture on congruence-preserving sequences

摘录 / EXCERPT: Let $(a_n)_{n\geq 0}$ be a sequence of integers that preserves congruences, meaning that ... for all natural integers n and k. Such a sequence is called a pseudo-polynomial; a se...

conjecture | open | MathDB 页面 | 原文线索

V2-005666 | MATHDB-367199

The prime-characterization conjecture for the maximum cycle-displacement function

摘录 / EXCERPT: Prime-characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005667 | MATHDB-367209

The four-squares conjecture for generalized Pell sequences

摘录 / EXCERPT: Let a, b and d be positive integers such that d is not a square, let $\alpha=a+b^{2}\sqrt{d}$ have norm ... and let $\varepsilon=(t+u\sqrt{d})/2$ be a unit in...

conjecture | open | MathDB 页面 | 原文线索

V2-005668 | MATHDB-367210

The four-squares conjecture for unrestricted unit-power Pell sequences

摘录 / EXCERPT: Unrestricted-power four-squares conjecture. There are at most four distinct integer squares among the y_k'. If $|N_{\alpha}|$ is a prime power or a perfect square, then there are...

conjecture | open | MathDB 页面 | 原文线索

V2-005669 | MATHDB-367214

The rational-point counting conjecture for curved submanifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be a compact submanifold of $\mathbb{R}^n$ of dimension m, with codimension $k=n-m$, and let $N_{\mathcal{M}}(Q,\delta)$ denote the number of rational points of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005670 | MATHDB-367221

Coefficient relations in mixed Berndt-type integral formulas

摘录 / EXCERPT: For each integer $m\ge 1$, let $c_{2,m}$, $e_{2,m}$, $d_{1,m}$, and $f_{1,m}$ be the numbers appearing in Theorems and. Coefficient relations. The following identities hold: ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-005671 | MATHDB-367224

Equivariant p-adic Artin conjecture for graduated orders

摘录 / EXCERPT: Let p be an odd rational prime, let K be a totally real number field, and let $\mathcal L/K$ be an admissible one-dimensional p-adic Lie extension. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005672 | MATHDB-367226

Baragar's conjecture on connectivity of Markoff graphs

摘录 / EXCERPT: Let p be a prime, and let $\mathcal{G}_p$ be the Markoff graph over $\mathbb{F}_p$: its vertices are the nonzero triples in $\mathbb{F}_p^3$ satisfying ... and two vertices are joi...

conjecture | open | MathDB 页面 | 原文线索

V2-005673 | MATHDB-367240

The toric regulator and Sreekantan regulator comparison conjecture

摘录 / EXCERPT: Toric-Sreekantan regulator conjecture. For each prime ℓ, the valuation of the toric regulator at ℓ is the Sreekantan regulator tensored with ${\mathbb Q}_\ell$. This conj...

conjecture | open | MathDB 页面 | 原文线索

V2-005674 | MATHDB-367243

Twisted arithmetic transfer conjecture for regular semisimple intersections

摘录 / EXCERPT: Let $L_0=\mathbb Z_p^n$ and $L_0^\vee=p^{-1}\mathbb Z_p^t\oplus\mathbb Z_p^{n-t}$, and let $\mathrm{GL}(L_0,L_0^\vee)$ be the corresponding parahoric subgroup. For a regular semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-005675 | MATHDB-367270

Linnik's conjecture on small coordinates in representations by three squares

摘录 / EXCERPT: Linnik's conjecture. For each such n, there exists a representation with

conjecture | open | MathDB 页面 | 原文线索

V2-005676 | MATHDB-367290

Larsen's conjecture on ranks over fixed fields of finitely generated Galois subgroups

摘录 / EXCERPT: Let E/K be an elliptic curve over a number field K, and let $G_K:=\operatorname{Gal}(\overline{K}/K)$. Let G be a topologically finitely generated subgroup of G_K, and writ...

conjecture | open | MathDB 页面 | 原文线索

V2-005677 | MATHDB-367303

Jutila's conjecture on moments of quadratic character sums

摘录 / EXCERPT: Jutila's conjecture. For all $k=1,2,\dots$ and $X\geq 3$, $Y\geq 1$, there are coefficients $c_1(k)$ and $c_2(k)$ depending only on k such that

conjecture | open | MathDB 页面 | 原文线索

V2-005678 | MATHDB-367355

Squarefree-value conjecture for integral multivariate polynomials

摘录 / EXCERPT: Let $P\in\mathbb{Z}[X_{1},\ldots,X_{n}]$ have total degree $d\geq 2$. For $m>1$, define ... Given $B_j\in\mathbb{R}$ with $B_j\geq1$ and $B=\prod_{j=1}^{n}[0,B_j]\cap\mathbb{Z}^{n}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-005679 | MATHDB-367370

Graham's rearrangement conjecture for subsets of prime cyclic groups

摘录 / EXCERPT: Let p) be prime and let $a_1,a_2,\dots,a_d$ be distinct nonzero elements of $\mathbb{Z}_p$. A set of elements is rearrangeable if its elements can be ordered so that all partial...

conjecture | open | MathDB 页面 | 原文线索

V2-005680 | MATHDB-367376

Eventual projective-integral equivalence conjecture for irrational parameters

摘录 / EXCERPT: Let $\alpha,\beta\in\mathbb{R}\setminus\mathbb{Q}$ be irrational. Say that they are eventually $\operatorname{PGL}_2(\mathbb{Z})$-equivalent if there exists $m\in\mathbb{N}$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-005681 | MATHDB-367382

The divisor correlation conjecture for shifted divisor functions

摘录 / EXCERPT: Let $k,l\geqslant 2$ be natural numbers, let $h\in\mathbb{Z}\setminus\{0\}$, and let $d_k(n)$ and $d_l(n)$ denote the k-fold and l-fold divisor functions. Divisor correlation c...

conjecture | open | MathDB 页面 | 原文线索

V2-005682 | MATHDB-367389

The generality conjecture for lattices in two-dimensional semistable representations

摘录 / EXCERPT: Let p be a prime and let f be a positive integer. Consider 2-dimensional semi-stable non-crystalline representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-005683 | MATHDB-367390

The conjecture that the sieve parameter satisfies $\widetilde{M}(k)=k$

摘录 / EXCERPT: Let $\widetilde{M}(k)$ denote the parameter in Theorem. Conjecture on $\widetilde{M}(k)$. One can take $\widetilde{M}(k)=k$ for all $k\geq 1$ in Theorem. This would improve the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-005684 | MATHDB-367391

The prime-power plus almost-prime conjecture

摘录 / EXCERPT: Let $k\geq 1$ be odd, and let N be a sufficiently large even number. Prime-power plus almost-prime conjecture. Then N can be expressed as the sum of a kth power of a prime an...

conjecture | open | MathDB 页面 | 原文线索

V2-005685 | MATHDB-367392

The prime-square plus prime conjecture for nonsquares

摘录 / EXCERPT: Let $N>24476$ be a non-square satisfying $N\equiv 0,2\pmod{6}$. Nonsquare prime-square representation conjecture. Then there exist primes p and q such that ... By Hua's theorem,...

conjecture | open | MathDB 页面 | 原文线索

V2-005686 | MATHDB-367394

General conjecture on general-position subsets of finite affine spaces

摘录 / EXCERPT: General conjecture. For every integer $d d45a 2$, there is a positive real number $C=C(d)$ such that, as the prime power q tends to infinity, with high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-005687 | MATHDB-367395

The Clergyman's Conjecture for anti-k-naccis

摘录 / EXCERPT: Let X_n be the sequence of anti-k-naccis, defined as sums of k consecutive missing numbers. A sequence is k-automatic when its base-k value sequence can be generated by a...

conjecture | open | MathDB 页面 | 原文线索

V2-005688 | MATHDB-367397

The generalized periodic characterization conjecture for S_{1,g,g+1}

摘录 / EXCERPT: For $g\geq 2$, let $S_{1,g,g+1}$ be the greedy 3-subsumfree sequence beginning with $1,g,g+1$, where each subsequent term is the smallest larger integer that is not the sum of thre...

conjecture | open | MathDB 页面 | 原文线索

V2-005689 | MATHDB-367407

Conjecture on decomposition and residual irreducibility of compatible systems

摘录 / EXCERPT: Compatible-systems conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005690 | MATHDB-367415

The Browkin-Brzeziński strong n-conjecture for subsum-free integer tuples

摘录 / EXCERPT: For a non-zero integer m, let $\operatorname{rad}(m)$ be its largest square-free positive divisor. For $n\geq 3$ and $a=(a_1,\ldots,a_n)\in\mathbb{Z}^n$ with all $a_i\neq 0$, def...

conjecture | open | MathDB 页面 | 原文线索

V2-005691 | MATHDB-367436

A modulo 192 congruence family for partitions with distinct even parts

摘录 / EXCERPT: Let $ped(n)$ denote the number of partitions of n in which even parts are distinct and odd parts are unrestricted. Let $p\geq5$ be a prime satisfying ... and let t be a positive...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005692 | MATHDB-367441

Conjecture on sums of a 5-potent and an n-potent in finite fields

摘录 / EXCERPT: Let q be a prime power and let n satisfy the normal conditions for the Restricted Problem. An element $a\in\mathbb{F}_q$ is 5-potent if it is a fifth power, and n-potent if i...

conjecture | open | MathDB 页面 | 原文线索

V2-005693 | MATHDB-367454

Montgomery's ℓ^q conjecture for Dirichlet polynomials

摘录 / EXCERPT: Let M_{Dir} be the matrix associated with Dirichlet polynomials $D(t)=\sum_{n=N}^{2N}b_n e^{i t\log n}$, with $|b_n|\leq1$, and suppose $N\leq T\leq N^{O(1)}$. Montgomery's...

conjecture | open | MathDB 页面 | 原文线索

V2-005694 | MATHDB-367466

Stopple's asymptotic counting conjecture for zeta-derivative zero types

摘录 / EXCERPT: Assume the notation $N_0'(T)$, $N_1(T)$, and $N_2(T)$ denotes the counting functions up to height T for zeros of $\zeta'(s)$ of types 0, 1, and 2, respectively. Stopple's z...

conjecture | open | MathDB 页面 | 原文线索

V2-005695 | MATHDB-367467

Stopple's Z-curve conjecture for zeros of the zeta derivative

摘录 / EXCERPT: Let a Z-curve be one of the level curves described in the source for $\zeta'(s)$, and classify zeros of $\zeta'(s)$ by types 0, 1, and 2. Write a zero as...

conjecture | open | MathDB 页面 | 原文线索

V2-005696 | MATHDB-367468

Stopple's type 2 zero curvature estimate for zeros of the zeta derivative

摘录 / EXCERPT: Let $\rho'=\beta'+i\gamma'$ be a sequence of type 2 zeros of $\zeta'(s)$, and let the sum range over the other zeros $\lambda'\ne\rho'$ of $\zeta'(s)$. Stopple's type 2 curvature e...

conjecture | open | MathDB 页面 | 原文线索

V2-005697 | MATHDB-367470

Wang-Lazebnik-Thomason's cycle-length conjecture for Wenger graphs

摘录 / EXCERPT: Let $W_m(q)$ denote the Wenger graph with parameter m over the finite field of order q. Wang-Lazebnik-Thomason's conjecture. For every $m\ge 2$ and every prime power $q\ge 3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-005698 | MATHDB-367472

Kodess's isomorphism conjecture for monomial digraphs

摘录 / EXCERPT: Let $D(q;m,n)=D_2(\mathbb F_q;X^mY^n)$ be a monomial digraph over the field with q elements. Kodess's conjecture. For a prime power q, the digraphs $D(q;m_1,n_1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005699 | MATHDB-367473

Lazebnik-Ustimenko-Woldar's girth conjecture for Wenger-type graphs

摘录 / EXCERPT: Let $D(k,q)$ be the graph defined by the parameters k and the prime power q. Lazebnik-Ustimenko-Woldar's conjecture. For every prime power $q\ge 4$, the graph has girth ... The s...

conjecture | open | MathDB 页面 | 原文线索

V2-005700 | MATHDB-367474

Lazebnik-Ustimenko-Woldar's diameter upper-bound conjecture

摘录 / EXCERPT: Let $C\!D(k,q)$ be a connected component of $D(k,q)$, with $k\ge 2$ and q a prime power. Lazebnik-Ustimenko-Woldar's conjecture. There is a positive constant C such that ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-005701 | MATHDB-367475

Schneider's exact-diameter conjecture for components of $D(k,q)$

摘录 / EXCERPT: Let $C\!D(k,q)$ be a connected component of $D(k,q)$. Schneider's conjecture. For every prime power q, $\operatorname{diam}(C\!D(3,q))=6$. For $k>3$ odd, the diameter is $k+5$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-005702 | MATHDB-367489

Kim's anticyclotomic weak main conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\operatorname{\mathbb{Q}}$ with conductor N_E, and let $p\geq 5$ be a prime at which E has good supersingular reduction. Let K be an imagin...

conjecture | open | MathDB 页面 | 原文线索

V2-005703 | MATHDB-367502

The Two Conjecture for quadratic Dirichlet characters

摘录 / EXCERPT: Let χ_1 and χ_2 be nontrivial, primitive Dirichlet characters modulo q_1 and q_2, respectively, with $\chi_1\chi_2(-1)=1$. Let S_{χ_1,χ_2} be their generalize...

conjecture | open | MathDB 页面 | 原文线索

V2-005704 | MATHDB-367503

The Generalized Two Conjecture for quadratic number fields

摘录 / EXCERPT: Let χ_1 and χ_2 be nontrivial, primitive Dirichlet characters modulo q_1 and q_2, respectively, with $\chi_1\chi_2(-1)=1$. Let F_{χ_1,χ_2} be the smallest num...

conjecture | open | MathDB 页面 | 原文线索

V2-005705 | MATHDB-367505

Kolyvagin's nontriviality conjecture for modular forms

摘录 / EXCERPT: Kolyvagin's conjecture. The set of Kolyvagin classes is nontrivial:

conjecture | open | MathDB 页面 | 原文线索

V2-005706 | MATHDB-367506

Strict Kolyvagin nontriviality conjecture for non-ordinary modular forms

摘录 / EXCERPT: Strict Kolyvagin conjecture. The strict Kolyvagin set is nontrivial:

conjecture | open | MathDB 页面 | 原文线索

V2-005707 | MATHDB-367527

Growth-ratio conjecture for Birch-Erdős representation functions

摘录 / EXCERPT: Let $f_{p,q}(n)$ denote the function considered in the Birch-Erdős theorem. Let $q>p>1$ be relatively prime integers, and let $ell\ge 2$ be an integer. Growth-ratio conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005708 | MATHDB-367528

Parity-density conjecture for the Birch-Erdős function at $(p,q)=(2,3)$

摘录 / EXCERPT: Let $f_{2,3}(n)$ be the Birch-Erdős representation function for the parameters $p=2$ and $q=3$. Parity-density conjecture. ... The preceding results establish that both even and odd...

conjecture | open | MathDB 页面 | 原文线索

V2-005709 | MATHDB-367529

Sign-change conjecture for Birch-Erdős representation functions

摘录 / EXCERPT: Let $f_{p,q}(n)$ be the Birch-Erdős representation function, with p and q relatively prime integers satisfying $q>p>2$. Sign-change conjecture. The sign of ... changes infinitely...

conjecture | open | MathDB 页面 | 原文线索

V2-005710 | MATHDB-367530

Equality-frequency conjecture for Birch-Erdős representation functions

摘录 / EXCERPT: Let $f_{p,q}(n)$ be the Birch-Erdős representation function. Equality-frequency conjecture. ... for infinitely many n. This conjecture accompanies the proposed sign-change phenomen...

conjecture | open | MathDB 页面 | 原文线索

V2-005711 | MATHDB-367542

The short-interval divisor conjecture for arbitrary fixed scales

摘录 / EXCERPT: Let $n\geq 1$ be an integer, let $\mathcal{D}_n$ be the set of positive divisors of n, and define ... Fix real numbers θ and ϵ satisfying $0<\epsilon<\theta<1$. Sho...

conjecture | open | MathDB 页面 | 原文线索

V2-005712 | MATHDB-367543

The conjectural structure of the second homology of ${\rm SL}_2(\mathbb{Z}[1/n])$

摘录 / EXCERPT: Let $n=p_1\cdots p_l$, with $l>1$, be the prime decomposition of n such that $1<r_{p_1}\leq\cdots\leq r_{p_l}$; if $r_{p_j}=r_{p_{j+1}}$ for some j, assume that $p_j<p_{j+1}$....

conjecture | open | MathDB 页面 | 原文线索

V2-005713 | MATHDB-367550

Erdős-Graham growth conjecture for rational Egyptian underapproximations

摘录 / EXCERPT: Rational Erdős-Graham conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-005714 | MATHDB-367553

Nath-Saikia-Sarma's quadratic congruence conjecture for 2-regular 3-tuple partitions

摘录 / EXCERPT: Let $T_{\ell,k}(n)$ denote the number of k-tuple partitions of n whose components are ℓ-regular partitions, and write $T_2(n)=T_{2,3}(n)$. Let $p\geq 5$ be a prime satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-005715 | MATHDB-367562

Isomorphism conjecture for torus-link and circular-group presentations

摘录 / EXCERPT: Torus-link/circular-group isomorphism conjecture. For all $n,m\in\mathbb{N}^*$, this morphism is an isomorphism. The claim generalises the previously established coprime case; the...

conjecture | open | MathDB 页面 | 原文线索

V2-005716 | MATHDB-367591

Montgomery's simplicity conjecture for the zeros of the Riemann zeta-function

摘录 / EXCERPT: Let $N^*(T)$ be the multiplicity-weighted zero count ... where m_ρ is the multiplicity of the zero ρ. Montgomery's simplicity conjecture. As $T\to\infty$, ... Since...

conjecture | open | MathDB 页面 | 原文线索

V2-005717 | MATHDB-367596

Eswarathasan-Levine conjecture on the finiteness of divisible harmonic numbers

摘录 / EXCERPT: Eswarathasan-Levine conjecture. The set J_p is finite for all primes p.

conjecture | open | MathDB 页面 | 原文线索

V2-005718 | MATHDB-367597

Eswarathasan-Levine conjecture on infinitely many harmonic primes

摘录 / EXCERPT: Eswarathasan-Levine conjecture. There are infinitely many harmonic primes.

conjecture | open | MathDB 页面 | 原文线索

V2-005719 | MATHDB-367634

The smooth-value asymptotic for polynomial values

摘录 / EXCERPT: Let $F(x)\in\mathbb{Z}[x]$, and let $\Psi_F^*(x,y)$ denote the counting function for the relevant integers whose values under F satisfy the smoothness condition encoded by the so...

conjecture | open | MathDB 页面 | 原文线索

V2-005720 | MATHDB-367663

Eventual periodicity of combinatorial zero mutable Laurent polynomial counts

摘录 / EXCERPT: Eventual-periodicity conjecture. 1. For each $a\in\mathbb N$ there exists $b_0\in\mathbb N$ such that for all $b\geq b_0$, the number $|\textup{ZMLP}_{\textup{comb}}(a,b)|$ depends...

conjecture | open | MathDB 页面 | 原文线索

V2-005721 | MATHDB-367671

abc conjecture for a Galois number field

摘录 / EXCERPT: The abc conjecture for a Galois number field. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-005722 | MATHDB-367675

Further generalized cubic partition congruences modulo squared primes

摘录 / EXCERPT: Further generalized cubic partition congruence conjecture.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005723 | MATHDB-367676

Modular-three congruences for overcubic partitions

摘录 / EXCERPT: Overcubic partition congruence conjecture modulo powers of three.

conjecture | open | MathDB 页面 | 原文线索

V2-005724 | MATHDB-367695

Waldhausen's chromatic reformulation of the Lichtenbaum-Quillen conjecture

摘录 / EXCERPT: Let $\mathcal{O}$ be a nice ring, let ℓ be prime, and let $\mathbb{S}/\ell$ denote the mod-ℓ sphere spectrum. Let L_1^f denote finite-height localization, let v_1 be...

conjecture | open | MathDB 页面 | 原文线索

V2-005725 | MATHDB-367696

Rognes' algebraic K-theory red-shift conjecture

摘录 / EXCERPT: Fix a prime p. Let R be a suitable $\mathbb{E}_1$-ring of fp-type n, where fp-type is the chromatic finiteness notion described in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005726 | MATHDB-367710

The reversible-prime conjecture in every base

摘录 / EXCERPT: Fix a base $b\geq 2$ and let $\overleftarrow{n}$ denote the integer obtained by reversing the base-b digits of n. A reversible prime is a prime number whose digital reverse is...

conjecture | open | MathDB 页面 | 原文线索

V2-005727 | MATHDB-367741

Conjecture on the Hilbert symbol and the coefficients $U_k^{(u)}$

摘录 / EXCERPT: The conjecture. For $0\leq k\leq \left\lceil\frac{j-i_0}{2}\right\rceil$, the coefficients satisfy the stated congruences for $U_k^{(u)}$ and U_k', and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-005728 | MATHDB-367745

The Duffin-Schaeffer-type conjecture for multiple zeta-star values

摘录 / EXCERPT: Let $f\colon \mathbb{N}\rightarrow [1,+\infty)$, and let B_f be the set of real numbers $\alpha\in [1,+\infty)$ such that ... for infinitely many $r,k_1,\ldots,k_r$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-005729 | MATHDB-367746

Integral equivariant Iwasawa main conjecture for arbitrary sets of primes

摘录 / EXCERPT: Let F) be a totally real field, let p be a prime, let H_{∞}/F be the relevant infinite extension with Galois group $\mathcal{G}$, and let S, S_{ram}, and S_p denot...

conjecture | open | MathDB 页面 | 原文线索

V2-005730 | MATHDB-367798

The mock-modularity conjecture for BPS invariants of type II limits

摘录 / EXCERPT: Consider a type II$_{b}$ limit associated with a Tyurin degeneration of the form $V_0=V_1\cup_Z V_2$ and a special Lagrangian 3-cycle Γ_0 dual to an element of...

conjecture | open | MathDB 页面 | 原文线索

V2-005731 | MATHDB-367815

Liouville number stratification conjecture

摘录 / EXCERPT: Let $\mathbb{R}_{S_n}$ be the reals at definability level S_n, let $\mathbb{R}_{S_\omega}=\bigcup_{n\in\mathbb{N}}\mathbb{R}_{S_n}$, and let $\mathbb{A}_{S_n}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005732 | MATHDB-367846

Conjecture on primitives of the greedy convolution of length 4

摘录 / EXCERPT: Let M denote the primitives of rank 4 for the greedy convolution of length 4. The rank-one and rank-two primitives are the objects considered below, and divisibility and memb...

conjecture | open | MathDB 页面 | 原文线索

V2-005733 | MATHDB-367864

Leopoldt conjecture for Anderson t-modules

摘录 / EXCERPT: Let E be an Anderson t-module, let P be a prime of A, and define ... and ... Also write $\mathscr{U}(E;\mathscr{O}_L)=\exp_E(U(E;\mathscr{O}_L))$. Leopoldt conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005734 | MATHDB-367866

Vanishing-order bound for P-adic L-series of Anderson t-modules

摘录 / EXCERPT: Let E be the Anderson t-module under consideration over L/K, let P be a prime of A, and let Ω_E be the period lattices associated with E. Write r_{Ω_E} fo...

conjecture | open | MathDB 页面 | 原文线索

V2-005735 | MATHDB-367867

Conjecture on the minimal orders of ordered and additive factorizations

摘录 / EXCERPT: Minimal-order conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-005736 | MATHDB-367880

ABC conjecture for coprime integer triples

摘录 / EXCERPT: ABC conjecture. For every $\epsilon>0$, there is a constant C_ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005737 | MATHDB-367901

Symmetry conjecture for the special values of L_{+}

摘录 / EXCERPT: Let L_{+} be the function appearing in the paper, and let $\mathscr{C}$ be its associated constant. Symmetry conjecture. The values $L_{+}(2k)$ satisfy the following symmetry: .....

conjecture | open | MathDB 页面 | 原文线索

V2-005738 | MATHDB-367925

Gonek-Hughes-Farmer conjecture for maximum Dirichlet L-values

摘录 / EXCERPT: Let $q>1$, let $\mathcal{X}$ be the group of Dirichlet characters modulo q, and let $L(\chi,s)$ denote the associated Dirichlet L-function. Gonek-Hughes-Farmer conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-005739 | MATHDB-367928

Erdős-Graham pseudo-greedy gap conjecture

摘录 / EXCERPT: Pseudo-greedy gap conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-005740 | MATHDB-367947

Classification conjecture for prime algebraic tangles

摘录 / EXCERPT: Classification conjecture for prime algebraic tangles. Two prime algebraic tangles are isotopic if and only if they are related by a finite sequence of twist, flype, ring, and elem...

conjecture | open | MathDB 页面 | 原文线索

V2-005741 | MATHDB-367949

The Franušić-Jadrijević conjecture on Diophantine quadruples in commutative rings

摘录 / EXCERPT: Let $\mathcal{R}$ be a commutative ring with unity and let $n\in\mathcal{R}\setminus\{0\}$. A Diophantine quadruple with property $D(n)$ in $\mathcal{R}$ is a set of four nonzero e...

conjecture | open | MathDB 页面 | 原文线索

V2-005742 | MATHDB-367952

The four-squares conjecture for unrestricted unit-power sequences

摘录 / EXCERPT: The unrestricted four-squares conjecture. There are at most four distinct integer squares among the y_k'. If $|N_{\alpha}|$ is a prime power or a perfect square, then there are a...

conjecture | open | MathDB 页面 | 原文线索

V2-005743 | MATHDB-367972

Wolstenholme's higher-power congruences for Gaussian integers

摘录 / EXCERPT: Wolstenholme's higher-power congruences. For every such k and p,

conjecture | open | MathDB 页面 | 原文线索

V2-005744 | MATHDB-367973

Conjectural coefficient pattern for Gaussian integer polynomials

摘录 / EXCERPT: Conjectural coefficient pattern. If $p=4k+3>3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-005745 | MATHDB-367974

Lucas congruence for Gaussian binomial coefficients

摘录 / EXCERPT: Lucas congruence for Gaussian binomial coefficients. For positive integers A,B,C,D for which these coefficients are defined,

conjecture | open | MathDB 页面 | 原文线索

V2-005746 | MATHDB-368027

Atkin-O'Brien partition congruence conjecture for powers of 13

摘录 / EXCERPT: Let $P(N)$ be defined by $P(N)=p(n)$ when $N=24n-1$, and $P(N)=0$ if $N<-1$, if $N\not\equiv -1\pmod{24}$, or if N is nonintegral. Let $\left(\frac{a}{b}\right)$ denote the Jacob...

conjecture | open | MathDB 页面 | 原文线索

V2-005747 | MATHDB-368052

Murmuration conjecture for elliptic curves ordered by naive height

摘录 / EXCERPT: For integers A,B, let $E_{A,B}$ be the elliptic curve ... with no prime p satisfying both $p^4\mid A$ and $p^6\mid B$. Define its naive height by ... and write $N(E)$ for its condu...

conjecture | open | MathDB 页面 | 原文线索

V2-005748 | MATHDB-368059

Conjectural essential spectral gap bound for twisted Selberg zeta functions

摘录 / EXCERPT: Essential spectral gap conjecture. For every $\omega\in\mathcal{H}^1(X)$,

conjecture | open | MathDB 页面 | 原文线索

V2-005749 | MATHDB-368060

Conjecture on knot Floer homology of prime fibered positive knots

摘录 / EXCERPT: Let K be a prime fibered positive knot of genus g, and let $\widehat{HFK}_{-1}(K,g-1)$ denote its knot Floer homology in Alexander grading $g-1$ and Maslov grading -1, with c...

conjecture | open | MathDB 页面 | 原文线索

V2-005750 | MATHDB-368076

The even-iterate frequency conjecture for the Collatz map

摘录 / EXCERPT: Let $T:\mathbb N\to\mathbb N$ be the Collatz function, let $T^{(k)}$ denote its k-fold iterate, and fix $n_0\in\mathbb N$ with $n_0\geq 1$. The even-iterate frequency conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-005751 | MATHDB-368077

The digit-frequency conjecture for the rational-base sequence

摘录 / EXCERPT: Let $\operatorname{sat}(k)$ be the sequence defined in the paper, and let $\langle\operatorname{sat}(k)\rangle$ denote its associated word over the alphabet $\{1,2\}$. The digit-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-005752 | MATHDB-368078

The normality conjecture for the limiting rational-base word

摘录 / EXCERPT: For each $k\geq 1$, let $\langle\operatorname{sat}(k)\rangle$ be the associated word, and define the infinite word ... An infinite word over a finite alphabet is normal if every word...

conjecture | open | MathDB 页面 | 原文线索

V2-005753 | MATHDB-368083

The zeta-function conjecture

摘录 / EXCERPT: Let Ω be a lattice in $\mathbb{C}$ with algebraic invariants g_2,g_3. Let $p\in\mathbb{C}\smallsetminus\Omega$ with algebraic $\zeta(p)$, and let α be an algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-005754 | MATHDB-368088

Goldbach's conjecture for prime-like distributions

摘录 / EXCERPT: Generalization of Goldbach's conjecture. Let Q be a subset of natural numbers whose distribution is similar to primes. There exists $N_Q\in\mathbb{N}$ such that, for any even int...

conjecture | open | MathDB 页面 | 原文线索

V2-005755 | MATHDB-368091

The Skolem conjecture for simple rational linear recurrence binary sequences

摘录 / EXCERPT: Let $\boldsymbol{u}=(u_n)_{n\in\mathbb{Z}}$ be a simple rational LRBS, meaning a simple rational linear recurrence binary sequence, taking values in $\mathbb{Z}[1/b]$ for some inte...

conjecture | open | MathDB 页面 | 原文线索

V2-005756 | MATHDB-368099

The complete-primality conjecture for primitive ideals of Witt and Virasoro enveloping algebras

摘录 / EXCERPT: Let $\g$ be one of $\W{-1}$, W, or $\Vir$, and let $\Ua(\g)$ be its enveloping algebra. Complete-primality conjecture. Every primitive ideal of $\Ua(\g)$ is completely prime. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005757 | MATHDB-368110

Geometric C_1 transfer for algebraically maximal valued fields

摘录 / EXCERPT: Let (K,v) be an algebraically maximal valued field with divisible value group and perfect residue field k. A field is called geometrically C_1 if every smooth projective sepa...

conjecture | open | MathDB 页面 | 原文线索

V2-005758 | MATHDB-368111

Kazashi and Sloan's number-theoretic conjecture for unshifted lattice rules

摘录 / EXCERPT: Let n be an odd prime. For $1\leq\kappa\leq n-1$, define ... where $r(j,n)$ is the unique integer congruent to j modulo n in $\{-(n-1)/2,\ldots,(n-1)/2\}$. Kazashi and Sloan's...

conjecture | open | MathDB 页面 | 原文线索

V2-005759 | MATHDB-368133

The arithmetic point-count conjecture for Berglund-Hübsch Calabi-Yau orbifolds

摘录 / EXCERPT: Let A be an $n\times n$ Berglund-Hübsch matrix over $\mathbb F_p$, with $n\ge 3$, and let G be a subgroup of G_A such that $J\in G\cap G^T$. Assume ... Let X_A be the hypersu...

conjecture | open | MathDB 页面 | 原文线索

V2-005760 | MATHDB-368153

Koblitz's prime-order conjecture for CM elliptic curves

摘录 / EXCERPT: Koblitz's conjecture. There exists a constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005761 | MATHDB-368154

Artin's holomorphy conjecture

摘录 / EXCERPT: Let L/K be a Galois extension with Galois group ... For every irreducible character $\chi\colon G\to\mathbb{C}$, let $L(s,\chi)$ denote the associated Artin L-function. Artin's h...

conjecture | open | MathDB 页面 | 原文线索

V2-005762 | MATHDB-368157

Kohen's first-zero bound conjecture for univariate constant term sequences

摘录 / EXCERPT: Let P be a univariate Laurent polynomial and let p be prime. For the constant term sequence $A_p(n)=\operatorname{ct}[P^n]\bmod p$, suppose there exists some $n\in\mathbb{N}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-005763 | MATHDB-368158

Kagey-Rajesh characterization of the equality g(n)=2n

摘录 / EXCERPT: For a non-negative integer n, let $g(n)$ be the smallest integer $s\geq n$ for which there are distinct integers $a_1,\ldots,a_t\in[n,s]$, with $a_1=n$, such that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005764 | MATHDB-368159

The prime-power sharpness conjecture for k(m)

摘录 / EXCERPT: Let $k(m)$ be the maximum number of distinct values of $g_m(n)$ attained at a fixed endpoint in the paper's construction, and let $d(m)$ denote the number of positive divisors of...

conjecture | open | MathDB 页面 | 原文线索

V2-005765 | MATHDB-368160

The finiteness conjecture for simultaneous T_m and T'_m values

摘录 / EXCERPT: For each $m\geq2$ and integer n, let $T_m(n)$ and $T'_m(n)$ be the auxiliary quantities defined in the paper. Finiteness conjecture. For each $m\geq2$, there are only finitely ma...

conjecture | open | MathDB 页面 | 原文线索

V2-005766 | MATHDB-368215

Localised Iwasawa main conjecture for a twisted modular form

摘录 / EXCERPT: Let β be a height-one prime ideal of the Iwasawa algebra Λ, and retain the notation $H^1_{\operatorname{IW}}(\mathbb Q_\infty,T\otimes\mathcal O_d(\chi))$, X_∞...

conjecture | open | MathDB 页面 | 原文线索

V2-005767 | MATHDB-368216

Kurihara-type conjecture for twisted Kurihara numbers

摘录 / EXCERPT: Let $E/\mathbb Q$ be an elliptic curve, let $p\geq 5$ be prime, and let χ be a character in the setting of the twisted Kurihara numbers. For n in the indexing set...

conjecture | open | MathDB 页面 | 原文线索

V2-005768 | MATHDB-368263

The subgroup-counting conjecture with the arithmetic-function exponent

摘录 / EXCERPT: Let R be a finite group, and let $T\leq R$ be a subgroup. Define ... For a non-zero integer n, let $\lambda(n)$ denote the number of prime divisors of n, counted with multiplic...

conjecture | open | MathDB 页面 | 原文线索

V2-005769 | MATHDB-368276

Erdős's conjecture on the smallest prime divisors of Sierpiński sequences

摘录 / EXCERPT: Let k be a Sierpiński number, meaning that $k\cdot 2^n+1$ is composite for every positive integer n. For each positive integer n, let the smallest prime divisor of...

conjecture | open | MathDB 页面 | 原文线索

V2-005770 | MATHDB-368289

One-modulus formula for refined tree coefficients

摘录 / EXCERPT: Let $\boldsymbol\gamma=(\gamma_1,\ldots,\gamma_n)$ be an ordered collection of charges, let $\hat{\boldsymbol\gamma}$ denote the corresponding hatted charge data, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005771 | MATHDB-368290

Higher-supersymmetry refined generating-function modularity conjecture

摘录 / EXCERPT: Let $\mathfrak Y$ be a compactification with higher supersymmetry, let $\gamma=(p^0,p^A,q_A,q_0)$ be an electromagnetic charge vector, and let ... After restricting to the sublattice...

conjecture | open | MathDB 页面 | 原文线索

V2-005772 | MATHDB-368377

Cesàro limit conjecture for the order-two abundancy index

摘录 / EXCERPT: Main conjecture. The Cesàro mean of the centered partial sums should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-005773 | MATHDB-368399

Nonexistence conjecture for odd Kalita-Saikia near superperfect numbers

摘录 / EXCERPT: Let a Kalita-Saikia near superperfect number mean a number that is both a Kalita-Saikia number and a near superperfect number. Nonexistence conjecture. There are no odd Kalita-Saik...

conjecture | open | MathDB 页面 | 原文线索

V2-005774 | MATHDB-368412

The refined Beilinson-Bloch-Kato conjecture for modular forms

摘录 / EXCERPT: Let $f\in S_k(\Gamma_0(N))$ be a newform, let $p\ge 3$ be a prime, and suppose that ρ_f has large image. Let $\widetilde{\boldsymbol{\delta}}^{\mathrm{min},\dagger}$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-005775 | MATHDB-368413

The Iwasawa main conjecture for Kato's Euler system

摘录 / EXCERPT: Let $\mathbb{Q}_\infty$ be the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$, let Λ be its Iwasawa algebra, and assume that ρ_f has large image. For a height-one...

conjecture | open | MathDB 页面 | 原文线索

V2-005776 | MATHDB-368414

The canonical-period Tamagawa conjecture for Kurihara numbers

摘录 / EXCERPT: Let $f\in S_k(\Gamma_0(N))$ be a newform with $k\ge 2$ and $p\ge 3$, with $p\nmid N$ and ρ_f of large image. Suppose that f is ordinary at p and p-distinguished, or that...

conjecture | open | MathDB 页面 | 原文线索

V2-005777 | MATHDB-368415

The Kato-system Tamagawa conjecture

摘录 / EXCERPT: Let $f\in S_k(\Gamma_0(N))$ be a newform with $k\ge 2$ and $p\ge 3$, and suppose that ρ_f has large image. Let N_0 be the prime-to-p part of N. The Kato-system Tamagawa...

conjecture | open | MathDB 页面 | 原文线索

V2-005778 | MATHDB-368416

The minimal Kurihara-number Tamagawa conjecture

摘录 / EXCERPT: Let $f\in S_k(\Gamma_0(N))$ be a newform with $k\ge 2$ and $p\ge 3$. Assume that ρ_f has large image, that f is not congruent modulo π to any anomalous ordinary CM modul...

conjecture | open | MathDB 页面 | 原文线索

V2-005779 | MATHDB-368424

Orthogonal quasimodularity conjecture for Gromov-Witten potentials of Enriques surfaces

摘录 / EXCERPT: Let $M=U\oplus U(2)\oplus E_8(-2)$ be the Enriques lattice, let $\mathcal D$ be the relevant orthogonal domain, and let $\mathcal H_{-2}\subset\mathcal D$ be the norm -2 Noether-...

conjecture | open | MathDB 页面 | 原文线索

V2-005780 | MATHDB-368438

Conrad-Edixhoven-Stein conjecture for rational torsion on $J_1(p)$

摘录 / EXCERPT: Let $p\geq 5$ be a prime, and let $X_1(p)$ be the modular curve with Jacobian $J_1(p)$. The rational cuspidal subgroup is the subgroup generated by divisor classes given by differe...

conjecture | open | MathDB 页面 | 原文线索

V2-005781 | MATHDB-368465

Equivalent odd-subspace formulation of the Chowla-Milnor conjecture

摘录 / EXCERPT: Equivalent Chowla-Milnor formulation. One has

conjecture | open | MathDB 页面 | 原文线索

V2-005782 | MATHDB-368473

The Carlitz conjecture on even-degree exceptional polynomials

摘录 / EXCERPT: Let q be a prime power. A polynomial $f(X)\in\mathbb{F}_q[X]$ is exceptional if it induces a bijection on $\mathbb{F}_{q^k}$ for infinitely many positive integers k. Carlitz co...

conjecture | solved | MathDB 页面 | 原文线索

V2-005783 | MATHDB-368486

Wang-Williams conjecture on free algebras of unitary modular forms

摘录 / EXCERPT: Let E be an imaginary quadratic field with odd discriminant $-D$, let $(L,\langle,\rangle)$ be a Hermitian lattice over $\mathcal{O}_E$ of signature $(1,n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-005784 | MATHDB-368495

Rational torsion generation conjecture for J_1(p)

摘录 / EXCERPT: Let p be a prime, and let $J_1(p)$ be the Jacobian of the modular curve $X_1(p)$. A rational cusp difference is a divisor class represented by the difference of two rational cusp...

conjecture | open | MathDB 页面 | 原文线索

V2-005785 | MATHDB-368496

Strangeness-dimension conjecture for prime level newforms

摘录 / EXCERPT: For each prime p, let $\operatorname{strdim}(p)$ denote the strangeness dimension defined in the surrounding discussion. Strangeness-dimension conjecture. The following two forms...

conjecture | open | MathDB 页面 | 原文线索

V2-005786 | MATHDB-368497

Positive-even-rank dimension conjecture for prime level newforms

摘录 / EXCERPT: For each prime p, let $\operatorname{perdim}(p)$ be the number of weight-2 newforms f for $\Gamma_0(p)$ such that $w(f)=-f$ and $L(f,1)=0$. Positive-even-rank dimension conje...

conjecture | open | MathDB 页面 | 原文线索

V2-005787 | MATHDB-368521

Simple algebraic-group closure conjecture over the reals and p-adics

摘录 / EXCERPT: Let $\mathcal{A}$ be a connected commutative algebraic group over $\mathbb{Q}$, let $\mathcal{L}\subset\mathcal{A}(\mathbb{Q})$ be finitely generated, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005788 | MATHDB-368530

Finite generation and Cohen-Macaulayness of matrix-valued Hilbert modular forms

摘录 / EXCERPT: Let $r,c\geq 1$ be integers, let F be a totally real field, and let $\rho:\Gamma_F\rightarrow\operatorname{GL}_r(\mathbb{C})$ be a complex representation. The graded space...

conjecture | open | MathDB 页面 | 原文线索

V2-005789 | MATHDB-368531

Dieulefait-González-Lario congruence conjecture for modular forms of CM type mod \ell

摘录 / EXCERPT: Let $f(q)=\sum_{n=1}^\infty a_nq^n\in S_2(\Gamma_0(N))^{\operatorname{new}}$ be a normalized newform of CM type mod ℓ by an imaginary quadratic field K. A newform has comple...

conjecture | open | MathDB 页面 | 原文线索

V2-005790 | MATHDB-368542

Finiteness conjecture for exceptional heavenly elliptic curves over quadratic fields

摘录 / EXCERPT: For every real number $B\geq 1$, let $\mathcal{R}_{B}$ be the set of ordered pairs (ℓ,K) such that K is a quadratic field, $\ell\geq B$ is prime, and there exists an ellipti...

conjecture | open | MathDB 页面 | 原文线索

V2-005791 | MATHDB-368565

Strong ironing property for rolling-ball optimizers

摘录 / EXCERPT: Let $f,g:\mathbb{R}^d\to\mathbb{R}$ be two continuous functions, and let $\Gamma,\Gamma^\prime$ be their respective graphs. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005792 | MATHDB-368567

Generalized Kauffman-Harary conjecture

摘录 / EXCERPT: Let K be a prime alternating knot, and let D be an alternating diagram of K without nugatory crossings, so that D is a minimal diagram. Let $X^{(2)}_K$ denote the double co...

conjecture | open | MathDB 页面 | 原文线索

V2-005793 | MATHDB-368580

Synchronization conjecture under the $(*)$ condition

摘录 / EXCERPT: For a prime p, let $O_p(G)$ be the intersection of all Sylow p-subgroups of G. Say that G has $(*)_p$ if there exist two Sylow p-subgroups whose intersection is contained...

conjecture | open | MathDB 页面 | 原文线索

V2-005794 | MATHDB-368581

Giannelli's degree-preserving refinement of the McKay conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Write $\operatorname{Irr}_{p'}(G)$ for the irreducible complex characters of G whose degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-005795 | MATHDB-368582

Inductive Giannelli-McKay condition for quasisimple groups

摘录 / EXCERPT: Let S be a quasisimple group with cyclic center, let p be a prime, let $P\in\operatorname{Syl}_p(S)$, and set $A=\operatorname{Aut}(S)_P$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005796 | MATHDB-368603

Correspondence between non-Miyawaki Hermitian eigenforms and non-Yoshida algebraic modular forms

摘录 / EXCERPT: Main correspondence conjecture. There is a one-to-one correspondence between such F and G. It has the following properties: the degree-six zeta function $L(F;s)$ equals the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-005797 | MATHDB-368609

Wang's transformation conjecture for Dwork's p-adic hypergeometric function

摘录 / EXCERPT: Wang's transformation conjecture. If p is an odd prime, then

conjecture | open | MathDB 页面 | 原文线索

V2-005798 | MATHDB-368612

Nonvanishing principal-minor conjecture for Fourier matrices of square-free order

摘录 / EXCERPT: Square-free Fourier principal-minor conjecture. If N is square-free, then all principal minors of ${\mathcal{F}}_N$ are nonzero.

conjecture | open | MathDB 页面 | 原文线索

V2-005799 | MATHDB-368650

Regular genus formula for the n-dimensional torus

摘录 / EXCERPT: Let n be an integer with $n\geq 5$, and let $\Gamma^\prime$ be the $(n+1)$-regular colored graph constructed from the colored triangulation described in the paper. It has...

conjecture | open | MathDB 页面 | 原文线索

V2-005800 | MATHDB-368659

Larcher-Stockinger conjecture on non-Poissonian pair correlations

摘录 / EXCERPT: Larcher-Stockinger conjecture. If for almost all α the pair correlations of $(a_n\alpha)_{n \in \mathbb{N}} \mod 1$ are not Poissonian, then the pair correlations of this se...

conjecture | open | MathDB 页面 | 原文线索

V2-005801 | MATHDB-368660

Conjecture on Poissonian correlations for prime rotations

摘录 / EXCERPT: Prime-rotation conjecture. The prime rotation sequence $(p_n\alpha)_{n \in \mathbb{N}} \mod 1$ has Poissonian correlations of all orders and, consequently, has Poissonian gap distr...

conjecture | open | MathDB 页面 | 原文线索

V2-005802 | MATHDB-368664

The bounded descent conjecture for the recursive divisor-sum sequence

摘录 / EXCERPT: Let $\mathfrak{R}$ be the recursive map on positive integers defined by ... where σ is the sum-of-divisors function. Bounded descent conjecture. There exists a constant...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005803 | MATHDB-368684

Half-of-primes conjecture for primes outside a two-generated numerical semigroup

摘录 / EXCERPT: Let a and b be integers with $b>a\ge 1$ and $\gcd(a,b)=1$. Set $S(a,b)=ab-a-b$, and let $\pi^*(a,b)$ be the number of primes not belonging to the numerical semigroup...

conjecture | open | MathDB 页面 | 原文线索

V2-005804 | MATHDB-368688

Polynomiality conjecture for double cosets of parabolic linear groups

摘录 / EXCERPT: Let n be a positive integer, let $\lambda,\mu\vdash n$, and let q be a prime power. Write $\operatorname{GL}_n(\mathbb F_q)$ for the general linear group and...

conjecture | open | MathDB 页面 | 原文线索

V2-005805 | MATHDB-368689

The p-adic Littlewood conjecture

摘录 / EXCERPT: Let p be a prime and let $b1$ be an irrational real number. Let $\lvert q\rvert_p=p^{-v_p(q)}$ denote the p-adic absolute value of the integer q, where...

conjecture | open | MathDB 页面 | 原文线索

V2-005806 | MATHDB-368690

The 2-adic Littlewood conjecture in continued-fraction form

摘录 / EXCERPT: For an irrational real number $\alpha=[a_0;a_1,a_2,\ldots]$, let $M(\alpha)=\sup_{n\geq 1}a_n(\alpha)$, where $a_n(\alpha)$ are its continued-fraction partial quotients. The 2-adic...

conjecture | open | MathDB 页面 | 原文线索

V2-005807 | MATHDB-368691

The B-variant of the 2-adic Littlewood conjecture

摘录 / EXCERPT: For an irrational real number $\alpha=[a_0;a_1,a_2,\ldots]$, let $B(\alpha)=\limsup_{n\to\infty}a_n(\alpha)$, where $a_n(\alpha)$ are its continued-fraction partial quotients. The...

conjecture | open | MathDB 页面 | 原文线索

V2-005808 | MATHDB-368694

Khintchine's theorem with a moving target

摘录 / EXCERPT: Khintchine's moving-target conjecture. If $psi:\mathbb{N}\tomathbb{R}_{geq 0}$ is decreasing and $sum_{q=1}^{infty}psi(q)=infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-005809 | MATHDB-368704

Conjecture on highly-composite Mersenne indices

摘录 / EXCERPT: Let N be an index of a highly-composite Mersenne number if $\tau(2^N-1)>\tau(2^m-1)$ for every integer m with $1\leqslant m<N$. Highly-composite Mersenne index conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-005810 | MATHDB-368732

The non-generic Galois group conjecture for random integer matrices

摘录 / EXCERPT: Let $M_n(T)$ be the number of integer $n\times n$ matrices A such that the characteristic polynomial χ_A of A does not have Galois group the full S_n. The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-005811 | MATHDB-368737

Langlands correspondence for reductive groups over finite fields

摘录 / EXCERPT: Let G be a reductive group over the finite field k, let $G^{\vee}$ be its dual group, and let ℓ be a prime distinct from the characteristic of k. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-005812 | MATHDB-368754

Effective non-vanishing conjecture for the first two moments without twist

摘录 / EXCERPT: Effective non-vanishing conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-005813 | MATHDB-368773

The Harris-Venkatesh conjecture for weight-one newforms

摘录 / EXCERPT: Let $f=\sum_n a_nq^n$ be a newform of weight 1 and level $\Gamma_1(N)$. Let ρ_f be its Deligne-Serre representation, let ${\mathrm{Ad}}(\rho_f)$ be the trace-free adjoint re...

conjecture | open | MathDB 页面 | 原文线索

V2-005814 | MATHDB-368787

Non-negligible probability of obtaining a prime ideal by random pre-processing

摘录 / EXCERPT: Non-negligible-prime-ideal conjecture. The proportion of such coefficient vectors is inverse-polynomial in d and $\log(B)$:

conjecture | open | MathDB 页面 | 原文线索

V2-005815 | MATHDB-368810

The frequency conjecture for Bézout coefficients of random integer polynomials

摘录 / EXCERPT: For positive integers m,n,H, let ... where $\mathrm{H}(h)$ is the maximum of the absolute values of the coefficients of h and $L(h)$ is its leading coefficient. For coprime integ...

conjecture | open | MathDB 页面 | 原文线索

V2-005816 | MATHDB-368811

The asymptotic inequality conjecture for Bézout coefficients and resultants

摘录 / EXCERPT: For positive integers m,n,H, let ... where $\mathrm{H}(h)$ is the maximum absolute value of the coefficients of h and $L(h)$ is its leading coefficient. For coprime integer polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-005817 | MATHDB-368815

The elementary-abelian p-group complexity conjecture

摘录 / EXCERPT: Let p be prime, let $G=(C_p)^n$, and let $\binom{n}{k}_p$ denote the Gaussian binomial coefficient, counting the subgroups of C_p^n isomorphic to C_p^k. Let $\mathfrak{c}(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-005818 | MATHDB-368817

The stable Carlsson conjecture for manifolds

摘录 / EXCERPT: Stable Carlsson conjecture. There exists a constant C such that, for every prime $p>C$,

conjecture | open | MathDB 页面 | 原文线索

V2-005819 | MATHDB-368831

The conjecture that bounded-above Lüroth sets with digits 3 and 4 cover modulo 1

摘录 / EXCERPT: Let $L_{\leq k}$ denote the Lüroth set whose digits are at most k, for $k\in\mathbb{N}\setminus\{1\}$. For subsets of $\mathbb{R}$, write $A+B=\{a+b:a\in A,b\in B\}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-005820 | MATHDB-368833

The conjecture that continued-fraction sets with matched bounds cover modulo 1

摘录 / EXCERPT: For $k\in\mathbb{N}$, let $F_{\leq k}$ and $F_{\geq k}$ denote the continued-fraction sets whose partial quotients are respectively bounded above by k and bounded below by k. W...

conjecture | open | MathDB 页面 | 原文线索

V2-005821 | MATHDB-368834

The conjecture that badly and well approximable numbers cover modulo 1 by addition

摘录 / EXCERPT: Let $F_{\leq k}$ be the continued-fraction set with partial quotients bounded above by k, and let $\bigcup_{k\in\mathbb{N}}F_{\leq k}$ be the set of numbers with bounded partial...

conjecture | open | MathDB 页面 | 原文线索

V2-005822 | MATHDB-368860

The Clergyman's conjecture for anti-recurrence sequences

摘录 / EXCERPT: An anti-recurrence sequence is the increasing sequence generated by a positive integral linear form, by selecting the integers represented by that form without repetition. A linear...

conjecture | open | MathDB 页面 | 原文线索

V2-005823 | MATHDB-368861

The Clergyman's conjecture for linear-form anti-recurrence sequences

摘录 / EXCERPT: Let $\mathbf a=(a_1,\dots,a_k)$ be a positive integral vector of dimension $k>1$, and let $f(\mathbf x)=\mathbf a\cdot\mathbf x$ be its linear form. Let A_n be the anti-recurrence...

conjecture | open | MathDB 页面 | 原文线索

V2-005824 | MATHDB-368934

A 7-adic recurrence congruence for generalized Frobenius partitions with parameter 6

摘录 / EXCERPT: The recurrence congruence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005825 | MATHDB-368950

Huneke-Lyubeznik finiteness conjecture for associated primes of local cohomology

摘录 / EXCERPT: Huneke-Lyubeznik conjecture. For every $i \ge 0$, the local cohomology module $H^i_I(R)$ has finitely many associated prime ideals.

conjecture | open | MathDB 页面 | 原文线索

V2-005826 | MATHDB-368951

Lyubeznik's finiteness conjecture for associated primes of Lyubeznik functors

摘录 / EXCERPT: Lyubeznik's conjecture. For every Lyubeznik functor $T(\,\cdot\,)$ on R, the module $T(R)$ has finitely many associated prime ideals.

conjecture | open | MathDB 页面 | 原文线索

V2-005827 | MATHDB-368986

Companion-zero conjecture for the smooth divisor-count indicator

摘录 / EXCERPT: For a prime p, a companion zero at p is a real zero $x\neq p$ of a smooth prime indicator with $|x-p|<1$. A family $x_p^{(\pm)}(\kappa)$ is a prime-zero approximant if...

conjecture | open | MathDB 页面 | 原文线索

V2-005828 | MATHDB-368987

Asymmetric companion-zero conjecture for the smooth divisor-sum indicator

摘录 / EXCERPT: For $x>0$ and steepness parameter $\kappa>0$, define the smooth divisor-sum indicator by ... For each odd prime $p\ge 3$, the asymmetric companion-zero conjecture. For sufficiently l...

conjecture | open | MathDB 页面 | 原文线索

V2-005829 | MATHDB-368990

Lower-bound conjecture for the Diophantine exponent of elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve defined over $\mathbb{Q}$, let $r=\operatorname{rank}(E(\mathbb{Q}))\geq1$, and let σ_E denote its Diophantine exponent. Elliptic-curve exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-005830 | MATHDB-368996

Infinite-families conjecture for sums of two cubes

摘录 / EXCERPT: Let $c_1\in\mathbb{Z}_{\geq 0}$, and let $A(c_1),B(c_1),C(c_1),D(c_1)$ be the integer-valued functions defined by the symbolic generator in Theorem 3.1. The generator produces solu...

conjecture | open | MathDB 页面 | 原文线索

V2-005831 | MATHDB-368997

Divisor-bound conjecture for the symbolic generator

摘录 / EXCERPT: Let $\Delta_n\in\mathbb{Z}_{>0}$, and let $r\in\mathbb{Z}_{>0}$ be a divisor of Δ_n. Consider the parametrized formulas for A,B,C,D constructed in Sections 2.1 and 2.2, an...

conjecture | open | MathDB 页面 | 原文线索

V2-005832 | MATHDB-369000

Prime tableaux from coarsest positroidal subdivisions

摘录 / EXCERPT: Let $T \in \operatorname{SSYT}(k,[n])$ be a tableau with no frozen factors. Write $T=S^{\cup r}$ with r maximal, where $S=\underline{T}$ has no further common frozen factor. A po...

conjecture | open | MathDB 页面 | 原文线索

V2-005833 | MATHDB-369024

The p-modular Hadamard conjecture

摘录 / EXCERPT: Let p be an odd prime, and let a p-modular Hadamard matrix of order d mean the matrix notion used in the paper. The p-modular Hadamard conjecture. For all but finitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-005834 | MATHDB-369025

Paley I automorphism conjecture for finite-field SICs

摘录 / EXCERPT: Let $q>11$ be a prime power with $q\equiv3\bmod4$, let $d=q+1$, and let H be the Paley I Hadamard matrix of order d. Write $\tilde H$ for the associated object and...

conjecture | open | MathDB 页面 | 原文线索

V2-005835 | MATHDB-369026

Division-line conjecture for Pythagoras numbers of integers in biquadratic fields

摘录 / EXCERPT: Division-line conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005836 | MATHDB-369033

Uniqueness of the degree greater than one morphism from $X_0(p)$

摘录 / EXCERPT: Let p be a prime. Write $X_0(p)^+$ for the quotient of the modular curve $X_0(p)$ by its usual involution. Uniqueness conjecture. Up to isomorphism, the quotient map ... is the onl...

conjecture | open | MathDB 页面 | 原文线索

V2-005837 | MATHDB-369046

Diamond-Sasaki geometric modularity conjecture

摘录 / EXCERPT: Let F be a totally real field in which p is unramified, let G_F be its absolute Galois group, and let Σ_v denote the embeddings associated with a place $v\mid p$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005838 | MATHDB-369058

Coates-Sujatha fine Selmer group conjecture

摘录 / EXCERPT: Let F be a number field, let p be an odd prime, and let F_cyc be the cyclotomic $\mathbb{Z}_p$-extension of F. For an elliptic curve E over F, write...

conjecture | open | MathDB 页面 | 原文线索

V2-005839 | MATHDB-369079

Fenn-Rourke-Sanderson torsion conjecture for dihedral quandle homology

摘录 / EXCERPT: Let p be an odd prime, and let R_p be the dihedral quandle of order p. Write $\operatorname{H}^{\rm R}_n(\operatorname{R}_p)$ for its rack homology. Fenn-Rourk...

conjecture | open | MathDB 页面 | 原文线索

V2-005840 | MATHDB-369094

Determinant lower-bound conjecture for GCD tensors

摘录 / EXCERPT: Let $S=\\{s_1,\ldots,s_n\\}$ be an ordered set of distinct positive integers. Let $\mathcal{T}_{[S]}$ denote the mth-order, n-dimensional GCD tensor on S, and let Φ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-005841 | MATHDB-369106

The prime-minus-one sum-product conjecture

摘录 / EXCERPT: Write $\mathbb{P}$ for the set of primes. Prime-minus-one sum-product conjecture. For every $a\in\mathbb{N}$ there exist $x,y\in\mathbb{N}$ with $x>y>a$ such that ... This asks for s...

conjecture | open | MathDB 页面 | 原文线索

V2-005842 | MATHDB-369114

Density conjecture for non-trivial isotrivial elliptic fibrations

摘录 / EXCERPT: Let k be a number field. A non-trivial isotrivial elliptic fibration is an elliptic fibration over $\mathbb{P}^1$ that is isotrivial but not trivial. Density conjecture. The k-...

conjecture | open | MathDB 页面 | 原文线索

V2-005843 | MATHDB-369137

The degree-monotone McKay bijection conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime number, and let P be a Sylow p-subgroup of G. Write $N_G(P)$ for the normalizer of P in G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005844 | MATHDB-369138

The degree-monotone height-zero block bijection conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let B be a p-block of G, and let b be its Brauer correspondent. Write $\operatorname{Irr}_0(B)$ for the set of height-zero ir...

conjecture | open | MathDB 页面 | 原文线索

V2-005845 | MATHDB-369144

Realization of all genus 2 reduction types over the rationals

摘录 / EXCERPT: Realization conjecture. All 54 reduction types of the genus 2 classification are realised over $\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-005846 | MATHDB-369157

The dimension conjecture for T-prime ideals

摘录 / EXCERPT: Dimension conjecture. Every T-prime ideal admits a dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-005847 | MATHDB-369213

Weighted metric theorem for approximation by uniformly distributed sequences

摘录 / EXCERPT: Let $n\in\mathbb N$, let $\omega=(\boldsymbol{\omega}_j)_{j\geq 1}$ be a sequence in $[0,1]^n$, and let $\Psi=(\psi_1,\ldots,\psi_n)$ be an n-tuple of non-increasing functions. W...

conjecture | open | MathDB 页面 | 原文线索

V2-005848 | MATHDB-369227

Montgomery's large value conjecture for Dirichlet polynomials

摘录 / EXCERPT: Montgomery's large value conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-005849 | MATHDB-369229

Block orthogonality conjecture for principal blocks of \mathbb{Z}_\ell-spetses

摘录 / EXCERPT: Let $\mathbb{G}=(W,\Lambda)$ be a simply connected $\mathbb{Z}_\ell$-spets such that $\ell>2$ is very good for $\mathbb{G}$ and coprime with $|W|$. Let $q\in1+\ell\mathbb{Z}_\ell$,...

conjecture | open | MathDB 页面 | 原文线索

V2-005850 | MATHDB-369237

Associated-prime conjecture for second powers of Kneser graph neighborhood ideals

摘录 / EXCERPT: Let $G=K(n,k)$ be the Kneser graph of k-element subsets of an n-element set, and let $NI(G)$ denote its closed neighborhood ideal in the corresponding polynomial ring with homo...

conjecture | open | MathDB 页面 | 原文线索

V2-005851 | MATHDB-369241

Finiteness of multiply gleeful numbers for every exponent at least 3

摘录 / EXCERPT: Let $f_k(n)$ denote the number of gleeful representations of the integer n with parameter k. The universal finiteness conjecture. For every integer $k\ge 3$, there are finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-005852 | MATHDB-369242

Infinitely many simultaneous duplicates for exponents 2 and 3 or 4

摘录 / EXCERPT: For integers $k<k'$ let $f_k(n)$ and $f_{k'}(n)$ count the corresponding gleeful representations of n. The simultaneous-duplicate conjecture. If $k'=3$ or 4, then there are inf...

conjecture | open | MathDB 页面 | 原文线索

V2-005853 | MATHDB-369243

Finiteness of simultaneous duplicates for separated exponents

摘录 / EXCERPT: Let $k<k'$ be positive integer parameters, and let $f_k(n)$ count the gleeful representations of n with parameter k. The separated-exponent finiteness conjecture. If $k\ge 3$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-005854 | MATHDB-369249

Asymptotic Mahler measure conjecture for Gaussian periods

摘录 / EXCERPT: Let α_n be the Gaussian-period algebraic integer defined in the paper, and let $\mathrm{m}(\alpha_n)$ denote its Mahler measure. For a property of odd integers n, say that...

conjecture | open | MathDB 页面 | 原文线索

V2-005855 | MATHDB-369251

Specific case of the prime tuplets conjecture

摘录 / EXCERPT: Specific case of prime tuplets conjecture. The number of integers q with $x\leqslant q<2x$ for which every qr_j+1 is prime satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-005856 | MATHDB-369269

Finiteness conjecture for equal radicals of consecutive products

摘录 / EXCERPT: Let k and ℓ be integers greater than 1, and let m and n be integers. Define ... where $\operatorname{rad}(r)$ is the largest squarefree divisor of r. Finiteness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-005857 | MATHDB-369315

The Igusa-function formula conjecture for the middle symplectic local zeta function

摘录 / EXCERPT: For a positive integer n, let q be the residue-field cardinality and let s be a complex variable. Let $\mathcal{Z}_{n,n,\mathfrak{o}}^{\mathsf{C}}(s)$ denote the symplectic l...

conjecture | open | MathDB 页面 | 原文线索

V2-005858 | MATHDB-369324

Asymptotic-density-zero conjecture for dimensions of weight-2 newspaces

摘录 / EXCERPT: Asymptotic-density-zero conjecture. The set A has asymptotic density 0 in the natural numbers. The corresponding density statement is known for the dimensions...

conjecture | open | MathDB 页面 | 原文线索

V2-005859 | MATHDB-369327

The discriminant conjecture for the field generated by a root of $f_{r,p}$

摘录 / EXCERPT: Let p be the odd prime and let θ be the algebraic number considered in the preceding discriminant computation, with $\mathbb{Q}(\theta)$ its generated number field. Discri...

conjecture | open | MathDB 页面 | 原文线索

V2-005860 | MATHDB-369360

Infinitude conjecture for square Delta-primitive numbers

摘录 / EXCERPT: ... -primitive numbers. The preceding argument establishes that there are infinitely many ... -primitive is used in the source for the relevant class of integers.

conjecture | open | MathDB 页面 | 原文线索

V2-005861 | MATHDB-369361

The conjecture that no cyclic codes have both constant relative distance and dual distance

摘录 / EXCERPT: Let q be a fixed prime power. An infinite family of cyclic codes over $\\F_q$ has constant relative distance if its relative distance is bounded below by a positive constant, and...

conjecture | open | MathDB 页面 | 原文线索

V2-005862 | MATHDB-369362

The conjecture that asymptotically good cyclic codes do not exist

摘录 / EXCERPT: Let q be a fixed prime power. A family of cyclic codes over $\\F_q$ is asymptotically good if it has positive limiting rate and positive limiting relative distance. Cyclic-code a...

conjecture | open | MathDB 页面 | 原文线索

V2-005863 | MATHDB-369383

The range conjecture for supra-unitary rational parameters

摘录 / EXCERPT: For $b\in\mathbb{N}\setminus\{1\}$, let $a\in\{1,\ldots,b-1\}$ be relatively prime to b, and let $s\in\mathbb{N}$. For the function f_α appearing in the paper, the range c...

conjecture | open | MathDB 页面 | 原文线索

V2-005864 | MATHDB-369384

The positive irrational range conjecture

摘录 / EXCERPT: Let $\alpha>0$ be irrational, and let f_α be the function defined in the paper. The positive irrational range conjecture. ... Together with the paper's established upper bound...

conjecture | open | MathDB 页面 | 原文线索

V2-005865 | MATHDB-369385

The unit-interval range conjecture for irrational parameters

摘录 / EXCERPT: Let $\alpha\in(0,1)$ be irrational, and let f_α be the function defined in the paper. The unit-interval range conjecture. ... This is presented as the assertion that every irr...

conjecture | open | MathDB 页面 | 原文线索

V2-005866 | MATHDB-369386

The golden-ratio characterization by constant range

摘录 / EXCERPT: Let $\alpha>0$ be irrational, let f_α be the function defined in the paper, and let φ denote the golden ratio. The golden-ratio range conjecture. ... The authors state...

conjecture | open | MathDB 页面 | 原文线索

V2-005867 | MATHDB-369400

The Beal conjecture for generalized Fermat equations

摘录 / EXCERPT: Consider the generalized Fermat equation ... A solution $(a,b,c)\in\mathbb{Z}^3$ is primitive if $\operatorname{gcd}(a,b,c)=1$ and trivial if $abc=0$. The Beal Prize conjecture. If...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005868 | MATHDB-369401

The Fontaine-Mazur modularity conjecture over totally real fields

摘录 / EXCERPT: Let K be a totally real field and let $p\geq3$ be prime. A continuous representation ... is geometric if it satisfies the geometricity conditions of Fontaine and Mazur. The Fontain...

conjecture | open | MathDB 页面 | 原文线索

V2-005869 | MATHDB-369413

Prime-divisor conjecture for the double-discriminant constant

摘录 / EXCERPT: Double-discriminant constant conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-005870 | MATHDB-369439

Maeda's conjecture for normalized Hecke eigenforms

摘录 / EXCERPT: Maeda's conjecture. The Galois conjugates satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-005871 | MATHDB-369440

Conjecture on linear independence of twisted periods with different indices

摘录 / EXCERPT: Let K be the weight, S_K the space of cusp forms under consideration, and $r_{\ell,\chi}$ the twisted period indexed by ℓ and a primitive Dirichlet character χ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005872 | MATHDB-369459

The EMCHS heuristic prime gap bound

摘录 / EXCERPT: Let p_n denote the nth prime, and let δ and ϵ be parameters in the ranges considered by the Enhanced Multidimensional Chaotic Heuristic Sieve (EMCHS), including...

conjecture | open | MathDB 页面 | 原文线索

V2-005873 | MATHDB-369460

The bounded-chaos conjecture for a rigorous EMCHS outcome

摘录 / EXCERPT: Let χ be the chaotic map used in the EMCHS model, let $y_0\in[0,1]$, and let $\rho(y)$ denote the proposed invariant density. For a suitable class of functions g, bounded-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-005874 | MATHDB-369471

Prime smoothness and prescribed odd-prime valuations

摘录 / EXCERPT: Let $n>0$ be an integer and let $x\in\{0,1\}^n$ be a vector of bits of length n. For each $1\leq k\leq n$, let q_k denote the kth odd prime. Prime smoothness and prescribed v...

conjecture | open | MathDB 页面 | 原文线索

V2-005875 | MATHDB-369477

The Jiang-Rallis integral point-count conjecture

摘录 / EXCERPT: The Jiang-Rallis integral point-count conjecture. The number of such triples is p^2-1. This conjecture gives the remaining finite-field count needed to evaluate one of the sub-in...

conjecture | open | MathDB 页面 | 原文线索

V2-005876 | MATHDB-369495

Eisenbud-Goto conjecture for homogeneous prime ideals

摘录 / EXCERPT: Let K be an algebraically closed field, let $R=K[x_1,\dots,x_N]$, and let $P\subset (x_1,\ldots,x_N)^2$ be a homogeneous prime ideal. Write $\operatorname{reg}(P)$ for its Castel...

conjecture | open | MathDB 页面 | 原文线索

V2-005877 | MATHDB-369532

Beresnevich-Yang conjecture for simultaneous approximation on nondegenerate curves

摘录 / EXCERPT: Let $\mathbf{M}_{n,1}$ be the set of nondegenerate curves in $\mathbb{R}^n$; that is, curves $\mathcal{C}=\{\mathbf{f}(x): x\in I,\ \mathbf{f}:I\to \mathbb{R}^n\}$ such that they a...

conjecture | open | MathDB 页面 | 原文线索

V2-005878 | MATHDB-369534

Nonabelian Cohen-Lenstra distribution conjecture for pro-prime-to-roots-of-unity class groups

摘录 / EXCERPT: Let $Q={\mathbb Q}$ or $\mathbb F_q(t)$, and let μ_Q be the group of roots of unity contained in Q. Let Γ be the finite group and Γ_∞ the cyclic subgroup g...

conjecture | open | MathDB 页面 | 原文线索

V2-005879 | MATHDB-369543

Conjecture on asymptotic decorrelation of quadratic forms of Hecke eigenforms

摘录 / EXCERPT: Let f_1,f_2,f_3,f_4 be L^2-normalized Hecke eigenforms of even weights k_1,k_2,k_3,k_4, so that ... Assume that $k_1+k_2=k_3+k_4$. Quadratic-form decorrelation conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-005880 | MATHDB-369544

The inert quadratic order conjecture for finitized Coh zeta functions

摘录 / EXCERPT: The inert quadratic order conjecture. For each $m\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-005881 | MATHDB-369554

The non-CM Fourier coefficient conjecture for odd discriminant level

摘录 / EXCERPT: Let k be an integer with $k\geq 3$, and let $D=p_1\cdots p_\ell$ be an odd discriminant. Let $S_k^{\mathrm{Ell}}(\Gamma_1(D),\omega_{E/\mathbb{Q}})$ denote the space of elliptic...

conjecture | open | MathDB 页面 | 原文线索

V2-005882 | MATHDB-369557

Galois refinement of the nilpotent block conjecture

摘录 / EXCERPT: Let p be a prime and let B be a p-block of a finite group. An irreducible character in B is almost p-rational if, writing c for its conductor, p^2 does not divide c...

conjecture | open | MathDB 页面 | 原文线索

V2-005883 | MATHDB-369558

Boston's branch-group conjecture for just-infinite pro-p groups

摘录 / EXCERPT: Let p be a prime, let T_p be the rooted p-ary tree, and let W_p be its full automorphism group. A pro-p group is just-infinite if all its non-trivial closed normal subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-005884 | MATHDB-369560

Carlsson's toral rank conjecture

摘录 / EXCERPT: Let p be a prime, let $G=({\Bbb Z}/p)^k$, and let X be a finite free G-complex. Carlsson's toral rank conjecture. One has ... This conjecture gives a lower bound on the total m...

conjecture | open | MathDB 页面 | 原文线索

V2-005885 | MATHDB-369612

Classification conjecture for quadratic permutation systems over finite fields of odd characteristic

摘录 / EXCERPT: Let q be an odd prime power and let $n\geq 3$. A quadratic system is a map ... where each $f_i\in\mathbb{F}_q[x_1,\dots,x_n]$ has degree at most 2. Classification conjecture. Suc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005886 | MATHDB-369623

Main conjecture on limited prime power sums

摘录 / EXCERPT: Let n be a natural number greater than 23. A prime power is a number P^k where P is prime and $k\geq 2$ is an integer. Main conjecture on limited prime power sums. Every na...

conjecture | open | MathDB 页面 | 原文线索

V2-005887 | MATHDB-369626

Markov Uniqueness conjecture for Markov triples

摘录 / EXCERPT: Markov Uniqueness conjecture. If $\max(a,b,c)=\max(a',b',c')$, then there exists a permutation $\sigma\in\mathfrak{S}_3$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-005888 | MATHDB-369636

Buzzard's slope conjecture for modular forms

摘录 / EXCERPT: Let $p>3$ and N be as above, and suppose that p is $\Gamma_0(N)$-regular, meaning that the conditions on the eigenvalues of T_p in weights up to k_p hold as specified above...

conjecture | open | MathDB 页面 | 原文线索

V2-005889 | MATHDB-369637

Bateman-Horn conjecture for admissible irreducible integer polynomials

摘录 / EXCERPT: Let $f_1,\ldots,f_k$ be irreducible integer polynomials satisfying the admissibility condition. Let $P(x)$ denote the number of integers $n\leq x$ for which all $f_i(n)$ are prime....

conjecture | open | MathDB 页面 | 原文线索

V2-005890 | MATHDB-369642

Prime characterization conjecture for the reduced Dressian of the cyclic matroid

摘录 / EXCERPT: Let $n\geqslant 7$ be an integer, and let ${\mathcal T}_n$ denote the cyclic matroid used in the paper. Write $\operatorname{\underline{Dr}}_{{\mathcal T}_n}$ for its reduced Dress...

conjecture | open | MathDB 页面 | 原文线索

V2-005891 | MATHDB-369651

Gras's p-rationality conjecture for number fields

摘录 / EXCERPT: Gras's conjecture. For $p\gg 0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-005892 | MATHDB-369678

Hecke realization conjecture for motives of cusp forms

摘录 / EXCERPT: Let $M^n_{\mathrm{cusp}}=M(\overline{\mathcal{M}}_{1,n})_{\varepsilon}[n]$ be the motive associated to the space of cusp forms for $\Gamma=\operatorname{SL}_2(\mathbb{Z})$ of weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-005893 | MATHDB-369684

Universal-enveloping-algebra conjecture for the balanced multiple q-zeta Lie algebra

摘录 / EXCERPT: Let $\mathfrak{bm}_0$ be the space of all $\Phi\in\mathbb{Q}\langle\mathcal{B}\rangle$ satisfying $(\Phi\mid b_0)=0$, $(\Phi\mid b_k)=0$ for $k=2,4,6$, primitiveness for the balanc...

conjecture | open | MathDB 页面 | 原文线索

V2-005894 | MATHDB-369688

Finiteness conjecture for missed square and half-integer areas of primitive triangles

摘录 / EXCERPT: Finiteness conjecture. The sets

conjecture | open | MathDB 页面 | 原文线索

V2-005895 | MATHDB-369701

Greenberg's Iwasawa main conjecture for CM fields

摘录 / EXCERPT: Let ${\mathcal{K}}$ be a CM field with maximal totally real subfield ${\mathcal{F}}$, let p be an odd prime unramified in ${\mathcal{F}}$ such that every prime above p splits i...

conjecture | open | MathDB 页面 | 原文线索

V2-005896 | MATHDB-369709

Dickson-Hardy-Littlewood prime tuples conjecture for two linear forms

摘录 / EXCERPT: Let $W\geq 1$ be fixed, and let b_1,b_2 be fixed distinct integers coprime to W. For each prime p, let $\nu_p(W\cdot+b_1,W\cdot+b_2)$ be the number of distinct residue classe...

conjecture | open | MathDB 页面 | 原文线索

V2-005897 | MATHDB-369711

Aryan-Madhavani-Parikh-Sengupta-Zhang finiteness conjecture for 2-near perfect numbers

摘录 / EXCERPT: Let $n=2^k p^m$, where p is an odd prime, and suppose that n is 2-near perfect when ... for distinct positive divisors d_1,d_2 of n. Finiteness conjecture. There are only fin...

conjecture | open | MathDB 页面 | 原文线索

V2-005898 | MATHDB-369731

Four-polynomial generation conjecture for the natural numbers

摘录 / EXCERPT: Let p_1,p_2,p_3,p_4 be the four polynomials constructed in the paper, evaluated at $x,y,z\in\mathbb{N}^*$. The relevant generated parameter is a natural number $q\geq1$. Four-pol...

conjecture | open | MathDB 页面 | 原文线索

V2-005899 | MATHDB-369733

The weak AACM conjecture for fundamental units of real quadratic fields

摘录 / EXCERPT: Suppose $d>0$ is a square-free integer and write $d=pm$ for an odd prime p and integer m. If $d\equiv 1\left(\bmod{\,4}\right)$, set $\delta=1$; otherwise let $\delta=2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005900 | MATHDB-369752

Local association of index-prime orders in real quadratic fields

摘录 / EXCERPT: Let $p\in\mathbb N$ be prime, let $K=\mathbb Q[\sqrt{p}]$, and let R_p be the order of index p in K. An order is locally associated when it has the local association property...

conjecture | open | MathDB 页面 | 原文线索

V2-005901 | MATHDB-369766

Kato's main conjecture without the p-adic zeta function

摘录 / EXCERPT: Let f be the modular form under consideration, let $G_ fty$ be the relevant Galois group, and let $\mathcal{O}$ be the coefficient ring. For $\natural\in\{\sharp,\flat\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005902 | MATHDB-369773

Conjecture for moments of derivatives of even orthogonal Hecke L-functions

摘录 / EXCERPT: Let q be prime, let $H_2^*(q)$ be the set of primitive Hecke eigenforms of weight 2 for $\Gamma_0(q)$, and let $L_f(s)=\sum_{n=1}^{\infty}\lambda_f(n)n^{-s}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-005903 | MATHDB-369813

Corrected rank conjecture for twin prime elliptic curves

摘录 / EXCERPT: Let $p\geq7$ and let $(p,p-2)$ be a twin-prime pair. Define the twin-prime elliptic curve by ... Write $r(E_p)$ for its Mordell-Weil rank. Corrected twin-prime rank conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-005904 | MATHDB-369814

Modularity conjecture for the listed lift-dual rank-four Nahm sums

摘录 / EXCERPT: For each triple (A,B,C) listed in Tables 1-4 and Table 5 (the table labels are those used in the source), let $\mathcal{DL}_i(A,B,C)$ be its lift-dual triple and let the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-005905 | MATHDB-369818

The Polycirculant Conjecture for elusive permutation groups

摘录 / EXCERPT: Let G be a finite transitive permutation group. Call G elusive if it contains no derangements of prime order. The 2-closure of G on its permutation domain is the largest su...

conjecture | open | MathDB 页面 | 原文线索

V2-005906 | MATHDB-369853

Stevenhagen's conjecture on the negative Pell equation

摘录 / EXCERPT: Let d' range over square-free positive integers, and let $\mathcal F$ be the set of square-free positive integers all of whose odd prime divisors are congruent to $1\pmod 4$. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-005907 | MATHDB-369898

Etingof's generalized Benson conjecture for semisimplified representation categories

摘录 / EXCERPT: Generalized Benson conjecture. Every prime dividing the order of G' is less than p.

conjecture | open | MathDB 页面 | 原文线索

V2-005908 | MATHDB-369899

Etingof's generalized Benson conjecture for induced prime-to-\p symmetries

摘录 / EXCERPT: Generalized Benson conjecture. If $(\mathcal{C},G')$ is an induced p'-symmetry of V, then every prime dividing the order of G' is less than p.

conjecture | open | MathDB 页面 | 原文线索

V2-005909 | MATHDB-369908

Hung-Tiep's conductor-degree conjecture for character fields

摘录 / EXCERPT: Let G be a finite group and let χ be an irreducible complex character of G. Let $\mathbb{Q}(\chi)$ be the field generated by the values of χ, and let $c(\chi)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-005910 | MATHDB-369909

The cyclotomic-integer conductor-length conjecture

摘录 / EXCERPT: Let α be a cyclotomic integer, meaning an algebraic integer expressible as a sum of roots of unity. Let $l(\alpha)$ be the smallest number of roots of unity occurring in any...

conjecture | open | MathDB 页面 | 原文线索

V2-005911 | MATHDB-369923

Skolem's local-global conjecture for purely exponential Diophantine equations

摘录 / EXCERPT: Consider the purely exponential Diophantine equation ... where the variables x_{ij} are positive integers. Skolem's conjecture. If this equation has no solution, then there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-005912 | MATHDB-369932

Compatibility conjecture for Mazur-Rubin local constants and epsilon constants

摘录 / EXCERPT: Let p be an odd prime, let L be a finite extension of $\mathbb Q_p$, and let T_1 and T_2 be symplectic self-dual $\mathcal O_L$-representations of $G_{\mathbb Q_p}$. Put .....

conjecture | open | MathDB 页面 | 原文线索

V2-005913 | MATHDB-369938

Graham's valid-ordering conjecture over finite prime fields

摘录 / EXCERPT: Let p be prime. A valid ordering of a subset $S\subseteq\mathbb{F}_p\setminus\{0\}$ is an ordering $g_1,g_2,\ldots,g_n$ of its elements such that the partial sums ... are all disti...

conjecture | open | MathDB 页面 | 原文线索

V2-005914 | MATHDB-369940

The shifted-prime largest-prime-factor extremes conjecture

摘录 / EXCERPT: Let a be the fixed shift appearing in the statement, let $P^{+}(m)$ denote the largest prime factor of m, and let $\boldsymbol{\vartheta}$ and $\boldsymbol{\Theta}$ be real num...

conjecture | open | MathDB 页面 | 原文线索

V2-005915 | MATHDB-369948

Bukh's corrected conjecture on sums of linear transformations

摘录 / EXCERPT: Let $\mathcal{L}_0,\ldots,\mathcal{L}_k\in\operatorname{Mat}_d(\mathbb{Z})$ be irreducible and coprime matrices, and let A be a finite subset of $\mathbb{Z}^d$. Bukh's corrected...

conjecture | open | MathDB 页面 | 原文线索

V2-005916 | MATHDB-369949

Effective mass equidistribution conjecture for holomorphic Hecke forms

摘录 / EXCERPT: Let $\Gamma\subseteq\operatorname{SL}_2(\operatorname{\mathbb{Z}})$ be a Fuchsian group of the first kind, let $f\in S_k(\Gamma)$ be a normalized Hecke cusp form, and let μ_f...

conjecture | open | MathDB 页面 | 原文线索

V2-005917 | MATHDB-369950

Nelson's quadratic Hecke-sum conjecture

摘录 / EXCERPT: Let $f\in H_k(R(1))$, let $\psi\in C_0^\infty(0,\infty)$, and let $q\in\operatorname{\mathbb{Q}}[x]$ be an irreducible quadratic integer-valued polynomial. Nelson's quadratic Hecke...

conjecture | open | MathDB 页面 | 原文线索

V2-005918 | MATHDB-369987

The CM Iwasawa main conjecture for Katz's p-adic L-function

摘录 / EXCERPT: Let K/F be a p-ordinary CM quadratic extension for an odd prime p, and let h_K^- be its relative class number. Fix embeddings...

conjecture | open | MathDB 页面 | 原文线索

V2-005919 | MATHDB-370002

Long Game Strategy conjecture for the Ordered Zeckendorf Game

摘录 / EXCERPT: Let n be the initial number of copies of F_1 in the Ordered Zeckendorf game, and let the Long Game Strategy (LGS) be the priority-based strategy that performs switch moves, com...

conjecture | open | MathDB 页面 | 原文线索

V2-005920 | MATHDB-370005

Lehrer's conjecture on character-degree polynomials of unitriangular groups

摘录 / EXCERPT: Let q be a power of an odd prime, let $U_n(q)$ be the unitriangular group of degree n over the finite field ${\mathbb F}_q$, and let $e\in {\mathbb N}$. Lehrer's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-005921 | MATHDB-370006

Isaacs's strengthened conjecture on unitriangular-group character counts

摘录 / EXCERPT: Let q be a power of an odd prime, let $U_n(q)$ be the unitriangular group of degree n over the finite field ${\mathbb F}_q$, and let $N_{n,e}(q)$ denote the number of irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-005922 | MATHDB-370026

Artin's primitive root conjecture

摘录 / EXCERPT: Artin's primitive root conjecture. The set of a-rooted primes is infinite.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005923 | MATHDB-370037

The semiprimeness conjecture for bounded skew power series rings

摘录 / EXCERPT: Semiprimeness conjecture. The ring $Q^+[[x;sigma,delta]]$ is (semi)prime.

conjecture | open | MathDB 页面 | 原文线索

V2-005924 | MATHDB-370042

Strong approximation conjecture for Markoff surfaces

摘录 / EXCERPT: Let p be a prime and let $\mathbf{M}_k$ be the Markoff surface ... Let $\mathcal{E}_k$ denote the union of the exceptional orbits in $\mathbf{M}_k(\mathbb{F}_p)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-005925 | MATHDB-370043

Divisibility conjecture for generic Markoff-surface orbits

摘录 / EXCERPT: Let p be a prime, and let ... be a Markoff surface over $\mathbb{F}_p$. Call a level k generic when it contains no orbits associated with A_4, S_4, or A_5. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005926 | MATHDB-370044

Universality of lower-order density terms for composite levels

摘录 / EXCERPT: Lower-order universality conjecture. As long as the prime factors of N tend to infinity sufficiently fast relative to the reciprocal of the error scale being considered, the lowe...

conjecture | open | MathDB 页面 | 原文线索

V2-005927 | MATHDB-370054

Optimality conjecture for the Mayer-Erdős phenomenon on similarly ordered Farey fractions

摘录 / EXCERPT: Let $\frac{a_1}{b_1},\frac{a_2}{b_2},\ldots$ be the Farey sequence of order n, and let $f(n)$ denote the least positive integer r such that some pair of fractions r positions...

conjecture | open | MathDB 页面 | 原文线索

V2-005928 | MATHDB-370137

Gouvêa's generalization of Mazur's dimension conjecture

摘录 / EXCERPT: Let p be a prime, K a number field, S a finite set of primes of K, and $\overline{\rho}:G_{K,S}\to \operatorname{GL}_{n}(\mathbb{F})$ a continuous absolutely irreducible re...

conjecture | open | MathDB 页面 | 原文线索

V2-005929 | MATHDB-370161

Minimalist conjecture for central vanishing of level-one cusp forms

摘录 / EXCERPT: Minimalist central-vanishing conjecture. If $k\equiv2\, (\mathrm{mod}\ 4)$, then $m(f)=1$. When $k\equiv2\, (\mathrm{mod}\ 4)$, the functional equation forces central vanis...

conjecture | open | MathDB 页面 | 原文线索

V2-005930 | MATHDB-370162

Uniform lower bound for the L^α norms of unimodular polynomials

摘录 / EXCERPT: Let $\mathbf{c}$ be an unimodular sequence and let $(P_{\mathbf{c},n})$ be its associated sequence of analytic polynomials. For $\alpha>2$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-005931 | MATHDB-370163

Hajela-Smith conjecture for exponential sums of the Liouville function

摘录 / EXCERPT: Let $\boldsymbol{\lambda}(n)$ be the Liouville function, defined by $\boldsymbol{\lambda}(1)=1$ and $\boldsymbol{\lambda}(n)=(-1)^r$ when r is the number of prime factors of n counted with multiplicit...

conjecture | open | MathDB 页面 | 原文线索

V2-005932 | MATHDB-370164

$\boldsymbol{\lambda}$-L^p conjecture for Liouville exponential sums

摘录 / EXCERPT: Let $\boldsymbol{\lambda}(n)$ be the Liouville function, and for $N\in\mathbb{N}$ define ... For a fixed exponent p, let $\|L_N\|_p$ denote its L^p norm on the unit circle....

conjecture | open | MathDB 页面 | 原文线索

V2-005933 | MATHDB-370165

The shifted Lonely Runner Conjecture

摘录 / EXCERPT: Given pairwise distinct positive velocities $v_1,\dotsc,v_n \in \mathbb{R}$ and arbitrary starting points $s_1,\dotsc,s_n \in \mathbb{R}$, the distance from s_j+tv_j to the neare...

conjecture | open | MathDB 页面 | 原文线索

V2-005934 | MATHDB-370166

Frey-Mazur conjecture

摘录 / EXCERPT: Frey-Mazur conjecture. If $E[p]$ and $E'[p]$ are isomorphic as Galois modules for some prime $p>17$, then E and E' are $\mathbb{Q}$-isogenous.

conjecture | open | MathDB 页面 | 原文线索

V2-005935 | MATHDB-370170

Tao's shifted convolution conjecture for generalised divisor functions

摘录 / EXCERPT: Let $d_k(n)$ denote the number of ordered factorizations of n into k positive integers, and let $d_l(n)$ be defined similarly. Let $\boldsymbol{\epsilon>0}$ and $k,l\ge 2$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-005936 | MATHDB-370171

The triple convolution conjecture for generalised divisor functions

摘录 / EXCERPT: For integers $k,l,m\ge 2$, define ... For $\epsilon>0$ and $0<h\le x^{1-\epsilon}$, the triple convolution conjecture. ... as $x\to\infty$, where $\nabla_{h,k,l,m}$ is the explicit Eul...

conjecture | open | MathDB 页面 | 原文线索

V2-005937 | MATHDB-370183

Littlewood's conjecture

摘录 / EXCERPT: Let (α,β) be a pair of real numbers, and let $\epsilon>0$. The system ... uses solutions $(p,r,q)\in\mathbb{Z}\times\mathbb{Z}\times\mathbb{N}$ for an unbounded set of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-005938 | MATHDB-370184

Uniform Littlewood conjecture

摘录 / EXCERPT: Let (α,β) be a pair of real numbers, and let $\epsilon>0$. Consider the system ... A pair is called multiplicatively singular if, for every $\epsilon>0$, this system has a...

conjecture | open | MathDB 页面 | 原文线索

V2-005939 | MATHDB-370196

Vojta's weak conjecture for integral points

摘录 / EXCERPT: Vojta's weak conjecture. For every $varepsilon>0$, there is a constant C_ε such that

conjecture | open | MathDB 页面 | 原文线索

V2-005940 | MATHDB-370197

Eventual 4-periodicity conjecture for C_p-Mackey functors

摘录 / EXCERPT: Let p be prime and let M be a C_p-Mackey functor. A projective resolution of M is eventually 4-periodic if its differentials repeat with period 4 from some point onward...

conjecture | open | MathDB 页面 | 原文线索

V2-005941 | MATHDB-370215

Kurepa's left-factorial conjecture

摘录 / EXCERPT: For each integer $n\geq 2$, let the left factorial $\mathbf{!n}$ be the sum of the factorials below n, and let $n!$ denote the usual factorial. Kurepa's conjecture. The two facto...

conjecture | open | MathDB 页面 | 原文线索

V2-005942 | MATHDB-370216

Bounded-gcd conjecture for shifted Kurepa factorial sequences

摘录 / EXCERPT: Bounded-gcd conjecture. For every Kurepa factorial sequence, $\mathcal{F}_n(a)$ is bounded above by 2, with

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-005943 | MATHDB-370218

Heim-Neuhauser's Hurwitz conjecture for D'Arcais polynomials

摘录 / EXCERPT: For $n\geq 1$, let $P_n^{\sigma}(x)$ be the D'Arcais polynomial defined by ... A polynomial is Hurwitz if the real parts of all its zeros are negative. Heim-Neuhauser's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-005944 | MATHDB-370237

Tightness conjecture for the torsion of random hypertrees

摘录 / EXCERPT: Tightness conjecture. For every prime p, $\Gamma_{n,p}$ is tight:

conjecture | open | MathDB 页面 | 原文线索

V2-005945 | MATHDB-370241

The Taelman-Leopoldt conjecture for Drinfeld modules

摘录 / EXCERPT: Let K be a global Q-field, let E be a Drinfeld A-module over $\mathcal{O}_K$, and let $\mathfrak{p}$ be a prime of A. Write $S_{\mathfrak{p}}$ for the set of places of...

conjecture | open | MathDB 页面 | 原文线索

V2-005946 | MATHDB-370260

Hughes-Pearce-Crump complex discrete moment conjecture for zeta'

摘录 / EXCERPT: Hughes-Pearce-Crump conjecture. For $\Re(k)>-3$,

conjecture | open | MathDB 页面 | 原文线索

V2-005947 | MATHDB-370261

Hughes-Keating-O'Connell conjecture for discrete moments of zeta'

摘录 / EXCERPT: Hughes-Keating-O'Connell conjecture. For any fixed k with $\Re(k)>-3/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-005948 | MATHDB-370273

Conjecture on exceptional pairs for representable prime squares

摘录 / EXCERPT: Let a_1 and a_2 be integers with $a_2>a_1>40$ and $\gcd(a_1,a_2)=1$. Let π_{2,a_1,a_2} denote the number of prime squares $p^2\le g_{a_1,a_2}$ representable as...

conjecture | open | MathDB 页面 | 原文线索

V2-005949 | MATHDB-370274

Conjecture on the growth of the prime-power threshold

摘录 / EXCERPT: Let $g(k)$ be the least positive integer such that for every pair of integers satisfying $g(k)<a_1<a_2$, there is a prime power $p^k\le g_{a_1,a_2}$ of the form $p^k=a_1x+a_2y$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-005950 | MATHDB-370288

Dasappa et al.'s congruence conjecture for the restricted partition function K

摘录 / EXCERPT: Let $K(n)$ be defined by its generating function ... where $f_k=(q^k;q^k)_\infty$ and $(a;q)_\infty=\prod_{i\geq 0}(1-aq^i)$. Dasappa et al.'s conjecture. For all $n\geq 0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-005951 | MATHDB-370289

The integral index-ratio conjecture

摘录 / EXCERPT: Let n) be a positive integer with divisors $d_1<d_2<\cdots<d_{\tau(n)}$, and let k be its index ratio, meaning that the sum of the even-indexed divisors equals k times the su...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005952 | MATHDB-370290

The prime index-ratio divisor-pair conjecture

摘录 / EXCERPT: Let p be a prime number, and let n be a p-index ratio number, with divisors $d_1<d_2<\cdots<d_{\tau(n)}$. Prime index-ratio divisor-pair conjecture. Then ... Theorem 9 proves t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005953 | MATHDB-370291

The uniqueness conjecture for subunit index-ratio classes

摘录 / EXCERPT: Let G_k denote the set of positive integers whose index ratio is k. Uniqueness conjecture for subunit index-ratio classes. If $k<1$, then G_k is either empty or has only one...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-005954 | MATHDB-370301

Leading-term conjecture for volumes of (1,k)-annuli

摘录 / EXCERPT: Let $V_{\mathbb{A}_{1,k}}$ denote the volume of the moduli space of $(1,k)$-annuli, viewed as a polynomial in π^2. Here k is a positive integer. Leading-term conjecture for a...

conjecture | open | MathDB 页面 | 原文线索

V2-005955 | MATHDB-370302

Sixth-last-term conjecture for volumes of (1,k)-annuli

摘录 / EXCERPT: Let $V_{\mathbb{A}_{1,k}}$ denote the volume of the moduli space of $(1,k)$-annuli, with $k\geqslant 12$. Sixth-last-term conjecture for annular volumes. The sixth-last term of...

conjecture | open | MathDB 页面 | 原文线索

V2-005956 | MATHDB-370309

Asymptotic scarcity of Legendre cordial labelings for complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. For an integer m, let $\mathbb{J}(n,m)$ be the set of primes $p\leqslant m$ for which K_n is a Legendre cordial graph modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-005957 | MATHDB-370313

The alternating torus-knot 3D index conjecture

摘录 / EXCERPT: Let $p,q>0$ be coprime integers with $pq eq 0\pmod{2}$, and let $\mathcal{T}_{(p,q)}$ be a 1-efficient ideal triangulation of the complement of the (p,q) torus knot. The altern...

conjecture | open | MathDB 页面 | 原文线索

V2-005958 | MATHDB-370339

The prime-power formula for the minimal period of the generalized digit map

摘录 / EXCERPT: Let $M = \prod_{p|M} p^{a_p}$ and $N = K \cdot \prod_{p|M} p^{b_p}$ be integers with $a_p, b_p \ge 1$ and $\gcd(K,M) = 1$. For each prime factor $p|M$, define ... The sequence $b(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-005959 | MATHDB-370371

The associated-prime persistence conjecture for homological shift ideals

摘录 / EXCERPT: Let S be the polynomial ring in the paper, let $I\subset S$ be a polymatroidal ideal, and let $\operatorname{HS}_i(I^k)$ denote the ith homological shift ideal of I^k. Associ...

conjecture | open | MathDB 页面 | 原文线索

V2-005960 | MATHDB-370378

Uniqueness conjecture for companion zeros of the Fejér-Dirichlet indicator

摘录 / EXCERPT: Let $q>1$ be fixed and let $p\geq 5$ be an odd prime. Theorem provides a real zero of $\mathfrak F(\cdot,q)$ in $(p-1,p)$. Uniqueness conjecture. For every such q and p, this z...

conjecture | open | MathDB 页面 | 原文线索

V2-005961 | MATHDB-370382

TMF-valued TQFT conjecture for four-dimensional invariants

摘录 / EXCERPT: Let $\mathcal{Z}$ denote the invariants of closed 3- and 4-manifolds and the proposed maps associated to cobordisms, valued in modules over TMF. TMF-valued TQFT co...

conjecture | open | MathDB 页面 | 原文线索

V2-005962 | MATHDB-370383

Theta-function image conjecture for positive-definite even unimodular forms

摘录 / EXCERPT: Let $Q\colon\mathbb{Z}^d\to\mathbb{Z}$ be positive definite and unimodular, let b be the corresponding even bilinear form, and let $\Theta_Q=\Theta_b$ be its theta function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-005963 | MATHDB-370385

TMF-homology conjecture for boundary conditions of supersymmetric sigma models

摘录 / EXCERPT: Let X be the target of a 3-dimensional $\mathcal{N}=1$ supersymmetric sigma model, and let X_+ denote X with a disjoint basepoint. Sigma-model boundary-condition conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-005964 | MATHDB-370404

Colliot-Thélène's local-to-global conjecture for zero-cycles

摘录 / EXCERPT: Colliot-Thélène's conjecture. The complex above is exact.

conjecture | open | MathDB 页面 | 原文线索

V2-005965 | MATHDB-370411

The E-infinity refinement conjecture for the DGAs S_{2p^l-2}

摘录 / EXCERPT: Let p be the prime appearing in the definition of the DGAs S_{2n}, and let l be a positive integer. The E-infinity refinement conjecture. For each $l>0$, the DGA S_{2p^l-2}...

conjecture | open | MathDB 页面 | 原文线索

V2-005966 | MATHDB-370413

The asymptotic generalized Fermat conjecture

摘录 / EXCERPT: Let K be a number field, and let A,B,C be non-zero elements of $\mathcal{O}_K$. Let Ω be the subgroup of roots of unity inside $\mathcal{O}_K^{\times}$. Suppose ... for an...

conjecture | open | MathDB 页面 | 原文线索

V2-005967 | MATHDB-370436

Kato's residue conjecture for local fields

摘录 / EXCERPT: Kato's residue conjecture. The map Δ_j is an isomorphism for all $r\geq 1$ and $j\geq 0$. This conjecture concerns comparison of Kato homology on a regular proper flat model...

conjecture | open | MathDB 页面 | 原文线索

V2-005968 | MATHDB-370461

Nilpotent Floquet-Bloch prime orbit asymptotic conjecture

摘录 / EXCERPT: Let φ_t be a smooth, transitive and weakly mixing Anosov flow on a compact manifold N. Let Γ be a finitely generated torsion-free nilpotent group, let...

conjecture | open | MathDB 页面 | 原文线索

V2-005969 | MATHDB-370463

Lamzouri's maximum-order conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let $L(s,\chi)$ be the Dirichlet L-function, and let the maximum range over primitive nontrivial Dirichlet characters χ modulo integers $q\leq Q$. Lamzouri's maximum-order c...

conjecture | open | MathDB 页面 | 原文线索

V2-005970 | MATHDB-370482

Saikia's parity and 2-adic congruence conjecture for overpartition prime-tuples

摘录 / EXCERPT: Let $\overline{p}_{q}(n)$ denote the number of overpartition q-tuples of n, where q is prime. Saikia's conjecture. For all $n\geq 0$ and primes q, ... These congruences exten...

conjecture | open | MathDB 页面 | 原文线索

V2-005971 | MATHDB-370483

Four remaining 2-adic congruences for overpartition tuples with odd parts

摘录 / EXCERPT: Let $\overline{OPT}_k(n)$ denote the number of overpartition k-tuples of n with odd parts. The four-congruence conjecture. For all $i\geq 1$, $n\geq 0$, and odd $r>0$, ... These...

conjecture | open | MathDB 页面 | 原文线索

V2-005972 | MATHDB-370484

Das-Saikia-Sarma 3-adic and 2-adic congruence conjecture for odd-part overpartition tuples

摘录 / EXCERPT: Let $\overline{OPT}_k(n)$ denote the number of overpartition k-tuples of n with odd parts. Das-Saikia-Sarma's first conjecture. For all $i,j\geq 1$, all $n\geq 0$, and positive...

conjecture | open | MathDB 页面 | 原文线索

V2-005973 | MATHDB-370485

Das-Saikia-Sarma second congruence conjecture for odd-part overpartition tuples

摘录 / EXCERPT: Let $\overline{OPT}_k(n)$ denote the number of overpartition k-tuples of n with odd parts. Das-Saikia-Sarma's second conjecture. For all $i\geq 1$, all $n\geq 0$, and integers...

conjecture | open | MathDB 页面 | 原文线索

V2-005974 | MATHDB-370512

Gross-Prasad-assisted large-prime Selmer vanishing conjecture for symmetric powers

摘录 / EXCERPT: Let F be a totally real number field, let r be a positive integer, and let A be an elliptic curve over F with no complex multiplication over $\ovl F$. Suppose there exist a...

conjecture | open | MathDB 页面 | 原文线索

V2-005975 | MATHDB-370526

Liu-Sharifzadeh conjecture for type I groups

摘录 / EXCERPT: Let G be a finite Abelian group of type I, meaning that its order is divisible by a prime congruent to 2 modulo 3, and let $\mu(G)$ and $f_{\max}(G)$ denote the maximum size...

conjecture | open | MathDB 页面 | 原文线索

V2-005976 | MATHDB-370546

Finite-exception conjecture for surjectivity of the maps $\tau^{\mathfrak p}$

摘录 / EXCERPT: Let K be a totally real number field of degree n. For an odd prime p unramified in $K/\mathbb{Q}$ and a place $\mathfrak{p}$ above p, let $\tau^{\mathfrak p}$ be the map ap...

conjecture | open | MathDB 页面 | 原文线索

V2-005977 | MATHDB-370554

Congruence conjecture for the second-order mock theta function coefficients

摘录 / EXCERPT: Let $a(m)$ be defined by the paper's generating function, and suppose that m is a positive integer such that ... Write the prime factorization of $4m+3$ as ... where each...

conjecture | open | MathDB 页面 | 原文线索

V2-005978 | MATHDB-370567

The arc-index-Turaev-genus inequality for non-split prime links

摘录 / EXCERPT: Arc-index-Turaev-genus conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-005979 | MATHDB-370612

The finite-field planar exceptional-set conjecture

摘录 / EXCERPT: Let p be prime, let $X\subset\mathbb F_p^2$, and let $E_s(X)$ denote the exceptional set of projection directions with projected dimension at most s. Finite-field exceptional-s...

conjecture | open | MathDB 页面 | 原文线索

V2-005980 | MATHDB-370617

Finiteness conjecture for rationality fields of newforms

摘录 / EXCERPT: Let $f=\sum a_nq^n\in S_k(N)=S_k(\Gamma_0(N))$ be a newform, and let $K_f=\mathbb Q((a_n)_n)$ be its rationality field, namely the number field generated by its Fourier coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-005981 | MATHDB-370646

Stability conjecture for nontrivial direct product graph pairs

摘录 / EXCERPT: A graph pair (Γ,Σ) is nontrivial if Γ and Σ are coprime connected twin-free graphs and exactly one of them is bipartite. A graph pair is stable if it has...

conjecture | open | MathDB 页面 | 原文线索

V2-005982 | MATHDB-370658

Analytic continuation conjecture for the topo-symmetric extension zeta function

摘录 / EXCERPT: Analytic continuation conjecture. The function $\zeta_{\mathrm{ts}}(s;G,H)$ admits an analytic continuation to the complex plane, with poles reflecting the arithmetic structure of...

conjecture | open | MathDB 页面 | 原文线索

V2-005983 | MATHDB-370716

Ramírez Alfonsín-Skałba asymptotic conjecture for primes in a numerical semigroup

摘录 / EXCERPT: Let $S_0(a,b)$ be the numerical semigroup generated by a and b, let $g_{0,a,b}$ be its Frobenius number, and let $\pi_{0,a,b}$ count the primes in $S_0(a,b)$ not exceeding...

conjecture | open | MathDB 页面 | 原文线索

V2-005984 | MATHDB-370717

Chen-Zhu conjecture on primes in the gaps of a numerical semigroup

摘录 / EXCERPT: Let $G_0(a,b)$ be the set of gaps of the numerical semigroup generated by a and b, let $g_{0,a,b}$ be its Frobenius number, and let $\pi^*_{0,a,b}$ be the number of primes in...

conjecture | open | MathDB 页面 | 原文线索

V2-005985 | MATHDB-370768

Gross's conjecture on Fourier coefficients of quaternionic modular forms

摘录 / EXCERPT: Let G be the group supporting quaternionic modular forms, and let $\mathbb X$ be the representation of its Levi subgroup $M\cong\operatorname{GL}_2$ identified with...

conjecture | open | MathDB 页面 | 原文线索

V2-005986 | MATHDB-370776

Lang's conjecture on canonical heights of elliptic-curve points

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve with j-invariant j_E and minimal discriminant Δ_E. Let $\hat h$ denote the canonical height and h the relevant height on ration...

conjecture | open | MathDB 页面 | 原文线索

V2-005987 | MATHDB-370784

Agashe's conjecture on Tamagawa factors and level lowering

摘录 / EXCERPT: Let A be the optimal elliptic curve associated to the weight-2 newform f of level N, and for each prime $p\mid N$ let $c_p(A)$ be its Tamagawa number. Agashe's Tamagawa-lev...

conjecture | open | MathDB 页面 | 原文线索

V2-005988 | MATHDB-370785

Conjecture on the floor-function sum for a product of powers of primes modulo 4

摘录 / EXCERPT: Let p and q be distinct primes, let $\alpha,\beta\geq 1$, and define ... Also set $h^*(-p)=h(-p)$ for $p\geq 7$ and $h^*(-3)=1/3$. The mixed-congruence class-number conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-005989 | MATHDB-370786

Conjecture on the floor-function sum for two primes congruent to 3 modulo 4

摘录 / EXCERPT: Let p and q be distinct primes with $p\equiv q\equiv3\pmod{4}$, let $\alpha,\beta\geq1$, and define ... Set $h^*(-p)=h(-p)$ for $p\geq7$ and $h^*(-3)=1/3$. The same-congruence cl...

conjecture | open | MathDB 页面 | 原文线索

V2-005990 | MATHDB-370787

Conjecture for the floor-function sum and class numbers of odd squarefree integers

摘录 / EXCERPT: Let n be an odd squarefree positive integer, and let $h^*(-p)=h(-p)$ for $p\geq7$ and $h^*(-3)=1/3$. The odd-squarefree class-number conjecture. If $n\equiv1\pmod{4}$, then ... If...

conjecture | open | MathDB 页面 | 原文线索

V2-005991 | MATHDB-370788

Möbius inversion formula for class numbers in terms of the floor-function sum

摘录 / EXCERPT: The Möbius-inversion class-number conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-005992 | MATHDB-370789

Asymptotic distribution conjecture for the floor-function sum

摘录 / EXCERPT: Let $f(n)$ denote the sum defined earlier in the paper. The limiting-values conjecture. As $n\to\infty$, ... The paper motivates these limits through numerical plots and says t...

conjecture | open | MathDB 页面 | 原文线索

V2-005993 | MATHDB-370801

Conjectured bounds for real-variable Kloosterman sums in dimensions 3 and 4

摘录 / EXCERPT: Let $B_{d,n}(r)$ and $\widetilde{B}_{d,n}(r)$ be the functions appearing in the radial Fourier interpolation formulas for dimensions $d=3,4$, with n an appropriate positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-005994 | MATHDB-370818

The zero-free line conjecture for Miller-basis modular forms

摘录 / EXCERPT: For a modular form in the Miller basis, let its associated Faber polynomial be the polynomial whose zeros correspond to zeros of the modular form on the relevant boundary component...

conjecture | open | MathDB 页面 | 原文线索

V2-005995 | MATHDB-370833

Wang-Hu-Long's irrationally elliptic Reeb-orbit conjecture

摘录 / EXCERPT: Let Σ^{2n-1} bound a convex domain in $\mathbb{R}^{2n}$, with $n\geq 2$, and let $\mathcal{T}(\Sigma)$ denote the set of prime closed Reeb orbits on Σ. A closed Reeb...

conjecture | open | MathDB 页面 | 原文线索

V2-005996 | MATHDB-370835

The torsion polynomial conjecture for lens spaces

摘录 / EXCERPT: Torsion-polynomial conjecture. For every prime $p\ge3$, the map $q\mapsto \tau(p,q)$ is constant precisely on the orbits $\{\pm q^{\pm1}\}$. Equivalently, $\tau(p,q)$ distinguishes...

conjecture | open | MathDB 页面 | 原文线索

V2-005997 | MATHDB-370850

Conjecture on termination of the multidimensional Zeckendorf conversion algorithm

摘录 / EXCERPT: Let k be the length of the vector $\vec{\mathbf{c}}=(c_1,\ldots,c_k)$, and let the algorithm in the proof of Theorem main be the conversion algorithm for $\vec{\mathbf{c}}$-repre...

conjecture | open | MathDB 页面 | 原文线索

V2-005998 | MATHDB-370862

McCullough-Wanderley Q-classification conjecture for essential Markoff triples

摘录 / EXCERPT: Let q be a prime power and let an essential triple $(x,y,z)\in\mathbb{F}_q^3$ be the image under $(A,B)\mapsto(\operatorname{tr}A,\operatorname{tr}B,\operatorname{tr}(AB))$ of a...

conjecture | open | MathDB 页面 | 原文线索

V2-005999 | MATHDB-370863

Ross-Selinger asymptotic T-count conjecture for single-qubit Z rotations

摘录 / EXCERPT: Ross-Selinger conjecture. The asymptotic scaling of the T-count required to approximate $R_z(\theta)$ within error ε is

conjecture | open | MathDB 页面 | 原文线索

V2-006000 | MATHDB-370866

The two-or-infinity conjecture for prime closed geodesics on the two-sphere

摘录 / EXCERPT: Let F be a Finsler metric on S^2, and call a closed geodesic prime if it is not an iterate of a shorter closed geodesic. Two-or-infinity conjecture for Finsler metrics. Every F...

conjecture | open | MathDB 页面 | 原文线索

V2-006001 | MATHDB-370887

Conjecture on generators with nonsquare quadratic translate over finite fields

摘录 / EXCERPT: Let p be an odd prime, and write $(\mathbb{Z}/p\mathbb{Z})^{*}$ for the multiplicative group of nonzero residue classes modulo p. A residue class $\overline{k}$ is a generator...

conjecture | open | MathDB 页面 | 原文线索

V2-006002 | MATHDB-370888

Conjecture on generators with nonsquare quadratic translate over finite fields of odd characteristic

摘录 / EXCERPT: Let $\mathbb{K}$ be a finite field of characteristic different from 2, and let $\mathbb{K}^{*}$ denote its multiplicative group. An element x is a generator if it generates...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006003 | MATHDB-370918

The polylogarithmic expression conjecture for twisted partial zeta values

摘录 / EXCERPT: Let k be as in the preceding theorem, let $\calpha,\cbeta$ be the parameters in the narrow ideal-class partial zeta value $\czeta(k,(\calpha,\cbeta),\cscrB)$, and let $\cscrB^*$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-006004 | MATHDB-370933

Strong uniform boundedness conjecture for Brauer groups of K3 surfaces

摘录 / EXCERPT: Strong uniform boundedness conjecture. There is a constant $B=B(d,L)$, independent of X, such that

conjecture | open | MathDB 页面 | 原文线索

V2-006005 | MATHDB-370934

Strong uniform boundedness conjecture for Brauer groups of K3 surfaces without fixed Picard lattice

摘录 / EXCERPT: Strong uniform boundedness conjecture. If X is a K3 surface over a number field of bounded degree d, then

conjecture | open | MathDB 页面 | 原文线索

V2-006006 | MATHDB-370937

The generalized Mahler conjecture for rational bases

摘录 / EXCERPT: Let $p>q\geq 1$ be coprime integers with $p\leq q^2$. A positive real number x is a $Z_{p/q}$-number if the sequence of fractional parts ... is contained in $[0,1/q)$. The generali...

conjecture | open | MathDB 页面 | 原文线索

V2-006007 | MATHDB-370938

The rational-base equidistribution conjecture for iterated ceiling maps

摘录 / EXCERPT: Let $p>q\geq 1$ be coprime integers and define $T_{p/q}(x)=\left\lceil\frac{p}{q}x\right\rceil$. The rational-base equidistribution conjecture. For every $n\in\mathbb N_{>0}$ and e...

conjecture | open | MathDB 页面 | 原文线索

V2-006008 | MATHDB-370939

Akiyama's conjecture on triple expansions in rational bases

摘录 / EXCERPT: Let $p>q$ be coprime integers. An expansion of a positive real number x in rational base p/q is an infinite word satisfying the rational-base language and valuation conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-006009 | MATHDB-370941

Universal termination conjecture for the proposed algorithm

摘录 / EXCERPT: Universal termination conjecture. There exist universal functions $S(a,b,c)$ and $T(a,b,c,p,k,n)$ such that, when the proposed algorithm is guided by this pair of functions, it ter...

conjecture | open | MathDB 页面 | 原文线索

V2-006010 | MATHDB-370942

Two-solutions conjecture for Type-I transcendental Diophantine equations

摘录 / EXCERPT: Two-solutions conjecture. The equation has at most two solutions (x,y) in positive integers; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-006011 | MATHDB-370956

Baragar-Bourgain-Gamburd-Sarnak strong approximation conjecture for the Markoff equation

摘录 / EXCERPT: Baragar-Bourgain-Gamburd-Sarnak conjecture. The Markoff equation $\mathbb{X}$ satisfies strong approximation at every prime p. Equivalently, $\operatorname{Out}(\Pi)$ acts transi...

conjecture | open | MathDB 页面 | 原文线索

V2-006012 | MATHDB-370957

Unbounded denominators conjecture for noncongruence modular forms

摘录 / EXCERPT: Unbounded denominators conjecture. The Fourier coefficients of f have unbounded denominators. Equivalently, for every positive integer n, the Fourier coefficients of nf are n...

conjecture | open | MathDB 页面 | 原文线索

V2-006013 | MATHDB-370983

Zagier's dimension conjecture for multiple zeta values

摘录 / EXCERPT: Let $\zeta(s_1,\ldots,s_r)$ be a multiple zeta value with $s_1>1$ and $s_2,\ldots,s_r\ge1$, and let its weight be $s_1+\cdots+s_r$. For $k\ge1$, define ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-006014 | MATHDB-371020

Logarithmic step bound for the large steps algorithm

摘录 / EXCERPT: Let $k\geq 3$ and let $\vec{\mathbf{v}}\in \mathbb{Z}^{k-1}\setminus\{\vec{\mathbf{0}}\}$. In the Large Steps Algorithm, let j_{lsb} denote the bound on the largest index...

conjecture | open | MathDB 页面 | 原文线索

V2-006015 | MATHDB-371025

Pontiveros's conjecture on dense sets with incomplete sums of dilates

摘录 / EXCERPT: Let p be a prime, let k be a positive integer, and let $\lambda_1,\dots,\lambda_k$ be nonzero integers viewed in $\mathbb{F}_p$. For a set $A\subseteq\mathbb{F}_p$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-006016 | MATHDB-371027

The Newton-polytope determination conjecture for points on hypersurfaces

摘录 / EXCERPT: Let C_f be the hypersurface defined by a polynomial f, let H_f be its Newton polytope, let $G(H)$ be the greatest common divisor of all coordinates of all elements of H, an...

conjecture | open | MathDB 页面 | 原文线索

V2-006017 | MATHDB-371028

Bhargava's conjecture on the lowest odd-degree point of hyperelliptic curves

摘录 / EXCERPT: Let g be the genus of a hyperelliptic curve. Consider locally soluble hyperelliptic curves that have points of odd degree. Bhargava's conjecture. As the family is ordered in the...

conjecture | open | MathDB 页面 | 原文线索

V2-006018 | MATHDB-371037

Levesley-Salp-Velani conjecture for very well approximable points in the Cantor set

摘录 / EXCERPT: Levesley-Salp-Velani conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006019 | MATHDB-371038

Bugeaud-Durand dimension conjecture for rationally approximable points in the Cantor set

摘录 / EXCERPT: Bugeaud-Durand conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006020 | MATHDB-371051

The simultaneous-primes conjecture for $9q^2-2$ and $9q^2+2$

摘录 / EXCERPT: Let q be a prime satisfying $q>3$ and $q\equiv 3\pmod 4$. Simultaneous-primes conjecture. There are infinitely many numbers $3q$ such that both $9q^2-2$ and $9q^2+2$ are prime. T...

conjecture | open | MathDB 页面 | 原文线索

V2-006021 | MATHDB-371062

Odd-primary algebraic transfer injectivity conjecture in ranks at most four

摘录 / EXCERPT: Odd-primary transfer conjecture. The transfer homomorphism $Tr_h^{\mathscr A_p}(\mathbb F_p)$ is one-to-one for any odd prime p and $1\leq h\leq 4$. This extends the known low-ra...

conjecture | open | MathDB 页面 | 原文线索

V2-006022 | MATHDB-371079

The prime-increment conjecture for the LCM recurrence

摘录 / EXCERPT: Prime-increment conjecture. For every $n\geq 2$, we have $b_n\in\{1\}\cup\mathbb{P}$, where $\mathbb{P}$ denotes the set of prime numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-006023 | MATHDB-371084

Sun's perfect-power repulsion conjecture for the partition function

摘录 / EXCERPT: Sun's perfect-power repulsion conjecture. The partition function $p(n)$ repels perfect powers in the following senses: (i) for every $n\ge2$ and $k\ge2$, $p(n)\neq m^k$ for all int...

conjecture | open | MathDB 页面 | 原文线索

V2-006024 | MATHDB-371092

Full generic escape-of-mass conjecture for quadratic Laurent series

摘录 / EXCERPT: Let q be a prime power, let $\Theta(t)\in\mathbb{F}_q(\!(t^{-1})\!)$ be quadratic irrational, and let $P(t)\in\mathbb{F}_q[t]$ be irreducible. For $n,k\in\mathbb{N}$, define ... Fu...

conjecture | open | MathDB 页面 | 原文线索

V2-006025 | MATHDB-371093

Garrett-Rogers conjecture on automaticity of number walls

摘录 / EXCERPT: Let q be a positive power of a prime, and let $\textbf{S}$ be an automatic sequence over $\mathbb{F}_q$. Let $W_q(\textbf{S})$ denote the number wall associated with $\textbf{S}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006026 | MATHDB-371133

Prime-graph-complement invariance conjecture for PSL(2, q)-solvable groups

摘录 / EXCERPT: Let p and q be prime powers such that $\operatorname{PSL}(2,q)$ and $\operatorname{PSL}(2,p)$ are K_4-groups. Suppose one of the following holds: p and q are primes with...

conjecture | open | MathDB 页面 | 原文线索

V2-006027 | MATHDB-371157

The supersingular-prime congruence for elliptic-curve poles

摘录 / EXCERPT: Let $C/\mathbb{Q}$ be an elliptic curve with $j(C)\notin\{0,1728\}$. A prime p is supersingular for C when $a_p(C)=0$ (for $p\geq5$), where $a_p(C)=p+1-|C(\mathbb{F}_p)|$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006028 | MATHDB-371158

The good-prime congruence for meromorphic modular forms with elliptic-curve poles

摘录 / EXCERPT: Let $C/\mathbb{Q}$ be an elliptic curve with $j(C)\notin\{0,1728\}$, and write ... where $k\in\{4,6,8,10,14\}$. Let p be a good prime of C, with $v_p(j(C))=0=v_p(j(C)-1728)$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-006029 | MATHDB-371159

The CM magnetic divisibility conjecture for higher-order poles

摘录 / EXCERPT: Let $G_{k,D}^{(r)}$ denote the integral linear combination of $E_k/(j-j(\alpha_D))^i$ for $1\leq i\leq r$ having a pole of order r at the CM point of discriminant D. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006030 | MATHDB-371160

The complementary-pole coefficient-product divisibility conjecture

摘录 / EXCERPT: Let $G_{k,D}^{(r)}$ be the CM higher-order-pole forms, let $k\in\{4,6,8,10,14\}$, $1\leq r\leq k-1$, and let $D<-4$ be a discriminant with class number 1. The complementary-pole...

conjecture | open | MathDB 页面 | 原文线索

V2-006031 | MATHDB-371161

Generalized Andrews mod 8 identity for indexed partitions

摘录 / EXCERPT: Generalized Andrews identity.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006032 | MATHDB-371163

The parity conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over a number field F. Write $\begin{displaystyle}\operatorname{rk}(E/F)\end{displaystyle}$ for its algebraic rank and $w(E/F)$ for its global root n...

conjecture | open | MathDB 页面 | 原文线索

V2-006033 | MATHDB-371166

Triply-transitivity conjecture for collinearity graphs of elliptic polar spaces

摘录 / EXCERPT: Let q be a prime power, and let the collinearity graph of the polar space $\mathcal{Q}^{-}(5,q)$ be the graph whose vertices are the points of $\mathcal{Q}^{-}(5,q)$, with two ve...

conjecture | open | MathDB 页面 | 原文线索

V2-006034 | MATHDB-371190

Corvaja-Zannier weak Hilbert property conjecture

摘录 / EXCERPT: Corvaja-Zannier conjecture. Under these hypotheses, Y has WHP potentially over K.

conjecture | open | MathDB 页面 | 原文线索

V2-006035 | MATHDB-371191

Fibration conjecture for the weak Hilbert property

摘录 / EXCERPT: Fibration conjecture for WHP. (a) Let Ξ be the set of all $z\in Z(K)$ such that Y_z has WHP over K. If Ξ is not strongly thin in Z, then Y has WHP over K. (b) If...

conjecture | open | MathDB 页面 | 原文线索

V2-006036 | MATHDB-371193

Weak Hilbert property conjecture for geometric fibrations

摘录 / EXCERPT: Weak Hilbert property conjecture for geometric fibrations. In the situation above, Y has WHP over K.

conjecture | open | MathDB 页面 | 原文线索

V2-006037 | MATHDB-371194

Weak Hilbert property conjecture for Néron models of elliptic curves

摘录 / EXCERPT: Néron-model WHP conjecture. Then E has WHP over K.

conjecture | open | MathDB 页面 | 原文线索

V2-006038 | MATHDB-371199

Period Count for Fibonacci recurrences modulo primes

摘录 / EXCERPT: Let p be prime, and let $\#(p)$ count distinct periods modulo p of the Fibonacci recurrence and its parity transform over all initial conditions...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006039 | MATHDB-371200

Self-similarity of Fibonacci period structures at prime powers

摘录 / EXCERPT: For the Fibonacci recurrence and its parity transform modulo prime powers, Self-Similarity at Prime Powers. At each transition $p^k\to p^{k+1}$, all existing periods are preserved,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006040 | MATHDB-371202

Probability distribution of Fibonacci sequence minima

摘录 / EXCERPT: Consider the Fibonacci recurrence $a_n=a_{n-1}+a_{n-2}$ and its chiral recurrence $a_n=-a_{n-1}+a_{n-2}$ with random integer initialization $(a_0,a_1)\in\mathbb Z^2$. Let F_n and...

conjecture | open | MathDB 页面 | 原文线索

V2-006041 | MATHDB-371213

Threshold conjecture for positivity of partial sums of a random multiplicative function

摘录 / EXCERPT: Let f be a random completely multiplicative function taking values in $\\{1,-1\\}$, with the values $f(p)$ chosen independently and uniformly at random over the primes. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006042 | MATHDB-371214

Smooth-support mean-value conjecture for completely multiplicative functions

摘录 / EXCERPT: Let x be sufficiently large. Consider a real-valued completely multiplicative function f satisfying $|f(n)|\leq 1$ for every positive integer n, and suppose that $f(p)=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-006043 | MATHDB-371228

The disjoint zero-block conjecture for the floor-function differences

摘录 / EXCERPT: Let $n\geq 1$ be an odd integer, and define ... Suppose that $\delta:=d_n(\lambda)\geq 2$ for some $\lambda\geq 1$. Disjoint zero-block conjecture. Each such λ corresponds to...

conjecture | open | MathDB 页面 | 原文线索

V2-006044 | MATHDB-371243

Shunia's closed-form formula for nontrivial integer roots

摘录 / EXCERPT: Let $n,m\in\mathbb{N}$ satisfy $n>2$, $\lfloor\log_2(n)\rfloor+1\geq m>1$, and suppose there is no $k\in\mathbb{N}$ such that $k^m=n$. Then Shunia's conjecture. ... The formula gives...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006045 | MATHDB-371260

Class number formula conjecture for split primes 7 and 11

摘录 / EXCERPT: Class number formula conjecture. For $n>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-006046 | MATHDB-371266

Leading-pole cancellation conjecture for the holomorphic modular ambiguity

摘录 / EXCERPT: Leading-pole cancellation conjecture. The choice

conjecture | open | MathDB 页面 | 原文线索

V2-006047 | MATHDB-371294

Almost all continued-fraction strings have no nontrivial symmetries

摘录 / EXCERPT: Almost-symmetry-free strings conjecture. For any integer $n\ge4$,

conjecture | open | MathDB 页面 | 原文线索

V2-006048 | MATHDB-371295

Asymptotic count of length-four continued-fraction strings with nontrivial symmetries

摘录 / EXCERPT: Length-four symmetry-count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006049 | MATHDB-371319

Baker-Montgomery conjecture on real zeros of quadratic Dirichlet L-function derivatives

摘录 / EXCERPT: Let $d53d_d(d70e_1,d70e_2)$ be the number of real zeros of $L'(s,\chi_d)$ on the interval $[d70e_1,d70e_2]$, where χ_d is the primitive quadratic character attached to the fun...

conjecture | open | MathDB 页面 | 原文线索

V2-006050 | MATHDB-371344

Prime-diagonal rank conjecture for graphical Hermite simplices

摘录 / EXCERPT: Let G be the topological ordering of a graph on $[n]$, and let B be the matrix over ${\mathbb{Z}}/p{\mathbb{Z}}$ given by ... Let r_p be the rank of B over...

conjecture | open | MathDB 页面 | 原文线索

V2-006051 | MATHDB-371351

Craig-van Ittersum-Ono conjecture on prime-detecting quasi-modular forms

摘录 / EXCERPT: Let $q=e^{2\pi i\tau}$, and let Ω be the subset of the graded ring of integer-weight quasi-modular forms of full level consisting of forms ... whose Fourier coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-006052 | MATHDB-371387

Prime degree conjecture for branched-cover branch data

摘录 / EXCERPT: Let d be a positive integer, and let $\mathcal{D}$ be a collection of nontrivial partitions of d. A collection satisfying the Riemann-Hurwitz formula is called a candidate bran...

conjecture | open | MathDB 页面 | 原文线索

V2-006053 | MATHDB-371417

The uniqueness conjecture for extensions of D(4)-triples

摘录 / EXCERPT: Let $n\neq0$ be an integer. A set of distinct positive integers is a $D(n)$-tuple if the product of any two distinct elements, increased by n, is a perfect square. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-006054 | MATHDB-371423

Sharp upper-bound conjecture for alternating products in the modular group

摘录 / EXCERPT: Sharpness conjecture. The number of such solutions is at most

conjecture | open | MathDB 页面 | 原文线索

V2-006055 | MATHDB-371439

Tait's conjecture on minimal crossing diagrams of links

摘录 / EXCERPT: Tait's conjecture. A reduced alternating diagram of L has minimal crossing number, and if L is prime then no non-alternating diagram has minimal crossing number.

conjecture | open | MathDB 页面 | 原文线索

V2-006056 | MATHDB-371441

Generation conjecture for genuine equivariant stable cohomology operations

摘录 / EXCERPT: Let $\Gmr$ be a finite group and let p be a prime. For each i, let $\Ecal^{(p^i)}_{\Gmr,p}$ be the collection of integral $\uprho_{\Gmr}$-stable $\Hmr\underline{\FF}_p$-Euleria...

conjecture | open | MathDB 页面 | 原文线索

V2-006057 | MATHDB-371464

The generalized arithmetic-progression divisor conjecture

摘录 / EXCERPT: Let n be a positive integer. The n-divisor property means that every integer from 1 through n divides at least one member of a specified product. The generalized arithmetic...

conjecture | open | MathDB 页面 | 原文线索

V2-006058 | MATHDB-371465

The most general divisor-set conjecture

摘录 / EXCERPT: Let n be a positive integer, and let divisibility of a product mean that each integer in $\{1,\ldots,n\}$ divides that product. The most general divisor-set conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-006059 | MATHDB-371466

The residue-class formulation of the divisor-set conjecture

摘录 / EXCERPT: Let n be a positive integer, and let $\mathbb{Z}^{\widetilde{O}(n^{n^{1/3}})}$ denote the integers in the indicated size range. The residue-class formulation. There exists a set...

conjecture | open | MathDB 页面 | 原文线索

V2-006060 | MATHDB-371486

Infinitude of prime and lucky numbers of every fractional order

摘录 / EXCERPT: Under the restricted factorization described in the source, assign a fractional order $\operatorname{ord}(p)$ or $\operatorname{ord}(l)$ recursively to primes p and lucky numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-006061 | MATHDB-371506

Non-interpretability of multiplication for Ulam expansions

摘录 / EXCERPT: Let (a,b) be coprime, and let $\mathcal{U}_{a,b}$ denote the expansion of $(\mathbb{N},+,0,1)$ by the predicate for the associated Ulam sequence. Non-interpretability of multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-006062 | MATHDB-371507

Tameness under bounded gaps for Ulam expansions

摘录 / EXCERPT: Let (a,b) be coprime, let $U(a,b)$ be the associated Ulam sequence, and write its consecutive gaps as g_k. Condition (R3) means that there exists B such that all gaps g_k a...

conjecture | open | MathDB 页面 | 原文线索

V2-006063 | MATHDB-371508

Ulam dichotomy for expansions of Presburger arithmetic

摘录 / EXCERPT: Let (a,b) be coprime, and let $\mathcal{U}_{a,b}$ be the expansion of $(\mathbb{N},+,0,1)$ by the predicate for the associated Ulam sequence. Ulam dichotomy. For every coprime...

conjecture | open | MathDB 页面 | 原文线索

V2-006064 | MATHDB-371509

Tameness of all Ulam expansions

摘录 / EXCERPT: Let (a,b) be coprime, let $\mathcal{U}_{a,b}$ denote the expansion of $(\mathbb{N},+,0,1)$ by the associated Ulam predicate, and let $\operatorname{Th}(\mathcal{U}_{a,b})$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-006065 | MATHDB-371522

The power difference set conjecture for multiplicative subgroups of finite fields

摘录 / EXCERPT: Let p be prime, and let K be the multiplicative subgroup of ℓ-th powers in $\mathbb{F}_p$. A power difference set conjecture asserts that if K forms a difference set of...

conjecture | open | MathDB 页面 | 原文线索

V2-006066 | MATHDB-371523

Marques-Trojovský conjecture on the Bonse inequality threshold function

摘录 / EXCERPT: Marques-Trojovský conjecture. The function $\Psi(x)$ satisfies: (i) $\Psi(x)=21$ for all $x\in[0.9,1.3]$; (ii) $\Psi(x)=149$ for all $x\in[0.5,0.8]$; (iii) $\Psi(x)=59{,}875$ for a...

conjecture | open | MathDB 页面 | 原文线索

V2-006067 | MATHDB-371524

Conjecture on the asymptotic expansion of the Bonse threshold parameters

摘录 / EXCERPT: Asymptotic expansion conjecture. There exists a constant $\kappa>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006068 | MATHDB-371554

Generalized Elliott-Halberstam conjecture for shifted convolutions

摘录 / EXCERPT: Let $x\geq2$, $q\in\mathbb{N}$, and $a,h\in\mathbb{Z}$ with $(a,q)=1$. Define ... where $\mathbf{1}_{(a(a+h),q)=1}$ equals 1 when both a and $a+h$ are coprime to q, and equals...

conjecture | open | MathDB 页面 | 原文线索

V2-006069 | MATHDB-371555

González Robert-et al. twisted lim sup conjecture

摘录 / EXCERPT: González Robert-et al. conjecture. For almost every sequence ${\bm \omega}$ in $\mathbb{R}^{d}$, for every d-tuple of non-increasing functions ${\bf r}=(r_1,\ldots,r_d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-006070 | MATHDB-371565

Gica's conjecture for odd quadratic-polynomial values in the square-free case

摘录 / EXCERPT: Let $d ot\for all?$.

conjecture | open | MathDB 页面 | 原文线索

V2-006071 | MATHDB-371566

Conjecture for odd quadratic-polynomial values in the nonsquare-free case

摘录 / EXCERPT: Let $d ot\for all?$.

conjecture | open | MathDB 页面 | 原文线索

V2-006072 | MATHDB-371567

Conjecture for even quadratic-polynomial values in the square-free real quadratic case

摘录 / EXCERPT: Conjecture. $M'_{\mathrm{even}}(d)\leq 2$ if and only if $d\in\{2,6,10,14,22,26,30,38,42,62,110,122,182,278,362,398\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-006073 | MATHDB-371579

Serre-Swinnerton-Dyer exceptional-prime conjecture for the eta-quotient newform $E_4(z)\Delta(z)$

摘录 / EXCERPT: Serre-Swinnerton-Dyer conjecture. The prime $\ell=59$ is an exceptional prime of type III for

conjecture | open | MathDB 页面 | 原文线索

V2-006074 | MATHDB-371584

Arithmetic Mabuchi period conjecture for K-polystable pairs

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{D})$ be an arithmetic log pair over $\mathbb{Z}$ such that $-\mathcal{K}_{(\mathcal{X},\mathcal{D})}$ is a relatively ample $\mathbb{Q}$-line bundle, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006075 | MATHDB-371610

Maximum-loneliness formulation of the Lonely Runner Conjecture

摘录 / EXCERPT: Let $V=\{v_1,v_2,\dots,v_n\}$ be a set of n distinct positive integers, and define the maximum loneliness by ... where $\mathbf{T}=\mathbf{R}/\mathbf{Z}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006076 | MATHDB-371625

Infinitude conjecture for quartic Fermat near misses

摘录 / EXCERPT: Quartic near-miss infinitude conjecture. If this equation has an integer solution, then it has infinitely many integer solutions. The conjecture is suggested by the paper's non-thi...

conjecture | open | MathDB 页面 | 原文线索

V2-006077 | MATHDB-371630

Adelic Dirac-Hilbert-Pólya conjecture for the Riemann zeros

摘录 / EXCERPT: Let $(\mathcal{D}_{\mathrm{arith}},\{\eta_p\})$ be a chiral adelic Dirac system, where the coefficient functions η_p are even, and let ξ_{Dirac} denote its spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-006078 | MATHDB-371654

Toda-prime lower bound for certain square-free multiples of 3

摘录 / EXCERPT: Let $T(n)$ be the set of Toda primes of n, where a Toda prime is an odd prime p satisfying ... Write $t(n)=|T(n)|$. Toda-prime lower-bound conjecture. Let n be an odd, square-f...

conjecture | open | MathDB 页面 | 原文线索

V2-006079 | MATHDB-371655

Bernoulli-denominator conjecture for Toda-prime sets

摘录 / EXCERPT: For an even integer $2m$, let $P(2m)$ be the set of primes p such that $p-1\mid 2m$, so that ... where D_{2m} is the denominator of the Bernoulli number B_{2m}. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006080 | MATHDB-371664

Montgomery's pair correlation conjecture for the nontrivial zeros of the Riemann zeta function

摘录 / EXCERPT: Let $\rho=\beta+i\gamma$ and $\rho'=\beta'+i\gamma'$ range over the nontrivial zeros of the Riemann zeta function. For fixed $\lambda>0$ and $T\to\infty$, consider pairs with posit...

conjecture | open | MathDB 页面 | 原文线索

V2-006081 | MATHDB-371714

Conjecture on the Steiner diameter and Steiner (k,k')-radius of trees

摘录 / EXCERPT: Let n, k, and $k^{\prime}$ be integers with $n\geq k\geq3$ and $k>k^{\prime}\geq1$, and let T be a tree with order n. The Steiner diameter-radius conjecture. ... The paper pr...

conjecture | open | MathDB 页面 | 原文线索

V2-006082 | MATHDB-371721

Mascia-Rinaldo-Romeo primality conjecture for polyomino ideals

摘录 / EXCERPT: Mascia-Rinaldo-Romeo conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-006083 | MATHDB-371729

Sun's classification conjecture for suitable two-variable linear polynomials

摘录 / EXCERPT: Let $P(X,Y,Z,W)\in\mathbb{Z}[X,Y,Z,W]$ be called suitable if, for every $n\in\mathbb{N}$, there exist $x,y,z,w\in\mathbb{N}$ such that ... where $\mathcal{S}=\{x^2:x\in\mathbb{Z}\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-006084 | MATHDB-371751

Uniform normality preservation under rational multiplication for dynamically generated basic sequences

摘录 / EXCERPT: Let $Q=(q_n)_{n\geq 1}$ be a dynamically generated basic sequence, and let $\mathcal{UN}(Q)$ denote the set of uniformly Q-normal numbers. Uniform normality multiplication conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-006085 | MATHDB-371791

The reducibility conjecture for falling-factorial polynomials

摘录 / EXCERPT: Reducibility conjecture. If $x^{\underline{k}}-a!$ is reducible over $\mathbb{Z}$, then $(k,a)=(3,6)$, $(k,a)=(4,7)$, or there exists an integer t such that $a=t!-1$ and $k=t!-t$...

conjecture | open | MathDB 页面 | 原文线索

V2-006086 | MATHDB-371792

Equidistribution conjecture for fractional parts of factorial roots

摘录 / EXCERPT: For a positive integer A, let ... where the set is counted as a multiset when cardinalities are computed. Equidistribution conjecture. For any interval $I\subset[0,1]$, ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-006087 | MATHDB-371796

Poisson Tail Conjecture for gaps between consecutive primes

摘录 / EXCERPT: Let p_n denote the nth prime, and let $\varepsilon>0$. Poisson Tail Conjecture. For $1\leq H\leq(\log X)^{2-\varepsilon}$, the two gap-counting quantities satisfy ... ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-006088 | MATHDB-371801

Chebyshev-like bias conjecture for entry points of primes in Lucas sequences

摘录 / EXCERPT: Chebyshev-like bias conjecture. (Weak) If U is a regular Lucas sequence with $D>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-006089 | MATHDB-371815

Aigner's ordering conjectures for Markov numbers

摘录 / EXCERPT: Aigner's conjectures. The following ordering properties should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-006090 | MATHDB-371825

The density conjecture for {-1,1}-perfect numbers

摘录 / EXCERPT: Let $\mathcal{A}$ denote the density of the abundant positive integers, and let the density of the positive $\{-1,1\}$-perfect numbers mean the corresponding natural density when i...

conjecture | open | MathDB 页面 | 原文线索

V2-006091 | MATHDB-371826

The nonsquare odd abundant numbers conjecture for {-1,1}-perfect numbers

摘录 / EXCERPT: Let n be a positive odd integer. Call n abundant if $\sigma(n) \geq 2n$, and call it nonsquare if it is not a perfect square. Nonsquare odd abundant numbers conjecture. Every n...

conjecture | open | MathDB 页面 | 原文线索

V2-006092 | MATHDB-371831

Frobenius's uniqueness conjecture for Markov triples

摘录 / EXCERPT: A Markov number is a positive integer occurring in a Markov triple, that is, a triple of positive integers satisfying the Markov equation. A Markov triple is ordered here as...

conjecture | open | MathDB 页面 | 原文线索

V2-006093 | MATHDB-371832

Uniqueness conjecture for generalized Markov values

摘录 / EXCERPT: Let $\mathcal M_{0,0,0}$ denote the classical discrete Markov spectrum, and let $\mathcal L(\alpha)$ denote the Markov-Lagrange value associated with an irrational α. Two ir...

conjecture | open | MathDB 页面 | 原文线索

V2-006094 | MATHDB-371833

Injectivity conjecture for Markov tree fraction labelings

摘录 / EXCERPT: Fix $\sigma\in \mathfrak S_3$ and consider the Markov tree $\mathrm{M}\mathbb T(0,0,0,\sigma)$. For any irreducible fractions $t,s\in[0,1]$, let n_t and n_s be Markov numbers w...

conjecture | open | MathDB 页面 | 原文线索

V2-006095 | MATHDB-371834

Generalized injectivity conjecture for GM-tree fraction labelings

摘录 / EXCERPT: Fix $\sigma\in \mathfrak S_3$, and let k_1,k_2,k_3 be nonnegative integers. Consider the generalized Markov tree $\mathrm{M}\mathbb T(k_1,k_2,k_3,\sigma)$. For any irreducible fr...

conjecture | open | MathDB 页面 | 原文线索

V2-006096 | MATHDB-371841

Das et al.'s congruence conjecture for generalized overcubic partitions

摘录 / EXCERPT: Das et al.'s conjecture. For all $n\geq 0$ and $i\geq 1$, these five congruences hold.

conjecture | open | MathDB 页面 | 原文线索

V2-006097 | MATHDB-371846

The generalized Fermat finiteness conjecture over the integers

摘录 / EXCERPT: Let $A,B,C\in\mathbb{Z}$ satisfy $\gcd(A,B,C)=1$. For prime exponents p,q,r with ... a non-trivial primitive integer solution of $Ax^p+By^q+Cz^r=0$ is a solution with $xyz\ne0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006098 | MATHDB-371864

The optimal linear bound for zero-sum Ramsey numbers of forests

摘录 / EXCERPT: Let p be prime, let $n_p=n(p)$ be a positive integer, and let F be a forest on $n\geq n_p$ vertices such that $p\mid e(F)$. Here $R(F,\mathbb{Z}_p)$ denotes the least order of...

conjecture | open | MathDB 页面 | 原文线索

V2-006099 | MATHDB-371865

A linear upper bound on zero-sum Ramsey numbers of graphs

摘录 / EXCERPT: Let p be prime, let $c_p=c(p)$ be an integer, and let G be a graph on n vertices such that $p\mid E(G)$. The graph linear-bound conjecture. There is an integer $c_p=c(p)$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-006100 | MATHDB-371908

Polynomial separation conjecture for capacitary holomorphic functions

摘录 / EXCERPT: Polynomial separation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-006101 | MATHDB-371923

Zagier-duality modularity conjecture for type D_k Cartan matrices

摘录 / EXCERPT: Let $\mathcal{C}(D_k)$ be the Cartan matrix of the Dynkin diagram of type D_k, and let $f_{A,b,C}(q)$ denote the corresponding Nahm sum. Zagier-duality conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-006102 | MATHDB-371927

The prime-values conjecture for Drinfeld-module auxiliary polynomials

摘录 / EXCERPT: Let A be the coefficient ring and let ... be a Drinfeld module. For each prime polynomial P, let $g_{P,\phi}$ denote the auxiliary polynomial associated with P and ϕ. Prim...

conjecture | open | MathDB 页面 | 原文线索

V2-006103 | MATHDB-371943

Canonical period comparison under totally real quadratic base change

摘录 / EXCERPT: Let F be a totally real number field, let f be a Hilbert newform of parallel even weight over F, let F' be a totally real quadratic extension of F, and let f' be the ba...

conjecture | open | MathDB 页面 | 原文线索

V2-006104 | MATHDB-371944

Vanishing of the $4\mu4$-invariant after totally real base change

摘录 / EXCERPT: Let F be a totally real number field, let K be a quadratic imaginary extension of F, let F' be a finite totally real extension of F, and let $f_{F'}$ be a Hilbert modular...

conjecture | open | MathDB 页面 | 原文线索

V2-006105 | MATHDB-371948

Existence conjecture for a projective-geometry triple-array family

摘录 / EXCERPT: Existence conjecture. For any prime power q, there is a $\left(\frac{q^3-1}{q-1},\frac{q^4-1}{q-1},\frac{(q^4-1)(q^3-1)}{(q^2-1)(q-1)}\right)$-triple array.

conjecture | open | MathDB 页面 | 原文线索

V2-006106 | MATHDB-371962

Infinitesimal containment conjectures for congruence completions

摘录 / EXCERPT: In the first case, let $\Gamma=\operatorname{SL}_2(\mathbb{Z})$ and let p be any prime. In the second case, let Γ be a cocompact arithmetic lattice of the second type in...

conjecture | open | MathDB 页面 | 原文线索

V2-006107 | MATHDB-371983

Median and mode conjecture for crosscap number and unoriented genus of prime alternating knots

摘录 / EXCERPT: Median and mode conjecture. For any $c\geq 12$, the medians and modes for γ and Γ among all c-crossing prime alternating knots equal

conjecture | open | MathDB 页面 | 原文线索

V2-006108 | MATHDB-371985

Classification conjecture for prime alternating links with crosscap number three

摘录 / EXCERPT: Crosscap-three classification conjecture. For each even crossing number $2k\geq 6$, there are exactly

conjecture | open | MathDB 页面 | 原文线索

V2-006109 | MATHDB-371986

Link-knot defect proportion comparison conjecture

摘录 / EXCERPT: Defect proportion comparison conjecture. For all even $c\geq 4$ and all odd $c\geq 11$, the proportion of c-crossing prime alternating links L with $\gamma(L)>\Gamma(L)$ is lar...

conjecture | open | MathDB 页面 | 原文线索

V2-006110 | MATHDB-371993

Rational Angle Conjecture for maximal volume ideal polyhedra

摘录 / EXCERPT: Let $\mathcal{T}$ be a realizable triangulation of the sphere, and consider the volume-maximizing ideal polyhedron with combinatorial type $\mathcal{T}$. Rational Angle Conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-006111 | MATHDB-372016

Iizuka's successive quadratic fields conjecture

摘录 / EXCERPT: Let p be a prime number and let m be a positive integer. For an integer d, consider the $m+1$ successive quadratic fields ... Here, “real (or imaginary) quadratic” means that t...

conjecture | open | MathDB 页面 | 原文线索

V2-006112 | MATHDB-372018

Hensley's dimension criterion for Zaremba's conjecture

摘录 / EXCERPT: Hensley's conjecture. One has $\delta_A>1/2$ if and only if $D_A\supset\mathbb{N}_{\gg 1}$, meaning that Zaremba's conjecture holds for all sufficiently large N.

conjecture | open | MathDB 页面 | 原文线索

V2-006113 | MATHDB-372057

Modular transformation conjecture for the genus-two signature

摘录 / EXCERPT: Let $\sigma_2(q/p)$ denote the genus-two signature at a rational number q/p. For coprime odd integers $0<q<p$, consider the transformed fraction $q/(2q+p)$. Modular transformatio...

conjecture | open | MathDB 页面 | 原文线索

V2-006114 | MATHDB-372071

Algebraic independence of distinct-prime valuation generating functions at independent points

摘录 / EXCERPT: Let $T_p(z)$ denote the valuation generating function associated with a prime p. Let $p_1,p_2,\dots,p_m$ be distinct primes, and let $\alpha_1,\alpha_2,\dots,\alpha_m$ be non-zer...

conjecture | open | MathDB 页面 | 原文线索

V2-006115 | MATHDB-372073

The local equivariant Tamagawa number conjecture for Tate motives

摘录 / EXCERPT: Let p be an odd prime, let K/k be a finite abelian extension of number fields with Galois group G, and let S contain $S_p\cup S_\infty\cup S_{\operatorname{ram}}(K/k)$, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-006116 | MATHDB-372084

Conjecture C_n on short cubic character sums

摘录 / EXCERPT: Conjecture C_n. For every $\varepsilon>0$, there exists $\delta=\delta(n)>0$ such that, for all M,

conjecture | open | MathDB 页面 | 原文线索

V2-006117 | MATHDB-372087

Mocanu's Jacobi-form isomorphism conjecture for the lattices D_3, D_5, and D_7

摘录 / EXCERPT: Mocanu's conjecture. For every $k \geq 2$, there are Hecke-equivalent isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-006118 | MATHDB-372098

Inductive Galois Alperin weight conjecture

摘录 / EXCERPT: Fix a prime p, a sufficiently large p-modular system $(\mathcal{K},\mathcal{O},F)$ with finite residue field, and a finite abelian group...

conjecture | open | MathDB 页面 | 原文线索

V2-006119 | MATHDB-372116

Lehmer-type conjecture for adelic heights above the energy minimum

摘录 / EXCERPT: Let K be a number field, let μ be an adelic measure defined over K, let h_μ be its associated height on $\mathbb{P}^1(\overline{K})$, and let $\mathcal{L}(\mu)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-006120 | MATHDB-372120

The extended g-vector classification conjecture for real prime modules

摘录 / EXCERPT: Let $\mathcal{Q}^{\leq \xi}_I$ be the principal quiver of $\Gamma^{\leq \xi}_I$, let $\mathcal{C}^{\leq \xi}_I$ be its associated cluster category, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006121 | MATHDB-372123

The Erdős-Ingham nonvanishing conjecture for finite reciprocal Dirichlet sums

摘录 / EXCERPT: Let S be a finite subset of the integers at least 2, and let t be a real number. Erdős-Ingham conjecture. ... The paper's concluding remarks explain that the elementary approac...

conjecture | open | MathDB 页面 | 原文线索

V2-006122 | MATHDB-372133

Pappas-Rapoport flatness conjecture for orthogonal spin local models

摘录 / EXCERPT: Let F be a complete discretely valued field with ring of integers $\mathcal{O}$, uniformizer π, and residue field k of characteristic $p>2$. Let $V=F^{2n}$ with a split non...

conjecture | open | MathDB 页面 | 原文线索

V2-006123 | MATHDB-372134

Flatness conjecture for spin local models of split even orthogonal groups

摘录 / EXCERPT: Let F be a discretely valued field with ring of integers $\CO$, and let G be the split even orthogonal group of rank $2n$. Let $\CL$ be a self-dual lattice chain, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006124 | MATHDB-372135

Flatness conjecture for pseudo-maximal spin local models

摘录 / EXCERPT: Let F be a discretely valued field with ring of integers $\CO$, let k be its residue field, and let $n\geq 1$. For a non-empty subset $I\subset[0,n]$, let Λ_I be the as...

conjecture | open | MathDB 页面 | 原文线索

V2-006125 | MATHDB-372136

Generalized Hedtke-Murthy conjecture for cyclic normal subgroups of prime index

摘录 / EXCERPT: Let G be a finite group with a cyclic normal subgroup of prime index p, and let (S,T,U) be a triple product property triple of G. Let $\rho(G)=\frac{\beta(G)}{|G|}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-006126 | MATHDB-372139

Cohen's classification conjecture for sums of a 5-potent and an n-potent

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field of order $q=p^v$, where p is prime and v is a positive integer. For an integer $r>1$, call $a\in\mathbb{F}_q$ r-potent if $a^r=a$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-006127 | MATHDB-372148

Higher-weight image conjecture for normalized generalized Dedekind sums

摘录 / EXCERPT: Let χ_1 and χ_2 be quadratic characters, let k be the weight, and let $\widetilde{S}_{\chi_1,\chi_2,k}(\Gamma_1(q_1q_2))$ denote the image of the normalized generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-006128 | MATHDB-372182

The minimal-discrepancy Nash valuation conjecture for 3-fold terminal singularities

摘录 / EXCERPT: Let (X,p) be a 3-fold terminal singularity with Gorenstein index $m\geq 1$. An exceptional prime divisor E over X has discrepancy $a(E,X)$, and its associated valuation is de...

conjecture | open | MathDB 页面 | 原文线索

V2-006129 | MATHDB-372183

The csc determinant conjecture for odd primes

摘录 / EXCERPT: Cosecant determinant conjecture. The quantity c_p is an integer. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-006130 | MATHDB-372185

Interior-region decay conjecture for the Fourier interpolation basis functions

摘录 / EXCERPT: Let $f_n(x)$ be the Fourier interpolation basis functions, and let $\varepsilon,\delta>0$. Consider the region $\varepsilon n\le x\le n-\sqrt n$. Interior-region decay conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-006131 | MATHDB-372186

Polylogarithmic growth conjecture for the auxiliary function Psi

摘录 / EXCERPT: Polylogarithmic growth conjecture. There exists $k>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006132 | MATHDB-372230

Watanabe-Yoshida conjecture on Hilbert-Kunz multiplicities of simple singularities

摘录 / EXCERPT: Let p be a prime number and define ... Let E_n denote the Euler number, equivalently the number of linear extensions of the zigzag poset of size n. Watanabe-Yoshida conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-006133 | MATHDB-372235

The dense rational non-freeness conjecture

摘录 / EXCERPT: Let $G_{\alpha}=\left\langle \begin{bmatrix}1&1\\0&1\end{bmatrix},\begin{bmatrix}1&0\\\alpha&1\end{bmatrix}\right\rangle<\operatorname{SL}_2(\mathbb{C})$, and call a parameter non-f...

conjecture | open | MathDB 页面 | 原文线索

V2-006134 | MATHDB-372247

Emerton's equidimensionality conjecture for the big Hecke algebra

摘录 / EXCERPT: Emerton's conjecture. The ring $\mathbb{T}$ is equidimensional of Krull dimension 4.

conjecture | open | MathDB 页面 | 原文线索

V2-006135 | MATHDB-372248

Zariski density of modular points on deformation components

摘录 / EXCERPT: Zariski-density conjecture. If C contains an irreducible one-dimensional point, then the set of modular points contained in C is Zariski dense.

conjecture | open | MathDB 页面 | 原文线索

V2-006136 | MATHDB-372260

The symmetric 4-term amicable-pair conjecture

摘录 / EXCERPT: Let $A=2^n abcd$ and $B=2^n efgh$ be a symmetric higher-order amicable pair, with a,b,c,d,e,f,g,h denoting the relevant prime factors. Define x and y through the normalized e...

conjecture | open | MathDB 页面 | 原文线索

V2-006137 | MATHDB-372267

The divisibility conjecture for equiconstructible equilateral triangles

摘录 / EXCERPT: Let (a,b,c) be the side lengths of the triangular tile, and say that a positive real number X is equiconstructible by (a,b,c) when an equilateral triangle with side length...

conjecture | open | MathDB 页面 | 原文线索

V2-006138 | MATHDB-372268

The tile-count divisibility conjecture for $(2\alpha,2\beta,\alpha+\beta)$ targets

摘录 / EXCERPT: Let (a,b,c) be the side lengths of the triangular tile, with angles determined by a,b,c, and let T be a triangle with angles $(2\alpha,2\beta,\alpha+\beta)$. Tile-count divis...

conjecture | open | MathDB 页面 | 原文线索

V2-006139 | MATHDB-372269

The converse ideal-trapezoid tileability conjecture

摘录 / EXCERPT: Let $ABCD$ be an ideal trapezoid, and let (a,b,c) be a triple of pairwise coprime integers satisfying ... Write x and y for the side parameters of the ideal trapezoid as in the...

conjecture | open | MathDB 页面 | 原文线索

V2-006140 | MATHDB-372292

Codimension-two refined estimate conjecture

摘录 / EXCERPT: Codimension-two conjecture. For all sufficiently large even $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-006141 | MATHDB-372302

Harper's conjecture on moments of Möbius character and zeta sums

摘录 / EXCERPT: Harper's conjecture. For all $0\leq q\leq 1$ and any fixed $A>0$, for large prime r,

conjecture | open | MathDB 页面 | 原文线索

V2-006142 | MATHDB-372318

Hooley's Hypothesis R for incomplete Kloosterman sums

摘录 / EXCERPT: Let $ u_1< u_2$ be positive integers, let n be a positive integer, let b be an integer coprime to n, and let $\bar a$ denote the multiplicative inverse of a modulo n. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-006143 | MATHDB-372328

Toroczkai's existence conjecture for prime gap graphs

摘录 / EXCERPT: Let p_n be the n-th prime, with $p_0=1$, and let ... be the first n prime gap sequence. A prime gap graph on $n\geq 2$ vertices is a simple graph whose vertex degrees are exact...

conjecture | open | MathDB 页面 | 原文线索

V2-006144 | MATHDB-372356

Erdős's eventual-time conjecture for quadratic character sums

摘录 / EXCERPT: Erdős's eventual-time conjecture. There exists a constant $C_\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006145 | MATHDB-372380

A two-sided inequality for the zeta function, digamma function, and cotangent

摘录 / EXCERPT: Let $B_1(x)$ be the first Bernoulli polynomial, let ψ denote the digamma function, and let $\gamma=-\psi(1)$ be the Euler-Mascheroni constant. Define ... For $0<x<1$, the inequa...

conjecture | open | MathDB 页面 | 原文线索

V2-006146 | MATHDB-372402

The optimal upper-bound conjecture for Farey sequences of $\Gamma_0(p^n)$

摘录 / EXCERPT: Let p be a prime and let $n\geq 3$. A Farey sequence of $\Gamma_0(p^n)$ is denoted by $\{e_i\}$. Optimal upper-bound conjecture. The group $\Gamma_0(p^n)$ has a Farey sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-006147 | MATHDB-372421

Euclidean-dual distance conjecture for the cyclic code generated by $Q_n(x)Q_1(x)$

摘录 / EXCERPT: Let n be a composite positive integer, let q be a prime power with $\operatorname{char}(\mathbb{F}_q)\nmid n$, and let $\mathcal{C}_{n,1}$ be the cyclic code over...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006148 | MATHDB-372456

Borwein-Girgensohn parity conjecture for multiple zeta values

摘录 / EXCERPT: Let $r\textgreater 1$ and let ${\boldsymbol{\sl{k}}}=(k_1,\ldots,k_r)\in(\mathbb{Z}_{>0})^r$ with $k_r>1$. The Borwein-Girgensohn parity conjecture. The multiple zeta value...

conjecture | open | MathDB 页面 | 原文线索

V2-006149 | MATHDB-372457

Nonexistence of nontrivial prime gradings on matrix algebras

摘录 / EXCERPT: Nonexistence conjecture. No nontrivial G-grading Γ on $M_n(K)$ satisfies the primeness property for graded central polynomials.

conjecture | open | MathDB 页面 | 原文线索

V2-006150 | MATHDB-372517

Generalized Fermat's conjecture for compatible systems of endomorphisms

摘录 / EXCERPT: Generalized Fermat's conjecture. If there is $N_1\geqslant N_0$ such that $Y_N(K)$ is finite for every integer $N\geqslant N_1$, then there exists $N_2\geqslant N_0$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-006151 | MATHDB-372518

Multi-indexed generalized Fermat's conjecture

摘录 / EXCERPT: Multi-indexed generalized Fermat's conjecture. If there is $I_1\geqslant I_0$ such that $Y_I(K)$ is finite for every $I\geqslant I_1$, then there exists $I_2\geqslant I_0$ such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-006152 | MATHDB-372545

Henry's inequality conjecture for the Riemann zeta and digamma functions

摘录 / EXCERPT: Let $\b0$ denote the Euler-Mascheroni constant, and define ... For $0<s<1$, the Riemann zeta function $\zeta(s)$ and the digamma function $\psi(s)=\frac{\Gamma'(s)}{\Gamma(s)}$ satis...

conjecture | open | MathDB 页面 | 原文线索

V2-006153 | MATHDB-372553

The central binomial coefficient divisibility conjecture

摘录 / EXCERPT: Let n be an integer greater than 4, and consider the central binomial coefficient $\binom{2n}{n}$. Divisibility conjecture. The central binomial coefficient $\binom{2n}{n}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-006154 | MATHDB-372554

Erdős's valuation growth conjecture for central binomial coefficients

摘录 / EXCERPT: Erdős's valuation growth conjecture. The p-adic valuation of the central binomial coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006155 | MATHDB-372562

Ball's weak cylinder conjecture for point sets in finite affine 3-space

摘录 / EXCERPT: Let $S\subseteq\mathbb{F}_p^3$ be a set of p^2 points, where p is prime. The cylinder condition means that S is the union of p pairwise disjoint parallel lines, and hence h...

conjecture | open | MathDB 页面 | 原文线索

V2-006156 | MATHDB-372563

Ball's strong cylinder conjecture for p-divisible sets in finite affine 3-space

摘录 / EXCERPT: Let $S\subseteq\mathbb{F}_p^3$ be a set of p^2 points, where p is prime. The set S is p-divisible if every affine hyperplane in $\mathbb{F}_p^3$ intersects S in a number...

conjecture | open | MathDB 页面 | 原文线索

V2-006157 | MATHDB-372564

Rédei's additive decomposition conjecture in three-dimensional prime vector spaces

摘录 / EXCERPT: Let $A,B\subseteq\mathbb{F}_p^3$ be subsets with $0\in A$ and $0\in B$. The notation $A+B$ denotes the sumset $\{a+b:a\in A,\ b\in B\}$, and $\langle A\rangle$ denotes the linear s...

conjecture | open | MathDB 页面 | 原文线索

V2-006158 | MATHDB-372573

Granville-Pappalardi conjecture for prime values with power bases

摘录 / EXCERPT: Let $a,b\geqslant 2$ be integers, and choose maximal positive integers k and ℓ such that $a=A^k$ and $b=B^\ell$. Granville-Pappalardi's power-base conjecture. There are infi...

conjecture | open | MathDB 页面 | 原文线索

V2-006159 | MATHDB-372582

Quadratic lower-bound conjecture for k-Pisano-Legendre primes

摘录 / EXCERPT: Quadratic lower-bound conjecture. For every such a and b, there exists a real number c with $0<c<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006160 | MATHDB-372583

Infinitude conjecture for admissible Pisano-Legendre indices

摘录 / EXCERPT: Let a and b be integers with $(a,b)\neq(0,0)$, and let $\chi(a,b)$ be the set of integers k for which a k-Pisano-Legendre prime relative to (a,b) exists. Infinitude conje...

conjecture | open | MathDB 页面 | 原文线索

V2-006161 | MATHDB-372584

Infinitely many zero-Pisano-Legendre primes conjecture

摘录 / EXCERPT: Infinitely many zero-Pisano-Legendre primes conjecture. If $0\in\chi(a,b)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-006162 | MATHDB-372622

Inverse-logarithmic asymptotic conjecture for the spectral error

摘录 / EXCERPT: Let κ parametrize the operators $D_{\log}^{(\kappa,\kappa)}$, and let $\mathcal{E}(\kappa)$ denote the numerical error between their eigenvalues and the imaginary parts of t...

conjecture | open | MathDB 页面 | 原文线索

V2-006163 | MATHDB-372631

Extremal Sylow p-subgroups conjecture for p-singular families of general linear groups

摘录 / EXCERPT: Let $p\in\mathbb{N}$ be prime, let q be a power of p, and let $\mathrm{GL}_{n,q}$ denote the general linear group over the field with q elements. A family is p-singular if...

conjecture | open | MathDB 页面 | 原文线索

V2-006164 | MATHDB-372632

Erdős's conjecture on powers of 2 with digit omission in ternary

摘录 / EXCERPT: For a positive integer m, write its ternary expansion $(m)_3$ using digits from $\{0,1,2\}$. Erdős's conjecture. There are only finitely many powers of 2 whose ternary expansio...

conjecture | open | MathDB 页面 | 原文线索

V2-006165 | MATHDB-372647

Keating-Snaith conjecture for mixed moments of zeta derivatives

摘录 / EXCERPT: Let $n_{1}>n_{2}\geq 0$ and $l\geq 1$, $h\geq 0$ be integers with $h\leq l$. Let c_l be the arithmetic factor in the zeta-moment asymptotic, and let $a_{h,l}(n_{1},n_{2})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006166 | MATHDB-372661

Schinzel's conjecture on the number of equal-sum-and-product solutions

摘录 / EXCERPT: Schinzel's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006167 | MATHDB-372670

Kurihara's nonvanishing conjecture for modular symbols

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be the elliptic curve in the setup, let p be an odd prime satisfying the irreducibility and Manin-constant hypotheses, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006168 | MATHDB-372684

Infinite admissible lengths conjecture for consecutive prime sums

摘录 / EXCERPT: Let p_n denote a prime number, and for each integer $k\geq 1$ define ... where p_{n+i} are consecutive primes. Infinite admissible lengths conjecture. For every prime number...

conjecture | open | MathDB 页面 | 原文线索

V2-006169 | MATHDB-372689

The prime-between-consecutive-cubes conjecture

摘录 / EXCERPT: For each integer $n\geq 1$, consider the interval $(n^3,(n+1)^3)$.
Prime-between-consecutive-cubes conjecture. For all $n\geq 1$, there exists a prime between n^3 and $(n+1)^3$....

conjecture | open | MathDB 页面 | 原文线索

V2-006170 | MATHDB-372723

Generic exponential lower-bound conjecture for the Diophantine parameter of random isometries

摘录 / EXCERPT: Let $R_{\mathrm{DPV}}(n,\theta)$ denote the parameter appearing in the Diophantine condition $\mathrm{DPV}(n,R)$ for an angle $\theta\in[0,2\pi)$. The paper proves, when...

conjecture | open | MathDB 页面 | 原文线索

V2-006171 | MATHDB-372731

The binary-prefix characterization of as a generalized eigenvalue of LCM and GCD matrices

摘录 / EXCERPT: Let $T=\{1,\dots,n\}$ with $n>3$. Write ${\bf L}_T$ and ${\bf G}_T$ for the LCM and GCD matrices indexed by T, respectively, and call λ a g-eigenvalue of ${\bf L}_T$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-006172 | MATHDB-372732

Verstraëte's asymptotic conjecture for polynomial product representations

摘录 / EXCERPT: Verstraëte's conjecture. For some positive constant $\rho=\rho(k,h)$ depending only on k and h, either

conjecture | open | MathDB 页面 | 原文线索

V2-006173 | MATHDB-372734

The weak form of Szpiro's conjecture

摘录 / EXCERPT: Weak form of Szpiro's conjecture. There exists a constant $s>0$ such that, for mutually coprime integers A, B, and C satisfying $A+B=C$,

conjecture | open | MathDB 页面 | 原文线索

V2-006174 | MATHDB-372760

Merca-Ono-Tsai finiteness conjecture for near-perfect powers of partition numbers

摘录 / EXCERPT: Merca-Ono-Tsai's finiteness conjecture. For fixed integers $k>1$ and $d\geq 0$, there are at most finitely many n for which $\Delta_k(n)\leq d$. This finiteness assertion is used...

conjecture | open | MathDB 页面 | 原文线索

V2-006175 | MATHDB-372762

Merca-Ono-Tsai stabilization conjecture for partition numbers

摘录 / EXCERPT: For each fixed integer $d\geq 0$, define ... and let $M_k(d)$ be the largest n for which $p(n)$ is within d of a kth power. Merca-Ono-Tsai's stabilization conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-006176 | MATHDB-372763

Equidistribution conjecture for roots of partition numbers

摘录 / EXCERPT: For each positive integer n, let $p(n)$ denote the partition function, and let $\{r\}$ denote the fractional part of a real number r. The equidistribution conjecture. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-006177 | MATHDB-372770

The Brauer-Siegel conjecture for families of number fields

摘录 / EXCERPT: Brauer-Siegel conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-006178 | MATHDB-372786

The anti-conjecture to unknotting additivity

摘录 / EXCERPT: Anti-conjecture to the Unknotting Additivity Conjecture. For every non-trivial knot K, there is a symbiont knot $K^\prime$ for which

conjecture | refuted | MathDB 页面 | 原文线索

V2-006179 | MATHDB-372810

Limit g-vectors and prime non-real basis elements for Gr(3,9) and Gr(4,8)

摘录 / EXCERPT: Limit-vector and prime-element conjecture. All limit g-vectors for $\mathbb{C}[\operatorname{Gr}(3,9)]$ and $\mathbb{C}[\operatorname{Gr}(4,8)]$ can be obtained by the braid-group...

conjecture | open | MathDB 页面 | 原文线索

V2-006180 | MATHDB-372811

The uniqueness conjecture for extending Diophantine triples

摘录 / EXCERPT: Uniqueness conjecture for extensions. Then

conjecture | open | MathDB 页面 | 原文线索

V2-006181 | MATHDB-372814

The prime-field planar induced matching conjecture

摘录 / EXCERPT: Let $\mathrm{IM}(2,q)$ denote the maximum size of an induced matching in the point-line incidence graph of $\mathbb{F}_q^2$. Prime-field planar induced matching conjecture. There e...

conjecture | open | MathDB 页面 | 原文线索

V2-006182 | MATHDB-372828

Neumann-Praeger conjecture on Kronecker classes

摘录 / EXCERPT: Let G be a finite group with subgroups U and U', and let $n=|G:U'|$. For a subgroup V of G, write $V^g=g^{-1}Vg$ and let $\bigcup_{g\in G}V^g$ denote the union of its con...

conjecture | open | MathDB 页面 | 原文线索

V2-006183 | MATHDB-372841

Pointwise global Katz-Sarnak conjecture for $\mathcal M_g$ and $\mathcal A_g$

摘录 / EXCERPT: For each prime l, let $v_l(t)$ denote the local factor defined from the trace distribution in $\operatorname{GSp}_{2g}(\mathbb{Z}_l)$, with $v_p(t)=1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006184 | MATHDB-372845

Projective Fourier coefficient vanishing conjecture for quaternionic modular forms on G

摘录 / EXCERPT: Projective Fourier coefficient vanishing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-006185 | MATHDB-372846

Saito-Kurokawa type lift conjecture for quaternionic Hecke eigenforms

摘录 / EXCERPT: Let h be a cuspidal Hecke eigenform on $\mathbf{SL}_{2}(\mathbb{Z})$ of weight $2\ell-n+2$, and let F_h be the function on $\mathbf{G}(\mathbb{A})$ whose constant term along...

conjecture | open | MathDB 页面 | 原文线索

V2-006186 | MATHDB-372849

Refinement of the Craig-van Ittersum-Ono conjecture for prime-detecting quasimodular forms

摘录 / EXCERPT: Let N be a positive integer. Let $\beta\widetilde{E}(N)$ denote the space of quasimodular Eisenstein forms on $\Gamma_{0}(N)$, let Ω_{N} denote the space of prime-detectin...

conjecture | open | MathDB 页面 | 原文线索

V2-006187 | MATHDB-372856

Generalised Euclid-Mullin conjecture

摘录 / EXCERPT: Let $a,m\in\mathbb{N}$ satisfy ... Let $\operatorname{GEM}(a,m)$ be the class of polynomials introduced in the paper, and let $\operatorname{GEM}_1(f;a,m)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006188 | MATHDB-372889

The mod-4 valuation law for Gaussian power sums

摘录 / EXCERPT: Let p be an odd prime and define ... with $v_p(x+yi):=\min\{v_p(x),v_p(y)\}$. The mod-4 valuation conjecture. For $1\le n\le p-2$, ... and for $4\le n\le p-2$, ... These formulas are s...

conjecture | open | MathDB 页面 | 原文线索

V2-006189 | MATHDB-372890

Odd-multiple valuation conjecture for inert Gaussian power sums

摘录 / EXCERPT: Let $p\ge7$ be a prime with $p\equiv3\pmod4$, and write $n=r(p-1)$ with $r\ge1$. Odd-multiple valuation conjecture. ... Equivalently, among multiples of $p-1$, ... for $p\equiv3\pmod4$...

conjecture | open | MathDB 页面 | 原文线索

V2-006190 | MATHDB-372904

Infinite five-square progressions over quadratic extensions of quadratic fields

摘录 / EXCERPT: Let K_1 be a quadratic extension of $\mathbb{Q}(\sqrt{-2})$ and K_2 a quadratic extension of $\mathbb{Q}(\sqrt{2})$. An arithmetic progression of squares is non-constant if its...

conjecture | open | MathDB 页面 | 原文线索

V2-006191 | MATHDB-372919

Conjecture on multiplicative-order scarcity and exponential-sum cancellation

摘录 / EXCERPT: Exponential-sum and small-order conjecture. There exists a non-increasing mapping $f:\mathbb{N}\to\mathbb{R}_+$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-006192 | MATHDB-372932

The first limiting conjecture for eta-derivative determinants

摘录 / EXCERPT: The first limiting conjecture. Except for countably many values of a,

conjecture | open | MathDB 页面 | 原文线索

V2-006193 | MATHDB-372933

The root-concentration conjecture for eta-derivative polynomials

摘录 / EXCERPT: The root-concentration conjecture. Except for countably many values of a, for every positive integer l and non-negative integer m, the maximal distance between the roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-006194 | MATHDB-372934

The coefficient-ratio conjecture for eta-derivative polynomials

摘录 / EXCERPT: The coefficient-ratio conjecture. Except for countably many values of a, for every positive integer l, non-negative integer m, and integer n satisfying $0<n\leq m+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-006195 | MATHDB-372935

The Taylor-coefficient limiting ratio conjecture for eta derivatives

摘录 / EXCERPT: The Taylor-coefficient limiting ratio conjecture. Except for countably many values of a,

conjecture | open | MathDB 页面 | 原文线索

V2-006196 | MATHDB-372936

The minor-ratio conjecture for eta-derivative determinants

摘录 / EXCERPT: The minor-ratio conjecture. If $d_0=f_0=0$ and both omitted sets are finite, then, except for countably many values of a,

conjecture | open | MathDB 页面 | 原文线索

V2-006197 | MATHDB-372940

The parity conjecture for the counting function $N_k(n)$

摘录 / EXCERPT: Parity conjecture. The number $N_k(n)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006198 | MATHDB-372961

Max formula for cyclic covering dimensions of invariant direct sums

摘录 / EXCERPT: Let q be a prime power, let σ be an operator with $\sigma^n=\operatorname{id}$, and let W_1 and W_2 be σ-invariant subspaces satisfying ... Write $h_{\sigma}(V)$ for...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006199 | MATHDB-372972

Erdős-Kac conjecture for integer parts of polynomials with an irrational coefficient

摘录 / EXCERPT: Erdős-Kac conjecture for integer parts of polynomials. The random variable X_N converges in distribution to the standard Gaussian $\mathcal{N}(0,1)$ as $N\rightarrow\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-006200 | MATHDB-372973

Quantitative Erdős-Kac conjecture for Beatty sequences

摘录 / EXCERPT: Quantitative Erdős-Kac conjecture for Beatty sequences. The distance between X_N and the standard Gaussian $\mathcal{N}(0,1)$ is bounded above by

conjecture | open | MathDB 页面 | 原文线索

V2-006201 | MATHDB-372974

Quantitative Erdős-Kac conjecture for irreducible quadratic polynomials

摘录 / EXCERPT: Quantitative Erdős-Kac conjecture for irreducible quadratics. The Kolmogorov distance between X_N and the standard Gaussian $\mathcal{N}(0,1)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-006202 | MATHDB-372980

Miller et al.'s conjecture on infinite square-free digit walks

摘录 / EXCERPT: Miller et al.'s conjecture. There exists an infinite sequence of square-free integers

conjecture | open | MathDB 页面 | 原文线索

V2-006203 | MATHDB-373009

The Lang-Vojta conjecture for integral points

摘录 / EXCERPT: The Lang-Vojta conjecture. Under these hypotheses, every set of (D,S)-integral points in X is Zariski degenerate. These problems are central in Diophantine geometry and, for th...

conjecture | open | MathDB 页面 | 原文线索

V2-006204 | MATHDB-373024

Erdős-Turán conjecture for irreducible-in-variables two-variable polynomials

摘录 / EXCERPT: Erdős-Turán conjecture for two-variable polynomials. There exists a constant c depending only on f such that

conjecture | open | MathDB 页面 | 原文线索

V2-006205 | MATHDB-373030

Infinitely many rational elliptic curves with 2-torsion over quadratic fields

摘录 / EXCERPT: Let p be a prime satisfying ... For an elliptic curve $E/\mathbb Q$, write $E(\mathbb Q(\sqrt{-p}))_{\mathrm{tors}}$ for its torsion subgroup over $\mathbb Q(\sqrt{-p})$. Existence...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006206 | MATHDB-373031

Bachman-Bao-Wu's coefficient-solubility conjecture for ternary cyclotomic polynomials

摘录 / EXCERPT: For an odd prime p, define ... where q and r range over primes. Bachman-Bao-Wu's conjecture. For every odd prime p and every integer h with $1\le h\le M(p)$, the equation .....

conjecture | open | MathDB 页面 | 原文线索

V2-006207 | MATHDB-373034

Mock modularity conjecture for log Gromov-Witten series on the mirror to $\mathbb{P}^2$

摘录 / EXCERPT: Let (Y,D) be the rational elliptic surface mirror to $\mathbb{P}^2$. For each $r\in\mathbb{Z}_{>0}$ and each integer c_1 with $-r<c_1\leq 0$, let $h^{GW}_{r,c_1}(\tau)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-006208 | MATHDB-373035

The dimension conjecture for multiple Eisenstein series

摘录 / EXCERPT: Let $\mathcal{E}_k$ be the weight-k component of the algebra of multiple Eisenstein series. Define the generating series for modular forms and cusp forms by ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-006209 | MATHDB-373062

Non-coprime-power obstruction to tree-pair representations of Bieri-Strebel groups

摘录 / EXCERPT: Non-coprime-power conjecture. This Bieri-Strebel group does not have a well-defined tree-pair representation.

conjecture | open | MathDB 页面 | 原文线索

V2-006210 | MATHDB-373076

The abc conjecture for coprime non-zero integers

摘录 / EXCERPT: For a non-zero integer a, let $N(a)$ be its algebraic radical, defined by ... For every $[?]$...

conjecture | open | MathDB 页面 | 原文线索

V2-006211 | MATHDB-373083

The optimal Sturm bound conjecture for quasimodular forms

摘录 / EXCERPT: Optimal Sturm bound conjecture. The form f is determined uniquely by its first $\dim \widetilde{M}_{2k}$ Fourier coefficients. Furthermore, for every integer $m\geq 2$, the reduc...

conjecture | open | MathDB 页面 | 原文线索

V2-006212 | MATHDB-373091

Han-Ren conjecture on the length of MDS elliptic codes

摘录 / EXCERPT: Let q be a prime power, let $E/\mathbb{F}_q$ be an elliptic curve, and let P be a set of pairwise distinct points of $E(\mathbb{F}_q)$. Let D be an $\mathbb{F}_q$-rational di...

conjecture | open | MathDB 页面 | 原文线索

V2-006213 | MATHDB-373095

Monotonicity conjecture for extremal depth-one quasimodular forms

摘录 / EXCERPT: Let w be an even integer with $w \ge 6$ and $w \ne 8,10$, and let $X_{w,1}$ denote the extremal quasimodular form of weight w and depth 1. Monotonicity conjecture. For all ev...

conjecture | open | MathDB 页面 | 原文线索

V2-006214 | MATHDB-373096

Complete positivity conjecture for the quasimodular forms $Y_{w,2}$

摘录 / EXCERPT: Let $X_{w,2}(z)$ be the depth-two extremal quasimodular form of weight w, and define ... A quasimodular form is completely positive when all coefficients in the relevant complete-p...

conjecture | open | MathDB 页面 | 原文线索

V2-006215 | MATHDB-373102

Convergence conjecture for non-degenerate manifolds in affine subspaces

摘录 / EXCERPT: Let $\mathcal L\subset\mathbb{R}^n$ be an affine subspace satisfying Hypothesis D. Let $\mathcal M\subset\mathcal L$ be a smooth manifold which is non-degenerate when viewed as a m...

conjecture | open | MathDB 页面 | 原文线索

V2-006216 | MATHDB-373151

The odd-prime level bound using the last two invariant factors of the walk matrix

摘录 / EXCERPT: Let G be a controllable graph, let p be an odd prime factor of $\det W(G)$, and let d_{n-1} and d_n be the last two invariant factors of the walk matrix $W(G)$. Write v_p...

conjecture | open | MathDB 页面 | 原文线索

V2-006217 | MATHDB-373170

Deep congruences between twins of newforms

摘录 / EXCERPT: Let f be an admissible newform of weight $k>2$ and level $\Gamma_0(p)$, and suppose that p splits completely in its Hecke field K_f. Write $\varepsilon(f)$ for its Atkin--Leh...

conjecture | open | MathDB 页面 | 原文线索

V2-006218 | MATHDB-373171

Calegari--Stein opposite Atkin--Lehner sign conjecture

摘录 / EXCERPT: Let $k<p-1$, and let f and g be two newforms of level $\Gamma_0(p)$. Write $\varepsilon(f)$ and $\varepsilon(g)$ for their Atkin--Lehner signs, and say that the forms are congr...

conjecture | open | MathDB 页面 | 原文线索

V2-006219 | MATHDB-373177

Prime-order divisor conjecture for automorphisms of Kimura Hadamard matrices

摘录 / EXCERPT: Let H be a Kimura Hadamard matrix (KHM) of order $4(2k)+4$, and let $(R,S)\in\operatorname{Aut}(H)$ be an automorphism of prime order p. Prime-order divisor conjecture. Then .....

conjecture | open | MathDB 页面 | 原文线索

V2-006220 | MATHDB-373183

Tate's symplecticity conjecture for the Brauer group of a surface

摘录 / EXCERPT: Let X be a smooth, proper, geometrically connected surface over a finite field of characteristic p. Write ... for its Brauer group, let $\operatorname{Br}(X)_{\nd}$ be the quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-006221 | MATHDB-373215

Conjecture on the smallest positive zero of the transformed polynomial \widetilde\Xi_n

摘录 / EXCERPT: Let $(\Lambda_n)_{n\ge1}$ be the family of polynomials defined by the paper's explicit-polynomial construction, and set ... For each n, let $0<\beta_n<1$ denote the smallest positi...

conjecture | open | MathDB 页面 | 原文线索

V2-006222 | MATHDB-373240

Chebyshev-Artin primitive-root conjecture

摘录 / EXCERPT: For a prime p, let R_p denote the relevant Chebyshev residue structure, and let ω_a be the element associated with $a\in R_p$. A Chebyshev primitive root is an element...

conjecture | open | MathDB 页面 | 原文线索

V2-006223 | MATHDB-373253

Serre's modularity conjecture for weight two over number fields

摘录 / EXCERPT: Let K be a number field, let p be a rational prime unramified in K, and let G_K denote the absolute Galois group of K. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006224 | MATHDB-373254

Eichler-Shimura-type conjecture for weight two eigenforms over number fields

摘录 / EXCERPT: Let K be a number field, let $\mathfrak N$ be an ideal of $\mathcal O_K$, and let $\mathfrak f$ be a non-trivial, new, weight two complex eigenform over K of level...

conjecture | open | MathDB 页面 | 原文线索

V2-006225 | MATHDB-373258

Lupu's conjectural formula for $H(a,b)$

摘录 / EXCERPT: Let $a,b\in\mathbb{Z}_{\geq 0}$, and let $H(a,b)$ denote the quantity used in Lupu's Zagier-type formula for multiple t-values. Lupu's conjecture for $H(a,b)$. For nonnegative in...

conjecture | open | MathDB 页面 | 原文线索

V2-006226 | MATHDB-373259

Lupu's conjectural formula for $T(a,b)$

摘录 / EXCERPT: Let $a,b\in\mathbb{Z}_{\geq 0}$, and define ... where $t(s_1,\ldots,s_r)$ is the multiple t-value over positive odd integers. Lupu's conjecture for $T(a,b)$. For nonnegative intege...

conjecture | open | MathDB 页面 | 原文线索

V2-006227 | MATHDB-373266

Conjecture on pole cancellation in the Nekrasov instanton sum with adjoint matter

摘录 / EXCERPT: Let p,q be positive coprime integers, let $b^2=p/q$, and let $\boldsymbol{Y}$ range over tuples of partitions with corresponding terms $Z_{\vec{Y}}$. Let ϵ_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-006228 | MATHDB-373276

Graham's conjecture on valid orderings in cyclic groups of prime order

摘录 / EXCERPT: Let p be a prime, and let $A\subseteq\mathbb{Z}_p\setminus\{0\}$. An ordering $a_1,\ldots,a_{|A|}$ of A is valid if its partial sums ... are pairwise distinct. Graham's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-006229 | MATHDB-373290

Generalized Sárközy conjecture on additive decompositions of multiplicative subgroups

摘录 / EXCERPT: Let $d\geq 2$ be fixed, let q be a prime power with $q\equiv 1\pmod d$, and let $G\subseteq {\mathbb F}_q^*$ be the multiplicative subgroup of index d. Generalized Sárközy conj...

conjecture | open | MathDB 页面 | 原文线索

V2-006230 | MATHDB-373291

Lev and Sonn's difference-set conjecture for finite-field squares

摘录 / EXCERPT: Let p be a prime, and let $\mathcal{R}_p$ denote the set of nonzero squares in ${\mathbb F}_p$. Lev and Sonn's conjecture. For all sufficiently large primes p, there is no subs...

conjecture | open | MathDB 页面 | 原文线索

V2-006231 | MATHDB-373292

KYY's multiplicative decomposition conjecture for shifted subgroups

摘录 / EXCERPT: Let $d\geq 2$ be fixed, let q be a sufficiently large prime power with $q\equiv 1\pmod d$, and let $G\subseteq {\mathbb F}_q^*$ be the multiplicative subgroup of index d. For s...

conjecture | open | MathDB 页面 | 原文线索

V2-006232 | MATHDB-373295

Serre's uniformity question for elliptic-curve Galois representations

摘录 / EXCERPT: Serre's uniformity question. There exists a bound C such that for every non-CM elliptic curve $E/\mathbb{Q}$ and every prime $\ell>C$, the representation $\rho_{E,\ell}$ is surje...

open_problem | open | MathDB 页面 | 原文线索

V2-006233 | MATHDB-373308

The extremal spanning-tree conjecture for C_4-free graphs

摘录 / EXCERPT: Let q be a prime power and let $n=q^2+q+1$. For an n-vertex C_4-free graph G, write $\tau(G)$ for its number of spanning trees, and let $\mathrm{st}(n,C_4)$ denote the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-006234 | MATHDB-373310

Conjecture on Euler liars and Carmichael or Sophie Germain pseudoprimes

摘录 / EXCERPT: Let n be an odd squarefree composite integer, and let Euler liars in $\mathbb{Z}_n^*$ mean the elements a satisfying the Euler probable-prime congruence...

conjecture | open | MathDB 页面 | 原文线索

V2-006235 | MATHDB-373320

Legendre's conjectural prime recursion for a Mills-type construction

摘录 / EXCERPT: Define $q_1=2$, and for each positive integer n, let q_{n+1} be the least prime exceeding q_n^2. Legendre's recursive prime-gap assertion. Then ... for all n. This assertion...

conjecture | open | MathDB 页面 | 原文线索

V2-006236 | MATHDB-373331

The Dirichlet L-function analogue of the Hughes-Keating-O'Connell moment conjecture

摘录 / EXCERPT: Dirichlet L-function moment conjecture. For all $r\in\mathbb{C}$ with $\Re(r)>-3/2$, there is a constant $C_{q,r}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006237 | MATHDB-373332

Ng's large-deviation conjecture for the summatory Möbius function

摘录 / EXCERPT: Let $M(x)=\sum_{n\leq x}\mu(n)$ denote the summatory Möbius function. Ng's large-deviation conjecture. There exists a constant $B>0$ such that ... This conjecture concerns the precis...

conjecture | open | MathDB 页面 | 原文线索

V2-006238 | MATHDB-373333

The Linear Independence Conjecture for ordinates of zeta zeros

摘录 / EXCERPT: Linear Independence Conjecture. This set is linearly independent over $\mathbb{Q}$. The conjecture is used in the paper's large-deviation arguments for the summatory functions unde...

conjecture | open | MathDB 页面 | 原文线索

V2-006239 | MATHDB-373335

Sarnak's volume-approximation conjecture for polynomial level sets

摘录 / EXCERPT: Let $F=(F_1,\ldots,F_p):\mathbb{R}^n\to\mathbb{R}^p$ be a polynomial map of degree d, let $K\subseteq\mathbb{R}^n$ be a compact set satisfying (P1), and let $T>0$ and α be...

conjecture | open | MathDB 页面 | 原文线索

V2-006240 | MATHDB-373352

Pappas-Rapoport-Smithling topological flatness conjecture for naive local models

摘录 / EXCERPT: Let F/F_0 be a quadratic extension, let (V,ϕ) be a symmetric space of dimension $2n+2$, and let $I\subset [0,n]$ be non-empty. Let $\RM^\naive_I$ be the naive local model ov...

conjecture | open | MathDB 页面 | 原文线索

V2-006241 | MATHDB-373353

Pappas-Rapoport flatness conjecture for spin local models at parahoric level

摘录 / EXCERPT: Let F/F_0 be a quadratic extension, let (V,ϕ) be a symmetric space of dimension $2n+2$, and let $I\subset [0,n]$ be non-empty. Let $\RM^\pm_I$ be the spin local model over...

conjecture | open | MathDB 页面 | 原文线索

V2-006242 | MATHDB-373377

The existence of composite weakly almost-prime and almost-prime numbers

摘录 / EXCERPT: For each positive integer n, let ... where $1=d_1<d_2<\dots<d_k=n$ are all divisors of n. A positive integer n is weakly almost-prime if $n\mid T_n(x)$ for every integer x; i...

open_problem | open | MathDB 页面 | 原文线索

V2-006243 | MATHDB-373378

The infinitude of composite almost-prime numbers

摘录 / EXCERPT: A positive integer n is almost-prime if it is square-free and satisfies $n\mid T_n(x)$ for every integer x, where ... and $1=d_1<d_2<\dots<d_k=n$ are all divisors of n. Infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-006244 | MATHDB-373379

The distribution of prime divisors in composite almost-prime numbers

摘录 / EXCERPT: Let n be a composite almost-prime number, and write $\omega(n)$ for the number of its prime divisors. Distribution conjecture. There exists an $\epsilon<1$ such that ... is satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-006245 | MATHDB-373414

Prime-restricted asymptotic formula for the maximum packing number

摘录 / EXCERPT: Let $\mathbb{P}(x)$ be the set of prime numbers not exceeding x, and let $M_{\mathbb{P}(n)}(n)$ denote the corresponding prime-restricted maximum packing number. Prime-restricted...

conjecture | open | MathDB 页面 | 原文线索

V2-006246 | MATHDB-373428

The inert-prime congruence for the AESZ operator

摘录 / EXCERPT: Let $A_p(z)$ be the quantity defined from the holomorphic period and its truncated period for the displayed Calabi-Yau operator, and let p be a rational prime. Inert-prime congru...

conjecture | open | MathDB 页面 | 原文线索

V2-006247 | MATHDB-373439

Mock Jacobi correspondence for dual \widehat{Z}-series of four-fibered Seifert homology spheres

摘录 / EXCERPT: Let $\Phi_p^{[j]}(q)^\vee$ be the vector of mock modular forms dual under $q \to 1/q$ to the linear combination of false theta functions $\Phi^{[j]}_p(q)$ associated to the...

conjecture | open | MathDB 页面 | 原文线索

V2-006248 | MATHDB-373457

Finite-width characterization of locally-finite distributive lattices with few ideal components

摘录 / EXCERPT: Let $\mathcal{L}$ be a locally-finite distributive lattice, and let $\mathcal{P}$ be its set of prime filters. Let $\mathcal{I}$ be the lattice of ideals of $\mathcal{P}$. A lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-006249 | MATHDB-373502

Conjecture on the large-length variance of left truncations

摘录 / EXCERPT: For a base $b\geq 2$ and a positive integer ℓ, let Trun^L_b denote the number of left truncations and let $\mathrm{Var}_{b^\ell}(\mathrm{Trun}^L_b)$ be its variance...

conjecture | open | MathDB 页面 | 原文线索

V2-006250 | MATHDB-373503

Conjecture on the large-length variance of polynomial left truncations

摘录 / EXCERPT: Let q be a prime power, let $b\in\mathbb F_q[T]$ have degree $m\geq 1$, and let $\mathrm{Var}_{b^\ell}(\mathrm{Trun}^L_{q,b})$ be the variance defined over polynomials of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-006251 | MATHDB-373504

Conjectural overpartition congruence modulo 19

摘录 / EXCERPT: For a positive integer n, let $\overline{p}(n)$ denote the number of overpartitions of n, and let $\left(\frac{n}{p}\right)$ denote the Legendre symbol for an odd prime p. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-006252 | MATHDB-373506

The Information Conservation conjecture for the essential prime and fractal zero sets

摘录 / EXCERPT: Let P_{ess} be the essential fractal prime set, with information measure $\iota(P_{\text{ess}})=-\zeta(1/2)$, and let Z_F be the fractal zero set constructed from the po...

conjecture | open | MathDB 页面 | 原文线索

V2-006253 | MATHDB-373518

Appleby's conjecture on algebraic-unit normalized overlaps

摘录 / EXCERPT: Let a SIC-POVM in $\mathbb{C}^d$ have normalized overlaps $e^{i\theta_{jl}}$, where the angles θ_{jl} are defined by the normalized inner products of its fiducial vectors. A...

conjecture | open | MathDB 页面 | 原文线索

V2-006254 | MATHDB-373521

Conder's conjecture on simple groups with cosets of p-elements

摘录 / EXCERPT: Let S be a simple group, let p be a prime, and let P be a Sylow p-subgroup of S. A group element is a p-element if its order is a power of p. Conder's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-006255 | MATHDB-373534

OEIS conjecture that the Hofstadter consecutive-sum sequence omits infinitely many integers

摘录 / EXCERPT: Let $(a_n)_{n\ge1}$ be the strictly increasing sequence defined by ... and, for $k\ge3$, let a_k be the least integer greater than a_{k-1} that can be written as a sum of at leas...

conjecture | open | MathDB 页面 | 原文线索

V2-006256 | MATHDB-373547

Monomial congruence counting conjecture

摘录 / EXCERPT: Let a,b,c be non-zero integers with a,b,c setwise coprime, and let $d\geqslant 1$ be another integer. For a prime $q\geqslant 3$, define $N_{a,b,c,d}(L,M,N;q)$ to count triples...

conjecture | open | MathDB 页面 | 原文线索

V2-006257 | MATHDB-373563

The information-rate bound for fiber-57 returns in Collatz orbits

摘录 / EXCERPT: Let $(n_0,n_1,\ldots)$ be a Collatz orbit with $n_0>1$. Let $\tau_1<\tau_2<\cdots$ be successive fiber-57 return times, set $q_j=\lfloor n_{\tau_j}/64\rfloor$, and let I_r be t...

conjecture | open | MathDB 页面 | 原文线索

V2-006258 | MATHDB-373580

Finiteness conjecture for prime critically frustrated signed graphs

摘录 / EXCERPT: A signed graph (G,σ) is a graph with a positive or negative sign on each edge. Its frustration index $l(G,\sigma)$ is the minimum number of negative edges among signatures s...

conjecture | open | MathDB 页面 | 原文线索

V2-006259 | MATHDB-373595

The Hausdorff-dimension conjecture for the exceptional set Θ

摘录 / EXCERPT: Hausdorff-dimension conjecture for $Theta$. The set $Theta$ in Theorem $$ has Hausdorff dimension at least $2/3$.

conjecture | open | MathDB 页面 | 原文线索

V2-006260 | MATHDB-373615

Convergence to Haar measure for elliptic-curve Frobenius types

摘录 / EXCERPT: Let q be a prime power, let r be the degree of the L-polynomial, and let mu_q denote the empirical measure on the unit circle obtained by assigning weight...

conjecture | open | MathDB 页面 | 原文线索

V2-006261 | MATHDB-373621

The Borwein-Bailey-Girgensohn series conjecture

摘录 / EXCERPT: Main conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006262 | MATHDB-373622

Meromorphic continuation conjecture for the function Phi

摘录 / EXCERPT: Meromorphic continuation conjecture. The function $\Phi(s)$ admits a meromorphic continuation to

conjecture | open | MathDB 页面 | 原文线索

V2-006263 | MATHDB-373623

Erdős-Sárközy-Stewart conjecture on prime factors of subset sums

摘录 / EXCERPT: For a finite set $A e\textstyle\varnothing$ of positive integers, let $\Sigma^*(A)$ denote the set of nonempty subset sums, and let $P(A)$ be the greatest prime factor of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006264 | MATHDB-373624

Erdős-Sárközy-Stewart quadratic prime-factor conjecture for subset sums

摘录 / EXCERPT: For a finite set $A\ne\textstyle\varnothing$ of positive integers, let $\Sigma^*(A)$ denote the set of nonempty subset sums, and let $P(A)$ be the greatest prime factor of...

conjecture | open | MathDB 页面 | 原文线索

V2-006265 | MATHDB-373630

The dimension conjecture for finite multiple zeta values

摘录 / EXCERPT: Dimension conjecture. The sequence $(d_w)_{w=0}^{\infty}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006266 | MATHDB-373639

Modularity conjecture for higher Heegner cycles

摘录 / EXCERPT: Modularity conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-006267 | MATHDB-373640

Principality conjecture for algebraic-principal Heegner cycles

摘录 / EXCERPT: Let $\mathrm{Pin}^{\mathrm{alg}}$ be the subgroup of algebraic principal cycles and let Pin be the subgroup of principal cycles in the group of Heegner cycles. Principal...

conjecture | open | MathDB 页面 | 原文线索

V2-006268 | MATHDB-373641

Arithmetic modularity conjecture for higher Heegner cycles

摘录 / EXCERPT: Arithmetic modularity conjecture. The function $\widehat{\phi}_\kappa$ is a cusp form. For $\kappa=1$, the source states that arithmetic modularity on the modular curve $X_0(N)$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-006269 | MATHDB-373643

The fixed-constant-term analogue of van der Waerden's conjecture

摘录 / EXCERPT: Let ... be a polynomial with fixed outer coefficients $a_n=1$ and $a_0=b$, and let $E_{n,b}(H)$ denote the number of such polynomials of height at most H whose Galois group is not...

conjecture | open | MathDB 页面 | 原文线索

V2-006270 | MATHDB-373676

The exponential lower-bound conjecture for prime 2-ball passing patterns

摘录 / EXCERPT: For a fixed integer $k\geq 1$, let $P'(2,n,k)$ denote the number of prime 2-ball, k-hand passing patterns of length n. As n tends to infinity, Exponential lower-bound conject...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006271 | MATHDB-373678

The size bound for the set of mutual divisibility solutions

摘录 / EXCERPT: Size-bound conjecture. For any integer $n>1$,

open_problem | open | MathDB 页面 | 原文线索

V2-006272 | MATHDB-373695

Realization and arithmetic compatibility conjecture for rank 2 hypergeometric motives

摘录 / EXCERPT: Realization and compatibility conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-006273 | MATHDB-373740

Fiorilli's generalized Skewes-number growth conjecture for quadratic residue races

摘录 / EXCERPT: For an integer q, let $\varrho(q)$ denote the number of square roots of 1 modulo q, let $\operatorname{rad}(q)$ be its radical, and let $x_{q;R,NR}$ be the generalized Skewes...

conjecture | open | MathDB 页面 | 原文线索

V2-006274 | MATHDB-373745

Rational multiples of powers of pi among symmetrized Mordell-Tornheim zeta values

摘录 / EXCERPT: Let $\overline{\zeta}_n$ denote the symmetrized Mordell-Tornheim zeta value of order n, and let $n\geq 1$. A value is a rational multiplicity of π^n if it has the form...

conjecture | open | MathDB 页面 | 原文线索

V2-006275 | MATHDB-373746

Monomial-count conjecture for symmetrized Mordell-Tornheim zeta values

摘录 / EXCERPT: Let $\overline{\zeta}_n$ be the symmetrized Mordell-Tornheim zeta value of order n, and let $p(n)$ denote the partition function. A monomial has total weight n when the sum of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006276 | MATHDB-373747

Rational-span conjecture for zeta values and symmetrized Mordell-Tornheim values

摘录 / EXCERPT: Let $\zeta(n)$ denote the Riemann zeta value at the integer n, and let $\overline{\zeta}_m$ denote the symmetrized Mordell-Tornheim zeta value of order m. Rational-span conject...

conjecture | open | MathDB 页面 | 原文线索

V2-006277 | MATHDB-373749

Chinburg's weak and strong conjectures for Mahler measures

摘录 / EXCERPT: Chinburg's conjectures. For every fundamental discriminant $-f<0$, there exists a bivariate rational function $P\in\mathbf{Z}(x,y)$ and a rational number $r\in\mathbf{Q}^{\times}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006278 | MATHDB-373753

The integer complexity conjecture for 2-, 3-, and 5-smooth numbers with bounded 5-adic exponent

摘录 / EXCERPT: Let a,b,c be nonnegative integers satisfying $a+b+c>0$ and $c<6$. For a positive integer n, let $\|n\|$ denote its integer complexity, namely the least number of ones needed to...

conjecture | open | MathDB 页面 | 原文线索

V2-006279 | MATHDB-373767

The zeta-function identity for Gorenstein orders

摘录 / EXCERPT: Zeta-function identity for Gorenstein orders. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-006280 | MATHDB-373817

Projective type II thin-set counting conjecture

摘录 / EXCERPT: Let $M\subset\mathbb{P}^{n-1}(\mathbb{Q})$ be a thin set of type II, and let $N_{\mathbb{P}^{n-1}}(M,B)$ denote the number of points of M of height at most B. Projective type I...

conjecture | open | MathDB 页面 | 原文线索

V2-006281 | MATHDB-373818

The Analytic Lindelöf Hypothesis for ordinary Dirichlet series

摘录 / EXCERPT: Let $L(s)=\sum a_n n^{-s}$ be an ordinary Dirichlet series with entire continuation of order at most 1, or meromorphic continuation with finitely many poles. Define the order fun...

conjecture | open | MathDB 页面 | 原文线索

V2-006282 | MATHDB-373819

The refined Sharp Kahane conjecture for ordinary Dirichlet series

摘录 / EXCERPT: Let $L(s)=\sum a_n n^{-s}$ be an ordinary Dirichlet series with entire continuation of order at most 1, and let $\mu(\sigma)$ denote its order function on vertical lines. Refined...

conjecture | open | MathDB 页面 | 原文线索

V2-006283 | MATHDB-373820

Cantor-weighted Hurwitz zeta alpha-second-moment conjecture

摘录 / EXCERPT: Let ν be the Cantor measure used in the paper and let $\widetilde\nu$ be its pushforward to $(0,1)$. Let $\zeta(s,\alpha)$ denote the Hurwitz zeta function. Cantor-weighted Hur...

conjecture | open | MathDB 页面 | 原文线索

V2-006284 | MATHDB-373870

The generalized Lindelöf hypothesis for Dirichlet L-functions

摘录 / EXCERPT: Let $\chi\pmod{q}$ be a Dirichlet character, and write $s=\sigma+it$. Generalized Lindelöf hypothesis. For every $\varepsilon>0$, if $\sigma\geq 1/2$ and $|s-1|\geq 1/2$, then ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-006285 | MATHDB-373880

The approximate uniform sampling conjecture for gaps in survival intervals

摘录 / EXCERPT: Let p_k be a prime, let $\mathcal{G}(p_k^{\#})$ be the corresponding Eratosthenes-sieve cycle, and let $\Delta H(p_k)=[p_k^2,p_{k+1}^2]$ be its interval of survival. The populati...

conjecture | open | MathDB 页面 | 原文线索

V2-006286 | MATHDB-373899

The square-freeness frequency conjecture for symmetric binary walk matrices

摘录 / EXCERPT: Square-freeness frequency conjecture. For every prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-006287 | MATHDB-373935

Miller and Scheinerman's conjecture on zeros in the character table of the symmetric group

摘录 / EXCERPT: Let S_N be the symmetric group on N letters, and consider its character table. The proportion of entries in this table that are equal to zero is measured among all entries. Mil...

conjecture | open | MathDB 页面 | 原文线索

V2-006288 | MATHDB-373942

Campana's Orbifold Mordell conjecture

摘录 / EXCERPT: Orbifold Mordell conjecture. Assume that

conjecture | open | MathDB 页面 | 原文线索

V2-006289 | MATHDB-373943

Campana's generalized Lang conjecture

摘录 / EXCERPT: Campana's generalized Lang conjecture. If $\dim X \geq 1$, then $(\mathcal{X},\Delta)(\mathcal{O}_{K,S})$ is not dense in X.

conjecture | open | MathDB 页面 | 原文线索

V2-006290 | MATHDB-373985

Generalized Dynkin-diagram modularity conjecture for Nahm quadruples

摘录 / EXCERPT: Let X and Y be Dynkin diagrams of type $ABCDEFGT$, with Cartan matrices $C(X)$ and $C(Y)$. Let $D(X)$ be the diagonal matrix whose j-th diagonal entry is α_j^2/β^2...

conjecture | open | MathDB 页面 | 原文线索

V2-006291 | MATHDB-373994

The Euler-factor conjecture for Calabi-Yau differential operators

摘录 / EXCERPT: Let $\mathcal{L}$ be a Calabi-Yau operator arising as a Picard-Fuchs operator for a Calabi-Yau motive X_φ. Let $p\geq5$, and let φ be such that...

conjecture | open | MathDB 页面 | 原文线索

V2-006292 | MATHDB-374013

The Golomb return conjecture for threshold positions

摘录 / EXCERPT: Let G be Golomb's sequence, let $S(k)=\sum_{n=1}^{k}G(n)$, and let $N_r(n)$ denote the relevant multiplicity for order r; define $M(r)=\sup_n N_r(n)$. For each k, let j_k b...

conjecture | open | MathDB 页面 | 原文线索

V2-006293 | MATHDB-374014

Mundici's conjecture on sums of squared Farey gaps

摘录 / EXCERPT: Mundici's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006294 | MATHDB-374033

Prime-Zeta spectral correspondence

摘录 / EXCERPT: Prime-Zeta spectral correspondence. As $N\to\infty$, the local statistics of $\{\widetilde{\lambda}_k\}$ asymptotically approximate those of the normalized ordinates $\{\gamma_n\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006295 | MATHDB-374038

Vojta's Main Conjecture

摘录 / EXCERPT: Vojta's Main Conjecture. Given $\epsilon > 0$, there exists a Zariski-closed subset $Z=Z_\epsilon eq X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006296 | MATHDB-374055

Rigidity conjecture for hosting sets of prime knots

摘录 / EXCERPT: Let K) and L be prime knots, and let $S(K)$ and $S(L)$ denote their hosting sets. Hosting-set rigidity conjecture. If ... then $K=L$. This asks whether a prime knot is determined...

conjecture | open | MathDB 页面 | 原文线索

V2-006297 | MATHDB-374061

Mordell's conjecture on rational points of curves

摘录 / EXCERPT: Let k be a number field and let X be a smooth projective curve of genus $g\ge 2$ defined over k. Mordell's conjecture. The set of rational points $X(k)$ is finite. This is th...

conjecture | open | MathDB 页面 | 原文线索

V2-006298 | MATHDB-374065

Logarithmic diameter conjecture for randomly oriented divisor graphs

摘录 / EXCERPT: Let ${\mathcal D}_\rho(N)$ be the randomly oriented divisor graph with orientation parameter ρ, and let $\operatorname{Diameter}({\mathcal D}_\rho(N))$ denote its diameter. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-006299 | MATHDB-374072

The characterization of strongly matchable numbers by M-numbers

摘录 / EXCERPT: For a positive integer n, let $D(n)$ denote its set of divisors and let $\tau(n)=|D(n)|$. Say that n is strongly matchable if every coprime arithmetic progression of $\tau(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-006300 | MATHDB-374084

Navarro's Galois Alperin weight conjecture for finite groups

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, let $\Gamma={\rm Gal}(k/\mathbb{F}_p)$, let G be a finite group, and let $\mathcal{O}_G$ be its p-orbit c...

conjecture | open | MathDB 页面 | 原文线索

V2-006301 | MATHDB-374085

The Galois Alperin weight conjecture for finite category algebras

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, let $\Gamma={\rm Gal}(k/\mathbb{F}_p)$, and let $\mathcal{C}$ be a finite category with p-orbit category...

conjecture | open | MathDB 页面 | 原文线索

V2-006302 | MATHDB-374086

The blockwise Galois Alperin weight conjecture for finite EI-categories

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, let $\Gamma={\rm Gal}(k/\mathbb{F}_p)$, and let $\mathcal{C}$ be a finite EI-category, meaning that every end...

conjecture | open | MathDB 页面 | 原文线索

V2-006303 | MATHDB-374087

The inductive Alperin weight condition for finite groups

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, let Γ be the Galois group of $k/\mathbb{F}_p$, and let G be a finite group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006304 | MATHDB-374088

The inductive Alperin weight condition for finite categories

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, and let $\mathcal{C}$ be a finite category. Let $\operatorname{Aut}(\mathcal{C})$ denote its automorphism gro...

conjecture | open | MathDB 页面 | 原文线索

V2-006305 | MATHDB-374089

The combined Galois-automorphism Alperin weight conjecture for finite categories

摘录 / EXCERPT: Let p be a prime, let k be an algebraic closure of $\mathbb{F}_p$, let $\Gamma={\rm Gal}(k/\mathbb{F}_p)$, and let $\mathcal{C}$ be a finite category. If $\mathcal{C}$ is a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-006306 | MATHDB-374110

Deines' conjecture on semi-stable non-isogenous discriminant twins

摘录 / EXCERPT: Two elliptic curves E_1 and E_2 over $\mathbb{Q}$ are discriminant twins if they have the same minimal discriminant and the same conductor. They are semi-stable if they have no...

conjecture | open | MathDB 页面 | 原文线索

V2-006307 | MATHDB-374119

The conjecture that transcendental elements of the Mahler-value set are S-numbers

摘录 / EXCERPT: Let $\mathbf M$ be the set of values under consideration in the paper. For a complex number ξ and a positive integer d, define $w_d(\xi)$ as the supremum of the real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-006308 | MATHDB-374134

Classical abc conjecture for the algebraic radical

摘录 / EXCERPT: Classical abc conjecture. For every $\varepsilon>0$, there is a constant $K(\varepsilon)$ depending only on ε such that whenever a, b, and c are three coprime non...

conjecture | open | MathDB 页面 | 原文线索

V2-006309 | MATHDB-374141

Three-way equivalence for LP-sharp sphere packing dimensions

摘录 / EXCERPT: Three-way equivalence. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-006310 | MATHDB-374146

Twisted Linnik-Selberg conjecture for Salié sums

摘录 / EXCERPT: Let $T(m,n;c)$ denote the Salié sum for odd positive c, and let θ be the inverse of L modulo c when it exists. For every $epsilon>0$ there is a $delta>0$ such that, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-006311 | MATHDB-374172

Nonvanishing conjecture for the first Fourier coefficient of twisted divisor-function forms

摘录 / EXCERPT: Nonvanishing conjecture. The first Fourier coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006312 | MATHDB-374173

Non-eigenform conjecture for the forms $\mathcal{F}_{D,k,e}$

摘录 / EXCERPT: Non-eigenform conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-006313 | MATHDB-374212

Erdős's conjecture on backward consecutive integers

摘录 / EXCERPT: Let $\omega(n)$ denote the number of distinct prime factors of n. Erdős's conjecture. There are infinitely many n such that ... for all integers $1\leq k<n$. This is Erdős Proble...

conjecture | open | MathDB 页面 | 原文线索

V2-006314 | MATHDB-374213

Erdős's conjecture on prime factors of backward shifts

摘录 / EXCERPT: Let $\omega(n)$ denote the number of distinct prime factors and let $\Omega(n)$ count prime factors with multiplicity. Erdős's conjecture. For $\varepsilon>0$, there are infinitely...

conjecture | refuted | MathDB 页面 | 原文线索

V2-006315 | MATHDB-374214

Conjecture on short intervals containing integers with many prime factors

摘录 / EXCERPT: For $x>0$, define ... where $\log_2 n=\log\log n$. Short-interval conjecture. For $\varepsilon>0$, there is a constant C such that, for sufficiently large x, ... and ... This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-006316 | MATHDB-374215

Conjecture on short intervals containing integers with many distinct prime factors

摘录 / EXCERPT: Let $\omega(n)$ denote the number of distinct prime factors and define ... where $\log_2 n=\log\log n$ and $\log_3 n=\log\log\log n$. Short-interval prime-factor conjecture. For some...

conjecture | open | MathDB 页面 | 原文线索

V2-006317 | MATHDB-374261

Ito's conjecture on the mean absolute value of elliptic Dedekind sums

摘录 / EXCERPT: Let $K_2(X)$ be the Farey set of elements a/c in the class-number-one imaginary quadratic field with discriminant parameter $D=2$ and $|c|^2<X$, and let ... For the normalized elli...

conjecture | open | MathDB 页面 | 原文线索

V2-006318 | MATHDB-374265

The sharp sum-product conjecture for dense subsets of finite fields

摘录 / EXCERPT: Let p be prime and let $A\subset\mathbb{F}_p$ satisfy $|A|=o(p)$ and $|A|\geq p^{1-c}$ for some sufficiently small constant $c>0$. Sharp sum-product conjecture. Then ... This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-006319 | MATHDB-374274

Generalized k-Markov uniqueness conjecture for all labels

摘录 / EXCERPT: Let $\mathcal{R}$ be the set of labels used for generalized k-Markov numbers, and let $m^{(k)}_{(p,q)}$ denote the generalized k-Markov number attached to $(p,q)\in\mathcal{R}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006320 | MATHDB-374293

Dimension conjecture for Schur squares of HRS codes

摘录 / EXCERPT: Let p, m, r, s, and t be positive integers with p prime, let $\boldsymbol{\alpha}=(\alpha_1,\ldots,\alpha_r)\in(\mathbb{F}_{p^m})^r$ have pairwise distinct entries, and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006321 | MATHDB-374306

The degree bound conjecture for field invariants of cyclic prime-group representations

摘录 / EXCERPT: Let $G=\mathbb{Z}/p\mathbb{Z}$ with p an odd prime, let $\Bbbk$ be a field of characteristic different from p, and let V be a representation of G. Let m be the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-006322 | MATHDB-374307

Pairwise coprime CRT modulus candidate-growth conjecture

摘录 / EXCERPT: Let N be the signal length, and let m_1, m_2, and m_3 be pairwise coprime moduli satisfying $m_1,m_2,m_3=\Theta(\sqrt{N})$. For a k-sparse signal, apply 3-view CRT gating...

conjecture | open | MathDB 页面 | 原文线索

V2-006323 | MATHDB-374314

Zaremba's conjecture

摘录 / EXCERPT: For positive integers $a_1,\ldots,a_\ell\geq1$, define the continued fraction ... For a positive integer A, let Q_A be the set of denominators of reduced fractions in $(0,1)$ adm...

conjecture | open | MathDB 页面 | 原文线索

V2-006324 | MATHDB-374315

The codimension-two spin-genus representation conjecture for quadratic forms

摘录 / EXCERPT: Let $q:\mathbb{Z}^m\to\mathbb{Z}$ and $Q:\mathbb{Z}^n\to\mathbb{Z}$ be integral quadratic forms with $m<n$, and let ... Assume that q is primitively represented by an element of th...

conjecture | open | MathDB 页面 | 原文线索

V2-006325 | MATHDB-374336

Kim's conjecture on eventual equality of nonabelian Chabauty sets

摘录 / EXCERPT: Kim's conjecture. For $n\gg 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006326 | MATHDB-374358

The one-prime coincidence conjecture for vanishing elliptic-curve constants

摘录 / EXCERPT: Let $\mathcal{E}$ be a non-CM elliptic curve over $\mathbb{Q}$, let j be a positive integer, and let $\mathcal{C}_{\mathcal{E},j}$ denote the constant considered in the paper. A...

conjecture | open | MathDB 页面 | 原文线索

V2-006327 | MATHDB-374359

The no-large-prime coincidence conjecture for elliptic curves

摘录 / EXCERPT: The no-large-prime coincidence conjecture. If $p\geq 5$, then there are no non-CM elliptic curves $\mathcal{E}$ over $\mathbb{Q}$ with a p-coincidence. This is a more focused for...

conjecture | open | MathDB 页面 | 原文线索

V2-006328 | MATHDB-374381

Universal-constant conjecture for optimal balanced discrepancy

摘录 / EXCERPT: Let $k\geq 4$, and let $\delta^*(k)$ denote the optimal balanced discrepancy for the module-lattice sign-selection problem. Write δ^* for the constant value proposed by the...

conjecture | open | MathDB 页面 | 原文线索

V2-006329 | MATHDB-374391

The anatomy-of-integers conjecture for products of integers in short intervals

摘录 / EXCERPT: Anatomy-of-integers conjecture. For a fixed integer $s\geqslant 1$ and any integers $2\leqslant h<u$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-006330 | MATHDB-374402

Denef's holomorphy conjecture for twisted topological zeta functions

摘录 / EXCERPT: Let $F:(\mathbb{C}^{n+1},\mathbf{0})\to(\mathbb{C},\mathbf{0})$ define a germ of an isolated hypersurface singularity. For $x\in F^{-1}(0)$, let $\mathcal{F}_{F,x}$ be the Milnor f...

conjecture | open | MathDB 页面 | 原文线索

V2-006331 | MATHDB-374404

Isaacs-Navarro-Tiep conjecture on Sylow character degrees

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. Isaacs-Navarro-Tiep conjecture. Then ... This strengthens one of the preceding conjectures...

conjecture | open | MathDB 页面 | 原文线索

V2-006332 | MATHDB-374406

Conjecture on character degrees and block heights

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a p-block of G with defect group D. Write $\operatorname{ht}(B)$ for the set of heights of the irreducible chara...

conjecture | open | MathDB 页面 | 原文线索

V2-006333 | MATHDB-374431

The $N=16n+6$ divisor-sum congruence conjecture

摘录 / EXCERPT: Let $N=16n+6$ for some nonnegative integer n. Define $\a0\sigma_0(m)$ to be the number of positive divisors of m. The $N=16n+6$ congruence conjecture. ... The authors report comp...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006334 | MATHDB-374432

The coefficient characterization conjecture for divisor-sum congruences

摘录 / EXCERPT: The coefficient characterization conjecture. Under this assumption, $A\equiv0\pmod4$ and $B\equiv2\pmod4$.

conjecture | open | MathDB 页面 | 原文线索

V2-006335 | MATHDB-374435

Transcendence of the prime-indexed sequence modulo primes

摘录 / EXCERPT: Let p_n denote the nth prime, and define ... Let $\mathcal{C}_{\mathcal{A}}$ denote the subring of algebraic elements of $\mathcal{A}$. Transcendence conjecture. ... The paper prov...

conjecture | open | MathDB 页面 | 原文线索

V2-006336 | MATHDB-374456

The Fourier uniformity conjecture for the Liouville function

摘录 / EXCERPT: Let $\lambda:\mathbb{N}\to\{+1,-1\}$ be the Liouville function, defined by $\lambda(n)=(-1)^{\Omega(n)}$, where $\Omega(n)$ counts prime factors with multiplicity. Let $H=H(X)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-006337 | MATHDB-374497

Universal exponent-one blocker conjecture for Master-Hits

摘录 / EXCERPT: Let $(a,b,m,n)\in\mathcal{M}$ be a Master-Hit, and let f_1 and $\mathcal{P}(a,b,m,n)$ denote the associated integer and set of primes. For a prime ℓ, write $v_\ell(f_1)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-006338 | MATHDB-374498

Basis-form conjecture for blockers of Master-Hits

摘录 / EXCERPT: Let $(a,b,m,n)\in\mathcal{M}$ be a Master-Hit, and call a prime divisor of f_1 that satisfies the relevant coprimality condition with $\mathcal{P}(a,b,m,n)$ a blocker. Basis-form...

conjecture | open | MathDB 页面 | 原文线索

V2-006339 | MATHDB-374499

Conjecture E1 on canonical sums of two squares for Master-Hits

摘录 / EXCERPT: Conjecture E1. For every Master-Hit $(a,b,m,n)\in\mathcal{M}$, the integer $\xi^2+\eta^2$ is not a perfect square. Equivalently, (ξ,η) are never the legs of a Pythagorean tr...

conjecture | open | MathDB 页面 | 原文线索

V2-006340 | MATHDB-374500

Strict semi-scaledness conjecture for single-blocker Master-Hits

摘录 / EXCERPT: Strict semi-scaledness conjecture. Every primitive Master-Hit (a,b,m,n) with exactly one exponent-one blocker satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006341 | MATHDB-374536

Zaremba's conjecture on bounded partial quotients

摘录 / EXCERPT: Every rational number $a/q\in(0,1)$ has a continued-fraction expansion ... where the $a_i\geq 1$ are its partial quotients. Zaremba's conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-006342 | MATHDB-374539

Laradji-Mignotte-Tzanakis conjecture for $5x^2+q^{2\alpha}=y^5$

摘录 / EXCERPT: Laradji-Mignotte-Tzanakis conjecture. If q is not of the form

conjecture | open | MathDB 页面 | 原文线索

V2-006343 | MATHDB-374590

Generalized Markov-Hurwitz uniqueness conjecture

摘录 / EXCERPT: Generalized Markov-Hurwitz uniqueness conjecture. If the two solutions have the same first coordinate, then their remaining coordinates agree:

conjecture | open | MathDB 页面 | 原文线索

V2-006344 | MATHDB-374599

Exponential upper-bound conjecture for the strong Tic-Tac-Toe threshold

摘录 / EXCERPT: Let q be a prime power. In the strong $(m,n)_q$-Tic-Tac-Toe game on the affine space $4\mathbb{F}_q^m4$, let $T(n,q)$ denote the smallest dimension m for which the first player...

conjecture | open | MathDB 页面 | 原文线索

V2-006345 | MATHDB-374601

Vanishing conjecture for LIP cohomology of almost basic sets

摘录 / EXCERPT: An almost basic set is a set of the type defined in the paper, and $H^k(U,\mathcal{LIP})$ denotes its degree-k cohomology with coefficients in the sheaf of locally integer-valued...

conjecture | open | MathDB 页面 | 原文线索

V2-006346 | MATHDB-374615

The truncation invariance conjecture for matroid zeta-function coefficients

摘录 / EXCERPT: Truncation invariance conjecture. For every integer k with $0\leqslant k<\operatorname{rk}(M)$,

conjecture | open | MathDB 页面 | 原文线索

V2-006347 | MATHDB-374616

The leading-term conjecture for the Möbius inversion of matroid topological zeta functions

摘录 / EXCERPT: Leading-term conjecture. The Taylor expansion of $Y_M(s)$ at $s=0$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006348 | MATHDB-374631

The frequency conjecture for ratios arising from $p_{m-1}+1$

摘录 / EXCERPT: Let p_m denote the m-th prime, set $n_m=p_{m-1}+1$, and let L_m be the largest prime factor of n_m when $L_m>m$; define $R_m=n_m/L_m$. For a fixed positive integer r, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-006349 | MATHDB-374635

Huang-Theriault conjecture on mod-p^r hyperbolicity

摘录 / EXCERPT: Let X be a simply-connected finite CW-complex. For a prime p and $r\geq 1$, say that X is mod-p^r hyperbolic if the number of p^r-torsion summands in the homotopy group...

conjecture | open | MathDB 页面 | 原文线索

V2-006350 | MATHDB-374667

Ternary-partition numerator conjecture at minus one

摘录 / EXCERPT: Let $\mathcal T(n)$ be the set of ternary partitions of n, meaning partitions whose parts are powers of 3, and define ... Ternary-partition evaluation conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-006351 | MATHDB-374668

Ternary-partition numerator conjecture at one

摘录 / EXCERPT: Let $\mathcal T(n)$ be the set of ternary partitions of n, meaning partitions whose parts are powers of 3, and define ... Let Δ denote the operator used in the source. Ter...

conjecture | open | MathDB 页面 | 原文线索

V2-006352 | MATHDB-374691

The reduced p-section subnormalizer conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let $P\in\operatorname{Syl}_p(G)$, and let $x\in P$. Let $R_p(x)$ be a set of representatives for the conjugacy classes in the reduce...

conjecture | open | MathDB 页面 | 原文线索

V2-006353 | MATHDB-374693

The picky Evseev condition

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let $P\in\operatorname{Syl}_p(G)$, let $H=\mathbf{N}_G(P)$, and let $\mathcal P$ be the set of picky elements of P. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-006354 | MATHDB-374694

The mixed-order strong subnormalizer conjecture

摘录 / EXCERPT: Let G be a finite group and let $x\in G$. Write $x=\prod_p x_p$ for the decomposition of x into its p-parts. Assume that, for every prime p dividing $o(x)$, the pair...

open_problem | open | MathDB 页面 | 原文线索

V2-006355 | MATHDB-374703

Katz-Sarnak density conjecture for families of L-functions

摘录 / EXCERPT: Let $L(s)$ be an L-function with nontrivial zeros written as $\rho=\frac12+i\gamma$, let $\tilde{\gamma}=\frac{\log c(L)}{2\pi}\gamma$, and for an even Schwartz function ϕ o...

conjecture | open | MathDB 页面 | 原文线索

V2-006356 | MATHDB-374704

Keating-Snaith conjecture on logarithmic central values

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of L-functions of type of symmetry $\mathrm{SO(even)}$ or Sp, and let $\mathcal{F}_X$ be a finite truncation increasing to $\mathcal{F}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006357 | MATHDB-374722

The E''-family H-moment conjecture

摘录 / EXCERPT: $E”$-family H-moment conjecture. The filtered H-moment conjecture remains true after every occurrence of $E'_{Gamma,(Gamma_v)_{v otin T}}(D,Q)$ is replaced by...

conjecture | open | MathDB 页面 | 原文线索

V2-006358 | MATHDB-374757

Carvajal-Rojas-Schwede-Tucker conjecture on positivity of the limit F-signature

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a KLT singularity essentially of finite type over $\mathbb{C}$. For every sufficiently large prime p, let $\mathscr{s}_p(R)$ denote the F-signature of...

conjecture | open | MathDB 页面 | 原文线索

V2-006359 | MATHDB-374769

The global resolvent trace conjecture for the Riemann zeta function

摘录 / EXCERPT: Let h be a test function with Fourier transform $\hat h$, and let ρ run over the zeros of the Riemann zeta function $\zeta(s)$. Resolvent trace identity. The global resolven...

conjecture | open | MathDB 页面 | 原文线索

V2-006360 | MATHDB-374772

Perego's conjecture on moduli spaces of stable sheaves on K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface with a Kähler class ω and a Mukai vector ... where $\xi\in \mathrm{NS}(S)$, $r>1$ is prime to ξ, and ω is v-generic. Let $M_v(S,\omega)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-006361 | MATHDB-374775

Dickson's conjecture on simultaneous prime values of linear polynomials

摘录 / EXCERPT: Let $k\in\boldsymbol{N}_{geq 1}$ and let $bar f=(f_i:i<k)$, where $f_i(x)=a_i x+b_i$ with a_i,b_i non-negative integers and $a_igeq 1$. Assume that no integer $ngeq 1$ divides al...

conjecture | open | MathDB 页面 | 原文线索

V2-006362 | MATHDB-374787

Permutation conjecture for prime-exponent LCMs of distinct covering systems

摘录 / EXCERPT: A distinct covering system is a covering system whose moduli are distinct integers greater than 1. Let $m\geq 2$, let a, b, and c be positive integers, and let L denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-006363 | MATHDB-374795

The Generalized Riemann Hypothesis for Dirichlet L-functions

摘录 / EXCERPT: Let χ be a Dirichlet character modulo $q greater than or equal to 1$, and let ${\mathcal N}(\frac{1}{2},\infty,\chi)$ denote the number of nontrivial zeros of the associated D...

conjecture | open | MathDB 页面 | 原文线索

V2-006364 | MATHDB-374809

Rossmann's pole-existence conjecture for algebra zeta functions

摘录 / EXCERPT: Let $\mathfrak{O}$ be the coefficient ring and let A be a finite-dimensional $\mathfrak{O}$-algebra. Write $\zeta_A(t)$ and $\zeta^{\lhd}_A(t)$ for its subalgebra and ideal zeta...

conjecture | open | MathDB 页面 | 原文线索

V2-006365 | MATHDB-374810

Rossmann's simple-pole conjecture for nilpotent algebra zeta functions

摘录 / EXCERPT: Let $\mathfrak{O}$ be the coefficient ring, let A be an n-dimensional nilpotent $\mathfrak{O}$-algebra, and write $\zeta_A(t)$ and $\zeta^{\lhd}_A(t)$ for its subalgebra and id...

conjecture | open | MathDB 页面 | 原文线索

V2-006366 | MATHDB-374816

The conjecture on the classification of unitary perfect numbers

摘录 / EXCERPT: The classification conjecture. The five integers above are the only unitary perfect numbers. No further unitary perfect number is known; this is the classical Subbarao-Warren probl...

conjecture | open | MathDB 页面 | 原文线索

V2-006367 | MATHDB-374817

The finiteness conjecture for the even Higgs-exponent obstruction set

摘录 / EXCERPT: Let H_{even} be the set of even integers m for which every prime divisor of 2^m+1 is 3-Higgs. A prime is 3-Higgs when every prime divisor q of $r-1$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-006368 | MATHDB-374818

Hybrid semigroup-friable shifted-prime conjecture

摘录 / EXCERPT: Let $\mathcal{S}_3^{(\leq 3)}$ be the bounded-exponent semigroup used in the paper, and let $\operatorname{ord}_r(2)$ denote the multiplicative order of 2 modulo a prime r. Hyb...

conjecture | open | MathDB 页面 | 原文线索

V2-006369 | MATHDB-374819

Semigroup log-mass conjecture

摘录 / EXCERPT: Let $\mathcal{P}_3$ be the set of odd primes used in the paper whose shifted-prime closure has the stated 3-Higgs property, and let $\mathcal{S}_3^{(\leq 3)}$ be the associated bou...

conjecture | open | MathDB 页面 | 原文线索

V2-006370 | MATHDB-374820

Divisor-transference conjecture for $\Phi_{4p}(2)$

摘录 / EXCERPT: Let $\mathcal{P}_3$ and $\mathcal{S}_3^{(\leq 3)}$ be as used in the paper, and define ... ... Here $\Phi_{4p}(2)$ is the value of the cyclotomic polynomial Φ_{4p} at 2. Divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-006371 | MATHDB-374821

Sublogarithmic growth conjecture for the bounded-exponent semigroup

摘录 / EXCERPT: Let $A(x)=\#\bigl(\mathcal{S}_3^{(\leq 3)}\cap[1,x]\bigr)$, where $\mathcal{S}_3^{(\leq 3)}$ is the bounded-exponent semigroup associated with the prime set $\mathcal{P}_3$. Sublog...

conjecture | open | MathDB 页面 | 原文线索

V2-006372 | MATHDB-374822

Divisor mod-16 equidistribution conjecture for $\Phi_{4p}(2)$

摘录 / EXCERPT: Let $\mathcal{P}_3$ be the prime set used in the paper, let $\Phi_{4p}(2)$ be the cyclotomic value, and let $\omega(N)$ denote the number of distinct prime divisors of N. Divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-006373 | MATHDB-374824

Improved error-term conjecture for the limit form of the trace formula

摘录 / EXCERPT: Improved error-term conjecture. In all terms above, the error term $o(X)$ can be replaced by $O(X^{1/2+\varepsilon})$ for any $\varepsilon>0$, where the implied constant only depen...

conjecture | open | MathDB 页面 | 原文线索

V2-006374 | MATHDB-374826

The distance-three vertex conjecture for Pythagorean walks

摘录 / EXCERPT: Let $\mathcal{G}$ be the graph of Pythagorean walks, with distinguished vertex $O(0,0)$, and let graph distance be measured from $O(0,0)$. A vertex is a symmetric counterpart of a...

conjecture | open | MathDB 页面 | 原文线索

V2-006375 | MATHDB-374843

Generalized Erdős-Sierpiński conjecture for the sum-of-divisors function

摘录 / EXCERPT: Let k be a positive integer, let n range over positive integers, and let $\sigma(n)$ denote the sum of the positive divisors of n. Generalized Erdős-Sierpiński conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-006376 | MATHDB-374846

Quadratic polynomial primitive-divisor density conjecture

摘录 / EXCERPT: Let $\phi\in\mathbb{Z}[x]$ be quadratic, with infinite critical orbit and all iterates irreducible. For an integer sequence $a_n=\phi(a_{n-1})$, let $P(a_n)$ denote the largest pri...

conjecture | open | MathDB 页面 | 原文线索

V2-006377 | MATHDB-374850

Strong Dynamical Wieferich Prime Conjecture

摘录 / EXCERPT: Let $b\in\frac{1}{2}\mathbb{Z}$ and let $\phi\in\mathbb{Z}[x]$ be separable and quadratic. Assume that the forward orbit $\{\phi^n(b):n=1,2,\ldots\}$ is infinite. Strong Dynamical...

conjecture | open | MathDB 页面 | 原文线索

V2-006378 | MATHDB-374861

Finite-order conjecture for rational dynamics of the prime-representing map

摘录 / EXCERPT: Let ... where $\{x\}=x-\left\lfloor x\right\rfloor$. For a rational number $x\geq 2$, define its order under $\mathcal{T}$ to be the least non-negative integer n such that...

conjecture | open | MathDB 页面 | 原文线索

V2-006379 | MATHDB-374862

Hcabdlog's conjecture

摘录 / EXCERPT: Let $b\geq 2$ be a fixed base. For a positive integer, write $\overleftarrow{n}$ for its digit reversal in base b. Then p_1 and p_2 are primes, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006380 | MATHDB-374863

Two reversed primes in prime bases

摘录 / EXCERPT: Let b be a fixed prime, and let $\overleftarrow{p}$ denote the digit reversal of a prime p in base b. Two-reversed-prime conjecture. Every sufficiently large even integer N...

conjecture | open | MathDB 页面 | 原文线索

V2-006381 | MATHDB-374864

Three reversed primes in base 10

摘录 / EXCERPT: Work in base $b=10$, and write $\overleftarrow{p}$ for the digit reversal of a prime p. Base-10 ternary reversed-prime conjecture. Every sufficiently large odd integer N can be...

conjecture | open | MathDB 页面 | 原文线索

V2-006382 | MATHDB-374865

Asymptotic Hcabdlog's conjecture

摘录 / EXCERPT: Let $b\geq 2$ be fixed. Define ... and ... Also let $\#\mathfrak{B}(N)$ denote the number of admissible integer representations encoded by the source's set $\mathfrak{B}(N)$. Asymptoti...

conjecture | open | MathDB 页面 | 原文线索

V2-006383 | MATHDB-374866

Pure reversed-prime circle-method asymptotic

摘录 / EXCERPT: Pure reversed-prime asymptotic conjecture. For every $A>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006384 | MATHDB-374867

Existence of the Nnamlerinchs constant

摘录 / EXCERPT: For a fixed base $b\geq 2$, define the Nnamlerinchs constant $\overleftarrow{K}_b$ to be the smallest number greater than 1 such that every sufficiently large integer can be writ...

conjecture | open | MathDB 页面 | 原文线索

V2-006385 | MATHDB-374868

Logarithmic upper bound for the Nnamlerinchs constant

摘录 / EXCERPT: For a fixed base $b\geq 2$, let $\overleftarrow{K}_b$ be the smallest number greater than 1 such that every sufficiently large integer is a sum of at most $\overleftarrow{K}_b$ r...

conjecture | open | MathDB 页面 | 原文线索

V2-006386 | MATHDB-374878

Moretó's picky conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $x\in G$ be a picky p-element, meaning that x lies in a unique Sylow p-subgroup. Let $P\in\operatorname{Syl}_p(G)$ cont...

conjecture | open | MathDB 页面 | 原文线索

V2-006387 | MATHDB-374879

Moretó's subnormalizer conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $x\in G$ be a p-element. Write $\operatorname{Irr}^x(G)$ for the irreducible complex characters of G that are nonzero at...

conjecture | open | MathDB 页面 | 原文线索

V2-006388 | MATHDB-374889

Kuperberg's uniform Poisson conjecture for primes in short intervals

摘录 / EXCERPT: Let $\pi(t)$ denote the number of primes at most t. For a real parameter $\lambda=(\log x)^{o(1)}$ and an integer $k\ll (\log_2 x)^2$, count integers $n\leqslant x$ according to...

conjecture | open | MathDB 页面 | 原文线索

V2-006389 | MATHDB-374890

Folklore equality conjecture for the maximal prime gap constant

摘录 / EXCERPT: Folklore equality conjecture. The inequality is in fact an equality:

conjecture | open | MathDB 页面 | 原文线索

V2-006390 | MATHDB-374929

Uniqueness of the -primitive decomposition for fixed square-free part

摘录 / EXCERPT: Let $s\in\mathbb{N}$ be square-free, and let $P(s)$ be the set of Δ-primitives with square-free part s. A natural number n has a Δ-primitive decomposition if it c...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006391 | MATHDB-374930

Infinitude of -primitive squares

摘录 / EXCERPT: A natural number is called Δ-primitive when it has the Δ property but is not a nontrivial square multiple of another number with the Δ property. Infinitude con...

conjecture | open | MathDB 页面 | 原文线索

V2-006392 | MATHDB-374931

The two-prime -primitive number conjecture

摘录 / EXCERPT: Let p and q be distinct primes, and let $x,y\in\mathbb{N}$. Consider natural numbers of the form ... A natural number is Δ-primitive if it has the Δ property and is...

conjecture | open | MathDB 页面 | 原文线索

V2-006393 | MATHDB-374940

Igusa-Denef-Loeser Strong Monodromy Conjecture

摘录 / EXCERPT: Strong Monodromy Conjecture. 1. If $f\in\mathbb{Q}[x_1,\ldots,x_n]$, then, for all but finitely many primes p, every pole s_0 of $Z_{f,p}(s)$ has $\operatorname{Re}(s_0)$ as a...

conjecture | open | MathDB 页面 | 原文线索

V2-006394 | MATHDB-374970

Geometrization conjecture for holomorphic conformal field theories

摘录 / EXCERPT: Geometrization conjecture. If k is a positive integer divisible by 4, every group-like element of

conjecture | open | MathDB 页面 | 原文线索

V2-006395 | MATHDB-374977

Uniform exponent conjecture for sums of d-th roots

摘录 / EXCERPT: Let $d\geq 2$, $k\geq 1$, and $\varepsilon>0$. For $N\geq 1$ and $\beta\in\mathbb R/\mathbb Z$, consider the distance to the nearest integer, denoted by $\|\cdot\|$. Uniform expone...

conjecture | open | MathDB 页面 | 原文线索

V2-006396 | MATHDB-374978

Taylor-cancellation conjecture for sums of d-th roots

摘录 / EXCERPT: Let $d\geq 2$ and $k\geq 1$. Let A_i be positive integers, let u_i,v_i be integers, let $\lambda\ne 0$ be a real constant, and let $L(M)$ be an integer-valued linear polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-006397 | MATHDB-374995

Goldmakher's conjecture on logarithmic sums of multiplicative functions

摘录 / EXCERPT: Goldmakher's conjecture. Let $f\in\mathcal{F}$. Then for any $1\leq y\leq x$,

conjecture | open | MathDB 页面 | 原文线索

V2-006398 | MATHDB-374996

The infinitude of prime Mersenne numbers

摘录 / EXCERPT: Infinitude conjecture. There are infinitely many integers n such that $M(n)$ is prime.

conjecture | open | MathDB 页面 | 原文线索

V2-006399 | MATHDB-374997

Bateman-Selfridge-Wagstaff conjecture on Mersenne-prime forms

摘录 / EXCERPT: Bateman-Selfridge-Wagstaff conjecture. If two of these three statements are true, then the third is also true.

conjecture | open | MathDB 页面 | 原文线索

V2-006400 | MATHDB-374998

Dynamical characterization of infinitely many Mersenne primes

摘录 / EXCERPT: Dynamical Mersenne-prime conjecture. For every $n_0>0$, there exists a prime integer $n>n_0$ such that, for every prime q with $1<q\leq \left\lfloor\sqrt{M(n)}\right\rfloor$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-006401 | MATHDB-375033

Divisibility of the class number of the quadratic field k' by 16

摘录 / EXCERPT: Let p be a prime, let k' be the quadratic field and ε' a unit of k' occurring in the algorithm, and let $h_{k'}$ denote its class number. Let η_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-006402 | MATHDB-375042

Bege's conjecture on sums of Apostol's Möbius functions in arithmetic progressions

摘录 / EXCERPT: For integers $k\geq 2$ and $q\geq 1$, define the error term $E_{k,q}(x)$ by ... where ... Here $\varphi(q)$ is Euler's totient function, $\theta(q)$ is the number of positive squarefre...

conjecture | open | MathDB 页面 | 原文线索

V2-006403 | MATHDB-375048

The 1/2-conjecture for q-binomial coefficients with fractional index

摘录 / EXCERPT: For $r\in\mathbb{Q}^+$ and an integer $k\geq 1$, define the integer trace of the generalized q-binomial coefficient by retaining its integer exponents, and write ... For power seri...

conjecture | open | MathDB 页面 | 原文线索

V2-006404 | MATHDB-375053

Wang-Qin-Xia's stability conjecture for nontrivial graph pairs

摘录 / EXCERPT: Let (Γ,Σ) be a nontrivial graph pair, meaning that Γ and Σ are coprime connected twin-free graphs and exactly one of them is bipartite. Assume that the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-006405 | MATHDB-375056

Conjecture on the logarithmic density of non-prime endomorphisms of the free group

摘录 / EXCERPT: Conjecture on the non-prime density. The logarithmic density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006406 | MATHDB-375080

Watanabe-Yoshida's conjecture on the minimum Hilbert-Kunz multiplicity of singularities

摘录 / EXCERPT: Let p be an odd prime and let ... which is a singular ring of characteristic p and dimension $d-1$. For a Noetherian local ring $(R,\mathfrak{m})$ of characteristic p, write...

conjecture | open | MathDB 页面 | 原文线索

V2-006407 | MATHDB-375085

Moretó's subnormaliser conjecture

摘录 / EXCERPT: Let G be a finite group and let ℓ be a prime number. For an ℓ-element $x\in G$, define the subnormaliser ... Let $\operatorname{Irr}^x(G)$ denote the set of complex irred...

conjecture | open | MathDB 页面 | 原文线索

V2-006408 | MATHDB-375105

Chen's conjecture on positive-density shifted Romanoff type sumsets

摘录 / EXCERPT: Chen's conjecture. If $r_1,\ldots,r_t$ are positive integers with $r_1^{-1}+\cdots+r_t^{-1}\geq 1$, then $\mathcal{R}_{\mathcal{A}}\cap(\mathcal{R}_{\mathcal{A}}+2)$ has positive l...

conjecture | open | MathDB 页面 | 原文线索

V2-006409 | MATHDB-375123

The Costas polynomial characterization conjecture for finite fields

摘录 / EXCERPT: Let $q=p^n$ be a prime power and let $\mathbb{F}_q$ be the finite field with q elements. A mapping g has fixed point 0 and a cycle of length $q-1$ if its permutation action c...

conjecture | open | MathDB 页面 | 原文线索

V2-006410 | MATHDB-375138

Polynomial-scale density conjecture for ideal degree-three Prouhet-Tarry-Escott solutions

摘录 / EXCERPT: Polynomial-scale density conjecture. The full counting function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006411 | MATHDB-375143

Hong's divisibility conjecture for gcd-closed sets satisfying condition G

摘录 / EXCERPT: Let a and b be positive integers with $a\mid b$, and let S be a finite gcd-closed set of positive integers. For each $x\in S$, let $G_S(x)$ denote the set of greatest-type di...

conjecture | open | MathDB 页面 | 原文线索

V2-006412 | MATHDB-375153

A bounded-limit conjecture for the generalized quality metric

摘录 / EXCERPT: For an abc-triple (a,b,c), let $q_C(a,b,c;\alpha,\beta)$ be the alternative quality metric considered in the source, with parameters α and β. Generalized quality m...

conjecture | open | MathDB 页面 | 原文线索

V2-006413 | MATHDB-375158

Transcendence of simple geodesics in finite modular covers

摘录 / EXCERPT: Let $\Gamma=\operatorname{PSL}_2(\mathbb{Z})$, let $\Gamma^\prime\subset\Gamma$ be a finite-index subgroup corresponding to a finite cover $\mathbf{M}^\prime\to\mathbf{M}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006414 | MATHDB-375159

Ferraro's multiplicity conjecture for relative zeta functions

摘录 / EXCERPT: Let $R:=A$, $B:=B_A$, and $H:=H_A$. For a positive degree d, let $\mathcal{I}(d)$ be the set of ideals in B whose norm has degree d, and define the analytic function...

conjecture | open | MathDB 页面 | 原文线索

V2-006415 | MATHDB-375186

McCranie's prime-quadruplet characterization of jump-condition solutions

摘录 / EXCERPT: McCranie's conjecture. Every composite integer n satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-006416 | MATHDB-375256

Power-ratio conjecture for Euler's, Dedekind's and sum-of-divisors functions

摘录 / EXCERPT: Let $\varphi(n)$, $\psi(n)$ and $\sigma(n)$ denote the Euler totient, Dedekind psi and sum-of-divisors functions, respectively. For integers $k\geq1$ and $n\geq2$, use...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006417 | MATHDB-375257

Fully powered mixed-ratio conjecture for arithmetic functions

摘录 / EXCERPT: Let $\varphi(n)$, $\psi(n)$ and $\sigma(n)$ denote the Euler totient, Dedekind psi and sum-of-divisors functions, respectively. For integers $k\geq1$ and $n\geq2$, form the three q...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006418 | MATHDB-375282

The function-field analogue of Vojta's stronger conjecture

摘录 / EXCERPT: Function-field Vojta conjecture. Let $\mathcal{A}$ be an ample line bundle on $\mathcal{X}$ and let r be a positive integer. For every $\epsilon>0$, there is a proper Zariski-clo...

conjecture | open | MathDB 页面 | 原文线索

V2-006419 | MATHDB-375319

Hardy-Littlewood prime k-tuples conjecture

摘录 / EXCERPT: Let $h_1,h_2,\ldots,h_k$ be distinct positive even integers such that the numbers in the sequence $\mathbb{P}_k=(p,p+h_1,\ldots,p+h_k)$ do not form a complete residue class modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-006420 | MATHDB-375320

HypothesiX residue-pairing bound conjecture

摘录 / EXCERPT: For a squarefree integer Q with $6\mid Q$, let U_Q be the set of residues $r\bmod Q$ such that both r and $r+2$ are coprime to Q. Define the residue-pairing bound ... where .....

conjecture | open | MathDB 页面 | 原文线索

V2-006421 | MATHDB-375321

The parity problem in sieve theory

摘录 / EXCERPT: Let A be a set whose elements are all products of an odd number of primes, or all products of an even number of primes. Parity problem. Without injecting additional ingredients,...

open_problem | open | MathDB 页面 | 原文线索

V2-006422 | MATHDB-375322

Pairing-defect formulation of the parity problem

摘录 / EXCERPT: Let $\Delta_Q(x)$ denote the pairing defect function introduced in the paper for the residue-pairing framework associated with the residue-pairing bound $B_Q(x)$. Pairing-defect pa...

open_problem | open | MathDB 页面 | 原文线索

V2-006423 | MATHDB-375331

Associated-prime extension conjecture for separated stars on edge ideals

摘录 / EXCERPT: Associated-prime extension conjecture. Every associated prime P of $R/(l,f_1,\ldots,f_q)$ is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-006424 | MATHDB-375353

Optimal convergence rate for the modular distribution of minimal residues

摘录 / EXCERPT: Let $F_N(r)$ denote the empirical distribution of $q_{\min}\bmod \ell$, let $\pi_{\mathrm{perm}}(r)$ be the permitted-residue distribution, and let $C_{\mathrm{opt}}(k,\ell)$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-006425 | MATHDB-375354

Square-root cancellation conjecture for the oscillatory phase average

摘录 / EXCERPT: Let $A_N(\gamma)$ be the oscillatory phase average associated with the centred layer moment, let X be the relevant scale, and let R be the corresponding correlation quantity. S...

open_problem | open | MathDB 页面 | 原文线索

V2-006426 | MATHDB-375363

Nonvanishing conjecture for distinct multiple zeta-star values

摘录 / EXCERPT: Let s satisfy $\operatorname{Re}(s)>1$, and let $(k_1,\ldots,k_p)$ and $(l_1,\ldots,l_q)$ be distinct finite sequences of positive integers. Write $H_s^\star(k_1,\ldots,k_p)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-006427 | MATHDB-375442

Asymptotic counting conjecture for q-matroids with fixed rank

摘录 / EXCERPT: Asymptotic counting conjecture. For all $r\geq 2$ and prime powers q, we have

conjecture | open | MathDB 页面 | 原文线索

V2-006428 | MATHDB-375443

Asymptotic counting conjecture for q-matroids of rank r on a 2r-dimensional space

摘录 / EXCERPT: Asymptotic counting conjecture. For each prime power q, we have

conjecture | open | MathDB 页面 | 原文线索

V2-006429 | MATHDB-375494

Square-prime short-interval conjecture

摘录 / EXCERPT: Square-prime short-interval conjecture. For all $x \geqslant 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-006430 | MATHDB-375532

Beilinson's regulator conjecture for smooth projective curves

摘录 / EXCERPT: Let X be a smooth, projective, geometrically connected curve of genus g defined over $\mathbb{Q}$. Let $K_4^{(3)}(X)$ denote the third Adams eigenspace of...

conjecture | open | MathDB 页面 | 原文线索

V2-006431 | MATHDB-375538

Bunyakovsky's conjecture on prime values of polynomials

摘录 / EXCERPT: Bunyakovsky's conjecture. The value $f(n)$ should be prime for infinitely many positive integers n.

conjecture | open | MathDB 页面 | 原文线索

V2-006432 | MATHDB-375539

Weight conjecture for minimal nilpotent hypertoric characters

摘录 / EXCERPT: Let $N\geq 1$, and let $V(\Delta)$ be the boundary hypertoric vertex operator superalgebra for the minimal nilpotent orbit closure of $\mathfrak{sl}_N$. Weight conjecture. After re...

conjecture | open | MathDB 页面 | 原文线索

V2-006433 | MATHDB-375543

Douady's optimality conjecture for the Bruno condition

摘录 / EXCERPT: Douady's optimality conjecture. If θ is not a Bruno number, then no polynomial of degree at least 2 can have a Siegel disk of rotation number θ.

conjecture | open | MathDB 页面 | 原文线索

V2-006434 | MATHDB-375552

Ng's conjecture for the second ratio moment at the zeros of the zeta function

摘录 / EXCERPT: Let the non-trivial zeros of the Riemann zeta function be written as $\rho=\tfrac12+i\gamma$. Assume the Riemann Hypothesis and that these zeros are simple. Ng's conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-006435 | MATHDB-375553

Pearce-Crump's Dirichlet reciprocal derivative moment conjecture

摘录 / EXCERPT: Let χ be a primitive Dirichlet character modulo $q>1$, and let $L(s,\chi)$ be its Dirichlet L-function, with non-trivial zeros $\rho=\tfrac12+i\gamma$. Assume the Generalise...

conjecture | open | MathDB 页面 | 原文线索

V2-006436 | MATHDB-375554

Pearce-Crump's family-averaged Dirichlet ratio moment conjecture

摘录 / EXCERPT: Let $q\not\equiv2\pmod4$, and let $\varphi^*(q)$ denote the number of primitive characters modulo q. For each primitive character χ modulo q, write the non-trivial zeros o...

conjecture | open | MathDB 页面 | 原文线索

V2-006437 | MATHDB-375560

Velani's conjecture for dyadic approximation on the middle-third Cantor set

摘录 / EXCERPT: Let C be the middle-third Cantor set and let ... W2:=left{xin[0,1]: |2^nx|<(n)text{ for infinitely many }ninright}. ... is monotonic, then ... This is a Khintchine-type zeroone law for...

conjecture | open | MathDB 页面 | 原文线索

V2-006438 | MATHDB-375570

Gawron-Miska-Ulas conjecture on the unboundedness of power-series coefficients

摘录 / EXCERPT: Let $F(x)=\prod_{n=0}^{\infty}(1-x^{2^n})$ and, for an integer m, define ... Thus $t_m(n)$ is the coefficient of x^n in $F_m(x)$. Gawron-Miska-Ulas conjecture. For each integer...

conjecture | open | MathDB 页面 | 原文线索

V2-006439 | MATHDB-375571

Cao-Wang's unresolved modular rank four Nahm sum identity

摘录 / EXCERPT: Cao-Wang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006440 | MATHDB-375572

The even and odd Nahm sum identities for $\hat{Z}_e(q)$ and $\hat{Z}_o(q)$

摘录 / EXCERPT: The even and odd Nahm sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006441 | MATHDB-375573

Shi-Wang's triple Nahm sum identity

摘录 / EXCERPT: Shi-Wang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006442 | MATHDB-375580

Conjecture on optimal cancellation for prime-power Gauss sums

摘录 / EXCERPT: Let p be a prime and $k\ge 1$ an integer. Let $\chi\colon \mathbb{F}_q^\times\to\mathbb{C}^\times$ be a non-trivial character of order $o=\operatorname{ord}(\chi)$, where $o=p^k$...

conjecture | open | MathDB 页面 | 原文线索

V2-006443 | MATHDB-375603

Bogomolov property conjecture for modular Galois representations

摘录 / EXCERPT: Let $f\in S_k(\Gamma_1(N))$ be a normalized eigenform, and let $\rho_{f,v}$ denote its Galois representation at a non-Archimedean place v and $\widehat{\rho}_f$ the product of th...

conjecture | open | MathDB 页面 | 原文线索

V2-006444 | MATHDB-375605

Rémond's conjecture on heights in divisible hull extensions

摘录 / EXCERPT: Let $\mathbb{K}$ be a number field, let $\Gamma\leq\mathbb{K}^{\times}$ be a non-torsion finitely generated subgroup, and define its divisible hull by ... Put...

conjecture | open | MathDB 页面 | 原文线索

V2-006445 | MATHDB-375613

Finite sign law for Ramanujan's third-order mock theta function $\rho(q)$

摘录 / EXCERPT: The finite sign law. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006446 | MATHDB-375614

The odd square-plane conjecture for transversal difference numbers

摘录 / EXCERPT: Odd square-plane conjecture. For every odd prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-006447 | MATHDB-375615

The higher-rank square-modulus conjecture for transversal difference numbers

摘录 / EXCERPT: Higher-rank square-modulus conjecture. For every odd prime p and every $r\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-006448 | MATHDB-375629

Weighted dual quadratic large sieve conjecture

摘录 / EXCERPT: Weighted dual quadratic large sieve conjecture. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006449 | MATHDB-375630

Ideal moment conjecture for sums of square roots modulo one

摘录 / EXCERPT: Let $r\geq 1$ be fixed. For $h\sim H$, define the moment $M_{2r}(H,n)$ as in the source. Ideal moment conjecture. For every fixed $\delta>0$ and every $\varepsilon>0$, one expects...

conjecture | open | MathDB 页面 | 原文线索

V2-006450 | MATHDB-375633

Kaplansky's conjecture on von Neumann regular rings

摘录 / EXCERPT: Kaplansky's conjecture. R is von Neumann regular if and only if R is semiprime and each prime factor ring of R is von Neumann regular.

conjecture | open | MathDB 页面 | 原文线索

V2-006451 | MATHDB-375635

The profinite Schinzel hypothesis H over S-integers

摘录 / EXCERPT: Extended Schinzel hypothesis H. If

conjecture | open | MathDB 页面 | 原文线索

V2-006452 | MATHDB-375636

The finite-field Bateman-Horn conjecture for one polynomial

摘录 / EXCERPT: Finite-field Bateman-Horn conjecture. As $X\to\infty$ through powers of q,

conjecture | open | MathDB 页面 | 原文线索

V2-006453 | MATHDB-375637

The finite-field Bateman-Horn conjecture for several polynomials

摘录 / EXCERPT: Finite-field Bateman-Horn conjecture. As $X\to\infty$ through powers of q,

conjecture | open | MathDB 页面 | 原文线索

V2-006454 | MATHDB-375640

de Jong's conjecture on finite geometric monodromy in characteristic p

摘录 / EXCERPT: Let k be a finite field of characteristic $\ell\ne p$, let X be a geometrically connected smooth curve over k, and let ... be the exact sequence for the arithmetic fundamental...

conjecture | open | MathDB 页面 | 原文线索

V2-006455 | MATHDB-375661

Parity conjecture for coefficients of the odd triangular partition series

摘录 / EXCERPT: Let $T_o(q)=\sum_{n\geq 0}t_o(n)q^n$, and let n be a nonnegative integer. Parity conjecture. If $8n+9$ has a prime divisor $p\equiv 5,7\pmod{8}$ to odd exponent, equivalently if...

conjecture | open | MathDB 页面 | 原文线索

V2-006456 | MATHDB-375662

Hasler's ternary structural infinitude conjecture for binomial congruences

摘录 / EXCERPT: Hasler's structural infinitude conjecture. The equation $\binom{3n}{n} \equiv 3^n \pmod{n}$ has infinitely many solutions of the form $n=3^k\cdot p$, where k is a positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-006457 | MATHDB-375666

Dasgupta-Károlyi-Serra-Szegedy conjecture for distinct products in finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group with $|G|>1$, and let p be the smallest prime divisor of $|G|$. Let $k<p$, let $a_1,\ldots,a_k\in G$ be pairwise distinct, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006458 | MATHDB-375677

The no-solution conjecture for the plus exponential Diophantine equation

摘录 / EXCERPT: No-solution conjecture for the plus equation. This equation has no solutions in positive integers for $n\geq4$. The conjecture is proposed as an analogue of Cohn's conjectures for...

conjecture | open | MathDB 页面 | 原文线索

V2-006459 | MATHDB-375678

The cubic plus-equation solution conjecture

摘录 / EXCERPT: Cubic plus-equation conjecture. The only positive integer solutions are

conjecture | open | MathDB 页面 | 原文线索

V2-006460 | MATHDB-375733

Geometric Bombieri--Lang's special-locus conjecture

摘录 / EXCERPT: Let K/k be a finitely generated extension of fields of characteristic zero, with k algebraically closed in K. Define the algebraic special set $\operatorname{Sp}(X)$ as the Z...

conjecture | open | MathDB 页面 | 原文线索

V2-006461 | MATHDB-375742

Conjecture on non-full-weight cyclic codes over \(\mathbb F_3\)

摘录 / EXCERPT: Let $n>3$ be a prime, and let $\operatorname{ord}_n(3)$ denote the multiplicative order of 3 modulo n. Set ... Let $\mathcal N(n,3)$ denote the family of cyclic codes over...

conjecture | open | MathDB 页面 | 原文线索

V2-006462 | MATHDB-375761

Heath-Brown's conjecture on primes in arithmetic progressions

摘录 / EXCERPT: Let $A\geq 2$, and let b and l be coprime integers. Heath-Brown's conjecture. There exists a prime $p\equiv b\pmod l$ such that ... This conjecture is used as a conditional input...

conjecture | open | MathDB 页面 | 原文线索

V2-006463 | MATHDB-375766

Deng-Espinosa's cubic order zeta-function conjecture

摘录 / EXCERPT: Let R be a Gorenstein order over $\mathbb{Z}$. Let $R^\vee=\operatorname{Hom}(R,\mathbb{Z})$, and define Yun's Dedekind zeta function by ... Here $L(s,R)$ is the cubic-order L-fu...

conjecture | open | MathDB 页面 | 原文线索

V2-006464 | MATHDB-375772

Lubin's conjecture for height-one p-adic dynamical systems

摘录 / EXCERPT: Lubin's conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-006465 | MATHDB-375794

Quasi-polynomial continued-fraction conjecture for $g(N)=\sqrt{R_N/2}$

摘录 / EXCERPT: Let R_N be the quantity defined in the paper, let $g(N)=\sqrt{R_N/2}$, and let N_k and Q_{2k+1} be the sequences defined above. Write $\operatorname{PQ}_j(g)$ for the j-th...

conjecture | open | MathDB 页面 | 原文线索

V2-006466 | MATHDB-375800

The characterization of equality in the two-color reciprocal Rado number

摘录 / EXCERPT: Let $f_2(k)$ be the smallest positive integer n such that every 2-coloring of $\{1,2,\ldots,n\}$ has a monochromatic solution, with the variables not necessarily distinct, to ... C...

conjecture | open | MathDB 页面 | 原文线索

V2-006467 | MATHDB-375813

Generalized Pythagorean partition-regularity conjecture

摘录 / EXCERPT: Generalized Pythagorean partition-regularity conjecture. The equation

conjecture | open | MathDB 页面 | 原文线索

V2-006468 | MATHDB-375843

Kloosterman L^1 anti-concentration conjecture

摘录 / EXCERPT: Kloosterman L^1 anti-concentration conjecture. For every $0<\alpha<\frac12$, there exists a constant $c_\alpha>0$ such that, for every odd prime power q, if...

conjecture | open | MathDB 页面 | 原文线索

V2-006469 | MATHDB-375871

Cohen-Lenstra conjecture for Sylow subgroups of quadratic class groups

摘录 / EXCERPT: Cohen-Lenstra conjecture. The proportion of discriminants $D\in S_X^{\pm}$ for which the Sylow p-subgroup of the class group of $\mathbb{Q}(\sqrt D)$ is isomorphic to B satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-006470 | MATHDB-375876

Andrews-El Bachraoui's parity conjecture for the odd companion series

摘录 / EXCERPT: Let $t_o(n)$ be defined by ... where q is a complex number with $|q|<1$ and $(a;q)_n$ denotes the standard q-Pochhammer symbol. Andrews-El Bachraoui's conjecture. If $8n+9$ has a...

conjecture | open | MathDB 页面 | 原文线索

V2-006471 | MATHDB-375878

Conjecture on congruences for overcubic partition tuples

摘录 / EXCERPT: Let $n\geq 0$, $i\geq 3$, and $k\geq 0$. The overcubic partition tuple function $\overline{b}_m(r)$ is considered at the indicated arithmetic progressions. Congruence conjecture. F...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006472 | MATHDB-375892

Conjecture A on the refined abc bound

摘录 / EXCERPT: For coprime positive integers a and b, put $c=a+b$ and define ... where $\gamma(n)=\prod_{p\mid n}p$ is the radical of n. Conjecture A. There exists a real number C_1 such th...

conjecture | open | MathDB 页面 | 原文线索

V2-006473 | MATHDB-375893

The H-refined abc conjecture

摘录 / EXCERPT: For $n\in\mathbb{N}$, let $\gamma(n)=\prod_{p\mid n}p$ and let $\omega(n)$ denote the number of distinct prime factors of n. Define ... The convention $0^0=1$ is used when $n=1$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-006474 | MATHDB-375896

Quartic large sieve upper bound

摘录 / EXCERPT: Let $\Xi_4(A,B)$ denote the quartic large sieve quantity for parameters $A,B\geq 1$, and let $\varepsilon>0$. Quartic large sieve upper-bound conjecture. For any $A,B\geq 1$, ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-006475 | MATHDB-375903

Sun's conjecture on the partial sums of $(-2)^{\Omega(n)}$

摘录 / EXCERPT: Sun's conjecture. For all $x\geq 3078$,

conjecture | open | MathDB 页面 | 原文线索

V2-006476 | MATHDB-375912

Folklore modularity conjecture for ADET Cartan-matrix Nahm sums

摘录 / EXCERPT: Let X and Y be Dynkin diagrams of ADET type, with Cartan matrices $\mathcal{C}(X)$ and $\mathcal{C}(Y)$, ranks $r(X)$ and $r(Y)$, and Coxeter numbers $h(X)$ and $h(Y)$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-006477 | MATHDB-375916

Rump's prime-order classification conjecture for quasilinear cycle sets

摘录 / EXCERPT: Rump's prime-order conjecture. If A has size p, then there is an isomorphism of quasilinear cycle sets

conjecture | open | MathDB 页面 | 原文线索

V2-006478 | MATHDB-375928

Shifted Möbius Elliott-Halberstam conjecture

摘录 / EXCERPT: Let $0<\theta<1$ be fixed, let $h\neq0$ be a fixed even integer, and let $\Lambda(n)$, $\mu(n)$ and $\varphi(q)$ denote the von Mangoldt function, Möbius function and Euler totient...

conjecture | open | MathDB 页面 | 原文线索

V2-006479 | MATHDB-375929

Möbius-twisted Elliott-Halberstam conjecture

摘录 / EXCERPT: Let $0<\theta<1$ be fixed, let N be a sufficiently large natural number, and let $\Lambda(n)$, $\mu(n)$ and $\varphi(q)$ denote the von Mangoldt function, Möbius function and Eul...

conjecture | open | MathDB 页面 | 原文线索

V2-006480 | MATHDB-375930

Diagonal Möbius-logarithmic Elliott-Halberstam conjecture

摘录 / EXCERPT: Let $0<\theta<1$ be fixed, let N be a sufficiently large natural number, and let g be either $g(n)=\mu(n)$ or $g(n)=\mu(n)\log(n)$. Let $\Lambda(n)$ and $\varphi(q)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-006481 | MATHDB-375931

Gonek-Hejhal conjecture at k=1

摘录 / EXCERPT: Gonek-Hejhal conjecture at $k=1$. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006482 | MATHDB-375947

Kellner's conjecture on the term count and Bell-polynomial values of the polynomials ψ_ν

摘录 / EXCERPT: Let $\mathcal{P}(n)$ be the partition function, and define ... Let $\mathbb{B}_n$ be the complete Bell polynomial, and define the alternating harmonic number ... For the polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-006483 | MATHDB-375953

Rubin-Stark conjecture on the integrality of Rubin-Stark elements

摘录 / EXCERPT: Rubin-Stark conjecture. The Rubin-Stark element satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006484 | MATHDB-375967

The Selberg-Linnik bound for twisted Kloosterman sums

摘录 / EXCERPT: Selberg-Linnik bound. For every $m,n\neq0$ and every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006485 | MATHDB-375968

The conjecture on uncorrelated classical and modular-symbol Kloosterman sums

摘录 / EXCERPT: For $m,n\neq0$, define ... and let $r(S,S^*)(x)$ be the Pearson correlation of the sequences $S(m,n;c)$ and $S^*(m,n;c)$ over the dyadic window $W_x=\{N\mid c:x<c\leq2x\}$. Uncorrela...

conjecture | open | MathDB 页面 | 原文线索

V2-006486 | MATHDB-375975

Gath's conjecture for Cygan--Korányi ball discrepancy

摘录 / EXCERPT: Gath's conjecture. The optimal order of the error term is $2q-1$; that is, the infimum of the exponents κ for which $|\mathcal E_q(t)|\lesssim t^\kappa$ holds equals $2q-1$....

conjecture | open | MathDB 页面 | 原文线索

V2-006487 | MATHDB-375980

Cancellative elements in bend quotients of tropical principal prime ideals

摘录 / EXCERPT: Cancellative-elements conjecture. An element of $S[\mathcal{M}]/\operatorname{Bend}(J)$ is cancellative if and only if it is the image of a term of $S[\mathcal{M}]$.

conjecture | open | MathDB 页面 | 原文线索

V2-006488 | MATHDB-375989

The exact factor-complexity formula for the $(3/2)^n$ steering word

摘录 / EXCERPT: Let $T=(t_n)_{n\ge0}$ be the steering word considered in the paper, and let $p_T(k)$ denote the number of distinct factors of length k occurring in T. Exact complexity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-006489 | MATHDB-376010

Diamantis-Rolen odd-part containment conjecture for derivative period polynomials

摘录 / EXCERPT: Diamantis-Rolen odd-part containment conjecture. There is a real number $b=b(f,m,k)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006490 | MATHDB-376048

Consecutive-value conjecture for supersingular elliptic curves

摘录 / EXCERPT: For each $n\in\mathbb{Z}_{>0}$, there should exist a prime p such that, for every integer m with $1\leq m\leq n$, there is a supersingular elliptic curve E defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-006491 | MATHDB-376049

Surjectivity conjecture for the minimal isogeny degree on primes

摘录 / EXCERPT: Let $\mathbb{P}$ be the set of prime numbers, and let $\delta\colon\mathbb{P}\to\mathbb{Z}_{>0}$ assign to each prime p the value $\delta(p)$. Prime-value surjectivity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-006492 | MATHDB-376050

Infinitude conjecture for wisde primes

摘录 / EXCERPT: Wisde-prime infinitude conjecture. The set $\mathcal{W}$ is infinite. The paper gives computational evidence for many such primes and notes that this is equivalent to the infinitud...

conjecture | open | MathDB 页面 | 原文线索

V2-006493 | MATHDB-376058

Heim-Neuhauser's log-concavity challenge for D'Arcais polynomials

摘录 / EXCERPT: Let $P_n^\sigma(X)=\sum_{k=0}^n p_n^\sigma(k)X^k$ be the D'Arcais polynomial defined by ... and call a sequence $(a_k)_{k\in\mathbb{N}}$ log-concave at $k\geq 1$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-006494 | MATHDB-376059

Abdesselam's log-concavity conjecture for subgroup-counting polynomials

摘录 / EXCERPT: Let $g_\ell(n)$ denote the number of subgroups of index n in $\mathbb{Z}^\ell$, and write ... Call a sequence $(a_k)_{k\in\mathbb{N}}$ log-concave at $k\geq 1$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-006495 | MATHDB-376067

The exact gcd conjecture for the degree-five Carlitz-Wieferich factor

摘录 / EXCERPT: Let $q=19^3$, let ... and let $[5]=T^{q^5}-T$ and M_5 be Thakur's polynomial. The polynomial $P\in\mathbf{F}_{19^3}[T]$ is the degree-five c-Wieferich prime exhibited in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-006496 | MATHDB-376068

The quadratic inverse sieve conjecture

摘录 / EXCERPT: Let N be a positive integer and let $A\subseteq [N]$. For each prime p, define the set of occupied residue classes ... A rational quadratic is the image of a quadratic polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-006497 | MATHDB-376069

Green--Harper's inverse quadratic large sieve conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a set of positive integers. For each prime p, let $\mathcal A_p$ denote the set of residue classes occupied by $\mathcal A$ modulo p. A rational quadratic i...

conjecture | open | MathDB 页面 | 原文线索

V2-006498 | MATHDB-376079

Cramér-Granville conjecture on prime gaps

摘录 / EXCERPT: Let $\mathcal{P}=\{p_1,p_2,\ldots\}$ be the set of prime numbers, enumerated in increasing order. Cramér-Granville conjecture. For some $\kappa>1$, ... Cramér proposed the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-006499 | MATHDB-376080

Cramér-Granville conjecture for good-prime gaps

摘录 / EXCERPT: Let $\mathcal{P}_{\mathrm{good}}=\{g_1,g_2,\ldots\}$ be the subset of good primes, enumerated in increasing order, where good primes are the primes outside the paper's exceptional...

conjecture | open | MathDB 页面 | 原文线索

V2-006500 | MATHDB-376088

Erdős-Turán conjecture on largest prime factors of consecutive integers

摘录 / EXCERPT: Let $P^+(n)$ denote the largest prime factor of n, with $P^+(1)=1$. Erdős-Turán conjecture. As x tends to infinity, ... This conjecture concerns the expected symmetry between the...

conjecture | open | MathDB 页面 | 原文线索

V2-006501 | MATHDB-376089

Erdős-Pomerance conjecture on independent largest prime factors

摘录 / EXCERPT: Let $P^+(n)$ denote the largest prime factor of n. For $a,b\in[0,1]$, define ... Let $\rho(t)$ denote the Dickman-de Bruijn function. Erdős-Pomerance conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-006502 | MATHDB-376090

De Koninck-Doyon conjecture on orderings of largest prime factors

摘录 / EXCERPT: Fix an arbitrary integer $k\geq 2$, and let $a_1,a_2,\ldots,a_k$ be any permutation of $0,1,\ldots,k-1$. Let $P^+(m)$ denote the largest prime factor of m. De Koninck-Doyon conje...

conjecture | open | MathDB 页面 | 原文线索

V2-006503 | MATHDB-376092

The parity conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Parity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006504 | MATHDB-376101

The mod 8 congruence conjecture for the 2-color overpartition function

摘录 / EXCERPT: Let $\overline{p}_{\ell}(n)$ denote the 2-color overpartition function, where ℓ is a positive integer, and let $n\ge0$. The mod 8 congruence conjecture. For all...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006505 | MATHDB-376102

Higher-modulus congruences for the 2-color overpartition function

摘录 / EXCERPT: Let $\overline{p}_{6}(n)$ and $\overline{p}_{8}(n)$ denote the 2-color overpartition functions with indicated color parameters, and let $n\ge0$. Higher-modulus congruence conject...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006506 | MATHDB-376103

Conjectured arithmetic congruences for the 9-color overpartition function

摘录 / EXCERPT: Let $\overline{p}_{9}(n)$ denote the 9-color overpartition function, and let $n\ge0$ and $\alpha\ge1$. Arithmetic congruence conjecture. For all $n\ge0$ and $\alpha\ge1$, ... These...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006507 | MATHDB-376106

Friedlander-Iwaniec Conjecture 2 on automorphic summatory errors

摘录 / EXCERPT: Friedlander-Iwaniec Conjecture 2. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006508 | MATHDB-376108

Consistency of the Boolean Prime Ideal Theorem with no Vitali set

摘录 / EXCERPT: Consistency conjecture. It is consistent with ZF plus the Axiom of Dependent Choices (DC) that BPI holds and there is no Vitali set in the real line.

conjecture | open | MathDB 页面 | 原文线索

V2-006509 | MATHDB-376150

Bernoulli numerator divisibility and representation by a quadratic form

摘录 / EXCERPT: Bernoulli-quadratic-form conjecture. For every prime p,

conjecture | open | MathDB 页面 | 原文线索

V2-006510 | MATHDB-376156

Erdős-Granville-Pomerance-Spiro conjecture on preimages under the sum-of-proper-divisors function

摘录 / EXCERPT: Erdős-Granville-Pomerance-Spiro conjecture. If $\mathcal{A}$ has asymptotic density zero, then $s^{-1}(\mathcal{A})$ also has asymptotic density zero.

conjecture | open | MathDB 页面 | 原文线索

V2-006511 | MATHDB-376172

The infinitude conjecture for prime products in $E^*_{Y^2}$

摘录 / EXCERPT: Infinitude conjecture. $E^*_{Y^2}$ contains infinitely many numbers of the form $\pi_1\pi_2$ for such primes π_1 and π_2.

conjecture | open | MathDB 页面 | 原文线索

V2-006512 | MATHDB-376185

Conjecture on the relational complexity of the primitive action of $PGL_2(p)$

摘录 / EXCERPT: Let p be a prime greater than 7, let $G=PGL_2(p)$, and let Ω be the permutation domain of the action under consideration. The relational-complexity conjecture asserts th...

conjecture | open | MathDB 页面 | 原文线索

V2-006513 | MATHDB-376190

Low-Roberts conjecture on connected simple Hartke magic graphs

摘录 / EXCERPT: Let $p\geq 3$ be prime. A Hartke $\mathbb{Z}_p$-magic graph is a graph with the Hartke magic-labeling property over $\mathbb{Z}_p$; its order is its number of vertices. Low-Roberts...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006514 | MATHDB-376193

Andrews' divergence conjecture for the coefficients of $v_1(q)$

摘录 / EXCERPT: Andrews' divergence conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-006515 | MATHDB-376194

Andrews' same-sign triple conjecture for coefficients of $v_1(q)$

摘录 / EXCERPT: Andrews' same-sign triple conjecture. There is an infinite sequence with these indicated values and growth condition such that $V_1(N_k),V_1(N_k+1),V_1(N_k+2)$ all have the same si...

conjecture | open | MathDB 页面 | 原文线索

V2-006516 | MATHDB-376195

Andrews' local-minimum conjecture for coefficients of $v_1(q)$

摘录 / EXCERPT: Andrews' local-minimum conjecture. The three numbers

conjecture | open | MathDB 页面 | 原文线索

V2-006517 | MATHDB-376216

Prime Hankel representation for the prime-cyclotomic gamma function

摘录 / EXCERPT: Let $\Gamma_{\mathbb P}^{\mathrm{cyc}}$ be the prime-cyclotomic gamma function. A prime Hankel representation should consist of an oriented contour...

conjecture | open | MathDB 页面 | 原文线索

V2-006518 | MATHDB-376217

Centered subcritical Vaughan moment conjecture

摘录 / EXCERPT: Centered subcritical Vaughan moment conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006519 | MATHDB-376218

Mascia et al.'s primality conjecture for polyominoes

摘录 / EXCERPT: Mascia et al.'s conjecture. $\mathcal{P}$ is prime if and only if $\mathcal{P}$ contains no zig-zag walk.

conjecture | open | MathDB 页面 | 原文线索

V2-006520 | MATHDB-376223

Prime affine direction conjecture for additive codes

摘录 / EXCERPT: Prime affine direction conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006521 | MATHDB-376230

Calinescu-Milas-Penn conjecture on modularity of principal tadpole Nahm sums

摘录 / EXCERPT: Let $T_r=A_{2r}/\mathbb{Z}_2$ be the tadpole Dynkin diagram, and let $\mathbf{T}_r$ be its Cartan matrix. For $\mathbf{x}_r=(x_1,\ldots,x_r)$, define the tadpole Nahm sum ... The sum...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006522 | MATHDB-376231

Milas-Wang conjecture on a Wronskian formula for principal tadpole Nahm sums

摘录 / EXCERPT: Let $T_r=A_{2r}/\mathbb{Z}_2$ be the tadpole Dynkin diagram, and let $\mathcal{T}_r(\mathbf{x}_r;q)$ denote its tadpole Nahm sum. The sum $\mathcal{T}_r(1,\ldots,1;q)$ is the princ...

conjecture | open | MathDB 页面 | 原文线索

V2-006523 | MATHDB-376241

Product identities for the four dual Nahm sums of Zagier's Example 12

摘录 / EXCERPT: For $|q|<1$, define $J_m=(q^m;q^m)_\infty$ and $J_{a,m}=(q^a,q^{m-a},q^m;q^m)_\infty$. Let the two dual Nahm sums be the two triple sums displayed below, with denominator factors...

conjecture | open | MathDB 页面 | 原文线索

V2-006524 | MATHDB-376265

Equidistribution conjecture for the signs of square-free representation differences

摘录 / EXCERPT: For each odd square-free integer t, let $\Delta(t)=r_{Q_1}(t)-r_{Q_2}(t)$, and define ... ... and $N_{\ne 0}(X)=N_+(X)+N_-(X)$. Sign equidistribution conjecture. As $X\to\infty$, ... E...

conjecture | open | MathDB 页面 | 原文线索

V2-006525 | MATHDB-376269

Sárközy's multiplicative irreducibility conjecture for shifted quadratic residues

摘录 / EXCERPT: Let p be a sufficiently large prime, and let $\mathcal R_p$ denote the set of nonzero squares in $\mathbb F_p$. For $\lambda\in\mathbb F_p^*$, consider the shifted set...

conjecture | open | MathDB 页面 | 原文线索

V2-006526 | MATHDB-376270

Kim-Yip-Yoo's multiplicative irreducibility conjecture for shifted subgroups

摘录 / EXCERPT: Let $d\geq 2$ be fixed. Let $q\equiv 1\pmod d$ be a sufficiently large prime power, and let G be the multiplicative subgroup of $\mathbb F_q$ of index d. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-006527 | MATHDB-376310

Erdős-Selfridge odd covering problem

摘录 / EXCERPT: Erdős-Selfridge odd covering problem. Does there exist a covering system of $\mathbb{Z}$ whose moduli are odd, distinct, and greater than 1?

open_problem | open | MathDB 页面 | 原文线索

V2-006528 | MATHDB-376347

Cube finite-place mean conjecture

摘录 / EXCERPT: Cube finite-place mean conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-006529 | MATHDB-376351

The ideal-prime conjecture for convexity

摘录 / EXCERPT: Ideal-prime conjecture. If the 6-net σ is prime, its ideal neoplatonic realization $\Delta(\sigma,\infty)$ is convex.

conjecture | open | MathDB 页面 | 原文线索

V2-006530 | MATHDB-376352

The volume-maximizing conjecture for prime 6-nets

摘录 / EXCERPT: Volume-max conjecture. For a prime 6-net σ, its ideal neoplatonic realization maximizes volume among all ideal geodesic 2-cycles with combinatorics σ.

conjecture | open | MathDB 页面 | 原文线索

V2-006531 | MATHDB-376373

Hardy-Littlewood's binary Goldbach asymptotic conjecture

摘录 / EXCERPT: Let $r_2'(N)$ denote the number of representations of N as a sum of two primes, and let $\mathfrak{S}(N)$ be the singular series ... For even integers N, let $r_2(N)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-006532 | MATHDB-376387

The integral-lattice conjecture for Schur MacMahon series

摘录 / EXCERPT: Let $\mathcal{M}_{\mathbb{Z}}:=\mathcal{M}\cap\mathbb{Z}[[q]]$ be the integral lattice of quasimodular forms, and let $\mathcal{G}_{\mathbb{Z}}:=\sum_\lambda\mathbb{Z}\,g(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-006533 | MATHDB-376388

The nonvanishing conjecture for Schur Eisenstein transition coefficients

摘录 / EXCERPT: Let $\nu\subseteq\lambda$ be partitions, let $d_\lambda(\nu)$ be the corresponding transition coefficient, and let $F_{\lambda\nu}$ be the associated element of the symmetric-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-006534 | MATHDB-376389

The bounded-weight Schur MacMahon spanning conjecture

摘录 / EXCERPT: Let $F_{\leq k}\mathcal{M}_{\mathbb{Z}}:=F_{\leq k}\mathcal{M}\cap\mathbb{Z}[[q]]$ be the integral quasimodular forms of filtered weight at most k, and let $g(\lambda)$ be the Sc...

conjecture | open | MathDB 页面 | 原文线索

V2-006535 | MATHDB-376409

The abc-conjecture for a number field

摘录 / EXCERPT: The abc-conjecture for a number field. For every $\kappa>1$ and every $x\in K$ distinct from 0 and 1,

conjecture | open | MathDB 页面 | 原文线索

V2-006536 | MATHDB-376410

The weak abc-conjecture for a number field

摘录 / EXCERPT: Let K be a number field, let M_K be its set of places, and define the height h and conductor cond as in the preceding number-field abc formula...

conjecture | open | MathDB 页面 | 原文线索

V2-006537 | MATHDB-376411

The canonical form conjecture for minimally dependent shifted elements

摘录 / EXCERPT: Let Λ be a finitely generated subgroup of $\overline{\mathbb{Q}}^\times$ containing ζ_{m-1}, with $m\ge 2$. Say that algebraic numbers are multiplicatively dependent...

conjecture | open | MathDB 页面 | 原文线索

V2-006538 | MATHDB-376412

Universal dominance conjecture for the residue class $-1 \pmod N$

摘录 / EXCERPT: Universal dominance conjecture. For any modulus $N\ge 3$, the quantity $\log L(1,\chi_{1,a})$ achieves its unique absolute minimum over all non-principal residue classes...

conjecture | open | MathDB 页面 | 原文线索

V2-006539 | MATHDB-376415

Rémond's Bogomolov property conjecture for radical extensions

摘录 / EXCERPT: Let $\mathbb{K}$ be a number field, and let $\tilde\Gamma\subseteq\overline{\mathbb{Q}}^\times$ be a subgroup of finite rank. Write $\tilde\Gamma^{\mathrm{div}}$ for its divisible...

conjecture | open | MathDB 页面 | 原文线索

V2-006540 | MATHDB-376416

Log-integrability conjecture for Möbius spectral densities

摘录 / EXCERPT: Let ρ be the density of a potential spectral measure of the Möbius function $\bm\mu$ with respect to Lebesgue measure. Log-integrability conjecture. The density ρ is log-...

conjecture | open | MathDB 页面 | 原文线索

V2-006541 | MATHDB-376417

Möbius-\(\mathcal{B}\)-free topological entropy question

摘录 / EXCERPT: Let $\mathcal{B}$ be the defining set of the $\mathcal{B}$-free system, and let $X_{\mathcal{B},3}$ denote the Möbius-$\mathcal{B}$-free dynamical system with shift map S. Möbius...

open_problem | open | MathDB 页面 | 原文线索

V2-006542 | MATHDB-376461

Göttsche's generating-function conjecture for the Kotschick-Morgan wall-crossing term

摘录 / EXCERPT: Let X be a 4-manifold with $b_+(X)=1$ and $b_1(X)=0$, let $\xi\in H^2(X,\mathbb Z)$ satisfy $\xi^2<0$, and let $\lambda_{[X]}(q)$ be the power-series factor associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-006543 | MATHDB-376500

Strong arithmetic Shafarevich conjecture for semistable curves

摘录 / EXCERPT: Let $K(C)$ be a field as above, and let ρ be an ideal of the field F^{nr}. Strong arithmetic Shafarevich conjecture. There exists a finite extension F of K and...

conjecture | open | MathDB 页面 | 原文线索

V2-006544 | MATHDB-376501

Geometric bounded-components form of the arithmetic Shafarevich conjecture

摘录 / EXCERPT: Let C' be a semistable curve over F, and let C_L be the model associated with a finite nonramified extension L of $F(C')$. Geometric Shafarevich conjecture. There is a numb...

conjecture | open | MathDB 页面 | 原文线索

V2-006545 | MATHDB-376508

Biran's Pell-equation conjecture for Seshadri-like constants

摘录 / EXCERPT: Let $N>9$ be square-free, and let d/m be the fundamental solution of the Pell equation $d^2-Nm^2=1$. Write $(d;m^{\times N})$ for the vector with first entry d and N further...

conjecture | open | MathDB 页面 | 原文线索

V2-006546 | MATHDB-376524

The modular identity predicted by mirror symmetry for Schoen's Calabi-Yau threefold

摘录 / EXCERPT: Modular mirror-symmetry identity.

conjecture | open | MathDB 页面 | 原文线索

V2-006547 | MATHDB-376545

The asymptotic arithmetic fibration conjecture for L-targets

摘录 / EXCERPT: Asymptotic arithmetic fibration conjecture. The set of ${\cal L}$-targets of V forms an asymptotic arithmetic ${\cal L}$-fibration.

conjecture | open | MathDB 页面 | 原文线索

V2-006548 | MATHDB-376547

The height comparison conjecture for primitive fibers

摘录 / EXCERPT: Height comparison conjecture. There exist a family of v-adic metrics on K_Y and constants $c_2>c_1>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006549 | MATHDB-376619

Structural decomposition conjecture for minimal pseudo-codewords

摘录 / EXCERPT: Let q be an even prime power and let $\mathbf{H}=\mathbf{H}_{\operatorname{PG}(2,q)}$. A component is significant if, for most lines passing through its point, its value is the s...

conjecture | open | MathDB 页面 | 原文线索

V2-006550 | MATHDB-376722

Exceptional-factor conjecture for Calabi-Yau hypersurface zeta functions

摘录 / EXCERPT: Let a family of Calabi-Yau hypersurfaces have Hodge numbers $h^{2,1}$ and $h^{2,1}_{\mathrm{poly}}$, and let $R_{\mathrm{excep}}(t)$ denote a factor of the numerator of its zeta fu...

conjecture | open | MathDB 页面 | 原文线索

V2-006551 | MATHDB-376723

Singularity-degeneration conjecture for Calabi-Yau zeta functions

摘录 / EXCERPT: Singularity-degeneration conjecture. The degeneration of $\operatorname{deg}(R_{\mathbf{v}})$ for each piece of the zeta function of a Calabi-Yau manifold at a singular point in mo...

conjecture | open | MathDB 页面 | 原文线索

V2-006552 | MATHDB-376724

Degeneration conjecture for isolated A1 singularities

摘录 / EXCERPT: Let $R_{\mathbf{v}}(t)$ be a contribution to the zeta function associated with a monomial class $\mathbf{v}$, and let $\operatorname{deg}(R_{\mathbf{v}})$ denote its rational-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-006553 | MATHDB-376725

Degeneration-conservation conjecture for coincident singularities

摘录 / EXCERPT: Let $R_{\mathbf{v}}(t)$ be a contribution to the zeta function, and let the degree degeneration measure the decrease in $\operatorname{deg}(R_{\mathbf{v}})$ caused by a singularity...

conjecture | open | MathDB 页面 | 原文线索

V2-006554 | MATHDB-376727

Toric splitting conjecture for Calabi-Yau threefold zeta functions

摘录 / EXCERPT: Let X be a Calabi-Yau threefold with $h^{1,1}(X)\neq h^{1,1}_{\mathrm{toric}}(X)$, and set ... Let χ be a character depending on the defining equation of a singular locus. Tor...

conjecture | open | MathDB 页面 | 原文线索

V2-006555 | MATHDB-376782

The CFKRS moment conjecture for the Riemann zeta function

摘录 / EXCERPT: The Riemann zeta function is defined by $[... ELLIPSIZATION...$ and continued meromorphically to the complex plane. For complex shifts $[... ELLIPSIZATION...$, define the $2k$-poin...

conjecture | open | MathDB 页面 | 原文线索

V2-006556 | MATHDB-376793

Universality conjecture for prime plane curves and alternating knots

摘录 / EXCERPT: Let α'_p be the number of closed prime self-intersecting curves with p crossings, and let $\alpha”_p$ be the number of prime alternating knots with p crossings. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006557 | MATHDB-376808

Kurlberg-Rudnick rate conjecture for Hecke quantum unique ergodicity

摘录 / EXCERPT: Let p be prime, let $\rho_{1/p}(A)$ be the quantized operator, and let ${\rm T}_A$ be its Hecke torus. For a character $\chi:{\rm T}_A\@ifnextchar^{\t@@}{\t@@^{}}\mathbb C^*$, le...

conjecture | open | MathDB 页面 | 原文线索

V2-006558 | MATHDB-376891

PL-manifold conjecture for the nerve of the meta-tree poset

摘录 / EXCERPT: For $n\ge 2$, let $MT(n)$ be the set of isomorphism classes of meta-trees with n tails, and define a partial order $<$ on $MT(n)$ by declaring ${\bf T}<{\bf T}^{\prime}$ when the...

conjecture | open | MathDB 页面 | 原文线索

V2-006559 | MATHDB-376919

The determinant characterization conjecture for achiral links

摘录 / EXCERPT: Let L be an achiral link, and let n be an even natural number. The determinant characterization conjecture for achiral links. An even natural number n occurs as the determina...

conjecture | open | MathDB 页面 | 原文线索

V2-006560 | MATHDB-376920

The determinant characterization conjecture for prime alternating achiral knots

摘录 / EXCERPT: Let n be an odd natural number. The determinant characterization conjecture for prime alternating achiral knots. Then n is the determinant of a prime alternating achiral knot i...

conjecture | open | MathDB 页面 | 原文线索

V2-006561 | MATHDB-376921

The spanning-tree characterization conjecture for planar self-dual graphs

摘录 / EXCERPT: Let G be a planar self-dual graph without cut vertex, and let n be an odd natural number. The spanning-tree characterization conjecture for planar self-dual graphs. Then n is...

conjecture | open | MathDB 页面 | 原文线索

V2-006562 | MATHDB-376937

The prime-level double-shuffle conjecture for multiple polylogarithms

摘录 / EXCERPT: Let $N>1$, and consider multiple polylogarithms evaluated at N-th roots of unity. The double-shuffle and distribution relations are families of relations among these values. Prim...

conjecture | open | MathDB 页面 | 原文线索

V2-006563 | MATHDB-376958

Holomorphy conjecture for the twisted topological zeta function

摘录 / EXCERPT: Let $f:X\to\bb A^1_{\bb C}$ be as in the construction of the twisted topological zeta function, let $e\geq 1$, and write $Z^{(e)}_{{\rm top}}(s)$ for that rational function. A mono...

conjecture | open | MathDB 页面 | 原文线索

V2-006564 | MATHDB-377010

The modularity conjecture for rigid Calabi-Yau threefolds over Q

摘录 / EXCERPT: The modularity conjecture. Any rigid Calabi-Yau threefold X defined over ${\mathbb Q}$ is modular: its L-series coincides, up to finitely many Euler factors, with the L-serie...

conjecture | open | MathDB 页面 | 原文线索

V2-006565 | MATHDB-377011

Ulmer's semisimplicity conjecture for the Hecke operator at level Np

摘录 / EXCERPT: Ulmer's conjecture. The action of U_p on $S_k(\Gamma_0(Np),\mathbb Q_p)$ is semisimple. In particular, the polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-006566 | MATHDB-377017

The diagonal dihedral Lie algebra conjecture for prime cyclotomic level

摘录 / EXCERPT: Let p be a prime number, let $\boldsymbol{}[p]$ be the group of p-th roots of unity, let ${D}^{\Delta}_{\bullet}(\boldsymbol{}[p])$ be the diagonal, depth-equals-weight part of...

conjecture | open | MathDB 页面 | 原文线索

V2-006567 | MATHDB-377019

The prime-level diagonal modular-complex conjecture

摘录 / EXCERPT: Let p be a prime, let $E_m(N)$ be the arithmetic coefficient module defined in the source, let $M^*_{(m)}$ be the dual modular complex, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006568 | MATHDB-377030

A strong form of Goldbach's conjecture

摘录 / EXCERPT: A strong form of Goldbach's conjecture. For every even integer $n\geq 6$,

conjecture | open | MathDB 页面 | 原文线索

V2-006569 | MATHDB-377031

Exponential form of the bounding functions for Goldbach partitions

摘录 / EXCERPT: Exponential bounding-functions conjecture. The lower and upper bounding functions of $G(n)$ can be expressed respectively in the form

conjecture | open | MathDB 页面 | 原文线索

V2-006570 | MATHDB-377087

Deligne-Ihara structural conjectures for Galois image Lie algebras

摘录 / EXCERPT: Deligne-Ihara conjectures. The embedding $\varPsi_l$ is an isomorphism for every prime l; the Lie algebras $\mathfrak g^l_\bullet$ and $\mathfrak D_\bullet$ are free; and their g...

conjecture | open | MathDB 页面 | 原文线索

V2-006571 | MATHDB-377161

Intersection-dimension conjecture ID-1

摘录 / EXCERPT: Let $(A,\mathfrak{n})$ be a quasi-unmixed local ring with prime ideals P and Q such that $\sqrt{P+Q}=\mathfrak{n}$. ID-1 conjecture. If ... and A/P is analytically unramified,...

conjecture | open | MathDB 页面 | 原文线索

V2-006572 | MATHDB-377162

Intersection-dimension conjecture ID-2

摘录 / EXCERPT: Let $(A,\mathfrak{n})$ be a quasi-unmixed local ring with prime ideals P and Q such that $\sqrt{P+Q}=\mathfrak{n}$. ID-2 conjecture. If ... and both A/P and A/Q are analytica...

conjecture | open | MathDB 页面 | 原文线索

V2-006573 | MATHDB-377173

Integrality conjecture for TQFT invariants and cut number

摘录 / EXCERPT: Let M be an oriented connected closed three-manifold, possibly equipped with a banded trivalent colored graph, and let p be a prime. Define ... For a connected three-manifold M...

conjecture | open | MathDB 页面 | 原文线索

V2-006574 | MATHDB-377177

Köhler-Roessler relative equivariant fixed-point formula conjecture

摘录 / EXCERPT: Relative fixed-point formula conjecture. The diagram

conjecture | open | MathDB 页面 | 原文线索

V2-006575 | MATHDB-377190

Holomorphic anomaly equation for elliptic-surface Gromov-Witten series

摘录 / EXCERPT: For $g\geq 0$ and $n\geq 1$, let $Z_{g;n}(q):=\sum_dN_{g,d;n}q^d$, where $N_{g,d;n}:=\sum_{(\beta,\sigma)=d,\,(\beta,F)=n}N_g(\beta)$. Let $E_2(q),E_4(q),E_6(q)$ be Eisenstein seri...

conjecture | open | MathDB 页面 | 原文线索

V2-006576 | MATHDB-377241

The even-degree extension of Langmann's irreducibility theorem

摘录 / EXCERPT: Let $H(t,X)\in\mathbb Q[t,X]$ be a homogeneous polynomial of degree n that is not a proper power. Consider the specialization $H(\bar t,X)-1$ for integers $\bar t$. Even-degree i...

conjecture | open | MathDB 页面 | 原文线索

V2-006577 | MATHDB-377252

Greenberg's height-two annihilator conjecture for multiple $[200~mathbb{Z}_p$-extensions

摘录 / EXCERPT: Let p be a prime, let K be a number field, and let $\widetilde{K}$ be the compositum of all $\mathbb{Z}_p$-extensions of K. Let X be the Galois group of the maximal abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-006578 | MATHDB-377254

Constant-order structure conjecture for Reshetikhin-Turaev theory

摘录 / EXCERPT: Let p be a fixed prime with $p\geq 3$. Let ${\cal V}^I_p$ denote the integral TQFT, and let $\dov {\cal V}^{FN}_p$ denote the corresponding Fibonacci-type TQFT; the tensor produc...

conjecture | open | MathDB 页面 | 原文线索

V2-006579 | MATHDB-377290

Finite-dimensional representation conjecture for rational Cherednik algebras of type A

摘录 / EXCERPT: Finite-dimensional representation conjecture. The algebra ${\mathsf{H}}_c$ has a nonzero finite-dimensional representation V if and only if r is coprime to n; moreover, in th...

conjecture | open | MathDB 页面 | 原文线索

V2-006580 | MATHDB-377308

Margulis's bounded-degree Mahler measure conjecture

摘录 / EXCERPT: Margulis's conjecture. There is a function $\ell:\mathbb N\to\mathbb R^{>0}$ such that $m(p)\geq1+\ell\bigl(d(p)\bigr)$ for every non-cyclotomic monic polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-006581 | MATHDB-377334

Cameron's prime-element conjecture for permutation group algebras

摘录 / EXCERPT: Cameron's prime-element conjecture. If G has no finite orbits, then ε is a prime element in $A(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-006582 | MATHDB-377337

The integer-coordinate conjecture for Heron simplices

摘录 / EXCERPT: Integer-coordinate conjecture. Every Heron simplex can be represented with integer coordinates.

conjecture | open | MathDB 页面 | 原文线索

V2-006583 | MATHDB-377338

The integer-distances to integer-coordinates embedding conjecture

摘录 / EXCERPT: Embedding conjecture. One can find a Euclidean motion T such that

conjecture | open | MathDB 页面 | 原文线索

V2-006584 | MATHDB-377339

The divisibility embedding conjecture

摘录 / EXCERPT: Divisibility embedding conjecture. There is a set $N\subset\mathbb{Z}^n$ such that $k\cdot N$ is congruent to M.

conjecture | open | MathDB 页面 | 原文线索

V2-006585 | MATHDB-377340

The prime divisibility embedding conjecture

摘录 / EXCERPT: Prime divisibility embedding conjecture. There is a Euclidean motion T such that

conjecture | open | MathDB 页面 | 原文线索

V2-006586 | MATHDB-377345

Khovanov's alternating-knot refinement

摘录 / EXCERPT: Let L be a prime alternating knot, and let $s(L)$ and ${\text{\it Kh}}'(L)$ be the integer and Laurent polynomial from Khovanov's structural conjecture. The signature of L is t...

conjecture | open | MathDB 页面 | 原文线索

V2-006587 | MATHDB-377347

Jacobi transformation conjecture for VOA characters

摘录 / EXCERPT: Let V be a rational vertex operator algebra of CFT type, let M be an irreducible V-module, and let ${\mathfrak h}$ and L° be the spaces defined in the source, with...

conjecture | open | MathDB 页面 | 原文线索

V2-006588 | MATHDB-377365

Primitive powers conjecture for integer matrices

摘录 / EXCERPT: Let $r\geq 2$ and let $A\in \operatorname{Mat}_r(\mathbf Z)$ be nonsingular. Define $\gcd(A-I)$ as the greatest common divisor of the entries of $A-I$, and call A primitive when...

conjecture | open | MathDB 页面 | 原文线索

V2-006589 | MATHDB-377391

Weak finitarity conjecture for Pisot units

摘录 / EXCERPT: Let β be a Pisot unit, meaning an algebraic integer greater than 1 whose other algebraic conjugates have modulus less than 1 and whose norm is a unit. Call β weakly...

conjecture | open | MathDB 页面 | 原文线索

V2-006590 | MATHDB-377403

Serre's companion form conjecture for tame ramification of modular Galois representations

摘录 / EXCERPT: Let $f=\sum a_nq^n$ be an ordinary newform of type (k,ϵ) for $\Gamma_1(N)$, defined over a field E of characteristic p, with $(p,N)=1$, and let ... be the Galois represe...

conjecture | open | MathDB 页面 | 原文线索

V2-006591 | MATHDB-377420

Poonen's singular-locus density conjecture for arithmetic Bertini

摘录 / EXCERPT: Singular-locus density conjecture. There exists a constant $c>0$ such that, whenever d and p are sufficiently large,

conjecture | open | MathDB 页面 | 原文线索

V2-006592 | MATHDB-377421

The EKG prime-neighbor conjecture

摘录 / EXCERPT: Let $(a(n))_{n\geq 1}$ be the EKG sequence defined by $a(1)=1$, $a(2)=2$, and, for $n\geq 3$, letting $a(n)$ be the smallest natural number not already in the sequence such that...

conjecture | open | MathDB 页面 | 原文线索

V2-006593 | MATHDB-377424

The sub-four-thirds exponent conjecture for Frobenius numbers

摘录 / EXCERPT: Let (a,b,c) be an admissible triplet, meaning a triplet of positive integers satisfying the admissibility conditions for the three-variable Frobenius problem, and let $f(a,b,c)$...

conjecture | open | MathDB 页面 | 原文线索

V2-006594 | MATHDB-377446

Schmutz Schaller's counting-function conjecture for square and hexagonal lattices

摘录 / EXCERPT: Schmutz Schaller's counting-function conjecture. For every x, one has

conjecture | open | MathDB 页面 | 原文线索

V2-006595 | MATHDB-377447

The weighted counting-function conjecture for square and hexagonal lattices

摘录 / EXCERPT: Weighted counting-function conjecture. For $x\geq 8$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-006596 | MATHDB-377448

The reciprocal counting-function conjecture for square and hexagonal lattices

摘录 / EXCERPT: Reciprocal counting-function conjecture. For every x, one has

conjecture | open | MathDB 页面 | 原文线索

V2-006597 | MATHDB-377450

The torus analogue of Mazur's conjecture

摘录 / EXCERPT: The torus analogue of Mazur's conjecture. The connected component of the identity of the closure of the image of F in $\Gamma\backslash S({\mathbb R})$ equals the connected compo...

conjecture | open | MathDB 页面 | 原文线索

V2-006598 | MATHDB-377451

Schneider-Lang-Ramachandra's exponential conjecture

摘录 / EXCERPT: Schneider-Lang-Ramachandra's exponential conjecture. Under these assumptions, the determinant of M is nonzero.

conjecture | open | MathDB 页面 | 原文线索

V2-006599 | MATHDB-377465

Flatness conjecture for the naive symplectic local model

摘录 / EXCERPT: Let $\mathbf M^{\mathrm{naive}}_{GSp_{2g}}$ denote the naive local model for the symplectic similitude group GSp_{2g} in the ramified setting considered in the paper. Flatness co...

conjecture | open | MathDB 页面 | 原文线索

V2-006600 | MATHDB-377535

Displaced discrete-moment conjecture for the Riemann zeta function

摘录 / EXCERPT: Let $N(T)$ denote the number of non-trivial zeros $\tfrac12+{\mathrm{i}}\gamma_n$ with $0<\gamma_n\leq T$, and set ... For k with $\operatorname{Re}(k)>-1/2$, let $a(k)$ be the ari...

conjecture | open | MathDB 页面 | 原文线索

V2-006601 | MATHDB-377536

Conditional displaced fourth-moment conjecture for the Riemann zeta function

摘录 / EXCERPT: Let $N(T)$ denote the number of non-trivial zeros $\tfrac12+{\mathrm{i}}\gamma_n$ with $0<\gamma_n\leq T$, and set ... Let F_2 be the function specified by the paper's formula for...

conjecture | open | MathDB 页面 | 原文线索

V2-006602 | MATHDB-377579

Koszulness of annihilator and extension-cohomology modules

摘录 / EXCERPT: Let ℓ be a prime number and let F be a field containing a primitive ℓth root of unity. For $u\in H^1(G_F,\mathbb Z/\ell)$, let ... be its annihilator, and for a cyclic ex...

conjecture | open | MathDB 页面 | 原文线索

V2-006603 | MATHDB-377588

The sumset obstruction conjecture for prime triple sumsets

摘录 / EXCERPT: The sumset obstruction conjecture. If, for some κ, one has

conjecture | open | MathDB 页面 | 原文线索

V2-006604 | MATHDB-377629

Quantum induction irreducibility and unitarity conjecture for orthogonal groups

摘录 / EXCERPT: Assume $p^{\prime}+q\geq 2n+1$, $p+q^{\prime}\geq 2n+1$, $p+q\equiv p^{\prime}+q^{\prime}\pmod 2$, and $p^{\prime}+q^{\prime}\geq p+q$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006605 | MATHDB-377630

Quantum induction irreducibility and unitarity conjecture for metaplectic groups

摘录 / EXCERPT: Assume $n+n^{\prime}+1\geq p+q$ and $n^{\prime}\geq n$. Let $\mathcal Q(2n;p,q;2n^{\prime})(\pi)$ be the quantum-induced module defined from the relevant oscillator representation....

conjecture | open | MathDB 页面 | 原文线索

V2-006606 | MATHDB-377635

The lifting, bounded-conductor, purity and integrality conjecture for compatible systems

摘录 / EXCERPT: Galois-theoretic conjecture. The system satisfies all of the following:

conjecture | open | MathDB 页面 | 原文线索

V2-006607 | MATHDB-377684

Poonen's finiteness conjecture for modular curves of fixed genus

摘录 / EXCERPT: Let $X_1(N)$ be the usual modular curve over $\mathbb{Q}$, and call a curve X over $\mathbb{Q}$ modular if there is a nonconstant morphism $X_1(N)\to X$ over $\mathbb{Q}$ for som...

conjecture | open | MathDB 页面 | 原文线索

V2-006608 | MATHDB-377694

Canonical left tilde-S-region conjecture on completely prime ideals

摘录 / EXCERPT: Canonical left $\tilde{S}$-region conjecture. The ideal $J(\lambda)$ is completely prime.

conjecture | open | MathDB 页面 | 原文线索

V2-006609 | MATHDB-377696

Kurano-Roberts symbolic-power intersection conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a regular local ring, and let $\mathfrak{p}$ and $\mathfrak{q}$ be prime ideals in R such that ... Here $\mathfrak{p}^{(n)}$ denotes the nth symbolic po...

conjecture | open | MathDB 页面 | 原文线索

V2-006610 | MATHDB-377702

Coleman's unramified torsion conjecture for curves with good reduction

摘录 / EXCERPT: Let $p\geq 5$ be a prime number, and suppose that $K/{\mathbb Q}_p$ is an unramified finite extension. Let X/K be a curve of genus $g\geq 2$, embedded in its Jacobian via a K-r...

conjecture | open | MathDB 页面 | 原文线索

V2-006611 | MATHDB-377703

Generalization of the prime-power characteristic-polynomial formula

摘录 / EXCERPT: The prime-power generalization conjecture. For every integer $k\geq0$, there exists such a polynomial $c_k(t)$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-006612 | MATHDB-377705

Characteristic-polynomial congruences at prime powers congruent to 2 modulo 3

摘录 / EXCERPT: Prime-power congruence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006613 | MATHDB-377757

Conjecture on the distribution of irregular pairs by index difference

摘录 / EXCERPT: Let M be a fixed positive integer and let c be an integer with $0\le c<M$. An irregular pair is a pair (p,t) such that p is prime, t is even, and p divides the Bernoull...

conjecture | open | MathDB 页面 | 原文线索

V2-006614 | MATHDB-377769

Fundamental groups of homology 4-spheres and complements of contractible submanifolds

摘录 / EXCERPT: Fix a prime p, and for a group or space X let $\beta_i^p(X)$ denote the dimension of $H_i(X,\mathbf{Z}_p)$. A homology 4-sphere is a 4-manifold with the homology of the 4-spher...

conjecture | open | MathDB 页面 | 原文线索

V2-006615 | MATHDB-377797

Dyadic nondivisibility conjecture for multiple harmonic sums

摘录 / EXCERPT: Let l be a positive integer and let ${\boldsymbol{s}}\in\mathbb N^l$. Define $J({\boldsymbol{s}}|2)$ to be the set of nonnegative integers n for which the numerator of the mult...

conjecture | open | MathDB 页面 | 原文线索

V2-006616 | MATHDB-377798

Reserved-set classification conjecture for multiple harmonic sums

摘录 / EXCERPT: Let ${\boldsymbol{s}}\in\mathbb N^l$. Let $RJ({\boldsymbol{s}})$ be the reserved divisibility set, and let $RJ_1({\boldsymbol{s}})$ and $RJ_2({\boldsymbol{s}})$ denote the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-006617 | MATHDB-377824

Cyclic-group Lehmer constant conjecture

摘录 / EXCERPT: For an integer $n\geqslant2$, let $\rho(n)$ denote the smallest prime number that does not divide n. For the cyclic group $\mathbb{Z}_{\!/n}$, let $\lambda(\mathbb{Z}_{\!/n})$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-006618 | MATHDB-377897

The gap conjecture for products of faithful characters in finite p-groups

摘录 / EXCERPT: Let G be a finite p-group, where p is an odd prime number, and let $\chi, \psi \in \operatorname{Irr}(G)$ be faithful characters. Let $\eta(\chi,\psi)$ denote the number of d...

conjecture | open | MathDB 页面 | 原文线索

V2-006619 | MATHDB-377914

Conjectures for two-cycles with arbitrary and primitive-root conditions

摘录 / EXCERPT: Two-cycle counting conjectures. The paper conjectures

conjecture | open | MathDB 页面 | 原文线索

V2-006620 | MATHDB-377915

Average-order conjecture for arbitrary two-cycle counts

摘录 / EXCERPT: Average two-cycle conjecture. For every $C>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006621 | MATHDB-377919

The short-interval prime conjecture

摘录 / EXCERPT: For a real number $\theta>0$ and sufficiently large x, the interval $[x,x+x^\theta]$ contains a prime number. The short-interval prime conjecture. Given $\theta>0$, for all suffi...

conjecture | open | MathDB 页面 | 原文线索

V2-006622 | MATHDB-377921

Conjectural local statistics for shifted zeta-zero differences

摘录 / EXCERPT: Assume the Riemann Hypothesis, let γ and γ' range over imaginary parts of non-trivial zeros of $\zeta(s)$, and let h be the shift parameter. For fixed $\alpha>0$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-006623 | MATHDB-377922

Conjectural pair correlation in short positive intervals

摘录 / EXCERPT: Assume the Riemann Hypothesis and let γ,γ' range over imaginary parts of non-trivial zeros of $\zeta(s)$. For $0<\alpha<\beta\ll\log T$, the short-interval pair-correla...

conjecture | open | MathDB 页面 | 原文线索

V2-006624 | MATHDB-377937

Surjectivity of the genus function for weight-2 newforms on \(\Gamma_0(N)\)

摘录 / EXCERPT: Let $g_0^+(N,2)$ denote the dimension of the space of weight-2 newforms on $\Gamma_0(N)$, equivalently the genus of the corresponding new modular curve. Surjectivity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-006625 | MATHDB-377948

Denef-Loeser's topological monodromy conjecture

摘录 / EXCERPT: Let h be a complex polynomial and let $Z_{\mathrm{top},0}(h,s)$ be its local topological zeta function at the origin. If s_0 is a pole, write it in the source's notation as...

conjecture | open | MathDB 页面 | 原文线索

V2-006626 | MATHDB-377962

Holzer and König's maximal two-generated transformation monoid conjecture

摘录 / EXCERPT: For $n e 7$ be an integer, choose coprime integers $k,l e 2$ with $k+l=n$, and let $U_{k,l}$ be the transformation monoid defined by the two-cycle permutation construction above. H...

conjecture | open | MathDB 页面 | 原文线索

V2-006627 | MATHDB-377976

The Han-Monsky lower-bound and rigidity conjecture for Hilbert-Kunz multiplicity

摘录 / EXCERPT: Let $d\geq1$ be an integer and let $p>2$ be prime. Define ... Let $(A,\mathfrak m,k)$ be a d-dimensional unmixed local ring with $k=\overline{\mathbb F_p}$. Han-Monsky lower-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-006628 | MATHDB-377977

The higher-dimensional Hilbert-Kunz lower-bound conjecture

摘录 / EXCERPT: Use the notation above: let $d\geq1$, let $p>2$ be prime, define ... and let $(A,\mathfrak m,k)$ be a d-dimensional unmixed local ring with $k=\overline{\mathbb F_p}$. Higher-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-006629 | MATHDB-377990

Parity conjecture for the Möbius function on shifted primes

摘录 / EXCERPT: Let $S=(q_1,\dots,q_t)$ be a finite sequence of primes and define $\nu_S(n)=(\nu_{q_1}(n),\dots,\nu_{q_t}(n))$. Say that n is S-squarefree if $q^2\nmid n$ for every prime...

conjecture | open | MathDB 页面 | 原文线索

V2-006630 | MATHDB-377991

Möller's conjecture on scaled averages of cyclotomic coefficients

摘录 / EXCERPT: Let $a_n(k)$ denote the coefficient of x^k in the cyclotomic polynomial of order n, and let $M_n(a_n(k))$ denote the limiting mean value of $a_n(k)$ as n varies. For $k\ge1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006631 | MATHDB-378000

Multiplier-group subgroup conjecture for algebraic quasiperiodic flows

摘录 / EXCERPT: Let T^n be the n-torus, let ϕ be a quasiperiodic flow on T^n generated by a constant vector field X, and let F be a real algebraic number field of degree n over...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006632 | MATHDB-378001

Multiplier-group equality conjecture for real algebraic number fields

摘录 / EXCERPT: Let F be a real algebraic number field of degree n over $\mathbb{Q}$, and let T^n denote the n-torus. The multiplier-group equality conjecture. There exists a quasiperiodic...

conjecture | open | MathDB 页面 | 原文线索

V2-006633 | MATHDB-378002

Multiplier-group distinction conjecture for algebraic and transcendental quasiperiodic flows

摘录 / EXCERPT: Let ϕ be a quasiperiodic flow on T^n generated by a constant vector field X. Call ϕ algebraic if it is F-algebraic for a real algebraic number field F of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-006634 | MATHDB-378004

Conjecture on the roots of the polynomials alpha_l and beta_l

摘录 / EXCERPT: Root-location conjecture. All the roots of the polynomials $\alpha_l(m)$ and $\beta_l(m)$ lie on the line $\operatorname{Re}(m)=-1/2$. This conjecture is based on extensive numeric...

conjecture | open | MathDB 页面 | 原文线索

V2-006635 | MATHDB-378017

The conductor class group conjecture for associated graded rings

摘录 / EXCERPT: Let R be a Cohen--Macaulay ring, let $\mathfrak{p}$ be a prime ideal of finite projective dimension, and let $\mathrm G=\bigoplus_{t\geq 0}\mathfrak{p}^t/\mathfrak{p}^{t+1}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-006636 | MATHDB-378018

Normality conjecture for associated graded rings of prime ideals

摘录 / EXCERPT: Let R be a Cohen--Macaulay ring and let $\mathfrak{p}$ be a prime ideal of finite projective dimension. Write ... for the associated graded ring. Assume that $\mathrm G$ is a Cohen...

conjecture | open | MathDB 页面 | 原文线索

V2-006637 | MATHDB-378040

Equivariant Birch and Swinnerton-Dyer relation

摘录 / EXCERPT: Let G be a finite abelian group and let $V\to W$ be a G-cover of smooth projective geometrically connected curves of positive genus over ${\bf Q}$. Set $A={\rm Jac}(V)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-006638 | MATHDB-378045

Approximate multiplication exceptional-set conjecture

摘录 / EXCERPT: Approximate multiplication exceptional-set conjecture. The set $E(r)$ is finite. The expanding condition is necessary, and the paper proves the conjecture for rationals with denomi...

conjecture | open | MathDB 页面 | 原文线索

V2-006639 | MATHDB-378052

Power-residue density conjecture for non-CM newforms

摘录 / EXCERPT: Let $f=\mathfrak{a}0\sum a_nq^n$ be a newform with rational Fourier coefficients, and let $m\geq 2$ be an integer. Define $\delta_m(f)$ as the relative density of the primes...

conjecture | open | MathDB 页面 | 原文线索

V2-006640 | MATHDB-378055

Generalized Ramanujan-Selberg conjecture for unramified cuspidal representations of GL(n)

摘录 / EXCERPT: Let π be a cuspidal automorphic representation of $GL(n)$ that is unramified at a prime p, and let $\alpha_{p,j}$ be the quantities appearing in its local Euler product. Gene...

conjecture | open | MathDB 页面 | 原文线索

V2-006641 | MATHDB-378085

Neiderreiter's logarithmic discrepancy conjecture for arithmetic permutations

摘录 / EXCERPT: Let p) be prime, let $\mathbb{Z}_p^\times$ denote the nonzero residue classes modulo p, and let ψ_k be the arithmetic permutation defined in the surrounding discussion. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-006642 | MATHDB-378086

Strong logarithmic discrepancy conjecture for arithmetic permutations

摘录 / EXCERPT: Let p be prime, let ψ_k be the arithmetic permutation considered in the paper, and let $D(\psi_k)$ denote its discrepancy. Strong logarithmic discrepancy conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-006643 | MATHDB-378087

Uniform fourth-root bound conjecture for incomplete power sums

摘录 / EXCERPT: Let p be prime and define ... Here a is understood modulo p, $k\in\mathbb{Z}_p^\times$, and M is in the range used for incomplete sums in the paper. Uniform power-sum conject...

conjecture | open | MathDB 页面 | 原文线索

V2-006644 | MATHDB-378088

Bounded-average-partial-quotient approximation conjecture

摘录 / EXCERPT: Fix an integer $B\geq2$, and let $\mathcal{C}_B$ be the set of irrational numbers whose partial quotients are bounded in average by B. For each n, consider rational points...

conjecture | open | MathDB 页面 | 原文线索

V2-006645 | MATHDB-378103

The large-order conjecture for Kummer field generators

摘录 / EXCERPT: Let q be a prime power and let $\textbf{F}_{q^n}$ be a finite field. Let $g\in\textbf{F}_{q^n}$ satisfy the conditions in Theorem, and suppose that $n\geq\log q$. Large-order con...

conjecture | open | MathDB 页面 | 原文线索

V2-006646 | MATHDB-378113

Atkin-Swinnerton-Dyer congruence conjecture for noncongruence cusp forms

摘录 / EXCERPT: Let Γ be a noncongruence subgroup of $SL_2(\mathbb Z)$ with finite index, let $k\geq 2$, and let $S_k(\Gamma)$ have dimension d. Denote by X_{Γ} the modular curve a...

conjecture | open | MathDB 页面 | 原文线索

V2-006647 | MATHDB-378118

Real Bredon cohomology comparison conjecture for algebraic groups

摘录 / EXCERPT: Let G be an algebraic group over $\mathbb R$, let ... and let p be a prime number. Regard $G(\mathbb C)$ as a Γ-space and let $G(\mathbb C)^{\operatorname{top}}$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-006648 | MATHDB-378119

Buzzard's classical 2-adic Newton polygon conjecture for level-1 cusp forms

摘录 / EXCERPT: Let $k\geq12$ be even, let $S_k=S_k(\Gamma_0(1),\mathbf{Q})$ be the space of level-1 cusp forms of weight k, and let $m=\dim S_k$. For a polynomial with rational coefficients,...

conjecture | open | MathDB 页面 | 原文线索

V2-006649 | MATHDB-378120

Mazur's tame potential semistability conjecture for classical modular points

摘录 / EXCERPT: Let $\overline{\Omega}$ be the closure of the crystalline locus in the deformation space $X(\overline{\rho})$, and let $\rho \in \overline{\Omega}$ be a classical modular point. Sa...

conjecture | open | MathDB 页面 | 原文线索

V2-006650 | MATHDB-378135

Conjecture on Smarandache cube-digital subsequences

摘录 / EXCERPT: Smarandache cube-digital subsequence conjecture. All terms of the Smarandache cube-digital subsequence are of the form $D \times 10^{3k}$ where $D \in \left\{0,1,8\right\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006651 | MATHDB-378180

Conjectures for fixed-point counts of the discrete logarithm map

摘录 / EXCERPT: Fixed-point counting conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-006652 | MATHDB-378185

Average fixed-point count conjectures

摘录 / EXCERPT: Average fixed-point conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-006653 | MATHDB-378186

Average auxiliary-count conjecture

摘录 / EXCERPT: Average auxiliary-count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006654 | MATHDB-378187

The conjecture on the error in estimated numbers of discrete logarithm fixed points

摘录 / EXCERPT: Let p be prime, let g and h range over the parameters defining the fixed-point counting function, and let $N_{, g \mathop\mathsf{ANY}, h \mathop\mathsf{ANY}}(p)$ denote the c...

conjecture | open | MathDB 页面 | 原文线索

V2-006655 | MATHDB-378190

Orbit cardinality formulas for primitive square-tiled surfaces in H(2)

摘录 / EXCERPT: Let n be an odd integer, and consider the two orbits of primitive n-square-tiled surfaces in ${\mathcal{H}}(2)$, denoted orbit A and orbit B. For a prime p, write...

conjecture | open | MathDB 页面 | 原文线索

V2-006656 | MATHDB-378191

Elliptic-point count conjecture for prime square-tiled surfaces in H(2)

摘录 / EXCERPT: Let n be prime, and consider the elliptic points associated with primitive n-square-tiled surfaces in ${\mathcal{H}}(2)$. Elliptic-point count conjecture. For prime n, the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-006657 | MATHDB-378252

Langlands' conjecture with finite-field Laurent-series coefficients

摘录 / EXCERPT: Langlands' conjecture. Let ρ be as in de Jong's conjecture. Then there exists a non-zero cuspidal Hecke eigen-form f_ρ with eigenvalues corresponding to ρ.

conjecture | open | MathDB 页面 | 原文线索

V2-006658 | MATHDB-378257

Amoroso-Zannier's relative Lehmer conjecture

摘录 / EXCERPT: Amoroso-Zannier's conjecture. There exists a constant $c(K)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006659 | MATHDB-378259

The vanishing conjecture for Morita-Mumford classes of surface bundles

摘录 / EXCERPT: Let B be the base of a surface bundle, let p be a prime, and let κ_i denote its Morita-Mumford classes. Vanishing conjecture. For $s\geq 1$ and $v\geq 0$, one has ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-006660 | MATHDB-378340

Loeser's generalized monodromy conjecture for multivariable zeta functions

摘录 / EXCERPT: Loeser's generalized monodromy conjecture. For almost all finite places $\mathfrak P$, if H is a polar hyperplane in $\mathcal P(Z_\phi(F_{\mathfrak P},\chi;s))$, with ϕ res...

conjecture | open | MathDB 页面 | 原文线索

V2-006661 | MATHDB-378356

Breuil-Mézard conjecture for tame determinant types

摘录 / EXCERPT: Fix an odd prime p, a finite extension E of ${\mathbb Q}_p$ with residue field ${\mathbf k}_E$, and a potentially crystalline representation...

conjecture | open | MathDB 页面 | 原文线索

V2-006662 | MATHDB-378357

Refinement of the modular-degree parity theorem for Neumann-Setzer curves

摘录 / EXCERPT: Let $p=u^2+64$ be prime, and let E_0 be the Neumann-Setzer elliptic curve of conductor p. The modular degree of E_0 is the degree of its optimal modular parametrization from...

conjecture | open | MathDB 页面 | 原文线索

V2-006663 | MATHDB-378359

Parity conjecture for modular degrees of non-Neumann-Setzer curves

摘录 / EXCERPT: Let p be prime and let E be an optimal elliptic curve quotient of $J_0(p)$. A Neumann-Setzer curve is the special curve family considered in the paper. The non-Neumann-Setzer m...

conjecture | open | MathDB 页面 | 原文线索

V2-006664 | MATHDB-378369

The integrality conjecture for the regulator element ${\mathfrak S}_{K/k}$

摘录 / EXCERPT: Let k be a totally real number field, let p be an odd prime splitting completely in k, and let K be as in the paper. Write $G=\operatorname{Gal}(K/k)$ and let $\delta(K/k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-006665 | MATHDB-378370

Rubin's Conjecture A' for higher Stark regulators

摘录 / EXCERPT: Let K^+/k be the totally real subextension under consideration, let $G^+=\operatorname{Gal}(K^+/k)$, let $S_0=S_{\rm ram}(K/k)$, and let $U_{S_0}(K^+)$ be the group of global...

conjecture | open | MathDB 页面 | 原文线索

V2-006666 | MATHDB-378371

The refined Rubin-Stark reciprocity conjecture

摘录 / EXCERPT: Let p be an odd prime splitting completely in k, let K/k and the other notation be as in the paper, and let $G=\operatorname{Gal}(K/k)$. Let $\eta^+_{K/k}$ be the image of th...

conjecture | open | MathDB 页面 | 原文线索

V2-006667 | MATHDB-378374

The p-adic convergence conjecture for elliptic divisibility sequences

摘录 / EXCERPT: Let ${\mathcal W}=(W_n)_{n\ge0}$ be an elliptic divisibility sequence and let p be a prime. For an exponent $N=N_p\ge1$, consider the subsequences indexed by mp^{kN} for each...

conjecture | open | MathDB 页面 | 原文线索

V2-006668 | MATHDB-378384

The fractal intersection conjecture

摘录 / EXCERPT: The fractal intersection conjecture. The closure $\overline{Z(\bar K)\cap F}^{\,\mathrm{Zar}}$ is a union of finitely many points and finitely many components B_i such that...

conjecture | open | MathDB 页面 | 原文线索

V2-006669 | MATHDB-378389

Conjecture on the number of zeta-function zeros in strips for hyperbolic rational maps

摘录 / EXCERPT: Let f be a hyperbolic rational map, let $Z(s)$ denote its dynamical zeta function, let $\\{\mu_j\}$ be the zeros of Z counted with multiplicity, and let δ be the dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-006670 | MATHDB-378391

Density conjecture for sums of distinct powers

摘录 / EXCERPT: Density conjecture for sums of distinct powers. If every pair $a_1,a_2\in A$ satisfies $\gcd\{a_1,a_2\}=1$ and

conjecture | open | MathDB 页面 | 原文线索

V2-006671 | MATHDB-378392

Heuristic exponent conjecture for $(2,1,2)$-numbers

摘录 / EXCERPT: Heuristic exponent conjecture for $(2,1,2)$-numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-006672 | MATHDB-378393

Asymptotic conjecture for $(2,k,k)$-numbers

摘录 / EXCERPT: Asymptotic conjecture for $(2,k,k)$-numbers. For each k,

conjecture | open | MathDB 页面 | 原文线索

V2-006673 | MATHDB-378397

Signature-zero prime-determinant linking-form conjecture

摘录 / EXCERPT: Let K be a knot, let $\sg(K)$ denote its signature, let $D=\dt(K)$ be its prime determinant, let H_1 denote the first homology group of the double branched cover of K, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-006674 | MATHDB-378421

Denef-Sargos polar multiplicity conjecture for stable faces

摘录 / EXCERPT: Let f be a singular complex analytic germ at $0\in\mathbb{C}^{n}$ with $f(0)=0$, and let φ be a Schwarz function on $\mathbb{C}^{n}$ with sufficiently small support and...

conjecture | open | MathDB 页面 | 原文线索

V2-006675 | MATHDB-378446

Rank-torsion determination conjecture for prime knots

摘录 / EXCERPT: Rank-torsion conjecture. The knots K_1 and K_2 have the same ranks of the Khovanov homology if and only if they have the same torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-006676 | MATHDB-378453

The S-integer conjecture for left orderability of arithmetic groups

摘录 / EXCERPT: Let G be a ${\mathord{\mathbb{Q}}}$-simple algebraic ${\mathord{\mathbb{Q}}}$-group, let $\{p_1,\ldots,p_n\}$ be a nonempty set of prime numbers, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-006677 | MATHDB-378461

Equidistribution of prime values over allowable congruence classes

摘录 / EXCERPT: Let $m\ge2$ be an integer, and let $f(X)\in\mathbb Z[X]$ represent infinitely many primes. A congruence class modulo m is allowable if every integer r in it satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-006678 | MATHDB-378463

Existence of a limiting density for primitive roots among polynomial primes

摘录 / EXCERPT: Primitive-root density conjecture. The quotient $R_g(f;x)$ tends to a limit as x tends to infinity; denote this conjectural density by $\delta_g(f)$. This predicts that the relat...

conjecture | open | MathDB 页面 | 原文线索

V2-006679 | MATHDB-378464

Griffin's dream for primitive roots in polynomial prime values

摘录 / EXCERPT: Let f be a polynomial representing infinitely many distinct primes, let $g\in G$, and let $c_g(f)$ denote the number of primes represented by f for which g is a primitive roo...

conjecture | open | MathDB 页面 | 原文线索

V2-006680 | MATHDB-378483

Integral-closure conjecture for the plus and minus Hecke algebras

摘录 / EXCERPT: Integral-closure conjecture. If $k<p-1$, then $\mathbf{T}^{+}$ and $\mathbf{T}^{-}$ are integrally closed. Equivalently, every congruence between distinct eigenforms in...

conjecture | open | MathDB 页面 | 原文线索

V2-006681 | MATHDB-378484

Fricke-sign conjecture for theta congruences in weights 2 and 4

摘录 / EXCERPT: Fricke-sign conjecture. If $f\in S_2(\overline{\mathbf{Z}}_p)$ and $g\in S_4(\overline{\mathbf{Z}}_p)$ are eigenforms satisfying $\theta f\equiv g\pmod p$, then the w_p-eigenvalu...

conjecture | open | MathDB 页面 | 原文线索

V2-006682 | MATHDB-378485

Lario-Rio conjecture on minimal conductors of mod 3 representations

摘录 / EXCERPT: Let $\mathbf{P}\overline{\rho}: \mathrm{Gal}(\overline{\mathbf Q}/\mathbf Q) \rightarrow \mathrm{PGL}_2(\mathbf F_3)$ be an irreducible representation. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-006683 | MATHDB-378491

Character-sum conjecture for the extended-support problem

摘录 / EXCERPT: Let $P\geq 1$, let $k\leq P$ be a nonsquare, so that the character $\left(\frac{k}{\cdot}\right)$ is nonprincipal, and set $H=P^{2/3+\delta}$. Let $e(x)=\exp(2\pi i x)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006684 | MATHDB-378494

The prime-power annihilation conjecture for polynomial rings over the integers

摘录 / EXCERPT: Prime-power annihilation conjecture. For every prime power $q=p^e$, there exists $k\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006685 | MATHDB-378516

Infinitely many lengths with exactly 2^e very odd sequences

摘录 / EXCERPT: For each natural number n, let $S(n)$ denote the number of very odd binary sequences of length n. For an integer $e\geq 1$, consider the level set of lengths satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-006686 | MATHDB-378517

Asymptotic order of the number of lengths with 2^e very odd sequences

摘录 / EXCERPT: Let $N_{2^e}(x)$ denote the number of integers $n\leq x$ for which $S(n)=2^e$, and let $r_{\omega_1}(e)$ be the parameter used in the paper to state the preceding bounds. Suppose t...

conjecture | open | MathDB 页面 | 原文线索

V2-006687 | MATHDB-378530

Rational-function lower-bound conjecture

摘录 / EXCERPT: Rational-function lower-bound conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-006688 | MATHDB-378531

Exceptional sine-sum conjecture

摘录 / EXCERPT: Let q and n be positive integers, and let $p_1,\ldots,p_n$ be distinct integers relatively prime to q such that $q>(7/4)^n$ and ... Assume that for every $k\in\{1,\ldots,n\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006689 | MATHDB-378539

Baker and Wantz's 2-rank conjecture for Buekenhout-Metz unitals

摘录 / EXCERPT: Let q be an odd prime power, let β be a primitive element of $\mathbb F_{q^2}$, and define ... for $r\in\mathbb F_q$. Set ... and let $\mathcal U_{\beta}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-006690 | MATHDB-378552

Auil's equality conjecture for the iterative and natural Beurling approximations

摘录 / EXCERPT: Let B_n be the natural Beurling approximation defined by ... and let $\widetilde{B}_n$ be the approximating function obtained from the paper's iterative sequence...

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V2-006691 | MATHDB-378553

Auil's conjecture that the iterative sequence enumerates the Möbius numbers

摘录 / EXCERPT: Let $\mathbb K=\{k_i\}_{i\in\mathbb N}$ be the set of integers produced by the paper's iterative construction, and let ... be the set of square-free, or Möbius, numbers. Auil's enume...

conjecture | open | MathDB 页面 | 原文线索

V2-006692 | MATHDB-378555

The representation-count formula for binary quadratic form numbers

摘录 / EXCERPT: Let a ${\mathcal B}$-number be a positive integer represented by the binary quadratic form a^2+ab+b^2. Suppose that such a number is written in the general form ... where x has n...

conjecture | open | MathDB 页面 | 原文线索

V2-006693 | MATHDB-378584

Lexicographically least extremal permutation conjecture

摘录 / EXCERPT: Let $n>2$ be prime, and define a permutation g of $\mathbb Z_n$ by ... Order permutations lexicographically, and let $\Psi(n)$ be the minimum number of collinear triples in a permu...

conjecture | open | MathDB 页面 | 原文线索

V2-006694 | MATHDB-378593

Harvey-Moore modularity conjecture for fiberwise rational curves

摘录 / EXCERPT: Let X be a tamed $K3$-fibered Calabi-Yau threefold, and let n_C denote the virtual number of rational curves in the fiberwise class $C\in H_2(X,\mathbf Z)_f$. Suppose that the...

conjecture | open | MathDB 页面 | 原文线索

V2-006695 | MATHDB-378614

The Δ-conjecture for irregular primes and Bernoulli numbers

摘录 / EXCERPT: Δ-conjecture. The following properties are equivalent:

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V2-006696 | MATHDB-378615

The conjecture that the marginal ideal sum is prime and perfect

摘录 / EXCERPT: Primality and perfection conjecture. If Δ is any simplicial complex, then

conjecture | open | MathDB 页面 | 原文线索

V2-006697 | MATHDB-378618

The torus-knot invariant formula conjecture

摘录 / EXCERPT: Let $T(p,q)$ be a torus knot of type (p,q), where p and q are coprime integers. Let $\rho_{j,k}\colon \pi T(p,q)\to\operatorname{SO}(3)$ send the standard generators a and...

conjecture | open | MathDB 页面 | 原文线索

V2-006698 | MATHDB-378632

Isospectrality determines the zeta functions of canonical models

摘录 / EXCERPT: Canonical-model zeta conjecture. If M_{K_1} and M_{K_2} are isospectral, then the Hasse-Weil zeta functions of the canonical models of $M_{\tilde{K}_1}$ and $M_{\tilde{K}_2}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-006699 | MATHDB-378644

Conjecture on absolute moments of primes in short intervals

摘录 / EXCERPT: Let X be large, let h be a positive real parameter, and let $\psi(x)=\sum_{n\leq x}\Lambda(n)$ be Chebyshev's function. For fixed $\epsilon>0$ and $\lambda>0$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-006700 | MATHDB-378645

Conjecture on absolute moments of primes in relative short intervals

摘录 / EXCERPT: Let X be large, let δ be a positive real parameter, and let $\psi(x)=\sum_{n\leq x}\Lambda(n)$ be Chebyshev's function. For fixed $\epsilon>0$ and $\lambda>0$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-006701 | MATHDB-378658

The affine surface hypothesis for polynomials with irreducible leading form

摘录 / EXCERPT: Let $\delta\geq 2$. For a polynomial $g\in\mathbb{Z}[T_1,\ldots,T_\nu]$, let $\mathfrak{h}(g)$ denote its homogeneous part of maximal degree, and let $\mathcal{J}_\delta$ be the se...

conjecture | open | MathDB 页面 | 原文线索

V2-006702 | MATHDB-378663

Bacher's conjecture on Bernoulli-number numerator quotients

摘录 / EXCERPT: Bacher's conjecture. Then $A(p)=(l-1)p+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-006703 | MATHDB-378667

Golyshev's modularity conjecture for the D3 equation of Fano threefolds

摘录 / EXCERPT: Golyshev's conjecture. The family of counting equations for X contains an equation

conjecture | open | MathDB 页面 | 原文线索

V2-006704 | MATHDB-378763

The 3x+1 conjecture

摘录 / EXCERPT: $(3x+1)$-Conjecture. For every $x\in\Pi$, there is an integer n such that

conjecture | open | MathDB 页面 | 原文线索

V2-006705 | MATHDB-378766

The quadratic exponential-sum bound and its extremal polynomials

摘录 / EXCERPT: Let $m\geq 3$ be odd, let n be a non-negative integer, and let f be a quadratic polynomial in n variables modulo m. Write $q=2\cos\left(\frac{\pi}{2m}\right)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006706 | MATHDB-378768

Characterization of the Barnes multiple zeta-function limit

摘录 / EXCERPT: Let $t=\varphi(s)$ be a meromorphic function on $\mathbb{C}$, and let $\zeta_{q,r}(s,t,z;\boldsymbol{\omega})$ and $\zeta_r(s,z;\boldsymbol{\omega})$ denote the corresponding q-a...

conjecture | open | MathDB 页面 | 原文线索

V2-006707 | MATHDB-378770

Linear-density conjecture for primitive divisors in quadratic polynomial sequences

摘录 / EXCERPT: Let β be a nonzero integer and define the sequence $P=(P_n)$ by ... A term P_n has a primitive divisor if it has a prime divisor that divides no earlier nonzero term of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-006708 | MATHDB-378771

The Shimura correspondence central-value formula for level p^2

摘录 / EXCERPT: Let p be the prime and let f be the weight-2 eigenform introduced in the construction above. Fix $\sigma\in\{+,-\}$, and let $c_f(d)$ be the coefficient of q^d in ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006709 | MATHDB-378775

The converse between semigroup property and integer normed pairings for quadratic forms

摘录 / EXCERPT: Let $f:\mathbb{R}^2\to\mathbb{R}$ be an integer quadratic form, meaning that its coefficients are integers. The form f has semigroup property if the product of any two values rep...

conjecture | open | MathDB 页面 | 原文线索

V2-006710 | MATHDB-378781

Borwein-Choi conjecture on representations by binary quadratic forms

摘录 / EXCERPT: Borwein-Choi conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006711 | MATHDB-378866

Friedlander-Iwaniec twisted incomplete Kloosterman-sum conjecture

摘录 / EXCERPT: Let $a,q\geq2$ be integers with $(a,q)=1$ and q not a perfect square, and let $H,K\geq1$ be reals. Define $e(u):=e^{2\pi i u}$, let $\left(\frac{\cdot}{\cdot}\right)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-006712 | MATHDB-378867

Twisted quadratic Kloosterman-sum bound with additive twists

摘录 / EXCERPT: Let $a,q\geq2$ be integers with $(a,q)=1$ and q not a perfect square. Let $H,K\geq1$ and λ,μ be reals. Define $e(u):=e^{2\pi i u}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-006713 | MATHDB-378868

The relative undecidability conjecture for finiteness of integral points

摘录 / EXCERPT: Relative undecidability conjecture. The finiteness problem for integral points is undecidable relative to the existence problem.

conjecture | open | MathDB 页面 | 原文线索

V2-006714 | MATHDB-378876

Browning-Heath-Brown conjecture on rational points of non-singular hypersurfaces

摘录 / EXCERPT: Let $n\geqslant3$, let $F\in\mathbb{Z}[X_0,\ldots,X_n]$ be a non-singular form of degree $d\geqslant2$, and define ... For $B\geqslant1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006715 | MATHDB-378880

Oscillation conjecture for recurrence sequences without positive dominating roots

摘录 / EXCERPT: Let $(a(n))$ be a recurrence sequence, and let its dominating characteristic roots be the characteristic roots of maximal modulus among those occurring with nonzero coefficient in...

conjecture | open | MathDB 页面 | 原文线索

V2-006716 | MATHDB-378902

The lower-bound conjecture for the zero-sum-free sequence constant

摘录 / EXCERPT: Let $n>1$ be a positive integer, and let p be the smallest prime divisor of n. For positive integers n and k, let ... where S ranges over sequences in $\mathbb{Z}_n$ of len...

conjecture | open | MathDB 页面 | 原文线索

V2-006717 | MATHDB-378916

Twisted Kirillov-Reshetikhin crystal conjecture for constructed perfect crystals

摘录 / EXCERPT: Twisted Kirillov-Reshetikhin crystal conjecture. The perfect $U_q^{\prime}(\widehat{\mathfrak{g}})$-crystal $\widehat{\mathcal{B}}^{i,s}$ is isomorphic to the conjectural crystal b...

conjecture | open | MathDB 页面 | 原文线索

V2-006718 | MATHDB-378925

Weil's Tamagawa number conjecture for simply connected groups

摘录 / EXCERPT: Weil's conjecture. If G is simply connected, then

conjecture | open | MathDB 页面 | 原文线索

V2-006719 | MATHDB-378931

The conjecture on the least exponent for double almost squares

摘录 / EXCERPT: Let $g(\theta)$ be the least exponent such that, for some $c_1,c_2>0$, every interval of the form ... contains an integer $n=a_1b_1=a_2b_2$, where ... and all four integers lie in ... an...

conjecture | open | MathDB 页面 | 原文线索

V2-006720 | MATHDB-378932

The average Hardy-Littlewood prime k-tuple conjecture

摘录 / EXCERPT: Let $\Lambda(n)$ be the von Mangoldt function, and for distinct integers $d_1,\ldots,d_k$ with $1\leq d_i\leq H$, let $\mathcal{D}=\{d_1,\ldots,d_k\}$. Define $E_k(x;\mathcal{D})$...

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V2-006721 | MATHDB-378933

The conjecture on the 19th and 20th lattice invariants

摘录 / EXCERPT: Let v be the invariant assigned to the lattices indexed by $1,\ldots,24$, with the indices 19 and 20 referring to the corresponding entries in the preceding table. Conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-006722 | MATHDB-378981

Oppenheim's conjecture on lattices with positive norm minimum

摘录 / EXCERPT: Let $n\geqslant 3$ and let $\Lambda\subset\mathbb{R}^n$ be an n-dimensional lattice with positive norm minimum $N(\Lambda)>0$. A lattice is called algebraic when it is similar, m...

conjecture | open | MathDB 页面 | 原文线索

V2-006723 | MATHDB-378982

Sail formulation of Oppenheim's conjecture

摘录 / EXCERPT: Let $n\geqslant 3$, let $\Lambda\subset\mathbb{R}^n$ be an n-dimensional lattice, and let Π be a sail generated by Λ. A sail's facets and edge stars are its facets an...

conjecture | open | MathDB 页面 | 原文线索

V2-006724 | MATHDB-379003

Vasyunin's conjecture on high-dimensional cyclic quotient families

摘录 / EXCERPT: Vasyunin's conjecture. Up to a permutation of the coordinates in $T^{(2d+1)}$, the corresponding points lie in the subtorus

conjecture | open | MathDB 页面 | 原文线索

V2-006725 | MATHDB-379007

The prime-divisor conjecture for twisted Borcherds product CM values

摘录 / EXCERPT: Let F be the real quadratic field and let K be a CM extension of F as in the preceding construction. Write $\mathcal{O}_F$ for the ring of integers of F, let $\mathfrak I$...

conjecture | open | MathDB 页面 | 原文线索

V2-006726 | MATHDB-379033

The elementary type conjecture for finitely generated pro-p Galois groups

摘录 / EXCERPT: Let $p>2$ be a prime and let F be a field containing a primitive p-th root of unity. Write $F(p)$ for the maximal p-extension of F, let $G=\operatorname{Gal}(F(p)/F)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006727 | MATHDB-379063

The limsup conjecture for the zeta zero-counting error term

摘录 / EXCERPT: Let $S(t)$ denote the error term in the number of zeros of the Riemann zeta-function up to height t. Limsup conjecture. ... The conjecture is motivated by arguments analogous to th...

conjecture | open | MathDB 页面 | 原文线索

V2-006728 | MATHDB-379064

The maximum critical-value conjecture for families of L-functions

摘录 / EXCERPT: Let $\mathcal F$ be a family of L-functions, and for $F\in\mathcal F$ let $c(F)$ denote its conductor. Assume the family is ordered by conductor and has approximately D members...

conjecture | open | MathDB 页面 | 原文线索

V2-006729 | MATHDB-379065

The prime-model conjecture for the maximum of the truncated zeta-function

摘录 / EXCERPT: Let $Z_X(s)$ denote the truncated zeta-function used in the paper, and let $A>0$ be fixed. Truncated-zeta maximum conjecture. If $2<X<\log^A T$, then, as $T\to\infty$, ... The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-006730 | MATHDB-379071

The fundamental-germ isomorphism conjecture for nonstandard integers

摘录 / EXCERPT: Let $r,s\in\mathbb R\setminus\mathbb Q$ be irrational numbers, and let ${}^{\ast}\mathbb Z_r$ and ${}^{\ast}\mathbb Z_s$ denote their associated groups of nonstandard diophantine a...

conjecture | open | MathDB 页面 | 原文线索

V2-006731 | MATHDB-379106

The Girth Conjecture for Heegaard splittings

摘录 / EXCERPT: Let M be a closed orientable 3-manifold and let $H\cup H^{\prime}$ be a Heegaard splitting of M. The algebraic girth is defined using the minimum girth of a nontrivial relato...

conjecture | open | MathDB 页面 | 原文线索

V2-006732 | MATHDB-379107

The primitive curve conjecture for Heegaard splittings

摘录 / EXCERPT: Let M be a closed orientable 3-manifold and let $H\cup H^{\prime}$ be a Heegaard splitting of M. A closed curve in ∂H is primitive in H if it does not represent a...

conjecture | open | MathDB 页面 | 原文线索

V2-006733 | MATHDB-379184

The equality conjecture for the homotopy-exponent valuation bound

摘录 / EXCERPT: Let p be any prime, let $\alpha,l\in\mathbb{N}$, and let $n,r\in\mathbb{Z}$ with $n\ge 2p^{\alpha}-1$. The theorem gives a lower bound for the p-adic valuation of the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-006734 | MATHDB-379185

Adams's cofibration-generation conjecture for the thick subcategory C_2

摘录 / EXCERPT: Let p be an odd prime, let $\mathcal C_2$ be the thick subcategory of finite p-local spectra of type at least 2, and let $V(1)$ denote the Smith-Toda complex. Adams's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-006735 | MATHDB-379186

Adams's refined generation conjecture for type-2 atomic spectra

摘录 / EXCERPT: Let p be an odd prime, let $V(1)$ be the Smith-Toda complex, and let X be a type-2 atomic spectrum. For a finite p-torsion atomic spectrum X, define $e(X)=k$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-006736 | MATHDB-379188

The spacing conjecture for reduced fractions with square moduli

摘录 / EXCERPT: Spacing conjecture for square moduli.

conjecture | open | MathDB 页面 | 原文线索

V2-006737 | MATHDB-379189

The higher-power modulus spacing conjecture

摘录 / EXCERPT: Higher-power modulus spacing conjecture. The local multiplicity of $S_{k,Q}$ at scale Q^{-k-1} should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-006738 | MATHDB-379194

Parker's conjectural coordinator polynomial for cyclotomic lattices of order twice a prime

摘录 / EXCERPT: Let p be an odd prime, let $\zeta_{2p}=e^{2\pi i/(2p)}$, and let $h_{2p}(x)$ be the coordinator polynomial of $\mathbb Z[\zeta_{2p}]$ with respect to the set of all $2p$th roots...

conjecture | open | MathDB 页面 | 原文线索

V2-006739 | MATHDB-379220

Conjecture on moments of the derivative of the Riemann zeta-function

摘录 / EXCERPT: Let k be a positive integer, let $\theta(t)$ be the Riemann-Siegel theta function, and define Hardy's function by $Z(t)=e^{i\theta(t)}zeta(1/2+it)$. Let a_k be the arithmetic f...

conjecture | open | MathDB 页面 | 原文线索

V2-006740 | MATHDB-379238

The algebraicity-multiplier group conjecture for quasiperiodic flows

摘录 / EXCERPT: Algebraicity-multiplier group conjecture. A quasiperiodic flow ψ is F-algebraic if and only if M_ψ is a finite index subgroup of ${\mathfrak o}_F^*$.

conjecture | open | MathDB 页面 | 原文线索

V2-006741 | MATHDB-379243

Conjecture for the first two terms in the quadratic-twist vanishing ratio

摘录 / EXCERPT: Let E be an elliptic curve of conductor Q, let $q\nmid Q$ be prime, and let $S^{\pm}(X)$ be the relevant families of fundamental discriminants with sign $\pm$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-006742 | MATHDB-379261

The local monodromy conjecture for topological zeta functions

摘录 / EXCERPT: Let $f:({\mathbb C}^{n+1},0)\to({\mathbb C},0)$ be a germ of an analytic function, and let $Z_{top,0}(f,s)\in{\mathbb Q}(s)$ be its local topological zeta function. A pole s_0 of...

conjecture | open | MathDB 页面 | 原文线索

V2-006743 | MATHDB-379264

The Birch--Swinnerton-Dyer formula for the false Tate representations

摘录 / EXCERPT: Let p be an odd prime, let $K=\mathbb{Q}(\mu_p)$ and let F be the false Tate extension layer used in the paper. Let $E/\mathbb{Q}$ be an elliptic curve with $E(K)$ finite, let...

conjecture | open | MathDB 页面 | 原文线索

V2-006744 | MATHDB-379265

The Coates--Sujatha module conjecture for false Tate curve extensions

摘录 / EXCERPT: Let p be an odd prime, let $F_\infty=\mathbb{Q}(\mu_{p^\infty},\sqrt[p^n]{m}:n\geq1)$, let $G=\operatorname{Gal}(F_\infty/\mathbb{Q})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006745 | MATHDB-379307

Minimal-prime classification conjecture for permanental ideals

摘录 / EXCERPT: Let $M_{m,n}=(x_{i,j})$ be a generic $m\times n$ matrix over a field k, and let $I_d(m,n)\subset k[x_{i,j}]$ be generated by the permanents of its $d\times d$ submatrices. A mini...

conjecture | open | MathDB 页面 | 原文线索

V2-006746 | MATHDB-379311

The duality cofiber-sequence conjecture for the K(2)-local sphere

摘录 / EXCERPT: Let p be a prime and let ℓ be a generator of $\mathbb{Z}_p^\times$ when p is odd, or of $\mathbb{Z}_2^\times/\{\pm 1\}$ when $p=2$. Write $Q(\ell)_{K(2)}$ for the $K(2)$-l...

conjecture | open | MathDB 页面 | 原文线索

V2-006747 | MATHDB-379344

Vanishing conjecture for high tensor-power Selmer groups

摘录 / EXCERPT: Vanishing conjecture for high tensor powers. One has $Sel^0_T(V_n)=0$ for $n\gg0$. This conjecture concerns the disappearance of globally unramified Selmer classes in sufficiently...

conjecture | open | MathDB 页面 | 原文线索

V2-006748 | MATHDB-379359

Swinnerton-Dyer's refined error-term conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let $S\subseteq\mathbb{P}^d$ be a possibly singular del Pezzo surface, let $U\subseteq S$ be obtained by deleting exceptional divisors, and let ρ_S be the relevant Picard rank...

conjecture | open | MathDB 页面 | 原文线索

V2-006749 | MATHDB-379369

Conjectural moments of the truncated Hadamard factor

摘录 / EXCERPT: Let $Z_X(s)$ be the truncated Hadamard factor, let $X,T\to\infty$ with $X=O((\log T)^{2-\epsilon})$, and let $k>-1/2$ be fixed. Let G be the Barnes G-function and let γ...

conjecture | open | MathDB 页面 | 原文线索

V2-006750 | MATHDB-379396

The 6n alternating binomial-sum divisibility conjecture

摘录 / EXCERPT: The 6n divisibility conjecture. For all positive r, s, t, and n, $S_{6}(r,s,t;n)$ is divisible by both

conjecture | open | MathDB 页面 | 原文线索

V2-006751 | MATHDB-379397

The binomial alternating-sum gcd conjecture

摘录 / EXCERPT: The alternating-sum gcd conjecture. For all positive m and n,

conjecture | open | MathDB 页面 | 原文线索

V2-006752 | MATHDB-379399

The tmf Tate-spectrum non-splitting conjecture at the prime 3

摘录 / EXCERPT: Let tmf denote the connective topological modular forms spectrum, let Σ_3 be the symmetric group on three letters, and let $\mathit{tmf}^{t\Sigma_3}$ denote its T...

conjecture | open | MathDB 页面 | 原文线索

V2-006753 | MATHDB-379400

The higher-prime Hopf-algebra conjecture for eo_{p-1}

摘录 / EXCERPT: Let p be a prime, let eo_{p-1} be the proposed connective form of EO_{p-1}, and let $a_2,\dots,a_{p-1}$ be exterior generators with ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006754 | MATHDB-379426

Positive-characteristic freeness conjecture for Weyl-algebra bimodules

摘录 / EXCERPT: Let R be a finitely generated smooth commutative algebra over $\mathbb Z$, let $g\in\operatorname{Aut}(P_{n,R})$, and for each sufficiently large prime p let $M_{g,p}$ be a bim...

conjecture | open | MathDB 页面 | 原文线索

V2-006755 | MATHDB-379437

The Bloch-Kato formula for modular forms

摘录 / EXCERPT: For each prime p, let $V_p:=V_{f,\mathfrak{p}}$ be the p-adic Galois representation attached to a newform f of weight $2k-2$, let $T_p:=T_{f,\mathfrak{p}}$ be a stable lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-006756 | MATHDB-379462

Conjecture on the zeros of the divisor polynomials of Andrews-Gordon series

摘录 / EXCERPT: Conjecture on the zeros of $F(\mathcal{F}_k,x)$. The polynomial $F(\mathcal{F}_k,x)$ has simple roots, all of which are real and lie in the interval $[0,1728]$.

conjecture | open | MathDB 页面 | 原文线索

V2-006757 | MATHDB-379463

Parity conjecture for β_{-N} when $a_E(2)$ is even

摘录 / EXCERPT: Let E be an elliptic curve with conductor N_E, and let $a_E(2)$ denote its eigenvalue at 2. For a prime $N\equiv 3\pmod{4}$, let β_{-N} be the associated integer. Parit...

conjecture | open | MathDB 页面 | 原文线索

V2-006758 | MATHDB-379464

Parity conjecture for the square-index coefficient β_{-4N}

摘录 / EXCERPT: Let E be an elliptic curve with conductor N_E, and let β_{D} denote the coefficient associated with a fundamental discriminant or its square multiple as in the paper. Squ...

conjecture | open | MathDB 页面 | 原文线索

V2-006759 | MATHDB-379465

Conjecture on the parity of Tate-Shafarevich groups of quadratic twists

摘录 / EXCERPT: Let E be an elliptic curve as above, let N be a prime with $N\equiv 3\pmod{4}$, and suppose $\beta_{-N}\ne 0$. Let E^{-N} denote the quadratic twist of E associated with th...

conjecture | open | MathDB 页面 | 原文线索

V2-006760 | MATHDB-379466

Parity conjecture for β_{-4N} and β_{-4} when $N\equiv 1\pmod{4}$

摘录 / EXCERPT: Let N be a prime congruent to 1 modulo 4, and let β_{-4N} and β_{-4} be the associated integers. Parity-matching conjecture. If ... then β_{-4N} and...

conjecture | open | MathDB 页面 | 原文线索

V2-006761 | MATHDB-379483

Hecke-Fricke characterization conjecture for cusp forms

摘录 / EXCERPT: Let N be a positive integer, let k be a weight, and let χ_0 be the trivial character modulo N. Let $\mathcal H=\{x+iy\in\mathbb C:y>0\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-006762 | MATHDB-379484

Hecke-Fricke converse theorem for degree-2 L-functions

摘录 / EXCERPT: Hecke-Fricke converse theorem. If the Dirichlet series satisfies the stated functional equation and Euler product, then it equals $L(s,f)$ for some

conjecture | open | MathDB 页面 | 原文线索

V2-006763 | MATHDB-379492

Asymptotic conjecture for non-degenerate systems of linear equations in primes

摘录 / EXCERPT: Let s,t be positive integers with $t\geqslant s+2$. An $s\times t$ integer matrix $\mathbf{A}$ is non-degenerate if it has rank s and every non-zero vector in its row span over...

conjecture | open | MathDB 页面 | 原文线索

V2-006764 | MATHDB-379493

Möbius orthogonality conjecture for nilsequences

摘录 / EXCERPT: Let G/Γ be a k-step nilmanifold, let $T_g(x\Gamma)=xg\Gamma$, and let $(F(T_g^n\cdot x))_{n\in\mathbb{N}}$ be a k-step nilsequence. The Möbius orthogonality conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-006765 | MATHDB-379496

Torus-link reformulated colored HOMFLY polynomial structure conjecture

摘录 / EXCERPT: Let $T(rl,kl)$ be a torus link, where r and k are relatively prime, and let $n=n_1+\cdots+n_l$. Let $f_{\lambda^1,\dots,\lambda^l}(t,\nu)$ be the reformulated colored HOMFLY po...

conjecture | open | MathDB 页面 | 原文线索

V2-006766 | MATHDB-379524

Becker's conjecture on the Mahler classification of automatic real numbers

摘录 / EXCERPT: Let ξ be an irrational real number whose base-b expansion is generated by a finite automaton; equivalently, its digit sequence is k-automatic for some integer base...

conjecture | open | MathDB 页面 | 原文线索

V2-006767 | MATHDB-379551

Large sieve conjecture for square moduli

摘录 / EXCERPT: Let S be the set of squares, and let $S_t(Q/t)$ be the subset defined earlier in the paper. For $M\in\mathbb{Z}$, $N\in\mathbb{N}$, and a complex sequence (a_n), one has ... wher...

conjecture | open | MathDB 页面 | 原文线索

V2-006768 | MATHDB-379579

Miller's conjecture on odd-prime Miller groups

摘录 / EXCERPT: Miller's conjecture. If G is a Miller p-group for an odd prime p, then G is special.

conjecture | open | MathDB 页面 | 原文线索

V2-006769 | MATHDB-379594

Completeness conjecture for triangles satisfying Pick's formula almost correctly

摘录 / EXCERPT: Let $p,q \in \mathbb{Z}$ be coprime integers with $1<q<p^2$, with p odd and q even. Let $\Delta(p,q)$ denote the triangle defined in the paper, and suppose that Pick's formula...

conjecture | open | MathDB 页面 | 原文线索

V2-006770 | MATHDB-379600

Igusa's conjecture on uniform exponential-sum bounds for homogeneous polynomials

摘录 / EXCERPT: Igusa's conjecture. There exists a constant c such that for every prime p and every positive integer m,

conjecture | open | MathDB 页面 | 原文线索

V2-006771 | MATHDB-379602

Asymptotic ratio conjecture for prime webs

摘录 / EXCERPT: Asymptotic ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006772 | MATHDB-379633

The field-of-definition conjecture for the zeros of h_χ

摘录 / EXCERPT: Let N be a prime level, let χ be the character used to define h_χ, and let $B_{m,\chi}$ denote the m-th generalised Bernoulli number attached to χ. Write ... For e...

conjecture | open | MathDB 页面 | 原文线索

V2-006773 | MATHDB-379683

Random power-residue symbol conjecture for CM elliptic curves

摘录 / EXCERPT: Let E_t^d be the CM elliptic curve and let w_d denote the order of the relevant unit group. For a prime p, write $\left(\frac{a_p(E_t^d)}{p}\right)_m$ for the mth power res...

conjecture | open | MathDB 页面 | 原文线索

V2-006774 | MATHDB-379695

Conjectural strengthenings on curves without abelian points

摘录 / EXCERPT: Let K be any number field and let E be any elliptic curve over K. Write K^{ab} for the maximal abelian extension of K. Conjectural strengthenings. The foll...

conjecture | open | MathDB 页面 | 原文线索

V2-006775 | MATHDB-379731

Conjectured error term for class numbers of orders in quartic fields

摘录 / EXCERPT: Class-number asymptotic conjecture. Under the conditions of Theorem 1, as $x\to\infty$ one has

conjecture | open | MathDB 页面 | 原文线索

V2-006776 | MATHDB-379750

Miyawaki-Ikeda lifting conjecture

摘录 / EXCERPT: Let k and n be natural numbers with $k-n$ even. Let $f\in S_{2k}(\Gamma_1)$ be a normalized Hecke eigenform, and let $F_{2n}\in S_{k+n}(\Gamma_{2n})$ be the Ikeda lift of f....

conjecture | open | MathDB 页面 | 原文线索

V2-006777 | MATHDB-379751

Endoscopic contribution conjecture for Siegel modular forms of genus two

摘录 / EXCERPT: Let (l,m) be regular. The endoscopic contribution $e_{\rm endo}({\mathcal A}_2,\mathbb V_{l,m})$ is defined using the cohomology of the moduli space ${\mathcal A}_2$ of principal...

conjecture | open | MathDB 页面 | 原文线索

V2-006778 | MATHDB-379752

The extension of Theorem B for complete Padovan sequences

摘录 / EXCERPT: Let $p \geq 5$ be a prime with $\varrho_p=3$, let α, β, and γ be the roots of X^3-X-1 in $\mathbb F_p$, and define ... A Φ_3-sequence $(a_n)_n$ is complete...

conjecture | open | MathDB 页面 | 原文线索

V2-006779 | MATHDB-379761

Green-Konyagin-Littlewood conjecture for the Fourier algebra norm

摘录 / EXCERPT: Let p be a prime number and let $A\subset\mathbb{Z}/p\mathbb{Z}$ have density bounded away from 0 and 1 by an absolute constant. Green-Konyagin-Littlewood conjecture. Then .....

conjecture | open | MathDB 页面 | 原文线索

V2-006780 | MATHDB-379763

Shioda's conjecture on transcendental lattices at supersingular primes

摘录 / EXCERPT: Shioda's conjecture. If $\mathfrak{p}$ is a supersingular prime, then the two lattices T_Z and $T_{\mathfrak{p}}$ are similar.

conjecture | open | MathDB 页面 | 原文线索

V2-006781 | MATHDB-379789

Green-Tao Möbius and nilsequences conjecture

摘录 / EXCERPT: Let $s\geqslant 1$. An s-step nilmanifold is a quotient $G/\Gamma=(G/\Gamma,d_{G/\Gamma})$ with smooth metric, and an s-step nilsequence is a sequence of the form…

conjecture | solved | MathDB 页面 | 原文线索

V2-006782 | MATHDB-379790

Dickson's generalised Hardy-Littlewood conjecture for affine-linear forms

摘录 / EXCERPT: Let N,d,t,L be positive integers, let $\Psi=(\psi_1,\ldots,\psi_t)$ be a system of affine-linear forms with $\|\Psi\|_N\leqslant L$, and let $K\subset[-N,N]^d$ be a convex body.…

conjecture | open | MathDB 页面 | 原文线索

V2-006783 | MATHDB-379835

Explicit Waldspurger-type formula for level p^2 Shimura lifts

摘录 / EXCERPT: Let p be a prime, let f be the modular form in the construction, let $\widetilde{\mathscr{O}}$ and $\mathscr{O}'$ be the relevant orders, and write … Let…

conjecture | open | MathDB 页面 | 原文线索

V2-006784 | MATHDB-379850

Traverso's isogeny cutoff conjecture for p-divisible groups

摘录 / EXCERPT: Traverso's isogeny cutoff conjecture. The isogeny cutoff satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-006785 | MATHDB-379857

Sharp upper-bound conjecture for Fermat quotients

摘录 / EXCERPT: Let v_p denote the p-adic valuation. The variables x and q are integers with $x>1$, and p and q are primes in the second assertion. Sharp upper-bound conjecture for Fer…

conjecture | open | MathDB 页面 | 原文线索

V2-006786 | MATHDB-379860

Oppenheim's algebraicity conjecture for linear forms

摘录 / EXCERPT: Oppenheim's conjecture. The lattice

conjecture | open | MathDB 页面 | 原文线索

V2-006787 | MATHDB-379892

Zeros of Eisenstein series for the Fricke group \(\Gamma_0^{*}(5)\)

摘录 / EXCERPT: Let $k \geqslant 4$ be an even integer. For the Eisenstein series $E_{k,5}^{*}(z)$ associated with the Fricke group $\Gamma_0^{*}(5)$, let $\mathbb{F}^{*}(5)$ be the fundamental do…

conjecture | open | MathDB 页面 | 原文线索

V2-006788 | MATHDB-379907

The conjectured optimal bound for completely multiplicative functions with mean value zero

摘录 / EXCERPT: Conjectured optimal bound. For $1/\sqrt e\leqslant \alpha\leqslant 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-006789 | MATHDB-379908

Finite-plane Besicovitch set cardinality conjecture

摘录 / EXCERPT: For an odd prime power q, a Besicovitch set in ${\mathbb{F}_q}^2$ is a subset containing a line in every direction. Finite-plane Besicovitch set conjecture. The smallest Besicovi…

conjecture | open | MathDB 页面 | 原文线索

V2-006790 | MATHDB-379917

Minkowski improvement conjecture for ideal classes in totally real fields

摘录 / EXCERPT: Fix an integer $d \geq 3$. For a totally real number field K of degree d, let $\operatorname{disc}(K)$ denote its discriminant, and let an ideal class mean a class of fractiona…

conjecture | open | MathDB 页面 | 原文线索

V2-006791 | MATHDB-379922

Solymosi's affine-cube conjecture for squares

摘录 / EXCERPT: An affine cube of dimension d in $\mathbb Z$ is a set … where $b_0,\ldots,b_d$ are nonzero integers. Solymosi's conjecture. There exists an integer $d>0$ such that no affine cube…

conjecture | open | MathDB 页面 | 原文线索

V2-006792 | MATHDB-379923

The congruence-spacing conjecture for square roots modulo integers

摘录 / EXCERPT: For integers a and b, consider positive solutions of $x^2\equiv a\pmod b$. The congruence-spacing conjecture. There exists a constant N such that, for every a and b, ther…

conjecture | open | MathDB 页面 | 原文线索

V2-006793 | MATHDB-379924

The bounded-residue-set conjecture

摘录 / EXCERPT: Let $d\geq1$ be an integer. For every prime power q, let $\Omega(q)\subset\mathbb Z/q\mathbb Z$ contain at most d residues, and define $\Omega(b)$ for general b by the Chines…

conjecture | open | MathDB 页面 | 原文线索

V2-006794 | MATHDB-379925

The discriminant-controlled polynomial congruence conjecture

摘录 / EXCERPT: Fix an integer $d\geq2$. For a monic polynomial $f(x)\in\mathbb Z[x]$, consider integers b such that whenever p^2 divides b, the prime p does not divide the discriminant of…

conjecture | open | MathDB 页面 | 原文线索

V2-006795 | MATHDB-379930

The bounded lattice-points-in-arcs conjecture for circles

摘录 / EXCERPT: The circle-arc conjecture. The number of these lattice points in any arc of length $R^{1-\epsilon}$ is bounded uniformly in R. The source states the equivalent short-interval for...

conjecture | open | MathDB 页面 | 原文线索

V2-006796 | MATHDB-379931

The bounded lattice-points-in-diagonal-arcs conjecture

摘录 / EXCERPT: The diagonal-arc conjecture. The number of these lattice points in an arc of length $R^{1-\epsilon}$ around the diagonal is bounded uniformly in R. The source presents this as th...

conjecture | open | MathDB 页面 | 原文线索

V2-006797 | MATHDB-379932

Bourgain's short-interval circle congruence conjecture

摘录 / EXCERPT: Bourgain's circle congruence conjecture. There exist $\delta>0$ and an integer $m>0$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-006798 | MATHDB-379933

The bounded short-arc lattice-point conjecture

摘录 / EXCERPT: For an integer n, count integer points on the circle $a^2+b^2=n$ whose second coordinate lies in a short interval. The short-arc lattice-point conjecture. For every $\alpha<1/2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006799 | MATHDB-379945

The irreducibility conjecture for the numerator of the binary-form Poincaré series

摘录 / EXCERPT: Let $n\ge3$, and write the Poincaré series in lowest terms as ... where $A_n,B_n\in\mathbf Z[t]$ are relatively prime and $A_n(0)=B_n(0)=1$. Irreducibility conjecture. The polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-006800 | MATHDB-379949

Faber-Pandharipande conjecture on the prime valuations of intersection-number denominators

摘录 / EXCERPT: Faber-Pandharipande conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006801 | MATHDB-379996

The finitely generated Drinfeld-module Siegel conjecture for integral points

摘录 / EXCERPT: Let K be a global function field, let M_K be its set of places, let $S\subseteq M_K$ be finite, and let ϕ be a Drinfeld module over K. A point γ is S-integral f...

open_problem | open | MathDB 页面 | 原文线索

V2-006802 | MATHDB-380016

The exact asymptotic formula conjecture for Turán's problem 10

摘录 / EXCERPT: Exact asymptotic formula conjecture. One can choose $A(\alpha)=\underline A(\alpha)=\overline A(\alpha)$ in Theorem $u77$. This would determine the true asymptotic constant for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-006803 | MATHDB-380017

The exact asymptotic formula conjecture for unit-circle Turán sums

摘录 / EXCERPT: Exact unit-circle asymptotic conjecture. One can choose $B(\alpha)=\underline B(\alpha)=\overline B(\alpha)$ in Theorem avs. This conjecture would close the gap between the known...

conjecture | open | MathDB 页面 | 原文线索

V2-006804 | MATHDB-380018

The equality of unrestricted and unit-circle asymptotic constants conjecture

摘录 / EXCERPT: For $\alpha>0$, let $A(\alpha)$ be the asymptotic constant conjectured for ... and let $B(\alpha)$ be the corresponding constant when $|z_k|=1$. Equality of asymptotic constants conj...

conjecture | open | MathDB 页面 | 原文线索

V2-006805 | MATHDB-380024

Congruence conjecture for the normalized generating function of vacuum modules

摘录 / EXCERPT: Let $L(k\Lambda_0)$ denote the level-k vacuum module, let $\mathcal{F}_{L(k\Lambda_0)}(q)$ be its normalized generating function, and let $p=2k+3$ be a prime with $p\geq 5$. Cong...

conjecture | open | MathDB 页面 | 原文线索

V2-006806 | MATHDB-380025

Zero-location conjecture for the normalized generating function of vacuum modules

摘录 / EXCERPT: Zero-location conjecture. For every $k\neq 2i^2-2$, $i\geq 2$, the zeros of $G(\mathcal{F}_{L(k\Lambda_0)},j)$ are simple and lie inside the interval $[0,1728]$. This is a numerica...

conjecture | open | MathDB 页面 | 原文线索

V2-006807 | MATHDB-380026

Interpolated rational-sum approximation conjecture

摘录 / EXCERPT: Let α be a real number, let $N\geq 1$ be real, and let $0\leq\beta\leq 1$. The integers a_1,a_2 and positive integers q_1,q_2 satisfy $1\leq q_1,q_2\leq N$. Interpolated...

conjecture | open | MathDB 页面 | 原文线索

V2-006808 | MATHDB-380027

Rationally conditioned two-rational approximation conjecture

摘录 / EXCERPT: Let $\epsilon>0$ be any small positive real number and let $N\geq 1$. Suppose that α has a rational approximation a/q with integers a,q, $1\leq q\leq N^2$, $(a,q)=1$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-006809 | MATHDB-380028

Short-factor congruence conjecture

摘录 / EXCERPT: Let $\epsilon>0$ be any small positive real number, let q be a positive integer, and let c be an integer with $(c,q)=1$. Short-factor congruence conjecture. The congruence ... ha...

conjecture | open | MathDB 页面 | 原文线索

V2-006810 | MATHDB-380029

Coprime-factor congruence conjecture in short intervals

摘录 / EXCERPT: Let θ satisfy $1/2<\theta\leq 1$. Let q be a positive integer and let c be an integer with $(c,q)=1$. Coprime-factor congruence conjecture in short intervals. There is a...

conjecture | open | MathDB 页面 | 原文线索

V2-006811 | MATHDB-380033

Conjecture that cohomology free vector fields are toral Diophantine flows

摘录 / EXCERPT: Cohomology-free vector field conjecture. Any cohomology free vector field is smoothly conjugated to a Diophantine vector field on a torus.

conjecture | open | MathDB 页面 | 原文线索

V2-006812 | MATHDB-380047

Bazzi-Mitter conjecture on hyperelliptic-curve point bounds

摘录 / EXCERPT: For an odd prime p and each non-empty subset $S\subset GF(p)$, let X_S be the hyperelliptic curve defined by ... Write $B(c,p)$ for the assertion that $|X_S(GF(p))|\leq c\,p$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-006813 | MATHDB-380048

The QQR code duality conjecture

摘录 / EXCERPT: Let $p>5$ be an odd prime, let $Q,N\subset GF(p)^\times$ be the quadratic residues and non-residues, and let C_{NQ} be the quasi-quadratic residue code ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-006814 | MATHDB-380049

The formally self-dual extension conjecture for QQR codes

摘录 / EXCERPT: Let $p>5$ be an odd prime, let $C_{NQ}\subset {\mathbb F}^{2p}$ be the quasi-quadratic residue code, and let C' be the code spanned by C_{NQ} and the all-ones codeword. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006815 | MATHDB-380052

Conjecture on the limiting distribution of random Cayley graph diameters

摘录 / EXCERPT: Let q be prime, let $g_1,\ldots,g_k$ be independent uniformly random generators of the cyclic group ${\mathbb Z}_q$, and let $Diam(q,k)$ denote the diameter of the resulting dire...

conjecture | open | MathDB 页面 | 原文线索

V2-006816 | MATHDB-380059

Algebraic independence conjecture for odd zeta values

摘录 / EXCERPT: Algebraic independence conjecture for odd zeta values. The numbers $\pi,\zeta(3),\zeta(5),\zeta(7),\zeta(9),\ldots$ are algebraically independent over $\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-006817 | MATHDB-380066

Three-dimensional independence conjecture for low-depth multiple zeta values

摘录 / EXCERPT: Define the multiple zeta values by ... for $s_1\geq 2$ and $s_2,\ldots,s_p\geq 1$. Low-depth independence conjecture. Among the numbers 1, $\zeta(2)$, $\zeta(3)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006818 | MATHDB-380067

Three-dimensional independence conjecture for selected multiple zeta values

摘录 / EXCERPT: Define the multiple zeta values by ... for $s_1\geq 2$ and $s_2,\ldots,s_p\geq 1$. Selected-value independence conjecture. Among the numbers 1, $\zeta(2)$, $\zeta(3)$, $\zeta(2,3)$...

conjecture | open | MathDB 页面 | 原文线索

V2-006819 | MATHDB-380068

Depth-three symmetry conjecture for multiple zeta-value series

摘录 / EXCERPT: Let $n\geq 0$ be an integer. Let $\sigma\in\{2,3\}$ be the element satisfying $\sigma\equiv n\pmod 2$, and let $\widetilde{\sigma}$ be the other element of $\{2,3\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006820 | MATHDB-380069

Hoffman's conjecture for symmetric triple series

摘录 / EXCERPT: Let $A=3$ and let $n\geq 0$ be an integer. Define ... For a polynomial $P(K_1,K_2,K_3)$ in any of the four families specified in Theorem ... sum{k1geq k2geq k3geq 1}frac{P(K1,K2,K3)}{(...

conjecture | open | MathDB 页面 | 原文线索

V2-006821 | MATHDB-380107

The genus-two Rankin-product lifting conjecture to genus four

摘录 / EXCERPT: Let f and g be Siegel modular forms of genus 2, of weights $k>4$ and $l=k-2$, with Satake parameters $(\alpha_0,\alpha_1,\alpha_2)$ and $(\beta_0,\beta_1,\beta_2)$, respectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-006822 | MATHDB-380108

The even-genus Rankin-product lifting conjecture

摘录 / EXCERPT: Let $m\geq 1$, and let f and g be Siegel modular forms of genus $2m$, of weights $k>2m$ and $l=k-2m$, with Satake parameters $(\alpha_0,\alpha_1,\ldots,\alpha_{2m})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006823 | MATHDB-380111

The local Serre weight conjecture at a ramified prime

摘录 / EXCERPT: Let $\mathfrak p\mid p$ be a place of F, with ramification index e, residue degree s, and inertia group $I_{\mathfrak p}$. Let $W_{\mathfrak p}^?(\rho)$ denote the predicted...

conjecture | open | MathDB 页面 | 原文线索

V2-006824 | MATHDB-380131

The conjecture on infinitely many cubic-order approximations by rationals of the form a/b^2

摘录 / EXCERPT: Conjecture on cubic-order a/b^2-approximations. There exists a constant $c(\alpha)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006825 | MATHDB-380155

The IFG field curve-index existence conjecture

摘录 / EXCERPT: IFG field curve-index conjecture. If K is an IFG field and $I\mid 2g-2$, then there exists a curve $C_{/K}$ of genus g with $I(C)=I$.

conjecture | open | MathDB 页面 | 原文线索

V2-006826 | MATHDB-380156

The SNC index specialization conjecture

摘录 / EXCERPT: SNC index specialization conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-006827 | MATHDB-380208

The polylogarithmic prime-gap conjecture

摘录 / EXCERPT: Polylogarithmic prime-gap conjecture. There exists an absolute constant $k>1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006828 | MATHDB-380211

Chain-homotopy conjecture for the maps g_1^j and g_2^j on dihedral rack complexes

摘录 / EXCERPT: Chain-homotopy conjecture. The maps g_1^j and g_2^j are chain homotopic. The source notes that these maps are chain maps for the stated prime orders; proving their chain homoto...

conjecture | open | MathDB 页面 | 原文线索

V2-006829 | MATHDB-380234

Power-operation conjecture for cyclic A-infinity structures

摘录 / EXCERPT: Let R be an even commutative S-algebra, let $x\in R_d$, and let p be a prime such that n is invertible in R_* for every $n<p$. Let $\bar c_n(x)\in\pi_{p(d+2)-2}R/(p,x)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-006830 | MATHDB-380254

Prime-divisor density conjecture for stable critically infinite quadratic polynomials

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x]$ be a quadratic polynomial. Call f stable if every iterate f^n is irreducible over $\mathbb{Q}$, and call f critically infinite if the forward orbit of...

conjecture | open | MathDB 页面 | 原文线索

V2-006831 | MATHDB-380257

Dickson's conjecture for affine sublattices

摘录 / EXCERPT: Let L be an affine sublattice of $\mathbb{Z}^t$. For each prime p, let $\alpha_p(L)$ be the local factor defined from the image L_p of L in $\mathbb{Z}_p^t$; a local obstru...

conjecture | open | MathDB 页面 | 原文线索

V2-006832 | MATHDB-380258

Quantitative Dickson conjecture for sublattices

摘录 / EXCERPT: Let L be a sublattice of $\mathbb{Z}^t$, let N be large, let $K\subseteq[-N,N]^t$ be convex with volume $\gg N^t$, and let $a\in\mathbb{Z}^t$ have size $O(N)$. Assume that all...

conjecture | open | MathDB 页面 | 原文线索

V2-006833 | MATHDB-380261

Conjecture on the limiting distribution of the centred sums over zeta-zero cycles

摘录 / EXCERPT: Limiting-distribution conjecture. For every $a>0$ there is a Borel function $f_a:\mathbb{C}\to\mathbb{R}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-006834 | MATHDB-380262

Conjecture on the limiting distribution of the continuous zeta-zero orbit function

摘录 / EXCERPT: Let S denote the multiset of ordinates of the nontrivial zeros of the Riemann zeta function. For $a>0$, define the continuous function $H_a:\mathbb{R}_{>0}\to\mathbb{C}$ by ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-006835 | MATHDB-380279

Meromorphic-power conjecture for Selberg zeta-functions in the index-six cusp case

摘录 / EXCERPT: Meromorphic-power conjecture. Then there is an integer N such that

conjecture | open | MathDB 页面 | 原文线索

V2-006836 | MATHDB-380305

The characterization of sums of consecutive integers

摘录 / EXCERPT: Consecutive-sum conjecture. A natural number is a sum of consecutive integers if and only if it is not a power of 2. This characterizes exactly which natural numbers admit a repr...

conjecture | open | MathDB 页面 | 原文线索

V2-006837 | MATHDB-380306

The factorisation conjecture for the overconvergent U operator

摘录 / EXCERPT: Let $p\in\{2,3,5\}$, let r range over some open interval containing $\frac{1}{2}$, and let $\mathcal{S}_k(r)$ be the space on which the overconvergent U operator acts. A factor...

conjecture | open | MathDB 页面 | 原文线索

V2-006838 | MATHDB-380307

Gouvêa-Mazur spectral expansion conjecture, strong form

摘录 / EXCERPT: Let p be prime, let $r\in(\frac{1}{p+1},\frac{p}{p+1})$, and let h be an r-overconvergent modular function on the region $X_0(1)_{\ge p^{-r}}$. Its spectral expansion is the...

conjecture | open | MathDB 页面 | 原文线索

V2-006839 | MATHDB-380311

The natural-density conjecture for primitive divisors of n^2+b

摘录 / EXCERPT: Let b be an integer such that $-b$ is not an integer square, and define ... Natural-density conjecture. As x tends to infinity, ... This would identify the natural density of indic...

conjecture | open | MathDB 页面 | 原文线索

V2-006840 | MATHDB-380330

The prime pure alternance graph conjecture

摘录 / EXCERPT: A bicoloured alternance graph is called pure and prime in the sense developed in the paper; connectedness is with respect to the underlying graph. The white one-point graph and the...

conjecture | open | MathDB 页面 | 原文线索

V2-006841 | MATHDB-380332

The Paley complementation-set exclusivity conjecture

摘录 / EXCERPT: Let G be the Paley graph of prime order p, namely $G=\operatorname{Cay}(\mathbb{Z}_p,S)$ where S is the set of quadratic residues modulo p. Let $a,b\in\mathbb{Z}_p^*$ have...

conjecture | open | MathDB 页面 | 原文线索

V2-006842 | MATHDB-380373

Finiteness of local torsion primes for non-CM elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over ${\mathbf Q}$, and let $d\geq 1$ be fixed. A prime p is a local torsion prime for E if there exists an extension $K/{\mathbf Q}_p$ of degree a...

conjecture | open | MathDB 页面 | 原文线索

V2-006843 | MATHDB-380379

K. Saito's nonnegative eta-product conjecture

摘录 / EXCERPT: K. Saito's conjecture. The Fourier coefficients of $S_N(\tau)$ are nonnegative. This conjecture concerns the positivity of coefficients of eta products associated with elliptic roo...

conjecture | open | MathDB 页面 | 原文线索

V2-006844 | MATHDB-380395

The corrected Ruelle zeta-function rationality conjecture

摘录 / EXCERPT: Corrected Ruelle zeta-function conjecture. For any $x_1,\dots,x_k\in\mathbb F_q\setminus\{0,1,t\}$ with $k\ge 1$, the series

conjecture | open | MathDB 页面 | 原文线索

V2-006845 | MATHDB-380398

The tensor-functor realization conjecture for finite-field motives

摘录 / EXCERPT: Tensor-functor realization conjecture. For every prime p there exists a tensor functor

conjecture | open | MathDB 页面 | 原文线索

V2-006846 | MATHDB-380411

Morris's minimal-generator extension conjecture for non-FC families

摘录 / EXCERPT: Minimal-generator extension conjecture. If $\mathcal{B}$ is not FC, then $\mathcal{B}^{\prime}$ is not FC either. This conjecture proposes that replacing a minimal generator by...

conjecture | open | MathDB 页面 | 原文线索

V2-006847 | MATHDB-380433

Conjecture on approximation by rationals with prime denominators

摘录 / EXCERPT: Prime-denominator approximation conjecture. For every real ξ, there exists a constant $c(\xi)$ such that the inequality

conjecture | open | MathDB 页面 | 原文线索

V2-006848 | MATHDB-380470

The pre-Bloch group parity conjecture

摘录 / EXCERPT: Let F be an algebraically closed field. Here $\mathfrak{p}_n(F)$ denotes the primitive part of the homology group associated with $\operatorname{GL}_n(F)$, and l is a prime. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-006849 | MATHDB-380472

Exactness conjecture for the homological t-groups

摘录 / EXCERPT: Let F be an infinite field, let k be either $\mathbb{Q}$ or $\mathbb{Z}/l$ for a prime l, and assume $K_2^M(F)\otimes k=0$. Let $\mathfrak{t}_r^{(k)}(F)$ denote the homologic...

conjecture | open | MathDB 页面 | 原文线索

V2-006850 | MATHDB-380504

Hare-Smyth zero-endpoint interval conjecture for the monic integer transfinite diameter

摘录 / EXCERPT: Zero-endpoint interval conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006851 | MATHDB-380505

Maximal obstruction conjecture for short intervals

摘录 / EXCERPT: Maximal obstruction conjecture. If I has a maximal obstruction $m(I)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-006852 | MATHDB-380518

The Taniyama-Shimura conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Let $\bold Q$ denote the field of rational numbers, and call an elliptic curve over $\bold Q$ modular if it admits a holomorphic map from some modular curve $X_0(N)$ onto it. Taniy...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006853 | MATHDB-380519

The semistable Taniyama-Shimura conjecture

摘录 / EXCERPT: Let $\bold Q$ denote the field of rational numbers, and call an elliptic curve over $\bold Q$ semistable if it is semistable at every prime, meaning that at each prime it has eithe...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006854 | MATHDB-380520

The semistable modular lifting conjecture

摘录 / EXCERPT: Let $G_{\bold Q}=\operatorname{Gal}(\bar{\bold Q}/\bold Q)$, let p be an odd prime, and let $\bar{\varepsilon}_p:G_{\bold Q}\to\bold F_p^\times$ be the character describing the a...

conjecture | open | MathDB 页面 | 原文线索

V2-006855 | MATHDB-380521

Mazur's conjecture on modular deformation liftings

摘录 / EXCERPT: Let p be an odd prime, let k be a finite field of characteristic p, and let deformation data be a pair $\mathcal D=(\Sigma,t)$ consisting of a finite set of primes and one of...

conjecture | open | MathDB 页面 | 原文线索

V2-006856 | MATHDB-380522

Mazur's $R=\bold T$ conjecture

摘录 / EXCERPT: Let p, the finite field k, the deformation data $\mathcal D$, the absolutely irreducible type-$\mathcal D$ representation $\bar\rho:G_{\bold Q}\to\operatorname{GL}_2(k)$, and t...

conjecture | open | MathDB 页面 | 原文线索

V2-006857 | MATHDB-380548

Milnor's conjecture on dilogarithm values at roots of unity

摘录 / EXCERPT: For an integer $N\geq 3$, let D_2 denote the Bloch-Wigner dilogarithm, and consider the real numbers $D_2(e^{2\pi\sqrt{-1}j/N})$ indexed by integers j relatively prime to N w...

conjecture | open | MathDB 页面 | 原文线索

V2-006858 | MATHDB-380558

Thurston's indecomposable-implies-geometric conjecture

摘录 / EXCERPT: Let M^{3} be a closed, prime, atoroidal 3-manifold. A locally homogeneous Riemannian metric is a Riemannian metric such that for any two points p and q there is an isometry...

conjecture | open | MathDB 页面 | 原文线索

V2-006859 | MATHDB-380563

Vanishing conjecture for the congruence subgroup homology

摘录 / EXCERPT: Let k be the field appearing in the power-series congruence subgroup, let n,m be positive integers, let p be a prime, and write $C_{n,m}(k[[t_1,t_2,\dots,t_m]])$ for the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-006860 | MATHDB-380577

The domination conjecture for perturbative PSU(n)-invariants

摘录 / EXCERPT: Domination conjecture. The perturbative invariant $\tau^{PSU(n)}(M)$ dominates the quantum invariants $\tau_r^{SU(n)}(M)$ for every positive integer r, not necessarily prime.

conjecture | open | MathDB 页面 | 原文线索

V2-006861 | MATHDB-380611

Howe's duality conjecture for reductive dual pairs

摘录 / EXCERPT: Let G) and $G^{\prime}$ be mutual centralizers forming a reductive dual pair inside a symplectic group over a local field F. Let ω be an oscillator representation of the...

conjecture | open | MathDB 页面 | 原文线索

V2-006862 | MATHDB-380612

Conjecture on the instanton partition function for rational surfaces

摘录 / EXCERPT: Let X be a rational surface, let v_0 denote the relevant charge, let $Z_{v_0}(\tau)$ be its partition function, let $\widetilde{Z}_{v_0}^{int}(\tau)$ be the corresponding integ...

conjecture | open | MathDB 页面 | 原文线索

V2-006863 | MATHDB-380631

Hamada's p-rank conjecture for designs with Singer parameters

摘录 / EXCERPT: Let ${\mathcal D}$ be a symmetric design with Singer parameters ... where $q=p^s$ and p is prime. Write ${\rm rank}_p{\mathcal D}$ for the dimension over ${\mathbb F}_p$ of the cod...

conjecture | open | MathDB 页面 | 原文线索

V2-006864 | MATHDB-380655

Singly exponential generic decidability conjecture for Diophantine prefixes

摘录 / EXCERPT: A Diophantine prefix is generically decidable when an algorithm decides the corresponding positive-integer Diophantine sentences on the restricted collection of inputs specified by...

conjecture | open | MathDB 页面 | 原文线索

V2-006865 | MATHDB-380657

The nef anticanonical bundle converse to the Weak Lang Conjecture

摘录 / EXCERPT: The nef anticanonical bundle converse. If the anticanonical bundle $-K_X$ of X is nef, then for some finite extension K' of K the set $X(K')$ of K'-rational points of X i...

conjecture | open | MathDB 页面 | 原文线索

V2-006866 | MATHDB-380690

Finite-ray prediction accuracy hypothesis for prime numbers

摘录 / EXCERPT: Let r_{p_0} be a ray with $p_0\in\overline{C}$. Let l denote the number of correct decimal digits available in the floating-point arithmetic, and let $n^*(l)$ be a finite index...

conjecture | open | MathDB 页面 | 原文线索

V2-006867 | MATHDB-380695

Voronoi candidate conjecture for modular symbols

摘录 / EXCERPT: Let ${\mathbf v}=[v_1,\dots,v_n]$ be a modular symbol with $\|{\mathbf v}\|>1$. Define ... For a top-dimensional cone σ in the Voronoi decomposition containing...

conjecture | open | MathDB 页面 | 原文线索

V2-006868 | MATHDB-380696

LLL candidate conjecture for modular symbols

摘录 / EXCERPT: Let ${\mathbf v}=[v_1,\dots,v_n]$ be a modular symbol with $\|{\mathbf v}\|>1$, and define ... Let E be the standard basis of ${\mathbb Z}^n$. The quadratic form $b({\mathbf v})$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-006869 | MATHDB-380697

Reduction conjecture for Hecke cycles below the cohomological dimension

摘录 / EXCERPT: Let ξ and ξ' be the cycles associated with Algorithm, and suppose $n\leq4$ and $\|\xi\|>1$. For each modular symbol ${\mathbf v}$ in step A of the algorithm, choose...

conjecture | open | MathDB 页面 | 原文线索

V2-006870 | MATHDB-380714

The associated-prime conjecture for initial ideals of lattice ideals

摘录 / EXCERPT: Let S be a polynomial ring, let $I_{\mathcal L}$ be a lattice ideal in S, and let M be its initial ideal with respect to some term order. For a prime ideal P in S, write...

conjecture | open | MathDB 页面 | 原文线索

V2-006871 | MATHDB-380771

The polynomial bound on integer zeros of straight-line computable polynomials

摘录 / EXCERPT: Let a straight-line program for a polynomial in one variable be a list ... where each s_i, for $i\geq 2$, is one of s_j+s_k, s_j-s_k, or s_js_k for some $j,k<i$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-006872 | MATHDB-380790

Periodic triangle sequences and algebraic powers

摘录 / EXCERPT: Let n be a positive integer and let $\bigtriangleup$ denote the region of ordered n-tuples used by the triangle algorithm. Let $0 \leq \alpha_n \leq \cdots \leq \alpha_1 < 1$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-006873 | MATHDB-380791

Algebraicity from eventually periodic triangle sequences

摘录 / EXCERPT: Let $(\alpha_1,\ldots,\alpha_n)$ be an n-tuple of real numbers in $\bigtriangleup$, and suppose that its triangle sequence is eventually periodic. Algebraicity conjecture for per...

conjecture | open | MathDB 页面 | 原文线索

V2-006874 | MATHDB-380805

The classical absolute-value conjecture for Xi zeroes

摘录 / EXCERPT: Let $\Xi(\chi,t)=\Lambda(\chi,1/2+it)$ for an abelian character χ over a number field. The classical absolute-value conjecture. For every $e\geq0$, there are at most two zeroe...

conjecture | open | MathDB 页面 | 原文线索

V2-006875 | MATHDB-380806

The characteristic-p Generalized Simplicity Conjecture

摘录 / EXCERPT: Let $\overline{k}$ be a fixed algebraic closure of k, let $k^{\rm sep}$ be its separable closure, and let $G=\operatorname{Gal}(k^{\rm sep}/k)$. Let $\rho:G\to\mathbf C_\infty^*$...

conjecture | open | MathDB 页面 | 原文线索

V2-006876 | MATHDB-380826

The folklore conjecture on surjectivity and second cohomology over extensions of the cyclotomic field

摘录 / EXCERPT: Let p be a prime with $p\geqslant 3$, let F^{cyc} be the cyclotomic extension of F, and let F^S be the maximal extension of F unramified outside the specified set S. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-006877 | MATHDB-380846

The congruence ABC conjecture for N

摘录 / EXCERPT: Congruence ABC conjecture for N. For each $\epsilon>0$, there exists a constant C_ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006878 | MATHDB-380932

Critical invariant-circle breakdown threshold conjecture for twist maps

摘录 / EXCERPT: Let ρin$(0,1)$ be a Diophantine number, and let F be a twist map of the form given in the source. Denote by $\Lambda(\rho)$ the parameter threshold at which the invariant ci...

conjecture | open | MathDB 页面 | 原文线索

V2-006879 | MATHDB-380961

The reduced decomposition-matrix conjecture for spin symmetric groups

摘录 / EXCERPT: Let m be a positive integer, let p be an odd prime, and let ${\rm DP}(m)$ denote the relevant set of strict partitions of m. Let ${\rm DPR}_p(m)$ denote the indexing set for...

conjecture | open | MathDB 页面 | 原文线索

V2-006880 | MATHDB-380986

The conjecture that Vassiliev invariants distinguish prime unoriented knots

摘录 / EXCERPT: A Vassiliev invariant is a finite-type invariant of knots; a knot is prime if it cannot be expressed as a nontrivial connected sum, and unoriented knots are considered without a ch...

conjecture | open | MathDB 页面 | 原文线索

V2-006881 | MATHDB-380989

The tropical rank conjecture for binary phylogenetic tree models

摘录 / EXCERPT: Let τ be a binary tree on binary random variables, let f be the associated multilinear map, and let I_f be its prime ideal. For a $2 \times \cdots \times 2$ table, flatten...

conjecture | open | MathDB 页面 | 原文线索

V2-006882 | MATHDB-381020

Generalized Siegel conjecture for elliptic curves

摘录 / EXCERPT: Let C be an elliptic curve defined over $\mathbb Q$, embedded in projective space, and let x_n be a fixed coordinate in a coprime integral representative $[x_0,\ldots,x_N]$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-006883 | MATHDB-381032

Rubin's integral Stark conjecture

摘录 / EXCERPT: Let $(L/K,S,T)$ be a basic triple, let $G=\operatorname{Gal}(L/K)$, and let $r\geq 0$ be an integer. Let $U_{S,T}$ be the group of S-units of L that are congruent to 1 at all...

conjecture | open | MathDB 页面 | 原文线索

V2-006884 | MATHDB-381037

The exceptional-set power conjecture for arithmetic zeta functions

摘录 / EXCERPT: Exceptional-set power conjecture. If $\gamma\notin\mathbb{Q}$, then there exists $B\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006885 | MATHDB-381053

Density-zero conjecture for central vanishing of elliptic-curve twists

摘录 / EXCERPT: Let $p>2$ be a prime, let $E/{\mathbb Q}$ be an elliptic curve, and let Σ_p denote the set of Dirichlet characters of order p. For $S\subset\Sigma_p$, say that S has den...

conjecture | open | MathDB 页面 | 原文线索

V2-006886 | MATHDB-381072

Ratio of differences of fourth powers conjecture

摘录 / EXCERPT: A number t is represented as a ratio of two differences of fourth powers if there exist integers or rational numbers a,b,c,d such that ... with $c^4-d^4\neq 0$. Ratio of differen...

conjecture | open | MathDB 页面 | 原文线索

V2-006887 | MATHDB-381139

The weight-six generation conjecture for integral modular-form rings

摘录 / EXCERPT: Let $N\geq 1$. Write $M(\Gamma_0(N),\mathbb{Z}[\frac{1}{6N}])$ for the graded algebra of modular forms for $\Gamma_0(N)$ with coefficients in $\mathbb{Z}[\frac{1}{6N}]$. Weight-six...

conjecture | open | MathDB 页面 | 原文线索

V2-006888 | MATHDB-381158

Conjecture on determinants of trimidiation matrices

摘录 / EXCERPT: Determinant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006889 | MATHDB-381166

The exponent-pair conjecture for weak Carmichael numbers

摘录 / EXCERPT: Let e,f,e',f' be integers with $1\le e\le f$, $e+f\ge 3$, and likewise $1\le e'\le f'$ and $e'+f'\ge 3$. Let ${\mathcal P}_w(e,f)$ denote the set of pairs (p,q) of distinct pri...

conjecture | open | MathDB 页面 | 原文线索

V2-006890 | MATHDB-381192

The prime-power vanishing conjecture for algebraic monodromy of arrangements

摘录 / EXCERPT: Let ${\pazocal{A}}$ be an arrangement of rank at least 3. For each prime p and integer $s\ge 1$, let ${e}_{p^s}({\pazocal{A}})$ denote the corresponding elementary divisor mult...

conjecture | open | MathDB 页面 | 原文线索

V2-006891 | MATHDB-381216

Regularity conjecture for cycles of reduced forms of discriminant a^2+4

摘录 / EXCERPT: Let a cycle of reduced forms have caliber l, discriminant a^2+4, and sum n. Regularity conjecture. There is an integer $r\geq 0$ such that ... If $r>0$, the cycle consists of i...

conjecture | open | MathDB 页面 | 原文线索

V2-006892 | MATHDB-381218

The conjecture that the modified Erdős-Ginzburg-Ziv constant of Z_p^3 equals $9p-8$

摘录 / EXCERPT: Let p be a prime, and let $s_1(Z_p^3)$ denote the least integer ℓ such that every sequence of length ℓ in the finite abelian group Z_p^3 contains a zero-sum subsequen...

conjecture | open | MathDB 页面 | 原文线索

V2-006893 | MATHDB-381222

Lebesgue-Nagell-Ramanujan conjecture for 274837012705

摘录 / EXCERPT: Conjecture for 274837012705. Every solution (x,y,n) has one of the following forms:

conjecture | open | MathDB 页面 | 原文线索

V2-006894 | MATHDB-381225

Generalized $c4$-Li criterion for the class $aS^{\sharp \flat}(\sigma_0,\sigma_1)$

摘录 / EXCERPT: Generalized τ-Li criterion. Non-vanishing of F in the half-plane $\operatorname{Re}(s)>\tau/2$ is equivalent to growth of $\lambda_F(n,\tau)$ as $\mathcal{C}_F\tau n\log n$...

conjecture | open | MathDB 页面 | 原文线索

V2-006895 | MATHDB-381267

Finiteness conjecture for logarithmic class groups of real quadratic fields

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many primes ℓ for which the ℓ-group of logarithmic classes of K is nontrivial.

conjecture | open | MathDB 页面 | 原文线索

V2-006896 | MATHDB-381280

Keskin-Siar-Karaatli conjecture for a quadratic Diophantine equation

摘录 / EXCERPT: Let n be a nonnegative integer, and let k be an integer. Consider the Diophantine equation ... A positive integer solution means $x,y>0$, and a positive odd integer solution mean...

conjecture | open | MathDB 页面 | 原文线索

V2-006897 | MATHDB-381296

The theta-kernel weight divisibility conjecture

摘录 / EXCERPT: Let p be a prime, let n, j, and k be positive integers with $j<n$, and let $F\in M_k(\Gamma^{(n)}_{1}(N))_{\mathbb{Z}_{(p)}}$ with $k\in\mathbb{Z}_{\ge 1}$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-006898 | MATHDB-381299

Bilu-Luca conjecture on distinct number fields in fibers

摘录 / EXCERPT: Bilu-Luca conjecture. There exists a real number $c>0$ such that, for every sufficiently large integer N, at least cN of the number fields...

conjecture | open | MathDB 页面 | 原文线索

V2-006899 | MATHDB-381300

The explicit formula conjecture for Farey transfer-operator eigenfunctions

摘录 / EXCERPT: Let q be a parameter and let $\mathbf{E}_{q}(y)$ denote the function introduced through the Farey transfer operators. For the previously considered cases $q=1$ and $q=\frac12$, o...

conjecture | open | MathDB 页面 | 原文线索

V2-006900 | MATHDB-381308

Polynomial lower-bound conjecture for sums of roots of unity

摘录 / EXCERPT: Polynomial lower-bound conjecture. For given $n\ge 1$ there exist constants $c(n)>0$ and $\lambda(n)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006901 | MATHDB-381309

Tsaknias's local-type conjecture for non-CM newform Galois orbits

摘录 / EXCERPT: Let p be a prime and let $n\geq 1$. Consider non-CM newform Galois orbits of level p^n, weight k, and trivial Nebentypus. For an inertial local type τ of conductor p^n...

conjecture | open | MathDB 页面 | 原文线索

V2-006902 | MATHDB-381327

The genericity conjecture for hyperbolic knots

摘录 / EXCERPT: Let n be a positive integer, and consider all prime knots with n or fewer crossings. The genericity conjecture for hyperbolic knots. The percentage of these knots that are hype...

conjecture | open | MathDB 页面 | 原文线索

V2-006903 | MATHDB-381331

Folklore generic Newton polygon conjecture for triangular exponential sums

摘录 / EXCERPT: Let Δ be a triangle with vertices at $(0,0)$, $\mathbf P_1=(a_1,b_1)$, and $\mathbf P_2=(a_2,b_2)$. Assume the hypotheses on the prime p from the main theorem. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-006904 | MATHDB-381352

The asymptotic conjecture for connected functional graphs of quadratic polynomials

摘录 / EXCERPT: Let p be a prime, and let I_p denote the number of parameters a for which the functional graph associated with the quadratic polynomial f_a is connected. The computations u...

conjecture | open | MathDB 页面 | 原文线索

V2-006905 | MATHDB-381354

Gross's interpolation conjecture for p-adic Artin L-functions

摘录 / EXCERPT: Fix an odd prime p, a field isomorphism $\iota:\mathbb{C}\simeq\mathbb{C}_p$, a character $\psi\in\operatorname{Irr}_{\mathbb{C}_p}(\mathcal{G})$, and a finite set of places S...

conjecture | open | MathDB 页面 | 原文线索

V2-006906 | MATHDB-381386

Homological stability conjecture for spaces of polynomial maps

摘录 / EXCERPT: Homological stability conjecture. The cohomology group

conjecture | open | MathDB 页面 | 原文线索

V2-006907 | MATHDB-381400

The modularity conjecture for elliptic curves and QM surfaces over imaginary quadratic fields

摘录 / EXCERPT: Let K be an imaginary quadratic field. A Bianchi newform over K is a weight-2 cuspidal eigenform of level $[?]$ with rational Hecke eigenvalues; here $[?]$ denotes the ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-006908 | MATHDB-381429

The polynomial Erdős-Wagstaff density conjecture

摘录 / EXCERPT: Let $P(x)$ be a polynomial, and let $\bar{\mathbf{d}}$ denote upper density. For a positive integer T, consider the set of integers n for which $p-1$ divides $P(n)$ for some pr...

conjecture | open | MathDB 页面 | 原文线索

V2-006909 | MATHDB-381431

Stolz-Teichner's geometric description of topological modular forms

摘录 / EXCERPT: Stolz-Teichner conjecture. The degree-n topological modular forms of X are

conjecture | open | MathDB 页面 | 原文线索

V2-006910 | MATHDB-381435

Integral Prasanna-Venkatesh conjecture for the adjoint regulator

摘录 / EXCERPT: Let f be the modular form, F the field, p and k the prime and weight from Theorem 5.11, and let $u_2(f_F)$, Ω_f^+, Ω_f^-, and $R_{f,\alpha}$ be respectively t...

conjecture | open | MathDB 页面 | 原文线索

V2-006911 | MATHDB-381439

Almost-everywhere Weyl-sum exponent conjecture

摘录 / EXCERPT: Almost-everywhere Weyl-sum exponent conjecture. For $\alpha\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-006912 | MATHDB-381462

Rational Stark conjecture for a character at s = 0

摘录 / EXCERPT: Rational Stark conjecture. There exists an element $u_\chi\in(k\otimes_{\mathbb{Z}}U_{v,S})^{\chi^{-1}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-006913 | MATHDB-381463

The phantom-term conjecture for Sturm's operator

摘录 / EXCERPT: Phantom-term conjecture. Sturm's operator produces phantom terms on all the facets of the cone not perpendicular to each of the short simple roots. The phantom terms arise as Maass...

conjecture | open | MathDB 页面 | 原文线索

V2-006914 | MATHDB-381482

A q-congruence for the central trinomial coefficient

摘录 / EXCERPT: Let p be a prime with $p\geq 5$. Define the q-analogue of the trinomial coefficient by ... where $[n]_q=(1-q^n)/(1-q)$ and $\lfloor x\rfloor$ denotes the integral part of the rea...

conjecture | open | MathDB 页面 | 原文线索

V2-006915 | MATHDB-381483

Polynomial bound conjecture for maximal independent generating sets

摘录 / EXCERPT: Let G be a finite group, let $m(G)$ be the largest size of a minimal generating set of G, and let ... where $\pi(G)$ is the set of prime divisors of $|G|$ and $d_p(G)$ is the min...

conjecture | open | MathDB 页面 | 原文线索

V2-006916 | MATHDB-381498

Polynomial-growth conjecture for the least ternary height representative

摘录 / EXCERPT: For a natural number h, let n_h be the smallest ternary integer, if it exists, such that $A(n_h)=h$. Polynomial-growth conjecture for the least ternary height representative. T...

conjecture | open | MathDB 页面 | 原文线索

V2-006917 | MATHDB-381514

Generalized perfect-number conjecture for Mersenne exponents

摘录 / EXCERPT: Generalized perfect-number conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-006918 | MATHDB-381515

Beukers' conjecture on solutions of the generalized Lebesgue-Ramanujan-Nagell equation

摘录 / EXCERPT: Let D be a fixed positive nonsquare integer, let p be a fixed prime with $\gcd(D,p)=1$, and let $N(-D,2)$ denote the number of solutions (x,n) in $\mathbb{N}^2$ of ... The exce...

conjecture | open | MathDB 页面 | 原文线索

V2-006919 | MATHDB-381516

Cohn's conjecture on the exponent in the equation $a^n+b^n=z^2$

摘录 / EXCERPT: Let a and b be integers with $2\leq a<b$, and consider the equation ... Here (z,n) denotes a solution of the equation. Cohn's conjecture. Every solution (z,n) satisfies ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-006920 | MATHDB-381531

Unbounded-orbit conjecture for the quadratic Collatz-type recursion

摘录 / EXCERPT: Let the quadratic Collatz-type recursion be the recursion referred to as $(q1)$, with orbits (x_n) under its iteration. Unbounded-orbit conjecture. The recursion has an unbounded...

conjecture | open | MathDB 页面 | 原文线索

V2-006921 | MATHDB-381532

Unbounded-orbit conjecture for orbits containing numbers of the form 2^m-1

摘录 / EXCERPT: Let an orbit of the quadratic Collatz-type recursion be an orbit of the recursion referred to as $(q1)$, and let $m\geq 3$. 2^m-1 orbit conjecture. Orbits containing an odd numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-006922 | MATHDB-381547

The square Frobenius number conjecture for consecutive generators differing by two

摘录 / EXCERPT: Let $a\geqslant 3$ be an odd integer, and let $\langle a,a+2\rangle$ denote the numerical semigroup generated by a and $a+2$. Let $(u_n)_{n\geqslant 1}$ be the recursive sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-006923 | MATHDB-381549

The Banks-Martin conjecture analogue for primitive sets over function fields

摘录 / EXCERPT: Banks-Martin conjecture analogue. For each $q\geq 5$, the inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-006924 | MATHDB-381553

Mautner's cleanness conjecture for 0-cuspidal pairs

摘录 / EXCERPT: Let G be a group, let $\Bbbk$ be a field of characteristic l, and let $(C,\mathcal{E})\in\mathscr{I}(G)$ be a 0-cuspidal pair. A prime l is rather good for G if it is goo...

conjecture | open | MathDB 页面 | 原文线索

V2-006925 | MATHDB-381559

Mazur's uniform Mordell-Lang conjecture for curves

摘录 / EXCERPT: Mazur's uniform Mordell-Lang conjecture. Let $g\ge 2$ be an integer. Then there exists a constant $c(g)\ge 1$ such that, for every smooth curve C of genus g defined over F, e...

conjecture | open | MathDB 页面 | 原文线索

V2-006926 | MATHDB-381568

Moment conjecture for k-free numbers in short arithmetic progressions

摘录 / EXCERPT: Let $X>1$, let q be a positive integer, and let a range over residue classes relatively prime to q. Define the error term $E(X,q,a)$ by ... where ... For $ell\geq 1$, define the...

conjecture | open | MathDB 页面 | 原文线索

V2-006927 | MATHDB-381573

Noncongruence and odd-density conjecture for eta-quotient coefficients

摘录 / EXCERPT: Noncongruence and odd-density conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-006928 | MATHDB-381580

Conjectured invariants of the Shtuka surface for level 9amma_0(4(0))

摘录 / EXCERPT: Conjectured invariants of S_q. If q is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-006929 | MATHDB-381584

Eventual monotonicity conjecture for generalized partition ratios

摘录 / EXCERPT: Let $A\subset\mathbb{Z}_{>0}$, let $p^{(k)}(n)$ be the generalized partition functions associated with A, and define ... Assume that A has properties $P_{k+1},P_{k+2},\dots$. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-006930 | MATHDB-381606

Veech's conjecture on the Pinsker-factor criterion for Sarnak's conjecture

摘录 / EXCERPT: Veech's conjecture. Condition $\pi_0\perp L^2(\Pi(\kappa))$ for each Furstenberg system $\kappa\in V_S(\boldsymbol{\mu})$ is equivalent to Sarnak's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-006931 | MATHDB-381611

Near-square classification conjecture for binary recurrence sequences

摘录 / EXCERPT: Let a and b_{1} be relatively prime positive integers and put $b=-b_{1}^{2}$. Define the binary recurrence sequence by $u_{0}(a,b)=0$, $u_{1}(a,b)=1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006932 | MATHDB-381635

The four-cycle conjecture for the abelian loop

摘录 / EXCERPT: Four-cycle conjecture. There are infinitely many such quartets.

conjecture | open | MathDB 页面 | 原文线索

V2-006933 | MATHDB-381639

Existence of nontrivial μ-Sondow numbers above |μ|

摘录 / EXCERPT: Let ... mathfrak{S}{$mu} denote the set of ... mu $in $\mathbb{Z}$, ... The paper explains that $1\in\mathfrak{S}_{$mu}$ for every ... mu-Sondow number other than $1...

conjecture | open | MathDB 页面 | 原文线索

V2-006934 | MATHDB-381655

The conjecture on the values of n for which M(n)=2

摘录 / EXCERPT: Conjecture. The only values of $n\in\mathbb{N}_{\geq 3}$ such that $M(n)=2$ are

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006935 | MATHDB-381656

Finiteness conjecture for odd integers with equal binary digit sums 14 or 15

摘录 / EXCERPT: Finiteness conjecture. Let $k\in\{14,15\}$. Then the number of odd integers n with $s(n)=s(n^2)=k$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-006936 | MATHDB-381669

Bak's inhomogeneous dyadic approximation conjecture in Cantor's set

摘录 / EXCERPT: Throughout, let K be the middle-third Cantor set, let μ be its natural probability measure, and for $\psi:\mathbb R\to[0,\infty)$ and $y\in\mathbb R$ define ... Here $\|x\|$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-006937 | MATHDB-381676

The regularity conjecture for D(4)-quadruples

摘录 / EXCERPT: Regularity conjecture. Any $D(4)$-quadruple is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-006938 | MATHDB-381680

Dummit-Ford-Kisilevsky-Sands conjecture for Gauss factorials

摘录 / EXCERPT: Let G_n denote the Gauss factorial, and call a prime p 1-exceptional for $m=3$ when the corresponding exceptional congruence holds. Dummit-Ford-Kisilevsky-Sands conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-006939 | MATHDB-381685

The Linear AFL conjecture

摘录 / EXCERPT: Let M be the Rapoport-Zink space of strict π-divisible O_F-modules with quasi-isogeny to $\mathbb{X}$, and let $Z_1,Z_2\to M$ be the closed immersions associated with the m...

conjecture | open | MathDB 页面 | 原文线索

V2-006940 | MATHDB-381697

Monotone subsequences conjecture for fractional-part sequences involving the Euler-Gompertz constant

摘录 / EXCERPT: Monotone subsequences conjecture. There exists a subsequence of the first sequence that is monotone non-decreasing and a subsequence of the second sequence that is monotone non-inc...

conjecture | open | MathDB 页面 | 原文线索

V2-006941 | MATHDB-381698

Ma's conjecture on squares in exponential expressions

摘录 / EXCERPT: Ma's conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-006942 | MATHDB-381707

Effective Lang conjecture for partial quotients of cubic irrationals

摘录 / EXCERPT: Effective Lang conjecture. There exists an absolute constant C such that, for every real cubic irrational number x, all partial quotients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-006943 | MATHDB-381721

The prime-cycle clique extremal conjecture

摘录 / EXCERPT: Let p be a prime number, and let $\mathcal C_{\geq p}^{\mathrm{prime}}$ be the family of all cycles of length at least p. For integers n and r satisfying $r<p$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-006944 | MATHDB-381729

Positselski's uniform symbol-length conjecture

摘录 / EXCERPT: Let p) be a prime, let $\bar G$ be a pro-p group, let n be a positive integer, and let $\bar\omega\in H^n(\bar G,\mathbb{F}_p)$. For a field F containing a root of unity of...

conjecture | open | MathDB 页面 | 原文线索

V2-006945 | MATHDB-381736

The index divisor free prime conjecture

摘录 / EXCERPT: Index divisor free prime conjecture. Except when $(d,k)=(27,3)$, there exists an index divisor free prime for (d,k).

conjecture | open | MathDB 页面 | 原文线索

V2-006946 | MATHDB-381737

Erdős-Selfridge conjecture on prime divisors of consecutive products

摘录 / EXCERPT: Erdős-Selfridge conjecture. If $n+k\geq p^{(k)}$, then there is a prime greater than k which divides $(n+1)\cdots(n+k)$ to the first power.

conjecture | open | MathDB 页面 | 原文线索

V2-006947 | MATHDB-381738

The index divisor conjecture for products of consecutive integers

摘录 / EXCERPT: Index divisor conjecture. There exists a prime $p>k$ which divides Δ. Moreover, the index at which p occurs in the product $n(n-1)\cdots(n-k)$ does not divide the power to...

conjecture | open | MathDB 页面 | 原文线索

V2-006948 | MATHDB-381740

Nonexistence conjecture for cobalancing numbers with coefficients $a=x^2-1$ and $b=1$

摘录 / EXCERPT: Nonexistence conjecture. The equation has no integer solutions (x,n,m) for $x>1$ and $n\geq 1$ when $x\equiv 0,1,$ or $3\pmod 4$. Equivalently, the corresponding Diophantine equa...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-006949 | MATHDB-381753

Classification conjecture for irreducible λ-quiddities on the subgroup generated by the golden ratio

摘录 / EXCERPT: Let $\varphi=\frac{1+\sqrt{5}}{2}$ be the golden ratio, and let $\langle\varphi\rangle=\{k\varphi:k\in\mathbb{Z}\}$ be the subgroup of $\mathbb{C}$ generated by φ. An irred...

conjecture | open | MathDB 页面 | 原文线索

V2-006950 | MATHDB-381771

Shell-wise p-adic Duffin-Schaeffer zero-one law

摘录 / EXCERPT: Shell-wise p-adic Duffin-Schaeffer conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-006951 | MATHDB-381796

Symplectic variance conjecture for sums of divisor functions over quadratic residues

摘录 / EXCERPT: Let $d_k(n)$ denote the k-fold divisor function, and define ... For $x^{1/k}\leq y$ and primes p with $y\leq p\leq 2y$, write $\mathrm{Var}_{p\in[y,2y]}$ for the variance over th...

conjecture | open | MathDB 页面 | 原文线索

V2-006952 | MATHDB-381797

Orthogonal sector variance conjecture for norm-form divisor sums over Gaussian ideals

摘录 / EXCERPT: The orthogonal sector variance conjecture. If $x\leq K^\ell$, then there is a constant $a^O_\ell\in\mathbb Q$, depending on ℓ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-006953 | MATHDB-381809

Barker-Gelvin stable unital basis conjecture for source algebras

摘录 / EXCERPT: Let $\mathcal{O}$ be a complete local noetherian commutative ring with algebraically closed residue field of prime characteristic p. Let G be a finite group with $p\mid |G|$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-006954 | MATHDB-381823

The edge-map conjecture for integral modular forms on $\mathcal{M}_0(n)$

摘录 / EXCERPT: For every positive integer n, let $\operatorname{TMF}_0(n)$ denote the associated spectrum of topological modular forms and let $\operatorname{MF}_0(n)_0$ denote the degree-zero...

conjecture | open | MathDB 页面 | 原文线索

V2-006955 | MATHDB-381852

Hong's power LCM matrix size-threshold conjecture

摘录 / EXCERPT: Let t be a positive integer, let $S=\{x_1,x_2,\ldots,x_n\}$ be a GCD-closed set of positive integers, and let the power LCM matrix be the matrix ... The Hong conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-006956 | MATHDB-381901

The exact formula conjecture for primitive representations by x^2+216y^2

摘录 / EXCERPT: Let m be a positive integer with $m\equiv 1\pmod{24}$. Suppose that all prime divisors of m are congruent to 1, 5, 7, or 11 modulo 24, and write ... where...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-006957 | MATHDB-381902

The parity conjecture for even and odd 3-regular partitions on \mathcal P-progressions

摘录 / EXCERPT: Let $b_{3,e}(n)$ and $b_{3,o}(n)$ denote the numbers of 3-regular partitions of n having, respectively, an even and an odd number of parts. Let $n\geqslant 0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-006958 | MATHDB-381920

Bétermin's conjecture on minimizers of the Lennard-Jones lattice energy

摘录 / EXCERPT: Let A denote the lattice-area parameter in the Lennard-Jones minimization problem ... The triangular lattice, rhombic lattices, a unique minimizer, and rectangular lattices are the...

conjecture | open | MathDB 页面 | 原文线索

V2-006959 | MATHDB-381924

Infinitude conjecture for prime exponents in values of $3^{3+6c'}+1$

摘录 / EXCERPT: Infinitude conjecture. There are infinitely many values of $c'\not\equiv 3\pmod 7$ such that every prime other than 2 and 7 occurring in the factorization of $3^{3+6c'}+1$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-006960 | MATHDB-381942

Stolz-Teichner's TMF classification conjecture for two-dimensional supersymmetric quantum field theories

摘录 / EXCERPT: The group $cpi_d\operatorname{TMF}$ is the degree-d part of topological modular forms, and cpi_0 denotes connected components of the space of theories. A two-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-006961 | MATHDB-381943

Generalized fiber sum formula for free hypermultiplet theories

摘录 / EXCERPT: Let X_1 and X_2 be smooth, spin, closed, oriented four-manifolds, and let ... be their generalized fiber sum, also smooth, spin, closed, and oriented. Let d_1 and d_2 be thei...

conjecture | open | MathDB 页面 | 原文线索

V2-006962 | MATHDB-381970

Polynomial interpolation conjecture for constant terms of Hecke-group modular forms

摘录 / EXCERPT: Let $\mathcal{K}$ and $\overline{\mathcal{K}}$ be the two families of meromorphic modular forms in the source, and let $A_{\mathcal{K},k,m}(0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-006963 | MATHDB-381971

2-adic valuation conjecture for Ā at k=2 and powers of 2

摘录 / EXCERPT: Let $A_{\overline{\mathcal K},k,m}(0)$ denote the relevant constant term. 2-adic valuation conjecture at $k=2$. For every integer $a\geq2$, ... This is an empirical special case of t...

conjecture | open | MathDB 页面 | 原文线索

V2-006964 | MATHDB-381972

Odd-prime valuation conjecture for Ā at k=p

摘录 / EXCERPT: Let $A_{\overline{\mathcal K},k,m}(0)$ denote the relevant constant term. Odd-prime valuation conjecture. If $p>2$ is prime and a is a positive integer, then ... This is an empiric...

conjecture | open | MathDB 页面 | 原文线索

V2-006965 | MATHDB-381973

2-adic valuation conjecture when k has one binary digit

摘录 / EXCERPT: Let $d_2(k)$ be the sum of the binary digits of k, let $a=\operatorname{ord}_2(m)$, and let $o=\operatorname{ord}_2(A_{\overline{\mathcal K},k,m}(0))$. Binary-digit valuation con...

conjecture | open | MathDB 页面 | 原文线索

V2-006966 | MATHDB-381974

Catalan-number conjecture for 2-adic orders with two binary digits

摘录 / EXCERPT: Catalan valuation conjecture. Then $t=C_{1,n}$. The source presents this as an observed pattern in its empirical study of constant-term valuations; no resolution is given.

conjecture | open | MathDB 页面 | 原文线索

V2-006967 | MATHDB-381975

2-adic valuation conjecture for two binary digits and m congruent to 2 modulo 4

摘录 / EXCERPT: Let $d_2(k)$ be the sum of the binary digits of k. Two-digit valuation conjecture. If $d_2(k)=2$, $m=4j+2$ for $j=1,2,\ldots$, and $a=\operatorname{ord}_2(m)=1$, then ... This is a...

conjecture | open | MathDB 页面 | 原文线索

V2-006968 | MATHDB-381976

3-adic valuation conjecture when m is divisible by 3

摘录 / EXCERPT: Let $A_{\overline{\mathcal K},k,m}(0)$ denote the relevant constant term, and let $d_3(k)$ be the sum of the base-3 digits of k. 3-adic valuation conjecture. If $m\equiv0\pmod3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-006969 | MATHDB-381977

Valuation conjecture for the K-family at prime parameters

摘录 / EXCERPT: Let $A_{\mathcal K,k,m}(0)$ denote the constant term for the second family of meromorphic modular forms. K-family prime-parameter conjecture. The following assertions hold: if...

conjecture | open | MathDB 页面 | 原文线索

V2-006970 | MATHDB-381978

Conjecture on the factorization of constant-term polynomials for the Hecke group

摘录 / EXCERPT: Let $\overline{h}_k(x)=\nu_k\,p_{k,1}(x)\,p_{k,2}(x)\cdots p_{k,\alpha}(x)=\nu_k\,\widetilde{p}_k(x)$, where each $p_{k,n}$ is monic and ν_k is rational. Write the OEIS sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-006971 | MATHDB-382000

The prime-cyclic minimal-set formulation of the Lemke-Kleitman conjecture

摘录 / EXCERPT: Let p be a prime, let $\mathbb{Z}_p$ be the cyclic group of order p, and let $\mathcal A(\mathbb{Z}_p)$ denote the set of minimal zero-sum sequences over this group. A set...

conjecture | open | MathDB 页面 | 原文线索

V2-006972 | MATHDB-382018

Yang-Revin-Vdovin conjecture on conjugates detecting the pi-radical

摘录 / EXCERPT: Let π be a fixed subset of the set $\mathbb{P}$ of all primes with $\pi'=\mathbb{P}\setminus\pi\ne\varnothing$. A π-group is a finite group whose order has all prime diviso...

conjecture | open | MathDB 页面 | 原文线索

V2-006973 | MATHDB-382043

Positivity conjecture for toroidal Casimir energy in unitary bosonic CFTs

摘录 / EXCERPT: Let $d\geq 3$ and let the theory be a unitary bosonic conformal field theory that is not a topological quantum field theory. Let $\mathcal{E}(\tau)$ denote its dimensionless modula...

conjecture | open | MathDB 页面 | 原文线索

V2-006974 | MATHDB-382046

Cantor families of periodic solutions near nonresonant spherical capillary frequencies

摘录 / EXCERPT: Let (n_0,j_0) be a pair of natural numbers with $n_0\geq2$, $j_0\geq1$, and set ... Suppose that the only natural number solution of the Diophantine equation ... is (j_0,n_0). Peri...

conjecture | open | MathDB 页面 | 原文线索

V2-006975 | MATHDB-382047

KAM families of quasi-periodic solutions for spherical water droplets

摘录 / EXCERPT: Let ... . KAM conjecture. If the Diophantine equation has only finitely many solutions, then a KAM type result holds for ... , depending on a parameter which takes value in a Cantor-ty...

conjecture | open | MathDB 页面 | 原文线索

V2-006976 | MATHDB-382062

Böcherer-Kikuta's generalization conjecture for mod p^m singular forms

摘录 / EXCERPT: Let n, k, and N be positive integers, let r be an even integer, and let p be a prime. Let χ be a quadratic Dirichlet character modulo N satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-006977 | MATHDB-382069

The nonexistence of simple parity congruences for even-regular partitions when 3 divides the odd part

摘录 / EXCERPT: Let m_0 be the odd part of an even integer m, and let $b_m(n)$ denote the coefficients of the corresponding m-regular partition generating function. Theorem gives congruences...

conjecture | open | MathDB 页面 | 原文线索

V2-006978 | MATHDB-382070

The even-regular partition density conjecture

摘录 / EXCERPT: Let $b_m(n)$ be the coefficients of the generating function for m-regular partitions. For even m, write ... where m_0 is odd. The odd density of a sequence is the limiting prop...

conjecture | open | MathDB 页面 | 原文线索

V2-006979 | MATHDB-382071

The eta-quotient parity master conjecture

摘录 / EXCERPT: Let ... be an eta-quotient shifted so that all powers of q are integral, and let δ_F be the odd density of its coefficients $c(n)$, when this density exists. A polynomial is...

conjecture | open | MathDB 页面 | 原文线索

V2-006980 | MATHDB-382072

The Ramanujan-Kolberg identity conjecture for multipartitions

摘录 / EXCERPT: Let $p_t(n)$ denote the number of t-multipartitions of n. Let a and t be odd integers, with $3\mid t$ whenever $3\mid a$. Define b by ... when $3\mid t$, and by...

conjecture | open | MathDB 页面 | 原文线索

V2-006981 | MATHDB-382076

Ma-Zha-She coset equation conjecture

摘录 / EXCERPT: Let p be an odd prime, let $H=\langle-1,2\rangle\subset\mathbb{F}_p^{\times}$, let ℓ_0 be the index parameter used for H, and let g be a generator of...

conjecture | open | MathDB 页面 | 原文线索

V2-006982 | MATHDB-382097

Nilpotent-factor conjecture for non-commuting graphs

摘录 / EXCERPT: Let p be a prime, and suppose that $G=P\times A$ is a finite group, where $P\in\operatorname{Syl}_p(G)$ is non-abelian and A is an abelian p'-group. Let Γ_K denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-006983 | MATHDB-382111

The natural-boundary conjecture for the Goldbach Dirichlet series

摘录 / EXCERPT: Let $G_2(n)=\sum_{k+m=n}\Lambda(k)\Lambda(m)$ for positive integers n, where Λ is the von Mangoldt function, and define ... Under the Riemann hypothesis, $\Phi_2(s)$ admits...

conjecture | open | MathDB 页面 | 原文线索

V2-006984 | MATHDB-382137

Dimension conjecture for same-prime-divisor approximation on generalized Cantor sets

摘录 / EXCERPT: Let $C(b,D)$ be a generalized Cantor set with Hausdorff dimension γ, where b and t have the same prime divisors, D contains at least one of 0 and $b-1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-006985 | MATHDB-382138

Zero-full Hausdorff-measure conjecture for independent prime divisors

摘录 / EXCERPT: Let $C(b,D)$ be a generalized Cantor set with Hausdorff dimension γ, with D containing at least one of 0 and $b-1$. Let $t\geq 2$ be an integer such that the sets of pri...

conjecture | open | MathDB 页面 | 原文线索

V2-006986 | MATHDB-382214

Univariate inhomogeneous Duffin-Schaeffer conjecture for systems of linear forms

摘录 / EXCERPT: Univariate inhomogeneous Duffin-Schaeffer conjecture for systems of linear forms. For almost every $\mathbf{x}\in \operatorname{Mat}_{n\times m}(\mathbb{R})$, there exist infinitel...

conjecture | open | MathDB 页面 | 原文线索

V2-006987 | MATHDB-382221

The Baker-Bilu obstruction conjecture for integral points on curves

摘录 / EXCERPT: Let X be a nice curve, meaning a smooth, projective, geometrically integral curve, of genus at least 2 over $\overline{\mathbb{Q}}$. For a point $x\in X(\overline{\mathbb{Q}})$...

conjecture | open | MathDB 页面 | 原文线索

V2-006988 | MATHDB-382243

GW/PT correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Let M be a Calabi-Yau threefold, and let β be a nonzero curve class in M. The symbols $\operatorname{Z_{GW}^{\prime}}(M;u)_\beta$ and $\operatorname{Z_{PT}}(M;q)_\beta$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-006989 | MATHDB-382264

Conrey-Farmer-Keating-Rubinstein-Snaith conjecture for moments of real Dirichlet L-functions

摘录 / EXCERPT: Conrey-Farmer-Keating-Rubinstein-Snaith conjecture. For some $\delta>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-006990 | MATHDB-382272

The orbit-representation conjecture for integral points of $X_{N,y}$

摘录 / EXCERPT: Let $N\in\mathbb{N}$ and $y\in\Gamma_N$. Let $\eta\in X_{N,y}(\mathbb{Z})$ be an integral point. For $\widehat{w}\in\Sigma W^0$ and $g\in G$, define the action $g\cdot\widehat{w}$...

conjecture | open | MathDB 页面 | 原文线索

V2-006991 | MATHDB-382273

Nonvanishing conjecture for critical p-adic L-functions

摘录 / EXCERPT: Throughout, let f_α be a θ-critical normalized eigenform, and let $L_{\mathrm{S},\alpha}^{[0]}(f,\xi)$ and $L_{\mathrm{K},\alpha}(f,\xi)$ denote the corresponding cri...

open_problem | open | MathDB 页面 | 原文线索

V2-006992 | MATHDB-382282

Extension of Corollary 12 to primes greater than 23

摘录 / EXCERPT: Extension conjecture. Corollary 12 also holds for primes $p>23$.

conjecture | open | MathDB 页面 | 原文线索

V2-006993 | MATHDB-382283

Greenberg-Mazur Iwasawa main conjecture for elliptic curves

摘录 / EXCERPT: Let p be an odd prime and E an elliptic curve with good ordinary reduction at p. Let $ ... be the cyclotomic Iwasawa algebra, and let $L_p(E)\in\Lambda$ be the p-adic $L...

conjecture | open | MathDB 页面 | 原文线索

V2-006994 | MATHDB-382295

Serre's conjecture on Artin representations for regular local rings

摘录 / EXCERPT: Serre's conjecture. Let ℓ be a prime number invertible in A. The function a_G is the character of a $\overline{\mathbb{Q}}_\ell$-representation of G, called the Artin re...

conjecture | open | MathDB 页面 | 原文线索

V2-006995 | MATHDB-382300

The weaker inhomogeneous Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $ $ be the inhomogeneous Duffin-Schaeffer statement referred to as , and let be the additional divergence condition referred to as . Weaker inhomogeneous Duffin-Schaeffer conje...

conjecture | open | MathDB 页面 | 原文线索

V2-006996 | MATHDB-382301

The global hypoellipticity conjecture for the Vladimirov sub-Laplacian

摘录 / EXCERPT: Let $\mathbb{K}$ be a non-archimedean local field with ring of integers $\mathscr{O}_\mathbb{K}$, prime ideal $\mathfrak{p}=\textbf{p}\mathscr{O}_\mathbb{K}$, and residue field...

conjecture | open | MathDB 页面 | 原文线索

V2-006997 | MATHDB-382318

Refined moment conjecture for greatest common divisors in Lucas sequences

摘录 / EXCERPT: Refined moment conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-006998 | MATHDB-382330

The infinitude conjecture for lengths of corresponding sequences

摘录 / EXCERPT: Infinitude conjecture. For each positive integer $t\neq 2$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-006999 | MATHDB-382341

Hirschhorn-Sellers conjecture on the parity of 2^t-core partition numbers

摘录 / EXCERPT: Hirschhorn-Sellers conjecture. For all $n\geq 1$ with $3\nmid n$,

conjecture | open | MathDB 页面 | 原文线索

V2-007000 | MATHDB-382364

Chowla's conjecture for Liouville correlations

摘录 / EXCERPT: Let λ denote the Liouville function, and let $H=\{h_1,\ldots,h_d\}$ be a non-empty set of non-negative integers. Chowla's conjecture. The correlation of the Liouville funct...

conjecture | open | MathDB 页面 | 原文线索

V2-007001 | MATHDB-382381

Conjectural congruences for the k-elongated plane partition function modulo powers of 5

摘录 / EXCERPT: Let $d_k(n)$ denote the k-elongated plane partition function, and let $c,n\geq 0$. For a list of integers in an argument, interpret the corresponding congruence as holding for ea...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007002 | MATHDB-382425

Kollár's conjecture on diophantine subsets of the rational function field

摘录 / EXCERPT: Let D be a diophantine subset of $\mathbb{C}\left(z\right)$, and for each $n\in\mathbb{Z}_{\geq 1}$ let $\mathbb{C}\left[z\right]_n$ be the $(n+1)$-dimensional $\mathbb{C}$-vecto...

conjecture | open | MathDB 页面 | 原文线索

V2-007003 | MATHDB-382449

The conjecture on digital anomalies with exponent two

摘录 / EXCERPT: Digital-anomaly conjecture for $k=2$. The only digital anomalies (x,y,B,k) with $k=2$ are

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007004 | MATHDB-382460

The parking-number identity for arbitrary finite Coxeter systems

摘录 / EXCERPT: Let (W,S) be an irreducible finite Coxeter system with Coxeter number h, let $\vec{c}$ be a Coxeter word in S, let $p>0$ be an integer coprime to h, and let $J\subseteq S$....

conjecture | open | MathDB 页面 | 原文线索

V2-007005 | MATHDB-382461

Mersenne-triplet admissibility conjecture

摘录 / EXCERPT: Mersenne-triplet admissibility conjecture. For all $p\geq 2$, $(2^{p-1},2^{p}-1,1)_{\pmb{+}}$ is an admissible triplet of order one. It is strongly admissible only for $p=2$ and we...

conjecture | open | MathDB 页面 | 原文线索

V2-007006 | MATHDB-382462

Generalized Collatz convergence and cycle conjecture for the triplets $(2^{p+1},3\times2^p,2^p)_+$

摘录 / EXCERPT: Generalized Collatz convergence and cycle conjecture. For all $p\geq0$ with $p\ne2$, the triplet $(2^{p+1},3\times2^p,2^p)_+$ is strongly admissible of order one, with unique trivi...

conjecture | open | MathDB 页面 | 原文线索

V2-007007 | MATHDB-382467

Kionke's conjecture on the Weil abscissa of procyclic groups

摘录 / EXCERPT: Kionke's conjecture. The shifted partial Riemann zeta function $\zeta_S(s-1)$ has the same abscissa of convergence as the Weil representation zeta function $\zeta^W_{H_S}(s)$. This...

conjecture | open | MathDB 页面 | 原文线索

V2-007008 | MATHDB-382493

Conjectural duality for iterated q-integrals on the projective line minus four points

摘录 / EXCERPT: Let $N\geq 0$, let K be the field of fractions of $\mathbb{Q}[A,B,C,D,q]/(A-q^{N}D)$, and let W^{0} be the space containing words $(u_{1}v_{1})\cdots(u_{k}v_{k})$ with letters...

conjecture | open | MathDB 页面 | 原文线索

V2-007009 | MATHDB-382506

Gap-free floor conjecture for semiprime subset sums

摘录 / EXCERPT: Gap-free floor conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-007010 | MATHDB-382510

Random-point conjecture for sums of square roots modulo one

摘录 / EXCERPT: For integers $r\ge 1$ and real numbers $n\ge 1$, $0<\eta\le 1/2$, define ... Here $a_i,b_j\sim n$ denotes the same dyadic range as in the source, and $\lVert\cdot\rVert$ is distance...

conjecture | open | MathDB 页面 | 原文线索

V2-007011 | MATHDB-382512

The Extended 1-2-3 Conjecture of Pilz

摘录 / EXCERPT: Let A be a finite subset of the positive integers, and let n be a positive integer. For each positive integer k, write $k\boldsymbol{\bullet} A=\{ka:a\in A\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-007012 | MATHDB-382515

Congruences for Domb numbers and Lucas sequences

摘录 / EXCERPT: Congruences for Domb numbers and Lucas sequences. For every odd prime p: (i) if one of the Legendre symbols $(\frac{p}{3})$ and $(\frac{p}{13})$ is 1, then

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007013 | MATHDB-382570

Nonvanishing conjecture for the theta lift from quaternionic modular forms

摘录 / EXCERPT: Theta-lift nonvanishing conjecture. If $L(f,1)=0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-007014 | MATHDB-382870

Cyclotomic conjecture

摘录 / EXCERPT: For integers $n>2$ and $k>1$, let $F_{n,k}(x)=\Phi_n(1+x+\cdots+x^k)\in\mathbb{Q}[x]$, where Φ_n is the n-th cyclotomic polynomial. Determine, exactly in terms of n and...

conjecture | solved | MathDB 页面 | 原文线索

V2-007015 | MATHDB-382955

Godara-Sarkar conjecture for the small Davenport constant of Heisenberg groups

摘录 / EXCERPT: Godara and Sarkar conjectured that, for every odd prime p, if $H_{p^3}=\operatorname{UT}_3(\mathbb{F}_p)$ is the Heisenberg group of order p^3, then $\mathsf{d}(H_{p^3})=3p-3$,...

conjecture | special_case | MathDB 页面 | 原文线索

V2-007016 | MATHDB-382956

Tarizadeh’s Conjecture 5.8 on the maximality of purely-prime ideals

摘录 / EXCERPT: For every commutative ring R, every purely-prime ideal of R is purely-maximal; that is,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007017 | MATHDB-383166

Hung-Martínez-Navarro conjecture

摘录 / EXCERPT: For every finite group G, every prime p, and every Sylow p-subgroup $P\in\operatorname{Syl}_p(G)$, define...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007018 | MATHDB-383196

Gow-McGuire conjecture on primitive quartic polynomials

摘录 / EXCERPT: Gow--McGuire Conjecture 1. Let $q\ne 13$ be any odd prime power. Let $\alpha\in\mathbb{F}_{q^{2}}\setminus\mathbb{F}_{q}$ be fixed. Then there exists $\lambda\in\mathbb{F}_{q}$ suc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007019 | MATHDB-383197

Eaton-Moretó conjecture for principal blocks

摘录 / EXCERPT: For every finite group G and prime p, let $P\in\operatorname{Syl}_p(G)$ be nonabelian, and define $m(P)=\min\{\varphi(1):\varphi\in\operatorname{Irr}(P),\ \varphi(1)>1\}$. Then...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007020 | MATHDB-383205

ABC conjecture and its exceptional-set problem

摘录 / EXCERPT: For every $\varepsilon>0$, there are only finitely many triples $(a,b,c)\in\mathbb{Z}_{>0}^3$ such that $a+b=c$, $\gcd(a,b,c)=1$, and $c>\operatorname{rad}(abc)^{1+\varepsilon}$, w...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007021 | MATHDB-383276

Chowla’s conjecture

摘录 / EXCERPT: For every integer $k\ge 1$ and every collection of distinct natural numbers $h_1,\ldots,h_k$, where $\lambda(n)=(-1)^{\Omega(n)}$ is the Liouville function, one has...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007022 | MATHDB-383323

Erdős-Mauduit-Sárközy conjecture

摘录 / EXCERPT: Let $\mathcal{B}_n=\left\{f(x)=\sum_{i=0}^{n}\varepsilon_i x^i:\ \varepsilon_i\in\{0,1\},\ \varepsilon_n=1\right\}$. The Erdős-Mauduit-Sárközy conjecture asserts that there exists...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007023 | MATHDB-383409

RSA small-private-exponent problem

摘录 / EXCERPT: Determine the supremum of the exponents α for which the following holds: for every fixed $\varepsilon>0$, there is a deterministic polynomial-time algorithm that, given a ba...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007024 | MATHDB-383450

Coppersmith small-root problem

摘录 / EXCERPT: Given a modular polynomial equation $f(x_1,\ldots,x_n)\equiv 0\pmod N$ and the associated lattice-based Coppersmith method, determine the optimal asymptotic region of exponents...

open_problem | open | MathDB 页面 | 原文线索

V2-007025 | MATHDB-383490

Csordas-Dimitrov second-level concavity conjecture

摘录 / EXCERPT: Let Φ be the classical Jacobi-theta kernel in the Fourier representation of the Riemann Ξ-function. Define $s(t)=\Phi(\sqrt{t})$ and $f(t)=s'(t)^2-s(t)s''(t)$ for $t>0$. T...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007026 | MATHDB-383501

Chow-Rimanić polynomial-kernel covering conjecture

摘录 / EXCERPT: For every prime power q and integer $k\ge 0$, let $C_k(q)$ be the least cardinality of a family of nonzero polynomials over $\mathbb{F}_q$ whose associated codimension-k partia...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007027 | MATHDB-383537

Beluhov’s Conjecture on the Tchoukaillon Array

摘录 / EXCERPT: Let $T_{i,j}$ denote the entry in row i and column j of the Tchoukaillon array. Then, as $i+j\to\infty$ with $i,j\ge 0$, ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007028 | MATHDB-383714

OEIS A080170 conjecture

摘录 / EXCERPT: For every integer $k\ge 2$, the binomial-coefficient gcd quantity $D(k)$ specified in OEIS A080170 satisfies $D(k)=1$ if and only if...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007029 | MATHDB-383771

On Artin's conjecture over an algebraically closed field of prime characteristic (problem 1)

摘录 / EXCERPT: If A is a connected graded domain over an algebraically closed field of prime characteristic, is $Q_{\text{gr}}(A)_0$ finite over its centre?

open_problem | open | MathDB 页面 | 原文线索

V2-007030 | MATHDB-383779

Points on Elliptic Curves over Finite Fields (problem 1)

摘录 / EXCERPT: Is there an elliptic curve over $\mathbb{Z}$ such that, for every prime p (or just infinitely many primes p), it has exactly p points over $\mathbb{F}_p$?

open_problem | open | MathDB 页面 | 原文线索

V2-007031 | MATHDB-384430

Expectations of Characteristic Polynomials (problem 6)

摘录 / EXCERPT: Calculate $\mathbb{E}\left|\frac{Z\prime}{Z}\right|^{2k}_{\theta+i\varepsilon}$ - is this related to Painlev'e?

open_problem | open | MathDB 页面 | 原文线索

V2-007032 | MATHDB-384431

Number Theory and Zeta (problem 1)

摘录 / EXCERPT: Is the value distribution of $|\zeta(1/2+it)|$ related to Painlev'e?

open_problem | open | MathDB 页面 | 原文线索

V2-007033 | MATHDB-384432

Number Theory and Zeta (problem 2)

摘录 / EXCERPT: Where is the arithmetic part in $P(s)$, the spacing distribution between zeros of the zeta function, for $\zeta(1/2+it)$? Can arithmetic corrections be found?

open_problem | open | MathDB 页面 | 原文线索

V2-007034 | MATHDB-384433

Number Theory and Zeta (problem 3)

摘录 / EXCERPT: Find the small s expansion for $E_N(0,s)$ - coefficients are polynomials in N. Find this expansion for all number theoretic ensembles in the literature.

open_problem | open | MathDB 页面 | 原文线索

V2-007035 | MATHDB-384436

Within random matrix theory, how does one distinguish the diagonal and off-diagonal contributions (from the point of view of Number Theory)?

摘录 / EXCERPT: Within random matrix theory, how does one distinguish the diagonal and off-diagonal contributions (from the point of view of Number Theory)? Is there a statistic one can write down...

open_problem | open | MathDB 页面 | 原文线索

V2-007036 | MATHDB-384909

Roth with common difference $p-1$

摘录 / EXCERPT: For all $C>0$ and sufficiently large N, if $A\subset [N]$ with $|A|\geq N/\log^C N$, then there exists x and p prime such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007037 | MATHDB-385588

The Problems (problem 22)

摘录 / EXCERPT: Let ΠK be the projection body of K. It is known that if K is a polytope and $\Pi \Pi K$ is homothetic to K, then K is a direct sum of planar polygons (in even dimensio...

open_problem | open | MathDB 页面 | 原文线索

V2-007038 | MATHDB-385726

Miscellaneous (problem 4)

摘录 / EXCERPT: Let f be a countable-to-1 aperiodic (i.e. acyclic) Borel function which is well founded as a binary relation (no infinite back-orbits). begin{enumerate}[a.] item Is there a Borel...

open_problem | open | MathDB 页面 | 原文线索

V2-007039 | MATHDB-385758

Iwahori-Matsumoto presentation

摘录 / EXCERPT: What is the “Iwahori-Matsumoto Presentation” of Iwahori-Hecke algebra for $\widetilde{G}_F$ for split G_F?

open_problem | open | MathDB 页面 | 原文线索

V2-007040 | MATHDB-385761

Steinberg representation

摘录 / EXCERPT: Define Steinberg representation of $\widetilde{G}_{F}$. What is the formal degree of Steinberg representation of $\widetilde{G}_{F}$ when G is split over F? What is analogue of...

open_problem | open | MathDB 页面 | 原文线索

V2-007041 | MATHDB-385764

In the real case, given a cover of ... , describe Brylinski-Deligne data.

摘录 / EXCERPT: In the real case, given a cover of $G_{\mathbb{R}}$, describe Brylinski-Deligne data.

open_problem | open | MathDB 页面 | 原文线索

V2-007042 | MATHDB-385766

Lurie Conjecture

摘录 / EXCERPT: Twisted Whittaker models by D. Gaitsgory

open_problem | open | MathDB 页面 | 原文线索

V2-007043 | MATHDB-385767

Covering group of tori

摘录 / EXCERPT: Let T be a (not necessary split) torus over a p--adic field. Construct $\widetilde{T}$ ... explicitly within Brylinski-Deligne framework.

open_problem | open | MathDB 页面 | 原文线索

V2-007044 | MATHDB-385768

Base change

摘录 / EXCERPT: Given a genuine irreducible representation $\widetilde{\pi}$ of $\widetilde{G}_F,$ describe $Lift_{L/F}(\widetilde{\pi})$ the base change lift of $\widetilde{\pi}.$ Note that...

open_problem | open | MathDB 页面 | 原文线索

V2-007045 | MATHDB-385769

Lifts (problem 2)

摘录 / EXCERPT: Does $Gal(L/F)$ act on $\widetilde{G}_L$?

open_problem | open | MathDB 页面 | 原文线索

V2-007046 | MATHDB-385770

Lifts (problem 3)

摘录 / EXCERPT: Make sense of transferring representation from $\widetilde{G}_F$ to $\widetilde{G'}_F,$ where two linear groups G_F and G'_F are inner forms each other.

open_problem | open | MathDB 页面 | 原文线索

V2-007047 | MATHDB-385771

Lifts (problem 4)

摘录 / EXCERPT: Is there a natural lifting from representations of G_F to $\widetilde{G'}_F$?

open_problem | open | MathDB 页面 | 原文线索

V2-007048 | MATHDB-385772

What is the local character identity for ... ?

摘录 / EXCERPT: What is the local character identity for $Mp(2n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-007049 | MATHDB-385776

Interpret formal degrees of representations of ... on level of Hecke algebra.

摘录 / EXCERPT: Interpret formal degrees of representations of $\widetilde{G}_{F}$ on level of Hecke algebra. (a la Opdam, Reeder, et al)

open_problem | open | MathDB 页面 | 原文线索

V2-007050 | MATHDB-385782

Jacobi forms (problem 1)

摘录 / EXCERPT: What is a quasi-Jacobi form? What are interesting examples? Generalize this notion to (quasi) automorphic forms. Generalize this notion to mock Siegel modular forms.

open_problem | open | MathDB 页面 | 原文线索

V2-007051 | MATHDB-385783

Is there a mock version of skew holomorphic Jacobi forms?

摘录 / EXCERPT: Is there a mock version of skew holomorphic Jacobi forms?

open_problem | open | MathDB 页面 | 原文线索

V2-007052 | MATHDB-385784

Are there connections between low index Mock modular Jacobi forms and finite groups (and geometry)?

摘录 / EXCERPT: Are there connections between low index Mock modular Jacobi forms and finite groups (and geometry)?

open_problem | open | MathDB 页面 | 原文线索

V2-007053 | MATHDB-385789

Specific functions (problem 1)

摘录 / EXCERPT: Can we calculate the leading terms of $F_g^B(q)$ at the orbifold point $z=0$ of the quintic ...

open_problem | open | MathDB 页面 | 原文线索

V2-007054 | MATHDB-385790

Specific functions (problem 2)

摘录 / EXCERPT: Do the coefficients of h_m have enumerative significance? Can we find h_m for $m \geq 9$?

open_problem | open | MathDB 页面 | 原文线索

V2-007055 | MATHDB-385797

Critical height conjecture

摘录 / EXCERPT: Let $f:\mathbb{P}^1 \to \mathbb{P}^1$, not Lattes. Define the critical height ... Does there exist $c_1,c_2 > 0$ depending on d,h_M such that for all...

open_problem | open | MathDB 页面 | 原文线索

V2-007056 | MATHDB-385802

field of definition vs field of moduli

摘录 / EXCERPT: Are all PCF maps defined over their field of moduli?

open_problem | open | MathDB 页面 | 原文线索

V2-007057 | MATHDB-385805

Algebraic proof of Thurston rigidity

摘录 / EXCERPT: Algebraic proof of infinitesimal Thurston Rigidity, i.e. if f is PCF non-Lattes and degree at least 2, then 1 is not an eigenvalue of...

open_problem | open | MathDB 页面 | 原文线索

V2-007058 | MATHDB-385806

Ramfication in pre-image towers

摘录 / EXCERPT: There exists a finite set of primes S such that all K_n are unramified outside of S. (Known for $\mathbb{P}^1$). begin{enumerate} item If not PCF and $X=\mathbb{P}^1$ let...

open_problem | open | MathDB 页面 | 原文线索

V2-007059 | MATHDB-385807

Detecting PCF maps

摘录 / EXCERPT: Let f be a PCF rational function defined over a number field. Is it “algebraically detectable” from $K_n = K(\mathop{Per}_n(f))$ or the tower of such extensions that f is PCF....

open_problem | open | MathDB 页面 | 原文线索

V2-007060 | MATHDB-385809

Notions of PCF

摘录 / EXCERPT: begin{enumerate} item Does (1) imply (2)? item Are PCF (1) or (2) Zariski dense in M_d^N? item Is there are more general definition than such that: a) “there are more of them”? (...

open_problem | open | MathDB 页面 | 原文线索

V2-007061 | MATHDB-385812

Entropy

摘录 / EXCERPT: begin{enumerate} item Let $f(z) = z^2 + c$, $c \in {\\mathbb{R}}$ and PCF with invariant $[a,b]$ with fixed point at one end. Which algebraic numbers arise as e^s where s is th...

open_problem | open | MathDB 页面 | 原文线索

V2-007062 | MATHDB-385813

Milnor's Characterisation of Lattes

摘录 / EXCERPT: begin{enumerate} item Does this hold over positive characteristic? item What do you get from the dual of Frobenius when inseparable? item What about characteristic 2 and 3 and...

open_problem | open | MathDB 页面 | 原文线索

V2-007063 | MATHDB-385816

Density results for PCF polynomials (problem 1)

摘录 / EXCERPT: Describe ... for p prime as $x\rightarrow \infty.$

open_problem | open | MathDB 页面 | 原文线索

V2-007064 | MATHDB-385817

What is the growth rate of ... for PCF and non PCF maps?

摘录 / EXCERPT: What is the growth rate of $N(x)$ for PCF and non PCF maps?

open_problem | open | MathDB 页面 | 原文线索

V2-007065 | MATHDB-385818

Density results for PCF polynomials (problem 3)

摘录 / EXCERPT: If $f(z)\in\mathbb{F}_q(T)$, what happens then? What happens with other fields?

open_problem | open | MathDB 页面 | 原文线索

V2-007066 | MATHDB-385819

Conjugacy Invariants (problem 1)

摘录 / EXCERPT: Is Pink's generating function Φ_w always rational for every $\rho(\text{Frob}_p)=w?$

open_problem | open | MathDB 页面 | 原文线索

V2-007067 | MATHDB-385820

Conjugacy Invariants (problem 2)

摘录 / EXCERPT: Cook up other conjugacy invariant things.

open_problem | open | MathDB 页面 | 原文线索

V2-007068 | MATHDB-385821

Geometric vs. Arithmetic representations (problem 1)

摘录 / EXCERPT: Is there any example where G does not have finite index in $G'?$

open_problem | open | MathDB 页面 | 原文线索

V2-007069 | MATHDB-385822

Geometric vs. Arithmetic representations (problem 2)

摘录 / EXCERPT: Understand the relation with PCFness. Note this is related to Problem 1.1. Perhaps start with $f(z)=z^2-1.$

open_problem | open | MathDB 页面 | 原文线索

V2-007070 | MATHDB-385823

Dynatomic Modular Curves (problem 1)

摘录 / EXCERPT: begin{enumerate} item Analogy between dynatomic and modular curves (See Silverman, ADS Section 4.1 and 4.2 for dynatomic polynomials). item What are other proxies for level structu...

open_problem | open | MathDB 页面 | 原文线索

V2-007071 | MATHDB-385824

Dynatomic Modular Curves (problem 2)

摘录 / EXCERPT: begin{enumerate} item For $p|N,$ $X^{\text{dyn}}(N)$ has bad reduction at p for the standard model, but evidence suggests there exists a different model with good reduction. Is t...

open_problem | open | MathDB 页面 | 原文线索

V2-007072 | MATHDB-385825

Boundary of M_d (problem 1)

摘录 / EXCERPT: Give an algebraic description of the boundary and maps in the boundary in characteristic p not necessarily $0.$ (Reference DeMarco). Try $d=3.$ Motivation: results for moduli cur...

open_problem | open | MathDB 页面 | 原文线索

V2-007073 | MATHDB-385826

Boundary of M_d (problem 2)

摘录 / EXCERPT: begin{enumerate} item (Want to look at a curve in M_2). For $f(z)=\frac{z^2-1}{z^2-c},$ there is a collection of PCF maps which approach $c=1$ on the curve. Compare the arboreal...

open_problem | open | MathDB 页面 | 原文线索

V2-007074 | MATHDB-385828

Bad reduction for models of critical orbit relation curves.

摘录 / EXCERPT: Bad reduction for models of critical orbit relation curves. e.g. $M_2.$

open_problem | open | MathDB 页面 | 原文线索

V2-007075 | MATHDB-385830

Dynatomic side (problem 2)

摘录 / EXCERPT: What do you need to know about K_f to figure out what f was (or how close to f can you get)?

open_problem | open | MathDB 页面 | 原文线索

V2-007076 | MATHDB-385832

The previous section's problem appropriately phrased for preimage extensions.

摘录 / EXCERPT: The previous section's problem appropriately phrased for preimage extensions.

open_problem | open | MathDB 页面 | 原文线索

V2-007077 | MATHDB-385833

Is there a deformation theory for arboreal Galois representations?

摘录 / EXCERPT: Is there a deformation theory for arboreal Galois representations?

open_problem | open | MathDB 页面 | 原文线索

V2-007078 | MATHDB-385836

Generalizing: Can you find an example in higher dimensions?

摘录 / EXCERPT: Generalizing: Can you find an example in higher dimensions?

open_problem | open | MathDB 页面 | 原文线索

V2-007079 | MATHDB-385837

Formal period n (problem 1)

摘录 / EXCERPT: Describe $K_n\cap K_{n'}.$ Prove that this is small.

open_problem | open | MathDB 页面 | 原文线索

V2-007080 | MATHDB-385840

What operations are there on these “representations”?

摘录 / EXCERPT: What operations are there on these “representations”? And do some of them correspond to operations on $f?$

open_problem | open | MathDB 页面 | 原文线索

V2-007081 | MATHDB-385845

Moments of divisor functions in short intervals

摘录 / EXCERPT: Evaluate ... for different ranges of H.

open_problem | open | MathDB 页面 | 原文线索

V2-007082 | MATHDB-385853

Can one prove the fourth moment formula starting from ...

摘录 / EXCERPT: Can one prove the fourth moment formula starting from ...

open_problem | open | MathDB 页面 | 原文线索

V2-007083 | MATHDB-385854

Is there any inclusion/exclusion structure to the diagonal or type I or type II sums?

摘录 / EXCERPT: Is there any inclusion/exclusion structure to the diagonal or type I or type II sums?

open_problem | open | MathDB 页面 | 原文线索

V2-007084 | MATHDB-385855

Why are the type II terms not already accounted for in the type I calculations?

摘录 / EXCERPT: Why are the type II terms not already accounted for in the type I calculations? Is there a way to arrange the type I calculations to reflect these new main terms?

open_problem | open | MathDB 页面 | 原文线索

V2-007085 | MATHDB-385857

Develop a heuristic for moments of ... -functions of automorphic forms.

摘录 / EXCERPT: Develop a heuristic for moments of L-functions of automorphic forms.

open_problem | open | MathDB 页面 | 原文线索

V2-007086 | MATHDB-385858

Other families of L-functions (problem 3)

摘录 / EXCERPT: Develop a heuristic for moments of imprimitive L-functions, or if your L-function factors into degree 1 factors, e.g. ... (or different powers).

open_problem | open | MathDB 页面 | 原文线索

V2-007087 | MATHDB-385859

Develop a heuristic for quadratic twists of ... -functions

摘录 / EXCERPT: Develop a heuristic for quadratic twists of L-functions

open_problem | open | MathDB 页面 | 原文线索

V2-007088 | MATHDB-385860

Other families of L-functions (problem 5)

摘录 / EXCERPT: The mechanism for computing moments in families like χ (mod q), $q\leq Q$, involve different orthogonality relations than $|m-n|=$small. Yet the answers are the same. Why?

open_problem | open | MathDB 页面 | 原文线索

V2-007089 | MATHDB-385866

Making the method more rigorous (problem 4)

摘录 / EXCERPT: Can we make the calculations uniform in k?

open_problem | open | MathDB 页面 | 原文线索

V2-007090 | MATHDB-385873

Develop a heuristic for ... Find connections with the work of Bogomolny-Keating and with the Ratios Conjectures.

摘录 / EXCERPT: Develop a heuristic for ... Find connections with the work of Bogomolny-Keating and with the Ratios Conjectures.

open_problem | open | MathDB 页面 | 原文线索

V2-007091 | MATHDB-385878

Study the “reverse mathematics" of nonstandard analysis.

摘录 / EXCERPT: Study the “reverse mathematics" of nonstandard analysis.

open_problem | open | MathDB 页面 | 原文线索

V2-007092 | MATHDB-385879

Is there a combinatorial result which implies ... comprehension and has a proof in nonstandard analysis?

摘录 / EXCERPT: Is there a combinatorial result which implies Π^1_1 comprehension and has a proof in nonstandard analysis?

open_problem | open | MathDB 页面 | 原文线索

V2-007093 | MATHDB-385880

Nonstandard Analysis (problem 3)

摘录 / EXCERPT: Give a nonstandard analysis proof of Szemer'{e}di's Theorem in the spirit of the original proof.

open_problem | open | MathDB 页面 | 原文线索

V2-007094 | MATHDB-385881

Study boundary amenable groups from the perspective of nonstandard analysis.

摘录 / EXCERPT: Study boundary amenable groups from the perspective of nonstandard analysis. 1. Give an “internal" algebraic description of boundary amenability. 2. Prove analogs of the results of...

open_problem | open | MathDB 页面 | 原文线索

V2-007095 | MATHDB-385882

Give a nonstandard analysis description of Bohr sets in ... .

摘录 / EXCERPT: Give a nonstandard analysis description of Bohr sets in ${\\mathbb{Z}}$.

open_problem | open | MathDB 页面 | 原文线索

V2-007096 | MATHDB-385883

Does the choice of specific ultrafilter have an affect on the resulting model of nonstandard analysis?

摘录 / EXCERPT: Does the choice of specific ultrafilter have an affect on the resulting model of nonstandard analysis? Can special properties of ultrafilters (e.g. idempotence, P-points, selecti...

open_problem | open | MathDB 页面 | 原文线索

V2-007097 | MATHDB-385884

Give a model theoretic proof of the existence of idempotent points in ... .

摘录 / EXCERPT: Give a model theoretic proof of the existence of idempotent points in ${^*}\mathbb{N}$.

open_problem | open | MathDB 页面 | 原文线索

V2-007098 | MATHDB-385885

Give a ZFC proof of the existence of a Hausdorff ultrafilter.

摘录 / EXCERPT: Give a ZFC proof of the existence of a Hausdorff ultrafilter.

open_problem | open | MathDB 页面 | 原文线索

V2-007099 | MATHDB-385886

Study partition regularity for degenerate norm form equations.

摘录 / EXCERPT: Study partition regularity for degenerate norm form equations.

open_problem | open | MathDB 页面 | 原文线索

V2-007100 | MATHDB-385887

Asymptotic Bases (problem 1)

摘录 / EXCERPT: Let A be an asymptotic basis of order 2 such that ... for some $c > 0$ and all $n \geq n_0$. Does A contain a minimal asymptotic basis of order 2?

open_problem | open | MathDB 页面 | 原文线索

V2-007101 | MATHDB-385888

Asymptotic Bases (problem 2)

摘录 / EXCERPT: Let A be an asymptotic basis of order 2 such that ... Does A contain a minimal asymptotic basis of order 2?

open_problem | open | MathDB 页面 | 原文线索

V2-007102 | MATHDB-385889

$B+C~$Conjecture (problem 1)

摘录 / EXCERPT: Suppose G is a countable amenable group and $A\subseteq G$ has positive Banach density. Are there infinite $B,C\subseteq G$ such that $B\cdot C\subseteq A$?

open_problem | open | MathDB 页面 | 原文线索

V2-007103 | MATHDB-385890

$B+C~$Conjecture (problem 2)

摘录 / EXCERPT: Suppose G is an NIP expansion of a countable amenable group. Let $A\subseteq G$ be definable, with positive upper Banach density. Are there infinite $B,C\subseteq G$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-007104 | MATHDB-385891

Structure in$~(A+B)\cdot (C+D)$ (problem 1)

摘录 / EXCERPT: Suppose $A\subseteq\mathbb{N}\times\mathbb{N}$ has positive Banach density, and set $\Delta=\{x\cdot y:(x,y)\in A-A\}$. Does Δ contain a nontrivial subgroup of $\mathbb{Z}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007105 | MATHDB-385892

Structure in$~(A+B)\cdot (C+D)$ (problem 2)

摘录 / EXCERPT: Find a quantitative version of the second theorem above (i.e. express the index of the subgroup in terms of the Banach density of A).

open_problem | open | MathDB 页面 | 原文线索

V2-007106 | MATHDB-385893

Near Arithmetic Progressions (problem 1)

摘录 / EXCERPT: For all $t\in\mathbb{N}$ and sufficiently large n, ...

open_problem | open | MathDB 页面 | 原文线索

V2-007107 | MATHDB-385894

Near Arithmetic Progressions (problem 2)

摘录 / EXCERPT: For all $t,k\in\mathbb{N}$ and sufficiently large n, ...

open_problem | open | MathDB 页面 | 原文线索

V2-007108 | MATHDB-385897

Explore security of obfuscation constructions that use current candidate multilinear maps.

摘录 / EXCERPT: Explore security of obfuscation constructions that use current candidate multilinear maps.

open_problem | open | MathDB 页面 | 原文线索

V2-007109 | MATHDB-385898

What is the security of d-local pseudorandom number generators?

摘录 / EXCERPT: What is the security of d-local pseudorandom number generators? This means that each bit of the string depends on at most d bits of the seed.

open_problem | open | MathDB 页面 | 原文线索

V2-007110 | MATHDB-385899

Can we construct local pseudorandom number generators from algebraic methods?

摘录 / EXCERPT: Can we construct local pseudorandom number generators from algebraic methods?

open_problem | open | MathDB 页面 | 原文线索

V2-007111 | MATHDB-385900

Can we construct a bilinear pairing which takes values in E(Fq) where the source groups are also Discrete-Log-Hard?

摘录 / EXCERPT: Can we construct a bilinear pairing which takes values in E(Fq) where the source groups are also Discrete-Log-Hard?

open_problem | open | MathDB 页面 | 原文线索

V2-007112 | MATHDB-385901

Does there exist a quantum-resistant bilinear map?

摘录 / EXCERPT: Does there exist a quantum-resistant bilinear map?

open_problem | open | MathDB 页面 | 原文线索

V2-007113 | MATHDB-385902

Can we construct multilinear maps from supersingular isogeny graphs?

摘录 / EXCERPT: Can we construct multilinear maps from supersingular isogeny graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-007114 | MATHDB-385904

Can we get multilinear maps from cohomological constructions?

摘录 / EXCERPT: Can we get multilinear maps from cohomological constructions? In particular, there is a natural multiplication map from $A[\ell]^{2g}$ into $\mu_\ell^{\otimes g}$ essentially given...

open_problem | open | MathDB 页面 | 原文线索

V2-007115 | MATHDB-385905

Can we find a pairing friendly elliptic curve with embedding degree ... ?

摘录 / EXCERPT: Can we find a pairing friendly elliptic curve with embedding degree 13? Destiny estimates suggest they exist, but how do we find them?

open_problem | open | MathDB 页面 | 原文线索

V2-007116 | MATHDB-385906

Is there a way to have direct bootstrapping from k-linear maps to (k+1)-linear maps that do not take advantage of specific properties of the map?

摘录 / EXCERPT: Is there a way to have direct bootstrapping from k-linear maps to (k+1)-linear maps that do not take advantage of specific properties of the map?

open_problem | open | MathDB 页面 | 原文线索

V2-007117 | MATHDB-385907

The intersection product on cycles defines a linear map.

摘录 / EXCERPT: The intersection product on cycles defines a linear map. Unfortunately, the target is ${\mathbb Z}$ where the discrete log problem is easy. Can we define an equivalence relation on...

open_problem | open | MathDB 页面 | 原文线索

V2-007118 | MATHDB-385908

Multilinear map constructions (problem 9)

摘录 / EXCERPT: Let X be an abelian surface over the algebraic closure of a finite field of characteristic different from ℓ. One candidate for a multilinear map might be the cup product ... Wou...

open_problem | open | MathDB 页面 | 原文线索

V2-007119 | MATHDB-385909

Find an algorithm for the approximate-gcd problem for general parameters.

摘录 / EXCERPT: Find an algorithm for the approximate-gcd problem for general parameters. In particular, given a set of integers $\{n_i=qp_i+r_i\}$ where the r_i are small, find q in polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-007120 | MATHDB-385910

Is there a CRT-style result for Approximate GCD?

摘录 / EXCERPT: Is there a CRT-style result for Approximate GCD? In particular, given X as a product of distinct primes, can we find any of the primes if we are allowed a polynomial number of samp...

open_problem | open | MathDB 页面 | 原文线索

V2-007121 | MATHDB-385911

Lower bounds (problem 1)

摘录 / EXCERPT: Let $h \in {\\mathbb{N}}$ be fixed. Is there a sequence $X_k \to \infty$ such that ... as $k \to \infty$? We could also consider variants for multiplicative functions f with values...

open_problem | open | MathDB 页面 | 原文线索

V2-007122 | MATHDB-385912

Lower bounds (problem 2)

摘录 / EXCERPT: Is there a bounded sequence $a(n)$ such that each $a(n)$ is computable in time $O(\log^A n)$ for some fixed $A > 0$ and such that ... for all X and some fixed $\epsilon > 0$? Can w...

open_problem | open | MathDB 页面 | 原文线索

V2-007123 | MATHDB-385913

Lower bounds (problem 3)

摘录 / EXCERPT: Can we obtain lower bounds for ... In particular, can we get a lower bound of the form $XH^{\frac{1}{2}-\epsilon}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007124 | MATHDB-385914

Lower bounds (problem 4)

摘录 / EXCERPT: Can we get a reasonable omega result for ... where $f: {\\mathbb{N}} \to \{\pm1\}$ is totally multiplicative? Anything better than $\log \log \log \log \log X$ would be interesting.

open_problem | open | MathDB 页面 | 原文线索

V2-007125 | MATHDB-385917

M\"obius and Liouville systems (problem 1)

摘录 / EXCERPT: Can a Liouville system be isomorphic to $(\mathbb{T}^2, m_{\mathbb{T}^2}, T)$, where $T:\mathbb{T}^2\to \mathbb{T}^2$ is given by...

open_problem | open | MathDB 页面 | 原文线索

V2-007126 | MATHDB-385918

M\"obius and Liouville systems (problem 2)

摘录 / EXCERPT: Is -1 in the spectrum of a Liouville system? Equivalently, can we show that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007127 | MATHDB-385920

Fourier uniformity (problem 1)

摘录 / EXCERPT: How much can we lower H in the local Fourier uniformity conjecture ...

open_problem | open | MathDB 页面 | 原文线索

V2-007128 | MATHDB-385923

Special correlations (problem 1)

摘录 / EXCERPT: Show that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007129 | MATHDB-385924

Special correlations (problem 2)

摘录 / EXCERPT: Show that ... and that ... for some $\epsilon > 0$. It would even be interesting to show that both sets $\{p: \lambda(p-1) = 1 \}$ and $\{p: \lambda(p-1) = -1 \}$ are infinite. For the...

open_problem | open | MathDB 页面 | 原文线索

V2-007130 | MATHDB-385925

Special correlations (problem 3)

摘录 / EXCERPT: What can we say about correlations with itself of ... with $\delta \to 0$? Here $p^+(n):= \max\{p: p|n\}$. Even upper bounds would be welcome. This kind of result would imply sharp m...

open_problem | open | MathDB 页面 | 原文线索

V2-007131 | MATHDB-385927

Can anything be done for correlations of unbounded functions?

摘录 / EXCERPT: Can anything be done for correlations of unbounded functions? What is the behavior in almost all intervals of ... with g an unbounded multiplicative function such that...

open_problem | open | MathDB 页面 | 原文线索

V2-007132 | MATHDB-385928

Generalizations (problem 1)

摘录 / EXCERPT: We have the following polynomial versions of the Sarnak and Chowla conjectures, where (X, μ, T) is a 0-entropy system, $f \in C(X)$, $x \in X$ and $p(t) \in {\\mathbb{Z}}[t]$...

open_problem | open | MathDB 页面 | 原文线索

V2-007133 | MATHDB-385929

Generalizations (problem 2)

摘录 / EXCERPT: Let $\mathcal{A} = \{a_1, a_2, \dots\}$ be a sequence of integers with $(a_i, a_j) = 1$ for $i \not= j$. Let $\pi(n):= (-1)^{\#\{a_j:a_j|n\}}$. Is it true that ... Can we find a set...

open_problem | open | MathDB 页面 | 原文线索

V2-007134 | MATHDB-385930

Generalizations (problem 3)

摘录 / EXCERPT: Suppose that for all fixed χ mod q and $t \in {\\mathbb{R}}$, $f:{\\mathbb{N}} \to {\\mathbb{C}}$ is multiplicative with ... Does there exists a subsequence $X_k \to \infty$ s...

open_problem | open | MathDB 页面 | 原文线索

V2-007135 | MATHDB-385932

Proof techniques and examples (problem 1)

摘录 / EXCERPT: Is is possible to show that ... with $H > X^\epsilon$ implies ...

open_problem | open | MathDB 页面 | 原文线索

V2-007136 | MATHDB-385935

Computability (problem 1)

摘录 / EXCERPT: Given $f\in \mathbb Z[x_1,\dots, x_n]$, define the following sets. ... ... begin{enumerate} item Can the Lebesgue measure of $B(f)$ be positive? item If so, what is the maximal compl...

open_problem | open | MathDB 页面 | 原文线索

V2-007137 | MATHDB-385936

Complexity (problem 1)

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, and let G be an ordered divisible group. Consider the Hahn field, defined as follows: ... If K and G are computabl...

open_problem | open | MathDB 页面 | 原文线索

V2-007138 | MATHDB-385937

Decidability (problem 1)

摘录 / EXCERPT: Given an elliptic curve E given by $y^2 = x^3 + Ax + B$: begin{enumerate} item Is there are algorithm that decides if $E(\mathbb Z) \setminus\{\infty\}$ is empty? item Is there a...

open_problem | open | MathDB 页面 | 原文线索

V2-007139 | MATHDB-385938

Decidability (problem 2)

摘录 / EXCERPT: begin{enumerate} item Is the first order theory of $\mathbb C(t)$ decidable in signature with 1 and t? item Does it help if $\frac{\partial}{\partial t_i}$ is added to the sign...

open_problem | open | MathDB 页面 | 原文线索

V2-007140 | MATHDB-385939

Decidability (problem 3)

摘录 / EXCERPT: Is the existential theory of $\{\text{complex holomorphic functions over $mathbb C$}\}$ in signature $0,1,+,\cdot, z$ decidable?

open_problem | open | MathDB 页面 | 原文线索

V2-007141 | MATHDB-385941

Decidability (problem 5)

摘录 / EXCERPT: Let p be a fixed prime. Define ... Does $(\mathbb Z, +, \mid_p, 0, 1)$ have a decidable existential theory?

open_problem | open | MathDB 页面 | 原文线索

V2-007142 | MATHDB-385943

Definability (problem 1)

摘录 / EXCERPT: begin{enumerate} item Is $\mathbb C[t_1,\dots, t_n]$ definable in $\mathbb C(t_1,\dots, t_n)$? item Is there a diophantine definition? end{enumerate}

open_problem | open | MathDB 页面 | 原文线索

V2-007143 | MATHDB-385944

Definability (problem 2)

摘录 / EXCERPT: Define a non-trivial valuation on $\mathbb C(t_1,\dots, t_n)$.

open_problem | open | MathDB 页面 | 原文线索

V2-007144 | MATHDB-385952

Problems related to Xiao-Zhu (problem 1)

摘录 / EXCERPT: Consider Xiao-Zhu specialized to Tate classes associated to Hilbert modular forms at an inert prime. In the generic case, i.e. when the Satake parameters are different, then there...

open_problem | open | MathDB 页面 | 原文线索

V2-007145 | MATHDB-385954

Can one upgrade the isomorphism of Hodge structures in the Ichino-Prasanna Theorem to come from “absolute Hodge” or even “motivated” cycles?

摘录 / EXCERPT: Can one upgrade the isomorphism of Hodge structures in the Ichino-Prasanna Theorem to come from “absolute Hodge” or even “motivated” cycles?

open_problem | open | MathDB 页面 | 原文线索

V2-007146 | MATHDB-385955

Can one prove an analogue of Ichino-Prasanna for inner forms of unitary groups?

摘录 / EXCERPT: Can one prove an analogue of Ichino-Prasanna for inner forms of unitary groups?

open_problem | open | MathDB 页面 | 原文线索

V2-007147 | MATHDB-385958

Can one realize base change functoriality from Tate cycles?

摘录 / EXCERPT: Can one realize base change functoriality from Tate cycles?

open_problem | open | MathDB 页面 | 原文线索

V2-007148 | MATHDB-385960

Can one use these methods to realize a ... -adic Jacquet-Langlands correspondence?

摘录 / EXCERPT: Can one use these methods to realize a p-adic Jacquet-Langlands correspondence?

open_problem | open | MathDB 页面 | 原文线索

V2-007149 | MATHDB-385961

Can these methods be used to prove anything about Nekovar-Scholl's plectic conjectures?

摘录 / EXCERPT: Can these methods be used to prove anything about Nekovar-Scholl's plectic conjectures?

open_problem | open | MathDB 页面 | 原文线索

V2-007150 | MATHDB-385964

In characteristic 0, the complex uniformization gives a classification of all Hecke-equivariant automorphic bundles.

摘录 / EXCERPT: In characteristic 0, the complex uniformization gives a classification of all Hecke-equivariant automorphic bundles. Can you classify all Hecke-equivariant automorphic bundles on t...

open_problem | open | MathDB 页面 | 原文线索

V2-007151 | MATHDB-385969

Decidability

摘录 / EXCERPT: Is reflectivity of a maximal arithmetic lattice a decidable problem? If so, can we find an efficient algorithm to solve this decision problem?

open_problem | open | MathDB 页面 | 原文线索

V2-007152 | MATHDB-385970

Vinberg's Algorithm (problem 2)

摘录 / EXCERPT: Suppose Γ is reflective. Given a bound for the volume of the fundamental domain, D_Γ, can we also find a bound on the number of walls in D_Γ?

open_problem | open | MathDB 页面 | 原文线索

V2-007153 | MATHDB-385971

Vinberg's Algorithm (problem 3)

摘录 / EXCERPT: If Γ is non-reflective, then $\operatorname{Aut}(P)$ contains an infinite-order element. How many walls of the fundamental domain to we need to find via Vinberg's algorithm...

open_problem | open | MathDB 页面 | 原文线索

V2-007154 | MATHDB-385972

Vinberg's Algorithm (problem 4)

摘录 / EXCERPT: Suppose Γ is non-reflective. After running Vinberg's algorithm to find part of the Coxeter polytope P, we can look for symmetries in the Coxeter diagram and deduce from th...

open_problem | open | MathDB 页面 | 原文线索

V2-007155 | MATHDB-385973

Classification of maximal reflective arithmetic lattices (problem 1)

摘录 / EXCERPT: Scharlau has recently classified all non-uniform lattices in dimensions $n=3$ and $n=4$ (possibly also $n = 5$?). It is known that such lattices do not exist when $n=20$ or when...

open_problem | open | MathDB 页面 | 原文线索

V2-007156 | MATHDB-385974

Can we classify all non-uniform maximal reflective arithmetic lattices when ... ?

摘录 / EXCERPT: Can we classify all non-uniform maximal reflective arithmetic lattices when $n=6$? How about just those that are congruence?

open_problem | open | MathDB 页面 | 原文线索

V2-007157 | MATHDB-385975

Can we classify non-uniform maximal reflective arithmetic lattices in dimensions ... ?

摘录 / EXCERPT: Can we classify non-uniform maximal reflective arithmetic lattices in dimensions $n \geq 10$? How about just those that are congruence?

open_problem | open | MathDB 页面 | 原文线索

V2-007158 | MATHDB-385976

For ... , there is only one known example of a non-uniform reflective arithmetic lattice.

摘录 / EXCERPT: For $n=19$, there is only one known example of a non-uniform reflective arithmetic lattice. Is it unique, which is the case for the lone example in dimension 21?

open_problem | open | MathDB 页面 | 原文线索

V2-007159 | MATHDB-385977

Create a central repository (similar to the LMFDB) for all known data relating to Coxeter polytopes in low-dimension.

摘录 / EXCERPT: Create a central repository (similar to the LMFDB) for all known data relating to Coxeter polytopes in low-dimension.

open_problem | open | MathDB 页面 | 原文线索

V2-007160 | MATHDB-385978

Quasi-reflective Groups

摘录 / EXCERPT: There are only finitely-many quasi-reflective arithmetic lattices in each dimension, and we have a complete list in dimension $n=3$. Can we classify all of them for $n > 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-007161 | MATHDB-385979

Quasi-reflective and Quasi-arithmetic groups (problem 2)

摘录 / EXCERPT: Do these quasi-reflective groups contain any salient features that might suggest an effective algorithm for detecting infinite-order symmetries in general (and hence an effective h...

open_problem | open | MathDB 页面 | 原文线索

V2-007162 | MATHDB-385980

Quasi-arithmetic Groups

摘录 / EXCERPT: To what extent are quasi-arithmetic groups like arithmetic reflection groups? Are there only finitely many maximal quasi-arithmetic reflective lattices?

open_problem | open | MathDB 页面 | 原文线索

V2-007163 | MATHDB-385981

Generalizations

摘录 / EXCERPT: What is the appropriate n-dimensional generalization of the Apollonian packing?

open_problem | open | MathDB 页面 | 原文线索

V2-007164 | MATHDB-385982

Super-Apollonian Generalizations

摘录 / EXCERPT: For which parameter values... begin{enumerate} item does $\Gamma(\mu,\lambda)$ preserve a quadratic form q? item is $\Gamma(\mu,\lambda)$ discrete? item is $\Gamma(\mu,\lambda)$...

open_problem | open | MathDB 页面 | 原文线索

V2-007165 | MATHDB-385983

Automorphism Loci (problem 1)

摘录 / EXCERPT: Given a subgroup G of ${\rm{PGL}}_{N+1}$, we define $\mathcal{M}_d^N[G]$ to be the Zariski closed subset defined by maps whose automorphism group contains an isomorphic copy of...

open_problem | open | MathDB 页面 | 原文线索

V2-007166 | MATHDB-385984

What is the class determined by ... in the homology?

摘录 / EXCERPT: What is the class determined by $\mathcal{M}_d^N[G]$ in the homology?

open_problem | open | MathDB 页面 | 原文线索

V2-007167 | MATHDB-385985

What are possible automorphism groups for maps at the boundary of ... ?

摘录 / EXCERPT: What are possible automorphism groups for maps at the boundary of $(\mathcal{M}_d^N)^{\rm{ss}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007168 | MATHDB-385986

Automorphism Loci (problem 4)

摘录 / EXCERPT: What is the essential dimension of $\mathcal{M}_d^N[G]$?

open_problem | open | MathDB 页面 | 原文线索

V2-007169 | MATHDB-385987

Automorphism Loci (problem 5)

摘录 / EXCERPT: Relate the singular locus of $\mathcal{M}_d^N$ to the automorphism locus ... which is where the conjugation action fails to be free.

open_problem | open | MathDB 页面 | 原文线索

V2-007170 | MATHDB-385988

Do “rigid” -- analogues of maps that are not flexible Lattès -- PCF maps form a subset of bounded height in ... ?

摘录 / EXCERPT: Do “rigid” -- analogues of maps that are not flexible Lattès -- PCF maps form a subset of bounded height in $\mathcal{M}_d^N(\overline{\Bbb{Q}})$?

open_problem | open | MathDB 页面 | 原文线索

V2-007171 | MATHDB-385989

Is there a countable Zariski dense (Galois invariant) subset of “dynamically interesting” maps in ... (with bounded height)?

摘录 / EXCERPT: Is there a countable Zariski dense (Galois invariant) subset of “dynamically interesting” maps in $\mathcal{M}_d^N(\overline{\Bbb{Q}})$ (with bounded height)?

open_problem | open | MathDB 页面 | 原文线索

V2-007172 | MATHDB-385990

What is the locus in ... of maps with “small” arboreal image?

摘录 / EXCERPT: What is the locus in $\mathcal{M}_d^N$ of maps with “small” arboreal image? More precisely, what can be said about the locus in $\mathcal{M}_d^N\times\Bbb{P}^N$ formed by (f,P) s...

open_problem | open | MathDB 页面 | 原文线索

V2-007173 | MATHDB-385991

Arithmetic Questions (problem 4)

摘录 / EXCERPT: For a map f defined over a field K, let $K^{\rm{pre}}_f$ be the extension of K generated by all preperiodic points of f. What can be said about $[K^{\rm{pre}}_f:K]$? What...

open_problem | open | MathDB 页面 | 原文线索

V2-007174 | MATHDB-385992

Arithmetic Questions (problem 5)

摘录 / EXCERPT: The Shafarevich dimension of $\mathcal{M}_d^N$ is defined as ... What is this dimension? Is it equal to $2d-2$ when $N=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-007175 | MATHDB-385993

Arithmetic Questions (problem 6)

摘录 / EXCERPT: Generalizations of dynamical local/global PCF (post-critically finite) $\Leftrightarrow$ PCB (post-critically bounded)?

open_problem | open | MathDB 页面 | 原文线索

V2-007176 | MATHDB-385994

Arithmetic Questions (problem 7)

摘录 / EXCERPT: Find an appropriate critical/canonical height on $\mathcal{M}_d^N$.

open_problem | open | MathDB 页面 | 原文线索

V2-007177 | MATHDB-385995

Find/prove dynamical analogues of the Szpiro conjecture.

摘录 / EXCERPT: Find/prove dynamical analogues of the Szpiro conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-007178 | MATHDB-385996

Arithmetic Questions (problem 9)

摘录 / EXCERPT: For $\alpha\in\bar{\Bbb{Q}}$, investigate $\max\{\hat{\lambda}_{\nu,f}(\sigma\alpha)\mid \sigma\in{\rm{Gal}}(\bar{\Bbb{Q}}/\Bbb{Q})\}$ ($\hat{\lambda}_{\nu,f}$ the local height ass...

open_problem | open | MathDB 页面 | 原文线索

V2-007179 | MATHDB-385997

Arithmetic Questions (problem 10)

摘录 / EXCERPT: Lift ${\rm{End}}^N_d(\overline{\Bbb{F}_p})$ to PCF elements of ${\rm{End}}^N_d(\overline{\Bbb{Q}_p})$.

open_problem | open | MathDB 页面 | 原文线索

V2-007180 | MATHDB-385998

Arithmetic Questions (problem 11)

摘录 / EXCERPT: Is there a bound for $[K_{\rm{def}}:K_{\rm{moduli}}]$ depending on N but not on d?

open_problem | open | MathDB 页面 | 原文线索

V2-007181 | MATHDB-385999

Arithmetic Questions (problem 12)

摘录 / EXCERPT: Study dynamical systems $\Bbb{P}^N\rightarrow\Bbb{P}^N$ whose field of moduli is contained in $\Bbb{R}$. When can they be defined over the reals?

open_problem | open | MathDB 页面 | 原文线索

V2-007182 | MATHDB-386000

In the definition of PCF maps in higher dimensions, is the “all the way down” condition needed?

摘录 / EXCERPT: In the definition of PCF maps in higher dimensions, is the “all the way down” condition needed?

open_problem | open | MathDB 页面 | 原文线索

V2-007183 | MATHDB-386001

PCF Maps in Higher Dimensions (problem 2)

摘录 / EXCERPT: Is the set of PCF maps in $\mathcal{M}_d^N$ Zariski dense for $N\geq 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-007184 | MATHDB-386002

PCF Maps in Higher Dimensions (problem 3)

摘录 / EXCERPT: For which $(N+1)\times (N+1)$ matrices A the map $\mathbf{X}\in\Bbb{P}^N\mapsto A\mathbf{X}^d\in\Bbb{P}^N$ is PCF for some $d\geq 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-007185 | MATHDB-386003

Geometry and Topology of Moduli Spaces (problem 1)

摘录 / EXCERPT: Is the moduli space $\mathcal{M}_d^N:={\rm End}}^N_d/{\rm PGL}}_{N+1}$ rational?

open_problem | open | MathDB 页面 | 原文线索

V2-007186 | MATHDB-386005

Geometry and Topology of Moduli Spaces (problem 3)

摘录 / EXCERPT: Does the gonality of $\mathcal{M}_d^N[n]$ tend to infinity as $n\to\infty$? How about the moduli spaces $\mathcal{M}_d^N[\mathcal{P}]$ where $\mathcal{P}$ is a portrait?

open_problem | open | MathDB 页面 | 原文线索

V2-007187 | MATHDB-386007

Compute Betti numbers and the fundamental group of ... .

摘录 / EXCERPT: Compute Betti numbers and the fundamental group of $\mathcal{M}_d^N(\Bbb{C})$.

open_problem | open | MathDB 页面 | 原文线索

V2-007188 | MATHDB-386008

Understand “dynamically significant” “fibrations” ... (singular fibers allowed), e.g.

摘录 / EXCERPT: Understand “dynamically significant” “fibrations” $\mathcal{M}_d^N\rightarrow\Bbb{A}^{m}$ (singular fibers allowed), e.g. those defined by multipliers.

open_problem | open | MathDB 页面 | 原文线索

V2-007189 | MATHDB-386009

Geometry and Topology of Moduli Spaces (problem 7)

摘录 / EXCERPT: Obtain an dynamical understanding of $(\mathcal{M}_d^N)^{\rm ss}}-\mathcal{M}_d^N$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-007190 | MATHDB-386010

What is the equivariant cohomology of the semi-stable locus ... ?

摘录 / EXCERPT: What is the equivariant cohomology of the semi-stable locus $(\mathcal{M}_d^N)^{\rm ss}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007191 | MATHDB-386011

Geometry and Topology of Moduli Spaces (problem 9)

摘录 / EXCERPT: What are some non-GIT compactifications of $\mathcal{M}_d^N$?

open_problem | open | MathDB 页面 | 原文线索

V2-007192 | MATHDB-386012

Geometry and Topology of Moduli Spaces (problem 10)

摘录 / EXCERPT: Understand different GIT quotients for $\mathcal{M}_d^N[n]$.

open_problem | open | MathDB 页面 | 原文线索

V2-007193 | MATHDB-386013

Geometry and Topology of Moduli Spaces (problem 11)

摘录 / EXCERPT: Are distinguished curves in $\mathcal{M}_d^N$ or $\mathcal{M}_d^N[n]$ (e.g. dynatomic curves) irreducible or smooth?

open_problem | open | MathDB 页面 | 原文线索

V2-007194 | MATHDB-386014

Generalization of Complex Dynamical Results to Higher Dimensions (problem 1)

摘录 / EXCERPT: What is the bifurcation measure $\mu_{\rm bif}}$ in $\mathcal{M}_d^N$ (over a field $\Bbb{C}_\nu$)?

open_problem | open | MathDB 页面 | 原文线索

V2-007195 | MATHDB-386016

Generalization of Complex Dynamical Results to Higher Dimensions (problem 3)

摘录 / EXCERPT: Self-similarity of p-adic bifurcation loci in 1-parameter families when $p<\text{ the degree}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007196 | MATHDB-386018

Shifted convolution sum via various delta methods

摘录 / EXCERPT: For a holomorphic or a Hecke--Maass cusp form f with the nth Fourier coefficient $\lambda_f(n)$, obtain cancellations in the summation ... via begin{enumerate} item Duke-Friedlande...

open_problem | open | MathDB 页面 | 原文线索

V2-007197 | MATHDB-386019

Level aspect subconvexity for GL(2)

摘录 / EXCERPT: Use a delta method to obtain level aspect subconvexity for GL(2) L-functions.

open_problem | open | MathDB 页面 | 原文线索

V2-007198 | MATHDB-386022

Subconvexity bound for trace function twists

摘录 / EXCERPT: Subconvexity bounds for $\rm GL(1)$ twists of a fixed $\rm GL(2)\times GL(2)$ L-function has been proved using a delta symbol by C. Raju [cite] in the archimedean aspect, and by...

open_problem | open | MathDB 页面 | 原文线索

V2-007199 | MATHDB-386023

Subconvexity for $\rm GL(3)\times GL(1)$ via averaging over a family

摘录 / EXCERPT: Can we obtain a subconvexity bound by averaging over a family, e.g. by bounding ...

open_problem | open | MathDB 页面 | 原文线索

V2-007200 | MATHDB-386024

Application to Quantum Unique Ergodicity

摘录 / EXCERPT: Obtaining a bound of the form ... for $|h|\ll 1$ and some $\delta>0$ would imply the result of Holowinsky and Soundararajan (conditionally on non-negativity of the central values...

open_problem | open | MathDB 页面 | 原文线索

V2-007201 | MATHDB-386025

$GL(3)$ Kuznetsov trace formula in the level aspect

摘录 / EXCERPT: Develop a Kuznetsov trace formula for automorphic forms for the group...

open_problem | open | MathDB 页面 | 原文线索

V2-007202 | MATHDB-386026

Subconvexity bound for Rankin-Selberg L-functions

摘录 / EXCERPT: Obtain a subconvexity bound for a $\rm GL_n\times GL_m$ Rankin--Selberg L-function when n and m are not coprime. The simplest first cases would be obtaining a subconvexity bo...

open_problem | open | MathDB 页面 | 原文线索

V2-007203 | MATHDB-386028

Asymptotic for second moment of $\rm GL_2\times GL_2$ in level aspect

摘录 / EXCERPT: Let $\mathcal{B}_k(p)$ be a basis for holomorphic or Hecke-Maass forms for $\Gamma_0(p)\subset SL_2(\mathbb{Z})$ of weight k. Obtain an asymptotic with a power saving error term...

open_problem | open | MathDB 页面 | 原文线索

V2-007204 | MATHDB-386033

Strong hybrid bounds for $L(1/2, f\times \chi)$

摘录 / EXCERPT: Prove strong hybrid bounds for other ranges of p and q.

open_problem | open | MathDB 页面 | 原文线索

V2-007205 | MATHDB-386034

Subconvexity bound $\rm GL_2$ in the `conductor dropping' case

摘录 / EXCERPT: Let f be a holomorphic or Hecke-Maass cusp form for $\rm GL_2$. Use a δ-method to prove the following subconvexity bounds for the following L-functions: begin{enumerate}...

open_problem | open | MathDB 页面 | 原文线索

V2-007206 | MATHDB-386037

Hardy-type theorem for $\rm GL_3$ L-functions

摘录 / EXCERPT: Use a δ-method to prove that there are infinitely many critical zeros for a $\rm GL_3(\mathbb{Z})$ L-functions.

open_problem | open | MathDB 页面 | 原文线索

V2-007207 | MATHDB-386049

Central derivative of Rankin--Selberg L-functions for weight-2 modular forms and congruent forms

摘录 / EXCERPT: Let f be a modular form of weight 2 and let χ be a character of the class group of an imaginary quadratic number field K. Formulate a direct connection ... for a modular f...

open_problem | open | MathDB 页面 | 原文线索

V2-007208 | MATHDB-386050

Non-vanishing of the central L-derivative for $\mathrm{U}(n) \times \mathrm{U}(n-1)$

摘录 / EXCERPT: Given an automorphic representation π of $\mathrm{U}(n)$, show that there exists an automorphic representation σ of $\mathrm{U}(n-1)$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007209 | MATHDB-386051

p-indivisibility of $L(\chi, 0)$

摘录 / EXCERPT: Fix a prime p. Let q be a prime and let χ be an odd character of $({\\mathbb{Z}}/q{\\mathbb{Z}})^\times$. How often is $L(\chi, 0)$ indivisible by p?

open_problem | open | MathDB 页面 | 原文线索

V2-007210 | MATHDB-386055

Central values and central derivatives (problem 2)

摘录 / EXCERPT: Is there a p-adic interpolation of $L'(\pi \otimes \sigma, \frac{1}{2})$?

open_problem | open | MathDB 页面 | 原文线索

V2-007211 | MATHDB-386056

Central values and central derivatives (problem 3)

摘录 / EXCERPT: Given an automorphic representation π of $\mathrm{U}(n)$, show that there exists an automorphic representation σ (with some restriction) of $\mathrm{U}(n-1)$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007212 | MATHDB-386057

Meta problem about values & derivatives

摘录 / EXCERPT: Given a subconvexity result ... 1) find an analogue of the result where the size of $L(\pi, \frac{1}{2})$ is replaced by some "arithmetic" analogue of size, or 2) find an analogue of...

open_problem | open | MathDB 页面 | 原文线索

V2-007213 | MATHDB-386058

Non-vanishing of products of twisted L-functions

摘录 / EXCERPT: Let f be a newform (corresponding to elliptic curve?) and let χ be a quadratic character. Does there exist a quadratic character ψ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007214 | MATHDB-386059

Construction of abelian surfaces for paramodular newforms

摘录 / EXCERPT: Given a paramodular newform F of weight 2 with rational coefficients, find a construction of the abelian surface to which it corresponds.

open_problem | open | MathDB 页面 | 原文线索

V2-007215 | MATHDB-386062

Arithmetic applications of horizontal equidistribution results

摘录 / EXCERPT: Let K be an imaginary quadratic number fields. There are equidistribution results for CM points with square-free order of conductor C as $C \rightarrow \infty$. Is there an int...

open_problem | open | MathDB 页面 | 原文线索

V2-007216 | MATHDB-386063

Computing Fourier coefficients from L-values

摘录 / EXCERPT: Let d be a positive integer and let F be a Siegel modular form with Fourier coefficients $a(F)$. In many cases, we know the (generalized) B"{o}cherer conjecture: ... Knowing this...

open_problem | open | MathDB 页面 | 原文线索

V2-007217 | MATHDB-386069

There were several questions proposed around this, including What should the predicted values of ... be?

摘录 / EXCERPT: There were several questions proposed around this, including What should the predicted values of a,b,c be? What is the "shape" of the set U that one should take? In the usual M...

open_problem | open | MathDB 页面 | 原文线索

V2-007218 | MATHDB-386074

Initial Problem Session (problem 6)

摘录 / EXCERPT: Skorobogatov constructed a bielliptic surface $X/\mathbb Q$ such that $X(\mathbb A_{\mathbb Q})^{\operatorname{Br}}\neq\emptyset$, but $X(\mathbb Q)=\emptyset$. It is known that...

open_problem | open | MathDB 页面 | 原文线索

V2-007219 | MATHDB-386081

Initial Problem Session (problem 13)

摘录 / EXCERPT: Compute $\operatorname{irr}(X/F)$ for a del Pezzo surface X over a general field F.

open_problem | open | MathDB 页面 | 原文线索

V2-007220 | MATHDB-386084

Initial Problem Session (problem 16)

摘录 / EXCERPT: Let $X_d\subset\mathbb P^3_k$ (k a number field) be a smooth surface of degree d. Are the degree $\le d-3$ points not Zariski dense? What is $\min\delta(X_d/k)$?

open_problem | open | MathDB 页面 | 原文线索

V2-007221 | MATHDB-386087

Initial Problem Session (problem 19)

摘录 / EXCERPT: Classify surfaces X/k of general type with $2\in\delta(X/k)$, possibly using geometric and/or dynamical constructions.

open_problem | open | MathDB 页面 | 原文线索

V2-007222 | MATHDB-386090

This isomorphism can be understood in terms of the respective moduli problems when evaluated on perfectoid rings.

摘录 / EXCERPT: This isomorphism can be understood in terms of the respective moduli problems when evaluated on perfectoid rings. Is there a geometric description of the two tower isomorphism for...

open_problem | open | MathDB 页面 | 原文线索

V2-007223 | MATHDB-386091

Is there a way to enhance the two tower isomorphism in terms of analytic stacks?

摘录 / EXCERPT: Is there a way to enhance the two tower isomorphism in terms of analytic stacks?

open_problem | open | MathDB 页面 | 原文线索

V2-007224 | MATHDB-386092

Power Operations (problem 1)

摘录 / EXCERPT: Can one use the $\text{GL}_n(\mathbf{Z}_p)$ action on $\mathcal{H}^{n-1}$ to understand power operations of $L_{K(0)} E_n$?

open_problem | open | MathDB 页面 | 原文线索

V2-007225 | MATHDB-386094

Rational Homotopy of Invariants (problem 1)

摘录 / EXCERPT: What is $\pi_*(E^{h\Gamma}) \otimes \mathbf{Q}$ for closed subgroups $\Gamma \subset \mathbf{G}_n$?

open_problem | open | MathDB 页面 | 原文线索

V2-007226 | MATHDB-386095

Rational Homotopy of Invariants (problem 2)

摘录 / EXCERPT: What is the connection between a choice of maximal torus of $\mathbf{G}_n$ of dimension n and there being n exterior generators? Can this be related to the identification (say...

open_problem | open | MathDB 页面 | 原文线索

V2-007227 | MATHDB-386096

Hopkins' Chromatic Splitting Conjecture (problem 1)

摘录 / EXCERPT: What are the iterated localizations ... where $t_0 < t_1 < \dots < t_n$?

open_problem | open | MathDB 页面 | 原文线索

V2-007228 | MATHDB-386097

Is ... a Huber ring? (Probably not.) Is it a solid ring?

摘录 / EXCERPT: Is $\pi_0(L_{K(n-1)} L_{K(n)} E_n)$ a Huber ring? (Probably not.) Is it a solid ring?

open_problem | open | MathDB 页面 | 原文线索

V2-007229 | MATHDB-386098

Hopkins' Chromatic Splitting Conjecture (problem 3)

摘录 / EXCERPT: Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007230 | MATHDB-386099

Hopkins' Chromatic Splitting Conjecture (problem 4)

摘录 / EXCERPT: In char p, ...

open_problem | open | MathDB 页面 | 原文线索

V2-007231 | MATHDB-386100

Approach to Fargues-Fontaine Curve using THH (problem 1)

摘录 / EXCERPT: There is an identification $\text{THH}(\mathbf{Z}_p) \cong t_{\ge 0}(\text{im}(J))^{tC_p}$. Does this isomorphism yield an integral version of the Fargues-Fontaine curve?

open_problem | open | MathDB 页面 | 原文线索

V2-007232 | MATHDB-386101

Is ... surjective? This is the last remaining case not covered by.

摘录 / EXCERPT: Is $\text{Pic}_{2,2} \rightarrow \text{Pic}^\text{alg}_{2,2}$ surjective? This is the last remaining case not covered by.

open_problem | open | MathDB 页面 | 原文线索

V2-007233 | MATHDB-386103

Mapping class group orbits in higher rank

摘录 / EXCERPT: What are some sufficient conditions for a point to have a Zariski dense orbit under the mapping class group on $M_\gamma(\Sigma_{g,n}, {\\textrm{SL}}_3)$? Here γ is a fixed...

open_problem | open | MathDB 页面 | 原文线索

V2-007234 | MATHDB-386104

Classification of MCG-finite localy systems of higher rank

摘录 / EXCERPT: Can we classify the finite orbts under the mapping class group in $M(\Sigma_{0,n}, \mathrm{SL}_3)$, in particular the case $n = 4$?

open_problem | open | MathDB 页面 | 原文线索

V2-007235 | MATHDB-386107

Geometric local systems on $\mathcal{M}_g$ and rigidity

摘录 / EXCERPT: Consider a local system $\mathbb{L}$ on $\mathcal{M}_g$ (or a space with some level structure) or equivalently a representation of $\mathrm{Mod}(\Sigma_g)$ (or correspondingly a re...

open_problem | open | MathDB 页面 | 原文线索

V2-007236 | MATHDB-386108

Higher Prym period maps

摘录 / EXCERPT: Is the higher Prym map unique in general?

open_problem | open | MathDB 页面 | 原文线索

V2-007237 | MATHDB-386116

Does the theorem still hold if we drop the adjective “simple” from the closed curve?

摘录 / EXCERPT: Does the theorem still hold if we drop the adjective “simple” from the closed curve?

open_problem | open | MathDB 页面 | 原文线索

V2-007238 | MATHDB-386117

Monodromy of surface bundles (problem 3)

摘录 / EXCERPT: Let X / K be a smooth curve over a field K which is one of the following: (1) a finite field $\mathbb{F}_q$ (2) a number field K (3) the function field of a curve over...

open_problem | open | MathDB 页面 | 原文线索

V2-007239 | MATHDB-386118

Conjecture of Katzarkov-Pantov-Simpson

摘录 / EXCERPT: Does the monodromy representation of a nonisotrivial family $\mathcal{C} \to B$ of curves of genus $g \ge 2$ (perhaps one arising from a sufficiently ample Lefschetz pencil on a su...

open_problem | open | MathDB 页面 | 原文线索

V2-007240 | MATHDB-386119

New examples of finite orbits of homomorphism $\pi_1(\Sigma_{g,n}) \to \mathrm{Mod}(\Sigma_{h})$

摘录 / EXCERPT: What are some examples of finite orbits under the mapping class group $\mathrm{Mod}(\Sigma_{g,n})$ of homomorphisms (considered up to conjugation)...

open_problem | open | MathDB 页面 | 原文线索

V2-007241 | MATHDB-386120

Isotrivial isogeny factors in families of Jacobians

摘录 / EXCERPT: Let $\mathcal{C} \to B$ be a nonisotrivial family of proper curves of genus g. What is the maximum dimension of an isotrivial isogeny factor in the Jacobian fibration of...

open_problem | open | MathDB 页面 | 原文线索

V2-007242 | MATHDB-386121

Counterexamples to the Putman-Wieland conjecture

摘录 / EXCERPT: Can we classify all counterexamples to the Putman-Wieland conjecture in genus $g = 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-007243 | MATHDB-386122

Hyperbolicity of surface bundles

摘录 / EXCERPT: Are there examples in higher dimension given as surface bundles over a surface? Can these be compact examples? In particular, for $g,h \ge 2$ does there exist an Σ_g-bundle...

open_problem | open | MathDB 页面 | 原文线索

V2-007244 | MATHDB-386126

Uniform stability

摘录 / EXCERPT: Fix a finite group G and an irreducible representation $\rho: G \to \operatorname{GL}_r(\mathbb C)$. One might be interested in understanding homological stability needed for mom...

open_problem | open | MathDB 页面 | 原文线索

V2-007245 | MATHDB-386127

Monodromy images

摘录 / EXCERPT: What is the image of the Burau representation specialized at other roots of unity? If one specializes to the case of a cyclic group, one may want to know exactly what the image of...

open_problem | open | MathDB 页面 | 原文线索

V2-007246 | MATHDB-386128

Moments of derivatives

摘录 / EXCERPT: What do the methods for computing moments of L function give if one tries to compute moments of derivatives. Even in the random matrix theory setting computing the distribution of...

open_problem | open | MathDB 页面 | 原文线索

V2-007247 | MATHDB-386129

Higher degree cyclic covers

摘录 / EXCERPT: If one starts from the number theoretic side, consider χ cyclic of order more than 2. It is natural to think about summing $L(1/2,\chi_d)^r L(1/2, \overline{\chi}_d)^s$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-007248 | MATHDB-386130

Hyperelliptic Torelli

摘录 / EXCERPT: Can you give a quadratic presentation of the Maltsev completion of hyperelliptic Torelli using the computation from Bergstrom-Diaconu-Petersen-Westerland?

open_problem | open | MathDB 页面 | 原文线索

V2-007249 | MATHDB-386131

Negative moments

摘录 / EXCERPT: Can one prove a version of the pair correlation conjecture using topological methods? Will sawin thinks this should be possible in some range. Can one use homological stability met...

open_problem | open | MathDB 页面 | 原文线索

V2-007250 | MATHDB-386132

Error terms

摘录 / EXCERPT: What kind of error terms do people expect for the moments? Diaconu suggests that there are infinitely many higher order terms. Could there be interesting secondary stable homology...

open_problem | open | MathDB 页面 | 原文线索

V2-007251 | MATHDB-386133

The moments conjecture

摘录 / EXCERPT: Assume the most optimistic bounds on unstable cohomology. can we get the full ratios conjecture without a limitation on the size of q. I.e. can we get it for fixed q instead of...

open_problem | open | MathDB 页面 | 原文线索

V2-007252 | MATHDB-386134

A conjecture about shifted L functions

摘录 / EXCERPT: Let $a = \{a_1, \ldots, a_k\}$. Define ... Then ... Observe ... In order to recover the certain moments you're interested in one needs to take $x = d^{|a|}$. In order to get the correct...

open_problem | open | MathDB 页面 | 原文线索

V2-007253 | MATHDB-386135

Twisted averages of L functions

摘录 / EXCERPT: Consider $\sum_d L(1/2, \chi_d) \cdot \phi(d)$ as d varies over polynomials over $\mathbb F_q$ generating quadratic fields and ϕ is some interesting arithmetic function. For...

open_problem | open | MathDB 页面 | 原文线索

V2-007254 | MATHDB-386136

Vanishing of L functions

摘录 / EXCERPT: As in the question by Jordan Ellenberg on twisted averages of L functions, take ϕ to be a quadratic character and try to compute $\sum_d L(1/2, \chi_d) \cdot \phi(d)$. Is th...

open_problem | open | MathDB 页面 | 原文线索

V2-007255 | MATHDB-386137

Elliptic Curves

摘录 / EXCERPT: Take E over $\mathbb F_q(t)$ containing nth roots of unity. one can study the analogous question. what is...

open_problem | open | MathDB 页面 | 原文线索

V2-007256 | MATHDB-386139

Epsilon factors

摘录 / EXCERPT: Fix a finite group G and an irreducible, nontrivial complex representation ρ. study ... where the sum is over G-extensions k over $\mathbb F_q(t)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-007257 | MATHDB-386140

Representation stability

摘录 / EXCERPT: Consider the ordered Hurwitz space where one marks branch points. is there representation stability? There is some work in progress by Ellenberg-Shusterman and Himes-Miller-Wilson-...

open_problem | open | MathDB 页面 | 原文线索

V2-007258 | MATHDB-386141

Rack Hurwitz spaces

摘录 / EXCERPT: Is there a common generalization of Malle's conjecture and the Davis-Schlank conjecture that gives asymptotics for c extensions for a rack c? The relevant homological stability...

open_problem | open | MathDB 页面 | 原文线索

V2-007259 | MATHDB-386142

Cohomology coming from quantum shuffle algebras

摘录 / EXCERPT: Note that racks give braided vector spaces, which give quantum shuffle algebras whose cohomology agrees with that of the Hurwitz space associated to c, Hur^c_n....

open_problem | open | MathDB 页面 | 原文线索

V2-007260 | MATHDB-386143

Finding examples of secondary stability

摘录 / EXCERPT: Can one find any example of secondary stable homology relating to some arithmetic statistics question? For example, can one understand the secondary stable homology in ordered conf...

open_problem | open | MathDB 页面 | 原文线索

V2-007261 | MATHDB-386144

Predicting secondary stability

摘录 / EXCERPT: Can one use the secondary term for counting cubic extensions to make some guess for the secondary stable homology of Hurwitz spaces parameterizing (simply branched) trigonal curves...

open_problem | open | MathDB 页面 | 原文线索

V2-007262 | MATHDB-386146

Unstable cohomology

摘录 / EXCERPT: Let c be a union of conjugacy classes in a group G. What can we say about the Frobenius action on the cohomology of Hur^c_n? What can we say about top degree c...

open_problem | open | MathDB 页面 | 原文线索

V2-007263 | MATHDB-386147

distribution of discriminants

摘录 / EXCERPT: This is a question suggested by Jordan Ellenberg at Arizona Winter School several years ago. Consider all monic polynomials over $\mathbb F_q$ all of fixed degree d. What is the...

open_problem | open | MathDB 页面 | 原文线索

V2-007264 | MATHDB-386289

Miscellaneous (problem 1)

摘录 / EXCERPT: Let $a_1, a_2, a_3, \ldots$ be a sequence of real numbers (a signal) transmitted in the presence of additive noise such that...

open_problem | open | MathDB 页面 | 原文线索

V2-007265 | MATHDB-386290

Miscellaneous (problem 2)

摘录 / EXCERPT: Let $X,Y \in \{0,1\}$ have joint distribution ... then $I(X;Y) = 1 - h(\alpha)$ for $0 \leq \alpha \leq 1/2$ where...

open_problem | open | MathDB 页面 | 原文线索

V2-007266 | MATHDB-386291

Miscellaneous (problem 3)

摘录 / EXCERPT: Given random variables $X, Y \in \mathbb{R}$ with $I(X; Y) \leq \alpha$ and $I(X+Y; X-Y) \leq \beta$. The question is, how close are X and Y to Gaussian? Meaning we would like...

open_problem | open | MathDB 页面 | 原文线索

V2-007267 | MATHDB-386293

Miscellaneous (problem 5)

摘录 / EXCERPT: For X and Y i.i.d. log-concave random variables, what is ... We expect that the maximum is achieved at $\lambda = 1/2$. It is known that this is not true without log-cavity. The...

open_problem | open | MathDB 页面 | 原文线索

V2-007268 | MATHDB-386709

Odd primary actions on Lubin--Tate theories

摘录 / EXCERPT: For an odd prime p, can one produce an understandable model for the C_{p^n}-action on a $E_{(p-1)mp^{n-1}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007269 | MATHDB-386727

Calculation of $\mathrm{TP}(\mathbb{Z})$

摘录 / EXCERPT: Fix a prime p. Let $j = \tau_{\geq 0} L_{K(1)} \mathbb{S}$. Then, $\mathrm{TP}(\mathbb{Z}_p) \simeq j^{tS^1}$, for the trivial S^1-action on j.

open_problem | open | MathDB 页面 | 原文线索

V2-007270 | MATHDB-387069

Miscellaneous (problem 6)

摘录 / EXCERPT: begin{enumerate}[label=alph)] item Relate $tr(T(m,n))$ to rings of quasi-invariants and to A. Wilson's conjecture on $\nabla_{p_1}^n$. item Describe $H^*(\text{Hilb}(x^{nd}=y^n))$...

open_problem | open | MathDB 页面 | 原文线索

V2-007271 | MATHDB-389406

Are there infinitely many Fibonacci primes

摘录 / EXCERPT: Are there infinitely many Fibonacci primes?

open_problem | open | MathDB 页面 | 原文线索

V2-007272 | MATHDB-390464

Chung-Graham-Spiro Fibonacci-walk gap-spectrum conjecture

摘录 / EXCERPT: Let $F_1=F_2=1$ and $F_{t+2}=F_{t+1}+F_t$ for $t\geq 1$. For each $n\geq 2$, write uniquely $n=aF_t+bF_{t-1}$ with $t\geq 2$ and $1\leq a\leq b\leq F_t$. Let $D=\{d_1<d_2<\cdots\}$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007273 | MATHDB-390465

D’Arcais-polynomial Hurwitz conjecture

摘录 / EXCERPT: Let $\sigma(k):=\sum_{d\mid k}d$, and define the D'Arcais polynomials $P_n^\sigma(X)$ by...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007274 | MATHDB-390596

Moore’s conjecture

摘录 / EXCERPT: For every simply connected finite CW complex X, if X is rationally hyperbolic, then there exists a finite set of primes P_X such that, for every prime $p\notin P_X$ and every...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007275 | MATHDB-390601

Spectrum problem for the refined Diophantine exponent

摘录 / EXCERPT: For a finite alphabet $\mathcal A$, determine the spectrum $\operatorname{Spec}_{\mathrm{rD}}(\mathcal A)=\{\operatorname{rD}(w):w\in\mathcal A^{\mathbb N}\}$ of the refined Diopha...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007276 | MATHDB-390641

Jouart conjecture

摘录 / EXCERPT: The supplied source describes the Jouart conjecture as asserting that the indices m for which $P(m)=3m+i(m)$ is prime exhibit a regular exclusion structure characterized by two f...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007277 | MATHDB-391640

Twin-Prime Conjectures

摘录 / EXCERPT: $\#\{p\in\mathbb{N}: p\text{ and }p+2\text{ are prime}\}=\infty.$

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007278 | MATHDB-391799

Conjectured formula for OEIS A091468

摘录 / EXCERPT: Let $a(n)$ denote the nth term of OEIS A091468. The conjecture is that, for every $n\ge 0$, $a(n)=3\,\mathrm{A003239}(n)-2$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007279 | MATHDB-391903

Chowla conjecture

摘录 / EXCERPT: Let $\lambda(n)=(-1)^{\Omega(n)}$ be the Liouville function. For every pair of distinct fixed nonnegative integers h_1,h_2, one has...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007280 | MATHDB-391913

Hermite’s periodicity problem for cubic irrationals; Karpenkov’s Problem 4

摘录 / EXCERPT: Determine whether Karpenkov's analytic extension of the $\sin^2$-algorithm is eventually periodic for every cubic irrationality of signature $(1,1)$, that is, for every real cubic...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-007281 | MATHDB-391922

Dyadic denominator law for the phase constants of the Jacobi zeros

摘录 / EXCERPT: Let $\kappa_r(A,B)\in\mathbb{Q}[A,B]$ be the r-th additive phase constant in the asymptotic phase equation for the zeros of Jacobi polynomials, with $A=\alpha^2$ and $B=\beta^2$....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007282 | MATHDB-391925

Conjecture on the asymptotic behavior of the rectangle partition function p(m,n)

摘录 / EXCERPT: Let $p(m,n)$ be the number of partitions of an $m\times n$ rectangle into integer-sided rectangular blocks, where partitions are identified when they consist of the same multiset o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007283 | MATHDB-391931

Eaton-Moretó conjecture

摘录 / EXCERPT: For every prime p, every finite group G, every p-block B of G, and every defect group D of B, one has $\operatorname{mh}(B)\leq \operatorname{mh}(D)$, where...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007284 | MATHDB-391988

Artin’s primitive-root conjecture for quadratic number fields

摘录 / EXCERPT: Let K be a quadratic number field and let $\alpha\in K^\times$ be neither a root of unity nor a square in K. Then the set...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007285 | MATHDB-392090

Maximum-load problem for linear hashing

摘录 / EXCERPT: For integers n and u with $u=n^{1+o(1)}$, and a prime $p>u$, define...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-007286 | MATHDB-392095

Riemann Hypothesis via Weil’s positivity criterion

摘录 / EXCERPT: Determine whether $\lambda^*(L)\ge 0$ for every $L>0$, where $\lambda^*(L)=\inf\left\{Q(f)/\lVert f\rVert_2^2:0\ne f,\ \operatorname{supp}f\subseteq[-L,L]\right\}$ and Q is Weil’...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-007287 | MATHDB-392109

Uniform Littlewood’s conjecture

摘录 / EXCERPT: For every pair $(x,y)\in\mathbb{R}^{2}$, where $\langle t\rangle$ denotes the distance from t to the nearest integer, one has...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007288 | MATHDB-392120

Wu-Zhan Problem 2 on generalized Riemann-type graph dimensions

摘录 / EXCERPT: Let g be a nonconstant real-valued trigonometric polynomial, and let $S(a;q)$ denote the associated square-class chirp functional at the rational point a/q, as defined in the W...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007289 | MATHDB-392265

Cusick’s conjecture

摘录 / EXCERPT: Let $s_2(n)$ denote the number of ones in the binary expansion of n, and for each integer $t\ge 1$ define c_t to be the natural density...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007290 | MATHDB-392275

Wang-Sun conjecture on quadratic representations of practical numbers

摘录 / EXCERPT: A positive integer m is called practical if every positive integer $r\le m$ can be expressed as a sum of distinct positive divisors of m. The Wang-Sun conjecture asserts that,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007291 | MATHDB-392284

Coates-Sujatha Conjectures A and B

摘录 / EXCERPT: Let K be a number field, let p be a prime, and let $R(E/L)$ denote the fine Selmer group of an elliptic curve E/K over an algebraic extension L/K. Write K^{cyc}...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007292 | MATHDB-392305

Refined Gan-Gross-Prasad conjecture for Fourier-Jacobi periods of degree-2 Siegel cusp forms

摘录 / EXCERPT: Let F be a degree-2 Siegel cuspidal Hecke eigenform, with associated automorphic representation π_F of $\mathrm{GSp}_4(\mathbb A)$, and let g be an associated half-integr...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007293 | MATHDB-392476

Pólya’s conjecture and related sign criteria

摘录 / EXCERPT: Let $\lambda(n)=(-1)^{\Omega(n)}$ be the Liouville function and define, for sufficiently large x, $f(x):=-\sum_{n\le x}\frac{\lambda(n)\log n}{\sqrt n}\log\frac{x}{n}$. Is...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007294 | MATHDB-392478

Ramassamy’s period conjecture for Euler up/down numbers

摘录 / EXCERPT: Let E_n be defined by $\sum_{n\ge 0}E_nz^n/n!=\sec z+\tan z$. For $q\ge 1$, define the minimal eventual period and preperiod by...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007295 | MATHDB-392734

Uniform distribution of multidimensional digital-sum sequences

摘录 / EXCERPT: For coprime bases $q_1,\ldots,q_d$ and irrational numbers $\alpha_1,\ldots,\alpha_d$, the sequence $\bigl(s_{q_1}(n)\alpha_1,\ldots,s_{q_d}(n)\alpha_d\bigr)_{n\ge 0}$ is uniformly...

open_problem | open | MathDB 页面 | 原文线索

V2-007296 | MATHDB-392755

Finite-cokernel twisting theorem for étale cohomology

摘录 / EXCERPT: Let K be a number field, let p be a prime, and let S be a finite set of places. Let K_S be the maximal extension of K unramified outside S, and let T_p be a finitely...

open_problem | open | MathDB 页面 | 原文线索

V2-007297 | MATHDB-392816

Modularity of rigid Calabi-Yau threefolds over \(\mathbb{Q}\)

摘录 / EXCERPT: Rigid Calabi-Yau threefolds defined over the field of rational numbers are equipped with 2-dimensional Galois representations, which are conjecturally equivalent to modular forms o...

open_problem | open | MathDB 页面 | 原文线索

V2-007298 | MATHDB-392821

Quillen’s conjecture for finite groups

摘录 / EXCERPT: For a finite group G and a prime p, if the geometric realization $|\mathcal{S}_p(G)|$ of the poset of nontrivial p-subgroups of G is contractible, then $O_p(G)\ne 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-007299 | MATHDB-392962

Localization of the break-decomposition domain for local systems on a p-adic punctured disc

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, complete for a non-archimedean norm with residue characteristic $p>0$. Let $\bbell\ne p$ be a prime, let...

open_problem | open | MathDB 页面 | 原文线索

V2-007300 | MATHDB-392964

Polynomial-time computation of the mod- \(\ell\) Galois representation attached to \(\Delta\)

摘录 / EXCERPT: Let ℓ be prime, and let V_{ℓ} be the two-dimensional mod-ℓ Galois representation associated with the level-one, weight-12 modular form ... Assume that the image of the...

open_problem | open | MathDB 页面 | 原文线索

V2-007301 | MATHDB-392969

Modularity of the arithmetic generating series from special divisors on a Shimura curve

摘录 / EXCERPT: Theorem 1.2. The Laurent series $\widehat{\phi}_1=\sum_t\widehat{Z}_t(v)q^t$, with $\widehat{Z}_0(v)=-[\widehat{\omega}]-(0,\log v+c)$, is a non-holomorphic modular form of weight...

open_problem | open | MathDB 页面 | 原文线索

V2-007302 | MATHDB-392971

Weight part of Serre’s conjecture for Hilbert modular forms over totally real fields

摘录 / EXCERPT: Let F be a totally real field, let p be unramified in F, and let $\rho:G_F\to\mathrm{GL}_2(\overline{\mathbb F}_p)$ be continuous, irreducible, and totally odd. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-007303 | MATHDB-392972

Ihara’s Lemma, version 1

摘录 / EXCERPT: Let M_k denote the module of k-valued modular forms on $U(B)$ fixed by the Iwahori subgroup at $v\in S'$, and let M_k^v denote the $k[\mathrm{GL}(n,F_v)]$-module generated by...

open_problem | open | MathDB 页面 | 原文线索

V2-007304 | MATHDB-393054

Riemann hypothesis for Epstein zeta functions of class number one

摘录 / EXCERPT: Let $Q(x,y)=ax^2+bxy+cy^2$ be a positive definite quadratic form with $a,b,c\in\mathbb{Z}$ and discriminant $D=b^2-4ac$. If the class number of Q is 1, is every nontrivial zero...

conjecture | open | MathDB 页面 | 原文线索

V2-007305 | MATHDB-393055

Ratios conjecture for the Riemann zeta function

摘录 / EXCERPT: Let $s=\tfrac12+it$, with $\operatorname{Re}\gamma,\operatorname{Re}\delta>0$ and $|\alpha|,|\beta|<\tfrac14$. The ratios conjecture of Conrey, Farmer, and Zirnbauer asserts that .....

conjecture | open | MathDB 页面 | 原文线索

V2-007306 | MATHDB-393060

Hardy-Littlewood quantitative conjecture for prime tuples

摘录 / EXCERPT: Let $H=\{h_1,\ldots,h_k\}$ be a finite set of distinct integer shifts such that, for every prime p, the elements of H do not occupy all residue classes modulo p. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-007307 | MATHDB-393063

A Burgess-like bound for twisted L-functions

摘录 / EXCERPT: The report states and proves the following theorem: let f be an arbitrary modular form, either holomorphic or Maass, of level D and with some nebentypus, and let χ be a pr...

open_problem | open | MathDB 页面 | 原文线索

V2-007308 | MATHDB-393064

Infinitude and asymptotic count of primes with a prime half predecessor

摘录 / EXCERPT: Let ... The conjecture is that there exists a strictly positive constant C such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007309 | MATHDB-393066

A sum involving the derivative of the Riemann zeta-function over simple zeros

摘录 / EXCERPT: The report states the theorem: if all zeros of the Riemann zeta-function are simple, then ... where the sum runs over all zeros $\rho=\beta+i\gamma$ of $\zeta(s)$.

open_problem | open | MathDB 页面 | 原文线索

V2-007310 | MATHDB-393067

Bound for reciprocal derivatives at the zeros of the Riemann zeta function

摘录 / EXCERPT: The report states the following theorem: if all zeros of the Riemann zeta function are simple, then ... where the sum is over all zeros $\rho=\beta+i\gamma$ of $\zeta(s)$. The theore...

open_problem | open | MathDB 页面 | 原文线索

V2-007311 | MATHDB-393069

Conditional sign changes in the prime-number-theorem error term

摘录 / EXCERPT: Assume the Generalized Riemann Hypothesis. For every arbitrarily small but fixed $\eta>0$ and sufficiently large N, if ... then there exist $n_1,n_2\in[N+1,2N]$ such that ... and .....

open_problem | open | MathDB 页面 | 原文线索

V2-007312 | MATHDB-393135

Rédei’s mod-\p direction theorem

摘录 / EXCERPT: Throughout, let $q=p^h$, where p is prime. If A is a set of q points of $AG(2,q)$ determining fewer than $(q+3)/2$ directions-where a direction is determined when some line i...

open_problem | open | MathDB 页面 | 原文线索

V2-007313 | MATHDB-393136

Higher-dimensional direction theorem for affine point sets

摘录 / EXCERPT: Let $q=p^h$, where p is prime, and let A be a set of q^{n-1} points of $\mathrm{AG}(n,q)$. Let N be the number of directions not determined by A. If $N\ge p^e q$ for some...

open_problem | open | MathDB 页面 | 原文线索

V2-007314 | MATHDB-393140

Hansen-Mullen conjecture for primitive polynomials

摘录 / EXCERPT: Let q be a prime power, let $a\in\mathbb{F}_q$, let $n\ge 2$, and fix an integer m with $0<m<n$. Then there exists a primitive polynomial ... of degree n with $a_m=a$, with the...

conjecture | open | MathDB 页面 | 原文线索

V2-007315 | MATHDB-393142

Main Conjecture for MDS codes

摘录 / EXCERPT: Let $q=p^h$ be a prime power and let $m(k-1;q)$ denote the maximum length of a q-ary MDS code of dimension k (equivalently, the maximum size of an arc in $\mathrm{PG}(k-1;q)$)....

conjecture | open | MathDB 页面 | 原文线索

V2-007316 | MATHDB-393149

Asymptotically good subfamilies of binary quadratic-residue codes

摘录 / EXCERPT: Let C_p be the binary quadratic-residue code of prime length $p\equiv \pm1\pmod 8$, defined over $\mathbb F_2$ using a primitive p-th root of unity, and let d_p denote its mi...

open_problem | open | MathDB 页面 | 原文线索

V2-007317 | MATHDB-393150

Asymptotics of the minimal distance of quadratic residue codes

摘录 / EXCERPT: Let $p \equiv 1 \pmod 8$ be prime, and let d_p denote the minimal distance of the binary quadratic residue code of length p. Then, for infinitely many primes p, ... If the gene...

open_problem | open | MathDB 页面 | 原文线索

V2-007318 | MATHDB-393167

Conjecture on extremal modular forms in Case (2) modulo a prime

摘录 / EXCERPT: Let f be the extremal modular form of weight $k=12\mu$. Suppose that a prime p is in Case (2), meaning that $p\nmid a_i$ for all $i\ge 1$ with $p\nmid i$, and that there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-007319 | MATHDB-393177

Saito’s conjecture on positivity of eta products for integers with at least two distinct prime divisors

摘录 / EXCERPT: For every positive integer h divisible by at least two distinct primes, all Fourier coefficients in the q-expansion of ... are non-negative, where $q=e^{2\pi i\tau}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-007320 | MATHDB-393181

Genus version of the basis problem at prime level

摘录 / EXCERPT: Under the preceding assumptions that q is prime with $q\equiv 1\pmod 4$ and that m is divisible by 4, for every ν satisfying $1<\nu<m-1$ one has ... Here...

open_problem | open | MathDB 页面 | 原文线索

V2-007321 | MATHDB-393196

Rota’s Conjecture for finite-field matroid representability

摘录 / EXCERPT: For each prime power q, the set of excluded minors for representability over $\mathrm{GF}(q)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-007322 | MATHDB-393197

Bounded inequivalent representations of k-coherent matroids

摘录 / EXCERPT: For any prime power q and integer $k \ge 2$, there exists a constant $c_{q,k}$ such that each k-coherent matroid has at most $c_{q,k}$ inequivalent representations over...

open_problem | open | MathDB 页面 | 原文线索

V2-007323 | MATHDB-393283

An equivariant analogue of Glauberman’s ZJ-Theorem

摘录 / EXCERPT: Let p be a prime and define $P=O_p(G;R)$ to be the unique maximal $R C_G(R)$-invariant p-subgroup of G, equivalently the intersection of all R-invariant Sylow p-subgroups...

open_problem | open | MathDB 页面 | 原文线索

V2-007324 | MATHDB-393285

Simple groups of simultaneous characteristic p- and characteristic q-type

摘录 / EXCERPT: Which finite simple groups G are of both characteristic p-type and characteristic q-type for distinct primes p and q? Here, for a prime r, characteristic r-type means...

open_problem | open | MathDB 页面 | 原文线索

V2-007325 | MATHDB-393286

Bicharacteristic classification theorem for finite simple groups

摘录 / EXCERPT: Theorem 2 (in progress). Let G be a K-proper simple group and let p be an odd prime such that $m_{2,p}(G)\ge 3$. Assume that G is of even type and of weak p-type. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-007326 | MATHDB-393342

Arithmetic dependence of tight closure on the prime characteristic

摘录 / EXCERPT: Let $R_{\mathbb Z}$ be a finitely generated ring extension of $\mathbb Z$, let $I\subseteq R_{\mathbb Z}$ be an ideal, and let $f\in R_{\mathbb Z}$. For each prime number p, set...

open_problem | open | MathDB 页面 | 原文线索

V2-007327 | MATHDB-393343

Arithmetic variation of a tight-closure containment on the Fermat septic

摘录 / EXCERPT: In $R=K[X,Y,Z]/(Z^7-X^7-Y^7)$, the containment $X^3Y^3\in (X^4,Y^4,Z^4)^*$ holds for infinitely many prime characteristics and does not hold for infinitely many prime characteristi...

open_problem | open | MathDB 页面 | 原文线索

V2-007328 | MATHDB-393397

Modularity of the coefficients of the constructed theories

摘录 / EXCERPT: The coefficients of the theories constructed are always modular forms.

open_problem | open | MathDB 页面 | 原文线索

V2-007329 | MATHDB-393408

Density of embeddings of the maximal extension unramified outside \(p\) and \(\infty\)

摘录 / EXCERPT: Let p be a prime, and let $\mathbb Q_{\{1,p\}}$ denote a maximal algebraic extension of $\mathbb Q$ unramified outside p and ∞. Determine whether the field embeddings...

open_problem | open | MathDB 页面 | 原文线索

V2-007330 | MATHDB-393410

Dense image of embeddings for extensions unramified outside two sets of places

摘录 / EXCERPT: Let E be a CM field, let F be its maximal totally real subfield, and let $v=uu'$ be a finite place of F that splits in E. Let $\ll$ be a prime number distinct from the resi...

open_problem | open | MathDB 页面 | 原文线索

V2-007331 | MATHDB-393420

Conjectural global Langlands correspondence for \(\mathrm{GL}_n\)

摘录 / EXCERPT: Let L be a number field, let n be a positive integer, let ℓ be a fixed prime, and fix an isomorphism $\iota:\overline{\mathbb{Q}}_{\ell}\cong\mathbb{C}$. The conjectural g...

open_problem | open | MathDB 页面 | 原文线索

V2-007332 | MATHDB-393471

At least two distinct prime closed geodesics on every Finsler 2-sphere

摘录 / EXCERPT: For every Finsler metric F on the 2-sphere S^2, there exist at least two distinct prime closed geodesics.

open_problem | open | MathDB 页面 | 原文线索

V2-007333 | MATHDB-393486

Cohen-Lenstra conjectured divisibility probabilities for quadratic class numbers

摘录 / EXCERPT: Cohen and Lenstra conjectured that, for a prime p, the probability that p divides the class numbers of imaginary quadratic fields is ... and the probability that p divides the...

conjecture | open | MathDB 页面 | 原文线索

V2-007334 | MATHDB-393494

Algebraic construction of points in a false Tate curve extension

摘录 / EXCERPT: Let $E/\bb Q$ be the elliptic curve $y^2+y=x^3-x^2$ and let m be divisible by 11 but by no anomalous prime for $E/\bb Q$. The report explains that $E/\bb Q(\sqrt[3^n]{m})$ shou...

open_problem | open | MathDB 页面 | 原文线索

V2-007335 | MATHDB-393529

Characterization of countably generated prime ideals in H^∞

摘录 / EXCERPT: The report states that a nonzero prime ideal I in H^∞ is countably generated if and only if either $I=\{f\in H^\infty:f(z_0)=0\}$ for some $z_0\in\mathbb{D}$, or I is th...

open_problem | open | MathDB 页面 | 原文线索

V2-007336 | MATHDB-393541

Classification of countably generated prime ideals in H^{∞}

摘录 / EXCERPT: Let H^{∞} be the algebra of bounded analytic functions on the open unit disk $\mathbb{D}$. A nonzero prime ideal I in H^{∞} is countably generated if and only if ei...

open_problem | open | MathDB 页面 | 原文线索

V2-007337 | MATHDB-393600

Finite generation and dimension of cohomology primitives

摘录 / EXCERPT: Let p be a prime, let G be a finite group, and let $C\cong (\mathbb{Z}/p)^c$ be a central subgroup of G. With $H^*(G)=H^*(G;\mathbb{Z}/p)$ and ... where $m:C\times G\to G$ is m...

open_problem | open | MathDB 页面 | 原文线索

V2-007338 | MATHDB-393604

Qd(p)-free rank-two finite groups have homotopy rank two

摘录 / EXCERPT: Let G be a finite group with $\operatorname{rk}(G)=2$. If, for every odd prime p, no subquotient of G is isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-007339 | MATHDB-393632

Permanent-cycle criterion for the Adams spectral sequence classes h_i

摘录 / EXCERPT: In the classical Adams spectral sequence at the prime 2, $E_2^{s,t}=\operatorname{Ext}_{A^*}^{s,t}(\mathbb F_2,\mathbb F_2)\Rightarrow \pi_{t-s}^*\otimes\mathbb Z_2$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-007340 | MATHDB-393798

Arithmetic pro-\p Elementary Type Conjecture

摘录 / EXCERPT: Let p be a prime and let F be a field of characteristic different from p containing the pth roots of unity. If $G_F(p)=\operatorname{Gal}(F(p)/F)$ is finitely generated as...

conjecture | open | MathDB 页面 | 原文线索

V2-007341 | MATHDB-393900

James’s conjecture for blocks of symmetric groups

摘录 / EXCERPT: Let F be a field of prime characteristic p, and let B be a block of $F S_n$ of weight strictly less than p (equivalently, with abelian defect group). If the decomposition m...

conjecture | open | MathDB 页面 | 原文线索

V2-007342 | MATHDB-393901

James’s Conjecture for weight-three blocks of symmetric groups

摘录 / EXCERPT: Let F be a field of prime characteristic p, let B be a block of FS_n of weight 3, and write the decomposition matrix of B as DJ, where D is the decomposition matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-007343 | MATHDB-393903

Holt's cohomology-dimension conjecture

摘录 / EXCERPT: For every finite group G, every prime p, and every faithful irreducible G-module M over a field of characteristic p, there is an absolute constant C such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-007344 | MATHDB-393905

Character-count inequality for blocks with one Brauer character

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a p-block of G with defect group D. If $l(B)=1$, is it true that $k(B)\leq k(D)$? Here $k(B)$ is the number of i...

open_problem | open | MathDB 页面 | 原文线索

V2-007345 | MATHDB-393908

Strengthened McKay Conjecture with local fields and Schur indices

摘录 / EXCERPT: Let p be a prime number, let G be any finite group, and let P be a Sylow p-subgroup of G. Then there exists a bijection ... such that: 1. f commutes with the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-007346 | MATHDB-393923

Torsion in endotrivial modules for symmetric groups

摘录 / EXCERPT: For the symmetric group $G=S_n$, with $p>2$ prime and k an algebraically closed field of characteristic p, the authors strongly suspect that, for $n\ge p^2$, the group $T(G)$ o...

open_problem | open | MathDB 页面 | 原文线索

V2-007347 | MATHDB-393925

MAKRODINU for unipotent blocks of finite reductive groups

摘录 / EXCERPT: The conjecture (MAKRODINU) holds under the following hypotheses: (1) $G=G(q)$ is a finite reductive group; (2) ℓ is a prime such that $\ell\nmid q$ and ℓ is excellent for...

open_problem | open | MathDB 页面 | 原文线索

V2-007348 | MATHDB-393961

Explicit poset isomorphism for H-stable prime ideals of quantum matrices

摘录 / EXCERPT: Give an explicit poset isomorphism, with respect to inclusion and the Bruhat order, between the poset of H-stable prime ideals in $O_q(M_n(\mathbb{C}))$ and the sub-poset of...

open_problem | open | MathDB 页面 | 原文线索

V2-007349 | MATHDB-393964

Cohen-Fischman semiprime smash product question

摘录 / EXCERPT: Is $A\#H$ semiprime whenever A is a semiprime algebra over a field k, H is a finite-dimensional semisimple Hopf algebra over k, and H acts on A?

open_problem | open | MathDB 页面 | 原文线索

V2-007350 | MATHDB-393965

Stability of the prime radical under arbitrary cosemisimple Hopf actions

摘录 / EXCERPT: Let k be a field of characteristic 0, let H be a finite-dimensional cosemisimple Hopf algebra over k, and let A be an arbitrary H-module k-algebra. Is the prime radic...

open_problem | open | MathDB 页面 | 原文线索

V2-007351 | MATHDB-393967

Asymptotic growth of Euler characteristics of coprime quiver moduli

摘录 / EXCERPT: Assume that Θ is generic. There exists a constant $C_Q\in\mathbb{R}$ such that, for large dimension vectors d coprime for Θ, ...

open_problem | open | MathDB 页面 | 原文线索

V2-007352 | MATHDB-393974

Prime ideals of the Iwasawa algebra of an almost simple uniform pro-[U+007F]p group

摘录 / EXCERPT: Let G be almost simple, meaning that G is a uniform pro-p group whose Lie algebra has no non-trivial ideals, so that $Z(\Omega_G)=\mathbb{F}_p$ and every nontrivial normal su...

open_problem | open | MathDB 页面 | 原文线索

V2-007353 | MATHDB-393975

Greenberg’s conjecture for the cyclotomic field $\mathbb{Q}(\mu_p)$ for $p<1000$

摘录 / EXCERPT: For every odd prime $p<1000$, let $F=\mathbb{Q}(\mu_p)$, let $\widetilde F$ be the compositum of all $\mathbb{Z}_p$-extensions of F, and let $X_{\widetilde F}$ be the Galois grou...

conjecture | open | MathDB 页面 | 原文线索

V2-007354 | MATHDB-393986

Denominators of Eisenstein classes for elliptic modular forms

摘录 / EXCERPT: For an even integer $k>0$, determine the smallest positive integers with respect to divisibility, $\04_{\rm DR}(k)$ and $\04_{\rm B}(k)$, such that ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-007355 | MATHDB-393989

Existence of mild pro-\p Galois groups over number fields other than \(\mathbb{Q}\)

摘录 / EXCERPT: Let K be a number field, let p be an odd prime, and let S be a finite set of primes of K containing no prime dividing p. Writing $K_S(p)$ for the maximal pro-p extensio...

open_problem | open | MathDB 页面 | 原文线索

V2-007356 | MATHDB-393991

Weak Greenberg conjecture in the totally split abelian case

摘录 / EXCERPT: Let p be an odd prime and let F be a totally real abelian number field such that $p\nmid [F:\mathbb{Q}]$ and p is totally split in F. Then the weak Greenberg conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-007357 | MATHDB-393993

Equivalence of Boston’s two conjectures for restricted-ramification pro-\p Galois groups

摘录 / EXCERPT: Let p be a prime, K a number field, and S a finite set of places of K with $S\cap S_p=\varnothing$. Assume that $G_{K,S}$ is infinite. Are the following two conjectures equ...

conjecture | open | MathDB 页面 | 原文线索

V2-007358 | MATHDB-393995

Finite Schur σ-groups with arbitrarily large derived length

摘录 / EXCERPT: The report asks whether there exist finite Schur $\c3$-groups with arbitrarily large derived length for some fixed prime p. It answers affirmatively for $p=3$: let F be the fre...

open_problem | open | MathDB 页面 | 原文线索

V2-007359 | MATHDB-393998

Main Conjecture for Modular Tower component branches

摘录 / EXCERPT: Given any MT component branch and any number field K, for $k\gg 0$, $H_k^{\prime\,\mathrm{rd}}(K)=\varnothing$.

conjecture | open | MathDB 页面 | 原文线索

V2-007360 | MATHDB-393999

Main Conjecture for Modular Tower component branches with $r=4$

摘录 / EXCERPT: Given any MT component branch $[U+007F]{H'_k}_{k=0}^{\infty}$, and any number field K, for $k\gg0$, $H_k^{\prime\,\mathrm{rd}}(K)=\varnothing$.

conjecture | open | MathDB 页面 | 原文线索

V2-007361 | MATHDB-394010

Equality of the log canonical threshold and F-threshold for infinitely many primes

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x_1,\ldots,x_n]$. For each prime p, let f_p be the image of f in $\mathbb{F}_p[x_1,\ldots,x_n]$, and let $\mathfrak m=(x_1,\ldots,x_n)$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-007362 | MATHDB-394054

Kimmerle’s Prime Graph Conjecture for Integral Group Rings

摘录 / EXCERPT: For every finite group G, the Gruenberg-Kegel (prime) graph satisfies ... where $V(\mathbb{Z}G)$ is the normalized unit group of the integral group ring $\mathbb{Z}G$; the vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-007363 | MATHDB-394070

The p-adic Beilinson conjecture for totally real number fields

摘录 / EXCERPT: Let k be a totally real number field, let $n\geq 2$ be odd, and let p be any prime. With F, $D_k^{1/2,p}$, $L_p(n,\omega_p^{1-n},k)$, $\operatorname{Eul}_p(n,k)$, and the p...

conjecture | open | MathDB 页面 | 原文线索

V2-007364 | MATHDB-394120

Lens spaces bounding rational homology balls and the ribbon conjecture for 2-bridge knots

摘录 / EXCERPT: Let $p>q>0$ be coprime integers. Define $R\subset\mathbb{Q}_{>0}$ as the smallest subset such that $f(R)\subseteq R$ and $g(R)\subseteq R$, where, for $p/q\in\mathbb{Q}_{>0}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-007365 | MATHDB-394246

Infinitude of primes in bounded arithmetic with the prime-counting function

摘录 / EXCERPT: Does $I\Delta_0(\pi)$ prove that there exist arbitrarily large prime numbers? Here $I\Delta_0(\pi)$ is $I\Delta_0$ augmented by the function $\pi(x)$, counting primes less than or...

open_problem | open | MathDB 页面 | 原文线索

V2-007366 | MATHDB-394261

Essential dimension of \(\mathrm{PGL}_p\) for primes \(p>3\)

摘录 / EXCERPT: For an algebraically closed field k and a prime $p>3$, is $\operatorname{ed}_k(\mathrm{PGL}_p)\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-007367 | MATHDB-394323

Main Conjecture on recovering function fields from relative absolute Galois groups

摘录 / EXCERPT: Let K and K' be fields satisfying the hypotheses of the Main Theorem: they are perfect, and K allows finite extensions of degree greater than 2 and prime to ${\rm char}(K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-007368 | MATHDB-394358

A q-Eulerian identity for Betti-number polynomials of Rees products

摘录 / EXCERPT: Let q be a prime power and let $n\ge 1$. Define $\beta_n(q,t)=\sum_{j=1}^{n}\beta_{n,j}(q)t^j$, where $\beta_{n,j}(q)$ is the top Betti number of the order complex of the induced...

open_problem | open | MathDB 页面 | 原文线索

V2-007369 | MATHDB-394464

Triangular-lattice formula for the planar Riesz-energy constant

摘录 / EXCERPT: For $s>2$, the asymptotic constant for minimal discrete Riesz s-energy in dimension 2 satisfies ... where $\zeta_L(s)$ is the zeta function of the planar triangular lattice...

open_problem | open | MathDB 页面 | 原文线索

V2-007370 | MATHDB-394482

Existence of a Diophantine quintuple

摘录 / EXCERPT: Does there exist a set S of five positive distinct integers such that, for every pair of distinct elements $x,y\in S$, the number $xy+1$ is a perfect square?

open_problem | open | MathDB 页面 | 原文线索

V2-007371 | MATHDB-394488

Uniqueness of the non-trivial diagonal Nagell-Ljunggren solution

摘录 / EXCERPT: For integers x,y, an odd prime p, and $e\in\{0,1\}$ satisfying ... the report conjectures that the displayed solution ... is the only non-trivial solution.

open_problem | open | MathDB 页面 | 原文线索

V2-007372 | MATHDB-394489

Removal of the class-number hypothesis in the prime-power cyclotomic generator problem

摘录 / EXCERPT: Let q be a power of a prime $p\ge 2$, let ζ_q be a primitive qth root of unity, and let h_q^+ be the class number of $\mathbb{Q}(\zeta_q+\overline{\zeta_q})$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-007373 | MATHDB-394493

Four-integer Diophantus-Euler square-product-plus-sum problem

摘录 / EXCERPT: Conjecture: There is no set $D\subseteq\mathbb Z$ with exactly four elements such that, for every pair of distinct elements $x,y\in D$, the integer $xy+x+y$ is a perfect square.

open_problem | open | MathDB 页面 | 原文线索

V2-007374 | MATHDB-394497

Algebraicity of simultaneous real and p-adic base-\[?] expansions

摘录 / EXCERPT: Let p be a prime number, let $(a_n)_{n\ge 1}$ be an infinite sequence taking values in $\{0,1,\ldots,p-1\}$, and set ... Prove that the real number α and the p-adic number...

open_problem | open | MathDB 页面 | 原文线索

V2-007375 | MATHDB-394536

Cohen-Lenstra-Gerth mean-value conjecture for squared narrow class groups

摘录 / EXCERPT: Let ell be a prime and let $k>0$. With C_D denoting the narrow class group of the quadratic field of fundamental discriminant D,...

conjecture | open | MathDB 页面 | 原文线索

V2-007376 | MATHDB-394537

Cohen-Lenstra-Gerth mean-value conjecture for the 2-part of quadratic class groups

摘录 / EXCERPT: Let $\ll$ be a prime. Then - $M^{-}\!\left(\u000ell^{\operatorname{rk}_{\u000ell}(C_D^{2})}\right)= 2$; -...

conjecture | open | MathDB 页面 | 原文线索

V2-007377 | MATHDB-394539

Kim’s nonabelian Chabauty dimension conjecture

摘录 / EXCERPT: Let X be a smooth and absolutely irreducible algebraic curve defined over $\mathbb{Q}$, with genus $g\ge 2$ and no points at infinity ($r=0$). Fix a prime number p at which...

conjecture | open | MathDB 页面 | 原文线索

V2-007378 | MATHDB-394541

A Hom-form of the pro-\p birational anabelian conjecture

摘录 / EXCERPT: Fix a prime number p and let k be a sub-p-adic field. For a regular function field $K\mid k$, let $G_{K\mid k}:=\ker(G_K\to G_k)$, let N_K be the kernel of the projection o...

conjecture | open | MathDB 页面 | 原文线索

V2-007379 | MATHDB-394543

Iwasawa-type growth formula for a one-prime-ramified \p-adic tower of function fields

摘录 / EXCERPT: Conjecture. Suppose F is any function field of characteristic p. Given an extension F_∞/F ramified only at one prime f, with...

open_problem | open | MathDB 页面 | 原文线索

V2-007380 | MATHDB-394567

Existence of a nice character satisfying Assumption A(F,p)

摘录 / EXCERPT: For every totally real number field F and every prime number $p\ge 3$, Assumption $A(F,p)$ holds: there exists a finite-order character...

open_problem | open | MathDB 页面 | 原文线索

V2-007381 | MATHDB-394568

Sharifi’s cyclotomic cup-product and modular-symbol conjecture

摘录 / EXCERPT: Let p be an odd prime. For every positive integer r, put $F_r=\mathbb{Q}(\zeta_{p^r})$. For integers u,v prime to p, there exists an isomorphism ... of...

conjecture | open | MathDB 页面 | 原文线索

V2-007382 | MATHDB-394570

Removing the hypotheses in Schmidt’s cofinality theorem for tame K(π,1)-rings

摘录 / EXCERPT: Conjecture. In Theorem 1, the condition $(*)$ can be removed. Equivalently, let k be a number field, let p be a prime number, assume p is odd or k is totally imaginary, let...

open_problem | open | MathDB 页面 | 原文线索

V2-007383 | MATHDB-394571

Complete splitting of primes outside the ramification set

摘录 / EXCERPT: Let k be a number field and let p be a prime number such that $p\ne 2$ or k is totally imaginary. Let S be a finite non-empty set of non-archimedean primes of k with...

open_problem | open | MathDB 页面 | 原文线索

V2-007384 | MATHDB-394576

Parity conjecture for elliptic curves admitting a prime-degree isogeny

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}$, let p be an odd prime, and let E/F be an elliptic curve admitting an F-isogeny of degree p. If $p=3$, assume that E has sem...

conjecture | open | MathDB 页面 | 原文线索

V2-007385 | MATHDB-394616

Global smooth linearization of circle diffeomorphisms with Diophantine rotation number

摘录 / EXCERPT: Any smooth diffeomorphism of the circle with a Diophantine rotation number is smoothly conjugated to a rotation.

open_problem | open | MathDB 页面 | 原文线索

V2-007386 | MATHDB-394621

Vanishing quadratic twists in square and nonsquare residue classes

摘录 / EXCERPT: For a fixed elliptic curve E and a prime p, when quadratic twists are restricted to arithmetic progressions modulo p, the ratio between the counts of twists with $L(E_d,1)=0$...

open_problem | open | MathDB 页面 | 原文线索

V2-007387 | MATHDB-394624

Generalised Riemann Hypothesis for Hecke L-functions

摘录 / EXCERPT: For every prime number q and every $f\in H^*_{\kappa}(q)$, all zeros of the completed Hecke L-function $\Lambda(f,s)$ lie on the critical line $\{s\in\mathbb{C}:\Re s=1/2\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-007388 | MATHDB-394626

Prime separation from the Eisenstein Hecke eigenvalues

摘录 / EXCERPT: For any Maass-Hecke eigenform f of Laplacian eigenvalue $\lambda=\frac14+r^2$, there is a prime $p<\frac{\sqrt{\lambda}}{2\pi}$ such that its Hecke eigenvalue λ_p differs...

open_problem | open | MathDB 页面 | 原文线索

V2-007389 | MATHDB-394627

Cohen-Lenstra density conjecture for real quadratic fields and prime discriminants

摘录 / EXCERPT: Let M be the set of discriminants of real quadratic number fields, and let $S\subset M$ consist of those whose class number is a power of 2. Let M' be the set of prime discri...

conjecture | open | MathDB 页面 | 原文线索

V2-007390 | MATHDB-394633

Moment asymptotic for regulators of rank-one quadratic twists

摘录 / EXCERPT: Let E have conductor N. For each prime $p\mid N$, fix a sign w_p such that $\prod_{p\mid N}w_p=w(E)$, where $w(E)$ is the sign of the functional equation of E. Set ... For th...

open_problem | open | MathDB 页面 | 原文线索

V2-007391 | MATHDB-394634

Rank verification for prime-conductor elliptic curves in the Stein-Watkins database

摘录 / EXCERPT: Every elliptic curve of prime conductor in the Stein-Watkins database has rank at least as large as predicted by the Birch-Swinnerton-Dyer conjecture. In particular, for each of th...

open_problem | open | MathDB 页面 | 原文线索

V2-007392 | MATHDB-394635

Generation of modular forms on \(\Gamma(\ell)\) by weight-one Eisenstein series

摘录 / EXCERPT: I conjecture that all modular forms of weight 3 and above on $\Gamma(\ell)$ are polynomials in the weight-one Eisenstein series.

open_problem | open | MathDB 页面 | 原文线索

V2-007393 | MATHDB-394659

Irreducibility of restrictions from $GL_n(q)$ to $SL_n(q)$

摘录 / EXCERPT: Let q be a prime power, and let ... be an irreducible $\mathbb FGL_n(q)$-representation, where the σ_i are ℓ'-elements. A partition λ is called ℓ-divisible...

open_problem | open | MathDB 页面 | 原文线索

V2-007394 | MATHDB-394660

Recognition of finite simple groups from the probabilistic zeta function

摘录 / EXCERPT: Conjecture: Suppose that G is a finite simple group and that H is a finite group with $\\operatorname{Frat}(H)=1$ and $P_G(s)=P_H(s)$. Then $G\cong H$.

open_problem | open | MathDB 页面 | 原文线索

V2-007395 | MATHDB-394755

Examples attaining the lower bound for the \b6-order

摘录 / EXCERPT: For every prime p, do there exist a field k of characteristic p, a finite group G, and a nonzero Tate cohomology class $[\zeta]\in \widehat{H}^{n}(G,k)$ of even degree n,...

open_problem | open | MathDB 页面 | 原文线索

V2-007396 | MATHDB-394810

Holomorphic lifting from $\mathrm{GSp}_{2m}\times\mathrm{GSp}_{2m}$ to GSp_{4m}

摘录 / EXCERPT: Conjecture 1. Let f and g be two Siegel modular forms of genus $2m$ and of weights $k>2m$ and $l=k-2m$. Then there exists a Siegel modular form F of genus $4m$ and of weight...

open_problem | open | MathDB 页面 | 原文线索

V2-007397 | MATHDB-394815

Paramodular Conjecture for rational abelian surfaces of prime conductor

摘录 / EXCERPT: Let p be a prime. There is a bijection between Hecke eigenforms with rational eigenvalues $f\in S_2^2(K(p))$ that are not Gritsenko lifts and isogeny classes of rational abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-007398 | MATHDB-394820

Products of Eisenstein series spanning modular forms at prime level

摘录 / EXCERPT: Claim 1. Let q be a prime. With $E_r(\tau)$ and $E_{r,\psi}(\tau)$ the Eisenstein series defined in the report, let ... where $4\le r\le k$ with r even, $1\le t\le k$, and ψ...

open_problem | open | MathDB 页面 | 原文线索

V2-007399 | MATHDB-394824

Duke-Imamoglu-Ikeda lift existence theorem

摘录 / EXCERPT: Let f be an elliptic modular form of weight $2k$ for the full modular group $\mathrm{SL}_2(\mathbb{Z})$. For $n\equiv k\pmod 2$, there exists a nonzero Siegel cusp form...

open_problem | open | MathDB 页面 | 原文线索

V2-007400 | MATHDB-394840

Classification of central Frattini extensions by small group rings

摘录 / EXCERPT: Fix a finite p-group H, where p) is an arbitrary prime, and let $k=\mathbb F_p$. Let $\mathcal F(H)$ be the collection of short exact sequences of finite p-groups ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-007401 | MATHDB-394845

Prime-graph coincidence for normalized units of integral group rings

摘录 / EXCERPT: Let G be a finite group. Do the Gruenberg-Kegel (prime) graphs of $V(\mathbb ZG)$ and G coincide?

open_problem | open | MathDB 页面 | 原文线索

V2-007402 | MATHDB-394860

Counting local symmetric algebras over finite fields

摘录 / EXCERPT: Let p be a prime and let k be the field with p elements. Count, estimate, or give bounds for the number of isomorphism classes of local symmetric k-algebras of dimension...

open_problem | open | MathDB 页面 | 原文线索

V2-007403 | MATHDB-394861

Minimal trellis realization for convolutional codes over finite rings

摘录 / EXCERPT: For convolutional codes over finite rings of the form $\mathbb{Z}_{p^r}$, construct a minimal trellis from a left-prime polynomial encoder and express the minimum number of trellis...

open_problem | open | MathDB 页面 | 原文线索

V2-007404 | MATHDB-394866

Prime divisors of automorphism groups of extremal Type II codes

摘录 / EXCERPT: If C is a binary self-dual doubly even extremal code of length $24m$ with $m>2$ and p is a prime satisfying $p \mid |\operatorname{Aut}(C)|$, then $p \le 12m$.

open_problem | open | MathDB 页面 | 原文线索

V2-007405 | MATHDB-394945

Yoshida-type liftings to level-2 Siegel modular forms

摘录 / EXCERPT: For the regular pair (l,m) considered in the report, let $\Gamma_2[2]=\ker\bigl(\operatorname{Sp}(4,\mathbb Z)\to\operatorname{Sp}(4,\mathbb Z/2)\bigr)$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-007406 | MATHDB-394946

Yoshida-type lifting at level 2

摘录 / EXCERPT: Let $f\in S_{l+m+4}(\Gamma_0(2))^{\mathrm{new}}$ and $g\in S_{l-m+2}(\Gamma_0(2))^{\mathrm{new}}$ be newforms. Then there exists a Siegel modular form...

open_problem | open | MathDB 页面 | 原文线索

V2-007407 | MATHDB-394948

Parity implication for Selmer groups of Hilbert modular forms

摘录 / EXCERPT: Let F be a totally real number field, let g be a cuspidal Hilbert modular eigenform over F of parallel weight k and trivial character, let K be a totally imaginary quadra...

open_problem | open | MathDB 页面 | 原文线索

V2-007408 | MATHDB-394949

Arithmetic intersection formula for Hirzebruch-Zagier divisors and CM cycles

摘录 / EXCERPT: Let $F=\mathbb{Q}(\sqrt D)$, where $D\equiv 1\pmod 4$ is prime, and let $K=F(\sqrt{\Delta})$ be a non-biquadratic quartic CM field with real quadratic subfield F. Assume that...

open_problem | open | MathDB 页面 | 原文线索

V2-007409 | MATHDB-394952

Supersingular Kottwitz-Rapoport strata conjecture

摘录 / EXCERPT: Let p be a prime and $g\ge 1$. For $0\le i\le \lfloor g/2\rfloor$, define ... where Adm is the admissible set of relative positions for the Kottwitz-Rapoport strat...

conjecture | open | MathDB 页面 | 原文线索

V2-007410 | MATHDB-394997

Hardy-Littlewood prime-tuples conjecture

摘录 / EXCERPT: Let $k\ge 1$ and let $H=\{h_1,\ldots,h_k\}$ be a set of distinct non-negative integers. For each prime p, let $\nu_H(p)$ be the number of distinct residue classes modulo p occu...

conjecture | open | MathDB 页面 | 原文线索

V2-007411 | MATHDB-395004

Range of validity for hybrid moments of the Riemann zeta-function

摘录 / EXCERPT: For fixed $k,\ell,m\in\mathbb N$ and $1\ll G\ll T$, determine the range of $G=G(T;k,\ell,m)$ for which ...

open_problem | open | MathDB 页面 | 原文线索

V2-007412 | MATHDB-395006

Almost-prime solutions to ternary quadratic equations

摘录 / EXCERPT: Let $g(x_1,x_2,x_3)$ be an indefinite anisotropic quadratic form with determinant $d(g)$, and let $t\ne 0$ be an integer such that $d(g)t$ is square-free. The reported result state...

open_problem | open | MathDB 页面 | 原文线索

V2-007413 | MATHDB-395010

Statistics of zeros in the local regime for hyperelliptic-curve zeta functions

摘录 / EXCERPT: For a fixed odd prime power q, let $\mathcal H_{2g+2,q}$ be the family of curves $C:y^2=Q(x)$ over $\mathbb F_q$, where Q is monic, square-free, and has degree $2g+2$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-007414 | MATHDB-395012

Lenstra’s conjecture on small nonvanishing Fermat quotients

摘录 / EXCERPT: For a prime p, define the Fermat quotient by $q_p(a)=\frac{a^{p-1}-1}{p}$, and let ℓ_p be the smallest positive integer a such that $q_p(a)\not\equiv 0\pmod p$. H. W. Lens...

conjecture | open | MathDB 页面 | 原文线索

V2-007415 | MATHDB-395013

Diophantine inequalities in function fields: Spencer’s main theorem

摘录 / EXCERPT: Let ... where ... then, for every fixed integer ... where [...]{N(P;lambda)} counts [...]{xinmathbb Fq[t]^s} satisfying [...]{langlelambda1x1^k+cdots+lambdasxs^krangle<widehattau} and [....

open_problem | open | MathDB 页面 | 原文线索

V2-007416 | MATHDB-395036

Sylow-subgroup characterization of the soluble radical

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Then $P\subseteq\operatorname{sol}(G)$ if and only if $\langle P,g\rangle$ is soluble for all...

open_problem | open | MathDB 页面 | 原文线索

V2-007417 | MATHDB-395109

Functional equation for ideal reduced zeta functions

摘录 / EXCERPT: Let L be a finite-dimensional torsion-free nilpotent Lie algebra over $\mathbb{C}[[t]]$ with a nice and simple basis $B=\{x_1,\ldots,x_d\}$. Let $0=Z_0\le Z_1\le\cdots\le Z_c=L$...

open_problem | open | MathDB 页面 | 原文线索

V2-007418 | MATHDB-395110

Uniformity of representation zeta functions for semisimple Lie lattices

摘录 / EXCERPT: Let L be a Lie lattice over $\bb Z$ such that $\bb Q\otimes_{\u0005bb Z}L$ is a semisimple $\bb Q$-Lie algebra of dimension d. For each prime p and each admissible...

open_problem | open | MathDB 页面 | 原文线索

V2-007419 | MATHDB-395111

The fake degree conjecture for algebra groups at odd primes

摘录 / EXCERPT: Let p be an odd prime and let $P=1+A$ be a finite algebra group, equipped with its classical Cayley map and associated co-adjoint orbit set $\mathcal O$. Then the fake degree con...

conjecture | open | MathDB 页面 | 原文线索

V2-007420 | MATHDB-395161

Prime matrix theorem for determinant-2 integral matrices

摘录 / EXCERPT: Fix any norm on $M_2(\mathbb{R})$. Let ... and let $\mathcal P$ denote the set of prime numbers. The Prime Matrix Theorem conjectures that there is a constant C such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007421 | MATHDB-395162

Prime entries in determinant-two matrices

摘录 / EXCERPT: Is the set of 2×2 matrices with determinant 2 and all four entries prime infinite? Equivalently, is ... infinite?

open_problem | open | MathDB 页面 | 原文线索

V2-007422 | MATHDB-395265

Almost-everywhere strong Diophantine condition for pairs in $SU(2)$

摘录 / EXCERPT: For almost every pair $(a,b)\in SU(2,\mathbb{R})$, there exists a constant $C(a,b)>0$ such that ... for every nontrivial word $w\in F_2$, where d is a Riemannian distance on...

open_problem | open | MathDB 页面 | 原文线索

V2-007423 | MATHDB-395277

Measurable conjugacy of area-preserving surface diffeomorphisms to Diophantine rotations

摘录 / EXCERPT: Let f be an area-preserving diffeomorphism of a surface M equipped with an area form. Can f be measurably conjugated to (S^1,R_β), where R_β is a rotation and...

open_problem | open | MathDB 页面 | 原文线索

V2-007424 | MATHDB-395326

Additive bases of vector spaces over prime fields

摘录 / EXCERPT: Let p be a prime. Is there a constant $r=r(p)$ such that the following holds? Let V be a finite-dimensional vector space over the field of order p, and let $B_1,\ldots,B_r$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-007425 | MATHDB-395339

Van Hamme’s p-adic supercongruence (0.2)

摘录 / EXCERPT: For every prime $p>2$, ... where $(a)_n$ is the Pochhammer symbol and $\left(\frac{\cdot}{p}\right)$ denotes the Legendre symbol.

open_problem | open | MathDB 页面 | 原文线索

V2-007426 | MATHDB-395340

A Ramanujan-type supercongruence labeled (0.4)

摘录 / EXCERPT: For every prime $p>2$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-007427 | MATHDB-395417

A four-point positive-characteristic Hurwitz-number formula

摘录 / EXCERPT: Let $p>0$ be prime, and let $C_i=e_i$ be the conjugacy class of a single cycle in S_p for $i=1,2,3,4$. Assume that p does not divide any ramification index and that...

open_problem | open | MathDB 页面 | 原文线索

V2-007428 | MATHDB-395522

u-invariant of function fields of curves over p-adic fields

摘录 / EXCERPT: Let p be an odd prime and let F be the function field of a curve over a p-adic field. Then $u(F)=8$.

open_problem | open | MathDB 页面 | 原文线索

V2-007429 | MATHDB-395647

Classification of torus-invariant prime ideals in the quantum Grassmannian

摘录 / EXCERPT: Let k be a field, let $q\in k\setminus\{0\}$ be not a root of unity, and fix integers $0<m\le n$. Let $A=\mathcal O_q(G_{m,n})$ be the quantum Grassmannian, and let $H=(k^*)^n$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-007430 | MATHDB-395660

Kostant’s embedding conjecture for finite groups in adjoint simple Lie groups

摘录 / EXCERPT: Let G be the adjoint group corresponding to a complex simple Lie algebra, and let h be its Coxeter number. If $2h+1$ is a power of a prime, then $L_2(2h+1)$ embeds in G; up t...

conjecture | open | MathDB 页面 | 原文线索

V2-007431 | MATHDB-395688

Brauer Height Zero Conjecture

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let B be a p-block of G with defect group D. Then all irreducible complex characters in B have height zero if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-007432 | MATHDB-395692

Gluing problem for an extraspecial fusion system

摘录 / EXCERPT: Let p be an odd prime, let P be an extraspecial p-group of order p^3 and exponent p, and let $\mathcal F$ be a saturated fusion system on P. Write $\mathcal F^c$ for th...

open_problem | open | MathDB 页面 | 原文线索

V2-007433 | MATHDB-395737

Lyubeznik’s conjecture on associated primes of local cohomology

摘录 / EXCERPT: For regular rings R, each local cohomology module $H^k_{\mathfrak a}(R)$ has finitely many associated prime ideals.

conjecture | open | MathDB 页面 | 原文线索

V2-007434 | MATHDB-395782

Indecomposability of generalized Severi-Brauer motives

摘录 / EXCERPT: Let D be a central division F-algebra of degree p^n, where p is prime, and let $M_{m,D}$ be the distinguished indecomposable summand of the Chow motive with $\mathbb{F}_p$-...

open_problem | open | MathDB 页面 | 原文线索

V2-007435 | MATHDB-395823

Existence of an infinite Smith group

摘录 / EXCERPT: There exists an infinite group Q such that, for every Hausdorff space X of finite covering dimension, every action of Q on X, and every prime p for which X has the same...

open_problem | open | MathDB 页面 | 原文线索

V2-007436 | MATHDB-395844

Two-variable Iwasawa main conjecture for an elliptic curve over an imaginary quadratic field

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve of conductor N, parametrized by a cuspidal Hecke eigenform $f\in S_2(\Gamma_0(N))$. Let p be a rational prime of either good ordinary or...

conjecture | open | MathDB 页面 | 原文线索

V2-007437 | MATHDB-395847

Modularity of all residual representations of type U(3)

摘录 / EXCERPT: In the setting of the report-where E is an imaginary quadratic field, p is an odd prime split in E, S is a finite c-stable set of places containing the places above p a...

open_problem | open | MathDB 页面 | 原文线索

V2-007438 | MATHDB-395853

Injectivity and image of the theta map for a semidirect-product Iwasawa algebra

摘录 / EXCERPT: Fix an odd prime p. Let $G=H\rtimes\Gamma$, where H is an abelian pro-p group and $\Gamma\cong\mathbb Z_p$. For a complete unramified extension $\mathcal O$ of $\mathbb Z_p$,...

open_problem | open | MathDB 页面 | 原文线索

V2-007439 | MATHDB-395854

Equivariant/noncommutative Iwasawa main conjecture (Conjecture 2.1)

摘录 / EXCERPT: Let p be a prime, let F be a totally real number field, and let F_∞/F be a totally real p-adic Lie Galois extension in which only finitely many primes of F ramify. A...

conjecture | open | MathDB 页面 | 原文线索

V2-007440 | MATHDB-395859

Gabber’s absolute purity theorem in étale cohomology

摘录 / EXCERPT: Let ℓ be a prime number and set $\Lambda=\mathbb{Z}/\ell^\nu\text{\mathbb{Z}}$ for some $\nu eq 0$. Let $i:Y\to X$ be a closed embedding of codimension c between regular sch...

open_problem | open | MathDB 页面 | 原文线索

V2-007441 | MATHDB-395882

Weil-polynomial conjecture for the limiting Frobenius on \(\ker N\)

摘录 / EXCERPT: For the matrices (F,N) associated with the degeneration of projective hypersurfaces described on the page, and for a suitable prime number p, the polynomial ... conjecturally has...

conjecture | open | MathDB 页面 | 原文线索

V2-007442 | MATHDB-395890

Cohen-Lenstra density conjecture for \-ranks of quadratic class groups

摘录 / EXCERPT: Conjecture 3. Let ℓ be a prime and $r\ge 0$. Then the density of fundamental discriminants D such that $\operatorname{rk}_{\ell}(C_D^2)=r$ is equal to -...

conjecture | open | MathDB 页面 | 原文线索

V2-007443 | MATHDB-395895

p-adic algebraicity and infinite order of generalized Heegner points

摘录 / EXCERPT: Assume that r is odd. Let $p=\mathfrak p\overline{\mathfrak p}$ be a rational prime that splits in $K/\mathbf Q$. Let $H\subset K^{\mathrm{ab}}$ be a finite extension of...

open_problem | open | MathDB 页面 | 原文线索

V2-007444 | MATHDB-395897

Oda’s conjecture on periods of Hilbert modular forms

摘录 / EXCERPT: Let $\Lambda_f := \left\{\int_\gamma \omega_f : \gamma \in H_r(X(\mathbb{C}),\mathbb{Z})\right\}$. Then Λ_f is a lattice in $\mathbb{C}$, and the elliptic curve...

conjecture | open | MathDB 页面 | 原文线索

V2-007445 | MATHDB-395905

Shimura’s conjecture on Petersson inner products of quaternionic modular forms

摘录 / EXCERPT: Let F be a totally real field, and let f be a Hilbert modular newform over F of weights $(k_1,\ldots,k_d)$ with $k_1\equiv\cdots\equiv k_d\pmod 2$. Let B be a quaternion al...

conjecture | open | MathDB 页面 | 原文线索

V2-007446 | MATHDB-395911

Unboundedness of p-Selmer groups in dihedral extensions

摘录 / EXCERPT: Let p be a prime number. As $E/\mathbb{Q}$ ranges over elliptic curves and $F/\mathbb{Q}$ ranges over Galois extensions with Galois group D_{2p}, the order of the p-Selmer gr...

open_problem | open | MathDB 页面 | 原文线索

V2-007447 | MATHDB-396006

Igusa’s monodromy conjecture for p-adic zeta functions

摘录 / EXCERPT: Let f be a non-constant polynomial in n variables over $\Q$. For all but finitely many prime numbers p, if s_0 is a pole of the p-adic Igusa zeta function ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-007448 | MATHDB-396007

Intrinsic definition of the topological zeta function

摘录 / EXCERPT: Find an “intrinsic” definition of the topological zeta function.

open_problem | open | MathDB 页面 | 原文线索

V2-007449 | MATHDB-396107

Minimizers of lattice zeta functions

摘录 / EXCERPT: Let C be a smooth convex body with center at o, let $[U+007F]\lVert\cdot\rVert_C$ be the corresponding norm on E^d, and define ... Determine, for a given $s>d$ or for all sufficiently...

open_problem | open | MathDB 页面 | 原文线索

V2-007450 | MATHDB-396261

Graded cellular basis for cyclotomic Khovanov-Lauda-Rouquier algebras of type A

摘录 / EXCERPT: Suppose that $\mathcal O$ is a commutative integral domain such that e is invertible in $\mathcal O$, $e=0$, or e is a non-zero prime number, and let $\mathscr R_n^\Lambda$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-007451 | MATHDB-396262

Blocks of cyclotomic Khovanov-Lauda-Rouquier algebras are graded symmetric

摘录 / EXCERPT: In the setting of Theorem 1-where O is a commutative integral domain such that e is invertible in O, $e=0$, or e is a non-zero prime number, and...

open_problem | open | MathDB 页面 | 原文线索

V2-007452 | MATHDB-396324

Goldschmidt-Sims conjecture for prime-degree finite amalgams

摘录 / EXCERPT: If both m and n are prime, then there are only finitely many faithful (m,n)-amalgams of finite groups.

conjecture | open | MathDB 页面 | 原文线索

V2-007453 | MATHDB-396358

Families of quantum minors generating torus-invariant prime ideals

摘录 / EXCERPT: What are the families of quantum minors that generate torus-invariant prime ideals?

open_problem | open | MathDB 页面 | 原文线索

V2-007454 | MATHDB-396361

Graded prime ideals of the algebra $R(\alpha,\beta)$

摘录 / EXCERPT: Let k be an uncountable algebraically closed field, let $\alpha,\beta\in k$ be algebraically independent over the prime subfield, and let ... where ... Is the augmentation ideal...

open_problem | open | MathDB 页面 | 原文线索

V2-007455 | MATHDB-396371

Claim about the Chow group of a norm-one group

摘录 / EXCERPT: Let k be a field of characteristic 0, let $p>2$ be a prime, let D be a central simple algebra over k of prime degree p, and let $G=\mathrm{SL}_{1,D}$. The report states t...

open_problem | open | MathDB 页面 | 原文线索

V2-007456 | MATHDB-396418

Diophantine approximation on the Hilbert modular group

摘录 / EXCERPT: Let K be a totally real number field, let V be its set of places, let O be its ring of integers, and let S be a proper subset of the infinite places. With the height H an...

open_problem | open | MathDB 页面 | 原文线索

V2-007457 | MATHDB-396419

Quantitative Diophantine approximation on a simply connected almost simple group

摘录 / EXCERPT: Let G be a connected simply connected almost simple algebraic K-group, and let S be a finite subset of the set V of places of K such that G is isotropic over...

open_problem | open | MathDB 页面 | 原文线索

V2-007458 | MATHDB-396422

Prime hypotenuses in thin Pythagorean-triple orbits

摘录 / EXCERPT: Let $Q(x,y,z)=x^2+y^2-z^2$, let $G=\mathrm{SO}_Q(\mathbb{R})$, let $v_0=(3,4,5)\in\mathbb{Z}^3$ be a primitive Pythagorean triple, and let $\Gamma<G(\mathbb{Z})$ be a discrete, fin...

open_problem | open | MathDB 页面 | 原文线索

V2-007459 | MATHDB-396487

Kuhn’s conjectural bound for the invariant e(P)

摘录 / EXCERPT: Fix a prime p and let P be a finite p-group. Let $C=C(P)$ be the maximal central elementary abelian p-subgroup of P, let $c(P)$ be its rank, and let $e(P)$ be the top deg...

open_problem | open | MathDB 页面 | 原文线索

V2-007460 | MATHDB-396505

Boston-Markin minimal-ramification conjecture

摘录 / EXCERPT: Let G be a finite nontrivial group and let g be a positive integer. A G-extension of $\Q$ ramified at g primes, including the infinite prime, exists if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-007461 | MATHDB-396602

Mod-[U+007F]p homology growth in arithmetic lattices in \(\mathrm{SL}_2(\mathbb C)\)

摘录 / EXCERPT: Let Γ be an arithmetic lattice in $\mathrm{SL}_2(\mathbb C)$ and let p be a prime number. Do the numbers ... tend to zero as $i\to\infty$ for any chain of normal subgroups...

open_problem | open | MathDB 页面 | 原文线索

V2-007462 | MATHDB-396622

Linearity of prime 3-manifold groups

摘录 / EXCERPT: Let N be a closed, oriented prime 3-manifold. Is $\[1m\pi_1(N)\[0m$ linear, i.e. is $\[1m\pi_1(N)\[0m$ isomorphic to a subgroup of $\[1m\mathrm{GL}(n,\mathbb{C})\[0m$ for some...

open_problem | open | MathDB 页面 | 原文线索

V2-007463 | MATHDB-396690

Necessity of the Diophantine condition for a Riesz basis

摘录 / EXCERPT: Let α be an irrational real number and let $I=[a,b)\subset\mathbb T=\mathbb R/\mathbb Z$. Define $\Lambda(\alpha,I)=\{n\in\mathbb Z:a\leq n\alpha<b\}$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-007464 | MATHDB-396781

Classical Mumford-Tate conjecture for the abelian part of a 1-motive

摘录 / EXCERPT: With the notation of the report, for the abelian part A of the fixed 1-motive M over the number field k and a prime $\[1m\ell\[0m$, the classical Mumford-Tate conjecture pr...

conjecture | open | MathDB 页面 | 原文线索

V2-007465 | MATHDB-396801

Inverse problem for planar polynomial vector fields with prescribed invariant curves

摘录 / EXCERPT: Given irreducible, pairwise relatively prime polynomials $f_1,\ldots,f_r$ in the stated planar (complex in general) setting, with $f=f_1\cdots f_r$, find all polynomial vector fiel...

open_problem | open | MathDB 页面 | 原文线索

V2-007466 | MATHDB-396802

Inverse problem for Darboux integrating factors

摘录 / EXCERPT: In the planar (complex, in general) polynomial differential-system setting, let $f_1,\ldots,f_r$ be irreducible and pairwise relatively prime polynomials, let $f=f_1\cdots f_r$, an...

open_problem | open | MathDB 页面 | 原文线索

V2-007467 | MATHDB-396838

Entringer-Tout Conjecture for prime-labeled trees

摘录 / EXCERPT: Every tree is prime, where a graph G of order n is prime if there is a bijection $f:V(G)\to [n]=\{1,\ldots,n\}$ such that the labels assigned to the endpoints of every edge of...

conjecture | open | MathDB 页面 | 原文线索

V2-007468 | MATHDB-396905

Tight colored Tverberg theorem

摘录 / EXCERPT: Let $r\ge 2$ be prime, let $d\ge 1$, and set $N:=(r-1)(d+1)$. Let Δ^N be an N-dimensional simplex whose vertex set is partitioned into $m+1$ color classes...

open_problem | open | MathDB 页面 | 原文线索

V2-007469 | MATHDB-396977

Mazur-Gouvêa conjecture on p-adic analytic families of modular eigenforms

摘录 / EXCERPT: Fix a prime p, an integer N relatively prime to p, a Dirichlet character $\chi:(\mathbb Z/(pN))^*\to\mathbb C^*$, and let S_k^α denote the slope-α subspace of c...

conjecture | open | MathDB 页面 | 原文线索

V2-007470 | MATHDB-397041

Indecomposability of the projective-free Lie module at prime-power degree

摘录 / EXCERPT: Let F be a field of characteristic $p>0$. For $k\geqslant 0$, set $n:=p^k$. Is the projective-free part $\mathrm{Lie}^{\mathrm{pf}}_{p}(n)$ of the Lie module indecomposable?

open_problem | open | MathDB 页面 | 原文线索

V2-007471 | MATHDB-397045

Existence of ordinary-character basic sets

摘录 / EXCERPT: For every finite group G and prime p, does there exist a basic set BS contained in $\operatorname{Irr}(G)$? Equivalently, can one always choose ordinary irreducible character...

open_problem | open | MathDB 页面 | 原文线索

V2-007472 | MATHDB-397048

Cyclic vertices of indecomposable Specht modules

摘录 / EXCERPT: Let F be a field of prime characteristic p, let $\[1m\lambda\u001b[0m$ be a partition of n, and let S^λ be the corresponding indecomposable Specht $F S_n$-module. The...

open_problem | open | MathDB 页面 | 原文线索

V2-007473 | MATHDB-397051

Vertices of an exceptional hook module for a symmetric group

摘录 / EXCERPT: For an odd prime p, let $n=pw$ with $w\equiv 1\pmod p$, and let $\lambda=(n-p+1,1^{p-1})$. Are the vertices of the irreducible symmetric-group module D^λ the Sylow p-su...

open_problem | open | MathDB 页面 | 原文线索

V2-007474 | MATHDB-397163

Zariski density of quantum mapping-class-group representations

摘录 / EXCERPT: Suppose that $g\ge 3$ and $p\ge 5$ is an odd prime. Then $\widetilde{\rho}_p(M_g)$ is a discrete Zariski-dense subgroup of G_p.

open_problem | open | MathDB 页面 | 原文线索

V2-007475 | MATHDB-397179

McCallum-Sharifi generation conjecture

摘录 / EXCERPT: For every odd prime p and every integer $r\ge 1$, let $A_r=\operatorname{Cl}(\mathbb{Q}(\zeta_{p^r}))\{p\}$ and let A_r^{-} be its minus part under complex conjugation. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-007476 | MATHDB-397180

Iwasawa main conjecture for CM fields

摘录 / EXCERPT: Let K/F be a totally imaginary quadratic extension of a totally real number field F, let $p>2$ be a rational prime, and assume that K is p-ordinary, meaning that every prim...

conjecture | open | MathDB 页面 | 原文线索

V2-007477 | MATHDB-397181

Kato’s main conjecture for a modular form

摘录 / EXCERPT: Let $f=\sum_{n\geq 1}a_nq^n\in S_k(\Gamma_0(N),\theta)$ be a normalized eigen-newform, let $F=\mathbb{Q}(a_n:n\geq 1)$, let p be an odd prime with $p\nmid N$, let $v\mid p$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-007478 | MATHDB-397211

Determination of wild-prime Euler factors for hypergeometric motives

摘录 / EXCERPT: For hypergeometric data consisting of disjoint tuples a and b of equal cardinality, satisfying the stated cyclotomic-product condition, and for the associated motive H(a,b;t) with...

open_problem | open | MathDB 页面 | 原文线索

V2-007479 | MATHDB-397214

Smallest prime supporting an even Galois representation unramified outside that prime

摘录 / EXCERPT: What is the smallest prime p for which there exists an absolutely irreducible continuous even Galois representation $\rho:G_{\mathbb Q}\to\mathrm{GL}_2(\mathbb F_p)$ unramified a...

open_problem | open | MathDB 页面 | 原文线索

V2-007480 | MATHDB-397220

Torsion-growth conjecture for Bianchi groups

摘录 / EXCERPT: Let G be a Bianchi group and let H denote hyperbolic 3-space. Then ... where the limit is taken over prime ideals $\mathfrak p$ of residue degree one.

conjecture | open | MathDB 页面 | 原文线索

V2-007481 | MATHDB-397229

Mordell-Weil problem for cubic surfaces

摘录 / EXCERPT: Let K be a field finitely generated over its prime subfield, and let S be a smooth cubic surface over K in $\b[?25l\mathbb{P}^3$. For a subset $B\subseteq S(K)$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-007482 | MATHDB-397231

Computing the admissible pair attached to a supercuspidal local component

摘录 / EXCERPT: Let f be a modular newform of weight $k\ge 2$ and level N, and let p be an odd prime such that the local component $\Pi_{f,p}$ is supercuspidal. Under the parametrization of...

open_problem | open | MathDB 页面 | 原文线索

V2-007483 | MATHDB-397232

Cohen-Lenstra conjecture for p-parts of imaginary quadratic class groups

摘录 / EXCERPT: Fix an odd prime p. For a fixed positive integer g, among the imaginary quadratic fields K such that the p-rank of A_K is g, ordered by discriminant, the probability th...

conjecture | open | MathDB 页面 | 原文线索

V2-007484 | MATHDB-397233

Heuristic frequency conjecture for p-class tower groups of imaginary quadratic fields

摘录 / EXCERPT: Fix an odd prime p. Suppose G is a finite p-group which is a Schur σ-group of generator rank $g\ge 1$ or, more generally, suppose c is a positive integer and G is...

conjecture | open | MathDB 页面 | 原文线索

V2-007485 | MATHDB-397257

PORC behavior of the number of groups of order p^{10}

摘录 / EXCERPT: Is the total number of groups of order p^{10}, as a function of the prime p, a PORC function?

open_problem | open | MathDB 页面 | 原文线索

V2-007486 | MATHDB-397288

Equidistribution for Siegel modular forms with level tending to infinity

摘录 / EXCERPT: Extend the equidistribution results above to the case where the level tends to infinity.

open_problem | open | MathDB 页面 | 原文线索

V2-007487 | MATHDB-397358

Dedekind zeta functions do not determine number fields up to isomorphism

摘录 / EXCERPT: There exist non-isomorphic number fields K and L with $\zeta_K=\zeta_L$.

open_problem | open | MathDB 页面 | 原文线索

V2-007488 | MATHDB-397371

Coordinates with strong rationality properties for height-2 Lubin-Tate deformation spaces

摘录 / EXCERPT: For each prime p, does there exist a coordinate on the Lubin-Tate deformation space of a one-dimensional formal group of height 2 with the strong rationality properties require...

open_problem | open | MathDB 页面 | 原文线索

V2-007489 | MATHDB-397488

Convergence of the Schottky-Klein prime-function product

摘录 / EXCERPT: Given a Schottky group $\Gamma=\langle g_1,\ldots,g_d\rangle$ associated with finitely many disjoint circles and its fundamental region F, and distinct $z,\xi\in F$, does the pro...

open_problem | open | MathDB 页面 | 原文线索

V2-007490 | MATHDB-397562

Self-extension criterion for prime quantum affine representations

摘录 / EXCERPT: For a general Lie algebra (mathfrak g), is a simple object V of the category $\widehat{\mathcal F}$ of finite-dimensional representations of the quantum affine algebra characteri...

open_problem | open | MathDB 页面 | 原文线索

V2-007491 | MATHDB-397581

Brauer-Robinson theorem on \-blocks of unipotent characters of \GL_n(q)

摘录 / EXCERPT: Let ell be a prime number. Assume ell and q are coprime, and let d be the order of q modulo ell. Then chi^{lambda} and chi^{lambda'} are in the same ell-block if and only if lambda...

open_problem | open | MathDB 页面 | 原文线索

V2-007492 | MATHDB-397591

Verification of the inductive conditions for all simple groups and primes

摘录 / EXCERPT: Do the inductive McKay and inductive blockwise Alperin weight conditions hold for every simple group and every prime?

open_problem | open | MathDB 页面 | 原文线索

V2-007493 | MATHDB-397657

Improving the Chalk-Erdős approximating function for coprime inhomogeneous approximation

摘录 / EXCERPT: Let ξ be an irrational real number and let y be a real number. The Chalk-Erdős theorem asserts that, for some absolute constant c, there are infinitely many coprime integer...

open_problem | open | MathDB 页面 | 原文线索

V2-007494 | MATHDB-397658

Erdős's conjecture on perfect powers in primitive arithmetic progressions

摘录 / EXCERPT: For positive integers x,d,k,y,n with $\\gcd(x,d)=1$, $k>3$, and $n\ge 2$, the Diophantine equation ... has no solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-007495 | MATHDB-397682

Equivariant-map criterion for convex equipartitions

摘录 / EXCERPT: For integers $d\ge 2$ and $n\ge 2$, an S_n-equivariant continuous map ... does not exist if and only if n is a prime power.

open_problem | open | MathDB 页面 | 原文线索

V2-007496 | MATHDB-397757

Injectivity of cable operations in localized knot concordance

摘录 / EXCERPT: If p and q are coprime positive integers, then the (p,q)-cable of K is smoothly concordant to the (p,q)-cable of J in a smooth $\mathbb{Z}[1/p]$-homology...

open_problem | open | MathDB 页面 | 原文线索

V2-007497 | MATHDB-397807

Connectedness and birational equivalence of fibers of the period map

摘录 / EXCERPT: We expect that the fibers of the period map p are connected and parametrize birationally equivalent prime Fano fourfolds of degree ten.

open_problem | open | MathDB 页面 | 原文线索

V2-007498 | MATHDB-397821

Constancy of p-rank on Kottwitz-Rapoport strata

摘录 / EXCERPT: In the fixed PEL setup of the report, for a rational prime $p\ne 2$, the p-rank should be constant on all Kottwitz-Rapoport strata if and only if the multichain L is complete.

open_problem | open | MathDB 页面 | 原文线索

V2-007499 | MATHDB-397822

Vector-valued modularity of the arithmetic theta series

摘录 / EXCERPT: Conjecturally, the formal power series $\widehat{\Theta}(\tau)=\sum_{m>0}\widehat{\Theta}(m)\in\widehat{\mathrm{Pic}}(M)[[q]]$ is a vector-valued modular form.

open_problem | open | MathDB 页面 | 原文线索

V2-007500 | MATHDB-397936

The reciprocity conjecture of Khare-Wintenberger

摘录 / EXCERPT: Assume that p is an odd prime, that F is totally real, and, for simplicity, that $[F(\mu_p):F]=2$. Let $F_\infty=F(\mu_{p^\infty})$ be the cyclotomic $\mathbb Z_p$-extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-007501 | MATHDB-397944

Buzzard-Diamond-Jarvis conjecture for totally real fields unramified at p

摘录 / EXCERPT: Let $p>2$ be prime, let F be a totally real field unramified at all places above p, and let $\bar r:G_F\to\operatorname{GL}_2(\mathbb F)$ be a modular Galois representation. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-007502 | MATHDB-397946

Question 1.1: Distribution of Selmer sequences of elliptic curves

摘录 / EXCERPT: Fix a prime p and a global field k. For each elliptic curve E over k, let ... be the short exact sequence Seq_E, and let $\mathcal{E}$ contain one representa...

open_problem | open | MathDB 页面 | 原文线索

V2-007503 | MATHDB-397947

Distributional model for Selmer and Shafarevich-Tate sequences

摘录 / EXCERPT: Conjecture 3.2. Fix a global field k and a prime p. In the model with $V=\mathbb{Z}_p^{2n}$, $W=\mathbb{Z}_p^n\times\{0\}$, and random $Z\in\operatorname{OGr}_n(\mathbb{Z}_p)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-007504 | MATHDB-397988

The L-space/left-orderability conjecture for prime 3-manifolds

摘录 / EXCERPT: Conjecture ([BGW]). Let W be a closed, connected, orientable, prime 3-manifold. Then W is LO if and only if it is NLS, where LO means that $\[4m\pi_1(W)\u001b[0m$ is left-ord...

conjecture | open | MathDB 页面 | 原文线索

V2-007505 | MATHDB-398001

Conjugation action in non-Abelian Iwasawa theory

摘录 / EXCERPT: Fix an odd prime number p. Let $k_\\infty=\\mathbb{Q}(\\mu_{p^\\infty})$, let $\\widetilde{k}$ be the maximal pro-p extension of $k_\\infty$ unramified outside p, and set ... T...

open_problem | open | MathDB 页面 | 原文线索

V2-007506 | MATHDB-398047

The Herman conjecture

摘录 / EXCERPT: In the neighbourhood of a diophantine elliptic fixed point, a real analytic symplectomorphism has a positive measure set of invariant tori.

conjecture | open | MathDB 页面 | 原文线索

V2-007507 | MATHDB-398119

Chabauty-Coleman bound at a prime of bad reduction

摘录 / EXCERPT: Let X be a smooth projective geometrically irreducible curve of genus $g\ge 2$ over $\mathbb{Q}$, and assume that the Mordell-Weil rank r of the Jacobian of X is less than...

open_problem | open | MathDB 页面 | 原文线索

V2-007508 | MATHDB-398331

Finiteness of torsion-prime sets for Cremona and affine automorphism groups

摘录 / EXCERPT: Are the sets $\operatorname{Tors}(\mathrm{Cr}_n)$, $\operatorname{Tors}(\operatorname{Aut}\mathbb{A}^n)$, and $\operatorname{Tors}(\operatorname{Aut}^*\mathbb{A}^n)$ finite?

open_problem | open | MathDB 页面 | 原文线索

V2-007509 | MATHDB-398332

Classification of completely prime primitive ideals outside type A

摘录 / EXCERPT: Classifying the completely prime primitive ideals of $U(\mathfrak g)$ remains unresolved outside type A.

open_problem | open | MathDB 页面 | 原文线索

V2-007510 | MATHDB-398468

Goldschmidt-Sims conjecture for regular trees

摘录 / EXCERPT: For a prime number $d\ge 3$, there are, up to conjugacy, only finitely many discrete subgroups $\Gamma<\operatorname{Aut}(T_d)$ such that $\Gamma\backslash T_d$ is a geometric edge...

conjecture | open | MathDB 页面 | 原文线索

V2-007511 | MATHDB-398483

Upper bound for Euclidean minima of abelian fields of odd prime-power conductor

摘录 / EXCERPT: Let p be an odd prime number, and let K be an abelian number field of conductor p^r. If $r\ge 2$, then ... where $n=[K:\mathbb{Q}]$, $M(K)$ is the Euclidean minimum of K, and...

open_problem | open | MathDB 页面 | 原文线索

V2-007512 | MATHDB-398487

Inverse Galois realization of finite orthogonal groups

摘录 / EXCERPT: Fix an odd integer $n\ge 5$, a prime $\ell\ge 5$, and a non-degenerate quadratic space (V,q) over $\mathbb{F}_{\ell}$ with $\dim_{\mathbb{F}_{\ell}}V=n$. Let $O(V)$ be the group...

open_problem | open | MathDB 页面 | 原文线索

V2-007513 | MATHDB-398492

Detection of minimized decomposition and inertia groups in pro-\ll Galois groups

摘录 / EXCERPT: Assume that ℓ is an odd prime. Let K be a field with $\operatorname{char}K\ne\ell$, let n be a positive integer or ∞, and write ... For $\Sigma\le G_K^{a,n}$ and all...

open_problem | open | MathDB 页面 | 原文线索

V2-007514 | MATHDB-398496

Euclidean minima of abelian number fields of odd prime-power conductor

摘录 / EXCERPT: Let p be an odd prime number, and let K be an abelian number field of conductor p^r. If $r\ge 2$, then ... where $n=[K:\mathbb{Q}]$ and d_K is the absolute value of the discr...

open_problem | open | MathDB 页面 | 原文线索

V2-007515 | MATHDB-398550

Rational points on the non-split Cartan modular curves

摘录 / EXCERPT: Show that for every prime number $p>37$, the only rational points on $X_{\\mathrm{ns}}^{+}(p)$ are the cusps and the CM-points.

open_problem | open | MathDB 页面 | 原文线索

V2-007516 | MATHDB-398554

Explicit generators of finite-index subgroups in Jacobians over function fields

摘录 / EXCERPT: Let p be a prime, let $F=\mathbb F_p(t)$, and let $r>0$ be an integer with $p\nmid r$. Let C be the smooth projective curve over F associated to ... let...

open_problem | open | MathDB 页面 | 原文线索

V2-007517 | MATHDB-398555

Polynomial formula for the Shafarevich-Tate group order

摘录 / EXCERPT: In the setup where p is prime, $F=\mathbb{F}_p(t)$, $r>0$ is not divisible by p, $d=p^f+1$ with $r\mid d$, $K=\mathbb{F}_p(\mu_d,u)$ and $u^d=t$, and J is the Jacobian of the...

open_problem | open | MathDB 页面 | 原文线索

V2-007518 | MATHDB-398556

Polynomial formula for the Tate-Shafarevich group in the Legendre case

摘录 / EXCERPT: In the setup above with p prime, $d=p^f+1$, $2\mid d$, $K=\mathbb{F}_p(\mu_d,u)$ with $u^d=t$, and J the Jacobian of $y^2=x(x+1)(x+t)$, one expects that there is a polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-007519 | MATHDB-398565

Iwasawa’s \bc = 0 conjecture

摘录 / EXCERPT: Let K be a number field and let l be a prime. Let K_∞/K be the cyclotomic $\mathbb{Z}_l$-extension, and let K_n/K be its degree l^n subextension. Iwasawa’s $\mu=0$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-007520 | MATHDB-398566

Computing zeta functions of low genus curves in average polynomial time

摘录 / EXCERPT: Let $X/\mathbb{Q}$ be a smooth projective curve of genus g, and let $X_p/\mathbb{F}_p$ denote its reduction modulo a prime p. For every prime p of good reduction, define...

open_problem | open | MathDB 页面 | 原文线索

V2-007521 | MATHDB-398620

Typical measure-theoretic growth of the Grigorchuk groups

摘录 / EXCERPT: Let p be a prime, let $\Omega_p=\{0,1,\ldots,p\}^{\mathbb N}$, and let $\Omega_{p,\infty}\subseteq\Omega_p$ consist of the sequences in which every symbol $0,1,\ldots,p$ occurs i...

open_problem | open | MathDB 页面 | 原文线索

V2-007522 | MATHDB-398650

Equidistribution of angles of prime ideals in \(\mathbf Q(i)\)

摘录 / EXCERPT: For $K=\mathbf Q(i)$, every prime ideal $\mathfrak p$ arising from a rational prime $p\equiv1\pmod 4$ has the form $(a+ib)$, with uniquely determined $a>0$ and $b>0$, and defines a...

open_problem | open | MathDB 页面 | 原文线索

V2-007523 | MATHDB-398684

Serre’s uniformity problem for the nonsplit Cartan modular curve

摘录 / EXCERPT: For every prime number $p>37$, the rational points on the modular curve $X^+_{\mathrm{ns}}(p)$ consist only of the cusps and the CM-points.

open_problem | open | MathDB 页面 | 原文线索

V2-007524 | MATHDB-398690

Highly biased prime number races for elliptic curves

摘录 / EXCERPT: For prime number races associated with the summatory function of the Frobenius traces $a_p(E)$ of elliptic curves over $\mathbb{Q}$, is it possible to find races that are highly bi...

open_problem | open | MathDB 页面 | 原文线索

V2-007525 | MATHDB-398693

Asymptotic formulae for nonclassical moments of the Riemann zeta function

摘录 / EXCERPT: Upgrade Theorem 1 to give an asymptotic formula for the moments $\displaystyle \int_T^{2T}\left|\zeta\!\left(\tfrac12+it\right)\right|^{2k}\,dt$ in any case when $k\ne 0,1,2$.

open_problem | open | MathDB 页面 | 原文线索

V2-007526 | MATHDB-398696

Reconciling rank conditions for prime solutions with Schmidt-Birch conditions

摘录 / EXCERPT: Can the rank conditions in the asymptotic formula for prime solutions of systems of integral polynomials be brought into agreement with the corresponding conditions of Schmidt and...

open_problem | open | MathDB 页面 | 原文线索

V2-007527 | MATHDB-398699

Consecutive prime-gap ratios

摘录 / EXCERPT: Let p_n denote the nth prime and let $d_n=p_{n+1}-p_n$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-007528 | MATHDB-398703

Intersections of polynomial images of multiplicative subgroups

摘录 / EXCERPT: For a prime p, estimate the size of $f(G)\cap g(H)$, namely $\#(f(G)\cap g(H))$, where $f,g\in\mathbb F_p[X]$ and $G,H\subseteq\mathbb F_p^*$ are multiplicative subgroups.

open_problem | open | MathDB 页面 | 原文线索

V2-007529 | MATHDB-398706

Bounded-error quantitative twin-prime conjecture

摘录 / EXCERPT: Let $\pi_2(x):=\#\{p\le x: p+2\text{ is prime}\}$. The report asks whether the stronger quantitative form ... holds, and requests a proof or disproof.

conjecture | open | MathDB 页面 | 原文线索

V2-007530 | MATHDB-398841

Finitude of quadratic coefficient fields of modular forms

摘录 / EXCERPT: Are there finitely many or infinitely many quadratic fields that occur as coefficient fields Q_f of modular forms f of arbitrary level and arbitrary weight?

open_problem | open | MathDB 页面 | 原文线索

V2-007531 | MATHDB-398842

Control of coefficient fields of modular forms

摘录 / EXCERPT: Control or predetermine the coefficient field Q_f associated with a modular form f.

open_problem | open | MathDB 页面 | 原文线索

V2-007532 | MATHDB-398843

Realization of projective symplectic groups over finite fields as Galois groups

摘录 / EXCERPT: Theorem 1 (Arias-de-Reyna, Dieulefait, Shin, Wiese). Let $n,d\in\mathbb{N}$. There exists a positive-density set $\mathcal{L}$ of rational primes such that, for every prime...

open_problem | open | MathDB 页面 | 原文线索

V2-007533 | MATHDB-398853

Real-variable proof of the equivalence of the Nyman-Báez-Duarte criterion and a quantitative prime-counting estimate

摘录 / EXCERPT: Is there a real-variable proof of the equivalence between ... and the assertion that, for every positive δ, ... can be achieved by a suitable choice of coefficients c_α,...

open_problem | open | MathDB 页面 | 原文线索

V2-007534 | MATHDB-398959

The sweep map on rational Catalan paths

摘录 / EXCERPT: Let (m,n) be a pair of relatively prime positive integers. Let $\operatorname{Grid}(m,n)$ be the $n\times m$ grid of labelled squares whose upper-left square has label...

open_problem | open | MathDB 页面 | 原文线索

V2-007535 | MATHDB-398979

Existence of a p-adic Wollmilchsau

摘录 / EXCERPT: For every prime p, the Teichmüller curve defined by the Wollmilchsau does not contain any p-adic origamis.

open_problem | open | MathDB 页面 | 原文线索

V2-007536 | MATHDB-399029

Goldie-type quotient rings for Hopf-module algebras beyond the center-finite case

摘录 / EXCERPT: Let H be a Hopf algebra over a field and let A be a noetherian H-semiprime H-module algebra. Without assuming that A is module-finite over its center, must A have an ar...

open_problem | open | MathDB 页面 | 原文线索

V2-007537 | MATHDB-399039

Gorensteinness of H-simple module algebras

摘录 / EXCERPT: Suppose that an H-module algebra A is H-simple, noetherian, and module-finite over its center, but not necessarily semiprime. Is it possible to conclude that A is Gorenstei...

open_problem | open | MathDB 页面 | 原文线索

V2-007538 | MATHDB-399040

Existence of finite H-orbits of prime ideals beyond the center-finite case

摘录 / EXCERPT: Can the existence of H-orbits of prime ideals be proved for some classes of H-module algebras other than the algebras module-finite over their centers, e.g., for PI algebras?

open_problem | open | MathDB 页面 | 原文线索

V2-007539 | MATHDB-399061

Injective lifting from pairs of elliptic modular forms

摘录 / EXCERPT: For integers $k\ge 3$ and even integers $j\ge 0$, there exists an injective lifting ...

open_problem | open | MathDB 页面 | 原文线索

V2-007540 | MATHDB-399062

Ibukiyama’s Shimura-type correspondence for Siegel modular forms of half-integral weight

摘录 / EXCERPT: For integers k,j with $k\ge 3$ and j even, there is a Hecke-equivariant isomorphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-007541 | MATHDB-399064

Restriction of meromorphic quadratic-form Poincaré series to Bianchi modular forms

摘录 / EXCERPT: For a discriminant $-D<0$, define the meromorphic quadratic-form Poincaré series ... Is $f_{k,-D}$ obtained by restricting a Bianchi modular form?

open_problem | open | MathDB 页面 | 原文线索

V2-007542 | MATHDB-399146

Generalized Beilinson-Flach classes and Rankin-Selberg Euler systems

摘录 / EXCERPT: Theorem (Kings-Loeffler-Zerbes). Let f and g be modular forms of weights $k+2$ and $k'+2$, respectively, and let j be an integer with $0\leq j\leq\min(k,k')$. Then there exis...

open_problem | open | MathDB 页面 | 原文线索

V2-007543 | MATHDB-399149

Generic non-triviality of logarithms of Heegner points modulo p

摘录 / EXCERPT: Let $p>3$ be an odd prime, let $K/\mathbb{Q}$ be an imaginary quadratic extension with ring of integers $\mathcal O$, and assume that p splits in K (ord). Let N...

open_problem | open | MathDB 页面 | 原文线索

V2-007544 | MATHDB-399228

Boston-Bush-Hajir distribution conjecture for odd-\p class-tower groups

摘录 / EXCERPT: For an odd prime p and a finite Schur σ-group H, the proportion of imaginary quadratic fields K for which $\operatorname{Gal}(K_{p,\mathrm{ur}}/K)$ is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-007545 | MATHDB-399237

Which closed smooth manifolds are isolated?

摘录 / EXCERPT: Which closed smooth manifolds are isolated, where a closed smooth manifold is called isolated if, for some prime p, it admits a $\mathbb{Z}/p$-action ρ such that the associa...

open_problem | open | MathDB 页面 | 原文线索

V2-007546 | MATHDB-399240

Regular points of Vogel’s plane for the universal adjoint character

摘录 / EXCERPT: Determine the points in Vogel’s plane for which the universal adjoint character ... is regular at finite x. For the displayed Diophantine family, with $k,n,m\ne -1$, the condition...

open_problem | open | MathDB 页面 | 原文线索

V2-007547 | MATHDB-399285

Topological Tverberg conjecture for non-prime-power numbers of parts

摘录 / EXCERPT: For every integer $d\ge 2$ and every integer $r\ge 2$ that is not a prime power, let $N=(r-1)(d+1)$. For every continuous map $f:\Delta_N\to\mathbb{R}^d$, do there exist points...

conjecture | open | MathDB 页面 | 原文线索

V2-007548 | MATHDB-399310

The Burnside space of the p-completed classifying space

摘录 / EXCERPT: Theorem 2 (The Burnside space of $B G^{\p}$). Let p be a prime and let G be a finite group. Then ... where the product runs over G-conjugacy classes $[Q]$ of p-subgroups Q...

open_problem | open | MathDB 页面 | 原文线索

V2-007549 | MATHDB-399358

Hardy-Littlewood prime-tuple analogue over finite polynomial rings

摘录 / EXCERPT: For any positive integers n and k, there exists a constant $c_{n,k}$ depending only on n and k such that, for every prime power q and any distinct...

open_problem | open | MathDB 页面 | 原文线索

V2-007550 | MATHDB-399362

Characterization of diophantine henselian valuation rings by the residue field

摘录 / EXCERPT: Let F be a field. An equicharacteristic henselian valuation ring with residue field F is diophantine if and only if, for every $F_1,F_2\equiv F$ and every non-trivial valuation...

open_problem | open | MathDB 页面 | 原文线索

V2-007551 | MATHDB-399367

Uniform definability of the canonical p-henselian valuation

摘录 / EXCERPT: Fix a prime p. Let $K(p)$ be the compositum of all Galois extensions of K of p-power degree, and let v_K^p denote the canonical p-henselian valuation on a field...

open_problem | open | MathDB 页面 | 原文线索

V2-007552 | MATHDB-399498

The Chromatic Splitting Conjecture at $n=p=2$

摘录 / EXCERPT: The strong form of the Chromatic Splitting Conjecture at chromatic level $n=2$ and prime $p=2$ cannot hold; the report presents this as refuted by Beaudry’s work.

conjecture | open | MathDB 页面 | 原文线索

V2-007553 | MATHDB-399551

McKay conjecture for the prime 2

摘录 / EXCERPT: For every finite group G and every Sylow 2-subgroup S of G, G and its normalizer $N_G(S)$ have the same number of irreducible characters of odd degree.

conjecture | open | MathDB 页面 | 原文线索

V2-007554 | MATHDB-399552

Dudas-Malle conjecture for unipotent parts of projective characters

摘录 / EXCERPT: In the report’s setup, let $G_n=GL(n,q)$, where $ \ell$ is a prime not dividing q. For the Hecke-algebra characters $ psi$, Kazhdan-Lusztig basis elements C_w, and correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-007555 | MATHDB-399555

Characters of distinct prime-coprime degrees

摘录 / EXCERPT: Theorem 7. Let G be a finite group, let $p\ne q$ be primes, and suppose that $\{p,q\}\ne\{5,7\}$. Then $\operatorname{Irr}_{p'}(G)=\operatorname{Irr}_{q'}(G)$ if and only if G...

open_problem | open | MathDB 页面 | 原文线索

V2-007556 | MATHDB-399556

Natural bijection for odd-prime McKay characters

摘录 / EXCERPT: Let G be a finite group, let $p>2$ be an odd prime, and let $P\in\operatorname{Syl}_p(G)$. Suppose that $N_G(P)=P$. Then, for any $\chi\in\operatorname{Irr}_{p'}(G)$, the restric...

open_problem | open | MathDB 页面 | 原文线索

V2-007557 | MATHDB-399563

Conjectured sharp bounds for the order of a finite group from its Brauer character degrees

摘录 / EXCERPT: Let p be a prime and let G be a finite group with $O_p(G)=1$. If $\beta(1)_p\le p$ for all $\beta\in\operatorname{IBr}_p(G)$, then the bounds in Theorem 5 should be sharpenable...

conjecture | open | MathDB 页面 | 原文线索

V2-007558 | MATHDB-399574

James's conjecture on rank polynomials for unipotent Specht modules

摘录 / EXCERPT: Let 0 0q be a prime power, let $G=\mathrm{GL}_n(q)$, let K be a field of characteristic different from the defining prime and containing a primitive p-th root of unity, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-007559 | MATHDB-399623

Splitting of the Künneth-Landweber sequence in oriented bordism

摘录 / EXCERPT: Let p be an odd prime and $n>1$. Writing $\widetilde{\Omega}_*$ for reduced oriented bordism, the iterated Künneth map ... has an Ω_*-linear retraction ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-007560 | MATHDB-399624

Positive scalar curvature for p-atoral manifolds with elementary abelian fundamental group

摘录 / EXCERPT: Let p be an odd prime. A closed oriented manifold M^d is called p-atoral if, for every $c_1,\ldots,c_d\in H^1(B\pi_1(M);\mathbb F_p)$, one has ... where $f:M\to B\pi_1(M)$ is t...

open_problem | open | MathDB 页面 | 原文线索

V2-007561 | MATHDB-399628

Existence of unstable Adams operations for p-local compact groups

摘录 / EXCERPT: Given a prime p, a p-local compact group $G=(S,\mathcal F,\mathcal L)$, and a prescribed p-adic degree ζ, does there exist an unstable Adams operation on G of degree...

open_problem | open | MathDB 页面 | 原文线索

V2-007562 | MATHDB-399632

Realization problem for modules of constant Jordan type

摘录 / EXCERPT: Let p be a prime, let $E\cong (\mathbb Z/p)^r$ be an elementary abelian p-group of rank r, and let k be an algebraically closed field of characteristic p. Determine which...

open_problem | open | MathDB 页面 | 原文线索

V2-007563 | MATHDB-399640

Extension criterion for one-dimensional modules and endotrivial modules

摘录 / EXCERPT: Let k be an algebraically closed field of prime characteristic p, let G be a finite group whose order is divisible by p, let S be a Sylow p-subgroup of G, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-007564 | MATHDB-399727

Berger conjecture for simply connected manifolds with all geodesics closed

摘录 / EXCERPT: If M is a simply connected Riemannian manifold all of whose geodesics are closed, then all the prime geodesics have the same length.

conjecture | open | MathDB 页面 | 原文线索

V2-007565 | MATHDB-399735

Explicit Chern class formula on the Bloch group

摘录 / EXCERPT: Let F be the field occurring in the construction, let p be a prime with $p\nmid w_F:=|\mu_F|$, let ζ_p be a primitive pth root of unity, and let H be the maximal exte...

open_problem | open | MathDB 页面 | 原文线索

V2-007566 | MATHDB-399738

Completeness of the computation of elliptic curves of prime conductor up to 10^{12}

摘录 / EXCERPT: The report states that the described methods find models for all elliptic curves over $\mathbb{Q}$ with prime conductor at most 10^{10}, and conjecturally do so for all such curv...

open_problem | open | MathDB 页面 | 原文线索

V2-007567 | MATHDB-399741

Distribution of Shafarevich-Tate groups of elliptic curves of fixed rank

摘录 / EXCERPT: Let $r\in\mathbb Z_{\ge 0}$ and let p be prime. Ranging over finite abelian p-groups G equipped with a nondegenerate alternating pairing $[\ ,\ ]$, what is the probability ……

open_problem | open | MathDB 页面 | 原文线索

V2-007568 | MATHDB-399747

Rational points on the curves $C_p: 5y^2=4x^p+1$

摘录 / EXCERPT: Let p be an odd prime. If $p\le 19$, or, assuming GRH, if $p\le 53$, then the only rational points on $C_p: 5y^2=4x^p+1$ are ∞, $(1,1)$, and $(1,-1)$.

open_problem | open | MathDB 页面 | 原文线索

V2-007569 | MATHDB-399928

Counterexamples to the topological Tverberg conjecture

摘录 / EXCERPT: Let $r\ge 6$ be an integer that is not a prime power, let $k\ge 3$ be an integer, and let $N=(r-1)(rk+2)$. Then there exists a continuous map $F:\Delta_N\to\mathbb{R}^{rk+1}$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-007570 | MATHDB-399929

Counterexample to an r-fold van Kampen-Flores statement

摘录 / EXCERPT: Let $r \ge 6$ be an integer that is not a prime power and let $k \ge 3$ be an integer. Then, for any N, there exists a continuous map $f:\Delta^N \to \mathbb{R}^{rk}$ such that,...

open_problem | open | MathDB 页面 | 原文线索

V2-007571 | MATHDB-399958

Kodaira dimension and rationality of complex ball quotients from four-folds of K3[2]-type

摘录 / EXCERPT: For the quasi-projective moduli spaces of four-folds of $K3^{[2]}$-type with a non-symplectic automorphism of prime order $p\ge 3$, represented as complex ball quotients...

open_problem | open | MathDB 页面 | 原文线索

V2-007572 | MATHDB-400028

Simultaneous prime avoidance in Gabber's altered desingularization

摘录 / EXCERPT: Can Gabber's theorem be strengthened so that the degree $\deg(f)$ of the desingularizing alteration is not divisible by two (or more) fixed primes that are invertible on X?

open_problem | open | MathDB 页面 | 原文线索

V2-007573 | MATHDB-400031

Non-archimedean Ax-Lindemann theorem

摘录 / EXCERPT: Let p be a prime number and let F be a finite extension of $ \mathbb{Q}_p$. For $1\leq i\leq n$, let Γ_i be an arithmetic Schottky subgroup of $\operatorname{PGL}(2,F)$...

open_problem | open | MathDB 页面 | 原文线索

V2-007574 | MATHDB-400212

A two-dimensional Furstenberg ×2 ×3 density question

摘录 / EXCERPT: Let $1,x,y$ be multiplicatively independent over $\\mathbb{Q}$. Is it true that, if a,b are relatively prime integers, the set ... is dense in $[0,1]^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-007575 | MATHDB-400219

Extension of the infimum property of Diophantine-approximation spectra

摘录 / EXCERPT: Does the following property, proved in the report for $n=3$, extend to every dimension $n>3$: for every positive integer m and every linear map $T:\mathbb{R}^n\to\mathbb{R}^m$, t...

open_problem | open | MathDB 页面 | 原文线索

V2-007576 | MATHDB-400265

Paramodular modularity of the weight-two form at level 713

摘录 / EXCERPT: For the case $k=2$, $N=713$ on F_1, the corresponding antisymmetric paramodular form is conjecturally modular with respect to a known abelian surface $A/\Q$ of rank 1 and condu...

open_problem | open | MathDB 页面 | 原文线索

V2-007577 | MATHDB-400268

Geometric meaning of the modular form F_Λ

摘录 / EXCERPT: Understand a geometric meaning of the modular form F_Λ. In [5], [6], for some M, F_Λ appears as the elliptic genus of some bundle over a K3 surface. Is a similar...

open_problem | open | MathDB 页面 | 原文线索

V2-007578 | MATHDB-400269

Finiteness of lattices carrying reflective modular forms of bounded vanishing order

摘录 / EXCERPT: For every fixed natural number $N>0$, there are, up to scaling, only finitely many integral lattices L of signature $(2,n)$ with $n\ge 4$ that carry a reflective modular form of...

open_problem | open | MathDB 页面 | 原文线索

V2-007579 | MATHDB-400431

Uniform rationality of graded ideal zeta functions

摘录 / EXCERPT: For every $c\in\mathbb N$ and $d\in\mathbb N_{\ge 2}$, there exists a rational function $W^{\mathrm{gr}}_{c,d}(X,Y)\in\mathbb Q(X,Y)$ such that, for almost all primes p and every...

open_problem | open | MathDB 页面 | 原文线索

V2-007580 | MATHDB-400435

Higman’s conjecture on conjugacy classes of unitriangular groups

摘录 / EXCERPT: For each fixed n, the number of conjugacy classes in $U_n(q)$, as a function of q (where q is a power of a prime p), is an integral polynomial in q. Here $U_n(q)$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-007581 | MATHDB-400439

Spectral halo conjecture, Newton-polygon scaling

摘录 / EXCERPT: Let p be a prime, let N be a positive integer coprime to p, and let $\Gamma=\Gamma_1(N)\cap\Gamma_0(p)$. Set $\Lambda=\mathbf Z_p[[\mathbf Z_p^\times]]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-007582 | MATHDB-400526

Contact-locus characterization for Nash adjacencies

摘录 / EXCERPT: Let p_{i_0} be any free point among the infinitely near points $[U+007F]p_i[U+007F]_{i\in I}$ blown up in order to obtain the minimal model of a prime divisor E. We conjecture that ... for a c...

open_problem | open | MathDB 页面 | 原文线索

V2-007583 | MATHDB-400527

Valuative characterization of deformations associated with divisors over $(\mathbb{C}^2,0)$

摘录 / EXCERPT: Let E and F be divisors above $O\in\mathbb{C}^2$, and let S be the minimal model where $E+F$ appear. For a divisor $E=\sum_i a_iD_i$ with prime components D_i, define...

open_problem | open | MathDB 页面 | 原文线索

V2-007584 | MATHDB-400547

Removing the \log log\ factor in multiplicative Diophantine approximation

摘录 / EXCERPT: It is known that the sets ... and ... have full Hausdorff dimension. Can the factor $\log\log q$ be dropped in these assertions?

open_problem | open | MathDB 页面 | 原文线索

V2-007585 | MATHDB-400570

Diophantine model in an HTP-generic subring of \(\mathbb{Q}\)

摘录 / EXCERPT: Does there exist an HTP-generic subring of $\mathbb{Q}$ within which there is a diophantine model of $(\mathbb{Z}, +, \cdot)$?

open_problem | open | MathDB 页面 | 原文线索

V2-007586 | MATHDB-400573

Decidability of the existential theory of \(\mathbb{F}_q[t]\) with powers of \(t\)

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, with $q=p^n$ for a prime p. Is the existential theory of $\mathbb{F}_q[t]$ in the language of addition, divisibility, and the predicate for...

open_problem | open | MathDB 页面 | 原文线索

V2-007587 | MATHDB-400575

Irreducibility of polynomials over number fields is diophantine

摘录 / EXCERPT: Let K be a number field. For fixed $r,d\ge 0$, the set ... is diophantine.

open_problem | open | MathDB 页面 | 原文线索

V2-007588 | MATHDB-400576

Finite-sum characterization of the p-adic Kochen ring

摘录 / EXCERPT: For a prime number p, define ... Is there some integer n_p such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-007589 | MATHDB-400577

DPRM analogue for rings of integers of number fields

摘录 / EXCERPT: Is it true that, for every number field K, the Diophantine sets of its ring of integers O_K are exactly the listable sets of O_K?

open_problem | open | MathDB 页面 | 原文线索

V2-007590 | MATHDB-400584

Diophantine definability of \(\mathbb{F}_q[t]\) in \(\mathbb{F}_q(t)\)

摘录 / EXCERPT: Let q be a power of a prime. Is the ring $\mathbb{F}_q[t]$ diophantine over the rational function field $\mathbb{F}_q(t)$?

open_problem | open | MathDB 页面 | 原文线索

V2-007591 | MATHDB-400589

Two-quantifier Diophantine definition of \(\mathbb{N}\) in \(\mathbb{Z}\)

摘录 / EXCERPT: Find a 2-quantifier Diophantine definition of $\mathbb{N}$ in $\mathbb{Z}$.

open_problem | open | MathDB 页面 | 原文线索

V2-007592 | MATHDB-400593

Diophantine definability of degree equality in $\mathbb{C}[t]$

摘录 / EXCERPT: Consider the binary relation E on $\mathbb{C}[t]$ defined by ... Is E Diophantine (or definable) in $\mathbb{C}[t]$ over the ring language with parameters?

open_problem | open | MathDB 页面 | 原文线索

V2-007593 | MATHDB-400595

Decidability of the existential theory of \(\mathbb{F}_p(t)\)

摘录 / EXCERPT: For a fixed prime p, is $\operatorname{Th}_{\exists}(\mathbb{F}_p(t),L_{\mathrm{ring}})$ decidable?

open_problem | open | MathDB 页面 | 原文线索

V2-007594 | MATHDB-400618

Is the map \(\pi_S\) an isomorphism for sufficiently large sets of primes?

摘录 / EXCERPT: Let F be a global field, and let S contain all archimedean primes, at least two primes in total, and at least one nonarchimedean prime. For sufficiently large S, the report s...

open_problem | open | MathDB 页面 | 原文线索

V2-007595 | MATHDB-400620

Cuspidal cohomology for Gaussian congruence subgroups at split primes

摘录 / EXCERPT: Let $\Gamma:=\mathrm{SL}_2(\mathbb Z[i])$, and let p be a rational prime splitting as $p=\mathfrak y\cdot\overline{\mathfrak y}$ in $\mathbb Z[i]$, with $p\equiv5\pmod 8$. Let th...

open_problem | open | MathDB 页面 | 原文线索

V2-007596 | MATHDB-400628

Squarefree Order Conjecture for reductions of elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve of conductor N_E. For every prime $p\nmid N_E$, let $E_p/\mathbb{F}_p$ be the reduction of E modulo p. Then there exists an explicit c...

conjecture | open | MathDB 页面 | 原文线索

V2-007597 | MATHDB-400629

Andrade-Keating moment conjecture for quadratic Dirichlet L-functions over function fields

摘录 / EXCERPT: For fixed q (in the report, q is prime with $q\equiv 1\pmod 4$), as $g\to\infty$, conjecturally ... where $\mathcal H_{2g+1}$ is the ensemble of monic square-free polynomials of...

conjecture | open | MathDB 页面 | 原文线索

V2-007598 | MATHDB-400632

Cancellation in bilinear sums of normalized Kloosterman sums

摘录 / EXCERPT: For a prime $q\to\infty$, let $\mathcal M$ and $\mathcal N$ be intervals modulo q with cardinalities M and N, and let α_m,β_n satisfy $|\alpha_m|,|\beta_n|<1$. Def...

open_problem | open | MathDB 页面 | 原文线索

V2-007599 | MATHDB-400640

Linear Independence Conjecture for the ordinates of zeta zeros

摘录 / EXCERPT: The positive ordinates of the nontrivial zeros of the Riemann zeta-function $\zeta(s)$ are linearly independent over the rational numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-007600 | MATHDB-400666

Cohomology of level subgroups of the mapping class group

摘录 / EXCERPT: Let $\operatorname{Mod}_g(\ell)$ be the level-ℓ congruence subgroup of the mapping class group, for $g,\ell\ge 2$. If p is a prime dividing ℓ, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-007601 | MATHDB-400685

Regularity of non-degenerate homogeneous prime ideals versus multiplicity

摘录 / EXCERPT: Does there exist a bound on the regularity of non-degenerate homogeneous prime ideals purely in terms of the multiplicity?

open_problem | open | MathDB 页面 | 原文线索

V2-007602 | MATHDB-400695

Criterion for Gorenstein monomial curves of length five

摘录 / EXCERPT: Let $a_1,\dots,a_5$ be relatively prime positive integers defining a Gorenstein, non-complete-intersection monomial curve in $\mathbb{A}^5$. Can one give a criterion for such Goren...

open_problem | open | MathDB 页面 | 原文线索

V2-007603 | MATHDB-400745

Zagier’s Quantum Modularity Conjecture for colored Jones polynomials

摘录 / EXCERPT: For a knot $K\subset S^3$, let $J_{K,N}(q)\in\mathbb Z[q^{\pm1}]$ be its colored Jones polynomials, and for coprime integers a,c with $c>0$ define...

conjecture | open | MathDB 页面 | 原文线索

V2-007604 | MATHDB-400763

Green’s arithmetic triangle removal problem

摘录 / EXCERPT: The report states that Green’s problem is solved. For a prime p, let C_p be the constant defined in the report. If $0<\epsilon\leq 1/2$ and $\delta=(\epsilon/3)^{C_p}$, then wh...

open_problem | open | MathDB 页面 | 原文线索

V2-007605 | MATHDB-400764

Tower-type bound for Green’s popular progression-differences theorem

摘录 / EXCERPT: For a prime p, let $n_p(\varepsilon)$ be the least positive integer such that, for every $n\ge n_p(\varepsilon)$ and every subset $A\subseteq\mathbb{F}_p^n$ of density α,...

open_problem | open | MathDB 页面 | 原文线索

V2-007606 | MATHDB-401105

Herman’s analytic KAM-stability conjecture for elliptic fixed points

摘录 / EXCERPT: For analytic Hamiltonians, KAM stability holds in a neighborhood of an elliptic fixed point if its frequency vector is assumed to be Diophantine.

conjecture | open | MathDB 页面 | 原文线索

V2-007607 | MATHDB-401212

Differentiability of volumes along a prime divisor for big Kähler classes

摘录 / EXCERPT: Let X be a compact Kähler manifold, let $\alpha\in H^{1,1}(X,\mathbb{R})$ be big, and let Y be a prime divisor intersecting the Kähler locus $K(\alpha)$. Then ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-007608 | MATHDB-401221

Prime-value characterization for multiplicative functions with small partial sums

摘录 / EXCERPT: Let f be a multiplicative function satisfying $|\Lambda_f|\le D\Lambda$ and ... If $A>D$, then there exist real numbers $\gamma_1,\ldots,\gamma_k$, with $0\le k\le D$, such that ……

open_problem | open | MathDB 页面 | 原文线索

V2-007609 | MATHDB-401225

Distribution of local components in the unramified unitary dual

摘录 / EXCERPT: For each prime p, how do the points $\\pi_p \in \widehat{G(\mathbb{Q}_p)}^{\mathrm{ur}}$ distribute as $\\pi$ ranges over the multiset $\mathcal{F}_{\Omega}(t)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-007610 | MATHDB-401229

Burgess bound for Dirichlet L-functions

摘录 / EXCERPT: For a primitive Dirichlet character $\c7$ modulo M, with M assumed prime for simplicity, the note establishes ...

open_problem | open | MathDB 页面 | 原文线索

V2-007611 | MATHDB-401232

Conjecture on the variance of angles of Gaussian primes

摘录 / EXCERPT: Let $N=N(X)$ be the number of primes $p\le X$ with $p\equiv1\pmod 4$. For each such prime, write $p=a^2+b^2$ with $a>b>0$ and associate the angle $\theta_p\in(0,\pi/4)$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-007612 | MATHDB-401242

Reducing the prime set in the crossed-product characterization of the UCT problem

摘录 / EXCERPT: In the theorem characterizing the UCT problem, replace the requirement that the following conditions hold for every prime $p\in\{2,3\}$ by the requirement that they hold only for...

open_problem | open | MathDB 页面 | 原文线索

V2-007613 | MATHDB-401245

Cartan subalgebras and UCT for finite cyclic actions on \(\mathcal O_2\)

摘录 / EXCERPT: Let p be prime, and let $\alpha:\mathbb Z_p\curvearrowright\mathcal O_2$ be an outer strongly approximately inner action. Then $\mathcal O_2\rtimes_\alpha\mathbb Z_p$ satisfies t...

open_problem | open | MathDB 页面 | 原文线索

V2-007614 | MATHDB-401342

Polynomial equivalence of QC generic decoding and QC codeword weight

摘录 / EXCERPT: Let p be prime and $R_p=\mathbb{F}_2[x]/(x^p-1)$. Consider the following two problems for index-2 quasi-cyclic codes: Problem 2 (QC Generic Decoding): given $s,h\in R_p$ and an...

open_problem | open | MathDB 页面 | 原文线索

V2-007615 | MATHDB-401354

Removing the minimum in the character-sum moment bound

摘录 / EXCERPT: Is it possible to prove the sharper bound that, for a sufficiently large prime r, uniformly for $1\le x\le r$ and $0\le q\le 1$, ... with the minimum appearing in the proved Theore...

open_problem | open | MathDB 页面 | 原文线索

V2-007616 | MATHDB-401356

Keating-Snaith conjecture for moments of the Riemann zeta function

摘录 / EXCERPT: Conjecture (Keating-Snaith).

conjecture | open | MathDB 页面 | 原文线索

V2-007617 | MATHDB-401357

Farmer-Gonek-Hughes conjecture on the maximum size of the Riemann zeta function

摘录 / EXCERPT: Conjecture (Farmer, Gonek, Hughes): ...

conjecture | open | MathDB 页面 | 原文线索

V2-007618 | MATHDB-401407

FCC optimality for the three-dimensional Epstein zeta function

摘录 / EXCERPT: It is conjectured that the Epstein zeta function is minimized by the Face Centered Cubic (FCC) lattice in dimension $d=3$.

open_problem | open | MathDB 页面 | 原文线索

V2-007619 | MATHDB-401519

Super-expander families from finite special linear groups

摘录 / EXCERPT: For a prime power q, let $\cF_q$ be the finite field of order q. Let $(p_m)_{m\cin\cmathbb N}$ be a sequence of primes and let $(n_m)_{m\cin\cmathbb N}$ be a sequence of intege...

open_problem | open | MathDB 页面 | 原文线索

V2-007620 | MATHDB-401627

Schinzel’s hypothesis

摘录 / EXCERPT: Let $P_1(t),\ldots,P_n(t)\in\mathbb{Z}[t]$ be irreducible polynomials with positive leading coefficients such that the product $\prod_{i=1}^n P_i(t)$ has no fixed prime divisor for...

conjecture | open | MathDB 页面 | 原文线索

V2-007621 | MATHDB-401630

Decomposition of the external cohomology classes \(\rho^*\gamma_{n,s}\)

摘录 / EXCERPT: Fix a prime number p. For a profinite group G, write $H^i(G)=H^i(G,\mathbb Z/p)$, and let $G(i)=G(i,p)$ be its lower p-central filtration, defined by $G(1)=G$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-007622 | MATHDB-401717

Mod-\p criterion for analytic rank one of a CM elliptic curve

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a CM elliptic curve and let $p>3$ be an ordinary prime for E. Suppose that: 1. the mod-p Galois representation...

open_problem | open | MathDB 页面 | 原文线索

V2-007623 | MATHDB-401722

Images of GL_2-type Galois representations

摘录 / EXCERPT: Let p be a prime, let A be a complete local noetherian pro-p ring, and let $\rho:G_{\mathbb Q}\to\mathrm{GL}_2(A)$ be a continuous representation unramified outside a finite...

open_problem | open | MathDB 页面 | 原文线索

V2-007624 | MATHDB-401961

Donkin’s conjecture on the restriction of tilting modules

摘录 / EXCERPT: In the notation of the report: for every prime characteristic p and every p-alcove A in the fundamental box, Donkin conjectures that ... Here the displayed assertion is the rep...

conjecture | open | MathDB 页面 | 原文线索

V2-007625 | MATHDB-402014

Uniqueness of the homological prime lying over a triangular prime

摘录 / EXCERPT: Let T be a tensor-triangulated category, let T^c be its compact-rigid subcategory, and let $P\subset T^c$ be a triangular prime. The Yoneda functor...

open_problem | open | MathDB 页面 | 原文线索

V2-007626 | MATHDB-402062

MacWilliams-type identities for pivot and reverse-pivot partitions

摘录 / EXCERPT: Let q be a prime power and let $m\ge n>0$. For a code $\mathcal C\le \mathbb F_q^{n\times m}$, let $\mathcal C^\perp$ be its trace-product dual. If $\mathcal P=(P_i)_{i\in I}$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-007627 | MATHDB-402063

Generalized Hamada conjecture for subspace designs

摘录 / EXCERPT: Let $q=p^m$ be a prime power, let V be a vector space of dimension v over $\d53d_q$, and let $D=(V,\mathcal B)$ be a t-$(v,k,\lambda)_q$ subspace design, where...

conjecture | open | MathDB 页面 | 原文线索

V2-007628 | MATHDB-402074

Boltje-Külshammer Conjecture 1 on Wedderburn components of block algebras

摘录 / EXCERPT: Let p be a prime, let G be a finite group, and let b be a block idempotent of $\r\mathbb{Z}_pG$ of positive defect. For a semisimple $\r\mathbb{Q}_p$-algebra C and a simple...

conjecture | open | MathDB 页面 | 原文线索

V2-007629 | MATHDB-402075

Turull’s refined character-counting conjecture for block algebras

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let $K=\mathbb{Q}_p(\zeta)$, and let $\mathcal{O}=\mathbb{Z}_p[\zeta]$, where ζ is a root of unity of order $\exp(G)$. Let b...

conjecture | open | MathDB 页面 | 原文线索

V2-007630 | MATHDB-402078

Linear constituents of restrictions to Sylow subgroups of symmetric groups

摘录 / EXCERPT: Let p be a prime, let n be a natural number, let Pinoperatorname{Syl}p(Sn), and let chiinoperatorname{Irr}(Sn). Determine the linear constituents of the restriction chiP. Equivalen...

open_problem | open | MathDB 页面 | 原文线索

V2-007631 | MATHDB-402079

Finite-field bounds for relations in Hochschild cohomology of block algebras

摘录 / EXCERPT: Let k be a sufficiently large field of prime characteristic p, let G be a finite group, and let B be a block of kG with defect group P. Can one show that the relations...

open_problem | open | MathDB 页面 | 原文线索

V2-007632 | MATHDB-402081

Brauer’s height zero conjecture for two primes

摘录 / EXCERPT: Let G be a finite group and p,q two primes. Then G has a Sylow p-subgroup commuting with a Sylow q-subgroup if and only if all characters in $B_p(G)$ have degree prime to...

conjecture | open | MathDB 页面 | 原文线索

V2-007633 | MATHDB-402082

Prime Graph Question for normalized units in integral group rings

摘录 / EXCERPT: For every finite group G and every pair of distinct primes p and q, does $\[200m\mathbb{Z}G$ contain a normalized unit of order pq if and only if G contains an element of...

open_problem | open | MathDB 页面 | 原文线索

V2-007634 | MATHDB-402085

Alperin-McKay conjecture for special linear and unitary groups

摘录 / EXCERPT: Let $G=\operatorname{SL}_n(\varepsilon q)$, where q is a prime power and $\varepsilon\in\{\pm1\}$, with $\operatorname{SL}_n(-q)=\operatorname{SU}_n(q)$. If ℓ is a prime sat...

conjecture | open | MathDB 页面 | 原文线索

V2-007635 | MATHDB-402090

Explicit computation of the local invariant of an irreducible character

摘录 / EXCERPT: For a prime p, a finite group G, and $[U+007F]chi\in\operatorname{Irr}(G)$, how can one explicitly compute the local invariant $\operatorname{inv}(\chi)\in\mathbb{Q}/\mathbb{Z}$?

open_problem | open | MathDB 页面 | 原文线索

V2-007636 | MATHDB-402095

Fried conjecture for flows satisfying A.1-A.3

摘录 / EXCERPT: Let $(\phi^t)_{t\in\mathbb R}$ be a flow on a closed manifold X satisfying A.1-A.3 from the report, and let $R_\rho(s)$ be the associated Ruelle zeta function, defined for...

conjecture | open | MathDB 页面 | 原文线索

V2-007637 | MATHDB-402172

Coarse moduli spaces for HHN and HN strata of Higgs bundles

摘录 / EXCERPT: Given a compact Riemann surface Σ of genus g, a line bundle $L\to\Sigma$ of positive degree, and coprime rank r and degree d, can coarse moduli spaces be constructed f...

open_problem | open | MathDB 页面 | 原文线索

V2-007638 | MATHDB-402272

Explicit calculation of equivariant topological modular forms for finite cyclic groups

摘录 / EXCERPT: Can one explicitly calculate TMF^{C_n}, or at least its homotopy groups?

open_problem | open | MathDB 页面 | 原文线索

V2-007639 | MATHDB-402359

Relationships between toric degenerations from maximal prime cones

摘录 / EXCERPT: What is the relationship between the different toric degenerations arising from different maximal prime cones of the tropicalization textstyle$\operatorname{trop}(X)$, in particula...

open_problem | open | MathDB 页面 | 原文线索

V2-007640 | MATHDB-402448

CFKRS shifted-moment conjecture for the Riemann zeta function

摘录 / EXCERPT: Let A and B be sets of complex numbers such that, for every $\b1 eq A$ and $\b2 eq B$, $|\Re\alpha|\ll(\log T)^{-1}$, $|\Im\alpha|\ll T$, and similarly for β. Let ψ...

conjecture | open | MathDB 页面 | 原文线索

V2-007641 | MATHDB-402453

Unbounded long-sum discrepancy for completely multiplicative functions over \(\mathbb F_q[t]\)

摘录 / EXCERPT: Let q be a prime power and let $\mathcal M$ be the set of monic polynomials in $\mathbb F_q[t]$. Given a completely multiplicative function $f:\mathcal M\to\{-1,+1\}$, must it be...

open_problem | open | MathDB 页面 | 原文线索

V2-007642 | MATHDB-402458

Green’s covering problem for prime-modulus residue classes

摘录 / EXCERPT: We are interested in positive integers up to X, and set $H=X^{\theta}$, where θ is slightly greater than $1/2$, say $\theta=0.51$. Put $I=[H,2H]$. For each prime $p\in I$,...

open_problem | open | MathDB 页面 | 原文线索

V2-007643 | MATHDB-402460

Prime number theorem for Anzai skew products

摘录 / EXCERPT: Given an irrational real number α and a real-analytic 1-periodic function g with $\int_{\mathbb T} g(x)\,dx=0$, define ... If $T_{\alpha,g}$ is uniquely ergodic, then for ev...

open_problem | open | MathDB 页面 | 原文线索

V2-007644 | MATHDB-402463

Chowla’s k-point correlation conjecture over \(\mathbb{F}_q[T]\)

摘录 / EXCERPT: For an odd prime p, an integer $k\ge 1$, and a power q of p satisfying $q>p^{2k^{2}e^{2}}$, for any distinct $h_1,\ldots,h_k\in\mathbb{F}_q[T]$, ... Here $\mathbb{F}_q[T]^+$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-007645 | MATHDB-402468

Structure and properties of prime images of the structure algebra

摘录 / EXCERPT: Determine the structure of prime images kM/P of kM. When is kM/P a domain? When is kM/P a maximal order in its quotient ring? When does kM/P have some good homological pr...

open_problem | open | MathDB 页面 | 原文线索

V2-007646 | MATHDB-402542

Local Kudla-Rapoport conjecture for unitary Rapoport-Zink spaces

摘录 / EXCERPT: Let p be a prime, let F_0 be a finite extension of $\a0\mathbb{Q}_p$ with residue field $k=\mathbb{F}_q$ and uniformizer ϖ, and let F/F_0 be an unramified quadratic ex...

conjecture | open | MathDB 页面 | 原文线索

V2-007647 | MATHDB-402546

Shimura-type correspondence for vector-valued Siegel modular forms

摘录 / EXCERPT: For every even integer $j\ge 0$ and every odd integer $k\ge 3$, there should be an isomorphism of Hecke modules ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-007648 | MATHDB-402547

Original Harder conjecture on congruences

摘录 / EXCERPT: Let k,j be integers with $j\ge 0$ even and $k\ge 3$, and let f be a Hecke eigen elliptic cusp form of weight $2k+j-2$. Then there should exist a prime ideal $\mathfrak l$ divid...

conjecture | open | MathDB 页面 | 原文线索

V2-007649 | MATHDB-402548

Gukov’s conjecture that the plumbing invariant $Z(q)$ is a quantum modular form

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with N vertices, and let $M=(m_{jk})$ be the associated positive-definite symmetric integral matrix, with $m_{jk}=-1$ when vertices j,k are connected a...

conjecture | open | MathDB 页面 | 原文线索

V2-007650 | MATHDB-402558

Conjecture K on congruences between Klingen-Eisenstein and type-\(I) lifts

摘录 / EXCERPT: Let k,j,n be positive even integers with $k\ge 4$. Let $f\in S_{2k+j-2}(\mathrm{SL}_2(\mathbb Z))$ be a primitive form, and suppose that a “big prime” ideal $\mathfrak p$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-007651 | MATHDB-402560

Holomorphicity of modular forms arising from concomitants

摘录 / EXCERPT: Which concomitants give rise to holomorphic modular forms?

open_problem | open | MathDB 页面 | 原文线索

V2-007652 | MATHDB-402562

Modular forms for reductive groups without complex symmetric spaces

摘录 / EXCERPT: Suppose that G is a reductive $\mathbb{Q}$-group whose associated symmetric space does not have complex structure. Is there any notion of “modular forms” for G which have a Fou...

open_problem | open | MathDB 页面 | 原文线索

V2-007653 | MATHDB-402751

Simplicity of fusion systems of finite simple groups

摘录 / EXCERPT: Fix a prime p and a known finite simple group G such that $p\mid |G|$. Let $S\in\operatorname{Syl}_p(G)$ and set $\mathcal F=\mathcal F_S(G)$. Then one of the following holds:...

open_problem | open | MathDB 页面 | 原文线索

V2-007654 | MATHDB-402753

Realizability of equivariant mapping-cone chain complexes

摘录 / EXCERPT: Let $n\ge 1$, let p be prime, let $G=(\mathbb Z/p)^n$, and let K_* be the Koszul complex of the group ring $\mathbb F_pG$. For a cycle $w\in K_*$ of degree r, there exists a...

open_problem | open | MathDB 页面 | 原文线索

V2-007655 | MATHDB-402754

Equality of the dimension-function subgroup and the Bouc-kernel for odd primes

摘录 / EXCERPT: For odd prime p, does the equality $C_{ba^{+}}(G,p)=\ker\Psi_G$ hold?

open_problem | open | MathDB 页面 | 原文线索

V2-007656 | MATHDB-402777

Approximating optimal dual measures by modular forms

摘录 / EXCERPT: Can a suitable modular-form level N be found to approximate the strongest choices of $\bc$ in the dual bound on the bound for the CohnElkies linear-programming bound for sphere p...

open_problem | open | MathDB 页面 | 原文线索

V2-007657 | MATHDB-402779

Sharpness of a low-moment bound for a short zeta-function integral

摘录 / EXCERPT: Conjecture: The upper bound in Theorem 1 is sharp. Namely, for all sufficiently large T and $0\le q\le 1$, the quantity ... should have the same order of magnitude as ...

open_problem | open | MathDB 页面 | 原文线索

V2-007658 | MATHDB-402780

Harper’s conjecture that two multiplicative-chaos bounds are sharp

摘录 / EXCERPT: In the report, Harper states that both of the following bounds are sharp. For all sufficiently large T and uniformly for $0\le q\le 1$, ... For a large prime r, uniformly for...

conjecture | open | MathDB 页面 | 原文线索

V2-007659 | MATHDB-402790

General power-residue spins: automorphic interpretation and prime equidistribution

摘录 / EXCERPT: Define $\lambda_m(n;L/K)$ in general, for K,L,m as described above, and find an automorphic interpretation. Show equidistribution along primes.

open_problem | open | MathDB 页面 | 原文线索

V2-007660 | MATHDB-402821

Continuous generalization of Birch’s theorem beyond prime powers

摘录 / EXCERPT: For all integers $q\ge 2$ and $d\ge 1$, letting $n=q(d+1)-1$, every continuous map $f:\Delta_n\to\mathbb{R}^d$ has points $x_1,\ldots,x_q$ lying in q pairwise disjoint faces of...

open_problem | open | MathDB 页面 | 原文线索

V2-007661 | MATHDB-403001

Characterization of normal Sylow subgroups by constituents of a permutation character

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Then $P\trianglelefteq G$ if and only if $p\nmid\chi(1)$ for every irreducible constituent...

open_problem | open | MathDB 页面 | 原文线索

V2-007662 | MATHDB-403002

Does the character submatrix on p-elements detect abelian Sylow p-subgroups?

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let $P\in\operatorname{Syl}_p(G)$. Write $X_p(G)$ for the submatrix of the character table of G corresponding to the p-elemen...

open_problem | open | MathDB 页面 | 原文线索

V2-007663 | MATHDB-403006

Combinatorial description of Sylow branching coefficients

摘录 / EXCERPT: Let p be an odd prime, let $n\in\mathbb N$, and let $P_n\in\operatorname{Syl}_p(S_n)$. For each partition $\lambda\in\mathcal P(n)$, let $\chi^\lambda\in\operatorname{Irr}(S_n)$...

open_problem | open | MathDB 页面 | 原文线索

V2-007664 | MATHDB-403007

Combinatorial description of Sylow branching multiplicities for the trivial character

摘录 / EXCERPT: Let p be a prime, let $n\in\mathbb{N}$, and let $P_n\in\operatorname{Syl}_p(S_n)$. For each partition $\lambda\in\mathcal{P}(n)$, let $\chi^\lambda\in\operatorname{Irr}(S_n)$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-007665 | MATHDB-403013

Malle-Navarro conjecture on normal Sylow subgroups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, and let P be a Sylow p-subgroup of G. Then P is normal in G if and only if every irreducible constituent χ of the perm...

conjecture | open | MathDB 页面 | 原文线索

V2-007666 | MATHDB-403064

Minimal ramification conjecture for rational function fields over finite fields

摘录 / EXCERPT: Let G be a nontrivial finite group, let $q=p^\nu$ be a prime power, and let $r_{\mathbb F_q(T)}(G)$ be the minimal number of primes of $\mathbb F_q(T)$-including the prime at inf...

conjecture | open | MathDB 页面 | 原文线索

V2-007667 | MATHDB-403084

Invariance determination of the Fermat-curve relation for primes p ≥ 7

摘录 / EXCERPT: For the Fermat curve of exponent p, with $p\ge 7$ prime, is the formula for Δ_X also determined by invariance properties under the geometric and arithmetic automorphisms o...

open_problem | open | MathDB 页面 | 原文线索

V2-007668 | MATHDB-403254

Maximum number of primes in a very short interval

摘录 / EXCERPT: Let $S(y)$ be the largest cardinality of an admissible set of integers $0\le a_1<\cdots<a_k\le y$, where admissibility means that there is no prime p dividing one of...

open_problem | open | MathDB 页面 | 原文线索

V2-007669 | MATHDB-403263

Fisher’s conjecture on non-trivial prime-level congruences for primes at least 17

摘录 / EXCERPT: For every prime $p\ge 17$, the only non-trivial pairs of p-congruent elliptic curves over $\mathbb{Q}$ are the two examples described in the report-one symplectic pair, with cond...

conjecture | open | MathDB 页面 | 原文线索

V2-007670 | MATHDB-403265

Strong Malle conjecture for generalized and twisted Heisenberg groups

摘录 / EXCERPT: Let k be a number field, let $\cell\ell$ be a prime number, and let $G=H(\ell,d)$ or $G=\widetilde{H}(\ell,d)$ be a transitive subgroup of S_{ℓ^2}. Then the strong Malle con...

conjecture | open | MathDB 页面 | 原文线索

V2-007671 | MATHDB-403267

Rational points on the normalizer non-split Cartan modular curves

摘录 / EXCERPT: For each prime number $p\ge 3$, determine the set of rational points $X(\Gamma^+_{\mathrm{ns}}(p))(\mathbb Q)$, where $\Gamma_{\mathrm{ns}}(p)$ is a cyclic subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-007672 | MATHDB-403269

Poonen-Rains moment conjecture for Selmer groups of elliptic curves

摘录 / EXCERPT: Let p be prime. When elliptic curves $E/\mathbb{Q}$ are ordered by height, the mth moment of the size of $\operatorname{Sel}_p(E)$ is ...

conjecture | open | MathDB 页面 | 原文线索

V2-007673 | MATHDB-403274

Totally real prime-power discriminant fields with infinite unramified towers

摘录 / EXCERPT: For every prime number $p\ge 2$, does there exist a totally real finite extension $K/\mathbb{Q}$, unramified outside $\{p\}$, such that K admits an infinite unramified p-extens...

open_problem | open | MathDB 页面 | 原文线索

V2-007674 | MATHDB-403385

Terminating prime decompositions of knots in general 3-manifolds

摘录 / EXCERPT: Does every knot in a general 3-manifold admit a terminating prime decomposition into connected-sum summands? No: the knot $S^1\times\{\ast\}\subset S^1\times S^2$ can be decomposed...

open_problem | open | MathDB 页面 | 原文线索

V2-007675 | MATHDB-403487

Zariski density of Pythagorean triples with prime hypotenuse

摘录 / EXCERPT: Let $F(x,y,z)=x^2+y^2-z^2$, let $G=\mathrm{SO}^{\circ}_F(\mathbb{R})$, and let $\Gamma<G(\mathbb{Z})$ be geometrically finite with no unipotent elements other than the identity. Fo...

open_problem | open | MathDB 页面 | 原文线索

V2-007676 | MATHDB-403537

Stabilization of the dimension of Teichmüller modular forms

摘录 / EXCERPT: $\dim H^0(M_g,\lambda_g^{\otimes 12})=71$ for all $g\gg 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-007677 | MATHDB-403539

A more direct relation between harmonic-Maass coefficients and twisted central L-derivatives

摘录 / EXCERPT: Let p be prime, let $G\in S_2(\Gamma_0(p))$ be a newform with rational Fourier coefficients, let $g\in S_{3/2}(\Gamma_0(4p))$ correspond to G under the Shimura correspondence,...

open_problem | open | MathDB 页面 | 原文线索

V2-007678 | MATHDB-403540

Which weight-4 modular forms arise from Calabi-Yau threefolds?

摘录 / EXCERPT: Which weight-4 modular forms can be realised by a Calabi-Yau threefold?

open_problem | open | MathDB 页面 | 原文线索

V2-007679 | MATHDB-403576

Hyperbolic Kazhdan groups with large finite simple quotients

摘录 / EXCERPT: Let p be an odd prime and define ... Then G_p is infinite and hyperbolic. For each $p\ge 7$, it has Kazhdan’s property (T), whereas for $p=3$ it does not. For each $p\ge 3$, it...

open_problem | open | MathDB 页面 | 原文线索

V2-007680 | MATHDB-403599

Montgomery pair-correlation gap conjecture for the zeros of the Riemann zeta function

摘录 / EXCERPT: Let $\gamma_1\leq\gamma_2\leq\gamma_3\leq\cdots$ enumerate the positive ordinates of the real zeros of the Riemann Ξ-function. The report states the conjecture ...

conjecture | open | MathDB 页面 | 原文线索

V2-007681 | MATHDB-403661

Hausdorff-measure convergence theorem for Diophantine approximation on nondegenerate manifolds

摘录 / EXCERPT: Let $n>d\ge 1$ be integers, put $m=n-d$, let $s>0$ and $l\in\mathbb N$, let $\psi:\mathbb N\to\mathbb R_{\ge 0}$ be monotonic, let $U\subset\mathbb R^d$ be an open ball, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-007682 | MATHDB-403700

Koszulity and Hilbert series of Feigin-Odesskii elliptic algebras

摘录 / EXCERPT: For every integer pair of relatively prime integers $n>k\ge 1$ with $n\ge 3$, every complex elliptic curve E, and every point $\xi\in E$, is the Feigin-Odesskii elliptic algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-007683 | MATHDB-403716

Hopkins-Morel conjecture over the prime field \(\mathbb{F}_p\)

摘录 / EXCERPT: For a prime p, prove that the canonical map ... is an equivalence in $\mathrm{SH}(\mathbb{F}_p)$, where $x_1,x_2,\ldots$ are polynomial generators of the Lazard ring.

conjecture | open | MathDB 页面 | 原文线索

V2-007684 | MATHDB-403794

Mori’s question on smooth limits of prime-degree hypersurfaces

摘录 / EXCERPT: If $n\ge 3$, is every smooth projective limit of prime-degree hypersurfaces of dimension n in $\mathbb{P}^{n+1}_{\mathbb{C}}$ also a hypersurface in...

open_problem | open | MathDB 页面 | 原文线索

V2-007685 | MATHDB-403795

Conjecture C on smooth limits of prime-degree plane curves

摘录 / EXCERPT: Let p be a prime number that is not a Markov number. Any smooth limit of plane curves of degree p is a plane curve.

conjecture | open | MathDB 页面 | 原文线索

V2-007686 | MATHDB-403819

Fields of values of restrictions to Sylow p-subgroups

摘录 / EXCERPT: Let G be a finite group, let p be a prime, let $P\in\operatorname{Syl}_p(G)$, and let $\chi\in\operatorname{Irr}(G)$ have p'-degree. Suppose that the conductor of χ is...

open_problem | open | MathDB 页面 | 原文线索

V2-007687 | MATHDB-403820

Galois-equivariant McKay bijection preserving local fields of values

摘录 / EXCERPT: For a finite group G, a prime p, and a Sylow p-subgroup P, does there exist a bijection ... commuting with the G_p-action such that ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-007688 | MATHDB-403872

Smooth numbers represented by a sum of two almost-prime squares

摘录 / EXCERPT: There exist constants $C,D>0$ such that, for any function g satisfying $(\log n)^D\le g(n)\le n$, every sufficiently large integer n can be represented in the form ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-007689 | MATHDB-403874

Typical lower bound for prime values of random polynomials

摘录 / EXCERPT: Let $d\ge 1$ and $\varepsilon>0$ be fixed. Define ... which parametrizes $f_{\mathbf c}(t)=c_0t^d+\cdots+c_d$. Then, for 100% of degree-d polynomials $f\in\mathbb Z[t]$ with coe...

open_problem | open | MathDB 页面 | 原文线索

V2-007690 | MATHDB-403877

Bounds for sums of a prime and k-th powers

摘录 / EXCERPT: Let $H(k)$ be the smallest number s such that every sufficiently large natural number is the sum of a prime and s natural k-th powers. Let c_r, for $r=4$ or 5, be the uni...

open_problem | open | MathDB 页面 | 原文线索

V2-007691 | MATHDB-403878

Prime values of irreducible polynomials

摘录 / EXCERPT: Let $P\in\mathbb{Z}[X]$ be an irreducible polynomial with no fixed divisor. Are there infinitely many integers n such that $P(n)$ is a prime number?

open_problem | open | MathDB 页面 | 原文线索

V2-007692 | MATHDB-403879

Unconditional support for the harmonically weighted one-level density in a prime-level newform family

摘录 / EXCERPT: Fix an even weight k. Let ϕ be an even Schwartz function such that ... For prime N, the harmonically weighted one-level density of the low-lying zeros of the L-functions a...

open_problem | open | MathDB 页面 | 原文线索

V2-007693 | MATHDB-403881

Gonek’s conjecture on negative moments of the Riemann zeta-function

摘录 / EXCERPT: For fixed $k>0$, define ... where $0<\alpha\le 1$. Gonek’s conjecture predicts, uniformly for $1/\log T\le\alpha\le1$, ... and uniformly for $0<\alpha\le1/\log T$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-007694 | MATHDB-403895

Unconditional lower bound for the least quadratic nonresidue

摘录 / EXCERPT: For an odd prime p, let $n_2(p)$ denote the least positive quadratic nonresidue modulo p. Prove unconditionally that $n_2(p)=\Omega\bigl((\log p)\log\log p\bigr)$, or obtain a...

open_problem | open | MathDB 页面 | 原文线索

V2-007695 | MATHDB-403897

Divisibility of almost all character-table entries by a fixed prime power

摘录 / EXCERPT: For every fixed prime power p^r, as $n\to\infty$, almost all entries of the character table of S_n are divisible by p^r.

open_problem | open | MathDB 页面 | 原文线索

V2-007696 | MATHDB-404004

Along-Jaeger-Tarsi conjecture

摘录 / EXCERPT: For every prime p and every positive integer n, every invertible $n\times n$ matrix A over $\mathbb F_p$ admits a vector $v\in\mathbb F_p^n$ such that neither v nor Av ha...

conjecture | open | MathDB 页面 | 原文线索

V2-007697 | MATHDB-404016

Minimum size of an irredundant spanning-normal hyperplane cover

摘录 / EXCERPT: For a prime p and a positive integer n, determine ... Here a covering is irredundant if no proper subcollection still covers $\mathbb F_p^n$, and if...

open_problem | open | MathDB 页面 | 原文线索

V2-007698 | MATHDB-404018

Additive Basis conjecture of Alon, Linial, and Meshulam

摘录 / EXCERPT: For every prime $p\ge 3$, there exists an integer $k=k(p)$ such that, for every $n\ge 1$, the union (with multiplicities) of k linear bases of $\mathbb{F}_p^n$ is an additive bas...

conjecture | open | MathDB 页面 | 原文线索

V2-007699 | MATHDB-404032

Existence of subspace Steiner designs

摘录 / EXCERPT: For every prime power q and integers $s>r\ge 1$, there is a threshold $n_2(q,s)$ such that, whenever $n\ge n_2(q,s)$ and ... for every $0\le i\le r-1$, there exists an $(n,s,r)_q$-...

open_problem | open | MathDB 页面 | 原文线索

V2-007700 | MATHDB-404095

Conjecture 4: Vanishing conjecture for projective space in even degree

摘录 / EXCERPT: For any finite field F of characteristic p and any prime $\ invertible in F, one has ... for all integers $i,n\ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-007701 | MATHDB-404109

Existence and interpolation of a p-adic L-function for a Siegel eigenform

摘录 / EXCERPT: Let F be a holomorphic, cuspidal Siegel modular eigenform of weight (k_1,k_2) with $k_1\ge k_2\ge 2$, of prime-to-p level and ordinary at p. There exists a p-adic L-fun...

open_problem | open | MathDB 页面 | 原文线索

V2-007702 | MATHDB-404112

Nonvanishing of central twisted L-values for Dirichlet characters of fixed order

摘录 / EXCERPT: Fix a modular form f of weight $k\ge 2$. Fix $d\in\mathbb{Z}_{\ge 1}$, and let χ vary through all Dirichlet characters of order d, ordered by conductor. Define ... What is t...

open_problem | open | MathDB 页面 | 原文线索

V2-007703 | MATHDB-404113

Non-vanishing of central twists for characters of order greater than two

摘录 / EXCERPT: Fix a modular form f of weight $k\ge 2$ and an integer $d>2$. Let $\mathcal K_d(X)=\{\chi\text{ primitive of conductor }D,\ \operatorname{ord}(\chi)=d,\ D\le X\}$. It is conjectu...

open_problem | open | MathDB 页面 | 原文线索

V2-007704 | MATHDB-404118

Non-vanishing of Kato's Kolyvagin system

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve without CM, and let p be an odd prime of good ordinary reduction for E such that $E(\mathbb{Q})[p]=0$. Assume that the rational cyclotom...

open_problem | open | MathDB 页面 | 原文线索

V2-007705 | MATHDB-404120

Recovery of E_∞ as super-Hölder vectors

摘录 / EXCERPT: Let p be prime, $E=\mathbf F_p((X))$, $E_n=\mathbf F_p((X^{1/p^n}))$, $E_\infty=\bigcup_{n\ge 0}E_n$, and let $\widetilde E$ be the X-adic completion of E_∞. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-007706 | MATHDB-404125

Bloch-Kato prediction for the Selmer group of ρ_π

摘录 / EXCERPT: In the setup where p is prime, $n\ge 1$, $G_n=\mathrm{SO}(n,n+1)$ is split, π is a cuspidal automorphic representation of G_n that is discrete series at infinity, and Arthu...

open_problem | open | MathDB 页面 | 原文线索

V2-007707 | MATHDB-404126

The nonvanishing of Selmer groups of certain symplectic Galois representations

摘录 / EXCERPT: Fix a prime p. For an integer $n\ge 1$, let G_n denote the split orthogonal group $\mathrm{SO}(n,n+1)$. Let π be a cuspidal automorphic representation of G_n which is dis...

open_problem | open | MathDB 页面 | 原文线索

V2-007708 | MATHDB-404190

Kulkarni’s conjecture on generators of $\Gamma_0(N)$

摘录 / EXCERPT: For every prime N, the congruence subgroup $\Gamma_0(N)$ has a freely independent generating set with matrix representatives in $\mathrm{PSL}(2,\mathbb Z)$ such that the $(2,1)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-007709 | MATHDB-404202

Wild classification of compact equivariant Eilenberg-MacLane modules

摘录 / EXCERPT: For an odd prime p and $G=C_p$, the classification of compact $H\mathbb{F}_p$-modules is wild. Here $\mathbb{F}_p$ denotes the constant Mackey functor; in the stated sense, a wil...

open_problem | open | MathDB 页面 | 原文线索

V2-007710 | MATHDB-404272

Cabling torsion conjecture in the knot concordance group

摘录 / EXCERPT: For a knot K and relatively prime integers p,q, let $K_{p,q}$ denote its (p,q)-cable. Conjecture 1. The (p,q)-cable $K_{p,q}$ is torsion in the knot concordance group...

conjecture | open | MathDB 页面 | 原文线索

V2-007711 | MATHDB-404282

Frobenius pushforward as a strong generator

摘录 / EXCERPT: Let R be a commutative noetherian ring of prime characteristic $p>0$ such that its Frobenius map $F:R\to R$, $r\mapsto r^p$, is finite (that is, R is F-finite). Is F_*R a s...

open_problem | open | MathDB 页面 | 原文线索

V2-007712 | MATHDB-404288

Field of definition of torsion points on cyclic Fermat quotients

摘录 / EXCERPT: For a prime p, let $1\le k<p-1$, let X_k be the curve $y^p=x(x-1)^k$, and let $J_k=\operatorname{Jac}(X_k)$. If ζ is a primitive pth root of unity, define...

open_problem | open | MathDB 页面 | 原文线索

V2-007713 | MATHDB-404289

Congruence subgroup problem for the mapping class group

摘录 / EXCERPT: Let g,r be nonnegative integers satisfying $2g-2+r>0$, let k be algebraically closed of characteristic 0, and let P denote the set of all prime numbers. With...

open_problem | open | MathDB 页面 | 原文线索

V2-007714 | MATHDB-404293

Anderson-Ihara’s question on the pro-[U+007F]p outer Galois representation

摘录 / EXCERPT: For an odd prime p, is the field $\r\Omega^*$ associated with the kernel of the pro-p outer Galois representation $\rho_{0,3}:G_{\mathbb Q}\to\operatorname{Out}(\Pi_{0,3})$ equ...

open_problem | open | MathDB 页面 | 原文线索

V2-007715 | MATHDB-404296

Massey vanishing conjecture

摘录 / EXCERPT: For every field k, every prime p, and every integer $n\ge 3$, every defined n-fold Massey product of degree-one classes in mod-p Galois cohomology contains 0; equivalentl...

conjecture | open | MathDB 页面 | 原文线索

V2-007716 | MATHDB-404297

Formality of cochains of real projective profinite groups

摘录 / EXCERPT: Let G be a real projective profinite group and let p be a prime number. If $C^{\bullet}(G,\mathbb{F}_p)$ denotes the differential graded $\mathbb{F}_p$-algebra of continuous co...

open_problem | open | MathDB 页面 | 原文线索

V2-007717 | MATHDB-404299

Koszulity conjecture for Galois cohomology

摘录 / EXCERPT: For a prime number p and every field k containing a primitive pth root of unity, is the cohomology algebra $H^{\bullet}(\Gamma(k),\mathbb{F}_p)$ of the absolute Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-007718 | MATHDB-404317

Coleman conjecture for curves whose Jacobian has supersingular good reduction

摘录 / EXCERPT: Let $p>3$ be a prime, let k be a perfect field of characteristic p, let W be the ring of Witt vectors with coefficients in k, and let K be the fraction field of W. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-007719 | MATHDB-404357

The weak implies strong conjecture

摘录 / EXCERPT: If R is an excellent weakly F-regular ring of prime characteristic $p>0$, then R is strongly F-regular.

conjecture | open | MathDB 页面 | 原文线索

V2-007720 | MATHDB-404372

The Birkhoff problem for submodules of finite abelian p-groups of exponent p^6

摘录 / EXCERPT: For a prime p, describe the category $\operatorname{mono}(1\to 2,\operatorname{mod}\mathbb{Z}/p^6\mathbb{Z})$.

open_problem | open | MathDB 页面 | 原文线索

V2-007721 | MATHDB-404373

Similarity of the two length-six monomorphism categories

摘录 / EXCERPT: For a prime p and a field k, is the structure of $\operatorname{mono}(1\to2,\operatorname{mod}\,\mathbb{Z}/p^6\mathbb{Z})$ similar to the structure of...

open_problem | open | MathDB 页面 | 原文线索

V2-007722 | MATHDB-404382

Shalev-Smoktunowicz conjecture on right nilpotence of powerful skew braces

摘录 / EXCERPT: Let p be a prime number and let A be a powerful skew brace of abelian type of size p^n. Then A is right nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-007723 | MATHDB-404388

Construction of skew braces of prime-power order

摘录 / EXCERPT: Construct skew braces of prime-power order.

open_problem | open | MathDB 页面 | 原文线索

V2-007724 | MATHDB-404443

The P=W conjecture for coprime rank and degree

摘录 / EXCERPT: Let C be a smooth projective complex curve, let $n\in\mathbb N$ and $d\in\mathbb Z$ satisfy $\gcd(n,d)=1$, and let $M_{\mathrm{Dol}}=M_{\mathrm{Dol}}(n,d)$ be the coarse moduli s...

conjecture | open | MathDB 页面 | 原文线索

V2-007725 | MATHDB-404460

Ol’shanskiĭ’s finite-group exponent question

摘录 / EXCERPT: For a given number $k\ge 2$ and any prime number n, does there exist a number $N=N(k,n)$ such that every finite group generated by $A=\{a_1,\ldots,a_k\}$ has exponent at most n...

open_problem | open | MathDB 页面 | 原文线索

V2-007726 | MATHDB-404532

Kim’s Conjecture for smooth projective curves

摘录 / EXCERPT: Let $X/\mathbb{Q}$ be a smooth projective curve of genus at least 2 with $X(\mathbb{Q})\neq\varnothing$, let p be a prime of good reduction for X, and let $X(\mathbb{Q}_p)^n$...

conjecture | open | MathDB 页面 | 原文线索

V2-007727 | MATHDB-404548

Adams operations on truncated Brown-Peterson spectra

摘录 / EXCERPT: For a prime $\ell\neq p$, there exists an Adams operation ψ^{ℓ} on $BP\langle n\rangle$ making it an $E_1\otimes A_2$-MU^{ψ}-algebra, and the map...

open_problem | open | MathDB 页面 | 原文线索

V2-007728 | MATHDB-404601

Categorical Fargues-Scholze conjecture

摘录 / EXCERPT: For a quasi-split reductive group G over a finite extension F of $\mathbb{Q}_p$ and a prime $\ell\ne p$, and for a Whittaker datum (U,ψ), the functor ... from...

conjecture | open | MathDB 页面 | 原文线索

V2-007729 | MATHDB-404604

Spectral action on isocrystals: categorical local Langlands equivalence

摘录 / EXCERPT: Let $F=\mathbb{F}_q((t))$ be a local function field of residue characteristic $p>0$, let G be a connected reductive group over F, let $\ell\ne p$ be prime, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-007730 | MATHDB-404744

Quillen’s conjecture for the prime 2

摘录 / EXCERPT: For every finite group G, if $O_2(G)=1$, then the poset $\d49c_2(G)$ of nontrivial elementary abelian 2-subgroups of G is not $\d53e$-acyclic; equivalently, it satisfies (H-Q...

conjecture | open | MathDB 页面 | 原文线索

V2-007731 | MATHDB-404769

Converse to the Hernandez-Leclerc cluster-monomial and cluster-variable correspondence

摘录 / EXCERPT: For the Hernandez-Leclerc subcategory $\mathcal F_n$, every real irreducible representation is a cluster monomial, and every real prime irreducible representation is a cluster vari...

open_problem | open | MathDB 页面 | 原文线索

V2-007732 | MATHDB-404781

Class-breadth conjecture for finite groups of odd prime-power order

摘录 / EXCERPT: Does the class-breadth inequality $cl(G) \le br(G)+1$ hold for p-groups when p is an odd prime? Here $br(g)=|g^G|=[G:C_G(g)]$, and $br(G)$ denotes the breadth of G; $cl(G)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-007733 | MATHDB-404822

Bichromatic Splitting Conjecture

摘录 / EXCERPT: Fix integers $0\le t<n$ and a prime $p\gg n$. Then there is an isomorphism ...

conjecture | open | MathDB 页面 | 原文线索

V2-007734 | MATHDB-404823

Algebraic bichromatic chromatic splitting conjecture

摘录 / EXCERPT: Fix integers $0\le t<n$ and a prime $p\gg n$. Let Γ_n be a one-dimensional formal group of height n over $\mathbb F_p$, let $A_n:=W(\mathbb F_p)[[u_1,\ldots,u_{n-1}]]$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-007735 | MATHDB-404853

Multiplicity conjecture for Reeb flows

摘录 / EXCERPT: Let M be the boundary of a star-shaped domain in $\mathbb{R}^{2n}$, which is assumed to be smooth. Then the Reeb flow on M has at least n prime closed orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-007736 | MATHDB-404854

Hofer-Zehnder conjecture for Reeb flows on smooth star-shaped hypersurfaces

摘录 / EXCERPT: Let M be the smooth boundary of a star-shaped domain in $\mathbb{R}^{2n}$. The Hofer-Zehnder conjecture asserts that the Reeb flow on M has exactly n prime closed orbits when...

conjecture | open | MathDB 页面 | 原文线索

V2-007737 | MATHDB-404855

Theorem B: a centrally symmetric case of the Reeb Hofer-Zehnder conjecture

摘录 / EXCERPT: Assume that the Reeb flow on the boundary $M^{2n-1}\subset\mathbb{R}^{2n}$ of a centrally symmetric star-shaped domain has finitely many prime closed orbits and is dynamically conv...

conjecture | open | MathDB 页面 | 原文线索

V2-007738 | MATHDB-404899

Primality of an ultrapower of a diffuse type I algebra tensored with a II_1 factor

摘录 / EXCERPT: Let M be a $\rm II_1$ factor. Is an ultrapower of $L^{\infty}[0,1]\,\bar\otimes\,M$ prime? Here a tracial von Neumann algebra is prime if it is not isomorphic to a von Neumann te...

open_problem | open | MathDB 页面 | 原文线索

V2-007739 | MATHDB-404929

Combinatorial Denef formulae for ask zeta functions of arbitrary graphs

摘录 / EXCERPT: Is there a meaningful combinatorial formula (in the spirit of Theorem 1) for the functions $W^-_\Gamma(X,T)$ which is valid for all graphs on a given vertex set?

open_problem | open | MathDB 页面 | 原文线索

V2-007740 | MATHDB-404930

Rossmann’s conjecture for local submodule zeta functions

摘录 / EXCERPT: Whenever E is a nilpotent algebra of matrices in the setting above, the value at $s=0$ of ... does not depend on E.

conjecture | open | MathDB 页面 | 原文线索

V2-007741 | MATHDB-404931

Coincidence conjecture for reduced and topological subalgebra zeta functions

摘录 / EXCERPT: Let A be a not necessarily associative $\mathbb Z$-algebra whose underlying $\mathbb Z$-module is free of finite rank d, and let $Z_A^{\mathrm{red}}(T)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-007742 | MATHDB-404933

Siegel modular generating series from enumerative algebra

摘录 / EXCERPT: Find natural counting problems in enumerative algebra that yield Siegel modular forms, and formulate clear questions for the Siegel modular forms community regarding which properti...

open_problem | open | MathDB 页面 | 原文线索

V2-007743 | MATHDB-404934

Determination of half-integral-weight cusp forms by prime-indexed Fourier coefficients

摘录 / EXCERPT: Let $S_{k+\frac12}(4N)$ denote the space of holomorphic cusp forms of half-integral weight $k+\frac12$ for $[U+007F]Gamma_0(4N)$. For $f dots\in S_{k+\frac12}(4N)$, write...

open_problem | open | MathDB 页面 | 原文线索

V2-007744 | MATHDB-404943

Conjecture 5: hitting all restricted projective-plane lines

摘录 / EXCERPT: Let q be a large prime power, put $n=q^2+q+1$, let P be the set of n points of a projective plane of order q, and let $L_1,L_2,\dots,L_n$ be the sets of points of its lines...

conjecture | open | MathDB 页面 | 原文线索

V2-007745 | MATHDB-404946

Graham's rearrangement conjecture over \(\mathbb F_p\)

摘录 / EXCERPT: For every prime p, every subset $S\subseteq \mathbb F_p\setminus\{0\}$ admits a valid ordering: there is an ordering $s_1,\ldots,s_{|S|}$ of the elements of S such that the par...

conjecture | open | MathDB 页面 | 原文线索

V2-007746 | MATHDB-404952

Lower bound for sums of dilates by an algebraic number

摘录 / EXCERPT: Let $f(x)\in\mathbb Z[x]$ be the minimal polynomial of an algebraic number λ, with coprime coefficients, and suppose that its full complex factorisation is...

open_problem | open | MathDB 页面 | 原文线索

V2-007747 | MATHDB-404997

Modular resurgence and quantum modular forms

摘录 / EXCERPT: Let $f(q)$ be a q-series with $q=e^{2\pi i z}$. If its asymptotic expansion $\phi(z)$ as $z\to 0$ is modular resurgent, then $f(q)$ is a holomorphic quantum modular form for some...

open_problem | open | MathDB 页面 | 原文线索

V2-007748 | MATHDB-405016

Primes with prime reverse in a fixed base

摘录 / EXCERPT: For a fixed integer base $b\ge 2$, are there infinitely many prime numbers p whose base-b reverse $\overleftarrow{p}$ is also prime?

open_problem | open | MathDB 页面 | 原文线索

V2-007749 | MATHDB-405019

Even-order logarithmic Chowla conjecture

摘录 / EXCERPT: For every even integer $k\ge 4$ and any distinct integers $h_1,\ldots,h_k$, prove that ... where $\lambda(n)=(-1)^{\Omega(n)}$ is the Liouville function and $\Omega(n)$ counts prime...

conjecture | open | MathDB 页面 | 原文线索

V2-007750 | MATHDB-405026

Positive density of sums of three 3-full numbers

摘录 / EXCERPT: Recall that a positive integer n is 3-full if $p\mid n$ implies $p^3\mid n$ for every prime p. Let A be the set of numbers which are the sum of three 3-full numbers. Does A...

open_problem | open | MathDB 页面 | 原文线索

V2-007751 | MATHDB-405082

Nontriviality of the constructed higher-dimensional QCAs

摘录 / EXCERPT: Prove that the QCAs constructed for the $SU(p)_1$ Walker-Wang model, for every odd prime p, cannot be realized as constant-depth circuits.

open_problem | open | MathDB 页面 | 原文线索

V2-007752 | MATHDB-405137

Conner-Floyd fixed-point conjecture for odd cyclic prime-power actions

摘录 / EXCERPT: If an odd cyclic p-group acts on a closed, connected, orientable, positive-dimensional smooth manifold, its fixed-point set cannot consist of a single point.

conjecture | open | MathDB 页面 | 原文线索

V2-007753 | MATHDB-405140

Free actions of odd-prime groups Q_d(p) on products of two spheres

摘录 / EXCERPT: For an odd prime p, can $Q_d(p)=((\mathbb{Z}/p)\times(\mathbb{Z}/p))\rtimes \mathrm{SL}_2(p)$ act freely on a finite CW-complex X with $X\simeq S^n\times S^m$ for some dimensio...

open_problem | open | MathDB 页面 | 原文线索

V2-007754 | MATHDB-405147

Fraenkel’s conjecture

摘录 / EXCERPT: For every integer $m\ge 3$, suppose that the positive integers admit a partition $\mathbb{Z}_{>0}=B(\alpha_1,\beta_1)\sqcup\cdots\sqcup B(\alpha_m,\beta_m)$ into Beatty sequences...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007755 | MATHDB-405167

Asymptotically optimal list size of random linear codes

摘录 / EXCERPT: For every prime power q and every $p\in(0,1-1/q)$, let $H_q(p)$ denote the q-ary entropy. As $\varepsilon\to0$ with $1-H_q(p)-\varepsilon>0$, a random linear code...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007756 | MATHDB-405179

Birch’s theorem and the variable-bound problem for the smooth Hasse principle

摘录 / EXCERPT: For each integer $d\geq 5$, determine the least integer $B(d)$ such that the following holds: for every homogeneous form $F\in\mathbb{Q}[x_1,\ldots,x_n]$ of degree d, let...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-007757 | MATHDB-405231

Fejér-Riesz inequality problem for Dirichlet series

摘录 / EXCERPT: Does there exist an absolute constant $C>0$ such that every Dirichlet polynomial $f(s)=\sum_{n=1}^{N}a_n n^{-s}$ satisfies ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007758 | MATHDB-405247

Kravitz’s denominator conjecture for the Lonely Runner spectrum

摘录 / EXCERPT: For every sextuple of distinct positive integers $v_1,\ldots,v_6$, define $\operatorname{ML}(v_1,\ldots,v_6)=\max_{t\in\mathbb{R}}\min_{1\leq i\leq 6}\|t v_i\|$, where...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-007759 | MATHDB-405257

Subconvexity problem for symmetric-square L-functions in the level aspect

摘录 / EXCERPT: For a family of tempered cuspidal automorphic representations π of $\mathrm{GL}_2(\mathbb{Q})$ with bounded archimedean parameters and conductor $q(\pi)$ tending to infinity, d...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007760 | MATHDB-405262

Erdős-Sárközy question

摘录 / EXCERPT: For a finite set $A\subseteq\mathbb{Z}_{>0}$, define its subset-sum set by $H(A)=\left\{\sum_{a\in B}a:B\subseteq A\right\}$. Let $g_3(n)$ be the least integer N such that there...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007761 | MATHDB-405263

Genčev-Rucki conjecture

摘录 / EXCERPT: For every integer $r>1$, the multiple Apéry-like series $ \mathcal{A}_r$ introduced by Genčev and Rucki, involving central binomial coefficients and finite multiple harmonic star s...

conjecture | solved | MathDB 页面 | 原文线索

V2-007762 | MATHDB-405266

Syracuse conjecture (Collatz conjecture)

摘录 / EXCERPT: For every positive integer $n\in\mathbb{Z}_{>0}$, define the Collatz map $T:\mathbb{Z}_{>0}\to\mathbb{Z}_{>0}$ by $T(n)=n/2$ if n is even and $T(n)=3n+1$ if n is odd. The conje...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007763 | MATHDB-405268

Han-Xiong 1/2-conjecture

摘录 / EXCERPT: Let $r\in\mathbb{Q}_{>0}$ and $k\in\mathbb{Z}_{\ge 1}$. If $\operatorname{Tr}_{\mathbb{Z}}\!\binom{r}{k}_q$ denotes the integer-exponent part (the integer trace) of the Gaussian bi...

conjecture | claimed_progress | MathDB 页面 | 原文线索

B02 代数 / Algebra | 4,312 条

V2-007764 | MATHDB-52

Non-unimodal Artinian Gorenstein algebras in embedding dimension 4

摘录 / EXCERPT: It remains open whether a standard graded Artinian Gorenstein algebra of embedding dimension 4 can have a non-unimodal Hilbert function; the analogous phenomenon is impossible in d...

open_problem | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-007765 | MATHDB-58

Weak Lefschetz property for codimension-three Artinian Gorenstein algebras

摘录 / EXCERPT: Conjecture. Every standard graded Artinian Gorenstein algebra of embedding dimension three over a field of characteristic zero has the weak Lefschetz property.

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-007766 | MATHDB-315467

Existence of a non-sofic group

摘录 / EXCERPT: Open question 3.8: Is every group sofic?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007767 | MATHDB-315483

Explicit minimal-dimensional counterexample to Deré's real-form conjecture

摘录 / EXCERPT: Can one construct explicitly a complex Lie algebra isomorphic to its conjugate but not definable over $\mathbb R$, and determine the smallest two-step nilpotent dimension where thi...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007768 | MATHDB-315493

Two-variable Factorial Conjecture

摘录 / EXCERPT: If $\mathcal L(x^a y^b)=a!b!$ and $\mathcal L(f^m)=0$ for every positive integer m, must a polynomial $f\in\mathbb C[x,y]$ be zero?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007769 | MATHDB-315495

Principal-minor-one matrices under taking powers

摘录 / EXCERPT: For which commutative rings is the property that all principal minors equal 1 inherited by every integer power of a matrix?

open_problem | open | MathDB 页面 | 原文线索

V2-007770 | MATHDB-315524

Log-concavity of codimension-three pure O-sequences

摘录 / EXCERPT: For a pure O-sequence $h=(h_0,\ldots,h_e)$ of codimension three and type two, is $h_i^2\geq h_{i-1}h_{i+1}$ for every interior index i?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007771 | MATHDB-315614

Solubilizer Conjecture A.1

摘录 / EXCERPT: If G is nonsolvable and $\operatorname{Sol}_G(x)\cap\operatorname{Sol}_G(y)$ is nonempty, must that intersection contain a nontrivial normal subgroup of G?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007772 | MATHDB-315724

Mathieu property for compact connected Lie groups

摘录 / EXCERPT: Let G be a compact connected Lie group. If f and h are finite-type complex-valued functions such that ... for all $P\in\mathbb{N}$, then ... for all sufficiently large...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-007773 | MATHDB-315859

Crouzeix's conjecture

摘录 / EXCERPT: Let $n \ge 1$ and let $A \in \mathbb{C}^{n \times n}$. Write ... for the field of values (numerical range) of A, a compact convex subset of $\mathbb{C}$, and let $\|\cdot\|$ denote...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007774 | MATHDB-315864

Goncharov conjecture

摘录 / EXCERPT: Let F be a field. For a field E write $E^{\times}_{\mathbb{Q}}:=E^{\times}\otimes_{\mathbb{Z}}\mathbb{Q}$, and let $\mathbb{Q}[\mathbb{P}^{1}_{E}]$ be the $\mathbb{Q}$-vector s...

conjecture | open | MathDB 页面 | 原文线索

V2-007775 | MATHDB-315865

Green's conjecture

摘录 / EXCERPT: Let C be a smooth irreducible complex projective curve of genus $g\ge 3$ that is not hyperelliptic, and let K denote a canonical divisor on C. For a divisor $D=\sum_P m_P P$...

conjecture | open | MathDB 页面 | 原文线索

V2-007776 | MATHDB-315868

Williamson conjecture

摘录 / EXCERPT: For a positive integer n, call a matrix $M=(m_{i,j})_{i,j=0}^{n-1}$ with entries $m_{i,j}\in\{+1,-1\}$ circulant if $m_{i,j}$ depends only on $j-i \bmod n$, i.e. there is a seque...

conjecture | open | MathDB 页面 | 原文线索

V2-007777 | MATHDB-315869

Ryser's conjecture on circulant Hadamard matrices

摘录 / EXCERPT: Let n be a positive integer and let $a=(a_{0},a_{1},\dots ,a_{n-1})$ with $a_{j}\in\{-1,1\}$ for every j. Let $H=\operatorname{circ}(a_{0},a_{1},\dots ,a_{n-1})$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-007778 | MATHDB-315870

Barker sequence conjecture

摘录 / EXCERPT: Let N be a positive integer and let $a_1,a_2,\dots,a_N$ be a sequence with $a_j\in\{+1,-1\}$ for all j. For $1\le v<N$ define the aperiodic autocorrelation coefficients ... Call...

conjecture | open | MathDB 页面 | 原文线索

V2-007779 | MATHDB-315877

Köthe conjecture

摘录 / EXCERPT: Let R be an associative ring, not necessarily commutative and not necessarily possessing a multiplicative identity. An element $a \in R$ is nilpotent if $a^{n} = 0$ for some inte...

conjecture | open | MathDB 页面 | 原文线索

V2-007780 | MATHDB-315878

Monomial conjecture

摘录 / EXCERPT: Let A be a Noetherian local ring of Krull dimension d, and let $x_1,\dots,x_d$ be a system of parameters for A, that is, $x_1,\dots,x_d$ lie in the maximal ideal of A and t...

conjecture | open | MathDB 页面 | 原文线索

V2-007781 | MATHDB-315884

Wild problem

摘录 / EXCERPT: A wild problem is a class of problems where the various parts that specify the problem can change, or keep changing, and there is no single "right" outcome to aim for. Every move o...

open_problem | open | MathDB 页面 | 原文线索

V2-007782 | MATHDB-316153

Suita conjecture

摘录 / EXCERPT: In mathematics, the Suita conjecture is a conjecture related to the theory of the Riemann surface, the boundary behavior of conformal maps, the theory of Bergman kernel, and the th...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007783 | MATHDB-316180

Nagata's conjecture

摘录 / EXCERPT: In algebra, Nagata's conjecture states that Nagata's automorphism of the polynomial ring k[x,y,z] is wild. The conjecture was proposed by Nagata and proved by Ualbai U. Umirbaev an...

conjecture | open | MathDB 页面 | 原文线索

V2-007784 | MATHDB-316213

Lafforgue's theorem

摘录 / EXCERPT: In mathematics, Lafforgue's theorem, due to Laurent Lafforgue, completes the Langlands program for general linear groups over algebraic function fields, by giving a correspondence...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-007785 | MATHDB-316234

Milnor conjecture

摘录 / EXCERPT: Let F be a field with $\operatorname{char} F \neq 2$. Define the Milnor K-ring of F as the graded ring ... where the tensor algebra is taken over $\mathbb{Z}$ on the abelian gr...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-007786 | MATHDB-316245

The arbitrary-degree elimination conjecture for the Rees algebra

摘录 / EXCERPT: Arbitrary-degree elimination conjecture. For arbitrary n, L has projective dimension two and specializes to the defining ideal of the Rees algebra $\mathcal R$.

conjecture | open | MathDB 页面 | 原文线索

V2-007787 | MATHDB-316259

The full Denis-Mordell-Lang conjecture for Drinfeld modules

摘录 / EXCERPT: Let g be a positive integer, and let $\phi_1:A\to K\{\tau\},\dots,\phi_g:A\to K\{\tau\}$ be Drinfeld modules. An algebraic $(\phi_1,\dots,\phi_g)$-submodule of $\mathbb{G}_a^g$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-007788 | MATHDB-316310

Mixed commutativity conjecture for the local integrals of motion

摘录 / EXCERPT: Let ${\cal I}_n$ and ${\されcal I}_n^*$ be the local integrals of motion for the deformed Virasoro algebra, defined by contour integrals for generic ${\rm Re}(s)>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-007789 | MATHDB-316311

Specialized commutativity conjecture for the nonlocal integrals of motion

摘录 / EXCERPT: At the specialization $s=2$, let ${\cal G}_n^{DV}$ denote the nonlocal integrals of motion for the deformed Virasoro algebra, defined by the multiple contour integrals in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-007790 | MATHDB-316312

Mixed commutativity conjecture for specialized local and nonlocal integrals

摘录 / EXCERPT: At $s=2$, let ${\cal I}_n^{DV}$ and ${\cal G}_m^{DV}$ be respectively the specialized local and nonlocal integrals of motion for the deformed Virasoro algebra. Mixed specialized co...

conjecture | open | MathDB 页面 | 原文线索

V2-007791 | MATHDB-316313

Specialization conjecture for the nonlocal integrals of motion

摘录 / EXCERPT: Let ${\cal G}_n(s)$ be the nonlocal integrals of motion for the deformed Virasoro algebra at parameter s, and let ${\cal G}_n^{DV}$ be the specialized nonlocal integrals defined...

conjecture | open | MathDB 页面 | 原文线索

V2-007792 | MATHDB-316317

η-invariance of the local integrals of motion

摘录 / EXCERPT: Let ${\cal I}_n$, for $n=1,2,\ldots$, be the Local Integrals of Motion for the deformed W-algebra $W_{q,t}(\widehat{sl_N})$, defined by analytic continuation when necessary. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-007793 | MATHDB-316319

Zanello's Gorenstein Interval Conjecture

摘录 / EXCERPT: Let $(1,h_1,\ldots,h_e)$ be a Gorenstein h-vector, so $h_e=1$ and the entries are symmetric. Fix an index i and a positive integer α. Suppose that both ... and ... are Goren...

conjecture | open | MathDB 页面 | 原文线索

V2-007794 | MATHDB-316324

Chisini's conjecture on generic coverings of the projective plane

摘录 / EXCERPT: Chisini's conjecture. Given two generic coverings $f:S\longrightarrow\mathbb{P}^2_{\mathbb{C}}$ and $f':S'\longrightarrow\mathbb{P}^2_{\mathbb{C}}$, both of degree at least 5, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-007795 | MATHDB-316335

The extended double-shuffle completeness conjecture for Euler sums

摘录 / EXCERPT: Extended double-shuffle conjecture. The induced map φ_R is injective; equivalently, the algebra of Euler sums is isomorphic to R_{EDS}.

conjecture | open | MathDB 页面 | 原文线索

V2-007796 | MATHDB-316361

The inverse shuffle-product conjecture for rational and algebraic power series over the binary field

摘录 / EXCERPT: Let σ be the quadratic form on $\overline{\mathbb F_2}[[X]]$ considered in the paper, which restricts to a bijection on $1+X\overline{\mathbb F_2}[[X]]$. A series is rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-007797 | MATHDB-316382

Brauer-Siegel conjecture for constant algebraic tori

摘录 / EXCERPT: Let K be the function field in the preceding setting, and let $T=T_0\times_{\mathbb F_q}K$, where T_0 is a fixed $\mathbb F_q$-torus. Put $d=\dim T$,...

conjecture | open | MathDB 页面 | 原文线索

V2-007798 | MATHDB-316398

The Hilbert coefficient bounds conjecture

摘录 / EXCERPT: Hilbert coefficient bounds conjecture. For $i=0,1,\ldots,n-s$,

conjecture | open | MathDB 页面 | 原文线索

V2-007799 | MATHDB-316399

Boij-Söderberg convex-hull conjecture for Betti diagrams

摘录 / EXCERPT: Boij-Söderberg convex-hull conjecture. The set $\bar B_{m,M}$ is the convex hull of $\Pi_{m,M}$.

conjecture | open | MathDB 页面 | 原文线索

V2-007800 | MATHDB-316414

The involutive antiautomorphism conjecture for tridiagonal pairs

摘录 / EXCERPT: Assume $\mathbb F$ is algebraically closed. Let V be a vector space over $\mathbb F$ with finite positive dimension, and let A,A^* be a tridiagonal pair on V. The involutive...

conjecture | open | MathDB 页面 | 原文线索

V2-007801 | MATHDB-316416

The parameter-array existence and uniqueness conjecture for tridiagonal pairs

摘录 / EXCERPT: Assume $\mathbb F$ is algebraically closed. Let d be a nonnegative integer, and let ... be a sequence of scalars from $\mathbb F$. A parameter array of a tridiagonal pair is the se...

conjecture | open | MathDB 页面 | 原文线索

V2-007802 | MATHDB-316417

The parameter-array isomorphism conjecture for tridiagonal pairs

摘录 / EXCERPT: Assume $\mathbb F$ is algebraically closed. The parameter-array isomorphism conjecture. Two tridiagonal pairs over $\mathbb F$ are isomorphic if and only if they have a parameter a...

conjecture | open | MathDB 页面 | 原文线索

V2-007803 | MATHDB-316474

Migliore-Nagel-Römer equality characterization conjecture

摘录 / EXCERPT: Let V be a finite set, let $R=\Bbbk[V]$ be a standard graded polynomial ring, and let I be a homogeneous ideal of height c. Let $e(R/I)$ be the Hilbert-Samuel multiplicity, a...

conjecture | solved | MathDB 页面 | 原文线索

V2-007804 | MATHDB-316519

The refined p-adic Beilinson conjecture for motives

摘录 / EXCERPT: The refined p-adic Beilinson conjecture. There are $e(n,M_\pi^E),e_p(n,M_\pi^E)\in(E\otimes_{\mathbb Q}\mathbb Q)^*$ such that the archimedean and p-adic regulator formulas hold,...

conjecture | open | MathDB 页面 | 原文线索

V2-007805 | MATHDB-316520

Zagier's conjecture on higher polylogarithms and K-theory

摘录 / EXCERPT: Zagier's conjecture. For $n\geq2$, there is an injection

conjecture | open | MathDB 页面 | 原文线索

V2-007806 | MATHDB-316535

Burns-Flach integrality conjecture for equivariant elliptic-curve L-values

摘录 / EXCERPT: Burns-Flach integrality conjecture. Assuming the rationality conjecture, one has

conjecture | open | MathDB 页面 | 原文线索

V2-007807 | MATHDB-316556

Internal zonotopal space interpolation conjecture for box splines

摘录 / EXCERPT: Let X be a unimodular list of vectors, let $Z(X)$ be its zonotope, and let ${\cal Z}_-(X)$ be the set of integer points in the interior of $Z(X)$. Let M_X be the box spline ass...

conjecture | open | MathDB 页面 | 原文线索

V2-007808 | MATHDB-316567

Third-degree Hilbert function formula for the symmetric quotient

摘录 / EXCERPT: Third-degree Hilbert function conjecture. If $d\geq 4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-007809 | MATHDB-316585

Finite generation conjecture for Galois groups of PAC extensions

摘录 / EXCERPT: Finite generation conjecture. If M is PAC over K, then

conjecture | open | MathDB 页面 | 原文线索

V2-007810 | MATHDB-316601

Finiteness conjecture for monomial bases of the centres of symmetric-group and Hecke algebras

摘录 / EXCERPT: Let S_n be the symmetric group, let ${\mathbb Z}S_n$ be its integral group algebra, and let ${\mathcal H}_n$ be the corresponding Iwahori--Hecke algebra. Write $Z(A)$ for the cen...

conjecture | open | MathDB 页面 | 原文线索

V2-007811 | MATHDB-316617

The symmetric shifting obstruction to embeddability in even-dimensional spheres

摘录 / EXCERPT: Symmetric shifting obstruction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-007812 | MATHDB-316643

Generalized quadratic recurrence integrality conjecture

摘录 / EXCERPT: Let $b(n)_{n=1}^\infty$ be generated by ... where $i<k-i<k$, $j<k-j<k$, and $b(l)=1$ for all $l<k$. Consider the recurrence with initial terms $b(l)=1$ for $0\leq l\leq k$. Generaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-007813 | MATHDB-316662

Breuil-Schneider's invariant norm conjecture for locally algebraic representations

摘录 / EXCERPT: Let L'/L be a finite Galois extension of p-adic fields and let K be a coefficient field. Fix an integer $d\geq 1$, an object (r,N,V) of $\mathrm{WD}_{L'/L}$ of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-007814 | MATHDB-316666

The toric g-vector conjecture for non-simplicial polytopes

摘录 / EXCERPT: Toric g-vector conjecture. The toric g-vector of a non-simplicial polytope is an M-sequence, i.e. it satisfies Macaulay inequalities.

conjecture | open | MathDB 页面 | 原文线索

V2-007815 | MATHDB-316685

Regularity-bounded Stanley decomposition conjecture

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ and let M be a finitely generated ${\mathbb Z}^n$-graded S-module. A Stanley decomposition of M is a finite direct sum ... Here $m_iK[Z_i]$ denotes a...

conjecture | open | MathDB 页面 | 原文线索

V2-007816 | MATHDB-316688

Amenability and right common multiples in group algebras

摘录 / EXCERPT: Let G be a group and let $\mathbb K G$ denote the group algebra of G over a field $\mathbb K$. Amenability and common-multiple conjecture. The following are equivalent: G is...

conjecture | open | MathDB 页面 | 原文线索

V2-007817 | MATHDB-316716

Polini-Ulrich conjecture on colon ideals of parameter ideals

摘录 / EXCERPT: Polini-Ulrich conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-007818 | MATHDB-316732

Existence of gluings for algebraic 2-Calabi-Yau realizations

摘录 / EXCERPT: Let n_1 and n_2 be positive integers, let B_1 and B_2 be antisymmetric integer matrices of sizes $n_1\times n_1$ and $n_2\times n_2$, and let B_{21} be an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-007819 | MATHDB-316743

Bounded covering-degree conjecture for Calabi-Yau threefold fibrations

摘录 / EXCERPT: Let $f:X\rightarrow B$ be a proper surjective morphism from a normal algebraic threefold X onto $B\simeq \mathbb{P}^1$, whose general fibre is a surface with Kodaira dimension ze...

conjecture | open | MathDB 页面 | 原文线索

V2-007820 | MATHDB-316766

The refined non-commutative main conjecture for an elliptic curve

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p, let $K=\mathbb{Q}(E(p))$, and let G be its relevant Galois group, assumed to have no element of order...

conjecture | open | MathDB 页面 | 原文线索

V2-007821 | MATHDB-316771

Uniqueness of Chevalley bases for finite-dimensional modular Lie superalgebras

摘录 / EXCERPT: Let $p>2$, and let a finite-dimensional Lie algebra or Lie superalgebra have an indecomposable Cartan matrix normalized as in the source. A Chevalley basis uniqueness conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-007822 | MATHDB-316779

Jannsen's generalisation of the Tate conjecture for K-theory

摘录 / EXCERPT: Let Y be proper over a finite field k, of dimension d, and let $\Ybar$ be its base change to an algebraic closure. Let $\Kp{q}{m}Y$ denote the relevant K-theoretic group and...

conjecture | open | MathDB 页面 | 原文线索

V2-007823 | MATHDB-316783

The q-Schur Rock block conjecture

摘录 / EXCERPT: Let $q\in k^\times$, and let p be the least natural number such that ... Let n,w be natural numbers with $n\geq w$. Let ${\cal Q}_p(n,w)$ be the quasi-hereditary subquotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-007824 | MATHDB-316785

The Hecke Rock block conjecture for Schiver doubles

摘录 / EXCERPT: Let $q\in k^\times$, and let p be the least natural number such that ... Let n,w be natural numbers with $n\geq w$. Let ${\cal D}_{A_{p-1}}(n)$ be the Schiver double associated t...

conjecture | open | MathDB 页面 | 原文线索

V2-007825 | MATHDB-316789

The Super KSh conjecture for simple modular Lie superalgebras in characteristic greater than 5

摘录 / EXCERPT: Super KSh conjecture. The examples obtained by applying the KSh procedure exhaust all simple finite dimensional Lie superalgebras over algebraically closed fields of characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-007826 | MATHDB-316790

The amended KSh conjecture for modular Lie algebras and Lie superalgebras

摘录 / EXCERPT: Amended KSh conjecture. These constructions yield all $\mathbb{Z}$-graded simple finite dimensional examples of Lie algebras and Lie superalgebras in characteristic $p>0$, while th...

conjecture | open | MathDB 页面 | 原文线索

V2-007827 | MATHDB-316802

Existence and functoriality of heat kernels on abstract Wiener groups

摘录 / EXCERPT: Let $\mathbf H\subset\mathbf G$ be an abstract Wiener group with Lie algebras $\mathbf h\subset\mathbf g$, and let $\nu_{t/N}^{\mathbf h\subset\mathbf g}$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-007828 | MATHDB-316803

Left-right invariance criterion for abstract Wiener-group heat kernels

摘录 / EXCERPT: Let $\mathbf H\subset\mathbf G$ be an abstract Wiener group with Lie algebra $\mathbf h$, let $\mathbf H_0$ be the identity component of $\mathbf H$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-007829 | MATHDB-316809

Minimal-model Chern-ratio conjecture for random surfaces

摘录 / EXCERPT: Minimal-model Chern-ratio conjecture. There exist minimal nonsingular projective surfaces X_0 of general type whose Chern ratios

conjecture | open | MathDB 页面 | 原文线索

V2-007830 | MATHDB-316829

Coincidence of tagged-triangulation and cluster frieze patterns

摘录 / EXCERPT: Coincidence conjecture. For every tagged triangulation $\widetilde{\mathcal T}$ of $S_{\odot}$, the frieze patterns coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-007831 | MATHDB-316830

Realization of type D_N frieze patterns by tagged triangulations

摘录 / EXCERPT: Realization conjecture. Every frieze pattern of type D_N is of the form $F(\widetilde{\mathcal T})$ for some tagged triangulation $\widetilde{\mathcal T}$ of $S_{\odot}$.

conjecture | open | MathDB 页面 | 原文线索

V2-007832 | MATHDB-316831

The Two Osculating Points Problem

摘录 / EXCERPT: Let C be a smooth complete complex curve, let V define a morphism $\varphi_V:C\to\bP^n$ that is locally toric, and let n_1,n_2 be positive integers with $n_1+n_2=n+2$. Suppos...

open_problem | open | MathDB 页面 | 原文线索

V2-007833 | MATHDB-316849

de Gier's stationary distribution conjecture for half-turn-invariant FPL link patterns

摘录 / EXCERPT: Let $A_{\textsc{HT}}(N;\pi)$ be the number of half-turn-invariant FPLs of size N with half-turn-invariant link pattern π, and let $A_{\textsc{HT}}(N)$ be the total number of...

conjecture | open | MathDB 页面 | 原文线索

V2-007834 | MATHDB-316860

Unimodality conjecture for typical weight multiplicity polynomials of \(\mathfrak{gl}(n,m)\)

摘录 / EXCERPT: Let $P^+(n,m)$ be the set of dominant weights, let ρ be the relevant Weyl vector, and let $K_{\lambda,\mu}(q)$ denote the q-analog of the weight multiplicity for...

conjecture | open | MathDB 页面 | 原文线索

V2-007835 | MATHDB-316901

Base-change compatibility of the new t-structure

摘录 / EXCERPT: Let A be an associative algebra and consider ... Equip it with the two t-structures described in the source: new_1, obtained by tensoring the new t-structure on the affi...

conjecture | open | MathDB 页面 | 原文线索

V2-007836 | MATHDB-316902

The renormalized affine-flag localization conjecture

摘录 / EXCERPT: Let $ Op$ be the space of opers with nilpotent singularity, let $\tN/\cG$ be the nilpotent-cone quotient, and let ... be the ind-completed categorical base change. Let ... be the exten...

conjecture | open | MathDB 页面 | 原文线索

V2-007837 | MATHDB-316903

Exactness of global sections in the new t-structure

摘录 / EXCERPT: Let $\Gamma_\Fl:\bD_{\mathrm{ren}}(\fD(\Fl^{\operatorname{aff}}_G)_\crit\operatorname{mod})\to\bD_{\mathrm{ren}}(\hg_\crit\operatorname{mod}_\nilp)$ be the global-sections functor,...

conjecture | open | MathDB 页面 | 原文线索

V2-007838 | MATHDB-316930

Partition conjecture relating characteristic posets and regularity

摘录 / EXCERPT: Let K be a field, $S=K[x_1,\ldots,x_n]$, and let $J\subset I$ be monomial ideals. Let $P_{I/J}$ be the characteristic poset of I/J, and let $\operatorname{reg}(I/J)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-007839 | MATHDB-316938

The non-commutative parameter algebra conjecture

摘录 / EXCERPT: Non-commutative parameter algebra conjecture. Parameter algebras should be non-commutative. This proposal is motivated by the fact that the commutativity relations among the parame...

conjecture | open | MathDB 页面 | 原文线索

V2-007840 | MATHDB-316939

The chain-complex deformation proposal

摘录 / EXCERPT: Let a deformation problem originally concern modules over a parameter algebra, with flatness of the universal family needed to construct the deformation. Over non-commutative rings...

conjecture | open | MathDB 页面 | 原文线索

V2-007841 | MATHDB-316942

The nonexistence of a sufficient decomposition in a generalized function algebra

摘录 / EXCERPT: Nonexistence conjecture. There exists $u\in{\rm Gen}_{\mathscr{S}}(\mathbb{R}^d)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-007842 | MATHDB-316953

Ito-Terwilliger endpoint algebra conjecture for tridiagonal systems

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let V be a finite-dimensional vector space, and let $\Phi=(A;\{E_i\}_{i=0}^d;A^*;\{E_i^*\}_{i=0}^d)$ be a tridiagonal system. Let $\mathcal{D}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-007843 | MATHDB-316954

The finite-field quadratic-growth conjecture for simple algebras

摘录 / EXCERPT: Let k be a finite field and let A be a finitely generated k-algebra of quadratic growth. Finite-field quadratic-growth conjecture. Then A is not simple. The conjecture is m...

conjecture | open | MathDB 页面 | 原文线索

V2-007844 | MATHDB-316956

Kashiwara-Nakashima-Okado conjecture on positive geometric crystals and perfect crystals

摘录 / EXCERPT: Let G and $\mathfrak g=\langle \mathfrak t,e_i,f_i\rangle_{i\in I}$ be the affine Kac-Moody group and algebra associated with a generalized Cartan matrix, and let B^-_w be the...

conjecture | open | MathDB 页面 | 原文线索

V2-007845 | MATHDB-316967

Evans' Lex-Plus-Powers conjecture for graded Betti numbers

摘录 / EXCERPT: Let $S=\Bbbk[x_{1},\ldots,x_{n}]$, let $P=(x_{1}^{e_{1}},\ldots,x_{r}^{e_{r}})$, and let $F=(f_{1},\ldots,f_{r})$ be a homogeneous regular sequence with $\deg f_i=e_i$. Let I be...

conjecture | open | MathDB 页面 | 原文线索

V2-007846 | MATHDB-316968

Monomial-ideal reduction form of the Lex-Plus-Powers conjecture

摘录 / EXCERPT: Let $S=\Bbbk[x_{1},\ldots,x_{n}]$, let $(f_{1},\ldots,f_{r})$ be a regular sequence with $\deg f_i=e_i$, and set ... Let I be the homogeneous ideal containing this regular sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-007847 | MATHDB-316969

Gasharov-Hibi-Peeva conjecture on Betti numbers over a power quotient

摘录 / EXCERPT: Let $S=\Bbbk[x_{1},\ldots,x_{n}]$, let $P=(x_{1}^{e_{1}},\ldots,x_{r}^{e_{r}})$, and set $R=S/P$. Let I be a homogeneous ideal containing P, and let L be the lex-plus-P ide...

conjecture | open | MathDB 页面 | 原文线索

V2-007848 | MATHDB-316972

The Nomura-Terwilliger conjecture on the Terwilliger algebra corner

摘录 / EXCERPT: Nomura-Terwilliger conjecture. With respect to this notation, the following hold: (i) $E^*_0\mathcal{T}E^*_0$ is generated by $E^*_0\mathcal{D}E^*_0$; and (ii) the elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-007849 | MATHDB-316992

Sharifi's modular-symbol isomorphism conjecture

摘录 / EXCERPT: Let (p,k) be an irregular pair with k even and $k\geq12$, let $F={\bf Q}(\mu_p)$, and let $A_F^{(1-k)}$ be the corresponding eigenspace of the p-part of the class group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-007850 | MATHDB-316993

Algebraic packing property conjecture for uniform clutters

摘录 / EXCERPT: Let $\mathcal C$ be a uniform clutter with incidence vectors $v_1,\ldots,v_q$, and let P be the associated polytope. The clutter satisfies the packing property if every minor sat...

conjecture | open | MathDB 页面 | 原文线索

V2-007851 | MATHDB-316999

The Rouquier-family refinement and dimension conjecture for complex reflection groups

摘录 / EXCERPT: Let W be a complex reflection group satisfying the generic Hecke algebra hypothesis. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-007852 | MATHDB-317008

The complete-symmetric-polynomial regular-sequence conjecture for three variables

摘录 / EXCERPT: Let $h_k(3)$ be the complete symmetric polynomial of degree k in three variables, and for $A=\{a,b,c\}$ with $a<b<c$ let $h_A(3)$ denote the set of polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-007853 | MATHDB-317018

Bertola-Prats Ferrer conjecture on Harish-Chandra integrals as nilpotent integrals

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple Lie algebra over $\mathbb C$, let $\mathfrak h$ be a Cartan subalgebra, let K be the maximal compact group in $\exp(\mathfrak g)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-007854 | MATHDB-317044

Negativity characterization of Cohen-Macaulayness via the first Hilbert coefficient

摘录 / EXCERPT: Negativity characterization. For a parameter ideal J,

conjecture | open | MathDB 页面 | 原文线索

V2-007855 | MATHDB-317068

Barad's factorization conjecture for the type B Temperley-Lieb Gram determinant

摘录 / EXCERPT: Barad's factorization conjecture. The determinant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-007856 | MATHDB-317072

Herzog-Huneke-Srinivasan multiplicity conjecture

摘录 / EXCERPT: Let $S={\bf k}[x_1,\ldots,x_n]$ be a polynomial ring over a field, and let N be a finitely generated graded S-module with a minimal graded free resolution of length l. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-007857 | MATHDB-317073

Simplicial-complex criterion for multiplicity upper-bound equality

摘录 / EXCERPT: Let Γ be a $(d-1)$-dimensional simplicial complex. Let $Q_0,\ldots,Q_{d-1}$ and $c_0,\ldots,c_{d-1}$ be the quantities defined in the paper, and let $A(j,Q_i)$ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-007858 | MATHDB-317092

The Yoneda-convolution conjecture for half-density sheaves

摘录 / EXCERPT: Let $\mathcal{S}_{n-k,k}$ be the Slodowy slice, let Y be the corresponding Springer fiber, and let A range over all irreducible components of Y. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-007859 | MATHDB-317094

The telescope conjecture for module categories

摘录 / EXCERPT: Let $Lambda$ be an artin algebra, and let $(\mathcal{A},\mathcal{B})$ be a hereditary cotorsion pair in the category of all left $Lambda$-modules. A cotorsion pair is hereditary wh...

conjecture | open | MathDB 页面 | 原文线索

V2-007860 | MATHDB-317095

Finite-type reformulation of the telescope conjecture for module categories

摘录 / EXCERPT: Let Λ be an artin algebra, and let $(\mathcal{A},\mathcal{B})$ be a hereditary cotorsion pair in the category of all left Λ-modules. A cotorsion pair is of finite t...

conjecture | open | MathDB 页面 | 原文线索

V2-007861 | MATHDB-317115

The group-theoretic Torelli conjecture for curves over finite fields

摘录 / EXCERPT: Let $k=\overline{\mathbb F}_p$ with $p\geq 3$, and let C and $\tilde{C}$ be smooth projective curves over k of genus at least 2, with Jacobians J and $\tilde{J}$. Suppose t...

conjecture | open | MathDB 页面 | 原文线索

V2-007862 | MATHDB-317116

The four-configuration Torelli conjecture for curves

摘录 / EXCERPT: Let C be a curve of genus $\mathsf g(C)\geq 2$. An R_4-configuration is the configuration defined earlier in the paper, and a curve realizes an R_4-string when it carries suc...

conjecture | open | MathDB 页面 | 原文线索

V2-007863 | MATHDB-317134

Martsinkovsky's quasi-homogeneity conjecture for the fundamental module

摘录 / EXCERPT: Let A be a normal analytic algebra of Krull dimension two over an algebraically closed field of characteristic zero. The quasi-homogeneity conjecture. The isomorphism ... is equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-007864 | MATHDB-317139

Batyrev and Popov's quadratic generation conjecture for Cox rings of del Pezzo surfaces

摘录 / EXCERPT: Let X be a del Pezzo surface. Let I_X be the defining ideal in a presentation ${\rm Cox}(X)\cong k[\mathcal G]/I_X$, where $\mathcal G$ is a minimal homogeneous generating set...

conjecture | open | MathDB 页面 | 原文线索

V2-007865 | MATHDB-317145

The Böttcher-Wenzel conjecture for matrix commutators

摘录 / EXCERPT: Let X and Y be two $n\times n$ matrices, let $[X,Y]=XY-YX$ be their commutator, and let $||\cdot||$ denote the Hilbert-Schmidt norm. Böttcher-Wenzel conjecture. ... The conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-007866 | MATHDB-317150

Weyman's minimal-generators conjecture for nilpotent orbit ideals

摘录 / EXCERPT: Let λ be a partition, and let U_λ denote Weyman's generating set for the ideal ${\mathcal{J}}_\lambda$ associated with the nilpotent orbit determined by λ....

conjecture | open | MathDB 页面 | 原文线索

V2-007867 | MATHDB-317172

The finite-by-cyclic assembly conjecture for algebraic K-theory

摘录 / EXCERPT: Let G be a discrete group and R a ring. Let $\mathcal{\mathrm{fbc}}$ denote the family of finite-by-cyclic subgroups, $\mathcal{\mathrm{all}}$ the family of all subgroups, and...

conjecture | open | MathDB 页面 | 原文线索

V2-007868 | MATHDB-317173

The positive-coordinate formula for iterated Nil-groups

摘录 / EXCERPT: Positive-coordinate Nil-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-007869 | MATHDB-317212

Weibel's K-dimensional conjecture

摘录 / EXCERPT: Weibel's K-dimensional conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-007870 | MATHDB-317221

Generalized Sato-Tate conjecture for curves

摘录 / EXCERPT: Let C be a curve of genus g, and let $\bar{L}_p(T)$ denote its unitarized L-polynomial. Let H be an infinite compact subgroup of $USp(2g)$. Generalized Sato-Tate conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-007871 | MATHDB-317223

Weak continuity conjecture for generic discrete series characters

摘录 / EXCERPT: Weak continuity conjecture. A generic family δ of irreducible discrete series characters has weakly continuous limits to every point $k\in\overline{U}$.

conjecture | open | MathDB 页面 | 原文线索

V2-007872 | MATHDB-317238

Minimal kinetic-energy conjecture for blowup solutions to the focusing energy-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Minimal kinetic-energy conjecture. If $E_*<\|\nabla W\|_2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-007873 | MATHDB-317270

A tensor-product divisibility criterion for common rational composition

摘录 / EXCERPT: Common-composition criterion. There exists a rational function $g\inK(t)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-007874 | MATHDB-317287

The cluster algebra realization conjecture for ${d49c}_J$

摘录 / EXCERPT: Let J be a subset of the index set of simple roots, and let ${\cal A}_J$ be the cluster algebra constructed from the corresponding subcategory, contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-007875 | MATHDB-317289

The dual canonical basis conjecture for rigid preprojective modules

摘录 / EXCERPT: Let Λ be the preprojective algebra, and let M be a finite-dimensional rigid Λ-module, meaning that its self-extension group vanishes. Let φ_M be the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-007876 | MATHDB-317292

Yamazaki's generalized Birch-Tate conjecture for tori

摘录 / EXCERPT: Let K be a totally real number field and let T be a torus over K that is split by a totally real field. Let $X(T)=\operatorname{Hom}(\mathbb{G}_m,T)$ be the cocharacter group...

conjecture | open | MathDB 页面 | 原文线索

V2-007877 | MATHDB-317347

Kashiwara-Toledano Laredo crystal commutor conjecture

摘录 / EXCERPT: Kashiwara-Toledano Laredo's conjecture. For any symmetrizable Kac-Moody algebra $\mathfrak{g}$, σ^{KT} is a coboundary structure on $Cryst(\mathfrak{g})$.

conjecture | open | MathDB 页面 | 原文线索

V2-007878 | MATHDB-317367

Conjecture on null pairs in universal Lie nilpotent associative algebras

摘录 / EXCERPT: Let A be an associative algebra, and let $L_1(A)=A$ and $L_{k+1}(A)=[A,L_k(A)]$ denote its lower central series. A pair (i,j) is called null when, for every associative algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-007879 | MATHDB-317372

Uniqueness conjecture for the triangulordinary filtration on the eigencurve

摘录 / EXCERPT: Let $\mathscr{E}^0$ be the semistable, distinct-eigenvalue locus of the eigencurve, let $\mathscr{V}^0$ be its rank-2 family of Galois representations, and set ... For each...

conjecture | open | MathDB 页面 | 原文线索

V2-007880 | MATHDB-317414

Jie-Tai Yu's degree estimate for commutators

摘录 / EXCERPT: Jie-Tai Yu's conjecture. If neither $\operatorname{deg}(f)$ nor $\operatorname{deg}(g)$ divides the other, then

conjecture | open | MathDB 页面 | 原文线索

V2-007881 | MATHDB-317415

Makar-Limanov and Jie-Tai Yu's fractional-power conjecture

摘录 / EXCERPT: Makar-Limanov and Jie-Tai Yu's conjecture. For every $m>n$ not divisible by n, the formal power series

conjecture | open | MathDB 页面 | 原文线索

V2-007882 | MATHDB-317450

Compatibility conjecture for the Grossman-Larson coproduct and rooted-tree coproduct

摘录 / EXCERPT: Compatibility conjecture. For any rooted tree t,

open_problem | open | MathDB 页面 | 原文线索

V2-007883 | MATHDB-317462

Dimension conjecture for q,t-Fuss-Catalan numbers of complex reflection groups

摘录 / EXCERPT: Let W be a well-generated complex reflection group. The q,t-Fuss-Catalan number is defined by ... where $M^{(m)}$ is the associated W-module, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-007884 | MATHDB-317489

Free dendriform trialgebra conjecture for colored parking-function quasi-symmetric functions

摘录 / EXCERPT: Free dendriform trialgebra conjecture. The algebra ${\bf PQSym}^{(l)}$ is a free dendriform trialgebra.

conjecture | open | MathDB 页面 | 原文线索

V2-007885 | MATHDB-317503

The projective-dimension-two equality conjecture for Betti-diagram semigroups

摘录 / EXCERPT: Projective-dimension-two equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-007886 | MATHDB-317533

Ring-specific Oort conjecture for cyclic covers

摘录 / EXCERPT: Ring-specific Oort conjecture. The cover f lifts over R. This is a stronger, ring-specific form of Oort's conjecture because it specifies the mixed-characteristic ring over whi...

conjecture | open | MathDB 页面 | 原文线索

V2-007887 | MATHDB-317534

Arithmetic form of the ring-specific Oort conjecture

摘录 / EXCERPT: Arithmetic ring-specific Oort conjecture. There exist $l>0$ and a normal, C-Galois, regular branched cover $Y\rightarrow D$ such that Y_k is reduced, $L_m=M$ for $m\gg0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-007888 | MATHDB-317548

Zamolodchikov periodicity conjecture for Dynkin Y-systems

摘录 / EXCERPT: Zamolodchikov periodicity conjecture. Every solution of this system is periodic in t, with period dividing $2(h+h')$. The periodicity conjecture is a central structural statement...

conjecture | open | MathDB 页面 | 原文线索

V2-007889 | MATHDB-317554

Yau-Zaslow norm-dependence conjecture for genus-zero K3 BPS counts

摘录 / EXCERPT: Let S be a K3 surface, let $\beta\in H^2(S,\mathbb{Z})$ be an effective algebraic class, and let $r_{0,\beta}$ denote its genus-zero BPS count, defined through the reduced Gromov...

conjecture | open | MathDB 页面 | 原文线索

V2-007890 | MATHDB-317564

Brownawell-Yu conjecture on algebraic independence of Drinfeld logarithms

摘录 / EXCERPT: Let ρ be a Drinfeld $\mathbb{F}_q[t]$-module defined over $\overline{k}$. Suppose $F_{\delta_1},\dots,F_{\delta_s}$ are quasi-periodic functions for $\mathbb{C}_\infty$-linear...

conjecture | open | MathDB 页面 | 原文线索

V2-007891 | MATHDB-317600

The pseudo-arithmetical characterization conjecture

摘录 / EXCERPT: Pseudo-arithmetical characterization conjecture. R is pseudo-arithmetical if and only if the zero ideal is locally irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-007892 | MATHDB-317605

The elementary generation conjecture for $SL_2(\mathbb Z[t,t^{-1}])$

摘录 / EXCERPT: Let $\mathbb Z[t,t^{-1}]$ be the Laurent polynomial ring in one variable over the integers, let $SL_2(\mathbb Z[t,t^{-1}])$ be its special linear group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-007893 | MATHDB-317606

The quaternionic non-finite-generation conjecture for anisotropic groups

摘录 / EXCERPT: Let R and F be as in the Krull-dimension-three non-finite-generation conjecture, so R is a finitely generated integral domain of Krull dimension at least 3 and F is its f...

conjecture | open | MathDB 页面 | 原文线索

V2-007894 | MATHDB-317660

Kottwitz-style component conjecture for parahoric bundle moduli

摘录 / EXCERPT: Assume that $\operatorname{char}(k)\neq 2$, that $\pi:\widetilde X\rightarrow X$ is a possibly ramified double cover with involution σ, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-007895 | MATHDB-317689

Revised Malle conjecture for number fields

摘录 / EXCERPT: Revised Malle conjecture. Fixing N and k, one has

conjecture | refuted | MathDB 页面 | 原文线索

V2-007896 | MATHDB-317693

Glaz's existence problem for a non-Noetherian Cohen-Macaulay notion

摘录 / EXCERPT: Glaz's existence problem. Can one define non-Noetherian Cohen-Macaulay rings so that: (i) the definition agrees with the usual definition for Noetherian rings; (ii) coherent regula...

open_problem | open | MathDB 页面 | 原文线索

V2-007897 | MATHDB-317709

Freeness conjecture for the degree-one subalgebra of the noncommutative Connes-Kreimer algebra

摘录 / EXCERPT: Freeness conjecture. The $\fontfamily{U}\fontshape{n}\selectfont{As}^2$-subalgebra of NCK generated by elements of degree 1 is a free algebra over the operad of two compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-007898 | MATHDB-317731

Generating-function conjecture for Thom polynomials of nilpotent algebras

摘录 / EXCERPT: Let Q be a $[?]$-dimensional, commutative, nilpotent algebra with $\deg(\operatorname{tp}_Q(l))=\mu\cdot l+\gamma$. Write ... and let RES denote the iterated resid...

conjecture | open | MathDB 页面 | 原文线索

V2-007899 | MATHDB-317735

Minimal-intersection conjecture for truncated symmetric-function modules

摘录 / EXCERPT: Let Λ be the graded ring of symmetric functions, with positive-degree ideal Λ_+. For $i\leq j$, define the Λ-modules ... and ... Thus A is included in B. Mi...

conjecture | open | MathDB 页面 | 原文线索

V2-007900 | MATHDB-317736

The semisimple similarity number conjecture for matrix sequences

摘录 / EXCERPT: Semisimple similarity number conjecture. For arbitrary $m\times m$ sequences, not necessarily semisimple, $\mathcal{A}\sim\mathcal{B}$ if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-007901 | MATHDB-317743

Surjectivity conjecture for the graded evaluation map of hyperelliptic Abelian functions

摘录 / EXCERPT: Surjectivity conjecture. The map ev^{gr} is surjective. Equivalently, $\text{gr}^{KP}\,A$ is generated as a ${\cal D}$-module by $1\in \text{gr}^{KP}_0A$ and the elements...

conjecture | open | MathDB 页面 | 原文线索

V2-007902 | MATHDB-317745

Odd-dimensional antisymmetric solution conjecture for noncommutative determinants

摘录 / EXCERPT: Let n be a positive integer, and let A and B be the matrices occurring in the identity $\operatorname{eq.further.1}$, with parameters h_1 and q_i. A matrix is antisymmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-007903 | MATHDB-317756

The hyperlogarithmic realization conjecture

摘录 / EXCERPT: Let ${\cal H}_n(E)$ be the space of hyperlogarithmic symbols and let δ_n be its coproduct. Suppose $r_{{\cal H}_n}$ denotes the realization map referred to in the paper's pr...

conjecture | open | MathDB 页面 | 原文线索

V2-007904 | MATHDB-317776

General global maximum conjecture for the rank-two likelihood model

摘录 / EXCERPT: Let $n\ge2$ and let s,t satisfy $0<t<s$. For positive u and positive integer block dimensions, write $(u)_{l_1\times l_2}$ for the $l_1\times l_2$ matrix all of whose entries e...

conjecture | open | MathDB 页面 | 原文线索

V2-007905 | MATHDB-317777

Alternative general global maximum conjecture for the rank-two likelihood model

摘录 / EXCERPT: Let $n\ge2$ and let s,t satisfy $0<s\le t$. Let $L(P)$ be the likelihood function defined in the source by equation (newLP). Alternative likelihood conjecture. The matrix ... is a...

conjecture | open | MathDB 页面 | 原文线索

V2-007906 | MATHDB-317779

Type A restriction is weakly multiplicity-free

摘录 / EXCERPT: Let ${\mathscr{H}}$ be the Hecke algebra of $(W,S)=({\mathcal{S}}_n,\{s_1,\ldots,s_{n-1}\})$, and let $J\subseteq S$ satisfy ... A restriction is weakly multiplicity-free when there...

conjecture | open | MathDB 页面 | 原文线索

V2-007907 | MATHDB-317781

Full Reeh-Schlieder conjecture for the free Dirac field

摘录 / EXCERPT: Let M be a globally hyperbolic spin spacetime with a non-compact Cauchy surface. Let ${\mathbf{B}}$ denote the causal locally covariant theory used for the C^*-algebraic approa...

conjecture | open | MathDB 页面 | 原文线索

V2-007908 | MATHDB-317842

Local higher-weight crystalline lift conjecture

摘录 / EXCERPT: Let $K/\mathbb{Q}_p$ be a finite extension with residue field k_K, absolute ramification degree e, and let $\overline{\rho}_K:G_K\to\operatorname{GL}_2(\overline{\mathbb{F}}_p)$...

conjecture | open | MathDB 页面 | 原文线索

V2-007909 | MATHDB-317847

The Mamma conjecture for the fundamental additive invariant

摘录 / EXCERPT: Mamma conjecture. Given a group G, the fundamental additive invariant ${\nobreak\boldsymbol{\mathbf{E}}}^{\mathsf{fund}}$ has the ${\mathcal V}C$-assembly property when the base...

conjecture | open | MathDB 页面 | 原文线索

V2-007910 | MATHDB-317848

The Mamma conjecture for K-theory with dg-category coefficients

摘录 / EXCERPT: Mamma conjecture. Given a group G, the functors $K(-;{\mathcal B})$ have the ${\mathcal V}C$-assembly property for all strictly finite dg cells ${\mathcal B}$, when the base ring...

conjecture | open | MathDB 页面 | 原文线索

V2-007911 | MATHDB-317884

Representational assembly conjecture for geometric fields

摘录 / EXCERPT: Representational assembly conjecture. The representational assembly map

conjecture | open | MathDB 页面 | 原文线索

V2-007912 | MATHDB-317885

Representational assembly conjecture for finite-characteristic fields

摘录 / EXCERPT: Finite-characteristic representational assembly conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-007913 | MATHDB-317935

The GKM-pair character conjecture for Braden-MacPherson sheaves

摘录 / EXCERPT: Let ${\widehat{\mathcal G}}$ be the affine moment graph, let $w\in\mathcal V={\widehat{\mathcal W}}$, and let (K,w) be a GKM-pair. For the Braden-MacPherson sheaf...

conjecture | open | MathDB 页面 | 原文线索

V2-007914 | MATHDB-317949

The Grassmannian Euler-characteristic conjecture for affine quivers

摘录 / EXCERPT: Let Q be an affine quiver of affine type, let $ \lambda displaystyle\in \mathbb P^1_0(Q)$, and let E be a quasi-simple module in an exceptional tube. For a dimension vector...

conjecture | open | MathDB 页面 | 原文线索

V2-007915 | MATHDB-317963

Nikolaev's torsion conjecture for CM elliptic curves

摘录 / EXCERPT: Let $E_{CM}\cong E(K)$ be an elliptic curve with complex multiplication. Let $E_{tors}(K)$ be the finite-order subgroup of $E(K)$. If $K'\supset K$ is a non-trivial extension obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-007916 | MATHDB-317965

Blasius-Panchishkin conjecture on twisted critical values of automorphic L-functions

摘录 / EXCERPT: Let Π be a regular algebraic cuspidal representation of ${\rm GL}_n({\mathbb A}_{\mathbb Q})$. For an even Dirichlet character χ, viewed as a Hecke character, the finite...

conjecture | open | MathDB 页面 | 原文线索

V2-007917 | MATHDB-317966

Period relations conjecture for Rankin-Selberg motives

摘录 / EXCERPT: Let Π and Σ be regular algebraic cuspidal automorphic representations of ${\rm GL}_n({\mathbb A})$ and ${\rm GL}_{n-1}({\mathbb A})$, respectively, with associated highe...

conjecture | open | MathDB 页面 | 原文线索

V2-007918 | MATHDB-317967

Taelman's class number conjecture for abelian t-modules

摘录 / EXCERPT: Let K be a finite separable extension of $\mathbf{F}_q(t)$, let $A=\mathbf{F}_q[t]$, and let K_∞ and O_K denote the completions and the ring of integers used in the defi...

conjecture | open | MathDB 页面 | 原文线索

V2-007919 | MATHDB-318017

Takemura's conjecture on the support of the Heun root-counting measure

摘录 / EXCERPT: Let $Q(z)$ have roots a_1,a_2,a_3, and let μ be the limiting root-counting measure from the Shapiro-Tater conjecture. For $i\in\{1,2,3\}$, let γ_i be the curve consist...

conjecture | open | MathDB 页面 | 原文线索

V2-007920 | MATHDB-318037

Completeness conjecture for the lattice of coideal subalgebras of odd-dimensional Kac algebras

摘录 / EXCERPT: Let n be odd. The Kac algebra $K\!D(n)$ has lattice of coideal subalgebras $\operatorname{I}(K\!D(n))$, described in Theorem theorem.lattice.nodd, with an exceptional set...

conjecture | open | MathDB 页面 | 原文线索

V2-007921 | MATHDB-318056

Hain and Looijenga's compactly supported tautological duality conjecture

摘录 / EXCERPT: Hain and Looijenga's conjecture. These intersection pairings are perfect (Poincaré duality), and $R^{\bullet}_c(M_g)$ is a free $R^{\bullet}(M_g)$-module of rank one.

conjecture | open | MathDB 页面 | 原文线索

V2-007922 | MATHDB-318065

Faithful Galois action on components of moduli of minimal surfaces

摘录 / EXCERPT: Faithful-action conjecture. There is a minimal surface of general type S such that S and its conjugate S^{ϕ} have non-isomorphic fundamental groups. In particular, S and...

conjecture | open | MathDB 页面 | 原文线索

V2-007923 | MATHDB-318066

Faithful Galois action on unmarked triangle curves

摘录 / EXCERPT: Triangle-curve faithfulness conjecture. The absolute Galois group $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully on the set of isomorphism classes of unmark...

conjecture | open | MathDB 页面 | 原文线索

V2-007924 | MATHDB-318110

The minimal-generator conjecture for the diagonal ideal

摘录 / EXCERPT: Let I be the radical ideal defining the diagonal locus of $(\mathbb{C}^2)^n$, and let M_{d_1,d_2} denote its quotient by the ideal generated by the coordinate variables, in bid...

conjecture | solved | MathDB 页面 | 原文线索

V2-007925 | MATHDB-318111

The conjectural generating set for the diagonal ideal

摘录 / EXCERPT: Let Λ be the set of integer sequences $\lambda_1\geq\cdots\geq\lambda_{n-1}\geq\lambda_n=0$ satisfying $\lambda_i\leq n-i$ for all i. For each $\lambda\in\Lambda$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-007926 | MATHDB-318140

Vasconcelos's flatness criterion for algebras

摘录 / EXCERPT: Vasconcelos's conjecture. A is R-flat if and only if its n-fold tensor power

conjecture | open | MathDB 页面 | 原文线索

V2-007927 | MATHDB-318161

Monotonicity conjecture for invariants of unconstrained hidden Markov models

摘录 / EXCERPT: Let $\mathbf{f}_{n,l}$ be the unconstrained hidden Markov model with l hidden states and strings of length n, and let $d(n,l)$ denote the maximum degree of its invariants. Mono...

conjecture | open | MathDB 页面 | 原文线索

V2-007928 | MATHDB-318200

The orthogonal-polynomial version of the Duistermaat-van der Kallen conjecture

摘录 / EXCERPT: Let $B\subset\mathbb R^n$, let $w(z)$ be a weight function, and let $\{u_\alpha\mid\alpha\in\mathbb N^n\}$ be a sequence of orthogonal polynomials over B. Let $\mathcal M$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-007929 | MATHDB-318202

Graded Buchsbaum-Eisenbud-Horrocks rank conjecture

摘录 / EXCERPT: Let k be a field, let $S=k[x_1,\dots,x_n]$ be the polynomial ring with its usual grading, and let M be a graded Cohen-Macaulay S-module of codimension c. If ... is its graded...

conjecture | open | MathDB 页面 | 原文线索

V2-007930 | MATHDB-318212

The solitary wave conjecture for focusing mass-critical NLS

摘录 / EXCERPT: Let $d\ge 1$, and consider the focusing mass-critical nonlinear Schrödinger equation. Let Q denote its ground state, and suppose that a global solution with initial data...

conjecture | open | MathDB 页面 | 原文线索

V2-007931 | MATHDB-318215

Square-root growth conjecture for ranks of generic six-in-nine distributions

摘录 / EXCERPT: Let H be a distribution with co-rank s and rank r, so that its derived distribution satisfies $[H,H]=T$ and hence $s<r(r-1)/2$. Square-root growth conjecture. The minimal pos...

conjecture | open | MathDB 页面 | 原文线索

V2-007932 | MATHDB-318298

The vanishing conjecture for differential operators with constant coefficients

摘录 / EXCERPT: Let $z=(z_1,, z_n)$ be n commuting variables, let $\mathbb C[z]$ be the polynomial algebra, and write $\partial=(\partial_1,\dots,\partial_n)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-007933 | MATHDB-318300

Makar-Limanov's extension conjecture for free subalgebras

摘录 / EXCERPT: Let R be an algebra without free subalgebras of rank two, and let S be an extension of R obtained by extending the field K. Makar-Limanov's conjecture. The algebra S does...

conjecture | open | MathDB 页面 | 原文线索

V2-007934 | MATHDB-318358

Kashiwara-Vergne conjecture

摘录 / EXCERPT: Let ${\bf k}={\mathbb R}$ or ${\mathbb C}$, let $\G$ be a finite-dimensional ${\bf k}$-Lie algebra, and let $\hat{\mathfrak f}_{2}$ denote the completed free Lie algebra on x,y....

conjecture | open | MathDB 页面 | 原文线索

V2-007935 | MATHDB-318371

Weak Bass conjecture

摘录 / EXCERPT: Let G be a group, and let $K_{0}(\mathbb{Z}G)$ be the Grothendieck group of finitely generated projective $\mathbb{Z}G$-modules. Write ϵ for the augmentation trace and...

conjecture | open | MathDB 页面 | 原文线索

V2-007936 | MATHDB-318387

F-isomorphism conjecture for sub-Hopf algebra invariants of the Steenrod algebra

摘录 / EXCERPT: Let $\mathcal{A}$ be the mod-2 Steenrod algebra, let D be the relevant normal sub-Hopf algebra, and let $\mathcal{E}$ be the family of sub-Hopf algebras used in the paper. For...

conjecture | open | MathDB 页面 | 原文线索

V2-007937 | MATHDB-318388

Stagewise F-isomorphism conjecture for the sub-Hopf algebras D(m)

摘录 / EXCERPT: For each $m\geq 1$, let $D(m-1)\subseteq D(m)$ be consecutive members of the family of normal sub-Hopf algebras $D(m)$, and let $P_{D(m)}H_*(BV_n)$ denote the $D(m)$-primitive elem...

conjecture | open | MathDB 页面 | 原文线索

V2-007938 | MATHDB-318403

Schiffmann's conjecture on canonical bases and the centre of cyclic-quiver Hall algebras

摘录 / EXCERPT: Let H_n be the Ringel-Hall algebra of the cyclic quiver with n vertices, let $b_{(u,cdots, u)}$ and $b_{(u,cdots, u)}^*$ denote the canonical and dual canonical basis element...

conjecture | open | MathDB 页面 | 原文线索

V2-007939 | MATHDB-318416

Babelon-Bernard-Smirnov's surjectivity conjecture for the KdV evaluation map

摘录 / EXCERPT: Let ${\cal D}$ be the ring of commuting KdV derivations, let A be the space of KdV fields, and let $\bar S_{2n}$ be defined by ... Using the identification...

conjecture | open | MathDB 页面 | 原文线索

V2-007940 | MATHDB-318452

Alper's étale-local quotient conjecture for Artin stacks

摘录 / EXCERPT: Alper's étale-local quotient conjecture. There exists an étale neighborhood $S'\to S$ with $s'\mapsto s$ and an étale, representable morphism

conjecture | open | MathDB 页面 | 原文线索

V2-007941 | MATHDB-318598

The positivity conjecture for cluster algebras

摘录 / EXCERPT: Positivity conjecture. For any cluster algebra $\mathcal{A}$, any seed Σ, and any cluster variable x, the Laurent polynomial $[x]_{\Sigma}^{\mathcal{A}}$ has coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-007942 | MATHDB-318606

The coset-covering function conjecture for finite modules over Dedekind domains

摘录 / EXCERPT: Let R be a Dedekind domain satisfying the assumptions above, let $\mathfrak{m}_i$ be maximal ideals of R, and let n_i be positive integers. Let M be the finite direct sum .....

conjecture | open | MathDB 页面 | 原文线索

V2-007943 | MATHDB-318607

The local coset-covering function conjecture for Dedekind domains

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a local Dedekind domain with finite residue field, and let $n_1,\ldots,n_l$ be positive integers. For the finite R-module ... write $\phi(M)$ for its cose...

conjecture | open | MathDB 页面 | 原文线索

V2-007944 | MATHDB-318608

Auroux-Donaldson-Katzarkov-Yotov conjecture for branch-curve fundamental groups

摘录 / EXCERPT: Let L be an ample line bundle on a smooth projective surface S, and let $X_m\subset \mathbb{P}^{N(m)}$ be the image of the embedding of S by the complete linear system...

conjecture | open | MathDB 页面 | 原文线索

V2-007945 | MATHDB-318655

Stronger congruence conjecture for projective congruence data

摘录 / EXCERPT: Let a projective congruence datum that is also Galois have exponent N and global dimension n. Stronger congruence conjecture. If $n\equiv 2\pmod{4}$, then ... and ... The conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-007946 | MATHDB-318663

Linear independence of zigzag words in the algebra T

摘录 / EXCERPT: Fix a nonnegative integer d, a sequence $(\{\theta_i\}_{i=0}^d;\{\theta^*_i\}_{i=0}^d)$ of scalars in $\mathbb{K}$ satisfying the distinctness condition and the stated recurrence...

conjecture | open | MathDB 页面 | 原文线索

V2-007947 | MATHDB-318665

Factorization conjecture for shapes of tridiagonal pairs

摘录 / EXCERPT: Let (A,A^*) be a tridiagonal pair over $\mathbb{K}$ with diameter d, and let $\{\rho_i\}_{i=0}^d$ denote its shape, where ρ_i is the corresponding shape multiplicity. Shap...

conjecture | open | MathDB 页面 | 原文线索

V2-007948 | MATHDB-318676

Evans-Sturmfels-Uhler conjecture on commuting birth and death ideals

摘录 / EXCERPT: Let E be a grid, and let I^{E} denote its commuting birth and death ideal, formed as the sum of the ideals associated with all induced squares of E. Evans-Sturmfels-Uhler con...

conjecture | open | MathDB 页面 | 原文线索

V2-007949 | MATHDB-318749

The Hilbert-Kunz function conjecture for the nodal cubic

摘录 / EXCERPT: Let $f=x^{3}+y^{3}+xyz$ be the element of $(\mathbb{Z}/2)[x,y,z]$ defining a nodal cubic. Let ϕ_f denote the associated function, and let ϕ_0 be the function defined recu...

conjecture | open | MathDB 页面 | 原文线索

V2-007950 | MATHDB-318772

The image conjecture for commuting differential operators

摘录 / EXCERPT: Let $t\in\mathbb C$, and let ... be the image of the commuting differential operators $(\xi_i-t\partial_i)$. A subspace $\mathcal M$ of a commutative algebra $\mathcal A$ is a Mathie...

conjecture | open | MathDB 页面 | 原文线索

V2-007951 | MATHDB-318773

The reformulated image conjecture for the deformed polynomial algebra

摘录 / EXCERPT: For $t\in\mathbb C$, let $\mathcal B_t[\xi,z]$ be the algebra on the vector space $\mathbb C[\xi,z]$ with product $\ast_t$, and set ... A subspace of a commutative algebra is a Mathi...

conjecture | open | MathDB 页面 | 原文线索

V2-007952 | MATHDB-318775

The vanishing conjecture for quadratic differential operators

摘录 / EXCERPT: Let K be a field of characteristic zero, let $P(z)\in {\mathcal A}_K[z]$, and let Λ be a quadratic differential operator on ${\mathcal A}_K[z]$. Assume ... for every intege...

conjecture | open | MathDB 页面 | 原文线索

V2-007953 | MATHDB-318776

The eventual-to-uniform vanishing conjecture

摘录 / EXCERPT: Let K be a field of characteristic zero, let ${\mathcal A}_K[\xi,z]$ denote the polynomial algebra in the variables ξ and z, and let $P(\xi,z)\in {\mathcal A}_K[\xi,z]$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-007954 | MATHDB-318803

The one-dimensionality conjecture for invariant forms on classical Lie algebras

摘录 / EXCERPT: One-dimensionality conjecture. The space of symmetric bilinear invariant forms on L is 1-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-007955 | MATHDB-318849

The rationality conjecture for equivariant Artin L-function leading terms

摘录 / EXCERPT: Let F'/F be the finite extension and let $G_{F'/F}$ be its Galois group. For the Tate motive $\mathbb{Q}(m)_{F'/F}$, let $L^*(\mathbb{Q}(m)_{F'/F})$ be its equivariant Artin L-...

conjecture | open | MathDB 页面 | 原文线索

V2-007956 | MATHDB-318852

The Koszul resolution conjecture for the basic deformation-quantization bimodule

摘录 / EXCERPT: Let X be a finite-dimensional vector space, let $A=\wedge(X)=\mathrm S(X[-1])$ with generators of degree 1, and let $B=\mathrm S(X^*)$ concentrated in degree 0. Let K be th...

conjecture | open | MathDB 页面 | 原文线索

V2-007957 | MATHDB-318854

Converse to Borel-Tits for anisotropic reductive groups

摘录 / EXCERPT: Converse to Borel-Tits. Every split spherical BN-pair for $G(k)$ is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-007958 | MATHDB-318856

Zigzag-word basis conjecture for the tridiagonal-pair algebra

摘录 / EXCERPT: Let T be the algebra generated by the standard generators $\{e_i\}_{i=0}^d$ and $\{e_i^*\}_{i=0}^d$. A word in these generators is zigzag (or ZZ) when it is alternating and sat...

conjecture | open | MathDB 页面 | 原文线索

V2-007959 | MATHDB-318857

Basis conjecture for the principal right ideal of the tridiagonal-pair algebra

摘录 / EXCERPT: Let T be the tridiagonal-pair algebra, let e_0^* be its distinguished starred generator, and let ZZ words be the zigzag words in the standard generators. Principal-ideal basi...

conjecture | open | MathDB 页面 | 原文线索

V2-007960 | MATHDB-318858

Faithfulness conjecture for the action on the principal right ideal

摘录 / EXCERPT: Let $R=\mathbb{F}[x_1,\ldots,x_d]$, let Te_0^* be the principal right ideal with its right R-module structure induced by the map μ, and let $\operatorname{End}_R(Te_0^*)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-007961 | MATHDB-318859

Free-module basis conjecture for the principal right ideal

摘录 / EXCERPT: Let $R=\mathbb{F}[x_1,\ldots,x_d]$, let Te_0^* be the principal right ideal, and call a ZZ word feasible when it is nontrivial, ends in e_0^*, and has mutually distinct indic...

conjecture | open | MathDB 页面 | 原文线索

V2-007962 | MATHDB-318860

Rank conjecture for principal submodules of the tridiagonal-pair algebra

摘录 / EXCERPT: Let $R=\mathbb{F}[x_1,\ldots,x_d]$, and let e_i,e_i^* be the standard generators of T. Principal-submodule rank conjecture. For $0\le i\le d$, the R-submodules $e_i^*Te_0^*$...

conjecture | open | MathDB 页面 | 原文线索

V2-007963 | MATHDB-318880

Zagier's K-theoretic polylogarithm conjecture

摘录 / EXCERPT: Zagier's K-theoretic conjecture. There is a homomorphism $B_m(F)\longrightarrow K_{2m-1}(F)$ with finite kernel and cokernel such that the composition with $R_m(F)$ equals the map...

conjecture | open | MathDB 页面 | 原文线索

V2-007964 | MATHDB-318887

Conjecture for the semicanonical basis of an affine cluster algebra

摘录 / EXCERPT: Let Q be an affine quiver. Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be the initial cluster and coefficient tuples, and let $\mathcal A(Q,\boldsymbol{x},\boldsymbol{y})$ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-007965 | MATHDB-318910

Degree conjecture for the constant-reciprocation simplified model

摘录 / EXCERPT: Degree conjecture for the constant-reciprocation model. The ideal $I_{\tilde{\mathcal C}_n}$ is generated in degrees 2, 3, and 4, and the binomials are of the forms described...

conjecture | open | MathDB 页面 | 原文线索

V2-007966 | MATHDB-318911

Minimal generation conjecture for the edge-reciprocation simplified model

摘录 / EXCERPT: Minimal generation conjecture for the edge-reciprocation model. For $n\geq4$, the ideal $I_{\tilde{\mathcal E}_n}$ is minimally generated by homogeneous binomials of degrees 2 an...

conjecture | open | MathDB 页面 | 原文线索

V2-007967 | MATHDB-318912

Lifting and overlapping conjecture for minimal Markov bases of the p1 models

摘录 / EXCERPT: Lifting and overlapping conjecture. Minimal Markov and Gröbner bases for the p_1 models can be obtained from Markov and Gröbner bases of $I_{\mathcal A_n}$ by repeated lifting an...

conjecture | open | MathDB 页面 | 原文线索

V2-007968 | MATHDB-318950

Finite-dimensionality conjecture for the non-commuting Harmonic system

摘录 / EXCERPT: Finite-dimensionality conjecture. The non-commuting Harmonic system Har_n is finite dimensional for every n.

conjecture | open | MathDB 页面 | 原文线索

V2-007969 | MATHDB-318951

Eventual fullness conjecture for alternating non-commutative harmonics

摘录 / EXCERPT: Eventual fullness conjecture. The solution space of this linear system is all of $Alt_n^{(d)}$ for sufficiently large d.

conjecture | open | MathDB 页面 | 原文线索

V2-007970 | MATHDB-318958

The injectivity conjecture for the map $\bar{\varphi}$

摘录 / EXCERPT: Fix non-negative integers d_1,d_2, let M_{d_1,d_2} be the corresponding quotient space of alternating bihomogeneous polynomials, and let … be the induced linear map, where...

conjecture | open | MathDB 页面 | 原文线索

V2-007971 | MATHDB-318961

The Keum-Naie surface characterization conjecture

摘录 / EXCERPT: Keum-Naie surface characterization conjecture. Then S is a Keum-Naie surface.

conjecture | open | MathDB 页面 | 原文线索

V2-007972 | MATHDB-318962

Manetti's equality-case conjecture for the Severi inequality

摘录 / EXCERPT: Manetti's equality-case conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-007973 | MATHDB-319017

The BMR freeness and symmetrizing-form conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: Let W be an irreducible complex reflection group, let A be the coefficient ring, and write ${\mathcal{H}}(W,{\mathbf u})$ for its cyclotomic Hecke algebra. Let $B(W)$ be the br...

conjecture | open | MathDB 页面 | 原文线索

V2-007974 | MATHDB-319018

The canonical section conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: Let W be an irreducible complex reflection group, let $B(W)$ be its braid group, and let ${\mathcal{H}}(W,{\mathbf u})$ be its cyclotomic Hecke algebra with symmetrizing form t...

conjecture | open | MathDB 页面 | 原文线索

V2-007975 | MATHDB-319055

Bazhanov-Kirillov-Reshetikhin dilogarithm identity for simply laced Y-systems

摘录 / EXCERPT: Let X_r be a simply laced Dynkin diagram of finite type with rank r and index set I, and let $ell a geq 2$ be an integer. Let $L(x)$ be the Rogers dilogarithm function. For p...

conjecture | open | MathDB 页面 | 原文线索

V2-007976 | MATHDB-319056

Ravanini-Tateo-Valleriani periodicity conjecture for simply laced Y-systems

摘录 / EXCERPT: Let X_r and $X'_{r'}$ be simply laced Dynkin diagrams of finite type, with index sets I and I', Coxeter numbers h and h', and ranks r and r'. A family of positive rea...

conjecture | open | MathDB 页面 | 原文线索

V2-007977 | MATHDB-319073

Breuil's locally analytic vectors conjecture for crystabelian representations

摘录 / EXCERPT: Let V be an irreducible crystabelian representation of $G_{{\mathbb {Q}}_p}$ with Hodge-Tate weights $(0,k-1)$ for an integer $k\geq 2$. Let (α,β) be the associated pa...

conjecture | open | MathDB 页面 | 原文线索

V2-007978 | MATHDB-319079

Unitary preservation under restriction for rational Cherednik algebras

摘录 / EXCERPT: Let W be a complex reflection group with reflection representation $\mathfrak{h}$, and let W' be a parabolic subgroup of W, namely the stabilizer of a point in $\mathfrak{h}$...

conjecture | open | MathDB 页面 | 原文线索

V2-007979 | MATHDB-319085

Theta-pairing sign conjecture for odd-dimensional hypersurface singularities

摘录 / EXCERPT: Theta-pairing sign conjecture. If n is odd, then the pairing

conjecture | open | MathDB 页面 | 原文线索

V2-007980 | MATHDB-319124

The class number formula conjecture for the Carlitz module

摘录 / EXCERPT: The class number formula conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-007981 | MATHDB-319129

Central generation of Whittaker vectors in standard Whittaker modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra with a fixed triangular decomposition ... and set $\mathfrak{n}=\mathfrak{n}_+$. Let $\lambda\in(\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])^*$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-007982 | MATHDB-319131

Exhaustion and annihilator classification for nonzero Whittaker modules of the Witt algebra

摘录 / EXCERPT: Let $\mathfrak{w}_n$ be the Witt algebra with positive nilpotent subalgebra $\mathfrak{n}_+$, and let $L(\lambda,\mu)$ be the modules constructed in Theorem 91 from parameters sati...

conjecture | open | MathDB 页面 | 原文线索

V2-007983 | MATHDB-319146

Conjectured negative answer to the degree problem via equal admissible groups

摘录 / EXCERPT: Let the degree problem ask whether equivalent number fields must have the same degree. Degree-problem conjecture. The degree problem has a negative answer. Moreover, there exist a...

conjecture | open | MathDB 页面 | 原文线索

V2-007984 | MATHDB-319155

Bass's complex Hattori-Stallings vanishing conjecture

摘录 / EXCERPT: Let G be a group, let P be a finitely generated projective $\mathbb C G$-module, and let $s\in G$ have infinite order. The Bass vanishing conjecture. The Hattori-Stallings trac...

conjecture | open | MathDB 页面 | 原文线索

V2-007985 | MATHDB-319156

Purity conjecture for Lusztig perverse sheaves on curves

摘录 / EXCERPT: Let $X=X_0\otimes\overline{\mathbb{F}_q}$ be a smooth projective curve obtained from a curve X_0 over $\mathbb{F}_q$, and let F be the geometric Frobenius acting on X and its...

conjecture | open | MathDB 页面 | 原文线索

V2-007986 | MATHDB-319157

Trace-isomorphism conjecture for the completed Hall algebra

摘录 / EXCERPT: Let X be a smooth projective curve over $\mathbb{C}$, let $\widehat{\mathfrak{U}}_X$ be the completed algebra generated by the canonical-basis elements associated with Lusztig pe...

conjecture | open | MathDB 页面 | 原文线索

V2-007987 | MATHDB-319159

Graded-universality conjecture for Hall categories of nongeneric curves

摘录 / EXCERPT: Let $k=\mathbb{C}$, let X be a smooth projective curve of genus g, let T^a_g be the torus associated with the Frobenius-eigenvalue parameter space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-007988 | MATHDB-319173

The fractal Weyl upper bound for resonances on geometrically finite surfaces

摘录 / EXCERPT: Let Γ be a geometrically finite discrete group of isometries of $\mathbb{H}^2$ such that $X=\Gamma\backslash\mathbb{H}^2$ is a smooth surface of infinite area. Let $R(X)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-007989 | MATHDB-319190

Fialowski-Penkava's stratification conjecture for moduli spaces of algebras

摘录 / EXCERPT: Let n be a fixed finite dimension, and consider the moduli space of complex Lie or associative algebras of dimension n. A jump deformation is a deformation that is equivalent t...

conjecture | open | MathDB 页面 | 原文线索

V2-007990 | MATHDB-319223

The formal cell-zeta dimension conjecture

摘录 / EXCERPT: Let $n=\ell+3$, let d_k be defined by ... and consider formal cell-zeta values on ${\mathfrak{M}}_{0,n}$ modulo all linear relations from dihedral and modular shuffle relations. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-007991 | MATHDB-319224

The formal cell-zeta and formal multiple-zeta algebra conjecture

摘录 / EXCERPT: Let ${\cal FC}$ be the formal cell-zeta algebra and ${\cal FZ}$ the formal multiple-zeta algebra defined using formal iterated-integral symbols modulo the regularised double-shuffl...

conjecture | open | MathDB 页面 | 原文线索

V2-007992 | MATHDB-319229

The Laurent positivity conjecture for cluster variables

摘录 / EXCERPT: Conjecture de positivité de Laurent. Les variables d'amas s'écrivent comme des polynômes de Laurent à coefficients entiers positifs en les variables de tout amas.

conjecture | open | MathDB 页面 | 原文线索

V2-007993 | MATHDB-319252

The asymptotic hole-size conjecture for global function fields

摘录 / EXCERPT: Let K be a function field of genus g with exact constant field $\mathbb{F}_q$, and let S be a non-empty finite set of degree-one places of K. Let S' be a finite set of pl...

conjecture | open | MathDB 页面 | 原文线索

V2-007994 | MATHDB-319253

The maximal-hole conjecture for reduced divisors

摘录 / EXCERPT: Let K be a function field of genus g with exact constant field $\mathbb{F}_q$, and let S be a non-empty finite set of degree-one places of K. Assume that S contains at le...

conjecture | open | MathDB 页面 | 原文线索

V2-007995 | MATHDB-319260

Finiteness conjecture for algebraically primitive Teichmüller curves in genus three

摘录 / EXCERPT: A Teichmüller curve in ${\mathcal M}_{3}$ is algebraically primitive when its trace field has degree 3. Finiteness conjecture. There are only finitely many algebraically primitiv...

conjecture | open | MathDB 页面 | 原文线索

V2-007996 | MATHDB-319261

Finiteness conjecture for cylinder-width triples in the hyperelliptic stratum

摘录 / EXCERPT: Let C be an algebraically primitive Teichmüller curve in the hyperelliptic stratum ${\Omega\mathcal M}_{3}(4)^{{\rm hyp}}$, and let $(r_1:r_2:r_3)$ be the projectivized triple of...

conjecture | open | MathDB 页面 | 原文线索

V2-007997 | MATHDB-319277

Beheshti-Eisenbud containment conjecture for powers of general linear ideals

摘录 / EXCERPT: Beheshti-Eisenbud conjecture. For $q\gg0$,

conjecture | open | MathDB 页面 | 原文线索

V2-007998 | MATHDB-319283

Bloch--Kato vanishing conjecture for the adjoint representation

摘录 / EXCERPT: Let F be the totally real field, let ρ_Π be the Galois representation attached to the automorphic representation Π, and set $V={\rm Ad}'(\rho_\Pi)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-007999 | MATHDB-319320

Multiplicativity conjecture for layered resultants

摘录 / EXCERPT: Let f, g, and h be layered polynomials for which the layered resultants below are defined, and let L be the layer structure; write...

open_problem | open | MathDB 页面 | 原文线索

V2-008000 | MATHDB-319345

Wood's conjecture on harmonic polynomials for Steenrod operators

摘录 / EXCERPT: Wood's conjecture. The graded vector space V_n is isomorphic to the graded regular representation of the symmetric group. In particular, it has dimension $n!$.

conjecture | open | MathDB 页面 | 原文线索

V2-008001 | MATHDB-319371

Multiplicity bound from containment of powers

摘录 / EXCERPT: Suppose that $\mathfrak{a}$ and J are $\mathfrak{m}$-primary ideals in a d-dimensional Noetherian local or Noetherian graded k-algebra $(R,\mathfrak{m})$, where k is a fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-008002 | MATHDB-319393

The algebraic-parameter non-renormalization conjecture for quadratic polynomials

摘录 / EXCERPT: Let c be an algebraic number, and consider the quadratic polynomial ... A quadratic polynomial is infinitely renormalizable if it is renormalizable at infinitely many times. The al...

conjecture | open | MathDB 页面 | 原文线索

V2-008003 | MATHDB-319399

Universality conjecture for the strongly co-Stone hull of a residuated lattice

摘录 / EXCERPT: Let A be a residuated lattice, let $\tilde{A}$ be its strongly co-Stone hull, and let $\epsilon:A\to\tilde{A}$ be the canonical morphism. For a subset $X\subseteq A$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-008004 | MATHDB-319413

The periodicity conjecture for pairs of Dynkin diagrams

摘录 / EXCERPT: The periodicity conjecture. All solutions to this system are periodic in t of period dividing $2(h+h')$. This conjecture predicts a uniform periodicity phenomenon for algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-008005 | MATHDB-319417

The zigzag-word basis conjecture for the algebra $\tilde T$

摘录 / EXCERPT: Fix an integer $d\geq 0$ and a sequence $p\in\operatorname{Feas}(d,\mathbb F)$. Let $\tilde T=\tilde T(p,\mathbb F)$ be the algebra defined in the paper, and let zigzag words mean...

conjecture | open | MathDB 页面 | 原文线索

V2-008006 | MATHDB-319418

The injectivity conjecture for the homomorphism from $\tilde T$ to $\mathcal A\otimes T$

摘录 / EXCERPT: Let $\tilde T$ and T be the algebras defined in the paper, and let $\mathcal A$ be the algebra acting on finite-dimensional irreducible modules described in the surrounding resul...

conjecture | open | MathDB 页面 | 原文线索

V2-008007 | MATHDB-319424

The eigencurve intersection conjecture for one-dimensional local representations

摘录 / EXCERPT: Eigencurve intersection conjecture. If $x\in\mathcal{E}$ is such that $\pi(\rho_{x,l},N_x)$ is one dimensional, then there exist irreducible components...

conjecture | open | MathDB 页面 | 原文线索

V2-008008 | MATHDB-319425

The geometric level-raising conjecture for the eigencurve

摘录 / EXCERPT: Geometric level-raising conjecture. If $\mathcal{Z}\subset\mathcal{E}$ is a generically principal-series at l component and there exists $x\in\mathcal{Z}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-008009 | MATHDB-319454

Forster-Swan-type conjecture for generators of finite algebras

摘录 / EXCERPT: Generator-number conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008010 | MATHDB-319498

Schinzel's bound on the number of irreducible factors of Kf

摘录 / EXCERPT: Let f be the polynomial considered in the source, let Kf denote its associated polynomial vocabulary description, let $\Omega_1(f)$ be the number of irreducible factors of Kf...

conjecture | open | MathDB 页面 | 原文线索

V2-008011 | MATHDB-319510

p-parity conjecture for self-dual twists

摘录 / EXCERPT: p-parity conjecture for self-dual twists.

conjecture | open | MathDB 页面 | 原文线索

V2-008012 | MATHDB-319513

Mutation invariance of generic cluster characters

摘录 / EXCERPT: Mutation-invariance conjecture. Under these assumptions, the diagram identifying cluster monomials with generic characters via ϕ commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-008013 | MATHDB-319515

The product-of-chains conjecture for distributive cross section lattices

摘录 / EXCERPT: Let Λ be a distributive cross section lattice of the J-irreducible monoid of the pair (G_0,ρ), where G_0 is a simple algebraic group of one of the types A_{n-1},...

conjecture | open | MathDB 页面 | 原文线索

V2-008014 | MATHDB-319605

The characteristic-two conjecture for the weak Lefschetz property of monomial complete intersections

摘录 / EXCERPT: Let K be a field of characteristic 2, let d be a positive integer, and set ... The characteristic-two conjecture. The algebra A has the weak Lefschetz property if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-008015 | MATHDB-319636

The transcendence conjecture for the KZ associator

摘录 / EXCERPT: Transcendence Conjecture. The morphism φ_{KZ} is injective. This asserts algebraic independence of the coordinate functions of the associator in the sense encoded by the co...

conjecture | open | MathDB 页面 | 原文线索

V2-008016 | MATHDB-319637

The torsion-group conjecture for associated Lie algebras

摘录 / EXCERPT: Let G be a group, and let its associated Lie algebra be defined from the relevant group-theoretic filtration. Torsion-group conjecture. If G is a torsion group, then its associ...

conjecture | open | MathDB 页面 | 原文线索

V2-008017 | MATHDB-319638

The subalgebra conjecture for thinned self-similar group algebras

摘录 / EXCERPT: Let G be a self-similar group as described in the source, let ${\mathscr A}(G)$ be its thinned algebra with augmentation ideal ϖ, and define its associated graded algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008018 | MATHDB-319639

The quotient conjecture for Lie algebras of self-similar groups

摘录 / EXCERPT: Let G be a self-similar group as described in the source, and let ${\mathscr L}^\Bbbk(G)$ be its associated Lie algebra over the field $\Bbbk$. Let ${\mathscr L}(G)$ be the self-...

conjecture | open | MathDB 页面 | 原文线索

V2-008019 | MATHDB-319669

The shift homomorphism conjecture for divided-power generators

摘录 / EXCERPT: Let p be the characteristic of the field $\mathbb{K}$, let $X=\{e_{i,i+1}^{(p^k)}\mid 1\le i\le n-1,\\ k\in\mathbb{N}_0\}$, and write $a_{ik}=e_{i,i+1}^{(p^k)}$. Equip X with t...

conjecture | open | MathDB 页面 | 原文线索

V2-008020 | MATHDB-319670

The characteristic-two Gröbner basis conjecture for \(\mathfrak{U}_{\mathbb{K}}(sl^+_n)\)

摘录 / EXCERPT: Assume $p=2$ and $n\ge4$. For a sequence $i=(i_1,\dots,i_l)$, let $a_{i,k}=a_{i_1,k}\dots a_{i_l,k}$; write $i..j$ for the consecutive sequence from i to j, and let $L_l[m]$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-008021 | MATHDB-319693

Unimodularity of the right transformation in a diagonal reduction

摘录 / EXCERPT: Let A_* be a G-algebra, let $A=\operatorname{Quot}(A_*)$, and let ... where R_* is a G-algebra. For a matrix $M\in R^{p\times p}$, there are square matrices U,V,D with entr...

conjecture | open | MathDB 页面 | 原文线索

V2-008022 | MATHDB-319695

Asymptotic unimodality conjecture for level Hilbert functions

摘录 / EXCERPT: For positive integers e and t, consider level Hilbert functions of codimension r, socle degree e, and type t. Asymptotic unimodality conjecture. For fixed positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-008023 | MATHDB-319696

Uniqueness conjecture for maximal solvable extensions of nilpotent Lie algebras

摘录 / EXCERPT: Let ${\mathfrak n}$ be a complex nilpotent Lie algebra that is not characteristically nilpotent. Let ${\mathfrak s}$ and $\tilde{{\mathfrak s}}$ be solvable Lie algebras with nilra...

conjecture | open | MathDB 页面 | 原文线索

V2-008024 | MATHDB-319716

Vanishing conjecture for the Whitehead group

摘录 / EXCERPT: Vanishing conjecture for the Whitehead group. The Whitehead group of a torsionfree group vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-008025 | MATHDB-319748

Eisenbud-Green-Harris conjecture on quadratic Artinian algebras

摘录 / EXCERPT: Let K be a field, and consider the h-vectors of quadratic Artinian K-algebras and the f-vectors of simplicial complexes. Eisenbud-Green-Harris conjecture. Every h-vector...

conjecture | open | MathDB 页面 | 原文线索

V2-008026 | MATHDB-319754

Gabber-type maximal Cohen-Macaulay conjecture for reflexive modules

摘录 / EXCERPT: Generalized Gabber conjecture. If $[N]=0$ in $\overline G(R)_{\mathbb Q}$, then $\operatorname{Hom}_R(N,N)$ is a maximal Cohen-Macaulay R-module if and only if N is free.

conjecture | open | MathDB 页面 | 原文线索

V2-008027 | MATHDB-319755

Tor-rigidity conjecture for local complete intersections

摘录 / EXCERPT: Tor-rigidity conjecture. The pair (M,N) is Tor-rigid: for any $i>0$, the condition $\operatorname{Tor}_i^R(M,N)=0$ forces $\operatorname{Tor}_j^R(M,N)=0$ for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-008028 | MATHDB-319779

The classical Bass conjecture

摘录 / EXCERPT: Let G be a discrete group. For a finitely generated projective $C{Z}G$-module P, let r_P denote its Hattori-Stallings rank, written as $r_P=sum_{[s]in [G]}r_P(s)[s]$, where...

conjecture | solved | MathDB 页面 | 原文线索

V2-008029 | MATHDB-319780

The complex Bass conjecture

摘录 / EXCERPT: Let G be a discrete group, and let ... be the Hattori-Stallings trace, where $[G]$ denotes the set of conjugacy classes of elements of G. The $C{C}G$ Bass conjecture. The image o...

conjecture | open | MathDB 页面 | 原文线索

V2-008030 | MATHDB-319839

Jacquet-Langlands preservation conjecture for endo-classes

摘录 / EXCERPT: Let ${\rm G}={\rm GL}_m(D)$ and ${\rm H}={\rm GL}_{md}(F)$, where D is a central division algebra over the non-archimedean local field F of reduced degree d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008031 | MATHDB-319858

Recursive formula conjecture for the deformed Gaiotto-state inner product

摘录 / EXCERPT: Recursive formula conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-008032 | MATHDB-319889

General genus and codimension conjecture for Birkhoff-stratum curves

摘录 / EXCERPT: General Birkhoff-stratum conjecture. The stratum $\Sigma_{1,2,\dots,n}$ contains a codimension-n subspace $W_{1,2,\dots,n}$ closed under pointwise multiplication; algebraically i...

conjecture | open | MathDB 页面 | 原文线索

V2-008033 | MATHDB-319900

The D-multidissection basis conjecture for $V^D(n)$

摘录 / EXCERPT: For $n\geq 2$, label the vertices of the centrally symmetric polygon ${\Bbb P}_{2n}$ clockwise by $1,2,\dots,n,\bar{1},\bar{2},\dots,\bar{n}$. Let $V^D(n)$ be the quotient space de...

conjecture | open | MathDB 页面 | 原文线索

V2-008034 | MATHDB-319906

Frenkel-Gross comparison conjecture for Kloosterman connections

摘录 / EXCERPT: Let $k={\mathbb C}$, let G be almost simple, and let $\mathcal G$ be a quasi-split form of G on ${\mathbb P}^1_{\backslash\{0,\infty\}}$ determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-008035 | MATHDB-320004

Fomin-Zelevinsky's Laurent-positivity conjecture for chamber minors

摘录 / EXCERPT: Let M be an $n\times n$ matrix with entries in $\mathbb R$ that is totally positive, meaning that all minors of M are positive. For a commutation class of double wiring diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-008036 | MATHDB-320059

Suslin's conjecture on universal vanishing of SK_1

摘录 / EXCERPT: Let F be a field, let A be a central simple F-algebra, and for every extension L/F write $\operatorname{SK}_1(A\otimes_F L)$ for the reduced Whitehead group. Suslin's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-008037 | MATHDB-320102

The coniveau equality for differential forms

摘录 / EXCERPT: Let F be the ambient field over k, let $\Omega^j_{F|k}$ be the space of j-forms over k, and let N_j denote the level filtration defined by subfields of transcendence degr...

conjecture | open | MathDB 页面 | 原文线索

V2-008038 | MATHDB-320103

Galois invariance of the Det-part of K_1-groups of Iwasawa algebras

摘录 / EXCERPT: Let L be either an arbitrary Galois extension L^0 of finite absolute ramification index over $\mathbb{Q}_p$ or the completion of such an extension, let $S=\mathcal{O}_L$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-008039 | MATHDB-320144

Existence and isolated singularities of maximal modification algebras

摘录 / EXCERPT: Let R be a normal 3-CY$-$ ring with rational singularities. A module M is called maximal modifying (MM) if it is an MM module in the sense of the paper, and an R-order has...

conjecture | open | MathDB 页面 | 原文线索

V2-008040 | MATHDB-320158

Breuil's lattice conjecture for semistable representations

摘录 / EXCERPT: Let $D=(D,\varphi_D,N,\operatorname{Fil}^\bullet D_K)$ be a weakly admissible filtered (φ,N)-module such that $\operatorname{gr}^w D_K=0$ for all $w\notin[0,p-1]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008041 | MATHDB-320168

The lifting conjecture for endomorphisms of Weyl algebras

摘录 / EXCERPT: Let A_n be the n'th Weyl algebra over ${\mathbb C}$, let $A_n^{(t)}$ be the quantized Weyl algebra over the parameter ring used in the paper, and let $\operatorname{End}(A_n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-008042 | MATHDB-320170

The isotropic-subspace conjecture for hyperplane arrangements

摘录 / EXCERPT: Let A be a hyperplane arrangement in $\mathbb{C}\mathbb{P}^n$, and let V be an n-isotropic subspace of its Orlik-Solomon algebra of dimension $d\geq n$ that is not contained...

conjecture | open | MathDB 页面 | 原文线索

V2-008043 | MATHDB-320176

Conjectures on the finite Fomin-Stanley algebra and the j-basis

摘录 / EXCERPT: Let W be a Weyl group, let $\kappa:{\mathbb A}\to({\mathbb A}_{\mathrm{af}})_0$ be given by ... and let ${\mathbb B}$ be the finite Fomin-Stanley algebra, the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-008044 | MATHDB-320188

Vershik's conjecture on finite dimensionality of generic quadratic algebras

摘录 / EXCERPT: Let $n\in{\mathbb N}$ with $n\geqslant 3$. Consider a generic quadratic ${\mathbb C}$-algebra, meaning a quadratic algebra in n generators whose defining relations are generic in...

conjecture | open | MathDB 页面 | 原文线索

V2-008045 | MATHDB-320201

Operator biflatness of Fourier algebras

摘录 / EXCERPT: Let G be a locally compact group and let ${\mathrm{A}(G)}$ denote its Fourier algebra. Operator biflatness conjecture. ${\mathrm{A}(G)}$ is always operator biflat. This would ext...

conjecture | open | MathDB 页面 | 原文线索

V2-008046 | MATHDB-320237

The parity-shift Serre functor conjecture for projective-injective modules

摘录 / EXCERPT: Parity-shift Serre functor conjecture. Under the assumption $\Pi V\cong V$, the functor Π is the Serre functor on

conjecture | open | MathDB 页面 | 原文线索

V2-008047 | MATHDB-320242

Endomorphism-algebra conjecture for objects in the diagrammatic q-Schur category

摘录 / EXCERPT: Let $lambda\rightarrow\ru(n,d)$ be arbitrary, and let $mu\rightarrow\ru(n,d)$ be obtained from $lambda$ by moving all zero entries to the end while preserving the relative order of...

conjecture | open | MathDB 页面 | 原文线索

V2-008048 | MATHDB-320243

Freeness conjecture for morphism spaces in the diagrammatic q-Schur category

摘录 / EXCERPT: Let $lambda\rightarrow\ru(n,d)$ be a weight and let $mathbf i$ and $mathbf j$ be sequences for which the displayed 1-morphisms are defined. The morphism space is...

conjecture | open | MathDB 页面 | 原文线索

V2-008049 | MATHDB-320246

Representation stability for graded Torelli Lie algebras of surfaces

摘录 / EXCERPT: Let $\mathcal{I}_{g,1}$ be the Torelli group of a genus-g surface with one boundary component, and let $\operatorname{gr}(\mathcal{I}_{g,1})$ be the graded rational Lie algebra a...

conjecture | open | MathDB 页面 | 原文线索

V2-008050 | MATHDB-320255

Lusztig's excellent-element conjecture for unipotent classes

摘录 / EXCERPT: Lusztig's excellent-element conjecture. Then $u_w\in {\mathcal{O}}_C$.

conjecture | open | MathDB 页面 | 原文线索

V2-008051 | MATHDB-320278

The degree-\b2n\b11 consequence conjecture for n-ary intermolecular recombination

摘录 / EXCERPT: Degree-$2n-1$ consequence conjecture. Every polynomial identity satisfied by n-ary intermolecular recombination in degree $2n-1$ is a consequence of complete symmetry in degree...

conjecture | open | MathDB 页面 | 原文线索

V2-008052 | MATHDB-320279

The finite-generation conjecture for identities of n-ary intermolecular recombination

摘录 / EXCERPT: Finite-generation conjecture. Every polynomial identity satisfied by n-ary intermolecular recombination, with no restriction on the degree, is a consequence of complete symmetry...

conjecture | open | MathDB 页面 | 原文线索

V2-008053 | MATHDB-320281

Odd-variable equal-power weak Lefschetz conjecture

摘录 / EXCERPT: Odd-variable equal-power WLP conjecture. The ring R/I fails the weak Lefschetz property if and only if $d>1$. Furthermore, when $n=3$, so that R has seven variables, R/I fail...

conjecture | open | MathDB 页面 | 原文线索

V2-008054 | MATHDB-320283

Generic powers of linear forms fail the weak Lefschetz property

摘录 / EXCERPT: Generic-powers WLP conjecture. If $n\ge r+1\ge 5$, then S/I fails the weak Lefschetz property for all $t\gg0$.

conjecture | open | MathDB 页面 | 原文线索

V2-008055 | MATHDB-320284

Failure of the weak Lefschetz property for almost complete intersections of squares

摘录 / EXCERPT: Let $A_{r,2}$ denote the quotient algebra formed from r generic linear forms in the construction used in the paper, with the generators squared, and let WLP denote the weak Lefsc...

conjecture | open | MathDB 页面 | 原文线索

V2-008056 | MATHDB-320285

Universality conjecture for the word u_N in the 0-Hecke algebra

摘录 / EXCERPT: Universality conjecture. The word u_N is universal.

conjecture | open | MathDB 页面 | 原文线索

V2-008057 | MATHDB-320287

The even-dimensional complete intersection torsion-freeness conjecture

摘录 / EXCERPT: Complete intersection hypersurface conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008058 | MATHDB-320350

Asymptotic regularity bound for symbolic powers of reduced ideal sheaves

摘录 / EXCERPT: Asymptotic regularity conjecture. There is a constant e such that for all $p\geq 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-008059 | MATHDB-320358

Kunita's exchange conjecture for purely nondeterministic hidden Markov models

摘录 / EXCERPT: Kunita's exchange conjecture. If the hidden process is purely nondeterministic and the observations are nondegenerate, then $$ holds true.

conjecture | open | MathDB 页面 | 原文线索

V2-008060 | MATHDB-320359

The C-arithmeticity conjecture for automorphic representations

摘录 / EXCERPT: The C-arithmeticity conjecture. The representation cpi is C-arithmetic if and only if it is C-algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-008061 | MATHDB-320360

The conjecture on L-algebraic automorphic representations and Galois representations

摘录 / EXCERPT: The Galois-representation existence conjecture. If π is L-algebraic, then there is a finite set S containing all infinite places, all places dividing p, and all places wh...

conjecture | open | MathDB 页面 | 原文线索

V2-008062 | MATHDB-320361

The strong Galois-representation conjecture for L-algebraic automorphic representations

摘录 / EXCERPT: The strong Galois-representation conjecture. There is a continuous representation

conjecture | open | MathDB 页面 | 原文线索

V2-008063 | MATHDB-320362

Clozel's arithmeticity and Galois-representation conjecture for algebraic representations of GLn

摘录 / EXCERPT: Clozel's conjecture. There is a number field $E\subset\mathbb{C}$ over which π_f is defined, and the strong Galois-representation conjecture holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-008064 | MATHDB-320393

The law of large numbers conjecture for Demazure modules of affine $b[0msl_2$

摘录 / EXCERPT: Fix a dominant integral weight Λ and a sequence $(w^{(k)})$ in W^{aff} such that $l(w^{(k)}) \to \infty$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008065 | MATHDB-320420

Jacobi-Trudi determinant formula for projective covers

摘录 / EXCERPT: Suppose that $\mathfrak{g}$ is of type B_n, C_n or D_{n+1} and that $\lambda\in P^+$ satisfies $\lambda(h_i)=0$ for $i\ge n$. Let Ψ_λ be the root subset defined fr...

conjecture | open | MathDB 页面 | 原文线索

V2-008066 | MATHDB-320424

The Minkowski space conjecture on celestial polynomial independence

摘录 / EXCERPT: Let $\xi_1,\dotsc,\xi_n\subset\mathbb{R}^{3,1}$ be n non-intersecting world lines in Minkowski space, and let $x_i\in\xi_i$ be distinct events. For $i\ne j$, let u_{ij} be the...

conjecture | open | MathDB 页面 | 原文线索

V2-008067 | MATHDB-320433

Injectivity conjecture for the endomorphism algebra of divided power 1-morphisms

摘录 / EXCERPT: Injectivity conjecture. The natural map is injective. This conjecture identifies the explicitly presented algebra H_i^n with a subalgebra of the endomorphism algebra; the source...

conjecture | open | MathDB 页面 | 原文线索

V2-008068 | MATHDB-320503

The free Lie-module generator conjecture for the cyclic Lie sub-operad of PreLie

摘录 / EXCERPT: Let PreLie be the PreLie operad and let M be the sub-operad generated by the image of CycLie under the injective map λ. The composition...

conjecture | open | MathDB 页面 | 原文线索

V2-008069 | MATHDB-320504

The free-operad identification conjecture for the cyclic Lie sub-operad of PreLie

摘录 / EXCERPT: Let PreLie be the PreLie operad, let CycLie be the cyclic Lie module, and let M be the sub-operad generated by the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-008070 | MATHDB-320507

Strong Butler conjecture on stability of dual spans

摘录 / EXCERPT: Strong Butler conjecture. The bundle $D_V(E)$ is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-008071 | MATHDB-320508

Nonemptiness conjecture for coherent systems with $n+1$ sections

摘录 / EXCERPT: $n+1$-section nonemptiness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008072 | MATHDB-320543

Consequences of the simple product conjecture for compatible families

摘录 / EXCERPT: Let $\mathscr B(\infty)$ be the crystal indexing the dual canonical basis. A family is called compatible or strongly compatible as in the source, and $b^{[m]}$ and $S_m(b)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-008073 | MATHDB-320602

The higher-rank quasi-isometry classification conjecture for function-field arithmetic groups

摘录 / EXCERPT: Let K be a global function field, let S be a nonempty set of valuations of K, and let $\mathbf{G}$ be a connected, absolutely simple, K-isotropic algebraic K-group of adj...

conjecture | open | MathDB 页面 | 原文线索

V2-008074 | MATHDB-320621

Kontsevich's noncommutative Laurent phenomenon conjecture

摘录 / EXCERPT: Kontsevich's conjecture. For any $r_1,r_2\in\mathbb{Z}_{>0}$, all these iterations are given by noncommutative Laurent polynomials in x and y.

conjecture | open | MathDB 页面 | 原文线索

V2-008075 | MATHDB-320630

Kazhdan-Lusztig nonnegativity conjecture

摘录 / EXCERPT: Nonnegativity conjecture. The coefficients of $P_{x,w}(q)$ are nonnegative for every Coxeter group W and all $x\leq w$.

conjecture | open | MathDB 页面 | 原文线索

V2-008076 | MATHDB-320675

Waldschmidt's conjecture on algebraic independence of Weierstrass values

摘录 / EXCERPT: Let $\Lambda\subset\mathbb{C}$ be a lattice, let $\lambda\in\Lambda\setminus\{0\}$, and let $\omega\in\mathbb{C}\setminus(\mathbb{Q}\lambda\cup\Lambda)$. Waldschmidt's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-008077 | MATHDB-320677

The finite coradical filtration conjecture for co-Frobenius Hopf algebras

摘录 / EXCERPT: Let H be a co-Frobenius Hopf algebra, meaning that it has a nonzero left integral (equivalently, any of the equivalent conditions characterizing co-Frobenius Hopf algebras). Its...

conjecture | solved | MathDB 页面 | 原文线索

V2-008078 | MATHDB-320693

Gimenez-Sengupta-Srinivasan conjecture on Betti numbers of arithmetic monomial curves

摘录 / EXCERPT: Let $({\bf m})=m_0,\ldots,m_n$ be an arithmetic sequence of positive integers, and let $C({\bf m})\subset\mathbb A_k^{n+1}$ be the associated affine monomial curve, with homogeneou...

conjecture | open | MathDB 页面 | 原文线索

V2-008079 | MATHDB-320765

Large-gap conjecture for algebraically integrable differential operators

摘录 / EXCERPT: Let L be an algebraically integrable differential operator of order $n\ge 2$, with gaps $q_1,\dots,q_{n-1}$ in its local indices, and call L fully cyclically symmetric when it...

conjecture | open | MathDB 页面 | 原文线索

V2-008080 | MATHDB-320766

Rational large-gap conjecture for algebraically integrable differential operators

摘录 / EXCERPT: Let L be a rational algebraically integrable differential operator with gaps $q_1,\dots,q_{n-1}$, and call it fully cyclically symmetric when it has the stated cyclic symmetry. R...

conjecture | open | MathDB 页面 | 原文线索

V2-008081 | MATHDB-320767

Classification conjecture for third-order one-pole operators

摘录 / EXCERPT: Consider algebraically integrable third-order differential operators L with one pole, in the notation of the classification subsection, and let q and r be the associated gap...

conjecture | open | MathDB 页面 | 原文线索

V2-008082 | MATHDB-320768

Calogero-Moser critical-point conjecture for symmetric elliptic operators

摘录 / EXCERPT: Let $\ell\in\{3,4,6\}$, and let L be an order-ℓ operator invariant under the group $\mathbb Z_\ell$ on an elliptic curve with the corresponding symmetry. Assume that its pol...

conjecture | open | MathDB 页面 | 原文线索

V2-008083 | MATHDB-320769

Rational Calogero-Moser critical-point conjecture for cyclic operators

摘录 / EXCERPT: Let $\ell\ge 2$ and let $L=\partial^\ell+a_2(z)\partial^{\ell-2}+\dots+a_\ell(z)$ be a rational differential operator invariant under $\mathbb Z_\ell$, with coefficients vanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-008084 | MATHDB-320772

Makar-Limanov and J.-T. Yu's fractional-power conjecture

摘录 / EXCERPT: Makar-Limanov and J.-T. Yu's conjecture. The series $g^{m/n}$ has a monomial of positive degree containing a negative power of an indeterminate in X.

conjecture | open | MathDB 页面 | 原文线索

V2-008085 | MATHDB-320773

Arbitrary-field fractional-power conjecture

摘录 / EXCERPT: Arbitrary-field fractional-power conjecture. Then $g^{m/n}\in K((X))$ has a monomial of positive degree containing a negative power of an indeterminate in X.

conjecture | open | MathDB 页面 | 原文线索

V2-008086 | MATHDB-320813

Affine orbit-closure conjecture for reparametrisation groups

摘录 / EXCERPT: Affine orbit-closure conjecture. This point has stabiliser $\mathbb{G}_{k,p}'$ in $\mathrm{SL}(n)$, and the closure of its $\mathrm{SL}(n)$-orbit is the union of its orbit and fini...

conjecture | open | MathDB 页面 | 原文线索

V2-008087 | MATHDB-320814

Finite-generation conjecture for jet-differential invariant fibres

摘录 / EXCERPT: Finite-generation conjecture. Every fibre $\mathcal{O}((J_{k,p})_x)^{\mathbb{G}_{k,p}'}$ of $E_{k,p}^n$ is a finitely generated graded complex algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-008088 | MATHDB-320864

Infinitely many rational curves on projective K3 surfaces

摘录 / EXCERPT: Let X be a projective K3 surface over an algebraically closed field. An integral rational curve on X is an integral curve whose normalization is a rational curve. The rational-...

conjecture | open | MathDB 页面 | 原文线索

V2-008089 | MATHDB-320875

The field-independent monodromy morphism conjecture for complex reflection groups

摘录 / EXCERPT: Let W be an arbitrary complex (pseudo-)reflection group, let B be its braid group, and let $\mathbb{k}$ be a field of characteristic 0. Let $\mathcal{T}$ be the infinitesima...

conjecture | open | MathDB 页面 | 原文线索

V2-008090 | MATHDB-320920

The constant-term conjecture for W-algebra extensions of (2,p) minimal models

摘录 / EXCERPT: Constant-term conjecture. This constant term equals

conjecture | open | MathDB 页面 | 原文线索

V2-008091 | MATHDB-320925

Genus monotonicity for differential-operator embeddings of smooth curves

摘录 / EXCERPT: Let X and Y be smooth curves, and write $\mathcal{D}(X)$ and $\mathcal{D}(Y)$ for their rings of differential operators. Let $Q(\mathcal{D}(Y))$ denote the quotient division ri...

conjecture | open | MathDB 页面 | 原文线索

V2-008092 | MATHDB-320926

Noetherianity conjecture for quadratic-growth algebras with locally nilpotent derivations

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, and let A be a finitely generated k-algebra that is a domain of quadratic growth. Suppose that A does not sati...

conjecture | open | MathDB 页面 | 原文线索

V2-008093 | MATHDB-320941

Uniqueness of minimal orders for even Clifford algebras

摘录 / EXCERPT: Let Q be a conic bundle and let $\operatorname{Cl}_0(Q)$ be its even Clifford algebra. Consider orders Morita equivalent to $\operatorname{Cl}_0(Q)$ and having the same rank and...

conjecture | open | MathDB 页面 | 原文线索

V2-008094 | MATHDB-321031

The central special-elements conjecture for higher algebraic K-theory

摘录 / EXCERPT: Fix a strictly negative integer r, a finite Galois extension F/k with group G, a character χ as in the setup, and a finite set of places S as above. Let $\emb{F}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-008095 | MATHDB-321100

Chow-theoretic injectivity conjecture for products of K3 surfaces

摘录 / EXCERPT: Let S be an algebraic $K3$ surface, let m be any integer, and let $P\in CH^p(S^m)_\mathbb{Q}$ be a weighted-degree-k polynomial expression in the classes $pr_i^*c_1(L_s)$, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-008096 | MATHDB-321111

Clozel's algebraicity conjecture for inner forms of general linear groups

摘录 / EXCERPT: Clozel's conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-008097 | MATHDB-321163

The fully faithful graded Grothendieck group conjecture for Leavitt path algebras

摘录 / EXCERPT: Let K_0^{gr} denote the graded Grothendieck group functor. Consider the category whose objects are Leavitt path algebras and whose morphisms are graded homomorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-008098 | MATHDB-321273

Kontsevich's positivity conjecture for non-commutative Laurent polynomials

摘录 / EXCERPT: Let $K=k(x,y)$ be the skew field of rational functions in non-commutative variables x and y, where k is $\mathbb{Q}$, $\mathbb{Q}(q)$, or any field containing either one. For...

conjecture | open | MathDB 页面 | 原文线索

V2-008099 | MATHDB-321289

The Jacobian-minor primaryness conjecture for ideals generated by quadrics

摘录 / EXCERPT: Let J be an ideal generated by quadrics in a polynomial ring S, and let $r={\rm ht}(J)\geq 2$. Let Θ be the Jacobian matrix of J, let $I_r(\Theta)$ denote its ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-008100 | MATHDB-321300

The dimension conjecture for electrical Lie algebras

摘录 / EXCERPT: Let D be a Dynkin diagram and let $\mathfrak e_D$ be its electrical Lie algebra. Let R^+ denote the set of positive roots in the root system of D. Dimension conjecture. The d...

conjecture | open | MathDB 页面 | 原文线索

V2-008101 | MATHDB-321301

The finite-dimensionality conjecture for electrical Lie algebras

摘录 / EXCERPT: Let D be a Dynkin diagram, and let $\tilde e_D$ be the electrical Lie algebra associated with D. Finite-dimensionality conjecture. The Lie algebra $\tilde e_D$ is finite dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-008102 | MATHDB-321322

Koszulity parity conjecture for totally associative n-ary operads

摘录 / EXCERPT: Let ${t\mathcal{A}ss}^n_d$ be the operad for degree-d totally associative n-algebras. The Koszul property is considered for arbitrary arity n and degree d. Koszulity parity...

conjecture | open | MathDB 页面 | 原文线索

V2-008103 | MATHDB-321323

Koszulity parity conjecture for partially associative n-ary operads

摘录 / EXCERPT: Let ${p\mathcal{A}ss}^n_d$ be the operad for degree-d partially associative n-ary algebras. The Koszul duality isomorphisms identify ${p\mathcal{A}ss}^n_d$ with the Koszul dual...

conjecture | open | MathDB 页面 | 原文线索

V2-008104 | MATHDB-321324

Gap-length conjecture for minimal models of partially associative n-ary operads

摘录 / EXCERPT: Let ${p\mathcal{A}ss}^n_d$ be the operad for degree-d partially associative n-ary algebras, in the non-Koszul parity case $n \not\equiv d \pmod{2}$. A gap in a minimal model me...

conjecture | open | MathDB 页面 | 原文线索

V2-008105 | MATHDB-321333

Multiple-cover conjecture for sheaf-counting invariants on K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $\widetilde{H}(S,\mathbb{Z})=\mathbb{Z}\oplus H^2(S,\mathbb{Z})\oplus\mathbb{Z}$ be its Mukai lattice, and let v be an algebraic class in this lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-008106 | MATHDB-321335

Duflo's conjecture on invariant Poisson and associative centers

摘录 / EXCERPT: Duflo's conjecture. With these notations, there is an algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-008107 | MATHDB-321363

The reciprocity-image lower-bound conjecture for CM fields

摘录 / EXCERPT: Let K be a CM field of degree $2g$, and let S be a primitive CM type for K. Let L be the normal closure of K, and let $r_K:L^{\times}\to K^{\times}$ be the reciprocity mo...

conjecture | open | MathDB 页面 | 原文线索

V2-008108 | MATHDB-321413

Minimality conjecture for trace reductions of three 3x3 matrices

摘录 / EXCERPT: Let R be an evaluation map obtained from the trace reduction referred to in the source, and let the chosen monomials be the monomials used to construct the corresponding generati...

conjecture | open | MathDB 页面 | 原文线索

V2-008109 | MATHDB-321414

Conjectured bound for the first relation degree of trace algebras

摘录 / EXCERPT: Let n and d denote the parameters of the trace algebra considered in the source, and let $N^1(n,d)$ denote the quantity measuring the relevant first relation degree. Bound conj...

conjecture | open | MathDB 页面 | 原文线索

V2-008110 | MATHDB-321418

Pell-equation lower-bound conjecture for Seshadri constants on Picard-rank-one surfaces

摘录 / EXCERPT: Pell-equation lower-bound conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-008111 | MATHDB-321459

Vanishing conjecture for the K-theoretic obstruction of geometrically ruled surfaces

摘录 / EXCERPT: Soit K un corps global et soit V une surface projective, lisse et géométriquement réglée sur K. Avec $\cyr{X}\mathcal N(V)$ défini comme le noyau de la restriction de...

conjecture | open | MathDB 页面 | 原文线索

V2-008112 | MATHDB-321495

Trace-zero form rigidity conjecture for totally real quartic fields

摘录 / EXCERPT: Trace-zero form rigidity conjecture. If an isomorphism of quadratic modules

conjecture | open | MathDB 页面 | 原文线索

V2-008113 | MATHDB-321523

The finiteness-length conjecture for polynomial points of simple groups

摘录 / EXCERPT: Finiteness-length conjecture. The group

conjecture | open | MathDB 页面 | 原文线索

V2-008114 | MATHDB-321554

The maximal-dimension conjecture for 1-spec matrix spaces

摘录 / EXCERPT: Let V be a 1-spec subspace of $\operatorname{M}_n(\mathbb{K})$ such that ... Assume $n>4$. A linear subspace W of $\operatorname{M}_n(\mathbb{K})$ has a trivial spectrum if no...

conjecture | open | MathDB 页面 | 原文线索

V2-008115 | MATHDB-321557

Cyclotomic Lie algebra conjecture for multiple polylogarithms

摘录 / EXCERPT: Let ${\cal Z}(N)$ be the algebra of multiple polylogarithm values at N-th roots of unity, filtered by weight, and let $UC_{\bullet}(N)^{\vee}$ denote the graded dual of the unive...

conjecture | open | MathDB 页面 | 原文线索

V2-008116 | MATHDB-321567

Algebraic non-contractibility criterion for weighted barycentric spaces

摘录 / EXCERPT: Let Σ be a compact surface with singular points $p_1,\ldots,p_m$ and ordered weights $-1<\alpha_1\leq\cdots\leq\alpha_m<0$. For $\rho>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008117 | MATHDB-321602

The Gorensteinness conjecture for rings in \(\mathbf{G}(r)\)

摘录 / EXCERPT: Let R be a local ring in the class $\mathbf{G}(r)$, where $r\ge2$. Gorensteinness conjecture. If R is in $\mathbf{G}(r)$ for some $r\ge2$, then R is Gorenstein. The only know...

conjecture | open | MathDB 页面 | 原文线索

V2-008118 | MATHDB-321606

The conjecture on central simple minimal nonabelian Lie algebras of types \(\mathsf B\)--\(\mathsf D\)

摘录 / EXCERPT: Let L be a central simple minimal nonabelian Lie algebra, and consider Lie algebras of types $\mathsf B$--$\mathsf D$, with type $\mathsf D_4$ treated as the stated exception. Ty...

conjecture | open | MathDB 页面 | 原文线索

V2-008119 | MATHDB-321618

Localization conjecture for braid representations of simple objects

摘录 / EXCERPT: Let X be a simple object in a braided fusion category $\mathcal{C}$. For each n, let W_n^X be the minimal faithful module associated with the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-008120 | MATHDB-321632

The KP-BSO compatibility conjecture for multilinear identities

摘录 / EXCERPT: KP-BSO compatibility conjecture. If the field $\mathbb{F}$ has characteristic 0 or $p>d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-008121 | MATHDB-321643

The weak Lefschetz property conjecture for level monomial ideals

摘录 / EXCERPT: Let R be the polynomial ring in the variables used to define the ideal, over a field K of characteristic zero. Let $0<\alpha\leq\beta\leq\gamma\leq2(\alpha+\beta)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008122 | MATHDB-321647

Degree bound conjecture for Bernstein-Sato polynomials of homogeneous polynomials

摘录 / EXCERPT: Let $P(x_1,\ldots,x_n)\not\equiv0$ be a homogeneous polynomial in n variables with coefficients in a field K of characteristic 0, and let $b(s)$ be its Bernstein-Sato polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-008123 | MATHDB-321657

Asymptotic existence conjecture for pure Betti diagrams

摘录 / EXCERPT: Let ${\bf d}$ be a degree sequence, and let $\pi({\bf d})$ denote the smallest integer diagram on the ray $t\pi({\bf d})$ for $t>0$. Such integer diagrams are $m\pi({\bf d})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-008124 | MATHDB-321662

Coates-Sinnott conjecture on higher Stickelberger annihilators

摘录 / EXCERPT: Let F/K be an abelian CM extension of a totally real number field K, let ${\bf f}$ be the conductor of F/K, and let $\Theta_n({\bf b},{\bf f})$ be the higher Stickelberger el...

conjecture | open | MathDB 页面 | 原文线索

V2-008125 | MATHDB-321674

Stillman's conjecture on projective dimension of polynomial-ring quotients

摘录 / EXCERPT: Let K be a field and let $R=K[x_1,\ldots,x_N]$ be a polynomial ring. For fixed positive integers n and $d_1,\ldots,d_n$, let $I\subseteq R$ be an ideal generated by n homogen...

conjecture | open | MathDB 页面 | 原文线索

V2-008126 | MATHDB-321675

Castelnuovo-Mumford regularity bound for ideals

摘录 / EXCERPT: Let R be a polynomial ring over a field, and let $I\subseteq R$ be an ideal with minimal homogeneous generators whose number and degrees are fixed. The regularity-bound conjectur...

open_problem | open | MathDB 页面 | 原文线索

V2-008127 | MATHDB-321676

Razmyslov-type upper-bound conjecture for nilpotency degrees

摘录 / EXCERPT: Upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008128 | MATHDB-321709

Weak Lefschetz conjecture for Gorenstein algebras presented by quadrics

摘录 / EXCERPT: Let R/I be an artinian Gorenstein algebra presented by quadrics, with socle degree at least 3, and suppose that R is defined over a field of characteristic zero. The WLP conj...

conjecture | open | MathDB 页面 | 原文线索

V2-008129 | MATHDB-321710

The $h_2=r$ conjecture for artinian Gorenstein algebras

摘录 / EXCERPT: Let R/I be an artinian Gorenstein algebra of codimension $r\geq 3$ and socle degree e, and assume that $h_2=r$. The $h_2=r$ conjecture. Then ... Furthermore, if $e\geq 4$, the id...

conjecture | open | MathDB 页面 | 原文线索

V2-008130 | MATHDB-321754

The X-basis conjecture for the Ding-Iohara Fock space

摘录 / EXCERPT: Let m be a positive integer, let ${\boldsymbol \lambda}=(\lambda^{(1)},\ldots,\lambda^{(m)})$ be an m-tuple of partitions, and let $\mathcal{F}_{\mathbf{u}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008131 | MATHDB-321756

The higher-level Whittaker-vector expansion conjecture

摘录 / EXCERPT: Let $\boldsymbol{\alpha}=(\alpha_1,\ldots,\alpha_m)$ and $\boldsymbol{\beta}=(\beta_1,\ldots,\beta_m)$ be parameter tuples, and let $\left|G;\Lambda,\boldsymbol{\alpha}\right>$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008132 | MATHDB-321807

The local crystalline-lift conjecture for Serre weights

摘录 / EXCERPT: Let $K/{\mathbb{Q}}_p$ be a finite totally ramified extension, and let ${\overline{\rho}}:G_K\to\operatorname{GL}_2({\overline{{\mathbb{F}}}}_p)$ be a continuous representation. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-008133 | MATHDB-321808

The GHS conjecture on crystalline Serre weights

摘录 / EXCERPT: Let $K/{\mathbb{Q}}_l$ be a finite extension, and let $\overline{\rho}:G_K\to\operatorname{GL}_2(\overline{{\mathbb{F}}}_l)$ be a continuous representation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008134 | MATHDB-321812

Conjecture on fundamental groups of embedded-degeneratable surfaces with insufficient singular points

摘录 / EXCERPT: Structure conjecture. If $m\geq 1$ and $\ell-m>n-1$, then one can associate a graph T and an Artin group $A(T)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008135 | MATHDB-321813

Conjecture on Artin-group quotients for simply-degeneratable surfaces

摘录 / EXCERPT: Artin-group quotient conjecture. Under these assumptions, one can associate a graph T and an Artin group $A(T)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008136 | MATHDB-321832

Chuang-Miyachi's Koszul duality conjecture for affine category O blocks

摘录 / EXCERPT: Let $e,\ell,m>0$, let $\nu\in\mathbb{Z}^{\ell}(m)$, and let $\mathcal{O}^{\nu}_{\mu,a}$ denote the block defined by ... with $a\in\mathbb{N}$. Chuang-Miyachi's conjecture. The blocks...

conjecture | open | MathDB 页面 | 原文线索

V2-008137 | MATHDB-321833

Varagnolo-Vasserot's quotient conjecture for Schur categories

摘录 / EXCERPT: Fix $e,\ell,m>0$ and let $\nu\in\mathbb{N}^{\ell}(m)$. Let $\mathcal{O}^{\nu}(n)$ be the affine category block decomposition introduced in the source, let $\mathbf{S}^{\nu}(n)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-008138 | MATHDB-321902

The global epsilon constant conjecture

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with Galois group G. Let $T\Omega^{\rm loc}(L/K,1)$ be the element of $K_0(\mathbb{Z}[G],\mathbb{Q})$ constructed from the...

conjecture | open | MathDB 页面 | 原文线索

V2-008139 | MATHDB-321928

Bruinier-Ono integrality conjecture for CM-values of a weak Maass form

摘录 / EXCERPT: Bruinier and Ono's integrality conjecture. For the Maass form $P_p(z)$ and the CM points α_Q occurring in the formula for $p(n)$, the number

conjecture | open | MathDB 页面 | 原文线索

V2-008140 | MATHDB-321944

Primary binomial ideals in positive characteristic are mesoprimary

摘录 / EXCERPT: Let $\Bbbk$ be an algebraically closed field of arbitrary characteristic, let Q be a finitely generated commutative monoid, and let $\Bbbk[Q]$ be its monoid algebra. Mesoprimary...

conjecture | open | MathDB 页面 | 原文线索

V2-008141 | MATHDB-321952

The conformal block inequality for symmetric divisors on \overline{M}_{0,9}

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra. Let $D:=\operatorname{Sym}^{\ell}(\mathfrak{g})\otimes\mathbb{Q}[S_9]}\mathbb{Q}[\overline{M}_{0,9}]$ be a symmetrized conformal block divisor on $\overline{M}_{0,9}$ associat...

conjecture | open | MathDB 页面 | 原文线索

V2-008142 | MATHDB-321979

The conformal unit embedding conjecture for finite torsion-free Lie conformal algebras

摘录 / EXCERPT: Let $H=\Bbbk[T]$, and let L be a finite Lie conformal algebra that is torsion-free as an H-module. An embedding conjecture. L can be embedded into an associative conformal al...

conjecture | open | MathDB 页面 | 原文线索

V2-008143 | MATHDB-322045

Prediction for the average fiber size of full-monodromy covers

摘录 / EXCERPT: Full-monodromy fiber prediction. The average number of points in the fiber over z should be

conjecture | open | MathDB 页面 | 原文线索

V2-008144 | MATHDB-322074

Filippov's identity-basis conjecture for the simple 7-dimensional Malcev algebra

摘录 / EXCERPT: Let the simple 7-dimensional Malcev algebra be the algebra under consideration, and let the Filippov h-polynomial be the degree-5 polynomial identity associated with it. Fili...

conjecture | open | MathDB 页面 | 原文线索

V2-008145 | MATHDB-322078

The recursive demipotent conjecture for meet semi-lattices

摘录 / EXCERPT: Let L be a meet semi-lattice with a fixed linear extension, and let D range over the valid diagrams of L, meaning subsets containing the minimal elements and stable under joi...

conjecture | open | MathDB 页面 | 原文线索

V2-008146 | MATHDB-322086

Caprace-Marquis conjecture on split spherical BN-pairs of anisotropic groups

摘录 / EXCERPT: Caprace-Marquis conjecture. Every split spherical BN-pair of $G(k)$ is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-008147 | MATHDB-322087

Caprace-Marquis weaker conjecture on virtually trivial split spherical BN-pairs

摘录 / EXCERPT: Caprace-Marquis weaker conjecture. Every split spherical BN-pair of $G(k)$ is virtually trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-008148 | MATHDB-322127

Projectivity conjecture for cyclic local modules over endomorphism rings

摘录 / EXCERPT: Let ${\bf R}$ be a Noetherian local ring, let E be a local ${\bf R}$-module, and let ${\Lambda}=\operatorname{End}_{\bf R}(E)$. Suppose that E is ${\Lambda}$-cyclic and has fin...

conjecture | open | MathDB 页面 | 原文线索

V2-008149 | MATHDB-322157

The monomial-ordering transcendence-degree conjecture

摘录 / EXCERPT: Let R be a ring equipped with a monomial ordering ... trdeg of R. Monomial-ordering transcendence-degree conjecture. The statements denoted by TheoremtMain, equations(tMainA) and...

conjecture | open | MathDB 页面 | 原文线索

V2-008150 | MATHDB-322170

The Bocci-Harbourne-Essential containment conjecture

摘录 / EXCERPT: Let K be an algebraically closed field, let $R=K[\mathbb{P}^{N}]$, and let $I\subseteq R$ be a homogeneous ideal. Bocci-Harbourne-Essential conjecture. For every r, ... The state...

conjecture | open | MathDB 页面 | 原文线索

V2-008151 | MATHDB-322171

The refined Chudnovsky conjecture for symbolic initial degrees

摘录 / EXCERPT: Let K be an algebraically closed field, let I be the radical ideal of a finite set of points in $\mathbb{P}^{N}$, and let $\alpha(I)$ denote the least degree of a nonzero form...

conjecture | open | MathDB 页面 | 原文线索

V2-008152 | MATHDB-322224

Non-flatness conjecture for rings of integer-valued polynomials

摘录 / EXCERPT: Non-flatness conjecture. The domain $\operatorname{Int}(D)$ is not flat over D for either of these choices of D; in particular, there exists a local, Noetherian, one-dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-008153 | MATHDB-322231

Squarefree conjecture for initial ideals of modular lattice ideals

摘录 / EXCERPT: Let L be a modular lattice, and let I_L denote its lattice ideal. For a monomial order $<$, write $\ini_<(I_L)$ for the initial ideal. Squarefree conjecture. For every monomial...

conjecture | open | MathDB 页面 | 原文线索

V2-008154 | MATHDB-322241

Potential diagonalizability conjecture for mod p representations

摘录 / EXCERPT: Let $K/\mathbb Q_p$ be a finite extension, and let ${\bar{r}}:G_K\to\operatorname{GL}_n(\overline{\mathbb F}_p)$ be continuous. Potential diagonalizability conjecture. The represen...

conjecture | open | MathDB 页面 | 原文线索

V2-008155 | MATHDB-322275

Cohen-Macaulayness from the first Buchsbaum-Rim coefficient

摘录 / EXCERPT: Let $({\bf R},{\mathfrak m})$ be a Noetherian local ring with $\dim {\bf R}\geq 2$, and let $U\subseteq {\mathfrak m}{\bf R}^r$ be a parameter module of ${\bf R}^r$, where $r>0$. H...

conjecture | open | MathDB 页面 | 原文线索

V2-008156 | MATHDB-322276

Hochster-Seibert-Schenck conjecture on powers of general linear forms

摘录 / EXCERPT: Hochster-Seibert-Schenck conjecture. The quotient R/I fails the WLP for all sufficiently large values of t.

conjecture | open | MathDB 页面 | 原文线索

V2-008157 | MATHDB-322297

Categorical rationality conjecture for fibrations in intersections of two four-dimensional quadrics

摘录 / EXCERPT: Let $X \to \mathbb{P}^1$ be a fibration in complete intersections of two four-dimensional quadrics over an algebraically closed field of characteristic zero. Write $\mathcal{A}_X$...

conjecture | open | MathDB 页面 | 原文线索

V2-008158 | MATHDB-322302

Refined Lehrer-Solomon conjecture for finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group of rank n, let $\mathcal R$ be a set of representatives of its conjugacy classes, and let $C_W(w)$ be the centralizer of w. Let ρ_W be th...

conjecture | open | MathDB 页面 | 原文线索

V2-008159 | MATHDB-322303

Conjecture B on top-degree Coxeter characters

摘录 / EXCERPT: Let W be a finite Coxeter group of rank n, let W_L denote a parabolic subgroup generated by a subset L of a Coxeter generating set, and call a conjugacy class cuspidal if i...

conjecture | open | MathDB 页面 | 原文线索

V2-008160 | MATHDB-322323

A-H conjecture on stabilization of chains induced by monomials

摘录 / EXCERPT: Let $k\in {\mathbb P}$ and let $f\in\mathbb{K}[y_1,\ldots,y_k]$ be a monomial. For each $n\geq k$, define $\phi_n:R_n\to T_n$ by ... and let $I_n=\ker\phi_n$ be the corresponding inv...

conjecture | open | MathDB 页面 | 原文线索

V2-008161 | MATHDB-322356

Canonical-form conjecture for natural reorderings of finite connected quandles

摘录 / EXCERPT: Let Q be a finite connected quandle with elements $1,2,\ldots,n$ naturally ordered, profile $\{1,\ell_1,\ell_2,\ldots,\ell_k\}$, and $1\leq\ell_1\leq\ell_2\leq\cdots\leq\ell_k$....

conjecture | open | MathDB 页面 | 原文线索

V2-008162 | MATHDB-322358

The semistar-counting conjecture for integrally closed domains

摘录 / EXCERPT: Semistar-counting conjecture. The domain D is a principal ideal domain (PID) if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008163 | MATHDB-322404

Positivity conjecture for the bases of cluster algebras from unpunctured surfaces

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra from an unpunctured surface with at least two marked points, and let $\mathcal{B}^{\circ}$ and $\mathcal{B}$ be the two bases constructed in...

conjecture | open | MathDB 页面 | 原文线索

V2-008164 | MATHDB-322405

Atomicity conjecture for the basis of cluster algebras from unpunctured surfaces

摘录 / EXCERPT: Let $\mathcal{A}$ be the cluster algebra under consideration and let $\mathcal{B}$ be the basis constructed in the paper. A basis is atomic when an element $a\in\mathcal{A}^{+}$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-008165 | MATHDB-322408

Avramov's eventual monotonicity conjecture for Betti numbers

摘录 / EXCERPT: Avramov's conjecture. The sequence of Betti numbers $(\beta_i(M))_{i\geqslant 0}$ is eventually non-decreasing; that is, $\beta_i(M)\leqslant\beta_{i+1}(M)$ for all $i\gg 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-008166 | MATHDB-322417

The categorical equivalence conjecture for equivariant algebras and Zariski structures

摘录 / EXCERPT: Categorical equivalence conjecture. The functor $\mathsf{Alg}(k)^{op} \rightarrow \mathsf{Zar}$ is an equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-008167 | MATHDB-322418

The noncommutative equivariant algebra conjecture for Zariski structures

摘录 / EXCERPT: Noncommutative equivariant algebra conjecture. The Zariski structure $\operatorname{nSpec} A$ is non-classical.

conjecture | open | MathDB 页面 | 原文线索

V2-008168 | MATHDB-322425

Sign-coherence and constant-term conjectures for cluster-algebra matrices and F-polynomials

摘录 / EXCERPT: Sign-coherence and constant-term conjectures. (i) Every column of C' is a nonzero vector, and its nonzero components are either all positive or all negative. (ii) Every F-polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-008169 | MATHDB-322426

The full exceptional collection conjecture for homogeneous spaces

摘录 / EXCERPT: Let $\BG$ be a semisimple algebraic group and let $\BP\subset\BG$ be a parabolic subgroup. The homogeneous space $\BG/\BP$ has a bounded derived category of coherent sheaves...

conjecture | open | MathDB 页面 | 原文线索

V2-008170 | MATHDB-322428

The one-generation conjecture for the block algebra

摘录 / EXCERPT: Let $\SB$ be an exceptional block, and let $A_\SB$ be the associated A_∞ algebra. One-generation conjecture. The algebra $A_\SB$ is one-generated as an A_∞ algebra, a...

conjecture | open | MathDB 页面 | 原文线索

V2-008171 | MATHDB-322430

The surface-group sum-of-squares conjecture

摘录 / EXCERPT: Let G be a surface group, and let $\mathbb{C}[G]^h$ denote the hermitian elements of its group algebra. Write $\Sigma^2\mathbb{C}[G]$ for the cone of finite sums of hermitian squ...

conjecture | open | MathDB 页面 | 原文线索

V2-008172 | MATHDB-322436

Syzygy Hilbert-series conjecture for the alternating ideal

摘录 / EXCERPT: Syzygy Hilbert-series conjecture. This Hilbert series equals

conjecture | open | MathDB 页面 | 原文线索

V2-008173 | MATHDB-322461

The four-variable power-sum regular-sequence conjecture from computer calculations

摘录 / EXCERPT: Let p_i be the power sum symmetric polynomial of degree i in $S=K[x_1,x_2,x_3,x_4]$. Let $A=\{a,b,n\}$ with $a<b<n$. The four-variable power-sum conjecture. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-008174 | MATHDB-322463

Conca-Krattenthaler-Watanabe's regular-sequence conjecture for four power sums

摘录 / EXCERPT: Let $A\subset\mathbb{N}^*$ have $|A|=4$, say $A=\{a_1,a_2,a_3,a_4\}$, with $\gcd(A)=1$, and let $p_A(4)$ denote the corresponding four power sum symmetric polynomials in four varia...

conjecture | open | MathDB 页面 | 原文线索

V2-008175 | MATHDB-322471

Nonalgebraicity conjecture for products involving multiple matrix or Grassmann factors

摘录 / EXCERPT: Let K be the ground field, let E denote the Grassmann algebra, and let R be a PI-algebra with ... where each R_i is one of K, E, $M_2(K)$, or $E\otimes_K E$, and at least...

conjecture | open | MathDB 页面 | 原文线索

V2-008176 | MATHDB-322518

Cluster-variable connectivity conjecture

摘录 / EXCERPT: A cluster variable is a variable occurring in a cluster, a seed is a cluster together with its exchange data, and the exchange graph is the graph whose vertices are seeds and whose...

conjecture | open | MathDB 页面 | 原文线索

V2-008177 | MATHDB-322519

Face connectivity conjecture for cluster algebras

摘录 / EXCERPT: Let ${\mathcal X}$ be a set of cluster variables. A seed is a cluster together with its exchange data, and the exchange graph is the graph whose vertices are seeds and whose edges...

conjecture | open | MathDB 页面 | 原文线索

V2-008178 | MATHDB-322520

Rigidity conjecture for extended exchange matrices

摘录 / EXCERPT: Let u and v be vertices of the exchange graph of a cluster algebra with principal coefficients. Two extended exchange matrices are equivalent when there is a bijection of their...

conjecture | open | MathDB 页面 | 原文线索

V2-008179 | MATHDB-322521

Rigidity conjecture for the matrices H^v

摘录 / EXCERPT: Let u and v be vertices of the exchange graph of a cluster algebra with principal coefficients, and let H^u and H^v denote the associated matrices. Rigidity conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-008180 | MATHDB-322523

Strong $\mathbf g$-vector support conjecture

摘录 / EXCERPT: A cluster monomial is a monomial in cluster variables from one seed, its support is the set of variables with nonzero exponent, and its $\mathbf g$-vector is the product of the...

conjecture | open | MathDB 页面 | 原文线索

V2-008181 | MATHDB-322524

Maximal-monomial conjecture for F-polynomials

摘录 / EXCERPT: An F-polynomial is the polynomial in the initial coefficient variables obtained from a cluster variable by specializing the initial cluster variables to 1. Maximal-monomial con...

conjecture | open | MathDB 页面 | 原文线索

V2-008182 | MATHDB-322526

The ν-conjecture relating $\mathbf g$- and denominator vectors

摘录 / EXCERPT: Assume B is acyclic. Let $\nu:V\to V^*$ and $\eta:V^*\to V$ be the mutually inverse maps defined from the bilinear forms E and F in the source. For a cluster variable x, le...

conjecture | open | MathDB 页面 | 原文线索

V2-008183 | MATHDB-322527

Integer-basis conjecture for denominator vectors

摘录 / EXCERPT: Assume B is acyclic. Let $\mathrm{Ex}_0(B)$ be the vertices of the exchange graph, let $I(v)$ index the cluster at v, and let $\mathbf d(x)$ denote the denominator vector of a...

conjecture | open | MathDB 页面 | 原文线索

V2-008184 | MATHDB-322529

Strong denominator-vector support conjecture

摘录 / EXCERPT: Let the strong $\mathbf g$-vector support conjecture be the assertion concerning equal $\mathbf g$-vectors, supports, and mutation sequences that do not exchange variables in the c...

conjecture | open | MathDB 页面 | 原文线索

V2-008185 | MATHDB-322585

The small-degree characteristic conjecture for the strong Lefschetz property

摘录 / EXCERPT: Let $R=K[x_0,\ldots,x_n]$ be a polynomial ring over a field K, let $I_{\underline{d}}=(x_0^{d_0},\ldots,x_n^{d_n})$, and let t be the socle degree of $R/I_{\underline{d}}$. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-008186 | MATHDB-322591

Explicit $W^{(2)}_N$ subalgebras of extended $\W_{n,\ell}$-algebras

摘录 / EXCERPT: Let $\mathfrak{W}_{n,\ell}$ be an extended algebra constructed from $\widehat{\mathfrak{gl}}(1\mid 1)$, and let $W^{(2)}_N$ denote the Feigin-Semikhatov algebra at level K. Expli...

conjecture | open | MathDB 页面 | 原文线索

V2-008187 | MATHDB-322642

Threshold scattering and blow-up conjecture for the 3D radial combined NLS

摘录 / EXCERPT: Threshold scattering and blow-up conjecture. If $K(u_0)\geq 0$, then $I=\mathbb{R}$ and u scatters in both time directions as $t\to\pm\infty$ in H^1; if $K(u_0)<0$, then u bl...

conjecture | open | MathDB 页面 | 原文线索

V2-008188 | MATHDB-322671

The anticyclotomic main conjecture for the Rankin-Selberg representation

摘录 / EXCERPT: Let $\rho_\Lambda:=\rho_p(\pi)|_{G_{\mathcal K}}\otimes\lambda\varepsilon_\Lambda$ be the rank-two Λ-adic Galois representation, with the local conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-008189 | MATHDB-322677

The Invertibles Conjecture for group rings

摘录 / EXCERPT: Let K be a division ring and let G be a group. A one-sided invertible element of $K[G]$ is an element having either a left or a right inverse. The Invertibles Conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-008190 | MATHDB-322678

The Zero Divisors Conjecture for group rings

摘录 / EXCERPT: Let K be a division ring and let G be a group. The Zero Divisors Conjecture. If $K[G]$ contains zero divisors, then G has torsion. The source states that the Invertibles Conj...

conjecture | open | MathDB 页面 | 原文线索

V2-008191 | MATHDB-322715

Generalized Kac-Wakimoto conjecture for modified dimensions

摘录 / EXCERPT: Generalized Kac-Wakimoto conjecture. Every simple ${\mathfrak{g}}$-supermodule J is ambidextrous. If L is another simple supermodule satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-008192 | MATHDB-322719

Coreduced modules are strongly coreduced

摘录 / EXCERPT: Coreduced-module conjecture. A coreduced G-module is strongly coreduced.

conjecture | open | MathDB 页面 | 原文线索

V2-008193 | MATHDB-322779

Yu's conjecture on the degree of the Jacobian bracket

摘录 / EXCERPT: Let $k[x_1,\ldots,x_n]$ be a polynomial ring, and let f and g be algebraically independent polynomials whose homogeneous components of maximal degree are algebraically dependen...

conjecture | open | MathDB 页面 | 原文线索

V2-008194 | MATHDB-322790

Sub-irreducibility is equivalent to quotient-irreducibility for finite semirings

摘录 / EXCERPT: Let $(R,+,\cdot)$ be a finite, simple, additively idempotent semiring, and let $(M,+)$ be a finite idempotent R-semimodule. A semimodule is sub-irreducible if it is non-quasitriv...

conjecture | open | MathDB 页面 | 原文线索

V2-008195 | MATHDB-322791

Classification conjecture for finite simple semirings with irreducible semimodules

摘录 / EXCERPT: Let $(R,\vee,\circ)$ be a finite simple additively idempotent semiring with an absorbing greatest element, and suppose it possesses a finite idempotent irreducible semimodule witho...

conjecture | open | MathDB 页面 | 原文线索

V2-008196 | MATHDB-322810

Tractability conjecture for finite-domain CSPs

摘录 / EXCERPT: Let ${\mathfrak B}$ be a finite structure with finite relational signature, and let ${\mathfrak C}$ be the expansion of the core of ${\mathfrak B}$ by a constant symbol for each el...

conjecture | open | MathDB 页面 | 原文线索

V2-008197 | MATHDB-322832

The differential-determinant membership conjecture for differential operators

摘录 / EXCERPT: Let ${\mathcal A}={\mathbb C}[a^{(n)},b^{(n)},d^{(n)}\mid n\in{\mathbb Z}_+]/\{ad-b^2\}$ be the differential algebra described above, and let A be the matrix under consideration....

conjecture | open | MathDB 页面 | 原文线索

V2-008198 | MATHDB-322839

Monotonicity conjecture for the positive margins property

摘录 / EXCERPT: Monotonicity conjecture. Assume that (G,d) does not have the positive margins (or interior point) property. If $d'_v\ge d_v$ for all $v\in V$, then (G,d') does not have the pos...

conjecture | open | MathDB 页面 | 原文线索

V2-008199 | MATHDB-322840

The algebraic independence conjecture for logarithms

摘录 / EXCERPT: Let $\mathcal{L}$ be the $\mathbb Q$-vector space of logarithms of algebraic numbers: ... For $n\geq 2$, let ${\sf LD}_n^{\mathbb Q}$ denote the relation of n elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-008200 | MATHDB-322872

The Mathieu moonshine conjecture for twisted elliptic genera

摘录 / EXCERPT: Twisted elliptic-genus module conjecture. For $k\equiv0,3\pmod4$ and $k\geq-1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-008201 | MATHDB-322900

Andruskiewitsch-Schneider's generation conjecture for finite-dimensional pointed Hopf algebras

摘录 / EXCERPT: Assume that k is an algebraically closed field of characteristic zero. A Hopf algebra is pointed when its coradical is spanned by grouplike elements; its grouplike and skew primi...

conjecture | solved | MathDB 页面 | 原文线索

V2-008202 | MATHDB-322901

Wang-Zhang-Zhuang's classification conjecture for Hopf algebras satisfying (H2)

摘录 / EXCERPT: Wang-Zhang-Zhuang's conjecture. Then H is isomorphic to one of the Hopf algebras in the family constructed in Construction WZZeg. The surrounding results show that these algebras...

conjecture | open | MathDB 页面 | 原文线索

V2-008203 | MATHDB-322927

Representability conjecture for graded automorphism functors of N-Lie conformal superalgebra forms

摘录 / EXCERPT: Let $N\in\{1,2,3,4\}$, let $\mathcal{A}$ be a complex N-Lie conformal superalgebra, and let $\mathcal{L}$ be an $\widehat{\mathcal{S}}/\mathcal{R}$-form of...

conjecture | open | MathDB 页面 | 原文线索

V2-008204 | MATHDB-322929

The distinguished involution conjecture for leading coefficients of finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group with weight function L, let ${\operatorname{Irr}}(W)$ denote its irreducible representations, and let $c_{w,E}$ and f_E be the leading coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-008205 | MATHDB-322930

The unique nonnegative constituent conjecture for left cells

摘录 / EXCERPT: Unique nonnegative constituent conjecture. For each left cell ${\mathfrak{C}}$ of W, there is a unique $E\in{\mathcal{S}}_L(W)$ such that $m({\mathfrak{C}},E)>0$; moreover, for t...

conjecture | open | MathDB 页面 | 原文线索

V2-008206 | MATHDB-322943

Non-algebraicity conjecture for rational-function composition generating functions

摘录 / EXCERPT: Non-algebraicity conjecture. The generating function $D(z)$ is not algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-008207 | MATHDB-322951

Point-independence conjecture for tempered D-modules on Bun_G

摘录 / EXCERPT: Point-independence conjecture. The subcategory $\Dmod^x_{\on{temp}}(\Bun_G)\subset\Dmod(\Bun_G)$ is independent of the choice of the point $x\in X$. This is deduced in the source a...

conjecture | open | MathDB 页面 | 原文线索

V2-008208 | MATHDB-322989

The classical Lie algebra conjecture on centralizers and sheets

摘录 / EXCERPT: Classical Lie algebra sheet conjecture. If there is only one sheet S passing through μ, then

conjecture | open | MathDB 页面 | 原文线索

V2-008209 | MATHDB-323070

Lund's conjecture on non-algebraic quadrature domains

摘录 / EXCERPT: Let $n>2$. A quadrature domain in $\mathbb{R}^n$ is a domain satisfying a quadrature identity for the relevant class of functions, and an algebraic domain is one whose boundary is...

conjecture | open | MathDB 页面 | 原文线索

V2-008210 | MATHDB-323071

PBW conjecture on differential algebraicity of symmetric operad series

摘录 / EXCERPT: A symmetric PBW operad is a symmetric operad with binary generators whose quadratic relations form a Gröbner basis. Its exponential generating series is the formal power series ass...

conjecture | open | MathDB 页面 | 原文线索

V2-008211 | MATHDB-323072

Differential-algebraicity conjecture for generating series of associative operads

摘录 / EXCERPT: An associative operad is an operad in the associative-operad class considered in the source. Its exponential generating series records the dimensions of its components. Associative...

conjecture | open | MathDB 页面 | 原文线索

V2-008212 | MATHDB-323130

The trace bound for complex conjugation in RLAESDC Galois representations

摘录 / EXCERPT: Trace bound conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-008213 | MATHDB-323134

The Selmer near-companion converse conjecture

摘录 / EXCERPT: Let K be a number field, let $n\geq 1$, and let E_1 and E_2 be elliptic curves over K. They are n-Selmer near-companions if their n-Selmer groups have uniformly bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-008214 | MATHDB-323142

Previdi's delooping conjecture for exact categories

摘录 / EXCERPT: Let $\mathcal{A}$ be an exact category. Write $S(\mathcal{A})$ for the geometric realization of the simplicial category $iS_{\bullet}(\mathcal{A})$ given by Waldhausen's...

conjecture | open | MathDB 页面 | 原文线索

V2-008215 | MATHDB-323147

K-stability conjecture for nonsingular del Pezzo surfaces in finite characteristic

摘录 / EXCERPT: Let S be a nonsingular del Pezzo surface over an algebraically closed field of finite characteristic. K-stability conjecture for nonsingular del Pezzo surfaces. The surface S i...

conjecture | open | MathDB 页面 | 原文线索

V2-008216 | MATHDB-323151

Multiplicativity conjecture for the two-one map

摘录 / EXCERPT: Multiplicativity conjecture for the two-one map. The map X is an algebra homomorphism with respect to the $1/2$-harmonic product.

conjecture | open | MathDB 页面 | 原文线索

V2-008217 | MATHDB-323166

The dendriform realization conjecture for refined flow idempotents

摘录 / EXCERPT: Let $\mathscr{F}_t$ be the global series introduced from the parameterized flow series, and let $\mathsf{F}_t=-(1-t)\mathbf{N}_t\mathbf{T}\mathbf{P}_t$ be its dendriform candidate....

conjecture | open | MathDB 页面 | 原文线索

V2-008218 | MATHDB-323168

The descent-algebra idempotent conjecture for the series Z

摘录 / EXCERPT: Let Z be the dendriform image of the series $\mathscr{Z}$, let Z_n be its homogeneous component of degree n, and let b be the flow variable. Z-series desc...

conjecture | open | MathDB 页面 | 原文线索

V2-008219 | MATHDB-323170

Characteristic-p dimensions of multihomogeneous components of B_2

摘录 / EXCERPT: Let $A_n(\mathbb F_p)$ be the free associative algebra on n generators over the finite field $\mathbb F_p$, and let $B_2(A_n(\mathbb F_p))[m_1,\ldots,m_n]$ denote its multihomoge...

conjecture | open | MathDB 页面 | 原文线索

V2-008220 | MATHDB-323171

Vanishing range for torsion in lower-central-series quotients

摘录 / EXCERPT: Let $A_n(\mathbb Z)$ be the free associative algebra on n generators, and let $B_\ell(A_n(\mathbb Z))[m_1,\ldots,m_n]$ denote the multihomogeneous component of multidegree...

conjecture | open | MathDB 页面 | 原文线索

V2-008221 | MATHDB-323206

Weak Horrocks conjecture for finite-dimensional modules

摘录 / EXCERPT: Weak Horrocks conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008222 | MATHDB-323210

Walker's Drinfeld-center conjecture for rectangular and annular relation categories

摘录 / EXCERPT: Let Λ be a $\mathbb{C}$-linear category. Let $\Lambda^{\mathcal R}$ and $\Lambda^{\mathcal A}$ be, respectively, the rectangular and annular versions of the corresponding l...

conjecture | open | MathDB 页面 | 原文线索

V2-008223 | MATHDB-323211

Jones's planar-algebraic Walker conjecture for finite-depth subfactors

摘录 / EXCERPT: Let $N\subset M$ be a finite-depth subfactor, let P be its associated subfactor planar algebra, and let the N-N-bimodule category generated by $_N L^2(M)_M$ be the category w...

conjecture | open | MathDB 页面 | 原文线索

V2-008224 | MATHDB-323230

Krull-Schmidt property for categorifications of the BMW skein and q-Schur algebras

摘录 / EXCERPT: Krull-Schmidt property. In the case of the categorifications of $\operatorname{Skein}_q(n,N)$ and $S_q(n,\Delta_N)$, the category $\mathcal{X}'$ has the Krull-Schmidt property.

conjecture | open | MathDB 页面 | 原文线索

V2-008225 | MATHDB-323238

The local Oort conjecture for cyclic groups

摘录 / EXCERPT: Local Oort conjecture. The local lifting problem can always be solved when G is cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-008226 | MATHDB-323239

The good-reduction lifting conjecture for cyclic covers of the projective line

摘录 / EXCERPT: Let K be the ground field, let $X=\mathbb{P}^1_K$, and let $\bar{x}_0$ be the distinguished branch point with residue field κ_0. For a character...

conjecture | open | MathDB 页面 | 原文线索

V2-008227 | MATHDB-323291

The strong nilpotency index conjecture over reduced noncommutative rings

摘录 / EXCERPT: Let M be a square matrix of size m whose entries are functions from a set S to a reduced noncommutative ring R. For $v\in S$, write $M|_v$ for the matrix obtained by evalua...

conjecture | open | MathDB 页面 | 原文线索

V2-008228 | MATHDB-323313

Morita's conjecture on the abelianization of derivations of a free Lie algebra

摘录 / EXCERPT: Let F_n be the free group of rank n, let $\mathcal{L}_n$ be the free Lie algebra associated with the lower central series of F_n, and let $\mathrm{Der}^+(\mathcal{L}_n)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-008229 | MATHDB-323315

Drensky-Yu's Poisson-bracket degree conjecture

摘录 / EXCERPT: Let k be a field of characteristic 0, and let f and g be algebraically independent polynomials in $k[x_1,\ldots,x_n]$. Assume that the homogeneous components of maximal deg...

conjecture | open | MathDB 页面 | 原文线索

V2-008230 | MATHDB-323386

Boundedness conjecture for the invariant c of product-quotient surfaces

摘录 / EXCERPT: Let S be a regular product-quotient surface with invariants p_g, q, and $c:=\gamma+p_g\in\mathbb{N}$, where c is half the codimension of the subspace of $H^2(S,\mathbb{C})$...

conjecture | open | MathDB 页面 | 原文线索

V2-008231 | MATHDB-323387

Conjecture on the negative part of the canonical divisor

摘录 / EXCERPT: Let S be a regular product-quotient surface with quotient model X, and let $H^2(X)$ be viewed inside $H^2(S)$. Write the Zariski decomposition of the canonical divisor as ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-008232 | MATHDB-323393

Coates-Sinnott conjecture for number fields

摘录 / EXCERPT: Let K/k be an abelian extension of number fields with Galois group G. Let S be a finite set of primes of k containing the infinite primes and the primes that ramify in...

conjecture | open | MathDB 页面 | 原文线索

V2-008233 | MATHDB-323394

Linnik's conjecture on counting number fields by discriminant

摘录 / EXCERPT: For a positive integer d, let $\Delta(d,X)$ denote the number of degree-d field extensions F of $\mathbb{Q}$ in an algebraic closure $\overline{\mathbb{Q}}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-008234 | MATHDB-323427

The normality conjecture for toric rings arising from cyclic polytopes

摘录 / EXCERPT: Let Q be the affine semigroup associated with the cyclic polytope construction in the paper, and let $K[Q]$ be its semigroup algebra over the field K. Write n and d for the...

conjecture | open | MathDB 页面 | 原文线索

V2-008235 | MATHDB-323494

The interval conjecture for lengths of maximal green sequences

摘录 / EXCERPT: Let Q be a cluster quiver. For $l\in\mathbb{Z}_{\geq 0}$, let $\mathrm{green}_l(Q)$ denote the set of maximal green sequences of Q of length l. Interval conjecture. The set .....

conjecture | open | MathDB 页面 | 原文线索

V2-008236 | MATHDB-323513

Khare-Wintenberger reciprocity conjecture for Frobenius and ray class lines

摘录 / EXCERPT: Khare-Wintenberger reciprocity conjecture. The Frobenius line M_Q and the ray class line N_Q are equal. This conjecture concerns the relationship between Frobenius and ray clas...

conjecture | open | MathDB 页面 | 原文线索

V2-008237 | MATHDB-323516

The subclosedness conjecture for categories associated to preprojective algebra quotients

摘录 / EXCERPT: Let Q be a quiver, let W be its Weyl group, let Π be the associated preprojective algebra, and for $w\in W$ let I_w be the corresponding ideal. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008238 | MATHDB-323517

The affine description conjecture for subcategories from preprojective quotients

摘录 / EXCERPT: Suppose that Q is an affine quiver, let W be its Weyl group, let Π be the associated preprojective algebra, and let I_w be the ideal associated to $w\in W$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008239 | MATHDB-323518

The reduced Ext-quiver characterization conjecture for subcategories from preprojective quotients

摘录 / EXCERPT: Let Q be a quiver, let W be its Weyl group, let Π be the associated preprojective algebra, and for $w\in W$ let I_w be the corresponding ideal. For a full subcategory...

conjecture | open | MathDB 页面 | 原文线索

V2-008240 | MATHDB-323528

Polynomial-ring quasimonomial valuation conjecture for graded sequences

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero and let $R=k[x_1,\ldots,x_m]$. Let $\mathfrak q$ be a nonzero ideal and let $\mathfrak a_\bullet$ be a graded sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-008241 | MATHDB-323531

The Strong Conjecture for algebraic and noncrossing parking spaces

摘录 / EXCERPT: Let W be a real reflection group of rank n with Coxeter number h, let V be its reflection representation, and let $k\geq 1$. Let $\theta_1,\dots,\theta_n$ be a homogeneous...

conjecture | open | MathDB 页面 | 原文线索

V2-008242 | MATHDB-323564

Universal-element conjecture for central simple Lie algebras

摘录 / EXCERPT: Universal-element conjecture. All central simple Lie algebras of the same type can be assembled from the same universal elements, which are Lie algebra structures over $\mathbb Z$.

conjecture | open | MathDB 页面 | 原文线索

V2-008243 | MATHDB-323591

Augmented down-up algebra grading conjecture for $\hat{\mathbb T}$

摘录 / EXCERPT: The grading conjecture. The degree-zero component has basis

conjecture | open | MathDB 页面 | 原文线索

V2-008244 | MATHDB-323592

Algebra grading conjecture for generalized uniform-poset algebras

摘录 / EXCERPT: The generalized grading conjecture. The corresponding algebra S has a direct-sum algebra grading

conjecture | open | MathDB 页面 | 原文线索

V2-008245 | MATHDB-323646

Positivity conjecture for graph LP algebras

摘录 / EXCERPT: Let ${\mathcal A}_\Gamma$ be a graph LP algebra and let $t_{\mathcal S}$ be a seed. Write ${\mathcal L}(t_{\mathcal S})$ for the Laurent polynomial ring associated with this seed....

conjecture | open | MathDB 页面 | 原文线索

V2-008246 | MATHDB-323647

Finite-type conjecture for linear LP algebras

摘录 / EXCERPT: A linear LP algebra is an LP algebra whose exchange polynomials are linear in the relevant cluster variables, allowing the coefficients and constant terms described in the paper. F...

conjecture | open | MathDB 页面 | 原文线索

V2-008247 | MATHDB-323648

Simple-polytope conjecture for linear LP algebras

摘录 / EXCERPT: Let ${\mathcal A}$ be a linear LP algebra of rank r. Simple-polytope conjecture. There exists a convex simple polytope $P({\mathcal A})$ of dimension r such that its vertices a...

conjecture | open | MathDB 页面 | 原文线索

V2-008248 | MATHDB-323660

Specialness conjecture for affine spaces associated with inflated conjugacy classes

摘录 / EXCERPT: Let G be an algebraic group, let P be a parabolic subgroup with unipotent radical N, and let C be a conjugacy class in P/N. Let D be the inflation of C to P, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-008249 | MATHDB-323715

Non-generation of the derived category of the Barlow surface by an exceptional collection

摘录 / EXCERPT: Let S be the Barlow surface, a smooth minimal surface of general type with $p_g(S)=q(S)=0$ and $K_S^2=1$, and let $D^b\operatorname{Coh}(S)$ denote its bounded derived category o...

conjecture | open | MathDB 页面 | 原文线索

V2-008250 | MATHDB-323722

Gorenstein and palindromic Hilbert-series conjecture for Cherednik quotients

摘录 / EXCERPT: Gorenstein and palindromicity conjecture. If τ is trivial, then $M_c(S_n,\mathfrak{h},\tau)/J_c(S_n,\mathfrak{h},\tau)$ is Gorenstein for all n, both when $c\in\mathbf{F}_p$...

conjecture | open | MathDB 页面 | 原文线索

V2-008251 | MATHDB-323731

The Read-inspired superalgebra isomorphism for the spin Weyl group of type D

摘录 / EXCERPT: Read-inspired isomorphism conjecture. There is a superalgebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-008252 | MATHDB-323740

Sturmfels-Sullivant degree-bound conjecture for the 3-Kimura model

摘录 / EXCERPT: Sturmfels-Sullivant's 3-Kimura conjecture. The ideal of the 3-Kimura model is generated in degree at most 4 for any tree T.

conjecture | open | MathDB 页面 | 原文线索

V2-008253 | MATHDB-323753

Minh-Petitot polynomiality conjecture for the algebra of multiple zeta values

摘录 / EXCERPT: Let $\mathcal{Z}$ denote the $\mathbb{Q}$-algebra generated by all multiple zeta values. Minh-Petitot conjecture. The algebra $\mathcal{Z}$ is a polynomial algebra over...

conjecture | open | MathDB 页面 | 原文线索

V2-008254 | MATHDB-323768

Existence and uniqueness conjecture for triplets of pure free squarefree complexes

摘录 / EXCERPT: Existence and uniqueness conjecture. For each balanced degree triplet (A,B,C) of type n, there exists such a triplet of complexes. Moreover, its Betti numbers are uniquely dete...

conjecture | open | MathDB 页面 | 原文线索

V2-008255 | MATHDB-323771

The strong form of Goncharov's conjecture for multiple zeta values

摘录 / EXCERPT: Let $\mathfrak{Z}$ be the $\mathbb{Q}$-algebra generated by all multiple zeta values, and let $\mathfrak{Z}_w$ be the $\mathbb{Q}$-vector space spanned by the multiple zeta values...

conjecture | open | MathDB 页面 | 原文线索

V2-008256 | MATHDB-323772

The classification conjecture for compatible algebras with straightening laws on a disjoint union of chains

摘录 / EXCERPT: Let P be a finite poset that is a disjoint union of chains. Write ${\mathcal J}(P)$ for the poset of order ideals of P, and let $K[\mathcal O(P)]$, $K[\mathcal C(P^)]$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008257 | MATHDB-323779

Abstract generalized Lax conjecture for generalized Clifford algebras

摘录 / EXCERPT: Let h be a hyperbolic polynomial in direction e, let $\mathcal A_e(h)$ be its generalized Clifford algebra, and let $\Sigma^2\mathcal A_e(h)$ denote the cone of sums of hermiti...

conjecture | open | MathDB 页面 | 原文线索

V2-008258 | MATHDB-323789

Existence of Galois representations attached to torsion Hecke classes over imaginary quadratic fields

摘录 / EXCERPT: Existence conjecture. There exists a continuous Galois representation

conjecture | open | MathDB 页面 | 原文线索

V2-008259 | MATHDB-323846

Entropy criterion for birational maps from cluster recurrences

摘录 / EXCERPT: Let $b_{1,j+1}$ be the coefficients defining a birational map ... , and let $[x]_+=\max(x,0)$. Define ... Let $\lambda_{\max}$ be the root of largest magnitude among the roots of these...

conjecture | open | MathDB 页面 | 原文线索

V2-008260 | MATHDB-323848

Spectral-radius refinement of symbolic periodicity

摘录 / EXCERPT: Spectral-radius refinement conjecture. There are two possibilities: (1) if $|\lambda_{\max}|=\rho({\bf M}_\pm)>\rho({\bf M}_\mp)\geq1$, then $k=1$ and the symbol is repeated,...

conjecture | open | MathDB 页面 | 原文线索

V2-008261 | MATHDB-323862

Fontaine-Mazur-Langlands conjecture for geometric Galois representations

摘录 / EXCERPT: Let F be a number field and let ... be an irreducible geometric Galois representation, meaning that it is almost everywhere unramified and de Rham at every place above ℓ. Font...

conjecture | open | MathDB 页面 | 原文线索

V2-008262 | MATHDB-323863

Clozel's infinity-type descent conjecture

摘录 / EXCERPT: Clozel's infinity-type descent conjecture. Conjecturally, the assertion proved under that hypothesis should hold for all isobaric C- or L-algebraic automorphic representations,...

conjecture | open | MathDB 页面 | 原文线索

V2-008263 | MATHDB-323864

Hodge-Tate symmetry conjecture for geometric Galois representations

摘录 / EXCERPT: Let F be a number field and let ... be an irreducible geometric Galois representation. For an embedding $\tau\colon F\hookrightarrow\overline{\mathbb{Q}}_\ell$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-008264 | MATHDB-323918

Morrison-Peters principal graph conjecture for subfactors of index between 5 and $3+\sqrt{5}$

摘录 / EXCERPT: Morrison-Peters conjecture. Any subfactor with index in the range $(5,3+\sqrt{5})$ has principal graphs (A_∞,A_∞), the pair of graphs represented in the source by...

conjecture | open | MathDB 页面 | 原文线索

V2-008265 | MATHDB-323919

Completeness conjecture for Class I toral symmetric pairoid quasigradings

摘录 / EXCERPT: Let $(\mathfrak{g},[,])$ be a simple Lie algebra equipped with a toral symmetric pairoid quasigrading of Class I. The listed examples are those in Examples 140.95-140.140, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008266 | MATHDB-323920

Classification conjecture for Class II bi-Lie structures

摘录 / EXCERPT: Let $(\mathfrak{g},[,],[,]')$ be a bi-Lie structure of Class II. There should exist a Cartan subalgebra $\mathfrak{h}$ and a basis of the root system...

conjecture | open | MathDB 页面 | 原文线索

V2-008267 | MATHDB-323934

The Eisenbud-Green-Harris conjecture for complete intersections

摘录 / EXCERPT: Let $A=B=K[X_1,\dots,X_n]$, let $\mathfrak a=(f_1,\dots,f_r)$ be generated by a homogeneous regular sequence with degrees $d_1\leq\cdots\leq d_r$, and let $R=A/\mathfrak a$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-008268 | MATHDB-323945

The monomial almost-complete-intersection Rees algebra conjecture

摘录 / EXCERPT: Let I be a monomial ideal of $k[x_1,\ldots,x_n]$. Suppose that I is an almost complete intersection of finite colength, meaning that it has finite colength and one more minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-008269 | MATHDB-323953

Formal GAGA conjecture for good moduli spaces with affine diagonal

摘录 / EXCERPT: Formal GAGA conjecture. If $\mathscr{X}$ has affine diagonal, then the completion functor is an equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-008270 | MATHDB-323998

The finite maximal-path conjecture for coclass graphs of nilpotent semigroups

摘录 / EXCERPT: Let $r\in\mathbb N_0$ and let K be an arbitrary field. The graph $\mathcal G_{r,K}$ has rooted-tree components, and a maximal infinite path is an infinite path starting at the ro...

conjecture | open | MathDB 页面 | 原文线索

V2-008271 | MATHDB-324006

Inverse Power Method conjecture on high-accuracy evaluation of the first eigenvector coordinate

摘录 / EXCERPT: Inverse Power Method conjecture. After K iterations of the Inverse Power Method, the coordinate X_1 is evaluated with high relative accuracy; more specifically, its relative ac...

conjecture | open | MathDB 页面 | 原文线索

V2-008272 | MATHDB-324040

Numerical Chow-group vanishing conjecture for local complete intersections

摘录 / EXCERPT: Numerical Chow-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008273 | MATHDB-324041

Chow-theoretic theta-pairing positivity conjecture

摘录 / EXCERPT: Chow-theoretic theta conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008274 | MATHDB-324042

Theta-pairing positivity and Chern-character equality conjecture

摘录 / EXCERPT: Theta-positivity conjecture. For every finitely generated A-module M,

conjecture | open | MathDB 页面 | 原文线索

V2-008275 | MATHDB-324068

Cyclotomic expansion conjecture for orbit pure braid groups

摘录 / EXCERPT: Cyclotomic expansion conjecture. There exists a filtration-preserving algebra morphism

conjecture | open | MathDB 页面 | 原文线索

V2-008276 | MATHDB-324085

The kernel characterization conjecture for the methorious-operator extension

摘录 / EXCERPT: Let Φ be an umbral character set for an integro-differential algebra $\mathcal{F}$ with left extensible coefficient algebra $\mathcal{E}$. Let $K\mathcal{E}[\partial]_{\Phi}$...

conjecture | open | MathDB 页面 | 原文线索

V2-008277 | MATHDB-324115

Local finiteness of Grothendieck monoids of profinitary exact categories

摘录 / EXCERPT: Let $\mathscr A$ be a profinitary exact category, and let $\Gamma_{\mathscr A}$ be its Grothendieck monoid. A monoid is locally finite if, for every $\gamma\in\Gamma_{\mathscr A}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-008278 | MATHDB-324136

Sign-coherence conjecture for principal coefficients

摘录 / EXCERPT: Let B be an exchange matrix, and consider an extended exchange matrix in a Y-pattern with initial exchange matrix B and principal coefficients. The rows indexed $n+1$ through...

conjecture | open | MathDB 页面 | 原文线索

V2-008279 | MATHDB-324137

Positivity of the nonnegative orthant as a mutation-fan cone

摘录 / EXCERPT: Nonnegative-orthant conjecture. The nonnegative orthant $(\mathbb R_{ge0})^n$ is a B-cone. This assertion is equivalent, for a given B, to the sign-coherence conjecture for pri...

conjecture | open | MathDB 页面 | 原文线索

V2-008280 | MATHDB-324138

Mutation compatibility conjecture for g-vectors

摘录 / EXCERPT: Let $\mathbb{T}_n$ be the n-regular tree indexing seeds, let t_0 and t_1 be adjacent vertices in direction k, and let B_0 and B_1 be exchange matrices with...

conjecture | open | MathDB 页面 | 原文线索

V2-008281 | MATHDB-324141

Linear independence conjecture for g-vectors

摘录 / EXCERPT: Let B be an exchange matrix and let $t_1,\ldots,t_m$ be vertices of the n-regular tree $\mathbb T_n$. Let $i_1,\ldots,i_m\in[n]$ and let $c_1,\ldots,c_m\in\mathbb Z$. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-008282 | MATHDB-324142

Affine-type positive basis conjecture for universal geometric cluster algebras

摘录 / EXCERPT: Let B be an exchange matrix of affine type, meaning that it is mutation equivalent to an acyclic exchange matrix whose Cartan companion is of affine type. Let $\boldsymbol{g}$-ve...

conjecture | open | MathDB 页面 | 原文线索

V2-008283 | MATHDB-324145

The mutation-equivariance conjecture for g-vectors

摘录 / EXCERPT: Let $\mathbb{T}_n$ be the n-regular tree indexing seeds, let $t_0\stackrel{k}{\text{---\!---}}t_1$ be an edge, and let B_0 and B_1 be exchange matrices with $B_1=\mu_k(B_0)$....

conjecture | open | MathDB 页面 | 原文线索

V2-008284 | MATHDB-324220

The K-theoretic Kaplansky and unit conjectures

摘录 / EXCERPT: K-theoretic Kaplansky and unit conjectures. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008285 | MATHDB-324225

The primary decomposition conjecture for ideals in acyclic cluster algebras

摘录 / EXCERPT: Primary decomposition conjecture. For all $\mathbf{a}\in\mathbb{N}^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-008286 | MATHDB-324231

Mirror-direction invariance for tensor-diagram cluster structures

摘录 / EXCERPT: Mirror-direction conjecture. Reversing the direction-specifically, replacing the clockwise direction in the definition of special invariants by the counterclockwise direction-does...

conjecture | open | MathDB 页面 | 原文线索

V2-008287 | MATHDB-324232

Web-basis conjecture for cluster monomials

摘录 / EXCERPT: Cluster-monomial web-basis conjecture. All cluster monomials are web invariants.

conjecture | open | MathDB 页面 | 原文线索

V2-008288 | MATHDB-324233

Three-term exchange-relation criterion for web invariants

摘录 / EXCERPT: Three-term cluster conjecture. Then z and z' are cluster variables, and the displayed relation is an exchange relation.

conjecture | open | MathDB 页面 | 原文线索

V2-008289 | MATHDB-324234

Three-term non-cluster criterion for web invariants

摘录 / EXCERPT: Three-term non-cluster conjecture. Then z is not a cluster variable.

conjecture | open | MathDB 页面 | 原文线索

V2-008290 | MATHDB-324235

Forest-diagram characterization of cluster monomials

摘录 / EXCERPT: Forest-presentation conjecture. The web invariant z is a cluster monomial if and only if $z=[D]$ for some forest diagram D.

conjecture | open | MathDB 页面 | 原文线索

V2-008291 | MATHDB-324278

The component-module conjecture for Drinfeld modules

摘录 / EXCERPT: Component-module conjecture. There exists an integer $N\geq 0$ and an ideal $\mathfrak{a}\subseteq A$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008292 | MATHDB-324279

Burnham-Rosen-Sidman-Vermeire conjecture on graph curves and Betti tables

摘录 / EXCERPT: Let G be a graph with d vertices, genus g, and girth $b3$. Let $\overline{G}$ be the corresponding graph curve, with Betti numbers $b_{i,j}(\overline{G})$. Burnham-Rosen-Sidm...

conjecture | open | MathDB 页面 | 原文线索

V2-008293 | MATHDB-324280

Bridge-count conjecture for graph curves

摘录 / EXCERPT: Let G be a graph and let $\overline{G}\subset \mathbb{P}^{n}$ be its graph curve. Write $b_{i,j}(\overline{G})$ for the graded Betti numbers, and call an edge of G a bridge if...

conjecture | open | MathDB 页面 | 原文线索

V2-008294 | MATHDB-324354

The every-cluster supertoricity conjecture

摘录 / EXCERPT: A cluster is a cluster in the cluster algebra under consideration. Every-cluster supertoricity conjecture. Every cluster is supertoric. The claim is a broad structural assertion in...

conjecture | open | MathDB 页面 | 原文线索

V2-008295 | MATHDB-324359

Tree-extension conjecture for Schubert-cell decompositions of quiver Grassmannians

摘录 / EXCERPT: Let T be a tree extension of S, and let M be a T-module such that M_α is an isomorphism for all arrows α in $T-S$. Let M_S be the restriction of M to S,...

conjecture | open | MathDB 页面 | 原文线索

V2-008296 | MATHDB-324367

Fomin-Zelevinsky sign-coherence conjecture for c- and d-vectors

摘录 / EXCERPT: Sign-coherence conjecture. (i) Any c-vector of $\mathcal{A}_\bullet(B)$ is a nonzero vector, and its components are either all non-negative or all non-positive. (ii) Any non-init...

conjecture | open | MathDB 页面 | 原文线索

V2-008297 | MATHDB-324368

Root conjecture for c-vectors of cluster algebras

摘录 / EXCERPT: Root conjecture. Every c-vector of the cluster algebra $\mathcal{A}_\bullet(B)$ is a root of $\Delta(A(B))$.

conjecture | open | MathDB 页面 | 原文线索

V2-008298 | MATHDB-324375

The algebraic-extension characterization via Stallings graph coverings

摘录 / EXCERPT: Let $\mathbf{F}$ be a free group, and let $H\leq J\leq\mathbf{F}$. For a basis X of $\mathbf{F}$, write $H\leq_{{\scriptscriptstyle \overset{\twoheadrightarrow}{X}}}J$ when the c...

conjecture | open | MathDB 页面 | 原文线索

V2-008299 | MATHDB-324376

The free-extension characterization of algebraic extensions

摘录 / EXCERPT: Let $H\leq J$ be subgroups of a free group $\mathbf{F}$. Let $\mathbf{F}'$ be a free extension of $\mathbf{F}$, meaning that $\mathbf{F}\leq\mathbf{F}'$ and $\mathbf{F}'$ is free,...

conjecture | open | MathDB 页面 | 原文线索

V2-008300 | MATHDB-324418

The scalewise birational extension conjecture for completed local domains

摘录 / EXCERPT: Scalewise birational extension conjecture. There exists an ideal $H\subset\hat R$ with $H\cap R=(0)$ and a valuation $\hat\nu_-$ centered at $\hat R/H$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008301 | MATHDB-324479

The Second Brauer-Thrall conjecture for maximal Cohen-Macaulay modules

摘录 / EXCERPT: Second Brauer-Thrall conjecture for MCM modules. If R has unbounded CM type and k is infinite, then R has strongly unbounded CM type.

conjecture | open | MathDB 页面 | 原文线索

V2-008302 | MATHDB-324504

Overconvergence conjecture for F-analytic representations

摘录 / EXCERPT: Let K be the finite extension of the maximal unramified extension of a finite extension of $\mathbb{Q}_p$ considered in the paper, let F be its coefficient field, and let V b...

conjecture | open | MathDB 页面 | 原文线索

V2-008303 | MATHDB-324527

Existence of an exotic crossed-product duality for every coaction

摘录 / EXCERPT: Let (A,δ) be a coaction of a locally compact group G. Say that (A,δ) satisfies E-crossed-product duality when ... where K_δ is the kernel ideal of the Kataya...

conjecture | open | MathDB 页面 | 原文线索

V2-008304 | MATHDB-324528

Dual coaction duality for exotic crossed products

摘录 / EXCERPT: Let (B,α) be an action of a locally compact group G on a C^*-algebra B, and let $B\rtimes_{\alpha,E}G$ be its E-crossed product. Denote by $\hat\alpha_E$ the dual co...

conjecture | open | MathDB 页面 | 原文线索

V2-008305 | MATHDB-324545

Main conjecture for Kähler-Einstein cubic surfaces

摘录 / EXCERPT: Let $\overline{\mathcal{M}_3}^{ALG}:=\mathbb{P}^{19}//SL(4,\mathbb{C})$ be the algebraic compactification of the moduli space of cubic surfaces, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008306 | MATHDB-324546

Algebraic compactification conjecture for KE degree-two Del Pezzo surfaces

摘录 / EXCERPT: Let $\overline{\mathcal{M}}^{GH}_2$ denote the Gromov-Hausdorff compactification of degree-two Kähler-Einstein Del Pezzo surfaces. Degree-two compactification conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-008307 | MATHDB-324619

Orthogonal-conjugacy conjecture for infinitesimal Hecke algebra representations

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let B be its Artin group with generators $\mathfrak{s}$ associated to $s\in S$, let $\mathbbm{k}$ be a field of characteristic 0, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-008308 | MATHDB-324624

The lattice W-algebra centralizer conjecture

摘录 / EXCERPT: Let $\mathcal{W}'_{\mathfrak{q},N}$ be the associative algebra in the XXZ representation generated by $W\!B^{\alpha}_j$, with $1\leq j\leq N-2p+2$ and $\alpha\in\{0,\pm\}$, togethe...

conjecture | open | MathDB 页面 | 原文线索

V2-008309 | MATHDB-324628

Kashina's exponent divisibility conjecture for semisimple Hopf algebras

摘录 / EXCERPT: Let H be a semisimple Hopf algebra over $\mathbb{C}$. Kashina's conjecture. The exponent of H divides $\operatorname{dim}(H)$. This conjecture is a divisibility analogue of the...

conjecture | open | MathDB 页面 | 原文线索

V2-008310 | MATHDB-324636

Pardue's characteristic-independence conjecture for p-Borel-fixed ideals

摘录 / EXCERPT: Pardue's conjecture. The $\mathbb{N}$-graded Betti table of $I(S\otimes_{\mathbb Z}\ell)$ is independent of $\operatorname{char}\ell$ (equivalently, of ℓ) for all fields...

conjecture | open | MathDB 页面 | 原文线索

V2-008311 | MATHDB-324644

Split-factor reformulation of the Eisenbud-Green-Harris conjecture

摘录 / EXCERPT: Let $S=k[x_1,\dots,x_n]$ be a polynomial ring over a field k. Let I be a homogeneous ideal containing a regular sequence $f_1,\dots,f_n$ with degrees ... A degree-a_i form is s...

conjecture | open | MathDB 页面 | 原文线索

V2-008312 | MATHDB-324693

R=T for non-Eisenstein new Hecke algebras

摘录 / EXCERPT: Let $\mathfrak{m}$ be a maximal ideal of $\mathbf{T}^{\mathrm{new}}_{\Sigma}$, let $k=\mathbf{T}_{\Sigma}/\mathfrak{m}$ be a field of characteristic p, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-008313 | MATHDB-324713

Conjecture on Borel subalgebra conjugacy classes in the Jacobson-Witt algebra

摘录 / EXCERPT: Let $\mathfrak{g}=W_n$ be the Jacobson-Witt algebra of rank n over the field $\mathbb{F}$, where W_n is the derivation algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-008314 | MATHDB-324725

Koszul-duality conjecture for new identities of corresponding operations

摘录 / EXCERPT: Let Dias and $\mathsf{Dend}=\mathsf{Dias}^!$ be the dual operads of associative dialgebras and dendriform algebras, respectively. For $n\leq 3$, multilinear identities o...

conjecture | open | MathDB 页面 | 原文线索

V2-008315 | MATHDB-324747

Strong positivity conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra, and let a cluster monomial mean a monomial in cluster variables all belonging to the same cluster. Strong positivity conjecture. Any cluster...

conjecture | open | MathDB 页面 | 原文线索

V2-008316 | MATHDB-324767

Direct finiteness conjecture for group algebras of countable groups

摘录 / EXCERPT: Let F be a field and G a countable group. The group algebra $F(G)$ is directly finite, meaning that for any $x,y\in F(G)$, $xy=1$ implies $yx=1$. Direct finiteness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-008317 | MATHDB-324768

Tateo's dilogarithm identities for RSG/SG Y-systems

摘录 / EXCERPT: Tateo's dilogarithm identities.

conjecture | open | MathDB 页面 | 原文线索

V2-008318 | MATHDB-324782

Finite-generation equivalence for $\a\d$-bundles over $\a\d$-adic period rings

摘录 / EXCERPT: Finite-generation conjecture. The following tensor categories are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-008319 | MATHDB-324813

Lie triple isomorphism decomposition conjecture for generalized matrix algebras

摘录 / EXCERPT: Let $\mathcal{G}=\left[\smallmatrix A & M\\ N & B \endsmallmatrix \right]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008320 | MATHDB-324818

The dual F-signature conjecture for F-injectivity

摘录 / EXCERPT: Let $(R, \mathfrak{m}, k)$ be a reduced Cohen-Macaulay F-finite local ring of characteristic $p>0$, and let ω_R be a canonical module. The dual F-signature conjecture. R...

conjecture | open | MathDB 页面 | 原文线索

V2-008321 | MATHDB-324830

Strong Broadhurst-Kreimer and Zagier conjecture for the linearized double shuffle algebra

摘录 / EXCERPT: Let $\mathfrak{ls}$ be the Lie algebra of solutions to the linearized double shuffle relations, let S denote the space of even period polynomials, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008322 | MATHDB-324842

The base conjecture for graded retracts of monomial quotients

摘录 / EXCERPT: Base conjecture. Every graded algebra retract of S/I has a base.

conjecture | open | MathDB 页面 | 原文线索

V2-008323 | MATHDB-324852

Positive-characteristic extension of the semisimple Hopf action theorems

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let K be a semisimple Hopf algebra that is not necessarily cosemisimple. Consider the conclusions of Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-008324 | MATHDB-324859

Keel's strict nefness conjecture for surfaces over finite fields

摘录 / EXCERPT: Let X be a smooth projective surface over $\bar{\mathbb F}_p$, and let L be a strictly nef line bundle on X, meaning that $L\cdot C>0$ for every irreducible curve...

conjecture | open | MathDB 页面 | 原文线索

V2-008325 | MATHDB-324860

Nef effectiveness conjecture for surfaces over finite fields

摘录 / EXCERPT: Let X be a smooth projective surface over $\bar{\mathbb F}_p$, and let L be a nef line bundle on X. The Iitaka dimension $\kappa(L)$ is the maximum dimension of the images of...

conjecture | open | MathDB 页面 | 原文线索

V2-008326 | MATHDB-324861

Pseudoeffective divisor effectiveness conjecture for surfaces over finite fields

摘录 / EXCERPT: Let X be a smooth projective surface over $\bar{\mathbb F}_p$, and let D be a pseudoeffective $\mathbb Q$-divisor, meaning that its numerical class lies in the closure of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-008327 | MATHDB-324921

Characteristic-two dimension conjecture for 2-spec matrix spaces

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic 2 with more than 2 elements, and let $\mathcal{V}$ be a 2-spec subspace of $\operatorname{M}_n(\mathbb{K})$. Characteristic-two...

conjecture | open | MathDB 页面 | 原文线索

V2-008328 | MATHDB-324922

Characteristic-two dimension conjecture for 1-star-spec matrix spaces

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic 2 with more than 2 elements, and let $\mathcal{V}$ be a $1^\star$-spec subspace of $\operatorname{M}_n(\mathbb{K})$. Characteristi...

conjecture | open | MathDB 页面 | 原文线索

V2-008329 | MATHDB-324923

Characteristic-two classification conjecture for maximal 1-star-spec spaces

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic 2 with more than 2 elements. Let $\mathcal{V}$ be a $\overline{1}^\star$-spec subspace of $\operatorname{M}_n(\mathbb{K})$ with .....

conjecture | open | MathDB 页面 | 原文线索

V2-008330 | MATHDB-324924

Characteristic-two classification conjecture for maximal 2-spec spaces

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic 2 with more than 2 elements. Let $\mathcal{V}$ be a $\overline{2}$-spec subspace of $\operatorname{M}_n(\mathbb{K})$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-008331 | MATHDB-324925

Handelman's conjecture that every *-regular ring is unit-regular

摘录 / EXCERPT: A ring is *-regular when it is von Neumann regular and its involution is proper; it is unit-regular when for every element a there is a unit u such that $a=aua$. Handelman's...

conjecture | open | MathDB 页面 | 原文线索

V2-008332 | MATHDB-324946

Chen-Kane-Wales regular-sequence conjecture for complete symmetric functions

摘录 / EXCERPT: Let $A=\{a,b,c\}$ with $a<b<c$, and let h_d denote the complete homogeneous symmetric polynomial of degree d in three variables. Chen-Kane-Wales conjecture. The polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-008333 | MATHDB-324958

MPSW's eta-pairing vanishing conjecture for complete intersections

摘录 / EXCERPT: MPSW's eta-pairing vanishing conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-008334 | MATHDB-324967

Dao's theta-pairing vanishing conjecture for even-dimensional isolated singularities

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a local ring whose completion $\widehat R$ is a hypersurface in an unramified regular local ring. Assume that R is an even-dimensional isolated singular...

conjecture | open | MathDB 页面 | 原文线索

V2-008335 | MATHDB-324968

Rigidity of modules over one-dimensional complete intersection domains

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a one-dimensional complete intersection domain. A finitely generated R-module is rigid if its higher Tor with every relevant partner vanishes as in the...

conjecture | open | MathDB 页面 | 原文线索

V2-008336 | MATHDB-325075

The fixed-determinant coherent-systems dimension conjecture

摘录 / EXCERPT: Let C be a smooth projective curve of genus g, let r,d,k be integers, let $\mathcal{L}$ be a line bundle on C of degree d, and let $G(\alpha;r,\mathcal{L},k)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-008337 | MATHDB-325077

Clozel-Buzzard-Gee conjecture on C-algebraicity and fields of rationality

摘录 / EXCERPT: Let G be a connected reductive group over $\mathbb{Q}$ and let π be an automorphic representation of $G(\mathbb{A})$. The field of rationality $\mathbb{Q}(\pi)$ is the field...

conjecture | open | MathDB 页面 | 原文线索

V2-008338 | MATHDB-325081

The invariant-theoretic characterization conjecture for generic three-dimensional Sklyanin algebras

摘录 / EXCERPT: Let $A=A(a,b,c)$ be a generic algebra, and let G be a finite subgroup of $\operatorname{Aut}(A)$. Here cci, gci, and nci denote the corresponding notions of complete intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-008339 | MATHDB-325103

The resolution conjecture for Artinian trivariate monomial ideals

摘录 / EXCERPT: Let $S=\Bbbk[x,y,z]$, let $\mathfrak{m}=\brc{x,y,z}$, let $I\subseteq\mathfrak{m}^2$ be an $\mathfrak{m}$-primary Artinian monomial ideal minimally generated by n monomials, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008340 | MATHDB-325104

The complete classification conjecture for Artinian trivariate monomial ideals

摘录 / EXCERPT: Let $S=\Bbbk[x,y,z]$, let $\mathfrak{m}=\brc{x,y,z}$, let $I\subseteq\mathfrak{m}^2$ be an $\mathfrak{m}$-primary Artinian monomial ideal minimally generated by n monomials, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008341 | MATHDB-325105

Semisimplicity conjecture for finite-dimensional universal enveloping algebras of anti-Jordan triple systems

摘录 / EXCERPT: Let S be a simple finite-dimensional anti-Jordan triple system, and let $U(S)$ denote its universal enveloping algebra. Semisimplicity conjecture. If $U(S)$ is finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-008342 | MATHDB-325126

Finite-field equality conjecture for nilpotency degrees

摘录 / EXCERPT: Let ${\mathbb F}$ be a finite field and ${\mathbb L}$ an infinite field such that ... Let $C_{n,d}^{\mathbb F}$ and $C_{n,d}^{\mathbb L}$ denote the nilpotency degrees of the relativ...

conjecture | open | MathDB 页面 | 原文线索

V2-008343 | MATHDB-325133

The generator conjecture for regular local rings

摘录 / EXCERPT: Let A be a commutative noetherian ring with unit and let $0\leq p\leq\dim A$. Denote by $\operatorname{\mathcal{M}}^p_A$ the category of finitely generated A-modules whose supp...

conjecture | open | MathDB 页面 | 原文线索

V2-008344 | MATHDB-325137

Existence and properties of the Bloch-Kato submodule

摘录 / EXCERPT: Bloch-Kato submodule conjecture. There is an R-submodule, called the Bloch-Kato submodule,

conjecture | open | MathDB 页面 | 原文线索

V2-008345 | MATHDB-325204

Local dependence of the mod p Jacquet-Langlands correspondence

摘录 / EXCERPT: Let $\mathfrak{m}$ be a maximal ideal of the Hecke algebra, with associated Galois representation $\bar{\rho}$. Let $\mathbf{F}[\mathfrak{m}]$ denote the corresponding Hecke-isotyp...

conjecture | open | MathDB 页面 | 原文线索

V2-008346 | MATHDB-325207

Laumon's purity conjecture for compactified Jacobians

摘录 / EXCERPT: Let C be a projective geometrically integral algebraic curve over k whose singularities are all planar, and let $\overline{\operatorname{Jac}}_{C}$ be its compactified Jacobian...

conjecture | open | MathDB 页面 | 原文线索

V2-008347 | MATHDB-325228

The Tor inequality for general intersections of homogeneous ideals

摘录 / EXCERPT: Let $S=K[x_1,dots,x_n]$ be a standard graded polynomial ring, and let $I,J\subset S$ be homogeneous ideals. Write I^{lex} for the lex ideal with the same Hilbert functio...

conjecture | open | MathDB 页面 | 原文线索

V2-008348 | MATHDB-325259

The finite-dimensionality conjecture for noetherian universal enveloping algebras

摘录 / EXCERPT: Let ${\Bbbk}$ be a field of characteristic 0, let L be a Lie algebra over ${\Bbbk}$, and let $U(L)$ denote its universal enveloping algebra.
Finite-dimensionality conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-008349 | MATHDB-325263

Hopf 2-algebra conjecture for the categorified Hall algebra

摘录 / EXCERPT: Let Q be a simply laced quiver, let $\mathcal{A}={\rm Rep}(Q)$ be the category of finite-dimensional representations over a finite field $\mathbb{F}_q$, and let $\mathcal{A}_0$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-008350 | MATHDB-325276

Cohen-Macaulay and levelness conjecture for $X_i^{(m)}$

摘录 / EXCERPT: Cohen-Macaulay and levelness conjecture. When $\operatorname{char}(K)>i$ and $n>2i$, $X_i^{(m)}$ is Cohen-Macaulay and its coordinate ring is a level algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-008351 | MATHDB-325290

Fernández de Bobadilla-Luengo-Melle-Hernández-Némethi conjecture on Alexander polynomials of rational cuspidal curves

摘录 / EXCERPT: Fernández de Bobadilla-Luengo-Melle-Hernández-Némethi conjecture. For every $j=0,dots,d-3$,

conjecture | open | MathDB 页面 | 原文线索

V2-008352 | MATHDB-325298

Fiebig's conjecture on moment-graph sheaves and the Kazhdan-Lusztig basis

摘录 / EXCERPT: Let $\widehat{W}^{\circ}$ be the finite subset of an affine Weyl group used in the moment-graph construction, let $\mathcal{B}(w)$ be the indecomposable object associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-008353 | MATHDB-325299

Hall-Littlewood and Rogers-Szegő polynomial identity

摘录 / EXCERPT: Let $M=(M_1,dots,M_m)\in\mathbb Z_{+}^m$ and set $m_0(\lambda):=\infty$. For a partition λ, write λ' for its conjugate, $m_j(\lambda)$ for the multiplicity of the p...

conjecture | open | MathDB 页面 | 原文线索

V2-008354 | MATHDB-325301

Herzog-Srinivasan periodicity conjecture for monomial-curve Betti numbers

摘录 / EXCERPT: Let K be a field, let ${\mathbf a}=(a_1<\cdots<a_n)$ be a sequence of positive integers, and let $I({\mathbf a}+j)$ be the defining ideal of the affine monomial curve parametrize...

conjecture | open | MathDB 页面 | 原文线索

V2-008355 | MATHDB-325305

The Artin-Schreier-Witt counting conjecture for noncyclic abelian p-groups

摘录 / EXCERPT: Let F be a global function field of characteristic p, and let G be a finite noncyclic abelian p-group. Write $Z(F,G;X)$ for the number of Artin-Schreier-Witt extensions...

conjecture | open | MathDB 页面 | 原文线索

V2-008356 | MATHDB-325318

Weak closedness of indecomposable traces for AF-algebras of distributive lattices

摘录 / EXCERPT: Let Γ be a countable distributive lattice, realized as the lattice of finite ideals of a partially ordered set, and let $A(\Gamma)$ be the AF-algebra associated with the H...

conjecture | open | MathDB 页面 | 原文线索

V2-008357 | MATHDB-325321

The generic nonsymmetric matrix equation conjecture for cubic systems

摘录 / EXCERPT: Let K be the base field, let $n=3$, and let B,C,D be generic 3×3 matrices. Consider Eq., the nonsymmetric generic matrix equation in the unknown matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-008358 | MATHDB-325322

The commutation conjecture for generic commuting matrix equations

摘录 / EXCERPT: Let K be a field, let g be a nonzero polynomial in the noncommuting indeterminates $u_1,\ldots,u_{k+1}$ such that ... for every $i\leq k$, the indeterminate u_i appears exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-008359 | MATHDB-325324

Gonericity conjecture for canonical curves

摘录 / EXCERPT: Gonericity conjecture. In characteristic zero, C is $\ell+2$-generic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008360 | MATHDB-325325

Geometric characterization conjecture for goneric curves

摘录 / EXCERPT: Geometric characterization conjecture. The curve C is d-gonepic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008361 | MATHDB-325347

The Factorial Conjecture in dimension m

摘录 / EXCERPT: Factorial Conjecture. For all $f\in{\mathbb C}^{[m]}$,

conjecture | open | MathDB 页面 | 原文线索

V2-008362 | MATHDB-325348

The Strong Factorial Conjecture

摘录 / EXCERPT: Strong Factorial Conjecture. $F_\cap^{[m]}={\mathbb C}^{[m]}$; equivalently, every polynomial belongs to the strong factorial set. This conjecture is explicitly stated as stronger...

conjecture | open | MathDB 页面 | 原文线索

V2-008363 | MATHDB-325376

The geometric-solution conjecture for the complex volume of hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot, and let x be an algebraic solution of the periodicity equations associated with the braid-induced triangulation of $S^3\setminus K$. For the jth o...

conjecture | open | MathDB 页面 | 原文线索

V2-008364 | MATHDB-325392

The CY-reduction conjecture for three-dimensional rational Gorenstein singularities

摘录 / EXCERPT: Let R be a 3-dimensional Gorenstein normal normal domain over $\mathbb{C}$ with rational singularities, so $\mathop{\underline{\rm CM}}\nolimits R$ is a 2-CY triangulated cat...

conjecture | open | MathDB 页面 | 原文线索

V2-008365 | MATHDB-325419

Finite-generation converse for Frobenius operators on Q-Gorenstein rings

摘录 / EXCERPT: Finite-generation converse conjecture. If the order of the canonical module is a multiple of the characteristic of R, then $\mathcal F(E)$ is not a finitely generated ring extens...

conjecture | open | MathDB 页面 | 原文线索

V2-008366 | MATHDB-325481

Pommerening's invariant-minor conjecture for unipotent subgroups of special linear groups

摘录 / EXCERPT: Let k be a field, let U be a unipotent subgroup of $\operatorname{SL}_n(k)$ normalised by a maximal torus, and let the invariant minors be the minors described in the source th...

conjecture | open | MathDB 页面 | 原文线索

V2-008367 | MATHDB-325497

Finite-index conjecture for quadratic polynomial arboreal representations

摘录 / EXCERPT: Let $f\in\mathbb{Q}[x]$ be a quadratic polynomial. Its iterates are irreducible when every polynomial f^n is irreducible, and f is post-critically infinite when the orbit of th...

conjecture | open | MathDB 页面 | 原文线索

V2-008368 | MATHDB-325498

Quotient-freeness conjecture for relational wiring-diagram algebras

摘录 / EXCERPT: Let $\mathcal{S}$ be the singly typed wiring diagrams operad. For a set $A\in\operatorname{Ob}({\bf Set})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008369 | MATHDB-325524

The odd axial-puncture criterion for regularity of the bi-adjacency matrix

摘录 / EXCERPT: The odd axial-puncture criterion.

conjecture | open | MathDB 页面 | 原文线索

V2-008370 | MATHDB-325525

The characteristic bound conjecture for type-two monomial algebras

摘录 / EXCERPT: Characteristic bound conjecture. Then R/I has the weak Lefschetz property in every characteristic $p>\frac{1}{2}(a+b+c)$.

conjecture | open | MathDB 页面 | 原文线索

V2-008371 | MATHDB-325532

Gabber's torsion-freeness conjecture for punctured spectra

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a local complete intersection ring of dimension 3. Let ... be the punctured spectrum of R. Gabber's conjecture. The group $\operatorname{Pic}(U_R)$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-008372 | MATHDB-325553

Galois-orbit conjecture for rational intrinsic data

摘录 / EXCERPT: Fix a set of fusion rules, and consider all modular categories having those fusion rules. For a modular category $\mathcal C$, let $\operatorname{ID}(\mathcal C)$ denote its fusion...

conjecture | open | MathDB 页面 | 原文线索

V2-008373 | MATHDB-325679

Belkale-Kumar-type characterization of the saturated tensor semigroup

摘录 / EXCERPT: Belkale-Kumar-type conjecture. Let $(\lambda_1,\dots,\lambda_s,\mu)\in P_+^{s+1}$, assume that none of the λ_j is W-invariant, and assume $\mu-\sum_{j=1}^s\lambda_j\in Q$...

conjecture | open | MathDB 页面 | 原文线索

V2-008374 | MATHDB-325788

The equality conjecture for hypergeometric logarithmic quotients

摘录 / EXCERPT: Let $n\ne2$ and let $a_i,b_i\in\mathbb{Q}$ satisfy $0<a_i,b_i<1$ for $i=1,2,\ldots,n$. Let $F(a\mid z)$ be the generalized hypergeometric solution and let $G(a\mid z)$ be its first...

conjecture | refuted | MathDB 页面 | 原文线索

V2-008375 | MATHDB-325791

Characteristic-polynomial reversal conjecture for pseudo-inverses of supertropical matrices

摘录 / EXCERPT: Characteristic-polynomial reversal conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008376 | MATHDB-325796

Nilpotence conjecture for the Jacobson radical of small profinite rings

摘录 / EXCERPT: Nilpotence conjecture. The Jacobson radical $J(R)$ is nilpotent. In particular, R has an open nilpotent ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-008377 | MATHDB-325797

Open null ideal conjecture for small profinite rings

摘录 / EXCERPT: Null ideal conjecture. A small profinite ring has an open null ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-008378 | MATHDB-325850

Brumer's conjecture for abelian CM-extensions

摘录 / EXCERPT: Let K/k be a finite abelian Galois CM-extension of number fields with Galois group G, and let S be a finite set of places of k containing all infinite places and all finite...

conjecture | open | MathDB 页面 | 原文线索

V2-008379 | MATHDB-325863

Smoothness conjecture for iterated orbit-closure blow-ups

摘录 / EXCERPT: Let (G,V) be a strongly prehomogeneous vector space, meaning an algebraic group G acts linearly on a finite-dimensional vector space V with finitely many or...

conjecture | open | MathDB 页面 | 原文线索

V2-008380 | MATHDB-325887

The determinant-lattice conjecture for elliptic units

摘录 / EXCERPT: Let Λ be the Iwasawa algebra in the source, let $Q(\Lambda)$ be its total ring of fractions, and let Δ^∞ be the determinant module constructed from the inverse...

conjecture | open | MathDB 页面 | 原文线索

V2-008381 | MATHDB-325939

The symmetric denominator implication for compatible cluster variables

摘录 / EXCERPT: Symmetric denominator implication. If, for every cluster $\mathbf{x}$ containing x, ${\mathcal L}(x',\mathbf{x})$ has no x in its denominator, then, for every cluster...

conjecture | open | MathDB 页面 | 原文线索

V2-008382 | MATHDB-325940

The weight-system coefficient conjecture for ${\mathfrak{sl}}_2$

摘录 / EXCERPT: Weight-system coefficient conjecture. For any chord diagram d with $2k$ chords,

conjecture | open | MathDB 页面 | 原文线索

V2-008383 | MATHDB-325949

Conjecture on automorphic points in residual deformation spaces

摘录 / EXCERPT: Let $r:G_{F,S}\rightarrow \mathrm{GL}_d(\overline{\mathbb{F}}_p)$ be an automorphic residual representation, and let $Y_d^\bot(r):=Y_d(r)\cap Y_d^\bot$. An automorphic point is a p...

conjecture | open | MathDB 页面 | 原文线索

V2-008384 | MATHDB-325954

The common horizontal tangent multiplicity conjecture for resultants

摘录 / EXCERPT: Let f_1,f_2 define two affine plane curves, and let g generate their elimination ideal in the y-coordinate, while R denotes their resultant with respect to x. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-008385 | MATHDB-325975

Postnikov-Shapiro's graded Betti-number conjecture for power ideals

摘录 / EXCERPT: Let $R=k[x_1,\ldots,x_n]$ be a polynomial ring over a field k of characteristic 0 or sufficiently large. For a linear degree function ϕ satisfying ... let I_ϕ be the mo...

conjecture | open | MathDB 页面 | 原文线索

V2-008386 | MATHDB-325980

Conjecture on the tricategory of commutative algebras and cospans

摘录 / EXCERPT: Let $\mathcal{Z}$ be a braided monoidal category satisfying the assumptions required for the bicategory construction. Take objects to be commutative algebras in $\mathcal{Z}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008387 | MATHDB-325987

The hypersurface conjecture for Gorenstein rings of countable Cohen-Macaulay type

摘录 / EXCERPT: Let R be a standard graded Gorenstein ring of countable Cohen-Macaulay type, meaning that it has only countably many indecomposable graded Cohen-Macaulay modules up to shifts. Hy...

conjecture | open | MathDB 页面 | 原文线索

V2-008388 | MATHDB-325998

The compositum conjecture for the fourth p-Zassenhaus layer

摘录 / EXCERPT: Fourth-layer compositum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008389 | MATHDB-325999

The kernel n-unipotent conjecture for maximal pro-p Galois groups

摘录 / EXCERPT: Kernel n-unipotent conjecture. For every integer $n\geq 3$, the group $G_F(p)$ has the kernel n-unipotent property.

conjecture | open | MathDB 页面 | 原文线索

V2-008390 | MATHDB-326019

The characteristic-sequence conjecture for affine crystallographic arrangements

摘录 / EXCERPT: Characteristic-sequence conjecture. The characteristic sequence of $\hat R$ is one of

conjecture | open | MathDB 页面 | 原文线索

V2-008391 | MATHDB-326026

Infinite-dimensionality conjecture for generalized Fomin-Kirillov algebras

摘录 / EXCERPT: Let n be a positive integer, let $\alpha_{ijk},\beta_{ijk},\gamma_{ij},\lambda_{ijkl}\in\{1,-1\}$, and let $A(\alpha,\beta,\gamma,\lambda)$ be the algebra generated by x_{ij} f...

conjecture | open | MathDB 页面 | 原文线索

V2-008392 | MATHDB-326039

Drinfeld's smooth categorical compactification conjecture for coherent D-modules

摘录 / EXCERPT: Drinfeld's conjecture. The DG enhancement of $D^b_{\operatorname{coh}}(\mathcal D_X)$ has a smooth categorical compactification.

conjecture | open | MathDB 页面 | 原文线索

V2-008393 | MATHDB-326065

Kac-Wakimoto principal-admissible character transformation conjecture

摘录 / EXCERPT: Let $\widehat{\mathfrak{g}}$ be an affine Lie superalgebra, let $\varphi:\widehat{\mathfrak{g}}\to\widehat{\mathfrak{g}}$ be a compatible homomorphism, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008394 | MATHDB-326071

Web criterion for compatibility of cluster and coefficient variables

摘录 / EXCERPT: Web compatibility conjecture. Two cluster or coefficient variables are compatible if and only if their product is a web invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-008395 | MATHDB-326072

Color-independence conjecture for surface cluster structures

摘录 / EXCERPT: Color-independence conjecture. Changing all colors of marked points on any subset of boundary components does not change the cluster type of $\mathcal{A}(S_{g,\sigma})$.

conjecture | open | MathDB 页面 | 原文线索

V2-008396 | MATHDB-326073

Bivariant web-and-forest and compatibility conjecture

摘录 / EXCERPT: Bivariant web conjecture. In any such cluster algebra: all cluster variables are bi-web invariants; a bi-web invariant is a cluster monomial if and only if it can also be written a...

conjecture | open | MathDB 页面 | 原文线索

V2-008397 | MATHDB-326084

Prasad-Takloo-Bigheash local distinction conjecture

摘录 / EXCERPT: Let F_v be a local field, let D_v be a quaternion division algebra over F_v, and let H_v and H'_v be the relevant subgroups of $\operatorname{GL}_n(D_v)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008398 | MATHDB-326095

The radicality conjecture for patch ideals of K-orbit closures

摘录 / EXCERPT: Let $I_{\gamma,\beta}$ be the patch ideal defining a neighborhood of the K-orbit closure Y_{γ} at the point indexed by β. Patch-ideal radicality conjecture. The idea...

conjecture | open | MathDB 页面 | 原文线索

V2-008399 | MATHDB-326096

The Cohen-Macaulay and Hilbert-polynomial conjectures for K-orbit closures

摘录 / EXCERPT: Hilbert-polynomial conjecture. The following assertions hold: (i) $\operatorname{gr}_{\mathfrak m_p}\mathcal O_{p,Y_{\gamma}}$ is Cohen-Macaulay; (ii)...

conjecture | open | MathDB 页面 | 原文线索

V2-008400 | MATHDB-326097

The reduced tangent-cone conjecture for K-orbit closures

摘录 / EXCERPT: Let $p\in Y_{\gamma}$ be a point of the K-orbit closure, and let $\operatorname{gr}_{\mathfrak m_p}\mathcal O_{p,Y_{\gamma}}$ be the associated graded local ring. Reduced tangent...

conjecture | open | MathDB 页面 | 原文线索

V2-008401 | MATHDB-326121

Rationality conjecture for primitive algebraic compactifications of $\mathbb{C}^2$

摘录 / EXCERPT: Let $\bar X$ be a primitive algebraic compactification of $\mathbb{C}^2$ formed by minimally resolving the singularities of a curve-germ at a point on the line L_∞ at infini...

conjecture | open | MathDB 页面 | 原文线索

V2-008402 | MATHDB-326144

Infinite algebraic entropy conjecture for automorphisms of exponentially growing groups

摘录 / EXCERPT: Let G be a finitely generated group such that $h_{alg}(id_G)=\infty$, equivalently, G has exponential growth. Infinite algebraic entropy conjecture. Then $h_{alg}(\phi)=\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-008403 | MATHDB-326160

Width bound and equality characterization for the defining ideal of a numerical semigroup tangent cone

摘录 / EXCERPT: Let H be a numerical semigroup, let I_H^* denote the defining ideal of its tangent cone, let $\mu(H)$ be the number of minimal generators of H, and let $\width(H)$ be the dif...

conjecture | open | MathDB 页面 | 原文线索

V2-008404 | MATHDB-326161

Interval-completion bound for the defining ideal of a numerical semigroup tangent cone

摘录 / EXCERPT: Let $H=\langle a_1,\dots,a_r\rangle$ be a numerical semigroup, and let $\widetilde H$ be its interval completion, the numerical semigroup generated by all integers in the interval...

conjecture | open | MathDB 页面 | 原文线索

V2-008405 | MATHDB-326186

Strongly Koszul toric rings of 0-1 polytopes are trivial

摘录 / EXCERPT: Let $\mathcal{P}$ be a $(0,1)$-polytope, and let $k[\mathcal{P}]$ be the toric ring generated by $\mathcal{P}$. A ring R is trivial if it can be constructed from polynomial rings...

conjecture | open | MathDB 页面 | 原文线索

V2-008406 | MATHDB-326188

The minimality conjecture for product-quotient surfaces

摘录 / EXCERPT: Let S be a product-quotient surface, with geometric genus $p_g(S)$ and invariant $4gamma4$. Minimality conjecture. S is minimal if and only if ... The paper proves this when...

conjecture | open | MathDB 页面 | 原文线索

V2-008407 | MATHDB-326189

The bounded-gamma conjecture for product-quotient surfaces

摘录 / EXCERPT: Let S be a product-quotient surface of general type, let X be its quotient model, and let $p_g(S)$ and $q(S)$ denote its geometric genus and irregularity. Let $\gamma(X)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-008408 | MATHDB-326193

Existence and non-existence conjecture for ${\cal A}\cT_{n,\omega_U}$

摘录 / EXCERPT: Let ${\cal A}\cT_{n,\omega_U}$ denote the fusion category associated with the construction considered in the paper, indexed by an integer n and a parameter ω_U. The preced...

conjecture | open | MathDB 页面 | 原文线索

V2-008409 | MATHDB-326208

The Clifford algebra isomorphism conjecture for geometric Waldspurger periods

摘录 / EXCERPT: Let X be a smooth projective curve and let E be an irreducible SL_2-local system on X such that, for every $d\ts 0$ and every covering $\phi:Y\to X$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-008410 | MATHDB-326211

The free group algebra conjecture for division rings

摘录 / EXCERPT: Let D be a division ring with center k. Let a free group algebra over k mean the group algebra over k of a free group embedded in D. Free group algebra conjecture. If D...

conjecture | open | MathDB 页面 | 原文线索

V2-008411 | MATHDB-326215

Upper semicontinuity conjecture for the number of moving curve ideal generators

摘录 / EXCERPT: For $d\geq 1$, let $\mathbb{V}_d$ be the set of triples of degree-d homogeneous polynomials defining proper parametrizations, and let $n_0({\mathcal K})$ denote the relevant inva...

conjecture | open | MathDB 页面 | 原文线索

V2-008412 | MATHDB-326216

Generic maximality conjecture for moving curve ideal generators

摘录 / EXCERPT: For $d\geq 1$, let $\mathbb{V}_d$ be the set of triples of degree-d homogeneous polynomials defining proper parametrizations. Let $\mathbb{W}_d$ be the open subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-008413 | MATHDB-326243

Contraction algebra conjecture for analytic types of threefold flops

摘录 / EXCERPT: Suppose that $Y\to Y_{\mathrm{con}}$ and $Z\to Z_{\mathrm{con}}$ are flopping contractions of a single curve in smooth projective 3-folds, to points p and q respectively. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-008414 | MATHDB-326251

Computational conjecture on rex moves with a common rightmost generator

摘录 / EXCERPT: Let W be the Coxeter group, let $s\in S$, and let $\underline{w}$ and $\underline{w'}$ be reduced expressions, each ending in s. A rex move is a morphism between reduced expres...

conjecture | open | MathDB 页面 | 原文线索

V2-008415 | MATHDB-326264

The rank formula conjecture for the third lower central series quotient

摘录 / EXCERPT: Let $A=\mathbb{Z}\langle x,y\rangle/(x^m+y^m)$, and let $N_3(A)$ denote the third lower central series quotient, with $N_3(A)[d]$ its degree-d component. Rank formula conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-008416 | MATHDB-326302

The wild McKay point-counting conjecture for quotient singularities

摘录 / EXCERPT: Point-counting realization of the wild McKay conjecture. If the residue field κ is finite, then

conjecture | open | MathDB 页面 | 原文线索

V2-008417 | MATHDB-326312

Vanishing Haantjes tensor conjecture for dispersionless systems from Novikov algebras

摘录 / EXCERPT: Let $\mathbb{A}_n$ be a Novikov algebra, and let the associated equations of hydrodynamic type be the dispersionless systems constructed from that algebra, such as the system descr...

conjecture | open | MathDB 页面 | 原文线索

V2-008418 | MATHDB-326324

Weinberger-Yu conjecture for finite-part independence in reduced operator K-theory

摘录 / EXCERPT: Let G be a group with torsion elements $g_1,g_2,\ldots,g_k$ of distinct positive orders $d_1,\ldots,d_k$. For each g_i, let p_i be the idempotent corresponding to g_i, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008419 | MATHDB-326357

Finiteness conjecture for genera of algebraic groups

摘录 / EXCERPT: Let G be an absolutely almost simple simply connected algebraic group over a finitely generated field K of characteristic zero, or over a field of good characteristic relative...

conjecture | open | MathDB 页面 | 原文线索

V2-008420 | MATHDB-326358

Borel-Tits conjecture on abstract homomorphisms of algebraic groups

摘录 / EXCERPT: Let $\mathbf{G}$ and $\mathbf{G}'$ be algebraic groups defined over infinite fields k and k', respectively. Let $\mathbf{G}^+$ denote the subgroup of $\mathbf{G}(k)$ generated...

conjecture | open | MathDB 页面 | 原文线索

V2-008421 | MATHDB-326391

Diagram-independence conjecture for boundary algebras of marked surfaces

摘录 / EXCERPT: Boundary-algebra independence conjecture. The boundary algebra of a weak Postnikov diagram on (X,M) does not depend on the choice of diagram up to isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-008422 | MATHDB-326396

The converse conjecture for good nilpotent elements

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a simple Lie algebra and let e be a nilpotent element of ${\mathfrak{g}}$. A nilpotent element e is good when it satisfies the hypothesis of Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-008423 | MATHDB-326458

Karpenko's anisotropy conjecture for orthogonal involutions

摘录 / EXCERPT: Let A be a central simple algebra over a field F, and let ${\mathcal F}_{A,o}$ denote the function field associated with A and orthogonal involutions. An orthogonal involutio...

conjecture | open | MathDB 页面 | 原文线索

V2-008424 | MATHDB-326469

Collective contraction conjecture for empty ovals

摘录 / EXCERPT: Let C_m be any smooth real plane curve of degree m, and let its empty ovals be the components whose interiors contain no other real components. Collective contraction conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-008425 | MATHDB-326471

Ragsdale's conjecture for \(M\)-curves

摘录 / EXCERPT: Let C be an M-curve of even degree $m=2k$, with p even ovals and n odd ovals, and let $\chi=p-n$ be the Euler characteristic of the Ragsdale membrane. Ragsdale's M-curve...

conjecture | open | MathDB 页面 | 原文线索

V2-008426 | MATHDB-326493

Classification conjecture for arithmetic discrete groups from rigid DAHA modules

摘录 / EXCERPT: Let $\mathcal{H}$ and $\mathcal{H}^{\checkmark}$ denote the two DAHA-type algebras considered in the paper, and let $\mathcal{B}_q$ be the associated braid-group representation. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-008427 | MATHDB-326525

Cyclotomic Hilbert-series conjecture for finite-dimensional graph algebras

摘录 / EXCERPT: Cyclotomic Hilbert-series conjecture. The Hilbert series $\mathcal H_G(t)$ is a product of cyclotomic polynomials.

conjecture | open | MathDB 页面 | 原文线索

V2-008428 | MATHDB-326534

Narrow ring class field reciprocity conjecture for Darmon points

摘录 / EXCERPT: Let K/F be a quadratic extension, let $\mathcal{O}$ be an order in K, let $K_{\mathcal{O}}^+$ be the associated narrow ring class field, and let $P_{q,b}$ be the Darmon point a...

conjecture | open | MathDB 页面 | 原文线索

V2-008429 | MATHDB-326595

Benois's trivial zeros conjecture for semistable Galois representations

摘录 / EXCERPT: Let M be a critical motive with nonzero complex value $L(0,M)$, let V be a semistable representation of $G_{\mathbb{Q}_p}$, and let D be a regular filtered (ϕ,N)-submodu...

conjecture | open | MathDB 页面 | 原文线索

V2-008430 | MATHDB-326614

Conjecture on regularity of powers of cycle edge ideals

摘录 / EXCERPT: Let C_k be a k-cycle and let $I=I(C_k)$ be its edge ideal. Cycle edge-ideal powers conjecture. The regularity of I^q is a linear function for $q\geq 2$. This conjecture is mo...

conjecture | open | MathDB 页面 | 原文线索

V2-008431 | MATHDB-326651

The complex Leavitt path algebra star-isomorphism conjecture

摘录 / EXCERPT: Let E and F be directed graphs, and let $L_{\a0\mathbb{C}}(E)$ and $L_{\a0\mathbb{C}}(F)$ denote their Leavitt path algebras over the complex numbers, with the complex-conjugat...

conjecture | open | MathDB 页面 | 原文线索

V2-008432 | MATHDB-326652

The complex Leavitt path algebra to graph C-star algebra conjecture

摘录 / EXCERPT: Let E and F be directed graphs, let $L_{\mathbb{C}}(E)$ and $L_{\mathbb{C}}(F)$ be their Leavitt path algebras, and let $C^*(E)$ and $C^*(F)$ be the associated graph C^*-alge...

conjecture | open | MathDB 页面 | 原文线索

V2-008433 | MATHDB-326653

The complex Leavitt path algebra ring-isomorphism conjecture

摘录 / EXCERPT: Let E and F be directed graphs, and let $L_{\mathbb{C}}(E)$ and $L_{\mathbb{C}}(F)$ be their Leavitt path algebras over the complex numbers, while $C^*(E)$ and $C^*(F)$ are the...

conjecture | open | MathDB 页面 | 原文线索

V2-008434 | MATHDB-326654

The generalized strong isomorphism conjecture for Leavitt path algebras

摘录 / EXCERPT: Let K be a field with a fixed involution, and let E and F be directed graphs. Write $L_K(E)$ and $L_K(F)$ for their Leavitt path algebras, equipped with the induced involutio...

conjecture | open | MathDB 页面 | 原文线索

V2-008435 | MATHDB-326758

The generalized Iitaka conjecture for klt pairs

摘录 / EXCERPT: Generalized Iitaka conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008436 | MATHDB-326760

The analytic descent conjecture for Frobenius modules

摘录 / EXCERPT: Let T be a finite free $\mathbb{Z}_p$-module equipped with a continuous G_K-action. For $r>0$, define ... Let $D^r_{K_\infty}(T)^{\operatorname{an}}$ be the set of uniformly r-...

conjecture | open | MathDB 页面 | 原文线索

V2-008437 | MATHDB-326772

The symmetric-shifting conjecture for 2-dimensional complexes embeddable in 4-space

摘录 / EXCERPT: Let K be a 2-dimensional simplicial complex, and let K^s denote its symmetric algebraic shifting. Symmetric-shifting conjecture. If K is embeddable in $\mathbb{R}^4$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-008438 | MATHDB-326773

The balanced-shifting embeddability conjecture

摘录 / EXCERPT: Let K be a d-dimensional balanced simplicial complex, meaning that its vertices are colored with $d+1$ colors so that every edge has vertices of different colors. Let K^b den...

conjecture | open | MathDB 页面 | 原文线索

V2-008439 | MATHDB-326776

The balanced Kalai-Sarkaria conjecture

摘录 / EXCERPT: Let K be a d-dimensional balanced complex, and let $<$ be a $(2,2,\ldots,2)$-admissible order, meaning that the least two vertices of each color class form an initial segment o...

conjecture | open | MathDB 页面 | 原文线索

V2-008440 | MATHDB-326779

The balanced linkless-embeddability conjecture

摘录 / EXCERPT: Let K be a d-dimensional balanced simplicial complex, and let $<$ be a $(3,\ldots,3)$-admissible order. Let $K^{b,<}$ denote the corresponding balanced shift. Balanced linkless...

conjecture | open | MathDB 页面 | 原文线索

V2-008441 | MATHDB-326810

Gubeladze's nilpotence conjecture for monoid algebras

摘录 / EXCERPT: Let k be a commutative ring, let M be an abelian monoid, and let $\mathfrak c=(c_1,c_2,\ldots)$ be a sequence of integers with $c_i\ge 2$. For each $c\ge 2$, the dilation...

conjecture | open | MathDB 页面 | 原文线索

V2-008442 | MATHDB-326826

The kernel filtration conjecture for maximal pro-p Galois groups

摘录 / EXCERPT: Let F be a field of characteristic $\not=p$ containing a primitive pth root of unity, and let $G=G_F(p)$ be the Galois group of the maximal pro-p-extension of F. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008443 | MATHDB-326829

Gekhtman-Shapiro-Vainshtein correspondence conjecture for simple Lie groups

摘录 / EXCERPT: Gekhtman-Shapiro-Vainshtein correspondence conjecture. Let $\mathcal{G}$ be a simple complex Lie group. For any Belavin-Drinfeld triple $T=(\Gamma_{1},\Gamma_{2},\gamma)$ there exi...

conjecture | open | MathDB 页面 | 原文线索

V2-008444 | MATHDB-326844

Hermitian Kazhdan-Lusztig polynomial sign conjecture

摘录 / EXCERPT: Let $X=X({\mathcal O},{\mathcal L})$ be a standard module for the affine Hecke algebra with Hermitian Kazhdan-Lusztig polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-008445 | MATHDB-326920

Existence of rigid dualizing DG modules for tractable commutative DG rings

摘录 / EXCERPT: Let A be a tractable commutative DG ring, with traction $\mathbb{K} \to A$. A rigid dualizing DG module over A relative to $\mathbb{K}$ is a pair (R,ρ) consisting of a dua...

conjecture | open | MathDB 页面 | 原文线索

V2-008446 | MATHDB-326921

Rigidity criterion for dualizing DG modules

摘录 / EXCERPT: Let A be a tractable commutative DG ring with traction $\mathbb{K}\to A$, and let (M,ρ) be a rigid DG module over A relative to $\mathbb{K}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008447 | MATHDB-326922

Functorial construction of rigid dualizing DG modules

摘录 / EXCERPT: Functorial rigidity conjecture. In the pseudo-finite case, R_B has an induced rigidifying isomorphism ρ_B relative to $\mathbb{K}$; in the essentially smooth case, R_B lik...

conjecture | open | MathDB 页面 | 原文线索

V2-008448 | MATHDB-326933

Stability conjecture for decompositions of Betti diagrams of powers of monomial ideals

摘录 / EXCERPT: Stability conjecture. There is a k_0 such that, for all $k>k_0$, there are translations of pure diagrams $\pi_1(k),\ldots,\pi_m(k)$ such that every decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-008449 | MATHDB-326938

Stratification of the category of graded current-algebra modules

摘录 / EXCERPT: Let $\mathbf{g}$ be a finite-dimensional Lie algebra, and let $\mathcal{I}(\mathbf{g}\otimes\mathbb{C}[x])$ denote the category of graded finitely generated $\mathbf{g}$-locally fi...

conjecture | open | MathDB 页面 | 原文线索

V2-008450 | MATHDB-326939

Excellent-filtration conjecture for integrable affine Lie algebra modules

摘录 / EXCERPT: Let $\hat{\mathbf{g}}$ be the affine Lie algebra associated with $\mathbf{g}$, and let $\mathcal{M}_{k,\lambda}$ be an integrable $\hat{\mathbf{g}}$-module of level k and highest...

conjecture | open | MathDB 页面 | 原文线索

V2-008451 | MATHDB-326940

Stratification conjecture for the Lie algebra of vector fields $L_0(n)$

摘录 / EXCERPT: Let $L_0(n)$ be the Lie algebra of vector fields appearing in the paper's vector-fields examples, and let $\mathcal{I}(L_0(n))$ be its associated category of graded modules....

conjecture | open | MathDB 页面 | 原文线索

V2-008452 | MATHDB-326941

Adjoint-comparison conjecture for projectives in the super current algebra category

摘录 / EXCERPT: Let $\mathfrak{g}=\mathbf{g}\otimes\mathbb{C}[x,\xi]$, and let $\lambda,\mu\in\mathsf{P}_{\Delta}$ be incomparable integral dominant weights. Let $\overline{\imath}_{\lambda}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008453 | MATHDB-326943

Main conjecture on non-constant families of covers with specified ramification

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let d be a positive integer, let $S=\{P_1,\ldots,P_n\}$ be a set of points on $\mathbb{P}^1_k$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008454 | MATHDB-327001

Kulakova-Lando conjecture on the \mathfrak{sl}_2 weight system and chord diagrams

摘录 / EXCERPT: Kulakova-Lando conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008455 | MATHDB-327015

Minimal total Markov basis size conjecture for contingency-table matrices

摘录 / EXCERPT: Let M be the matrix in the equation defining the space of tables, and let A, B, and C denote the index sets used to specify the table and its conditionals. A total Markov b...

conjecture | open | MathDB 页面 | 原文线索

V2-008456 | MATHDB-327055

Conjecture on minimal graded free resolutions of codimension two complete intersections

摘录 / EXCERPT: Let S be the polynomial ring and let $A(f,g)$ and $B(f,g)$ be the graded algebras associated with a codimension two complete intersection defined by forms f and g of degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-008457 | MATHDB-327059

Uniform multipole approximation for scattered fields with boundary continuity

摘录 / EXCERPT: Let the obstacles be planar scatterers, and let the scattered field be defined in the exterior of the obstacles and on their boundaries. Assume that the scattered field is analytic...

conjecture | open | MathDB 页面 | 原文线索

V2-008458 | MATHDB-327067

Weak Gorensteinness conjecture for Eisenstein Hecke algebras

摘录 / EXCERPT: Let $\mathfrak{h}$ and $\mathfrak{H}$ be the Eisenstein components of the cuspidal and full Hecke algebras, respectively. The ideals I and $I_{\mathfrak{H}}$ are their Eisenstein...

conjecture | open | MathDB 页面 | 原文线索

V2-008459 | MATHDB-327068

Greenberg's finiteness conjecture for the plus Iwasawa module

摘录 / EXCERPT: For general $N>1$, let X^+ denote the plus part of the relevant cyclotomic Iwasawa module. Greenberg's conjecture. The module X^+ is finite. This is a finiteness form of Greenb...

conjecture | open | MathDB 页面 | 原文线索

V2-008460 | MATHDB-327127

The central norm criterion for homothety of Frobenius forms

摘录 / EXCERPT: Let R be a Frobenius algebra with an ideal $\mathfrak m$ such that ... and let B and B' be forms on R related by $B'(r,s)=B(r,su)$ for some $u\in U(R)$. Suppose that B has...

conjecture | open | MathDB 页面 | 原文线索

V2-008461 | MATHDB-327148

The partial local p-th power conjecture for Galois extensions

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a Galois extension of degree n with Galois group G, and let $\eta\in K^{\times}$ be such that the $\mathbb{Z}[G]$-module generated by η has...

conjecture | open | MathDB 页面 | 原文线索

V2-008462 | MATHDB-327157

The bijection between labeled Stokes triangulations and extended seeds

摘录 / EXCERPT: Let $(\mathbf{S},\mathbf{M},\mathbf{A})$ be a marked bordered surface with a labeled Stokes triangulation T^0, and let $B^0=B(T^0)$. Let $\mathrm{LST}(T^0)$ be the set of labeled...

conjecture | open | MathDB 页面 | 原文线索

V2-008463 | MATHDB-327171

Modified Xiao's conjecture for the irregularity of fibrations

摘录 / EXCERPT: Modified Xiao's conjecture. Every non-trivial fibration satisfies

conjecture | refuted | MathDB 页面 | 原文线索

V2-008464 | MATHDB-327179

Solvability conjecture for arithmetic Kac-Witt algebras in composite dimension

摘录 / EXCERPT: Let n be the dimension parameter, let G be the group indexing the choices of σ, and let $\mathrm{KW}_{b,\sigma}^{\mathscr{A}}(\xi)$ denote the corresponding arithmetic K...

conjecture | open | MathDB 页面 | 原文线索

V2-008465 | MATHDB-327225

Failure of additivity conjecture for sofic entropy

摘录 / EXCERPT: Failure of additivity conjecture. There is such an exact sequence for which

conjecture | open | MathDB 页面 | 原文线索

V2-008466 | MATHDB-327226

Bowen's f-invariant entropy addition conjecture

摘录 / EXCERPT: Let $r\in\mathbb N$, and let A, B, and C be $Z(F_r)$-modules in an exact sequence ... Suppose that the actions of F_r on $(\widehat{A},\lambda_{\widehat{A}})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-008467 | MATHDB-327234

Aljadeff-author's conjecture on graded Gelfand-Kirillov dimension

摘录 / EXCERPT: Let G be a finite abelian group and let A be a G-graded PI-algebra. For each $k\geq 1$, write $\operatorname{GKdim}_k^G(A)$ for the k-th $\mathbb{Z}_2$-graded Gelfand-Kiril...

conjecture | open | MathDB 页面 | 原文线索

V2-008468 | MATHDB-327248

Briançon-Skoda number polynomial-extension conjecture

摘录 / EXCERPT: Polynomial-extension conjecture. The Briançon-Skoda number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-008469 | MATHDB-327249

Negative first Hilbert coefficient criterion for Cohen-Macaulayness

摘录 / EXCERPT: Negative Hilbert coefficient conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008470 | MATHDB-327250

The j_1 criterion for regular sequences

摘录 / EXCERPT: j_1 regular-sequence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008471 | MATHDB-327252

Almost Cohen-Macaulayness of Rees algebras of monomial almost complete intersections

摘录 / EXCERPT: Monomial Rees-algebra conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008472 | MATHDB-327253

Recursive elimination-equation conjecture for three forms

摘录 / EXCERPT: Recursive elimination conjecture. For arbitrary n,

conjecture | open | MathDB 页面 | 原文线索

V2-008473 | MATHDB-327254

Negative first Hilbert coefficient criterion via faithful big Cohen-Macaulay modules

摘录 / EXCERPT: Faithful big Cohen-Macaulay conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008474 | MATHDB-327271

HKOTT's perfectness conjecture for Kirillov-Reshetikhin crystals

摘录 / EXCERPT: HKOTT's perfectness conjecture. The crystal $B^{r,s}$ is perfect if and only if $\frac{s}{c_r}$ is an integer. If $B^{r,s}$ is perfect, its level is $\frac{s}{c_r}$.

conjecture | open | MathDB 页面 | 原文线索

V2-008475 | MATHDB-327290

Quasi-polynomiality conjecture for profiles of bounded-signature structures

摘录 / EXCERPT: Quasi-polynomiality conjecture. If φ_R is bounded by some polynomial, then φ_R is eventually a quasi-polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-008476 | MATHDB-327302

The inverse-map conjecture over the rational function field

摘录 / EXCERPT: Let P_{θ} and Y_{θ} be the modules defined in the function-field setting, and let ... be the maps constructed in the paper. Inverse-map conjecture over...

conjecture | open | MathDB 页面 | 原文线索

V2-008477 | MATHDB-327323

The product-order minimum conjecture for bidegrees of curves

摘录 / EXCERPT: Let C/k be a curve, and let ... where $\simeq$ denotes birational equivalence. Product-order minimum conjecture. The set $S_{1,1}(C)$ admits a minimum with respect to the product o...

conjecture | open | MathDB 页面 | 原文线索

V2-008478 | MATHDB-327355

The centralizer index conjecture for rational maps with automorphisms

摘录 / EXCERPT: Let K be a field, let $\phi\in K(x)$, and let T_∞ be its pre-image tree rooted at 0. Let $A(\phi)$ be the subgroup of $\operatorname{Aut}(T_\infty)$ generated by the act...

conjecture | open | MathDB 页面 | 原文线索

V2-008479 | MATHDB-327356

The family conjecture for quadratic rational maps

摘录 / EXCERPT: Let $\phi(x)=b(x^2+1)/x$ with $b\in\mathbb{Q}$ and $b\notin\{0,\pm 1/2\}$. Let T_∞ be the pre-image tree rooted at 0, let $G_\infty(\phi)$ be its arboreal Galois group, an...

conjecture | open | MathDB 页面 | 原文线索

V2-008480 | MATHDB-327357

The general symmetry conjecture for quadratic rational maps

摘录 / EXCERPT: Let K be a global field of characteristic 0 or greater than 2, and let $\phi(x)\in K(x)$ have degree 2. Assume that ϕ is not post-critically finite and that 0 is not...

conjecture | open | MathDB 页面 | 原文线索

V2-008481 | MATHDB-327361

The associated-form conjecture for invariants of degree-\n(m-2) forms

摘录 / EXCERPT: Associated-form conjecture. For any ${\tt I}\in{\mathcal I}_n^m$ there exists an absolute invariant $\mathbf I$ of forms of degree $n(m-2)$ on $\mathbb C^n$ such that, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-008482 | MATHDB-327372

The conjecture on infinite delta-invariants of reductive groups

摘录 / EXCERPT: Let G be an infinite reductive group over a field of positive characteristic, and let G^{0} denote its connected component. Assume that G^{0} is not a torus. Infinite delta-i...

conjecture | open | MathDB 页面 | 原文线索

V2-008483 | MATHDB-327373

The discriminant formula for the quantized plane at roots of unity

摘录 / EXCERPT: Let A_q be the quantized algebra generated by x and y with relation $yx=qxy$, where q is a primitive nth root of unity, and let $C(A_q)$ denote its center. For elements...

conjecture | open | MathDB 页面 | 原文线索

V2-008484 | MATHDB-327382

Existence of a bounded-degree twisted group algebra for a nondegenerate grading

摘录 / EXCERPT: Let W be an algebra over an algebraically closed field F of characteristic zero satisfying a polynomial identity of degree n. Suppose that W is nondegenerately graded by a...

conjecture | open | MathDB 页面 | 原文线索

V2-008485 | MATHDB-327391

The cubic-generator conjecture for graded artinian Gorenstein algebras

摘录 / EXCERPT: Let $A=S/I$ be a graded artinian Gorenstein algebra with Hilbert function $\{1,g-2,g-2,1\}$. A cubic-generator conjecture asserts that I has $0,1,\ldots,g-5$ or $g-3$ cubic minim...

conjecture | open | MathDB 页面 | 原文线索

V2-008486 | MATHDB-327420

A Breuil-Mézard conjecture for discrete series extended types

摘录 / EXCERPT: Let K be the local field and let G_K be its absolute Galois group. Let $\mathbb{F}$ be the coefficient field, let Γ be the relevant Grothendieck group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008487 | MATHDB-327451

Uniqueness of maximal-dimensional commutative subalgebras for n congruent to 3 modulo 4

摘录 / EXCERPT: Let E be the Grassmann algebra and let $A_1,A_2\subseteq E$ be maximal-dimensional commutative subalgebras. Uniqueness conjecture. If $n=4k+3$, then $A_1\cong A_2$. This asks whe...

conjecture | open | MathDB 页面 | 原文线索

V2-008488 | MATHDB-327454

Artin's unirationality conjecture for supersingular K3 surfaces

摘录 / EXCERPT: Let X be a supersingular K3 surface over an algebraically closed field. Artin's conjecture. The surface X is unirational. The paper proves this conjecture, resolving Artin's qu...

conjecture | open | MathDB 页面 | 原文线索

V2-008489 | MATHDB-327456

Degree-two Cohen-Macaulay Hirsch conjecture

摘录 / EXCERPT: Degree-two Cohen-Macaulay Hirsch conjecture. If S/I is Cohen-Macaulay, then I is Hirsch.

conjecture | open | MathDB 页面 | 原文线索

V2-008490 | MATHDB-327470

The Kazhdan-graded structure conjecture for finite W-superalgebras

摘录 / EXCERPT: Let ${\frak {g}}$ be a Lie superalgebra with reductive even part ${\frak {g}}_{\bar 0}$, let W_χ be its finite W-superalgebra, and let Gr_KW_χ denote its Kazhdan-graded a...

conjecture | open | MathDB 页面 | 原文线索

V2-008491 | MATHDB-327472

The generator-commutator conjecture for regular finite W-superalgebras

摘录 / EXCERPT: Let $\frak g$ be a basic classical Lie superalgebra, let χ be regular, and let W_χ be the associated finite W-superalgebra. Let π denote the relevant quotient or repr...

conjecture | open | MathDB 页面 | 原文线索

V2-008492 | MATHDB-327480

Conjecture on the kernel quotient map for specialized BMW algebras

摘录 / EXCERPT: For $n\geq 3$, let $\mathcal{B}_n^{\pm}$ be the specialized algebra, let $\mathcal{I}_n^{\pm}$ be its defining ideal, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008493 | MATHDB-327481

Extension conjecture for the exceptional Markov trace

摘录 / EXCERPT: Let t_3^0 be the Markov trace on the specialized algebra described by the conditions $t_3^0(1)=t_3^0(s_1)=0$ and $t_3^0(s_1s_2)=1$. Markov-trace extension conjecture. The trace...

conjecture | open | MathDB 页面 | 原文线索

V2-008494 | MATHDB-327497

The levelness conjecture for the power ideal algebra $R_{2,n,d}$

摘录 / EXCERPT: Let $R_{2,n,d}$ be the graded algebra under consideration, with socle ${\rm Soc}(R_{2,n,d})$ and graded component $[R_{2,n,d}]_{2d-2}$. Levelness conjecture. $R_{2,n,d}$ is a level...

conjecture | open | MathDB 页面 | 原文线索

V2-008495 | MATHDB-327498

The injectivity-surjectivity conjecture for the maps $\mu_{i,\mathbf{h}}$

摘录 / EXCERPT: Let $\mu_{i,\mathbf{h}}$ be the maps defined in Definition, indexed by i and a vector $\mathbf{h}$, and let $\mathrm{wt}(\mathbf{h})$ denote the weight of $\mathbf{h}$. Let k,...

conjecture | open | MathDB 页面 | 原文线索

V2-008496 | MATHDB-327499

The graded-piece characterization conjecture for $R_{i,\mathbf{h}}$

摘录 / EXCERPT: Graded-piece characterization conjecture. One has $R_{i,\mathbf{h}}\neq 0$ if and only if $\mathbf{h}\in\mathcal{H}_j$; moreover, for $\mathbf{h}\in\mathcal{H}_j$,

conjecture | open | MathDB 页面 | 原文线索

V2-008497 | MATHDB-327511

Moreno-Socias's generic weakly-grevlex ideal conjecture

摘录 / EXCERPT: Moreno-Socias's conjecture. There is a non-empty Zariski-open subset $U\subseteq A_{d_1}\times\dots\times A_{d_s}$ such that every $(f_1,\dots,f_s)\in U$ generates an ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-008498 | MATHDB-327536

Lewis-Reiner-Stanton conjecture on invariants modulo Frobenius powers

摘录 / EXCERPT: Lewis-Reiner-Stanton conjecture. The G-fixed subalgebra Q^G has Hilbert series

conjecture | open | MathDB 页面 | 原文线索

V2-008499 | MATHDB-327539

Landweber-Stong depth conjecture for finite general linear groups

摘录 / EXCERPT: Let H be a subgroup of $GL_n({\mathbb F}_q)$ acting on $S={\mathbb F}_q[\mathbf{x}]$, and let $D_{n,j}$ denote the Dickson polynomials. Landweber-Stong conjecture. The depth of t...

conjecture | open | MathDB 页面 | 原文线索

V2-008500 | MATHDB-327540

Regularity conjecture for amenable d-sequences

摘录 / EXCERPT: Regularity conjecture for amenable d-sequences. If ${\mathbf x}$ is an amenable d-sequence relative to M,

conjecture | refuted | MathDB 页面 | 原文线索

V2-008501 | MATHDB-327548

Finite-cover conjecture for A^1-fibrations in smooth affine threefold families

摘录 / EXCERPT: Let $f:Y\to T$ be a smooth morphism from a smooth affine threefold Y onto a smooth affine curve T, and suppose that every closed fiber Y_t has an $\mathbb{A}^1$-fibration of...

conjecture | open | MathDB 页面 | 原文线索

V2-008502 | MATHDB-327557

Divisor criterion for the square-root class of Belyi functions

摘录 / EXCERPT: Square-root class conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008503 | MATHDB-327577

Duality conjecture for proper commutative group stacks

摘录 / EXCERPT: Duality conjecture. Then $D(G)$ is algebraic, proper, flat and finitely presented, with finite diagonal, and G is dualizable.

conjecture | open | MathDB 页面 | 原文线索

V2-008504 | MATHDB-327609

Premet's conjugacy conjecture for Cartan subalgebras of restricted Lie algebras

摘录 / EXCERPT: Premet's conjecture. There exists a nonempty Zariski open subset $V\subseteq{\mathfrak g}$ consisting of regular elements such that, for any $u,v\in V$, the Cartan subalgebras

conjecture | open | MathDB 页面 | 原文线索

V2-008505 | MATHDB-327655

Bijection between standard pairs and web basis indices

摘录 / EXCERPT: Let ${\mathcal{M}}$ be a positroid, and let an ordered pair (I,J) of k-element subsets be ${\mathcal{M}}$-standard if $I=\pi_k(x)$ and $J=\pi_k(y)$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-008506 | MATHDB-327687

Taylor's conjecture on the class-invariant homomorphism for CM elliptic curves

摘录 / EXCERPT: Taylor's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008507 | MATHDB-327720

Dixmier's degree-sequence conjecture for hsops of binary-form invariants

摘录 / EXCERPT: Let I be the ring of invariants of binary forms of degree n under the action of $\mathrm{SL}(2,\mathbf{C})$. A homogeneous system of parameters (hsop) is an algebraically indep...

conjecture | open | MathDB 页面 | 原文线索

V2-008508 | MATHDB-327722

Stanley's squarefree quotient conjecture at Stanley depth d+1

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$. Let $I\subset S$ be minimally generated by squarefree monomials $f_1,\ldots,f_r$ of degree d, and let E be a set of squarefree monomials of degree at...

conjecture | open | MathDB 页面 | 原文线索

V2-008509 | MATHDB-327736

Uniform Markov degree conjecture for iterated toric fiber products

摘录 / EXCERPT: Uniform Markov degree conjecture. The maximum Markov degree of the displayed iterated toric fiber product is bounded by a constant independent of $r_1,\dots,r_s$. This general conj...

conjecture | open | MathDB 页面 | 原文线索

V2-008510 | MATHDB-327865

Perrin-Riou's conjecture on the Beilinson-Kato class

摘录 / EXCERPT: Perrin-Riou's conjecture. The element $\frak{z}_0^{\textup{BK}}$ is non-trivial if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008511 | MATHDB-327894

The phantom-category conjecture for Barlow surfaces and rational blow-downs

摘录 / EXCERPT: Phantom-category conjecture. The derived categories of the Barlow surface and of the rational blow-down described in the source contain a phantom category as a semiorthogonal piece...

conjecture | open | MathDB 页面 | 原文线索

V2-008512 | MATHDB-327909

The lattice formulation of Stanley's conjecture

摘录 / EXCERPT: Lattice formulation of Stanley's conjecture. For all finite semilattices L, the following inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-008513 | MATHDB-327933

The disguised residual intersection conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d, let I be an ideal satisfying the sliding depth condition, and let $\mathfrak{a}\subseteq I$ define an alge...

conjecture | open | MathDB 页面 | 原文线索

V2-008514 | MATHDB-327960

Linear recurrence conjecture for the A_{1,N-1} T_z-systems

摘录 / EXCERPT: Linear recurrence conjecture. The iterates satisfy a constant-coefficient linear recurrence relation of order $4(N-1)(N-2)$. For odd N, it has the form

conjecture | open | MathDB 页面 | 原文线索

V2-008515 | MATHDB-327974

The density conjecture for uncyclic matrices

摘录 / EXCERPT: Let $q\geqslant 2$ be a finite field size and let $n\geqslant 1$. Write $\operatorname{unc}(n,q)$ for the number of uncyclic matrices in $\operatorname{M}(n,q)$, where an uncyclic...

conjecture | open | MathDB 页面 | 原文线索

V2-008516 | MATHDB-327992

The noncommutative p-adic L-function interpolation conjecture for elliptic curves

摘录 / EXCERPT: Let K_∞/K be a strongly admissible p-adic Lie extension with Galois group G, let E be an elliptic curve, and let $\Lambda(G)$ be the Iwasawa algebra. Let R be the se...

conjecture | open | MathDB 页面 | 原文线索

V2-008517 | MATHDB-328016

The nonvanishing solution conjecture for invertible complex matrices

摘录 / EXCERPT: Let $A=(a_{i,j})_{1\leq i,j\leq n}$ be a complex invertible matrix, and let $x_1,\dots,x_n$ be complex variables. Nonvanishing solution conjecture. The system ... has a solution...

conjecture | open | MathDB 页面 | 原文线索

V2-008518 | MATHDB-328017

The classification conjecture for non-nilpotent evolution algebras satisfying condition P

摘录 / EXCERPT: Let E be an n-dimensional non-nilpotent evolution algebra satisfying condition P. Let ES_1 denote the one-dimensional evolution algebra used in the source, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008519 | MATHDB-328032

Rank monotonicity conjecture for induced representations of Clifford algebras

摘录 / EXCERPT: Rank monotonicity conjecture. The obtained representation of C_f has rank no less than the rank of the initial representation of C_g.

conjecture | open | MathDB 页面 | 原文线索

V2-008520 | MATHDB-328033

Generation conjecture for tensor products of cyclic algebras

摘录 / EXCERPT: Generation conjecture. The algebra A is generated by X and Y.

conjecture | open | MathDB 页面 | 原文线索

V2-008521 | MATHDB-328034

Maximality conjecture for a family of p^k-central spaces

摘录 / EXCERPT: Maximality conjecture. The p^k-central space V is maximal with respect to inclusion.

conjecture | open | MathDB 页面 | 原文线索

V2-008522 | MATHDB-328038

Nilpotence conjecture for solutions of generic matrix commutator equations

摘录 / EXCERPT: Nilpotence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008523 | MATHDB-328097

The affine W-algebra action conjecture for non-simply laced affine Lie algebras

摘录 / EXCERPT: Let $\cal G$ be an affine Lie algebra with connected Dynkin diagram, let $\cal G^{\vee}$ be its Langlands dual, and let $G_{\cal G^{\vee}}$ be the semisimple simply connected group...

conjecture | open | MathDB 页面 | 原文线索

V2-008524 | MATHDB-328099

The integral Verma-to-Fock embedding conjecture

摘录 / EXCERPT: Let ${}_\mathbf A M(\boldsymbol a)$ be the $\mathbf A_T$-form of the Verma module generated by $|\boldsymbol a\rangle$, and let ${}_\mathbf A F(\boldsymbol a)$ be the $\mathbf A_T$...

conjecture | solved | MathDB 页面 | 原文线索

V2-008525 | MATHDB-328113

The quantization correspondence conjecture for nilpotent algebraic groups

摘录 / EXCERPT: Let G be an affine connected nilpotent algebraic group over $\mathbb{C}$, and let $\mathfrak{g}$ be its Lie algebra. A Hopf 2-cocycle on the coordinate Hopf algebra of G is con...

conjecture | open | MathDB 页面 | 原文线索

V2-008526 | MATHDB-328116

The Alexander polynomial coefficient inequality for rational cuspidal curves

摘录 / EXCERPT: Let C be a rational cuspidal curve of degree d with critical points $z_1,\ldots,z_n$. Let $K_1,\ldots,K_n$ be the corresponding links of singular points, with Alexander polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-008527 | MATHDB-328119

Breuning's local equivariant epsilon constant conjecture

摘录 / EXCERPT: Let N/K be a Galois extension of local number fields, let G be its Galois group, and let $R_{N/K}$ be Breuning's element of $K_0(\mathbb Z_p[G],\mathbb Q_p)$ incorporating loca...

conjecture | open | MathDB 页面 | 原文线索

V2-008528 | MATHDB-328128

Morrison-Peters conjecture on subfactor planar algebras below index $3+\sqrt{5}$

摘录 / EXCERPT: Morrison-Peters conjecture. There are exactly 2 non-Temperley-Lieb subfactor planar algebras with index in $(5,3+\sqrt{5})$: the unique $\mathfrak{su}(2)_5$ subfactor planar alge...

conjecture | open | MathDB 页面 | 原文线索

V2-008529 | MATHDB-328129

Enumeration conjecture for non-Temperley-Lieb subfactor planar algebras at index $3+\sqrt{5}$

摘录 / EXCERPT: Enumeration conjecture. At index $3+\sqrt{5}$, there are exactly 13 non-Temperley-Lieb subfactor planar algebras: the Bisch-Jones Fuss-Catalan A_3*A_4 subfactor planar algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008530 | MATHDB-328134

Polynomiality conjecture for absolutely indecomposable bundles of a Harder-Narasimhan type

摘录 / EXCERPT: Fix $g\geq 0$. For a smooth projective curve X of genus g over a finite field and a Harder-Narasimhan type $(\alpha_1,\ldots,\alpha_t)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008531 | MATHDB-328135

Kac-type conjecture for the spherical Hall Lie algebra

摘录 / EXCERPT: Let $A_{g,r,d}(0)$ be the constant term of the polynomial counting absolutely indecomposable vector bundles of rank r and degree d, and let $\mathfrak{h}^{\mathrm{sph}}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-008532 | MATHDB-328136

Hilbert-series conjecture for the nonnegative spherical Hall algebra

摘录 / EXCERPT: Let $\mathcal{H}^{\mathrm{sph},\geq 0}_{\nu}$ be the subspace of the spherical Hall algebra component $\mathcal{H}^{\mathrm{sph}}_{\nu}$ consisting of functions supported on...

conjecture | open | MathDB 页面 | 原文线索

V2-008533 | MATHDB-328139

The first $gamma,delta$ conjecture for endomorphisms of the Weyl algebra

摘录 / EXCERPT: The first γ,δ conjecture. For every endomorphism f of A_1, there exist involutions γ and δ, each of γ and δ is conjugate to α, su...

conjecture | open | MathDB 页面 | 原文线索

V2-008534 | MATHDB-328140

The first g,h conjecture for endomorphisms of the Weyl algebra

摘录 / EXCERPT: The first g,h conjecture. For every endomorphism f of A_1, there exist g and h, both automorphisms, both anti-automorphisms or one is an automorphism and the other is an...

conjecture | open | MathDB 页面 | 原文线索

V2-008535 | MATHDB-328141

The second γ,δ conjecture for endomorphisms of the Weyl algebra

摘录 / EXCERPT: The second γ,δ conjecture. For every endomorphism f of A_1, there exist involutions γ and δ, such that f is a γ,δ-endomorphism of A_1.

conjecture | open | MathDB 页面 | 原文线索

V2-008536 | MATHDB-328191

One-singularity curve auto-arc structure conjecture

摘录 / EXCERPT: Let C be a connected reduced curve over k with exactly one singular point p, and let $\bar C\to C$ be its normalization of degree e. Let $\mathfrak{l}_{P_i(n)}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-008537 | MATHDB-328194

Yoshinaga's conjecture on free arrangements

摘录 / EXCERPT: Let an arrangement be called free if its module of logarithmic derivations is free, inductively free if it belongs to the inductively free class, and rigid if its moduli space is z...

conjecture | open | MathDB 页面 | 原文线索

V2-008538 | MATHDB-328200

The density conjecture for generators of projective algebras

摘录 / EXCERPT: Let R be the ring of algebraic integers in a number field K, and let A be an R-algebra that is finitely generated and projective as an R-module. For each maximal ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-008539 | MATHDB-328228

Vojta's linear bound conjecture for curves on surfaces

摘录 / EXCERPT: Let X be a smooth projective surface, let C be any curve on X, let g denote its geometric genus, and let $k_C=K_X\cdot C$. Vojta's conjecture. There exist constants A and...

conjecture | open | MathDB 页面 | 原文线索

V2-008540 | MATHDB-328229

Vojta's $4+\epsilon$ conjecture for curves on surfaces

摘录 / EXCERPT: Let X be a smooth projective surface, let C be any curve on X, let g denote its geometric genus, and let $k_C=K_X\cdot C$. Vojta's conjecture. For every real number...

conjecture | open | MathDB 页面 | 原文线索

V2-008541 | MATHDB-328243

Cameron-Glynn algebra isomorphism conjecture for age-inexhaustible structures

摘录 / EXCERPT: Let $\mathcal{M}$ be an age-inexhaustible relational structure, meaning that any two finite substructures of $\mathcal{M}$ have disjoint copies within $\mathcal{M}$. Let the Camero...

conjecture | open | MathDB 页面 | 原文线索

V2-008542 | MATHDB-328297

The Taylor-to-SD(wedge) conjecture for oriented trees

摘录 / EXCERPT: The Taylor-to-SD(wedge) conjecture. For every oriented tree $\mathbb H$, either $\operatorname{\mathbf{alg}}\mathbb H$ is not Taylor or it is $\mathrm{SD}(\wedge)$. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-008543 | MATHDB-328351

The conjectural equivalence of simple Bernstein blocks for related general linear groups

摘录 / EXCERPT: Conjectural equivalence of Bernstein blocks. There is an equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-008544 | MATHDB-328380

Quasi-split generalization of the spherical-module theorem for rational Cherednik algebras

摘录 / EXCERPT: Let G be a quasi-split group in the generality considered in the paper, let $a\in\mathfrak{c}(F)^{\textup{rs}}_{\nu}$, and use the perverse and Chern filtrations, the affine Spri...

conjecture | open | MathDB 页面 | 原文线索

V2-008545 | MATHDB-328393

Subalgebra conjecture for nonlocal and classical Lie symmetries

摘录 / EXCERPT: Let $\mathcal{G}$ be the Lie group of nonlocal symmetry diffeomorphisms associated with the integrable map (respectively,), and let and denote the corresponding continuous dynamic...

conjecture | open | MathDB 页面 | 原文线索

V2-008546 | MATHDB-328396

Cherednik-Orr conjecture on nonsymmetric Macdonald polynomials and PBW characters

摘录 / EXCERPT: Cherednik-Orr conjecture. Assume that $\la$ is an antidominant weight. Then

conjecture | open | MathDB 页面 | 原文线索

V2-008547 | MATHDB-328400

Conjecture on the bispectral algebra for a three-by-three step-up example

摘录 / EXCERPT: Bispectral algebra conjecture. The algebra of all such $\Theta(x)$ is the algebra of polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-008548 | MATHDB-328401

Upper semi-continuity conjecture for Hilbert-Kunz multiplicity

摘录 / EXCERPT: Upper semi-continuity conjecture. If R is an excellent ring, then the function

conjecture | open | MathDB 页面 | 原文线索

V2-008549 | MATHDB-328411

Jones's centralizer-index conjecture for arboreal Galois groups

摘录 / EXCERPT: Let ϕ be a rational function, let $T(\phi)$ be its rooted preimage tree, let $A(\phi)$ be the group of Möbius transformations commuting with ϕ and fixing 0, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008550 | MATHDB-328412

Jones-Manes centralizer-index conjecture for quadratic rational functions

摘录 / EXCERPT: Let $\phi(x)\in\mathbb{Q}(x)$ be a rational function of the form ... where $k_0\in\mathbb{Z}\setminus\{0\}$. Let $C(\phi)$ be the centralizer associated with the nontrivial Möbius sy...

conjecture | open | MathDB 页面 | 原文线索

V2-008551 | MATHDB-328442

The Hankel gradient ideal colon conjecture

摘录 / EXCERPT: Let k be the base field, let R be the polynomial ring containing the entries of the generic $m\times m$ Hankel matrix $\mathcal{H}_m$, and let J be the gradient ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-008552 | MATHDB-328443

The linear-type conjecture for the Hankel determinant gradient ideal

摘录 / EXCERPT: Let R be the polynomial ring containing the entries of the generic $m\times m$ Hankel matrix $\mathcal{H}_m$, and let J be the gradient ideal of its determinant. Linear-type co...

conjecture | open | MathDB 页面 | 原文线索

V2-008553 | MATHDB-328487

Rains-Vazirani's bar-operator conjecture for quasiparabolic sets

摘录 / EXCERPT: Rains-Vazirani's bar-operator conjecture. If X is bounded below, then each of the modules

conjecture | open | MathDB 页面 | 原文线索

V2-008554 | MATHDB-328506

Abhyankar's conjecture for affine curves

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$. Let X be a smooth projective curve of genus g over k, let B be a non-empty set of r points of X, and l...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-008555 | MATHDB-328530

The pole conjecture for eigenvectors of Lannes' T-functor

摘录 / EXCERPT: Let $K^{red}(\mathcal U)$ be the Grothendieck ring of finite direct sums of indecomposable reduced injective unstable modules, and let v be a virtual module in...

conjecture | open | MathDB 页面 | 原文线索

V2-008556 | MATHDB-328584

Genus bound for low-degree curves on complete intersection threefolds

摘录 / EXCERPT: Let $X=Z(f_1,\dots,f_{n-3})\subset\mathbb{P}^n$ be a complete intersection threefold, where f_i has degree k_i, and let $C\subset X$ be a curve of degree d and genus g. The...

conjecture | open | MathDB 页面 | 原文线索

V2-008557 | MATHDB-328585

Bayer-Macrì-Stellari Castelnuovo inequality for curves on complete intersection threefolds

摘录 / EXCERPT: Let $X=Z(f_1,\dots,f_{n-3})\subset\mathbb{P}^n$ be a complete intersection threefold, where f_i has degree k_i, and let $C\subset X$ be a curve of degree d and genus $g(C)$....

conjecture | open | MathDB 页面 | 原文线索

V2-008558 | MATHDB-328587

Pivotal symmetric ideal generated by the recursive idempotent

摘录 / EXCERPT: Let $\mathsf{D}_{{\mathfrak S}(n)}$ be the diagrammatic representation-theoretic object associated with the symmetric group ${\mathfrak S}(n)$, and let $\mathsf{P}^{(n)}$ be its id...

conjecture | open | MathDB 页面 | 原文线索

V2-008559 | MATHDB-328605

The Drinfeld moduli conjecture for special p-divisible groups

摘录 / EXCERPT: Let F be the relevant base field, let O_D be the specified maximal order, and fix an r-special p-divisible group $({\mathbb {X}},\iota_{{\mathbb {X}}})$ over an algebraic c...

conjecture | open | MathDB 页面 | 原文线索

V2-008560 | MATHDB-328654

F-trivialization conjecture for complete local rings

摘录 / EXCERPT: Let R be a complete d-dimensional local ring of positive characteristic, and let M be a d-dimensional R-module. Write $\mathbf{F}_{*}^nM$ for the n-fold Frobenius trans...

conjecture | open | MathDB 页面 | 原文线索

V2-008561 | MATHDB-328655

The very small maximal Cohen-Macaulay module conjecture

摘录 / EXCERPT: Very small MCM conjecture. Every complete local ring admits a very small MCM.

conjecture | open | MathDB 页面 | 原文线索

V2-008562 | MATHDB-328667

The protected chiral algebra of the T_n SCFT for n at least 3

摘录 / EXCERPT: The Tn chiral algebra conjecture. The protected chiral algebra of the T_n SCFT for any $n\geqslant3$ is a $\mathcal{W}$-algebra with the following generators: three sets of...

conjecture | open | MathDB 页面 | 原文线索

V2-008563 | MATHDB-328668

Genus-zero class S chiral algebras

摘录 / EXCERPT: Genus-zero chiral algebra conjecture. The protected chiral algebra is a $\mathcal{W}$-algebra with singlet generators W_i, $i=1,\ldots,n$, of dimension $i+1$, together with addit...

conjecture | open | MathDB 页面 | 原文线索

V2-008564 | MATHDB-328677

Huneke's uniform Artin-Rees conjecture

摘录 / EXCERPT: Let R be an excellent Noetherian ring of finite Krull dimension, and let $N\subseteq M$ be finitely generated R-modules. Huneke's uniform Artin-Rees conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-008565 | MATHDB-328678

Hübl's conjecture on symbolic powers and regular local rings

摘录 / EXCERPT: Hübl's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-008566 | MATHDB-328683

Cyclic monodromy conjecture for Bethe vectors

摘录 / EXCERPT: Let $\fg$ be the Lie algebra in the Gaudin model, and let the theorem referred to in the source as Theorem assert the cyclic monodromy property for the associated Bethe vectors. Cy...

conjecture | open | MathDB 页面 | 原文线索

V2-008567 | MATHDB-328686

Kamnitzer-Weekes monodromy conjecture for shift-of-argument algebras

摘录 / EXCERPT: Let the fundamental group of the De Concini-Procesi wonderful compactification act on the eigenvectors of the shift-of-argument algebra in a finite-dimensional irreducible represen...

conjecture | open | MathDB 页面 | 原文线索

V2-008568 | MATHDB-328691

The graded Amitsur conjecture for codimension growth

摘录 / EXCERPT: Graded Amitsur conjecture. There exists a graded PI-exponent

conjecture | open | MathDB 页面 | 原文线索

V2-008569 | MATHDB-328692

General desuspension conjecture for derived infinitesimal bimodules

摘录 / EXCERPT: Let $\mathbb K$ be a commutative ring, let $E_m^{\mathbb{1}}\to\mathcal O$ be a morphism of differential graded operads, and suppose that $\mathcal O$ is augmented over...

conjecture | open | MathDB 页面 | 原文线索

V2-008570 | MATHDB-328723

Tor-vanishing conjecture for modules of Loewy length at most two

摘录 / EXCERPT: Tor-vanishing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008571 | MATHDB-328759

Table-locus equations conjecture for all entries of Zhao's table

摘录 / EXCERPT: Fix $Q=(u,u-r)$ with $u>r\ge2$, put $B=J_Q$, and let $E(\mu,\nu)$ and $X_s(k_1,k_2)$ be the coordinate expressions defined in the source. All-table loci equations conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-008572 | MATHDB-328787

The isolated log canonical and dense F-pure type conjecture

摘录 / EXCERPT: Let $x\in X$ be an n-dimensional normal $\mathbb{Q}$-Gorenstein singularity over an algebraically closed field of characteristic zero, and suppose that x is an isolated non-log...

conjecture | open | MathDB 页面 | 原文线索

V2-008573 | MATHDB-328788

The Hilbert-Kunz multiplicity lower-bound and rigidity conjecture

摘录 / EXCERPT: Let k be a field of characteristic $p\geq 3$, let $d\geq 2$, and set ... For an arbitrary d-dimensional unmixed local ring $(A,\mathfrak{m})$ of characteristic p with residue f...

conjecture | open | MathDB 页面 | 原文线索

V2-008574 | MATHDB-328794

Equidimensional equality criterion for Serre intersection multiplicity

摘录 / EXCERPT: Let A be a regular local ring. Let M and N be equidimensional, finitely generated A-modules such that ... and $\ell(M\otimes_A N)<\infty$. Write $\operatorname{gr}A$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-008575 | MATHDB-328797

Conjecture on frieze counts for exceptional Dynkin types

摘录 / EXCERPT: Frieze-count conjecture. The numbers of friezes of types E_6, E_7, E_8 and F_4 are 868, 4400, 26592 and 112, respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-008576 | MATHDB-328798

Conjectural bound for values of Dynkin-diagram friezes

摘录 / EXCERPT: Unitary-frieze bound conjecture. The value of a frieze at any node of a Dynkin diagram is less than the maximal value of that node over the set of unitary friezes.

conjecture | open | MathDB 页面 | 原文线索

V2-008577 | MATHDB-328823

The Stokes triangulation-seed correspondence conjecture

摘录 / EXCERPT: Stokes triangulation-seed correspondence conjecture. We have $T=T'$ if and only if $(B',x',y')=(B,x,y)$. This conjecture asserts that the seed obtained from signed mutations record...

conjecture | open | MathDB 页面 | 原文线索

V2-008578 | MATHDB-328824

Denominator-matrix recursion for source-sink moves

摘录 / EXCERPT: Source-sink denominator recursion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008579 | MATHDB-328825

Sigma action on denominator matrices for source-sink moves

摘录 / EXCERPT: Sigma denominator-matrix conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008580 | MATHDB-328826

Signed-column conjecture for denominator matrices

摘录 / EXCERPT: Signed-column conjecture. For all $t\in\mathbb{T}_n$, the matrix $D_t^{B_0;t_0}$ has signed columns.

conjecture | open | MathDB 页面 | 原文线索

V2-008581 | MATHDB-328839

The wpo characterization conjecture for strongly algebraic permutation classes

摘录 / EXCERPT: Strongly algebraic characterization conjecture. A permutation class is strongly algebraic if and only if it is wpo. The source proves that every strongly algebraic class is wpo and...

conjecture | open | MathDB 页面 | 原文线索

V2-008582 | MATHDB-328882

Conjectural existence of Galois representations attached to ordinary Hecke algebras

摘录 / EXCERPT: Let F be the number field, S a finite set of finite places containing those dividing p, U a product subgroup, and ${\mathbb T}^{S}_\text{ord}(U)$ the ordinary Hecke algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008583 | MATHDB-328883

Determinant condition for Taylor-Wiles Galois lifts

摘录 / EXCERPT: Let Q be an auxiliary Taylor-Wiles set, let $\overline{\rho}_{\mathfrak m}$ be the residual representation, let μ be the specified character, and let $r_{{\mathfrak m},Q}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-008584 | MATHDB-328926

The asymptotic upper-bound conjecture for medial quandles

摘录 / EXCERPT: Asymptotic upper-bound conjecture. The number of isomorphism classes of medial quandles of order n is at most

conjecture | open | MathDB 页面 | 原文线索

V2-008585 | MATHDB-328981

Graded *-identity representability conjecture

摘录 / EXCERPT: Let F be a field of characteristic zero, and let G be a finite abelian group. A G-graded finitely generated associative PI-algebra with graded involution has a proper graded...

conjecture | open | MathDB 页面 | 原文线索

V2-008586 | MATHDB-328982

Classification conjecture for finite-dimensional graded *-simple algebras

摘录 / EXCERPT: Classification conjecture. The algebra A is $*$-graded simple if and only if it is isomorphic as a graded $*$-algebra either to the G-graded simple algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008587 | MATHDB-328983

Finite-dimensional superalgebra realization conjecture for identities with superinvolution

摘录 / EXCERPT: Finite-dimensional realization conjecture. There exists a finite-dimensional associative superalgebra $C=C_{\bar{0}} \oplus C_{\bar{1}}$ over F with superinvolution that satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-008588 | MATHDB-328984

Finite-dimensional Grassmann-envelope conjecture for identities with involution

摘录 / EXCERPT: Grassmann-envelope conjecture. Any associative F-algebra with involution satisfies the same $*$-identities as $E(C)$ for some associative superalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008589 | MATHDB-329022

The one-dimensional transverse Killing quotient conjecture for locally homogeneous pp-waves

摘录 / EXCERPT: Let $({\cal M},\mathrm{g})$ be an indecomposable locally homogeneous pp-wave of dimension greater than 3. Let V denote the parallel null vector field of the pp-wave, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008590 | MATHDB-329023

Katthän's lattice gluing conjecture for Stanley projective dimension

摘录 / EXCERPT: Lattice gluing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-008591 | MATHDB-329054

Conjecture on the exceptional three-part Cohen-Macaulay arrangements

摘录 / EXCERPT: Let $b>c$ and let q,r,s be positive integers. For the partition ... let X_λ denote the associated subspace arrangement. Three-part non-Cohen-Macaulayness conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-008592 | MATHDB-329055

Conjecture on Cohen-Macaulayness and exceptional parameters for two-part arrangements

摘录 / EXCERPT: Let s be a positive integer and let b be a complex parameter. For the partition ... let Y_λ denote the associated quotient arrangement. Generic Cohen-Macaulayness conject...

conjecture | open | MathDB 页面 | 原文线索

V2-008593 | MATHDB-329068

The unit-cube Pierce-Birkhoff conjecture

摘录 / EXCERPT: Let $PWP(n)_u$ be the set of continuous piecewise-polynomial functions from $[0,1]^n$ to $[0,1]$, let $ISD(n)_u$ be the set of continuous functions on $[0,1]^n$ expressible as fini...

conjecture | open | MathDB 页面 | 原文线索

V2-008594 | MATHDB-329093

The asymptotic lower-bound conjecture for Nullstellensatz certificates of graph-coloring ideals

摘录 / EXCERPT: Asymptotic certificate-degree conjecture. For every field $\mathbb K$ and every positive integer m, there exists a constant k_0 with the following property: for each $k>k_0$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-008595 | MATHDB-329116

Uniform branch-bound conjecture for torsion curves

摘录 / EXCERPT: Let $\mathcal{C}$ be the curve parameterizing pairs of Legendre coordinates with simultaneous torsion, and let $\mathcal{Z}$ be the corresponding parameter space. Uniform branch-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-008596 | MATHDB-329123

Poisson distribution conjecture for point counts of random curves

摘录 / EXCERPT: Point-count distribution conjecture. For every nonnegative integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-008597 | MATHDB-329137

The formal Knizhnik-Zamolodchikov associator conjecture

摘录 / EXCERPT: Let $A(M^{\delta}_0)$ be the cooperadic algebra of Brown's compactified moduli spaces, let $\boldsymbol{\mathcal{Z}}_{\scriptstyle{\mathrm{ns}}}(A(M^{\delta}_0))$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-008598 | MATHDB-329147

Algebraic boundary-point density conjecture for the Mandelbrot set

摘录 / EXCERPT: Let ${\mathcal M}$ be the parameter set under consideration, and let $\partial {\mathcal M}$ denote its boundary. A point of $\partial {\mathcal M}$ is algebraic if it is algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-008599 | MATHDB-329168

Lara Rodríguez and Thakur's conjecture on zeta-like multizeta values

摘录 / EXCERPT: For a multizeta value $\zeta_{A}(s_{1},\ldots,s_{r})$ of weight $w:=\sum_{i=1}^{r}s_i$, call it zeta-like if the ratio $\zeta_{A}(s_{1},\ldots,s_{r})/\zeta_{A}(w)$ is algebraic ove...

conjecture | open | MathDB 页面 | 原文线索

V2-008600 | MATHDB-329196

Boundary valuation conjecture for theta bases

摘录 / EXCERPT: Boundary valuation conjecture. The proposition stated in the source holds for every $v\in\mathcal A_{\operatorname{prin}}^{\operatorname{trop}}(\mathbb Z)$; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-008601 | MATHDB-329261

Exception conjecture for $p_2p_1^{d-2}$

摘录 / EXCERPT: Let p_k denote the power-sum symmetric function of degree k, and let an n-exception be a homogeneous symmetric polynomial whose associated polarization module has the excepti...

conjecture | open | MathDB 页面 | 原文线索

V2-008602 | MATHDB-329262

Lower-bound conjecture for derivatives of symmetric polynomials

摘录 / EXCERPT: Let f be a homogeneous symmetric polynomial of degree d in $\mathbb{R}[x_{11},\ldots,x_{1n}]$, and let $E_{1,1}^{(2)}$ denote the second polarization operator used in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-008603 | MATHDB-329264

Frobenius-series conjecture for the degree-two monomial family

摘录 / EXCERPT: Let $\mathcal{T}_2$ be the family of all monomials of total degree 2 in $x_{11},\ldots,x_{1n}$, and let $\mathcal{M}_{\mathcal{T}_2}$ be its associated polarization module. Frobe...

conjecture | open | MathDB 页面 | 原文线索

V2-008604 | MATHDB-329265

Frobenius-series conjecture for the degree-three monomial family

摘录 / EXCERPT: Let $\mathcal{T}_3$ be the family of all monomials of total degree 3 in $x_{11},\ldots,x_{1n}$, and let $\mathcal{M}_{\mathcal{T}_3}$ be its associated polarization module. Frobe...

conjecture | open | MathDB 页面 | 原文线索

V2-008605 | MATHDB-329267

Embedding conjecture for polarization modules

摘录 / EXCERPT: Let f be a homogeneous symmetric polynomial of degree d, and let h_m denote the complete homogeneous symmetric function of degree m as used in the source. Embedding conject...

conjecture | open | MathDB 页面 | 原文线索

V2-008606 | MATHDB-329268

h-positivity conjecture for Hilbert series of polarization modules

摘录 / EXCERPT: Let f be any homogeneous symmetric polynomial, and let $\mathcal{M}_f({\bold q})$ be the Hilbert series of its polarization module. h-positivity conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-008607 | MATHDB-329295

The two-variable Beilinson-Flach element main conjecture

摘录 / EXCERPT: Let $\mathcal{K}$ be the imaginary quadratic field in the source, let $\Lambda_\mathcal{K}$ be its two-variable Iwasawa algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008608 | MATHDB-329328

Moduli tracking conjecture under involutive mutation

摘录 / EXCERPT: Let I be the index set and let M be the module in the moduli-tracking setup. Let $\upnu_I$ denote mutation at I, and assume $\upnu_I\upnu_I M\cong M$. Moduli tracking conject...

conjecture | open | MathDB 页面 | 原文线索

V2-008609 | MATHDB-329329

Special MM generators and the three-dimensional Auslander-McKay correspondence

摘录 / EXCERPT: Consider non-Gorenstein rational 3-fold singularities whose minimal models have fibres of dimension at most one. A special MM generator is the proposed non-Gorenstein analogue of...

conjecture | open | MathDB 页面 | 原文线索

V2-008610 | MATHDB-329342

Shuffle-stuffle relations conjecture for multiple zeta values

摘录 / EXCERPT: Let MZVs be the convergent multiple zeta values, with relations generated by products of their iterated-integral and nested-sum descriptions. The resulting identities are called sh...

conjecture | open | MathDB 页面 | 原文线索

V2-008611 | MATHDB-329343

Weight-separation and transcendence conjecture for multiple polylogarithms

摘录 / EXCERPT: Let $\mathcal{A}_n$ be the vector space spanned by all multiple polylogarithmic functions (MPLs) of weight n, with $\mathcal{A}_0=\mathbb{Q}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008612 | MATHDB-329350

Generalized M-polynomial product conjecture for graded Lie algebra components

摘录 / EXCERPT: Generalized M-polynomial product conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008613 | MATHDB-329351

The t=-1 phenomenon for M-polynomials of graded Lie algebra components

摘录 / EXCERPT: The t=-1 phenomenon conjecture. The value $\EuScript M_{\Delta(1)}(-1)$ equals the number of upper ideals I satisfying $I^*=I$.

conjecture | open | MathDB 页面 | 原文线索

V2-008614 | MATHDB-329352

Orlik's orbit-average conjecture for reverse operators on 1-standard gradings

摘录 / EXCERPT: Orlik's orbit-average conjecture. The following assertions hold: (i) ${\mathsf{ord}}(\mathfrak X_{\Delta(1)})=d_1+1$; (ii) the average size of the antichains in every...

conjecture | open | MathDB 页面 | 原文线索

V2-008615 | MATHDB-329353

Extra-special orbit and Lagrangian-ideal conjecture

摘录 / EXCERPT: Extra-special orbit conjecture. The number of $\mathfrak X_{\Delta(1)}$-orbits is $\#\Pi_I$, every orbit has size $h-1$, and, if h is even, every orbit contains a unique Lagrangi...

conjecture | open | MathDB 页面 | 原文线索

V2-008616 | MATHDB-329363

Full faithfulness of the extended Whittaker coefficient functor

摘录 / EXCERPT: Let G be a connected reductive group with connected center. Write $\type{\coeff}_{G,\ext}: \Dmod(\Bun_G)\to\Wh(G,\ext)$ for the extended Whittaker coefficient functor from D-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-008617 | MATHDB-329368

Rationality conjecture for linear relations among brackets

摘录 / EXCERPT: Let a collection consist of mono- or bi-brackets, viewed as elements of the relevant bracket algebra over $\mathbb Q$ and $\mathbb C(q)$. Rationality conjecture for bracket relatio...

conjecture | open | MathDB 页面 | 原文线索

V2-008618 | MATHDB-329411

Hochster's finiteness conjecture for rank-one maximal Cohen-Macaulay modules

摘录 / EXCERPT: Let R be a local normal domain over an algebraically closed field. The divisor class group $\operatorname{Cl}(R)$ is the group of rank-one reflexive classes. Hochster's finitenes...

conjecture | open | MathDB 页面 | 原文线索

V2-008619 | MATHDB-329412

Strict-henselization finiteness conjecture for rank-one maximal Cohen-Macaulay modules

摘录 / EXCERPT: Let R be a local normal domain, and let R^{sh} denote its strict henselization. The group $\operatorname{Cl}(R^{sh})$ is the divisor class group of the strict henselization. St...

conjecture | open | MathDB 页面 | 原文线索

V2-008620 | MATHDB-329429

Monodromy transposition conjecture for the spectral branching points

摘录 / EXCERPT: Monodromy conjecture. For any $\sigma_{i,j}\in\Sigma_n$ and any sufficiently large positive base point B, the monodromy corresponding to $\mathcal P_{i,j}$ is the simple transpos...

conjecture | open | MathDB 页面 | 原文线索

V2-008621 | MATHDB-329443

The p-nilary group-ring conjecture for fields of positive characteristic

摘录 / EXCERPT: Let F be a field with $\operatorname{char}(F)>0$ and let G be a group. For every nontrivial finite normal subgroup H of G such that $|H|=0$ in F, the p-nilary group-ring...

conjecture | open | MathDB 页面 | 原文线索

V2-008622 | MATHDB-329463

The exceptional symmetric-group factorization conjecture for bismash products

摘录 / EXCERPT: The exceptional factorization conjecture. The bismash product H has an irreducible module with Schur indicator -1.

conjecture | open | MathDB 页面 | 原文线索

V2-008623 | MATHDB-329465

The non-exceptional symmetric-group factorization conjecture for Schur indicators

摘录 / EXCERPT: Non-exceptional factorization conjecture. The Schur indicator of every irreducible H-module is 0 or 1.

conjecture | open | MathDB 页面 | 原文线索

V2-008624 | MATHDB-329490

Hrubeš's rank conjecture for anticommuting matrix families

摘录 / EXCERPT: Let $e_{1},\dots,e_{k}$ be an anticommuting family of $n\times n$ complex matrices, meaning that $e_i e_j=-e_j e_i$ for every distinct i,j, and let $\mathrm{rk}(A)$ denote matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-008625 | MATHDB-329504

The Huneke-Jorgensen-Wiegand tensor-product Tor-vanishing conjecture

摘录 / EXCERPT: Huneke-Jorgensen-Wiegand conjecture. If $M\otimes_R N$ is a $(c+1)$st syzygy and M has rank, must

open_problem | open | MathDB 页面 | 原文线索

V2-008626 | MATHDB-329505

The Moore-Piepmeyer-Spiroff-Walker eta-pairing conjecture for complete intersections

摘录 / EXCERPT: Moore-Piepmeyer-Spiroff-Walker conjecture. If R is a complete intersection of codimension $c\geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-008627 | MATHDB-329512

Nonvanishing of specializations of the algebraic anticyclotomic p-adic L-function

摘录 / EXCERPT: Let $\mathbf f$ be the Hida family over K, let $L_p^{\rm alg}(\mathbf f/K)\in\mathbb I[[\Gamma_\infty]]$ be its algebraic anticyclotomic p-adic L-function, and let κ d...

conjecture | open | MathDB 页面 | 原文线索

V2-008628 | MATHDB-329542

Bardet-Faugère-Salvy-Yelin genericity conjecture for semi-regular sequences

摘录 / EXCERPT: Let a semi-regular sequence be a sequence of homogeneous elements in the relevant polynomial-ring setting, and let the number of variables tend to infinity. Bardet-Faugère-Salvy-Ye...

conjecture | open | MathDB 页面 | 原文线索

V2-008629 | MATHDB-329543

Bardet-Faugère-Salvy genericity conjecture for binary algebraic systems

摘录 / EXCERPT: For integers $n,m,d_1,\dots,d_m$, let $E(n,m,d_1,\dots,d_m)$ be the set of algebraic systems of m equations of degrees $d_1,\dots,d_m$ in n variables over ${\mathbb F}_2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008630 | MATHDB-329544

Genericity conjecture for degree-d semi-regular sequences of length n

摘录 / EXCERPT: For integers n and d with $1\leq d\leq n$, let $\pi(n,d)$ be the proportion of sequences of degree d and length n in n variables that are semi-regular. Degree-d genericit...

conjecture | open | MathDB 页面 | 原文线索

V2-008631 | MATHDB-329545

Fixed-degree genericity conjecture for semi-regular sequences

摘录 / EXCERPT: Fix an integer $d>1$. For integers n and m, let $\pi(n,m,d)$ be the proportion of sequences of degree d and length m in n variables that are semi-regular. Fixed-degree ge...

conjecture | open | MathDB 页面 | 原文线索

V2-008632 | MATHDB-329552

The quadratic-polynomial arboreal Galois rigidity conjecture

摘录 / EXCERPT: For $a\in\mathbb{Z}$, let $\phi_a(x)=x^2+a$, let T_n denote the rooted binary tree of height n, let T_∞ denote the infinite rooted binary tree, and let $G_n(\phi_a)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008633 | MATHDB-329555

Conca-Krattenthaler-Watanabe power-sum regular-sequence conjecture

摘录 / EXCERPT: Conca-Krattenthaler-Watanabe's conjecture. The polynomials $p_a(3),p_b(3),p_c(3)$ form a regular sequence in S. Conca et al. verified some special cases, but the conjecture remai...

conjecture | open | MathDB 页面 | 原文线索

V2-008634 | MATHDB-329561

Scheimbauer’s factorization-algebra comparison conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a nice $\use@mathgroup \M@U \symAMSb{E}_{n}$-monoidal ∞-category. Let $\mathfrak{Alg}_{n}^{\textup{FA}}(\mathcal{C})$ denote Scheimbauer’s...

conjecture | open | MathDB 页面 | 原文线索

V2-008635 | MATHDB-329601

Park-Shahabi signed main conjecture

摘录 / EXCERPT: Let E/F be the CM elliptic curve in the paper, let F^{cyc} be the cyclotomic extension, and let Λ_{cyc} be its Iwasawa algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008636 | MATHDB-329602

Park-Shahabi Rubin-Stark determinant and nonvanishing conjecture

摘录 / EXCERPT: Let $\mathcal{S}^{E,p}_{\infty}$ be the cyclic Λ-module generated by the tower of Rubin-Stark elements after localization and twisting, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008637 | MATHDB-329633

Akbari-Mohammadian-Radjavi-Raja's commuting graph diameter conjecture

摘录 / EXCERPT: Let $\mathbb{F}$ be a field and let $\operatorname{Mat}_n(\mathbb{F})$ be the algebra of $n\times n$ matrices over $\mathbb{F}$. Let $\Gamma(\mathbb{F},n)$ be the graph whose verti...

conjecture | open | MathDB 页面 | 原文线索

V2-008638 | MATHDB-329648

Even-parameter real-point conjecture for the system L_m

摘录 / EXCERPT: Even-parameter real-point conjecture. Each point $(g_0,g_1,\dots,g_{m-1},h)\in L_m\cap(\mathbb{R}^*\times(\mathbb{S}^1)^m)$ satisfies $g_0^2\geqslant1/(m+1)$. In fact, each such po...

conjecture | open | MathDB 页面 | 原文线索

V2-008639 | MATHDB-329742

Existence of a quasi-bialgebra with preantipode that is not quasi-Hopf

摘录 / EXCERPT: A quasi-bialgebra is an algebraic structure with coproduct, counit, and associator satisfying the quasi-bialgebra axioms; a preantipode is a map satisfying the three preantipode ax...

conjecture | open | MathDB 页面 | 原文线索

V2-008640 | MATHDB-329750

The equality conjecture for the first formal Galois group injection

摘录 / EXCERPT: Let q be a parameter and let the inclusion in Proposition 5.24i be the inclusion of the corresponding formal Galois functors. Equality conjecture. If q is not a root of unity,...

conjecture | open | MathDB 页面 | 原文线索

V2-008641 | MATHDB-329751

The bijectivity conjecture for the third formal Galois group injection

摘录 / EXCERPT: Let A be a commutative $L^{\natural}$-algebra, and let the injection in Proposition 6.8m be the functorial map from $\mathcal{CF}_{L/k}(A)$ to the formal group $\hat{G}_3(A)$. Bi...

conjecture | open | MathDB 页面 | 原文线索

V2-008642 | MATHDB-329752

The noncommutative formal Galois group bijectivity conjecture

摘录 / EXCERPT: Noncommutative formal Galois group conjecture. If q is not a root of unity, the injection in Proposition 6.8o is a bijection. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-008643 | MATHDB-329755

Local geometric Langlands functoriality conjecture for the bimodule K($\boldsymbol{\mathcal{X}}$)

摘录 / EXCERPT: Let G and H be split reductive connected groups over $\mathbf{k}$, let $\check{G}$ and $\check{H}$ be their dual groups, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008644 | MATHDB-329756

Geometric Howe correspondence conjecture for K($\boldsymbol{\mathcal{X}}$)

摘录 / EXCERPT: Let (G,H) be a dual split reductive pair over $\mathbf{k}$, let $\mathcal{W}$ be the graded Weil category, and let $(D\mathcal{W})^{I_{G}\times I_{H}}$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-008645 | MATHDB-329757

Howe correspondence conjecture for dual general linear pairs

摘录 / EXCERPT: Let $G=\mathbf{GL}_{n}$ and $H=\mathbf{GL}_{m}$ with $n\leq m$. Let $K(\mathcal{X})$ and $K(DP_{I_{H}\times I_{G}}(\Pi(F)))$ be the associated bimodules under the extended affine H...

conjecture | open | MathDB 页面 | 原文线索

V2-008646 | MATHDB-329800

Recursive Fröberg conjecture for adjoining a generic form

摘录 / EXCERPT: Let $S=\Bbbk[x_1,\ldots,x_n]$, let $J\subset S$ be generated by generic forms, and let $f\in S$ be a generic form of degree e. Recursive Fröberg conjecture. For every integer j...

conjecture | open | MathDB 页面 | 原文线索

V2-008647 | MATHDB-329832

The left-cell characterization by the enhanced right descent invariant

摘录 / EXCERPT: Let (W,S) be a Coxeter system with weight function p, and let $\tau^{\Delta_2,\mathcal{R}^p}$ denote the invariant associated with the strongly Kazhdan-Lusztig-admissible pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-008648 | MATHDB-329834

Iterated center of an indecomposable bimodule is trivial

摘录 / EXCERPT: Bimodule center conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008649 | MATHDB-329905

Maximal dimension conjecture for Kummer spaces in tensor products of cyclic algebras

摘录 / EXCERPT: Let d be a positive integer, let F be an infinite field of characteristic 0 containing a primitive dth root of unity, and let ... be a division algebra, where each factor is...

conjecture | open | MathDB 页面 | 原文线索

V2-008650 | MATHDB-329955

The powers-of-generic-forms conjecture for equal degrees

摘录 / EXCERPT: Let $R=\mathbb{C}[x_1,\ldots,x_n]$, with $g_1,\ldots,g_k$ generic forms of degree $d\geq2$, and let $m\in\mathbb{N}$. Set ... For a power series $F(t)$, write $\lceil F(t)\rceil$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-008651 | MATHDB-329991

Exceptional Dynkin zigzag intrinsic formality conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be a ground field of characteristic zero, let Γ be a tree of type E_6, E_7, or E_8, and let A_Γ be the corresponding zigzag algebra. Exceptiona...

conjecture | open | MathDB 页面 | 原文线索

V2-008652 | MATHDB-330017

Hilbert depth equals Stanley depth for sums of maximal ideals and free modules

摘录 / EXCERPT: Let $R=\mathds{K}[X_1,\ldots,X_n]$ be a polynomial ring in an arbitrary positive number of variables, let $\mathfrak m=(X_1,\ldots,X_n)$ be its homogeneous maximal ideal, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008653 | MATHDB-330037

The strong Koszul conjecture for finite Coxeter arrangements

摘录 / EXCERPT: Let $F\subset\mathbb{C}^n$ be a finite Coxeter arrangement. A reduced equation of F is a strongly Koszul free divisor when the principal symbols of the logarithmic differential o...

conjecture | open | MathDB 页面 | 原文线索

V2-008654 | MATHDB-330145

Mutation invariance of maximal green sequences for surface quivers

摘录 / EXCERPT: Mutation-invariance conjecture. If Q exhibits a maximal green sequence, then Q' also exhibits a maximal green sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-008655 | MATHDB-330168

The Lie algebra arithmetic transfer conjecture

摘录 / EXCERPT: Lie algebra arithmetic transfer conjecture. (a) There exists $\phi'\in C_c^\infty(\mathfrak{s})$ transferring to $(\mathbf{1}_{\mathfrak{k}_0},0)$ such that, whenever matching...

conjecture | open | MathDB 页面 | 原文线索

V2-008656 | MATHDB-330184

The generalized right-tight conjecture on essential socles

摘录 / EXCERPT: Let R be a ring, let $E(R_R)$ be the injective envelope of the right regular module, and call R right generalized R-tight when every cyclic submodule of $E(R_R)$ embeds in a...

conjecture | open | MathDB 页面 | 原文线索

V2-008657 | MATHDB-330212

Calabi-Yau contraction isomorphism conjecture for smooth dg algebras

摘录 / EXCERPT: Calabi-Yau contraction isomorphism conjecture. Under the formality assumption, the induced morphism of dg k-modules

conjecture | open | MathDB 页面 | 原文线索

V2-008658 | MATHDB-330230

Galois conjugacy conjecture for periodic cycles of quadratic polynomials

摘录 / EXCERPT: Galois conjugacy conjecture. There are some $m\in\mathbb{Z}/g\mathbb{Z}$ and some nontrivial $\tau\in\operatorname{Gal}(K/\mathbb{Q})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008659 | MATHDB-330239

Hilbert-function criterion for freeness of plane curves

摘录 / EXCERPT: Let $C:f=0$ be a plane curve of degree d. Let $ct(f)$ and $st(f)$ denote the coincidence and stability thresholds of its Milnor algebra, and let $T=3(d-2)$. Freeness criterion co...

conjecture | open | MathDB 页面 | 原文线索

V2-008660 | MATHDB-330243

Douglas-Reinbacher-Yau strong Bogomolov inequality for simply connected surfaces

摘录 / EXCERPT: Douglas-Reinbacher-Yau conjecture. The Chern classes should satisfy the improved Bogomolov inequality

conjecture | refuted | MathDB 页面 | 原文线索

V2-008661 | MATHDB-330308

The elliptic Broadhurst-Kreimer conjecture for geometric derivations

摘录 / EXCERPT: Let $\mathfrak{d}$ be the depth-graded Lie algebra embedded in the B-graded Lie algebra of derivations of the free Lie algebra $\mathbb L(a,b)$ preserving Θ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008662 | MATHDB-330328

Noether's maximal rank conjecture for linear series on general curves

摘录 / EXCERPT: Noether's maximal rank conjecture. The maps μ_m have maximal rank for all m; that is, each μ_m is either injective or surjective. The conjecture is a central open problem...

conjecture | open | MathDB 页面 | 原文线索

V2-008663 | MATHDB-330349

Severi equality characterization for irregular surfaces

摘录 / EXCERPT: Let X be a minimal surface of general type and maximal Albanese dimension. The Severi equality characterization states that ... if and only if the canonical model of X is a flat...

conjecture | open | MathDB 页面 | 原文线索

V2-008664 | MATHDB-330350

Manetti's geography inequality for irregular surfaces

摘录 / EXCERPT: Let X be a minimal surface of general type and maximal Albanese dimension with irregularity $q\geq 4$. Manetti's geography conjecture. One should have ... This inequality concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-008665 | MATHDB-330379

The conjectural classification of exceptional mod-ll Galois images over the rationals

摘录 / EXCERPT: Exceptional-image classification conjecture. $G_E(\ell)$ is either equal to $\mathbf{GL}_2(\ell)$ or conjugate to one of the 63 groups listed in the specified tables.

conjecture | open | MathDB 页面 | 原文线索

V2-008666 | MATHDB-330389

Conjecture on dependent eigenvectors and the quasi-inverse matrix

摘录 / EXCERPT: Let A be a nonsingular matrix with n distinct eigenvalues. Suppose that the eigenvectors of A are dependent, and let A^∇ denote its quasi-inverse matrix. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008667 | MATHDB-330390

Conjecture on eigenvector independence for quasi-inverse matrices

摘录 / EXCERPT: Let A be a nonsingular matrix, and let A^∇ denote its quasi-inverse matrix. Suppose that A^∇ has n distinct eigenvalues. Quasi-inverse eigenvector conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-008668 | MATHDB-330417

Transcendence conjecture for continued fractions with k-regular terms

摘录 / EXCERPT: Let $k\geq 2$ be an integer, and let $(a_n)_{n\geq 0}$ be a k-regular sequence of positive integers, meaning that its k-kernel ... is contained in a finitely generated module. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-008669 | MATHDB-330423

Normalized supercharacter formula for integrable osp(3|2) modules

摘录 / EXCERPT: Let $L(\Lambda)$ be an integrable nonzero-level module over the affine Lie superalgebra in the surrounding section, with $\Lambda\in\Omega_k$ or $\Lambda\in\Omega'_k$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-008670 | MATHDB-330431

Disjointness conjecture for splitting fields of factorial polynomials

摘录 / EXCERPT: Let f_n be the polynomial defined in the paper and let $\operatorname{Spl}(f_n)$ denote its splitting field over $\mathbb{Q}$. Let $n_1\ne n_2$ be positive integers. Splitting-fi...

conjecture | open | MathDB 页面 | 原文线索

V2-008671 | MATHDB-330456

The Lie derivation property conjecture for generalized matrix algebras

摘录 / EXCERPT: Let $\mathcal{G}$ be a generalized matrix algebra and let $\mathcal{M}$ be a faithful $\mathcal{A}$-$\mathcal{B}$-bimodule, with projection maps $\pi_{\mathcal{A}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008672 | MATHDB-330496

The Bernstein twisted extended-quotient conjecture

摘录 / EXCERPT: Let F) be a non-archimedean local field and G a connected reductive algebraic group over F. For a Levi subgroup L of G, write $W(G,L)=N_G(L)/L$. For an inertial equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-008673 | MATHDB-330497

The L-packets conjecture for Bernstein components

摘录 / EXCERPT: Let F be a non-archimedean local field, G a connected reductive algebraic group over F, and ${\mathfrak s}=[L,\omega]_G$ an inertial equivalence class. Let $T_{\mathfrak s}$...

conjecture | open | MathDB 页面 | 原文线索

V2-008674 | MATHDB-330542

The maximal rank conjecture for multiplication maps of general linear series

摘录 / EXCERPT: Maximal rank conjecture. The multiplication maps μ_m have maximal rank for all m.

conjecture | open | MathDB 页面 | 原文线索

V2-008675 | MATHDB-330571

Existence of an integral domain with non-plethory integer-valued polynomial ring

摘录 / EXCERPT: Let D be an integral domain. Non-plethory-domain conjecture. There exists an integral domain D such that ${\operatorname{Int}}(D)$ is not an (idempotent) D-plethory. The prec...

conjecture | open | MathDB 页面 | 原文线索

V2-008676 | MATHDB-330577

A conjectural formula for the first coefficient of the maximum likelihood degree polynomial

摘录 / EXCERPT: Let $m\geq 2$, and let Λ_m be the object whose coefficients are $\lambda_1,\dots,\lambda_{m-1}$. Conjectural formula. For $m\geq 2$, the first coefficient satisfies ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-008677 | MATHDB-330582

Conjecture on bimodules of maximal amenable subalgebras of free group factors

摘录 / EXCERPT: Let $t\in(1,\infty]$ and let $N\subseteq L(\mathbb{F}_{t})$ be a maximal amenable von Neumann subalgebra. Regard the following as N-N bimodules, where...

conjecture | open | MathDB 页面 | 原文线索

V2-008678 | MATHDB-330609

Castelnuovo-Mumford regularity and projective dimension for ideals of poset homomorphisms

摘录 / EXCERPT: Let P and Q be posets with s and t connected components, respectively. Write $\mathfrak{a}(P)$ and $\mathfrak{a}(Q)$ for the invariants appearing in the proposed formulas,...

conjecture | open | MathDB 页面 | 原文线索

V2-008679 | MATHDB-330610

Gersten's conjecture for regular local rings

摘录 / EXCERPT: Let $X=\operatorname{Spec}(R)$ be the spectrum of a regular local ring R. For each integer l, let $\mathrm{M}_l(X)$ be the category of coherent sheaves on X whose supports ha...

conjecture | open | MathDB 页面 | 原文线索

V2-008680 | MATHDB-330618

The non-commutative extension of the matrix power-sum theorem

摘录 / EXCERPT: Non-commutative power-sum conjecture. The theorem for finite commutative rings remains true for non-commutative R: $S_k^d(R)=0$ unless $d=2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-008681 | MATHDB-330650

The product-cluster reduction lemma for LP algebras

摘录 / EXCERPT: Let two variables belong to different clusters of an LP algebra, and let the base ring be the coefficient ring in which the algebra is defined. Product-cluster reduction lemma. The...

open_problem | open | MathDB 页面 | 原文线索

V2-008682 | MATHDB-330651

Surjectivity conjecture for word maps on $PSL_2(\mathbb C)$

摘录 / EXCERPT: Surjectivity conjecture. For every non-identity word $w\in F(x,y)$, the word map

conjecture | open | MathDB 页面 | 原文线索

V2-008683 | MATHDB-330662

Exchange-graph isomorphism and seed-exchange-matrix reconstruction conjecture

摘录 / EXCERPT: Let $d4d0$ and $d4d0'$ be connected cluster algebras, each of finite type or skew-symmetric finite mutation type. Let $d4e2=(\mathbf{x},B)$ and $d4e2'=(\mathbf{x}',B')$ be seeds of...

conjecture | open | MathDB 页面 | 原文线索

V2-008684 | MATHDB-330703

The conjecture on Frobenius-twisted unstable-module extensions

摘录 / EXCERPT: The Frobenius-extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008685 | MATHDB-330731

The subadditive-labelling criterion for Zamolodchikov integrability

摘录 / EXCERPT: Let Q be a recurrent quiver: its underlying graph is bipartite, and mutating at its black vertices followed by mutating at its white vertices returns the same quiver. A positive...

conjecture | open | MathDB 页面 | 原文线索

V2-008686 | MATHDB-330732

The weakly subadditive-labelling criterion for Arnold-Liouville integrability

摘录 / EXCERPT: Let Q be a recurrent quiver, meaning that its underlying graph is bipartite and mutating at its black vertices followed by its white vertices returns the same quiver. A positive...

conjecture | open | MathDB 页面 | 原文线索

V2-008687 | MATHDB-330733

Integrability of upper cluster algebra elements for recurrent quivers

摘录 / EXCERPT: Let Q be a recurrent quiver and let $\mu_-\mu_+$ be its associated mutation map. Let the associated upper cluster algebra be the intersection of the Laurent polynomial rings of i...

conjecture | open | MathDB 页面 | 原文线索

V2-008688 | MATHDB-330734

Single-tensor-diagram representations of cluster variables

摘录 / EXCERPT: Let $\mathfrak T^*$ be an initial seed of a cluster algebra, and let X be a cluster variable in the cluster algebra generated from that seed.
Single-tensor-diagram conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-008689 | MATHDB-330736

Novikov's conjecture for algebraic K-theory

摘录 / EXCERPT: Novikov conjecture for K-theory. The assembly map

conjecture | open | MathDB 页面 | 原文线索

V2-008690 | MATHDB-330754

Elementary type conjecture for finitely generated Demushkin pro-p groups

摘录 / EXCERPT: Let K be a field such that its maximal pro-p Galois quotient $G_K(p)$ is a finitely generated pro-p Demushkin group of rank at least 3. Elementary type conjecture for Demus...

conjecture | open | MathDB 页面 | 原文线索

V2-008691 | MATHDB-330757

Universal twined elliptic genus conjecture for K3 sigma models

摘录 / EXCERPT: Let Λ be the Leech lattice, and let $V^{s\natural}$ be the canonically-twisted module considered in the paper. A supersymmetry-preserving automorphism of a supersymmetric n...

conjecture | open | MathDB 页面 | 原文线索

V2-008692 | MATHDB-330765

Conjectural presentation of the extended triple affine Artin group

摘录 / EXCERPT: Let $\dddot{X}_n$ be a double affine Coxeter diagram, let $B(X_n)$ be the Artin group associated to its finite subdiagram, and let $\mho_{01}$, $\mho_{02}$, and $\mho_{03}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-008693 | MATHDB-330766

Conjectural presentation of the extended double affine Artin group

摘录 / EXCERPT: Let $\ddot{X}_n$ be a double affine Coxeter diagram with finite nodes labelled $1,\ldots,n$. Let $B(X_n)$ be the Artin group of the finite subdiagram, and let $\mho$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008694 | MATHDB-330781

Kedlaya-Sutherland's Sato-Tate conjecture for Jacobians of curves

摘录 / EXCERPT: Let C be an absolutely irreducible nonsingular projective curve of genus g over $\mathbb{Q}$, and let $\operatorname{Jac} C$ be its Jacobian. Assume that ... For a curve with goo...

conjecture | open | MathDB 页面 | 原文线索

V2-008695 | MATHDB-330817

Determinant-module formulation of the Iwasawa main conjecture

摘录 / EXCERPT: Determinant-module Iwasawa main conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008696 | MATHDB-330845

Ding's index-generalized Löewy length conjecture for Gorenstein local rings

摘录 / EXCERPT: Let $(R,\mathfrak{m},k)$ be a Gorenstein local ring of dimension d. The Ding conjecture. ... This conjecture concerns the relationship between Auslander's index and generalized Löe...

conjecture | open | MathDB 页面 | 原文线索

V2-008697 | MATHDB-330866

Lusztig's conjecture on the twisted involution module and a Hecke algebra ideal

摘录 / EXCERPT: Lusztig's conjecture. With $\mathcal{H}^{\mathbb Q(u)}=\mathbb Q(u)\otimes_{\mathcal A}\mathcal H_u$, there is a unique isomorphism of $\mathcal H^{\mathbb Q(u)}$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-008698 | MATHDB-330898

Brüstle-Dupont-Perotin's interval conjecture for maximal green sequence lengths

摘录 / EXCERPT: Let Q be a cluster quiver, meaning a finite connected quiver without loops or oriented cycles. A maximal green sequence is a green mutation sequence that cannot be extended by an...

conjecture | open | MathDB 页面 | 原文线索

V2-008699 | MATHDB-330899

Surjectivity conjectures for special automorphisms and Keller maps

摘录 / EXCERPT: Let R be a $\mathbb{Z}$-algebra and let k be any field. Suppose there is a surjective ring homomorphism $R\longrightarrow k$, and let π denote the induced map on polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-008700 | MATHDB-330900

Lifting strong Keller maps through a surjective ring homomorphism

摘录 / EXCERPT: Let R be a $\mathbb{Z}$-algebra and let k be a field of characteristic p. Suppose there is a surjective ring homomorphism $R\longrightarrow k$, and let π be the induced m...

conjecture | open | MathDB 页面 | 原文线索

V2-008701 | MATHDB-330902

The linear-eigenvalue-growth conjecture for the Hermitian-Toeplitz matrix T

摘录 / EXCERPT: Linear-eigenvalue-growth conjecture. There exists an upper bound on the largest eigenvalue of $\mathbf{T}$ that grows linearly with N:

conjecture | open | MathDB 页面 | 原文线索

V2-008702 | MATHDB-330911

Positive-characteristic weak Lefschetz conjecture for Artinian monomial algebras of type two

摘录 / EXCERPT: Let K be a field and let $R=K[x,y,z]$. An Artinian monomial ideal $I\subset R$ gives an Artinian monomial algebra R/I, whose type is the dimension of its socle. The algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008703 | MATHDB-330935

Kaneko-Zagier isomorphism conjecture for finite multiple zeta values

摘录 / EXCERPT: Let $\FMZ_{w,1}$ be the $\Q$-vector space generated by finite multiple zeta values of weight w and superbity one, and let $\MZV_w$ be the corresponding weight-w space of multip...

conjecture | open | MathDB 页面 | 原文线索

V2-008704 | MATHDB-330966

Conjecture on $\mathscr{C}^{N,1,1}$ and Xu's conformal-inclusion category

摘录 / EXCERPT: The $\mathscr{C}^{N,1,1}$ conjecture. The unitary fusion category $\mathscr{C}^{N,1,1}$ is isomorphic to Xu's bimodule category for this conformal inclusion. The source verifies th...

conjecture | open | MathDB 页面 | 原文线索

V2-008705 | MATHDB-330975

Surjectivity isomorphism conjecture for endomorphisms of the identity

摘录 / EXCERPT: Let B be the graded algebra under consideration, let $B_\psi=B/\langle b-\psi(b)\mid b\in B\rangle$, and let C be the graded vector space with basis v_d for...

conjecture | open | MathDB 页面 | 原文线索

V2-008706 | MATHDB-331041

Cooper-Embree-Ha-Ha-Hoefel lower-bound conjecture for monomial ideals

摘录 / EXCERPT: Let I be a monomial ideal, and let $\operatorname{big-height}(I)$ denote the maximum of the heights of the associated primes $P\in\operatorname{Ass}(I)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008707 | MATHDB-331050

The even-index lower central series product noncontainment conjecture

摘录 / EXCERPT: Even-index product conjecture. If $i,j\geq 2$ are both even, then in general

conjecture | open | MathDB 页面 | 原文线索

V2-008708 | MATHDB-331051

The Weil-generic vanishing conjecture for lower central series quotients

摘录 / EXCERPT: Weil-generic vanishing conjecture. For Weil generic P of degree d,

conjecture | open | MathDB 页面 | 原文线索

V2-008709 | MATHDB-331058

Kaplansky-Lvov multilinear conjecture for matrix algebras

摘录 / EXCERPT: Let f be a multilinear polynomial with complex coefficients, and let $d\in\mathbb{N}$. Consider the set of values obtained by evaluating f on $M_d(\mathbb{C})$. Kaplansky-Lvov...

conjecture | open | MathDB 页面 | 原文线索

V2-008710 | MATHDB-331089

Mináč-Tân's Kernel n-Unipotent Conjecture

摘录 / EXCERPT: Let F be a field containing a primitive p-th root of unity, let $G=G_F(p)$ be the maximal pro-p quotient of its absolute Galois group, and let $n\geq 3$ be an integer. A pro-...

conjecture | open | MathDB 页面 | 原文线索

V2-008711 | MATHDB-331096

GCD representatives for compressed zero-divisor graph classes in quotients of UFDs

摘录 / EXCERPT: GCD-representative conjecture. For every $a\in D$,

conjecture | open | MathDB 页面 | 原文线索

V2-008712 | MATHDB-331097

Compressed zero-divisor graph description for ideals in UFDs

摘录 / EXCERPT: Compressed-graph structure conjecture. An edge joins the vertices represented by $\"]\phi(\prod_i d_i)$ and $\"]\phi(\prod_i b_i)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008713 | MATHDB-331098

Compressed-graph invariance for similarly factored ideals in UFD quotients

摘录 / EXCERPT: Compressed-graph invariance conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-008714 | MATHDB-331114

The Hecke-algebra structure conjecture for Bernstein components

摘录 / EXCERPT: Let G be a reductive p-adic group, let $\mathfrak{s}=[L,\omega]_G$ be an inertial equivalence class, let $R_{\mathfrak{s}}$ be the root system defined by the poles of the Haris...

conjecture | open | MathDB 页面 | 原文线索

V2-008715 | MATHDB-331116

Flath-Towber conjectures on defining equations of reductive algebraic groups

摘录 / EXCERPT: Flath-Towber conjectures. The homomorphism μ is surjective, and the ideal $\ker\mu$ in $\mathcal S^+\otimes_k\mathcal S^-$ is generated by the invariant subspace $(\ker\mu)^G$....

conjecture | open | MathDB 页面 | 原文线索

V2-008716 | MATHDB-331121

Main conjecture for cuspidal automorphic modules

摘录 / EXCERPT: Let G_n be a classical group over F and a pure inner F-form of an F-quasisplit classical group G_n^*. For $\pi\in{\mathcal {A}}_{\mathrm{cusp}}(G_n)$ in the global Arthur...

conjecture | open | MathDB 页面 | 原文线索

V2-008717 | MATHDB-331160

Conjecture on nonsingular lifts of non-triangular definite matrices

摘录 / EXCERPT: Let B be a non-triangular definite matrix in $T^\ell T^u$, where T^ℓ and T^u are the lower- and upper-triangular submonoids generated by elementary matrices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008718 | MATHDB-331213

Generalized Gersten injectivity conjecture for abstract blow-up squares

摘录 / EXCERPT: Let A be a Noetherian local $\mathbb Q$-algebra, let $I\subseteq A$ be an ideal, let $n\in\mathbb Z$, and consider an abstract blow-up square ... Assume that X is regular. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-008719 | MATHDB-331236

Sartori-Stroppel's conjecture on supersymmetric generators of the walled Brauer center

摘录 / EXCERPT: Sartori-Stroppel's conjecture. For every $\delta\in\mathbb C$, the center of $B_{r,s}(\delta)$ is generated by the supersymmetric polynomials in $L_1,\ldots,L_{r+s}$.

conjecture | open | MathDB 页面 | 原文线索

V2-008720 | MATHDB-331248

Caro's conjecture on direct images of overconvergent isocrystals

摘录 / EXCERPT: Caro's conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-008721 | MATHDB-331249

Rational permutation braid conjecture for type D_n

摘录 / EXCERPT: Rational permutation braid conjecture. Every element of ${\boldsymbol D}_c$ is a rational permutation braid.

conjecture | open | MathDB 页面 | 原文线索

V2-008722 | MATHDB-331269

Existence of sheaves realizing connected seminormal partial normalizations

摘录 / EXCERPT: Existence conjecture. There exists a sheaf $\mathcal I\in\overline J_{\mathrm{C}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008723 | MATHDB-331279

Acyclicity conjecture for the involutive quotient bialgebra

摘录 / EXCERPT: Let X be a set with an involutive solution σ of the Yang-Baxter equation, satisfying $\sigma(x,y)^2=(x,y)$. Let B be the differential bialgebra associated to this soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-008724 | MATHDB-331287

Classical affine Schur-Weyl kernel conjecture

摘录 / EXCERPT: Let $E[AS_m]$ be the group algebra of the affine symmetric group, let $\chi:E[AS_m]\to\operatorname{End}_{\operatorname{Sym}(\underline{x})}(R^m_n)$ be the specialization at $q=1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-008725 | MATHDB-331307

The level-rank duality Hilbert-series conjecture for SH^c modules

摘录 / EXCERPT: Let N and M be the parameters of the level-rank duality, let X be the partially ordered set defined by the relations in the source, and let $\mathcal{A}(X)$ be the set of all...

conjecture | open | MathDB 页面 | 原文线索

V2-008726 | MATHDB-331388

The Main Conjecture on AIM loops

摘录 / EXCERPT: The Main Conjecture. The quotient $Q/N(Q)$ is an abelian group and $Q/Z(Q)$ is a group. In particular, Q is nilpotent of class at most 3.

conjecture | open | MathDB 页面 | 原文线索

V2-008727 | MATHDB-331400

Aliev-Smyth's conjecture on isolated torsion points and Newton polytopes

摘录 / EXCERPT: Let $f\in\mathbb{C}[X_1,\ldots,X_n]$, let $Z(f)\subset\mathbb{G}_{\mathrm{m}}^n$ be its zero set, and let Δ be the Newton polytope of f. An isolated torsion point of...

conjecture | open | MathDB 页面 | 原文线索

V2-008728 | MATHDB-331405

Coefficient-field enlargement conjecture for compatible lisse sheaves

摘录 / EXCERPT: Let X, E, λ, and $\mathcal{E}$ be as in the compatibility conjecture for de Rham lisse sheaves, and let conditions (i) and (ii) denote respectively the coefficient cond...

conjecture | open | MathDB 页面 | 原文线索

V2-008729 | MATHDB-331421

Uniform discrete-spectrum bound conjecture

摘录 / EXCERPT: Let $\mathbf G$ be a connected reductive linear algebraic group over $\mathbb Q$, let $\eta:\mathbf G\to\operatorname{GL}(n)$ be a faithful rational representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008730 | MATHDB-331449

The epi-morphism kernel generator conjecture

摘录 / EXCERPT: Let k be a field of characteristic zero, and let ... be an epimorphism of k-algebras. Set $I=\ker(\varphi)$, and let $\mu(I)$ denote the minimal number of generators of I. Epi-...

conjecture | open | MathDB 页面 | 原文线索

V2-008731 | MATHDB-331450

The highest weight equivalence conjecture for rational Cherednik category \(\mathcal{O}\)

摘录 / EXCERPT: Let $\mathcal{P}_\ell(n)$ be the poset of ℓ-tuples of diagrams, embedded in $\mathbb{Z}^{\mathbf{s}}$ as described, and let $\mathcal{O}_{\kappa,\mathbf{s}}(n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008732 | MATHDB-331452

Jacquet-Langlands invariance conjecture for automorphic L-invariants

摘录 / EXCERPT: Let F be a totally real field, let ${\mathcal P}$ be a place at which both quaternion algebras split, and let G_1 and G_2 be the multiplicative groups of two quaternion algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-008733 | MATHDB-331465

Etingof-Kim-Ma conjecture on products of commutator ideals

摘录 / EXCERPT: Etingof-Kim-Ma conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008734 | MATHDB-331483

Conjecture on differential algebraicity of four permutation-class generating functions

摘录 / EXCERPT: For a permutation $\mathcal{C}$, write $F_{\mathcal{C}}(x)$ for its generating function. A formal power series is differentially algebraic if there exist $k\geq 0$ and a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-008735 | MATHDB-331508

Polynomiality conjecture for symmetric invariants of matrix semi-direct products

摘录 / EXCERPT: Let $n>m>k\geqslant 1$, and let $\mathfrak q$ and V be the Lie algebra and representation considered in the paper, with G acting on V^*. Let ${\bf F}_I$ denote the explicitly...

conjecture | open | MathDB 页面 | 原文线索

V2-008736 | MATHDB-331536

Niv's characteristic-polynomial conjecture for supertropical adjoint matrices

摘录 / EXCERPT: Let $(\mathcal{G},\ast,0,\leq)$ be an ordered Abelian group, let $\mathcal{S}=\mathcal{G}^{(0)}\cup\mathcal{G}^{(1)}\cup\{\varepsilon\}$ be the associated supertropical semifield,...

conjecture | open | MathDB 页面 | 原文线索

V2-008737 | MATHDB-331547

Upper-bound intersection conjecture for noncommutative polygons

摘录 / EXCERPT: For each $n\ge 2$, let $\mathcal{U}_n$ be the subalgebra generated by all x_{ij}, $i\ne j$, and $x_{i,i^\pm}^{-1}$, and let $\mathcal{A}_\Delta$ be the algebra associated with a...

conjecture | open | MathDB 页面 | 原文线索

V2-008738 | MATHDB-331548

Monoid parametrization conjecture for units of the noncommutative polygon algebra

摘录 / EXCERPT: Let $\mathbb{T}_n$ be the group generated by the symbols t_{ij} subject to the triangular relations, and let $\mathcal{A}_n$ be the associated algebra with unit group...

conjecture | open | MathDB 页面 | 原文线索

V2-008739 | MATHDB-331549

Divisibility conjecture for surface localization monoids

摘录 / EXCERPT: Let Σ be a marked surface, let Δ be a triangulation, and let $\mathcal{A}_\Sigma$ be the associated algebra. The natural homomorphism ... comes from the inclusion of...

conjecture | open | MathDB 页面 | 原文线索

V2-008740 | MATHDB-331607

Skalit's tangent-cone criterion for intersection multiplicity

摘录 / EXCERPT: Skalit's tangent-cone conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008741 | MATHDB-331652

Positive-characteristic dense-orbit conjecture over transcendental fields

摘录 / EXCERPT: Positive-characteristic dense-orbit conjecture. There exists a point $x\in X(k)$ whose orbit is Zariski dense in X.

conjecture | open | MathDB 页面 | 原文线索

V2-008742 | MATHDB-331660

Okounkov-Bezrukavnikov filtration conjecture for rational Cherednik category O

摘录 / EXCERPT: Let H_m be the rational Cherednik algebra, let $\mathcal{O}_m$ be its category $\mathcal{O}$, let $M_m(\lambda)$ be a Verma module, and let s^m_λ be the stable-basis vect...

conjecture | open | MathDB 页面 | 原文线索

V2-008743 | MATHDB-331670

Giambruno's polynomial exponent conjecture for basic PI algebras

摘录 / EXCERPT: Let F be an algebraically closed field of characteristic zero. Let A be a finite-dimensional basic algebra with Wedderburn-Malcev decomposition ... where $J(A)$ is the Jacobson r...

conjecture | open | MathDB 页面 | 原文线索

V2-008744 | MATHDB-331731

The degree-four rational curve characterization by three non-flexes

摘录 / EXCERPT: Let $C\subset\mathbb P^2$ be a plane rational curve of degree four. Consider the family of curves given by ... where $\beta(\beta-1)(\beta^2+\beta+1)\ne0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008745 | MATHDB-331750

The conjecture that real-closure equals strong 2-closure

摘录 / EXCERPT: Let G be a permutation group. Its real closure is the closure defined using the real permutation module, and its strong 2-closure is the group preserving the relevant strong or...

conjecture | open | MathDB 页面 | 原文线索

V2-008746 | MATHDB-331762

The weak local negativity conjecture for plane curves

摘录 / EXCERPT: Weak local negativity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008747 | MATHDB-331766

The saturation conjecture for additive polynomial mappings

摘录 / EXCERPT: Let K be the global function field in the setup, let $F\in\operatorname{End}_{\mathbb{F}_p}(\mathbb{G}_{a/K}^d)$ be an additive polynomial mapping, and let κ and δ...

conjecture | open | MathDB 页面 | 原文线索

V2-008748 | MATHDB-331797

Flatness and Cohen-Macaulayness conjecture for deformed arc-space algebras

摘录 / EXCERPT: Let $\mathcal{C}$ be the cone in the source, let $\mathpzc{Arc}_q^N$ be the mapping space associated with the deformation $\mathbb{C}[z,q]/(z^{N+1}-q^{N+1})$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-008749 | MATHDB-331798

Cohen-Macaulay and Gorenstein conjecture for finite arc algebras

摘录 / EXCERPT: Let $J[\delta,(1)^N]$ and $A[\delta,(1)^N]$ be the finite arc algebras introduced in the source. Finite arc-algebra singularity conjecture. The algebra $J[\delta,(1)^N]$ is Cohen-M...

conjecture | open | MathDB 页面 | 原文线索

V2-008750 | MATHDB-331811

Uniqueness conjecture for fully faithful representations of unitary fusion categories in bimodules

摘录 / EXCERPT: Let $\mathcal{C}$ be a unitary fusion category, and let R be a hyperfinite factor which is either of type ${\rm II}_1$ or ${\rm III}_1$. A fully faithful representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-008751 | MATHDB-331817

Ravenel's Local Conjecture for local formal module Ext groups

摘录 / EXCERPT: Let $K/\mathbb{Q}_p$ be a finite field extension with ring of integers A, uniformizer π, and residue field $\mathbb{F}_q$. Ravenel's Local Conjecture. ... where I_n is the id...

conjecture | open | MathDB 页面 | 原文线索

V2-008752 | MATHDB-331848

Extension of the main theorem to all symmetrizable Cartan matrices

摘录 / EXCERPT: Let C be a symmetrizable generalized Cartan matrix, let D be a symmetrizer of C, and let $H=H(C,D,\Omega)$, $\mathcal{M}=\mathcal{M}(H)$, and $\mathfrak{n}=\mathfrak{n}(C)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-008753 | MATHDB-331869

The reciprocity conjecture for Kolyvagin determinants

摘录 / EXCERPT: Let T be the p-adic Galois representation and let $\mathfrak{K}(T)$ be the Λ-module of Kolyvagin determinants. For an even Dirichlet character θ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008754 | MATHDB-331900

Conjecture on canonical grading of general Artinian Gorenstein algebras

摘录 / EXCERPT: Canonical-grading conjecture. A general algebra of type (n,d) is canonically graded if and only if (n,d) belongs to the following list:

conjecture | open | MathDB 页面 | 原文线索

V2-008755 | MATHDB-331928

The weak Birch and Swinnerton-Dyer conjecture for elliptic curves over number fields

摘录 / EXCERPT: Weak BSD conjecture. The function $L(E,s)$ has an analytic continuation to $\mathbb{C}$, and the analytic rank and algebraic rank of E coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-008756 | MATHDB-331931

The generalized Mahler-measure growth conjecture for divisor divisibility sequences

摘录 / EXCERPT: Growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008757 | MATHDB-332047

Relative distinction conjecture for complex points of real reductive groups

摘录 / EXCERPT: Let G be a reductive algebraic group over $\mathbb{R}$, and let π be an irreducible representation of $G(\mathbb{C})$. Complex-real distinction conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-008758 | MATHDB-332060

Unique-factorisation conjecture for germ-like generalized power series

摘录 / EXCERPT: Let $K((\mathbb{R}^{\leq 0}))$ be the ring of generalized power series, let J be the germ ideal, and let v_J denote the order-value. A series $a\in K((\mathbb{R}^{\leq 0}))$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-008759 | MATHDB-332088

Vanishing conjecture for Carlitz finite zeta values

摘录 / EXCERPT: Let $n>0$ and $s\geq 0$ be integers. Write the base-q expansion of n as $n=n_0+n_1q+\cdots+n_rq^r$ and set $\ell_q(n):=\sum_i n_i$. For the finite zeta value of level s and e...

conjecture | open | MathDB 页面 | 原文线索

V2-008760 | MATHDB-332089

Transcendence conjecture for the Carlitz period analogue

摘录 / EXCERPT: Let K be the rational function field used in the Carlitzian setting, and let $\widehat{\pi}$ be the associated Carlitz period analogue. Transcendence conjecture. The element...

conjecture | open | MathDB 页面 | 原文线索

V2-008761 | MATHDB-332161

Schur-Brauer version of van der Waerden's theorem for semimodules

摘录 / EXCERPT: Let $G=B_1\cup\dotsm\cup B_q$ be a finite partition of a semimodule $(G,\pmb{+})$ over a semiring $(R,+,\cdot)$. For a subset $F\subseteq R$, write $Fb=\{fb:f\in F\}$. Schur-Brauer...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-008762 | MATHDB-332162

Oyama's derivation relation conjecture for finite and symmetrized multiple zeta values

摘录 / EXCERPT: Let $\mathfrak H^1$ be the relevant noncommutative polynomial algebra of words, let $z=x+y$, and let ∂_l be the derivation indexed by $l\in\mathbb Z_{\geq1}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-008763 | MATHDB-332172

Log Bloch conjecture for smooth algebraic surfaces

摘录 / EXCERPT: Let U be a smooth algebraic surface, let $p_g(U)$ denote its geometric genus, and let $T(U)$ be the Albanese kernel, namely the kernel of the Albanese morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-008764 | MATHDB-332176

Oort's conjecture on lifting cyclic Galois covers

摘录 / EXCERPT: Oort's conjecture. Every cyclic Galois cover of k-curves lifts to characteristic zero.

conjecture | open | MathDB 页面 | 原文线索

V2-008765 | MATHDB-332177

Strong Oort conjecture for cyclic-by-p groups

摘录 / EXCERPT: Strong Oort conjecture. A cyclic-by-p group G is an Oort group for k if and only if G is of the form C_m, D_{2p^n}, or A_4, with the last case occurring only if $p=2$...

conjecture | open | MathDB 页面 | 原文线索

V2-008766 | MATHDB-332184

Schnetz's cosmic Galois closure conjecture for massless phi-four theory

摘录 / EXCERPT: Schnetz's cosmic Galois closure conjecture. The space V is itself closed under the action of the cosmic Galois group $\mathcal{C}_{0,0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-008767 | MATHDB-332194

Linearity and parity conjecture for Rouquier complexes of Mikado braids

摘录 / EXCERPT: Let $({\mathcal{W}},{\mathcal{S}})$ be a Coxeter system with length function ℓ, let $A\in\mathscr{B}(\Phi^+)$, and let w_A be the associated Mikado braid. Set $\beta:=w_A$,...

conjecture | open | MathDB 页面 | 原文线索

V2-008768 | MATHDB-332195

The dimension-three two-dimensional subalgebra conjecture for Lie p-algebras

摘录 / EXCERPT: Two-dimensional subalgebra conjecture. L contains a two-dimensional subalgebra.

conjecture | open | MathDB 页面 | 原文线索

V2-008769 | MATHDB-332209

The period conjecture for reduction of p-adic multiple zeta values

摘录 / EXCERPT: Let $\operatorname{red}_{\zeta}$ be the map obtained by reducing p-adic multiple zeta values modulo sufficiently large primes, from the $\mathbb{Q}$-algebra of p-adic multiple...

conjecture | open | MathDB 页面 | 原文线索

V2-008770 | MATHDB-332256

The quadratic resolvent counting conjecture

摘录 / EXCERPT: Quadratic resolvent counting conjecture. There is a constant β_n with $0<\beta_n<1/2$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008771 | MATHDB-332266

Blackadar-Handelman density conjecture for dimension functions

摘录 / EXCERPT: Blackadar-Handelman density conjecture. The set $LDF(A)$ is dense in $DF(A)$.

conjecture | open | MathDB 页面 | 原文线索

V2-008772 | MATHDB-332267

Blackadar-Handelman Choquet simplex conjecture for dimension functions

摘录 / EXCERPT: Blackadar-Handelman Choquet simplex conjecture. The affine space $DF(A)$ is a Choquet simplex for any C^*-algebra A.

conjecture | open | MathDB 页面 | 原文线索

V2-008773 | MATHDB-332291

Migliore-Nagel injective conjecture for Artinian Gorenstein algebras presented by quadrics

摘录 / EXCERPT: Let A be an Artinian Gorenstein algebra of socle degree at least three, presented by quadrics over a field $\mathbb{K}$ of characteristic zero. For $L\in A_1$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-008774 | MATHDB-332292

Migliore-Nagel weak Lefschetz conjecture for Artinian Gorenstein algebras

摘录 / EXCERPT: Let $A=\bigoplus_i A_i$ be an Artinian Gorenstein algebra presented by quadrics over a field $\mathbb{K}$ of characteristic zero. The Weak Lefschetz Property means that there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-008775 | MATHDB-332304

Derived Morita reconstruction conjecture for infinity-contraction algebras

摘录 / EXCERPT: Infinity-contraction algebra reconstruction conjecture. The completions of the stalks at p and q are isomorphic if and only if A^∞_{con} and B^∞_{con} are derived...

conjecture | open | MathDB 页面 | 原文线索

V2-008776 | MATHDB-332347

Nakayama automorphism conjecture for connected Hopf algebras

摘录 / EXCERPT: Let H be a connected Hopf algebra of finite Gelfand-Kirillov dimension, with antipode S. The Nakayama automorphism conjecture. The Nakayama automorphism of the coalgebra H is...

conjecture | open | MathDB 页面 | 原文线索

V2-008777 | MATHDB-332376

The lifting-Oort-group conjecture for automorphism groups of cyclic curves

摘录 / EXCERPT: Let X be a smooth cyclic curve over a field k of characteristic p. Let N be a cyclic normal subgroup of $G=\operatorname{Aut}_{k}(X)$ such that the quotient X/N has genus...

conjecture | open | MathDB 页面 | 原文线索

V2-008778 | MATHDB-332416

Gorensteinness conjecture for Rees algebras of ideals

摘录 / EXCERPT: Gorensteinness conjecture. If $\mathcal{R}(I)$ is an almost Gorenstein graded ring, then A is a Gorenstein ring.

conjecture | open | MathDB 页面 | 原文线索

V2-008779 | MATHDB-332428

The effective bounded negativity conjecture for blowups of the projective plane

摘录 / EXCERPT: Effective bounded negativity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-008780 | MATHDB-332517

The tagged mapping class group conjecture for orbifold cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra arising from a triangulation of an orbifold $\mathcal{O}$. Let $\operatorname{MCG}_{\bowtie}(\mathcal{O})$ denote the tagged mapping class gr...

conjecture | open | MathDB 页面 | 原文线索

V2-008781 | MATHDB-332550

Fayers' degree bound conjecture for cyclotomic Hecke algebra decomposition numbers

摘录 / EXCERPT: Let ${\boldsymbol\lambda},{\boldsymbol\mu}\in\mathscr{P}_{n}$. For the polynomials $d_{{\boldsymbol\lambda}{\boldsymbol\mu}}(q)$ and the defect...

conjecture | open | MathDB 页面 | 原文线索

V2-008782 | MATHDB-332551

Completely graded decomposition conjecture for semisimple complex algebras

摘录 / EXCERPT: Let A be a finite-dimensional semisimple complex algebra of the form specified in the source, with multiplicities $b(n)$ in its decomposition. A connected grading of A is compl...

conjecture | open | MathDB 页面 | 原文线索

V2-008783 | MATHDB-332555

Hardt-Price gap-independent approximation conjecture for the noisy power method

摘录 / EXCERPT: Let $\boldsymbol{A}$ be a matrix, let $\boldsymbol{A}_k$ be its best rank-k approximation, and let $\boldsymbol{U}_k$ contain the top-k left singular vectors of...

conjecture | open | MathDB 页面 | 原文线索

V2-008784 | MATHDB-332563

The equivariant local epsilon-constant conjecture for potentially semistable representations

摘录 / EXCERPT: Let K be a finite extension of ${\mathbb Q}_p$, let N/K be a finite Galois extension with Galois group G, and let T be a ${\mathbb Z}_p[G_K]$-lattice such that...

conjecture | open | MathDB 页面 | 原文线索

V2-008785 | MATHDB-332564

The vanishing conjecture for the modified equivariant local epsilon term

摘录 / EXCERPT: Let K be a finite extension of ${\mathbb Q}_p$, let N/K be a finite Galois extension with Galois group G, and retain the notation $\tilde R_{N/K}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008786 | MATHDB-332586

Breuil's locally analytic socle conjecture for global modular Galois representations

摘录 / EXCERPT: Breuil's conjecture. For a character $\chi:T(L)\rightarrow E^{\times}$,

conjecture | solved | MathDB 页面 | 原文线索

V2-008787 | MATHDB-332613

Open-or-finite orbit conjecture for the boundary of the preimage tree

摘录 / EXCERPT: Let K be a number field, let $\phi\in K(z)$ be a rational function, and let $\alpha\in\mathbb{P}^1(K)$. Let T be the rooted tree of iterated preimages of α, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008788 | MATHDB-332627

The graph-generalized Orlik-Terao conjecture

摘录 / EXCERPT: The graph-generalized Orlik-Terao conjecture. For any simple connected graph Γ, there exists an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-008789 | MATHDB-332633

Strong union-closed sets conjecture for matrices

摘录 / EXCERPT: Let F be a reduced union-closed matrix, let F^T be its transpose, and let $\overline{F}$ be the characteristic matrix of the complements of the represented sets. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-008790 | MATHDB-332653

Frequency-module conjecture for Jacobi matrices of Cantor-type measures

摘录 / EXCERPT: Let $K(\gamma)$ be the Cantor-type set determined by γ, and let $H_{\mu_{K(\gamma)}}$ have recurrence-coefficient sequence $(a_n)_{n=1}^\infty$. An almost periodic sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-008791 | MATHDB-332658

The pertinency-reflection-number inequality

摘录 / EXCERPT: Pertinency-reflection-number conjecture. For every finite subgroup G of $\operatorname{Aut}_{gr}(R)$,

conjecture | open | MathDB 页面 | 原文线索

V2-008792 | MATHDB-332663

The non-trivially valued mutation-periodicity conjecture

摘录 / EXCERPT: The non-trivially valued mutation-periodicity conjecture. Corollary ... and ... , also hold when Q is non-trivially valued.

conjecture | open | MathDB 页面 | 原文线索

V2-008793 | MATHDB-332696

The $c_2=-z_2$ maximum-weight MZV conjecture

摘录 / EXCERPT: Let G be a primitive log-divergent graph with loop number h_G and $c_2(G)=-z_2$. $c_2=-z_2$ conjecture. ... Here $\mathcal{Z}_2$ is the algebra of multiple polylogarithmic period...

conjecture | open | MathDB 页面 | 原文线索

V2-008794 | MATHDB-332698

The derivation constraints for $c_2=-z_3$ ϕ^4 periods

摘录 / EXCERPT: Let G be a primitive log-divergent ϕ^4 graph with loop number h_G and $c_2(G)=-z_3$, and let δ_m be the derivation with respect to the modified parity basis in the...

conjecture | open | MathDB 页面 | 原文线索

V2-008795 | MATHDB-332705

The non-isotrivial superelliptic-curve conjecture for polynomials

摘录 / EXCERPT: Non-isotrivial superelliptic-curve conjecture. There exist integers $\ell\geq2$ and $m\geq1$ such that $\gcd(\ell,p)=1$ and $C_{\ell,m}(\phi)$ is a non-isotrivial curve of genus at...

conjecture | open | MathDB 页面 | 原文线索

V2-008796 | MATHDB-332741

Power-sum conjecture for matrix rings over finite commutative rings

摘录 / EXCERPT: Let $d>1$ and let R be a finite commutative ring. For $k\geq 1$, write $S_k(\mathbb{M}_d(R))=\sum_{A\in\mathbb{M}_d(R)}A^k$. The unique element $e\in R\setminus\{0\}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-008797 | MATHDB-332788

The open-orbit D-module equivalence conjecture

摘录 / EXCERPT: Let G/P and G/Q be the two homogeneous spaces in the setup, with $Z_0\subset G/P\times G/Q$ the open G-orbit and $j:Z_0\to G/P\times G/Q$ its inclusion. Write δ_{Z_0}...

conjecture | open | MathDB 页面 | 原文线索

V2-008798 | MATHDB-332789

The Hodge-module open-orbit equivalence conjecture

摘录 / EXCERPT: Hodge-module open-orbit equivalence conjecture. The object $\mathcal{T}$ is the kernel of an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-008799 | MATHDB-332791

The dense-open line-bundle description of the graded kernel

摘录 / EXCERPT: Let ... and let $\vec{\operatorname{gr}}(\mathcal{T})$ be the associated graded object of the Hodge-module kernel $\mathcal{T}$. Dense-open line-bundle conjecture. There exists a den...

conjecture | open | MathDB 页面 | 原文线索

V2-008800 | MATHDB-332844

Watanabe's bold conjecture on embeddings into quadratic complete intersections

摘录 / EXCERPT: Let $\mathbb{F}$ be a field and let $R=\mathbb{F}[x_1,\ldots,x_n]$ have its standard grading. A complete intersection algebra is a quotient $R/\langle f_1,\ldots,f_n\rangle$ by a h...

conjecture | open | MathDB 页面 | 原文线索

V2-008801 | MATHDB-332849

Fomin-Kirillov divisibility conjecture for rectangular staircase permutations

摘录 / EXCERPT: Fomin-Kirillov divisibility conjecture. The polynomial $FK(w,\ell)$ divides $FK(w,\ell+1)$ in $\mathbb{Q}[x]$, with quotient

conjecture | open | MathDB 页面 | 原文线索

V2-008802 | MATHDB-332882

Linear resolution conjecture for ideals generated by products of code forms

摘录 / EXCERPT: Let $\mathbb K$ be a field of characteristic 0, let $\mathcal C \e R$ be an $[n,k]$-linear code represented by linear forms $cell_1,cell_2,\ldots,cell_n$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-008803 | MATHDB-332883

Colon ideal conjecture for generalized star configurations

摘录 / EXCERPT: Let $\mathbb K$ be a field of characteristic 0, let $\mathcal C=(cell_1,\ldots,cell_n)\subset R=\mathbb K[x_1,\ldots,x_k]$ be an $[n,k]$-linear code with no zero columns, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008804 | MATHDB-332909

Weak Schanuel conjecture

摘录 / EXCERPT: Let $\alpha_1,\ldots,\alpha_n$ be non-zero algebraic numbers such that the numbers $\log \alpha_1,\ldots,\log \alpha_n$ are $\mathbb{Q}$-linearly independent. Weak Schanuel conject...

conjecture | open | MathDB 页面 | 原文线索

V2-008805 | MATHDB-332910

Osorno formula for Ganter-Kapranov 2-characters of crossed modules

摘录 / EXCERPT: Let $\mathcal{K}=(H\xrightarrow{\partial}G)$ be a crossed module whose fundamental group $\pi_1(\mathcal{K})$ is finite. Let Θ be a 2-representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-008806 | MATHDB-332991

The weight-monodromy conjecture for motives

摘录 / EXCERPT: Weight-monodromy conjecture. This representation is pure of weight w; namely, the eigenvalues of $\sigma(v)$ on the r-th graded piece of the monodromy filtration are Weil numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-008807 | MATHDB-333042

Conjecture on Weyl groupoids for cyclic-group Nichols algebras

摘录 / EXCERPT: Let C_n be the cyclic group of order n, and let $\mathcal{B}_{C_n}$ denote the Nichols algebra with the braiding considered in the paper. Weyl-groupoid conjecture. The Weyl gro...

conjecture | open | MathDB 页面 | 原文线索

V2-008808 | MATHDB-333043

Conjecture on finite-dimensional elementary Fomin-Kirillov Nichols algebras

摘录 / EXCERPT: Let $G=G(m,p,n)$ be a complex reflection group in the non-exceptional series, and let $\mathcal{B}_G$ be its elementary Fomin-Kirillov Nichols algebra. Finite-dimensionality conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-008809 | MATHDB-333044

Conjecture on finite-dimensional Fomin-Kirillov algebras for non-exceptional Coxeter groups

摘录 / EXCERPT: Let G be a non-exceptional indecomposable Coxeter group, and let $\mathcal{E}_G$ be its Fomin-Kirillov algebra. Finite-dimensionality conjecture. Up to isomorphism, the only fini...

conjecture | open | MathDB 页面 | 原文线索

V2-008810 | MATHDB-333049

Minkowski-sum lattice-point formula for G_2 representations

摘录 / EXCERPT: Let $\lambda=m_1\varpi_1+m_2\varpi_2\in\mathcal{P}_+$, and let $\boldsymbol{G_2^{\varpi_1}}(1)$ and $\boldsymbol{G_2^{\varpi_2}}(1)$ be the polytopes defined by the displayed inequ...

conjecture | open | MathDB 页面 | 原文线索

V2-008811 | MATHDB-333052

Strong local Langlands correspondence in families

摘录 / EXCERPT: With the notation above, let $R_\nu^{\operatorname{inv}}$ denote the subalgebra of $\mathcal G_\nu$-invariant elements of R_ν. Strong local Langlands in families. There is an i...

conjecture | open | MathDB 页面 | 原文线索

V2-008812 | MATHDB-333100

The characteristic-ideal equality for generalized Heegner-cycle Selmer modules

摘录 / EXCERPT: Let $(f,K,\mathfrak p)$ be an admissible triple. Let K_∞ be the anticyclotomic $\mathbb Z_p$-extension of K, let $\Lambda={\mathcal O}_\mathfrak p[[\Gamma_\infty]]$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-008813 | MATHDB-333103

The conjecture that every integrable system admits an integrable rational recursion operator

摘录 / EXCERPT: Let ${\mathcal W} \subset {\mathcal V}$ be an integrable system. An integrable rational recursion operator is a rational operator $L=AB^{-1}$ satisfying the integrability condition...

conjecture | open | MathDB 页面 | 原文线索

V2-008814 | MATHDB-333115

Lemmermeyer's reflection conjecture for quartic fields and cubic resolvents

摘录 / EXCERPT: Let L be a number field with Galois group S_4 or A_4 over $\mathbb{Q}$. Let K_1 be a quartic subfield and K_2 its cubic resolvent. For a number field K, write...

conjecture | open | MathDB 页面 | 原文线索

V2-008815 | MATHDB-333142

Makar-Limanov's symmetric free subalgebra conjecture for division rings with involution

摘录 / EXCERPT: Makar-Limanov's involutional conjecture. If D is infinite dimensional over k and finitely generated as a division k-algebra, then there exist two symmetric elements in D wh...

conjecture | open | MathDB 页面 | 原文线索

V2-008816 | MATHDB-333192

The deformation conjecture for letterplace ideals

摘录 / EXCERPT: Let P be a finite poset, and let $L(2,P)\subseteq \Bbbk[x_{[2]\times P}]$ be the letterplace ideal. A rigid deformation conjecture. The ideal $L(2,P)$ deforms to a rigid ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-008817 | MATHDB-333221

The quiver and mutation conjecture for rank two Lie groups

摘录 / EXCERPT: Quiver and mutation conjecture. For every semisimple, simply connected, complex Lie group G, there exists a quiver Q_G and quiver mutations $\mu_G^{\operatorname{rot}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008818 | MATHDB-333229

NS Jack-superpolynomial expansion for the admissible superpartition family

摘录 / EXCERPT: Let $\bar\Gamma=\Gamma_{2a-l,l}$ be the superpartition defined in the source, and let $\mathcal X_k(\bar\Gamma)$ be the set obtained by applying all allowed products of the operato...

conjecture | open | MathDB 页面 | 原文线索

V2-008819 | MATHDB-333268

Harbourne, Schenck and Seceleanu's asymptotic weak Lefschetz conjecture

摘录 / EXCERPT: Let K be a field of characteristic 0, let $R=K[x_1,\dots,x_r]$, and let $\ell_1,\dots,\ell_n$ be general linear forms. For a positive integer t, set ... and let $A=R/I$. Harbou...

conjecture | open | MathDB 页面 | 原文线索

V2-008820 | MATHDB-333300

Ax-Schanuel conjecture for the first Painlevé equation

摘录 / EXCERPT: Let (x_i,y_i), for $i=1,\ldots,n$, be solutions of the two-variable first Painlevé equation ... Write $\bar{x}=(x_1,\ldots,x_n)$, $\bar{y}=(y_1,\ldots,y_n)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008821 | MATHDB-333346

The conjectural description of the Grothendieck group of divided power algebras

摘录 / EXCERPT: Let $\bD$ be the divided power algebra over $\bk$, let R be the relevant grading ring, and let $\cC_n$ be the category of finitely generated $\bk$-modules supported on...

conjecture | open | MathDB 页面 | 原文线索

V2-008822 | MATHDB-333347

Generalized-minor realization of regular cluster variables for Coxeter double Bruhat cells

摘录 / EXCERPT: Let Q be a finite valued quiver without oriented cycles, and let $G^{c,c^{-1}} \subset G$ be the associated Coxeter double Bruhat cell. The algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008823 | MATHDB-333348

Level-zero-minor conjecture for regular cluster variables in affine type

摘录 / EXCERPT: Let B be an acyclic skew-symmetrizable matrix whose Cartan companion A is of affine type. Let $x_{i;t} \in \mathcal{A}_{\widetilde{B}_{dp}}$ be a regular cluster variable, and...

conjecture | open | MathDB 页面 | 原文线索

V2-008824 | MATHDB-333378

The braided tensor equivalence conjecture for Temperley-Lieb and Virasoro categories

摘录 / EXCERPT: Temperley-Lieb-Virasoro braided equivalence conjecture. There is an equivalence of braided tensor categories

conjecture | open | MathDB 页面 | 原文线索

V2-008825 | MATHDB-333396

Fortuny-Grau-Oller-Marcén-Rúa conjecture on power sums of matrices over finite commutative rings

摘录 / EXCERPT: Fortuny-Grau-Oller-Marcén-Rúa conjecture. All such power sums vanish unless all of the following conditions hold: $d=2$; $|R|\equiv 2\pmod 4$ and $1<k\equiv 0,\pm1\pmod 6$; and the...

conjecture | open | MathDB 页面 | 原文线索

V2-008826 | MATHDB-333397

Blockwise unimodality conjecture for Frobenius seaweed spectra

摘录 / EXCERPT: Let A be any top or bottom block in a Frobenius seaweed of $\mathfrak{sl}(n)$ with $n\geq 2$. For each eigenvalue i, let d_i be its multiplicity in the multiset of eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-008827 | MATHDB-333405

The module-category pullback conjecture for commutative semiring infinity-categories

摘录 / EXCERPT: Let $\mathcal{R}\in\operatorname{CAlg}_{\operatorname{Fin}}$ and let $\mathcal{L}$ be a cyclic commutative algebra over $\mathcal{R}$, meaning a commutative Fin-al...

conjecture | open | MathDB 页面 | 原文线索

V2-008828 | MATHDB-333425

The crystalline and shortness conjecture for Galois representations attached to automorphic representations

摘录 / EXCERPT: Crystalline and shortness conjecture. If π is unramified at $\mathfrak{q}\mid p$, then

conjecture | open | MathDB 页面 | 原文线索

V2-008829 | MATHDB-333458

Rigidity of graded noetherian down-up algebras under semisimple Hopf algebra actions

摘录 / EXCERPT: Rigidity conjecture. The graded noetherian down-up algebras are rigid with respect to semisimple Hopf algebra actions.

conjecture | open | MathDB 页面 | 原文线索

V2-008830 | MATHDB-333461

De Rham geometric Langlands conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, let $G^{\vee}$ be its Langlands dual group, and let $\mathcal Conn_{G^{\vee}}(X)$ be the de Rham...

conjecture | open | MathDB 页面 | 原文线索

V2-008831 | MATHDB-333503

Schinzel's exponential degree conjecture for rational fibers

摘录 / EXCERPT: Let X be a smooth geometrically irreducible projective curve over $\mathbb{Q}$, and let $t\in\mathbb{Q}(X)$ be a non-constant rational function with...

conjecture | open | MathDB 页面 | 原文线索

V2-008832 | MATHDB-333528

Schinzel's Strong Diversity Conjecture for number fields in parametric families

摘录 / EXCERPT: Let X be a curve over $\mathbb{Q}$, let $tmin \mathbb{Q}(X)$ be a non-constant $\mathbb{Q}$-rational function of degree at least 2, and let $P_1,ldots,P_N$ be the points in the...

conjecture | open | MathDB 页面 | 原文线索

V2-008833 | MATHDB-333534

The generalized center decomposition conjecture for derived planar nearrings

摘录 / EXCERPT: Generalized center decomposition conjecture. The generalized center satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-008834 | MATHDB-333622

The almost Gorenstein conjecture for Rees algebras of p_g-ideals

摘录 / EXCERPT: Let A be the two-dimensional normal local ring under consideration, let $I\subset A$ be a p_g-ideal, and let $\mathcal{R}(I)=\bigoplus_{n\geq 0}I^nt^n$ denote its Rees algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-008835 | MATHDB-333645

Noncommutative Auslander correspondence for semisimple Hopf actions

摘录 / EXCERPT: Let H be a semisimple Hopf algebra and let A be a noetherian Artin-Schelter regular, connected graded algebra generated in degree one. Suppose that A is a graded H-module a...

conjecture | open | MathDB 页面 | 原文线索

V2-008836 | MATHDB-333655

Cyclotomic Gaudin algebra spectrum conjecture

摘录 / EXCERPT: Let $\lambda_0\in\mathfrak{h}^{*,\nu}$ be the dominant ν-invariant weight satisfying $\langle\check\alpha_i,\lambda_0\rangle\in\mathbb{Q}$ for all $i\in I$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008837 | MATHDB-333702

The Krull-ring conjecture for integer-valued polynomial rings

摘录 / EXCERPT: Let R be a Krull ring, and let $\operatorname{Int}(R)$ denote its ring of integer-valued polynomials. A PVMR is the corresponding Prüfer-v-multiplication ring notion for rings wi...

conjecture | open | MathDB 页面 | 原文线索

V2-008838 | MATHDB-333705

The noncommutative main conjecture for the adjoint Selmer group

摘录 / EXCERPT: Adjoint Selmer-category conjecture. The dual Selmer group $\mathrm{Sel}^\ast_{E_\infty}(\mathrm{Ad}^0(\rho))$ is in the category...

conjecture | open | MathDB 页面 | 原文线索

V2-008839 | MATHDB-333730

Brenti's unimodality conjecture for standard graded Gorenstein domains

摘录 / EXCERPT: Let A be a standard graded Gorenstein integral domain, and write its Hilbert series as $H(A;z)=h(A;z)/(1-z)^{\dim(A)}$, where $h(A;z)$ is the h-polynomial. Brenti's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-008840 | MATHDB-333747

The Harish-Chandra-type conjecture for rational Cherednik algebras

摘录 / EXCERPT: Harish-Chandra-type conjecture. There exists a subalgebra $H_{1,\mathbf c}'\subseteq H_{1,\mathbf c}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-008841 | MATHDB-333753

Algebraic-solution relation for q-deformed conformal blocks

摘录 / EXCERPT: Algebraic-solution relation. For $s=\pm1$,

conjecture | open | MathDB 页面 | 原文线索

V2-008842 | MATHDB-333762

The Pfister involution conjecture in characteristic two

摘录 / EXCERPT: Pfister involution conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-008843 | MATHDB-333784

Ginzburg-Rallis global conjecture for the exterior-cube L-function

摘录 / EXCERPT: Let k be a global field with ring of adeles ${\boldsymbol A}$, and let D range over quaternion algebras over k. For an irreducible cuspidal automorphic representation π o...

conjecture | open | MathDB 页面 | 原文线索

V2-008844 | MATHDB-333798

The Kadison-Kaplansky conjecture for torsion-free groups

摘录 / EXCERPT: Kadison-Kaplansky conjecture. The algebra $C^*_r\Gamma$ contains no non-trivial projections. The conjecture is presented as a consequence of surjectivity of the Baum-Connes assembl...

conjecture | open | MathDB 页面 | 原文线索

V2-008845 | MATHDB-333816

The formal-power-series Multiplicative Rule of Three

摘录 / EXCERPT: Let $R[[x,y]]$ be the ring of formal power series in x and y with coefficients in a ... gi=1+alpha{i1}xui+alpha{i2}(xui)^2+alpha{i3}(xui)^3+cdots,qquad hi=1+beta{i1}yvi+beta{i2}(...

conjecture | open | MathDB 页面 | 原文线索

V2-008846 | MATHDB-333817

Failure of the Multiplicative Rule of k for linear factors

摘录 / EXCERPT: Let $R=\mathcal{A}/I$ be a quotient ring, and let g_S and h_S denote the products associated with a subset S in the definition of the Multiplicative Rule of k. Failure of t...

conjecture | open | MathDB 页面 | 原文线索

V2-008847 | MATHDB-333838

The PBW conjecture for mock-Lie algebras

摘录 / EXCERPT: Let L be a mock-Lie algebra, and let $U(L)$ denote its universal enveloping algebra. The Poincaré-Birkhoff-Witt theorem holds for L when the ordered monomials in a basis of L...

conjecture | open | MathDB 页面 | 原文线索

V2-008848 | MATHDB-333869

The lattice-point conjecture for F-polynomials

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite-type cluster algebra and let ${{\sf F}_{\beta}(\mathbf{y})}$ be an F-polynomial associated to a positive root β. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008849 | MATHDB-333888

Clozel's motive-association conjecture for regular algebraic cuspidal representations

摘录 / EXCERPT: Let K be the quadratic imaginary field under consideration. Let Π be a regular algebraic cuspidal representation of $GL_n(\mathbb{A}_K)$, let $\mathbb{Q}(\Pi)$ be its rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-008850 | MATHDB-333889

Automorphic Deligne conjecture for critical Rankin-Selberg values

摘录 / EXCERPT: Let K be the quadratic imaginary field under consideration. Let Π and Π' be regular algebraic cuspidal representations of $GL_n(\mathbb{A}_K)$ and $GL_{n'}(\mathbb{A}_K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-008851 | MATHDB-333897

Buzzard-Diamond-Jarvis tensor-product conjecture in the definite case

摘录 / EXCERPT: Let F be a totally real field, let D be a quaternion algebra over F, and let $\overline{r}:G_F\to\mathrm{GL}_2(\mathbb F)$ be a two-dimensional continuous totally odd absolut...

conjecture | open | MathDB 页面 | 原文线索

V2-008852 | MATHDB-333923

Existence and uniqueness conjecture for G-twisted Jacobian algebras

摘录 / EXCERPT: Let f be an invertible polynomial and let G be a commutative group acting as in the preceding definitions. Write ${\rm Jac}'(f,G)$ for a G-twisted Jacobian algebra of f, an...

conjecture | open | MathDB 页面 | 原文线索

V2-008853 | MATHDB-334024

Existence of descent for cuspidal automorphic representations

摘录 / EXCERPT: Let E/K be a cyclic extension of global fields, and let $\rho: \Gamma_K \to \widehat{G}(\overline{\mathbb{Q}}_l)$ be a continuous, almost everywhere unramified homomorphism with...

conjecture | open | MathDB 页面 | 原文线索

V2-008854 | MATHDB-334025

The band-operation conjecture for dual canonical basis elements

摘录 / EXCERPT: Let V be the defining representation in the tensor-diagram setting, let $R_{a,b}(V)$ be the relevant invariant ring, and let $\operatorname{Inv}(V)$ denote the invariant algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-008855 | MATHDB-334033

Fukaya-Kato's pseudo-isomorphism conjecture for the reciprocity map

摘录 / EXCERPT: Let θ be the character indexing the modules in the construction, let Λ and R be the coefficient rings, let $\mathfrak{h}$ be the Hecke algebra, let I be the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-008856 | MATHDB-334034

Torsion quotient conjecture for the residual Selmer group

摘录 / EXCERPT: Torsion quotient conjecture. The $(\Lambda\otimes_R\mathfrak{h}/I)$-module L_{ϵ} is torsion as a Λ-module. This is the first of the expected conjectures in the resid...

conjecture | open | MathDB 页面 | 原文线索

V2-008857 | MATHDB-334036

Main conjecture for the Selmer group of a modular representation

摘录 / EXCERPT: Let $epsilon$ be the relevant odd character, let $Lambda$ be the Iwasawa algebra, let R be the coefficient ring, let $\mathfrak{h}$ be the Hecke algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-008858 | MATHDB-334111

Fomin-Zelevinsky's conjecture on constant terms of F-polynomials

摘录 / EXCERPT: Let Σ be a cluster algebra with principal coefficients, and let $F(x)$ denote the F-polynomial of a cluster variable x, obtained by evaluating its Laurent expansion at t...

conjecture | open | MathDB 页面 | 原文线索

V2-008859 | MATHDB-334136

The Taylor-Wiles-level Galois lifting conjecture

摘录 / EXCERPT: Taylor-Wiles-level lifting conjecture. If $\overline{\rho}_{\mathfrak{m}}$ is absolutely irreducible, there exists a continuous lifting

conjecture | open | MathDB 页面 | 原文线索

V2-008860 | MATHDB-334139

The mixed-characteristic Q-sequence conjecture

摘录 / EXCERPT: Let R be a complete local domain of mixed characteristic. A Q-sequence is a sequence with the quasilength property used in the definition of robust algebras. Mixed-characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-008861 | MATHDB-334140

The mixed-characteristic colon-capturing conjecture for tight closure

摘录 / EXCERPT: Let $(R,\operatorname{\mathfrak{m}})$ be a d-dimensional complete local domain of mixed characteristic p. Assume that $p,x_2,\dots,x_d$ is a system of parameters. Set ... and, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-008862 | MATHDB-334153

The Delta Conjecture for generalized coinvariant algebras

摘录 / EXCERPT: Let $\mathcal{LD}_n$ be the set of labeled Dyck paths of size n. For $P\in\mathcal{LD}_n$, let $a_i(P)$, $\ell_i(P)$, $\operatorname{area}(P)$, $\operatorname{dinv}(P)$, $d_i(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-008863 | MATHDB-334178

Kala's ring-theoretic conjecture related to Abhyankar's construction

摘录 / EXCERPT: Let $A=\mathbb{Z}[T_1,\dots,T_n]$, and let $A^+=\mathbb{N}[T_1,\dots,T_n]$. Let B be a subring of A and I an ideal of B. Write $QF(B)$ for the quotient field of B. Kala's...

conjecture | open | MathDB 页面 | 原文线索

V2-008864 | MATHDB-334222

Minimal dimension conjecture for Malcev algebras of the second type

摘录 / EXCERPT: Minimal dimension conjecture. Every Malcev algebra of the second type of dimension <23 over a field of characteristic 0 is an algebra of the first type.

conjecture | open | MathDB 页面 | 原文线索

V2-008865 | MATHDB-334248

The n-GCD conjecture for polynomial automorphisms

摘录 / EXCERPT: Let $F:\mathbb{C}[x_1,\ldots,x_n] \to \mathbb{C}[x_1,\ldots,x_n]$ be a $\mathbb{C}$-algebra endomorphism with invertible Jacobian. Write $F_i=F(x_i)$, and let l_i be the degree o...

conjecture | open | MathDB 页面 | 原文线索

V2-008866 | MATHDB-334270

Stability conjecture for non-orthogonal coarse-grid correction in RN AMG

摘录 / EXCERPT: Let A be nonsingular. Define the non-orthogonal coarse-grid correction using transfer operators ... The correction is stable in the sense that ... for some constant C independent o...

conjecture | open | MathDB 页面 | 原文线索

V2-008867 | MATHDB-334272

Curtis's weak conjecture on the Bernstein center and finite-type deformation algebras

摘录 / EXCERPT: Let F be a non-archimedean local field, let $G_n=\operatorname{GL}_n(F)$, let Z_n be the Bernstein center of the category of smooth representations of G_n, and let ν be a...

conjecture | open | MathDB 页面 | 原文线索

V2-008868 | MATHDB-334287

Monotonicity conjecture for algebraic stringy Euler numbers of Kawamata log-terminal pairs

摘录 / EXCERPT: Let (X,Δ) be a Kawamata log-terminal pair, and let $e^{\rm str}_{\rm alg}(X,\Delta)$ denote its algebraic stringy Euler number. Monotonicity conjecture. The algebraic string...

conjecture | open | MathDB 页面 | 原文线索

V2-008869 | MATHDB-334301

Carlitz-Schanuel conjecture over Tate algebras

摘录 / EXCERPT: Carlitz-Schanuel conjecture over Tate algebras. The displayed transcendence-degree inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-008870 | MATHDB-334302

Operator-theoretic generalization of Denis-Schanuel's conjecture

摘录 / EXCERPT: Operator-theoretic generalization of Denis-Schanuel's conjecture. The asserted regular μ-independence holds.

conjecture | open | MathDB 页面 | 原文线索

V2-008871 | MATHDB-334303

Regular μ-independence of Carlitz zeta values and the Carlitz period

摘录 / EXCERPT: Regular-independence conjecture. Every finite subset of this set is regularly μ-independent over K_s.

conjecture | open | MathDB 页面 | 原文线索

V2-008872 | MATHDB-334304

The Carlitz-exponential μ-transcendence dichotomy

摘录 / EXCERPT: Dichotomy conjecture. Either f or $\exp_C(f)$ is regularly μ-transcendental over K_s.

conjecture | open | MathDB 页面 | 原文线索

V2-008873 | MATHDB-334315

The Effective Gonality Conjecture for syzygies of curves

摘录 / EXCERPT: Let C be a smooth complex algebraic curve of genus g and gonality k, and let L be a very ample line bundle on C with ... Write $h^0(L)=\dim H^0(C,L)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008874 | MATHDB-334341

The CM conjecture for arithmetic shtukas with one paw

摘录 / EXCERPT: Let C be the ambient complete algebraically closed field, and let $G=(V,\mathcal{L})$ be a shtuka with one paw: V is a finite-dimensional $\mathbb{Q}_p$-vector space and...

conjecture | open | MathDB 页面 | 原文线索

V2-008875 | MATHDB-334362

The non-bidirectionality conjecture for subatomic domain properties

摘录 / EXCERPT: Let the classes of rings in Figure 1 be the unique factorization domains, atomic domains, semi-atomic domains, almost atomic domains, quasi-atomic domains, Furstenberg domains, s...

conjecture | open | MathDB 页面 | 原文线索

V2-008876 | MATHDB-334363

The completeness conjecture for the hierarchy of subatomic domains

摘录 / EXCERPT: Let Figure 1 be the directed diagram of implications among unique factorization domains, atomic domains, semi-atomic domains, almost atomic domains, quasi-atomic domains, Fursten...

conjecture | open | MathDB 页面 | 原文线索

V2-008877 | MATHDB-334389

Hrushovski's finite-basis conjecture for radical well-mixed difference ideals

摘录 / EXCERPT: Let K be a difference field and let R be a finitely difference generated difference algebra over K. A difference ideal in R is radical well-mixed if it is both radical and...

conjecture | open | MathDB 页面 | 原文线索

V2-008878 | MATHDB-334404

The bounded t-structure conjecture for negative K-theory

摘录 / EXCERPT: Let E be a small stable ∞-category equipped with a bounded t-structure. Write $\mathrm{K}_{-n}(E)$ for its negative nonconnective K-groups. Bounded t-structure vanishi...

conjecture | open | MathDB 页面 | 原文线索

V2-008879 | MATHDB-334405

The heart-equivalence conjecture for K-theory of bounded t-structures

摘录 / EXCERPT: Heart-equivalence conjecture. This natural map should be an equivalence of nonconnective K-theory spectra. The assertion would identify the K-theory of a boundedly t-structur...

conjecture | open | MathDB 页面 | 原文线索

V2-008880 | MATHDB-334413

Conjecture on zero spacing near an algebraic singularity for regular measures

摘录 / EXCERPT: Let μ be a finite Borel measure supported on a compact subset K of the real line and assume that μ is regular in the sense of Stahl and Totik on K. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008881 | MATHDB-334423

Radical decomposition conjecture for the colon ideal of the Jacobian ideal

摘录 / EXCERPT: Assume $r=m-2$, and retain the notation for the degenerated generic matrix, the Jacobian ideal J, the colon ideal $J:I$, the submatrices M_j,N_j, and the determinants...

conjecture | open | MathDB 页面 | 原文线索

V2-008882 | MATHDB-334424

Cohen-Macaulay decomposition conjecture for the maximal degeneration

摘录 / EXCERPT: Let R be the polynomial ring of entries of the degenerated generic matrix, let J be the Jacobian ideal of its determinant, let I be the ideal of submaximal minors, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008883 | MATHDB-334432

The minimal mutation-infinite quiver conjecture for bounded exchange-graph components

摘录 / EXCERPT: Let Q be a minimal mutation-infinite quiver. Let $\widehat{\Psi}(Q)$ denote the subgraph associated with quivers admitting maximal green sequences, as in Theorem. Minimal mutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-008884 | MATHDB-334433

The finiteness conjecture for direct sums of quivers

摘录 / EXCERPT: Let Q and Q' be quivers such that $\Psi(Q)$ and $\Psi(Q')$ are finite and non-empty, and let $Q\oplus Q'$ denote their direct sum. Direct-sum finiteness conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-008885 | MATHDB-334466

Hain-Looijenga perfect-pairing and freeness conjecture for compactly supported tautological classes

摘录 / EXCERPT: Hain-Looijenga conjecture. These pairings are perfect for every $k\geq0$; in addition, $R_c^*(M_g)$ is a free $R^*(M_g)$-module of rank one. This conjecture proposes a form of dual...

conjecture | open | MathDB 页面 | 原文线索

V2-008886 | MATHDB-334467

Universal-Jacobian generation conjecture for tautological relations on fiber powers

摘录 / EXCERPT: Let $\mathcal{C}$ be the universal curve over $\mathcal{M}_{g,1}$, let $\mathcal{C}^n$ be its n-fold fiber product, and let $\mathcal{R}_n$ be the polynomial algebra generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-008887 | MATHDB-334478

The sharp/flat Beilinson-Flach element main conjecture

摘录 / EXCERPT: Let K be as in the source, let Λ_K be the corresponding two-variable Iwasawa algebra, and let $\mathcal X^{\hat\bullet\forall}$ and $\mathcal X^{\bullet0}$ be the dual Se...

conjecture | open | MathDB 页面 | 原文线索

V2-008888 | MATHDB-334482

Semiduality conjecture for arithmetic groups over function fields

摘录 / EXCERPT: Let $\mathcal{O}_S$ be a ring of S-integers in a global function field K of characteristic p, and let $\mathbf G$ be a noncommutative, absolutely almost simple algebraic K-...

conjecture | open | MathDB 页面 | 原文线索

V2-008889 | MATHDB-334512

Universal enveloping triassociative algebra conjecture for Lie trialgebras

摘录 / EXCERPT: Let $F\colon \mathbf{TriAss} \to \mathbf{TriLie}$ be the functor obtained from the operations in Lemma, sending a triassociative algebra to its associated Lie trialgebra. For a Lie...

conjecture | open | MathDB 页面 | 原文线索

V2-008890 | MATHDB-334513

Arity-seven identity completeness conjecture for Jordan trialgebras

摘录 / EXCERPT: Let $\mathbb{F}$ be a field of characteristic 0 or characteristic $p>7$. A multilinear polynomial identity of arity at most 7 is an identity involving at most seven distinct va...

conjecture | open | MathDB 页面 | 原文线索

V2-008891 | MATHDB-334514

Arity-eight new identity conjecture for Jordan trialgebras

摘录 / EXCERPT: Consider the Jordan product and diproduct in every triassociative algebra. Their defining identities are those of Jordan trialgebras; also consider the linearizations of the Glenni...

conjecture | open | MathDB 页面 | 原文线索

V2-008892 | MATHDB-334527

The graded clock condition characterization for gentle one-cycle algebras

摘录 / EXCERPT: Graded clock condition conjecture. A graded gentle one-cycle algebra satisfies the graded clock condition if and only if it is derived equivalent to $\Gamma(p,q,0)$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-008893 | MATHDB-334567

Linear bound for the invariant-subspace Helly number

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite family of linear operators $\mathbb K^d\to\mathbb K^d$, where $\mathbb K$ is an arbitrary field and $d\geqslant 2$. Define $HI(\mathbb K^d)$ to be the...

conjecture | open | MathDB 页面 | 原文线索

V2-008894 | MATHDB-334593

Levelness conjecture for four general uniform powers in three variables

摘录 / EXCERPT: Let $R=k[x,y,z]$ and let ... where $L_1,\dots,L_4$ are general linear forms. Levelness conjecture. The quotient R/I is level with Cohen-Macaulay type k. This predicts a precise s...

conjecture | open | MathDB 页面 | 原文线索

V2-008895 | MATHDB-334594

Strong Lefschetz conjecture for squares in quotients by powers of general linear forms

摘录 / EXCERPT: Let $R=k[x,y,z]$ and let $A=R/I$ be any Artinian quotient generated by powers of general linear forms. For a general linear form L, multiplication by L^2 is the graded map ... fo...

conjecture | open | MathDB 页面 | 原文线索

V2-008896 | MATHDB-334622

Gerko's conjecture on chains of semidualizing modules

摘录 / EXCERPT: Let $(R,\mathfrak{m},k)$ be an Artinian local ring, with all modules finite. Let $[C_n]\vartriangleleft\cdots\vartriangleleft[C_1]\vartriangleleft[C_0]$ be a chain in...

conjecture | open | MathDB 页面 | 原文线索

V2-008897 | MATHDB-334629

The focus conjecture for the support of the algebraic spectrum

摘录 / EXCERPT: Let $b=u+iv$ with $v\ne 0$, and let Γ_b be the real rational cubic with a node at the origin defined by ... where $x=\operatorname{Re}z$ and $y=\operatorname{Im}z$. Its three...

conjecture | open | MathDB 页面 | 原文线索

V2-008898 | MATHDB-334638

Jacquet-Rallis Fundamental Lemma for unitary Lie algebras

摘录 / EXCERPT: Let $\mathfrak{s}(E_0)$ be the symmetric Lie algebra and let $\mathfrak{u}(J_0)$ and $\mathfrak{u}(J_1)$ be the Lie algebras of the corresponding unitary groups. Let $\Omega(y)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008899 | MATHDB-334640

Arithmetic Fundamental Lemma for unitary Lie algebras

摘录 / EXCERPT: Let $\mathfrak{s}(E_0)$ be the symmetric Lie algebra and let $\mathfrak{u}(J_1)$ be the Lie algebra of the odd unitary group. Let $\Omega(y)$ and ∂O_y denote the transfer fact...

conjecture | open | MathDB 页面 | 原文线索

V2-008900 | MATHDB-334644

A polynomial bound for rigidity of symbolic powers

摘录 / EXCERPT: Let R be a d-dimensional polynomial ring over a field, and let I be an ideal of R. Suppose there is a polynomial function f of degree at most d, whose coefficients depe...

conjecture | open | MathDB 页面 | 原文线索

V2-008901 | MATHDB-334698

Fundamental conjectures on exchange graphs and seeds of cluster algebras

摘录 / EXCERPT: Exchange graph and seed conjectures. (a) The exchange graph of a cluster algebra with rank n only depends on the initial exchange matrix; (b) every seed is uniquely determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-008902 | MATHDB-334720

The Fontaine-Mazur-Langlands automorphy conjecture for conjugate self-dual representations

摘录 / EXCERPT: Fontaine-Mazur-Langlands conjecture. There exists an algebraic essentially conjugate self-dual cuspidal automorphic representation π of GL_m/K such that...

conjecture | open | MathDB 页面 | 原文线索

V2-008903 | MATHDB-334729

Compatibility conjecture for Poisson and Dunn-Lurie additivity

摘录 / EXCERPT: Compatibility conjecture. This diagram is commutative. The question is whether the Dunn-Lurie additivity functor is compatible with the Poisson additivity functor under the formali...

conjecture | open | MathDB 页面 | 原文线索

V2-008904 | MATHDB-334754

Property (star) for acyclic cluster seeds

摘录 / EXCERPT: Let a seed be a seed of a cluster algebra, and let Property $(\star)$ denote the property defined earlier in the paper. Acyclic-seed conjecture. Property $(\star)$ holds for seeds...

conjecture | open | MathDB 页面 | 原文线索

V2-008905 | MATHDB-334803

The finite-dimensionality conjecture for categorical representability dimension two

摘录 / EXCERPT: Let T be a category, and let $\operatorname{rdim}\mathsf{T}$ denote its representability dimension. Finite-dimensionality conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-008906 | MATHDB-334804

Orlov's categorical representability conjecture for rational surfaces

摘录 / EXCERPT: Let S be a surface over a field k, and let $\operatorname{rdim}(S)$ denote its categorical representability dimension. Orlov's conjecture. The surface S is k-rational if an...

conjecture | open | MathDB 页面 | 原文线索

V2-008907 | MATHDB-334814

Generalized Eulerian duality conjecture for holonomic Weyl-algebra modules

摘录 / EXCERPT: Generalized Eulerian duality conjecture. The shifted module $^\sharp M^\dagger(+n)$ is a left generalized Eulerian $A_n(K)$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-008908 | MATHDB-334815

Stabilization conjecture for the filtered Kashiwara-Vergne Lie algebras

摘录 / EXCERPT: For each integer $k\geq 1$, let $\widehat{\mathfrak{krv}}^{(k)}_2$ and $\widehat{\mathfrak{krv}}^{(k+1)}_2$ denote the successive filtered Kashiwara-Vergne Lie algebras. Stabilizat...

conjecture | open | MathDB 页面 | 原文线索

V2-008909 | MATHDB-334818

Infinite-dimensional Nichols algebras for specified Yetter-Drinfeld modules

摘录 / EXCERPT: Let $M\langle xy,x)\rangle$, $M\langle y,xy)\rangle$, and $W^{b_1,-1}$ denote the Yetter-Drinfeld modules introduced in the paper, and let $\mathfrak{B}(-)$ denote their Nichols al...

conjecture | open | MathDB 页面 | 原文线索

V2-008910 | MATHDB-334864

Derivation conjecture for the algebra of multiple t-values

摘录 / EXCERPT: Let QSym be the algebra of quasisymmetric functions, let $A_-:\operatorname{QSym}\to\operatorname{QSym}$ be the derivation defined by $A_-(1)=0$ and ... let...

conjecture | open | MathDB 页面 | 原文线索

V2-008911 | MATHDB-334865

Fibonacci dimension conjecture for multiple t-values

摘录 / EXCERPT: Let $\mathcal T$ be the $\mathbb Q$-subalgebra of $\mathbb R$ generated by 1 and the multiple t-values, and let $\mathcal T_n$ be the subspace of $\mathcal T$ spanned by all mu...

conjecture | open | MathDB 页面 | 原文线索

V2-008912 | MATHDB-334873

Characteristic-independence conjecture for level h-vectors

摘录 / EXCERPT: Characteristic-independence conjecture. An h-vector is level in some characteristic if and only if it is level in every characteristic.

conjecture | open | MathDB 页面 | 原文线索

V2-008913 | MATHDB-334883

Almost Cohen-Macaulayness conjecture for Rees algebras of monomial almost complete intersections

摘录 / EXCERPT: Let I be a monomial ideal of $k[x_1,\ldots,x_n]$ that is an almost complete intersection of finite colength, and let $\mathbf R=k[x_1,\ldots,x_n]$. Its Rees algebra is...

conjecture | open | MathDB 页面 | 原文线索

V2-008914 | MATHDB-334919

The model-complete-core CSP dichotomy conjecture

摘录 / EXCERPT: Model-complete-core CSP dichotomy conjecture. One of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-008915 | MATHDB-335010

Primitive automorphic base-change and descent conjecture for perfect central extensions

摘录 / EXCERPT: Let E/F be a Galois extension of number fields and let n be an integer such that ... and, for every divisor $m\mid n$, there are no nontrivial irreducible representations...

conjecture | open | MathDB 页面 | 原文线索

V2-008916 | MATHDB-335018

The generalized Cartan matrix module-algebra conjecture for Hecke-Hopf algebras

摘录 / EXCERPT: Let $A=(a_{ij})$, $i,j\in I$, be a generalized Cartan matrix, and let $W=W_A$ be the corresponding crystallographic Coxeter group, with ... for $i\ne j$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008917 | MATHDB-335054

The basis conjecture for the handlebody Hecke algebra

摘录 / EXCERPT: Let $H_{g,n}(q)$ be the Hecke algebra associated with the handlebody braid group, let $G_{g,n}$ be the corresponding group, and let $\Sigma_{g,n}$ be the set of elements ... where ea...

conjecture | open | MathDB 页面 | 原文线索

V2-008918 | MATHDB-335098

Countability conjecture for decomposable transcendental Hodge structures of surface families

摘录 / EXCERPT: Countability conjecture. There are at most countably many integral polarized Hodge structures $\mathcal{T}_{S_b}$, arising from the fibers S_b of f, which are decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-008919 | MATHDB-335110

The idempotent conjecture for images of LF derivations

摘录 / EXCERPT: Idempotent conjecture. If $e\in\operatorname{Im}\delta$ is an idempotent, then

conjecture | open | MathDB 页面 | 原文线索

V2-008920 | MATHDB-335111

The Laurent polynomial Mathieu-subspace conjecture

摘录 / EXCERPT: Let K be a field of characteristic zero, and let $z=(z_1,z_2,\dots,z_n)$ be n commuting indeterminates. For a commutative K-algebra ${\mathcal A}$ and a K-subspace V of...

conjecture | open | MathDB 页面 | 原文线索

V2-008921 | MATHDB-335135

The maximal rank conjecture for general Brill-Noether curves

摘录 / EXCERPT: Maximal Rank Conjecture. The rank of this restriction map is

conjecture | open | MathDB 页面 | 原文线索

V2-008922 | MATHDB-335141

Spectral transitive-symmetry conjecture for simple Jordan algebras

摘录 / EXCERPT: Let S be a finite-dimensional convex compact set that is spectral, meaning that all of its states have orthogonal decompositions with a common spectrum, and suppose that the symm...

conjecture | open | MathDB 页面 | 原文线索

V2-008923 | MATHDB-335142

Symmetric spectral state-space conjecture for formally real Jordan algebras

摘录 / EXCERPT: Let a state space be a convex set whose states admit spectral decompositions, and call it symmetric when any n-frame can be transformed into any other n-frame by an affine map...

conjecture | open | MathDB 页面 | 原文线索

V2-008924 | MATHDB-335143

Uniqueness conjecture for maximal immediate extensions of valued differential fields

摘录 / EXCERPT: Uniqueness conjecture. If K has small derivation and $\operatorname{res}(K)$ is linearly surjective, then K has the uniqueness property.

conjecture | open | MathDB 页面 | 原文线索

V2-008925 | MATHDB-335163

The q-compatibility characterization conjecture for the Monster

摘录 / EXCERPT: Let $\mathfrak{m}$ be the Monster Lie algebra, and let G be the group of q-compatible automorphisms of $\mathfrak{m}$. The Monster group $\mathbb{M}$ is a subgroup of G. The q-...

conjecture | open | MathDB 页面 | 原文线索

V2-008926 | MATHDB-335176

Conjectures on Cohen-Macaulay residual intersections and their canonical modules

摘录 / EXCERPT: Let R be a Cohen-Macaulay local or $^*\!$local ring, and let I be a strongly Cohen-Macaulay ideal, or more generally an ideal satisfying sliding depth. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-008927 | MATHDB-335202

Murthy's complete intersection conjecture

摘录 / EXCERPT: Murthy's complete intersection conjecture. For every ideal I in A,

conjecture | open | MathDB 页面 | 原文线索

V2-008928 | MATHDB-335204

Malle-Adam-Malle conjecture for 2-ranks of class groups

摘录 / EXCERPT: Let K be a number field whose Galois closure has Galois group S_n, and suppose that K has odd degree n and signature (r_1,r_2). Let C_K be its class group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008929 | MATHDB-335205

The average 2-torsion size of narrow class groups

摘录 / EXCERPT: Let K be a number field whose Galois closure has Galois group S_n, and suppose that K has odd degree n and signature (r_1,r_2). Let C_K^+ denote the narrow class group,...

conjecture | open | MathDB 页面 | 原文线索

V2-008930 | MATHDB-335206

The predicted unit signature-rank distribution for odd-degree S_n-fields

摘录 / EXCERPT: Let K be a number field with signature (r_1,r_2) whose Galois closure has Galois group S_n, where n is odd. Let E_K be the unit group of K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008931 | MATHDB-335224

Fomin-Zelevinsky's C-matrix conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal A_{t_0}$ be the cluster algebra with principal coefficients at Σ_{t_0}, and let ... be a seed of $\mathcal A_{t_0}$ obtained from Σ_{t_0} by iterated muta...

conjecture | open | MathDB 页面 | 原文线索

V2-008932 | MATHDB-335247

The trivial idempotents conjecture for torsion-free group rings

摘录 / EXCERPT: Let $\Bbbk$ be a field of characteristic zero and let G be a torsion-free group. An idempotent in the group ring $\Bbbk[G]$ is an element p satisfying $p^2=p$. The trivial idem...

conjecture | open | MathDB 页面 | 原文线索

V2-008933 | MATHDB-335310

Positive grading is unnecessary for the bimodule 3-Calabi-Yau conclusion

摘录 / EXCERPT: Let (Q,W) be a quiver with potential, and consider the conclusions of the bimodule 3-Calabi-Yau result [?]. In particular, these include the corresponding conclusions for the q...

conjecture | open | MathDB 页面 | 原文线索

V2-008934 | MATHDB-335345

Conjecture on long screening lattices and Cartan-like roots

摘录 / EXCERPT: Let $Lambda$ be a fractional lattice with short basis $alpha_i^{short}$, and let $Lambda^{long}$ be the integral sublattice with associated long basis $alpha_i^{long}$, chosen so t...

conjecture | open | MathDB 页面 | 原文线索

V2-008935 | MATHDB-335349

Kazhdan-Lusztig correspondence conjecture for logarithmic conformal field theories

摘录 / EXCERPT: Let $\mathcal{W}^{log}$ be the proposed logarithmic conformal field theory, and let the smaller long-screening kernel be the corresponding subalgebra. Kazhdan-Lusztig correspondenc...

conjecture | open | MathDB 页面 | 原文线索

V2-008936 | MATHDB-335378

K-theory conjecture for tensor powers of the Leavitt algebra

摘录 / EXCERPT: For $k\geq 1$, let L_k be the quotient algebra identified in the paper with the tensor power $L(1,3)^{\otimes k}$ of the Leavitt algebra. Let $K_0(L_k)$ and $K_1(L_k)$ denote its...

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V2-008937 | MATHDB-335385

The coefficient-7 conjecture for the first three steps of monomial resolutions

摘录 / EXCERPT: Let $I\subseteq S$ be a monomial ideal, and let $\mathbb{F}_{\bullet}$ be its minimal free resolution. A basis of $\mathbb{F}_{\bullet}$ can be chosen so that the differential on...

conjecture | open | MathDB 页面 | 原文线索

V2-008938 | MATHDB-335403

MTWZ finite multiple zeta value algebraicity conjecture

摘录 / EXCERPT: Let $\mathbb{N}$ denote the positive integers. For each $m,n\in\mathbb{N}$, let $R_n^{(m,1)}$ and $S_n^{(m,1)}$ be the finite multiple harmonic sum quantities defined in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-008939 | MATHDB-335448

Regular-sequence conjecture for Mishchenko-Fomenko algebras of nilpotent centralizers

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra, let e be a nilpotent element, and write $\mathfrak{g}^{e}$ for its centralizer. Let $\mathcal{F}_\xi(\mathfrak{g}^{e})$ be the Mish...

conjecture | open | MathDB 页面 | 原文线索

V2-008940 | MATHDB-335458

Silverman's conjecture on common divisors of elliptic sequences

摘录 / EXCERPT: Let $K=k(C)$ be the function field of a smooth projective curve C over an algebraically closed field k of characteristic 0. Let E_1/K and E_2/K be elliptic curves, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-008941 | MATHDB-335467

The generalized Herbrand-Ribet conjecture

摘录 / EXCERPT: Let K be a CM Galois extension of $\mathbb Q$, let K^+ be its totally real subfield, and let $\tau\in\operatorname{Gal}(K/K^+)$ be the nontrivial element. Let ρ be an irre...

conjecture | open | MathDB 页面 | 原文线索

V2-008942 | MATHDB-335491

The small shuffle algebra action conjecture for the entire K-theory

摘录 / EXCERPT: Let S be the surface under consideration, let $\CV_\emph{sm}$ be its small shuffle algebra, and let $K_\CM$ denote the entire K-theory group in place of the tautological part...

conjecture | open | MathDB 页面 | 原文线索

V2-008943 | MATHDB-335493

Cospectral cluster quiver mutation conjecture

摘录 / EXCERPT: Cospectral cluster quiver mutation conjecture. The quivers $Q',Q”\in\operatorname{Mut}(Q)$ are cospectral if and only if there exists a cluster quiver $R\in\operatorname{Mut}(Q)$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-008944 | MATHDB-335555

The De Concini-Procesi compactification conjecture for Bethe subalgebras of Yangians

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a semisimple Lie algebra, let $Y({\mathfrak{g}})$ be its Yangian, and let $M_{\mathfrak{g}}$ be the De Concini-Procesi compactification of the regular part...

conjecture | open | MathDB 页面 | 原文线索

V2-008945 | MATHDB-335570

The converse to the pure-projective characterization of elementary torsion pairs

摘录 / EXCERPT: Let $(\mathcal{T},\mathcal{F})$ be an elementary torsion pair in $\operatorname{Mod}\textrm{-}{R}$, where R is a ring. The torsion class $\mathcal{T}$ is said to have enough pure...

conjecture | open | MathDB 页面 | 原文线索

V2-008946 | MATHDB-335594

AFO's stability conjecture for normal bundles of general canonical curves

摘录 / EXCERPT: Let C be a general non-hyperelliptic smooth projective curve of genus $g\geq 7$, canonically embedded as $C\hookrightarrow\mathbb{P}^{g-1}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008947 | MATHDB-335635

Extension of the maximal-rank corollary beyond the bound on b

摘录 / EXCERPT: Let A be the Artinian algebra and let $\times L^b$ denote multiplication by the b-th power of a general linear form as in Corollary. The corollary currently assumes $b\leq t$....

conjecture | open | MathDB 页面 | 原文线索

V2-008948 | MATHDB-335682

The virtualization conjecture for Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Let $B^{r,s}$ be a Kirillov--Reshetikhin crystal for an affine type, and let ϕ and $(\gamma_a)_{a\in I}$ be the folding and scaling data defining virtual crystals. Define ... Vi...

conjecture | open | MathDB 页面 | 原文线索

V2-008949 | MATHDB-335697

Huneke-Mustaţă-Takagi-Watanabe conjecture on F-thresholds and multiplicities

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a d-dimensional Noetherian local ring of characteristic $p>0$. Let $J\subseteq\mathfrak m$ be an ideal generated by a system of parameters, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008950 | MATHDB-335704

Nagata's conjecture on Zariski chambers of the exceptional ray

摘录 / EXCERPT: Let X be the blow-up of ${\mathbb P}^2$ at a set of n general points, let H be the hyperplane class, and let $\mathbb E=E_1+\cdots+E_n$ be the exceptional divisor. Nagata's c...

conjecture | open | MathDB 页面 | 原文线索

V2-008951 | MATHDB-335705

Szemberg's Pell-equation lower bound for Seshadri constants

摘录 / EXCERPT: Szemberg's conjecture. For a very general point $x\in X$,

conjecture | open | MathDB 页面 | 原文线索

V2-008952 | MATHDB-335719

A sharper loss-of-orthogonality bound for MGS-HA

摘录 / EXCERPT: Let A be the symmetric positive-definite matrix defining the non-standard inner product, let Z be the input matrix with m rows, let n be the number of columns, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008953 | MATHDB-335734

The Hopf-algebraic bijection conjecture for Catalan-avoiding up-down permutations and prographs

摘录 / EXCERPT: Bijection and Hopf-algebra conjecture. There exists a bijection between $A_{2n}(1234)$ and the prographs of $PC(n)$ that preserves the Schützenberger involution and can be extended...

conjecture | open | MathDB 页面 | 原文线索

V2-008954 | MATHDB-335738

Conjecture on symbolic-square generators and the square of the transformed fat ideal

摘录 / EXCERPT: Conjecture on the transformed ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-008955 | MATHDB-335748

Equivalence conjecture for principal blocks of $D(2|1;\zeta)$ category $\mathcal O$

摘录 / EXCERPT: Principal-block equivalence conjecture. The principal blocks $\mathcal B_0$ for all parameters ζ are equivalent as highest weight categories.

conjecture | open | MathDB 页面 | 原文线索

V2-008956 | MATHDB-335770

The finite-index criterion for arboreal Galois groups of quadratic rational maps

摘录 / EXCERPT: Let K be the field from which the rational map is defined, let $\phi\in K(x)$ be a rational function of degree 2, and let $\mathcal{T}$ be the complete rooted 2-ary tree atta...

conjecture | open | MathDB 页面 | 原文线索

V2-008957 | MATHDB-335777

Triviality conjecture for linear Boltzmann weights

摘录 / EXCERPT: A biquasile is an algebraic structure with operations satisfying the axioms introduced in the paper, and a linear Boltzmann weight is a weight of the form defined in Proposition. A...

conjecture | open | MathDB 页面 | 原文线索

V2-008958 | MATHDB-335801

The basis conjecture for the mixed Hecke algebra on two fixed strands

摘录 / EXCERPT: Basis conjecture. The set Λ_n is a $\mathbb{Z}[q^{\pm 1}]$-linear basis for the algebra $\mathrm{H}_{2,n}(q)$.

conjecture | open | MathDB 页面 | 原文线索

V2-008959 | MATHDB-335859

Bayer-Fluckiger et al.'s conjecture on totally decomposable orthogonal involutions

摘录 / EXCERPT: Bayer-Fluckiger et al.'s conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-008960 | MATHDB-335860

The bi-bracket reduction conjecture

摘录 / EXCERPT: For $s_1,\ldots,s_l\geq 1$ and $r_1,\ldots,r_l\geq 0$, a bi-bracket is the q-series ... Let $\operatorname{\mathcal{MD}}$ be the algebra generated by brackets, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008961 | MATHDB-335868

Fomin-Zelevinsky sign-coherence conjecture for g-vectors

摘录 / EXCERPT: Let $\mathcal A(\Sigma)$ be a cluster algebra with principal coefficients at $\Sigma=(X,Y,B)$, where B is an $n\times n$ sign-skew-symmetric integer matrix. For a cluster X' an...

conjecture | open | MathDB 页面 | 原文线索

V2-008962 | MATHDB-335869

Fomin-Zelevinsky recurrence conjecture for g-vectors

摘录 / EXCERPT: Fix a regular tree $\mathbb T_n$. For vertices $t_1,t_2\in\mathbb T_n$ joined by an edge labeled k, write $t_1\overset{k}{--}t_2$, let $B^2=\mu_k(B^1)$, and for $a\in[1,n]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-008963 | MATHDB-335887

Algebraicity conjecture for cluster generating functions of generalized consecutive patterns

摘录 / EXCERPT: Let $m\geq4$ and let $\tau=134\ldots(m-1)2m$ be the consecutive pattern of length m. Let $T(x)$ be the cluster generating function specified by the preceding conjectural hypergeo...

conjecture | open | MathDB 页面 | 原文线索

V2-008964 | MATHDB-335894

Algebraic ODP volume gap conjecture

摘录 / EXCERPT: Algebraic ODP volume gap conjecture. The infimum of the normalized volumes of valuations centered at p satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-008965 | MATHDB-335911

The PVMD power-series complete integral-closure conjecture

摘录 / EXCERPT: Power-series complete integral-closure conjecture. $D[[X]]$ is integrally closed if and only if $D[[X]]$ is completely integrally closed, and this holds if and only if D is compl...

conjecture | open | MathDB 页面 | 原文线索

V2-008966 | MATHDB-335912

The Archimedean characterization conjecture for PVMDs

摘录 / EXCERPT: Archimedean characterization conjecture. D is completely integrally closed if and only if D is Archimedean.

conjecture | open | MathDB 页面 | 原文线索

V2-008967 | MATHDB-335933

FFRT conjecture for invariant rings

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero and let W be a finite dimensional k-representation for a reductive group G. Put ... Here FFRT means that the num...

conjecture | open | MathDB 页面 | 原文线索

V2-008968 | MATHDB-335935

Levasseur-Stafford simplicity conjecture for differential operators on invariant rings

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero and let W be a finite dimensional k-representation for a reductive group G. Put ... Here $D_k(R)$ denotes the ri...

conjecture | open | MathDB 页面 | 原文线索

V2-008969 | MATHDB-335961

The modified-weight characterization of kappa-hyperelliptic semigroups

摘录 / EXCERPT: Modified-weight characterization. Whenever $g \gg \ka$, $\sss$ is $\ka$-hyperelliptic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-008970 | MATHDB-335965

The representation claim for Heyting algebras with an E-pair

摘录 / EXCERPT: Let $\mathfrak{A}$ and $\mathfrak{B}$ be Heyting algebras such that $\mathfrak{A}\preccurlyeq\mathfrak{B}$. Let $a\in|\mathfrak{A}|$, and let $(a,a^{\ast})$ be an $\mathcal{E}$-pai...

conjecture | open | MathDB 页面 | 原文线索

V2-008971 | MATHDB-335966

Characteristic-independent regularity bound for ideals

摘录 / EXCERPT: Characteristic-independence conjecture. The upper bound

conjecture | open | MathDB 页面 | 原文线索

V2-008972 | MATHDB-335978

The quadratic Cohen-Macaulay Hirsch conjecture

摘录 / EXCERPT: The quadratic Cohen-Macaulay Hirsch conjecture. If S/I is Cohen-Macaulay, then I is Hirsch.

conjecture | open | MathDB 页面 | 原文线索

V2-008973 | MATHDB-336096

Uniqueness of the pair-of-pants A-infinity structure

摘录 / EXCERPT: Let Π_n be the n-dimensional pair of pants, let L_I be the relevant objects, and consider the algebra $\bigoplus_{I,J}HW^*(L_I,L_J)$. Assume its A_∞-structure is com...

conjecture | open | MathDB 页面 | 原文线索

V2-008974 | MATHDB-336135

Derived isomorphism conjecture for finite-dimensional Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}_1$ and $\mathfrak{g}_2$ be finite-dimensional Lie algebras over $\mathbb{C}$, and let $\mathfrak{U}\mathfrak{g}_1$ and $\mathfrak{U}\mathfrak{g}_2$ denote their e...

conjecture | open | MathDB 页面 | 原文线索

V2-008975 | MATHDB-336146

The three-stage surjectivity conjecture for quadratic arboreal representations

摘录 / EXCERPT: Let $\phi_c(x)=x^2+c$ with $c\in\mathbb{Z}$. Write $G_n(\phi_c)$ for the Galois group at the nth stage of the arboreal representation and $G_\infty(\phi_c)$ for its full image, v...

conjecture | open | MathDB 页面 | 原文线索

V2-008976 | MATHDB-336157

Maximal adelic image conjecture for hyperelliptic curves

摘录 / EXCERPT: Let $g\geq 2$ and let $H\subset\mathcal S_{2g+2}$ be a closed subgroup. Write $H(2)$ for its reduction modulo 2 subgroup, identified with the natural symmetric-group subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-008977 | MATHDB-336230

F-regularity conjecture for nearly commuting matrix components

摘录 / EXCERPT: F-regularity conjecture for nearly commuting matrices. The rings

conjecture | open | MathDB 页面 | 原文线索

V2-008978 | MATHDB-336231

F-purity conjecture for the nearly commuting-matrix ring

摘录 / EXCERPT: F-purity conjecture. The quotient ring R/I is F-pure for all n.

conjecture | open | MathDB 页面 | 原文线索

V2-008979 | MATHDB-336233

Regular-sequence conjecture for nearly commuting matrices

摘录 / EXCERPT: Regular-sequence conjecture. The displayed collection is a regular sequence on R/I and hence is part of a system of parameters on R/J and R/Q.

conjecture | open | MathDB 页面 | 原文线索

V2-008980 | MATHDB-336297

Equality of local and ordinary derivations for genetic Volterra algebras

摘录 / EXCERPT: Local-derivation conjecture. The category of local derivations of A coincides with the category of derivations of A.

conjecture | open | MathDB 页面 | 原文线索

V2-008981 | MATHDB-336385

Bremner-Madariaga conjecture on the quadraticity of the second Veronese power of PreLie

摘录 / EXCERPT: Let PreLie be the operad of pre-Lie algebras, and let $\operatorname{PreLie}^{[2]}$ denote its second Veronese power. The preceding computations show that all rela...

conjecture | open | MathDB 页面 | 原文线索

V2-008982 | MATHDB-336432

The exponent-separation conjecture for characteristic Betti numbers

摘录 / EXCERPT: Let M be an ideal in $S=k[x_1,\ldots,x_n]$ with minimal generators. For indices $i<j$, suppose that no variable among...

conjecture | open | MathDB 页面 | 原文线索

V2-008983 | MATHDB-336433

The almost-generic nondominant-part conjecture for characteristic Betti numbers

摘录 / EXCERPT: Let M be minimally generated by $\{m_1,\ldots,m_q,n_1,\ldots,n_p\}$, where the m_i are dominant and the n_i are nondominant, and let $M'=(n_1,\ldots,n_p)$. Almost-generic non...

conjecture | open | MathDB 页面 | 原文线索

V2-008984 | MATHDB-336434

The projective-dimension-two conjecture for monomial ideals

摘录 / EXCERPT: Let M be a monomial ideal minimally generated by more than one monomial. Suppose that some minimal generator divides the least common multiple of every pair of generators. Projec...

conjecture | open | MathDB 页面 | 原文线索

V2-008985 | MATHDB-336488

Diagonalizability conjecture for premagic matrices

摘录 / EXCERPT: Let $\mathbf{M}$ be a premagic matrix, and let its eigenvectors be the vectors associated with its eigenvalues. Diagonalizability conjecture. $\mathbf{M}$ is diagonalizable, and it...

conjecture | open | MathDB 页面 | 原文线索

V2-008986 | MATHDB-336491

Ene-Herzog-Hibi regularity conjecture for binomial edge ideals

摘录 / EXCERPT: Let G be a graph, let J_G be its binomial edge ideal, and let $\operatorname{in}_{<_{\mathrm{lex}}}(J_G)$ be its initial ideal with respect to the lexicographic order. Ene-Herz...

conjecture | open | MathDB 页面 | 原文线索

V2-008987 | MATHDB-336534

Seshadri constant conjecture for many points on a surface

摘录 / EXCERPT: Let X be a smooth projective surface and let L be a nef divisor on X. Let $\epsilon(n;X,L)$ denote the relevant multipoint Seshadri constant for n points. Seshadri constant...

conjecture | open | MathDB 页面 | 原文线索

V2-008988 | MATHDB-336553

Interpolation property for the p-adic L-function

摘录 / EXCERPT: Let $K_1(Q_\Lambda[G])$ be the first algebraic K-group of the total quotient algebra $Q_\Lambda[G]$, let Nrd denote the reduced norm, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008989 | MATHDB-336554

Non-commutative Iwasawa main conjecture for the Selmer complex

摘录 / EXCERPT: Let $\Lambda[G]$ be the Iwasawa algebra, let $Q_\Lambda[G]$ be its total quotient algebra, let ∂ be the boundary map in relative K-theory, and let $\mathrm{SC}(U,T,T^0)$...

conjecture | open | MathDB 页面 | 原文线索

V2-008990 | MATHDB-336570

Characterization of strong singularity by a real submatrix

摘录 / EXCERPT: Let $\text{$text{$\mathversion{bold}$ A $}$}$\in\mathbb{IR}^{n\times n}$ be an interval matrix. A submatrix is real when all of its entries are degenerate intervals, and $\text{$te...

conjecture | open | MathDB 页面 | 原文线索

V2-008991 | MATHDB-336577

Pflueger's effective-weight conjecture for numerical semigroups

摘录 / EXCERPT: Pflueger's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-008992 | MATHDB-336582

Nonexistence of finite-dimensional local models for nonabelian quotient surface singularities

摘录 / EXCERPT: Let $G < \operatorname{GL}(2, \mathbb{C})$ be a nonabelian finite subgroup and let $R=\mathbb{C}\llbracket x,y\rrbracket^G$. A finite-dimensional local $\mathbb{C}$-algebra is an a...

conjecture | open | MathDB 页面 | 原文线索

V2-008993 | MATHDB-336625

Full-twist prediagonalization conjecture for finite Coxeter systems

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let $\operatorname{FT}(W)$ be the Rouquier complex of the full-twist braid, and let λ range over the two-sided cells of W. Full-tw...

conjecture | open | MathDB 页面 | 原文线索

V2-008994 | MATHDB-336626

Full-twist diagonalization conjecture for Coxeter systems

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let $\operatorname{FT}(W)$ denote the Rouquier complex of its full-twist braid, and let λ range over the two-sided cells of W. An idempot...

conjecture | open | MathDB 页面 | 原文线索

V2-008995 | MATHDB-336628

Brumer's conjecture for abelian extensions

摘录 / EXCERPT: Let L/K be an abelian extension of number fields with Galois group G. Let S be a finite set of places of K containing all archimedean places and all places ramified in...

conjecture | open | MathDB 页面 | 原文线索

V2-008996 | MATHDB-336635

The composite-(2,11) Galois image conjecture

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbf{Q}$, let $\rho_{E,2}$ and $\rho_{E,11}$ denote its mod-2 and mod-11 Galois representations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008997 | MATHDB-336636

The composite-(2,13) Galois image conjecture

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbf{Q}$, let $\rho_{E,2}$ and $\rho_{E,13}$ denote its mod-2 and mod-13 Galois representations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-008998 | MATHDB-336637

Independence of the cluster-model property for triangulated-surface quivers

摘录 / EXCERPT: Let $\mathbb{T}$ be a triangulation of a not necessarily orientable surface, and suppose that there is a rigid potential W_I on the quiver-with-potential $\Diamond_I(\mathbb{T})$...

conjecture | open | MathDB 页面 | 原文线索

V2-008999 | MATHDB-336660

Berenstein-Kazhdan's presentation conjecture for the algebra ${\bf D}(W)$

摘录 / EXCERPT: Berenstein-Kazhdan's presentation conjecture. The algebra ${\bf D}(W)$ is generated by the D_s subject to the idempotent relations $D_s^2=D_s$, the braid relations

conjecture | open | MathDB 页面 | 原文线索

V2-009000 | MATHDB-336676

Kontsevich-Kanel-Belov correspondence conjecture for Weyl and Poisson automorphisms

摘录 / EXCERPT: Kontsevich-Kanel-Belov correspondence conjecture. The automorphism groups are isomorphic:

conjecture | open | MathDB 页面 | 原文线索

V2-009001 | MATHDB-336681

Kippenhahn's conjecture on hermitian matrix algebras

摘录 / EXCERPT: Kippenhahn's conjecture. The algebra generated by H and K cannot be the full algebra $M_{2n}(\mathbb{C})$.

conjecture | open | MathDB 页面 | 原文线索

V2-009002 | MATHDB-336703

A finite-matrix bound from the weighted Bernstein-Walsh estimate

摘录 / EXCERPT: Finite-matrix convergence conjecture. There is a (modest) constant $C\in\mathbb R$ and a technique of associating a measure σ with compact support and continuous potential t...

conjecture | open | MathDB 页面 | 原文线索

V2-009003 | MATHDB-336712

Hsiang's finiteness-obstruction conjecture for finitely dominated classifying spaces

摘录 / EXCERPT: Let Γ be a group and let $B\Gamma$ denote its classifying space. Assume that $B\Gamma$ is finitely dominated, and denote its finiteness obstruction by $\sigma(B\Gamma)$. Hsi...

conjecture | solved | MathDB 页面 | 原文线索

V2-009004 | MATHDB-336720

Generation conjecture for finite-dimensional pointed Hopf algebras

摘录 / EXCERPT: Let L be a cosemisimple Hopf algebra and let A be a pointed Hopf algebra with diagram $\mathcal R$ and infinitesimal braiding $\mathcal V=\mathcal R^1$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-009005 | MATHDB-336754

The tropical web-function conjecture for four flags

摘录 / EXCERPT: Let n be a positive integer and let a,b,c,d satisfy $1\leq a,b,c,d<n$, $a+b>n$, and $a+b+c+d=2n$. Let F be the cluster-algebra function on...

conjecture | open | MathDB 页面 | 原文线索

V2-009006 | MATHDB-336759

The naive-algebra presentation conjecture for Fukaya Hall composition subalgebras

摘录 / EXCERPT: Let (S,M) be a graded marked surface with a full arc system A, and suppose that (S,M) has enough marked intervals, meaning that for each disk (D_k^2,M_k) in the decompositi...

conjecture | open | MathDB 页面 | 原文线索

V2-009007 | MATHDB-336811

Existence of residual Galois representations attached to ordinary Hecke maximal ideals

摘录 / EXCERPT: Let $m\geq 1$ and let $\mathfrak{m}_m$ be a maximal ideal of ${\bf h}_{\rm ord}(K_{\infty,\infty}^m;\mathcal{O})$. A Galois representation $\overline{\rho}_{\mathfrak{m}}$ is under...

conjecture | open | MathDB 页面 | 原文线索

V2-009008 | MATHDB-336884

Characterization of local rings whose only strong Rees ideals are powers of the maximal ideal

摘录 / EXCERPT: Strong Rees property characterization conjecture. Then $\dim A=2$, and either A is a regular local ring or

conjecture | open | MathDB 页面 | 原文线索

V2-009009 | MATHDB-336906

Claire Glanois's kernel-image conjecture for period-polynomial maps

摘录 / EXCERPT: Let $\mathbf W_{N,r}$ be the space of restricted even period polynomials, and let $\varphi^{(r)}_j$ be the maps on the polynomial spaces used in the preceding lemmas. For $r\geq3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-009010 | MATHDB-336913

B1 conjecture on positive bi-brackets

摘录 / EXCERPT: Let $\mathcal Z_q$ be the bi-bracket algebra, let $\mathcal Z_q^+$ be the subspace spanned by positive bi-brackets, and let $\operatorname{Fil}^{W,L}_{w,l}$ denote the weight-and-l...

conjecture | open | MathDB 页面 | 原文线索

V2-009011 | MATHDB-336914

B2 conjecture on expressing bi-brackets by brackets

摘录 / EXCERPT: Let $\mathcal Z_q$ be the bi-bracket algebra and let $\mathcal Z_q^{\circ}$ be the bracket subalgebra. B2 conjecture. Every bi-bracket is a linear combination of brackets; equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-009012 | MATHDB-336922

van der Put-Singer inverse problem conjecture for difference Galois groups

摘录 / EXCERPT: van der Put-Singer inverse problem conjecture. If G/G° is cyclic, then G is the Galois group of a linear difference equation

conjecture | open | MathDB 页面 | 原文线索

V2-009013 | MATHDB-336979

Quadratic-growth tau-sequence conjecture for octagon posets

摘录 / EXCERPT: Octagon-poset tau-sequence conjecture. For each octagon poset P, the R-system associated with $G(P)$ admits a strong τ-sequence consisting of irreducible Laurent polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-009014 | MATHDB-337024

Fiber-type conjecture for Rees algebras of ideals with alternating basic entry sequences

摘录 / EXCERPT: Let I be the ideal considered in the alternating case, with an alternating basic entry sequence as in the preceding construction, and let $\mathcal{R}_R(I)$ denote its Rees algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-009015 | MATHDB-337028

Generation conjecture for the polar linearised double shuffle Lie algebra

摘录 / EXCERPT: Generation conjecture. The Lie algebra $\mathfrak{p}\mathfrak{ls}$ is generated by $\mathfrak{p}\mathfrak{ls}_1$.

conjecture | open | MathDB 页面 | 原文线索

V2-009016 | MATHDB-337029

Surjectivity conjecture for the elliptic and polar linearised double shuffle Lie algebras

摘录 / EXCERPT: Surjectivity conjecture. The map is an isomorphism:

conjecture | open | MathDB 页面 | 原文线索

V2-009017 | MATHDB-337058

The weighted Bounded Negativity Conjecture

摘录 / EXCERPT: Weighted Bounded Negativity Conjecture. There exists a nonnegative integer $b_w\in\mathbb Z$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-009018 | MATHDB-337094

Nilpotence modulo 2 of the matrices associated with run-length encoding

摘录 / EXCERPT: Nilpotence conjecture. For all $a\geqslant 0$, $b\geqslant 0$, and $k\geqslant 0$, there exists an integer $N\geqslant 0$ such that every entry of

conjecture | open | MathDB 页面 | 原文线索

V2-009019 | MATHDB-337095

The conjectured predecessor containment for three general binary forms

摘录 / EXCERPT: Let $d=2s+1\geq 7$. Given three general forms f_1,f_2,f_3 of degree d in R, let $I=(f_1,f_2,f_3)$, let $\delta(I)$ be the ideal generated by the associated degree-$d-2$ forms...

conjecture | open | MathDB 页面 | 原文线索

V2-009020 | MATHDB-337101

Gras's Borel-Cantelli conjecture for p-rationality of real Galois fields

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a real Galois extension with Galois group G, and let $\eta\in E_K$ be a Minkowski unit, meaning a unit generating a sub-G-module of finite index in E_K....

conjecture | open | MathDB 页面 | 原文线索

V2-009021 | MATHDB-337142

Kaplansky's unit conjecture for torsion-free group algebras

摘录 / EXCERPT: Let $\mathbb{F}$ be a field and G a torsion-free group. A unit of the group algebra $\mathbb{F}[G]$ is called trivial when it is a non-zero scalar multiple of a group element. Ka...

conjecture | open | MathDB 页面 | 原文线索

V2-009022 | MATHDB-337153

Scalar proportionality of the SVV and HM symmetrizing forms

摘录 / EXCERPT: Let $\mathscr{R}_{\beta}^{\Lambda}$ be a cyclotomic quiver Hecke algebra, with homogeneous symmetrizing forms $\operatorname{Tr}^{\text{SVV}}$ and $\operatorname{Tr}^{\text{HM}}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-009023 | MATHDB-337167

The Sharifi-inspired Hecke annihilation conjecture for the map \(\varpi\)

摘录 / EXCERPT: Let N and p be as in the construction above, and let ... be the homomorphism induced by the Manin-symbol map. Let T_n denote the Hecke operators and let $\langle d\rangle$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-009024 | MATHDB-337171

The nonvanishing conjecture for the Eisenstein elements \(e_0^1\) and \(e_0^0\)

摘录 / EXCERPT: Let L be the set of supersingular lambda invariants, let M' be the free module generated by the symbols $[\lambda]$ for $\lambda\in L$, and let e_0^0 and e_0^1 be the eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-009025 | MATHDB-337172

The Sharifi-inspired Hecke annihilation conjecture for \(\varpi\) away from \(p\)

摘录 / EXCERPT: Let ... be the homomorphism induced by the Manin-symbol map. Let T_n denote the Hecke operators and let $\langle d\rangle$ denote the dth diamond operator. Sharifi-inspired away-...

conjecture | open | MathDB 页面 | 原文线索

V2-009026 | MATHDB-337178

Narayana-number formula for the determinant's linear Betti numbers

摘录 / EXCERPT: The Betti-number conjecture. The Betti number $\beta_{r,r+1}$ is given by

conjecture | open | MathDB 页面 | 原文线索

V2-009027 | MATHDB-337190

Mutation compatibility conjecture for arc representations

摘录 / EXCERPT: Let (Σ,M) be a marked surface, and let σ and τ be tagged triangulations related by a flip at the tagged arc i, so that $f_i(\sigma)=\tau$. For a tagged arc i...

conjecture | open | MathDB 页面 | 原文线索

V2-009028 | MATHDB-337194

Artin's holomorphy conjecture for nontrivial representations

摘录 / EXCERPT: Let L/k be a finite normal extension of number fields with $\operatorname{Gal}(L/k)\simeq G$, let ρ be a complex representation of G, and let $L(s,\rho,L/k)$ be the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-009029 | MATHDB-337195

Langlands' strong Artin reciprocity conjecture

摘录 / EXCERPT: Let L/k be a finite Galois extension of number fields with $\operatorname{Gal}(L/k)\simeq G$, and let ρ be an m-dimensional complex representation of G. Let $L(s,\rho)$...

conjecture | open | MathDB 页面 | 原文线索

V2-009030 | MATHDB-337223

Gersten's acyclicity conjecture for regular local rings

摘录 / EXCERPT: Let $X = \operatorname{Spec} A$, where $(A,\mathfrak{m}_A,L)$ is a regular local ring. For each $n \geq 0$, let $\mathcal{G}_n(X)$ be the Gersten complex associated with the codime...

conjecture | open | MathDB 页面 | 原文线索

V2-009031 | MATHDB-337225

Gersten's arithmetic DVR conjecture

摘录 / EXCERPT: Let (R,pR,k) be a discrete valuation ring essentially smooth over $\mathbb{Z}$, with residue field k. Gersten's Arithmetic DVR Conjecture. For every $n \geq 0$, the transfer ma...

conjecture | open | MathDB 页面 | 原文线索

V2-009032 | MATHDB-337307

The double shuffle algebra action on the K-theory of moduli spaces

摘录 / EXCERPT: Let S be a smooth projective surface, and fix rank and Chern classes so that $\CM$ denotes the corresponding moduli space of stable sheaves on S. Let $K_\CM$ be its K-theory....

conjecture | open | MathDB 页面 | 原文线索

V2-009033 | MATHDB-337308

Conjecture on the commutation relations of the geometric operators

摘录 / EXCERPT: Let (n,k) and (n',k') be lattice points, and let $e_{n,k}$ and $e_{n',k'}$ be the corresponding geometric operators. Let Δ_* denote pushforward along the diagonal, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009034 | MATHDB-337318

Classical Farrell-Jones conjecture for regular group rings

摘录 / EXCERPT: Let k be a regular ring, let G be a torsionfree group, let $\mathrm{B}G$ denote its classifying space, and let $\mathbf{K}(k)$ denote the algebraic K-theory spectrum of k....

conjecture | open | MathDB 页面 | 原文线索

V2-009035 | MATHDB-337329

The $fM_H(G)$ conjecture for Selmer groups over admissible extensions

摘录 / EXCERPT: Let E be an elliptic curve over a number field K with good and ordinary reduction at all primes dividing p. Let K_∞/K be an admissible Galois extension, with...

conjecture | open | MathDB 页面 | 原文线索

V2-009036 | MATHDB-337374

The symmetric-group conjecture for Galois groups of bounded-height polynomials

摘录 / EXCERPT: Let f be as in Theorem 1, namely a random polynomial in the family specified there, and let its Galois group be viewed as a subgroup of S_n. Symmetric-group conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-009037 | MATHDB-337400

Tasaka's isomorphism conjecture for the depth-three kernel

摘录 / EXCERPT: Tasaka's isomorphism conjecture. The map $\widetilde{\xi}$ is an isomorphism. The conjecture concerns the depth-three case and is attributed to Tasaka. The paper does not report a...

conjecture | open | MathDB 页面 | 原文线索

V2-009038 | MATHDB-337426

Conjecture that the quadrics generate the gaussoid ideal J_n

摘录 / EXCERPT: Quadratic generation conjecture. The ideal J_n is generated by its quadrics, listed explicitly in Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-009039 | MATHDB-337467

Surjectivity of linear higher Chow cycles for number fields

摘录 / EXCERPT: For a number field k, let $\mathrm{CH}^n(k,2n-1)_{\mathbb{Q}}$ be the simplicial higher Chow group and let $\mathrm{LCH}^n(k,2n-1)_{\mathbb{Q}}$ be the image of the canonical hom...

conjecture | open | MathDB 页面 | 原文线索

V2-009040 | MATHDB-337471

The vanishing of the Iwasawa bc-invariant for totally real fields

摘录 / EXCERPT: Vanishing of the Iwasawa μ-invariant. For every totally real field F, the Galois group over F_{cyc} of the maximal abelian p-extension of F_{cyc} unra...

conjecture | open | MathDB 页面 | 原文线索

V2-009041 | MATHDB-337472

Uniform version of the Tame Fontaine-Mazur conjecture

摘录 / EXCERPT: Let ${\rm K}$ be a number field, and let Γ be a uniform pro-p group of dimension $d>2$, so that Γ is infinite. A Galois extension ${\rm L}/{\rm K}$ is finitely and...

conjecture | open | MathDB 页面 | 原文线索

V2-009042 | MATHDB-337474

Fontaine-Mazur unramified conjecture for representations

摘录 / EXCERPT: Let K be a number field, and let ... denote the Galois group of the maximal unramified extension of K. Fontaine-Mazur's unramified conjecture. Every continuous Galois representat...

conjecture | open | MathDB 页面 | 原文线索

V2-009043 | MATHDB-337489

Dao-Takahashi's finite-radius conjecture for resolving subcategories

摘录 / EXCERPT: Let R be a Cohen-Macaulay local ring. Let $\operatorname{\mathcal{X}}$ be a resolving subcategory of finitely generated R-modules with finite radius. Dao-Takahashi's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-009044 | MATHDB-337490

Just-infinite GK-dimension conjecture for the positive Witt enveloping algebra

摘录 / EXCERPT: Let ${\rm U}(W_+)$ be the universal enveloping algebra of the positive Witt algebra W_+ over a field of characteristic zero. The algebra ${\rm U}(W_+)$ has just infinite Gelfand-...

conjecture | open | MathDB 页面 | 原文线索

V2-009045 | MATHDB-337491

Ascending-chain conjecture for two-sided ideals of the positive Witt enveloping algebra

摘录 / EXCERPT: Let W_+ be the positive Witt algebra over a field of characteristic zero, and let ${\rm U}(W_+)$ be its universal enveloping algebra. The two-sided ideals of ${\rm U}(W_+)$ satis...

conjecture | open | MathDB 页面 | 原文线索

V2-009046 | MATHDB-337519

The elementary derivation conjecture for trinomial algebras

摘录 / EXCERPT: Elementary derivation conjecture. All $\deg$-homogeneous locally nilpotent derivations of a trinomial algebra are elementary.

conjecture | open | MathDB 页面 | 原文线索

V2-009047 | MATHDB-337527

The levelness conjecture for a family of s-lecture hall polytopes

摘录 / EXCERPT: Let ${\boldsymbol s}\in\mathbb{Z}_{\geq 1}^n$ be a sequence, and let $\mathbf{P}_n^{({\boldsymbol s})}$ denote the corresponding s-lecture hall polytope. Levelness conjecture. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-009048 | MATHDB-337569

Migliore-Miro-Roig conjecture on ghost terms in resolutions of generic forms

摘录 / EXCERPT: Migliore-Miro-Roig conjecture. The minimal free resolution of S/I has no redundant term. Consequently, the minimal free resolution is the minimum one consistent with the Hilbert...

conjecture | open | MathDB 页面 | 原文线索

V2-009049 | MATHDB-337570

Iarrobino's conjecture on powers of generic linear forms

摘录 / EXCERPT: Iarrobino's conjecture. Outside these exceptions, the algebras

conjecture | open | MathDB 页面 | 原文线索

V2-009050 | MATHDB-337572

Crucial-Lundqvist-Nevo lattice-path conjecture for two generic forms in an exterior algebra

摘录 / EXCERPT: Crucial-Lundqvist-Nevo conjecture. The Hilbert series of $\Lambda(V)/(f,g)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-009051 | MATHDB-337573

Bigraded Fröberg conjecture for generic forms on \(\mathbb P^1\times\mathbb P^1\)

摘录 / EXCERPT: Bigraded Fröberg conjecture. The Hilbert series of $R/(f_1,\ldots,f_r)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-009052 | MATHDB-337594

Centrality conjecture for CPA-structures on free-nilpotent Lie algebras

摘录 / EXCERPT: Let $F_{g,c}$ be the free-nilpotent Lie algebra with $g\ge 2$ generators and nilpotency class $c\ge 3$. A CPA-structure is central when its products take values in the center of...

conjecture | open | MathDB 页面 | 原文线索

V2-009053 | MATHDB-337599

Existence of strongly stable ideals with prescribed regularity and h-polynomial degree

摘录 / EXCERPT: Existence conjecture. There exists a strongly stable ideal I of S such that

conjecture | open | MathDB 页面 | 原文线索

V2-009054 | MATHDB-337626

A Bloch-Kato nonvanishing implication for polarized Hecke characters of weight minus one

摘录 / EXCERPT: Let E be a quadratic imaginary field, let χ be an algebraic Hecke character of E, and let $\chi_p:G_E\to K$ be its p-adic realization. Suppose that χ is polarized a...

conjecture | open | MathDB 页面 | 原文线索

V2-009055 | MATHDB-337633

Last-row Schur module conjecture for the Veronese surface

摘录 / EXCERPT: Last-row Schur module conjecture. The last nonzero syzygy space is

conjecture | open | MathDB 页面 | 原文线索

V2-009056 | MATHDB-337673

Broadhurst's algebra-basis conjecture for MZV(6)

摘录 / EXCERPT: Let $B^6=\{B^6_n,n\geq1\}$ be the Broadhurst set, where the sets B^6_n are formed from iterated integrals indexed by Lyndon words in the letters 0, ξ_3^2, and $\xi_2=-1$, s...

conjecture | open | MathDB 页面 | 原文线索

V2-009057 | MATHDB-337676

Moreno-Socías's conjecture for generic complete intersections

摘录 / EXCERPT: Moreno-Socías's conjecture for generic complete intersections. Then $\operatorname{in}(I)$ is almost reverse lexicographic.

conjecture | open | MathDB 页面 | 原文线索

V2-009058 | MATHDB-337698

Non-torsion primitive-point conjecture for large almost simple Galois extensions

摘录 / EXCERPT: Let C/K be a smooth, projective, hyperelliptic curve of genus $g\geq 1$ with a K-rational point P_0, and embed C into its Jacobian J_C using P_0. Let L/K be a finite...

conjecture | open | MathDB 页面 | 原文线索

V2-009059 | MATHDB-337699

Non-torsion points for the Osada polynomial family

摘录 / EXCERPT: Let $n\geq 5$ be odd, let $0\neq d\in\mathbb Z$ be chosen so that $g(X)=X^n-X-1+d^2$ has distinct roots, and let $C/\mathbb Q$ be the smooth projective hyperelliptic curve defined...

conjecture | open | MathDB 页面 | 原文线索

V2-009060 | MATHDB-337714

Existence of balanced cell representations for affine Hecke algebras

摘录 / EXCERPT: An affine Hecke algebra is an algebra associated with an affine Weyl group and a choice of parameters; a balanced system of cell representations is a system of representations sati...

conjecture | open | MathDB 页面 | 原文线索

V2-009061 | MATHDB-337715

Existence of the Plancherel-to-cell map for affine Hecke algebras

摘录 / EXCERPT: Let Π range over the classes of representations appearing in the Plancherel formula, and let the two-sided cells be the cells of the affine Hecke algebra. A Plancherel-to-cell...

conjecture | open | MathDB 页面 | 原文线索

V2-009062 | MATHDB-337716

Plancherel formula and Lusztig's a-function conjecture

摘录 / EXCERPT: Let Π be a class of representations appearing in the Plancherel theorem, and let Ω be the map from such classes to two-sided cells; write $\Gamma=\Omega(\Pi)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009063 | MATHDB-337728

The intermediate value conjecture for hyperseries

摘录 / EXCERPT: Let $\textbf{Hy}$ be the intended field of hyperseries, equipped with field operations, differentiation, and composition. Let Φ be a partial function from $\textbf{Hy}$ into i...

conjecture | open | MathDB 页面 | 原文线索

V2-009064 | MATHDB-337730

The generalized Lech inequality for mixed multiplicities

摘录 / EXCERPT: Let $(R,{\tt\mathfrak m})$ be a Noetherian local ring of dimension d, and let s_i be the unsigned Stirling numbers of the first kind defined by ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-009065 | MATHDB-337731

The generalized Lech inequality for products with the maximal ideal

摘录 / EXCERPT: Let $(R,{\mathfrak m})$ be a Noetherian local ring of dimension $d\geq4$, and let $\lambda(R/I)$ denote the length of R/I. The product-form generalized Lech inequality. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-009066 | MATHDB-337732

The mixed-multiplicity bound for full ideals

摘录 / EXCERPT: Let $(R,{\mathfrak m})$ be a regular local ring of dimension d, and let t_i be defined by ... An ideal is ${\mathfrak m}$-full if it has the corresponding maximal-ideal fullness...

conjecture | open | MathDB 页面 | 原文线索

V2-009067 | MATHDB-337734

Doro's conjecture on the commutative center of Moufang loops

摘录 / EXCERPT: Doro's conjecture. If $\operatorname{Nuc}(M)=1$, then $\operatorname{C}(M)$ is normal in M.

conjecture | open | MathDB 页面 | 原文线索

V2-009068 | MATHDB-337739

Categorification conjecture for the m-tensor quiver Schur algebra

摘录 / EXCERPT: For positive integers $r_1,\dots,r_m$ with $l=r_1+\cdots+r_m$, let $S_n(\mathbf{r})$ be the p-DG m-tensor quiver Schur algebra ... and let $\mathcal{D}^c(S_n(\mathbf{r}))$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-009069 | MATHDB-337750

Manin-Mumford conjecture for abelian uniformizable T-modules

摘录 / EXCERPT: Let A be the coefficient ring and let B be a subring of A. Let $\mathcal{A}=(\mathbb{G}_{a}^{m},\Phi)$ be an abelian uniformizable T-module, meaning an abelian uniformizabl...

conjecture | open | MathDB 页面 | 原文线索

V2-009070 | MATHDB-337783

The comparison conjecture for canonical and bi-canonical degree

摘录 / EXCERPT: Comparison conjecture. In general,

conjecture | open | MathDB 页面 | 原文线索

V2-009071 | MATHDB-337784

The hyperplane-section inequality for bi-canonical degree

摘录 / EXCERPT: Hyperplane-section conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009072 | MATHDB-337820

The positive-isotopy conjecture for scannable algebraic divides

摘录 / EXCERPT: Let D and D' be link-equivalent scannable algebraic divides of the same minimal index. Let $\beta(D)$ and $\beta(D')$ be their associated positive braids; two positive braids a...

conjecture | open | MathDB 页面 | 原文线索

V2-009073 | MATHDB-337891

Dimitrov-Wan genus stability conjecture for ordinary algebraic-geometric towers

摘录 / EXCERPT: Genus stability conjecture. There is a polynomial $G(x)\in\mathbb{Q}[x]$ of degree at most ${\rm dim}(G)+1$, depending on the tower, such that for all sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-009074 | MATHDB-337910

The isolated-singularity tensor product conjecture for maximal Cohen-Macaulay modules

摘录 / EXCERPT: Let R be a local complete intersection domain, and let $M,N\in\operatorname{mod} R$. Assume that R is an isolated singularity. Tensor-product maximal Cohen-Macaulay conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-009075 | MATHDB-337911

The one-dimensional tensor-product torsion-freeness conjecture

摘录 / EXCERPT: Let R be a one-dimensional local complete intersection domain, and let $M,N\in\operatorname{mod} R$. Tensor-product torsion-freeness conjecture. If $M\otimes_R N$ is torsion-free...

conjecture | open | MathDB 页面 | 原文线索

V2-009076 | MATHDB-337917

Local mod p Langlands determination conjecture for ordinary representations

摘录 / EXCERPT: Let F be a CM field in which p is unramified, let w be a place of F above p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009077 | MATHDB-337918

Weight elimination conjecture for ordinary mod p Galois representations

摘录 / EXCERPT: Let $\overline{r}:G_F\rightarrow\mathrm{GL}_n(\mathbf{F})$ be a continuous automorphic Galois representation with $\overline{r}|_{G_{F_w}}\cong\overline{\rho}_0$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-009078 | MATHDB-337943

Marin-Wagner's kernel-generation conjecture for the Links-Gould representation

摘录 / EXCERPT: Let $b1\in\mathbb{C}\setminus\mathbb{Q}$, let $H_n(\alpha)$ be the cubic Hecke algebra through which the braid-group representation factors, and let r_2 and r_3 be the relation...

conjecture | open | MathDB 页面 | 原文线索

V2-009079 | MATHDB-338046

Rush-Shi's cyclic sieving conjecture for exceptional minuscule posets

摘录 / EXCERPT: Let P be one of the two minuscule posets associated to an exceptional Lie algebra of type E, and let $k\in\mathbb{Z}_{\geq 0}$. Write ${\mathsf{PP}}^k(P)$ for the plane partiti...

conjecture | open | MathDB 页面 | 原文线索

V2-009080 | MATHDB-338049

The braid group and exchange graph conjecture for quivers with superpotential

摘录 / EXCERPT: Let (Q,W) be a quiver with superpotential of degree N, and let Γ be its associated Ginzburg differential graded algebra. Write $\operatorname{CT}(Q,W)$ for the cluster b...

conjecture | open | MathDB 页面 | 原文线索

V2-009081 | MATHDB-338072

Degree-three recovery thresholds for heterogeneous MRA

摘录 / EXCERPT: Heterogeneous MRA recovery conjecture. Generic unique recovery is possible at degree 3 if $\mathcal{U}>Kp+K-1$, while generic list recovery is possible at degree 3 precisely if...

conjecture | open | MathDB 页面 | 原文线索

V2-009082 | MATHDB-338085

Herzog-Srinivasan conjectures on generators of initial ideals of numerical semigroups

摘录 / EXCERPT: Let H be a numerical semigroup minimally generated by $a_1<\dots<a_n$. Let $\operatorname{width}(H)=a_n-a_1$, let I_H^* denote the ideal of initial forms associated to the defi...

conjecture | open | MathDB 页面 | 原文线索

V2-009083 | MATHDB-338127

Fröberg-Iarrobino conjecture on Hilbert functions of generic power ideals

摘录 / EXCERPT: Let S be a polynomial ring in n variables over $\mathbb C$, and let $\ell_1,\ldots,\ell_r$ be generic linear forms. For a positive integer d, set...

conjecture | open | MathDB 页面 | 原文线索

V2-009084 | MATHDB-338128

Nicklasson's conjecture on powers of generic forms

摘录 / EXCERPT: Let S be a polynomial ring over $\mathbb C$, and let $g_1,\ldots,g_r$ be generic forms of degree $d>1$. Nicklasson's conjecture. The ideal ... has the same Hilbert series as an ide...

conjecture | open | MathDB 页面 | 原文线索

V2-009085 | MATHDB-338129

The cyclotomic numerical semigroup conjecture

摘录 / EXCERPT: Let $\mathcal S=\langle s_1,\ldots,s_k\rangle$ be a numerical semigroup and let $k[\mathcal S]=k[x^{s_1},\ldots,x^{s_k}]$ be its semigroup ring. Write its Hilbert series as...

conjecture | open | MathDB 页面 | 原文线索

V2-009086 | MATHDB-338131

The Hilbert-series conjecture for two quadratic forms in an exterior algebra

摘录 / EXCERPT: Let E be an exterior algebra over $\mathbb C$ with n generators, let $f,g\in E$ be generic quadratic forms, and let S be the corresponding commutative polynomial ring with ge...

conjecture | open | MathDB 页面 | 原文线索

V2-009087 | MATHDB-338176

Consistency criterion for torsion-free nilpotent group presentations

摘录 / EXCERPT: Consistency conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009088 | MATHDB-338177

Degree conjecture for reduced multiplication and powering polynomials

摘录 / EXCERPT: Degree conjecture. For every relevant n,

conjecture | open | MathDB 页面 | 原文线索

V2-009089 | MATHDB-338196

The confluence-relations conjecture for multiple zeta values

摘录 / EXCERPT: Let $\mathcal{A}^{0}$ be the algebra of admissible words, let $L:\mathcal{A}^{0}\to\mathbb{R}$ be the map whose values are multiple zeta values, and let $\mathcal{I}_{\rm CF}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-009090 | MATHDB-338240

Categorical cyclotomic Hecke algebra conjecture for framed matrix factorizations

摘录 / EXCERPT: Categorical cyclotomic Hecke algebra conjecture. The homomorphism ϕ^{ct} is an isomorphism. The map is known to be injective when $r=1$, while the asserted isomorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-009091 | MATHDB-338292

The potential conjecture for monoidal categorification

摘录 / EXCERPT: Let $\mathcal{C}$ be a rigid monoidal abelian category with a system of renormalized r-matrices, and let $(\{\mathcal{F}_i\}_{i \in I}, \widetilde{B})$ be a monoidal seed, where...

conjecture | open | MathDB 页面 | 原文线索

V2-009092 | MATHDB-338295

The wreath-product conjecture for monodromy of reduced irreducible tuples

摘录 / EXCERPT: Let A be a tuple in the setting of Step (1) of Classification, and let B be reduced and irreducible of mixed volume d. Write j for the map occurring in that classification,...

conjecture | open | MathDB 页面 | 原文线索

V2-009093 | MATHDB-338312

Comes-Kujawa's basis conjecture for cyclotomic oriented Brauer-Clifford supercategories

摘录 / EXCERPT: Fix nonnegative integers a,b, a field $\beta5$ of characteristic not two, and parameters $\mathbf{u}=(u_1,\ldots,u_b)\in(\kappa^*)^b$. Let ... and let ℓ be the degree of...

conjecture | open | MathDB 页面 | 原文线索

V2-009094 | MATHDB-338385

The nilpotent characterization conjecture for Θ-positive unipotents

摘录 / EXCERPT: Let G be a semisimple Lie group with a Θ-positive structure, with nilpotent Lie algebra $\mathfrak{u}_\Theta$ and positivity cones c°_β for $\beta\in\Theta$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-009095 | MATHDB-338389

Non-vanishing conjecture for twisted central critical values

摘录 / EXCERPT: Non-vanishing conjecture. There is an algebraic Hecke character χ of $\operatorname{GL}_1(\mathbb A_F)$, with conjugate self-dual archimedean component χ_∞, such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-009096 | MATHDB-338391

Finite-dimensional semisimplicity conjecture for universal partial Hopf algebras

摘录 / EXCERPT: Let H be a Hopf algebra and let H_{par} denote its universal partial Hopf algebra. Assume that H is finite dimensional and semi-simple. Finite-dimensional semisimplicity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-009097 | MATHDB-338392

The excellent-set conjecture for Serre conditions of simplicial complexes

摘录 / EXCERPT: Excellent-set conjecture. If $A\subseteq\Delta$ is excellent and Δ satisfies (S_{ℓ}), then Δ_A satisfies (S_{ℓ}).

conjecture | open | MathDB 页面 | 原文线索

V2-009098 | MATHDB-338414

Solomon-Terao Hilbert-series characterizations of free arrangements

摘录 / EXCERPT: Solomon-Terao Hilbert-series conjectures. The arrangement ${\mathcal{A}}$ is free if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-009099 | MATHDB-338415

Top-degree and socle-dimension conjecture for Solomon-Terao algebras

摘录 / EXCERPT: Top-degree and socle-dimension conjecture. The top nonzero degree and its dimension satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-009100 | MATHDB-338426

The analytic Iwasawa main conjecture for geometric p-adic L-functions

摘录 / EXCERPT: Let $V=R_f^*\otimes R_g^*$, let $\mathbb{V}^{\dag}$ be its analytic Iwasawa deformation, and let $\mathbb{D}_\lambda$ be the (φ,Γ)-submodule associated with the indica...

conjecture | open | MathDB 页面 | 原文线索

V2-009101 | MATHDB-338432

The maximal-path ideal conjecture for staged trees

摘录 / EXCERPT: Maximal-path ideal conjecture. The kernel of φ is exactly the ideal generated by maximal paths,

conjecture | open | MathDB 页面 | 原文线索

V2-009102 | MATHDB-338436

The Centralizer Conjecture for the Weyl algebra over an integral domain

摘录 / EXCERPT: The Centralizer Conjecture over D. Suppose $A,B,w\in A_1(D)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-009103 | MATHDB-338438

Schoutens's conjecture on very small maximal Cohen-Macaulay modules

摘录 / EXCERPT: Schoutens's conjecture. Every complete local ring admits a very small maximal Cohen-Macaulay module.

conjecture | open | MathDB 页面 | 原文线索

V2-009104 | MATHDB-338439

Conjecture on unbounded ranks of small maximal Cohen-Macaulay modules

摘录 / EXCERPT: Unbounded-rank small maximal Cohen-Macaulay conjecture. For every integer N, there exists a complete local domain R that does not admit small maximal Cohen-Macaulay modules of...

conjecture | open | MathDB 页面 | 原文线索

V2-009105 | MATHDB-338447

The idempotent bound for affine subspaces of generic algebras

摘录 / EXCERPT: Let A be a generic algebra, and let Π be a k-dimensional affine subspace of A. An idempotent is an element $e\in A$ satisfying $e^2=e$. The idempotent bound. The affine s...

conjecture | open | MathDB 页面 | 原文线索

V2-009106 | MATHDB-338502

The BMM symmetrising trace conjecture for complex reflection groups

摘录 / EXCERPT: Let W be a complex reflection group with generic Hecke algebra $\mathcal{H}(W)$ over $\mathbb{Z}[\mathbf{u},\mathbf{u}^{-1}]$. Let $B(W)$ be its braid group, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009107 | MATHDB-338504

Free algebra existence in W-pretoposes with WISC

摘录 / EXCERPT: A W-pretopos is a category with the structural properties implicit in the paper, and WISC denotes the weak internal axiom of choice used there. Free algebra existence conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-009108 | MATHDB-338505

Failure of initial algebras for monic polynomials with reductions

摘录 / EXCERPT: Let a topos with natural number object be a topos equipped with a natural number object, and let a monic polynomial with reductions be as considered in the paper. Initial-algebra f...

conjecture | open | MathDB 页面 | 原文线索

V2-009109 | MATHDB-338507

Algebraic model structures on the effective topos

摘录 / EXCERPT: Let the two awfs's in the paper's section on the effective topos be the specified algebraic weak factorisation systems. Effective-topos model-structure conjecture. These two awfs's...

conjecture | open | MathDB 页面 | 原文线索

V2-009110 | MATHDB-338508

Algebraic model structures on cubical assemblies

摘录 / EXCERPT: Let the two awfs's in the paper's section on cubical assemblies be the specified algebraic weak factorisation systems. Cubical-assembly model-structure conjecture. These two awfs's...

conjecture | open | MathDB 页面 | 原文线索

V2-009111 | MATHDB-338509

Algebraic model structure on the Kleene-Vesley topos

摘录 / EXCERPT: Let the awfs in the paper's section on functional realizability be the specified awfs on the Kleene-Vesley topos. Kleene-Vesley model-structure conjecture. This awfs forms the triv...

conjecture | open | MathDB 页面 | 原文线索

V2-009112 | MATHDB-338522

Greenberg's conjectural formula for the exceptional-zero \mathcal{L}-invariant

摘录 / EXCERPT: Let V be a p-adic ordinary exceptional Galois representation, and assume that hypotheses S, T, and U are satisfied. Let $L_p(V,S)$ denote the conjectured p-adic L-function...

conjecture | open | MathDB 页面 | 原文线索

V2-009113 | MATHDB-338534

Hibi's lexicographic or reverse lexicographic quadratic Gröbner basis conjecture

摘录 / EXCERPT: Hibi's conjecture. If I_A has a quadratic Gröbner basis, then I_A has either a lexicographic or a reverse lexicographic quadratic Gröbner basis.

conjecture | open | MathDB 页面 | 原文线索

V2-009114 | MATHDB-338563

BPS Lie algebra identification conjecture for quivers

摘录 / EXCERPT: Let Q be a quiver, let $\bar{\mathfrak{n}}_Q$ be the graded BPS Lie algebra arising from the filtration on $H_*(T^*\mathcal{M}_Q)$, and let $\widetilde{\mathfrak{n}}_Q$ be the po...

conjecture | open | MathDB 页面 | 原文线索

V2-009115 | MATHDB-338564

Identification conjecture for geometric and Borcherds Lie algebras of quivers

摘录 / EXCERPT: Let Q be a quiver, let $\mathbf{g}_Q$ be the graded Lie algebra constructed from Maulik-Okounkov classical R-matrices, let $\bar{\mathfrak{g}}_Q$ denote the geometric Lie algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-009116 | MATHDB-338580

Jantzen filtration intertwining conjecture for the KZ functor

摘录 / EXCERPT: Let W be a finite real reflection group with rational Cherednik algebra parameter space $\mathfrak{p}$, let $\mathfrak{p}_{\mathbb{R}}$ be its real form, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009117 | MATHDB-338581

Local structure conjecture for Dunkl weight functions

摘录 / EXCERPT: Let W be a finite real reflection group acting on $\mathfrak{h}$, let $c\in\mathfrak{p}_{\mathbb{R}}$ be real, and let $\lambda\in\operatorname{Irr}(W)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-009118 | MATHDB-338586

The universal ring and pro-Hall embedding conjecture for fully residually free pro-p groups

摘录 / EXCERPT: Let $\mathbb{F}$ be a free pro-p group. Let the universal class of $\mathbb{Z}_p$ consist of the relevant $\mathbb{Z}_p$-rings in the language with constants, and let a fully res...

conjecture | open | MathDB 页面 | 原文线索

V2-009119 | MATHDB-338588

Grothendieck-Witt theory and products of exact categories with duality

摘录 / EXCERPT: Product conjecture for Grothendieck-Witt spectra. There is a canonical equivalence of spectra

conjecture | open | MathDB 页面 | 原文线索

V2-009120 | MATHDB-338607

Breuil-Schneider conjecture in the generic case

摘录 / EXCERPT: Let F be a finite extension of ${\mathbb Q}_p$, let E be a sufficiently large coefficient field, and write $G=GL_n(F)$. Let π be an absolutely irreducible generic represent...

conjecture | open | MathDB 页面 | 原文线索

V2-009121 | MATHDB-338614

The singular cyclic case: infinitely many cluster variables in each occurring degree

摘录 / EXCERPT: Let $a\geq 3$ and let ... The graded cluster algebra $\mathcal{A}(\underline{x},B,(a,2,a))$ is the graded cluster algebra associated with this singular cyclic matrix and grading. Sin...

conjecture | open | MathDB 页面 | 原文线索

V2-009122 | MATHDB-338615

The $\widetilde{E}_7$ degree-$\pm(2,1)$ uniqueness conjecture

摘录 / EXCERPT: Let ... be the graded cluster algebra considered for affine type $\widetilde{E}_7$, with two-dimensional degree vectors. $\widetilde{E}_7$ degree-$\pm(2,1)$ conjecture. The degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-009123 | MATHDB-338616

The marked-surface quadrilateral bijection conjecture for X-seed patterns

摘录 / EXCERPT: Let (S,M) be a marked surface and let $\mathcal{S}$ be an $\mathcal X$-seed pattern from (S,M). For every triangulation T of (S,M) and every $\gamma\in T$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009124 | MATHDB-338618

Dau et al.'s algebraic GM-MDS conjecture

摘录 / EXCERPT: Algebraic GM-MDS conjecture. If $S_1,\ldots,S_k\subset[n]$ satisfy the MDS condition and $|S_i|=k-1$ for every i, then the polynomials in $P(\mathcal{S})$ are linearly independen...

conjecture | open | MathDB 页面 | 原文线索

V2-009125 | MATHDB-338644

Minimal-dimension conjecture for the constructed Grassmann-algebra subalgebra

摘录 / EXCERPT: Let $G(4k+9)$ be the Grassmann algebra, let G_0 denote its even component, and let $M_{\mathcal{C}}$ be the subspace associated with the constructed maximal commutative algebraic...

open_problem | open | MathDB 页面 | 原文线索

V2-009126 | MATHDB-338658

Kaledin-Lehn formality conjecture for polystable sheaves on K3 surfaces

摘录 / EXCERPT: Let X be a projective K3 surface with a generic polarization H, and let F be an H-polystable coherent sheaf on X. Kaledin-Lehn's formality conjecture. The differential gr...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-009127 | MATHDB-338680

Pardue's Conjecture E on generic initial ideals

摘录 / EXCERPT: Let K be a field, let $R=K[x_1,\ldots,x_n]$, and let $I=(f_1,\ldots,f_n)$ be a generic homogeneous ideal of type $(n;d_1,\ldots,d_n)$. Let $\operatorname{in}(I)$ be the initial i...

conjecture | open | MathDB 页面 | 原文线索

V2-009128 | MATHDB-338683

The rigid-reflection correspondence for rank-3 Coxeter groups

摘录 / EXCERPT: Fix an integer $m\geq 2$. Let ... with $m_{13}=m_{31}=\infty$, and let $\mathcal H(m)$ be the rank-2 Kac-Moody algebra with Cartan matrix ... Write roots of its root lattice as $[a,b]$...

conjecture | open | MathDB 页面 | 原文线索

V2-009129 | MATHDB-338702

Polarized Donovan-Wemyss conjecture for contraction algebras

摘录 / EXCERPT: Polarized Donovan-Wemyss conjecture. The spaces X and X' are formally isomorphic at their singular points if and only if $\Lambda(Y,\Phi)$ and $\Lambda(Y',\Psi)$ are isomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-009130 | MATHDB-338703

Quasi-homogeneity conjecture for contraction algebra potentials

摘录 / EXCERPT: Quasi-homogeneity conjecture. The threefold X is quasi-homogeneous if and only if the potential Φ is quasi-homogeneous. The source introduces this as a conjectural relation...

conjecture | open | MathDB 页面 | 原文线索

V2-009131 | MATHDB-338708

Conjecture on radical maximal subalgebras of type E_8 in characteristic five

摘录 / EXCERPT: Let G be an algebraic group of type E_8 with Lie algebra $\mathfrak{g}=\operatorname{Lie}(G)$ over an algebraically closed field of characteristic 5. Let M be a maximal sub...

conjecture | open | MathDB 页面 | 原文线索

V2-009132 | MATHDB-338709

Conjecture on Cartan- and Melikyan-type maximal subalgebras in E_8

摘录 / EXCERPT: Let G be an algebraic group of type E_8 with Lie algebra $\mathfrak{g}=\operatorname{Lie}(G)$ over a field of characteristic 5. Let M be a maximal subalgebra of Cartan or M...

conjecture | open | MathDB 页面 | 原文线索

V2-009133 | MATHDB-338710

Conjecture on the E_8 Weisfeiler graded Lie algebra in characteristic two

摘录 / EXCERPT: Let $p=2$, and let $\mathfrak{g}$ be of type E_8 with maximal subalgebra M_8 and M_8-invariant subspace L'_{-1} as in Theorem none8p2. For the Weisfeiler filtration...

conjecture | open | MathDB 页面 | 原文线索

V2-009134 | MATHDB-338711

Conjecture on the E_7 Weisfeiler graded Lie algebra in characteristic two

摘录 / EXCERPT: Let $p=2$, and let $\mathfrak{g}$ be of type E_7 with maximal subalgebra M_7 and M_7-invariant subspace L'_{-1} as in Theorem none7p2. For the Weisfeiler filtration...

conjecture | open | MathDB 页面 | 原文线索

V2-009135 | MATHDB-338723

The local derivation conjecture for maximal solvable Lie algebras

摘录 / EXCERPT: Let L be a maximal solvable Lie algebra with nilradical N. A local derivation on L is a linear map such that, for every $x\in L$, there exists a derivation D_x of L with...

conjecture | open | MathDB 页面 | 原文线索

V2-009136 | MATHDB-338795

Greenberg-Benois conjecture on trivial zeros and ℓ-invariants

摘录 / EXCERPT: Greenberg-Benois conjecture. If $\mathcal{L}_p(S,V)$ has e_0+t trivial zeros, then S^{e_0+t+r} divides it exactly and

conjecture | open | MathDB 页面 | 原文线索

V2-009137 | MATHDB-338815

Affine-space conjecture for moduli of cocycles on pointed Hopf algebras

摘录 / EXCERPT: Affine-space conjecture. The moduli space of 2-cocycles up to equivalence is the disjoint union of affine spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-009138 | MATHDB-338816

Conjecture on numerical ranges of composition operators with elliptic symbols

摘录 / EXCERPT: Numerical-range conjecture. The numerical range of C_φ on $H^2(D)$ is the interior of the convex hull of an algebraic curve of class p and degree p^2-p. Moreover, the r...

conjecture | open | MathDB 页面 | 原文线索

V2-009139 | MATHDB-338821

Berenstein-Geiss-Muller conjecture on maximal green sequences and upper cluster algebras

摘录 / EXCERPT: Berenstein-Geiss-Muller conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-009140 | MATHDB-338824

The Frobenius form conjecture for primitive axial algebras of Monster type

摘录 / EXCERPT: Let A be a primitive axial algebra of Monster type, with set of axes and Miyamoto group acting on A. Frobenius form conjecture. Every such algebra admits a Frobenius form which...

conjecture | open | MathDB 页面 | 原文线索

V2-009141 | MATHDB-338825

The Norton-Sakuma classification conjecture for dihedral axial algebras

摘录 / EXCERPT: Let A be a dihedral axial algebra of Monster type over a field of characteristic 0. The nine Norton-Sakuma algebras are the nine classified dihedral algebras arising in this se...

conjecture | open | MathDB 页面 | 原文线索

V2-009142 | MATHDB-338833

Higher-dimensional Huneke-Wiegand tensor product conjecture

摘录 / EXCERPT: Higher-dimensional Huneke-Wiegand conjecture. If $M\otimes_R M^*$ is reflexive, then M is free.

conjecture | open | MathDB 页面 | 原文线索

V2-009143 | MATHDB-338835

The admissible-monomial prediction for the maps \(\Phi\)

摘录 / EXCERPT: Admissible-monomial prediction. If ω is a weight vector, then

conjecture | open | MathDB 页面 | 原文线索

V2-009144 | MATHDB-338852

Permutation gauging conjecture for dyslectic modules in type A_1

摘录 / EXCERPT: Let N be a positive integer, and let ${\mathcal{C}}$ be the modular tensor category of dyslectic $\mathbf{1} \oplus ((4N-1)\Lambda_1)$-modules in...

conjecture | open | MathDB 页面 | 原文线索

V2-009145 | MATHDB-338853

Greenberg's fixed-point conjecture for abelian cyclotomic extensions

摘录 / EXCERPT: Greenberg's conjecture. The characteristic polynomial of the Λ-module $\mathcal C^T_S(K_{\infty})$ is not divisible by ω. Equivalently, its fixed submodule is finite...

conjecture | open | MathDB 页面 | 原文线索

V2-009146 | MATHDB-338869

The equivalence conjecture for factorizable line bundles

摘录 / EXCERPT: The equivalence conjecture. The map above is an isomorphism of Picard groupoids. This predicts that factorizable line bundles on the Ran affine Grassmannian are completely paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-009147 | MATHDB-338878

Existence and empirical construction of linear Pfaffian representations for plane curves

摘录 / EXCERPT: Proposed conjecture. Every such polynomial admits a linear Pfaffian R-representation. Moreover, when $d\geq 5$, such a representation can be found using the empirical method desc...

conjecture | open | MathDB 页面 | 原文线索

V2-009148 | MATHDB-338953

Stability conjecture for the genus-free boundary subring

摘录 / EXCERPT: Let $\mathcal{G}_n$ be the formal graph algebra on graphs with n legs, without genus assignments, and let ... be the homomorphism sending a graph to its genus-free boundary class.....

conjecture | open | MathDB 页面 | 原文线索

V2-009149 | MATHDB-338963

Jones's index conjecture for quadratic rational maps

摘录 / EXCERPT: Let K be a global field, let $f\in K(x)$ have degree two, and let T_∞ be the infinite rooted binary tree of preimages of a root α under f. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009150 | MATHDB-339002

The alternating-form symmetric-operator equality conjecture

摘录 / EXCERPT: Alternating-form symmetric-operator equality conjecture. There exists a maximal partially complete b-singular flag $\mathcal{F}$ of V such that

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-009151 | MATHDB-339006

Positive-characteristic purity conjecture for general K3 unions of scrolls

摘录 / EXCERPT: Let a be a positive integer, let $\overline{\mathbb{F}}$ be an algebraically closed field of characteristic p, and let $e=(e_1,e_2)\in\overline{\mathbb{F}}^2$ be general. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009152 | MATHDB-339021

The Gorenstein Interval Conjecture for Gorenstein Hilbert functions

摘录 / EXCERPT: Let A be an artinian standard graded algebra over a field k, and write its Hilbert function as $h=(1,h_1,\dots,h_e)$, where e is the socle degree. A Hilbert function is Goren...

conjecture | open | MathDB 页面 | 原文线索

V2-009153 | MATHDB-339025

Maximality conjecture for the curves C_p

摘录 / EXCERPT: Let $C_p=V(Q,P)$ be the curve over $\mathbb{F}_{p^2}$ defined by ... For any p with $p\equiv 2\pmod{3}$, the maximality conjecture. The curve C_p over $\mathbb{F}_{p^2}$ is maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-009154 | MATHDB-339073

Margulis's arithmeticity conjecture for discrete subgroups containing horospherical lattices

摘录 / EXCERPT: Let G be a semisimple real algebraic Lie group of real rank at least 2, and let U be a non-trivial horospherical subgroup of G. Let Γ be a discrete Zariski dense sub...

conjecture | open | MathDB 页面 | 原文线索

V2-009155 | MATHDB-339087

Classification conjecture for simple finite-dimensional Lie algebras with nontrivial Hom-Lie structures

摘录 / EXCERPT: Classification conjecture. If L admits a nontrivial Hom-Lie structure, then L is isomorphic either to a 3-dimensional simple algebra, or, in the case of positive characterist...

conjecture | open | MathDB 页面 | 原文线索

V2-009156 | MATHDB-339094

The diminishing-action characterization of lower-dimensional amoebas

摘录 / EXCERPT: Diminishing-action conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009157 | MATHDB-339207

The adequacy conjecture for the j-function's Ax-Schanuel inequality

摘录 / EXCERPT: Let C be an algebraically closed field with $\operatorname{td}(C/\mathbb{Q})=\aleph_0$, let $\mathfrak{C}$ be the class of E_j-fields with constant field C, and let U be it...

conjecture | open | MathDB 页面 | 原文线索

V2-009158 | MATHDB-339258

BPS conjecture on augmentation quotients of dihedral quandle rings

摘录 / EXCERPT: BPS conjecture. begin{enumerate} item If $n>1$ is an odd integer, then

conjecture | open | MathDB 页面 | 原文线索

V2-009159 | MATHDB-339311

The SHGH conjecture for blow-ups of the projective plane

摘录 / EXCERPT: Let X be a composite of blow-ups of $\mathbb P^2$ at points $p_1,\ldots,p_n$ in very general position. A (-1)-rational curve is a rational curve with self-intersection -1. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-009160 | MATHDB-339313

The common-unitary normal-form conjecture for GLT algebras

摘录 / EXCERPT: Common-unitary normal-form conjecture. Any algebra $\mathfrak A\subseteq\mathfrak C_D$ induced by such $\mathscr A$ and s admits a unitary sequence $\{U_n\}_n$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-009161 | MATHDB-339314

Berman's cyclic-module characterization of infinity-operads

摘录 / EXCERPT: Let Fin^{op} be the opposite category of finite sets, and let Burn^{eff} denote the effective Burnside 2-category, or spans of finite sets. A cyclic...

conjecture | open | MathDB 页面 | 原文线索

V2-009162 | MATHDB-339326

Extension of the mixed double-dimer construction to all self-intersecting contours

摘录 / EXCERPT: Let ... be the three-dimensional space of toric cluster variables described in the source. For self-intersecting contours, consider subgraphs with vertices of multiplicity one and tw...

conjecture | open | MathDB 页面 | 原文线索

V2-009163 | MATHDB-339357

The SHGH vanishing conjecture for blow-ups of the projective plane

摘录 / EXCERPT: Let X be the blow-up of $\mathbb P^2$ at n generic points, and let C be a curve on X. The SHGH vanishing conjecture asserts that ... for every curve C on X. The source id...

conjecture | open | MathDB 页面 | 原文线索

V2-009164 | MATHDB-339405

Generic initial-chain conjecture for squarefree monomial ideals

摘录 / EXCERPT: Assume $c=1$. Let $\mathcal I=(I_n)_{n\ge 1}$ be an $\operatorname{Inc}(\mathbb{N})^i$-invariant chain of squarefree monomial ideals. For each n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009165 | MATHDB-339428

Parametric algebraic formula for the excursions of a Gessel-like model

摘录 / EXCERPT: The parametric algebraic-form conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009166 | MATHDB-339440

Perrin-Riou-Howard Heegner point main conjecture

摘录 / EXCERPT: Perrin-Riou-Howard's Heegner point main conjecture. The Λ-modules $\mathrm{Sel}_{\mathrm{Gr}}(K,\mathbf{T})$ and $\mathrm{Sel}_{\mathrm{Gr}}(K,\mathbf{A})$ have rank 1 an...

conjecture | open | MathDB 页面 | 原文线索

V2-009167 | MATHDB-339441

Bertolini-Darmon-Prasanna anticyclotomic main conjecture

摘录 / EXCERPT: Bertolini-Darmon-Prasanna's anticyclotomic main conjecture. The module $X^{N^-}_{\emptyset,0}$ is Λ-torsion, and

conjecture | open | MathDB 页面 | 原文线索

V2-009168 | MATHDB-339475

Positive structure constants conjecture for the surface skein basis

摘录 / EXCERPT: Let $\mathcal{S}$ be an oriented surface of finite type, let $\mathcal{S}\mathrm{Web}_q$ be its skein algebra, and let $\mathcal{S}\mathrm{B}_T$ be the basis used in the paper. Pos...

conjecture | open | MathDB 页面 | 原文线索

V2-009169 | MATHDB-339508

Bessel-module conjecture for relevant inner forms of unitary groups

摘录 / EXCERPT: Let G_n be a unitary group of F-rank ${\mathfrak r}$ with $2{\mathfrak r}< {\mathfrak n}$, and let G_n^* be its F-quasisplit pure inner form. Define ... For a representation...

conjecture | open | MathDB 页面 | 原文线索

V2-009170 | MATHDB-339554

Minimality conjecture for three-variable GT-systems

摘录 / EXCERPT: Let $d\ge 3$ be an integer and let $1\le n<m\le d-1$ satisfy $\operatorname{gcd}(n,m,d)=1$. Let $I^d_{0,n,m}\subset k[x,y,z]$ be the associated GT-system. Minimality conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-009171 | MATHDB-339560

Schinzel-Zassenhaus-Boyd conjecture on the house

摘录 / EXCERPT: Schinzel-Zassenhaus-Boyd conjecture. There is a constant $T>1$ such that if α is not a root of unity, then

conjecture | open | MathDB 页面 | 原文线索

V2-009172 | MATHDB-339576

Donaldson-Sun's algebraic determination conjecture for metric tangent cones

摘录 / EXCERPT: Let (X,d_∞) be a Gromov-Hausdorff limit as in the construction of Donaldson and Sun, let $x\in X$ be a point, let $C=C_xX$ be a metric tangent cone, and let W be the affin...

conjecture | open | MathDB 页面 | 原文线索

V2-009173 | MATHDB-339586

Kobayashi's signed Iwasawa main conjecture for elliptic curves

摘录 / EXCERPT: Let Γ_{Cyc} be the cyclotomic Galois group, let E be an elliptic curve, and let E_K denote its base change to the extension K. For signs...

conjecture | open | MathDB 页面 | 原文线索

V2-009174 | MATHDB-339604

Weak Lefschetz conjecture for ideals generated by powers and squarefree monomials

摘录 / EXCERPT: Let R be the polynomial ring in r variables over a field of positive characteristic, and let $I_{r,k,d}$ be the ideal ... Consider the Artinian algebra $R/I_{r,k,d}$. Weak Lefsch...

conjecture | open | MathDB 页面 | 原文线索

V2-009175 | MATHDB-339608

General weighted-homogeneous polynomial ideal conjecture

摘录 / EXCERPT: General weighted-homogeneous polynomial ideal conjecture. The ideal I contains no polynomial of standard degree $3m-3$. If true, the construction in the paper would imply the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-009176 | MATHDB-339629

The BCM-rationality characterization of rational singularities

摘录 / EXCERPT: BCM-rationality conjecture. R is BCM-rational if and only if R has rational singularities.

conjecture | open | MathDB 页面 | 原文线索

V2-009177 | MATHDB-339656

The dimension formula for the comtrans operad

摘录 / EXCERPT: Let $\mathcal{CT}$ be the symmetric weight-graded comtrans operad, and let $\mathcal{CT}(w)$ denote its component of weight w. In the free comtrans algebra, the multilinear subsp...

conjecture | open | MathDB 页面 | 原文线索

V2-009178 | MATHDB-339657

Finite-dimensionality conjecture for enveloping comtrans algebras

摘录 / EXCERPT: Let C be a finite-dimensional comtrans algebra, without assuming that it is special, and let $U(C)$ denote its enveloping algebra. Finite-dimensionality conjecture. If C is fin...

conjecture | open | MathDB 页面 | 原文线索

V2-009179 | MATHDB-339666

Non-self-similarity conjecture for a Lie superalgebra and its analogue

摘录 / EXCERPT: Let $\mathbf R=\operatorname{Lie}(v_0,v_1)$ be the Lie superalgebra whose generators v_0,v_1 are defined by the pivot construction referenced in the source. A Lie superalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-009180 | MATHDB-339667

Non-self-similarity conjecture for the associative superalgebra generated by pivot elements

摘录 / EXCERPT: Let $A=\operatorname{Alg}(v_0,v_1)$ be the associative superalgebra corresponding to the Lie superalgebra $\mathbf R=\operatorname{Lie}(v_0,v_1)$, where v_0,v_1 are defined by th...

conjecture | open | MathDB 页面 | 原文线索

V2-009181 | MATHDB-339676

Equality of projective dimensions for discrete modules

摘录 / EXCERPT: Let $\mathfrak{A}$ be an order in a finite-dimensional semisimple algebra, and let M be a discrete object of $\mathsf{LCA}_{\mathfrak{A}}$. Projective resolutions of M in...

conjecture | open | MathDB 页面 | 原文线索

V2-009182 | MATHDB-339756

The generalized Onsager presentation conjecture

摘录 / EXCERPT: Generalized Onsager presentation conjecture. The Lie algebra $\widetilde{\mathfrak k}$ is isomorphic to $\mathfrak k$.

conjecture | open | MathDB 页面 | 原文线索

V2-009183 | MATHDB-339757

The centre-generation conjecture for weak generalized Satake subalgebras

摘录 / EXCERPT: Centre-generation conjecture. A single element of

conjecture | open | MathDB 页面 | 原文线索

V2-009184 | MATHDB-339780

The class $\mathcal{P}'$ equality conjecture for cluster and upper cluster algebras

摘录 / EXCERPT: Let Q be a quiver in the class $\mathcal{P}'$, let $(\mathbf{x},\mathbf{y},Q)$ be a seed with quiver Q, and let $\mathcal{A}$ and $\mathcal{U}$ be the associated cluster algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-009185 | MATHDB-339781

The class $\mathcal{P}$ ground-ring equality conjecture for cluster and upper cluster algebras

摘录 / EXCERPT: Let Q be a quiver in the class $\mathcal{P}$, and let $\mathcal{A}$ and $\mathcal{U}$ be the cluster algebra and upper cluster algebra associated with the cluster algebra defined...

conjecture | open | MathDB 页面 | 原文线索

V2-009186 | MATHDB-339816

Double-shuffle kernel conjecture for refined symmetric multiple zeta values

摘录 / EXCERPT: Let $\mathfrak{h}=\bigoplus_{k=-1}^{\infty}\mathfrak{h}_{k}$ be graded by ... and write $\mathfrak{h}_{k}^{0}=\mathfrak{h}_{k}\cap\mathfrak{h}^{0}$. For $u,v,w\in\mathfrak{h}^{0}$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-009187 | MATHDB-339858

Teissier's first conjecture on mixed multiplicities

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring of dimension d, and let I and J be $\mathfrak{m}$-primary ideals. For $0\leq i\leq d$, let $e_i(I\mid J)$ denote their mixed...

conjecture | open | MathDB 页面 | 原文线索

V2-009188 | MATHDB-339867

Strong-coupling scattering-matrix convergence conjecture for leaky curves

摘录 / EXCERPT: Let Γ be a C^4-smooth, asymptotically straight planar curve. Let $\Sigma_{\alpha,\Gamma}(k)$ be the on-shell scattering matrix at energy k^2, and let S_Γ be the o...

conjecture | open | MathDB 页面 | 原文线索

V2-009189 | MATHDB-339907

Existence of p-bases for higher Frobenius sandwiches

摘录 / EXCERPT: Existence conjecture for higher Frobenius sandwiches. There should exist a $p^\bullet$-basis of R over S.

conjecture | open | MathDB 页面 | 原文线索

V2-009190 | MATHDB-339916

Algebraicity conjecture for Kreweras and reverse Kreweras walks

摘录 / EXCERPT: Let $G(t;a,b)$ denote the generating function for the corresponding model, with parameters t, a, and b. Models 19 and 20 are the Kreweras and reverse Kreweras walks, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-009191 | MATHDB-339940

Overring noetherianity conjecture for graded maximal orders

摘录 / EXCERPT: Let T be the algebra in the paper's main theorems, let $Q_{\operatorname{gr}}(T)$ denote its graded quotient ring, and consider graded overrings...

conjecture | open | MathDB 页面 | 原文线索

V2-009192 | MATHDB-339988

Stückrad-Vogel conjecture on multiplicity ratios

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated R-module. Let $e(I,M)$ be the Hilbert-Samuel multiplicity of M with respect to I, and se...

conjecture | open | MathDB 页面 | 原文线索

V2-009193 | MATHDB-340079

The self-dual W-graph conjecture

摘录 / EXCERPT: Self-dual W-graph conjecture. For any irreducible self-dual representation, there exists a self-dual W-graph defined over K. If there exists a W-graph $\Gamma=(X,I,\mu)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-009194 | MATHDB-340133

Fully supersingular Bloch--Beilinson conjecture

摘录 / EXCERPT: Fully supersingular Bloch--Beilinson conjecture. Numerical and algebraic equivalence coincide on ${\rm CH}^{i}(X)_{\mathbb{Q}}$, so $\operatorname{Griff}^{i}(X)_{\mathbb{Q}}=0$; th...

conjecture | open | MathDB 页面 | 原文线索

V2-009195 | MATHDB-340143

Kohl-Olsen levelness conjecture for lecture hall simplices

摘录 / EXCERPT: Let ${\boldsymbol s}\in\mathbb{Z}_{\geq 1}^n$ be a sequence. Suppose there exists ${\boldsymbol c}\in\mathbb{Z}^n$ such that ... for every $j>1$, with $c_1=1$. Kohl-Olsen's levelness...

conjecture | open | MathDB 页面 | 原文线索

V2-009196 | MATHDB-340178

The finiteness conjecture for centralizers of simple derivations in two variables

摘录 / EXCERPT: The finiteness conjecture. If D is a simple derivation of $K[x_1,x_2]$, then

conjecture | open | MathDB 页面 | 原文线索

V2-009197 | MATHDB-340179

The trivial-centralizer conjecture for simple Shamsuddin derivations

摘录 / EXCERPT: The trivial-centralizer conjecture. If D is a simple Shamsuddin derivation of $K[x]$, then

conjecture | open | MathDB 页面 | 原文线索

V2-009198 | MATHDB-340180

The converse conjecture for simplicity and trivial centralizers of Shamsuddin derivations

摘录 / EXCERPT: The converse conjecture. The derivation D is simple if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-009199 | MATHDB-340181

The translation-centralizer conjecture for simple polynomial derivations

摘录 / EXCERPT: Let K be a field of characteristic zero and let $K[x]=K[x_1,\ldots,x_n]$. Consider the derivation ... Let $\operatorname{Aut}(K[x])_D$ denote the subgroup of polynomial automorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-009200 | MATHDB-340215

The Breuil-Mézard cycle-map conjecture with fixed determinant

摘录 / EXCERPT: Let K be a non-archimedean local field of residue characteristic l, let $\mathbb{F}$ be a finite field of characteristic $p\ne l$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009201 | MATHDB-340242

The PBW decomposition conjecture for the current algebra

摘录 / EXCERPT: Let $\mathcal A_q$ be the current algebra over $\mathbb F$, with generators $\mathcal W_{-k}$, $\mathcal W_{k+1}$, $\mathcal G_{k+1}$, and $\mathcal{\tilde G}_{k+1}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-009202 | MATHDB-340243

The center and q-Onsager factorization conjecture for the current algebra

摘录 / EXCERPT: Let $\mathcal A_q$ be the current algebra, let $Z(\mathcal A_q)$ denote its center, let $\Delta_{k+1}\in Z(\mathcal A_q)$, and let $\langle\mathcal W_0,\mathcal W_1\rangle$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-009203 | MATHDB-340244

The ideal-center direct-sum conjecture for the current algebra

摘录 / EXCERPT: Let J be the two-sided ideal of $\mathcal A_q$ generated by $\mathcal W_0$ and $\mathcal W_1$, and let $Z(\mathcal A_q)$ denote the center. Ideal-center direct-sum conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-009204 | MATHDB-340245

The lower-generator uniqueness conjecture for the q-Onsager algebra

摘录 / EXCERPT: Let $\mathcal O_q$ be the q-Onsager algebra with generators A,B, and let $\mathcal J$ be its two-sided ideal generated by A and B. Lower-generator uniqueness conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-009205 | MATHDB-340246

The tensor-product decomposition conjecture for the current algebra

摘录 / EXCERPT: Let $\mathcal A_q$ be the current algebra, let $\mathcal O_q$ be the q-Onsager algebra with generators A,B, let $\{\tilde g_{k+1}\}_{k\in\mathbb N}\subseteq\mathcal J$ be the seq...

conjecture | open | MathDB 页面 | 原文线索

V2-009206 | MATHDB-340249

Livingston's linear independence conjecture for logarithms of sine values

摘录 / EXCERPT: Livingston's conjecture. In either case, the displayed numbers are linearly independent over the field of algebraic numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-009207 | MATHDB-340285

The Lecture Hall cone initial-subalgebra conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$. There exist a $\mathbb{Z}^2$-graded polynomial ring $\Bbbk[x_1,\dots,x_m]$ for some $m\in\mathbb{N}$, a graded subalgebra $A_n\subseteq\Bbbk[x_1,\dots,x_m]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-009208 | MATHDB-340401

Assem-Schiffler-Shramchenko unistructurality conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal A(\mathcal S)$ be a cluster algebra contained in an ambient field $\mathcal F$, generated by cluster variables grouped into overlapping clusters. The set of cluster v...

conjecture | open | MathDB 页面 | 原文线索

V2-009209 | MATHDB-340437

Severi's maximal rank conjecture for general Brill-Noether curves

摘录 / EXCERPT: Severi's maximal rank conjecture. These restriction maps have maximal rank, meaning that each is either injective or surjective. Equivalently, the dimension of the space of degree-...

conjecture | open | MathDB 页面 | 原文线索

V2-009210 | MATHDB-340439

The inverse-map conjecture for b and $\varpi_{N,G}$

摘录 / EXCERPT: The inverse-map conjecture for b and $\varpi_{N,G}$. The two displayed maps are inverse to each other. This is proposed as an analogue of Sharifi's conjecture; the source does no...

conjecture | open | MathDB 页面 | 原文线索

V2-009211 | MATHDB-340440

Sharifi's refined isomorphism and compatibility conjecture

摘录 / EXCERPT: Let I and I_{∞} be the relevant Eisenstein ideals, let G be the group of diamond-operator coinvariants, let χ_b be the character associated with the extension class...

conjecture | open | MathDB 页面 | 原文线索

V2-009212 | MATHDB-340463

Murai-Nevo's multidegree algebra conjecture for the cd-index

摘录 / EXCERPT: Let $\mathbf{k}$ be a field, let P be a Gorenstein* poset of rank $n+1$, and let M be a degree-n cd-monomial. Define the multidegree of M by replacing each c with...

conjecture | open | MathDB 页面 | 原文线索

V2-009213 | MATHDB-340525

Parity conjecture for the seaweed index of odd-part partitions

摘录 / EXCERPT: Let $p=(\lambda_1,\ldots,\lambda_l)$ be a partition of n into odd parts, and let $\operatorname{ind}_S(p)$ denote its associated seaweed-index statistic. Parity conjecture. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-009214 | MATHDB-340532

Stillman's conjecture on uniform projective dimension

摘录 / EXCERPT: Let $d_1,\dots,d_r$ be positive integers. For homogeneous polynomials $f_1,\dots,f_r$ of degrees $\deg(f_i)=d_i$ in a polynomial ring ... where n is arbitrary, consider the quotien...

conjecture | open | MathDB 页面 | 原文线索

V2-009215 | MATHDB-340574

Sakuma's conjecture for 2-generated axial algebras of Monster type

摘录 / EXCERPT: Let A be a 2-generated axial algebra of Monster type over a field of characteristic 0. Sakuma's conjecture. The algebra A is one of the nine Norton-Sakuma algebras. A class...

conjecture | open | MathDB 页面 | 原文线索

V2-009216 | MATHDB-340598

Polynomial Stillman bound conjecture for projective dimension

摘录 / EXCERPT: Polynomial Stillman-bound conjecture. For all characteristics, the projective dimension of R/I is at worst

conjecture | open | MathDB 页面 | 原文线索

V2-009217 | MATHDB-340600

The covering conjecture for Aomoto-Gel'fand hypergeometric systems

摘录 / EXCERPT: Covering conjecture. On each member of this open covering, the system $E(n,m)$ is represented locally by a GKZ system; equivalently, the hypergeometric $\mathcal{D}$-modules of...

conjecture | open | MathDB 页面 | 原文线索

V2-009218 | MATHDB-340601

Bass-Quillen-Suslin conjecture on completing unimodular vectors

摘录 / EXCERPT: Let R be a local ring. Assume either that R is a regular local ring or that $1/r!\in R$. Let $T=(T_1,\ldots,T_n)$ and let v be a unimodular vector over $R[T]$ of length $r+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-009219 | MATHDB-340610

The compatible-seed existence conjecture for monoidal categorifications

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra and $\mathcal{C}$ an Artinian monoidal categorification of $\mathcal{A}$. Assume that the simple objects of $\mathcal{C}$ are parametrized by...

conjecture | open | MathDB 页面 | 原文线索

V2-009220 | MATHDB-340707

The cellular tree normal form conjecture for quiver roots

摘录 / EXCERPT: Cellular tree normal form conjecture. The root α admits a cellular tree normal form, with c_i cells of dimension i.

conjecture | open | MathDB 页面 | 原文线索

V2-009221 | MATHDB-340710

Hassanzadeh-Naeliton conjecture for algebraic residual intersections

摘录 / EXCERPT: Let R be a Cohen-Macaulay local ring of dimension d, let I be an ideal satisfying sliding depth, and let s be the residual-intersection parameter. An algebraic s-residual...

conjecture | open | MathDB 页面 | 原文线索

V2-009222 | MATHDB-340711

Kitt ideal equality conjecture for algebraic residual intersections

摘录 / EXCERPT: Let R be a Cohen-Macaulay ring, let $\mathfrak a\subseteq I$ be ideals, and write ... Assume that J is an algebraic s-residual intersection. The Kitt ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-009223 | MATHDB-340717

Hilbert's fourteenth problem for the zip-form algebra

摘录 / EXCERPT: Let k be the ground field and let $R_{\rm zip}$ be the k-algebra of functions associated with the zip datum. Hilbert finite-generation conjecture. The k-algebra $R_{\rm zip}$...

conjecture | refuted | MathDB 页面 | 原文线索

V2-009224 | MATHDB-340735

Generalized Ihara's lemma for unitary similitude groups

摘录 / EXCERPT: Let $F=F^+E$ be a CM field with $E/{\mathbb Q}$ quadratic imaginary, and let $\overline G$ be the associated unitary similitude group. Let S be a finite set of places and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009225 | MATHDB-340763

Li-Nadler's colimit conjecture for traces of affine Hecke categories

摘录 / EXCERPT: Let G be a simply-connected reductive algebraic group, let LG be its loop group, and let C be an affine alcove. For each face I of C, let G_I be the Levi of the parahor...

conjecture | open | MathDB 页面 | 原文线索

V2-009226 | MATHDB-340786

Kleinian singularity classification conjecture for full-support Harish-Chandra bimodules

摘录 / EXCERPT: Let $Y=\mathbb{C}^2/\Gamma$, where Γ is a finite subgroup of $\operatorname{SL}_2(\mathbb{C})$, and let $\mathfrak{h}_\Gamma^*$ be the Cartan space associated with the ADE t...

conjecture | open | MathDB 页面 | 原文线索

V2-009227 | MATHDB-340812

Domokos and Zubor's dimension conjecture for maximal commutative subalgebras of Grassmann algebras

摘录 / EXCERPT: Let F be a field of characteristic different from two, and let $G(n)$ be the Grassmann algebra ... A maximal commutative subalgebra is a commutative subalgebra maximal with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-009228 | MATHDB-340821

Twisted Type D factorization conjecture for even rank

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{\geq4}$ be even, and let R be the regular algebra of the diagonal Tannakian subcategory of $\mathcal{C}(\mathfrak{so}_{2n},4)$ of rank 2. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009229 | MATHDB-340823

The infinite-dimensionality conjecture for integrable evolution equations

摘录 / EXCERPT: Let $\mathcal{E}$ be the infinite prolongation of a $\textup{(1+1)}$-dimensional evolution equation. For $p\in\mathbb{Z}_{\ge 0}$ and $a\in\mathcal{E}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009230 | MATHDB-340840

The non-free conjecture for unramified pro-p Galois groups over cyclotomic extensions

摘录 / EXCERPT: Non-free conjecture. The group $\mathcal{X}(k_\infty)$ is not a non-abelian free pro-p group.

conjecture | open | MathDB 页面 | 原文线索

V2-009231 | MATHDB-340860

The rational nearly cuspidal curve bound for the Jacobian module

摘录 / EXCERPT: Let $C:f=0$ be a plane curve, and let $J_f e S$ be the Jacobian ideal generated by the partial derivatives of f. Let I_f be the saturation of J_f with respect to the maximal...

conjecture | open | MathDB 页面 | 原文线索

V2-009232 | MATHDB-340879

Precise pullback conjecture for local Langlands correspondences and Hecke algebras

摘录 / EXCERPT: Precise pullback conjecture. For every $(\phi,\rho)\in\Phi_e(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-009233 | MATHDB-340903

Finite-generation conjecture for conformal blocks of simple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, and let X be a stable curve in $\overline{\mathcal{M}}_{g,n}$. Write $\mathbb{V}_{X,\mathfrak{g}}^{\dagger}$ for the algebra of confor...

conjecture | open | MathDB 页面 | 原文线索

V2-009234 | MATHDB-340906

Broué-Malle-Rouquier freeness conjecture for generic Hecke algebras

摘录 / EXCERPT: Let $W\subset GL(V)$ be a complex reflection group, let $R(W)$ be its associated Laurent polynomial ring, and let $\mathcal{H}(W)$ be the associated generic Hecke algebra over...

conjecture | open | MathDB 页面 | 原文线索

V2-009235 | MATHDB-340940

The Schinzel-Zassenhaus conjecture on houses of algebraic integers

摘录 / EXCERPT: Let α be an algebraic integer of degree d, with conjugates $\alpha=\alpha_1,\ldots,\alpha_d$. Its house is $\house{\alpha}=\max_{1\leq i\leq d}|\alpha_i|$. An algebraic in...

conjecture | open | MathDB 页面 | 原文线索

V2-009236 | MATHDB-340961

Malara-Migliore-Nagel conjecture on the weak Lefschetz property

摘录 / EXCERPT: Let K be a field of characteristic zero, let $R=K[x_0,\ldots,x_n]$, and let $L\in R$ be a general linear form. Consider the ideal ... The weak Lefschetz property (WLP) for an Artin...

conjecture | open | MathDB 页面 | 原文线索

V2-009237 | MATHDB-340977

Langlands' analytic continuation and functional equation conjecture for automorphic L-functions

摘录 / EXCERPT: Let F be a number field, let $\mathbb{A}$ be its ring of adeles, let G be a connected linear algebraic group defined over F, let π be an automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-009238 | MATHDB-340987

An elementary generators-and-relations proof of the isomorphism defined by ϑ

摘录 / EXCERPT: The paper considers the map ϑ arising in its comparison between the relevant generator-and-relation description and algebraic K-theory. Elementary proof conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-009239 | MATHDB-340988

An elementary correspondence between double exact sequences and Bass--Swan generators

摘录 / EXCERPT: A double exact sequence is one of the generator types used in the paper's description of the relevant K-theory group, while a Bass--Swan generator is the corresponding generator...

conjecture | open | MathDB 页面 | 原文线索

V2-009240 | MATHDB-341015

The two-thirds nonnegativity rule for twisted Verlinde algebras

摘录 / EXCERPT: Let k be an even positive integer, and let $\mathfrak{g}$ be a twisted affine Lie algebra not of type $A_{2\ell}^{(2)}$. Let P_k^+ be the set of dominant weights at level k,...

conjecture | open | MathDB 页面 | 原文线索

V2-009241 | MATHDB-341032

Creutzig-Huang-Yang modularity conjecture for admissible affine categories

摘录 / EXCERPT: Creutzig-Huang-Yang conjecture. The ribbon tensor category structure on $\mathcal{O}_{\ell,\,\operatorname{ord}}$ is modular except in the following cases:

conjecture | open | MathDB 页面 | 原文线索

V2-009242 | MATHDB-341065

Kontsevich's injectivity conjecture for the evaluation map on formal periods

摘录 / EXCERPT: Let $\tilde{\mathcal{P}}$ be the algebra of formal periods generated by symbols (X,Y,i,ω,σ), modulo bilinearity, functoriality, and boundary relations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009243 | MATHDB-341091

Ground state conjecture for energy-critical NLS

摘录 / EXCERPT: Ground state conjecture. The solution extends uniquely to a global solution $u\in C(\mathbb{R}:\dot{H}^1(\mathbb{R}^d))$ of

conjecture | open | MathDB 页面 | 原文线索

V2-009244 | MATHDB-341107

Torsion-freeness conjecture for algebraic cobordism of Rost motives

摘录 / EXCERPT: Algebraic cobordism torsion-freeness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009245 | MATHDB-341130

Chenevier's density conjecture for polarized global Galois representations

摘录 / EXCERPT: Chenevier's conjecture. Assume that there exists a regular conjugate self-dual algebraic cuspidal automorphic representation π of $\operatorname{GL}_n(\mathbb{A}_E)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-009246 | MATHDB-341154

The central-commutator nilpotence conjecture for two endomorphisms

摘录 / EXCERPT: Central-commutator nilpotence conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-009247 | MATHDB-341222

The Calabi-Yau-$\mathbb{X}$ orbit-category hull conjecture

摘录 / EXCERPT: Let $\Gamma^{\mathbb{X}}_{\mathbf{T}}$ be the Calabi-Yau-$\mathbb{X}$ Ginzburg algebra, let $\Gamma^{N}_{\mathbf{T}}$ be its Calabi-Yau-N reduction, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009248 | MATHDB-341227

Generalized Kruskal conjecture for product-vector sums

摘录 / EXCERPT: Generalized Kruskal conjecture. Under these hypotheses, a nontrivial zero-sum subcollection must exist. This relaxes general position to lower bounds on the dimensions of the subsy...

conjecture | solved | MathDB 页面 | 原文线索

V2-009249 | MATHDB-341238

Treger's decomposition conjecture for reflexive modules

摘录 / EXCERPT: Treger's conjecture. R is a cone such as

conjecture | open | MathDB 页面 | 原文线索

V2-009250 | MATHDB-341239

Wolffhardt's projective-dimension conjecture for symmetric algebras

摘录 / EXCERPT: Wolffhardt's conjecture. If $\operatorname{Sym}_R(M)$ is a UFD, then

conjecture | open | MathDB 页面 | 原文线索

V2-009251 | MATHDB-341259

K-theory non-detection conjecture for stable infinity-categories

摘录 / EXCERPT: K-theory non-detection conjecture. There is no natural number n such that, for all such $\mathcal{C}$ and $\mathcal{D}$, an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-009252 | MATHDB-341260

Truncated K-theory invariance conjecture for stable n-categories

摘录 / EXCERPT: Truncated K-theory conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009253 | MATHDB-341268

Green-red balance conjecture for framed quiver mutation sequences

摘录 / EXCERPT: Let Q be a framed quiver and let $v_1,v_2,\ldots$ be a mutation sequence. For each mutation index i, let r_i denote the associated term and let M be the set of relevant mon...

conjecture | open | MathDB 页面 | 原文线索

V2-009254 | MATHDB-341270

Polynomiality conjecture for deformed F-polynomials

摘录 / EXCERPT: Let Q be a framed quiver and let $v_1,v_2,\ldots$ be a mutation sequence. Polynomiality conjecture for deformed F-polynomials. The deformed F-polynomial S_n is a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-009255 | MATHDB-341271

Periodic convergence conjecture for deformed F-polynomials

摘录 / EXCERPT: Let Q be a framed quiver and consider the periodic mutation sequence ... Assume that performing the mutations $v_1,\ldots,v_p$ on the base quiver returns it to the original quiver....

conjecture | open | MathDB 页面 | 原文线索

V2-009256 | MATHDB-341332

The exceptional Serre-constant set conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Let $\mathcal{A}$ be the set of Serre constants of non-CM elliptic curves over $\mathbb Q$, and let $\mathcal{A}_{\infty}$ be the subset of those constants attained by infinitely m...

conjecture | open | MathDB 页面 | 原文线索

V2-009257 | MATHDB-341347

Bahturin's Amitsur-type conjecture for Hopf-module codimensions

摘录 / EXCERPT: Let F be a field of characteristic 0, let H be a Hopf algebra over F, and let A be a finite-dimensional associative H-module algebra, with codimensions ... where P_n^H...

conjecture | open | MathDB 页面 | 原文线索

V2-009258 | MATHDB-341365

The non-finite generation conjecture for identities of mutation algebras

摘录 / EXCERPT: Non-finite generation conjecture. The operad ideal $\mathcal{K}$ is not finitely generated. Equivalently, no finite set of polynomial identities generates all identities satisfied...

conjecture | open | MathDB 页面 | 原文线索

V2-009259 | MATHDB-341375

Mizuno's exponent characteristic-polynomial conjecture for Y-systems

摘录 / EXCERPT: Let X_r be a finite type Dynkin diagram and let ℓ be a positive integer with $\ell greater than or equal to 2$. Let $J(\eta)$ be the Jacobian matrix at the unique...

conjecture | open | MathDB 页面 | 原文线索

V2-009260 | MATHDB-341376

The branching-function conjecture for partition q-series of mutation loops

摘录 / EXCERPT: Let $\ell greater than or equal to 2$, let $\gamma=\gamma(X_r,\ell)$ be the mutation loop, and let H_{γ} be its associated group. For $\sigma eed in H_{\gamma}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009261 | MATHDB-341394

Minimum distance conjecture for complete intersection ideals

摘录 / EXCERPT: Let $I\subset S:=K[t_1,\ldots,t_s]$ be a complete intersection graded ideal of dimension 1, generated by forms $f_1,\ldots,f_c$ with $c=s-1$, where $d_i=\deg(f_i)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009262 | MATHDB-341395

Two special cases of the minimum distance conjecture for complete intersections

摘录 / EXCERPT: Let $\mathbb{X}$ be a finite set of reduced points in $\mathbb{P}^{s-1}$, and suppose that $I=I(\mathbb{X})$ is a complete intersection generated by $f_1,\ldots,f_c$, where $c=s-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-009263 | MATHDB-341409

The 2-Calabi-Yau conjecture for multiplicative preprojective algebras

摘录 / EXCERPT: Let Q be a connected non-Dynkin quiver and let $q\in (\mathbb{C}^\times)^{Q_0}$. Write $\Lambda^q(Q)$ for the multiplicative preprojective algebra of Q with parameter q. 2-Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-009264 | MATHDB-341510

Free associative Białynicki-Birula linearization conjecture

摘录 / EXCERPT: Let $F_n=\mathbb{K}\langle x_1,\ldots,x_n\rangle$ be the free associative algebra, and let $\mathbb{T}_{n-1}$ act effectively on F_n. Free associative Białynicki-Birula conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-009265 | MATHDB-341536

The DAHA-Jones correspondence for double-torus knots

摘录 / EXCERPT: Let $\boldsymbol{c}$ be a simple closed curve on the genus-two surface $\Sigma_{2,0}$, and let $\overline{P}_n(q_u,x_u,x_d;\boldsymbol{c})$ denote its reduced DAHA invariant. The pa...

conjecture | open | MathDB 页面 | 原文线索

V2-009266 | MATHDB-341588

Leites' classification conjecture for simple Lie superalgebras in positive characteristic

摘录 / EXCERPT: Let k_p be the ground field of characteristic p, with $p\geq 7$, and let $\mathfrak{g}$ be a finite-dimensional simple Lie superalgebra over k_p. The examples of Proposition...

conjecture | open | MathDB 页面 | 原文线索

V2-009267 | MATHDB-341602

Itoh's conjecture on the third normal Hilbert coefficient

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be an analytically unramified Gorenstein local ring of dimension $d\geq 3$, and let I be a parameter ideal. The normal Hilbert coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-009268 | MATHDB-341625

The minimal degree conjecture for standard identities of matrix Grassmann algebras

摘录 / EXCERPT: Let R be a commutative unital ring, let $m,n\geq 1$, and let E^{m} be the m-generated Grassmann algebra ... For $k\geq 1$, the standard polynomial is ... It is a standard identit...

conjecture | open | MathDB 页面 | 原文线索

V2-009269 | MATHDB-341652

Unbounded resurgence of fiber products

摘录 / EXCERPT: Unbounded fiber-product resurgence conjecture. There exists a homogeneous polynomial ideal I such that

conjecture | open | MathDB 页面 | 原文线索

V2-009270 | MATHDB-341705

Wiegold's conjecture for nilpotent Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a nilpotent Lie algebra over a field $\mathbb{F}$. For an element $x\in\mathfrak{g}$, define its breadth $b(x)$ by ... where $C_{\mathfrak{g}}(x)$ is the Lie ce...

conjecture | open | MathDB 页面 | 原文线索

V2-009271 | MATHDB-341734

Finite F-representation type for invariant rings of reductive groups

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let G be a reductive group, and let W be a finite-dimensional k-representation of G. The invariant ring...

conjecture | open | MathDB 页面 | 原文线索

V2-009272 | MATHDB-341736

The full symmetric Galois group conjecture for rational plane curves

摘录 / EXCERPT: Let $\mathbf{d}$ be a degree, let P be a set of $3\mathbf{d}-1$ points of $\mathbb{P}^2(\mathbb{Q})$ in general position, let C_P be the set of $N_{\mathbf{d}}$ complex rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-009273 | MATHDB-341780

Carlisle-Kuhn polynomiality conjecture for the Grothendieck ring of unstable modules

摘录 / EXCERPT: Let $K^n(\operatorname{\mathcal{U}})$ be the free abelian group generated by isomorphism classes of direct summands of H^*V_n, where $V_n=(\operatorname{\mathbb{F}}_p)^n$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-009274 | MATHDB-341784

Realization by reflections for mutation classes from unpunctured surfaces or orbifolds

摘录 / EXCERPT: Let a mutation class be originating from an unpunctured surface or orbifold if it arises from an unpunctured surface or orbifold in the sense of the paper. A realization by reflect...

conjecture | open | MathDB 页面 | 原文线索

V2-009275 | MATHDB-341797

Goncharov-Shen's cluster Donaldson-Thomas transformation conjecture

摘录 / EXCERPT: Let $\mathcal{C}_{\mathfrak{g},\Sigma}$ be the mutation class associated to an admissible pair $(\Sigma,\mathfrak{g})$. Let $\mathbf{w}_0=(w_0,\ldots,w_0)\in W(\mathfrak{g})^p$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-009276 | MATHDB-341816

Properness conjecture for commuting maps on incidence algebras

摘录 / EXCERPT: Let $(X,\leqslant)$ be a locally finite pre-ordered set, let $\mathcal{R}$ be a 2-torsion-free commutative ring with unity, and let $I(X,\mathcal{R})$ denote the incidence algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-009277 | MATHDB-341834

Junker-Koenigsmann conjecture on ample fields

摘录 / EXCERPT: A field K is ample if every nonsingular curve over K has either no K-points or infinitely many. Junker-Koenigsmann's conjecture. Every infinite field with finitely generated...

conjecture | open | MathDB 页面 | 原文线索

V2-009278 | MATHDB-341864

General Noetherianity conjecture for pointwise finitely generated $VIC(\mathbb{Z})$-modules

摘录 / EXCERPT: A $VIC(\mathbb{Z})$-module is a functor from the category $VIC(\mathbb{Z})$; it is finitely generated if it is generated by finitely many elements under the morphisms of this categ...

conjecture | open | MathDB 页面 | 原文线索

V2-009279 | MATHDB-341881

The radicality conjecture for Specht ideals

摘录 / EXCERPT: Let K be a field, let $R=K[x_1,\ldots,x_n]$, and let $I^{\rm Sp}_\lambda$ denote the Specht ideal associated to a partition λ of n. Radicality conjecture. The Specht id...

conjecture | open | MathDB 页面 | 原文线索

V2-009280 | MATHDB-341882

The characteristic bound conjecture for Cohen-Macaulay Specht ideals

摘录 / EXCERPT: Let K be a field, let $R=K[x_1,\ldots,x_n]$, and let $\lambda=(\lambda_1,\ldots,\lambda_l)$ be a partition of n satisfying condition (2) or (3) of the stated classification pro...

conjecture | open | MathDB 页面 | 原文线索

V2-009281 | MATHDB-341883

The linear-resolution conjecture for two-row Specht ideals

摘录 / EXCERPT: Let K be a field, let $R=K[x_1,\ldots,x_{2a+1}]$, and let $I^{\rm Sp}_{(a,a,1)}$ be the Specht ideal associated to the partition $(a,a,1)$. Linear-resolution conjecture. If ... the...

conjecture | open | MathDB 页面 | 原文线索

V2-009282 | MATHDB-341895

The integral-closure locality conjecture for torsion-free tensor products

摘录 / EXCERPT: Let R be a one-dimensional local domain whose integral closure $\overline R$ is finitely generated as an R-module. Let M and N be finitely generated R-modules, neither of...

conjecture | open | MathDB 页面 | 原文线索

V2-009283 | MATHDB-341920

Relative Serre functor conjecture for Soergel bimodules of finite Coxeter systems

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system with longest element w_0, and choose a realization $\mathfrak{h}$ of W. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009284 | MATHDB-341939

Uniqueness of clusters from their F-matrices

摘录 / EXCERPT: F-matrix uniqueness conjecture. If clusters $\mathbf{z}$ and $\mathbf{z'}$ in $\operatorname{\mathcal{A}}$ have $F_{\mathbf{z}}=F_{\mathbf{z'}}$, then $\mathbf{z}=\mathbf{z'}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009285 | MATHDB-341949

Conjecture on Gröbner bases and linear resolutions of products of maximal-minor ideals

摘录 / EXCERPT: The Gröbner-basis and linear-resolution conjecture. The initial ideal satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-009286 | MATHDB-341952

The order conjecture for the symbol $[0,0,1]$

摘录 / EXCERPT: For each integer $N\ge 2$, consider the symbol $[0,0,1]$ in the module $\mathcal B_3(\mathbb Z/N\mathbb Z)$. The order conjecture. The element has order 1, equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-009287 | MATHDB-341968

Naive gluing conjecture for the comparison map

摘录 / EXCERPT: Let (S,M,A) be a graded marked surface with a full arc system A. Say that it has enough marked intervals when ... and, for each k, the inclusion $\iota_k:(D^2_k,M_k)\to(S,M)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-009288 | MATHDB-341986

Lusztig's P1-P15 conjectures for Coxeter groups with complete graph

摘录 / EXCERPT: Let (W,S) be a Coxeter group with weight function L, Hecke algebra structure constants $h_{x,y,z}$, Kazhdan-Lusztig $\mathbf{a}$-function, and sets $\mathcal{D}_{\geq N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009289 | MATHDB-341990

Polynomiality conjecture for the positive Weyl-group invariant algebra of type $A_{2l}^{(2)}$

摘录 / EXCERPT: Polynomiality conjecture. For each $\tau\in\mathbb{H}$, $\widetilde{Th}^+\vert_{Y_\tau}$ is a polynomial ring over $\mathbb{C}$ generated by

conjecture | open | MathDB 页面 | 原文线索

V2-009290 | MATHDB-342001

The Ext-enhanced monoidal structure conjecture for free-monodromic tilting sheaves

摘录 / EXCERPT: Let $\mathsf{FM}_2^{\mathrm{Ext}}$ be the Ext-enhanced category of free-monodromic complexes, and let $\mathrm{Tilt}_2^{\mathrm{Ext}}$ be the full subcategory consisting of...

conjecture | open | MathDB 页面 | 原文线索

V2-009291 | MATHDB-342036

Tate's full faithfulness conjecture for motives

摘录 / EXCERPT: Let F be a number field and let $\operatorname{Rep}_{\mathbb Q}_\ell(G_F)$ be the category of ℓ-adic representations of $G_F=\operatorname{Gal}(\overline F/F)$. Consider t...

conjecture | open | MathDB 页面 | 原文线索

V2-009292 | MATHDB-342046

The deformation conjecture for irregular weight-one Galois representations

摘录 / EXCERPT: Assume that g is an irregular weight-one eigenform at p, with g_α its p-stabilisation. Let $S_1^{(p)}(N,\chi)[[g_\alpha]]_0$ be the quotient of the generalised eigensp...

conjecture | open | MathDB 页面 | 原文线索

V2-009293 | MATHDB-342059

The enrichment-module equivalence for symmetric monoidal Lawvere theories

摘录 / EXCERPT: Let $\mathcal{L}$ be a Lawvere theory admitting a symmetric monoidal structure compatible with finite products. Consider the ∞-categories of...

conjecture | open | MathDB 页面 | 原文线索

V2-009294 | MATHDB-342063

Nonsingularity conjecture for elliptic inflectionary curves

摘录 / EXCERPT: Let $k\geq 1$, and let $\overline{\mathcal{C}(1,k)}$ be the projective plane inflectionary curve defined by the homogenized inflection polynomial $P_{1,k}$, with singular points...

conjecture | open | MathDB 页面 | 原文线索

V2-009295 | MATHDB-342064

Support and coefficient symmetry conjecture for elliptic inflection polynomials

摘录 / EXCERPT: For $k\geq 1$, let $P_{1,k}(x,\lambda)$ be the inflection polynomial, and define the polygons ... Write $\operatorname{Supp}(P_{1,k})$ for its exponent support. Support and coefficie...

conjecture | open | MathDB 页面 | 原文线索

V2-009296 | MATHDB-342080

Algebraic entropy criterion for integrability of birational maps

摘录 / EXCERPT: A birational map is assumed to admit an algebraic entropy, measuring the asymptotic growth of the degrees of its iterates. Algebraic entropy conjecture. The condition that algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-009297 | MATHDB-342151

Positive-characteristic groupwise isomorphism conjectures for universal enveloping algebras

摘录 / EXCERPT: Groupwise positive-characteristic isomorphism conjectures. The following three statements should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-009298 | MATHDB-342155

The geometric Langlands categorification conjecture for Hall algebras

摘录 / EXCERPT: Let X be a smooth projective complex curve and let $\mathbf{Bun}(X)_\mathsf{dR}$ and $\mathbf{Coh}(X_\mathsf{dR})$ denote the de Rham moduli stacks of bundles and coherent sheave...

conjecture | open | MathDB 页面 | 原文线索

V2-009299 | MATHDB-342199

ML degree formula for the MA(1) model

摘录 / EXCERPT: Let $MA(1)$ be the moving-average model with $n>1$ sample points, and let its maximum-likelihood degree be the number of complex critical points of the likelihood function for gene...

conjecture | open | MathDB 页面 | 原文线索

V2-009300 | MATHDB-342215

The graded Grothendieck-group classification conjecture for finite Leavitt path algebras

摘录 / EXCERPT: Let E_1 and E_2 be finite graphs and F a field. Write $M^{\operatorname{gr}}_{E_i}$ for the graded graph monoid and $K_0^{\operatorname{gr}}(L_F(E_i))$ for the graded Grothen...

conjecture | open | MathDB 页面 | 原文线索

V2-009301 | MATHDB-342219

Mazur-Rubin-Sano conjecture for Rubin-Stark elements

摘录 / EXCERPT: Mazur-Rubin-Sano conjecture. In the module

conjecture | open | MathDB 页面 | 原文线索

V2-009302 | MATHDB-342255

Dual-morphism conjecture for the explicit higher-degree construction

摘录 / EXCERPT: Dual-morphism conjecture. The linear map $\psi:M(W^*)\rightarrow M(V^*)$ associated to

conjecture | open | MathDB 页面 | 原文线索

V2-009303 | MATHDB-342318

Asymptotic sign coherence conjecture

摘录 / EXCERPT: Asymptotic sign coherence conjecture. There exists a sequence of strict sign vectors

conjecture | open | MathDB 页面 | 原文线索

V2-009304 | MATHDB-342327

Mazur-Rubin-Sano exceptional zero conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let k_{Γ}/k be a $\mathbb{Z}_p$-extension with Galois group Γ, let $\Gamma_n:=\operatorname{Gal}(k_n/k)$, and put $L_\Gamma:=k_\Gamma L$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009305 | MATHDB-342357

Gross's higher Stark conjecture

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with group G, let S contain all archimedean places, and let $r>1$ be an integer. For a complex character χ of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-009306 | MATHDB-342358

The higher refined p-adic class number formula conjecture

摘录 / EXCERPT: Let L/K be the Galois extension and $G=\operatorname{Gal}(L/K)$, with $G^+=G/\langle\tau\rangle$, and let e_r denote the relevant idempotent. Let $L_{p,S}(r)$ be the central el...

conjecture | open | MathDB 页面 | 原文线索

V2-009307 | MATHDB-342394

Vertical-to-horizontal trace essential-surjectivity conjecture

摘录 / EXCERPT: Let $\operatorname{vTr}(\operatorname{SBim})$ and $\operatorname{hTr}(\operatorname{SBim})$ denote the vertical and horizontal traces of the category of Soergel bimodules, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009308 | MATHDB-342435

Hazrat-Roozbeh's graph-algebra classification conjecture

摘录 / EXCERPT: Let E and F be finite graphs, let k be a field, let $C^*(E)$ and $C^*(F)$ denote their graph C^*-algebras, let $L_k(E)$ and $L_k(F)$ denote their Leavitt path algebras, and...

conjecture | open | MathDB 页面 | 原文线索

V2-009309 | MATHDB-342458

Herzog-Hibi conjecture on depth functions of powers of homogeneous ideals

摘录 / EXCERPT: Let S be a standard graded algebra over a field k. For a homogeneous ideal $Q\subseteq S$, its depth function is the function $n\mapsto\operatorname{depth} S/Q^n$ for $n\geq 1$...

conjecture | open | MathDB 页面 | 原文线索

V2-009310 | MATHDB-342476

The idempotent truncation conjecture for blob algebras and Soergel endomorphism algebras

摘录 / EXCERPT: Let Λ_n be the indexing set of weights, let W be the relevant Weyl group, and let $\mathcal{A}^w$ denote the orbit or alcove associated with $w\in W$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-009311 | MATHDB-342497

The bibar-cube conjecture for perverse sheaves on symmetric powers

摘录 / EXCERPT: Let A be a primitive bialgebra in a braided monoidal abelian category, and let ${\mathbb{B}\mathbb{B}}_n(A)$ denote its nth bibar-cube, viewed as a double representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-009312 | MATHDB-342498

Generalized derivative Thomae formula for singular half-periods

摘录 / EXCERPT: Generalized derivative Thomae formula. The following identity holds:

conjecture | open | MathDB 页面 | 原文线索

V2-009313 | MATHDB-342578

The Milnor-Moore conjecture for A_∞ algebras

摘录 / EXCERPT: Let A be an A_∞ algebra over a characteristic-zero field with a cocommutative, conilpotent coproduct Δ that is a strict A_∞ morphism ... Let $\overline\Delta$...

conjecture | open | MathDB 页面 | 原文线索

V2-009314 | MATHDB-342579

Connectedness conjecture for the crystalline deformation-ring fibre

摘录 / EXCERPT: Let $\mathbb{F}$ be a finite field of characteristic p, let K be a finite extension of $\mathbb{Q}_p$ with absolute Galois group G_K, and let $V_{\mathbb{F}}$ be an absolutel...

conjecture | open | MathDB 页面 | 原文线索

V2-009315 | MATHDB-342598

Classification conjecture for right cyclic pivotal unitary Temperley-Lieb-Jones modules

摘录 / EXCERPT: Let Γ be the fixed data defining the unitary $*$-2-category $\operatorname{TLJ}(\Gamma)$. A MW-type bipartite balanced Γ-fair graph is a bipartite balanced Γ-f...

conjecture | open | MathDB 页面 | 原文线索

V2-009316 | MATHDB-342599

Jones's surjectivity conjecture for the arboreal representation of x^2+t

摘录 / EXCERPT: Let k be a field of characteristic different from 2, and let t be transcendental over k. For the polynomial ... let its arboreal representation be the natural Galois action o...

conjecture | open | MathDB 页面 | 原文线索

V2-009317 | MATHDB-342609

The lambda-algebra cycle conjecture for higher Lannes-Zarati stems

摘录 / EXCERPT: Let A be the mod p Steenrod algebra, let M be an unstable A-module, and let $\Lambda\otimes M^\#$ be the lambda-algebra complex computing...

conjecture | open | MathDB 页面 | 原文线索

V2-009318 | MATHDB-342610

The positive Singer-construction containment conjecture

摘录 / EXCERPT: Let A be the mod p Steenrod algebra, let M be an unstable A-module, and let $P_s=H^*(B(\mathbb Z/p)^s)$. Let $\mathscr{R}_sM$ be the s-th Singer construction and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009319 | MATHDB-342640

The maximal-subfields algebraicity conjecture for division rings

摘录 / EXCERPT: Let D be a division ring with center F. A subfield K of D is maximal if it is maximal among subfields of D. Maximal-subfields algebraicity conjecture. If every maximal su...

conjecture | open | MathDB 页面 | 原文线索

V2-009320 | MATHDB-342641

Mahdavi-Hezavehi's weak Kurosh problem

摘录 / EXCERPT: Let D be a non-commutative division ring algebraic over its center. A division subring is centrally finite if it has finite dimension over its center. Mahdavi-Hezavehi's weak Kur...

open_problem | open | MathDB 页面 | 原文线索

V2-009321 | MATHDB-342643

Generic Summand Conjecture for Bessel modules

摘录 / EXCERPT: Generic Summand Conjecture. There exists a cuspidal realization ${\mathcal{C}}_\pi$ with first occurrence index ℓ_0 and a suitable k-rational orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-009322 | MATHDB-342693

Genericity conjecture for the favorable spectral-data locus

摘录 / EXCERPT: Let X be an algebraic surface, and let $\mathscr B_X^{\Diamond}(k)$ be the distinguished subset of the postulated Hitchin base $\mathscr B_X(k)$ defined in the surrounding discus...

conjecture | open | MathDB 页面 | 原文线索

V2-009323 | MATHDB-342710

Absolute irreducibility conjecture for compatible systems attached to automorphic representations

摘录 / EXCERPT: Let F be the number field and $\bar F$ its algebraic closure, with absolute Galois group $\Gamma_F=\operatorname{Gal}(\bar F/F)$. Let π be a regular algebraic, conjugate sel...

conjecture | open | MathDB 页面 | 原文线索

V2-009324 | MATHDB-342753

Petukhov-Shestakov conjecture on the just infinite GK-dimension of the positive Witt enveloping algebra

摘录 / EXCERPT: Let $\mathbb K$ be a field of characteristic zero, and let W_+ be the positive Witt algebra with basis $\{e_n:n\geqslant 1\}$ and Lie bracket $[e_i,e_j]=(j-i)e_{i+j}$. Its univer...

conjecture | open | MathDB 页面 | 原文线索

V2-009325 | MATHDB-342755

The Euclidean mechanical systems kernel conjecture

摘录 / EXCERPT: Euclidean kernel conjecture. Consider all smooth potentials $V\in C^{\infty}(\mathbb{R}^d)$ with arbitrary d. Then

conjecture | open | MathDB 页面 | 原文线索

V2-009326 | MATHDB-342789

The EZ submodule conjecture

摘录 / EXCERPT: Let $Z_{p,m}$ and $Z_{p,m}'$ be the submodules defined above, and let E_1 be the first member of the corresponding filtration of submodules. EZ submodule conjecture. With the not...

conjecture | open | MathDB 页面 | 原文线索

V2-009327 | MATHDB-342807

The power-of-\q conjecture for multiplicity-one weights

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra of rank r, and let λ and μ be weights such that the multiplicity $m(\lambda,\mu)$ of μ in the irreducible representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-009328 | MATHDB-342851

Formality conjecture for dg multiplicative preprojective algebras

摘录 / EXCERPT: Formality conjecture. If Q is connected and not Dynkin, then

conjecture | open | MathDB 页面 | 原文线索

V2-009329 | MATHDB-342853

Strong free product property conjecture for non-Dynkin quivers

摘录 / EXCERPT: Strong free product property conjecture. The map σ' is a linear isomorphism, equivalently $(L_Q,r,\sigma,kQ_0[t,(t+q)^{-1}])$ satisfies the strong free product property.

conjecture | open | MathDB 页面 | 原文线索

V2-009330 | MATHDB-342884

Herzog's convergence conjecture for Stanley depths of powers

摘录 / EXCERPT: Let $S=\mathbb{K}[x_1,\dots,x_n]$ and let I be a monomial ideal of S. Denote the Stanley depth of a module by ${\rm sdepth}$. Herzog's convergence conjecture. The sequence ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-009331 | MATHDB-342934

Tanisaki quotient conjecture for the modules $V_n^=({\mathbf a})$

摘录 / EXCERPT: Let $r\leq n$ be nonnegative integers, put $k=n-r$, and let ${\mathbf a}\in({\mathbb Z}_{\geq0})^r$. Define $V_n^=({\mathbf a})$ as the associated graded quotient for the component...

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V2-009332 | MATHDB-343011

Bertram-Feinberg-Mukai conjecture for rank-two canonical bundles

摘录 / EXCERPT: Set $\rho_{g,k}:=3g-3-\binom{k+1}{2}$. Let C be a general curve of genus $g\geq 2$, and consider the moduli space of stable rank-two vector bundles on C with canonical determin...

conjecture | open | MathDB 页面 | 原文线索

V2-009333 | MATHDB-343030

Geramita-Harbourne-Migliore conjecture on powers of star-configuration ideals

摘录 / EXCERPT: Let ${\mathcal A}=\{H_1,\ldots,H_s\}$ be a generic collection of $s\geq n+1$ hyperplanes in $\mathbb P^n$, with defining linear forms $\ell_1,\ldots,\ell_s$. For $1\leq c\leq n$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-009334 | MATHDB-343094

The synchronicity conjecture for cluster x- and y-variables

摘录 / EXCERPT: Let $(\mathbf{x},B)$ and $(\mathbf{y},B)$ be, respectively, a seed without coefficients and a Y-seed with coefficients in a universal semifield, sharing the same skew-symmetrizab...

conjecture | open | MathDB 页面 | 原文线索

V2-009335 | MATHDB-343095

Fomin-Zelevinsky's exchange-graph conjecture for cluster patterns

摘录 / EXCERPT: Let $\mathbf{\Sigma}=\{\Sigma_t=(\mathbf{x}_t,\mathbf{y}_t,B_t)\mid t\in\mathbb{T}_n\}$ be a cluster pattern with coefficients in any semifield $\mathbb{P}$. An unlabeled cluster i...

conjecture | open | MathDB 页面 | 原文线索

V2-009336 | MATHDB-343096

Fomin-Zelevinsky's principal-coefficients periodicity and finiteness conjecture

摘录 / EXCERPT: Let $\widetilde{\mathbf{\Sigma}}[t_0]=\{\widetilde{\Sigma}_t=(\widetilde{\mathbf{x}}_t,\widetilde{\mathbf{y}}_t,B_t)\mid t\in\mathbb{T}_n\}$ be a cluster pattern with principal coe...

conjecture | open | MathDB 页面 | 原文线索

V2-009337 | MATHDB-343097

Laurent positivity conjecture for F-polynomials of generalized cluster algebras

摘录 / EXCERPT: Let $\mathbf{\Gamma}(\mathbf{B}_R,t_0)=\{\Gamma_t=(\mathbf{F}_t(\mathbf{u},\mathbf{v}),G_t,C_t,\mathbf{v}_t,B_t)\mid t\in\mathbb{T}_n\}$ be the FGC-pattern of $\mathbf{B}_R$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-009338 | MATHDB-343108

Miasnikov-Ventura-Weil algebraic-extension conjecture for free groups

摘录 / EXCERPT: Miasnikov-Ventura-Weil conjecture. If the morphism

conjecture | open | MathDB 页面 | 原文线索

V2-009339 | MATHDB-343109

Parzanchevski-Puder revised algebraic-extension conjecture for free groups

摘录 / EXCERPT: Parzanchevski-Puder conjecture. If, for every free extension F' of F_Y and every basis X of F', the morphism

conjecture | refuted | MathDB 页面 | 原文线索

V2-009340 | MATHDB-343110

Miasnikov-Ventura-Weil conjecture for free groups of rank greater than two

摘录 / EXCERPT: Rank-greater-than-two conjecture. The original Miasnikov-Ventura-Weil conjecture holds for $H\leq K\leq F_Y$ whenever $|Y|>2$.

conjecture | open | MathDB 页面 | 原文线索

V2-009341 | MATHDB-343147

Separable essential finite-generation conjecture for valuation rings

摘录 / EXCERPT: Let $(K,\nu)\subset (L,\omega)$ be in situation, with valuation rings $\mathcal{O}_\nu$ and $\mathcal{O}_\omega$. Assume in addition that the field extension $L|K$ is separable. Se...

conjecture | open | MathDB 页面 | 原文线索

V2-009342 | MATHDB-343160

The derived isomorphism conjecture for enveloping algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ and $\mathfrak{g'}$ be finite-dimensional Lie algebras over $\mathbb{C}$. Write $U(\mathfrak{g})$ for the enveloping algebra of $\mathfrak{g}$, and likewise for...

conjecture | open | MathDB 页面 | 原文线索

V2-009343 | MATHDB-343202

Inverse-eigenvector product conjecture for Gram matrices

摘录 / EXCERPT: Inverse-eigenvector product conjecture. There exists an inverse eigenvector α such that

conjecture | open | MathDB 页面 | 原文线索

V2-009344 | MATHDB-343226

Degree-product bound for AS regular Hopf invariants

摘录 / EXCERPT: Degree-product conjecture. There is a minimal generating set of A^H such that the product of the degrees of its generators is less than or equal to $\dim_{\Bbbk} H$.

conjecture | open | MathDB 页面 | 原文线索

V2-009345 | MATHDB-343232

The analytic spread bound for ideals in Noetherian local rings

摘录 / EXCERPT: Symbolic analytic spread conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-009346 | MATHDB-343256

The functional-equation characterization of the $gl_{1|2}$ transfer-matrix spectrum

摘录 / EXCERPT: Let $T_{1}^{(K)}(\lambda)$ be the transfer matrix of the general $gl_{1|2}$-graded Yang-Baxter algebra with twisted boundary conditions. Let $t_{1}(\lambda \mid \{x_a\})$ be the po...

conjecture | open | MathDB 页面 | 原文线索

V2-009347 | MATHDB-343260

Nonexistence of finite-dimensional DG algebra phantoms

摘录 / EXCERPT: Let $\mathscr R$ be a smooth finite-dimensional DG algebra, and let $\mathscr{P}\!\mathit{erf}\!\operatorname{--}\mathscr R$ denote its DG category of perfect modules. A phantom is...

conjecture | open | MathDB 页面 | 原文线索

V2-009348 | MATHDB-343261

Asymptotic Lech's inequality

摘录 / EXCERPT: Asymptotic Lech's inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-009349 | MATHDB-343262

The Hilbert-Kunz strengthening of asymptotic Lech's inequality

摘录 / EXCERPT: Hilbert-Kunz strengthening.

conjecture | open | MathDB 页面 | 原文线索

V2-009350 | MATHDB-343263

The splitting type stratification conjecture for Brill-Noether loci

摘录 / EXCERPT: The splitting type stratification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009351 | MATHDB-343275

Ordinary implies ordinary Galois representations

摘录 / EXCERPT: Let F) be an imaginary quadratic field, let $\mathfrak{m}$ be a non-Eisenstein maximal ideal of residue characteristic $p>2$ associated to a residual Galois representation...

conjecture | open | MathDB 页面 | 原文线索

V2-009352 | MATHDB-343340

Demailly-Green-Griffiths-Lang-Vojta equivalence conjecture

摘录 / EXCERPT: Demailly-Green-Griffiths-Lang-Vojta conjecture. The following statements are equivalent: X is algebraically hyperbolic over k; X is bounded over k; for all $n\geq 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009353 | MATHDB-343350

Countable-dimensionality conjecture for locally finite Steinitz numbers

摘录 / EXCERPT: Countable-dimensionality conjecture. If τ is a locally finite Steinitz number, A is a unital locally matrix algebra, and $\mathbf{n}(A)=\tau$, then A is countable dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-009354 | MATHDB-343358

Expanding eigenbasis conjecture for recursive eigenvector matrices

摘录 / EXCERPT: Let A be a matrix of dimension at most 7, and repeatedly construct eigenvector matrices X_i whose columns have unit $\mathbb{L}_2$ norm. A matrix property holds for almost al...

conjecture | open | MathDB 页面 | 原文线索

V2-009355 | MATHDB-343359

Expanding orthogonal similarity variant conjecture

摘录 / EXCERPT: Expanding orthogonal similarity variant conjecture. For almost all matrices of dimension at most 7, repeatedly applying this eigenvector construction and random orthogonal or uni...

conjecture | open | MathDB 页面 | 原文线索

V2-009356 | MATHDB-343383

The vanishing-cycle W-algebra action conjecture for framed quiver moduli spaces

摘录 / EXCERPT: Vanishing-cycle W-algebra action conjecture. There is a $W_\kappa({\mathfrak gl}_r)$-action on this direct sum identifying it with the W-vacuum module.

conjecture | open | MathDB 页面 | 原文线索

V2-009357 | MATHDB-343397

The specific character-web conjecture for $mathfrak{sl}_3$ Grassmannians

摘录 / EXCERPT: Character-web conjecture. If $\operatorname{ch}(T)$ is a cluster monomial, then

conjecture | open | MathDB 页面 | 原文线索

V2-009358 | MATHDB-343398

The compatibility product conjecture for Grassmannian characters

摘录 / EXCERPT: Compatibility product conjecture. Compatible cluster variables satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-009359 | MATHDB-343413

The graded-character conjecture for KR-modules of types BCD

摘录 / EXCERPT: Let I be the set of Dynkin labels, let $I_<\subset I$ be the labels of the short roots in types B_N and C_N, and let $t_a=2$ for $a\in I_<$ and $t_a=1$ otherwise. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009360 | MATHDB-343439

Explicit basis conjecture for invariant differential forms and multiderivations of coincidental reflection groups

摘录 / EXCERPT: Let W be a coincidental complex reflection group, let $S(V^*)$ be the symmetric algebra on the dual of its reflection representation, and choose basic invariants $f_1,\ldots,f_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-009361 | MATHDB-343445

Chang-Schiffler's conjecture on cluster automorphisms

摘录 / EXCERPT: Let $\mathcal A$ be a cluster algebra, and let $f:\mathcal A\rightarrow\mathcal A$ be a $\mathbb{Z}$-algebra homomorphism. A cluster automorphism is an automorphism satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-009362 | MATHDB-343463

Generating equivalence relations on graph C*-algebras by structure-preserving moves

摘录 / EXCERPT: Generating-moves conjecture. In all six cases, the equivalence relation is generated by the moves that leave the graphs invariant in the relevant sense. Explicitly,

conjecture | open | MathDB 页面 | 原文线索

V2-009363 | MATHDB-343472

Non-algebraicity conjecture for generating functions of non-super-strict partitions

摘录 / EXCERPT: Non-algebraicity conjecture. If μ is not super-strict, meaning that μ contains two nonzero parts which differ by 1, then the generating function for μ is not algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-009364 | MATHDB-343492

The ordered-partition Hilbert-series formula for the super-diagonal coinvariant ring

摘录 / EXCERPT: Let $n\geq1$. For an ordered set partition Π of $[n]$, let $|\Pi|$ be its number of blocks, let $\tau(\Pi)$ be its associated permutation, and let $w_\Pi(c)$ be the schedule nu...

conjecture | open | MathDB 页面 | 原文线索

V2-009365 | MATHDB-343493

The candidate basis conjecture for the super-diagonal coinvariant ring

摘录 / EXCERPT: Let $n\geq1$. For each ordered set partition Π of $[n]$, let $\mathcal{B}_\Pi$ be the set of monomials constructed from the associated permutation, Grassmann monomial, and sche...

conjecture | open | MathDB 页面 | 原文线索

V2-009366 | MATHDB-343494

The homogeneous-component nonemptiness conjecture for the super-diagonal coinvariant ring

摘录 / EXCERPT: Let a, b, and c be non-negative integers. Consider the homogeneous component of order n in SDR_n with x-degree a, y-degree b, and z-degree c. H...

conjecture | open | MathDB 页面 | 原文线索

V2-009367 | MATHDB-343523

The nilpotent singular-support conjecture for lisse Hecke actions

摘录 / EXCERPT: Let G be a reductive group, let X be a smooth proper curve, and let $\Dmod(\Bun_G)$ carry its Hecke action. Let $\Dmod_{\on{Nilp}}(\Bun_G)$ be the subcategory with nilpotent si...

conjecture | open | MathDB 页面 | 原文线索

V2-009368 | MATHDB-343524

Laumon's conjecture on loose Hecke eigensheaves

摘录 / EXCERPT: Laumon's conjecture on loose Hecke eigensheaves. A loose Hecke eigensheaf has a nilpotent singular support. This is presented as a special case of the preceding inclusion conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-009369 | MATHDB-343525

The ULA module characterization conjecture for local systems

摘录 / EXCERPT: ULA module characterization conjecture. This functor induces an equivalence onto

conjecture | open | MathDB 页面 | 原文线索

V2-009370 | MATHDB-343534

Chebyshev positivity conjecture for negative-Euler-characteristic surfaces

摘录 / EXCERPT: Positivity conjecture. Both $(\hat{T}_n)$ and (S_n) are positive on every surface Σ with negative Euler characteristic.

conjecture | open | MathDB 页面 | 原文线索

V2-009371 | MATHDB-343570

Bloch-Kato conjecture for unitary Galois representations

摘录 / EXCERPT: Let π be an automorphic representation with associated Galois representation ρ_π, let χ be a character, and let $\mathcal{K}$ be the relevant number field. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009372 | MATHDB-343616

The Legendrian BMW algebra isomorphism conjecture

摘录 / EXCERPT: Let $\operatorname{BMW}^{\mathfrak{Leg}}_n$ be the Legendrian BMW algebra and let $\operatorname{BMW}^{\infty}_n$ be the specialized BMW algebra, with...

conjecture | open | MathDB 页面 | 原文线索

V2-009373 | MATHDB-343631

The Beilinson-Drinfeld regular-singularity conjecture for Hecke eigensheaves

摘录 / EXCERPT: Beilinson-Drinfeld regular-singularity conjecture. The D-module Δ^0_λ is irreducible and has regular singularities on each connected component of...

conjecture | open | MathDB 页面 | 原文线索

V2-009374 | MATHDB-343638

Auto-equivalence and cluster automorphism group isomorphism conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a 2-Calabi-Yau triangulated category with cluster structure, let T be a cluster tilting object, and let ϕ be a cluster map sending cluster tilting obj...

conjecture | open | MathDB 页面 | 原文线索

V2-009375 | MATHDB-343639

Quiver determines the potential up to right equivalence conjecture

摘录 / EXCERPT: Let (Q,W) be a non-degenerate quiver with potential, and let (Q',W') be mutation equivalent to (Q,W). Quiver-determines-potential conjecture. If $Q'\cong Q$, then (Q',W') i...

conjecture | open | MathDB 页面 | 原文线索

V2-009376 | MATHDB-343659

Alilooee-Banerjee-Beyarslan-Hà regularity conjecture for powers of edge ideals

摘录 / EXCERPT: Let G be a graph, let $I(G)$ be its edge ideal, and let ${\rm reg}(I(G))$ denote its Castelnuovo-Mumford regularity. For each integer $s\geq 1$, Alilooee-Banerjee-Beyarslan-Hà's...

conjecture | open | MathDB 页面 | 原文线索

V2-009377 | MATHDB-343673

Full generic arboreal Galois group conjecture for rational functions

摘录 / EXCERPT: Let k be a number field, and let $\phi(z)\in k(z)$ be a rational function of degree $d\geq 2$, defined over k. Let $L=k(t)$ be the function field, and define the generic arbore...

conjecture | open | MathDB 页面 | 原文线索

V2-009378 | MATHDB-343678

Polynomiality conjecture for the ML degree of linear covariance models

摘录 / EXCERPT: ML-degree polynomiality conjecture. For $m\geq 3$, the ML degree is a polynomial of degree $m-1$ in n. In the cases stated in the source, for $m=3$ it equals

conjecture | open | MathDB 页面 | 原文线索

V2-009379 | MATHDB-343713

Classification conjecture for maximally generically transitive actions

摘录 / EXCERPT: Let G be a connected group of finite Morley rank acting faithfully, definably and transitively on a set X of Morley rank n, and suppose the action is generically $(n+2)$-tran...

conjecture | open | MathDB 页面 | 原文线索

V2-009380 | MATHDB-343715

Transitive-module identification conjecture

摘录 / EXCERPT: Let (G,V) be a module of finite Morley rank, with G connected and simple modulo its at most finite centre. Suppose that G is transitive on $V\setminus\{0\}$ and generically...

conjecture | open | MathDB 页面 | 原文线索

V2-009381 | MATHDB-343716

Highly generically transitive module conjecture

摘录 / EXCERPT: Let (G,V) be a faithful module of finite Morley rank, and suppose that V has Morley rank n and G is generically n-transitive. Highly generically transitive module conject...

conjecture | open | MathDB 页面 | 原文线索

V2-009382 | MATHDB-343741

Skewing identities for hook-indexed modules

摘录 / EXCERPT: For each positive integer n, let $\mathcal{S}_\rho$ denote the graded Frobenius characteristic associated with the hook shape ρ, let $e_1^\perp$ be the operator adjoint to m...

conjecture | open | MathDB 页面 | 原文线索

V2-009383 | MATHDB-343753

Folklore conjecture on the algebra of multizeta values in positive characteristic

摘录 / EXCERPT: Let $\overline{\mathfrak{Z}}$ be the $\overline{\mathbb{Q}}$-algebra generated by multizeta values, and let $\overline{\mathfrak{Z}}_w$ be the $\overline{\mathbb{Q}}$-vector space...

conjecture | open | MathDB 页面 | 原文线索

V2-009384 | MATHDB-343760

Centre-determines-cancellation conjecture for affine noetherian domains

摘录 / EXCERPT: Let k be an uncountable algebraically closed field of characteristic zero, and let A be an affine noetherian domain over k. Write $Z(A)$ for the centre of A. An algebra is...

conjecture | open | MathDB 页面 | 原文线索

V2-009385 | MATHDB-343782

Conjecture on distinct critical points for two-dimensional linear Gaussian covariance models

摘录 / EXCERPT: Distinct-solutions conjecture. For generic values of A, B, and S, the score equations of $\mathcal{M}_{A,B}$ with sample covariance matrix S have $2n-3$ distinct solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-009386 | MATHDB-343805

Conjecture on semilattice-like terms in bounded width algebras

摘录 / EXCERPT: Semilattice-like terms conjecture. Every bounded width algebra has terms w,s satisfying these identities.

conjecture | open | MathDB 页面 | 原文线索

V2-009387 | MATHDB-343806

Minority-pair conjecture for minimal bounded width algebras

摘录 / EXCERPT: Minority-pair conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009388 | MATHDB-343807

Recursive subalgebra conjecture for minimal bounded width algebras

摘录 / EXCERPT: Recursive subalgebra conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-009389 | MATHDB-343808

Binary-invariants conjecture for minimal bounded width algebras

摘录 / EXCERPT: Binary-invariants conjecture. Among minimal bounded width algebras, $\mathbb{A}$ is determined up to term equivalence by $\operatorname{Inv}_2(\mathbb{A})$.

conjecture | open | MathDB 页面 | 原文线索

V2-009390 | MATHDB-343874

The conjecture that the modified Bloch group computes indecomposable K3

摘录 / EXCERPT: Modified Bloch group conjecture. The homomorphism $\psi_{F,\mathrm{tf}}$ is an isomorphism. Extensive computations for imaginary quadratic fields motivate the claim that the modifi…

conjecture | open | MathDB 页面 | 原文线索

V2-009391 | MATHDB-343909

The Galois group conjecture for the polynomial Q_n

摘录 / EXCERPT: Let $n\geq 2$, let Q_n be the polynomial studied in the paper, and write … For a polynomial with rational coefficients, let $\operatorname{Gal}(Q_n)$ denote its Galois group and…

conjecture | open | MathDB 页面 | 原文线索

V2-009392 | MATHDB-343911

The elliptic Hall algebra quotient conjecture for the punctured torus Homflypt skein

摘录 / EXCERPT: Let $\mathrm{Sk}(T^2-D^2)$ be the Homflypt skein algebra of the punctured torus, and let $\mathcal{E}_{\sigma,\bar\sigma}$ be the elliptic Hall algebra with generators…

conjecture | open | MathDB 页面 | 原文线索

V2-009393 | MATHDB-343912

The braid-to-tangle skein isomorphism conjecture for the punctured torus

摘录 / EXCERPT: Let $\mathrm{BSk}_n(T^2,*)$ be the braid skein algebra of the punctured torus with n strands, and let $\mathrm{Sk}_n(T^2,*)$ be the corresponding full framed Homflypt tangle skei…

conjecture | open | MathDB 页面 | 原文线索

V2-009394 | MATHDB-343916

Gaiotto's factorization-category conjecture for affine Lie superalgebras

摘录 / EXCERPT: Assume $N>M$ and let $c\neq0$ be a non-rational number. Let $\Gr_{\GL_N}$ denote the affine Grassmannian, $U_{M,N}(\bF)$ the relevant unipotent subgroup with character $\chi_{M,N}$…

conjecture | open | MathDB 页面 | 原文线索

V2-009395 | MATHDB-343917

Degenerate supergroup conjecture for the geometric Langlands limit

摘录 / EXCERPT: Let $N>M$, let $\Gr_{\GL_N}$ be the affine Grassmannian, and let $\ul\GL(M|N)$ be the group introduced in the source as the appropriate extension governing the $c\to\infty$ limit.…

conjecture | open | MathDB 页面 | 原文线索

V2-009396 | MATHDB-343975

Free-field realization conjecture for the \(x^{N_3}y^{N_2}z^{N_4}w^{N_1}\) algebra

摘录 / EXCERPT: Let $\mathcal{L}^{(\alpha)}$ be the elementary Miura operator for $\alpha=x,y,z,w$, with parameter N_α, and let $x^{N_3}y^{N_2}z^{N_4}w^{N_1}$ denote the algebra obtained fr…

conjecture | open | MathDB 页面 | 原文线索

V2-009397 | MATHDB-343976

Bosonization conjecture for the \(CY^3_{m,n}\) pseudo-differential operators

摘录 / EXCERPT: Let $\mathcal{L}^{(x)},\mathcal{L}^{(y)},\mathcal{L}^{(z)},\mathcal{L}^{(w)}$ be the elementary pseudo-differential operators associated with the four coordinate functions on…

conjecture | open | MathDB 页面 | 原文线索

V2-009398 | MATHDB-343981

B. Shapiro's zero-location conjecture for generalized polynomial recurrences

摘录 / EXCERPT: Shapiro's conjecture. Every zero of every $P_n(z)$ that is not a zero of $A(z)$ or $B(z)$ lies on $\mathcal C$. This generalizes Tran's zero-location conjecture to recurrences with…

conjecture | open | MathDB 页面 | 原文线索

V2-009399 | MATHDB-343982

Sign conjecture for Shapiro's generalized recurrence curve

摘录 / EXCERPT: Sign conjecture. On the portion of this curve containing the zeros, the following inequalities hold: if k is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-009400 | MATHDB-344017

Rudolph's conjecture on the linear independence of algebraic knots

摘录 / EXCERPT: A knot is algebraic if it arises as the link of an isolated singularity of a complex curve. Let $\mathcal{C}$ denote the smooth knot concordance group. Rudolph's conjecture. The se...

conjecture | open | MathDB 页面 | 原文线索

V2-009401 | MATHDB-344032

Easy Coefficients Conjecture for rotation symmetric Boolean functions

摘录 / EXCERPT: Easy Coefficients Conjecture. The recursion for the weights of any rotation symmetric function has easy coefficients, attached to a multiset of algebraic integers. At least in the...

conjecture | open | MathDB 页面 | 原文线索

V2-009402 | MATHDB-344117

Conjecture on disjoint positive root sets in folded cluster structures

摘录 / EXCERPT: Let $D^{\perp}\simeq\mathbf{D}_{2m}$ with $m\geq 2$, let $1\leq l\leq m-1$, and let $\mathcal{P}$ be the power set of $\{1,\cdots,l\}$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-009403 | MATHDB-344148

The fibration conjecture for composed Schubert problems

摘录 / EXCERPT: Let $\underline{\lambda}$ and $\underline{\rho}$ be nontrivial Schubert problems, and let $\underline{\lambda}\circ\underline{\rho}$ be their composition. A composition is fibered...

conjecture | open | MathDB 页面 | 原文线索

V2-009404 | MATHDB-344159

The algebraic Iwasawa-theoretic Birch-Swinnerton-Dyer conjecture

摘录 / EXCERPT: Assume $\text{\russ\char88}(E/\mathbb Q)$ is finite, and use the notation v_ξ, L_ℓ, $\operatorname{Tam}(E)$, $R_\omega^{\rm Boc}$, and Q^{r-1} from the paper. Algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-009405 | MATHDB-344199

The anticyclotomic p-adic Birch-Swinnerton-Dyer conjecture for $L_{\mathfrak p}(f)$

摘录 / EXCERPT: Let $E/\mathbf{Q}$ be an elliptic curve, K an imaginary quadratic field, and let J be the augmentation ideal in the anticyclotomic Iwasawa algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009406 | MATHDB-344200

The square-root anticyclotomic p-adic Birch-Swinnerton-Dyer conjecture

摘录 / EXCERPT: Let $E/\mathbf{Q}$ be an elliptic curve, K an imaginary quadratic field, and let J be the augmentation ideal in the anticyclotomic Iwasawa algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009407 | MATHDB-344260

Sparsity conjecture for post-critically finite maps with fixed tail-length

摘录 / EXCERPT: Sparsity conjecture. Let $d\ge3$ and $n\ge2$. For every fixed tail-length $\ell\ge1$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-009408 | MATHDB-344271

Subfactorizability conjecture for three-dimensional fusion bialgebras of type II

摘录 / EXCERPT: Let $\b5$ be the parameter of the one-parameter family of three-dimensional canonical fusion bialgebras in case II, with $\b5\rfloor 3$. A fusion bialgebra is subfactorizable if it...

conjecture | open | MathDB 页面 | 原文线索

V2-009409 | MATHDB-344277

Conjecture on logarithmic integrals with powers of \log(1-x) and \log(1+x)

摘录 / EXCERPT: Logarithmic-integral algebraicity conjecture. For every $n\geq 1$, this value belongs to the algebra over $\mathbb{Q}$ generated by π^2, $\log 2$, and $\zeta(n)$ for odd n. T...

conjecture | open | MathDB 页面 | 原文线索

V2-009410 | MATHDB-344287

Nandi's partition identities for level 4 standard modules of $A_2^{(2)}$

摘录 / EXCERPT: Nandi's conjecture. The following partition sets are equinumerous by weight, with the indicated product-side sets:

conjecture | open | MathDB 页面 | 原文线索

V2-009411 | MATHDB-344293

Pellarin's expansion conjecture for Tate-algebra zeta values

摘录 / EXCERPT: Pellarin's expansion conjecture. For all $n\in\mathbb{N}^*$ and $\Sigma\subset\mathbb{N}^*$ satisfying $|\Sigma|\equiv n\pmod{q-1}$, there is a unique expansion

conjecture | open | MathDB 页面 | 原文线索

V2-009412 | MATHDB-344294

Harmonic-product conjecture for the algebra of Frobenius-twisted zeta values

摘录 / EXCERPT: Harmonic-product conjecture. The only $\mathbb{F}_p$-algebraic relations in $\mathbb{I}$ are those coming from the harmonic product.

conjecture | open | MathDB 页面 | 原文线索

V2-009413 | MATHDB-344312

Noncommutative Lie algebraic structures on the full vector shift

摘录 / EXCERPT: Full-shift noncommutativity conjecture. The full vector shift X is not compatible with a nontrivial Lie algebraic subshift structure.

conjecture | open | MathDB 页面 | 原文线索

V2-009414 | MATHDB-344410

Equality of the relative and dual F-rational signatures

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be an F-rational local ring. Write $\operatorname{s_{rel}}(R)$ for its relative F-rational signature and $\operatorname{s_{dual}}(R)$ for its dual F-rational...

conjecture | open | MathDB 页面 | 原文线索

V2-009415 | MATHDB-344468

Universal conductor invariance of wild ramification for nearby cycles

摘录 / EXCERPT: Let S be a strictly henselian trait with algebraically closed residue field, and let X, x, $\mathcal F$, and $\mathcal F'$ be as in Theorem: X is a strictly local S-schem...

conjecture | open | MathDB 页面 | 原文线索

V2-009416 | MATHDB-344476

Russell-Tymoczko positivity conjecture for the transitioning matrix

摘录 / EXCERPT: Let $\mathcal{T}(n,n)$ be the set of standard Young tableaux of shape (n,n), let $\{v_T:T\in\mathcal{T}(n,n)\}$ and $\{w_S:S\in\mathcal{T}(n,n)\}$ be the tableau and web bases of...

conjecture | open | MathDB 页面 | 原文线索

V2-009417 | MATHDB-344516

Hodge classes are motivated conjecture

摘录 / EXCERPT: Let k be an algebraically closed subfield of $\mathbf C$, let AM be the category of André motives over k, let $M\in\operatorname{AM}$, and let $r(M)$ be its re...

conjecture | open | MathDB 页面 | 原文线索

V2-009418 | MATHDB-344532

The conjugate mixed complete-homogeneous and power-sum basis conjecture

摘录 / EXCERPT: Conjugate mixed basis conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-009419 | MATHDB-344533

The compatibility-degree exchangeability conjecture

摘录 / EXCERPT: Let $\mathcal{A}(B)$ be a cluster algebra, let x,x' be cluster variables, and let $(x\parallel x')$ denote their compatibility degree. Compatibility-degree exchangeability conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-009420 | MATHDB-344540

Equality of characteristic polynomials for tropical and ordinary presentation matrices

摘录 / EXCERPT: Let $w\text{?}$ be a point of $b2X_{(t_0)}(b2R^{\mathrm{trop}})$, and let $b3$ be a path representing a mutation loop. For the sign sequence $b5_{b3}(w)$, consider the matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-009421 | MATHDB-344541

Completeness conjecture for finite-field matrix rings

摘录 / EXCERPT: Completeness conjecture. Every ring of the form

conjecture | open | MathDB 页面 | 原文线索

V2-009422 | MATHDB-344590

Converse to the finite-global-dimension criterion for the singularity functor

摘录 / EXCERPT: With notation as above, let R be a commutative ring, let A be the algebra occurring in the construction of the singularity functor, and let M be the corresponding maximal Coh...

conjecture | open | MathDB 页面 | 原文线索

V2-009423 | MATHDB-344591

Conjecture on derived contraction algebras determining stable endomorphism dgas

摘录 / EXCERPT: Let A be an algebra with idempotent e, let M be the corresponding maximal Cohen-Macaulay module, and write ... for the derived quotient. Assume that A/AeA is an Artinian loca...

conjecture | open | MathDB 页面 | 原文线索

V2-009424 | MATHDB-344607

Equality of real Schur representation sets for finite-dimensional Jacobian algebras

摘录 / EXCERPT: Let A be a finite-dimensional Jacobian algebra. Consider the three sets of real Schur representations: all real Schur representations; those constructed from the stated Schur-rep...

conjecture | open | MathDB 页面 | 原文线索

V2-009425 | MATHDB-344610

Conjecture on the gap between the smallest Salem numbers

摘录 / EXCERPT: Let θ_n^{-1} denote the relevant Perron-number sequence, and let a Salem number be a real algebraic integer greater than 1 whose conjugates satisfy the usual Salem-number...

conjecture | open | MathDB 页面 | 原文线索

V2-009426 | MATHDB-344634

Identification of the sheaf-theoretic and KZ functors

摘录 / EXCERPT: Let $\mathcal{C}_0$ and $\mathcal{B}_0$ be the source and target categories, let $\mathcal{M}:\mathcal{C}_0\longrightarrow\mathcal{B}_0$ be the sheaf-theoretic functor, and let Φ_*, KZ, and Ψ^* be...

conjecture | solved | MathDB 页面 | 原文线索

V2-009427 | MATHDB-344660

The generalized-action PI-exponent conjecture for H-nice Lie algebras

摘录 / EXCERPT: Generalized-action PI-exponent conjecture. If L is H-nice, then there exist constants $C\in\mathbb{R}$, $t\in\frac{\mathbb{Z}}{2}$, and $d\in\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-009428 | MATHDB-344676

Minimality conjecture for generators of HTC-identifiable graph model ideals

摘录 / EXCERPT: Let G be an HTC-identifiable graph, let $\mathcal{M}(G)$ denote its Gaussian structural equation model, and let $J(\mathcal{M}(G))$ be the corresponding model ideal. The generato...

conjecture | open | MathDB 页面 | 原文线索

V2-009429 | MATHDB-344732

L-function degree and slope stability conjecture for algebraic-geometric \(\mathbb{Z}_p\)-towers

摘录 / EXCERPT: Let $L(\chi_n,s)$ be the L-function of a primitive character of order p^n, let $\ell(n)$ be its degree, let $\ell_\alpha(n)$ count the slopes equal to a fixed rational number...

conjecture | open | MathDB 页面 | 原文线索

V2-009430 | MATHDB-344755

Integrality of central characters for FC sets

摘录 / EXCERPT: Integrality conjecture. $cech{\mathsf{C}}{alpha}$ is always an algebraic integer.

conjecture | open | MathDB 页面 | 原文线索

V2-009431 | MATHDB-344765

Expected dimension conjecture for Prym-Brill-Noether loci of generic Prym curves

摘录 / EXCERPT: Let C be a curve of genus g, and let ... be a generic Prym curve, meaning an unramified double cover. Let $V^r(C,f)$ denote the Prym-Brill-Noether locus of rank-r linear series...

conjecture | open | MathDB 页面 | 原文线索

V2-009432 | MATHDB-344798

The Hilbert-Kunz multiplicity and F-threshold inequality conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m},K)$ be a local ring. Let $I\subseteq R$ be an ideal generated by a part of a system of parameters $(f_1,\ldots,f_\ell)$, and set $\overline{R}=R/I$. Let J be...

conjecture | open | MathDB 页面 | 原文线索

V2-009433 | MATHDB-344801

Kontsevich's Weyl-Poisson automorphism conjecture

摘录 / EXCERPT: Let $W_{n,\mathbb{C}}$ be the n-th Weyl algebra over $\mathbb{C}$, and let $P_{n,\mathbb{C}}$ be the n-th commutative Poisson algebra. Kontsevich's conjecture. The automorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-009434 | MATHDB-344802

Kontsevich-Kanel-Belov rational Weyl-Poisson correspondence conjecture

摘录 / EXCERPT: Let $W_n(\mathbb{Q})$ be the n-th Weyl algebra over $\mathbb{Q}$ and let $P_n(\mathbb{Q})$ be the polynomial Poisson algebra in $2n$ variables with its canonical Poisson structur...

conjecture | open | MathDB 页面 | 原文线索

V2-009435 | MATHDB-344804

Poisson torus-action linearization conjecture

摘录 / EXCERPT: For $n\geq 1$, let $P_n=\mathbb{K}[z_1,\ldots,z_{2n}]$ be the commutative Poisson algebra with bracket ... An action is effective if its kernel is trivial, and regular if it is algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-009436 | MATHDB-344805

Białynicki-Birula linearization conjecture for free associative algebras

摘录 / EXCERPT: Let $F_n=\mathbb{K}\langle x_1,\ldots,x_n\rangle$ be the free associative algebra, and let $\mathbb{T}_{n-1}$ be the $(n-1)$-dimensional algebraic torus. Free-associative torus-act...

conjecture | open | MathDB 页面 | 原文线索

V2-009437 | MATHDB-344811

The canonical-module ideal conjecture for polarizations of Artinian monomial ideals

摘录 / EXCERPT: Let I be an Artinian monomial ideal, let J be a polarization of I, and let $\Delta(J)$ be the associated simplicial complex. Let $k[\Delta(J)]$ denote its Stanley-Reisner rin...

conjecture | open | MathDB 页面 | 原文线索

V2-009438 | MATHDB-344852

Hayashi's cycle-divisibility conjecture for finite indecomposable quandles

摘录 / EXCERPT: A finite rack X has profile $1^{m_0}l_1^{m_1}l_2^{m_2}\cdots l_k^{m_k}$ when its inner automorphism has m_0 cycles of length 1, m_1 cycles of length l_1, and so on. Hayas...

conjecture | open | MathDB 页面 | 原文线索

V2-009439 | MATHDB-344856

Rank conjecture for the image of the delta map

摘录 / EXCERPT: Let $\mathcal{A}_k^{\rm even}$ be the set of even-weight admissible compositions used in the paper, let δ be the map defined there, and let ${\rm Im}(\delta)$ denote its ima...

conjecture | open | MathDB 页面 | 原文线索

V2-009440 | MATHDB-344860

Automorphic-Galois comparison conjecture for \a3-invariants

摘录 / EXCERPT: Let $\rho_{\pi,\mathfrak{p}}\colon\operatorname{Gal}(\overline{\mathbb{Q}}_p/F_{\mathfrak{p}})\to\widehat{G'}(\Omega)$ be the local Galois representation attached to π, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009441 | MATHDB-344887

Foil diagrams attain the upper bound on figure-eight descendants

摘录 / EXCERPT: A free foil diagram with n crossings is a free knot diagram, and an algebraic free diagram is a free knot diagram arising from an algebraic tangle. Foil figure-eight upper-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-009442 | MATHDB-344907

GL-equivalence conjecture for sandpile groups of Cayley graphs

摘录 / EXCERPT: Let two Cayley graphs have generator multiplicities, and let GL-equivalence mean equivalence under the natural action of the relevant general linear group on the generators. GL-e...

conjecture | open | MathDB 页面 | 原文线索

V2-009443 | MATHDB-344936

The Grothendieck-Serre conjecture for reductive groups

摘录 / EXCERPT: Let R be a regular local ring containing a field k, and let G be a smooth reductive algebraic group over k. Write K for the field of fractions of R. Grothendieck-Serre...

conjecture | open | MathDB 页面 | 原文线索

V2-009444 | MATHDB-344937

The versal torsor conjecture for reductive groups over algebraically closed fields

摘录 / EXCERPT: Let k be an algebraically closed field, let G be a connected reductive linear algebraic group over k, let R be a regular local ring containing k, and let ... be a G-torso...

conjecture | open | MathDB 页面 | 原文线索

V2-009445 | MATHDB-344995

Koenigsmann's ample-field conjecture

摘录 / EXCERPT: Let F be an infinite field with finitely generated absolute Galois group. Koenigsmann's conjecture. Then F is ample. This conjecture concerns which fields are ample and is pres...

conjecture | open | MathDB 页面 | 原文线索

V2-009446 | MATHDB-345040

Dominance conjecture for nonzero Lie algebra word maps

摘录 / EXCERPT: Dominance conjecture. Then φ_{w} is dominant. This extends the stated Lie-algebra analogue of Borel's theorem beyond the explicitly assumed $\mathfrak{sl}_{2}$ condition; t...

conjecture | open | MathDB 页面 | 原文线索

V2-009447 | MATHDB-345050

Extension of the Rankin-Cohen algebra proposition to arbitrary graded weights

摘录 / EXCERPT: Generalized Rankin-Cohen proposition. The proposition concerning Rankin-Cohen algebras holds for such a graded algebra; it is not necessary that $M_0={\sf k}.1$. The paper identifi...

conjecture | open | MathDB 页面 | 原文线索

V2-009448 | MATHDB-345054

Simplicity criterion for indecomposable exact algebras

摘录 / EXCERPT: Let $\mathcal{C}$ be the tensor category in the source, and let $A\in\mathcal{C}$ be an indecomposable algebra object. An algebra is called exact when it satisfies the exactness co...

conjecture | open | MathDB 页面 | 原文线索

V2-009449 | MATHDB-345094

Abelian arboreal extension classification for polynomials

摘录 / EXCERPT: Let K be a number field, let $\phi(x)\in K[x]$ be a polynomial of degree $d\geq2$, and let $\alpha\in K$ be non-exceptional for ϕ. Write $K_\infty(\phi,\alpha)$ for the fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-009450 | MATHDB-345163

n-ary coherence conjecture for polyadic tensor categories

摘录 / EXCERPT: n-ary coherence conjecture. If the n-ary associator $\scalebox{1.15}{\textsf{\upshape A}}^{\left(2n-1\right)\otimes}$ satisfies such n-ary coherence conditions that this isom...

conjecture | open | MathDB 页面 | 原文线索

V2-009451 | MATHDB-345164

Arity-reducibility conjecture for polyadic tensor categories with units

摘录 / EXCERPT: Let a polyadic (n-ary) tensor category be equipped with a unit object and unitors. Arity-reducibility conjecture. It should be arity-reducible to a binary category, with its n-...

conjecture | open | MathDB 页面 | 原文线索

V2-009452 | MATHDB-345165

Existence conjecture for arity-nonreducible polyadic groupal categories

摘录 / EXCERPT: Let a polyadic nonunital non-strict groupal category be a polyadic non-strict semigroupal category equipped with a querfunctor and quertors satisfying the groupal coherence conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-009453 | MATHDB-345170

Fomin-Kirillov's infinite-dimensionality conjecture

摘录 / EXCERPT: Let $\EuScript{E}_m$ be the Fomin-Kirillov algebra over a field $\mathbf{k}$ associated with the symmetric group $\mathbb{S}_m$. Fomin-Kirillov's conjecture. The algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-009454 | MATHDB-345171

Infinite-dimensionality conjecture for the Nichols-Woronowicz algebra of the symmetric group

摘录 / EXCERPT: Let $\mathbb{S}_m$ be the symmetric group, and let $\EuScript{B}_{\mathbb{S}_m}$ be its Nichols-Woronowicz algebra model for Schubert calculus. Infinite-dimensionality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-009455 | MATHDB-345172

Integral conjecture for finite-dimensional Nichols-Woronowicz algebras

摘录 / EXCERPT: Integral conjecture. The element

conjecture | open | MathDB 页面 | 原文线索

V2-009456 | MATHDB-345180

The residual-intersection conjecture for powers of ideals

摘录 / EXCERPT: Residual-intersection conjecture. If I is unmixed and $J:I$ is a geometric $(\ell-1)$-residual intersection of I, then, for all $\rho\gg0$,

conjecture | open | MathDB 页面 | 原文线索

V2-009457 | MATHDB-345181

The self-duality conjecture for powers of canonical modules of analytic spread one

摘录 / EXCERPT: Canonical-module self-duality conjecture. If ω_R has analytic spread 1, then some power of ω_R, regarded as a fractional ideal, is ω_R-self-dual up to a shif...

conjecture | open | MathDB 页面 | 原文线索

V2-009458 | MATHDB-345203

Exponential decay conjecture for incomplete selected inversion errors

摘录 / EXCERPT: Let H be the matrix and let E denote the error matrix in the notation of the paper's selected-inversion setup. Write $\mathrm{level}(i,j)$ for the level of fill between indices...

conjecture | open | MathDB 页面 | 原文线索

V2-009459 | MATHDB-345204

Exponential decay conjecture for incomplete inversion errors

摘录 / EXCERPT: Let A be the matrix, let B be the output of incomplete selected inversion, and let F denote the error matrix in the notation of the paper's selected-inversion setup. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009460 | MATHDB-345218

LI conjecture on Artin L-function central zeros

摘录 / EXCERPT: Let $L/\mathbb Q$ be a number-field extension and let $L_0/\mathbb Q$ be its Galois closure. Let χ_0 denote the trivial character, and classify irreducible characters of...

conjecture | open | MathDB 页面 | 原文线索

V2-009461 | MATHDB-345246

The correspondence between traversals and comonad algebras

摘录 / EXCERPT: Let $\mathbf{S} \to [\mathbf{C},\mathbf{C}]$ be the monoidal action of applicative functors such that the higher-order functor … is a comonad. Consider families of morphisms … sati...

conjecture | open | MathDB 页面 | 原文线索

V2-009462 | MATHDB-345290

Tran's conjecture on roots of polynomial recurrence sequences

摘录 / EXCERPT: Tran's conjecture. Every zero of every polynomial $P_n(z)$ lies on $\mathcal C$, and the zeros become dense in $\mathcal C$ as $n\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-009463 | MATHDB-345310

Freeness conjecture for pattern Hopf algebras of associative presheaves

摘录 / EXCERPT: Let h be an associative presheaf, and let $\mathcal{A}(h)$ denote its pattern Hopf algebra. Freeness conjecture. The pattern Hopf algebra $\mathcal{A}(h)$ is free. This would est...

conjecture | open | MathDB 页面 | 原文线索

V2-009464 | MATHDB-345318

Multi-trek separation conjecture for higher-order moment tensors

摘录 / EXCERPT: Let G be a directed acyclic graph with node set V, and let $k>3$. For subsets $S_1,\ldots,S_k\subseteq V$ with … write $\mathcal{N}^{(k)}_{S_1,\ldots,S_k}$ for the subtensor of...

conjecture | open | MathDB 页面 | 原文线索

V2-009465 | MATHDB-345433

Freudenburg's conjugacy conjecture for maximal Lie subalgebras

摘录 / EXCERPT: Freudenburg's conjugacy conjecture. Every maximal Lie subalgebra $\mathcal{A}\subset\operatorname{LND}(\mathbb{K}[x_1,\ldots,x_n])$ is conjugate to $\mathfrak{T}$. This is the uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-009466 | MATHDB-345438

The conjecture on moduli spaces of semiorthogonal decompositions for families of dg categories

摘录 / EXCERPT: Moduli-space conjecture. There exists an étale algebraic space $\operatorname{\mathsf{SOD}}_{\mathcal{D}/U}$ over U with a functorial bijection

conjecture | open | MathDB 页面 | 原文线索

V2-009467 | MATHDB-345450

Holomorphic reflexivity of exponential analytic functionals

摘录 / EXCERPT: Exponential analytic functional conjecture. The algebra

conjecture | open | MathDB 页面 | 原文线索

V2-009468 | MATHDB-345490

Rubin-type main conjecture for supersingular Iwasawa modules

摘录 / EXCERPT: Let F_f be a height 2 and self-dual formal group, and let $\mathbb{U}'$ and $\mathbb{X}'$ be the torsion $\Lambda(G_{\infty},\mathcal{O}_{L_p})$-modules occurring in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-009469 | MATHDB-345494

Purely Wild Inertia Conjecture

摘录 / EXCERPT: Purely Wild Inertia Conjecture. The subgroup P occurs as the inertia group at a point above ∞ in a connected G-Galois cover of $\mathbb{P}^1$ branched only at ∞ if...

conjecture | open | MathDB 页面 | 原文线索

V2-009470 | MATHDB-345495

Generalized Inertia Conjecture

摘录 / EXCERPT: Generalized Inertia Conjecture. There is a connected G-Galois cover of $\mathbb{P}^1$ étale away from B such that I_i occurs as an inertia group above x_i for...

conjecture | open | MathDB 页面 | 原文线索

V2-009471 | MATHDB-345496

Generalized Purely Wild Inertia Conjecture

摘录 / EXCERPT: Generalized Purely Wild Inertia Conjecture. There is a connected G-Galois cover of $\mathbb{P}^1$ étale away from B such that P_i occurs as an inertia group above x_i for...

conjecture | open | MathDB 页面 | 原文线索

V2-009472 | MATHDB-345524

Formality conjecture for the cyclic bar complex of the Soergel Hecke category

摘录 / EXCERPT: Let W be a Coxeter group with realization V over $\mathbb{C}$, and let $\mathcal{C}(W)$ denote the corresponding Soergel Hecke category. Let Tr be its universa...

conjecture | open | MathDB 页面 | 原文线索

V2-009473 | MATHDB-345534

Improved lower-bound conjecture for Hilbert-Kunz multiplicities

摘录 / EXCERPT: Let $(A,\mathfrak{m}, k)$ be a formally unmixed non-regular local ring of dimension $d \geq 1$ with algebraically closed residue field. Let $m \geq 1$ be an integer, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009474 | MATHDB-345585

Brick occurrence characterization of simple objects

摘录 / EXCERPT: Let W be a Weyl group, let $w\in W$, and let $\mathcal{F}(w)$ be its torsion-free class. A brick is an object with division endomorphism ring. Brick occurrence characterization....

conjecture | open | MathDB 页面 | 原文线索

V2-009475 | MATHDB-345587

Positive characteristic form of Buzzard-Gee's conjecture on Galois representations attached to Hecke eigenforms

摘录 / EXCERPT: Let G be the reductive group and L its reflex field from the setup, let K be the level, let p be the coefficient characteristic, and let $\Sigma_{K,p}$ contain the places w...

conjecture | open | MathDB 页面 | 原文线索

V2-009476 | MATHDB-345602

ACCP conjecture for cyclic Puiseux monoids

摘录 / EXCERPT: ACCP conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009477 | MATHDB-345628

Angiono-Azevedo-Heckenberger conjecture on finite root systems

摘录 / EXCERPT: Angiono-Azevedo-Heckenberger conjecture. If $\mathscr{B}_{\mathfrak{q}}$ has finite Gelfand-Kirillov dimension, then its root system is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-009478 | MATHDB-345828

Derived contraction algebra conjecture for stable endomorphism dgas

摘录 / EXCERPT: Let R be a Gorenstein singularity, let M be a maximal Cohen-Macaulay R-module, and set $A:=\operatorname{End}_R(R\oplus M)$. Let $e\in A$ be the idempotent corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-009479 | MATHDB-345834

Chari's conjecture for truncated local Weyl modules

摘录 / EXCERPT: Let P^+ be the dominant integral weights of $\mathfrak{sl}_n$, let $\mathcal{P}_{\lambda}$ be the set of decompositions of $\lambda in P^+$ into two dominant weights, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009480 | MATHDB-345878

Algebraic formality conjecture for Poincaré duality algebras

摘录 / EXCERPT: Let V be a differential Poincaré duality algebra with $H^0(V)=\operatorname{Span}\{1\}$ and $H^1(V)=0$, and suppose that V is formal as a differential graded algebra. A Poinc...

conjecture | open | MathDB 页面 | 原文线索

V2-009481 | MATHDB-345914

S-duality conjecture for SU(r) Vafa-Witten invariants on surfaces

摘录 / EXCERPT: S-duality conjecture for Vafa-Witten invariants. The two partition functions satisfy the S-duality transformation formula

conjecture | open | MathDB 页面 | 原文线索

V2-009482 | MATHDB-345932

Benson-Etingof-Ostrik conjecture on fiber functors to ${\rm Ver}_{p^\infty}$

摘录 / EXCERPT: Let $\mathbf{k}$ be an algebraically closed field of characteristic $p>0$. A symmetric tensor category of moderate growth is a symmetric tensor category over $\mathbf{k}$ satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-009483 | MATHDB-345940

Ross-symbol form of the weak Beilinson conjecture for hypergeometric elliptic curves

摘录 / EXCERPT: Let $\alpha\in{\mathbb Q}\setminus\{0,1\}$ and let X_α be the elliptic curve in the preceding construction. Let $\xi_{\mathrm{Ross}}\in K_2(X_\alpha)^{(2)}$ be the higher Ro...

conjecture | open | MathDB 页面 | 原文线索

V2-009484 | MATHDB-345941

Weak Beilinson conjecture for the transcendental motive of a hypergeometric K3 surface

摘录 / EXCERPT: Let $\alpha\in{\mathbb R}\setminus\{0,1\}$, let X_α be the associated smooth compactification, and let $\Delta_\alpha^+=\frac12(F_\infty+1)\Delta_\alpha$ be the real-Frobeni...

conjecture | open | MathDB 页面 | 原文线索

V2-009485 | MATHDB-345958

Universal coequalizer conjecture for nearby cycles of automorphic categories

摘录 / EXCERPT: Let $C\rightsquigarrow C_0$ be a degeneration of a smooth complex projective curve to a nodal curve, with marked points $S\rightsquigarrow S_0$ disjoint from the nodes. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009486 | MATHDB-345963

Chen's polymorphism dichotomy conjecture for the surjective constraint satisfaction problem

摘录 / EXCERPT: Let A be a finite domain, let Γ be a constraint language on A, and let $\operatorname{SCSP}(\Gamma)$ denote the surjective constraint satisfaction problem over Γ....

conjecture | open | MathDB 页面 | 原文线索

V2-009487 | MATHDB-345988

The class-number-one zetalike multizeta-value conjecture

摘录 / EXCERPT: The class-number-one zetalike conjecture. The multizeta values

conjecture | open | MathDB 页面 | 原文线索

V2-009488 | MATHDB-346071

Linear-independence conjecture for the matrix associated with Paley-graph clique bounds

摘录 / EXCERPT: In the setting of the preceding discussion, let M be the set of integers m satisfying ... and let $A_{m,n}$ be the matrix defined in the paper, with $L(n)$ its associated index s...

conjecture | open | MathDB 页面 | 原文线索

V2-009489 | MATHDB-346113

Phillips' radius of comparison conjecture for tensor products with C(X)

摘录 / EXCERPT: Let A be a stably finite unital C-algebra and let X be a compact metric space. The radius of comparison, denoted by ${\operatorname{rc}}$, measures comparison in the Cuntz semi...

conjecture | open | MathDB 页面 | 原文线索

V2-009490 | MATHDB-346163

Dimension conjecture for the ordinary deformation ring

摘录 / EXCERPT: Let R_ρ^{ord} be the ordinary deformation ring attached to the Galois representation ρ and let Λ_E be the ordinary weight algebra, with...

conjecture | open | MathDB 页面 | 原文线索

V2-009491 | MATHDB-346197

Sturmfels-Uhler polynomiality conjecture for the maximum likelihood degree

摘录 / EXCERPT: Let V be a vector space, let $\boldsymbol{S}^2V$ denote the space of symmetric matrices on V, and let $c6(n,a)$ be the maximum likelihood degree associated with symmetric matri...

conjecture | open | MathDB 页面 | 原文线索

V2-009492 | MATHDB-346241

The projective homogeneous space Hasse principle over p-adic curve function fields

摘录 / EXCERPT: Projective homogeneous space Hasse principle. If $Y(F_\nu)\neq\varnothing$ for every $\nu\in\Omega_F$, then $Y(F)\neq\varnothing$.

conjecture | open | MathDB 页面 | 原文线索

V2-009493 | MATHDB-346242

The principal homogeneous space Hasse principle over p-adic curve function fields

摘录 / EXCERPT: Principal homogeneous space Hasse principle. If $Y(F_\nu)\neq\varnothing$ for every $\nu\in\Omega_F$, then $Y(F)\neq\varnothing$.

conjecture | open | MathDB 页面 | 原文线索

V2-009494 | MATHDB-346318

The strong Artin conjecture

摘录 / EXCERPT: Let L/k be a Galois extension of number fields with ${\mathrm {Gal}}(L/k) \cong G$. Let ρ be a complex representation of G of dimension m. Strong Artin conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-009495 | MATHDB-346358

Alternating-sum decomposition conjecture for the modules $V\!h_k(n)$

摘录 / EXCERPT: Let $Lie^{(2)}$ be the symmetric-function species used in the plethysm below, and let $V\!h_k(n)$ be the S_n-module whose Frobenius characteristic is the degree-n term of...

conjecture | open | MathDB 页面 | 原文线索

V2-009496 | MATHDB-346408

Regular-sequence conjecture for the generic Hankel gradient ideal

摘录 / EXCERPT: Let $\mathcal H_m$ be the fully generic $m\times m$ Hankel matrix, let $J=J_m[0]$ be its gradient ideal, and let $x_{m+3},\ldots,x_{2m-1}$ be the indicated variables. Regular-seque...

conjecture | open | MathDB 页面 | 原文线索

V2-009497 | MATHDB-346413

The good-differential conjecture for the diagrammatic Hecke category

摘录 / EXCERPT: Let S_n be the symmetric group, and equip its diagrammatic Hecke category with either of the good differentials d or $\bar{d}$. After specialization to characteristic p, this...

conjecture | open | MathDB 页面 | 原文线索

V2-009498 | MATHDB-346474

The dg-quiver Schur algebra quasi-isomorphism conjecture

摘录 / EXCERPT: Let ${}_{dg}Q_b^{\underline{N}},d_N)$ be the dg-quiver Schur algebra defined as the dg-endomorphism algebra of the direct sum of the dg-enhanced modules...

conjecture | open | MathDB 页面 | 原文线索

V2-009499 | MATHDB-346475

The dg-enhanced KLRW-to-quiver Schur algebra isomorphism conjecture

摘录 / EXCERPT: Let $T_b^{\lambda,\underline{N}}$ be the dg-enhanced KLRW algebra and let ${}_{dg}Q_b^{\underline{N}},d_N)$ be the dg-quiver Schur algebra, with the two maps of graded dg-algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-009500 | MATHDB-346489

Goldfeld's minimalist conjecture for elliptic curves

摘录 / EXCERPT: Let E range over elliptic curves over $\mathbb{Q}$, ordered by naive height, and let rank mean the algebraic or analytic rank as appropriate. Goldfeld's minimalist conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-009501 | MATHDB-346490

Park-Poonen-Voight-Wood conjecture on the distribution of elliptic-curve ranks

摘录 / EXCERPT: Let elliptic curves over $\mathbb{Q}$ be ordered by height, and let algebraic rank denote Mordell-Weil rank. Park-Poonen-Voight-Wood conjecture. (i) The proportion of elliptic curv...

conjecture | open | MathDB 页面 | 原文线索

V2-009502 | MATHDB-346494

Equivariant multiplicity conjecture for flag minors

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra of finite simply-laced type, let N be the maximal unipotent subgroup associated with $\mathfrak{g}$, let Φ_{+} be its set of positive roo...

conjecture | open | MathDB 页面 | 原文线索

V2-009503 | MATHDB-346495

Characterization of flag minors by equivariant multiplicities

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra of finite simply-laced type, let W be its Weyl group, let w_0 be the longest element of W, let N be the associated maximal unipotent sub...

conjecture | open | MathDB 页面 | 原文线索

V2-009504 | MATHDB-346499

The asymptotic gonality-point equality conjecture for binary curves

摘录 / EXCERPT: Gonality-point equality conjecture. Fix $\gamma\geq 2$. For g sufficiently large,

conjecture | open | MathDB 页面 | 原文线索

V2-009505 | MATHDB-346504

Globality conjecture for dual pairs of Malcev algebras and Moufang loops

摘录 / EXCERPT: Globality conjecture. Every dual pair $(M,\mathcal{L})$ of Malcev algebras and their corresponding Moufang loops is global.

conjecture | open | MathDB 页面 | 原文线索

V2-009506 | MATHDB-346525

Polynomial limit of the Riemann theta function for rational node-free degenerations

摘录 / EXCERPT: Polynomial-limit conjecture. In the limit, the Riemann theta function converges to a polynomial in ${\bf z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009507 | MATHDB-346533

Strong Fröberg conjecture for Hilbert series of general ideals

摘录 / EXCERPT: Let $\mathcal{S}$ be the polynomial ring in $n+1$ variables, and let $f_1,\ldots,f_s$ be general forms of degrees $d_1,\ldots,d_s$. Set ... For a power series $P=\sum_{i\geq0}a_it^i$...

conjecture | open | MathDB 页面 | 原文线索

V2-009508 | MATHDB-346543

Écalle-Chapoton conjecture on L-classes in planar binary trees

摘录 / EXCERPT: Écalle-Chapoton conjecture. Any Lie element of ${\bf PBT}$ is a linear combination of L-classes.

conjecture | open | MathDB 页面 | 原文线索

V2-009509 | MATHDB-346547

The Erdős-Burgess constant finiteness conjecture for commutative rings

摘录 / EXCERPT: Let R be a commutative unitary ring. Write $J(R)$ for its Jacobson radical, and let $\textsc{I}(\mathcal{S}_R)$ denote the Erdős-Burgess constant of the multiplicative semigroup...

conjecture | open | MathDB 页面 | 原文线索

V2-009510 | MATHDB-346565

Algebraic form of the cyclotomic numerical semigroup conjecture

摘录 / EXCERPT: Let $\operatorname{\mathbbm{k}}$ be an algebraically closed field, and let R be a graded domain of Krull dimension one over $\operatorname{\mathbbm{k}}$. Call R cyclotomic when...

conjecture | open | MathDB 页面 | 原文线索

V2-009511 | MATHDB-346594

The unbounded class-group rank conjecture for fixed-degree number fields

摘录 / EXCERPT: Let $d>1$ and $n>1$ be two integers. For a number field K, write $\operatorname{Cl}(K)$ for its ideal class group and let $\operatorname{rk}_n M$ denote the maximal integer r s...

conjecture | open | MathDB 页面 | 原文线索

V2-009512 | MATHDB-346604

The inverse-generator conjecture for finite-type cluster configuration spaces

摘录 / EXCERPT: Inverse-generator conjecture. If D is not of type A_1, then the ring

conjecture | open | MathDB 页面 | 原文线索

V2-009513 | MATHDB-346621

Thue-Morse continued-fraction algebraicity conjecture

摘录 / EXCERPT: Let a,b be two distinct elements of $\mathbb{F}_2[z]\backslash\mathbb{F}_2$, and let $\mathbf{u}(z)$ be the (a,b)-Thue-Morse sequence. Thue-Morse continued-fraction algebraicit...

conjecture | open | MathDB 页面 | 原文线索

V2-009514 | MATHDB-346672

Breit-Wigner series convergence conjecture for super-exponentially decreasing potentials

摘录 / EXCERPT: Breit-Wigner convergence conjecture. The series $B(V)$ converges if and only if V is compactly supported.

conjecture | open | MathDB 页面 | 原文线索

V2-009515 | MATHDB-346674

Breit-Wigner divergence alternative under Froese's hypotheses

摘录 / EXCERPT: Let V be a super-exponentially decreasing potential, let $\widehat V$ have completely regular growth, and let $B(V)$ be its Breit-Wigner series. Assume that V is not compactly...

conjecture | open | MathDB 页面 | 原文线索

V2-009516 | MATHDB-346774

The derived Langlands conjecture for the principal Bernstein block

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$ or of $\mathbb{F}_p((t))$, let $\mathbb{G}$ be a split reductive group over F, and set $G=\mathbb{G}(F)$. Let C be a characteris...

conjecture | open | MathDB 页面 | 原文线索

V2-009517 | MATHDB-346779

Strong Molino conjecture for singular Riemannian foliations

摘录 / EXCERPT: Let $\mathcal{F}$ be a complete singular Riemannian foliation, and let $\mathscr{C}_{\mathcal{F}}$ be the locally constant sheaf of Lie algebras associated with its structural theo...

conjecture | open | MathDB 页面 | 原文线索

V2-009518 | MATHDB-346833

The minor ideal generation conjecture for the matrix A

摘录 / EXCERPT: Minor ideal generation conjecture. For all $e\geq 0$, the ideal $\operatorname{Min}_e(A)$ is generated by $M(e,r-1-t_2+e)$.

conjecture | open | MathDB 页面 | 原文线索

V2-009519 | MATHDB-346834

The weak Fitting-ideal generation conjecture

摘录 / EXCERPT: Weak Fitting-ideal generation conjecture. The union of the sets

conjecture | open | MathDB 页面 | 原文线索

V2-009520 | MATHDB-346859

Monotonicity of maximum likelihood degree under facial submatrices

摘录 / EXCERPT: Let $A \in \mathbb{Z}^{n \times r}$ and $\overline{A} \in \mathbb{Z}^{n \times s}$ be a facial submatrix of A. The facial-submatrix maximum likelihood degree conjecture. The maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-009521 | MATHDB-346889

The type II projection-lattice and locally measurable-ring conjecture

摘录 / EXCERPT: Let M and N be von Neumann algebras of type II. Write $\mathcal{P}(M)$ and $\mathcal{P}(N)$ for their projection lattices, and let $LS(M)$ and $LS(N)$ denote their algebras of...

conjecture | open | MathDB 页面 | 原文线索

V2-009522 | MATHDB-346926

Fraser-Sherman-Bennett conjecture on source-target quasi-cluster equivalence

摘录 / EXCERPT: Let G be a reduced plabic graph, and let Σ_G^S and Σ_G^T denote respectively its source and target seeds. A quasi-cluster transformation is the relevant equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-009523 | MATHDB-346938

Invariance of the SU(r) partition function under tensoring

摘录 / EXCERPT: Tensoring invariance conjecture. For any such (S,H), c_1, and $b3$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-009524 | MATHDB-346988

Eventual linearity of regularity for FI- and OI-modules

摘录 / EXCERPT: Let c be a positive integer. Let $\mathbf{M}$ be a finitely generated graded OI-module over $\mathbf{P}=({\mathbf{X}}^{\operatorname{OI},1})^{\otimes c}$, or a f...

conjecture | open | MathDB 页面 | 原文线索

V2-009525 | MATHDB-346989

Eventual linearity of projective dimension for FI- and OI-modules

摘录 / EXCERPT: Let c be a positive integer. Let $\mathbf{M}$ be a finitely generated OI-module over $\mathbf{P}=({\mathbf{X}}^{\operatorname{OI},1})^{\otimes c}$, or a finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-009526 | MATHDB-347019

The comparison conjecture for canonical and bidual canonical degrees

摘录 / EXCERPT: Comparison conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-009527 | MATHDB-347020

The hyperplane-section inequality for bidual canonical degree

摘录 / EXCERPT: Hyperplane-section conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-009528 | MATHDB-347021

The codimension-three equigenerated monomial almost complete intersection SLP classification conjecture

摘录 / EXCERPT: SLP classification conjecture. The equigenerated monomial almost complete intersections of codimension three with the SLP are precisely those given by

conjecture | open | MathDB 页面 | 原文线索

V2-009529 | MATHDB-347022

The WLP failure criterion for codimension-three equigenerated monomial almost complete intersections

摘录 / EXCERPT: Let $a\geq b\geq c$, set $d=a+b+c$, and define ... The weak Lefschetz failure conjecture. The algebra $R_{a,b,c}$ fails the weak Lefschetz property if and only if both of the followi...

conjecture | open | MathDB 页面 | 原文线索

V2-009530 | MATHDB-347026

The generic tensor-rank conjecture for three-dimensional quaternionic factors

摘录 / EXCERPT: Let T be a tensor in ... The generic tensor-rank conjecture. The generic rank of T is 3. The preceding discussion suggests this quaternionic analogue of the complex generic-ran...

conjecture | open | MathDB 页面 | 原文线索

V2-009531 | MATHDB-347079

Nonvanishing determinant conjecture for the Atrahasis code construction

摘录 / EXCERPT: Let n, k, d, and t be integers such that $n-1\geq d\geq k\geq 2$ and $d\leq t(d-k+1)$. Let ... The main theorem establishes that, when $\alpha\leq 3003$, there exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-009532 | MATHDB-347106

Indecomposability of Newton polytopes of F-polynomials as shard polytopes

摘录 / EXCERPT: Let a finite-type cluster algebra be given, and let an F-polynomial be one of its cluster-algebra F-polynomials. Its Newton polytope is the convex hull of the exponent vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-009533 | MATHDB-347144

The property R_{∞} conjecture for finitely presented S-arithmetic subgroups of Borel groups

摘录 / EXCERPT: Let $\mathbf{B}$ be a Borel subgroup of a split, connected, reductive, non-commutative linear algebraic group $\mathbf{G}$, all defined over a global field. An S-arithmetic subgr...

conjecture | open | MathDB 页面 | 原文线索

V2-009534 | MATHDB-347158

Main Borcherds-algebra conjecture for BPS Lie algebras

摘录 / EXCERPT: Fix a Serre subcategory $\mathcal{S}$, a stability condition $\upzeta$, and a slope $\uptheta$. Let $\operatorname{\mathcal{S}ta}^{G,\upzeta}_{\Pi_Q,\uptheta}$ be the direct sum of...

conjecture | open | MathDB 页面 | 原文线索

V2-009535 | MATHDB-347161

Hiskens's impasse-surface avoidance conjecture for static-load power systems

摘录 / EXCERPT: Suppose Assumption 1 and Assumption 2 hold. Consider a differential-algebraic equation system with all loads purely static. Let $al V_t$ denote the generator buses and $al V_L$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-009536 | MATHDB-347179

The density conjecture for the exceptional quivers X6 and X7

摘录 / EXCERPT: Density conjecture for X_6 and X_7. The quiver X_6 is cluster-$\mathbf{g}$-dense, and the quiver X_7 is half cluster-$\mathbf{g}$-dense.

conjecture | open | MathDB 页面 | 原文线索

V2-009537 | MATHDB-347202

The Breuil-Mézard conjecture for potentially semistable deformation rings

摘录 / EXCERPT: Let $K/\mathbb{Q}_p$ be a finite extension with ring of integers $\mathcal{O}_K$ and residue field k, let G_K be its absolute Galois group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009538 | MATHDB-347205

Conjecture on Hironaka decompositions of Lorentz-invariant algebras for n=d+1

摘录 / EXCERPT: Hironaka decomposition conjecture. The Hironaka decomposition is

conjecture | open | MathDB 页面 | 原文线索

V2-009539 | MATHDB-347265

Langlands' functoriality conjecture for Galois representations

摘录 / EXCERPT: Let E/F be a finite extension, let $\rho:\operatorname{Gal}(E/F)\rightarrow GL_n(\mathbb{C})$ be a representation, and let $L(\rho,s)$ denote its Artin L-function. Langlands' f...

conjecture | open | MathDB 页面 | 原文线索

V2-009540 | MATHDB-347276

The affine twisted zastava Ext-algebra conjecture

摘录 / EXCERPT: Let $\mathfrak g_{\operatorname{aff},\varsigma}$ be the affine Lie algebra associated with $\mathfrak g',\sigma$, and ζ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009541 | MATHDB-347279

Regularity bound by the number of maximal cliques for binomial edge ideals

摘录 / EXCERPT: The clique-count regularity conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-009542 | MATHDB-347280

Bank's hypertranscendency conjecture for perturbations of the gamma function

摘录 / EXCERPT: Let h be an entire function, let Γ denote the Euler gamma function, and let $\mathcal{M}_0$ be the field of meromorphic functions y satisfying $T(r,y)=o(r)$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-009543 | MATHDB-347316

The parabolic symmetrising trace conjecture for complex reflection groups

摘录 / EXCERPT: Let W be a complex reflection group, let I be an intersection of reflecting hyperplanes of W, and let W_I be the corresponding parabolic subgroup. Let $d4d5(W)$ be the gene...

conjecture | open | MathDB 页面 | 原文线索

V2-009544 | MATHDB-347317

Broué-Malle-Michel's symmetrising trace conjecture for generic Hecke algebras

摘录 / EXCERPT: Let W be a complex reflection group, let $R(W)$ be its ring of definition, and let $\mathcal{H}(W)$ be its generic Hecke algebra. Let $B(W)$ be the braid group, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009545 | MATHDB-347364

Nonfinite basis conjecture for polynomial identities of matrix algebras in characteristic two

摘录 / EXCERPT: Let K be an infinite field of characteristic 2, and let $M_2(K)$ denote the algebra of 2×2 matrices. A polynomial identity basis is a finite set of polynomial identitie...

conjecture | open | MathDB 页面 | 原文线索

V2-009546 | MATHDB-347365

Nonfinite basis conjecture for the three-dimensional cross-product Lie algebra

摘录 / EXCERPT: Let K be an infinite field of characteristic 2, and let K^3 be the three-dimensional simple Lie algebra equipped with the cross product. Nonfinite basis conjecture. The polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-009547 | MATHDB-347366

Formanek's minimal-degree conjecture for central polynomials

摘录 / EXCERPT: Let K be a field of characteristic 0, and let $M_d(K)$ denote the algebra of $d\times d$ matrices. A central polynomial for $M_d(K)$ is a polynomial that takes central values o...

conjecture | open | MathDB 页面 | 原文线索

V2-009548 | MATHDB-347378

Characteristic-dependence conjecture for syzygies of Veronese embeddings

摘录 / EXCERPT: Let $r\geq 7$, and let d be sufficiently large. Consider the d-uple embedding of $\mathbb P^r$ and its Betti table. Characteristic-dependence conjecture. For any $d\gg 0$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-009549 | MATHDB-347390

Canonical truncation conjecture for parabolic contractions in type C

摘录 / EXCERPT: Let $\operatorname{\mathfrak{q}}^C$ be a parabolic contraction in type C, and let $\operatorname{\mathfrak{q}}^C_\Lambda$ denote its canonical truncation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009550 | MATHDB-347391

Factorization conjecture for semi-invariants in type C

摘录 / EXCERPT: Factorization conjecture. There is a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-009551 | MATHDB-347392

Polynomiality criterion for semi-invariants in type C

摘录 / EXCERPT: Let $\operatorname{\mathfrak{q}}^C$ be a standard parabolic contraction in type C, and let $\operatorname{Sy}(\operatorname{\mathfrak{q}}^C)$ denote its algebra of semi-invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-009552 | MATHDB-347393

Weight-semigroup group conjecture for semi-invariants of parabolic contractions

摘录 / EXCERPT: Let $\operatorname{\mathfrak{q}}$ be a parabolic contraction, and let Λ be the semigroup of weights of its semi-invariant algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-009553 | MATHDB-347414

Algebraic freeness conjecture for molecules with spherical boundary

摘录 / EXCERPT: Let $\mathcal{S}$ be the class of molecules with spherical boundary. Algebraic freeness conjecture. The class $\mathcal{S}$ of molecules with spherical boundary is algebraically fr...

conjecture | open | MathDB 页面 | 原文线索

V2-009554 | MATHDB-347442

Bethe-vector eigenvalue conjecture for transfer matrices of the super Yangian

摘录 / EXCERPT: Let M be a finite-dimensional irreducible $\mathrm{Y}(\mathfrak{gl}_{m|n})$-module of highest ℓ-weight $\boldsymbol\zeta=(\zeta_i(u))_{i\in\bar I}^s$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009555 | MATHDB-347462

The multiplicity-sequence criterion for integral dependence

摘录 / EXCERPT: Let $I\subset J$ be arbitrary ideals in an equidimensional and universally catenary Noetherian local ring. The multiplicity sequence is the sequence of normalized leading coefficie...

conjecture | open | MathDB 页面 | 原文线索

V2-009556 | MATHDB-347463

The isomorphism conjecture for surface quiver groups and extended affine Weyl groups

摘录 / EXCERPT: Isomorphism conjecture. The map φ is an isomorphism, equivalently

conjecture | open | MathDB 页面 | 原文线索

V2-009557 | MATHDB-347481

The diagonal-maximality conjecture for inverse Vandermonde matrices

摘录 / EXCERPT: Let $b>1$, and set ... For each n, let $M_b(n)$ denote the largest absolute value of an entry of the inverse Vandermonde matrix, and let $c_{i,j,n}$ denote its entries. Diagonal-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-009558 | MATHDB-347496

The main OIT conjecture for modular-tower monodromy

摘录 / EXCERPT: Let O be a modular-tower orbit with geometric monodromy G_O, and for each J' let $\mathcal G(\overline{\mathbf S}_{J'})$ denote the isomorphism class of the corresponding dec...

conjecture | open | MathDB 页面 | 原文线索

V2-009559 | MATHDB-347509

Injectivity after restriction to compact open subgroups

摘录 / EXCERPT: Compact-subgroup injectivity conjecture. If $\pi\in\mathfrak{Rep}^{\textnormal{ladm}}(G)$ is injective, then $\pi|_K$ is injective in $\mathfrak{Rep}^\infty(K)$. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-009560 | MATHDB-347523

The rank-matrix characterization conjecture for codimension-three Artinian Gorenstein algebras

摘录 / EXCERPT: Let M) be an upper triangular matrix of size $d+1$ with non-negative entries. For an upper triangular matrix, let $\operatorname{diag}(i,M)$ denote its i-th diagonal vector, an...

conjecture | open | MathDB 页面 | 原文线索

V2-009561 | MATHDB-347535

The Dicksonianity conjecture for polynomial-growth graded Lie algebras

摘录 / EXCERPT: Dicksonianity conjecture. Then $\mathfrak g$ is Dicksonian.

conjecture | open | MathDB 页面 | 原文线索

V2-009562 | MATHDB-347577

Rationality hypothesis for noetherian connected graded algebras with balanced dualizing complexes

摘录 / EXCERPT: Let A be a noetherian connected graded algebra with a balanced dualizing complex. Hypothesis ... . Rationality over $\mathbb{Q}$ is known for many classes of graded algebras, inclu...

conjecture | open | MathDB 页面 | 原文线索

V2-009563 | MATHDB-347578

The trimmed exponential Galois group conjecture

摘录 / EXCERPT: Trimmed exponential Galois group conjecture. For every $m>n\geq 1$, the polynomial $f_{n,m}$ is irreducible and

conjecture | open | MathDB 页面 | 原文线索

V2-009564 | MATHDB-347586

Etingof's conjecture on restriction of unitary representations

摘录 / EXCERPT: Let W be a type B complex reflection group, let $\mathcal{O}_c(W)$ be its rational Cherednik algebra category $\mathcal{O}$ at parameter c, and let $W'\subset W$ be a parabol...

conjecture | open | MathDB 页面 | 原文线索

V2-009565 | MATHDB-347612

Isomorphism conjecture for the zeta and lambda realizations

摘录 / EXCERPT: For $n\in\mathbb{N}^*$ and $\Sigma\in\mathfrak{S}$, let $\boldsymbol{C}_{n,\Sigma}$ be the corresponding space of composition arrays, and let $\mathcal{Z}_{n,\Sigma}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009566 | MATHDB-347613

Grading conjecture for the lambda-degree of the lambda-value algebra

摘录 / EXCERPT: Let $\mathcal{L}(K)$ be the K-algebra generated by the lambda values, equipped with the lambda-degree induced by the lambda-type of composition arrays and taking values in the mo...

conjecture | open | MathDB 页面 | 原文线索

V2-009567 | MATHDB-347693

The reduction to $k+1$ variables conjecture for subspaces without pure powers

摘录 / EXCERPT: Reduction to $k+1$ variables conjecture. For a generic linear form $lin A_1$, either $bar W$ contains no power of a linear form, or

conjecture | open | MathDB 页面 | 原文线索

V2-009568 | MATHDB-347705

Categorical geometric Langlands correspondence

摘录 / EXCERPT: Categorical geometric Langlands correspondence. There is an equivalence of DG categories

conjecture | open | MathDB 页面 | 原文线索

V2-009569 | MATHDB-347734

The positivity conjecture for 1-bounded entropy

摘录 / EXCERPT: Let $r>1$ and let Q be a von Neumann subalgebra of the free group factor $L({\mathbb F}_r)$. Assume that Q is diffuse and nonamenable. Positivity conjecture for 1-bounded entro...

conjecture | open | MathDB 页面 | 原文线索

V2-009570 | MATHDB-347749

The mod p Langlands lifting and automorphy conjecture for global fields

摘录 / EXCERPT: Let F be a global field of characteristic different from p. Let G be a connected reductive group, and let G' be the split group over F whose root datum is dual to that of...

conjecture | open | MathDB 页面 | 原文线索

V2-009571 | MATHDB-347836

The regular-at-infinity lifting conjecture for Galois representations

摘录 / EXCERPT: Regular-at-infinity lifting conjecture. Every continuous representation $\bar{\rho}:G_K\to\operatorname{GL}_n(k)$ that is regular at infinity lifts to an irreducible geometric repr...

conjecture | open | MathDB 页面 | 原文线索

V2-009572 | MATHDB-347839

The universal framed prodeformation conjecture

摘录 / EXCERPT: Let A be a connective dga and let S be a one-dimensional A-module. Let $E\coloneqq {\mathrm{\normalfont\mathbb{R}}}\mathrm{End}_A(S)$ be the derived endomorphism dga of S....

conjecture | open | MathDB 页面 | 原文线索

V2-009573 | MATHDB-347879

Quadratic generation from monomial-freeness of J_1

摘录 / EXCERPT: Let J_1 and J_2 be the ideals under consideration, with J_1 monomial-free. Quadratic-generation conjecture. If J_1 is monomial-free, then J_2 is quadratically generated....

conjecture | open | MathDB 页面 | 原文线索

V2-009574 | MATHDB-347920

Andruskiewitsch-Angiono-Heckenberger finite-root-system conjecture for Nichols algebras

摘录 / EXCERPT: Let Γ be an abelian finitely generated group, let V be a Yetter-Drinfeld module of diagonal type over Γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009575 | MATHDB-347990

Central-measure support conjecture for the pascalized coupled Young graph

摘录 / EXCERPT: Let Θ be the coupled Young graph, let $\mathcal{P}(\Theta)$ be its pascalization, and let $A_{(\mathcal{C}_H,\delta)}(\infty)$ be the associated inductive limit algebra at t...

conjecture | open | MathDB 页面 | 原文线索

V2-009576 | MATHDB-348102

Deformation conjecture for F-purity in Q-Gorenstein rings

摘录 / EXCERPT: Deformation conjecture for F-purity. Then R is F-pure.

conjecture | open | MathDB 页面 | 原文线索

V2-009577 | MATHDB-348112

The algebraic Cheeger-Gromoll splitting conjecture

摘录 / EXCERPT: Let $((X,D),L)$ be an open K-polystable polarized dlt log Calabi-Yau pair, with exactly two connected components of $\operatorname{Supp}(\lfloor D\rfloor)$, interpreted as having t...

conjecture | open | MathDB 页面 | 原文线索

V2-009578 | MATHDB-348126

The periodic Higgs-bundle characterization of Beauville's algebraic numbers

摘录 / EXCERPT: Let $\rho_{ obreak\baselineskip\lambda}$ be the local system associated with $ obreak\lambda\in\mathbb{P}^1\setminus\{0,1,\infty\}$, and let $(E_{\lambda},\theta_{\lambda})$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-009579 | MATHDB-348157

Taylor's strict local-global compatibility conjecture for cuspidal representations of ${\mathrm{GL}}_{2n}$

摘录 / EXCERPT: Let Π be a cuspidal automorphic form on ${\mathrm{GL}}_{2n}({\mathbb A}_{\mathbb Q})$ with infinitesimal character χ_H, where H is a multiset of distinct integers. A str...

conjecture | open | MathDB 页面 | 原文线索

V2-009580 | MATHDB-348163

Realizability conjecture for unitary 2-shaded planar algebras by multifactor inclusions

摘录 / EXCERPT: Let $\cP_\bullet$ be a unitary 2-shaded planar algebra. A finite-index homogeneous connected $\rm II_1$ multifactor inclusion is an inclusion $A\subset B$ with finite Jones index a...

conjecture | open | MathDB 页面 | 原文线索

V2-009581 | MATHDB-348164

Harbourne-Schenck-Seceleanu conjecture on the WLP for powers of general linear forms

摘录 / EXCERPT: Harbourne-Schenck-Seceleanu conjecture. If $r+1\geq n\geq5$, then $R_{n,r,d}$ fails the WLP for $d\gg0$.

conjecture | open | MathDB 页面 | 原文线索

V2-009582 | MATHDB-348211

Power-series extension formula for absorbing numbers

摘录 / EXCERPT: Power-series extension conjecture. For every ideal I of R,

conjecture | open | MathDB 页面 | 原文线索

V2-009583 | MATHDB-348230

Existence of DODDs for arbitrary matrices in sufficiently large dimensions

摘录 / EXCERPT: DODD existence conjecture. The $d\times d$ 0-inflation of any matrix $\mathbf{X}\in\mathbb{R}^{m\times n}$ admits a DODD for some sufficiently large $d\geq m,n$. Thus every such ma...

conjecture | open | MathDB 页面 | 原文线索

V2-009584 | MATHDB-348284

Nestedness Conjecture for the oper stratification

摘录 / EXCERPT: Nestedness Conjecture. The stratification is nested: there is a partial order on the index set $\{\alpha\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-009585 | MATHDB-348285

Oper Stratum Conjecture

摘录 / EXCERPT: Let the de Rham moduli space $\mathbb{M}_{\mathrm{dR}}(X,r)$ be equipped with its oper stratification, whose strata are indexed by α. Oper Stratum Conjecture. The oper strat...

conjecture | open | MathDB 页面 | 原文线索

V2-009586 | MATHDB-348301

Dimension-preserving leak-removal conjecture for output connectable compartmental models

摘录 / EXCERPT: Let $\mathcal{M}=(G,In,Out,V)$ represent a linear compartmental model. Assume that G is output connectable and $|Out|=1$. Assume the dimension of the image of the coefficient map...

conjecture | open | MathDB 页面 | 原文线索

V2-009587 | MATHDB-348336

Optimal Gonality Conjecture for optimal curves

摘录 / EXCERPT: Optimal Gonality Conjecture. Every optimal curve C over $\mathbb{F}_q$ of genus g has gonality at most $\operatorname{geomgon}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009588 | MATHDB-348337

Asymptotic point-count conjecture for curves of fixed gonality

摘录 / EXCERPT: Asymptotic point-count conjecture. For sufficiently large genus g, one has

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-009589 | MATHDB-348384

Zhao's locally finite derivation conjecture for Mathieu-Zhao spaces

摘录 / EXCERPT: LFED conjecture. Let K be a field of characteristic zero and $\mathcal{A}$ a K-algebra. Then for every locally finite derivation or $\mathcal{E}$-derivation δ of...

conjecture | open | MathDB 页面 | 原文线索

V2-009590 | MATHDB-348385

Zhao's locally nilpotent derivation conjecture for theta-Mathieu-Zhao spaces

摘录 / EXCERPT: LNED conjecture. Let K be a field of characteristic zero, let $\mathcal{A}$ be a K-algebra, and let δ be a locally nilpotent derivation or $\mathcal{E}$-derivation of...

conjecture | open | MathDB 页面 | 原文线索

V2-009591 | MATHDB-348386

The LFED conjecture for locally finite derivations and E-derivations

摘录 / EXCERPT: Let K be a field of characteristic zero, let $\mathcal{A}$ be a K-algebra, and let δ be a locally finite derivation or E-derivation of $\mathcal{A}$. A K-subspace M...

conjecture | open | MathDB 页面 | 原文线索

V2-009592 | MATHDB-348387

The LNED conjecture for locally nilpotent derivations and E-derivations

摘录 / EXCERPT: Let K be a field of characteristic zero, let $\mathcal{A}$ be a K-algebra, and let δ be a locally nilpotent derivation or E-derivation of $\mathcal{A}$. A ϑ-id...

conjecture | open | MathDB 页面 | 原文线索

V2-009593 | MATHDB-348392

Isaacs criterion for formal conjugacy class sizes of commutative Grothendieck rings

摘录 / EXCERPT: Let $\mathcal{F}$ be a complex commutative Grothendieck ring with simultaneously diagonalized fusion matrices having eigenvalues $(\lambda_{i,j})$. Let (c_j) be the eigenvalues o...

conjecture | open | MathDB 页面 | 原文线索

V2-009594 | MATHDB-348433

Two-global-statement realizability conjecture for gaussoids

摘录 / EXCERPT: Two-global-statement realizability conjecture. All minimal gaussoid extensions of at most two global conditional-independence statements are realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-009595 | MATHDB-348482

The conjecture that all Galois orders are principal Galois orders

摘录 / EXCERPT: A Galois order is an algebraic object in the setting above, and a principal Galois order is a Galois order satisfying the additional principalness property established for the clas...

conjecture | open | MathDB 页面 | 原文线索

V2-009596 | MATHDB-348504

Laurent positivity conjecture for cluster ensemble invariant bases

摘录 / EXCERPT: Let Q be the quiver defining the cluster ensemble with $\mathcal A$-space and $\mathcal X$-space, and let Γ act on these spaces with subgroup $\Gamma_\circ$. A corresponde...

conjecture | open | MathDB 页面 | 原文线索

V2-009597 | MATHDB-348506

Converse of the balancedness criterion for regular staged trees

摘录 / EXCERPT: Let a staged tree be regular when, for every pair of vertices in the same stage and every pair of outgoing edges, the corresponding subtree monomials satisfy the required equality...

conjecture | open | MathDB 页面 | 原文线索

V2-009598 | MATHDB-348532

The variable-exponent central power-difference conjecture for rings

摘录 / EXCERPT: Variable-exponent central power-difference conjecture. Under these conditions, R is commutative.

conjecture | open | MathDB 页面 | 原文线索

V2-009599 | MATHDB-348554

Vasconcelos's conjecture on the conormal module

摘录 / EXCERPT: Let I be an ideal of a ring R, and set $S=R/I$. The modules S and I/I^2 are considered over R and S, respectively. Vasconcelos's conormal-module conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-009600 | MATHDB-348555

Vasconcelos's finiteness conjecture for semidualizing modules

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a local ring. A finitely generated R-module C is semidualizing if the homothety morphism ... is an isomorphism and $\operatorname{Ext}_{R}^{i}(C,C)=0$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-009601 | MATHDB-348557

The generation conjecture for invariants of current Lie superalgebras

摘录 / EXCERPT: Generation conjecture. Let $T(B)$ be the algebra generated by $T^k(a_1,\ldots,a_k)$ where $a_1,a_2,\ldots,a_k\in B$ and $k\in\mathbb{Z}_{+}$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-009602 | MATHDB-348569

Four-line generation conjecture for line-arrangement ideals

摘录 / EXCERPT: Four-line generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009603 | MATHDB-348591

Fröberg-Lundqvist conjecture for bihomogeneous ideals on ${\mathbb P}^1\times{\mathbb P}^1$

摘录 / EXCERPT: Let R be the bihomogeneous coordinate ring of ${\mathbb P}^1\times{\mathbb P}^1$, let $I_W=\langle f_0,f_1,f_2\rangle$ be generated by three bihomogeneous polynomials of bidegree...

conjecture | open | MathDB 页面 | 原文线索

V2-009604 | MATHDB-348606

Semisimplicity conjecture for periodic parabolic Higgs and de Rham bundles

摘录 / EXCERPT: Let C be a smooth projective algebraic curve over $\mathbb{C}$. A periodic parabolic Higgs/de Rham bundle means a periodic object of the corresponding parabolic Higgs or de Rham...

conjecture | open | MathDB 页面 | 原文线索

V2-009605 | MATHDB-348622

The finitely bounded homogeneous CSP dichotomy conjecture

摘录 / EXCERPT: Finitely bounded homogeneous CSP dichotomy conjecture. Exactly one of the following holds: either $\operatorname{Pol}(\mathbb A)$ has a uniformly continuous minion homomorphism to...

conjecture | open | MathDB 页面 | 原文线索

V2-009606 | MATHDB-348630

Nonsingularity conjecture for t-wise intersection matrices

摘录 / EXCERPT: For a positive integer n, write $[n]=\{1,\ldots,n\}$. For subsets $I_1,\ldots,I_t\subseteq[n]$, define their weight by ... and write...

conjecture | open | MathDB 页面 | 原文线索

V2-009607 | MATHDB-348632

Cohen-Macaulay descent under tensoring with a perfect module

摘录 / EXCERPT: Cohen-Macaulay descent conjecture. If $M\otimes_R N$ is a Cohen-Macaulay R-module, then N is a maximal Cohen-Macaulay R-module. This conjecture asks whether Cohen-Macaulaynes...

conjecture | open | MathDB 页面 | 原文线索

V2-009608 | MATHDB-348735

Generation conjecture for the Lie algebra of Kirchhoff differences

摘录 / EXCERPT: Let $\mathcal L_n$ be the Lie algebra generated by the Kirchhoff differences $\varkappa_{ij}$ in the group algebra $\mathbb C[S_n]$, with $1\leq i<j\leq n$. Generation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-009609 | MATHDB-348738

The unital antilinear contraction conjecture for uniform algebras

摘录 / EXCERPT: Unital uniform-algebra conjecture. If $\|\theta_\alpha\|\leq 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-009610 | MATHDB-348860

Grothendieck-Teichmüller conjecture

摘录 / EXCERPT: Let $\widehat{\operatorname{GT}}$ be the profinite Grothendieck-Teichmüller group and let $G_\mathbb{Q}$ be the absolute Galois group of $\mathbb{Q}$. Grothendieck-Teichmüller conj...

conjecture | open | MathDB 页面 | 原文线索

V2-009611 | MATHDB-348888

C-group Galois representation conjecture for Hecke eigenvalues

摘录 / EXCERPT: C-group Galois representation conjecture. There exists an admissible representation

conjecture | open | MathDB 页面 | 原文线索

V2-009612 | MATHDB-348894

Bound conjecture for Wilf numbers of semimodules

摘录 / EXCERPT: Bound conjecture. There exists a semimodule Δ minimally generated by $[0,g]$ for a gap g of Γ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-009613 | MATHDB-348914

The coloured-cycle symmetry conjecture for inverse linear forms

摘录 / EXCERPT: Coloured-cycle symmetry conjecture. All binomial linear forms in $I(\mathcal{L}^{-1})$ are induced by symmetries.

conjecture | open | MathDB 页面 | 原文线索

V2-009614 | MATHDB-348921

Galois invariance and descent of graded Higgs bundles

摘录 / EXCERPT: Galois-invariance conjecture. A generalized representation is $\operatorname{Gal}(\bar k/k)$-invariant if and only if the corresponding Higgs bundle is graded and defined over k...

conjecture | open | MathDB 页面 | 原文线索

V2-009615 | MATHDB-348922

Galois invariance implies gradedness for associated Higgs bundles

摘录 / EXCERPT: Gradedness conjecture. Then (E,θ) is graded.

conjecture | open | MathDB 页面 | 原文线索

V2-009616 | MATHDB-348939

The regularity bound for the stabilization point of the minimum distance function

摘录 / EXCERPT: Regularity bound conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-009617 | MATHDB-348953

Free abelianity conjecture for fundamental groups of Galois covers

摘录 / EXCERPT: Free abelianity conjecture. The group $\pi_1(X_{\operatorname{Gal}})$ is a free abelian group.

conjecture | open | MathDB 页面 | 原文线索

V2-009618 | MATHDB-348992

The flagged Schur module character conjecture for northwest diagrams

摘录 / EXCERPT: Flagged Schur module character conjecture. For every northwest diagram D,

conjecture | open | MathDB 页面 | 原文线索

V2-009619 | MATHDB-349000

Linear-resolution conjecture for maximal minors of Jacobian dual matrices

摘录 / EXCERPT: Let $S=k[x_1,\dots,x_n]$ be a polynomial ring, let $I=(f_1,\dots,f_r)$ be a linearly presented $\operatorname{\mathfrak{m}}$-primary ideal of S, and let Θ be the Jacobian...

conjecture | open | MathDB 页面 | 原文线索

V2-009620 | MATHDB-349038

Acyclicity conjecture for the mutation graph of positive braids

摘录 / EXCERPT: A positive braid word is a word in the braid generators with only positive exponents, and its mutation graph has vertices given by equivalence classes of Demazure weaves and edges...

conjecture | open | MathDB 页面 | 原文线索

V2-009621 | MATHDB-349039

Associahedron conjecture for mutation graphs of torus braids

摘录 / EXCERPT: Let β be the $(n,2)$ torus braid, let Δ denote the positive half-twist, and consider the mutation graph whose vertices are equivalence classes of Demazure weaves...

conjecture | open | MathDB 页面 | 原文线索

V2-009622 | MATHDB-349047

Equivariant Iwasawa main conjecture for de Rham (φ,Γ)-modules

摘录 / EXCERPT: Equivariant Iwasawa main conjecture. The equivariant analogue of Nakamura’s formula should link the determinant of the Perrin-Riou-Nakamura big exponential map with the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-009623 | MATHDB-349049

Fei's saturation conjecture for principal-coefficient cluster variable Newton polytopes

摘录 / EXCERPT: Let $\mathcal{A}(\Sigma)$ be a cluster algebra with principal coefficients at the seed Σ. A Newton polytope is the convex hull of the exponent vectors of the monomials occur...

conjecture | open | MathDB 页面 | 原文线索

V2-009624 | MATHDB-349050

The boundary-frozen saturation characterization for surface cluster algebras

摘录 / EXCERPT: Let (S,M) be a marked surface and let $\mathcal{A}^{\mathrm{bd}}$ be the cluster algebra from (S,M) with boundary frozen variables. For a triangulation T and an arc γ,...

conjecture | open | MathDB 页面 | 原文线索

V2-009625 | MATHDB-349090

Algebraic independence conjecture for automatic numbers in independent bases

摘录 / EXCERPT: Let $r\geq 1$ be an integer. Let $b_1,\ldots,b_r$ be multiplicatively independent positive integers, and, for every i, $1\leq i\leq r$, let ξ_i be a real number that is autom...

conjecture | open | MathDB 页面 | 原文线索

V2-009626 | MATHDB-349201

Fintushel-Stern's Montgomery-Yang problem

摘录 / EXCERPT: A pseudo-free differentiable circle action on the 5-dimensional sphere is an action whose non-free orbits are isolated orbit types. Fintushel-Stern's Montgomery-Yang problem. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-009627 | MATHDB-349211

Extension of the upper bound for Hilbert coefficients of Gorenstein modules

摘录 / EXCERPT: Let M be a finitely generated graded Gorenstein R-module, minimally generated by homogeneous elements of degree zero. Let $s=\operatorname{codim} M$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009628 | MATHDB-349233

Makar-Limanov's descent conjecture for free associative subalgebras

摘录 / EXCERPT: Makar-Limanov's conjecture. If $F\otimes_K A$ contains a free K-algebra on at least two free generators, then A also contains a free K-algebra on the same number of free gene...

conjecture | open | MathDB 页面 | 原文线索

V2-009629 | MATHDB-349254

Winnie-the-Pooh conjecture for orthogonal decompositions of $sl(n)$

摘录 / EXCERPT: Winnie-the-Pooh conjecture. The Lie algebra $sl(n)$ has an orthogonal decomposition if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-009630 | MATHDB-349273

Gradient criterion for redundant generators of vanishing ideals

摘录 / EXCERPT: Let $X\subset\mathbb{R}^n$ and let $\mathcal{R}_n$ denote the polynomial ring in n variables. For a set $G\subset\mathcal{R}_n$, write $G^{\mathrm{deg}(g)-1}$ for the relevant de...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-009631 | MATHDB-349335

Linear-type conjecture for Jacobian ideals of central hyperplane arrangements

摘录 / EXCERPT: Let $\mathbb K$ be a field with $\operatorname{char}(\mathbb K)\nmid m$, let $R=\mathbb K[x_1,\ldots,x_n]$, and let $f\in R$ be the defining polynomial of a central hyperplane arra...

conjecture | open | MathDB 页面 | 原文线索

V2-009632 | MATHDB-349443

Stable Harbourne-Huneke conjecture for radical ideal containments

摘录 / EXCERPT: Let $I\subseteq k[x_0,\dots,x_n]$ be a homogeneous radical ideal of big height h, and let $\mathcal{M}=(x_0,\dots,x_n)$ be the graded maximal ideal. Stable Harbourne-Huneke conje...

conjecture | open | MathDB 页面 | 原文线索

V2-009633 | MATHDB-349446

The bounded negativity conjecture for plane curves with rational components

摘录 / EXCERPT: Let K be a fixed algebraically closed ground field. A multiplicity sequence is a positive integer sequence $(d,m_1,\ldots,m_r)$ arising from a reduced plane curve of degree d,...

conjecture | open | MathDB 页面 | 原文线索

V2-009634 | MATHDB-349447

The bounded negativity conjecture for plane curves with bounded genus

摘录 / EXCERPT: Let K be a fixed algebraically closed ground field. A multiplicity sequence is a positive integer sequence $(d,m_1,\ldots,m_r)$ arising from a reduced plane curve, with the m_i...

conjecture | open | MathDB 页面 | 原文线索

V2-009635 | MATHDB-349457

Hida's simplicial concentration conjecture

摘录 / EXCERPT: Let Λ be the Hida-Iwasawa algebra, let R_h be the universal minimal ordinary deformation ring, let $\mathcal{R}_h$ be the associated simplicial deformation ring, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-009636 | MATHDB-349458

Deformation-theoretic non-abelian Leopoldt conjecture

摘录 / EXCERPT: Let R_h be the universal minimal ordinary deformation ring and let Λ be the Hida-Iwasawa algebra. Deformation-theoretic non-abelian Leopoldt conjecture. ... This is identif...

conjecture | open | MathDB 页面 | 原文线索

V2-009637 | MATHDB-349495

Holik-Kara's strengthened Yu conjecture

摘录 / EXCERPT: Let F and G be algebraically independent polynomials in $\mathbb{C}[x_1,\ldots,x_n]$. Assume that each generates its own centralizer in $\mathbb{C}[x_1,\ldots,x_n]$, that...

conjecture | open | MathDB 页面 | 原文线索

V2-009638 | MATHDB-349511

Kochetov's PI conjecture for smash products of hyperalgebras

摘录 / EXCERPT: Kochetov's conjecture. The smash product $H\#\mathbb{F}G$ is PI if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-009639 | MATHDB-349512

Nefness conjecture for the diagonal class on a very general self-product of a curve

摘录 / EXCERPT: Let C be a smooth projective curve of genus g over $\mathbb C$. Denote by f_1 and f_2 the classes of the fibers of the two projections, and by δ the class of the dia...

open_problem | open | MathDB 页面 | 原文线索

V2-009640 | MATHDB-349514

Bahri-Xu linear algebraic conjecture

摘录 / EXCERPT: Let $x_1,\dots,x_p$ with $p\geq 2$ be distinct vectors in $\mathbb R^m$. Define the symmetric matrix $A=(a_{ij})_{1\leq i,j\leq p}$ by ... For $k=1,\dots,p$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009641 | MATHDB-349521

Finite test set conjecture for universal quadratic forms

摘录 / EXCERPT: Let F be a totally real number field, let $ \mathcal{O}_F^+$ denote its totally positive algebraic integers, and let an $ \mathcal{O}_F$-lattice represent an element when its ass...

conjecture | open | MathDB 页面 | 原文线索

V2-009642 | MATHDB-349586

Lando-Yang conjecture on primitive projections of chord diagrams

摘录 / EXCERPT: Lando-Yang conjecture. The projection $ u(D)$ is a linear combination of connected Jacobi diagrams with at most $\tau(D)$ legs. In particular, for any Lie algebra bfg and its ass...

conjecture | open | MathDB 页面 | 原文线索

V2-009643 | MATHDB-349600

The Iwasawa-theoretic Rubin-Stark congruence conjecture

摘录 / EXCERPT: Let K_∞/k be the relevant $mathds{Z}_p$-extension with Galois group containing Γ, let Σ and T be the prescribed sets of places, and let $V'\setminus V=W$ hav...

conjecture | open | MathDB 页面 | 原文线索

V2-009644 | MATHDB-349660

The algebraic-closure comparison conjecture for multilinear rank

摘录 / EXCERPT: Let $\mathbf k$ be a field, let $\bar{\mathbf k}$ be its algebraic closure, and let P be a multilinear $\mathbf k$-polynomial of degree $d\geq 2$. Write $r_{\mathbf k}(P)$ for it...

conjecture | open | MathDB 页面 | 原文线索

V2-009645 | MATHDB-349662

The algebraic-closure comparison conjecture for ranks of subspaces

摘录 / EXCERPT: Let $\mathbf k$ be a field, let $\bar{\mathbf k}$ be its algebraic closure, let $V_1,\ldots,V_d$ be $\mathbf k$-vector spaces, and let $L\subset (V_1\otimes\cdots\otimes V_d)^\vee$...

conjecture | open | MathDB 页面 | 原文线索

V2-009646 | MATHDB-349798

The infinity-categorical structure conjecture for HOM-simplicial sets of $\Omega B As$-algebras

摘录 / EXCERPT: The infinity-categorical structure conjecture. The simplicial sets $\mathrm{HOM}_{\mathsf{\Omega B As\text{-}Alg}}(A,B)_\bullet$ are ∞-categories.

conjecture | open | MathDB 页面 | 原文线索

V2-009647 | MATHDB-349813

The DMT conjecture for regular representations of central division algebras

摘录 / EXCERPT: Let ${\mathcal D}$ be an index n ${\mathbb Q}$-central division algebra, let $\Lambda\subset {\mathcal D}$ be an order, and let ψ_{reg} denote its regular representation in...

conjecture | open | MathDB 页面 | 原文线索

V2-009648 | MATHDB-349815

Generic nonvanishing conjecture for the variational-equation Galois parameters

摘录 / EXCERPT: Let λ and E be the parameters of the axisymmetric Bianchi IX cosmological model, and let $\gamma_1,\gamma_2\in\mathbb{C}$ be the constants appearing in the differential G...

conjecture | open | MathDB 页面 | 原文线索

V2-009649 | MATHDB-349890

Mayer's refinement of Scholz's conjecture for ramified cubic extensions

摘录 / EXCERPT: Let L be a non-Galois totally real cubic field with ramified Galois closure N, conductor $f>1$ over a real quadratic field K, and type α, with $U_N=U_0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009650 | MATHDB-349919

The Isaacs conjecture for spherical fusion categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a spherical fusion category over $\mathbb{C}$, and let $s\geq 0$ be rational. If A is its fusion ring, a category is s-Isaacs when, for every character...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-009651 | MATHDB-349928

The non-admissible collapsing isomorphism conjectures for exceptional W-algebras

摘录 / EXCERPT: Let E_6, E_7, E_8, and F_4 denote the corresponding simple exceptional Lie algebras, and let the labels such as $2A_2$, A_2+3A_1, E_6, and $\widetilde A_2$ denote nilpo...

conjecture | open | MathDB 页面 | 原文线索

V2-009652 | MATHDB-349932

Uniqueness of minimal reducing subalgebras for nilpotent elements

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple Lie algebra, let $f\in\mathfrak g$ be a non-zero nilpotent element, and let $\mathfrak s$ be an $\mathfrak{sl}_2$-triple containing f. A reducin...

conjecture | open | MathDB 页面 | 原文线索

V2-009653 | MATHDB-349949

Yanagawa's radicality conjecture for Specht ideals

摘录 / EXCERPT: Fix a positive integer n, a field ${\mathbb F}$, and let $R={\mathbb F}[x_1,\ldots,x_n]$. For a partition $\lambda\vdash n$, let $\mathfrak{a}(\lambda)$ denote the Specht ideal g...

conjecture | open | MathDB 页面 | 原文线索

V2-009654 | MATHDB-349950

Perfection monotonicity conjecture for two-rowed Specht ideals

摘录 / EXCERPT: Let n and k be integers satisfying $1\leq k<k+1\leq n-k$, and let $\mathfrak{a}(n+1,j+1,j+1)$ denote the Specht ideal associated with the partition $(n+1,j+1,j+1)$. Perfection...

conjecture | open | MathDB 页面 | 原文线索

V2-009655 | MATHDB-350064

Stark's Galois-equivariance conjecture for special values and regulators

摘录 / EXCERPT: Stark's conjecture. For every automorphism α of $\mathbb{C}$,

conjecture | open | MathDB 页面 | 原文线索

V2-009656 | MATHDB-350065

Weak exceptional zero conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let $\Lambda=\mathcal{O}_p[[\Gamma]]$, let $(U_\infty')^\chi$ be the χ-component of the inverse limit of the p-unit modules, let ε_∞^χ be the inverse-lim...

conjecture | open | MathDB 页面 | 原文线索

V2-009657 | MATHDB-350066

Positivity conjecture for pre-canonical bases in type A

摘录 / EXCERPT: Let Φ be a root system of type A_n, let X^+ denote the dominant weights, and let $\mathbf{N}^i$ be the pre-canonical bases of the spherical Hecke algebra. Positivity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-009658 | MATHDB-350153

Mazur-Tate conjecture for modular elements

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, let $F/\mathbb{Q}$ be a finite real abelian extension with Galois group G, let $r={\rm rank}(E(\mathbb{Q}))$, and let I_F be the...

conjecture | open | MathDB 页面 | 原文线索

V2-009659 | MATHDB-350167

Positselski's pro-ℓ Bogomolov conjecture

摘录 / EXCERPT: Positselski's pro-ℓ Bogomolov conjecture. The maximal pro-ℓ Galois group of $\sqrt[\ell^\infty]{\mathbb{K}}$ is a free pro-ℓ group.

conjecture | open | MathDB 页面 | 原文线索

V2-009660 | MATHDB-350205

Lex-plus-powers conjecture for graded Betti numbers

摘录 / EXCERPT: Let $R=K[x_1,\ldots,x_n]$, and let I be a homogeneous ideal containing a regular sequence of degrees $2\leq a_1\leq\ldots\leq a_n$. Suppose that there exists a lex-plus-powers id...

conjecture | open | MathDB 页面 | 原文线索

V2-009661 | MATHDB-350238

The maximal-index conjecture for halidon rings of integers modulo n

摘录 / EXCERPT: Let $R=\mathbb{Z}_{n}$, where ... with primes $p_{1}<p_{2}<p_{3}<\cdots<p_{k}$, including 2. Write $\psi(n)$ for the quantity appearing in the conjecture. Maximal-index conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-009662 | MATHDB-350246

Fomin-Zelevinsky's Laurent positivity conjecture for F-polynomials

摘录 / EXCERPT: Let $\mathbf{\Sigma}$ be any cluster pattern with initial point $t_0\in \mathbb{T}_n$, and let $F_{i;t}(\mathbf{y})$ be an F-polynomial in that pattern. Fomin-Zelevinsky's Lauren...

conjecture | open | MathDB 页面 | 原文线索

V2-009663 | MATHDB-350247

The abstract-index admissibility conjecture

摘录 / EXCERPT: Let $p \in \mathbb{P} \cup \{0\}$, and let $(\mathcal{G},\mathcal{H},\theta)$ be a p-embedding of abstract indices. The abstract-index admissibility conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-009664 | MATHDB-350255

Matúš's rational-point conjecture for discrete conditional independence models

摘录 / EXCERPT: Matúš's rational-point conjecture. For any discrete conditional independence model $\mathcal{C}$, there exists a distribution

conjecture | open | MathDB 页面 | 原文线索

V2-009665 | MATHDB-350260

The Classification Conjecture for contraction algebras of smooth threefold flops

摘录 / EXCERPT: The Classification Conjecture. The complete local cDV singularities are isomorphic if and only if their contraction algebras have equivalent bounded derived categories:

conjecture | open | MathDB 页面 | 原文线索

V2-009666 | MATHDB-350261

The Realisation Problem for contraction algebras

摘录 / EXCERPT: Realisation Problem. If $f\in\mathbb{C}\langle\!\langle x,y\rangle\!\rangle$ satisfies $\operatorname{GKdim}\EuScript{J}\mathrm{ac}(f)\leq 1$, then $\EuScript{J}\mathrm{ac}(f)$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-009667 | MATHDB-350313

Gras's conjecture on the torsion of maximal pro-p Galois groups

摘录 / EXCERPT: Let K be a number field, let K_p/K be the maximal pro-p extension unramified outside p, and write ... where $\mathcal T_{K,p}=Tor(G_{K,p}^{ab})$ is the torsion subgroup. Gras...

conjecture | open | MathDB 页面 | 原文线索

V2-009668 | MATHDB-350338

Extraneous-mode conjecture for the type B twisted W-algebra module

摘录 / EXCERPT: Let r be a positive integer, let $\epsilon(i)$ denote the twist shift for the strong generators, and define the positive modes ... Set $S'=\{\mathsf{W}^{2r-1}_{1/2}\}\subset S_+$....

conjecture | open | MathDB 页面 | 原文线索

V2-009669 | MATHDB-350339

Extraneous-mode conjecture for the type C twisted W-algebra module

摘录 / EXCERPT: Let S_+ be the subset of positive modes in the twisted mode expansion ... Set $S'=\{\widetilde{\mathsf{W}}^{r+1}_{1/2}\}\in S_+$. Type C extraneous-mode conjecture. The subset S'...

conjecture | open | MathDB 页面 | 原文线索

V2-009670 | MATHDB-350345

Completeness and faithfulness conjecture for Wronskian Bethe equations

摘录 / EXCERPT: Let $\mathcal{H}$ be a cyclic representation of the Yangian $\mathcal{Y}(\mathfrak{so}_{2r})$, with highest-weight vector a cyclic vector. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009671 | MATHDB-350370

The superspace coinvariant problem

摘录 / EXCERPT: Let Ω_n be the superspace algebra with commuting variables $x_1,\ldots,x_n$ and anticommuting variables $\theta_1,\ldots,\theta_n$, acted on by the symmetric group...

open_problem | open | MathDB 页面 | 原文线索

V2-009672 | MATHDB-350373

Iyengar-Ma-Walker length-multiplicity conjecture

摘录 / EXCERPT: Let R be a local ring, and let M be a nonzero R-module of finite projective dimension. Iyengar-Ma-Walker length-multiplicity conjecture. ... This is one of two multiplicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-009673 | MATHDB-350374

Iyengar-Ma-Walker Dutta multiplicity conjecture

摘录 / EXCERPT: Let R be a local ring with maximal ideal, and let F be a short complex supported on the maximal ideal. Write $\chi_{\infty}(F)$ for the Dutta multiplicity of F. Iyengar-Ma-Wa...

conjecture | open | MathDB 页面 | 原文线索

V2-009674 | MATHDB-350407

The linearity conjecture for support loci of Ext modules

摘录 / EXCERPT: Let cM be a regular holonomic cD_X-module, and let $W\to X$ be a relative compact open subset. Suppose that, locally on W, ... for some $n\le l\le n+r$. For a support locus...

conjecture | open | MathDB 页面 | 原文线索

V2-009675 | MATHDB-350408

Budur's weak conjecture on Bernstein-Sato zero loci

摘录 / EXCERPT: Let B_F be the Bernstein-Sato ideal associated with the tuple F, and let $Z(B_F)\subseteq\mathbb C^r$ denote its zero locus. Budur's weak conjecture. Each irreducible component...

conjecture | open | MathDB 页面 | 原文线索

V2-009676 | MATHDB-350438

The cluster adjacency conjecture for positroid tiles of the $m=2$ amplituhedron

摘录 / EXCERPT: Let $Z_{\hat{G}(\mathcal{T})}$ be a positroid tile of the amplituhedron $\mathcal{A}_{n,k,2}(Z)$. A facet is a codimension-one boundary piece, and each facet lies on a hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-009677 | MATHDB-350462

The partition characterization of the Andrews-Gordon companion identities

摘录 / EXCERPT: Let $r\geq 2$ and $1\leq i\leq r$. Let $\mathcal{C}_{r,i}$ be the set of partitions $\lambda=(\lambda_1,\dots,\lambda_s)$ with at most $i-1$ parts equal to 1 and satisfying eithe...

conjecture | open | MathDB 页面 | 原文线索

V2-009678 | MATHDB-350479

The Dutta-multiplicity bound for short complexes

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a local ring, let $e(R)$ denote its Hilbert-Samuel multiplicity, and let F be a short complex supported on the maximal ideal, meaning a non-exact complex...

conjecture | open | MathDB 页面 | 原文线索

V2-009679 | MATHDB-350480

The Lech-type Dutta-multiplicity conjecture for complete local rings

摘录 / EXCERPT: Let R be a complete local ring, and let F be a short complex in the perfect-complex category, meaning a short complex in $\operatorname{perf} R$ supported on the maximal ideal....

conjecture | open | MathDB 页面 | 原文线索

V2-009680 | MATHDB-350495

The generalized GWA GK-dimension conjecture

摘录 / EXCERPT: Let R be an affine $\Bbbk$-algebra and let $W:=R(\mathbf x,\mathbf y,\sigma,h)$ be a generalized Weyl algebra of degree m. Write ... Assume that R is a finitely generated modul...

conjecture | open | MathDB 页面 | 原文线索

V2-009681 | MATHDB-350563

Multiplicativity conjecture for discriminant algebras and norm functors

摘录 / EXCERPT: Let Δ be the discriminant algebra operation, sending rank-n algebras $\mathcal{A}\to\mathcal{B}$ to quadratic algebras $\mathcal{A}\to\Delta_{\mathcal{B}/\mathcal{A}}$. Le...

open_problem | open | MathDB 页面 | 原文线索

V2-009682 | MATHDB-350599

Nonvanishing leading-term conjecture for the elliptic graph-complex classes

摘录 / EXCERPT: For $n\geq0$, let t_{2n} be the loop-free graph with $2n+2$ vertices described in the source, and let $\delta_{2n}\in\mathfrak{r}^{\mathrm{non-fr}}_{(1)}$ be Enriquez's derivatio...

conjecture | open | MathDB 页面 | 原文线索

V2-009683 | MATHDB-350649

The characteristic-profile scattering conjecture for the Einstein-Maxwell-Klein-Gordon system

摘录 / EXCERPT: Let $\phi_{\mathcal H^+}$ be a suitable finite-energy scalar-field profile on the event horizon of a spherically symmetric Einstein-Maxwell-Klein-Gordon black hole, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009684 | MATHDB-350658

The scaling-group conjecture for unimodular solvable algebraic groups over $\mathbb{Q}_n$

摘录 / EXCERPT: Scaling-group conjecture. The scaling group of G is a subgroup, possibly trivial, of

conjecture | open | MathDB 页面 | 原文线索

V2-009685 | MATHDB-350722

BMM symmetrising trace conjecture for generic Hecke algebras

摘录 / EXCERPT: BMM symmetrising trace conjecture. There exists a symmetrising trace $\tau:\mathcal{H}(W)\to R$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-009686 | MATHDB-350733

The maximal-complement conjecture for local and 2-local automorphisms of solvable Leibniz algebras

摘录 / EXCERPT: Let L be a solvable Leibniz algebra with a given nilradical, and suppose that the dimension of a complementary space to the nilradical is maximal. Maximal-complement conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-009687 | MATHDB-350741

The regularity formula for squarefree powers of edge ideals of forests

摘录 / EXCERPT: Forest regularity conjecture. For every $1\leq k\leq \operatorname{mat}(G)$,

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V2-009688 | MATHDB-350805

Conjecture that a trivial Kazhdan-Lusztig polynomial makes Theta a q-power

摘录 / EXCERPT: Let W be the Weyl group, let $x,y,w\in W$, let ℓ denote the length function, and let $\Theta(x,y,w)$ be the polynomial defined in the paper. Assume ... and also...

conjecture | open | MathDB 页面 | 原文线索

V2-009689 | MATHDB-350809

The converse categorification conjecture for the categorical Heisenberg algebra

摘录 / EXCERPT: Converse categorification conjecture. If $\bicat{F}_{{\cat V}}$ categorifies $F_{{\cat V}}$, then π is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-009690 | MATHDB-350810

The symmetric-power criterion for surjectivity of the categorical Heisenberg map

摘录 / EXCERPT: Symmetric-power criterion conjecture. If the canonical morphism above is an isomorphism, then π is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-009691 | MATHDB-350887

Primality conjecture for weakly connected collections of cells

摘录 / EXCERPT: Let $\mathcal{P}$ be a weakly connected collection of cells, and let $I_{\mathcal{P}}$ denote its polyomino ideal. A zig-zag walk is the particular sequence of inner intervals of...

conjecture | open | MathDB 页面 | 原文线索

V2-009692 | MATHDB-350893

Simplicity conjecture for nonzero central quotients of the Virasoro enveloping algebra

摘录 / EXCERPT: Let $V\!ir$ be the Virasoro algebra, let z denote its central element, and let $\operatorname{U}(V\!ir)$ be its universal enveloping algebra. The Virasoro central-quotient simpli...

conjecture | open | MathDB 页面 | 原文线索

V2-009693 | MATHDB-350897

Surjectivity of Kirillov's orbit method for Virasoro and Witt algebras

摘录 / EXCERPT: Let $\mathfrak g$ be one of $V\!ir$, W, or $W_{\geq -1}$. For a local function $\chi\in\mathfrak g^*$, choose a polarization and form the annihilator of the induced module; write...

conjecture | open | MathDB 页面 | 原文线索

V2-009694 | MATHDB-350916

Greenberg-Ochiai's main conjecture for general Galois deformations

摘录 / EXCERPT: Let $\mathscr{R}$ be a complete Noetherian local domain with finite residue field, let $\mathscr{T}$ be a finite-rank free $\mathscr{R}$-module with continuous $G_{\mathbb{Q}}$-act...

conjecture | open | MathDB 页面 | 原文线索

V2-009695 | MATHDB-350953

Chinburg's strong Stark conjecture

摘录 / EXCERPT: Let K be a number field, let L/K be a finite Galois extension with group G, let S be a finite set of places of K containing S_{∞} and all places ramified in L/K,...

conjecture | open | MathDB 页面 | 原文线索

V2-009696 | MATHDB-350954

The equivariant Tamagawa number conjecture for $(h^0(\operatorname{Spec}(L)),\mathbb{Z}[G])$

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with Galois group G, and let S be a finite set of places of K containing all archimedean places and all places ramifie...

conjecture | open | MathDB 页面 | 原文线索

V2-009697 | MATHDB-350964

The odd-width admissible-partition character identity for $C_{(1)}_\ell$

摘录 / EXCERPT: The odd-width character identity. The principally specialized character

conjecture | open | MathDB 页面 | 原文线索

V2-009698 | MATHDB-350991

Betti-origin deformation conjecture for the annular Hecke functor

摘录 / EXCERPT: Let $\sf{H}_W$ be the Hecke category, let $\sf{AH}:\sf{H}_W^{\operatorname{op}}\to\mathsf{Mod}_2^b(\bb{A}_W)$ be the annular Hecke functor, and let $\bb{A}_{W,\epsilon}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009699 | MATHDB-351013

Budur's linear-factor generation conjecture for Bernstein-Sato ideals

摘录 / EXCERPT: Let $F=f_1,\dots,f_\ell$ be a tuple of elements of $\mathbb{C}\{x_1,\dots,x_d\}$, and let $B_F\subseteq\mathbb{C}[s_1,\dots,s_\ell]$ be its Bernstein-Sato ideal. A linear polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-009700 | MATHDB-351025

Cluster-variable conjecture for dual canonical basis immanants

摘录 / EXCERPT: Fix m, and let G^{w_0,w_0} denote the big open double Bruhat cell in SL_m. For $v\in S_n$, let $\operatorname{Imm}_v X(R,C)$ denote the associated dual canonical basis immana...

conjecture | open | MathDB 页面 | 原文线索

V2-009701 | MATHDB-351099

Goulden-Jackson character nonnegativity conjecture

摘录 / EXCERPT: Goulden-Jackson conjecture. For every $\theta\in\Theta$, the value of the linear extension of χ^{λ} to the group algebra satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-009702 | MATHDB-351120

Damiani's PBW-basis conjecture for the q-Onsager algebra

摘录 / EXCERPT: Let O_q be the q-Onsager algebra, with alternating generators $\{W_{-i}\}_{i\in\mathbb N}$, $\{\tilde G_{j+1}\}_{j\in\mathbb N}$, and $\{W_{k+1}\}_{k\in\mathbb N}$. A linear or...

conjecture | open | MathDB 页面 | 原文线索

V2-009703 | MATHDB-351121

Centralizer conjecture for the alternating generators $\tilde G_{k+1}$ of the q-Onsager algebra

摘录 / EXCERPT: Let O_q be the q-Onsager algebra, and let $\tilde G_{k+1}$, $k\in\mathbb N$, be its alternating generators. For $x\in O_q$, consider the subalgebra generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-009704 | MATHDB-351135

The symmetrising-form conjecture for generic Hecke algebras

摘录 / EXCERPT: Symmetrising-form conjecture. The form $t_{W,\mathbf u}$ takes values in A and is a symmetrising form on $\mathcal H(W,\mathbf u)$. This is the Hecke-algebra conjecture attribute...

conjecture | open | MathDB 页面 | 原文线索

V2-009705 | MATHDB-351137

The conjectured coinvariant dimensions in the degree $2^{s+4}+2^{s+1}-3$ family

摘录 / EXCERPT: Coinvariant-dimension conjecture. This space is trivial if $s=2$ and has dimension 1 if $s\neq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-009706 | MATHDB-351138

The conjectured kernel-invariant dimensions for Kameko's squaring map

摘录 / EXCERPT: Let $k=\mathbb Z_2$, let $P_4=k[x_1,x_2,x_3,x_4]$, and write $Q^{\otimes 4}=k\otimes_A P_4$. For a positive integer s, Kameko's squaring operation is the $kGL_4(k)$-module epimor...

conjecture | open | MathDB 页面 | 原文线索

V2-009707 | MATHDB-351139

The conjectured coinvariant dimensions in the degree $2^{s+3}+2^s-2$ family

摘录 / EXCERPT: Coinvariant-dimension conjecture. Its dimension is

conjecture | open | MathDB 页面 | 原文线索

V2-009708 | MATHDB-351155

Elagin-Lunts diagonal dimension conjecture for curves

摘录 / EXCERPT: Let k be a field, and let X be a smooth projective curve over k. Write $\mathrm{Ddim}(X)$ for the least integer n such that there exist $G,H\in D_{\operatorname{perf}}(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-009709 | MATHDB-351181

Regularity conjecture for Jordan-Lie inner ideals of simple locally finite associative algebras

摘录 / EXCERPT: Let A be a simple locally finite associative algebra over $\mathbb{F}$, let $L=[A,A]$, and let B be a Jordan-Lie inner ideal of L. Regularity conjecture. Every Jordan-Lie inn...

conjecture | open | MathDB 页面 | 原文线索

V2-009710 | MATHDB-351201

The refined Oort conjecture for cyclic covers of curves

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$. Let $\phi: Z \xrightarrow{} X$ be a cyclic Γ-cover of curves over k, and let $\phi_1: Y \xrightarrow{} X$...

conjecture | open | MathDB 页面 | 原文线索

V2-009711 | MATHDB-351202

The local refined Oort conjecture for cyclic extensions

摘录 / EXCERPT: Let k be the residue field of a discrete valuation ring R of characteristic zero, and let $k[[z_2]]/k[[x]]$ be a cyclic Γ-extension with Galois sub-extension...

conjecture | open | MathDB 页面 | 原文线索

V2-009712 | MATHDB-351233

Chan-Young-Zhang's graded-isolated-singularity conjecture for cyclic invariants

摘录 / EXCERPT: Let $\operatorname{\Bbbk}$ be an algebraically closed field of characteristic zero, let $A=\operatorname{\Bbbk}_{-1}[x_0,\dots,x_{n-1}]$ with $n\geqslant 2$, and let $G=C_n$ act on...

conjecture | open | MathDB 页面 | 原文线索

V2-009713 | MATHDB-351242

The exchange-property characterization for sums of two square-zero endomorphisms

摘录 / EXCERPT: Exchange-property conjecture. The endomorphism u is the sum of two square-zero endomorphisms of V if and only if u has the exchange property. Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-009714 | MATHDB-351243

The similarity characterization for products of two involutions

摘录 / EXCERPT: Involution-product conjecture. The endomorphism u is the product of two involutions in $\operatorname{GL}(V)$ if and only if u is similar to its inverse.

conjecture | open | MathDB 页面 | 原文线索

V2-009715 | MATHDB-351249

Local Ohno-Nakagawa duality conjecture for traced quartic rings

摘录 / EXCERPT: Let K be a 2-adic local field and let $0\leq t\leq s\leq e=v_K(2)$. Let $C_1\subseteq C_2$ be orders in a cubic K-algebra R with $C_2/C_1\cong\mathcal{O}_K/\pi^{s-t}$ cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-009716 | MATHDB-351259

Stable-category invariance of codimension for geometric complete intersections

摘录 / EXCERPT: Stable-category codimension conjecture. If $\operatorname{\underline{CM}}_{\leq r}(A)$ is triangle equivalent to $\operatorname{\underline{CM}}_{\leq r}(B)$, then $m=n$.

conjecture | open | MathDB 页面 | 原文线索

V2-009717 | MATHDB-351260

Equivalent image-classification form of the L'vov-Kaplansky conjecture

摘录 / EXCERPT: Equivalent L'vov-Kaplansky conjecture. The image of p satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-009718 | MATHDB-351261

Generalized Kaplansky conjecture for simple finite-dimensional algebras

摘录 / EXCERPT: Generalized Kaplansky conjecture. The evaluation of any multilinear polynomial over A is a vector subspace of A.

conjecture | open | MathDB 页面 | 原文线索

V2-009719 | MATHDB-351262

Octonion-valued multilinear polynomial image conjecture

摘录 / EXCERPT: Octonion image conjecture. The image of this evaluation is a real vector subspace of $\mathbb{O}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009720 | MATHDB-351287

Hida's codimension conjecture for nearly ordinary Hecke algebras

摘录 / EXCERPT: Hida's codimension conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-009721 | MATHDB-351313

Nonexistence of vector bundles realizing a multiple of stacky height on elliptic curves

摘录 / EXCERPT: Height-multiple nonexistence conjecture. There is no vector bundle $\mathscr W$ on $\overline{\mathscr M}_{1,1}$ for which there exists an integer n such that

conjecture | open | MathDB 页面 | 原文线索

V2-009722 | MATHDB-351318

Algebraicity-or-density dichotomy for the Hodge locus

摘录 / EXCERPT: Algebraicity-or-density conjecture. If $\operatorname{HL}(S, {\mathbb V}^{\otimes})_{\operatorname{typ}}$ is empty, then $\operatorname{HL}(S, {\mathbb V}^{\otimes})$ is algebraic;...

conjecture | open | MathDB 页面 | 原文线索

V2-009723 | MATHDB-351319

Algebraicity of the Hodge locus in level at least three

摘录 / EXCERPT: Algebraicity conjecture in high level. If ${\mathbb V}$ has level at least 3, then $\operatorname{HL}(S, {\mathbb V}^{\otimes})$ is algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-009724 | MATHDB-351324

The dimension formula for the largest matchable subspace

摘录 / EXCERPT: The dimension formula for the largest matchable subspace. If $\langle AB\rangle\neq A$ and A is not matched to B, then

conjecture | open | MathDB 页面 | 原文线索

V2-009725 | MATHDB-351331

Pullback injectivity conjecture for forgetful charts of prestable curves

摘录 / EXCERPT: Let $(g,n)\neq(1,0)$, and let $d\geq 0$ be fixed. For each $m\geq 0$, consider the forgetful morphism ... It forgets the last m markings without stabilizing the curve. Pullback inj...

conjecture | open | MathDB 页面 | 原文线索

V2-009726 | MATHDB-351378

Pure matching curves and finite discrete indecomposable modules

摘录 / EXCERPT: Assume that the ground ring k is a field. Let $\mathscr{J}_\mathcal{T}$ and $\mathscr{G}_\mathcal{T}$ be the algebras appearing in the construction, and let ... be the inclusion of...

conjecture | open | MathDB 页面 | 原文线索

V2-009727 | MATHDB-351379

Almost slice-fullness conjecture for finite-dimensional left-Hilbert modules

摘录 / EXCERPT: Let $\mathcal{H}$ be a left-Hilbert module that is finite-dimensional as a vector space. Let X be a measure space, let $\mathbb{F}$ be the underlying scalar field, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009728 | MATHDB-351383

The nonzero-characteristic canonical subgroup conjecture for $T_{\boldsymbol{n},\boldsymbol{w}}$ cluster algebras

摘录 / EXCERPT: Let $T_{\boldsymbol{n},\boldsymbol{w}}$ be a quiver of the indicated type, let $\chi(T_{\boldsymbol{n},\boldsymbol{w}})$ be its associated quantity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009729 | MATHDB-351426

Divisibility of the characteristic polynomial by a common divisor

摘录 / EXCERPT: Let $A=\mathbb{K}[p,q]$ be the subalgebra generated by p and q, let $\chi_{p,q}(x)$ be their characteristic polynomial, and let $a(x)$ be a polynomial of degree at least two di...

conjecture | open | MathDB 页面 | 原文线索

V2-009730 | MATHDB-351427

Pure-derivative conjecture for derivations of polynomial subalgebras

摘录 / EXCERPT: Let $A\subseteq\mathbb{K}[x]$ be a subalgebra with spectrum containing α, let $C=\{\alpha_1,\ldots,\alpha_s\}$ be its spectrum, and let $\alpha_j\sim\alpha$ mean that...

conjecture | open | MathDB 页面 | 原文线索

V2-009731 | MATHDB-351428

Dimension conjecture for ordinary-derivative derivations

摘录 / EXCERPT: Let $A\subseteq\mathbb{K}[x]$ be a finite-codimension subalgebra, let k_α be the dimension of the vector space of all α-derivations of A, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009732 | MATHDB-351429

Derivations of intersections with disjoint spectra

摘录 / EXCERPT: Intersection-derivation conjecture. The set of derivations of A is the union of the derivations of A_1 and A_2, restricted to A. This conjecture concerns how derivations be...

conjecture | open | MathDB 页面 | 原文线索

V2-009733 | MATHDB-351430

Recursive construction of higher-order alpha-derivations

摘录 / EXCERPT: Higher-order derivation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-009734 | MATHDB-351445

Basic compatibility conjecture for puncture weights of cluster variables

摘录 / EXCERPT: Let P be the weight-vector set used for the cluster algebra $\mathscr{A}(\mathcal{M})$, let $\mathbb{M}_\circ$ be the set of punctures, and let pi_p denote projection to the we...

conjecture | open | MathDB 页面 | 原文线索

V2-009735 | MATHDB-351447

Tagged cluster monomial compatibility conjecture

摘录 / EXCERPT: Let $T_1,\dots,T_s$ be tensor diagrams such that each $[T_i]$ is a cluster variable of $\mathscr{A}(\mathcal{M})$. Let $varphi_i$ be a tagging of T_i, with invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-009736 | MATHDB-351448

Arborization conjecture for tagged non-elliptic SL3 webs

摘录 / EXCERPT: Let $\mathcal{M}$ be one of the moduli spaces considered in the paper, and let $[(T,varphi)]inmathscr{A}(\mathcal{M})$ be a tagged non-elliptic ${\rm SL}_3$ web invariant. The arbo...

conjecture | open | MathDB 页面 | 原文线索

V2-009737 | MATHDB-351452

The generalized Jacobian conjecture on algebraic closedness

摘录 / EXCERPT: Let k be a field of characteristic 0, let $n>1$, and let $r\in\{2,\dots,n\}$. For polynomials $f_1,f_2,\dots,f_r\in k[x_1,\dots,x_n]$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-009738 | MATHDB-351475

Aprodu-Farkas-Ortega stability conjecture for canonical curves

摘录 / EXCERPT: Let C be a general smooth canonical curve of genus $g\geq 7$, embedded in $\mathbb{P}^{g-1}$ by its canonical linear system, and let $\mathcal{N}_{C/\mathbb{P}^{g-1}}$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-009739 | MATHDB-351490

Bijection conjecture for pseudo-involutions fixing stable root spaces

摘录 / EXCERPT: Bijection conjecture. The assignment $(X,\tau,\chi) \mapsto \theta(X,\tau,\chi)$ induces a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-009740 | MATHDB-351555

Linear-ordering conjecture for reflection representations of skew-symmetrizable matrices

摘录 / EXCERPT: Let B be a skew-symmetrizable matrix indexed by $mathcal I={1,ldots,n}$. For a mutation sequence $\boldsymbol{w}$, let $r_i^{\boldsymbol{w}}$ be the associated reflections, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009741 | MATHDB-351569

The integral conformal circuit decomposition conjecture

摘录 / EXCERPT: Let $W a a$ be a rational linear subspace. For $z a a a W\cap\mathbb{Z}^n$, a conformal circuit decomposition is a representation of z as a sum of conformal circuit vectors. A ve...

conjecture | open | MathDB 页面 | 原文线索

V2-009742 | MATHDB-351608

The n-adic normal number conjecture for irrational algebraic numbers

摘录 / EXCERPT: Let α be an irrational algebraic number and let $n>1$ be an integer. An n-adic expansion of α is a representation $\sum_{l=0}^{\infty}a_ln^l$ with $0\le a_l<n$ sati...

conjecture | open | MathDB 页面 | 原文线索

V2-009743 | MATHDB-351615

The characteristic-three realization conjecture for generalized Cartan matrices

摘录 / EXCERPT: Let A be a purely even, symmetrizable, indecomposable, generalized Cartan matrix of size n with nonzero determinant whose off-diagonal entries lie between -2 and 0, inclusi...

conjecture | open | MathDB 页面 | 原文线索

V2-009744 | MATHDB-351616

Stronger Buchsbaum-Eisenbud-Horrocks conjecture for syzygy ranks

摘录 / EXCERPT: Stronger Buchsbaum-Eisenbud-Horrocks conjecture for syzygy ranks.

conjecture | open | MathDB 页面 | 原文线索

V2-009745 | MATHDB-351617

Local Buchsbaum-Eisenbud-Horrocks conjecture

摘录 / EXCERPT: Local Buchsbaum-Eisenbud-Horrocks conjecture. For all i,

conjecture | open | MathDB 页面 | 原文线索

V2-009746 | MATHDB-351633

C*-algebraic Novak's conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra. For $x\in\mathcal{A}$, define the C-algebraic cosine by ... where $e^x=\sum_{m=0}^{\infty}x^m/m!$. Write $\mathcal{A}_{\mathrm{sa}}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-009747 | MATHDB-351654

Symmetric-or-not-2-transitive dichotomy for Schubert Galois groups

摘录 / EXCERPT: Let ${\mu^\bullet}$ be a Schubert problem on the Grassmannian $\operatorname{Gr}(m,n)$, and let its Schubert Galois group act on the solutions of the problem. Schubert Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-009748 | MATHDB-351683

Malle-type conjecture for counting algebraic tori by finite monodromy group

摘录 / EXCERPT: Malle-type conjecture for algebraic tori. For every $n\geq 1$ and every finite subgroup $1\neq H\leq\operatorname{GL}_n(\mathbb{Z})$, there are positive integers $a(H),b(H)$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-009749 | MATHDB-351690

The band basis and common triangular basis conjecture

摘录 / EXCERPT: Band-triangular basis conjecture. The (quantized) band basis coincides with the common triangular basis after localization at the frozen variables.

conjecture | open | MathDB 页面 | 原文线索

V2-009750 | MATHDB-351739

Cancellation equivalence for symmetric Poisson algebras and enveloping algebras

摘录 / EXCERPT: Let $\mathfrak g$ be a finite-dimensional Lie algebra. Its symmetric Poisson algebra is denoted by $PS(\mathfrak g)$, and its universal enveloping algebra by $U(\mathfrak g)$. Canc...

conjecture | open | MathDB 页面 | 原文线索

V2-009751 | MATHDB-351740

Cancellation equivalence for Poisson algebras and their quantizations

摘录 / EXCERPT: Let Z be a Poisson algebra, and let A be a quantization of Z. Quantization cancellation conjecture. Z is Poisson cancellative if and only if A is cancellative. This propo...

conjecture | open | MathDB 页面 | 原文线索

V2-009752 | MATHDB-351788

Isomorphism conjecture for three deformed W-algebras of type $A_{2N}^{(2)}$

摘录 / EXCERPT: Isomorphism conjecture. The three algebras are isomorphic as associative algebras:

conjecture | open | MathDB 页面 | 原文线索

V2-009753 | MATHDB-351822

Structural characterization of Lie triple higher derivations on triangular algebras

摘录 / EXCERPT: Let $\mathfrak{A}=\operatorname{Tri}(\mathcal{A},\mathcal{M},\mathcal{B})$ be a triangular algebra, and let $\mathcal{L}=\{L_n\}_{n\in\mathbb{N}}$ be a sequence of $\mathcal{R}$-li...

open_problem | open | MathDB 页面 | 原文线索

V2-009754 | MATHDB-351823

Properness conjecture for Lie triple higher derivations on triangular algebras

摘录 / EXCERPT: Let $\mathfrak{A}=\operatorname{Tri}(\mathcal{A},\mathcal{M},\mathcal{B})$ be a triangular algebra, and let $\mathcal{L}=\{L_n\}_{n\in\mathbb{N}}$ be a Lie triple higher derivation...

open_problem | open | MathDB 页面 | 原文线索

V2-009755 | MATHDB-351827

Countability conjecture for minor equivalence classes of finite-domain clones

摘录 / EXCERPT: Countability conjecture. The minor equivalence relation on clones on finite domains has only countably many classes.

conjecture | open | MathDB 页面 | 原文线索

V2-009756 | MATHDB-351828

Three-element reduction conjecture for minor-preserving maps

摘录 / EXCERPT: Three-element reduction conjecture. Every clone on three elements that does not admit a minor-preserving map from ${\mathbf I}$ has a minor-preserving map to one of the following t...

conjecture | open | MathDB 页面 | 原文线索

V2-009757 | MATHDB-351865

Partial Schutzenberger action conjecture for the affine cactus group

摘录 / EXCERPT: Let $\overline{M_{0,n+2}^{comp}}$ be the compact real moduli space, and let the eigenlines of Bethe subalgebras on a tensor product of Kirillov-Reshetikhin representations form the...

conjecture | open | MathDB 页面 | 原文线索

V2-009758 | MATHDB-351866

Bender-Knuth monodromy conjecture for skew tableaux

摘录 / EXCERPT: Let J_{n+1} be the cactus group associated with the split real moduli space, and let the caterpillar points index the eigenbases of Bethe subalgebras on a tensor product of skew...

conjecture | open | MathDB 页面 | 原文线索

V2-009759 | MATHDB-351870

The total degree bounds conjecture for pseudo-reflection groups

摘录 / EXCERPT: Let G be a pseudo-reflection group. Write $\mathcal{SH}_G^{i,k}$ for the component of x-degree i and θ-degree k, let Δ_G be the discriminant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009760 | MATHDB-351883

Noncommutative torsion conjecture for Selmer groups over strongly admissible extensions

摘录 / EXCERPT: Let F_∞/F be a strongly admissible p-adic Lie extension: it is Galois, contains the cyclotomic $\mathbb{Z}_p$-extension, is unramified outside finitely many primes, and...

conjecture | open | MathDB 页面 | 原文线索

V2-009761 | MATHDB-351891

Non-commutative Moore sequence conjecture for idèle class groups

摘录 / EXCERPT: Non-commutative Moore sequence conjecture. One expects an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-009762 | MATHDB-351905

The Unreasonable Slightness Conjecture for elementary matrix groups

摘录 / EXCERPT: Let $dim=3$, 4, or 5. Let $\mathcal{A}$ be either an imaginary quadratic field when $dim=3$, a rational definite quaternion algebra equipped with an orthogonal involution...

conjecture | open | MathDB 页面 | 原文线索

V2-009763 | MATHDB-351936

Rigidity conjecture for regularity of intermediate ideals of edge ideals

摘录 / EXCERPT: Let G be a simple graph and let I be its edge ideal. For each integer $s\geq 1$, let J be an intermediate ideal in $\operatorname{Inter}(I^s,I^{(s)})$, the set of monomial id...

conjecture | open | MathDB 页面 | 原文线索

V2-009764 | MATHDB-351956

Banaszak-Kedlaya's algebraic Sato-Tate conjecture

摘录 / EXCERPT: Algebraic Sato-Tate conjecture. There is an algebraic subgroup $\operatorname{AST}(A)$ of GSp_{2g} over $\mathbb Q$, called the algebraic Sato-Tate group, such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-009765 | MATHDB-351971

Entropy coincidence conjecture for cluster algebras and periodic Laurent-property maps

摘录 / EXCERPT: Let ${\cal A}$ be a cluster algebra, or let φ be a map with the Laurent property obtained from a cluster algebra with periodicity in the sense of the cited constructions. L...

conjecture | open | MathDB 页面 | 原文线索

V2-009766 | MATHDB-351990

Universality of the Galois action on the thrice-punctured projective line

摘录 / EXCERPT: Universality conjecture. Any irreducible $\overline{\mathbb{Q}}_p$-representation of G_F that is almost everywhere unramified and de Rham at places above p can be established a...

conjecture | open | MathDB 页面 | 原文线索

V2-009767 | MATHDB-352024

Exel's Effros-Hahn conjecture for Steinberg algebras

摘录 / EXCERPT: Let K be a field, let $\mathcal{G}$ be an ample groupoid, and let $u\in\mathcal{G}^{(0)}$. Write $A_K(\mathcal{G})$ for the Steinberg algebra, $\mathcal{G}_u$ for the isotropy gr...

conjecture | open | MathDB 页面 | 原文线索

V2-009768 | MATHDB-352027

Stability conjecture for E8 weak Jacobi-form generator counts

摘录 / EXCERPT: For each index t, let $d_{k,t}$ be the number of weight-k generators of the free module $J_{*,E_8,t}^{\operatorname{w},W(E_8)}$ of $W(E_8)$-invariant weak Jacobi forms. Stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-009769 | MATHDB-352039

The nearby-cycles identification for visible stable polar representations

摘录 / EXCERPT: Let (G,V) be a visible stable polar representation for a connected reductive group G. Let $\pi\colon V\to V/\!/G$ be the quotient map, let P be its nearby cycles sheaf, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009770 | MATHDB-352069

Hochster's small Cohen-Macaulay conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a complete local ring. A small Cohen-Macaulay module over R is a nonzero finitely generated R-module S such that $\operatorname{depth}(S)=\dim(R)$....

conjecture | open | MathDB 页面 | 原文线索

V2-009771 | MATHDB-352099

Neguț's Kac-polynomial formula for the graded dimension of the shuffle algebra

摘录 / EXCERPT: Let Q be a quiver, let ${\mathcal B}_{\b0}$ be the slope-zero algebra graded by dimension vectors, and write ... where $A_{Q,\boldsymbol{n}}(t)$ is the Kac polynomial. Define ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-009772 | MATHDB-352114

The Furstenberg-boundary criterion for simplicity of the reduced group C*-algebra

摘录 / EXCERPT: Let G be a second countable locally compact group. Furstenberg-boundary simplicity conjecture. If some point of the Furstenberg boundary of G has a trivial stabilizer, then ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-009773 | MATHDB-352115

The boundary-representation C*-simplicity conjecture

摘录 / EXCERPT: Let G be a second countable locally compact group and let $\partial_F G$ be its Furstenberg boundary. For $z\in\partial_F G$, assume that z is a continuity point of the stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-009774 | MATHDB-352131

The Strong Bass conjecture

摘录 / EXCERPT: Let G be a group, let $r \colon K_0\mathbb{Z}G\rightarrow HH_0(\mathbb{Z}G)$ be the Hattori-Stallings trace map, and write the image of a finitely generated projective module P...

conjecture | open | MathDB 页面 | 原文线索

V2-009775 | MATHDB-352173

Geometric irreducibility and genus conjecture for Weierstrass inflectionary curves

摘录 / EXCERPT: Let $n=2\ell$, let F have characteristic zero or sufficiently positive characteristic, and let $\mathcal{C}^{\ell}_m\subset\mathbb{P}(1,2,1)$ be the inflectionary curve derived f...

conjecture | open | MathDB 页面 | 原文线索

V2-009776 | MATHDB-352176

D4 inflectionary curve genus conjecture

摘录 / EXCERPT: Let $n=2\ell$, assume that $\operatorname{char}(F)$ is zero or sufficiently positive, and let $\mathcal{C}_m$ be the m-th inflectionary curve from $y^n=x^5+x^3+sx$. D4 genus conj...

conjecture | open | MathDB 页面 | 原文线索

V2-009777 | MATHDB-352216

Super lambda-length conjecture for peripheral arcs in an annulus

摘录 / EXCERPT: Let p_n be the sequence used in the source and define $w_n=p_{2n-4}$. In the annulus with a marked point on each boundary, consider the peripheral arcs described in the source: a...

conjecture | open | MathDB 页面 | 原文线索

V2-009778 | MATHDB-352249

Algebraicity conjecture for the auxiliary series of the seven Weyl models

摘录 / EXCERPT: Consider the seven Weyl models in Table ... . Weyl-model algebraicity conjecture. For every one of these seven models, $A(x,y)$ is algebraic; consequently, $C(x,y)$ is D-finite. In a...

conjecture | open | MathDB 页面 | 原文线索

V2-009779 | MATHDB-352286

Lusztig's double-coset conjecture for minimally dominant elements

摘录 / EXCERPT: Let W be the Weyl group of a maximal torus contained in a Borel subgroup B of a twisted reductive group over an algebraically closed field. Let w be a minimal-length element...

conjecture | open | MathDB 页面 | 原文线索

V2-009780 | MATHDB-352287

Conjugacy invariance under root-avoiding simple reflection shifts

摘录 / EXCERPT: Let G be a twisted reductive group over an algebraically closed field, with Borel subgroup B. Let W be its corresponding Weyl group, let $w\in W$, and let s_i be a simple r...

conjecture | open | MathDB 页面 | 原文线索

V2-009781 | MATHDB-352288

Orbit-profile invariance conjecture for Nichols algebras

摘录 / EXCERPT: Orbit-profile invariance conjecture. Equality of the orbit profiles should preserve the stated dimension, or Gelfand-Kirillov dimension, under the conditions intended in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-009782 | MATHDB-352324

Cluster-adjacency conjecture for facets of amplituhedron positroid tiles

摘录 / EXCERPT: Cluster-adjacency conjecture. Each element of $\operatorname{Facet}(Z_{\pi})$ is a cluster variable for $\operatorname{Gr}_{m,n}$, and these cluster variables are pairwise compatib...

conjecture | open | MathDB 页面 | 原文线索

V2-009783 | MATHDB-352335

The finite-root-system conjecture for Nichols algebras of diagonal type

摘录 / EXCERPT: Let V be a braided vector space of diagonal type, and let $\mathscr{B}(V)$ be its Nichols algebra. The root system of $\mathscr{B}(V)$ is the root system arising from its restric...

conjecture | open | MathDB 页面 | 原文线索

V2-009784 | MATHDB-352393

Morier-Genoud--Ovsienko's unimodality conjecture for q-deformed rationals

摘录 / EXCERPT: Let M be an indecomposable type $\mathbb{A}$ module, and let h be the linearization of its submodule lattice that sends a submodule L to its total dimension ... Equivalently, i...

conjecture | open | MathDB 页面 | 原文线索

V2-009785 | MATHDB-352401

No broken diamond in hexagon conjecture for modified crystal operator paths

摘录 / EXCERPT: No broken diamond in hexagon conjecture. The paths E_i and E_{i+1} in the graph induced by the modified crystal operators do not contain any edges that give way to Stembridge E...

conjecture | open | MathDB 页面 | 原文线索

V2-009786 | MATHDB-352432

CLRW's Prym-Brill-Noether dimension conjecture for general k-gonal curves

摘录 / EXCERPT: Let $\pi \colon \widetilde{C} \to C$ be an unramified double cover of a general k-gonal curve, with $k \geq 3$. Let $\ell = \left\lceil \frac{k}{2} \right\rceil$ and define ... Her...

conjecture | open | MathDB 页面 | 原文线索

V2-009787 | MATHDB-352504

Fomin-Zelevinsky's tropical formula for g-vectors

摘录 / EXCERPT: Let $F_{i;v}$ be the F-polynomial associated with the cluster variable $x_{i;v}$, and let $\mathbf{g}(x_{i;v})=(g_1,\dots,g_m)$ be its g-vector. Write $u_1,\dots,u_m$ for tropica...

conjecture | open | MathDB 页面 | 原文线索

V2-009788 | MATHDB-352505

Finite-type cluster algebras satisfy property T0*

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra of finite cluster type, with property T0 as defined in the paper. Finite-type T0 conjecture. Property T0 is satisfied for any cluster algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-009789 | MATHDB-352619

Lara Rodríguez-Thakur rationality conjecture for the ratios $\alpha_{n,1}$

摘录 / EXCERPT: For each of the four class-number-one base rings A in Examples (i)-(iv), define ... The source introduces the corresponding variables X and Y as powers of θ and η i...

conjecture | open | MathDB 页面 | 原文线索

V2-009790 | MATHDB-352650

Length bound for strong Gröbner bases over rings

摘录 / EXCERPT: Let $\mathcal{P}$ be the polynomial algebra under consideration, and let $\mathcal{G}\subset\mathcal{P}\setminus\{0\}$ be a finite set whose polynomials involve monomials of length...

conjecture | open | MathDB 页面 | 原文线索

V2-009791 | MATHDB-352658

Morava K-theory dimension formula conjecture for real Grassmannians

摘录 / EXCERPT: Let $n\geq 1$, let ϵ be 0 or 1, and let $l\geq 0$. Set $m=2^{n+1}-\epsilon+2l$, and let $k_n(\operatorname{Gr}_d({\mathbb R}^m))$ denote the dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-009792 | MATHDB-352664

Fine Selmer torsion conjecture for Hida deformations

摘录 / EXCERPT: Retain the Hida-family settings of the source. Let $\mathcal{A}$ be the big Galois representation, let F_∞ be a $\mathbb{Z}_{p}$-extension of a number field F, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-009793 | MATHDB-352685

Nuclear images of sub-sℓ-monoids of negative cones

摘录 / EXCERPT: Nuclear-image conjecture. The nuclear images of sub-sℓ-monoids of negative cones of ℓ-groups are precisely the integral sℓ-monoids.

conjecture | open | MathDB 页面 | 原文线索

V2-009794 | MATHDB-352686

Finite nuclear images of conuclear images of negative cones

摘录 / EXCERPT: Finite nuclear-image conjecture. The finite nuclear images of conuclear images of negative cones of ℓ-groups are precisely the finite integral residuated lattices.

conjecture | open | MathDB 页面 | 原文线索

V2-009795 | MATHDB-352726

The categorical half-twist conjecture

摘录 / EXCERPT: Categorical half-twist conjecture. If w lies in the cell λ, then

conjecture | open | MathDB 页面 | 原文线索

V2-009796 | MATHDB-352765

The characterization of toric vanishing ideals for colored Gaussian graphical models

摘录 / EXCERPT: Toric vanishing ideal conjecture. $I_{\mathcal{G}}$ is toric if and only if $\mathcal{G}$ is an RCOP block graph.

conjecture | open | MathDB 页面 | 原文线索

V2-009797 | MATHDB-352804

The multiplicative lower bound conjecture for joint covering gonality of curves

摘录 / EXCERPT: Let X and Y be very general curves of genera $g_X,g_Y\geq 1$, and let $\operatorname{cov.gon}(X,Y)$ denote their joint covering gonality. The multiplicative covering-gonality c...

conjecture | open | MathDB 页面 | 原文线索

V2-009798 | MATHDB-352840

Herzog-Hibi-Ohsugi conjecture on powers of cover ideals of chordal graphs

摘录 / EXCERPT: Herzog-Hibi-Ohsugi conjecture. If G is chordal, then

conjecture | open | MathDB 页面 | 原文线索

V2-009799 | MATHDB-352867

The symmetric-support conjecture for general polynomial orbits

摘录 / EXCERPT: Let K be a field, let $d4d0 \subseteq (\mathbb{Z}_{\geq 0})^n$ be a support set, and let $f \in K^\mathcal{A}$. Denote by k the minimal number of strictly positive entries amon...

conjecture | open | MathDB 页面 | 原文线索

V2-009800 | MATHDB-352871

The minus component of the equivariant Tamagawa number conjecture in determinant form

摘录 / EXCERPT: Let H/F be a finite abelian CM-extension, let $G=\operatorname{Gal}(H/F)$, and let Σ,Σ' be finite sets of places satisfying the hypotheses used to define the perfect...

conjecture | open | MathDB 页面 | 原文线索

V2-009801 | MATHDB-352877

Bijectivity conjecture for orthogonal maps fixing the radical

摘录 / EXCERPT: Let R be a commutative ring and let (M,q) be a quadratic module over R. Write $M^{\perp}$ for the radical of the polar form of q, and let an orthogonal map mean an R-line...

conjecture | open | MathDB 页面 | 原文线索

V2-009802 | MATHDB-352878

Twisted-center unit conjecture for Clifford algebras

摘录 / EXCERPT: Let A be a commutative ring with unit in which $2\neq0$, let (N,Q) be a quadratic module over A, and set $\mathcal{C}:=\mathcal{C}(N,Q)$. Let e be the idempotent used to fo...

conjecture | open | MathDB 页面 | 原文线索

V2-009803 | MATHDB-352947

The non-noetherianity conjecture for infinite-dimensional subalgebras of the Witt algebra

摘录 / EXCERPT: Let $W_{\geq -1}$ be the Lie algebra of polynomial vector fields on the affine line with degrees at least -1, and let $U(-)$ denote the universal enveloping algebra. Non-noetheria...

conjecture | open | MathDB 页面 | 原文线索

V2-009804 | MATHDB-352948

The nested-subalgebra classification conjecture for infinite-codimension Witt subalgebras

摘录 / EXCERPT: Let $W_{\geq -1}$ be the Lie algebra of polynomial vector fields on the affine line with degrees at least -1, let $\Bbbk$ be the base field, and let $L(f)$ and $L(f,g)$ be the sub...

conjecture | open | MathDB 页面 | 原文线索

V2-009805 | MATHDB-352949

Buchweitz-Greuel-Schreyer conjecture on minimal MCM rank

摘录 / EXCERPT: Let P be a regular local ring with algebraically closed residue field and let $f\in P$. Define ... where $\operatorname{Sing}(P/f)$ denotes the singular locus of P/f. Buchweitz-G...

conjecture | open | MathDB 页面 | 原文线索

V2-009806 | MATHDB-352974

Lech's conjecture on Hilbert-Samuel multiplicities

摘录 / EXCERPT: Lech's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-009807 | MATHDB-352983

The Hodge-Tate weight conjecture for crystalline Breuil-Kisin modules

摘录 / EXCERPT: Assume $K=K_{{\mathrm{cyc}}}\cap K_\infty$. Let $\mathfrak M$ be a crystalline Breuil-Kisin module of rank d, let $\lambda:=\frac{\xi}{E([\pi^{\flat}])}$, and let $\widetilde A$...

conjecture | open | MathDB 页面 | 原文线索

V2-009808 | MATHDB-352984

The contracted ideal monomiality conjecture

摘录 / EXCERPT: Let N be a positive integer, and let $\mathfrak{J}(N)=J(N)\cap k[x,y]$ be the contracted ideal. The contracted ideal monomiality conjecture. ... The preceding theorem proves contai...

conjecture | open | MathDB 页面 | 原文线索

V2-009809 | MATHDB-352985

The many-variable containment conjecture for symmetric ideals

摘录 / EXCERPT: The many-variable containment conjecture. For any $r\geq 2$ and $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-009810 | MATHDB-353005

The conjecture on polynomial class-sum bases for centers of Hecke algebras

摘录 / EXCERPT: Polynomial class-sum basis conjecture. There exists a choice of a basis $\{b_w:w\in W\}$ of H and a choice of conjugacy class representatives $\{w_C:C\in\operatorname{Cl}(W)\}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-009811 | MATHDB-353011

Gerstenhaber algebra conjecture for Ext groups of commuting central monoid-comonoid pairs

摘录 / EXCERPT: Let ${\mathscr{C}}$ be a monoidal category fulfilling a few mild conditions. Let Z be a braided commutative monoid in the left weak monoidal centre of ${\mathscr{C}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009812 | MATHDB-353036

The Débes-Deschamps conjecture on finite split embedding problems

摘录 / EXCERPT: Let k be a Hilbertian field. A finite split embedding problem over k is an epimorphism ... where G is finite, L/k is Galois, and there is an embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-009813 | MATHDB-353041

Exceptional-collection conjecture for rational homogeneous spaces

摘录 / EXCERPT: Let G be a semisimple algebraic group and let $\mathrm{P}\subset \mathrm{G}$ be a parabolic subgroup. Let ${\mathbf D^{\mathrm{b}}}(\mathrm{G}/\mathrm{P})$ denote the bo...

conjecture | open | MathDB 页面 | 原文线索

V2-009814 | MATHDB-353063

Eventual sign periodicity conjecture for the weight two Euler characteristic of moduli spaces of curves

摘录 / EXCERPT: Let ${\mathcal M}_g$ denote the moduli space of smooth curves of genus g, and let $\chi_2({\mathcal M}_g)$ be its weight two compactly supported Euler characteristic. Eventual si...

conjecture | open | MathDB 页面 | 原文线索

V2-009815 | MATHDB-353108

The Galois-conjugate description of the principal-vein Master Teapot

摘录 / EXCERPT: The principal-vein Master Teapot conjecture. The set $\Upsilon_{p/q}$ is given by the closure of the pairs consisting of $e^{h(\theta)}$ and its Galois conjugates, as displayed abo...

conjecture | open | MathDB 页面 | 原文线索

V2-009816 | MATHDB-353149

Conjecture on Fuchsian types of three Nakayama algebras

摘录 / EXCERPT: Fuchsian-type conjecture. The following three statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-009817 | MATHDB-353188

Finiteness conjecture for omega-primality of algebraic cyclic free semirings

摘录 / EXCERPT: Omega-primality finiteness conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-009818 | MATHDB-353191

The cluster variable positivity conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be a cluster algebra with an initial cluster X. Positivity conjecture. Every cluster variable of ${\mathcal A}$ is a Laurent polynomial in the cluster variable...

conjecture | open | MathDB 页面 | 原文线索

V2-009819 | MATHDB-353208

MV-basis cluster conjecture for the unipotent coordinate ring

摘录 / EXCERPT: MV-basis cluster conjecture. The MV basis for $\mathbb C[N]$ contains all cluster monomials. Moreover, every element of the MV basis is g-pointed in each cluster.

conjecture | open | MathDB 页面 | 原文线索

V2-009820 | MATHDB-353257

Polynomial codimension conjecture for Jordan net orbits

摘录 / EXCERPT: Polynomial codimension conjecture. The codimensions of $A^{(2)}_{r,k_1,k_2}$, $A^{(3)}_{k_1,k_2,k_3}$, and $B^{(2)}_{k,\ell_1,\ell_2}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-009821 | MATHDB-353258

Nonexistence conjecture for the degeneration $B^{(1)}_3\not\to C_{6,6}$

摘录 / EXCERPT: Nonexistence conjecture. For $n=6$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-009822 | MATHDB-353291

Conservativity conjecture for transcendental motives of surfaces

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p\geq 0$, let X be a smooth projective surface over k, and let $t_{2}(X)$ denote its transcendental Chow motive with...

conjecture | open | MathDB 页面 | 原文线索

V2-009823 | MATHDB-353350

Non-finite generation conjecture for cluster algebras of positive-genus surfaces

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a surface of genus g with n punctures, and let $\mathcal{A}(\Sigma_{g,n})$ denote its cluster algebra. Non-finite generation conjecture. For $g\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-009824 | MATHDB-353351

Inequality conjecture between cluster and upper cluster algebras

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a surface of genus g with n punctures, and let $\mathcal{A}(\Sigma_{g,n})$ and $\mathcal{U}(\Sigma_{g,n})$ denote its cluster and upper cluster algebras,...

conjecture | open | MathDB 页面 | 原文线索

V2-009825 | MATHDB-353352

Roger-Yang injectivity conjecture for the curve algebra representation

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a punctured surface, let $\mathcal{C}(\Sigma_{g,n})_{\mathbb{C}}$ be its complexified curve algebra, and let $\mathcal{T}^{d}(\Sigma_{g,n})$ be its decorated...

conjecture | open | MathDB 页面 | 原文线索

V2-009826 | MATHDB-353353

Curve and upper cluster algebra comparison conjecture for punctured surfaces

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a surface of genus g with n punctures, let $\mathcal{C}(\Sigma_{g,n})$ be its curve algebra, let $\mathcal{C}(\Sigma_{g,n})'$ be the subalgebra generated...

conjecture | open | MathDB 页面 | 原文线索

V2-009827 | MATHDB-353396

Conjecture on the greenness of the mutation sequence for \(\Gamma_{\overline{w}}\)

摘录 / EXCERPT: Greenness conjecture. The sequence of vertices of $\Gamma_{\overline{w}}$ mutated by $\mu_\bullet$ is a green sequence with respect to $\Gamma_{\overline{w}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009828 | MATHDB-353458

Free action conjecture for the Duflo-Serganova functor on Kac-modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a type I Lie superalgebra, let $u\neq 0$ satisfy $[u,u]=0$, and let $\mathbb{O}\subseteq\widetilde{G_u}$ be the purely odd supergroup with Lie superalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-009829 | MATHDB-353463

The domain conjecture for completed central reductions of enveloping algebras

摘录 / EXCERPT: Let L be the coefficient field, let $\mathfrak{h}$ be a semisimple L-Lie algebra, let $\mathfrak{h}_0\subset\mathfrak{h}$ be an $\mathcal{O}$-Lie lattice, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009830 | MATHDB-353518

Completeness of the intermediate-growth scale for finitely generated Lie PI-algebras

摘录 / EXCERPT: Let L be a finitely generated Lie PI-algebra. Write $\gamma_L(n)$ for its growth function, and let $\ln^{(q)}n$ denote the q-fold iterated natural logarithm. Intermediate-growt...

conjecture | open | MathDB 页面 | 原文线索

V2-009831 | MATHDB-353522

Strong Lefschetz conjecture for Jacobian algebras of smooth hypersurfaces

摘录 / EXCERPT: Let $S=\mathbb{C}[x_0,\ldots,x_n]$ be the graded polynomial ring in $n+1$ variables, with $n\geq 2$, and let $f\in S_d$ be a homogeneous polynomial such that the hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-009832 | MATHDB-353542

Universal Smith normal form conjecture for the Adinkra Laplacian

摘录 / EXCERPT: Let L be the Laplacian of an Adinkra, and let $\hat{L}$ be its polynomial lift over $\mathbb{Z}[x]$. Write the Smith normal form of L as a tuple of invariant factors, with mult...

conjecture | open | MathDB 页面 | 原文线索

V2-009833 | MATHDB-353598

The geometric-to-strict compatibility conjecture

摘录 / EXCERPT: Geometric-to-strict compatibility conjecture. Any semisimple geometric representation

conjecture | open | MathDB 页面 | 原文线索

V2-009834 | MATHDB-353599

The strict compatibility and automorphy conjecture

摘录 / EXCERPT: Strict compatibility and automorphy conjecture. Any weakly compatible system of Galois representations is strictly compatible, and is in addition automorphic: there is an algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-009835 | MATHDB-353600

Nilpotent support conjecture for type I graded Lie algebras

摘录 / EXCERPT: Let G be the connected reductive group with involution θ, let $K=G^\theta$, and let $\mathfrak g_1$ be the degree-one component of the associated graded Lie algebra. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-009836 | MATHDB-353627

Squarefree initial ideal conjecture for stretched lattice join-meet ideals

摘录 / EXCERPT: Let $k\geq 3$ and let $(n_1,\ldots,n_k)\neq(1,\ldots,1)$. Let $L_k(n_1,\ldots,n_k)$ be the stretched finite lattice with chains ... for $i=1,\ldots,k$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009837 | MATHDB-353645

Locally analytic period rings as power-series rings in the cyclotomic case

摘录 / EXCERPT: Let K be a p-adic field, let K_∞/K be an infinitely ramified p-adic Lie extension containing a cyclotomic extension, and let d be the dimension of its Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-009838 | MATHDB-353650

Uniqueness conjecture for non-abelian simple A-H Frobenius monoids

摘录 / EXCERPT: Uniqueness conjecture. Non-degenerate standard A-H Frobenius monoids differ only by their scalars. More precisely, but less generally, all non-abelian simple A-H F. monoids are iso...

conjecture | open | MathDB 页面 | 原文线索

V2-009839 | MATHDB-353730

Weighted Erdős-Burgess lower-bound conjecture for finite rings

摘录 / EXCERPT: Weighted Erdős-Burgess lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009840 | MATHDB-353740

Universality conjecture for the universal bulk-boundary system

摘录 / EXCERPT: Let $\mathcal T$ be a Poisson BV theory on N, and let $\mathcal Z(\mathcal T)$ be its universal bulk theory. Let ... be the respective projections. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-009841 | MATHDB-353763

Uniform boundedness conjecture for semisimple algebraic groups

摘录 / EXCERPT: Let G be a semisimple linear algebraic group over an infinite field k. Write $G^+(k)$ for the subgroup generated by the subgroups $R_u(Q)(k)$, where Q ranges over all minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-009842 | MATHDB-353798

Characteristic-p Lefschetz conjecture for codimension-three Artinian Gorenstein algebras

摘录 / EXCERPT: Let A be an Artinian Gorenstein algebra of codimension three, with socle degree j_A, over an infinite field k of characteristic $p>j_A$. For a general enough linear...

conjecture | open | MathDB 页面 | 原文线索

V2-009843 | MATHDB-353877

Infinite-level A2 identity for H_(infinity,a,b)

摘录 / EXCERPT: Let $a,b\in\mathbb{Z}_{\geq 0}\cup\\{\infty\\}$, and let $H_{(\infty,a,b)}(1,q)$ be the specialization at $z=1$ of the function defined from the infinite (z,q) sum...

conjecture | open | MathDB 页面 | 原文线索

V2-009844 | MATHDB-353878

Seed truncation conjecture for finite-level A2 identities

摘录 / EXCERPT: Let $m\in 3k+\{-1,0,1\}$, let $c=(c_0,c_1,c_2)$ be a composition of $m-3$, and let $H_{(c_0,c_1,c_2)}$ and $S_m(\rho\mid\sigma)$ be the finite-level functions and truncated (z,q)...

conjecture | open | MathDB 页面 | 原文线索

V2-009845 | MATHDB-353905

Hanes's asymptotic Lech inequality conjecture

摘录 / EXCERPT: Let $(R,\operatorname{\mathfrak{m}})$ be a Noetherian local ring of dimension $d\geq 1$. Write $\widehat{R}$ for its completion, $\operatorname{Spec}\widehat{R}$ for its spectrum,...

conjecture | open | MathDB 页面 | 原文线索

V2-009846 | MATHDB-353906

Weak Hanes conjecture for boundedly generated ideals

摘录 / EXCERPT: Let $d\geq 2$ and $N\geq d$ be positive integers. Let $(R,\operatorname{\mathfrak{m}})$ be a Noetherian local ring of dimension d, and let I be an $\operatorname{\mathfrak{m}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-009847 | MATHDB-353955

Shimura's conjecture on the algebraic independence of period symbols

摘录 / EXCERPT: Let K) be a CM field of degree $2n$, and let $\{\sigma_i\}_{i=1}^n$ be a CM type of K. The associated Shimura period symbols $p_K(\sigma_i,\operatorname{id})$ are elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-009848 | MATHDB-353956

Yoshida's conjecture on CM period symbols and pi

摘录 / EXCERPT: Let K be a CM field that is Galois over $\mathbb{Q}$, and let Φ be a CM type of K. The quantities $p_K(\operatorname{id},\sigma)$ are Shimura period symbols, and...

conjecture | open | MathDB 页面 | 原文线索

V2-009849 | MATHDB-353966

Harbourne's symbolic-power containment conjecture

摘录 / EXCERPT: Let $\mathbb{k}$ be a field, let $R=\mathbb{k}[x_1,\ldots,x_d]$, and let $I\subseteq R$ be a radical homogeneous ideal of bigheight h. Harbourne's conjecture. One has ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-009850 | MATHDB-353971

Smirnov's stability conjecture for Hilbert-Samuel coefficients of Frobenius powers

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a Cohen-Macaulay local ring of dimension d, and let I be an $\mathfrak m$-primary ideal. The ideal I is stable if it satisfies $I^2=QI$ for a minimal...

conjecture | refuted | MathDB 页面 | 原文线索

V2-009851 | MATHDB-354001

Persistence Conjecture for arithmetic hyperbolicity of algebraic stacks

摘录 / EXCERPT: Persistence Conjecture. If X is arithmetically hyperbolic over k, then X is absolutely arithmetically hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-009852 | MATHDB-354012

The constancy conjecture for the regulator on the Lie coalgebra

摘录 / EXCERPT: Let $\mathbb L(F)$ be the Lie coalgebra of multiple polylogarithms over a field F, with cobracket δ, and let $\mathbb L^{\operatorname{symb}}(\mathbb C)$ be its symbolic v...

conjecture | open | MathDB 页面 | 原文线索

V2-009853 | MATHDB-354023

Coates-Sujatha's generalized $\b5=0$ conjecture

摘录 / EXCERPT: For each $i=0,\ldots,p-2$, let $\mathbb{H}^2(j_*T_{f,i}(k-r))$ be the finitely generated torsion $\Lambda^{(i)}$-module appearing in Kato's main conjecture. Coates-Sujatha's genera...

conjecture | open | MathDB 页面 | 原文线索

V2-009854 | MATHDB-354049

Bley-Burns-Hahn conjecture on Galois-Gauss sums

摘录 / EXCERPT: Let L/K be a weakly ramified Galois extension of number fields of odd degree. Write $\mathfrak{a}_{L/K}$ for the canonical pre-image associated with the stable-isomorphism class...

conjecture | open | MathDB 页面 | 原文线索

V2-009855 | MATHDB-354056

The projective-dimension conjecture for initial ideals of $L_2(n_1,1)$

摘录 / EXCERPT: Let n_1 be a positive integer, and let $R_{n_1,1}$ be the ring and $I_{L_2(n_1,1)}$ the ideal appearing in the paper, with $\mathrm{in}_{\prec}(I_{L_2(n_1,1)})$ denoting its init...

conjecture | open | MathDB 页面 | 原文线索

V2-009856 | MATHDB-354076

The stable Harbourne conjecture for radical ideals

摘录 / EXCERPT: Let I be a radical ideal of big height c in a regular ring. For each positive integer n, let $I^{(n)}$ denote the nth symbolic power of I. Stable Harbourne conjecture. On...

conjecture | open | MathDB 页面 | 原文线索

V2-009857 | MATHDB-354079

Gorenstein conjecture for the edge algebra of Γ_{3k+2}

摘录 / EXCERPT: Let $R_{\Gamma_{3k+2}}$ be the edge algebra associated with the graph Γ_{3k+2}. Gorenstein conjecture for $R_{\Gamma_{3k+2}}$. The algebra $R_{\Gamma_{3k+2}}$ is Gorenstein...

conjecture | open | MathDB 页面 | 原文线索

V2-009858 | MATHDB-354129

Merino's Lissajous decomposition conjecture for Chebyshev polynomials

摘录 / EXCERPT: Merino's conjecture. This curve is the union of a finite number of Lissajous curves. The conjecture concerns the decomposition of the Chebyshev-defined curve into parametrized Liss...

conjecture | open | MathDB 页面 | 原文线索

V2-009859 | MATHDB-354153

Intrinsic equivalence under free-group basis changes

摘录 / EXCERPT: Basis-change conjecture. The subshifts $X_\pm$ and $Y_\pm$ are intrinsically equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-009860 | MATHDB-354154

Intrinsic equivalence under recognizable homomorphisms of free groups

摘录 / EXCERPT: Recognizable-image conjecture. The subshifts $X_\pm$ and $Y_\pm$ are intrinsically equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-009861 | MATHDB-354160

The spetsial-parameter coefficient conjecture for canonical basis vectors

摘录 / EXCERPT: Let $\mathbf{s}=(s+1,s,\ldots,s)$ be the spetsial multicharge, and let λ be a cylindric l-partition. Write ... and let $m(\lambda)$ be the multipartition selected by the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-009862 | MATHDB-354208

Hall-Littlewood positivity conjecture for orbit harmonics

摘录 / EXCERPT: Let $Z\subset\mathbb{C}^n$ be any finite S_n-stable set such that $\mathbb{C}[x_1,\dots,x_n]/\operatorname{gr}\mathbf{I}(Z)$ is a graded S_n-module, where $\mathbf{I}(Z)$ is th...

conjecture | open | MathDB 页面 | 原文线索

V2-009863 | MATHDB-354210

Fields Group superspace harmonic-generation conjecture

摘录 / EXCERPT: Let $d_j:\Omega_n\to\Omega_n$ be defined for $j\geq1$ by ... and let $\delta_n\in\mathbb{C}[x_1,\dots,x_n]$ be the Vandermonde determinant. Let SR_n be the superspace coinvariant a...

conjecture | open | MathDB 页面 | 原文线索

V2-009864 | MATHDB-354238

Vogan's orbit method conjecture for real reductive groups

摘录 / EXCERPT: Let $G_{\mathbb R}$ be the real points of a connected reductive algebraic group, and let $\mathsf{Cov}^{i\mathbb R}(G_{\mathbb R})$ denote the real co-adjoint covers of...

conjecture | open | MathDB 页面 | 原文线索

V2-009865 | MATHDB-354255

Rank-two realization conjecture for finite Weyl groupoids

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and let ${\bf q}\in\mathbb{C}^{\{1,2\}^d}$ be a d-dimensional tensor defining a Weyl groupoid through the generalized Rosso condition. Rank-two realization c...

conjecture | open | MathDB 页面 | 原文线索

V2-009866 | MATHDB-354259

The LFED Conjecture on images of locally finite derivations

摘录 / EXCERPT: The LFED Conjecture. Images of locally finite K-derivations and K-$\mathcal{E}$-derivations of K-algebras are Mathieu subspaces.

conjecture | open | MathDB 页面 | 原文线索

V2-009867 | MATHDB-354336

Burns's Fitting ideal conjecture for Selmer modules

摘录 / EXCERPT: Let H/F be a finite abelian CM extension, let $R=\mathbf{Z}[1/2][G]^-$, and define the Selmer module ... with contragredient G-action. Let $\Theta_{S,T}$ be the smoothed Stickelb...

conjecture | open | MathDB 页面 | 原文线索

V2-009868 | MATHDB-354373

Invertibility conjecture for powers of lattices in rank-n algebras

摘录 / EXCERPT: Lattice-power invertibility conjecture. The lattice $\mathfrak c^{n-1}$ and every higher power $\mathfrak c^m$ with $m\geq n-1$ are invertible in their common endomorphism ring.

conjecture | open | MathDB 页面 | 原文线索

V2-009869 | MATHDB-354376

Fourier-Dumitrescu fusion-product conjecture for restricted local Weyl modules

摘录 / EXCERPT: Let $W_1,\dots,W_r$ be the modules considered in the preceding construction, let $\mathcal{A}(\lambda_1,\dots,\lambda_r)$ be the associated parameter algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009870 | MATHDB-354415

Speculation on twisted Gelfand-Graev representations and shifted SSV representations

摘录 / EXCERPT: Speculation on the shifted representation. For every $z\in P$, there is an isomorphism of $\mathcal{H}_I$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-009871 | MATHDB-354421

The radical splitting conjecture for connected algebraic groups

摘录 / EXCERPT: Let G be a connected algebraic group over a field k. Radical splitting conjecture. One has ... This would follow from a positive answer to Tits's question of whether torsors unde...

conjecture | open | MathDB 页面 | 原文线索

V2-009872 | MATHDB-354493

The singular orbit conjecture for trinomial hypersurfaces

摘录 / EXCERPT: Let X be a trinomial hypersurface, and let ${\mathbb T}$ act on X by the torus action considered in the paper. Singular orbits are points or subsets of X consisting entirely...

conjecture | open | MathDB 页面 | 原文线索

V2-009873 | MATHDB-354503

Quéguiner-Mathieu-Semenov-Zainoulline conjecture on the J-invariant after generic splitting

摘录 / EXCERPT: Let (A,σ) be a central simple algebra with orthogonal involution and trivial discriminant, and let $G=\operatorname{PGO}^+(A,\sigma)$ be the connected component of its autom...

conjecture | open | MathDB 页面 | 原文线索

V2-009874 | MATHDB-354504

Existence of non-central idempotents in simple Lie algebras of characteristic 2

摘录 / EXCERPT: Non-central idempotent conjecture. Every finite-dimensional simple Lie algebra over a field of characteristic 2 has a non-central idempotent, and hence a non-degenerate...

conjecture | open | MathDB 页面 | 原文线索

V2-009875 | MATHDB-354511

Weak Fontaine-Mazur conjecture for unramified pro-p extensions

摘录 / EXCERPT: Let K be a number field. An unramified pro-p-extension is a Galois pro-p extension of K unramified at every place. Weak Fontaine-Mazur conjecture. Every unramified pro-p-...

conjecture | open | MathDB 页面 | 原文线索

V2-009876 | MATHDB-354512

Uniform weak Fontaine-Mazur conjecture

摘录 / EXCERPT: Let K be a number field and let L/K be an everywhere unramified Galois pro-p extension. A pro-p group is uniform if it is a uniform pro-p group in the sense of Lazard. Un...

conjecture | open | MathDB 页面 | 原文线索

V2-009877 | MATHDB-354523

Conjecture on the Noether different and the Frobenius annihilator ideal

摘录 / EXCERPT: Let S be the graded ring, T its degree-zero subring, $S^e=S\boxtimes_T S$, and let $\mathbb D$ denote the relevant graded dualizing object. Let $\mathfrak m$ be the homogeneous...

conjecture | open | MathDB 页面 | 原文线索

V2-009878 | MATHDB-354528

Abhyankar's conjecture for the affine line

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let $G=p(G)$ be a finite quasi-p group. Abhyankar's conjecture. There exists a Galois extension $K/k(T)$ ram...

conjecture | open | MathDB 页面 | 原文线索

V2-009879 | MATHDB-354536

Rigidity conjecture for post-Lie structures on reductive and semisimple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ and $\mathfrak{n}$ be Lie algebras, with $\mathfrak{g}$ reductive and $\mathfrak{n}$ semisimple. Suppose that the pair $(\mathfrak{g},\mathfrak{n})$ admits a pos...

conjecture | open | MathDB 页面 | 原文线索

V2-009880 | MATHDB-354582

Scattering fan-mutation fan coincidence conjecture

摘录 / EXCERPT: Let B be an exchange matrix. The scattering fan-mutation fan coincidence conjecture. The scattering fan $\operatorname{ScatFan}^T\!(B)$ coincides with the mutation fan...

conjecture | open | MathDB 页面 | 原文线索

V2-009881 | MATHDB-354630

The noncommutative Auslander conjecture for semisimple Hopf actions

摘录 / EXCERPT: Noncommutative Auslander conjecture. The map φ is an isomorphism of graded algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-009882 | MATHDB-354684

Local inverse Simpson quasi-isomorphism conjecture

摘录 / EXCERPT: Let (M,θ_M,ϕ_M) be an enhanced log Higgs module over $R[\frac{1}{p}]$, and let $V(M,\theta_M,\phi_M)$ be the associated representation of Γ. Local inverse Simpson...

conjecture | open | MathDB 页面 | 原文线索

V2-009883 | MATHDB-354712

Kollár's canonical algebra conjecture for rational surface singularities

摘录 / EXCERPT: Kollár's conjecture. The canonical algebra is a finitely generated $\mathcal{O}_{\mathcal{X}}$-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-009884 | MATHDB-354757

The remainder vanishing conjecture for Jacobian pairs

摘录 / EXCERPT: The remainder vanishing conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-009885 | MATHDB-354758

The divisibility and support descent conjecture

摘录 / EXCERPT: The divisibility and support descent conjecture. The following implications hold: for $w=(0,1)$ and $i\in[m(n-m)(a-1)/a+1,m(n-m)]$, the i-th w-divisibility conditions imply the...

conjecture | open | MathDB 页面 | 原文线索

V2-009886 | MATHDB-354790

Palindromicity and sign-pattern conjecture for a graded Euler characteristic

摘录 / EXCERPT: For $n\geq 0$, define ... where $S^o[n,k]$ are the ordered type A q-Stirling polynomials. Palindromicity and sign-pattern conjecture. The polynomial $F_n(q)$ is palindromic after...

conjecture | open | MathDB 页面 | 原文线索

V2-009887 | MATHDB-354799

Locality compatibility of the universal enveloping ideal

摘录 / EXCERPT: Let $(\mathfrak{g},\top_{\mathfrak{g}},[\,\])$ be a locality Lie algebra, let $\mathcal{T}_\top(\mathfrak{g})$ be its locality tensor algebra, and let $J_\top(\mathfrak{g})$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-009888 | MATHDB-354812

Bardakov-Passi-Singh conjecture on augmentation quotients of dihedral quandles

摘录 / EXCERPT: Let $\mathbb{R}_n=\{a_0,a_1,\cdots,a_{n-1}\}$ denote the dihedral quandle of order n, and let $\Delta^k(\mathbb{R}_n)$ denote the kth power of its augmentation ideal. Bardakov-...

conjecture | open | MathDB 页面 | 原文线索

V2-009889 | MATHDB-354834

Haiman's h-positivity conjecture for Kazhdan-Lusztig characters

摘录 / EXCERPT: Haiman's conjecture. For any $w\in S_n$, the dual Frobenius character $\operatorname{ch}(q^{\frac{\ell(w)}{2}}C'_w)$ is h-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-009890 | MATHDB-354837

The conjecture for $\mathbf{Q}_{m,n}^{k}(1)$

摘录 / EXCERPT: Let M and N be positive integers, and define ... Let $L_{M,N}$ be the symmetric function associated with the M,N-path tuples, let $\mathbf{Q}_{m,n}$ be the elliptic Hall algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-009891 | MATHDB-354858

Jones-Lee refinement for cyclic reductions in arithmetic progressions

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$ of conductor N_E, and let $a,n\in\mathbb{N}$ satisfy $\gcd(a,n)=1$. Let $\pi_{E,a,n}(x)$ count the primes $p\leq x$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-009892 | MATHDB-354868

Injectivity conjecture for evaluation maps of linearized polynomials

摘录 / EXCERPT: Injectivity conjecture. If $1\le r\le q-1$, then the evaluation map ϵ_r is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-009893 | MATHDB-354910

Block shuffle relation for refined symmetric multiple zeta values

摘录 / EXCERPT: Let $\mathfrak{X}_{\mathrm{ev}}'$, $\mathfrak{X}_{\mathrm{ev}}^+$, $\mathfrak{X}_{\mathrm{od}}$, $\mathcal{H}$, and L_B° be the parity-graded spaces and block-regularization...

conjecture | open | MathDB 页面 | 原文线索

V2-009894 | MATHDB-354913

Refined Chern-Simons representation conjecture for the spherical DAHA module

摘录 / EXCERPT: Let q and t satisfy the shortening condition defining the $(k+1)$-dimensional module $\repV_{k+1}$ of the spherical double affine Hecke algebra, with basis indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-009895 | MATHDB-354914

Argyres-Douglas modular representation conjecture for the DAHA module

摘录 / EXCERPT: Let $\scD^{(1)}_{\ell}$ be the ℓ-dimensional module singled out among the four modules in the shortening regime, with basis indexed by $0\leq j\leq \ell-1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-009896 | MATHDB-354961

Shangguan's generic Vandermonde full-rank conjecture

摘录 / EXCERPT: Shangguan's full-rank conjecture. The matrix $M_{G,(J_1,\ldots,J_t)}$ has full column rank.

conjecture | open | MathDB 页面 | 原文线索

V2-009897 | MATHDB-354969

The full Perazzo conjecture for minimal Artinian Gorenstein Hilbert functions

摘录 / EXCERPT: The full Perazzo conjecture. The Hilbert vector H is minimal in $\mathcal{AG}(r,d)$; equivalently, if $\hat{H}\preceq H$ is a comparable Artinian Gorenstein Hilbert vector, then...

conjecture | open | MathDB 页面 | 原文线索

V2-009898 | MATHDB-354970

The strong full Perazzo conjecture for strongly minimal Hilbert functions

摘录 / EXCERPT: The strong full Perazzo conjecture. The Hilbert vector H is strongly minimal in $\mathcal{AG}(r,d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-009899 | MATHDB-354980

Top-degree cocenter basis conjecture for cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Let $\Lambda\in P^+$ and $\alpha\in Q^+$. Let $\mathscr{R}_{\alpha}^{\Lambda}$ be the cyclotomic quiver Hecke algebra over a ground field K, let $d_{\Lambda,\alpha}$ denote the r...

conjecture | open | MathDB 页面 | 原文线索

V2-009900 | MATHDB-355012

Hazrat's full-faithfulness conjecture for the graded Grothendieck-group functor

摘录 / EXCERPT: Let ℓ be a field. Consider the category of unital Leavitt path ℓ-algebras with graded homomorphisms modulo inner automorphisms, and the category of preordered abelian gro...

conjecture | open | MathDB 页面 | 原文线索

V2-009901 | MATHDB-355051

Local-global compatibility conjecture for the Galois representations

摘录 / EXCERPT: Let F, S, T, Q, Σ, the Hecke algebras $\mathbb{T}_{\Sigma,Q}$ and $\widetilde{\mathbb{T}}^S(K_{\Sigma,Q})$, and the universal deformation rings $R_{\Sigma,Q}$ be as...

conjecture | open | MathDB 页面 | 原文线索

V2-009902 | MATHDB-355059

Kaplansky-Lvov conjecture for multilinear polynomial images on matrix algebras

摘录 / EXCERPT: Let p be a multilinear polynomial in m variables over a field k, and let Φ_p be the associated polynomial map on $M_n(k)$. Write $\mathfrak{sl}_n(k)$ for the set of trac...

conjecture | open | MathDB 页面 | 原文线索

V2-009903 | MATHDB-355060

Fagundes-de Mello conjecture on multilinear polynomial images in upper triangular matrix algebras

摘录 / EXCERPT: Let $T_m(K)$ denote the algebra of upper triangular $m\times m$ matrices over a field K, and let p be a multilinear polynomial in noncommutative variables. Fagundes-de Mello co...

conjecture | open | MathDB 页面 | 原文线索

V2-009904 | MATHDB-355061

Two-summand conjecture for polynomial images in upper triangular matrix algebras

摘录 / EXCERPT: Two-summand conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-009905 | MATHDB-355192

The index bound conjecture for Galois images of elliptic curves

摘录 / EXCERPT: Index bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-009906 | MATHDB-355193

The conjectural list of adelic image indices for non-CM elliptic curves

摘录 / EXCERPT: Adelic image index conjecture. The index

conjecture | open | MathDB 页面 | 原文线索

V2-009907 | MATHDB-355200

The weekly green path conjecture for cluster mutation classes

摘录 / EXCERPT: Weekly green path conjecture. There is a weekly green path from v_0 to v. In particular, the products

conjecture | open | MathDB 页面 | 原文线索

V2-009908 | MATHDB-355251

The tensor-product embedding conjecture for algebraic central division algebras

摘录 / EXCERPT: Tensor-product embedding conjecture. The algebra $\widetilde R$ is K-isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-009909 | MATHDB-355336

The left-cell counting conjecture for type AIII Schur algebras

摘录 / EXCERPT: Let $n=2r+1$, let $\boldsymbol{c}_{\lambda}^{\jmath}$ be the two-sided cell associated with a special partition λ of $2d+1$ with at most n parts, and consider semistandar...

conjecture | open | MathDB 页面 | 原文线索

V2-009910 | MATHDB-355360

Multiplicity-one conjecture for common eigenspaces of alternating elements

摘录 / EXCERPT: Multiplicity-one conjecture. For every irreducible $\mathbb T$-module, the common eigenspaces for each of these four families all have dimension one. This conjecture would give a u...

conjecture | open | MathDB 页面 | 原文线索

V2-009911 | MATHDB-355361

The Hadamard-closure conjecture for the subalgebra of a Q-polynomial graph

摘录 / EXCERPT: Let $\Gamma=(X,\mathcal R)$ be a finite, undirected, connected graph without loops or multiple edges, with diameter $D\geq 1$, and let A be its adjacency matrix. An ordering...

conjecture | open | MathDB 页面 | 原文线索

V2-009912 | MATHDB-355369

The classification conjecture for multiplicatively idempotent congruence-simple semirings

摘录 / EXCERPT: Classification conjecture. Every multiplicatively idempotent congruence-simple semiring is finite and isomorphic to one of the semirings $\mathbb{S}_1$, $\dots$, $\mathbb{S}_8$.

conjecture | open | MathDB 页面 | 原文线索

V2-009913 | MATHDB-355370

The conjecture on finitely generated multiplicatively divisible commutative semirings

摘录 / EXCERPT: Multiplicative idempotence conjecture. If S is multiplicatively divisible, then S is multiplicatively idempotent.

conjecture | open | MathDB 页面 | 原文线索

V2-009914 | MATHDB-355392

Peirce-spectrum conjecture for complex algebras of odd dihedral quandles

摘录 / EXCERPT: Let R_n be a dihedral quandle of odd order n, let $\mathbb{C}[R_n]$ be its complex quandle algebra, and let $\sigma(\mathbb{C}[R_n])$ denote its Peirce spectrum. Peirce-spectru...

conjecture | open | MathDB 页面 | 原文线索

V2-009915 | MATHDB-355398

Kac's constant-term conjecture for loopless quivers

摘录 / EXCERPT: Let $\Gamma=(\Gamma_0,\Gamma_1)$ be a finite quiver with no loops, let $\alpha\in\mathbb{N}^n$ be a dimension vector, and let $A_\Gamma(\alpha,q)$ be its Kac polynomial. Let the Ka...

conjecture | open | MathDB 页面 | 原文线索

V2-009916 | MATHDB-355417

Diagonal reduction conjecture for strong commutativity preservers of incidence algebras

摘录 / EXCERPT: Let X be a finite connected poset, let $I(X,K)$ be its incidence algebra over K, and let $D(X,K)$ denote the diagonal subalgebra. A map $\theta:I(X,K)\to I(X,K)$ is a strong co...

conjecture | open | MathDB 页面 | 原文线索

V2-009917 | MATHDB-355445

Chevalley-Eilenberg coalgebra embedding conjecture

摘录 / EXCERPT: Let $\Bbbk$ be the base ring. Write $\operatorname{Lie}_{\Bbbk}$ for the ∞-category of Lie algebras, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-009918 | MATHDB-355446

Chevalley-Eilenberg embedding conjecture for perfect Lie algebras

摘录 / EXCERPT: Let $\operatorname{Lie}^{\operatorname{perf}}_{\Bbbk}\coloneqq\operatorname{Alg}_{\operatorname{Lie}}(\operatorname{Perf}_{\Bbbk})$ be the ∞-category of Lie algebras whose u...

conjecture | open | MathDB 页面 | 原文线索

V2-009919 | MATHDB-355447

Fully faithful enhanced Chevalley-Eilenberg group functor conjecture

摘录 / EXCERPT: Let $\operatorname{Grp}(\operatorname{Lie}_{\Bbbk})$ be the ∞-category of group objects in Lie algebras, and let ... be the Tate-realization functor, denoted in the source by...

conjecture | open | MathDB 页面 | 原文线索

V2-009920 | MATHDB-355448

Fully faithful Chevalley-Eilenberg functor for representations

摘录 / EXCERPT: Fix a Lie algebra $\mathfrak{g}$ whose underlying $\Bbbk$-module is perfect. Let $\operatorname{LMod}_{\operatorname{U}(\mathfrak{g})}$ be the ∞-category of left modules ove...

conjecture | open | MathDB 页面 | 原文线索

V2-009921 | MATHDB-355470

Bad-tree generation conjecture for graded identities of upper triangular Lie algebras

摘录 / EXCERPT: Let G be a group and let $(UT_n^{(-)},\eta)$ be an elementary G-grading, where $UT_n^{(-)}$ denotes the Lie algebra of upper triangular $n\times n$ matrices. A tree μ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-009922 | MATHDB-355471

One-kill-coarsening codimension conjecture for upper triangular Lie algebras

摘录 / EXCERPT: Let η_1 and η_2 be gradings on $UT_n^{(-)}$. Say that η_1 and η_2 correspond to a 1-kill-coarsening when passing from one grading to the other moves exactly one...

conjecture | open | MathDB 页面 | 原文线索

V2-009923 | MATHDB-355507

Nilperiod rings are NI-rings

摘录 / EXCERPT: A ring R is nilperiod if every nilpotent element of R is eventually annihilated by a power map in the sense established in the paper, and it is an NI-ring if its set of nilpote...

conjecture | open | MathDB 页面 | 原文线索

V2-009924 | MATHDB-355521

Tensor indecomposability conjecture for existentially closed factors

摘录 / EXCERPT: Let $\mathcal E$ denote the class of existentially closed II$_1$ factors. Tensor indecomposability conjecture. For every $P\in\mathcal E$, there do not exist II$_1$ factors A and...

conjecture | open | MathDB 页面 | 原文线索

V2-009925 | MATHDB-355522

von Neumann algebraic von Neumann problem

摘录 / EXCERPT: A II$_1$ factor is nonamenable if it is not amenable, and $L(\mathbb F_2)$ denotes the group II$_1$ factor of the free group on two generators. von Neumann algebraic von Neumann pr...

conjecture | open | MathDB 页面 | 原文线索

V2-009926 | MATHDB-355523

Canonical McDuff decomposition conjecture for existentially closed factors

摘录 / EXCERPT: Let $\mathcal E$ denote the class of existentially closed II$_1$ factors. A McDuff factor P admits only the canonical McDuff decomposition if, for every tensor decomposition ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-009927 | MATHDB-355524

Cartan-free conjecture for existentially closed factors

摘录 / EXCERPT: Let $\mathcal E$ denote the class of existentially closed II$_1$ factors. A Cartan subalgebra is a maximal abelian subalgebra whose normalizer generates the factor. Cartan-free con...

conjecture | open | MathDB 页面 | 原文线索

V2-009928 | MATHDB-355549

Nonnegativity of quiver Catalan numbers

摘录 / EXCERPT: Let Q be a locally acyclic quiver, and let χ_Q denote its quiver Catalan number, defined from the point count of a minimal extension when that point count is polynomial. Non...

conjecture | open | MathDB 页面 | 原文线索

V2-009929 | MATHDB-355582

Dimension bound for t-spread lexsegment ideals

摘录 / EXCERPT: Let $S=K[x_1,\dots,x_n]$ be a polynomial ring, and let I be a t-spread lexsegment ideal generated in degree $d\ge 2$, meaning that its minimal monomial generators have degree...

conjecture | open | MathDB 页面 | 原文线索

V2-009930 | MATHDB-355615

The spectral conjecture for finite-dimensional real *-algebras

摘录 / EXCERPT: Let $(\mathcal A,*)$ be an associative, unital, finite-dimensional $*$-algebra over $\mathbb R$. The spectral monoid $\overline{\operatorname{Her}}(\mathcal A,*)$ is the monoid of...

conjecture | open | MathDB 页面 | 原文线索

V2-009931 | MATHDB-355654

Caviglia-VandeBogert-Weyman conjecture on licci ideals and Dynkin formats

摘录 / EXCERPT: Let R be a Gorenstein local ring and let $\mathfrak{f}=(1,m,m+n-1,n)$ be a resolution format realized by some grade 3 perfect ideal in R. A perfect ideal is licci if it belon...

conjecture | open | MathDB 页面 | 原文线索

V2-009932 | MATHDB-355655

Conjecture on licci ideals and nonzero structure maps

摘录 / EXCERPT: Let I be a perfect ideal of height 3 in a Gorenstein local ring R with infinite residue field. For a structure map $w^{(i)}_{j,k}(J)$, write “nonzero modulo the maximal ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-009933 | MATHDB-355658

Optimality conjecture for the natural orthogonal complement in hidden Markov model reduction

摘录 / EXCERPT: Let $cmathcal{E}_cperp$ be the natural orthogonal-complement effective subspace defined above, and let $coverline{\bm p}$ be the vector defined in Corollary corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-009934 | MATHDB-355675

Habegger's height conjecture for modularly related points

摘录 / EXCERPT: Let $C\subset \mathbb{A}^2$ be an irreducible algebraic curve defined over $\overline{\mathbb{Q}}$ that is not special. Let $s=(s_1,s_2)$ be a point of $C(\overline{\mathbb{Q}})$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-009935 | MATHDB-355676

The strong Rickard conjecture for functor categories

摘录 / EXCERPT: Let $\mathcal{X}$ and $\mathcal{Y}$ be essentially small idempotent complete additive categories. Write $\operatorname{Mod}-\mathcal{X}$ for the category of modules over...

conjecture | open | MathDB 页面 | 原文线索

V2-009936 | MATHDB-355721

Conjectured equivalence for commuting-pair counts of quiver endomorphism algebras

摘录 / EXCERPT: Let Q be an acyclic quiver and let $\mathbf{d}\in\mathbb{Z}_{\geq 0}^{Q_0}$. For each $\mathbf{e}\leq\mathbf{d}$, let $[Q,\mathbf{e}]$ denote the relevant commuting-pair count in...

conjecture | open | MathDB 页面 | 原文线索

V2-009937 | MATHDB-355790

Character-occurrence finiteness conjecture for elliptic curves over the maximal abelian extension

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, and let X be the set of even characters of $\operatorname{Gal}(\mathbb{Q}^{\operatorname{ab}}/\mathbb{Q})$. Character-occurrence f...

conjecture | open | MathDB 页面 | 原文线索

V2-009938 | MATHDB-355822

One-parameter subgroup conjecture for the heat equation symmetry pseudogroup

摘录 / EXCERPT: Let G be the point-symmetry pseudogroup under consideration, with splitting $G=G^{\rm ess}\ltimes G^{\rm lin}$ and Lie algebras $\mathfrak g$ and $\mathfrak g^{\rm ess}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-009939 | MATHDB-355824

Periodicity conjecture for the Delta-triangle

摘录 / EXCERPT: Let $p\geq2$ be an integer, and let $P=\Delta(p)$, where ... Let $f\in\mathbb{K}^{\widehat{P}}$ be a $\mathbb{K}$-labeling such that $R^pf\neq\operatorname{\bot}$. Set $a=f(0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009940 | MATHDB-355825

Periodicity conjecture for the Nabla-triangle

摘录 / EXCERPT: Let $p\geq2$ be an integer, and let $P=\nabla(p)$, where ... Use the same notation as for the Delta-triangle: f is a $[?]$-labeling of $\widehat{P}$, R is birational rowmotion, a...

conjecture | open | MathDB 页面 | 原文线索

V2-009941 | MATHDB-355826

Nathan Williams's periodicity conjecture for trapezoids

摘录 / EXCERPT: Let p be an integer with $p>1$, let $s\in\mathbb{N}$, and let P be the subposet ... of $[p]\times[p]$. Let $f\in\mathbb{K}^{\widehat{P}}$ be a $\mathbb{K}$-labeling such that...

conjecture | open | MathDB 页面 | 原文线索

V2-009942 | MATHDB-355926

Three mod-9 identities from the $\mathrm{D}_4^{(3)}$ principal-subspace sum

摘录 / EXCERPT: Let q be a formal variable, let $m_1,m_2\geq 0$, and define the theta-product notation by $\theta(a_1,\ldots,a_r;Q)=\prod_{i=1}^r(a_i;Q)_\infty(Q/a_i;Q)_\infty$. The mod-$9$ iden...

conjecture | open | MathDB 页面 | 原文线索

V2-009943 | MATHDB-355949

Superpotential conjecture for fundamental group algebras of acyclic circle bundles

摘录 / EXCERPT: Let X be a compact, acyclic, orientable circle bundle of dimension n, and let $k[pi_1(X)]$ denote its fundamental group algebra over a field k. A superpotential algebra is an...

conjecture | open | MathDB 页面 | 原文线索

V2-009944 | MATHDB-355984

The tridiagonal-relations conjecture for circular Hessenberg pairs

摘录 / EXCERPT: Tridiagonal-relations conjecture. There exists a sequence of scalars $\beta,\gamma,\gamma^*,\varrho,\varrho^*$ in $\mathbb{F}$ such that both tridiagonal relations hold.

conjecture | open | MathDB 页面 | 原文线索

V2-009945 | MATHDB-355992

The comparison conjecture for canonical and bi-canonical degrees

摘录 / EXCERPT: Comparison conjecture. In general,

conjecture | open | MathDB 页面 | 原文线索

V2-009946 | MATHDB-356012

Cheong-Huang-Kaplan conjecture on cokernels of polynomial functions of random matrices

摘录 / EXCERPT: Fix finitely many monic polynomials $P_{1}(t),\dots,P_{l}(t)\in\mathbb{Z}_{p}[t]$ whose images in $\mathbb{F}_{p}[t]$ modulo p are distinct and irreducible. For $1\leq j\leq l$,...

conjecture | open | MathDB 页面 | 原文线索

V2-009947 | MATHDB-356053

The commutative gyrogroup conjecture

摘录 / EXCERPT: A gyrogroup is a set equipped with a binary operation and the associated gyroautomorphisms satisfying the gyrogroup axioms; it is commutative when its binary operation satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-009948 | MATHDB-356065

The tau-tilting finiteness and brick-discreteness conjecture

摘录 / EXCERPT: Tau-tilting finiteness-brick-discreteness conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-009949 | MATHDB-356118

Conjecture on the normalized depth function of squarefree powers

摘录 / EXCERPT: Normalized-depth conjecture. The function $g_I(k)$ is nonincreasing.

conjecture | open | MathDB 页面 | 原文线索

V2-009950 | MATHDB-356137

Twisted linear period conjecture for central simple algebras

摘录 / EXCERPT: Twisted linear period conjecture. If P_χ is not identically zero, then $L(\frac{1}{2},\pi_{0,E}\otimes\chi)\ne0$ and $L(s,\pi_0,\wedge^2\otimes\chi|_{\mathbb{A}_F^\times})$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-009951 | MATHDB-356147

Conjectural algebraic form of the torus one-point block recursion

摘录 / EXCERPT: Let $K_N(\Delta,\Delta_1,\beta)$ be defined by the recursion in the source, and let P_N be the quantity occurring there, $Q_N:=P_N-N$, and B_s the Laurent-polynomial basis defi...

conjecture | open | MathDB 页面 | 原文线索

V2-009952 | MATHDB-356148

Universal null-operator conjecture for affine $\mathfrak{sl}_2$

摘录 / EXCERPT: Let $U(\widehat{\mathfrak{sl}}_2)$ be the enveloping algebra of the affine Lie algebra, let W_1+W_2 be the subspace being quotiented out, and let $\hat{T}^0_1$ be the level-one n...

conjecture | open | MathDB 页面 | 原文线索

V2-009953 | MATHDB-356149

Generation conjecture for nonzero-charge affine $\mathfrak{sl}_2$ null vectors

摘录 / EXCERPT: Let E^Q be the eigenspace of null operators of charge $Q\in\mathbb{Z}$ under the adjoint action of J^0_0, so that $[J^0_0,J^Q]=QJ^Q$. For each nonzero charge, operators can be...

conjecture | open | MathDB 页面 | 原文线索

V2-009954 | MATHDB-356151

Coupled-soliton decomposition conjecture for affine general linear Lie superalgebra cellular automata

摘录 / EXCERPT: Let v be a coupled soliton. A coupled-soliton decomposition conjecture asserts that there exist positive integers d and $\widetilde{t}$ such that, whenever w is an uncoupled...

conjecture | open | MathDB 页面 | 原文线索

V2-009955 | MATHDB-356152

Soliton separation conjecture for affine general linear Lie superalgebra cellular automata

摘录 / EXCERPT: Let p be any state, and let T_{∞} be the time-evolution map. A soliton separation conjecture asserts that there exist a positive integer $\widetilde{t}$, solitons...

conjecture | open | MathDB 页面 | 原文线索

V2-009956 | MATHDB-356153

Soliton content identity conjecture for affine general linear Lie superalgebra cellular automata

摘录 / EXCERPT: Let $v=v_1\otimes v_2\otimes\cdots\otimes v_{\mathfrak{d}}\in(B^{r,1})^{\otimes\mathfrak{d}}$ be a soliton. For each factor v_j, let k_j be the value of k assigned by the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-009957 | MATHDB-356168

Vorst's conjecture on K-regularity and regularity

摘录 / EXCERPT: Vorst's conjecture. If R is K_{d+1}-regular, then R is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-009958 | MATHDB-356178

Zhao's Image Conjecture for commuting differential operators

摘录 / EXCERPT: Let $\mathbb{C}[z]$ be a polynomial ring and let $\mathbb{D}[z]$ be the set of differential operators Φ of $\mathbb{C}[z]$ of the form … for some $h(z)\in\mathbb{C}[z]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-009959 | MATHDB-356179

The locally nilpotent epsilon-derivation conjecture

摘录 / EXCERPT: Let K be a field of characteristic 0, let $\mathcal A$ be a K-algebra, and let δ be a locally nilpotent K-derivation or K-ε-derivation of $\mathcal A$....

conjecture | open | MathDB 页面 | 原文线索

V2-009960 | MATHDB-356200

Characteristic-two conjecture for non-descending-hereditary reductive pairs

摘录 / EXCERPT: Characteristic-two conjecture. If H is non-G-dh, then $p=2$.

conjecture | open | MathDB 页面 | 原文线索

V2-009961 | MATHDB-356251

Conjecture on attainable double negation degrees of bounded BCK-algebras

摘录 / EXCERPT: Attainability conjecture for double negation degrees. For each $n\geq 3$, every possible double negation degree is attained by some bounded BCK-algebra. Consequently, every rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-009962 | MATHDB-356310

The determinantal-ring conjecture on n-sem dualizing modules

摘录 / EXCERPT: The determinantal-ring conjecture. If $0\neq[M]\in\operatorname{Cl}(R)$, then M is exactly $(m+n-2t+1)$-semidualizing. Hence

conjecture | open | MathDB 页面 | 原文线索

V2-009963 | MATHDB-356334

The differential resultant conjecture for commuting matrix ordinary differential operators

摘录 / EXCERPT: Let K be a differential field whose field of constants C is algebraically closed and has characteristic zero. Let L and B be commuting matrix ordinary differential operator...

conjecture | open | MathDB 页面 | 原文线索

V2-009964 | MATHDB-356356

Conjecture on algebraic independence of non-CM elliptic-curve periods

摘录 / EXCERPT: Conjecture sur les périodes des courbes elliptiques sans multiplication complexe. Les quatre périodes de E sont algébriquement indépendantes.

conjecture | open | MathDB 页面 | 原文线索

V2-009965 | MATHDB-356393

Mesyan conjecture on images of multilinear polynomials

摘录 / EXCERPT: Let K be a field, let $n\geq 2$ and $m\geq 1$ be integers, and let $f(x_1,\dots,x_m)$ be a nonzero multilinear polynomial in $K\langle x_1,\dots,x_m\rangle$. Mesyan conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-009966 | MATHDB-356394

Strong Mesyan-type conjecture for multilinear polynomial images

摘录 / EXCERPT: Let K be a field, let $n\geq 2$ be an integer, and let $f(x_1,\dots,x_m)$ be a multilinear polynomial. Say that f satisfies $(S1)$ when it is a central polynomial, so its image...

conjecture | open | MathDB 页面 | 原文线索

V2-009967 | MATHDB-356398

Finiteness conjecture for maximal green sequences of acyclic quivers

摘录 / EXCERPT: Let Q be a quiver with no oriented cycles. A maximal green sequence (MGS) is a sequence of green mutations starting with an extended exchange matrix $\tilde{B}={B\over I}$ and en...

conjecture | open | MathDB 页面 | 原文线索

V2-009968 | MATHDB-356434

Chen's few-subpowers expressive-rate conjecture

摘录 / EXCERPT: Let $Gamma$ be a constraint language, and let $r(n)$ be the logarithm of the number of distinct n-variable relations definable by primitive positive formulas over $Gamma$. Chen's...

conjecture | open | MathDB 页面 | 原文线索

V2-009969 | MATHDB-356438

Bounded relational-width term conjecture for finite idempotent algebras

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite idempotent algebra, meaning that every basic operation satisfies $f(x,\ldots,x)=x$. Bounded relational-width term conjecture. The algebra $\mathbb{A}$...

conjecture | open | MathDB 页面 | 原文线索

V2-009970 | MATHDB-356439

Minimal Taylor algebra structural conjecture

摘录 / EXCERPT: Let $\mathbb{A}$ be a minimal Taylor algebra, meaning a Taylor algebra whose clone is minimal in the sense used by the source, and suppose that $\mathbb{A}$ is generated by two ele...

conjecture | open | MathDB 页面 | 原文线索

V2-009971 | MATHDB-356491

The nonassociative loop-switch strategy conjecture

摘录 / EXCERPT: Let $\mathcal{L}=(\{1,a,b,c,d\},\ast)$ be the smallest loop that is not a group, with multiplication operation $\ast$. Loop-switch strategy conjecture. There exists a nontrivial ad...

conjecture | open | MathDB 页面 | 原文线索

V2-009972 | MATHDB-356534

Existence conjecture for the PI-exponent of finite-dimensional algebras with involution

摘录 / EXCERPT: Let A be a finite-dimensional algebra with involution $*\colon A\to A$. Its $*$-codimensions are denoted by $c_n^*(A)$, and when the limit exists its $*$-PI-exponent is ... Existen...

conjecture | open | MathDB 页面 | 原文线索

V2-009973 | MATHDB-356535

Realization conjecture for PI-exponents of algebras with involution

摘录 / EXCERPT: Let $\alpha\geq 1$ be a real number. An algebra with involution A_α has a $*$-PI-exponent, when the limit exists, defined by ... Realization conjecture for $*$-PI-exponents. F...

conjecture | open | MathDB 页面 | 原文线索

V2-009974 | MATHDB-356573

Cherednik-Danilenko conjecture for compactified Jacobian Poincaré polynomials

摘录 / EXCERPT: Let $(r_1,s_1),\ldots,(r_{\ell},s_{\ell})$ be a sequence of characteristic pairs for an algebraic curve. Starting with $f_{\ell+1}:=p_1\in\Lambda$, recursively apply the operators...

conjecture | open | MathDB 页面 | 原文线索

V2-009975 | MATHDB-356574

The height-rank conjecture for non-simple polyominoes

摘录 / EXCERPT: Height-rank conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-009976 | MATHDB-356575

The König-type conjecture for non-simple thin polyominoes

摘录 / EXCERPT: König-type conjecture. The ideal $I_{\mathcal{P}}$ is of König type.

conjecture | open | MathDB 页面 | 原文线索

V2-009977 | MATHDB-356579

Existence of a weakly fadelian ring that is not fadelian

摘录 / EXCERPT: Existence conjecture. There is a weakly fadelian ring which is not fadelian.

conjecture | open | MathDB 页面 | 原文线索

V2-009978 | MATHDB-356627

Conjecture on bracket width of vector fields on non-rational smooth plane affine curves

摘录 / EXCERPT: Bracket-width conjecture. The bracket width of $\operatorname{Vec}(C)$ is exactly two.

conjecture | open | MathDB 页面 | 原文线索

V2-009979 | MATHDB-356631

Eto's determinant conjecture for row-factorization matrices

摘录 / EXCERPT: Let S be a numerical semigroup with embedding dimension $\operatorname{e}(S)=e$, and let $\operatorname{RF}(\operatorname{F})$ denote a row-factorization matrix for the Frobenius...

conjecture | open | MathDB 页面 | 原文线索

V2-009980 | MATHDB-356697

Reciprocal-branch-point conjecture for the genus-two family J_2(\lambda,1)

摘录 / EXCERPT: Let $J_2(\lambda,1)$ be the genus-two family of translation surfaces in the stratum $\mathcal{H}(1,1)$, and let a,b,c be complex parameters. A genus-two hyperelliptic curve is gi...

conjecture | open | MathDB 页面 | 原文线索

V2-009981 | MATHDB-356699

The axial central extension conjecture for Jordan algebras

摘录 / EXCERPT: Let F be a field of characteristic different from 2. Let J be a Jordan algebra over F generated by idempotents, and let a $\mathcal{J}(\frac{1}{2})$-axial central extension...

conjecture | open | MathDB 页面 | 原文线索

V2-009982 | MATHDB-356704

Volume-rigidity conjecture for punctured surface triangulations

摘录 / EXCERPT: A triangulation of a compact connected surface without boundary is a simplicial complex representing the surface; removing a single triangle means deleting that triangular face and...

conjecture | open | MathDB 页面 | 原文线索

V2-009983 | MATHDB-356705

Volume-rigidity conjecture for punctured sphere triangulations

摘录 / EXCERPT: Let $d\ge 3$, and let K be a triangulation of the $(d-1)$-sphere. Removing a single $(d-1)$-simplex means deleting that facet and its interior. Higher-dimensional volume-rigidity...

conjecture | open | MathDB 页面 | 原文线索

V2-009984 | MATHDB-356724

Alahmadi-Alsulami-Jain-Zelmanov equivalence conjecture for algebra growth functions

摘录 / EXCERPT: Alahmadi-Alsulami-Jain-Zelmanov conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-009985 | MATHDB-356788

Monomial positivity conjecture for H-graph ideals

摘录 / EXCERPT: Let P be a $(\mathbf{3}+\mathbf{1})$-free poset, let I_H^P be the corresponding H-graph ideal, and let $\mathfrak{m}^P_\lambda(\mathbf{u})$ be the elements defined in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-009986 | MATHDB-356789

Diagram-positivity conjecture for noncommutative monomial symmetric functions

摘录 / EXCERPT: Let I_H be the H-graph ideal, and let $\mathfrak{m}_\lambda$ be the corresponding noncommutative monomial symmetric-function element. Say that an element is diagram positive mo...

conjecture | open | MathDB 页面 | 原文线索

V2-009987 | MATHDB-356819

Coll-Mayers-Mayers conjecture on the seaweed index generating function

摘录 / EXCERPT: Let $G(q)$ be the q-series ... and let $e(n)$ and $o(n)$ denote respectively the numbers of partitions of n into odd parts whose index is even and odd. Coll-Mayers-Mayers conject...

conjecture | open | MathDB 页面 | 原文线索

V2-009988 | MATHDB-356845

Commutant elements are commutators in regular Lie algebras

摘录 / EXCERPT: Let L be a non-nilpotent regular Lie algebra with a nondegenerate symmetric invariant bilinear form. An element $x\in L$ is called a commutant element if x belongs to the commu...

conjecture | open | MathDB 页面 | 原文线索

V2-009989 | MATHDB-356851

Fomin-Kirillov nonnegative Schubert-polynomial conjecture in the Fomin-Kirillov algebra

摘录 / EXCERPT: Let $\mathcal{E}_n$ be the Fomin-Kirillov algebra generated by $\mathbf{x}_{ab}$ for $1\leq a<b\leq n$, and define the commuting Dunkl elements ... Let $\mathfrak{S}_w$ denote the Sc...

conjecture | open | MathDB 页面 | 原文线索

V2-009990 | MATHDB-356863

McNamara's center conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: McNamara's center conjecture. The center $Z(\mathscr{H}_{n,K}^\Lambda)$ is precisely the set of symmetric polynomials in $\mathcal{L}_1,\ldots,\mathcal{L}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-009991 | MATHDB-356864

The center conjecture for cyclotomic KLR algebras of type A

摘录 / EXCERPT: Center conjecture. The center of $\mathscr{R}_{\alpha,K}^{\Lambda}$ consists precisely of the symmetric elements in $\mathscr{R}_{\alpha,K}^{\Lambda}$.

conjecture | open | MathDB 页面 | 原文线索

V2-009992 | MATHDB-356865

Chavli-Pfeiffer's integral basis conjecture for Hecke algebra centers

摘录 / EXCERPT: Chavli-Pfeiffer's integral basis conjecture. There exist an R-basis $\{\mathbf{b}_w\mid w\in W\}$ of $\mathscr{H}(W)$ and conjugacy-class representatives...

conjecture | open | MathDB 页面 | 原文线索

V2-009993 | MATHDB-356893

The tangle and planar quasi-partition algebra isomorphism conjecture

摘录 / EXCERPT: Let ... , let ... and ... such that ... The conjecture proposes that these algebras are isomorphic for a suitable choice of parameter. The supplied text gives no proof or resolution.

conjecture | open | MathDB 页面 | 原文线索

V2-009994 | MATHDB-356906

The structure conjecture for the free Jordan superalgebra $J(D_1|D_2)$

摘录 / EXCERPT: Let $J(D_1|D_2)$ be the free Jordan superalgebra with $D_1+D_2\geq 1$. Let G be the acting algebraic group, let ${\mathcal R}^s(G)$ be the relevant super representation ring, let...

conjecture | open | MathDB 页面 | 原文线索

V2-009995 | MATHDB-356907

The graded-dimension conjecture for free Jordan superalgebra components

摘录 / EXCERPT: Let $R={\mathbb Z}[x]/(x^2-1)$, and let $a_n^s=\overline{\dim}J_n(D_1|D_2)$ and $b_n^s=\overline{\dim}{\mathcal B}^s_n(J(D_1|D_2))$ be sequences in R. Write ... Define...

conjecture | open | MathDB 页面 | 原文线索

V2-009996 | MATHDB-356908

The generating-function equation for graded dimensions of the free Jordan superalgebra

摘录 / EXCERPT: Let $R={\mathbb Z}[x]/(x^2-1)$, let $a_n^s=\overline{\dim}J_n(D_1|D_2)$, and write ... Define ... and ... The generating-function conjecture. The sequence a_n^s is the unique solution...

conjecture | open | MathDB 页面 | 原文线索

V2-009997 | MATHDB-356910

Jaulent's strong cyclotomic conjecture for S-split T-ramified Iwasawa modules

摘录 / EXCERPT: Strong cyclotomic conjecture. The characteristic polynomial of the Λ-module $\mathcal C^{T_K}_{S_K}(K_\infty)$ is not divisible by $\omega=\gamma-1$. Equivalently, its fixe...

conjecture | open | MathDB 页面 | 原文线索

V2-009998 | MATHDB-356927

Extension of uniqueness of signed τ-exceptional sequences to all finite-dimensional algebras

摘录 / EXCERPT: Uniqueness conjecture. The statement of Theorem $$ holds for all finite-dimensional algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-009999 | MATHDB-356929

Index equality for periodic contractions of semisimple Lie algebras

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple Lie algebra and let ϑ be a periodic automorphism of order m, inducing the grading ... Let $\mathfrak g_{(0)}$ be the contraction of...

conjecture | open | MathDB 页面 | 原文线索

V2-010000 | MATHDB-356931

Representability conjecture for finitely generated PI-algebras with square-zero Jacobson radical

摘录 / EXCERPT: Let A be a finitely generated PI-algebra over the field of rational numbers, and let J be its Jacobson radical. Representability conjecture. If ... then A is representable. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-010001 | MATHDB-356979

The strongly-algebraic characterization conjecture for permutation classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a permutation class. It is strongly algebraic if $\mathcal{C}$ and every subclass of $\mathcal{C}$ have algebraic generating functions, and it is well-quasi-or...

conjecture | open | MathDB 页面 | 原文线索

V2-010002 | MATHDB-357003

Tau-rigidity conjecture for sums of an indecomposable syzygy and its involutive translate

摘录 / EXCERPT: Tau-rigidity conjecture. If M is an indecomposable syzygy, then $M\oplusc3M$ is $c4$-rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-010003 | MATHDB-357074

Boston-Markin conjecture for totally real tamely ramified extensions

摘录 / EXCERPT: Boston-Markin conjecture. There exists a totally real tamely ramified G-extension $K/\mathbb{Q}$ such that the number of finite primes of $\mathbb{Q}$ ramified in K is...

conjecture | open | MathDB 页面 | 原文线索

V2-010004 | MATHDB-357075

Boston-Markin conjecture over the rational function field

摘录 / EXCERPT: Function-field Boston-Markin conjecture. There exists a regular G-extension $K/\mathbb{F}_q(T)$, split completely at infinity, such that the number of ramified primes of...

conjecture | open | MathDB 页面 | 原文线索

V2-010005 | MATHDB-357077

Localized quantitative Boston-Markin conjectures over function fields

摘录 / EXCERPT: Localized function-field Boston-Markin conjectures. There exists a positive real number $\delta_q^{G,C}$ such that, as $n_1,\dots,n_{d_\lhd(G)}\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-010006 | MATHDB-357111

The periodic mapping-torus volume conjecture for Kauffman bracket intertwiners

摘录 / EXCERPT: Let S be an oriented surface with negative Euler characteristic, let φ be a periodic diffeomorphism, and let $[\gamma]$ be a φ-invariant smooth character with...

conjecture | open | MathDB 页面 | 原文线索

V2-010007 | MATHDB-357140

Kanade-Russell's ideal-membership conjecture for cylindric recurrences

摘录 / EXCERPT: Fix $m=3k+\{-1,0,1\}$ with $k\geq3$. Let $\mathcal S_m$ be the set of relations obtained by writing the coupled q-difference equations for the profiles c with $|c|+3=m$ in term...

conjecture | open | MathDB 页面 | 原文线索

V2-010008 | MATHDB-357141

Finite truncation conjecture for the Kanade-Russell recurrence ideal

摘录 / EXCERPT: Fix $m=3k+\{-1,0,1\}$ with $k\geq3$. Let $\mathcal S_m$ be the set of relations obtained from the coupled q-difference equations in terms of the functions $S_m(\rho\mid\sigma)$....

conjecture | open | MathDB 页面 | 原文线索

V2-010009 | MATHDB-357142

The dimension conjecture for the kernel of Kameko's squaring operation

摘录 / EXCERPT: Let $P_k=\mathbb F_2[x_1,\ldots,x_k]$ be the polynomial algebra, let $Q P_k$ denote its quotient by the image of the positive-degree Steenrod algebra, and let ... be Kameko's squarin...

conjecture | open | MathDB 页面 | 原文线索

V2-010010 | MATHDB-357146

Uniqueness conjecture for sufficiently positive Manin components

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a Fano fibration over an algebraically closed field κ of characteristic 0. Let A be the affine space of section classes and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010011 | MATHDB-357176

Maximal analytic spread implies linear type for linear free divisors

摘录 / EXCERPT: Maximal-spread linear-type conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-010012 | MATHDB-357201

The pointed Hopf algebra Dixmier-Moeglin equivalence conjecture

摘录 / EXCERPT: Let H be an affine noetherian pointed Hopf $\mathbb{C}$-algebra, and let $G(H)$ denote its group of grouplike elements. The pointed Dixmier-Moeglin equivalence conjecture. The fo...

conjecture | open | MathDB 页面 | 原文线索

V2-010013 | MATHDB-357244

Categorified perverse-Hodge symmetry for integrable systems

摘录 / EXCERPT: Let $\pi_M:M\to B$ be an integrable system, let Ω_M^k be the sheaf of Kähler differentials, and let P_k be the Hodge module from the decomposition theorem, with associated...

conjecture | open | MathDB 页面 | 原文线索

V2-010014 | MATHDB-357272

Additivity conjecture for generalized Jordan derivations on triangular rings

摘录 / EXCERPT: Let A and B be rings and let $\mathcal{T}$ be a triangular ring satisfying the following conditions: for $a\in A$, $aA=0$ implies $a=0$; and for $b\in B$, $bB=0$ implies $b=0$....

conjecture | open | MathDB 页面 | 原文线索

V2-010015 | MATHDB-357289

Generic logarithmic Lie-length conjecture for \mathfrak{su}(n)

摘录 / EXCERPT: Let S be a random Lie-generating set of the special unitary Lie algebra $\mathfrak{su}(n)$, and let $\operatorname{Lie}\ell(S)$ be the least bracket length needed for brackets in...

conjecture | open | MathDB 页面 | 原文线索

V2-010016 | MATHDB-357315

Classification of rank-two simple Lie algebras in Verlinde categories

摘录 / EXCERPT: Let $p>0$, let ${\rm Ver}_p^+$ be the relevant Verlinde category, and let $\mathfrak{g}$ be a simple Lie algebra in ${\rm Ver}_p^+$ containing a simple Lie algebra $\mathfrak{s}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-010017 | MATHDB-357343

The theta-function preservation conjecture under the DT transformation

摘录 / EXCERPT: Let DT denote the DT transformation, and let theta functions be the theta functions associated with the cluster or scattering data under consideration. Theta-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-010018 | MATHDB-357355

Abelianness of group gradings that are coarsenings of abelian gradings

摘录 / EXCERPT: Let L be a Lie algebra with a group grading Γ that is a coarsening of an abelian grading of L. Abelianness conjecture. Then Γ is abelian. The paper states that no...

conjecture | open | MathDB 页面 | 原文线索

V2-010019 | MATHDB-357363

The shortest-cycle characterization of connected shelves

摘录 / EXCERPT: Let S be a shelf, with its operation and connectedness defined in the paper. A shortest-cycle characterization of connected shelves. There exists a shortest cycle that covers all...

conjecture | open | MathDB 页面 | 原文线索

V2-010020 | MATHDB-357364

The connected idempotent shelf polynomial conjecture

摘录 / EXCERPT: Connected idempotent shelf polynomial conjecture. If S is idempotent, then

conjecture | open | MathDB 页面 | 原文线索

V2-010021 | MATHDB-357370

The two-element projection quotient conjecture for clone-minimal dispersive algebras

摘录 / EXCERPT: Two-element projection quotient conjecture. For every $a\ne b\in\mathbb{A}$, there is a surjective homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010022 | MATHDB-357371

The three-element generated subalgebra conjecture for clone-minimal dispersive algebras

摘录 / EXCERPT: Three-element generated subalgebra conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-010023 | MATHDB-357398

Surjectivity of the first Coleman map and injectivity of the Coleman map

摘录 / EXCERPT: Let L be the field appearing above, let G_L be its absolute Galois group, and let $\tau:G_L\to o_L^\times$, $\cK_1:\widehat{U}\to o_\infty$, and $\cK:\widehat{U}\to\cC$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-010024 | MATHDB-357417

Costello's twisted holography principle

摘录 / EXCERPT: Twisted holography principle. These two algebras should be Koszul dual:

conjecture | open | MathDB 页面 | 原文线索

V2-010025 | MATHDB-357435

The infinitude conjecture for Galois number fields with trivial relative class group

摘录 / EXCERPT: For a number field K Galois over $\mathbb{Q}$, define the relative class group ... Here $Cl(K)$ is the ideal class group and $Po(K)$ is the Pólya group of K. Infinitude conjectur...

conjecture | solved | MathDB 页面 | 原文线索

V2-010026 | MATHDB-357463

Rouquier's faithfulness conjecture for the 2-braid-group categorification

摘录 / EXCERPT: Let $B_{(W,S)}$ be the generalized braid group associated with the Coxeter system (W,S), and let 2-${\mathcal B}r$ be the monoidal category generated by the Rouquier complexes...

conjecture | open | MathDB 页面 | 原文线索

V2-010027 | MATHDB-357484

The small Cohen-Macaulay conjecture

摘录 / EXCERPT: A small Cohen-Macaulay module over a local ring is a finitely generated module whose depth equals the dimension of the ring. The small Cohen-Macaulay conjecture. Every complete loc...

conjecture | open | MathDB 页面 | 原文线索

V2-010028 | MATHDB-357515

de Klerk-Laurent hypercube quadratic-module degree conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$ be even, and let $\mathrm{B}^{n}=[-1,1]^n$. Write $\mathcal{Q}(\mathrm{B}^{n})_n$ for the degree-n truncation of the quadratic module generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-010029 | MATHDB-357516

Genericity conjecture for sphere packings

摘录 / EXCERPT: Let $G=(V,E)$ be the contact graph of a sphere packing P with generic radii, where generic means that the radii form an algebraically independent set. Regard P as a bar-and-joi...

conjecture | open | MathDB 页面 | 原文线索

V2-010030 | MATHDB-357538

Log-liftability conjecture for log del Pezzo pairs with standard coefficients

摘录 / EXCERPT: Work over an algebraically closed field k of characteristic $p>0$. A pair (X,Δ) is log liftable if it admits the lifting property for the pair and its boundary divisors sp...

conjecture | open | MathDB 页面 | 原文线索

V2-010031 | MATHDB-357561

Finite universal Sagbi basis conjecture for the Grassmannian coordinate ring

摘录 / EXCERPT: Let $G(m,n)$ denote the Grassmannian, and let a universal Sagbi basis mean a finite generating set that is a Sagbi basis for every monomial order. Finite universal Sagbi basis conj...

conjecture | open | MathDB 页面 | 原文线索

V2-010032 | MATHDB-357570

Bigraded refinement of skein-module Langlands duality

摘录 / EXCERPT: Let G be the group appearing in the skein construction, let Z denote the relevant central subgroup, and let π_1 denote the corresponding fundamental-group twist. For a clos...

conjecture | open | MathDB 页面 | 原文线索

V2-010033 | MATHDB-357574

Rognes' connectivity conjecture for common basis complexes

摘录 / EXCERPT: Given a PID R, let $\operatorname{CB}_n(R)$ denote Rognes' common basis complex: its vertices are the proper nonzero summands of R^n, and a collection $\{V_0,\dots,V_p\}$ spans...

conjecture | open | MathDB 页面 | 原文线索

V2-010034 | MATHDB-357612

The Structural Rank Conjecture for logarithms of algebraic numbers

摘录 / EXCERPT: Structural Rank Conjecture. For every $M\in M_{m\times n}(\mathscr L+\mathbf Q)$, its rank equals its structural rank.

conjecture | open | MathDB 页面 | 原文线索

V2-010035 | MATHDB-357623

The conjecture that the unit-group ring is trivial for non-big non-imaginary-quadratic fields

摘录 / EXCERPT: Let K be a non-big field, meaning an algebraic extension of $\mathbb{Q}$ that is not big, and let R_K denote the ring constructed from the unit group of K. Triviality conject...

conjecture | open | MathDB 页面 | 原文线索

V2-010036 | MATHDB-357682

Wilkinson's complete-pivoting growth-factor bound conjecture

摘录 / EXCERPT: Let $g\big[\mathbf{CP}_n(\mathbb{R})\big]$ denote the maximum growth factor for complete pivoting over real $n\times n$ matrices. Wilkinson's conjecture. ... with equality achieved o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010037 | MATHDB-357683

Complete-pivoting threshold conjecture for the maximum growth factor

摘录 / EXCERPT: For each positive integer n, let $g\big[\mathbf{CP}_n(\mathbb{R})\big]$ be the maximum growth factor for complete pivoting over real $n\times n$ matrices. Complete-pivoting thres...

conjecture | open | MathDB 页面 | 原文线索

V2-010038 | MATHDB-357684

Superlinear growth conjecture for complete pivoting

摘录 / EXCERPT: Superlinear growth conjecture. The maximum growth factor is superlinear:

conjecture | open | MathDB 页面 | 原文线索

V2-010039 | MATHDB-357687

Motivated Grothendieck period conjecture in terms of periods and motivated Galois groups

摘录 / EXCERPT: Motivated Grothendieck period conjecture. The motive M satisfies MGPC if Ω^{And}_M is connected and

conjecture | open | MathDB 页面 | 原文线索

V2-010040 | MATHDB-357701

Uniqueness failure conjecture for additive actions on degenerate hypersurfaces

摘录 / EXCERPT: Additive-action multiplicity conjecture. There are at least two non-equivalent induced additive actions on X.

conjecture | open | MathDB 页面 | 原文线索

V2-010041 | MATHDB-357737

Strong Elliott conjecture for K-pure real-rank-zero C*-algebras

摘录 / EXCERPT: Let A be a separable, nuclear, K-pure ${\rm C}^*$-algebra of real rank zero and stable rank one. Let $({\rm\underline{K}}(A),{\rm\underline{K}}(A)_+,\Sigma A)_{\rm \Lambda}$ cons...

conjecture | open | MathDB 页面 | 原文线索

V2-010042 | MATHDB-357738

Weak Elliott conjecture via the total Cuntz semigroup

摘录 / EXCERPT: Let A be a unital, separable, nuclear ${\rm C}^*$-algebra of real rank zero and stable rank one, and let $\underline{\mathrm{Cu}}_u(A)$ denote its total Cuntz semigroup invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-010043 | MATHDB-357763

Kanade-Russell conjecture for the complete family of Andrews-Schilling-Warnaar identities

摘录 / EXCERPT: Let a,b,k be integers such that $0\leqslant a,b\leqslant k$, let $\tau\in\{-1,0,1\}$, and set ... For weakly decreasing nonnegative integer sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-010044 | MATHDB-357773

The quadraticity-quadratic Gröbner basis conjecture for abelian group projections

摘录 / EXCERPT: Quadraticity-quadratic Gröbner basis conjecture. The algebra $\operatorname{A}(X_{n,d}^G)$ is quadratic if and only if $\operatorname{I}(X_{n,d}^G)$ has a quadratic Gröbner basis.

conjecture | open | MathDB 页面 | 原文线索

V2-010045 | MATHDB-357782

The BR-linkage conjecture for formats $(2,6,5,1)$ and $(2,5,5,2)$

摘录 / EXCERPT: Let R be a commutative local or graded ring, and let M be a perfect module with a minimal free resolution of one of the formats ... A module is in the BR-linkage class of a Buchs...

conjecture | open | MathDB 页面 | 原文线索

V2-010046 | MATHDB-357786

The conjectured colon formula for the graded core of a monomial ideal

摘录 / EXCERPT: Colon conjecture. If I and h are as in the theorem preceding the conjecture, then

conjecture | open | MathDB 页面 | 原文线索

V2-010047 | MATHDB-357789

The hom- or e-vanishing mutation conjecture for nondegenerate Jacobi-finite QPs

摘录 / EXCERPT: Let $(Q,\mathcal{S})$ be a nondegenerate Jacobi-finite QP. For a pair (δ,ϵ) of δ-vectors, call it hom-vanishing when...

conjecture | solved | MathDB 页面 | 原文线索

V2-010048 | MATHDB-357790

The saturation conjecture for nondegenerate Jacobi-finite QPs

摘录 / EXCERPT: Let $(Q,\mathcal{S})$ be a nondegenerate Jacobi-finite quiver with potential. A pair $(\delta,\check\epsilon)$ has the saturation property when...

conjecture | open | MathDB 页面 | 原文线索

V2-010049 | MATHDB-357826

Bozec-Schiffmann positivity conjecture for absolutely cuspidal polynomials

摘录 / EXCERPT: Let Q be a quiver and let $\mathbf{d}\in\mathbf{N}^{Q_0}$ be a dimension vector. The polynomial $C^{\mathrm{abs}}_{Q,\mathbf{d}}(q)$ is the absolutely cuspidal polynomial associa...

conjecture | open | MathDB 页面 | 原文线索

V2-010050 | MATHDB-357835

The SI-sequence conjecture for smooth ACM curves

摘录 / EXCERPT: SI-sequence conjecture. The sequence $AV_{X,1}$ is an SI-sequence shifted by 1. Its last nonzero term is

conjecture | open | MathDB 页面 | 原文线索

V2-010051 | MATHDB-357863

Self-similarity conjecture for open subgroups of norm-one groups

摘录 / EXCERPT: Self-similarity conjecture. Assume that $n\geq 2$ and $p\geq d(n^2-1)$. Then no open subgroup of $SL_1^{ne}(D)$ is self-similar.

conjecture | open | MathDB 页面 | 原文线索

V2-010052 | MATHDB-357869

Depth lower bound for powers of the path ideal of a cycle graph

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$, let $J_{n,m}$ be the path ideal of the cycle graph on n vertices with paths of length m, let d be the parameter defined by the paper for n and...

conjecture | open | MathDB 页面 | 原文线索

V2-010053 | MATHDB-357887

Deligne-style algebraicity conjecture for GSp4 times GL2 L-values

摘录 / EXCERPT: Let $ 0\pi$ be a cuspidal automorphic representation of $ 0\operatorname{GSp}_4$ of weight (k_1,k_2), and let $ 0\sigma$ be a cuspidal automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-010054 | MATHDB-357917

The Beilinson-Bloch-Kato conjecture for geometric Galois representations

摘录 / EXCERPT: Let E be the number field in the paper, let $\rho\colon G_E\to \operatorname{GL}_n(\overline{\mathbf{Q}}{}_p)$ be an irreducible geometric representation of weight -1, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010055 | MATHDB-357945

Fontaine's conjecture for torsion crystalline representations

摘录 / EXCERPT: Let K be a complete discrete valuation field of mixed characteristic $(0,p)$ with perfect residue field k, and let $G_K=\operatorname{Gal}(\overline{K}/K)$. Fix a positive inte...

conjecture | open | MathDB 页面 | 原文线索

V2-010056 | MATHDB-357954

Orlov's perfect-dg-category phantom conjecture

摘录 / EXCERPT: A phantom is a nonzero admissible subcategory whose Grothendieck group vanishes. Let $\mathcal{R}$ be a smooth finite-dimensional dg-algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010057 | MATHDB-357967

Lvov-Kaplansky conjecture on multilinear polynomial images of matrix algebras

摘录 / EXCERPT: Let $n\geq 2$ be an integer, let K be a field, and let $K\langle X\rangle$ be the free associative algebra over K on a countable set of noncommutative variables. A multilinear...

conjecture | open | MathDB 页面 | 原文线索

V2-010058 | MATHDB-357968

Fagundes-Mello conjecture on multilinear polynomial images of upper triangular matrix algebras

摘录 / EXCERPT: Let K be a field and let UT_n denote the algebra of all $n\times n$ upper triangular matrices over K. Let f be a multilinear polynomial, and let $f(UT_n)$ denote its image...

conjecture | open | MathDB 页面 | 原文线索

V2-010059 | MATHDB-357983

The generic linear-form reduction conjecture for subspaces without powers

摘录 / EXCERPT: Let $A={\mathbb{C}}[x_1,\dots,x_n]$ be a polynomial ring, let A_d denote its degree-d component, and let $W\subseteq A_d$ be a k-dimensional subspace with $k\le d-1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010060 | MATHDB-357984

Shuffle Hopf algebra conjecture for multiple zeta values in positive characteristic

摘录 / EXCERPT: Let the MZV's in positive characteristic be the multiple zeta values over the coefficient field used in the source, viewed as a set carrying the relevant shuffle product. Shuffle H...

conjecture | open | MathDB 页面 | 原文线索

V2-010061 | MATHDB-358025

Bostan-Weil-Yurkevich conjecture on integral Pochhammer transforms

摘录 / EXCERPT: Let $(c_n)_{n\geq 0}$ be the sequence discussed in the paper, and for $(u,v)\in\mathbb{Q}^2$ define $\widetilde c_n=w^n(u)_n(v)_n c_n$, where $(u)_n$ and $(v)_n$ are Pochhammer sym...

conjecture | open | MathDB 页面 | 原文线索

V2-010062 | MATHDB-358044

Factorization conjecture for the determinant of the Painlevé coefficient matrix

摘录 / EXCERPT: Let $\nu\geqslant 3$, let $\tilde f_0,\tilde f_1,\ldots,\tilde f_{\nu-1}$ be algebraically independent variables, and let $F_\nu(z)$ be the matrix introduced in the paper. Regard...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010063 | MATHDB-358070

Popa's W*-superrigidity conjecture for property (T) groups

摘录 / EXCERPT: Let G be an ICC group with property (T), let H be a group, and let $\operatorname{L}(G)$ and $\operatorname{L}(H)$ denote their group von Neumann algebras. An isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-010064 | MATHDB-358151

The relative wrapped Fukaya category completion conjecture

摘录 / EXCERPT: Let $X\to Y$ be a rank-one algebraic torus fibration with discriminant divisor $D\subset Y$, and let s denote the invertible variable associated with the S^1 action on X. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010065 | MATHDB-358169

Drinfeld's gerbe isomorphism conjecture for truncated Barsotti-Tate groups

摘录 / EXCERPT: Drinfeld's gerbe isomorphism conjecture. These two gerbes are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-010066 | MATHDB-358172

The weak Malle conjecture over the rationals

摘录 / EXCERPT: Let G be a transitive subgroup of S_n. Let $\mathcal F_n(X;G)$ be the set of degree-n number fields K satisfying ... where $\hat K$ is the Galois closure of K over...

conjecture | open | MathDB 页面 | 原文线索

V2-010067 | MATHDB-358184

Filaseta-Moy conjecture on the Galois groups of the sextic binomial polynomials

摘录 / EXCERPT: Let ... be the degree-six truncation of the binomial expansion, and let $\mathfrak{S}_6$ denote the full symmetric group on six symbols. Filaseta-Moy conjecture. The Galois group of...

conjecture | open | MathDB 页面 | 原文线索

V2-010068 | MATHDB-358188

The minor-homomorphism conjecture for totally symmetric and Mal'cev clones

摘录 / EXCERPT: Let E_k be a k-element set with $k\geq 2$, and let $\mathcal S$ be a clone over E_k. Say that $\mathcal S$ satisfies $\operatorname{TS}(n)$ for every $n\geq 2$ and satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-010069 | MATHDB-358208

Geometric Breuil--Mézard conjecture for \(\mathrm{GSp}_4\) deformation rings

摘录 / EXCERPT: Let G_K be the absolute Galois group of K, let $\overline{\rho}:G_K\rightarrow\mathrm{GSp}_4(\mathbf{F})$ be continuous, and let $R_{\overline{\rho}}^\square$ be its framed def...

conjecture | open | MathDB 页面 | 原文线索

V2-010070 | MATHDB-358211

Gee--Herzig--Savitt's tame weight conjecture

摘录 / EXCERPT: Let F, χ, and $\overline r$ be as in the paper, with $\overline r$ automorphic of some weight and level. Let $W(\overline r)$ be its set of modular Serre weights, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010071 | MATHDB-358212

Breuil--Mézard weight conjecture without a tameness hypothesis

摘录 / EXCERPT: Let $\overline r$ be as in the paper, and let $W_{\mathrm{gen}}(\overline r)$ be the subset of generic modular Serre weights. Let $W^{\mathrm{BM}}_{\mathrm{gen}}(\overline r_p)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-010072 | MATHDB-358214

The acyclicity criterion for semiorthogonal indecomposability of minimal surfaces

摘录 / EXCERPT: Acyclicity criterion. The surface X is not semiorthogonally indecomposable if and only if $\mathcal{O}_X$ is acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-010073 | MATHDB-358240

Uniqueness conjecture for commuting operators in the $U_{\mathfrak{h}_5}$ algebra

摘录 / EXCERPT: Uniqueness conjecture. Up to canonical transformation, there are no other non-trivial commuting operators in the $U_{\mathfrak{h}_5}$ algebra of degree 2 and 3 in p other tha...

conjecture | open | MathDB 页面 | 原文线索

V2-010074 | MATHDB-358260

Plectic points and the algebraic rank of quadratic twists

摘录 / EXCERPT: Let A/F be a modular elliptic curve, let E/F be a quadratic CM extension of class number one, let A^E/F be the quadratic twist of A by E/F, and let ψ be an optimal e...

conjecture | open | MathDB 页面 | 原文线索

V2-010075 | MATHDB-358271

Equivalence of short pp-definitions and few subpowers

摘录 / EXCERPT: Let Γ be a constraint language. It has short pp-definitions if every n-ary relation pp-definable from Γ is definable by a primitive positive formula whose length is...

conjecture | open | MathDB 页面 | 原文线索

V2-010076 | MATHDB-358307

Graded-ring conjecture for absorbing ideals

摘录 / EXCERPT: Graded-ring conjecture. For any ideal α of a graded ring R,

conjecture | open | MathDB 页面 | 原文线索

V2-010077 | MATHDB-358314

Milnor's Galois-orbit conjecture for unicritical preperiodic parameters

摘录 / EXCERPT: Milnor's Galois-orbit conjecture. These values of c form one Galois orbit under the action of the absolute Galois group of $\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010078 | MATHDB-358318

Conjecture on the isolated-singularity injection for Kloosterman motives

摘录 / EXCERPT: Isolated-singularity conjecture. The morphism above is an isomorphism for any n and λ when the matrix $N(V_\lambda)+E(V_\lambda)$ is invertible. This would imply unramifi...

conjecture | open | MathDB 页面 | 原文线索

V2-010079 | MATHDB-358363

Simis' Cremona-map criterion for normal Rees algebras

摘录 / EXCERPT: Simis' reformulation. The map F is a Cremona map if and only if the Rees algebra $S[Iz]$ is normal.

conjecture | open | MathDB 页面 | 原文线索

V2-010080 | MATHDB-358365

The algebraic packing problem for squarefree monomial ideals

摘录 / EXCERPT: Algebraic packing conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-010081 | MATHDB-358366

Normality conjecture for packing-property Rees algebras

摘录 / EXCERPT: Packing-property normality conjecture. The Rees algebra $S[Iz]$ is normal.

conjecture | open | MathDB 页面 | 原文线索

V2-010082 | MATHDB-358382

Folklore Koszul self-duality conjecture for operads and right modules

摘录 / EXCERPT: Let O be a termwise Σ-finite, termwise cofibrant operad and let R be a termwise Σ-finite, termwise cofibrant right module in $(\mathrm{Sp},\wedge)$. Write K for...

conjecture | open | MathDB 页面 | 原文线索

V2-010083 | MATHDB-358424

Spectral decomposition conjecture for Hecke operators on half-densities

摘录 / EXCERPT: Let C be a smooth proper curve over a non-archimedean local field K, let G be a split reductive group, and let $H(C)$ be the commutative algebra of Hecke operators acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-010084 | MATHDB-358462

Cai-Fan rationality and meromorphy conjecture for spherical characters

摘录 / EXCERPT: Let E be a coefficient field embeddable into $\mathbb C$, let R be a finitely reduced Noetherian E-algebra, let Σ be a fixed Zariski-dense subset of closed points of...

conjecture | open | MathDB 页面 | 原文线索

V2-010085 | MATHDB-358465

Fagundes-Mello conjecture on images of multilinear polynomials on upper triangular matrix algebras

摘录 / EXCERPT: Let K be a field, let $n\geq 2$ be an integer, and let p be a multilinear polynomial in noncommuting variables. Let $T_n(K)$ denote the algebra of all $n\times n$ upper triangu...

conjecture | open | MathDB 页面 | 原文线索

V2-010086 | MATHDB-358488

Speciality conjecture for transposed Poisson algebras

摘录 / EXCERPT: Speciality conjecture. Every transposed Poisson algebra is special.

conjecture | open | MathDB 页面 | 原文线索

V2-010087 | MATHDB-358514

Arithmetic fundamental lemma for the spherical Hecke algebra

摘录 / EXCERPT: Let $\varphi'\in\mathcal{H}_{K'^{{{{{{{{{\flat}}}}}}}}}}\otimes\mathcal{H}_{K'}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010088 | MATHDB-358518

The eigenvalue decomposition conjecture for Toeplitz matrices generated by $g_{r,s}$

摘录 / EXCERPT: The eigenvalue decomposition conjecture. The positive eigenvalues satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-010089 | MATHDB-358522

Projection conjecture for cluster variables of Gr(3,n)

摘录 / EXCERPT: Given $n \geq n' \geq 3$ and a subset $I \subseteq [n]$ with $|I|=n'$, define the projection ... by retaining the columns indexed by I. Projection conjecture. The element x is a...

conjecture | open | MathDB 页面 | 原文线索

V2-010090 | MATHDB-358523

Improved dimension dependence for approximate matchings

摘录 / EXCERPT: Let M and N be persistence modules that are ϵ-interleaved, with $\epsilon\geq 0$. Let ... Here $B(M)$ is the relevant neighborhood of M defined in the paper, and an...

conjecture | open | MathDB 页面 | 原文线索

V2-010091 | MATHDB-358524

Matching conjecture for upset-decomposable modules

摘录 / EXCERPT: Let M and N be upset decomposable persistence modules, where an upset decomposable module is a direct sum of upset modules. The upset matching conjecture. There is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-010092 | MATHDB-358582

The bi-canonical degree comparison conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Cohen-Macaulay local ring that has a canonical ideal $\mathcal{C}$. The canonical degree and bi-canonical degree are denoted by $\operatorname{cdeg}(R)$...

conjecture | open | MathDB 页面 | 原文线索

V2-010093 | MATHDB-358583

The hyperplane-section conjecture for the bi-canonical degree

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Cohen-Macaulay local ring with a canonical ideal $\mathcal{C}$. Let x be an element of R that is regular modulo $\mathcal{C}$, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-010094 | MATHDB-358592

Conjecture on the maximal number of terms in the averaged Baker-Campbell-Hausdorff expansion

摘录 / EXCERPT: Term-count conjecture. For $d\geq L$, the maximal possible number of terms satisfies $Q_{L,d}=Q_{L,L}$ and coincides with the number of terms obtained by expanding R_L in the ave...

conjecture | open | MathDB 页面 | 原文线索

V2-010095 | MATHDB-358597

Categorical graph minor conjecture

摘录 / EXCERPT: Let $cmathcal{GM}$ be the category of graphs and minor morphisms. A category is Noetherian over a ring if its associated module categories satisfy the relevant ascending-chain fini...

conjecture | open | MathDB 页面 | 原文线索

V2-010096 | MATHDB-358600

Strong 1-boundedness characterized by vanishing first L^2-Betti number

摘录 / EXCERPT: Let (M,τ) be a tracial von Neumann algebra, and let A be a weak*-dense, finitely presented, algebraically sofic, unital $*$-subalgebra of M. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-010097 | MATHDB-358628

Cossu-Zanardo conjecture on factorizations of quadratic-ring matrices

摘录 / EXCERPT: Cossu-Zanardo conjecture. There exist $a,b,c\in\mathbb{Z}[\sqrt{D}]$ with $a(1-a)=bc$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010098 | MATHDB-358629

Cossu-Zanardo necessity conjecture for idempotent factorizations

摘录 / EXCERPT: Cossu-Zanardo necessity conjecture. If this matrix can be written as a product of two idempotent matrices, then it must satisfy the Cossu-Zanardo conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010099 | MATHDB-358650

The COP-Farrell-Jones conjecture for Hecke algebras

摘录 / EXCERPT: COP-Farrell-Jones conjecture for Hecke algebras. For every uniformly regular ring R with $\mathbb{Q} \subseteq R$, the projection

conjecture | open | MathDB 页面 | 原文线索

V2-010100 | MATHDB-358716

The Unipotent Wilf Conjecture for unipotent numerical monoids

摘录 / EXCERPT: Let G be a unipotent linear algebraic group, and let $M=G_{\mathbb N}$. Let S be a unipotent numerical monoid in M, with relative conductor $\mathtt{c}_M(S)$, relative sporad...

conjecture | open | MathDB 页面 | 原文线索

V2-010101 | MATHDB-358742

Iwasawa's main conjecture in the coherent case

摘录 / EXCERPT: Let $\mathbb{L}$ be the coefficient field, let $\mathcal{E}/E$ be a $mathtt{P}$-extension, and let $\mathbb{W}_Pi$ be the Galois representation attached to Pi. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-010102 | MATHDB-358743

Incoherent Rankin-Selberg p-adic height nonvanishing conjecture

摘录 / EXCERPT: Assume Hypothesis $ref{hy:galois}$, let $\mathbb{V}_Pi$ be the associated Galois representation, and let $\mathscr{L}^1_\mathcal{E}(Pi)$ be the height-valued p-adic L-function....

conjecture | open | MathDB 页面 | 原文线索

V2-010103 | MATHDB-358744

Iwasawa's main conjecture via the Rankin-Selberg p-adic height

摘录 / EXCERPT: Assume that Pi is incoherent and Hypothesis $ref{hy:galois}$ holds. Let $\mathcal{E}/E$ be a $mathtt{P}$-extension. Height-form main conjecture. The following are equivalent: (i)...

conjecture | open | MathDB 页面 | 原文线索

V2-010104 | MATHDB-358769

Equivariant injection conjecture for the matrix coinvariant algebra

摘录 / EXCERPT: Equivariant injection conjecture. For all $0<d<n-1$, there exists a linear injection

conjecture | open | MathDB 页面 | 原文线索

V2-010105 | MATHDB-358790

Nagata's theorem extension and the Grothendieck-Remak property for tensor categories

摘录 / EXCERPT: Let k be a field and let $\mathcal{C}$ be a tensor category of moderate growth. Write $\mathsf{Alg}\mathcal{C}$ for its category of algebra objects, and let A^0 denote the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-010106 | MATHDB-358791

The maximal nilpotency conjecture for Ver_{p^∞}

摘录 / EXCERPT: Let $\mathcal{C}$ be a tensor category. For every simple object $L\ne\mathbbm{1}$, require that $\operatorname{Sym} L$ be finite, and for every non-split monomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-010107 | MATHDB-358796

The finite-type conjecture for finitely generated algebras in positive characteristic

摘录 / EXCERPT: Finite-type conjecture. Every finitely generated algebra in $mathcal C$ is of finite type.

conjecture | open | MathDB 页面 | 原文线索

V2-010108 | MATHDB-358801

Novik-Zheng higher affine-stress reconstruction conjecture

摘录 / EXCERPT: Let $d\geq 4$, let $2\leq i\leq d/2$, and let P be a simplicial d-polytope. Assume that P has no missing faces of dimension at least $d-i+1$. Equivalently, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010109 | MATHDB-358810

The parity-dependent spectrum conjecture for repeated-part type-A seaweeds

摘录 / EXCERPT: Parity-dependent spectrum conjecture. The seaweed $\mathfrak{g}$ is Frobenius, and the set of distinct eigenvalues in its spectrum is the collection of integers in

conjecture | open | MathDB 页面 | 原文线索

V2-010110 | MATHDB-358811

The recursive spectrum conjecture for repeated-part type-A seaweeds

摘录 / EXCERPT: Recursive spectrum conjecture. The distinct eigenvalues in the spectrum of $\mathfrak{g}_{k,r}$ are precisely the integers in $[-k+1,k]$, and $\mathfrak{g}_{k,r}$ has the log-conca...

conjecture | open | MathDB 页面 | 原文线索

V2-010111 | MATHDB-358812

The log-concave spectrum conjecture for Frobenius maximal parabolic type-A seaweeds

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Frobenius maximal parabolic type-A seaweed, meaning a Frobenius seaweed subalgebra of $\mathfrak{sl}(n)$ obtained from a maximal parabolic construction. Its...

conjecture | open | MathDB 页面 | 原文线索

V2-010112 | MATHDB-358825

The SHGH conjecture for plane blowups

摘录 / EXCERPT: Let X be a very general blowup of $\mathbb{P}^2$ at $m\geq 10$ points $p_1,\dots,p_m$. Let H denote the pullback of the class of a line, let E_i denote the exceptional diviso...

conjecture | open | MathDB 页面 | 原文线索

V2-010113 | MATHDB-358828

Oort's conjecture on direct sums of Newton polygons of curves

摘录 / EXCERPT: Oort's conjecture. If ξ_i occurs on $\mathcal{M}_{g_i}$ for $i=1,2$, then $\xi_1\oplus\xi_2$ occurs on $\mathcal{M}_{g_1+g_2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010114 | MATHDB-358834

Binary weak nilpotence conjecture for the identities $w_n=0$ through $w_{2n-3}=0$

摘录 / EXCERPT: Binary weak nilpotence conjecture. For each n, the identities $w_n=w_{n+1}=\cdots=w_{2n-3}=0$ imply the identity $w_{2n-2}=0$, and hence imply weak nilpotence of index n.

conjecture | open | MathDB 页面 | 原文线索

V2-010115 | MATHDB-358835

Nilpotence-index-15 conjecture for multiplication algebras under $w_5=0$

摘录 / EXCERPT: Nilpotence-index-15 conjecture. The associative algebras $L(A)$ and $R(A)$ are nilpotent; in both cases, the nilpotence is of index 15 and is not less in general.

conjecture | open | MathDB 页面 | 原文线索

V2-010116 | MATHDB-358836

Symmetric weak nilpotence conjecture for the identities $t_n=0$

摘录 / EXCERPT: Symmetric weak nilpotence conjecture. The identity $t_4=0$ does not imply $t_5=0$, and hence does not imply the index-4 weak nil property; the identities $t_4=t_5=0$ imply the we...

conjecture | open | MathDB 页面 | 原文线索

V2-010117 | MATHDB-358837

The 7-Engel conjecture for the identity $t_4=0$

摘录 / EXCERPT: 7-Engel conjecture. The identity $t_4=0$ implies the 7-Engel identity.

conjecture | open | MathDB 页面 | 原文线索

V2-010118 | MATHDB-358838

Ternary multiplication nilpotence conjecture under $w_7^{(3)}=0$

摘录 / EXCERPT: Ternary multiplication nilpotence conjecture. The associative algebras $L(A)$ and $R(A)$ are nilpotent; in both cases, the nilpotence is of index 13 and is not less in general.

conjecture | open | MathDB 页面 | 原文线索

V2-010119 | MATHDB-358839

Finiteness conjecture for Gröbner bases of odd-arity ternary identities

摘录 / EXCERPT: Ternary Gröbner-basis finiteness conjecture. For each $d\in\mathbb{N}$, the reduced Gröbner basis for the nonsymmetric operad encoding the identity $w_{2d+1}^{(3)}=0$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-010120 | MATHDB-358840

Finiteness conjecture for Gröbner bases of k-ary identities

摘录 / EXCERPT: k-ary Gröbner-basis finiteness conjecture. For all $k\in\mathbb{N}$ and $d\equiv1\pmod{k-1}$, the reduced Gröbner basis for the nonsymmetric operad encoding the identity...

conjecture | open | MathDB 页面 | 原文线索

V2-010121 | MATHDB-358885

The analytic-torsion orthogonal complement conjecture for FI-objects

摘录 / EXCERPT: Let $\FIVAnly$ be the ∞-category of analytic $\FI$-objects, and let $\FIVTors$ be the full sub-∞-category of torsion $\FI$-objects. For full subcategories...

conjecture | open | MathDB 页面 | 原文线索

V2-010122 | MATHDB-358919

Panja-Prasad's two-summand conjecture for upper triangular matrix algebras

摘录 / EXCERPT: Let K be an algebraically closed field, let $n\geq 2$ and $m\geq 1$ be integers, and let $T_n(K)$ denote the algebra of $n\times n$ upper triangular matrices over K. For...

conjecture | open | MathDB 页面 | 原文线索

V2-010123 | MATHDB-358928

Odoni's conjecture on surjective arboreal representations

摘录 / EXCERPT: Let K be a Hilbertian field of characteristic 0, and let $n\geq 2$. For a polynomial $f\in K[x]$, write $f^{-k}(0)$ for the set of k-th preimages of 0, and let $[S_n]^k$ de...

conjecture | refuted | MathDB 页面 | 原文线索

V2-010124 | MATHDB-358973

Derived equivalence conjecture for Azumaya algebras on determinantal Calabi-Yau threefolds

摘录 / EXCERPT: Let $\widehat{X}$ be a small resolution of $X_{\vec{d}}$, let $S_{\vec{d},1}$ be the relevant singular locus, and let $f:\widehat{X}\to\mathbb{P}^3$ be the morphism in the construc...

conjecture | open | MathDB 页面 | 原文线索

V2-010125 | MATHDB-359001

The non-continuity conjecture for the Le Potier function on irregularity-zero surfaces

摘录 / EXCERPT: Le Potier non-continuity conjecture. The function $\Phi_{S,H}$ is not continuous at 0.

conjecture | open | MathDB 页面 | 原文线索

V2-010126 | MATHDB-359033

The Clifford-algebra resolution conjecture for the Calabi-Yau double cover

摘录 / EXCERPT: Let X be the base of the Calabi-Yau double cover Y, let Cl_{0} be the even part of the sheaf of Clifford algebras obtained from the quadratic GLSM superpotential, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-010127 | MATHDB-359041

Ideal Generation Conjecture for general points

摘录 / EXCERPT: Let S be the homogeneous coordinate ring of $\mathbb{P}^n$, let Z be a general set of r points, and let $I(Z)$ be its vanishing ideal. Let d be the Hilbert regularity of...

conjecture | open | MathDB 页面 | 原文线索

V2-010128 | MATHDB-359105

Herzog-Stamate conjecture on Betti numbers of tangent cones

摘录 / EXCERPT: Let $H = \langle a_1, a_2, \ldots, a_n \rangle$ be a numerical semigroup, and let I_H^* be the defining ideal of the tangent cone of the semigroup ring $K[H]$. Denote by ... the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-010129 | MATHDB-359245

The Fundamental Lemma for central simple algebras

摘录 / EXCERPT: Let D be a central simple algebra and let G and G' be the groups in the Guo-Jacquet comparison, with test functions $f_D=1_{O_D^\times}$ and f'_D as defined from the standa...

conjecture | open | MathDB 页面 | 原文线索

V2-010130 | MATHDB-359322

Strict dyadic Swan-exponent inequality for irreducible orthogonal representations

摘录 / EXCERPT: Strict inequality conjecture. Assume $p=2$. For any irreducible orthogonal Galois representation ρ of G which is not tame, one has

conjecture | open | MathDB 页面 | 原文线索

V2-010131 | MATHDB-359341

The depth-two Apéry-set conjecture for simplicial affine semigroup rings

摘录 / EXCERPT: Let $\mathcal{S}$ be the simplicial affine semigroup under consideration, let $\Bbbk[\mathcal{S}]$ be its semigroup ring, let E be the distinguished generating set, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-010132 | MATHDB-359342

Embedding conjecture for functors with diagonal-preserving liftings

摘录 / EXCERPT: Embedding conjecture. If F has a diagonal-preserving lifting, then F can be embedded in a weak pullback-preserving functor.

conjecture | open | MathDB 页面 | 原文线索

V2-010133 | MATHDB-359424

The reduction-number conjecture for asymptotic fullness

摘录 / EXCERPT: Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring with infinite residue field and ... For a minimal reduction I of $\mathfrak{m}$, let $n_3(I)$ denote the asymptotic fullness...

conjecture | open | MathDB 页面 | 原文线索

V2-010134 | MATHDB-359448

Dao and Takahashi's infinite-radius conjecture for resolving subcategories

摘录 / EXCERPT: Let R be a Cohen--Macaulay local ring, and let $\mathcal{X}$ be a resolving subcategory of $\operatorname{mod} R$. A module is maximal Cohen--Macaulay if its depth equals...

conjecture | open | MathDB 页面 | 原文线索

V2-010135 | MATHDB-359454

Levelness conjecture for simple thin path polyominoes with singleton maximal intervals

摘录 / EXCERPT: Let $\mathcal{P}$ be a simple thin polyomino, meaning a path polyomino with no interval containing more than two cells, and suppose that every maximal interval consists of a single...

conjecture | open | MathDB 页面 | 原文线索

V2-010136 | MATHDB-359469

Cowling's conjecture on non-archimedean rank-one groups

摘录 / EXCERPT: Let F be a local field of characteristic 0, let $\mathbf{G}$ be an almost simple, connected and simply connected F-algebraic group, and set $G=\mathbf{G}(F)$. In Theorem, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-010137 | MATHDB-359495

Conjecture on orientation reversal for even-winding satellite patterns

摘录 / EXCERPT: Let P be a pattern with winding number, the algebraic intersection number of the pattern with a meridional disk of the solid torus, a nonzero even integer. For a knot K, write...

conjecture | open | MathDB 页面 | 原文线索

V2-010138 | MATHDB-359541

Bertolini-Darmon's derived anticyclotomic p-adic Birch and Swinnerton-Dyer conjecture

摘录 / EXCERPT: Let E/Q be an elliptic curve, let K be an imaginary quadratic field satisfying the Heegner hypothesis, and let K_infty/K be the anticyclotomic Z_p-extension with Galois gro...

conjecture | open | MathDB 页面 | 原文线索

V2-010139 | MATHDB-359565

The stable second Brauer-Thrall conjecture for τ-tilting infinite algebras

摘录 / EXCERPT: Stable Brauer-Thrall II' conjecture. There exist $\bd\in K_0(A)^+$ and $\theta\in\K_0(A)^*$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010140 | MATHDB-359655

The type C rack conjecture for finite Nichols algebras

摘录 / EXCERPT: Let $\mathcal{O}$ be a finite rack of type C, let $\mathfrak{q}$ be a faithful 2-cocycle on $\mathcal{O}$, and let $\mathscr{B}(\mathcal{O},\mathfrak{q})$ denote the associated N...

conjecture | open | MathDB 页面 | 原文线索

V2-010141 | MATHDB-359730

Affine Hecke algebra conjecture for metaplectic type theory

摘录 / EXCERPT: Let $\widetilde{G}$ be the metaplectic cover and $(\widetilde{J},\widetilde{\lambda})$ the type constructed in the paper. Let $\widetilde{M}$ be the associated Levi subgroup,...

conjecture | open | MathDB 页面 | 原文线索

V2-010142 | MATHDB-359785

Amitsur's conjecture on codimension growth of associative PI algebras

摘录 / EXCERPT: Amitsur's conjecture. The limit exists and belongs to $\mathbb Z_+$.

conjecture | open | MathDB 页面 | 原文线索

V2-010143 | MATHDB-359805

The Z_p^2-extension cotorsion conjecture for multi-signed Selmer groups

摘录 / EXCERPT: Let F be an imaginary quadratic field, let F_∞ be its unique Z_p^2-extension, and let $\Lambda(\Omega)$ denote the Iwasawa algebra associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-010144 | MATHDB-359874

Linear convergence of the Collatz method for dual number matrices

摘录 / EXCERPT: Collatz convergence-rate conjecture. What is the condition such that both sequences $\{\underline{\lambda}_k\}$ and $\{\overline{\lambda}_k\}$ converge to the same dual number...

open_problem | open | MathDB 页面 | 原文线索

V2-010145 | MATHDB-359880

Invertibility conjecture for the left partially dualized quasi-Hopf algebra

摘录 / EXCERPT: Let H be a Hopf algebra, let B be a left coideal subalgebra, and let $(H/B^+H)^\ast\#B$ be the associated left partially dualized quasi-Hopf algebra. In Remark on invertibility...

conjecture | open | MathDB 页面 | 原文线索

V2-010146 | MATHDB-359897

Hecke-algebra analogue of the somewhere-to-below shuffle commutator bounds

摘录 / EXCERPT: Hecke deformation conjecture. Corollary 28 and Corollary 29 both seem to hold in $\mathcal{H}$ when every t_ℓ is replaced by $t_\ell^{\mathcal{H}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010147 | MATHDB-359948

The tropical F-polynomial identity for nondegenerate quivers with potential

摘录 / EXCERPT: Let δ and $\check{\epsilon}$ be weights for a quiver with potential, and write $\check{f}_{\delta}$ and $f_{\check{\epsilon}}$ for the dual and ordinary tropical F-polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-010148 | MATHDB-359949

The mutation formula for Kashiwara data in classical examples

摘录 / EXCERPT: Let $\boldsymbol{i}$ be a reduced expression of the longest element ω_0, and let $\operatorname{trop}(\Delta,\mathcal{S})$ denote the tropical points of a seed. For a sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-010149 | MATHDB-359950

BK-biperfectness of theta and triangular bases

摘录 / EXCERPT: Let $\overline{\mathcal{C}}(\Delta)$ be the upper cluster algebra, and let a basis be BK-biperfect when it satisfies the BK-biperfectness conditions for the weak upper normal cryst...

conjecture | open | MathDB 页面 | 原文线索

V2-010150 | MATHDB-359972

Simon’s conjecture for monomial ideals

摘录 / EXCERPT: Simon’s conjecture for monomial ideals. If $G(I)\subset G(\mathfrak{m}^d)$, then I can be extended to $\mathfrak{m}^d$ by linear quotients.

conjecture | open | MathDB 页面 | 原文线索

V2-010151 | MATHDB-359974

Catanese's paracanonical system conjecture for surfaces

摘录 / EXCERPT: Catanese's conjecture. The dimension of the paracanonical system satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010152 | MATHDB-360001

The top Gallai-polynomial conjecture via second-order Eulerian numbers

摘录 / EXCERPT: For $n>1$, let $Q_{n,n-1}(q)$ be the coefficient appearing in the conjectured Hilbert-series expansion of the Gallai algebra $\mathcal G_{n,k}$. Let $E(n-1,j)$ denote the second-or...

conjecture | open | MathDB 页面 | 原文线索

V2-010153 | MATHDB-360130

Cascini-Spicer's algebraic integrability conjecture for foliated triples

摘录 / EXCERPT: Let $(X,\mathcal{F},B)$ be a $\mathbb{Q}$-factorial projective foliated triple, such that $\mathcal{F}$ is algebraically integrable, B is a $\mathbb{Q}$-divisor, (X,B) is klt,...

conjecture | open | MathDB 页面 | 原文线索

V2-010154 | MATHDB-360132

The ACSS-F-dlt equivalence conjecture for generalized foliated quadruples

摘录 / EXCERPT: Let $(X,\mathcal{F},B,\mathbf{M})$ be a generalized foliated quadruple. Here ACSS denotes the technical condition defined in the source, and F-dlt denotes foliated divisorial log t...

conjecture | open | MathDB 页面 | 原文线索

V2-010155 | MATHDB-360138

Uniqueness of the adjacent cluster under mutation

摘录 / EXCERPT: Let $(\mathbf x, \mathbf p, B)$ denote a seed in the cluster algebra with coefficients, or let $(\mathbf x,B)$ denote a seed in the coefficient-free cluster algebra. A cluster is t...

conjecture | open | MathDB 页面 | 原文线索

V2-010156 | MATHDB-360172

The maximal-minor F-threshold formula for large determinantal rings

摘录 / EXCERPT: Let $R=S/I_t(X)$, where X is an $n\times m$ matrix of indeterminates, $t=m\leq n$, and $c^{\mathfrak{m}}(\mathfrak{m})$ denotes the F-threshold of the homogeneous maximal ideal....

conjecture | open | MathDB 页面 | 原文线索

V2-010157 | MATHDB-360173

The square determinantal-ring F-threshold conjecture

摘录 / EXCERPT: Let $R=S/I_t(X)$ be the determinantal ring associated with an $n\times n$ matrix of indeterminates, and let $c^{\mathfrak{m}}(\mathfrak{m})$ denote the F-threshold of its homogeneo...

conjecture | open | MathDB 页面 | 原文线索

V2-010158 | MATHDB-360212

Cyclic Galois group conjecture for generic spectral covers

摘录 / EXCERPT: Let X be a compact Riemann surface of genus g_X, and let $L\to X$ be a holomorphic line bundle of positive degree. Fix a generic section ... with distinct zeros, and consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-010159 | MATHDB-360281

The square-root conjecture for characteristic-polynomial coefficients of special D-partitions

摘录 / EXCERPT: Square-root conjecture. Then $c_{2k_i}=(a_{k_i})^2$, where a_{k_i} is a polynomial on the Lie algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-010160 | MATHDB-360321

Generalized Haiman ideal Hilbert-series conjecture

摘录 / EXCERPT: Assume $d_1\leq\cdots\leq d_n$, set $d_{ij}=\min(d_i,d_j)$, and define ... Let $\mathfrak m=(t_1,\ldots,t_n,x_1,\ldots,x_n)$. Generalized Haiman ideal Hilbert-series conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-010161 | MATHDB-360335

Full runner removal conjecture for Ariki-Koike algebras

摘录 / EXCERPT: Let $\bm\lambda,\bm\mu\in\mathcal{P}^r$ lie in a block B of $\mathcal{H}_{r,n}$, with $\bm\mu$ e-multiregular. Obtain $\bm\lambda^{+\varnothing}$ and $\bm\mu^{+\varnothing}$ fr...

conjecture | open | MathDB 页面 | 原文线索

V2-010162 | MATHDB-360380

HMTW's F-threshold and multiplicity conjecture

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a noetherian local ring of dimension d containing a positive characteristic field. Let J be an ideal generated by a full system of parameters, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010163 | MATHDB-360430

Pantone-Bóna asymptotic conjecture for permutations avoiding A_{5,5}

摘录 / EXCERPT: Let $A_{5,5}=12354,12453,13452,23451$ be the set of patterns consisting of the four length-five permutations that begin with an increasing subsequence of length four and end in an...

conjecture | open | MathDB 页面 | 原文线索

V2-010164 | MATHDB-360431

Non-algebraicity conjecture for the generating function of A_{5,5}-avoiding permutations

摘录 / EXCERPT: Let $A_{5,5}=\{12354,12453,13452,23451\}$ and define the generating function ... where $\operatorname{Av}_n(A_{5,5})$ is the set of permutations of length n avoiding every pattern...

conjecture | open | MathDB 页面 | 原文线索

V2-010165 | MATHDB-360468

L'vov-Kaplansky conjecture for multilinear noncommutative polynomials

摘录 / EXCERPT: L'vov-Kaplansky conjecture. If f is multilinear, then its image in the matrix algebra $M_n(K)$ is a vector space for every $n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010166 | MATHDB-360469

Existence of monomial ideals with the strong Lefschetz property

摘录 / EXCERPT: Let $R=k[x_1,\dots,x_n]$ where k is a field of characteristic zero. Fix $n,d\geq 3$ and let μ lie in the interval $\left[n,\binom{n+d-1}{d}\right]$. A monomial ideal I mini...

conjecture | open | MathDB 页面 | 原文线索

V2-010167 | MATHDB-360485

The mock plectic regulator conjecture

摘录 / EXCERPT: Let K be a quadratic imaginary field, let $E/\mathbb Q$ be an elliptic curve, let $T_p(E)$ be its p-adic Tate module, and let δ_∞ and $\log_E^{a_p}$ be the maps def...

conjecture | open | MathDB 页面 | 原文线索

V2-010168 | MATHDB-360531

Freeness conjecture for the symmetric algebra of a nilradical

摘录 / EXCERPT: Let $\mathfrak g$ be a simple Lie algebra, let $\mathfrak n$ be a nilradical in $\mathfrak g$, and let U be the maximal unipotent subgroup associated with the chosen Borel subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-010169 | MATHDB-360563

Dichotomy conjecture for content in commutative rings

摘录 / EXCERPT: Dichotomy conjecture. The content of R satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010170 | MATHDB-360640

Nonnegativity conjecture for Hilbert coefficients of quadratic algebras

摘录 / EXCERPT: Nonnegativity conjecture. If S_n/I is a quadratic algebra, then $e_i(I)\ge0$ for all i.

conjecture | open | MathDB 页面 | 原文线索

V2-010171 | MATHDB-360641

Conjectural Hilbert series for generic quadratic ideals in four and five variables

摘录 / EXCERPT: Generic quadratic Hilbert-series conjecture. For the indicated values of n,r, the following formulas hold:

conjecture | open | MathDB 页面 | 原文线索

V2-010172 | MATHDB-360642

Hilbert-coefficient conjecture for powers of the square of the homogeneous maximal ideal

摘录 / EXCERPT: Square-of-the-maximal-ideal conjecture. The coefficients satisfy:

conjecture | open | MathDB 页面 | 原文线索

V2-010173 | MATHDB-360643

Conjectural Hilbert series for four-variable squarefree quadratic monomial extensions

摘录 / EXCERPT: Let $S_n=k[x_1,\ldots,x_n]$. For a set M of squarefree quadratic monomials, let $I_{n,M}$ be the monomial ideal generated by all squares x_i^2 together with the monomials in...

conjecture | open | MathDB 页面 | 原文线索

V2-010174 | MATHDB-360648

Migliore-Miró-Roig-Nagel conjecture on powers of general linear forms

摘录 / EXCERPT: Migliore-Miró-Roig-Nagel conjecture. If $n=3$, then R/I fails the WLP if and only if $d\geq 3$; if $n\geq 4$, then R/I fails the WLP if and only if $d>1$. The conjecture concer...

conjecture | open | MathDB 页面 | 原文线索

V2-010175 | MATHDB-360701

Computed deviations from Fröberg's conjectured Hilbert series

摘录 / EXCERPT: For generic linear forms l_i, let ... and let ... Here $F_{n,r,d}(t)$ is the Hilbert series conjectured by Fröberg for generic degree-d forms in the cases under consideration. The...

conjecture | open | MathDB 页面 | 原文线索

V2-010176 | MATHDB-360731

Generalized Becker finiteness conjecture for higher Pythagoras numbers

摘录 / EXCERPT: Let R be a commutative ring with identity. For a positive integer s, let $p_{2s}(R)$ denote the least number of $2s$-th powers needed in the relevant representation defining th...

conjecture | open | MathDB 页面 | 原文线索

V2-010177 | MATHDB-360749

Buchsbaum-Eisenbud conjectural resolution for Schur powers of a determinantal cokernel

摘录 / EXCERPT: Buchsbaum-Eisenbud conjectural resolution. A resolution of $\operatorname{coker}(\varphi^*_{p-1,1})$ should always start out as

conjecture | open | MathDB 页面 | 原文线索

V2-010178 | MATHDB-360796

Scalarity conjecture for the root-of-unity knot invariant

摘录 / EXCERPT: Let $q=\omega$ be a root of unity of order $N\geq 1$, let $H_{c,t}$ be the $4N$-dimensional Nichols f-algebra with scaling automorphism $\phi_t x_i=t_i x_i$, and let $R_{c,t}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-010179 | MATHDB-360797

Scalarity conjecture for the finite-dimensional Yetter-Drinfeld knot invariant

摘录 / EXCERPT: Assume that q is not a root of unity, that (q,t_1,t_2) satisfies $t_1t_2q^n=1$ for some integer $n\geq 1$, and that $(1-t_1)(1-t_2)\ne 0$. Let Y_n be the $4n$-dimensional rig...

conjecture | open | MathDB 页面 | 原文线索

V2-010180 | MATHDB-360820

Taketomi-Tamaru conjecture on congruent automorphism orbits and Ricci solitons

摘录 / EXCERPT: Let $\mathfrak{l}$ be an n-dimensional Lie algebra, and let L be the connected and simply connected Lie group with Lie algebra $\mathfrak{l}$. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-010181 | MATHDB-360840

Buzzard-Gee conjecture for Galois representations attached to automorphic forms

摘录 / EXCERPT: Buzzard-Gee conjecture. The following are true. There exists a map

conjecture | open | MathDB 页面 | 原文线索

V2-010182 | MATHDB-360842

Integral Galois representation conjecture for non-Eisenstein Hecke ideals

摘录 / EXCERPT: Integral Galois representation conjecture. There exists a unique continuous representation

conjecture | open | MathDB 页面 | 原文线索

V2-010183 | MATHDB-360846

Construction of torsion Galois representations

摘录 / EXCERPT: Let $K\subset G(\mathbf{A}_{F^+}^{\infty})$ be any good subgroup and let $\mathfrak{m}\subset \mathbf{T}^T(K,\lambda,\tau)$ be a maximal ideal. Let $G_{F,T}$ denote the Galois grou...

conjecture | open | MathDB 页面 | 原文线索

V2-010184 | MATHDB-360877

Integral and compatible Satake parameters for discrete automorphic representations

摘录 / EXCERPT: Let G be a connected reductive group over a global function field K, let $U\subset G(\mathbf A_K)$ be open compact, let E be a noetherian $\mathbb Z[\frac1p]$-algebra embedde...

conjecture | open | MathDB 页面 | 原文线索

V2-010185 | MATHDB-360915

Caruso-David-Mézard's genetic decomposition conjecture for deformation rings

摘录 / EXCERPT: Genetic decomposition conjecture. There exists a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-010186 | MATHDB-360930

Broué-Malle-Michel specialization conjecture for Deligne-Lusztig endomorphism algebras

摘录 / EXCERPT: Let $\mathbb{G}$ be a generic finite reductive group, let $e>0$, and let $(\mathbb{L},\lambda)$ be a Φ_e-cuspidal pair. Let $H_{W_{\mathbb{G},\mathbb{L},\lambda}}(\vec{u})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-010187 | MATHDB-361015

The strong π-regularity conjecture for pseudo π-polar π-UU rings

摘录 / EXCERPT: Let R be a ring. A ring is pseudo π-polar if it satisfies the pseudo π-polar condition, and π-UU if every unit $u\in U(R)$ has $1-u^n\in J(R)$ for some integer...

conjecture | open | MathDB 页面 | 原文线索

V2-010188 | MATHDB-361052

The formal multiple Eisenstein series spanning conjecture

摘录 / EXCERPT: Let $\mathcal{G}^{\!f}$ be the commutative $\mathbb{Q}$-algebra of formal multiple Eisenstein series, with elements $G^{\!f}(k_1,\dots,k_r)$ and double-indexed elements...

conjecture | open | MathDB 页面 | 原文线索

V2-010189 | MATHDB-361054

The analogue of the Deligne-Ihara conjecture for once-punctured CM elliptic curves

摘录 / EXCERPT: Let I be the index set of multi-indices used to define the characters $kappa_{\boldsymbol{m}}$, let g be the associated graded Lie algebra, and let $chi^{\boldsymbol{m}}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-010190 | MATHDB-361092

Vasconcelos' Vanishing Conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring of positive dimension, and suppose that R is unmixed. Let Q be a parameter ideal of R. Vasconcelos' Vanishing Conjecture. Th...

conjecture | solved | MathDB 页面 | 原文线索

V2-010191 | MATHDB-361093

Vasconcelos' uniformity conjecture for Chern coefficients

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring, let I be an $\mathfrak{m}$-primary ideal, and let $\mathcal{A}$ be an I-good filtration. Vasconcelos' uniformity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-010192 | MATHDB-361094

Vasconcelos' positivity conjecture for normal Hilbert coefficients

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be an analytically unramified Noetherian local ring of positive dimension, and let I be an $\mathfrak{m}$-primary ideal. Vasconcelos' positivity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-010193 | MATHDB-361095

The j-multiplicity regular-sequence conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated unmixed R-module. Let $\mathbf{x}=x_1,\ldots,x_m$ be an amenable partial system of parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-010194 | MATHDB-361114

The converse conjecture for symbolic F-thresholds of radical ideals

摘录 / EXCERPT: Converse conjecture. If I is symbolic F-split, then

conjecture | open | MathDB 页面 | 原文线索

V2-010195 | MATHDB-361176

Realization of small ML degrees for the triple Segre product

摘录 / EXCERPT: Small-ML-degree realization conjecture. The resulting scaled Segre embedding has ML degree d. This gives a proposed construction realizing every degree in the indicated range; th...

conjecture | open | MathDB 页面 | 原文线索

V2-010196 | MATHDB-361179

The pentagon relation for Baxter operators in the punctured-torus skein algebra

摘录 / EXCERPT: Let T^2-D be the once-punctured torus, and let $\widehat{\operatorname{Sk}}(T^2-D)$ be its completed skein algebra. For $\mathbf{x},\mathbf{y}\in\mathbb{Z}^2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010197 | MATHDB-361184

Rational identifiability conjecture for inverse Gaussian and gamma mixtures

摘录 / EXCERPT: For k-mixtures of the inverse Gaussian distribution or the gamma distribution, consider the moments through order $3k$, and let rational identifiability mean that a general momen...

conjecture | open | MathDB 页面 | 原文线索

V2-010198 | MATHDB-361195

Finiteness conjecture for Picard rank jumps in Lefschetz pencils

摘录 / EXCERPT: Picard-rank finiteness conjecture. The Picard number of $\mathcal{Y}_s$ is greater than or equal to 2 for at most finitely many values of $s\in\mathbb{P}^1(\mathbb{C})$.

conjecture | open | MathDB 页面 | 原文线索

V2-010199 | MATHDB-361233

Jahan-Zheng's conjecture on componentwise linear quotients

摘录 / EXCERPT: Let $S=K[x_1,\dots,x_n]$ be a polynomial ring over a field K, and let $I\subset S$ be a monomial ideal. For each $j\geq 0$, let $I_{\langle j\rangle}$ be the monomial ideal gener...

conjecture | open | MathDB 页面 | 原文线索

V2-010200 | MATHDB-361255

Okounkov's Kac polynomial character conjecture for the Maulik-Okounkov Lie algebra

摘录 / EXCERPT: Let Q be an arbitrary finite quiver and fix a dimension vector $\mathbf{d}\in\mathbf{N}^{Q_0}$. Let $\mathtt{a}_{Q,\mathbf{d}}(q)\in\mathbf{Z}[q]$ be Kac's polynomial, counting i...

conjecture | open | MathDB 页面 | 原文线索

V2-010201 | MATHDB-361432

The fullness conjecture for graded Grothendieck groups of Leavitt path algebras

摘录 / EXCERPT: Hazrat's fullness conjecture. There exists a unital $\mathbb{Z}$-graded k-homomorphism $\psi:L(E)\rightarrow L(F)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010202 | MATHDB-361433

The equivariant K-theory classification conjecture for graph C*-algebras

摘录 / EXCERPT: Equivariant K-theory classification conjecture. There is an order-preserving $\mathbb{Z}[x,x^{-1}]$-module isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010203 | MATHDB-361434

The Morita, singularity-category and shift-equivalence classification conjecture

摘录 / EXCERPT: Morita-singularity classification conjecture. The following conditions are equivalent: 1. There is an isomorphism of partially ordered $\mathbb{Z}[x,x^{-1}]$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-010204 | MATHDB-361451

The fiber-type classification conjecture for path ideals of cycles

摘录 / EXCERPT: Fiber-type classification conjecture. (1) If

conjecture | open | MathDB 页面 | 原文线索

V2-010205 | MATHDB-361485

Antieau-Gepner-Heller theorem of the heart conjecture for bounded t-structures

摘录 / EXCERPT: Antieau-Gepner-Heller's theorem of the heart conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-010206 | MATHDB-361490

Huneke's conjecture on tight Hilbert coefficients and F-rationality

摘录 / EXCERPT: Let $(R,\frak{m})$ be an excellent Noetherian local ring of characteristic $p>0$ such that its completion $\widehat{R}$ is reduced and satisfies (S_2). For a parameter ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-010207 | MATHDB-361617

Minimal content conjecture for p-adic L-functions in Hida families

摘录 / EXCERPT: Let $\bT_{\mathfrak{m}}$ be the relevant ordinary Hecke algebra, let $\mathcal{I}_{\mathfrak{m}}$ be its congruence ideal, and let $L_p^+(\mathfrak{m},\omega^j)$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-010208 | MATHDB-361628

Bucher-Machacek's red-size conjecture for connected quivers

摘录 / EXCERPT: Let Q be a connected quiver on n vertices that does not admit a reddening sequence. The red size $\operatorname{Red}(Q)$ is the maximal number of red vertices obtained after mu...

conjecture | open | MathDB 页面 | 原文线索

V2-010209 | MATHDB-361629

Bucher-Machacek's mutation-invariance conjecture for unrestricted red size

摘录 / EXCERPT: Let Q be a quiver, and let $\operatorname{uRed}(Q)$ denote its unrestricted red size, namely the maximal number of red vertices obtained after an arbitrary mutation sequence. Two...

conjecture | open | MathDB 页面 | 原文线索

V2-010210 | MATHDB-361648

Classification conjecture for algebras of Jordan type

摘录 / EXCERPT: Classification conjecture. Every algebra of Jordan type is either a Jordan algebra, a Matsuo algebra, or a factor of a Matsuo algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-010211 | MATHDB-361772

Generalized opposite-category Farrell-Jones conjecture for td-groups

摘录 / EXCERPT: Generalized opposite-category Farrell-Jones conjecture. The td-group G satisfies this conjecture if, for every Hecke category with G-support, the displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-010212 | MATHDB-361777

The Dao-number regularity conjecture for ideals

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be the local ring under consideration, let $I\subset R$ be an ideal, let $\mathcal{R}(\mathfrak{m})=\bigoplus_{k\ge0}\mathfrak{m}^k$ be the Rees algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-010213 | MATHDB-361866

Palindromicity conjecture for polar degrees of binary no-three-way interaction models

摘录 / EXCERPT: Let $X_{(2,2,k)}^3$ be the no-three-way interaction model of two binary random variables and a third one with state space k. Its polar degrees form a sequence. Palindromicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-010214 | MATHDB-361884

Strong Malle's conjecture for extensions with Galois group G

摘录 / EXCERPT: Let K be a number field, let $G\subset S_n$ act transitively on $\{1,\dots,n\}$, and let $M(K,G;B)$ count degree-n extensions L/K with Galois closure of group G and discrim...

conjecture | open | MathDB 页面 | 原文线索

V2-010215 | MATHDB-361896

Totally ramified subfield conjecture for p-algebras

摘录 / EXCERPT: Let K be a complete discrete valued field of characteristic $p>0$ with residue field k, and let A be a p-algebra over K, meaning a central simple algebra whose center has...

conjecture | open | MathDB 页面 | 原文线索

V2-010216 | MATHDB-361979

Conjecture that rank 2 webs give all rank 2 cluster variables

摘录 / EXCERPT: Let $\mathcal{W}$ be the set of rank 2 webs $W(R,S,T,V)$ on the Grassmannian $\operatorname{Gr}(k,n)$, where R,S,T are cyclically separated subsets and V is disjoint from the...

conjecture | open | MathDB 页面 | 原文线索

V2-010217 | MATHDB-361989

The generating-ideal conjecture for weak polynomial identities of the Weyl algebra

摘录 / EXCERPT: Generating-ideal conjecture. The ideal of all weak polynomial identities for the pair $(\mathsf{A}_1,\mathsf{V})$ is equal to $\mathcal{I}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010218 | MATHDB-361994

Sufficient ampleness conjecture for maximal infinitesimal variation

摘录 / EXCERPT: Sufficient ampleness conjecture. Any sufficiently ample line bundle L on S has maximal infinitesimal variation.

conjecture | open | MathDB 页面 | 原文线索

V2-010219 | MATHDB-361997

Equivalence between lax algebras for parametrised endofunctors and iterated monads

摘录 / EXCERPT: Equivalence conjecture. The assignment sending $(X,(P,f))$ to the corresponding lax algebra for F^κ,

conjecture | open | MathDB 页面 | 原文线索

V2-010220 | MATHDB-362021

Regularity formula for t-path ideals of unicyclic graphs

摘录 / EXCERPT: Regularity conjecture for t-path ideals. The displayed formula should hold. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-010221 | MATHDB-362059

The (S_2)-condition conjecture for edge rings of triangular cactus graphs

摘录 / EXCERPT: Let G be a triangular cactus graph, meaning a cactus graph whose blocks are all 3-cycles, and let its diameter be the maximum distance between two vertices of G. The (S_2)-...

conjecture | open | MathDB 页面 | 原文线索

V2-010222 | MATHDB-362083

The equal-rank conjecture for maximal cluster subalgebras

摘录 / EXCERPT: Equal-rank conjecture. The rank of $\mathcal{A}(\Sigma_{1})$ is the same as the rank of $\mathcal{A}(\Sigma_{2})$.

conjecture | open | MathDB 页面 | 原文线索

V2-010223 | MATHDB-362084

The ordered reduced-subalgebra conjecture for cluster algebras

摘录 / EXCERPT: Ordered reduced-subalgebra conjecture. Let $H_{1},H_{2},\ldots,H_{s}\in 2^{\operatorname{Aut}\mathcal{A}}\backslash\operatorname{ker}\phi$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-010224 | MATHDB-362102

Maximum-likelihood degree formula for generic polygon Wachspress models

摘录 / EXCERPT: Wachspress ML-degree conjecture. For a generic n-gon Q, the maximum likelihood degree is

conjecture | open | MathDB 页面 | 原文线索

V2-010225 | MATHDB-362121

The quasi-reductivity characterization of contact Lie poset algebras

摘录 / EXCERPT: Quasi-reductivity conjecture. An index-one Lie poset algebra of type A, B, C, or D is contact if and only if it is quasi-reductive.

conjecture | open | MathDB 页面 | 原文线索

V2-010226 | MATHDB-362134

The equivariant classification conjecture for graph C^*-algebras

摘录 / EXCERPT: Equivariant classification conjecture. There is an order-preserving $\mathbb Z[x,x^{-1}]$-module isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010227 | MATHDB-362146

The eigenvalue conjecture for permutation-like matrices

摘录 / EXCERPT: Let PL_d denote the set of permutation-like matrices of degree d, and let $A\in PL_d$ be a permutation-like matrix with $d>1$. A permutation-like matrix eigenvalue conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-010228 | MATHDB-362216

CMPP's product formula conjecture for admissible colored partitions of type $C_l^{(1)}$

摘录 / EXCERPT: CMPP's conjecture. The generating function for Λ-admissible colored partitions is

conjecture | open | MathDB 页面 | 原文线索

V2-010229 | MATHDB-362296

Vertical flip duality conjecture for W-superalgebra cosets

摘录 / EXCERPT: Let $N,n,m\in\mathbb{Z}$ satisfy $N>n,m\geq2$, and define $k^\sharp=-k-N+n+m$ and $\ell^\sharp=-\ell-N+n+m$. Let $\operatorname{Com}(A,B)$ denote the commutant of A in B. Verti...

conjecture | open | MathDB 页面 | 原文线索

V2-010230 | MATHDB-362315

The Gotzmann threshold polynomiality conjecture for powers of x_2

摘录 / EXCERPT: Gotzmann threshold polynomiality conjecture. For all $n \ge 3$, $\tau_n(x_2^d)$ is a polynomial of degree 2^{n-2} in d with dominant term

conjecture | open | MathDB 页面 | 原文线索

V2-010231 | MATHDB-362353

Okada's conjecture on canonical wavefront sets of unipotent representations

摘录 / EXCERPT: Let $G=\mathrm G(F)$, where $\mathrm G$ is a connected reductive algebraic group defined over F and inner to a split group. Let ${}^{K}\mathrm{WF}(\pi)$ denote the unramified can...

conjecture | open | MathDB 页面 | 原文线索

V2-010232 | MATHDB-362364

The a-generated nilpotent-or-solvable algebraicity conjecture

摘录 / EXCERPT: Let $G\subseteq\operatorname{Aut}(X)$ be a-generated, meaning that $G=\langle Y\rangle$ for an irreducible algebraic subset $Y\subseteq\operatorname{Aut}(X)$ containing the identit...

conjecture | open | MathDB 页面 | 原文线索

V2-010233 | MATHDB-362405

Critical-weight theta-block conjecture for intermediate Lie algebras

摘录 / EXCERPT: For each intermediate Lie algebra in the intermediate exceptional series, form the theta block from its associated positive roots and weights, using the construction described in t...

conjecture | open | MathDB 页面 | 原文线索

V2-010234 | MATHDB-362406

Critical-weight and lattice-index theta-block conjecture for intermediate Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}_I$ be an intermediate Lie algebra between $\mathfrak{g}_1$ and $\mathfrak{g}_2$, and let the associated theta block be constructed from the positive roots and wei...

conjecture | open | MathDB 页面 | 原文线索

V2-010235 | MATHDB-362407

Moreno-Frías-Rosales conjecture for numerical semigroup semimodules

摘录 / EXCERPT: For a numerical semigroup S, let $g(S)$ denote its genus and, for each $r\in\mathbb{N}_0$, let ... where $\widetilde{\Delta}=\Delta\cup\{0\}$ is the numerical semigroup associated...

conjecture | open | MathDB 页面 | 原文线索

V2-010236 | MATHDB-362438

Garsia-Procesi modules are weakly quasisymmetric characteristic compatible

摘录 / EXCERPT: Let $\falR_{\fmu}$ denote the Garsia-Procesi module associated with a partition $\fmu$, and let $\fY={x^{\alpha}\mid \alpha\text{ is sub-Yamanouchi}}$ be its specified monomial bas...

conjecture | open | MathDB 页面 | 原文线索

V2-010237 | MATHDB-362458

The Novikov kernel conjecture for universal geometric operations

摘录 / EXCERPT: Consider the universal map $\Phi_{{\text{\rm\tiny geo}}}^{\mathrm{Novikov}}$ induced by $\Phi_{{\text{\rm\tiny geo}}}$ after taking the operad quotient ... in the construction given...

conjecture | open | MathDB 页面 | 原文线索

V2-010238 | MATHDB-362461

Guralnick's conjecture on monodromy groups of large-degree indecomposable coverings

摘录 / EXCERPT: Let $f:X\rightarrow \mathbb P^1$ be an indecomposable covering of degree n, with X of genus $g_X=g$, and let $\operatorname{Mon}(f)$ denote its monodromy group. For each intege...

conjecture | open | MathDB 页面 | 原文线索

V2-010239 | MATHDB-362477

The split-octonion extension of Theorem 5.1

摘录 / EXCERPT: Let O be a split-octonion algebra, and let $f:O\rightarrow O$ be an additive map. Write $O^\times$ for the set of invertible elements of O. Split-octonion extension conjecture....

open_problem | open | MathDB 页面 | 原文线索

V2-010240 | MATHDB-362478

The additive-map factorization conjecture for alternative division rings

摘录 / EXCERPT: Additive-map factorization conjecture. There exists a fixed element $q\in D$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010241 | MATHDB-362507

Sols and Sols' conjectural bound for sections of rank-two bundles

摘录 / EXCERPT: Sols and Sols' conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-010242 | MATHDB-362527

The converse to E1-Morita invariance of module categories

摘录 / EXCERPT: The converse Morita conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-010243 | MATHDB-362531

Direct-sum decomposition conjecture for the associated separable algebra

摘录 / EXCERPT: Separable-algebra decomposition conjecture. There is a separable algebra $B'\in\CM$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010244 | MATHDB-362532

Fusion conjecture for one-dimensional condensable algebras

摘录 / EXCERPT: Condensable-algebra fusion conjecture. Their relative tensor product satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010245 | MATHDB-362534

Center-realization conjecture for 2-Morita equivalent condensable algebras

摘录 / EXCERPT: Center-realization conjecture. An indecomposable subalgebra

conjecture | open | MathDB 页面 | 原文线索

V2-010246 | MATHDB-362535

Nonfree local modules detect algebra automorphisms

摘录 / EXCERPT: Automorphism-detection conjecture. If M is not free, then M cannot be fixed by all algebra automorphisms of A.

conjecture | open | MathDB 页面 | 原文线索

V2-010247 | MATHDB-362563

Hilbert-series conjecture for the differential closure of the central zonotopal ideal

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a rank r central multiarrangement in $\mathbb{C}^n$ with $|{\mathcal{A}}|=m$. Define ... Here Ω_n is the superspace algebra, λ_L is the linear...

conjecture | open | MathDB 页面 | 原文线索

V2-010248 | MATHDB-362568

Base-point-freeness conjecture for lc algebraically integrable foliations

摘录 / EXCERPT: Base-point-freeness conjecture. Under these hypotheses, the divisor $K_{\mathcal{F}}+B+A$ is semi-ample$/U$, and in the Cartier case its sufficiently high multiples are globally ge...

conjecture | open | MathDB 页面 | 原文线索

V2-010249 | MATHDB-362572

The h-reducedness conjecture for the normalization quotient

摘录 / EXCERPT: Let R be a commutative ring with no nilpotent elements, and let $\overline{R}$ denote its integral closure. The quotient $\overline{R}/R$ is called h-reduced when it has the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-010250 | MATHDB-362574

Ikeda's conductor avoidance conjecture

摘录 / EXCERPT: Let R be an analytically unramified ring, let $\mathfrak{C}$ denote its conductor ideal, and let $\underline{x}$ be a regular sequence in R. Ikeda's conductor avoidance conject...

conjecture | open | MathDB 页面 | 原文线索

V2-010251 | MATHDB-362628

Sullivant's saturation conjecture for Gaussian DAG models

摘录 / EXCERPT: Sullivant's saturation conjecture. The vanishing ideal satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010252 | MATHDB-362652

Deterministic sub-distributive plasma conjecture

摘录 / EXCERPT: Deterministic sub-distributive plasma conjecture. If R is a deterministic plasma with a sub-distributive monoidal structure, then $\hat{H}R$ is canonically an $\mathbb{F}_1$-alge...

conjecture | open | MathDB 页面 | 原文线索

V2-010253 | MATHDB-362656

Almkvist's unimodality conjecture for two-variable Hilbert functions

摘录 / EXCERPT: Almkvist's conjecture. For each m, the Hilbert function $H(m,n)$ is unimodal for all $n\geq 11$. Moreover, if m is even, then $H(m,n)$ is unimodal for all n.

conjecture | open | MathDB 页面 | 原文线索

V2-010254 | MATHDB-362657

The Lefschetz and Hodge-Riemann strengthening of Almkvist's conjecture

摘录 / EXCERPT: Almkvist-Lefschetz strengthening. (i) For each m, the algebra $A(m,n)$ satisfies the SLP for all $n\geq 11$. (ii) Moreover, if m is even, then $A(m,n)$ satisfies the SLP for al...

conjecture | open | MathDB 页面 | 原文线索

V2-010255 | MATHDB-362677

Bracket-width conjecture for formal current Lie algebras

摘录 / EXCERPT: Bracket-width conjecture. The bracket width of $\mathfrak g\otimes_{G_a}G_a[[t]]$ is equal to 2 if $\mathfrak g$ is of type $\mathrm A_n$ or $\mathrm C_n$ $(n\ge 2)$, and to 1...

conjecture | open | MathDB 页面 | 原文线索

V2-010256 | MATHDB-362707

Weak Frey-Mazur conjecture for elliptic curves over number fields

摘录 / EXCERPT: Weak Frey-Mazur conjecture. There is a constant N_{E_0}, depending only on E_0, such that if $p>N_{E_0}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-010257 | MATHDB-362730

Rationality conjecture for the abelian part of an ell-adic representation

摘录 / EXCERPT: Let K and $E\subset\overline{\mathbb{Q}}_\ell$ be number fields, and let $\rho_\ell:\operatorname{Gal}_K\to\mathrm{GL}_n(\overline{\mathbb{Q}}_\ell)$ be a semisimple E-rational...

conjecture | open | MathDB 页面 | 原文线索

V2-010258 | MATHDB-362748

The Okrasa-Rzążewski conjecture on indecomposable graph cores

摘录 / EXCERPT: Let H be a connected core on $k\geq 3$ vertices. Write E_H for the edge relation of H, let $\langle E_H\rangle$ denote the relational clone generated by E_H, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010259 | MATHDB-362772

Two-generation conjecture for traceless matrices over the binary field

摘录 / EXCERPT: Two-generation conjecture. If $n \neq 4$, then $\operatorname{\mathfrak{sl}}_n(\mathbf{F}_2)$ is 2-generated.

conjecture | open | MathDB 页面 | 原文线索

V2-010260 | MATHDB-362801

The Virasoro conjecture for stable sheaves on simply connected surfaces

摘录 / EXCERPT: Virasoro conjecture. For every $k\geq -1$ and every $D\in\mathbb D^S$,

conjecture | open | MathDB 页面 | 原文线索

V2-010261 | MATHDB-362820

The automorphism-group conjecture for the exceptional function fields

摘录 / EXCERPT: Automorphism-group conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010262 | MATHDB-362834

Ritter-Weiss non-smoothed equivariant main conjecture with uniqueness

摘录 / EXCERPT: Non-smoothed equivariant main conjecture with uniqueness. There exists a unique element $\zeta_S(L^+_\infty/K)\in K_1(\mathcal Q(\mathcal G))$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010263 | MATHDB-362835

Equivariant Artin conjecture for the smoothed zeta element

摘录 / EXCERPT: Equivariant Artin conjecture for the smoothed zeta element. The element ζ_S^T lies in the image of the natural map

conjecture | open | MathDB 页面 | 原文线索

V2-010264 | MATHDB-362837

Triangular multi-lump solutions from rational curves with two cusps

摘录 / EXCERPT: Multi-lump construction conjecture. The constructions from rational curves with cuspidal singularities should generalize to give similar real regular M-lump solutions for all...

conjecture | open | MathDB 页面 | 原文线索

V2-010265 | MATHDB-362861

Center-generation conjecture for the universal Drinfeld-Yetter algebra

摘录 / EXCERPT: Let $\mathfrak{U}_{\mathsf{DY}}^1$ be the universal Drinfeld-Yetter algebra, with unit-labelled element r_1^{id}. Center-generation conjecture. The center of...

conjecture | open | MathDB 页面 | 原文线索

V2-010266 | MATHDB-362905

Gouvea's dimension conjecture for universal deformation rings

摘录 / EXCERPT: Gouvea's dimension conjecture. In the presentation proposition for the universal deformation ring, equality always holds in the Krull-dimension bound; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-010267 | MATHDB-362921

Lovett-Adiprasito-Kazhdan-Ziegler partition-rank conjecture

摘录 / EXCERPT: Lovett-Adiprasito-Kazhdan-Ziegler conjecture. There exists a constant C_d such that for any finite field and multilinear form f we have

conjecture | open | MathDB 页面 | 原文线索

V2-010268 | MATHDB-362923

Benson's generation conjecture for cochains on compact Lie groups

摘录 / EXCERPT: Benson's generation conjecture. The category $\mathcal{D}^b(C^*BG)$ is generated as a thick subcategory of $\mathcal{D}(C^*BG)$ by the objects $\operatorname{Hom}_{C_*G}(k,X)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-010269 | MATHDB-362925

Deformation to the associated graded algebra via an injective well-ordered valuation

摘录 / EXCERPT: Let A be a domain equipped with an injective well-ordered valuation ν, and let $gr A$ denote the associated graded algebra of the filtration on A induced by ν. A family...

conjecture | open | MathDB 页面 | 原文线索

V2-010270 | MATHDB-362945

Cellular structure conjecture for the infinite Apéry resolution

摘录 / EXCERPT: Let S be a numerical semigroup, let R be its numerical semigroup algebra, and consider the infinite Apéry resolution of the residue field over R. Cellular structure conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-010271 | MATHDB-362966

Malle-Bhargava heuristics for prescribed local conditions

摘录 / EXCERPT: Malle-Bhargava heuristics. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-010272 | MATHDB-362988

The generalized Quillen-Lichtenbaum conjecture for Artin L-functions

摘录 / EXCERPT: Let F'/F be a G-Galois extension of number fields, where G is finite, and let $\rho\colon G\to \mathrm{GL}_N(\mathcal{O}_E)$ be an E-linear Galois representation for a numb...

conjecture | open | MathDB 页面 | 原文线索

V2-010273 | MATHDB-363058

Jorgensen-Leuschke's canonical-module Betti-number conjecture

摘录 / EXCERPT: Let R be a Cohen-Macaulay local ring with canonical module ω. Jorgensen-Leuschke conjecture. The inequality ... implies that R is Gorenstein. The paper states this as an o...

open_problem | open | MathDB 页面 | 原文线索

V2-010274 | MATHDB-363069

The vanishing conjecture for equal-rank Lie superalgebra invariants of slice virtual knots

摘录 / EXCERPT: Vanishing conjecture. For all $m\geq 1$, if K is a slice virtual knot, then

conjecture | open | MathDB 页面 | 原文线索

V2-010275 | MATHDB-363109

Base-case conjecture for centrally extended orbits in finite models

摘录 / EXCERPT: Let e be a nilpotent element in a classical Lie algebra $\mathfrak{g}$, with partition λ, and let $\lambda^\intercal$ be its dual partition. Form λ_{red}...

conjecture | open | MathDB 页面 | 原文线索

V2-010276 | MATHDB-363116

Dao-Celikbas conjecture on the third Dao number and reduction numbers

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Cohen-Macaulay local ring with infinite residue field and dimension at least 2. A minimal reduction I of $\mathfrak{m}$ is a reduction minimal under...

conjecture | open | MathDB 页面 | 原文线索

V2-010277 | MATHDB-363161

Conjecture on lexicographic monotonicity of edges in exterior shifting

摘录 / EXCERPT: Lexicographic edge conjecture. If $\sigma<_{\mathrm{lex}}\sigma'$, then

conjecture | open | MathDB 页面 | 原文线索

V2-010278 | MATHDB-363183

The infinite-codimension conjecture for products of principal series

摘录 / EXCERPT: Let K be a field, let P_α denote the principal series of order type ω^α, and let R_α denote the corresponding set of products of principal series appear...

conjecture | open | MathDB 页面 | 原文线索

V2-010279 | MATHDB-363196

The equivalence conjecture for c6-modules, B-pairs and vector bundles

摘录 / EXCERPT: Let R be a ring and consider the corresponding $c6$-module functors, ${\square\square\square}B$-pair functors, and vector bundle functors over ${\square\square\square}FF$ functor...

conjecture | open | MathDB 页面 | 原文线索

V2-010280 | MATHDB-363320

Asymptotic strictness conjecture for the type of numerical monoids

摘录 / EXCERPT: Let $\mathscr{T}(e,m)$ and $\mathscr{D}(e,m)$ denote the functions defined in the paper for the type and the relevant upper bound of numerical monoids. Asymptotic strictness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-010281 | MATHDB-363325

The Yangian identification conjecture for cotangent Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional differential graded Lie algebra and set ... Let $\mathcal A_\hbar(\mathfrak{d},\gamma)$ be the Hopf algebra constructed in the source, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010282 | MATHDB-363345

The \(\mathfrak{M}_H(G)\)-conjecture for arithmetic Iwasawa modules

摘录 / EXCERPT: $\mathfrak{M}_H(G)$-conjecture. The Iwasawa module $X(E/F_\infty)$ belongs to the category $\mathfrak{M}_H(G)$. This is the non-commutative Iwasawa-theoretic expectation that arith...

conjecture | open | MathDB 页面 | 原文线索

V2-010283 | MATHDB-363390

The hereditary atomicity conjecture for integral domains

摘录 / EXCERPT: Hereditary atomicity conjecture. Every hereditarily atomic domain satisfies the ACCP.

conjecture | open | MathDB 页面 | 原文线索

V2-010284 | MATHDB-363411

Stability of symmetrically preconditioned conjugate gradient

摘录 / EXCERPT: Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be the least-squares matrix, and let $\boldsymbol{R}$ be computed from $\boldsymbol{A}$ by sketching and $\textsf{QR}$ factorizing the...

conjecture | open | MathDB 页面 | 原文线索

V2-010285 | MATHDB-363421

The multiplicative Thom-Sebastiani conjecture for Bernstein-Sato roots of ideals

摘录 / EXCERPT: Multiplicative Thom-Sebastiani conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-010286 | MATHDB-363440

The Karoubi-generation conjecture for affine Soergel trace categories

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and let $\operatorname{Tr}(\mathrm{ASBim}_n)^{\operatorname{Kar}}$ be the idempotent completion of the trace category of affine Soergel bimodules. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-010287 | MATHDB-363491

Conjecture on the signs in the leading-matrix realization

摘录 / EXCERPT: Sign conjecture. In Theorem, one has

conjecture | open | MathDB 页面 | 原文线索

V2-010288 | MATHDB-363500

Herzog's canonical trace conjecture for codimension-two Cohen-Macaulay algebras

摘录 / EXCERPT: Let S be either a regular local ring with residue class field K or a regular positively graded K-algebra. Let I be a perfect ideal of grade two, and let A be an Hilbert-B...

conjecture | open | MathDB 页面 | 原文线索

V2-010289 | MATHDB-363530

The non-square grid conjecture on the weak Lefschetz property

摘录 / EXCERPT: Non-square grid conjecture. The Artinian algebra A fails to have the weak Lefschetz property.

conjecture | open | MathDB 页面 | 原文线索

V2-010290 | MATHDB-363600

The free odd generation conjecture for the double shuffle Lie algebra

摘录 / EXCERPT: Free odd generation conjecture. The double shuffle Lie algebra $\mathfrak{dm}$ is a free Lie algebra with exactly one generator in each odd weight $w\geq 3$:

conjecture | open | MathDB 页面 | 原文线索

V2-010291 | MATHDB-363602

Ihara-Kaneko-Zagier extended double shuffle conjecture

摘录 / EXCERPT: Ihara-Kaneko-Zagier conjecture. All algebraic relations in the algebra $\mathcal{Z}$ of multiple zeta values are consequences of the extended double shuffle relations.

conjecture | open | MathDB 页面 | 原文线索

V2-010292 | MATHDB-363614

The non-algebraic-basis conjecture for pi

摘录 / EXCERPT: Let $E\!L^{alg}$ be the field of algebraically based numbers, obtained from $\mathbb{Q}$ by repeatedly adjoining algebraic numbers, exponentials, and logarithms of previously gener...

conjecture | open | MathDB 页面 | 原文线索

V2-010293 | MATHDB-363615

The non-exponential-algebraicity conjecture for pi

摘录 / EXCERPT: A complex number is exponentially algebraic if it is a root of a non-degenerate exponential-logarithmic system of equations. Non-exponential-algebraicity conjecture. ... This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-010294 | MATHDB-363639

Lando's uniqueness conjecture for the sl(2)-weight system

摘录 / EXCERPT: Lando's conjecture. There exists a unique 4-invariant that extends the $\mathfrak{sl}(2)$-weight system.

conjecture | open | MathDB 页面 | 原文线索

V2-010295 | MATHDB-363686

Cyclic versus purely inseparable totally ramified maximal subfields of p-algebras

摘录 / EXCERPT: Let K be a complete discrete valued field of characteristic p with residue field k, and let A be a p-algebra over K. A maximal subfield of A is totally ramified cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-010296 | MATHDB-363753

Colliot-Thélène-Parimala-Suresh local-global conjecture for projective homogeneous spaces

摘录 / EXCERPT: Let F be the function field of a p-adic curve, let G be a connected linear algebraic group over F, and let Z be a projective homogeneous space under G. Colliot-Thélène-...

conjecture | open | MathDB 页面 | 原文线索

V2-010297 | MATHDB-363779

The Dynkin-range licci conjecture for perfect ideals

摘录 / EXCERPT: Let I be a grade c perfect ideal in a local Noetherian ring R. Let d denote the deviation of I and t the minimal number of generators of $\operatorname{Ext}^c(R/I,R)$....

conjecture | open | MathDB 页面 | 原文线索

V2-010298 | MATHDB-363821

Minimality conjecture for Hilbert depth differences of squarefree monomial ideals

摘录 / EXCERPT: Minimality conjecture. If $q<q(d)$ or $n<n(d)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-010299 | MATHDB-363835

The F4-type Galois representation counting conjecture

摘录 / EXCERPT: The F4-type Galois representation counting conjecture. The number of equivalence classes of such representations with $\mathrm{HT}(\rho)=\mathrm{HT}(a,b,c,d)$ and $a,b,c,d\geq 1$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-010300 | MATHDB-363851

The harmonic algebra algebraic-properties conjecture

摘录 / EXCERPT: Let P be a lattice polytope, and let ${\mathcal H}_P\subseteq\mathbb{R}[y_0,\mathbf{y}]$ be its harmonic algebra, with bigraded ideal $\overline{{\mathcal H}}_P$. Harmonic algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-010301 | MATHDB-363858

Zamolodchikov-Ravanini-Tateo-Valleriani periodicity and dilogarithm conjecture for Y-systems

摘录 / EXCERPT: Zamolodchikov-Ravanini-Tateo-Valleriani conjecture. The solution is periodic with

conjecture | open | MathDB 页面 | 原文线索

V2-010302 | MATHDB-363882

Pantone's algebraicity conjecture for inversion sequences avoiding 000 and 102

摘录 / EXCERPT: Let $\mathcal{I}(000,102)$ be the class of inversion sequences avoiding the patterns 000 and 102, and let $F(x)$ be its ordinary generating function. Pantone's algebraicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-010303 | MATHDB-363883

Pantone's algebraicity conjecture for inversion sequences avoiding 010 and 102

摘录 / EXCERPT: Let $\mathcal{I}(010,102)$ be the class of inversion sequences avoiding the patterns 010 and 102, and let $F(x)$ be its ordinary generating function. Pantone's algebraicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-010304 | MATHDB-363897

Order-to-multiplet conjecture for Weyl-Heisenberg SICs

摘录 / EXCERPT: Fix a positive integer $d\neq 3$, let $\Delta_d=(d+1)(d-3)$, and write $\Delta_d=f^2\Delta_0$, where $\Delta_0=\operatorname{disc}(\mathcal O_{K_d})$ and $K_d=\Q(\sqrt{\Delta_d})$....

conjecture | open | MathDB 页面 | 原文线索

V2-010305 | MATHDB-363901

Galois-field conjecture for Weyl-Heisenberg SICs

摘录 / EXCERPT: Fix $d\geq1$, $d\neq3$, and let S be a d-dimensional Weyl-Heisenberg line-SIC with associated fields $\fieldtrip{S}$, $\fieldproj{S}$, and $\fieldeproj{S}$. Galois-field conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-010306 | MATHDB-363903

Multiplet count conjecture for Weyl-Heisenberg SICs

摘录 / EXCERPT: Fix $d\in\mathbb Z_{>0}$, $d\neq3$, set $\Delta=\Delta_d=(d+1)(d-3)$ and $K=\Q(\sqrt{\Delta})$. Let $\mathcal O_\Delta$ and $\mathcal O_K$ denote the quadratic order of discriminan...

conjecture | open | MathDB 页面 | 原文线索

V2-010307 | MATHDB-363935

Polynomial closure conjecture for the sets $\mathcal{A}_n$

摘录 / EXCERPT: For each integer $n\geq 1$, let $\mathcal{A}_n$ be the set defined in the paper, viewed as a subset of the algebraic integers $\overline{\mathbb{Z}}$. A subset is polynomially clos...

conjecture | open | MathDB 页面 | 原文线索

V2-010308 | MATHDB-363956

Non-commutativity and linguization of Oshida-Rule

摘录 / EXCERPT: Non-commutativity and linguization of Oshida-Rule. It is not possible to prove from the supplied formulation that this identity holds using matrices A; equivalently, it is not po...

open_problem | open | MathDB 页面 | 原文线索

V2-010309 | MATHDB-363992

The modular construction conjecture for type A_1 automorphic Galois representations

摘录 / EXCERPT: Let K be a totally real field and let $\{\rho_{\pi, \lambda}: \mathrm{Gal}_{K} \to \mathrm{GL}_n(\overline E_{\lambda})\}_{\lambda}$ be the compatible system of K defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-010310 | MATHDB-364017

Modular Kraus entropic conjecture for Parseval frames

摘录 / EXCERPT: Let $\mathcal{E}$ be a Hilbert C^*-module over a commutative unital C^*-algebra $\mathcal{A}$. Let $\{\tau_j\}_{j=1}^\infty$ and $\{\omega_k\}_{k=1}^\infty$ be two Parseval fra...

conjecture | open | MathDB 页面 | 原文线索

V2-010311 | MATHDB-364043

Hilbert function conjecture for general linear exact zero divisors

摘录 / EXCERPT: Let R be a standard graded algebra, and let $d>0$. A pair (ℓ,Q) of exact zero divisors means that ℓ is a general linear form, Q is an element of degree $d-1$, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-010312 | MATHDB-364045

Closure conjecture for graded gentle algebras under derived equivalence

摘录 / EXCERPT: Let $\mathbf S=(S,\Sigma,\eta)$ be a graded smooth surface with stops, and let $\mathcal W(\mathbf S)$ denote its partially wrapped Fukaya category. A formal generator is a generat...

conjecture | open | MathDB 页面 | 原文线索

V2-010313 | MATHDB-364097

Maximal-order uniqueness conjecture for maximal doubly even codes

摘录 / EXCERPT: Let $C\subset \mathbb{F}_2^n$ be a maximal doubly even binary code, and let $I=(i_0,i_1,\ldots,i_{n-1})$. Consider the Clifford algebra $\mathbb{C}_n$ and the rational Clifford alg...

conjecture | open | MathDB 页面 | 原文线索

V2-010314 | MATHDB-364100

Involution-stable maximal-order conjecture

摘录 / EXCERPT: Let $\mathbb{C}_n$ be the Clifford algebra with generators $i_1,\ldots,i_{n-1}$, and let $\mathbb{Z}[i_1,\ldots,i_{n-1}]$ be the indicated Clifford order. A maximal order is closed...

conjecture | open | MathDB 页面 | 原文线索

V2-010315 | MATHDB-364101

Doubly even code order conjecture

摘录 / EXCERPT: Let C be a doubly even binary code of length n, and let Λ_C be the inverse image of C under the natural map ... Let $\mathbb{Z}[\frac{1}{2}\Lambda_C]$ denote the ring g...

conjecture | open | MathDB 页面 | 原文线索

V2-010316 | MATHDB-364124

Decompletion conjecture for compact p-adic Lie groups

摘录 / EXCERPT: Let G be a compact p-adic Lie group, and let M and the associated period rings and locally analytic-vector functor be as in Theorem 1. The Lie algebra $\mathrm{Lie}(G)$ is no...

conjecture | open | MathDB 页面 | 原文线索

V2-010317 | MATHDB-364128

Geometrically C_1 transfer principle for algebraically maximal valued fields

摘录 / EXCERPT: Let (K,v) be an algebraically maximal valued field with divisible value group and perfect residue field k. A field is geometrically C_1 if every smooth projective separably r...

conjecture | open | MathDB 页面 | 原文线索

V2-010318 | MATHDB-364149

Conjecture on the smallest orbit of stable vectors for stable gradings with s_0=0

摘录 / EXCERPT: Let θ be a stable grading on the dual Lie algebra $\widehat{\mathfrak g}$, with associated parahoric subgroup P. Let $p:\widehat{\mathfrak g}_1\to\widehat{\mathfrak g}(1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-010319 | MATHDB-364188

Contractibility conjecture for the principal stability component of \(\mathbb P^3\)

摘录 / EXCERPT: Contractibility conjecture. The principal connected component is the union of the geometric and algebraic stability conditions and is contractible:

conjecture | open | MathDB 页面 | 原文线索

V2-010320 | MATHDB-364319

The \texorpdfstring{Π_k}{Pi-k} conjecture for fields and permutation matrices

摘录 / EXCERPT: Let F be a field and let k be a positive integer. The paper uses the condition Π_k for a field, although its full definition is not included in the supplied excerpt....

conjecture | open | MathDB 页面 | 原文线索

V2-010321 | MATHDB-364331

The general r-spin monodromy conjecture for algebraic surfaces

摘录 / EXCERPT: Let X be a smooth simply-connected algebraic surface, and let L be an ample line bundle on X. Let $k \gg 0$ be sufficiently large, and suppose that a general smooth section o...

conjecture | open | MathDB 页面 | 原文线索

V2-010322 | MATHDB-364346

Finite Hirsch length and nuclear dimension for elementary amenable groups

摘录 / EXCERPT: Finite Hirsch length conjecture. The group G has finite Hirsch length if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-010323 | MATHDB-364366

The algebraic Matrix Coefficient Conjecture for matrices of logarithms

摘录 / EXCERPT: Let $\mathscr{L}$ be the $\mathbf{Q}$-vector space of logarithms of algebraic numbers, and let $\mathscr{L}_p$ be its p-adic analogue. Consider an $n\times n$ matrix whose entrie...

conjecture | open | MathDB 页面 | 原文线索

V2-010324 | MATHDB-364385

Chabert-Halberstadt's Pólya group conjecture for S4-fields

摘录 / EXCERPT: Let K be an S_4-field. The Chabert-Halberstadt conjecture. ... Here $\operatorname{Po}(K)$ is the Pólya group of K, and $\operatorname{Cl}(K)$ is its ideal class group. The con...

conjecture | open | MathDB 页面 | 原文线索

V2-010325 | MATHDB-364386

Chabert-Halberstadt's pre-Pólya group conjecture for A5-fields

摘录 / EXCERPT: Chabert-Halberstadt's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010326 | MATHDB-364403

Hazrat's shift-equivalence conjecture for Cuntz-Krieger and Leavitt path algebras

摘录 / EXCERPT: Hazrat's conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010327 | MATHDB-364409

Non-degeneracy conjecture for cyclic quiver potentials

摘录 / EXCERPT: Let $C(W_n^I)$ be the potential associated with the cyclic quiver construction, for parameters $m,n\in\mathbb{Z}_n$. Non-degeneracy conjecture. For any $m,n\in\mathbb{Z}_n$, the po...

conjecture | open | MathDB 页面 | 原文线索

V2-010328 | MATHDB-364438

Griffiths conjecture on period images

摘录 / EXCERPT: Griffiths conjecture. The period image $\mathlcal P$ is quasi-projective and φ is induced by an algebraic morphism; L is ample on $\mathlcal P$; and there exists a Baily-...

conjecture | solved | MathDB 页面 | 原文线索

V2-010329 | MATHDB-364459

The O-O conjecture for splitting types of primitive covers

摘录 / EXCERPT: Let $\alpha\colon C\to\mathbb{P}^1$ be a degree k cover, and let $\vec{a}=(a_1,\ldots,a_{k-1})$ be defined by ... The tuple $\vec{a}$ is the splitting type, or the scrollar invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-010330 | MATHDB-364483

The bounded indecomposable-syzygy conjecture for local Golod rings

摘录 / EXCERPT: Let $(R,\mathfrak{m},k)$ be a local Golod ring of embedding codimension e. An indecomposable-syzygy conjecture asserts that there is a set of at most $e+1$ indecomposable R-mod...

conjecture | open | MathDB 页面 | 原文线索

V2-010331 | MATHDB-364488

The super Artin basis conjecture for complex reflection groups

摘录 / EXCERPT: Super Artin basis conjecture. The set ${\mathcal S}{\mathcal A}_{m,n}$ is a basis for $\operatorname{SRG}_{m,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010332 | MATHDB-364502

Exactness of the Chain Complex Construction Procedure

摘录 / EXCERPT: Exactness conjecture. The chain complex obtained by applying the Chain Complex Construction Procedure results in a free resolution of the cokernel of the initial map. That is, it i...

conjecture | open | MathDB 页面 | 原文线索

V2-010333 | MATHDB-364514

The stronger bounded negativity conjecture for blow-ups of the plane

摘录 / EXCERPT: Stronger bounded negativity conjecture. If C is an integral curve in $\operatorname{Bl}_{n}\mathbb{P}^2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-010334 | MATHDB-364546

Filtration submodule and annihilation conjecture for general linear invariants

摘录 / EXCERPT: Filtration submodule and annihilation conjecture. For each $1\leq k<\min(m,n)$, $\mathcal{F}_{n,k}$ is an $\mathcal{A}$-submodule and a ${\mathcal{D}}_n$-submodule of...

conjecture | open | MathDB 页面 | 原文线索

V2-010335 | MATHDB-364554

Nonnegativity conjecture for Schubert expansions of transparent diagrams

摘录 / EXCERPT: Let D be a diagram, and write its flagged Schur character in the Schubert-polynomial basis as ... where the sum is over all permutations and c^w_D are integers. A diagram is call...

conjecture | open | MathDB 页面 | 原文线索

V2-010336 | MATHDB-364558

Antipode formula for the coproduct of the socle

摘录 / EXCERPT: Antipode formula for the socle. The coproduct of the socle element should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-010337 | MATHDB-364590

Gras's freeness conjecture for maximal pro-p extensions unramified outside p

摘录 / EXCERPT: Let K be a number field, let S_p be the set of p-adic primes of K, and let $\Gamma_{S_p}=\operatorname{Gal}(K_{S_p}/K)$, where K_{S_p} is the maximal pro-p extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-010338 | MATHDB-364597

Symmetric monoidality of the trivial module functor for orthogonal epimorphisms

摘录 / EXCERPT: Symmetric monoidality conjecture. For OEpi,

conjecture | open | MathDB 页面 | 原文线索

V2-010339 | MATHDB-364598

Product rule for Weiss derivatives

摘录 / EXCERPT: Product-rule conjecture. There is an equivalence of right $K(\mathsf{OEpi})$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-010340 | MATHDB-364599

Mixed chain rule for Weiss derivatives

摘录 / EXCERPT: Mixed chain-rule conjecture. There is an equivalence of right $K(\mathsf{OEpi})$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-010341 | MATHDB-364600

Bimodule-valued mixed chain rule for Weiss derivatives

摘录 / EXCERPT: Bimodule chain-rule conjecture. For functors

conjecture | open | MathDB 页面 | 原文线索

V2-010342 | MATHDB-364602

Perrin-Riou's Heegner point main conjecture for multiplicative primes

摘录 / EXCERPT: Let K be an imaginary quadratic field, let K_∞ be the relevant anticyclotomic extension, and let Λ be its Iwasawa algebra. Let $\mathfrak{X}$ denote the relevant P...

conjecture | open | MathDB 页面 | 原文线索

V2-010343 | MATHDB-364638

Degree bound for entries of Kruithof limits

摘录 / EXCERPT: Let A be a positive $m \times n$ matrix, let V be a positive $m \times 1$ matrix, and let W be a positive $1 \times n$ matrix whose entry sums agree. Let x be the top-left...

conjecture | open | MathDB 页面 | 原文线索

V2-010344 | MATHDB-364641

Extension of the fundamental-variable braid formula to arbitrary generalized Cartan matrices

摘录 / EXCERPT: Let C be a generalized Cartan matrix, and let ${\overline{\mathcal{U}}}(\dot{{\mathbf f}t})$ denote the algebra whose fundamental variables are considered in Theorem. Arbitrary-t...

conjecture | open | MathDB 页面 | 原文线索

V2-010345 | MATHDB-364661

Coates-Sujatha pseudo-nullity conjecture for fine Selmer groups

摘录 / EXCERPT: Coates-Sujatha's pseudo-nullity conjecture. The module

conjecture | open | MathDB 页面 | 原文线索

V2-010346 | MATHDB-364691

Almost-equal Kronecker blocks conjecture for semi-direct sums of \(\operatorname{sl}(n)\)

摘录 / EXCERPT: Let $\mathfrak{g} = \operatorname{sl}(n) \ltimes \left(\mathbb{C}^n\right)^k$ be the semi-direct sum given by the standard representation, with $k<n$ and ... Assume $1<r<k-1$. Almost...

conjecture | open | MathDB 页面 | 原文线索

V2-010347 | MATHDB-364734

Subtraction-free characterization conjecture for bounded cluster-variable ratios

摘录 / EXCERPT: Let $\mathcal{D}$ be a full-rank cluster algebra of finite type, and consider ratios of products of its cluster variables on the totally positive locus. Such a ratio is called boun...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010348 | MATHDB-364772

Finiteness conjecture for nonmodular orbifold Q-subcanonical Shimura curves

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many orbifold $\mathbb{Q}$-subcanonical Shimura curves, up to isomorphism, over $\mathbb{C}$ that are not modular curves.

conjecture | open | MathDB 页面 | 原文线索

V2-010349 | MATHDB-364774

Finiteness conjecture for Q-subcanonical Hurwitz curves

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many $\mathbb{Q}$-subcanonical Hurwitz curves.

conjecture | open | MathDB 页面 | 原文线索

V2-010350 | MATHDB-364775

Genericity conjecture for Q-subcanonical points on odd cyclic coverings of hyperelliptic curves

摘录 / EXCERPT: Genericity conjecture. The generic X has no $\mathbb{Q}$-subcanonical points.

conjecture | open | MathDB 页面 | 原文线索

V2-010351 | MATHDB-364798

Conjecture on cocharacter multiplicities with an empty fourth partition

摘录 / EXCERPT: Cocharacter multiplicity conjecture. The multiplicity $m_{\langle\lambda\rangle}$ is nonzero if and only if either $w_1\leq 2$, or $w_1\geq 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010352 | MATHDB-364799

Conjecture on the complete support of the *-cocharacter of M_{1,2}(F)

摘录 / EXCERPT: Cocharacter support conjecture. The multiplicity $m_{\langle\lambda\rangle}$ is nonzero if and only if one of the following holds: (i) $\lambda(3)=\lambda(4)=\emptyset$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010353 | MATHDB-364811

Fomin-Zelevinsky's denominator conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra with trivial coefficients, and fix a cluster $\mathbf{x}=(x_1,\dots,x_n)$. For every cluster monomial z, write its Laurent expansion unique...

conjecture | open | MathDB 页面 | 原文线索

V2-010354 | MATHDB-364868

The uniform-generator conjecture for the graded ideal of an induced orbit

摘录 / EXCERPT: Let $\fg$ be a classical Lie algebra of rank n, let $\fl=\fg(a)\times\prod_{i=1}^{k}\fgl_{b_i}$ be a Levi subalgebra with $a\neq0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010355 | MATHDB-364972

Perfectoid purity under flat local base change

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring and let $(R,\mathfrak{m})\subseteq(S,\mathfrak{n})$ be a flat local homomorphism. The flat base-change conjecture. If R is perfe...

conjecture | open | MathDB 页面 | 原文线索

V2-010356 | MATHDB-364973

Lim-perfectoid purity implies perfectoid purity and injectivity

摘录 / EXCERPT: Let R be a ring for which the notions of lim-perfectoid pure, perfectoid pure, lim-perfectoid injective, and perfectoid injective are defined. The lim-perfectoid comparison conje...

conjecture | open | MathDB 页面 | 原文线索

V2-010357 | MATHDB-364990

The 6abc regular-sequence conjecture for three power sums

摘录 / EXCERPT: Let K be a field of characteristic zero, let $p_m=x_1^m+x_2^m+x_3^m$, and let a,b,c be integers satisfying $0<a<b<c$ and $\operatorname{gcd}(a,b,c)=1$. The 6abc conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-010358 | MATHDB-364991

The four-power-sum regular-sequence conjecture

摘录 / EXCERPT: Let K be a field of characteristic zero, let $S=K[x_1,x_2,x_3,x_4]$, and write $p_m=x_1^m+x_2^m+x_3^m+x_4^m$. Let $A=\{a_1,a_2,a_3,a_4\}$ be a set of positive integers with...

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V2-010359 | MATHDB-365091

The dimension bound for four-generated primitive-axis Jordan-type algebras

摘录 / EXCERPT: Let A be a $\mathcal{J}(\alpha,\beta)$-algebra generated by four primitive axes. Here, $\mathcal{J}(\alpha,\beta)$ denotes the class of algebras under consideration with eigenval...

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V2-010360 | MATHDB-365095

Mizuno's exponent formula for finite type Y-systems

摘录 / EXCERPT: Mizuno's conjecture. The following identity holds:

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V2-010361 | MATHDB-365107

Generalized branched-annulus compactification conjecture for Coxeter groups

摘录 / EXCERPT: Let W be a Coxeter group, let $\delta\in W$ be a Coxeter element, and let ${\normalfont\textsc{Br}}_{W,\delta}^{m}$ be the bisimplicial complex constructed with algebraic cell la...

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V2-010362 | MATHDB-365135

The additive reachability conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be the stable category of the Frobenius category $\mathcal{F}$, let T be a cluster-tilting object, let Σ denote the suspension, and let...

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V2-010363 | MATHDB-365137

The monoidal-additive correspondence conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra with monoidal categorification $\mathscr{C}_\ell$ and additive categorification $(\mathcal{F},\widehat{T})$, let $\mathcal{C}$ be the stable...

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The compatibility conjecture for real modules

摘录 / EXCERPT: Let $\mathcal{A}$ be as above, let $M_1,M_2\in\mathcal{P}_\ell^+$ be dominant monomials such that $L(M_1)$ and $L(M_2)$ are real, and let $\mathbf{g}_{M_1},\mathbf{g}_{M_2}$ be the...

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Schweizer's torsion conjecture for rank-two Drinfeld modules

摘录 / EXCERPT: Schweizer's Drinfeld-module torsion conjecture.

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V2-010366 | MATHDB-365182

Poonen's uniform boundedness conjecture for torsion of Drinfeld modules

摘录 / EXCERPT: Fix q, $r\geq 1$, and $d\geq 1$. Let L range over extensions of F of degree at most d, and let ϕ range over rank-r Drinfeld A-modules over L. Poonen's Drinfeld-m...

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V2-010367 | MATHDB-365186

Bosonic and fermionic dimension versus subdimension conjecture

摘录 / EXCERPT: Let x be an object in a 2-rig. Its bosonic dimension is n when $[?]$; its fermionic dimension is n when $[?]$. Dimension versus subdimension conjecture. If x has bosonic di...

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V2-010368 | MATHDB-365214

Drton-Sturmfels-Sullivant conjecture on the ML-degree of cycle graphs

摘录 / EXCERPT: Let C_n be the cycle graph on n vertices, and let $\MLdeg(C_n)$ denote the number of critical points of the Gaussian log-likelihood function for generic sample covariance data,...

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V2-010369 | MATHDB-365246

The fruitfulness conjecture for ordered Q_n quivers and their surface gluings

摘录 / EXCERPT: The ordered-quiver fruitfulness conjecture. The quivers Q_n provide fruitful seeds, and any quiver obtained by gluing them according to an ideal triangulation of a bordered surfa...

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V2-010370 | MATHDB-365247

Optimal nilpotency-index bound for the Fibonacci restricted Lie algebra

摘录 / EXCERPT: Let $0\ne a\in\mathbf L$ be written in the basis as $a=\sum_{j=n}^{m}r_{j-3}v_j$, with $r_j\in R$. The preceding result gives the bound $a^{2^{m-n+2}}=0$. Optimality conjecture. Th...

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V2-010371 | MATHDB-365248

Nonexistence of the normalized growth limit for the Fibonacci Lie algebra

摘录 / EXCERPT: Let $\mathcal L$ be the Fibonacci Lie algebra, let $\gamma_{\mathcal L}(n)$ denote its regular growth function, and let $\lambda=(1+\sqrt 5)/2$. Growth-limit conjecture. The normal...

conjecture | open | MathDB 页面 | 原文线索

V2-010372 | MATHDB-365249

Two-line-family conjecture for the standard monomials of the Fibonacci Lie algebra

摘录 / EXCERPT: Let W_N be the set of standard monomials of length N in the Fibonacci Lie algebra, and regard their multidegrees as points in the plane. Two-line-family conjecture. For every f...

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V2-010373 | MATHDB-365250

Growth problem for the characteristic-zero Fibonacci Lie algebra

摘录 / EXCERPT: Let K be a field of characteristic zero, let $R=K[t_i\mid i\ge 0]$, and let $L=\operatorname{Lie}(v_1,v_2)\subset\operatorname{Der}R$. Characteristic-zero growth problem. Determi...

open_problem | open | MathDB 页面 | 原文线索

V2-010374 | MATHDB-365251

Non-nilpotence conjecture for the associative hull of the Fibonacci Lie algebra

摘录 / EXCERPT: Let $p=2$, let R be the truncated polynomial ring from the construction, let $\mathcal L=\operatorname{Lie}(v_1,v_2)\subset\operatorname{Der}R$ be the Fibonacci Lie algebra, and...

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V2-010375 | MATHDB-365253

Infinite-presentation conjecture for the Fibonacci restricted Lie algebra

摘录 / EXCERPT: Let $p=2$ and let $\mathbf L=\operatorname{Lie}_p(v_1,v_2)$ be the Fibonacci restricted Lie algebra. Infinite-presentation conjecture. The algebra $\mathbf L$ is infinitely present...

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V2-010376 | MATHDB-365255

Finite endomorphic presentation conjecture for Fibonacci Lie algebras

摘录 / EXCERPT: Let $p=2$, let $\mathcal L=\operatorname{Lie}(v_1,v_2)$ be the Fibonacci Lie algebra, and let $\mathbf L=\operatorname{Lie}_p(v_1,v_2)$ be its Fibonacci restricted version. The sou...

conjecture | open | MathDB 页面 | 原文线索

V2-010377 | MATHDB-365307

The identity-generation conjecture for mutations of perm algebras

摘录 / EXCERPT: Identity-generation conjecture. Every multilinear identity in the mutation of a $\mathfrak{perm}$ algebra over a field of characteristic zero is a consequence of

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V2-010378 | MATHDB-365433

The centralizer quotient conjecture in equivariant motives

摘录 / EXCERPT: Let G be a reductive algebraic group, $H\subset G$ the centralizer of a one-parameter subgroup λ of G, and $\Gamma\subset G$ a finite cyclic group such that H is inva...

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V2-010379 | MATHDB-365447

Conjectural structure of the Iwasawa module in a real quadratic case

摘录 / EXCERPT: Let $K=\mathbb{Q}(\sqrt{D})$, where $D=pq$ with $p\equiv 5\pmod 8$ and $q\equiv 1\pmod 8$, and suppose … Assume also … Let X_{∞} be the Iwasawa module of the cyclotomic…

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V2-010380 | MATHDB-365473

The triviality conjecture for extensions of Taft algebras

摘录 / EXCERPT: Let n be a positive integer, let ω be the parameter in the Taft algebra, and define … An extension consists of a Frobenius automorphism ϕ and an element θ; cal…

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V2-010381 | MATHDB-365519

The L_{∞}-algebra conjecture for the BV double complex

摘录 / EXCERPT: L_{∞}-algebra conjecture. There exists an L_{∞} structure on the complex $({\bf F}^{\bullet}_{{\bf b}^-},\mathcal{Q})$, such that its bilinear operation is the symmet…

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V2-010382 | MATHDB-365546

Donovan-Wemyss conjecture for contraction algebras

摘录 / EXCERPT: Donovan-Wemyss conjecture. The base singularities are isomorphic if and only if the contraction algebras are isomorphic:

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V2-010383 | MATHDB-365547

Geometric Realisation Conjecture for Jacobi algebras

摘录 / EXCERPT: Geometric Realisation Conjecture. Every f whose Jacobi algebra satisfies

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V2-010384 | MATHDB-365631

The Krull-dimension conjecture for reciprocal complements

摘录 / EXCERPT: Krull-dimension conjecture. The inequality

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V2-010385 | MATHDB-365632

Zelmanov's strongness conjecture for Engel Lie algebras

摘录 / EXCERPT: Let n and p be integers with $n<p$, and let L be an n-Engel Lie algebra over $\mathbb{F}_p$. For every $x e L$, let $I(x)$ be the ideal generated by x; call L k-stron…

conjecture | open | MathDB 页面 | 原文线索

V2-010386 | MATHDB-365633

The commutativity conjectures for strong Lie algebras

摘录 / EXCERPT: Let n be a positive integer, let L be an n-strong Lie algebra over a field with at least n elements, and let $A(L)$ be its universal enveloping algebra. For a monomial…

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V2-010387 | MATHDB-365657

Quadratic commutativity conjecture for D8 qq-characters

摘录 / EXCERPT: Let $\mathsf{T}_{\four}^{K}(x)$ denote the D8 qq-character and let $\mathsf{f}^{K_{1}|K_{2}}_{\four\four}(x_{1}/x_{2})$ be the corresponding one-loop factor. D8 quadratic commuta…

conjecture | open | MathDB 页面 | 原文线索

V2-010388 | MATHDB-365720

Gaitsgory's factorization conjecture for local systems on the formal punctured disc

摘录 / EXCERPT: Gaitsgory's factorization conjecture. There exists a fully faithful functor

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V2-010389 | MATHDB-365794

Minimality conjecture for the pole-order integer in separated variables

摘录 / EXCERPT: Minimality conjecture. If $\mathrm{F}(p)$ is trivial, then the integer k in equation (mult) is minimal such that

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V2-010390 | MATHDB-365820

Initial-ideal decomposition conjecture for almost complete intersections

摘录 / EXCERPT: Initial-ideal decomposition conjecture. The initial ideal decomposes as

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V2-010391 | MATHDB-365837

The Virasoro-type reduction conjecture for adjacent nilpotent orbits

摘录 / EXCERPT: Let $\mathbb{O}$ be a nilpotent orbit of $\mathfrak{g}$, with representative $f\in\mathfrak{g}$ satisfying … and assume that the nonzero semisimple element of…

conjecture | open | MathDB 页面 | 原文线索

V2-010392 | MATHDB-365893

Ghosh's regular-sequence conjecture for the Casas-Alvero problem

摘录 / EXCERPT: Let K be a field, let n be a positive integer, and set $R=K[x_1,\ldots,x_{n-1}]$. For $j\in\{1,\ldots,n-1\}$, let $\Phi_j:R\to R$ be given by $\Phi_j(x_i)=x_i-x_j$ for…

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V2-010393 | MATHDB-365906

Goncharov's classical polylogarithm depth conjecture

摘录 / EXCERPT: Let F be an infinite field. The truncated cobracket … is obtained by omitting the $\mathcal{L}_1(F)\wedge\mathcal{L}_{n-1}(F)$ component of the cobracket. Goncharov's classical p…

conjecture | open | MathDB 页面 | 原文线索

V2-010394 | MATHDB-365907

Goncharov's trilogarithm isomorphism conjecture

摘录 / EXCERPT: Let F be an infinite field. Let $B_3(F)$ be Goncharov's Bloch group defined using the stated relations, and let $\mathcal{L}^{\mathrm{f}}_3(F)$ be the weight-three component of t...

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V2-010395 | MATHDB-365912

The conjectured relative-error threshold for Gaussian trace estimation

摘录 / EXCERPT: Let m be the sampling number, let $mu=r_{\mathrm{eff}}(A)$ for a nonzero symmetric positive semidefinite matrix A, and let $\varepsilon_{\mathrm{rel}}$ be the threshold appeari...

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V2-010396 | MATHDB-365913

The conjectured absolute-error threshold for Gaussian trace estimation

摘录 / EXCERPT: Let m be the sampling number, let $\lambda=\lVert A\rVert_2$ and $\phi=\lVert A\rVert_F$ for a symmetric matrix A, and let $\varepsilon_{\mathrm{abs}}$ be the threshold appeari...

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V2-010397 | MATHDB-365933

The local-global compatibility conjecture for the Hecke-algebra Galois representation

摘录 / EXCERPT: Let E be the quadratic field under consideration, let $\mathfrak{m}\subset h$ be a non-Eisenstein maximal ideal, and write $\mathbf{T}:=h_\mathfrak{m}$. Let ... be the Galois repre...

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V2-010398 | MATHDB-366026

Maximality divisibility conjecture for generalized van der Geer-van der Vlugt curves

摘录 / EXCERPT: Maximality divisibility conjecture. If $\overline{C}_{R,r}$ is $\mathbb{F}_{p^{2f}}$-maximal, then

conjecture | open | MathDB 页面 | 原文线索

V2-010399 | MATHDB-366069

Cactus-to-toggle factorization conjecture for minuscule crystals

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let λ be a minuscule dominant weight, and let w_0^J be the corresponding Weyl group element. Write $C_{\mathfrak{g}}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-010400 | MATHDB-366161

Conjecture on the explicit irregular Hodge module structure of Frenkel-Gross connections

摘录 / EXCERPT: Let $\check G$ be the Langlands dual group, let $V\in\operatorname{Rep}(\check G)$, and let $\nabla_{\check G}^{\mathrm{H}}(V)$ be the rescalable integrable mixed twistor module as...

conjecture | open | MathDB 页面 | 原文线索

V2-010401 | MATHDB-366190

The correspondence conjecture for Rota-Baxter operators on the Sweedler Hopf algebra

摘录 / EXCERPT: Let H_4 be the Sweedler Hopf algebra and let $H^{(-)}_4$ denote its underlying Lie algebra with the commutator bracket. A Rota-Baxter operator on $H^{(-)}_4$ is a linear map sati...

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V2-010402 | MATHDB-366191

Hussain-Ma-Yau-Zuo's contact invariance conjecture for higher Nash blowup algebras

摘录 / EXCERPT: Let f and g be germs in the convergent power series ring $\mathbb C\{\mathbf{x}\}$, with $f(\mathbf{0})=g(\mathbf{0})=0$. Two germs are contact equivalent at the origin if on...

conjecture | open | MathDB 页面 | 原文线索

V2-010403 | MATHDB-366195

The monodromy subcover conjecture for splitting Jacobians

摘录 / EXCERPT: Let X, Y, $\bar{Y}$, and W be the curves in the diagram ... For a suitable choice of a subcover W, the algebraic correspondence ... induces the isogeny in the paper's hiding el...

conjecture | open | MathDB 页面 | 原文线索

V2-010404 | MATHDB-366245

The volume-invariant algebra conjecture for cyclic polytopes

摘录 / EXCERPT: Let d and n be positive integers, let C^d_n be the cyclic d-polytope with n vertices, and let Inv^d_{d+2} denote the algebra of degree-d volume invariants fo...

conjecture | open | MathDB 页面 | 原文线索

V2-010405 | MATHDB-366275

Margulis-Platonov conjecture on normal subgroups of arithmetic groups

摘录 / EXCERPT: Let k be a number field and let $\mathbf{G}$ be a k-group. Let T be the (possibly empty) set of finite places of k at which $\mathbf{G}$ is k_v-anisotropic, and write ... V...

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V2-010406 | MATHDB-366276

The critical-norm scattering conjecture for defocusing NLS in Euclidean space

摘录 / EXCERPT: Consider the defocusing nonlinear Schrödinger equation ... in $\mathbb{R}^d$, with $d\geq 1$ and $\alpha\geq 4/d$. Set $s_c=d/2-2/\alpha$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010407 | MATHDB-366277

The critical-norm scattering conjecture for defocusing NLS outside a convex obstacle

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be the exterior domain of a smooth, compact, strictly convex obstacle, and consider the defocusing nonlinear Schrödinger equation with Dirichlet bou...

conjecture | open | MathDB 页面 | 原文线索

V2-010408 | MATHDB-366330

Canonical Jordan block structure conjecture for regular differential-algebraic equations

摘录 / EXCERPT: Let $E,F:\mathcal I\rightarrow\mathbb{R}^{m\times m}$ be sufficiently smooth, and let $\{E,F\}$ be a regular pair with index $\mu\geq 2$. Write r for the dimension of the differe...

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V2-010409 | MATHDB-366385

The one-parameter family conjecture for the even Haagerup subfactor planar algebra

摘录 / EXCERPT: The even part of the Haagerup subfactor planar algebra $H\mathcal{P}$ is a subfactor planar algebra lying in a one-parameter family of subfactor planar algebras. Haagerup family co...

conjecture | open | MathDB 页面 | 原文线索

V2-010410 | MATHDB-366418

The two-dimensional shuffle-generator conjecture for conjugation invariants

摘录 / EXCERPT: Let $\mathsf{ConjInv}$ be the algebra of conjugation invariants, and call a homogeneous element a generator at level n if it is one of the homogeneous generators used for this sh...

conjecture | open | MathDB 页面 | 原文线索

V2-010411 | MATHDB-366420

The shuffle-of-areas closure conjecture

摘录 / EXCERPT: Shuffle-of-areas closure conjecture. If all i_j are distinct, then

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V2-010412 | MATHDB-366439

Braided monoidal structure conjecture for type A Soergel bimodule complexes

摘录 / EXCERPT: Let m and n denote objects indexing Rouquier complexes, and let $m\boxtimes n\rightarrow n\boxtimes m$ be the corresponding Rouquier complexes. Let $\tau_{m,n}$ be the naturali...

conjecture | open | MathDB 页面 | 原文线索

V2-010413 | MATHDB-366454

Palka's height bound conjecture for singular del Pezzo surfaces

摘录 / EXCERPT: Let $\overline{X}$ be a singular del Pezzo surface. The height bound conjecture. If $\operatorname{char}\mathscr{k}\neq 2,3$, then ... The height is a bounded invariant intended to o...

conjecture | open | MathDB 页面 | 原文线索

V2-010414 | MATHDB-366458

The commutative loop conjecture for integer spaces of sigma-sets

摘录 / EXCERPT: Commutative loop conjecture. The structure $(3^{A},\oplus)$ satisfies:

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V2-010415 | MATHDB-366466

Deformation invariance of the Bounded Negativity Conjecture

摘录 / EXCERPT: Deformation-invariance conjecture. The Bounded Negativity Conjecture is true for X if and only if it is true for Y.

conjecture | open | MathDB 页面 | 原文线索

V2-010416 | MATHDB-366468

The bijection between cluster-tilting objects and d-angulations

摘录 / EXCERPT: Let (S,M,D) be a d-angulation of a marked surface, let (Q,W) be its associated quiver with superpotential, and let Γ be the associated d-dimensional Ginzburg algebra...

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V2-010417 | MATHDB-366480

Equivalent linear-algebraic formulation of the weighted initial-ideal conjecture

摘录 / EXCERPT: Equivalent formulation. The map $f\cdot:R_w\to M_{w+3m}$ is an isomorphism for every $w\bsin\mathbb{N}$. This is presented as an equivalent linear-algebraic formulation of the weig...

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V2-010418 | MATHDB-366495

The elliptic-mutation conjecture for elliptic isomonodromic flows

摘录 / EXCERPT: Consider elliptic cluster algebras associated with elliptic generalizations of cluster mutations, and elliptic isomonodromic deformation flows. Elliptic-mutation conjecture. Ellipt...

conjecture | open | MathDB 页面 | 原文线索

V2-010419 | MATHDB-366504

The derivation construction conjecture for transposed Poisson n-Lie algebras

摘录 / EXCERPT: Let $n\geq 2$ be an integer. Let $(L,\cdot,[\cdot,\ldots,\cdot])$ be a transposed Poisson n-Lie algebra, and let D be a derivation of both $(L,\cdot)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010420 | MATHDB-366539

Conjecture on Hall-monomial constants in the free Lie algebra of rank 2

摘录 / EXCERPT: Hall-monomial constants conjecture. If M is a Hall monomial that is a constant under the derivation δ, then

conjecture | open | MathDB 页面 | 原文线索

V2-010421 | MATHDB-366540

Conjecture that constants equal the algebra U^(1) in the free Lie algebra of rank 2

摘录 / EXCERPT: Constants-equality conjecture. The algebra of constants is exactly $U^{(1)}$:

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V2-010422 | MATHDB-366580

Harbourne-Janssen-Nagel-Schenck conjecture on negative curves on blowups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb{F}_e$ be the e-th Hirzebruch surface, with C_e denoting its negative section. Let $P=\{p_1,\dots,p_r\}\subset \mathbb{F}_e$ be r points in very general position,...

conjecture | open | MathDB 页面 | 原文线索

V2-010423 | MATHDB-366619

Non-isomorphism conjecture for cyclotomic restricted rational Cherednik algebras

摘录 / EXCERPT: Let m be even, let $\frac{m}{p}$ be odd, and set $k=\mathbb{Q}(\sqrt[m]{1})$. Let $\underline{\overline{H}}_{\underline{c}}(\mu(G(m,p,n)))$ and $\overline H_c(G(m,p,n))$ be the t...

conjecture | open | MathDB 页面 | 原文线索

V2-010424 | MATHDB-366646

The counterexample conjecture for symbolic-power asymptotic regularity

摘录 / EXCERPT: Let $R={\Bbbk}[x,y,a,b]$ be the polynomial ring from the preceding construction, let $S=R[z]$, and set $I=Q\cap(f,z)\subseteq S$. The source specifies that...

conjecture | open | MathDB 页面 | 原文线索

V2-010425 | MATHDB-366647

Characteristic-two regularity upper-bound conjecture

摘录 / EXCERPT: Assume that ${\Bbbk}$ has characteristic 2, let $R={\Bbbk}[x,y,a,b]$, set $Q=(x^3,y^3)$ and $f=xya-(x^2+y^2)b$, and let $n=2^s$ for some $s\in\mathbb N$. Characteristic-two upper...

conjecture | open | MathDB 页面 | 原文线索

V2-010426 | MATHDB-366676

Conjecture on polyhomogeneity for scattering solutions with infinitely regular data

摘录 / EXCERPT: Let $N\in\mathbb N_{\geq1}$, let ϕ solve $\Box_{\mathbb R^{2N+1,1}}\phi=0$ with no incoming radiation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010427 | MATHDB-366681

Trivial-action conjecture for automorphisms of surfaces

摘录 / EXCERPT: Let S be a smooth projective surface, and let $G\subset\mathrm{Aut}(S)$ be a group of automorphisms. Assume that every element of G acts trivially on $H^{2,0}(S)$. Trivial-acti...

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V2-010428 | MATHDB-366710

Rotation invariance conjecture for Newton-Okounkov bodies of plabic graphs

摘录 / EXCERPT: Let G be a plabic graph, and let rG denote any rotation of G. Its associated Newton-Okounkov body is denoted by Δ_G. Rotation invariance conjecture. All rotations of a...

conjecture | open | MathDB 页面 | 原文线索

V2-010429 | MATHDB-366774

Herzog's last theorem

摘录 / EXCERPT: Herzog's last theorem. The following conditions are equivalent:

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V2-010430 | MATHDB-366776

Equality of stable and unstable algebra K-theory ranks for mixing Smale spaces

摘录 / EXCERPT: Let (X,φ) be a mixing Smale space, and let S and U denote its stable and unstable algebras, respectively. Stable-unstable rank conjecture. One has ... This equality predi...

conjecture | open | MathDB 页面 | 原文线索

V2-010431 | MATHDB-366789

Linear weighted bounded negativity conjecture

摘录 / EXCERPT: Let X be a smooth projective surface over an algebraically closed field of characteristic zero. Let C be a reduced and irreducible curve on X, and let D be a big and nef di...

conjecture | open | MathDB 页面 | 原文线索

V2-010432 | MATHDB-366821

Categorical splitting conjecture for non-degenerate braided tensor subcategories

摘录 / EXCERPT: Categorical splitting conjecture. There is an equivalence of braided tensor categories

conjecture | open | MathDB 页面 | 原文线索

V2-010433 | MATHDB-366884

Level-one exceptional theta-lift conjecture for automorphic representations of F4

摘录 / EXCERPT: Let n be such that there is a cuspidal Hecke eigenform of weight $2n+12$ for $\operatorname{SL}_{2}(\mathbb{Z})$, and let π be the associated level-one algebraic automorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-010434 | MATHDB-366913

Unconstrained Hilbert-series conjecture for Koszul Gorenstein algebras

摘录 / EXCERPT: Koszul Roller Coaster conjecture. There exists such an algebra satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-010435 | MATHDB-366922

Weak p-adic Schanuel conjecture

摘录 / EXCERPT: Let $\alpha_1,\ldots,\alpha_n$ be nonzero algebraic numbers. Weak p-adic Schanuel conjecture. If $\log_p(\alpha_1),\ldots,\log_p(\alpha_n)$ are linearly independent over...

conjecture | open | MathDB 页面 | 原文线索

V2-010436 | MATHDB-366923

Weak p-adic four exponentials conjecture

摘录 / EXCERPT: Let $\mathbb{L}$ be the $\overline{\mathbb{Q}}$-linear subspace of $\overline{\mathbb{Q}}_p$ generated by 1 and the p-adic logarithms of nonzero algebraic numbers. Let N be a...

conjecture | open | MathDB 页面 | 原文线索

V2-010437 | MATHDB-366994

The Generalized Hom-Dixmier Conjecture for higher-order hom-associative Weyl algebras

摘录 / EXCERPT: Let K be a field of characteristic zero, let $k,k'\in K^n$ each have exactly m nonzero elements, and let A_n^k and $A_n^{k'}$ be the corresponding higher-order hom-associativ...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-010438 | MATHDB-367004

Gross-Zagier algebraicity conjecture for higher Green's functions at CM points

摘录 / EXCERPT: Let d_1 and d_2 be discriminants such that one of them is fundamental when k is even. For any CM points z_1 and z_2 of discriminants d_1 and d_2, let H be the compo...

conjecture | open | MathDB 页面 | 原文线索

V2-010439 | MATHDB-367006

Nakai's conjecture on higher derivations and regularity

摘录 / EXCERPT: Let k be a field of characteristic zero and let A be a k-algebra of finite type. For each integer $q\ge 1$, let $Der^q(A)$ denote the module of differential operators of orde...

conjecture | open | MathDB 页面 | 原文线索

V2-010440 | MATHDB-367012

Converse characterization for Cohen-Macaulay powers of weighted trees

摘录 / EXCERPT: Let G_ω be a weighted tree. Proposition gives necessary conditions for $I(G_\omega)^n$ to be Cohen-Macaulay for all $n\geqslant 1$: G has a perfect matching...

conjecture | open | MathDB 页面 | 原文线索

V2-010441 | MATHDB-367047

Zilber-Pink conjecture for Picard numbers in Lefschetz pencils

摘录 / EXCERPT: Let $\mathcal{Y}\to\mathbb{P}^1$ be a Lefschetz pencil of degree-d surfaces in $\mathbb{P}^3$. Picard-number finiteness conjecture. If $d\geq 5$, the Picard number of...

conjecture | open | MathDB 页面 | 原文线索

V2-010442 | MATHDB-367099

The K3 rational curves conjecture

摘录 / EXCERPT: Let (X,H) be a projective K3 surface over an algebraically closed field, where H is a polarization. The K3 rational curves conjecture. X contains infinitely many integral rat...

conjecture | open | MathDB 页面 | 原文线索

V2-010443 | MATHDB-367164

Duality compatibility conjecture for the filtered nearby-cycle object

摘录 / EXCERPT: Let Ψ be the nearby-cycle object with filtrations F and G, let $\mathbb{D}$ denote duality, and let b be the isomorphism relating the dual nearby cycles to...

conjecture | open | MathDB 页面 | 原文线索

V2-010444 | MATHDB-367175

The generalized Amitsur exponent conjecture for finite actions

摘录 / EXCERPT: Let W be a unital F-algebra over a field of characteristic zero, and let A be a W-algebra. Say that the action of W on A is finite when the image of the action has fini...

conjecture | open | MathDB 页面 | 原文线索

V2-010445 | MATHDB-367218

Kuznetsov's tensor-product conjecture for Macaulay rings

摘录 / EXCERPT: Let S be a Macaulay ring, let x be a variable not appearing in S, and let $n>0$. Kuznetsov's tensor-product conjecture. Then $S\otimes_K K[x]/(x^n)$ is a Macaulay ring. This...

conjecture | open | MathDB 页面 | 原文线索

V2-010446 | MATHDB-367219

Adjusted tensor-product conjecture for Macaulay rings

摘录 / EXCERPT: Let S be a Macaulay ring, let x be a variable not appearing in S, and let n be strictly larger than the largest degree of any element of S. Adjusted tensor-product conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-010447 | MATHDB-367295

Nisnevich's conjecture on torsors over punctured regular local rings

摘录 / EXCERPT: Let R be a regular local ring of dimension $d\geq 1$, let m be its maximal ideal, and let $u\in m\setminus m^2$. Let G be a strictly isotropic reductive group over R. Nisne...

conjecture | open | MathDB 页面 | 原文线索

V2-010448 | MATHDB-367296

Quillen's question on projective modules over a punctured regular local ring

摘录 / EXCERPT: Let R be a regular local ring with maximal ideal m, and let $f\in m\setminus m^2$. Quillen's question. Every finitely generated projective R_f-module is free. This is the mod...

open_problem | open | MathDB 页面 | 原文线索

V2-010449 | MATHDB-367329

Interior boundary-cone conjecture for Virasoro conformal-block divisors

摘录 / EXCERPT: Let $V=\text{Vir}_{2,2k+1}$, and let $\{W_{a_i}\}_{i=1}^n$ be a set of nontrivial simple V-modules, with $a_i\neq 1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010450 | MATHDB-367402

The support tau-tilting enumeration conjecture for gentle arbor quivers

摘录 / EXCERPT: The support tau-tilting enumeration conjecture. The number of support-τ-tilting objects for the gentle quiver-with-relations G_t is equal to the number of elements of P_t.

conjecture | open | MathDB 页面 | 原文线索

V2-010451 | MATHDB-367439

Lusztig's boundedness conjecture for positively weighted Coxeter groups

摘录 / EXCERPT: Lusztig's boundedness conjecture. The number $N_L(W)$ is a bound for any positively weighted Coxeter group (W,S,L) of finite rank. This is a weighted generalization of Xi's bound...

conjecture | open | MathDB 页面 | 原文线索

V2-010452 | MATHDB-367496

Global distinction conjecture for unitary symmetric pairs

摘录 / EXCERPT: Let F/F_0 be a quadratic extension, let F' be the quadratic étale algebra associated with the unitary symmetric pair, and let π_0 be a cuspidal automorphic representation o...

conjecture | open | MathDB 页面 | 原文线索

V2-010453 | MATHDB-367637

Recursive description of the integers $c_j(a,b)$

摘录 / EXCERPT: Recursive description of the integers c_j. If $b-j\leq rq$, then $c_j(a,b)=rq$. Otherwise,

conjecture | open | MathDB 页面 | 原文线索

V2-010454 | MATHDB-367638

Davis-Erman-Martinova conjecture on nonstandard Veronese subrings

摘录 / EXCERPT: Let S be a nonstandard graded polynomial ring, and for a positive integer e let $S^{(e)}$ denote its e-th Veronese subring. A graded algebra is nonstandard Koszul when it has...

conjecture | open | MathDB 页面 | 原文线索

V2-010455 | MATHDB-367640

The q-deformed random-to-random commutativity conjecture

摘录 / EXCERPT: Assume $q\neq0$. Let $\operatorname{\mathcal{H}}_n^{\operatorname{aff}}$ be the affine Hecke algebra containing $\operatorname{\mathcal{H}}_n$ and the Laurent polynomial ring in...

conjecture | open | MathDB 页面 | 原文线索

V2-010456 | MATHDB-367645

The Galois-width conjecture for Gaussian-mixture moment maps

摘录 / EXCERPT: Galois-width conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010457 | MATHDB-367650

Socle quotient conjecture for Leavitt path algebras of Kronecker squares

摘录 / EXCERPT: Let Q be a quiver, let $\widehat{Q}$ denote its Kronecker square, and let $L_{\mathbb K}(Q)$ be the Leavitt path algebra over a field $\mathbb K$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010458 | MATHDB-367651

Finite-quiver socle quotient conjecture for Leavitt path algebras

摘录 / EXCERPT: Let Q be a finite quiver, let $\widehat{Q}$ denote its Kronecker square, and let $L_{\mathbb K}(Q)$ be the Leavitt path algebra over a field $\mathbb K$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010459 | MATHDB-367712

The partially strict unimodal criterion for 2-admissible I-Gorenstein sequences

摘录 / EXCERPT: Let $A=k[... /J$ be an AG k-algebra. Let $H=(h0,h1,ldots,hu)$ be a 2-admissible sequence, where a sequence is **partially strict unimodal** if...

conjecture | open | MathDB 页面 | 原文线索

V2-010460 | MATHDB-367713

The uniqueness conjecture for symmetric decompositions of 2-admissible I-Gorenstein sequences

摘录 / EXCERPT: Let $A=k[\![x,y]\!]/J$ be an AG k-algebra, let I be an ideal in A, and let H be a 2-admissible $G(I)$-Gorenstein sequence. A symmetric decomposition is a decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-010461 | MATHDB-367721

Positivity conjecture for super lambda-lengths

摘录 / EXCERPT: Consider the once-punctured torus with a spin structure induced from the cyclic orientation of the two triangles in the fundamental domain, clockwise on one triangle and counter-cl...

conjecture | open | MathDB 页面 | 原文线索

V2-010462 | MATHDB-367764

Dispersion-manageability conjecture for non-scattering data

摘录 / EXCERPT: Non-scattering-data dispersion-manageability conjecture. Given such u_0, both of the following assertions hold: (i) for fixed t_0,t_1, there exist $\gamma_+=\gamma_+(u_0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010463 | MATHDB-367807

The six-multiplication maximum-degree conjecture

摘录 / EXCERPT: Let Π_d be the polynomial set associated with degree-d approximation conditions, and let $\overline{\Pi^*_{2^6}}$ denote the closure of the polynomial set obtainable with six...

conjecture | open | MathDB 页面 | 原文线索

V2-010464 | MATHDB-367808

The seven-multiplication maximum-degree conjecture

摘录 / EXCERPT: Let Π_d be the polynomial set associated with degree-d approximation conditions, and let $\overline{\Pi^*_{2^7}}$ denote the closure of the polynomial set obtainable with sev...

conjecture | open | MathDB 页面 | 原文线索

V2-010465 | MATHDB-367831

Koszulness and confluence conjecture for the family of associative-coassociative properads

摘录 / EXCERPT: Conjecture. For every $a\in\mathbb{C}^4$, the following are equivalent: the properad $\mathcal{P}_a$ induces a confluent system; the morphism φ_a is a bijection in weight...

conjecture | open | MathDB 页面 | 原文线索

V2-010466 | MATHDB-367863

The injectivity conjecture for Anderson t-module exponentials

摘录 / EXCERPT: Injectivity conjecture. The P-adic L-series is non-zero if and only if the exponential map $\exp_E:L_\infty^d\rightarrow L_\infty^d$ is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-010467 | MATHDB-367865

Independence conjecture for the vanishing order of P-adic L-series

摘录 / EXCERPT: Let E be the Anderson t-module under consideration, let P vary over primes of A, and define the order of vanishing of the P-adic L-series at $z=1$ by ... to be the greate...

conjecture | open | MathDB 页面 | 原文线索

V2-010468 | MATHDB-367869

The common-ground conjecture on strong standard canonical form for regular DAEs

摘录 / EXCERPT: The common-ground conjecture. The pair $\{E,F\}$ is transformable into strong standard canonical form, and the Jordan normal form of its constant nilpotent matrix N consists exac...

conjecture | open | MathDB 页面 | 原文线索

V2-010469 | MATHDB-367943

The A2 obstruction conjecture for reddening sequences

摘录 / EXCERPT: Let Q be a quiver of rank at least 4, and let $[A_2]$ denote the finite-type A_2 quiver. Say that $[A_2]$ embeds into the mutation class of Q when it occurs there under the...

conjecture | open | MathDB 页面 | 原文线索

V2-010470 | MATHDB-367953

Fundamental theorem conjecture for the special linear supergroup

摘录 / EXCERPT: Let $\widetilde{\mathcal{O}}^{\mathrm{SL}(r|s)}$ denote the relevant superalgebra, and let $Y_{\underline{i}|\underline{\hat{\mu}}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-010471 | MATHDB-367960

Conjecture on minimal generators of the annihilator of the complete symmetric form

摘录 / EXCERPT: Minimal-generator conjecture. If d is odd, then $\operatorname{Ann}(H_{2d+1})$ has no minimal generators in degree $d+2$, and

conjecture | open | MathDB 页面 | 原文线索

V2-010472 | MATHDB-367961

Containment conjecture for initial annihilator ideals of generic forms

摘录 / EXCERPT: Initial-ideal containment conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-010473 | MATHDB-367997

The uniform Galois-group conjecture for reductions of D-finite series

摘录 / EXCERPT: Let $f(x)\in\mathbb{Q}[[x]]$ be a D-finite series and let n be the minimal order of a differential operator that annihilates $f(x)$. Let K be a number field and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010474 | MATHDB-368028

Mazur-Rubin's conjecture on Selmer near-companion elliptic curves

摘录 / EXCERPT: Mazur-Rubin's conjecture. If E_1 and E_2 are p-Selmer near-companions over K, then there exists a G_K-module isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010475 | MATHDB-368057

Divisibility conjecture for exponents of modular-unit basis elements

摘录 / EXCERPT: Let N be a positive integer, let L be the level-dependent integer used for the Galois action, and let $\mathcal{D}_N$ and $F_{m,h}$ be the indexing set and modular-unit basis i...

conjecture | open | MathDB 页面 | 原文线索

V2-010476 | MATHDB-368072

Generic quadratic non-commutative algebra Hilbert-series conjecture

摘录 / EXCERPT: Let B be a non-commutative algebra generated by n degree-one elements with r homogeneous quadratic relations; this is an algebra of type (n,r). For a power series...

conjecture | open | MathDB 页面 | 原文线索

V2-010477 | MATHDB-368073

Generic Lie algebra Hilbert-series conjecture

摘录 / EXCERPT: Let a Lie algebra of type (n,r) be generated by n degree-one elements with r homogeneous quadratic relations. For a power series, let $[A(z)]$ denote the truncation operation...

conjecture | open | MathDB 页面 | 原文线索

V2-010478 | MATHDB-368082

The split semi-elliptic Lindemann--Weierstrass conjecture

摘录 / EXCERPT: Let Ω be a lattice in $\mathbb{C}$ with algebraic invariants g_2,g_3. Let s and n be nonnegative integers. Let $t_1,\dots,t_s$ be $\mathbb{Q}$-linearly independent alg...

conjecture | open | MathDB 页面 | 原文线索

V2-010479 | MATHDB-368095

The Weyl-algebra factorization conjecture for the negative Witt algebra

摘录 / EXCERPT: Let $\W{-1}$ denote the negative Witt algebra, let $\Ua(\W{-1})$ be its enveloping algebra, and let A_k denote the relevant k-th Weyl-algebra target. For each n, write...

conjecture | open | MathDB 页面 | 原文线索

V2-010480 | MATHDB-368096

The partial-converse conjecture for primitive ideals of local functions

摘录 / EXCERPT: Let $\g$ be one of $\W{-1}$, W, or $\Vir$, and let χ and η be one-point local functions on $\g$. Let $P(\chi)$ and $P(\eta)$ denote their associated Poisson primitive i...

conjecture | open | MathDB 页面 | 原文线索

V2-010481 | MATHDB-368097

The inclusion conjecture for primitive ideals of local functions

摘录 / EXCERPT: Let $\g$ be one of $\W{-1}$, W, or $\Vir$, and let χ and η be local functions on $\g$. Let $P(\chi)$ and $P(\eta)$ denote the associated Poisson primitive ideals, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-010482 | MATHDB-368098

The strong Dixmier-map surjectivity conjecture for Witt and Virasoro algebras

摘录 / EXCERPT: Let $\g$ be one of $\W{-1}$, W, or $\Vir$. Let $\operatorname{P.Prim}\operatorname{S}(\g)$ denote the Poisson primitive spectrum of the symmetric algebra and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010483 | MATHDB-368100

The ACC conjecture for primitive ideals of Witt and Virasoro enveloping algebras

摘录 / EXCERPT: Let $\g$ be one of $\W{-1}$, W, or $\Vir$, and let $\Ua(\g)$ be its enveloping algebra. ACC conjecture. The algebra $\Ua(\g)$ satisfies the ascending chain condition on primitive...

conjecture | open | MathDB 页面 | 原文线索

V2-010484 | MATHDB-368112

The strong Lefschetz property conjecture for zero-dimensional complete intersections

摘录 / EXCERPT: Let $S=\mathbb{C}[x_1,\ldots,x_n]$ with $n\geq 3$, and let $f_j\in S_{d_j}$ be homogeneous polynomials of degrees $d_j\geq 2$ such that ... is a zero-dimensional complete intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-010485 | MATHDB-368141

Paz's length conjecture for full matrix algebras

摘录 / EXCERPT: Let $\mathbb{F}$ be an arbitrary field, and let $\ell(\mathrm{M}_n(\mathbb{F}))$ denote the length of the full matrix algebra $\mathrm{M}_n(\mathbb{F})$. Paz's length conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-010486 | MATHDB-368142

Proper stratification conjecture for KLR algebras of affine quivers

摘录 / EXCERPT: Proper stratification conjecture. The preorder of is properly stratifying for $A_0(\vecn)$ in arbitrary characteristic.

conjecture | open | MathDB 页面 | 原文线索

V2-010487 | MATHDB-368150

The norm relations test for positive rank of elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, let $F / \mathbb{Q}$ be a finite Galois extension with $G = \operatorname{Gal}(F / \mathbb{Q})$, and let ρ be an irreducible re...

conjecture | open | MathDB 页面 | 原文线索

V2-010488 | MATHDB-368151

The norm relations test over arbitrary number fields

摘录 / EXCERPT: Let E / L be an elliptic curve, let F / L be a Galois extension with Galois group G, and let ρ be an irreducible representation of G. Suppose that ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-010489 | MATHDB-368177

The uri multiplicity threshold shuffle conjecture

摘录 / EXCERPT: Uri multiplicity threshold shuffle conjecture. The uri multiplicities satisfy the threshold shuffle identities from the source's definition. The supplied text gives no resolution s...

conjecture | open | MathDB 页面 | 原文线索

V2-010490 | MATHDB-368195

The complementary-vector conjecture for pure O-sequences

摘录 / EXCERPT: Let $(a_0,\ldots,a_d)$ be a pure O-sequence with $a_d\neq 0$. Its complementary vector is ... An M-vector, or O-sequence, is the Hilbert function of a finite-dimensional standa...

conjecture | open | MathDB 页面 | 原文线索

V2-010491 | MATHDB-368231

Odd-reflection isomorphism conjecture for the super Yangian of $D(2,1;\lambda)$

摘录 / EXCERPT: Let $D(2,1;\lambda)$ be the exceptional Lie superalgebra with Dynkin diagrams $G(D_{\lambda})$ and $G(D_{\lambda}^{\circ})$ related by the odd reflection s_{α_2}, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010492 | MATHDB-368262

Field-of-definition conjecture for superspecial genus-4 hyperelliptic curves

摘录 / EXCERPT: Let H_a be a genus-4 hyperelliptic curve over an algebraic closure of $\mathbb{F}_p$, with parameter a. Field-of-definition conjecture. If H_a is superspecial, then...

conjecture | open | MathDB 页面 | 原文线索

V2-010493 | MATHDB-368284

The FSEANI conjecture for the real and complex fields

摘录 / EXCERPT: Let $n\in\mathbb{N}$. A field $\mathbb{K}$ is an n-FSEANI if the parametric family of systems of equations given by (EE) has a non-zero solution in $\mathbb{K}$ for every choice...

conjecture | open | MathDB 页面 | 原文线索

V2-010494 | MATHDB-368379

The LinKaw*-Drop1 equality conjecture

摘录 / EXCERPT: Let LinKaw^* and $\mathsf{Drop1}$ be the two families of relations defined in the source, with ... Here $\mathcal{H}$ is the source's word algebra and ϕ, $\ast$, x,...

conjecture | open | MathDB 页面 | 原文线索

V2-010495 | MATHDB-368380

Arcara-Miles conjecture on destabilising line bundles on surfaces

摘录 / EXCERPT: Let X be a smooth projective surface with a fixed Kähler class $[\omega_0]$ and a line bundle L. The stability condition $\sigma_{[\omega_0]}$ is the stability condition suppor...

conjecture | open | MathDB 页面 | 原文线索

V2-010496 | MATHDB-368425

Tame approximation conjecture for Grunwald realizations

摘录 / EXCERPT: Tame approximation conjecture. There exists a Galois extension $L/\mathbf{Q}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010497 | MATHDB-368429

The three-variable Iwasawa-Greenberg main conjecture for Rankin-Selberg families

摘录 / EXCERPT: Iwasawa-Greenberg main conjecture. The ${\mathbb{I}}_{\mathcal F,\mathcal G}[[\Gamma_{\mathrm{cyc}}]]$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-010498 | MATHDB-368433

Cohen-Macaulayness conjecture for the ASM embedding $A\mapsto 1\oplus A$

摘录 / EXCERPT: Let $A\in\operatorname{ASM}(n)$ be an alternating sign matrix, and let $1\oplus A\in\operatorname{ASM}(n+1)$ be the direct sum of the 1×1 identity matrix with A. Cohen-Mac...

conjecture | open | MathDB 页面 | 原文线索

V2-010499 | MATHDB-368441

Type bound for almost symmetric generalized numerical semigroups

摘录 / EXCERPT: Type-bound conjecture. The type of S can be at most three.

conjecture | open | MathDB 页面 | 原文线索

V2-010500 | MATHDB-368510

The q-bi-bracket and multiple q-zeta value conjectures

摘录 / EXCERPT: Let ${\rm q}{\mathcal {BD}}$ and ${\rm q}{\mathcal {MD}}$ denote the corresponding q-bi-bracket and multiple-divisor-value algebras, and let ${\bf qMZV}$ denote the algebra of mu...

conjecture | open | MathDB 页面 | 原文线索

V2-010501 | MATHDB-368541

Finiteness conjecture for heavenly elliptic curves over quadratic fields up to geometric isomorphism

摘录 / EXCERPT: Finiteness conjecture. The set $\overline{\mathcal{H}}_{7}$ is finite. The conjecture concerns heavenly elliptic curves over quadratic fields modulo $\overline{\mathbb{Q}}$-isomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-010502 | MATHDB-368604

Fields conjecture on the symmetric-group module structure of the superspace coinvariant ring

摘录 / EXCERPT: Let Ω_n be the superspace ring with diagonal $\mathfrak{S}_n$-action, let $SR_n=\Omega_n/SI_n$ be its superspace coinvariant ring, and let $\mathbb{F}[\mathcal{OP}_n]$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-010503 | MATHDB-368631

Weimar-Woods conjecture on isomorphism and normalization of graded contractions

摘录 / EXCERPT: Weimar-Woods conjecture. Two graded contractions ε and ε' are isomorphic if and only if they are equivalent via normalization:

conjecture | open | MathDB 页面 | 原文线索

V2-010504 | MATHDB-368655

Xiao's conjecture on the genus of canonically fibered surfaces

摘录 / EXCERPT: Let $f:X\dashrightarrow C$ be a canonically fibered surface, meaning that X and C are irreducible, smooth and projective over $\mathbb{C}$ with dimensions 2 and 1, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-010505 | MATHDB-368695

Strong rationality equivalence for unitary minimal W-algebras

摘录 / EXCERPT: Let $W_k(\mathfrak{g},f)$ be a unitary minimal W-algebra, with corresponding conformal net denoted by $\mathcal{A}_{W_k(\mathfrak{g},f)}$. The algebra is strongly rational when i...

conjecture | open | MathDB 页面 | 原文线索

V2-010506 | MATHDB-368707

Trace equality conjecture for one-dimensional graded fiber products

摘录 / EXCERPT: Let A and B be the graded rings and $R=A\times_T B$ the graded fiber product considered above, with canonical modules ω_R and $\omega_{A,B}:=\omega_A\oplus\omega_B$. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-010507 | MATHDB-368726

Rigidity conjecture for Hopf actions on Weyl Poisson algebras

摘录 / EXCERPT: Let H be a finite-dimensional Hopf algebra acting on a Weyl Poisson algebra. Rigidity conjecture. The H-action must factor through a group algebra. This conjecture asks whether...

conjecture | open | MathDB 页面 | 原文线索

V2-010508 | MATHDB-368727

Integrability conjecture for the cluster maps $\psi_{D_{2N}}$

摘录 / EXCERPT: Integrability conjecture. The discrete dynamics induced by the iteration of cluster maps $\psi_{D_{2N}}$ is a discrete integrable system.

conjecture | open | MathDB 页面 | 原文线索

V2-010509 | MATHDB-368772

Strong Watanabe-Yoshida conjecture for Hilbert-Kunz multiplicity

摘录 / EXCERPT: Let $d\geq 2$. Set ... where ... Let $(R,\mathfrak{m},k)$ be an unmixed non-regular local ring of characteristic p and dimension d. Here, unmixed means that...

conjecture | open | MathDB 页面 | 原文线索

V2-010510 | MATHDB-368776

The kernel conjecture for Jacobian derivations

摘录 / EXCERPT: Kernel conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010511 | MATHDB-368777

The Second Kernel Conjecture

摘录 / EXCERPT: Let K be a field of characteristic zero and let $K[T]=K[t_1,\dots,t_n]$. A derivation D of a ring is locally nilpotent if for every a there is an integer $m(a)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-010512 | MATHDB-368786

The licci classification conjecture for (c,d,t) triplets

摘录 / EXCERPT: Fix integers $c\geq 3$, $d\geq 1$, and $t\geq 1$. For a perfect ideal, the codimension, deviation, and type are respectively c, d, and t; associate to the triplet the diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-010513 | MATHDB-368823

Generalized Elashvili conjecture for relative modality in generalized Takiff Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra, and let $\mathbf{x}$ be an element of $\mathfrak{g}_\infty$. For natural numbers m and n with $m\leq n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-010514 | MATHDB-368848

Non-negativity conjecture for the coefficients C

摘录 / EXCERPT: Non-negativity conjecture. The coefficients

conjecture | open | MathDB 页面 | 原文线索

V2-010515 | MATHDB-368855

Trinh-Xue's conjecture on intersections of Hecke-algebra blocks

摘录 / EXCERPT: Trinh-Xue's conjecture. There exists a choice of complete HC-parametrisation $\Psi=\{\Psi_{{\mathbf L},\lambda}\}$ such that, for every such pair $({\mathbf L},\lambda)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010516 | MATHDB-368858

Explicit local Langlands conjecture for toral supercuspidal representations

摘录 / EXCERPT: Let G be a connected simple algebraic group, let $\hg$ be the associated affine Kac-Moody algebra at critical level, and let $(J,\tphi)$ be a K-type for the loop group obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-010517 | MATHDB-368880

Coordinate-basis expansion and linear independence of Bethe vectors

摘录 / EXCERPT: Bethe-vector basis conjecture. For every $|\mathbf{x}\rangle\in\mathbb{V}$,

conjecture | open | MathDB 页面 | 原文线索

V2-010518 | MATHDB-368894

Perverse character sheaf conjecture for loop Lie algebras

摘录 / EXCERPT: Let G be a reductive group, let $\xi=(M,\lambda,L,\mathcal{O},\mathcal{L})$ be a polar-cuspidal datum for G, and let $\mathbf{Q}\subset LM$ be a parahoric subgroup containing...

conjecture | open | MathDB 页面 | 原文线索

V2-010519 | MATHDB-368921

Skorobogatov's conjecture for algebraic K3 surfaces

摘录 / EXCERPT: Skorobogatov's conjecture. The Brauer-Manin obstruction is the only obstruction to the Hasse principle for algebraic K3 surfaces over number fields.

conjecture | open | MathDB 页面 | 原文线索

V2-010520 | MATHDB-369047

Crystalline liftability conjecture for irregular weights

摘录 / EXCERPT: Let K be the finite unramified extension of $\mathbb Q_p$ used in the paper, let G_K be its absolute Galois group, and let Σ be the corresponding set of embeddings. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010521 | MATHDB-369052

Unidentifiability conjecture for the mammillary model with input 2 and output 3

摘录 / EXCERPT: Let $n\geq 5$. Let $\mathcal M=(G,\textit{In},\textit{Out},\textit{Leak})$ be an n-compartment mammillary model with $\textit{In}=\{2\}$, $\textit{Out}=\{3\}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010522 | MATHDB-369075

Residual finiteness conjecture for adjoint groups of quandles

摘录 / EXCERPT: A quandle is an algebraic structure with its standard quandle operation, and a quandle is residually finite if distinct elements can be separated by homomorphisms to finite quandle...

conjecture | open | MathDB 页面 | 原文线索

V2-010523 | MATHDB-369087

Converse conjecture for Selmer near-companion elliptic curves

摘录 / EXCERPT: Let K be the number field underlying the absolute Galois group G_K, let $n\geq 2$, and let $\mathsf E_1$ and $\mathsf E_2$ be elliptic curves over K. They are n-Selmer near...

conjecture | open | MathDB 页面 | 原文线索

V2-010524 | MATHDB-369148

Coates-Sujatha fine Selmer conjecture for elliptic curves

摘录 / EXCERPT: Coates-Sujatha's conjecture. For every elliptic curve E over a number field K, $Y(E/\textbf{K})$ is a finitely generated $\mathbb{Z}_p$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-010525 | MATHDB-369163

The v-number conjecture for crown graphs

摘录 / EXCERPT: Let $G=C_{n,n}$ with $n \ge 4$. The graph G is the crown graph obtained from the complete bipartite graph on two parts of size n by deleting a perfect matching, and J_G denot...

conjecture | open | MathDB 页面 | 原文线索

V2-010526 | MATHDB-369214

The Tschirnhausen realization conjecture for primitive covers

摘录 / EXCERPT: The Tschirnhausen realization conjecture. The tuple $(e_1,\dots,e_{d-1})$ is achieved as the scrollar invariants of a smooth irreducible primitive cover $C \to \mathbb{P}^1$ of deg...

conjecture | open | MathDB 页面 | 原文线索

V2-010527 | MATHDB-369215

Várilly-Alvarado and Vakil's limiting measure conjecture for scrollar invariants

摘录 / EXCERPT: Várilly-Alvarado and Vakil's limiting conjecture. For all $B \subseteq \mathbb{R}^{d-1}$, the limit $\lim_{g \to \infty}\pi^{\textnormal{geo}}(B,g)$ exists and depends only on the...

conjecture | open | MathDB 页面 | 原文线索

V2-010528 | MATHDB-369216

Várilly-Alvarado and Vakil's Grothendieck-ring measure conjecture

摘录 / EXCERPT: Várilly-Alvarado and Vakil's Grothendieck-ring measure conjecture. For all B, the limit $\lim_{g\to\infty}\pi^{\textnormal{gr}}(B,g)$ converges to a measure...

conjecture | open | MathDB 页面 | 原文线索

V2-010529 | MATHDB-369217

Várilly-Alvarado and Vakil's density-function conjecture for scrollar invariants

摘录 / EXCERPT: Várilly-Alvarado and Vakil's density-function conjecture. The quantities $\rho^{\textnormal{geo}}(x)$ and $\rho^{\textnormal{gr}}(x)$ exist and are independent of the choice of...

conjecture | open | MathDB 页面 | 原文线索

V2-010530 | MATHDB-369279

The automorphic Galois group conjecture for global Langlands correspondences

摘录 / EXCERPT: Automorphic Galois group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010531 | MATHDB-369297

The reduced projective class group conjecture for torsionfree groups

摘录 / EXCERPT: Let G be a group and let $\widetilde{K}_0(\mathbb{Z}G)$ denote its reduced projective class group. The reduced projective class group conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-010532 | MATHDB-369298

The regular-ring projective class group conjecture for torsionfree groups

摘录 / EXCERPT: Let G be a group and let R be a regular ring, meaning that R is Noetherian and every R-module has a finite-dimensional projective resolution. The regular-ring projective cl...

conjecture | open | MathDB 页面 | 原文线索

V2-010533 | MATHDB-369300

The Whitehead group conjecture for torsionfree groups

摘录 / EXCERPT: Let G be a group and let $\operatorname{Wh}(G)$ denote its Whitehead group. The Whitehead group conjecture. If G is torsionfree, then $\operatorname{Wh}(G)$ vanishes. This woul...

conjecture | open | MathDB 页面 | 原文线索

V2-010534 | MATHDB-369301

The negative K-theory conjecture for torsionfree groups

摘录 / EXCERPT: Let R be a regular ring and let G be a group. The negative K-theory conjecture. If G is torsionfree, then ... This is the negative-degree part of the Farrell-Jones program for...

conjecture | open | MathDB 页面 | 原文线索

V2-010535 | MATHDB-369302

The algebraic K-theory Farrell-Jones conjecture for torsionfree groups

摘录 / EXCERPT: Let G be a group, let R be a regular ring, and let BG be a classifying space for G. The spectrum $\mathbf{K}(R)$ represents algebraic K-theory, and the assembly map is th...

conjecture | open | MathDB 页面 | 原文线索

V2-010536 | MATHDB-369353

Zywina's agreeable-closure classification for rational non-CM j-invariants

摘录 / EXCERPT: Let $j \in \Q$ be a non-CM rational j-invariant. Write $\cG_j$ for the associated adelic Galois-image subgroup, let $(\cG_j)_{\operatorname{agr}}$ be its agreeable closure, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-010537 | MATHDB-369389

Hedden-Pinzón-Caicedo's winding-zero rank-expansion conjecture

摘录 / EXCERPT: Let $P\subset S^1\times D^2$ be a pattern, and let its winding number be the algebraic intersection number of P with a meridional disk of $S^1\times D^2$. A pattern is rank expan...

conjecture | open | MathDB 页面 | 原文线索

V2-010538 | MATHDB-369415

The change-of-basis coefficients between the dual canonical and MV bases

摘录 / EXCERPT: Let L be a finite-dimensional simple module over the relevant KLR algebra, let d_L be the corresponding vector in the dual canonical basis of ${\mathbb C}[N]$, and let b_Z be...

conjecture | open | MathDB 页面 | 原文线索

V2-010539 | MATHDB-369417

Monoidality of the cyclic functor from shifted category O to KLR modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie-theoretic datum of the paper, let $\mathcal O_{\mathrm{sh}}$ be the shifted category O, and let $R=\bigoplus_{\nu\in Q_+}R_\nu$ be the associated KLR...

conjecture | open | MathDB 页面 | 原文线索

V2-010540 | MATHDB-369474

Recursive formula for fundamental models of degree $2n-2$

摘录 / EXCERPT: For each n, let Δ_n denote the relevant n-state model space, and let a_n be the number of fundamental models in Δ_n of degree $2n-1$. For $n\geq 3$, Proposition...

conjecture | open | MathDB 页面 | 原文线索

V2-010541 | MATHDB-369488

The rhombic period-lattice conjecture for trigonal curves

摘录 / EXCERPT: Let $\mathcal{V}$ be a trigonal curve whose finite branch points $\{(e_i,d_i)\}$ are real or occur in complex-conjugate pairs. Let ω and η be its period matrices, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010542 | MATHDB-369489

The finite-subspace reality conjecture for trigonal sigma-functions

摘录 / EXCERPT: Let $\mathcal{V}$ be a trigonal curve whose finite branch points are real or occur in complex-conjugate pairs. Let $\wp_{i,j}$ and $\wp_{i,j,k}$ denote the associated Kleinian func...

conjecture | open | MathDB 页面 | 原文线索

V2-010543 | MATHDB-369515

The faithfulness conjecture for the triangular functor

摘录 / EXCERPT: Let Σ be a connected oriented marked surface other than a sphere with 4 punctures or a projective plane with 2 punctures, and let ${\bf F}:{\bf TSurf}^t\to{\bf Grp}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-010544 | MATHDB-369516

The cluster-automorphism generation conjecture for noncommutative surface algebras

摘录 / EXCERPT: Let Σ be a marked surface, let ${\mathcal A}_\Sigma$ be its noncommutative cluster algebra, and let $I_{p,1}(\Sigma)$ denote the set of the relevant once-punctured indices....

conjecture | open | MathDB 页面 | 原文线索

V2-010545 | MATHDB-369586

The Dixmier-Moeglin conjecture for noetherian pointed Hopf algebras

摘录 / EXCERPT: Dixmier-Moeglin conjecture for pointed Hopf algebras. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010546 | MATHDB-369587

The finite-Krull-dimension conjecture for noetherian group algebras

摘录 / EXCERPT: Finite-Krull-dimension conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-010547 | MATHDB-369599

Finite-dimensional representability conjecture for generalized polynomial identities

摘录 / EXCERPT: Finite-dimensional representability conjecture. There exists a finite-dimensional W-(super)algebra B such that

conjecture | open | MathDB 页面 | 原文线索

V2-010548 | MATHDB-369639

The dominant solvable plabic tangle quasi-cluster conjecture

摘录 / EXCERPT: Let $(G,\mathbf{D})$ be a dominant solvable plabic tangle, where G has n outer boundary vertices and inner disks $D^{(1)},\dots,D^{(\ell)}$, and let m-VRCs denote the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-010549 | MATHDB-369641

The global-branch positivity conjecture for higher-intersection promotions

摘录 / EXCERPT: Let $(G,\mathbf{D})$ be a tangle admitting a brushing, with $\dim\Pi_G=km$ and positive intersection number $\operatorname{IN}_m(G)>0$. Let $F_{\widehat{(G,\mathbf{D})}}$ be the as...

conjecture | open | MathDB 页面 | 原文线索

V2-010550 | MATHDB-369650

Mináč-Tân's Massey vanishing conjecture for pro-p absolute Galois groups

摘录 / EXCERPT: Massey vanishing conjecture. G_K satisfies the Massey vanishing property.

conjecture | open | MathDB 页面 | 原文线索

V2-010551 | MATHDB-369655

Compact-group specialness conjecture for equivariant Kasparov theory

摘录 / EXCERPT: Let G be a compact group. The equivariant Kasparov theory KK^G is said to be very special when it has the special properties defined in the surrounding discussion. Compact-grou...

conjecture | open | MathDB 页面 | 原文线索

V2-010552 | MATHDB-369683

Depth-graded double shuffle conjecture for multiple zeta values

摘录 / EXCERPT: Let $\mathcal{Z}$ be the algebra of multiple zeta values, equipped with its depth filtration, and let $\operatorname{gr}_D\mathcal{Z}$ be the associated depth-graded algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010553 | MATHDB-369699

The extension-monad conjecture for rings and their additive reducts

摘录 / EXCERPT: Extension-monad conjecture. The extension monad of a ring has the same underlying set as the extension monad of its additive reduct.

conjecture | open | MathDB 页面 | 原文线索

V2-010554 | MATHDB-369700

The dibinary-algebra extension-monad conjecture for the beta reduct

摘录 / EXCERPT: Dibinary extension-monad conjecture. The extension monad of $\mathbb{A}$ has the same underlying set as the extension monad of its β reduct.

conjecture | open | MathDB 页面 | 原文线索

V2-010555 | MATHDB-369707

Infinite collision conjecture over the algebraic closure of a finite field

摘录 / EXCERPT: Infinite collision conjecture. The set $C(\alpha_1,\alpha_2;\beta)$ is infinite. The conjecture reverses the expectation stated immediately beforehand, which predicted finiteness w...

conjecture | open | MathDB 页面 | 原文线索

V2-010556 | MATHDB-369732

Eager's holomorphic Seiberg duality conjecture

摘录 / EXCERPT: Eager's conjecture. One can construct quantizations of these theories such that there is an equivalence of 2-dimensional holomorphic factorization algebras

conjecture | open | MathDB 页面 | 原文线索

V2-010557 | MATHDB-369745

Deligne-Drinfeld-Ihara, Alekseev-Torossian and Schneps-Raphael isomorphism conjecture

摘录 / EXCERPT: Let the four Lie algebras in the inclusions referred to in the source be graded by weight, and let the associated graded Lie algebras for the depth filtration be included as in the...

conjecture | open | MathDB 页面 | 原文线索

V2-010558 | MATHDB-369746

Broadhurst-Kreimer conjecture for the linearized Kashiwara-Vergne Lie algebra

摘录 / EXCERPT: Let $\widehat{\mathfrak{lkv}}_2$ be the linearized Kashiwara-Vergne Lie algebra, bigraded by weight and depth. Let $\mathrm{BK}_{W,D}$ be the coefficients defined by ... with s,t f...

conjecture | open | MathDB 页面 | 原文线索

V2-010559 | MATHDB-369806

Three-legs PT3 qq-character collision conjecture

摘录 / EXCERPT: Three-legs PT3 qq-character collision conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010560 | MATHDB-369852

DeMarco's orbit-closure conjecture for pairs of marked points

摘录 / EXCERPT: Let $f_t(x)$ be a non-isotrivial algebraic family of rational functions of degree greater than 1 over $\mathbb C$, parametrized by $t\in\mathbb A^1_{\mathbb C}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010561 | MATHDB-369855

Galois-module structure conjecture for quadratic twists

摘录 / EXCERPT: Let $E/\mathbb Q$ be an elliptic curve of rank 1, let $\mathcal F_b$ be the twist family specified by the local data b, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-010562 | MATHDB-369866

Hussain-Ma-Yau-Zuo's non-existence conjecture for negative derivations

摘录 / EXCERPT: Hussain-Ma-Yau-Zuo's conjecture. The graded algebra

conjecture | open | MathDB 页面 | 原文线索

V2-010563 | MATHDB-369869

The cocenter basis conjecture for Hecke algebras of Coxeter groups

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let R be the coefficient ring, and let $\mathcal{H}$ be its Hecke algebra. The cocenter is the quotient $\mathcal{H}/[\mathcal{H},\mathcal{H}]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-010564 | MATHDB-369877

Lück's determinant conjecture for all discrete groups

摘录 / EXCERPT: Let Γ be a discrete group, let $L(\Gamma)$ be its group von Neumann algebra, and let τ be the standard trace on $L(\Gamma)$. For $A\in M_n(\mathbb{Z}[\Gamma])$, write ……

conjecture | open | MathDB 页面 | 原文线索

V2-010565 | MATHDB-369896

Homogeneous realization conjecture for Grassmann algebra gradings

摘录 / EXCERPT: Homogeneous realization conjecture. Every $\mathbb{Z}_{2}$-grading of E is $\mathbb{Z}_{2}$-graded isomorphic to some homogeneous $\mathbb{Z}_{2}$-grading of E.

conjecture | open | MathDB 页面 | 原文线索

V2-010566 | MATHDB-369931

Kato's local epsilon conjecture

摘录 / EXCERPT: Let (R,T,ζ) be a triple as in the paper's setup, where T is an R-representation of $G_{\mathbb Q_p}$ and ζ is a $\mathbb Z_p$-basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-010567 | MATHDB-369969

Independence conjecture for Auslander regular incidence algebras

摘录 / EXCERPT: Let K be a field of characteristic 0, and let P be a poset. Denote its incidence algebra by KP, and let independent and the Coxeter permutation be the corresponding notions...

conjecture | open | MathDB 页面 | 原文线索

V2-010568 | MATHDB-369990

Asymptotic Lech consequence for isolated singularities

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a local ring. Suppose its completion $\widehat{R}$ has an isolated singularity, meaning that $\widehat{R}_P$ is regular for every...

conjecture | open | MathDB 页面 | 原文线索

V2-010569 | MATHDB-369991

Lim-stable singularities are of log canonical type

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a normal local domain essentially of finite type over a field of characteristic zero. Suppose that R is lim-stable, for example, semistable. Lim-stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-010570 | MATHDB-369992

Stirling-number refinement of Lech's inequality for monomial ideals

摘录 / EXCERPT: Stirling-number Lech conjecture. The first displayed inequality, together with the equivalent reindexing given in the source, holds for every $\mathfrak{m}$-primary monomial ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-010571 | MATHDB-370062

Evenness of non-inner almost inner derivations of quasireductive Lie superalgebras

摘录 / EXCERPT: Let $L=L_\mathbf{0}\oplus L_\mathbf{1}$ be a quasireductive Lie superalgebra, and let an almost inner derivation of L be a derivation that is almost inner, while an inner derivat...

conjecture | open | MathDB 页面 | 原文线索

V2-010572 | MATHDB-370172

Christ's Higgs-cosingularity equivalence conjecture with Hom-bijections

摘录 / EXCERPT: Christ's conjecture. The restriction of Φ to ${\mathcal H}$ is an equivalence of k-linear categories

conjecture | open | MathDB 页面 | 原文线索

V2-010573 | MATHDB-370185

Simultaneously bounded and dense orbits for two diagonal flows

摘录 / EXCERPT: Let $d\geq 3$, let $X_d=\mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z})$, and let $v^1\nev^2$ be two nonzero vectors in the diagonal subalgebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-010574 | MATHDB-370234

The discreteness conjecture for real cluster c-vectors

摘录 / EXCERPT: Let $B\in{\bf SC}$ be a real exchange matrix with skew-symmetrizer $D=\mathrm{diag}(d_1,\dots,d_n)$, where ${\bf SC}$ denotes the class of skew-symmetrizable matrices satisfying th...

conjecture | open | MathDB 页面 | 原文线索

V2-010575 | MATHDB-370235

The combined standard-hypothesis and discreteness conjecture for real cluster matrices

摘录 / EXCERPT: Let ${\bf SC}$ be the class of real skew-symmetrizable exchange matrices satisfying the sign-coherent property, and let ${\bf B}(B)$ denote the mutation-equivalence class of B. T...

conjecture | open | MathDB 页面 | 原文线索

V2-010576 | MATHDB-370236

Bhatt-Hochster-Ma conjecture on lim Cohen-Macaulay sequences

摘录 / EXCERPT: Bhatt-Hochster-Ma conjecture. The ring R has a lim Cohen-Macaulay sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-010577 | MATHDB-370281

The semi-elliptic Lindemann-Weierstrass conjecture

摘录 / EXCERPT: Let Ω be a lattice in $\mathbb{C}$ with algebraic invariants. Let $t_1,\dots,t_s$ be $\mathbb{Q}$-linearly independent algebraic numbers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010578 | MATHDB-370287

Vanishing conjecture for holomorphic forms on \overline{\mathcal{M}}_{3,n}

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{3,15}$ and $\overline{\mathcal{M}}_{3,16}$ denote the moduli spaces of stable genus-3 curves with, respectively, 15 and 16 marked points. Vanishi...

conjecture | open | MathDB 页面 | 原文线索

V2-010579 | MATHDB-370292

Coleman's finiteness conjecture for curves with special-type Jacobians

摘录 / EXCERPT: Let g be a sufficiently large genus, and let a smooth projective curve of genus g have a Jacobian of ST type, as in the source. Coleman's conjecture. Only finitely many smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-010580 | MATHDB-370297

The integral-closedness criterion for integer-valued polynomial rings on algebras

摘录 / EXCERPT: Integral-closedness criterion. If $\operatorname{Int}_K(A)$ is integrally closed, then $\operatorname{Int}_K(A)$ is Prüfer.

conjecture | open | MathDB 页面 | 原文线索

V2-010581 | MATHDB-370317

Primality conjecture for permanental ideals

摘录 / EXCERPT: Let X be an $m\times n$ generic, symmetric, or Hankel matrix, and let $\Bbbk$ be a field of characteristic $p\neq 2$. Write $P_t(X)$ for the ideal generated by the $t\times t$ pe...

conjecture | open | MathDB 页面 | 原文线索

V2-010582 | MATHDB-370318

F-purity conjecture for the 3×4 third permanental ideal

摘录 / EXCERPT: Let X be a 3×4 generic matrix of indeterminates, and let $\Bbbk$ be a field of characteristic $p>2$. F-purity conjecture for the 3×4 third permanental ideal. The q...

conjecture | open | MathDB 页面 | 原文线索

V2-010583 | MATHDB-370325

Efrat's arithmetical Elementary Type Conjecture

摘录 / EXCERPT: Efrat's arithmetical Elementary Type Conjecture. There is such a decomposition where, for each $1\leq i\leq n$, one of the following holds: $G_i\cong\mathbb{Z}_p$; G_i is the dec...

conjecture | open | MathDB 页面 | 原文线索

V2-010584 | MATHDB-370326

Elementary Type Conjecture in Lego form

摘录 / EXCERPT: Elementary Type Conjecture in Lego form. Every cyclotomic pro-p pair of Galois type and of finite rank is in ET_p.

conjecture | open | MathDB 页面 | 原文线索

V2-010585 | MATHDB-370327

Demuškin-to-p-adic type conjecture

摘录 / EXCERPT: Demuškin-to-p-adic type conjecture. If $\mathcal{G}_F(p)$ is a Demuškin cyclotomic pro-p pair of rank at least 3, then $\mathcal{G}_F(p)$ has p-adic type.

conjecture | open | MathDB 页面 | 原文线索

V2-010586 | MATHDB-370328

Demuškin maximal pro-p Galois group conjecture

摘录 / EXCERPT: Demuškin maximal pro-p Galois group conjecture. If $G_F(p)$ is a pro-p Demuškin group of rank at least 3, then F is arithmetically Demuškin.

conjecture | open | MathDB 页面 | 原文线索

V2-010587 | MATHDB-370372

The direct-product conjecture for clean graphs over rings

摘录 / EXCERPT: Direct-product conjecture. The clean graphs satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-010588 | MATHDB-370391

Logarithmic Kazhdan-Lusztig conjecture for screening subalgebras and Nichols algebras

摘录 / EXCERPT: Logarithmic Kazhdan-Lusztig conjecture. In good cases, there is an equivalence of braided tensor categories

conjecture | open | MathDB 页面 | 原文线索

V2-010589 | MATHDB-370410

The existence of non-quasi-isomorphic DGAs equivalent as MU-algebras

摘录 / EXCERPT: Let A and B be differential graded algebras (DGAs), and let $\MU$ denote the complex cobordism spectrum. The MU-algebra equivalence conjecture. There exist DGAs A and B tha...

conjecture | open | MathDB 页面 | 原文线索

V2-010590 | MATHDB-370421

Polishchuk-Positselski rationality conjecture for Koszul algebras

摘录 / EXCERPT: Polishchuk-Positselski conjecture. The Hilbert series of A is a rational function.

conjecture | open | MathDB 页面 | 原文线索

V2-010591 | MATHDB-370422

Khoroshkin-Piontkovski differential-algebraicity conjecture for Koszul symmetric operads

摘录 / EXCERPT: Khoroshkin-Piontkovski conjecture. The Hilbert series of $\mathcal{P}$ is differential algebraic over $\mathbb{Z}[t]$.

conjecture | open | MathDB 页面 | 原文线索

V2-010592 | MATHDB-370423

Order-one differential-algebraicity conjecture for binary-generated Koszul symmetric operads

摘录 / EXCERPT: Order-one differential-algebraicity conjecture. The Hilbert series of $\mathcal{P}$ is differential algebraic of order 1 over $\mathbb{Z}[t]$.

conjecture | open | MathDB 页面 | 原文线索

V2-010593 | MATHDB-370433

Rapid generation conjecture for finite simple Lie algebras

摘录 / EXCERPT: Rapid generation conjecture. There exist absolute constants C,D such that, for every nonabelian finite simple Lie algebra $\operatorname{\mathfrak{g}}$ over $\mathbf{F}_p$ and ev...

conjecture | open | MathDB 页面 | 原文线索

V2-010594 | MATHDB-370470

The Euler conductor noncontainment conjecture for isolated hypersurface singularities

摘录 / EXCERPT: Let $R=k[[x_1,\ldots,x_n]]$ define an isolated hypersurface singularity, with $f\in R$ and k a field of characteristic zero. Let J_f be the Jacobian ideal and let I_f denote...

conjecture | open | MathDB 页面 | 原文线索

V2-010595 | MATHDB-370498

Cohen-Macaulayness conjecture for Rees algebras of complementary edge ideals

摘录 / EXCERPT: Let G be a finite simple graph, and let $I_c(G)$ denote its complementary edge ideal in the polynomial ring associated to G. Write $\mathcal{R}(I_c(G))$ for the Rees algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-010596 | MATHDB-370530

The one-sided fundamental semigroup conjecture for property (T) group factors

摘录 / EXCERPT: Let G be an icc property (T) group. For a II_1 factor $\mathcal M$, let $\mathcal F_s(\mathcal M)$ be the set of all $t\in\mathbb R_+$ for which there exists a $*$-emb...

conjecture | open | MathDB 页面 | 原文线索

V2-010597 | MATHDB-370549

Equivalence between the noncommutative krv1 equation and coefficient identities

摘录 / EXCERPT: Let k be the coefficient field, let $\mathfrak{fr}_k(x_0,x_1)$ be the completed free Lie algebra on x_0,x_1, and let ψ_{451}, ψ_{123}, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010598 | MATHDB-370571

Conjecture on lifting the Green witness to an L_{∞} algebra

摘录 / EXCERPT: Let $(F,Q,(-,-),W)$ be a free BV field theory, and suppose its Green's witness W is obtained by truncating an L_{∞} algebra to its underlying cochain complex. Green-witnes...

conjecture | open | MathDB 页面 | 原文线索

V2-010599 | MATHDB-370572

Conjectural factorization-algebra enhancement of the first law of black-hole thermodynamics

摘录 / EXCERPT: Let M be a spacetime, let $g \in [\mathrm{Lor}(M)/\mathrm{Diff}^{+}(M)]$ be an asymptotically flat and stationary black-hole solution, and let $\mathscr{E}_g$ be the differential...

conjecture | open | MathDB 页面 | 原文线索

V2-010600 | MATHDB-370583

Explicit adjoint formulas for parabolic Soergel bimodule categories

摘录 / EXCERPT: Explicit adjoint formulas conjecture. The adjoints satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-010601 | MATHDB-370621

Kedlaya's prediction on bounded-support p-adic algebraic numbers

摘录 / EXCERPT: Let the support of a p-adic Hahn expansion be the ordered set of exponents with nonzero coefficients, and let its order type be the order type of that set. Kedlaya's prediction....

conjecture | open | MathDB 页面 | 原文线索

V2-010602 | MATHDB-370701

The Schneider-Ollivier surjectivity conjecture for the canonical morphism

摘录 / EXCERPT: Schneider-Ollivier's surjectivity conjecture. The canonical morphism $\mathbb{X}^*\to\mathcal{H}$ is a surjection.

conjecture | open | MathDB 页面 | 原文线索

V2-010603 | MATHDB-370737

Bhargava's asymptotic conjecture for degree-d number fields with Galois group S_d

摘录 / EXCERPT: Let $N_d(X)$ denote the number of number fields of degree d having discriminant with absolute value at most X. Let $q(n,k)$ denote the number of partitions of n into at most...

conjecture | open | MathDB 页面 | 原文线索

V2-010604 | MATHDB-370742

The generalized Jacobian conjecture for quadratic mappings

摘录 / EXCERPT: Let A be a commutative binary complex algebra, meaning that its binary operation is symmetric. Generalized Jacobian conjecture for quadratic mappings. If A is Engel, then A i...

conjecture | open | MathDB 页面 | 原文线索

V2-010605 | MATHDB-370900

The lifting conjecture for exact equivariant Morita classes

摘录 / EXCERPT: Lifting conjecture. The set X is a complete set of exact H-equivariant Morita equivalence class representatives.

conjecture | open | MathDB 页面 | 原文线索

V2-010606 | MATHDB-370914

The adelic Mumford-Tate conjecture for Hodge-maximal motives

摘录 / EXCERPT: Let M be a motive over a number field F with Betti realization M_B, Mumford-Tate group G_{MT}, and adelic Galois representation ... Assume that the Betti realization M_B is...

conjecture | open | MathDB 页面 | 原文线索

V2-010607 | MATHDB-371080

The monomial glicci conjecture

摘录 / EXCERPT: Let $\mathbb K$ be a field, let $R=\mathbb K[x_1,\ldots,x_n]$, and let $I\subseteq R$ be a Cohen--Macaulay monomial ideal. Monomial glicci conjecture. The ideal I is glicci, mean...

conjecture | open | MathDB 页面 | 原文线索

V2-010608 | MATHDB-371088

Stafford's cyclicity conjecture for torsion Weyl-algebra modules

摘录 / EXCERPT: Let A_n be the nth Weyl algebra, and let a finitely generated A_n-module be torsion when it has rank zero. A module is cyclic if it is generated by one element. Stafford's co...

conjecture | open | MathDB 页面 | 原文线索

V2-010609 | MATHDB-371167

Farrell-Jones conjecture for K_0 of torsion-free group rings

摘录 / EXCERPT: Let G be a torsion-free group and let R be a regular ring. Farrell-Jones conjecture for K_0. The map induced by the inclusion $R\to RG$ is an isomorphism ... This is described...

conjecture | open | MathDB 页面 | 原文线索

V2-010610 | MATHDB-371168

Coherence of group algebras of surface-by-cyclic groups

摘录 / EXCERPT: Surface-by-cyclic group-algebra coherence conjecture. The group algebra $\mathbb{Q}[G]$ is coherent. The source identifies the virtual surface-group case as difficult and open, mot...

conjecture | open | MathDB 页面 | 原文线索

V2-010611 | MATHDB-371180

Tachikawa's commutative conjecture on canonical modules

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be a Noetherian local ring with a canonical module K_R. Tachikawa's conjecture. If ... for all $i>0$, then R is Gorenstein. This is the commutative versio...

conjecture | open | MathDB 页面 | 原文线索

V2-010612 | MATHDB-371242

D-algebraicity conjecture for Catalan subsequences

摘录 / EXCERPT: Let $(C_n)_{n\in\mathbb{N}}$ be the Catalan numbers, and let $d\in\mathbb{N}\!\setminus\!\{0\}$. A sequence is D-algebraic of order 2 and degree $2d$ if it satisfies a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-010613 | MATHDB-371285

The winning-strategy conjecture for Ideal Chomp on polynomial algebras

摘录 / EXCERPT: Winning-strategy conjecture. Player A has a winning strategy on $\bar{K}[x_1,\dots,x_n]$ for any $n\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-010614 | MATHDB-371301

Conjecture on algebraicity, diagonals and global boundedness for linear-coefficient recurrences

摘录 / EXCERPT: The algebraicity-diagonal-global-boundedness conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010615 | MATHDB-371318

The characterization of Z-stable graph algebras by regularity and graph conditions

摘录 / EXCERPT: Graph-algebra characterization conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010616 | MATHDB-371326

Hyperoctahedral Galois group conjecture for sorted binomial polynomials

摘录 / EXCERPT: Let $P_n(z)$ be the sorted binomial polynomial. For odd $n=2m+1$, the source identifies the relevant target group as the mth hyperoctahedral group. Hyperoctahedral Galois group c...

conjecture | open | MathDB 页面 | 原文线索

V2-010617 | MATHDB-371328

The formal extended double shuffle conjecture for multiple zeta values

摘录 / EXCERPT: Let $\mathcal{Z}^f$ be the quotient of the shuffle algebra by the two-sided ideal generated by the formal extended double shuffle relations, and write $\zeta^f(k_1,\ldots,k_r)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-010618 | MATHDB-371329

Kaneko's finite multiple zeta algebra isomorphism conjecture

摘录 / EXCERPT: Kaneko's conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-010619 | MATHDB-371371

Phantom-category existence and deformation conjecture for critical blowups of Hirzebruch surfaces

摘录 / EXCERPT: Let n be a nonnegative integer, set $d(n)=6+\max\\{3,n\\}$, and let X be obtained by blowing up $\mathbf{F}_n$ at $d(n)$ general points. Let $\mathcal{C}_n$ be a subcategory of...

conjecture | open | MathDB 页面 | 原文线索

V2-010620 | MATHDB-371385

The c4-tilting finiteness conjecture for symmetric algebras

摘录 / EXCERPT: Let A be a finite-dimensional symmetric algebra. The algebra A is τ-tilting finite if it has only finitely many isomorphism classes of support τ-tilting modules, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010621 | MATHDB-371390

Sporadic Wigner-Heisenberg algebra conjecture

摘录 / EXCERPT: Sporadic Wigner-Heisenberg conjecture. The only finite-dimensional, non-nilpotent, and solvable Lie subalgebra of $\hat{A}_1$ that is not covered by the two families...

conjecture | open | MathDB 页面 | 原文线索

V2-010622 | MATHDB-371395

Refined Bachmann conjecture for formal qMZV filtrations

摘录 / EXCERPT: Refined Bachmann conjecture. For all $z,d,w\in\mathbb{Z}_{>0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-010623 | MATHDB-371400

Inductive refined Bachmann conjecture for the case of fewer zeros than depth

摘录 / EXCERPT: Fix $z,d,w\in\mathbb{Z}_{>0}$ with $z<d$, and assume that ... for all $z'\leq z$, $d'<d$, and $w'<w$. For every $\mathbf{k}=(k_1,\ldots,k_d)\in\mathbb{Z}_{>0}^d$ and every word...

conjecture | open | MathDB 页面 | 原文线索

V2-010624 | MATHDB-371435

The regularity conjecture for antiunitary representations

摘录 / EXCERPT: Let G be a Lie group with Lie algebra ${\mathfrak g}$, let $h\in{\mathfrak g}$ be an Euler element, and let G_{τ_h} denote the subgroup associated with the involution determ...

conjecture | open | MathDB 页面 | 原文线索

V2-010625 | MATHDB-371472

Hopf-divisibility conjecture for rational tangle relations

摘录 / EXCERPT: Let $[n_1,n_2,\dots,n_r]$ and $\epsilon[n_1,n_2,\dots,n_r]$ be two diagrams of rational links, where $\epsilon=1$ if r is odd and $\epsilon=-1$ otherwise. The Rational Tangle Alg...

conjecture | open | MathDB 页面 | 原文线索

V2-010626 | MATHDB-371512

Green's conjecture on graded Betti numbers of curves

摘录 / EXCERPT: Let X be a curve of genus g. For a divisor D on X of degree d, let g_d^r denote a subspace of the Riemann-Roch space $L(D)$ of dimension $r+1$, and let $\beta_{p,2}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-010627 | MATHDB-371517

Zariski-dense orbit conjecture

摘录 / EXCERPT: Zariski-dense orbit conjecture. There exists $x\in X(k)$ whose forward orbit under f is well-defined and Zariski-dense in X.

conjecture | open | MathDB 页面 | 原文线索

V2-010628 | MATHDB-371551

Elgin-Reading-Stella's conjectures on cluster scattering wall-function coefficients

摘录 / EXCERPT: Elgin-Reading-Stella's conjectures. For $i,j>0$, the coefficient $\tau(i,j)$ satisfies all of the following: it is a polynomial in b, c, and g; this polynomial has g as a f...

conjecture | open | MathDB 页面 | 原文线索

V2-010629 | MATHDB-371651

Budzik's conjecture on superalgebra invariant multiplicities

摘录 / EXCERPT: Let k and ℓ be nonnegative integers, let λ be a partition, and let $m_\lambda(k,\ell)$ denote the multiplicity associated with the hook Schur decomposition for the s...

conjecture | open | MathDB 页面 | 原文线索

V2-010630 | MATHDB-371678

Connes' character rigidity conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be a higher-rank lattice and let a character mean the character object considered in the source. Connes' character rigidity conjecture. Every character of Γ is...

conjecture | open | MathDB 页面 | 原文线索

V2-010631 | MATHDB-371713

Qureshi-Rinaldo-Romeo conjecture on simple polyominoes

摘录 / EXCERPT: Let $\mathcal{P}$ be a simple polyomino, meaning a polyomino without holes, and let $K[\mathcal{P}]$ be its associated coordinate ring. Qureshi-Rinaldo-Romeo conjecture. The switch...

conjecture | open | MathDB 页面 | 原文线索

V2-010632 | MATHDB-371722

Rinaldo-Romeo thinness conjecture for rook polynomials of polyominoes

摘录 / EXCERPT: Rinaldo-Romeo conjecture. The polyomino $\mathcal{P}$ is thin if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-010633 | MATHDB-371736

Ngô's compact-support conjecture for the -Schwartz space

摘录 / EXCERPT: Assume that ρ factors through $\widehat{G}(\mathbb{C})\rtimes\operatorname{Gal}(E/F)$ for a finite Galois extension E/F, and let M_ρ be Ngô's reductive monoid attached...

conjecture | open | MathDB 页面 | 原文线索

V2-010634 | MATHDB-371739

Zywina's conjecture on possible adelic indices of non-CM elliptic curves

摘录 / EXCERPT: Zywina's conjecture. The adelic index satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010635 | MATHDB-371774

Perrin-Riou's local Tamagawa-number conjecture

摘录 / EXCERPT: Let V be a crystalline representation of G_ℓ over K_λ, and let $T\subseteq V$ be a G_ℓ-stable $\mathcal{O}_\lambda$-lattice. Let ω and ω^* be latt...

conjecture | open | MathDB 页面 | 原文线索

V2-010636 | MATHDB-371819

Functorial Goldman-Turaev formality conjecture for framed surfaces

摘录 / EXCERPT: Let $\boldsymbol{\widehat{\mathbf{PaB}}}^f$ be the prounipotent completion of the operad of parenthesized framed braids, let $\boldsymbol{\widehat{\mathbf{PaB}}}^f_g$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-010637 | MATHDB-371839

Classicality detection conjecture via the Fontaine operator

摘录 / EXCERPT: Let $\mathbb{T}^S(K^p,\mathcal{O})_{\mathfrak{m}}$ be the localized Hecke algebra, let $\varphi:\mathbb{T}^S(K^p,\mathcal{O})_{\mathfrak{m}}\to\mathcal{O}$ be an eigensystem, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-010638 | MATHDB-371840

Breuil-Mézard conjecture for two-dimensional Galois representations

摘录 / EXCERPT: Let $L/\mathbb{Q}_p$ be a finite extension, let E be a coefficient field, let W_L be the set of Serre weights, and let $\overline{\rho}_L:G_L\to\mathrm{GL}_2(\mathbb{F})$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-010639 | MATHDB-371855

The conjecture on Galois representations attached to automorphic representations of GSp4

摘录 / EXCERPT: Let G be the group under consideration, let F be the underlying number field, and let $\pi=\pi_\infty\otimes\pi_f$ be an automorphic representation of $G(\mathbb A)$. For a spl...

conjecture | open | MathDB 页面 | 原文线索

V2-010640 | MATHDB-371874

Christophersen's problem on automorphism groups of local algebras

摘录 / EXCERPT: Let $A=\mathbb{C}\oplus\mathfrak{m}$ be a finite-dimensional local algebra of dimension n, where $\mathfrak{m}$ is its maximal ideal, and let $\operatorname{Aut}(A)^{\circ}$ deno...

open_problem | open | MathDB 页面 | 原文线索

V2-010641 | MATHDB-371915

Taelman's conjecture for finite uniformizable t-modules

摘录 / EXCERPT: Let $K={\mathbb F}_q(\theta)$, let F be a finite extension of K, let ${\cal O}_F$ be an integral closure of A in F, and let $F_{\infty}=F\otimes_K K_{\infty}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010642 | MATHDB-371926

The squarefree-value conjecture for Drinfeld module sequences

摘录 / EXCERPT: Let A be the polynomial ring in the paper, let ϕ be the given Drinfeld module, and let $a\in A$ be the fixed base. A polynomial has no square factors when it is squarefree, a...

conjecture | open | MathDB 页面 | 原文线索

V2-010643 | MATHDB-371942

Finite simple CKL-algebras conjecture

摘录 / EXCERPT: Let X be a finite CKL-algebra, meaning an L-algebra satisfying the CKL axioms. Call X simple if it has no nontrivial ideals, and call it linear if its underlying order is linea...

conjecture | open | MathDB 页面 | 原文线索

V2-010644 | MATHDB-372029

The basis and generation conjecture for the Norton subalgebra of the bilinear forms graph

摘录 / EXCERPT: Let $\Gamma=(X,\mathcal R)$ be the bilinear forms graph, let $x,y\in X$ be adjacent vertices, and let $U=U(x,y)$ be the subspace defined earlier. Let $E=E_1$ be the primitive idemp...

conjecture | open | MathDB 页面 | 原文线索

V2-010645 | MATHDB-372032

The local splitting conjecture for very critical extensions

摘录 / EXCERPT: Let D be the filtered φ-module under consideration, let σ range over the relevant embeddings, and let I range over subsets that are very critical for σ. L...

conjecture | open | MathDB 页面 | 原文线索

V2-010646 | MATHDB-372037

The solvability-idempotent conjecture for complex evolution algebras

摘录 / EXCERPT: Let $\mathcal{E}$ be a complex evolution algebra. An element $u\in\mathcal{E}$ is an idempotent if $u^2=u$. If $\mathcal{E}$ has natural basis $\{e_1,\dots,e_n\}$ and structure mat...

conjecture | open | MathDB 页面 | 原文线索

V2-010647 | MATHDB-372047

Intermediate reduced degrees for the int* degree of a polytope

摘录 / EXCERPT: Let ${\mathcal P} \subset {\mathbb R}^n$ be a normal polytope of dimension n with $({\mathcal P} \setminus \partial {\mathcal P}) \cap {\mathbb Z}^n \neq \emptyset$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-010648 | MATHDB-372082

Bonanzinga-Eliahou conjecture on Gotzmann thresholds of powers of x_2

摘录 / EXCERPT: Bonanzinga-Eliahou's conjecture. For $n\geq 3$, the Gotzmann threshold $\tau_n(x_2^d)$ is a polynomial of degree 2^{n-2} in d with the same dominant term as the $(n-2)$-iterate...

conjecture | open | MathDB 页面 | 原文线索

V2-010649 | MATHDB-372189

Existence of the Minkowski dimension of arboreal Galois images

摘录 / EXCERPT: Let K be a number field, let $f:{\mathbb P}^1\to{\mathbb P}^1$ be a rational map of degree $d\geq2$ defined over K, and let $\alpha\in{\mathbb P}^1(K)$. Minkowski-dimension exi...

conjecture | open | MathDB 页面 | 原文线索

V2-010650 | MATHDB-372190

Andrews-Petsche conjecture on abelian arboreal Galois images

摘录 / EXCERPT: Let K be a number field, let K^{ab} be its maximal abelian extension, let $f:{\mathbb P}^1\to{\mathbb P}^1$ be a polynomial map of degree $d\geq2$ defined over K, an...

conjecture | open | MathDB 页面 | 原文线索

V2-010651 | MATHDB-372301

Inverse-image conjecture for the drop 1 operator and the Hoffman subring

摘录 / EXCERPT: Inverse-image conjecture. For any word w in $\mathcal{H}^0$ that cannot be written in the form $w_1\tau(w_2)$ for words w_1,w_2 in $\mathcal{H}^{2,3}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-010652 | MATHDB-372376

Involution compatibility for positive-characteristic multiple zeta values

摘录 / EXCERPT: Involution compatibility conjecture. For every $\mathfrak{s}\in\mathcal{I}$,

conjecture | open | MathDB 页面 | 原文线索

V2-010653 | MATHDB-372377

The induced involution conjecture for v-adic multiple zeta values

摘录 / EXCERPT: Induced involution conjecture. The map ι induces an R-algebra involution on $\mathcal{Z}_{v,R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010654 | MATHDB-372395

Normalization conjecture for the Kac-Moody unipotent automorphism group

摘录 / EXCERPT: Normalization conjecture. The group $\mathcal H$ normalizes $\widehat{\mathcal{U}}$. This is one of the conjectures proposed for extensions of Scott's definition in the Kac-Moody setting;...

conjecture | open | MathDB 页面 | 原文线索

V2-010655 | MATHDB-372396

Decomposition conjecture for the completed Kac-Moody automorphism group

摘录 / EXCERPT: Let $\widehat{\mathfrak g}$ be the completed Kac-Moody algebra, with graded pieces $\mathfrak g_k$. Define ... and let $\widehat{\mathcal{U}}=\widehat{\mathcal{U}}_1$ and $\mathcal H$ be as above. D...

conjecture | open | MathDB 页面 | 原文线索

V2-010656 | MATHDB-372405

The BPS Lie algebra conjecture for symmetric quivers with potential

摘录 / EXCERPT: Let Q be a symmetric quiver with potential W. Let $\mathfrak{g}_{Q,\mathsf{W}}$ be the Lie superalgebra defined from the classical R-matrix, and write ... The positive...

conjecture | open | MathDB 页面 | 原文线索

V2-010657 | MATHDB-372468

The strong Lefschetz conjecture for Milnor algebras

摘录 / EXCERPT: Strong Lefschetz conjecture. For any smooth $F\in S_d$, any integer $k\in[0,T/2)$, and any generic linear form $\ell\in S_1$, the induced multiplication map is an isomorphism. The...

conjecture | open | MathDB 页面 | 原文线索

V2-010658 | MATHDB-372510

The contraction-algebra BPS-Gopakumar-Vafa conjecture

摘录 / EXCERPT: Let A be the contraction algebra of a rational flopping curve C, namely a Jacobi algebra of the type defined in the cited work. For every $d\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010659 | MATHDB-372653

Grunewald-O'Halloran conjecture on degenerations of nilpotent Lie algebras

摘录 / EXCERPT: Grunewald-O'Halloran conjecture. Every complex finite-dimensional nilpotent Lie algebra of dimension greater than two is the degeneration of another Lie algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-010660 | MATHDB-372673

Main conjecture for Heegner points

摘录 / EXCERPT: Let Λ^{ac} be the anticyclotomic Iwasawa algebra, let $\mathbf{T}$ be the big Galois representation, and suppose ${\mathrm{Sel}}_{\mathrm{Gr}}(\mathbf{T})$ is torsion-free of...

conjecture | open | MathDB 页面 | 原文线索

V2-010661 | MATHDB-372713

Weyl-algebra subalgebra graph-admissibility conjecture

摘录 / EXCERPT: Let A_n be the complex n-mode Weyl algebra, and let $\mathfrak{g}$ be a finite-dimensional Lie subalgebra of A_n. Weyl-algebra subalgebra conjecture. Every finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-010662 | MATHDB-372714

Center-representing minimal graph conjecture for Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional minimal-graph-admissible Lie algebra with non-trivial center $\mathcal{Z}(\mathfrak{g})$. Center-representing graph conjecture. The Lie a...

conjecture | open | MathDB 页面 | 原文线索

V2-010663 | MATHDB-372715

Solvable ideal subset conjecture for centers of Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional minimal-graph-admissible Lie algebra associated with a minimal graph $G(V,E)$, and let $\mathcal{Z}(\mathfrak{g})$ denote its center. Sol...

conjecture | open | MathDB 页面 | 原文线索

V2-010664 | MATHDB-372716

Kernel-intersection criterion for graph-admissible Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional minimal-graph-admissible Lie algebra associated with a minimal graph $G(V,E)$, with $\dim(\mathfrak{g})\geq 2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010665 | MATHDB-372717

Graph criterion for semisimplicity of Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional graph-admissible Lie algebra associated with the labeled directed graph $G(V,E)$. Graph semisimplicity conjecture. The Lie algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-010666 | MATHDB-372718

Fine-grained magma-graph representation conjecture for Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex finite-dimensional Lie algebra, and let $(\mathcal{M},\delta)$ be an abelian magma such that $\mathfrak{g}$ is $\mathcal{M}$-magma-graded, with the...

conjecture | open | MathDB 页面 | 原文线索

V2-010667 | MATHDB-372719

Finest-granularity magma-graph faithfulness conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex finite-dimensional Lie algebra and $(\mathcal{M},\delta)$ an abelian magma such that $\mathfrak{g}$ is $\mathcal{M}$-magma-graded with finest granul...

conjecture | open | MathDB 页面 | 原文线索

V2-010668 | MATHDB-372720

Two-step nilpotent minimal-graph-admissibility conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional k-step nilpotent Lie algebra. Nilpotent minimal-graph-admissibility conjecture. If $k\leq 2$, then $\mathfrak{g}$ is minimal-graph-admi...

conjecture | open | MathDB 页面 | 原文线索

V2-010669 | MATHDB-372721

Chebyshev classification of positive fixed-point proportion

摘录 / EXCERPT: Let $f\in \mathbb{C}[x]$ be a complex polynomial, and let $G_\infty(\mathbb{C},f,t)$ denote its geometric iterated Galois group over a transcendental parameter t. A polynomial is...

conjecture | open | MathDB 页面 | 原文线索

V2-010670 | MATHDB-372722

Positive fixed-point proportion dichotomy for polynomial iterated Galois groups

摘录 / EXCERPT: Let K be a field and let $f\in K[x]$ be a polynomial of degree $d\geq 2$. Assume either that $\operatorname{char}(K)=0$ or that $\operatorname{char}(K)$ does not divide the local...

conjecture | open | MathDB 页面 | 原文线索

V2-010671 | MATHDB-372724

The Koszul-dual duality conjecture for finite-dimensional minimal L_∞-algebras

摘录 / EXCERPT: Koszul-dual duality conjecture. The derived linear dual of the dg $\operatorname{CE}_\bullet(\mathfrak{g})$-comodule...

conjecture | open | MathDB 页面 | 原文线索

V2-010672 | MATHDB-372787

Johnson's determinantal identity for contiguous minors of Toeplitz matrices

摘录 / EXCERPT: Johnson's conjecture. For every $n\ge 2$ and every Toeplitz matrix $A\in M_n(\mathbb{R})$ satisfying $A+A^{\top}=2J_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-010673 | MATHDB-372790

Symmetric-subspace closure conjecture for the Norton algebra

摘录 / EXCERPT: Symmetric-subspace closure conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-010674 | MATHDB-372791

Symmetric generation conjecture for the Norton algebra

摘录 / EXCERPT: Symmetric generation conjecture. The subspace $\omega^\perp\cap {\rm Sym}(S)$ is the subalgebra of the Norton algebra EV generated by $E\hat x+E\hat y$.

conjecture | open | MathDB 页面 | 原文线索

V2-010675 | MATHDB-372795

Hyperbolic polynomial branch and extension conjecture

摘录 / EXCERPT: Hyperbolic polynomial branch and extension conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010676 | MATHDB-372800

Kährström's conjecture for involutions in category O

摘录 / EXCERPT: Let Inv_n be the set of involutions in $\mathrm S_n$, let $L_z^{(n)}$ be the simple highest weight module indexed by z, let $\theta_x^{(n)}$ be the projective functor...

conjecture | open | MathDB 页面 | 原文线索

V2-010677 | MATHDB-372818

Sookdeo's conjecture on S-integral points in backward orbits

摘录 / EXCERPT: Let K be a number field with algebraic closure $\overline{K}$, let S be a finite set of places of K containing all archimedean places, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010678 | MATHDB-372823

The reduced Boston conjecture for simple-type pro-p groups

摘录 / EXCERPT: Reduced Boston conjecture. 1. If G is a uniform p-adic analytic pro-p group of simple type in characteristic zero and is hereditarily just-infinite, then there is no pro-p...

conjecture | open | MathDB 页面 | 原文线索

V2-010679 | MATHDB-372832

Linear independence conjecture for non-commutative polynomial classes

摘录 / EXCERPT: Linear independence conjecture. The subset $\{\langle f_i\rangle\}_{i=1}^r$ of $X(A)$ is $\mathbb Z$-linearly independent.

conjecture | open | MathDB 页面 | 原文线索

V2-010680 | MATHDB-372868

The implication from a6-set theoretic basis to the positive basis property

摘录 / EXCERPT: Expected implication. If H has a Φ-set theoretic basis, then H has the positive basis property.

conjecture | open | MathDB 页面 | 原文线索

V2-010681 | MATHDB-372895

The elliptic Deligne-Ihara conjecture for complex multiplication

摘录 / EXCERPT: Let X be the once-punctured elliptic curve under consideration, let K be its complex multiplication field, let I_K be the specified index set, and let $\boldsymbol{m}$ range...

conjecture | open | MathDB 页面 | 原文线索

V2-010682 | MATHDB-372896

The Deligne-Ihara conjecture for the thrice-punctured projective line

摘录 / EXCERPT: Let $T=\mathbb{P}^{1}_{\mathbb{Q}}-\{0,1,\infty\}$, and let $\mathfrak{t}=\bigoplus_{m>0}\mathfrak{t}_m$ be the graded Lie algebra over $\mathbb{Z}_p$ associated with the lower cen...

conjecture | open | MathDB 页面 | 原文线索

V2-010683 | MATHDB-372908

Kneser-Tits conjecture for simply connected isotropic almost simple groups

摘录 / EXCERPT: Let k be the base field, and let G be a simply connected, almost k-simple, k-isotropic algebraic group over k. Let $G(k)^+$ denote the abstract subgroup of $G(k)$ generat...

conjecture | open | MathDB 页面 | 原文线索

V2-010684 | MATHDB-372938

The Strong Maximal Rank Conjecture

摘录 / EXCERPT: The Strong Maximal Rank Conjecture. The locus of line bundles $L\in W^r_dC$ for which ϕ^k_L is not of maximal rank has dimension

conjecture | open | MathDB 页面 | 原文线索

V2-010685 | MATHDB-372959

The determinantal basis criterion for extending generalized spline modules

摘录 / EXCERPT: Let (G,β) be an edge-labeled graph over a GCD domain R, and let $F_1,F_2,\ldots,F_n\in\hat{R}_G$ be splines. The determinant of these splines is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-010686 | MATHDB-372966

The conjecture that the first L^2-Betti numbers agree over the von Neumann algebra and division ring

摘录 / EXCERPT: Let G be a finitely presented group and let F be a field. Denote by $b^{(2)}_1(G;\mathcal N(G))$ the first L^2-Betti number with coefficients in the group von Neumann algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-010687 | MATHDB-372969

Conjecture on the Courant contact model and the cubic model

摘录 / EXCERPT: Let the Courant contact model be the local L_∞-algebra construction associated to the relevant Courant algebroid, and let the cubic model be the constrained-field model intr...

conjecture | open | MathDB 页面 | 原文线索

V2-010688 | MATHDB-372982

Geometric Budur-type conjecture for Bernstein-Sato ideals

摘录 / EXCERPT: Let X be the underlying space, let $\mathcal{M}$ be a regular holonomic $\mathscr{D}_X$-module, and let $\mathcal{N}_0$ be the associated module. For a monoid ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-010689 | MATHDB-372993

The conjecture on unobstructed contraction algebras and smooth centres

摘录 / EXCERPT: Unobstructedness and smooth-centre conjecture. The point $[A]$ is smooth in $\mathcal{M}_d$, equivalently, A is unobstructed; moreover, the dimension of the centre is constant in...

conjecture | open | MathDB 页面 | 原文线索

V2-010690 | MATHDB-372999

Koszul filtration conjecture for quadratic binomial Gröbner bases

摘录 / EXCERPT: Koszul filtration conjecture. The quotient $S/(G)$ has a Koszul filtration. Moreover, if (G) is toric, then $S/(G)$ has a monomial Koszul filtration.

conjecture | open | MathDB 页面 | 原文线索

V2-010691 | MATHDB-373028

Unlikely Newton polygons and Ekedahl-Oort types for abelian covers

摘录 / EXCERPT: Let $g \geq 18$, and let m be the least common multiple of the exponents of all of the Galois groups of abelian covers of $\mathbf{P}^1$ branched at three points of genus g. Un...

conjecture | open | MathDB 页面 | 原文线索

V2-010692 | MATHDB-373036

The sl2-algebra conjecture for multiple Eisenstein series

摘录 / EXCERPT: The $\mathfrak{sl}_2$-algebra conjecture for multiple Eisenstein series. These maps should be well-defined $\mathbb{Q}$-linear maps on $\mathcal{E}$, should be derivations, and sho...

conjecture | open | MathDB 页面 | 原文线索

V2-010693 | MATHDB-373037

The derivative and relations conjecture for multiple Eisenstein series

摘录 / EXCERPT: The derivative and relations conjecture for multiple Eisenstein series. For every $w\in\mathfrak{H}^{\geq2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-010694 | MATHDB-373038

The derivative conjecture for formal multiple Eisenstein series

摘录 / EXCERPT: Let $\mathcal{E}^f$ be the formal multiple Eisenstein-series algebra, let $\mathcal{G}^f$ be the ambient formal algebra, and let $\theta=-\mathcal{D}\circ\varphi$ on the relevant w...

conjecture | open | MathDB 页面 | 原文线索

V2-010695 | MATHDB-373039

The conjecture for multiple Eisenstein-diamond series

摘录 / EXCERPT: Let $\mathfrak{H}^0$ be the admissible word space, let $\mathfrak{H}^0_\ast$ denote its harmonic-product algebra, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-010696 | MATHDB-373063

Vemulapalli's scrollar-invariant polytope conjecture for primitive covers

摘录 / EXCERPT: Let $k\geq 3$, let $g\geq 0$, and consider the polytope ... A scrollar-invariant tuple is a tuple of positive integers $\mathbf e=(e_1,\dots,e_{k-1})$ with $e_1+\dots+e_{k-1}=g+k-1$....

conjecture | open | MathDB 页面 | 原文线索

V2-010697 | MATHDB-373075

Product bounds conjecture for Hilbert-depth estimates

摘录 / EXCERPT: Product bounds conjecture. The following inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-010698 | MATHDB-373081

The finite-length Gorenstein conjecture

摘录 / EXCERPT: Let R be a commutative Noetherian local ring and let M be a finitely generated R-module. Assume that M has finite length and that $\operatorname{Gdim}(M)$, the Gorenstein d...

conjecture | open | MathDB 页面 | 原文线索

V2-010699 | MATHDB-373105

Lafforgue's rational-structure conjecture for the excursion algebra action

摘录 / EXCERPT: Let X be a curve with compactification $\ol{X}$, let $\sG$ be a reductive group, and let G be its Langlands dual. Let ... where $\Bun_G^{\on{level}}$ is the moduli space of G-b...

conjecture | open | MathDB 页面 | 原文线索

V2-010700 | MATHDB-373122

Symmetric-group log-concavity conjecture for the ideals I(n) and J(n)

摘录 / EXCERPT: For each positive integer n, let $I(n)$ and $J(n)$ be the ideals in the polynomial ring ${\mathbb C}[{\mathbf x}_{\binom{[n]}{2}}]$ on variables indexed by the two-element subset...

conjecture | open | MathDB 页面 | 原文线索

V2-010701 | MATHDB-373168

The root-count conjecture for colored mutations of admissible Dynkin biagrams

摘录 / EXCERPT: Colored-mutation root-count conjecture. The number of Γ-mutations in one period is the total number of roots in the root systems associated to the Γ-components,...

conjecture | open | MathDB 页面 | 原文线索

V2-010702 | MATHDB-373169

Regularity and Gröbner basis conjecture for powers of binary-form ideals

摘录 / EXCERPT: Let U be a two-dimensional vector space, let a,b be positive integers, and write $S=\operatorname{Sym}(\operatorname{Sym}^b U)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010703 | MATHDB-373175

Bjerkevik's stability conjecture for the pruning distance

摘录 / EXCERPT: Bjerkevik's conjecture. On persistence modules of bounded pointwise dimension, the pruning distance is stable and Lipschitz equivalent to the interleaving distance. More precisely,...

conjecture | open | MathDB 页面 | 原文线索

V2-010704 | MATHDB-373179

The Additivity conjecture for braided and symmetric pseudomonoids

摘录 / EXCERPT: Let $\mathcal{C}$ be a Gray monoid. A braided and symmetric pseudomonoid additivity conjecture. Every braiding on $\mathcal{C}$ induces an equivalence ... of 2-categories, and every...

conjecture | open | MathDB 页面 | 原文线索

V2-010705 | MATHDB-373192

Invariance of the number of minimal generators by degree for S-subalgebras

摘录 / EXCERPT: Let $K\langle X_d\rangle$ be the free associative algebra on X_d, equipped with the operation $\circ$, and let $(F,\circ)$ be an S-subalgebra. A minimal generating system is a...

conjecture | open | MathDB 页面 | 原文线索

V2-010706 | MATHDB-373201

Non-differential-algebraicity conjecture for Av(4231, 4123, 4312)

摘录 / EXCERPT: Let $\operatorname{Av}(4231,4123,4312)$ be the permutation class avoiding 4231, 4123, and 4312, and let $F(x)=\sum_{n\geq 0}a_nx^n$ be its ordinary generating function. Non-di...

conjecture | open | MathDB 页面 | 原文线索

V2-010707 | MATHDB-373231

The formal-dissection realization conjecture for derived-equivalent gentle and skew-gentle algebras

摘录 / EXCERPT: Let A be a graded gentle or skew-gentle algebra, and let $\mathbf S=(S,\Sigma,\eta)$ be the graded surface associated to A. For a formal dissection Δ of $\mathbf S$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010708 | MATHDB-373248

Right Kan extension conjecture for the universal finitary localizing invariant

摘录 / EXCERPT: Let $\mathcal{U}: \mathrm{CAlg}(\mathrm{Pr}^L_{\mathrm{dual}}) \rightarrow \mathrm{CAlg}(\mathrm{Mot})$ be the universal finitary localizing invariant, and consider its restriction...

conjecture | open | MathDB 页面 | 原文线索

V2-010709 | MATHDB-373275

Fröberg's conjecture on Hilbert series of generic ideals

摘录 / EXCERPT: Let $k[x_1,\dots,x_n]$ be a polynomial ring, and let $I=(f_1,\dots,f_r)$ be a generic ideal whose generators satisfy $\deg f_i=d_i$. For a power series $F(T)$, write $[F(T)]$ for i...

conjecture | open | MathDB 页面 | 原文线索

V2-010710 | MATHDB-373280

Modularity conjecture for the rank-2 attractor at $\phi=33\pm8\sqrt{17}$

摘录 / EXCERPT: Modularity conjecture at $\phi_{\pm}$. The semisimplification of the representation of G_K is isomorphic to the restriction to G_K of

conjecture | open | MathDB 页面 | 原文线索

V2-010711 | MATHDB-373287

Conjecture on generators of the ideal I^{μ}_N in the parabolic Hecke algebra

摘录 / EXCERPT: Generator conjectures. The element X^{μ}_N generates I^{μ}_N; the element Y^{μ}_N generates I^{μ}_N; and

conjecture | open | MathDB 页面 | 原文线索

V2-010712 | MATHDB-373294

Galois Images Parametrization Conjecture

摘录 / EXCERPT: Galois Images Parametrization Conjecture. There exist finitely many explicit open subgroups $G_1,\ldots,G_n$ of $\operatorname{GL}_2(\widehat{\mathbb{Z}})$ such that, for every non...

conjecture | open | MathDB 页面 | 原文线索

V2-010713 | MATHDB-373296

Sutherland-Zywina explicit uniformity conjecture

摘录 / EXCERPT: Explicit uniformity conjecture. Either

conjecture | open | MathDB 页面 | 原文线索

V2-010714 | MATHDB-373297

Zywina's generalized exceptional j-invariant conjecture

摘录 / EXCERPT: Zywina's generalized conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-010715 | MATHDB-373319

Higher-level runner-removal conjecture for graded decomposition numbers

摘录 / EXCERPT: Fix a subdivision datum and let $\boldsymbol{\lambda}=(\boldsymbol{\lambda}^{(1)},\dots,\boldsymbol{\lambda}^{(\ell)})$ and $\boldsymbol{\mu}$ be ℓ-partitions in...

conjecture | open | MathDB 页面 | 原文线索

V2-010716 | MATHDB-373321

Holographic tilting principle for perfectoid boundary carriers

摘录 / EXCERPT: Let X be a perfectoid boundary carrier equipped with a condensed sheaf/module $\mathcal{F}$ of sources and operators, and let $X^{\flat}$ be its tilt. Assume the holographic gene...

conjecture | open | MathDB 页面 | 原文线索

V2-010717 | MATHDB-373347

Malle's number field counting conjecture

摘录 / EXCERPT: Let k be a number field and G a transitive permutation group of degree n. Write $\mathcal{F}_{\disc,k}(G;X)$ for the set of G-extensions L/k with $|\disc(L/k)|\leq X$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-010718 | MATHDB-373356

Linear deformation conjecture for filiform compatible Lie algebras

摘录 / EXCERPT: Linear deformation conjecture. The algebra $\mathcal{L}$ is a linear deformation of the algebra $\mathcal{L}_s$ for certain $s \in \mathbb{N}^t$.

conjecture | open | MathDB 页面 | 原文线索

V2-010719 | MATHDB-373357

Dimension bound for solvable compatible extensions of $\mathcal{L}_s$

摘录 / EXCERPT: Dimension-bound conjecture. There are no solvable compatible Lie extensions of $\mathcal{L}_s$ of dimension greater than $\underline{n_s}+5$.

open_problem | open | MathDB 页面 | 原文线索

V2-010720 | MATHDB-373373

Quasi-periodic Poincaré-Bertrand formula

摘录 / EXCERPT: Let γ_0 denote the portion of γ lying in the strip $[-d/2,d/2) \times \mathbb{R}$ and define ... These are the quasi-periodic Hilbert transform and Laplace double-layer...

conjecture | open | MathDB 页面 | 原文线索

V2-010721 | MATHDB-373381

GHL monoidal categorification conjecture for shifted category O

摘录 / EXCERPT: Assume that ${\mathfrak{g}}$ is simply laced. Let $\mathcal{A}$ be the infinite-rank cluster algebra embedded in $K_0(\mathcal{O}^{\mathrm{sh}})$ by ... and let $I^{-1}\circ i$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-010722 | MATHDB-373429

Algebraic independence of logarithms of distinct primes

摘录 / EXCERPT: Algebraic-independence conjecture. The numbers $\ln p_1,\ldots,\ln p_m$ are algebraically independent. Equivalently, ψ is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-010723 | MATHDB-373445

Ihara-Kaneko-Zagier conjecture on algebraic relations among multiple zeta values

摘录 / EXCERPT: Ihara-Kaneko-Zagier conjecture. All algebraic relations in the algebra $\mathcal{Z}$ are obtained from the comparison of the extended maps

conjecture | open | MathDB 页面 | 原文线索

V2-010724 | MATHDB-373446

Stabilizer interpretation conjecture for the depth-graded multiple-zeta-value algebra

摘录 / EXCERPT: Stabilizer interpretation conjecture. There is an algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010725 | MATHDB-373572

The anti-canonical cylinder criterion for weighted del Pezzo hypersurfaces

摘录 / EXCERPT: Let S be a quasi-smooth, well-formed del Pezzo hypersurface of degree d in the weighted projective space ... where S is defined over an algebraically closed field $\Bbbk$ of ch...

conjecture | open | MathDB 页面 | 原文线索

V2-010726 | MATHDB-373619

Quadratic generation conjecture for join-meet binomial algebras of simple planar distributive lattices

摘录 / EXCERPT: Quadratic generation conjecture. The defining ideal $I_{{\mathcal R}_K(L)}$ is generated by quadrics if and only if at most one interval is $D_{2^{2}\cdot 3^{2}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-010727 | MATHDB-373662

Sign-coherence and normalized c-vectors conjecture for real rank-3 exchange matrices

摘录 / EXCERPT: Sign-coherence and normalization conjecture. (a) For every ${\bf w}\in\mathcal{T}$, all C-patterns ${\bf C}(B^{\bf w})$ and ${\bf C}((B^{\bf w})^{\top})$ are sign-coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-010728 | MATHDB-373698

The principal ideal theorem dimension conjecture for differential domains

摘录 / EXCERPT: Let (K,∂) be a differential field of characteristic zero. Let A be a differentially finitely generated differential K-domain, let $f\in A$ be a non-unit, and let P b...

conjecture | open | MathDB 页面 | 原文线索

V2-010729 | MATHDB-373711

Square-root error conjecture for the counting norm-trace problem in finite semisimple algebras

摘录 / EXCERPT: Let B be a semisimple algebra of dimension n over $\q$. For $a\in \q^*$, let $x\in B^*$ be regular and let $r\geq 2$ be an integer. Writing ... so that...

conjecture | open | MathDB 页面 | 原文线索

V2-010730 | MATHDB-373731

Cohen-Macaulayness and type of the symmetric algebra of the inversion-factor ideal

摘录 / EXCERPT: Let R be the polynomial ring in the variables ${\bf x}$, and let $D_1,\ldots,D_d$ be the inversion factors associated to the basic representatives of a general linear...

conjecture | open | MathDB 页面 | 原文线索

V2-010731 | MATHDB-373738

Dimension formula for tied Party-Hecke algebras

摘录 / EXCERPT: Let $T\mathcal{P}_n$ be the tied party monoid, and let $\mathcal{EP}_n(p,q)$ be the tied Party-Hecke algebra generated by $J_1,\ldots,J_{n-1}$, $F_1,\ldots,F_{n-1}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010732 | MATHDB-373748

Makhlouf-Silvestrov conjecture for infinitesimal deformations of $\mathfrak{sl}_2$

摘录 / EXCERPT: Makhlouf--Silvestrov conjecture. The resulting Hom--Lie algebra $(V,[\cdot,\cdot]_t,\alpha_t)$ is in fact a Lie algebra: the bracket $[\cdot,\cdot]_t$ satisfies the ordinary Jacobi...

conjecture | open | MathDB 页面 | 原文线索

V2-010733 | MATHDB-373765

The reductive local structure conjecture for algebraic stacks

摘录 / EXCERPT: Reductive local structure conjecture. Under these hypotheses, such an étale morphism exists.

conjecture | open | MathDB 页面 | 原文线索

V2-010734 | MATHDB-373851

The coprimality conjecture for reductions of two elliptic curves

摘录 / EXCERPT: Let E_1 and E_2 be non-CM elliptic curves over $\mathbb{Q}$. For $x\geq 1$, define ... where N_{E_i} denotes the conductor of E_i for $i=1,2$. Coprimality conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-010735 | MATHDB-373867

Copositivity conjecture for promotion maps

摘录 / EXCERPT: Let $\widetilde{\varphi}:\operatorname{Gr}(4,n)\to\operatorname{Gr}(4,n')$ be a promotion map. Promotion-map copositivity conjecture. The map $\widetilde{\varphi}$ is copositive, n...

conjecture | open | MathDB 页面 | 原文线索

V2-010736 | MATHDB-373868

Rational parametrization conjecture for Landau-map fibers

摘录 / EXCERPT: Fix a rational Landau diagram $\mathcal{L}_1$ in the setting of the factorization theorem, and let ψ_1 be its Landau map with external data $\mathbf{M}^{(1)}$. Rational parame...

conjecture | open | MathDB 页面 | 原文线索

V2-010737 | MATHDB-373869

Cluster factorization conjecture for outerplanar Landau diagrams

摘录 / EXCERPT: Let $\mathcal{L}=G_u\cup H_u^{\triangle}$ be a planar Landau diagram, where G is an outerplanar graph, and let $\Delta_{\mathcal{L}}$ denote its LS discriminant. Cluster Factoriz...

conjecture | open | MathDB 页面 | 原文线索

V2-010738 | MATHDB-373885

Coble's restriction-map problem for automorphisms of Coble surfaces

摘录 / EXCERPT: Let X be a Coble surface with Coble curve $C\simeq\mathbb{P}^1$, and let ... be the restriction homomorphism. Coble's restriction-map problem. Is $\ker(\rho)$ trivial for a general...

open_problem | open | MathDB 页面 | 原文线索

V2-010739 | MATHDB-373887

Algebraic values of the continued-fraction set of multiple zeta-star values

摘录 / EXCERPT: Algebraicity conjecture.

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V2-010740 | MATHDB-373888

Integral algebraic values in the sets of multiple zeta-star values

摘录 / EXCERPT: Integral algebraicity conjecture. For every integer $q\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-010741 | MATHDB-373931

Equivariant positivity conjecture for Chern-Mather classes of symmetric-subgroup orbit closures

摘录 / EXCERPT: Let G be a reductive algebraic group with involution θ, let $K=G^{\theta}_0$ be the identity component of the fixed-point group, and let $B\supset H$ be a θ-stable...

conjecture | open | MathDB 页面 | 原文线索

V2-010742 | MATHDB-373948

Type C fourth-order ODE conjecture for superelliptic polynomial families

摘录 / EXCERPT: Let $r\geq 3$ and let $j_0\in\{-r+1,\ldots,-2\}$. For each such initial condition, let $\{P_{j_0,n}(c)\}$ be the corresponding polynomial family. Type C ODE conjecture. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-010743 | MATHDB-373971

Ragunathan-Van Tuyl conjecture on the Weak Lefschetz Property of van der Waerden rings

摘录 / EXCERPT: Ragunathan-Van Tuyl's conjecture. Fix an integer $k\geq 3$. If k is odd, then there exists an m such that $A(n,k)$ fails to have the WLP for every $n\geq m$. If k is even, th...

conjecture | open | MathDB 页面 | 原文线索

V2-010744 | MATHDB-374059

Vemulapalli-Vishwanath conjecture on primitive cover scrollar invariants

摘录 / EXCERPT: Let d be a degree and let $\vec{e}=(e_1,\ldots,e_{d-1})$ be a $(d-1)$-tuple of scrollar invariants. A cover is primitive if it does not factor through a nontrivial proper subcove...

conjecture | open | MathDB 页面 | 原文线索

V2-010745 | MATHDB-374076

Greenberg's semisimplicity conjecture for the cyclotomic Selmer module

摘录 / EXCERPT: Let A be as in the cyclotomic Iwasawa setting, let $X(A/F_\mathrm{cyc})$ have a pseudo-isomorphism with elementary factors including...

conjecture | open | MathDB 页面 | 原文线索

V2-010746 | MATHDB-374095

Kähler package conjecture for corrected perverse objects of conifold degenerations

摘录 / EXCERPT: Let $\pi:\mathcal X\to\Delta$ be a projective conifold degeneration, and let $\mathcal P$ be the corrected perverse object constructed from nearby and vanishing cycles. Suppose the...

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V2-010747 | MATHDB-374137

Derived Interleaving Metric conjecture

摘录 / EXCERPT: Let P be a poset and let M and N be persistence modules over P. Let $d_I(M,N)$ denote their interleaving distance, and let $d_{DG}(M,N)$ denote the derived convolution metr...

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V2-010748 | MATHDB-374201

The quantized Lagrangian correspondence conjecture for the de Rham geometric Langlands correspondence

摘录 / EXCERPT: Let $\mathbf{L_H(d)}$ be the Fourier transform associated with the Lagrangian correspondence $\mathbb L_H(d)$, and let $\mathbf{q\text{-}L_H(d)}$ be its quantization to a functor .....

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V2-010749 | MATHDB-374291

The Jacobian-colon-integral-closure conjecture for isolated hypersurface singularities

摘录 / EXCERPT: Let K be a field of arbitrary characteristic and let ... Let $f\in R$ define an isolated hypersurface singularity, let $j(f)$ be its Jacobian ideal, and write $j(f):f$ for the idea...

conjecture | open | MathDB 页面 | 原文线索

V2-010750 | MATHDB-374292

Hassanzadeh-Nejad-Simis conjecture on the Jacobian colon ideal

摘录 / EXCERPT: Let k be a field of characteristic zero and let ... Let $f\in R$ define an isolated hypersurface singularity, with Jacobian ideal $j(f)$ and Tjurina ideal $tj(f)$. Hassanzadeh-Neja...

conjecture | open | MathDB 页面 | 原文线索

V2-010751 | MATHDB-374330

Klein-Elias-Hogancamp conjecture for Soergel bimodules

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let $\mathcal{H}$ be a two-sided cell, and let d be a Duflo involution in $\mathcal{H}$ with $\mathtt{a}(d)=a$. Let $\widetilde{\mathrm{C}}_d$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-010752 | MATHDB-374374

The negative-curve characterization of instability for line bundles

摘录 / EXCERPT: Let X be a smooth complex projective surface, let $B,omega\bigin\operatorname{NS}(X)\otimes\mathbb{R}$ with $omega$ ample, and let...

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V2-010753 | MATHDB-374383

Sharp growth-factor bound for accretive-dissipative matrices

摘录 / EXCERPT: Let $A=B+iC\in\mathcal A_n$, where B and C are the real and imaginary parts of an accretive-dissipative matrix, and suppose that ... Here κ denotes the condition number an...

conjecture | open | MathDB 页面 | 原文线索

V2-010754 | MATHDB-374387

Hanusa-Nath positivity conjecture for affine type C_3^{(1)} core abaci

摘录 / EXCERPT: Let $\mathbb{N}$ denote the nonnegative integers, and let $\mathcal{C}_0$ be the set of core abaci of affine type $C_3^{(1)}$ with charge 0. The preceding proposition identifies...

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V2-010755 | MATHDB-374396

Ultra-Euclidean functions are preserved by tilde transformation

摘录 / EXCERPT: Let R be a Euclidean domain, let $f:R\setminus\{0\}\to\mathbb{Z}_0^+$ be an ultra-Euclidean function, and let $\widetilde{f}$ denote the associated transformed function. Preserva...

conjecture | open | MathDB 页面 | 原文线索

V2-010756 | MATHDB-374397

Classification of domains admitting an ultra-Euclidean function

摘录 / EXCERPT: Let R be a Euclidean domain, and suppose that R admits an ultra-Euclidean function. Ultra-Euclidean classification conjecture. Then either R is a field or there is a field...

conjecture | open | MathDB 页面 | 原文线索

V2-010757 | MATHDB-374398

Kim-Yoo's representation-theoretic conjecture for two-row genomic Schur functions

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\lambda_2)\vdash n$ with $\ell(\lambda)=2$, and set $l_\lambda=\max\{\lambda_1,\lambda_2+1\}$. For $l_\lambda\le m\le n$, let $\mathsf{Par}(\lambda;m)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-010758 | MATHDB-374400

Nakai's conjecture for finitely generated algebras

摘录 / EXCERPT: Nakai's conjecture. If $Der_k^n(A)$ is generated by $Der_k^1(A)$ for every $n\geq 2$, then A is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-010759 | MATHDB-374401

Weak Nakai conjecture for finitely generated algebras

摘录 / EXCERPT: Weak Nakai conjecture. If $der_k^q(A)=Der_k^q(A)$ for each integer $q\geq 1$, then A is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-010760 | MATHDB-374447

The characteristic-three classification conjecture for simple transposed Poisson algebras

摘录 / EXCERPT: Characteristic-three classification conjecture. Every simple finite-dimensional non-trivial transposed Poisson algebra over $\mathbb{F}$ is isomorphic to $\mathcal{W}_n(q)$ for som...

conjecture | open | MathDB 页面 | 原文线索

V2-010761 | MATHDB-374472

Determinantal presentation conjecture for the transfer elimination ideal

摘录 / EXCERPT: Let $n=qp$ and let $I=\langle f_0,f_1,\ldots,f_{p-1}\rangle\cap R^{S_n}$ be the elimination ideal, where the f_i are the polynomials obtained by collecting the coefficients of th...

conjecture | open | MathDB 页面 | 原文线索

V2-010762 | MATHDB-374526

The Poisson n-Lie algebra construction conjecture

摘录 / EXCERPT: Poisson n-Lie algebra construction conjecture. Then $(\mathcal A, \cdot, [-,\dots,-])$ forms a Poisson n-Lie algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-010763 | MATHDB-374555

Properness conjecture for the completed relative Ginzburg dg algebra

摘录 / EXCERPT: Let $\underline{w}_0$ be a reduced expression of the longest element and let $[a,b]$ be a finite integer interval. Let $\widehat{\mathbf{\Gamma}}^{[a,b]}(\underline{w}_0)$ be the c...

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V2-010764 | MATHDB-374556

Properness conjecture for relative Ginzburg dg algebras

摘录 / EXCERPT: Let $\underline{w}_0$ be a reduced expression of the longest element and let $[a,b]$ be a finite integer interval. Let $\mathbf{\Gamma}^{[a,b]}(\underline{w}_0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010765 | MATHDB-374584

Adiprasito-Kazhdan-Ziegler stability conjecture for partition rank

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $d\geq 2$ be an integer, and let ... Here $\operatorname{PR}_{\mathbb{K}}(f)$ denotes the partition rank of f, and $\overline{\mathbb{K}}$ is an al...

conjecture | open | MathDB 页面 | 原文线索

V2-010766 | MATHDB-374737

The Frobenius-Witt cotangent complex over the field with one element

摘录 / EXCERPT: Conjecture on the cotangent complex over the field of one element. There exists a functor

conjecture | open | MathDB 页面 | 原文线索

V2-010767 | MATHDB-374747

Lang-Rohrlich conjecture for algebraic values of the Gamma function

摘录 / EXCERPT: Lang-Rohrlich conjecture. The value $\Gamma(z)$ is algebraic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-010768 | MATHDB-374752

Ananyan-Hochster's quadratic bound conjecture for projective dimension

摘录 / EXCERPT: Let S be a polynomial ring over a field, and let $I\subseteq S$ be an ideal generated by n quadrics and of height h. Ananyan-Hochster's quadratic bound conjecture. One has .....

conjecture | open | MathDB 页面 | 原文线索

V2-010769 | MATHDB-374771

The weak Serre weight conjecture for globally defined weights

摘录 / EXCERPT: Let $G(k)$ be the finite reductive group in the local setup, let $\overline{\rho}_0$ be the local residual representation, and let $\overline{r}$ be its suitable globalization. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-010770 | MATHDB-374790

The SLP conjecture for Hartshorne-Rao modules of general unions of lines

摘录 / EXCERPT: Let $C\subset\mathbb P^3$ be the union of r general lines. The strong Lefschetz property conjecture. Then $M(C)$ has SLP. Here $M(C)$ denotes the Hartshorne-Rao module of C, an...

conjecture | open | MathDB 页面 | 原文线索

V2-010771 | MATHDB-374849

Generic quadratic arboreal surjectivity conjecture over the rational function field

摘录 / EXCERPT: Let $K=\mathbb{F}_p(t)$, let T be the complete infinite rooted binary tree, and let $f_t=x^2+t\in K[x]$. Write $G_{f_t}(0)$ for the arboreal Galois group associated with the orb...

conjecture | open | MathDB 页面 | 原文线索

V2-010772 | MATHDB-374859

The 45312-avoidance conjecture for reversal factorizations of Kazhdan-Lusztig basis elements

摘录 / EXCERPT: 45312-avoidance conjecture. If $w \in \mathfrak{S}_n$ does not avoid the pattern 45312, then $\widetilde{C}_{w}(q)$ has no reversal factorization.

conjecture | open | MathDB 页面 | 原文线索

V2-010773 | MATHDB-374888

The p-stability criterion for strongly F-regular rings

摘录 / EXCERPT: The p-stability criterion. The system $\mathfrak{b}_\bullet$ is p-stable if

conjecture | open | MathDB 页面 | 原文线索

V2-010774 | MATHDB-374910

The Stratified Strong Maximal Rank Conjecture for quadrics of bounded rank

摘录 / EXCERPT: Fix positive integers g,r,d with $r\geq 3$ and $\rho(g,r,d)\geq 0$, and let C be a general curve of genus g. For a linear series $\ell=(L,V)\in G^r_d(C)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-010775 | MATHDB-374918

The gamma-cap conjecture for Poisson fields of two variables

摘录 / EXCERPT: Let $K=K\{\{\mathbf{f}\}\}$ be a Poisson field, let ${}^1\Gamma_0(K)$ be the 1-valuation gamma-cap, and let $\operatorname{fht}(K)$ denote the flag height. Gamma-cap conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-010776 | MATHDB-374951

Fujimoto's polynomial general-position conjecture for even integers

摘录 / EXCERPT: Fujimoto conjecture. These polynomials are in general position: any m polynomials chosen from them are linearly independent over $\mathbb{C}$. This conjecture is the algebraic in...

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V2-010777 | MATHDB-374990

Frobenius-parabolic conjecture for simple Lie algebras

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra. A Lie subalgebra is Frobenius when its index is zero; let ${\EuScript M}_{\sf Fr}$ denote the corresponding maximal dimension invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-010778 | MATHDB-374991

Centralizer-centre conjecture for strange nilpotent orbits

摘录 / EXCERPT: Let G be an algebraic group with Lie algebra ${\mathfrak g}$, let ${\mathcal N}$ be the nilpotent cone, and let $e\in{\mathcal N}$. Write ${\mathfrak g}^e$ for the centraliser of...

conjecture | open | MathDB 页面 | 原文线索

V2-010779 | MATHDB-374993

Strange-orbit conjecture for distinguished partitions in ${\mathfrak{sl}}_n$

摘录 / EXCERPT: For every $n\geqslant3$, let ${\mathcal O}(\boldsymbol{\lambda})\subset{\mathfrak{sl}}_n$ be the unique nilpotent orbit with ... where ${\mathfrak b}$ is a Borel subalgebra and...

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V2-010780 | MATHDB-375013

Conjectural Dickson-module generation for excellent polynomials

摘录 / EXCERPT: Fix $m\geq 0$. Let $S=\mathbb{F}_q[x_1,\ldots,x_n]$, let $A_m^{(n)}$ be the set of $(n+1,m)$-excellent polynomials, meaning polynomials f for which $\delta_{n+1}(f)$ is a polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-010781 | MATHDB-375020

Quadratic growth conjecture for the Heisenberg algebra

摘录 / EXCERPT: Quadratic growth conjecture.

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V2-010782 | MATHDB-375031

The nested 2-restricted Specht classification conjecture

摘录 / EXCERPT: Let I be the index set of the affine type ${\tt A}_1^{(1)}$, let $\theta\in\mathbb{Z}_{\geq0}I$, and let $\mathcal{NP}_{\mathrm{2res}}(\theta)$ be the set of skew diagrams...

conjecture | open | MathDB 页面 | 原文线索

V2-010783 | MATHDB-375032

The affine type A Kleshchev skew-Specht classification conjecture

摘录 / EXCERPT: Let $e\geq2$, let $i\in\mathbb{Z}_e$, and let $R_\theta({\tt A}_{e-1}^{(1)})$ be the KLR algebra for a root-lattice element θ. Let $\mathcal{MP}_{\mathrm K}^{(i,i)}(\theta)$...

conjecture | open | MathDB 页面 | 原文线索

V2-010784 | MATHDB-375043

The general-field moduli conjecture for intrinsic torsion subgroups

摘录 / EXCERPT: Let k be any field and let E be an elliptic curve over k with ... Write $E(k)_\mathrm{tors}^\mathrm{is}$ for the intrinsic subgroup, and let H be the subgroup associated with...

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V2-010785 | MATHDB-375052

The conjecture identifying the band basis with the common triangular basis

摘录 / EXCERPT: Let the skein algebra of a marked surface have frozen variables, and consider its localization at those frozen variables. The skein algebra carries a (quantized) band basis and a c...

conjecture | open | MathDB 页面 | 原文线索

V2-010786 | MATHDB-375088

Göksel's Markov group containment conjecture

摘录 / EXCERPT: Let f be the post-critically finite quadratic polynomial under consideration, and let $G_n(f)$ and $M_n(f)$ denote, respectively, its level-n Galois group and even Markov group...

conjecture | open | MathDB 页面 | 原文线索

V2-010787 | MATHDB-375108

Separating/nonseparating algebraicity principle for exterior Cauchy transforms

摘录 / EXCERPT: Separating/nonseparating algebraicity principle. If γ is separating, then the exterior Cauchy transform is algebraic. If γ is nonseparating, then the exterior Cauchy...

conjecture | open | MathDB 页面 | 原文线索

V2-010788 | MATHDB-375110

Generic non-algebraicity conjecture for smooth Weierstrass cubics

摘录 / EXCERPT: Smooth Weierstrass cubic period-test conjecture. For a generic smooth real Weierstrass cubic, the compact oval is nonseparating and the exterior Cauchy transform of the domain boun...

conjecture | open | MathDB 页面 | 原文线索

V2-010789 | MATHDB-375134

Primality conjecture for ideals defining two-row Springer fiber components

摘录 / EXCERPT: Let n be a positive integer, let $\boldsymbol{\rho}$ be a standard Young tableau of shape $(n-\rho,\rho)$, and let $\mathcal{I}_{\boldsymbol{\rho}}$ be the associated ideal. Prim...

conjecture | open | MathDB 页面 | 原文线索

V2-010790 | MATHDB-375142

Küronya-Pintye conjecture on regularity and integral closure

摘录 / EXCERPT: Küronya-Pintye conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-010791 | MATHDB-375145

Arzhantsev's conjecture that every semisimple Lie algebra is an iUA-Lie ring

摘录 / EXCERPT: A Lie algebra is semisimple if it has no nonzero solvable ideals, and an iUA-Lie ring is a Lie ring for which every commutator-preserving injection into another Lie ring is additiv...

conjecture | open | MathDB 页面 | 原文线索

V2-010792 | MATHDB-375151

Cantarini-Kac classification conjecture for simple Jordan superalgebras

摘录 / EXCERPT: Let F be a field of zero characteristic, and let J be a simple Jordan superalgebra over F with nonzero odd part. The possible superalgebras include $H(A,*)$, where $(A,*)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-010793 | MATHDB-375179

Uniqueness conjecture for tetrahedral genus-zero dessins

摘录 / EXCERPT: Uniqueness conjecture. For each such passport, its realization as a dessin d'enfant is unique.

conjecture | open | MathDB 页面 | 原文线索

V2-010794 | MATHDB-375180

Peterson-Thom conjecture on unique maximal amenable extensions

摘录 / EXCERPT: Peterson-Thom conjecture. There exists a unique maximal amenable von Neumann subalgebra $\mathcal{P}$ of $\mathcal{L}(\mathbb{F}_r)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010795 | MATHDB-375181

Peterson-Thom conjecture for hyperbolic groups

摘录 / EXCERPT: Peterson-Thom conjecture for hyperbolic groups. If $\mathcal{A}\capmathcal{B}$ is diffuse, then the von Neumann algebra they generate is amenable:

conjecture | open | MathDB 页面 | 原文线索

V2-010796 | MATHDB-375217

Safronov's root-of-unity skein conjecture

摘录 / EXCERPT: Safronov's root-of-unity skein conjecture. There is a line bundle $\mathcal{L}_q$ on $\mathrm{Loc}_{G^\vee}(M)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010797 | MATHDB-375297

Double-duality and exact-sequence conjecture for triangular t-modules

摘录 / EXCERPT: Double-duality and exact-sequence conjecture. For every triangular $\mathbf{t}$-module Υ,

conjecture | open | MathDB 页面 | 原文线索

V2-010798 | MATHDB-375298

Cartier--Nishi duality conjecture for two-dimensional triangular t-modules

摘录 / EXCERPT: Cartier--Nishi duality conjecture. The Cartier--Nishi theorem holds for Υ.

conjecture | open | MathDB 页面 | 原文线索

V2-010799 | MATHDB-375310

Eisenbud-Huneke-Ulrich conjecture on powers of ideals with linear syzygies

摘录 / EXCERPT: Eisenbud-Huneke-Ulrich conjecture. If I is $\mathfrak{m}$-primary and the resolution of I is linear for p steps, where $p\geq 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-010800 | MATHDB-375323

Haglund-Remmel-Wilson valley Delta conjecture

摘录 / EXCERPT: Let Λ be the algebra of symmetric functions over $\mathbb{Q}(q,t)$, let $\{\widetilde H_\mu\}$ be the modified Macdonald basis, and define Δ'_f by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-010801 | MATHDB-375409

Burity-Simis-Tohăneanu minimal-reduction conjecture for Jacobian ideals

摘录 / EXCERPT: Let k be a field, let $R=k[x_1,\ldots,x_n]$, and let $\mathcal A$ be a central hyperplane arrangement with defining linear forms whose product is f. Let $I\subseteq R$ be the i...

conjecture | open | MathDB 页面 | 原文线索

V2-010802 | MATHDB-375411

Rationality of isotypic Hilbert series for nonzero symmetric ideals

摘录 / EXCERPT: Let $K[X]$ be the polynomial ring in countably many variables, with the symmetric group acting by permuting variables. A homogeneous ideal $I\subset K[X]$ is symmetric if it is inv...

conjecture | open | MathDB 页面 | 原文线索

V2-010803 | MATHDB-375426

The division-ring embedding conjecture for torsion-free group algebras

摘录 / EXCERPT: Let G be a torsion-free group and let $\mathbb{K}$ be a field. The division-ring embedding conjecture. The group algebra $\mathbb{K}[G]$ embeds into a division ring. This is a st...

conjecture | open | MathDB 页面 | 原文线索

V2-010804 | MATHDB-375454

Parity conjecture for the unique invariant theta characteristic on modular curves

摘录 / EXCERPT: Parity conjecture. The unique invariant theta characteristic on $X(p)$ is always even.

conjecture | open | MathDB 页面 | 原文线索

V2-010805 | MATHDB-375483

Greenberg's conjecture for totally real fields

摘录 / EXCERPT: Greenberg's conjecture. The module $X(k^\mathrm{c}_\infty)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-010806 | MATHDB-375485

Greenberg's generalized conjecture for maximal multiple extensions

摘录 / EXCERPT: Greenberg's generalized conjecture. The module $X(\widetilde{k})$ is pseudo-null as a $\mathbb{Z}_p\llbracket\operatorname{Gal}(\widetilde{k}/k)\rrbracket$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-010807 | MATHDB-375486

Ozaki's non-freeness conjecture for unramified pro-p Galois groups

摘录 / EXCERPT: Ozaki's non-freeness conjecture. The group $\mathcal{G}(k^\mathrm{c}_\infty)$ is not a non-abelian free pro-p-group.

conjecture | open | MathDB 页面 | 原文线索

V2-010808 | MATHDB-375488

Kollár-Villarino conjecture on finite Pythagoras numbers and even powers

摘录 / EXCERPT: Kollár-Villarino conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010809 | MATHDB-375497

The proper containment conjecture for identities of principal subalgebras

摘录 / EXCERPT: Proper containment conjecture. The ideal of polynomial identities of the path algebra is properly contained in that of its principal subalgebra:

conjecture | open | MathDB 页面 | 原文线索

V2-010810 | MATHDB-375504

Stabilizing solution conjecture for the line-graph structural space

摘录 / EXCERPT: Let H be the stabilizing positive semidefinite solution of the reduced discrete algebraic Riccati equation under Assumption, and let ${\mathcal V}_{\mathcal L}$ denote the struct...

conjecture | open | MathDB 页面 | 原文线索

V2-010811 | MATHDB-375568

Mazur's torsionness conjecture for the elliptic-curve Selmer group

摘录 / EXCERPT: Mazur's conjecture. The Λ-module ${\rm Sel}_{E}(K_\infty)[p^\infty]^{\lor}$ is a finitely generated torsion Λ-module.

conjecture | open | MathDB 页面 | 原文线索

V2-010812 | MATHDB-375638

Khare-Moon finiteness conjecture for mod p Galois representations over number fields

摘录 / EXCERPT: Let K be a number field, let n be a positive integer, let $\mathfrak{R}$ be a nonzero ideal of the ring of integers of K, and let G_K be the absolute Galois group of K. F...

conjecture | open | MathDB 页面 | 原文线索

V2-010813 | MATHDB-375639

Moon-Taguchi finiteness conjecture for geometric mod p representations over function fields

摘录 / EXCERPT: Let K be a global function field of characteristic different from p, with absolute Galois group G_K. Let n be a positive integer and let $\mathfrak{R}$ be an effective divi...

conjecture | open | MathDB 页面 | 原文线索

V2-010814 | MATHDB-375672

The proposed superspace model for colored faces of wreath products

摘录 / EXCERPT: Proposed colored-face model. The ideal $SI'_{n,r}$ is stable under ${\mathbb{Z}}_r\wr{\mathfrak S}_n$, and there is a module isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-010815 | MATHDB-375700

The proposed Rogers-Ramanujan type character identity for type $A^{(2)}_{6n+3}$

摘录 / EXCERPT: Rogers-Ramanujan type character identity. For $n\geq 1$ and $0\leq b\leq 2n$, the identity

conjecture | open | MathDB 页面 | 原文线索

V2-010816 | MATHDB-375708

Positivity conjectures for the canonical basis of the pseudo-centralizer

摘录 / EXCERPT: Positivity conjectures. The Laurent polynomials $Q_{x,w}\in\mathbb{Z}[v,v^{-1}]$ have non-negative coefficients for all $x\in W_{\mathrm{aff}}$ and $w\in W_{\mathrm{aff}}^+$, and t...

conjecture | open | MathDB 页面 | 原文线索

V2-010817 | MATHDB-375719

Orthogonal splitting type dimension conjecture for general covers

摘录 / EXCERPT: Orthogonal splitting type dimension conjecture. Its dimension is

conjecture | open | MathDB 页面 | 原文线索

V2-010818 | MATHDB-375804

Geiss-Hernandez-Leclerc's cluster categorification conjecture for shifted category O

摘录 / EXCERPT: Let $\mathcal{O}_{sh}^{\mathbb Z}$ be the integral finite-length subcategory of shifted category $\mathcal{O}_{sh}$, and let $\mathscr A$ be the cluster algebra associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-010819 | MATHDB-375805

Frenkel-Hernandez-Geiss-Hernandez-Leclerc conjecture on Q-variables and simple modules

摘录 / EXCERPT: Let $(i,a)\in I\times_2\mathbb Z$, let $w\in W$, and let $Q_{w\varpi_i^{\vee},a}$ be the corresponding Q-variable. Let χ_{ℓ} denote the ℓ-character and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010820 | MATHDB-375811

Generation conjecture for the two-segment piecewise-linear path ideal

摘录 / EXCERPT: Let $T(\mathbb R^d)$ be the tensor algebra equipped with the left and right halfshuffle products, and let $\mathcal{I}(\mathbf{PL}^d_{\leq 2})$ denote the ideal associated with piec...

conjecture | open | MathDB 页面 | 原文线索

V2-010821 | MATHDB-375844

The equivalence of zero Jordan product determined and zero product determined algebras

摘录 / EXCERPT: Equivalence conjecture. A is zJpd if and only if it is zpd.

conjecture | open | MathDB 页面 | 原文线索

V2-010822 | MATHDB-375862

Equivalence between twisted bimodules and loosely discrete opfibrations

摘录 / EXCERPT: Let $\lMod$ be the 2-category of loosely discrete opfibrations and $\TwMod$ the 2-category of twisted bimodules, with the 2-functor $\twsection{-}: \lMod \to \TwMod$ given by t...

conjecture | open | MathDB 页面 | 原文线索

V2-010823 | MATHDB-375869

Bounded-support conjecture for p-adic algebraic Hahn series

摘录 / EXCERPT: Bounded-support conjecture. The support $\operatorname{Supp}(f)$ must be a finite set.

conjecture | open | MathDB 页面 | 原文线索

V2-010824 | MATHDB-375919

The optimistic semisimple Serre-weight conjecture

摘录 / EXCERPT: Let $\text{undefined}:\text{undefined}\to{}^LG(\text{undefined})$ be a semisimple L-parameter. Let $W^{\text{undefined}}(\text{undefined})$ be the set of Serre weights arising fr...

conjecture | open | MathDB 页面 | 原文线索

V2-010825 | MATHDB-375991

Product formula for the (p,q,h)-deformed binomial coefficients

摘录 / EXCERPT: Let m and k be nonnegative integers, let p and q be parameters, and let Z_p be the generator appearing in the (p,q)-deformed generalized Weyl algebra. Write $[i]_{p,q}$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010826 | MATHDB-375996

Clozel's existence conjecture for Galois representations attached to algebraic automorphic representations

摘录 / EXCERPT: Let F be a number field, let $n\geq 1$, and let $G_F=\operatorname{Gal}(\overline{F}/F)$. Let π be a cuspidal, algebraic automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-010827 | MATHDB-376022

Folding conjecture for equivariant foam categories and non-simply laced webs

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simply laced simple Lie algebra and let τ be a diagram automorphism. Let $\mathfrak{g}_\tau$ be the non-simply laced Lie algebra Langlands dual to the...

conjecture | open | MathDB 页面 | 原文线索

V2-010828 | MATHDB-376063

Lincs completion-final INC coalgebra comparison conjecture

摘录 / EXCERPT: Let $\mathsf{T}_{\mathsf{INC}}$ denote the tangent endofunctor on $\mathsf{INC}(\mathcal{C})$, and suppose either preceding existence theorem applies, so that the universal Lincs c...

conjecture | open | MathDB 页面 | 原文线索

V2-010829 | MATHDB-376152

The conjecture that join-meet ideals of nonmodular lattices are not linearly related

摘录 / EXCERPT: Let L be a nonmodular lattice, and let I_L denote its join-meet ideal. An ideal is linearly related when its linear syzygies generate all of its syzygies. Nonmodular-lattice jo...

conjecture | open | MathDB 页面 | 原文线索

V2-010830 | MATHDB-376168

Schoen's conjecture on products of curves dominating very general K3 surfaces

摘录 / EXCERPT: Let S be a very general K3 surface. A dominant rational map from the product of two smooth projective curves to S is a rational map $C\times D\dashrightarrow S$ whose image is...

conjecture | open | MathDB 页面 | 原文线索

V2-010831 | MATHDB-376173

Deré's real definability conjecture for complex Lie algebras

摘录 / EXCERPT: Deré's conjecture. If L is isomorphic to its complex conjugate, then L is defined over $\mathbb{R}$; equivalently, there exists a real Lie algebra $L_{\mathbb{R}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010832 | MATHDB-376180

Eisenbud-Green-Harris conjecture for homogeneous ideals containing regular sequences

摘录 / EXCERPT: Eisenbud-Green-Harris conjecture. The ideal I has the same Hilbert function as an ideal containing $x_1^{a_1},\dots,x_n^{a_n}$. This conjecture predicts that the Hilbert function...

conjecture | open | MathDB 页面 | 原文线索

V2-010833 | MATHDB-376200

Adjamagbo's separable Jacobian conjecture

摘录 / EXCERPT: Adjamagbo's separable Jacobian conjecture. Assume that

conjecture | open | MathDB 页面 | 原文线索

V2-010834 | MATHDB-376243

Wan's product-trace conjecture for finite semisimple algebras

摘录 / EXCERPT: Wan's conjecture. For $a\in\mathbb F_q^{\times}$, regular $x\in B^{\times}$, and integer $r\ge2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-010835 | MATHDB-376259

Integral closedness of ideals with nearly Gorenstein extended Rees algebras

摘录 / EXCERPT: Integral-closedness conjecture. If $\mathcal{R}'(I)$ is nearly Gorenstein, then I is integrally closed.

conjecture | open | MathDB 页面 | 原文线索

V2-010836 | MATHDB-376297

The Leading Terms Ideals Conjecture for rational monomial orders

摘录 / EXCERPT: Let R be a strongly discrete coherent ring. For $n\a0\geq 1$, let $R[X_1,\ldots,X_n]$ be a polynomial ring, and let $\prec$ be a rational monomial order on its monomials. For a f...

conjecture | open | MathDB 页面 | 原文线索

V2-010837 | MATHDB-376313

Mendez-Pinto-Villarreal conjecture for Simis monomial ideals

摘录 / EXCERPT: Let S be a polynomial ring and let I be a monomial ideal in S. An ideal is Simis when its symbolic Rees algebra is generated by its elements of degree one; a monomial ideal h...

conjecture | open | MathDB 页面 | 原文线索

V2-010838 | MATHDB-376326

N-bonacci growth conjecture for iterated Wronskian brackets

摘录 / EXCERPT: Let $d\geqslant 1$ be the dimension of the base $\mathbb{R}^d$, let $k\geqslant 1$ be a differential order, and set ... Let W_d^k denote the Wronskian N-ary bracket on...

conjecture | open | MathDB 页面 | 原文线索

V2-010839 | MATHDB-376338

Lima-Pereira-Nuño-Ballesteros-Orefice-Okamoto-Tomazella Tor-length conjecture

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a regular local ring of dimension n. Let I be an ideal generated by an R-regular sequence of length $k\leq n$, and let J be an ideal generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-010840 | MATHDB-376339

Local Tor-length conjecture for regular sequences

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a regular local ring of dimension n. Let I be an ideal generated by an R-regular sequence of length $k\leq n-1$, and let J be an ideal generated b...

conjecture | open | MathDB 页面 | 原文线索

V2-010841 | MATHDB-376340

Purity conjecture for tensor products of almost strictly pure t-modules

摘录 / EXCERPT: Let ϕ and ψ be two almost strictly pure t-modules over K, and let $\phi\otimes\psi$ denote their tensor product. Choose a $K\{\tau\}$-basis of the corresponding t-m...

conjecture | open | MathDB 页面 | 原文线索

V2-010842 | MATHDB-376356

Transcendence conjecture for twisted special L-values over finite extensions

摘录 / EXCERPT: Let $\mathbf{A}=\mathbb{F}_q[t]$, $A=\mathbb{F}_q[\theta]$, and $K=\operatorname{Frac}(A)=\mathbb{F}_q(\theta)$. Let E be a finite extension of K and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010843 | MATHDB-376371

Depth-one reduction conjecture for the derivative sums in the f_6 family

摘录 / EXCERPT: Depth-one reduction conjecture. Every derivative sum in the f_6 family belongs to the subalgebra of the MZV algebra generated by the single zeta values $\zeta(m)$; no irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-010844 | MATHDB-376439

Conjecture on maximal compact quotients of affine-line fundamental groups

摘录 / EXCERPT: Let $K={\bf F}_q(T)$, let G/K be an isotropic, semisimple, simply-connected algebraic group, and let Z be its center. Let ${\bf A}^{\mathrm{fin}}_K$ denote the finite adeles of...

conjecture | open | MathDB 页面 | 原文线索

V2-010845 | MATHDB-376441

Koszulness conjecture for the residual tetragonal canonical algebra

摘录 / EXCERPT: Let C be a non-hyperelliptic curve with a base-point free tetragonal system T, presented as a complete intersection on a three-dimensional rational normal scroll as in the prec...

conjecture | open | MathDB 页面 | 原文线索

V2-010846 | MATHDB-376442

Pencil decomposition conjecture for residual tetragonal line bundles

摘录 / EXCERPT: Let C be a non-hyperelliptic curve of genus g with a base-point free tetragonal system T, and suppose C has the complete-intersection presentation on a three-dimensional ra...

conjecture | open | MathDB 页面 | 原文线索

V2-010847 | MATHDB-376444

Residue formula conjecture for Verlinde intersection numbers

摘录 / EXCERPT: Let G be a group, let g denote the genus of C, let $\gt$ be the Cartan subalgebra of G, let $\M$ denote the moduli space under consideration, and let P be a Weyl-symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-010848 | MATHDB-376467

The finite-dimensional presentation conjecture for the elliptic Bloch group

摘录 / EXCERPT: Let E be a cubic curve in P^2. For a point $p\in P^2$ and three lines l_1,l_2,l_3 through p, let $A_i:=l_i\cap E$ and define ... Let $\tilde R^*_3(E)\subset I^4_E$ be the sub...

conjecture | open | MathDB 页面 | 原文线索

V2-010849 | MATHDB-376479

Faber's strong conjecture for the compactly supported tautological algebra

摘录 / EXCERPT: Faber's strong conjecture. These pairings are perfect for $k=0,1,2,\dots$, and $\mathcal R_c^\bullet({\mathcal M}_g)$ is a free $\mathcal R^\bullet({\mathcal M}_g)$-module of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-010850 | MATHDB-376484

The existence conjecture for log canonical models

摘录 / EXCERPT: Existence conjecture for log canonical models. The graded ring of $\cal O_S$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-010851 | MATHDB-376579

The borderline Lifshits-tail asymptotic for algebraically decaying single-impurity potentials

摘录 / EXCERPT: Borderline Lifshits-tail conjecture. As $E\downarrow0$,

conjecture | open | MathDB 页面 | 原文线索

V2-010852 | MATHDB-376661

Approximation of smooth periodic potentials by algebraic-geometrical potentials

摘录 / EXCERPT: Approximation property. Any smooth potential can be approximated by algebraic-geometrical ones with the same periods.

open_problem | open | MathDB 页面 | 原文线索

V2-010853 | MATHDB-376703

Cluster fan polytopality conjecture

摘录 / EXCERPT: Cluster fan polytopality conjecture. The simplicial fan generated by the clusters is the normal fan of a simple n-dimensional convex polytope $P(\Phi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-010854 | MATHDB-376704

Cluster complex face-lattice conjecture

摘录 / EXCERPT: Cluster complex face-lattice conjecture. This poset is the face lattice of a simple n-dimensional convex polytope $P(\Phi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-010855 | MATHDB-376708

Rationality and pole-order conjecture for higher-genus BCOV functions

摘录 / EXCERPT: Let x and y be the algebraic coordinates used for the two-parameter model, and let dis_0 denote the discriminant component defined by ... For $g\geq 2$, define the propagators...

conjecture | open | MathDB 页面 | 原文线索

V2-010856 | MATHDB-376733

The Borcherds invariant-subalgebra conjecture for orbifolded M-theory in D=1

摘录 / EXCERPT: Let ${\mathfrak{e}}_{10}$ be the hyperbolic Kac-Moody algebra, and let $\mathcal{U}_{2}^{{\mathbbm{Z}}_{n}}$ be the automorphism defining the orbifold action. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010857 | MATHDB-376857

Coincidence of fine gradings and fine group gradings on simple complex Lie algebras

摘录 / EXCERPT: Let L be a simple complex Lie algebra. A grading of L is called fine if it cannot be properly refined by another grading, and a fine group grading is a fine grading whose index...

conjecture | open | MathDB 页面 | 原文线索

V2-010858 | MATHDB-376877

The strong quantization conjecture for Lie normalizers

摘录 / EXCERPT: Let ${\mathfrak b}$ be a finite-dimensional basic algebra, and let ${\mathfrak n}({\mathfrak b})$ be its Lie normalizer in $P({\mathfrak b})$. Strong quantization conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-010859 | MATHDB-376897

The tameness conjecture for homomorphisms of polytopal algebras

摘录 / EXCERPT: Tameness conjecture. Any homomorphism in $\operatorname{Pol}(k)$ is obtained by a sequence of these constructions, starting from the identity mapping $k\to k$. Moreover, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-010860 | MATHDB-376899

Temperley-Lieb relations for derived cap and cup functors

摘录 / EXCERPT: For integers n and k, let $D^b(\mathcal O_{k,n-k})$ be the bounded derived category of the relevant parabolic category, and define functors ... ... by...

conjecture | open | MathDB 页面 | 原文线索

V2-010861 | MATHDB-376901

Temperley-Lieb relations for parabolic cap and cup functors

摘录 / EXCERPT: Parabolic Temperley-Lieb relations conjecture. These functors should satisfy the natural isomorphisms corresponding to the defining relations of the Temperley-Lieb category, with...

conjecture | open | MathDB 页面 | 原文线索

V2-010862 | MATHDB-376914

Kato's weak Tamagawa number conjecture for constructible K-theory elements

摘录 / EXCERPT: Kato's weak Tamagawa number conjecture. There is a subspace $H^{\operatorname{constr}}_{{\cal M}}$ such that $r_{{\cal D}}$ and r_p restricted to it are isomorphisms and H^2_p...

conjecture | open | MathDB 页面 | 原文线索

V2-010863 | MATHDB-376924

The infinitude conjecture for observable non-reductive subgroups

摘录 / EXCERPT: Let G be an algebraic group and $H\subseteq G$ an observable subgroup. Write $W(H)=N_G(H)/H$ for the corresponding quotient of the normalizer. The subgroup H is reductive if it...

conjecture | open | MathDB 页面 | 原文线索

V2-010864 | MATHDB-376938

The modular-complex conjecture for cyclotomic Galois Lie algebras

摘录 / EXCERPT: Let $M^*_{(m)}$ be the modular complex, let $\Gamma_1(m;N)$ be the indicated congruence subgroup, let ${\rm V}_m$ be the representation appearing in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010865 | MATHDB-376961

Deligne's generation conjecture for the Ihara graded Galois Lie algebra

摘录 / EXCERPT: Deligne's conjecture. The $\mathbb Q_\ell$-form

conjecture | open | MathDB 页面 | 原文线索

V2-010866 | MATHDB-376976

The odd-degree annihilator correction conjecture

摘录 / EXCERPT: Odd-degree annihilator correction conjecture. For odd $d>3$,

conjecture | open | MathDB 页面 | 原文线索

V2-010867 | MATHDB-376977

The generic Hilbert series conjecture for ideals in the square-free algebra

摘录 / EXCERPT: Let V_n be an n-dimensional vector space, let $\mathfrak{S}(V_n)$ denote the square-free algebra, and let I_n be a generic ideal with generators of degrees $d_1,\dots,d_r$. W...

conjecture | open | MathDB 页面 | 原文线索

V2-010868 | MATHDB-377015

The dihedral Lie algebra containment conjecture for cyclotomic Galois symmetries

摘录 / EXCERPT: Let N be a positive integer, let $\boldsymbol{}[N]$ denote the group of N-th roots of unity, let ${\rm Gr}{\cal G}_{\bullet \bullet}^{(l)}(\boldsymbol{}[N])$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-010869 | MATHDB-377016

The dihedral Lie algebra equality conjecture for the trivial group

摘录 / EXCERPT: Let ${D}_{\bullet \bullet}:={D}_{\bullet \bullet}(\{e\})$ be the dihedral Lie algebra for the trivial group, let $\xi=\xi_{\{e\}}$ be its embedding, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010870 | MATHDB-377018

The modular-complex description of the graded Galois Lie algebra

摘录 / EXCERPT: Let L_m be the rank-m lattice used to define the modular complex, let $V_m:=L_m\otimes\mathbb{Q}$, let $M^*_{(m)}$ be the dual modular complex, and let $S^{w-m}V_m$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-010871 | MATHDB-377020

The refined dihedral Lie coalgebra description conjecture

摘录 / EXCERPT: Let ${\cal C}(G)$ be the refined dihedral Lie coalgebra for a finite commutative group G, let ${C}_{\bullet}(G)$ be its graded dual, and let ${\rm Gr}^W{\cal G}^{(l)}$ be the wei...

conjecture | open | MathDB 页面 | 原文线索

V2-010872 | MATHDB-377026

Existence of a lifted Markov trace for cubic Hecke algebras

摘录 / EXCERPT: Let $Q(x)=x^3-\alpha x^2-\beta x-1$, and let $H(Q,n)=\mathbb{C}[B_n]/(Q(b_j);\,j=1,\dots,n-1)$ be the cubic Hecke algebra. The invariant $I_{(\alpha,\beta)}$ is defined by a Markov...

conjecture | open | MathDB 页面 | 原文线索

V2-010873 | MATHDB-377051

The well-defined operation conjecture for unimodular row classes

摘录 / EXCERPT: Operation conjecture. The operation $*$ is well defined and gives the expected group operation on $\operatorname{Um}_{2,n}(A)/E_n(A)$ for $n\geq4$ and $d\leq2n-6$.

conjecture | open | MathDB 页面 | 原文线索

V2-010874 | MATHDB-377085

Deligne-Ihara freeness conjecture for the stable derivation algebra

摘录 / EXCERPT: Let $\mathfrak D_\bullet=\bigoplus_{w>2}\mathfrak D_w$ be the stable derivation algebra, a filtered graded Lie algebra over $\mathbb Q$. The conjecture concerns its generators in t...

conjecture | open | MathDB 页面 | 原文线索

V2-010875 | MATHDB-377088

Algebraic-relations conjecture for multiple zeta values and the Drinfel'd associator

摘录 / EXCERPT: Let the multiple zeta values be the coefficients associated with the Drinfel'd associator, and let Property II refer to the listed relations of that associator, denoted in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-010876 | MATHDB-377090

Chekanov's augmentation-independence conjecture for first-order Poincaré-Chekanov polynomials

摘录 / EXCERPT: Let (A,∂) be the differential graded algebra of a Legendrian knot, let ε be an augmentation of A, and let the first-order Poincaré-Chekanov polynomial be the...

conjecture | open | MathDB 页面 | 原文线索

V2-010877 | MATHDB-377139

Goncharov's algebraic structure conjecture for multiple zeta values

摘录 / EXCERPT: Let ${\cal Z}$ be the commutative algebra over $\mathbb Q$ spanned by multiple zeta values, and let ${\cal Z}_k$ denote its weight-k subspace. Let ${\cal F}(3,5,...)_{\bullet}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-010878 | MATHDB-377140

Goncharov's Hodge multiple-zeta Hopf algebra conjecture

摘录 / EXCERPT: Let ${\cal Z}^{\cal H}_{\bullet}$ be the graded Hopf algebra of framed Hodge-Tate structures generated by the Hodge multiple-zeta structures, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010879 | MATHDB-377141

Goncharov's cyclotomic period conjecture

摘录 / EXCERPT: Let N be a positive integer. Let ${\cal Z}^{\cal H}_{\bullet}(\mu_N)$ be the commutative graded Hopf algebra generated by framed Hodge-Tate structures associated with multiple po...

conjecture | open | MathDB 页面 | 原文线索

V2-010880 | MATHDB-377160

Symbolic-power intersection conjecture SP-2

摘录 / EXCERPT: SP-2 conjecture. The intersection of symbolic powers satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-010881 | MATHDB-377172

Sturmfels's conjecture on exceptional parameters for A-hypergeometric systems

摘录 / EXCERPT: Sturmfels's conjecture. The exceptional set is empty if and only if I_A is Cohen-Macaulay. For Cohen-Macaulay I_A, the rank equals $\operatorname{vol}(A)$ for every parameter,...

conjecture | open | MathDB 页面 | 原文线索

V2-010882 | MATHDB-377223

Tempered solution ind-sheaves of holonomic D-modules are real-constructible

摘录 / EXCERPT: Real-constructibility conjecture. The ind-sheaf $Sol^t(\mathcal{M})$ belongs to $D^b_{\mathbb{R}\text{-c}}(\mathrm{I}[\mathbb{C}_X])$. This is presented as a consequence of the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-010883 | MATHDB-377224

Regularity conjecture for tempered solution ind-sheaves

摘录 / EXCERPT: Regularity conjecture. The module $\mathcal{M}$ is regular holonomic if and only if $Sol^t(\mathcal{M})$ is regular. The text notes that the only-if direction follows from the prec...

conjecture | open | MathDB 页面 | 原文线索

V2-010884 | MATHDB-377235

Classification conjecture for fermion representations of affine Krichever-Novikov algebras

摘录 / EXCERPT: Let Σ be the underlying compact Riemann surface of genus g, let $\mathfrak{g}$ be the finite-dimensional Lie algebra used to construct the affine Krichever-Novikov algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-010885 | MATHDB-377242

Smallness equivalence conjecture for differential associative pseudoalgebras

摘录 / EXCERPT: Smallness equivalence conjecture. A unital pseudoalgebra $\operatorname{Diff} A$ is small if and only if A is a small X^{cop}-simple algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-010886 | MATHDB-377244

Hatayama et al.'s identification of transfer matrices with Kirillov-Reshetikhin q-characters

摘录 / EXCERPT: Transfer-matrix q-character conjecture. The identification

conjecture | open | MathDB 页面 | 原文线索

V2-010887 | MATHDB-377329

General algebraic-form conjecture for generating functions counting occurrences of 321

摘录 / EXCERPT: Let $s_n((3,2,1),r)$ denote the number of permutations of length n containing exactly r occurrences of the pattern $(3,2,1)$, and let ... For each r, let P_r and Q_r denote...

conjecture | open | MathDB 页面 | 原文线索

V2-010888 | MATHDB-377335

Cameron's integral-domain conjecture for permutation group algebras

摘录 / EXCERPT: Cameron's integral-domain conjecture. If G has no finite orbits, then $A(G)$ is an integral domain.

conjecture | open | MathDB 页面 | 原文线索

V2-010889 | MATHDB-377363

The finite-type dimension conjecture for Nichols algebras of Cartan type

摘录 / EXCERPT: Let (V,c) be a braided vector space with $V=\mathbb C x_1\oplus\cdots\oplus\mathbb C x_{\theta}$ and ... Assume that $q_{ii}=-1$ for every i and $q_{ij}\in\{\pm1\}$ for all i,j...

conjecture | open | MathDB 页面 | 原文线索

V2-010890 | MATHDB-377364

The D_4 and affine D_4 dimension conjecture for Nichols algebras

摘录 / EXCERPT: Let (V,c) and the matrix (A_{ij}) be as above: $V=\mathbb C x_1\oplus\cdots\oplus\mathbb C x_{\theta}$, $c(x_i\otimes x_j)=q_{ij}x_j\otimes x_i$, $q_{ii}=-1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-010891 | MATHDB-377367

The center conjecture for the parabolic category $\mathcal{O}^{n,n}$

摘录 / EXCERPT: Let $\mathcal{O}^{n,n}$ be the parabolic highest weight category for $\mathfrak{sl}_{2n}$, and let $A_{n,n}$ be its finite-dimensional algebra model. Let $e\in A_{n,n}$ be the idem...

conjecture | open | MathDB 页面 | 原文线索

V2-010892 | MATHDB-377369

The center isomorphism conjecture for parabolic algebras

摘录 / EXCERPT: Let A_{κ} be a finite-dimensional algebra with $A_{\kappa}\text{-mod}\cong\mathcal{O}^{\kappa}$, and let $e\in A_{\kappa}$ be the maximal idempotent such that the left...

conjecture | open | MathDB 页面 | 原文线索

V2-010893 | MATHDB-377380

Surjectivity of the cyclotomic Milnor-to-Quillen K-theory map

摘录 / EXCERPT: Surjectivity conjecture. The natural map is surjective. This conjecture concerns whether Milnor K-theory accounts for the relevant p-adic Quillen K-theory under Vandiver's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-010894 | MATHDB-377383

Universality conjecture for curves over finite algebraic closures

摘录 / EXCERPT: Universality conjecture. Every such curve C is universal.

conjecture | open | MathDB 页面 | 原文线索

V2-010895 | MATHDB-377405

Galois-cover fundamental group conjecture

摘录 / EXCERPT: Let X be a simply connected complex surface, let $f_k:X\to\mathbb{CP}^2$ be a generic projection defined by sections of $L^{\otimes k}$, let $\tilde{X}_k$ be the associated Galoi...

conjecture | open | MathDB 页面 | 原文线索

V2-010896 | MATHDB-377427

Annihilator-generator conjecture for powers of central arrangement polynomials

摘录 / EXCERPT: Annihilator-generator conjecture. For any central arrangement Q,

conjecture | open | MathDB 页面 | 原文线索

V2-010897 | MATHDB-377438

Winkelmann's isogeny criterion for algebraic multiplicative quotients

摘录 / EXCERPT: Let $\alpha_1,\alpha_2\in\mathbb C$ be algebraic numbers with ... Let ... Two nonzero complex numbers are multiplicatively dependent if there are integers $(m,n)\ne(0,0)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-010898 | MATHDB-377478

Langlands-Shelstad fundamental lemma for orthogonal Lie algebras

摘录 / EXCERPT: Let ${\mathfrak g}=\operatorname{so}(2n+1)$ and let ${\mathfrak h}=\operatorname{so}(2k+1)\times\operatorname{so}(2n-2k+1)$. Let Y and Z be regular semisimple elements, and sup...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-010899 | MATHDB-377480

Stanley's log-concavity conjecture for h-vectors of graded Cohen-Macaulay domains

摘录 / EXCERPT: Stanley's conjecture. The sequence $(h_0,h_1,\dots,h_s)$ is log-concave, meaning that $h_{i-1}h_{i+1}\le h_i^2$ for $2\le i\le s-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-010900 | MATHDB-377487

The resultant formula for parametrized surfaces with non-local-complete-intersection base points

摘录 / EXCERPT: The resultant formula. If $V_1(A)=\emptyset$ and $V_2(A)\ne\emptyset$, then, up to a nonzero constant,

conjecture | open | MathDB 页面 | 原文线索

V2-010901 | MATHDB-377489

Kac's conjecture for coherent sheaves on weighted projective lines

摘录 / EXCERPT: Let $\mathbb{X}_{\mathbf{p},\underline{\lambda}}$ be a weighted projective line, let $\mathfrak{Lg}$ be its corresponding loop Kac-Moody algebra, and use the identification...

conjecture | open | MathDB 页面 | 原文线索

V2-010902 | MATHDB-377521

Conjugate Bailey-pair conjecture for affine A-type string functions

摘录 / EXCERPT: Let $\ell\in\mathbb{Z}_{+}$, $k\in Q\cap P_{+}$, and $\eta\in\mathcal{P}$ satisfy $\eta_1\leq n-1$ and $|\eta|\equiv\ell\pmod n$. Let $\Lambda\in P_{+}^{N}$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-010903 | MATHDB-377522

Nonnegative-coefficient conjecture for affine configuration sums

摘录 / EXCERPT: Let $P^{(p,p')}$ be the parameter set used for the affine weights, let P_{+}^{r} denote the dominant weights at level r, and let $X_{\eta,\Lambda,\Lambda'}(q)$ be the configura...

conjecture | open | MathDB 页面 | 原文线索

V2-010904 | MATHDB-377523

Vanishing conjecture for affine configuration sums

摘录 / EXCERPT: Let $P^{(p,p')}$ be the parameter set used for the affine weights, and let $X_{\eta,\Lambda,\Lambda'}(q)$ be the corresponding configuration sum. Vanishing conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-010905 | MATHDB-377528

Litherland's radical conjecture for quadratic quandle invariants

摘录 / EXCERPT: Let K be a link. Let $A_p(K)$ be its Alexander module, let $A'_p(K)$ be the submodule of elements annihilated by h, and let $\operatorname{Rep}'(K,\mathbb{E}_p)$ be the annihil...

conjecture | open | MathDB 页面 | 原文线索

V2-010906 | MATHDB-377533

Optimal alternating factorizations in groups with Lie algebra sl(2)

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geq3$. Let $A_n(s)$ and $A(n;r,s,t)$ denote the alternating factorizations used in the paper, let s_n be the associated switching parameters, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010907 | MATHDB-377539

The non-abelian Iwasawa Main Conjecture

摘录 / EXCERPT: Non-abelian Main Conjecture. The element $\delta_p(G_\infty,M,k)$ is induced by a generator

conjecture | open | MathDB 页面 | 原文线索

V2-010908 | MATHDB-377540

The non-abelian zeta-distribution main conjecture for Artin motives

摘录 / EXCERPT: Zeta-distribution main conjecture. There exist $f,g\in\Lambda$ such that $(\operatorname{rn}(f_ng_n^{-1}))_n$ is the zeta distribution $\mathcal{L}_S(G_\infty,M^\lor,1-k)$ and the...

conjecture | open | MathDB 页面 | 原文线索

V2-010909 | MATHDB-377541

Injectivity conjecture for the free quasi-Lie algebra's 2-torsion map

摘录 / EXCERPT: Let H be a free abelian group, and let $L(H)$ and $L'(H)$ denote the free Lie algebra and free quasi-Lie algebra on H, respectively. For $l\geq 1$, write $L_l(H)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010910 | MATHDB-377542

Isomorphism conjecture for the tree-to-quasi-Lie map

摘录 / EXCERPT: Let H be a free abelian group, let $L'(H)$ be the free quasi-Lie algebra on H, and let $\mathcal A_k^t(H)$ be the abelian group generated by labelled planar binary trees of the...

conjecture | open | MathDB 页面 | 原文线索

V2-010911 | MATHDB-377562

The Frobenius conjecture for biseparable corings

摘录 / EXCERPT: Let A be a ring and let ${\mathcal C}$ be an A-coring, meaning an A-bimodule equipped with counit and comultiplication maps satisfying the coring axioms. The coring...

conjecture | open | MathDB 页面 | 原文线索

V2-010912 | MATHDB-377595

Simplicity descent conjecture for differential operator rings

摘录 / EXCERPT: Let k be any field, and let $R\hookrightarrow S$ be an inclusion of k-algebras that splits as a map of R-modules. Simplicity descent conjecture. If $D_k(S)$ is a simple ring,...

conjecture | open | MathDB 页面 | 原文线索

V2-010913 | MATHDB-377596

The divergence criterion for polynomial inverse maps

摘录 / EXCERPT: Divergence criterion. The map $F(z)$ is polynomial if and only if $G(z)$ is polynomial. The source presents this as an algebraic formulation equivalent to the Jacobian conjecture,...

conjecture | open | MathDB 页面 | 原文线索

V2-010914 | MATHDB-377599

Injectivity of evaluation on infinitely many roots of unity

摘录 / EXCERPT: Let Z be an infinite subset of the set of all roots of unity over $\mathbb{Q}$, and let ... be the homomorphism induced by evaluating polynomials at the elements of Z. Injectivit...

conjecture | open | MathDB 页面 | 原文线索

V2-010915 | MATHDB-377600

Units of the cyclotomic completion of the integer polynomial ring

摘录 / EXCERPT: Let $\mathbb Z[q]^{\mathbb N}$ denote the cyclotomic completion of $\mathbb Z[q]$, and let $U(R)$ be the multiplicative group of units of a ring R. Unit-group conjecture. ... The c...

conjecture | open | MathDB 页面 | 原文线索

V2-010916 | MATHDB-377618

The core formula conjecture for ideals satisfying residual conditions

摘录 / EXCERPT: Core formula conjecture. The core of I should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-010917 | MATHDB-377638

The Linear Artin Approximation Theorem

摘录 / EXCERPT: Linear Artin Approximation Theorem. Then the system has a true solution $\underline{U}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-010918 | MATHDB-377644

Finite flat complete-intersection conjecture for deformation rings

摘录 / EXCERPT: Let S be the fixed set of primes and let $S'=s'_1,\dots,s'_n\}$ be a ramified auxiliary set whose existence is guaranteed by Theorem 1. The ring $R_{S\cup S'}$ is a finite flat c...

conjecture | open | MathDB 页面 | 原文线索

V2-010919 | MATHDB-377672

Sturmfels's regularity conjecture for unions of subspaces

摘录 / EXCERPT: Sturmfels's conjecture. The Castelnuovo-Mumford regularity of I is at most d.

conjecture | solved | MathDB 页面 | 原文线索

V2-010920 | MATHDB-377706

Embedding conjecture for torsion-free associative conformal algebras

摘录 / EXCERPT: Let C be an associative conformal algebra that is ∂-torsion free, meaning that no nonzero element of C is annihilated by a nonzero polynomial in ∂. A unital c...

conjecture | open | MathDB 页面 | 原文线索

V2-010921 | MATHDB-377732

Compatible-systems conjecture for l-adic representations

摘录 / EXCERPT: Compatible-systems conjecture. The following should hold: (1) every continuous semisimple de Rham representation $R:G_{\mathbb{Q}}\to GL_n(\overline{\mathbb{Q}}_l)$ unramified at a...

conjecture | open | MathDB 页面 | 原文线索

V2-010922 | MATHDB-377733

Global functional-equation conjecture for l-adic representations

摘录 / EXCERPT: Functional-equation conjecture. One has $m_p^R=m_{w-p}^R$ for all p, hence $m_{w/2}\equiv\dim V\pmod 2$, and: (1) $L(\iota R,s)$ extends to an entire function except for a single...

conjecture | open | MathDB 页面 | 原文线索

V2-010923 | MATHDB-377734

Automorphic-to-Galois compatibility conjecture for GL(n)

摘录 / EXCERPT: Automorphic-to-Galois compatibility conjecture. There is an irreducible geometric strongly compatible system $\mathcal{R}$ of l-adic representations such that

conjecture | open | MathDB 页面 | 原文线索

V2-010924 | MATHDB-377736

Non-zero-divisor conjecture for Vogel's algebra Lambda

摘录 / EXCERPT: Let Λ be the graded commutative algebra formed from the antisymmetric subspace of connected tri-/univalent graphs with three labeled univalent vertices, and let Λ_0...

conjecture | open | MathDB 页面 | 原文线索

V2-010925 | MATHDB-377737

Filtration-independence conjecture for connected graph diagram spaces

摘录 / EXCERPT: Let $\hat {\cal B}_{m}^{\,u}$ be the diagram space equipped with the filtration whose quotient at (f,o,e) is ${\cal G}_{f,o,e}^{\,}\,\hat {\cal B}_{m}^{\,u}$. Let $T(m,u)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-010926 | MATHDB-377738

Vanishing conjecture for filtration quotients of connected graph diagrams

摘录 / EXCERPT: Let $\hat {\cal B}_{m}^{\,u}$ be the diagram space with filtration quotients ${\cal G}_{f,o,e}^{\,}\,\hat {\cal B}_{m+2f}^{\,u}$, where f,o,e are the filtration parameters. Vanis...

conjecture | open | MathDB 页面 | 原文线索

V2-010927 | MATHDB-377746

Flatness conjecture for the Borel Zastava projection

摘录 / EXCERPT: Let $\fg$ be a Kac-Moody algebra, $\fb$ a Borel subgroup, $\bC$ the curve, and μ a dominant coweight. Let $\on{Maps}^\mu(\bC,\CG_{\fg,\fb})$ be the corresponding space of maps,...

conjecture | open | MathDB 页面 | 原文线索

V2-010928 | MATHDB-377748

Kawamata's adjoint-ideal conjecture for Gorenstein section rings

摘录 / EXCERPT: Kawamata's conjecture in the Fano case. One has

conjecture | open | MathDB 页面 | 原文线索

V2-010929 | MATHDB-377773

Gorenstein conjecture for tautological rings of moduli spaces of pointed curves

摘录 / EXCERPT: Gorenstein conjecture. The tautological rings of the filtration of $\overline{M}_{g,n}$ are finite-dimensional Gorenstein algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-010930 | MATHDB-377801

The Isomorphism Conjecture in algebraic K-theory

摘录 / EXCERPT: Isomorphism Conjecture in algebraic K-theory. The map from the colimit over finite subgroups

conjecture | open | MathDB 页面 | 原文线索

V2-010931 | MATHDB-377841

Direct-summand conjecture for quasi-free Hilbert modules

摘录 / EXCERPT: Direct-summand conjecture. The modules $\mathcal{R}_1$ and $\mathcal{R}_2$ are quasi-free of ranks k_1 and k_2, respectively, and

conjecture | open | MathDB 页面 | 原文线索

V2-010932 | MATHDB-377853

Kac's polynomiality and root-multiplicity conjectures

摘录 / EXCERPT: Kac's conjectures. For every root $d\in\mathbf{N}I$, there exists a polynomial $P_d(t)$ with nonnegative integer coefficients such that

conjecture | open | MathDB 页面 | 原文线索

V2-010933 | MATHDB-377855

The formality conjecture for the chiral deformation algebra of SK

摘录 / EXCERPT: Let K be a finitely generated free ${\cal D}_X$-module, and let SK denote the associated chiral algebra. Let def_{SK} be its deformation pro-$*$-Lie algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010934 | MATHDB-377908

The Cachazo-Douglas-Seiberg-Witten conjecture for simple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra over $\mathbb{C}$, let g be its dual Coxeter number, and let $A=\wedge(\mathfrak{g}\oplus\mathfrak{g})/I$, where I...

conjecture | open | MathDB 页面 | 原文线索

V2-010935 | MATHDB-377925

Simplified presentation conjecture for tangent conic-line arrangements

摘录 / EXCERPT: Simplified presentation conjecture. The presentation has the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-010936 | MATHDB-378056

Cassels's conjecture on Maass forms with eigenvalue one quarter

摘录 / EXCERPT: Let ϕ be a Maass form with Laplace eigenvalue $\frac14$. Cassels's conjecture. Must ϕ arise from a Galois representation? The source presents this as an open conjectural...

open_problem | open | MathDB 页面 | 原文线索

V2-010937 | MATHDB-378062

Generalized coinvariant-algebra isomorphism for finite Coxeter groups

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let $BE(W,S)$ be the associated algebra, and let ${\bf R}[\theta_s\mid s\in S]$ denote the subalgebra generated by its Dunkl elements. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-010938 | MATHDB-378063

Nonnegative monomial-basis conjecture for Bernstein-Eilenberg algebras

摘录 / EXCERPT: Let (W,S) be a crystallographic Coxeter system. Let $BE(W,S)$ be the associated algebra, let X_w be the Bernstein-Gelfand-Gelfand polynomial indexed by $w\in W$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010939 | MATHDB-378098

Herzog-Srinivasan upper multiplicity conjecture for arbitrary graded algebras

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ be a polynomial ring with its standard grading, let $I\subset S$ be a graded ideal, and let $R=S/I$ be a standard graded K-algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-010940 | MATHDB-378105

Non-crossing families and labeling sets of plabic diagrams

摘录 / EXCERPT: Let $[1\dots n]$ be labeled cyclically, and let a non-crossing family be a family of pairwise non-crossing k-subsets, where non-crossing means that no chord with endpoints in...

conjecture | open | MathDB 页面 | 原文线索

V2-010941 | MATHDB-378134

Exchange graph conjectures for cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}(\mathcal{S})$ be a cluster algebra with seed $(\tilde{\mathbf{x}},\tilde B)$, principal part B, rank n, clusters, and exchange graph whose vertices are seeds a...

conjecture | open | MathDB 页面 | 原文线索

V2-010942 | MATHDB-378151

Toën's full faithfulness conjecture for commutative ring spectra

摘录 / EXCERPT: Let Sp^{Σ} be the category of symmetric spectra with its positive model structure, and let $Comm(Sp^{\Sigma})$ be the model category of commutative ring spectra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-010943 | MATHDB-378152

Toën's comparison conjecture for commutative ring spectra

摘录 / EXCERPT: Let Sp^{Σ} be the category of symmetric spectra and let $Comm(Sp^{\Sigma})$ be the model category of commutative ring spectra. Let LSp^{Σ} be the tensor Segal categor...

conjecture | open | MathDB 页面 | 原文线索

V2-010944 | MATHDB-378171

Etingof's conjecture that the elements $\widehat p_i$ generate the invariant algebra

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra over $\mathbb{C}$, let B be the algebra used in the paper, and set $E=(B\otimes B)^{\mathfrak{g}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010945 | MATHDB-378172

Etingof's degree-g relation conjecture for the \widehat p_i

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra with dual Coxeter number g. Let $p_1,\ldots,p_r$ be homogeneous generators of $(S\mathfrak{g})^{\mathfrak{g}}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-010946 | MATHDB-378173

The conjecture that the ideal L is generated by the higher \widehat p_i

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra, let $E=(B\otimes B)^{\mathfrak{g}}$, and let $p_1,\ldots,p_r$ and $\widehat p_1,\ldots,\widehat p_r$ be as above. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010947 | MATHDB-378174

Existence conjecture for Buchsbaum Stanley-Reisner rings with linear resolution

摘录 / EXCERPT: Existence conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-010948 | MATHDB-378176

The converse conjecture for F-pure thresholds and Segre-product tangent cones

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a d-dimensional F-finite F-pure non-$\mathbb{Q}$-Gorenstein normal local ring of characteristic $p>0$. Let $\mathrm{c}(\mathfrak{m})$ denote its F-pure...

conjecture | open | MathDB 页面 | 原文线索

V2-010949 | MATHDB-378218

The cluster-variable parametrization conjecture for simply-laced cluster algebras

摘录 / EXCERPT: Let $C=\{u_1,\ldots,u_n\}$ be a cluster in a cluster algebra of simply-laced type and rank n, with associated quiver Q_C and category $\operatorname{Mod} Q_C$ defined by the sh...

conjecture | open | MathDB 页面 | 原文线索

V2-010950 | MATHDB-378222

Mixed Tate motives over dual numbers conjecture

摘录 / EXCERPT: Let $F_{\varepsilon}=F[\varepsilon]/(\varepsilon^2)$, and let ${\cal Q}_{\bullet}(F_{\varepsilon})$ be the Lie coalgebra combining the non-Euclidean and Euclidean scissor-congruenc...

conjecture | open | MathDB 页面 | 原文线索

V2-010951 | MATHDB-378223

Higher additive Dehn map conjecture

摘录 / EXCERPT: Assume that F is an algebraically closed field. Let $\beta_n(F)$ be the additive higher Bloch module and ${\cal E}_n(F)$ the Euclidean F-scissor congruence group. Higher additi...

conjecture | open | MathDB 页面 | 原文线索

V2-010952 | MATHDB-378239

Cluster-category correspondence conjecture for hereditary algebras

摘录 / EXCERPT: Cluster-category correspondence conjecture. There is a one-to-one correspondence between the cluster variables of $\operatorname{\mathcal A}(H)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-010953 | MATHDB-378240

Exchange-relation conjecture for cluster-category triangles

摘录 / EXCERPT: Exchange-relation conjecture. The exchange relation in $\operatorname{\mathcal A}(H)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-010954 | MATHDB-378254

Geller's conjecture on injectivity of K_2 for one-dimensional local domains

摘录 / EXCERPT: Let A be a local one-dimensional domain with field of fractions F. Geller's conjecture. The ring A is regular if and only if the map ... is injective. The conjecture gives a cr...

conjecture | open | MathDB 页面 | 原文线索

V2-010955 | MATHDB-378255

Artinian Geller conjecture for subalgebras of principal ideal algebras

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, and let A and B be K-algebras essentially of finite type over K. Assume that B is a finite-dimensional pr...

conjecture | open | MathDB 页面 | 原文线索

V2-010956 | MATHDB-378256

Berger's conjecture on torsion in Kähler differentials

摘录 / EXCERPT: Let A be a one-dimensional local domain essentially of finite type over a field K of characteristic zero, and let $\Omega_{A/K}$ denote its module of Kähler differentials. Berg...

conjecture | open | MathDB 页面 | 原文线索

V2-010957 | MATHDB-378279

Expected Hilbert-series formula for relatively compressed algebras over general complete intersections

摘录 / EXCERPT: Let $R=k[x_1,\ldots,x_n]$, let ${\mathfrak c}\subset R$ be a general complete intersection of type $(d_1,\ldots,d_c)$, and let $A=R/I$ be relatively compressed with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-010958 | MATHDB-378281

Non-Koszul ghost-term linkage conjecture for relatively compressed level algebras

摘录 / EXCERPT: Let $R=k[x_1,\ldots,x_n]$ and let ${\mathfrak c}\subset R$ be a general complete intersection with fixed generator degrees. Let $A=R/I$ be a relatively compressed level quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-010959 | MATHDB-378282

No-ghost-term conjecture for equal-degree complete intersections

摘录 / EXCERPT: Let $R=k[x_1,\ldots,x_n]$ and let ${\mathfrak c}\subset R$ be a complete intersection generated by forms all of the same degree. Let A be a relatively compressed level Artinian q...

conjecture | open | MathDB 页面 | 原文线索

V2-010960 | MATHDB-378338

Generalized Springer correspondence conjecture for graded Hecke algebras of type B_n

摘录 / EXCERPT: Let R_0 be a root system of type B_n, let $\bbH$ be the associated graded Hecke algebra with long-root label $k_1\neq 0$ and short-root label $k_2=mk_1$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-010961 | MATHDB-378349

Stable affine character conjecture for deformed universal characters

摘录 / EXCERPT: Let R be a dominant sequence of rectangles and let λ be a partition. For each of the four stable families indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-010962 | MATHDB-378350

Stable fermionic-formula transpose duality conjecture

摘录 / EXCERPT: Let R be a dominant sequence of rectangles, let λ be a partition, and let $\diamondsuit$ denote one of the four stable families. Write R^t and λ^t for the trans...

conjecture | open | MathDB 页面 | 原文线索

V2-010963 | MATHDB-378372

The canonical dimension formula for special orthogonal groups

摘录 / EXCERPT: Let SO_{2m-1} and SO_{2m} denote the special orthogonal groups of the indicated dimensions, and let $\operatorname{cd}(G)$ denote the canonical di...

conjecture | open | MathDB 页面 | 原文线索

V2-010964 | MATHDB-378383

Quinn's Hyperelementary Induction Conjecture

摘录 / EXCERPT: Let G be a group, and let $\overline{{\mathcal H}}$ be the class of possibly infinite hyperelementary groups. Say that a group satisfies induction for a class of groups when the...

conjecture | open | MathDB 页面 | 原文线索

V2-010965 | MATHDB-378387

Central extensions preserve bounded locality for finite-type Lie conformal algebras

摘录 / EXCERPT: Central-extension bounded-locality conjecture. The locality function $S_{\widehat{\goth L},\,\cal G\cup\cal M}$ is also uniformly bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-010966 | MATHDB-378388

The Verlinde algebra description of the Thom class for loop-group representations

摘录 / EXCERPT: The Verlinde algebra isomorphism conjecture. For every homogeneous degree z^n with $n\neq 0$, the map

conjecture | open | MathDB 页面 | 原文线索

V2-010967 | MATHDB-378438

K-theory computation for Col-divisible polytopes

摘录 / EXCERPT: Let R be a commutative ring and let P be a Col-divisible polytope of arbitrary dimension. Write $K_i(R,P)$ for its associated K-groups, and let ${\mathfrak c}(P)\leq\dim P$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-010968 | MATHDB-378452

The real-rank conjecture for left orderability of arithmetic groups

摘录 / EXCERPT: Let G be a ${\mathord{\mathbb{Q}}}$-simple algebraic ${\mathord{\mathbb{Q}}}$-group, and let $G_{\mathord{\mathbb{Z}}}$ denote its integral points. A subgroup is left orderable i...

conjecture | open | MathDB 页面 | 原文线索

V2-010969 | MATHDB-378468

Shapiro's conjecture on split decompositions of algebras with involution

摘录 / EXCERPT: Shapiro's conjecture. (A,σ) admits a decomposition as a tensor product of r quaternion algebras with involution in which each quaternion algebra is split.

conjecture | open | MathDB 页面 | 原文线索

V2-010970 | MATHDB-378528

Majid's conjecture on the quadratic differential algebra of simply-laced Weyl groups

摘录 / EXCERPT: Let W be a Weyl group, let ${\cal C}$ be its set of reflections, and let Λ_w be the left-invariant differential algebra associated with this differential structure. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-010971 | MATHDB-378529

Completeness conjecture for the relations of the Weyl-group algebra \(\Lambda_w(B_n)\)

摘录 / EXCERPT: Let B_n be the Weyl group of type B_n. Let $\Lambda_w(B_n)$ be its left-invariant differential algebra, and let $\Lambda_{quar}(B_n)$ be the quotient of the quadratic algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-010972 | MATHDB-378566

Rationality conjecture for simply connected quotient surfaces

摘录 / EXCERPT: Let S be a projective surface with quotient singularities such that ... and let S^0 denote its smooth locus. Assume that S^0 is simply connected. Rationality conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-010973 | MATHDB-378580

The shape-vector binomial bound for tridiagonal pairs

摘录 / EXCERPT: Let A,A^* be a tridiagonal pair on a finite-dimensional vector space V over an algebraically closed field $\mathbb K$. Let d be its diameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010974 | MATHDB-378583

The generating-set conjecture for Leonard systems

摘录 / EXCERPT: Let Φ be a Leonard system with primitive idempotents $E_0,E_1,\ldots,E_d$ and dual primitive idempotents $E^*_0,E^*_1,\ldots,E^*_d$ in its associated algebra $\mathcal A$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-010975 | MATHDB-378587

The mod-p compatibility conjecture for crystalline representations

摘录 / EXCERPT: Let V be a crystalline representation of ${\rm G}_{{\mathbb Q}_p}$, let T be a Galois-invariant lattice, and write $V\bmod p$ for the semisimplification of T/pT. Let $\Pi(V)$...

conjecture | open | MathDB 页面 | 原文线索

V2-010976 | MATHDB-378616

The radicality conjecture for hierarchical-model ideals

摘录 / EXCERPT: Radicality conjecture. The ideal I_{Δ} is radical if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-010977 | MATHDB-378627

Lehmer inequality for Drinfeld modules

摘录 / EXCERPT: Let K be a finitely generated field, let A be the coefficient ring of a Drinfeld module, and let $\phi:A\rightarrow K\{\tau\}$ be a Drinfeld module. For a point...

conjecture | open | MathDB 页面 | 原文线索

V2-010978 | MATHDB-378628

Nuttall's bounded-spurious-poles conjecture for algebraic functions

摘录 / EXCERPT: Let f be an algebraic function analytic at ∞, and let $[n/n]$ denote its diagonal Padé approximants. A spurious pole is a pole of a Padé approximant in the sense defined i...

conjecture | open | MathDB 页面 | 原文线索

V2-010979 | MATHDB-378643

Segre's conjecture on reduced linear systems

摘录 / EXCERPT: Segre's conjecture. If $\mathcal L$ is non-empty and reduced, then it is non-special.

conjecture | open | MathDB 页面 | 原文线索

V2-010980 | MATHDB-378672

The algebraicity criterion for generalized double Laurent series

摘录 / EXCERPT: Algebraicity criterion. The generalized double Laurent series is algebraic over $\mathbb{F}_q(t,u)$ if and only if: (a) for some positive integers a and b,...

conjecture | open | MathDB 页面 | 原文线索

V2-010981 | MATHDB-378690

Unconditional S-tuple extension of the mean value theorem for quadratic extensions

摘录 / EXCERPT: Let S be a finite set of places of a number field k containing all archimedean places, and let $L_S=(L_v)_{v\in S}$ be an unconditional S-tuple of separable quadratic algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-010982 | MATHDB-378694

Garcia-Stillman-Sturmfels conjecture for the binary hidden naive Bayes model

摘录 / EXCERPT: Garcia-Stillman-Sturmfels conjecture. The ideal $\mathfrak a(2;2,2,\dots,2)$ is generated by the 3×3 minors of all two-dimensional flattenings arising from bipartitions of...

conjecture | open | MathDB 页面 | 原文线索

V2-010983 | MATHDB-378697

Huneke-Srinivasan multiplicity conjecture

摘录 / EXCERPT: Huneke-Srinivasan multiplicity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-010984 | MATHDB-378727

Nilpotent-orbit filtration conjecture for affine Hecke algebras

摘录 / EXCERPT: Let G be a connected reductive algebraic group over ${\mathbb{C}}$ with simply-connected derived group, let $\tilde{W}_a$ be its extended affine Weyl group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-010985 | MATHDB-378743

The solvable-radical sequence conjecture for finite-dimensional Lie algebras

摘录 / EXCERPT: Let L be a finite-dimensional Lie algebra. For a sequence $\overrightarrow{u}=\overrightarrow{u}(x,y)$, write $L(\overrightarrow{u})$ for the corresponding ideal and let the solv...

conjecture | open | MathDB 页面 | 原文线索

V2-010986 | MATHDB-378750

Stabilization conjecture for kernels of monomial-induced maps

摘录 / EXCERPT: Let R_n be the polynomial ring and let Q_n denote the kernel considered above, with the symmetric group acting by permuting the indices. For a monomial f, consider the result...

conjecture | open | MathDB 页面 | 原文线索

V2-010987 | MATHDB-378791

Asymptotic density conjecture for cyclic Drinfeld-module structures

摘录 / EXCERPT: Let $L=F_{q^{n}}$ be a finite field, let P be the A-characteristic of L, let $m=[L,A/P]$, and let $d=\operatorname{deg}P$. Let $C(d,m,q)$ and $C_{0}(d,m,q)$ denote the two qu...

conjecture | open | MathDB 页面 | 原文线索

V2-010988 | MATHDB-378824

Canonical Frobenius subalgebra conjecture for flat associative deformations

摘录 / EXCERPT: Let B be a local commutative algebra over k with maximal ideal ${\mathfrak m} \subset B$, and let $\widetilde{A}$ be a flat associative algebra over B such that...

conjecture | open | MathDB 页面 | 原文线索

V2-010989 | MATHDB-378831

Jöllenbeck's resolution conjecture for monomial rings

摘录 / EXCERPT: Let $S=k[x_1,\ldots,x_n]$, let $\mathfrak a=\langle m_1,\ldots,m_l\rangle$ be a monomial ideal, and set $A=S/\mathfrak a$. Let R be the graded commutative algebra constructed fro...

conjecture | open | MathDB 页面 | 原文线索

V2-010990 | MATHDB-378832

Jöllenbeck's strong gcd-condition conjecture for Golod monomial rings

摘录 / EXCERPT: Let $S=k[x_1,\ldots,x_n]$, let $\mathfrak a=\langle m_1,\ldots,m_l\rangle\subset S$ be a monomial ideal, and set $A=S/\mathfrak a$. The ideal $\mathfrak a$ satisfies the strong gcd...

conjecture | open | MathDB 页面 | 原文线索

V2-010991 | MATHDB-378886

The resonance formula for the linear strand of an arrangement's Orlik-Solomon algebra

摘录 / EXCERPT: Let ${\mathcal A}$ be a complex hyperplane arrangement, let $A=H^*(X({\mathcal A}),\Bbbk)$ be its Orlik-Solomon algebra over a field of characteristic 0, and let E be the exter...

conjecture | open | MathDB 页面 | 原文线索

V2-010992 | MATHDB-378924

Harder's Tamagawa number conjecture for split semisimple groups

摘录 / EXCERPT: Harder's conjecture. If ${\mathcal G}$ is split, the Tamagawa number of G equals the number of connected components of the moduli space of ${\mathcal G}$-torsors on X.

conjecture | open | MathDB 页面 | 原文线索

V2-010993 | MATHDB-378956

Square-free initial ideals conjecture for generic row-linear matrices

摘录 / EXCERPT: Let t_{ij} be distinct variables over a field K with $1\leq i\leq m$ and $1\leq j\leq n$. Let $L=(L_{ij})$ be an $m\times n$ matrix with ... where $a_{ijk}\in K$ for all i,j,k....

conjecture | open | MathDB 页面 | 原文线索

V2-010994 | MATHDB-378957

Lexicographic initial ideals conjecture for row-linear determinantal ideals

摘录 / EXCERPT: Let $L=(L_{ij})$ be an $m\times n$ matrix whose entries have the form ... with $a_{ijk}\in K$ for all i,j,k. Assume that, for each i, the forms $L_{i1},\dots,L_{in}$ are linearly...

conjecture | open | MathDB 页面 | 原文线索

V2-010995 | MATHDB-378968

The $G_{j,\alpha}^{B_n}$ elementary-symmetric-function conjecture

摘录 / EXCERPT: The $G_{j,\alpha}^{B_n}$ elementary-symmetric-function conjecture. For every $\alpha\in\mathbb{Q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-010996 | MATHDB-378988

The Manin-Mumford conjecture for Drinfeld modules

摘录 / EXCERPT: Let K be a function field of transcendence degree 1 over $\mathbb{F}_p$, and let $\phi:A\rightarrow K\{\tau\}$ be a Drinfeld module. Let $n\ge 1$ and let X be an irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-010997 | MATHDB-378993

The determinant conjecture for cluster monomials and MV-polynomials

摘录 / EXCERPT: The determinant conjecture. Every cluster monomial is an MV-polynomial that is naturally expressible as a determinant: if ψ is a cluster monomial, then there is a Kostant pict...

conjecture | open | MathDB 页面 | 原文线索

V2-010998 | MATHDB-378994

The allowable-path determinant conjecture for acyclic Kostant pictures

摘录 / EXCERPT: The allowable-path determinant conjecture. For every such Kostant picture,

conjecture | open | MathDB 页面 | 原文线索

V2-010999 | MATHDB-378995

The geometric interpretation conjecture for cluster-monomial MV-polynomials

摘录 / EXCERPT: The geometric interpretation conjecture. The function χ_1 is well-defined and

conjecture | open | MathDB 页面 | 原文线索

V2-011000 | MATHDB-378998

Conjecture on socle elements in codimension-3 graded algebras

摘录 / EXCERPT: Let ${\bf H}$ be a Hilbert function of the form ... with $h_d<h_{d+2}$. Socle-element conjecture. (a) Any graded algebra of codimension 3 with Hilbert function ${\bf H}$ has a socl...

conjecture | open | MathDB 页面 | 原文线索

V2-011001 | MATHDB-379004

The quasiharmonic deformation conjecture for singular vectors

摘录 / EXCERPT: Let W be a real reflection group with reflection representation V, let $H(W)$ be the universal rational Cherednik algebra, and let $\mathcal{QH}\subset H(W)$ be the space of un...

conjecture | open | MathDB 页面 | 原文线索

V2-011002 | MATHDB-379005

The flatness conjecture for universal quasiharmonics at constant parameter

摘录 / EXCERPT: Let W be a real reflection group of rank ℓ, with Coxeter number h and basic invariant degrees $d_1,\dots,d_\ell$. Let r satisfy $r\equiv d_i-1\pmod h$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-011003 | MATHDB-379006

The W-goodness finite-dimensional quotient conjecture

摘录 / EXCERPT: Let W be a real reflection group with reflection representation V. A positive integer r is W-good if, for every parameter c with $|c|=r$, the Cherednik algebra $H_c(W)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-011004 | MATHDB-379011

The extended multiplicity conjecture for Cohen-Macaulay graded algebras

摘录 / EXCERPT: Let $R=k[x_1,\dots,x_n]$ be a standard graded polynomial ring over a field, let $I\subset R$ be a graded ideal, and set $c=\operatorname{codim} R/I$. For a minimal graded free reso...

conjecture | open | MathDB 页面 | 原文线索

V2-011005 | MATHDB-379012

The multiplicity conjecture for finitely generated graded torsion modules

摘录 / EXCERPT: Let $R=k[x_1,\dots,x_n]$ be a standard graded polynomial ring over a field, and let N be a finitely generated graded torsion R-module. Set $c=n-\dim N$ and, for a minimal grade...

conjecture | open | MathDB 页面 | 原文线索

V2-011006 | MATHDB-379019

Bj1rner and Swartz's g-vector conjecture for doubly CohenMacaulay complexes

摘录 / EXCERPT: Let X be a doubly CohenMacaulay, or 2-CM, complex, and let $g(X)$ denote its g-vector. An M-sequence is the f-vector of an order ideal of monomials. Bjrner and Swartz's...

conjecture | open | MathDB 页面 | 原文线索

V2-011007 | MATHDB-379029

The nonnegativity conjecture for cluster variables

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra of rank n with a seed whose cluster and frozen variables are $x_1,\dots,x_m$. By the Laurent phenomenon, every cluster variable is a Lauren...

conjecture | solved | MathDB 页面 | 原文线索

V2-011008 | MATHDB-379030

The bipartite Cambrian congruence conjecture for transformed cluster fans

摘录 / EXCERPT: Let R_u and R_v be regions of the Coxeter arrangement, and let the transformed cluster fan be the fan whose maximal cones are under consideration. Write $u\equiv v$ when u an...

conjecture | solved | MathDB 页面 | 原文线索

V2-011009 | MATHDB-379045

Positivity conjecture for exceptional Hall coefficients

摘录 / EXCERPT: Positivity conjecture. For any object M and any exceptional object K of ${\mathcal C}$, the integer $r(M,K)$ is nonnegative.

conjecture | open | MathDB 页面 | 原文线索

V2-011010 | MATHDB-379046

The Huneke-Srinivasan and Herzog-Srinivasan multiplicity bounds

摘录 / EXCERPT: Huneke-Srinivasan-Herzog-Srinivasan conjecture. If R/I is Cohen-Macaulay, then

conjecture | open | MathDB 页面 | 原文线索

V2-011011 | MATHDB-379060

The augmentation-ideal conjecture for universal enveloping algebras of simple Lie triple systems

摘录 / EXCERPT: Let V be a simple Lie triple system and let $U(V)$ be its universal enveloping algebra. The augmentation-ideal conjecture. The only proper ideal of $U(V)$ is its augmentation ide...

conjecture | open | MathDB 页面 | 原文线索

V2-011012 | MATHDB-379067

The cyclic module conjecture for finite Gorenstein injective dimension

摘录 / EXCERPT: Let R be a local ring, and let M be a nonzero cyclic R-module of finite G-injective dimension. Cyclic module conjecture. If such an M exists, then R is Gorenstein. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-011013 | MATHDB-379101

Ehrenpreis conjecture for the mapping class group of a surface solenoid

摘录 / EXCERPT: Let $\widehat{\Sigma}$ be the algebraic universal cover of a closed surface Σ, with Teichmüller space $\mathcal{T}(\widehat{\Sigma})$. Let ${\sf M}(\widehat{\Sigma},\ell)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-011014 | MATHDB-379102

L^1 Ehrenpreis conjecture for the mapping class group of a surface solenoid

摘录 / EXCERPT: Let $\widehat{\Sigma}$ be the algebraic universal cover of a closed surface Σ, let ℓ be a base leaf, and let ${\sf M}(\widehat{\Sigma},\ell)$ be its mapping class group...

conjecture | open | MathDB 页面 | 原文线索

V2-011015 | MATHDB-379110

Kuzmin's conjecture on the nilpotency class of nil algebras

摘录 / EXCERPT: Kuzmin's conjecture. The exact value of the nilpotency class is

conjecture | open | MathDB 页面 | 原文线索

V2-011016 | MATHDB-379117

Anick's conjecture on wild automorphisms

摘录 / EXCERPT: Let K be a field of characteristic 0, and let ... be the Anick automorphism of $K\langle x,y,z\rangle$. An automorphism is wild if it does not belong to the group generated by el...

conjecture | open | MathDB 页面 | 原文线索

V2-011017 | MATHDB-379126

Fukaya-Kato equivariant local Tamagawa-number conjecture

摘录 / EXCERPT: Let Λ be the Iwasawa algebra of the relevant p-adic Lie group, let $\widetilde\Lambda$ be its unramified coefficient extension, and let $\mathbb T$ be the associated big...

conjecture | open | MathDB 页面 | 原文线索

V2-011018 | MATHDB-379163

Integral lower-central-series conjecture for the pure cactus group

摘录 / EXCERPT: Let Γ_n be the pure cactus group, let Γ_n^k denote its lower central series, and let $\mathcal{L}_n$ be the associated graded Lie algebra after quotienting by 2-tor...

conjecture | open | MathDB 页面 | 原文线索

V2-011019 | MATHDB-379164

Rational Malcev conjecture for the pure cactus group

摘录 / EXCERPT: Let $\widehat{\Gamma_n}$ be the prounipotent completion of the pure cactus group Γ_n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011020 | MATHDB-379165

Injectivity conjecture for the map from the cactus Lie algebra to the braid Lie algebra

摘录 / EXCERPT: Let L_n be the quadratic Lie algebra associated with the pure cactus group, let $\mathbf{L}_n$ be the corresponding braid Lie algebra, and let $\xi_n:L_n\to\mathbf{L}_n$ be defin...

conjecture | open | MathDB 页面 | 原文线索

V2-011021 | MATHDB-379169

Conjecture on p-reductivity of Bergman submodules and orthogonal complements

摘录 / EXCERPT: Let Ω be a bounded, strongly pseudoconvex domain in $\mathbb{C}^m$ with smooth boundary, let $B^2(\Omega)$ be its Bergman space, and let $\mathcal{S}\subseteq B^2(\Omega)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-011022 | MATHDB-379170

Finite-multiplicity extension of the p-reductivity conjecture

摘录 / EXCERPT: Let Ω be a bounded, strongly pseudoconvex domain in $\mathbb{C}^m$, and consider submodules of $B^2(\Omega)\otimes\mathbb{C}^k$. Finite-multiplicity p-reductivity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-011023 | MATHDB-379177

The row-module dual canonical basis conjecture for shifted Yangians

摘录 / EXCERPT: Let $\boldsymbol{\boldsymbol{\text?}}$ be a tensor product parameter with associated free abelian group $S^{\boldsymbol{\boldsymbol{\text?}}}(V_{\mathbb Z})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011024 | MATHDB-379179

Order-polynomial construction of commutative descent algebras for finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group, let $w\in W$, and let $\Omega_W(w;t)$ denote the order polynomial associated with W and w. Assume that t_W is a linear function depending o...

conjecture | open | MathDB 页面 | 原文线索

V2-011025 | MATHDB-379206

Langlands correspondence for \(\mathrm{GL}_2\) over imaginary quadratic fields

摘录 / EXCERPT: Langlands correspondence for GL_2/K. The representation ρ_{λ} exists as above, is potentially semistable at every place v, and its associated representation...

conjecture | open | MathDB 页面 | 原文线索

V2-011026 | MATHDB-379229

Splitting conjecture for the wild inertia extension of a local field

摘录 / EXCERPT: Let $K=\mathbb{Q}_q$ be unramified over $\mathbb{Q}_p$, with maximal unramified extension K^nr and separable algebraic closure K^s. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-011027 | MATHDB-379230

Wild inertia abelianization conjecture for local fields

摘录 / EXCERPT: Let $K=\mathbb{Q}_q$ be unramified over $\mathbb{Q}_p$, let $\overline{k}$ be an algebraic closure of its residue field, and let $P=\operatorname{Gal}(K^s/K^\mathrm{tr})$ be the wi...

conjecture | open | MathDB 页面 | 原文线索

V2-011028 | MATHDB-379258

Holomorphic anomaly conjecture for the Gromov-Witten potential of a Calabi-Yau A-infinity category

摘录 / EXCERPT: For a Calabi-Yau A_∞ category, let the Gromov-Witten potential define a left ideal in the sheaf of Weyl algebras associated to its periodic cyclic chain complex, and let thi...

conjecture | open | MathDB 页面 | 原文线索

V2-011029 | MATHDB-379259

Non-simply-laced fusion conjecture for Weyl modules

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a simple Lie algebra, let $\lambda=\lambda_1+\cdots+\lambda_n$ be a sum of dominant integral weights, and let $c_1,\ldots,c_n$ be pairwise distinct complex...

conjecture | open | MathDB 页面 | 原文线索

V2-011030 | MATHDB-379260

Semi-infinite fusion conjecture for affine highest-weight modules

摘录 / EXCERPT: Let $\widehat{\mathfrak{g}}$ be an affine Kac-Moody algebra, let $\Lambda=m\Lambda_0+\lambda$ be a dominant integral weight, and let $V(\Lambda)$ and $V(\lambda)$ denote the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-011031 | MATHDB-379267

Kernel conjecture for restricted natural operations of quadratic Koszul algebras

摘录 / EXCERPT: Let ${\cal P}$ be a quadratic Koszul operad. Let ${\cal S}^*_{{\cal P}_\infty}$ be the suboperad of natural operations generated by the restricted class of operations generalizing...

conjecture | open | MathDB 页面 | 原文线索

V2-011032 | MATHDB-379286

The rational Novikov conjecture for assembly maps

摘录 / EXCERPT: Let π be a group, and let A^L_π and A^K_π denote the assembly maps in L-theory and K-theory, respectively. Their rationalizations are obtained by tensoring with the...

conjecture | open | MathDB 页面 | 原文线索

V2-011033 | MATHDB-379297

Perrin-Riou's equivariant epsilon-isomorphism conjecture

摘录 / EXCERPT: $C_{\mathrm{EP}}(L/K,V)$. The map $\delta_{V,L/K}$ sends $\Delta_{\mathrm{EP}}(L/K,T)$ onto $\mathbf{Z}_p[G]_{V,L/K}$.

conjecture | open | MathDB 页面 | 原文线索

V2-011034 | MATHDB-379300

Takahashi's converse conjecture for contravariantly finite G-projective modules

摘录 / EXCERPT: Let R be a commutative Noetherian Henselian local ring with maximal ideal $\mathfrak m$ and residue class field k, and let $\mathrm{mod}\,R$ denote the category of finitely gen...

conjecture | open | MathDB 页面 | 原文线索

V2-011035 | MATHDB-379301

The quasitriangular Malcev isomorphism conjecture

摘录 / EXCERPT: For each n, let ${\mathfrak{qtr}}_n$ be the Lie algebra generated by the elements r_{ij}, subject to the classical Yang-Baxter relations, and let ${\mathbf{QTr}}_n$ be the grou...

conjecture | open | MathDB 页面 | 原文线索

V2-011036 | MATHDB-379308

Monomial conjecture for maximal permanents of a generic matrix

摘录 / EXCERPT: Let $I_d(d,2d-1)$ be the ideal generated by the maximal permanents of the generic $d\times(2d-1)$ matrix $M_{d,2d-1}$ over a field k. Monomial conjecture for maximal permanents....

conjecture | open | MathDB 页面 | 原文线索

V2-011037 | MATHDB-379310

De Loera's determinantal refinement of Kahn's conjecture

摘录 / EXCERPT: Let $M_{d,d}$ be the generic $d\times d$ matrix over a field k, and let I be the ideal of $d\times d$ permanents of the $d\times2d$ matrix $(M_{d,d}\mid M_{d,d})$. De Loera's r...

conjecture | open | MathDB 页面 | 原文线索

V2-011038 | MATHDB-379371

The freeness conjecture for the parking-function dual Hopf algebra

摘录 / EXCERPT: Let ${\bf PQSym}^*$ denote the dual Hopf algebra of parking functions, realized as a dendriform trialgebra with operations induced from the dendriform trialgebra structure on...

conjecture | open | MathDB 页面 | 原文线索

V2-011039 | MATHDB-379382

Maximal-order conjecture for completed group algebras of dihedral-free groups

摘录 / EXCERPT: Let G be a compact p-adic analytic group. Write Ω_G for its completed group algebra over the coefficient ring used in the paper. Call G dihedral-free if every orbital...

conjecture | open | MathDB 页面 | 原文线索

V2-011040 | MATHDB-379401

Serre's p-adic Stark conjecture at $s=1$

摘录 / EXCERPT: Let $E/\mathbb Q$ be the number field and $\bar G$ its relevant Galois group, let $E_0=\{x\in E:\operatorname{Tr}_{E/\mathbb Q}(x)=0\}$, and let μ_∞ and μ_p be the arc...

conjecture | open | MathDB 页面 | 原文线索

V2-011041 | MATHDB-379402

The Fukaya-Kato local epsilon-isomorphism conjecture

摘录 / EXCERPT: Let $L/\mathbb Q_p$ be finite, let V be a finite-dimensional L-representation of $G_{\mathbb Q_p}$, and let T be a Galois-stable $\mathcal O_L$-lattice. Let G be the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-011042 | MATHDB-379406

Arthur functoriality conjecture for Levi subgroups

摘录 / EXCERPT: Let G be a semisimple algebraic group over ${\mathbb Q}$, let L be the Levi subgroup associated with a θ-stable parabolic subalgebra $\mathfrak{q}$, let τ be a unit...

conjecture | open | MathDB 页面 | 原文线索

V2-011043 | MATHDB-379416

The conjecture on full exceptional collections for projective homogeneous spaces

摘录 / EXCERPT: Let G be a semisimple algebraic group and let G/P be a projective homogeneous space of G. A collection of vector bundles on G/P is exceptional if its derived endomorphism c...

conjecture | open | MathDB 页面 | 原文线索

V2-011044 | MATHDB-379432

Yasuda's local smoothness conjecture for higher Nash blowups

摘录 / EXCERPT: Let R be a local complete Noetherian domain with coefficient field k, and set $X:=\operatorname{Spec}R$. For every $n\in\mathbb{N}_0$, consider the n-th Nash blowup...

conjecture | open | MathDB 页面 | 原文线索

V2-011045 | MATHDB-379444

The degree conjecture for Bruhat-graph sheaves

摘录 / EXCERPT: Let $({\mathcal W},{\mathcal S})$ be a Coxeter system with length function l, and let ${\mathscr B}(x)^y$ be the stalk of the sheaf ${\mathscr B}(x)$ at $y\in{\mathcal W}$. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-011046 | MATHDB-379445

The genericity conjecture for Bruhat-graph sheaf inclusions

摘录 / EXCERPT: Let $({\mathcal W},{\mathcal S})$ be a Coxeter system with length function l, and let ${\mathscr B}(x)^{[y]}\hookrightarrow {\mathscr B}(x)^y$ be the inclusion of the costalk int...

conjecture | open | MathDB 页面 | 原文线索

V2-011047 | MATHDB-379446

The Hard Lefschetz decomposition conjecture for Bruhat-graph sheaves

摘录 / EXCERPT: Choose a generic line in $V^\ast$ and let $\rho:S\to{\mathbb R}[T]$ be the corresponding graded surjection, with T of degree 2. For an S-module M, set...

conjecture | open | MathDB 页面 | 原文线索

V2-011048 | MATHDB-379447

The Hard Lefschetz isomorphism conjecture for Bruhat-graph sheaves

摘录 / EXCERPT: Choose a generic line and the associated graded algebra ${\mathbb R}[T]$, with T of degree 2, and let ${\mathscr B}(x)^{\delta y}_T$ be the corresponding module. The Hard Lefsc...

conjecture | open | MathDB 页面 | 原文线索

V2-011049 | MATHDB-379448

The direct-sum-of-uniforms conjecture for finitely generated injective modules

摘录 / EXCERPT: Direct-sum-of-uniforms conjecture. E is a direct sum of uniform modules.

conjecture | open | MathDB 页面 | 原文线索

V2-011050 | MATHDB-379456

Small-height equidistribution conjecture for Drinfeld modules

摘录 / EXCERPT: Let $\phi:A\rightarrow K\{\tau\}$ be a Drinfeld module of generic characteristic, let $g\geq 1$, and let $\{x_k\}\subset\mathbb{G}_a^g(K^{\operatorname{sep}})$ be a strict sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-011051 | MATHDB-379499

The general ideal-lattice formula for commutative rings

摘录 / EXCERPT: The paper considers the ideal-theoretic identities established above for a commutative ring and natural number n. General ideal-lattice formula conjecture. The theorem above hold...

conjecture | open | MathDB 页面 | 原文线索

V2-011052 | MATHDB-379519

Generic p-torsion conjecture for smooth and hyperelliptic curves

摘录 / EXCERPT: Let $0\leq f<g$, let ${\cal M}_g$ be the moduli space of smooth curves of genus g, and let ${\cal H}_g$ be the hyperelliptic locus when $p\geq3$. Write $V_{g,f}$ for the locus of...

conjecture | open | MathDB 页面 | 原文线索

V2-011053 | MATHDB-379543

Strong and weak conjectures on essential specializations and the discriminant ideal

摘录 / EXCERPT: Strong and weak conjectures. The strong conjecture asserts that all essential specializations are singular. The weak conjecture asserts

conjecture | open | MathDB 页面 | 原文线索

V2-011054 | MATHDB-379563

The comonotone classification conjecture for nullity classes

摘录 / EXCERPT: Comonotone classification conjecture. If f is comonotone, then $N(f)$ is an aisle.

conjecture | open | MathDB 页面 | 原文线索

V2-011055 | MATHDB-379564

The exchange-graph independence conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}(\mathbf{x},\mathbf{y},B)$ be a cluster algebra with exchange matrix B. Its exchange graph is the graph whose vertices are seeds and whose edges join seeds relate...

conjecture | open | MathDB 页面 | 原文线索

V2-011056 | MATHDB-379567

The unique maximal-monomial conjecture for F-polynomials

摘录 / EXCERPT: Fix B^0,t_0,t, and let $F_{\ell;t}^{B^0;t_0}(y_1,\dots,y_n)$ be the associated F-polynomial. Unique maximal-monomial conjecture. Each such polynomial has a unique monomial of m...

conjecture | open | MathDB 页面 | 原文线索

V2-011057 | MATHDB-379568

The h-vector and g-vector relation conjecture

摘录 / EXCERPT: Let $t_0\xrightarrow{k}t_1$ be adjacent vertices with $B^1=\mu_k(B^0)$, and let h_k,h'_k and g_k be the integers defined from the relevant F-polynomial specializations and th...

conjecture | open | MathDB 页面 | 原文线索

V2-011058 | MATHDB-379569

The g-vector formula from a single F-polynomial

摘录 / EXCERPT: Fix B^0,t_0,ℓ,t such that $F_{\ell;t}^{B^0;t_0}\neq1$, and write $\mathbf{g}_{\ell;t}^{B^0;t_0}=(g_1,\dots,g_n)$. Let $\operatorname{Trop}(u_1,\dots,u_n)$ denote the tropical...

conjecture | open | MathDB 页面 | 原文线索

V2-011059 | MATHDB-379571

The denominator-vector conjecture for cluster variables

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra with initial seed at t_0, initial cluster $(x_{1;t_0},\dots,x_{n;t_0})$, and let $\mathbf{d}_{\ell;t}^{B^0;t_0}=(d_1,\dots,d_n)$ be the den...

conjecture | open | MathDB 页面 | 原文线索

V2-011060 | MATHDB-379574

The g-vector parametrization conjecture for cluster monomials

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra with initial seed at t_0, and let $\mathbf{g}_{\mathbf{a};t}^{B^0;t_0}$ be the $\mathbf{g}$-vector of a cluster monomial $x_{\mathbf{a};t}$...

conjecture | open | MathDB 页面 | 原文线索

V2-011061 | MATHDB-379575

The g-vector transition conjecture

摘录 / EXCERPT: Let $t_0\xrightarrow{k}t_1$ be adjacent vertices of $\mathbb{T}_n$, let $B^1=\mu_k(B^0)$, and fix $t\in\mathbb{T}_n$ and $\mathbf{a}\in\mathbb{Z}_{\geq0}^n$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-011062 | MATHDB-379576

The denominator-vector formula from F-polynomials

摘录 / EXCERPT: Fix B^0,t_0,ℓ,t such that the cluster variable $x_{\ell;t}$ is not in the initial cluster at t_0, and write its denominator vector as...

conjecture | open | MathDB 页面 | 原文线索

V2-011063 | MATHDB-379581

Rep(m): Galois representations attached to regular automorphic representations of U(m)

摘录 / EXCERPT: Let E be the quadratic imaginary field in the paper, let G_E be its absolute Galois group, and let π be an automorphic representation of $\operatorname{U}(m)$ with regular...

conjecture | open | MathDB 页面 | 原文线索

V2-011064 | MATHDB-379582

Fine deformation conjecture for self-dual Galois representations

摘录 / EXCERPT: Let E be the number field in the deformation problem, let V be an absolutely irreducible crystalline p-adic representation with the stated self-duality and distinct local dat...

conjecture | open | MathDB 页面 | 原文线索

V2-011065 | MATHDB-379589

The stabilizer conjecture for braid monodromy and algebraic monodromy

摘录 / EXCERPT: Let f be a singular function, with braid monodromy group acting by the Artin representation on its algebraic monodromy homomorphism. Stabilizer conjecture. The braid monodromy gr...

conjecture | open | MathDB 页面 | 原文线索

V2-011066 | MATHDB-379590

The Dynkin-diagram conjecture for braid monodromy groups

摘录 / EXCERPT: Let f be a singular function and let a Dynkin diagram encode its algebraic monodromy. Consider the braid subgroup associated to such a Dynkin diagram, allowing conjugation in the...

conjecture | open | MathDB 页面 | 原文线索

V2-011067 | MATHDB-379715

Bellon-Viallet algebraic-entropy conjecture

摘录 / EXCERPT: Let f be a rational map, and let its algebraic entropy be the exponential growth rate of the degrees of its iterates. Bellon-Viallet conjecture. The algebraic entropy of every ra...

conjecture | open | MathDB 页面 | 原文线索

V2-011068 | MATHDB-379718

Existence and realization of the Henselization for almost commutative local rings

摘录 / EXCERPT: Let A be an almost commutative separated local ring. Denote by $\bar{A}$ the intersection of all local Henselian rings H in the completion $\hat{A}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-011069 | MATHDB-379726

Freeness conjecture for the word quasi-symmetric functions trialgebra

摘录 / EXCERPT: Freeness conjecture. The dendriform trialgebra ${\bf WQSym}$ is free.

conjecture | open | MathDB 页面 | 原文线索

V2-011070 | MATHDB-379797

Span-intersection conjecture for upper and lower roots

摘录 / EXCERPT: Let W be a finite Coxeter group of rank n, let c be a Coxeter element, and let $(\alpha_1,\ldots,\alpha_k,\beta_1,\ldots,\beta_{n-k})$ be a c-cluster, where the α_i...

conjecture | open | MathDB 页面 | 原文线索

V2-011071 | MATHDB-379798

Upper-root description of the Cambrian cluster-noncrossing bijection

摘录 / EXCERPT: Let W be a finite Coxeter group, let c be a Coxeter element, and let a c-cluster C have upper roots among its roots. Associate to C the intersection of the hyperplanes or...

conjecture | open | MathDB 页面 | 原文线索

V2-011072 | MATHDB-379802

Scattering conjecture for critical gKdV

摘录 / EXCERPT: Scattering conjecture for critical gKdV. There exist functions

conjecture | open | MathDB 页面 | 原文线索

V2-011073 | MATHDB-379803

Scattering conjecture for critical NLS

摘录 / EXCERPT: Scattering conjecture for critical NLS. There exist functions

conjecture | open | MathDB 页面 | 原文线索

V2-011074 | MATHDB-379805

The multiplicity conjecture for standard graded algebras

摘录 / EXCERPT: Multiplicity conjecture. The inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-011075 | MATHDB-379806

The graded one-relator coherence conjecture

摘录 / EXCERPT: Let A be a connected finitely generated ${\mathbb N}$-graded algebra over a field k, with $A_0=k$. The algebra A is graded coherent if every finitely generated graded ideal i...

conjecture | open | MathDB 页面 | 原文线索

V2-011076 | MATHDB-379807

Bondal's coherence conjecture for regular algebras

摘录 / EXCERPT: Let A be a regular graded algebra, meaning that it has finite global dimension and satisfies the Gorenstein condition ... The algebra A is graded coherent if every finitely gener...

conjecture | open | MathDB 页面 | 原文线索

V2-011077 | MATHDB-379808

The Anick wildness conjecture

摘录 / EXCERPT: The Anick wildness conjecture. The automorphism ν_1 of $K\langle x,y,z\rangle$ is wild. The conjecture was later proved by Umirbaev, who established that the Anick automorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-011078 | MATHDB-379809

The Strong Anick conjecture on wild coordinates

摘录 / EXCERPT: Let $K\langle x,y,z\rangle$ be the free associative algebra over a field K. A coordinate is called wild if it is the image of a generator under a wild automorphism and no tame au...

conjecture | open | MathDB 页面 | 原文线索

V2-011079 | MATHDB-379814

Existence conjecture for ordinary rank-2 Drinfeld modules with prescribed Frobenius matrix

摘录 / EXCERPT: Let A be the coefficient ring, let Φ be a rank-2 Drinfeld A-module with characteristic ideal χ_{Φ}, and let P be the characteristic polynomial datum appearing...

conjecture | open | MathDB 页面 | 原文线索

V2-011080 | MATHDB-379815

Asymptotic cyclicity conjecture for rank-2 Drinfeld modules

摘录 / EXCERPT: Asymptotic cyclicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-011081 | MATHDB-379833

Zariski's cancellation conjecture for polynomial algebras

摘录 / EXCERPT: Let R be an algebra over a field K, let z be an indeterminate, and let $x_1,\dots,x_n$ be commuting indeterminates. Zariski's cancellation conjecture. If ... then ... The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-011082 | MATHDB-379834

Cancellation conjecture for free associative algebras

摘录 / EXCERPT: Let R be an algebra over a field K, let z be a free generator, and let $x_1,\dots,x_n$ be noncommuting free generators. Write $K\langle x_1,\dots,x_n\rangle$ for the free ass...

conjecture | open | MathDB 页面 | 原文线索

V2-011083 | MATHDB-379846

Bruns-Gubeladze tameness conjecture for polytopal algebra homomorphisms

摘录 / EXCERPT: Let k be a field and let $\operatorname{Pol}(k)$ denote the category of polytopal algebras over k. A homomorphism is called tame if it is obtained from the identity mapping...

conjecture | open | MathDB 页面 | 原文线索

V2-011084 | MATHDB-379872

Hartley's conjecture on group identities in units of group algebras

摘录 / EXCERPT: Hartley's conjecture. If the unit group $U(FG)$ satisfies a group identity, then FG satisfies a polynomial identity.

conjecture | open | MathDB 页面 | 原文线索

V2-011085 | MATHDB-379875

Infinitely many indecomposable totally reflexive modules over non-Gorenstein local rings

摘录 / EXCERPT: Let R be a non-Gorenstein local ring. A finitely generated R-module is totally reflexive if it has G-dimension zero. Infinitely many indecomposable totally reflexive modules co...

conjecture | open | MathDB 页面 | 原文线索

V2-011086 | MATHDB-379876

Schreyer's singular-locus conjecture for countable Cohen-Macaulay type

摘录 / EXCERPT: Let R be an analytic Cohen-Macaulay local ${\mathcal C}$-algebra of countable Cohen-Macaulay type. The Schreyer conjecture. R has at most a one-dimensional singular locus, name...

conjecture | solved | MathDB 页面 | 原文线索

V2-011087 | MATHDB-379877

Takahashi's infinitude conjecture for indecomposable totally reflexive modules

摘录 / EXCERPT: Let R be a non-Gorenstein local ring, and suppose that there is a nonfree totally reflexive R-module. The Takahashi conjecture. There are infinitely many isomorphism classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-011088 | MATHDB-379893

Nonexistence conjecture for Buchsbaum homogeneous integral domains of minimal multiplicity

摘录 / EXCERPT: Nonexistence conjecture. There is no Buchsbaum homogeneous integral domain over k with minimal multiplicity of degree $q \ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-011089 | MATHDB-379944

The conjecture on the polynomial part of the invariant ideal

摘录 / EXCERPT: Let $n\ge3$ and let $I_n=\langle p_n,q_n\rangle\subset\mathbf Z[z,t]$, with p_n and q_n as defined in the paper. The intersection $I_n\cap\mathbf Z[t]$ is an ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-011090 | MATHDB-379980

Upper semicontinuity of Euler-Koszul multiplicities

摘录 / EXCERPT: Let A be fixed, let L be a filtration, let τ be the corresponding face or stratum, and let $\mu^{L,\tau}_{A,0}(\beta)$ denote the multiplicity of the associated component...

conjecture | open | MathDB 页面 | 原文线索

V2-011091 | MATHDB-379984

The q-tetrahedron action conjecture for classical-parameter graphs

摘录 / EXCERPT: Let $Gamma$ be a distance-regular graph with classical parameters (D,b,α,β), where $b\ne1$, and suppose $Gamma$ is not a dual polar graph. Let T be its Terwilliger alg...

conjecture | open | MathDB 页面 | 原文线索

V2-011092 | MATHDB-379985

The q-tetrahedron action conjecture for type I Q-polynomial graphs

摘录 / EXCERPT: Let $Gamma$ be a Q-polynomial distance-regular graph with eigenvalues $theta_i$ and dual eigenvalues $theta_i^*$. For $2\leq i\leq D-1$, define ... Assume these expressions are equ...

conjecture | open | MathDB 页面 | 原文线索

V2-011093 | MATHDB-379997

The conjecture that the partial cluster algebra equals the unipotent coordinate ring

摘录 / EXCERPT: Let G be of simply-laced type A, D, or E, let J be the chosen subset of simple roots, let K be its complementary subset, and let N_K be the corresponding unipotent su...

conjecture | open | MathDB 页面 | 原文线索

V2-011094 | MATHDB-380000

Equidimensionality conjecture for restricted nilpotent centralizer intersections

摘录 / EXCERPT: Let G be a reductive group over a field k of good characteristic, with Lie algebra ${\mathfrak g}$. Let ${\cal N}_1$ be the restricted nilpotent cone, let $e\in{\cal N}_1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-011095 | MATHDB-380010

Conjecture on binomial denominators for trace-algebra Poincaré series

摘录 / EXCERPT: Binomial-denominator conjecture. The denominators $D(C_{n,2};s,t)$ and $D(T_{n,2};s,t)$ can be written as products of binomials $1-s^a t^b$, where a and b are nonnegative integ...

conjecture | open | MathDB 页面 | 原文线索

V2-011096 | MATHDB-380011

Conjecture on vanishing numerator values for trace algebras

摘录 / EXCERPT: Vanishing-numerator conjecture. For $n\ge5$,

conjecture | open | MathDB 页面 | 原文线索

V2-011097 | MATHDB-380012

Conjectural relation between denominators of two trace algebras

摘录 / EXCERPT: Denominator-relation conjecture. For all n,

conjecture | open | MathDB 页面 | 原文线索

V2-011098 | MATHDB-380013

Conjectural equality of diagonal numerator-denominator gcds

摘录 / EXCERPT: Diagonal-gcd conjecture. For all n,

conjecture | open | MathDB 页面 | 原文线索

V2-011099 | MATHDB-380035

The Tamari-interval conjecture for inverse images of non-crossing trees

摘录 / EXCERPT: Let T be a non-crossing tree of degree n, not necessarily based, viewed as an element of the operad of moulds $ obreak\operatorname{Mould}$. Let Dend denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-011100 | MATHDB-380051

Villarreal's determinant conjecture for packing-property clutters

摘录 / EXCERPT: Villarreal's determinant conjecture. If $\alpha_0(\mathcal C')=\beta_1(\mathcal C')$ for all minors $\mathcal C'$ of $\mathcal C$ and $x^{v_1},\dots,x^{v_q}$ have degree $d\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-011101 | MATHDB-380060

Structural properties conjecture for character sheaves

摘录 / EXCERPT: Let G be a connected unipotent algebraic group, let e be a minimal idempotent of ${\mathscr D}_G(G)$, let ${\mathcal M}_e^{perv}$ be the full subcategory of perverse objects in...

conjecture | open | MathDB 页面 | 原文线索

V2-011102 | MATHDB-380063

Easy-group characterization conjecture

摘录 / EXCERPT: Let H be an algebraic group over the ground field, and call it easy if every h lies in the neutral connected component of its centralizer. The easy-group characterization conje...

conjecture | open | MathDB 页面 | 原文线索

V2-011103 | MATHDB-380076

Panyushev-Premet-Yakimova conjecture on symmetric invariants of centralisers

摘录 / EXCERPT: Let $\mathfrak g$ be a finite-dimensional reductive Lie algebra of rank l over an algebraically closed field of characteristic zero. For $x\in\mathfrak g$, let $\mathfrak g_x$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-011104 | MATHDB-380104

Cyclic-word presentation conjecture for the free associative Lie algebra

摘录 / EXCERPT: Let A_n be the free associative algebra on generators $x_1,\dots,x_n$, regarded as a Lie algebra with bracket $[a,b]=a\cdot b-b\cdot a$. For a monomial...

conjecture | open | MathDB 页面 | 原文线索

V2-011105 | MATHDB-380106

Kernel ideal conjecture for the two-differential form map

摘录 / EXCERPT: Let A_n be the free associative algebra and let $\varphi(d,D):A_n\to\Omega(d,D)^{even}_*$ be the algebra map sending each generator x_i to x_i. For $m_1,m_2,m_3,m_4\in A_n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-011106 | MATHDB-380129

Existence and compatibility conjecture for type D affine Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Type D affine crystal conjecture. If $B^{r,s}$ exists, then

conjecture | open | MathDB 页面 | 原文线索

V2-011107 | MATHDB-380130

Structural properties conjecture for type D Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Let $B^{r,s}$ be a crystal of type $D_n^{(1)}$. Let $B(\Lambda)$ denote the classical D_n-crystal of highest weight Λ, and let the complement of a partition Λ in...

conjecture | open | MathDB 页面 | 原文线索

V2-011108 | MATHDB-380138

Orevkov's conjecture on empty outer ovals of M-curves with deep nests

摘录 / EXCERPT: Let A be an M-curve with a deep nest, meaning a nest with at least three levels of nesting, of arbitrary degree. Orevkov's conjecture. A must have some empty outer ovals. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-011109 | MATHDB-380141

The entropicity conjecture for idempotent algebras with the generalized entropic property

摘录 / EXCERPT: Entropicity conjecture. Every idempotent algebra (A,f) with the generalized entropic property is entropic.

conjecture | open | MathDB 页面 | 原文线索

V2-011110 | MATHDB-380180

Artin-Shioda conjecture on unirationality of supersingular K3 surfaces

摘录 / EXCERPT: Artin-Shioda conjecture. Every supersingular $K3$ surface is unirational.

conjecture | open | MathDB 页面 | 原文线索

V2-011111 | MATHDB-380188

The high-level genus conjecture for modular-tower components

摘录 / EXCERPT: High-level genus conjecture. For large k, all components of $\overline{\mathcal H}_k$ have genus exceeding 1.

conjecture | open | MathDB 页面 | 原文线索

V2-011112 | MATHDB-380198

Functoriality conjecture for Drinfeld doubles in the Calabi-Yau case

摘录 / EXCERPT: Calabi-Yau Drinfeld-double conjecture. The assignment

conjecture | open | MathDB 页面 | 原文线索

V2-011113 | MATHDB-380214

The tame-base-extension limit conjecture for Borger's Artin conductor

摘录 / EXCERPT: Let E be the local field under consideration and let G_E denote its absolute Galois group. Let ρ be a representation of G_E with finite local monodromy. For a positive i...

conjecture | open | MathDB 页面 | 原文线索

V2-011114 | MATHDB-380221

The slope conjecture for relative irregularity of fibred surfaces

摘录 / EXCERPT: Let $f\colon S \longrightarrow B$ be a fibration of genus g, and let q_f denote its relative irregularity, with $q_f<g-1$. The slope $s(f)$ is the ratio of the relevant relativ...

conjecture | open | MathDB 页面 | 原文线索

V2-011115 | MATHDB-380223

Boij-Söderberg decomposition conjecture for Cohen-Macaulay Betti diagrams

摘录 / EXCERPT: Boij-Söderberg conjecture. The Betti diagram of any Cohen-Macaulay R-module is a non-negative linear combination of pure diagrams, and every pure diagram is a rational multiple o...

conjecture | open | MathDB 页面 | 原文线索

V2-011116 | MATHDB-380224

Zanello's multiplicity bounds from the sign changes of the third difference

摘录 / EXCERPT: Zanello's conjecture. If R/I is level of codimension three, then

conjecture | open | MathDB 页面 | 原文线索

V2-011117 | MATHDB-380260

Multilinear Kashiwara-Vergne conjecture

摘录 / EXCERPT: Let $\mathfrak g$ be a finite-dimensional Lie algebra, let n be a positive integer, and let $x_1,\ldots,x_n\in\mathfrak g$. Define ... For each i, let $\partial_{x_i}F_i$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-011118 | MATHDB-380288

TL-pfaffinant Schur Q-positivity conjecture

摘录 / EXCERPT: Let P_n be the algebra containing the element f, let Pfaf'_D denote the diagram pfaffinants, and let $A_{\lambda/\mu}$ be a generalized Q-Jacobi-Trudi matrix assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-011119 | MATHDB-380294

Rubin-Stark conjecture for function fields

摘录 / EXCERPT: Let Gamma be the Galois group in the setup, let U be the relevant group of S-units, and let $\mathcal{R}_{eta}$ be the regulator map associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-011120 | MATHDB-380296

Gross's refined class number formula conjecture

摘录 / EXCERPT: Let K' be the relevant extension, let H' be its Galois group, let $I(H')$ be the augmentation ideal, and let $r_{K'}$ be the rank of the $S(K')$-unit group. Let $\theta_{H'}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-011121 | MATHDB-380308

Crystallinity conjecture for the local Galois representation attached to an unramified automorphic representation

摘录 / EXCERPT: Crystallinity conjecture. If π is unramified at p, then

conjecture | open | MathDB 页面 | 原文线索

V2-011122 | MATHDB-380310

Takahashi's finite-length module conjecture for Cohen-Macaulayness

摘录 / EXCERPT: Takahashi's conjecture. Then R is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-011123 | MATHDB-380342

The standard-expression criterion for left Noetherian differential operator rings

摘录 / EXCERPT: Let S be an affine semigroup with standard expression, and let S_2 be the semigroup appearing in the standard expressions of S and S_2 referenced in the source. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-011124 | MATHDB-380356

The strong cluster-map conjecture for unipotent cells

摘录 / EXCERPT: Strong cluster-map conjecture. For every $w\in W$, the coordinate ring $\mathbb{C}[U^w]$ is a cluster algebra modelled by the strong cluster map φ_w for the standard compon...

conjecture | open | MathDB 页面 | 原文线索

V2-011125 | MATHDB-380372

Intersection-dimension conjecture for even-dimensional admissible hypersurfaces

摘录 / EXCERPT: Intersection-dimension conjecture. If $\dim R$ is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-011126 | MATHDB-380381

The converse Latin-quandle conjecture for the quandle polynomial

摘录 / EXCERPT: Let Q be a finite quandle, and let $qp_Q(s,t)$ denote its quandle polynomial. A quandle is Latin when its operation is left-invertible, in addition to the right-invertibility req...

conjecture | open | MathDB 页面 | 原文线索

V2-011127 | MATHDB-380385

Zhang's stratiform length conjecture for division algebras

摘录 / EXCERPT: Let K be a field and let A be a finitely generated K-algebra that is a domain of GK dimension d. A chain ... of division subalgebras has stratiform length m when each D_i...

conjecture | open | MathDB 页面 | 原文线索

V2-011128 | MATHDB-380386

The division-subalgebra finite-dimensionality conjecture

摘录 / EXCERPT: Let K be a field and let A be a finitely generated K-algebra that is a domain of finite GK dimension. Let $Q(A)$ be its quotient division algebra, and let D be a division s...

conjecture | open | MathDB 页面 | 原文线索

V2-011129 | MATHDB-380392

Elashvili's conjecture on indices of centralizers

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra over an algebraically closed field of characteristic 0, and let $x\in\mathfrak{g}$. By $C_{\mathfrak{g}}(x)$ denote the centralizer...

conjecture | open | MathDB 页面 | 原文线索

V2-011130 | MATHDB-380416

The algebraicity conjecture for hereditary ordered graph speeds

摘录 / EXCERPT: Algebraicity conjecture. Every $c\in\mathcal{S}$ is either an integer or an irrational algebraic number; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-011131 | MATHDB-380453

The exchange-graph conjecture for cluster algebras

摘录 / EXCERPT: Let ${\mathcal A}$ be a cluster algebra with initial exchange matrix B. Its exchange graph is the quotient of the regular tree of seeds by simultaneous permutations of cluster va...

conjecture | open | MathDB 页面 | 原文线索

V2-011132 | MATHDB-380454

The seed-and-adjacency conjecture for cluster algebras

摘录 / EXCERPT: Let ${\mathcal A}$ be a cluster algebra of rank n, with clusters consisting of n cluster variables. Its exchange graph has clusters as vertices, up to permutation of cluster va...

conjecture | open | MathDB 页面 | 原文线索

V2-011133 | MATHDB-380455

Gordon-Martino conjecture on Calogero-Moser and Kazhdan-Lusztig cells

摘录 / EXCERPT: Gordon-Martino conjecture. There is a natural identification of the $CM_{\bf c}$-partition and the $KL_{\bf L}$-partition, induced by attaching a $KL_{\bf L}$-cell to an irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-011134 | MATHDB-380460

Crystal-basis conjecture for the type-D module V_θ

摘录 / EXCERPT: Let I be the parameter set with involution θ, and assume that no $i\in I$ satisfies $\theta(i)=i$. Let V_θ be the type-D module constructed in the source, with oper...

conjecture | open | MathDB 页面 | 原文线索

V2-011135 | MATHDB-380461

Global-crystal-basis conjecture for the type-D module V_θ

摘录 / EXCERPT: Assume that no $i\in I$ satisfies $\theta(i)=i$, and use the lattice L_θ, its bar-conjugate L_θ^-, and the integral module $(V_\theta)_\mathbf A$ defined from the bar...

conjecture | open | MathDB 页面 | 原文线索

V2-011136 | MATHDB-380462

Crystal and global-basis conjecture for type-D affine Hecke representations

摘录 / EXCERPT: Let $I=\{p^n\mathbin{;}n\in\mathbb Z_{\mathrm{odd}}\}$, with neither 1 nor -1 in I, and let $\theta(a)=a^{-1}$. Let $K_{D,I}$ be the Grothendieck group of finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-011137 | MATHDB-380476

The core formula for ideals generated by general forms

摘录 / EXCERPT: Core formula conjecture. The core of I is

conjecture | open | MathDB 页面 | 原文线索

V2-011138 | MATHDB-380485

Bass's conjecture for commutative integral domains

摘录 / EXCERPT: Let R be a commutative integral domain and G a group. For $g\in G$, write $\operatorname{HS}_{RG}(P)(g)$ for the value at the conjugacy class (g) of the Hattori-Stallings ran...

conjecture | open | MathDB 页面 | 原文线索

V2-011139 | MATHDB-380486

Bass's conjecture for fields of characteristic zero

摘录 / EXCERPT: Let F be a field of characteristic zero and G a group. Let $\operatorname{class}_F(G)_f$ denote the indicated F-vector space of finite conjugacy classes, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011140 | MATHDB-380499

Root-system realization conjecture for extended restricted roots

摘录 / EXCERPT: Suppose that $\tilde{\Phi}$ is not of type $\mathsf{B}\mathsf{C}$. Let $\tilde{\mathbf A}$ be the Cartan matrix whose realization is given by the vector space ${}^e\!{\mathfrak s}$...

conjecture | open | MathDB 页面 | 原文线索

V2-011141 | MATHDB-380517

Miller-Villarreal conjectures on generators of Gorenstein ideals

摘录 / EXCERPT: Miller-Villarreal conjectures. One always has

conjecture | open | MathDB 页面 | 原文线索

V2-011142 | MATHDB-380549

Ramakrishnan's injectivity conjecture for the Bloch regulator

摘录 / EXCERPT: Let F be a subfield of $\mathbb{C}$ with embedding $\sigma:F\hookrightarrow\mathbb{C}$. Let $\mathcal B(F)$ and $\mathcal B(\mathbb{C})$ denote the corresponding Bloch groups, an...

conjecture | open | MathDB 页面 | 原文线索

V2-011143 | MATHDB-380551

Bloch regulator injectivity conjecture

摘录 / EXCERPT: Let $\mathcal B(\mathbb C)$ be the Bloch group, defined as the kernel of the complex Dehn invariant, and let ... be the Bloch regulator map. Bloch regulator injectivity conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-011144 | MATHDB-380582

Fermionic formula for restricted one-dimensional sums

摘录 / EXCERPT: Let n and l be positive integers, and let $X^{(l)}_{ u}(\rho)$, $X^{(l)'}_{ u}(\rho)$, $F^{(l)}_{ u}(q)$, and $F^{(l)'}_{ u}(q)$ denote the restricted one-dimensional sums and...

conjecture | open | MathDB 页面 | 原文线索

V2-011145 | MATHDB-380716

Lian-Yau mirror-moonshine conjecture for algebraic K3 surfaces

摘录 / EXCERPT: Let $z(q)$ be the mirror map for a one-parameter deformation of an algebraic K3 surface arising from an orbifold construction whose Picard-Fuchs equation is third order. A Lian-Yau...

conjecture | open | MathDB 页面 | 原文线索

V2-011146 | MATHDB-380763

The Drinfeld-associator generation conjecture for unramified Tate periods

摘录 / EXCERPT: Let P be the algebra of periods, let $P_{\mathbb{Z},\mathrm{Tate}}$ be the subalgebra generated by $(2\pi i)^{\pm1}$ and periods of mixed Tate motives unramified over...

conjecture | open | MathDB 页面 | 原文线索

V2-011147 | MATHDB-380764

The torsor compatibility conjecture for period maps

摘录 / EXCERPT: Let P be the algebra of periods and let T be one of the corresponding torsors of isomorphisms, with the period construction giving a map from the period algebra to T. In the...

conjecture | open | MathDB 页面 | 原文线索

V2-011148 | MATHDB-380779

The W-algebra conjecture for the generalized Witten potential

摘录 / EXCERPT: Let $Z(\mathbf{t})$ be the exponential of the KdV_r potential function. Let W_r^+ denote the relevant algebra of differential operators, containing the Virasoro genera...

conjecture | open | MathDB 页面 | 原文线索

V2-011149 | MATHDB-380801

The orbit conjecture for real reduced double Bruhat cells

摘录 / EXCERPT: Let G be a split semisimple algebraic group with Weyl group W, and let $L^{u,v}=N\overline{u}N\cap B_-vB_-$ be the reduced double Bruhat cell for $u,v\in W$. For a reduced word...

conjecture | solved | MathDB 页面 | 原文线索

V2-011150 | MATHDB-380860

The orbit-degree decomposition conjecture for finite-field algebras

摘录 / EXCERPT: Orbit-degree decomposition conjecture. Then S_n splits as a direct sum of fields of degree $|\mathcal O_i|$ for all $i\in\{0,\dots,k\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-011151 | MATHDB-380915

The quotient conjecture for equivariant K-theory

摘录 / EXCERPT: Quotient conjecture. This composition is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-011152 | MATHDB-380939

Morales-Ramis-type conjecture for Newton systems at Darboux points

摘录 / EXCERPT: Differential Galois integrability conjecture. The identity component of the differential Galois group of the k-th order variational equations is Abelian for every...

conjecture | open | MathDB 页面 | 原文线索

V2-011153 | MATHDB-380960

Descent of the chord-diagram star product for simple groups

摘录 / EXCERPT: Let $\mathcal{M}^{G}$ be the moduli space associated with a simple group G, and let the star product be the product defined by equation on the chord-diagram algebra. Descent conj...

conjecture | open | MathDB 页面 | 原文线索

V2-011154 | MATHDB-380970

Nonnegativity conjecture for Schubert polynomials in Dunkl elements

摘录 / EXCERPT: Let $\widetilde{\cal G}_n^0$ be the quotient of $\widetilde{\cal G}_n$ by the relations $[ij]^2=0$. Let $\widetilde\theta_1,\ldots,\widetilde\theta_n$ be the Dunkl elements, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-011155 | MATHDB-380971

Finite-generation conjecture for ${\cal E}_n^t$

摘录 / EXCERPT: Let ${\cal E}_n^t$ be the quadratic algebra with central parameters $t_{ij}=[ij]^2$, and let $\mathbb{Z}[t]=\mathbb{Z}[t_{ij},\ 1\leq i<j\leq n]$. Finite-generation conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-011156 | MATHDB-380974

Finite-generation conjecture for ${\cal B}_n^t$

摘录 / EXCERPT: Let ${\cal B}_n^t$ be the quotient of the braid algebra ${\cal B}_n$ with parameters t_{ij}, and let $\mathbb{Q}[t]=\mathbb{Q}[t_{ij},\ 1\leq i<j\leq n]$. Finite-generation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-011157 | MATHDB-380976

The mixing-index conjecture for affine Demazure modules

摘录 / EXCERPT: Let B be a perfect crystal, let $\{w^{(k)}\}$ be a sequence of Weyl group elements, and let $\lambda=l\Lambda_0$ be the level-l dominant integral weight for which the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-011158 | MATHDB-380991

The generating conjecture for homogeneous invariants of the 2-state GM+I model

摘录 / EXCERPT: Generating conjecture. The ideal of homogeneous invariants for the 2-state GM+I model on T is generated by those 3×3 minors of edge flattenings that do not involve the va...

conjecture | open | MathDB 页面 | 原文线索

V2-011159 | MATHDB-381006

Finite generation of the quantization parameter semigroup

摘录 / EXCERPT: Let $\mathfrak{h}$ be the Cartan subalgebra and let Λ denote the set of parameters for which the corresponding quantization exists, viewed as a semigroup under the semigrou...

conjecture | open | MathDB 页面 | 原文线索

V2-011160 | MATHDB-381009

The vanishing conjecture for homogeneous Hessian-nilpotent quartics

摘录 / EXCERPT: Let $z=(z_1,z_2,\dots,z_n)$, let $D_i=\frac{\partial}{\partial z_i}$, and let $\Delta_n=\sum_{i=1}^n D_i^2$ be the Laplace operator on the polynomial algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-011161 | MATHDB-381029

Finite generation conjecture for intermediate groups of SL_2 over higher-dimensional domains

摘录 / EXCERPT: Finite generation conjecture. The group Γ is never finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-011162 | MATHDB-381034

The WLP characterization conjecture for level Artinian almost complete intersections

摘录 / EXCERPT: Let $R=K[x,y,z]$, and let ... be a level Artinian almost complete intersection, where $t>0$ and $0\leq\alpha\leq\beta\leq\gamma$. Assume that K is algebraically closed of character...

conjecture | open | MathDB 页面 | 原文线索

V2-011163 | MATHDB-381039

Herzog's minimality conjecture for exterior generic annihilator numbers

摘录 / EXCERPT: Herzog's minimality conjecture. For $1\leq i\leq n$ and $j\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-011164 | MATHDB-381043

Conjecture on the integer algebraic spectrum of the spin chain

摘录 / EXCERPT: Integer-spectrum conjecture. The algebraic eigenvectors span the three lowest levels of the spin chain. When $M\geq 1$, the fourth algebraic energy, $\mathsf{E}=3$, is the fifth low...

conjecture | open | MathDB 页面 | 原文线索

V2-011165 | MATHDB-381070

Compatibility conjecture for valuation ideals under tree morphisms

摘录 / EXCERPT: Let $\Delta_{\ell-1}$ and Δ_ℓ be the value groups appearing in the tight extensions, let $\hat{\mathcal P}”_{\beta\ell}$ and $\hat{\mathcal P}'_{\beta\ell}$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-011166 | MATHDB-381082

Fomin-Zelevinsky's conjectures on cluster algebras

摘录 / EXCERPT: Fomin-Zelevinsky's conjectures. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-011167 | MATHDB-381086

Goodearl's conjecture on Jacobson radicals of affine algebras of GK-dimension 2

摘录 / EXCERPT: Goodearl's conjecture. If R is an affine algebra of GK-dimension 2, then its Jacobson radical $J(R)$ is nil.

conjecture | open | MathDB 页面 | 原文线索

V2-011168 | MATHDB-381093

Classification of two-dimensional Leibniz triple systems

摘录 / EXCERPT: Classification conjecture. Every 2-dimensional Leibniz triple system over F is isomorphic to one of the systems 1-5.

conjecture | open | MathDB 页面 | 原文线索

V2-011169 | MATHDB-381118

Interior conjecture for fixed points of finitely generated Galois subgroups on curves

摘录 / EXCERPT: Let K be an infinite field, let X be a curve over K, let K^{sep} be a separable closure of K, and let G_K be the absolute Galois group of K. For...

conjecture | open | MathDB 页面 | 原文线索

V2-011170 | MATHDB-381122

The Buzzard-Diamond-Jarvis conjecture for unitary groups

摘录 / EXCERPT: Fix an imaginary CM field F in which p is unramified, let F^+ be its maximal totally real subfield, and set ... Let $({\mathbb Z}^2_+)^S_0$ consist of tuples a satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-011171 | MATHDB-381150

Bruns-Gubeladze conjecture on graded algebra retracts of polytopal algebras

摘录 / EXCERPT: Let k be a field, let A be a polytopal algebra over k, and let B be a graded algebra retract of A. Bruns-Gubeladze conjecture. B is again a polytopal algebra over k....

conjecture | open | MathDB 页面 | 原文线索

V2-011172 | MATHDB-381175

Classification of linear period 1 seed polynomials with positive coefficients

摘录 / EXCERPT: Let n be a positive integer, and let P be a linear polynomial in $x_1,\ldots,x_n$ with positive coefficients. Say that P generates a period 1 seed when it generates a seed of...

conjecture | open | MathDB 页面 | 原文线索

V2-011173 | MATHDB-381183

Polynomial-growth conjecture for divisors and mapping-class-group orbits on fully ramified Z_n curves

摘录 / EXCERPT: Let X be the Z_n curve associated to the family in the source, with parameters p and q, and let g be its genus. A divisor of degree g is non-special when it is not spec...

conjecture | open | MathDB 页面 | 原文线索

V2-011174 | MATHDB-381205

The finiteness conjecture for radical mixed difference ideals

摘录 / EXCERPT: Let k be a σ-field and R a finitely σ-generated k-σ-algebra. A σ-ideal is radical if it is radical as an ordinary ideal, and mixed if, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-011175 | MATHDB-381260

Auslander's conjecture for semisimple Hopf actions on AS regular algebras

摘录 / EXCERPT: Auslander's conjecture. The skew group ring $A\#G$ is naturally isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-011176 | MATHDB-381279

The counting conjecture for S_n-fields with local conditions

摘录 / EXCERPT: Let K be an S_n-field for $n\geq 3$ and signature (r_1,r_2). Let $\mathcal S$ be a finite set of local conditions, and let $|\mathcal S|$ denote the product of their local de...

conjecture | open | MathDB 页面 | 原文线索

V2-011177 | MATHDB-381281

Rayleigh-Ritz mixed-type majorization error bounds conjecture

摘录 / EXCERPT: Rayleigh-Ritz mixed-type majorization error bounds conjecture. The following weak majorization inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-011178 | MATHDB-381302

Extension of the Auslander theorem for noetherian graded down-up algebras

摘录 / EXCERPT: Extension conjecture. Theorem ${\rm{}}$ also holds when $\beta=-1$ and $\alpha\neq 2$. This would extend the noncommutative Auslander theorem to the remaining parameter cases for n...

conjecture | open | MathDB 页面 | 原文线索

V2-011179 | MATHDB-381338

Virtual Vafa-Witten formula for rank 2 sheaves on arbitrary surfaces

摘录 / EXCERPT: Virtual Vafa-Witten formula. $\overline{\chi}_{-y}^{\mathrm{vir}}(M)$ equals the coefficient of $x^{\operatorname{vd}(M)}$ in

conjecture | open | MathDB 页面 | 原文线索

V2-011180 | MATHDB-381358

The generalized Lax conjecture in algebraic form

摘录 / EXCERPT: Let p be a polynomial hyperbolic with respect to $e\in \mathbb{R}^n$, and let $\Lambda_+(p,e)$ denote its hyperbolicity cone. A polynomial has a definite determinantal representa...

conjecture | open | MathDB 页面 | 原文线索

V2-011181 | MATHDB-381368

Local intersection multiplicity conjecture for cuspidal curves and their 2-Hessians

摘录 / EXCERPT: Let C be a cuspidal plane algebraic curve and let H_2 be its 2-Hessian curve. At a point $p\in C$, write m for the multiplicity of C at p, δ for its delta invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-011182 | MATHDB-381411

The wreath product construction for Jordan superalgebras and Poisson superalgebras

摘录 / EXCERPT: Wreath product conjecture. The algebra J is a Jordan superalgebra. In this case, J should be called the wreath product of J_1 with H.

open_problem | open | MathDB 页面 | 原文线索

V2-011183 | MATHDB-381432

The algebraic trivial-zero conjecture for Artin representations

摘录 / EXCERPT: Let H be the number field occurring in the Artin representation ρ, let $\mathbb{Q}_\infty$ be the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011184 | MATHDB-381442

The finiteness-length formula for Abels groups over infinitely generated rings

摘录 / EXCERPT: Let R be a finitely generated commutative ring with unity that is infinitely generated as a $\b7\mathbb{Z}$-module. For $n\b7geq 2$, let $\b7\mathbf{A}_n(R)$ be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-011185 | MATHDB-381470

Uniqueness of Chevalley bases for finite-dimensional Lie (super)algebras

摘录 / EXCERPT: Let $p>2$ and let a finite-dimensional Lie algebra or Lie superalgebra have an indecomposable Cartan matrix normalized as in the source's normalization convention. A Chevalley basi...

conjecture | open | MathDB 页面 | 原文线索

V2-011186 | MATHDB-381479

The orthogonal Kronecker modules conjecture for generic Harder-Narasimhan filtrations

摘录 / EXCERPT: Let $\mathbb{F}_e$ be a Hirzebruch surface and let H_m be an arbitrary polarization. Let $\mathbf{v}\in K(\mathbb{F}_e)$ be a character such that there are $H_{\lceil m\rceil}$-p...

conjecture | open | MathDB 页面 | 原文线索

V2-011187 | MATHDB-381543

Zariski density conjecture for homogeneous polynomials on the rock-paper-scissors algebra

摘录 / EXCERPT: Let $\tilde{\mathfrak M}$ be the rock-paper-scissors algebra with unit over a field $\mathbb F$, and let p be a homogeneous polynomial in one variable whose sum of coefficients i...

conjecture | open | MathDB 页面 | 原文线索

V2-011188 | MATHDB-381565

Tate's Brumer-Stark conjecture

摘录 / EXCERPT: Let F be a number field, let H/F be a finite abelian extension with Galois group G, and let S and T be disjoint finite sets of places of F such that S contains the ar...

conjecture | open | MathDB 页面 | 原文线索

V2-011189 | MATHDB-381596

Conjecture on nonvanishing determinant conditions for lattice codes

摘录 / EXCERPT: Let $n\geq m$, let $\mathcal{L}$ be a $2mn$-dimensional lattice in $M_n(\mathbb{C})$, and suppose that ... Here $\Delta_m(\mathcal{L})$ denotes the infimum of the products of the sma...

conjecture | open | MathDB 页面 | 原文线索

V2-011190 | MATHDB-381634

The virtual-resolution minimality conjecture for multigraded regular modules

摘录 / EXCERPT: Virtual-resolution minimality conjecture. The virtual resolution of $M(\mathbf d)$ from Proposition is the minimal free resolution of $M_{\geq\mathbf d}(\mathbf d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-011191 | MATHDB-381648

Polynomial degree bound for Betti numbers of graded quotients

摘录 / EXCERPT: Polynomial Betti-bound conjecture. All Betti numbers of S/I are bounded, as functions of d, by polynomials of degree at most $p-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-011192 | MATHDB-381652

Crystal conjecture on radicality of join-meet ideals

摘录 / EXCERPT: Let $L_k(n_1,n_2,\ldots,n_k)$ be the lattice considered in the paper, and let $I_{L_k(n_1,n_2,\ldots,n_k)}$ denote its join-meet ideal in the associated polynomial ring. Crystal co...

conjecture | open | MathDB 页面 | 原文线索

V2-011193 | MATHDB-381693

The regularity-depth equality conjecture for chessboard complexes

摘录 / EXCERPT: Let $\Delta_{m,n}$ be a chessboard complex with $n\geq m\geq 1$, and let $S/{\mathcal F}(\Delta_{m,n})$ denote its facet ring. Regularity-depth equality conjecture. ... The preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-011194 | MATHDB-381696

Shank-Wehlau-Broer conjecture on modular invariant rings

摘录 / EXCERPT: Shank-Wehlau-Broer conjecture. If S^G is a direct summand of S as an S^G-module, then S^G is a polynomial ring.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-011195 | MATHDB-381702

The nontrivial-invariant conjecture for discrete-time $\mathfrak{sl}_{2}(\mathbb{R})$ coalgebra systems

摘录 / EXCERPT: Let $d>3$ and let f be the function appearing in the discrete-time $\mathfrak{sl}_{2}(\mathbb{R})$ coalgebra system. Let C denote its Casimir function. An invariant is called t...

conjecture | open | MathDB 页面 | 原文线索

V2-011196 | MATHDB-381730

The Pro-p Elementary Type Conjecture

摘录 / EXCERPT: Let F be a field containing a root of unity of order p, and let $G_F(p)$ be its maximal pro-p Galois group. Let $\mathfrak{B}$ be the class of standard building blocks and le...

conjecture | open | MathDB 页面 | 原文线索

V2-011197 | MATHDB-381732

Comparison conjecture for analytic Lubin-Tate (φ_L,Γ_L)-modules

摘录 / EXCERPT: Let L be a finite extension of $\mathbb{Q}_p$, let K be a coefficient field containing L, and let V be an L-analytic o_L-linear representation of G_L. Let M be the...

conjecture | open | MathDB 页面 | 原文线索

V2-011198 | MATHDB-381733

Perfectness conjecture for analytic (φ_L,Γ_L)-modules from $\mathcal{R}_L$

摘录 / EXCERPT: Let L be a finite extension of $\mathbb{Q}_p$, let K be a coefficient field containing L, and let M_0 be an L-analytic (φ_L,Γ_L)-module over $\mathcal{R}_L$....

conjecture | open | MathDB 页面 | 原文线索

V2-011199 | MATHDB-381734

Comparison conjecture for étale Lubin-Tate (φ_L,Γ_L)-modules

摘录 / EXCERPT: Let V be an o_L-linear representation of G_L, let $M_{\mathrm{LT}}=\mathbb{D}(V)$ be an étale L-analytic (φ_L,Γ_L)-module over $\mathbf{A}_L$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011200 | MATHDB-381749

The Carlitz zeta value Fitting-ideal conjecture for negative twists

摘录 / EXCERPT: For an integer n written as $n=\delta+h(q-1)$ by Euclidean division by $q-1$, define ... and let $P_n(x)$ be ... Let $\zeta_C^*(-n)$ be the first nonzero coefficient in the expansion...

conjecture | open | MathDB 页面 | 原文线索

V2-011201 | MATHDB-381768

Tensor product functoriality conjecture for RAECSDC representations

摘录 / EXCERPT: Let $r\geq 1$. Let F be a CM field, and let (π_1,ψ_1) and (π_2,ψ_2) be RAECSDC automorphic representations of $\mathrm{GL}_2({\mathbf A}_F)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-011202 | MATHDB-381774

The flattening surjectivity conjecture for differential-module Betti cones

摘录 / EXCERPT: Let $S=\mathbb{k}[x_1,\ldots,x_n]$ be a polynomial ring over a field $\mathbb{k}$. For an integer a, let $\mathrm{BS}_{\mathrm{mod}}(S)$ be the rational cone of Betti tables of graded finite...

conjecture | open | MathDB 页面 | 原文线索

V2-011203 | MATHDB-381775

The differential-module Boij-Söderberg decomposition conjecture

摘录 / EXCERPT: Let $S=\mathbb{k}[x_1,\ldots,x_n]$ be a polynomial ring over a field $\mathbb{k}$. A degree sequence is the sequence of generating degrees of a pure resolution, and folding such a r...

conjecture | open | MathDB 页面 | 原文线索

V2-011204 | MATHDB-381776

The categorical duality conjecture for degree-zero differential-module Betti vectors

摘录 / EXCERPT: Let $S=\mathbb{k}[x_1,\ldots,x_n]$ and $A=\mathbb{k}[t]$. Let $\mathbb{B}$ be the vector space of Betti vectors, and let $\mathcal{E}$ be a vector bundle on $\mathbb{P}^{n-1}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-011205 | MATHDB-381777

The facet conjecture for degree-zero differential-module Betti cones

摘录 / EXCERPT: Let $S=\mathbb{k}[x_1,\ldots,x_n]$ and $A=\mathbb{k}[t]$. Let $\operatorname{D^b_{DM}}(S)$ and $\operatorname{D^b_{DM}}(A)$ denote the relevant bounded derived categories of differe...

conjecture | open | MathDB 页面 | 原文线索

V2-011206 | MATHDB-381818

Gordienko-Kochetov conjecture on ordinary and differential exponents

摘录 / EXCERPT: Let L be a Lie F-algebra acting on a finite-dimensional algebra A by derivations, so that A is an L-algebra. Let $c_n(A)$ and $c_n^L(A)$ denote the ordinary and different...

conjecture | open | MathDB 页面 | 原文线索

V2-011207 | MATHDB-381859

The finite GK-dimension conjecture for Nichols algebras of diagonal type

摘录 / EXCERPT: Let (V,c) be a braided vector space of diagonal type over an algebraically closed field of characteristic zero, and let $\mathfrak{B}(V)$ denote its Nichols algebra. The finite G...

conjecture | open | MathDB 页面 | 原文线索

V2-011208 | MATHDB-381884

Bergeron-Garsia Science Fiction conjecture for intersections of Garsia-Haiman modules

摘录 / EXCERPT: Science Fiction conjecture. The bigraded $\mathfrak{S}_n$-module $\bigcap_{i=1}^{k}V_{\mu^{(i)}}$ has dimension $\frac{n!}{k}$, and its Frobenius characteristic is

conjecture | open | MathDB 页面 | 原文线索

V2-011209 | MATHDB-381897

Stability of Grassmannian cluster variables under increasing relabellings

摘录 / EXCERPT: Let $T\in\operatorname{SSYT}(k,[n])$ be a tableau whose distinct entries are $a_1<\ldots<a_r$. Let $n'\ge n$ and let $f:\{a_1,\ldots,a_r\}\to[n']$ satisfy $f(a_1)<\ldots<f(a_r)$; w...

conjecture | open | MathDB 页面 | 原文线索

V2-011210 | MATHDB-381911

The exact-one failure conjecture for the weak Lefschetz property

摘录 / EXCERPT: Exact-one failure conjecture. If there is a complete intersection $J\in CI(d,d,d,d)$ that fails WLP by more than one, then there exists a complete intersection $I\in CI(d,d,d,d)$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-011211 | MATHDB-381982

Capparelli-Meurman-Primc-Primc odd-modulus conjecture

摘录 / EXCERPT: Let $k_0,\ldots,k_n$ be non-negative integers, let E_1 be the set of parts, and let $\mathcal{Q}^{k_0,\ldots,k_n}_{\mathds{1}}$ be the partitions into parts in E_1 whose freque...

conjecture | open | MathDB 页面 | 原文线索

V2-011212 | MATHDB-381986

Mumford's conjecture on invariant complements in symmetric powers

摘录 / EXCERPT: Mumford's conjecture. For some $d\in \mathbb{Z}_{+}$, the submodule

conjecture | open | MathDB 页面 | 原文线索

V2-011213 | MATHDB-381997

The representation conjecture for $[?]$-r-consequence

摘录 / EXCERPT: Let $L gin\mathbf{ExtInt}$, let Σ_L be the set of all L-theories, and set ... Here $\mathbf{LT}_L[D]$ is the Lindenbaum-Tarski algebra relative to D, and $\mathcal{M}_L$-r...

conjecture | open | MathDB 页面 | 原文线索

V2-011214 | MATHDB-381998

The representation conjecture for $L^\ast$-consequence

摘录 / EXCERPT: Let $L\in\mathbf{ExtInt}$ and let $\mathcal{M}_L$ be the class of Lindenbaum-Tarski algebras associated with all L-theories. Define $\Vdash_{L^\ast}$ by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-011215 | MATHDB-382010

Ma's multiplicity conjecture for perfect ideals

摘录 / EXCERPT: Ma's multiplicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-011216 | MATHDB-382011

The naive generator bound for large truncations of perfect ideals

摘录 / EXCERPT: The naive generator bound. If

conjecture | open | MathDB 页面 | 原文线索

V2-011217 | MATHDB-382012

The adjusted generator bound for perfect ideals

摘录 / EXCERPT: The adjusted generator bound. If

conjecture | open | MathDB 页面 | 原文线索

V2-011218 | MATHDB-382051

Vanishing of the first big Cohen-Macaulay Hilbert coefficient and BCM-rationality

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring such that its completion $\widehat{R}$ is reduced and satisfies Serre's S_2 condition. Let B be a balanced big Cohen-Macaulay...

conjecture | open | MathDB 页面 | 原文线索

V2-011219 | MATHDB-382143

Gross's regulator conjecture for imaginary quadratic extensions

摘录 / EXCERPT: Let k be an imaginary quadratic field, let F/k be a finite abelian extension with Galois group G, let $r<0$, and let $ onzerochar\chi\in\widehat{G}$ be a character. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-011220 | MATHDB-382237

Conjectural presentation of the tamely ramified Eisenstein module

摘录 / EXCERPT: Let G be a reductive group satisfying the stated restriction that, for every root $\check{\alpha}\in\Lambda^{\vee}$, the map $\Lambda\to\mathbb{Z}$ given by...

conjecture | open | MathDB 页面 | 原文线索

V2-011221 | MATHDB-382277

The conjecture on the quotient loop-model module without through-lines

摘录 / EXCERPT: Quotient loop-model module without through-lines conjecture. For generic $\mathfrak q$, the scaling limit of the quotient loop-model module without through-lines is the direct sum...

conjecture | open | MathDB 页面 | 原文线索

V2-011222 | MATHDB-382278

The conjecture on loop-model modules with through-lines

摘录 / EXCERPT: Loop-model modules with through-lines conjecture. For $j>0$ and $2j$ through-lines, the scaling limit of $\mathscr W_{j,1}$ is the direct sum displayed above. This extends the prop...

conjecture | open | MathDB 页面 | 原文线索

V2-011223 | MATHDB-382302

Gaudin monodromy conjectures for crystal tensor products

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let $V(\lambda_1),\ldots,V(\lambda_n)$ be irreducible representations, write...

conjecture | open | MathDB 页面 | 原文线索

V2-011224 | MATHDB-382336

Classification conjecture for Frobenius automorphisms of extended structures on cyclic group algebras

摘录 / EXCERPT: Classification conjecture. The only possibilities for ϕ are

conjecture | open | MathDB 页面 | 原文线索

V2-011225 | MATHDB-382340

Heegner Point Main Conjecture for potentially ordinary elliptic curves at Eisenstein primes

摘录 / EXCERPT: Let K be the imaginary quadratic field, let $(\tilde{f},\chi_\varepsilon)$ be a Heegner pair, and let Λ be the relevant Iwasawa algebra. Let $\bT_\varepsilon$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-011226 | MATHDB-382354

The volume-free semi-algebraic chamber decomposition conjecture for K-stability

摘录 / EXCERPT: The wall-crossing setup concerns a log-bounded family $\widetilde{\mathcal{G}}$ subject to a volume condition $(\clubsuit)$, under which Theorem ... still holds without the volume co...

conjecture | open | MathDB 页面 | 原文线索

V2-011227 | MATHDB-382379

The bounded perfection exponent conjecture for Anderson modules

摘录 / EXCERPT: Let E be an Anderson A-module over a field R, and let $b(E)$ be an exponent such that the residue-in-τ pairing factors through $R^{1/q^{b(E)}}$. Let $w(E)$ range over th...

conjecture | open | MathDB 页面 | 原文线索

V2-011228 | MATHDB-382436

Hopf deformation conjecture for the zero-slope subalgebra and the positive BPS Lie algebra

摘录 / EXCERPT: Let $\mathcal{B}_{\mathbf{0}}^{+}$ be the slope subalgebra of the preprojective K-theoretic Hall algebra, and let $\mathfrak{n}_{Q}^{BPS}$ denote the positive half of the BPS Lie...

conjecture | open | MathDB 页面 | 原文线索

V2-011229 | MATHDB-382469

The finite basis conjecture for polynomial identities of 2 × 2 matrix algebras in characteristic 2

摘录 / EXCERPT: Finite basis conjecture. The algebra $M_2(K)$ does not have a finite basis of its polynomial identities.

conjecture | open | MathDB 页面 | 原文线索

V2-011230 | MATHDB-382486

The free-algebra pullback conjecture for elementary embeddings

摘录 / EXCERPT: Let $k_1,\ldots,k_n$ be elementary embeddings in $\mathcal{E}_{\lambda}$ such that $\mathcal{A}_{k_1,\ldots,k_n}$ is free, and let $\ell,p\in\mathcal{A}_{k_1,\ldots,k_n}$. The pull...

conjecture | open | MathDB 页面 | 原文线索

V2-011231 | MATHDB-382517

The asymptotic two-value conjecture for the Hilbert depth of $I_{n,n}$

摘录 / EXCERPT: Let S be the polynomial ring and $I_{n,n}$ the special class of squarefree monomial ideals considered in the paper, and let $\operatorname{hdepth}(S/I_{n,n})$ denote the Hilbert...

conjecture | open | MathDB 页面 | 原文线索

V2-011232 | MATHDB-382534

The exceptional-set characterization of Cohen-Macaulay toric ideals

摘录 / EXCERPT: Exceptional-set characterization. The exceptional set $\mathcal{E}(A)$ is empty if and only if the toric ideal I_A is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-011233 | MATHDB-382550

The core formula conjecture for ideals

摘录 / EXCERPT: Core formula conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-011234 | MATHDB-382568

Non-liftability of wild z-automorphisms from polynomial to free associative algebras

摘录 / EXCERPT: Let K be a field of characteristic 0, let $K[x,y,z]$ be the commutative polynomial algebra, and let $K\langle x,y,z\rangle$ be the free associative algebra. An automorphism is...

conjecture | open | MathDB 页面 | 原文线索

V2-011235 | MATHDB-382574

Vanishing of Hochster's theta pairing on even-dimensional isolated hypersurfaces

摘录 / EXCERPT: Let R be an admissible hypersurface, meaning that $\widehat{R}$ is a quotient of an unramified or equicharacteristic regular local ring by a nonzero element, with an isolated sin...

conjecture | open | MathDB 页面 | 原文线索

V2-011236 | MATHDB-382582

Inverse Galois problem: prove that M23 is a Galois group over Q.

摘录 / EXCERPT: M_{23} is one of the five Mathieu groups and one of the 26 sporadic finite simple groups. It has order ... and admits a natural action on 23 points. The problem asks whether th...

open_problem | claimed_solved | Shaowu Zhang | MathDB 页面 | 原文线索

V2-011237 | MATHDB-382584

F-coherent rings and regular rings

摘录 / EXCERPT: Let R be a complete local domain containing an algebraically closed field. If the perfection of R , R_{perf} (union of all p^{e} th roots, in the sense of positive characte...

open_problem | open | Shravan Patankar | MathDB 页面 | 原文线索

V2-011238 | MATHDB-382585

Bhatt-Iyengar-Ma Tor-Vanishing Problem

摘录 / EXCERPT: Question. If

open_problem | claimed_solved | Shravan Patankar | MathDB 页面 | 原文线索

V2-011239 | MATHDB-383034

Atkinson-Lloyd conjecture for primitively intransitive spaces of linear operators

摘录 / EXCERPT: Let U and V be finite-dimensional vector spaces over a field $\mathbb{F}$. Let $\mathcal{S}$ be a linear subspace of $\mathrm{Hom}(U,V)$. We say that it is intransitive when...

open_problem | claimed_progress | Clément de Seguins Pazzis | MathDB 页面 | 原文线索

V2-011240 | MATHDB-383349

Bi-UF Positive Conjecture

摘录 / EXCERPT: For every subsemiring $S\subseteq\mathbb{R}_{\ge 0}$ with $1\in S$, if the additive monoid $(S,+)$ and the nonzero multiplicative monoid $(S\setminus\{0\},\cdot)$ are both factoria...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-011241 | MATHDB-383770

Factors of cocycle twists of Sklyanin algebras (problem 1)

摘录 / EXCERPT: Let S be a 4-dimensional Sklyanin algebra, not finite over its centre. Odesskii defines a twist by the Klein four-group K_4 acts diagonally on $S \otimes_k M_2(k) = M_2(S)$. Le...

open_problem | open | MathDB 页面 | 原文线索

V2-011242 | MATHDB-383772

Colin's problems (problem 1)

摘录 / EXCERPT: Let A be a graded algebra over a finite field satisfying Artin's conditions which is birationally PI, i.e. $Q_{\text{gr}}(A)_0$ is PI. Find "weak" conditions $(*)$ so that the de...

open_problem | open | MathDB 页面 | 原文线索

V2-011243 | MATHDB-383773

Etingof's Problem (problem 1)

摘录 / EXCERPT: Let A be a filtered deformation of a commutative finitely generated domain A_0, where A_0 is a k-algebra and $\text{char } k = p$. Is A PI?

open_problem | open | MathDB 页面 | 原文线索

V2-011244 | MATHDB-383774

Sub-division rings of the Weyl algebra (problem 1)

摘录 / EXCERPT: Suppose that D is a sub-division ring of $Q(A_1)$, where A_1 is the first Weyl algebra. Is it necessarily the case that either D is isomorphic to $Q(A_1)$ or that D is PI?

open_problem | open | MathDB 页面 | 原文线索

V2-011245 | MATHDB-383775

Algebras birational to the Sklyanin algebra (problem 1)

摘录 / EXCERPT: Suppose that D is a division ring of transcendence degree 2 with a unique valuation ν such that $\mathcal{O}_\nu/\mathfrak{m}_\nu = k(E)$, where E is an elliptic curve. Sup...

open_problem | open | MathDB 页面 | 原文线索

V2-011246 | MATHDB-383776

Division rings of GK-transcendence 2 (problem 1)

摘录 / EXCERPT: Suppose that D is a finitely generated division algebra over a finite field k of GK-transcendence degree 2. Is D PI (or finite over its centre)?

open_problem | open | MathDB 页面 | 原文线索

V2-011247 | MATHDB-383777

Does this result have a noncommutative analogue?

摘录 / EXCERPT: Does this result have a noncommutative analogue? More precisely, does a similar result hold if we replace R by $k\langle x_1,\dots, x_d \rangle /(r_1, \dots, r_s)$?

open_problem | open | MathDB 页面 | 原文线索

V2-011248 | MATHDB-383778

Generalising a result of Schmidt-van den Dries (problem 2)

摘录 / EXCERPT: Does a version of the above result hold for finitely presented modules M over a finitely presented k-algebra A? That is, is torsion in M detected in bounded degree?

open_problem | open | MathDB 页面 | 原文线索

V2-011249 | MATHDB-383781

Sklyanin algebras and cocycle twists (problem 2)

摘录 / EXCERPT: Suppose A is an AS regular algebra which is Koszul, Auslander regular, Cohen-Macaulay etc. and that there exists a homogeneous central element Ω. If...

open_problem | open | MathDB 页面 | 原文线索

V2-011250 | MATHDB-383782

Sklyanin algebras and cocycle twists (problem 3)

摘录 / EXCERPT: If R_Ω is not a domain, is $Q_{\text{gr}}(R_\Omega)_0$ isomorphic to $M_2(D_{\text{Skl}})$, where D_{Skl} is the Sklyanin division algebra.

open_problem | open | MathDB 页面 | 原文线索

V2-011251 | MATHDB-383783

Completions of division rings (Todo: ask Matt) (problem 1)

摘录 / EXCERPT: Let D be a division ring of transcendence degree 2 over k, where $D = Q_{\text{gr}}(A)_0$ for some graded algebra A. Assume D has a valuations ν and that A has cubic...

open_problem | open | MathDB 页面 | 原文线索

V2-011252 | MATHDB-383784

When is A "geometrically X"? (problem 1)

摘录 / EXCERPT: Let R be a commutative domain (finitely generated over $\mathbb{Z}$?) and let A be a finitely presented R-algebra. If X is some property, we say that A is "geometrically X"...

open_problem | open | MathDB 页面 | 原文线索

V2-011253 | MATHDB-383785

Do the elliptic algebras of Feigin and Odesskii ... have any interesting factor rings of GK dimension 3, that are not a factor ring of a twisted homogeneous coordinate ring?

摘录 / EXCERPT: Do the elliptic algebras of Feigin and Odesskii $Q_{n,k}(E,\tau)$ have any interesting factor rings of GK dimension 3, that are not a factor ring of a twisted homogeneous coordinat...

open_problem | open | MathDB 页面 | 原文线索

V2-011254 | MATHDB-383786

Factor rings and σ-invariant valuations (problem 1)

摘录 / EXCERPT: Let A be a graded domain of GK dimension 3 (generated in degree 1?), so that $Q_{\text{gr}}(A) = D[x,x^{-1};\sigma]$. Is there a relation between: 1. Factor rings of A which ar...

open_problem | open | MathDB 页面 | 原文线索

V2-011255 | MATHDB-383787

Questions about GK 3 graded domains (problem 1)

摘录 / EXCERPT: Let A be a graded domain of GK dimension 3. Must A have a positive dimensional space of points? Must A have any points? What about fat points instead?

open_problem | open | MathDB 页面 | 原文线索

V2-011256 | MATHDB-383788

Questions about GK 3 graded domains (problem 2)

摘录 / EXCERPT: Does $D = Q_{\text{gr}}(A)_0$ necessarily have a valuation?

open_problem | open | MathDB 页面 | 原文线索

V2-011257 | MATHDB-383789

Does every GK 2 module map to a fat point?

摘录 / EXCERPT: Does every GK 2 module map to a fat point?

open_problem | open | MathDB 页面 | 原文线索

V2-011258 | MATHDB-383790

On sub-division rings (problem 1)

摘录 / EXCERPT: Let $D = Q(A)$ where A is a finitely generated k-algebra which is a domain of GK dimension 2. Let $D' \subseteq D$ be a sub-division ring. Does there exist A' of finite GK di...

open_problem | open | MathDB 页面 | 原文线索

V2-011259 | MATHDB-383792

Classification of Affine Anosov action

摘录 / EXCERPT: Let $G=\mathbb{Z}^{k}\times \mathbb{R}^{l}$, $k+l\ge 2$. begin{enumerate} item Which affine actions of G are Anosov? item Are there any non-standard ones? item Which ones have ra...

open_problem | open | MathDB 页面 | 原文线索

V2-011260 | MATHDB-383799

Irreducible lattices in product of rank one groups

摘录 / EXCERPT: Let $\Gamma=\textrm{SL}_{2}(\mathbb{Z}[\sqrt{2}])$. Γ acts on 4-torus by automorphism using $\mathbb{Q}$-irreducible representation defined over $\mathbb{Q}$. Is it locall...

open_problem | open | MathDB 页面 | 原文线索

V2-011261 | MATHDB-383802

Blow up procedure

摘录 / EXCERPT: Let $G=\mathbb{Z},\mathbb{Z}^{k} (k\ge 2)$ or a higher rank lattice Γ. Let $\alpha:G\to \textrm{Aut}(\mathbb{T}^{n})$ be a linear Anosov action. We can blow up along a perio...

open_problem | open | MathDB 页面 | 原文线索

V2-011262 | MATHDB-383805

Cocycle over projective action

摘录 / EXCERPT: Let Γ be a lattice in $\textrm{SL}_{n}(\mathbb{R})$ with $n\ge 3$. Let ... be a cocycle over a projective action on $\mathbb{R}P^{n-1}$ by Γ. Is β measurably coho...

open_problem | open | MathDB 页面 | 原文线索

V2-011263 | MATHDB-383814

Orbit equivalence for IMG

摘录 / EXCERPT: 1. Can we find $\varphi \in Homeo(\partial T)$ such that the orbit of the group generated by φ and $IMG(f)$ are orbit equivalent? 2. Can we make φ explicit?

open_problem | open | MathDB 页面 | 原文线索

V2-011264 | MATHDB-383815

Groups of subshifts not containing consecutive letters

摘录 / EXCERPT: begin{enumerate} item Describe relations in $G_\mathcal{F}$. item Can $\mathcal{H}_\mathcal{F}$ be algebraically interpreted as an invariant inside $G_\mathcal{F}$? item What is te...

open_problem | open | MathDB 页面 | 原文线索

V2-011265 | MATHDB-383816

Defining relations for subgroups of full group of a shift

摘录 / EXCERPT: Is it true that for all finitely generated subgroup $H \leq F(\Omega)$, H is either elementary amenable or for any K finitely presented subgroup such that...

open_problem | open | MathDB 页面 | 原文线索

V2-011266 | MATHDB-383817

Entropy of subshifts of finite type

摘录 / EXCERPT: begin{enumerate} item Given a recursively presented and finitely generated infinite amenable group G, what is E_G? item Does there exist a recursively presented and finitely ge...

open_problem | open | MathDB 页面 | 原文线索

V2-011267 | MATHDB-383819

Commutator width of Thompson groups

摘录 / EXCERPT: begin{enumerate} item Prove that ρ_1 and ρ_2 are bounded item Prove that $\rho_1 = 1$ and $\rho_2 = 3$ item Is there a finitely presented simple group with commutator wid...

open_problem | open | MathDB 页面 | 原文线索

V2-011268 | MATHDB-383821

Growths functions of repetitive graphs

摘录 / EXCERPT: What functions can be realized as growths functions of repetitive graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-011269 | MATHDB-383822

Nekrashevych C*-algebras

摘录 / EXCERPT: When is the Nekrashevych C^*-algebra of a contracting group simple?

open_problem | open | MathDB 页面 | 原文线索

V2-011270 | MATHDB-385013

Does every nuclear C-algebra belong to the bootstrap class?

摘录 / EXCERPT: Does every nuclear C-algebra belong to the bootstrap class?

open_problem | open | MathDB 页面 | 原文线索

V2-011271 | MATHDB-385014

Do all nuclear C-algebras satisfy the UCT (Universal Coefficient Theorem)?

摘录 / EXCERPT: Do all nuclear C-algebras satisfy the UCT (Universal Coefficient Theorem)?

open_problem | open | MathDB 页面 | 原文线索

V2-011272 | MATHDB-385021

Is every exact C-algebra a subalgebra of a nuclear C-algebra?

摘录 / EXCERPT: Is every exact C-algebra a subalgebra of a nuclear C-algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011273 | MATHDB-385027

Is the isomorphism relation in the following classes of separable C-algebras: ... complete analytic?

摘录 / EXCERPT: Is the isomorphism relation in the following classes of separable C-algebras: begin{itemize} item all item nuclear item exact item simple item simple exact item nuclear...

open_problem | open | MathDB 页面 | 原文线索

V2-011274 | MATHDB-385170

Is there a CAT(0) group with 2 visual boundaries so that one is locally connected and the other is not?

摘录 / EXCERPT: Is there a CAT(0) group with 2 visual boundaries so that one is locally connected and the other is not? Is there an algebraic condition on a CAT(0) group that characterizes when th...

open_problem | open | MathDB 页面 | 原文线索

V2-011275 | MATHDB-385186

Group von Neumann Algebras (problem 2)

摘录 / EXCERPT: Take $BS(m,n)=\langle a,t: ta^nt^{-1}=a^n \rangle$ with $n>|m|>2 $ and define ... $L(G(m,n))$ is known to be a type III$_{|m|/n}$. begin{enumerate} item Does the isomorphism class de...

open_problem | open | MathDB 页面 | 原文线索

V2-011276 | MATHDB-385187

Group von Neumann Algebras (problem 3)

摘录 / EXCERPT: Let $\Gamma $ be an icc property (T) group. Show that $\operatorname{Out}(L(\Gamma)) $ is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-011277 | MATHDB-385188

Group von Neumann Algebras (problem 4)

摘录 / EXCERPT: Describe the groups $\Gamma $ which $W^* $ embed into free groups, i.e.$L(\Gamma)\hookrightarrow L(\mathbb{F}_n)$.

open_problem | open | MathDB 页面 | 原文线索

V2-011278 | MATHDB-385190

Group von Neumann Algebras (problem 6)

摘录 / EXCERPT: Let Γ be a non-amenable icc group with $\beta_1^{(2)}(\Gamma)>0 $. Then $L(\Gamma) $ cannot be generated by two property (T) subfactors with diffuse intersection.

open_problem | open | MathDB 页面 | 原文线索

V2-011279 | MATHDB-385192

Structural Properties of von Neumann Algebras (problem 2)

摘录 / EXCERPT: If $\beta_1^{(2)}(\Gamma)\neq 0 $, then does it follow that $\Gamma $ is emph{Cartan rigid}.

open_problem | open | MathDB 页面 | 原文线索

V2-011280 | MATHDB-385193

Structural Properties of von Neumann Algebras (problem 3)

摘录 / EXCERPT: Let G is a totally disconnected group and $K\subseteq G $ compact. begin{enumerate} item When do the K invariant vectors in $L^2(K\backslash G) \subset L^2(G)$ generate...

open_problem | open | MathDB 页面 | 原文线索

V2-011281 | MATHDB-385194

Structural Properties of von Neumann Algebras (problem 4)

摘录 / EXCERPT: begin{enumerate} item Find two trace-scaling automorphisms of $L(\mathbb{F}_\infty)\bar\otimes \mathbb{B}(\ell^2) $ that are not outer conjugate. item Find two non-isomorphic type...

open_problem | open | MathDB 页面 | 原文线索

V2-011282 | MATHDB-385195

Structural Properties of von Neumann Algebras (problem 5)

摘录 / EXCERPT: Classify Cartan subalgebras of a II1 factor M when M does not have a unique Cartan.

open_problem | open | MathDB 页面 | 原文线索

V2-011283 | MATHDB-385197

C* Algebras (problem 2)

摘录 / EXCERPT: Find torsion-free simple groups such that $\Gamma_1\not\cong \Gamma_2 $ and $C^*_r(\Gamma_1)\cong C^*_r(\Gamma_2) $.

open_problem | open | MathDB 页面 | 原文线索

V2-011284 | MATHDB-385198

C* Algebras (problem 3)

摘录 / EXCERPT: Is $C^*_r(\mathbb{F}_\infty) $ finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-011285 | MATHDB-385199

C* Algebras (problem 4)

摘录 / EXCERPT: Is $\left(\mathbb{Z}/2\mathbb{Z} \right)^{(\mathbb{F}_2)}\rtimes (\mathbb{F}_2\times\mathbb{F}_2) $ C* super-rigid?

open_problem | open | MathDB 页面 | 原文线索

V2-011286 | MATHDB-385259

Algebraic properties (problem 10)

摘录 / EXCERPT: Which Artin groups have property R_∞?

open_problem | open | MathDB 页面 | 原文线索

V2-011287 | MATHDB-386881

Is there a positive real number which is a cyclotomic integer and is largest amongst its Galois conjugates, but which is not realized as the dimension of an object in a fusion category?

摘录 / EXCERPT: Is there a positive real number which is a cyclotomic integer and is largest amongst its Galois conjugates, but which is not realized as the dimension of an object in a fusion cate...

open_problem | open | MathDB 页面 | 原文线索

V2-011288 | MATHDB-386888

Is there an upper bound on the supertransitivity of a subfactor planar algebra?

摘录 / EXCERPT: Is there an upper bound on the supertransitivity of a subfactor planar algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011289 | MATHDB-386943

List of all question (problem 1)

摘录 / EXCERPT: For condensable algebras $A_1,A_2, \ldots, A_k \in \mathcal{C}$, sequentially condense in the following order ... What automorphisms do we get by this process? Can this yield the Flo...

open_problem | open | MathDB 页面 | 原文线索

V2-011290 | MATHDB-386951

Find a short algebraic definition of a tetracategory (preferably, something less than 10 pages long).

摘录 / EXCERPT: Find a short algebraic definition of a tetracategory (preferably, something less than 10 pages long). Current known definition can be found in 'Spans in 2-categories: A monoidal tr...

open_problem | open | MathDB 页面 | 原文线索

V2-011291 | MATHDB-386964

List of all question (problem 22)

摘录 / EXCERPT: Given two $(3+1)$D SPTs $\mathcal{A},\mathcal{B}$ with the same group G and an $\mathcal{A}$-$\mathcal{B}$ bimodule category $\mathcal{M}$ representing the boundary between them,...

open_problem | open | MathDB 页面 | 原文线索

V2-011292 | MATHDB-386969

How to define the skeleton of fusion ... -categories and their bimodules?

摘录 / EXCERPT: How to define the skeleton of fusion 2-categories and their bimodules? What is the generalization of $6j$-symbols?

open_problem | open | MathDB 页面 | 原文线索

V2-011293 | MATHDB-386985

For Rep(...), construct module categories over it.

摘录 / EXCERPT: For Rep($SU(2)$), construct module categories over it. Study its Morita dual and the associated data.

open_problem | open | MathDB 页面 | 原文线索

V2-011294 | MATHDB-386986

Provide a classification of Lagrangian submodules of a module with an anti-Hermitian form (over a commutative ring).

摘录 / EXCERPT: Provide a classification of Lagrangian submodules of a module with an anti-Hermitian form (over a commutative ring).

open_problem | open | MathDB 页面 | 原文线索

V2-011295 | MATHDB-386989

Find an example of an exact algebra that is not separable.

摘录 / EXCERPT: Find an example of an exact algebra that is not separable.

open_problem | open | MathDB 页面 | 原文线索

V2-011296 | MATHDB-386990

Provide an intrinsic characterization of exact algebras in tensor categories.

摘录 / EXCERPT: Provide an intrinsic characterization of exact algebras in tensor categories.

open_problem | open | MathDB 页面 | 原文线索

V2-011297 | MATHDB-386992

Can you find graph planar algebra embeddings for known near-group categories in small dimensions and spot a pattern?

摘录 / EXCERPT: Can you find graph planar algebra embeddings for known near-group categories in small dimensions and spot a pattern?

open_problem | open | MathDB 页面 | 原文线索

V2-011298 | MATHDB-387061

Can we categorify the Elliptic Hall algebra?

摘录 / EXCERPT: Can we categorify the Elliptic Hall algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011299 | MATHDB-387062

Can we categorify "simplier" Hall algebras such as the Hall algebra on the affine line?

摘录 / EXCERPT: Can we categorify "simplier" Hall algebras such as the Hall algebra on the affine line?

open_problem | open | MathDB 页面 | 原文线索

V2-011300 | MATHDB-387063

Categorification (problem 4)

摘录 / EXCERPT: Relate categorification of Elliptic Hall algebra to the trace of the affine Hecke category and to End$\bigg(\displaystyle \bigoplus_n SBim_n\bigg)$.

open_problem | open | MathDB 页面 | 原文线索

V2-011301 | MATHDB-387075

The group has the express goal of finding ways in which current computer algebra systems can interact with formal theorem provers, such as Lean, and vice versa.

摘录 / EXCERPT: The group has the express goal of finding ways in which current computer algebra systems can interact with formal theorem provers, such as Lean, and vice versa. The guiding motivat...

open_problem | open | MathDB 页面 | 原文线索

V2-011302 | MATHDB-391930

Stone’s conjecture on fully semimonotone Q_0-matrices

摘录 / EXCERPT: For every integer $n\ge 1$ and every matrix $A\in E_0^f\cap Q_0$ of order n, all principal minors of A are nonnegative; equivalently, $A\in P_0$, that is, $\det A[I,I]\ge 0$ fo...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-011303 | MATHDB-392306

Polynomial-extension preservation problem for the strong finite type property

摘录 / EXCERPT: For every commutative ring R, if every ideal $I\subseteq R$ contains a finitely generated ideal $J\subseteq I$ and an integer $n\ge 1$ such that $I^n\subseteq J$, then every idea...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-011304 | MATHDB-392323

Two-sidedness conjecture for null ideals of finite rings

摘录 / EXCERPT: For every finite associative unital ring R, let $N(R):=\{f(x)\in R[x]\mid f(a)=0\text{ for every }a\in R\}$ be the null ideal of R. The conjecture asserts that $N(R)$ is a two-...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-011305 | MATHDB-392501

Linear independence of polynomial compositions conjecture

摘录 / EXCERPT: For every fixed integer $m\ge 1$ and every collection of pairwise distinct nonconstant polynomials $f_1,\ldots,f_m\in\mathbb{C}[x]$, there exists an integer d_0 such that, for ev...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-011306 | MATHDB-392673

An analogue of Lie’s third theorem in the general context

摘录 / EXCERPT: Find necessary and sufficient conditions for a Lie algebra to be “integrable” to a Lie group in the general context described in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-011307 | MATHDB-392724

Complete reducibility and strong reductivity

摘录 / EXCERPT: Let G be a connected reductive linear algebraic group defined over an algebraically closed field k, and let H be a closed subgroup of G. Then H is G-completely reducibl...

open_problem | open | MathDB 页面 | 原文线索

V2-011308 | MATHDB-392725

Normal subgroups of completely reducible subgroups

摘录 / EXCERPT: Let G be a connected reductive linear algebraic group over an algebraically closed field k. Let H be a closed subgroup of G with closed normal subgroup N. If H is G-c...

open_problem | open | MathDB 页面 | 原文线索

V2-011309 | MATHDB-392749

Theorem 1.1 on the layers of the slice filtration

摘录 / EXCERPT: Let k be a perfect field and let $E\in SH(k)$. The layer ν_nE has a natural structure of an HZ-module. For the sphere spectrum $\mathbb{1}$, there is a canonical isomorphis...

open_problem | open | MathDB 页面 | 原文线索

V2-011310 | MATHDB-392784

Genus bound for curves in the algebraic part of an entire-curve current

摘录 / EXCERPT: Assume that for some positive $\[1m\alpha\[0m$ one has $T[\varphi] \geq \alpha[C]$. Must the geometric genus of the curve C be strictly less than 2?

open_problem | open | MathDB 页面 | 原文线索

V2-011311 | MATHDB-392794

Effective bound for multiple linear systems on an algebraic surface

摘录 / EXCERPT: Let X be an algebraic surface, let A be a nef and big divisor, let L be an arbitrary divisor, and let $k\ge 0$ be an integer. Define ... Assume either $n>k+B(A,L)$, or...

open_problem | open | MathDB 页面 | 原文线索

V2-011312 | MATHDB-392795

Effective Matsusaka’s Big Theorem for algebraic surfaces

摘录 / EXCERPT: In the setting of the report, if H is ample on the algebraic surface X, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-011313 | MATHDB-392930

Direct summands of serial modules

摘录 / EXCERPT: A module is serial if it is a direct sum of uniserial modules, and uniserial if its lattice of submodules is totally ordered. The question was whether every direct summand of a ser...

open_problem | open | MathDB 页面 | 原文线索

V2-011314 | MATHDB-392931

Pure-projective modules over uniserial rings

摘录 / EXCERPT: Is every pure-projective module over a uniserial ring a direct sum of finitely presented modules?

open_problem | open | MathDB 页面 | 原文线索

V2-011315 | MATHDB-392932

Krull-Gabriel dimension and domestic representation type

摘录 / EXCERPT: It is conjectured that the Krull-Gabriel dimension of a finite-dimensional algebra is finite if and only if the algebra is of domestic (or finite) representation type.

open_problem | open | MathDB 页面 | 原文线索

V2-011316 | MATHDB-392954

Characterization of scalar functions preserving the domain of a noncommutative derivation

摘录 / EXCERPT: In the setup of a separable rearrangement-invariant function space $E(0,\infty)$ over a semifinite von Neumann algebra (M,τ), let $\gamma^E=\{\gamma_t^E\}_{t\in\mathbb R}$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-011317 | MATHDB-392970

Generating series of special divisors on a Shimura curve

摘录 / EXCERPT: Let B be an indefinite quaternion division algebra over $\mathbb{Q}$, let $\mathcal{O}_B$ be a fixed maximal order, and let $\mathcal{M}$ be the associated moduli problem of pair...

open_problem | open | MathDB 页面 | 原文线索

V2-011318 | MATHDB-393010

Holomorphic structures in the low-discriminant non-algebraic case

摘录 / EXCERPT: Let X be a non-algebraic complex surface and let E be a differentiable rank-r complex vector bundle on X with $c_1(E)\in NS(X)$. Writing ... and ... determine whether bundles s...

open_problem | open | MathDB 页面 | 原文线索

V2-011319 | MATHDB-393011

Determination of the sets of holomorphic-bundle discriminants

摘录 / EXCERPT: For a non-algebraic complex surface X, determine, for $r\in\mathbb N$ and $c\in\operatorname{NS}(X)$, the set ... where ... In particular, determine its smallest element...

open_problem | open | MathDB 页面 | 原文线索

V2-011320 | MATHDB-393031

Grothendieck’s representation-theoretic question

摘录 / EXCERPT: Let $\operatorname{Rep}_A(\Gamma)$ denote the category of finitely generated A-modules with a Γ-action. Suppose that $u:\Gamma_1\to\Gamma_2$ is a homomorphism of finitely...

open_problem | open | MathDB 页面 | 原文线索

V2-011321 | MATHDB-393038

Coefficient ring of the unstabilized bivariant theory

摘录 / EXCERPT: Determine the coefficient ring $R=kk^{\mathrm{alg}}(\mathbb{C};\mathbb{C})$ for the bivariant K-theory kk^{alg} defined on arbitrary locally convex algebras.

open_problem | open | MathDB 页面 | 原文线索

V2-011322 | MATHDB-393082

Equivalence between nilpotent connections and representations of a quotient Lie algebra

摘录 / EXCERPT: Under the setting described in the report, the functor F induces an equivalence of categories between $\mathrm{Vec}_{\mathrm{nil}}$ and the category of nilpotent representations...

open_problem | open | MathDB 页面 | 原文线索

V2-011323 | MATHDB-393089

Weak Zagier conjecture

摘录 / EXCERPT: Let F be an algebraic number field. For every $k\ge 1$, there should exist a $ ...

conjecture | open | MathDB 页面 | 原文线索

V2-011324 | MATHDB-393146

Existence of limiting orbit-length distributions for birational maps over finite fields

摘录 / EXCERPT: Let L be a two-dimensional birational algebraic map with coefficients in an algebraic number field, reduced modulo the suitable primes p for which it acts on the corresponding...

open_problem | open | MathDB 页面 | 原文线索

V2-011325 | MATHDB-393147

Limiting orbit-length distribution for algebraically integrable maps

摘录 / EXCERPT: Conjecture 1(i): If L is algebraically integrable, then $D(x)$ is a step function with steps at $1/n$, for $n=1,2,\ldots$.

open_problem | open | MathDB 页面 | 原文线索

V2-011326 | MATHDB-393152

Computing isomorphisms of algebraic function fields

摘录 / EXCERPT: Let $F^{(1)}/k$ and $F^{(2)}/k$ be algebraic function fields of transcendence degree one and genus g over a perfect constant field k, with k algebraically closed in both fiel...

open_problem | open | MathDB 页面 | 原文线索

V2-011327 | MATHDB-393224

Existence of an Artin algebra that is not virtually Gorenstein

摘录 / EXCERPT: Find an Artin algebra which is not virtually Gorenstein.

open_problem | open | MathDB 页面 | 原文线索

V2-011328 | MATHDB-393227

Double Poisson structure on doubled-quiver path algebras

摘录 / EXCERPT: Let k be a field, let Q be a finite quiver, and let $\bar Q$ be its associated double quiver. The report states that the path algebra $k\bar Q$ has a natural double Poisson str...

open_problem | open | MathDB 页面 | 原文线索

V2-011329 | MATHDB-393228

Finitistic dimension conjecture

摘录 / EXCERPT: For an artin algebra A, define ... The finitistic dimension conjecture asserts that $\operatorname{fin.dim}(A)<\infty$ for every artin algebra A.

conjecture | open | MathDB 页面 | 原文线索

V2-011330 | MATHDB-393229

Finitistic dimension for artin algebras of left representation distance 3

摘录 / EXCERPT: For an artin algebra A with left representation distance $\operatorname{lrep.dis}(A)=3$, is the finitistic dimension conjecture true; that is, is...

open_problem | open | MathDB 页面 | 原文线索

V2-011331 | MATHDB-393265

Genericity of algebraic observability for continuous-time polynomial systems

摘录 / EXCERPT: Conjecture 1. Algebraic observability is a generic property for continuous-time polynomial systems (f,h) on $\mathbb{R}^n$ of bounded degrees.

open_problem | open | MathDB 页面 | 原文线索

V2-011332 | MATHDB-393270

The 56-dimensional module for $E_7(q)$ is not a $2F$-module

摘录 / EXCERPT: Let $G=E_7(q)$. Then the irreducible $\F_qG$-module of dimension 56 is not a $2F$-module.

open_problem | open | MathDB 页面 | 原文线索

V2-011333 | MATHDB-393277

Normal subgroups of G-completely reducible subgroups

摘录 / EXCERPT: Let G be a connected reductive algebraic group over an algebraically closed field k of characteristic $p>0$, and let $N \trianglelefteq H \leq G$ be closed subgroups (with the...

open_problem | open | MathDB 页面 | 原文线索

V2-011334 | MATHDB-393279

A nonsolvable quotient of a quaternion division algebra by a pure quaternion

摘录 / EXCERPT: There exists a quaternion division algebra D of characteristic zero, with center K, and a noncentral pure quaternion $e\in D$ satisfying $e^2\in K$, such that the quotient...

open_problem | open | MathDB 页面 | 原文线索

V2-011335 | MATHDB-393296

Richardson elements and parabolics given by an $\mathfrak{sl}_2$-triple

摘录 / EXCERPT: Let G be a semisimple connected, simply connected algebraic group over $\mathbb C$, and let $P\subset G$ be a parabolic subgroup with Lie algebra grading...

open_problem | open | MathDB 页面 | 原文线索

V2-011336 | MATHDB-393298

Kirillov-Reshetikhin crystals and affine Demazure crystals

摘录 / EXCERPT: For the untwisted affine Kac-Moody algebra $\widehat{\mathfrak g}$ associated with a simple complex Lie algebra $\mathfrak g$, let $c_k^\vee=a_k/a_k^\vee$ and let $l\in\mathbb N$....

open_problem | open | MathDB 页面 | 原文线索

V2-011337 | MATHDB-393315

Moving Lemma for the modified algebraic cobordism group

摘录 / EXCERPT: The canonical map $\vartheta_X:\Omega_*(X)_D\longrightarrow\Omega_*(X)$ is an isomorphism.

open_problem | open | MathDB 页面 | 原文线索

V2-011338 | MATHDB-393324

Nonnegativity of coefficients in stretching quasi-polynomials

摘录 / EXCERPT: Let λ,μ,ν be highest weights of a Lie algebra of type A_r, B_r, C_r, or D_r. For $n\in\mathbb Z_{>0}$, let $C^{n\nu}_{n\lambda,n\mu}$ denote the corresponding C...

open_problem | open | MathDB 页面 | 原文线索

V2-011339 | MATHDB-393340

Abhyankar’s sharper two-variable Jacobian-conjecture case

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, and let $f,g\in k[X,Y]$ form a jacobian pair, meaning that $J(f,g)=f_Xg_Y-f_Yg_X\in k^\times$. After the stated nor...

conjecture | open | MathDB 页面 | 原文线索

V2-011340 | MATHDB-393341

The two-dimensional Jacobian conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, and let $f,g\in k[X,Y]$. If the Jacobian $J(f,g)=f_Xg_Y-f_Yg_X$ is a nonzero constant, must $k[f,g]=k[X,Y]$?

conjecture | open | MathDB 页面 | 原文线索

V2-011341 | MATHDB-393349

Free algebraic \[?25lC^+\[?25h-actions on \[?25l\mathbb{C}^3\[?25h are translations

摘录 / EXCERPT: Every free algebraic $\mathbb{C}_+$-action on $\mathbb{C}^3$ is a translation in a suitable polynomial coordinate system.

open_problem | open | MathDB 页面 | 原文线索

V2-011342 | MATHDB-393351

Computing extension groups of rank-one MCM modules

摘录 / EXCERPT: Let k be a field, let $X=(x_{ij})$ be the generic $(n\times n)$-matrix over $S=k[x_{ij}]$, and set $R=S/(\det X)$, $L=\operatorname{coker}X$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-011343 | MATHDB-393402

A Baum-Connes assembly map and reduced C*-algebra for loop groups

摘录 / EXCERPT: Let G be a simply connected compact Lie group, and let $H=LG$ denote the loop group, with LG understood in the report as the universal central extension of the group of pointed...

open_problem | open | MathDB 页面 | 原文线索

V2-011344 | MATHDB-393421

Pseudo-nullity conjecture for the fine class-group module

摘录 / EXCERPT: In the setup of the report, let K be an imaginary quadratic field, let E/K be an elliptic curve with complex multiplication by K, and set $F_1=K(E[p^{\infty}])$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-011345 | MATHDB-393480

Computing maximal orders in quaternion algebras

摘录 / EXCERPT: Let F be a fixed number field, specified in bits by an irreducible polynomial over $\[?]Q$. Given a quaternion algebra $A=(a,b/F)$, encoded by $a,b\in F^*$, find a maximal order...

open_problem | open | MathDB 页面 | 原文线索

V2-011346 | MATHDB-393481

Integrality and algebraicity of factorial-ratio generating functions

摘录 / EXCERPT: Let $(\gamma_\nu)_{\nu\in\mathbb N}$ be a finitely supported sequence of integers, not identically zero, and define ... Assume that γ is regular, meaning ... and set ... Call...

open_problem | open | MathDB 页面 | 原文线索

V2-011347 | MATHDB-393561

Helgason conjecture for joint eigenfunctions on a symmetric space

摘录 / EXCERPT: Let $\mathcal D(X)$ be the algebra of invariant differential operators on X, and define ... The Helgason conjecture says that functions in $\mathcal E_\chi(X)$ are the Poisson tran...

conjecture | open | MathDB 页面 | 原文线索

V2-011348 | MATHDB-393587

Weak unperforation of the ordered K_0-group

摘录 / EXCERPT: Is the ordered group $K_0(A)$ of a simple real rank zero C^*-algebra A necessarily weakly unperforated?

open_problem | open | MathDB 页面 | 原文线索

V2-011349 | MATHDB-393594

Equality of the L^2-Betti numbers of a group and its group von Neumann algebra

摘录 / EXCERPT: Let $\a9$ be a discrete group and let $L\a9$ be the associated finite von Neumann algebra with its canonical trace. Is it true that $\b2_k^{(2)}(L\a9,\c4)=\b2_k^{(2)}(\a9)$?

open_problem | open | MathDB 页面 | 原文线索

V2-011350 | MATHDB-393595

When does a weakly dense *-subalgebra generate the whole von Neumann algebra?

摘录 / EXCERPT: Let 4M4 be a finite von Neumann algebra equipped with a faithful trace 4tau4, and let 4Asubseteq M4 be a weakly dense 4ast4-subalgebra. Let 4MA4 denote the smallest subalgebra of 4...

open_problem | open | MathDB 页面 | 原文线索

V2-011351 | MATHDB-393599

Steenrod’s problem for $p=2$

摘录 / EXCERPT: The report states the following classification. Suppose that P^* is a graded polynomial algebra over $\mathbb F_2$ in finitely many variables. If P^* occurs as...

open_problem | open | MathDB 页面 | 原文线索

V2-011352 | MATHDB-393617

Density theorem for singly degenerate Kleinian groups

摘录 / EXCERPT: Let Γ be a singly degenerate Kleinian group without parabolics. Then Γ is an algebraic limit of quasifuchsian groups.

open_problem | open | MathDB 页面 | 原文线索

V2-011353 | MATHDB-393658

Absolute algebraic connectivity optimization

摘录 / EXCERPT: For the undirected graph $G=(N,E)$, maximize the second-smallest Laplacian eigenvalue λ_2 over all nonnegatively weighted adjacency matrices having total weight one.

open_problem | open | MathDB 页面 | 原文线索

V2-011354 | MATHDB-393678

Coefficients of a max-algebraic characteristic polynomial

摘录 / EXCERPT: Given $A\in\mathbb{R}^{n\times n}$ and $k<n$, find a $k\times k$ principal submatrix of A whose optimal assignment problem value is maximal; denote this maximum by δ_k.

open_problem | open | MathDB 页面 | 原文线索

V2-011355 | MATHDB-393773

Rigid Lie algebra admitting nontrivial $L_\$-algebra deformations

摘录 / EXCERPT: Find a Lie algebra L which is rigid as a Lie algebra but admits deformations as an L_∞-algebra.

open_problem | open | MathDB 页面 | 原文线索

V2-011356 | MATHDB-393799

Free-product description of the absolute Galois group of the totally S-adic field

摘录 / EXCERPT: Let K be a countable Hilbertian field of characteristic 0, and let S be a finite set of inequivalent absolute values of K. For each $v\in S$, let $\widehat K_v$ be the comp...

open_problem | open | MathDB 页面 | 原文线索

V2-011357 | MATHDB-393800

Quasi-freeness of the absolute Galois group of a two-variable Laurent-series field

摘录 / EXCERPT: Let k be a field and let $K=k((x,t))$. Then the absolute Galois group $G_K=\operatorname{Gal}(K^{\mathrm{sep}}/K)$ is quasi-free of rank $\operatorname{card} K$; equivalently, ev...

open_problem | open | MathDB 页面 | 原文线索

V2-011358 | MATHDB-393803

Birational anabelian reconstruction for smooth projective curves

摘录 / EXCERPT: Conjecture. Call a field almost arbitrary (a.a.) if it is neither separably closed nor real closed and, in characteristic $p>0$, its absolute Galois group is not pro-p. Let K b...

open_problem | open | MathDB 页面 | 原文线索

V2-011359 | MATHDB-393843

Free non-commutative graded subalgebras in Ext algebras

摘录 / EXCERPT: Under the report’s standing assumptions that $(R,\mathfrak m,k)$ is a complete local ring with algebraically closed residue field, let $E=\operatorname{Ext}_R^*(k,k)$. It is conjec...

open_problem | open | MathDB 页面 | 原文线索

V2-011360 | MATHDB-393845

Are all Artin algebras virtually Gorenstein?

摘录 / EXCERPT: Are all Artin algebras virtually Gorenstein, i.e. for every Artin algebra Λ, does $P^{\prec\infty}_\Lambda=I^{\prec\infty}_\Lambda$ hold, where...

open_problem | open | MathDB 页面 | 原文线索

V2-011361 | MATHDB-393850

Ravenel’s Boolean algebra conjecture with Morava K-theories

摘录 / EXCERPT: Does the Boolean algebra theorem remain valid when the generators $\langle T(n)\rangle$ for $n\ge 0$ are replaced by $\langle K(n)\rangle$? Namely, is the Boolean subalgebra genera...

conjecture | open | MathDB 页面 | 原文线索

V2-011362 | MATHDB-393853

Derived equivalence of noncommutative crepant resolutions in dimension three

摘录 / EXCERPT: Let Λ be a module-finite algebra with $d=3$. If $M_i\in\operatorname{ref}\Lambda$ for $i=1,2$ gives a noncommutative crepant resolution...

open_problem | open | MathDB 页面 | 原文线索

V2-011363 | MATHDB-393854

Hopkins-Neeman classification of thick subcategories of perfect complexes

摘录 / EXCERPT: Let R be a commutative noetherian ring, let D be the derived category of R-modules, and let P be the full subcategory of D consisting of complexes isomorphic to perfect c...

open_problem | open | MathDB 页面 | 原文线索

V2-011364 | MATHDB-393858

Gap-labelling theorem for aperiodic systems

摘录 / EXCERPT: For the aperiodic-system algebra $A_P=C_P(\mathbb{R}^d)\rtimes\mathbb{R}^d$, the gap-labelling group $\tau_*(K_0(A_P))$ is the countable subgroup of $\mathbb{R}$ generated by the r...

open_problem | open | MathDB 页面 | 原文线索

V2-011365 | MATHDB-393890

Farrell-Jones conjecture for algebraic K-theory for word-hyperbolic groups

摘录 / EXCERPT: Let R be an associative ring with unit and let G be a word-hyperbolic group. Then the Farrell-Jones conjecture for algebraic K-theory with coefficients in R is true for G...

conjecture | open | MathDB 页面 | 原文线索

V2-011366 | MATHDB-393892

Bass Conjecture consequence for groups in \(\mathcal F\)

摘录 / EXCERPT: Suppose that $G\in\mathcal F$, where $\mathcal F$ is the family of groups for which the Farrell-Jones Conjecture for algebraic K-theory with arbitrary ring coefficients holds. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-011367 | MATHDB-393899

Classification of simple modules for cyclotomic Hecke algebras of type $G(r,p,n)$

摘录 / EXCERPT: For integers $r,p,n>2$ with $p\mid r$, let $d=r/p$, let $A=\mathbb{C}[x_1^{\pm1},\ldots,x_d^{\pm1},q^{\pm1}]$, let $K=\operatorname{Frac}(A)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-011368 | MATHDB-393909

Kato’s strengthened Lusztig character conjecture

摘录 / EXCERPT: Let G be as in the report, over an algebraically closed field of characteristic $p>0$. For every regular restricted dominant weight $\lambda\in X_1^+\cap X_{\mathrm{reg}}^+$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-011369 | MATHDB-393911

Conjecture 3.6 on modified reduced modules

摘录 / EXCERPT: Assume the ambient hypotheses that $p>2$ is odd and, if G has a component of type G_2, $p>3$. Assume further that $p>h$. For each $\[1m\lambda\in X^+_{\mathrm{reg}}\u001b[0m$,...

conjecture | open | MathDB 页面 | 原文线索

V2-011370 | MATHDB-393916

Hemmer’s signed Young module conjecture

摘录 / EXCERPT: Suppose $U\in\operatorname{mod}\text{-}k\Sigma_d$ is indecomposable and self-dual with a Specht (and hence dual Specht) filtration. Then $U\cong Y(\lambda\mid p\mu)$ for some parti...

conjecture | open | MathDB 页面 | 原文线索

V2-011371 | MATHDB-393917

Self-extensions of modules with Specht and dual Specht filtrations

摘录 / EXCERPT: Suppose $U\in \operatorname{mod}\text{-}k\Sigma_d$ is indecomposable with both Specht and dual Specht filtrations. Must ...

open_problem | open | MathDB 页面 | 原文线索

V2-011372 | MATHDB-393918

Induced-character realization for primitive idempotents in the type-\ B Mantaci\Reutenauer algebra

摘录 / EXCERPT: Let W_n be the Coxeter group of type B_n, let $\operatorname{Bip}(n)$ denote the set of bipartitions of n, and for each $\lambda\in\operatorname{Bip}(n)$ let C_λ be t...

open_problem | open | MathDB 页面 | 原文线索

V2-011373 | MATHDB-393956

Countability of height-one primes

摘录 / EXCERPT: Let k be an uncountable field and let A be a finitely generated Noetherian k-algebra of finite GK dimension. Suppose the number of height-one primes of A is countable. Is i...

open_problem | open | MathDB 页面 | 原文线索

V2-011374 | MATHDB-393959

Determine the inner automorphism in the antipode formula for noetherian AS-Gorenstein Hopf algebras

摘录 / EXCERPT: Let A be a noetherian AS-Gorenstein Hopf algebra of injective dimension d, with the notation of Corollary 9. If τ^ℓ_π and τ^r_π are respectively the left and r...

open_problem | open | MathDB 页面 | 原文线索

V2-011375 | MATHDB-393963

Rank-at-most-three 2-Calabi-Yau categories are acyclic

摘录 / EXCERPT: Assume that k is algebraically closed. Let $\d49e$ be a 2-Calabi-Yau category admitting a cluster-tilting object T. The rank of $\d49e$ is the number of pairwise indecomposab...

open_problem | open | MathDB 页面 | 原文线索

V2-011376 | MATHDB-393966

Group-theoreticality of semisimple Hopf algebras

摘录 / EXCERPT: Is every semisimple Hopf algebra over $\mathbb{C}$ group-theoretical?

open_problem | open | MathDB 页面 | 原文线索

V2-011377 | MATHDB-393968

Serre functor on the bounded derived category of equivariant perverse sheaves

摘录 / EXCERPT: Determine the Serre functor of $D^b(\operatorname{Perv}_B(G/B))$, where $G=\operatorname{SL}(n,\mathbb{C})$, B is the Borel subalgebra of all upper-triangular matrices, and...

open_problem | open | MathDB 页面 | 原文线索

V2-011378 | MATHDB-393970

Existence of an infinite-dimensional quadratic nil algebra

摘录 / EXCERPT: Does there exist an infinite-dimensional quadratic nil algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011379 | MATHDB-393982

Kernel of the arboreal quotient for \(f(x)=(x+1)^2-2\)

摘录 / EXCERPT: For the polynomial $f(x)=(x+1)^2-2$, the iterates generate fields whose associated Galois group is a quotient of the Galois group of the maximal pro-2 extension of $\mathbb{Q}$ u...

open_problem | open | MathDB 页面 | 原文线索

V2-011380 | MATHDB-393983

Dense settledness of quadratic arboreal Galois images

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x]$ be quadratic, and suppose that every iterate f^n is separable. Let G_f be the image of the associated arboreal representation...

open_problem | open | MathDB 页面 | 原文线索

V2-011381 | MATHDB-393984

Structure of maximal pro-[U+007F]p extensions of higher-dimensional local fields

摘录 / EXCERPT: Let K be an N-dimensional local field, meaning that its residue field is an $(N-1)$-dimensional local field. What can be said about the structure of the Galois group...

open_problem | open | MathDB 页面 | 原文线索

V2-011382 | MATHDB-393987

Serre’s congruence subgroup conjecture

摘录 / EXCERPT: For a connected, simply connected, absolutely almost simple algebraic group G over a global field k and a finite nonempty set S of places containing the archimedean places, t...

conjecture | open | MathDB 页面 | 原文线索

V2-011383 | MATHDB-393994

Galois realizability of the groups G_n as 3-class-tower groups

摘录 / EXCERPT: For $n\ge 2$, do the groups G_n arise as Galois groups of 3-class towers?

open_problem | open | MathDB 页面 | 原文线索

V2-011384 | MATHDB-394011

Geometric weak approximation for Del Pezzo surfaces of degree at most 3

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let B be a smooth connected curve over k, and let X be a Del Pezzo surface over $k(B)$ of degree $d\le 3$. Doe...

open_problem | open | MathDB 页面 | 原文线索

V2-011385 | MATHDB-394021

Field of definition and Galois orbit of a Beauville surface

摘录 / EXCERPT: Let S be a Beauville surface. Then S is defined over a number field.nn1. What is a field of definition of S?n2. What is the...

open_problem | open | MathDB 页面 | 原文线索

V2-011386 | MATHDB-394022

Galois orbit of a Beauville surface

摘录 / EXCERPT: Let S be a Beauville surface. What is the $\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)$-orbit of the point $[S]$ in the moduli space?

open_problem | open | MathDB 页面 | 原文线索

V2-011387 | MATHDB-394033

Intrinsicness of the Luna stratification

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let $G\to\mathrm{GL}(V)$ be a representation of a reductive linear algebraic group on a finite-dimensional vector s...

open_problem | open | MathDB 页面 | 原文线索

V2-011388 | MATHDB-394034

Index reduction for genus 1 curves

摘录 / EXCERPT: Let k be a field, let C/k be a genus 1 curve, and let A/k be a central simple algebra. Is the stronger index-reduction formula ... valid? Equivalently, can the index be compu...

open_problem | open | MathDB 页面 | 原文线索

V2-011389 | MATHDB-394037

Zero cycles of degree one and rational points on principal homogeneous spaces

摘录 / EXCERPT: Let k be a field and let G be a connected linear algebraic group defined over k. Let X be a principal homogeneous space for G defined over k which admits a zero cycle o...

open_problem | open | MathDB 页面 | 原文线索

V2-011390 | MATHDB-394042

Determine the relation module from approximations to polynomial roots

摘录 / EXCERPT: Determine Λ from decimal or p-adic approximations to the roots, without constructing G, or assuming that the characteristic polynomial is irreducible.

open_problem | open | MathDB 页面 | 原文线索

V2-011391 | MATHDB-394043

Determining the algebraic hull from \fe\\\\

摘录 / EXCERPT: The second problem is to determine $g(X)$ from $\fe\\\\$, again using approximations to the roots.

open_problem | open | MathDB 页面 | 原文线索

V2-011392 | MATHDB-394045

Arrow-class algebra and the descent algebra

摘录 / EXCERPT: For every finite Coxeter system (W,S), the image $\[?]$ of the arrow class algebra $\[?]$ under $\[?]$ is isomorphic to the descent algebra $\[?]$.

open_problem | open | MathDB 页面 | 原文线索

V2-011393 | MATHDB-394066

Vorst’s conjecture in characteristic zero

摘录 / EXCERPT: Let R be a commutative ring essentially of finite type over a field F of characteristic 0. If R is $K_{1+\dim(R)}$-regular-that is, for every $r\ge 0$, the pullback maps...

conjecture | open | MathDB 页面 | 原文线索

V2-011394 | MATHDB-394071

p-adic Beilinson conjecture for Artin motives

摘录 / EXCERPT: Let E be a number field, let $k/\mathbb{Q}$ be a finite Galois extension with group G, let π be an idempotent in $E[G]$, and let ρ be the representation of G on...

conjecture | open | MathDB 页面 | 原文线索

V2-011395 | MATHDB-394132

Hypercyclicity of the direct sum $T\oplus T$

摘录 / EXCERPT: Let X be a separable Fréchet space and let $T\in L(X)$ be hypercyclic, where $L(X)$ denotes the algebra of continuous linear operators on X. Must $T\oplus T$ be hypercyclic on...

open_problem | open | MathDB 页面 | 原文线索

V2-011396 | MATHDB-394259

Realization of centreless Lie tori by extended affine Lie algebras

摘录 / EXCERPT: Does every centreless Lie torus L arise from an extended affine Lie algebra E? If yes, how can one reconstruct E from L?

open_problem | open | MathDB 页面 | 原文线索

V2-011397 | MATHDB-394260

Isomorphism classification for the class M_2

摘录 / EXCERPT: Determine necessary and sufficient conditions for two algebras in M_2 to be isomorphic, where M_2 is the class of Lie algebras satisfying conditions (1)-(4) of Theorem 1: for a...

open_problem | open | MathDB 页面 | 原文线索

V2-011398 | MATHDB-394262

Kac-Moody Lie algebra with all Verma modules reducible

摘录 / EXCERPT: Does there exist a Kac-Moody Lie algebra such that all its Verma modules are reducible?

open_problem | open | MathDB 页面 | 原文线索

V2-011399 | MATHDB-394292

Cancellation for finitely generated 2-dimensional UFDs without a unit hypothesis

摘录 / EXCERPT: Let k be an algebraically closed field, and let A and B be finitely generated 2-dimensional unique factorization domains over k. If $A^{[1]} \cong B^{[1]}$, where $A^{[1]}$...

conjecture | open | MathDB 页面 | 原文线索

V2-011400 | MATHDB-394304

Fatou-Bieberbach neighborhoods in complements of algebraic subsets

摘录 / EXCERPT: For any point z in the complement of an algebraic subset X of $\mathbb{C}^n$ of codimension at least 2, there is a neighborhood U of z in $\\mathbb{C}^n \setminus X$ that...

open_problem | open | MathDB 页面 | 原文线索

V2-011401 | MATHDB-394312

Birational equivalence of a representation and its dual

摘录 / EXCERPT: Over an algebraically closed field k of characteristic zero, are there a connected semisimple algebraic group G and a finite-dimensional algebraic G-module V with trivial s...

open_problem | open | MathDB 页面 | 原文线索

V2-011402 | MATHDB-394313

Birational equivalence of a symmetric-power module and its twist

摘录 / EXCERPT: Let $G=SL_d/\mu_d$. Let V be the d-th symmetric power of k^d endowed with the natural action of G. Let σ be the automorphism of G induced by the automorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-011403 | MATHDB-394315

Cyclic quotient-group conjecture for two-dimensional reductive-group quotients

摘录 / EXCERPT: Suppose a complex reductive algebraic group G acts algebraically on $\C^n$ and the categorical quotient $\C^n // G$ is two-dimensional. In the situation where $\C^n // G$ is isom...

conjecture | open | MathDB 页面 | 原文线索

V2-011404 | MATHDB-394317

Zaidenberg’s variable-sum problem

摘录 / EXCERPT: Let R be a commutative ring with unity, and let $x_1,\ldots,x_m$ and $y_1,\ldots,y_n$ be disjoint sets of indeterminates. If $p\in R[x_1,\ldots,x_m]$ and $q\in R[y_1,\ldots,y_n]$...

open_problem | open | MathDB 页面 | 原文线索

V2-011405 | MATHDB-394322

Recovering fields from absolute Galois groups

摘录 / EXCERPT: If K is finitely generated over $\a0\mathbb{Q}$ and K' is a field with $G_{K'} \cong G_K$, then K' has a henselian valuation with residue field K and divisible value group.

open_problem | open | MathDB 页面 | 原文线索

V2-011406 | MATHDB-394324

Decidability of the theory of finite-dimensional \(\mathfrak{sl}_2(k)\)-modules

摘录 / EXCERPT: Determine whether the theory of finite-dimensional modules over the universal enveloping algebra U of $\mathfrak{sl}_2(k)$ is decidable.

open_problem | open | MathDB 页面 | 原文线索

V2-011407 | MATHDB-394331

Cotorsion modules over non-coherent rings

摘录 / EXCERPT: Are there non-coherent rings over which every cotorsion module is pure-injective?

open_problem | open | MathDB 页面 | 原文线索

V2-011408 | MATHDB-394332

Characterization of the co-X-dcc property

摘录 / EXCERPT: Let R be a ring and let X be a class of left R-modules axiomatized by pp implications. Then $R\text{-Mod}$ has the co-X-dcc property if and only if all cokernels of pure-in...

open_problem | open | MathDB 页面 | 原文线索

V2-011409 | MATHDB-394412

Non-algebraic patchworked pseudoholomorphic curves in \(\Sigma_2\)

摘录 / EXCERPT: For every integer $d\ge 3$, there exists a smooth real pseudoholomorphic patchworked curve of bidegree $(d,0)$ in Σ_2 that is not $\mathcal L$-isotopic to any real algebraic...

open_problem | open | MathDB 页面 | 原文线索

V2-011410 | MATHDB-394413

Nash-isomorphic irreducible models for real algebraic sets

摘录 / EXCERPT: Every real algebraic set of positive dimension is Nash isomorphic to an irreducible real algebraic set.

open_problem | open | MathDB 页面 | 原文线索

V2-011411 | MATHDB-394418

Pierce-Birkhoff conjecture (abstract version)

摘录 / EXCERPT: Let R be a real closed field and let $A=R[x_1,\ldots,x_n]$. A function $f:R^n\to R$ is piecewise polynomial if R^n is covered by finitely many closed semialgebraic sets P_i s...

conjecture | open | MathDB 页面 | 原文线索

V2-011412 | MATHDB-394483

The abc conjecture for the number field K

摘录 / EXCERPT: Let K be an algebraic number field of degree n, and let M_K be its set of places. For $a,b,c\in K^*$, define ... where the valuations are normalized in the usual way. Define .....

conjecture | open | MathDB 页面 | 原文线索

V2-011413 | MATHDB-394499

Function-field analogue of Hermite’s constant

摘录 / EXCERPT: Let K be a finite algebraic extension of $\\mathbb{F}_q(X)$, with constant field exactly $\mathbb{F}_q$, and let g be the genus of K. For $A\in\operatorname{GL}_n(K_A)$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-011414 | MATHDB-394505

Annihilators distinguishing finite-dimensional irreducible modules over a finite W-algebra

摘录 / EXCERPT: We conjecture that two irreducible finite-dimensional H_χ-modules V_1 and V_2 are isomorphic if and only if ...

open_problem | open | MathDB 页面 | 原文线索

V2-011415 | MATHDB-394506

Reduction of Courant brackets from reduction data

摘录 / EXCERPT: Let G be a Lie group with Lie algebra $\mathfrak g$, let M carry the natural Courant algebroid $TM\oplus T^*M$ defined using a closed 3-form H, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-011416 | MATHDB-394507

True Hopf algebra conjecture for deformation quantization of Poisson-Lie groups

摘录 / EXCERPT: In the case of a Poisson-Lie group, and in the absence of anomalies, the deformation quantization produced by the described procedure is conjectured to yield a true Hopf algebra if...

conjecture | open | MathDB 页面 | 原文线索

V2-011417 | MATHDB-394512

Deformation quantization of principal fiber bundles

摘录 / EXCERPT: For every principal fiber bundle $\operatorname{pr}:P\to M$ with structure Lie group G and every star product $\star$ on M, does there exist a right $\star$-module structure...

open_problem | open | MathDB 页面 | 原文线索

V2-011418 | MATHDB-394529

Inverse problem of differential Galois theory over \(\mathbb{R}(z)\)

摘录 / EXCERPT: Given any linear algebraic group G over $\mathbb{R}$, there exists a Picard-Vessiot extension $E/\mathbb{R}(z)$ whose differential Galois group is isomorphic to G.

open_problem | open | MathDB 页面 | 原文线索

V2-011419 | MATHDB-394530

Generic inverse problem in differential Galois theory

摘录 / EXCERPT: Given a linear algebraic group G over an algebraically closed field of constants C, does there exist a differential field F with field of constants C and a Picard-Vessiot e...

open_problem | open | MathDB 页面 | 原文线索

V2-011420 | MATHDB-394531

Accessory parameter problem in positive characteristic

摘录 / EXCERPT: Let $k=\bar{k}$ be an algebraically closed field of positive characteristic p, let $X=\mathbf{P}^1_k$, and choose a parameter t on X. Consider differential operators...

open_problem | open | MathDB 页面 | 原文线索

V2-011421 | MATHDB-394533

Differential Abhyankar Conjecture

摘录 / EXCERPT: Let $K\ne\mathbb{F}_p$ be an algebraically closed field of characteristic $p>0$. Let X be a smooth projective curve over K, let $S\subset X$ be a finite subset, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011422 | MATHDB-394538

Grothendieck’s Section Conjecture for projective hyperbolic curves over \(\mathbb Q\)

摘录 / EXCERPT: Let X be a smooth absolutely irreducible projective algebraic curve over $\mathbb Q$, with genus $g\ge 2$, and fix a rational point $x_0\in X(\mathbb Q)$. Then the Kummer map ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-011423 | MATHDB-394569

Sharifi’s reciprocity-map and two-variable p-adic L-function conjecture (Conjecture 2)

摘录 / EXCERPT: There exists an isomorphism $\phi:X_K^-\xrightarrow{\sim}Y^-/IY^-$ of Λ-modules such that, letting...

conjecture | open | MathDB 页面 | 原文线索

V2-011424 | MATHDB-394587

Existence of multi-output Boolean functions with maximal algebraic immunity

摘录 / EXCERPT: For all integers $1 \le m \le n$, there exists a function $f:\{0,1\}^n \to \{0,1\}^m$ whose algebraic immunity is $d^*(n,m)$, where $d^*(n,m)$ is the least integer d such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-011425 | MATHDB-394662

Conceptual proof for the double-centralizer results

摘录 / EXCERPT: Provide a conceptual proof of the results above, in particular Theorem 1: let G be a simple algebraic group of exceptional type over an algebraically closed field k of characte...

open_problem | open | MathDB 页面 | 原文线索

V2-011426 | MATHDB-394668

Integral-domain conjecture for the orbit algebra

摘录 / EXCERPT: Let G be a permutation group on a set $\d4aa$. For each $n\ge 0$, let V_n be the complex vector space of all G-invariant functions from the set of n-element subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-011427 | MATHDB-394688

Malcev Lie algebras of global 1-formal groups with positive deficiency

摘录 / EXCERPT: Let G be a 1-formal group with positive deficiency. If G is global, then $E_G \cong E_{\pi_1(M)}$, where M is either a smooth complex algebraic curve or the complement of a...

open_problem | open | MathDB 页面 | 原文线索

V2-011428 | MATHDB-394704

Weyl’s law for cusp forms on congruence quotients

摘录 / EXCERPT: Let G be a split adjoint semisimple algebraic group over $\mathbb Q$, let $G=G(\mathbb R)$, let $K\subset G$ be a maximal compact subgroup, and let $\Gamma\subset G(\mathbb Q)$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-011429 | MATHDB-394706

Occurrence of logarithmic terms in the blown-up distance expansion

摘录 / EXCERPT: For semi-algebraic X, do logarithmic terms actually occur in the expansion at Z of the polyhomogeneous conormal functions $ \pi^*I$ and $ \pi^*d$?

open_problem | open | MathDB 页面 | 原文线索

V2-011430 | MATHDB-394731

Aubert-Baum-Plymen geometric conjecture: equivalence of the component ideal with the ordinary quotient algebra

摘录 / EXCERPT: Let G be a reductive p-adic group, and let a connected component of the primitive ideal spectrum of its Hecke algebra H_G be given. The conjecture asserts that the ideal in...

conjecture | open | MathDB 页面 | 原文线索

V2-011431 | MATHDB-394752

Hopkins’ conjecture on finite-subgroup-equivariant generators for Morava E-theory

摘录 / EXCERPT: Conjecture (Hopkins). If $G\subset S_n$ is a finite subgroup, then there is a G-equivariant isomorphism ... where S denotes the symmetric-algebra functor, M_n is placed in degr...

conjecture | open | MathDB 页面 | 原文线索

V2-011432 | MATHDB-394759

Existence of self-similar solutions with algebraic decay

摘录 / EXCERPT: For the self-similar coagulation equation (6) in the report, the authors conjecture that solutions with algebraic decay also exist.

open_problem | open | MathDB 页面 | 原文线索

V2-011433 | MATHDB-394764

Dimension of the tautological ring of a general Jacobian

摘录 / EXCERPT: Let C be a general smooth irreducible curve of genus g over an algebraically closed field k, let J be its Jacobian, and let R be the smallest subring of...

open_problem | open | MathDB 页面 | 原文线索

V2-011434 | MATHDB-394765

Structure of the tautological ring of a general Jacobian

摘录 / EXCERPT: Let C) be a smooth irreducible curve of genus g over an algebraically closed field, let J be its Jacobian, and let $R=\bigoplus_{i,j}R_i^{(j)}$ be the tautological ring in...

open_problem | open | MathDB 页面 | 原文线索

V2-011435 | MATHDB-394769

Faithfulness of the absolute Galois action on rigid surfaces

摘录 / EXCERPT: Is the action of $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ faithful on rigid surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-011436 | MATHDB-394774

Algebraic fundamental groups of surfaces with small K^2

摘录 / EXCERPT: Let S be a surface with $K^2<3\c(S)$ having no irregular étale cover. Is it true that, up to a finite number of exceptions, $\c\c_1^{\mathrm{alg}}(S)$ is a subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-011437 | MATHDB-394817

Automorphic realization of a motive via a smallest connected reductive group

摘录 / EXCERPT: Let E and F be number fields, and let M be a motive over F with coefficients in E. Find a connected reductive algebraic group G defined over F, an automorphic represe...

open_problem | open | MathDB 页面 | 原文线索

V2-011438 | MATHDB-394822

Congruence primes of Ikeda lifts under special-value hypotheses

摘录 / EXCERPT: Let k,n be positive even integers, let $f\in S_{2k-n}(\Gamma_1)$ be a primitive form, and let K be an algebraic number field containing the Hecke fields of all primitive forms...

open_problem | open | MathDB 页面 | 原文线索

V2-011439 | MATHDB-394833

General structure of moving-hypersurface ideals for parametrized surfaces

摘录 / EXCERPT: Understand the structure, in the general case, of the ideal of relations defining the Rees algebra associated with a parametrized surface; the report states that the surface case i...

open_problem | open | MathDB 页面 | 原文线索

V2-011440 | MATHDB-394843

Brauer-Witt analogue for projective Schur algebras

摘录 / EXCERPT: Every element of the projective Schur group $PS(k)$ may be represented by a radical, and even by a radical abelian, algebra over k.

open_problem | open | MathDB 页面 | 原文线索

V2-011441 | MATHDB-394846

Invariant basis number for skew group algebras of fields

摘录 / EXCERPT: Let K be a field and let $t:G\to\operatorname{Aut}(K)$ be an action of a group G on K. Does the skew group algebra K_tG have the invariant basis number property; that is, d...

open_problem | open | MathDB 页面 | 原文线索

V2-011442 | MATHDB-394848

Modular group algebra isomorphism under a normal-complement hypothesis

摘录 / EXCERPT: Let G be a finite p-group and let k be the field with p elements. Suppose that G has a normal complement in the unit group $U(kG)$ of the group algebra kG, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-011443 | MATHDB-394859

Derived-series subgroups from commutator powers of augmentation ideals

摘录 / EXCERPT: Let G be a free group. For a commutative ring R, let $\Delta(G)=\Delta_R(G)$ denote the augmentation ideal of RG, set $\delta_1(RG)=[\Delta(G),\Delta(G)]_{RG}$, and define...

open_problem | open | MathDB 页面 | 原文线索

V2-011444 | MATHDB-394943

Conjecture (lemme fondamental) for unramified endoscopic Hecke algebras

摘录 / EXCERPT: Let F be a non-Archimedean local field of characteristic zero, let G be a reductive group over F, and let $(H,s,\hat\xi)$ be an endoscopic datum for G. Suppose that G and...

conjecture | open | MathDB 页面 | 原文线索

V2-011445 | MATHDB-394947

Extension of Theorem A without hypotheses (a) and (b)

摘录 / EXCERPT: In the setup of the report-where B_0 is an indefinite rational quaternion algebra, F is a real quadratic field with $\mathcal O_F=\mathbb Z[\sqrt{\Delta}]$, all primes dividing...

open_problem | open | MathDB 页面 | 原文线索

V2-011446 | MATHDB-394971

Global dimensions of endomorphism rings for hereditary Artin algebras

摘录 / EXCERPT: Let Λ be a hereditary Artin algebra and let $d\ge 3$ be in $\mathbb{N}\cup\{\infty\}$. There exists a Λ-module M, both a generator and a cogenerator, such that...

open_problem | open | MathDB 页面 | 原文线索

V2-011447 | MATHDB-394976

Algebra isomorphism for the coherent-sheaf Ext algebra

摘录 / EXCERPT: Conjecture: There is an isomorphism of algebras ... where, in the surrounding discussion, A and B range over crossingless matchings of $2k$ points.

open_problem | open | MathDB 页面 | 原文线索

V2-011448 | MATHDB-394978

Injectivity of the tilting-module map ψ

摘录 / EXCERPT: Let Q be a finite quiver without oriented cycles, let k be a ground field, and let $[U+007F]\operatorname{Tilt} Q$ denote the set of isomorphism classes of tilting modules in...

open_problem | open | MathDB 页面 | 原文线索

V2-011449 | MATHDB-394979

Generalized-standard component inequality over arbitrary fields

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field K, let $ \mathcal{C}$ be a generalized standard family of almost cyclic coherent components of the Auslander-Reiten quiver...

open_problem | open | MathDB 页面 | 原文线索

V2-011450 | MATHDB-394980

Parameterization of simple modules for affine cellular algebras

摘录 / EXCERPT: Let A be an affine cellular algebra with a cell chain $0=J_0\subset J_1\subset\cdots\subset J_n=A$ such that J_j/J_{j-1} has sandwich matrix $(\psi^{(j)}_{st})$. Then there is...

open_problem | open | MathDB 页面 | 原文线索

V2-011451 | MATHDB-395038

Separability-free version of Proposition 6

摘录 / EXCERPT: Does there exist a reductive algebraic group G over an algebraically closed field k of characteristic $p>0$ and a subgroup $H\leq G$, with no separability assumption on H, su...

open_problem | open | MathDB 页面 | 原文线索

V2-011452 | MATHDB-395051

Is the natural epimorphism for the Frobenius-algebra skein module an isomorphism?

摘录 / EXCERPT: For each closed oriented genus-g surface Σ_g, there is a natural epimorphism of algebras ... where $T(A)$ is the tensor algebra of A and $\big(\!\big(\cdot\big)\!\big)$ de...

open_problem | open | MathDB 页面 | 原文线索

V2-011453 | MATHDB-395194

Modularity of 2-adic Galois representations

摘录 / EXCERPT: Let $\rho:G_{\mathbb Q}\to\mathrm{GL}_2(\mathbb Q_2)$ be a continuous odd representation, and let $\bar\rho$ be its reduction, defined up to semisimplification. Assume that...

open_problem | open | MathDB 页面 | 原文线索

V2-011454 | MATHDB-395203

Extension of Theorem 4 to infinite spaces

摘录 / EXCERPT: Determine to what extent Theorem 4 extends from finite T_{0}-spaces to infinite spaces X. Theorem 4 asserts that every separable nuclear C^{*}-algebra over a finite T_{0}-s...

open_problem | open | MathDB 页面 | 原文线索

V2-011455 | MATHDB-395206

Permanence of the Bost conjecture under passage to closed subgroups

摘录 / EXCERPT: Let G be a locally compact second countable Hausdorff group and let H be a closed subgroup of G. If the Bost conjecture with arbitrary separable C^*-algebra coefficients is...

conjecture | open | MathDB 页面 | 原文线索

V2-011456 | MATHDB-395209

Simplicity and unique trace for reduced group C*-algebras

摘录 / EXCERPT: Does there exist a group Γ with amenable radical $R_a(\Gamma)=\{e\}$ for which the reduced C-algebra $C_r^*(\Gamma)$ is not simple or does not have a unique tracial state?

open_problem | open | MathDB 页面 | 原文线索

V2-011457 | MATHDB-395210

No faithful trace on the full group C*-algebra of \(\mathrm{SL}_4(\mathbb Z)\)

摘录 / EXCERPT: There is no faithful tracial state (faithful trace) on the full C-algebra $C^*(\mathrm{SL}_4(\mathbb Z))$.

open_problem | open | MathDB 页面 | 原文线索

V2-011458 | MATHDB-395212

Fiberwise KK-equivalence of continuous fields of C*-algebras

摘录 / EXCERPT: Let A and B be separable nuclear continuous $C(X)$-algebras over a finite-dimensional compact metrizable space X. If $\sigma \in KK_X(A,B)$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-011459 | MATHDB-395215

Recovery of the Cuntz semigroup from K-theory and traces for $\mathcal{Z}$-stable algebras

摘录 / EXCERPT: Let A be a unital simple exact finite C^*-algebra which absorbs the Jiang-Su algebra $\mathcal{Z}$ tensorially. Let $W(A)$ be the Cuntz semigroup, let $V(A)$ be the semigroup o...

open_problem | open | MathDB 页面 | 原文线索

V2-011460 | MATHDB-395219

Spectra of MASAs in the CAR algebra

摘录 / EXCERPT: Which compact metric spaces can occur as the spectra of MASAs of the CAR algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011461 | MATHDB-395221

K1-injectivity of unital properly infinite C*-algebras

摘录 / EXCERPT: Is every unital properly infinite C-algebra K1-injective; equivalently, is the canonical map $U(A)/U_0(A)\to K_1(A)$ injective for every unital properly infinite C-algebra A?

open_problem | open | MathDB 页面 | 原文线索

V2-011462 | MATHDB-395222

Stability of proper infiniteness under matrix amplification

摘录 / EXCERPT: For a unital C^*-algebra A, if $M_n(A)$ is properly infinite for some n, must A itself be properly infinite?

open_problem | open | MathDB 页面 | 原文线索

V2-011463 | MATHDB-395223

The Effros-Ruan conjecture for bilinear forms on C*-algebras

摘录 / EXCERPT: The Effros-Ruan conjecture states that if A and B are C^*-algebras and $u:A\times B\to\mathbb C$ is a jointly completely bounded bilinear form, then there exist states...

conjecture | open | MathDB 页面 | 原文线索

V2-011464 | MATHDB-395330

Classification of algebraic irreducible GKZ-systems

摘录 / EXCERPT: For $\[?m\alpha\in\mathbb{Q}^r$[0m, classify the irreducible GKZ-systems whose complete set of solutions is algebraic over $\[?m\mathbb{C}(v_1,\ldots,v_N)$[0m; in particular, obtai...

open_problem | open | MathDB 页面 | 原文线索

V2-011465 | MATHDB-395331

Algebraicity criterion for irreducible GKZ systems

摘录 / EXCERPT: Let $\b1\in\mathbb{Q}^r$ and suppose that the GKZ-system is irreducible. Let D be the common denominator of the coordinates of $\b1$. Then the solution set of the GKZ-system cons...

open_problem | open | MathDB 页面 | 原文线索

V2-011466 | MATHDB-395332

Classification of algebraic GKZ-hypergeometric systems

摘录 / EXCERPT: For general GKZ-hypergeometric systems, classify the irreducible systems that possess a complete set of solutions algebraic over $\mathbb{C}(v_1,\ldots,v_N)$; equivalently, classif...

open_problem | open | MathDB 页面 | 原文线索

V2-011467 | MATHDB-395373

Atiyah conjecture for group von Neumann algebras

摘录 / EXCERPT: Let Γ be a discrete group such that the set $\{|F|\mid F\subset\Gamma\text{ is a finite subgroup}\}$ is bounded, and let d be the least common multiple of these orders. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-011468 | MATHDB-395411

Cofinite universal property of the Borel-Serre partial compactification for arithmetic groups with torsion

摘录 / EXCERPT: Let G be a semisimple linear algebraic group defined over $\mathbb{Q}$, let $X=G(\mathbb{R})/K$ be its symmetric space of noncompact type, and let $\Gamma\subset G(\mathbb{Q})$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-011469 | MATHDB-395412

Rational Novikov conjecture for mapping class groups

摘录 / EXCERPT: Assume $2g-2+n>0$. The rational Novikov conjecture in algebraic K-theory holds for the mapping class group $\operatorname{Mod}_{g,n}$ of the orientable surface $S_{g,n}$ of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-011470 | MATHDB-395415

Popov’s conjecture on Kostant-Weierstrass sections for θ-groups

摘录 / EXCERPT: Let G be a reductive algebraic group over $\a0\mathbb C$ and let $\a0\theta:G\to G$ be a semisimple automorphism of finite order. With the associated grading...

conjecture | open | MathDB 页面 | 原文线索

V2-011471 | MATHDB-395416

Existence of Kostant-Weierstrass sections for Vinberg θ-groups

摘录 / EXCERPT: Let G be a reductive algebraic group over an algebraically closed field k, let $\theta:G\to G$ be a semisimple automorphism of finite order m (with $p\nmid m$ when...

open_problem | open | MathDB 页面 | 原文线索

V2-011472 | MATHDB-395421

Classification of full automorphism groups of curves of fixed genus

摘录 / EXCERPT: For a fixed genus $g\ge 2$, classify all groups that occur as full automorphism groups $\operatorname{Aut}_k(C)$ of connected nonsingular projective curves C over an algebraicall...

open_problem | open | MathDB 页面 | 原文线索

V2-011473 | MATHDB-395470

Maximally sparse polynomials have solid amoebas

摘录 / EXCERPT: Let $V_f\subset(\mathbb{C}^*)^n$ be an algebraic hypersurface defined by a maximally sparse polynomial f, meaning that the only monomials of f are those whose exponent vectors...

open_problem | open | MathDB 页面 | 原文线索

V2-011474 | MATHDB-395511

Semi-freeness of the absolute Galois group of a rational function field over an ample field

摘录 / EXCERPT: Let K be an ample field and x a variable. Then $\\operatorname{Gal}(K(x))$ is semi-free.

open_problem | open | MathDB 页面 | 原文线索

V2-011475 | MATHDB-395521

Factorization for rational connected linear algebraic groups in a patching setup

摘录 / EXCERPT: Let T be a complete discrete valuation ring with uniformizer t, let $\widehat{X}$ be a smooth projective T-curve with closed fibre X, and suppose $U_1,U_2\subseteq X$ cover...

open_problem | open | MathDB 页面 | 原文线索

V2-011476 | MATHDB-395523

Period-index bounds for central simple algebras

摘录 / EXCERPT: For a field F, does there exist an integer n depending only on F such that, for every central simple algebra A over F, $\operatorname{ind}(A)\mid \operatorname{per}(A)^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-011477 | MATHDB-395524

Almost all elliptic curves over suitable number fields have surjective adelic Galois representation

摘录 / EXCERPT: Let $k\ne\mathbb{Q}$ be a number field such that $k\cap\mathbb{Q}^{\mathrm{cyc}}=\mathbb{Q}$, where $\mathbb{Q}^{\mathrm{cyc}}$ is the cyclotomic extension of $\mathbb{Q}$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-011478 | MATHDB-395528

Normal subgroups of affine-curve fundamental groups in positive characteristic

摘录 / EXCERPT: Let $\pi_1(C)$ be the algebraic fundamental group of a smooth connected affine curve C defined over an algebraically closed field k of characteristic $p>0$ and of countable car...

open_problem | open | MathDB 页面 | 原文线索

V2-011479 | MATHDB-395530

Classification of unrestricted almost rational automorphisms of the Nottingham group

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$. An automorphism $\sigma\in\operatorname{Aut}_k(k[[t]])$ is almost rational with respect to β if there is an...

open_problem | open | MathDB 页面 | 原文线索

V2-011480 | MATHDB-395532

Specialization of a Galois cover to a prescribed extension

摘录 / EXCERPT: Given a field extension E/K that is Galois of group $H\subset G$, and a K-Galois cover $f:X\to B$ of group G, does there exist an unramified point $t_0\in B(K)$ such that the...

open_problem | open | MathDB 页面 | 原文线索

V2-011481 | MATHDB-395533

Specialization of a Galois cover over a henselian ring with big residue field

摘录 / EXCERPT: Let $(A,\mathfrak m)$ be a henselian ring as above, let K be its quotient field, and let $\kappa=A/\mathfrak m$ be its residue field. Let $f:X\to B$ be a K-Galois cover with gr...

open_problem | open | MathDB 页面 | 原文线索

V2-011482 | MATHDB-395534

Extension of the ramification theorem beyond the stated hypotheses

摘录 / EXCERPT: Let $f:Y\to X$ be a three-point G-Galois cover of the Riemann sphere, and suppose that a p-Sylow subgroup $P\subset G$ is cyclic of order p^n. Let $K/\mathbb{Q}$ be the field...

open_problem | open | MathDB 页面 | 原文线索

V2-011483 | MATHDB-395597

Universality of the Hilbert-module approach for dilations

摘录 / EXCERPT: Let T be a Markov semigroup on a unital C^*-algebra or von Neumann algebra B, and suppose that T has a general dilation to an E_0-semigroup or automorphism semigroup...

open_problem | open | MathDB 页面 | 原文线索

V2-011484 | MATHDB-395653

Unitarity classification for rational Cherednik algebra representations

摘录 / EXCERPT: Given the rational Cherednik algebra data from the report-a finite subgroup $W\subseteq GL(h)$, parameters $t=1$ and $c=(c_s)_{s\in\mathcal T}$ constant on W-conjugacy classes of...

open_problem | open | MathDB 页面 | 原文线索

V2-011485 | MATHDB-395655

Tensor-product simplicity in the category C_1

摘录 / EXCERPT: For the category C_1 associated with a simple Lie algebra $\g$ of type A_n, D_n, or E_n, a tensor product of cluster simple objects is simple if and only if all the objects...

open_problem | open | MathDB 页面 | 原文线索

V2-011486 | MATHDB-395658

Classification and decomposition of finite affine restrictions

摘录 / EXCERPT: Classify the quadratic subalgebras $\mathfrak a\subset\mathfrak g$ such that level-1 $\widehat{\mathfrak{so}}(\mathfrak p)$-modules restrict to $\widehat{\mathfrak a}$ with finite...

open_problem | open | MathDB 页面 | 原文线索

V2-011487 | MATHDB-395674

Occurrence of stable constant Jordan type [3][1]

摘录 / EXCERPT: Let $E\cong (\mathbb Z/p)^r$ with $r\ge 2$, and let k be an algebraically closed field of characteristic $p\ge 5$. Does there exist a finitely generated kE-module M such that...

open_problem | open | MathDB 页面 | 原文线索

V2-011488 | MATHDB-395675

Exactness of the extended permutation resolution for Specht modules

摘录 / EXCERPT: Fix a natural number r and a commutative ring k. For each partition $\lambda\in\Lambda(r)$ of r, let $\widetilde{C}^{\lambda}_{*}$ be the chain complex C^{λ}_{*} cons...

open_problem | open | MathDB 页面 | 原文线索

V2-011489 | MATHDB-395676

Cellular structures on type-\B Hecke algebras from unequal-parameter Kazhdan\\Lusztig theory

摘录 / EXCERPT: For the type-B Hecke algebra H_n over $A=\mathbb{Z}[Q,Q^{-1},q,q^{-1}]$ with parameters Q and q, let ξ be a positive irrational number and let r be the unique natural n...

open_problem | open | MathDB 页面 | 原文线索

V2-011490 | MATHDB-395677

Theorem 3 on the extension algebra for an elementary abelian group

摘录 / EXCERPT: Let $G=(C_p)^m$ and let k be a field of characteristic $p>2$. Set ... Then E is generated by elements γ_X of degree 2, indexed by subgroups $X\leq G$ with $|X|=p$, toget...

open_problem | open | MathDB 页面 | 原文线索

V2-011491 | MATHDB-395678

Representation type of modules of constant Jordan type

摘录 / EXCERPT: Is the category of modules with constant Jordan type of wild representation type?

open_problem | open | MathDB 页面 | 原文线索

V2-011492 | MATHDB-395679

Algebraicity versus periodicity for modules of an elementary abelian group

摘录 / EXCERPT: For an elementary abelian group of order p^2, an absolutely indecomposable module whose dimension is divisible by p is algebraic if and only if it is periodic.

open_problem | open | MathDB 页面 | 原文线索

V2-011493 | MATHDB-395680

Vertices of simple symmetric-group modules labelled by two-part partitions in characteristic 2

摘录 / EXCERPT: For $p=2$, given a $(2$-regular) partition $\lambda=(n-m,m)$ of n, determine the vertices of the simple $F S_n$-module D^λ, where F is an algebraically closed field of...

open_problem | open | MathDB 页面 | 原文线索

V2-011494 | MATHDB-395681

General version of James’ conjecture for Hecke algebras of exceptional type

摘录 / EXCERPT: Assume that ℓ does not divide $|W|$, and that k is the residue field of a discrete valuation ring $\mathcal O$ whose quotient field K is sufficiently large and has charact...

conjecture | open | MathDB 页面 | 原文线索

V2-011495 | MATHDB-395684

Broué’s abelian defect group conjecture for the non-principal 3-block of the Harada-Norton group

摘录 / EXCERPT: Assume that $p=3$, $G=HN$ is the Harada-Norton sporadic simple group, and A is a non-principal block algebra of $ OG$ with defect group $P=C_3\times C_3$ (in fact, A is the uni...

conjecture | open | MathDB 页面 | 原文线索

V2-011496 | MATHDB-395685

De Visscher-Donkin conjecture on simple-top Young modules

摘录 / EXCERPT: Conjecture 7. Let λ be a partition of r. Then $\operatorname{Top}(Y^\lambda)$ is simple if and only if there exists an integer $n\ge l(\lambda)$ such that $I_n(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-011497 | MATHDB-395701

Integrability of the Kac-Moody central extension of the smooth based loop group

摘录 / EXCERPT: Let G be a compact, simple, simply connected Lie group, let $\frak g$ be its Lie algebra, and let $\Omega^\infty G=C^\infty_*(S^1,G)$ be the smooth based loop group. Choose...

open_problem | open | MathDB 页面 | 原文线索

V2-011498 | MATHDB-395703

Classification of analytic Hilbert modules arising from Bergman-space cokernels

摘录 / EXCERPT: Classify the analytic Hilbert modules ... that appear in the preceding theorem, where $\Omega\subset\mathbb C^d$ is a strictly convex domain with smooth boundary, m,n are positive...

open_problem | open | MathDB 页面 | 原文线索

V2-011499 | MATHDB-395704

Trace-class classification for homogeneous quotient modules on the bidisk

摘录 / EXCERPT: Let $H^2=H^2(\boldsymbol D^2)$, and let $[p]$ denote the invariant submodule generated by a homogeneous polynomial p in the class described in the report: $p=p_1p_2$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-011500 | MATHDB-395711

Structure of the algebra commuting doubly with a Bergman-space multiplication operator

摘录 / EXCERPT: For a finite Blaschke product B, let M_B denote multiplication by B on the Bergman space $L_a^2(D)$, and let A_B be the algebra of operators on $L_a^2(D)$ that doubly commu...

open_problem | open | MathDB 页面 | 原文线索

V2-011501 | MATHDB-395724

Boij-Söderberg decomposition of Betti tables

摘录 / EXCERPT: Let $S=k[x_1,\dots,x_n]$ be a standard graded polynomial ring over a field. A finitely generated graded S-module M is pure of type $d=(d_0,\dots,d_m)$ if, in its minimal graded...

open_problem | open | MathDB 页面 | 原文线索

V2-011502 | MATHDB-395727

Whether every Koszul algebra is LG-quadratic

摘录 / EXCERPT: Is every Koszul algebra LG-quadratic? Here, an LG-quadratic algebra is one of the form $R\cong P/(I+L)$, where P is a polynomial ring, I has a quadratic Gröbner basis, and L...

open_problem | open | MathDB 页面 | 原文线索

V2-011503 | MATHDB-395728

Free resolutions over commutative Koszul algebras: Theorem 5

摘录 / EXCERPT: Let Q be a finitely generated Koszul K-algebra and J a homogeneous ideal with $0\neq J\subseteq(Q_+)^2$. For $R=Q/J$ and $c=\operatorname{Rate}R$, one has: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-011504 | MATHDB-395729

Canonical minimal representatives of finitely generated unit Frobenius modules

摘录 / EXCERPT: Let R be a regular and F-finite ring, and let $\d540$ be a finitely generated unit $R[F]$-module. Then there is a unique minimal $\d540$-module (M,γ) which generates...

open_problem | open | MathDB 页面 | 原文线索

V2-011505 | MATHDB-395730

Canonical minimal \b3-sheaf representative

摘录 / EXCERPT: For a regular, F-finite ring R and a finitely generated unit $R[F]$-module $\mathcal{M}$, there exists a unique minimal γ-module N that generates $\mathcal{M}$. In the...

open_problem | open | MathDB 页面 | 原文线索

V2-011506 | MATHDB-395734

Regularity bounds for Ext modules of graded ideals

摘录 / EXCERPT: For which graded ideals $I\subset S$ does ... hold?

open_problem | open | MathDB 页面 | 原文线索

V2-011507 | MATHDB-395735

Divisor of the symmetric algebra in implicitization

摘录 / EXCERPT: Let A and B be polynomial rings over the same field, in respectively n and $n+1$ variables. Let $I\subset A$ be generated by $n+1$ forms of the same degree $d>0$ defining a r...

open_problem | open | MathDB 页面 | 原文线索

V2-011508 | MATHDB-395738

Equality in the Green-Lazarsfeld index of Veronese algebras

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ be a polynomial ring over a field K of characteristic 0, with $n\ge 3$, and let $S^{(c)}=\bigoplus_{i\ge 0}S_{ic}$ be its c-th Veronese subring, whe...

open_problem | open | MathDB 页面 | 原文线索

V2-011509 | MATHDB-395778

The Procesi-Schacher conjecture for algebras with involution

摘录 / EXCERPT: For a central simple algebra A over a formally real field F, equipped with an involution $\d70e$, every totally $\d70e$-positive element $a\in\operatorname{Sym}(A,\u0001d70e)$...

conjecture | open | MathDB 页面 | 原文线索

V2-011510 | MATHDB-395784

Bounded generation by restricted elements in the trialitarian case

摘录 / EXCERPT: Let F be a field, let K/F be a cubic étale extension that is a field, and let D be a quaternion division algebra over K satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-011511 | MATHDB-395785

Kneser-Tits problem for isotropic trialitarian groups

摘录 / EXCERPT: Let F be a field and let G/F be a semisimple simply connected trialitarian group (that is, of absolute type D_4 with associated cubic étale algebra a field). If G is isotro...

open_problem | open | MathDB 页面 | 原文线索

V2-011512 | MATHDB-395786

Bounded generation of the restricted-norm subgroup in the trialitarian case

摘录 / EXCERPT: Let F be an infinite perfect field with $\\operatorname{char}(F)\\ne 2$, let K/F be a cubic étale extension that is a field, and let D be a quaternion division algebra over...

open_problem | open | MathDB 页面 | 原文线索

V2-011513 | MATHDB-395829

McClure’s conjecture on partial coalgebra structures on chains

摘录 / EXCERPT: The chains of a PL space form a partial co-algebra: the structure maps are generalized diagonal maps, and the natural transformation is given by the cartesian product of chains. Du...

conjecture | open | MathDB 页面 | 原文线索

V2-011514 | MATHDB-395846

Structure and Zariski closure of geometric Galois representations

摘录 / EXCERPT: Can we describe the Zariski closure of X^{g} in X?

open_problem | open | MathDB 页面 | 原文线索

V2-011515 | MATHDB-395856

Crystalline comparison conjecture for associated étale sheaves and filtered F-isocrystals

摘录 / EXCERPT: Let $p>0$, let k be a perfect field of characteristic p, and let K be a complete discrete valuation field with residue field k and ring of integers O_K. Fix an algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-011516 | MATHDB-395858

Serre’s E8 splitting conjecture

摘录 / EXCERPT: Let G be a simple algebraic group over $\\mathbb{Q}$ such that $G_{\\mathbb{R}}$ is a compact Lie group of type E_8. Let K be a field of characteristic 0. If G_K splits,...

conjecture | open | MathDB 页面 | 原文线索

V2-011517 | MATHDB-395892

Modular constructions of algebraic points on elliptic curves

摘录 / EXCERPT: Given a fixed elliptic curve $E/\Q$, which points in $E(\overline{\u0000Q})$ can be accounted for by a “modular construction”?

open_problem | open | MathDB 页面 | 原文线索

V2-011518 | MATHDB-395894

Algebraicity and Galois equivariance of complex Chow-Heegner points

摘录 / EXCERPT: Conjecture 5. Let H be a subfield of K^{ab} and let $\Delta_\varphi \in \operatorname{CH}^{r+1}(X_r)_0(H)$ be a generalised Heegner cycle defined over H. After fixin...

open_problem | open | MathDB 页面 | 原文线索

V2-011519 | MATHDB-395896

Algebraicity of ATR-cycle points (Conjecture 9)

摘录 / EXCERPT: Let F be a totally real field of degree $r+1$, with ordered real embeddings $v_0,v_1,\ldots,v_r$, and let E/F be an elliptic curve. Assume the setup of the report, including th...

conjecture | open | MathDB 页面 | 原文线索

V2-011520 | MATHDB-395898

Computing Frobenius conjugacy classes from a polynomial and its Galois group

摘录 / EXCERPT: Suppose $f(t)\in\mathbb{Z}[t]$ has degree n, with roots $\alpha_1,\ldots,\alpha_n$, splitting field $K=\mathbb{Q}(\alpha_1,\ldots,\alpha_n)$, and Galois group...

open_problem | open | MathDB 页面 | 原文线索

V2-011521 | MATHDB-395901

Nonsolvable Galois extensions ramified only at 3 and 5

摘录 / EXCERPT: There exists a finite, nonsolvable Galois extension K of $\mathbb{Q}$ which is ramified only at $p=3$, and one ramified only at $p=5$.

open_problem | open | MathDB 页面 | 原文线索

V2-011522 | MATHDB-395907

Algebraicity of quaternionic Stark-Heegner points

摘录 / EXCERPT: Let F be a totally real number field of narrow class number one with ring of integers $\d4de_F$, let N be a square-free ideal of $\d4de_F$, and let E/F be an elliptic curve o...

open_problem | open | MathDB 页面 | 原文线索

V2-011523 | MATHDB-395996

Toric local uniformization via formal embeddings

摘录 / EXCERPT: Given a noetherian excellent equicharacteristic local domain R with algebraically closed residue field $k=R/\mathfrak m$, does there exist a formal embedding of...

open_problem | open | MathDB 页面 | 原文线索

V2-011524 | MATHDB-395998

Formal normal form and Stokes filtration of an exponentially twisted Gauss-Manin connection

摘录 / EXCERPT: Let b[?25l(E,∇) be a free $\mathcal O_\Delta[x]$-module equipped with a meromorphic connection on $\Delta\times\mathbb A^1$, with logarithmic poles along...

open_problem | open | MathDB 页面 | 原文线索

V2-011525 | MATHDB-396004

Arc-closure equality for ideals in a Noetherian regular local ring

摘录 / EXCERPT: Let k be an algebraically closed field, let A be a Noetherian regular local k-algebra, and let $\\mathfrak a\subset A$ be an ideal. Set $X=\operatorname{Spec}(A/\mathfrak a)$...

open_problem | open | MathDB 页面 | 原文线索

V2-011526 | MATHDB-396066

Algebraic growth rate of Poincaré series for zonal Kleinian groups

摘录 / EXCERPT: Let G be a geometrically finite, essentially free, zonal Kleinian group acting on the $(N+1)$-dimensional hyperbolic space $\\mathbb{H}$, and let $r_{\max}$ denote the maximal ra...

open_problem | open | MathDB 页面 | 原文线索

V2-011527 | MATHDB-396150

Conjecture 6: Strongly stable fields are algebraically closed

摘录 / EXCERPT: Conjecture 6. Each infinite, strongly stable field is algebraically closed.

conjecture | open | MathDB 页面 | 原文线索

V2-011528 | MATHDB-396151

Conjecture 8 on exchange of generics in stable fields

摘录 / EXCERPT: Assume that K is an infinite, stable, saturated field with an additional structure. Are the following conditions equivalent? $(1')$ K is algebraically closed. $(2')$ For every...

conjecture | open | MathDB 页面 | 原文线索

V2-011529 | MATHDB-396153

Universal absolute-connectedness bound for connected perfect linear algebraic groups

摘录 / EXCERPT: Does there exist a universal natural number N such that every connected perfect linear algebraic group over an algebraically closed field is N-absolutely connected?

open_problem | open | MathDB 页面 | 原文线索

V2-011530 | MATHDB-396162

Rationality of quotients by connected linear algebraic groups

摘录 / EXCERPT: Let G be a connected linear algebraic group over $\a0\mathbb{C}$ and let V be a finite-dimensional complex representation of G. Writing V/G for any birational model of the...

open_problem | open | MathDB 页面 | 原文线索

V2-011531 | MATHDB-396166

Connected-component status of the $K_S^2=3$ Burniat deformation locus

摘录 / EXCERPT: For $K_S^2=3$, the general deformation of a Burniat surface is a Galois covering with group $(\mathbb{Z}/2\mathbb{Z})^2$ of a cubic surface with at least three singular points, wit...

open_problem | open | MathDB 页面 | 原文线索

V2-011532 | MATHDB-396180

Convergence of adaptive finite-element approximations for control-constrained optimal control

摘录 / EXCERPT: Under the standard adaptive iteration $\mathrm{SOLVE}\to\mathrm{ESTIMATE}\to\mathrm{MARK}\to\mathrm{REFINE}$, and under very mild assumptions on the modules of (7), the resulting s...

open_problem | open | MathDB 页面 | 原文线索

V2-011533 | MATHDB-396193

Commutativity of the semiclassical function algebra in the holonomy-loop construction

摘录 / EXCERPT: Whether the function algebra, should it emerge from the construction in a semi-classical limit, will be commutative, or whether it will pick up a noncommutative factor stemming fro...

open_problem | open | MathDB 页面 | 原文线索

V2-011534 | MATHDB-396230

Algebraic proof of complete reducibility for semisimple continuous groups

摘录 / EXCERPT: Find an algebraic proof of the full reducibility of representations of semisimple continuous groups.

open_problem | open | MathDB 页面 | 原文线索

V2-011535 | MATHDB-396235

Naimark’s problem for C*-algebras

摘录 / EXCERPT: Are the only C-algebras for which all irreducible representations are equivalent the algebras of compact operators?

open_problem | open | MathDB 页面 | 原文线索

V2-011536 | MATHDB-396239

The C*-algebra of a vector bundle

摘录 / EXCERPT: Let X be a compact metrizable space and let $E,F\in\mathrm{Vect}(X)$ be complex vector bundles of rank at least 2. Then $\mathcal O_E$ embeds as a unital $C(X)$-subalgebra of...

open_problem | open | MathDB 页面 | 原文线索

V2-011537 | MATHDB-396242

Innerness of automorphisms of the Calkin algebra

摘录 / EXCERPT: Is every automorphism of the Calkin algebra inner?

open_problem | open | MathDB 页面 | 原文线索

V2-011538 | MATHDB-396249

Kadison-Kastler conjecture for separable non-nuclear C*-algebras

摘录 / EXCERPT: Do sufficiently close separable non-nuclear C-subalgebras of a common mathcal{B}(H) have to be unitarily conjugate?

conjecture | open | MathDB 页面 | 原文线索

V2-011539 | MATHDB-396257

Modular branching conjecture for the affine Hecke algebra H_n

摘录 / EXCERPT: The modular branching conjecture for H_n is the statement that $\widetilde e_iL_\psi=L_{\widetilde e_i\psi}$, where $\widetilde e_i$ on the left-hand side is the module-theoretic...

conjecture | open | MathDB 页面 | 原文线索

V2-011540 | MATHDB-396264

Finite-dimensional irreducible modules for finite W-algebras of even-multiplicity nilpotent orbits

摘录 / EXCERPT: For 4mathfrak g=mathfrak{sp}{2n}(mathbb C)u00024 or 4mathfrak g=mathfrak{so}{2n}(mathbb C)u00024, and for an even-multiplicity nilpotent element 4einmathfrak gu00024 (that is, all...

open_problem | open | MathDB 页面 | 原文线索

V2-011541 | MATHDB-396267

Khovanov-Lauda conjecture for cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Let $H(\Gamma)$ be the quiver Hecke algebra associated with the Dynkin diagram Γ of a symmetrizable Kac-Moody algebra $\mathfrak g$, and let $H^\Lambda(\Gamma)$ be its cyclo...

conjecture | open | MathDB 页面 | 原文线索

V2-011542 | MATHDB-396271

Generalized Kac-Wakimoto conjecture for basic classical Lie superalgebras

摘录 / EXCERPT: Let g be a basic classical Lie superalgebra, let J be a simple g-supermodule which admits an ambidextrous trace, and let Lin IJ be a simple g-supermodule. Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-011543 | MATHDB-396310

Canonical basis for the module $V(\lambda)$

摘录 / EXCERPT: It is conjectured that the explicit module $V(\lambda)$ over the explicit algebra B admits a canonical basis.

open_problem | open | MathDB 页面 | 原文线索

V2-011544 | MATHDB-396355

Duflo’s conjecture for homogeneous spaces

摘录 / EXCERPT: Let $\mathfrak h\subset\mathfrak g$ be an inclusion of finite-dimensional Lie algebras over a field k of characteristic zero. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-011545 | MATHDB-396356

PBW-type isomorphism for an inclusion of Lie algebras

摘录 / EXCERPT: For an inclusion of Lie algebras $\mathfrak h\subset\mathfrak g$, under what assumption do we have an isomorphism of $\mathfrak h$-modules ... The report states that this holds preci...

open_problem | open | MathDB 页面 | 原文线索

V2-011546 | MATHDB-396362

Explicit description of cuspidal generalized weight-module categories

摘录 / EXCERPT: What is an explicit description of the category of cuspidal (generalized) weight modules?

open_problem | open | MathDB 页面 | 原文线索

V2-011547 | MATHDB-396363

Integral regular blocks of generalized weight \(\mathfrak{sl}_n\)-modules

摘录 / EXCERPT: Let $n>2$, and let $\widehat{\mathcal C}$ be the category of finitely generated, cuspidal, generalized weight $\mathfrak{sl}_n$-modules. Every integral regular block of...

open_problem | open | MathDB 页面 | 原文线索

V2-011548 | MATHDB-396365

Finite Gelfand-Kirillov dimension of affine Hopf algebras

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. What are necessary and sufficient conditions for an affine (that is, finitely generated) Hopf algebra H over k...

open_problem | open | MathDB 页面 | 原文线索

V2-011549 | MATHDB-396379

Verlinde algebra realization by a modular functor

摘录 / EXCERPT: For a compact Lie group G acting on itself by conjugation and a twisting τ, the equivariant twisted K-theory group $K_G^{* ,\tau}(G)$ is conjectured, in general, to be the V...

open_problem | open | MathDB 页面 | 原文线索

V2-011550 | MATHDB-396405

Cones of Hilbert functions

摘录 / EXCERPT: For the polynomial ring $S=k[x_0,\dots,x_n]$, describe the convex cone generated by the Hilbert functions of finitely generated $\mathbb{N}$-graded S-modules M that are generat...

open_problem | open | MathDB 页面 | 原文线索

V2-011551 | MATHDB-396408

Torsion-growth conjecture for symmetric-power local systems

摘录 / EXCERPT: Let Γ be a torsion-free cocompact arithmetic subgroup of $\operatorname{SL}(2,\mathbb{C})$ derived from a quaternion division algebra over an imaginary quadratic field, let...

conjecture | open | MathDB 页面 | 原文线索

V2-011552 | MATHDB-396427

Orbit-closure conjecture for maximal split tori

摘录 / EXCERPT: Let G be a real $\mathbb{R}$-split semisimple algebraic group of rank >1, let Γ be an irreducible lattice in G, let D be a maximal $\mathbb{R}$-split torus of G, a...

conjecture | open | MathDB 页面 | 原文线索

V2-011553 | MATHDB-396457

Continuous Lie quasi-states on classical Lie algebras

摘录 / EXCERPT: Describe the space of continuous Lie quasi-states, modulo the linear functionals, on classical Lie algebras. The simplest open cases are $\mathfrak{sl}(4,\mathbb{R})$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-011554 | MATHDB-396480

Carlson-Friedlander-Pevtsova conjecture for stable constant Jordan type

摘录 / EXCERPT: The report states the conjecture that if a finitely generated module M over kE, with $E\cong (\mathbb Z/p)^r$, has stable constant Jordan type $[2,1^j]$, then $j\ge r-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-011555 | MATHDB-396495

Auslander-Reiten conjecture for stable equivalences

摘录 / EXCERPT: For finite-dimensional algebras that are stably equivalent, the number of isomorphism classes of non-projective simple modules is the same for both algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-011556 | MATHDB-396561

Zilber’s almost-internality conjecture for non-locally modular strongly minimal sets

摘录 / EXCERPT: Zilber conjectured that any non-locally modular strongly minimal set X is almost internal to an algebraically closed field: there should be a definable algebraically closed field...

conjecture | open | MathDB 页面 | 原文线索

V2-011557 | MATHDB-396570

Quasi-monomiality of valuations computing asymptotic log canonical thresholds

摘录 / EXCERPT: Suppose that $X = \mathbb{A}^n_k$, with k an algebraically closed field of characteristic zero. Suppose that $\nu \in \operatorname{Val}_X$ has center at one point and transcende...

open_problem | open | MathDB 页面 | 原文线索

V2-011558 | MATHDB-396586

Bergeron-Venkatesh torsion-growth conjecture for congruence covers

摘录 / EXCERPT: Let B be a quaternion division algebra over an imaginary quadratic field, and let $\mathfrak{o}_B$ be a maximal order. Then $\mathfrak{o}_B^\times$ embeds into...

conjecture | open | MathDB 页面 | 原文线索

V2-011559 | MATHDB-396587

Algebraic Eigenvalue Conjecture

摘录 / EXCERPT: The report states that the Algebraic Eigenvalue Conjecture is solved by the following result: if μ is an integer measure, then (1) the atoms of μ are located at algebraic i...

conjecture | open | MathDB 页面 | 原文线索

V2-011560 | MATHDB-396588

Approximate subgroups of almost simple algebraic groups

摘录 / EXCERPT: Let G be an almost simple algebraic group over an algebraically closed field k, and let $d=\dim G$. Then there is a constant $C=C(d)>0$ such that, if $A\subset G(k)$ is a K-a...

open_problem | open | MathDB 页面 | 原文线索

V2-011561 | MATHDB-396632

Derived-bracket V-data for simultaneous deformations of an associative-algebra morphism

摘录 / EXCERPT: For a morphism of associative algebras $\Phi:U\to V$, with $L=\bigoplus_i L^i$, $L^i=T^{i+1}(U\oplus V)^*\otimes(U\oplus V)$, $\mathfrak a=\bigoplus_i\mathfrak a^i$,...

open_problem | open | MathDB 页面 | 原文线索

V2-011562 | MATHDB-396728

Classification of counterexamples to the loop-algebra property in two variables

摘录 / EXCERPT: In the two-dimensional case, the conjecture is that the only counterexamples are Margaux-like, so that they can be described by an invertible module over a 2-loop Azumaya algebra w...

open_problem | open | MathDB 页面 | 原文线索

V2-011563 | MATHDB-396729

Classification of anisotropic multiloop algebras via commuting automorphisms

摘录 / EXCERPT: Theorem 3.2. Let σ be an anisotropic n-tuple of commuting finite-order automorphisms of $\operatorname{Aut}(\mathfrak g)$, meaning that the common centralizer of all the...

open_problem | open | MathDB 页面 | 原文线索

V2-011564 | MATHDB-396732

Non-conjugacy of maximal split tori and MADs in a twisted loop setting

摘录 / EXCERPT: Let k be a field of characteristic 0, let $R=k[t_1^{\7},t_2^{\7}]$, and let D be the quaternion algebra over R generated by T_1,T_2 with relations $T_1^2=t_1$,...

open_problem | open | MathDB 页面 | 原文线索

V2-011565 | MATHDB-396734

A generalization of Fulton’s conjecture for arbitrary groups

摘录 / EXCERPT: Let G be any connected reductive complex algebraic group with Borel subgroup B and maximal torus $H\subset B$, and let $P\supseteq B$ be a standard parabolic subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-011566 | MATHDB-396739

Ext-groups for affine Hecke algebras and their Schwartz completions

摘录 / EXCERPT: Let H be an affine Hecke algebra with positive parameters, let $\[200m\pi,\[0m\pi'$ be finite-dimensional tempered H-modules, and let S be the Schwartz completion of H. The...

open_problem | open | MathDB 页面 | 原文线索

V2-011567 | MATHDB-396740

Euler-Poincaré pairings for arbitrary irreducible affine Hecke-algebra modules

摘录 / EXCERPT: Compute the Euler-Poincaré pairing $EP_H(\π,\π')$ for arbitrary irreducible H-modules $\π,\π'$, where H is an affine Hecke algebra with positive parameters.

open_problem | open | MathDB 页面 | 原文线索

V2-011568 | MATHDB-396741

Temperedness of invariant measures on quasi-affine homogeneous spaces

摘录 / EXCERPT: Assume that G and $H\subset G$ are algebraic groups and that $Z=G/H$ is unimodular and quasi-affine. Conjecture: there exists a rational G-module V and an embedding $Z\to V$...

open_problem | open | MathDB 页面 | 原文线索

V2-011569 | MATHDB-396742

Hodge-filtration signature conjecture for real-infinitesimal-character Harish-Chandra modules

摘录 / EXCERPT: If V is an irreducible Harish-Chandra module with real infinitesimal character, then the nonzero $\mathfrak{u}_{\mathbb R}$-invariant Hermitian form is nondegenerate on each...

conjecture | open | MathDB 页面 | 原文线索

V2-011570 | MATHDB-396750

Cyclic Deligne conjecture

摘录 / EXCERPT: For a Frobenius algebra A, the Hochschild cochains $CH^*(A,A)$ carry a chain-level action of a chain model of the framed little-discs operad, realized via the Cacti operad; conse...

conjecture | open | MathDB 页面 | 原文线索

V2-011571 | MATHDB-396757

GL$(n,\mathbb{Z})$-module structure of the Johnson homomorphism image or cokernel

摘录 / EXCERPT: For a free group F_n of rank $n\ge 2$, let $\tau_{k,\mathbb{Q}}:=\tau_k\otimes\operatorname{id}_{\mathbb{Q}}$ be the rational k-th Johnson homomorphism of...

open_problem | open | MathDB 页面 | 原文线索

V2-011572 | MATHDB-396792

Classification of invariant rings that are finitely generated

摘录 / EXCERPT: Let K be an algebraically closed field, let G be a linear algebraic group over K, let V be a finite-dimensional K-vector space with a linear G-action, or let X be an...

open_problem | open | MathDB 页面 | 原文线索

V2-011573 | MATHDB-396794

Lie’s conjecture on coordinate realizations of transitive Lie algebras of vector fields

摘录 / EXCERPT: Given a finite-dimensional complex transitive Lie algebra L of vector fields in $\mathbb{R}^n$, does there exist a coordinate system $y_1,\dots,y_n$ in which every coefficient of...

conjecture | open | MathDB 页面 | 原文线索

V2-011574 | MATHDB-396797

Smale’s polynomial-growth problem for classical polynomial Li\nard equations

摘录 / EXCERPT: Does the maximal number of limit cycles $N(1,n)$ for classical polynomial Linard equations, namely equation (1) with $g(x)=x$, grow at most according to an algebraic law of the for...

open_problem | open | MathDB 页面 | 原文线索

V2-011575 | MATHDB-396803

Determining invariant algebraic curves of a planar polynomial vector field

摘录 / EXCERPT: Given a planar polynomial vector field $X=P\,\partial/\partial x+Q\,\partial/\partial y$, find all invariant algebraic curves of X.

open_problem | open | MathDB 页面 | 原文线索

V2-011576 | MATHDB-396850

Z-stability of separable simple unital nuclear C*-algebras

摘录 / EXCERPT: Is every separable, simple, unital, and nuclear C-algebra A Z-stable, i.e. isomorphic to $A \otimes Z$, where Z is the Jiang-Su algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011577 | MATHDB-396852

Borel reduction from homeomorphism of compact Polish spaces to C*-algebra isomorphism

摘录 / EXCERPT: The homeomorphism relation for compact Polish spaces is Borel reducible to the isomorphism relation for unital nuclear simple separable C^*-algebras. In particular, isomorphism i...

open_problem | open | MathDB 页面 | 原文线索

V2-011578 | MATHDB-396858

Classification under weaker assumptions than Maharam’s theorem

摘录 / EXCERPT: Can a characterization analogous to Maharam’s classification be obtained for pairs (B,μ) in which B is an arbitrary Boolean algebra, not necessarily σ-complete, and...

open_problem | open | MathDB 页面 | 原文线索

V2-011579 | MATHDB-396946

Precise connection between cluster algebras and canonical bases

摘录 / EXCERPT: What is the precise connection between cluster algebras and canonical bases?

open_problem | open | MathDB 页面 | 原文线索

V2-011580 | MATHDB-396949

Generic basis conjecture for cluster algebras from quivers with potentials

摘录 / EXCERPT: Let 8B,mathbf{x}u00029 be a seed with $B\in\mathbb{Z}^{m\times m}$ antisymmetric and $\mathbf{x}=(x_1,\ldots,x_m)$ a transcendence basis of $\mathbb{C}(x_1,\ldots,x_m)$. Fix a gene...

conjecture | open | MathDB 页面 | 原文线索

V2-011581 | MATHDB-396956

Generic basis conjecture beyond the Buan-Iyama-Reiten-Scott setting

摘录 / EXCERPT: The authors of [11] conjecture that the set ... is a basis of the associated cluster algebra A_Q even outside the setting of Example 2.

conjecture | open | MathDB 页面 | 原文线索

V2-011582 | MATHDB-396959

Indecomposable tree modules for every root

摘录 / EXCERPT: Let $k=\\mathbb{C}$ and let $Q=(Q_0,Q_1)$ be a quiver without oriented cycles. Does there exist an indecomposable tree module for every root $d\in\mathbb{N}^{Q_0}$?

open_problem | open | MathDB 页面 | 原文线索

V2-011583 | MATHDB-396965

Conjectural equivalence of algebraic and arithmetic automorphic representations

摘录 / EXCERPT: One might conjecture that an automorphic representation π is L-algebraic if and only if it is L-arithmetic, and C-algebraic if and only if it is C-arithmetic.

open_problem | open | MathDB 页面 | 原文线索

V2-011584 | MATHDB-396974

Infinitely many cuspidal automorphic representations containing a prescribed archimedean type

摘录 / EXCERPT: Let G be a semisimple algebraic group defined over a number field k with G_∞ noncompact. Let $L\subset G(\mathbb A_f)$ be an open-compact subgroup, let...

open_problem | open | MathDB 页面 | 原文线索

V2-011585 | MATHDB-396975

Nontriviality of a prescribed Bernstein component of cuspidal automorphic forms

摘录 / EXCERPT: Let G be a semisimple algebraic group over a number field k, and let $T\subset V_f$ be a finite nonempty set of places such that G is unramified over k_v for every...

open_problem | open | MathDB 页面 | 原文线索

V2-011586 | MATHDB-397006

Sullivan’s realization theorem for minimal Sullivan algebras satisfying Poincaré duality

摘录 / EXCERPT: The report states that the realization question is completely answered in the simply connected case. Let A be a minimal Sullivan algebra satisfying Poincaré duality with respect...

open_problem | open | MathDB 页面 | 原文线索

V2-011587 | MATHDB-397040

Indecomposable Specht modules in characteristic 2

摘录 / EXCERPT: Which Specht $F S_n$-modules are indecomposable when F has characteristic 2?

open_problem | open | MathDB 页面 | 原文线索

V2-011588 | MATHDB-397042

Blocks of the symmetric-group centralizer algebra

摘录 / EXCERPT: Assume that a0(R,F,k) is a sufficiently large p-modular system, with R a discrete valuation ring, F its field of fractions of characteristic 0, and k its residue field...

open_problem | open | MathDB 页面 | 原文线索

V2-011589 | MATHDB-397043

Centers of centralizer algebras for symmetric groups

摘录 / EXCERPT: Let $S_l\leq S_n$ be the subgroup of permutations fixing every number larger than l, and write $\mathbb{Z}S_n^{S_l}=\{a\in\mathbb{Z}S_n:ah=ha\ \text{for all }h\in S_l\}$. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-011590 | MATHDB-397044

Hill’s conjecture on invariant factors of Cartan matrices of Hecke-algebra blocks

摘录 / EXCERPT: For $\ell=p^r$, Hill computed the invariant factors of the Cartan matrix associated with an individual block of the Iwahori-Hecke algebra H_n when $r\le p$, and conjectured that...

conjecture | open | MathDB 页面 | 原文线索

V2-011591 | MATHDB-397046

Foulkes’ Conjecture

摘录 / EXCERPT: For $m,n\in\mathbb{N}$ with $m<n$, let $H^{(m^n)}$ be the permutation module for S_{mn} acting on set partitions of a set of size mn into n sets of size m, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011592 | MATHDB-397107

Semisimplicity of the division closure for groups with bounded torsion

摘录 / EXCERPT: Let Γ be a group whose finite subgroups have uniformly bounded orders, and let D_Γ be the division closure of $\mathbb{Z}\Gamma$ in the algebra U_Γ of affiliat...

open_problem | open | MathDB 页面 | 原文线索

V2-011593 | MATHDB-397121

Explicit computation of multi-relative K-groups of truncated polynomial algebras

摘录 / EXCERPT: The report states that, when $p\nmid a_i$ for every $1\le i\le n$, the multi-relative algebraic K-groups satisfy ... ... where m is the unique integer such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-011594 | MATHDB-397122

Vanishing of negative algebraic K-groups, reduced K-theory, and the Whitehead group

摘录 / EXCERPT: Suppose that G is torsionfree. Then $K_n(\mathbb{Z}G)=0$ for $n\leq -1$, $\widetilde{K}_0(\mathbb{Z}G)=0$, and $\mathrm{Wh}(G)=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-011595 | MATHDB-397128

Bass conjecture on finite generation of algebraic K-groups

摘录 / EXCERPT: For any finitely generated commutative algebra R over the integers, the groups $K'_i(R)$ are finitely generated abelian groups, where $K'_i(R)$ denotes the K-theory of finitely g...

conjecture | open | MathDB 页面 | 原文线索

V2-011596 | MATHDB-397148

Presentation of the mapping class group Steinberg module from a single generator

摘录 / EXCERPT: Give a presentation for the Steinberg module $\operatorname{St}(\Sigma^1_g)$ based on the single generator from [Broa07, Theorem 4.2].

open_problem | open | MathDB 页面 | 原文线索

V2-011597 | MATHDB-397149

Module structure of the Steinberg module

摘录 / EXCERPT: Understand the structure of the Steinberg module $\mathrm{St}(\Sigma_g^1)=\widetilde{H}_{2g-2}(\mathcal{C};\mathbb{Z})$ of the mapping class group $\mathrm{Mod}(\Sigma_g^1)$ as a m...

open_problem | open | MathDB 页面 | 原文线索

V2-011598 | MATHDB-397150

Faithfulness of the Steinberg module action

摘录 / EXCERPT: Is $\operatorname{St}(\Sigma^1_g)$ a faithful $\operatorname{Mod}(\Sigma_g)$-module for $g\ge 1$? If not, what is the kernel of the action?

open_problem | open | MathDB 页面 | 原文线索

V2-011599 | MATHDB-397156

Spectral and scattering theory for natural elliptic operators on Riemann moduli space

摘录 / EXCERPT: For the Riemann moduli space $\mathcal M_g$ with $g>1$, investigate the spectral and scattering properties of the Laplacians and other natural elliptic operators associated with th...

open_problem | open | MathDB 页面 | 原文线索

V2-011600 | MATHDB-397170

Mapping class group as the automorphism group of a Lie algebra

摘录 / EXCERPT: Conjecture. The MCG is the automorphism group of the Lie algebra.

open_problem | open | MathDB 页面 | 原文线索

V2-011601 | MATHDB-397171

Automorphic higher-order module and an adelic tensor product

摘录 / EXCERPT: In the example $G=\mathrm{SL}_2(\mathbb{Q})$ and $\Gamma=\mathrm{SL}_2(\mathbb{Z})$, the resulting module V for $\mathrm{SL}_2(\mathbb{A})$ injects into ... It is conjectured that...

open_problem | open | MathDB 页面 | 原文线索

V2-011602 | MATHDB-397176

Existence of a weakly admissible filtration with prescribed semisimplified Frobenius

摘录 / EXCERPT: Given a φ-module with fixed semisimplified Frobenius Φ^{ss}, does there exist a weakly admissible filtration on it?

open_problem | open | MathDB 页面 | 原文线索

V2-011603 | MATHDB-397278

Which subalgebras admit a SAGBI basis?

摘录 / EXCERPT: Fix a total order $<$ on $\mathbb{Z}^n$ respecting addition. Determine which subalgebras $A\subseteq\mathbb{C}[x_1,\ldots,x_n]$ possess a finite SAGBI basis; that is, a finite set...

open_problem | open | MathDB 页面 | 原文线索

V2-011604 | MATHDB-397353

Pure states in the local-global principle for Hilbert C^*-modules

摘录 / EXCERPT: Conjecture 3. If L is a proper submodule of the Hilbert A-module E, then there exists a pure state ω on A such that $\iota_\omega(L)^\perp\ne\{0\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-011605 | MATHDB-397355

Entropy-Fuglede-Kadison determinant equality without von Neumann invertibility

摘录 / EXCERPT: It is conjectured that, for a countable amenable group Γ and $f\in\mathbb Z[\Gamma]$, the hypothesis in Theorem 1 that f be invertible in the group von Neumann algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-011606 | MATHDB-397356

Algebraic K-theory Novikov conjecture for group algebras over Schatten class operators

摘录 / EXCERPT: For the ring of Schatten class operators and every group G, the associated algebraic K-theory assembly map is rationally injective. Consequently, the algebraic K-theory Novik...

conjecture | open | MathDB 页面 | 原文线索

V2-011607 | MATHDB-397365

Descent from algebraic K-theory of MU to algebraic K-theory of the sphere

摘录 / EXCERPT: Can $K(S)$ be recovered from $K(MU)$ by descent along the Hopf-Galois extension $S\to MU$, in the sense of [9]?

open_problem | open | MathDB 页面 | 原文线索

V2-011608 | MATHDB-397513

Decidability of the theory of countable Boolean algebras under direct sum

摘录 / EXCERPT: Is the first-order theory of the class of countable Boolean algebras with the direct-sum operation decidable? The report answers no: Theorem 1 states that this theory is 1-equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-011609 | MATHDB-397521

Proving that Artinian implies Noetherian without proving finite length

摘录 / EXCERPT: Let R be a commutative ring with identity. Recall that R is Noetherian if it satisfies the ascending chain condition on its ideals, Artinian if it satisfies the descending chai...

open_problem | open | MathDB 页面 | 原文线索

V2-011610 | MATHDB-397533

Upper bound separating FETI algebraic and discretization errors

摘录 / EXCERPT: The report states that it proves the upper bound ... where $\lvert\cdot\rvert_b$ is the norm associated with the FETI preconditioner and $[u_N]$ is the jump of u_N at the interface...

open_problem | open | MathDB 页面 | 原文线索

V2-011611 | MATHDB-397552

Valuation cone as a fundamental domain in characteristic different from 2

摘录 / EXCERPT: For an algebraically closed field k with $\operatorname{char} k\ne 2$, the valuation cone $V(G/K)$ is the fundamental domain of a finite reflection group attached to G/K.

open_problem | open | MathDB 页面 | 原文线索

V2-011612 | MATHDB-397564

Wakimoto modules associated to jet characters

摘录 / EXCERPT: Let $\mathfrak h_n$, $\mathfrak b_n$, and $\mathfrak g_n$ denote the n-jets of $\mathfrak h$, $\mathfrak b$, and $\mathfrak g$, respectively. For a character...

open_problem | open | MathDB 页面 | 原文线索

V2-011613 | MATHDB-397565

Categorical decomposition of critical-level affine Kac-Moody modules

摘录 / EXCERPT: Using the center Z of the abelian category $\widehat{\mathfrak g}_{\mathrm{crit}}\text{-mod}$, express this abelian category as a direct product.

open_problem | open | MathDB 页面 | 原文线索

V2-011614 | MATHDB-397569

Kostant Clifford algebra conjecture

摘录 / EXCERPT: Let $\mathfrak g$ be a complex simple Lie algebra and $C(\mathfrak g)$ its Clifford algebra. Choose a triangular decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-011615 | MATHDB-397570

An Analogue Enveloping Algebra Conjecture

摘录 / EXCERPT: Let V be a finite-dimensional $\mathfrak g$-module and let V_0 be its zero-weight subspace. Regard $V\otimes U(\mathfrak g)$ as a $U(\mathfrak g)$-bimodule via ... Let Φ^+ b...

conjecture | open | MathDB 页面 | 原文线索

V2-011616 | MATHDB-397582

Computing homomorphism spaces between Specht modules

摘录 / EXCERPT: Given a field F and Specht modules S^λ and S^μ for the symmetric group S_n, compute the homomorphism space $\operatorname{Hom}_{F S_n}(S^\lambda,S^\mu)$ in general.

open_problem | open | MathDB 页面 | 原文线索

V2-011617 | MATHDB-397606

Description of orbits in a \theta-group

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple Lie algebra over $\mathbb C$ with adjoint group G, and let $\mathfrak g=\bigoplus_{i\in\mathbb Z/m\mathbb Z}\mathfrak g_i$ be a grading, with...

open_problem | open | MathDB 页面 | 原文线索

V2-011618 | MATHDB-397610

Panyushev’s polynomiality conjecture for contractions of simple Lie algebras

摘录 / EXCERPT: Let r $\mathfrak g=\mathfrak g_0\oplus\mathfrak g_1$ be a symmetric decomposition of a simple non-Abelian Lie algebra $\mathfrak g=\operatorname{Lie}G$ over an algebraically closed...

conjecture | open | MathDB 页面 | 原文线索

V2-011619 | MATHDB-397617

Popov’s conjecture on Kostant-Weierstrass sections

摘录 / EXCERPT: For a classical Lie algebra r$\mathfrak g$ over a field rk with r$\operatorname{char} k\neq 2$, and for any finite-order automorphism rθ defining the grading r...

conjecture | open | MathDB 页面 | 原文线索

V2-011620 | MATHDB-397634

When do the two full semigroup C*-algebra constructions coincide?

摘录 / EXCERPT: Given a subsemigroup P of a group G, for which semigroups is the canonical homomorphism $C^*(P)\to C_s^*(P)$ an isomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-011621 | MATHDB-397635

K-theory decomposition for semigroup C*-algebras

摘录 / EXCERPT: Let P be a subsemigroup of a countable group G such that P has independent constructible ideals and $P\subseteq G$ is Toeplitz. If G satisfies the Baum-Connes conjecture fo...

open_problem | open | MathDB 页面 | 原文线索

V2-011622 | MATHDB-397636

Necessity of the determinant-sign condition for Leavitt path algebra classification

摘录 / EXCERPT: Is the determinant-sign hypothesis necessary in the following assertion? Suppose E and F are finite graphs for which $C^*(E)$ and $C^*(F)$ (equivalently, $L_{\mathbb C}(E)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-011623 | MATHDB-397637

Necessity of the determinant-sign hypothesis in Leavitt path algebra classification

摘录 / EXCERPT: Let E and F be finite graphs such that $L_K(E)$ and $L_K(F)$ are purely infinite and simple. If there is an isomorphism $\varphi:K_0(L_K(E))\to K_0(L_K(F))$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-011624 | MATHDB-397643

Higher-quiver cellular resolution conjecture for consistent toric algebras

摘录 / EXCERPT: Conjecture: The statement of Theorem 0.2 holds for every consistent toric algebra of global dimension equal to $n=\dim X$. Namely, the higher quiver $\Delta\subset (S^1)^n$ exists,...

conjecture | open | MathDB 页面 | 原文线索

V2-011625 | MATHDB-397661

Upper bound for the generalized Schanuel constant

摘录 / EXCERPT: For a number field k and a nonzero algebraic number θ, the generalized constant $g_k(\theta)$ defined in the preceding discussion satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-011626 | MATHDB-397662

Exceptional units conjecture for algebraic integers

摘录 / EXCERPT: For a positive integer d, let ... where, for a nonzero algebraic integer α, with $K=\mathbb{Q}(\alpha)$ and ring of integers $\mathcal O_K$, $E_0(\alpha)=0$ if $\alpha-1$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-011627 | MATHDB-397937

Crystalline comparison conjecture of Fontaine

摘录 / EXCERPT: Let K be a mixed-characteristic $(0,p)$ local field with perfect residue field k, let $K\subset\overline K$ be an algebraic closure, and let $\mathcal O_K$ be the ring of integ...

conjecture | open | MathDB 页面 | 原文线索

V2-011628 | MATHDB-397941

Traverso’s isomorphism-number conjecture for p-divisible groups

摘录 / EXCERPT: Let k be an algebraically closed field of positive characteristic p, and let D be a p-divisible group over k of positive height r. Write (M,φ,ϑ) for its...

conjecture | open | MathDB 页面 | 原文线索

V2-011629 | MATHDB-397945

Exhaustivity of Faltings’s p-adic Simpson correspondence for curves

摘录 / EXCERPT: Let X be a smooth proper curve over K. Are all semistable Higgs modules of slope zero obtained, via Faltings’s p-adic Simpson correspondence, from “true” p-adic representatio...

open_problem | open | MathDB 页面 | 原文线索

V2-011630 | MATHDB-398043

The Nash problem for surfaces

摘录 / EXCERPT: The Nash mapping is bijective for any surface defined over an algebraically closed field of characteristic 0.

open_problem | open | MathDB 页面 | 原文线索

V2-011631 | MATHDB-398048

Alexander polynomials of plane-curve complements

摘录 / EXCERPT: Let $C\subset\mathbb{C}^2$ be an irreducible algebraic curve transverse to the line at infinity, and let $\pi_1=\pi_1(\mathbb{C}^2\setminus C)$. Define $\Delta_C(t)$ as the charact...

open_problem | open | MathDB 页面 | 原文线索

V2-011632 | MATHDB-398050

Classification of finite-dimensional Nichols algebras over groups

摘录 / EXCERPT: Classify finite-dimensional Nichols algebras over groups.

open_problem | open | MathDB 页面 | 原文线索

V2-011633 | MATHDB-398054

Infinite-dimensionality of the Fomin-Kirillov algebras

摘录 / EXCERPT: For an integer $n\ge 3$, let E_n be the algebra generated by $x_{(ij)}$ for $1\le i<j\le n$, with relations ... ... for $1\le i<j<k\le n$, and ... for pairwise distinct i,j,k,l. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-011634 | MATHDB-398055

Quadraticity and identification of the Nichols algebra associated with the Fomin-Kirillov algebra

摘录 / EXCERPT: For the braided vector space (V_n,c) defined in the report, it is conjectured that the Nichols algebra $\mathfrak{B}(V_n)$ is quadratic and that $\mathfrak{B}(V_n)=\mathcal{E}_n$...

open_problem | open | MathDB 页面 | 原文线索

V2-011635 | MATHDB-398059

Classification of finite-dimensional Nichols algebras over non-abelian groups

摘录 / EXCERPT: Let V be an absolutely irreducible Yetter-Drinfeld module over a group G. Assume that G is generated by the support of V. If the Nichols algebra $B(V)$ is finite-dimensiona...

open_problem | open | MathDB 页面 | 原文线索

V2-011636 | MATHDB-398072

Converse criterion for freeness of the piecewise-polynomial module

摘录 / EXCERPT: Let Σ be the polyhedral fan and C the chain complex considered in the report. Does freeness of $C_0(\Sigma)$ as a $\operatorname{Sym}(N)_{\mathbb Q}$-module imply that, fo...

open_problem | open | MathDB 页面 | 原文线索

V2-011637 | MATHDB-398074

Extension of a pentagon-arrangement automorphism to an $E_\$-coalgebra or fundamental-group automorphism

摘录 / EXCERPT: Let A, B, C, D, E be the vertices of a regular pentagon. Consider the real affine arrangement consisting of the lines $AB, CE, AC, DE, AD, BC, AE, BD$ (four pairs of parallel lin...

open_problem | open | MathDB 页面 | 原文线索

V2-011638 | MATHDB-398094

Stability of tensorial Jiang-Su absorption under small perturbations

摘录 / EXCERPT: Does there exist $\[1m\delta > 0$[0m, such that whenever A is a separable C^*-algebra which absorbs $\[1mZ$[0m tensorially and B is another C^*-algebra with $d(A,B)<\delta$...

open_problem | open | MathDB 页面 | 原文线索

V2-011639 | MATHDB-398100

Conjecture 7: centrally large subalgebras and tracial \ce\-stability

摘录 / EXCERPT: Let A be an infinite dimensional simple separable unital C^*-algebra, and let $B\subset A$ be a centrally large subalgebra. If B is tracially $\mathcal{Z}$-stable in the sens...

conjecture | open | MathDB 页面 | 原文线索

V2-011640 | MATHDB-398103

Conjecture 6: classification of outer poly-\a7Z actions on Kirchberg algebras

摘录 / EXCERPT: Let G be a poly-$\mathbb Z$ group and let α and β be outer actions of G on a unital Kirchberg algebra A. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-011641 | MATHDB-398104

Complete stable-isomorphism invariant for graph C*-algebras with finitely many ideals

摘录 / EXCERPT: Find an algebraic invariant that will be a complete stable isomorphism invariant for the class of graph C^*-algebras with finitely many ideals.

open_problem | open | MathDB 页面 | 原文线索

V2-011642 | MATHDB-398105

Complete isomorphism invariant for graph C*-algebras with finitely many ideals

摘录 / EXCERPT: Find an algebraic invariant that will be a complete isomorphism invariant for the class of graph C-algebras with finitely many ideals.

open_problem | open | MathDB 页面 | 原文线索

V2-011643 | MATHDB-398106

Classification under the Dadarlat-Elliott K-theoretic hypotheses

摘录 / EXCERPT: Let $d\in\{0,1\}$. Is filtrated K-theory a complete invariant for separable, stable continuous $C([0,1])$-algebras whose fibers are stable Kirchberg algebras satisfying the UCT,...

open_problem | open | MathDB 页面 | 原文线索

V2-011644 | MATHDB-398107

Existence of phantom Cuntz-Krieger algebras

摘录 / EXCERPT: Does there exist a C^*-algebra A such that: 1. A is unital, separable, nuclear, and purely infinite; 2. A has real rank zero; 3. $X=\operatorname{Prim}(A)$ is a finite spac...

open_problem | open | MathDB 页面 | 原文线索

V2-011645 | MATHDB-398109

Whether every $\l$-open C*-algebra is $\l$-closed

摘录 / EXCERPT: Every $\l$-open C-algebra is $\l$-closed.

open_problem | open | MathDB 页面 | 原文线索

V2-011646 | MATHDB-398110

Robert’s conjecture for tensoring with the Razak-Jacelon algebra

摘录 / EXCERPT: If A and B are separable nuclear C^*-algebras, then … where $T(A)$ denotes the cone of lower semicontinuous traces on A_+ with values in $[0,\infty]$, the isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-011647 | MATHDB-398112

Dadarlat-Toms question on tensor powers of characterless C*-algebras

摘录 / EXCERPT: Question 11 (Dadarlat-Toms). Does $\mathcal Z$ embed unitally into $\bigotimes_{n=1}^{\infty} D$ whenever D is a unital C^*-algebra without characters?

open_problem | open | MathDB 页面 | 原文线索

V2-011648 | MATHDB-398115

Toms-Winter-type classification conjecture in the UCT class

摘录 / EXCERPT: Conjecture. For a simple, separable, unital, nonelementary, nuclear C^*-algebra A in the UCT class, the following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-011649 | MATHDB-398188

Intersection criterion for algebraic closures in a free group

摘录 / EXCERPT: Let F be a nonabelian free group. For finite tuples $\bar a,\bar b,\bar c\in F$, … if and only if … and …

open_problem | open | MathDB 页面 | 原文线索

V2-011650 | MATHDB-398192

Algebraic closedness of regular fields

摘录 / EXCERPT: Each regular field is algebraically closed.

open_problem | open | MathDB 页面 | 原文线索

V2-011651 | MATHDB-398242

Complex or algebraic structures on higher-dimensional Donaldson-Uhlenbeck compactifications

摘录 / EXCERPT: Do the higher-dimensional analogues of the Donaldson-Uhlenbeck compactifications admit natural complex or even algebraic structures?

open_problem | open | MathDB 页面 | 原文线索

V2-011652 | MATHDB-398243

Artin’s conjecture for supersingular K3 surfaces

摘录 / EXCERPT: For a K3 surface over an algebraically closed field of characteristic $p>2$, if the height h of its formal Brauer group is ∞, then its Picard number satisfies $\rho=22$.

conjecture | open | MathDB 页面 | 原文线索

V2-011653 | MATHDB-398245

Pixton’s conjectural description of tautological relations

摘录 / EXCERPT: Let R be the ideal in the strata algebra $S_{g,n}$ generated by the pullbacks and pushforwards of the special classes $R(g,n,r)$ defined as sums over dual graphs. Conjecture: R...

open_problem | open | MathDB 页面 | 原文线索

V2-011654 | MATHDB-398246

Pixton’s conjecture on the completeness of tautological relations

摘录 / EXCERPT: Conjecture: Let R be the ideal in the strata algebra $S_{g,n}$ generated by the classes $R(g,n,r)$ described as sums over dual graphs. Then R is the ideal of tautological relat...

conjecture | open | MathDB 页面 | 原文线索

V2-011655 | MATHDB-398248

Other sections of a height-twelve quartic del Pezzo fibration

摘录 / EXCERPT: Let k be an algebraically closed field with $\operatorname{char}(k)\ne 2$, and let $\pi:X\to\mathbb{P}^{1}$ be a smooth quartic del Pezzo surface fibration whose fibers are compl...

open_problem | open | MathDB 页面 | 原文线索

V2-011656 | MATHDB-398253

Variational crystalline Hodge conjecture for Fourier-Mukai correspondences of K3 surfaces

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic different from 2, and let W denote the Witt vectors of k. Let X and Y be K3 surfaces over k, with lifts X/W...

conjecture | open | MathDB 页面 | 原文线索

V2-011657 | MATHDB-398287

Non-stability of archimedean quadratic modules in dimension at least two

摘录 / EXCERPT: Let $M= \Sigma+\Sigma g_1+\cdots+\Sigma g_r\subseteq\mathbb{R}[x_1,\ldots,x_n]$ be the quadratic module associated with $K=\{u\in\mathbb{R}^n:g_i(u)\ge 0, \ i=1,\ldots,r\}$. Under...

open_problem | open | MathDB 页面 | 原文线索

V2-011658 | MATHDB-398288

Helton-Nie conjecture in dimension two

摘录 / EXCERPT: The Helton-Nie conjecture holds in dimension two: every convex semi-algebraic subset of $\mathbb{R}^2$ is a linear projection of a spectrahedron.

conjecture | open | MathDB 页面 | 原文线索

V2-011659 | MATHDB-398295

Matrix Positivstellensatz implication

摘录 / EXCERPT: Let A be a real or complex unital $*$-algebra, with involution written $a\mapsto a^+$. Let $\mathcal R$ be a fixed separating family of $*$-representations of A, and define ... L...

open_problem | open | MathDB 页面 | 原文线索

V2-011660 | MATHDB-398327

Rationality versus stable rationality in Noether’s problem

摘录 / EXCERPT: Let k be a field, let G/k be a linear algebraic group, and let V be a finite-dimensional generically free representation of G over k. Assume that Noether’s problem for...

open_problem | open | MathDB 页面 | 原文线索

V2-011661 | MATHDB-398333

Borel-Tits conjecture for abstract homomorphisms of algebraic groups

摘录 / EXCERPT: Let G and G' be algebraic groups defined over infinite fields k and k', respectively. For an abstract homomorphism $\rho:G(k)\to G'(k')$ such that $\rho(G^+)$ is Zariski-de...

conjecture | open | MathDB 页面 | 原文线索

V2-011662 | MATHDB-398337

Kazhdan-Lusztig positivity conjecture

摘录 / EXCERPT: For a Coxeter system (W,S) with Hecke algebra H and Kazhdan -Lusztig basis ... the conjecture asserted: 1. $P_{y,x}\in\mathbb N[q]$; 2. if $C'_xC'_y=\sum_z\mu^z_{xy}C'_z$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-011663 | MATHDB-398339

Soergel's conjecture on indecomposable Soergel bimodules

摘录 / EXCERPT: Soergel conjectured that the basis of the Hecke algebra given by indecomposable Soergel bimodules categorifies the Kazhdan-Lusztig basis; the report states that Elias and Williamso...

conjecture | open | MathDB 页面 | 原文线索

V2-011664 | MATHDB-398428

Recollements induced by finitely generated partial tilting modules of projective dimension greater than one

摘录 / EXCERPT: For a finitely generated partial tilting module over a ring B of projective dimension greater than one, does the recollement induced by the partial tilting module have a left-han...

open_problem | open | MathDB 页面 | 原文线索

V2-011665 | MATHDB-398429

Fully faithful derived Hom functors and compact objects

摘录 / EXCERPT: Let A and B be ordinary associative unital algebras and let ${}_B T_A$ be a bimodule. Assuming that B lies in the essential image of...

open_problem | open | MathDB 页面 | 原文线索

V2-011666 | MATHDB-398431

Recollements of module categories and Kuhn’s conjecture

摘录 / EXCERPT: Any recollement of $\operatorname{Mod}\text{-}A$ such that all the terms are equivalent to module categories is equivalent to a typical recollement.

conjecture | open | MathDB 页面 | 原文线索

V2-011667 | MATHDB-398432

Derived Jordan-Hölder property for derived module categories

摘录 / EXCERPT: For a ring A, does A satisfy (JH), meaning that $D(A)$ has a finite stratification that is unique up to ordering and equivalence?

open_problem | open | MathDB 页面 | 原文线索

V2-011668 | MATHDB-398433

Tilting modules arising from universal localisation for hereditary rings

摘录 / EXCERPT: Let A be a hereditary ring and let T be a tilting A-module. The following are equivalent: (1) T arises from universal localisation. (2) T is strongly minimal. (3) T is...

open_problem | open | MathDB 页面 | 原文线索

V2-011669 | MATHDB-398435

Faithful Galois action on Beauville surfaces

摘录 / EXCERPT: The absolute Galois group $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully on the discrete subset of the moduli space M of surfaces of general type correspo...

open_problem | open | MathDB 页面 | 原文线索

V2-011670 | MATHDB-398440

Non-simplicity of the planar Cremona group

摘录 / EXCERPT: For any algebraically closed field F, the group $\operatorname{Cr}_F(2)$ contains proper normal subgroups. In fact, one can construct an explicit birational transformation g o...

open_problem | open | MathDB 页面 | 原文线索

V2-011671 | MATHDB-398485

Random Galois extensions of Hilbertian fields are almost surely Hilbertian

摘录 / EXCERPT: Let K be a countable Hilbertian field and let L/K be a Galois extension. For a fixed positive integer n, the set of...

open_problem | open | MathDB 页面 | 原文线索

V2-011672 | MATHDB-398488

Prescribed coefficient ring in the local lifting problem for cyclic groups

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic p, let G be a finite cyclic group with $p^3\mid |G|$, let $K/k((t))$ be a G-Galois extension, and let A be the in...

open_problem | open | MathDB 页面 | 原文线索

V2-011673 | MATHDB-398493

Quantitative specialization solving unramified Grunwald problems

摘录 / EXCERPT: Assume $G\ne\{1\}$ is a regular Galois group over $\mathbb{Q}$, and fix a regular realization $F/\mathbb{Q}(T)$ of G. Then there exist an integer $m_0>0$ and a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-011674 | MATHDB-398498

Pfister involutions in characteristic two

摘录 / EXCERPT: Let (A,σ) be a non-metabolic central simple algebra with orthogonal involution over a field F of characteristic 2. Are the following equivalent? 1. (A,σ) is isomo...

open_problem | open | MathDB 页面 | 原文线索

V2-011675 | MATHDB-398558

Modularity of Kac-Wakimoto characters for $n>m$

摘录 / EXCERPT: For the Kac-Wakimoto characters associated with irreducible highest-weight $s\ell(m,n)^{\wedge}$-modules, with $n,m\in\mathbb{N}$ and $n>m$, can any modularity statement be establi...

open_problem | open | MathDB 页面 | 原文线索

V2-011676 | MATHDB-398563

Unboundedness Conjecture for full number fields

摘录 / EXCERPT: For a finite set of primes P, let $F_P(m)$ be the number of full degree-m number fields ramified within P, where a degree-m number field is full if its associated Galois gr...

conjecture | open | MathDB 页面 | 原文线索

V2-011677 | MATHDB-398628

Toms-Winter conjecture in the stably finite case

摘录 / EXCERPT: For a separable unital simple nuclear stably finite infinite-dimensional C^*-algebra A, the following are equivalent: (1) $A \cong A \otimes \mathcal{Z}$; (2) A has strict co...

conjecture | open | MathDB 页面 | 原文线索

V2-011678 | MATHDB-398632

Distribution of a selfadjoint polynomial in free variables

摘录 / EXCERPT: Let $x_1=x_1^*,\ldots,x_m=x_m^*$ be free with respect to a state φ on a unital C^*-algebra A, and let p be a selfadjoint polynomial in m non-commuting variables. Ho...

open_problem | open | MathDB 页面 | 原文线索

V2-011679 | MATHDB-398636

Finite nuclear dimension versus \[math]\[0m\[math]\mathcal Z\[0m\[math]\[0m-stability beyond the known cases

摘录 / EXCERPT: For a C[math][0m[math]^[0m-algebra [math]A[0m with finite nuclear dimension and with no ideal having type I representations, does [math]A[0m absorb the Jiang[math][0m[math]text{--}...

open_problem | open | MathDB 页面 | 原文线索

V2-011680 | MATHDB-398637

Classification of crossed-product \ II$_1$ factors

摘录 / EXCERPT: Given a trace-preserving action of a countable group Γ on a tracial von Neumann algebra (B,τ) and the crossed product $M=B\rtimes\Gamma$, the main question is to decide...

open_problem | open | MathDB 页面 | 原文线索

V2-011681 | MATHDB-398644

K-theory formula for reduced semigroup crossed products

摘录 / EXCERPT: Theorem 2. Suppose that $P\subseteq G$ satisfies the (weak) Toeplitz condition and the independence condition. Suppose further that G acts on the C^*-algebra A such that G...

open_problem | open | MathDB 页面 | 原文线索

V2-011682 | MATHDB-398681

Kaplansky’s direct finiteness conjecture for group algebras

摘录 / EXCERPT: For every countable discrete group Γ and every field K, the group algebra $K\Gamma$ is directly finite; equivalently, for all $a,b\in K\Gamma$, $ab=1$ implies $ba=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-011683 | MATHDB-398758

Whether g-vectors determine the partial order on support τ-tilting modules

摘录 / EXCERPT: We conjecture that g-vectors determine the partial order on $s\tau\text{-tilt}\Lambda$.

open_problem | open | MathDB 页面 | 原文线索

V2-011684 | MATHDB-398759

Classification of τ-rigid-finite algebras

摘录 / EXCERPT: Classify finite-dimensional (basic) algebras Λ over the algebraically closed field k that are τ-rigid-finite, meaning that there are only finitely many isomorphism c...

open_problem | open | MathDB 页面 | 原文线索

V2-011685 | MATHDB-398763

Equality of the skein algebra and cluster algebra for triangulated surfaces

摘录 / EXCERPT: For 0$ 0\Sigma$ any surface with a triangulation that is not a closed surface with exactly one puncture, 0$ 0\operatorname{Sk}(\Sigma)=\mathcal{A}(\Sigma)$. 0

open_problem | open | MathDB 页面 | 原文线索

V2-011686 | MATHDB-398811

Classifiable conjugacy for automorphisms of an infinite-dimensional simple C*-algebra

摘录 / EXCERPT: Determine whether there exists an infinite-dimensional simple separable unital C-algebra A such that conjugacy in $\operatorname{Aut}(A)$ is classifiable.

open_problem | open | MathDB 页面 | 原文线索

V2-011687 | MATHDB-398821

Rationality of the conjugation-invariant function field

摘录 / EXCERPT: Let k be an algebraically closed field with $\\operatorname{char} k=0$, and let G be simple. Is $k(G)$ rational over $k(G)^G$ for the conjugation action of G on itself? The r...

open_problem | open | MathDB 页面 | 原文线索

V2-011688 | MATHDB-398888

Periodicity of a finite-dimensional algebra from periodicity of its simple modules

摘录 / EXCERPT: An important open problem asks whether an algebra A is periodic over its enveloping algebra if all simple A-modules are periodic.

open_problem | open | MathDB 页面 | 原文线索

V2-011689 | MATHDB-398889

Graded coherence of higher preprojective algebras

摘录 / EXCERPT: It is conjectured that the higher preprojective algebra of every higher representation-infinite algebra Λ is always graded coherent.

open_problem | open | MathDB 页面 | 原文线索

V2-011690 | MATHDB-398894

Are all wild algebras controlled wild?

摘录 / EXCERPT: Let k be an algebraically closed field and let $\fLambda$ be a finite-dimensional k-algebra. If $\fLambda$ is wild, must it be controlled wild? Here, $\fLambda$ is controlled w...

open_problem | open | MathDB 页面 | 原文线索

V2-011691 | MATHDB-398897

Periodicity conjecture for finite-dimensional algebras

摘录 / EXCERPT: Let K be an algebraically closed field and let A be a finite-dimensional associative K-algebra with identity. If every simple module in the category $\operatorname{mod} A$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-011692 | MATHDB-398899

A grading for Brauer algebras

摘录 / EXCERPT: Does an analogue of the graded/categorified picture for symmetric groups work for Brauer algebras; in particular, is there geometric representation-theoretic structure that provide...

open_problem | open | MathDB 页面 | 原文线索

V2-011693 | MATHDB-398900

Bijection between Weyl-group elements and quotient-closed subcategories

摘录 / EXCERPT: Let Q be a Dynkin quiver and let k be an algebraically closed field. Let W be the Weyl group associated to Q. Using the fixed AR-word, define $f(X)$ for a full additive sub...

open_problem | open | MathDB 页面 | 原文线索

V2-011694 | MATHDB-398934

An E_2-operad model for iterated cobar constructions

摘录 / EXCERPT: Can we find an E_2 operad such that $S_*(\Omega^2X)$ is equivalent to $\Omega S_*(\Omega X)$ as E_2-algebras, and therefore to $\Omega^2S_*(X)$ as well?

open_problem | open | MathDB 页面 | 原文线索

V2-011695 | MATHDB-398996

Are all convex semialgebraic sets projections of spectrahedra?

摘录 / EXCERPT: Is it true that all convex semialgebraic sets are projections of spectrahedra (i.e., solution sets of linear matrix inequalities)?

open_problem | open | MathDB 页面 | 原文线索

V2-011696 | MATHDB-398997

A denominator Positivstellensatz for the Weyl algebra

摘录 / EXCERPT: Is every hermitian element of the Weyl algebra which is represented by a positive symmetric operator in the Schrödinger representation a sum of squares with denominator?

open_problem | open | MathDB 页面 | 原文线索

V2-011697 | MATHDB-399011

Truncated realizability for basic semi-algebraic sets of generalized functions

摘录 / EXCERPT: Can one generalize the results already obtained for the full realizability problem on basic semi-algebraic sets of generalized functions to the truncated case? In fact, it would be...

open_problem | open | MathDB 页面 | 原文线索

V2-011698 | MATHDB-399015

Finite-dimensional attainment in free polynomial optimization

摘录 / EXCERPT: Let $p,g_1,\\ldots,g_r\in\mathbb{R}\langle X\rangle$ be symmetric, and suppose the free quadratic module $QM(g)$ contains $N-\sum_i X_i^2$ for some $N\in\mathbb{N}$. Is the infimum...

open_problem | open | MathDB 页面 | 原文线索

V2-011699 | MATHDB-399026

Characterization of reflection Hopf algebras

摘录 / EXCERPT: Over an algebraically closed field of characteristic zero, completely characterize the semisimple Hopf algebras H acting inner-faithfully and grading-preservingly on degree-one-g...

open_problem | open | MathDB 页面 | 原文线索

V2-011700 | MATHDB-399030

Noetherianity of the bosonization associated to the standard filtration

摘录 / EXCERPT: Let H be a Hopf algebra with bijective antipode. Let H_0 be its coradical, let $H[0]$ be the subalgebra generated by H_0, and define the standard filtration by...

open_problem | open | MathDB 页面 | 原文线索

V2-011701 | MATHDB-399031

Equivalence of finiteness properties for a Hopf algebra and its bosonized graded form

摘录 / EXCERPT: Let H be a Hopf algebra with bijective antipode. Let $H_{[0]}$ be the Hopf coradical, generated as an algebra by the coradical H_0, and define the standard filtration by...

open_problem | open | MathDB 页面 | 原文线索

V2-011702 | MATHDB-399032

Closed constructions of finite-dimensional Nichols algebras by diagram folding

摘录 / EXCERPT: Is there a closed construction of the known finite-dimensional Nichols algebras (arbitrary rank, arbitrary characteristic) from diagonal Nichols algebras?

open_problem | open | MathDB 页面 | 原文线索

V2-011703 | MATHDB-399034

Hopf-ideal decomposition relative to the maximal pointed Hopf subalgebra

摘录 / EXCERPT: For each Hopf algebra H over an algebraically closed field, let H' be its maximal pointed Hopf subalgebra, introduced using a generalized direct-sum decomposition of coalgebras...

open_problem | open | MathDB 页面 | 原文线索

V2-011704 | MATHDB-399035

Converse to gauge invariance of Frobenius-Schur indicator values

摘录 / EXCERPT: Let C be the category of modules of a semisimple Hopf algebra H, generated by its irreducible modules $V_1,V_2,\ldots,V_n$. Let m be the exponent of H, and let $\nu_j(V_i)$...

open_problem | open | MathDB 页面 | 原文线索

V2-011705 | MATHDB-399036

Classification of reductive affine algebraic supergroups

摘录 / EXCERPT: Classify all super-commutative Hopf superalgebras A over an algebraically closed field k with $\\operatorname{char} k \\ne 2$ such that: (a) A is finitely generated as an alg...

open_problem | open | MathDB 页面 | 原文线索

V2-011706 | MATHDB-399037

Bound on the PI degree of the dual Hopf algebra

摘录 / EXCERPT: If a cosemisimple Hopf algebra H over k acts inner faithfully on a PI domain of PI degree d, must $\\operatorname{PIdeg}(H^*) \\le d^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-011707 | MATHDB-399038

Braided Hopf algebra structures on noetherian Artin-Schelter regular algebras

摘录 / EXCERPT: Let A be a noetherian Artin-Schelter regular algebra. Is A a braided Hopf algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011708 | MATHDB-399041

Constructing finite-dimensional Hopf actions on Artin-Schelter regular algebras

摘录 / EXCERPT: Given an Artin-Schelter regular algebra A, is there a natural way of constructing finite-dimensional Hopf algebras H that act on A inner faithfully?

open_problem | open | MathDB 页面 | 原文线索

V2-011709 | MATHDB-399042

Abelianness of the double commutant algebra for finite Blaschke products

摘录 / EXCERPT: For a Blaschke product ϕ of finite order, the algebra $A_\phi=\{M_\phi,M_\phi^*\}'$ is abelian.

open_problem | open | MathDB 页面 | 原文线索

V2-011710 | MATHDB-399043

Analytic disks in the maximal ideal space of the Dirichlet multiplier algebra

摘录 / EXCERPT: Does $M(M(\\mathcal D_0))$ contain a nontrivial analytic disk?

open_problem | open | MathDB 页面 | 原文线索

V2-011711 | MATHDB-399045

Characters over the origin for multiplier algebras of properly embedded discs

摘录 / EXCERPT: Let $f:\mathbb D\to V\subset\mathbb B_\infty$ be a proper embedding that extends continuously to $\overline{\mathbb D}$, and assume that $f(0)=0$. Is the character of point evaluat...

open_problem | open | MathDB 页面 | 原文线索

V2-011712 | MATHDB-399048

Arveson-Douglas Conjecture

摘录 / EXCERPT: Let H^2_d be the Drury-Arveson space on the unit ball $B_d\subset\mathbb{C}^d$. Assume that $M\subset H^2_d$ is generated, as a module over $\mathbb{C}[z_1,\ldots,z_d]$, by finit...

conjecture | open | MathDB 页面 | 原文线索

V2-011713 | MATHDB-399073

Finiteness of finite quadratic modules with vanishing cusp-form space

摘录 / EXCERPT: Let $r_0\in\mathbb{Z}_{\ge 0}$. There exist only finitely many isomorphism classes of finite quadratic modules A with minimal number of generators $\le r_0$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-011714 | MATHDB-399084

Nonvanishing of the finite-dimensional perfect derived subcategory for AS Gorenstein algebras

摘录 / EXCERPT: Let k be an algebraically closed field, and let $A=k\\oplus A_1\\oplus A_2\\oplus\\cdots$ be a finitely generated $\\mathbb N$-graded k-algebra that is Artin-Schelter Gorenstei...

open_problem | open | MathDB 页面 | 原文线索

V2-011715 | MATHDB-399087

Algebraic sliceness of the closures of \((\sigma_1\sigma_2^{-1})^k\)

摘录 / EXCERPT: Let $k\not\equiv 0\pmod 3$ and $\alpha_k=(\sigma_1\sigma_2^{-1})^k$. Then the knot $\widehat{\alpha_k}$ is algebraically slice if and only if k is odd.

open_problem | open | MathDB 页面 | 原文线索

V2-011716 | MATHDB-399095

A possible q-version of the boundary map for presimplicial modules

摘录 / EXCERPT: Let k be a commutative ring and let C be a chain complex arising from a presimplicial module, with $ ... $ The report asks whether it is useful, or has already been used, to cons...

open_problem | open | MathDB 页面 | 原文线索

V2-011717 | MATHDB-399100

Algebraic version of the exponential BKK theorem in two variables

摘录 / EXCERPT: For a quasialgebraic system of exponential sums in two variables with supports $\Lambda=(\Lambda_1,\Lambda_2)$, Newton polygons Δ_1,Δ_2, and $\Delta=\Delta_1+\Delta_2$,...

open_problem | open | MathDB 页面 | 原文线索

V2-011718 | MATHDB-399131

Gonality conjecture for weight-one syzygies of algebraic curves

摘录 / EXCERPT: Let C be a smooth complex projective curve of genus $g\ge 2$, and let L be a very ample line bundle on C of sufficiently large degree, with $r=h^0(C,L)-1$. If $K_{p,1}(C;L)$...

conjecture | open | MathDB 页面 | 原文线索

V2-011719 | MATHDB-399143

Conjectural algebraicity and vanishing of the generalized Kato class

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve and let $\varrho=\varrho_1\otimes\varrho_2$, where ϱ_1 and ϱ_2 are odd, irreducible, two-dimensional representations of...

open_problem | open | MathDB 页面 | 原文线索

V2-011720 | MATHDB-399148

Non-liftability of automorphism groups of K3 surfaces in positive characteristic

摘录 / EXCERPT: Let X be a K3 surface defined over an algebraically closed field k of characteristic $p>0$, where $p>2$ if X is Artin-supersingular. 1. Assume that either the Picard number o...

open_problem | open | MathDB 页面 | 原文线索

V2-011721 | MATHDB-399238

Positivity of the Chebyshev basis for surface skein algebras

摘录 / EXCERPT: For an oriented compact surface Σ (possibly with boundary), and for general t, the basis b_T of the skein algebra $S(\Sigma)$ associated with the collection...

open_problem | open | MathDB 页面 | 原文线索

V2-011722 | MATHDB-399332

Heber’s Standardness Conjecture for Einstein solvable Lie algebras

摘录 / EXCERPT: Every metric solvable Einstein Lie algebra $(\mathfrak g,\langle\cdot,\cdot\rangle)$ is standard: if $\mathfrak n$ is its nilradical and $\mathfrak a:=\mathfrak n^\perp$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-011723 | MATHDB-399334

Isomorphism classes of Lie algebras with a prescribed index set

摘录 / EXCERPT: Let $\Lambda\subseteq\Theta_n=\{(i,j,k):1\le i<j<k\le n\}$. Let $\mathcal L_\Lambda$ denote the set of Lie algebras whose Lie brackets, relative to a basis $\{x_i\}_{i=1}^n$, have...

open_problem | open | MathDB 页面 | 原文线索

V2-011724 | MATHDB-399357

Total blow-up along a valuation

摘录 / EXCERPT: For a regular local ring of dimension $\97\e 3$ and a valuation, the total blow-up along the valuation may not be a valuation ring. What are its algebraic properties, and under whi...

open_problem | open | MathDB 页面 | 原文线索

V2-011725 | MATHDB-399359

Maximality under infinite algebraic extensions

摘录 / EXCERPT: Assume that (K,v) is a maximal field and let $(L|K,v)$ be an infinite algebraic extension. Suppose that $L|K$ contains an infinite separable subextension, or that ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-011726 | MATHDB-399361

d-Henselian fields with few constants are d-algebraically maximal

摘录 / EXCERPT: Conjecture: If a valued differential field K with small derivation and residue field of characteristic 0 is d-henselian and has few constants, then K is d-algebraically maxim...

open_problem | open | MathDB 页面 | 原文线索

V2-011727 | MATHDB-399363

Kontsevich-Zagier conjecture for periods

摘录 / EXCERPT: Let $X_1,X_2\subseteq\mathbb{R}^n$ be semialgebraic sets, and let $f_i:X_i\to\mathbb{R}$ be semialgebraic functions. If $\int_{X_1} f_1=\int_{X_2} f_2$, does it follow that one int...

conjecture | open | MathDB 页面 | 原文线索

V2-011728 | MATHDB-399364

Kontsevich-Zagier conjecture for semialgebraic integrals

摘录 / EXCERPT: Let $X_1,X_2\subseteq\mathbb{R}^n$ be semialgebraic sets, meaning Boolean combinations of inequalities $g(x_1,\ldots,x_n)\ge 0$ with $g\in\mathbb{Q}[x_1,\ldots,x_n]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011729 | MATHDB-399372

The expected coefficient ring for cyclic local liftings

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic p, let G be cyclic with $p^3\mid |G|$, and let $k[[z]]/k[[t]]$ be a G-Galois extension. The local Oort theorem guar...

open_problem | open | MathDB 页面 | 原文线索

V2-011730 | MATHDB-399405

Decidability of the 0-generated subalgebra of the polymodal provability algebra

摘录 / EXCERPT: Let M be the Lindenbaum Boolean algebra of Peano arithmetic, equipped with unary operators $d_n:M\to M$ for $n=0,1,\ldots$, where d_n sends the equivalence class of a sentence...

open_problem | open | MathDB 页面 | 原文线索

V2-011731 | MATHDB-399423

Commutant question for the left and right Brownian representations

摘录 / EXCERPT: Whether the von Neumann algebras generated by the left and right Brownian representations of $H(G)$ are commutants of each other, where G is a compact connected Lie group and...

open_problem | open | MathDB 页面 | 原文线索

V2-011732 | MATHDB-399426

Euclidean-invariant integral representations of Schwinger functionals

摘录 / EXCERPT: Given a symmetric, Euclidean-invariant functional S on the Borchers-Uhlmann algebra over $\mathcal{S}(\mathbb{R}^d)$, when does there exist an integral representation of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-011733 | MATHDB-399461

Isomorphism of the current algebra, the q-Onsager algebra, and a coideal subalgebra

摘录 / EXCERPT: For q not a root of unity, show that the infinite-dimensional algebra A_q, the q-Onsager algebra, and the coideal subalgebra of $U_q(\widehat{\mathfrak{sl}}_2)$ described abo...

open_problem | open | MathDB 页面 | 原文线索

V2-011734 | MATHDB-399470

Functorial integration of finite-dimensional real Leibniz algebras

摘录 / EXCERPT: Does there exist a functor J from the category of all finite-dimensional real Leibniz algebras, with bracket-preserving linear maps as morphisms, to the category of pointed manif...

open_problem | open | MathDB 页面 | 原文线索

V2-011735 | MATHDB-399471

Existence of bimodule deformations from semi-Poisson brackets

摘录 / EXCERPT: In the general Poisson case, is the existence of a semi-Poisson bracket enough to obtain a bimodule deformation?

open_problem | open | MathDB 页面 | 原文线索

V2-011736 | MATHDB-399480

Stabilization of nonlinear controllable systems by trigonometric-polynomial feedback

摘录 / EXCERPT: Assume that the Lie algebra generated by the vector fields satisfies ... Is it possible to construct, for some $N\ge 0$ and $\omega>0$, a time-varying feedback law ... such that the tr...

open_problem | open | MathDB 页面 | 原文线索

V2-011737 | MATHDB-399502

Farrell-Jones Isomorphism Conjecture in algebraic K-theory

摘录 / EXCERPT: For a group G and an associative ring with unit R, the Farrell-Jones Conjecture predicts that the assembly map ... is a weak equivalence, where $\mathbf{K}$ denotes non-connectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-011738 | MATHDB-399511

Determinant of a Witt-vector torsion complex

摘录 / EXCERPT: Given a perfect ring R and a map $f:M\to N$ of finite projective $W(R)$-modules such that $f[1/p]$ is an isomorphism, let $Q=\operatorname{coker}(f)$. Then Q is a finite $W(R)$...

open_problem | open | MathDB 页面 | 原文线索

V2-011739 | MATHDB-399513

Ind-projectivity of the mixed-characteristic affine Grassmannian

摘录 / EXCERPT: Let Perf be the category of perfect $\mathbf F_p$-algebras. For $R\in\mathrm{Perf}$, define $\mathrm{Gr}(R)$ to be the set of isomorphism classes of pairs (E,φ),...

open_problem | open | MathDB 页面 | 原文线索

V2-011740 | MATHDB-399519

Elementary proof of the generic-vanishing decomposition without Hodge modules

摘录 / EXCERPT: Is it really necessary to use Hodge modules in this context? More specifically, is there an elementary proof, without Hodge modules, of the decomposition result for higher direct i...

open_problem | open | MathDB 页面 | 原文线索

V2-011741 | MATHDB-399532

Weak Verlinde formula beyond rationality

摘录 / EXCERPT: Assuming there is a special simple module M_r with lowest conformal weight, is the weak Verlinde formula valid beyond rationality; that is, do the asymptotic dimensions ... give a...

open_problem | open | MathDB 页面 | 原文线索

V2-011742 | MATHDB-399557

Kawanaka’s wave-front-set conjecture for generalised Gelfand-Graev characters

摘录 / EXCERPT: Let G be a connected reductive algebraic group over $\overline{\mathbb F}_p$, with $p>0$ good for G, and let $F:G\to G$ be a Frobenius endomorphism such that G^F is finite....

conjecture | open | MathDB 页面 | 原文线索

V2-011743 | MATHDB-399559

Conjecture A on graded Cartan invariants of symmetric groups and Hecke algebras

摘录 / EXCERPT: Let $H_n(\xi)$ be the Iwahori-Hecke algebra of type A_{n-1} over a field containing a primitive ℓ-th root of unity ξ, let $A=\mathbb{Z}[v,v^{-1}]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-011744 | MATHDB-399561

Conjecture A on graded Cartan matrices of Hecke algebras

摘录 / EXCERPT: Let $A=\mathbb{Z}[v,v^{-1}]$. For the graded Cartan matrix $C_vH_n(\xi)$ of the Iwahori-Hecke algebra $H_n(\xi)$, where ξ is a primitive ℓ-th root of unity, one has ... wher...

conjecture | open | MathDB 页面 | 原文线索

V2-011745 | MATHDB-399568

Vertices of hook-labelled simple modules of symmetric groups

摘录 / EXCERPT: Let $p>2$, let F be a field of characteristic p, and let $n\in\mathbb{N}$ satisfy $n\equiv p\pmod{p^2}$. Then the vertices of the simple FS_n-module $D^{(n-p+1,1^{p-1})}$ are...

open_problem | open | MathDB 页面 | 原文线索

V2-011746 | MATHDB-399569

Reduction of the torsion endo-trivial-module problem for groups with dihedral Sylow 2-subgroups

摘录 / EXCERPT: In order to determine the structure of $T_T(G)$, is it true that the problem can be reduced essentially to the case where $O_{p'}(G)=1$? Here the question concerns the case $p=2$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-011747 | MATHDB-399570

Tilting modules and the anti-spherical module

摘录 / EXCERPT: Let G be semisimple and simply connected over an algebraically closed field k of characteristic p, with p larger than the Coxeter number of G. Let Rep_0...

open_problem | open | MathDB 页面 | 原文线索

V2-011748 | MATHDB-399612

Equality of the equivariant Brieskorn lattice and Gauss-Manin system

摘录 / EXCERPT: The conjecture is that the submodule H_B of the Gauss-Manin system generated by the classes $[p_\lambda\omega]$ of the Plücker coordinates satisfies $H_B=GW_q^{\mathrm{eq}}$.

open_problem | open | MathDB 页面 | 原文线索

V2-011749 | MATHDB-399617

Comparison of the Kähler seminorm with the weight function

摘录 / EXCERPT: Assume that k is discretely valued with a uniformizer π_k, that X is a smooth algebraizable Berkovich space over k, that $\phi\in\omega_X^{\otimes m}$, and that $x\in X$...

open_problem | open | MathDB 页面 | 原文线索

V2-011750 | MATHDB-399619

Real algebraic surfaces of tridegree $(4,4,2)$ with large first Betti number

摘录 / EXCERPT: The existence of a nonsingular real algebraic surface X of degree $(4,4,2)$ in $(\mathbb{CP}^1)^3$ satisfying $b_1(\mathbb{R}X)\ge h^{1,1}(X)+6$ is still open.

open_problem | open | MathDB 页面 | 原文线索

V2-011751 | MATHDB-399622

Rickard’s tensor-product question for stable equivalences of Morita type

摘录 / EXCERPT: Let K be a field, and let A,B,C,D be finite-dimensional K-algebras. If A and B are stably equivalent of Morita type and C and D are stably equivalent of Morita type,...

open_problem | open | MathDB 页面 | 原文线索

V2-011752 | MATHDB-399638

Bilax monoidal structure for a functor induced by a stable equivalence of Morita type

摘录 / EXCERPT: Let A and B be K-algebras, and let (M,N,φ,ψ) be a stable equivalence of Morita type, where M is an $A\otimes B^{\mathrm{op}}$-module and N is a...

open_problem | open | MathDB 页面 | 原文线索

V2-011753 | MATHDB-399645

Action of a coideal subalgebra on half-integral blocks of category $\mathcal O$

摘录 / EXCERPT: Theorem 1. On the direct sum of the blocks in $\mathcal O(\mathfrak g)$ for half-integral weights there is an action of the coideal subalgebra B_{2n}, where Letzter’s generators...

open_problem | open | MathDB 页面 | 原文线索

V2-011754 | MATHDB-399647

Monoidal categorification conjecture for the category \(\mathcal O^+\)

摘录 / EXCERPT: The category $\mathcal O^+$ is a monoidal categorification of the infinite-rank cluster algebra $A(Q)$.

conjecture | open | MathDB 页面 | 原文线索

V2-011755 | MATHDB-399656

The multiplicative eigenvalue problem for a compact simply connected group

摘录 / EXCERPT: Let G be a simple, connected, simply-connected complex algebraic group, with Borel subgroup B, maximal torus $H\subset B$, Weyl group W, and fundamental alcove … where...

open_problem | open | MathDB 页面 | 原文线索

V2-011756 | MATHDB-399674

Generation of the tail algebra by the tail C*-algebra

摘录 / EXCERPT: Is T_φ generated as a von Neumann algebra by D_φ?

open_problem | open | MathDB 页面 | 原文线索

V2-011757 | MATHDB-399680

Rosenberg’s conjecture on quasidiagonality of amenable group C*-algebras

摘录 / EXCERPT: For every countable discrete group G, if G is amenable, then its reduced group C-algebra $C^*_{\lambda}(G)$ is quasidiagonal.

conjecture | open | MathDB 页面 | 原文线索

V2-011758 | MATHDB-399687

Nonabelian determination of truncated Galois groups by Krasner hyperfields

摘录 / EXCERPT: For a local field K and $u>1$, does the hyperfield $K/(1+\mathfrak m_K^u)$ determine the quotient...

open_problem | open | MathDB 页面 | 原文线索

V2-011759 | MATHDB-399737

Computing the local Galois representation of a hyperelliptic curve

摘录 / EXCERPT: Given a hyperelliptic curve C/K over a local field K, in the form $C:y^2=c\prod_{r\in R}(x-r)$ with $R\subset\overline K$, $|R|=2g+1$ or $2g+2$, and residue characteristic diff...

open_problem | open | MathDB 页面 | 原文线索

V2-011760 | MATHDB-399852

Approximation of holomorphic foliations by Nash algebraic foliations

摘录 / EXCERPT: Can holomorphic foliations be approximated by Nash algebraic foliations uniformly on compact subsets of any polynomially convex open subset of $\mathbb{C}^N$?

open_problem | open | MathDB 页面 | 原文线索

V2-011761 | MATHDB-399911

Existence of an algebraic surface of Boy type

摘录 / EXCERPT: Does there exist an algebraic surface of the form of Boy’s surface?

open_problem | open | MathDB 页面 | 原文线索

V2-011762 | MATHDB-399932

Stanley’s depth conjecture

摘录 / EXCERPT: Is the Stanley depth of every finitely generated graded S-module M bounded below by its depth; that is, does $\operatorname{sdepth} M \ge \operatorname{depth} M$ hold for every...

conjecture | open | MathDB 页面 | 原文线索

V2-011763 | MATHDB-399957

Rationality of the ball-quotient moduli spaces

摘录 / EXCERPT: For the algebraic moduli space represented by the complex ball quotient $M^{\rho,\xi}_{K(T)}/G \cong \Omega/\Gamma$ in Theorem 3, in which cases is it rational?

open_problem | open | MathDB 页面 | 原文线索

V2-011764 | MATHDB-399961

Dimension of the locus covered by rational curves under deformation

摘录 / EXCERPT: When deforming $X_0=S^{[k]}_{\\varepsilon}$ keeping the class of the rational curves algebraic, does (some component of) the locus they cover keep its dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-011765 | MATHDB-400030

Hyperbolic Ax-Lindemann theorem

摘录 / EXCERPT: Let $j:\mathbb{H}^n\to\mathbb{C}^n$ be the morphism $(z_1,\ldots,z_n)\mapsto (j(z_1),\ldots,j(z_n))$, where j is the elliptic modular function. Let V be an irreducible algebrai...

open_problem | open | MathDB 页面 | 原文线索

V2-011766 | MATHDB-400070

Affine-linear threshold for rational curves of prescribed gonality

摘录 / EXCERPT: Is there an affine linear function f such that for all $g \geq f(\gamma)$ there are algebraic curves defined over the rational numbers with genus g and gonality γ? Note...

open_problem | open | MathDB 页面 | 原文线索

V2-011767 | MATHDB-400088

Algebras with finite separable-equivalence classes up to Morita equivalence

摘录 / EXCERPT: Which algebras (other than block algebras) have the property that their separable equivalence class consists of finitely many Morita equivalence classes?

open_problem | open | MathDB 页面 | 原文线索

V2-011768 | MATHDB-400090

Dependence of tensor-product modules on the coproduct

摘录 / EXCERPT: Let k be a field, let A be a k-algebra, and let $\Delta:A\to A\otimes_k A$ be a coproduct on A. For left A-modules M and N, how does the A-module...

open_problem | open | MathDB 页面 | 原文线索

V2-011769 | MATHDB-400094

Rouquier’s conjecture on BV algebras of Koszul Calabi-Yau algebras

摘录 / EXCERPT: Let A be a Koszul Calabi-Yau algebra, and let $A^!$ be its Koszul dual algebra (which is necessarily a graded symmetric algebra). Then there is an isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-011770 | MATHDB-400110

Existence of global monomial degrees for all simple finite-dimensional Lie algebras

摘录 / EXCERPT: Let $\mathfrak g$ be a simple finite-dimensional complex Lie algebra with triangular decomposition $\mathfrak g=\mathfrak n^+\oplus\mathfrak h\oplus\mathfrak n^-$. Let Δ^+ b...

open_problem | open | MathDB 页面 | 原文线索

V2-011771 | MATHDB-400117

Finite generation of monoids arising from birational sequences

摘录 / EXCERPT: Let G be a complex semisimple algebraic group, let $B=TU$ be a Borel subgroup with unipotent radical U, and let U^{-} be the unipotent radical of the opposite Borel subgroup....

open_problem | open | MathDB 页面 | 原文线索

V2-011772 | MATHDB-400135

Finite generation of the symbolic Rees algebra of a Fermat ideal

摘录 / EXCERPT: Let $n\ge 3$ be an integer, let k be a field containing n distinct nth roots of 1, and let $R=k[x,y,z]$. For the Fermat ideal ... is the symbolic Rees algebra ... finitely gene...

open_problem | open | MathDB 页面 | 原文线索

V2-011773 | MATHDB-400145

Zero-divisors cup-length formula for Orlik-Solomon algebras

摘录 / EXCERPT: Let 4mathcal A be a central arrangement, let $A=H^*(X_{\mathcal A},\mathbb C)$ be its Orlik-Solomon algebra, and construct the squarefree monomial ideal $I_\Delta\subset E'$u0004 f...

open_problem | open | MathDB 页面 | 原文线索

V2-011774 | MATHDB-400188

Triviality of minuscule modifications arising from the Gross-Messing period map

摘录 / EXCERPT: Assume $E=\\mathbb{Q}_p$. Let H be a p-divisible group over $\\mathbb{F}_p$, let $D\\in\\varphi\\mathrm{Mod}_L$ be its rational covariant Dieudonné module, and set...

open_problem | open | MathDB 页面 | 原文线索

V2-011775 | MATHDB-400190

Étale-local existence of B-structures on G-bundles

摘录 / EXCERPT: Let k be an algebraically closed field, let G be a split reductive group over k, let X be a connected smooth projective curve over k, and let $B\subset G$ be a Borel subg...

open_problem | open | MathDB 页面 | 原文线索

V2-011776 | MATHDB-400200

Scholze-Weinstein description of the infinite-level Lubin-Tate tower

摘录 / EXCERPT: Let $H_0/\mathbb F_p$ be the connected p-divisible group of dimension 1 and height n corresponding to the isocrystal b. For every perfectoid $\mathbb Q_p$-algebra (R,R^+)...

open_problem | open | MathDB 页面 | 原文线索

V2-011777 | MATHDB-400202

Intersection of the rings of E-values and G-values

摘录 / EXCERPT: Conjecture: Let E be the ring of values taken at algebraic points by E-functions, and let G be the ring of values taken at algebraic points by analytic continuations of G-funct...

open_problem | open | MathDB 页面 | 原文线索

V2-011778 | MATHDB-400203

Algebraic values of E-functions

摘录 / EXCERPT: If an E-function has coefficients in a number field K, then at an algebraic point α it takes either a transcendental value or a value in $K(\alpha)$. In particular, there...

open_problem | open | MathDB 页面 | 原文线索

V2-011779 | MATHDB-400204

Algebraic values of transcendental Mahler functions

摘录 / EXCERPT: How can one determine the algebraic points at which a transcendental function $f_1(z)$ takes algebraic values?

open_problem | open | MathDB 页面 | 原文线索

V2-011780 | MATHDB-400216

Strong Lehmer conjecture for the height of algebraic numbers

摘录 / EXCERPT: Let h denote the absolute logarithmic Weil height on a fixed algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$. Northcott's theorem implies that there is a function...

conjecture | open | MathDB 页面 | 原文线索

V2-011781 | MATHDB-400224

Weak Diversity Conjecture

摘录 / EXCERPT: Let X be an algebraic curve over $\b[?]$ with genus g, let $t\in\mathbb{Q}(X)$ be a non-constant rational function of degree $\nu\ge 2$, and fix an algebraic closure...

conjecture | open | MathDB 页面 | 原文线索

V2-011782 | MATHDB-400225

Strong Diversity Conjecture (Schinzel)

摘录 / EXCERPT: Let X be an algebraic curve over $\mathbb{Q}$ of genus g, let $t\in\mathbb{Q}(X)$ be a non-constant rational function of degree $\nu\ge 2$, and, for every positive integer n...

conjecture | open | MathDB 页面 | 原文线索

V2-011783 | MATHDB-400232

Lehmer's problem for algebraic numbers

摘录 / EXCERPT: Does there exist a positive real constant c such that, for every nonzero algebraic number α of degree d that is not a root of unity, the usual logarithmic Weil height sa...

open_problem | open | MathDB 页面 | 原文线索

V2-011784 | MATHDB-400332

Quadratic Siegel's lemma for a small isotropic vector

摘录 / EXCERPT: Let E be a rigid adelic space of dimension n over an algebraic extension $K/\mathbb{Q}$, equipped with a quadratic form q. If q is isotropic, then for every $\varepsilon>0$...

open_problem | open | MathDB 页面 | 原文线索

V2-011785 | MATHDB-400333

Quadratic Siegel's lemma avoiding an algebraic set

摘录 / EXCERPT: Let K be a Siegel field, let $q:K^n\to K$ be a quadratic form with $q\ne0$, and let I be an ideal of $K[X_1,\ldots,X_n]$ generated by polynomials of total degree at most M. L...

open_problem | open | MathDB 页面 | 原文线索

V2-011786 | MATHDB-400342

Rational isomorphism of hermitian forms over maximal orders

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field F, and assume that $2\in R^{\times}$. Let A be a maximal R-order in a separable F-algebra, and let $\sigma:A\to A$...

open_problem | open | MathDB 页面 | 原文线索

V2-011787 | MATHDB-400354

Continuity problem in algebraic K-theory

摘录 / EXCERPT: For an I-adically complete ring A, is the canonical map $K(A)\to\operatorname{holim}_{s}K(A/I^{s})$ a weak equivalence after profinite completion?

open_problem | open | MathDB 页面 | 原文线索

V2-011788 | MATHDB-400372

Distinguished model of the algebraic intermediate Jacobian

摘录 / EXCERPT: Question 3. Does $J^{2n+1}_a(X_C)$ admit a distinguished model J over K, in the sense that the Abel-Jacobi map ... is $\operatorname{Aut}(\mathbb{C}/K)$-equivariant?

open_problem | open | MathDB 页面 | 原文线索

V2-011789 | MATHDB-400413

Vanishing consequence of the Ausoni-Rognes red-shift conjectures

摘录 / EXCERPT: For a commutative ring spectrum R with $K(n+i)_*R=0$ for all $i\ge 1$, one expects that $K(n+i)_*K(R)=0$ for all $i\ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-011790 | MATHDB-400417

Survival of all powers of v_{n-1} in iterated algebraic K-theory

摘录 / EXCERPT: To prove the red-shift conjecture for $\mathbb F_p$, show that all powers of the generator v_{n-1} of connective Morava K-theory survive in the map...

open_problem | open | MathDB 页面 | 原文线索

V2-011791 | MATHDB-400418

Malle’s conjecture for number fields with specified Galois group

摘录 / EXCERPT: For a fixed integer n and a transitive subgroup $G\leq S_n$, define ... Malle conjectures that there exist constants a, b, and C such that, as $X\to\infty$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-011792 | MATHDB-400424

Nontriviality of local algebras for integrable models with bound states

摘录 / EXCERPT: Assume that strong commutativity holds for a suitable extension $\widetilde\phi_-$. Define ... and define $\mathcal A(W_R-y)$ similarly. For ... show that this intersection is nontrivi...

open_problem | open | MathDB 页面 | 原文线索

V2-011793 | MATHDB-400433

Efficient generation of the Hecke algebra in condensation

摘录 / EXCERPT: Find an efficient method for identifying a small set of group elements S such that eSe generates the Hecke algebra eke, where e is the idempotent obtained as the sum of the...

open_problem | open | MathDB 页面 | 原文线索

V2-011794 | MATHDB-400453

UCT for finite-nuclear-dimension C*-algebras

摘录 / EXCERPT: Does every unital, simple, separable, nuclear C^*-algebra with finite nuclear dimension satisfy the Universal Coefficient Theorem (UCT)?

open_problem | open | MathDB 页面 | 原文线索

V2-011795 | MATHDB-400456

Self-absorption of the Razak-Jacelon algebra

摘录 / EXCERPT: Is the algebra W self-absorbing, i.e. is $W\cong W\otimes W$?

open_problem | open | MathDB 页面 | 原文线索

V2-011796 | MATHDB-400457

Blackadar-Kirchberg Problem: quasidiagonality of stably finite nuclear C*-algebras

摘录 / EXCERPT: Are all stably finite nuclear C^*-algebras quasidiagonal?

open_problem | open | MathDB 页面 | 原文线索

V2-011797 | MATHDB-400458

Quasidiagonality of the hyperfinite \II_1\ factor

摘录 / EXCERPT: Is the hyperfinite II_1 factor R quasidiagonal as a C^*-algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-011798 | MATHDB-400460

Characterization of the class \(\mathcal N_1\)

摘录 / EXCERPT: Which C^*-algebras belong to the class $\mathcal N_1$?

open_problem | open | MathDB 页面 | 原文线索

V2-011799 | MATHDB-400461

Examples of II_1 factors with s-MASAs but no Cartan subalgebras

摘录 / EXCERPT: Produce examples of $\a0\mathrm{II}_1$ factors that have s-MASAs but do not have Cartan subalgebras.

open_problem | open | MathDB 页面 | 原文线索

V2-011800 | MATHDB-400476

Diagonal-preserving classification of unital graph C^*-algebras by graph moves

摘录 / EXCERPT: Conjecture: For all unital graph C^*-algebras, one graph can be transformed into another using only the classical graph moves-thus avoiding the Cuntz splice and the new move desc...

open_problem | open | MathDB 页面 | 原文线索

V2-011801 | MATHDB-400477

Injectivity of the canonical Hecke-algebra C*-completion map for $(\mathrm{SL}_2(\mathbb{Q}_p),\mathrm{SL}_2(\mathbb{Z}_p))$

摘录 / EXCERPT: For the Hecke pair $(\mathrm{SL}_2(\mathbb{Q}_p),\mathrm{SL}_2(\mathbb{Z}_p))$, the canonical surjection ... is not injective. Here G is the totally disconnected completion associa...

open_problem | open | MathDB 页面 | 原文线索

V2-011802 | MATHDB-400478

Non-injectivity of the canonical Hecke-algebra C*-completion map for n=3

摘录 / EXCERPT: For the pair $(G,H)=(\mathrm{SL}_3(\mathbb{Q}_p),\mathrm{SL}_3(\mathbb{Z}_p))$, the canonical surjection ... is not injective.

open_problem | open | MathDB 页面 | 原文线索

V2-011803 | MATHDB-400479

Tail C*-algebra inclusion for symmetric states

摘录 / EXCERPT: For every symmetric state $\psi\in SS(A)$ on the universal unital free product $A=\ast_{1}^{\infty}A_0$, does the tail C^*-algebra satisfy $D_\psi\subseteq \pi_\psi(A)$?

open_problem | open | MathDB 页面 | 原文线索

V2-011804 | MATHDB-400519

Smallest integration-stable algebra containing subanalytic and oscillatory functions

摘录 / EXCERPT: What is the smallest $\mathcal C$-algebra $\mathcal E$ of functions containing all subanalytic functions and all complex exponentials e^{ih} with h subanalytic, and stable unde...

open_problem | open | MathDB 页面 | 原文线索

V2-011805 | MATHDB-400531

Strict monotonicity of the algebraic stringy Euler number under Mori flips

摘录 / EXCERPT: Let $g:X\dashrightarrow X^+$ be a Mori flip. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-011806 | MATHDB-400550

First-order determination of finitely generated commutative rings

摘录 / EXCERPT: For any finitely generated commutative ring R, there is a first-order sentence in the language of rings that determines R, among the finitely generated commutative rings, up to...

open_problem | open | MathDB 页面 | 原文线索

V2-011807 | MATHDB-400562

Northcott property of fields generated by algebraic numbers of bounded degree

摘录 / EXCERPT: For a positive integer d, let $\mathbb{Q}^{(d)}$ be the compositum of all extensions of $\mathbb{Q}$ of degree at most d. Does $\mathbb{Q}^{(d)}$ have Property (N), meaning tha...

open_problem | open | MathDB 页面 | 原文线索

V2-011808 | MATHDB-400591

Uniqueness of the predicate for the rational integers in an ultrapower

摘录 / EXCERPT: Let A be a ring of totally real algebraic integers, let L be the ring language, let I be a set, let $\mathcal F$ be an ultrafilter on I, and put $R=A^I/\mathcal F$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-011809 | MATHDB-400597

Decidability of the existential theory of rigid entire-function rings

摘录 / EXCERPT: Let k be an algebraically closed field, complete for a non-trivial absolute value $|{-}|$. Let A_k be the ring of power series in the variable t with infinite radius of conve...

open_problem | open | MathDB 页面 | 原文线索

V2-011810 | MATHDB-400606

Converse to triple linkage versus the u-invariant

摘录 / EXCERPT: Let F be a field of characteristic different from 2. If $u(F)\le 4$, must every set of three quaternion algebras over F be linked; equivalently, must F have triple linkage?

open_problem | open | MathDB 页面 | 原文线索

V2-011811 | MATHDB-400611

Connected components of real homogeneous spaces

摘录 / EXCERPT: Let G be a connected, simply connected, absolutely simple algebraic group over $\d53d$, and let $H\subset G$ be a connected, simply connected, semisimple $\d53d$-subgroup. For th...

open_problem | open | MathDB 页面 | 原文线索

V2-011812 | MATHDB-400622

Koecher reduction theory and computations with the Voronoï cell complex

摘录 / EXCERPT: Consider Koecher reduction theory in the Voronoï setting for general number fields F. It is known that perfect forms do not come from F, but are to be defined over the Galois c...

open_problem | open | MathDB 页面 | 原文线索

V2-011813 | MATHDB-400669

Characterization of pseudo-Anosov stretch factors by bi-Perron units

摘录 / EXCERPT: Every bi-Perron algebraic unit is a pseudo-Anosov stretch factor, where a bi-Perron number is an algebraic unit whose Galois conjugates, except possibly its reciprocal if that reci...

open_problem | open | MathDB 页面 | 原文线索

V2-011814 | MATHDB-400682

Buchsbaum-Eisenbud-Horrocks conjecture: lower bounds for individual Betti numbers

摘录 / EXCERPT: Let M be a module of finite length over a regular local ring of dimension d. The ith Betti number of M is conjectured to satisfy ...

conjecture | open | MathDB 页面 | 原文线索

V2-011815 | MATHDB-400684

Hochster’s Direct Summand Conjecture

摘录 / EXCERPT: Let R be a regular ring, and let $f:R\to S$ be a finite injective map. Then f admits an R-module splitting.

conjecture | open | MathDB 页面 | 原文线索

V2-011816 | MATHDB-400693

Weak Horrocks Conjecture

摘录 / EXCERPT: Assume $(R,\mathfrak m,k)$ is a local ring of dimension d. If M is a non-zero R-module of finite length and finite projective dimension, then ... where $\beta_i(M)$ is the i-...

conjecture | open | MathDB 页面 | 原文线索

V2-011817 | MATHDB-400697

Characterization of multigraded regularity by free Cox complexes

摘录 / EXCERPT: Let M be a B-saturated S-module. The regularity of M contains $d\in\mathbb{Z}^r$ if and only if $M(d)$ has a free Cox complex ... such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-011818 | MATHDB-400700

Golod homomorphisms in the omitted even-socle case for compressed level local Artinian rings

摘录 / EXCERPT: Let k be algebraically closed, and let $(R,\mathfrak m,k)$ be a compressed level local Artinian ring with embedding dimension e, socle dimension $c>1$, and even socle degree...

open_problem | open | MathDB 页面 | 原文线索

V2-011819 | MATHDB-400708

Adjoint of the Haagerup tensor-product functor

摘录 / EXCERPT: Let A and B be C^*-algebras, and let F be a C^*-correspondence from A to B. Consider the functor ... and the functor ... where F^* is the operator-theoretic adjoint B...

open_problem | open | MathDB 页面 | 原文线索

V2-011820 | MATHDB-400732

Extending surreal measure and integration beyond tame definable categories

摘录 / EXCERPT: Can one extend the measure and integration theory on the field of surreal numbers beyond the semialgebraic and globally subanalytic category?

open_problem | open | MathDB 页面 | 原文线索

V2-011821 | MATHDB-400736

Semidefinite approximations of the polynomial abscissa

摘录 / EXCERPT: Let $Q\subset\mathbb{R}^n$ be compact and semialgebraic, and let $p(q,s)=\sum_{k=0}^m p_k(q)s^k$ be a monic parameterized polynomial as described in the report. If...

open_problem | open | MathDB 页面 | 原文线索

V2-011822 | MATHDB-400749

Quantization of spectral curves and DQ-modules

摘录 / EXCERPT: Let Σ be a compact Riemann surface of genus $g\ge 2$, let (E,ϕ) be a Higgs bundle of rank $r\ge 2$ on Σ, and let $\widehat{\mathcal W}_{T^*\Sigma}(0)$ be the can...

open_problem | open | MathDB 页面 | 原文线索

V2-011823 | MATHDB-400779

Every semi-algebraic graph property is easily testable

摘录 / EXCERPT: Every semi-algebraic graph property is easily testable.

open_problem | open | MathDB 页面 | 原文线索

V2-011824 | MATHDB-400840

Support Conjecture for generalized preprojective algebras

摘录 / EXCERPT: For the setup above with $K=\mathbb{C}$ and $\Pi=\Pi(C,D)$, prove that ... is an ideal in $\widetilde{\mathcal{M}}(\Pi)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-011825 | MATHDB-400841

Classification of quasi-tilted algebras with a prescribed Euclidean cluster-tilted relation-extension

摘录 / EXCERPT: Given a cluster-tilted algebra B of Euclidean type, find all quasi-tilted algebras C such that $B=\widetilde{C}$, where $\widetilde{C}$ is the relation-extension of C. The re...

open_problem | open | MathDB 页面 | 原文线索

V2-011826 | MATHDB-400868

Helton-Nie conjecture on semidefinite representations

摘录 / EXCERPT: Every convex semialgebraic set $K\subseteq\mathbb{R}^n$ has a semidefinite representation; that is, there exist $m\ge 0$, a matrix size, and real symmetric matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-011827 | MATHDB-400966

Sharp posterior phase transition for overfitted hidden Markov models

摘录 / EXCERPT: For an overfitted hidden Markov model with true state number $K^*<K$, determine-possibly using techniques from algebraic statistics-the posterior behavior of the extra states, in p...

open_problem | open | MathDB 页面 | 原文线索

V2-011828 | MATHDB-400974

Description of simple and tilting representation coefficients

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic p, let G be a connected reductive K-algebraic group, and let X^+ be the dominant weights. For...

open_problem | open | MathDB 页面 | 原文线索

V2-011829 | MATHDB-400975

Tilting character formula in terms of the p-canonical basis

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic p, let G be a connected reductive K-algebraic group with Coxeter number h, and let W_f be its finite Weyl group...

open_problem | open | MathDB 页面 | 原文线索

V2-011830 | MATHDB-400976

Characters of indecomposable tilting modules and symmetric-group decomposition numbers

摘录 / EXCERPT: Determine the characters of indecomposable tilting modules for GL_m, equivalently determine all decomposition numbers for symmetric groups for partitions with at most...

open_problem | open | MathDB 页面 | 原文线索

V2-011831 | MATHDB-400980

Theorem 2 on cocharacter-closed orbits

摘录 / EXCERPT: Let $v\in V$ and suppose that G_v is k-defined. Then: 1. If $G\cdot v$ is Zariski-closed, then $G(k)\cdot v$ is cocharacter-closed over k. 2. Let k'/k be an algebraic field...

open_problem | open | MathDB 页面 | 原文线索

V2-011832 | MATHDB-400982

Relation between the GHKK canonical basis and the Kashiwara-Lusztig canonical basis

摘录 / EXCERPT: Let G be a simple, simply connected, simply laced algebraic group over $\bb C$, and let $U\subset G$ be its unipotent radical. Gross-Hacking-Keel-Kontsevich construct a canonical...

open_problem | open | MathDB 页面 | 原文线索

V2-011833 | MATHDB-400985

Description of the saturated tensor cone for symmetrizable Kac-Moody algebras

摘录 / EXCERPT: Let $(\lambda_1,\lambda_2,\mu)\in P^3_{+,\mathbb Q}$ with $\lambda_1(c)>0$ and $\lambda_2(c)>0$. Then $(\lambda_1,\lambda_2,\mu)\in SQ(\mathfrak g)$ if and only if: (1)...

open_problem | open | MathDB 页面 | 原文线索

V2-011834 | MATHDB-400987

Kurosh problem for finitely generated algebraic algebras

摘录 / EXCERPT: Is every finitely generated algebraic algebra A over a field F finite-dimensional?

open_problem | open | MathDB 页面 | 原文线索

V2-011835 | MATHDB-401037

Prasad’s relative local Langlands multiplicity conjecture

摘录 / EXCERPT: Let E/F be a quadratic extension of local fields, let G be a reductive algebraic group over F, and let G^{op} be the quasi-split opposition group obtained by twist...

conjecture | open | MathDB 页面 | 原文线索

V2-011836 | MATHDB-401038

Stable-discrete version of Prasad’s conjecture for Galois pairs

摘录 / EXCERPT: Let F be a p-adic field, let E/F be a quadratic extension, let H be a connected reductive group over F, and set $G:=R_{E/F}H$, so that $G(F)=H(E)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-011837 | MATHDB-401040

Distinction of the generalized Steinberg representation

摘录 / EXCERPT: For $\pi=\mathrm{St}$, the generalized Steinberg representation of $G(F)$, and χ a character of Galois type, the report states that ...

open_problem | open | MathDB 页面 | 原文线索

V2-011838 | MATHDB-401047

ABV-packet conjecture for local Arthur packets

摘录 / EXCERPT: Let F be a p-adic field and let G be a quasi-split connected reductive algebraic group over F. For an Arthur parameter ψ of G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-011839 | MATHDB-401056

Generalized Shalika trace formula for arbitrary n

摘录 / EXCERPT: Let F be a p-adic field and let D/F be a division algebra. Set $G(F)=\mathrm{GL}_{2n}(D)$, let $P=MU$ be the standard parabolic subgroup of type (n,n), and let $H_0(F)$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-011840 | MATHDB-401059

Induction maps on cocenters of Hecke algebras of p-adic groups

摘录 / EXCERPT: Let G be a connected reductive p-adic group and let M be a standard Levi subgroup. Construct an induction map $\bar i_M$ on the cocenter of the Hecke algebra of M such that...

open_problem | open | MathDB 页面 | 原文线索

V2-011841 | MATHDB-401186

Relation between Milnor invariants of primes and Galois elements

摘录 / EXCERPT: Is there any relation between $\mu_l(l)$ and $\mu^{(l)}(g;l)$ for $l=2,3$?

open_problem | open | MathDB 页面 | 原文线索

V2-011842 | MATHDB-401226

Effective Plancherel equidistribution of unramified local components

摘录 / EXCERPT: Let G be a split semisimple algebraic group defined over $\[1m\mathbb{Z}\[0m$, with suitable maximal compact subgroups $K_\infty\subseteq G(\mathbb{R})$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-011843 | MATHDB-401241

The Universal Coefficient Theorem problem for separable nuclear C*-algebras

摘录 / EXCERPT: Do all separable, nuclear C^*-algebras satisfy the universal coefficient theorem of Rosenberg-Schochet?

open_problem | open | MathDB 页面 | 原文线索

V2-011844 | MATHDB-401246

Purely infinite simplicity of ample groupoid C*-algebras

摘录 / EXCERPT: What are the necessary and sufficient conditions on an ample groupoid to ensure that the corresponding groupoid C^*-algebra is purely infinite simple?

open_problem | open | MathDB 页面 | 原文线索

V2-011845 | MATHDB-401247

Converse between pure infiniteness of Steinberg algebras and groupoid C*-algebras

摘录 / EXCERPT: For an ample Hausdorff groupoid, if the corresponding groupoid C^*-algebra is purely infinite simple, is the associated Steinberg algebra necessarily algebraically purely infinit...

open_problem | open | MathDB 页面 | 原文线索

V2-011846 | MATHDB-401248

Universal property Γ for simple nuclear C*-algebras

摘录 / EXCERPT: It is open whether all simple, separable, unital and nuclear C-algebras have property Γ.

open_problem | open | MathDB 页面 | 原文线索

V2-011847 | MATHDB-401249

Finite-dimensional norm attainment and residual finite-dimensionality

摘录 / EXCERPT: For a C^*-algebra, if the elements that attain their norm under some finite-dimensional representation form a dense subset, does the algebra have a separating family of finite-di...

open_problem | open | MathDB 页面 | 原文线索

V2-011848 | MATHDB-401251

Inner conjugacy of diagonal MASAs under outer automorphisms of graph C*-algebras

摘录 / EXCERPT: Let E be a finite graph such that $C^*(E)$ is purely infinite and simple, and let D_E be its diagonal MASA. For which outer automorphisms α of $C^*(E)$ are the two MASAs...

open_problem | open | MathDB 页面 | 原文线索

V2-011849 | MATHDB-401252

Inner conjugacy of diagonal MASAs under localized automorphisms of graph C*-algebras

摘录 / EXCERPT: Let E be a finite graph such that $C^*(E)$ is purely infinite and simple, and let D_E be its diagonal MASA. Determine, for outer localized automorphisms α of $C^*(E)$, w...

open_problem | open | MathDB 页面 | 原文线索

V2-011850 | MATHDB-401253

Cartan subalgebra of the CAR algebra with connected spectrum

摘录 / EXCERPT: Does the CAR algebra $M_{2^\\infty}$ have a Cartan subalgebra with connected spectrum?

open_problem | open | MathDB 页面 | 原文线索

V2-011851 | MATHDB-401254

Non-\C*-diagonal Cartan subalgebras in simple infinite-dimensional C*-algebras

摘录 / EXCERPT: Do all simple infinite-dimensional C^*-algebras with Cartan subalgebras also admit a Cartan subalgebra that is not a C^*-diagonal?

open_problem | open | MathDB 页面 | 原文线索

V2-011852 | MATHDB-401259

Chen’s approximable-algebra question

摘录 / EXCERPT: Let $B=\bigoplus_{m\ge 0}B_m$ be an integral graded algebra with $B_0=k$ a field, where all B_m are finite-dimensional over k, $B_m\ne 0$ for all sufficiently large m, and, f...

open_problem | open | MathDB 页面 | 原文线索

V2-011853 | MATHDB-401296

Non-birational GPK3 double mirrors and the birational Torelli problem

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let V be a 5-dimensional k-vector space, and set $W=\bigwedge^2V$. For $g_1,g_2\in\operatorname{PGL}(W)$, defi...

open_problem | open | MathDB 页面 | 原文线索

V2-011854 | MATHDB-401380

Generic models and redundant sentences for algebraic theories

摘录 / EXCERPT: Does the generic model of an algebraic theory T satisfy every T-redundant sentence, where a sentence is T-redundant if adding it to T yields no new geometric implications?

open_problem | open | MathDB 页面 | 原文线索

V2-011855 | MATHDB-401474

Lower-bound part of the Malle conjecture (LB-Malle)

摘录 / EXCERPT: For the counting function $N(G,y)$ of Galois extensions E/k with group G and discriminant norm $\left|N_{k/\mathbb{Q}}(d_{E})\right|\le y$, does there exist a positive constant...

conjecture | open | MathDB 页面 | 原文线索

V2-011856 | MATHDB-401480

The Beckmann-Black problem over \(\mathbb Q\)

摘录 / EXCERPT: Can every finite Galois extension $F/\mathbb Q$ be lifted to a $\mathbb Q$-regular Galois extension $E/\mathbb Q(T)$ (possibly depending on $F/\mathbb Q$) with the same Galois grou...

open_problem | open | MathDB 页面 | 原文线索

V2-011857 | MATHDB-401488

Number and degrees of coverings describing reducible specializations

摘录 / EXCERPT: Let $f(t,x)\in\mathbb{Q}[t,x]$ be irreducible, with $n=\deg_x f$, and assume that $\mathbb{Q}$ is algebraically closed in the Galois closure of f over $\mathbb{Q}(t)$. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-011858 | MATHDB-401493

Galois-equivariant identification of configuration-space fundamental groupoids with the profinite parenthesized braid operad

摘录 / EXCERPT: For each n, there is a set P_n of $\Q$-rational tangential basepoints of $\\operatorname{Conf}_n$ and a $G_{\mathbb Q}$-equivariant isomorphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-011859 | MATHDB-401509

Bost's conjecture

摘录 / EXCERPT: For all locally compact groups G and all G-C^*-algebras B, the map ... is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-011860 | MATHDB-401526

Noetherianity of module categories over twisted commutative algebras

摘录 / EXCERPT: If A is a finitely generated twisted commutative algebra over a noetherian coefficient ring, is the module category $\operatorname{Mod} A$ locally noetherian?

open_problem | open | MathDB 页面 | 原文线索

V2-011861 | MATHDB-401534

Gorenstein property of Solomon-Terao algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a hyperplane arrangement and let $\eta\in S_2$. When is the Solomon-Terao algebra $ST(\mathcal{A},\eta)$ Gorenstein?

open_problem | open | MathDB 页面 | 原文线索

V2-011862 | MATHDB-401555

Brauer-Manin obstruction for torsors over \(\mathbb{C}((x,y))\)

摘录 / EXCERPT: For a torsor Z under a connected linear algebraic group G over $K=\mathbb{C}((x,y))$, is the Brauer-Manin obstruction the only obstruction to the local-global principle; equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-011863 | MATHDB-401563

Properness of the global-to-local map over finitely generated fields

摘录 / EXCERPT: Let K be a field. When can one equip K with a natural set V of discrete valuations such that, for a given absolutely almost simple adjoint algebraic K-group G, the natura...

open_problem | open | MathDB 页面 | 原文线索

V2-011864 | MATHDB-401569

The alternative Clifford algebra of a nondegenerate ternary quadratic form

摘录 / EXCERPT: Let q be a nondegenerate ternary quadratic form over an arbitrary field F. Then $C_{\mathrm{alt}}(q)$ is an octonion algebra over a polynomial ring in one variable. More precis...

open_problem | open | MathDB 页面 | 原文线索

V2-011865 | MATHDB-401576

Bipartite versus tripartite Kirchberg tensor-product relation

摘录 / EXCERPT: Does Kirchberg’s conjecture ... for the full group C^*-algebra $C^*(F_\infty)$ of the free group with countably many generators imply that the analogous triple tensor-product relat...

open_problem | open | MathDB 页面 | 原文线索

V2-011866 | MATHDB-401578

Existence of Lévy-Khintchine decompositions for generating functionals

摘录 / EXCERPT: Given an augmented algebra (A,ϵ) and a generating functional ψ on A, is it always possible to write $\psi=\psi_G+\psi_R$ as a sum of two generating functionals such...

open_problem | open | MathDB 页面 | 原文线索

V2-011867 | MATHDB-401583

Quantitative stability of Petz recovery under near-equality in the data processing inequality

摘录 / EXCERPT: Let ρ and σ be density matrices, let Φ be restriction to a subalgebra $\mathcal N$, and write $\rho_{\mathcal N}=\Phi(\rho)$ and $\sigma_{\mathcal N}=\Phi(\sigma)$...

open_problem | open | MathDB 页面 | 原文线索

V2-011868 | MATHDB-401620

Bridgeland’s Hall-algebra realization of the whole quantized enveloping algebra

摘录 / EXCERPT: Let $\vec Q$ be a quiver without oriented cycles, let $\operatorname{Rep}_{\mathbb F_q}(\vec Q)$ be its category of finite-dimensional representations over $\mathbb F_q$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-011869 | MATHDB-401622

Just infinite Gelfand-Kirillov dimension for the full Witt algebra

摘录 / EXCERPT: Let W be the full Witt algebra over $\r\mathbb{C}$ with basis $\r\{e_n:n\in\mathbb{Z}\}$ and Lie bracket $[e_i,e_j]=(j-i)e_{i+j}$. Does $U(W)$ have just infinite Gelfand-Kirillov...

open_problem | open | MathDB 页面 | 原文线索

V2-011870 | MATHDB-401624

Chebotarev density theorem in very short intervals over number fields

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a finite Galois extension with Galois group G, let $C\subseteq G$ be a conjugacy class, and let $\pi_C(x;E)$ count the primes $p\le x$ unramified in E who...

open_problem | open | MathDB 页面 | 原文线索

V2-011871 | MATHDB-401625

Chebotarev density theorem in short intervals over number fields

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a finite Galois extension with Galois group G, and let $C\subseteq G$ be a conjugacy class. For every fixed $\varepsilon>0$, with $y=x^{\varepsilon}$, does...

open_problem | open | MathDB 页面 | 原文线索

V2-011872 | MATHDB-401638

Removing the characteristic restriction in bounded-ramification embedding problems

摘录 / EXCERPT: Does Theorem 1 remain valid if the hypothesis $\operatorname{char}(K_0)\nmid |H|$ is removed, while retaining that K/K_0 is a finite Galois extension of global fields, that the k...

open_problem | open | MathDB 页面 | 原文线索

V2-011873 | MATHDB-401640

Freeness of the absolute differential Galois group of $k(x)$

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, and let $F=k(x)$ with derivation $\partial=d/dx$. Is the absolute differential Galois group Γ_F free as a pr...

open_problem | open | MathDB 页面 | 原文线索

V2-011874 | MATHDB-401641

Differential inverse Galois problem over Laurent series fields

摘录 / EXCERPT: If $K=k((t))$ has characteristic zero, then every linear algebraic group over K is a differential Galois group over $K(x)$ with respect to $\\partial=d/dx$.

open_problem | open | MathDB 页面 | 原文线索

V2-011875 | MATHDB-401643

Freeness of the absolute differential Galois group in the countable case

摘录 / EXCERPT: Let k be an algebraically closed field of countably infinite transcendence degree over $\a0\boldsymbol{Q}$. For $F=k(x)$ with derivation $\a0d/dx$, the absolute differential Galo...

open_problem | open | MathDB 页面 | 原文线索

V2-011876 | MATHDB-401644

Monodromy of pulled-back hyperplane sections

摘录 / EXCERPT: Let k be algebraically closed, let C be a smooth projective irreducible curve over k, and let $f_1,\ldots,f_n\in k(C)$. Let $\varphi:C\to\mathbb{P}^n$ be the morphism extendi...

open_problem | open | MathDB 页面 | 原文线索

V2-011877 | MATHDB-401648

Realizing the profinite completion of a surface group as an absolute Galois group

摘录 / EXCERPT: Let $g\ge 2$ and ... Does there exist a field F such that $\operatorname{Gal}(F)\cong\widehat{\Sigma}_g$, the profinite completion of Σ_g?

open_problem | open | MathDB 页面 | 原文线索

V2-011878 | MATHDB-401699

Maximal A_k-singularity on a plane curve of fixed degree

摘录 / EXCERPT: Let $C\subset\mathbb{A}^2(\mathbb{C})$ be an algebraic curve, not necessarily reduced, of degree d. An A_k-singularity means that, near the singular point, the curve is analyti...

open_problem | open | MathDB 页面 | 原文线索

V2-011879 | MATHDB-401701

Families of maximal A_k-singularities for bidegree $(3,b)$

摘录 / EXCERPT: Does there exist a family, for unbounded increasing integers b, of not necessarily reduced algebraic curves in $\mathbb A^2(\mathbb C)$ of bidegree $(3,b)$ having a maximal A_k...

open_problem | open | MathDB 页面 | 原文线索

V2-011880 | MATHDB-401712

Hilbert’s conjecture on the resolvent degree of the universal polynomial

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0. Let $P_n=\mathbb A^n\setminus V(\operatorname{disc})$ be the space of monic separable polynomials of degree n, and...

conjecture | open | MathDB 页面 | 原文线索

V2-011881 | MATHDB-401740

Characterization of C-links in the 3-sphere

摘录 / EXCERPT: Which link types arise as the intersection of an algebraic curve $V \subset \mathbb{C}^2$ with the unit sphere $S^3 = \{(z,w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1\}$?

open_problem | open | MathDB 页面 | 原文线索

V2-011882 | MATHDB-401744

Self-adjointness and regularity of sums of operators in Hilbert C^*-modules

摘录 / EXCERPT: Let S,T be self-adjoint and regular operators in the Hilbert B-module E. Assume: 1. There are constants $C_0,C_1,C_2>0$ such that ... for all $x\in F(S,T)$, where ... 2. There is...

open_problem | open | MathDB 页面 | 原文线索

V2-011883 | MATHDB-401745

The \\-form of the Baum\Connes conjecture with coefficients

摘录 / EXCERPT: For a -algebra A, the localized BaumConnes morphism ... defined by $\mu_\tau(x\otimes[\tau])=\mu_{\mathbb R}(x)\otimes[\tau]_r$, is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-011884 | MATHDB-401747

The minimal exact crossed product

摘录 / EXCERPT: The report presents the claim that the exotic group algebra associated to the minimal exact crossed-product functor satisfies $C_E^*(G):=\mathbb{C}\rtimes_E G=C_r^*(G)$.

open_problem | open | MathDB 页面 | 原文线索

V2-011885 | MATHDB-401751

Examples showing failure of the Koszul-module vanishing bound in small characteristic

摘录 / EXCERPT: In the setup above, with $n\ge 3$, V an n-dimensional vector space over a field k, and $K\subseteq \bigwedge^2 V$, generate examples in characteristic satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-011886 | MATHDB-401850

Stable-range isomorphism of the two Torelli-group graded Lie algebras

摘录 / EXCERPT: For any $k\ne 2$, we have $t_g(k)\cong m_g(k)$, where g is understood to be sufficiently large, in a stable range for the property under consideration.

open_problem | open | MathDB 页面 | 原文线索

V2-011887 | MATHDB-401854

Cluster superalgebras and the Laurent phenomenon

摘录 / EXCERPT: What is the correct notion of a cluster superalgebra, including spin structures? A key question is to explore whether an appropriate Laurent phenomenon holds.

open_problem | open | MathDB 页面 | 原文线索

V2-011888 | MATHDB-401963

Socle of the Grothendieck-group module for category \(\mathcal O\)

摘录 / EXCERPT: Let $\mathfrak g=\mathfrak{gl}(m|n)$, let $\mathcal O_{m|n}$ be the category $\mathcal O$ of integral weights, and let $K_{m|n}$ be its complexified reduced Grothendieck group. Set...

open_problem | open | MathDB 页面 | 原文线索

V2-011889 | MATHDB-401968

Classification of Gelfand-Tsetlin modules for \(\mathfrak{gl}_n\)

摘录 / EXCERPT: Classify the finitely generated $U(\mathfrak{gl}_n)$-modules on which the Gelfand-Tsetlin subalgebra Γ, generated by the centers of...

open_problem | open | MathDB 页面 | 原文线索

V2-011890 | MATHDB-401969

Isomorphism between two non-trivial W-algebras

摘录 / EXCERPT: Conjecture: Let f belong to the nilpotent orbit of $\mathfrak{sl}_7$ associated with the partition $(3,2^2)$, and let f' belong to the nilpotent orbit of $\mathfrak{sl}_4$ asso...

open_problem | open | MathDB 页面 | 原文线索

V2-011891 | MATHDB-401971

Schieder’s bialgebra conjecture

摘录 / EXCERPT: Let G be a reductive complex algebraic group, with a chosen Borel subgroup $B\supset T$ and opposite unipotent radicals N and N^-. Let $G^\vee$ be the Langlands dual group, a...

conjecture | open | MathDB 页面 | 原文线索

V2-011892 | MATHDB-402043

Brown-Goodearl Question

摘录 / EXCERPT: Is every noetherian Hopf algebra Gorenstein, equivalently, does every noetherian Hopf algebra have finite self-injective dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-011893 | MATHDB-402049

Determine the inner automorphism in the generalized Radford formula

摘录 / EXCERPT: Let H be an AS-Gorenstein Hopf algebra, with the notation of Corollary 6 for the antipode S, the character $\\chi$, and the left and right winding automorphisms. Corollary 6 st...

open_problem | open | MathDB 页面 | 原文线索

V2-011894 | MATHDB-402050

Finite-type property under categorical duality

摘录 / EXCERPT: For every finite tensor category $\mathscr C$ of finite type over a base field k and every exact $\mathscr C$-module category $\mathscr M$, the dual category...

open_problem | open | MathDB 页面 | 原文线索

V2-011895 | MATHDB-402052

Nichols-Zoeller-type projectivity theorem for infinite-dimensional weak Hopf algebras

摘录 / EXCERPT: Let W_1 and W_2 be two noetherian weak Hopf algebras satisfying the general hypothesis as in the previous theorems, and homogeneous of the same Gelfand-Kirillov dimension. Supp...

open_problem | open | MathDB 页面 | 原文线索

V2-011896 | MATHDB-402070

Further applications of Galois-action invariants for MRD codes

摘录 / EXCERPT: It is still an open problem to understand whether the sequences of θ-dimensions and θ-increments, and the associated maps S_i^θ, can characterize other familie...

open_problem | open | MathDB 页面 | 原文线索

V2-011897 | MATHDB-402098

Universal pseudo-arithmeticity of hyperbolic lattices

摘录 / EXCERPT: Let $\Gamma \subset \operatorname{PO}(n,1)$ be a lattice with $n \ge 4$. Let K be its trace field and let G be its ambient algebraic K-group. The lattice Γ is called p...

open_problem | open | MathDB 页面 | 原文线索

V2-011898 | MATHDB-402146

Equivalence between p-divisible groups and divided prismatic Dieudonné modules

摘录 / EXCERPT: For a quasi-regular semiperfectoid ring R, the natural functor ... is an equivalence of categories. Here $\mathrm{BT}(R)$ is the category of p-divisible groups over R, and...

open_problem | open | MathDB 页面 | 原文线索

V2-011899 | MATHDB-402157

Fundamental group of the space of simple real divisors in a fixed linear equivalence class

摘录 / EXCERPT: Let X be a non-singular projective real algebraic curve, and let $A=P_1+\33+P_d$, with $P_1,\33,P_d\in RX$, be a simple real divisor. Let $\operatorname{Emb}(A,X)$ be the space f...

open_problem | open | MathDB 页面 | 原文线索

V2-011900 | MATHDB-402175

Equivalence between compact A-branes and finite-dimensional spherical DAHA modules

摘录 / EXCERPT: The authors hope to demonstrate an equivalence between compact A-branes and finite-dimensional representations of the spherical double affine Hecke algebra $\ddot H$ associated wit...

open_problem | open | MathDB 页面 | 原文线索

V2-011901 | MATHDB-402203

Vorst’s conjecture

摘录 / EXCERPT: Assume all rings are unital and commutative. A ring A is called K_n-regular if, for every positive integer m, the canonical map ... is an isomorphism. Let A be an algebra ess...

conjecture | open | MathDB 页面 | 原文线索

V2-011902 | MATHDB-402263

Determine the odd-primary Adem relations for free spectral Lie algebras

摘录 / EXCERPT: Determine the Adem relations among the operations on free spectral Lie algebras over $\mathbb F_p$ for odd primes p.

open_problem | open | MathDB 页面 | 原文线索

V2-011903 | MATHDB-402268

Classification of localizing subcategories of module categories over Noetherian ring spectra

摘录 / EXCERPT: Given a commutative Noetherian ring spectrum R (that is, $\\pi_*R$ graded Noetherian), can we give a classification of the localizing subcategories of $\operatorname{Mod} R$?

open_problem | open | MathDB 页面 | 原文线索

V2-011904 | MATHDB-402279

Classification of full homomorphisms into the norm ultrapower

摘录 / EXCERPT: Let A and B be separable, unital, nuclear C^*-algebras, with B $\mathcal Z$-stable, and let $\phi,\psi:A\to B_\omega$ be full $*$-homomorphisms that agree on traces. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-011905 | MATHDB-402283

Whether the full free-group C*-algebra has the WEP

摘录 / EXCERPT: Does $A=C^ast(F_2)$ (or $A=C^ast(F_\infty)$), which is known to have the local lifting property (LLP), also have the weak expectation property (WEP)?

open_problem | open | MathDB 页面 | 原文线索

V2-011906 | MATHDB-402286

Characterization of groups with strongly RFD group C*-algebras

摘录 / EXCERPT: Can we characterize which discrete groups have strongly RFD group C^*-algebras, where “strongly RFD” means that all quotients are RFD?

open_problem | open | MathDB 页面 | 原文线索

V2-011907 | MATHDB-402287

Range of Elliott invariants for orbit-breaking equivalence-relation C^*-algebras

摘录 / EXCERPT: It is not known whether the C^*-algebras of such equivalence relations can exhaust the Elliott invariant.

open_problem | open | MathDB 页面 | 原文线索

V2-011908 | MATHDB-402292

The rank problem for separable unital simple exact C^*-algebras

摘录 / EXCERPT: Given a separable, unital, simple, exact C^*-algebra A, and given $f\in \operatorname{LAff}(T(A))_{++}$, is there an $a\in (A\otimes\mathcal{K})_+$ with $d_\tau([a])=f(\tau)$ f...

open_problem | open | MathDB 页面 | 原文线索

V2-011909 | MATHDB-402369

Epsilon dichotomy for linear models for tempered representations

摘录 / EXCERPT: Let E/F be a quadratic extension of non-Archimedean local fields of characteristic 0, and let D be a quaternion algebra over F containing an embedded copy of E. Put...

open_problem | open | MathDB 页面 | 原文线索

V2-011910 | MATHDB-402378

ABPS Conjecture for Bernstein components of reductive p-adic groups

摘录 / EXCERPT: Let F be a local non-Archimedean field, let G be the group of F-rational points of a connected reductive algebraic group, and let $s=[L,\sigma]_G$ be an inertial class. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-011911 | MATHDB-402380

Group-theoretic realization of the c-operation on Jacquet modules

摘录 / EXCERPT: The report states the following result. Let C be a Chevalley involution of G with $C(a)=a^{-1}$ for $a\in A$ and $C(B)=B^{-}$. Define $f^{c}(g)=f(C(g)^{-1})$ on the Hecke algeb...

open_problem | open | MathDB 页面 | 原文线索

V2-011912 | MATHDB-402392

The UCT problem for nuclear C*-algebras

摘录 / EXCERPT: Does every nuclear C-algebra satisfy the Universal Coefficient Theorem (UCT) for K-theory?

open_problem | open | MathDB 页面 | 原文线索

V2-011913 | MATHDB-402410

Zimmer cocycle super-rigidity theorem

摘录 / EXCERPT: Let G be a simply connected semisimple Lie group all of whose simple factors have real rank at least 2. Let H be a real algebraic group, and let $G\curvearrowright (X,\mu)$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-011914 | MATHDB-402424

Classification of graph planar algebra embeddings

摘录 / EXCERPT: What are all graphs Γ with embeddings $\operatorname{PA}(N\subset M)\hookrightarrow \operatorname{GPA}(\Gamma)$?

open_problem | open | MathDB 页面 | 原文线索

V2-011915 | MATHDB-402430

Generation of subfactor planar algebras for symmetric-group inclusions

摘录 / EXCERPT: Are the subfactor planar algebras for $S_2\times S_{n-2}\subset S_n$ generated by their 2-boxes?

open_problem | open | MathDB 页面 | 原文线索

V2-011916 | MATHDB-402431

Generation by 2-boxes for the $S_2\times S_3\subset S_5$ subfactor planar algebra

摘录 / EXCERPT: Is the subfactor planar algebra associated with the inclusion $R\rtimes(S_2\times S_3)\subset R\rtimes S_5$ generated by its 2-boxes?

open_problem | open | MathDB 页面 | 原文线索

V2-011917 | MATHDB-402482

Kneser-Tits conjecture for isotropic simple simply connected groups

摘录 / EXCERPT: For a simple, simply connected, isotropic algebraic group G over a field K, the abstract group $G(K)$ coincides with the normal subgroup $G(K)^+$ generated by the unipotent rad...

conjecture | open | MathDB 页面 | 原文线索

V2-011918 | MATHDB-402484

Local-to-global existence for homogeneous spaces over a field

摘录 / EXCERPT: Let k be a field, let G be an affine algebraic k-group, and let X be a homogeneous space for G. Must $X(k((t)))\ne\varnothing$ imply $X(k)\ne\varnothing$?

open_problem | open | MathDB 页面 | 原文线索

V2-011919 | MATHDB-402486

Automorphism group of a Tits web

摘录 / EXCERPT: Let $M=(X,(U_x)_{x earrow X})$ be a Moufang set with nonabelian root groups, constructed from an algebraic group of relative rank 1 or from a Frobenius twist of a group of mixed...

open_problem | open | MathDB 页面 | 原文线索

V2-011920 | MATHDB-402552

Algebraicity of RM values of rigid meromorphic cocycles

摘录 / EXCERPT: If J is a rigid meromorphic cocycle, then there is a finite extension H_J of $\bb Q$-the “field of definition” of J-for which $J[\tau]$ belongs to the compositum of H_J and...

open_problem | open | MathDB 页面 | 原文线索

V2-011921 | MATHDB-402569

Existence and weak functoriality of big Cohen-Macaulay algebras

摘录 / EXCERPT: Theorem 1. Big Cohen-Macaulay algebras exist and are weakly functorial.

open_problem | open | MathDB 页面 | 原文线索

V2-011922 | MATHDB-402574

Non-complete direct sums of conic modules yielding NC(C)Rs

摘录 / EXCERPT: In the setting above, in which case is there some (non-complete) direct sum B of conic modules such that $\operatorname{End}_R(B)$ is a NC(C)R?

open_problem | open | MathDB 页面 | 原文线索

V2-011923 | MATHDB-402586

Characterization of quasi-ordinary singularities by the invariant κ(f)

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, and let $f\in K[[x_1,\ldots,x_d]][z]$ be a Weierstrass polynomial satisfying $f(0)=0$. The hypersurface singularity...

open_problem | open | MathDB 页面 | 原文线索

V2-011924 | MATHDB-402601

Perfectness of tropical Dolbeault duality for non-archimedean curves

摘录 / EXCERPT: Let K be an algebraically closed complete non-archimedean field and let X be a smooth projective curve over K. Let $\mathcal{X}$ be a strictly semistable model of X, with i...

open_problem | open | MathDB 页面 | 原文线索

V2-011925 | MATHDB-402658

Isomorphism problem for free group factors

摘录 / EXCERPT: Are the group von Neumann algebras $L(F_n)$ and $L(F_m)$ isomorphic for distinct integers $n,m\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-011926 | MATHDB-402660

A computationally efficient procedure for convexifying free semialgebraic sets

摘录 / EXCERPT: Devise a (computationally efficient) procedure that determines which non-convex free semialgebraic sets can be analytically transformed into convex ones.

open_problem | open | MathDB 页面 | 原文线索

V2-011927 | MATHDB-402666

Equality of free Stein dimension and non-microstates free entropy dimension

摘录 / EXCERPT: Does $\sigma(X)=\delta^{*}(X)$ hold for every tuple $X=(x_{1},\ldots,x_{n})$ of self-adjoint operators generating a tracial von Neumann algebra (M,τ)?

open_problem | open | MathDB 页面 | 原文线索

V2-011928 | MATHDB-402687

Existence of exotic group C^*-algebras for non-amenable locally compact groups

摘录 / EXCERPT: Let G be a non-amenable locally compact group. Does there always exist an exotic group C^*-algebra A of G? Here A is a completion of $C_c(G)$ for a C^*-norm satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-011929 | MATHDB-402694

Property R_∞ for finitely presented non-virtually-polycyclic soluble S-arithmetic groups

摘录 / EXCERPT: Let $\Gamma \le B$ be an S-arithmetic subgroup of a Borel subgroup of a split, connected, reductive, non-commutative linear algebraic group G defined over a global field. If...

open_problem | open | MathDB 页面 | 原文线索

V2-011930 | MATHDB-402701

Isomorphism of a Leavitt path algebra with that of its Cuntz splice

摘录 / EXCERPT: In the setting of purely infinite simple Leavitt path algebras associated with finite graphs, is the Leavitt path algebra of a graph isomorphic to the Leavitt path algebra of its C...

open_problem | open | MathDB 页面 | 原文线索

V2-011931 | MATHDB-402702

α-equivariant E-theory groups

摘录 / EXCERPT: Given a G-C^*-algebra A, an H-C^*-algebra B, and a group homomorphism $\alpha:G\to H$, do there exist “α-equivariant” E-theory groups $E^\alpha_*(A,B)$ that gene...

open_problem | open | MathDB 页面 | 原文线索

V2-011932 | MATHDB-402705

Extension of cyclic cocycles from the group algebra

摘录 / EXCERPT: When does a cyclic cocycle of $\mathbb{C}\Gamma$ extend to a cyclic cocycle of A?

open_problem | open | MathDB 页面 | 原文线索

V2-011933 | MATHDB-402755

Quasiprojectivity of images of mixed period maps

摘录 / EXCERPT: Let X be a separated algebraic space of finite type over $\cmathbb{C}$, and let $\varphi:X\to M$ be the period map associated to an admissible variation of graded-polarized integ...

open_problem | open | MathDB 页面 | 原文线索

V2-011934 | MATHDB-402756

Griffiths’ quasiprojectivity conjecture for images of period maps

摘录 / EXCERPT: Let X be an algebraic space, let M be a moduli space of polarized integral pure Hodge structures, and let $\phi:X\to M$ be the period map corresponding to a polarized integral...

conjecture | open | MathDB 页面 | 原文线索

V2-011935 | MATHDB-402775

Approximation of Hecke-Laplace eigenfunctions by their period integral

摘录 / EXCERPT: Let F be an imaginary quadratic extension of $\\mathbb{Q}$, let D be a quaternion algebra over $\\mathbb{Q}$ which is not split at $\\infty$ and not split over F, and let f...

open_problem | open | MathDB 页面 | 原文线索

V2-011936 | MATHDB-402799

Local criteria for detecting interval summands in bipersistence modules

摘录 / EXCERPT: For a pointwise finite-dimensional representation M of $\d53d^2$, find local, easily checkable conditions that guarantee the existence of interval summands in the direct-sum deco...

open_problem | open | MathDB 页面 | 原文线索

V2-011937 | MATHDB-402809

Injective dimension of the Jacobson radical

摘录 / EXCERPT: For every connected, non-semisimple Artin algebra A, with J its Jacobson radical, the injective dimension of J equals the global dimension of A: ...

open_problem | open | MathDB 页面 | 原文线索

V2-011938 | MATHDB-402810

Auslander-Reiten question on Cohen-Macaulay Artin algebras

摘录 / EXCERPT: For a Cohen-Macaulay Artin algebra A with Cohen-Macaulay dimension $d>0$, does $\operatorname{CM}(A)=\Omega^{d}(\operatorname{mod}\text{-}A)$ imply that A is Iwanaga-Gorenstein...

open_problem | open | MathDB 页面 | 原文线索

V2-011939 | MATHDB-402811

Existence of non-weakly Gorenstein finite-dimensional algebras over finite fields

摘录 / EXCERPT: Does there exist a finite-dimensional non-weakly Gorenstein algebra over a finite field?

open_problem | open | MathDB 页面 | 原文线索

V2-011940 | MATHDB-402812

Systematic construction of non-weakly Gorenstein algebras

摘录 / EXCERPT: Let A be a connected, non-semisimple symmetric Artin algebra, and let X and M be finitely generated right A-modules. If $\\operatorname{Ext}^{\ell}_A(X,M)\neq 0$ for some...

open_problem | open | MathDB 页面 | 原文线索

V2-011941 | MATHDB-402813

Euler-characteristic analogue for submodule Grassmannians over other fields

摘录 / EXCERPT: For finite-dimensional hereditary algebras over fields other than algebraically closed or finite fields, in particular in characteristic zero, is there an appropriate analogue of t...

open_problem | open | MathDB 页面 | 原文线索

V2-011942 | MATHDB-402843

Stratification of syzygy modules related to the $Q(a)$ decomposition

摘录 / EXCERPT: Consider a minimal resolution of an AG $A=R/I$; can we find some stratification on the syzygy modules, related to the $Q(a)$ decomposition?

open_problem | open | MathDB 页面 | 原文线索

V2-011943 | MATHDB-402845

Center-to-center decomposition for Gorenstein algebras with a power-sum dual generator

摘录 / EXCERPT: Does a Gorenstein algebra with dual generator $\sum_i \ell_i^{d_i}$ with the SLP decompose center-to-center by always choosing the variables as m in the operations...

open_problem | open | MathDB 页面 | 原文线索

V2-011944 | MATHDB-402847

Irreducible components of families of local Artinian Gorenstein algebras

摘录 / EXCERPT: Given a Gorenstein sequence T, is the family $\operatorname{Gor}(T)$ an irreducible algebraic set? If not, how many irreducible components does it have?

open_problem | open | MathDB 页面 | 原文线索

V2-011945 | MATHDB-402848

Strong Lefschetz property for height-three Artinian Gorenstein algebras

摘录 / EXCERPT: Is it true in general that a height-three Artinian Gorenstein algebra is strong Lefschetz?

open_problem | open | MathDB 页面 | 原文线索

V2-011946 | MATHDB-402856

Asymptotic behavior of quadratic Artinian Gorenstein algebras

摘录 / EXCERPT: Let $\mathcal{QG}(r)$ be the family of standard graded Artinian Gorenstein algebras presented by quadrics, of socle degree 4 and codimension r, and define ... Conjecture: ...

open_problem | open | MathDB 页面 | 原文线索

V2-011947 | MATHDB-402857

Conjectural minimum second Hilbert-function entry in socle degree four

摘录 / EXCERPT: For an integer $m\ge 3$, set $r=m+\binom{m+2}{3}$. Among standard graded Artinian Gorenstein K-algebras of codimension r and socle degree 4, let $\mu(r)$ denote the least pos...

open_problem | open | MathDB 页面 | 原文线索

V2-011948 | MATHDB-402921

Left-bracketed areas as a basis for the area algebra

摘录 / EXCERPT: Let $\mathcal A(\mathbb R^d)$ be the area algebra spanned by all iterated bracketings of the operation $\operatorname{area}(a,b)=a\succ b-b\succ a$ starting from the letters...

open_problem | open | MathDB 页面 | 原文线索

V2-011949 | MATHDB-402968

Virtual algebraic fibring and vanishing of higher L^2-Betti numbers

摘录 / EXCERPT: Let G be a RFRS group of type $\mathrm{FP}_n(\mathbb{Q})$. Then G admits a finite-index subgroup H and an epimorphism $\phi:H\to\mathbb{Z}$ with kernel of type...

open_problem | open | MathDB 页面 | 原文线索

V2-011950 | MATHDB-403054

Density-zero specialization sets for regular Galois extensions of genus at least two

摘录 / EXCERPT: Let $E/\mathbb{Q}(T)$ be a $\mathbb{Q}$-regular Galois extension with group G, and assume that the genus g of the function field E satisfies $g\ge 2$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-011951 | MATHDB-403055

Density of local specializations versus global specializations

摘录 / EXCERPT: Let $E/\mathbb{Q}(T)$ be a $\mathbb{Q}$-regular Galois extension with group G, and assume that the genus g of the function field E satisfies $g\ge 2$. Let $S(G,x)$ be the set...

open_problem | open | MathDB 页面 | 原文线索

V2-011952 | MATHDB-403056

Hasse-principle violations in twists of abelian Galois covers

摘录 / EXCERPT: Let $f:X\to\mathbb P^1_{\mathbb Q}$ be a $\mathbb Q$-regular Galois cover with abelian Galois group G and $r\ge 7$ branch points. Assume the abc-conjecture holds and that f has...

open_problem | open | MathDB 页面 | 原文线索

V2-011953 | MATHDB-403066

Birational section conjecture

摘录 / EXCERPT: Let k be a finitely generated field extension of $ \mathbb{Q}$, let X/k be a smooth, proper, geometrically connected curve, and let $K=k(X)$. Choose algebraic closures...

conjecture | open | MathDB 页面 | 原文线索

V2-011954 | MATHDB-403070

An \ad\ic Galois analogue of the polylogarithmic coefficient formula

摘录 / EXCERPT: For the fixed data and notation above, does ... hold as collections of functions on G_K? Here c_w^z are the Magnus coefficient functions of the ℓ-adic Galois associator,...

open_problem | open | MathDB 页面 | 原文线索

V2-011955 | MATHDB-403073

Surjectivity of the Galois representation into the Grothendieck-Teichmüller group

摘录 / EXCERPT: Is the canonical morphism $\rho:\operatorname{Gal}(\mathbb{Q})\to\widehat{\mathrm{GT}}$ surjective? Equivalently, is ρ an isomorphism, given that its injectivity follows from...

open_problem | open | MathDB 页面 | 原文线索

V2-011956 | MATHDB-403075

Equality of the absolute Galois group and the profinite Grothendieck-Teichmüller group

摘录 / EXCERPT: Is the inclusion $G_{\cmathbb Q}\subseteq\widehat{GT}$ an equality?

open_problem | open | MathDB 页面 | 原文线索

V2-011957 | MATHDB-403082

Grothendieck-Teichmüller group versus the absolute Galois group

摘录 / EXCERPT: The Grothendieck-Teichmüller group is conjecturally isomorphic to the absolute Galois group of $\mathbb{Q}$, i.e. $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.

open_problem | open | MathDB 页面 | 原文线索

V2-011958 | MATHDB-403086

Pop-Topaz minimalistic Neukirch-Uchida reconstruction result

摘录 / EXCERPT: For Galois extensions $K\mid K_0$ and $L\mid L_0$ satisfying Hypothesis $(H)_\Sigma$-namely, with $\mu_\Lambda\subset F$, with $\mathbb{Q}_p^{\mathrm{ab}}\subset F_v$ for almost al...

open_problem | open | MathDB 页面 | 原文线索

V2-011959 | MATHDB-403099

Conjecture 1 on Galois representations attached to maximal ideals of the unramified Hecke algebra

摘录 / EXCERPT: Let $m\subset T_S(K)$ be a maximal ideal. Then there exists a continuous semisimple Galois representation $\bar\rho_m:G_{F,S}\to \mathrm{GL}_d(T_S(K)_m/m)$ such that: 1. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-011960 | MATHDB-403100

Local-global compatibility at Taylor-Wiles primes

摘录 / EXCERPT: There exists an $\mathcal O[\Delta_Q]$-algebra homomorphism ... that maps the characteristic polynomial of $\rho_{S\cup Q}(\operatorname{Fr}_v)$ to $P_v(X)$ for every...

open_problem | open | MathDB 页面 | 原文线索

V2-011961 | MATHDB-403107

Uniqueness of the k-semisimplification

摘录 / EXCERPT: Let G be a possibly non-connected reductive linear algebraic group over a field k, and let H be a closed k-subgroup of G. A k-semisimplification of H is a subgroup...

open_problem | open | MathDB 页面 | 原文线索

V2-011962 | MATHDB-403111

Local-global principle for projective homogeneous spaces over function fields

摘录 / EXCERPT: Suppose k is discretely valued. If V/F is a projective homogeneous space over a connected linear algebraic group G/F, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-011963 | MATHDB-403112

Bijections in the conjectural diagram for cuspidal character sheaves

摘录 / EXCERPT: Vilonen and Xue propose a conjectural diagram relating cuspidal character sheaves on g1, via the nearby-cycle construction and Fourier transform, to simple modules of Hecke algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-011964 | MATHDB-403115

Representability of homomorphism functors for all target algebraic groups

摘录 / EXCERPT: Characterize the algebraic groups G such that, for every algebraic group H, the functor $\operatorname{Hom}_{\mathrm{gp}}(G,H)$ is representable.

open_problem | open | MathDB 页面 | 原文线索

V2-011965 | MATHDB-403117

Non-redundancy criterion for string-cone inequalities

摘录 / EXCERPT: Let G be a simple, simply connected, simply-laced algebraic group over $\mathbb C$, let U be the unipotent radical of a Borel subgroup of G, and let $\mathbf{i}$ be a reduced...

open_problem | open | MathDB 页面 | 原文线索

V2-011966 | MATHDB-403119

Injectivity of the Hall-algebra map for quivers with potential

摘录 / EXCERPT: Conjecture 3. The $\mathbb{N}Q_0$-graded algebra homomorphism $\psi:\mathbf{U}^+\to K(Q,W)$ is injective.

open_problem | open | MathDB 页面 | 原文线索

V2-011967 | MATHDB-403253

Distribution of Galois-module structures in a restricted quadratic-field family

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve of rank 1 with full rational 2-torsion and no cyclic 4-isogeny. Restrict to quadratic fields $F=\mathbb{Q}(\sqrt d)$, with d square-...

open_problem | open | MathDB 页面 | 原文线索

V2-011968 | MATHDB-403258

Smallest Galois module containing all quadratic twists

摘录 / EXCERPT: Given $n>1$ and $H>0$, what is the smallest $G_{\\mathbb Q}$-module N such that, for any positive integer $d<H$, there is an equivariant embedding ...

open_problem | open | MathDB 页面 | 原文线索

V2-011969 | MATHDB-403260

Finiteness of degree-\d points on curves

摘录 / EXCERPT: Let k be a number field and let X/k be a nice (smooth, proper, geometrically irreducible) algebraic curve. Given an integer d, is the number of degree-d points on X finit...

open_problem | open | MathDB 页面 | 原文线索

V2-011970 | MATHDB-403264

Completeness of the list of exceptional \-adic Galois images

摘录 / EXCERPT: Conjecture: Aside from the -adic images known to arise for infinitely many mathbb{Q}-isomorphism classes of elliptic curves $E/\mathbb{Q}\$, the list of 22 exceptional images that...

open_problem | open | MathDB 页面 | 原文线索

V2-011971 | MATHDB-403271

Presentations of pro-\C completions of unramified Galois groups

摘录 / EXCERPT: Let Γ and Q be as described in the report, let C be a finite set of finite Γ-groups, and let K/Q be a Galois extension with Galois group Γ that is split...

open_problem | open | MathDB 页面 | 原文线索

V2-011972 | MATHDB-403323

Spherical part of the local Springer action conjecture

摘录 / EXCERPT: For every algebraically closed field K and $\\gamma\in C^\bullet(K)$, the action of the spherical part $\mathbb{Q}_\ell[X_*(T)]^W$ of $\widetilde W$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-011973 | MATHDB-403325

Langlands' conjecture on automorphic L-functions

摘录 / EXCERPT: Let k be a number field, let A be its ring of adeles, let G be a k-split reductive algebraic group, and let $G^{\vee}(\mathbb{C})$ be its complex dual group. For any finite...

conjecture | open | MathDB 页面 | 原文线索

V2-011974 | MATHDB-403402

Criteria for finite generation of initial algebras

摘录 / EXCERPT: Let $k[\boldsymbol{x}]=k[x_1,\dots,x_n]$ be a polynomial ring over a field k, let $\preceq$ be a monomial order, and let A be a k-subalgebra of $k[\boldsymbol{x}]$. Determine...

open_problem | open | MathDB 页面 | 原文线索

V2-011975 | MATHDB-403403

Polynomial multiplicity stability for symmetric semi-algebraic sets

摘录 / EXCERPT: Let R be a real closed field, let $\Lambda_n=R[X_1,\ldots,X_n]^{\mathop{\rm Sym}(n)}$ with the graded maps $\Lambda_{n+1}\to\Lambda_n$ obtained by setting $X_{n+1}=0$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-011976 | MATHDB-403409

Finite Gelfand-Kirillov dimension for Nichols algebras of diagonal type

摘录 / EXCERPT: Assume that G is abelian and that V is a finite-dimensional Yetter-Drinfeld module over $\mathbb{C}G$. If V is of diagonal type, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-011977 | MATHDB-403410

Conjecture on Nichols algebras over conjugacy classes of type C

摘录 / EXCERPT: Let G be a group and let $\cmathcal{O}$ be a finite conjugacy class of type C. Then $\operatorname{GK\text{-}dim}\mathscr{B}(V)=\infty$ for any...

conjecture | open | MathDB 页面 | 原文线索

V2-011978 | MATHDB-403415

Geometric interpretation of polynomials characterizing free Nichols algebras

摘录 / EXCERPT: When $\mathcal V$ is the category of Yetter-Drinfeld modules over a finite abelian group, the condition for a Nichols algebra to be a free algebra-that is, the $N=1$ case-has been...

open_problem | open | MathDB 页面 | 原文线索

V2-011979 | MATHDB-403420

Universal algebraic construction of coideals via the Swiss-cheese operad

摘录 / EXCERPT: Tamarkin gave a construction of Hopf-algebraic deformations of universal enveloping algebras quantizing Lie bialgebras, based on formality of the E_2-operad. Can one give a simil...

open_problem | open | MathDB 页面 | 原文线索

V2-011980 | MATHDB-403424

Nichs algebra with nonabelian support, infinite dimension, and finite GK-dimension

摘录 / EXCERPT: Find a genuine Nichols algebra over a non-abelian group whose support is not abelian, has infinite dimension, and has finite Gelfand-Kirillov dimension. Here "genuine" means exclud...

open_problem | open | MathDB 页面 | 原文线索

V2-011981 | MATHDB-403438

Semicontinuity of completed fibre dimensions

摘录 / EXCERPT: Let k be an arbitrary field, let A be a Noetherian ring, set $R:=A[[x_1,\\ldots,x_n]]$, and let M be a finitely generated R-module. For $\mathfrak p\in\operatorname{Spec}A$...

open_problem | open | MathDB 页面 | 原文线索

V2-011982 | MATHDB-403466

Three-dimensional Jacobian algebras and the 3-WLP

摘录 / EXCERPT: Let K be a field of characteristic 0, let $S=K[x_1,x_2,x_3]$ with its standard grading, and let A be a central hyperplane arrangement in K^3. Then $S/J(A)$ has the 3-weak...

open_problem | open | MathDB 页面 | 原文线索

V2-011983 | MATHDB-403489

Pole order versus extension dimension for generic kernels of Kirillov-Reshetikhin modules

摘录 / EXCERPT: For any Kirillov-Reshetikhin (KR) modules V and W, let $o(V,W)$ denote the pole order at $z=1$ of the normalized R-matrix $R_{V,W}(z)$ for simple $U_q(L\mathfrak g)$-modules....

open_problem | open | MathDB 页面 | 原文线索

V2-011984 | MATHDB-403499

Unitarity under a codimension-four condition for unipotent Harish-Chandra modules

摘录 / EXCERPT: Let G be a real reductive Lie group. The report defines “unipotent” irreducible Harish-Chandra modules as those annihilated by a unipotent ideal, and notes that some members of t...

open_problem | open | MathDB 页面 | 原文线索

V2-011985 | MATHDB-403501

Morton-Samuelson conjecture for the DAHA-tangle skein map

摘录 / EXCERPT: The homomorphism φ from the double affine Hecke algebra $\ddot H_{q,t}(m)$ to the tangle skein algebra of the thickened torus is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-011986 | MATHDB-403624

Braided tensor structure for the full category of \sl_2\-modules at admissible level

摘录 / EXCERPT: Let k be an admissible level for $\sl_2\$, and let $\mathcal C_k(\mathfrak{sl}_2)$ denote the category of modules considered in the report beyond the ordinary modules, including...

open_problem | open | MathDB 页面 | 原文线索

V2-011987 | MATHDB-403626

The nature and unification of the algebras behind deformed W-algebras

摘录 / EXCERPT: Determine the nature of the algebras discussed above and whether these examples are part of a single construction.

open_problem | open | MathDB 页面 | 原文线索

V2-011988 | MATHDB-403632

Minimality of the virtual resolution $G_\bullet$

摘录 / EXCERPT: For the complex $G_\bullet$ constructed from a finitely generated $\mathbb{Z}^r$-graded S-module M and a degree d, is $G_\bullet$ the minimal free resolution of the truncatio...

open_problem | open | MathDB 页面 | 原文线索

V2-011989 | MATHDB-403663

Rosen’s continued-fraction problem for algebraic number fields

摘录 / EXCERPT: Find more general definitions of continued-fraction expansions for an algebraic number field K such that every element of K is represented by a finite expansion and elements ou...

open_problem | open | MathDB 页面 | 原文线索

V2-011990 | MATHDB-403670

Determining the Northcott or Bogomolov property for infinite algebraic extensions

摘录 / EXCERPT: Given an infinite algebraic extension $L/\mathbb{Q}$, decide whether it satisfies the Northcott property (N)-for every real $T>0$, $\{\alpha\in L:h(\alpha)<T\}$ is finite-or the Bo...

open_problem | open | MathDB 页面 | 原文线索

V2-011991 | MATHDB-403698

Cotangent DG module for commutative DG rings

摘录 / EXCERPT: The conjectural description of the cotangent DG module $L_{B/A}$ associated to a homomorphism $A\to B$ of commutative DG rings is believed to coincide with the cotangent complex of...

open_problem | open | MathDB 页面 | 原文线索

V2-011992 | MATHDB-403701

Simplicity of nonzero central-charge Virasoro enveloping-algebra quotients

摘录 / EXCERPT: If $\lambda\ne 0$, is $U(\operatorname{Vir})/(z-\lambda)$ simple?

open_problem | open | MathDB 页面 | 原文线索

V2-011993 | MATHDB-403702

Characterization of nontrivial Poisson cores in the Virasoro symmetric algebra

摘录 / EXCERPT: For which $\c7\roulettein \d4e9\mathrm{ir}^*$ does $\operatorname{PCore}(\mathfrak m_\chi)\supsetneq (z-\chi(z))$? The report answers: exactly when $\c7$ is local.

open_problem | open | MathDB 页面 | 原文线索

V2-011994 | MATHDB-403708

Chromatic height shift for commutative ring spectra

摘录 / EXCERPT: For every $A\\in\\mathrm{CAlg}(\\mathrm{Sp})$ with $\\operatorname{height}(A)\\ge 1$, one has $\\operatorname{height}(K(A))=\\operatorname{height}(A)+1$.

open_problem | open | MathDB 页面 | 原文线索

V2-011995 | MATHDB-403709

Chromatic height shift for algebraic K-theory

摘录 / EXCERPT: For a commutative algebra spectrum $A\in\operatorname{CAlg}(\mathrm{Sp})$ with $\operatorname{height}(A)\ge 1$, one has ... where ... (and equivalently, when finite,...

open_problem | open | MathDB 页面 | 原文线索

V2-011996 | MATHDB-403777

Which nilpotent Lie groups arise as model boundaries of complex domains?

摘录 / EXCERPT: Under the interpretation discussed in the report, a nilpotent Lie group G whose Lie algebra decomposes as $ \mathfrak g=\mathfrak h\oplus\mathfrak n$, where $\mathfrak h$ is a re...

open_problem | open | MathDB 页面 | 原文线索

V2-011997 | MATHDB-403811

Characterizing the C*-algebraic pairs arising in Theorem 1

摘录 / EXCERPT: What C-algebraic properties characterize the pairs of C-algebras that appear in Theorem 1?

open_problem | open | MathDB 页面 | 原文线索

V2-011998 | MATHDB-403812

Preservation of classifiability under crossed products by nonamenable groups

摘录 / EXCERPT: Let G be a nonamenable countable discrete group acting on a classifiable C^*-algebra A. When is $A \rtimes G$ again classifiable?

open_problem | open | MathDB 页面 | 原文线索

V2-011999 | MATHDB-403816

Isomorphism of q-Gaussian algebras with free group factors

摘录 / EXCERPT: Is the q-Gaussian von Neumann algebra $\Gamma_q(\mathbb{R}^d)=W^*(X_1,\ldots,X_d)$ isomorphic to the free group factor for every $-1<q<1$?

open_problem | open | MathDB 页面 | 原文线索

V2-012000 | MATHDB-403818

Positivity of pre-canonical-basis decompositions in affine type A

摘录 / EXCERPT: For the pre-canonical bases N^i_λ of the spherical Hecke algebra in affine type A, where λ is a dominant weight and i is a positive integer, the decomposition o...

open_problem | open | MathDB 页面 | 原文线索

V2-012001 | MATHDB-403828

Broué’s abelian defect conjecture for double covers of symmetric groups

摘录 / EXCERPT: For odd characteristic p, the superblocks of the twisted group algebras $S(n)=K\widehat{S}_n/(z+1)$ corresponding to weight spaces of the basic representation $V(-\Lambda_0)$ who...

conjecture | open | MathDB 页面 | 原文线索

V2-012002 | MATHDB-403834

Closed formula for endomorphism algebras of Specht modules in characteristic 2

摘录 / EXCERPT: Find a closed formula for the dimension of the endomorphism algebra of Specht modules in characteristic 2.

open_problem | open | MathDB 页面 | 原文线索

V2-012003 | MATHDB-403864

The p-adic non-abelian Hodge correspondence for rigid spaces

摘录 / EXCERPT: Let X be a connected proper smooth rigid space over a complete algebraically closed non-archimedean extension K of $\$Qp$, with a fixed base point $xin X(K)...

open_problem | open | MathDB 页面 | 原文线索

V2-012004 | MATHDB-403942

Explicit character formulas for standard modules of all types and levels

摘录 / EXCERPT: Find explicit expressions for the characters of standard modules of all types at all levels.

open_problem | open | MathDB 页面 | 原文线索

V2-012005 | MATHDB-403943

A q-version of the parking-space recursion

摘录 / EXCERPT: Give a q-version of Theorem 2, for instance via Rouquier’s shift functors for Cherednik algebras, or by generalizing the Lie-theoretic q-Kreweras numbers of [12], or via the fr...

open_problem | open | MathDB 页面 | 原文线索

V2-012006 | MATHDB-404062

Explicit separation between matrix-algebra and \\-factor trace positivity

摘录 / EXCERPT: Find a simple and explicit example of a noncommutative polynomial that is trace positive in all matrix algebras but has negative trace in a II$_1$-factor.

open_problem | open | MathDB 页面 | 原文线索

V2-012007 | MATHDB-404107

Period-index conjecture

摘录 / EXCERPT: Let K be a field of finite transcendence degree d over an algebraically closed field k. For all $\b[?]$ no: the report states: For all $\b[?]$

conjecture | open | MathDB 页面 | 原文线索

V2-012008 | MATHDB-404111

Modularity of almost all elliptic curves over Galois imaginary CM fields

摘录 / EXCERPT: Let F be an imaginary CM field that is Galois over $\\mathbb{Q}$ and such that $\zeta_5 \notin F$. Then 100% of Weierstrass equations defined over F, ordered by their height...

open_problem | open | MathDB 页面 | 原文线索

V2-012009 | MATHDB-404123

Deligne’s conjecture on critical values of L-functions for Hecke characters

摘录 / EXCERPT: Let L and T be number fields, and let χ be a critical algebraic Hecke character of L with values in T. With $L(\chi,s)$ and the period $c^+(\chi)$ defined as in the re...

conjecture | open | MathDB 页面 | 原文线索

V2-012010 | MATHDB-404200

Coherence of the stacks of equivariant formal groups

摘录 / EXCERPT: Conjecture 1. The stacks of equivariant formal groups $\mathcal{M}_{FG}^{G}$ are coherent in the sense that coherent sheaves on them (corresponding to comodules whose underlying mo...

open_problem | open | MathDB 页面 | 原文线索

V2-012011 | MATHDB-404204

Chromatic Milnor-Moore conjecture: full faithfulness of the universal enveloping functor

摘录 / EXCERPT: Conjecture A. The universal enveloping algebra functor ... is fully faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-012012 | MATHDB-404267

Isomorphism type of the concordance group G

摘录 / EXCERPT: Let $\mathcal R$ be the commutative semiring of graded chain complexes over $\mathbb Z[x]$ consisting of grading-shifted free modules of finite total rank, modulo the equivalence r...

open_problem | open | MathDB 页面 | 原文线索

V2-012013 | MATHDB-404294

The Deligne-Ihara conjecture

摘录 / EXCERPT: The Soulé elements $\{\sigma_m\}_{m\ge 3,\,m\text{ odd}}$ freely generate $\mathfrak g\otimes_{\mathbb Z_p}\mathbb Q_p$ as a graded Lie algebra over $\mathbb Q_p$.

conjecture | open | MathDB 页面 | 原文线索

V2-012014 | MATHDB-404310

Group-theoretical specialization conjecture for admissible fundamental groups

摘录 / EXCERPT: For pointed stable curves over algebraically closed fields of characteristic $p>0$, open continuous homomorphisms between admissible fundamental groups completely determine the top...

conjecture | open | MathDB 页面 | 原文线索

V2-012015 | MATHDB-404314

Resolution of non-singularities for stable models of hyperbolic curves

摘录 / EXCERPT: For every hyperbolic curve X over an algebraic closure of $\cmathbb{Q}_p$ and every closed point x of the stable model $\coverline{X}$ of X, there exists a finite étale cover...

open_problem | open | MathDB 页面 | 原文线索

V2-012016 | MATHDB-404347

Primitive-element decompositions for positive and negative renormalization

摘录 / EXCERPT: In regularity structures, Hopf algebras also play an important role. Is it possible to obtain a similar decomposition for positive and/or negative renormalization? If so, what woul...

open_problem | open | MathDB 页面 | 原文线索

V2-012017 | MATHDB-404365

Nakai’s Conjecture for finitely generated characteristic-zero algebras

摘录 / EXCERPT: Nakai’s Conjecture asks whether the converse to Grothendieck’s smooth-case result holds: if the ring of differential operators $D_{R\mid k}$ is generated as an R-algebra by the...

conjecture | open | MathDB 页面 | 原文线索

V2-012018 | MATHDB-404371

Right Calabi-Yau structures for DG algebras in the Derived Auslander-Iyama Correspondence

摘录 / EXCERPT: Let Λ be a basic finite-dimensional algebra that is twisted $(d+2)$-periodic with respect to the Nakayama automorphism ν of Λ. Let A be any DG algebra that co...

open_problem | open | MathDB 页面 | 原文线索

V2-012019 | MATHDB-404377

Is the parametrization map \(\Phi\) an isomorphism?

摘录 / EXCERPT: Assume that Λ is a finite-dimensional representation-finite $\mathbb C$-algebra, written $\Lambda\cong \mathbb C Q/I$, where I is admissible, and let n be the number of...

open_problem | open | MathDB 页面 | 原文线索

V2-012020 | MATHDB-404379

Scaling of rigid parts under canonical decomposition

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field and let $\theta e K_0(\operatorname{proj} A)$. Is the rigid part of $m\theta$ always equal to m times the rigid part of...

open_problem | open | MathDB 页面 | 原文线索

V2-012021 | MATHDB-404387

Trivialization theory for classes of L-algebras

摘录 / EXCERPT: Develop a trivialization theory for classes of L-algebras that relates an L-algebra X to a trivialization X_{tr}. Find relations between $S(X)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-012022 | MATHDB-404411

Graded bivariant K-theory and the classification of Leavitt path algebras

摘录 / EXCERPT: Let ℓ be a field and let E,F be two finite primitive graphs. The following conditions are equivalent: 1. There exists a $\mathbb{Z}$-equivariant isomorphism of ordered group...

open_problem | open | MathDB 页面 | 原文线索

V2-012023 | MATHDB-404449

Improving intersection bounds for regular semisimple conjugacy classes

摘录 / EXCERPT: Let G be a linear algebraic group of rank r defined over a finite field F. Let $g\in G(F)$ be a regular semisimple element, and let $A\subset G(F)$ be a set generating $G(F)$...

open_problem | open | MathDB 页面 | 原文线索

V2-012024 | MATHDB-404465

Mixing weighted backward shifts without hypercyclic algebras on unrooted directed trees

摘录 / EXCERPT: Characterize the unrooted directed trees A supporting a mixing weighted backward shift $B_\lambda:X\to X$ with no hypercyclic algebra, where $X=\ell^p(A)$, $1\le p<\infty$, or...

open_problem | open | MathDB 页面 | 原文线索

V2-012025 | MATHDB-404478

TF-equivalence classes and canonical decompositions in Grothendieck groups

摘录 / EXCERPT: Let A be a finite-dimensional algebra over an algebraically closed field k. For each $\theta\in K_0(\operatorname{proj}A)$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-012026 | MATHDB-404494

Isomorphism of the free group algebras L_{F_2} and L_{F_3}

摘录 / EXCERPT: It is an open problem whether L_{F_2} is isomorphic to L_{F_3}, where F_n is the free noncommutative group with n generators.

open_problem | open | MathDB 页面 | 原文线索

V2-012027 | MATHDB-404495

Peterson-Thom conjecture

摘录 / EXCERPT: Let $r\in\mathbb N$ with $r>1$, let F_r be the free group with r free generators, and let $L(F_r)$ be the free group von Neumann algebra generated by the left regular represent...

conjecture | open | MathDB 页面 | 原文线索

V2-012028 | MATHDB-404514

Iyama-Wemyss derived-equivalence conjecture for minimal models and maximal modification algebras

摘录 / EXCERPT: In the three-dimensional setting, Iyama and Wemyss expected that a minimal model of $\operatorname{Spec} R$ and a maximal modification algebra of R are derived equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-012029 | MATHDB-404528

Structure of the outer automorphism group of an absolute Galois group

摘录 / EXCERPT: What is the structure of the group $\operatorname{Out}(G_K)$?

open_problem | open | MathDB 页面 | 原文线索

V2-012030 | MATHDB-404534

Quasiprojective linear Shafarevich theorem

摘录 / EXCERPT: Theorem. Let X be a connected normal complex algebraic space whose fundamental group admits a faithful finite-dimensional complex linear representation ... Then there is a partial...

open_problem | open | MathDB 页面 | 原文线索

V2-012031 | MATHDB-404535

Existence of a Shafarevich morphism

摘录 / EXCERPT: Let X be a connected normal complex algebraic space. Let $\rho:\pi_1(X^{\mathrm{an}})\to\mathrm{GL}_r(\mathbb{C})$ be a nonextendable finite-dimensional complex linear representa...

open_problem | open | MathDB 页面 | 原文线索

V2-012032 | MATHDB-404546

π-finite p-space descent for localized algebraic K-theory

摘录 / EXCERPT: The functor $K_{T(n+1)}:\mathrm{Cat}^{L}_{f_n}\to\mathrm{Sp}_{T(n+1)}$ commutes with $[U+007F]pi$-finite p-space-indexed (co)limits.

open_problem | open | MathDB 页面 | 原文线索

V2-012033 | MATHDB-404547

Asymptotic L_{f_1}-locality in the Lichtenbaum-Quillen question

摘录 / EXCERPT: Is the fiber of the localization map $K(\mathcal O_F[1/p])\longrightarrow L_{f_1}K(\mathcal O_F[1/p])$ bounded above, where K denotes p-complete algebraic K-theory?

open_problem | open | MathDB 页面 | 原文线索

V2-012034 | MATHDB-404558

Classification of Yetter-Drinfeld modules yielding finite-dimensional Nichols algebras

摘录 / EXCERPT: Classify Yetter-Drinfeld modules over groups that provide finite-dimensional Nichols algebras.

open_problem | open | MathDB 页面 | 原文线索

V2-012035 | MATHDB-404563

Finite GK-dimension and finite root systems for Nichols algebras of diagonal type

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, and let q be a braiding matrix of diagonal type. For the corresponding Nichols algebra B_q, one has ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-012036 | MATHDB-404593

Existence of a bilinear form for the E_{11} tensor-hierarchy representation

摘录 / EXCERPT: Does the representation carried by the infinite collection of field strengths F^I, as defined through the tensor-hierarchy superalgebra $T(\mathfrak e_{11})$, admit a $K(E_{11})$...

open_problem | open | MathDB 页面 | 原文线索

V2-012037 | MATHDB-404596

Hyperbolic Kac-Moody generalization of the Sugawara construction

摘录 / EXCERPT: Does there exist a hyperbolic Kac-Moody generalization of the Sugawara construction, together with a corresponding hyperbolic generalization of the Virasoro algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-012038 | MATHDB-404598

A substitute for the Casimir element in the universal enveloping algebra of $\mathfrak{k}(A)$

摘录 / EXCERPT: Another open question is whether there exists a substitute for the Casimir element within the universal enveloping algebra of $\mathfrak{k}(A)$ or a completion thereof.

open_problem | open | MathDB 页面 | 原文线索

V2-012039 | MATHDB-404751

Multiplicity conjecture for generic direct summands

摘录 / EXCERPT: Let G be a simply connected simple algebraic group over an algebraically closed field k of characteristic $p>0$, with p at least the Coxeter number h. Let $W_{\rm aff}^{+}$...

conjecture | open | MathDB 页面 | 原文线索

V2-012040 | MATHDB-404753

Lie algebras of proper closed normal subgroups

摘录 / EXCERPT: Let $G\subseteq \operatorname{Aut}(X)$ be a connected closed subgroup and $N\subsetneq G$ a proper normal and closed subgroup. Does it follow that...

open_problem | open | MathDB 页面 | 原文线索

V2-012041 | MATHDB-404755

Conjectural crystal description of the Grothendieck-group representation

摘录 / EXCERPT: The crystal of type C defined in the report is conjecturally isomorphic to the crystal of the $\mathfrak{g}$-module $K({}^{0}\mathscr{O}^{\rho})$.

open_problem | open | MathDB 页面 | 原文线索

V2-012042 | MATHDB-404771

Harish-Chandra center for affine Kac-Moody algebras in positive characteristic

摘录 / EXCERPT: Let G be a connected reductive group over an algebraically closed field F of characteristic p, with p larger than the relevant Coxeter numbers, and let $\chat{\mathfrak g}$...

open_problem | open | MathDB 页面 | 原文线索

V2-012043 | MATHDB-404773

Balmer spectrum of the stable category of finite-dimensional representations of a quasireductive supergroup

摘录 / EXCERPT: Let G be a complex quasireductive supergroup, with G_0 having connected identity component reductive, and let $\mathfrak g$ be its Lie superalgebra. Define ... For...

open_problem | open | MathDB 页面 | 原文线索

V2-012044 | MATHDB-404774

Crystal model for the Grothendieck-group representation

摘录 / EXCERPT: Let cmathfrak{g} be the complex simple Lie algebra whose Cartan matrix is the unfolded symmetric Cartan matrix cmathbf{C}. The crystal of type cmathbf{C} defined in Proposition 4.4...

open_problem | open | MathDB 页面 | 原文线索

V2-012045 | MATHDB-404790

Finding a dense set of G-submodules

摘录 / EXCERPT: Given a subgroup $G \leq \mathrm{GL}(V)$ by a generating set, where V is a finite-dimensional vector space over a finite field, find a set S of G-submodules of V such that...

open_problem | open | MathDB 页面 | 原文线索

V2-012046 | MATHDB-404821

Expected embedding of Harish-Chandra bimodules into the differential Hecke category

摘录 / EXCERPT: There is an exact and fully faithful embedding of monoidal categories ...

open_problem | open | MathDB 页面 | 原文线索

V2-012047 | MATHDB-404893

Characterization of exact algebras in finite tensor categories

摘录 / EXCERPT: Let $\mathcal C$ be a finite tensor category and let A be an algebra in $\mathcal C$. Then A is exact-that is, for every object $M\in \mathcal C_A$ and every projective object...

open_problem | open | MathDB 页面 | 原文线索

V2-012048 | MATHDB-404894

Rigidity of module categories from the local-module subcategory

摘录 / EXCERPT: If A is a commutative algebra in a Grothendieck-Verdier category $\mathcal C$ and the category $\mathcal C_A^{\mathrm{loc}}$ of local A-modules is rigid, under what conditions...

open_problem | open | MathDB 页面 | 原文线索

V2-012049 | MATHDB-404898

Tensor-factor uniqueness for finite tensor products of SAW*-algebras

摘录 / EXCERPT: Suppose that $m,n\in\mathbb{N}$ and that A_i, for $i<m$, and B_j, for $j<n$, are SAW*-algebras such that ... Can one conclude that $m=n$? If so, is there a permutation π o...

open_problem | open | MathDB 页面 | 原文线索

V2-012050 | MATHDB-404901

Agreement of essential and reduced groupoid C^*-algebras

摘录 / EXCERPT: For a non-Hausdorff étale groupoid G with Hausdorff unit space, determine when the essential groupoid C^*-algebra $C^*_{\mathrm{ess}}(G)$ agrees with the classical reduced grou...

open_problem | open | MathDB 页面 | 原文线索

V2-012051 | MATHDB-404902

When do essential and reduced groupoid C*-algebras agree?

摘录 / EXCERPT: For a non-Hausdorff étale groupoid G, determine when the essential groupoid C-algebra $C^*_{\mathrm{ess}}(G)$ agrees with the reduced groupoid C-algebra $C^*_r(G)$.

open_problem | open | MathDB 页面 | 原文线索

V2-012052 | MATHDB-404903

Strict comparison for the C*-algebra generated by free semicircular elements

摘录 / EXCERPT: Does the C-algebra generated by r free semicircular elements have strict comparison for every integer $r\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-012053 | MATHDB-404908

Cantor-spectrum diagonals in AF algebras

摘录 / EXCERPT: In an AF algebra, are all Cantor-spectrum C-diagonals AF diagonals?

open_problem | open | MathDB 页面 | 原文线索

V2-012054 | MATHDB-404911

Local lifting property for surface-group C*-algebras

摘录 / EXCERPT: Does the full group C-algebra $C^*(\Gamma)$ have Kirchberg's local lifting property (LLP) when Γ is the fundamental group of a general closed surface?

open_problem | open | MathDB 页面 | 原文线索

V2-012055 | MATHDB-404913

Elementary equivalence of reduced free-group C*-algebras

摘录 / EXCERPT: For free groups F_m and F_n on $m,n\ge 2$ generators, the reduced group C-algebras $C_r^*(F_m)$ and $C_r^*(F_n)$ have the same first-order theory as C-algebras (equivalently, h...

open_problem | open | MathDB 页面 | 原文线索

V2-012056 | MATHDB-404914

Lifting the continuous part of an induced action on a C*-algebra ultrapower

摘录 / EXCERPT: Let $\mathbb Z\curvearrowright A$ be a continuous action on a separable C^*-algebra A. Let $\mathcal U$ be a non-principal ultrafilter on $\mathbb N$. Does the continuous part...

open_problem | open | MathDB 页面 | 原文线索

V2-012057 | MATHDB-404915

Isomorphism among three von Neumann algebra ultraproducts

摘录 / EXCERPT: Let $\mathcal U$ be a non-principal ultrafilter on $\mathbb N$. Consider $L(\mathrm{SL}_2(\mathbb Z))^{\mathcal U}$, $\prod_{\mathcal U} M_n$, and $L(\mathbb F_2)^{\mathcal U}$. Ar...

open_problem | open | MathDB 页面 | 原文线索

V2-012058 | MATHDB-404971

A noncommutative Marcinkiewicz theorem for grouplike-valued random variables

摘录 / EXCERPT: Characterise the elements of the form $\log_{\otimes}\mathbb{E}[X]$, where X is a random variable on $G(T((V)))$, the grouplike elements of the tensor-series algebra. In particul...

open_problem | open | MathDB 页面 | 原文线索

V2-012059 | MATHDB-404973

The Nash problem for surfaces in positive characteristic

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let X be a normal algebraic surface over k, and let $f:Y\to X$ be its minimal resolution. For each exceptional...

open_problem | open | MathDB 页面 | 原文线索

V2-012060 | MATHDB-404974

Wedge Lifting Problem for Surface Singularities

摘录 / EXCERPT: Let X be an algebraic surface over an algebraically closed field k, and let $f:Y\to X$ be its minimal resolution. Given a wedge $\Phi:\operatorname{Spec} k[[s,t]]\to X$ whose s...

open_problem | open | MathDB 页面 | 原文线索

V2-012061 | MATHDB-404975

Derived arc and jet space functors

摘录 / EXCERPT: Let $R_\bullet$ be an animated k-algebra, and let $P_\bullet\to R_\bullet$ be a cofibrant replacement. The left derived functors of the classical arc and m-jet functors are com...

open_problem | open | MathDB 页面 | 原文线索

V2-012062 | MATHDB-404981

Complete invariant for indecomposable reflexive modules on quotient surface singularities

摘录 / EXCERPT: For a quotient surface singularity (X,x), which invariant must be added to the Chern polynomial-namely, the rank and first Chern class-to obtain a complete invariant of isomorphi...

open_problem | open | MathDB 页面 | 原文线索

V2-012063 | MATHDB-405010

Explicit description of higher idempotent completions of Soergel bimodule 2-categories

摘录 / EXCERPT: Describe explicitly the 2-categories $\operatorname{Kar}^2\mathrm{SBim}$ and $\operatorname{Kar}^2\mathrm{sSBim}$; equivalently, find and classify the objects and data constituting...

open_problem | open | MathDB 页面 | 原文线索

V2-012064 | MATHDB-405029

Finite length of permutation modules for homogeneous structures

摘录 / EXCERPT: Suppose that M is a structure which is homogeneous in a finite relational language and $G=\operatorname{Aut}(M)$. Let $k\in\mathbb{N}$ and consider G acting on $W=M^k$. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-012065 | MATHDB-405095

Algebraic de Rham complex and effective periods

摘录 / EXCERPT: Let D^∞ be the infinite product of closed unit discs, let $\mathcal O_{\mathrm{alg}}(D^\infty)$ be the sub-$\mathbb Q$-algebra of $\mathcal O(D^\infty)$ consisting of holomo...

open_problem | open | MathDB 页面 | 原文线索

V2-012066 | MATHDB-405105

Whether every amenable trace is quasidiagonal

摘录 / EXCERPT: Is every amenable trace on a C-algebra quasidiagonal?

open_problem | open | MathDB 页面 | 原文线索

V2-012067 | MATHDB-405110

Completely bounded approximation property for high-rank lattice von Neumann algebras

摘录 / EXCERPT: Let $d\ge 2$. Does $L^p\bigl(L_{\mathrm{PSL}_{2d-1}(\mathbb Z)}\bigr)$ have the completely bounded approximation property for some $p\ne 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-012068 | MATHDB-405114

Socle conjecture for the Aizenbud-Lapid operation in type A_n

摘录 / EXCERPT: Let Q be of type A_n, let $\mathcal C$ be the set of irreducible components under consideration, and let $\pi(C)$ denote the corresponding simple Hecke-algebra module. The repo...

conjecture | open | MathDB 页面 | 原文线索

V2-012069 | MATHDB-405115

Classification of Auslander regular linear Nakayama algebras

摘录 / EXCERPT: Classify all Auslander regular linear Nakayama algebras and give an explicit enumeration of them.

open_problem | open | MathDB 页面 | 原文线索

V2-012070 | MATHDB-405117

Equality of configuration spaces defined by universal F-polynomials and u-equations

摘录 / EXCERPT: Let Λ be a finite-dimensional basic $\mathbb{C}$-algebra of finite representation type, with $\widetilde{M}_\Lambda$, M_Λ, $\widetilde{U}_\Lambda$, and U_Λ...

open_problem | open | MathDB 页面 | 原文线索

V2-012071 | MATHDB-405200

Schumaker’s lower-bound equality conjecture for C^1 cubic spline spaces

摘录 / EXCERPT: For every nondegenerate planar triangulation $\mathcal{T}$, the dimension of the C^1 piecewise-cubic spline space satisfies $\dim S_3^1(\mathcal{T})=P_{\mathcal{T}}(1,3)$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-012072 | MATHDB-405214

Pérez-García-Verstraete-Wolf-Cirac conjecture

摘录 / EXCERPT: For every field $\mathbb{F}$, every integer $n\geq 1$, and every family $S\subseteq M_n(\mathbb{F})$, if there exists an integer $\ell\geq 1$ such that...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-012073 | MATHDB-405223

Moscariello-Sammartano bounded-type problem for numerical semigroups

摘录 / EXCERPT: Does there exist a function $f:\mathbb{N}\to\mathbb{N}$ such that, for every numerical semigroup S with embedding dimension 4, the type satisfies $t(S)\le f\bigl(\eta(S)\bigr)$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-012074 | MATHDB-405249

Strong shift equivalence problem for the Baker matrices and Ashley’s eight-by-eight matrix

摘录 / EXCERPT: Given two square nonnegative integer matrices $A\in\mathbb{N}^{m\times m}$ and $B\in\mathbb{N}^{n\times n}$, determine whether they are strong shift equivalent: that is, whether th...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-012075 | MATHDB-405260

Matkowski’s conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be the coefficient field. Matkowski’s conjecture asserts that every two-variable formal power series $M\in\mathbb{K}[[x,y]]$ satisfying the weak associativity equa...

conjecture | claimed_solved | MathDB 页面 | 原文线索

B03 群论 / Group Theory | 4,219 条

V2-012076 | MATHDB-315615

Solubilizer Conjecture A.13

摘录 / EXCERPT: If $\operatorname{Sol}_G(x)$ is a proper subgroup of a nonsolvable finite group G, must the intersection of all its conjugates lie in the hypercenter of G?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-012077 | MATHDB-315616

Solubilizer Conjecture A.16

摘录 / EXCERPT: If $\operatorname{Sol}_G(x)$ is a proper subgroup of a nonsolvable finite group G, must its intersection with its normalizer be metabelian?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-012078 | MATHDB-315890

Isomorphism problem of Coxeter groups

摘录 / EXCERPT: Let S be a finite set. A Coxeter matrix over S is a symmetric matrix $M=(m_{st})_{s,t\in S}$ with entries in $\{1,2,3,\dots\}\cup\{\infty\}$ such that $m_{ss}=1$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-012079 | MATHDB-315891

Problems in loop theory and quasigroup theory

摘录 / EXCERPT: In mathematics, especially abstract algebra, loop theory and quasigroup theory are active research areas with many open problems. As in other areas of mathematics, such problems ar...

open_problem | open | MathDB 页面 | 原文线索

V2-012080 | MATHDB-315895

Kazhdan-Lusztig conjectures

摘录 / EXCERPT: Let W be a Coxeter group with generating set S, let $\ell(w)$ denote the length of $w \in W$ (the smallest number of elements of S in a product expressing w), and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012081 | MATHDB-316202

Hanna Neumann conjecture

摘录 / EXCERPT: In the mathematical subject of group theory, the Hanna Neumann conjecture is a statement about the rank of the intersection of two finitely generated subgroups of a free group. The...

conjecture | open | MathDB 页面 | 原文线索

V2-012082 | MATHDB-316203

Density theorem

摘录 / EXCERPT: In the mathematical theory of Kleinian groups, the density conjecture of Lipman Bers, Dennis Sullivan, and William Thurston, later proved independently by Namazi & Souto (2012) and...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-012083 | MATHDB-316223

Spherical space form conjecture

摘录 / EXCERPT: In geometric topology, the spherical space form conjecture states that a finite group acting on the 3-sphere is conjugate to a group of isometries of the 3-sphere.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-012084 | MATHDB-316260

Conjecture on generators of the kernel of the representation functor

摘录 / EXCERPT: Let the diagrammatic category and its representation functor be as in the paper. The kernel is understood as a pivotal category ideal, and the elements described in Theorems...

conjecture | open | MathDB 页面 | 原文线索

V2-012085 | MATHDB-316279

Enomoto-Kashiwara's crystal-basis conjecture for symmetric crystals

摘录 / EXCERPT: Assume that I is an index set with an involution θ and that there is no $i\in I$ such that $\theta(i)=i$. Let $V_\theta(\lambda)$ be the representation defined in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-012086 | MATHDB-316302

Parabolic Brundan-Kazhdan-Lusztig conjecture

摘录 / EXCERPT: Let $\mathfrak m$ be a tuple of positive integers, let $n\in\mathbb N\cup\{\infty\}$, and let $\mathcal O_{{\bf m}|n}^{+,\Delta}$ be the category of modules with Verma flags, with...

conjecture | open | MathDB 页面 | 原文线索

V2-012087 | MATHDB-316303

General super duality conjecture

摘录 / EXCERPT: Let ${\bf m}$ be a tuple of positive integers, and let $\mathcal O_{{\bf m}|\infty}^{++}$ and $\mathcal O_{{\bf m}+\infty}^{++}$ denote the two categories appearing in the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-012088 | MATHDB-316323

The Beauville structure conjecture for finite simple nonabelian groups

摘录 / EXCERPT: Beauville structure conjecture. All finite simple nonabelian groups except $\mathfrak{A}_5$ admit an unmixed Beauville structure.

conjecture | open | MathDB 页面 | 原文线索

V2-012089 | MATHDB-316326

The component-hive bijection conjecture for convolution fibres

摘录 / EXCERPT: Component-hive bijection conjecture. The generic value of H on each component of $\mathcal{G}r_{\lambda^1 \cdots \lambda^k}$ is a k-hive, and this gives a bijection between the...

conjecture | open | MathDB 页面 | 原文线索

V2-012090 | MATHDB-316372

Successor conjecture for maximal projective-degree partitions

摘录 / EXCERPT: For each n, let $\lambda^{(n)}_{g\max}\models n$ be a partition of n maximizing its projective degree g^λ. For a partition $\mu\models n$, define its set of 1-success...

conjecture | open | MathDB 页面 | 原文线索

V2-012091 | MATHDB-316373

Conjectural sums of minimal permutation representation invariants for groups of orders p^4 and p^5

摘录 / EXCERPT: For a finite group G, let $4\Delta(G)4$ denote the invariant used in the paper, and let the sums range over isomorphism classes of groups of the indicated order. Numerical-sum co...

conjecture | open | MathDB 页面 | 原文线索

V2-012092 | MATHDB-316375

Conjecture on non-degeneracy of bilinear forms for Brauer algebra modules

摘录 / EXCERPT: Non-degeneracy conjecture. (i) If $n=2k+1$, then the bilinear form on the $B_n(\hat{z})$-module $S^{(1)}$ determined by the referenced construction is non-degenerate if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-012093 | MATHDB-316376

Conjecture on semisimplicity criteria for Brauer algebras

摘录 / EXCERPT: Semisimplicity conjecture. (i) If $n=2k+1$, then $B_n(\hat{z})$ is semisimple only if $\kappa\mathfrak{S}_n$ is semisimple and $p_{2k-1}(\hat{z})\ne 0$. (ii) If $n=2k$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-012094 | MATHDB-316384

Lurie's conjecture on the twisted Whittaker category

摘录 / EXCERPT: Let G be a reductive group, G its Langlands dual group, and let $W^c(Gr_G):=D^cmod(Gr_G)^{N((t)),chi}$ denote the twisted Whittaker category, where c is the twisting paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-012095 | MATHDB-316396

Rigidity conjecture for braided symmetric and exterior cubes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let V^q be a simple $U_q(\mathfrak{g})$-module, and let V be its classical limit. The degree-three braided symmetric and exterior po...

conjecture | refuted | MathDB 页面 | 原文线索

V2-012096 | MATHDB-316403

Non-mixed dual distance for dually uniquely ergodic trees

摘录 / EXCERPT: Let $T\in\partial cv_N$ be an $\mathbb R$-tree with dense orbits, and suppose that its dual lamination $L(T)$ is uniquely ergodic; then T is dually uniquely ergodic. In this case...

conjecture | open | MathDB 页面 | 原文线索

V2-012097 | MATHDB-316413

The factorisation conjecture for shapes of tridiagonal pairs

摘录 / EXCERPT: Let $\lbrace \rho_i\rbrace_{i=0}^d$ denote the shape of a tridiagonal pair over a field $\mathbb F$. Then there exists a nonnegative integer N and positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-012098 | MATHDB-316429

Frenkel-Gaitsgory conjecture for graded opers

摘录 / EXCERPT: Frenkel-Gaitsgory conjecture. The category $\KLlam$ is semisimple for every $\slam\in \bar P^+$, and, for each $\lambda\in \Paf{\operatorname{crit}}$, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-012099 | MATHDB-316430

Finite-generation and generic Azumaya conjecture for elliptic Hecke algebras

摘录 / EXCERPT: Let W be a group acting on V, let X be the associated space, and let $[?]$ be the relevant parameter data with $\tau_{H,m}$ satisfying ... for every H. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012100 | MATHDB-316438

Expected action of the type D descent algebra on D_n-Lie monomials

摘录 / EXCERPT: Let D_n be the Weyl group of type D with descent algebra Σ_{D_n}, let $\mathcal{C}(n)$ denote the relevant class of compositions, and let $P^{(\kappa)}$ and B_ν be t...

conjecture | open | MathDB 页面 | 原文线索

V2-012101 | MATHDB-316443

Maximal nilpotent quotient conjecture for generalised Fabrykowski-Gupta groups

摘录 / EXCERPT: For $p\geq3$, let Γ_p be the generalised Fabrykowski-Gupta group in the ascending finite L-presented family. A maximal nilpotent quotient is a nilpotent quotient through w...

conjecture | open | MathDB 页面 | 原文线索

V2-012102 | MATHDB-316484

Dynnikov-Wiest conjecture on linear-length sigma-consistent braid representatives

摘录 / EXCERPT: Let B_n be the braid group with generating set $\Sigma=\{\sigma_1,\ldots,\sigma_{n-1}\}$, and let $l_\Sigma(w)$ denote the Σ-length of a braid w. A braid word is...

conjecture | open | MathDB 页面 | 原文线索

V2-012103 | MATHDB-316489

Parity conjecture for random Cayley graphs of ${\rm PGL}_2(p)$

摘录 / EXCERPT: Let $G={\rm PGL}_2(p)$, with p varying over primes, and consider random 4-regular Cayley graphs over G. Parity conjecture. The girth of such graphs is almost surely even. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-012104 | MATHDB-316490

Asymptotic girth conjecture for random Cayley graphs of ${\rm PGL}_2(p)$

摘录 / EXCERPT: Let $G={\rm PGL}_2(p)$, with p varying over primes, and consider random 4-regular Cayley graphs over G. Asymptotic girth conjecture. The girth is ... for some constant c sati...

conjecture | open | MathDB 页面 | 原文线索

V2-012105 | MATHDB-316491

Local short-generator conjecture for expanders of ${\rm PSL}_2({\mathbb F}_p)$

摘录 / EXCERPT: Local short-generator conjecture. There is a constant C, independent of p, such that the ball of radius C in the generating set Σ_p contains two elements g,h with

conjecture | open | MathDB 页面 | 原文线索

V2-012106 | MATHDB-316492

Free-amoeba diversification conjecture

摘录 / EXCERPT: Consider the toy genetics process in which an amoeba has a DNA configuration of length n, and each generation produces two offspring through independently activated crossover sym...

conjecture | open | MathDB 页面 | 原文线索

V2-012107 | MATHDB-316494

Nontriviality conjecture for short words in symmetric two-groups

摘录 / EXCERPT: Let W_n be the symmetric 2-group, and let w be a word of length 2^m. Short-word nontriviality conjecture. If $m<n$, then ... This conjecture asserts that no word below the na...

conjecture | open | MathDB 页面 | 原文线索

V2-012108 | MATHDB-316537

Generalization of the eigencone and invariant restriction conjecture for symmetric subgroups

摘录 / EXCERPT: Let G be a connected simply-connected, semisimple complex algebraic group, let σ be a diagram automorphism of G, and let K be its fixed subgroup. Fix a standard parabo...

conjecture | open | MathDB 页面 | 原文线索

V2-012109 | MATHDB-316577

Shareshian's EL-labeling conjecture for solvable subgroup lattices

摘录 / EXCERPT: Let G be a finite solvable group, and let $L(G)$ denote its subgroup lattice. Choose a chief series ... A complement to a subgroup N is a subgroup H with $HN=G$ and $H\cap N=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-012110 | MATHDB-316579

Minimality conjecture for the classification of degenerate gentle two-cycle algebras

摘录 / EXCERPT: Let $\Lambda_0(p,r)$, for $p\in\mathbb{N}_+$ and $r\in[0,p-1]$, and $\Lambda_0'(p,0)$, for $p\in\mathbb{N}_+$, be the algebras in the classification of degenerate gentle two-cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-012111 | MATHDB-316580

Silver-Williams representation-shift characterization of fibered knots

摘录 / EXCERPT: Let k be a knot, let cpi be its knot group, and for each finite group ${\Sigma}$ let h_{Σ} denote the topological entropy of the representation shift associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-012112 | MATHDB-316595

Existence of a non-w-sofic group

摘录 / EXCERPT: A group G is w-sofic if there exists $\alpha>0$ such that, for every finite set $\Phi\subset G$ and every $\epsilon>0$, there is a finite group with a bi-invariant metric admitti...

conjecture | open | MathDB 页面 | 原文线索

V2-012113 | MATHDB-316598

The induction conjecture for string modules over dihedral 2-groups

摘录 / EXCERPT: Induction conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012114 | MATHDB-316620

Rigidity conjecture for indecomposable tensor representations of classical locally finite Lie algebras

摘录 / EXCERPT: Let $\mathfrak g$ be one of the classical locally finite Lie algebras $\mathfrak{gl}_\infty$, $\mathfrak{sl}_\infty$, $\mathfrak{sp}_\infty$, or $\mathfrak{so}_\infty$, and let M...

conjecture | open | MathDB 页面 | 原文线索

V2-012115 | MATHDB-316622

3-dimensional homotopy covering conjecture

摘录 / EXCERPT: Let a P3R group be a finitely presented group whose universal cover is proper homotopy equivalent to a 3-manifold. An end is semi-stable when its fundamental pro-group at infinity...

conjecture | open | MathDB 页面 | 原文线索

V2-012116 | MATHDB-316630

Stabilizer characterization conjecture for faces of permutation polytopes

摘录 / EXCERPT: Stabilizer characterization conjecture. If $P(H)\preceq P(G)$ is a face, then

conjecture | open | MathDB 页面 | 原文线索

V2-012117 | MATHDB-316639

Reductivity conjecture for fiber-preserving linear maps

摘录 / EXCERPT: Reductivity conjecture. The group H_0 is reductive.

conjecture | open | MathDB 页面 | 原文线索

V2-012118 | MATHDB-316683

Weak Fulton conjecture on the equivariant nef cone of the moduli space of stable pointed curves

摘录 / EXCERPT: Let $\overline{\mathbf{M}}_{0,n}$ be the moduli space of stable n-pointed genus-zero curves, let A_{α} be the divisor considered in the paper, and let p_i denote the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-012119 | MATHDB-316709

The boundedness-multiplicity-free module conjecture for reductive subalgebras

摘录 / EXCERPT: Let $\mathfrak{k}\subset\mathfrak{g}$ be a reductive in $\mathfrak{g}$ subalgebra. A $(\mathfrak{g},\mathfrak{k})$-module is multiplicity free if every simple $\mathfrak{k}$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-012120 | MATHDB-316711

Classification conjecture for GHC-regular representations in anti-de Sitter space

摘录 / EXCERPT: Let $\rho: \Gamma\to\operatorname{SO}_{0}(2,n)$ be a GHC-regular representation. A representation is quasi-Fuchsian if it is a deformation of the inclusion of a lattice in...

conjecture | open | MathDB 页面 | 原文线索

V2-012121 | MATHDB-316721

Kerov's unified expansion conjecture for Kerov polynomial components

摘录 / EXCERPT: Let K_r be the Kerov polynomial and write its component of weight $r-2k+1$ as $K_{r,r-2k+1}$. For a partition $ u$, let $l(u)$ be its length, let $m_i(u)$ be the multiplicity o...

conjecture | open | MathDB 页面 | 原文线索

V2-012122 | MATHDB-316741

Lubotzky's filtration conjecture for free-by-linear groups

摘录 / EXCERPT: Let ... be a split exact sequence of groups, where F_n is a free group on n letters and Γ admits a finite-dimensional faithful linear representation. If the conjugation ac...

conjecture | open | MathDB 页面 | 原文线索

V2-012123 | MATHDB-316745

Conjecture on the cyclic module generated by a power of the permanent

摘录 / EXCERPT: Permanent cyclic-module conjecture. The displayed isomorphism holds. This conjecture predicts the irreducible decomposition of the cyclic module generated by a power of the permane...

conjecture | open | MathDB 页面 | 原文线索

V2-012124 | MATHDB-316754

The HKOTY conjecture on equality of the M-sum and N-sum

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra. For dominant integral λ and multiplicity data $\mathbf n$, let $M_{\lambda,\mathbf n}^{(k)}$ and $N_{\lambda,\mathbf n}^{(k)}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-012125 | MATHDB-316781

Donovan's finiteness conjecture for blocks with a fixed defect group

摘录 / EXCERPT: Let k have characteristic p. Let P be a p-group, and let B_P be the set of blocks with defect group P. For a block B, write $Phi(B)$ for its module category up to equ...

conjecture | open | MathDB 页面 | 原文线索

V2-012126 | MATHDB-316784

The arbitrary-characteristic Rock block isomorphism conjecture

摘录 / EXCERPT: Let $(K,{\cal O},k)$ be a p-modular system, let w be a natural number, let $\rho=\rho(p,w)$ be a Rouquier core, and let $v=wp+r$. In $k\Sigma_v$, let e be the specified idemp...

conjecture | open | MathDB 页面 | 原文线索

V2-012127 | MATHDB-316817

Lusztig's conjecture on multiplicities of modular representations

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>h$, let $G^\vee$ be the Langlands-dual connected simple algebraic group over k, and let $X^\vee$ be its cocharacter...

conjecture | open | MathDB 页面 | 原文线索

V2-012128 | MATHDB-316818

Bergeron's dimension conjecture for lattice diagrams with multiple holes

摘录 / EXCERPT: Let μ be a partition of $n+k$, let (i,j) be a cell, and let s be the cardinality of the shadow of (i,j) in μ. With the notation of the preceding theorem, let...

conjecture | open | MathDB 页面 | 原文线索

V2-012129 | MATHDB-316844

Vanishing asymptotic conjecture for second cohomology ratios

摘录 / EXCERPT: Asymptotic cohomology conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012130 | MATHDB-316853

The faithfulness conjecture for Krammer representations of one-class reflection groups

摘录 / EXCERPT: Faithfulness conjecture. If W admits only one conjugacy class of reflections, then R is faithful for $m\notin\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-012131 | MATHDB-316873

Distinct cellular-automata classes over the integers and the free group

摘录 / EXCERPT: Let A be an alphabet, let $\mathbb{Z}$ be the infinite cyclic group, and let $\mathbb{F}_2$ be the free group on two generators. Write $CA(A,G)$ for the class of cellular automat...

conjecture | open | MathDB 页面 | 原文线索

V2-012132 | MATHDB-316880

The Genericity Conjecture for connected groups of finite Morley rank

摘录 / EXCERPT: Genericity Conjecture. The union of the connected definable nilpotent subgroups of G contains a definable generic subset of G.

conjecture | open | MathDB 页面 | 原文线索

V2-012133 | MATHDB-316910

The product replacement graph connectivity conjecture

摘录 / EXCERPT: Connectivity conjecture. If $k\geq d(G)+1$, then $\tilde \Gamma_k(G)$ is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-012134 | MATHDB-316914

General version of James' conjecture for exceptional Hecke algebras

摘录 / EXCERPT: Let W be a finite Weyl group, let $\mathcal H_k(W,\xi)$ be its Hecke algebra over a field k with parameter ξ, and let $e<\infty$ satisfy the standing assumptions...

conjecture | open | MathDB 页面 | 原文线索

V2-012135 | MATHDB-316915

Cellular-basis and W-graph isomorphism conjecture for Hecke algebras

摘录 / EXCERPT: Let $\mathcal H$ be the Hecke algebra in the paper, let Λ index its modules, and for each $\lambda\in\Lambda$ suppose that a W-graph affords an $\mathcal H$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-012136 | MATHDB-316926

Bessel-model Gross-Prasad conjecture for GL_{2n} and $GL_n(D)$

摘录 / EXCERPT: Let D be a quaternion division algebra over a local field k, and let K be a quadratic separable algebra over k. Let π be an irreducible admissible generic representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-012137 | MATHDB-316935

The Kirillov-Reshetikhin completeness conjecture

摘录 / EXCERPT: Let $\mathfrak g$ be a Lie algebra and let $\mathcal H=\bigotimes_{j=1}^N V_j$ be a tensor product of Kirillov-Reshetikhin modules. Write ... and let $M_{\lambda,\mathbf n}$ be the n...

conjecture | open | MathDB 页面 | 原文线索

V2-012138 | MATHDB-316940

The image-equals-projective conjecture for Kostant's problem

摘录 / EXCERPT: Let W be the Weyl group, let $\boldsymbol{R}$ be a right cell, and let $\widehat{\boldsymbol{R}}=\{x\in W:x\leq_{\mathtt{R}}w\text{ for some }w\in\boldsymbol{R}\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012139 | MATHDB-316952

Parameter-array classification conjecture for sharp tridiagonal systems

摘录 / EXCERPT: Let $\mathbb{K}$ be a field and let Φ be a sharp tridiagonal system on a finite-dimensional vector space V. Its parameter array is the scalar data attached to its eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-012140 | MATHDB-316957

Generic multiplicity conjecture for standard modules

摘录 / EXCERPT: Let ${\widehat{\mathcal W}}$ be the affine Weyl group, let $Z'(x\cdot 0)$ be the standard module and $L'(w\cdot 0)$ the corresponding simple module, and let A_y denote the alcove...

conjecture | open | MathDB 页面 | 原文线索

V2-012141 | MATHDB-316958

Moment-graph character formula conjecture for intersection sheaves

摘录 / EXCERPT: Let ${\widehat{\mathcal G}}_k$ be the affine moment graph over the field k, let $\mathscr B_k(w)$ be the intersection sheaf attached to $w\in{\widehat{\mathcal W}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012142 | MATHDB-316973

The discrete-family conjecture for simply connected Gorenstein differential graded algebras

摘录 / EXCERPT: Let A be a simply connected Gorenstein differential graded algebra of finite type. Write H^eA and H^*A for its cohomology in degree e and in a...

conjecture | open | MathDB 页面 | 原文线索

V2-012143 | MATHDB-316977

The smooth-volume-preservation conjecture for generic random groups

摘录 / EXCERPT: Let Γ be a group that is generic in an appropriate model for random groups, let M be a compact manifold, and consider an action of Γ on M. A smooth volume form is...

conjecture | open | MathDB 页面 | 原文线索

V2-012144 | MATHDB-316988

Finite-dimensionality criterion for highest weight modules of finite W-algebras

摘录 / EXCERPT: Let $ab ab$ be a W_0-orbit in ab^*, and choose $ab 't\inab$ such that ... for every $ab\in\Phi_0^+$. Let $L(\Lambda,e)$ be the irreducible highest weight $U(\mathfrak g,e)$-modul...

conjecture | open | MathDB 页面 | 原文线索

V2-012145 | MATHDB-316989

Whittaker-category equivalence for finite W-algebras

摘录 / EXCERPT: Let ab be a W_0-orbit in ab^*, and let $M(\Lambda,\chi)$ and $L(\Lambda,\chi)$ be the standard and irreducible objects in the category $\mathcal O(\chi)$, while...

conjecture | open | MathDB 页面 | 原文线索

V2-012146 | MATHDB-316997

Higher symmetric-group transvectant redundancy conjecture

摘录 / EXCERPT: The standard representation of the permutation group $\mathfrak S_d$ is denoted by $V_{(d-1,1)}$, and the irreducible representations indexed by partitions $(d-2,2)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012147 | MATHDB-317003

Mazorchuk's conjecture on mixed modules over the Virasoro algebra

摘录 / EXCERPT: Mazorchuk's conjecture. There are no irreducible mixed modules over the Virasoro algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-012148 | MATHDB-317006

Frenkel-Mukhin conjecture for the q-character algorithm

摘录 / EXCERPT: Frenkel-Mukhin conjecture. For any dominant monomial m_+, the algorithm never fails and stops after finitely many steps. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-012149 | MATHDB-317014

Polynomiality conjecture for unitriangular conjugacy classes

摘录 / EXCERPT: Unitriangular polynomiality conjecture. For every positive integer m, $\rho_{(1^m)}(q)$ is a polynomial in q with rational coefficients. This is the special case corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-012150 | MATHDB-317016

Existence of equivariant bases for link chord-diagram spaces

摘录 / EXCERPT: Let ${\cal A}^m_n$ denote the corresponding degree-n chord-diagram space for links with m components, with the induced symmetric-group action on components. An equivariant basi...

conjecture | open | MathDB 页面 | 原文线索

V2-012151 | MATHDB-317031

The C_π inheritance conjecture for subgroups containing a π-Hall subgroup

摘录 / EXCERPT: C_π inheritance conjecture. Then $A\in C_\pi$.

conjecture | open | MathDB 页面 | 原文线索

V2-012152 | MATHDB-317061

Ginzburg's ideal-cell correspondence conjecture

摘录 / EXCERPT: Let G be a reductive group with nilpotent cone $\mathfrak N$, let ${\mathfrak O}$ be a G-orbit in $\mathfrak N$, and let Ω be the corresponding two-sided Kazhdan-Lusztig...

conjecture | open | MathDB 页面 | 原文线索

V2-012153 | MATHDB-317084

Travkin's bimodule Kazhdan-Lusztig cell conjecture for microlocal cells

摘录 / EXCERPT: Bimodule-microlocal cell conjecture. The bimodule Kazhdan-Lusztig cells coincide with the two-sided microlocal cells.

conjecture | open | MathDB 页面 | 原文线索

V2-012154 | MATHDB-317085

Travkin's character-sheaf formula conjecture for mirabolic Kazhdan-Lusztig basis elements

摘录 / EXCERPT: Mirabolic character-sheaf formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012155 | MATHDB-317086

Travkin's asymptotic mirabolic bimodule conjecture

摘录 / EXCERPT: Asymptotic mirabolic bimodule conjecture. The based bimodule

conjecture | open | MathDB 页面 | 原文线索

V2-012156 | MATHDB-317091

The coherent-sheaf category conjecture for the two-block Springer construction

摘录 / EXCERPT: Let $\mathcal{C}$ be the heart of the t-structure on coherent sheaves on the smooth compactification Z_n, and let A be the corresponding highest-weight algebra. Coherent-shea...

conjecture | open | MathDB 页面 | 原文线索

V2-012157 | MATHDB-317107

The colored RSK multiplicity-free decomposition conjecture for wreath products

摘录 / EXCERPT: Let $\mathbb{Z}_r\wr S_n$ be the wreath product, and let absolute involutions be mapped to shapes by the colored RSK. For each fixed number of cycles, consider the submodule spanne...

conjecture | open | MathDB 页面 | 原文线索

V2-012158 | MATHDB-317110

Conjectural radical presentations of fusion ideals for types B and D

摘录 / EXCERPT: Conjectural radical presentation. For $\mathfrak{g}$ of type B_r with $r\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-012159 | MATHDB-317132

Acyclicity of the wall-crossing submodule of principal indecomposable modules

摘录 / EXCERPT: Acyclicity conjecture. If $w\in W_p$ and $w\cdot\lambda\in X(T)_+$, then $(Q_{w\cdot\lambda})_s$ is acyclic for $\mathop{\strut^{s}\Theta}$.

conjecture | open | MathDB 页面 | 原文线索

V2-012160 | MATHDB-317135

Karras' tame representation-type conjecture for normal surface singularities

摘录 / EXCERPT: Let $(A,\mathfrak{m})$ be a normal surface singularity over the algebraically closed field $k=A/\mathfrak{m}$ of characteristic zero. The tame representation-type conjecture. The r...

conjecture | open | MathDB 页面 | 原文线索

V2-012161 | MATHDB-317136

The finite-discrete-tame-wild trichotomy conjecture for Cohen-Macaulay modules

摘录 / EXCERPT: Let $(A,\mathfrak{m})$ be a local Cohen-Macaulay ring of Krull dimension two. The finite-discrete-tame-wild trichotomy conjecture. The representation type of the category of Cohen-...

conjecture | open | MathDB 页面 | 原文线索

V2-012162 | MATHDB-317137

Closedness conjecture for wild Cohen-Macaulay representation type in families

摘录 / EXCERPT: Let $X\to T$ be a flat family of two-dimensional surface singularities, and define ... The closedness conjecture. The subset B is Zariski-closed in T. Such a closedness result wo...

conjecture | open | MathDB 页面 | 原文线索

V2-012163 | MATHDB-317147

Geck's version of James' conjecture for Iwahori-Hecke algebras

摘录 / EXCERPT: Let W be a finite group with generating set S, let $k=\mathbb{F}_\ell$ be the field of ℓ elements, and let $\theta_\ell^\zeta:R\to k$ be the ring homomorphism sending...

conjecture | open | MathDB 页面 | 原文线索

V2-012164 | MATHDB-317159

Canonical Plancherel factorization conjecture for nonrational conformal blocks

摘录 / EXCERPT: Let ${\mathbb U}$ parametrize the representations under consideration, and let $d\mu_{12}$ be the measure appearing in the factorization of conformal blocks. A representation is ca...

conjecture | open | MathDB 页面 | 原文线索

V2-012165 | MATHDB-317167

Approximation and exact computation conjecture for finite-group link homomorphism invariants

摘录 / EXCERPT: Finite-group link-invariant conjecture. (a) There exists an FPRAS for computing $H_L(G)$ for any finite group G. (b) If G is solvable, then there is a polynomial-time algorithm...

conjecture | open | MathDB 页面 | 原文线索

V2-012166 | MATHDB-317184

Non-finite generation of nonuniform lattices in right-angled hyperbolic buildings

摘录 / EXCERPT: Let X be a right-angled hyperbolic building, let $G=\operatorname{Aut}(X)$, and let Γ be a nonuniform lattice in G. Non-finite generation conjecture. The group Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-012167 | MATHDB-317185

Non-finite presentability of nonuniform lattices in locally finite polyhedral complexes

摘录 / EXCERPT: Let X be a locally finite polyhedral complex, let $G=\operatorname{Aut}(X)$, and let Γ be a nonuniform lattice in G. Non-finite presentability conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-012168 | MATHDB-317205

The Farahat-Higman polynomiality conjecture for Iwahori-Hecke structure constants

摘录 / EXCERPT: Polynomiality conjecture. There exists a polynomial $f_{\lambda \mu}^\nu$ in one variable with coefficients in $\mathbb Q[\xi]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012169 | MATHDB-317206

Generation conjecture for the universal Hecke center by the elements E_r

摘录 / EXCERPT: Generation conjecture. The $\mathbb B[\xi]$-algebra $\mathcal K$ is generated by E_r, $r=1,2,\ldots$.

conjecture | open | MathDB 页面 | 原文线索

V2-012170 | MATHDB-317208

van Dijk's Gelfand-pair conjecture for connected complex symmetric pairs

摘录 / EXCERPT: van Dijk's conjecture. Any connected symmetric pair (G,H) over $\mathbb{C}$ such that G/H is connected is a Gelfand pair.

conjecture | open | MathDB 页面 | 原文线索

V2-012171 | MATHDB-317209

van Dijk's GK-pair conjecture for connected complex symmetric pairs

摘录 / EXCERPT: van Dijk's GK conjecture. If $F=\mathbb{C}$, any connected symmetric pair is a GK pair.

conjecture | open | MathDB 页面 | 原文线索

V2-012172 | MATHDB-317210

The conjecture that every good symmetric pair is a GK pair

摘录 / EXCERPT: Good-pair GK conjecture. Every good symmetric pair is a GK pair.

conjecture | open | MathDB 页面 | 原文线索

V2-012173 | MATHDB-317213

Specht-module cohomology nonvanishing conjecture

摘录 / EXCERPT: Let k be the field of characteristic p, let d be a positive integer, and let $\lambda\vdash d$ be a partition. Write $H^0(d)$ for the relevant induced module and $L(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-012174 | MATHDB-317274

Iyama's derived equivalence conjecture for maximal l-orthogonal modules

摘录 / EXCERPT: Let Λ be an algebra, and let M_1 and M_2 be maximal l-orthogonal modules in $\operatorname{mod}\nolimits\Lambda$, meaning that for a Λ-module M one has...

conjecture | open | MathDB 页面 | 原文线索

V2-012175 | MATHDB-317276

The parabolic Euler-character conjecture for ortho-symplectic Lie superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{spo}(2n|2m+1)$, and let $\mathfrak{p}$ be the maximal parabolic subalgebra obtained by removing the simple root ϵ_m, with Levi subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-012176 | MATHDB-317307

The regular-equation extension conjecture

摘录 / EXCERPT: The regular-equation extension conjecture. Every regular equation in a group G has a solution in some group extending G. This extends the Gerstenhaber-Rothaus theorem from fini...

conjecture | open | MathDB 页面 | 原文线索

V2-012177 | MATHDB-317308

Glebsky-Rivera Martínez's weak soficity conjecture

摘录 / EXCERPT: A weakly sofic group is a subgroup of a metric ultraproduct of an arbitrary family of finite groups equipped with bi-invariant metrics. Glebsky-Rivera Martínez's weak soficity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-012178 | MATHDB-317339

Auslander-Reiten's generalized Nakayama conjecture

摘录 / EXCERPT: Auslander-Reiten's generalized Nakayama conjecture. Any indecomposable injective Λ-module appears as a direct summand in the minimal injective resolution of Λ.

conjecture | open | MathDB 页面 | 原文线索

V2-012179 | MATHDB-317389

Exactness conjecture for the Felder complex after taking coinvariants

摘录 / EXCERPT: Felder exactness conjecture. The Felder complex remains exact after taking coinvariants with respect to $\mathcal{A}(p)[{\mathbf{u}}]^-$; equivalently, the displayed complex is exa...

conjecture | open | MathDB 页面 | 原文线索

V2-012180 | MATHDB-317406

The quantum multiplication conjecture for Laumon spaces

摘录 / EXCERPT: The quantum multiplication conjecture. The isomorphism $\Psi:V_{\ul{d}}\to\fV_{\ul{d}}$ carries the operator $M_{\\D_k}$ of quantum multiplication by $c_1(D_k)$ to the operator...

conjecture | open | MathDB 页面 | 原文线索

V2-012181 | MATHDB-317407

The integral shift-of-argument quotient conjecture for Laumon spaces

摘录 / EXCERPT: Let $\fQ_{\ul{d}}$ be the Laumon space, let $\ul{\\A}{}_{\mu}$ be the integral form of the shift-of-argument subalgebra, and let $\fv_{\ul{d}}$ be the corresponding module vector....

conjecture | open | MathDB 页面 | 原文线索

V2-012182 | MATHDB-317408

The equivariant cohomology module conjecture for Laumon spaces

摘录 / EXCERPT: Let $\bar B:=\,'B/(\hbar=0)=\bigoplus_{\ul{d}}H^\bullet_G(\CQ_{\ul{d}})$, and let $\mathsf U:=(\BC[\fgl_n]\rtimes U(\fgl_n))\otimes\BC[\ft]$ act on it. Let $\fg_{\leq 0}$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-012183 | MATHDB-317409

The geometric Eisenstein-series stalk conjecture

摘录 / EXCERPT: Let sB be the geometric Eisenstein-series constructible complex on Bun, with stalk sB_η at the GL_n-bundle indexed by a dominant weight η. Let $\CN$ be the nilpoten...

conjecture | open | MathDB 页面 | 原文线索

V2-012184 | MATHDB-317411

Polynomiality conjecture for weight multiplicities of Weyl modules

摘录 / EXCERPT: Polynomiality conjecture. For $d>0$, $\dim W^d(\lambda)^{\lambda-\mu}$ is a polynomial in $\lambda_1,\dots,\lambda_{{\rm rk}({\mathfrak g})}$, of degree $\mu^i d$ in the variable...

conjecture | open | MathDB 页面 | 原文线索

V2-012185 | MATHDB-317412

Weyl-module filtration conjecture for truncated polynomial current algebras

摘录 / EXCERPT: Weyl-module filtration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012186 | MATHDB-317438

Minimal smooth group actions on the circle are Lebesgue-ergodic

摘录 / EXCERPT: Let a finitely generated group act smoothly and minimally on the circle, meaning that every orbit is dense. Let Lebesgue measure denote the standard Lebesgue measure on the circle....

conjecture | open | MathDB 页面 | 原文线索

V2-012187 | MATHDB-317439

Hector's conjecture on exceptional minimal sets

摘录 / EXCERPT: Let a finitely generated group of C^2 circle diffeomorphisms admit an exceptional minimal set Λ, meaning a minimal closed invariant subset of the circle that is homeomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-012188 | MATHDB-317466

Trivial-representation kernel conjecture for rational Cherednik modules

摘录 / EXCERPT: Let W be a real reflection group, let $\mathbf{R}[\mathbf{x},\mathbf{y}]$ be graded by degree in $\mathbf{x}$ minus degree in $\mathbf{y}$, and let … be the natural surjection of...

conjecture | open | MathDB 页面 | 原文线索

V2-012189 | MATHDB-317470

Uniqueness of connected promotion operators on tensor products of type A crystals

摘录 / EXCERPT: Let $B:=B(\lambda^{1}) \otimes \cdots \otimes B(\lambda^{\ell})$ be a tensor product of classical highest weight crystals of type A_n with $n\ge 2$. A promotion operator is an af...

conjecture | open | MathDB 页面 | 原文线索

V2-012190 | MATHDB-317493

The quasi-isometry conjecture for finitely generated torsionfree nilpotent groups

摘录 / EXCERPT: Folk conjecture. If G_1 and G_2 are quasi-isometric, then L_1 and L_2 are isomorphic as Lie groups.

conjecture | open | MathDB 页面 | 原文线索

V2-012191 | MATHDB-317501

Kudla-Rallis multiplicity-one conjecture for local zeta integrals

摘录 / EXCERPT: Kudla-Rallis multiplicity-one conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012192 | MATHDB-317524

Ramakrishnan's Schur algebra conjecture for local Langlands correspondence

摘录 / EXCERPT: Let D be a division algebra of index r over a non-archimedean local field k, let π be an irreducible representation of $\operatorname{GL}_m(D)$, and let σ_{π} be...

conjecture | open | MathDB 页面 | 原文线索

V2-012193 | MATHDB-317526

The eventual rotation-quasimorphism formula for scl on a punctured torus

摘录 / EXCERPT: Let F_2 be the free group on generators a,b, and let $w\in [F_2,F_2]$ be arbitrary. Let S be a hyperbolic once-punctured torus with $\pi_1(S)\cong F_2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012194 | MATHDB-317540

Premet's uniqueness conjecture for primitive ideals of W-algebras

摘录 / EXCERPT: Premet's uniqueness conjecture. For any primitive $\mathcal J\in\mathfrak{Id}_{\mathbb O}(\mathcal U)$, the set of primitive ideals...

conjecture | open | MathDB 页面 | 原文线索

V2-012195 | MATHDB-317559

Distinction criterion for generic induced representations of $GL(n,K)$

摘录 / EXCERPT: Distinction conjecture. The representation π is distinguished if and only if, after reordering the Δ_i, there is an integer r between 1 and $t/2$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012196 | MATHDB-317609

The rank inequality for units of the Burnside ring

摘录 / EXCERPT: Let G be a finite group. Let $\Omega(G)$ be the Burnside ring, let $\Omega^*(G)$ denote its unit group, and let $\Omega_2(G)$ be the subring of monomial representations induced f...

conjecture | open | MathDB 页面 | 原文线索

V2-012197 | MATHDB-317613

Černý's conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group and let $\mathcal{Delta}$ be a generating set. The automaton $(G,\mathcal{Delta})$ is the Cayley graph of G with respect to $\mathcal{Delta}$; call G a Černý group if every sync...

conjecture | open | MathDB 页面 | 原文线索

V2-012198 | MATHDB-317614

Černý's conjecture for cyclic groups

摘录 / EXCERPT: Let G be a finite cyclic group. Call G a Černý group if the relevant synchronizing automata containing its Cayley graph admit synchronizing words satisfying the Černý bound. Če...

conjecture | open | MathDB 页面 | 原文线索

V2-012199 | MATHDB-317615

Černý's conjecture for abelian groups

摘录 / EXCERPT: Let G be a finite abelian group. Call G a Černý group if the relevant synchronizing automata containing its Cayley graph admit synchronizing words satisfying the Černý bound. Č...

conjecture | open | MathDB 页面 | 原文线索

V2-012200 | MATHDB-317616

Černý's conjecture for dihedral groups

摘录 / EXCERPT: Let G be a finite dihedral group. Call G a Černý group if the relevant synchronizing automata containing its Cayley graph admit synchronizing words satisfying the Černý bound....

conjecture | open | MathDB 页面 | 原文线索

V2-012201 | MATHDB-317633

Nilpotency conjecture for generators of cyclotomic Khovanov-Lauda algebras at quantum characteristic zero

摘录 / EXCERPT: Let R^{Λ}_{α} be the cyclotomic Khovanov-Lauda algebra, with generators $y_1,\ldots,y_d$, and let ${\Lambda}$ be a dominant weight of level l. Assume that the quantu...

conjecture | open | MathDB 页面 | 原文线索

V2-012202 | MATHDB-317679

Contracting isometry conjecture for groups acting on non-affine buildings

摘录 / EXCERPT: Let (W,S) be an irreducible non-spherical and non-affine Coxeter system with S finite, let X be a building of type (W,S), and let G be a group acting on X by automorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-012203 | MATHDB-317690

The stable Morita-equivalence conjecture for rational representations of GL_2

摘录 / EXCERPT: Let F be the field in which rational representations of $GL_2(F)$ are considered, and let $\underleftarrow{\mathcal{C}}_p(F)\operatorname{-mod}$ denote the corresponding module c...

conjecture | open | MathDB 页面 | 原文线索

V2-012204 | MATHDB-317700

Lower-central-series quotient conjecture for the group G

摘录 / EXCERPT: Let G be the group considered in the paper, let H_i be the associated upper-triangular subgroups, and let $\lambda_i(G)$ and $\gamma_i(G)$ denote the terms of the lower exponen...

conjecture | open | MathDB 页面 | 原文线索

V2-012205 | MATHDB-317761

Converses to the median-space characterizations of Kazhdan and Haagerup groups

摘录 / EXCERPT: Let G be a group acting by isometries on a median space X. Proposition FHFM states that if G is a Kazhdan group, then every action of G on a median space is bounded, while...

conjecture | open | MathDB 页面 | 原文线索

V2-012206 | MATHDB-317767

Existence and uniqueness conjecture for interpolating representations

摘录 / EXCERPT: Let $\g$ be a simple Lie algebra, let P' be the specified set of allowed weights, and let $\mathcal U_{q,t}(\g)$ be the interpolating quantum group. Let $\Pi(\chi(\lambda))$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-012207 | MATHDB-317768

Classical positivity conjecture for Langlands dual characters

摘录 / EXCERPT: Let $\g$ be a simple Lie algebra, let P^+ be the set of dominant integral weights, let P' be the specified sublattice, and let Π be the Langlands duality character map. Cla...

conjecture | open | MathDB 页面 | 原文线索

V2-012208 | MATHDB-317769

Normal-crystal realization conjecture for Langlands duality

摘录 / EXCERPT: Let $\g$ be a simple Lie algebra, let P' be the specified set of weights, and let $\widetilde{\mathcal B}(\lambda)$ be the deformed subset of the crystal associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-012209 | MATHDB-317783

Zimmer's low-dimensional action conjecture

摘录 / EXCERPT: Let $n>2$, let M be a manifold, and let $\rho:SL(n,\mathbb Z)\to\operatorname{Diff}(M)$ be a homomorphism. Zimmer's conjecture. The image of ρ is finite when $\dim(M)<n-1$;...

conjecture | open | MathDB 页面 | 原文线索

V2-012210 | MATHDB-317784

The affine-action classification conjecture

摘录 / EXCERPT: Let $\Gamma<SL(n,\mathbb R)$ be a lattice. An action of Γ on a compact manifold is affine-connection preserving when it preserves both a volume form and an affine connection...

conjecture | open | MathDB 页面 | 原文线索

V2-012211 | MATHDB-317786

The circle-action conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be a higher-rank lattice acting continuously on the circle S^1. The circle-action conjecture. Every such action should be finite. This is part of the low-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-012212 | MATHDB-317787

Zimmer's generalized low-dimensional action conjecture

摘录 / EXCERPT: Let G be a semisimple Lie group with property (T) and let Γ be a lattice in G. Define d as the least dimension of an infinite-image linear representation of Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-012213 | MATHDB-317788

The boundary-of-quotients conjecture for rank-one lattices

摘录 / EXCERPT: Let Γ be a cocompact lattice in $Sp(1,n)$ or F_4^{-20}, and let Γ' be a Gromov-hyperbolic quotient of Γ. Write $\partial\Gamma'$ and $\partial\Gamma$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-012214 | MATHDB-317790

The property-(T) invariant-metric conjecture

摘录 / EXCERPT: Let Γ be a group with property (T) acting smoothly on a compact manifold M and preserving volume. Let $d(\Gamma)$ be the smallest dimension in which Γ has an infi...

conjecture | open | MathDB 页面 | 原文线索

V2-012215 | MATHDB-317791

Gromov's generic-group conjecture for smooth actions

摘录 / EXCERPT: A random-group model is a probabilistic model producing finitely generated groups. Gromov's generic-group conjecture. There should exist a model for random groups in which a generi...

conjecture | open | MathDB 页面 | 原文线索

V2-012216 | MATHDB-317792

The irreducible product-action conjecture

摘录 / EXCERPT: Let $\Gamma=\Gamma_1\times\Gamma_2$, where an action is irreducible when each factor acts ergodically on the underlying measure space. Assume that Γ_1 has property (T) and...

conjecture | open | MathDB 页面 | 原文线索

V2-012217 | MATHDB-317798

The positive-characteristic higher-rank finite-quotient conjecture

摘录 / EXCERPT: Let G be a semisimple algebraic group defined over a field k of positive characteristic, and let $\Gamma<G$ be a lattice. Assume every simple factor of G has k-rank at leas...

conjecture | open | MathDB 页面 | 原文线索

V2-012218 | MATHDB-317820

Conjecture on cohomological realization of regular functions on nilpotent orbit covers

摘录 / EXCERPT: Cohomological realization conjecture. For each $\chi\in\widehat{A({\mathcal O})}$, there is a representation ξ of P_e satisfying $\xi|_{G^e}=\chi$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012219 | MATHDB-317824

Conjecture on graph isomorphisms and noncongruent equispectral volumes

摘录 / EXCERPT: Graph-isomorphism conjecture. Then $V_1\sim V_2$ does not hold unless every g_i is an automorphism of the tile t, when $\mathbf{G}_1$ and $\mathbf{G}_2$ are isomorphic groups.

conjecture | open | MathDB 页面 | 原文线索

V2-012220 | MATHDB-317830

Atiyah's rationality conjecture for von Neumann dimensions

摘录 / EXCERPT: Let G be a group, let m,n be positive integers, and let $A\in M(m\times n;\mathbb C G)$. Right multiplication by A induces a bounded G-operator ... with $x\mapsto xA$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-012221 | MATHDB-317845

The subexponential subgroup growth conjecture for pro-p groups

摘录 / EXCERPT: Subexponential subgroup growth conjecture. If G has subexponential subgroup growth and

conjecture | open | MathDB 页面 | 原文线索

V2-012222 | MATHDB-317868

Conjecture on asymptotically CAT(0) graphs and hyperbolicity

摘录 / EXCERPT: A graph is called $delta$-CAT(0) when its geodesic triangles satisfy the corresponding CAT(0) comparison condition up to an additive constant, and it is asymptotically CAT(0) when...

conjecture | open | MathDB 页面 | 原文线索

V2-012223 | MATHDB-317869

Finiteness conjecture for subdirect products of non-abelian limit groups

摘录 / EXCERPT: Let $L_1,\ldots,L_n$ be non-abelian limit groups with $n\geq 2$, let $D=L_1\times\cdots\times L_n$, and let $H\leq D$ be a subdirect product that intersects each factor non-trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-012224 | MATHDB-317902

The lattice action conjecture for lattices in special linear groups

摘录 / EXCERPT: Let Γ be a lattice in $\mathop{\mathrm{SL}}\nolimits(n,\mathbb{R})$, with $n\geq 3$. Lattice action conjecture. Γ has no faithful action on S^1. This asks whether W...

conjecture | open | MathDB 页面 | 原文线索

V2-012225 | MATHDB-317903

The lattice left-orderability conjecture for lattices in special linear groups

摘录 / EXCERPT: Let Γ be a lattice in $\mathop{\mathrm{SL}}\nolimits(n,\mathbb{R})$, with $n\geq 3$. A group is left orderable if it admits a left-invariant total order. Left-orderability c...

conjecture | open | MathDB 页面 | 原文线索

V2-012226 | MATHDB-317904

The high-rank lattice action conjecture

摘录 / EXCERPT: Let G be a connected semisimple Lie group with finite center and real rank at least 2, and let Γ be an irreducible lattice in G. High-rank lattice action conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-012227 | MATHDB-317905

The non-cocompact rank-three lattice action conjecture

摘录 / EXCERPT: Let Γ be a non-cocompact lattice in either $\mathop{\mathrm{SL}}\nolimits(3,\mathbb{R})$ or $\mathop{\mathrm{SL}}\nolimits(3,\mathbb{C})$. Non-cocompact rank-three conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-012228 | MATHDB-317932

Higher order Borel conjecture

摘录 / EXCERPT: Let G be a Lie group with maximal compact subgroup K, Lie algebra ${\mathfrak g}$, arithmetic subgroup Γ, and let E be a finite-dimensional representation of G. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012229 | MATHDB-317936

The Lefschetz condition conjecture for Braden-MacPherson sheaves

摘录 / EXCERPT: Let $w\in{\widehat{\mathcal W}}$ and suppose that $\operatorname{char} K>N(w)$. For every $x\le w$, let ... be the indicated map between the costalk and stalk of the reduced Braden-M...

conjecture | open | MathDB 页面 | 原文线索

V2-012230 | MATHDB-317938

Closure of surface-subgroup-free right-angled Artin graphs under complete graph amalgamation

摘录 / EXCERPT: Let $A(\Gamma)$ denote the right-angled Artin group associated to a graph Γ, and let $\mathcal{N}$ be the class of graphs such that $A(\Gamma)$ does not contain a closed hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-012231 | MATHDB-317987

Extension of the coefficient formula to all totally nonnegative loop-group elements

摘录 / EXCERPT: Let X be an element of the totally nonnegative part of the loop group $GL_n(\mathbb{R}((t)))$, denoted $GL_n(\mathbb{R}((t)))_{\geq 0}$. Corollary gives a coefficient formula for...

conjecture | open | MathDB 页面 | 原文线索

V2-012232 | MATHDB-318004

The Torelli group commuting-or-free conjecture

摘录 / EXCERPT: Torelli group commuting-or-free conjecture. Either g and h commute, or the subgroup they generate is a free group:

conjecture | open | MathDB 页面 | 原文线索

V2-012233 | MATHDB-318015

Symmetry conjecture for the two-factor fermionic sum

摘录 / EXCERPT: Let $\mathfrak g$ be a Kac-Moody algebra, let β be a positive-root-lattice element, and let $X^{\lambda_1,\lambda_2}_\beta$ be the bilinear fermionic quantity defined from th...

conjecture | open | MathDB 页面 | 原文线索

V2-012234 | MATHDB-318028

Extension and compatibility conjecture for equivariant homology of spectra

摘录 / EXCERPT: Let G be a finite group. Write $\operatorname{\sf StHom}^{naive}(G)$ and $\operatorname{\sf StHom}(G)$ for the categories of naive and genuine G-spectra, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012235 | MATHDB-318043

The Donkin-Koppinen filtration conjecture for the polynomial algebra

摘录 / EXCERPT: Let R_{pol} be the algebra of polynomial invariants, with generators C_r, $\sigma_i(C_{00})$, $\sigma_j(C_{11})$, $\sigma_n(C_{11})$, and $Ber(C)$ as in the preceding construct...

conjecture | open | MathDB 页面 | 原文线索

V2-012236 | MATHDB-318044

The generation conjecture for the supersymmetric polynomial algebra

摘录 / EXCERPT: Let A_s be the algebra of supersymmetric polynomials and let $A_s(p)$ be the subalgebra defined in the paper. Generation conjecture for A_s. The algebra A_s is generated over...

conjecture | open | MathDB 页面 | 原文线索

V2-012237 | MATHDB-318045

Four-conjugate intersection conjecture for solvable Hall subgroups

摘录 / EXCERPT: Four-conjugate intersection conjecture. There exist elements $x,y,z,t\in G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012238 | MATHDB-318046

Feigin-Frenkel's conjecture on restricted Verma module multiplicities

摘录 / EXCERPT: Feigin-Frenkel's conjecture. The restricted linkage principle holds: for $\lambda,\mu\in\Lambda$, $[\overline\Delta(\lambda):L(\mu)]\ne 0$ implies

conjecture | open | MathDB 页面 | 原文线索

V2-012239 | MATHDB-318067

Beauville's existence conjecture for finite nonabelian simple groups

摘录 / EXCERPT: Beauville existence conjecture. Every finite nonabelian simple group occurs except $\mathfrak U_{5}$.

conjecture | open | MathDB 页面 | 原文线索

V2-012240 | MATHDB-318073

Universal Verma module conjecture for the Laumon-space representation of \(\widehat{\mathfrak{gl}}_n\)

摘录 / EXCERPT: Kuznetsov's conjecture. a) The operators a_m, $m\in\mathbb Z$, and the operators $\fe_i,\fh_i,\ff_i$ satisfy the relations of the Lie algebra $\widehat{\mathfrak{gl}}_n$, and equ...

conjecture | open | MathDB 页面 | 原文线索

V2-012241 | MATHDB-318074

Integrable extension conjecture for the \(\widehat{\mathfrak{sl}}_n\)-module \(V(\mu)\)

摘录 / EXCERPT: Integrable extension conjecture. The $\widehat{\mathfrak{sl}}_n$-module $V(\mu)$ extends to the irreducible integrable $\widehat{\mathfrak{gl}}_n$-module with highest weight μ.

conjecture | open | MathDB 页面 | 原文线索

V2-012242 | MATHDB-318080

The GK-pair conjecture for connected complex symmetric pairs

摘录 / EXCERPT: GK-pair conjecture. Any connected symmetric pair over $\mathbb{C}$ is a GK-pair.

conjecture | open | MathDB 页面 | 原文线索

V2-012243 | MATHDB-318081

The GK-pair conjecture for good symmetric pairs

摘录 / EXCERPT: GK-pair conjecture for good pairs. Every good symmetric pair is a GK-pair.

conjecture | open | MathDB 页面 | 原文线索

V2-012244 | MATHDB-318088

The bi-automaticity conjecture for Shephard groups

摘录 / EXCERPT: Shephard groups' bi-automaticity conjecture. Every Shephard group is bi-automatic. The conjecture extends analogous conjectures for Artin and Coxeter groups. The paper proves it fo...

conjecture | open | MathDB 页面 | 原文线索

V2-012245 | MATHDB-318094

Gelfand-Graev interpolating-family conjecture for tensor products of representations of $GL(n,q)$

摘录 / EXCERPT: Gelfand-Graev interpolating-family conjecture. There exists such a family for which

conjecture | open | MathDB 页面 | 原文线索

V2-012246 | MATHDB-318122

Haiman's trivial-representation kernel conjecture

摘录 / EXCERPT: Haiman's kernel conjecture. The kernel of this surjection does not contain a copy of the trivial representation. For $m=1$, the source attributes the conjecture to Haiman; the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-012247 | MATHDB-318136

The center and Azumaya-locus conjecture for infinitesimal Cherednik algebras

摘录 / EXCERPT: Let H_c be the infinitesimal Cherednik algebra associated with $\mathfrak{g}=\mathfrak{gl}_n$ and parameter c over a field of characteristic p, equipped with its standard fil...

conjecture | open | MathDB 页面 | 原文线索

V2-012248 | MATHDB-318164

Conjectural irreducibility criterion for the Lawrence-Krammer representation

摘录 / EXCERPT: Let $G_1,\ldots,G_{n-1}$ be the matrices describing the actions of the BMW generators $g_1,\ldots,g_{n-1}$ on the spanning elements of a vector space W, and define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-012249 | MATHDB-318197

The growth-rate conjecture for Thompson's group F

摘录 / EXCERPT: Growth-rate conjecture. The growth rate is

conjecture | open | MathDB 页面 | 原文线索

V2-012250 | MATHDB-318207

Langlands dual character conjecture for simple quantum affine representations

摘录 / EXCERPT: Langlands dual character conjecture. There exists an irreducible representation V^L of $U_t(\widehat{\mathfrak{g}}^L)$ of highest weight $\Pi(\lambda)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012251 | MATHDB-318243

Generalized Dipper-James-Murphy conjecture for Kleshchev multipartitions

摘录 / EXCERPT: Let $\mathcal{P}_r(n)$ be the set of r-partitions of n, let $\mathcal{K}_r(n)$ be the set of Kleshchev r-partitions with respect to $(e;\mathbf{Q})$, and let an r-partition...

conjecture | open | MathDB 页面 | 原文线索

V2-012252 | MATHDB-318248

Morita super-equivalence conjecture for reduced enveloping superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a restricted Lie superalgebra, let $\chi\in\mathfrak{g}_{\bar{0}}^*$ be a p-character with Jordan decomposition $\chi=\chi_s+\chi_n$, and let ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012253 | MATHDB-318264

The order-2, order-3, and order-7 exponent conjecture for POS-groups

摘录 / EXCERPT: POS-group conjecture. Under these assumptions, $|G|=42$.

conjecture | open | MathDB 页面 | 原文线索

V2-012254 | MATHDB-318289

The Uglov-Takemura conjecture for the affine Gelfand-Tsetlin module

摘录 / EXCERPT: Let $V(\mu)$ be the $\mathbb{C}$-linear span of the vectors $\langle\widetilde{\ul{d}}\rangle$ for $\widetilde{\ul{d}}\in D(\mu)$, equipped with the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-012255 | MATHDB-318291

The finiteness characterization of type $\mathfrak F\mathrm{FP}_\infty$

摘录 / EXCERPT: Let G be a group. Write $\mathfrak F\mathrm{FP}_\infty$ for the finiteness property relative to the family of finite subgroups, and write FP_∞ for the ordinary fi...

conjecture | open | MathDB 页面 | 原文线索

V2-012256 | MATHDB-318294

The classification conjecture for towers of algebras with one-dimensional degree-one part

摘录 / EXCERPT: Classification conjecture. In the weak version, the pair $(K(A),G(A))$, together with its distinguished bases, should be isomorphic to one of the examples in Section 4. In the stro...

conjecture | open | MathDB 页面 | 原文线索

V2-012257 | MATHDB-318295

Uniqueness conjecture for Ginzburg-Rallis models

摘录 / EXCERPT: Uniqueness conjecture for Ginzburg-Rallis models. The Ginzburg-Rallis model on $V_{\mathbb D}$ is unique up to scalar:

conjecture | open | MathDB 页面 | 原文线索

V2-012258 | MATHDB-318366

Talelli's reformulated conjecture on actions on sphere times Euclidean space

摘录 / EXCERPT: Let Γ be a torsion-free group acting smoothly and properly discontinuously on $S^n\times\mathbb{R}^k$. Talelli's reformulated conjecture. If Γ is a torsion-free group...

conjecture | open | MathDB 页面 | 原文线索

V2-012259 | MATHDB-318370

Existence conjecture for very good Sylow twists

摘录 / EXCERPT: Let $(\mathbf G,F)$ be a classical group with regular number d, and let a very good Sylow d-twist mean a Sylow d-twist having the maximal extensibility property described abo...

conjecture | open | MathDB 页面 | 原文线索

V2-012260 | MATHDB-318396

Vardi's conjecture on transversals in cyclic groups

摘录 / EXCERPT: Vardi's conjecture. There exist real constants $0<c_1<c_2<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012261 | MATHDB-318399

Nonnegativity conjecture for the coefficients of the orthogonal Weingarten polynomial

摘录 / EXCERPT: Let $P_n(d)$ be the polynomial defined by the orthogonal Weingarten-function formula in the preceding discussion, for a positive integer n. Coefficient nonnegativity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-012262 | MATHDB-318427

The reducibility criterion for Weyl modules of type A

摘录 / EXCERPT: The reducibility criterion. Let F be a weight element of $\mathcal U^-$ and $\omega\in X^+(T)$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-012263 | MATHDB-318468

Principal-filtration conjecture for short Kostka-Foulkes analogues

摘录 / EXCERPT: Principal-filtration conjecture. If $\mu\in\mathfrak X_{+,H}$ satisfies the vanishing conditions of Theorem, then

conjecture | open | MathDB 页面 | 原文线索

V2-012264 | MATHDB-318474

The key conjecture on quantum group representation rings and DAHA

摘录 / EXCERPT: Key conjecture. The commutative algebra $K_0(Rep_q\,G)$ can be canonically identified with the algebra of W-invariants in this polynomial representation of DAHA. For generic q,...

conjecture | open | MathDB 页面 | 原文线索

V2-012265 | MATHDB-318476

Kerov's nonnegative coefficients conjecture for Kerov polynomials

摘录 / EXCERPT: Kerov's conjecture. The coefficients of the Kerov polynomials K_k are nonnegative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-012266 | MATHDB-318481

Serre's version of the Centre Conjecture for spherical buildings

摘录 / EXCERPT: Let G be a connected group, let $X=X(G,k)$ be its spherical Tits building over k, and identify X with its geometric realization. A subcomplex Y of X is convex if, wheneve...

conjecture | open | MathDB 页面 | 原文线索

V2-012267 | MATHDB-318488

Proficiency conjecture for finite simple groups and their universal covers

摘录 / EXCERPT: Let S be a finite simple group and let $\tilde{S}$ be its universal cover. A finite group is proficient when it has the relevant profinite presentation property discussed in the...

conjecture | open | MathDB 页面 | 原文线索

V2-012268 | MATHDB-318491

Linearity conjecture for the natural extension of the holomorph of a rank-two free group

摘录 / EXCERPT: Let F_2 and F_3 be free groups of ranks 2 and 3, respectively. Let G be the group in the natural extension ... where $\operatorname{Hol}(F_2)$ acts naturally on F_3. Line...

conjecture | open | MathDB 页面 | 原文线索

V2-012269 | MATHDB-318492

Linearity conjecture for split extensions by free groups with homologically trivial action

摘录 / EXCERPT: Let G be a linear group, and let ... be a split exact sequence, where F_n is a finitely generated free group. Split-extension linearity conjecture. If G acts trivially on the h...

conjecture | open | MathDB 页面 | 原文线索

V2-012270 | MATHDB-318493

Linearity conjecture for free-by-cyclic groups

摘录 / EXCERPT: Free-by-cyclic linearity conjecture. The group Γ is linear. This extends the paper’s corollary for extensions of F_2 by $\mathbb Z$ to arbitrary free groups; the source pr...

conjecture | open | MathDB 页面 | 原文线索

V2-012271 | MATHDB-318508

Conjecture on holomorphic polydifferentials for abelian p-extensions

摘录 / EXCERPT: Holomorphic polydifferential module conjecture. The regular representation of G occurs $\Gamma_{p^n-1}(m)+(g_E-1)(2m-1)$ times in the representation of G on $\Omega_F(m)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-012272 | MATHDB-318522

Paracompactness conjecture for orbit spaces of proper group actions

摘录 / EXCERPT: Paracompactness conjecture. If X is a paracompact proper G-space, then the orbit space X/G is paracompact.

conjecture | open | MathDB 页面 | 原文线索

V2-012273 | MATHDB-318523

Invariant-metric conjecture for metrizable proper group actions

摘录 / EXCERPT: Invariant-metric conjecture. The topology of X is metrizable by a G-invariant metric.

conjecture | open | MathDB 页面 | 原文线索

V2-012274 | MATHDB-318546

The symmetric-submodule conjecture for imprimitive reflection groups

摘录 / EXCERPT: Symmetric-submodule conjecture. The module $Sym(r,p,n)^*$ is isomorphic, as a $G(r,p,n)$-module, to the direct sum of all unsplit irreducible representations and exactly one repres...

conjecture | open | MathDB 页面 | 原文线索

V2-012275 | MATHDB-318547

Existence of a non-locally nilpotent Engel group

摘录 / EXCERPT: Let G be an Engel group, meaning that there is an integer n such that ... for all $x,y\in G$. Engel-group conjecture. There exists an Engel group that is not locally nilpotent. T...

conjecture | open | MathDB 页面 | 原文线索

V2-012276 | MATHDB-318548

Converse representation conjecture for finite Dyck-like lattices

摘录 / EXCERPT: Converse representation conjecture. Every finite Dyck-like lattice can be represented as a sublattice of a lattice of Dyck-like paths of suitable length.

conjecture | open | MathDB 页面 | 原文线索

V2-012277 | MATHDB-318549

The subgroup-breadth bound for finite p-groups

摘录 / EXCERPT: Subgroup-breadth bound.

conjecture | open | MathDB 页面 | 原文线索

V2-012278 | MATHDB-318605

The cut-support conjecture for mixed Witt algebra modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra under consideration, graded by $\mathbb{Z}^n$, and let V be a mixed $\mathfrak{g}$-module: there are weights λ and $\lambda+\alpha$...

conjecture | open | MathDB 页面 | 原文线索

V2-012279 | MATHDB-318616

Specht-dimension inequality for regularization and regularization-class minima

摘录 / EXCERPT: Let λ be a partition, and let $\mathcal{S}\lambda$ and $\mathcal{R}\lambda$ denote the two partition operations used in the source, with $S^{\mathcal{S}\lambda}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012280 | MATHDB-318619

The interchange-process comparison inequality

摘录 / EXCERPT: Let $k\geq 2$, let $\mathcal S_k$ be the symmetric group on k letters, and let $g:\mathcal S_k\to\mathbb R$ be any function. For nonnegative numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-012281 | MATHDB-318638

The characterization of plain groups by finite convergent presentations

摘录 / EXCERPT: Plain-group characterization conjecture. The following assertions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-012282 | MATHDB-318677

Convexity conjecture for thick generalized affine buildings

摘录 / EXCERPT: Let X be an affine building modeled over $\mathbb{A}=\mathbb{A}(\mathrm{R},\Lambda,T)$, where T is a proper translation subgroup of Λ^n. Let A be an apartment identif...

conjecture | open | MathDB 页面 | 原文线索

V2-012283 | MATHDB-318683

The Cartan invariant conjecture for higher order peak algebras

摘录 / EXCERPT: Cartan invariant conjecture. The Cartan invariant

conjecture | open | MathDB 页面 | 原文线索

V2-012284 | MATHDB-318692

Universality conjecture for rank-two Soergel morphism formulas

摘录 / EXCERPT: Rank-two universality conjecture. There exist rational functions $\{\lambda_i^{s,r}\}_{i\in I}$ in two variables and compositions $\{g_i^{s,r}\}_{i\in I}$ of the basic morphisms, p...

conjecture | open | MathDB 页面 | 原文线索

V2-012285 | MATHDB-318700

Explicit boundary-value formula for character sheaves on the semistable locus

摘录 / EXCERPT: Let A' be a character sheaf on G^1, let I be the set indexing the relevant standard Levi subgroups, and let $J\subset I$. Write $Z_{J,\delta;1}$ for the corresponding boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-012286 | MATHDB-318702

Givental integral representation for the ground-state eigenfunction of the BC_n Toda chain

摘录 / EXCERPT: The ground-state BC_n Toda conjecture. The ground-state eigenfunction has the integral representation

conjecture | open | MathDB 页面 | 原文线索

V2-012287 | MATHDB-318703

Givental integral representation for generic-eigenvalue eigenfunctions of the BC_n Toda chain

摘录 / EXCERPT: The generic-spectrum BC_n Toda conjecture. The corresponding eigenfunction is

conjecture | open | MathDB 页面 | 原文线索

V2-012288 | MATHDB-318706

Semi-freeness conjecture for absolute Galois groups of fully Hilbertian fields

摘录 / EXCERPT: Let K be a fully Hilbertian field, and let G_K denote its absolute Galois group. A profinite group of infinite rank m is semi-free if every finite split embedding problem for...

conjecture | open | MathDB 页面 | 原文线索

V2-012289 | MATHDB-318863

The weight conjecture for free products of three non-trivial groups

摘录 / EXCERPT: For a group G, its weight is the minimum number of elements whose normal closure is G. A free product is a group of the form $G_1*G_2*G_3$, where each factor is a group. Weight...

open_problem | open | MathDB 页面 | 原文线索

V2-012290 | MATHDB-318940

Uniform boundedness conjecture for global log canonical thresholds

摘录 / EXCERPT: Let $n\geqslant 1$ and let $\bar{G}\subset\operatorname{Aut}(\mathbb{P}^{n})$. Uniform threshold conjecture. There exists a real number $\alpha\in\mathbb{R}$, independent of n an...

conjecture | open | MathDB 页面 | 原文线索

V2-012291 | MATHDB-318960

The group-class conjecture for Reidemeister numbers and generalized fixed points

摘录 / EXCERPT: Let G be a group, let $\phi:G\to G$ be an automorphism, let $R(\phi)$ denote its Reidemeister number, let $C^*(G)$ be the group C^*-algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012292 | MATHDB-318980

The cohomological component conjecture for generalized PRV components

摘录 / EXCERPT: Cohomological component conjecture. $V(\nu')$ is a cohomological component of $V(\lambda')\otimes V(\mu')$ if and only if $V(\nu')$ is a generalized PRV component of...

conjecture | open | MathDB 页面 | 原文线索

V2-012293 | MATHDB-318998

Interlacing-pattern conjecture for discrete series of unitary groups

摘录 / EXCERPT: Interlacing-pattern conjecture. There exists a bijective correspondence between $I(a_.,b_.)$ and discrete series representations (π_1,π_2) of relevant pairs (G,H) with thes...

conjecture | open | MathDB 页面 | 原文线索

V2-012294 | MATHDB-319003

The center conjecture for convex subcomplexes of spherical buildings

摘录 / EXCERPT: Center conjecture. At least one of the following two possibilities holds:

conjecture | open | MathDB 页面 | 原文线索

V2-012295 | MATHDB-319004

Bergeron-Clozel conjecture on infinitesimal characters of Arthur packets

摘录 / EXCERPT: Bergeron and Clozel's conjecture. If θ is an Arthur parameter, then every element of $\Pi(\theta)$ has the infinitesimal character of θ. This predicts that the infini...

conjecture | open | MathDB 页面 | 原文线索

V2-012296 | MATHDB-319016

The BMR dimension and representation conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: Let W be an irreducible complex reflection group and let A be the coefficient ring with fraction field ... Write ${\mathcal{H}}(W,{\mathbf u})$ for its cyclotomic Hecke algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-012297 | MATHDB-319035

Higher level James conjecture for adjustment matrices

摘录 / EXCERPT: Let F be a field of characteristic $p>0$, let ${\Lambda}$ be a dominant weight, and let ${\alpha}$ index a block with defect ${\operatorname{def}}({\alpha})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012298 | MATHDB-319063

Infinite abelianization conjecture for congruence subgroups of \operatorname{Aut}^+(F_2)

摘录 / EXCERPT: Let F_2 be the free group of rank 2, let G be a non-trivial finite group, and let ... be an epimorphism. Write $\Gamma^+(G,\pi)$ for the associated subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-012299 | MATHDB-319065

Strong factorization conjecture for analytic vectors of F-representations

摘录 / EXCERPT: Let G be a real Lie group and let (π,E) be an F-representation, with E^ω denoting the space of analytic vectors and $\Pi(\varphi)$ the operator associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-012300 | MATHDB-319069

Proper 3-realizability conjecture for finitely generated one-relator groups

摘录 / EXCERPT: Let G be a finitely generated one-relator group. Theorem 1.1 establishes proper 3-realizability under the hypothesis that G has finitely many ends. Proper 3-realizability con...

conjecture | open | MathDB 页面 | 原文线索

V2-012301 | MATHDB-319112

Core semisparsity conjecture for quotient representations of polytopes

摘录 / EXCERPT: Core semisparsity conjecture. If N is semisparse in W, then $\operatorname{Core}(W,N)$ is also semisparse.

conjecture | open | MathDB 页面 | 原文线索

V2-012302 | MATHDB-319126

The 3-transitive derangement graph maximal independent set conjecture

摘录 / EXCERPT: Let G be a 3-transitive permutation group of degree n acting on a set Ω, and let Γ_G be its derangement graph. An independent set is a set of vertices containin...

conjecture | open | MathDB 页面 | 原文线索

V2-012303 | MATHDB-319128

Sustained simplicity of the nilpotent socle of simple Whittaker modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra with a fixed triangular decomposition ... Let L be a simple Whittaker module for the Whittaker pair $(\mathfrak{g},\mathfrak{n}_+)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012304 | MATHDB-319130

Generic uniqueness and annihilator conjecture for Whittaker modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be in the setup of the preceding construction, let $\mu\in\mathfrak{h}^*$ be such that the Verma module $M(\mu)$ is simple, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012305 | MATHDB-319145

Schacher's Sylow metacyclic conjecture for rational admissibility

摘录 / EXCERPT: Let G be a finite group. A group is Sylow metacyclic if all of its Sylow subgroups are metacyclic. Schacher's conjecture. The group G is Q-admissible if and only if G is Sy...

conjecture | open | MathDB 页面 | 原文线索

V2-012306 | MATHDB-319149

The no-orbit-shortening conjecture for reduced Hurwitz components

摘录 / EXCERPT: Let $\bar O_k$ be a component at level k, and let ${{\mathcal H}}^{{\text{rd}}}_{\bar O_k}$ denote the corresponding reduced Hurwitz-space component. Orbit shortening is the redu...

conjecture | open | MathDB 页面 | 原文线索

V2-012307 | MATHDB-319158

Universality conjecture for generic-curve Hall algebras

摘录 / EXCERPT: Let X and X' be generic smooth projective curves of the same genus g, and let $\mathcal{K}_X$ and $\mathcal{K}_{X'}$ be their graded Grothendieck algebras over...

conjecture | open | MathDB 页面 | 原文线索

V2-012308 | MATHDB-319164

Kim's tiling conjecture for minimal diskbusting words

摘录 / EXCERPT: Let F be a free group, and let $w\in F$ be a minimal diskbusting word, meaning that w is diskbusting and minimal in the sense used for diskbusting words. Tiling conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-012309 | MATHDB-319165

The virtual polygonality conjecture for diskbusting words

摘录 / EXCERPT: Let F be a free group, and let $w\in F$ be a diskbusting word. Virtual polygonality conjecture. The word w is virtually polygonal, meaning that some finite-index cover or finit...

conjecture | open | MathDB 页面 | 原文线索

V2-012310 | MATHDB-319174

Classical-group Beauville structure conjecture with prescribed hyperbolic types

摘录 / EXCERPT: Classical-group Beauville structure conjecture. If the Lie rank of G is large enough, then G admits an unmixed Beauville structure $(x_1,y_1;x_2,y_2)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012311 | MATHDB-319183

Biautomaticity conjecture for compact nonpositively curved triangle-square complexes

摘录 / EXCERPT: Let K be a compact nonpositively curved triangle-square complex, meaning a complex built from triangles and squares with nonpositive curvature, and let $\pi_1(K)$ denote its fund...

conjecture | open | MathDB 页面 | 原文线索

V2-012312 | MATHDB-319184

Nonexistence of intrinsically aperiodic flats in compact triangle-square complexes

摘录 / EXCERPT: Let K be a compact nonpositively curved triangle-square complex, let $\widetilde K$ be its universal cover, and let F be an intrinsically aperiodic flat. Intrinsically aperiodi...

conjecture | open | MathDB 页面 | 原文线索

V2-012313 | MATHDB-319185

Finiteness conjecture for thoroughly crumpled planes

摘录 / EXCERPT: Let K be a compact nonpositively curved triangle-square complex, and let a thoroughly crumpled plane mean the type of flat plane defined in the source. Thoroughly crumpled planes...

conjecture | open | MathDB 页面 | 原文线索

V2-012314 | MATHDB-319186

Restriction equivalence for non-semisimple blocks of Deligne's categories

摘录 / EXCERPT: Fix $d\in\mathbb{Z}_{\geq 0}$. Let $\mathcal{B}_d$ be the nontrivial block in $\underline{\operatorname{Rep}}(S_d;F)$ containing $L(\varnothing)$, and let ... be the restriction func...

conjecture | open | MathDB 页面 | 原文线索

V2-012315 | MATHDB-319196

The lexicographic free-group conjecture for finitely presented \Lambda-free groups

摘录 / EXCERPT: Lexicographic free-group conjecture. G is $\mathbb{Z}^k$-free for an appropriate $k\in\mathbb{N}$, where $\mathbb{Z}^k$ is ordered lexicographically.

conjecture | open | MathDB 页面 | 原文线索

V2-012316 | MATHDB-319232

Basis conjecture for conformal blocks of tensor products of vector representations

摘录 / EXCERPT: Basis conjecture. These 2^{N-1} polynomials form a basis of the space of conformal blocks $\CB_{\zz}^2(N,N)$ for any choice of pairwise different z_i's.

conjecture | open | MathDB 页面 | 原文线索

V2-012317 | MATHDB-319233

Hrushovski's nilpotent control conjecture for approximate subgroups of special linear groups

摘录 / EXCERPT: Let $A\subseteq \operatorname{SL}_n(\mathbb{C})$ be a K-approximate group, and say that a set B K'-controls A when the source's approximate-group control relation holds. Hr...

conjecture | open | MathDB 页面 | 原文线索

V2-012318 | MATHDB-319256

Conjecture on directions and the hyperbolic boundary of totally disconnected groups

摘录 / EXCERPT: Directions and boundary conjecture. The map that assigns to each element of non-trivial scale its attracting boundary point defines an injection from the set of directions of G i...

conjecture | open | MathDB 页面 | 原文线索

V2-012319 | MATHDB-319304

Weak $(1,1)$ maximal inequality for ergodic free-group actions

摘录 / EXCERPT: Let F be a finitely generated free group, and let $w\mapsto T_w$ be an ergodic action of F on a probability space $(X,\mathscr B,\mu)$. For $n\geq1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-012320 | MATHDB-319322

Riordan-number conjecture for invariant tensors of the three-dimensional representation

摘录 / EXCERPT: Let V be the three-dimensional representation of $\rm SL(2)$, equivalently the vector representation of $\rm SO(3)$, and let $N(\otimes^r V)$ denote its invariant tensors equippe...

conjecture | open | MathDB 页面 | 原文线索

V2-012321 | MATHDB-319339

The Carter-Payne homomorphism conjecture for Specht modules

摘录 / EXCERPT: The Carter-Payne homomorphism conjecture. If $\gamma<e$, then right multiplication by $L_{\lambda\mu}$ induces a non-zero $\mathscr{H}_n$-homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-012322 | MATHDB-319347

Kashiwara's tensor-product conjecture for finite affine crystals

摘录 / EXCERPT: Kashiwara's conjecture. Every good finite-dimensional affine crystal, that is, every good crystal arising from a $U_q(\widehat{sl_n})$-module, is a tensor product of Kirillov-Reshe...

conjecture | open | MathDB 页面 | 原文线索

V2-012323 | MATHDB-319350

The linearisation conjecture for finite group actions on the 4-sphere

摘录 / EXCERPT: Linearisation conjecture. Every smooth finite group action on $\mathbb S^4$ with an isolated fixed point is smoothly conjugate to a linear action.

conjecture | open | MathDB 页面 | 原文线索

V2-012324 | MATHDB-319351

Thurston's quadratic Dehn function conjecture for $SL(n;\mathbb{Z})$

摘录 / EXCERPT: Thurston's conjecture. When $n\geq 4$, $SL(n;\mathbb{Z})$ has a quadratic Dehn function.

conjecture | open | MathDB 页面 | 原文线索

V2-012325 | MATHDB-319397

The realization conjecture for minimal balanced NAH-graphs

摘录 / EXCERPT: Realization conjecture. Every minimal balanced NAH-graph is the minimal NAH-graph for some NAH-manifold. Equivalently, every bisimilarity class of NAH-graphs which contains a balan...

conjecture | open | MathDB 页面 | 原文线索

V2-012326 | MATHDB-319405

Conjecture on expressible elements in cyclic groups

摘录 / EXCERPT: Let A be a cyclic group of order m. For $\boldsymbol{\pi}=(\pi_1,\ldots,\pi_m)\in(\mathbf{Z}/m\mathbf{Z})^m$ with $\pi_1\ge\cdots\ge\pi_m$, an element $x\in A$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-012327 | MATHDB-319408

Schur-functor decomposition conjecture for lower central series quotients

摘录 / EXCERPT: Let $T(V)$ be the tensor algebra of a vector space V, and let $B_k(T(V))$ be its kth lower central series quotient. For a Young diagram λ, let $\mathcal{G}_\lambda$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-012328 | MATHDB-319409

Euler-operator injection conjecture for lower central series quotients

摘录 / EXCERPT: Euler-operator injection conjecture. The operator D is an injection

conjecture | open | MathDB 页面 | 原文线索

V2-012329 | MATHDB-319423

The minuscule-weight criterion for derived equivalence

摘录 / EXCERPT: Let $\la=(\la_1,\dots,\la_\ell)$ be the sequence of weights defining the category $\fQ_\la$, let $\EuScript{Q}_\la$ be the heart of its perverse t-structure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012330 | MATHDB-319444

Mariño's SO/Sp Kauffman-invariant integrality conjecture

摘录 / EXCERPT: Let ${\mathcal K}$ be a knot, let ${\mathcal G}_R({\mathcal K})$ denote its Kauffman invariant in representation R, and let ${\mathcal H}_{(R_1,R_2)}({\mathcal K})$ denote the HO...

conjecture | open | MathDB 页面 | 原文线索

V2-012331 | MATHDB-319447

The spectrum conjecture for finite-group equivariant KK-theory

摘录 / EXCERPT: Spectrum conjecture. For every finite group G, the natural map

conjecture | open | MathDB 页面 | 原文线索

V2-012332 | MATHDB-319461

Lehrer-Shoji's exterior-power conjecture for Springer fibres

摘录 / EXCERPT: Lehrer-Shoji's conjecture. Suppose that e is a regular nilpotent in a Levi subalgebra. Then, for every $i=0,1,\ldots,\ell$,

conjecture | open | MathDB 页面 | 原文线索

V2-012333 | MATHDB-319463

The primitive-character cohomology conjecture for the Lubin-Tate hypersurface

摘录 / EXCERPT: Primitive-character cohomology conjecture. As a virtual module for $\operatorname{Gal}(\overline{\mathbf F}_q/{\mathbf F}_{q^h})$, one has $\dim\mathcal{H}_\psi=(-1)^{h-1}$, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-012334 | MATHDB-319475

Uniform boundedness conjecture for global log canonical thresholds of projective spaces

摘录 / EXCERPT: Uniform boundedness conjecture. There is an $\alpha\in\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012335 | MATHDB-319520

Gersten's area-undistortion conjecture for abelian-by-cyclic groups

摘录 / EXCERPT: Let $M\in GL(k,\mathbb{Z})$ and let $k\geq 3$. Consider the semidirect product $\mathbb{Z}^k\rtimes_M\mathbb{Z}$, where M acts on $\mathbb{Z}^k$. The subgroup $\mathbb{Z}^k$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-012336 | MATHDB-319521

Heisenberg area-distortion conjecture

摘录 / EXCERPT: Let $\mathcal{H}^3$ and $\mathcal{H}^5$ denote the three- and five-dimensional Heisenberg groups. The area distortion of $\mathcal{H}^3$ in $\mathcal{H}^5$ measures the filling are...

conjecture | open | MathDB 页面 | 原文线索

V2-012337 | MATHDB-319522

Eigenvalue model conjecture for volume distortion

摘录 / EXCERPT: Let M be a matrix with at least one eigenvalue off the unit circle, and consider the semidirect product $\mathbb{Z}^k\rtimes_M\mathbb{Z}$. Let N be the diagonal matrix whose di...

conjecture | open | MathDB 页面 | 原文线索

V2-012338 | MATHDB-319523

Higher-dimensional Gersten conjecture for volume undistortion

摘录 / EXCERPT: Let M be a square matrix of size at least m with nonzero determinant, let $k<m$, and write $\Gamma_M=\mathbb{Z}^m\rtimes_M\mathbb{Z}$. The subgroup $\mathbb{Z}^m$ is k-volume...

conjecture | open | MathDB 页面 | 原文线索

V2-012339 | MATHDB-319526

Bahturin-Zaicev codimension-growth inequality for finite group gradings

摘录 / EXCERPT: Bahturin-Zaicev conjecture. One has

conjecture | solved | MathDB 页面 | 原文线索

V2-012340 | MATHDB-319527

Equality of the parking-function and Gelfand-pair polynomials

摘录 / EXCERPT: Equality conjecture. The polynomials $C_n(q)$ and $S_n(q)$ are identical for all $n\geq 1$. This would identify the dimension-generating polynomial arising from the multiplicity-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-012341 | MATHDB-319553

The group-algebra linear-independence conjecture for maximal subgroups

摘录 / EXCERPT: Maximal-subgroup invariant-vector conjecture. There exist vectors $\xi_i\in l(G)$, $1\leq i\leq n$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-012342 | MATHDB-319554

Aschbacher-Guralnick's conjecture on conjugacy classes of maximal subgroups

摘录 / EXCERPT: Aschbacher-Guralnick conjecture. The number of conjugacy classes of maximal subgroups of G is at most the number of conjugacy classes of G.

conjecture | open | MathDB 页面 | 原文线索

V2-012343 | MATHDB-319563

Casselman's first conjecture on Iwahori intertwining operators

摘录 / EXCERPT: Let $\hat{\Phi}$ be a simply-laced root system, let $u\leqslant v$ in its Bruhat order, and let $S(u,v)=\{\alpha\in\hat{\Phi}^+\mid u\leqslant vr_\alpha<v\}$. Let M_v be the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-012344 | MATHDB-319564

Casselman's second conjecture on inverse Iwahori coefficients

摘录 / EXCERPT: Let $\hat{\Phi}$ be a simply-laced root system, let $u\leqslant v$ in its Bruhat order, and let $S'(u,v)=\{\alpha\in\hat{\Phi}^+\mid u\leqslant ur_\alpha<v\}$. Let $\tilde m(u,v)$...

conjecture | open | MathDB 页面 | 原文线索

V2-012345 | MATHDB-319567

Classification conjecture for discrete and tame Cohen-Macaulay representation type

摘录 / EXCERPT: Let k be an algebraically closed field and let A be a reduced complete Cohen-Macaulay non-isolated surface singularity. Representation-type classification conjecture. 1. If A...

conjecture | open | MathDB 页面 | 原文线索

V2-012346 | MATHDB-319570

Amenability from intersecting minimal ideals of the Stone-Čech compactification

摘录 / EXCERPT: Amenability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-012347 | MATHDB-319571

Maximum-attainment conjecture for minimal invariant convex sets

摘录 / EXCERPT: Maximum-attainment conjecture. If this intersection is nonempty, then every function in $\mathcal{C}$ attains its maximum at every point in the intersection.

conjecture | open | MathDB 页面 | 原文线索

V2-012348 | MATHDB-319607

The Chern-class upper-bound conjecture for compact Lie groups

摘录 / EXCERPT: Let G be a compact Lie group. Let $e(G)$ denote the invariant defined in the paper, let $n(G)$ be the minimal dimension over $\mathbb C$ of a faithful complex representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-012349 | MATHDB-319608

The elementary-abelian Chern-class bound

摘录 / EXCERPT: Let C be an elementary abelian p-group of rank c, and let ρ be a faithful n-dimensional complex representation of C. Let $e(\rho)$ be the top degree of the indecompo...

conjecture | open | MathDB 页面 | 原文线索

V2-012350 | MATHDB-319611

The generation and derived-subalgebra conjecture for reachable nilpotent elements

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra, let ${\mathcal N}$ be its nilpotent cone, and let $e\in{\mathcal N}$ be reachable. Write ${\mathfrak g}^e$ for the centralizer of e,...

conjecture | open | MathDB 页面 | 原文线索

V2-012351 | MATHDB-319648

The classification conjecture for degenerate Verma modules over E(5,10)

摘录 / EXCERPT: Let $L=E(5,10)$, let $\mathfrak g_0\simeq \mathfrak{sl}_5(\mathbb C)$ be its degree-zero component, and let $F(n_1,n_2,n_3,n_4)$ be the finite-dimensional irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-012352 | MATHDB-319649

The completeness conjecture for morphisms between degenerate Verma modules over E(5,10)

摘录 / EXCERPT: Let $E(5,10)$ be the exceptional Lie superalgebra, and consider its degenerate (minimal) Verma modules. The source constructs morphisms of degrees 2, 3, and 5 in addition to...

conjecture | open | MathDB 页面 | 原文线索

V2-012353 | MATHDB-319657

The module-theoretic reformulation of Oliver's conjecture

摘录 / EXCERPT: The module-theoretic reformulation. If V is an F-module, then there exists $1\neq g\in\Omega_1(Z(G))$ such that the minimal polynomial of the action of g on V divides

conjecture | open | MathDB 页面 | 原文线索

V2-012354 | MATHDB-319658

The strong form of Oliver's conjecture for the subgroup Y(S)

摘录 / EXCERPT: Strong form of Oliver's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012355 | MATHDB-319659

The module-theoretic reformulation of the strong Oliver conjecture

摘录 / EXCERPT: Module-theoretic strong conjecture. If V is an F-module, then there are quadratic elements in $\Omega_1(Z(G))$.

conjecture | open | MathDB 页面 | 原文线索

V2-012356 | MATHDB-319660

The weakly closed quadratic subgroup conjecture

摘录 / EXCERPT: Weakly closed quadratic subgroup conjecture. If there is an elementary abelian subgroup $1\neq E\leq G$ that is both quadratic on V and weakly closed in $C_G(E)$ with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-012357 | MATHDB-319661

Double-coset characterization conjecture for simple-group actions

摘录 / EXCERPT: Consider the double-coset G-spaces of the form $K\backslash L/\Gamma$, where L is a semisimple Lie group with an irreducible lattice Γ and a compact subgroup K, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-012358 | MATHDB-319725

The Killing-form nondegeneracy conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite simple nonabelian group, and let K be the Killing form for the universal differential calculus on G. Killing-form nondegeneracy conjecture. The form K is...

conjecture | open | MathDB 页面 | 原文线索

V2-012359 | MATHDB-319726

The conjugacy-class Killing-form conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite simple nonabelian group and let ${\mathcal C}\subseteq G\setminus\{e\}$ be a conjugacy class closed under inversion. Let K be the Killing form on the complex...

conjecture | open | MathDB 页面 | 原文线索

V2-012360 | MATHDB-319727

The irreducibility conjecture for non-involution classes in finite simple groups

摘录 / EXCERPT: Let G be a finite nonabelian simple group and let ${\mathcal C}\subseteq G\setminus\{e\}$ be a conjugacy class not consisting of involutions. Let K be the associated Killing-fo...

conjecture | open | MathDB 页面 | 原文线索

V2-012361 | MATHDB-319728

The signature conjecture for conjugacy-class Killing forms

摘录 / EXCERPT: Let G be a finite nonabelian simple group and let ${\mathcal C}\subseteq G\setminus\{e\}$ be a conjugacy class closed under inverses. Let K be the Killing form on the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-012362 | MATHDB-319729

The reducible-class conjecture for Killing forms on symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group with $n>4$, and let K denote the Killing form associated with a conjugacy class. Symmetric-group reducibility conjecture. The conjugacy classes w...

conjecture | open | MathDB 页面 | 原文线索

V2-012363 | MATHDB-319732

Premet's orbit-fiber conjecture for finite-dimensional irreducible W-algebra modules

摘录 / EXCERPT: Premet's conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-012364 | MATHDB-319733

Premet's one-dimensional representation conjecture for W-algebras

摘录 / EXCERPT: Premet's conjecture. Every W-algebra has a one-dimensional representation, equivalently a two-sided ideal of codimension 1.

conjecture | open | MathDB 页面 | 原文线索

V2-012365 | MATHDB-319734

Permutation-character conjecture for equivariant Ehrhart series

摘录 / EXCERPT: Permutation-character conjecture. If $\varphi[t]$ is effective, then $\varphi[1]$ is a permutation representation.

conjecture | open | MathDB 页面 | 原文线索

V2-012366 | MATHDB-319735

Trivial-representation conjecture for equivariant Ehrhart coefficients

摘录 / EXCERPT: Trivial-representation conjecture. If $h_i^*>0$, then the trivial representation occurs with non-zero multiplicity in φ_i.

conjecture | open | MathDB 页面 | 原文线索

V2-012367 | MATHDB-319808

Equivariant mirror symmetry for stringy invariants of dual reflexive hypersurfaces

摘录 / EXCERPT: Suppose that a finite group G acts linearly on a lattice M of rank d via a homomorphism ... Let P and P^* be polar, G-invariant, reflexive polytopes, and let X and...

conjecture | open | MathDB 页面 | 原文线索

V2-012368 | MATHDB-319838

Embedding-type conjecture for simple strata with intersecting simple characters

摘录 / EXCERPT: For $i=1,2$, let $[\Lambda,n,m,\beta_i]$ be a simple stratum in a simple central F-algebra A. Assume that the intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-012369 | MATHDB-319844

The totally decomposable word conjecture

摘录 / EXCERPT: The totally decomposable word conjecture. The following are equivalent: w is totally decomposable; w is uniquely universal; w is universal; and $w(X,A)=B$ has a solution in t...

conjecture | open | MathDB 页面 | 原文线索

V2-012370 | MATHDB-319845

The word-polynomial local-solvability conjecture

摘录 / EXCERPT: The word-polynomial local-solvability conjecture. If w is not totally decomposable, then the equation

conjecture | open | MathDB 页面 | 原文线索

V2-012371 | MATHDB-319923

Ramanujan conjecture for SU(3) bigraphs

摘录 / EXCERPT: Let G be the p-adic group and I an Iwahori subgroup, and let Γ be a cocompact lattice in G. Consider the space of I-invariant vectors ... An irreducible unitary...

conjecture | open | MathDB 页面 | 原文线索

V2-012372 | MATHDB-319930

Irreducibility conjecture for the degree-two nilpotency series of finite simple groups

摘录 / EXCERPT: Irreducibility conjecture. If G is a finite simple group, then $R_2(G,s)$ is irreducible in $\mathcal{R}$. The examples in the paper show irreducibility for particular finite sim...

conjecture | open | MathDB 页面 | 原文线索

V2-012373 | MATHDB-319955

Lehrer-Isaacs conjecture on character-degree polynomials for unitriangular groups

摘录 / EXCERPT: Lehrer-Isaacs conjecture. For fixed n and e, $N_{n,e}(q)$ can be expressed as a polynomial in q with integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-012374 | MATHDB-319965

The finiteness-property conjecture for full subdirect products of curve groups

摘录 / EXCERPT: Let G be a subgroup of a direct product D of smooth curve groups, and let ϕ be the associated map to the product of the quotient groups, as in the surrounding construction...

conjecture | open | MathDB 页面 | 原文线索

V2-012375 | MATHDB-319966

The tensor-power occurrence conjecture for semisimple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra and let $V(\lambda)$ be an irreducible $\mathfrak{g}$-module of dimension r, with highest weight λ. Let $\Pi(\lambda)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-012376 | MATHDB-319980

Generic-variable basis conjecture for acyclic cluster algebras

摘录 / EXCERPT: Let Q be an acyclic quiver. Let $\mathcal A(Q)$ be its acyclic cluster algebra and let $\mathcal G(Q)$ be the set of generic variables associated with Q. Generic-variable basis...

conjecture | open | MathDB 页面 | 原文线索

V2-012377 | MATHDB-320029

Noncrossing conjecture for 3-separating subspaces of Cayley graphs

摘录 / EXCERPT: Let G be a finitely generated group, and let $\mathscr{C}(G)$ be its Cayley graph. A subset $Y\subseteq\mathscr{C}(G)$ is 3-separating when its complement has at least three deep...

conjecture | open | MathDB 页面 | 原文线索

V2-012378 | MATHDB-320057

The vector-valued quantum unique ergodicity conjecture

摘录 / EXCERPT: Let X be the relevant hyperbolic three-manifold, let ρ_m be the representation defining the homogeneous bundle $X\times_K\rho_m$, and let $\{s_n\}$ be a sequence of L^2-no...

conjecture | open | MathDB 页面 | 原文线索

V2-012379 | MATHDB-320061

Extension of the non-uniform lattice Hausdorff-distance lemma

摘录 / EXCERPT: Let $G=SL_n({\mathbb R})$ and let Γ be a discrete Zariski dense subgroup of G that is not cocompact. The lemma holds for any such subgroup: every group at finite Hausdorff...

conjecture | open | MathDB 页面 | 原文线索

V2-012380 | MATHDB-320091

The abelian-by-finite conjecture for small profinite groups

摘录 / EXCERPT: Abelian-by-finite conjecture. Each small profinite group is abelian-by-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-012381 | MATHDB-320092

The three intermediate conjectures for small profinite groups

摘录 / EXCERPT: Intermediate conjectures. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-012382 | MATHDB-320110

The strengthened group-module conjecture

摘录 / EXCERPT: Strengthened group-module conjecture. For maximal subgroups K_i, $1\leq i\leq n$, set

conjecture | open | MathDB 页面 | 原文线索

V2-012383 | MATHDB-320119

Serre's character conjecture for local automorphism groups

摘录 / EXCERPT: Serre's character conjecture. The function a_G is a character of G. The supplied text gives no resolution status.

conjecture | open | MathDB 页面 | 原文线索

V2-012384 | MATHDB-320125

Newelski's abelian-by-finite conjecture for small profinite groups

摘录 / EXCERPT: Let H be a small profinite group, with its ambient action denoted by G when written as (H,G). Newelski's abelian-by-finite conjecture. Every small profinite group is abelian-...

conjecture | open | MathDB 页面 | 原文线索

V2-012385 | MATHDB-320143

Conjecture on the multiplicity of the trivial K-type in Dirac cohomology

摘录 / EXCERPT: Multiplicity conjecture. A unitary representation with infinitesimal character $\rho/2$ has Dirac cohomology consisting of the trivial K-type with multiplicity...

conjecture | open | MathDB 页面 | 原文线索

V2-012386 | MATHDB-320156

The cyclic-group Toeplitz permanent conjecture

摘录 / EXCERPT: Let n be a positive integer, let ${\mathcal C}_n$ be the cyclic group of order n, and for $l\geq n$ let $\hat{\mathcal M}_{{\mathcal C}_n,l}$ be the $l\times l$ Toeplitz matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-012387 | MATHDB-320165

Tikaradze's analogue of the Kac-Weisfeiler conjecture

摘录 / EXCERPT: Tikaradze's analogue of the Kac-Weisfeiler conjecture. Suppose that A is a nonnegatively filtered $\bold{k}$-algebra such that $\operatorname{gr} A$ is a finitely generated commu...

conjecture | open | MathDB 页面 | 原文线索

V2-012388 | MATHDB-320172

The F-A characterization conjecture for finitely generated groups

摘录 / EXCERPT: F-A characterization conjecture. G is F-A if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-012389 | MATHDB-320182

Wakamatsu Tilting Conjecture

摘录 / EXCERPT: Let A be a basic finite dimensional k-algebra and let T be a finitely generated A-module. Call T a Wakamatsu tilting module if $\operatorname{Ext}_A^i(T,T)=0$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-012390 | MATHDB-320194

Quantum folding conjecture for admissible diagram automorphisms

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let σ be an admissible diagram automorphism, and let $\mathfrak{g}_+$, $\mathfrak{n}$, $R(w_\circ)$, $\iota_\mathbf{i}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-012391 | MATHDB-320207

Semisimplicity conjecture for convolution complexes

摘录 / EXCERPT: Let D, V, and V' be as above, and let $K_1\cdot K_2\cdot\ldots\cdot K_m$ be a convolution complex in ... where each K_a is either $\mathscr E^{(n),\Omega}_{\bullet}$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-012392 | MATHDB-320217

K-theoretic deformation conjecture for extended affine Hecke algebras

摘录 / EXCERPT: Let $\mathcal R$ be a root datum with affine Weyl group W^e, let Γ be the finite group acting in the extended affine Hecke algebra, and let q be a positive parameter fun...

conjecture | open | MathDB 页面 | 原文线索

V2-012393 | MATHDB-320219

The decomposable arrangement group embedding conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be a decomposable arrangement, and let $G({\mathcal A})$ denote its arrangement group. Embedding conjecture. The group $G({\mathcal A})$ embeds in a product of f...

conjecture | open | MathDB 页面 | 原文线索

V2-012394 | MATHDB-320232

The signed Hecke-module duality conjecture for quasiparabolic subgroups

摘录 / EXCERPT: Let W be a Coxeter group and let $H\subset W$ be a quasiparabolic subgroup. If (X,hgt) is a scaled W-set, write $X^-=(X,-\operatorname{hgt})$. Let $H_W(q)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-012395 | MATHDB-320234

The injectivity conjecture for strongly typical regular central quotients

摘录 / EXCERPT: Injectivity conjecture. Assume λ is strongly typical and regular. Then $(U/I)^{\mathrm{ad}}$ is a direct sum of injective finite-dimensional modules.

conjecture | open | MathDB 页面 | 原文线索

V2-012396 | MATHDB-320235

The bimodule direct-summand conjecture for strongly typical regular quotients

摘录 / EXCERPT: Bimodule direct-summand conjecture. Assume λ is strongly typical and regular. Then the bimodule U/I is a direct summand of

conjecture | open | MathDB 页面 | 原文线索

V2-012397 | MATHDB-320236

The decomposition conjecture for self-associate strongly typical modules

摘录 / EXCERPT: Decomposition conjecture. Assume λ is strongly typical and regular, and $V\cong\Pi V$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-012398 | MATHDB-320238

Exterior-algebra conjecture for invariant Springer cohomology

摘录 / EXCERPT: Invariant exterior-algebra conjecture. Except when $e=F_4(a_3)$ in type F_4 or $e=E_8(a_7)$ in type E_8, this algebra is an exterior algebra on the subspace

conjecture | open | MathDB 页面 | 原文线索

V2-012399 | MATHDB-320244

Grothendieck-group realization of irreducible representations by categorical modules

摘录 / EXCERPT: Let $\lambda\rightarrow\Lambda^+(n,d)$ be dominant, let $\mathcal V_{\lambda}$ be the categorical module associated with λ, and let $\dot{\mathcal V}_{\lambda}$ be its Karoubi...

conjecture | open | MathDB 页面 | 原文线索

V2-012400 | MATHDB-320252

The Engel-word surjectivity conjecture for finite simple groups

摘录 / EXCERPT: Engel-word surjectivity conjecture. For every $n\in\mathbb N$, the map e_n is surjective for every finite simple non-abelian group G.

conjecture | open | MathDB 页面 | 原文线索

V2-012401 | MATHDB-320254

The non-power word surjectivity conjecture for PSL(2,q)

摘录 / EXCERPT: Non-power word surjectivity conjecture. If w is not a power word, then, for $G=\operatorname{PSL}(2,q)$,

conjecture | open | MathDB 页面 | 原文线索

V2-012402 | MATHDB-320256

Geck-Malle polynomiality and special-piece conjecture

摘录 / EXCERPT: Geck-Malle conjecture. (a) All entries of P and Λ belong to $\mathbb{Z}[u]$, and the polynomials occurring in P have non-negative coefficients. (b) If W is the Weyl g...

conjecture | open | MathDB 页面 | 原文线索

V2-012403 | MATHDB-320308

The generic-degree conjecture for decomposition numbers

摘录 / EXCERPT: Let W be a complex reflection group. Let $\theta:q\mapsto\xi$ be a specialization of ${\mathcal H}_q(W)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012404 | MATHDB-320309

The real Schur-element ratio conjecture for blocks

摘录 / EXCERPT: Let W be a complex reflection group, let $\theta:q\mapsto\xi$ be a specialization of ${\mathcal H}_q(W)$, and let s_χ denote the Schur element associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-012405 | MATHDB-320313

The coarse filling conjecture for S-arithmetic groups

摘录 / EXCERPT: Let K) be a global field, meaning a global function field or a number field. Let G be an absolutely almost simple K-isotropic K-group, let S be a non-empty finite set of...

conjecture | open | MathDB 页面 | 原文线索

V2-012406 | MATHDB-320339

Injectivity conjecture for homomorphisms between global Weyl modules

摘录 / EXCERPT: Let $A=\mathcal R_{k,\ell}$ for some $k,\ell\in\mathbb Z_+$. For $\bos=(s_1,\ldots,s_n)\in\mathbb Z_+^n$, let $W_A(\bos)$ denote the corresponding global Weyl module, and let P^+...

conjecture | open | MathDB 页面 | 原文线索

V2-012407 | MATHDB-320377

Isaacs-Navarro's block conjecture

摘录 / EXCERPT: Let b be a p-block of a finite group G, let P be a defect group of b, and let $H=N_G(P)$. Let $e\in\operatorname{Bl}(H)$ be the Brauer correspondent of b. For an intege...

conjecture | open | MathDB 页面 | 原文线索

V2-012408 | MATHDB-320457

The open amalgams conjecture for completely metrisable groups

摘录 / EXCERPT: Open amalgams conjecture. Every completely metrisable topological group has property (OA).

conjecture | open | MathDB 页面 | 原文线索

V2-012409 | MATHDB-320471

Lower escape-rate conjecture for the Hanoi towers group

摘录 / EXCERPT: Let $\mathcal{H}$ be the Hanoi towers group, equipped with a finite generating set, and let X_n be a simple symmetric random walk on $\mathcal{H}$. Here $|X_n|_{\mathcal{H}}$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-012410 | MATHDB-320476

The piecewise hereditary characterization of orbit categories equal to cluster categories

摘录 / EXCERPT: Piecewise hereditary characterization. The orbit category coincides with the cluster category for Λ if and only if Λ is piecewise hereditary.

conjecture | open | MathDB 页面 | 原文线索

V2-012411 | MATHDB-320550

Affine quantum Schur-Weyl reciprocity over arbitrary fields

摘录 / EXCERPT: Let $\mathbb F$ be any field, and let affine quantum Schur-Weyl reciprocity mean the mutual centralizer property for the affine Hecke and affine quantum Schur algebra actions on th...

conjecture | open | MathDB 页面 | 原文线索

V2-012412 | MATHDB-320563

Hivert-Thiéry conjecture on the Hilbert series of q-harmonics

摘录 / EXCERPT: Let n be a positive integer, let q be a complex number, and let ... where ... and let $\mathcal{H}_{\mathbf{x};q}$ be the space of polynomials annihilated by all $D_{k;q}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-012413 | MATHDB-320593

The L-equivalence conjecture for compact locally symmetric spaces

摘录 / EXCERPT: The L-equivalence conjecture. The lattices Γ_1 and Γ_2 are L-equivalent in G if and only if the compact locally symmetric spaces $\Gamma_1\backslash G/K$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012414 | MATHDB-320626

Asymptotic expansion conjecture for the sequence $s(n)$

摘录 / EXCERPT: Asymptotic expansion conjecture. There exists a sequence of rational polynomials $A_0,A_1,\dots$, with $A_i(x)\in\mathbb Q[x]$ and $\operatorname{deg} A_i=i$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-012415 | MATHDB-320637

The square-root factorial bound for Kronecker multiplicities

摘录 / EXCERPT: Let n be a positive integer, and let $\lambda,\mu,\rho\vdash n$ be partitions of n. The Kronecker multiplicity $\kappa(\lambda,\mu,\rho)$ is the multiplicity of the irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-012416 | MATHDB-320647

Central-character conjecture for dual conjugacy classes

摘录 / EXCERPT: Let $G_{\mathbb C}$ be a complex group with real form $G_{\mathbb R}$, and let $\mathfrak C_{\mathbb C}$ be a unipotent conjugacy class, or a semisimple conjugacy class obtained by...

conjecture | open | MathDB 页面 | 原文线索

V2-012417 | MATHDB-320702

Derived geometrization conjecture for affine Hecke algebras

摘录 / EXCERPT: Let G be the reductive group, let $\bmu$ be a character, and let $\bJ$ and $\cM$ be the corresponding proalgebraic group and character sheaf. Write $\sHg^\der$ for the twisted eq...

conjecture | open | MathDB 页面 | 原文线索

V2-012418 | MATHDB-320703

Parabolic principal-series and Satake conjecture

摘录 / EXCERPT: Let $\bmu$ be a parabolic character, let L be the associated Levi subgroup, and let Π^c, $\sW^c$, and $\sH^c$ be the corresponding principal-series family, induced representa...

conjecture | open | MathDB 页面 | 原文线索

V2-012419 | MATHDB-320704

Parabolic geometric Satake conjecture

摘录 / EXCERPT: Let $\bmu$ be a parabolic character, and let $\bJ^c$ and $\cM^c$ be the associated proalgebraic group and character sheaf. Let $\sHg^c$ be the abelian category of twisted...

conjecture | open | MathDB 页面 | 原文线索

V2-012420 | MATHDB-320705

Parabolic convolution action conjecture

摘录 / EXCERPT: Let $\bmu$ be a parabolic character, and let $\sHg^c$ and $\sWg^c$ be the associated twisted equivariant perverse-sheaf categories. Convolution is the product operation on these ca...

conjecture | open | MathDB 页面 | 原文线索

V2-012421 | MATHDB-320706

Parabolic central functor conjecture

摘录 / EXCERPT: Let $\sHg^c$ and $\sHg$ be the parabolic and original twisted equivariant perverse-sheaf categories, let $\sWg$ be the corresponding module category, and let $\sHg^{c,\der}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012422 | MATHDB-320708

Etingof's positivity conjecture for the finite-dimensional inner product

摘录 / EXCERPT: Etingof's positivity conjecture. The inner product $(,)$ is symmetric and positive definite, and in particular nondegenerate.

conjecture | solved | MathDB 页面 | 原文线索

V2-012423 | MATHDB-320709

Etingof's nondegeneracy conjecture for the q-deformed inner product

摘录 / EXCERPT: Etingof's q-nondegeneracy conjecture. If q is not a root of unity, then the form $(,)_q$ is nondegenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-012424 | MATHDB-320710

Etingof's simplicity conjecture for symplectic reflection algebras

摘录 / EXCERPT: Etingof's simplicity conjecture. The algebra $H_{c,k}(\Gamma_n)$ is simple if and only if (λ,k) belongs to none of the hyperplanes $E_{m,N}$ or $H_{\alpha,m,N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-012425 | MATHDB-320711

Etingof's aspherical-locus conjecture

摘录 / EXCERPT: Etingof's aspherical-locus conjecture. The aspherical locus is the union of the hyperplanes $E_{m,N}$ for $1\le N\le m-1$ and the hyperplanes $H_{\alpha,m,N}$ satisfying the stated...

conjecture | open | MathDB 页面 | 原文线索

V2-012426 | MATHDB-320712

Etingof's filtration conjecture for irrational parameters

摘录 / EXCERPT: Etingof's irrational-parameter filtration conjecture. For either filtration, there exists an isomorphism of vector spaces

conjecture | open | MathDB 页面 | 原文线索

V2-012427 | MATHDB-320713

Etingof's finite-dimensional representation-count conjecture

摘录 / EXCERPT: Etingof's representation-count conjecture. The number of isomorphism classes of finite-dimensional irreducible representations of $H_{c,k}(\Gamma_n)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-012428 | MATHDB-320714

Etingof's refined filtration conjecture for non-integral rational parameters

摘录 / EXCERPT: Etingof's refined filtration conjecture. For either filtration, there exists an isomorphism of vector spaces

conjecture | open | MathDB 页面 | 原文线索

V2-012429 | MATHDB-320715

Etingof's refined finite-dimensional representation-count conjecture

摘录 / EXCERPT: Etingof's refined representation-count conjecture. The number of isomorphism classes of finite-dimensional irreducible representations of $H_{c,k}(\Gamma_n)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-012430 | MATHDB-320735

Hiss-Lübeck-Malle-plus conjecture on the planar embedding of the Brauer tree

摘录 / EXCERPT: Let G be a finite reductive group whose principal ℓ-block has cyclic defect, and let h_0 be the relevant order of the roots of unity occurring in the block. Let q^δ...

conjecture | open | MathDB 页面 | 原文线索

V2-012431 | MATHDB-320736

Broué's geometric extension conjecture for the normalizer of a torus

摘录 / EXCERPT: Let G be a finite reductive group, let ℓ satisfy the preceding assumptions (B1)-(B3), let T_w be the associated torus, let b be the principal ℓ-block of G, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-012432 | MATHDB-320770

Etingof's conjecture on the graded pieces of rational double affine Hecke algebra representations

摘录 / EXCERPT: Let Γ_n be the object whose (i,j)-graded piece is denoted by ${\rm gr}}_{i,j}(\Gamma_n)$. Fix a rational parameter h with denominator $m>1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-012433 | MATHDB-320818

Conjecture on exponential Kirillov functions for unitriangular groups

摘录 / EXCERPT: Exponential Kirillov-function conjecture. The irreducible characters and exponential Kirillov functions of $\mathrm{UT}_n(q)$ coincide if and only if $n\leq 6p$.

conjecture | open | MathDB 页面 | 原文线索

V2-012434 | MATHDB-320823

The extended formula conjecture for degree-refined unitriangular character counts

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field with $q>1$ elements, and let $N_{n,e}(q)$ denote the number of irreducible characters of $\mathrm{UT}_n(q)$ of degree q^e. For a positive int...

conjecture | open | MathDB 页面 | 原文线索

V2-012435 | MATHDB-320835

The Moura-Pereira conjecture on modules associated with minimal affinizations

摘录 / EXCERPT: Let $\lie g$ be a simple Lie algebra, let P^+ denote its dominant integral weights, and let $M(\lambda)$ and $T(\lambda)$ be the graded $\lie g[t]$-modules defined in the paper f...

conjecture | open | MathDB 页面 | 原文线索

V2-012436 | MATHDB-320871

Farahat-Higman-type projection conjecture for finite general linear groups

摘录 / EXCERPT: For a positive integer n, let $\mathscr{B}_{\infty}(\mathbb{F}_q)$ be the filtered projective-limit algebra of split isomorphisms described in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012437 | MATHDB-320882

General block-size braid-group action conjecture

摘录 / EXCERPT: General block-size braid-group action conjecture. The braid-group action for general $m\times m$ matrix blocks should have the same form as and, with

conjecture | open | MathDB 页面 | 原文线索

V2-012438 | MATHDB-320918

Short product representation and expansion conjecture for ideal class groups

摘录 / EXCERPT: Let $G=\operatorname{cl}(\mathcal{O})$ be the ideal class group of an order $\mathcal{O}$ in an imaginary quadratic field, with cardinality h. Let S_k be the sequence of ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-012439 | MATHDB-320968

Olshanskii-Osin-Sapir conjecture on Floyd boundaries and relative hyperbolicity

摘录 / EXCERPT: Olshanskii-Osin-Sapir conjecture. If G has non-trivial Floyd boundary, then G is hyperbolic relative to a collection of proper subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-012440 | MATHDB-320969

Equivalent NRH subgroup conjecture for Floyd boundaries

摘录 / EXCERPT: NRH subgroup conjecture. If G is hyperbolic relative to a collection of NRH proper subgroups, then G acts geometrically finitely on its Floyd boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-012441 | MATHDB-320987

Density criterion for piecewise hereditary algebras

摘录 / EXCERPT: Let Λ be a τ_2-finite algebra of global dimension at most 2. Let $\mathcal{D}^{\rm b}(\Lambda)$ be the bounded derived category of Λ-modules, let...

conjecture | open | MathDB 页面 | 原文线索

V2-012442 | MATHDB-321056

The proposed Ext-algebra presentation for logarithmic model modules

摘录 / EXCERPT: Fix s with $1\,{\leqslant}\, s\,{\leqslant}\, p-1$. Let $\mathrm{Ext}^\bullet_s$ be the associative Yoneda algebra formed from the extension spaces between the specified irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-012443 | MATHDB-321079

Hernandez-Leclerc conjecture on cluster algebras and quantum affine representations

摘录 / EXCERPT: Hernandez-Leclerc conjecture. The assignment

conjecture | open | MathDB 页面 | 原文线索

V2-012444 | MATHDB-321107

Refined topological mirror symmetry conjecture for Hitchin moduli spaces

摘录 / EXCERPT: Let $\Gamma=H^1(C,\mathbb{Z}_n)$ act on the relevant Hitchin moduli space $\mathcal{M}^d$, and let $\hat\Gamma$ be its character group. Write $E_\kappa(\mathcal{M}^d;u,v)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-012445 | MATHDB-321113

Mohar's subgroup-embedding conjecture for multi-ended Cayley graphs

摘录 / EXCERPT: Let $G=Cay(\Gamma,S)$ be a d-regular Cayley graph with more than one end. An embedding here means a topological embedding of graphs. Mohar's subgroup-embedding conjecture. The gr...

conjecture | open | MathDB 页面 | 原文线索

V2-012446 | MATHDB-321115

General k-contractibility conjecture for multi-ended Cayley graphs

摘录 / EXCERPT: Let $G=Cay\langle s_1,\ldots,s_k\mid\mathcal R\rangle$ be a Cayley graph with more than one end. Call a presentation k-contractible if it has the property defined in the paper: i...

conjecture | open | MathDB 页面 | 原文线索

V2-012447 | MATHDB-321118

Accessible Cayley graph tree-amalgamation conjecture

摘录 / EXCERPT: Let G be an accessible Cayley graph. A tree amalgamation is the graph operation used in the paper to assemble graphs along a tree of identifying sets, and a subdivision is obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-012448 | MATHDB-321132

Lehrer-Solomon decomposition conjecture for Whitney homology

摘录 / EXCERPT: Let W be a finite real reflection group acting on V, let $\mathcal{O}_X$ be a W-orbit of intersection subspaces, and let ${{\operatorname{WH}}}_{\mathcal{O}_X}$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-012449 | MATHDB-321133

Eigenvalue formula for the standard representation of symmetrized shuffling operators

摘录 / EXCERPT: Let $\mathfrak{S}_n$ act on its group algebra $\mathbb{R}\mathfrak{S}_n$, and consider the $\chi^{(n-1,1)}$-isotypic component. For $k=1,2,\ldots,n-1$, let $\nu_{(k,1^{n-k})}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-012450 | MATHDB-321148

Ballmann-Buyalo closing lemma for CAT(0) groups

摘录 / EXCERPT: Closing lemma. If $\Lambda G=\partial X$ and

conjecture | open | MathDB 页面 | 原文线索

V2-012451 | MATHDB-321149

Strongly collapsible boundary characterization of rank-one CAT(0) groups

摘录 / EXCERPT: Strong-collapse rank-one conjecture. The boundary $\partial_\infty X$ is strongly collapsible if and only if G has rank one.

conjecture | open | MathDB 页面 | 原文线索

V2-012452 | MATHDB-321150

Incompressible-set covering conjecture for higher-rank CAT(0) groups

摘录 / EXCERPT: Incompressible-covering conjecture. If G has higher rank, then

conjecture | open | MathDB 页面 | 原文线索

V2-012453 | MATHDB-321160

Conjectural characterization of equivariant quantization by invariant differential operators

摘录 / EXCERPT: The equivariant quantization conjecture. The quantization has the following properties: (1) it does not exist or is not unique if and only if, for some $1\leq l\leq k$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-012454 | MATHDB-321164

The conjugacy class size multiplicity conjecture

摘录 / EXCERPT: Let G be a finite group. The multiplicity of a conjugacy class size of G is the number of distinct conjugacy classes having that size. Say that $|G|$ is m-bounded if...

conjecture | open | MathDB 页面 | 原文线索

V2-012455 | MATHDB-321250

The algebraic ideal-generation conjecture for the Scott-Wiegold reduction

摘录 / EXCERPT: Let $\mathbb F$ be any field, and let ... Let $J\lhd A$ be the ideal generated by $\{u,v,1-x^2,1-y^2\}$. Let M be the matrix defined earlier in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012456 | MATHDB-321291

The weak real version of Brauer's conjecture

摘录 / EXCERPT: Weak real version of Brauer's conjecture. The number ${\rm k}_{rv}(B)$ is bounded above by the number of G-real elements of D.

conjecture | open | MathDB 页面 | 原文线索

V2-012457 | MATHDB-321303

The Euler-characteristic geometric Satake equivalence conjecture

摘录 / EXCERPT: Let G be a reductive group, let $G^\vee$ be its Langlands dual group, and let $\mathrm{Gr}(G^\vee)$ denote the affine Grassmannian. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012458 | MATHDB-321347

Infinite self-extensions from a nonzero first self-extension

摘录 / EXCERPT: Let Λ be an artin algebra and let S be a simple Λ-module. Strong no loop conjecture. If ... then $\operatorname{Ext}^i_\Lambda(S,S)\neq 0$ for infinitely many integ...

conjecture | open | MathDB 页面 | 原文线索

V2-012459 | MATHDB-321366

Semisimplicity conjecture for the Cohen-Wales algebra

摘录 / EXCERPT: Semisimplicity conjecture. The CGW algebra is semisimple except possibly if r is a root of unity or if

conjecture | open | MathDB 页面 | 原文线索

V2-012460 | MATHDB-321381

Existence conjecture for the local Langlands correspondence in families

摘录 / EXCERPT: Let E be a finite extension of $\mathbb{Q}_p$, let A be a reduced p-torsion-free complete Noetherian local ring with maximal ideal $\mathfrak m$ and finite residue field k...

conjecture | open | MathDB 页面 | 原文线索

V2-012461 | MATHDB-321382

Interpolation conjecture for the local Langlands correspondence in families

摘录 / EXCERPT: Let S be a finite set of non-archimedean places of E, let A be as above, and for each $v\in S$ let $\rho_v:G_{E_v}\rightarrow\operatorname{GL}_n(A)$ be continuous. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-012462 | MATHDB-321383

Special-fibre surjectivity conjecture for local Langlands families

摘录 / EXCERPT: Let S be a finite set of non-archimedean places of E, let A be a complete Noetherian local ring with residue field k, let $G=\prod_{v\in S}\operatorname{GL}_n(E_v)$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-012463 | MATHDB-321396

The Lehrer-Solomon conjecture for finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group generated by a set S of simple reflections. Let Φ_S and Ψ_S denote the characters of the top components of the group algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-012464 | MATHDB-321397

The inductive Lehrer-Solomon conjecture for parabolic subgroups

摘录 / EXCERPT: Let W be a finite Coxeter group generated by S, let $L\subseteq S$, and let W_L be the corresponding parabolic subgroup. Let $\mathcal{C}_L$ be the set of cuspidal conjugacy...

conjecture | open | MathDB 页面 | 原文线索

V2-012465 | MATHDB-321426

The soluble poly-context-free group conjecture

摘录 / EXCERPT: Soluble poly-context-free group conjecture. Every finitely generated soluble group is poly-context-free if and only if it is virtually abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-012466 | MATHDB-321427

The bounded finite-subgroup conjecture for poly-context-free groups

摘录 / EXCERPT: Bounded finite-subgroup conjecture. If a group G is poly-context-free, then G does not have arbitrarily large finite subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-012467 | MATHDB-321440

The virtually cyclic intermediary class conjecture

摘录 / EXCERPT: Let $\mathfrak{F}$ be the class of finite or virtually cyclic groups. Kropholler's class ${{\scriptstyle\bf H}}\mathfrak{F}$ and the generalized class...

conjecture | open | MathDB 页面 | 原文线索

V2-012468 | MATHDB-321474

The profinite primitivity conjecture for free groups

摘录 / EXCERPT: Let $\mathbf{F}_{k}$ be the free group of rank k, and let $\widehat{\mathbf{F}}_{k}$ be its free profinite completion. An element is primitive if it belongs to a basis of the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-012469 | MATHDB-321489

The virtually symmetric-type bounded multiplicity conjecture for Zuckerman modules

摘录 / EXCERPT: Virtually symmetric-type bounded multiplicity conjecture. If $\mathfrak{q}$ is of virtually symmetric type, then the restriction

conjecture | open | MathDB 页面 | 原文线索

V2-012470 | MATHDB-321490

The equivalence of infinitesimal and unitary discrete decomposability

摘录 / EXCERPT: Discrete decomposability equivalence conjecture. The following two conditions on (G,G',π) are equivalent: (i) the restriction $\pi|_{G'}$ is infinitesimally discretely decompos...

conjecture | open | MathDB 页面 | 原文线索

V2-012471 | MATHDB-321494

Blaubeuren's conjecture on finitely generated images of profinite groups

摘录 / EXCERPT: Blaubeuren's conjecture. A profinite group cannot have an infinite finitely generated image.

conjecture | open | MathDB 页面 | 原文线索

V2-012472 | MATHDB-321517

Universality conjecture for Gaussian fluctuations of random partitions

摘录 / EXCERPT: The parameters ω_n define models of random partitions, and ω_n converges to 0 at speed $1/\sqrt{n}$ as n tends to infinity. The fluctuations of the partitions are d...

conjecture | open | MathDB 页面 | 原文线索

V2-012473 | MATHDB-321518

Finiteness equivalence for Thompson-type groups of a Cantor-algebra

摘录 / EXCERPT: Finiteness equivalence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012474 | MATHDB-321571

Thurston-Waldhausen virtual positive Betti number conjecture

摘录 / EXCERPT: Let $M=\Gamma\setminus G/K$ be an oriented finite-volume hyperbolic n-manifold, where $G=SO(n,1)$ and $K=SO(n)$. Write $\beta_1(M_0)=\dim H_1(M_0,\mathbb{R})$. Thurston-Waldhause...

conjecture | open | MathDB 页面 | 原文线索

V2-012475 | MATHDB-321574

Gaboriau's fixed price conjecture

摘录 / EXCERPT: Let Γ be a countable group acting ergodically and essentially freely, by measure-preserving Borel automorphisms, on a standard Borel probability space X. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012476 | MATHDB-321610

Triviality conjecture for the bounded cohomology of the stable mapping class group

摘录 / EXCERPT: Stable bounded-cohomology conjecture. The bounded cohomology of Γ_{∞} is trivial:

conjecture | open | MathDB 页面 | 原文线索

V2-012477 | MATHDB-321619

Finite-image conjecture for braid representations of simple objects

摘录 / EXCERPT: Let X be a simple object in a braided fusion category, and let (ρ_X,W_n^X) be the associated sequence of braid-group representations on the minimal faithful modules of the t...

conjecture | open | MathDB 页面 | 原文线索

V2-012478 | MATHDB-321621

The extension-graph conjecture for right-angled Artin groups

摘录 / EXCERPT: Extension-graph conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012479 | MATHDB-321651

Euclidean isoperimetric conjecture for higher-rank arithmetic groups

摘录 / EXCERPT: Let ${\bf G}$ be as above and let $k({\bf G},S)$ be the Euclidean rank. For a group of type F_n, an Euclidean m-dimensional isoperimetric inequality means that fillings of cell...

conjecture | open | MathDB 页面 | 原文线索

V2-012480 | MATHDB-321659

Adduced representation formula for unitary representations of $GL(n,\mathbb{R})$

摘录 / EXCERPT: Adduced representation conjecture. The adduced representation of π is the parabolic product of the adduced representations of its basic factors:

conjecture | open | MathDB 页面 | 原文线索

V2-012481 | MATHDB-321678

Dwyer quotient conjecture for the Grigorchuk group

摘录 / EXCERPT: Let ${\mathfrak G}$ be the Grigorchuk group, and let $M_c({\mathfrak G})$ denote its c-th Dwyer quotient. Dwyer quotient conjecture. ... with $m\in{\mathbb N}_0$. The conjecture pr...

conjecture | open | MathDB 页面 | 原文线索

V2-012482 | MATHDB-321679

Dwyer quotient conjecture for the twisted twin of the Grigorchuk group

摘录 / EXCERPT: Let $\bar{\mathfrak G}$ be the twisted twin of the Grigorchuk group, and let $M_c(\bar{\mathfrak G})$ denote its c-th Dwyer quotient. Dwyer quotient conjecture. ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-012483 | MATHDB-321684

Shafarevich's freeness conjecture for the absolute Galois group of the maximal abelian extension

摘录 / EXCERPT: Let $\mathbb{Q}_{\rm ab}$ denote the maximal abelian extension of $\mathbb{Q}$, and let $G_{\mathbb{Q}_{\rm ab}}$ be its absolute Galois group. Shafarevich's conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-012484 | MATHDB-321693

Polynomiality conjecture for tensor-power multiplicities in B_2

摘录 / EXCERPT: Polynomiality conjecture. For each i, the multiplicity function $M^{\omega _i}(\mu,p)$ is a polynomial in p with coefficients in $\mathbf{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-012485 | MATHDB-321694

Equality of upper-boundary multiplicities for the two fundamental B_2 modules

摘录 / EXCERPT: Upper-boundary multiplicity conjecture. The multiplicities of the corresponding upper-diagonal and upper-line weights coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-012486 | MATHDB-321695

Vanishing of the second diagonal in vector tensor powers of B_2

摘录 / EXCERPT: Second-diagonal vanishing conjecture. Its multiplicity is zero:

conjecture | open | MathDB 页面 | 原文线索

V2-012487 | MATHDB-321707

Nontriviality conjecture for higher Prym representations

摘录 / EXCERPT: Let $\Sigma_{g,n}^p$ be an orientable surface of genus g with n boundary components and p punctures. Let $K<\pi_1(\Sigma_{g,n}^p)$ be a finite-index characteristic subgroup,...

conjecture | open | MathDB 页面 | 原文线索

V2-012488 | MATHDB-321712

Puig's conjecture on source algebras of blocks

摘录 / EXCERPT: Puig's conjecture. Given a finite p-group P, there are only finitely many isomorphism classes of interior P-algebras that are source algebras of a block of FG; here G var...

conjecture | open | MathDB 页面 | 原文线索

V2-012489 | MATHDB-321715

Kropholler's conjecture on finite-dimensional models for proper actions

摘录 / EXCERPT: Let $\mathfrak{F}$ be the class of finite groups. A group G belongs to ${\scriptstyle \mathbf H}_{1}\mathfrak{F}$ if it admits a finite-dimensional contractible G-...

conjecture | open | MathDB 页面 | 原文线索

V2-012490 | MATHDB-321732

Allcock's CAT(0) conjecture for finite Coxeter group branched covers

摘录 / EXCERPT: Let W be a finite Coxeter group acting isometrically on $\mathbb{C}^n$, and let Δ be the union of the hyperplanes fixed by the reflections in W. The universal Coxeter br...

conjecture | open | MathDB 页面 | 原文线索

V2-012491 | MATHDB-321773

The weak bounded-type conjecture for geometrically tight Kleinian groups

摘录 / EXCERPT: A geometrically tight Kleinian group is a Kleinian group whose convex hull has the tightness property, and a group is of weakly bounded type when it satisfies the weak bounded type...

conjecture | open | MathDB 页面 | 原文线索

V2-012492 | MATHDB-321774

The tightness criterion for bounded type of Kleinian groups

摘录 / EXCERPT: Let G be a Kleinian group of divergence type, let $\Lambda(G)$ be its limit set, and let $H(\Lambda(G))$ be the convex hull of that limit set. Tightness criterion conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-012493 | MATHDB-321802

Non-formality conjecture for the algebras E_m^n

摘录 / EXCERPT: Let E_m^n denote the Ext algebra considered in the paper, equipped with its induced algebra structure and viewed as a differential graded algebra up to quasi-isomorphism. Non-for...

conjecture | open | MathDB 页面 | 原文线索

V2-012494 | MATHDB-321809

Fayers-Mathas conjecture on irreducible doubly-singular Specht modules in positive characteristic

摘录 / EXCERPT: Fayers-Mathas conjecture. The $\mathcal{H}_n$-module S^λ is irreducible if and only if λ is a $2p$-core and λ or λ' is an FM-partition.

conjecture | open | MathDB 页面 | 原文线索

V2-012495 | MATHDB-321817

Rationality characterization conjecture for conjugacy-class generating functions

摘录 / EXCERPT: Let G be a word-hyperbolic group. For each r, define ... and define its generating function by ... Here $|C|_c$ is the minimum word length among elements of the conjugacy class C...

conjecture | open | MathDB 页面 | 原文线索

V2-012496 | MATHDB-321836

Finite-polynomial parametrisation conjecture for modular character degrees

摘录 / EXCERPT: Let G be a finite group of Lie type with Weyl group ${\mathbf{W}}$ and Frobenius-induced automorphism γ, defined over a field with parameter q, and let k be a field of...

conjecture | open | MathDB 页面 | 原文线索

V2-012497 | MATHDB-321843

Conjectural reduced Hilbert series for low-rank orthogonal groups

摘录 / EXCERPT: Conjectural Hilbert-series formulas. The reduced Hilbert series are

conjecture | open | MathDB 页面 | 原文线索

V2-012498 | MATHDB-321887

The n-$(n+1)$-$(n+2)$ conjecture for fibre products

摘录 / EXCERPT: The n-$(n+1)$-$(n+2)$ conjecture. If N_1 is of type F_n, both Γ_1 and Γ_2 are of type F_{n+1}, and Q is of type...

conjecture | open | MathDB 页面 | 原文线索

V2-012499 | MATHDB-321913

Persistent points on the $h_0=0$ line

摘录 / EXCERPT: Persistence conjecture on the $h_0=0$ line. The points $(0,r_0)$ for $4\le r_0\le \sigma+2$ are persistent points for all $\sigma\ge2$.

conjecture | open | MathDB 页面 | 原文线索

V2-012500 | MATHDB-321914

Triangular gap conjecture for skeletal signatures

摘录 / EXCERPT: Triangular gap conjecture. For $\sigma\ge9$, let E_σ be the line with slope -3 passing through $(1,\sigma-1)$ and let D_σ be the line with slope -4 passing throug...

conjecture | open | MathDB 页面 | 原文线索

V2-012501 | MATHDB-321915

The conjectural pattern on the $h_0=2$ line

摘录 / EXCERPT: Conjecture on the $h_0=2$ line. The point $(2,1)$ is sporadic. The point $(2,\frac{2}{3}\sigma-4)$ is persistently missing for all $\sigma\ge7$. All points $(2,r_0)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-012502 | MATHDB-321916

The conjectural pattern on the $h_0=3$ line

摘录 / EXCERPT: Conjecture on the $h_0=3$ line. The point $(3,1)$ is sporadic. The points $(3,\frac{2}{3}\sigma-7)$ and $(3,\frac{2}{3}\sigma-8)$ are persistently missing for all $\sigma\ge18$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-012503 | MATHDB-321917

Sporadicity of points with second coordinate one

摘录 / EXCERPT: Sporadicity conjecture. For any $h_0\ge2$, the point $(h_0,1)$ is sporadic.

conjecture | open | MathDB 页面 | 原文线索

V2-012504 | MATHDB-321918

Eventual persistence away from the second-coordinate-one line

摘录 / EXCERPT: Eventual persistence conjecture. Any point (h_0,r_0) with $h_0\ge2$ and $r_0\ge2$ is persistent for all $\sigma\ge\sigma_0$ for some σ_0.

conjecture | open | MathDB 页面 | 原文线索

V2-012505 | MATHDB-322012

The local Langlands conjecture for the metaplectic group Mp_{2n}

摘录 / EXCERPT: Let F be a local field with $F\neq\mathbb C$ and $\operatorname{char}(F)\neq 2$, let Mp_{2n} be the metaplectic group, and let ψ be a nontrivial continuous additive charac...

conjecture | open | MathDB 页面 | 原文线索

V2-012506 | MATHDB-322032

The integral-type oper equivalence and exactness conjecture

摘录 / EXCERPT: Let $O\in Op_{\mathrm{int}}$ be an oper of integer type with residue x, and let $e\in {\mathcal N}$ be a nilpotent element such that $d+e\frac{dt}{t}$ is gauge equivalent to the...

conjecture | open | MathDB 页面 | 原文线索

V2-012507 | MATHDB-322041

Conjecture on equality of the induced and original relation subgroups

摘录 / EXCERPT: Let G be an invariantly L-presented group with finite-index subgroup ${\mathcal U}$, and let F, ${\mathcal L}$ and $\tilde{\mathcal L}$ denote the associated free group and r...

conjecture | open | MathDB 页面 | 原文线索

V2-012508 | MATHDB-322042

Finite-index subgroup conjecture for ascending L-presentations

摘录 / EXCERPT: Let G be a group and let ${\mathcal U}$ be a subgroup of finite index in G. An ascending L-presentation is an L-presentation whose substitution set consists of injective en...

conjecture | open | MathDB 页面 | 原文线索

V2-012509 | MATHDB-322043

The normal-rank bound for torsion-free groups

摘录 / EXCERPT: Let G be a torsion free discrete group. Write ${\rm nrk}(G)$ for its normal rank, the least number of elements needed to normally generate G. Normal-rank bound. ... The authors p...

conjecture | open | MathDB 页面 | 原文线索

V2-012510 | MATHDB-322098

The rank-gradient chain-independence conjecture

摘录 / EXCERPT: Let Γ be a finitely generated group, and let (Γ_i) and (Δ_i) be normal chains in Γ with trivial intersection. For a chain (Γ_i), define the rank...

conjecture | open | MathDB 页面 | 原文线索

V2-012511 | MATHDB-322113

Conjecture on graded local Weyl modules and support-independent dimension

摘录 / EXCERPT: Let $\lambda\in P_0^+$, where P_0^+ denotes the set of dominant weights relevant to the twisted current algebra, and let a graded local Weyl module and the associated graded modu...

conjecture | open | MathDB 页面 | 原文线索

V2-012512 | MATHDB-322122

One-stabilization conjecture for generating graphs of finite groups

摘录 / EXCERPT: One-stabilization conjecture. The tuples $(t_1,\ldots,t_n,1)$ and $(s_1,\ldots,s_n,1)$ are connected in $\bar X_{n+1}(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-012513 | MATHDB-322174

The braid-group action conjecture for link Floer chain complexes

摘录 / EXCERPT: Braid-group action conjecture. The two automorphisms $\rho(\sigma_i)$ and $\operatorname{Id}+\Psi_i\Phi_i$ in $\operatorname{Aut}_{N(K(\mathcal{A}_I))}(\mathit{CFL}(Y,L,p))$ are eq...

conjecture | open | MathDB 页面 | 原文线索

V2-012514 | MATHDB-322178

Spherical category O correspondence conjecture for cyclotomic rational Cherednik algebras

摘录 / EXCERPT: Let $p\in\mathfrak{c}_1^{sph}$, and let $\iota_p^L:\textsf{Irrep}(W)\xrightarrow{\sim}\textsf{Irrep}({\bf U}_p^0)$ be the bijection determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-012515 | MATHDB-322215

Babai-Seress quasipolynomial diameter conjecture for transitive permutation groups

摘录 / EXCERPT: Let G be a permutation group acting transitively on a set Ω of degree $n=|\Omega|$. For a generating set A of G, let $\operatorname{diam}(G)$ be the maximum diameter o...

conjecture | open | MathDB 页面 | 原文线索

V2-012516 | MATHDB-322242

Multiplicativity conjecture for the Bernstein-Zelevinsky derivative

摘录 / EXCERPT: Let G_n denote the relevant general linear group, and let $\mathcal{M}_{\infty}^{d_i}(G_{n_i})$ be the class of smooth representations of degree d_i for $i=1,2$. Set ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012517 | MATHDB-322268

The order-determination conjecture for non-commuting graphs

摘录 / EXCERPT: Order-determination conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-012518 | MATHDB-322282

Derived generalized Auslander-Reiten conjecture

摘录 / EXCERPT: Let R be a left Noetherian ring, let ${\sf D}^b(R)$ be the bounded derived category of finitely generated left R-modules, and write $X\perp Y$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-012519 | MATHDB-322284

Dolgachev's essential-dimension and Cremona-dimension inequality

摘录 / EXCERPT: Dolgachev's inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-012520 | MATHDB-322330

Koszulity conjecture for cyclotomic quiver Schur algebras

摘录 / EXCERPT: Let $A^{\underline{\boldsymbol{\nu}}}$ be the cyclotomic quiver Schur algebra associated with the multicharge $\underline{\boldsymbol{\nu}}$. Koszulity conjecture. The cyclotomic q...

conjecture | open | MathDB 页面 | 原文线索

V2-012521 | MATHDB-322335

Primality conjecture for indecomposable rigid dual semicanonical basis elements

摘录 / EXCERPT: Let C be a symmetric generalized Cartan matrix, let $\mathfrak{g}=\mathfrak{n}_-\oplus\mathfrak{h}\oplus\mathfrak{n}$ be its associated Lie algebra, and let ${\mathcal S}^*$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-012522 | MATHDB-322350

Hardness conjecture for immanants of large-depth Young diagrams

摘录 / EXCERPT: Let $\lambda(n)$ be any family of Young diagrams, where the depth of $\lambda(n)$ is the number of rows in the diagram. Assume that, for some constant $\delta>0$, the depth is...

conjecture | open | MathDB 页面 | 原文线索

V2-012523 | MATHDB-322361

Compact lifting conjecture for discriminant-preserving symmetries

摘录 / EXCERPT: Let f be the multigerm under consideration, let $p:\mathcal{A}_f\to\widetilde{\operatorname{Inv}}(D(f))$ be the target-action homomorphism, and let H_0 be a compact linear subg...

conjecture | open | MathDB 页面 | 原文线索

V2-012524 | MATHDB-322390

The largeness conjecture for hyperbolic 3-manifold groups

摘录 / EXCERPT: Let Γ be the fundamental group of a hyperbolic 3-manifold. A group is large if it has a finite-index subgroup that surjects onto a nonabelian free group. Largeness conject...

conjecture | open | MathDB 页面 | 原文线索

V2-012525 | MATHDB-322393

The positive-displacement parabolic conjecture for visibility

摘录 / EXCERPT: Let M be a tame, finite volume, negatively curved manifold, and let $\widetilde{M}$ be its universal cover. A parabolic element is an element of $\pi_1(M)$ acting parabolically o...

conjecture | open | MathDB 页面 | 原文线索

V2-012526 | MATHDB-322410

Lusztig's conjecture on frills and base sets

摘录 / EXCERPT: Let G be a group and let χ be an object for which the associated sets Y^λ and Y^0 are defined. Let the frills of Y^λ denote the additional structure on...

conjecture | open | MathDB 页面 | 原文线索

V2-012527 | MATHDB-322431

The unique-sign conjecture for Kazhdan-Lusztig two-sided cells

摘录 / EXCERPT: Unique-sign conjecture. There is a unique $\lambda_0\in\Lambda$ such that $E^{\lambda_0}\sim_L {\mathfrak{c}}$ and c_{w,λ_0} has the same sign for all $w\in {\mathfrak{c}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-012528 | MATHDB-322451

Non-negativity conjecture for deformed Verlinde expansion coefficients

摘录 / EXCERPT: Non-negativity conjecture. These expansion coefficients are always polynomials with non-negative coefficients. The statement is presented as an observation based on sample computat...

conjecture | open | MathDB 页面 | 原文线索

V2-012529 | MATHDB-322456

The lct criterion for exceptional quotient singularities

摘录 / EXCERPT: The lct criterion. The singularity $\mathbb{C}^{n+1}/G$ is exceptional if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-012530 | MATHDB-322499

Center conjecture for cyclotomic KLR algebra weight spaces

摘录 / EXCERPT: Let $\Lambda=\sum_{i\in I}\lambda_i\Lambda_i\in X^+$ and let $R^\Lambda(\alpha)$ be the summand of the cyclotomic KLR algebra corresponding to $\alpha\in I^m$. Assume that α...

conjecture | open | MathDB 页面 | 原文线索

V2-012531 | MATHDB-322510

Kostant's filtration conjecture for the odd Harish-Chandra projection

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{n}_+\oplus\mathfrak{h}\oplus\mathfrak{n}_-$ be a Lie algebra with triangular decomposition, let $\operatorname{Cl}(\mathfrak{g})$ be its Clifford algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-012532 | MATHDB-322540

Arthur's unitarity conjecture for integral special unipotent representations

摘录 / EXCERPT: Arthur's unitarity conjecture. The representation ρ is unitary.

conjecture | solved | MathDB 页面 | 原文线索

V2-012533 | MATHDB-322551

Bounded-conjugate product-growth conjecture for finite simple groups of Lie type

摘录 / EXCERPT: Let r be a positive integer, let G be a finite simple group of Lie type of rank r, and let S be a generating set of G. For $g\in G$, write $S^g=g^{-1}Sg$, and for a posit...

conjecture | open | MathDB 页面 | 原文线索

V2-012534 | MATHDB-322566

Polynomial q-torus embedding conjecture for $GL_q(N)$

摘录 / EXCERPT: Let a_{mn} be the matrix elements arising in the Gauss-Lusztig decomposition of $GL_q^+(N,\mathbb{R})$, and let U_i,V_i generate copies of the algebra of q-tori. An expressio...

conjecture | open | MathDB 页面 | 原文线索

V2-012535 | MATHDB-322567

Harmonic-analysis decomposition question for positive representations of $U_q(\mathfrak{gl}(N,\mathbb{R}))$

摘录 / EXCERPT: Let $U_{q}(\mathfrak{gl}(N,\mathbb{R}))$ act by left and right regular representations on the space $L^2(GL_q^+(N,\mathbb{R}))$, and let positive principal series representations m...

open_problem | open | MathDB 页面 | 原文线索

V2-012536 | MATHDB-322581

Nonexistence of admissible vectors for monomial representations of unimodular exponential solvable Lie groups

摘录 / EXCERPT: Let G be a unimodular exponential solvable Lie group, and let τ be a monomial representation of G. A vector η is admissible if the associated wavelet transform...

conjecture | open | MathDB 页面 | 原文线索

V2-012537 | MATHDB-322634

Realization conjecture for finite groups as Galois groups of coverings

摘录 / EXCERPT: Let $k(x)$ be the function field of the projective line, and let E be a finite regular Galois extension of $k(x)$ with Galois group G. Equivalently, the inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-012538 | MATHDB-322644

Weak equivalence conjecture for the adjoint action on the Lie algebra

摘录 / EXCERPT: Let G be an adjoint simple algebraic group with Lie algebra $\mathfrak{g}$. Let $L^2(\mathfrak{g})$ carry the representation induced by the adjoint action of G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012539 | MATHDB-322645

The injective splint conjecture for injection fans

摘录 / EXCERPT: Let Δ be a root system with an injective splint ... where $\Delta_{\mathfrak a}$ is a root subsystem corresponding to a regular reductive subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-012540 | MATHDB-322651

Conjecture on bi-shapes occurring in products of predecessor relations

摘录 / EXCERPT: Let $(\gamma|\lambda)$ be a bi-shape occurring in $(S_t)_\mu$. Suppose that there exists a 1-predecessor μ' of μ containing a t-bi-predecessor $(\alpha|\beta)$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-012541 | MATHDB-322667

Conjecture that strongly 2-dependent groups are stable

摘录 / EXCERPT: Stability conjecture for strongly 2-dependent groups. Every strongly 2-dependent group is stable; equivalently, if G is a group such that

conjecture | open | MathDB 页面 | 原文线索

V2-012542 | MATHDB-322672

Realizability conjecture for finite abelian NHDM symmetry groups

摘录 / EXCERPT: Let N be the number of Higgs doublets in an NHDM, and let G be a finite abelian group. A finite group is realizable in NHDM if it can occur as a symmetry group of an NHDM poten...

conjecture | open | MathDB 页面 | 原文线索

V2-012543 | MATHDB-322682

Hirzebruch formula conjecture for signatures of quotient spaces

摘录 / EXCERPT: Hirzebruch formula conjecture. The signature satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-012544 | MATHDB-322692

The F4 partition-function conjecture

摘录 / EXCERPT: Let Φ be a root system of type F_4, let W be its Weyl group, let ℓ be the Weyl-group length function, let ω_2, ω_3, and ω_4 be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-012545 | MATHDB-322697

Rigid-triple conjecture for exceptional finite groups

摘录 / EXCERPT: Rigid-triple conjecture. The triple (C_0,C_1,C_{∞}) is strictly rigid in $\widehat{G}(\mathbb{F}_{\ell})$: the equation

conjecture | open | MathDB 页面 | 原文线索

V2-012546 | MATHDB-322721

The non-Möbius-like universal-circle action conjecture for quasigeodesic flows

摘录 / EXCERPT: The non-Möbius-like action conjecture. The action of $\pi_1(M)$ on the universal circle $\partial \overline{P}$ is not Möbius-like.

conjecture | open | MathDB 页面 | 原文线索

V2-012547 | MATHDB-322741

Bogomolov's cohomologically trivial factorization conjecture

摘录 / EXCERPT: Let $G={\mathfrak G}_{\ell}$ be an ℓ-Sylow subgroup of the absolute Galois group of a function field over an algebraically closed field. Set ... where $G^{(1)}=[G,G]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012548 | MATHDB-322780

Alon's divisibility conjecture for semicharacters of finite groups

摘录 / EXCERPT: Let G be a finite group. A semicharacter of G is a function $f:G\rightarrow\mathbb{C}^{*}$ satisfying $f(ab)=f(a)f(b)$ whenever $a,b\in G$ commute. The semicharacters form an a...

conjecture | open | MathDB 页面 | 原文线索

V2-012549 | MATHDB-322782

Minimal realization conjecture for quantum affine algebras

摘录 / EXCERPT: Let an algebra be generated by $E_{i,r}\1_\lambda$ and $F_{i,r}\1_\lambda$, where $\1_\lambda$ are mutually orthogonal idempotents indexed by $\lambda\in\widehat{X}$, $i\in I$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-012550 | MATHDB-322785

Double affine braid group action conjecture for 2-representations

摘录 / EXCERPT: Let $\widehat{\widehat{\mathfrak{g}}}$ be the double-affine algebra in the sense of the paper, and let $B_{\widehat{\widehat{\mathfrak{g}}}}$ be its double affine braid group. A do...

conjecture | open | MathDB 页面 | 原文线索

V2-012551 | MATHDB-322799

The hyper aperiodic extension conjecture for partial colorings

摘录 / EXCERPT: Hyper aperiodic extension conjecture. Then y can be extended to a hyper aperiodic point in k^G if and only if $\overline{[y_*]} \cap k^G$ consists of aperiodic points.

conjecture | open | MathDB 页面 | 原文线索

V2-012552 | MATHDB-322800

The partial coloring extension conjecture

摘录 / EXCERPT: Partial coloring extension conjecture. Then y can be extended to a k-coloring if and only if $\overline{[y]} \cap k^G$ consists of aperiodic points.

conjecture | open | MathDB 页面 | 原文线索

V2-012553 | MATHDB-322831

Ringel's cluster-additive function conjecture for Dynkin diagrams

摘录 / EXCERPT: Let $\Gamma = \mathbb{Z} \Delta$, where Δ is one of the Dynkin diagrams $\mathbb{A}_n$, $\mathbb{D}_n$, $\mathbb{E}_6$, $\mathbb{E}_7$, and $\mathbb{E}_8$, and let f be cl...

conjecture | open | MathDB 页面 | 原文线索

V2-012554 | MATHDB-322850

Nonnegative plethystic characteristic conjecture for induced supercharacters of unitriangular groups

摘录 / EXCERPT: Let $\mathbb F_q$ be a fixed finite field, let U_n be the group of unipotent upper-triangular matrices over $\mathbb F_q$, let $G_n=GL_n(\mathbb F_q)$, and let χ be a superc...

conjecture | open | MathDB 页面 | 原文线索

V2-012555 | MATHDB-322851

Generic simplicity conjecture for Dirichlet tilings in hyperbolic 3-space

摘录 / EXCERPT: Generic simplicity conjecture. For generic choice of x, the tiling $\mathcal D_x$ is simple, meaning that its dual cell-complex is a simplicial complex.

conjecture | open | MathDB 页面 | 原文线索

V2-012556 | MATHDB-322852

Generic simplicity conjecture for Dirichlet complexes of hyperbolic groups

摘录 / EXCERPT: Generic simplicity conjecture. For generic $x\in\mathbb H^n$, the Dirichlet complex $\mathcal D_x(\Gamma)$ is simple.

conjecture | open | MathDB 页面 | 原文线索

V2-012557 | MATHDB-322853

Simplicity conjecture for cyclic loxodromic Dirichlet tilings

摘录 / EXCERPT: Cyclic loxodromic simplicity conjecture. The same conclusion holds for all cyclic loxodromic subgroups of $PO(n,1)$ with $n\geq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-012558 | MATHDB-322862

Kegel's derived-length conjecture for products of nilpotent groups

摘录 / EXCERPT: Let $a,b\in\mathbf{N}$. Kegel's conjecture. There is a function ... such that every finite group G that is a product of two nilpotent groups of respective nilpotency classes a an...

conjecture | open | MathDB 页面 | 原文线索

V2-012559 | MATHDB-322864

Classification conjecture for simple modules over Green biset functors

摘录 / EXCERPT: Let A be a Green biset functor. For a group H, let $I_A(H)$ be the submodule of $A(H\times H)$ generated by composites $a\circ b$, where $a\in A(H\times K)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-012560 | MATHDB-322898

Conjecture on Auslander-Reiten components outside the finitely presented representations

摘录 / EXCERPT: The authors' conjecture. The connected components of $\text{$mathitGamma$}_{{\rm rep}(Q)}$ containing representations not in $\overline{\mystrut(5,0){\rm rep\hspace{-1pt}}}\,(Q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-012561 | MATHDB-322947

Hrushovski's pseudofinite fixed-point conjecture for generic automorphisms

摘录 / EXCERPT: Hrushovski's fixed-point conjecture. The group of fixed points of α is pseudofinite, that is, an infinite model of the theory of finite groups.

conjecture | open | MathDB 页面 | 原文线索

V2-012562 | MATHDB-322948

Fixed-point implication for the Algebraicity Conjecture

摘录 / EXCERPT: Fixed-point implication conjecture. If the group of fixed points of α is pseudofinite, then G is isomorphic to a Chevalley group over an algebraically closed field.

conjecture | open | MathDB 页面 | 原文线索

V2-012563 | MATHDB-322960

Defect-zero block monotonicity conjecture for alternating groups

摘录 / EXCERPT: Defect-zero block monotonicity conjecture. The number of defect zero p-blocks of A_n is strictly less than the number of defect zero q-blocks of A_n. This conjecture is mot...

conjecture | open | MathDB 页面 | 原文线索

V2-012564 | MATHDB-323078

The norm inequality for dimensions of projective indecomposable modules

摘录 / EXCERPT: The norm inequality. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-012565 | MATHDB-323079

The Brauer character degree-square inequality and normal Sylow subgroup criterion

摘录 / EXCERPT: The Brauer degree-square conjecture. One always has

conjecture | open | MathDB 页面 | 原文线索

V2-012566 | MATHDB-323093

Generator-and-relation conjecture for the modules $M_\bm$

摘录 / EXCERPT: Generator-and-relation conjecture. The following $\g^a$-module $M_\bm$ is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-012567 | MATHDB-323094

Dyck-path monomial basis conjecture for the modules $M_\bm$

摘录 / EXCERPT: Monomial basis conjecture. The elements

conjecture | open | MathDB 页面 | 原文线索

V2-012568 | MATHDB-323098

The Gap Conjecture for finitely generated group growth

摘录 / EXCERPT: Let G be a finitely generated group with growth function $\gamma_G(n)$. Write $f\prec g$ when f is strictly bounded above by g in the relevant growth comparison. Gap Conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-012569 | MATHDB-323101

Grigorchuk's conjecture $G^*(\beta)$ for non-virtually-nilpotent groups

摘录 / EXCERPT: Let G be a finitely generated group with growth function $\gamma_G(n)$, and let β be the parameter used in the stronger form $G^*(\beta)$ of the parameterized Gap Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-012570 | MATHDB-323104

Non-integral orbit-stabilizer conjecture for finite-dimensional W-algebra modules

摘录 / EXCERPT: Let $\mathbb{O}$ be a special nilpotent orbit, let λ be a dominant representative of a regular central character, and let W_λ be its integral Weyl group. Let W_I...

conjecture | open | MathDB 页面 | 原文线索

V2-012571 | MATHDB-323160

Fixed-time marginal identity for the discrete-time particle system

摘录 / EXCERPT: Let $\boldsymbol{X}(n)$ be the particle configuration at time n, let $\boldsymbol{0}$ denote the densely packed initial configuration, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012572 | MATHDB-323192

Existence of completely hyperbolic nondegenerate actions on closed manifolds

摘录 / EXCERPT: Existence conjecture. Every closed smooth n-dimensional manifold admits a completely hyperbolic nondegenerate action of $\mathbb{R}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-012573 | MATHDB-323236

The fiber cardinality conjecture for quantum Gelfand-Tsetlin modules

摘录 / EXCERPT: Let $U_q(\mathfrak{gl}_N)$ be the quantum enveloping algebra and let Γ_q be its Gelfand-Tsetlin subalgebra. For any $\mathbf{m}\in \operatorname{Specm}\Gamma_q$, the fiber o...

conjecture | open | MathDB 页面 | 原文线索

V2-012574 | MATHDB-323276

Isomorphism conjecture for the Links-Gould quotient algebras

摘录 / EXCERPT: Let H_n be the generic cubic Hecke algebra, and define ... Let LG_n be the Links-Gould centralizer algebra. There are natural quotient maps $A_n\twoheadrightarrow LG_n$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-012575 | MATHDB-323279

Vershik-Peres speed exponent conjecture for random walks on finitely generated groups

摘录 / EXCERPT: Vershik-Peres conjecture. For every $\beta\in (1/2,3/4)$, there exists a random walk on a finitely generated group such that

conjecture | open | MathDB 页面 | 原文线索

V2-012576 | MATHDB-323318

UGA VIGRE conjecture on the complexity of Specht modules

摘录 / EXCERPT: Let S^λ be a Specht module in a block B of weight w. Say that a partition is $p\times p$ if both it and its conjugate are of the form $p\tau$, equivalently, if its Youn...

conjecture | open | MathDB 页面 | 原文线索

V2-012577 | MATHDB-323319

Stability conjecture for Ext between twisted two-part simple modules

摘录 / EXCERPT: Let k be a field of characteristic $p>2$, let $\lambda,\mu\vdash d$ be partitions, and write $D^{p\lambda}$ and $D^{p\mu}$ for the corresponding simple modules of symmetric group...

conjecture | open | MathDB 页面 | 原文线索

V2-012578 | MATHDB-323320

Mullineux twist-stability conjecture

摘录 / EXCERPT: Let λ be a partition and let m denote the Mullineux bijection on p-regular partitions. Mullineux twist-stability conjecture. If ... then ... The conjecture asserts that a r...

conjecture | open | MathDB 页面 | 原文线索

V2-012579 | MATHDB-323321

Eventual Mullineux twist-stability conjecture

摘录 / EXCERPT: Let $\lambda\vdash d$, let m be the Mullineux bijection, and let τ denote a partition. Eventual Mullineux stability conjecture. There exist integers $1\leq a<b$ such that ……

conjecture | open | MathDB 页面 | 原文线索

V2-012580 | MATHDB-323326

Existence of essentially pure diagrams for supersingular modules

摘录 / EXCERPT: Let $M_{\chi,\mathbf{J}}$ be a supersingular module, and let X be the diagram's underlying structure with associated homology modules $H_0(X,\mathcal{D}_{\chi,\mathbf{J}})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012581 | MATHDB-323333

Hiß-Lübeck-Malle conjecture on the shape and embedding of Brauer trees

摘录 / EXCERPT: Let Γ be the Brauer tree of the principal Φ_h-block. The characters χ_{m_ζ} arise from the Frobenius eigenspaces in the rational ℓ-adic cohomology, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-012582 | MATHDB-323338

The near-boundary formula for the equivariant Kirkman generating function

摘录 / EXCERPT: Let W be a real reflection group acting irreducibly on $V=\mathbb{C}^n$, with exponents $(e_1,\ldots,e_n)$ indexed so that the Coxeter number is $h=e_n+1$. Define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012583 | MATHDB-323356

Character conjecture for least affinizations of generic parabolic Verma modules

摘录 / EXCERPT: Let ${\mathfrak g}$ be of type ABCF or $\mathrm G$, with index set I, positive roots Δ^+, and fundamental weights ω_j. Let $J\subseteq I$ be a subdiagram...

conjecture | open | MathDB 页面 | 原文线索

V2-012584 | MATHDB-323357

Character conjecture for least affinizations with separated poles and zeros

摘录 / EXCERPT: Separated-poles-and-zeros character conjecture. The character of $L(\boldsymbol f)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-012585 | MATHDB-323362

The Schur-index-one conjecture for umbral groups

摘录 / EXCERPT: Let $G^{(\ell)}$ be an umbral group, and let ϱ be an irreducible representation of $G^{(\ell)}$ of type n. The Schur index of ϱ is the least positive integer s...

conjecture | open | MathDB 页面 | 原文线索

V2-012586 | MATHDB-323364

The umbral module doublet conjecture

摘录 / EXCERPT: Let $\Lambda=\{2,3,4,5,7,13\}$ and let $K^{(\ell)}_{r,-D/4\ell}$ be the corresponding representation of $G^{(\ell)}$. Say that a G-module is a doublet if it is isomorphic to the...

conjecture | open | MathDB 页面 | 原文线索

V2-012587 | MATHDB-323369

Howe duality conjecture for local theta correspondence

摘录 / EXCERPT: Howe duality conjecture. The set $\mathcal R_{\omega_{U,V}}(U,V)$ is the graph of a bijection between $\mathcal R_{\omega_{U,V}}(U)$ and $\mathcal R_{\omega_{U,V}}(V)$.

conjecture | open | MathDB 页面 | 原文线索

V2-012588 | MATHDB-323407

The complement conjecture for (cf-)ISLE classes

摘录 / EXCERPT: Let ${\mathscr{G}}$ be a class of finite groups, and let ${\mathscr{G}}^c$ denote its complement among the finite groups. A class is (cf-)ISLE if it has the (core-free) ISLE closur...

conjecture | open | MathDB 页面 | 原文线索

V2-012589 | MATHDB-323425

Classification conjecture for irreducible quasi-finite lowest-weight quantum toroidal modules

摘录 / EXCERPT: Let $\mathcal E_n$ denote the quantum toroidal algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012590 | MATHDB-323441

Character-eigenspace vanishing conjecture for finite group actions on surfaces

摘录 / EXCERPT: Let G be a finite group acting on a smooth projective complex surface S with $q=0$. Let $\chi:G\rightarrow\{1,-1\}$ be a character. For a G-representation V, write V^χ...

conjecture | open | MathDB 页面 | 原文线索

V2-012591 | MATHDB-323454

Shalen's rank-drop conjecture for finite covers of hyperbolic 3-manifolds

摘录 / EXCERPT: Let N be a compact, orientable hyperbolic 3-manifold and let $\widetilde N$ be any finite cover of N. Write $\operatorname{rk}(\pi)$ for the minimal number of generators of a...

conjecture | open | MathDB 页面 | 原文线索

V2-012592 | MATHDB-323470

Nonexistence of sink and source morphisms for regular modules

摘录 / EXCERPT: Let $n\ge2$, let Λ be in the setting of the paper, and let $\mathscr{R}$ denote the category of n-regular modules inside $\mathscr{C}^{0}$. A sink morphism is a right alm...

conjecture | open | MathDB 页面 | 原文线索

V2-012593 | MATHDB-323471

Nonvanishing conjecture for regular modules

摘录 / EXCERPT: Let Λ be an n-representation infinite algebra, and let $\mathscr{R}$ denote its category of n-regular modules. Nonvanishing conjecture. The category $\mathscr{R}$ is no...

conjecture | open | MathDB 页面 | 原文线索

V2-012594 | MATHDB-323472

Representation dimension conjecture for n-hereditary algebras

摘录 / EXCERPT: Let Λ be an n-hereditary algebra. Its representation dimension $\operatorname{repdim}\Lambda$ measures the homological complexity of Λ through the global dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-012595 | MATHDB-323473

Dimension conjectures for n-regular, module and derived categories

摘录 / EXCERPT: Let Λ be an n-representation infinite algebra. Write $\mathscr{R}$ for its category of n-regular modules, and let $\mathsf{D}^{\rm b}(\operatorname{mod}\Lambda)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-012596 | MATHDB-323512

The deviation conjectures for finite groups and subgroups

摘录 / EXCERPT: Deviation conjectures. With $\Delta=\Delta(G,H)$, one has $\Delta=1$ if and only if $H=G$, the index $[G:H]$ divides Δ, and Δ divides $|G|$. These conjectures propose...

conjecture | open | MathDB 页面 | 原文线索

V2-012597 | MATHDB-323545

Bux-Wortman conjecture on distortion dimension of S-arithmetic subgroups

摘录 / EXCERPT: Let $\mathbf G$ be a reductive group defined over a global field $\mathbb K$, let S be a finite set of valuations, and let ... be endowed with a left-invariant metric. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012598 | MATHDB-323548

The intrinsic description conjecture for the Fock space on nilpotent orbits

摘录 / EXCERPT: Let $\mathcal{X}_\lambda$ be the nilpotent orbit associated with the parameter λ, let $\mathcal{O}(\mathcal{X}_\lambda)$ denote its holomorphic functions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012599 | MATHDB-323559

The E-DES permutation group conjecture

摘录 / EXCERPT: Let the E-DES permutations act on the set of size 3^{18}, and let the group they generate be the subgroup of the symmetric group on this set generated by these permutations. E-DE...

conjecture | open | MathDB 页面 | 原文线索

V2-012600 | MATHDB-323560

Pawale's Noether number conjecture for non-abelian semidirect products

摘录 / EXCERPT: Let p and q be odd primes with q dividing $p-1$, and let $Z_p\rtimes Z_q$ denote the non-abelian semidirect product of cyclic groups of orders p and q. For a finite group...

conjecture | open | MathDB 页面 | 原文线索

V2-012601 | MATHDB-323568

The polar-action conjecture for nonnegatively curved compact manifolds

摘录 / EXCERPT: A polar action is an isometric group action admitting a section, a submanifold meeting every orbit orthogonally. Let a compact simply connected manifold have nonnegative sectional...

conjecture | open | MathDB 页面 | 原文线索

V2-012602 | MATHDB-323570

Isotypeness determines finitely generated free groups up to isomorphism

摘录 / EXCERPT: Let F_n be a free group of rank $n>1$, and let H be a group. Two groups are isotypic if they have the same types of tuples with respect to the relevant language. Free-group iso...

conjecture | open | MathDB 页面 | 原文线索

V2-012603 | MATHDB-323577

Matumoto's elementary homomorphism conjecture for scalar generalized Verma modules

摘录 / EXCERPT: Matumoto's conjecture. An arbitrary nontrivial homomorphism between scalar generalized Verma modules is a composition of elementary homomorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-012604 | MATHDB-323640

The converse characterization of cluster-roots for 2-representation-finite algebras

摘录 / EXCERPT: Let Λ be a 2-representation-finite algebra, and let Φ be its Coxeter transformation. A root is Φ-positive if ... for every $m\in\mathbb{Z}$. The cluster-root conj...

conjecture | open | MathDB 页面 | 原文线索

V2-012605 | MATHDB-323669

The first key lemma for the lattice model of the Weil representation

摘录 / EXCERPT: Let $L\subseteq L_1$ be a sublattice, let S_L be the associated subspace in the lattice model, and let $J_1(L)^\circ$ and $J_2(M)^\circ$ be the compact subgroups defined from the...

conjecture | open | MathDB 页面 | 原文线索

V2-012606 | MATHDB-323670

The second key lemma for the lattice model of the Weil representation

摘录 / EXCERPT: Let $L\subseteq L_1$ be a sublattice, let A, $B(L)$, K_1, $H_1(L)$, $L^\perp$, and ψ_1^w be as defined in the lattice model, and let r_2 be the associated exponent. Seco...

conjecture | open | MathDB 页面 | 原文线索

V2-012607 | MATHDB-323678

Manturov's freeness conjecture for the virtual braid group action on the free virtual quandle

摘录 / EXCERPT: Let FVQ_n be the free virtual quandle on generators $x_1,x_2,\ldots,x_n$, and let VB_n be the virtual braid group acting on $FVQ_n^{\times n}$ by the virtual quandle action. Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-012608 | MATHDB-323699

Hodge-Riemann bilinear relation for the unitary invariant hermitian form

摘录 / EXCERPT: Hodge-Riemann conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012609 | MATHDB-323718

Geometric Broué conjecture for unipotent blocks

摘录 / EXCERPT: Let $\chi_1,\dots,\chi_s$ be the unipotent ordinary characters in a unipotent ℓ-block B of kG with abelian defect group. Let B' be the Brauer correspondent of B, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-012610 | MATHDB-323721

Degree-sum conjecture for singular ideals of symmetric-group Cherednik modules

摘录 / EXCERPT: Degree-sum conjecture. The sum of the degrees in a minimal generating set of $J_{c+1}(S_n,\mathfrak{h},\tau)$ is exactly $n!$ greater than the corresponding sum for...

conjecture | open | MathDB 页面 | 原文线索

V2-012611 | MATHDB-323735

The conjecture identifying algebraic automorphic representations with their Arthur packets

摘录 / EXCERPT: Packet conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-012612 | MATHDB-323745

Thin invariant exhaustion conjecture for infinite groups

摘录 / EXCERPT: Thin invariant exhaustion conjecture. Every infinite group G admits a thin invariant exhaustion of $\ell^p G$ for all $p>2$.

conjecture | open | MathDB 页面 | 原文线索

V2-012613 | MATHDB-323757

Quotient description of the ordinary subobject classifier

摘录 / EXCERPT: Let $\mathcal{V}(\mathcal{H})$ be the context category of the Hilbert space $\mathcal{H}$, let $\mathcal{V}_f(\mathcal{H})$ be the same category with the group action forgotten, le...

conjecture | open | MathDB 页面 | 原文线索

V2-012614 | MATHDB-323773

The orbit-equivalent primitive closed permutation groups conjecture

摘录 / EXCERPT: Orbit-equivalent groups conjecture. G and H belong to the list above. This would strengthen the main theorem by removing its assumption that the groups are not 2-transitive. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-012615 | MATHDB-323774

The strong orbit-equivalence rigidity conjecture

摘录 / EXCERPT: Strong orbit-equivalence rigidity conjecture. If G and H are strongly orbit-equivalent, then

conjecture | open | MathDB 页面 | 原文线索

V2-012616 | MATHDB-323781

Surface subgroup conjecture for uniform lattices in semisimple Lie groups

摘录 / EXCERPT: Soient G un groupe de Lie réel semi-simple connexe de centre trivial et sans facteur compact, K un sous-groupe compact maximal de G et $X=G/K$ l'espace symétrique associé. So...

conjecture | open | MathDB 页面 | 原文线索

V2-012617 | MATHDB-323792

Infinite generation and odd-degree generators of the symplectic commutative graph algebra

摘录 / EXCERPT: Let $H_*(\mathfrak{c}^+_\infty)^{\mathrm{Sp}}$ be the symplectic-invariant homology algebra, let $\mathcal{A}^+(\phi)$ be the subalgebra generated by the positive-degree elements o...

conjecture | open | MathDB 页面 | 原文线索

V2-012618 | MATHDB-323803

Breuil-Schneider conjecture for locally algebraic representations of \(\mathrm{GL}_2\)

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$, let E be a finite extension of $\mathbb{Q}_p$ containing the images of all embeddings $\sigma:F\hookrightarrow E$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012619 | MATHDB-323809

Lackenby's arbitrary rank-gradient values conjecture

摘录 / EXCERPT: Lackenby's conjecture. For every real number $\alpha>0$ there exists a finitely generated group Γ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012620 | MATHDB-323860

Adler-Prasad multiplicity-one conjecture for restrictions of representations

摘录 / EXCERPT: Let F be a number field, let v be a place of F, and let G and $\widetilde{G}$ be quasi-split groups related by a central covering as in the source. For an irreducible admis...

conjecture | open | MathDB 页面 | 原文线索

V2-012621 | MATHDB-323866

Farb-Shalen conjecture on actions of arithmetic special linear groups on manifolds

摘录 / EXCERPT: Let $n>2$, let M be a compact smooth manifold of dimension r, and let H be a finite-index subgroup of $\mathrm{SL}_{n}(\mathbb{Z})$. A smooth action of H on M is said to...

conjecture | open | MathDB 页面 | 原文线索

V2-012622 | MATHDB-323885

Symmetric-group action conjecture for Heegaard Floer link homology

摘录 / EXCERPT: Let $T_n\subset S^3$ be the (n,n)-torus link. Fix a Heegaard diagram $\mathcal{D}_1=(\Sigma,\mathbf{\alpha},\mathbf{\beta},z_1,\dots,z_n,w_1,\dots,w_n)$ and a permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-012623 | MATHDB-323886

Irreducibility conjecture for symmetric-group summands of torus-link homology

摘录 / EXCERPT: Let $T_n=T(n,n)\subset S^3$ and let $v_{C,n}$ be the lattice point where the C^{th} and $(C+1)^{th}$ Alexander-support hypercubes meet. Assume the Heegaard Floer link homology is...

conjecture | open | MathDB 页面 | 原文线索

V2-012624 | MATHDB-323901

Rational hyperbolicity of simply connected hyperbolic polar manifolds

摘录 / EXCERPT: Let M be a simply connected polar manifold, meaning a Riemannian manifold with an isometric polar group action, and suppose that M is hyperbolic in the sense that its section h...

conjecture | open | MathDB 页面 | 原文线索

V2-012625 | MATHDB-323906

Bieri-Groves FP_m conjecture for metabelian groups

摘录 / EXCERPT: Bieri-Groves FP_m conjecture. G is of type FP_m if and only if A is m-tame as a $\mathbb{Z}Q$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-012626 | MATHDB-323907

The homological dimension-Hirsch length conjecture for elementary amenable groups

摘录 / EXCERPT: Let G be an elementary amenable group and let k be a non-zero commutative ring. The homological dimension-Hirsch length conjecture. If G has no k-torsion, then ... Equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-012627 | MATHDB-323908

The local-freeness conjecture for homological dimension one

摘录 / EXCERPT: Local-freeness conjecture. Every group G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-012628 | MATHDB-323909

The cohomological-dimension conjecture for elementary amenable groups

摘录 / EXCERPT: Cohomological-dimension conjecture. The cohomological dimension $\operatorname{cd}_{k}(G)$ is finite if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-012629 | MATHDB-323910

The vanishing conjecture for free rational modules in locally finite extensions

摘录 / EXCERPT: Vanishing conjecture. For every free $\mathbb QG$-module F,

conjecture | open | MathDB 页面 | 原文线索

V2-012630 | MATHDB-323974

Component-cluster cardinality and rigidity conjecture for basic algebras

摘录 / EXCERPT: Component-cluster conjecture. For any basic algebra Λ:

conjecture | open | MathDB 页面 | 原文线索

V2-012631 | MATHDB-324026

The Engel-value conjecture for verbal subgroups of residually finite groups

摘录 / EXCERPT: Let w be a group-word and let n be a positive integer. For a group G, write $w(G)$ for the verbal subgroup generated by all values of w in G. An element is n-Engel if,...

conjecture | open | MathDB 页面 | 原文线索

V2-012632 | MATHDB-324047

The GIM criterion from split representations and antisymmetric absolute involutions

摘录 / EXCERPT: Let $G(r,p,q,n)$ be the group under consideration, and let a generalized involution model (GIM) mean a generalized involution model for $G(r,p,q,n)$ with respect to the inverse tra...

conjecture | open | MathDB 页面 | 原文线索

V2-012633 | MATHDB-324048

The self-dual GIM conjecture for $G(r,p,q,n)$

摘录 / EXCERPT: Let $G(r,p,q,n)$ be the group under consideration. Say that $G(r,p,q,n)$ is self-dual when $G(r,p,q,n)\cong G(r,q,p,n)$, and let a generalized involution model (GIM) mean a general...

conjecture | open | MathDB 页面 | 原文线索

V2-012634 | MATHDB-324049

The finite-exception GIM classification conjecture for $G(r,p,q,n)$

摘录 / EXCERPT: Let $G(r,p,q,n)$ be the group under consideration, let a generalized involution model (GIM) mean a generalized involution model with respect to the inverse transpose automorphism,...

conjecture | open | MathDB 页面 | 原文线索

V2-012635 | MATHDB-324058

The symplectic-resolution conjecture for affine-Grassmannian slices

摘录 / EXCERPT: Let G be the reductive group and let $\Grlmbar$ be the affine-Grassmannian slice under consideration. A symplectic resolution is a resolution of singularities whose pullback of t...

conjecture | open | MathDB 页面 | 原文线索

V2-012636 | MATHDB-324065

The Hodge filtration compatibility conjecture for the affine Hecke equivalence

摘录 / EXCERPT: Let ${\mathcal{MH}}_{U\check{\ }}({G\check{\ }}/{B\check{\ }})$ be the category of mixed Hodge modules on ${ G\check{\ }}/{B\check{\ }}$ equivariant with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-012637 | MATHDB-324067

The parabolic-Whittaker geometric equivalence

摘录 / EXCERPT: Let P be a parabolic subgroup of G, let Q be another parabolic subgroup, and let $D_{IW_Q}({{\mathcal F}\ell}_P)$ be the corresponding category of partial Whittaker sheaves o...

conjecture | open | MathDB 页面 | 原文线索

V2-012638 | MATHDB-324070

Krull-Gabriel dimension characterization for discrete derived categories

摘录 / EXCERPT: Krull-Gabriel dimension conjecture. The Krull-Gabriel dimension of $\operatorname{Ab}(\operatorname{per} A)$ equals 0 or 1 if and only if $\operatorname{per} A$ is a discrete d...

conjecture | open | MathDB 页面 | 原文线索

V2-012639 | MATHDB-324077

The fixed-point conjecture for almost connected locally compact groups on buildings

摘录 / EXCERPT: Let G be an almost connected locally compact group. Recall that a group has property $\operatorname{(FB)}$ when every measurable action of G on a finite-rank building stabilise...

conjecture | open | MathDB 页面 | 原文线索

V2-012640 | MATHDB-324081

Existence of integral one-dimensional modules for finite W-algebras

摘录 / EXCERPT: Let $\mathbb{O}$ be a special nilpotent orbit, and let A and $\mathcal W$ denote the groups and finite W-algebra associated with the orbit as in the paper. The three exceptional...

conjecture | open | MathDB 页面 | 原文线索

V2-012641 | MATHDB-324094

Cohen-Macaulay induction conjecture for B-algebras

摘录 / EXCERPT: Let G be a simple algebraic group, let B be a Borel subgroup of G, and let R be a B-algebra. Say that R is Cohen-Macaulay when its underlying ring has the Cohen-Macaula...

conjecture | open | MathDB 页面 | 原文线索

V2-012642 | MATHDB-324146

Integrality conjecture for Fourier transforms of nilpotent coadjoint orbits

摘录 / EXCERPT: Let G be a reductive Lie group, let $\mathfrak{g}$ be its Lie algebra, and let $\mathcal{O}\subset \mathfrak{g}^*$ be a nilpotent coadjoint orbit. Let $H\subset G$ be a Cartan su...

conjecture | open | MathDB 页面 | 原文线索

V2-012643 | MATHDB-324171

Classification and law of large numbers for ergodic central measures on infinite unipotent matrices

摘录 / EXCERPT: Classification and law of large numbers for ergodic central measures. Let ϑ be an ergodic central measure on $\mathbb U$. There exist nonincreasing sequences of nonnegati...

conjecture | open | MathDB 页面 | 原文线索

V2-012644 | MATHDB-324195

The centralizer-coset odd-order conjecture

摘录 / EXCERPT: Let G be a finite group, let T be a Sylow 2-subgroup of G, and let $t\in Z(T)$ be an involution. For $g\in G$, the coset $C_G(t)g$ is a coset of the centralizer of t. Cen...

conjecture | open | MathDB 页面 | 原文线索

V2-012645 | MATHDB-324196

The finite-simple-group centralizer-coset conjecture

摘录 / EXCERPT: Let G be a finite simple group and let $t\in G$ be an involution. For $g\in G$, the coset $C_G(t)g$ is a coset of the centralizer of t. Finite-simple-group centralizer-coset co...

conjecture | open | MathDB 页面 | 原文线索

V2-012646 | MATHDB-324201

De Concini-Kac-Procesi divisibility conjecture for quantum group modules

摘录 / EXCERPT: De Concini-Kac-Procesi conjecture. The integer

conjecture | open | MathDB 页面 | 原文线索

V2-012647 | MATHDB-324202

Induction conjecture for irreducible quantum-group modules

摘录 / EXCERPT: Induction conjecture. There exists $W\in{\rm Spec}({\cal U}_{\chi_{L'}}(L'))$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-012648 | MATHDB-324207

Bounded-diameter conjecture for commuting graphs of centerless finite groups

摘录 / EXCERPT: Let G be a finite group with trivial centre, and let its commuting graph be the graph whose vertices are the non-central elements of G, with two distinct vertices adjacent exac...

conjecture | open | MathDB 页面 | 原文线索

V2-012649 | MATHDB-324210

The Kostka-support conjecture for minimal recurrences of stretched skew Schur polynomials

摘录 / EXCERPT: The Kostka-support conjecture. For sufficiently large r, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-012650 | MATHDB-324212

Completeness conjecture for physically relevant SO(3)-invariant Jordan multialgebras

摘录 / EXCERPT: Let $\mathcal T$ be a real finite-dimensional representation of $SO(3)$ of the form ... where W_{0}, W_{1}, and W_{2} are respectively the weight 0, natural, and symmetric tr...

conjecture | open | MathDB 页面 | 原文线索

V2-012651 | MATHDB-324219

The dimension dichotomy conjecture for finite-dimensional simple modules

摘录 / EXCERPT: Dimension dichotomy conjecture. Either all finite-dimensional simple modules over A are 1-dimensional, or else A has finite-dimensional simple modules of arbitrarily large di...

conjecture | open | MathDB 页面 | 原文线索

V2-012652 | MATHDB-324224

Yvonne-Rouquier conjecture on multiplicities for wreath products

摘录 / EXCERPT: Let $l\geq 1$, let $\mathcal{P}^l_n$ denote the set of l-multipartitions of n, and let $\Delta(\boldsymbol{\lambda})$ and $L(\boldsymbol{\mu})$ denote respectively the standard...

conjecture | open | MathDB 页面 | 原文线索

V2-012653 | MATHDB-324226

The algebraic-geometric rank equivalence for CAT(0) groups

摘录 / EXCERPT: Let X be a complete $\mathrm{CAT}(0)$ space and let G be a group acting geometrically on X, meaning properly and cocompactly by isometries. A geodesic line in X has rank on...

conjecture | open | MathDB 页面 | 原文线索

V2-012654 | MATHDB-324237

Conjecture on four-term irredundant generating sequences of PSL(2,p)

摘录 / EXCERPT: Let $G=\mathrm{PSL}(2,p)$), and let $\iota_n(G)$ denote the set of divisors occurring as the relevant invariant for irredundant generating sequences of length n in the paper. Fou...

conjecture | open | MathDB 页面 | 原文线索

V2-012655 | MATHDB-324288

Asymptotic growth conjecture for trivial words in Baumslag-Solitar groups

摘录 / EXCERPT: Let $\mathrm{BS}(N,N)$ be the Baumslag-Solitar group, and let $g_{n,0}$ denote the number of trivial words of length n in it. Write μ_N for the exponential growth rate and...

conjecture | open | MathDB 页面 | 原文线索

V2-012656 | MATHDB-324289

Non-D-finiteness conjecture for Baumslag-Solitar generating functions

摘录 / EXCERPT: Let $\mathrm{BS}(1,M)$ be the Baumslag-Solitar group with $M>1$. Let $G(z;q)$ be the bivariate generating function for the trivial-word enumeration considered in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012657 | MATHDB-324290

D-finiteness conjecture for Baumslag-Solitar generating functions

摘录 / EXCERPT: Let $\mathrm{BS}(N,M)$ be a Baumslag-Solitar group, and let $G(z;q)$ be the generating function for trivial-word enumeration considered in the paper. Write $[q^0]G(z;q)$ for its co...

conjecture | open | MathDB 页面 | 原文线索

V2-012658 | MATHDB-324293

The irreducible arithmetic-group conjecture for faithful actions on the line

摘录 / EXCERPT: Let Γ be an irreducible arithmetic group, meaning that no finite-index subgroup of Γ is a direct product of two infinite groups. A faithful action on $\mathbb{R}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-012659 | MATHDB-324294

The algebraic version of the irreducible arithmetic-group conjecture

摘录 / EXCERPT: Let Γ be a group. A total order $\prec$ on Γ is left-invariant if $a\prec b$ implies $ca\prec cb$ for all $a,b,c\in\Gamma$. The algebraic version of the conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-012660 | MATHDB-324296

The noncocompact \operatorname{SL}(3) arithmetic-subgroup conjecture

摘录 / EXCERPT: Let Γ be a noncocompact arithmetic subgroup of either $\operatorname{SL}(3,\mathbb{R})$ or $\operatorname{SL}(3,\mathbb{C})$. A faithful action on $\mathbb{R}$ means an acti...

conjecture | open | MathDB 页面 | 原文线索

V2-012661 | MATHDB-324300

The double-pole bound for twisted partial standard L functions

摘录 / EXCERPT: Double-pole conjecture. The twisted partial standard L function $L^S(\pi\otimes\chi,s)$ can have at most a double pole.

conjecture | open | MathDB 页面 | 原文线索

V2-012662 | MATHDB-324301

The pole-order conjecture for the E_8 Eisenstein series

摘录 / EXCERPT: Pole-order conjecture. The possible poles of $E(h,s)$ are as follows. If χ is trivial, then $E(h,s)$ can have a double pole at $\frac{10}{17}$, $\frac{11}{17}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-012663 | MATHDB-324315

The coisotropic orbit characterization for symplectic representations

摘录 / EXCERPT: Let $({\mathfrak m},V,\omega)$ be a symplectic representation of the Lie algebra ${\mathfrak m}$. A subspace $W\subseteq V$ is coisotropic when $W^{\perp_\omega}\subseteq W$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-012664 | MATHDB-324327

The center conjecture for infinitesimal Cherednik algebras of symplectic type

摘录 / EXCERPT: The center conjecture. The center of $H_\zeta(\mathfrak{sp}_{2n})$ is

conjecture | solved | MathDB 页面 | 原文线索

V2-012665 | MATHDB-324328

The minimal-generator characterisation of uniform pro-p groups

摘录 / EXCERPT: Let $p\geq 3$ and let G be a torsion-free pro-p group of finite rank. Write $d(G)$ for the minimal number of topological generators of G, and let $\dim(G)$ denote its dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-012666 | MATHDB-324350

Conjecture on equality of support, quantum orbit and minimal invariant subset

摘录 / EXCERPT: Let $\mathcal{G}$ be a compact quantum group acting on a compact Hausdorff space X. For $x\theta X$, let $\rm Orb_x$ denote the orbit of x, let $\rm supp\,\mu_x$ denote the suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-012667 | MATHDB-324361

Tensor-product multiplicity monotonicity for simple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, let P^+ be its set of dominant integral weights, and let $P^+(\lambda,2)$ denote the pairs of dominant integral wei...

conjecture | open | MathDB 页面 | 原文线索

V2-012668 | MATHDB-324372

James conjecture for affine Khovanov-Lauda-Rouquier algebras

摘录 / EXCERPT: James conjecture. Let $\alpha\in Q_+$, and let L_K be an irreducible $R_\alpha(K)$-module which factors through $R_\alpha^{\boldsymbol{\Lambda}}(K)$. Then reduction modulo p of...

conjecture | open | MathDB 页面 | 原文线索

V2-012669 | MATHDB-324373

Cuspidal short exact sequence conjecture in non-simply laced affine type

摘录 / EXCERPT: Cuspidal short exact sequence conjecture. For non-symmetric C, let $\rho\in\Phi_+^{\mathrm{re}}$, and let (β,γ) be a real minimal pair for ρ. Then there...

conjecture | open | MathDB 页面 | 原文线索

V2-012670 | MATHDB-324379

The H_1-semistability conjecture for finitely presented groups

摘录 / EXCERPT: Let G be a 1-ended finitely presented group, and let $\varepsilon(G)$ denote its end. Write $\operatorname*{pro}$-$H_{1}(\varepsilon(G);\mathbb{Z})$ for the inverse system of fir...

conjecture | open | MathDB 页面 | 原文线索

V2-012671 | MATHDB-324380

The CAT(0) Semistability Conjecture

摘录 / EXCERPT: Let G be a 1-ended CAT(0) group, meaning a group acting geometrically on a proper CAT(0) space. Its end invariant $\operatorname*{pro}$-$\pi_{1}(\varepsilon(G))$ is defined throu...

conjecture | open | MathDB 页面 | 原文线索

V2-012672 | MATHDB-324381

The Borel Conjecture for universal covers

摘录 / EXCERPT: Let M and N be closed aspherical manifolds with isomorphic fundamental groups. Their universal covers are contractible open manifolds. The Weak Borel Conjecture. The universal...

conjecture | open | MathDB 页面 | 原文线索

V2-012673 | MATHDB-324413

The geometric structure conjecture for Bernstein components

摘录 / EXCERPT: Let $\mathcal G$ be a quasi-split connected reductive p-adic group or an inner form of $\operatorname{GL}_n(F)$, and let $\mathfrak{s}$ be a Bernstein component of...

conjecture | open | MathDB 页面 | 原文线索

V2-012674 | MATHDB-324414

Tightness characterization of efficiency for irreducible handlebody automorphisms

摘录 / EXCERPT: Tightness conjecture. The automorphism f is efficient if and only if its associated pair of f-invariant laminations is tight.

conjecture | open | MathDB 页面 | 原文线索

V2-012675 | MATHDB-324432

Whittaker's conjecture on morphisms of homeomorphism groups

摘录 / EXCERPT: Let M be a compact manifold, and let $\mathrm{Homeo}_{0}(M)$ denote the identity component of its homeomorphism group. A group morphism is trivial if it maps every element to the...

conjecture | open | MathDB 页面 | 原文线索

V2-012676 | MATHDB-324433

Conjecture on morphisms from punctured-surface homeomorphism groups

摘录 / EXCERPT: Let S be a closed surface, let D be an open disc in S, and write $S-D$ for the surface obtained by removing D. Let $\mathrm{Homeo}_{0}(S-D)$ be the identity component of th...

conjecture | open | MathDB 页面 | 原文线索

V2-012677 | MATHDB-324434

Conjecture on actions of circle homeomorphisms on spheres and discs

摘录 / EXCERPT: Let $\mathbb{S}^{1}$ be the circle, $\mathbb{S}^{2}$ the sphere, and $\mathbb{D}^{2}$ the closed disc. Consider non-trivial actions of the group...

conjecture | open | MathDB 页面 | 原文线索

V2-012678 | MATHDB-324444

The quantum Langlands equivalence for the metaplectic symplectic group

摘录 / EXCERPT: Quantum Langlands conjecture. There is an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-012679 | MATHDB-324460

The finite-volume conjecture for outer space

摘录 / EXCERPT: Let X_n be outer space, let $\mathrm{Out}(F_n)$ act on it naturally, and let ds^2 and ds^2_{ε} be the two Riemannian metrics defined in the previous section. The qu...

conjecture | open | MathDB 页面 | 原文线索

V2-012680 | MATHDB-324489

Sharpness for fusion systems

摘录 / EXCERPT: Sharpness for fusion systems. For all $i\geq 1$ and $j\geq 0$, the higher limits vanish:

conjecture | open | MathDB 页面 | 原文线索

V2-012681 | MATHDB-324492

The conjecture that every quasi-minimal group is abelian

摘录 / EXCERPT: Let G be a quasi-minimal group, meaning an uncountable group in a countable language whose definable subsets are countable or co-countable. The quasi-minimal group conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-012682 | MATHDB-324520

Hyperbolicity conjecture for absolute Galois groups of number fields

摘录 / EXCERPT: Let k be a number field, and let $Gal(\bar{k}/k)$ denote its full Galois group. Absolute Galois group hyperbolicity conjecture. For any number field k, the profinite group...

conjecture | open | MathDB 页面 | 原文线索

V2-012683 | MATHDB-324530

Transfer of R-groups between p-adic inner forms of \(\operatorname{SL}_n\)

摘录 / EXCERPT: Transfer conjecture. For every discrete series representation $\sigma\in\Pi_\phi(M)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-012684 | MATHDB-324552

Injectivity conjecture for the profinite realization of knots

摘录 / EXCERPT: Profinite knot realization conjecture. The map h is injective. This is proposed as a knot-theoretic analogue of residual finiteness for braid groups. If it fails, the Kontsevich...

conjecture | open | MathDB 页面 | 原文线索

V2-012685 | MATHDB-324558

The Connes-Kasparov conjecture

摘录 / EXCERPT: Connes-Kasparov conjecture. The map $\widetilde{\mu}$ is an isomorphism of abelian groups, and

conjecture | open | MathDB 页面 | 原文线索

V2-012686 | MATHDB-324573

Conjecture that every word-hyperbolic group has property R

摘录 / EXCERPT: Let G be a word-hyperbolic group. An automatic structure for G has property R when its accepting automaton is both undirected and Perron-Frobenius, meaning that the adjacency m...

conjecture | open | MathDB 页面 | 原文线索

V2-012687 | MATHDB-324601

The full visibility conjecture for cocompact CAT(0) cube complexes

摘录 / EXCERPT: Full visibility conjecture. $\mathbf X$ is fully visible.

conjecture | open | MathDB 页面 | 原文线索

V2-012688 | MATHDB-324611

Kleshchev-Ram conjecture on decomposition numbers for Dynkin KLR algebras

摘录 / EXCERPT: Kleshchev-Ram conjecture. The decomposition numbers for KLR algebras associated to Dynkin quivers are trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-012689 | MATHDB-324637

Hamenstädt's quasi-isometric rigidity conjecture for purely real Heintze groups

摘录 / EXCERPT: A purely real Heintze group is a simply connected solvable Lie group of the form $N\rtimes\mathbf{R}$, where N is simply connected and nilpotent and the action of $\mathbf{R}$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-012690 | MATHDB-324657

Conjecture on functorial behavior of the decomposition number

摘录 / EXCERPT: Functorial decomposition-number conjecture. Without additional conditions, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-012691 | MATHDB-324668

Uniqueness conjecture for standardly embedded spherical subgroups

摘录 / EXCERPT: Standardly embedded subgroup uniqueness conjecture. Up to conjugation by an element of $N_L(K)$, a spherical subgroup $H\subset G$ standardly embedded in P is uniquely determined...

conjecture | open | MathDB 页面 | 原文线索

V2-012692 | MATHDB-324671

Kerov's conjecture on Macdonald-positive specializations

摘录 / EXCERPT: Assume $0<q,t<1$. A specialization ρ of the algebra of symmetric functions is Macdonald-positive if $P_\lambda(\rho)\ge 0$ for every partition λ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012693 | MATHDB-324717

Bien's conjecture on radical normal subgroups of division rings

摘录 / EXCERPT: Bien's conjecture. Every normal subgroup of D^* that is radical over K is commutative.

conjecture | open | MathDB 页面 | 原文线索

V2-012694 | MATHDB-324722

Weak amenability and C*-exactness for infinitely presented small cancellation groups

摘录 / EXCERPT: Let G be a finitely generated group defined by an infinite presentation satisfying the small cancellation condition $C'(1/10)$. Weak amenability and C^*-exactness (equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-012695 | MATHDB-324741

Kac's polynomiality conjecture for similarity classes of endomorphisms of finite torsion modules

摘录 / EXCERPT: Kac's conjecture. For every partition λ, there exists a monic polynomial $c_\lambda(q)$ of degree $|\lambda|$ whose coefficients are all non-negative integers such that, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-012696 | MATHDB-324744

Definable-envelope conjecture for nilpotent and soluble subgroups of NTP2 groups

摘录 / EXCERPT: Definable-envelope conjecture. If H is nilpotent (respectively, soluble), then there is a definable nilpotent (respectively, soluble) group finitely many of whose translates cove...

conjecture | open | MathDB 页面 | 原文线索

V2-012697 | MATHDB-324763

The quasi-isometric restriction conjecture for linear semidirect products

摘录 / EXCERPT: Let Γ be a finitely generated group, and let ρ be a linear representation of Γ on $\mathbf{R}^n$. Equip $\mathbf{R}^n\rtimes_\rho\Gamma$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-012698 | MATHDB-324774

The building-realization conjecture for exceptional Chevalley groups over finite fields

摘录 / EXCERPT: Let $\mathcal G$ be a Chevalley group of adjoint type τ, rank r, and dimension $n=r+2m$, with Lie algebra $\mathfrak g$. Let $\mathbb P[\Psi]$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-012699 | MATHDB-324779

Clozel's formula for higher homology of reductive p-adic groups

摘录 / EXCERPT: Let G be a reductive p-adic group. Choose a minimal parabolic subgroup $P_0\subset G$ with Levi decomposition $P_0=L_0U_0$. For each parabolic subgroup $P\supset P_0$ with Levi...

conjecture | open | MathDB 页面 | 原文线索

V2-012700 | MATHDB-324781

Lucchini-Detomi's monolithicity conjecture for sigma-elementary groups

摘录 / EXCERPT: Let G be a group with finite covering number ... sigma(G)$ is the smallest number of proper subgroups needed to cover G. Call G ** ... $sigma(G)<$sigma(G/N) ... sigma$-elementary g...

conjecture | open | MathDB 页面 | 原文线索

V2-012701 | MATHDB-324897

Crawley-Boevey's modification of the Second Brauer-Thrall Conjecture

摘录 / EXCERPT: Let A be an artin algebra of infinite representation type. For an A-module M, its endolength is its length as a module over its endomorphism ring. Brauer-Thrall II. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-012702 | MATHDB-324933

Pronormality conjecture for odd-index subgroups of finite simple groups

摘录 / EXCERPT: Pronormality conjecture. Every subgroup of odd index in a finite simple group is pronormal.

conjecture | open | MathDB 页面 | 原文线索

V2-012703 | MATHDB-324934

Hall subgroups of C_π-groups are inherited by overgroups

摘录 / EXCERPT: Overgroup inheritance conjecture. If $H\le M\le G$, then $M\in C_\pi$.

conjecture | open | MathDB 页面 | 原文线索

V2-012704 | MATHDB-324935

Hall subgroups of D_π-groups are inherited by overgroups

摘录 / EXCERPT: Overgroup inheritance conjecture. If $H\le M\le G$, then $M\in D_\pi$.

conjecture | open | MathDB 页面 | 原文线索

V2-012705 | MATHDB-324937

Hall subgroups with a Sylow tower are strongly pronormal

摘录 / EXCERPT: Sylow-tower strong pronormality conjecture. Every Hall subgroup having a Sylow tower is strongly pronormal.

conjecture | open | MathDB 页面 | 原文线索

V2-012706 | MATHDB-324938

Pronormal Hall subgroups are strongly pronormal

摘录 / EXCERPT: Pronormal-to-strongly-pronormal conjecture. Every pronormal Hall subgroup is strongly pronormal.

conjecture | open | MathDB 页面 | 原文线索

V2-012707 | MATHDB-324939

Hall subgroups are pronormal in their normal closure

摘录 / EXCERPT: Normal-closure pronormality conjecture. The Hall subgroup H is pronormal in its normal closure.

conjecture | open | MathDB 页面 | 原文线索

V2-012708 | MATHDB-324942

Product decomposition conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite non-abelian simple group, and let $S\subseteq G$ satisfy $|S|\geqslant 2$. A conjugate of S means a set gSg^{-1} with $g\in G$. Product Decomposition Conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-012709 | MATHDB-324943

Alternating-group conjugacy-class intersection conjecture

摘录 / EXCERPT: Let $G=A_n$ with n odd, let C be a conjugacy class of n-cycles, and let H be the centralizer in G of an element of C. Let $N_G(H)$ denote the normalizer of H in G,...

conjecture | open | MathDB 页面 | 原文线索

V2-012710 | MATHDB-324944

Maximally split torus intersection conjecture in $SL_n(2)$

摘录 / EXCERPT: Let $G=SL_n(2)$, let C be a conjugacy class of elements centralized by a maximally split torus, and let H be the centralizer in G of an element of C. Let $N_G(H)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-012711 | MATHDB-324950

Positivity and periodicity conjecture for the WZW fusion-ring elements $W^{(a)}_m$

摘录 / EXCERPT: Let I be the node set, let t_a and $h^{\vee}$ be the parameters associated with the Dynkin diagram, and let $W^{(a)}_m$ be the elements of the WZW fusion ring defined from the...

conjecture | open | MathDB 页面 | 原文线索

V2-012712 | MATHDB-324988

Buch's positivity conjecture for quiver coefficients

摘录 / EXCERPT: Buch's positivity conjecture. For Dynkin quivers,

conjecture | open | MathDB 页面 | 原文线索

V2-012713 | MATHDB-324989

Martino's conjecture on Calogero-Moser and Hecke families

摘录 / EXCERPT: Martino's conjecture. For every $b\in {\mathrm{Idem_{pr}}}({\bar Z}_c)$, there exists a central idempotent $b^{\mathcal H}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-012714 | MATHDB-324991

Calogero-Moser and Kazhdan-Lusztig family and character conjecture

摘录 / EXCERPT: Family and character conjecture. The partition of $\operatorname{Irr}(W)$ into Calogero-Moser c-families coincides with the partition into Kazhdan-Lusztig c-families, as conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-012715 | MATHDB-324996

Congruence-cover eigenvalue counting conjecture for geometrically finite groups

摘录 / EXCERPT: Let $G=\operatorname{SO}(n,1)^\circ$, let $\rho=\tfrac{n-1}{2}$, and let $\Gamma<G(\mathbb{Z})$ be geometrically finite and Zariski dense, with critical exponent δ. For...

conjecture | open | MathDB 页面 | 原文线索

V2-012716 | MATHDB-325037

Thompson's conjugacy-class product conjecture

摘录 / EXCERPT: Let G be a finite simple group, and let $C\subseteq G$ be a conjugacy class. Write $C^2=\{c_1c_2:c_1,c_2\in C\}$. Thompson's conjecture. Every finite simple group G has a conju...

conjecture | open | MathDB 页面 | 原文线索

V2-012717 | MATHDB-325038

Lie-to-group transfer conjecture for solvability-characterizing sequences

摘录 / EXCERPT: Let $\{v_n(x,y)\}$ be an Engel-like sequence of words in the free Lie algebra $\mathcal{W}_2$ that characterizes finite-dimensional solvable Lie algebras over fields of arbitrary c...

conjecture | open | MathDB 页面 | 原文线索

V2-012718 | MATHDB-325073

Conjecture on Property FW for irreducible lattices in higher-rank semisimple Lie groups

摘录 / EXCERPT: Let S be a connected semisimple Lie group with no compact simple factor, whose Lie algebra has real rank at least 2. An irreducible lattice is a lattice in S satisfying the u...

conjecture | open | MathDB 页面 | 原文线索

V2-012719 | MATHDB-325085

Fibred Farrell-Jones isomorphism conjecture

摘录 / EXCERPT: Let G be a discrete group and let $\mathcal F$ be a family of subgroups of G. For a group homomorphism $\varphi:K{\longrightarrow} G$, define the induced family ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012720 | MATHDB-325090

Nica's proper affine action conjecture for mapping class groups

摘录 / EXCERPT: Nica's conjecture. The mapping class group of a surface admits a proper affine isometric action on an L_p-space for some sufficiently large

conjecture | open | MathDB 页面 | 原文线索

V2-012721 | MATHDB-325122

Cluster monomial conjecture for real simple modules

摘录 / EXCERPT: Let $\mathcal{A}$ be the cluster algebra identified with the ring of truncated q-characters of the category $\mathcal{C}^-$. A simple $U_q(\widehat{\mathfrak{g}})$-module S is...

conjecture | open | MathDB 页面 | 原文线索

V2-012722 | MATHDB-325123

Geometric q-character formula for real simple modules

摘录 / EXCERPT: Let m be a dominant monomial in the variables $Y_{i,r}\in\mathbf{Y}^-$. Write ... with $z_{i,s}=1$ for $s>0$, and define ... The integer vector $g(m)\in\mathbb{Z}^{(W^-)}$ is the g...

conjecture | open | MathDB 页面 | 原文线索

V2-012723 | MATHDB-325124

The geometric R-group conjecture for non-tempered representations

摘录 / EXCERPT: Let G be a reductive p-adic group and let $\phi \in \Psi(G)$. Let M be a Levi subgroup of G such that the image of ϕ is contained in ${}^{L}M$ but not in any smaller L...

conjecture | open | MathDB 页面 | 原文线索

V2-012724 | MATHDB-325128

The universal-envelope conjecture for simple associative triple systems

摘录 / EXCERPT: Let $A_{p,q}$ ($p \le q$) be the simple associative triple system consisting of $(p+q) \times (p+q)$ block matrices of the form ... Let ω be one of the following trilinear ope...

conjecture | open | MathDB 页面 | 原文线索

V2-012725 | MATHDB-325297

Soergel's affine p-conjecture for affine Weyl groups

摘录 / EXCERPT: Let W be an affine Weyl group, let W° be the finite subset on which the affine construction is defined, let $\mathcal{B}$ be the corresponding category of Soergel bimodu...

conjecture | open | MathDB 页面 | 原文线索

V2-012726 | MATHDB-325334

Gorsky-Oblomkov-Rasmussen-Shende conjecture for rational Cherednik representations

摘录 / EXCERPT: Let $L_{\frac mn}$ be the finite-dimensional representation of the rational Cherednik algebra with parameter $c=\frac mn$, equipped with its natural grading and the extra filtratio...

conjecture | open | MathDB 页面 | 原文线索

V2-012727 | MATHDB-325337

Gordon-Stafford filtration conjecture for rational Cherednik representations

摘录 / EXCERPT: Let $L_{\frac mn}$ be the finite-dimensional representation of the rational Cherednik algebra, equipped with Gordon's generalized filtration, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012728 | MATHDB-325366

Isaacs's positivity conjecture for Lehrer's character-counting polynomials

摘录 / EXCERPT: Let $l_{n,c}(t)\in\mathbb Z[t]$ be the polynomial from Lehrer's conjecture, counting irreducible complex characters of $U_n(q)$ of degree q^c. Isaacs's positivity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-012729 | MATHDB-325367

Dipper-James standard basis conjecture for unipotent Specht modules

摘录 / EXCERPT: Let $\lambda\vdash n$. For each $\mathfrak s\in\operatorname{Std}(\lambda)$, let $r_{\mathfrak s}(t)\in\mathbb Z[t]$ and let $\mathcal B_{\mathfrak s}\subset S^\lambda$ be independ...

conjecture | open | MathDB 页面 | 原文线索

V2-012730 | MATHDB-325384

Sha-rigidity conjecture for algebraic and Kac-Moody groups

摘录 / EXCERPT: Let G be one of the groups in the following list: a split simple Chevalley group $\mathcal G(k)$ over a sufficiently large field k; the analogous group over a “good” ring; the...

conjecture | open | MathDB 页面 | 原文线索

V2-012731 | MATHDB-325427

Piecewise-quasilinearity of stable commutator length in free products of cyclic groups

摘录 / EXCERPT: Let $G_j^k = \langle a_j \rangle / \langle a_j^{o_{j,k}} \rangle = \mathbb Z/o_{j,k}\mathbb Z$ and let $G_k = *_j G_j^k$. Fix Γ such that $\Gamma \in B_1^H(G_k)$ for all k...

conjecture | open | MathDB 页面 | 原文线索

V2-012732 | MATHDB-325470

OD-characterizability conjecture for projective special linear groups over the binary field

摘录 / EXCERPT: Let $L_n(2)$ denote the projective special linear group over the binary field, for an integer $n\geqslant 2$. A finite group is OD-characterizable if it is uniquely determined, up...

conjecture | open | MathDB 页面 | 原文线索

V2-012733 | MATHDB-325471

OD-characterizability conjecture for automorphism groups of binary linear groups

摘录 / EXCERPT: Let $L_n(2)$ denote the projective special linear group over the binary field, and let ${\rm Aut}(L_n(2))$ be its automorphism group, for an integer $n\geqslant 2$. A finite group...

conjecture | open | MathDB 页面 | 原文线索

V2-012734 | MATHDB-325550

The conjectural restriction decompositions for type C_3 modules

摘录 / EXCERPT: Let $U_q(\mathfrak{g})$ be the finite quantum group associated with the type C_3 algebra, and let $\operatorname{Res}(V)$ denote restriction of a $U_q(\hat{\mathfrak{g}})$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-012735 | MATHDB-325574

Rank bound for Sylow subgroups admitting faithful simple endotrivial modules

摘录 / EXCERPT: Rank-bound conjecture. The Sylow p-subgroups of G have rank at most 2.

conjecture | open | MathDB 页面 | 原文线索

V2-012736 | MATHDB-325580

Half-class conjecture for non-nilpotent subsets

摘录 / EXCERPT: Half-class conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012737 | MATHDB-325581

Rank-bounded stabilization conjecture for nilpotency invariants

摘录 / EXCERPT: Rank-bounded stabilization conjecture. For every $r\in\mathbb{Z}^+$, there exists $\ell\in\mathbb{Z}^+$ such that, whenever G is a finite group of Lie type of rank r,

conjecture | open | MathDB 页面 | 原文线索

V2-012738 | MATHDB-325602

Uniqueness conjecture for the submodule series of the Weyl module

摘录 / EXCERPT: Let $G=\operatorname{SO}(2n+1,\mathbb{F})$ over a field $\mathbb{F}$ of characteristic 2, and let $V_\mathbb{F}(\lambda_k^\circ)$ be the Weyl module containing the submodules ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-012739 | MATHDB-325606

Horn's eigenvalue conjecture for sums of Hermitian matrices

摘录 / EXCERPT: Let n be a positive integer, and let S_r^n be the recursively defined set of triples of r-element subsets of $[n]=\{1,\dots,n\}$. Let λ,μ,ν be nonincreasing real...

conjecture | open | MathDB 页面 | 原文线索

V2-012740 | MATHDB-325607

Belkale-Kumar's diagram-automorphism restriction conjecture

摘录 / EXCERPT: Let G be a simply-connected semisimple complex algebraic group, let σ be a diagram automorphism of G, and let K be its fixed subgroup. Let $\hat{\Gamma}_s(G)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-012741 | MATHDB-325608

The simply-laced and regular-weight Littelmann path saturation conjecture

摘录 / EXCERPT: Let R be a root system, let $G^\vee$ be the corresponding group, let C_L be the cone defined by Littelmann paths, and let V_λ denote the irreducible $G^\vee$-representa...

conjecture | open | MathDB 页面 | 原文线索

V2-012742 | MATHDB-325637

Brauer's weak Donovan conjecture on Cartan invariants

摘录 / EXCERPT: Brauer's weak Donovan conjecture. There is a bound on the Cartan invariants of blocks of finite groups with defect group D that depends only on D.

conjecture | open | MathDB 页面 | 原文线索

V2-012743 | MATHDB-325731

Ext-duality and generic Ext-vanishing conjecture for general linear groups

摘录 / EXCERPT: Let F be a non-Archimedean local field, let π_1 and π_2 be irreducible admissible representations of ${\rm GL}_{n+1}(F)$ and ${\rm GL}_n(F)$, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012744 | MATHDB-325743

Total geodesicity conjecture for simplicial embeddings of pants graphs

摘录 / EXCERPT: Let Σ_{1} and Σ_{2} be compact, orientable surfaces, and let $\mathcal{P}(\Sigma_i)$ denote the pants graph of Σ_i. A simplicial embedding is a map ... that pres...

conjecture | open | MathDB 页面 | 原文线索

V2-012745 | MATHDB-325750

The dihedral-tensor conjecture for representations in the Johnson cokernel

摘录 / EXCERPT: Let H be the symplectic representation underlying the Johnson cokernel, let D_{2s} be the dihedral group of order $2s$, and write $[H^{\langle s\rangle}]_{D_{2s}}$ for the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-012746 | MATHDB-325754

Subgroup permutability probability conjecture for finite simple classical and alternating groups

摘录 / EXCERPT: Let G be a finite simple classical or alternating group. The probability that two subgroups of G permute is the subgroup permutability degree of G, denoted $\mathfrak{p}(G)$....

conjecture | open | MathDB 页面 | 原文线索

V2-012747 | MATHDB-325755

Subgroup permutability degree criterion for nonsimplicity

摘录 / EXCERPT: Let G be a finite group, and let $\mathfrak{p}(G)$ denote its subgroup permutability degree. Subgroup permutability degree criterion. If ... then G is not simple. This is present...

conjecture | open | MathDB 页面 | 原文线索

V2-012748 | MATHDB-325756

Block equivalence conjecture for orthosymplectic supergroups

摘录 / EXCERPT: Let m,n be natural numbers and fix a block ${\mathcal B}$ of atypicality k in ${\mathcal F}(\operatorname{SOSP}(m|2n))$. Let $\mathbb{H}_k^\infty$ be the limiting graded diagra...

conjecture | open | MathDB 页面 | 原文线索

V2-012749 | MATHDB-325760

Quasi-isometry implies commability for focal locally compact groups

摘录 / EXCERPT: Quasi-isometry-commability conjecture. The group H is commable to G.

conjecture | open | MathDB 页面 | 原文线索

V2-012750 | MATHDB-325807

The equal 2-part conjecture for noncentral real conjugacy classes

摘录 / EXCERPT: Equal 2-part conjecture. If the sizes of all noncentral real conjugacy classes of G have the same 2-part, then $\mathbf O^{2'}(G)$ is 2-nilpotent. In particular, G is solva...

conjecture | open | MathDB 页面 | 原文线索

V2-012751 | MATHDB-325811

Completeness conjecture for automorphism groups of austere right-angled Artin groups

摘录 / EXCERPT: Let A_Γ be a right-angled Artin group, and let Γ be austere, meaning that its defining graph has no nontrivial graph symmetries, no dominated vertices, and no separat...

conjecture | open | MathDB 页面 | 原文线索

V2-012752 | MATHDB-325882

Kobayashi's standard-form conjecture for reductive homogeneous spaces

摘录 / EXCERPT: Let G be a Lie group and let H be a reductive subgroup. A compact form of $H\backslash G$ is a compact quotient by a properly discontinuous group action. Suppose that L is a...

conjecture | open | MathDB 页面 | 原文线索

V2-012753 | MATHDB-325885

Rationality conjecture for the branching generating function of finite subgroups of $\mathbf{SL}_n\mathbb{C}$

摘录 / EXCERPT: Let $n\geq 2$ and let Γ be a subgroup of $\mathbf{SL}_n\mathbb{C}$. Let $\{\gamma_0,\ldots,\gamma_l\}$ be the equivalence classes of irreducible finite-dimensional complex r...

conjecture | open | MathDB 页面 | 原文线索

V2-012754 | MATHDB-325890

The KNS conjecture for Kirillov-Reshetikhin modules in type E

摘录 / EXCERPT: The KNS conjecture. The collection of real numbers $Q^{(i)}_k$ is a solution of the ℓ-restricted Q-system. Moreover, for every $i\in I$:

conjecture | open | MathDB 页面 | 原文线索

V2-012755 | MATHDB-325907

Lefschetz property conjecture for quantum cluster coefficients

摘录 / EXCERPT: Lefschetz property conjecture. Each $a_{\mathbf{n}}(q)$ is a positive integral combination of polynomials $P_{N,k}(q)$, where, for each $\mathbf{n}$, N and the parity of k are...

conjecture | open | MathDB 页面 | 原文线索

V2-012756 | MATHDB-325911

Cygan-Richardson criterion for globally hypoelliptic systems on nilmanifolds

摘录 / EXCERPT: Let N be a nilpotent Lie group of step r with Lie algebra $\mathfrak N$, let $\mathfrak N_j=[\mathfrak N,\mathfrak N_{j-1}]$ for $j=1,\dots,r$, let $\Gamma\setminus N$ be a com...

conjecture | open | MathDB 页面 | 原文线索

V2-012757 | MATHDB-325932

The categorical surjectivity conjecture for the depth-zero Bernstein center

摘录 / EXCERPT: The categorical surjectivity conjecture. This homomorphism is surjective. This conjecture would identify all depth-zero central Bernstein operators with classes arising from the ca...

conjecture | open | MathDB 页面 | 原文线索

V2-012758 | MATHDB-325933

The tame-local-system categorical surjectivity conjecture

摘录 / EXCERPT: Assume that a W-conjugacy class $[\theta]$ of tame local systems on T is Frobenius-invariant, so that it defines a subalgebra $Z_G^{\theta}\subset Z_G^0$ and a Frobenius-equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-012759 | MATHDB-325934

The central-sheaf conjecture for stable Bernstein centers

摘录 / EXCERPT: Assume that θ is a tame local system as in the finite-field situation, $V\in\operatorname{Rep}\check{G}$, and $\pi\in\operatorname{Irr}(G)^\theta$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012760 | MATHDB-325955

The center-fixing noninner automorphism conjecture for finite nonabelian p-groups

摘录 / EXCERPT: Let G be a finite nonabelian p-group, and let $Z(G)$ denote its center. An automorphism leaving the center elementwise fixed acts trivially on every element of $Z(G)$. The cent...

conjecture | open | MathDB 页面 | 原文线索

V2-012761 | MATHDB-325956

Strong Atiyah conjecture on the rationality of von Neumann dimensions

摘录 / EXCERPT: Let Γ be a discrete group, not necessarily amenable, and let $f\in M_{m,n}({\mathbb C}\Gamma)$ for some $m,n\in{\mathbb N}$. Let P_f denote the kernel of the orthogonal pr...

conjecture | open | MathDB 页面 | 原文线索

V2-012762 | MATHDB-325969

The umbral module factoring-through conjecture

摘录 / EXCERPT: Factoring-through conjecture. The $\mathbb{Q}$-graded G^X-module K^X_r is a faithful representation of G^X when $r\equiv0\pmod c$, and factors through $\bar G^X$ otherwise. T...

conjecture | open | MathDB 页面 | 原文线索

V2-012763 | MATHDB-325971

The type-n dual-pair conjecture for umbral modules

摘录 / EXCERPT: Let X be a Niemeier root system with Coxeter number m, and let D be a discriminant of H^X. Dual-pair conjecture. If ... for some integer λ, then the representation...

conjecture | open | MathDB 页面 | 原文线索

V2-012764 | MATHDB-325972

The doublet and sextet conjecture for umbral modules

摘录 / EXCERPT: Let X be a Niemeier root system with Coxeter number m. A representation is a doublet if it is the direct sum of two copies of one representation, and a sextet if it is the dire...

conjecture | open | MathDB 页面 | 原文线索

V2-012765 | MATHDB-326007

Minimal smooth circle actions are Lebesgue-ergodic

摘录 / EXCERPT: Let S^1 be the circle, let G be a finitely generated group acting smoothly and minimally on S^1, and let m denote Lebesgue measure on the circle. Minimal smooth circle-acti...

conjecture | open | MathDB 页面 | 原文线索

V2-012766 | MATHDB-326060

Ancestor vanishing conjecture for ADE surface singularities

摘录 / EXCERPT: Let G be a finite group of ADE type, let ρ be an irreducible two-dimensional representation of G, and let F_{ρ}^{1} be the associated Hodge bundle on...

conjecture | open | MathDB 页面 | 原文线索

V2-012767 | MATHDB-326064

Kac-Wakimoto affine character formula for partially integrable tame modules

摘录 / EXCERPT: Let $\widehat{\mathfrak{g}}$ be an affine Lie superalgebra, let $L(\Lambda)$ be a partially integrable tame $\widehat{\mathfrak{g}}$-module, and let $L^\#$ be the relevant lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-012768 | MATHDB-326091

Constructive recognition conjecture for untwisted Chevalley groups in characteristic 2

摘录 / EXCERPT: Let X be a black box group encrypting one of the untwisted Chevalley groups ${\rm G}(2^n)$, and let $u\in X$ be a given involution. Constructive recognition conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-012769 | MATHDB-326104

Twist invariance and triangular-basis conjecture for quantum cluster algebras

摘录 / EXCERPT: Assume the objects and notation of the preceding quantum cluster isomorphism conjecture, including the basis $\mathbf B_\mathbf{i}$ and the twist isomorphism $\eta_\mathbf{i}$. Twi...

conjecture | open | MathDB 页面 | 原文线索

V2-012770 | MATHDB-326118

The null-vector characterization of ideals in symplectic reflection algebras

摘录 / EXCERPT: Null-vector characterization conjecture. Each ideal of $H_{1,\eta}(A_{n-1})$ is the set of null-vectors of the degenerate bilinear form B_{sp} for some $\varkappa$-trace sp on...

conjecture | open | MathDB 页面 | 原文线索

V2-012771 | MATHDB-326130

Equicontinuous-or-expansive dichotomy for minimal Cantor pseudogroup actions

摘录 / EXCERPT: Let ${\mathcal G}_{{\mathcal F}}$ be a pseudogroup acting minimally on a Cantor space ${\mathfrak{X}}$. The action is equicontinuous-or-expansive dichotomy conjecture. either equic...

conjecture | open | MathDB 页面 | 原文线索

V2-012772 | MATHDB-326142

Families and Kazhdan-Lusztig cells for unequal parameters

摘录 / EXCERPT: Let W be a finite Coxeter group with weight function L. Lusztig's families partition the irreducible representations $\text{Irr}(W)$, and the Kazhdan-Lusztig preorder...

conjecture | open | MathDB 页面 | 原文线索

V2-012773 | MATHDB-326143

Geck's conjecture relating the preorders on irreducible representations

摘录 / EXCERPT: Let W be a finite Coxeter group with weight function L, and let $E,E'\in\text{Irr}(W)$. The preorder $\preceq_L$ is defined using standard representation-theoretic operations,...

conjecture | open | MathDB 页面 | 原文线索

V2-012774 | MATHDB-326169

The finite-order operator K-theory conjecture for finitely embeddable groups

摘录 / EXCERPT: Let G be a countable group. For an element $g\in G$ of finite order d, define ... in $\mathbb{Q}G$, and let $K_0^{\mathrm{fin}}(C^*(G))$ be the subgroup of $K_0(C^*(G))$ generate...

conjecture | open | MathDB 页面 | 原文线索

V2-012775 | MATHDB-326187

Block-theoretic analogue of Glauberman's Z* theorem

摘录 / EXCERPT: Let G be a group, let e be a block of the group algebra $ \mathcal O G$, and let (P,e_P) be an e-subpair of G. Let (D,e_D) be a maximal e-subpair containing (P,e_P)...

conjecture | open | MathDB 页面 | 原文线索

V2-012776 | MATHDB-326194

Translation-groupoid representation conjecture for orbifold groupoids

摘录 / EXCERPT: Translation-groupoid representation conjecture. Every orbifold groupoid can be represented by a translation groupoid with locally free group action.

conjecture | open | MathDB 页面 | 原文线索

V2-012777 | MATHDB-326230

The categorical Gelfand-Tsetlin basis conjecture for type A_n

摘录 / EXCERPT: Let $\boldsymbol{\rho}^s$ denote the functor ρ^s from $R_{n+1}^\boldsymbol{\rho}$-modules to $\boldsymbol{k}$-modules associated with $s\in\mathcal{S}(\lambda)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012778 | MATHDB-326250

Kazhdan-Lusztig conjecture for Verma-module multiplicities

摘录 / EXCERPT: Let W be the Weyl group of a complex semisimple Lie algebra, and let $h_{x,w}$ be the Kazhdan-Lusztig polynomials associated with W. Let category ${\mathcal O}$ be the correspo...

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V2-012779 | MATHDB-326261

Igusa-Liu-Paquette extension conjecture

摘录 / EXCERPT: Let S be a simple module over an artin algebra. Igusa-Liu-Paquette extension conjecture. If ... then ... for infinitely many integers i. This is presented as a stronger statement r...

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V2-012780 | MATHDB-326286

Existence of generalized Macmahon modules by analytic continuation

摘录 / EXCERPT: Fix non-negative integers $l_1,\dots,l_t,l'_1,\dots,l'_t$, and for sufficiently large k let $L_k=(l_1,\dots,l_t,0,\dots,0,l'_t,\dots,l'_1)$ be the vector with k components. Let...

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V2-012781 | MATHDB-326291

The conjecture on a non-rigid CAT(0) group with uncountably many boundaries

摘录 / EXCERPT: Non-rigidity conjecture. The group G will be a non-rigid CAT(0) group with uncountably many boundaries.

conjecture | open | MathDB 页面 | 原文线索

V2-012782 | MATHDB-326334

Finite-dimensionality conjecture for link invariants

摘录 / EXCERPT: Finite-dimensionality conjecture. The invariant $\Phi(L)$ is only finite-dimensional if all components of L are labeled with minuscule representations.

conjecture | open | MathDB 页面 | 原文线索

V2-012783 | MATHDB-326337

Anisotropic extension of the Walpole-Kelvin equivalence criterion

摘录 / EXCERPT: Let G be a subgroup of $\mathrm{SO}(3)$, let $\mathbb{T}$ be a tensor space with its G-representation, and let $\mathrm{C}\in\left[\mathbb{T}\otimes^S\mathbb{T}\right]^G$. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-012784 | MATHDB-326339

Cogrowth asymptotic conjecture for Baumslag-Solitar groups

摘录 / EXCERPT: Let $\mathrm{BS}(N,N)$ be the Baumslag-Solitar group, and let $g_{n,0}$ denote the number of trivial words of length n in its standard generating set. Let A_N and μ_N be th...

conjecture | open | MathDB 页面 | 原文线索

V2-012785 | MATHDB-326351

Pattern-avoiding permutation conjecture for multiplicities of maximal weights

摘录 / EXCERPT: Pattern-avoidance conjecture. The multiplicities satisfy

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V2-012786 | MATHDB-326359

Goodness and separability conjecture for non-CSP S-arithmetic groups

摘录 / EXCERPT: Let Γ be an S-arithmetic group, and suppose that Γ does not have the congruence subgroup property. If the ground field has characteristic zero, also assume this cha...

conjecture | open | MathDB 页面 | 原文线索

V2-012787 | MATHDB-326360

The minor conjecture for Cayley graphs of right groups

摘录 / EXCERPT: Let G be a group, let R_k be a right zero band with k elements, let $S=G\times R_k$, and let C be a generating set. Write $\pi_G(C)$ for the projection of C to G, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-012788 | MATHDB-326364

Hitchin's conjecture for simple Lie algebras

摘录 / EXCERPT: Hitchin's conjecture. For any primitive element $\omega \in P_d \subset \wedge^d(\mathfrak{g}^*)^\mathfrak{g}$, there exists an irreducible subspace $V_\omega\subset \mathfrak{g}$...

conjecture | open | MathDB 页面 | 原文线索

V2-012789 | MATHDB-326404

Benson-Carlson conjecture on free sphere-product actions

摘录 / EXCERPT: Benson-Carlson conjecture. One has $h(G)=r(G)$ for each finite group G. This conjecture relates the minimum number of sphere factors needed for a free action to the largest eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-012790 | MATHDB-326425

Babai-Godsil-Imrich-Lovász conjecture on generic GRRs

摘录 / EXCERPT: Babai-Godsil-Imrich-Lovász conjecture. The proportion of inverse-closed subsets S of R such that $\operatorname{Cay}(R,S)$ is a GRR goes to 1 as $n\to\infty$.

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V2-012791 | MATHDB-326481

Irreducibility conjecture for characteristic-zero Specht filtration factors

摘录 / EXCERPT: Let F be a field of characteristic zero, let $\boldsymbol{\blambda}$ be a partition label, and let $\boldsymbol{\bT}$ range over the equivalence classes $\boldsymbol{\bT}$ of sta...

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V2-012792 | MATHDB-326499

Bubboloni-Praeger-Spiga conjecture for the normal covering number of alternating groups

摘录 / EXCERPT: Normal covering conjecture for A_n.

conjecture | open | MathDB 页面 | 原文线索

V2-012793 | MATHDB-326521

Cellularity conjecture for the mirabolic Hecke algebra

摘录 / EXCERPT: Cellularity conjecture. The elements

conjecture | open | MathDB 页面 | 原文线索

V2-012794 | MATHDB-326545

Finite counterexample to free-product closure for automaton semigroups

摘录 / EXCERPT: Let S and T be finite semigroups, and let $S\star T$ denote their semigroup free product. An automaton semigroup is a semigroup recognized by a finite automaton acting on finit...

conjecture | open | MathDB 页面 | 原文线索

V2-012795 | MATHDB-326546

Non-automaton-monoid conjecture for the wreath product of the nonnegative integers with C2

摘录 / EXCERPT: Let $\mathbb{N}_0$ denote the additive monoid of nonnegative integers, let C_2 denote the cyclic group of order 2, and let $\mathbb{N}_0\wr C_2$ be their wreath product. An aut...

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V2-012796 | MATHDB-326547

The freeness conjecture for two-state reversible Mealy automata

摘录 / EXCERPT: Let a reversible Mealy automaton be a Mealy automaton whose transition maps are permutations, and let the semigroup generated by an automaton be the semigroup generated by its stat...

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V2-012797 | MATHDB-326573

The finiteness conjecture for ghost numbers of group algebras

摘录 / EXCERPT: Let G be a finite group, and let k be a field whose characteristic divides the order of G. The ghost number finiteness conjecture. The ghost number of kG is finite. The con...

conjecture | open | MathDB 页面 | 原文线索

V2-012798 | MATHDB-326579

The additive program-length conjecture for independent register permutations

摘录 / EXCERPT: Let A be an alphabet, let $1\leq k<n$, and let $g\in\mathrm{Sym}(A^k)$ and $h\in\mathrm{Sym}(A^{n-k})$. Define $f\in\mathrm{Sym}(A^n)$ by ... and ... Here $\mathcal{L}(u)$ denotes th...

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V2-012799 | MATHDB-326593

Ponomarev's non-embeddability conjecture for birational groups

摘录 / EXCERPT: Let $\mathcal N$ be the subgroup of the permutation group of $\mathbf Z$ generated by the transposition $\sigma=(1,2)$ and the translation τ defined by $\tau(i)=i+1$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-012800 | MATHDB-326601

Kobayashi's finite multiplicity conjecture for discretely decomposable restrictions

摘录 / EXCERPT: Kobayashi's conjecture. The multiplicity $m_{\tau}(\pi|_{G'})$ is finite for every $\tau\in\hat{G}'$. This is a finite-multiplicity assertion for discretely decomposable restrictio...

conjecture | open | MathDB 页面 | 原文线索

V2-012801 | MATHDB-326602

Januszewski's polynomial multiplicity-growth conjecture

摘录 / EXCERPT: Januszewski's conjecture. In the setting of Kobayashi's conjecture, $m_{\tau}(\pi|_{G'})$ grows at most polynomially in the norm of the infinitesimal character of τ, in the se...

conjecture | open | MathDB 页面 | 原文线索

V2-012802 | MATHDB-326607

Dominance-order conjecture for cuspidal classes of the symmetric group

摘录 / EXCERPT: Let λ and μ be partitions with only odd parts, and let $\mathcal{O}_\lambda$ and $\mathcal{O}_\mu$ be the corresponding cuspidal classes of the symmetric group. The dom...

conjecture | open | MathDB 页面 | 原文线索

V2-012803 | MATHDB-326615

Cohomological induction equivalence conjecture for generalized Harish-Chandra modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional Lie algebra, let $\mathfrak{k}\simeq\mathfrak{sl}(2)$ be a subalgebra, and let $\mathfrak{t}$ be a Cartan subalgebra of $\mathfrak{k}$. F...

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V2-012804 | MATHDB-326643

The flipping criterion for invertibility of morphisms of doubles

摘录 / EXCERPT: Flipping criterion. The morphism ψ is invertible if and only if it is flippable and sends the integral of ${\mathcal D}(G)$ to the integral of ${\mathcal D}(H)$.

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V2-012805 | MATHDB-326675

Chari-Fourier-Sagaki conjecture on generalized Schur positivity

摘录 / EXCERPT: Chari-Fourier-Sagaki conjecture. Under these conditions, there exists a surjective $\mathfrak{g}$-module map

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V2-012806 | MATHDB-326700

Regular-cycle proportion conjecture for primitive permutation groups

摘录 / EXCERPT: Regular-cycle proportion conjecture. There exists an absolute positive constant c such that, for every such (G,Ω) and every $g\in G$,

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V2-012807 | MATHDB-326708

Saito's finiteness conjecture for opposite series of hyperbolic groups

摘录 / EXCERPT: Let G be a group with a finite generating set S, and let $\Omega(G,S)$ be Saito's set associated with the word-growth series of (G,S). Saito's conjecture. If G is a word hy...

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V2-012808 | MATHDB-326718

Characterization conjecture for tensor modules over Mackey Lie algebras

摘录 / EXCERPT: Let V be a countable-dimensional vector space, let $\mathfrak{g}=\mathfrak{sl}(V,V_*)$, $\mathfrak{o}(V)$, or $\mathfrak{sp}(V)$, and let $\mathfrak{g}^M$ be the corresponding Ma...

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V2-012809 | MATHDB-326727

Multiplicity bound conjecture for symmetric pairs

摘录 / EXCERPT: Let (G,H) be a symmetric pair of real reductive groups. Suppose that G is quasi-split, let $B\subset G$ be a Borel subgroup, and let k be the number of open B-orbits on...

conjecture | open | MathDB 页面 | 原文线索

V2-012810 | MATHDB-326728

Multiplicity bound conjecture for generic representations of symmetric spaces

摘录 / EXCERPT: Let (G,H) be a symmetric pair of real reductive groups, with G quasi-split. Let $B\subset G$ be a Borel subgroup, let U be its unipotent radical, and let k be the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-012811 | MATHDB-326796

The property (A) conjecture for finitely generated free groups

摘录 / EXCERPT: Let F_r be a finitely generated free group, and let property (A) denote the property defined earlier in the source. Property (A) conjecture. Finitely generated free groups do not...

conjecture | open | MathDB 页面 | 原文线索

V2-012812 | MATHDB-326853

Minimal growth rate conjecture for Coxeter groups rigid up to diagram twisting

摘录 / EXCERPT: Let W be a Coxeter group rigid up to diagram twisting, meaning that its Coxeter generating systems are related by diagram twists. Let $comega_{\operatorname{Cox}}(W)$ denote the...

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V2-012813 | MATHDB-326897

Carroll-Chindris-Weyman conjecture on moduli spaces of tame algebras

摘录 / EXCERPT: Carroll-Chindris-Weyman conjecture. Let Λ be an algebra. Then the following conditions are equivalent:

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V2-012814 | MATHDB-326904

The finite-group or monomiality conjecture for group-type braided vector spaces

摘录 / EXCERPT: Finite-group or monomiality conjecture. Either G is finite, or there is a basis with respect to which c is monomial.

conjecture | open | MathDB 页面 | 原文线索

V2-012815 | MATHDB-326911

Strong imaginary Mackey conjecture

摘录 / EXCERPT: Let $\lambda,\mu\vDash n$ be compositions, and let M be an ${\mathscr S}_{\mu}$-module. Strong imaginary Mackey conjecture. The restriction of the induction of M decomposes as...

conjecture | open | MathDB 页面 | 原文线索

V2-012816 | MATHDB-326937

Stratification criterion for highest weight categories of graded Lie algebras

摘录 / EXCERPT: Let $\mathcal{I}(\mathfrak{g})$ be the highest weight category associated with a graded Lie algebra $\mathfrak{g}$ admitting an anti-involution. Suppose properties $(f1)$--$(f3)$ h...

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V2-012817 | MATHDB-326954

The Euclidean-space conjecture for negatively curved homogeneous Einstein manifolds

摘录 / EXCERPT: A homogeneous manifold is a manifold equipped with a transitive group action by isometries, and an Einstein manifold has Ricci tensor proportional to its metric. Its scalar curvatu...

conjecture | open | MathDB 页面 | 原文线索

V2-012818 | MATHDB-326984

Bass' conjecture on the Whitehead group

摘录 / EXCERPT: Bass' conjecture. If G is torsion-free, then $\mathit{Wh}(G)=0$. If G has torsion, then the inclusions of finite subgroups induce an injective group homomorphism

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V2-012819 | MATHDB-326989

McKay-Wanless conjecture on near-quadratic automorphism orders of quasigroups

摘录 / EXCERPT: Let $f(n)$ denote the maximum of the orders of automorphisms of quasigroups of order at most n. The preceding discussion gives $f(n)=O(n^2)$. McKay-Wanless conjecture. For every...

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V2-012820 | MATHDB-327036

Undecidability of profinite triviality for hyperbolic groups

摘录 / EXCERPT: Hyperbolic profinite-triviality conjecture. There is no algorithm that can determine whether or not a given hyperbolic group Γ has

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V2-012821 | MATHDB-327079

The maximal stably irreducible family generation conjecture

摘录 / EXCERPT: Fix a Weyl group element $w\in W$, write $m=\ell(w)$ for its length, and let a stably irreducible family be a set of irreducible modules closed under irreducible convolution up to...

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V2-012822 | MATHDB-327161

Rivin's rationality conjecture for spherical conjugacy growth series

摘录 / EXCERPT: Let G be a word hyperbolic group. Its spherical conjugacy growth series is denoted by $\widetilde{\sigma}(G)$. Rivin's conjecture. The series $\widetilde{\sigma}(G)$ is rational...

conjecture | open | MathDB 页面 | 原文线索

V2-012823 | MATHDB-327164

Infinite transitivity conjecture for Calogero-Moser spaces

摘录 / EXCERPT: Infinite transitivity conjecture. For each $n\geq 1$, the action of G on $\mathcal{C}_n$ is infinitely transitive.

conjecture | open | MathDB 页面 | 原文线索

V2-012824 | MATHDB-327165

Collective infinite transitivity conjecture for Calogero-Moser spaces

摘录 / EXCERPT: Collective infinite transitivity conjecture. The action of G on $\mathcal{C}$ is collectively infinitely transitive.

conjecture | open | MathDB 页面 | 原文线索

V2-012825 | MATHDB-327166

Connectedness conjecture for the stabilizers of Calogero-Moser spaces

摘录 / EXCERPT: Connectedness conjecture for the stabilizers. The groups G_n are connected and hence irreducible for all $n\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-012826 | MATHDB-327197

The normal generation bound for the first ℓ^2-Betti number

摘录 / EXCERPT: Let G be a torsionfree discrete group, and let $g_1,\dots,g_k$ be elements whose normal closure is G. The first ℓ^2-Betti number of G is denoted by $\beta_1^{(2)}(G)$. N...

conjecture | open | MathDB 页面 | 原文线索

V2-012827 | MATHDB-327214

Nepomechie-Wang completeness conjecture for highest weight Bethe states

摘录 / EXCERPT: Nepomechie-Wang completeness conjecture. Regular solutions and physical singular solutions of the Bethe ansatz equations provide all the highest weight vectors of $\mathfrak{H}_N$....

conjecture | open | MathDB 页面 | 原文线索

V2-012828 | MATHDB-327243

Kerov's classification conjecture for ergodic central measures over a finite field

摘录 / EXCERPT: Kerov's conjecture. Ergodic central measures on $\mathbb U$ are in one-to-one correspondence with triplets $(\boldsymbol\alpha;\boldsymbol\beta;\gamma)\in\mathbb R^{2\infty+1}$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-012829 | MATHDB-327295

Positivity conjecture for Alvis-Curtis duals of Deligne-Lusztig intersection cohomology

摘录 / EXCERPT: Let ${\mathbf G}$ be a connected reductive algebraic group with Steinberg endomorphism F, let $G={\mathbf G}^F$, let W be its Weyl group, and let Q_w denote the restriction t...

conjecture | open | MathDB 页面 | 原文线索

V2-012830 | MATHDB-327320

Tezuka's cohomology conjecture for loop groups and finite Chevalley groups

摘录 / EXCERPT: Let G be the group considered in the paper, let BG be its classifying space, and let $\mathcal{L}BG$ denote its free loop space. Write $G(\mathbb{F}_q)$ for the corresponding f...

conjecture | open | MathDB 页面 | 原文线索

V2-012831 | MATHDB-327322

Parker-Stewart stabilisation conjecture for the LHS spectral sequence

摘录 / EXCERPT: Parker-Stewart's conjecture. This spectral sequence stabilises at the E_2-page: $E_2^{ij}\cong E_\infty^{ij}$ for all i,j. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-012832 | MATHDB-327330

The centralizer conjecture for normal pairs of fusion systems

摘录 / EXCERPT: Let $((S,\mathcal{F}),(T,\mathcal{E}))$ be a normal pair of fusion systems realized by a normal pair (G,H) of finite groups. Write $O_{p'}(H)$ for the largest normal subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-012833 | MATHDB-327341

The character identity for lifting representations from G to M

摘录 / EXCERPT: Let G be the group with automorphism θ, let M be the associated subgroup, and let T_M be a maximal torus in M invariant under θ. Let ${\rm Nm}:T_M\to T_M$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-012834 | MATHDB-327353

The hyperfocal height-zero conjecture for nilpotent blocks

摘录 / EXCERPT: Let B be a p-block of a finite group with defect group P, let $\mathcal{F}$ be a fusion system of B on P, and let D be the hyperfocal subalgebra of a source algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-012835 | MATHDB-327403

The return-probability conjecture for Hilbertian compression

摘录 / EXCERPT: Let G be a group with a Cayley graph, let $P^{(n)}(e)$ denote the probability that the associated random walk returns to the identity after n steps, and let β denote the...

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V2-012836 | MATHDB-327438

The weight functor equivalence conjecture for Mackey Lie algebras

摘录 / EXCERPT: Let V and V_* be paired vector spaces with dual bases, let $\mathfrak{g}=\mathfrak{sl}(V,V_*)$, and let ${\mathbb T}_{\operatorname{End}(V)}$ and ${\mathbb T}_{\mathfrak{g}}$ d...

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V2-012837 | MATHDB-327439

Universal property conjecture for the restricted tensor category of Mackey Lie algebra modules

摘录 / EXCERPT: Let ${\mathbb T}_{\textsc{res}}$ be the full tensor subcategory of ${\mathfrak{g}^M}$-modules on the objects of ${\mathbb T}_{\operatorname{End}(V)}$. Let $\mathcal C$ be a tensor...

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V2-012838 | MATHDB-327471

The restriction-subquotient conjecture for irreducible W-superalgebra representations

摘录 / EXCERPT: Let $\cal A\simeq W_\chi$ be the image of W_χ in $\bar W_\chi^{\frak s}$, and consider irreducible representations of these associative algebras. Restriction-subquotient conje...

conjecture | open | MathDB 页面 | 原文线索

V2-012839 | MATHDB-327474

Equality of generalized commuting probabilities determines central quotient order

摘录 / EXCERPT: Let G_{1} and G_{2} be finite groups, and let $Pr^{2n}(G)$ denote the generalized commuting probability of G at level $2n$. Write $Z(G)$ for the center of G. The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-012840 | MATHDB-327505

Integral rigidity conjecture for equivariant stable model categories

摘录 / EXCERPT: Let G be a finite group, and let ${\mathscr C}$ be a cofibrantly generated, proper, G-equivariant stable model category. Write Sp_G^O for the category of ortho...

conjecture | open | MathDB 页面 | 原文线索

V2-012841 | MATHDB-327527

Categorical projectors

摘录 / EXCERPT: Categorical projectors. For every nontrivial partition λ,

conjecture | open | MathDB 页面 | 原文线索

V2-012842 | MATHDB-327528

Finite presentation of end terms of almost split sequences

摘录 / EXCERPT: Finite-presentation conjecture. The module Z is finitely presented.

conjecture | open | MathDB 页面 | 原文线索

V2-012843 | MATHDB-327559

Kaplansky's divisibility conjecture for semisimple Hopf algebras

摘录 / EXCERPT: Let H be a semisimple Hopf algebra, and let $\operatorname{Irr} {}_{H}^{}\mathcal{M}$ denote the set of isomorphism classes of irreducible finite-dimensional H-modules. Kaplans...

conjecture | open | MathDB 页面 | 原文线索

V2-012844 | MATHDB-327592

Macdonald's conjecture on regular torus characters of finite groups of Lie type

摘录 / EXCERPT: Macdonald's conjecture. For every such regular character θ, there exists an irreducible representation π_θ of $G({\mathbb F}_q)$ of dimension

conjecture | open | MathDB 页面 | 原文线索

V2-012845 | MATHDB-327624

Deformation conjecture for the W B_l-algebra

摘录 / EXCERPT: Let $U_{q,p}(B_l^{(1)})$ be the elliptic algebra with level-1 currents $e_j(z)$ and $f_j(z)$, and let ${\cal F}_{\lambda,\xi,\kappa}(\Lambda+\mu,\mu)$ denote the modules appearin...

conjecture | open | MathDB 页面 | 原文线索

V2-012846 | MATHDB-327626

Conjecture on joint spectral measures for rank-two graph pairs

摘录 / EXCERPT: Let Γ be a finite subgroup of $GL(2,\mathbb{Z})$, let $(\mathcal{G}^{\Gamma}_{\rho_1},\mathcal{G}^{\Gamma}_{\rho_2})$ be the associated pair of graphs, let $\mathbb{T}^2$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-012847 | MATHDB-327662

Finite-group quotient structure of fibers of wild jet maps

摘录 / EXCERPT: Fiber-structure conjecture. For $n\gg0$, the fiber of p_n over the image of γ is homeomorphic to a quotient of

conjecture | open | MathDB 页面 | 原文线索

V2-012848 | MATHDB-327686

The convex-hull conjecture for the Mathieu conjecture on $SU(2)$

摘录 / EXCERPT: Let $f=\sum_{i=1}^{k}A_i t^{\ell_i}_{m_i,n_i}$ be an $SU(2)$-finite function, with $A_i\neq0$ for every $1\leq i\leq k$. Let C be the closed convex hull of the points...

conjecture | open | MathDB 页面 | 原文线索

V2-012849 | MATHDB-327691

Generalised row-removal homomorphism conjecture for graded Specht modules

摘录 / EXCERPT: Let $\lambda,\mu\in\mathcal{P}^{l}_{n}$, $r\geq 0$, and $1\leq m\leq n$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-012850 | MATHDB-327710

Comparison conjecture for categorical Bockstein sequences

摘录 / EXCERPT: Let k be a complete discrete valuation ring with uniformizing element $l\in k$. Let $c\colon G\longrightarrow k^*$ be a continuous multiplicative character of a profinite group...

conjecture | open | MathDB 页面 | 原文线索

V2-012851 | MATHDB-327719

Triviality of actions of compactly supported diffeomorphism groups on the circle

摘录 / EXCERPT: Let M be a connected C^∞ manifold without boundary, let $r\geq 1$, and denote by ${\rm Diff}^r_c(M)_0$ the identity component of the group of compactly supported C^r dif...

conjecture | open | MathDB 页面 | 原文线索

V2-012852 | MATHDB-327740

Semisimplicity preservation for Harish-Chandra modules

摘录 / EXCERPT: Let $\mathcal{H}(\chi)_{\mathbb O}$ be the Harish-Chandra category associated with a nilpotent orbit $\mathbb O$, and let ${}^\chi \mathcal{H}^K_{\mathbb O}$ be the full subcategor...

conjecture | open | MathDB 页面 | 原文线索

V2-012853 | MATHDB-327777

The mixable Beauville group conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite simple group. A mixable Beauville group is a perfect group with elements $x_1,y_1,x_2,y_2 eq$ necessarily distinct such that $o(x_1)$ and $o(y_1)$ are even,...

conjecture | open | MathDB 页面 | 原文线索

V2-012854 | MATHDB-327873

Full stability-exceptional collections for Dynkin quivers

摘录 / EXCERPT: Let Q be a Dynkin quiver, let $D^b(Q)$ denote its bounded derived category, and let σ be a stability condition in $\mathrm{Stab}(D^b(Q))$. A full σ-exceptional coll...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-012855 | MATHDB-327874

Regularity-preserving orientations for quivers with loop underlying graph

摘录 / EXCERPT: Let Q be a quiver whose underlying graph is a loop, and let $\operatorname{Rep}_k(Q)$ denote its category of representations over k. A category is regularity preserving when it...

conjecture | open | MathDB 页面 | 原文线索

V2-012856 | MATHDB-327885

The Conway-group classification conjecture for symplectic automorphisms of manifolds of $K3^{[n]}$ type

摘录 / EXCERPT: Conway-group classification conjecture. There is a bijective correspondence between finite groups of symplectic automorphisms of manifolds of $K3^{[n]}$ type (for some n) and sub...

conjecture | open | MathDB 页面 | 原文线索

V2-012857 | MATHDB-327887

Classification of indecomposable injective smooth semilinear representations

摘录 / EXCERPT: Let k be a field, let $K=k(\Psi)$ with its standard $\mathop{\mathfrak{S}}\nolimits_{\Psi}$-action, and let $K[\binom{\Psi}{s}]$ denote the smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-012858 | MATHDB-327890

The arbitrary-type affine quasiheredity conjecture for KLR algebras

摘录 / EXCERPT: Let ${\tt C}$ be a generalized Cartan matrix and let α belong to the non-negative part of the corresponding root lattice. The KLR algebra $R_\alpha({\tt C})$ is defined for...

conjecture | open | MathDB 页面 | 原文线索

V2-012859 | MATHDB-327936

Thom's conjecture on discrete free subgroups in the hyperfinite full group

摘录 / EXCERPT: Thom's conjecture. There are no discrete free subgroups of rank 2 in the full group of the hyperfinite ergodic equivalence relation.

conjecture | open | MathDB 页面 | 原文线索

V2-012860 | MATHDB-327937

Stone-von Neumann conjecture for positive representations of the split real quantum Borel subalgebra

摘录 / EXCERPT: Let $\mathcal{U}_{q\widetilde{q}}(\mathfrak{b}_{\mathbb{R}})$ be the split real quantum Borel subalgebra, and let $\mathcal{P}=\mathcal{P}_{\vec{0}}$ denote the positive representa...

conjecture | open | MathDB 页面 | 原文线索

V2-012861 | MATHDB-327938

The decomposition conjecture for Gyoja's W-graph algebra

摘录 / EXCERPT: Decomposition conjecture. The family $(F^\lambda)_{\lambda\in\operatorname{Irr}(W)}$ satisfies all of the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-012862 | MATHDB-327961

Self-orthogonality conjecture for primitive idempotents in Noetherian algebras

摘录 / EXCERPT: Let A be a Noetherian k-algebra which is either semiperfect or positively graded, and let e be a primitive idempotent; in the positively graded case, assume that e has degr...

conjecture | open | MathDB 页面 | 原文线索

V2-012863 | MATHDB-327962

Self-orthogonality conjecture for finite-dimensional algebras with primitive idempotents

摘录 / EXCERPT: Let A be a finite-dimensional k-algebra, let e be a primitive idempotent, let S_e denote the associated simple module, and write $\Gamma=eAe$. Self-orthogonality conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-012864 | MATHDB-327963

Finiteness conjecture for Thompson groups of mock symmetric groups

摘录 / EXCERPT: Let $\Vmock$ be the generalized Thompson group built from the family of mock symmetric groups. Finiteness conjecture. The group $\Vmock$ is of type F_∞. This is exp...

conjecture | open | MathDB 页面 | 原文线索

V2-012865 | MATHDB-327966

The circle extension distinguishing number conjecture

摘录 / EXCERPT: Circle extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012866 | MATHDB-327984

The generalised tau-invariant conjecture for unequal parameter Hecke algebras

摘录 / EXCERPT: Generalised $\widetilde{\tau}^{\Delta}$-invariant conjecture. Two elements $y,w\in W$ belong to the same left cell if and only if y,w belong to the same two-sided cell and y,w...

conjecture | open | MathDB 页面 | 原文线索

V2-012867 | MATHDB-327994

Pyber's conjecture on uniformly quasirandom perfect groups of bounded commutator width

摘录 / EXCERPT: Pyber's conjecture. The class of perfect groups with commutator width at most n is Q.U.P.

conjecture | open | MathDB 页面 | 原文线索

V2-012868 | MATHDB-327999

The rank-one centralizer conjecture for admissible pairs and finite W-algebras

摘录 / EXCERPT: Rank-one centralizer conjecture. The e-admissible pairs are equivalent to one another. In particular, the associated finite W-algebras are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-012869 | MATHDB-328019

The exceptional-groups conjecture for strongly real Beauville groups

摘录 / EXCERPT: Let G be a non-abelian finite simple group. A strongly real Beauville group is a Beauville group admitting a Beauville structure with the strong reality property. Exceptional-gro...

conjecture | open | MathDB 页面 | 原文线索

V2-012870 | MATHDB-328020

Jones's Beauville conjecture for characteristically simple groups

摘录 / EXCERPT: Let G be a finite non-abelian characteristically simple group, meaning that G is isomorphic to H^k for a finite simple group H. A group is 2-generated if it has generators...

conjecture | open | MathDB 页面 | 原文线索

V2-012871 | MATHDB-328021

The direct-square conjecture for strongly real Beauville groups

摘录 / EXCERPT: Let H be a finite simple group of order greater than 3. Direct-square conjecture. The direct product $H\times H$ is a strongly real Beauville group. This is presented as a more...

conjecture | open | MathDB 页面 | 原文线索

V2-012872 | MATHDB-328022

The split-extension conjecture for strongly real Beauville groups

摘录 / EXCERPT: Let G be a simple group and let E be a split extension of G. A Beauville group is a group admitting a Beauville structure, and a strongly real Beauville group is one admittin...

conjecture | open | MathDB 页面 | 原文线索

V2-012873 | MATHDB-328024

The strongly real Beauville conjecture for finite simple groups

摘录 / EXCERPT: A strongly real Beauville group is a group admitting a Beauville structure satisfying the strong reality condition: there are an automorphism and conjugating elements that simultan...

conjecture | open | MathDB 页面 | 原文线索

V2-012874 | MATHDB-328025

The characteristically simple Beauville group conjecture

摘录 / EXCERPT: A characteristically simple group is a finite group isomorphic to H^k, where H is finite simple and k is a positive integer. The characteristically simple Beauville group con...

conjecture | open | MathDB 页面 | 原文线索

V2-012875 | MATHDB-328026

The irreducible cohomology conjecture for the complexes \mathcal X_a and \mathcal X_a^*

摘录 / EXCERPT: Let $\mathbb{T}^+$ be the category of representations under consideration, let $M\in\mathbb{T}^+$ be irreducible, and let $\mathcal X_a(M)$ and $\mathcal X_a^*(M)$ be the complexes...

conjecture | open | MathDB 页面 | 原文线索

V2-012876 | MATHDB-328027

The bicomplex and cone conjecture for \mathcal X_a\circ\mathcal X_b

摘录 / EXCERPT: Let $\mathcal X_a$ and $\mathcal X_b$ be the complexes of endofunctors, with terms $\mathcal X_a(n)$ and $\mathcal X_b(q)$, and let θ'_{ab} and $\theta”_{ab}$ be the morphis...

conjecture | open | MathDB 页面 | 原文线索

V2-012877 | MATHDB-328028

The bicomplex compatibility conjecture for \mathcal Z_{a,b}

摘录 / EXCERPT: Let ... and define ... Here the morphisms in the two displayed short exact sequences are the maps relating $\mathcal Z_{a,b}(n,q)$ to the corresponding compositions of $\mathcal X$- an...

conjecture | open | MathDB 页面 | 原文线索

V2-012878 | MATHDB-328035

Principal-subspace kernel presentation conjecture for higher-level standard modules

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{sl}(n+1)$, let Λ_i be the fundamental weights, and let $\bar{U(\mathfrak{n})}$, $\widetilde{U(\bar{\mathfrak{n}})\bar{\mathfrak{n}}_{+}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-012879 | MATHDB-328100

The rational span conjecture for the Verma module

摘录 / EXCERPT: Let $M(\boldsymbol a)$ be the Verma module, let ${}^{\mathbf R}M(\boldsymbol a)$ be its relevant $\mathbf R_T$-form, and let $\widehat W[\boldsymbol\lambda]$ be the operators index...

conjecture | open | MathDB 页面 | 原文线索

V2-012880 | MATHDB-328107

Equivalence conjecture for the Kirk-Livingston-Wang and quotient string-link representations

摘录 / EXCERPT: Let $\mathrm{pStr}(n)$ be the monoid of pure string links. The Kirk-Livingston-Wang representation is ... and let $p\mathcal W_{n,1/0}$ denote the corresponding quotient of the two-v...

conjecture | open | MathDB 页面 | 原文线索

V2-012881 | MATHDB-328137

Uniqueness conjecture for Markov numbers of rigid strings

摘录 / EXCERPT: Markov-number uniqueness conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-012882 | MATHDB-328146

Universality conjecture for Thompson's group V and coCF groups

摘录 / EXCERPT: Universality conjecture for V. The classes $\mathcal{V}$ and $co\mathcal{CF}$ are equal:

conjecture | open | MathDB 页面 | 原文线索

V2-012883 | MATHDB-328198

The power-of-two conjecture for separating class sums of W_n

摘录 / EXCERPT: Let W_n be the group considered above, and let a separating set be a collection of class sums whose eigenvalues distinguish the irreducible representations of the regular represe...

conjecture | open | MathDB 页面 | 原文线索

V2-012884 | MATHDB-328263

Wide-equivalence conjecture for the branching functor

摘录 / EXCERPT: Wide-equivalence conjecture. The functor Π^{λ} is injective on objects and therefore a wide equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-012885 | MATHDB-328275

Two-sided-cell conjecture for integral quantizations

摘录 / EXCERPT: Let $\mathcal{D}$ be an integral quantization of a conical symplectic resolution, let $\mathcal{I}$ index the simple objects of its category $\mathcal{O}$, and define two indices t...

conjecture | open | MathDB 页面 | 原文线索

V2-012886 | MATHDB-328276

Serre-functor conjecture for long twisting and shuffling functors

摘录 / EXCERPT: Let λ be an integral parameter and let ζ be a generic cocharacter. Define the long twist and long shuffle by ... These are endofunctors of the relevant bounded derived...

conjecture | open | MathDB 页面 | 原文线索

V2-012887 | MATHDB-328280

The O_2 Rokhlin-dimension conjecture for compact Lie group actions

摘录 / EXCERPT: The $\mathcal{O}_2$ Rokhlin-dimension conjecture. The action α has the Rokhlin property.

conjecture | open | MathDB 页面 | 原文线索

V2-012888 | MATHDB-328281

The action dimension conjecture for L^2-homology

摘录 / EXCERPT: For a discrete group G, let $L^2H_i(G)$ denote the L^2-homology of a contractible G-space, and let $\operatorname{actdim}(G)$ be the least dimension of a contractible manifol...

conjecture | open | MathDB 页面 | 原文线索

V2-012889 | MATHDB-328282

The cocompact action dimension conjecture for L^2-homology

摘录 / EXCERPT: For a discrete group G, let $L^2H_i(G)$ denote the L^2-homology of a contractible G-space, and let $\operatorname{cadim}(G)$ be the least dimension of a contractible G-mani...

conjecture | open | MathDB 页面 | 原文线索

V2-012890 | MATHDB-328283

The manifold form of the cocompact action dimension conjecture

摘录 / EXCERPT: Let $(M,\partial M)$ be an n-manifold with contractible components that admits a geometric group action, and let $L^2H_i(M)$ denote its L^2-homology. Manifold cocompact action...

conjecture | open | MathDB 页面 | 原文线索

V2-012891 | MATHDB-328331

Conjecture on Type 2 and Type 4 involution classes in finite orthogonal groups

摘录 / EXCERPT: Let $\operatorname{SO}(n,k)$ be a finite orthogonal group with n even. For Type 2 and Type 4 k-involutions, let $C_2(n,k)$ and $C_4(n,k)$ denote the numbers of isomorphy classe...

conjecture | open | MathDB 页面 | 原文线索

V2-012892 | MATHDB-328332

Conjecture on the number of involution isomorphy classes in symplectic groups

摘录 / EXCERPT: Let k be a field, let $n>2$, and let $C_i(2n,k)$ denote the number of isomorphy classes of involutions of Type i in $\operatorname{SP}(2n,k)$, for $1\leqslant i\leqslant 4$. Le...

conjecture | refuted | MathDB 页面 | 原文线索

V2-012893 | MATHDB-328339

Chenevier-Clozel's spectral expansion conjecture for a twisted orbital integral

摘录 / EXCERPT: Let G^0 be the group in the preceding setup, let θ_0 be its twisting automorphism, and let I_{δ_0} be the θ_0-twisted stabilizer of δ_0. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012894 | MATHDB-328370

Similarity classes over finite quotients and module isomorphism classes

摘录 / EXCERPT: Let R be a discrete valuation ring with residue field $\mathbf F_q$, let P be its maximal ideal, and let $M_m(R/P^n)$ denote the ring of $m\times m$ matrices over R/P^n. An...

conjecture | open | MathDB 页面 | 原文线索

V2-012895 | MATHDB-328375

Absorption conjecture for crossed products by compact group actions

摘录 / EXCERPT: Let G be a second-countable compact group, let A be a separable unital C^*-algebra, and let $\alpha\colon G\to\operatorname{Aut}(A)$ be an action of G on A with finite Ro...

conjecture | open | MathDB 页面 | 原文线索

V2-012896 | MATHDB-328381

Generating-function conjectures for spherical representations of types D and C

摘录 / EXCERPT: For a simple root system R, let $\mathfrak{L}_{\nu}(R)$ denote the irreducible spherical representation attached to R. Generating-function conjecture. The following identities...

conjecture | open | MathDB 页面 | 原文线索

V2-012897 | MATHDB-328425

Ringel's conjecture on indecomposable pure injective modules over domestic string algebras

摘录 / EXCERPT: Let A be a domestic string algebra, and let M be an infinite-dimensional indecomposable pure injective A-module. An infinite-dimensional band module means a Pr"ufer, adic, or...

conjecture | open | MathDB 页面 | 原文线索

V2-012898 | MATHDB-328509

Cyclic Sylow lifting conjecture

摘录 / EXCERPT: Let G be a finite group with a cyclic p-Sylow subgroup, and let $k[[z]]/k[[t]]$ be a G-extension. Cyclic Sylow lifting conjecture. For every such extension, the KGB obstructi...

conjecture | open | MathDB 页面 | 原文线索

V2-012899 | MATHDB-328522

Compatibility conjecture for ordinary and modular Harish-Chandra series in finite unitary groups

摘录 / EXCERPT: Let G be the finite unitary group and let X_μ,X_ν be the unipotent kG-modules labelled by partitions $\mu,\nu\in\mathcal{P}_n$, while Y_μ,Y_ν are the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-012900 | MATHDB-328523

2-core conjecture for weakly cuspidal unipotent modules

摘录 / EXCERPT: Let X_λ be the unipotent kG-module labelled by a partition $\lambda\in\mathcal{P}_n$, and let e be the parameter defining the relevant block-theoretic core. An e-core...

conjecture | open | MathDB 页面 | 原文线索

V2-012901 | MATHDB-328524

Finite-rank crystal-graph comparison conjecture for finite unitary groups

摘录 / EXCERPT: Let t be a non-negative integer, set $r=t(t+1)/2$ and $\iota=r\pmod 2$, and let $\widetilde{\mathcal{G}}_{\iota,q,\ell}^{t,\leq d}$ be the induced subgraph of the reindexed Haris...

conjecture | open | MathDB 页面 | 原文线索

V2-012902 | MATHDB-328525

Uglov crystal characterization conjecture for weak Harish-Chandra series

摘录 / EXCERPT: Let $\lambda\in\mathcal{P}$, let $t\in\mathbb{N}$ satisfy $\lambda_{(2)}=\Delta_t$, and set $\mu=\bar{\lambda}^{(2)}$. Assume that ℓ is sufficiently large, that e is odd, an...

conjecture | open | MathDB 页面 | 原文线索

V2-012903 | MATHDB-328527

The linear independence conjecture for Poincaré series of Mitchell-Priddy summands

摘录 / EXCERPT: Let L_λ be the indecomposable injective unstable module indexed by a partition λ, and let $P_\lambda(q)$ denote its Poincaré series. Linear independence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-012904 | MATHDB-328528

The eigenvalue and diagonalisation conjecture for Lannes' T-functor

摘录 / EXCERPT: Let $K^{red}(\mathcal U)$ be the Grothendieck ring of finite direct sums of indecomposable reduced injective unstable modules, and let $K^{red}_n(\mathcal U)$ be the subgroup gener...

conjecture | open | MathDB 页面 | 原文线索

V2-012905 | MATHDB-328551

The conjecture that virtual specialness is undecidable for non-positively curved cube complexes

摘录 / EXCERPT: Virtual specialness conjecture. Virtual specialness is an undecidable property of non-positively curved cube complexes.

conjecture | open | MathDB 页面 | 原文线索

V2-012906 | MATHDB-328560

The silting form of the Auslander-Reiten conjecture

摘录 / EXCERPT: Let k be a field, let A be a finite-dimensional k-algebra, and let $\mathsf{D}^{\rm b}(\mathsf{mod}\ A)$ be its bounded derived category. An object is presilting if it has no...

conjecture | open | MathDB 页面 | 原文线索

V2-012907 | MATHDB-328561

The Grothendieck-group form of the Tachikawa conjecture

摘录 / EXCERPT: Let k be a field, let A be a finite-dimensional k-algebra, and let n be the number of pairwise non-isomorphic simple A-modules. Let $\mathsf{D}^{\rm b}(\mathsf{mod}\ A)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-012908 | MATHDB-328562

The stable-category form of the Tachikawa conjecture

摘录 / EXCERPT: Let A be a finite-dimensional self-injective algebra, and let $\underline{\mathsf{mod}}A$ denote its stable module category. The Tachikawa conjecture. The stable category...

conjecture | open | MathDB 页面 | 原文线索

V2-012909 | MATHDB-328586

Recursive construction of central idempotents in partition algebras

摘录 / EXCERPT: Let D_r and D'_r be the partition-algebra objects containing the rank-one central idempotents $E(r)$ and $E'(r)$, respectively. Let h_i, s_i, and p_i denote the generator...

conjecture | open | MathDB 页面 | 原文线索

V2-012910 | MATHDB-328601

The kernel-intersection conjecture for Green correspondents of endotrivial modules

摘录 / EXCERPT: Kernel-intersection conjecture. One should have

open_problem | open | MathDB 页面 | 原文线索

V2-012911 | MATHDB-328661

Measure invariance of dimensions of polynomial-growth harmonic functions

摘录 / EXCERPT: Let $\mathbb{G}$ be a finitely generated group, and let μ and ν be courteous probability measures on $\mathbb{G}$. Measure-invariance conjecture. For every k, the dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-012912 | MATHDB-328665

Protasov-Slobodianiuk conjecture on partitions of infinite groups

摘录 / EXCERPT: Protasov-Slobodianiuk conjecture. Every infinite group G of cardinality $\varkappa$ has a partition as above such that

conjecture | open | MathDB 页面 | 原文线索

V2-012913 | MATHDB-328671

Prasad's conjecture on distinction of the Steinberg representation

摘录 / EXCERPT: Let F be a nonarchimedean local field with finite residual field, let E/F be a Galois quadratic extension, and let G be a connected reductive group defined over F. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012914 | MATHDB-328685

Simple-spectrum conjecture for boundary shift-of-argument algebras

摘录 / EXCERPT: Let $\fg$ be a semisimple Lie algebra and let the real points of the De Concini-Procesi compactification parameterize the corresponding commutative subalgebras of $U(\fg)$. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-012915 | MATHDB-328693

Jordan-cell rank conjecture for the vacuum tilting module

摘录 / EXCERPT: Jordan-cell rank conjecture. In the scaling limit of $\mathcal{T}_{0,\mathfrak{q}^2}$, states from limits of irreducible $\mathcal{X}_{j,P}$ subquotients should be involved in Jord...

conjecture | open | MathDB 页面 | 原文线索

V2-012916 | MATHDB-328696

Uniform compactness conjecture for normalized matrix coefficients on real spherical spaces

摘录 / EXCERPT: Let $Z=G/H$ be a real spherical space with compression cone A_Z^-. Let V be a Harish-Chandra module for $(\mathfrak{g},K)$, let V^∞ be its moderate-growth smooth globali...

conjecture | open | MathDB 页面 | 原文线索

V2-012917 | MATHDB-328702

The equivalence conjecture for CM finiteness, stable tilting, and derived dimension

摘录 / EXCERPT: Let $(R,\mathbb{L})$ be a Geigle-Lenzing complete intersection, let d be the dimension parameter, and let $A^{\rm CM}$ be its CM endomorphism algebra. A d-tilting object is an...

conjecture | open | MathDB 页面 | 原文线索

V2-012918 | MATHDB-328703

The tensor-product characterization of the Fano condition

摘录 / EXCERPT: Let $n\geq 3$ and let $p_1,\ldots,p_n\geq 2$ be integers. Set $\Lambda=\bigotimes_{i=1}^n k\mathbb{A}_{p_i-1}$. The tensor-product Fano conjecture. The following conditions are equ...

conjecture | open | MathDB 页面 | 原文线索

V2-012919 | MATHDB-328704

The Fano-VB finiteness conjecture for Geigle-Lenzing projective spaces

摘录 / EXCERPT: Let $\mathbb{X}$ be a Geigle-Lenzing projective space and let d be its dimension parameter. A d-tilting bundle on $\mathbb{X}$ is a tilting bundle whose endomorphism algebra ha...

conjecture | open | MathDB 页面 | 原文线索

V2-012920 | MATHDB-328706

Generalized Foulkes-type plethysm inequality

摘录 / EXCERPT: Let m,n be positive integers, let $ u$ be a partition of n, and let $ u'$ denote its conjugate partition. For a partition τ of mn, write $p^{\tau}_{\nu,(m)}$ for the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-012921 | MATHDB-328707

Newell-type plethysm equality

摘录 / EXCERPT: Let m,n be positive integers, let ν be a partition of n, and let ν' denote its conjugate partition. For a partition τ of mn, write $p^{\tau}_{\nu,(m)}$ for the c...

conjecture | open | MathDB 页面 | 原文线索

V2-012922 | MATHDB-328708

Two-step Newell-type plethysm equality

摘录 / EXCERPT: Let m,n be positive integers and let ν be a partition of n. For a partition τ of mn, write $p^{\tau}_{\nu,(m)}$ for the corresponding plethysm coefficient. Two-step...

conjecture | open | MathDB 页面 | 原文线索

V2-012923 | MATHDB-328739

The infinite-group partition conjecture for covering numbers

摘录 / EXCERPT: Partition conjecture. The group G can be partitioned into countably many subsets

conjecture | open | MathDB 页面 | 原文线索

V2-012924 | MATHDB-328742

Conjecture on algebraic Hilbert series for bounded symmetric-matrix modules

摘录 / EXCERPT: Bounded-module algebraicity conjecture. Every bounded $\operatorname{Sym}(\operatorname{Sym}^2(\mathbb{C}^{\infty}))$-module has an algebraic Hilbert series; equivalently, $H_M(t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-012925 | MATHDB-328754

Non-exact dimensionality of quasiconformal measures under divergence of the h-log series

摘录 / EXCERPT: Let X be a hyperbolic metric space and let $G\leq\operatorname{Isom}(X)$ be a geometrically finite group such that, for some $p\in\mathcal{L}^p$, the series associated with the h...

conjecture | open | MathDB 页面 | 原文线索

V2-012926 | MATHDB-328756

Table-locus complete-intersection conjecture

摘录 / EXCERPT: Fix $Q=(u,u-r)$ with $u>r\ge 2$, put $B=J_Q$, and let $\mathcal U_B$ be the maximal nilpotent subalgebra of the centralizer of B. For a partition P, let...

conjecture | open | MathDB 页面 | 原文线索

V2-012927 | MATHDB-328777

Ashrafi's conjecture on the chromatic number of full exponent group power graphs

摘录 / EXCERPT: Let G be a finite group. Its exponent is the least common multiple of the orders of its elements; G is a full exponent group if it contains an element whose order equals the ex...

conjecture | open | MathDB 页面 | 原文线索

V2-012928 | MATHDB-328784

Benjamini's conjecture on ends of traces of branching random walks

摘录 / EXCERPT: Benjamini's conjecture. The trace of a transient BRW on G has infinitely many ends.

conjecture | open | MathDB 页面 | 原文线索

V2-012929 | MATHDB-328791

Homological stability for automorphism groups of direct products

摘录 / EXCERPT: Let H and G be finitely generated groups, and let $F:U\! f\mathcal{G}\to \mathbb{Z}\operatorname{-Mod}$ be a coefficient system of finite degree. For each n, consider the hom...

conjecture | open | MathDB 页面 | 原文线索

V2-012930 | MATHDB-328803

Weyl-group invariance conjecture for geometric Eisenstein series

摘录 / EXCERPT: Weyl-group invariance conjecture. For every $w\in W$ and every $\mathcal S\in \operatorname{D}_{\zeta a}(\widetilde\operatorname{Bun}_T)^{reg}$, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-012931 | MATHDB-328827

Finiteness conjecture for LERF groups with Kazhdan property (T)

摘录 / EXCERPT: Let G be a countable group with Kazhdan property (T). Finiteness conjecture. The group G is LERF if and only if it is finite. By the preceding proposition, for groups with prop...

conjecture | open | MathDB 页面 | 原文线索

V2-012932 | MATHDB-328864

The Singer conjecture for Poincaré duality groups

摘录 / EXCERPT: Let Γ be an n-dimensional Poincaré duality group, and let $b_i^{(2)}(\Gamma)$ denote its ith ℓ^2-Betti number. Singer conjecture. The ℓ^2-Betti numbers vanish...

conjecture | open | MathDB 页面 | 原文线索

V2-012933 | MATHDB-328865

The finite-dimensional polynomial-growth harmonic functions conjecture for finitely generated groups

摘录 / EXCERPT: Let G be an infinite group that is either elementary amenable or non-amenable, and let μ be a symmetric, finitely supported generating probability measure on G. The space o...

conjecture | open | MathDB 页面 | 原文线索

V2-012934 | MATHDB-328897

Predicted socles and cosocles of exotic standard and costandard objects

摘录 / EXCERPT: Let G be the reductive group and let $\boldsymbol{X}$ be its weight lattice. For $lambda \in \boldsymbol{X}$, write $\operatorname{dom}(\lambda)$ for the dominant representative,...

conjecture | open | MathDB 页面 | 原文线索

V2-012935 | MATHDB-328914

Recognition-by-spectrum conjecture for finite simple classical groups

摘录 / EXCERPT: Recognition-by-spectrum conjecture. Every finite group isospectral to L is isomorphic to some group G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-012936 | MATHDB-328938

Derksen's degree-bound conjecture for syzygies of invariant rings

摘录 / EXCERPT: Let G be a finite group and V a finite-dimensional representation over a field k. Set $B=k[V]$, let $R=B^G$, and let I be the ideal of B generated by R_+. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012937 | MATHDB-328948

The PBW realization conjecture for the affine \(\mathfrak{sp}_{2n}\) basic representation

摘录 / EXCERPT: PBW realization conjecture. There is an isomorphism of $\mathfrak{sp}_{2n}^a\otimes t^{-1}\mathbb{C}[t^{-1}]$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-012938 | MATHDB-328956

Conjecture on the representation structure of electrical Lie algebras of type D

摘录 / EXCERPT: Let $\mathfrak{sp}_{2n}$ be the symplectic Lie algebra, let V_{ν} be its standard representation, let $V_{\mathbf{0}}$ be its trivial representation, and let V_{λ} be t...

conjecture | open | MathDB 页面 | 原文线索

V2-012939 | MATHDB-328988

The easy-representability conjecture for finitely presented qsf groups

摘录 / EXCERPT: Easy-representability conjecture. All finitely presented qsf groups are easy, that is, easily representable.

conjecture | open | MathDB 页面 | 原文线索

V2-012940 | MATHDB-329030

Finiteness conjecture for characters yielding universal mass formulas

摘录 / EXCERPT: Finiteness conjecture. For any finite group Γ (or at least any p-group), there are only finitely many characters χ for which the counting function c_χ has a univ...

conjecture | open | MathDB 页面 | 原文线索

V2-012941 | MATHDB-329059

The lower-order finite p-group cycle map conjecture

摘录 / EXCERPT: Cycle map conjecture. The map cl is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-012942 | MATHDB-329126

Finite separation conjecture for groupoid terms

摘录 / EXCERPT: Finite separation conjecture. Whenever two groupoid terms can be separated in an infinite groupoid, they can also be separated in a finite groupoid.

conjecture | open | MathDB 页面 | 原文线索

V2-012943 | MATHDB-329142

The finiteness-property conjecture for kernels in Lodha--Moore groups

摘录 / EXCERPT: Let $m\in\mathbb{N}\cup\{\infty\}$, let ψ be the character considered in the surrounding discussion, and let the ambient group be one of the Lodha--Moore groups G, G_y,...

conjecture | open | MathDB 页面 | 原文线索

V2-012944 | MATHDB-329155

The generalized 2-slope conjecture for minimal functions

摘录 / EXCERPT: Let $\pi\colon \mathbb R\to\mathbb R$ be a minimal function, meaning a minimal valid function for the single-row problem, and suppose that π is absolutely continuous and that i...

conjecture | open | MathDB 页面 | 原文线索

V2-012945 | MATHDB-329157

Kumar's rectangular partition conjecture for the Foulkes-Howe map

摘录 / EXCERPT: For vector spaces V, let $S^d(S^nV)$ and $S^n(S^dV)$ be the indicated symmetric powers, and let ... be the natural $\operatorname{GL}(V)$-module map. For a partition (d^n), write...

conjecture | open | MathDB 页面 | 原文线索

V2-012946 | MATHDB-329165

The finite-group or monomiality conjecture for group-type BVSs

摘录 / EXCERPT: Let (V,c) be a group-type BVS, and let $G=\langle g_i\rangle$ be the group generated by its associated operators. Assume that c is unitary and has finite order. Finite-group or...

conjecture | open | MathDB 页面 | 原文线索

V2-012947 | MATHDB-329170

Finiteness conjecture for orbit groups of finite automaton groups

摘录 / EXCERPT: Let G be a finite automaton group, and let $G_{\mathcal O}$ denote the orbit group associated with an orbit $\mathcal O$ of G. Orbit-group finiteness conjecture. For each finit...

conjecture | open | MathDB 页面 | 原文线索

V2-012948 | MATHDB-329172

Residual nilpotence conjecture for the groups $\widetilde{U_n}$ and $\widetilde{L_n}$

摘录 / EXCERPT: For $n\geq 3$, let $\widetilde{U_n}$ and $\widetilde{L_n}$ denote the groups defined by the corresponding lifted upper and lower triangular palindromic automorphism presentations;...

conjecture | open | MathDB 页面 | 原文线索

V2-012949 | MATHDB-329189

Aliniaeifard-Li-Nicholson conjecture on finite morphic p-groups

摘录 / EXCERPT: Aliniaeifard-Li-Nicholson conjecture. The only finite, morphic p-groups are the abelian, homocyclic p-groups, and the nonabelian group of order p^3 and exponent p, for an o...

conjecture | open | MathDB 页面 | 原文线索

V2-012950 | MATHDB-329240

Automorphic twisting conjecture for cohomological cuspidal representations

摘录 / EXCERPT: Let G be the reductive group in the source, let K be the chosen model of a maximal compact subgroup, and let M be a rational G-representation. For a factorizable automorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-012951 | MATHDB-329241

Galois invariance of cuspidal multiplicities

摘录 / EXCERPT: Let G be the reductive group in the source, let K be the chosen model of a maximal compact subgroup, and let Π be an irreducible cohomological cuspidal automorphic represen...

conjecture | open | MathDB 页面 | 原文线索

V2-012952 | MATHDB-329247

Promotion of pseudo-fibering conjecture

摘录 / EXCERPT: Promotion of pseudo-fibering conjecture. There are three possibilities: G is elementary; G is a Fuchsian group; or G is a closed hyperbolic 3-manifold group.

conjecture | open | MathDB 页面 | 原文线索

V2-012953 | MATHDB-329248

Convergence-group conjecture for mapping torus subgroups

摘录 / EXCERPT: Mapping torus convergence-group conjecture. For any p-A-like element ϕ of G, the mapping torus subgroup G_ϕ acts on S^2 as a convergence group.

conjecture | open | MathDB 页面 | 原文线索

V2-012954 | MATHDB-329249

Uniform convergence conjecture for mapping torus subgroups

摘录 / EXCERPT: Uniform convergence conjecture. This action is a uniform convergence group.

conjecture | open | MathDB 页面 | 原文线索

V2-012955 | MATHDB-329253

Conjecture on generic distinguished representations of general linear groups

摘录 / EXCERPT: Let F be the underlying field, and let τ be an irreducible generic representation of $\mathrm{GL}_n(F)$. A representation is distinguished when it has the distinguishedness...

conjecture | open | MathDB 页面 | 原文线索

V2-012956 | MATHDB-329263

Frobenius-series conjecture for the families A, B, and C

摘录 / EXCERPT: Let $\mathcal{A}=\{x_{1j}^d:1\leq j\leq n\}$, $\mathcal{B}=\{x_{1,i}^d-x_{1,j}^d:1\leq i<j\leq n\}$, and $\mathcal{C}=\{\prod_{a\in A}x_{1,a}:A\subseteq[n],\lvert A\rvert=d\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-012957 | MATHDB-329266

Completeness conjecture for degree-four and degree-five polarization modules

摘录 / EXCERPT: Let f be a homogeneous diagonally symmetric polynomial of degree 4 or 5, and let $\mathcal{M}_f$ be its polarization module. Completeness conjecture. Up to n-exceptions, th...

conjecture | open | MathDB 页面 | 原文线索

V2-012958 | MATHDB-329282

Truncated Weyl module-fusion product isomorphism conjecture

摘录 / EXCERPT: Let P^+ be the set of dominant integral weights. For $\lambda\in P^+$ and $N\geq1$, let $\operatorname{W}(\lambda,N)$ be the truncated Weyl module for...

conjecture | open | MathDB 页面 | 原文线索

V2-012959 | MATHDB-329414

Degree-and-type conjecture for representations of code loops

摘录 / EXCERPT: Degree-and-type conjecture. If V_{1} and V_{2} have the same degree and type, then they are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-012960 | MATHDB-329456

One-endedness conjecture for strongly aperiodic subshifts

摘录 / EXCERPT: Let G be an infinite, finitely generated group. A subshift of finite type over G is strongly aperiodic if it contains no periodic points, where a periodic point has nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-012961 | MATHDB-329478

Pride's folk conjecture on one-relator groups with torsion

摘录 / EXCERPT: A one-relator group with torsion is a group with a presentation $G=\langle X;R^n\rangle$, where R is not a proper power and $n>1$. Pride's folk conjecture. Apart from the complic...

conjecture | open | MathDB 页面 | 原文线索

V2-012962 | MATHDB-329479

Pride's JSJ conjecture for one-relator groups with torsion

摘录 / EXCERPT: Let G be a one-relator group with torsion that does not admit a free group as a free factor. Since one-relator groups with torsion are hyperbolic, G has a JSJ-decomposition, me...

conjecture | open | MathDB 页面 | 原文线索

V2-012963 | MATHDB-329507

The minimal-dimension conjecture for finite W-superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a basic Lie superalgebra over $\mathbb{C}$, let $e\in\mathfrak{g}_{\bar 0}$ be nilpotent, and set $d_1=\dim\mathfrak{g}_{\bar 1}-\dim\mathfrak{g}^e_{\bar 1}$....

conjecture | open | MathDB 页面 | 原文线索

V2-012964 | MATHDB-329547

Primeness conjecture for von Neumann algebras of groups in NC_1

摘录 / EXCERPT: Let $\Gamma\in NC_1$ and let $L(\Gamma)$ be its corresponding von Neumann algebra. For a nonzero projection $p\in L(\Gamma)$, let $B,C\subseteq pL(\Gamma)p$ be diffuse commuting vo...

conjecture | open | MathDB 页面 | 原文线索

V2-012965 | MATHDB-329558

The determinant formula conjecture for the matrices X_d

摘录 / EXCERPT: Determinant formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-012966 | MATHDB-329576

Stembridge's conjecture on stability of Kronecker coefficients

摘录 / EXCERPT: Let (α,β,γ) be a triple of partitions with $|\alpha|=|\beta|=|\gamma|$ and $g_{\alpha,\beta,\gamma}>0$. Such a triple is stable if, for every triple of partitions...

conjecture | open | MathDB 页面 | 原文线索

V2-012967 | MATHDB-329588

Converse root subgroup factorization conjecture for loop group elements

摘录 / EXCERPT: Converse root subgroup factorization conjecture. If g_1 is in the identity connected component of the sets in (I.1) and (I.3), then the converse holds: g_1 has a root subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-012968 | MATHDB-329598

Conjecture on the growth rates of simple and Verma modules in category O

摘录 / EXCERPT: Let $lambda\in\Lambda$, where $Lambda$ is the weight set indexing the simple modules $L(\lambda)$ and Verma modules $M(\lambda)$ in category ${\mathcal O}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-012969 | MATHDB-329616

The non-hyperbolicity conjecture for tessellation graphs with convex tiles

摘录 / EXCERPT: Non-hyperbolicity conjecture. Every tessellation graph of $\mathbb R^2\simeq\mathbb C$ with convex tiles is non-hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-012970 | MATHDB-329621

Stable presentation length equals half the stable complexity

摘录 / EXCERPT: Let M be a finite-volume hyperbolic 3-manifold. Write $T_{\infty}(M)$ for its stable presentation length and $\sigma_{\infty}(M)$ for its stable complexity. Stable presentation...

conjecture | open | MathDB 页面 | 原文线索

V2-012971 | MATHDB-329656

Marked-polytope invariance conjecture for nice (2,1)-presentations

摘录 / EXCERPT: Let G be a group admitting a nice $(2,1)$-presentation $\pi=\langle x,y\mid r\rangle$, and let $\mathcal{M}_\pi\subset H_1(G;\mathbb{R})$ be its marked polytope. Two subsets are...

conjecture | open | MathDB 页面 | 原文线索

V2-012972 | MATHDB-329678

The MIF conjecture for lacunary hyperbolic groups

摘录 / EXCERPT: Let G be a lacunary hyperbolic group, meaning a finitely generated group with at least one asymptotic cone that is an $\mathbb R$-tree. Suppose that G is not virtually cyclic a...

conjecture | open | MathDB 页面 | 原文线索

V2-012973 | MATHDB-329700

Peckness conjecture for edge posets of Boolean algebra quotients

摘录 / EXCERPT: Peckness conjecture. The edge poset of the quotient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-012974 | MATHDB-329783

Anderson-Mirković conjecture for the MV-polytope crystal structure

摘录 / EXCERPT: Anderson-Mirković conjecture. For every MV polytope P and every $j\in I$, $AM_j(P)$ is an MV polytope and

conjecture | refuted | MathDB 页面 | 原文线索

V2-012975 | MATHDB-329802

The exceptional Lie algebra trace-generation conjecture

摘录 / EXCERPT: Let G be one of the exceptional Lie groups, let (V,π) be respectively the 7-, 26-, 27-, 56-, or 248-dimensional representation for G_2, F_4, E_6, E_7, or...

conjecture | open | MathDB 页面 | 原文线索

V2-012976 | MATHDB-329814

Gateva-Ivanova's multipermutation conjecture for finite square-free solutions

摘录 / EXCERPT: Gateva-Ivanova's conjecture. Every finite nondegenerate involutive square-free set-theoretic solution $r:X\times X\to X\times X$ such that $|X|\geq2$ is a multipermutation solution...

conjecture | open | MathDB 页面 | 原文线索

V2-012977 | MATHDB-329838

Form-factor formulas for partial zero modes in GL(N)-invariant models

摘录 / EXCERPT: Let $T^{(1)}_{ij}[0]$ be the partial zero modes in a ${\rm GL}(N)$-invariant model. Bethe vectors and dual Bethe vectors are parametrized by sets $\bar t^k$ and cardinalities a_k...

conjecture | open | MathDB 页面 | 原文线索

V2-012978 | MATHDB-329863

The negative Houghton-group Sigma-invariant conjecture

摘录 / EXCERPT: Let $(H_n)_{n\in\mathbb{N}}$ be the Houghton groups. For $n\geq 2$, let $\chi_1,\dots,\chi_n$ be the characters defined by the eventual translation lengths, with...

conjecture | open | MathDB 页面 | 原文线索

V2-012979 | MATHDB-329874

Strong periodicity conjecture for finitely generated groups

摘录 / EXCERPT: Strong periodicity conjecture. A group is strongly periodic if and only if it is virtually cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-012980 | MATHDB-329875

Isaacs's fake degree conjecture for algebra groups

摘录 / EXCERPT: Let J be a finite-dimensional nilpotent algebra over a finite field F, let $G=1+J$ be the associated algebra group, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-012981 | MATHDB-329878

The fixed-cocharacter conjecture for strong involutions

摘录 / EXCERPT: Let G be a reductive group, let δ_0 be the specified automorphism, and let ξ be a strong involution for G with dominant regular integral infinitesimal cocharacter...

conjecture | open | MathDB 页面 | 原文线索

V2-012982 | MATHDB-329962

The standard Q-Koszul conjecture for forced gradings of modular representation algebras

摘录 / EXCERPT: Standard Q-Koszul conjecture. The algebra $\widetilde{\operatorname{gr}} A$ is standard Q-Koszul.

conjecture | open | MathDB 页面 | 原文线索

V2-012983 | MATHDB-329963

Conjecture II on Ext-dimension comparison for algebraic and quantum groups

摘录 / EXCERPT: Conjecture II. For all $\lambda,\mu\in\Gamma$ and all $n\geq 0$, the following equalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-012984 | MATHDB-329981

The generating-number conjecture for the quaternion group Q_8

摘录 / EXCERPT: Let k be a field of characteristic 2, and let Q_8 be the quaternion group of order 8. The generating number of the group algebra kQ_8 is the least generating length occur...

conjecture | open | MathDB 页面 | 原文线索

V2-012985 | MATHDB-329982

The generating-number conjecture for A_4

摘录 / EXCERPT: Let k be a field of characteristic 2, and let A_4 be the alternating group on four letters. The generating number of the group algebra kA_4 is the least generating length o...

conjecture | open | MathDB 页面 | 原文线索

V2-012986 | MATHDB-330003

The element-order divisibility matching conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group of order n, let C_n be the cyclic group of order n, and write $o(x)$ for the order of an element x. Element-order divisibility matching conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-012987 | MATHDB-330014

Tensor-product closure of positive representations

摘录 / EXCERPT: Let $\mathcal{P}_\lambda$ and $\mathcal{P}_\mu$ be positive representations of the split real quantum group $\mathcal{U}_{q\widetilde{q}}(\mathfrak{g}_{\mathbb{R}})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-012988 | MATHDB-330057

Wiegold's normal generation conjecture for finitely generated perfect groups

摘录 / EXCERPT: Let G be a finitely generated perfect group, meaning $G=[G,G]$, the commutator subgroup of G. Wiegold's normal generation conjecture. G can be normally generated by a single...

conjecture | open | MathDB 页面 | 原文线索

V2-012989 | MATHDB-330069

The genuine equivariant K-theory block decomposition conjecture

摘录 / EXCERPT: Genuine equivariant K-theory block decomposition conjecture. There is an equivalence of stable ∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-012990 | MATHDB-330077

The coherence and sectional-curvature conjecture for Coxeter groups

摘录 / EXCERPT: Let G be a Coxeter group with standard 2-complex X, and let $\overline X$ be the compression of a finite cover of X corresponding to a finite-index torsion-free subgroup. S...

conjecture | open | MathDB 页面 | 原文线索

V2-012991 | MATHDB-330095

Haase's face-subgroup conjecture for permutation polytopes

摘录 / EXCERPT: Haase's face-subgroup conjecture. If $H\leq G$ and $P(H)\preceq P(G)$ is a face, then there is a partition

conjecture | open | MathDB 页面 | 原文线索

V2-012992 | MATHDB-330169

The density principle for orbital integrals

摘录 / EXCERPT: Let $S(F_0)$ be the symmetric space and let $\mathfrak{s}(F_0)$ be its Lie algebra. Consider the orbital-integral distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-012993 | MATHDB-330174

Cartan-type non-maximal-height bound conjecture

摘录 / EXCERPT: Let $\mathfrak{X}(n,\underline{m})$ denote the minimal p-envelope of a simple Cartan-type Lie algebra $\mathsf{X}(n,\underline m)^{(2)}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-012994 | MATHDB-330183

Degree bound conjecture for transfer quotients of modular invariant rings

摘录 / EXCERPT: Let G be a finite group, let $\b5$ be a field of characteristic $p>0$, let V be a $\b5 G$-module, and let P be a Sylow-p-subgroup of G. For each subgroup $H\leq G$, let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-012995 | MATHDB-330223

The OD-characterizability conjecture for alternating groups

摘录 / EXCERPT: Let H be a finite group, and let $h_{\rm OD}(H)$ denote the number of isomorphism classes of groups G such that $|G|=|H|$ and ${\rm D}(G)={\rm D}(H)$. A group is OD-characteriz...

conjecture | open | MathDB 页面 | 原文线索

V2-012996 | MATHDB-330237

The Markov-invariance conjecture for the relative weight grading in symplectic Khovanov cohomology

摘录 / EXCERPT: Let $L_{\wp_{\circ}}$ be the distinguished Lagrangian submanifold used to define symplectic Khovanov cohomology, and let $\beta\times\mathrm{id}$ be a braid-group action on it. The...

conjecture | open | MathDB 页面 | 原文线索

V2-012997 | MATHDB-330254

The vanishing conjecture for Loewy-length-three summands

摘录 / EXCERPT: Let $E=(\mathbb{Z}/p)^2$ and let kE be its group algebra. For a kE-module M of constant Jordan type, let $\mathfrak{K}$ and $\mathfrak{I}$ denote the functors used to form th...

conjecture | open | MathDB 页面 | 原文线索

V2-012998 | MATHDB-330283

Cohomological dimension and boundary cohomology conjecture for relatively hyperbolic pairs

摘录 / EXCERPT: Let G be torsion-free and hyperbolic relative to a collection $\mathcal{H}$. Let $\operatorname{cd}(G,\mathcal{H})$ be the cohomological dimension of the pair $(G,\mathcal{H})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-012999 | MATHDB-330301

Classification conjecture for admissible anti-automorphisms of affine Hecke algebras

摘录 / EXCERPT: Classification conjecture. If κ is an admissible involutive anti-automorphism, then:

conjecture | open | MathDB 页面 | 原文线索

V2-013000 | MATHDB-330362

Algebraicity conjecture for archimedean Rankin-Selberg functionals

摘录 / EXCERPT: Let F be a totally real number field, let $n\geq 3$ in the Rankin-Selberg case, let χ be a critical quasi-character, and let ... be the non-zero equivariant functional obtaine...

conjecture | open | MathDB 页面 | 原文线索

V2-013001 | MATHDB-330381

Gukov-Stošic conjecture on symmetric and antisymmetric coloured HOMFLY homology

摘录 / EXCERPT: Let K be a knot, and let $H^{\Lambda^r}_{i,j,*}(K)$ and $H^{S^r}_{i,j,*}(K)$ denote, respectively, its antisymmetric- and symmetric-coloured HOMFLY homology groups, with indices...

conjecture | open | MathDB 页面 | 原文线索

V2-013002 | MATHDB-330398

Dual-pair conjecture for representations of type n

摘录 / EXCERPT: Let $H^{(\ell)}$ be the relevant class-number-valued object, let $G^{(\ell)}$ be the associated group, and let $K^{(\ell)}_{r,-D/(4\ell)}$ be the representation indexed by the disc...

conjecture | open | MathDB 页面 | 原文线索

V2-013003 | MATHDB-330402

Translation principle conjecture for the Dirac index

摘录 / EXCERPT: Translation principle conjecture. Assume that the $W_{\frak{g}}$-representation σ_K generated by the Weyl dimension formula for $\frak{k}$ corresponds to a nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-013004 | MATHDB-330429

The measure-independence conjecture for finite-dimensional harmonic functions

摘录 / EXCERPT: Let G be a finitely generated group, and let μ range over probability measures on G satisfying the standing assumptions. Write $H^k(G,\mu)$ for the space of μ-harmonic...

conjecture | open | MathDB 页面 | 原文线索

V2-013005 | MATHDB-330430

Ash-Pollack's cuspidal homology dimension conjecture for \(\operatorname{GL}_3(\mathbb Z)\)

摘录 / EXCERPT: Let g be even, let $\Psi_{(2g,g,0)}$ be the irreducible representation of $\operatorname{GL}_3(\mathbb Z)$ defined from the standard representation, and let s_w denote the dime...

conjecture | open | MathDB 页面 | 原文线索

V2-013006 | MATHDB-330478

Weak covering conjecture for infinite Cayley graphs

摘录 / EXCERPT: Let Γ be an infinite Cayley graph that is not quasi-isometric to $\mathbf Z$. A graph Δ is a cover of Γ if there is a covering map from Γ to Δ; w...

conjecture | refuted | MathDB 页面 | 原文线索

V2-013007 | MATHDB-330492

Wiltshire-Gordon's cochain-complex conjecture for Whitney representations

摘录 / EXCERPT: Let $n\geq 2$, and let $\{W_n^\bullet\}$ be the indicated sequence of S_n-representations. Wiltshire-Gordon's conjecture. Fixing $n\geq 2$, the S_n-representations...

conjecture | open | MathDB 页面 | 原文线索

V2-013008 | MATHDB-330493

Sharp stabilization conjecture for rank-selected homology of the partition lattice

摘录 / EXCERPT: Let Π_n be the partition lattice, let $S\subset\{1,2,\ldots,n-2\}$, and set $i=\max(S)$. Let $\beta_S(\Pi_n)$ denote the rank-selected homology S_n-representation. Sharp stab...

conjecture | open | MathDB 页面 | 原文线索

V2-013009 | MATHDB-330495

Quasi-polynomiality conjecture for cohomology of finitely generated \(\mathrm{FI}_d\)-modules

摘录 / EXCERPT: Let $\bk$ be an arbitrary field. Let $\FI_d$ be the category whose objects are finite sets and whose morphisms are injections with d-coloring of the complement: a morphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-013010 | MATHDB-330502

The age conjecture for counting G-fields

摘录 / EXCERPT: Let K be a number field, let G be a finite group, and let V be a finite-dimensional faithful representation of G over K. Let $N_{G,V,K}(B)$ count isomorphism classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-013011 | MATHDB-330504

The generalized Malle conjecture for linear group actions

摘录 / EXCERPT: Let K be a number field and let G act faithfully and linearly on $V=\mathbb{A}_K^d$, with the associated map $\overline{V}\to\overline{X}=\overline{V}/G$ étale in codimension o...

conjecture | open | MathDB 页面 | 原文线索

V2-013012 | MATHDB-330557

Polynomial-growth obstruction to amenability for left reversible semigroups

摘录 / EXCERPT: Let S be a left reversible semigroup if any two principal left ideals intersect, and call S finitely generated when a finite subset generates it. Its growth is polynomial when...

conjecture | open | MathDB 页面 | 原文线索

V2-013013 | MATHDB-330589

Unbounded rank super-expanders conjecture

摘录 / EXCERPT: A super-expander is a sequence of finite Cayley graphs whose expansion persists with respect to every uniformly convex Banach space, with suitable finite generating sets. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013014 | MATHDB-330601

Filtering subcategories stabilization conjecture for artinian algebras

摘录 / EXCERPT: Let A be an artinian algebra, and let ... be inclusions of subcategories satisfying (G3). Suppose that $n\geq 0$ and that either (G1) holds and every $M\in\mathcal{C}$ admits an ep...

conjecture | open | MathDB 页面 | 原文线索

V2-013015 | MATHDB-330613

The conjecture on type I groups acting on trees

摘录 / EXCERPT: Type I conjecture. Under these assumptions, G is a type I group.

conjecture | open | MathDB 页面 | 原文线索

V2-013016 | MATHDB-330620

Finite-presentability conjecture for generalised Wilson groups

摘录 / EXCERPT: A generalised Wilson group is an infinitely iterated wreath product of finite non-abelian simple permutation groups of the type considered in the paper. A profinite group is finite...

conjecture | open | MathDB 页面 | 原文线索

V2-013017 | MATHDB-330653

Pin's generalized synchronizing automaton conjecture

摘录 / EXCERPT: Let $A=(Q,\Sigma)$ be a finite automaton with $n=|Q|$ states. Its rank r is the minimum rank of its input words, where the rank of an input word is the cardinality of the image o...

conjecture | open | MathDB 页面 | 原文线索

V2-013018 | MATHDB-330674

Strong diameter conjecture for direct powers of finite groups

摘录 / EXCERPT: Let G be a finite group, let G^n denote its n-th direct power, and let $D(G^n)$ denote the diameter with respect to a generating set, using words in that set. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013019 | MATHDB-330675

Weak diameter conjecture for direct powers of finite groups

摘录 / EXCERPT: Let G be a finite group, let G^n denote its n-th direct power, and let A be a generating set of minimum size for G^n. Write ${\mathrm{rank}}(G)$ for the minimum size of a...

conjecture | open | MathDB 页面 | 原文线索

V2-013020 | MATHDB-330684

The non-arithmetic hyperbolic reflection group dimension conjecture

摘录 / EXCERPT: Non-arithmetic dimension conjecture. Theorem 1.1 is true without the arithmeticity assumption: there are no hyperbolic reflection groups in dimensions $n\geq 30$.

conjecture | open | MathDB 页面 | 原文线索

V2-013021 | MATHDB-330691

Effective constructibility of planar Cayley graphs

摘录 / EXCERPT: Let ${\mathcal P}$ be a general planar presentation, let $\operatorname{Cay}({\mathcal P})$ denote its Cayley graph, and let $n\in{\mathbb N}$. Effective constructibility conjectur...

conjecture | solved | MathDB 页面 | 原文线索

V2-013022 | MATHDB-330698

Higher-rank Casson-Lin linking-number conjecture for two-component links

摘录 / EXCERPT: Let $L=\ell_1\cup\ell_2$ be a two-component link in S^3, and let $h_{N,a}(L)$ denote its $SU(N)$ Casson-Lin invariant for an allowable tuple a. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013023 | MATHDB-330700

The locally discrete groups conjecture for analytic circle actions

摘录 / EXCERPT: Let $G\leq \operatorname{Diff}_+^\omega(\mathbf{S}^1)$ be a finitely generated locally discrete group. A locally discrete group's structure conjecture predicts both that the action...

conjecture | open | MathDB 页面 | 原文线索

V2-013024 | MATHDB-330730

Virtual solvability of amenable groups with the ascending chain condition

摘录 / EXCERPT: Let G be a finitely generated amenable residually finite group satisfying the ascending chain condition on closed subgroups. Virtual solvability conjecture. The group G is virt...

conjecture | open | MathDB 页面 | 原文线索

V2-013025 | MATHDB-330756

The noncommutative Shephard-Todd-Chevalley conjecture for AS regular algebras

摘录 / EXCERPT: Let A be an AS regular algebra, and let G be a finite group of graded automorphisms of A. A reflection of A is an automorphism of the kind defined in the surrounding discus...

conjecture | open | MathDB 页面 | 原文线索

V2-013026 | MATHDB-330783

The word-image conjecture for the projective linear group

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, let $w(x_1,\dots,x_m)$ be a nontrivial group word, and let $\operatorname{PSL}_2(K)$ denote the projective linear gr...

conjecture | open | MathDB 页面 | 原文线索

V2-013027 | MATHDB-330784

The degree-zero word reformulation for PSL_2

摘录 / EXCERPT: Degree-zero word reformulation. The image of w on G must be

conjecture | open | MathDB 页面 | 原文线索

V2-013028 | MATHDB-330794

The indecomposability conjecture for projective functors acting on simple modules

摘录 / EXCERPT: Indecomposability conjecture. The module $theta L$ is either indecomposable or zero.

conjecture | open | MathDB 页面 | 原文线索

V2-013029 | MATHDB-330835

Countability and countable-sum continuity for finite-dimensional commutative groupoid algebras

摘录 / EXCERPT: Countability and continuity conjecture. For every finite groupoid H and finite-dimensional commutative H-algebra A, $KK^H(A,B)$ is countable for all $B\in KK^H$ and commutes...

conjecture | open | MathDB 页面 | 原文线索

V2-013030 | MATHDB-330852

Distinct central characters for cuspidal pairs on a nilpotent orbit

摘录 / EXCERPT: Let G be any connected reductive group and suppose that $\boldsymbol{k}$ is big enough for G. For a fixed nilpotent orbit $\mathscr{O}$, consider the cuspidal pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-013031 | MATHDB-330853

Non-polynomiality conjecture for conjugacy classes of unitriangular groups

摘录 / EXCERPT: Let $U_n(q)$ be the group of upper triangular $n\times n$ matrices over the finite field $\mathbb F_q$ with ones on the diagonal, and let $k(U_n(q))$ denote its number of conjugacy...

conjecture | open | MathDB 页面 | 原文线索

V2-013032 | MATHDB-330854

Kirillov's lower-bound conjectures for small-field unitriangular groups

摘录 / EXCERPT: For $n\geq 1$, let $U_n(q)$ be the group of upper triangular $n\times n$ matrices over $\mathbb F_q$ with ones on the diagonal, and let $k(U_n(q))$ denote its number of conjugacy c...

conjecture | open | MathDB 页面 | 原文线索

V2-013033 | MATHDB-330863

Transitivity conjecture for the algebraic vector adelic Grassmannian actions

摘录 / EXCERPT: Transitivity conjecture. The group generated by Γ^{alg} and $(\Gamma')^{alg}$ acts transitively on $\mathcal{C}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-013034 | MATHDB-330878

The embeddability conjecture for diagram groups and right-angled Artin groups

摘录 / EXCERPT: Let $D(\mathcal{P},w)$ be a diagram group, and let $A(\mathcal{P},w)$ denote the associated right-angled Artin group. A diagram group is embeddable into a right-angled Artin group...

conjecture | open | MathDB 页面 | 原文线索

V2-013035 | MATHDB-330916

The hyperbolic Möbius-like action conjecture for closed hyperbolic 3-manifold groups

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold, and let $\pi_1(M)$ act on the circle S^1. An action is hyperbolic Möbius-like if every element is conjugate to a hyperbolic Möbius tr...

conjecture | open | MathDB 页面 | 原文线索

V2-013036 | MATHDB-330924

Well-definedness conjecture for the functions Δ_A

摘录 / EXCERPT: Well-definedness conjecture. The functions Δ_A are well-defined for every $d\geq 1$ and every $A\in M_d$, meaning that their values are independent of the specific admissibl...

conjecture | open | MathDB 页面 | 原文线索

V2-013037 | MATHDB-330928

Low-weight rotational eigenvector conjecture for 3^G subfactor planar algebras

摘录 / EXCERPT: Let G be a finite group, let $\cP_\bullet$ be a 3^G subfactor planar algebra, and for each $g\in G\setminus\{1\}$ let p_g be the projection in $\cP_{4,+}$ corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-013038 | MATHDB-330929

Yang-Baxter presentation conjecture for reduced 3^G subfactor planar algebras

摘录 / EXCERPT: Let G be a non-trivial finite group, let $\cP_\bullet$ be a 3^G subfactor planar algebra, and let $\cR_\bullet$ be its reduced subfactor planar algebra. A Yang-Baxter planar al...

conjecture | open | MathDB 页面 | 原文线索

V2-013039 | MATHDB-330930

Izumi's group-abelianity conjecture for 3^G subfactor planar algebras

摘录 / EXCERPT: Let G be a non-trivial finite group, and let a 3^G subfactor planar algebra be a subfactor planar algebra whose principal graph is a $3^{|G|}$ spoke graph. Let θ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-013040 | MATHDB-330931

Two-box generation conjecture for reduced 3^G subfactor planar algebras

摘录 / EXCERPT: Let G be a non-trivial finite group, let $\cR_\bullet$ be the reduced subfactor planar algebra of a 3^G subfactor planar algebra, and let $\cQ_\bullet$ be the planar subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-013041 | MATHDB-330932

Existence of a primitive ergodic element in a treeable equivalence relation

摘录 / EXCERPT: Primitive-element conjecture. There exists an ergodic element $f \in [\mathcal{R}]$ such that the subequivalence relation generated by f is primitive in ${\mathcal{R}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-013042 | MATHDB-330976

Chi-map isomorphism conjecture for wreath product algebras

摘录 / EXCERPT: For each $m\in\mathbb{N}$, let χ_m denote the algebra map defined in the paper. Chi-map isomorphism conjecture. The map χ_m is an isomorphism of algebras for all...

conjecture | open | MathDB 页面 | 原文线索

V2-013043 | MATHDB-330979

The algebraic Erdős-Ko-Rado conjecture for finite 2-transitive groups

摘录 / EXCERPT: Let G be a finite 2-transitive group. For each coset of a point stabiliser in G, let its characteristic vector be the corresponding element of the group algebra $\mathbb{C}G$...

conjecture | open | MathDB 页面 | 原文线索

V2-013044 | MATHDB-331025

Ordered-triangular extension conjecture for irreducible three-strand braid-group representations

摘录 / EXCERPT: Let ρ be an irreducible d-dimensional matrix representation of the braid group $\mathcal B_3$, with $\rho(\sigma_1)=A$ and $\rho(\sigma_2)=B$. The matrices are in ordered tr...

conjecture | open | MathDB 页面 | 原文线索

V2-013045 | MATHDB-331079

Inductive construction conjecture for discrete series of metaplectic and quaternionic orthogonal groups

摘录 / EXCERPT: Let $A_{p,q}(k)$ denote a middle-dimensional cohomologically induced representation associated with a Levi subgroup $\widetilde{U}(p,q)$ of $Mp_{2n}(\mathbb R)$ or with a Levi subg...

conjecture | open | MathDB 页面 | 原文线索

V2-013046 | MATHDB-331105

Classification of direct products of finite Coxeter groups as Beauville groups

摘录 / EXCERPT: Let K_1 and K_2 be finite Coxeter groups. A Beauville group is a finite group admitting a Beauville structure, and a Beauville structure is strongly real if it is inverted by a...

open_problem | open | MathDB 页面 | 原文线索

V2-013047 | MATHDB-331112

The enhanced local Langlands correspondence for inner forms

摘录 / EXCERPT: Let F be a local nonarchimedean field, let $(G,alpha)$ be an inner twist of a quasi-split F-group, let $\operatorname{Irr}(G)$ be the irreducible smooth representations of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-013048 | MATHDB-331113

The ABPS conjecture for Bernstein components

摘录 / EXCERPT: Let G be a reductive p-adic group, let $\mathfrak{s}=[L,\omega]_G$ be an inertial equivalence class, let $T_{\mathfrak{s}}$ be the corresponding complex torus, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013049 | MATHDB-331138

Cosocle decomposition conjecture for simple modules over cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Let A be a simple $R(\nu)$-module in $\operatorname{Rep}^{\Lambda_i}$ with $|\nu|\geq 1$. Let $r(A)$ and $\mathcal{R}(A)$ be as in the preceding theorem for trivial modules, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013050 | MATHDB-331154

Conjecture on the cosocle construction for simple cyclotomic KLR modules

摘录 / EXCERPT: Cosocle construction conjecture. Under the same hypotheses as the existence theorem,

conjecture | open | MathDB 页面 | 原文线索

V2-013051 | MATHDB-331227

Antolín-Ciobanu conjecture on formal growth in acylindrically hyperbolic groups

摘录 / EXCERPT: Let G be a finitely generated acylindrically hyperbolic group, meaning a group acting non-elementarily and acylindrically by isometries on a hyperbolic space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013052 | MATHDB-331231

The weakly aperiodic SFT characterization for finitely generated groups

摘录 / EXCERPT: Weak-aperiodicity conjecture. G has no weakly aperiodic SFT if and only if it is virtually cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-013053 | MATHDB-331232

The Geometric Gersten conjecture for translation-like actions

摘录 / EXCERPT: Geometric Gersten conjecture. If G is a finitely generated group that is not word-hyperbolic, then some Baumslag-Solitar group acts translation-like on G.

conjecture | open | MathDB 页面 | 原文线索

V2-013054 | MATHDB-331233

The one-relator translation-like action conjecture

摘录 / EXCERPT: One-relator action conjecture. Some nontrivial one-relator group acts translation-like on G.

conjecture | open | MathDB 页面 | 原文线索

V2-013055 | MATHDB-331258

Representation-category conjecture for superselection sectors of the quantum double model

摘录 / EXCERPT: Let G be a finite group. Let $4\Delta4$ be the category of localised and transportable endomorphisms of the quantum double model, and let $4\operatorname{Rep}_f \mathcal{D}(G)4$...

conjecture | open | MathDB 页面 | 原文线索

V2-013056 | MATHDB-331284

AGT conjecture for genus-one Liouville conformal blocks

摘录 / EXCERPT: AGT conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-013057 | MATHDB-331317

Ostrik's semisimplicity conjecture for quantum doubles

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, and let $U_q(\mathfrak{g})$ be its quantized enveloping algebra with quantum double $D(U_q(\mathfrak{g}))$. A finite-dimensional rep...

conjecture | open | MathDB 页面 | 原文线索

V2-013058 | MATHDB-331318

Ostrik's classification conjecture for simple quantum-double modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra. Let V_λ and V_μ denote the simple $U_q(\mathfrak{g})$-modules of dominant integral weights λ and μ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013059 | MATHDB-331319

Semisimplicity conjecture for finite-dimensional quantum-double modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra. Semisimplicity conjecture. The category of finite-dimensional $D(U_q(\mathfrak{g}))$-modules is semisimple. The source attributes th...

conjecture | open | MathDB 页面 | 原文线索

V2-013060 | MATHDB-331320

Ostrik's classification conjecture using the modules $V_{\lambda,\mu}$

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, let V_λ and V_μ be the simple $U_q(\mathfrak{g})$-modules of dominant integral weights λ and μ, and define .....

conjecture | open | MathDB 页面 | 原文线索

V2-013061 | MATHDB-331321

Local finiteness of highest-weight bivectors for semisimple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra and let G be the corresponding simply-connected algebraic group. Consider the $D(U_q(\mathfrak{g}))$-bimodule ... A highest-weight bi...

conjecture | open | MathDB 页面 | 原文线索

V2-013062 | MATHDB-331322

Peter-Weyl decomposition conjecture for the locally finite quantum-double bimodule

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra and G its corresponding simply-connected algebraic group. Let $\mathcal{H}_{\lambda,\mu}$ denote the bimodule constituents indexed...

conjecture | open | MathDB 页面 | 原文线索

V2-013063 | MATHDB-331323

Generation conjecture for the fundamental-weight summands

摘录 / EXCERPT: Fundamental-weight generation conjecture. The sum

conjecture | open | MathDB 页面 | 原文线索

V2-013064 | MATHDB-331324

Generator conjecture for the $\mathfrak{sl}_2$ bimodules $\mathcal{H}_{\lambda,\mu}$

摘录 / EXCERPT: Bimodule generation conjecture. The $D(H)$-bimodule $\mathcal{H}_{\lambda,\mu}$ is generated by $h_{\lambda,\mu}$. The conjecture is based on the examples computed in the paper, an...

conjecture | open | MathDB 页面 | 原文线索

V2-013065 | MATHDB-331348

The three-factor one-relator product non-triviality conjecture

摘录 / EXCERPT: Let A, B, and C be non-trivial groups, and let G be the quotient of their free product by the normal closure of one relator. Three-factor non-triviality conjecture. A one-r...

conjecture | open | MathDB 页面 | 原文线索

V2-013066 | MATHDB-331349

The commutativity conjecture for the identity component of a pseudofinite group's Bohr compactification

摘录 / EXCERPT: Commutativity conjecture. If G is a pseudofinite group, then $(bG)^{0}$ is commutative.

conjecture | open | MathDB 页面 | 原文线索

V2-013067 | MATHDB-331373

Wilton's inclusion/exclusion Hanna Neumann conjecture

摘录 / EXCERPT: Wilton's inclusion/exclusion Hanna Neumann conjecture. For all such H and K,

conjecture | open | MathDB 页面 | 原文线索

V2-013068 | MATHDB-331379

Mirabolic Schur-Weyl decomposition for $\mathrm{MT}_\mathbf{v}(2,d)$

摘录 / EXCERPT: Mirabolic Schur-Weyl decomposition. The decomposition of $\mathrm{MT}_\mathbf{v}(2,d)$ as a bimodule for $MU_\mathbf{v}(2,d)$ and R_d should be

conjecture | open | MathDB 页面 | 原文线索

V2-013069 | MATHDB-331380

The KKK Schur-Weyl duality conjecture for Dynkin quivers of type E

摘录 / EXCERPT: Let Q be a Dynkin quiver of finite type E_n. The functor $\mathcal{F}_Q^{(1)}$ is determined by $(U_q'(E_n^{(1)}),\phi_Q^{-1})$. KKK Schur-Weyl duality conjecture. The functor...

conjecture | open | MathDB 页面 | 原文线索

V2-013070 | MATHDB-331464

Localness and extension conjecture for Dolbeault cohomology homomorphisms

摘录 / EXCERPT: Localness and extension conjecture. Any continuous G'-homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013071 | MATHDB-331550

Lehnert's universality conjecture for Thompson's group V

摘录 / EXCERPT: Let V be R. Thompson's group, and call a group a $co\mathcal{CF}$ group if and only if it embeds as a finitely generated subgroup of $QAut(\mathscr{T}_{2,c})$. Lehnert's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-013072 | MATHDB-331551

Fang-Li-Wang-Xu conjecture on cubic graphical regular representations

摘录 / EXCERPT: Let G be a finite nonabelian simple group. A cubic graphical regular representation (GRR) of G is a Cayley graph $\mathrm{Cay}(G,S)$ of valency three whose automorphism group i...

conjecture | open | MathDB 页面 | 原文线索

V2-013073 | MATHDB-331552

Finiteness conjecture for finite simple groups without cubic GRRs

摘录 / EXCERPT: Let G range over finite nonabelian simple groups, and call G cubic-GRR-free if it has no cubic graphical regular representation. Finiteness conjecture. There are only finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-013074 | MATHDB-331576

Density conjecture for periodic derived orbit categories

摘录 / EXCERPT: Density conjecture. For each positive integer n, the embedding Δ is dense if and only if Λ is piecewise hereditary.

conjecture | open | MathDB 页面 | 原文线索

V2-013075 | MATHDB-331639

Gromov's quadratic Dehn function conjecture for higher-rank arithmetic groups

摘录 / EXCERPT: Let $\mathbf{G}$ be the algebraic group over the global field K and let S be the specified finite set of places, with ring of S-integers $\mathcal{O}_S$. Write ... for the geom...

conjecture | open | MathDB 页面 | 原文线索

V2-013076 | MATHDB-331665

Hodge-theoretic Koszul duality conjecture for Harish-Chandra sheaves

摘录 / EXCERPT: Let $G_{\mathbb R}$ be a real group with Cartan decomposition $\mathfrak g=\mathfrak k\oplus\mathfrak p$, let $\mathcal M_{\rm Ho}$ be the full subcategory of K-equivariant Hodge...

conjecture | open | MathDB 页面 | 原文线索

V2-013077 | MATHDB-331697

BGG resolution formula for $L_c((2n),\emptyset)$

摘录 / EXCERPT: Let $W=B_{2n}=G(2,1,2n)$ and let the parameter $c=1/(2n)$ be constant. Write Δ_i and Δ_{2n-i} for the standard-module sums specified in the preceding theorem, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013078 | MATHDB-331698

Algorithmic inverse decomposition-matrix conjecture for the principal block

摘录 / EXCERPT: Let B_0 be the principal block, let Λ_λ and $\bar{\Gamma}_\lambda$ be the poset and graph subsets defined by the algorithm, and let $d(\mu,\lambda)$ be the grading...

conjecture | open | MathDB 页面 | 原文线索

V2-013079 | MATHDB-331709

The converse to stability implying p-stability for connected real symmetric pairs

摘录 / EXCERPT: Converse stability conjecture. If (G,H,θ) is p-stable, then (G,H,θ) is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-013080 | MATHDB-331710

The naive kernel conjecture for rigid cocenter-representation duality

摘录 / EXCERPT: Let $\tilde H$ be an extended affine Hecke algebra over an algebraically closed field $\mathbf k$ of positive characteristic, with rigid cocenter...

conjecture | open | MathDB 页面 | 原文线索

V2-013081 | MATHDB-331711

The rigid determinant conjecture for extended affine Hecke algebras

摘录 / EXCERPT: Let $\tilde H$ be an extended affine Hecke algebra with generic parameter $\mathbf q$. Its rigid determinant is a polynomial ... For each parahoric subalgebra H_K, let $H_K^\sharp$...

conjecture | open | MathDB 页面 | 原文线索

V2-013082 | MATHDB-331725

The caret-partition tensor square conjecture

摘录 / EXCERPT: Let $m\geq 1$, and define the caret partition of $3m^2$ by ... Identify partitions of n with irreducible representations of S_n, and interpret $\gamma_m^{\otimes 2}$ as the tenso...

conjecture | open | MathDB 页面 | 原文线索

V2-013083 | MATHDB-331754

Bonnafé-Kemper conjecture on minimal generators of the invariant-theoretic kernel

摘录 / EXCERPT: Let $\mathbb{F}$ be the underlying field, let V be the representation and V^* its dual, and let G act on $V\oplus V^*$. Let S be the algebra and let ... be the algebra epimor...

conjecture | open | MathDB 页面 | 原文线索

V2-013084 | MATHDB-331777

Small representations for Kac-Moody groups

摘录 / EXCERPT: Small-representations conjecture. Kac-Moody groups possess a minimal unitary representation realizable automorphically. For $E_n(\mathbb{R})$, it is obtained by inducing from P_1...

conjecture | open | MathDB 页面 | 原文线索

V2-013085 | MATHDB-331802

McGovern's character formula conjecture for nilpotent orbit coordinate rings

摘录 / EXCERPT: McGovern's conjecture. For each nilpotent orbit $\mathcal{O}$, there are a fixed character μ, rational numbers c_w, and a subset $W_{\mathcal{O}}$ of W_G such that

conjecture | open | MathDB 页面 | 原文线索

V2-013086 | MATHDB-331826

Integral freeness conjecture for cuspidal algebras of imaginary root multiples

摘录 / EXCERPT: Let δ be the minimal positive imaginary root of the affine ADE root system, let $n\in\mathbb{Z}_{>0}$, and let k be either F or K. Write $C_{n\delta,\mathcal O}$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-013087 | MATHDB-331837

Parameter-scaling conjecture for locally free representations

摘录 / EXCERPT: Parameter-scaling conjecture. For all rank vectors $\mathbf{r}$,

conjecture | open | MathDB 页面 | 原文线索

V2-013088 | MATHDB-331838

Generalized Kac correspondence for tau-locally free modules

摘录 / EXCERPT: Let C be a symmetrizable Cartan matrix, D a symmetrizer, Ω an orientation, and $H=H(C,D,\Omega)$. Let τ_H denote the Auslander-Reiten translation, and call an H-m...

conjecture | open | MathDB 页面 | 原文线索

V2-013089 | MATHDB-331839

The graded-symmetry conjecture for the affine cyclotomic wreath product algebra

摘录 / EXCERPT: Graded-symmetry conjecture. The algebra $\mathcal{H}^{\mathbf{C}}_n(A)$ is graded symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-013090 | MATHDB-331840

The cyclic-cellularity conjecture for affine cyclotomic wreath product algebras

摘录 / EXCERPT: Cyclic-cellularity conjecture. If A is cyclic cellular, then so is $\mathcal{H}^{\mathbf{C}}_n(A)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013091 | MATHDB-331863

Breuil's socle conjecture for locally analytic representations of \(\mathrm{GL}_n(\mathbb{Q}_p)\)

摘录 / EXCERPT: Let ρ be an n-dimensional continuous representation of $\operatorname{Gal}(\overline{F}/F)$ over E, with $\rho_{\tilde v}$ crystalline and very regular for every...

conjecture | open | MathDB 页面 | 原文线索

V2-013092 | MATHDB-331876

The generalized degree-of-satisfiability conjecture for reasonable measures

摘录 / EXCERPT: Let G be a group, let $\{\mu_n\}_{n\in\mathbb{N}}$ be a sequence of “reasonable” measures on G, and let $\mathcal{E}\subseteq F_k$ be a set of equations in k variables. The...

conjecture | open | MathDB 页面 | 原文线索

V2-013093 | MATHDB-331877

The unrestricted degree of commutativity conjecture for finitely generated groups

摘录 / EXCERPT: Let G be a finitely generated group, and let X be a finite generating set. Define the degree of commutativity with respect to X by ... Here $\mathbb{B}_X(n)$ is the ball of rad...

conjecture | open | MathDB 页面 | 原文线索

V2-013094 | MATHDB-331883

The parity conjecture for Dirac cohomology

摘录 / EXCERPT: Let X be an irreducible $(\mathfrak{g},K)$-module, and let $H_D^+(X)$ and $H_D^-(X)$ be the even and odd parts of its Dirac cohomology. Dirac cohomology parity conjecture. ... A re...

conjecture | open | MathDB 页面 | 原文线索

V2-013095 | MATHDB-331884

The hypoellipticity conjecture for Dirac cohomology

摘录 / EXCERPT: Let $G(\mathbb{R})\supset K(\mathbb{R})$ be a real reductive group, let π be an irreducible admissible representation, and let X_π be its Harish-Chandra module. A represent...

conjecture | open | MathDB 页面 | 原文线索

V2-013096 | MATHDB-331889

Finite procyclic covering criterion for commutators

摘录 / EXCERPT: Let G be a profinite group whose commutator subgroup is finite-by-procyclic. For a subgroup K, write $\pi(K)$ for the set of primes dividing the orders of elements of K. Fini...

conjecture | open | MathDB 页面 | 原文线索

V2-013097 | MATHDB-331890

Countable covering conjecture for verbal values and group properties

摘录 / EXCERPT: Let Σ be a property of profinite groups such that every finite group is a Σ-group, the class of Σ-groups is closed under taking subgroups, quotients and extens...

conjecture | open | MathDB 页面 | 原文线索

V2-013098 | MATHDB-331891

Jaikin-Zapirain's profinite conciseness conjecture

摘录 / EXCERPT: Let w be a group-word and G a profinite group. Suppose that the set of w-values in G is countable. Profinite conciseness conjecture. Then the verbal subgroup $w(G)$ is fini...

conjecture | open | MathDB 页面 | 原文线索

V2-013099 | MATHDB-331892

Identity conjecture for words with countably many profinite values

摘录 / EXCERPT: Let w be a group-word and G a profinite group. Suppose that the set of w-values in G is countable. Identity conjecture. Then G satisfies a nontrivial group identity. The...

conjecture | open | MathDB 页面 | 原文线索

V2-013100 | MATHDB-331894

Locality conjecture for irreducible representations on frameworks

摘录 / EXCERPT: Locality conjecture. Every irreducible representation of $\mathsf{so}(2,3)$ is local on a framework M.

conjecture | open | MathDB 页面 | 原文线索

V2-013101 | MATHDB-331897

Dirac cohomology conjecture for irreducible rational Cherednik algebra modules

摘录 / EXCERPT: Let W be the reflection group, let $\mathfrak{h}$ be its reflection representation, let $\widetilde{W}$ be the associated pin double cover, and let χ be the spin module char...

conjecture | open | MathDB 页面 | 原文线索

V2-013102 | MATHDB-331898

Dirac cohomology conjecture for irreducible restricted rational Cherednik modules

摘录 / EXCERPT: Let $\overline{{\bf H}}$ be the restricted rational Cherednik algebra, let $\mathfrak{h}$ be the reflection representation of W, let $\widetilde{W}$ be the associated pin double...

conjecture | open | MathDB 页面 | 原文线索

V2-013103 | MATHDB-331899

Explicit q-binomial coefficients for finite-dimensional tensor product decompositions

摘录 / EXCERPT: Explicit coefficient conjecture. The coefficients $C_{\vec{k},\vec{m}}^{\vec{n}}$ can be represented explicitly by linear combinations of q-binomials.

conjecture | open | MathDB 页面 | 原文线索

V2-013104 | MATHDB-331915

Constancy of the leading-coefficient ratio for Gelfand-Kirillov subquotients

摘录 / EXCERPT: Leading-coefficient ratio conjecture. There exists a constant C, depending only on the Lie algebra $\mathfrak{g}$ and d, such that for all $M\in\mathcal{O}_\lambda^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-013105 | MATHDB-331933

Product monomial crystal conjecture for highest weights of truncated shifted Yangians

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple simply-laced Lie algebra, let λ and μ be dominant coweights, let $Y^\lambda_\mu(\mathbf R)$ be the corresponding truncated shifted Yan...

conjecture | open | MathDB 页面 | 原文线索

V2-013106 | MATHDB-331951

Zassenhaus's order conjecture for torsion units in integral group rings

摘录 / EXCERPT: Zassenhaus's order conjecture. There exists an element $g\in G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-013107 | MATHDB-331975

Discreteness and cone finiteness for triangulated categories

摘录 / EXCERPT: Let D be a Hom-finite Krull--Schmidt triangulated category and let H be the heart of a bounded t-structure. The category D is H-discre...

conjecture | open | MathDB 页面 | 原文线索

V2-013108 | MATHDB-331978

Displacement bound for purely loxodromic free Kleinian groups

摘录 / EXCERPT: Let $\Gamma=\langle\xi_1,\dots,\xi_n\rangle$ be a purely loxodromic, finitely generated free Kleinian group with $n\geq 2$, acting by isometries on hyperbolic three-space...

conjecture | open | MathDB 页面 | 原文线索

V2-013109 | MATHDB-332002

Three-involution bound for the first length parameter in dihedral groups

摘录 / EXCERPT: Let D_n be a dihedral group with an order n cyclic subgroup, and let S be a generating set composed of three involutions, none of which belongs to that cyclic subgroup. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013110 | MATHDB-332003

Flip-length bound by the maximum rotation length in dihedral groups

摘录 / EXCERPT: Let D_n be a dihedral group, let S be a generating set, and let $l_S(g)$ denote the length of an element g with respect to S. For an integer m with $0<m<n$, consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-013111 | MATHDB-332033

Amenable representation type and rank-metric closeness

摘录 / EXCERPT: The amenable representation-type conjecture. For every $\epsilon>0$ there exists $\delta>0$ such that, whenever M,N are R-modules satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-013112 | MATHDB-332042

Growth-cogrowth inequality for normal subgroups of free groups

摘录 / EXCERPT: Growth-cogrowth conjecture. For every non-trivial $N\trianglelefteq F_n$ we have

conjecture | open | MathDB 页面 | 原文线索

V2-013113 | MATHDB-332051

Bump-Nakasuji's factorization conjecture for inverse Casselman basis transition coefficients

摘录 / EXCERPT: Let R be a root system with Weyl group W, positive roots R^{+}, length function ℓ, and Bruhat order $\leq$. For $w,v\in W$ with $w\leq v$, define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013114 | MATHDB-332056

Quantum Humphreys conjecture for quantum-group modules

摘录 / EXCERPT: Quantum Humphreys conjecture. For every $\eta\in\operatorname{Spec}(Z_0^M(\mathfrak g))$ with $\pi_M(\eta)\in\mathcal O$, there exists an irreducible $U_{\eta}^{M}(\mathfrak g)$-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-013115 | MATHDB-332058

Integral commutativity conjectures for categorical induction and restriction functors

摘录 / EXCERPT: Let R be the integral coefficient ring, let $\overline R$ be the corresponding coefficient ring for the construction with e replaced by $e+1$, and let $O^{\nu,\Delta}_{\mu,R}$...

conjecture | open | MathDB 页面 | 原文线索

V2-013116 | MATHDB-332109

Church-Farb's stable cohomology conjecture for IA_n

摘录 / EXCERPT: Let IA_n be the IA-automorphism group of the free group F_n, and let $H^*(\mathrm{IA}_n;\mathbb{Q})$ denote its rational cohomology with the natural action of...

conjecture | open | MathDB 页面 | 原文线索

V2-013117 | MATHDB-332151

Same-order type characterization conjecture for nonabelian simple groups

摘录 / EXCERPT: Same-order type characterization conjecture. Then $G\cong S$.

conjecture | open | MathDB 页面 | 原文线索

V2-013118 | MATHDB-332245

Sommers' non-normal orbit-closure conjecture for non-abelian Lusztig quotients

摘录 / EXCERPT: Let $\mathcal{O}$ be a nilpotent orbit, let $S_{\mathcal{O}}$ be its special piece, and let $\overline{A}(\mathcal{O})$ be its Lusztig quotient. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013119 | MATHDB-332248

Panyushev's conjecture on orbit closures and dimensions in abelian nilradicals

摘录 / EXCERPT: Panyushev's conjecture. For the corresponding B-orbits, one has

conjecture | open | MathDB 页面 | 原文线索

V2-013120 | MATHDB-332260

Minimal-orbit-closure conjecture in the degree-two grading

摘录 / EXCERPT: Let G be a connected reductive algebraic group with Lie algebra $\mathfrak g$, let $\{e,h,f\}$ be an $\mathfrak{sl}_2$-triple, and write the associated grading as ... Let $B(h;0)$...

conjecture | open | MathDB 页面 | 原文线索

V2-013121 | MATHDB-332268

The local converse theorem for generic representations of p-adic general linear groups

摘录 / EXCERPT: Local converse conjecture. If π_1 and π_2 have the same central character and

conjecture | open | MathDB 页面 | 原文线索

V2-013122 | MATHDB-332269

Jacquet's conjecture on gamma factors for generic representations

摘录 / EXCERPT: Jacquet's conjecture. If π_1 and π_2 satisfy $\mathcal H_{\leq r}$, then $\pi_1\simeq\pi_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-013123 | MATHDB-332321

The first Zassenhaus conjecture for normalized units of integral group rings

摘录 / EXCERPT: Let G be a finite group. Write $\mathbb{Z}G$ for its integral group ring and ... for the group of normalized units, where ε is the augmentation map. First Zassenhaus co...

conjecture | open | MathDB 页面 | 原文线索

V2-013124 | MATHDB-332331

Takahashi's equivariant homological mirror symmetry conjecture for invertible polynomials

摘录 / EXCERPT: Let $f(x,y,z)$ be an invertible polynomial and let G be a finite group of diagonal symmetries containing the subgroup generated by the exponential grading operator. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013125 | MATHDB-332359

The LERF conjecture for groups with Kazhdan property (T)

摘录 / EXCERPT: Let G be a countable group with Kazhdan property (T). LERF conjecture. The group G is LERF if and only if it is finite. The preceding proposition establishes the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-013126 | MATHDB-332402

The hypothetical geometric Satake correspondence for the B_{dR}-affine Grassmannian

摘录 / EXCERPT: Let G be a quasi-split reductive group over E, let $\Gamma=\operatorname{Gal}(\overline E/E)$, and let L^+G act on the affine Grassmannian Gr. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013127 | MATHDB-332422

Asymptotic vanishing of normalized supercuspidal characters

摘录 / EXCERPT: Consider the set of pi in $\operatorname{Irr}^{\mathrm{sc}}(G)$ such that the central character of pi is unitary. For each fixed $\gamma\in G_{\mathrm{reg}}$, Asymptotic vanishi...

conjecture | open | MathDB 页面 | 原文线索

V2-013128 | MATHDB-332474

Classification conjecture for finite-dimensional irreducible type-1 comodules of the affine quantum group

摘录 / EXCERPT: Let $\widehat{SL_q(2)}$ be the quantum group defined in the paper, and let a type $\mathbf{1}$ comodule mean a comodule whose associated $\mathcal{T}$-comodule coaction acts semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-013129 | MATHDB-332475

Power-of-two dimension conjecture for finite-dimensional comodules of the FRT algebra

摘录 / EXCERPT: Let $\mathcal{A}_{ff}$ be the FRT algebra considered in the paper, and let a finite-dimensional comodule mean a finite-dimensional comodule over $\mathcal{A}_{ff}$. Power-of-two di...

conjecture | open | MathDB 页面 | 原文线索

V2-013130 | MATHDB-332487

Finkelberg-Mirković conjecture for Iwahori-constructible perverse sheaves

摘录 / EXCERPT: Let $\boldsymbol{G}^ atur$ be a complex connected reductive group Langlands-dual to $\boldsymbol{G}$, let β be a field of coefficients, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013131 | MATHDB-332518

Braverman-Kazhdan vanishing conjecture for gamma-sheaves

摘录 / EXCERPT: Let G be a reductive group with unipotent radical U of a Borel subgroup B, and let $T=B/U$. Let $\check G$ be the Langlands dual group, let ρ be a σ-positive repr...

conjecture | open | MathDB 页面 | 原文线索

V2-013132 | MATHDB-332526

Braden-Licata-Proudfoot-Webster symplectic duality conjecture for category $d4aa$

摘录 / EXCERPT: Let $d4aa$ be a category of special modules over $d4a5\operatorname{-mod}$ associated with a symplectic resolution, and let $d4aa^!$ be the corresponding category for its symplecti...

conjecture | open | MathDB 页面 | 原文线索

V2-013133 | MATHDB-332527

The nilpotent-group conjecture for the sum of element orders

摘录 / EXCERPT: Nilpotent-group conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-013134 | MATHDB-332547

Gan-Gross-Prasad local uniqueness conjecture for unitary groups

摘录 / EXCERPT: Let E/F be a quadratic extension of local fields of characteristic zero, let W be an n-dimensional hermitian space over E, and let $V=W\oplus^\perp e$ with $(e,e)=1$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-013135 | MATHDB-332568

Rouquier's conjecture on the KZ projective for rational Cherednik algebras

摘录 / EXCERPT: Rouquier's conjecture. For a general rational Cherednik algebra $H_c(W,\mathfrak{h})$, there exists a good set of parameter values...

conjecture | open | MathDB 页面 | 原文线索

V2-013136 | MATHDB-332582

Embedding criterion for right-angled Artin groups in Brin-Thompson groups

摘录 / EXCERPT: Let Γ be a finite simplicial graph, and let A_Γ be the right-angled Artin group defined by generators corresponding to the vertices of Γ, with two generators c...

conjecture | open | MathDB 页面 | 原文线索

V2-013137 | MATHDB-332624

Bonk-Kleiner's conformal-dimension characterization of Cannon's conjecture

摘录 / EXCERPT: Bonk-Kleiner's conjecture. The conformal dimension of $\partial\Gamma$ is attained by a metric in its conformal gauge.

conjecture | open | MathDB 页面 | 原文线索

V2-013138 | MATHDB-332628

The ungraded Orlik-Terao representation conjecture

摘录 / EXCERPT: Ungraded Orlik-Terao representation conjecture. There exists an isomorphism of S_n-representations

conjecture | open | MathDB 页面 | 原文线索

V2-013139 | MATHDB-332629

The Heaviside filtration conjecture for Orlik-Terao representations

摘录 / EXCERPT: Heaviside filtration conjecture. The graded representations M_n and D_n of S_n are both isomorphic to the associated graded representation of the Heaviside filtration on...

conjecture | open | MathDB 页面 | 原文线索

V2-013140 | MATHDB-332638

The drift conjecture for infinite finitely generated simple groups

摘录 / EXCERPT: Drift conjecture. The drift function satisfies

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-013141 | MATHDB-332640

The normalish subgroup conjecture for Thompson's group T

摘录 / EXCERPT: Normalish subgroup conjecture. Every normalish subgroup of T contains an embedded copy of R. Thompson's group F or an embedded non-abelian free subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-013142 | MATHDB-332649

The one-based/field-like dichotomy conjecture for full uniformly powerful pro-p groups

摘录 / EXCERPT: Let $\mathcal{G}=(G,I)$ be a uniformly powerful pro-p group that is full as a profinite group. For each n, let $P_n(G)$ denote the corresponding term in the lower p-series, a...

conjecture | open | MathDB 页面 | 原文线索

V2-013143 | MATHDB-332664

Finite-type Grothendieck-ring generation conjecture for quantum toroidal $\mathfrak{gl}(1)$

摘录 / EXCERPT: Let $\mathcal{O}^{fin}_{{\mathcal B}^\perp}$ be the category of finite-type modules, and let $K_0(\mathcal{O}^{fin}_{{\mathcal B}^\perp})$ denote its completed Grothendieck ring. F...

conjecture | open | MathDB 页面 | 原文线索

V2-013144 | MATHDB-332665

Cluster structure conjecture for finite-type quantum toroidal modules

摘录 / EXCERPT: Let $\mathcal{O}^{fin}_{{\mathcal B}^\perp}$ be the category of finite-type modules, let $R=\operatorname{Rep}\,\mathcal{O}^{fin}_{{\mathcal B}^\perp}$ be its Grothendieck ring, an...

conjecture | open | MathDB 页面 | 原文线索

V2-013145 | MATHDB-332671

Prest's decidability conjecture for theories of modules over finite-dimensional algebras

摘录 / EXCERPT: Let R be a finite-dimensional algebra over a suitably recursive field. The theory of R-modules is the collection of first-order sentences in the language of R-modules that ar...

conjecture | open | MathDB 页面 | 原文线索

V2-013146 | MATHDB-332710

Flat Closing Conjecture for proper CAT(0) spaces

摘录 / EXCERPT: Let X be a proper CAT(0) space with a proper cocompact action of a discrete group Γ. A flat $F\subset X$ is Γ-periodic if some subgroup $\Gamma_0\leq\Gamma$ preserv...

open_problem | open | MathDB 页面 | 原文线索

V2-013147 | MATHDB-332723

Truncated q-character relation for asymptotic representations

摘录 / EXCERPT: Truncated q-character relation. In $\operatorname{Frac}(K_0(\mathcal{O}^+))$,

conjecture | open | MathDB 页面 | 原文线索

V2-013148 | MATHDB-332729

Decoupling-block generation conjecture for restricted rational Cherednik algebras

摘录 / EXCERPT: Let $\overline{\mathbfsf{H}}$ be the restricted rational Cherednik algebra, with $\operatorname{DecGen}(\overline{\mathbfsf{H}})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013149 | MATHDB-332762

The characters conjecture for connected compact quantum groups

摘录 / EXCERPT: Let $G\subset U_N^+$ be a connected compact quantum group. Its central algebra $C(G)_{central}$ is the norm closure of the linear span of irreducible representation characters. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013150 | MATHDB-332767

Bobiński-Malicki conjecture on derived equivalences among the algebras A(p) and B(p,r)

摘录 / EXCERPT: Bobiński-Malicki conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-013151 | MATHDB-332783

The transfer conjecture for tame blocks of p-adic linear groups

摘录 / EXCERPT: The transfer conjecture. Suppose that ξ induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013152 | MATHDB-332784

Functoriality conjecture for integral blocks of p-adic groups

摘录 / EXCERPT: Let $\mathbf{G}'$ and $\mathbf{G}$ be groups of ${\rm GL}$-type with an L-homomorphism $\xi:{^{L}\mathbf{G}'}\rightarrow {^{L}\mathbf{G}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013153 | MATHDB-332785

Reduction of integral blocks to unipotent blocks

摘录 / EXCERPT: Let $\mathbf G$ be a group of ${\rm GL}$-type, let $\phi:I_F^{(\ell)}\rightarrow {^{L}\mathbf G}$ be an admissible parameter, and choose an extension...

conjecture | open | MathDB 页面 | 原文线索

V2-013154 | MATHDB-332786

Reduction of level-zero blocks to principal blocks

摘录 / EXCERPT: Let $\psi:P_F\rightarrow {^{L}\mathbf G}$ be a wild parameter, and let $\mathbf G_\psi$ be the associated tamely ramified group of ${\rm GL}$-type with...

conjecture | open | MathDB 页面 | 原文线索

V2-013155 | MATHDB-332790

The weight-graded orbit-closure conjecture

摘录 / EXCERPT: Let G/P and G/Q be the homogeneous spaces under consideration, with $Z_0,\ldots,Z_k$ the G-orbits in $G/P\times G/Q$, ordered by decreasing size. Let $j:Z_0\to G/P\times G/Q$...

conjecture | open | MathDB 页面 | 原文线索

V2-013156 | MATHDB-332912

2-Lusztig conjecture for Lie and Coxeter 2-representations

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional complex Lie algebra, let $\chi\in\mathfrak{g}^*$ be its nilpotent character, and let G be the associated finite component group....

conjecture | open | MathDB 页面 | 原文线索

V2-013157 | MATHDB-332913

2-McKay conjecture for 2-representations

摘录 / EXCERPT: 2-McKay conjecture. For each character χ, the 2-representations $\Theta_H(\chi)$ and $\Theta_N(\chi)$ of G are isomorphic. This is a 2-representation-theoretic strengthening...

conjecture | open | MathDB 页面 | 原文线索

V2-013158 | MATHDB-332998

Mulmuley-Sohoni padded-permanent border determinantal complexity conjecture

摘录 / EXCERPT: For $n>m$, let $X_{11}^{n-m}\mathrm{per}_m$ be the padded permanent, viewed as an element of $\mathsf{Sym}^n\mathbb{C}^{n^2}$, and let ${{\mathcal{D}}et}_n$ be the Euclidean (equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-013159 | MATHDB-332999

Mulmuley-Sohoni occurrence-obstruction conjecture

摘录 / EXCERPT: For $n>m$, let $Z_{n,m}$ be the orbit closure of the padded permanent $X_{11}^{n-m}\mathrm{per}_m$ in $\mathsf{Sym}^n\mathbb{C}^{n^2}$, and let ${{\mathcal{D}}et}_n$ be the orbit c...

conjecture | refuted | MathDB 页面 | 原文线索

V2-013160 | MATHDB-333022

Lapid's multiplicity-free width-two conjecture for products of ladder representations

摘录 / EXCERPT: Let ρ be the fixed supercuspidal representation defining the Bernstein component $\operatorname{Irr}_{[\rho]}$. Let ... be integers. Let $\pi\in \operatorname{Irr}_{[\rho]}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-013161 | MATHDB-333023

The center-diagonal coinvariant algebra conjecture for small quantum \(\mathfrak{sl}_m\)

摘录 / EXCERPT: The center-diagonal coinvariant algebra conjecture. In type A, there exists an S_m-action on $\mathsf z_0$ extending the action on C_m, commuting with the...

conjecture | open | MathDB 页面 | 原文线索

V2-013162 | MATHDB-333035

The Koopman C*-algebra conjecture for the Grigorchuk group

摘录 / EXCERPT: Let ${\mathcal G}$ be the 3-generated infinite torsion group of intermediate growth constructed by Grigorchuk, and let π be its Koopman representation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013163 | MATHDB-333061

The conjecture on regular equivalence classes of Haagerup monsters

摘录 / EXCERPT: Regular equivalence-class conjecture. There exists $2^{\aleph_0}$ regular equivalence classes of Haagerup monsters.

conjecture | open | MathDB 页面 | 原文线索

V2-013164 | MATHDB-333108

Containment of the amenable radical in random-walk stabilizers

摘录 / EXCERPT: Let G be a finitely generated group, and let $\boldsymbol{A_\mu}$ be the subgroup associated with a symmetric, aperiodic probability measure μ on G whose support generates...

conjecture | open | MathDB 页面 | 原文线索

V2-013165 | MATHDB-333109

Realization of the amenable radical by a random-walk subgroup

摘录 / EXCERPT: Let G be a finitely generated group. For a probability measure μ on G, let A_μ denote the associated subgroup, and let the amenable radical be the largest amenable norm...

conjecture | open | MathDB 页面 | 原文线索

V2-013166 | MATHDB-333141

Degree-of-commutativity conjecture for finitely generated groups

摘录 / EXCERPT: Let G be a finitely generated group and let S be a finite generating set for G. For $n\in\mathbb{N}$, write $\mathbb{B}_S(n)=\{g\in G:|g|_S\leqslant n\}$ and define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-013167 | MATHDB-333149

Prasad's Ext-vanishing conjecture for generic representations of general linear groups

摘录 / EXCERPT: Let F be the underlying nonarchimedean local field, let n be a positive integer, let π_1 be an irreducible generic representation of $GL(n+1,F)$, and let π_2 be an irre...

conjecture | open | MathDB 页面 | 原文线索

V2-013168 | MATHDB-333183

Complete reducibility of the level modules of the rank 3 Kac-Moody algebra

摘录 / EXCERPT: Complete reducibility conjecture. For each $m\in{\mathbb Z}$, ${\mathcal F}ib(m)$ completely reduces into a direct sum of one trivial module on level 0, one non-standard module o...

conjecture | open | MathDB 页面 | 原文线索

V2-013169 | MATHDB-333184

Direct-summand conjecture for the non-standard level modules

摘录 / EXCERPT: Direct-summand conjecture. For $0<|m|\leq 2$, the maximal submodule $U(m)$ is $\{0\}$, $V(m)=V^{\Lambda_m}$ is a direct summand of ${\mathcal F}ib(m)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-013170 | MATHDB-333185

Racah-Speiser recursion for the non-standard modules

摘录 / EXCERPT: Racah-Speiser recursion conjecture. For $m=\pm1,\pm2$, the weights of $P(V^{\Lambda_m}_-)$ follow

conjecture | open | MathDB 页面 | 原文线索

V2-013171 | MATHDB-333216

Thiagarajan-Badouel-Darondeau-Raoult conjecture for bounded flat grids

摘录 / EXCERPT: Let $\mathcal E$ be an event structure with event domain. A directed flat square grid is a directed square grid embedded isometrically in the event domain, with adjacent squares no...

conjecture | open | MathDB 页面 | 原文线索

V2-013172 | MATHDB-333220

Bridson-Wilton virtual specialness undecidability conjecture

摘录 / EXCERPT: Let Y be a finite nonpositively curved square complex. A square complex is virtually special if it has a finite-sheeted cover that is special. Bridson-Wilton's conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-013173 | MATHDB-333254

The orbit-method quantization conjecture for nilpotent orbit covers

摘录 / EXCERPT: Let G be a complex simple Lie group with Lie algebra $\mathfrak{g}$, and let K be a maximal compact subgroup of G with complexification $K_{\mathbb{C}}$. For a nilpotent orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-013174 | MATHDB-333255

Vogan's conjecture on special unipotent representations

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra of a complex semisimple Lie group. Let $\mathcal{O}$ be a special nilpotent orbit, let $\overline{A}(\mathcal{O})$ be its Lusztig quotient, an...

conjecture | open | MathDB 页面 | 原文线索

V2-013175 | MATHDB-333256

Achar-Sommers' conjecture on regular functions of nilpotent orbit covers

摘录 / EXCERPT: Achar-Sommers' conjecture. The maximal element in this expression is $h^{\vee}$, the semisimple element of a Jacobson-Morozov triple of $\mathcal{O}^{\vee}$. The paper states that...

conjecture | open | MathDB 页面 | 原文线索

V2-013176 | MATHDB-333258

Abrar's dominant dimension bound for Nakayama algebras

摘录 / EXCERPT: Abrar's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013177 | MATHDB-333261

Conjecture on dominant dimensions of endomorphism algebras

摘录 / EXCERPT: Dominant-dimension conjecture. The algebra B has dominant dimension n.

conjecture | refuted | MathDB 页面 | 原文线索

V2-013178 | MATHDB-333262

Conjecture that finite-dimensional algebras have property *

摘录 / EXCERPT: Dominant-dimension property conjecture. Every finite dimensional algebra with finite dominant dimension $n\geq 1$ has property $*$.

conjecture | open | MathDB 页面 | 原文线索

V2-013179 | MATHDB-333277

Grading conjecture for generalized minors on triple flag configurations

摘录 / EXCERPT: Let G be a reductive group, let ω_{i_l} be fundamental weights, and let u_l be the Weyl-group elements indexing the functions $\Delta_{u_l\omega_{i_l},\omega_{i_l}}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-013180 | MATHDB-333278

Edge-weight compatibility conjecture for cluster quivers on triple flag configurations

摘录 / EXCERPT: Let S_{AB,C,ω_k} denote the relevant exchange-matrix row or weight data associated with the edge AB in a triangle ABC, and let α_k be the corresponding simple roo...

conjecture | open | MathDB 页面 | 原文线索

V2-013181 | MATHDB-333297

The unicity of types conjecture for supercuspidal representations

摘录 / EXCERPT: Let $G=\mathbf{G}(F)$ be the group of rational points of a connected reductive algebraic group over a non-archimedean local field, and let π be a supercuspidal representation o...

conjecture | open | MathDB 页面 | 原文线索

V2-013182 | MATHDB-333315

Crystal structure on Coulomb-branch components

摘录 / EXCERPT: Let $\mathcal M(\mu,\lambda)$ be the Coulomb branch and let $\mathcal A$ be its attracting set. Consider the union, over the relevant weights μ, of the sets of irreducible comp...

conjecture | open | MathDB 页面 | 原文线索

V2-013183 | MATHDB-333318

Naive matching-function conjecture for parahoric subgroups

摘录 / EXCERPT: Let G be an arbitrary reductive group, let G^* be its quasisplit inner form, and let $J\subset G$ be an arbitrary parahoric subgroup. Put $L=J/J^+$ and let W_L be the Weyl gr...

conjecture | open | MathDB 页面 | 原文线索

V2-013184 | MATHDB-333377

The affine Temperley-Lieb quotient conjecture for induced subalgebras

摘录 / EXCERPT: Affine Temperley-Lieb quotient conjecture. There is an isomorphism of algebras

conjecture | open | MathDB 页面 | 原文线索

V2-013185 | MATHDB-333403

Dual canonical basis conjecture for acyclic quantum cluster algebras

摘录 / EXCERPT: Let $\mathcal A^q$ be a quantum cluster algebra associated with an acyclic quiver and a z-coefficient pattern. Let C denote Berenstein-Zelevinsky's triangular basis, and let th...

conjecture | open | MathDB 页面 | 原文线索

V2-013186 | MATHDB-333423

Computable d-Σ_2 Scott sentences for finitely presented groups

摘录 / EXCERPT: Let G be a finitely presented computable group. A computable d-Σ_2 Scott sentence is a Scott sentence for G that is the conjunction of a computable Σ_2 sentence...

conjecture | open | MathDB 页面 | 原文线索

V2-013187 | MATHDB-333437

Kervaire's conjecture on normal generation of free products

摘录 / EXCERPT: Kervaire's conjecture. For every non-trivial group G, the free product group $d53d * G$ cannot be normally generated by one element.

conjecture | open | MathDB 页面 | 原文线索

V2-013188 | MATHDB-333454

Conjecture on rank-4 chiral polytopes for projective special linear groups

摘录 / EXCERPT: Let $G\cong PSL(2,q)$ with $q=p^d$ and $d>1$. A chiral polytope existence conjecture asserts that G is the automorphism group of a chiral polytope of rank 4 if and only if one...

conjecture | open | MathDB 页面 | 原文线索

V2-013189 | MATHDB-333455

Conjecture on primitive elements and a square discriminant over finite fields

摘录 / EXCERPT: Primitive-element square conjecture. There exist such j_1 and j_2 for which

conjecture | open | MathDB 页面 | 原文线索

V2-013190 | MATHDB-333456

Conjectured intersection property for the constructed generators

摘录 / EXCERPT: Let α, β, and γ be the three generators constructed from the elements in the preceding construction, and let $\langle\alpha,\beta,\gamma\rangle$ denote the subg...

conjecture | open | MathDB 页面 | 原文线索

V2-013191 | MATHDB-333502

Plancherel dimension-ratio decay conjecture for subpartitions

摘录 / EXCERPT: Plancherel dimension-ratio decay conjecture. There exists $\alpha>0$ such that, for all s,

conjecture | open | MathDB 页面 | 原文线索

V2-013192 | MATHDB-333513

The naive conjecture relating circular orders and circle actions

摘录 / EXCERPT: Let G be a countable group. Write $\operatorname{CO}(G)$ for the space of circular orders on G, and let $\operatorname{Hom}(G,\operatorname{Homeo}_+(S^1))$ be the space of acti...

conjecture | open | MathDB 页面 | 原文线索

V2-013193 | MATHDB-333547

The finite W-algebra structure of noncommutative pfaffians

摘录 / EXCERPT: Let $\mathfrak{o}_N$ be the orthogonal Lie algebra with generators $F_{ij}=E_{ij}-E_{ji}$, and let PfF_I denote the noncommutative pfaffian of the skew-symmetric submatrix...

open_problem | open | MathDB 页面 | 原文线索

V2-013194 | MATHDB-333583

Relative Hodge-Riemann conjecture for Soergel bimodule products

摘录 / EXCERPT: Let W be the Weyl group under consideration, let $x_1,\dots,x_m\in W$, and let $B_{x_1}\dots B_{x_m}$ be equipped with its intersection form. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013195 | MATHDB-333592

Juan-Pineda-Leary conjecture on finite models for the virtually cyclic classifying space

摘录 / EXCERPT: Let G) be a group, and let ${\underline{\underline E}}G=E_{\mathcal{VCY}}(G)$ denote a G-CW-model for the classifying space for the family of virtually cyclic subgroups. The mo...

conjecture | open | MathDB 页面 | 原文线索

V2-013196 | MATHDB-333593

Finitely presented BVC groups are virtually cyclic

摘录 / EXCERPT: Let G be a group. It has BVC if there are virtually cyclic subgroups $V_1,\ldots,V_n$ such that every virtually cyclic subgroup of G is conjugate to a subgroup of some V_i. B...

conjecture | open | MathDB 页面 | 原文线索

V2-013197 | MATHDB-333595

Koszulity conjecture for the locally finite endomorphism algebra of a projective generator

摘录 / EXCERPT: Let $\mathcal B$ be a chosen block, let $P=\bigoplus_{\lambda\in\Lambda(\mathcal B)}P(\lambda)$ be a minimal projective generator of $\mathcal B$, and let ... denote its locally fini...

conjecture | open | MathDB 页面 | 原文线索

V2-013198 | MATHDB-333696

Larsen-Shalev conjecture for automorphic word maps

摘录 / EXCERPT: Larsen-Shalev conjecture. For each such word w, there exist constants $N(w),\eta(w)>0$ such that, for every nonabelian finite simple group S with $|S|\geq N(w)$,

conjecture | open | MathDB 页面 | 原文线索

V2-013199 | MATHDB-333697

Effective Larsen-Shalev conjecture for automorphic word maps

摘录 / EXCERPT: Effective Larsen-Shalev conjecture. The constants $N(w)$ and $\eta(w)$ can be computed algorithmically from the word w as input.

conjecture | open | MathDB 页面 | 原文线索

V2-013200 | MATHDB-333698

Algorithmic radical-boundedness conjecture for word maps

摘录 / EXCERPT: Algorithmic radical-boundedness conjecture. There exists an algorithm which, on input w, decides whether there is a function

conjecture | open | MathDB 页面 | 原文线索

V2-013201 | MATHDB-333736

Existence of the mod-l cycle homomorphism

摘录 / EXCERPT: Let $R_E(GL_n(\mathcal{O}_F))$ and $R_{\mathbb{F}}(GL_n(\mathcal{O}_F))$ be the Grothendieck groups of finite-length representations over E and $\mathbb{F}$, respectively, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-013202 | MATHDB-333742

Discrete Fuglede conjecture for spectral sets

摘录 / EXCERPT: Discrete Fuglede conjecture. A set $d4d0\subset\mathbb Z_M$ is a discrete spectral set if and only if it tiles $d4d1_M$ by translations. This is a finite analogue of Fuglede's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-013203 | MATHDB-333757

The conjecture that all finite-index subgroups of Thompson group F are 2-generated

摘录 / EXCERPT: Let F denote Thompson group F, and let a group be 2-generated if it has a generating set with two elements. The finite-index generation conjecture. Every finite-index subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-013204 | MATHDB-333776

Countability conjecture for horofunctions on Cayley graphs of polynomial growth

摘录 / EXCERPT: Countability conjecture. The set of horofunctions on G is countable.

conjecture | open | MathDB 页面 | 原文线索

V2-013205 | MATHDB-333785

Jiang's exterior-cube epsilon-factor conjecture for the Ginzburg-Rallis model

摘录 / EXCERPT: Let F be a local field and let π be an irreducible admissible representation of $\operatorname{GL}_6(F)$ satisfying the assumptions of Jiang's multiplicity-one conjecture, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-013206 | MATHDB-333795

The small pure resolution conjecture for pairs of partitions

摘录 / EXCERPT: Small pure resolution conjecture. For any pair of partitions $\lambda^{(0)}\subsetneq\lambda^{(1)}$, such a small pure resolution exists, equivariantly for both $\mathbf{GL}(V)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-013207 | MATHDB-333797

The strict congruence-topology hierarchy conjecture for free metabelian groups

摘录 / EXCERPT: Let F_2 be the free group of rank 2, let $\Phi_{2,r}$ denote its free solvable quotient of derived length r, let $C(\Phi_{2,r})$ be the kernel of the natural map from the aut...

conjecture | open | MathDB 页面 | 原文线索

V2-013208 | MATHDB-333818

Failure of the group-theoretic Rule of k

摘录 / EXCERPT: Fix integers $k\ge0$ and $N>k$. Let $g_1,\dots,g_N,h_1,\dots,h_N$ be generators of a group subject to the relations $g_Sh_S=h_Sg_S$ for all subsets S with $|S|\le k$. Failure of...

conjecture | open | MathDB 页面 | 原文线索

V2-013209 | MATHDB-333830

The center-surjectivity conjecture for cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Let A be a symmetric generalized Cartan matrix, and let Γ be its associated quiver with no loops. Let $\mathscr{R}_{\beta}(\Gamma)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013210 | MATHDB-333849

Strong eigencone functoriality along folds

摘录 / EXCERPT: Strong eigencone functoriality along folds. Along the folds of ϕ, one has

conjecture | open | MathDB 页面 | 原文线索

V2-013211 | MATHDB-333877

Centralizer rigidity conjecture for powers of interval exchange transformations

摘录 / EXCERPT: Centralizer rigidity conjecture. For every integer $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-013212 | MATHDB-333895

The modular GKO coset construction conjecture for affine $\mathfrak{sl}_2$

摘录 / EXCERPT: Modular GKO conjecture. If $\operatorname{char}{\mathbb F}>2\ell^2+\ell-3$, then the multiplicity-free decomposition above is valid over ${\mathbb F}$ as a direct sum of irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-013213 | MATHDB-333896

The irreducibility and extension conjecture for affine $\mathfrak{sl}_2$ Weyl modules

摘录 / EXCERPT: Affine Weyl-module conjecture. If $\operatorname{char}{\mathbb F}>2\ell^2+\ell-3$, then every $V(\lambda)$ with $\lambda\in X^+_\ell$ is irreducible, and for all...

conjecture | open | MathDB 页面 | 原文线索

V2-013214 | MATHDB-333905

The $K(\pi,1)$-conjecture for Artin groups

摘录 / EXCERPT: Let L be the nerve of an Artin group A_L, and let $\mathcal{S}(L)$ be the poset of simplices of L together with the empty simplex. The poset of groups over $\mathcal{S}(L)$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-013215 | MATHDB-333917

General irreducible-representation conjecture for proper biharmonic functions on SU(2)

摘录 / EXCERPT: For $n>1$, let $p,q\in{\mathbb C}^{n+1}$, and let π_n be the $(n+1)$-dimensional irreducible representation of $\mathrm{SU}(2)$. For fixed indices α and β, define...

conjecture | open | MathDB 页面 | 原文线索

V2-013216 | MATHDB-333930

Generalized boundary characterization for weighted paths

摘录 / EXCERPT: Let X be a set of infinitesimal paths with weight function $wg:X\longrightarrow\mathbb{R}^{+}$. Define the partition function ... and let $\mathcal{C}$ be a cone centered at 0. S...

conjecture | open | MathDB 页面 | 原文线索

V2-013217 | MATHDB-333939

Conjecture on contragredient invariance for distinguished representations of complex reductive groups

摘录 / EXCERPT: Let (G,H) be a real symmetric pair, where G is a complex connected reductive group. Let θ be the involution defining H, so that $H=G^{\theta}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013218 | MATHDB-333940

Lapid's L-packet invariance conjecture for distinguished representations

摘录 / EXCERPT: Let $\underline{G}$ be a connected algebraic group defined over a local field F, and let $\theta:\underline{G}\longrightarrow\underline{G}$ be an involution defined over F. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-013219 | MATHDB-333965

Exhaustion conjecture for rank-k representations of finite symplectic groups

摘录 / EXCERPT: Assume that $\dim(U)=k<n$. Let Sp be the finite symplectic group, let $O_{\beta+}$ and $O_{\beta-}$ correspond to the two isomorphism classes of inner products of rank k, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-013220 | MATHDB-334002

Exactness conjecture for the Casselman-Jacquet functor

摘录 / EXCERPT: Let $\mathfrak g$ be a Lie algebra, K a group acting compatibly on $\mathfrak g$, and MN the target subgroup. Let ... be the Casselman-Jacquet functor, and consider the subcatego...

conjecture | open | MathDB 页面 | 原文线索

V2-013221 | MATHDB-334003

Duality conjecture for the Casselman-Jacquet functor

摘录 / EXCERPT: Let $\mathfrak g$ be a Lie algebra with subgroups K, M, and N, and let θ determine the opposite Borel. Write ... for the duality equivalence given by taking...

conjecture | open | MathDB 页面 | 原文线索

V2-013222 | MATHDB-334004

Casselman-Jacquet identification conjecture

摘录 / EXCERPT: Let $V\in {\mathcal M}(\mathfrak g,K)$, and let $J(V)$ be the submodule of $\mathfrak n$-finite vectors in the $\theta(\mathfrak n)$-completion of V. Let ${\mathcal C}$ be the Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-013223 | MATHDB-334006

Geometric filtration conjecture for principal series

摘录 / EXCERPT: Let $\widetilde U=\widetilde U(\mathfrak g):=U(\mathfrak g)\otimes_{{\mathcal Z}(\mathfrak g)}S(\mathfrak h)$, and let ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-013224 | MATHDB-334018

Braverman-Kazhdan's exactness and induction conjecture for gamma sheaves

摘录 / EXCERPT: Braverman-Kazhdan's conjecture. The functor $\mathrm F_{G,\rho}$ is exact with respect to the perverse t-structure, and it commutes with induction functors. This conjecture conce...

conjecture | open | MathDB 页面 | 原文线索

V2-013225 | MATHDB-334019

Braverman-Kazhdan's acyclicity conjecture for gamma sheaves

摘录 / EXCERPT: Braverman-Kazhdan's acyclicity conjecture. The object $(\pi_U)_!\Psi_{G,\rho}$ is supported on $T=B/U\subset G/U$. Equivalently, for every $g\in G-B$,

conjecture | open | MathDB 页面 | 原文线索

V2-013226 | MATHDB-334020

Braverman-Kazhdan's acyclicity conjecture for gamma D-modules

摘录 / EXCERPT: Braverman-Kazhdan's acyclicity conjecture. The object $\operatorname{Av}_{U!}(\Psi_{G,\underline\lambda,c})$ is supported on $T=B/U\subset G/U$. This is the de Rham D-module form...

conjecture | open | MathDB 页面 | 原文线索

V2-013227 | MATHDB-334059

The 321-avoidance conjecture for products of ladder representations

摘录 / EXCERPT: Let $\pi_1,\pi_2\in\operatorname{Irr}_{\langle \rho \rangle}$ be ladder representations, written as $\pi_1=L(\mathfrak{m}^e_{\lambda_1,\mu_1})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013228 | MATHDB-334083

The Weyl-group bound for components of affine Springer fibers

摘录 / EXCERPT: Weyl-group component bound conjecture. For any regular semisimple $\gamma\in\mathfrak{g}$, the number of $\mathcal{T}$-orbits of components of...

conjecture | open | MathDB 页面 | 原文线索

V2-013229 | MATHDB-334097

The length conjecture for global integrals on GL_n

摘录 / EXCERPT: The length conjecture. If I satisfies the dimension equation and $I>1$, then I is zero for every choice of data. Equivalently, a nonzero global unipotent integral satisfying th...

conjecture | open | MathDB 页面 | 原文线索

V2-013230 | MATHDB-334113

The bound $\psi(G)\leq 2$ for finite groups of nonzero spread

摘录 / EXCERPT: Nonzero-spread bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013231 | MATHDB-334203

Cohomological realization conjecture for nilpotent orbit representations

摘录 / EXCERPT: Cohomological realization conjecture. For each $\chi\in\widehat{A(\mathcal{O})}$, there is a representation ξ of $P(e)$ satisfying $\xi\mid_{G(e)}=\chi$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-013232 | MATHDB-334217

Classification conjecture for finite filled groups

摘录 / EXCERPT: Classification conjecture. The group G is filled if and only if G is either an elementary abelian 2-group or one of

conjecture | open | MathDB 页面 | 原文线索

V2-013233 | MATHDB-334246

Hambleton-Taylor-Williams decomposition conjecture

摘录 / EXCERPT: Let G be a finite group, let R be a noetherian ring, let $X(G)$ be the set of isomorphism classes of rational irreducible representations of G, and for each $\rho\in X(G)$ le...

conjecture | open | MathDB 页面 | 原文线索

V2-013234 | MATHDB-334273

The local Langlands correspondence in families

摘录 / EXCERPT: Let A be a reduced complete Noetherian local $W(k)$-algebra, flat over $W(k)$, with residue field k, and let $\rho:G_F\to\operatorname{GL}_n(A)$ be a continuous n-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-013235 | MATHDB-334276

Conjecture on equal orders of two random permutations

摘录 / EXCERPT: Let σ and τ be two independent uniformly random permutations in S_k, and write $|\sigma|$ and $|\tau|$ for their orders. Equal-order conjecture. There is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-013236 | MATHDB-334280

The Sylow subgroup characterization of the FSZ property

摘录 / EXCERPT: FSZ Sylow conjecture. G is FSZ if and only if all of its Sylow subgroups are FSZ.

conjecture | open | MathDB 页面 | 原文线索

V2-013237 | MATHDB-334305

Mackey-Higson conjecture via coherent sheaf degeneration

摘录 / EXCERPT: Let $G_\mathbb{R}$ be a real reductive group with maximal compact subgroup K, let $G_{0,\mathbb{R}}$ be its Cartan motion group, and let $\mathcal{I}(Q,\phi)$ be a standard Haris...

conjecture | open | MathDB 页面 | 原文线索

V2-013238 | MATHDB-334322

Kosyak's commutant conjecture for Koopman representations

摘录 / EXCERPT: Let G act on a measure space X by α, let μ be a quasi-invariant measure, and let $\pi^{\alpha,\mu,X}$ be the associated Koopman representation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013239 | MATHDB-334334

First-level modules are preserved by Auslander-Reiten translation

摘录 / EXCERPT: First-level Auslander-Reiten conjecture. The module τM is also of the first level.

conjecture | open | MathDB 页面 | 原文线索

V2-013240 | MATHDB-334360

Mulmuley-Sohoni multiplicity conjecture for geometric complexity theory

摘录 / EXCERPT: The complexity classes ${\mathsf{VP}}_{ws}$ and ${\mathsf{VNP}}$ are considered through representation-theoretic multiplicities associated with them. Mulmuley-Sohoni multiplicity c...

conjecture | open | MathDB 页面 | 原文线索

V2-013241 | MATHDB-334369

Lusztig's original character formula conjecture

摘录 / EXCERPT: Let $G_{\Bbbk}$ be a semisimple algebraic group with weight lattice X, positive roots Φ_+, affine Weyl group $\Wa$, Coxeter number h, and p-dilated dot action $\pdot$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-013242 | MATHDB-334370

Lusztig's revised character formula conjecture

摘录 / EXCERPT: Let μ be p-regular, let $x\in\Wa$ satisfy $x\pdot\mu$ dominant, and let X_1^p denote the restricted region in the weight lattice. Let LCF denote Lusztig's...

conjecture | open | MathDB 页面 | 原文线索

V2-013243 | MATHDB-334372

Riche-Williamson conjecture on the Hecke action on the principal block

摘录 / EXCERPT: Let Rep_0 be the principal block of finite-dimensional algebraic representations, let $\mathcal H$ be the Hecke category, and let Θ_s be the wall-crossing f...

conjecture | open | MathDB 页面 | 原文线索

V2-013244 | MATHDB-334391

Guralnick's bounded first cohomology conjecture

摘录 / EXCERPT: Let k be a field, H a finite group, and M a finite-dimensional, absolutely irreducible, faithful kH-module. The first cohomology $H^1(H,M)$ is the quotient of the k-vecto...

conjecture | open | MathDB 页面 | 原文线索

V2-013245 | MATHDB-334392

Conjecture on the alternating or symmetric action of the Markoff automorphism group

摘录 / EXCERPT: Let $H(p)$ be the permutation group induced by the action of $\operatorname{Out}(\mathbf{F}_{2})$ on $\mathbb{Y}_{-2}(\mathbb{F}_{p})$, and let ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-013246 | MATHDB-334406

Dual Ore conjecture for distributive intervals of finite groups

摘录 / EXCERPT: Dual Ore conjecture. Every distributive interval $[H,G]$ of finite groups is linearly primitive; equivalently, there exists an irreducible complex representation V of G such th...

conjecture | open | MathDB 页面 | 原文线索

V2-013247 | MATHDB-334417

KST character-decay conjecture for supercuspidal representations

摘录 / EXCERPT: KST character-decay conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-013248 | MATHDB-334420

A geometric formula for sections of local systems on nilpotent orbits

摘录 / EXCERPT: Let G be a complex reductive group, let e be a nilpotent element with orbit ${\mathcal O}$, and let $A(e)$ be the component group of the centralizer of e. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013249 | MATHDB-334426

Local Gan-Gross-Prasad conjecture for tempered representations of real unitary groups

摘录 / EXCERPT: Let $\phi_n \in \Phi_\mathrm{temp}(\mathrm{U}_{n}(\mathbb{R}))$ and $\phi_{n+1} \in \Phi_\mathrm{temp}(\mathrm{U}_{n+1}(\mathbb{R}))$ have decompositions ... ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-013250 | MATHDB-334461

Bartholdi's conjecture on finitely presented or densely related automaton groups

摘录 / EXCERPT: Let G be a group generated by a finite state automaton. Bartholdi's conjecture. The group G is either finitely presented or densely related. This conjecture concerns the relati...

conjecture | open | MathDB 页面 | 原文线索

V2-013251 | MATHDB-334479

Burghelea's conjecture for periodic cyclic homology

摘录 / EXCERPT: Let G be a group. Write $\langle G \rangle^{\infty}$ for the conjugacy classes of infinite-order elements, and for $g\in G$ let C_g be its centralizer and...

conjecture | open | MathDB 页面 | 原文线索

V2-013252 | MATHDB-334480

Centralizer quotient conjecture for groups of finite asymptotic dimension

摘录 / EXCERPT: Let G be a discrete group with finite asymptotic dimension, and for $g\in G$ let $C_g<G$ denote its centralizer. Centralizer quotient conjecture. The quotient ... also has finite a...

conjecture | open | MathDB 页面 | 原文线索

V2-013253 | MATHDB-334504

Jiang's multiplicity dichotomy conjecture for the local Ginzburg-Rallis model

摘录 / EXCERPT: Let F be a local field, let D be the quaternion algebra over F, and let $m(\rho)$ and $m(\rho_D)$ denote the multiplicities for the Ginzburg-Rallis models of $GL_6(F)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013254 | MATHDB-334505

Exterior-cube epsilon-factor conjecture for the local Ginzburg-Rallis model

摘录 / EXCERPT: Let F be a local field, let π be an irreducible admissible representation of $GL_6(F)$ with central character χ^2, and let π_D be its local Jacquet-Langlands corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-013255 | MATHDB-334516

Conjecture A on decomposition numbers of defect-two characters

摘录 / EXCERPT: Conjecture A. One has $|\operatorname{IBr}(\chi^0)|\le p^2-1$ and $d_{\chi\varphi}\le p$ for every $\varphi\in\operatorname{IBr}(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013256 | MATHDB-334521

The unitary-group and symplectic distinction conjecture

摘录 / EXCERPT: Unitary-symplectic distinction conjecture. Some member of $\{\pi\}$ is distinguished by $\operatorname{Sp}_{2n}(F)$ if and only if $BC(\pi)$ is distinguished by...

conjecture | open | MathDB 页面 | 原文线索

V2-013257 | MATHDB-334523

The free-by-cyclic polytope conjecture

摘录 / EXCERPT: Let Γ be a free-by-cyclic group, and let $\mathfrak{G}(\Gamma)$ be the polytope group associated to $H_1(\Gamma;\mathbb{R})$. The polytope invariant $\mathcal{P}(\Gamma)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-013258 | MATHDB-334524

The two-dimensional classifying-space polytope conjecture

摘录 / EXCERPT: Let $\Gamma\ne\mathbb{Z}$ be an L^2-acyclic $\operatorname{Wh-AC}$-group of type F, meaning that it has a finite model for $K(\Gamma,1)$, and suppose that it admits a two-dimen...

open_problem | open | MathDB 页面 | 原文线索

V2-013259 | MATHDB-334525

The amenable-group vanishing conjecture for the polytope invariant

摘录 / EXCERPT: Let Γ be an L^2-acyclic $\operatorname{Wh-AC}$-group of type F. Let $\mathfrak{G}(\Gamma)$ be its polytope group and let $\mathcal{P}(\Gamma)\in\mathfrak{G}(\Gamma)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-013260 | MATHDB-334526

The marked-polytope description of the BNS invariant

摘录 / EXCERPT: Let Γ be a hyperbolic L^2-acyclic group of type F admitting a two-dimensional $K(\Gamma,1)$. Let $H^1(\Gamma;\mathbb{R})$ be the real character space, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013261 | MATHDB-334627

The conjectural character formula for discrete-series characters of \(\mathrm{SL}_\ell(F)\)

摘录 / EXCERPT: The conjectural character formula. The character value is given by

conjecture | open | MathDB 页面 | 原文线索

V2-013262 | MATHDB-334628

The coincidence conjecture for smooth and algebraic theta correspondences

摘录 / EXCERPT: Let G and G' be a reductive dual pair, let $\widetilde{G}$ and $\widetilde{G'}$ be their corresponding covering groups, and let $\omega_{G,G'}$ be the oscillator representation...

conjecture | open | MathDB 页面 | 原文线索

V2-013263 | MATHDB-334659

Vinberg's finite-generation conjecture for essential signatures

摘录 / EXCERPT: Vinberg's finite-generation conjecture. The semigroup Σ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-013264 | MATHDB-334660

Vinberg's saturation conjecture for essential signatures

摘录 / EXCERPT: Vinberg's saturation conjecture. The semigroup Σ is saturated, namely

conjecture | open | MathDB 页面 | 原文线索

V2-013265 | MATHDB-334661

Vinberg's polyhedral-inequality conjecture for essential signatures

摘录 / EXCERPT: Vinberg's polyhedral-inequality conjecture. There exist a family of subsets $M_{i}\subset\{1,\dots,N\}$ and a family of elements $l_{i}\in\mathfrak{t}_{\mathbb{Z}}$ such that the s...

conjecture | open | MathDB 页面 | 原文线索

V2-013266 | MATHDB-334669

Gersten's distortion-divergence conjecture for normal free subgroups

摘录 / EXCERPT: Gersten's conjecture. The distortion of F in G is equivalent to the divergence of G.

conjecture | open | MathDB 页面 | 原文线索

V2-013267 | MATHDB-334670

The linear-or-quadratic distortion conjecture for generalised Bestvina-Brady subgroups

摘录 / EXCERPT: Linear-or-quadratic distortion conjecture. The distortion of H_Φ in A_Γ is linear or quadratic.

conjecture | open | MathDB 页面 | 原文线索

V2-013268 | MATHDB-334671

Walker's piecewise quasilinearity conjecture for scl in free products of finite cyclic groups

摘录 / EXCERPT: Walker's conjecture. For any fixed chain c in F_n, ${\rm scl}(c_{\bm{o}})$ is piecewise quasilinear in $1/o_i$: there are some $p\in\mathbb{Z}_+$ and a finite partition of...

conjecture | open | MathDB 页面 | 原文线索

V2-013269 | MATHDB-334687

Calegari's spectral-gap conjecture for stable commutator length in free groups

摘录 / EXCERPT: Calegari's conjecture. For any integral chain c in a free group,

conjecture | open | MathDB 页面 | 原文线索

V2-013270 | MATHDB-334711

Sangroniz's finiteness conjecture for finite 2-groups with odd real class number

摘录 / EXCERPT: Let r be an odd natural number. A finite group element is real if it is conjugate to its inverse, and a conjugacy class is real if all its elements are real. Sangroniz's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-013271 | MATHDB-334712

Finiteness conjecture for 2-generated finite 2-groups with prescribed odd real class number

摘录 / EXCERPT: Let k be an odd number. A finite 2-group is 2-generated if it can be generated by two elements, and a real conjugacy class is a conjugacy class consisting of elements conjugate...

open_problem | open | MathDB 页面 | 原文线索

V2-013272 | MATHDB-334713

Rank-boundedness conjecture for finite 2-groups with few rational characters

摘录 / EXCERPT: Let G be a finite 2-group, let $\operatorname{rk}(G)$ denote its rank, and let q be the number of rational characters of G. Rank-boundedness conjecture. The rank...

conjecture | open | MathDB 页面 | 原文线索

V2-013273 | MATHDB-334739

Strictly outer actions satisfy the Intermediate Subfactor Property

摘录 / EXCERPT: Strictly outer action conjecture. Any strictly outer action $G\curvearrowright Q$ on a factor of any type satisfies the Intermediate Subfactor Property.

conjecture | open | MathDB 页面 | 原文线索

V2-013274 | MATHDB-334775

Reeder's Hilbert-series conjecture for small representations

摘录 / EXCERPT: Reeder's conjecture. For every small G-representation M,

conjecture | open | MathDB 页面 | 原文线索

V2-013275 | MATHDB-334776

Enhanced Reeder conjecture for small representations

摘录 / EXCERPT: Enhanced Reeder conjecture. For every small G-representation M, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013276 | MATHDB-334778

Kosyak's irreducibility criterion for Koopman representations

摘录 / EXCERPT: Let G act measurably on a space (X,bc) by $b1:G\to\operatorname{Aut}(X)$, where bc is G-quasi-invariant, meaning $bc^{\alpha_t}\simbc$ for every $t\in G$. The associated Ko...

conjecture | open | MathDB 页面 | 原文线索

V2-013277 | MATHDB-334779

The finite-field quasiregular irreducibility and equivalence conjecture

摘录 / EXCERPT: Let $m\in\mathbb N$. Set $X^m=B_m(\mathbb F_p)\backslash B^{\mathbb N}(\mathbb F_p)\simeq B^m(\mathbb F_p)$ and let $\mu_\alpha=\bigotimes_{k=1}^m\mu^k_\alpha$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-013278 | MATHDB-334790

Maximum-transversal conjecture for order-four latin hypercubes

摘录 / EXCERPT: Let $Q^n(\mathbb{Z}_4)$ and $Q^n(\mathbb{Z}_2^2)$ denote the n-dimensional latin hypercubes isotopic to the Cayley tables of the corresponding iterated groups. A transversal is a...

conjecture | open | MathDB 页面 | 原文线索

V2-013279 | MATHDB-334793

Ocneanu-type uniqueness conjecture for actions on strongly self-absorbing C*-algebras

摘录 / EXCERPT: Let $\mathcal{D}$ be a strongly self-absorbing $\mathrm{C^*}$-algebra, and let G be a countable, torsion-free amenable group. An action is pointwise strongly outer if every nontr...

conjecture | open | MathDB 页面 | 原文线索

V2-013280 | MATHDB-334881

The D-standardness conjecture for finite-dimensional algebra module categories

摘录 / EXCERPT: Let k be a field and let A be a finite-dimensional k-algebra. The module category $A\text{-}\operatorname{mod}$ is D-standard over k if every pseudo-identity on...

conjecture | open | MathDB 页面 | 原文线索

V2-013281 | MATHDB-334882

Thompson's q-power degree conjecture for unitriangular groups

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $n\geq 1$, and let $U_n(\mathbb{F}_q)$ be the group of unitriangular $n\times n$ matrices over $\mathbb{F}_q$. Thompson's q-power degree...

conjecture | open | MathDB 页面 | 原文线索

V2-013282 | MATHDB-334934

The graded symmetry conjecture for the Soergel algebra of a block

摘录 / EXCERPT: Graded symmetry conjecture. The algebra B_ξ is a graded symmetric algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-013283 | MATHDB-334935

The center conjecture for Whittaker coinvariants of \(\mathrm{GL}(m|n)\)

摘录 / EXCERPT: Center conjecture. The image of the canonical map

conjecture | open | MathDB 页面 | 原文线索

V2-013284 | MATHDB-334951

Burghelea's conjecture on periodic cyclic homology of group algebras

摘录 / EXCERPT: Burghelea's conjecture. If $K(G,1)$ has the homotopy type of a finite CW-complex, then

conjecture | open | MathDB 页面 | 原文线索

V2-013285 | MATHDB-334979

The eigenvalue bound conjecture for the operators $\rho_{(2)}$ and $\rho_{(3)}$

摘录 / EXCERPT: Let $\mathbb V_{g,d}$ be the relevant Floer vector space, and let (x,y) be a pair of simultaneous eigenvalues for the operators $\rho_{(2)}$ and $\rho_{(3)}$. The eigenvalue boun...

conjecture | open | MathDB 页面 | 原文线索

V2-013286 | MATHDB-334988

Local selfinjective endomorphism conjecture for finitistic Auslander algebras

摘录 / EXCERPT: Local selfinjective endomorphism conjecture. The algebra B is a finitistic Auslander algebra of dominant dimension 2.

conjecture | open | MathDB 页面 | 原文线索

V2-013287 | MATHDB-334989

Weak finitistic Auslander conjecture for one projective non-injective module

摘录 / EXCERPT: Weak finitistic Auslander conjecture. Then A is a weak finitistic Auslander algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-013288 | MATHDB-334990

Polynomial dominant-dimension conjecture for finitistic Auslander algebras

摘录 / EXCERPT: Polynomial dominant-dimension conjecture. There exists a polynomial function $f(n)$ such that every connected non-selfinjective algebra with n simple modules and dominant dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-013289 | MATHDB-334991

NASG tilting-finitistic conjecture

摘录 / EXCERPT: NASG tilting-finitistic conjecture. The algebra A is tilting-finitistic and therefore is a weak finitistic Auslander algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-013290 | MATHDB-334998

Brubaker-Bump-Friedberg conjecture on Hecke-module intertwiners for unique models

摘录 / EXCERPT: Brubaker-Bump-Friedberg conjecture. The map $\mathcal{L}$ is an $\mathcal{H}$-map from M to V_{ε} for some choice of linear character ε of $\mathcal{H}_0$...

conjecture | open | MathDB 页面 | 原文线索

V2-013291 | MATHDB-334999

Special-orbit conjecture for generalized Gelfand-Graev models

摘录 / EXCERPT: Special-orbit conjecture. The representation $\Gamma_{A,1}$ is a model for the universal principal series $\textup{ind}_B^G\chi_{\textup{univ}}^{-1}$ satisfying the diagram relatin...

conjecture | open | MathDB 页面 | 原文线索

V2-013292 | MATHDB-335013

Positivity conjecture for the Specht-to-web transition coefficients

摘录 / EXCERPT: Transition-coefficient positivity conjecture. For every tableau T' with corresponding web w', one has

conjecture | open | MathDB 页面 | 原文线索

V2-013293 | MATHDB-335043

Whittaker-W-algebra braided equivalence for generic level

摘录 / EXCERPT: Let G be the group associated with $\mathfrak{g}$, let Gr_G be its affine Grassmannian, and let $\operatorname{Whit}^c(\operatorname{Gr}_G)$ denote the Whittaker...

conjecture | open | MathDB 页面 | 原文线索

V2-013294 | MATHDB-335045

Generalized Burghelea Conjecture

摘录 / EXCERPT: Let G be a discrete group. For an element x in G, let G_x denote its centralizer, let $N_x=G_x/\langle x\rangle$ be the reduced centralizer, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013295 | MATHDB-335053

Puninski's conjecture on serial group rings with cyclic Sylow subgroups

摘录 / EXCERPT: Puninski's conjecture. The group ring FG is serial if and only if either G is p-solvable or the ring FH is serial.

conjecture | open | MathDB 页面 | 原文线索

V2-013296 | MATHDB-335073

Characterization of torus quotients

摘录 / EXCERPT: Let X be a compact complex space of dimension $n\geq 2$. Let $\widetilde{c}_2(X)$ denote the second orbifold Chern class of X, and let a finite group action be free in codimens...

conjecture | open | MathDB 页面 | 原文线索

V2-013297 | MATHDB-335086

Spectral decomposition conjecture for regular elliptic orbital integrals of $PGL(2,F)$

摘录 / EXCERPT: Let F be a local non-Archimedean field and let G be the group of F-points of a reductive group, with $\hat G_t$ denoting the set of tempered representations. For a regular el...

conjecture | open | MathDB 页面 | 原文线索

V2-013298 | MATHDB-335087

DeBacker's homogeneity conjecture for invariant distributions

摘录 / EXCERPT: Let F be a local non-Archimedean field, let $G=\operatorname{GL}(n,F)$ with Lie algebra $\mathfrak{g}$, and let $\mathcal B$ be its Bruhat-Tits building. Fix an irreducible smoot...

conjecture | open | MathDB 页面 | 原文线索

V2-013299 | MATHDB-335109

Thévenaz's solvability conjecture for finite groups and their largest B-group quotients

摘录 / EXCERPT: Let G be a finite group, and let $\beta(G)$ denote its largest quotient that is a B-group. Thévenaz's conjecture. The group $\beta(G)$ is solvable if and only if G is solvabl...

conjecture | open | MathDB 页面 | 原文线索

V2-013300 | MATHDB-335116

The fixed-point conjecture for the partition complex at elementary abelian subgroups

摘录 / EXCERPT: Let $\Cwiggle=C_{U\kern-.5pt\left(p^{k}\right)}\left(\Delta_{k}\right)/\left(\Delta_{k}\times S^{1}\right)$, where $C_{U\left(p^k\right)}(\Delta_k)$ is the centralizer of...

conjecture | open | MathDB 页面 | 原文线索

V2-013301 | MATHDB-335121

Lewis-Reiner-Stanton Hilbert-series conjecture for Frobenius-power quotients

摘录 / EXCERPT: Lewis-Reiner-Stanton conjecture. The Hilbert series is

conjecture | open | MathDB 页面 | 原文线索

V2-013302 | MATHDB-335123

Nonproportionality conjecture for the $n=k=m-1$ invariants

摘录 / EXCERPT: Let $S=\mathbb{F}_q[x_1,\ldots,x_n]$, let $Q=S/(x_1^{q^m},\ldots,x_n^{q^m})$, and let the invariant polynomials in Proposition be those constructed for the $n=k=m-1$ case. Let the...

conjecture | open | MathDB 页面 | 原文线索

V2-013303 | MATHDB-335124

Dickson-image conjecture for the $k=n$ term when $m=n+1$

摘录 / EXCERPT: Let $S=\mathbb{F}_q[x_1,\ldots,x_n]$, let $Q=S/(x_1^{q^m},\ldots,x_n^{q^m})$, and let $G=Gl_n(\mathbb{F}_q)$. Assume $m=n+1$ and consider the $k=n$ term of the conjectured Hilbert...

conjecture | open | MathDB 页面 | 原文线索

V2-013304 | MATHDB-335125

Dickson-image conjecture for the third Hilbert-series term when $q=n=2$

摘录 / EXCERPT: Let $S=\mathbb{F}_q[x_1,x_2]$, let $Q=S/(x_1^{q^m},x_2^{q^m})$, and let $G=Gl_2(\mathbb{F}_q)$. In the case $q=n=2$, consider the third term, referred to in the source as the $k=3$...

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V2-013305 | MATHDB-335159

Irreducibility conjecture for tensor products of \(\mathcal{U}_A\)-modules

摘录 / EXCERPT: Let $\mathcal{U}_A$ be the algebra in the paper, and let V_l and V_m be the modules defined by the representations π_x^l and π_y^m, with parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-013306 | MATHDB-335170

Lerman-Montgomery-Sjamaar conjecture on cotangent symplectic reductions

摘录 / EXCERPT: Let G and H be Lie groups and let X and Y be smooth manifolds on which G and H, respectively, act properly. Assume that the orbit spaces X/G and Y/H are diffeomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-013307 | MATHDB-335237

Finiteness conjecture for d-representation-finite dimensions

摘录 / EXCERPT: Finiteness conjecture. For a given finite-dimensional k-algebra A, there are only finitely many integers d such that A is d-representation-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-013308 | MATHDB-335238

Conjectural classification of scattered representations with Dirac cohomology

摘录 / EXCERPT: Scattered-representation classification conjecture. The set $\widehat{G}^{\mathrm{sd}}$ can be described as follows:

conjecture | open | MathDB 页面 | 原文线索

V2-013309 | MATHDB-335242

The finite-group representation conjecture for distributive intervals

摘录 / EXCERPT: Finite-group representation conjecture. If $[H,G]$ is a distributive interval, then there exists an irreducible complex representation V of G such that

conjecture | open | MathDB 页面 | 原文线索

V2-013310 | MATHDB-335250

The rational Whitehead injectivity conjecture

摘录 / EXCERPT: For a group G, let $\operatorname{Wh}(G)$ denote its Whitehead group, and let $\mathcal{F}\!\mathit{in}$ be the family of finite subgroups. The rational Whitehead injectivity con...

conjecture | open | MathDB 页面 | 原文线索

V2-013311 | MATHDB-335269

Nederhof's multiple-context-freeness conjecture for free abelian groups

摘录 / EXCERPT: Let $W(G)$ denote the word problem of a group G with respect to a finite symmetric generating set. In particular, $W(\mathbb{Z}^n)$ is the set of words over a finite generating s...

conjecture | open | MathDB 页面 | 原文线索

V2-013312 | MATHDB-335276

Bongartz-Ringel accessibility/no-gap conjecture

摘录 / EXCERPT: Bongartz-Ringel accessibility conjecture. Every indecomposable module of finite length over any algebra is accessible.

conjecture | open | MathDB 页面 | 原文线索

V2-013313 | MATHDB-335301

Nonabelian Cohen-Lenstra average conjecture for unramified extensions

摘录 / EXCERPT: Let G be a finite group and let G' be an admissible subgroup of $G\wr S_2$. Write c for the unique conjugacy class of order-two elements of G' outside $\ker\pi$ when G' i...

conjecture | open | MathDB 页面 | 原文线索

V2-013314 | MATHDB-335302

Lifting-invariant refinement of the nonabelian Cohen-Lenstra conjecture

摘录 / EXCERPT: Let Q be $\mathbb{Q}$ or $\mathbb{F}_q(t)$. Let G be a finite group and let G' be an admissible subgroup of $G\wr S_2$, with surjection $\pi:G'\to S_2$, and let c be the se...

conjecture | open | MathDB 页面 | 原文线索

V2-013315 | MATHDB-335305

Conjecture on the splitting of representations in Cases 4, 7 and 8

摘录 / EXCERPT: In the cases labeled Case 4 and Cases 7 and 8, let π denote the representation described in the surrounding case analysis. Splitting conjecture. Each such representation decomp...

conjecture | open | MathDB 页面 | 原文线索

V2-013316 | MATHDB-335306

Lapid's conjecture on L-packets of distinguished representations

摘录 / EXCERPT: Let G be a connected reductive group over a p-adic field F, let θ be an involution of G defined over F, and let $H=G^\theta$. Let π be an admissible irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-013317 | MATHDB-335323

The monstrous proposal for the bimonster quotient of the complex hyperbolic braid group

摘录 / EXCERPT: Let $P\Gamma$ act on the complex hyperbolic ball $\mathbb{B}^{13}$ with mirror arrangement $\EuScript{H}$, and let ... be the orbifold fundamental group based at τ. Let N be t...

conjecture | open | MathDB 页面 | 原文线索

V2-013318 | MATHDB-335329

Khovanov's symmetry conjecture for endomorphism algebras in parabolic category O

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex semisimple Lie algebra with fixed Borel subalgebra $\mathfrak{b}$, Cartan subalgebra $\mathfrak{h}\subset\mathfrak{b}$, and parabolic subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-013319 | MATHDB-335337

Heckenberger-Vocke conjecture on the structure of graded right coideal subalgebras

摘录 / EXCERPT: Heckenberger-Vocke conjecture. The map f is an injective homomorphism of $\mathbb{Z}$-graded right coideal subalgebras, and

conjecture | open | MathDB 页面 | 原文线索

V2-013320 | MATHDB-335338

Classification conjecture for Borel subalgebras of quantum groups

摘录 / EXCERPT: Classification conjecture. The algebra C is a Borel subalgebra if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-013321 | MATHDB-335351

Orbifold-Nichols algebra representation-category equivalence problem

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $ell=4$, and let $theta$ be a diagram automorphism. Consider an orbifold model of the logarithmic conformal field theory for...

open_problem | open | MathDB 页面 | 原文线索

V2-013322 | MATHDB-335357

Conjecture on Geck's and Murphy-type cellular bases for type B Hecke algebras

摘录 / EXCERPT: Fix a weighting $(s_1,s_2)\in \mathbb{Z}^2$ and set ... Consider Geck's cellular Kazhdan-Lusztig-type basis of the type B Hecke algebra and the graded cellular Murphy-type basis co...

conjecture | open | MathDB 页面 | 原文线索

V2-013323 | MATHDB-335376

The affine-or-almost-simple conjecture for point-primitive generalised quadrangles

摘录 / EXCERPT: Let $\mathcal{Q}$ be a thick finite generalised quadrangle, and let G be a collineation group of $\mathcal{Q}$ acting primitively on its point set. Affine-or-almost-simple conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-013324 | MATHDB-335390

Equal-sum partition conjecture for groups with at least three involutions

摘录 / EXCERPT: Let Γ be an Abelian group in $\mathcal{G}$, where $\mathcal{G}$ denotes the class used in the paper, and let g_0 be its identity element. Suppose that Γ has even or...

conjecture | open | MathDB 页面 | 原文线索

V2-013325 | MATHDB-335460

Existence conjecture for the model companion of finite group actions

摘录 / EXCERPT: Existence conjecture for finite group actions. If T^{mc} exists and G is finite, then T_G^{mc} exists.

conjecture | open | MathDB 页面 | 原文线索

V2-013326 | MATHDB-335598

The type $C^{(1)}_\ell$ Garnir-element conjecture for Specht modules

摘录 / EXCERPT: Let $\lambda\in \mathscr{P}^l_n$, and let $A\in [\lambda]$ be a Garnir node. Suppose $\mathbf{B}^A$ contains a bricks in the first row and b bricks in the second row. For each...

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V2-013327 | MATHDB-335667

Acyclic idempotent conjecture for Yoneda algebras

摘录 / EXCERPT: Acyclic idempotent conjecture. Assume that e is acyclic. If both A and Γ have finite global dimensions, then so does $Y(e)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013328 | MATHDB-335685

Navarro's refinement of the McKay conjecture

摘录 / EXCERPT: Navarro's refinement. There exists a bijection between the σ-invariant characters in

conjecture | open | MathDB 页面 | 原文线索

V2-013329 | MATHDB-335689

The non-self-intersection criterion for rigid modules over 2-complete quivers

摘录 / EXCERPT: Let R be the path algebra of a 2-complete quiver, and let f be the bijection from indecomposable modules corresponding to positive real roots to admissible curves on the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-013330 | MATHDB-335690

The non-self-crossing criterion for rigid modules over arbitrary acyclic quivers

摘录 / EXCERPT: Let R be the path algebra of an acyclic quiver, let g be the onto map from admissible curves to indecomposable modules corresponding to positive real roots, and let M be an i...

conjecture | open | MathDB 页面 | 原文线索

V2-013331 | MATHDB-335691

The curve characterization of real Schur roots

摘录 / EXCERPT: Let $\mathcal{Q}$ be an acyclic quiver, let $P_{\mathcal{Q}}$ be the relevant set of seeds, and for each $\sigma\in P_{\mathcal{Q}}$ let Γ_{σ} be the set of isotopy cla...

conjecture | open | MathDB 页面 | 原文线索

V2-013332 | MATHDB-335698

Independence of irreducibility from the classical-group representation

摘录 / EXCERPT: Irreducibility-independence conjecture. The irreducibility of

conjecture | open | MathDB 页面 | 原文线索

V2-013333 | MATHDB-335714

The dominant maximal weight count conjecture for affine type B

摘录 / EXCERPT: Let $\mathfrak g=B_n^{(1)}$, let $\ell\geq 2$, and let ${\rm max}^+(\lambda+\Lambda\mid\ell)$ denote the set of dominant maximal weights at the indicated level. Dominant maximal we...

conjecture | open | MathDB 页面 | 原文线索

V2-013334 | MATHDB-335766

The 2-group order conjecture for \(\langle\hat{S},\hat{T}^2\rangle\)

摘录 / EXCERPT: Consider the subgroup generated by the matrices $\hat{S}$ and $\hat{T}^2$ associated with $\operatorname{PSU}(2)_{4m+2}$. 2-group order conjecture. If $m+1=2^k$ for some integer...

conjecture | open | MathDB 页面 | 原文线索

V2-013335 | MATHDB-335779

The module-isomorphism conjecture for generalized coinvariant algebras

摘录 / EXCERPT: Let n,k,r be parameters, let $R_{n,k,r}$ be the graded S_n-module quotient defined from $I_{n,k,r}$, and let ... be the ideal obtained by replacing the homogeneous symmetric func...

conjecture | open | MathDB 页面 | 原文线索

V2-013336 | MATHDB-335780

The delta-operator formula for the Frobenius image of the diagonal coinvariant quotient

摘录 / EXCERPT: Let ${\mathbf{x}}_n=(x_1,\dots,x_n)$ and ${\mathbf{y}}_n=(y_1,\dots,y_n)$ be two sets of variables, and let $DR_{n,k}$ be the bigraded S_n-module quotient by the ideal generated...

conjecture | open | MathDB 页面 | 原文线索

V2-013337 | MATHDB-335805

Set-theoretical Cohen-Macaulay defect and reflection generation conjecture

摘录 / EXCERPT: Set-theoretical Cohen-Macaulay defect conjecture. Assume the notation and hypotheses above. Then

conjecture | open | MathDB 页面 | 原文线索

V2-013338 | MATHDB-335807

The conjecture on primitive formally dual subsets of finite cyclic groups

摘录 / EXCERPT: Let ${\mathbb{Z}}_N$ be the cyclic group of N elements. A subset $T\subseteq {\mathbb{Z}}_N$ is primitive if it is not contained in any proper coset of ${\mathbb{Z}}_N$, and a su...

conjecture | open | MathDB 页面 | 原文线索

V2-013339 | MATHDB-335809

Foliation conjecture for the toy Teichmüller space of the hexagon

摘录 / EXCERPT: Each point of $\operatorname{\mathcal{T}}(\hexagon)$ is represented by the closed upper half-plane $\overline{\mathbb{H}}\cup\{\infty\}$ with marked points...

conjecture | open | MathDB 页面 | 原文线索

V2-013340 | MATHDB-335810

Convex-prism decomposition conjecture for the toy Teichmüller space of the hexagon

摘录 / EXCERPT: Let $\operatorname{\mathcal{T}}(\hexagon)$ denote the toy Teichmüller space of the hexagon, equipped with its D_6 action and its geodesic structure. A convex triangular prism is...

conjecture | open | MathDB 页面 | 原文线索

V2-013341 | MATHDB-335864

The Atiyah Conjecture with rational coefficients

摘录 / EXCERPT: Let G be a torsion-free group, let m,n be positive integers, and let $A\in M_{m,n}(\mathbb{Q}G)$. Write $r_A\colon \mathcal{N}(G)^m\to\mathcal{N}(G)^n$ for right multiplication...

conjecture | open | MathDB 页面 | 原文线索

V2-013342 | MATHDB-335874

The single-degree duality conjecture for simple representations

摘录 / EXCERPT: Let G be a totally disconnected group in the setting of Proposition, let $\mathcal{H}$ be the Hecke algebra appearing there, and let $M\in\mathcal{M}_A(G)$ be a simple module. Si...

conjecture | open | MathDB 页面 | 原文线索

V2-013343 | MATHDB-335888

Projective restriction conjecture for the endomorphism algebra $\mathscr{X}_{m,n}$

摘录 / EXCERPT: Let $\mathscr{X}_{m,n}$ be the endomorphism algebra of the mixed tensor product considered in the paper, let $K(\lambda)$ be a projective module indexed by a bipartition λ,...

conjecture | open | MathDB 页面 | 原文线索

V2-013344 | MATHDB-335889

Unchanged simple-module restriction formulas at the special parameter $\theta=\delta/\gamma$

摘录 / EXCERPT: Let $\mathsf{qw}\kern-1.8pt\mathscr{B}_{m,n}$ be the quantum walled Brauer algebra with $\theta=\frac{\delta}{\gamma}$, let $\mathscr{C}\mathrm{r}_{m,n}$ denote the set of cross bi...

conjecture | open | MathDB 页面 | 原文线索

V2-013345 | MATHDB-335903

Valette's a-T-menability conjecture for wreath products

摘录 / EXCERPT: Let H and Q be discrete groups, and write $H\wr Q$ for their wreath product. A group is a-T-menable if it acts metrically properly on a Hilbert space by affine isometries. Vale...

conjecture | open | MathDB 页面 | 原文线索

V2-013346 | MATHDB-335917

Kiyota-Murai-Wada rationality conjecture for Cartan matrix eigenvalues

摘录 / EXCERPT: Let B be a block of a finite group G over an algebraically closed field of characteristic $p>0$, let $C\in\mathbb{Z}^{l\times l}$ be its Cartan matrix, and let D be a defect...

conjecture | open | MathDB 页面 | 原文线索

V2-013347 | MATHDB-335925

The Steinberg representation conjecture for real reductive groups

摘录 / EXCERPT: Let G be a real reductive group, and let St_G denote the Steinberg representation of $G({\mathbb R})$. Let $D_2: W_{\mathbb R}\rightarrow \operatorname{SL}_2({\mathbb C})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-013348 | MATHDB-335926

The transitivity conjecture for the Langlands-factor partial order

摘录 / EXCERPT: Let G be a reductive algebraic group over a local field F. For irreducible admissible representations of $G(F)$, define $\pi_1\geq\pi_2$ when π_1 appears as a Jordan-Hölder...

conjecture | open | MathDB 页面 | 原文线索

V2-013349 | MATHDB-335927

The low-rank reduction conjecture for Jordan-Hölder factors of real principal series

摘录 / EXCERPT: Low-rank reduction conjecture. An irreducible admissible representation π of $\operatorname{GL}_n({\mathbb R})$ occurs as a Jordan-Hölder factor of $Ps(\underline{\pi})$ if and...

conjecture | open | MathDB 页面 | 原文线索

V2-013350 | MATHDB-335934

p-uniformity conjecture for invariant rings

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero and let W be a finite dimensional k-representation for a reductive group G. Put ... A Frobenius summand means an...

conjecture | open | MathDB 页面 | 原文线索

V2-013351 | MATHDB-335936

Characteristic-independent Cohen-Macaulay conjecture for strongly critical weights

摘录 / EXCERPT: Let G be a reductive group with maximal torus H acting on a representation W with weights $\beta_1,\ldots,\beta_d\in X(H)$. Let ... and let ρ be half the sum of the positi...

conjecture | open | MathDB 页面 | 原文线索

V2-013352 | MATHDB-335937

Vogan's strong-stability basis conjecture

摘录 / EXCERPT: Let F be a local field, let G be a quasi-split connected reductive linear algebraic group over F, let $\lambda\in\Lambda({}^LG)$ be an infinitesimal parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013353 | MATHDB-335967

Characterization of groups with EDT0L and ET0L word problems

摘录 / EXCERPT: EDT0L and ET0L word-problem conjecture. A group has EDT0L word problem if and only if it is finite, and a group has ET0L word problem if and only if it is virtually free.

conjecture | open | MathDB 页面 | 原文线索

V2-013354 | MATHDB-335972

Semi-simple groups' monochromatic eigenvalue statistics conjecture

摘录 / EXCERPT: Semi-simple groups' eigenvalue statistics conjecture. As $p\to\infty$, for almost all choices of the generators, the distribution of its eigenvalues converges weakly to the Kesten-...

conjecture | open | MathDB 页面 | 原文线索

V2-013355 | MATHDB-335985

The Chevalley involution conjecture for Langlands-Vogan parameters

摘录 / EXCERPT: Let G be a reductive algebraic group over a local field F that is a pure innerform of a quasi-split group G_0 over F, equipped with a fixed triple (B_0,N_0,ψ_0) used t...

conjecture | open | MathDB 页面 | 原文线索

V2-013356 | MATHDB-335986

The MVW-type innerform conjecture for automorphisms of representations

摘录 / EXCERPT: Let G and G' be inner forms, and let ϕ and ϕ' be automorphisms arising from the corresponding adjoint groups. Suppose they correspond under the natural isomorphism of...

conjecture | open | MathDB 页面 | 原文线索

V2-013357 | MATHDB-336002

Humphreys' conjecture on Springer fiber graph duality for special nilpotent orbits

摘录 / EXCERPT: Let G be a simple algebraic group over an algebraically closed field, let $\mathfrak{g}=\operatorname{Lie}G$, and let $N\in\mathfrak{g}$ lie in a special nilpotent orbit. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013358 | MATHDB-336022

The logarithmic-logarithmic expansion conjecture for group representations

摘录 / EXCERPT: The logarithmic-logarithmic expansion conjecture. For every $\varepsilon>0$, there is a constant C_ε such that, for every finite group G and every nontrivial irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-013359 | MATHDB-336033

Cherlin's almost simple binary primitive group conjecture

摘录 / EXCERPT: Let G be an almost simple primitive permutation group acting on a set Ω, and suppose that its permutation action is binary. Cherlin's almost simple conjecture. Then ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-013360 | MATHDB-336084

The virtually free group conjecture for G-TCF

摘录 / EXCERPT: Let G be a finitely generated group. The theory G-TCF is the model companion theory for fields equipped with an action of G. A group is virtually free if it...

conjecture | open | MathDB 页面 | 原文线索

V2-013361 | MATHDB-336085

The locally virtually free group conjecture for arbitrary G-TCF

摘录 / EXCERPT: Let G be a group. The theory G-TCF is the model companion theory for fields equipped with an action of G. A group is locally virtually free if every finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-013362 | MATHDB-336098

Bardakov-Nasybullov-Neshchadim's nilpotency conjecture for groups with subgroup twisted conjugacy classes

摘录 / EXCERPT: Let G be a group, let $\operatorname{Aut}(G)$ denote its automorphism group, and for $\varphi\in\operatorname{Aut}(G)$ let $[e]_{\varphi}$ be the twisted conjugacy class of the i...

conjecture | open | MathDB 页面 | 原文线索

V2-013363 | MATHDB-336104

The quantum Thoma conjecture for compact quantum groups

摘录 / EXCERPT: Quantum Thoma conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-013364 | MATHDB-336142

Conjecture on chambers induced by tau-tilting pairs

摘录 / EXCERPT: Let A be an algebra. A chamber is a chamber in the wall and chamber structure of A, and a tau-tilting pair is a pair (M,P) consisting of a tau-tilting module and its projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-013365 | MATHDB-336143

Conjecture that the wall-and-chamber quiver equals the tau-tilting exchange graph

摘录 / EXCERPT: Let A be an algebra, and let $\mathfrak{G}_A$ be the quiver whose vertices are the chambers in the wall and chamber structure of A, with an arrow from the chamber...

conjecture | open | MathDB 页面 | 原文线索

V2-013366 | MATHDB-336144

Geck-Hiss conjectures on decomposition matrices of finite reductive groups

摘录 / EXCERPT: Let G be a finite reductive group with Weyl group W, defined over a field of characteristic p, and let k be a field of characteristic ℓ with $\ell \neq p$. Let d denote...

conjecture | open | MathDB 页面 | 原文线索

V2-013367 | MATHDB-336145

Dudas-Malle projectivity conjecture for intersection cohomology characters

摘录 / EXCERPT: Let G be a finite reductive group with Weyl group W, let F be the relevant Frobenius endomorphism, and let $\boldsymbol{T}$ be an F-stable torus. For $w \in W$, let Q_w...

conjecture | open | MathDB 页面 | 原文线索

V2-013368 | MATHDB-336205

The conjugacy separability growth conjecture for nilpotent groups

摘录 / EXCERPT: Conjugacy separability growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013369 | MATHDB-336278

Generalized Gerl conjecture for left reversible semigroups

摘录 / EXCERPT: Let S be a semigroup and let P be a probability measure on S whose support generates S. Define convolution by ... and let $P^{(n)}$ denote the n-fold convolution power. The...

conjecture | open | MathDB 页面 | 原文线索

V2-013370 | MATHDB-336279

Left reversible amenable monoid without maximal global cogrowth

摘录 / EXCERPT: Let S be a finitely generated monoid. Its global cogrowth is maximal when the global cogrowth rate equals the cardinality of a chosen finite generating set. Left amenable means t...

conjecture | open | MathDB 页面 | 原文线索

V2-013371 | MATHDB-336319

Nonnegativity conjecture for the coefficients of the orbital-measure polynomials

摘录 / EXCERPT: Let R_n and $\widehat R_n$ be the polynomials whose coefficients are denoted by r_κ and $\hat r_\kappa$, respectively. Nonnegativity conjecture. The coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-013372 | MATHDB-336389

Non-symmetric coefficient conjecture for order-dividing bijections from dihedral to cyclic groups

摘录 / EXCERPT: Let $D_{2n}=\langle s,r\mid s^2=r^n=1,\ srs=r^{-1}\rangle$ be a dihedral group and let $\mathbb{Z}_{2n}$ be the cyclic group of the same order. Suppose that ... is an order-dividing...

conjecture | open | MathDB 页面 | 原文线索

V2-013373 | MATHDB-336398

The invariant-theoretic conjecture for visible polar representations

摘录 / EXCERPT: Let G be an algebraic group acting on a vector space V. Assume that (G,V) is visible, meaning that it has finitely many nilpotent G-orbits, and polar, meaning that there ex...

conjecture | open | MathDB 页面 | 原文线索

V2-013374 | MATHDB-336405

Subag's Jantzen quotient conjecture

摘录 / EXCERPT: Let σ be a real structure of an algebraic family of Harish-Chandra pairs $(\boldsymbol{\mathfrak{g}},\boldsymbol{K})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013375 | MATHDB-336411

The compact induction conjecture for irreducible supercuspidal representations

摘录 / EXCERPT: Let G be a reductive group and let $\mathcal{M}(G)$ denote the representations of G. Let $\pi\in\mathcal{M}(G)$ be irreducible and supercuspidal. A subgroup K of G is compa...

conjecture | open | MathDB 页面 | 原文线索

V2-013376 | MATHDB-336425

Type D spin nilHecke algebra endomorphism conjecture

摘录 / EXCERPT: Type D spin nilHecke endomorphism conjecture. There is a graded algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013377 | MATHDB-336428

I-Todorov's conjecture on longest maximal green sequences

摘录 / EXCERPT: Let Λ be a cluster-tilted algebra of finite type, and let C be one of the associated tilted algebras. The Auslander-Reiten quiver of Λ is used to order the C-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-013378 | MATHDB-336429

Finer-graded Lefschetz inequality conjecture for free cyclic actions

摘录 / EXCERPT: Let Δ be an orientable $\mathbbm{k}$-homology manifold of dimension $d-1$ admitting a free action by the cyclic group $\mathbb{Z}/p\mathbb{Z}$. For the finer...

conjecture | open | MathDB 页面 | 原文线索

V2-013379 | MATHDB-336446

The affine parabolic Kazhdan-Lusztig conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: Let $e>h\ell$, let $\kappa\in I^\ell$ be an h-admissible multicharge, and let $\Bbbk$ be a field of characteristic $p\gg h\ell$. For $\lambda,\mu\in\mathscr{P}^{\ell}_{n}(h)$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-013380 | MATHDB-336447

The parabolic Kazhdan-Lusztig conjecture for Hecke algebras with separated multicharge

摘录 / EXCERPT: Let $e=\infty$ or, more generally, $e>n$, and let $\kappa\in I^\ell$ have no repeated entries. Let $\Bbbk$ be a field of characteristic $p\gg\ell$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013381 | MATHDB-336453

Non-negativity conjecture for Macdonald, Jack, Heckman-Opdam and Bessel coefficients

摘录 / EXCERPT: The coefficients $c_{\lambda,\mu}^{\nu}(\hat P; q,t)$, $c_{\lambda,\mu}^{\nu}(\hat J; \\,\theta)$, $c_{\ell,r}^{s}(\hat \mathcal F; \\,\theta)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013382 | MATHDB-336465

Derived dimension conjecture for tame derived representation type

摘录 / EXCERPT: Derived dimension conjecture. The derived dimension of A satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-013383 | MATHDB-336470

The representation-category equivalence for loop group nets

摘录 / EXCERPT: Let G be the compact, simply connected Lie group and k the level appearing in the representation categories. Let $\mathrm{Rep}(\cA_{G,k})$ be the category of representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-013384 | MATHDB-336471

The equivalence of local normality and positive energy for based loop groups

摘录 / EXCERPT: Let $\widetilde{\Omega G}$ be the centrally extended based loop group, and let H be a representation of it at level k. Call H locally normal when it belongs to the category o...

conjecture | open | MathDB 页面 | 原文线索

V2-013385 | MATHDB-336498

Olsson's conjecture for π-blocks

摘录 / EXCERPT: Olsson's conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-013386 | MATHDB-336506

Pyber's base size conjecture for primitive permutation groups

摘录 / EXCERPT: Let G be a finite primitive permutation group on a finite set S. A base is a subset $T\subseteq S$ satisfying $G_T=\{e\}$, and $b(G)$ denotes the minimum cardinality of a base....

conjecture | open | MathDB 页面 | 原文线索

V2-013387 | MATHDB-336518

Canonical identification of minimal and deformed maximal extensions

摘录 / EXCERPT: Let i be the inclusion of the relevant K-orbit, let ϕ be the corresponding local system, and let $i_!\phi$ and $i_+\phi$ denote its minimal and maximal extensions as...

conjecture | open | MathDB 页面 | 原文线索

V2-013388 | MATHDB-336551

The scattered-part cardinality conjecture for type A

摘录 / EXCERPT: Let N_{A_n} denote the cardinality of the scattered part of $\widehat{A}_n^{\mathrm{d}}$, the set of irreducible unitary representations for the complex simple group of type...

conjecture | open | MathDB 页面 | 原文线索

V2-013389 | MATHDB-336586

Elementary equivalence conjecture for finite-group coset expansions of non-abelian free groups

摘录 / EXCERPT: Let Q be a finite group, let F_{1} and F_{2} be non-abelian free groups, and for $i=1,2$ let $\pi_i:F_i\twoheadrightarrow Q$ be epimorphisms. Let $\mathcal{L}^{Q}$ be the lan...

conjecture | open | MathDB 页面 | 原文线索

V2-013390 | MATHDB-336600

The spread and generating graph equivalence conjecture

摘录 / EXCERPT: Let G be a finite group with $|G|\geqslant 4$, and let $\Gamma(G)$ be its generating graph. The spread of G is the largest integer k such that, for every set of k nonidenti...

conjecture | open | MathDB 页面 | 原文线索

V2-013391 | MATHDB-336602

Noether number conjecture for cyclic-by-cyclic semidirect products

摘录 / EXCERPT: Let m,n be positive integers, and let $C_n\ltimes_{\varphi} C_m$ be a semidirect product determined by a homomorphism $\varphi:C_n\to\operatorname{Aut}(C_m)$. Let $\beta(G)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-013392 | MATHDB-336661

Antolín, Martino and Ventura's meta-conjecture on degree of commutativity

摘录 / EXCERPT: Antolín, Martino and Ventura's meta-conjecture. For any “reasonable” sequence $M=(\mu_n)_{n=1}^\infty$ of probability measures on G, we have $\operatorname{dc}_M(G)>0$ if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-013393 | MATHDB-336669

Farb's genericity conjecture for pseudo-Anosov elements in mapping class groups

摘录 / EXCERPT: Let Mod be a mapping class group equipped with a finite generating set and its associated word metric. The pseudo-Anosov elements are elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-013394 | MATHDB-336670

Contracting-elements conjecture for mapping class groups on Cayley graphs

摘录 / EXCERPT: Let Mod be a mapping class group with a finite generating set, and let its Cayley graph be equipped with the corresponding word metric. An element of...

conjecture | open | MathDB 页面 | 原文线索

V2-013395 | MATHDB-336675

The right inner mapping group conjecture for simple right conjugacy closed loops

摘录 / EXCERPT: The right inner mapping group conjecture. The right inner mapping group of $(Q,\circ f)$ is exactly

conjecture | open | MathDB 页面 | 原文线索

V2-013396 | MATHDB-336677

Stable Prasad conjecture for square-integrable representations

摘录 / EXCERPT: Stable Prasad conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-013397 | MATHDB-336680

Cyclic groups minimize the number of connected components of power maps

摘录 / EXCERPT: Let G be a finite group of order n, let C_n denote the cyclic group of order n, and let $N(a,G)$ be the number of connected components of the functional graph $G(a,G)$ asso...

conjecture | open | MathDB 页面 | 原文线索

V2-013398 | MATHDB-336758

The twisted Drinfeld-Sokolov reduction conjecture for Weyl modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $\widehat{\mathfrak g}$ be its affine algebra, let d_{st} be the standard semi-infinite cohomology differential, let σ^λ be...

conjecture | open | MathDB 页面 | 原文线索

V2-013399 | MATHDB-336837

The averaging functional equation for opposite unipotent invariants

摘录 / EXCERPT: Let H be an algebraic group and define its shifted determinant line by ... For $H_1\subset H_2$, set ... These symbols also denote tensoring functors. Let $\CC$ be either the categor...

conjecture | open | MathDB 页面 | 原文线索

V2-013400 | MATHDB-336838

The geometric form of the Jacquet-functor adjointness conjecture

摘录 / EXCERPT: Let $\CC$ be either the category of $(\fg,K)$-modules with central character χ or the corresponding twisted $\mathcal D$-module category. Let $\operatorname{oblv}_{N^-}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013401 | MATHDB-336840

Kim's second adjointness conjecture for principal series functors

摘录 / EXCERPT: Let G be a reductive group with symmetric subgroup $K=G^\theta$, let P be a θ-minimal parabolic with unipotent radicals N and N^-, and let $M_K=K\cap P$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013402 | MATHDB-336856

Etingof's monodromy conjecture for Dunkl-operator eigenlines

摘录 / EXCERPT: Let W be a Weyl group, let C_Δ be the cactus group acting on W, and suppose that the family of algebras D_χ, parametrized by $\chi\in\mathfrak h^{reg}$, compactifie...

conjecture | open | MathDB 页面 | 原文线索

V2-013403 | MATHDB-336886

Minimal-dimensional representation conjecture for finite W-superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{g}_{\bar 0}+\mathfrak{g}_{\bar 1}$ be a basic Lie superalgebra over $\mathbb{C}$, let $e\in\mathfrak{g}_{\bar 0}$ be nilpotent, and let r be...

conjecture | open | MathDB 页面 | 原文线索

V2-013404 | MATHDB-336887

One-dimensional representation conjecture for refined W-superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a basic Lie superalgebra over $\mathbb{C}$, let $e\in\mathfrak{g}_{\bar 0}$ be nilpotent, and let χ be the corresponding linear function on...

conjecture | open | MathDB 页面 | 原文线索

V2-013405 | MATHDB-336905

Andersen's conjecture on modular and quantum tilting characters

摘录 / EXCERPT: Andersen's conjecture. Suppose Lusztig's conjecture holds for G in characteristic p. For $\lambda \in A_0$ and dominant $x \in \prescript{p}{}{W}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-013406 | MATHDB-336939

Conjecture on the upper rank of the endomorphism semigroup of a Brandt semigroup

摘录 / EXCERPT: Let B_n be the Brandt semigroup, and let ${\rm End}(B_n)$ denote its semigroup of endomorphisms. Write $r_4(S)$ for the upper rank of a semigroup S. Upper-rank conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013407 | MATHDB-336956

Quasi-flat model faithfulness and stationarity conjecture

摘录 / EXCERPT: Quasi-flat model conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013408 | MATHDB-336970

The almost simple group conjecture for the divisibility of Sylow subgroup numbers

摘录 / EXCERPT: Almost simple group conjecture. If S satisfies $\mathbf{DivSyl}(p)$, then every group L satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-013409 | MATHDB-336978

Indecomposability conjecture for shuffling functors on simple modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex semisimple Lie algebra of any type, let W be its Weyl group, let $L(y)$ be a simple module in the principal block of category $\mathcal{O}$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-013410 | MATHDB-336984

Determinant conjecture for globally positive group matrices

摘录 / EXCERPT: Let G be a group and let $A\in\operatorname{Mat}_n(\mathbb{Z}[G])$ be a globally positive matrix, meaning a matrix to which the stated positivity condition applies. Determinant C...

conjecture | open | MathDB 页面 | 原文线索

V2-013411 | MATHDB-336992

Embedding conjecture for automorphism groups of positive-entropy SFTs

摘录 / EXCERPT: Embedding conjecture. For every finite alphabet A, there is an embedding

conjecture | open | MathDB 页面 | 原文线索

V2-013412 | MATHDB-336996

The costable module resolution dimension conjecture

摘录 / EXCERPT: Costable module resolution dimension conjecture. Any non-injective costable module has infinite $\operatorname{add}(A \oplus D(A))$-resolution dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-013413 | MATHDB-336997

Generalized vulnerability conjecture for homomorphic encryption of classical matrix groups

摘录 / EXCERPT: Let q be odd, let n be a positive integer, and let G be any classical matrix group of dimension n over the finite field $\mathbb{F}_q$. Consider homomorphic encryption of...

conjecture | open | MathDB 页面 | 原文线索

V2-013414 | MATHDB-337018

The conjugacy-ratio conjecture for finitely generated groups

摘录 / EXCERPT: Conjugacy-ratio conjecture. The group G has positive conjugacy ratio if and only if G is virtually abelian:

conjecture | open | MathDB 页面 | 原文线索

V2-013415 | MATHDB-337046

The generalized oscillating tableaux-sl(n) web bijection conjecture

摘录 / EXCERPT: Let a generalized oscillating tableau have length k and n parts, with final component $(m,\ldots,m)$ for some $m\in\mathbb{Z}$. Consider $\operatorname{sl}(n)$ webs with k bo...

conjecture | open | MathDB 页面 | 原文线索

V2-013416 | MATHDB-337064

The Steinberg-type character formula for grade-zero standard modules

摘录 / EXCERPT: Let G be the reductive group, X^+ the dominant weights, p the characteristic, ρ the half-sum of the positive roots, and $\rho^$ the associated weight notation. For a reg…

conjecture | open | MathDB 页面 | 原文线索

V2-013417 | MATHDB-337117

The converse conjecture for (m,1)-almost hereditary algebras

摘录 / EXCERPT: Converse conjecture. The algebra A belongs to

conjecture | open | MathDB 页面 | 原文线索

V2-013418 | MATHDB-337174

Xia-Fang finiteness conjecture for cubic graphical regular representations

摘录 / EXCERPT: Xia-Fang finiteness conjecture. There are only finitely many finite nonabelian simple groups that have no cubic GRR.

conjecture | open | MathDB 页面 | 原文线索

V2-013419 | MATHDB-337175

Spiga's cubic GRR conjecture with one involution

摘录 / EXCERPT: Spiga's conjecture. Except for a finite number of cases and for the groups $ \operatorname{PSL}_2(q)$, every finite nonabelian simple group has a cubic GRR whose connection set con…

conjecture | open | MathDB 页面 | 原文线索

V2-013420 | MATHDB-337177

The K-intertwining conjecture for electroweak S matrices

摘录 / EXCERPT: Let ${\mathcal H}_k$ denote the Hilbert space associated with the incoming (k,l) scattering sector, and let ${\mathcal D}_l$ denote the corresponding outgoing space. An operator…

conjecture | open | MathDB 页面 | 原文线索

V2-013421 | MATHDB-337179

Representation-theoretic conjecture for the determinant's higher syzygies

摘录 / EXCERPT: The higher-syzygy representation conjecture. The degree $r+1$ component of F_r is isomorphic as a G_{det_n}-representation to

conjecture | open | MathDB 页面 | 原文线索

V2-013422 | MATHDB-337216

Lee-Lee conjecture on real Schur roots and simple loops

摘录 / EXCERPT: Lee-Lee conjecture. The reflections in W whose assigned class of loops contains a representative without self-intersections are precisely those corresponding to real Schur roots…

conjecture | open | MathDB 页面 | 原文线索

V2-013423 | MATHDB-337221

Wall's conjecture on maximal proper subgroups of finite groups

摘录 / EXCERPT: For a finite group G, let $\max(G)$ denote the number of maximal proper subgroups of G. Wall's conjecture. … This conjecture concerns a uniform bound on maximal proper subgroup…

conjecture | open | MathDB 页面 | 原文线索

V2-013424 | MATHDB-337229

The Whitehouse-module restriction conjecture for the LAnKe representations

摘录 / EXCERPT: For integers n and k, let $\rho_{n,k}$ be the representations in the LAnKe array, and let W_{k+1} denote the Whitehouse module. For each irreducible submodule of W_{k+1}, f…

conjecture | open | MathDB 页面 | 原文线索

V2-013425 | MATHDB-337276

Nekrashevych-Pete's conjecture on strongly scale-invariant groups

摘录 / EXCERPT: A finitely generated group Γ is strongly scale-invariant if there exists an embedding $\varphi:\Gamma\hookrightarrow\Gamma$ with image of finite index and such that … is fin…

conjecture | open | MathDB 页面 | 原文线索

V2-013426 | MATHDB-337300

Pole cancellation conjecture for Casselman and R-polynomial coefficients

摘录 / EXCERPT: Pole cancellation conjecture. The functions

conjecture | open | MathDB 页面 | 原文线索

V2-013427 | MATHDB-337351

The conjecture that O_3 is a 3-MCFL with the displayed grammar

摘录 / EXCERPT: The conjecture that O_3 is a 3-MCFL. The group O_3 is a 3-MCFL and is generated by the grammar G above.

conjecture | open | MathDB 页面 | 原文线索

V2-013428 | MATHDB-337356

Two-generator conjecture for alternating and symmetric groups

摘录 / EXCERPT: Two-generator conjecture. Two elements of order k suffice to generate Σ_n when k is even and to generate A_n when k is odd.

conjecture | open | MathDB 页面 | 原文线索

V2-013429 | MATHDB-337358

Canonical ordering conjecture for parameters of type 2i12 or 2r21

摘录 / EXCERPT: Let $\kappa=\{\alpha,\beta\}$, where α and β are orthogonal and interchanged by σ. Suppose κ is of type 2i12 or 2r21 for parameters γ and…

conjecture | open | MathDB 页面 | 原文线索

V2-013430 | MATHDB-337359

Conjectural Hecke matrix in the 1r1s case

摘录 / EXCERPT: In the 1r1s case, let $\mathcal M$ have basis $a_{(\gamma)},a_{(\gamma",+)},a_{(\gamma",-)}$, and let T_κ be the Hecke operator. Let q denote the parameter used in the Hec…

conjecture | open | MathDB 页面 | 原文线索

V2-013431 | MATHDB-337366

Bijection with the Newton-Okounkov polytope

摘录 / EXCERPT: The proposed bijection. The resulting point is the corresponding point in NO_G^r.

conjecture | open | MathDB 页面 | 原文线索

V2-013432 | MATHDB-337375

Embedding conjecture for hyperbolic structures from non-equivalent loxodromic elements

摘录 / EXCERPT: Embedding conjecture. Under these assumptions, $4\mathcal{H}(F_2)4$ embeds in $4\mathcal{H}(G)4$.

conjecture | open | MathDB 页面 | 原文线索

V2-013433 | MATHDB-337407

The supercenter conjecture for odd KLR algebras

摘录 / EXCERPT: Let $R_{{\mathfrak{b}}}(\nu)$ be the odd KLR algebra associated with a dimension vector ν, and let $Z(R_{{\mathfrak{b}}}(\nu))$ denote its supercenter. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013434 | MATHDB-337441

Spectral realization conjecture for hydrogen-atom solution spaces

摘录 / EXCERPT: Spectral realization conjecture. For almost every $\lambda\in\operatorname{spec}(H)$,

conjecture | open | MathDB 页面 | 原文线索

V2-013435 | MATHDB-337451

Navarro's self-normalizing Sylow 2-subgroup conjecture

摘录 / EXCERPT: Let G be a finite group and let $P\in\operatorname{Syl}_2(G)$. Let $\sigma\in\operatorname{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$ be the unique automorphism that fixes 2-ro...

conjecture | open | MathDB 页面 | 原文线索

V2-013436 | MATHDB-337457

Elementary-divisor conjecture for critical groups of induced representations

摘录 / EXCERPT: Let A be an abelian group of order r, let $\mathfrak{S}$ denote the tower of symmetric groups, and consider the differential tower of groups $A\wr\mathfrak{S}$. For $k\leq n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-013437 | MATHDB-337466

The hereditary abelian subcategory conjecture for the non-strong minimal right determiner example

摘录 / EXCERPT: Hereditary abelian subcategory conjecture. The category $\mathcal C'$ is a Hom-finite hereditary abelian category.

conjecture | open | MathDB 页面 | 原文线索

V2-013438 | MATHDB-337468

The relative Cannon conjecture

摘录 / EXCERPT: The relative Cannon conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013439 | MATHDB-337469

Nonexistence conjecture for doubly regular generalized quadrangles

摘录 / EXCERPT: Let $\mathcal{Q}$ be a generalized quadrangle of order $s>1$, and suppose a group of automorphisms acts regularly on both the point set and the line set of $\mathcal{Q}$. Nonexiste...

conjecture | open | MathDB 页面 | 原文线索

V2-013440 | MATHDB-337496

The admissible-polarization conjecture for the character formula

摘录 / EXCERPT: Admissible-polarization conjecture. The character formula

conjecture | open | MathDB 页面 | 原文线索

V2-013441 | MATHDB-337500

KMS conjecture on finite presentability of finitely generated Lambda-free groups

摘录 / EXCERPT: Let Λ be an ordered abelian group, and let G be a finitely generated group acting freely on a Λ-tree; such a group is called Λ-free. KMS conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-013442 | MATHDB-337506

Dixon's strong 2-generation conjecture for finite simple groups

摘录 / EXCERPT: Let (G_n) be any sequence of finite simple groups such that $|G_n|$ tends to infinity with n. Let $\mathbb{P}_2(G)$ denote the probability that two randomly chosen elements gen...

conjecture | open | MathDB 页面 | 原文线索

V2-013443 | MATHDB-337507

Conder's (2,3)- or (2,5)-generation conjecture

摘录 / EXCERPT: For a finite group G, being $(2,r)$-generated means that G is generated by an involution and an element of order r. Conder's conjecture. Every non-abelian finite simple group...

conjecture | open | MathDB 页面 | 原文线索

V2-013444 | MATHDB-337509

Liebeck-Shalev conjecture on epimorphic homomorphisms from Fuchsian groups

摘录 / EXCERPT: Let Γ be a Fuchsian group, and let G be a finite simple classical group. A randomly chosen homomorphism in ${\rm Hom}(\Gamma,G)$ is an epimorphism with a probability that...

conjecture | open | MathDB 页面 | 原文线索

V2-013445 | MATHDB-337511

Five-generation conjecture for maximal subgroups of almost simple groups

摘录 / EXCERPT: Let G be an almost simple group, meaning that $T\leq G\leq {\rm Aut}(T)$ for some non-abelian finite simple group T, and let H be a maximal subgroup of G. Write $d(H)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-013446 | MATHDB-337515

WPD conjecture for the cyclic splitting complex

摘录 / EXCERPT: Let $D53D$ be a finitely generated free group, let $D52CD66D61\operatorname{Out}(D53D)$ act on the cyclic splitting complex $D9D$. An element is loxodromic if it acts loxodromicall...

conjecture | open | MathDB 页面 | 原文线索

V2-013447 | MATHDB-337520

The Smoothness Conjecture for profinite groups

摘录 / EXCERPT: Smoothness Conjecture. If G is 1-smooth, then it has the weak Bloch-Kato property. In particular, it is smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-013448 | MATHDB-337521

Finite-order conjecture for the Coxeter transformation of cominuscule posets

摘录 / EXCERPT: Finite-order conjecture. The translation τ has finite order, specifically

conjecture | open | MathDB 页面 | 原文线索

V2-013449 | MATHDB-337523

The proper finite-index subgroup conjecture for $PD(3)$ groups

摘录 / EXCERPT: The proper finite-index subgroup conjecture. Every $PD(3)$ group contains a proper subgroup of finite index.

conjecture | open | MathDB 页面 | 原文线索

V2-013450 | MATHDB-337524

The abelian eighth-power subgroup conjecture for descendants of the candidate groups

摘录 / EXCERPT: Eighth-power subgroup conjecture. If G is any descendant of one of the 128 candidates for $Q_3(G_K)$, then G^8 has index at most 2^{40} and is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-013451 | MATHDB-337574

Leclerc's monoidal categorification conjecture for quantum affine algebras

摘录 / EXCERPT: Let $U_q\hat{\mathfrak{g}}$ be the quantum affine algebra associated with a Dynkin diagram of ADE type, with rank n, and let $\mathcal{C}_l$ be the category of finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-013452 | MATHDB-337588

Unique irreducible subrepresentation conjecture for products of regular ladder representations

摘录 / EXCERPT: Let π_1 and π_2 be a regular pair of ladder representations, and let $\mathcal{B}(\pi_1,\pi_2)$ denote the set of representations occurring in the relevant decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-013453 | MATHDB-337623

Perfectness conjecture for the closure of the compactly supported pure mapping class group

摘录 / EXCERPT: Perfectness conjecture. The group $\overline{\mathrm{PMap}}_c(S)$ is perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-013454 | MATHDB-337632

Dominant Schur module conjecture for Veronese syzygies

摘录 / EXCERPT: Let $K_{p,q}(\mathbb P^n,b;d)$ be the Koszul cohomology group of the d-uple Veronese embedding, let $\mathrm E_{p,q}(\mathbb P^n,b;d)$ be its monomial syzygy construction, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-013455 | MATHDB-337639

Farb's conjecture on geometric rank-one manifolds

摘录 / EXCERPT: Let M be a tame, complete, finite-volume n-manifold of bounded nonpositive curvature. A manifold has geometric rank one when it satisfies the rank-one condition used in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-013456 | MATHDB-337670

The common finite cover conjecture for special cube complexes

摘录 / EXCERPT: Common finite cover conjecture. There exist finite-sheeted covers

conjecture | open | MathDB 页面 | 原文线索

V2-013457 | MATHDB-337717

Generalisation of the affine-type inner product construction

摘录 / EXCERPT: Let the inner product be the construction given in Theorem, and let an affine type mean the type of an affine Weyl group or affine Hecke algebra. Inner product generalisation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-013458 | MATHDB-337751

Freeness conjecture for graded invariant modules over Dedekind domains

摘录 / EXCERPT: Let R be a Dedekind domain, let G be a finite group, and let $G\to \operatorname{GL}_n(R)$ be an R-representation. Write $R[x_1,\ldots,x_n]_d^G$ for the degree-d component...

conjecture | open | MathDB 页面 | 原文线索

V2-013459 | MATHDB-337763

Wise's cubulation conjecture for torsion-free one-relator groups

摘录 / EXCERPT: Wise's conjecture. Every torsion-free one-relator group with no subgroup isomorphic to $BS(m,n)$, for $m\neq\pm n$, acts freely on a CAT(0) cube complex.

conjecture | open | MathDB 页面 | 原文线索

V2-013460 | MATHDB-337773

The conjecture that all difference sets are Desarguesian

摘录 / EXCERPT: Let $\mathcal{D}$ be a difference set of order q, and let $\mathcal{T}(\mathcal{D})$ be the generalized triangle constructed from it. The difference set $\mathcal{D}$ is Desargue...

conjecture | open | MathDB 页面 | 原文线索

V2-013461 | MATHDB-337780

Stable rationality conjecture for quotients by free central extensions

摘录 / EXCERPT: Let A be a finite abelian group, let $F^c(A)$ be its free central extension, and let V be a faithful linear representation of $F^c(A)$ over an algebraically closed field k. F...

conjecture | open | MathDB 页面 | 原文线索

V2-013462 | MATHDB-337782

Central-extension conjecture for rational quotients of finite solvable groups

摘录 / EXCERPT: Let G be a finite solvable group. Central-extension rationality conjecture. There exists a central extension ... with a faithful representation V of G^c such that V/G^c is ra...

conjecture | open | MathDB 页面 | 原文线索

V2-013463 | MATHDB-337859

The Ext-degree formula conjecture for integrable sl(infinity)-modules

摘录 / EXCERPT: Let $\mathbb T_{\mathfrak{g},\mathfrak{k}}$ be the category under consideration, and let $\mathbf V^{\boldsymbol{\lambda},\boldsymbol{\mu}}$ denote its simple objects indexed by pa...

conjecture | open | MathDB 页面 | 原文线索

V2-013464 | MATHDB-337860

The socle and tensor filtration identification conjecture for K_m|n

摘录 / EXCERPT: Let $X_{\mathfrak{gl}(m|n)}$ be the set of elements used to define the Duflo-Serganova filtration, let $X^k_{\mathfrak{gl}(m|n)}$ consist of those of rank at most k, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-013465 | MATHDB-337874

Ngô's compatibility conjecture for the quantum Ngô action

摘录 / EXCERPT: Ngô's compatibility conjecture. The following diagram is commutative:

conjecture | open | MathDB 页面 | 原文线索

V2-013466 | MATHDB-337875

Ngô's fully faithful image conjecture for very central D-modules

摘录 / EXCERPT: Ngô's very central image conjecture. The Ngô functor defines a fully faithful braided monoidal functor on abelian categories, whose essential image is $\mathcal M(G)^G_{vc}$. Moreo...

conjecture | open | MathDB 页面 | 原文线索

V2-013467 | MATHDB-337939

Shareshian's subgroup-lattice homotopy conjecture

摘录 / EXCERPT: Let G be a finite group, and consider an open interval in the subgroup lattice of G. Shareshian's conjecture. Every such open interval has the same homotopy type as a wedge of...

conjecture | open | MathDB 页面 | 原文线索

V2-013468 | MATHDB-337970

Finite disjoint-union conjecture for finitely presented monoids of linear growth

摘录 / EXCERPT: Let S be a finitely presented monoid of linear growth. A free sandwich is a subset of the form $p\langle q\rangle r$, where p,q,r are words and $\langle q\rangle$ denotes the s...

conjecture | open | MathDB 页面 | 原文线索

V2-013469 | MATHDB-337982

Coherent preminimal non-minimality conjecture under a support condition

摘录 / EXCERPT: Let $\boldsymbol\omega\in\mathcal P_q^+$ be preminimal of order 2, let $k\in\partial I$ be the node such that $\boldsymbol\omega$ is not k-minimal, and let i_* be the opposit...

conjecture | open | MathDB 页面 | 原文线索

V2-013470 | MATHDB-338024

The elementary amenability conjecture for generalized basilica groups

摘录 / EXCERPT: Let X be a finite set, let X^* be the rooted tree given by the free monoid on X, and let $G\leq\operatorname{Aut}(X^*)$ be a generalized basilica group: namely, G admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-013471 | MATHDB-338073

Degree-three generic list recovery for spherical registration

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite set of nonzero spherical-harmonic frequencies, let V be the corresponding $\mathrm{SO}(3)$ representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013472 | MATHDB-338151

Ledet's essential dimension conjecture for cyclic p-groups

摘录 / EXCERPT: Let k be a field of characteristic $p>0$, let n be a natural number, and let C_{p^n} be a cyclic group of order p^n. Ledet's conjecture. ... This conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-013473 | MATHDB-338194

The Mackey formula for categories O of rational Cherednik algebras

摘录 / EXCERPT: Let W be a finite complex reflection group, and let W_a and W_b be parabolic subgroups. Write ${}^{a}W^{b}$ for a complete set of double coset representatives of...

conjecture | open | MathDB 页面 | 原文线索

V2-013474 | MATHDB-338195

Rubey's evacuation-reversal conjecture for vacillating tableaux

摘录 / EXCERPT: Rubey's conjecture. The vacillating tableau corresponding to $\varepsilon(T)$ is the reversal of the vacillating tableau corresponding to T.

conjecture | open | MathDB 页面 | 原文线索

V2-013475 | MATHDB-338202

Malle-Navarro's projective height zero conjecture

摘录 / EXCERPT: Malle-Navarro's projective height zero conjecture. Every irreducible character in B lying over λ has height 0 if and only if D/Z is abelian and λ extends to...

conjecture | open | MathDB 页面 | 原文线索

V2-013476 | MATHDB-338206

Kar-Nikolov's deficiency stabilisation conjecture

摘录 / EXCERPT: Let G) be a residually finite finitely presented group. For a finitely presented group G, its deficiency $\delta(G)$ is the maximum of $|X|-|R|$ over all presentations...

conjecture | open | MathDB 页面 | 原文线索

V2-013477 | MATHDB-338230

Injective-dimension conjecture for the Jacobson radical of an Artin algebra

摘录 / EXCERPT: Let A be an Artin algebra, let J denote its Jacobson radical, and write $\operatorname{gldim}(A)$ for its global dimension and $\operatorname{id}(J)$ for the injective dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-013478 | MATHDB-338250

Epsilon dichotomy conjecture for the Ginzburg-Rallis model

摘录 / EXCERPT: Let F be a local field, let β be a character of $F^{\times}$, and let Π be an irreducible generic representation of $\mathrm{GL}_6(F)$ with central character β^2....

conjecture | open | MathDB 页面 | 原文线索

V2-013479 | MATHDB-338256

Guarnieri-Vendramin conjecture for the number of braces of order p^2q

摘录 / EXCERPT: Guarnieri-Vendramin's conjecture. If p and q are primes such that $p<q$ and $p\nmid q-1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-013480 | MATHDB-338268

The decategorified blob versus Kazhdan-Lusztig conjecture

摘录 / EXCERPT: Let $P_1^l(n)$ be the set of l-tuples of non-negative integers summing to n. Let $\boldsymbol{\lambda},\boldsymbol{\mu}\in P_1^l(n)$ be regular, with $\boldsymbol{\mu}$ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-013481 | MATHDB-338269

The blob simple-module dimension and intersection-form conjecture

摘录 / EXCERPT: Let $P_1^l(n)$ be the set of l-tuples of non-negative integers summing to n, and let $P(\boldsymbol{\lambda})$ denote the orbit of $\boldsymbol{\lambda}$. For a regular...

conjecture | open | MathDB 页面 | 原文线索

V2-013482 | MATHDB-338291

The finite-type extension conjecture for coherent Satake categories

摘录 / EXCERPT: Finite-type extension conjecture. The category $\mathcal{P}_{coh}^{G(\mathcal{O}) \rtimes \mathbb{G}_m}(\mathrm{Gr}_{G_{Ad}})$ is a monoidal categorification of the quantum cluster...

conjecture | open | MathDB 页面 | 原文线索

V2-013483 | MATHDB-338314

King-Tollu-Toumazet polynomiality conjecture for stretched Littlewood-Richardson coefficients

摘录 / EXCERPT: Let λ, μ, and ν be partitions, and let $c^\nu_{\lambda,\mu}$ denote the Littlewood-Richardson coefficient. For positive integers N, write...

conjecture | open | MathDB 页面 | 原文线索

V2-013484 | MATHDB-338354

The homological non-triviality conjecture for actions of special automorphism groups of free groups

摘录 / EXCERPT: Homological non-triviality conjecture. When $n>k+1$, any non-trivial action of $\mathrm{SAut}(F_n)$ or $\mathrm{SL}_n(\mathbb{Z})$ on M^k by homeomorphisms is homologically non-t...

conjecture | open | MathDB 页面 | 原文线索

V2-013485 | MATHDB-338370

Generating-set invariance conjecture for forcing properties

摘录 / EXCERPT: Let Γ be a finitely generated, residually finite group, and let S_i for $i=1,2$ be finite generating sets with $\operatorname{Id}\notin S_i$ and $S_i^{-1}=S_i$. Set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013486 | MATHDB-338375

Non-kinetic bundles from nontrivial smooth SU(2)-actions

摘录 / EXCERPT: Let W be an arbitrary closed manifold on which $SU(2)$ acts smoothly and non-trivially. Let $s\colon BSU(2) \to B\operatorname{Diff}(W)$ be the induced map on classifying spaces,...

conjecture | open | MathDB 页面 | 原文线索

V2-013487 | MATHDB-338410

Universal cyclic thick Demazure module conjecture

摘录 / EXCERPT: Let P be the weight lattice, let D_c and $\mathbb D_c$ be the thick and local Demazure modules, and define ... with $\deg X_{i,j}=1$. Let $\widetilde{\mathfrak b}^-$ be the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-013488 | MATHDB-338411

Ext-vanishing conjecture for thick and local Demazure modules

摘录 / EXCERPT: Let Λ be a level-one fundamental weight, let $b,c\in P$, let $m\in\mathbf Q$, and let $p>0$. Let D_b and D_c be thick Demazure modules, let $\mathring{\mathbb D}_b$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013489 | MATHDB-338489

Bieri's deficiency-one HNN extension conjecture

摘录 / EXCERPT: Bieri's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013490 | MATHDB-338490

Friedl-Lück-Tillmann's amenable-group L^2-torsion polytope conjecture

摘录 / EXCERPT: Friedl-Lück-Tillmann's conjecture. If G is amenable but not virtually $\mathbb{Z}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-013491 | MATHDB-338501

The zero-t_0 limit of deformed hyperoctahedral q-Whittaker functions

摘录 / EXCERPT: Let $n\geq1$, and let the deformed hyperoctahedral q-Whittaker function depend on the parameter t_0. Let $\mathcal{P}^{(n)}$ be the polynomial defined by the recursion in. Zero...

conjecture | open | MathDB 页面 | 原文线索

V2-013492 | MATHDB-338518

Herzog's conjecture on the number of involutions

摘录 / EXCERPT: Herzog's conjecture. If $I_2(G)=I_2(H)$, then $|G|=|H|$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-013493 | MATHDB-338520

The generalized Height Zero Conjecture for theta-blocks

摘录 / EXCERPT: Let G be a finite group, let $N\trianglelefteq G$, and let $\theta\in\operatorname{Irr}(N)$ be G-invariant. Suppose that $B_\theta\subseteq\operatorname{Irr}(G\mid\theta)$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-013494 | MATHDB-338546

Profinite volume conjecture for 3-manifold groups

摘录 / EXCERPT: Profinite volume conjecture. Their volumes should agree:

conjecture | open | MathDB 页面 | 原文线索

V2-013495 | MATHDB-338565

Representation-theoretic positivity conjecture for absolutely cuspidal curve polynomials

摘录 / EXCERPT: Let g and r be integers, let $C_{g,r}^{\mathrm{abs}}$ be the absolutely cuspidal polynomial for curves of genus g and rank r, let $GSp(2g,\overline{\mathbb{Q}}_l)$ be the g...

conjecture | open | MathDB 页面 | 原文线索

V2-013496 | MATHDB-338567

Polyhedrality conjecture for graph functionals of compact Lie groups

摘录 / EXCERPT: Let G be a sscc Lie group and let S be a connected graph with k vertices and r edges. Write $d=\operatorname{rank}(G)$ and $l=\frac{\dim(G)-\operatorname{rank}(G)}{2}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013497 | MATHDB-338573

Simson's infinite tame-wild dichotomy conjecture for coalgebras

摘录 / EXCERPT: Simson's infinite tame-wild dichotomy conjecture. Every coalgebra C is either tame or wild, but not both.

conjecture | open | MathDB 页面 | 原文线索

V2-013498 | MATHDB-338574

Simson's Brauer-Thrall 3 conjecture for finite-dimensional algebras

摘录 / EXCERPT: Simson's Brauer-Thrall 3 conjecture. For any infinite cardinality λ, there is an indecomposable A-module M of dimension

open_problem | open | MathDB 页面 | 原文线索

V2-013499 | MATHDB-338575

The trace formula conjecture for multilinear representations of free PROs

摘录 / EXCERPT: Let $\mathcal T$ be the free PRO under consideration, let μ be a representation of $\mathcal T$ of dimension N, and let d denote the parameter appearing in the trace of the...

conjecture | open | MathDB 页面 | 原文线索

V2-013500 | MATHDB-338582

Asymptotic signature formula for irreducible modules of arbitrary support

摘录 / EXCERPT: Let W be a finite real reflection group, let $c\in\mathfrak{p}_{\mathbb{R}}$, and let $\lambda\in\operatorname{Irr}(W)$. Let $L_c(\lambda)$ be the corresponding irreducible ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-013501 | MATHDB-338583

Restricted homology conjecture for even n

摘录 / EXCERPT: Restricted homology conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-013502 | MATHDB-338584

Characteristic-three multistep homology conjecture

摘录 / EXCERPT: Characteristic-three homology conjecture. This sequence has nonzero homology if and only if $k=\lfloor n/2\rfloor$. In the exceptional case, H_k is isomorphic to the sign module....

conjecture | open | MathDB 页面 | 原文线索

V2-013503 | MATHDB-338585

Characteristic-five multistep homology conjecture

摘录 / EXCERPT: Characteristic-five homology conjecture. This sequence has nonzero homology if and only if $k\in\{\lfloor n/2\rfloor,\lfloor n/2\rfloor-1\}$. If $n=2m$ is even, then...

conjecture | open | MathDB 页面 | 原文线索

V2-013504 | MATHDB-338592

The unique p-rational height-zero character conjecture for Brauer correspondent blocks

摘录 / EXCERPT: Unique lifting conjecture. If B contains exactly one p-rational height-zero irreducible character, then $l(b)=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-013505 | MATHDB-338593

The Galois-invariance conjecture for odd-degree characters in principal 2-blocks

摘录 / EXCERPT: Galois-invariance conjecture. The normalizer $N_G(P)$ has a normal 2-complement if and only if all odd-degree irreducible characters in the principal 2-block of G are...

conjecture | open | MathDB 页面 | 原文线索

V2-013506 | MATHDB-338598

Integer-solution conjecture for commensurable 2-dimensional right-angled Artin groups

摘录 / EXCERPT: Let $\mathbb{G}(\Gamma_1)$ and $\mathbb{G}(\Gamma_2)$ be 2-dimensional right-angled Artin groups, meaning that their commutation graphs are triangle-free. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013507 | MATHDB-338600

Spin Kerov conjecture on coefficient positivity

摘录 / EXCERPT: Let K_{2k-1}^{spin} be the spin Kerov polynomial in the variables $\mathfrak{R}_2,\mathfrak{R}_4,\dots,\mathfrak{R}_{2k}$, defined by the identity ... for strict partition...

conjecture | open | MathDB 页面 | 原文线索

V2-013508 | MATHDB-338605

Bouc's nilpotence conjecture for the beta invariant of finite groups

摘录 / EXCERPT: Bouc's conjecture. $\beta(G)$ is nilpotent if and only if G is nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-013509 | MATHDB-338626

One-relator hyperbolization conjecture

摘录 / EXCERPT: Let G be a one-relator group whose presentation complex has negative immersions. One-relator hyperbolization conjecture. The group G is hyperbolic. The conjecture would imply t...

conjecture | open | MathDB 页面 | 原文线索

V2-013510 | MATHDB-338627

Relative hyperbolization conjecture for primitivity rank two one-relator groups

摘录 / EXCERPT: Let F be a free group, let $w\in F\setminus\{1\}$ have primitivity rank $\pi(w)=2$, and let $G=F/\langle\!\langle w\rangle\!\rangle$. Let $P<G$ be the peripheral subgroup supplie...

conjecture | open | MathDB 页面 | 原文线索

V2-013511 | MATHDB-338628

Conjecture on nonnegative coefficients in the highest-weight inner-product recursion

摘录 / EXCERPT: Let $|\lambda\rangle=E_{j_{n(w)}}^{-\cdots E_{j_1}^{-|\Lambda_s\rangle$ be a state in a finite-dimensional irreducible representation, and let n_k and Λ_i^k denote the coef...

conjecture | open | MathDB 页面 | 原文线索

V2-013512 | MATHDB-338632

Isomorphism conjecture for generalized and twisted multifold extensions

摘录 / EXCERPT: Let A be a piecewise hereditary algebra of tree type, let $\hat{A}$ be its repetitive category, and let ϕ be an automorphism of $\hat{A}$ with nonzero jump n. Write ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-013513 | MATHDB-338635

Extended defect group conjecture for real 2-regular elements

摘录 / EXCERPT: Let G be a finite group, let k be a field of characteristic 2, let M be an irreducible kG-module with Brauer character φ, and let g be a real 2-regular elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-013514 | MATHDB-338636

Even-indicator conjecture for weakly real principal indecomposable characters

摘录 / EXCERPT: Let Φ be a weakly real principal indecomposable character of a finite group. Even-indicator conjecture. The Frobenius-Schur indicator $\epsilon(\Phi)$ is even. The examples pr...

conjecture | open | MathDB 页面 | 原文线索

V2-013515 | MATHDB-338637

Strong matching conjecture for weakly real principal indecomposable characters

摘录 / EXCERPT: Let G be a finite group with w distinct weakly real 2-principal indecomposable characters $\Phi_1,\dots,\Phi_w$. Strong matching conjecture. Then G has w distinct symplec...

conjecture | open | MathDB 页面 | 原文线索

V2-013516 | MATHDB-338646

Weak Donovan conjecture on Morita-Frobenius numbers

摘录 / EXCERPT: Kessar's conjecture. There is $m(P)\in\mathbb{N}$ such that, if G is a finite group and B is a block of kG with defect groups isomorphic to P, then

conjecture | open | MathDB 页面 | 原文线索

V2-013517 | MATHDB-338706

Garcia-Ramos-Lupini conjecture on strongly outer group actions

摘录 / EXCERPT: Garcia-Ramos-Lupini conjecture. If G is amenable, then any two strongly outer actions of G on $\mathcal{D}$ are cocycle conjugate. If G is not amenable, then there exist unco...

conjecture | open | MathDB 页面 | 原文线索

V2-013518 | MATHDB-338776

Conjectural uncapped automorphisms in the E8 building over F2

摘录 / EXCERPT: E8(2) uncapped-automorphism conjecture. Let

conjecture | open | MathDB 页面 | 原文线索

V2-013519 | MATHDB-338777

The Whitehouse-type lifting conjecture for $Lie_n^{(2)}$

摘录 / EXCERPT: Let $Lie_n^{(2)}$ denote the relevant S_n-module, and let $Lie_{n-1}^{(2)}\uparrow^{S_n}$ denote induction from S_{n-1} to S_n. The Whitehouse-type lifting conjecture. The sy...

conjecture | open | MathDB 页面 | 原文线索

V2-013520 | MATHDB-338798

Local-global compatibility conjecture for cspan class="math-inline"cspan class="math-inline"GL_3(\mathbb Q_p)c/spanc/spanc

摘录 / EXCERPT: Local-global compatibility conjecture. The restriction morphism is bijective:

conjecture | open | MathDB 页面 | 原文线索

V2-013521 | MATHDB-338810

Injectivity conjecture for the Weyl group action on integrable quantum modules

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a symmetrizable Kac-Moody algebra, let $V\in\mathscr O^{int}_q({\mathfrak{g}})$ be an integrable highest weight $U_q({\mathfrak{g}})$-module, let I be the...

conjecture | open | MathDB 页面 | 原文线索

V2-013522 | MATHDB-338830

The atoroidal Poincaré duality group conjecture

摘录 / EXCERPT: Let G be a 3-dimensional Poincaré duality group. Call G atoroidal if it contains no subgroup isomorphic to $\mathbb Z\oplus\mathbb Z$. Atoroidal hyperbolicity conjecture. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-013523 | MATHDB-338861

Equivalence conjecture for the two methods of determining localized representations

摘录 / EXCERPT: Let Q be a singular point of the discriminant of an elliptic fibration. Method 1 and Katz-Vafa's method are two constructions of localized representations associated with Q. Lo...

conjecture | open | MathDB 页面 | 原文线索

V2-013524 | MATHDB-338876

Even-or-odd invariant forms give weakly hereditary filtrations

摘录 / EXCERPT: Let $\mathbf U$ be the quantum group and let $\mathbb P=P\otimes_R\mathcal F$ be as above, where $\mathcal F$ is a finite-dimensional $\mathbf U$-module. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013525 | MATHDB-338927

Foulkes's conjecture for symmetrized weight spaces

摘录 / EXCERPT: Let a and b be positive integers with $a<b$. For a polynomial representation W of GL_n of degree ab, consider the symmetrized weight spaces corresponding to the weights .....

conjecture | open | MathDB 页面 | 原文线索

V2-013526 | MATHDB-338946

Igusa-Todorov's conjecture on tilted algebras and maximal green sequences

摘录 / EXCERPT: Igusa-Todorov's conjecture. (a) The ordered B-modules form a maximal forward hom-orthogonal sequence of Λ-modules whose dimension vectors form the c-vectors of a maxima...

conjecture | open | MathDB 页面 | 原文线索

V2-013527 | MATHDB-338988

The relative McKay conjecture

摘录 / EXCERPT: Let G be a finite group, let $L\unlhd G$, let $P/L\in\operatorname{Syl}_p(G/L)$, and let $\nu\in\operatorname{Irr}(L)$ be P-invariant. Write $\operatorname{Irr}_{p'}(G\mid\nu)$...

conjecture | open | MathDB 页面 | 原文线索

V2-013528 | MATHDB-339026

The saturation conjecture for simply-laced simple groups

摘录 / EXCERPT: Let G be a simple, simply-connected, simply-laced group. Let $d4a2(G)$ be the saturated tensor cone of triples of dominant weights $(\lambda_1,\lambda_2,\lambda_3)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-013529 | MATHDB-339038

Non-semilinearity conjecture for the word problem of $F_2\times\mathbb{Z}$

摘录 / EXCERPT: Let F_2 be the free group of rank 2, let $\mathbb{Z}$ be the infinite cyclic group, and let MCF mean multiple context-free. The $F_2\times\mathbb{Z}$ word-problem conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-013530 | MATHDB-339042

The polylogarithmic-party interleaved group-product hardness conjecture

摘录 / EXCERPT: Polylogarithmic-party hardness conjecture. There is no protocol for the k-party t-tuple interleaved group product over G with $k=\log^c t$ parties and communication...

conjecture | open | MathDB 页面 | 原文线索

V2-013531 | MATHDB-339056

Minimal-generator conjecture for the generic ring

摘录 / EXCERPT: Minimal-generator conjecture. The ring $\hat R_{\mathrm{gen}}$ is generated by the six representations $W(\mu)$ corresponding to

conjecture | open | MathDB 页面 | 原文线索

V2-013532 | MATHDB-339057

Stronger minimal-generator conjecture when the first format parameter is one

摘录 / EXCERPT: Stronger minimal-generator conjecture. The ring $\hat R_{\mathrm{gen}}$ is generated by the four representations $W(\mu)$ corresponding to

conjecture | open | MathDB 页面 | 原文线索

V2-013533 | MATHDB-339097

Triviality conjecture for special modules

摘录 / EXCERPT: Let $\mathcal{T}_n^+$ be the relevant tensor category. An indecomposable module V in $\mathcal{T}_n^+$ is special if $\operatorname{sdim}(V)=1$, $V^*\cong V$, and $H^0(V)$ contai...

conjecture | open | MathDB 页面 | 原文线索

V2-013534 | MATHDB-339098

Berezin-determinant conjecture for invertible objects

摘录 / EXCERPT: Let $\overline{\mathcal{T}}_n$ be the quotient tensor category and let $\mathcal{T}_n$ be the original category. An object I of $\overline{\mathcal{T}}_n$ is invertible if it is...

conjecture | open | MathDB 页面 | 原文线索

V2-013535 | MATHDB-339099

Reduction conjecture for $GL(m|n)$ tensor categories

摘录 / EXCERPT: Let $m\geq n$, let $\mathcal{T}_{m|n}^+$ be the general tensor category, and let $L(\lambda)$ be an irreducible representation with $\operatorname{sdim}(L(\lambda))>0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013536 | MATHDB-339100

Triviality conjecture for the one-dimensional module I

摘录 / EXCERPT: Let I be the indecomposable module described in the source: it has superdimension 1, is self-dual under both dualities, occurs as a multiplicity-one summand of...

conjecture | open | MathDB 页面 | 原文线索

V2-013537 | MATHDB-339101

Parity-triviality conjecture for special modules

摘录 / EXCERPT: Let V be a special module in $\mathcal{T}_n^+$, meaning that V is indecomposable, $\operatorname{sdim}(V)=1$, $V^*\cong V$, and $H^0(V)$ contains the trivial module. Parity-tri...

conjecture | open | MathDB 页面 | 原文线索

V2-013538 | MATHDB-339159

Conjecture on multiplicity-free tempered restrictions and point reduced spaces

摘录 / EXCERPT: Let G be a reductive group with maximal compact subgroup K. Let $H<G$ be a θ-stable Cartan subgroup, and let ... be a cuspidal parabolic subgroup corresponding to H, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-013539 | MATHDB-339170

Asymptotic Volume Conjecture for finite-volume hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a finite volume hyperbolic 3-manifold with fundamental group G. For finite-index subgroups $K\leq G$, write $(H_1(K,\mathbb{Z}))_{tor}$ for the torsion subgroup of the...

conjecture | open | MathDB 页面 | 原文线索

V2-013540 | MATHDB-339254

The set-partition conjecture for simple transitive 2-representations

摘录 / EXCERPT: Let $n\geq 1$, and let $\mathcal{C}_n$ be the left cell 2-subcategory of projective functors for the star algebra considered in the paper. A simple transitive 2-representation...

conjecture | open | MathDB 页面 | 原文线索

V2-013541 | MATHDB-339292

Snevily's conjecture on transversals in abelian group subsquares

摘录 / EXCERPT: Let G be an abelian group and let $A,B\subseteq G$ be n-element sets. Form the $n\times n$ subsquare $S=A\times B$ of the multiplication table of G, with entries given by gro...

conjecture | open | MathDB 页面 | 原文线索

V2-013542 | MATHDB-339297

The Ichino-Ikeda and N. Harris branching coefficient formula

摘录 / EXCERPT: Ichino-Ikeda-Harris conjecture. The form $\mathcal{H}$ is nonzero, so the branching coefficient $\mathcal{L}(\Pi,\Sigma)$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-013543 | MATHDB-339315

Berman's genuine-naive equivariant homotopy duality conjecture

摘录 / EXCERPT: Let G be a finite group. Let $_G\text{Top}$ denote the ∞-category of genuine left G-spaces, and let Top^G denote the ∞-category of naive, or Borel, right...

conjecture | open | MathDB 页面 | 原文线索

V2-013544 | MATHDB-339316

Berman's genuine equivariant presentability conjecture

摘录 / EXCERPT: Let G be a finite group, let $\text{Burn}_G^{(\infty,2)}$ be the genuine equivariant Burnside $(\infty,2)$-category, and let Pr^{L} be the ∞-category of pr...

conjecture | open | MathDB 页面 | 原文线索

V2-013545 | MATHDB-339318

Stable-center single-cycle generation conjecture

摘录 / EXCERPT: Stable-center generation conjecture. The stable center

conjecture | open | MathDB 页面 | 原文线索

V2-013546 | MATHDB-339319

Product-type conjecture for one-cycle structure constants

摘录 / EXCERPT: One-cycle product conjecture. (1) For every irreducible polynomial $f\in\mathbb F_q[t]$ of degree d with constant term $-\xi a$, there exist $g,h\in G_d$ of modified types...

conjecture | open | MathDB 页面 | 原文线索

V2-013547 | MATHDB-339320

Integer-polynomial positivity conjecture for stable-center structure constants

摘录 / EXCERPT: Integer-polynomial positivity conjecture. There is a polynomial $ A_{{\boldsymbol{\lambda}}{\boldsymbol{\mu}}}^{{\boldsymbol{\nu}}}({\bf q})\in\mathbb Z[{\bf q}]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-013548 | MATHDB-339322

Detomi-Lucchini's conjecture on nonabelian sigma-elementary groups

摘录 / EXCERPT: A finite group G is $c3$-elementary if $c3(G)$ is finite and $c3(H)<c3(G)$ for every proper quotient H of G. A group is monolithic primitive if it is primitive and has a uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-013549 | MATHDB-339335

The stronger finite-space form of Webb's conjecture for elementary abelian p-subgroups

摘录 / EXCERPT: Stronger Webb conjecture. The finite space

conjecture | open | MathDB 页面 | 原文线索

V2-013550 | MATHDB-339378

Wise's non-positive immersions coherence conjecture

摘录 / EXCERPT: Wise's conjecture. The fundamental group π_1X is coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-013551 | MATHDB-339379

Hyperbolicity and virtual quasiconvex hierarchy for groups with branched good stackings

摘录 / EXCERPT: Hyperbolicity and hierarchy conjecture. If G has a good branched stacking, then G is hyperbolic and has a virtual quasiconvex hierarchy.

conjecture | open | MathDB 页面 | 原文线索

V2-013552 | MATHDB-339382

Vogan's conjecture on unipotent Harish-Chandra modules

摘录 / EXCERPT: Let $G_{\mathbb{R}}$ be a real reductive group with maximal compact subgroup K, complexified Lie algebra $\mathfrak{g}$, and associated decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-013553 | MATHDB-339388

The inertial-parameter conjecture for mgs representations

摘录 / EXCERPT: Let F be the function field of a smooth projective curve over a finite field, let G be a split semisimple group over F, and let u be a place. For a local representation...

conjecture | open | MathDB 页面 | 原文线索

V2-013554 | MATHDB-339401

Rank Rigidity Conjecture for proper geodesically complete CAT(0) spaces

摘录 / EXCERPT: Let X be a proper, geodesically complete $\operatorname{CAT}(0)$ space under a geometric action. A rank one axis is an axis in X that does not bound a flat half-plane. Rank Rig...

conjecture | open | MathDB 页面 | 原文线索

V2-013555 | MATHDB-339404

Prasad-Takloo-Bighash conjecture for distinction of discrete series

摘录 / EXCERPT: Let F be a non-archimedean local field of characteristic not 2, let E/F be a quadratic extension, let D be a central division F-algebra of dimension d^2, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013556 | MATHDB-339463

Amenable strongly transitive action on a totally ordered set

摘录 / EXCERPT: Let G be a group acting strongly transitively on a totally ordered set X by order-preserving bijections. Amenability conjecture. There exists an amenable group G admitting su...

conjecture | open | MathDB 页面 | 原文线索

V2-013557 | MATHDB-339481

Multiplicity conjecture for restriction across central isogenies

摘录 / EXCERPT: Let ${\mathrm G}_1$ and ${\mathrm G}_2$ be connected reductive groups over a local field k, and let $\lambda:{\mathrm G}_1\rightarrow {\mathrm G}_2$ be a k-homomorphism whose r...

conjecture | open | MathDB 页面 | 原文线索

V2-013558 | MATHDB-339489

Uncountably many maximal subgroups in hyperbolic triangle groups

摘录 / EXCERPT: Let Γ be a hyperbolic triangle group. A subgroup has infinite index if its index in Γ is infinite, and two subgroups are conjugate when one is obtained from the other...

conjecture | open | MathDB 页面 | 原文线索

V2-013559 | MATHDB-339525

Dijols-Prasad conjecture on Arthur packets and stable base change distinction

摘录 / EXCERPT: Let E/F be a quadratic extension of p-adic fields, let $U_{2n}(E/F)$ be the quasi-split unitary group in $2n$ variables, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013560 | MATHDB-339528

Distinction converse conjecture for supercuspidal representations

摘录 / EXCERPT: Let $E/{\mathcal F}$ be the quadratic extension implicit in the distinction notation, with $n\ge3$. Let π be a supercuspidal representation of $\operatorname{GL}_n(E)$, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-013561 | MATHDB-339564

Conjecture on the generators of the abelianisation of Gamma_n(1,n/2-1)

摘录 / EXCERPT: Let $\Gamma_n(1,n/2-1)$ be the cyclically presented group in the source, let $\Gamma_n(1,n/2-1)^\mathrm{ab}$ denote its abelianisation, and let $d(G)$ be the minimum number of gene...

conjecture | open | MathDB 页面 | 原文线索

V2-013562 | MATHDB-339574

Almost recognition by spectrum for specified simple classical groups

摘录 / EXCERPT: Vasil'ev-Grechkoseeva conjecture. Suppose that L is one of the following nonabelian simple groups:

conjecture | open | MathDB 页面 | 原文线索

V2-013563 | MATHDB-339575

The cohomological dimension formula for Bowditch boundaries

摘录 / EXCERPT: Cohomological dimension formula.

conjecture | open | MathDB 页面 | 原文线索

V2-013564 | MATHDB-339579

Equivariant minimizer conjecture for klt singularities

摘录 / EXCERPT: Let G be a group acting on a klt singularity $x\in (X,D)$, with x fixed by the action. Let $\widehat{\operatorname{vol}}_{(X,D),x}$ denote the normalized-volume function. Group...

conjecture | open | MathDB 页面 | 原文线索

V2-013565 | MATHDB-339596

Arthur-Barbasch-Vogan unitarity conjecture for special unipotent representations

摘录 / EXCERPT: Arthur-Barbasch-Vogan conjecture. X is unitarizable.

conjecture | open | MathDB 页面 | 原文线索

V2-013566 | MATHDB-339610

Putman-Wieland Prym-orbit conjecture

摘录 / EXCERPT: Let S_g be a surface of genus $g\geq 2$, and let a Prym representation be the linear representation of $\operatorname{Mod}(S_g)$ on the first homology of a finite characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-013567 | MATHDB-339623

Conjectured projective covers for admissible-level Virasoro minimal-model modules

摘录 / EXCERPT: Let $\mathcal{A}_1(u,v)$ be the Virasoro minimal model at admissible level, with irreducible modules $\mathcal{E}_{lambda;r,s}$ and $\mathcal{D}^{pm}_{r,s}$, spectral-flow automorp...

conjecture | open | MathDB 页面 | 原文线索

V2-013568 | MATHDB-339627

Manning-Wang dimension conjecture for Bowditch boundaries

摘录 / EXCERPT: Let $(G,\mathcal{P})$ be a type F relatively hyperbolic group pair, with Bowditch boundary $\partial(G,\mathcal{P})$, and let $\operatorname{cd}(G,\mathcal{P})$ denote the cohomo...

conjecture | open | MathDB 页面 | 原文线索

V2-013569 | MATHDB-339628

Friedberg-Gunnells-McNamara conjecture for Littelmann patterns of type G_2

摘录 / EXCERPT: Let G_2 have simple roots $\{\alpha_1,\alpha_2\}$, with α_2 the longer root. Let W be its Weyl group, Λ_W its weight lattice, and $\mathbb{C}[\Lambda_W]$ the ass...

conjecture | open | MathDB 页面 | 原文线索

V2-013570 | MATHDB-339664

The tameness conjecture for omega-categorical groups and rings

摘录 / EXCERPT: An omega-categorical structure is a structure whose theory has a unique countable model up to isomorphism. A group or ring is tame or supertame in the sense used in the paper's mod...

conjecture | open | MathDB 页面 | 原文线索

V2-013571 | MATHDB-339668

Restriction-injectivity conjecture for Kirillov-Shalika models

摘录 / EXCERPT: Let $G=\operatorname{GL}_{mn}$, let $\sigma\in\operatorname{Irr}G$ be m-homogeneous, and let D be the subgroup used to define the Kirillov-Shalika model. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013572 | MATHDB-339669

Mirabolic-invariance conjecture for bilinear forms on homogeneous representations

摘录 / EXCERPT: Let $G=\operatorname{GL}_{mn}$, let D be the subgroup used in the paper, and let $\sigma\in\operatorname{Irr}G$ be m-homogeneous. Consider bilinear forms on...

conjecture | open | MathDB 页面 | 原文线索

V2-013573 | MATHDB-339670

Uniqueness conjecture for mirabolic-invariant bilinear forms

摘录 / EXCERPT: Let $G=\operatorname{GL}_{mn}$, let D be the subgroup used in the paper, and let $\sigma,\sigma'\in\operatorname{Irr}G$ be m-homogeneous. Consider bilinear forms on...

conjecture | open | MathDB 页面 | 原文线索

V2-013574 | MATHDB-339697

Toda's Gepner-point existence conjecture for graded matrix factorizations

摘录 / EXCERPT: Let f be the homogeneous polynomial defining the graded matrix-factorization category $\operatorname{HMF}(f)$, and let h be its degree. Let Z_G be the central charge symbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-013575 | MATHDB-339709

Strong automorphism realisation conjecture for hyperbolic triangle groups

摘录 / EXCERPT: Strong automorphism realisation conjecture. The condition that at least one of p, q and r is ∞ can be omitted from the countable and strong automorphism realisation as...

conjecture | open | MathDB 页面 | 原文线索

V2-013576 | MATHDB-339715

Higher baby Verma quotient conjecture

摘录 / EXCERPT: Higher baby Verma quotient conjecture. Every irreducible $U_\chi^{[r]}(G)$-module is a homomorphic image of $Z_\chi^{[r]}(N,\lambda_r)$ for some irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-013577 | MATHDB-339740

Cubing conjecture for large sliding circuit set graphs

摘录 / EXCERPT: Cubing conjecture. If the sliding circuit set is large, then large regions of the sliding circuit set graph should be cubical.

conjecture | open | MathDB 页面 | 原文线索

V2-013578 | MATHDB-339771

Uniqueness and semi-strong self-absorption of strongly outer actions on strongly self-absorbing C*-algebras

摘录 / EXCERPT: Let $\mathcal{D}$ be a strongly self-absorbing C^*-algebra. Let Γ be a countable torsion-free amenable group. Uniqueness and semi-strong self-absorption conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-013579 | MATHDB-339787

Chas-Li-Maskit central limit conjecture for geometric lengths on a pair of pants

摘录 / EXCERPT: Let S) be the pair of pants whose boundary geodesics B_1, B_2, and B_3 have geometric lengths $4 ell_1$, $4 ell_2$, and $4 ell_3$. Let μ_n denote the uniform random dis...

conjecture | open | MathDB 页面 | 原文线索

V2-013580 | MATHDB-339790

Uniform expansion conjecture for bounded-rank finite simple groups

摘录 / EXCERPT: Let $r,k\in\mathbb{N}$. A finite simple group has rank at most r when its rank is bounded by r, and $Cay(G,S)$ denotes the Cayley graph generated by S. Uniform expansion conj...

conjecture | open | MathDB 页面 | 原文线索

V2-013581 | MATHDB-339796

The no-short-separating-arc conjecture for CAT(0) boundaries

摘录 / EXCERPT: Let G be a one-ended group acting geometrically on a CAT(0) space X, and let ∂X denote its boundary. Assume that ∂X has no finite cut. No-short-separating-a...

conjecture | open | MathDB 页面 | 原文线索

V2-013582 | MATHDB-339797

The 2π-circle splitting conjecture for CAT(0) boundaries

摘录 / EXCERPT: Let G be a one-ended group acting geometrically on a CAT(0) space X, and let ∂X denote its boundary. Assume that ∂X has no finite cut and that a circle of T...

conjecture | open | MathDB 页面 | 原文线索

V2-013583 | MATHDB-339802

Conjecture on fixing numbers of symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group on n letters, and let $\operatorname{fix}(S_n)$ denote the set of possible numbers of vertices that fix a graph whose automorphism group is S_n...

conjecture | open | MathDB 页面 | 原文线索

V2-013584 | MATHDB-339823

The tightly 3-interlacing conjecture for rank 2 modules

摘录 / EXCERPT: Fix integers (k,n) with $k\ge 3$. Let I and J be tightly 3-interlacing k-subsets, meaning that they are 3-interlacing and $|I\cap J|=k-3$. Let $L(I,J)$ denote the rank...

conjecture | open | MathDB 页面 | 原文线索

V2-013585 | MATHDB-339824

The exclusion of rank 2 modules with poset $(1^r,2)$

摘录 / EXCERPT: Let M be a rank 2 module whose poset is $(1^r,2)$, where $r\ge 4$. Exclusion conjecture. The module M is not rigid. Earlier results show that the poset condition is incompati...

conjecture | open | MathDB 页面 | 原文线索

V2-013586 | MATHDB-339825

The degree-2 real-root counting conjecture for rigid rank 2 modules

摘录 / EXCERPT: Assume $3\le k\le n/2$. Let $J_{k,n}$ be the associated Kac-Moody algebra, let α be a real root of degree 2, and let φ be the map from indecomposable modules to ro...

conjecture | open | MathDB 页面 | 原文线索

V2-013587 | MATHDB-339877

Amit's conjecture on word-map fibers in finite nilpotent groups

摘录 / EXCERPT: Let G be a finite nilpotent group, let $\overline{x}$ denote a tuple of variables, and let $w(\overline{x})$ be a word map on G. Write $\mu_{G,w}(1)$ for the probability that...

conjecture | open | MathDB 页面 | 原文线索

V2-013588 | MATHDB-339905

Williamson-Riche character formula conjecture for tilting modules

摘录 / EXCERPT: Let $\mathbf{G}$ be a reductive algebraic group whose derived subgroup is simply connected, let $\mathbf{B}\subset\mathbf{G}$ be a Borel subgroup, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013589 | MATHDB-339906

Algebraic-group properties of exponentials for over-restricted representations

摘录 / EXCERPT: Algebraic-group exponential conjecture. 1. The map

conjecture | open | MathDB 页面 | 原文线索

V2-013590 | MATHDB-339942

Artin's conjecture for supercharacter theories

摘录 / EXCERPT: Let K/F be a Galois extension of number fields with Galois group G. Let (X,K) be a supercharacter theory of G, and let $\operatorname{Sup}(G)$ be the set of supercharacters...

conjecture | open | MathDB 页面 | 原文线索

V2-013591 | MATHDB-339958

Conjecture on simple spectrum of Bethe subalgebras in the specified modules

摘录 / EXCERPT: Let $z_1,\ldots,z_k$ be real parameters satisfying ... Let $L(z_1,\ldots,z_k)$ and $L'(z_1,\ldots,z_k,\cdot)$ denote the modules used in the paper's restriction and multiplicity-spac...

conjecture | open | MathDB 页面 | 原文线索

V2-013592 | MATHDB-339961

Conjecture on counting unipotent irreducible Brauer characters

摘录 / EXCERPT: Counting conjecture. Suppose that p is good for $\mathbf{G}$. Then α_ℓ is the number of unipotent irreducible Brauer characters of G.

conjecture | open | MathDB 页面 | 原文线索

V2-013593 | MATHDB-339990

Large-degree distinguishing-number conjecture for symmetric-group actions

摘录 / EXCERPT: Let S_n act on a set, and let its distinguishing number be the least number of labels needed to eliminate all nonidentity elements of the action. The large-degree distinguishing-...

conjecture | open | MathDB 页面 | 原文线索

V2-013594 | MATHDB-339991

Abelianity conjecture for subgroup realization by labelings

摘录 / EXCERPT: Let Γ act on a set X. Say that a subgroup of Γ can be obtained through a labeling if it is the subgroup preserving the labeling. The abelianity conjecture. If every...

conjecture | open | MathDB 页面 | 原文线索

V2-013595 | MATHDB-339995

Maximal-consumer existence conjecture for partition consumption

摘录 / EXCERPT: Let $\lambda\vdash n$ be a partition, and say that a subgroup consumes a partition when it arises through the paper's consumption ordering. The maximal-consumer existence conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-013596 | MATHDB-339996

Exact-consumption realization conjecture for partitions

摘录 / EXCERPT: Let $\lambda\vdash n$ be a partition, and order partitions by the consumption ordering. The exact-consumption realization conjecture. For each partition $\lambda\vdash n$, there is...

conjecture | open | MathDB 页面 | 原文线索

V2-013597 | MATHDB-340052

The non-generic local Gan-Gross-Prasad conjecture for general linear groups

摘录 / EXCERPT: Let F be a field with Weil group W_F. For $r\in\mathbb{Z}_{>0}$, let an Arthur parameter for $GL_r(F)$ be a completely reducible representation ... where each ψ_i is an irre...

conjecture | open | MathDB 页面 | 原文线索

V2-013598 | MATHDB-340054

Quasi-isometry conjecture for separable horizontal surfaces in graph manifolds

摘录 / EXCERPT: Let N and M be two closed graph manifolds. Let $S \looparrowright N$ and $S' \looparrowright M$ be two separable, horizontal surfaces. Quasi-isometry conjecture. There is a qua...

conjecture | open | MathDB 页面 | 原文线索

V2-013599 | MATHDB-340106

Bialynicki-Birula-type linearization conjecture for torus actions on free algebras

摘录 / EXCERPT: Let F_n be the free associative algebra on n generators, and let $\mathbb{T}_{n-1}$ be the $(n-1)$-dimensional algebraic torus. An action is linearizable when it is conjugate t...

conjecture | open | MathDB 页面 | 原文线索

V2-013600 | MATHDB-340114

Bijection between strong torsion classes and maximal support pre-tilting modules

摘录 / EXCERPT: Let $(A,\mathcal{C})$ be an almost-directed d-homological pair, meaning that A is an algebra equipped with the subcategory $\mathcal{C}$, and suppose that ... A d-strong torsio...

conjecture | open | MathDB 页面 | 原文线索

V2-013601 | MATHDB-340162

Polya's conjecture for Dirichlet domains of cocompact group actions

摘录 / EXCERPT: Let M be a complete, contractible Riemannian manifold, and let a group Γ act properly, effectively, and cocompactly on M by isometries. A Dirichlet domain of Γ is...

conjecture | open | MathDB 页面 | 原文线索

V2-013602 | MATHDB-340188

The restricted critical-block conjecture for semi-infinite category O

摘录 / EXCERPT: Let $\widehat{\mathcal O}_{\overline\Lambda}^{{\operatorname{res}}}$ be a regular (in the sense of Feigin-Frenkel linkage) integral critical block. Let ... be the full subcategory co...

conjecture | open | MathDB 页面 | 原文线索

V2-013603 | MATHDB-340191

The subnormal subgroup algebraicity conjecture for division rings

摘录 / EXCERPT: Subnormal subgroup algebraicity conjecture. If N is non-central, then N is left algebraic (respectively, right algebraic) over K if and only if D is left algebraic (respect...

conjecture | open | MathDB 页面 | 原文线索

V2-013604 | MATHDB-340216

The existence conjecture for tempered types of \operatorname{GL}_n

摘录 / EXCERPT: Let $\mathtt{K}=\operatorname{GL}_n(\mathcal{O}_K)$ and let $B^{\mathrm{t}}(\operatorname{GL}_n)$ be the tempered Bernstein spectrum. For...

conjecture | open | MathDB 页面 | 原文线索

V2-013605 | MATHDB-340234

Linear conjugator conjecture for Morse elements

摘录 / EXCERPT: Linear conjugator conjecture. In every finitely generated group, the set of Morse elements satisfies the linear conjugator property.

conjecture | open | MathDB 页面 | 原文线索

V2-013606 | MATHDB-340259

The Siemons-Wagner conjecture on primitive permutation groups and subset orbit sizes

摘录 / EXCERPT: Let G be a primitive permutation group acting on a set Ω of cardinality $n\ge 8$. For a subset $S\subseteq \Omega$, write S^G for its orbit under G. Siemons-Wagner con...

conjecture | open | MathDB 页面 | 原文线索

V2-013607 | MATHDB-340265

Simple-spectrum conjecture for the quantised branching algebra

摘录 / EXCERPT: Let σ be an involution of ${\mathfrak g}$, let $H_1,\ldots,H_l$ be the chosen basic invariants with degrees $d_1,\ldots,d_l$, and let ϖ be the symmetrisation map. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013608 | MATHDB-340281

Conjecture on the joint spectrum of higher Gaudin Hamiltonians and \mathfrak{gl}_{M|N} spaces

摘录 / EXCERPT: Let $\boldsymbol\lambda=(\lambda^{(1)},\dots,\lambda^{(n)})$ be a typical sequence of polynomial $\mathfrak{gl}_{M|N}$ weights, let $\boldsymbol z=(z_1,\dots,z_n)$ be a sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-013609 | MATHDB-340282

The \mathfrak{gl}_{1|1} divisor parametrization conjecture for Gaudin eigenvectors

摘录 / EXCERPT: Consider a tensor product $L(\boldsymbol\lambda)$ of typical polynomial $\mathfrak{gl}_{1|1}$ modules, with singular subspace $L(\boldsymbol\lambda)^{\boldsymbol s}ing}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013610 | MATHDB-340283

The \mathfrak{gl}_{1|1} divisor parametrization conjecture for arbitrary modules

摘录 / EXCERPT: Let $\boldsymbol\lambda$ be a sequence of arbitrary $\mathfrak{gl}_{1|1}$ weights, and let $L(\boldsymbol\lambda)^{\boldsymbol s}ing}$ be its singular subspace. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013611 | MATHDB-340351

Platonic-case non-asphericity conjecture for relative presentations

摘录 / EXCERPT: Let $\mathcal{Q}$ be the relative presentation considered in the paper, with parameters $l>0$, $k\neq 0$ and non-trivial elements $g,h\in G$. Define ... Say that Case (P) holds whe...

conjecture | open | MathDB 页面 | 原文线索

V2-013612 | MATHDB-340352

AEJ-E asphericity conjecture for positive relative presentations

摘录 / EXCERPT: Let $l>0$ and $k>0$ with $l\neq k$, and let $\mathcal{Q}$ be the relative presentation considered in the paper. Define condition (AEJ-E) by the following alternatives: (i) $g=h^2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-013613 | MATHDB-340418

Braverman-Kazhdan's zeta-integral conjecture

摘录 / EXCERPT: Braverman-Kazhdan's zeta-integral conjecture. For every $f\in {\mathcal {S}}_{\rho}(G)$ and every $\varphi\in {\mathcal {C}}(\pi)$, the integral $Z(s,f,\varphi)$ is absolutely conv...

conjecture | open | MathDB 页面 | 原文线索

V2-013614 | MATHDB-340419

Braverman-Kazhdan's Fourier-transform conjecture

摘录 / EXCERPT: Let G be as above, let ρ be a finite-dimensional representation of ${}^{L}G$, and let ${\mathcal {S}}_{\rho}(G)$ be the proposed test-function space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013615 | MATHDB-340470

Cowling's conjecture on coefficient functions of Kunze-Stein groups

摘录 / EXCERPT: Let G be a Kunze-Stein group, and let $\pi:G\to B(H_\pi)$ be a unitary representation admitting a cyclic vector $\xi\in H_\pi$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-013616 | MATHDB-340481

The Morita-class count conjecture for weight-two blocks of symmetric groups

摘录 / EXCERPT: Weight-two Morita-class count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013617 | MATHDB-340482

The complementary Morita-class count conjecture for blocks of symmetric groups at p=2 and p=3

摘录 / EXCERPT: Complementary Morita-class count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013618 | MATHDB-340488

Ore's conjecture on commutators in finite non-abelian simple groups

摘录 / EXCERPT: Let G be a finite non-abelian simple group, and let $g\in G$. A commutator is an element of the form $xyx^{-1}y^{-1}$ for some $x,y\in G$. Ore's conjecture. Every element...

conjecture | open | MathDB 页面 | 原文线索

V2-013619 | MATHDB-340512

Lubotzky-Weiss conjecture on amenable and property (T) dense subgroups

摘录 / EXCERPT: Let K be a compact group, and let Λ_1 and Λ_2 be finitely generated subgroups dense in K. Lubotzky-Weiss conjecture. If Λ_1 is amenable and Λ_2...

conjecture | open | MathDB 页面 | 原文线索

V2-013620 | MATHDB-340515

The mutation-compatible semibrick pair completion conjecture

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra, and let $\mathcal{S}_p,\mathcal{S}_n\in\mathsf{sbrick}\Lambda$. A semibrick pair is $\mathcal{S}_p\sqcup\mathcal{S}_n[1]$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-013621 | MATHDB-340519

Zimmer's conjecture for lattice actions on manifolds

摘录 / EXCERPT: Zimmer's conjecture. If $d<\mathrm{rank}(G)$, then ρ has finite image.

conjecture | open | MathDB 页面 | 原文线索

V2-013622 | MATHDB-340520

Zimmer's conjecture on minimal dimensions and invariant metrics

摘录 / EXCERPT: Zimmer's conjecture. (1) If $\dim(M)<\min\{ d_{\mathrm{rep}}(G),d_{\mathrm{cpt}}(G),d_0(G)\}$, every homomorphism $\alpha:\Gamma\to\mathrm{Diff}(M)$ has finite image. (2) If...

conjecture | open | MathDB 页面 | 原文线索

V2-013623 | MATHDB-340521

Zimmer's projective-action classification conjecture

摘录 / EXCERPT: Zimmer's conjecture. Either $M=S^{n-1}$, or $M=\mathbb{R}P^{n-1}$ and α is C^{∞}-conjugate to the projective action on S^{n-1} or $\mathbb{R}P^{n-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-013624 | MATHDB-340540

The global conjugacy property conjecture for even Markov groups

摘录 / EXCERPT: Let W_n be the iterated wreath-product group acting on the rooted binary tree of height n, let $K_n=\operatorname{Ker}(\pi_n:W_n\twoheadrightarrow W_{n-1})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013625 | MATHDB-340571

Tzaban-Paolini-Shelah conjecture on the profinite-group cardinal invariant

摘录 / EXCERPT: Tzaban-Paolini-Shelah conjecture. For every infinite metrizable profinite group G,

conjecture | open | MathDB 页面 | 原文线索

V2-013626 | MATHDB-340614

The non-lattice conjecture for actual signature spaces

摘录 / EXCERPT: Non-lattice conjecture. There is no lattice structure on

conjecture | open | MathDB 页面 | 原文线索

V2-013627 | MATHDB-340639

Expansion conjecture for Lusztig's asymptotic Hecke algebra of reductive groups

摘录 / EXCERPT: Expansion conjecture. For any $w\in\tilde{W}$, the coefficient of any T_y in the expansion of t_w is a rational function $b_{w,y}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-013628 | MATHDB-340659

Morita equivalence conjecture for generalized Schur algebras and RoCK blocks

摘录 / EXCERPT: Let $\mathcal B$ be a weight d RoCK block of the usual q-Schur algebra $S_q(N,f)_\mathbb F$, with quantum characteristic $e>0$. Let $\ell=e-1$, and let Z be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-013629 | MATHDB-340671

Conjecture on the size of conjugacy classes of canonical idempotents in free inverse semigroups

摘录 / EXCERPT: Conjugacy-class cardinality conjecture. The conjugacy class of e has

conjecture | open | MathDB 页面 | 原文线索

V2-013630 | MATHDB-340678

Conjecture on t-exactness of the long intertwining transform

摘录 / EXCERPT: Let G be the reductive group, with U and U^- opposite maximal unipotent subgroups and T a maximal torus, and let ... be the Harish-Chandra transform and ... be the long intertw...

conjecture | open | MathDB 页面 | 原文线索

V2-013631 | MATHDB-340702

Essential nilpotency class conjecture for zero-entropy automorphism groups

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension $n\geq 1$ and Kodaira dimension $\kappa(X)$. Let $G\leq\operatorname{Aut}(X)$ be a subgroup of zero entropy. Choose a finite-index...

conjecture | open | MathDB 页面 | 原文线索

V2-013632 | MATHDB-340727

Uniqueness conjecture for strongly outer actions of torsion-free amenable groups

摘录 / EXCERPT: Let G be a countable torsion-free amenable group and let $d4ad4d$ be a strongly self-absorbing C^*-algebra. An action of G on $d4ad4d$ is strongly outer when each nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-013633 | MATHDB-340752

Marquis's fixed-point conjecture for locally elliptic actions on buildings

摘录 / EXCERPT: Marquis's conjecture. The group G has a global fixed point in $X\cup\partial X$.

conjecture | open | MathDB 页面 | 原文线索

V2-013634 | MATHDB-340784

Local embeddability of groups into finite Moufang loops

摘录 / EXCERPT: Local embeddability conjecture. Every group is locally embeddable into finite Moufang loops.

conjecture | open | MathDB 页面 | 原文线索

V2-013635 | MATHDB-340816

The general-position criterion for generalised Gelfand-Graev representations

摘录 / EXCERPT: Let G be a finite reductive group with Frobenius endomorphism F, let k be the underlying field of characteristic p, and let $ d\in\Delta$ index the corresponding grading of...

conjecture | open | MathDB 页面 | 原文线索

V2-013636 | MATHDB-340817

The integral criterion for special unipotent classes

摘录 / EXCERPT: Let G_0 be a connected reductive algebraic group over $\mathbb{C}$, with Lie algebra $\mathfrak{g}_0$, and let Δ index the relevant gradings. For $d\in\Delta$ and $i=1,2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-013637 | MATHDB-340820

The mixed Kazhdan-Lusztig equivalence for the Iwahori category

摘录 / EXCERPT: Let G be the reductive group, let I be an Iwahori subgroup, and let $\rep_{-\kappa}^I$ denote the corresponding category of Iwahori-equivariant modules at level $-\kappa$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013638 | MATHDB-340863

Uniqueness conjecture for Fourier-Jacobi models of the split group of type G2

摘录 / EXCERPT: Let k be a local field. Let $J={\mathrm{SL}}_2(k)\ltimes V$ be the subgroup of the split group ${\mathrm{G}}_2(k)$ described above, let $\widetilde{\mathrm{SL}}_2(k)$ be the meta...

conjecture | open | MathDB 页面 | 原文线索

V2-013639 | MATHDB-340867

The Lück approximation conjecture in the space of marked groups

摘录 / EXCERPT: Let F be a finitely generated free group, let M_k,M be normal subgroups of F, and identify $\operatorname{MG}(F)$ with the space of marked groups. Suppose that M_k converge...

conjecture | open | MathDB 页面 | 原文线索

V2-013640 | MATHDB-340868

The base-change conjecture for division closures of group algebras

摘录 / EXCERPT: Let L/K be an extension of subfields of $\mathbb{C}$, and let G be a group satisfying the strong Atiyah conjecture. Let $\mathcal D_{K[G]}$ and $\mathcal D_{L[G]}$ be, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-013641 | MATHDB-340870

The Suzuki functor conjecture on extension algebras

摘录 / EXCERPT: Let $\boldsymbol{c}$ denote the critical level, let $\mathsf{F}_{\mathbf{c}}$ be the Suzuki functor, and let $\mathbb{W}_{\mathbf{c}}(\lambda)$ and $\Delta_0(\lambda)$ be the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-013642 | MATHDB-340874

Linear isoperimetric function for universal covers of generalized Dehn complexes

摘录 / EXCERPT: Let X be a compact 2-complex with the generalized Dehn property, and let $\widetilde X$ denote its universal cover. Linear isoperimetric-function conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-013643 | MATHDB-340878

Borel's pullback conjecture for enhanced local Langlands correspondences

摘录 / EXCERPT: Borel's pullback conjecture. Suppose that a local Langlands correspondence exists for sufficiently large classes of representations of G and $\tilde G$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-013644 | MATHDB-340882

Abelian p-group reduction conjecture for automorphism groups

摘录 / EXCERPT: Abelian p-group reduction conjecture. $Aut_H(G)$ has a locally finite subgroup of power λ.

conjecture | open | MathDB 页面 | 原文线索

V2-013645 | MATHDB-340885

The faithfulness conjecture for inner-faithful semisimple Hopf actions

摘录 / EXCERPT: Faithfulness conjecture. If H acts inner-faithfully on A, then it also acts faithfully on A.

conjecture | open | MathDB 页面 | 原文线索

V2-013646 | MATHDB-340922

Atiyah's integrality conjecture for L^2-Betti numbers

摘录 / EXCERPT: Atiyah's integrality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013647 | MATHDB-340952

The converse nilpotence conjecture for supercharacter theories

摘录 / EXCERPT: Converse nilpotence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013648 | MATHDB-340960

Howe duality conjecture for the similitude theta correspondence

摘录 / EXCERPT: Let $\mathrm{GSp}^+(V_4)$ and $\mathrm{GSO}(U)$ be the similitude groups in the local Weil representation, and let $\widetilde{\Theta}(\pi^+)$ be the smooth theta lift associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-013649 | MATHDB-341009

The conjecture that the pure twin group PT_5 is free of rank 31

摘录 / EXCERPT: Let PT_n denote the pure twin group on n strands. In particular, the preceding results identify PT_4 as a free group, and the rank formula gives rank 31 for...

conjecture | open | MathDB 页面 | 原文线索

V2-013650 | MATHDB-341023

Happel-Unger conjecture on connectedness of tilting graphs

摘录 / EXCERPT: Happel-Unger conjecture. The graph $\mathcal{G}_t(\mathcal{H})$ is connected if and only if $\mathcal{H}$ contains no nonzero projective objects.

conjecture | open | MathDB 页面 | 原文线索

V2-013651 | MATHDB-341027

Etingof-Rains conjecture on Hilbert series of irreducible Cherednik-algebra quotients

摘录 / EXCERPT: Let $n=kp+r$ with $0\le r<p$. Define ... and ... Here $\mathcal{L}_{t,c}$ is the irreducible quotient of the polynomial representation of the rational Cherednik algebra, and c is gen...

conjecture | open | MathDB 页面 | 原文线索

V2-013652 | MATHDB-341099

Bi-orderability conjecture for fundamental groups of 3-manifolds

摘录 / EXCERPT: Bi-orderability conjecture. G is bi-orderable if and only if G has no generalized torsion element.

conjecture | open | MathDB 页面 | 原文线索

V2-013653 | MATHDB-341152

Atomic Kronecker coefficient upper-bound conjecture

摘录 / EXCERPT: Let $g_{\mu,\nu,\lambda}$ be a Kronecker coefficient and let $\tilde g_{\mu,\nu,\lambda}$ be the corresponding atomic Kronecker coefficient, with the analogous definition of atomic...

conjecture | open | MathDB 页面 | 原文线索

V2-013654 | MATHDB-341166

Multiplicity-one conjecture for the G_2 Fourier-Jacobi model

摘录 / EXCERPT: Let F be a p-adic field, let Π be an irreducible generic representation of $G_2(F)$, let ψ be a nontrivial additive character of F, and let $\widetilde I(\chi,\psi)$...

conjecture | open | MathDB 页面 | 原文线索

V2-013655 | MATHDB-341167

Generalized multiplicity-one conjecture for G_2 Fourier-Jacobi models

摘录 / EXCERPT: Let F be a p-adic field, let Π be an irreducible selfdual representation of $G_2(F)$, let ψ be a nontrivial additive character of F, and let $\widetilde\pi$ be an ir...

conjecture | open | MathDB 页面 | 原文线索

V2-013656 | MATHDB-341178

Generalised Saxl conjecture for staircase-like partitions

摘录 / EXCERPT: Let n be a positive integer, and let $\lambda\vdash n$ be a staircase-like partition, meaning a self-conjugate partition satisfying one of the three conditions described in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-013657 | MATHDB-341191

Conjecture on reversible Hadamard difference sets in $C_{2^n}\times C_{2^n}$

摘录 / EXCERPT: Reversible Hadamard difference set enumeration conjecture. The total number of reversible Hadamard difference sets in G_n of type A is

conjecture | open | MathDB 页面 | 原文线索

V2-013658 | MATHDB-341209

Quantization conjecture for the alpha invariant of compact group actions

摘录 / EXCERPT: Let (M,L) be a polarized manifold and let $\hat K$ be a compact subgroup of $\operatorname{Aut}(M)$. Suppose that L is $\hat K$-linearized. Let $H_{\hat K}^0(M,L^m)$ be the max...

conjecture | open | MathDB 页面 | 原文线索

V2-013659 | MATHDB-341216

Non-negative reduced series conjecture for coloured plane partitions

摘录 / EXCERPT: Reduced-series conjecture. There exists a factorisation

conjecture | open | MathDB 页面 | 原文线索

V2-013660 | MATHDB-341235

The ramification least-common-multiple conjecture for surface algebras

摘录 / EXCERPT: Ramification least-common-multiple conjecture. The least common multiple of the orders of σ_r and σ_s should have significant meaning in modular representation theory...

conjecture | open | MathDB 页面 | 原文线索

V2-013661 | MATHDB-341279

Watkins' GRR existence conjecture for sufficiently large groups

摘录 / EXCERPT: Watkins' conjecture. There is an integer n such that every group of cardinality at least n which is neither abelian of exponent greater than 2 nor generalized dicyclic admits...

conjecture | open | MathDB 页面 | 原文线索

V2-013662 | MATHDB-341285

Grigorchuk's modified Milnor problem

摘录 / EXCERPT: Grigorchuk's modified Milnor problem. A finitely presented group with vanishing algebraic entropy has at most polynomial growth.

conjecture | open | MathDB 页面 | 原文线索

V2-013663 | MATHDB-341415

Hudson's conjecture on trivial source Specht modules of 2-weight 2

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, let $\mathbb{F}$ be an algebraically closed field of characteristic p, and let S^λ denote the Specht module labelled by a par...

conjecture | open | MathDB 页面 | 原文线索

V2-013664 | MATHDB-341443

The elementary-abelian quotient conjecture for induced regular groups

摘录 / EXCERPT: Let G be a finite group, let $Z(G)$ denote its center, and call G induced regular when the induced non-centralizer graph on $G\setminus Z(G)$ is regular. A finite group is an e...

conjecture | open | MathDB 页面 | 原文线索

V2-013665 | MATHDB-341454

The index invariance conjecture for isomorphic open subgroups

摘录 / EXCERPT: Index invariance conjecture. If $H\simeq K$, then

conjecture | open | MathDB 页面 | 原文线索

V2-013666 | MATHDB-341458

Lusztig-Vogan cell conjecture for antispherical p-cells

摘录 / EXCERPT: Let $\boldsymbol{X}$ be the weight lattice, let $\boldsymbol{X}^+$ be its dominant cone, and let ${}^{\mathrm{f}}\hspace{-1pt}W_{\mathrm{ext}}^{\mathrm{f}}$ denote the subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-013667 | MATHDB-341475

Valette's conjecture on property RD for groups acting on symmetric spaces or affine buildings

摘录 / EXCERPT: Valette's conjecture. Property RD holds for any discrete group acting isometrically, properly and cocompactly either on a Riemannian symmetric space or on an affine building. Prope...

conjecture | open | MathDB 页面 | 原文线索

V2-013668 | MATHDB-341497

The conjectural A_7-crystal decomposition of $B^{1,s}$

摘录 / EXCERPT: Let $B^{1,s}$ be the Kirillov-Reshetikhin crystal of type $E_7^{(1)}$, and let η_i denote the fundamental weights of A_7. For $a,b,c,d\in\mathbb{Z}_{\geq 0}$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-013669 | MATHDB-341551

The first Grigorchuk group's 1-noise sensitivity conjecture

摘录 / EXCERPT: Let G be the first Grigorchuk group. The first Grigorchuk group's conjecture. The first Grigorchuk group is ℓ^1-noise sensitive. This is presented as a particular case of th...

conjecture | open | MathDB 页面 | 原文线索

V2-013670 | MATHDB-341552

The entropy noise sensitivity characterization conjecture

摘录 / EXCERPT: Let (G,μ) be a random walk, where G is a group and μ is its probability measure. The walk is entropy noise sensitive when its entropy noise sensitivity is defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-013671 | MATHDB-341573

Identification conjecture for ranked groups with an almost self-normalizing TI-subgroup

摘录 / EXCERPT: Let G) be a connected ranked group with involutions but with no infinite elementary abelian subgroup. Suppose that G has a definable, connected subgroup $C<G$ which is a TI-sub...

conjecture | open | MathDB 页面 | 原文线索

V2-013672 | MATHDB-341574

Nilpotent Borel subgroup conjecture for simple ranked groups

摘录 / EXCERPT: Let G be a simple ranked group, and let a Borel subgroup mean a definable, connected, soluble subgroup maximal among such subgroups. Nilpotent Borel subgroup conjecture. If all B...

conjecture | open | MathDB 页面 | 原文线索

V2-013673 | MATHDB-341575

Borovik-Burdges conjecture on strongly real elements and Cartan subgroups

摘录 / EXCERPT: Let G be a simple ranked group, and let C be a Cartan subgroup, that is, the centraliser of a maximal decent torus. An element is strongly real if it is a product of two involu...

conjecture | open | MathDB 页面 | 原文线索

V2-013674 | MATHDB-341615

Humphreys-Verma conjecture on lifting projective G_r-modules

摘录 / EXCERPT: Humphreys-Verma conjecture. For $\lambda\in X_r$, the G_r-module structure on $Q_r(\lambda)$ can be lifted to G.

conjecture | open | MathDB 页面 | 原文线索

V2-013675 | MATHDB-341630

Isometrisability implies Littlewood-function integrability

摘录 / EXCERPT: Let Γ be a discrete group. For $1\leq p\leq 2$, say that Γ is $(2,p)$-isometrisable when it admits the corresponding isometric representation property. Let q be the...

conjecture | open | MathDB 页面 | 原文线索

V2-013676 | MATHDB-341674

Hom-finiteness conjecture for homotopy categories of supergroup representations

摘录 / EXCERPT: Let G be an algebraic supergroup with reductive even part, let $\mathcal T=\operatorname{Rep}_k(G)$, and let $\mathcal C=\operatorname{Rep}(G)^\infty$. For objects...

conjecture | open | MathDB 页面 | 原文线索

V2-013677 | MATHDB-341675

Semisimplicity conjecture for negligible quotients of supergroup homotopy categories

摘录 / EXCERPT: Let G be a basic classical algebraic supergroup and let H be a subgroup satisfying $G_0\subset H\subset G$. Put $\mathcal T=\operatorname{Rep}_k(G)$, let $Ho\mathcal T$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-013678 | MATHDB-341683

Finiteness conjecture for DNA origami monoids

摘录 / EXCERPT: Let ${\mathcal O}_n$ be the origami monoid of degree n defined by the DNA-origami rewriting system. Finiteness conjecture. The order of ${\mathcal O}_n$ is finite for all n. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-013679 | MATHDB-341688

The ideal product decomposition conjecture for finite simple groups

摘录 / EXCERPT: Let G be a non-abelian finite simple group, and let $S_1,\dots,S_k$ be subsets of G. Ideal product decomposition conjecture. There exists c such that, whenever ... there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-013680 | MATHDB-341689

The normal product decomposition conjecture for finite simple groups

摘录 / EXCERPT: Let G be a non-abelian finite simple group, and let $S_1,\dots,S_k$ be normal subsets of G. Normal product decomposition conjecture. There exists c such that, whenever ... one...

conjecture | open | MathDB 页面 | 原文线索

V2-013681 | MATHDB-341724

Byott's solvability conjecture for regular subgroups

摘录 / EXCERPT: Byott's solvability conjecture. For every $n\in\mathbb{N}$, there do not exist finite groups G and N, both of order n, such that G is insolvable, N is solvable, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013682 | MATHDB-341754

Inclusion conjecture for quantum Grothendieck rings of categories O and C

摘录 / EXCERPT: Let $\mathscr{C}_{\mathbb{Z}}$ be the finite-dimensional representation category and let $\mathcal{O}^+_{\mathbb{Z}}$ be the category whose quantum Grothendieck ring is...

conjecture | open | MathDB 页面 | 原文线索

V2-013683 | MATHDB-341755

Fundamental-representation conjecture for quantum cluster variables

摘录 / EXCERPT: For each $(i,r)\in\hat{I}$, let $\tilde{\chi}_{i,r}\in K_t(\mathcal{O}^+_{\mathbb{Z}})$ be the quantum cluster variable obtained from the initial quantum seed by the prescribed mut...

conjecture | open | MathDB 页面 | 原文线索

V2-013684 | MATHDB-341770

Planar boundary conjecture for CAT(0) groups

摘录 / EXCERPT: Planar boundary conjecture. Every visual boundary of G is planar, and furthermore, G is virtually the fundamental group of a compact 3-manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-013685 | MATHDB-341793

The infinite-order element conjecture for actions on finite-dimensional CAT(0) complexes

摘录 / EXCERPT: Let G be a finitely generated group acting nontrivially on a finite-dimensional $\mathrm{CAT}(0)$ complex. Infinite-order element conjecture. The group G contains an element of...

conjecture | open | MathDB 页面 | 原文线索

V2-013686 | MATHDB-341801

The tensor-dual conjecture for the invariant gamma

摘录 / EXCERPT: Let G be a finite group, let k be a field of characteristic p dividing the order of G, and let M be a finite-dimensional kG-module. Write M^* for the dual module and...

conjecture | open | MathDB 页面 | 原文线索

V2-013687 | MATHDB-341802

The cosine-value conjecture for modules with gamma below two

摘录 / EXCERPT: Let G be a finite group, let k be a field of characteristic p dividing the order of G, and let M be a finite-dimensional kG-module. Cosine-value conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-013688 | MATHDB-341803

The eventual-recursion conjecture for tensor-power counts

摘录 / EXCERPT: Let G be a finite group, let k be a field of characteristic p dividing the order of G, and let M be a finite-dimensional kG-module. Let $\phi(n)=\mathsf{c}_n^G(M)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-013689 | MATHDB-341804

The Omega-algebraic eventual-recursion conjecture

摘录 / EXCERPT: Let G be a finite group, let k be a field of characteristic p dividing the order of G, and let M be an Omega-algebraic kG-module, meaning that the non-projective indeco...

conjecture | open | MathDB 页面 | 原文线索

V2-013690 | MATHDB-341807

Thompson's conjecture on the monster group's graded representation

摘录 / EXCERPT: Let $\tilde J=J-744$ be the normalised J-function, and let $\rho_ atural$ be a graded representation on an infinite-dimensional vector space ... whose graded parts $V^ atural_i$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-013691 | MATHDB-341808

Conway-Norton's Monstrous Moonshine conjecture

摘录 / EXCERPT: Let $d4dc$ be the monster group, let ... be an infinite-dimensional graded representation with finite-dimensional graded parts, and for each conjugacy class $[g]$ of $d4dc$ let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-013692 | MATHDB-341879

The SU(2)-cyclic rational homology 3-sphere conjecture

摘录 / EXCERPT: A closed oriented 3-manifold Y is an $SU(2)$-cyclic 3-manifold if every $SU(2)$-representation of $\pi_1(Y)$ has cyclic image. A rational homology 3-sphere is a 3-manifol...

conjecture | open | MathDB 页面 | 原文线索

V2-013693 | MATHDB-341892

Alexeev-Brion conjecture on integrality of string polytopes

摘录 / EXCERPT: Alexeev-Brion conjecture. For every reduced decomposition $\underline{w_0}$ and every $\lambda\in\Lambda^+$, the string polytope $\mathcal{Q}_{\underline{w_0}}(\lambda)$ is a latti...

conjecture | open | MathDB 页面 | 原文线索

V2-013694 | MATHDB-341893

Conjecture on integrality of standard string polytopes for classical groups

摘录 / EXCERPT: Integrality conjecture for standard string polytopes. The string polytope $\mathcal{Q}_{\underline{w_0}^{\mathrm{std}}}(\lambda)$ is a lattice polytope if and only if at least one...

conjecture | open | MathDB 页面 | 原文线索

V2-013695 | MATHDB-341933

Triangularity conjecture for RSK-standard representations

摘录 / EXCERPT: Let $\mathcal{M}$ be the set of multisegments. For a multisegment $\mathfrak{m}$, let $\Lambda(\mathfrak{m})$ be its RSK-standard representation, let $Z(\mathfrak{m})$ be the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-013696 | MATHDB-341934

Irreducibility conjecture for RSK-standard representation socles

摘录 / EXCERPT: Let $\mathfrak{m}$ be a multisegment, and let $\Lambda(\mathfrak{m})$ be its RSK-standard representation. Let $Z(\mathfrak{m})$ denote the irreducible representation attached to...

conjecture | open | MathDB 页面 | 原文线索

V2-013697 | MATHDB-341935

Iterated socle conjecture for RSK-standard representations

摘录 / EXCERPT: Let $cmathfrak{m}$ be a multisegment with RSK data ... For each ladder $cmathfrak{l}_i$, let $Z(cmathfrak{l}_i)$ be its irreducible representation, and define representations recursi...

conjecture | open | MathDB 页面 | 原文线索

V2-013698 | MATHDB-341936

RSK-standard basis conjecture for the Grothendieck subgroup

摘录 / EXCERPT: Let $cmathfrak{M}$ be the set of multisegments, let $cLambda(cmathfrak{m})$ be the RSK-standard representation attached to $cmathfrak{m}cincmathfrak{M}$, and let $cmathcal{R}'$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-013699 | MATHDB-341943

Genteelness conjecture for Jacobian algebras of generic quivers with potential

摘录 / EXCERPT: Let (Q,W) be a quiver with a generic potential, and suppose that its Jacobian algebra $J_{(Q,W)}$ is finite dimensional. Genteelness conjecture. The Jacobian algebra $J_{(Q,W)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-013700 | MATHDB-341967

Maximality conjecture for the $coperatorname{L}^1$ metric on full groups of graphings

摘录 / EXCERPT: Maximality conjecture. The $\mathrm L^1$ metric on the $\mathrm L^1$ full group of Φ is maximal.

conjecture | open | MathDB 页面 | 原文线索

V2-013701 | MATHDB-342009

Vogan's conjecture for complex nilpotent coadjoint orbits

摘录 / EXCERPT: Let G be a complex reductive Lie group, and let $X=G\cdot\lambda=G/G(\lambda)$ be a nilpotent coadjoint orbit. Suppose an irreducible G-equivariant local system on X is given...

conjecture | open | MathDB 页面 | 原文线索

V2-013702 | MATHDB-342010

Charney's CAT(0) conjecture for Artin groups

摘录 / EXCERPT: Let an Artin group be the group associated to a symmetric matrix (m_{ij}) with entries in $\{2,3,\dots\}\cup\{\infty\}$, generated by $s_1,\ldots,s_n$ subject to the braid relati...

conjecture | solved | MathDB 页面 | 原文线索

V2-013703 | MATHDB-342068

The hypersurface conjecture for Gorenstein rings of countable Cohen-Macaulay representation type

摘录 / EXCERPT: Hypersurface conjecture. If R has countable CM-representation type, then R is a hypersurface.

conjecture | open | MathDB 页面 | 原文线索

V2-013704 | MATHDB-342069

The dimension conjecture for Cohen-Macaulay categories of countable representation type

摘录 / EXCERPT: Dimension conjecture. If R has countable CM-representation type, then

conjecture | open | MathDB 页面 | 原文线索

V2-013705 | MATHDB-342073

Aschbacher's conjecture on simply connected p-subgroup complexes

摘录 / EXCERPT: Let G be a finite group such that $G=AF^*(G)$, where A is an elementary abelian p-subgroup of rank $r\geq 3$, and let $F^*(G)$ be the direct product of the A-conjugates of...

conjecture | open | MathDB 页面 | 原文线索

V2-013706 | MATHDB-342100

Refined local L-packet conjecture for unitary similitude groups

摘录 / EXCERPT: Let F be a local field, let $G=\operatorname{GU}_n$, let $\RU_n$ be the associated unitary group, and let $\phi\in\Phi_{\mathrm{bdd}}(G)$ be a bounded Langlands parameter. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013707 | MATHDB-342121

The equivalence conjecture for the functor Phi between principal W-algebra module categories

摘录 / EXCERPT: Equivalence conjecture. The functor $\Phi:\cal P^{n-m}\to W^\chi\text{-mod}$ defines an equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-013708 | MATHDB-342136

Theta representation as the least generic representation

摘录 / EXCERPT: Let $\overline{G}$ be an n-fold persistent covering group. Let $\Theta(\overline{G},\chi)=\pi_{\Gamma^-}$ be the theta representation associated to an unramified character χ...

conjecture | open | MathDB 页面 | 原文线索

V2-013709 | MATHDB-342150

The 0-1 coefficient conjecture for fully supported representations

摘录 / EXCERPT: 0-1 coefficient conjecture. One must have $n_i=0$ or 1 for every $1\leq i\leq l$.

conjecture | open | MathDB 页面 | 原文线索

V2-013710 | MATHDB-342163

Kantor's classification conjecture for flag-transitive generalized quadrangles

摘录 / EXCERPT: Kantor's conjecture. If $\mathcal{Q}$ is a finite flag-transitive generalized quadrangle and $\mathcal{Q}$ is not a classical generalized quadrangle, then, up to duality,...

conjecture | open | MathDB 页面 | 原文线索

V2-013711 | MATHDB-342167

The spherical-generator characterization of Gelfand and weak Gelfand type

摘录 / EXCERPT: Let Z_S be the additive monoid of highest-weight-vector weights for an irreducible quantum flag manifold $\mathcal{O}_q(G/L_S)$. Say that Z_S is generated by r elements if it...

conjecture | open | MathDB 页面 | 原文线索

V2-013712 | MATHDB-342174

Saxl's 2-modular tensor-square conjecture

摘录 / EXCERPT: Let $\Bbbk$ be a field of characteristic 2, let $n=k(k+1)/2$, and let $\rho(k)=(k,k-1,\dots,2,1)$. A partition is 2-regular if no positive part occurs at least twice, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013713 | MATHDB-342175

Saxl's arbitrary-rank symmetric p-core conjecture

摘录 / EXCERPT: For $n\in\mathbb{N}$, let ${\bf D}^{\mathbb{C}}(\lambda)$ denote the simple complex $\mathfrak{S}_n$-module labelled by the partition λ. A partition is a symmetric p-core...

conjecture | open | MathDB 页面 | 原文线索

V2-013714 | MATHDB-342176

Exceptional Specht-module decomposition conjecture for α_k

摘录 / EXCERPT: For $k\in\mathbb{N}$, let α_k be the partition defined earlier in the source, let α_k^R be its row-labelled constituent, let $\alpha_k^C=(2k+3,2k+1,2k-1,\dots,9,7,5)$...

conjecture | open | MathDB 页面 | 原文线索

V2-013715 | MATHDB-342177

Semisimplicity conjecture for the Specht modules indexed by β_k

摘录 / EXCERPT: For odd $k\in2\mathbb{N}+1$, let β_k be the partition defined earlier in the source, and let β_k^R and β_k^C be the two labels specified there. Let $w(\beta_k)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-013716 | MATHDB-342224

The sum-of-element-orders criterion for supersolvability

摘录 / EXCERPT: Supersolvability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013717 | MATHDB-342225

Universal irreducibility structure for Harish-Chandra parabolic induction

摘录 / EXCERPT: Let G be a reductive group, let $P=MN$ be a parabolic subgroup, and let $I^G_P(\sigma)$ be the parabolic induction of an irreducible admissible representation σ of M. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-013718 | MATHDB-342226

Goldberg-Jantzen type product formula for tempered generalized principal series

摘录 / EXCERPT: Let $I^G_P(\nu,\sigma)$ be a tempered generalized principal series, where $P=MN$ is a parabolic subgroup, σ is a unitary supercuspidal representation of M, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013719 | MATHDB-342249

Dixmier's HSOP conjecture for binary form invariants

摘录 / EXCERPT: Let S_d be the space of binary forms of degree d, let ${\rm Inv}_d$ be its ring of SL_2-invariants, and, for $d=2k$ with k even, let $\mathscr{H}_{n,p}(F)$ denote the coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-013720 | MATHDB-342253

The local Langlands correspondence of Vogan type

摘录 / EXCERPT: Let F be a p-adic field, let G^* be a quasi-split connected reductive algebraic group over F, and let G range over its pure inner forms. For each tempered parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-013721 | MATHDB-342286

The character formula for even-rank fork representations of orthosymplectic Lie superalgebras

摘录 / EXCERPT: Let m,n be nonnegative integers with $|m-n|$ even. Let p be a nonnegative integer and r an integer with $0\leq r\leq p$. Consider the representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-013722 | MATHDB-342319

Finiteness conjecture for tau-isospectrality of two lens spaces

摘录 / EXCERPT: Let $L=L(49;1,6,15)$ and $L'=L(49;1,6,20)$ be the given lens spaces, and let τ be an irreducible representation of $\operatorname{SO}(5)$. Finiteness conjecture. There are onl...

conjecture | open | MathDB 页面 | 原文线索

V2-013723 | MATHDB-342326

General braid group action conjecture for acyclic quivers

摘录 / EXCERPT: Let (Q,τ) be an acyclic $\imath$quiver, let W_{τ} be the restricted Weyl group, and let ${\rm Br}(W_{\tau})$ be the braid group generated by elements t_i for...

conjecture | open | MathDB 页面 | 原文线索

V2-013724 | MATHDB-342345

Strictness conjectures for hyperbolic structures on spherical Artin-Tits groups

摘录 / EXCERPT: Let $A=A_S$ be an irreducible Artin-Tits group of spherical type with rank at least 3. Let X_{abs}^A be the absorbable-element generating set, X_{NP}^A the union of normalize...

open_problem | open | MathDB 页面 | 原文线索

V2-013725 | MATHDB-342349

The NLB conjecture for nontrivial words

摘录 / EXCERPT: For a positive integer d, let $\operatorname{F}(X_1,\ldots,X_d)$ be the free group on $X_1,\ldots,X_d$, and let $w\in\operatorname{F}(X_1,\ldots,X_d)$ be a nontrivial word. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-013726 | MATHDB-342359

The Hall algebra group homomorphism conjecture for tau-tilting finite algebras

摘录 / EXCERPT: Let Λ be an arbitrary τ-tilting finite algebra. Let $G(\Lambda)$ be the group associated with the relations from 2-simple minded collections, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013727 | MATHDB-342406

Commutator-vanishing conjecture for homogeneous conjugacy-invariant pseudo-lengths

摘录 / EXCERPT: Let $\langle \alpha, \beta \rangle=\langle \alpha,\beta\rangle$ be the free group on two generators. A homogeneous, conjugacy-invariant pseudo-length function l on...

conjecture | open | MathDB 页面 | 原文线索

V2-013728 | MATHDB-342434

Malkoun's singular-value conjecture for Weyl-equivariant maps

摘录 / EXCERPT: Malkoun's singular-value conjecture. For every $\mathbf{x}\in X$,

conjecture | open | MathDB 页面 | 原文线索

V2-013729 | MATHDB-342453

Screening-operator identification conjecture for Fock representations

摘录 / EXCERPT: Let $\mathcal{F}_{\Lambda}$ be the Fock representation with screening operators ξ_i and η_i for $i\in\hat{I}^-\cup\{m\}$, and set ... For a weight Λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013730 | MATHDB-342494

Polynomial dimension-function conjecture for simple functors on finite p-rings

摘录 / EXCERPT: Soit A un p-anneau fini, c'est-à-dire un anneau fini dont la caractéristique est une puissance de p, et soit K un corps tel que $\mathrm{car}(K)\neq p$. Pour un foncteur...

conjecture | open | MathDB 页面 | 原文线索

V2-013731 | MATHDB-342515

Shalev's Engel word-map conjecture

摘录 / EXCERPT: Shalev's conjecture. For every finite simple non-abelian group G and every $n\geq 1$, the n-th Engel word map

conjecture | open | MathDB 页面 | 原文线索

V2-013732 | MATHDB-342516

Vaught's square-product conjecture

摘录 / EXCERPT: Vaught's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013733 | MATHDB-342541

Equivariant Jiang-Su stability conjecture for amenable group actions

摘录 / EXCERPT: Let Γ be a countable amenable group, and let A be a separable, simple, nuclear and $\mathcal{Z}$-stable $\mathrm{C^*}$-algebra. An action $\alpha:\Gamma\curvearrowright A$...

conjecture | open | MathDB 页面 | 原文线索

V2-013734 | MATHDB-342550

Triviality of the reduced horoboundary action for finitely generated nilpotent groups

摘录 / EXCERPT: Let G be a finitely generated nilpotent group endowed with a finite set of generators, and equip G with the corresponding word metric. The associated reduced horoboundary is th...

conjecture | open | MathDB 页面 | 原文线索

V2-013735 | MATHDB-342551

Finiteness conjecture for the closed geodesic norm on higher-rank locally symmetric spaces

摘录 / EXCERPT: Finiteness conjecture. The diameter of the closed geodesic norm is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-013736 | MATHDB-342561

Fourfold product stabilization conjecture for Schur functions

摘录 / EXCERPT: Fourfold product stabilization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013737 | MATHDB-342565

Trace class criterion for semidirect products with unipotent radicals

摘录 / EXCERPT: Trace class criterion. The group G is a trace class group if and only if H/L is compact.

conjecture | open | MathDB 页面 | 原文线索

V2-013738 | MATHDB-342575

The spin-chain equivalence conjecture for the infinite Temperley-Lieb algebra

摘录 / EXCERPT: Let w_{0},w_{1} be link states, let $\mathcal{S}(w_{0})$ and $\mathcal{S}(w_{1})$ be the associated spin-chain representations, and define ... by...

conjecture | open | MathDB 页面 | 原文线索

V2-013739 | MATHDB-342602

Coincidence of the splitting and module orders on faithfully balanced modules over a linearly oriented quiver

摘录 / EXCERPT: Let $fb (n)$ be the set of faithfully balanced $Lambda_n$-modules of composition length n, where $Lambda_n$ is the path algebra of the quiver $1\to2\todots\to n$. The relations...

conjecture | open | MathDB 页面 | 原文线索

V2-013740 | MATHDB-342629

Strong indecomposability conjecture for Bernstein components of Zelevinsky restrictions

摘录 / EXCERPT: Let $\mathfrak m$ be a multisegment for $n+1$, and let $\zeta(\mathfrak m)$ denote the associated Zelevinsky induced module. For the restriction to G_n, a Bernstein component is...

conjecture | open | MathDB 页面 | 原文线索

V2-013741 | MATHDB-342639

Weak Herstein-Scott conjecture for subnormal subgroups

摘录 / EXCERPT: Let D be a division ring and let D^* be its multiplicative group. A subgroup is non-central if it is not contained in the center of D. Weak Herstein-Scott conjecture. Every n...

conjecture | open | MathDB 页面 | 原文线索

V2-013742 | MATHDB-342685

Isomorphism conjecture for local and graded local Weyl modules

摘录 / EXCERPT: Let $\mathfrak a$ and $\mathfrak b$ be such that $U_{\mathbb Z}(\mathfrak a)$ and $U_{\mathbb Z}(\mathfrak b)$ have been defined. Let $\mathbf a=(a_1,\dots,a_n)\in\mathbb F^n$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013743 | MATHDB-342686

Higher twisted Drinfeld center equivalence conjecture

摘录 / EXCERPT: Let G be a finite group, let τ_h denote the transgression map, and let $\omega\in Z^{n+2}(G,\mathbf{k}^{\times})$. Suppose that a suitable notion of an n-category has been...

conjecture | open | MathDB 页面 | 原文线索

V2-013744 | MATHDB-342687

Classification conjecture for minimal modular extensions of 2Rep(G)

摘录 / EXCERPT: Let G be a finite group. A minimal modular extension of $\operatorname{2Rep}(G)$ is a unitary modular tensor 2-category equipped with a braided monoidal fully faithful embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-013745 | MATHDB-342697

The cofree representation characterization conjecture for connected reductive groups

摘录 / EXCERPT: The cofree representation characterization conjecture. begin{enumerate} item If V is cofree, then it is cnpure. item If G is semisimple and V is irreducible, then purity of...

conjecture | open | MathDB 页面 | 原文线索

V2-013746 | MATHDB-342718

Hernandez-Leclerc's cluster-monomial conjecture for real simple modules

摘录 / EXCERPT: Hernandez-Leclerc's conjecture. The cluster monomials of the cluster algebra are in bijection with the isomorphism classes of real simple objects in $\mathscr{C}^-$.

conjecture | open | MathDB 页面 | 原文线索

V2-013747 | MATHDB-342721

Zelevinsky's mutual-position conjecture for multisegment multiplicities

摘录 / EXCERPT: Let Π be a cuspidal support, let $\mathcal{O}(\Pi)$ denote the corresponding Zelevinsky line, and let $\mathbf{a}$ and $\mathbf{b}$ be multisegments belonging to the same...

conjecture | open | MathDB 页面 | 原文线索

V2-013748 | MATHDB-342748

Keedwell's conjecture on non-extendable near transversals in group latin squares

摘录 / EXCERPT: Let G be a finite non-Abelian group, and let $L(G)$ denote its latin square. A partial transversal is a collection of cells that intersects each row, column, and symbol class at...

conjecture | open | MathDB 页面 | 原文线索

V2-013749 | MATHDB-342752

The procyclic Engel-sink conjecture for profinite groups

摘录 / EXCERPT: Let G be a profinite group. An Engel sink of an element $g\in G$ is a subset $\mathcal E(g)$ such that, for every $x\in G$, all sufficiently long commutators $[x,g,g,\ldots,g]$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-013750 | MATHDB-342829

Uniqueness conjecture for the Moishezon element of the twistor-group action

摘录 / EXCERPT: Moishezon-element uniqueness conjecture. There is a unique Moishezon element in $\Gamma_{\mathcal X}$, namely the equivalence class of complex conjugation. The conjecture asserts t...

conjecture | open | MathDB 页面 | 原文线索

V2-013751 | MATHDB-342874

Finite-part assembly-map conjecture

摘录 / EXCERPT: Finite-part assembly conjecture. The group $\widetilde{\mathcal M}_{\gamma_1,\ldots,\gamma_n}$ has rank n, and every nonzero element of $K_0^{\mathrm{fin}}(C^*(\Gamma))$ lies out...

conjecture | open | MathDB 页面 | 原文线索

V2-013752 | MATHDB-342883

Wehrl-type L^{2s} conjecture for affine coherent states

摘录 / EXCERPT: Fix $\alpha>0$ and $s\geq1$. Let U be the affine-group representation on $L^2(\mathbb{R}_+)$, let $\eta_\alpha(k)=C(\alpha)k^\alpha e^{-k}$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-013753 | MATHDB-342890

Robinson's ordinary weight conjecture for fusion-system block invariants

摘录 / EXCERPT: Robinson's ordinary weight conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-013754 | MATHDB-342891

The polynomiality conjecture for local invariants of p-compact groups

摘录 / EXCERPT: Polynomiality conjecture. For each $d\geqslant 0$, the invariant $\mathbf{m}(\mathcal{F}(q),\alpha,d)$ is a rational polynomial in p^l.

conjecture | open | MathDB 页面 | 原文线索

V2-013755 | MATHDB-342903

The finite-group word-measure orbit conjecture of Shalev

摘录 / EXCERPT: Let $\mathbf{F}$ be the free group, and let $w_{1},w_{2}\in\mathbf{F}$. For a finite group G, write μ_{w} for the probability measure on G obtained by evaluating w on a H...

conjecture | open | MathDB 页面 | 原文线索

V2-013756 | MATHDB-342904

Wiegold's conjecture on Nielsen equivalence of finite simple group generating tuples

摘录 / EXCERPT: Wiegold's conjecture. If $r\geq 3$, then $\operatorname{Aut}(\mathbf{F}_{r})$ acts transitively on $\operatorname{Epi}(\mathbf{F}_{r},G)$. This is equivalent to asking whether all...

conjecture | open | MathDB 页面 | 原文线索

V2-013757 | MATHDB-342935

Superspace Vandermonde model for the Delta operator

摘录 / EXCERPT: Let $n=k+r$ and let ${\mathbf a}=(k-1,k-1,\dots,k-1)$ be a length-r sequence whose entries are all $k-1$. Let $\mathcal V_n({\mathbf a})$ be the smallest subspace containing the...

conjecture | open | MathDB 页面 | 原文线索

V2-013758 | MATHDB-342944

The classifying-space conjecture for outer actions on stable Kirchberg algebras

摘录 / EXCERPT: Classifying-space conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-013759 | MATHDB-342945

The classification conjecture for unital Kirchberg algebra actions

摘录 / EXCERPT: Classification conjecture for unital actions. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-013760 | MATHDB-342961

The presentation conjecture for the subgroup H_k of G_{k+1}^k

摘录 / EXCERPT: Let G_{k+1}^k be one of the groups under consideration, and let H_k be its subgroup generated by $b_1,\dots,b_k$. For distinct indices $i,j,m,l\in\{1,\dots,k\}$, consider the r...

conjecture | open | MathDB 页面 | 原文线索

V2-013761 | MATHDB-343042

Equal centralizer and commutator counts imply isoclinism

摘录 / EXCERPT: Centralizer-commutator conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-013762 | MATHDB-343043

The exponent conjecture for Schur multipliers of finite p-groups

摘录 / EXCERPT: Let G be a finite p-group, and write $M(G)=H_2(G,\mathbb{Z})$ for its Schur multiplier. Write $\operatorname{exp}(G)$ for the exponent of G. Exponent conjecture for finite...

conjecture | open | MathDB 页面 | 原文线索

V2-013763 | MATHDB-343046

The Lorentzian conjecture for weight multiplicities of \mathfrak{sl}_m-modules

摘录 / EXCERPT: Let Λ be the integral weight lattice of $\mathfrak{sl}_m(\mathbb{C})$, let $\mathrm{V}(\lambda)$ be the irreducible module with highest weight λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013764 | MATHDB-343047

The Lorentzian conjecture for normalized shifted characters

摘录 / EXCERPT: Let Λ be the weight lattice of $\mathfrak{sl}_m(\mathbb{C})$, let $\mathrm{V}(\lambda)$ be the irreducible module of highest weight λ, and define its shifted charac...

conjecture | open | MathDB 页面 | 原文线索

V2-013765 | MATHDB-343082

Isaacs-Keller-Meierfrankenfeld-Moreto's primitive character divisibility conjecture

摘录 / EXCERPT: Primitive character divisibility conjecture. The character degree $\chi(1)$ divides the size of some conjugacy class:

conjecture | open | MathDB 页面 | 原文线索

V2-013766 | MATHDB-343086

The finite simple group conjecture for pointed Hopf algebras

摘录 / EXCERPT: Let $\boldsymbol{G}$ be a finite simple non-abelian group. A complex Yetter-Drinfeld module V of $\boldsymbol{G}$ has an associated Nichols algebra $\mathfrak B(V)$. Finite simpl...

conjecture | open | MathDB 页面 | 原文线索

V2-013767 | MATHDB-343136

Finite generation conjecture for verbal subgroups

摘录 / EXCERPT: Finite generation conjecture. If

open_problem | open | MathDB 页面 | 原文线索

V2-013768 | MATHDB-343137

Open-subgroup triviality conjecture for word values

摘录 / EXCERPT: Open-subgroup triviality conjecture. If

open_problem | open | MathDB 页面 | 原文线索

V2-013769 | MATHDB-343227

Cannon-Kapovich-Kleiner-Haissinsky planarity conjecture for hyperbolic groups

摘录 / EXCERPT: Cannon-Kapovich-Kleiner-Haissinsky conjecture. If $b$partial G$ is planar, then G is virtually isomorphic to a Kleinian group.

conjecture | open | MathDB 页面 | 原文线索

V2-013770 | MATHDB-343247

Nearly-linear expected running time of ColorIsomorphic with WordReduction

摘录 / EXCERPT: Let the two inputs be transitive tuples of permutations in the symmetric group, with tuple length d, and choose the permutations uniformly at random. Nearly-linear running-time c...

conjecture | open | MathDB 页面 | 原文线索

V2-013771 | MATHDB-343306

Finiteness conjecture for indecomposable decreasing extensions

摘录 / EXCERPT: Finiteness conjecture for decreasing extensions. For every positive integer t, the set

conjecture | open | MathDB 页面 | 原文线索

V2-013772 | MATHDB-343323

Type 1 noncommutative Borsuk-Ulam conjecture for compact group actions

摘录 / EXCERPT: Let G be a nontrivial compact group acting freely on a unital C^*-algebra A. Define the join ... Equip $A\circledast C(G)$ with the diagonal action of G. Type 1 noncommutativ...

conjecture | open | MathDB 页面 | 原文线索

V2-013773 | MATHDB-343327

Malle-Robinson conjecture on simple modules in blocks of finite groups

摘录 / EXCERPT: Malle-Robinson conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013774 | MATHDB-343372

The generalised regularity classification for matrix semigroups over semirings

摘录 / EXCERPT: Generalised regularity classification. The displayed classification summarises the generalised regularity properties of $M_n(\mathfrak{L})$, $UT_n(\mathfrak{L})$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013775 | MATHDB-343394

Conjecture on recognition by spectrum for finite classical groups

摘录 / EXCERPT: Recognition conjecture for classical groups. Every finite group isospectral to L is an almost simple group with socle isomorphic to L.

conjecture | open | MathDB 页面 | 原文线索

V2-013776 | MATHDB-343409

The cardinality conjecture for automorphism groups of superextensions of monogenic semigroups

摘录 / EXCERPT: Let $\mathrm{M}_{r,m}$ denote a finite monogenic semigroup, let $\lambda(S)$ denote its superextension, and let $\operatorname{Aut}(G)$ denote the automorphism group of a group G...

conjecture | open | MathDB 页面 | 原文线索

V2-013777 | MATHDB-343410

The trivial-range conjecture for restriction automorphisms of superextensions

摘录 / EXCERPT: Trivial-range conjecture. The restriction operator R has trivial range. The conjecture concerns the restriction of automorphisms from the superextension of $\mathrm{M}_{r,m}$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-013778 | MATHDB-343411

The quantum Q-system conjecture for Macdonald operators of types BCD

摘录 / EXCERPT: Let G be one of the classical types D_N, B_N, or C_N, and let $M_{a,n}^{(G)}$ be the associated Macdonald difference operators, with the superscript G omitted below. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013779 | MATHDB-343412

The raising and lowering conjecture for q-Whittaker functions of types BCD

摘录 / EXCERPT: For a classical root system of type D_N, B_N, or C_N, let $M_{a,k}^{(G)}$ be the corresponding difference operators and let $\Pi_\lambda^{(G)}(q^{-1};\mathbf{x})$ be the dual...

conjecture | open | MathDB 页面 | 原文线索

V2-013780 | MATHDB-343417

Almost simple groups polynomial bound conjecture

摘录 / EXCERPT: Let X be a finite almost simple group, meaning that its socle $\operatorname{soc} X$ is a nonabelian simple group and...

conjecture | open | MathDB 页面 | 原文线索

V2-013781 | MATHDB-343426

Addition Theorem conjecture for algebraic entropy

摘录 / EXCERPT: Let S be a cancellative right amenable monoid acting on the abelian group A on the left by α. Let B be an α-invariant subgroup of A, and write α_B and...

conjecture | open | MathDB 页面 | 原文线索

V2-013782 | MATHDB-343491

Zabrocki's Frobenius-characteristic conjecture for the super-diagonal coinvariant ring

摘录 / EXCERPT: Let $n\geq1$. Let SDR_n be the super-diagonal coinvariant ring in commuting variables $x_1,\ldots,x_n$, $y_1,\ldots,y_n$ and Grassmann variables...

conjecture | open | MathDB 页面 | 原文线索

V2-013783 | MATHDB-343495

Odd-power Eisenstein module classification and enumeration of linear Mendelsohn triple systems

摘录 / EXCERPT: Odd-power classification conjecture. There is a quasigroup isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013784 | MATHDB-343515

Transcendence of conjugacy growth series for Baumslag-Solitar groups

摘录 / EXCERPT: For an integer $k\geq 2$, let $BS(1,k)=\langle a,t\mid tat^{-1}=a^k\rangle$, and let a finite generating set mean any finite generating set of this group. The Baumslag-Solitar tran...

conjecture | open | MathDB 页面 | 原文线索

V2-013785 | MATHDB-343516

Transcendence of conjugacy growth series outside virtually abelian groups

摘录 / EXCERPT: Let G be a finitely presented group that is not virtually abelian, and let a finite generating set mean any finite generating set of G. The general conjugacy-growth transcenden...

conjecture | open | MathDB 页面 | 原文线索

V2-013786 | MATHDB-343668

Cluster compatibility conjecture for decorated reduced words

摘录 / EXCERPT: Cluster compatibility conjecture. For any decorated reduced word for w_0, the set ${\bf X}_{\mathbb{I}}$ is a cluster for U mutation-equivalent to an initial cluster attached t...

conjecture | open | MathDB 页面 | 原文线索

V2-013787 | MATHDB-343670

Cluster conjecture for inverse-map functions on double Bruhat cells

摘录 / EXCERPT: Cluster conjecture for inverse-map functions. The set $\{f_1,\ldots,f_n\}$ is a cluster for $L^{e,w}$; equivalently, the fraction field generated by $\{f_1,\ldots,f_n\}$ is the fun...

conjecture | open | MathDB 页面 | 原文线索

V2-013788 | MATHDB-343683

Tanisaki's conjecture on characters of primitive ideal quotients

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let $U(\mathfrak{g})$ be its universal enveloping algebra, and let $\mathscr{I}_0$ be the set of primitive ideals with trivial central c...

conjecture | open | MathDB 页面 | 原文线索

V2-013789 | MATHDB-343706

Lazarus's Jordan-Hölder conjecture for universal unramified modules

摘录 / EXCERPT: Lazarus's conjecture. There is an equality of Jordan-Hölder multisets

conjecture | open | MathDB 页面 | 原文线索

V2-013790 | MATHDB-343708

Weak Ihara conjecture for cyclic automorphic modules

摘录 / EXCERPT: Weak Ihara conjecture. The $k[G(F_{v_0}^+)]$-module $\langle G(F_{v_0}^+)\cdot f\rangle$ is generic. This is explicitly presented as a weaker conjecture than the preceding weak Iha...

conjecture | open | MathDB 页面 | 原文线索

V2-013791 | MATHDB-343717

Pseudoreflection-group classification conjecture

摘录 / EXCERPT: Let (G,V) be a faithful, irreducible module of finite Morley rank, with G connected. A pseudoreflection subgroup is an abelian subgroup R such that ... and R acts transitivel...

conjecture | open | MathDB 页面 | 原文线索

V2-013792 | MATHDB-343719

Definable twist-and-tensor conjecture for modules of algebraic groups

摘录 / EXCERPT: Let (G,V) be a faithful, irreducible module of finite Morley rank, where $G=\mathbb{G}(\mathbb{K})$ is the group of $\mathbb{K}$-points of a simple algebraic group. Definable twi...

conjecture | open | MathDB 页面 | 原文线索

V2-013793 | MATHDB-343740

Modules conjecture for hook-indexed rectangular harmonic polynomials

摘录 / EXCERPT: Let $\rho=(a\,|\,b)$ be a hook shape, let $\mathcal{S}_\rho$ denote the corresponding graded Frobenius characteristic, and let $L_{\leq 2}$ retain the components indexed by partiti...

conjecture | open | MathDB 页面 | 原文线索

V2-013794 | MATHDB-343742

Hook components conjecture for alternant isotypic components

摘录 / EXCERPT: Let ρ be a partition, let n be its size, and let $(a\,|\,b)$ be a hook with $0\leq a\leq n-1$. Let $\mathcal{A}_\rho$ denote the alternant isotypic component associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-013795 | MATHDB-343743

Length conjecture for coefficients of hook modules

摘录 / EXCERPT: Let $\rho=(a\,|\,b)$ be a hook with $a\geq1$, let μ be a partition of n, and write $\bm{c}_{\rho\mu}$ for the corresponding symmetric-function coefficient. For a symmetric fu...

conjecture | open | MathDB 页面 | 原文线索

V2-013796 | MATHDB-343775

Unimodality conjecture for hook multiplicities of higher Lie characters

摘录 / EXCERPT: Unimodality conjecture. For every positive integers r and s, the sequence $m_1,m_2,\ldots$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-013797 | MATHDB-343790

Uniqueness of code-loop representations by degree and type

摘录 / EXCERPT: Let L be a code loop, and let V_{1} and V_{2} be representations of L. The degree-and-type uniqueness conjecture. If V_{1} and V_{2} have the same degree and type, then...

conjecture | open | MathDB 页面 | 原文线索

V2-013798 | MATHDB-343798

The l-adic Whittaker-sheaf equivalence conjecture

摘录 / EXCERPT: Let G be a reductive group, let U be the unipotent radical of a Borel subgroup, and let T be the corresponding maximal torus. Fix a non-degenerate homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-013799 | MATHDB-343799

The exactness conjecture for the non-linear Fourier transform

摘录 / EXCERPT: The exactness conjecture. The functor $\mathrm F_{G,\underline{\lambda}}$ is exact.

conjecture | open | MathDB 页面 | 原文线索

V2-013800 | MATHDB-343843

The eigenvalue conjecture for symmetric U_q(sl_N) Racah matrices

摘录 / EXCERPT: Let $U=U\left[\begin{matrix}R_1&R_2\\R_3&R_4\end{matrix}\right]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013801 | MATHDB-343847

The degree-zero word-map conjecture on GL2 modulo its center

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, let $w(x_1,\dots,x_m)$ be a group word whose degree in each variable x_i is 0, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-013802 | MATHDB-343885

Kim-Yu's conjecture on simple modules of regular Bernstein-block Hecke algebras

摘录 / EXCERPT: Kim-Yu's conjecture. There exists a bijection, determined by the choice of Σ, between the simple modules of ${\mathcal{H}}(G,\rho^{\vee})$ and those of...

conjecture | open | MathDB 页面 | 原文线索

V2-013803 | MATHDB-343888

The saturation conjecture for F-polynomials of cluster monomials

摘录 / EXCERPT: Let F_M denote the F-polynomial associated with a representation, and call its support saturated if every lattice point in the Newton polytope of F_M has a nonzero coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-013804 | MATHDB-344035

The DO and Schur-representation finiteness conjectures for algebras

摘录 / EXCERPT: DO and Schur-representation finiteness conjectures. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-013805 | MATHDB-344068

Invariable generation conjecture for finite classical groups

摘录 / EXCERPT: Invariable generation conjecture. There exists a positive constant $k>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-013806 | MATHDB-344071

Geck's conjecture on unipotent supports and generalized Gelfand-Graev representations

摘录 / EXCERPT: Geck's conjecture. For $d\in\Delta$, $d\in\Delta_{\mathbf k}$ if and only if either $\mathfrak g_d(1)=\{0\}$, or there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-013807 | MATHDB-344072

Geck's integral conjecture on special weighted Dynkin diagrams

摘录 / EXCERPT: Geck's integral conjecture. For $d\in\Delta$, $d\in\Delta_{\mathrm{spec}}$ if and only if either $\mathfrak g_{\mathbb Z,d}(1)=\{0\}$, or there exists a homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-013808 | MATHDB-344078

The projective-character submodule conjecture for the torus representation

摘录 / EXCERPT: Projective-character submodule conjecture. The subspace $\mathcal{P}$ is an $\mathcal{L}_{1,0}^{\mathrm{inv}}(\bar U_q)$-submodule of $\operatorname{SLF}(\bar U_q)$. The subspace i...

conjecture | open | MathDB 页面 | 原文线索

V2-013809 | MATHDB-344081

Gerbe duality conjecture for Gromov-Witten theory

摘录 / EXCERPT: Let G be a finite group, let Q act on the finite set of isomorphism classes of irreducible G-representations, and let $\mathcal{Y}$ be a gerbe with dual...

conjecture | open | MathDB 页面 | 原文线索

V2-013810 | MATHDB-344083

Quasi-isometric invariance of irreducibility and atoroidality

摘录 / EXCERPT: Quasi-isometric invariance of irreducibility and atoroidality. If $F\rtimes_\phi\mathbb Z$ and $F'\rtimes_\psi\mathbb Z$ are quasi-isometric, then ϕ is irreducible and atoroid...

conjecture | open | MathDB 页面 | 原文线索

V2-013811 | MATHDB-344094

Existence conjecture for connected higher Auslander Nakayama algebras

摘录 / EXCERPT: Let $n\geq 2$ and let k satisfy $2\leq k\leq 2n-2$. A connected Nakayama algebra is a connected finite-dimensional Nakayama algebra, and $\operatorname{gldim}(A)$ denotes its glo...

conjecture | open | MathDB 页面 | 原文线索

V2-013812 | MATHDB-344102

Finite-rank conjecture for profinite groups with finite-rank centralizers

摘录 / EXCERPT: Finite-rank conjecture. Then G has finite rank.

conjecture | open | MathDB 页面 | 原文线索

V2-013813 | MATHDB-344120

Broué's isotypy conjecture for blocks with abelian defect groups

摘录 / EXCERPT: Broué's isotypy conjecture. The block B is isotypic to its Brauer correspondent b_D in $\operatorname{N}_G(D)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013814 | MATHDB-344133

Minimality conjecture for weakly maximal subgroups of branch groups

摘录 / EXCERPT: Let G be a branch group and let $W\leq G$ be a weakly maximal subgroup. Write $\NR(W)$ for the non-rigidity tree of W, and let $\partial\NR(W)$ denote its boundary. Minimality...

conjecture | open | MathDB 页面 | 原文线索

V2-013815 | MATHDB-344142

Grigorchuk-Pansu conjecture in terms of Dirichlet-Poincaré profiles

摘录 / EXCERPT: Let G be a finitely generated group, and let $D\Lambda^p_G(n)$ denote its Dirichlet-Poincaré profile for $p\in[1,\infty)$. Grigorchuk-Pansu conjecture for Dirichlet-Poincaré prof...

conjecture | open | MathDB 页面 | 原文线索

V2-013816 | MATHDB-344143

The weakest dimension conjecture for free Jordan algebras

摘录 / EXCERPT: The weakest dimension conjecture. The sequence $(a_n)_{n\geq 1}$ is the unique solution of

conjecture | open | MathDB 页面 | 原文线索

V2-013817 | MATHDB-344144

Conjecture 1 for the character of free Jordan algebras

摘录 / EXCERPT: Conjecture 1. In ${\cal R}_{an}(GL(D))$,

conjecture | open | MathDB 页面 | 原文线索

V2-013818 | MATHDB-344145

The weak form of Conjecture 1 for free Jordan algebras

摘录 / EXCERPT: Weak form of Conjecture 1.

conjecture | open | MathDB 页面 | 原文线索

V2-013819 | MATHDB-344165

Mihalik's path-connectivity conjecture for right-angled Artin group boundaries

摘录 / EXCERPT: Mihalik's conjecture. The boundary ∂S_Γ is path connected if and only if Γ is a join.

conjecture | open | MathDB 页面 | 原文线索

V2-013820 | MATHDB-344166

Path-disconnected boundaries for iterated amalgams of CAT(0) groups

摘录 / EXCERPT: Generalized boundary conjecture. The group G acts on a CAT(0) space X such that ∂X is not path connected.

conjecture | open | MathDB 页面 | 原文线索

V2-013821 | MATHDB-344182

The $\forall\exists$-theory conjecture for virtually free groups

摘录 / EXCERPT: Let G and G' be virtually free groups. Their $\forall\exists$-theories are the theories consisting of first-order sentences with a universal quantifier block followed by an exi...

conjecture | open | MathDB 页面 | 原文线索

V2-013822 | MATHDB-344195

The Ding-Gao-Hjorth conjecture on essentially hyperfinite abelian-group actions

摘录 / EXCERPT: Ding-Gao-Hjorth conjecture. If $E\leq_B E^X_G$ and E is essentially countable, then E is essentially hyperfinite.

conjecture | open | MathDB 页面 | 原文线索

V2-013823 | MATHDB-344202

Constant-count conjecture for high-rank string C-group representations of symmetric groups

摘录 / EXCERPT: Constant-count conjecture. The number of string C-group representations of rank $n-i$ for S_n with $1\leq i\leq (n-3)/2$ is a constant independent of n.

conjecture | open | MathDB 页面 | 原文线索

V2-013824 | MATHDB-344213

The positive-theory conjecture for non-cyclic one-relator groups

摘录 / EXCERPT: Positive-theory conjecture. A non-cyclic one-relator group has trivial positive theory unless it is isomorphic to $BS(1,n)$ for some $n\in\mathbb Z$. This is the positive-theory an...

conjecture | open | MathDB 页面 | 原文线索

V2-013825 | MATHDB-344214

The positive theory of groups with independent WWPD actions

摘录 / EXCERPT: Let G be a group acting on a hyperbolic space. An action has independent WWPD when it satisfies the independent WWPD condition defined by Bestvina, Bromberg and Fujiwara; for act...

conjecture | open | MathDB 页面 | 原文线索

V2-013826 | MATHDB-344221

Wise's incoherence conjecture for free-by-free groups

摘录 / EXCERPT: Let $G=F_m\rtimes F_n$ be a semidirect product of free groups, where $m,n\geq 2$. A group is incoherent if it has a finitely generated subgroup that is not finitely presented. Wise...

conjecture | open | MathDB 页面 | 原文线索

V2-013827 | MATHDB-344267

The embedding conjecture for the higher rank Racah algebra and \(\mathfrak{sl}_n\)

摘录 / EXCERPT: The higher rank Racah algebra is the algebraic structure associated with the intermediate Casimir elements in an n-fold tensor product of $\mathfrak{su}(1,1)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013828 | MATHDB-344310

Character-projector conjecture for Hopf algebras with the Chevalley property

摘录 / EXCERPT: Character-projector conjecture. The elements p_i, for i in the set I of isomorphism classes of simple H-modules, are orthogonal idempotents and yield an isotypic decomposit...

conjecture | open | MathDB 页面 | 原文线索

V2-013829 | MATHDB-344404

Selberg Orthogonality Conjecture for cuspidal representations

摘录 / EXCERPT: Let $m,m'\geq 1$, let Π and Π' be unitary cuspidal representations of $\operatorname{GL}(m,\mathbb{A})$ and $\operatorname{GL}(m',\mathbb{A})$, respectively, and for each u...

conjecture | open | MathDB 页面 | 原文线索

V2-013830 | MATHDB-344438

Conjecture 6 on centralizer-like subgroups for the 1-Engel word

摘录 / EXCERPT: Let $G \times H$ be a group, let $(u,v) \in G \times H$, and let $(u,v)^{G \times H}$ denote the normal closure of (u,v) in $G \times H$. The associated subgroups...

conjecture | open | MathDB 页面 | 原文线索

V2-013831 | MATHDB-344446

Kropholler's conjecture on splittings over subgroups of codimension-1 subgroups

摘录 / EXCERPT: Let G) be a finitely generated group and let H be a subgroup. An H-almost invariant subset A is H-proper if it is not equivalent to either the empty set or G modulo fin...

conjecture | open | MathDB 页面 | 原文线索

V2-013832 | MATHDB-344447

Kropholler-Roller conjecture on splittings from codimension-1 subgroups

摘录 / EXCERPT: Let G be a finitely generated group and let H be a subgroup. An H-almost invariant subset A is H-proper if it is not equivalent to either the empty set or G modulo fini...

conjecture | open | MathDB 页面 | 原文线索

V2-013833 | MATHDB-344458

The highest abelian subgroup conjecture for Coxeter, Artin, and arrangement groups

摘录 / EXCERPT: Highest abelian subgroup conjecture. Any highest abelian subgroup in G is a Morse quasiflat.

conjecture | open | MathDB 页面 | 原文线索

V2-013834 | MATHDB-344507

The CG-group isoclinism conjecture

摘录 / EXCERPT: Let G and S be finite groups. Write $\operatorname{Cent}(G)$ for the set of centralizers of elements of G, and similarly for S. Two groups are isoclinic when there are isom...

conjecture | open | MathDB 页面 | 原文线索

V2-013835 | MATHDB-344529

Extension conjecture for partial Verma functors

摘录 / EXCERPT: Let chi be semisimple and lie in a Cartan subalgebra $\mathfrak{h}$, and let T be the corresponding torus. The T-action gives graded module categories...

conjecture | open | MathDB 页面 | 原文线索

V2-013836 | MATHDB-344543

The classification conjecture for pure-injective objects in homotopy categories of gentle algebras

摘录 / EXCERPT: Let cs be the homotopy category considered in the paper, and let $cSigma$-pure-injective objects be the objects classified by Theorem. Pure-injective classification conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-013837 | MATHDB-344553

Alternating-group conjecture for diamond-shaped tableau groups

摘录 / EXCERPT: Let G_λ be the group associated with a Young diagram λ, and let $\dim\lambda$ denote the dimension of the corresponding representation. Consider the diagrams...

conjecture | open | MathDB 页面 | 原文线索

V2-013838 | MATHDB-344554

Coxeter-group D_k conjecture for selected Young diagrams

摘录 / EXCERPT: Let G_λ be the group associated with a Young diagram λ, and let $\dim\lambda$ denote the dimension of the corresponding representation. For...

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V2-013839 | MATHDB-344555

Coxeter-group D_k conjecture for symmetric non-hook diagrams

摘录 / EXCERPT: Let λ be a symmetric Young diagram that is not a hook, let G_λ be the associated group, and let $\dim\lambda$ denote the dimension of the corresponding representati...

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V2-013840 | MATHDB-344600

Byott's solvability conjecture for finite groups with Hopf-Galois structures

摘录 / EXCERPT: Let Γ and Δ be finite groups of the same order. Write $e(\Gamma,\Delta)$ for the number of Hopf-Galois structures on a Galois extension with Galois group Γ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013841 | MATHDB-344619

The rank-two conjecture for homologically trivial finite group actions on 4-manifolds

摘录 / EXCERPT: Let M be a closed orientable 4-manifold with Euler characteristic $(M) \neq 0$, and let G be a finite group acting effectively, locally linearly, and homologically trivially...

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V2-013842 | MATHDB-344626

Stapledon's integrality conjecture for equivariant Ehrhart series

摘录 / EXCERPT: Let G be a finite group acting on a lattice polytope $P\subseteq\mathbb{R}^n$. Let M be the sublattice obtained by translating the affine span of P to the origin, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013843 | MATHDB-344627

Stapledon's positivity conjecture for coefficients of equivariant Ehrhart series

摘录 / EXCERPT: Let G be a finite group acting on a lattice polytope P, and let $H^*[z]=\sum_{j\geq 0}H^*_jz^j$ be its equivariant H^*-series. Let $h^*[z]$ be the ordinary h^*-polynomial o...

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V2-013844 | MATHDB-344629

Finite-seed conjecture for quantum unipotent cells of dominant minuscule Weyl group elements

摘录 / EXCERPT: Let W be the Weyl group, let $w\in W$ be dominant minuscule, and let $\mathcal{A}_q(\mathfrak{n}(w))$ be the associated cluster algebra of the quantum unipotent cell. The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-013845 | MATHDB-344633

Full faithfulness of the KZ functor on projective objects

摘录 / EXCERPT: Let cKZ be the functor from the relevant category of modules over the affine Hecke algebra to the corresponding category of modules over the algebra $cbbK$. A projective object m...

conjecture | open | MathDB 页面 | 原文线索

V2-013846 | MATHDB-344649

The bad-part density limit conjecture for cyclic and dihedral groups

摘录 / EXCERPT: Bad-part density limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013847 | MATHDB-344650

The bad-part density limit conjecture for non-abelian groups of order $3p$

摘录 / EXCERPT: Bad-part density limit conjecture for groups of order $3q_n$.

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V2-013848 | MATHDB-344652

The distinct-count conjecture for supercharacter theories of groups with a fixed number of conjugacy classes

摘录 / EXCERPT: Distinct-count conjecture for $\Gamma(k)$. The numbers $|Sup(G)|$ are distinct for all members G of $\Gamma(k)$ if and only if $k=6$. When $k=6$, the groups are

conjecture | open | MathDB 页面 | 原文线索

V2-013849 | MATHDB-344695

Diameter conjecture for McKay graphs of finite simple groups

摘录 / EXCERPT: Diameter conjecture. There is an absolute constant C such that for any finite non-abelian simple group G, and any nontrivial irreducible character α of G,

conjecture | open | MathDB 页面 | 原文线索

V2-013850 | MATHDB-344736

The proposed infinite-dimension criterion for equivariant dimensions

摘录 / EXCERPT: Let π be a discrete group with an action by a group Γ, let $\pi\rtimes\Gamma$ be the associated semidirect product, and let $\mathcal{F}$ and $\mathcal{G}$ denote the fa…

conjecture | open | MathDB 页面 | 原文线索

V2-013851 | MATHDB-344747

Orthosymplectic factorization-category equivalence

摘录 / EXCERPT: Let M and N be the parameters of the orthogonal setup, let $\bF$ be the Laurent series field, and let $U_{M,N}(\bF)$ be the unipotent group equipped with the character…

conjecture | open | MathDB 页面 | 原文线索

V2-013852 | MATHDB-344782

Invariant maximal-entropy measure conjecture for rational-map semigroups

摘录 / EXCERPT: Let S be a semigroup of rational maps. An invariant probability measure is a probability measure invariant under every element of S, and a non-injective element is a map in S…

conjecture | open | MathDB 页面 | 原文线索

V2-013853 | MATHDB-344783

Invariant maximal-entropy measure conjecture for free-subsemigroup-free rational-map semigroups

摘录 / EXCERPT: Let S be a finitely generated semigroup of rational maps, and say that a rank 2 free subsemigroup is a free subsemigroup generated by two elements. An invariant probability mea…

conjecture | open | MathDB 页面 | 原文线索

V2-013854 | MATHDB-344786

Quasi-isometric rigidity conjecture for purely real Heintze groups

摘录 / EXCERPT: A purely real Heintze group is a Heintze group $N\rtimes_\alpha\mathbb{R}$ whose defining derivation α has only real eigenvalues. Quasi-isometric rigidity conjecture. Two pu…

conjecture | open | MathDB 页面 | 原文线索

V2-013855 | MATHDB-344831

Farrokhi-Safa conjecture on relative commutativity degrees and central quotients

摘录 / EXCERPT: Farrokhi-Safa's conjecture. If $d(H,G)=d_k$, then $|H/Z(H,G)|$ is a product of at most k primes. Consequently, $|G/Z(G)|$ is a product of at most $|\mathcal{D}(G)|-1$ primes.

conjecture | open | MathDB 页面 | 原文线索

V2-013856 | MATHDB-344832

Farrokhi-Safa lower-bound conjecture for the number of commutativity degrees

摘录 / EXCERPT: Farrokhi-Safa's weaker conjecture. Every finite group G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-013857 | MATHDB-344834

Conjecture on the structure of nearly holomorphic automorphic forms on SL_2

摘录 / EXCERPT: The conjecture. (1) $\mathcal{N}(A \backslash G)_{(M,\tau)}\subset \mathcal{E}_0(M,\tau)$. (2) The action of $\mathcal{Z}$ on $\mathcal{N}(A \backslash G)$ is semisimple. (3) If…

conjecture | open | MathDB 页面 | 原文线索

V2-013858 | MATHDB-344835

The tight-pair global fixed-point conjecture

摘录 / EXCERPT: Let $(\operatorname{\mathcal{L}},G)$ be a tight pair, where G acts on the circle preserving the lamination system $\operatorname{\mathcal{L}}$. Tight-pair global fixed-point conj…

conjecture | open | MathDB 页面 | 原文线索

V2-013859 | MATHDB-344836

The pseudo-fibered triple promotion conjecture

摘录 / EXCERPT: Let $(G,\operatorname{\mathcal{L}}_1,\operatorname{\mathcal{L}}_2)$ be a pseudo-fibered triple. Suppose that G is finitely generated, torsion-free, and freely indecomposable. Pse…

conjecture | open | MathDB 页面 | 原文线索

V2-013860 | MATHDB-344837

The word-hyperbolicity conjecture for non-virtually-abelian pseudo-fibered groups

摘录 / EXCERPT: Let $(G,\operatorname{\mathcal{L}}_1,\operatorname{\mathcal{L}}_2)$ be a pseudo-fibered triple, with G finitely generated, torsion-free, and freely indecomposable as in the pseud…

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V2-013861 | MATHDB-344840

Friedl-Tillmann invariance conjecture for marked polytopes

摘录 / EXCERPT: Let G be a group admitting a nice $(2,1)$-presentation $f\pi=\langle x,y\mid r\rangle$, and let $\mathcal{M}_\pi\subset H_1(G;\mathbb{R})$ be its associated marked polytope. Frie…

conjecture | open | MathDB 页面 | 原文线索

V2-013862 | MATHDB-344851

Finite-center conjecture for AIP in CAT(0) groups

摘录 / EXCERPT: Finite-center conjecture. If G is a CAT(0) group and G and every finite-index subgroup of G have finite center, then G has AIP. This would clarify when the AIP hypothesis u...

conjecture | open | MathDB 页面 | 原文线索

V2-013863 | MATHDB-344862

The blockwise cyclic-defect conjecture

摘录 / EXCERPT: Let $p\in\{2,3\}$. Let G be a finite group and let B be a p-block of G with nontrivial defect group D. Let σ_1 be the Galois automorphism introduced in the paper,...

conjecture | open | MathDB 页面 | 原文线索

V2-013864 | MATHDB-344863

The 2-generated Sylow 2-subgroup conjecture

摘录 / EXCERPT: Let G be a finite group and let P be a Sylow 2-subgroup of G, with $4\mid |G|$. Let $B_0(G)$ be the principal 2-block, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013865 | MATHDB-344946

Sakellaridis-Venkatesh local factorization conjecture for the Shalika model

摘录 / EXCERPT: Let F_v be a local field, let σ_v be a tempered representation of the relevant group G_X, and let $\phi_{1,v},\phi_{2,v}$ be vectors in the associated representation. De...

conjecture | open | MathDB 页面 | 原文线索

V2-013866 | MATHDB-344991

Conder-Džambić-Jones conjecture on alternating quotients of amalgamated free products

摘录 / EXCERPT: Let A and B be finite groups, and let C be a subgroup of $A\cap B$ with index at least 2 in A and at least 3 in B. The Conder-Džambić-Jones conjecture. All but finite...

conjecture | open | MathDB 页面 | 原文线索

V2-013867 | MATHDB-344992

Conder's stronger conjecture on alternating quotients of amalgamated free products

摘录 / EXCERPT: Let A and B be finite groups, let C be a subgroup of $A\cap B$ of index at least 2 in A and at least 3 in B, and let K be the core of C in the amalgamated free pr...

conjecture | open | MathDB 页面 | 原文线索

V2-013868 | MATHDB-344993

Conjecture on half-arc-transitive coset graphs from the group H

摘录 / EXCERPT: Let $m\geqslant 7$ be an integer, let H be the group defined by the presentation and decomposition in the source, let $B=\langle a_1,\dots,a_{m-1}\rangle$, let...

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V2-013869 | MATHDB-345002

Monotonicity inequalities for Steinberg quotient coefficients

摘录 / EXCERPT: Let $\mathbb{X}^+$ be the dominant weights and let $lambda,mu,mu'\inmathbb{X}^+$. Write ... where $s(\mu)$ is the Weyl-orbit sum. Use the order relation $\uparrow$ on weights. Monoto...

conjecture | open | MathDB 页面 | 原文线索

V2-013870 | MATHDB-345003

Triangular decomposition conjecture for the classical limit of an quantum group

摘录 / EXCERPT: Let $\clUi_{\bbK_1}$ be the indicated algebra over $\bbK_1$, with subalgebras $\clUi_{-,\bbK_1}$, $\clUi_{0,\bbK_1}$, and $\clUi_{+,\bbK_1}$, and let $\frk$ be the reductive Lie al...

conjecture | open | MathDB 页面 | 原文线索

V2-013871 | MATHDB-345119

The Addition Theorem conjecture for locally finite groups

摘录 / EXCERPT: Locally finite-group Addition Theorem conjecture. If G is a locally finite group, then $AT(G)$ holds.

conjecture | open | MathDB 页面 | 原文线索

V2-013872 | MATHDB-345122

The positivity conjecture for orbit counts of pairs in finite abelian p-groups

摘录 / EXCERPT: Let $\underline{\lambda}=(\lambda_1^{\rho_1}>\lambda_2^{\rho_2}>\lambda_3^{\rho_3}>\ldots>\lambda_k^{\rho_k})\in\Lambda_0$ be a partition, and let $\mathcal R$ be a discrete valuat...

conjecture | open | MathDB 页面 | 原文线索

V2-013873 | MATHDB-345128

Generic simplicity conjecture for Takiff induced modules

摘录 / EXCERPT: Let $\mathfrak{T}$ be the Takiff Lie algebra under consideration, let $Q(0,\chi)$ denote the induced $\mathfrak{T}$-module associated with the trivial $\mathfrak{g}$-module and Tak...

conjecture | open | MathDB 页面 | 原文线索

V2-013874 | MATHDB-345129

Generic highest-weight realization conjecture for Takiff algebras

摘录 / EXCERPT: Let $\mathfrak{T}$ be a Takiff Lie algebra attached to a semisimple Lie algebra $\mathfrak{g}$. Let $\overline{Z(\mathfrak{g})}$ denote the Takiff part of the center used in the so...

conjecture | open | MathDB 页面 | 原文线索

V2-013875 | MATHDB-345146

Braid group automorphism conjecture for the universal b9c quantum group

摘录 / EXCERPT: Braid group automorphism conjecture. For every such i, there exist mutually inverse algebra automorphisms $\mathbf T_{i,e}'$ and $\mathbf T_{i,-e}”$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-013876 | MATHDB-345149

Asymptotic conjecture on bipartite Cayley index for groups

摘录 / EXCERPT: Asymptotic bipartite Cayley-index conjecture. There exists a positive integer n such that, if G is a group of order at least n and B is a subgroup of G having index 2,...

conjecture | open | MathDB 页面 | 原文线索

V2-013877 | MATHDB-345150

Bipartite DRR classification conjecture for finite groups

摘录 / EXCERPT: Let R be a finite group with a subgroup M of index 2. A bipartite DRR is a Cayley graph $\operatorname{Cay}(R,S)$, with $S\subseteq R\setminus M$, whose automorphism group is...

conjecture | open | MathDB 页面 | 原文线索

V2-013878 | MATHDB-345169

Finiteness-property characterization conjecture for twisted Brin-Thompson groups

摘录 / EXCERPT: Let G be an oligomorphic group of permutations of a set S, and let SV_G denote the associated twisted Brin-Thompson group. For $n\in\mathbb{N}$, write F_n fo...

conjecture | open | MathDB 页面 | 原文线索

V2-013879 | MATHDB-345210

Lévai-Pyber conjecture on cosets of bounded order in profinite groups

摘录 / EXCERPT: Let G be a profinite group, and let n be a positive integer. Suppose that the set of solutions of ... has positive Haar measure. Lévai-Pyber conjecture. Then G has an open subg...

conjecture | open | MathDB 页面 | 原文线索

V2-013880 | MATHDB-345215

Single-degree concentration conjecture for Drinfeld stratification cohomology

摘录 / EXCERPT: Let r divide n' and let $\theta \colon \mathbb T_h(\mathbb F_q) \cong \mathbb W_h^\times(\mathbb F_{q^n}) \to \overline{\mathbb Q}_\ell^\times$ be a character with trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-013881 | MATHDB-345240

Torsion-freeness conjecture for random cubical complex fundamental groups

摘录 / EXCERPT: Let Q be a random 2-dimensional cubical complex sampled as $Q\sim Q_2(n,p)$, where $p>1-(\tfrac 12)^{1/40}$. The fundamental group $\pi_1(Q)$ is the group associated with Q. To...

conjecture | open | MathDB 页面 | 原文线索

V2-013882 | MATHDB-345253

Derivative formula for the representations \(\rho_c(\tau)\)

摘录 / EXCERPT: Derivative conjecture. The representation $\rho_1(\tau)$ is $(rk,1)$. If F is non-archimedean, then for every $c>1$ the highest derivative of $\rho_c(\tau)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-013883 | MATHDB-345254

Intertwining-operator conjecture for \(\rho_c(\tau)\)

摘录 / EXCERPT: Intertwining-operator conjecture. For all $1<l\leq r$, $M(\zeta,w_{(k^{l})})$ is well defined at $\zeta=\zeta^{(l)}$ and has irreducible image. For all $c\geq1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-013884 | MATHDB-345275

Monotonicity conjecture for plethysm coefficients with a quadratic Schur function

摘录 / EXCERPT: Let ν, λ, and λ be partitions, with ν and λ arbitrary. Write $\nu\sqcup(1)$ for the partition obtained by adjoining a part 1 to ν, and similar...

conjecture | open | MathDB 页面 | 原文线索

V2-013885 | MATHDB-345289

The hyperplane fraction conjecture for the symmetric-group orbit

摘录 / EXCERPT: Let S_n act on $\mathbb{R}^n$ by permuting coordinates. For $n\geq 3$, let v range over vectors in $\mathbb{R}^n$ with distinct coordinates and not lying in the hyperplane ... Le...

conjecture | open | MathDB 页面 | 原文线索

V2-013886 | MATHDB-345321

Conjecture on normal right-angled Artin subgroups of mapping class groups

摘录 / EXCERPT: Normal RAAG conjecture. The subgroup N is isomorphic to one of the right-angled Artin subgroups of $\operatorname{Mod}(S)$ afforded by the authors' construction. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-013887 | MATHDB-345355

Single-orbit conjecture for wave-front sets of reductive-group representations

摘录 / EXCERPT: Let F be a non-Archimedean local field, let $\bf G$ be an algebraic reductive group over F, and let $G=\bf G(F)$. For $\pi\in\operatorname{Irr}(G)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013888 | MATHDB-345393

Topological Zimmer conjecture for low-dimensional compact manifolds

摘录 / EXCERPT: Topological Zimmer conjecture. Every such action factors through a finite group; equivalently, every homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-013889 | MATHDB-345394

Topological Zimmer conjecture for torsion-free finite-index subgroups on Euclidean spaces

摘录 / EXCERPT: Topological Zimmer conjecture for torsion-free subgroups. Every such action factors through a finite group.

conjecture | open | MathDB 页面 | 原文线索

V2-013890 | MATHDB-345395

Topological Zimmer conjecture for low-dimensional closed aspherical manifolds

摘录 / EXCERPT: Topological Zimmer conjecture for closed aspherical manifolds. Every such action factors through a finite group.

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V2-013891 | MATHDB-345400

Frenkel-Gaitsgory's affine Beilinson-Bernstein localization conjecture at the critical level

摘录 / EXCERPT: Frenkel-Gaitsgory's main conjecture. The induced functor

conjecture | open | MathDB 页面 | 原文线索

V2-013892 | MATHDB-345401

The left t-exactness conjecture for convolution at the critical level

摘录 / EXCERPT: Order-n convolution conjecture. The functor $\chi_{{\EuScript F}}$ is left t-exact up to shift.

conjecture | open | MathDB 页面 | 原文线索

V2-013893 | MATHDB-345488

Dovermann-Suh conjecture on inducing Smith exotic actions

摘录 / EXCERPT: Let G be an Oliver group and let N be a normal subgroup such that ... is a monomorphism. A finite group is an Oliver group if it satisfies the Oliver condition, and a Smith exoti...

conjecture | open | MathDB 页面 | 原文线索

V2-013894 | MATHDB-345489

Laitinen's conjecture on the primitivity invariant of finite groups

摘录 / EXCERPT: Let G be a finite group, and let $\operatorname{prim}(G)$ denote its primitivity invariant and $\operatorname{Sm}(G)$ its Smith invariant. Laitinen's conjecture. If ... then ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-013895 | MATHDB-345492

Ravenel's chromatic filtration conjecture for Burnside rings

摘录 / EXCERPT: Let G be a finite group and let $n\geqslant0$. The Ravenel ideal $\mathrm{J}_n(G)$ consists of virtual finite G-sets whose fixed-point sets have virtual cardinality zero for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-013896 | MATHDB-345493

The bivariant Ravenel filtration conjecture for the Burnside category

摘录 / EXCERPT: Let G and H be finite groups, and let $\mathrm{A}(G,H)$ be the abelian group of morphisms from G to H in the Burnside category. For $n\geqslant0$, let $\mathrm{J}_n(G,H)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-013897 | MATHDB-345556

Identity relating zonal-polynomial and orthogonal-group representation factors

摘录 / EXCERPT: Let n be a nonnegative integer, let γ be a partition of n, and let N be such that the quantities below are defined. For each partition λ of n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013898 | MATHDB-345584

Brick-sequence exact-sequence conjecture

摘录 / EXCERPT: Let W be a Weyl group, let $w\in W$, and let $B_1,\dots,B_l$ be a brick sequence of the torsion-free class $\mathcal{F}(w)$. Suppose that ... for $1\leq i<m<j\leq l$. Brick-sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-013899 | MATHDB-345586

Bounded-support conjecture for the restricted Weyl group action

摘录 / EXCERPT: Let $\mathfrak{g}$ be any simple real Lie algebra of rank r, and let $\lambda\in Q\cap\mathfrak{h}^+$ be a dominant weight, written in the basis of fundamental weights as ... Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-013900 | MATHDB-345599

The pseudo-bosonic realization conjecture for finite-dimensional Lie algebras

摘录 / EXCERPT: Let $\mathfrak{l}$ be a finite-dimensional Lie algebra. A Lie algebra is realizable by pseudo-bosonic operators if it admits generators represented by operators satisfying the requ...

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V2-013901 | MATHDB-345600

The Grigorchuk group projective spectrum conjecture

摘录 / EXCERPT: Let ${\mathcal G}$ be the Grigorchuk group, and let $p(R_\pi)\subset {\mathbb P}^4$ be the projective spectrum of the pencil ... where $z\in {\mathbb P}^4$. Let G be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-013902 | MATHDB-345605

The discreteness conjecture for spherical characters

摘录 / EXCERPT: Let ρ be an irreducible unitary representation of $\overline{\mathrm{GL}}$, and let $z(\rho)$ denote its spherical character. The preceding construction gives...

conjecture | open | MathDB 页面 | 原文线索

V2-013903 | MATHDB-345606

The discreteness-and-tail conjecture for unitary representations of $\overline{\mathrm{GL}}$

摘录 / EXCERPT: Fix k and let τ be an irreducible representation of $\mathrm{GL}(k,\mathbb F)$. Consider the triples (z,k,τ) that correspond to unitary representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-013904 | MATHDB-345614

The polynomial-exponent AND-weakness conjecture for finite solvable groups

摘录 / EXCERPT: Let G be a finite solvable group. An n-input G-program is a sequence of group elements selected according to the values of input bits, and it computes the n-input...

conjecture | open | MathDB 页面 | 原文线索

V2-013905 | MATHDB-345615

Quasipolynomial-time conjecture for equations over finite solvable groups

摘录 / EXCERPT: Let G be a finite solvable group. The problems $\textsc{EQN-SAT}(G)$ and $\textsc{EQN-ID}(G)$ ask, respectively, whether an equation over G has a satisfying assignment and whet...

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V2-013906 | MATHDB-345643

Conjectural operations on types for odd orthogonal groups

摘录 / EXCERPT: Odd orthogonal groups conjecture. The following operations have the stated effects on types:

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V2-013907 | MATHDB-345717

Conjecture on saturated fusion systems on groups of order p^6

摘录 / EXCERPT: Let $p\geqslant 5$, let S be a p-group of order p^6, and let $\mathcal{F}$ be a saturated fusion system on S such that $O_p(\mathcal{F})=1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013908 | MATHDB-345732

Hrushovski's nonexistence conjecture for groups containing a copy of \(\mathbb{Z}\times\mathbb{Z}\)

摘录 / EXCERPT: Let G be a group, and let $G-\mathrm{TCF}$ denote the theory of existentially closed G-fields, when such a theory exists. Hrushovski's nonexistence conjecture. If G has a sub...

conjecture | open | MathDB 页面 | 原文线索

V2-013909 | MATHDB-345793

Valette's conjecture on Coxeter quotients of automorphism groups

摘录 / EXCERPT: Let $\Gamma=(V,E)$ be a finite simplicial graph with edge-labeling $\varphi:E\rightarrow\mathbb{N}_{\geq 2}$, and let ... be the associated Coxeter group. Write...

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V2-013910 | MATHDB-345794

Konvalinka-Tewari conjecture on the parking space extension

摘录 / EXCERPT: Konvalinka-Tewari conjecture. The representation $\tau_{n,1}$ is isomorphic to σ_n. Furthermore, $\mathrm{Frob}(\tau_{n,1})$ is h-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-013911 | MATHDB-345795

Uniqueness of finite simple groups by the number of 2-element centralizers

摘录 / EXCERPT: Let G and H be finite simple groups. For a finite group K, let $2\text{-}\operatorname{Cent}(K)$ denote the set of centralizers of 2-element subsets of K, and let...

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V2-013912 | MATHDB-345804

The transfer conjecture for tensor-product invariants of special linear and spin groups

摘录 / EXCERPT: The transfer conjecture. If at least two of a_n,b_n,c_n are nonzero, then

conjecture | open | MathDB 页面 | 原文线索

V2-013913 | MATHDB-345806

The transfer conjecture for tensor-product invariants of symplectic and special linear groups

摘录 / EXCERPT: The symplectic transfer conjecture. If at least two of a_1,b_1,c_1 are nonzero and the central character of $\pi_1\otimes\pi_2\otimes\pi_3$ is trivial, then

conjecture | open | MathDB 页面 | 原文线索

V2-013914 | MATHDB-345815

Unique product conjecture for uniform pro-p groups

摘录 / EXCERPT: Unique product conjecture for uniform pro-p groups. If G is a uniform pro-p group, then G is a unique product group.

conjecture | open | MathDB 页面 | 原文线索

V2-013915 | MATHDB-345818

Stallings' equaliser conjecture for free groups

摘录 / EXCERPT: Stallings' equaliser conjecture. If g and h are homomorphisms with h injective, then

conjecture | open | MathDB 页面 | 原文线索

V2-013916 | MATHDB-345819

Inert equaliser conjecture for free-group homomorphisms

摘录 / EXCERPT: Inert equaliser conjecture. If S contains at least one injective homomorphism, then $\operatorname{Eq}(S)$ is inert in $F(\Sigma)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013917 | MATHDB-345831

Parameter-independence and associativity conjecture for fusion products

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple complex Lie algebra, let P^+ be its dominant integral weights, and let $a_1,\ldots,a_\mathbb C$ be pairwise distinct with...

conjecture | open | MathDB 页面 | 原文线索

V2-013918 | MATHDB-345832

Feigin's two-factor defining-relations conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple complex Lie algebra, let P^+ denote its dominant integral weights, and fix $\lambda_1,\lambda_2\in P^+$. Put $\lambda=\lambda_1+\lambda_2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-013919 | MATHDB-345833

Schur-positivity conjecture for the decomposition poset

摘录 / EXCERPT: Let $\lambda\in P^+$, let ... and order this set by $(\lambda_1,\lambda_2)\succeq(\mu_1,\mu_2)$ exactly when $(\lambda_1-\lambda_2)(h_\alpha)\leq(\mu_1-\mu_2)(h_\alpha)$ for every posit...

conjecture | open | MathDB 页面 | 原文线索

V2-013920 | MATHDB-345835

Four-head conjecture for finitely generated infinite groups

摘录 / EXCERPT: Four-head conjecture. If $G \geq \mathbb{Z}$ and G has decidable word problem, then $h(G)=3$; if $G \geq \mathbb{Z}$ and G has undecidable word problem, then $h(G)=4$; and if...

conjecture | open | MathDB 页面 | 原文线索

V2-013921 | MATHDB-345851

Subgroup closure conjecture for BCI-groups

摘录 / EXCERPT: Let G be a finite BCI-group, meaning that every bi-Cayley graph over G is a BCI-graph, and let $H\leq G$ be a subgroup. Subgroup closure conjecture. Then H is a BCI-group. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-013922 | MATHDB-345867

Weakly fair A_q(lambda) conjecture for the Dirac series of U(p,q)

摘录 / EXCERPT: Dirac-series conjecture. Any Dirac series member of $U(p,q)$ must be a weakly fair $A_{\mathfrak{q}}(\lambda)$ module satisfying $$.

conjecture | open | MathDB 页面 | 原文线索

V2-013923 | MATHDB-346000

The induced-subgraph degree conjecture for Cayley graphs

摘录 / EXCERPT: Let G be a finite group and let S be a symmetric set of non-identity elements generating G. Let $X=(G,S)$ be the corresponding undirected Cayley graph, and for $U\subseteq G$...

conjecture | open | MathDB 页面 | 原文线索

V2-013924 | MATHDB-346027

Conjecture on Green functions for Levi subgroups of type A

摘录 / EXCERPT: Let $\mathbf G$ be a group of type A, let $\mathbf L\leq\mathbf G$ be an F-stable Levi subgroup, and let $v\in\mathbf L_{\operatorname{uni}}^F$. Let u be the induced class of...

conjecture | open | MathDB 页面 | 原文线索

V2-013925 | MATHDB-346030

Gill-Pyber-Szabó product conjecture for normal subsets of finite simple groups

摘录 / EXCERPT: Let G be a non-abelian finite simple group, and let $S_1,\ldots,S_k$ be normal subsets of G, meaning unions of conjugacy classes. Gill-Pyber-Szabó's product conjecture. There e...

conjecture | open | MathDB 页面 | 原文线索

V2-013926 | MATHDB-346053

Spherical matrix-coefficient lower-bound conjecture

摘录 / EXCERPT: Let G be the ambient group, K a maximal compact subgroup, and $\Pi(G)_{\operatorname{sph}}$ the set of unitary irreducible spherical representations of G. For $g\in G$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-013927 | MATHDB-346079

Baumslag's conjectures for one-relator groups with torsion

摘录 / EXCERPT: Baumslag's conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-013928 | MATHDB-346114

The conjecture that character asymptotic exponents are invariants of reduced spaces

摘录 / EXCERPT: Let M_x be the reduced space parametrized by x, and suppose that the Fourier transform has a large-k asymptotic expansion ... Here $u(x)$ and $v(x)$ are the functions appearing...

conjecture | open | MathDB 页面 | 原文线索

V2-013929 | MATHDB-346123

Central-quotient dichotomy conjecture for irreducible Artin-Tits groups

摘录 / EXCERPT: Let A_Γ be an irreducible Artin-Tits group, meaning that its defining graph Γ does not decompose as a join of two non-empty subgraphs such that all edges between them...

conjecture | open | MathDB 页面 | 原文线索

V2-013930 | MATHDB-346124

Procongruence anabelian conjecture for congruence level structures

摘录 / EXCERPT: Let $\Bbbk$ be a sub-p-adic field with absolute Galois group $G_\Bbbk$. Let ${\mathcal M}^{\lambda}_\Bbbk$ and ${\mathcal M}^{\mu}_\Bbbk$ be moduli stacks of curves with congruen...

conjecture | open | MathDB 页面 | 原文线索

V2-013931 | MATHDB-346128

Hughes' conjecture on subgroups generated by non-\p elements

摘录 / EXCERPT: Hughes' conjecture. If $H_p(G)\neq 1$, then either $H_p(G)=G$ or

conjecture | open | MathDB 页面 | 原文线索

V2-013932 | MATHDB-346131

The higher double-dual section conjecture

摘录 / EXCERPT: Higher double-dual section conjecture. These maps satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-013933 | MATHDB-346158

Shumyatsky's conjecture on finite-rank profinite groups with bounded centralizers

摘录 / EXCERPT: Shumyatsky's conjecture. If $C_G(g)$ has finite rank for every nontrivial $g\in G$, and G is not a pro-p group, then G has finite rank.

conjecture | open | MathDB 页面 | 原文线索

V2-013934 | MATHDB-346222

The Kazhdan-Lusztig character conjecture at negative level

摘录 / EXCERPT: Let $\mathfrak g$ be a symmetrizable complex Kac-Moody algebra with Weyl group $\mathcal W$ and simple reflections $\mathcal S$. Fix a Weyl vector ρ, let...

conjecture | solved | MathDB 页面 | 原文线索

V2-013935 | MATHDB-346279

Leclerc's conjecture on products of real dual canonical basis elements

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a Kac--Moody algebra with symmetrizable Cartan datum, let $\mathsf{U_q}^{-}$ be the negative part of its quantized enveloping algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013936 | MATHDB-346282

Multiplicative reachability conjecture for common triangular bases

摘录 / EXCERPT: Let ${{\mathbf L}}$ be a common triangular basis of a quantum cluster algebra. A basis element $b\in{{\mathbf L}}$ is real if $b^{2}\in{{\mathbf L}}$. Multiplicative reachability c...

conjecture | open | MathDB 页面 | 原文线索

V2-013937 | MATHDB-346326

Lusztig's character conjecture for reductive algebraic groups

摘录 / EXCERPT: Lusztig's conjecture. Suppose $p\geq h$ and $x\bullet_p0\in\mathfrak{X}_+$, where $x\in W$ satisfies Jantzen's condition

conjecture | open | MathDB 页面 | 原文线索

V2-013938 | MATHDB-346340

Diagonal Supersymmetry conjecture for boson-fermion diagonal modules

摘录 / EXCERPT: Let n be a positive integer, let k and j index the commuting and anticommuting variable sets, and let $\mathcal{E}_n(\bm{q};\bm{z})$ be the generic Frobenius characteristic o...

conjecture | open | MathDB 页面 | 原文线索

V2-013939 | MATHDB-346349

The commutator-intersection conjecture for RCC loop folders

摘录 / EXCERPT: Commutator-intersection conjecture. If H is abelian and (G,H,T) is an RCC loop folder, then

conjecture | open | MathDB 页面 | 原文线索

V2-013940 | MATHDB-346353

Finiteness conjecture for semibricks and brick-finite algebras

摘录 / EXCERPT: Semibrick finiteness conjecture. If every semibrick in $\operatorname{mod}\Lambda$ is a finite set, then $\operatorname{mod}\Lambda$ is brick-finite, that is, Λ is τ-t...

conjecture | open | MathDB 页面 | 原文线索

V2-013941 | MATHDB-346367

Higher Specht basis conjecture for the generalized coinvariant ring $R_μ$

摘录 / EXCERPT: Higher Specht basis conjecture. The set of polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-013942 | MATHDB-346411

Hermiticity of Sugawara operators in the multipoint case

摘录 / EXCERPT: Hermiticity conjecture. The Sugawara operators $T(v)$ defined in Section CFT are Hermitian with respect to $\langle\cdot,\cdot\rangle_L$.

conjecture | open | MathDB 页面 | 原文线索

V2-013943 | MATHDB-346422

Generalized Amit conjecture for fibres of word maps

摘录 / EXCERPT: Let G be a finite nilpotent group, let w be a word in k variables, and define $N_w(g)$ to be the number of tuples $(g_1,\ldots,g_k)\in G^{(k)}$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-013944 | MATHDB-346428

A logarithmic bound for the irredundant base size of primitive groups

摘录 / EXCERPT: Logarithmic irredundant-base conjecture. There exists a constant $C>0$ such that, if G is not a primitive large-base group, then

conjecture | open | MathDB 页面 | 原文线索

V2-013945 | MATHDB-346448

Wavelet completeness conjecture for locally compact groups

摘录 / EXCERPT: Wavelet completeness conjecture. Characterize the locally compact groups G satisfying

open_problem | open | MathDB 页面 | 原文线索

V2-013946 | MATHDB-346473

The quasi-isomorphism conjecture for dg-cyclic modules

摘录 / EXCERPT: Let $G_{\rho}^{{\underline{N}}}$ and $Y_{\rho}^{{\underline{N}}}$ be the dg-enhanced and ordinary cyclic modules constructed from the tensor-product data, with differential d_N o...

conjecture | open | MathDB 页面 | 原文线索

V2-013947 | MATHDB-346496

Type A cluster-variable conjecture for strongly homogeneous modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type A_n, let I be the indexing set for the simple roots, and let $\mathcal{M}in_{0}^{+}$ denote the strict dominant minuscule elements of the Weyl gro...

conjecture | open | MathDB 页面 | 原文线索

V2-013948 | MATHDB-346497

Flag-minor conjecture for strict dominant minuscule modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra of finite simply-laced type, let W be its Weyl group, let N be the associated maximal unipotent subgroup, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-013949 | MATHDB-346513

Babai-Godsil conjecture on almost all Cayley graphs being GRRs

摘录 / EXCERPT: Babai-Godsil conjecture. If R is not generalised dicyclic or abelian of exponent greater than 2, then for almost all inverse-closed subsets S of R, $\mathop{\Gamma}(R,S)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-013950 | MATHDB-346531

Uniqueness conjecture for rank-one simple transitive 2-representations

摘录 / EXCERPT: Let D be the algebra of dual numbers, let $\mathscr{D}$ be the associated 2-category of projective bimodules, and for each positive integer k let $\mathcal{J}_k$ denote the two...

conjecture | open | MathDB 页面 | 原文线索

V2-013951 | MATHDB-346535

First Tachikawa conjecture

摘录 / EXCERPT: First Tachikawa conjecture. The algebra A is selfinjective if and only if this vanishing condition holds.

conjecture | open | MathDB 页面 | 原文线索

V2-013952 | MATHDB-346610

Exponent-square conjecture for the Schur multiplier

摘录 / EXCERPT: Let G be a finite group, and let $H_2(G,\mathbb{Z})$ denote its Schur multiplier. Exponent-square conjecture. ... This conjecture was proposed as a proposed replacement for Schur's...

conjecture | open | MathDB 页面 | 原文线索

V2-013953 | MATHDB-346652

The invariant centralizer conjecture for parabolic Schur algebras

摘录 / EXCERPT: Let $D(n,r)^V$ denote the subspace of the parabolic Schur algebra consisting of elements fixed by the relevant V-action, and let $\bbC\Psi(\mathfrak{S}_r)$ be the complex span of...

conjecture | open | MathDB 页面 | 原文线索

V2-013954 | MATHDB-346714

Lee-Lee conjecture on non-self-crossing admissible curves and real Schur roots

摘录 / EXCERPT: Lee-Lee conjecture. The set of roots associated to non-self-crossing admissible curves and the set of real Schur roots coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-013955 | MATHDB-346759

The non-tempered Gan-Gross-Prasad conjecture for Arthur type representations of general linear groups

摘录 / EXCERPT: Let F be a local field. For an Arthur type representation, let its associated Arthur parameter be a finite representation of $WD_F\times \mathrm{SL}_2(\mathbb C)$; two such param...

conjecture | open | MathDB 页面 | 原文线索

V2-013956 | MATHDB-346760

The Ext-branching conjecture for Arthur type representations of general linear groups

摘录 / EXCERPT: Let $G_n=\mathrm{GL}_n(F)$, and let π_M and π_N be Arthur type representations of G_{n+1} and G_n, respectively. For $k\geq 1$, write $\pi_M^{[k]}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013957 | MATHDB-346761

Andrews-Gordon type triple-sum conjecture for level 2 modules of $A^{(2)}_{13}$

摘录 / EXCERPT: The triple-sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-013958 | MATHDB-346776

The modified local Langlands family conjecture

摘录 / EXCERPT: Let $G={\rm GL}_n(F)$, let $X_{\check G}$ be the space of L-parameters (φ,N) for the principal block, and let $\widetilde{\mathcal{V}}_G$ be the family of smooth G-repres...

conjecture | open | MathDB 页面 | 原文线索

V2-013959 | MATHDB-346777

The maximal Cohen-Macaulay conjecture for the universal Hecke module

摘录 / EXCERPT: Let $X_{\check G}$ be the space of principal-block L-parameters, let $A_{\check G}$ be its coordinate ring, and let $\widetilde{\mathcal{M}}_G$ be the sheaf of modules over...

conjecture | open | MathDB 页面 | 原文线索

V2-013960 | MATHDB-346801

Folklore conjecture on the coincidence of smooth and algebraic theta correspondence

摘录 / EXCERPT: Folklore conjecture on theta correspondence. The inclusions

conjecture | open | MathDB 页面 | 原文线索

V2-013961 | MATHDB-346808

The HCIZ and BGW strong-coupling asymptotics conjecture

摘录 / EXCERPT: The HCIZ and BGW strong-coupling asymptotics conjecture. Given any $M\geq0$, there exists $\varepsilon_M>0$ such that, for every integer $k\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-013962 | MATHDB-346817

Horinaga's conjecture on nearly holomorphic automorphic forms

摘录 / EXCERPT: Horinaga's conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-013963 | MATHDB-346873

Huang's conjecture on parabolic induction from representations with nonzero Dirac cohomology

摘录 / EXCERPT: Let G be a complex classical Lie group, and let a unitary representation of G be given. A unitary representation has nonzero Dirac cohomology if its Dirac cohomology does not v...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-013964 | MATHDB-346903

The rank inequalities for symmetric-group actions on sl3 webs

摘录 / EXCERPT: Rank-inequality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-013965 | MATHDB-346911

The homomorphism conjecture for $d52f$-submaximal subgroups

摘录 / EXCERPT: Let $d52f$ be a class of finite groups, let A be an $d52f$-separable normal subgroup of a finite group G, and let $G\rightarrow G/A$ be the canonical epimorphism. The homomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-013966 | MATHDB-346912

The normalizer conjecture for $d52f$-submaximal subgroups

摘录 / EXCERPT: Let G be a finite group, let $A\trianglelefteq G$, and let H be an $d52f$-submaximal subgroup of G. The normalizer conjecture. Then H is $d52f$-submaximal in the normalizer...

conjecture | open | MathDB 页面 | 原文线索

V2-013967 | MATHDB-346916

The finitely generated groups twin-width conjecture

摘录 / EXCERPT: Finitely generated groups conjecture. For every group Γ generated by a finite set S, the class $F(\Gamma,S)$ has bounded twin-width.

conjecture | open | MathDB 页面 | 原文线索

V2-013968 | MATHDB-346936

Extension of the main theorems to exceptional Lie types

摘录 / EXCERPT: Let ${\mathfrak g}$ be a complex semisimple Lie algebra, and let the statements of Theorems 1 and 2 of the paper be understood in their previously established generality. Exception...

conjecture | open | MathDB 页面 | 原文线索

V2-013969 | MATHDB-346951

Induction isomorphism and vanishing conjecture for quantized flag manifolds

摘录 / EXCERPT: Induction isomorphism and vanishing conjecture. The map F is an isomorphism, and

conjecture | open | MathDB 页面 | 原文线索

V2-013970 | MATHDB-346952

Specialized induction isomorphism and vanishing conjecture

摘录 / EXCERPT: Let $t\in H(R)$, and define ... ... Let F_t be the map obtained from F by specialization and Frobenius reciprocity. Specialized induction isomorphism and vanishing conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-013971 | MATHDB-346977

Pointed representable fibrations and presheaves conjecture

摘录 / EXCERPT: Let $R(*)$ be the terminal object of $\mathrm{RepOrbSpc}$, and let $\mathrm{Rep}\{\mathbf B G\}$ be the full subcategory spanned by the classifying orbispaces of finite groups. Poi...

conjecture | open | MathDB 页面 | 原文线索

V2-013972 | MATHDB-346981

Compatibility with global classifying spaces of finite groups

摘录 / EXCERPT: Let G be a finite group. Write $\mathbf B G\in\mathrm{OrbSpc}$ for its classifying orbispace and $\mathbf B G\in\mathrm{GloSpc}$ for the corresponding global classifying space. C...

conjecture | open | MathDB 页面 | 原文线索

V2-013973 | MATHDB-346991

Novak-Rhoades equivariant log-concavity conjecture

摘录 / EXCERPT: Novak-Rhoades' conjecture. For any $n\geq3$, there exists an ${\mathfrak{S}}_n$-equivariant injection

conjecture | open | MathDB 页面 | 原文线索

V2-013974 | MATHDB-347004

Commutation of the nonabelian Fourier transforms on elliptic unipotent representations

摘录 / EXCERPT: Let G be the reductive p-adic group and let $\mathsf{R}_{\mathsf{un},\mathsf{ell}}$ denote its elliptic unipotent representation space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013975 | MATHDB-347013

Donovan's conjecture for blocks with fixed defect group

摘录 / EXCERPT: Donovan's conjecture. Amongst all finite groups G and blocks B of kG with defect group isomorphic to D, there are only finitely many Morita equivalence classes.

conjecture | open | MathDB 页面 | 原文线索

V2-013976 | MATHDB-347014

Bounded Morita Frobenius numbers for fixed defect groups

摘录 / EXCERPT: Bounded Morita Frobenius number conjecture. Amongst all finite groups G and blocks B of kG with defect group isomorphic to D, the Morita Frobenius number...

conjecture | open | MathDB 页面 | 原文线索

V2-013977 | MATHDB-347015

Weak Donovan conjecture on bounded Cartan invariants

摘录 / EXCERPT: Weak Donovan conjecture. There exists a constant $c(D)\in\mathbb N$ such that, if G is a finite group and B is a block of kG with defect group isomorphic to D, then the ent...

conjecture | open | MathDB 页面 | 原文线索

V2-013978 | MATHDB-347042

The isomorphism conjecture for deformed preprojective algebras of generalized Dynkin type

摘录 / EXCERPT: Let K be a field of characteristic different from 2, and let Δ be a generalized Dynkin type. Consider a deformed preprojective K-algebra of type Δ and the prepr...

conjecture | open | MathDB 页面 | 原文线索

V2-013979 | MATHDB-347045

Bader-Muchnik conjecture on L^2-irreducibility of Poisson boundaries

摘录 / EXCERPT: Let (Γ,τ) be a measured group, and let its Poisson boundary be (B,ν). For $p\in[1,\infty)$, the associated representation is ... for $\gamma\in\Gamma$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013980 | MATHDB-347089

Kåhrström's multiplicity-one conjecture for Duflo translates

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple complex Lie algebra with Weyl group W, let $d,y\in W$ with d a Duflo element, and let θ_d be the corresponding indecomposable projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-013981 | MATHDB-347101

Gheri's conjecture on the p-Frobenius ratio

摘录 / EXCERPT: Gheri's conjecture. The p-Frobenius ratio satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-013982 | MATHDB-347149

The unified denominator formula for untwisted quantum affine algebras

摘录 / EXCERPT: Let I be the index set of simple roots, let d_i be the symmetrizing integers, let r and $h^{\vee}$ be the parameters used in the denominator formula, and set $q_s=q^{1/r}$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-013983 | MATHDB-347157

Injectivity conjecture for the proposed BPS Lie algebra generators

摘录 / EXCERPT: Let J and $L^{\mathcal{S}}$ be the morphisms constructed from the proposed generators of the BPS Lie algebra associated with the preprojective CoHA, where $\mathcal{S}$ is a Serr...

conjecture | open | MathDB 页面 | 原文线索

V2-013984 | MATHDB-347162

P. Moravec's squared-exponent conjecture for the Schur multiplier

摘录 / EXCERPT: Let G be a finite group. Write $\operatorname{exp}(G)$ for the exponent of G, and let $H_2(G,\mathbb{Z})$ be its second homology group with integer coefficients. Moravec's squa...

conjecture | open | MathDB 页面 | 原文线索

V2-013985 | MATHDB-347193

The fusion-number characterization of nilpotent blocks

摘录 / EXCERPT: Fusion-number conjecture. For every block B of G,

conjecture | open | MathDB 页面 | 原文线索

V2-013986 | MATHDB-347251

The Shintani twisting conjecture for geometric Lusztig series

摘录 / EXCERPT: Let ${\mathbf G}$ be a connected reductive group with Frobenius root F. For an F^*-stable semisimple element s of the dual group, let ${\mathcal E}({\mathbf G}^F,(s))$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-013987 | MATHDB-347252

Compatibility of Jordan decomposition with Shintani twisting

摘录 / EXCERPT: Let ${\mathbf G}$ be a connected reductive group with Frobenius root F, let $s\in{\mathbf G}^{*F^*}$ be semisimple, and let $C_{{\mathbf G}^*}(s)$ satisfy condition $(*)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-013988 | MATHDB-347271

The upward-closure conjecture for non-multiplicity-free characters

摘录 / EXCERPT: Fix $D=\{d_1<d_2<\cdots<d_k\}$ and suppose $\alpha,\alpha^{\uparrow}\in{\sf Comp}_n(D)$, where ... for some j such that $\alpha_j+1\ne\alpha_i$ for every $i\ne j$. Say that a chara...

conjecture | open | MathDB 页面 | 原文线索

V2-013989 | MATHDB-347292

Linear product-of-characters conjecture for classical and alternating groups

摘录 / EXCERPT: For characters $\chi_1,\ldots,\chi_l$ of G, write ... when every irreducible character of G is a constituent of their product. Linear product-of-characters conjecture. There is a...

conjecture | open | MathDB 页面 | 原文线索

V2-013990 | MATHDB-347310

Commutativity conjectures for parabolic affine category O over a deformation ring

摘录 / EXCERPT: Let R be the deformation ring, let $\overline R$ be the corresponding deformation ring after replacing e by $e+1$, and let $O^{\nu,\Delta}_{\mu,R}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-013991 | MATHDB-347315

The parabolic freeness conjecture for complex reflection groups

摘录 / EXCERPT: Let W be a complex reflection group and let W_I be a parabolic subgroup of W. Let $d4d5(W)$ be the generic Hecke algebra of W, and let $d4d5_I(W)$ be the parabolic Hecke su...

conjecture | open | MathDB 页面 | 原文线索

V2-013992 | MATHDB-347318

Malle-Michel's lifting conjecture for Hecke algebra bases

摘录 / EXCERPT: Let W be a complex reflection group, let $B(W)$ be its braid group, let $\mathcal{H}(W)$ be its generic Hecke algebra, and let τ be a canonical symmetrising trace on...

conjecture | open | MathDB 页面 | 原文线索

V2-013993 | MATHDB-347332

Meiri-Puder's alternating-or-symmetric monodromy conjecture

摘录 / EXCERPT: Meiri-Puder's conjecture. For $\ell\geq 5$, Q_ℓ is the full alternating or symmetric group on $Y^*(\ell)$.

conjecture | open | MathDB 页面 | 原文线索

V2-013994 | MATHDB-347335

Automorphism conjecture for supercharacter theories with a non-coarsest restriction

摘录 / EXCERPT: Automorphism conjecture. If $\mathsf{S}_H\neq\mathsf{M}(H)$, then every subgroup of G is S-normal. In particular, S comes from automorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-013995 | MATHDB-347336

Automorphism refinement conjecture for supercharacter theories of $C_p\times C_p$

摘录 / EXCERPT: Automorphism refinement conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-013996 | MATHDB-347338

Finiteness conjecture for the V-like Lodha-Moore group

摘录 / EXCERPT: Finiteness conjecture. The group $V(G)$ and its commutator subgroup $[V(G),V(G)]$ are of type F_∞.

conjecture | open | MathDB 页面 | 原文线索

V2-013997 | MATHDB-347347

Parity of parabolic induction from 0-cuspidal pairs

摘录 / EXCERPT: Parity induction conjecture. The complex

conjecture | open | MathDB 页面 | 原文线索

V2-013998 | MATHDB-347376

Virtual torsion-freeness and hyperbolic embeddability of the groups H_W

摘录 / EXCERPT: Let H be the finitely generated subgroup of a hyperbolic group constructed in the paper, with a presentation whose relators lie in a set V. For a subset $W\subset V$, let H_W...

conjecture | open | MathDB 页面 | 原文线索

V2-013999 | MATHDB-347382

The character decomposition conjecture for higher-level q-oscillator modules of type B^{(1)}(0,n)

摘录 / EXCERPT: For $\lambda\in \mathscr{P}_+$ with $\ell(\lambda)\leq \min\{n,s/2\}$, define ... Let $V(\Lambda^{(s)}_\lambda)$ be the irreducible highest weight $U(osp_{1|2n})$-module with highest...

conjecture | open | MathDB 页面 | 原文线索

V2-014000 | MATHDB-347413

NPE characterization conjecture for finite simple groups

摘录 / EXCERPT: Let G be a group and S be a finite simple group. Let $ pe(H)$ denote the invariant used in the source, namely the combined data of the relevant element-order counts. NPE charac...

conjecture | open | MathDB 页面 | 原文线索

V2-014001 | MATHDB-347423

Ergodic-theoretic characterization conjecture for discrete Heisenberg groups

摘录 / EXCERPT: Let G be a discrete Heisenberg group, and let Theorem3.1's ergodic-theoretic characterization be denoted by Theorem; the associated results are Theorems,, and. Discrete Heisenber...

conjecture | open | MathDB 页面 | 原文线索

V2-014002 | MATHDB-347437

The ellipticity criterion for discrete series on real spherical spaces

摘录 / EXCERPT: Let ${\underline{G}}$ be a connected reductive group defined over $\mathbb{R}$, let $G={\underline{G}}(\mathbb{R})$, and let ${\underline{H}}\subset{\underline{G}}$ be an algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-014003 | MATHDB-347452

Artin groups' algebraic and $K(\pi,1)$ conjectures

摘录 / EXCERPT: Artin groups' algebraic and $K(\pi,1)$ conjectures. For every Artin group A_{Γ}: (1) A_{Γ} has solvable word and conjugacy problems; (2) A_{Γ} is torsion-free;...

conjecture | open | MathDB 页面 | 原文线索

V2-014004 | MATHDB-347456

Modulo-4 asymptotics for commutators of spectral projections of spin operators

摘录 / EXCERPT: Let J_x,J_y,J_z be the generators of an irreducible, unitary, n-dimensional representation of $SU(2)$, and set ... Write $o(1)$ for a quantity tending to zero as $n\to\infty$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-014005 | MATHDB-347484

The quasi-isometric rigidity conjecture for simply connected nilpotent Lie groups

摘录 / EXCERPT: Let G and H be simply connected nilpotent Lie groups. Quasi-isometric rigidity conjecture. The groups G and H are quasi-isometric if and only if they are isomorphic. This w...

conjecture | open | MathDB 页面 | 原文线索

V2-014006 | MATHDB-347485

The central-product Dehn function conjecture

摘录 / EXCERPT: Let $\mathfrak{k}$ and $\mathfrak{l}$ be nilpotent Lie algebras of step k and ℓ, respectively, with $k\geqslant \ell\geqslant 2$, and with one-dimensional centers...

conjecture | open | MathDB 页面 | 原文线索

V2-014007 | MATHDB-347499

The Hilbertianity conjecture for projective absolute Galois groups

摘录 / EXCERPT: Let $K\subset\overline{\mathbb Q}$ and suppose that the absolute Galois group G_K is a projective profinite group. The group G_K is profree if it is a free profinite group. Hil...

conjecture | open | MathDB 页面 | 原文线索

V2-014008 | MATHDB-347507

Embedding admissible representations into representations injective on an Iwahori subgroup

摘录 / EXCERPT: Embedding conjecture. For every $\pi\in\mathfrak{Rep}^{\textnormal{adm}}_{z_i=c_i}(G)$, there exists $A\in\mathfrak{Rep}^{\textnormal{adm}}_{z_i=c_i}(G)$ and a G-equivariant inje...

conjecture | open | MathDB 页面 | 原文线索

V2-014009 | MATHDB-347512

Finite word-width conjecture for irreducible lattices

摘录 / EXCERPT: Let S be a finite set of places, let G be a semisimple group over the corresponding local factors, let ... be an irreducible lattice in ... and let $w\in F_n$ be a word in the free...

conjecture | open | MathDB 页面 | 原文线索

V2-014010 | MATHDB-347551

Wavefront set conjecture for theta representations

摘录 / EXCERPT: Let F be p-adic with $p mid n$. Let $\tilde{G}$ be a persistent n-fold covering group. Let $ u\in X\otimes \mathbf{R}$ be exceptional and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014011 | MATHDB-347552

Wavefront set conjecture for constituents of regular unramified principal series

摘录 / EXCERPT: Let $\tilde{G}$ be a persistent n-fold cover, and let $\nu\in X\otimes\mathbf{R}$ be regular with $\Phi(\nu)\subset\Delta$. For $S\subset\Phi(\nu)$, let $W(S)$ be the Weyl subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-014012 | MATHDB-347564

The tempered local Langlands correspondence for rigid inner forms

摘录 / EXCERPT: Let G^{*} be a connected, reductive, quasi-split group over the local function field F, with finite central F-subgroup Z, and let G be an inner form of G^{*}. Fix a Whi...

conjecture | open | MathDB 页面 | 原文线索

V2-014013 | MATHDB-347568

Quotient structure conjecture for the loop Hecke algebra

摘录 / EXCERPT: Quotient structure conjecture. The structure of $L\!H_n/\chi^{(j+1)}$ is given by the $j\times j$ truncation of ${\mathcal M}^p_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-014014 | MATHDB-347573

Babai-Goodman bounded-orbit conjecture for automorphism groups of lattices

摘录 / EXCERPT: A finite group G is realized by a lattice L if $\operatorname{Aut}(L)\simeq G$. An orbit means an orbit of the action of $\operatorname{Aut}(L)$ on the elements of L. Babai-G...

conjecture | open | MathDB 页面 | 原文线索

V2-014015 | MATHDB-347581

The non-virtual-regularity conjecture for scale group actions

摘录 / EXCERPT: Let G be a compactly generated, topologically simple totally disconnected locally compact group, let $x\in G$ satisfy $s(x^{-1})>1$, and let $V_{--}\rtimes\langle x\rangle$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-014016 | MATHDB-347588

Stable-rank bound for special biserial algebras with a band

摘录 / EXCERPT: Let Λ be a special biserial algebra with at least one band, and let $\mathrm{st}(\Lambda)$ denote its stable rank. Stable-rank conjecture. The stable rank satisfies ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-014017 | MATHDB-347603

Perfect integrability of singular vectors for Gaudin Bethe algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let M be the finite-dimensional representation under consideration, let M^{sing} be its singular subspace, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014018 | MATHDB-347604

Perfect integrability for regular quasi-periodic Gaudin spin chains

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra with Cartan subalgebra $\mathfrak{h}$ and Cartan subgroup H, let M be the finite-dimensional representation under consideration, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014019 | MATHDB-347605

Perfect integrability for Gaudin algebras with arbitrary irregular parameter

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $\mu\in\mathfrak{g}^*\cong\mathfrak{g}$, and choose $\mu\in\mathfrak b_- = \mathfrak n_-\oplus\mathfrak h$ after conjugation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014020 | MATHDB-347618

The virtually Kleinian conjecture for relatively hyperbolic groups with planar boundary

摘录 / EXCERPT: Let G be a one-ended group and let $(G,\mathcal{P})$ be a relatively hyperbolic pair, with Bowditch boundary $\partial(G,\mathcal{P})$. Assume that this boundary is planar and ha...

conjecture | open | MathDB 页面 | 原文线索

V2-014021 | MATHDB-347646

Real and disjoint Casimir spectrum conjecture for parabolic positive representations

摘录 / EXCERPT: Let $\mathcal{P}_\lambda^J$ be an arbitrary parabolic positive representation of $\mathcal{U}_q(\mathfrak{g}_\mathbb{R})$, and let its Casimir operators have spectrum in the ambien...

conjecture | open | MathDB 页面 | 原文线索

V2-014022 | MATHDB-347698

Gottschalk's surjunctivity conjecture for cellular automata

摘录 / EXCERPT: Let G be a group and A a finite alphabet. A cellular automaton is a G-equivariant continuous map $T\colon A^G\to A^G$. Gottschalk's conjecture. If T is injective, then it i...

conjecture | open | MathDB 页面 | 原文线索

V2-014023 | MATHDB-347699

Free-product closure conjecture for dual surjunctive groups

摘录 / EXCERPT: Let G be a group, and let $\mathbb{Z}*G$ denote its free product with the infinite cyclic group. A group is dual surjunctive if every post-surjective cellular automaton over it i...

conjecture | open | MathDB 页面 | 原文线索

V2-014024 | MATHDB-347763

The Benson conjecture on tensor squares of odd-dimensional modules

摘录 / EXCERPT: Let G be a finite 2-group and let k be an algebraically closed field of characteristic 2. For an indecomposable kG-module M, let M^* denote its dual module. Benson's...

conjecture | open | MathDB 页面 | 原文线索

V2-014025 | MATHDB-347780

Residual torsion-free nilpotence criterion for genus one pretzel knots

摘录 / EXCERPT: Residual torsion-free nilpotence conjecture. The commutator subgroup of $\pi_1(M_J)$ is residually torsion-free nilpotent if and only if Δ_J is nontrivial.

conjecture | open | MathDB 页面 | 原文线索

V2-014026 | MATHDB-347818

Universal word-measure character bound for natural families of groups

摘录 / EXCERPT: Let $G=\{G(N)\}_N$ be a natural family of groups, and let $\chi=\{\chi_N\}_{N\geq N_0}$ be a stable irreducible character with $\chi_N\in\widehat{G(N)}$. Let $\pi(w)$ denote the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-014027 | MATHDB-347824

Normalizer covering number 11 conjecture

摘录 / EXCERPT: Let G be a group, and let $\sigma_n(G)$ denote its normalizer covering number, the least number of proper normalizers whose union is G, when such a covering exists. Normalizer...

conjecture | open | MathDB 页面 | 原文线索

V2-014028 | MATHDB-347844

Kervaire-Laudenbach-Howie conjecture on independent systems of equations

摘录 / EXCERPT: Kervaire-Laudenbach-Howie conjecture. Every independent system of equations over G has a solution in an overgroup of G.

conjecture | open | MathDB 页面 | 原文线索

V2-014029 | MATHDB-347854

The family-stabiliser tree conjecture

摘录 / EXCERPT: Family-stabiliser tree conjecture. The inequality $\operatorname{cd}_{\mathcal{F}}G\leqslant 1$ holds if and only if G acts on a tree whose stabilisers generate $\mathcal{F}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014030 | MATHDB-347855

Brown's conjecture on the geometric and virtual cohomological dimensions of groups

摘录 / EXCERPT: Brown's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014031 | MATHDB-347869

Fitting-series containment conjecture under centralizing fixed points

摘录 / EXCERPT: Let G be a solvable group and let A act on G by automorphisms with $(|A|,|G|)=1$. Let $F(G)$ be the Fitting subgroup, define $F_0(G)=1$, $F_1(G)=F(G)$, and let $F_{i+1}(G)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-014032 | MATHDB-347874

Gilman's conjecture on length-reducing rewriting systems

摘录 / EXCERPT: Let G be a group. A finite convergent length-reducing rewriting system for G is a presentation (Σ,T) by such a system, where each rewriting rule has a right-hand side of...

conjecture | open | MathDB 页面 | 原文线索

V2-014033 | MATHDB-347875

Madlener-Otto conjecture on length-reducing rewriting systems

摘录 / EXCERPT: Let G be a group. A finite convergent length-reducing rewriting system for G is a presentation (Σ,T) by such a system. Madlener-Otto conjecture. G admits a finite conv...

conjecture | open | MathDB 页面 | 原文线索

V2-014034 | MATHDB-347884

Spectrum conjecture for connected Nakayama algebras

摘录 / EXCERPT: Let $n\geq 2$ and let k satisfy $2\leq k\leq 2n-2$. A Nakayama algebra is a finite-dimensional algebra whose indecomposable modules are uniserial; it is connected when its underl...

conjecture | open | MathDB 页面 | 原文线索

V2-014035 | MATHDB-347889

Equivalence of the two presentations of the shifted $q=0$ affine algebra

摘录 / EXCERPT: Let $\dot{\mathcal{U}}_{0,N}(L\mathfrak{sl}_n)$ and $\dot{\mathcal{U}}'_{0,N}(L\mathfrak{sl}_n)$ be the two algebra presentations defined earlier, with generators...

conjecture | open | MathDB 页面 | 原文线索

V2-014036 | MATHDB-347925

The factorial-order conjecture for the reduced Grothendieck group of symmetric quotient singularities

摘录 / EXCERPT: Factorial-order conjecture. The reduced Grothendieck group of $\mathbb{C}[V]^{S_n}$ has order $n!$.

conjecture | open | MathDB 页面 | 原文线索

V2-014037 | MATHDB-347945

Generalized Ramanujan conjecture for $SO(n,1)$

摘录 / EXCERPT: Let $G=SO(n,1)$, and let $\widehat{G}^{\operatorname{sph}}_{\operatorname{Aut}}$ denote the automorphic dual of the spherical unitary dual of G. Generalized Ramanujan conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-014038 | MATHDB-347972

The characterization of partitions with Schaper number at least three

摘录 / EXCERPT: Schaper-number conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014039 | MATHDB-347973

Tran's conjecture on totally disconnected Morse boundaries of right-angled Coxeter groups

摘录 / EXCERPT: Let Λ be a graph, and let ... be its associated right-angled Coxeter group. A finitely generated group has a quasi-isometry invariant Morse boundary, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-014040 | MATHDB-347975

Li-Li formation conjecture for generalized Fitting subgroups

摘录 / EXCERPT: All groups considered are finite. Let $\mathfrak{U}$ denote the formation of all supersoluble groups, and let $\mathfrak{F}$ be a saturated formation containing $\mathfrak{U}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014041 | MATHDB-347993

Solvability from the average element order threshold of A_5

摘录 / EXCERPT: The average-order solvability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014042 | MATHDB-348017

Breuil-Paškūnas conjecture on the structure of the mod p Langlands representation

摘录 / EXCERPT: Let $\overline{\rho}=\overline{r}_v^{\vee}$ be generic in the sense of Breuil-Paškūnas, and let $\pi_v^D(\overline{r})$ denote the associated mod p Langlands representation. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-014043 | MATHDB-348022

The extended Horn inequalities conjecture for Newell-Littlewood numbers

摘录 / EXCERPT: Extended Horn inequalities conjecture. If $(\mu,\nu,\lambda)\in{\sf Par}_n^3$ satisfies the parity condition and every extended Horn inequality indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-014044 | MATHDB-348025

Conciseness conjecture for commutators of non-commutator words

摘录 / EXCERPT: Let $w=w(x_1,\dots,x_k)$ be a group-word, and let G_w denote the set of all values of w in a group G. The verbal subgroup $w(G)$ is the subgroup generated by G_w, and w i...

conjecture | open | MathDB 页面 | 原文线索

V2-014045 | MATHDB-348032

The AGS conjecture for real semisimple Lie algebra representations

摘录 / EXCERPT: Let $\mathfrak h$ be a real semisimple Lie algebra and V an orthogonal representation of $\mathfrak h$. AGS conjecture. If V has AGS in $\mathfrak h$, then ... The conjecture giv...

conjecture | open | MathDB 页面 | 原文线索

V2-014046 | MATHDB-348033

The AGS equivalence conjecture for complex semisimple Lie algebra representations

摘录 / EXCERPT: Let $\mathfrak h$ be a complex semisimple Lie algebra and V an orthogonal representation of $\mathfrak h$ over $\mathbb C$. AGS equivalence conjecture. One has ... This is the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-014047 | MATHDB-348039

The Hecke action conjecture for the principal block

摘录 / EXCERPT: Let G be a connected reductive algebraic group over an algebraically closed field $\Bbbk$ of characteristic p strictly bigger than the Coxeter number of G. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014048 | MATHDB-348069

Dixmier's unitarizability conjecture

摘录 / EXCERPT: Let G be a countable group. A representation of G on a Hilbert space is uniformly bounded if there is a constant K such that $\lvert \pi(g)\rvert<K$ for all $g\in G$, and G...

conjecture | open | MathDB 页面 | 原文线索

V2-014049 | MATHDB-348070

Hilburn's Coulomb-branch Levi restriction conjecture

摘录 / EXCERPT: Let H be a reductive group, let V be a representation of H, and let $\gamma:\mathbb{C}^\times\rightarrow H$ be a coweight. Let L_γ be the centralizer of γ, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-014050 | MATHDB-348103

The symmetric-group supercharacter theory conjecture

摘录 / EXCERPT: Symmetric-group supercharacter theory conjecture. For every $n\geqslant 7$, the symmetric group S_n has exactly 4 supercharacter theories.

conjecture | open | MathDB 页面 | 原文线索

V2-014051 | MATHDB-348116

The local Gan-Gross-Prasad multiplicity conjecture for special orthogonal groups

摘录 / EXCERPT: Let (W,V) be a pair of quadratic spaces over a local field F of characteristic zero, with W a subspace of V such that $W^{\perp}$ is odd-dimensional and split. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014052 | MATHDB-348118

Treumann-Venkatesh's linkage conjecture for functorial transfer

摘录 / EXCERPT: Treumann-Venkatesh's linkage conjecture. Linkage is compatible with functorial transfer: if π is linked to Π, then π transfers to Π under the Local Langlands corres...

conjecture | open | MathDB 页面 | 原文线索

V2-014053 | MATHDB-348162

Duflo's branching and moment-map conjecture for discrete series

摘录 / EXCERPT: Let G be a real almost algebraic group, let π be a discrete series representation attached to a coadjoint orbit $\mathcal{O}_{\pi}$, and let H be a closed almost algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-014054 | MATHDB-348176

Barbasch-Pandžić conjecture on Dirac series for complex simple Lie groups

摘录 / EXCERPT: Barbasch-Pandžić conjecture. If $\pi\in\widehat{G}$ has regular, half-integral infinitesimal character, then

conjecture | open | MathDB 页面 | 原文线索

V2-014055 | MATHDB-348177

Barbasch-Pandžić multiplicity-one conjecture for spin-lowest K-types

摘录 / EXCERPT: Barbasch-Pandžić multiplicity-one conjecture. For every $\pi\in\widehat{G}^d$, there is a unique spin-lowest K-type $V_{\mathfrak{k}}(\eta)$, and it occurs in π with multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-014056 | MATHDB-348218

Uniform torsion conjecture for metric ultraproducts

摘录 / EXCERPT: Let ${\mathcal G}^*_{\mathrm{met}}$ be a metric ultraproduct. A group has finite exponent, or is uniformly torsion, if there is a common positive integer annihilating every element...

conjecture | open | MathDB 页面 | 原文线索

V2-014057 | MATHDB-348227

Rigidity conjecture for finitely generated linear groups

摘录 / EXCERPT: A group G is isotypically rigid if every finitely generated group isotypic to G is isomorphic to G. Rigidity conjecture for finitely generated linear groups. Every finitely g...

conjecture | open | MathDB 页面 | 原文线索

V2-014058 | MATHDB-348228

Isotypic rigidity conjecture for Chevalley groups

摘录 / EXCERPT: Let $G_\Phi(R)$ be a Chevalley group associated with a root system Φ over a ring R, and let H be a group isotypic to $G_\Phi(R)$. Isotypic rigidity conjecture for Chevalle...

conjecture | open | MathDB 页面 | 原文线索

V2-014059 | MATHDB-348257

The Cartan-matrix monotonicity conjecture for τ-tilting counts

摘录 / EXCERPT: Cartan monotonicity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014060 | MATHDB-348258

The conjecture that wild Schur algebras in the exceptional cases are τ-tilting finite

摘录 / EXCERPT: Exceptional wild Schur algebra conjecture. All wild Schur algebras contained in $(\star)$ are τ-tilting finite.

conjecture | open | MathDB 页面 | 原文线索

V2-014061 | MATHDB-348263

The FF-subgroup conjecture for finite simple groups

摘录 / EXCERPT: FF-subgroup conjecture. All nontrivial subgroups of a finite simple group are FF-subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-014062 | MATHDB-348270

Vanishing probability for prescribed squares in random walks

摘录 / EXCERPT: Vanishing-squares conjecture. For every $a\in G$,

conjecture | open | MathDB 页面 | 原文线索

V2-014063 | MATHDB-348272

Generalized Tits conjecture for centers of spherical special subgroups

摘录 / EXCERPT: Let A be an Artin group associated to a Coxeter system (W,S). Let $\mathcal S$ be the set of irreducible, spherical subsets of S, and let RA be the right-ang...

conjecture | open | MathDB 页面 | 原文线索

V2-014064 | MATHDB-348288

Equality of directional group orders for two-step rules

摘录 / EXCERPT: Let $\mathcal Q$ be the class of two-step rules, and let G_e,G_n,G_w,G_s denote the corresponding groups for the four directions. Directional group-order conjecture. For every tw...

conjecture | open | MathDB 页面 | 原文线索

V2-014065 | MATHDB-348298

Centraliser structure conjecture for formal maps

摘录 / EXCERPT: Let K be a field of characteristic zero, let $\mathcal{G}$ be the group of formal maps under composition, and let $g\in\mathcal{G}$ be tangent to but not equal to the identity. F...

conjecture | open | MathDB 页面 | 原文线索

V2-014066 | MATHDB-348320

Undecidability conjecture for epimorphisms onto finitely generated virtually Abelian groups

摘录 / EXCERPT: Undecidability conjecture. The uniform epimorphism problem onto the class of all finitely generated virtually Abelian groups is not decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-014067 | MATHDB-348322

Conjectural classification of purely nonprincipal Morita classes with defect group $(C_2)^6$

摘录 / EXCERPT: Let $D=(C_2)^6$ and consider purely nonprincipal blocks with defect group D. For each group in the following list, there is one relevant Morita equivalence class of nonprincipal...

conjecture | open | MathDB 页面 | 原文线索

V2-014068 | MATHDB-348333

The ND conjecture for rational group algebras with nilpotent elements

摘录 / EXCERPT: ND conjecture. If $\mathbb{Q}[G]$ has some nonzero nilpotent elements and G has ND, then $\mathbb{Q}[G]$ has only one matrix component.

conjecture | open | MathDB 页面 | 原文线索

V2-014069 | MATHDB-348348

The semibrick-pair completion conjecture for tau-tilting finite algebras

摘录 / EXCERPT: Let Λ be a τ-tilting finite algebra, and let $\mathcal{D} \sqcup \mathcal{U}[1]$ be a semibrick pair, meaning a collection of bricks in the indicated degrees satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-014070 | MATHDB-348367

The rational relation number conjecture

摘录 / EXCERPT: Let α be a rational number with $|\alpha|<4$. A complex number is called a relation number when the associated group G_α is not free. Rational relation number conject...

conjecture | open | MathDB 页面 | 原文线索

V2-014071 | MATHDB-348397

Conjecture on cyclic subgroup separability and virtual compact specialness for graphs of groups

摘录 / EXCERPT: Cyclic subgroup separability and virtual specialness conjecture. The following conditions on G should be equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-014072 | MATHDB-348412

The extension of the codimension-one ideal theorem for finite W-algebras

摘录 / EXCERPT: Codimension-one ideal conjecture. Theorem $$ should hold for all nilpotent elements $e\in\mathfrak{g}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014073 | MATHDB-348445

Williams's conjecture on the abelianization of the groups $H(r,n,s)$

摘录 / EXCERPT: Let $H(r,n,s)=P(r,n,r+1,s,1)$, where $P(r,n,k,s,q)$ is the cyclically presented group under consideration. Williams's conjecture. For integers $r,sgeq 1$ and $ngeq 2$ satisfying .....

conjecture | open | MathDB 页面 | 原文线索

V2-014074 | MATHDB-348446

Conjecture that perfect groups $P(r,n,k,r-1,q)$ are of type $\tilde{\mathfrak{Z}}$

摘录 / EXCERPT: Type-$\tilde{\mathfrak{Z}}$ conjecture. If $P(r,n,k,r-1,q)$ is perfect, then it is of type $\tilde{\mathfrak{Z}}$. The preceding theorem establishes the corresponding perfectness c...

conjecture | open | MathDB 页面 | 原文线索

V2-014075 | MATHDB-348447

Conjecture I on the BNSR invariant of the weak commutativity quotient

摘录 / EXCERPT: Conjecture I. Under this identification,

conjecture | open | MathDB 页面 | 原文线索

V2-014076 | MATHDB-348558

Reeder's conjecture on graded multiplicities in the exterior algebra

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra over $\mathbb{C}$, fix a Cartan subalgebra $\mathfrak{h}$, and let Φ be the associated root system with Weyl group W. Choose posit...

conjecture | solved | MathDB 页面 | 原文线索

V2-014077 | MATHDB-348654

The two-colored Jones-Wenzl projector existence conjecture

摘录 / EXCERPT: Let (W,S) be a two-colored realization over a field $\Bbbk$, with colors s and t, and let JW_{n_s} and JW_{n_t} denote the corresponding Jones-Wenzl projectors. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-014078 | MATHDB-348714

Profile-count conjecture for rigid indecomposable rank 2 modules

摘录 / EXCERPT: Profile-count conjecture. The number of profiles of rigid indecomposable rank 2 modules in ${\rm CM}(B_{k,n})$ is $N_{k,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014079 | MATHDB-348715

Real canonical-profile conjecture in arbitrary rank

摘录 / EXCERPT: Real canonical-profile conjecture. Let $M\in {\rm CM}(B_{k,n})$ be rigid indecomposable and assume that $\varphi(M)$ is a real root in $J_{k,n}$. Then P_M is a cyclic permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-014080 | MATHDB-348716

Three-quasi-boxes rigidity conjecture for fully reduced rank 2 modules

摘录 / EXCERPT: Three-quasi-boxes rigidity conjecture. Then M is not rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-014081 | MATHDB-348736

Basis conjecture for the quotient of the Kirchhoff Lie algebra

摘录 / EXCERPT: Basis conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-014082 | MATHDB-348757

PRV conjecture for the fusion product

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let P_{ℓ} be the set of level-ℓ dominant weights, let W_{ℓ} be the affine Weyl group, and let $V(\lambda)$ denote the irr...

conjecture | open | MathDB 页面 | 原文线索

V2-014083 | MATHDB-348762

Cuntz's positivity conjecture for exterior-power fusion algebras

摘录 / EXCERPT: Let $d\geq 1$, let $1\leq n\leq d$, and let $I_{n,d}$ be the set of strictly increasing n-tuples $(i_1,\ldots,i_n)$ with $0\leq i_1<\cdots<i_n<d$. Set $\xi=\exp(i\pi/d)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-014084 | MATHDB-348767

Weighted-poset classification conjecture for the categories \mathcal{S}(W,S,S)

摘录 / EXCERPT: Weighted-poset equivalence conjecture. The categories $\mathcal{S}(W,S,S)$ and $\mathcal{S}(W',S',S')$ are equivalent if and only if their weighted posets are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-014085 | MATHDB-348775

Standard string-polytope lattice-point conjecture for classical groups

摘录 / EXCERPT: Standard string-polytope conjecture. The polytope $Q_{\underline{w_0}^\mathrm{std}}(\lambda)$ is a lattice polytope if and only if at least one of the following holds:...

conjecture | open | MathDB 页面 | 原文线索

V2-014086 | MATHDB-348777

Well-definedness and analytic behavior of the multiplicative Euler characteristic

摘录 / EXCERPT: Let ρ and π be admissible representations in the local-field setting, and let ... be a monomial resolution, with each $L(s,M_i,\pi)$ the product of the two-variable L-func...

conjecture | open | MathDB 页面 | 原文线索

V2-014087 | MATHDB-348836

The linear determinant multiplicity conjecture for two-dimensional topological theories

摘录 / EXCERPT: Let $s\ne 0$ be an integer, and let $\alpha(s)=(\beta_0,s,0,0,\dots)$, so that the generating function is $Z_{\alpha(s)}(T)=\beta_0+sT$. For arbitrary β_0 in characteristic z...

conjecture | open | MathDB 页面 | 原文线索

V2-014088 | MATHDB-348849

Wolf's semisimplicity conjecture for deformed Fomin-Kirillov algebras

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, let $\alpha_1,\alpha_2\in K$, and let $\mathcal{D}_4(\alpha_1,\alpha_2)$ be the quadratic K-algebra generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-014089 | MATHDB-348854

The local Gross-Prasad conjecture for GSpin(4) × GSpin(5)

摘录 / EXCERPT: Let F be a number field, let v be a place of F, let Π_v be an irreducible smooth representation of $G(F_v)$, and let $\Sigma_v=\Sigma_{1,v}\boxtimes\Sigma_{2,v}$ be an ir...

conjecture | open | MathDB 页面 | 原文线索

V2-014090 | MATHDB-348859

Drinfel'd-Grothendieck conjecture on the profinite Grothendieck-Teichmüller tower

摘录 / EXCERPT: Let $\widehat{\operatorname{GT}}$ be the profinite Grothendieck-Teichmüller group, defined using the first two levels of the genus-zero stage of the profinite Grothendieck-Teichmül...

conjecture | open | MathDB 页面 | 原文线索

V2-014091 | MATHDB-348863

Multiplicity-one conjecture for the Σ × σ period

摘录 / EXCERPT: Σ × σ multiplicity-one conjecture. With these notations,

conjecture | open | MathDB 页面 | 原文线索

V2-014092 | MATHDB-348870

Integrality conjecture for the Benson-Symonds invariant of symmetric-group permutation modules

摘录 / EXCERPT: Let n be a positive integer, let λ be a partition of n, and let M^λ be the corresponding permutation module of the symmetric group $\mathcal{S}_n$. The Benson-S...

conjecture | open | MathDB 页面 | 原文线索

V2-014093 | MATHDB-348878

Guba-Sapir conjecture on superadditivity of Dehn functions

摘录 / EXCERPT: Let G be a finitely presented group. A function is superadditive if $f(m+n)\geq f(m)+f(n)$ for all nonnegative integers m,n. Write $f\simeq g$ when the two functions are equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-014094 | MATHDB-348879

Conjecture on homological filling area and the Dehn function

摘录 / EXCERPT: Let H be a finitely presented group. Denote by FA_H its homological filling-area function and by δ_H its Dehn function, and write $f\preceq g$ for the usu...

conjecture | open | MathDB 页面 | 原文线索

V2-014095 | MATHDB-348896

Margalit's conjecture on the smallest finite non-cyclic quotient of the braid group

摘录 / EXCERPT: Let B_n be the braid group on n strands, and let Σ_n denote the finite quotient under consideration. A finite quotient is non-cyclic if it is not a cyclic group. Margali...

conjecture | open | MathDB 页面 | 原文线索

V2-014096 | MATHDB-348967

Uniform spectral gap conjecture for hyperbolic special groups

摘录 / EXCERPT: Uniform spectral gap conjecture. There is a uniform constant $B>0$ such that any hyperbolic C-special or A-special group has a spectral gap B for integral chains.

conjecture | open | MathDB 页面 | 原文线索

V2-014097 | MATHDB-348974

Generalized Hiraga-Ichino-Ikeda conjecture for tempered L-parameters

摘录 / EXCERPT: Let G be a connected reductive group over F. For a tempered L-parameter ϕ, let $\Pi_\phi(G)$ be its L-packet, let $\widetilde{\mathcal{S}}_\phi$ be the associated S-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-014098 | MATHDB-348980

The BNSR-invariant conjecture for the Stein groups F_S^r

摘录 / EXCERPT: BNSR-invariant conjecture. For any S and r,

conjecture | open | MathDB 页面 | 原文线索

V2-014099 | MATHDB-348985

Eigenvalue lifting for random transposition shuffles on generalized symmetric groups

摘录 / EXCERPT: Let $r,n\in\mathbb{N}$, let $G_{r,n}\cong\mathbb{Z}_r\wr S_n$ be the generalized symmetric group, and let an r-partition of n be a tuple of partitions whose sizes sum to n. E...

conjecture | open | MathDB 页面 | 原文线索

V2-014100 | MATHDB-349045

Aubert dual conjecture for representations parametrized by extended multi-segments

摘录 / EXCERPT: Let $\mathcal{E} = \cup_{\rho}\{([A_i,B_i]_{\rho}, l_i, \eta_i) \}_{i \in (I_\rho,>)}$ be an extended multi-segment, and let ... be the extended multi-segment defined from...

conjecture | open | MathDB 页面 | 原文线索

V2-014101 | MATHDB-349085

Rational growth conjecture for torus bundle groups

摘录 / EXCERPT: Rational growth conjecture. Every torus bundle group $G=\mathbb{Z}^2\rtimes_A\mathbb{Z}$ has rational growth with respect to some finite generating set.

conjecture | open | MathDB 页面 | 原文线索

V2-014102 | MATHDB-349086

Cuspidal character sheaves from local systems for stably graded Lie algebras

摘录 / EXCERPT: Let ${\mathfrak{g}}_1$ be the degree-one component of one of the stably graded Lie algebras considered here, let K act on ${\mathfrak{g}}_1$, and let ${\mathfrak{g}}_1^{rs}$ be i...

conjecture | open | MathDB 页面 | 原文线索

V2-014103 | MATHDB-349102

Tong-Viet's quasi-simple-group conjecture for Brauer character degrees

摘录 / EXCERPT: Let G be a finite quasi-simple group with centre $Z(G)$ of p'-order. Let $\operatorname{IBr}(G\mid\theta)$ denote the irreducible p-Brauer characters of G lying over...

conjecture | open | MathDB 页面 | 原文线索

V2-014104 | MATHDB-349105

Principal conjecture on fixed points of generic automorphisms

摘录 / EXCERPT: Let G be an infinite simple group of finite Morley rank, and let α be a generic automorphism of G. The fixed-point subgroup of α is $C_G(\alpha)$. Principal conje...

conjecture | open | MathDB 页面 | 原文线索

V2-014105 | MATHDB-349141

The Symbols Conjecture for cohomology of smooth pro-p groups

摘录 / EXCERPT: Symbols Conjecture. The cohomology algebra $H^*(G,\mathbb{F}_p)$ is generated in degree one. This conjecture sharpens the Smoothness Theorem by predicting a stronger structural pro...

conjecture | open | MathDB 页面 | 原文线索

V2-014106 | MATHDB-349182

Coherent-sheaf realization conjecture for smooth representations

摘录 / EXCERPT: Let K be a nonarchimedean local field, let d be a positive integer, and let $]_{ll]}$ be the coefficient ring. Let $L]_d(K)$-representations mean representations on $]_{ll]}$-m...

conjecture | open | MathDB 页面 | 原文线索

V2-014107 | MATHDB-349209

Non-decreasable K-types are u-small

摘录 / EXCERPT: For $0\leq j\leq s-1$, let $\Omega(j)$ be the set of K-types μ such that $\mu+2\rho_c\in \mathcal{C}^{(j)}$. A K-type $\mu\in\Omega(j)$ is non-decreasable if there is no...

conjecture | open | MathDB 页面 | 原文线索

V2-014108 | MATHDB-349215

Irreducibility conjecture for cell representations of the two Gelfand W-graphs of S_n

摘录 / EXCERPT: Let S_n be the symmetric group, and let $\Gamma^\mathbf{m}(S_n)$ and $\Gamma^\mathbf{n}(S_n)$ be the two Gelfand W-graphs considered in the paper. Their cell representations ar...

conjecture | open | MathDB 页面 | 原文线索

V2-014109 | MATHDB-349216

Lé vai-Pyber conjecture on profinite groups with many elements of bounded order

摘录 / EXCERPT: Let G be a profinite group, and let $X_n(G)$ denote the set of solutions of $x^n=1$ in G. Write ${\mathbf m}_G$ for the normalized Haar measure on G. Lé vai-Pyber conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-014110 | MATHDB-349217

The supremum conjecture for automorphism norm equations in finite groups

摘录 / EXCERPT: Supremum conjecture. For every positive integer n, one has $c_n<1$. This conjecture is known for $n=2$, where $c_2=\frac34$, and the paper proves it for $n=3$; it is presented as...

conjecture | open | MathDB 页面 | 原文线索

V2-014111 | MATHDB-349218

The solvable-group supremum conjecture for odd exponents

摘录 / EXCERPT: Solvable-group supremum conjecture. For every positive integer n, one has

conjecture | open | MathDB 页面 | 原文线索

V2-014112 | MATHDB-349231

Hypothesis T for finite groups and normal subgroups

摘录 / EXCERPT: Let G be a finite group and let $N\unlhd G$ be a normal subgroup. For an irreducible representation ρ of G, write $\ker\rho$ for its kernel; for a subgroup $H\subseteq G$,...

conjecture | open | MathDB 页面 | 原文线索

V2-014113 | MATHDB-349232

The intersection-multiplicity conjecture for field families

摘录 / EXCERPT: Let k be a number field, let G be a finite group, let $\pi\colon G\to S_n$ be a transitive faithful permutation representation with $n\geq2$, and let $N\unlhd G$ be normal. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-014114 | MATHDB-349241

The nuclear-rank conjecture for 2-special groups

摘录 / EXCERPT: Nuclear-rank conjecture. Every 2-special group has nuclear rank 0 or 2, and its valid children are always 2-special.

conjecture | open | MathDB 页面 | 原文线索

V2-014115 | MATHDB-349248

The polynomial relation conjecture for tensor powers of simple Drinfeld double modules

摘录 / EXCERPT: Let D_n be the Drinfeld double under consideration, let $\mathsf{V}(ell,r)$ be a simple D_n-module with $n e 2$ and $2ell e n+1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014116 | MATHDB-349292

Even and odd stretching conjectures for Newell-Littlewood coefficients

摘录 / EXCERPT: Even and odd stretching conjectures. If $| 0\boldsymbol{ u}”|+| 0\boldsymbol{ u}|+| 0\boldsymbol{ u}'|$ is even, then the assertions E(i)-E(v) in the source hold: positivity and mu...

conjecture | open | MathDB 页面 | 原文线索

V2-014117 | MATHDB-349294

Scott's fibering conjecture for irreducible compact 3-manifolds

摘录 / EXCERPT: Let M be an irreducible, compact 3-manifold with fundamental group G. Suppose that G contains a subnormal subgroup $N\neq\mathbb{Z}$, and that N is contained in a finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-014118 | MATHDB-349312

Bernon-Przebinda character-transfer conjecture

摘录 / EXCERPT: Bernon-Przebinda's conjecture. Up to a constant, ... on $\widetilde{{\rm G}'_{1}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014119 | MATHDB-349319

Weak Gersten conjecture for one-relator RG groups

摘录 / EXCERPT: Weak Gersten conjecture. Every one-relator RG group is hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-014120 | MATHDB-349407

Regularity conjecture for shuffled projective modules in category O

摘录 / EXCERPT: Let W be the Weyl group, let $\mathcal{O}_0$ be the principal block of category $\mathcal{O}$, and let P be a projective generator of $\mathcal{O}_0$. For a parabolic subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-014121 | MATHDB-349408

Regularity conjecture for shuffled tilting modules in category O

摘录 / EXCERPT: Let W be the Weyl group, let $\mathcal{O}_0$ be the principal block of category $\mathcal{O}$, and let T be a characteristic tilting module for $\mathcal{O}_0$. For a parabolic...

conjecture | open | MathDB 页面 | 原文线索

V2-014122 | MATHDB-349409

Projective-dimension conjecture for twisted tilting modules

摘录 / EXCERPT: Let W be the Weyl group with longest element w_0, let θ_x be the projective functor indexed by $x\in W$, and let Δ_z and ∇_z denote the standard and costan...

conjecture | open | MathDB 页面 | 原文线索

V2-014123 | MATHDB-349410

Projective-dimension conjecture for shuffled tilting modules

摘录 / EXCERPT: Let W be the Weyl group with longest element w_0, let C_x be the shuffling functor indexed by $x\in W$, and let T_y and P_{w_0y} denote the corresponding tilting...

conjecture | open | MathDB 页面 | 原文线索

V2-014124 | MATHDB-349422

Semistability conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Let G be a finitely generated group hyperbolic relative to a finite collection $\{P_1,\ldots,P_n\}$ of proper finitely generated subgroups. Semistability conjecture. If each...

conjecture | open | MathDB 页面 | 原文线索

V2-014125 | MATHDB-349459

The Picard group conjecture for blocks with TI defect groups

摘录 / EXCERPT: Picard group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014126 | MATHDB-349486

Centralizer-count conjecture for groups with quotient \(\mathbb{Z}_{p^n}\rtimes\mathbb{Z}_{p^n}\)

摘录 / EXCERPT: Centralizer-count conjecture. The number of distinct centralizers of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014127 | MATHDB-349587

Common reducibility conjecture for Kostka pairs

摘录 / EXCERPT: commonly reducible when $\lambda^{\bullet}$ and $\mu^{\bullet}$ are obtained by choosing common columns of λ and μ. Common reducibility conjecture. If $\lambda_1>r$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-014128 | MATHDB-349588

Bound on the largest partition part for Littlewood-Richardson Hilbert basis elements

摘录 / EXCERPT: Let λ, μ, and ν be partitions with at most r parts, and let ${\sf LR}_r^{\mathbb Z}$ be the semigroup of triples (λ,μ,ν) with positive Littlewood-Rich...

conjecture | open | MathDB 页面 | 原文线索

V2-014129 | MATHDB-349591

Permutation invariance of Hurwitz orbit sizes in G_6

摘录 / EXCERPT: Permutation-invariance conjecture. These two factorizations have Hurwitz orbits of equal size. The conjecture proposes that, in G_6, Hurwitz orbit size is unchanged by arbitraril...

conjecture | open | MathDB 页面 | 原文线索

V2-014130 | MATHDB-349603

Character-wise minus Gross-Kuz'min conjecture

摘录 / EXCERPT: Let K be a CM-field, let $p\ne2$, and let $R\subseteq K$ be a totally real subfield such that K/R is normal. Set $G=\operatorname{Gal}(K/R)$ and assume $K\cap R_\infty=R$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014131 | MATHDB-349642

The lower-central-series comparison conjecture for random Cayley graphs

摘录 / EXCERPT: Lower-central-series comparison conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014132 | MATHDB-349669

Farjoun's nilpotence-preservation conjecture for group localizations

摘录 / EXCERPT: Let L be a localization on the category of groups, meaning a functor equipped with a natural transformation from the identity satisfying the idempotence conditions for a localiza...

conjecture | open | MathDB 页面 | 原文线索

V2-014133 | MATHDB-349671

Farjoun's abelian-cokernel conjecture for group localizations

摘录 / EXCERPT: Let L be a localization on the category of groups, let G be a group, and let $G\to LG$ be the localization map. The cokernel of this map is the quotient of LG by the normal c...

conjecture | open | MathDB 页面 | 原文线索

V2-014134 | MATHDB-349672

Nikolov-Segal map conjecture for right exact localizations

摘录 / EXCERPT: Let L be a right exact localization on the category of groups, and let G be a nilpotent group. A homomorphism $f:H\to G$ is a Nikolov-Segal map if ... Here $[G,G]$ is the commuta...

conjecture | open | MathDB 页面 | 原文线索

V2-014135 | MATHDB-349734

Non-geometric Goldschmidt-Sims conjecture

摘录 / EXCERPT: Non-geometric Goldschmidt-Sims conjecture. There exists a function

conjecture | open | MathDB 页面 | 原文线索

V2-014136 | MATHDB-349743

Non-negativity conjecture for the coefficients of $A_g(n,q)$

摘录 / EXCERPT: For integers $g\ge 2$ and $n\ge 1$, let $A_g(n,q)$ be the polynomial associated in the paper with the counting of tuples of nilpotent matrices under simultaneous conjugation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-014137 | MATHDB-349793

Refined local Langlands compatibility conjecture for restriction and induction

摘录 / EXCERPT: Let $\overline{G}$ be a covering group with subgroup $\overline{G}_0$, let $\overline{G}^\dag=\overline{Z(G)}\cdot\overline{G}_0$, and let ϕ_π be an L-parameter of...

conjecture | open | MathDB 页面 | 原文线索

V2-014138 | MATHDB-349794

Metaplectic tensor product and restriction parameter conjecture

摘录 / EXCERPT: Let $\overline{M}_{\mathbf r}$ be the covering Levi subgroup with dual inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-014139 | MATHDB-349800

Przebinda's character correspondence conjecture via the Chc-star transform

摘录 / EXCERPT: Let $({\rm G},{\rm G}')$ be an irreducible dual pair in ${\rm Sp}(W)$ with ${\rm rk}({\rm G})\leq {\rm rk}({\rm G}')$. Let ${\rm Chc}^{*}$ be the map transferring invariant distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-014140 | MATHDB-349838

Model-theoretic construction conjecture for absolutely profinitely rigid groups

摘录 / EXCERPT: An absolutely profinitely rigid group is a group determined up to isomorphism by its profinite completion among the groups under consideration. Infinite-family construction conject...

conjecture | open | MathDB 页面 | 原文线索

V2-014141 | MATHDB-349860

The local Gross-Prasad multiplicity-one and epsilon-dichotomy conjectures

摘录 / EXCERPT: Let (G,H,ξ) be a Gross-Prasad triple attached to an admissible pair (W,V) over F. For a generic local L-parameter φ of G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014142 | MATHDB-349861

The Gross-Prasad parabolic-induction multiplicity reduction conjecture

摘录 / EXCERPT: Let (W,V) be an admissible pair. Let $\pi_V\in \Pi_{\mathrm{CW}}(\operatorname{SO}(V))$ and $\pi_W\in \Pi_{\mathrm{CW}}(\operatorname{SO}(W))$, possibly reducible, and suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-014143 | MATHDB-349862

The Gross-Prasad multiplicity comparison inequality

摘录 / EXCERPT: Let $\pi_V\in\Pi_{\mathrm{CW}}(\operatorname{SO}(V))$ and $\pi_W\in\Pi_{\mathrm{CW}}(\operatorname{SO}(W))$. Let X^+ be the space defining $\operatorname{GL}(X^+)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014144 | MATHDB-349920

Independence-of-ℓ conjecture for the spectral Bernstein center map

摘录 / EXCERPT: Let G be a reductive group over the local field E, let W_E be its Weil group, and let $\mathcal Z^{\mathrm{spec}}(G,\mathbb Q)$ be the natural rational form of the spectral B...

conjecture | open | MathDB 页面 | 原文线索

V2-014145 | MATHDB-349930

The generalized Sims conjecture for finite primitive groups

摘录 / EXCERPT: Generalized Sims conjecture. There exists a function $g:\mathbb{N}\to\mathbb{N}$ such that, if G is a finite primitive group with a suborbit β^{G_α} of length $d>1$, t...

conjecture | solved | MathDB 页面 | 原文线索

V2-014146 | MATHDB-349937

Filtration conjecture for standard projective modules

摘录 / EXCERPT: Filtration conjecture. For every $\lambda\in X(T)$, both $\widehat Q^I_{\chi}(\lambda)$ and $\widehat Q^{w^I}_{\chi}(\lambda^{w^I})$ have an $\mathcal L_\chi$-filtration.

conjecture | open | MathDB 页面 | 原文线索

V2-014147 | MATHDB-349938

Loewy-length conjecture for projective modules in standard Levi form

摘录 / EXCERPT: Projective-module Loewy-length conjecture. If χ has standard Levi form with $I=\Pi$ and λ is p-regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-014148 | MATHDB-349939

Conjecture on Loewy lengths of standard and projective modules

摘录 / EXCERPT: Loewy-length conjecture. Assume that χ has standard Levi form. When $\lambda\in X(T)$ is p-regular,

conjecture | open | MathDB 页面 | 原文线索

V2-014149 | MATHDB-349965

Compactness conjecture for Hecke operators

摘录 / EXCERPT: Let G be a reductive group, let X be a curve over a local field F, and let $\lambda\ne0$. Write $V_G(\lambda)$ for the dense domain on which the Hecke operators are initially...

conjecture | open | MathDB 页面 | 原文线索

V2-014150 | MATHDB-350001

The two-element base conjecture for saturated fusion systems

摘录 / EXCERPT: The two-element base conjecture for saturated fusion systems. Every saturated fusion system has a base of size 2.

conjecture | open | MathDB 页面 | 原文线索

V2-014151 | MATHDB-350028

Ostrik's fiber-functor conjecture for semisimple symmetric tensor categories

摘录 / EXCERPT: Ostrik's conjecture. The category $\mathcal{C}$ admits a fiber functor

conjecture | open | MathDB 页面 | 原文线索

V2-014152 | MATHDB-350029

The odd-torsion-free semisimplification conjecture for finite 2-groups

摘录 / EXCERPT: Benson's strengthened conjecture. One always has

conjecture | open | MathDB 页面 | 原文线索

V2-014153 | MATHDB-350030

Benson's normalizer-quotient conjecture for semisimplified representation categories

摘录 / EXCERPT: Benson's normalizer-quotient conjecture. The group Γ is a quotient of

conjecture | open | MathDB 页面 | 原文线索

V2-014154 | MATHDB-350061

Solvability and affinity conjectures for finite superconnected quandles

摘录 / EXCERPT: Superconnected quandle conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-014155 | MATHDB-350074

Isaacs-Navarro refinement of Späth's Character Triple Conjecture

摘录 / EXCERPT: Retain the notation above, and let $r(\vartheta)$ denote the residue of a character value as used in the source. Let Ω be a bijection from ${\mathcal{C}}^d(B,Z)_+/G$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-014156 | MATHDB-350101

Tame irreducible super Yangian modules are thin

摘录 / EXCERPT: Let $\mathrm{Y}(\mathfrak{gl}_{m|n})$ be the super Yangian. A finite-dimensional irreducible module is tame and thin according to the weight-space conditions for...

conjecture | open | MathDB 页面 | 原文线索

V2-014157 | MATHDB-350124

Macdonald module existence and eigenproperty conjecture

摘录 / EXCERPT: Let P be the weight lattice, let $\lambda=u_\lambda\lambda_{-}$ with λ_{-} antidominant and $u_\lambda\in W$ the shortest element in the finite Weyl group, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-014158 | MATHDB-350157

Global coherence conjecture for wavefront sets of automorphic representations

摘录 / EXCERPT: Global coherence of wavefront sets. There should exist a coadjoint orbit depending on π,

conjecture | open | MathDB 页面 | 原文线索

V2-014159 | MATHDB-350172

The bounded-length brick conjecture for tau-tilting infinite algebras

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra, and let $\operatorname{mod} \Lambda$ denote its category of finite-dimensional right Λ-modules. A module is a brick if its en...

conjecture | open | MathDB 页面 | 原文线索

V2-014160 | MATHDB-350184

Lusztig's parameter and label conjecture for Bernstein blocks

摘录 / EXCERPT: Lusztig's conjecture. (i) The q-parameters $q_\alpha,q_{\alpha *}$ are powers of q_F, except that for a short root α in a type B_n root system they can also be powers...

conjecture | open | MathDB 页面 | 原文线索

V2-014161 | MATHDB-350193

Li's vanishing conjecture for simple modules in the principal block

摘录 / EXCERPT: Li's conjecture. There exists $i\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014162 | MATHDB-350207

Donkin's tilting module conjecture

摘录 / EXCERPT: Let G be a reductive algebraic group over a field k of characteristic p, let G_rT be the corresponding Frobenius kernel together with a maximal torus, and let X_r denote...

conjecture | open | MathDB 页面 | 原文线索

V2-014163 | MATHDB-350212

Aroca-Cumplido finite-presentation conjecture for labeled braided Higman-Thompson groups

摘录 / EXCERPT: Let $d\geq 2$ and $r\geq 1$, let B_d be the braid group with d strings, and let $H\leq B_d$ be a finitely presented subgroup. The group $bV_{d,r}(H)$ is the labeled braided Hig...

conjecture | open | MathDB 页面 | 原文线索

V2-014164 | MATHDB-350234

Untangling conjecture for admissible qq-characters

摘录 / EXCERPT: Let $c7$ be an admissible qq-character with a dominant monomial. Its graph may be equipped with a supplement satisfying the compatibility condition around every square, and an in...

conjecture | open | MathDB 页面 | 原文线索

V2-014165 | MATHDB-350259

The bounded isomorphism conjecture for box spaces of residually finite groups

摘录 / EXCERPT: Let G be a residually finite group and let (G_n) be a nested sequence of finite-index normal subgroups with trivial intersection. The associated box space is the coarse disjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-014166 | MATHDB-350291

Conjecture on the asymptotic sharp threshold for Property (T) in random groups

摘录 / EXCERPT: Asymptotic threshold conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-014167 | MATHDB-350312

Hamiltonian construction conjecture for Pauli-group extensions of the quaternion group

摘录 / EXCERPT: Hamiltonian construction conjecture. Groups of this form may admit a construction of the Hamiltonian analogous to the construction of H_{eff}, such that passing from the larger g...

conjecture | open | MathDB 页面 | 原文线索

V2-014168 | MATHDB-350330

The Alvarez et al. missing-piece conjecture for locally discrete circle groups

摘录 / EXCERPT: Let $G\subset \operatorname{Diff}_+^\omega(\mathbb S^1)$ be a locally discrete, finitely generated subgroup. The dichotomy in the classification theorem stated immediately before t...

conjecture | open | MathDB 页面 | 原文线索

V2-014169 | MATHDB-350354

The double coinvariant algebra conjecture for the center of the small quantum group of type A

摘录 / EXCERPT: Let $\mathfrak{u}_0(\mathfrak{sl}_m)$ be the small quantum group at the relevant root of unity, let $\mathbf{z}(\mathfrak{u}_0(\mathfrak{sl}_m))$ denote its center, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014170 | MATHDB-350355

The invariant-center conjecture for semisimple small quantum groups

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra with Weyl group W, let $\mathfrak{u}_0(\mathfrak{g})$ be the corresponding small quantum group, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014171 | MATHDB-350356

The invariant-center conjecture for small quantum groups of type A

摘录 / EXCERPT: Let $\mathfrak{u}_q(\mathfrak{sl}_n)$ be the small quantum group of type A_{n-1}, let $\mathbf{z}(\mathfrak{u}_q(\mathfrak{sl}_n))$ be its center, and let the superscript...

conjecture | open | MathDB 页面 | 原文线索

V2-014172 | MATHDB-350357

The Coxeter divisibility conjecture for nontrivial center representations

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra with Coxeter number h, let $\mathfrak{u}_q(\mathfrak{g})$ be the corresponding small quantum group, and let L be an irreducible non-trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-014173 | MATHDB-350394

The permanent moment growth conjecture for complex Gaussian matrices

摘录 / EXCERPT: The permanent moment growth conjecture. The ratio $R_{k,t}$ is always smaller than 2 and quickly asymptotes to a constant C as both k and t increase. This conjecture is mot...

conjecture | open | MathDB 页面 | 原文线索

V2-014174 | MATHDB-350423

Exhaustion conjecture for simple bounded cuspidal affine modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be the basic classical Lie superalgebra, let $L(\mathfrak{g})$ be its loop superalgebra, and let $\widehat{\mathfrak{g}}$ be the corresponding affine Lie superal...

conjecture | open | MathDB 页面 | 原文线索

V2-014175 | MATHDB-350458

Positive rank gradient vanishes for residually finite groups satisfying a law

摘录 / EXCERPT: Let Γ be a finitely generated residually finite group satisfying a non-trivial group law. The rank gradient $RG(\Gamma)$ measures the asymptotic normalized growth of the min...

conjecture | open | MathDB 页面 | 原文线索

V2-014176 | MATHDB-350477

The finite Morley rank analogue of Glauberman's theorem for twisted subgroups

摘录 / EXCERPT: Finite Morley rank Glauberman conjecture. The group G is solvable.

conjecture | open | MathDB 页面 | 原文线索

V2-014177 | MATHDB-350481

Green-Liebeck code conjecture for conjugacy classes in symmetric groups

摘录 / EXCERPT: Let $n>2k$ and let j satisfy $2^j\leq k<2^{j+1}$. Let S_n be the symmetric group, let $X=x^{S_n}$ be the conjugacy class of x in S_n, and let Y_k denote the code subset f...

conjecture | open | MathDB 页面 | 原文线索

V2-014178 | MATHDB-350493

Non-perfectness conjecture for generating graphs of finite simple groups

摘录 / EXCERPT: Non-perfectness conjecture. The generating graph $\Gamma(G)$ is not perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-014179 | MATHDB-350494

Induced 5-hole conjecture for generating graphs of finite simple groups

摘录 / EXCERPT: Induced 5-hole conjecture. There exists a subset X of G such that the subgraph of $\Gamma(G)$ induced by X is a 5-hole.

conjecture | open | MathDB 页面 | 原文线索

V2-014180 | MATHDB-350522

Archimedean theta-packet conjecture

摘录 / EXCERPT: Let F be a real or complex local field, let G be an even orthogonal or unitary group over F, and let ψ be a local A-parameter with bounded image on the Weil group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014181 | MATHDB-350554

Unimodality conjecture for syzygy statistics of products of projective lines

摘录 / EXCERPT: Unimodality conjecture. If either d_1 or d_2 is sufficiently large, then each of the following functions is unimodal:

conjecture | open | MathDB 页面 | 原文线索

V2-014182 | MATHDB-350572

The measurable cocycle quotient conjecture for countable abelian group actions

摘录 / EXCERPT: Measurable cocycle quotient conjecture. For any measurable map $\omega\mapsto \rho_\omega$ from Ω to $Z^1_{<m}(G,X,S^1)$, there exists a set of positive measure...

conjecture | open | MathDB 页面 | 原文线索

V2-014183 | MATHDB-350582

The Gaudin algebra image conjecture for normalised polynomials

摘录 / EXCERPT: Let $p\in\Bbbk[t]$ be a normalised polynomial of degree n, let ψ_p be the homomorphism associated with p, and let ${\EuScript Z}(\widehat{\mathfrak q},t)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-014184 | MATHDB-350591

Equal-characteristic homological vanishing conjecture for VI_q modules

摘录 / EXCERPT: Let k be an arbitrary field, and let M be a VI_q module that is a subquotient of a module generated in degree d. For the representable kernel functor...

conjecture | open | MathDB 页面 | 原文线索

V2-014185 | MATHDB-350743

Sarnak-Xue multiplicity conjecture for unitary representations

摘录 / EXCERPT: Sarnak-Xue conjecture. The multiplicity of π satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014186 | MATHDB-350806

The WKB crystal structure conjecture for shift-of-argument eigenlines

摘录 / EXCERPT: Let $L(\lambda)$ be a finite-dimensional representation of $U(\mathfrak{gl}_n)$, let $u\in\widehat{\mathfrak h_{\rm reg}}(\mathbb{R})$, and let $E(\lambda,u)$ be the set of eigenli...

conjecture | open | MathDB 页面 | 原文线索

V2-014187 | MATHDB-350835

The conjecture on Morava character formulas for reduced weights

摘录 / EXCERPT: Character-formula conjecture. For every $w\in X^+_{\operatorname{red}}$, there exists such a γ for which

conjecture | open | MathDB 页面 | 原文线索

V2-014188 | MATHDB-350836

The general Morava character-formula conjecture

摘录 / EXCERPT: General character-formula conjecture. For general w, there exists a group homomorphism γ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014189 | MATHDB-350856

First Brauer-Thrall conjecture for finite-dimensional algebras

摘录 / EXCERPT: Let A be an algebra, and let τ-tilting theory be considered over a field $\mathbb{K}$. An algebra is of finite representation type when it has only finitely many isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-014190 | MATHDB-350859

Boston-Jones conjecture on settled elements in quadratic arboreal representations

摘录 / EXCERPT: Let $f(x)\in\mathbb{Q}[x]$ be a quadratic polynomial, let $\alpha\in\mathbb{Q}$, let T be the binary tree, and let ... be the associated arboreal representation. An element of its...

conjecture | open | MathDB 页面 | 原文线索

V2-014191 | MATHDB-350914

Madlener and Otto's plain group conjecture

摘录 / EXCERPT: Let G be a group. Say that G is plain if it is isomorphic to a free product of finitely many finite groups and finitely many copies of $\mathbb Z$. A finite convergent length-r...

conjecture | open | MathDB 页面 | 原文线索

V2-014192 | MATHDB-350936

Cubulation conjecture for closed hyperbolic manifolds of low dimension or simplest arithmetic type

摘录 / EXCERPT: Cubulation conjecture. Assume that either $n \leq 3$ or that M is arithmetic of simplest type. Then, for every $\lambda > 1$, there is a compact cubical model X such that there...

conjecture | open | MathDB 页面 | 原文线索

V2-014193 | MATHDB-350946

Cherlin's conjecture on finite primitive binary permutation groups

摘录 / EXCERPT: Let G be a permutation group on a set Ω. For a positive integer n and tuples $I=(\omega_1,\ldots,\omega_n)$ and J in Ω^n, write...

conjecture | open | MathDB 页面 | 原文线索

V2-014194 | MATHDB-350948

Generalized Knight conjecture for first-order sentences in random groups

摘录 / EXCERPT: Let $d<1/16$, and let Γ_ℓ be a random group of density d at length ℓ. A first-order sentence is almost surely true in this model if the probability that it holds i...

conjecture | open | MathDB 页面 | 原文线索

V2-014195 | MATHDB-350961

Countable-group slenderness coincidence conjecture

摘录 / EXCERPT: Countable-group slenderness coincidence conjecture. For every countable group, the classes of slender, inverse limit slender, n-slender, lch-slender and cm-slender groups coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-014196 | MATHDB-350969

The geometric model conjecture for dimer tree algebras

摘录 / EXCERPT: Let B be a dimer tree algebra, let $\mathcal{S}$ be its checkerboard polygon with $2N$ vertices, and let γ be a 2-diagonal. For the projective modules $P_0(\gamma)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014197 | MATHDB-350970

The tau-rigidity conjecture for syzygies over dimer tree algebras

摘录 / EXCERPT: Let B be a 2-Calabi-Yau tilted algebra whose quiver satisfies the dimer-tree condition, and let τ_B denote the Auslander-Reiten translation in $\operatorname{mod}B$. Tau-rig...

conjecture | open | MathDB 页面 | 原文线索

V2-014198 | MATHDB-350986

Anderson-Kogan conjecture on cluster monomials in the MV basis

摘录 / EXCERPT: Anderson-Kogan conjecture. The cluster monomials are contained in the MV basis.

conjecture | open | MathDB 页面 | 原文线索

V2-014199 | MATHDB-351046

Duality involution conjecture for Whittaker modules

摘录 / EXCERPT: Duality involution conjecture. If $X\in\mathcal{N}_\eta$ and $ \beta\in\mathbb{C}[W_\eta]$ is nonzero, then $X^{\vee_\beta}\in\mathcal{N}_\eta$ and

conjecture | open | MathDB 页面 | 原文线索

V2-014200 | MATHDB-351048

Injectivity conjecture for the Whittaker pairing map

摘录 / EXCERPT: Assume that $ \eta$ is a nondegenerate character, let W be the Weyl group, and let $ \mathcal{Z}_\mathfrak{g}$ be the category of modules used in the Whittaker induction functor...

conjecture | open | MathDB 页面 | 原文线索

V2-014201 | MATHDB-351049

Cyclic decomposition conjecture for Whittaker-induced modules

摘录 / EXCERPT: Let $ \mathfrak{g}$ be the Lie algebra under consideration, let $ Z(\mathfrak{g})$ be its center, and let U be a finite-dimensional $Z(\mathfrak{g})$-module. Let $I_\eta(U)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-014202 | MATHDB-351050

Contravariant-form and Whittaker-vector isomorphism conjecture

摘录 / EXCERPT: Let $ \eta$ be a character of the nilpotent subalgebra, let $U\in\mathcal{Z}_\mathfrak{g}$, and let $I_\eta(U)$ be the associated Whittaker-induced module. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-014203 | MATHDB-351053

The homotopy type conjecture for arithmetic manifolds

摘录 / EXCERPT: Let X be a symmetric space of non-compact type. An arithmetic X-manifold is an arithmetic manifold locally modeled on X, and a family has uniform homotopy complexity if there...

conjecture | open | MathDB 页面 | 原文线索

V2-014204 | MATHDB-351073

Single-block generation conjecture for q-character polynomials

摘录 / EXCERPT: Let J_a denote a unipotent Jordan block of size a. For a partition $\lambda=(\lambda_1\geq\cdots\geq\lambda_k)$, let J_λ be the unipotent Jordan matrix with block sizes...

conjecture | open | MathDB 页面 | 原文线索

V2-014205 | MATHDB-351074

Block-number filtration conjecture for q-character polynomial products

摘录 / EXCERPT: Block-number filtration conjecture. The only nonzero coefficients $R_{\lambda,\mu}^\nu(q)$ occur for partitions ν having at most $k+\ell$ parts. This conjecture asserts a filtr...

conjecture | open | MathDB 页面 | 原文线索

V2-014206 | MATHDB-351075

Product expansion conjecture for a single and a double Jordan block

摘录 / EXCERPT: For a positive integer a, let J_a be a unipotent Jordan block of size a, and let $J_{a,b}$ denote the unipotent Jordan matrix with Jordan block sizes a and b. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014207 | MATHDB-351076

Product expansion conjecture for three distinct Jordan block sizes

摘录 / EXCERPT: Let J_a denote a unipotent Jordan block of size a, let $J_{a,b,c}$ denote the unipotent Jordan matrix with the indicated block sizes, and let X_{J_λ} be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-014208 | MATHDB-351115

Conjecture on the unramified Langlands correspondence and elliptic pairings

摘录 / EXCERPT: Let G be a reductive group, let ${\mathcal C}(G^\vee)_{\mathsf{ss}}$ be the set of semisimple conjugacy classes in the dual group, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-014209 | MATHDB-351132

Zi's classification conjecture for simple transitive 2-representations

摘录 / EXCERPT: Let $\sc\text{B}\hspace{1.0pt}_{n}^{\operatorname{str}}$ be the finitary bicategory under consideration, and let simple transitive 2-representations of it be considered up to equ...

conjecture | open | MathDB 页面 | 原文线索

V2-014210 | MATHDB-351133

The principal-block dimension conjecture for -spetses

摘录 / EXCERPT: Principal-block dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014211 | MATHDB-351134

The projective-degree divisibility conjecture for principal blocks of spetses

摘录 / EXCERPT: Let $\mathbb{G}=(W\varphi^{-1},L)$ be as in the principal-block dimension conjecture, and suppose that W is an ℓ'-group, φ is trivial, and $q\equiv1\pmod\ell$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014212 | MATHDB-351136

The freeness conjecture for Yokonuma type algebras

摘录 / EXCERPT: Let $\mathcal Y$ be the Yokonuma type algebra attached to the finite ℓ-adic reflection group under consideration, defined over its base ring $\widehat A$. Let T be the finit...

conjecture | open | MathDB 页面 | 原文线索

V2-014213 | MATHDB-351141

Morse-element characterization for irreducible Artin-Tits groups of spherical type

摘录 / EXCERPT: Let A be an irreducible Artin-Tits group of spherical type, let $Z(A)$ denote its center, and let $a\in A$. An element a is Morse if its orbit in a Cayley graph of A is a Mor...

conjecture | open | MathDB 页面 | 原文线索

V2-014214 | MATHDB-351151

The finitely generated subgroup recognition conjecture for compact manifolds

摘录 / EXCERPT: Let M and N be compact connected C^p manifolds, and let $\Diff^p_c(M)_0$ and $\Diff^p_c(N)_0$ denote their groups of compactly supported C^p diffeomorphisms diffeotopic to...

conjecture | open | MathDB 页面 | 原文线索

V2-014215 | MATHDB-351192

The τ-perpendicular subcategory inheritance conjecture

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra. Let $\mathcal W\subseteq\mathsf{mod}\Lambda$ be a τ-perpendicular subcategory of $\mathsf{mod}\Lambda$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014216 | MATHDB-351207

Gurevich-Howe exhaustion conjecture for low U-rank representations of GL(n)

摘录 / EXCERPT: Gurevich-Howe exhaustion conjecture. Suppose $k<\left\lfloor\frac{n}{2}\right\rfloor$. Then, up to twist by a character, every irreducible representation of U-rank k of GL_n...

conjecture | open | MathDB 页面 | 原文线索

V2-014217 | MATHDB-351209

Benson's conjecture on the semisimplification of finite 2-group representations

摘录 / EXCERPT: Benson's conjecture. If $\bold G$ is a finite 2-group, then $\Gamma_{\bold G}=1$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-014218 | MATHDB-351210

Benson's conjecture for semisimplified representations of finite groups

摘录 / EXCERPT: Benson's conjecture. This natural surjective homomorphism is an isomorphism:

conjecture | open | MathDB 页面 | 原文线索

V2-014219 | MATHDB-351211

Benson's tensor-power conjecture for finite 2-groups

摘录 / EXCERPT: Let $\bold G$ be a finite 2-group, let V be a nontrivial indecomposable representation of $\bold G$ over a field $\bold k$ of characteristic 2, and let m be an odd positive...

conjecture | open | MathDB 页面 | 原文线索

V2-014220 | MATHDB-351223

Newell-Littlewood saturation conjecture

摘录 / EXCERPT: NL-saturation conjecture. The stated equivalence holds for every $(\lambda,\mu,\nu)\in({\operatorname{Par}}_n)^3$. This is the proposed analogue of the Littlewood-Richardson satura...

conjecture | open | MathDB 页面 | 原文线索

V2-014221 | MATHDB-351224

Extended Horn cone conjecture for Newell-Littlewood coefficients

摘录 / EXCERPT: Extended Horn cone conjecture. If $(\lambda,\mu,\nu)\in\overline{{\operatorname{EH}}}(n)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014222 | MATHDB-351276

Tait's perfect-difference-set extension conjecture

摘录 / EXCERPT: Tait's perfect-difference-set extension conjecture. The set S is contained in some perfect difference set S' of size $p+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-014223 | MATHDB-351299

Simplicity conjecture for generic radical central characters

摘录 / EXCERPT: Generic-character simplicity conjecture. The module $Q(0,\chi)$ is simple.

conjecture | open | MathDB 页面 | 原文线索

V2-014224 | MATHDB-351326

Gromov's boundedness conjecture for degree-two cohomology

摘录 / EXCERPT: Let X be a closed Riemannian manifold, and let $\alpha \in H^2(X;\mathbb{R})$. The class α is $\widetilde{d}$-bounded if its pullback to the universal cover...

conjecture | open | MathDB 页面 | 原文线索

V2-014225 | MATHDB-351332

The Dirac-index dichotomy conjecture

摘录 / EXCERPT: Let G be an equal-rank real reductive group with maximal compact subgroup K, and let $\widetilde K$ be the spin double cover of K. For an irreducible unitary...

conjecture | open | MathDB 页面 | 原文线索

V2-014226 | MATHDB-351333

The HP-condition unitarity conjecture for cohomological induction

摘录 / EXCERPT: Let G be a real reductive group with Lie algebra $\mathfrak{g}$ and maximal compact subgroup K, and let π be an irreducible unitary $(\mathfrak{g},K)$ module with infinites...

conjecture | open | MathDB 页面 | 原文线索

V2-014227 | MATHDB-351371

Borovik-Cherlin conjecture on generically transitive actions on connected abelian groups

摘录 / EXCERPT: Borovik-Cherlin's conjecture. The group V has the structure of an n-dimensional vector space over an algebraically closed field F of Morley rank 1, and

conjecture | open | MathDB 页面 | 原文线索

V2-014228 | MATHDB-351372

Borovik-Cherlin projective linearity conjecture for generically multiply transitive actions

摘录 / EXCERPT: Borovik-Cherlin's projective linearity conjecture. The pair (G,Ω) is equivalent to

conjecture | open | MathDB 页面 | 原文线索

V2-014229 | MATHDB-351382

Stable-vector embedding conjecture for dual Vinberg representations of type F_4

摘录 / EXCERPT: Let $\mathcal{G}$ be a group of type F_4, let x_0 be the point used to define the dual Vinberg representations, let h be the Coxeter number of $\mathcal{G}$, and for each pos...

conjecture | open | MathDB 页面 | 原文线索

V2-014230 | MATHDB-351391

The irreducible-component conjecture for symmetric Seesaw representations

摘录 / EXCERPT: Irreducible-component conjecture. If I is an irreducible component of $R(\mathcal{A},V)^{\langle-,-\rangle,\varepsilon}$ and $M,N\in I$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014231 | MATHDB-351392

The type C induction conjecture for Borel orbit closures

摘录 / EXCERPT: Type C induction conjecture. In type C, Borel orbit closures are induced by type A.

conjecture | open | MathDB 页面 | 原文线索

V2-014232 | MATHDB-351401

Commanding conjecture for virtually special cubulated groups

摘录 / EXCERPT: Let $G\curvearrowright X$ be a virtually special cubulated group, and let a collection of subgroups of G be almost malnormal if distinct subgroups in the collection have finite-i...

conjecture | open | MathDB 页面 | 原文线索

V2-014233 | MATHDB-351431

Semisimplicity conjecture for the category KL_k

摘录 / EXCERPT: Semisimplicity conjecture. The category KL_k is semisimple if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014234 | MATHDB-351439

Six-cycle spanning conjecture for the F4 diagrammatic category

摘录 / EXCERPT: Six-cycle spanning conjecture. For $m,n\in\mathbb{N}$, these component-planar graphs span

conjecture | open | MathDB 页面 | 原文线索

V2-014235 | MATHDB-351458

The centerless-group conjecture for the multiple holomorph

摘录 / EXCERPT: Let G be a group, and let $T(G)$ denote its multiple holomorph quotient. A group is elementary 2-abelian if all of its elements have order dividing 2, so that it is automatic...

conjecture | open | MathDB 页面 | 原文线索

V2-014236 | MATHDB-351459

The insolvable-group conjecture for the multiple holomorph

摘录 / EXCERPT: Let G be a finite insolvable group with no solvable normal subgroup as a direct factor, and let $T(G)$ denote its multiple holomorph quotient. A group is elementary 2-abelian i...

conjecture | open | MathDB 页面 | 原文线索

V2-014237 | MATHDB-351477

Maximal wave front partition conjecture for negative representations

摘录 / EXCERPT: Maximal wave front partition conjecture. If $G_{n^*}=\operatorname{Sp}_{2n^*}$ or O_{2n^*}, then

conjecture | open | MathDB 页面 | 原文线索

V2-014238 | MATHDB-351483

The quasi-freeness conjecture for graded symmetric-group harmonics

摘录 / EXCERPT: Let $\mathcal{R}_{k,n}$ and p_{α} be as in Section 2, with $d(\alpha)$ denoting the degree of p_{α}. Let $\mathcal{H}_{k,n}$ be the graded space of S_n-harmonics th...

conjecture | open | MathDB 页面 | 原文线索

V2-014239 | MATHDB-351497

Zhuang-Gao conjecture on the Gao constant

摘录 / EXCERPT: Let G be a finite group. Write ${\sf d}(G)$ for the smallest integer such that every sequence over G of length at least ${\sf d}(G)+1$ has a nonempty product-one subsequence, a...

conjecture | open | MathDB 页面 | 原文线索

V2-014240 | MATHDB-351519

Vanishing conjecture for higher localised \(\widehat{A}\)-genera

摘录 / EXCERPT: Vanishing conjecture for higher localised $\widehat{A}$-genera. If M admits a Γ-invariant Riemannian metric of positive scalar curvature, then for every closed m-form...

conjecture | open | MathDB 页面 | 原文线索

V2-014241 | MATHDB-351549

The cap product equality conjecture for right-angled Artin groups

摘录 / EXCERPT: Let A_Γ be a right-angled Artin group associated to a finite simple graph Γ. For a second homology class $\alpha\in H_2(A_\Gamma)$, let M_α denote the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-014242 | MATHDB-351558

Unitarity conjecture for special unipotent representations

摘录 / EXCERPT: Arthur's unitarity conjecture. Every special unipotent representation of $G_{\RR}$ is unitary.

conjecture | open | MathDB 页面 | 原文线索

V2-014243 | MATHDB-351559

Vogan's representation conjecture for real reductive groups

摘录 / EXCERPT: Vogan's conjecture. There is an isomorphism of K-representations

conjecture | open | MathDB 页面 | 原文线索

V2-014244 | MATHDB-351560

Cohen-Macaulay conjecture for Vogan's local-system sections

摘录 / EXCERPT: Cohen-Macaulay conjecture. The $\mathbb{C}[\OO_K]$-module

conjecture | open | MathDB 页面 | 原文线索

V2-014245 | MATHDB-351561

Refined BVLS duality conjecture

摘录 / EXCERPT: Refined BVLS duality conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-014246 | MATHDB-351629

Generation of finite simple groups by two Sylow subgroups

摘录 / EXCERPT: Let G be a finite simple group, and let r and s be primes dividing $|G|$. A Sylow-subgroup generation conjecture. There exist a Sylow r-subgroup P and a Sylow s-subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-014247 | MATHDB-351636

Finite cyclic quotient conjecture for C^1 actions of finite index subgroups

摘录 / EXCERPT: Finite cyclic quotient conjecture. Any such C^1 action of Γ on the circle factors through an action of a finite cyclic group.

conjecture | open | MathDB 页面 | 原文线索

V2-014248 | MATHDB-351735

Deligne-Simpson conjecture for Coxeter connections

摘录 / EXCERPT: Let G be a simple complex group with Lie algebra $\operatorname{\mathfrak{g}}$. Fix a Coxeter G-formal type $\mathscr{A}$ of slope r/h with $\gcd(r,h)=1$. A nilpotent orbit t...

conjecture | open | MathDB 页面 | 原文线索

V2-014249 | MATHDB-351769

DLS generation conjecture for outer automorphisms of special groups

摘录 / EXCERPT: Let G be a special group. A DLS automorphism is one of the automorphisms appearing in Theorems and; those appearing in Theorem are the relevant ones for...

conjecture | open | MathDB 页面 | 原文线索

V2-014250 | MATHDB-351816

The join expression conjecture for intersections of Verma modules

摘录 / EXCERPT: Let (W,S) be a Weyl group with Bruhat order. For $w\in W$, let $\mathbf{JM}(w)$ be the distinguished set of Bruhat-maximal join-irreducible elements whose join is w, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014251 | MATHDB-351864

Cyclic-vector conjecture for Bethe subalgebras in tame Yangian representations

摘录 / EXCERPT: Let $X\in\overline{M_{0,n+2}}$ and let $B(X)$ be the corresponding Bethe subalgebra of $Y(\mathfrak{gl}_n)$. A representation of $Y(\mathfrak{gl}_n)$ is tame if it has the form...

conjecture | open | MathDB 页面 | 原文线索

V2-014252 | MATHDB-351873

Fullness conjecture for webs of type P in positive characteristic

摘录 / EXCERPT: Let $\mathbbm{k}$ be a field, and let $\text{\boldmath$a$} \in \mathbbm{Z}^{r}$ and $\text{\boldmath$b$} \in \mathbbm{Z}^{s}$ be the parameters indexing the relevant objects, with...

conjecture | open | MathDB 页面 | 原文线索

V2-014253 | MATHDB-351886

The principal finite-bundle conjecture for the PCVP

摘录 / EXCERPT: Let Γ be a finite group and (E,X,π) be a principal Γ-bundle, where $(X,\widetilde{\mathbb{Z}}_X)$ is a ringed, connected, compact, Hausdorff, and PCVP space. Supp...

conjecture | open | MathDB 页面 | 原文线索

V2-014254 | MATHDB-351920

Polo's Schubert-character positivity conjecture

摘录 / EXCERPT: Let $\alpha,\beta\in\mathbb{N}^m$, and let κ_α and κ_β be Demazure characters. For a lower order ideal $\Gamma\subset\mathbb{N}^m$, let κ_Γ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-014255 | MATHDB-351946

The characteristic-cycle formula for parabolic Verma modules

摘录 / EXCERPT: Let G be the reductive group in the construction, with Borel subgroup B and parabolic subgroup P, and let $\mathscr W$ be its Weyl group. For $w\in\mathscr W$, let $w^{\max}$...

conjecture | open | MathDB 页面 | 原文线索

V2-014256 | MATHDB-351954

Gross-Reeder local Langlands parametrization conjecture for square-integrable representations

摘录 / EXCERPT: Gross-Reeder's parametrization conjecture. For every $\varphi\in\mathcal{D}_F(G)$, there exists a finite subset $\Pi_{\varphi}\subset\operatorname{Irr}_2(G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014257 | MATHDB-351955

Subgroup dichotomy for cyclically hyperbolic cubulable groups

摘录 / EXCERPT: Let G be a group acting geometrically on a CAT(0) cube complex. A group is cyclically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space with a lo...

conjecture | open | MathDB 页面 | 原文线索

V2-014258 | MATHDB-351960

Compatibility of close-field correspondences with local Langlands parameters

摘录 / EXCERPT: Let F and $F^\sharp$ be n-close local fields, let G be a split connected reductive group, and let ${\mathcal{A}}^n(G/F)$ denote the equivalence classes of irreducible admissi...

conjecture | open | MathDB 页面 | 原文线索

V2-014259 | MATHDB-351965

The Oh-Park denominator conjecture for Kirillov-Reshetikhin modules

摘录 / EXCERPT: Let I index the simple roots, let d_i be the symmetrizing integers, let r and $h^\vee$ be the parameters used in the paper, and let $\widetilde{c}_{ij}(u)$ be the coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-014260 | MATHDB-351966

The unrestricted generic-kernel divisor conjecture

摘录 / EXCERPT: Let $\mathfrak{O}(V^{(i)}_k,V^{(j)}_l)$ be the divisor of the normalized R-matrix denominator, and let $\dim_{q^{-1}}\mathop{\mathrm{ext}}^1_{\widetilde{\Pi}}$ denote the graded...

conjecture | open | MathDB 页面 | 原文线索

V2-014261 | MATHDB-351967

The shifted generic-kernel extension conjecture

摘录 / EXCERPT: Let $V^{(i)}_{k,q^p}$ denote the spectral shift of the Kirillov-Reshetikhin module $V^{(i)}_k$, and let $q^pK^{(i)}_k$ denote the corresponding grading shift of its generic kernel....

conjecture | open | MathDB 页面 | 原文线索

V2-014262 | MATHDB-351970

Rubin's conjecture on saturated transfer systems for cyclic groups

摘录 / EXCERPT: Fix positive integers $r_1,\ldots,r_k$, and let $p_1,\ldots,p_k$ be distinct primes. A saturated transfer system on a finite group is a transfer system satisfying the two-out-of-th...

conjecture | open | MathDB 页面 | 原文线索

V2-014263 | MATHDB-351997

The folklore conjecture on unramified extensions of quadratic number fields

摘录 / EXCERPT: Folklore conjecture. For any finite group G, there exist infinitely many quadratic number fields K such that K possesses a Galois extension with Galois group G unramified a...

conjecture | open | MathDB 页面 | 原文线索

V2-014264 | MATHDB-352048

Finite-index free-product conjecture for rational-map semigroups

摘录 / EXCERPT: Finite-index free-product conjecture. The semigroup S contains a finite-index subsemigroup which is a free product of its right amenable subsemigroups.

conjecture | open | MathDB 页面 | 原文线索

V2-014265 | MATHDB-352049

Free-product conjecture for cancellative rational-map semigroups

摘录 / EXCERPT: Cancellative free-product conjecture. The semigroup S is a free product of its amenable subsemigroups.

conjecture | open | MathDB 页面 | 原文线索

V2-014266 | MATHDB-352055

Finkelberg-Mirković conjecture for the principal block

摘录 / EXCERPT: Let G be as in the paper, let Gr° be the connected component of the affine Grassmannian containing the base point, and let I_{u} be the preimage of t...

conjecture | open | MathDB 页面 | 原文线索

V2-014267 | MATHDB-352056

Equivalence conjecture for the image of $\Phi_{y,A}$

摘录 / EXCERPT: Let A be a subset of the affine simple reflections and let y be an element of ${}^A W_{\mathrm{ext}}^S$; denote by $\mathsf{C}_{y,A}$ the Serre subcategory of...

conjecture | open | MathDB 页面 | 原文线索

V2-014268 | MATHDB-352087

The suborbit-index quantization conjecture for finite transitive groups

摘录 / EXCERPT: Suborbit-index quantization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014269 | MATHDB-352103

Symplectic duality cohomology and fixed-point coordinate-ring conjecture

摘录 / EXCERPT: Let $X\to X_0$ and $X^!\to X_0^!$ be symplectic dual conical symplectic resolutions, and let a torus $\mathbb{T}$ act on $X_0^!$. Denote by $H^*(X)$ the cohomology of X and by th...

conjecture | solved | MathDB 页面 | 原文线索

V2-014270 | MATHDB-352109

The conjecture that the Koszul complex mapping-cone theorem holds without acyclicity

摘录 / EXCERPT: Acyclicity conjecture. The theorem still holds if the acyclic assumption on Λ is dropped.

conjecture | open | MathDB 页面 | 原文线索

V2-014271 | MATHDB-352128

Non-unisingularity conjecture for the conjugate Specht module

摘录 / EXCERPT: Non-unisingularity conjecture. The module $\mathcal{S}^{(n-2,2)'}$ does not afford 1 as an eigenvalue on the class of such permutations. Every other element of S_n has 1 as a...

conjecture | open | MathDB 页面 | 原文线索

V2-014272 | MATHDB-352218

Gelfand-Kirillov conjecture for Lie algebras of linear algebraic groups

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra of a linear algebraic group over $\mathbb{C}$. For $r,s\in\mathbb{Z}_+$, let $\mathcal{D}_{r,s}=\mathcal{D}_r\otimes\mathbb{C}[y_1,\dots,y_s]$...

conjecture | open | MathDB 页面 | 原文线索

V2-014273 | MATHDB-352224

Connectedness and equidimensionality conjecture for enhanced Springer fibers

摘录 / EXCERPT: Connectedness and equidimensionality conjecture. The fiber ${{}^{\textsf{e}}\hskip-2pt\pi}_{\hskip-2pt\lambda[q]}^{-1}(Y,u)$ is connected, and all of its irreducible components hav...

conjecture | open | MathDB 页面 | 原文线索

V2-014274 | MATHDB-352236

Lusztig's denominator conjecture for affine Hecke algebras

摘录 / EXCERPT: Lusztig's denominator conjecture. For all $x,w\in W_{\mathrm{aff}}$, $a_{x,w}$ is a rational function of $\mathbf{q}$; its denominator is independent of x and, as a function of...

conjecture | open | MathDB 页面 | 原文线索

V2-014275 | MATHDB-352237

Cellwise Kazhdan-Lusztig classification at roots of the Poincaré polynomial

摘录 / EXCERPT: Let $\tilde{W}$ be an affine or extended affine Weyl group for which Lusztig's denominator conjecture holds, and let P_W be the Poincaré polynomial of its finite Weyl group. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-014276 | MATHDB-352248

The Tor property conjecture for U_k and partial Dyck path loci

摘录 / EXCERPT: Let $S=\mathbb{C}[x_1,\ldots,x_n]$, let U_k be the open locus defined in the paper, and let $U_{\pi'}=U_{\pi,l}$ be the smooth locus associated with a partial Dyck path...

conjecture | open | MathDB 页面 | 原文线索

V2-014277 | MATHDB-352269

The center conjecture for the affine partition algebra

摘录 / EXCERPT: Let k be a positive integer, let $\mathcal{A}_{2k}^{\text{aff}}$ be the affine partition algebra, and let z_l and p_n denote the central elements introduced in the paper for...

conjecture | open | MathDB 页面 | 原文线索

V2-014278 | MATHDB-352283

Uniform Fitting-height conjecture for fixed-point-free automorphisms satisfying polynomial identities

摘录 / EXCERPT: Uniform Fitting-height conjecture. For every $f(x)\in\mathbb{Z}[x]\setminus\{0\}$, there exist integers $k(f(x))>0$ and $m(f(x))$ such that, whenever G is a finite solvable group...

conjecture | open | MathDB 页面 | 原文线索

V2-014279 | MATHDB-352322

The local rigid-inner-form packet conjecture

摘录 / EXCERPT: Let F_v be a local completion of the global function field F, let G be a connected reductive group over F_v, and let W_{F_v}' be the Weil-Deligne group. For a tempered...

conjecture | open | MathDB 页面 | 原文线索

V2-014280 | MATHDB-352330

Stable-character expectation bound for word measures on general linear groups

摘录 / EXCERPT: Let K be a finite field with $|K|=q$, let $\mathbb{F}$ be a free group, and let $w\in\mathbb{F}$. Let $\mathcal{R}$ be the algebra of character-like functions on the groups...

conjecture | open | MathDB 页面 | 原文线索

V2-014281 | MATHDB-352361

The conjecture that every infinite cofinite Zariski group is abelian

摘录 / EXCERPT: Let G be an infinite group in the class ${\mathscr C}$ of groups with cofinite Zariski topology. Main conjecture. The group G is necessarily abelian; consequently, every infini...

conjecture | open | MathDB 页面 | 原文线索

V2-014282 | MATHDB-352384

Conjecture on character differences associated with Foulkes' conjecture

摘录 / EXCERPT: Let $\rho^{(a)}_b$ and $\chi^{(r)}$ denote the characters used in the source, let $\boxtimes$ denote the indicated outer product, and let $\operatorname{Char}(S_N)$ be the cone of...

conjecture | open | MathDB 页面 | 原文线索

V2-014283 | MATHDB-352407

Relative modularity conjecture for the category \(\mathcal{D}^{\aleph}\) of \(U_{\xi}^H\mathfrak{sl}(2|1)\)-modules

摘录 / EXCERPT: Let $U_{\xi}^H\mathfrak{sl}(2|1)$ be the quantum group under consideration, and let $\mathcal{D}^{\aleph}$ be the category defined from its module category by the construction desc...

conjecture | open | MathDB 页面 | 原文线索

V2-014284 | MATHDB-352417

The conjecture that the affine Hecke cocenter maps isomorphically to the p-adic cocenter

摘录 / EXCERPT: Let ${\bf H}=S_c^\infty(G)$ be the algebra of locally constant compactly supported functions on the p-adic group, let ${\bf J}$ be the corresponding algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014285 | MATHDB-352430

Beeker-Lazarovich boundary characterization conjecture

摘录 / EXCERPT: Beeker-Lazarovich conjecture. Under these assumptions,

open_problem | open | MathDB 页面 | 原文线索

V2-014286 | MATHDB-352431

Boundary Steenrod homology conjecture for hyperbolic-group boundaries

摘录 / EXCERPT: Boundary Steenrod homology conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014287 | MATHDB-352447

Hierarchical hyperbolicity conjecture for parabolically geometrically finite subgroups

摘录 / EXCERPT: Let S be a surface and let $G<\operatorname{Mod}(S)$ be parabolically geometrically finite, meaning that G is relatively hyperbolic relative to a possibly trivial collection of...

conjecture | open | MathDB 页面 | 原文线索

V2-014288 | MATHDB-352470

Main conjecture on computable functions on the space of marked groups

摘录 / EXCERPT: Main conjecture. Any Λ_{WP}-computable function defined on $\mathcal{G}_{WP}$, with image in any recursive metric space, is continuous.

conjecture | open | MathDB 页面 | 原文线索

V2-014289 | MATHDB-352472

Non-semi-decidability of LEF and isolated groups

摘录 / EXCERPT: LEF and isolated-groups conjecture. The set of LEF groups is not Λ_{WP}-co-semi-decidable, and the set of isolated groups is not Λ_{WP}-semi-decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-014290 | MATHDB-352473

Recognizability conjecture for marked isolated groups

摘录 / EXCERPT: Marked-isolated-groups conjecture. Marked isolated groups are the only marked groups that can be recognized for Λ_{WP}; equivalently, they are the only marked groups whose...

conjecture | open | MathDB 页面 | 原文线索

V2-014291 | MATHDB-352474

Solvable-word-problem preformation conjecture

摘录 / EXCERPT: Solvable-word-problem preformation conjecture. There exists a group H, non-isomorphic to G, with solvable word problem and rank at most $k+1$, such that G preforms H.

conjecture | open | MathDB 页面 | 原文线索

V2-014292 | MATHDB-352475

Finite groups as the only abstractly recognizable groups

摘录 / EXCERPT: Finite-recognizability conjecture. Finite groups are the only abstractly recognizable groups in $(\mathcal{G}_{WP},d,\nu_{WP})$.

conjecture | open | MathDB 页面 | 原文线索

V2-014293 | MATHDB-352477

Non-co-semi-decidability of LEF groups

摘录 / EXCERPT: LEF non-co-semi-decidability conjecture. The set of LEF groups is not Λ_{WP}-co-semi-decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-014294 | MATHDB-352506

Dual Gottschalk conjecture for post-surjective cellular automata

摘录 / EXCERPT: Let G be a group and let A be a finite set. Let $\tau\colon A^G\to A^G$ be a cellular automaton. It is post-surjective if, for every $x,y\in A^G$ such that y is asymptotic to...

conjecture | open | MathDB 页面 | 原文线索

V2-014295 | MATHDB-352536

Ingalls-Paquette self-orthogonality conjecture for primitive idempotents

摘录 / EXCERPT: Ingalls-Paquette's self-orthogonality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014296 | MATHDB-352608

Conjecture on transcendence of conjugacy growth series

摘录 / EXCERPT: Let G be a finitely presented group that is not virtually abelian, and let its conjugacy growth series be the formal power series whose coefficients are the values of the conjuga...

conjecture | open | MathDB 页面 | 原文线索

V2-014297 | MATHDB-352627

Nonvanishing conjecture for minimal generic tensor invariants

摘录 / EXCERPT: Let $\ell,m,n\transactionin eq 0$?

conjecture | open | MathDB 页面 | 原文线索

V2-014298 | MATHDB-352680

Nonexistence conjecture for simple sharply 2-transitive groups of finite Morley rank

摘录 / EXCERPT: Nonexistence conjecture. There is no simple sharply 2-transitive group of finite Morley rank.

conjecture | open | MathDB 页面 | 原文线索

V2-014299 | MATHDB-352703

Humphreys' conjecture on blocks for reduced enveloping algebras

摘录 / EXCERPT: Let G be a reductive algebraic group over an algebraically closed field of characteristic $p>0$, let ${\mathfrak g}$ be its Lie algebra, and let $\chi\in{\mathfrak g}^{*}$ be nil...

conjecture | open | MathDB 页面 | 原文线索

V2-014300 | MATHDB-352718

Kazhdan-Yom Din's asymptotic Schur orthogonality conjecture

摘录 / EXCERPT: Asymptotic Schur orthogonality relations. There exist $\mathbf{d}(V)\in{\mathbb Z}_{\ge 0}$ and $\mathbf{f}(V)\in{\mathbb R}_{>0}$ such that, for all $v_1,v_2,v_3,v_4\in V$,

conjecture | open | MathDB 页面 | 原文线索

V2-014301 | MATHDB-352725

The p-distinguished involution conjecture

摘录 / EXCERPT: The p-distinguished involution conjecture. In each left p-cell there is a unique involution d such that

conjecture | open | MathDB 页面 | 原文线索

V2-014302 | MATHDB-352731

Burgers-equation conjecture for character-measure limit shapes

摘录 / EXCERPT: Let $\mu_{n,k}(\lambda\mid\{x_i\}_{i=1}^n,\{y_j\}_{j=1}^k)$ be the character measure on partitions $\lambda\subseteq k^n$ defined using Schur functions, and take...

conjecture | open | MathDB 页面 | 原文线索

V2-014303 | MATHDB-352743

Generalized injectivity conjecture for minimal multisegments

摘录 / EXCERPT: Let ρ be a fixed cuspidal representation, let $d4c$ be an irreducible representation in $d4c\mathrm{rr}_{\rho}$, and let $\mathfrak n\in\mathrm{Mult}_{\rho}$ be minimal to...

conjecture | open | MathDB 页面 | 原文线索

V2-014304 | MATHDB-352759

Ghys's conjecture on nilpotent subgroups of finite diffeomorphism groups

摘录 / EXCERPT: Let M be a compact differentiable manifold, and let G be a finite subgroup of the diffeomorphism group of M. Ghys's conjecture. There is a constant I such that G has a ni...

conjecture | open | MathDB 页面 | 原文线索

V2-014305 | MATHDB-352760

Spherical Fundamental Local Equivalence

摘录 / EXCERPT: Let G be a reductive group, let $\check{G}$ be its Langlands dual group, and let κ and $\check{\kappa}$ be corresponding quantum parameters. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014306 | MATHDB-352783

Abhyankar's inertia conjecture

摘录 / EXCERPT: Abhyankar's inertia conjecture. The pair (G,I) is realizable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014307 | MATHDB-352794

Baer's conjecture on Noetherian group rings

摘录 / EXCERPT: Baer's conjecture. If $\mathbb{Z}G$ is Noetherian, then G is polycyclic-by-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-014308 | MATHDB-352817

Highest-weight compatibility conjecture for the Heckman-Opdam equivalence

摘录 / EXCERPT: Let c be a parameter for which the Heckman-Opdam functor F is an equivalence between the highest weight categories $\mathcal{O}_c$ and $\mathcal{O}_{\sigma(c)}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014309 | MATHDB-352823

Strong nonvanishing conjecture for normalized intertwining operators

摘录 / EXCERPT: Strong nonvanishing conjecture. The operator $M^*(s,\rho_c(\tau_a)\otimes \rho_d(\tau_b))$ is nonzero for every $s\in\mathbb{C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014310 | MATHDB-352828

The small cancellation torsion subgroup conjecture

摘录 / EXCERPT: Small cancellation torsion subgroup conjecture. Every torsion subgroup of G is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-014311 | MATHDB-352852

Nonvanishing conjecture for the normalized intertwining operator $M^*_{c,a}$

摘录 / EXCERPT: Let $M^*_{c,a}(s,\tau,\sigma_{\bar{r}})$ be the normalized intertwining operator introduced above, with $s\in\mathbb{C}$. Nonvanishing conjecture. The operator...

conjecture | open | MathDB 页面 | 原文线索

V2-014312 | MATHDB-352864

Nilpotent-piece conjecture for simple algebraic groups of type E7 and E8

摘录 / EXCERPT: Let G be a simple algebraic group of type E_7 or E_8 over an algebraically closed field of bad characteristic. Let the nilpotent pieces be defined with respect to weighted Dy...

conjecture | open | MathDB 页面 | 原文线索

V2-014313 | MATHDB-352884

Thick embedding conjecture for the solvable Heisenberg extension

摘录 / EXCERPT: Let H) be the Heisenberg group, and let $H\rtimes_{\psi}\mathbb{R}$ be its semidirect product with $\mathbb{R}$, where the action is given by ... An embedding is ε-thic...

conjecture | open | MathDB 页面 | 原文线索

V2-014314 | MATHDB-352892

The finite-index Anosov representation conjecture for cyclic doubles

摘录 / EXCERPT: Finite-index Anosov representation conjecture. There exists a finite-index subgroup Γ' of Γ containing w such that

conjecture | open | MathDB 页面 | 原文线索

V2-014315 | MATHDB-352924

Minimal-dimension conjecture for mapping class group actions on spheres

摘录 / EXCERPT: Let S_g be a closed orientable surface of genus $g\geq 3$, let $\Mod(S_g)$ denote its mapping class group, and let $\Homeo^+(\mathbb{S}^n)$ denote the group of orientation-preser...

conjecture | open | MathDB 页面 | 原文线索

V2-014316 | MATHDB-352966

Mirror-support and multiplicity conjectures for reductive Hitchin systems

摘录 / EXCERPT: Let $c\in C$ and let $\mathrm G$ be a reductive group. For $\mu\in X_+(\mathrm G^\vee)$, let ρ_μ be the corresponding irreducible representation of the Langlands dual group,...

conjecture | open | MathDB 页面 | 原文线索

V2-014317 | MATHDB-353003

The BMR freeness conjecture for Hecke algebras of complex reflection groups

摘录 / EXCERPT: BMR freeness conjecture. The algebra H is a free R-module of rank $|W|$.

conjecture | open | MathDB 页面 | 原文线索

V2-014318 | MATHDB-353004

The BMM symmetrising trace conjecture for Hecke algebras of complex reflection groups

摘录 / EXCERPT: BMM symmetrising trace conjecture. There exists a linear map $\tau:H\rightarrow R$ such that: (1) τ is a symmetrising trace, meaning that $\tau(h_1h_2)=\tau(h_2h_1)$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-014319 | MATHDB-353080

The Calogero-Moser and Kazhdan-Lusztig two-sided-cell conjecture

摘录 / EXCERPT: Let W be a finite real reflection group with parameter c. A Calogero-Moser two-sided c-cell is denoted by Γ, and let $\operatorname{Irr}_\Gamma^{c,\mathrm{CM}}(W)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014320 | MATHDB-353098

The bosonic and fermionic polynomial degree conjecture

摘录 / EXCERPT: Let k indistinguishable bosons have $G={\rm SU}(N)$ with $N={{n+k-1}\choose{k}}$, and let k indistinguishable fermions have $G={\rm SU}(K)$ with $K={{n}\choose{k}}$. In both ca...

conjecture | open | MathDB 页面 | 原文线索

V2-014321 | MATHDB-353101

Strong irreducibility conjecture for the quantum pseudo-Kähler plane representation

摘录 / EXCERPT: Let $\mathcal{C}_{\mathbf{q}}$ and $\mathcal{C}_{1/\mathbf{q}^\vee}$ be the two Hopf algebras represented on the Hilbert space $\mathscr{H}$ by $mathpi$, with the corresponding ope...

conjecture | open | MathDB 页面 | 原文线索

V2-014322 | MATHDB-353121

The TF-class canonical decomposition conjecture

摘录 / EXCERPT: TF-class canonical decomposition conjecture. For each $\theta\in K_0(\operatorname{\mathsf{proj}} A)$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-014323 | MATHDB-353122

The wall-dimension conjecture for TF equivalence classes

摘录 / EXCERPT: Wall-dimension conjecture. For each $\theta\in K_0(\operatorname{\mathsf{proj}} A)$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-014324 | MATHDB-353143

Martino's conjecture on Calogero-Moser and Rouquier families

摘录 / EXCERPT: Let W be a complex reflection group, let ${\mathcal{C}}$ be its parameter space, and let $\kappa:{\mathcal{C}}\rightarrow{\mathcal{K}}$ identify Calogero-Moser parameters with He...

conjecture | open | MathDB 页面 | 原文线索

V2-014325 | MATHDB-353145

Gordon-Martino conjecture for Calogero-Moser and Rouquier families

摘录 / EXCERPT: Let W be a complex reflection group, let ${\mathcal{C}}$ be its Calogero-Moser parameter space, and let $\kappa:{\mathcal{C}}\rightarrow{\mathcal{K}}$ be the parameter correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-014326 | MATHDB-353150

Fuchsian Nakayama category classification conjecture

摘录 / EXCERPT: Fuchsian-category conjecture. The following properties hold:

conjecture | open | MathDB 页面 | 原文线索

V2-014327 | MATHDB-353169

Prasad's multiplicity formula for Galois distinction

摘录 / EXCERPT: Let E/F be a quadratic extension with nontrivial Galois element $sigma$. Let G be a reductive group over F, let $G(E)$ be its base change, and let G^{op} be the qu...

conjecture | open | MathDB 页面 | 原文线索

V2-014328 | MATHDB-353170

Prasad's Whittaker-model expectation for distinguished representations

摘录 / EXCERPT: Let E/F be a quadratic extension, let G be a reductive group over F, and let N_0 be the unipotent subgroup associated with a fixed Whittaker datum. Let π be an irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-014329 | MATHDB-353207

Module-structure conjecture for perfect bases and affine Grassmannian cohomology

摘录 / EXCERPT: Module-structure conjecture. For every preprojective algebra module M of dimension vector ν, the direct sum

conjecture | open | MathDB 页面 | 原文线索

V2-014330 | MATHDB-353209

MV--theta basis coincidence conjecture for the unipotent coordinate ring

摘录 / EXCERPT: MV--theta coincidence conjecture. The MV and theta bases for $\mathbb C[N]$ coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-014331 | MATHDB-353267

The Amit-Ashurst conjecture for finite nilpotent groups

摘录 / EXCERPT: Amit-Ashurst conjecture. Let G be a finite nilpotent group. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014332 | MATHDB-353268

The non-exoticity conjecture for generalised block fusion systems

摘录 / EXCERPT: Non-exoticity conjecture. The generalised block fusion system $\mathcal F_{(P,e_P)}(H,G,b)$ is non-exotic.

conjecture | open | MathDB 页面 | 原文线索

V2-014333 | MATHDB-353292

Lower bounds for conjugacy depth functions of wreath products

摘录 / EXCERPT: Let A be a finitely generated abelian group and let G be a conjugacy separable group with separable cyclic subgroups. For groups with conjugacy depth function...

conjecture | open | MathDB 页面 | 原文线索

V2-014334 | MATHDB-353307

Higgs's conjecture on distinct character degrees above an invariant character

摘录 / EXCERPT: Higgs's conjecture. If all members of $\operatorname{Irr}(G\mid\theta)$ have distinct degrees, then $\operatorname{Irr}(G\mid\theta)$ has exactly one member; equivalently, there is...

conjecture | open | MathDB 页面 | 原文线索

V2-014335 | MATHDB-353308

Compatibility-frame representation conjecture for finite lattices

摘录 / EXCERPT: Compatibility-frame representation conjecture. There is a compatibility frame $(X,\vartriangleleft)$ with

conjecture | open | MathDB 页面 | 原文线索

V2-014336 | MATHDB-353310

The Grothendieck-group realization conjecture for the affine Hecke trace

摘录 / EXCERPT: Let $\mathcal{A}$ be the algebra generated by the elements $R_{\mathbf d}$ over $\mathbb{Z}[q_1^{\pm1},q_2^{\pm1}]$, and specialize its parameters by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014337 | MATHDB-353335

The divisibility conjecture for $T_y(j)$ in normal mixed Cayley graphs

摘录 / EXCERPT: The divisibility conjecture. For every $1\leq j\leq h$,

conjecture | open | MathDB 页面 | 原文线索

V2-014338 | MATHDB-353339

Tensor-product fixed-point conjecture for Coulomb branches

摘录 / EXCERPT: Let $W=W^1\oplus W^2$ be a decomposition inducing dominant coweights λ^1 and λ^2, let $\mathcal M^\nu(\lambda,\mu)$ be the corresponding deformation of the Coulomb...

conjecture | open | MathDB 页面 | 原文线索

V2-014339 | MATHDB-353368

Joseph's conjectures on limit points of commuting probabilities

摘录 / EXCERPT: Joseph's conjectures. One should have $\ell\in\mathbb{Q}$, $x_i\geq\ell$ for all but finitely many i, and $\ell\in S$. Equivalently, the second conjecture says that S is well-o...

conjecture | open | MathDB 页面 | 原文线索

V2-014340 | MATHDB-353381

Conjecture on low-dimensional irreducible characters of the symmetric group

摘录 / EXCERPT: Low-dimensional character conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014341 | MATHDB-353382

The orthogonal and exceptional groups conjecture on the action of the longest Weyl element

摘录 / EXCERPT: Assume that $\mathfrak{g}^{\mathbb{C}}$ is of type B_r for some $r\geq 1$, of type D_r for some $r\geq 3$, or of type E_6, E_7, E_8, F_4, or G_2. Let λ be a d...

conjecture | open | MathDB 页面 | 原文线索

V2-014342 | MATHDB-353403

Conjecture on the socle of tensor products of uniserial modules

摘录 / EXCERPT: Let $\mathfrak{g}_m=\mathfrak{sl}(2)\ltimes V(m)$, and let $V_1=V(a)\oplus V(b)$ and $V_2=V(c)\oplus V(d)$ be the socle decompositions of two uniserial $\mathfrak{g}_m$-modules. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-014343 | MATHDB-353407

Nielsen realization for aspherical closed manifolds with hyperbolic fundamental group

摘录 / EXCERPT: Let M be an aspherical closed manifold with hyperbolic fundamental group π. Let $j\colon G\to \operatorname{Out}(\pi)$ be an embedding of a finite group G into the outer au...

conjecture | open | MathDB 页面 | 原文线索

V2-014344 | MATHDB-353459

The invariant-dimension conjecture for regular representations

摘录 / EXCERPT: Let G and H be finite groups. Let V and V' be the regular representations of G and H, respectively. For a finite group K, write $\varphi_K(d)$ for the number of eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-014345 | MATHDB-353485

Non-residual-finiteness conjecture for random BMW groups

摘录 / EXCERPT: A BMW group is a group acting on a product of two regular trees, and a BMW group of degree (m,n) has tree factors of degrees m and n. In the range $n>m^5$, consider the rando...

conjecture | open | MathDB 页面 | 原文线索

V2-014346 | MATHDB-353500

P. Hall's profinite conciseness conjecture

摘录 / EXCERPT: Let $w(x_1,\dots,x_k)$ be a word in k variables, and let G be a profinite group. Write $w\{G\}$ for the set of word-values of w in G, and $w(G)=\langle w\{G\}\rangle$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-014347 | MATHDB-353515

Bapat-Sunder positive-definite-function conjecture

摘录 / EXCERPT: Let $\mathcal{S}_n$ be the symmetric group and let $c:\mathcal{S}_n\to\mathbb{C}$ satisfy ... for every complex-valued function x on $\mathcal{S}_n$. For $A=[a_{i,j}]\in{\cal H}_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-014348 | MATHDB-353517

Drury's partition-support conjecture for Schur power eigenvalues

摘录 / EXCERPT: Let Π be a partition of n, and let its Ferrers diagram be the associated diagram. For $A\in{\cal H}_n$, let the block of the Schur power matrix $\pi(A)$ associated with Π...

conjecture | open | MathDB 页面 | 原文线索

V2-014349 | MATHDB-353523

Higher Rank Flat Closing conjecture

摘录 / EXCERPT: Let X be a locally compact geodesically complete CAT(0) space with Tits diameter π, and suppose that X supports a geometric group action $\Gamma\curvearrowright X$. If X...

conjecture | open | MathDB 页面 | 原文线索

V2-014350 | MATHDB-353524

Lytchak's boundary-open-set conjecture for CAT(0) spaces

摘录 / EXCERPT: Let X be a locally compact CAT(0) space with a geometric group action. Write $\partial_T X$ for its Tits boundary. Lytchak's conjecture. Then $\partial_T X$ contains a relatively...

conjecture | open | MathDB 页面 | 原文线索

V2-014351 | MATHDB-353531

Type D_m cactus-generator conjecture for the crystal $B(n\omega_1)$

摘录 / EXCERPT: Type D_m cactus-generator conjecture. The action of the cactus group on $B(n\omega_1)$ is generated by the actions of cactus generators corresponding to one-element and two-eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-014352 | MATHDB-353544

Strengthened Saxl conjecture for symmetric and alternating squares

摘录 / EXCERPT: Let $n=\tfrac{1}{2}k(k+1)$ and let $\rho_k=(k,\ldots,2,1)\vdash n$ be the staircase partition. Write $S^2([\rho_k])$ and $A^2([\rho_k])$ for the symmetric and alternating parts of...

conjecture | open | MathDB 页面 | 原文线索

V2-014353 | MATHDB-353545

Multiplicity-free classification conjecture for symmetric and alternating Kronecker squares

摘录 / EXCERPT: Let $n\in\mathbb N$ and let $\lambda\vdash n$. A character is multiplicity-free when every irreducible constituent occurs with multiplicity at most one. The partitions listed in pa...

conjecture | open | MathDB 页面 | 原文线索

V2-014354 | MATHDB-353546

Refinement of the Heide-Saxl-Tiep-Zalesski conjecture for split squares

摘录 / EXCERPT: For $n\ge10$, let $P(n)$ be the set of partitions of n, and let $[\lambda]$ denote the irreducible character of S_n indexed by λ. Refinement of the HSTZ conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-014355 | MATHDB-353566

D'Adderio-Iraci-Vanden Wyngaerd conjecture for the supersymmetric double coinvariant ring

摘录 / EXCERPT: D'Adderio-Iraci-Vanden Wyngaerd conjecture. The space $(TDR_n)_{i,j}$ vanishes whenever $i+j\geq n$. If $i+j<n$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014356 | MATHDB-353586

Saturation of Demazure characters in type E

摘录 / EXCERPT: Saturation conjecture. For every $w\in W$ and every dominant λ, $\operatorname{char}(V_\lambda^w)$ is saturated.

conjecture | open | MathDB 页面 | 原文线索

V2-014357 | MATHDB-353651

Completeness conjecture for the Gaudin model associated with \mathfrak{gl}(1|1)

摘录 / EXCERPT: Completeness conjecture. The operator $\mathscr H(x)$ has simple spectrum on $(L_{\bm\Lambda}(\bm b))^{\mathrm{sing}}$. There is a bijection between the monic divisors y of...

conjecture | open | MathDB 页面 | 原文线索

V2-014358 | MATHDB-353652

Virtual torsion-freeness conjecture for generalized Bestvina-Brady groups

摘录 / EXCERPT: Let L be a flag complex, let M be a covering of L, let $S\subseteq\mathbb Z$ be periodic, and let $G_L^M(S)$ denote the associated generalized Bestvina-Brady group. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-014359 | MATHDB-353653

Virtual specialness conjecture for periodic generalized Bestvina-Brady groups

摘录 / EXCERPT: Let L be a flag complex, let M be a covering of L, let $S\subseteq\mathbb Z$ satisfy $S+n=S$ for some $n>0$, and let $G_L^M(S)\rtimes n\mathbb Z$ be the associated CAT(0) cub...

conjecture | open | MathDB 页面 | 原文线索

V2-014360 | MATHDB-353654

Iterated semidirect-product conjecture for braid-orbit permutation groups

摘录 / EXCERPT: Iterated semidirect-product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014361 | MATHDB-353657

Disjunction conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Disjunction conjecture for Deligne-Lusztig cohomology.

conjecture | open | MathDB 页面 | 原文线索

V2-014362 | MATHDB-353701

Ciobanu-Fine-Rosenberger's Surface Group Conjecture A

摘录 / EXCERPT: Surface Group Conjecture A. Let G be a residually finite Mel'nikov group. Then G is a surface group or $\operatorname{BS}(1,n)$ for some non-zero integer n.

conjecture | open | MathDB 页面 | 原文线索

V2-014363 | MATHDB-353702

The infinite-index subgroup surface-group conjecture for one-relator groups

摘录 / EXCERPT: Infinite-index subgroup surface-group conjecture. Then G is a surface group.

conjecture | open | MathDB 页面 | 原文线索

V2-014364 | MATHDB-353707

Louder--Wilton conjecture on negative immersions and hyperbolicity

摘录 / EXCERPT: Louder--Wilton conjecture. Every one-relator group whose presentation complex has negative immersions is hyperbolic.

conjecture | refuted | MathDB 页面 | 原文线索

V2-014365 | MATHDB-353741

The conjectural formula for the coefficient \(\Phi^1[n,n-2]\)

摘录 / EXCERPT: Let $n\geq 4$, and let $\Phi^1[n,n-2]$ denote the coefficient indexed by $(n,n-2)$ in the relevant symmetric-function-valued series. For a partition λ, write χ_λ...

conjecture | open | MathDB 页面 | 原文线索

V2-014366 | MATHDB-353828

Existence of Levi decompositions in definably connected groups

摘录 / EXCERPT: Let G be a definably connected group in an arbitrary o-minimal structure. A Levi decomposition conjecture. G has a Levi decomposition in the sense of the definition given in th...

conjecture | open | MathDB 页面 | 原文线索

V2-014367 | MATHDB-353829

Pellegrini-Shumyatsky centralizer-coset conjecture

摘录 / EXCERPT: Let G be a finite non-abelian simple group, let $t\in G$ be an involution, and let $\mathcal{Z}_G(t)$ denote the centralizer of t. A coset of the centralizer is a set...

conjecture | open | MathDB 页面 | 原文线索

V2-014368 | MATHDB-353846

Babai-Kantor-Lubotzky conjecture on expanders from finite simple groups

摘录 / EXCERPT: Let G be a nonabelian finite simple group. A subset $S\subset G$ is symmetric when it is closed under taking inverses, and $\operatorname{Cay}(G,S)$ denotes the Cayley graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-014369 | MATHDB-353882

Tselekidis's asymptotic-dimension bound for Coxeter groups

摘录 / EXCERPT: Tselekidis's conjecture. The asymptotic dimension of the Coxeter group satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014370 | MATHDB-353914

Commutation conjecture for the four-point DT operator and the normalized R-matrix

摘录 / EXCERPT: Let F be the Fock space, let $\mathsf{RR}^{H}_{\mathsf{F},\mathsf{F}\otimes\mathsf{F}}$ denote the four-point operator in cohomological DT theory, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014371 | MATHDB-353941

Moseley-Proudfoot-Young conjecture on the cohomology of configuration spaces in SU_2

摘录 / EXCERPT: Let D_n be the cohomology algebra of the configuration spaces of n ordered points in SU_2 up to translations, and let M_n be the intersection cohomology ring of the hyperto...

conjecture | open | MathDB 页面 | 原文线索

V2-014372 | MATHDB-353942

Freeness conjecture for the R_n-module T_n

摘录 / EXCERPT: Let R_n be the algebra appearing in the Orlik-Terao construction, and let T_n be the R_n-module that is a submodule of the free module OT_n. Freeness conjecture for T_n....

conjecture | open | MathDB 页面 | 原文线索

V2-014373 | MATHDB-353970

Witzel's conjecture on the Σ-invariants of S-arithmetic Borel subgroups

摘录 / EXCERPT: Witzel's conjecture. The Σ-invariants of Γ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-014374 | MATHDB-354020

Non-biautomaticity conjecture for irreducible lattices in products with tree automorphism groups

摘录 / EXCERPT: Non-biautomaticity conjecture. The group Γ is not biautomatic.

conjecture | open | MathDB 页面 | 原文线索

V2-014375 | MATHDB-354021

The weakly-top group abelianization growth conjecture

摘录 / EXCERPT: Weakly-top group abelianization growth conjecture. There exists $\delta>0$ such that every weakly-top group G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014376 | MATHDB-354045

Non-inner amenability conjecture for Röver-Nekrashevych groups

摘录 / EXCERPT: Let G be a self-similar group of automorphisms of the infinite rooted n-ary tree, and let $V_n(G)$ be the Röver-Nekrashevych group obtained by combining V_n with G. Non-inn...

conjecture | open | MathDB 页面 | 原文线索

V2-014377 | MATHDB-354114

Geller-Misiurewicz conjecture on powers of coarsely conjugate maps

摘录 / EXCERPT: Geller-Misiurewicz's conjecture. If f and g are coarsely conjugate, then f^n and g^n are coarsely conjugate for every natural number n.

conjecture | open | MathDB 页面 | 原文线索

V2-014378 | MATHDB-354115

Geller-Misiurewicz conjecture on coarse conjugacy from mutual coarse semiconjugacies

摘录 / EXCERPT: Geller-Misiurewicz's conjecture. If there exist coarse equivalences $\varphi\colon X\to Y$ and $\psi\colon Y\to X$ such that $\varphi\circ f$ is close to $g\circ\varphi$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014379 | MATHDB-354127

Ladder generalization of the cusp irreducibility criterion

摘录 / EXCERPT: Ladder generalization conjecture. The same equivalences remain true when $[a,a]$ is replaced by any ladder ${\bf m}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014380 | MATHDB-354128

Snake-module irreducibility conjecture for quantum affine algebras

摘录 / EXCERPT: Snake-module irreducibility conjecture. The tensor-product irreducibility theorem remains true if either $L(M)$ or $L(M')$ is a snake module and the other is an arbitrary simple mo...

conjecture | open | MathDB 页面 | 原文线索

V2-014381 | MATHDB-354137

GL_n and SL_n affine Springer cyclicity conjecture

摘录 / EXCERPT: Let $\mathfrak g=\operatorname{Lie}(GL_n)$ or $\mathfrak g=\operatorname{Lie}(SL_n)$, and let $\gamma\in g(cO)$ be elliptic, regular semisimple, and topologically nilpotent. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014382 | MATHDB-354144

Cabanes--Enguehard transitivity conjecture for e-pairs

摘录 / EXCERPT: Cabanes--Enguehard conjecture. The relation $\leq_e$ is transitive and therefore coincides with $\ll_e$. The conjecture concerns the poset of e-pairs and would make the directly...

conjecture | open | MathDB 页面 | 原文线索

V2-014383 | MATHDB-354197

The strengthened trivial-intersection conjecture for principal blocks

摘录 / EXCERPT: Strengthened trivial-intersection conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014384 | MATHDB-354199

The strengthened Brauer height-zero conjecture for two primes

摘录 / EXCERPT: Strengthened Brauer height-zero conjecture. The following are equivalent: q does not divide $\chi(1)$ for every $\chi\in\operatorname{Irr}_p(B_p(G))$ and p does not divide...

conjecture | open | MathDB 页面 | 原文线索

V2-014385 | MATHDB-354203

The relative Alperin-McKay block-bijection conjecture

摘录 / EXCERPT: Let $N\trianglelefteq G$ be a normal subgroup, and let $P\in\operatorname{Syl}_p(N)$. The relative Alperin-McKay block-bijection conjecture. There is a bijection ... such that for ea...

open_problem | open | MathDB 页面 | 原文线索

V2-014386 | MATHDB-354218

The rough-isometric embedding obstruction for isotropic trees

摘录 / EXCERPT: Let $R=R((p_n),(q_n))$ and $R'=R((p_n'),(q_n'))$ be isotropic trees. Let ${\mathcal{X}}((p_n),(q_n),(p_n'),(q_n'))$ denote the associated set of obstructions to a rough isometric e...

conjecture | open | MathDB 页面 | 原文线索

V2-014387 | MATHDB-354229

Query-complexity conjecture for amenable group presentations

摘录 / EXCERPT: Let S be a finite generating set and let E be a finite set of relators such that ... is a finitely presented amenable group. Let the Følner function of Γ measure the sizes...

conjecture | open | MathDB 页面 | 原文线索

V2-014388 | MATHDB-354249

Hamiltonian double-ray conjecture for groups with at most two ends

摘录 / EXCERPT: Let G be a group and let $\operatorname{Cay}(G,S)$ be a Cayley graph of G with respect to a generating set S. A Hamiltonian double ray is a spanning double ray, that is, a co...

conjecture | open | MathDB 页面 | 原文线索

V2-014389 | MATHDB-354250

Hamiltonicity conjecture for one-ended generalized quasi-dihedral groups

摘录 / EXCERPT: Let $G=\langle S\rangle$ be a one-ended generalized quasi-dihedral group, and let $\operatorname{Cay}(G,S)$ denote its Cayley graph. A Hamiltonian double ray is a spanning double r...

conjecture | open | MathDB 页面 | 原文线索

V2-014390 | MATHDB-354252

Kahn's Stiefel-Whitney conjecture for finite 2-groups

摘录 / EXCERPT: Throughout, let G be a finite 2-group, let $d(G)$ denote its minimal number of generators, and let r_G be the real regular representation of G. Define ... where $w_i(r_G)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-014391 | MATHDB-354253

Bounded nilpotency class for finitely generated d-maximal 2-groups

摘录 / EXCERPT: Let $r\geq 1$ be an integer. A finite 2-group G is d-maximal if $d(H)<d(G)$ for every proper subgroup $H<G$, where $d(G)$ is the minimal number of generators. The bounded-cla...

conjecture | open | MathDB 页面 | 原文线索

V2-014392 | MATHDB-354254

Finiteness conjecture for finitely generated d-maximal pro-2 groups

摘录 / EXCERPT: Let G be a finitely generated pro-2 group, and write $d(G)=\dim_{\mathbb{F}_2}G/G^{2}[G,G]$. Call G d-maximal if $d(H)<d(G)$ for every proper closed subgroup $H<G$. The pro...

conjecture | open | MathDB 页面 | 原文线索

V2-014393 | MATHDB-354323

Folding conjecture for irreducible l-highest weight modules

摘录 / EXCERPT: Let $\mathcal{U}_q(\mathcal{L}\mathfrak{g})$ and $\mathcal{U}_q(\mathcal{L}\mathfrak{g}^{\sigma})$ be the non-twisted and twisted quantum affine algebras, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014394 | MATHDB-354324

Kobayashi's discrete-series infinitude conjecture for reductive homogeneous spaces

摘录 / EXCERPT: Let G/H be a reductive homogeneous space in Setting, and let $\operatorname{Disc}(G/H)$ denote its discrete series representations. Kobayashi's conjecture. One has the equivalenc...

conjecture | open | MathDB 页面 | 原文线索

V2-014395 | MATHDB-354326

The stable-spectrum conjecture for deformations of discontinuous groups

摘录 / EXCERPT: Let G/H be a reductive homogeneous space, and let Γ be a finitely generated discontinuous group for G/H with non-trivial continuous deformations up to inner automorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-014396 | MATHDB-354386

The one-ended decidable-word-problem characterization of strongly aperiodic SFT groups

摘录 / EXCERPT: Strongly aperiodic SFT conjecture. A finitely generated group admits a strongly aperiodic SFT if and only if it is 1-ended and has decidable word problem.

conjecture | open | MathDB 页面 | 原文线索

V2-014397 | MATHDB-354388

The center conjecture for cyclotomic KLR algebras

摘录 / EXCERPT: Let $\beta\in Q_n^+$ and let $\Lambda\in P^+$. Let $p_\beta^\Lambda:\mathscr{R}_\beta\twoheadrightarrow\mathscr{R}_\beta^\Lambda$ be the canonical surjection. The center...

conjecture | open | MathDB 页面 | 原文线索

V2-014398 | MATHDB-354404

Polynomiality conjecture for faithful dimensions over finite truncated valuation rings

摘录 / EXCERPT: Let $\mathfrak{g}$ be a nilpotent $\mathbb{Z}$-Lie algebra finitely generated as an abelian group. For a finite truncated valuation ring R with maximal ideal $\mathfrak{m}$, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-014399 | MATHDB-354458

Reineke's total stability conjecture for Dynkin quivers

摘录 / EXCERPT: Let Q be a finite Dynkin quiver and let $A=KQ$ be its path algebra. A slope function is a map $\mu:K_0(A)\setminus\{0\}\to\mathbb{R}$ of the form $\mu=\frac{\theta}{\kappa}$, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-014400 | MATHDB-354461

The affine-linearity conjecture for Dedekind-sum averages over Mersenne subgroups

摘录 / EXCERPT: Affine-linearity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014401 | MATHDB-354481

Antolín-Martino-Ventura conjecture on degree of commutativity in exponential-growth groups

摘录 / EXCERPT: Let G be a finitely generated group of exponential word growth, and let S be a finite generating set of G. Antolín-Martino-Ventura conjecture. Then ... irrespective of the choi...

conjecture | open | MathDB 页面 | 原文线索

V2-014402 | MATHDB-354524

Vogan's local Langlands conjecture for tempered representations

摘录 / EXCERPT: Let F be a non-archimedean local field, let G^* be a quasi-split connected reductive group over F, and let C be an algebraically closed field of characteristic 0. For a t...

conjecture | open | MathDB 页面 | 原文线索

V2-014403 | MATHDB-354530

The minimal ramification conjecture for finite groups over finite fields

摘录 / EXCERPT: Let G be a nontrivial finite group. Write $d(G)$ for the minimal number of generators of G, G^{ab} for its abelianization, and $p(G^{\mathrm{ab}})$ for the subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-014404 | MATHDB-354548

Categorical braid group action conjecture for Lusztig's operators

摘录 / EXCERPT: Categorical braid group action conjecture. The functor $\cal{T}_{i,+1}'$ extends to an autoequivalence of $\textsf{Com}(\dot{\cal{U}}_Q(\mathfrak{g}))$ such that the induced automo...

conjecture | open | MathDB 页面 | 原文线索

V2-014405 | MATHDB-354587

The double-cell containment conjecture for Harish-Chandra cells

摘录 / EXCERPT: Double-cell containment conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-014406 | MATHDB-354591

Unimodality conjecture for the Kronecker-coefficient differences

摘录 / EXCERPT: Unimodality conjecture. For every $1\leq i\leq\binom{n}{2}-1$, the sequence $d_{\nu,i}$ is symmetric and unimodal. This is proposed as a stronger conjecture implying equivariant lo...

conjecture | open | MathDB 页面 | 原文线索

V2-014407 | MATHDB-354592

Novak-Sagan equivariant log-concavity conjecture for representations of the symmetric group

摘录 / EXCERPT: Novak-Sagan's conjecture. The graded representation V_n is equivariantly log-concave. The conjecture reformulates a log-concavity conjecture for the distribution of permutations...

conjecture | open | MathDB 页面 | 原文线索

V2-014408 | MATHDB-354608

The center-cohomology isomorphism conjecture for the small quantum group

摘录 / EXCERPT: Center-cohomology isomorphism conjecture. The algebra embedding

conjecture | open | MathDB 页面 | 原文线索

V2-014409 | MATHDB-354611

The bundle conjecture for primitive spectra of quantum coordinate algebras

摘录 / EXCERPT: Bundle conjecture. For any semisimple Lie group G and $(w_1,w_2)\in W\times W$, there exists some $\mathbf{i}\in\mathcal{I}_{w_1,w_2}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-014410 | MATHDB-354646

Rank rigidity conjecture for locally compact CAT(0) spaces

摘录 / EXCERPT: Let X be a locally compact geodesically complete $\operatorname{CAT}(0)$ space, and let G be a discrete group acting properly and cocompactly on X by isometries. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-014411 | MATHDB-354688

Kostant's indecomposability conjecture for projective functors in type A

摘录 / EXCERPT: Let θ_x and θ_y be projective functors acting on the principal block $4\mathcal{O}_04$ of the BGG category for $4\mathfrak{g}4$, and let L_y be the simple module in...

conjecture | open | MathDB 页面 | 原文线索

V2-014412 | MATHDB-354690

Finite-dimensionality conjecture for irreducible representations of Coxeter groups

摘录 / EXCERPT: Let (W,S) be a Coxeter system, and consider its irreducible complex representations. Finite-dimensionality conjecture. All irreducible representations of (W,S) are finite-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-014413 | MATHDB-354693

BBASV's invariant-center isomorphism conjecture for the small quantum group

摘录 / EXCERPT: BBASV's conjecture. The embedding above is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-014414 | MATHDB-354696

Antonyan's orbit-reversal conjecture for sl_2 triples

摘录 / EXCERPT: Let e,f,h form an $\mathfrak{sl}_2$ triple, and let $\overleftarrow h$ denote the element with reverse characteristic to that of h. Suppose $\tilde e,\tilde f,\overleftarrow h$...

conjecture | open | MathDB 页面 | 原文线索

V2-014415 | MATHDB-354777

The regular-subgroup solubility conjecture for finite soluble groups

摘录 / EXCERPT: Let N be a finite soluble group, and let G be a regular subgroup of the holomorph ... of N, meaning that G acts regularly on the underlying set of N. Regular-subgroup solub...

conjecture | open | MathDB 页面 | 原文线索

V2-014416 | MATHDB-354778

Monomiality from factoriality of the monomial character monoid

摘录 / EXCERPT: Factoriality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014417 | MATHDB-354779

Explicit monomial algebra for $SL(2,2^n)$

摘录 / EXCERPT: $SL(2,2^n)$ monomial-algebra conjecture. The monomial algebra is

conjecture | open | MathDB 页面 | 原文线索

V2-014418 | MATHDB-354801

Stanley-Stembridge conjecture for the functions $E^{\nu'}_{\lambda/\mu}$

摘录 / EXCERPT: Let λ and μ be partitions with $\mu\subseteq\lambda$, and let $E^{\nu'}_{\lambda/\mu}$ be the symmetric functions obtained by expanding the associated Frobenius charact...

conjecture | open | MathDB 页面 | 原文线索

V2-014419 | MATHDB-354810

Thompson's conjecture on derangements of primitive permutation groups

摘录 / EXCERPT: Thompson's conjecture. For any two distinct elements α and β of Ω, there exists a derangement $g\in G$ such that $\alpha^g=\beta$.

conjecture | open | MathDB 页面 | 原文线索

V2-014420 | MATHDB-354819

Aizenbud-Gourevitch-Sayag conjecture on twisted parabolic induction of Gelfand pairs

摘录 / EXCERPT: Let G be a reductive group, and let $P=LU$ be a Levi decomposition of a parabolic subgroup of G, with L and U the corresponding Levi and unipotent subgroups. Let H_0 be a...

conjecture | open | MathDB 页面 | 原文线索

V2-014421 | MATHDB-354820

Finite-orbit infinite-image representations of free groups

摘录 / EXCERPT: Let F_N be the free group on $N>0$ generators, and let $\operatorname{Out}(F_N)$ act on conjugacy classes of representations of F_N. Conjecture on free-group representations. F...

conjecture | open | MathDB 页面 | 原文线索

V2-014422 | MATHDB-354896

Twisted Jacquet module conjecture for cuspidal representations of GL(2n)

摘录 / EXCERPT: Let F_n be the unique degree-n extension of F, let $G=\operatorname{GL}(2n,F)$, and let $A\in\operatorname{M}(n,F)$ be the matrix specified in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014423 | MATHDB-354979

Indecomposability conjecture for cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Let $A=(a_{ij})_{i,j\in I}$ be a symmetrizable Cartan matrix, let $\Lambda\in P^+$ be dominant, and let $\beta\in Q_n^+$. Let $\mathscr{R}_{\beta}^{\Lambda}$ be the associated cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-014424 | MATHDB-355035

Minimality conjecture for defect subgroups of contragredient quasireductive supergroups

摘录 / EXCERPT: Minimality conjecture. Up to conjugacy, a defect subgroup D of G is the unique minimal splitting subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-014425 | MATHDB-355122

Braverman-Kazhdan-Gaitsgory conjecture on the essential image of lifted parabolic restriction

摘录 / EXCERPT: Braverman-Kazhdan-Gaitsgory conjecture. The essential image of WRes consists of those sheaves descending to the coarse quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-014426 | MATHDB-355170

Dales-Zelazko conjecture on finitely generated maximal left ideals

摘录 / EXCERPT: Let A be an infinite-dimensional unital Banach algebra. Dales-Zelazko conjecture. Then A has a maximal left ideal which is not finitely generated. The conjecture concerns the e...

conjecture | open | MathDB 页面 | 原文线索

V2-014427 | MATHDB-355234

The stable algebraic Albanese cohomology conjecture for Torelli groups

摘录 / EXCERPT: Let $\mathcal{I}_g$, $\mathcal{I}_{g,1}$ and $\mathcal{I}_g^1$ be the Torelli groups of a closed surface, a surface with one boundary component, and a surface with one marked point...

conjecture | open | MathDB 页面 | 原文线索

V2-014428 | MATHDB-355237

The Torelli-to-IA generation conjecture

摘录 / EXCERPT: Let $\mathcal{I}_{g,1}$ be the Torelli group of a genus-g surface with one boundary component, let IA_{2g} be the IA-automorphism group of a free group of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-014429 | MATHDB-355246

Explicit decomposition-number formula for RoCK spin superblocks

摘录 / EXCERPT: Let $h=2n+1$, let γ be a w-Rouquier h-bar-core, and let α be a strict partition and β a restricted h-strict partition, both with h-bar-core γ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014430 | MATHDB-355308

Closure conjecture for products of N=3 character spaces

摘录 / EXCERPT: Let $m,m'\in\mathbf N$ and $j,j'\in\mathbf Z$. For $m\in\mathbf N$ and $j\in\mathbf Z$, let $\overset{N=3}{CH}{}^{(+)}[K(m),j]$ denote the $\mathbf C((q^{\frac12}))$-linear span of...

conjecture | open | MathDB 页面 | 原文线索

V2-014431 | MATHDB-355314

Conjecture on conforming heredity data and Ringel duals

摘录 / EXCERPT: Let A_R be a based quasi-hereditary algebra over R with conforming heredity data, and let A'_R be a Ringel dual of A_R. Let $T^A(n,d)_\mathbb{F}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014432 | MATHDB-355320

Classification conjecture for finite congruence-simple semirings without nontrivial square-zero elements

摘录 / EXCERPT: Let S be a congruence-simple semiring with a multiplicatively absorbing element w, and suppose that ... for every $x\in S\setminus\{w\}$. Classification conjecture. If S is fin...

conjecture | open | MathDB 页面 | 原文线索

V2-014433 | MATHDB-355337

The symbol invariance conjecture for type D Weyl-group elements

摘录 / EXCERPT: Let $W_{D_d}\subset W_{C_d}$, let s_0 be the simple reflection inducing the outer automorphism, and let $sym(w)$ denote the symbol attached to w. For symbols with...

conjecture | open | MathDB 页面 | 原文线索

V2-014434 | MATHDB-355349

Parabolic intersection conjecture for Artin-Tits groups

摘录 / EXCERPT: Let A_I be an Artin-Tits group. For $X\subseteq I$, the standard parabolic subgroup A_X is the subgroup generated by the standard generators indexed by X; a parabolic subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-014435 | MATHDB-355354

The Cauchy Harish-Chandra character correspondence conjecture

摘录 / EXCERPT: Cauchy Harish-Chandra character correspondence conjecture. Assuming the character conjecture referred to in the source, define

conjecture | open | MathDB 页面 | 原文线索

V2-014436 | MATHDB-355368

The relative bi-exactness conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Let G be a finitely generated group hyperbolic relative to a collection of subgroups $\{H_\mu\}_{\mu \in \Lambda}$ of G. Relative bi-exactness conjecture. If all subgroups...

conjecture | open | MathDB 页面 | 原文线索

V2-014437 | MATHDB-355394

Twisted Teleman vanishing conjecture for twisted conformal blocks

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra with a standard automorphism σ, and let $D_{c,\sigma}$ and D_c be the corresponding sets of dominant weights. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014438 | MATHDB-355459

Existence of limits for products of equal powers

摘录 / EXCERPT: Let G be a group equipped with the bi-invariant metric norm $\|\cdot\|$. For elements $g_1,\dots,g_n\in G$, consider the sequence of normalized norms of products of equal power...

conjecture | open | MathDB 页面 | 原文线索

V2-014439 | MATHDB-355461

Isometry between the asymptotic completion and the free-group completion

摘录 / EXCERPT: Restrict to the free group $G=F(A)$, let $\Theta\subseteq P(G)$ be a set of cyclically reduced representatives for the equivalence classes in $P(G)$, and let τ be the stable n...

conjecture | open | MathDB 页面 | 原文线索

V2-014440 | MATHDB-355463

The generic two-boundary XXZ module conjecture

摘录 / EXCERPT: Let $\mathfrak{q}\in\mathbb{C}\backslash\mathfrak{q}^{i\pi\mathbb{Q}}$, let $\mathfrak{q}^{\alpha_l},\mathfrak{q}^{\alpha_r}\in\mathbb{C}\backslash\{\pm\mathfrak{q}^{\mathbb{Z}}\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-014441 | MATHDB-355464

The root-of-unity semi-simplified vacuum-module conjecture

摘录 / EXCERPT: Let $\mathfrak{q}=e^{i\pi/p}$ with $p\in\mathbb{N}\backslash\{0,1\}$, and let $\alpha_l,\alpha_r\in\mathbb{C}\backslash\mathbb{Z}$ satisfy $\alpha_l+\alpha_r\notin\mathbb{Z}$. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-014442 | MATHDB-355569

Asymptotic Clifford-symplectic Howe-type duality

摘录 / EXCERPT: Fix t and let n be sufficiently large. Let $C_{\mathbf{1}_t}$ be the code space appearing in the construction, let T' be the associated symplectic space, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014443 | MATHDB-355581

Higher nilpotency-class identity conjecture for unitriangular groups

摘录 / EXCERPT: Let $k\geq 11$ and let $\mathcal{H}\subset \mathsf{UT}(n,\mathbb{Q})$ be any finite set of matrices. Write $\mathcal{I}=\{1,2,\ldots,k+1\}$, let $\operatorname{PI}^{\mathcal{I}}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-014444 | MATHDB-355583

Hiss-Schröer bounded multiplicity conjecture for finite classical groups

摘录 / EXCERPT: Hiss-Schröer's conjecture. The multiplicities $m(\pi,\sigma)$ of the irreducible constituents of $\pi\otimes\omega_\psi^\vee$ are bounded by a function of n independent of q.

conjecture | open | MathDB 页面 | 原文线索

V2-014445 | MATHDB-355604

Equivalence conjecture for a non-integral block of category O

摘录 / EXCERPT: Equivalence conjecture. There is a $\lambda\notin\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014446 | MATHDB-355621

Myrberg-set invariance for geometric CAT(0) actions

摘录 / EXCERPT: Let G act geometrically on two proper CAT(0) spaces X and Y, and let the corresponding Myrberg sets be the subsets of their respective boundaries consisting of Myrberg points...

conjecture | open | MathDB 页面 | 原文线索

V2-014447 | MATHDB-355640

Homogeneity conjecture for special strict partitions

摘录 / EXCERPT: Homogeneity conjecture for special strict partitions. Suppose $p=3$ and λ is a special strict partition, written as $\lambda=\nu+3\alpha$ with ν a 3-bar core. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-014448 | MATHDB-355653

Automatic continuity criterion for epimorphisms onto discrete groups

摘录 / EXCERPT: Automatic continuity conjecture. Every epimorphism from a locally compact Hausdorff group L to G is continuous if and only if G has no non-trivial finite normal subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-014449 | MATHDB-355656

Nonzero higher structure map conjecture for Dynkin-type ideals

摘录 / EXCERPT: Let R be a local Gorenstein ring and let $I\subset R$ be a perfect ideal of Dynkin type. A structure map $w^{(i)}_{j,k}(I)$ has strictly positive rank modulo the maximal ideal if...

conjecture | open | MathDB 页面 | 原文线索

V2-014450 | MATHDB-355659

Invariant subalgebra classification conjecture for Furstenberg-Poisson crossed products

摘录 / EXCERPT: Invariant subalgebra classification conjecture. Every Γ-invariant subalgebra of

conjecture | open | MathDB 页面 | 原文线索

V2-014451 | MATHDB-355660

Invariant subalgebra crossed-product conjecture for Furstenberg-Poisson boundaries

摘录 / EXCERPT: Invariant crossed-product conjecture. The subalgebra $\mathcal{M}$ is a crossed product of the form

conjecture | open | MathDB 页面 | 原文线索

V2-014452 | MATHDB-355680

The rough-isometry conjecture for non-special Heintze groups

摘录 / EXCERPT: Let S) be a Heintze group that is not among the special-type subgroups of $\operatorname{SO}(n,1)$ or $\operatorname{SU}(n,1)$ for any $n\geqslant 2$. Equip S with any left-inv...

conjecture | open | MathDB 页面 | 原文线索

V2-014453 | MATHDB-355681

Quasiisometric rigidity for finitely generated groups near Heintze and Sol-type groups

摘录 / EXCERPT: Let Γ be a finitely generated group. A group is virtually a lattice in a Lie group if it has a finite-index subgroup that is a lattice in that Lie group. The quasiisometric-...

conjecture | open | MathDB 页面 | 原文线索

V2-014454 | MATHDB-355706

The subgroup membership conjecture for the groups $B(P_{3,k})$

摘录 / EXCERPT: For each integer $k\geq 2$, let ... The subgroup membership conjecture for $B(P_{3,k})$. The subgroup membership problem is decidable in $B(P_{3,k})$ for every $k\geq 2$. The subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-014455 | MATHDB-355716

The word-map nonconstancy conjecture for simple compact Lie groups

摘录 / EXCERPT: Word-map nonconstancy conjecture. Every non-constant word defines a non-constant word map.

conjecture | open | MathDB 页面 | 原文线索

V2-014456 | MATHDB-355754

The weak-normal covering conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite non-abelian simple group. Write $\gamma(G)$ for its normal covering number and $\gamma_w(G)$ for its weak normal covering number, and let $m\in\mathbb{N}$. Weak...

conjecture | open | MathDB 页面 | 原文线索

V2-014457 | MATHDB-355755

The bounded normal covering conjecture for finite simple groups

摘录 / EXCERPT: Let c be a positive integer, and let G be a finite non-abelian simple group. Write $\gamma(G)$ for its normal covering number. Bounded normal covering conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-014458 | MATHDB-355776

Fried-Völklein conjecture on projective absolute Galois groups

摘录 / EXCERPT: Let $K\leq\bar{\mathbb Q}$ be Hilbertian, meaning that Hilbert's irreducibility theorem holds in K, and let G_K be its absolute Galois group. Assume that G_K is projective am...

conjecture | open | MathDB 页面 | 原文线索

V2-014459 | MATHDB-355783

The interlaced-pair conjecture for ghost distributions

摘录 / EXCERPT: Let $(\mathfrak{g},\mathfrak{k})$ be a supersymmetric pair, let $\mathfrak{a}$ be a chosen Cartan subspace, and let $\overline{\Delta}$ be its reduced root system with restricted W...

conjecture | open | MathDB 页面 | 原文线索

V2-014460 | MATHDB-355827

Equivariant Sugrue conjecture on homological dimension of rational G-sheaves

摘录 / EXCERPT: Let G be a profinite group acting on a G-space X. Suppose that X has finite Cantor--Bendixson rank and non-empty perfect hull. Equivariant Sugrue conjecture. The homologica...

conjecture | open | MathDB 页面 | 原文线索

V2-014461 | MATHDB-355848

The converse to Happel's AR-triangle theorem for complete local rings

摘录 / EXCERPT: Converse to Happel's theorem. If

conjecture | open | MathDB 页面 | 原文线索

V2-014462 | MATHDB-355857

LLRZ's support conjecture for triangular bases of rank-2 cluster algebras

摘录 / EXCERPT: LLRZ's support conjecture. The coefficient $e(p,q)$ of $C[a_1,a_2]$ is nonzero if and only if $D(p,q)\ge0$.

conjecture | open | MathDB 页面 | 原文线索

V2-014463 | MATHDB-355868

Alperin's Weight Conjecture for connected \ell-local compact groups

摘录 / EXCERPT: Alperin's Weight Conjecture. If $\mathcal F$ is connected, meaning every element of S is $\mathcal F$-conjugate to an element of T, then

conjecture | open | MathDB 页面 | 原文线索

V2-014464 | MATHDB-355955

The planar bipartite graph basis conjecture for $S^{(k,k,k,1^{n-3k})}$

摘录 / EXCERPT: Let A be the set of planar bipartite graphs embedded in a disk satisfying the following conditions: there are n boundary vertices, all in the same part of a bipartition; every...

conjecture | open | MathDB 页面 | 原文线索

V2-014465 | MATHDB-355959

The center conjecture for finite W-superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie superalgebra and e the nilpotent element defining the finite W-superalgebra $U(\mathfrak{g},e)$. Let $Z(U(\mathfrak{g}))$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014466 | MATHDB-355973

Generalization of the Frobenius-pushforward splitting theorem to semisimple adjoint groups

摘录 / EXCERPT: Generalization conjecture. Do the results of Theorem hold for any semisimple adjoint group G?

open_problem | open | MathDB 页面 | 原文线索

V2-014467 | MATHDB-355977

Shelah's automorphism conjecture for free algebras

摘录 / EXCERPT: Let F be freely generated by $\\{x_i:i<\omega+\omega\\}$, and let $\langle x_i:\omega\leq i<\omega+\omega\rangle_F$ denote the subalgebra generated by those elements. Shelah's au...

conjecture | open | MathDB 页面 | 原文线索

V2-014468 | MATHDB-356011

Conjecture on regularity over the cyclic group of order a power of two

摘录 / EXCERPT: Let $f\colon \mathbb{Z}/2^n\mathbb{Z}\to\{\pm1\}$ be a Boolean function. Say that f has (δ,k)-$\mathbb{R}$-regularity and that it has ϵ-$\mathbb{Z}/2^n\mathbb{Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-014469 | MATHDB-356016

The maximal and minimal local Arthur parameter conjecture

摘录 / EXCERPT: Maximal and minimal parameter conjecture. For any $psi_1,psi_2 e Psi(pi)$, one has $lambda_{phi_{psi_1}}=lambda_{phi_{psi_2}}$. Moreover, there are unique...

conjecture | open | MathDB 页面 | 原文线索

V2-014470 | MATHDB-356022

Knight's limiting-density conjecture for one-relator groups

摘录 / EXCERPT: For a fixed integer $m\geq 2$, consider groups given by presentations with a fixed m-tuple of generators and one relator, in the Kapovich-Schupp model. For an elementary first-or...

conjecture | open | MathDB 页面 | 原文线索

V2-014471 | MATHDB-356041

Donkin's filtration conjecture

摘录 / EXCERPT: Let G be a semisimple algebraic group over k, let $r\geq 1$, let M be a rational G-module, and let St_r denote the r-th Steinberg representation. A good...

conjecture | open | MathDB 页面 | 原文线索

V2-014472 | MATHDB-356050

Conjecture on orbit counts of external lines to the twisted cubic

摘录 / EXCERPT: Let $q\equiv\xi\pmod3$, where $\xi\in\\{1,-1,0\\}$. The lines of $\mathrm{PG}(3,q)$ are partitioned into orbits under the group action described in the paper, and...

conjecture | open | MathDB 页面 | 原文线索

V2-014473 | MATHDB-356051

Conjecture on orbit counts for square parameters in characteristic three

摘录 / EXCERPT: The square-parameter orbit conjecture. The results of Example 6 should hold for every $m\ge1$; in particular, the corresponding formulas there for the numbers of orbits of length...

conjecture | open | MathDB 页面 | 原文线索

V2-014474 | MATHDB-356064

The brick-infinite, brick-continuous and generic-brick equivalence conjecture

摘录 / EXCERPT: Brick-infinite equivalence conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-014475 | MATHDB-356069

The automorphism group conjecture for the substructure ordering of finite directed graphs

摘录 / EXCERPT: The automorphism group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014476 | MATHDB-356070

Integral rank-one conjecture for K-theoretic BPS spaces

摘录 / EXCERPT: Integral rank-one conjecture. The $\mathbb{K}$-module $\mathrm{P}(nd)_{nv}$ is free of rank one.

conjecture | open | MathDB 页面 | 原文线索

V2-014477 | MATHDB-356084

The uniform nondivergence criterion for isotropic algebraic group actions

摘录 / EXCERPT: Let $\boldsymbol{G}$ be a semisimple $\mathbb{Q}$-algebraic group, let G be its real points, let Γ be an arithmetic lattice in G, and set $X:=G/\Gamma$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014478 | MATHDB-356096

The lifting conjecture for transfer systems on athbb{S}\mathrm{L}_2(\mathbb{F}_p)athbb{S}]

摘录 / EXCERPT: Let $G = \operatorname{SL}_2(\mathbb{F}_p)$ with $p > 5$, $p \neq 11$, and $p \equiv \pm 3 \mod 8$. Fix an arbitrary transfer system $\mathcal{R}$ on G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014479 | MATHDB-356114

The Gap Conjecture for finite group actions on 3-manifolds

摘录 / EXCERPT: Gap Conjecture. If the moc of each element of F is subexponential, then the action is topologically conjugate to a smooth (C^∞) action.

conjecture | open | MathDB 页面 | 原文线索

V2-014480 | MATHDB-356117

The finite group action conjecture for 3-manifolds

摘录 / EXCERPT: Finite group action conjecture. If F acts on M with subexponential moc for each element of F, then F is topologically conjugate to a smooth action.

conjecture | open | MathDB 页面 | 原文线索

V2-014481 | MATHDB-356130

Weight-space description of the refined averaging formula

摘录 / EXCERPT: Let μ be a geometric dominant cocharacter such that ϕ_{T} is μ-regular, let $b \in B(G,\mu)_{\mathrm{un}}$ be unramified, and let $w \in W_{b}$. Let b_{T} be the HN-...

conjecture | open | MathDB 页面 | 原文线索

V2-014482 | MATHDB-356156

Conjectural equivariant K-theory formulas for non-simply laced affine Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be B_n with $n\geqslant 3$ or F_4, and let e be a subregular nilpotent element of $\mathfrak{g}^\vee$. Let $\widehat{W}$ be the affine Weyl group, embedded...

conjecture | open | MathDB 页面 | 原文线索

V2-014483 | MATHDB-356160

The FI-homology decomposition conjecture for injective cogenerators

摘录 / EXCERPT: Let $a,b\in\mathbb{N}$, let n be a homological degree, and let $\mathfrak{Crit}(a,b)$ denote the set of critical pairs (λ,μ) used in the proposition. For each such pair...

conjecture | open | MathDB 页面 | 原文线索

V2-014484 | MATHDB-356163

Four-character Fitting-index conjecture

摘录 / EXCERPT: Let G be a finite group, and let $\mathbf F(G)$ denote its Fitting subgroup. Write $\operatorname{Irr}(G)$ for the set of irreducible characters of G. Four-character Fitting-in...

conjecture | open | MathDB 页面 | 原文线索

V2-014485 | MATHDB-356164

Fitting-index and group-order bounds in terms of character-table zeros

摘录 / EXCERPT: Let G be a finite group, and let $m(G)$ denote the maximum number of zeros in a row of the character table of G. Let $\mathbf F(G)$ denote the Fitting subgroup. Fitting-index a...

conjecture | open | MathDB 页面 | 原文线索

V2-014486 | MATHDB-356166

Average character degree refinement of Gluck's conjecture

摘录 / EXCERPT: Let G be a finite group, let ${\mathbf F}(G)$ denote its Fitting subgroup, and let $\operatorname{Irr}(N)$ be the set of irreducible characters of a normal subgroup N. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014487 | MATHDB-356199

Conjecture on conjugacy-class intersections in closed surface groups

摘录 / EXCERPT: Let Γ_g be the Cayley graph of a closed surface group with respect to a standard generating set. For a conjugacy class $\mathfrak{c}$, let $\lVert\mathfrak{c}\rVert$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-014488 | MATHDB-356232

Conjecture on the UT-rank of the monoid M_5

摘录 / EXCERPT: Let $M_5 = \operatorname{Mon}\langle a,b\mid aba=ba\rangle$ be the one-relation monoid with defining relation $aba=ba$. Its UT-rank is the least n such that M_5 admits a faithf...

conjecture | open | MathDB 页面 | 原文线索

V2-014489 | MATHDB-356233

Conjecture on tropical representations of the monoid M_4

摘录 / EXCERPT: Let $M_4 = \operatorname{Mon}\langle a,b\mid aba^2=ba\rangle$ and let M_7 denote its anti-isomorphic one-relation monoid. A faithful tropical representation of a monoid is an inj...

conjecture | open | MathDB 页面 | 原文线索

V2-014490 | MATHDB-356267

Endoscopic character identity for covers of reductive groups

摘录 / EXCERPT: Let G be a reductive group with cover $G(F)_{x_G}$, let $(\dot s,\mathcal H)$ be a refined endoscopic datum for G, and let $H(F)_{x_H}$ be the corresponding endoscopic cover. L...

conjecture | open | MathDB 页面 | 原文线索

V2-014491 | MATHDB-356284

Weak cluster structure conjecture for infinite type fD cluster categories

摘录 / EXCERPT: Let $n\in\mathbb{N}_{>0}$, and let $\mathcal{C}(\mathbb{D}_{n,\infty})$ be an infinite type $\mathbb{D}$ cluster category whose clusters are weakly cluster tilting. A full subcateg...

conjecture | open | MathDB 页面 | 原文线索

V2-014492 | MATHDB-356285

Weak cluster structure conjecture for punctured infinite type fD cluster categories

摘录 / EXCERPT: Let $n\in\mathbb{N}_{>0}$, and let $\mathcal{C}(\mathbb{D}_{\overline{n,\infty}})$ be an infinite type $\mathbb{D}$ weak cluster category. A full subcategory is weakly cluster tilt...

conjecture | open | MathDB 页面 | 原文线索

V2-014493 | MATHDB-356315

The torsion conjecture for discrete CAT(0)-space isometry groups

摘录 / EXCERPT: Torsion conjecture. If Γ is torsion, then Γ is finite. An easier version is that if Γ is almost-cocompact, then it contains an infinite-order element.

conjecture | open | MathDB 页面 | 原文线索

V2-014494 | MATHDB-356326

Wise's conjecture on non-positive immersions for contractible complexes

摘录 / EXCERPT: Let X be a contractible complex. Wise's conjecture. The complex X has (weak) non-positive immersions. Here, non-positive immersions means that for every immersion...

conjecture | open | MathDB 页面 | 原文线索

V2-014495 | MATHDB-356328

The restriction conjecture for local Langlands packets of disconnected groups

摘录 / EXCERPT: Let $B\to A$ be a map of finite groups acting on a connected reductive group G, and write $G^A=G\rtimes A$ and $G^B=G\rtimes B$. Let π^A correspond to (ϕ,ρ^A) for...

conjecture | open | MathDB 页面 | 原文线索

V2-014496 | MATHDB-356389

The dominant weight polyhedron conjecture for arbitrary positive-level weights

摘录 / EXCERPT: Let $\mathfrak{g}$ be an untwisted affine Kac-Moody Lie algebra, let δ be its null root, and let $\lambda\in\mathcal{P}^+$ be a dominant integral weight of positive level. F...

conjecture | open | MathDB 页面 | 原文线索

V2-014497 | MATHDB-356390

Affine Kostant tensor product conjecture for untwisted affine Kac-Moody algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be an untwisted affine Kac-Moody Lie algebra, let ρ be the affine Weyl weight used in the tensor product $V(\rho)\otimes V(\rho)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014498 | MATHDB-356391

The slice-rank conjecture for group multiplication tensors

摘录 / EXCERPT: Let G be a finite group, let $M(G)$ denote its matching number, and define the minimum slice rank ... where the minimum is over algebraically closed fields k and $T_{k[G]}$ is th...

conjecture | open | MathDB 页面 | 原文线索

V2-014499 | MATHDB-356392

The quasirandom groups conjecture for modular group algebras

摘录 / EXCERPT: Fix $\varepsilon>0$. Let G be a finite group of order n, and say that G is n^ε-quasirandom when every nontrivial irreducible complex representation of G has dim...

conjecture | open | MathDB 页面 | 原文线索

V2-014500 | MATHDB-356411

Przytycki's phase-transition conjecture for random group actions on CAT(0) cube complexes

摘录 / EXCERPT: Let S be a finite set of generators, let $d\in(0,1)$ be a density, and consider a random group in the Gromov density model with density d and relator length tending to infinity...

conjecture | open | MathDB 页面 | 原文线索

V2-014501 | MATHDB-356451

Affine action conjecture for groups acting on the line with finitely many fixed points

摘录 / EXCERPT: Let $N\in \mathbb{N}$ and let $G\leq {\mathrm{Homeo}}(\mathbb{R})$ be a subgroup with at most N fixed points. Let $I\subset \mathbb{R}$ be a maximal interval without global fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-014502 | MATHDB-356455

Large-representation tail bound conjecture for random gate-sets

摘录 / EXCERPT: Let $\epsilon>0$, let $\mathcal{S}\subset U(d)$ be a symmetric Haar random gate-set containing at least 3 elements, and let λ be an irreducible representation with dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-014503 | MATHDB-356471

Salamanca-Riba-Vogan unitarity conjecture for Hermitian modules

摘录 / EXCERPT: Salamanca-Riba-Vogan's unitarity conjecture. The above bijection preserves unitarity.

conjecture | open | MathDB 页面 | 原文线索

V2-014504 | MATHDB-356472

Salamanca-Riba-Vogan convex-hull conjecture for unitarily small lowest K-types

摘录 / EXCERPT: Salamanca-Riba-Vogan's convex-hull conjecture. If X is unitary, then Λ must lie in

conjecture | open | MathDB 页面 | 原文线索

V2-014505 | MATHDB-356473

Vogan's fundamental parallelepiped conjecture for real reductive groups

摘录 / EXCERPT: Vogan's fundamental parallelepiped conjecture. If X is unitary, then

conjecture | open | MathDB 页面 | 原文线索

V2-014506 | MATHDB-356485

Dominant-dimension inequality for tensor powers and Schur algebras

摘录 / EXCERPT: Let K be a field, let $S_R(n,d)$ denote the Schur algebra over a commutative ring R, and let V be its natural module. For $n,d\in\mathbb{N}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-014507 | MATHDB-356486

The folklore conjecture that almost all finite groups are 2-groups

摘录 / EXCERPT: Let a finite group be a group with finitely many elements, and call it a 2-group when its order is a power of 2. Folklore conjecture. Almost all groups are 2-groups. This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-014508 | MATHDB-356488

The alternating-group wreath-product strategy conjecture

摘录 / EXCERPT: Let A_n be the alternating group on n letters, let C_3 be the cyclic group of order 3, and let $A_n\wr C_3$ denote their wreath product. Alternating-group wreath-product st...

conjecture | open | MathDB 页面 | 原文线索

V2-014509 | MATHDB-356494

The orbit-bound conjecture for shortest surjective strategies

摘录 / EXCERPT: Let K/H be the set of equivalence classes of K under the action of H, and let N denote the length of a surjective strategy. Orbit-bound conjecture. If $G\wr H$ has a surjec...

conjecture | open | MathDB 页面 | 原文线索

V2-014510 | MATHDB-356498

Polynomial lower-bound conjecture for least singular values of random matrix polynomials

摘录 / EXCERPT: Let $P(U_1,\ldots,U_k,U_1^*,\ldots,U_k^*)$ be a noncommutative nonzero $*$-polynomial with complex coefficients, let $\underline{\lambda}\neq 0$ be a signature, and let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-014511 | MATHDB-356500

Breuer-Guralnick-Kantor spread-one conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group. A proper quotient means a quotient G/N with $1\neq N\trianglelefteqslant G$, and $s(G)$ denotes the largest integer k such that any k nontrivial el...

conjecture | open | MathDB 页面 | 原文线索

V2-014512 | MATHDB-356523

Extension of the stretched Kostka quasi-polynomial degree formula to arbitrary types

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex semisimple Lie algebra, with root system Φ, positive roots Φ_+, simple roots $\alpha_1,\ldots,\alpha_r$, and corresponding fundamental wei...

conjecture | open | MathDB 页面 | 原文线索

V2-014513 | MATHDB-356552

The 2-power solubilizer subgroup conjecture

摘录 / EXCERPT: 2-power solubilizer subgroup conjecture. If $|\operatorname{Sol}_G(x)|=2^n$ for some $x\in G$, then $\operatorname{Sol}_G(x)$ is a subgroup of G.

conjecture | open | MathDB 页面 | 原文线索

V2-014514 | MATHDB-356554

The solubilizer divisibility conjecture for normalizers

摘录 / EXCERPT: Let G be a finite insoluble group, and let $x\in G$. The solubilizer of x is ... The cyclic subgroup generated by x is denoted by $\langle x\rangle$, and its normalizer in G...

conjecture | open | MathDB 页面 | 原文线索

V2-014515 | MATHDB-356618

Kirillov-Klyachko saturation conjecture for reduced Kronecker coefficients

摘录 / EXCERPT: Kirillov-Klyachko conjecture. The reduced Kronecker coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-014516 | MATHDB-356619

Representation-valued concavity conjecture for stable symmetric-group representations

摘录 / EXCERPT: Representation-valued concavity conjecture. The function V is concave under tensor multiplication: whenever $(\lambda+\mu)/2$ is a partition,

conjecture | open | MathDB 页面 | 原文线索

V2-014517 | MATHDB-356620

Sorting conjecture for stable symmetric-group representations

摘录 / EXCERPT: Sorting conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014518 | MATHDB-356621

Multipartition sorting conjecture for stable symmetric-group representations

摘录 / EXCERPT: Multipartition sorting conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014519 | MATHDB-356622

Byott's insolvability conjecture for realizable group pairs

摘录 / EXCERPT: Byott's conjecture. If (G,N) is realizable and G is insolvable, then N is also insolvable.

conjecture | open | MathDB 页面 | 原文线索

V2-014520 | MATHDB-356642

Herzog-Kaplan-Lev conjecture on products of cycle classes

摘录 / EXCERPT: Herzog-Kaplan-Lev conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-014521 | MATHDB-356683

Pre-RdLi-commutativity implies strong RdLi-commutativity

摘录 / EXCERPT: Let $sigma, sigma' in \mathrm{Irr}^{square}$ and let $pi in \mathrm{Irr}$. A triple $(sigma,sigma',pi)$ is pre-RdLi-commutative when it satisfies the pre-RdLi-commutativity conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-014522 | MATHDB-356684

Duality for strongly commutative triples of square-integrable representations

摘录 / EXCERPT: Let $sigma_1,sigma_2 in \mathrm{Irr}^{square}$ and let $pi in \mathrm{Irr}$. The corollary in the paper asserts that $(sigma_1,sigma_2,pi)$ is strongly RdLi-commutative if and only...

open_problem | open | MathDB 页面 | 原文线索

V2-014523 | MATHDB-356735

Integrality conjecture for polynomial eigenfunctions of the Nazarov-Sklyanin Lax operator

摘录 / EXCERPT: Integrality conjecture. For any λ, s, μ, and l in this expansion, one has

conjecture | open | MathDB 页面 | 原文线索

V2-014524 | MATHDB-356795

Quasi-isometric rigidity conjecture for completely solvable Lie groups

摘录 / EXCERPT: Quasi-isometric rigidity conjecture. S and S' are quasi-isometric if and only if they are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-014525 | MATHDB-356800

Lück's integral torsion conjecture for residually finite 2-acyclic groups

摘录 / EXCERPT: Let G) be an infinite residually finite 2-acyclic group of type VF. Write $\rho^{(2)}(G)$ for its 2-torsion and $\rho^\mathbb{Z}(G)$ for its integral torsion. Lück's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-014526 | MATHDB-356842

The Brauer-relation parity conjecture for Jacobians

摘录 / EXCERPT: Let X be a curve defined over a number field K, and let G be a finite group of K-automorphisms of X. Assume that $\Omega^{1}(\operatorname{Jac}_X)$ is self-dual as a G-...

conjecture | open | MathDB 页面 | 原文线索

V2-014527 | MATHDB-356844

Representation-theoretic decomposition conjecture for genomic Schur functions

摘录 / EXCERPT: Let λ be a partition with $\ell(\lambda)\le 2$, and let $\mathcal{E}_{\lambda;m}$, $\mathsf{Par}(\lambda;m)$, $\mathbf{G}_E$, and X_μ be the sets and 0-Hecke modules...

conjecture | open | MathDB 页面 | 原文线索

V2-014528 | MATHDB-356880

Savitskii's minimal-character refinement of Knutson's conjecture

摘录 / EXCERPT: Let G be a finite group. For virtual complex characters of strictly positive degree, define a pre-order by ... A character χ is minimal if $\varphi\leq\chi$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-014529 | MATHDB-356881

Knutson-type conjecture for complex finite-dimensional group algebras

摘录 / EXCERPT: Let G be a finite group, and let $\mathbb{C}[G]$ be its complex group algebra. A bialgebra is of Knutson type if every simple module over it has trivial Knutson index, meaning th...

conjecture | open | MathDB 页面 | 原文线索

V2-014530 | MATHDB-356890

The Adams conjecture for local theta correspondence and Arthur packets

摘录 / EXCERPT: Adams's conjecture. The representation $theta_{-alpha}(pi)$ is contained in the A-packet parametrized by

conjecture | open | MathDB 页面 | 原文线索

V2-014531 | MATHDB-356892

The isomorphism criterion for the map

摘录 / EXCERPT: Let $r>1$, let $d=1$, and let m and n be the parameters for the map ... is an isomorphism if and only if $m n$. No resolution or supporting result is provided in the supplied tex...

conjecture | open | MathDB 页面 | 原文线索

V2-014532 | MATHDB-356990

Non-isomorphism conjecture for quandles constructed from an infinite-type quandle

摘录 / EXCERPT: Non-isomorphism conjecture. The quandles $d4ac_n(Q)$ and $d4ac_m(Q)$ are not isomorphic whenever $n\neq m$ and $n,m\in\mathbb{Z}^+$.

conjecture | open | MathDB 页面 | 原文线索

V2-014533 | MATHDB-356992

The hyperbolic k-geodeticity conjecture

摘录 / EXCERPT: Hyperbolic k-geodeticity conjecture. A hyperbolic group G is k-geodetic for some positive integer k if and only if G is geodetic. Furthermore, if G is infinite and...

conjecture | open | MathDB 页面 | 原文线索

V2-014534 | MATHDB-356994

Church-Farb stability conjecture for IA-automorphism groups

摘录 / EXCERPT: Church-Farb stability conjecture. For every $i\ge0$, the family $\{H^i(\operatorname{IA}_n,\mathbb{Q})\}_n$ is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-014535 | MATHDB-356995

Katada's stable Albanese cohomology conjecture for IA-automorphism groups

摘录 / EXCERPT: Katada's stable Albanese cohomology conjecture. For every algebraic V, the map i^* is stably injective and stably identifies its image with the invariant Albanese cohomology; e...

conjecture | open | MathDB 页面 | 原文线索

V2-014536 | MATHDB-357002

Quantum eigenvalue character conjecture for reflection equation algebras

摘录 / EXCERPT: Let ${\cal L}(R)$ be the reflection equation algebra, let $V_{T_\lambda}=\operatorname{Im}P_{T_\lambda}(R)$ be an ${\cal L}(R)$-invariant subspace, and let μ_i, $1\leq i\leq N$...

conjecture | open | MathDB 页面 | 原文线索

V2-014537 | MATHDB-357025

Local stability conjecture for finite wreath products with the integers

摘录 / EXCERPT: Let Δ be a finite group. The wreath product $\Delta \wr \mathbb{Z}$ is locally stable. Finite-wreath-product local stability conjecture. Every finite wreath product of this...

conjecture | open | MathDB 页面 | 原文线索

V2-014538 | MATHDB-357055

Asymptotic Gross-Kunze inner-product conjecture

摘录 / EXCERPT: Asymptotic Gross-Kunze inner-product conjecture. For each λ there exists a strictly positive scalar c such that

conjecture | open | MathDB 页面 | 原文线索

V2-014539 | MATHDB-357191

The conjecture on zeros of characters in 3-groups of maximal class

摘录 / EXCERPT: The zeros conjecture. G has an irreducible character that vanishes at exactly $3^n-3^{n-1}+6$ elements if and only if G is a 3-group of maximal class with a maximal abelian s...

conjecture | open | MathDB 页面 | 原文线索

V2-014540 | MATHDB-357192

The minimal-zero conjecture for finite p-groups

摘录 / EXCERPT: Minimal-zero conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014541 | MATHDB-357196

The generic Artin-group properties conjecture

摘录 / EXCERPT: Generic Artin-group properties conjecture. Every Artin group A_{Γ} is torsion-free; if A_{Γ} is irreducible and non-spherical, then it has trivial centre and is acyli...

conjecture | open | MathDB 页面 | 原文线索

V2-014542 | MATHDB-357198

Benson's quasi-polynomial conjecture for tensor powers

摘录 / EXCERPT: Let V be an odd-dimensional indecomposable representation of a finite 2-group G over an algebraically closed field $\Bbbk$ of characteristic 2. Assuming the relevant tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-014543 | MATHDB-357199

The alternative-Hopf-structure conjecture for monomial modules

摘录 / EXCERPT: Let $G=\mathbb{Z}/2^r\mathbb{Z}\times\mathbb{Z}/2^s\mathbb{Z}$, and let $\Bbbk\alpha(r,s)$ be the Hopf algebra with the alternative comultiplication used to define the tensor produ...

conjecture | open | MathDB 页面 | 原文线索

V2-014544 | MATHDB-357239

The group-theoretic smooth unknotting conjecture for 2-knots

摘录 / EXCERPT: Let $(\varphi_1,\varphi_2,\varphi_3)\in\operatorname{Alg}^{(4,2)}$ be a spherical triple, meaning that its corresponding surface is a sphere, and suppose that it represents S^4,...

conjecture | open | MathDB 页面 | 原文线索

V2-014545 | MATHDB-357296

Jordan--Hölder criterion for bricks to determine maximal green sequences

摘录 / EXCERPT: Jordan--Hölder criterion. The maximal green sequences ... have the same set of bricks if and only if they are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-014546 | MATHDB-357349

Conjecture on the socle of tensor products of length-two uniserial modules

摘录 / EXCERPT: Let $V_1=V(a)\oplus V(b)$ and $V_2=V(c)\oplus V(d)$ be the socle decompositions of two uniserial $\mathfrak{sl}(2)\ltimes V(m)$-modules, as in the classification theorem, and assum...

conjecture | open | MathDB 页面 | 原文线索

V2-014547 | MATHDB-357384

Conjecture on the Frobenius-Schur indicator of projective indecomposable characters

摘录 / EXCERPT: Let B be a real, non-principal 2-block with defect pair (D,E), and let Φ be its unique projective indecomposable character. Let ϵ denote the Frobenius-Schur ind...

conjecture | open | MathDB 页面 | 原文线索

V2-014548 | MATHDB-357386

The self-orthogonal module conjecture for Iwanaga-Gorenstein algebras

摘录 / EXCERPT: Self-orthogonal module conjecture. Every self-orthogonal Λ-module has finite projective dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-014549 | MATHDB-357387

The Iwanaga-Gorenstein Wakamatsu tilting conjecture

摘录 / EXCERPT: Iwanaga-Gorenstein Wakamatsu tilting conjecture. Any Wakamatsu tilting Λ-module is tilting.

conjecture | open | MathDB 页面 | 原文线索

V2-014550 | MATHDB-357388

The modified upward-closedness conjecture for Wakamatsu tilting modules

摘录 / EXCERPT: Modified upward-closedness conjecture. If $T\in\mathcal{X}_U$ and $U\in\mathcal{Y}_T$, and U is tilting, then T is tilting.

conjecture | open | MathDB 页面 | 原文线索

V2-014551 | MATHDB-357389

The maximal self-orthogonal conjecture for Wakamatsu tilting modules

摘录 / EXCERPT: Maximal self-orthogonal conjecture. Every Wakamatsu tilting module is maximal self-orthogonal.

conjecture | open | MathDB 页面 | 原文线索

V2-014552 | MATHDB-357390

The weak maximal self-orthogonal conjecture for projectively Wakamatsu tilting modules

摘录 / EXCERPT: Weak maximal self-orthogonal conjecture. Every projectively Wakamatsu tilting module is maximal self-orthogonal.

conjecture | open | MathDB 页面 | 原文线索

V2-014553 | MATHDB-357391

The converse Wakamatsu tilting conjecture for maximal-size self-orthogonal modules

摘录 / EXCERPT: Converse Wakamatsu tilting conjecture. If $|T|=|\Lambda|$, then T is Wakamatsu tilting.

conjecture | open | MathDB 页面 | 原文线索

V2-014554 | MATHDB-357459

The finite-group realization conjecture for block fusion systems

摘录 / EXCERPT: Let F be an algebraically closed field of characteristic $p>0$, let G be a finite group, and let B be a block of FG with defect group D. The fusion system...

conjecture | open | MathDB 页面 | 原文线索

V2-014555 | MATHDB-357460

Blockwise Z*-conjecture

摘录 / EXCERPT: Let F be an algebraically closed field of characteristic $p>0$, let G be a finite group, and let B be a block of FG with fusion system $\mathcal{F}=\mathcal{F}_D(B)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-014556 | MATHDB-357461

Rouquier's conjecture on derived equivalence for blocks

摘录 / EXCERPT: Let F be an algebraically closed field of characteristic $p>0$, let G be a finite group, and let B be a block of FG with defect group D and fusion system...

conjecture | open | MathDB 页面 | 原文线索

V2-014557 | MATHDB-357462

Temperley-Lieb representation conjecture for graded crossingless-matching vectors

摘录 / EXCERPT: Let $F^{n}_{A_{2n-1}}$ be the relevant Fock space, and let v_c denote the graded vector associated with a crossingless matching c. Temperley-Lieb representation conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-014558 | MATHDB-357467

The generalized exponent conjecture for finite non-abelian simple groups

摘录 / EXCERPT: Generalized exponent conjecture. The generalized exponent of G is at most 3.

conjecture | open | MathDB 页面 | 原文线索

V2-014559 | MATHDB-357471

The torsion-free one-relator product conjecture

摘录 / EXCERPT: Let A and B be torsion-free groups, set $H=A\star B$, and let $\savebox{\@brx}{$m@th{\langle}$}% \mathopen{\copy\@brx\kern-0.5\wd\@brx\usebox{\@brx}} w\savebox{\@brx}{$m@th{\rang...

conjecture | open | MathDB 页面 | 原文线索

V2-014560 | MATHDB-357472

The Kervaire-Laudenbach conjecture

摘录 / EXCERPT: Let A be a group, let $H=A\star\mathbb Z$, and let $p:H\to\mathbb Z$ be the standard projection to the $\mathbb Z$ factor. For $w\in H$, write $\savebox{\@brx}{$m@th{\langle}...

conjecture | open | MathDB 页面 | 原文线索

V2-014561 | MATHDB-357473

The Kervaire conjecture for free products with an infinite cyclic factor

摘录 / EXCERPT: Let A be a nontrivial group, set $H=A\star\mathbb Z$, and let $\savebox{\@brx}{$m@th{\langle}$}% \mathopen{\copy\@brx\kern-0.5\wd\@brx\usebox{\@brx}} w\savebox{\@brx}{$m@th{\rangle...

conjecture | open | MathDB 页面 | 原文线索

V2-014562 | MATHDB-357474

The Freiheitssatz for proper powers in arbitrary free products

摘录 / EXCERPT: Let $\{A_\lambda\}$ be a family of groups and let $\star_\lambda A_\lambda$ be their free product. For a fixed factor A_λ, let w be an element of the free product that is...

conjecture | open | MathDB 页面 | 原文线索

V2-014563 | MATHDB-357490

Agrarian analogue of the virtually free-by-cyclic theorem

摘录 / EXCERPT: Let $\mathbb F$ be a skew-field. The groups and invariants in the paper include the agrarian cohomological dimension $\operatorname{cd}_{\mathbb F}$ and the agrarian Betti numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-014564 | MATHDB-357491

Bass's conjecture on Davenport and Gao constants of metacyclic groups

摘录 / EXCERPT: Let C_n be the cyclic group of order n, and let ... be a metacyclic group. For triples $(m,n,s)\in\mathbb N\times\mathbb N\times\mathbb Z$ with multiplicative order...

conjecture | open | MathDB 页面 | 原文线索

V2-014565 | MATHDB-357510

The d-complete-poset LE-cactus conjecture

摘录 / EXCERPT: Let P be a d-complete poset. A poset is LE-cactus when its Bender-Knuth group action on its linear extensions satisfies the cactus relations. The d-complete-poset LE-cactus conje...

conjecture | open | MathDB 页面 | 原文线索

V2-014566 | MATHDB-357573

Generalized Iwahori filtration conjecture for simple Lie algebras

摘录 / EXCERPT: Generalized Iwahori filtration conjecture. The space of functions $\Bbbk[\mathbf I]$ admits a filtration whose associated graded space is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-014567 | MATHDB-357583

The local L-factor conjecture for the d4e2-Schwartz space

摘录 / EXCERPT: Main conjecture. The following assertions hold: ${\mathcal{H}}(G(\sF))\subset{d4d2}_{\rho}(G(\sF))$; for every matrix coefficient $c(g)$ of an irreducible representation π, the...

conjecture | open | MathDB 页面 | 原文线索

V2-014568 | MATHDB-357584

The direct-sum convolution conjecture for local L-factor Schwartz spaces

摘录 / EXCERPT: Direct-sum convolution conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014569 | MATHDB-357585

The semi-infinite Goresky-MacPherson extension conjecture

摘录 / EXCERPT: Semi-infinite extension conjecture. If $\CF$ is the constant sheaf on $G_{\sO}$, then $f_{i_{!*}\CF}$ corresponds under the Satake isomorphism to...

conjecture | open | MathDB 页面 | 原文线索

V2-014570 | MATHDB-357586

The locality conjecture for the local L-factor Schwartz space

摘录 / EXCERPT: Locality conjecture. There exists a $G(\sF)\times G(\sF)$-equivariant sheaf ${\mathbf S}_{\rho}$ of complex vector spaces on $\overline{G}(\sF)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-014571 | MATHDB-357607

Dimension formula for symmetric spaces times regular trees

摘录 / EXCERPT: Let X be a symmetric space with trivial compact factor, and let $\ell\in\mathbb{N}$. Here T_3 denotes the regular tree of valence 3, and $\operatorname{corank}(X)$ is the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-014572 | MATHDB-357610

No regular map from the hyperbolic plane to a tree times subexponential-growth space

摘录 / EXCERPT: Regular-map obstruction conjecture. There is no regular map

conjecture | open | MathDB 页面 | 原文线索

V2-014573 | MATHDB-357618

The Cauchy-Harish-Chandra character conjecture for p-adic dual pairs

摘录 / EXCERPT: Let (G,G') be a reductive dual pair over a non-Archimedean local field, let Π' be an irreducible admissible representation of $\widetilde{\mathrm{G}'}$ occu...

conjecture | open | MathDB 页面 | 原文线索

V2-014574 | MATHDB-357633

The relative Grothendieck conjecture for pro-class Galois groups

摘录 / EXCERPT: Let K_i be number fields for $i=1,2$, let ${\mathcal C}_i$ be nontrivial full classes of finite groups, and let ... be an isomorphism. Here $\Sigma({\mathcal C}_i)$ denotes the ass...

conjecture | open | MathDB 页面 | 原文线索

V2-014575 | MATHDB-357644

Huppert's conjecture on groups determined by character degrees

摘录 / EXCERPT: Let G be a finite group, and let H be a finite non-abelian simple group. Denote by the character degrees of a group the degrees of its complex irreducible characters. Huppert's...

conjecture | open | MathDB 页面 | 原文线索

V2-014576 | MATHDB-357667

The Rouquier-block multiplicity formula for cyclotomic q-Schur algebras

摘录 / EXCERPT: The Rouquier-block multiplicity conjecture. The multiplicity of $L(\boldsymbol{mu})$ as a composition factor of $Delta(\boldsymbol{lambda})$ is

conjecture | open | MathDB 页面 | 原文线索

V2-014577 | MATHDB-357676

Random groups are asymptotically almost surely graphically discrete

摘录 / EXCERPT: Random-group graphical discreteness conjecture. Random groups in the few-relator model and Gromov's density model are graphically discrete asymptotically almost surely as the lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-014578 | MATHDB-357677

Free-by-cyclic graphical discreteness conjecture

摘录 / EXCERPT: Free-by-cyclic graphical discreteness conjecture. A one-ended, hyperbolic free-by-cyclic group $F_n\rtimes_{\phi}\mathbb{Z}$ with trivial JSJ decomposition is graphically discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-014579 | MATHDB-357678

Sphere-boundary graphical discreteness conjecture in dimensions at least four

摘录 / EXCERPT: Sphere-boundary graphical discreteness conjecture. Hyperbolic groups with boundary the n-sphere for $n\geq 4$ are graphically discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-014580 | MATHDB-357713

Coherence from vanishing second L2-Betti number

摘录 / EXCERPT: Let X be an aspherical two-complex and let $G=\pi_1(X)$ be its fundamental group. Say that G has trivial second L^2-Betti number when $b_2^{(2)}(G)=0$. Betti coherence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-014581 | MATHDB-357727

The equality conjecture for the GY and projective extended TQFT bimodules

摘录 / EXCERPT: Equality conjecture. If each component of F intersects M_1, then, as $\mathbb{Z}$-graded bimodules over $(A_{EF}(M_2),A_{EF}(M_1))$,

conjecture | open | MathDB 页面 | 原文线索

V2-014582 | MATHDB-357781

Polynomial boundary-growth conjecture for Euler-element representations

摘录 / EXCERPT: Let $(U,\mathcal H)$ be an irreducible unitary representation of the connected real simple Lie group G, let $h\in\fam\gothfam\tengoth g$ be an Euler element, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014583 | MATHDB-357793

Transitive-element conjecture for the property \(\mathcal{P}(H,G)\) in binary wreath products

摘录 / EXCERPT: Let W_n be the ambient binary wreath-product group acting on $\{1,\dots,2^n\}$. For $n\geq 1$, let $H\leq W_n$ be a subgroup containing a permutation $\sigma\in W_n$ that acts tr...

conjecture | open | MathDB 页面 | 原文线索

V2-014584 | MATHDB-357794

The free-subgroup conjecture for normal subgroups of Leavitt path algebra units

摘录 / EXCERPT: Free-subgroup conjecture. Every non-central normal subgroup of $L_K(E)^\times$ contains a non-cyclic free subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-014585 | MATHDB-357841

The Bezrukavnikov-Kazhdan-Varshavsky surjectivity conjecture for the depth zero Bernstein center

摘录 / EXCERPT: Assume G is split, semisimple, and simply connected over $F=\mathbb{F}_q((t))$. Let LG be the loop group, $\mathfrak A^{\operatorname{Fr}}(LG)$ the Frobenius-equivariant Grothe...

conjecture | open | MathDB 页面 | 原文线索

V2-014586 | MATHDB-357864

Naive Ramanujan conjecture for cuspidal representations of quasi-split groups

摘录 / EXCERPT: Let G be a quasi-split group and let ... be a cuspidal representation of G. A representation is tempered when its matrix coefficients have the required $L^{2+\epsilon}$ decay; fo...

conjecture | open | MathDB 页面 | 原文线索

V2-014587 | MATHDB-357865

Revised Ramanujan conjecture for globally generic cuspidal representations

摘录 / EXCERPT: Let G be a quasi-split group and let ... be a globally generic cuspidal representation of G. Revised Ramanujan conjecture. For almost all places v, the local representation...

conjecture | open | MathDB 页面 | 原文线索

V2-014588 | MATHDB-357877

The higher-rank charge-statistic conjecture for type C_3 crystals

摘录 / EXCERPT: Let $\mathcal{B}$ be a crystal of type C_3, let $\mathcal{B}_+(\lambda)$ denote the relevant positive part, and let ϵ_i and s_i be the crystal statistics and simple re...

conjecture | open | MathDB 页面 | 原文线索

V2-014589 | MATHDB-357888

The non-abelian Friedlander-Gordon-Tannenbaum conjecture

摘录 / EXCERPT: Non-abelian Friedlander-Gordon-Tannenbaum conjecture. If $k\geq 3$ and k divides $n-1$, then there exists an orthomorphism of G that fixes the identity element and permutes the...

conjecture | open | MathDB 页面 | 原文线索

V2-014590 | MATHDB-357897

Homomorphism conjecture for shard modules of the same dimension

摘录 / EXCERPT: Let β be a positive real root, and let B_1 and B_2 be shard modules of dimension β. Homomorphism conjecture. Then ... This conjecture would imply that the dimension v...

conjecture | open | MathDB 页面 | 原文线索

V2-014591 | MATHDB-357959

Self-orthogonal quasi-generator conjecture

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field k. For $n\in\mathbb{N}\cup\{0\}$, an n-quasi-generator is a module admitting the quasi-generator property of level n, and...

conjecture | open | MathDB 页面 | 原文线索

V2-014592 | MATHDB-357989

Adler-Mishra-Stevens conjecture on the Bernstein-block local Langlands correspondence

摘录 / EXCERPT: Adler-Mishra-Stevens conjecture. The local Langlands correspondence for L induces a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-014593 | MATHDB-358030

Stanley-Stembridge decomposition conjecture for immanant characters

摘录 / EXCERPT: Stanley-Stembridge decomposition conjecture. Every immanant character $\Gamma^\theta_{\mu/\nu}$ is a non-negative integral sum of Stanley-Stembridge characters.

conjecture | open | MathDB 页面 | 原文线索

V2-014594 | MATHDB-358047

Kazhdan-Lusztig realization conjecture for W-algebra hierarchies

摘录 / EXCERPT: Let $\mathfrak g$ be a simply laced simple Lie algebra, let w be an element of its Weyl group, and let f be the image of w under the Kazhdan-Lusztig map. Consider the w-rea...

conjecture | open | MathDB 页面 | 原文线索

V2-014595 | MATHDB-358052

Azevedo-Shumyatsky conciseness conjecture for commutators of non-commutator words

摘录 / EXCERPT: Azevedo-Shumyatsky conjecture. The word

conjecture | open | MathDB 页面 | 原文线索

V2-014596 | MATHDB-358053

Extension of the Azevedo-Shumyatsky conciseness conjecture to outer commutator words

摘录 / EXCERPT: Extension of the Azevedo-Shumyatsky conjecture. Theorems A and B hold for an arbitrary outer commutator word w.

conjecture | open | MathDB 页面 | 原文线索

V2-014597 | MATHDB-358069

Jones' conjecture on outer automorphisms of group von Neumann algebras

摘录 / EXCERPT: Let G be an ICC group with Kazhdan's property (T). Write $\operatorname{Char}(G)$ for the group of characters of G, $\operatorname{Out}(G)$ for its outer automorphism group, an...

conjecture | open | MathDB 页面 | 原文线索

V2-014598 | MATHDB-358145

Conjecture on the partition-function bound in Harada's conjecture

摘录 / EXCERPT: Partition-function bound conjecture. For $n\geq 7$,

conjecture | open | MathDB 页面 | 原文线索

V2-014599 | MATHDB-358159

Kazhdan-Yom Din asymptotic Schur orthogonality conjecture

摘录 / EXCERPT: Let G be a semisimple group over a local field, let K be a maximal compact subgroup, fix a Haar measure dg on G, and define ... Let (π,H) be a tempered irreducible unitar...

conjecture | open | MathDB 页面 | 原文线索

V2-014600 | MATHDB-358182

The non-lattice conjecture for SOL-like groups

摘录 / EXCERPT: Let S be a SOL-like group. Suppose $\operatorname{Isom}(S,g)$ does not admit a uniform lattice for any left invariant Riemannian metric g on S. The non-lattice conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-014601 | MATHDB-358183

Cornulier's quasi-isometric classification conjecture for solvable Lie groups

摘录 / EXCERPT: Let S_1 and S_2 be simply connected solvable Lie groups of real type, meaning that all eigenvalues of $\operatorname{ad}(X)$ are real for every X in their Lie algebras. Cornu...

conjecture | open | MathDB 页面 | 原文线索

V2-014602 | MATHDB-358189

Conjecture D on fields of values of p-height zero characters

摘录 / EXCERPT: Conjecture D. The set of fields of values of the p-height zero characters of finite groups is exactly $\mathcal F_p$.

conjecture | open | MathDB 页面 | 原文线索

V2-014603 | MATHDB-358190

Quasi-simple reduction conjecture for fields of values of p-height zero characters

摘录 / EXCERPT: Quasi-simple reduction conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014604 | MATHDB-358196

Verbal-width conjecture for higher-rank arithmetic groups

摘录 / EXCERPT: Let Γ be the arithmetic group specified by the local-to-global conjecture above, let G denote the corresponding algebraic group, let $\mathbb{A}_S$ denote the product of t...

conjecture | open | MathDB 页面 | 原文线索

V2-014605 | MATHDB-358226

Sambale's conjecture on involutions and projective indecomposable characters

摘录 / EXCERPT: Let B be a real, non-principal 2-block with defect pair (D,E) and a unique projective indecomposable character Φ. For an involution $x\in G$, write x^G for its conjuga...

conjecture | open | MathDB 页面 | 原文线索

V2-014606 | MATHDB-358272

Sharpness problem for the subgroup spectral sequence of a p-local finite group

摘录 / EXCERPT: Sharpness problem. For every $p>0$ and $q\geq 0$,

open_problem | open | MathDB 页面 | 原文线索

V2-014607 | MATHDB-358280

Finite horofunction-boundary orbit conjecture for polynomial-growth groups

摘录 / EXCERPT: Finite-orbit conjecture. For any Cayley graph of G, the horofunction boundary contains a finite orbit for the canonical group action.

conjecture | open | MathDB 页面 | 原文线索

V2-014608 | MATHDB-358281

Subexponential-growth horofunction-boundary orbit conjecture

摘录 / EXCERPT: Subexponential-growth finite-orbit conjecture. The horofunction boundary of $Gamma$ contains a finite orbit for the underlying group action.

conjecture | open | MathDB 页面 | 原文线索

V2-014609 | MATHDB-358288

Saxl's staircase tensor-square conjecture

摘录 / EXCERPT: Saxl's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014610 | MATHDB-358296

Handlebody-curve cohomology conjecture for symplectic local systems

摘录 / EXCERPT: Let $g\geq 2$, let $\lambda=(\lambda_1\geq\cdots\geq\lambda_g\geq0)$ be a partition, and let V_λ be the corresponding representation of $\mathrm{GL}(g)$, while...

conjecture | open | MathDB 页面 | 原文线索

V2-014611 | MATHDB-358331

Puig's weak conjecture on source algebra dimensions

摘录 / EXCERPT: Let G be a finite group, D a defect group, B a block algebra, and ${\cal B}$ the corresponding block. Write ... for a finitely generated algebra Λ over ${\cal O}$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-014612 | MATHDB-358332

Eaton-Kessar-Külshammer-Sambale weak Donovan conjecture

摘录 / EXCERPT: Let G be a finite group, D a defect group, B a block algebra, and ${\cal B}$ the corresponding block. Write ${\mathbb{F}}{\cal B}$ for the block algebra over ${\mathbb{F}}$....

conjecture | open | MathDB 页面 | 原文线索

V2-014613 | MATHDB-358333

Bounded multiplicities conjecture for pointed fusion systems

摘录 / EXCERPT: Let G be a finite group, D a defect group, B a block algebra, and ${\cal B}$ the corresponding block. Let ${\cal L}{\cal P}={\cal L}{\cal P}(B)$ be the pointed fusion system...

conjecture | open | MathDB 页面 | 原文线索

V2-014614 | MATHDB-358347

The non-increasing order cover conjecture for torsion classes

摘录 / EXCERPT: Non-increasing cover conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014615 | MATHDB-358394

The odd-modulus group-to-cyclic solution conjecture for linear systems

摘录 / EXCERPT: Let $d>1$ be an odd integer. A linear system over $\mathbb{Z}_d$ is a system of linear constraints with coefficients and right-hand sides in $\mathbb{Z}_d$; a solution in a group...

conjecture | open | MathDB 页面 | 原文线索

V2-014616 | MATHDB-358395

The degree bound for compositions of automorphisms of free groups

摘录 / EXCERPT: Let F_n be the free group of rank n, and let f,g be automorphisms of F_n. Write $\operatorname{deg}(f)$ and $\operatorname{deg}(g)$ for their degrees. Degree-composition co...

conjecture | open | MathDB 页面 | 原文线索

V2-014617 | MATHDB-358401

Cartan nonsingularity conjecture for monoids with controlled stabilizers

摘录 / EXCERPT: Cartan nonsingularity conjecture for controlled stabilizers. If $C(\mathbb C M)$ is nonsingular, then $C(kM)$ is nonsingular for every field k of characteristic p.

conjecture | open | MathDB 页面 | 原文线索

V2-014618 | MATHDB-358409

Sylow-subgroup generation conjecture for cochains on classifying spaces

摘录 / EXCERPT: Sylow-subgroup generation conjecture. The bounded derived category $\mathsf{D}^{b}(C^*BG)$ is generated by the object

conjecture | open | MathDB 页面 | 原文线索

V2-014619 | MATHDB-358428

Finite presentability conjecture for finitely generated non-elementary subgroups of \(\operatorname{PSL}(2,\mathbb{R})\)

摘录 / EXCERPT: Finite presentability conjecture. Then H is finitely presented.

conjecture | open | MathDB 页面 | 原文线索

V2-014620 | MATHDB-358456

Reiter's sharp decay conjecture for valid functions

摘录 / EXCERPT: Let $\Sigma_n\curvearrowright X_n$ be an asymptotically oligomorphic sequence of group actions, let G be a compact group, and let $\phi\in\widehat G\setminus\{\mathbf{1}\}$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-014621 | MATHDB-358507

Discrete series conjecture for real reductive spherical spaces

摘录 / EXCERPT: Discrete series conjecture. For every $I\subset S$, the complement of $i\mathfrak{a}_{I,\mathrm{conj}}^*$ in $i\mathfrak{a}_I^*$ has Lebesgue measure zero.

conjecture | open | MathDB 页面 | 原文线索

V2-014622 | MATHDB-358660

Hayes's amalgamated free product conjecture for MF groups

摘录 / EXCERPT: Let G and H be MF groups, and let A be a common amenable subgroup. The amalgamated free product $G *_A H$ is the group obtained by identifying the two copies of A in G an...

conjecture | open | MathDB 页面 | 原文线索

V2-014623 | MATHDB-358668

Tail Conjecture for highest weights of $gl(m|n)$ modules

摘录 / EXCERPT: Tail Conjecture. For any $\lambda\in\Lambda^+_{m|n}$,

conjecture | open | MathDB 页面 | 原文线索

V2-014624 | MATHDB-358669

Langlands duality conjecture for canonical-basis categories

摘录 / EXCERPT: Let G be a reductive group, let $G^\vee$ be its Langlands dual, and let $\mathfrak{g}$ and $\mathfrak{g}^\vee$ denote their adjoint representations. Let P and $P^\vee$ be the w...

conjecture | open | MathDB 页面 | 原文线索

V2-014625 | MATHDB-358670

Coherent geometric Satake conjecture for canonical-basis categories

摘录 / EXCERPT: Let G be a reductive group with Langlands dual $G^\vee$, let $\mathfrak{g}$ be the adjoint representation of G, and let P and $P^\vee$ be the weight lattices of G and...

conjecture | open | MathDB 页面 | 原文线索

V2-014626 | MATHDB-358693

Nebbia's CCR conjecture for boundary-transitive groups on semi-regular trees

摘录 / EXCERPT: Let T be a thick semi-regular tree, and let G be a closed subgroup of $\operatorname{Aut}(T)$ acting transitively on the boundary ∂T. A locally compact group G is C...

conjecture | open | MathDB 页面 | 原文线索

V2-014627 | MATHDB-358694

The ergodic commutant conjecture for sofic groups

摘录 / EXCERPT: Ergodic commutant conjecture. Every sofic group G admits an embedding $\pi:G\to\mathcal{S}$ such that the action

conjecture | open | MathDB 页面 | 原文线索

V2-014628 | MATHDB-358727

The Lie-algebra cohomology conjecture for nilpotent subalgebras

摘录 / EXCERPT: Lie-algebra cohomology conjecture. (a) If

conjecture | open | MathDB 页面 | 原文线索

V2-014629 | MATHDB-358772

The conjecture that separable permutations are exactly the bijective Weyl-group actions on KL bases

摘录 / EXCERPT: Let W be a simply-laced Weyl group with Dynkin diagram I, and let $(U,\mathbb{B})$ be a based W-representation arising from categorification. In the symmetric-group case, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014630 | MATHDB-358779

Geometric wave-front set singleton conjecture

摘录 / EXCERPT: Let F be a nonarchimedean local field, let G be a reductive algebraic group over F, and let $\bar{F}$ be an algebraic closure of F. For an irreducible admissible representa...

conjecture | refuted | MathDB 页面 | 原文线索

V2-014631 | MATHDB-358780

Equality of rational and Zariski wave-front sets

摘录 / EXCERPT: Let F be a nonarchimedean local field, let G be a reductive algebraic group over F, and let π be an irreducible admissible representation of $G(F)$. Let...

conjecture | refuted | MathDB 页面 | 原文线索

V2-014632 | MATHDB-358829

Selection-rule conjecture for Jack-Lax eigenfunctions

摘录 / EXCERPT: Let ${\eusm{Z}}_\mu$ denote the Lax eigenfunction indexed by a partition μ, and let $\mu\cup\nu$ denote the union of the partitions μ and ν. Selection-rule conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-014633 | MATHDB-358855

Bao-Wang's stability conjecture for imathcanonical bases

摘录 / EXCERPT: Let $\mathbf{U}^\imath$ be the imathquantum group with parameters ${\boldsymbol \varsigma}$ and ${\boldsymbol \kappa}$. For each $\lambda,\mu,\nu\in X^+$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014634 | MATHDB-358856

Bao-Wang's asymptotic stability conjecture for trivial submodules

摘录 / EXCERPT: Let $\lambda\in X^+$ satisfy $\overline{\lambda}=\overline{0}$, and let $w_0\in V(\lambda)$ be the vector, unique up to nonzero scalar multiple, whose $\mathbf{U}^\imath$-submodule...

conjecture | open | MathDB 页面 | 原文线索

V2-014635 | MATHDB-358861

Knutson's conjecture on virtual character inverses

摘录 / EXCERPT: Let G be a finite group, let $\operatorname{Irr}(G)$ denote its irreducible characters, let $\mathbb{Z}[\operatorname{Irr}(G)]$ be the group of virtual characters, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014636 | MATHDB-358862

The conjecture that every symmetric group is of Knutson type

摘录 / EXCERPT: Let S_n denote the symmetric group on n letters, and say that a group is of Knutson type when all of its irreducible characters are regular invertible. The paper establishes th...

conjecture | open | MathDB 页面 | 原文线索

V2-014637 | MATHDB-358879

Akbari-Ashrafi conjecture on connected reduced power graphs of simple groups

摘录 / EXCERPT: Let G be a non-abelian simple group. The Akbari-Ashrafi conjecture. The reduced power graph of G is connected only if G is isomorphic to an alternating group A_n. This conj...

conjecture | open | MathDB 页面 | 原文线索

V2-014638 | MATHDB-358887

Steinberg's non-singularity conjecture for Cartan matrices of monoid algebras

摘录 / EXCERPT: Let M be a finite monoid, and let k be a field. Write kM for the monoid algebra of M over k, and assume that the Cartan matrix of the monoid algebra $\mathbb C M$ is non-...

conjecture | open | MathDB 页面 | 原文线索

V2-014639 | MATHDB-358962

The tensor square conjecture for symmetric partitions

摘录 / EXCERPT: Let $g(\lambda,\lambda,\mu)$ be a Kronecker coefficient, and let $\mathbb{S}_\lambda$ denote the irreducible S_n-module indexed by the partition λ. Tensor square conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-014640 | MATHDB-358965

GCT multiplicity-obstruction conjecture of Mulmuley and Sohoni

摘录 / EXCERPT: Let $\mathrm{dc}(\mathrm{per}_m)$ be the determinantal complexity of the $m\times m$ permanent. For integers d,n,m, let $\gamma_{\lambda,d,n,m}$ and $\delta_{\lambda,d,n}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-014641 | MATHDB-358994

Conjecture on the maximal Whittaker orbit of a genuine representation of a covering general linear group

摘录 / EXCERPT: Maximal Whittaker-orbit conjecture. One has $\mathcal{O}^{r,n}\leqslant\mathcal{O}$ for every $\mathcal{O}\in\mathcal{N}_{\rm Wh}^{\rm max}(\pi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-014642 | MATHDB-359104

Conjecture on conjugacy classes of non-p-solvable groups

摘录 / EXCERPT: Non-p-solvable conjugacy-class conjecture. If G is non-p-solvable, then

conjecture | open | MathDB 页面 | 原文线索

V2-014643 | MATHDB-359107

The Forking Path Conjecture for reduced expression graphs

摘录 / EXCERPT: Forking Path Conjecture. For any $w\in S_n$, if p and q are complete paths in $Rex(w)$ with $p_a=q_a$ and $p_z=q_z$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014644 | MATHDB-359108

The Forking Path Conjecture for conflated expression graphs

摘录 / EXCERPT: Forking Path Conjecture for conflated expression graphs. For any $w\in S_n$, if p and q are complete paths in Γ_w with $p_a=q_a$ and $p_z=q_z$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014645 | MATHDB-359109

The refined Forking Path Conjecture for the longest element

摘录 / EXCERPT: Refined Forking Path Conjecture. If p and q are paths in $\Gamma_{w_{0,n}}$ that both pass through $\mathbf{s}$ and $\mathbf{t}$, with $p_a=q_a$ and $p_z=q_z$, then

conjecture | open | MathDB 页面 | 原文线索

V2-014646 | MATHDB-359134

The proposed structure of the irreducible standard representation for p/3 < c < p/2

摘录 / EXCERPT: Proposed structure. The quotient and its character and Hilbert polynomial are

conjecture | open | MathDB 页面 | 原文线索

V2-014647 | MATHDB-359215

Lambda-independence conjecture for BPS sheaves of deformed preprojective algebras

摘录 / EXCERPT: Let $Q=(Q_0,Q_1)$ be a quiver, let $\lambda\in\mathbf{C}^{Q_0}$, and let Π_Q^{λ} be the λ-deformed preprojective algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-014648 | MATHDB-359221

The quasi-heredity criterion for odd-rank algebras of type B

摘录 / EXCERPT: Let n and r be positive integers, let q be the parameter, let Q be the second parameter, and let $\mathcal{L}^n(2r-1)$ denote the algebra under consideration. Quasi-heredit...

conjecture | open | MathDB 页面 | 原文线索

V2-014649 | MATHDB-359311

Jespers-Sun's nilpotent decomposition conjecture for integral group rings

摘录 / EXCERPT: Jespers-Sun's conjecture. A finite group G has ND if and only if $\mathbb{Q}G$ has at most one matrix component.

conjecture | open | MathDB 页面 | 原文线索

V2-014650 | MATHDB-359316

Cohen-Harmse-Morrison-Wright shuffle-group conjecture

摘录 / EXCERPT: Cohen-Harmse-Morrison-Wright conjecture. The shuffle group $G_{k,kn}$ contains the alternating group A_{kn} if $k\geq3$, $(k,n)\neq\{4,2^f\}$ for any positive integer f, and...

conjecture | open | MathDB 页面 | 原文线索

V2-014651 | MATHDB-359378

The standard and costandard module conjecture for non-symmetric quantum loop groups

摘录 / EXCERPT: For each $w\in\mathbb{N}^\bullet$, let W be the corresponding framing data, and let $\widetilde{\mathfrak{M}}^\bullet(W)$ be the associated moduli space with potential...

conjecture | open | MathDB 页面 | 原文线索

V2-014652 | MATHDB-359392

The projective-index classification conjecture for finite linear group actions on \(\mathbb{C}^2\)

摘录 / EXCERPT: Projective-index classification conjecture. Every finite group with a linear action on $\mathbb{C}^2$ has one of these five projective-index sets and associated canonical groups.

conjecture | open | MathDB 页面 | 原文线索

V2-014653 | MATHDB-359399

The indecomposability conjecture for projective functor images of simple modules

摘录 / EXCERPT: Let S_n be the symmetric group, let $x,y\in S_n$, let L_y be the corresponding simple highest weight module, and let θ_x be the projective functor associated with x. I...

conjecture | open | MathDB 页面 | 原文线索

V2-014654 | MATHDB-359402

The induced-representation multiplicity formula for Bessel and Fourier-Jacobi models

摘录 / EXCERPT: Induced-representation multiplicity formula. In the Bessel cases, for possibly reducible induced representations

conjecture | open | MathDB 页面 | 原文线索

V2-014655 | MATHDB-359430

Frobenius-truncated I-adic character formula

摘录 / EXCERPT: Let $\boldsymbol{k}$ be a field of characteristic $p>0$, let $\overline R=R/\langle x_1^p,\ldots,x_n^p,y_1^p,\ldots,y_n^p\rangle$, and let $\overline I$ be the corresponding ideal....

conjecture | open | MathDB 页面 | 原文线索

V2-014656 | MATHDB-359431

Leading-term conjecture for p-semistandard tableaux

摘录 / EXCERPT: Let $\overline I$ be the ideal in the Frobenius-truncated ring, let $Tab^{(p)}_n(a,b)$ be the set of p-semi-standard tableaux of shape (a,b), and let M_T denote the leading m...

conjecture | open | MathDB 页面 | 原文线索

V2-014657 | MATHDB-359432

p-restrictedness conjecture for composition factors of truncated symmetric powers

摘录 / EXCERPT: Let $\overline R_{(a,b)}=T_p\operatorname{Sym}^a(\boldsymbol{k}^n)\otimes T_p\operatorname{Sym}^b(\boldsymbol{k}^n)$, where $T_p\operatorname{Sym}^d(\boldsymbol{k}^n)$ is the trunc...

conjecture | open | MathDB 页面 | 原文线索

V2-014658 | MATHDB-359436

Gutin's uniqueness conjecture for maximal self-adjoint decompositions

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite-dimensional $\ast$-algebra over $\mathbb{R}$, and let $H\in\mathcal{A}$ be self-adjoint, meaning $H^{\ast}=H$. Let $k\in\mathbb{N}$ be maximal among t...

conjecture | open | MathDB 页面 | 原文线索

V2-014659 | MATHDB-359487

The quotient conjecture for irreducible generalized root systems

摘录 / EXCERPT: Let R be an irreducible generalized root system (GRS) of rank at least 2. Quotient conjecture. Every irreducible GRS of rank at least 2 is equivalent to a quotient of a root...

conjecture | open | MathDB 页面 | 原文线索

V2-014660 | MATHDB-359503

Web-basis conjecture for flamingo Specht modules

摘录 / EXCERPT: Let $r \geq 3$ and let $\mathscr{S} \subset \Pi(n,d,r)$ denote the collection of set partitions that can be made noncrossing by applying at most $r-3$ adjacent transpositions. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014661 | MATHDB-359514

Lyndon-Ullman-Kim-Koberda conjecture on freeness of parabolic matrix groups

摘录 / EXCERPT: Lyndon-Ullman-Kim-Koberda conjecture. For every nonzero rational value of q in $(-4,4)$, the group Δ_q is not free.

conjecture | open | MathDB 页面 | 原文线索

V2-014662 | MATHDB-359528

Dense-orbit and finite-orbit equivalence for orthogonal and symplectic Grassmannians

摘录 / EXCERPT: Dense-orbit and finite-orbit equivalence. Unless

conjecture | open | MathDB 页面 | 原文线索

V2-014663 | MATHDB-359562

Divisibility and symmetry conjecture for links in Hopf algebras with the dual Chevalley property

摘录 / EXCERPT: Let H be a non-cosemisimple Hopf algebra over $\Bbbk$ with the dual Chevalley property. Let $\mathcal{M}$ denote the set of simple subcoalgebras of H, and let $\mathcal{P}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-014664 | MATHDB-359563

Demonet's conjecture on rational completeness of the g-vector fan

摘录 / EXCERPT: Demonet's conjecture. A is τ-tilting finite if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014665 | MATHDB-359564

The τ-reduced second Brauer-Thrall conjecture

摘录 / EXCERPT: The τ-reduced Brauer-Thrall II' conjecture. There exist $\bd\in\K_0(A)^+$ and a generically indecomposable $\cZ\in\Irr^\tau(A,\bd)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014666 | MATHDB-359566

The second Brauer-Thrall conjecture for tau-tilting infinite algebras

摘录 / EXCERPT: Second Brauer-Thrall conjecture. If A is tau-tilting infinite, then there exists a dimension $d\geq 0$ and infinitely many pairwise non-isomorphic bricks...

conjecture | open | MathDB 页面 | 原文线索

V2-014667 | MATHDB-359577

The \Sh-rigidity conjecture for Chevalley and related groups

摘录 / EXCERPT: The Sh-rigidity conjecture. The following groups G are Sh-rigid: (1) $G=G(k)$, the group of k-points of a split simple Chevalley group defined over a sufficiently large field...

conjecture | open | MathDB 页面 | 原文线索

V2-014668 | MATHDB-359583

The concise symmetric Kronecker coefficient conjecture

摘录 / EXCERPT: For a partition λ, let $g(\lambda,\mu,\nu)$ be the Kronecker coefficient and define the symmetric Kronecker coefficient by ... A counting function is concise if every positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-014669 | MATHDB-359630

Outer-automorphism compatibility conjecture for local Langlands reciprocity

摘录 / EXCERPT: Let G be a reductive group, let $\Out_F(G)$ be its group of F-rational outer automorphisms, let $\theta\in\Out_F(G)$, and let $\mathfrak w$ be a Whittaker datum. Outer-automorp...

conjecture | open | MathDB 页面 | 原文线索

V2-014670 | MATHDB-359640

Wreath Macdonald-Kostka inversion conjecture

摘录 / EXCERPT: Let $\lambda^\bullet,\mu^\bullet\in\mathbb{Y}^r$ and let $\alpha\in Q$ be sufficiently deep in its Weyl chamber. Let λ^*, μ^-, and α^- denote the involutions and...

conjecture | open | MathDB 页面 | 原文线索

V2-014671 | MATHDB-359641

Wreath Macdonald-Kostka alpha-independence conjecture at q equals t inverse

摘录 / EXCERPT: Let $\lambda^\bullet,\mu^\bullet\in\mathbb{Y}^r$ and let $\alpha\in Q$ index the wreath Macdonald-Kostka polynomial $\tilde{K}_{\lambda^\bullet,\mu^\bullet,\alpha}(q,t)$. Alpha-ind...

conjecture | open | MathDB 页面 | 原文线索

V2-014672 | MATHDB-359686

Lubotzky's baby Wiegold conjecture

摘录 / EXCERPT: Baby Wiegold conjecture. For $n\ge 3$ and G a finite simple group, $\operatorname{Aut}(F_n)$ has no fixed points on $\mathcal{N}_n(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-014673 | MATHDB-359687

The noncongruence-surjection conjecture for finite simple groups

摘录 / EXCERPT: Noncongruence-surjection conjecture. If G is a nonabelian finite simple group, then there exists a surjection $\varphi:F_2\to G$ such that Γ_φ is noncongruence. Equi...

conjecture | open | MathDB 页面 | 原文线索

V2-014674 | MATHDB-359715

Snake in the Snail conjecture for minimal snake modules

摘录 / EXCERPT: Snake in the Snail conjecture. For odd m, the extended T-systems appear in successive lines of the snail operator, and the component corresponding to

conjecture | open | MathDB 页面 | 原文线索

V2-014675 | MATHDB-359720

The modular invariant-theory conjecture for arbitrary representations

摘录 / EXCERPT: Let $G\subset GL(V)$ be a finite group, let F be a field, and let V be a finite-dimensional F-vector space. Let $S(V)$ be the symmetric algebra of V, and write $S(V)^G$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-014676 | MATHDB-359830

Quantum Springer conjecture for strongly equivariant quantum character modules

摘录 / EXCERPT: Let G) be a connected reductive group, let W be its Weyl group, and let ρ denote the relevant quantum Springer constructions appearing in the isomorphism and theorems cite...

conjecture | open | MathDB 页面 | 原文线索

V2-014677 | MATHDB-359831

Quantum generalized Springer decomposition conjecture

摘录 / EXCERPT: Let G be a connected reductive group. A $\mathtt{q}$-cuspidal datum is a G-conjugacy class of pairs (L,C), where L is an elliptic-pseudo Levi subgroup of G and C is a s...

conjecture | open | MathDB 页面 | 原文线索

V2-014678 | MATHDB-359856

The near-linear value conjecture for cyclic groups

摘录 / EXCERPT: Let $\mathrm{val}(\mathbb Z_n)$ denote the value parameter for the cyclic group $\mathbb Z_n$, as defined in the paper. The near-linear value conjecture. For every $\varepsilon>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-014679 | MATHDB-359876

Naive fair-counting adaptation of Malle's conjecture

摘录 / EXCERPT: Let G be a non-trivial finite group and let k be a number field. Write $\mathfrak{f}(K/k)$ for the product of the primes of k that ramify in K. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014680 | MATHDB-359910

The two-orders characterization conjecture for finite simple groups

摘录 / EXCERPT: Two-orders characterization conjecture. $G\cong S$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014681 | MATHDB-359911

The Vo-set two-invariant characterization conjecture for finite simple groups

摘录 / EXCERPT: Vo-set characterization conjecture. $G\cong S$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014682 | MATHDB-359958

Bruhat-order characterization for affine Weyl groups of type A

摘录 / EXCERPT: Let (λ_1,w_1) and (λ_2,w_2) be elements of the affine Weyl group written in semidirect-product form, where λ_1 and λ_2 lie in the root lattice and...

conjecture | open | MathDB 页面 | 原文线索

V2-014683 | MATHDB-359965

Howie's conjecture on solvability of non-singular systems over groups

摘录 / EXCERPT: Let $\{w_j=1\}_{j\in J}$ be a system of equations with variables $\{x_i\}_{i\in I}$ over a group G. The system is non-singular when the rows of exponent sums of the variables in...

conjecture | open | MathDB 页面 | 原文线索

V2-014684 | MATHDB-359967

The solubilizer probability bound conjecture

摘录 / EXCERPT: Let G be a finite group and $x\in G$. Define the solubilizer probability of x by ... where $\operatorname{Sol}_G(x)$ is the set of elements $y\in G$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-014685 | MATHDB-359988

Open L-parameter characterization of generic representations

摘录 / EXCERPT: Let π be an irreducible representation of a reductive p-adic group $\mathcal G(F)$, and assume that a local Langlands correspondence exists for the Bernstein block of...

conjecture | open | MathDB 页面 | 原文线索

V2-014686 | MATHDB-359989

The standard submodule conjecture for reductive p-adic groups

摘录 / EXCERPT: Let $\mathcal G(F)$ be a reductive p-adic group, let π_{st} be a standard representation of $\mathcal G(F)$, and let π be an irreducible subquotient of π_{st}. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-014687 | MATHDB-359991

Green-Héthelyi-Lilienthal reformulation of Oliver's p-group conjecture

摘录 / EXCERPT: Green-Héthelyi-Lilienthal reformulation. If V is an F-module, then there is an element $1\neq g\in\Omega_1(Z(G))$ such that the minimal polynomial of the action of g on V d...

conjecture | open | MathDB 页面 | 原文线索

V2-014688 | MATHDB-359997

The freeness conjecture for the tamely ramified Eisenstein Hecke module

摘录 / EXCERPT: Let W be the Weyl group, let Λ be the coweight lattice, and let $\widetilde{C}$ be the Hecke module generated by the relevant tamely ramified Eisenstein series. Freeness...

conjecture | open | MathDB 页面 | 原文线索

V2-014689 | MATHDB-359998

The many-point Hecke-module presentation conjecture for tamely ramified Eisenstein series

摘录 / EXCERPT: Let $S\subset\mathbb{P}^1(\mathbb{F}_q)$ be a nonempty finite set of tame ramification points, let C_{Eis} be the corresponding Eisenstein Hecke module, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014690 | MATHDB-360017

Physical rigidity conjecture for generalised Frenkel-Gross connections

摘录 / EXCERPT: A generalised Frenkel-Gross G-connection is a connection on the multiplicative group with an irregular singularity at infinity of slope $1/h$ and a regular singularity at 0. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-014691 | MATHDB-360034

Sarnak-Xue density hypothesis

摘录 / EXCERPT: Let $G\leq GL_n$ be a connected semisimple linear algebraic group over $\mathbb{Q}$ with non-compact $G_\infty=G(\mathbb{R})$. For $q\in\mathbb{N}$, let $\Gamma(q)$ be the level-...

conjecture | open | MathDB 页面 | 原文线索

V2-014692 | MATHDB-360067

Kazhdan-Laumon finite cohomological dimension conjecture

摘录 / EXCERPT: Let G be a semisimple algebraic group split over a finite field, let $\mathcal{A}$ be the abelian category of glued perverse sheaves in the Kazhdan-Laumon construction, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-014693 | MATHDB-360096

The integrality and self-duality conjecture for Clifford-group characters

摘录 / EXCERPT: Integrality and self-duality conjecture. The entries in the character table of $\mathcal{C}_n$ are integers, and hence all irreducible characters of $\mathcal{C}_n$ are self-dual.

conjecture | open | MathDB 页面 | 原文线索

V2-014694 | MATHDB-360097

The faithfulness conjecture for the matrix representation of Clifford groups

摘录 / EXCERPT: Faithfulness conjecture. The matrix representation $\mathcal{M}_{2^n}$ for $\mathcal{C}_n$ is faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-014695 | MATHDB-360098

The normal-subgroup conjecture for n-qubit Clifford groups

摘录 / EXCERPT: Normal-subgroup conjecture. When $n\geq 3$, the n-qubit Pauli group $\mathcal{P}_n$ is the only non-trivial proper normal subgroup of $\mathcal{C}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-014696 | MATHDB-360134

The virtual mapping-torus conjecture for locally quasiconvex hyperbolic groups

摘录 / EXCERPT: A group is locally quasiconvex if every finitely generated subgroup is quasiconvex, and a group is hyperbolic in the sense of Gromov. A fully irreducible partial endomorphism is a...

conjecture | open | MathDB 页面 | 原文线索

V2-014697 | MATHDB-360135

The negative immersions to local quasiconvexity conjecture

摘录 / EXCERPT: Let X be a compact 2-complex with negative immersions. Write π_1X for its fundamental group, and let local quasiconvexity and the finite-generation intersection property re...

conjecture | open | MathDB 页面 | 原文线索

V2-014698 | MATHDB-360137

The locally quasiconvex virtual mapping-torus conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ be a locally quasiconvex hyperbolic group. An immersion of finite directed height is an immersion $\psi:H\rightarrow F$ of graphs satisfying the finite directed-h...

conjecture | open | MathDB 页面 | 原文线索

V2-014699 | MATHDB-360156

Affine Weyl group description conjecture for the map to the fusion ring

摘录 / EXCERPT: Let P be the weight lattice, P^+ the dominant integral weights, $P_{\ell}\subset P^+$ the weights of level ell, and let W_{ell} be the affine Weyl group acting on the weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-014700 | MATHDB-360157

Explicit coefficient conjecture for the map to the fusion ring

摘录 / EXCERPT: Let P be the weight lattice, P^+ the dominant integral weights, $P_{\ell}\subset P^+$ the weights of level ell, and let W_{ell} be the affine Weyl group with length functio...

conjecture | open | MathDB 页面 | 原文线索

V2-014701 | MATHDB-360178

The classification conjecture for maximal subgroups containing the cyclic factor

摘录 / EXCERPT: The classification conjecture. One of the six cases in the table holds:

conjecture | open | MathDB 页面 | 原文线索

V2-014702 | MATHDB-360181

Two-sided Curto-Herrero conjecture for tuples of matrices

摘录 / EXCERPT: Two-sided Curto-Herrero conjecture. For every ${\underline{A}},{\underline{B}}\in M_n^m$, the ${\rm GL}_n$-orbits of ${\underline{A}}$ and ${\underline{B}}$ coincide if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014703 | MATHDB-360184

Quinn's equivariant and stratified rigidity conjecture

摘录 / EXCERPT: Let M^n be a manifold with a “nice” group action or a space with a “nice” stratification, and let N^n be another manifold or space that is equivariantly or stratified homotopy...

conjecture | open | MathDB 页面 | 原文线索

V2-014704 | MATHDB-360206

The uniform insolubility conjecture for finite non-abelian simple groups

摘录 / EXCERPT: Let N be a finite non-abelian characteristically simple group, let $a,b\in\operatorname{Aut}(N)$, and identify N with $\operatorname{Inn}(N)$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-014705 | MATHDB-360239

Bader-Caprace-Lécureux's virtual simplicity conjecture for exotic Euclidean building lattices

摘录 / EXCERPT: Let X be an irreducible, locally finite, exotic Euclidean building and let $\Gamma < \operatorname{Aut}(X)$ be a lattice. Bader-Caprace-Lécureux's conjecture. Then Γ is vi...

conjecture | open | MathDB 页面 | 原文线索

V2-014706 | MATHDB-360263

Conjecture on the essential nilpotency class of zero entropy groups

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension at least 1, and let $G\leq \operatorname{Aut}(X)$ be a zero entropy subgroup. The essential nilpotency class conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-014707 | MATHDB-360277

Uniqueness conjecture for genus-two minimal surfaces in spherical space-forms

摘录 / EXCERPT: Let a round spherical space-form be a quotient of the round 3-sphere by a finite group of isometries, and suppose its Heegaard genus is 2. An embedded minimal surface of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-014708 | MATHDB-360366

Hilbert's Sextic Conjecture for the Valentiner action

摘录 / EXCERPT: Hilbert's Sextic Conjecture. For any smooth, irreducible, generically free A_6-curve X,

conjecture | open | MathDB 页面 | 原文线索

V2-014709 | MATHDB-360409

The linear conjugacy-class bound for finite groups

摘录 / EXCERPT: Let G be a finite group, let $k(G)$ denote its number of conjugacy classes, and let $|G|$ denote its order. Linear conjugacy-class bound. There exists a constant $C>0$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-014710 | MATHDB-360415

The Fourier pairing conjecture for irreducible spetsial complex reflection groups

摘录 / EXCERPT: Let W be an irreducible spetsial complex reflection group, and write $\operatorname{Irr}(W)$ for its irreducible characters. Let $\operatorname{Deg}_\chi(q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014711 | MATHDB-360494

Ismagilov's irreducibility conjecture for regular representations

摘录 / EXCERPT: Ismagilov's conjecture. The right regular representation is irreducible if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014712 | MATHDB-360495

Irreducibility conjecture for Koopman's representations

摘录 / EXCERPT: Koopman's irreducibility conjecture. The representation $\pi^{\alpha,\mu,X}$ is irreducible if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-014713 | MATHDB-360574

Module-isomorphism conjecture for regular posets

摘录 / EXCERPT: Module-isomorphism conjecture. For $P,Q\in\sfRP_n$, if

conjecture | open | MathDB 页面 | 原文线索

V2-014714 | MATHDB-360575

Indecomposability conjecture for disconnected regular Schur labeled skew shape modules

摘录 / EXCERPT: Indecomposability conjecture. If $\sh(\tau_P)$ is disconnected and contains no disconnected ribbon, then the associated module $\sfM_P$ is indecomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-014715 | MATHDB-360579

Homological concentration conjecture for free Lie algebras in the Tits-Kantor-Koecher category

摘录 / EXCERPT: Let $\operatorname{\mathfrak{T}}$ be the category whose objects are completely reducible $\mathfrak{sl}_2$-modules that decompose as direct sums of copies of the trivial and adjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-014716 | MATHDB-360599

Elliptic Lie-algebra representation conjecture via Steinberg correspondences

摘录 / EXCERPT: Let R be the relevant elliptic root system, let δ_E and δ_{pt} be the indicated imaginary-root classes, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014717 | MATHDB-360601

Almost finiteness of minimal essentially free elementary amenable group actions

摘录 / EXCERPT: Let G be an elementary amenable group acting minimally and essentially freely on an infinite compact metrizable space X with the small boundary property. The action is almost f...

conjecture | open | MathDB 页面 | 原文线索

V2-014718 | MATHDB-360628

The amenability characterization of richly synergodic groups

摘录 / EXCERPT: Let G be a countably infinite group. Recall that G is richly synergodic if the synergodic actions are dense in $\mathrm{Erg}(G \times G,X,\mu)$ for every standard probability s...

conjecture | open | MathDB 页面 | 原文线索

V2-014719 | MATHDB-360633

Twining big algebra conjecture for endoscopic groups

摘录 / EXCERPT: Let $\G$ be a connected semisimple complex Lie group with a distinguished automorphism σ fixing a pinning, let $\G_\sigma=((\G^\vee)^\sigma_0)^\vee$ be the associated endosc...

conjecture | open | MathDB 页面 | 原文线索

V2-014720 | MATHDB-360666

Equivalence-relation conjecture for projective-functor linkage

摘录 / EXCERPT: Let L be a simple $\mathfrak{g}$-module, and let $\mathcal{X}_L$ be the class of simple modules linked to L by projective functors. Define $M\triangleright N$ when M occurs i...

conjecture | open | MathDB 页面 | 原文线索

V2-014721 | MATHDB-360667

Nonexistence conjecture for strange subquotients

摘录 / EXCERPT: Let L be a simple $\mathfrak{g}$-module and let V be a finite-dimensional $\mathfrak{g}$-module. A strange subquotient of $V\otimes_{\mathbb{C}}L$ is a non-simple subquotient h...

conjecture | open | MathDB 页面 | 原文线索

V2-014722 | MATHDB-360668

Equality conjecture for the Serre subcategories associated with a simple module

摘录 / EXCERPT: Let L be a simple $\mathfrak{g}$-module. Let $\mathscr{A}(L)$ be the abelian category of finitely generated subquotients of modules $V\otimes_{\mathbb{C}}L$ with V finite-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-014723 | MATHDB-360676

Schleimer's hyperbolicity conjecture for multiarc and curve graphs

摘录 / EXCERPT: Let Σ be a compact, connected, orientable surface, and let $\mathcal A$ be a multiarc and curve graph on Σ. A compact, essential subsurface $W\subset\Sigma$ is a witn...

conjecture | open | MathDB 页面 | 原文线索

V2-014724 | MATHDB-360689

GRV23's restricted composition-factor conjecture

摘录 / EXCERPT: Let $\mathbf{k}$ be a field of characteristic $p>0$, let $T_p\operatorname{Sym}^d(\mathbf{k}^n)$ be the truncated symmetric power functor, and let $L(\lambda)$ be a simple...

conjecture | open | MathDB 页面 | 原文线索

V2-014725 | MATHDB-360690

Navarro's blockwise Alperin-Glauberman conjecture

摘录 / EXCERPT: Let $G\unlhd \Gamma$ be finite groups and let B be a p-block of G. For every radical p-subgroup Q of Γ, let $\mathrm{dz}({\bf N}_\Gamma(Q)/Q\mid B)$ denote the set...

conjecture | open | MathDB 页面 | 原文线索

V2-014726 | MATHDB-360693

Neumann-Praeger bounded-index conjecture for equal conjugate unions

摘录 / EXCERPT: Neumann-Praeger conjecture. There is an integer function f such that these conditions imply

conjecture | open | MathDB 页面 | 原文线索

V2-014727 | MATHDB-360754

Kac-Weisfeiler's maximal simple-module dimension conjecture

摘录 / EXCERPT: Kac-Weisfeiler conjecture (KW1).

conjecture | open | MathDB 页面 | 原文线索

V2-014728 | MATHDB-360777

Stable-rank characterization of Drozd-Crawley-Boevey representation-type trichotomy

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra over a field $\mathcal{K}$, let $\mathrm{st}(\Lambda)$ be the stable rank at which the descending powers of the module-category radica...

conjecture | open | MathDB 页面 | 原文线索

V2-014729 | MATHDB-360784

Saturated Newton polytope conjecture for Kronecker products of Schur functions

摘录 / EXCERPT: For a symmetric function f, let $f(x_1,\ldots,x_k)$ denote its specialization obtained by setting $x_m=0$ for all $m\geq k+1$. A polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-014730 | MATHDB-360792

Stable character bounds for word measures on wreath products

摘录 / EXCERPT: Let G be a finite group, let F_r be the free group of rank r, and let $w\in F_r$. For a stable irreducible character $\chi=(\chi_n)_{n\in\mathbb{N}}$ of $G\wr S_{\bullet}$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-014731 | MATHDB-360817

Existence of infinite cubulable groups with fixed-point property (FW_n)

摘录 / EXCERPT: A group is cubulable if it acts properly and cocompactly on a median graph. For $n\geq 1$, (FW_n) denotes the fixed-point property for actions on median graphs of cubica...

conjecture | open | MathDB 页面 | 原文线索

V2-014732 | MATHDB-360818

Common cubulable quotients of acylindrically hyperbolic groups

摘录 / EXCERPT: A group is cubulable if it acts properly and cocompactly on a median graph, and it is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic sp...

conjecture | open | MathDB 页面 | 原文线索

V2-014733 | MATHDB-360822

The classification conjecture for finite groups with full extended Reidemeister spectrum

摘录 / EXCERPT: For a finite group G, let $k(G)$ denote its class number. A finite group has full extended Reidemeister spectrum when its extended Reidemeister spectrum contains every possible v...

conjecture | open | MathDB 页面 | 原文线索

V2-014734 | MATHDB-360828

Non-isomorphism conjecture for pure infinitely braided Thompson groups

摘录 / EXCERPT: Let $BF_n(\{1\})$ and $BF_n(\mathcal{P}_n)$ be the groups associated respectively with the trivial label group and with the pure braid group $\mathcal{P}_n$ in the construction of...

conjecture | open | MathDB 页面 | 原文线索

V2-014735 | MATHDB-360829

The Geometric Hanna Neumann conjecture

摘录 / EXCERPT: Geometric Hanna Neumann conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-014736 | MATHDB-360830

The finite abelian hierarchy conjecture for the L^2-Hall property

摘录 / EXCERPT: Finite abelian hierarchy conjecture. Then G should have the L^2-Hall property and satisfy the Geometric Hanna Neumann conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014737 | MATHDB-360855

Relative hyperbolicity conjecture for fully irreducible automorphisms of infinitely ended groups

摘录 / EXCERPT: Let G be the fundamental group of a non-sporadic graph of groups with finite edge stabilisers, and let $\Phi\in\operatorname{Aut}(G)$ be fully irreducible relative to this splitt...

conjecture | open | MathDB 页面 | 原文线索

V2-014738 | MATHDB-360888

The regular-representation congruence for twisted Alexander polynomials of knots

摘录 / EXCERPT: Regular-representation congruence. The twisted Alexander polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014739 | MATHDB-360912

Finite-dimensionality criterion for irreducible twisted super Yangian modules

摘录 / EXCERPT: Let ${\mathscr{B}_{\bm s,\bm \varepsilon}}$ be the twisted super Yangian of type AIII, let $V(\bm\mu(u))$ be an irreducible ${\mathscr{B}_{\bm s,\bm \varepsilon}}$-module, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014740 | MATHDB-360927

Leclerc's conjecture on the Aizenbud-Lapid binary operation

摘录 / EXCERPT: Leclerc's conjecture. For $C_1,C_2\in \operatorname{Comp}$,

conjecture | open | MathDB 页面 | 原文线索

V2-014741 | MATHDB-360933

Oblomkov-Yun quasi-split extension conjecture

摘录 / EXCERPT: Let $\mathbb{G}$ be a quasi-split generic group, and let the local systems, perverse filtrations, rational DAHA actions, and braid-group actions be defined as in the split case. Ob...

conjecture | open | MathDB 页面 | 原文线索

V2-014742 | MATHDB-360935

The effectiveness conjecture for equivariant Ehrhart polynomials

摘录 / EXCERPT: Let G be a finite group, let M be a lattice of rank d, and let $\rho:G\to\operatorname{Aff}(M)$ be an affine representation. Let $P\subset M_\mathbb{R}$ be a G-invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-014743 | MATHDB-360936

Stapledon's trivial-representation conjecture for equivariant Ehrhart coefficients

摘录 / EXCERPT: Let G be a finite group, let M be a lattice, let $\rho:G\to\operatorname{Aff}(M)$ be an affine representation, and let P be a G-invariant lattice polytope. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-014744 | MATHDB-361010

The Archimedean co-socle analogy conjecture for degenerate principal series

摘录 / EXCERPT: Archimedean co-socle analogy conjecture. The length of the maximal semi-simple quotient of $\operatorname{I}_{P,v_1}(s,\mathbf{1})$ equals that of...

conjecture | open | MathDB 页面 | 原文线索

V2-014745 | MATHDB-361012

Hanany-Puder character bound conjecture for word maps on symmetric groups

摘录 / EXCERPT: Hanany-Puder conjecture. For any $k\in\mathbb{Z}_{\geq2}$, any $\lambda\vdash k$, and any $w\in F_m$,

conjecture | open | MathDB 页面 | 原文线索

V2-014746 | MATHDB-361034

Hernandez-Leclerc's bijection conjecture for cluster monomials and real simple modules

摘录 / EXCERPT: Let $\mathscr{C}_{\ell}$ be the Hernandez-Leclerc monoidal subcategory and let $K_{0}(\mathscr{C}_{\ell})$ be its Grothendieck ring, identified with a cluster algebra. A simple mod...

conjecture | open | MathDB 页面 | 原文线索

V2-014747 | MATHDB-361149

Root enumerators for index-two subgroups of classical Weyl groups

摘录 / EXCERPT: Let W be a Weyl group of type A, B, or D, and let $H\leq W$ be a subgroup of index 2. For an integer k, define the k-th root enumerator of H by ... A root enumerator...

conjecture | open | MathDB 页面 | 原文线索

V2-014748 | MATHDB-361156

Conjecture on unipotent tensor squares for GL(n,q)

摘录 / EXCERPT: Let n be a positive integer, let $\mu=(\mu_1,\mu_2,\dots)$ be a partition of n, and let τ be a partition of n. For triples of partitions define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-014749 | MATHDB-361157

Dominance monotonicity conjecture for unipotent tensor squares

摘录 / EXCERPT: Let n be a positive integer, let μ and μ' be partitions of n, and let τ be a partition of n. Write ... For partitions of n, let $\mu'\trianglelefteq\mu$ mean tha...

conjecture | open | MathDB 页面 | 原文线索

V2-014750 | MATHDB-361159

The rigid refined local Langlands correspondence for rigid inner twists

摘录 / EXCERPT: Rigid refined local Langlands conjecture. There is a bijection, depending on $\mathfrak{w}$,

conjecture | open | MathDB 页面 | 原文线索

V2-014751 | MATHDB-361169

Locally finite-dimensional infinite Gabriel theorem conjecture

摘录 / EXCERPT: Locally finite-dimensional infinite Gabriel theorem conjecture. The category $\operatorname{Rep}_{\mathrm{lf}}(\Omega)$ is of unique type and is infinite Krull-Schmidt if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-014752 | MATHDB-361181

Algebraicity conjecture for extensions of algebraic \operatorname{GL}(n,\mathbb{Z})-representations

摘录 / EXCERPT: Let ... be a short exact sequence of $\operatorname{GL}(n,\mathbb{Z})$-representations, where M' and M'' are algebraic $\operatorname{GL}(n,\mathbb{Z})$-representations. Algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-014753 | MATHDB-361182

Stable cohomology conjecture for polynomial \operatorname{Aut}(F_n)-modules

摘录 / EXCERPT: Let F_n be the free group of rank n, let IA_n be the kernel of the natural map $\operatorname{Aut}(F_n)\to\operatorname{GL}(n,\mathbb{Z})$, and let M be a po...

conjecture | open | MathDB 页面 | 原文线索

V2-014754 | MATHDB-361191

Projectivity characterization for restrictions from G to H

摘录 / EXCERPT: Projectivity conjecture. The restriction $\pi|_H$ is projective if and only if π is supercuspidal.

conjecture | open | MathDB 页面 | 原文线索

V2-014755 | MATHDB-361219

The measure-equivalence conjecture for words in free groups

摘录 / EXCERPT: Measure-equivalence conjecture. If u and w are measure equivalent in all finite groups, then there exists $\phi\in\operatorname{Aut}(F)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014756 | MATHDB-361220

Mel'nikov's conjecture on residually finite one-relator groups

摘录 / EXCERPT: Mel'nikov's conjecture. An infinite and residually finite Mel'nikov group is either a free group, a surface group or a solvable Baumslag--Solitar group.

conjecture | open | MathDB 页面 | 原文线索

V2-014757 | MATHDB-361226

Treumann-Venkatesh admissibility and functoriality conjectures for Tate cohomology

摘录 / EXCERPT: Let G and H be the groups occurring in the Local Langlands correspondence, let $\boldsymbol{Z}[[\sigma]$ be the group ring of...

conjecture | open | MathDB 页面 | 原文线索

V2-014758 | MATHDB-361240

Conjugacy conjecture for basic regular exact Borel subalgebras

摘录 / EXCERPT: Conjugacy conjecture. There exists an invertible element $x\in R$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-014759 | MATHDB-361259

Strong Property (T) elementary action conjecture for Gromov-hyperbolic spaces

摘录 / EXCERPT: Strong Property (T) action conjecture. Any action of G by isometries on X is elementary.

conjecture | open | MathDB 页面 | 原文线索

V2-014760 | MATHDB-361294

The dimension-generating-series conjecture for free alternative algebras

摘录 / EXCERPT: Dimension-generating-series conjecture. Suppose $a(z)$ and $b(z)$ are the solutions of

conjecture | open | MathDB 页面 | 原文线索

V2-014761 | MATHDB-361296

Prasad-Takloo-Bighash epsilon dichotomy conjecture for linear periods

摘录 / EXCERPT: Let F be a finite extension of $\d075\mathbb{Q}_p$ or $F=\mathbb{R}$, let D be a central division algebra over F with $\dim_F(D)=d^2$, put $G=\mathrm{GL}_m(D)$ and $n=md$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-014762 | MATHDB-361297

Prasad-Takloo-Bighash reformulation via component-group characters

摘录 / EXCERPT: Assume F is non-archimedean. Let $\phi\colon WD_F\to\mathrm{GL}_n(\mathbb{C})$ be an L-parameter, let Π_ϕ be the packet across all inner forms of $\mathrm{GL}_n(F)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-014763 | MATHDB-361298

Geometric multiplicity formula for linear periods

摘录 / EXCERPT: Let F be as in the linear-period setting, let $G=\mathrm{GL}_m(D)$, let E/F be quadratic, let H be the centralizer of $E^\times$ in G, and let $m_\mathrm{geom}(\pi,\chi)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-014764 | MATHDB-361299

Geometric epsilon-factor conjecture for Shalika periods

摘录 / EXCERPT: Geometric epsilon-factor conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-014765 | MATHDB-361300

Local relative trace formula geometric expansion for linear periods

摘录 / EXCERPT: Let $G=\mathrm{GL}_m(D)$, let H be the subgroup defining the linear period, let ω be a central character, and let $\mathcal{C}(G(F),\omega)$ be the corresponding Schwartz-...

conjecture | open | MathDB 页面 | 原文线索

V2-014766 | MATHDB-361301

Twisted local relative trace formula geometric expansion

摘录 / EXCERPT: Let $\tilde{\mathbf G}$ be the twisted group in the Shalika-period setting, let ω be the specified central character, and let $\mathcal{C}(\tilde{G},\omega)$ be the strongly...

conjecture | open | MathDB 页面 | 原文线索

V2-014767 | MATHDB-361318

Ay's unimodality conjecture for rectangular Kronecker coefficients

摘录 / EXCERPT: Let $g(\lambda^{(1)},\ldots,\lambda^{(d)})$ denote the generalized Kronecker coefficient, where the arguments are partitions of the same integer, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-014768 | MATHDB-361319

Ay's generalized unimodality conjecture for Kronecker coefficients

摘录 / EXCERPT: Let $g(\lambda^{(1)},\ldots,\lambda^{(d)})$ be the generalized Kronecker coefficient, let $\rho_k(\lambda)=(k,\lambda)$, and apply ρ_k componentwise to tuples of partitions. L...

conjecture | open | MathDB 页面 | 原文线索

V2-014769 | MATHDB-361344

Ghasemabadi's vanishing-order characterization of finite simple groups

摘录 / EXCERPT: Let G be a finite group and let H be a finite non-abelian simple group. Write ${\rm Vo}(G)$ for the set of orders of the vanishing elements of G, where a vanishing element is...

conjecture | open | MathDB 页面 | 原文线索

V2-014770 | MATHDB-361411

Solvability and abelianness conjectures for finite superconnected quandles

摘录 / EXCERPT: Solvability and abelianness conjectures. (i) Every finite superconnected quandle is solvable. (ii) Every finite simple superconnected quandle is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-014771 | MATHDB-361442

Q-character formula for shifted quantum affine simple representations

摘录 / EXCERPT: Let I be the set of simple-root indices, let W be the Weyl group, and let $a\in\mathbb{C}^*$. For $i\in I$ and $w\in W$, let $Q_{w(\varpi_i),a}$ be the corresponding Q-variab...

conjecture | open | MathDB 页面 | 原文线索

V2-014772 | MATHDB-361443

Shi's recognition-by-spectrum conjecture for finite simple groups

摘录 / EXCERPT: Let G be a simple group and let H be a group. Write $|G|$ for the order of G and let $\omega(G)$ denote the set of element orders of G. Shi's conjecture. $H\cong G$ if and...

conjecture | open | MathDB 页面 | 原文线索

V2-014773 | MATHDB-361463

Singer conjecture for closed aspherical manifolds

摘录 / EXCERPT: Let M be a closed aspherical n-manifold. Singer's conjecture. The ell^2-Betti numbers satisfy ... for all $p\neq n/2$. The Singer conjecture is a major unresolved problem conce...

conjecture | open | MathDB 页面 | 原文线索

V2-014774 | MATHDB-361474

The almost finiteness conjecture for free amenable group actions

摘录 / EXCERPT: Let G be a countably infinite amenable discrete group acting freely on a finite-dimensional compact metrizable space X. Almost finiteness conjecture. The action of G on X i...

conjecture | open | MathDB 页面 | 原文线索

V2-014775 | MATHDB-361489

Skein algebras and quantized Coulomb branches for surfaces of genus at most one

摘录 / EXCERPT: Let $S=S_{g,n}$ be a surface of genus $g\leq 1$ with $n>0$ boundary components, and let $(\widetilde{G},N)$ be the group and representation associated to a pants decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-014776 | MATHDB-361494

The non-decreasing condition conjecture for generically tau-reduced components

摘录 / EXCERPT: Let Λ be the finite-dimensional algebra under consideration, and let ... be a generically τ-reduced component, with $|\Lambda|$ denoting the number of isomorphism classe...

conjecture | open | MathDB 页面 | 原文线索

V2-014777 | MATHDB-361498

The conjectural order of the unipotent centralizer quotient in finite classical groups

摘录 / EXCERPT: Let G be either ${\rm Sp}(V)$ or $\Omega(V)$, let u be the representative in ... J={j:bj=2}. ... . The centralizer quotient order conjecture. One has ... This conjecture specifies th...

conjecture | open | MathDB 页面 | 原文线索

V2-014778 | MATHDB-361532

The alternating-group multifold-factorization conjecture

摘录 / EXCERPT: Let A_n denote the alternating group of degree n, and let “multifold-factorizable” have the meaning defined in the paper. Alternating-group multifold-factorization conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-014779 | MATHDB-361533

The symmetric-group multifold-factorization conjecture

摘录 / EXCERPT: Let S_n denote the symmetric group of degree n, and let “multifold-factorizable” have the meaning defined in the paper. Symmetric-group multifold-factorization conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-014780 | MATHDB-361571

Torsion-almost-divisibility conjecture for finitely generated commutative semirings

摘录 / EXCERPT: Torsion-almost-divisibility conjecture. The semigroup $S(+)$ is torsion if and only if $S(+)$ is almost-divisible.

conjecture | open | MathDB 页面 | 原文线索

V2-014781 | MATHDB-361611

The supertight programme conjecture for odd type groups of finite Morley rank

摘录 / EXCERPT: Super tight programme conjecture. The group G is isomorphic to a Chevalley group $X(K)$ over an algebraically closed field K of characteristic different from 2.

conjecture | open | MathDB 页面 | 原文线索

V2-014782 | MATHDB-361637

Fundamental-module unitarity criterion by real rank

摘录 / EXCERPT: Let X be a fundamental module with real infinitesimal character Λ, and let $\alpha\in\Delta(\mathfrak g,\mathfrak h)$ be a simple root. Suppose that ... Here, X is said t...

conjecture | open | MathDB 页面 | 原文线索

V2-014783 | MATHDB-361639

The fundamentalness conjecture for special unipotent representations of unitary groups

摘录 / EXCERPT: Let $U(p,q)$ be a real unitary group, and let a special unipotent representation mean a representation in the special-unipotent class considered in the paper. A representation is f...

conjecture | open | MathDB 页面 | 原文线索

V2-014784 | MATHDB-361646

Buzzard-Gee conjecture on C- and L-arithmetic automorphic representations

摘录 / EXCERPT: Let G be a connected reductive algebraic group defined over a number field F, and let $\pi=\bigotimes_v\pi_v$ be an irreducible automorphic representation of $G(\mathbb{A}_F)$....

conjecture | open | MathDB 页面 | 原文线索

V2-014785 | MATHDB-361655

Infinite periodic asymptotic dimension of nonabelian free groups

摘录 / EXCERPT: Let F be a nonabelian free group, and let $\operatorname{pdim}(F)$ denote its periodic asymptotic dimension. Free-group periodic-dimension conjecture. Nonabelian free groups have...

conjecture | open | MathDB 页面 | 原文线索

V2-014786 | MATHDB-361674

The coarse separation characterization of non-elementary hyperbolic groups

摘录 / EXCERPT: Let G be a non-elementary hyperbolic group. A subspace has subexponential growth if its growth function is subexponential, and G is coarsely separable by subspaces of subexpone...

conjecture | open | MathDB 页面 | 原文线索

V2-014787 | MATHDB-361711

Parity-sensitive representation conjecture for nonnormal cycle-generated Cayley graphs

摘录 / EXCERPT: Let S_n be the symmetric group, and for $1\leq r<k\leq n-2$ let $C(n,k;r)$ be the set of k-cycles moving every point from 1 to r. Write $\rho_{(n-1,1)}$ for the standard re...

conjecture | open | MathDB 页面 | 原文线索

V2-014788 | MATHDB-361727

Simplicity conjecture for finite generalised singularity modules

摘录 / EXCERPT: Choose a depth-r parabolic filtration $\bm{\psi} \in \mathcal{P}^{(r)}_{\Phi}$ with depth-r Levi factor $\bm{\phi} \in \mathcal{L}^{(r)}_{\Phi}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014789 | MATHDB-361737

Deligne's conjecture for the classical exceptional series

摘录 / EXCERPT: Let $\mathsf{Exc}_{R,\lambda}$ be the classical exceptional diagram category. A rigid symmetric monoidal category is a symmetric monoidal category in which every object has a dual,...

conjecture | open | MathDB 页面 | 原文线索

V2-014790 | MATHDB-361816

Arad-Herzog conjecture on products of non-trivial conjugacy classes

摘录 / EXCERPT: Arad-Herzog conjecture. This product is not a conjugacy class.

conjecture | open | MathDB 页面 | 原文线索

V2-014791 | MATHDB-361817

Solvability conjecture for conjugacy-class powers yielding inverse classes

摘录 / EXCERPT: Solvability conjecture for inverse-class products. If

conjecture | open | MathDB 页面 | 原文线索

V2-014792 | MATHDB-361831

Poly-context-free characterization of lattices in products of trees

摘录 / EXCERPT: Poly-context-free lattice conjecture. A lattice in the product of trees is poly-context-free if and only if it is reducible.

conjecture | open | MathDB 页面 | 原文线索

V2-014793 | MATHDB-361850

The pairing-matrix finite-order reformulation of the zero divisor conjecture

摘录 / EXCERPT: Pairing-matrix conjecture. If the generators $a_1,\ldots,a_n$ and $b_1,\ldots,b_m$ of Γ_C are distinct, then Γ_C contains a nontrivial element of finite order. This i...

conjecture | open | MathDB 页面 | 原文线索

V2-014794 | MATHDB-361858

Orbit-method compatibility conjecture for Deligne-Lusztig representations

摘录 / EXCERPT: Orbit-method compatibility conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-014795 | MATHDB-361892

The Character Triple Conjecture

摘录 / EXCERPT: Let $G\unlhd A$ be finite groups, let $U\unlhd G$ be a p-subgroup, and let B be a p-block of G. For $d\geq 0$ and $\varphi\in\operatorname{Irr}(U)$, use the sets...

conjecture | open | MathDB 页面 | 原文线索

V2-014796 | MATHDB-361917

Geiss-Leclerc-Schröer's rank-vector conjecture for symmetrisable Cartan matrices

摘录 / EXCERPT: Geiss-Leclerc-Schröer's rank-vector conjecture. The vector α is a root of the Kac-Moody Lie algebra of type C if and only if α is the rank vector of some τ-loc...

conjecture | open | MathDB 页面 | 原文线索

V2-014797 | MATHDB-361932

Guarnieri-Vendramin's quaternion-brace counting conjecture

摘录 / EXCERPT: Let $m\geq 3$, and let $q(4m)$ denote the number of braces B whose multiplicative group $(B,\circ)$ is a generalised quaternion group of order $4m$. Guarnieri-Vendramin's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-014798 | MATHDB-361965

Associativity conjecture for monoidal Jantzen filtrations

摘录 / EXCERPT: Associativity conjecture. The $\mathbb{Z}[t^{\pm1/2}]$-module $K(\mathscr{C})_t$ with this operation is a $\mathbb{Z}[t^{\pm1/2}]$-algebra, and hence gives a not necessarily commut...

conjecture | open | MathDB 页面 | 原文线索

V2-014799 | MATHDB-361973

The Lück approximation conjecture

摘录 / EXCERPT: Let Γ be a group, let K be a field, and let ... be a descending chain of normal subgroups of finite index with trivial intersection. For a matrix A over $K[\Gamma]$, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-014800 | MATHDB-362019

Solvability conjecture for groups with nilpotentizer of order eight

摘录 / EXCERPT: Let G be a group and let $x\in G$. Suppose that the nilpotentizer $nil_G(x)$ has order 8 and generates a maximal subgroup of G. Solvability conjecture. Then G is solvable....

conjecture | open | MathDB 页面 | 原文线索

V2-014801 | MATHDB-362020

The centralizer-count and derived-order conjecture for finite groups

摘录 / EXCERPT: Centralizer-count and derived-order conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-014802 | MATHDB-362035

Refined local Langlands correspondence for p-adic quasi-split groups

摘录 / EXCERPT: Refined local Langlands correspondence for p-adic quasi-split groups. The following should hold: for every L-parameter φ, there is a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-014803 | MATHDB-362105

Conjecture on column lengths in the decomposition of gamma modules

摘录 / EXCERPT: Let $n,k\geq 1$, and let $\gamma_{n,k}$ be an $\mathfrak S_{k(n-1)+1}$-module such that ... where every irreducible constituent of $\gamma_{n,k}$ has at most $k-1$ columns. Column-le...

conjecture | open | MathDB 页面 | 原文线索

V2-014804 | MATHDB-362169

The finite-set correspondence conjecture for anyonic $SL(N,\mathbb C)$ flat connections

摘录 / EXCERPT: Finite-set correspondence conjecture. The connected components of the anyonic $SL(N,\mathbb C)$ flat-connection space are in one-to-one correspondence with the points of

conjecture | open | MathDB 页面 | 原文线索

V2-014805 | MATHDB-362183

Deloro-Wiscons conjecture on ranked quasi-Frobenius groups of odd type

摘录 / EXCERPT: Let G and C be ranked connected groups such that $C<G$, $[N_G(C):C]<\finite$, and $C\cap C^g=\{1\}$ for every $g\notin N_G(C)$. Suppose that $C<G$ is a ranked quasi-Frobenius g...

conjecture | open | MathDB 页面 | 原文线索

V2-014806 | MATHDB-362273

Linear and smooth oriented equivalence conjecture for finite-group representations

摘录 / EXCERPT: Let n be a positive integer, let Γ be a finite group, and let $\rho,\rho':\Gamma\to\operatorname{O}(n)$ be homomorphisms. Two such homomorphisms are H-equivalent if they...

conjecture | open | MathDB 页面 | 原文线索

V2-014807 | MATHDB-362274

Smooth centralizer property conjecture for finite orthogonal groups

摘录 / EXCERPT: Let G be a finite subgroup of $\operatorname{O}(n)$. Say that G has the amphichiral linear centralizer property (LCP) if its centralizer in $\operatorname{O}(n)$ contains an el...

conjecture | open | MathDB 页面 | 原文线索

V2-014808 | MATHDB-362287

Solvabilizer number of PSL2 over fields of characteristic two

摘录 / EXCERPT: Characteristic-two PSL2 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014809 | MATHDB-362289

The A5 composition-factor conjecture for solvabilizer number three

摘录 / EXCERPT: A5 composition-factor conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014810 | MATHDB-362317

Locally indicable groups are L2-subgroup rigid

摘录 / EXCERPT: A group is locally indicable if every nontrivial finitely generated subgroup admits a surjective homomorphism onto $\mathbb{Z}$. The locally indicable subgroup-rigidity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-014811 | MATHDB-362354

Conjecture on the intersection of Jiang and dual Arthur packets

摘录 / EXCERPT: Let G_n be the group considered in the paper, let ψ be a local Arthur parameter, let $\widetilde\Pi^J_\psi$ be the Jiang packet, let $\widehat\Pi_{\phi_{\widehat\psi}}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-014812 | MATHDB-362365

The a-generated derived-subgroup conjecture

摘录 / EXCERPT: Let $G\subseteq\operatorname{Aut}(X)$ be a-generated, meaning that it is generated by an irreducible algebraic subset containing the identity. The a-generated derived-subgroup conj...

conjecture | open | MathDB 页面 | 原文线索

V2-014813 | MATHDB-362402

Wavefront-set upper-bound conjecture for Langlands parameters

摘录 / EXCERPT: Let ${\mathbf G}^\omega({\mathsf k})$ be a reductive p-adic group, let $\pi(\lambda,s,{{\mathcal O}}^\vee,{{\mathcal L}}^\vee)$ be an irreducible representation in the L-packet...

conjecture | open | MathDB 页面 | 原文线索

V2-014814 | MATHDB-362421

Finkelberg-Fourier-Littelmann finite-generation conjecture for essential monomials

摘录 / EXCERPT: Finkelberg-Fourier-Littelmann conjecture. For every birational sequence S and monomial ordering $>$, the monoid $\Gamma(S,>,\lambda)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-014815 | MATHDB-362428

The hyperbolic asymptotic Cantor manifold conjecture

摘录 / EXCERPT: An n-asymptotic Cantor manifold is a metric space of asymptotic dimension n that cannot be coarsely separated by a subset of asymptotic dimension at most $n-2$. For each intege...

conjecture | open | MathDB 页面 | 原文线索

V2-014816 | MATHDB-362448

Platonov-Potapchik's primitive-unipotent conjecture

摘录 / EXCERPT: Platonov-Potapchik's conjecture. If all primitive elements of G are unipotent, then the group G generated by S is unipotent.

conjecture | open | MathDB 页面 | 原文线索

V2-014817 | MATHDB-362471

The definable finite-by-abelian-by-finite conjecture for omega-categorical pseudofinite groups

摘录 / EXCERPT: Let G be a countably infinite group that is both omega-categorical, meaning that its theory has a unique countable model up to isomorphism, and pseudofinite, meaning that it is i...

conjecture | open | MathDB 页面 | 原文线索

V2-014818 | MATHDB-362474

Maximal-Arthur-parameter conjecture for first occurrence on the theta tower

摘录 / EXCERPT: Let π be a representation of G and let $\psi^{\max}(\pi)$ denote its maximal Arthur parameter. Let $d^\pm(\pi,\psi^{\max}(\pi))$ be the corresponding going-up or going-down i...

conjecture | open | MathDB 页面 | 原文线索

V2-014819 | MATHDB-362545

The sign formula conjecture for higher level Coxeter unipotent representations

摘录 / EXCERPT: Let $(\mathbb{G},T,\theta,r)$ be general data for the representation R_{T_r,U_r}^{θ}. Write $p:=\operatorname{char}(\mathbb{F}_q)$, let $(-)_p$ denote the p-part of a posi...

conjecture | open | MathDB 页面 | 原文线索

V2-014820 | MATHDB-362551

Kac-Roan-Wakimoto-Arakawa conjecture on Drinfeld-Sokolov reduction

摘录 / EXCERPT: Kac-Roan-Wakimoto-Arakawa conjecture. The functor Ψ_f^- is exact, and the image of every simple module is either simple or zero. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-014821 | MATHDB-362562

Non-coherence conjecture for poly-free and poly-surface groups

摘录 / EXCERPT: Non-coherence conjecture. For all $0 \leq m \leq n-1$, G is not m-coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-014822 | MATHDB-362581

The infinitude conjecture for cyclic groups with the weak acyclic matching property

摘录 / EXCERPT: Following Aliabadi, an abelian group G has the weak acyclic matching property if, for every pair of finite nonempty subsets $A,B\subseteq G$ with $|A|=|B|$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014823 | MATHDB-362587

Tachikawa's first conjecture

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field. The condition that all higher extensions from its k-dual back to A vanish is ... Here $D(A)$ denotes the k-dual of A. Ta...

conjecture | open | MathDB 页面 | 原文线索

V2-014824 | MATHDB-362588

Conjecture on torsion-free Auslander-Reiten sequences and selfinjectivity

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field. An Auslander-Reiten sequence is called n-torsion-free when it has the torsion-freeness property of order n considered in t...

conjecture | open | MathDB 页面 | 原文线索

V2-014825 | MATHDB-362639

The tile-structure conjecture for Prim polynomials

摘录 / EXCERPT: Tile-structure conjecture. For fixed n and p, there is a finite list of valid tiles such that $\mathrm{Prim}(d)$ is the sum

conjecture | open | MathDB 页面 | 原文线索

V2-014826 | MATHDB-362640

The duality and stabilization conjecture for valid tiles

摘录 / EXCERPT: Duality and stabilization conjecture. (a) If a tile is valid for n, then appending a zero to its vector produces a valid tile for $n+1$. (b) If $c_{\mathrm{in}}\in\mathbb{Z}$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-014827 | MATHDB-362641

The END-tile parametrization conjecture

摘录 / EXCERPT: END-tile parametrization conjecture. The END tiles are precisely

conjecture | open | MathDB 页面 | 原文线索

V2-014828 | MATHDB-362642

The structural formula conjecture for cohomology characters

摘录 / EXCERPT: Structural formula conjecture. For $d\geq e>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-014829 | MATHDB-362709

Gross-Prasad-Rallis geometric genericity conjecture for L-packets

摘录 / EXCERPT: Let G be a connected reductive group over a local field F, and let ϕ be a local Langlands parameter. An open parameter is a Langlands parameter whose orbit is open in the...

conjecture | open | MathDB 页面 | 原文线索

V2-014830 | MATHDB-362710

The open-parameter conjecture for generic ABV-packets

摘录 / EXCERPT: Let G be any quasi-split reductive group over a p-adic field F, and let ϕ be a local Langlands parameter of G. An open parameter is a parameter whose orbit is open in...

conjecture | open | MathDB 页面 | 原文线索

V2-014831 | MATHDB-362787

Projected-limit conjecture for twisted q-characters

摘录 / EXCERPT: Fix $i\frac{in I}$, $a\frac{inmathbb{C}^*$, a Weyl group element $w\frac{in W$, and a monomial m independent of k. Let M_k be the Kirillov-Reshetikhin monomial specified in t...

conjecture | open | MathDB 页面 | 原文线索

V2-014832 | MATHDB-362788

Inductive-limit and projected-limit conjecture for twisted q-characters

摘录 / EXCERPT: Let W be the Weyl group, let $w\frac{in W$, fix $i\frac{in I$ and $a\frac{inmathbb{C}^*$, and let m be a monomial in the variables $Y_{j,b}$. Let $V_infty^w(m)$ be the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-014833 | MATHDB-362789

Character-transfer conjecture from twisted to untwisted Borel modules

摘录 / EXCERPT: Let $w\frac{in W$, let $bpsi=T_w(bpsi_{i,a}^{-1})$, and write the projected limit $bpsipi_{q,infty}^w$ as $c\times a$, with c its constant part and a its non-constant part. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014834 | MATHDB-362839

The residually finite group conjecture for chaotic almost minimal actions

摘录 / EXCERPT: Let G be a group. A faithful action of G is called chaotic almost minimal (CAM) if it has dense periodic points and satisfies the almost minimality condition defined for CAM sy...

conjecture | open | MathDB 页面 | 原文线索

V2-014835 | MATHDB-362875

The gap conjecture for exponential percolation dimension

摘录 / EXCERPT: Let G be a finitely generated group, and let ${}^{e}\operatorname{pdim}(G)$ denote its exponential percolation dimension. Gap conjecture. There is a $\beta>0$ such that ... for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-014836 | MATHDB-362876

Positive exponential percolation dimension for super-polynomial growth groups

摘录 / EXCERPT: Let G be a finitely generated group of super-polynomial growth, and let ${}^{e}\operatorname{pdim}(G)$ denote its exponential percolation dimension. Positive-dimension conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-014837 | MATHDB-362877

Quasi-isometry invariance of percolation dimensions

摘录 / EXCERPT: Let G and H be finitely generated groups, and write $\operatorname{pdim}(G)$, $\nu\dim(G)$, and ${}^{e}\operatorname{pdim}(G)$ for the three percolation-dimension notions intro...

conjecture | open | MathDB 页面 | 原文线索

V2-014838 | MATHDB-362878

Conjecture on transcendence of conjugacy growth series for non virtually abelian finitely presented groups

摘录 / EXCERPT: Let G be a finitely presented group with a finite generating set X. Its conjugacy growth series is the generating function ... where $c_{G,X}(n)$ counts the conjugacy classes hav...

conjecture | open | MathDB 页面 | 原文线索

V2-014839 | MATHDB-362893

The almost simple fixer-ratio conjecture

摘录 / EXCERPT: Almost simple fixer-ratio conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-014840 | MATHDB-362911

The NOP conjecture on fixed points for finitely generated torsion groups

摘录 / EXCERPT: Let G be a finitely generated group acting by isometries on a finite-dimensional CAT(0) complex. NOP conjecture. If the action has no global fixed point, then G contains an ele...

conjecture | open | MathDB 页面 | 原文线索

V2-014841 | MATHDB-362929

The scalar product formula for parahoric Deligne-Lusztig induction

摘录 / EXCERPT: The scalar product conjecture. For all $(\theta',\mathbb{T}'_r,\mathbb{B}'_r)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-014842 | MATHDB-362957

Periodicity conjecture for submodule lattices of hook Specht modules

摘录 / EXCERPT: Let $L(r)$ denote the function governing the period in the corresponding result for 2-part partitions. For integers n,m and fixed r, let $\lambda=(n-r,1^r)\vdash n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014843 | MATHDB-362972

Vogan's exhaustion conjecture for unitary Harish-Chandra modules of U(p,q)

摘录 / EXCERPT: Let $U(p,q)$ be the real reductive group of unitary matrices of signature (p,q), and let $A_\mathfrak{q}(\lambda)$ denote a cohomologically induced module in the weakly fair rang...

conjecture | open | MathDB 页面 | 原文线索

V2-014844 | MATHDB-362983

The Benjamini-Nekrashevych-Pete conjecture on strongly scale-invariant groups

摘录 / EXCERPT: Let G be a finitely generated group. A group is strongly scale-invariant if it admits an injective homomorphism $f:G\to G$ such that $f(G)$ is a proper finite-index subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-014845 | MATHDB-362998

Random-matrix beyond square-root cancellation conjecture for symmetric-power characters

摘录 / EXCERPT: Let $U(N)$ be the unitary group, let ν_N be its Haar probability measure, and let $\operatorname{Sym}^k\mathbb C^N$ be the kth symmetric power of the standard representation....

conjecture | open | MathDB 页面 | 原文线索

V2-014846 | MATHDB-363018

The modular plethystic isomorphism for two-row partitions

摘录 / EXCERPT: Let $M,N\in\mathbb{N}$, let $d\in\mathbb{N}_0$, let $\mathbb{F}$ be the underlying field, and let E be the natural 2-dimensional $\mathbb{F}$-representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-014847 | MATHDB-363107

Finite models for the fixed-point components of Springer fibers

摘录 / EXCERPT: Let (e,h,f) be an $\mathfrak{sl}_2$ triple in $\mathfrak{g}$, let T_e be the torus with Lie algebra $\mathbb{C}h$, and let Spr be the Springer fiber. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-014848 | MATHDB-363108

Linear independence of cell modules for distinguished nilpotents

摘录 / EXCERPT: Let e be a distinguished nilpotent element, so that $A_e=Q_e$, and let c be the corresponding two-sided cell. Let J_c be identified with...

conjecture | open | MathDB 页面 | 原文线索

V2-014849 | MATHDB-363115

Nonexistence of parallel no-HGS extensions of squarefree degree

摘录 / EXCERPT: Let L/K be a separable field extension of squarefree degree. Say that L/K has the parallel no-HGS property if it admits a Hopf-Galois structure while a parallel extension assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-014850 | MATHDB-363148

Marietti's invariance conjecture for parabolic Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let (W,S) and (W',S') be Coxeter systems, with $J\subseteq S$ and $J'\subseteq S'$. For $u,v\in W^J$ and $u',v'\in (W')^{J'}$ with $u\leqslant v$ and $u'\leqslant v'$, write .....

conjecture | open | MathDB 页面 | 原文线索

V2-014851 | MATHDB-363167

Finite-index closure failure for the submonoid membership problem

摘录 / EXCERPT: Let G be a group and let $H\leq G$ be a subgroup of index 2. Write $\operatorname{SMON}(K)$ for the submonoid membership problem in a group K. Finite-index submonoid membersh...

conjecture | open | MathDB 页面 | 原文线索

V2-014852 | MATHDB-363172

Optimal-size conjecture for realizing symmetric groups by singular ASMs

摘录 / EXCERPT: Optimal-size conjecture. For $n>2$, the least k for which SA_k contains a group isomorphic to S_n is

conjecture | open | MathDB 页面 | 原文线索

V2-014853 | MATHDB-363176

Rosenberger's Tits alternative conjecture for generalised triangle groups

摘录 / EXCERPT: Rosenberger's conjecture. If G is a generalised triangle group, then either G is virtually solvable or G contains a non-abelian free subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-014854 | MATHDB-363177

Nonexistence conjecture for trace polynomials in type $(2,4,2)$

摘录 / EXCERPT: Let m be an odd integer with $m\geq 5$, and let $W\in\mathbb{Z}_2*\mathbb{Z}_4$. Let $\tau_W(\lambda)$ denote the trace polynomial associated with W. The trace-polynomial nonex...

conjecture | open | MathDB 页面 | 原文线索

V2-014855 | MATHDB-363178

Chinello-Dat conjecture on endo-parameter blocks for inner forms

摘录 / EXCERPT: Let $({\rm G},h)$ be an inner form of a general linear group or of a classical group with $p eq 2$, and let $\frak{t}$ be an endo-parameter for ${\rm G}$. Let ${\rm R}$ be an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-014856 | MATHDB-363179

Conjectural classification of ps-idempotents by unrefined endo-parameters

摘录 / EXCERPT: Let ${\rm G}$ be a classical group, let $\mathfrak{Z}_{{\rm G},\overline{\mathbb{Z}}[1/p]}^{\operatorname{st}}$ denote the stable center, and let $\mathcal{EP}({\rm G})$ be the set...

conjecture | open | MathDB 页面 | 原文线索

V2-014857 | MATHDB-363180

Onn's weak conjecture on representations of general linear groups over local rings

摘录 / EXCERPT: Onn's weak conjecture. There is an isomorphism of group algebras

conjecture | open | MathDB 页面 | 原文线索

V2-014858 | MATHDB-363218

Regularity extensions of Pyber, Cameron and Vdovin's conjectures

摘录 / EXCERPT: Let G be a finite group with a core-free maximal subgroup, and let $m(G)$ be the minimal index of a core-free maximal subgroup. Let $R_{\rm max}(G)$, $R_{\rm ns}(G)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014859 | MATHDB-363219

Eventual soluble regularity-two conjecture for symmetric and alternating groups

摘录 / EXCERPT: Let $R_{\rm sol}(G)$ denote the regularity invariant obtained by restricting to soluble subgroups. Eventual soluble regularity-two conjecture. ... for all sufficiently large n. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-014860 | MATHDB-363220

Mixed nilpotent-soluble regularity conjecture for symmetric and alternating groups

摘录 / EXCERPT: Let G be either S_n or A_n with $n\geqslant 13$, and let (H,K) be a pair of subgroups of G, where H is nilpotent and K is soluble. A pair (H,K) is regular when it s...

conjecture | open | MathDB 页面 | 原文线索

V2-014861 | MATHDB-363232

The RoCK core block conjecture for the map eta

摘录 / EXCERPT: RoCK core block conjecture. If $\Lambda_+^{\boldsymbol{\kappa}}(\omega)$ is a θ-RoCK core block with...

conjecture | open | MathDB 页面 | 原文线索

V2-014862 | MATHDB-363242

The connectivity conjecture for the generating graph of finite groups

摘录 / EXCERPT: Let G be a finite group, and let $\bar\Delta(G)$ denote its generating graph, whose vertices represent the elements of G and whose edges join pairs that generate G. Generatin...

conjecture | open | MathDB 页面 | 原文线索

V2-014863 | MATHDB-363268

The Lie-Luck conjecture for asymptotic representation ranks

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{C}$ and let Γ be a finitely generated subgroup of $\mathsf{G}(\mathbb{C})$. Fix a central character χ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014864 | MATHDB-363269

The quantitative Lie-Luck conjecture for representation ranks

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{C}$, let Γ be a finitely generated subgroup of $\mathsf{G}(\mathbb{C})$, fix a central character χ, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-014865 | MATHDB-363270

Local theta-lift identities for archimedean long root A-packets of $\boldsymbol{\rm G}_2$

摘录 / EXCERPT: Let F be an archimedean local field, let G be the group of type G_2, and let K/F be the quadratic algebra used to define G'_J. Let $\chi:K^\times\to S^1$ be a un...

conjecture | open | MathDB 页面 | 原文线索

V2-014866 | MATHDB-363310

The higher Frobenius functor conjecture for elementary abelian p-groups

摘录 / EXCERPT: Higher Frobenius functor conjecture. The functors $\Phi^V_\mathcal{C}$ define an $\mathbf{O}_n$-functor. For certain natural choices of V, it detects fibering over...

conjecture | open | MathDB 页面 | 原文线索

V2-014867 | MATHDB-363311

The thick tensor ideal conjecture for faithful representations of elementary abelian p-groups

摘录 / EXCERPT: Thick tensor ideal conjecture. The full subcategory $I\subset\mathcal{D}\subset\operatorname{Rep} C_p^n$ remains a thick tensor ideal when considered in $\operatorname{Rep} C_p^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-014868 | MATHDB-363313

The higher Frobenius conjecture for the functors \Phi^\ulambda_\mathcal{C}

摘录 / EXCERPT: Higher Frobenius conjecture. (1) For arbitrary $M\in\operatorname{Rep}E$, the module $R_{\underline{\lambda}}(St_{n-1})\otimes M$ belongs to $\mathcal{D}_{\underline{\lambda}}$. (2...

conjecture | open | MathDB 页面 | 原文线索

V2-014869 | MATHDB-363324

Quasi-isometry invariance of higher-dimensional homological Dehn functions

摘录 / EXCERPT: Quasi-isometry invariance conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014870 | MATHDB-363357

The symmetric Young character containment conjecture for Frobenius-exact tensor categories

摘录 / EXCERPT: Symmetric Young containment conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-014871 | MATHDB-363422

Carroll-Penland weakly aperiodic SFT conjecture

摘录 / EXCERPT: Let G be a finitely generated group. A weakly aperiodic SFT on G is a subshift of finite type with an infinite orbit, while a strongly aperiodic SFT has trivial stabilizers. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-014872 | MATHDB-363442

The slope-product generation conjecture for affine Soergel traces

摘录 / EXCERPT: Let $n\in\mathbb{N}$, let $(m_i,n_i)\in\mathbb{Z}\times\mathbb{N}$, and let $d_i\geq 1$. For the affine Soergel trace category $\operatorname{Tr}(\mathrm{ASBim}_n)$, write $\star$...

conjecture | open | MathDB 页面 | 原文线索

V2-014873 | MATHDB-363448

Gaiotto's conjecture for the exceptional quantum supergroup $U_q(\mathfrak{f}(4))$

摘录 / EXCERPT: Let $e\in\mathfrak{sp}(6)$ be a nilpotent element of Jordan type $(3,3)$, let $\operatorname{PGL}(2)\subset Z_{\operatorname{PSp}(6)}(e)$ be a fixed maximal reductive subgroup, and...

conjecture | open | MathDB 页面 | 原文线索

V2-014874 | MATHDB-363462

Fields' character table sum lower-bound conjecture

摘录 / EXCERPT: Fields' conjecture. For all finite groups G,

conjecture | open | MathDB 页面 | 原文线索

V2-014875 | MATHDB-363476

Normal Sylow subgroup characterization by height-zero characters

摘录 / EXCERPT: Normal Sylow subgroup conjecture. G has a normal Sylow q-subgroup if and only if, for any $p\neq q$, q does not divide the degree of any p-height zero irreducible character...

conjecture | open | MathDB 页面 | 原文线索

V2-014876 | MATHDB-363478

The center conjecture for Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group with irreducible factor decomposition ... A factor A_{Γ_i} is spherical when its associated Coxeter group is finite. The center conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-014877 | MATHDB-363490

Conjecture on leading matrices of distinguished involutions

摘录 / EXCERPT: Leading-matrix conjecture. There exists an ordering of the basis $d401_{d706}$ of M_{d706} such that the leading matrices of the distinguished involutions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-014878 | MATHDB-363496

The right-ideal characterization of ultrametric-preserving monoids

摘录 / EXCERPT: Right-ideal characterization conjecture. If $1_{\mathbf{P_{PU}}}$ belongs to $\mathbf{A}$, then

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-014879 | MATHDB-363515

Strong integrality conjecture for weakly symmetric representations

摘录 / EXCERPT: Let G be a reductive group with maximal torus T, let V be a weakly symmetric representation of G, and let $\lambda\in X_*(T)$. Write G^{λ} for the centralizer of th...

conjecture | open | MathDB 页面 | 原文线索

V2-014880 | MATHDB-363516

The main conjecture on integral semisimple local Langlands correspondence

摘录 / EXCERPT: Let bbG be a connected reductive group over a nonarchimedean local field bbF with residue cardinality q. Write $(bbfR_{{}^LbbG,bbZ[bqsqrt{q}^{-1}]})^{bbhat{bbG}}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-014881 | MATHDB-363517

The tempered packet conjecture

摘录 / EXCERPT: Let bbG be an bbF-quasi-split connected reductive group and fix a Whittaker datum $(bbU,bbpsi)$. Tempered packet conjecture. Every tempered L-packet contains a unique $bbpsi$...

conjecture | open | MathDB 页面 | 原文线索

V2-014882 | MATHDB-363519

Parabolic-induction compatibility for semisimple local Langlands

摘录 / EXCERPT: Let bbM be a Levi subgroup of bbG, let $bbP=bbM\ltimesbbN$ be a parabolic subgroup, let $bbrho\inbbPi_{bbC}(bbM)$, and let $bbpi$ be an irreducible subquotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-014883 | MATHDB-363522

Galois equivariance of the semisimple local Langlands correspondence

摘录 / EXCERPT: Suppose the semisimple local Langlands correspondence $bbLL_{bbG}$ exists. Let $bbsigma$ be an automorphism of bbC fixing $bqsqrt q$, and let $(bbM,bbrho)$ be a supercuspidal-sup...

conjecture | open | MathDB 页面 | 原文线索

V2-014884 | MATHDB-363541

Steinberg tensor product conjecture for general linear supergroups in characteristic 2

摘录 / EXCERPT: Steinberg tensor product conjecture. For λ 4-restricted and $\mu=(\mu_1,\dots,\mu_n)$ a highest $\operatorname{GL}(n)$-weight,

conjecture | open | MathDB 页面 | 原文线索

V2-014885 | MATHDB-363542

The codegree isomorphism conjecture for finite simple groups

摘录 / EXCERPT: The codegree isomorphism conjecture. Let H be a finite nonabelian simple group and G a finite group. Then

conjecture | open | MathDB 页面 | 原文线索

V2-014886 | MATHDB-363543

The faithful character degree conjecture for elementary abelian extensions of simple groups

摘录 / EXCERPT: The faithful character degree conjecture. There exists a faithful irreducible character χ of G such that

conjecture | open | MathDB 页面 | 原文线索

V2-014887 | MATHDB-363544

The diameter conjecture for Cayley graphs generated by p-singular elements

摘录 / EXCERPT: Diameter conjecture. The diameter satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-014888 | MATHDB-363586

The SFT Medvedev-degree characterization of non-virtually-free groups

摘录 / EXCERPT: Let G be an infinite, finitely generated, recursively presented group. Write $\mathcal{\mathfrak{M}}_{\texttt{SFT}}(G)$ for the class of Medvedev degrees realized by nonempty G...

conjecture | open | MathDB 页面 | 原文线索

V2-014889 | MATHDB-363587

The virtually-free characterization by computable SFT degrees

摘录 / EXCERPT: Let G be a finitely generated group, let a G-SFT be a subshift of finite type over G, and let $0_{\mathcal{\mathfrak{M}}}$ denote the computable Medvedev degree. Virtually-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-014890 | MATHDB-363591

The center conjecture for affine web endomorphism algebras

摘录 / EXCERPT: Let m be a positive integer, let $\bLambda_{\text{st}}(m)$ denote the set of objects under consideration, and let $\mathpzc{Web}^\bullet$ be the affine web category. For...

conjecture | open | MathDB 页面 | 原文线索

V2-014891 | MATHDB-363592

The fullness conjecture for cyclotomic web functors

摘录 / EXCERPT: Let $\Bbbk$ be a field, let $\mathpzc{Web}_{\mathbf u}$ be the cyclotomic web category, let $W(\Lambda)$ be the associated W-algebra, and let $\widehat\Phi$ be the functor from...

conjecture | open | MathDB 页面 | 原文线索

V2-014892 | MATHDB-363593

The arbitrary-characteristic Morita equivalence conjecture

摘录 / EXCERPT: Let $\Bbbk$ be a field, let the Morita equivalence in Theorem be the equivalence associated with the cyclotomic web construction, and let the Morita equivalence in Theorem be the e...

conjecture | open | MathDB 页面 | 原文线索

V2-014893 | MATHDB-363633

Toral relative Cannon conjecture

摘录 / EXCERPT: Let (G,P) be a relatively hyperbolic group pair, meaning that G is relatively hyperbolic relative to the collection of subgroups P. Assume that its Bowditch boundary is homeo...

conjecture | open | MathDB 页面 | 原文线索

V2-014894 | MATHDB-363638

The random-group theory conjecture for densities below one-half

摘录 / EXCERPT: Let $0\leq d<\frac{1}{2}$, and let ψ be a first-order sentence in the language of groups. Let F_k be the free group of rank $k\geq 2$, and consider the density-d random-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-014895 | MATHDB-363688

Lyons-Peres-Sun-Zheng occupation-measure conjecture

摘录 / EXCERPT: Occupation-measure conjecture. The expected occupation measure of every ball satisfies the displayed $O(r^2)$ bound. This is known for groups of polynomial growth and for groups wi...

conjecture | open | MathDB 页面 | 原文线索

V2-014896 | MATHDB-363689

Conjecture on slowly varying spectral and isoperimetric profiles

摘录 / EXCERPT: Slow-variation conjecture. If G has superpolynomial growth, then Φ and Λ are both roughly slowly varying.

conjecture | open | MathDB 页面 | 原文线索

V2-014897 | MATHDB-363696

Codegrees conjecture for finite nonabelian simple groups

摘录 / EXCERPT: Let H be a finite nonabelian simple group, and let G be a finite group. Write ${\mathrm {cod}}(G)$ for the set of character codegrees of G. Codegrees conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-014898 | MATHDB-363725

The exceptional simple-factor conjecture for profinite groups with positive proportion of 2-elements

摘录 / EXCERPT: Let Σ be the infinite family of finite non-abelian simple groups ... Suppose that a profinite group G satisfies $P_2(G)>0$, where $P_2(G)$ denotes the proportion of 2-elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-014899 | MATHDB-363727

Bridson's polynomial Dehn function conjecture for subgroups of products of free groups

摘录 / EXCERPT: Let $F_{m_1},\ldots,F_{m_n}$ be finitely generated free groups, and let G be a finitely presented subgroup of their direct product. Its Dehn function measures the number of conju...

conjecture | open | MathDB 页面 | 原文线索

V2-014900 | MATHDB-363728

Bridson's polynomial Dehn function conjecture for finitely presented residually free groups

摘录 / EXCERPT: A group G is residually free if, for every $g\in G\setminus\{1\}$, there is a homomorphism $\phi\colon G\to F_2$ to the rank-two free group such that $\phi(g)\neq 1$. For a finit...

conjecture | open | MathDB 页面 | 原文线索

V2-014901 | MATHDB-363754

Amit-Vishne-Shalev conjecture on profinite rigidity of free-group words

摘录 / EXCERPT: Let $F_n=\langle x_1,\ldots,x_n\rangle$ be the free group on n generators. For a word $w\in F_n$, let $\operatorname{Pr}_w(g)$ be the probability that $w(g_1,\ldots,g_n)=g$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-014902 | MATHDB-363763

First-order torsion-rigidity conjecture for Coxeter groups

摘录 / EXCERPT: First-order torsion-rigidity conjecture. For any Coxeter group W, if G is finitely torsion-generated and elementarily equivalent to W, then G is Coxeter.

conjecture | solved | MathDB 页面 | 原文线索

V2-014903 | MATHDB-363781

Monoidal categorification conjecture for quantum cluster algebras of double Bott-Samelson cells

摘录 / EXCERPT: Monoidal categorification conjecture. If C is symmetric, the pair

conjecture | open | MathDB 页面 | 原文线索

V2-014904 | MATHDB-363808

Prasad's distinction conjecture for Iwahori-spherical discrete series

摘录 / EXCERPT: Prasad's conjecture. The representation π is H-distinguished if and only if both the restriction of ϕ to W_E is trivial and π is generic. Under the first condition...

conjecture | open | MathDB 页面 | 原文线索

V2-014905 | MATHDB-363849

Ext-quiver conjecture for descent algebras of type A

摘录 / EXCERPT: Let n and p be as in the descent algebra setting, let $\mathop{\Lambda}\nolimits^+_p(n)$ denote the indexing set of p-regular partitions, and let $Q_{n,p}$ be the Ext quiver o...

conjecture | open | MathDB 页面 | 原文线索

V2-014906 | MATHDB-363884

Hopficity conjecture for labeled Thompson groups

摘录 / EXCERPT: Hopficity conjecture. $V(G)$ is Hopfian if and only if G is Hopfian.

conjecture | open | MathDB 页面 | 原文线索

V2-014907 | MATHDB-363936

Alternating-group rank conjecture for nonsolvable groups of finite Morley rank

摘录 / EXCERPT: Let n be a positive integer, let H be a nonsolvable connected group of finite Morley rank, and suppose that $\operatorname{Alt}(n)$ acts definably and faithfully on H by auto...

conjecture | open | MathDB 页面 | 原文线索

V2-014908 | MATHDB-363978

Conjecture on Busemann points of free two-step nilpotent groups

摘录 / EXCERPT: Free two-step nilpotent Busemann-point conjecture. The set of Busemann points of $N_{2,r}$ with respect to the standard generating set is countable.

conjecture | open | MathDB 页面 | 原文线索

V2-014909 | MATHDB-364003

Galois-orbit conjecture for Shalika models of residual representations of inner forms of general linear groups

摘录 / EXCERPT: Let ${\mathbb D}$ be the central division algebra and let ${\mathrm{GL}}'_n({\mathbb A})$ denote the corresponding inner form of ${\mathrm{GL}}_n$. Let Π' be a cuspidal irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-014910 | MATHDB-364031

Stability conjecture for the depth-r Bernstein-center image

摘录 / EXCERPT: Let G be the group under consideration, let r be an integral depth, and let $\mathcal Z^{st,r}(G)=\mathcal Z^r(G)\cap\mathcal Z^{st}(G)$ denote the subspace of stable elements...

conjecture | open | MathDB 页面 | 原文线索

V2-014911 | MATHDB-364048

Vogan's fundamental parallelepiped conjecture for real reductive Lie groups

摘录 / EXCERPT: Vogan's FPP conjecture. If $\langle \Lambda, \beta^{\vee}\rangle >1$ for any simple root β, then π is non-unitary.

conjecture | open | MathDB 页面 | 原文线索

V2-014912 | MATHDB-364049

Vogan's strengthened FPP conjecture for Hermitian modules

摘录 / EXCERPT: Vogan's strengthened FPP conjecture. Suppose Λ does not lie inside FPP. Then π is not unitary up to level $\mathfrak{p}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014913 | MATHDB-364052

Generic character-reduced skein-module irreducibility and dimension conjecture for handlebodies

摘录 / EXCERPT: Let H_g be the genus g handlebody, let $\rho\in \mathfrak{X}_{{\rm SL}_3(\mathbb{C})}(H_g)$ be represented by an irreducible representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-014914 | MATHDB-364070

g-tame incidence algebras of finite posets are tame

摘录 / EXCERPT: Tameness conjecture for g-tame incidence algebras. If $\operatorname{Inc}_\K(P)$ is $\mathbf{g}$-tame, then $\operatorname{Inc}_\K(P)$ is tame.

conjecture | open | MathDB 页面 | 原文线索

V2-014915 | MATHDB-364072

The unit-group conjecture for representation rings of symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group on n letters, let $\mathcal{R}(S_n)$ denote its representation ring of real representations, and let $U(\mathcal{R}(S_n))$ be the group of units...

conjecture | open | MathDB 页面 | 原文线索

V2-014916 | MATHDB-364073

The Harrison-centre conjecture for representation rings of symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group on n letters, let $\mathcal{R}(S_n)$ denote its representation ring, and let f_{S_n} be the relevant element defining the Harrison centre...

conjecture | open | MathDB 页面 | 原文线索

V2-014917 | MATHDB-364127

Fundamentality conjecture for connected twistless large-type Artin graphs

摘录 / EXCERPT: Fundamentality conjecture. If Γ is connected and twistless, then Γ is fundamental.

conjecture | open | MathDB 页面 | 原文线索

V2-014918 | MATHDB-364158

The second brick-Brauer-Thrall conjecture

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field k. Write $\operatorname{brick}(A,d)$ for the set of isomorphism classes of bricks of dimension d, and call A brick-finite...

conjecture | open | MathDB 页面 | 原文线索

V2-014919 | MATHDB-364159

The semibrick conjecture

摘录 / EXCERPT: Let A be an algebra, and call a set of pairwise Hom-orthogonal bricks in $\operatorname{mod} A$ a semibrick. Call A brick-infinite if it has infinitely many bricks up to isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-014920 | MATHDB-364160

Demonet's conjecture on the tau-tilting fan

摘录 / EXCERPT: Let A be an algebra of rank n, and let $\mathcal{F}_A$ denote its τ-tilting fan in $\mathbb{R}^n$. Call A brick-infinite if it has infinitely many bricks up to isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-014921 | MATHDB-364161

The stable brick conjecture

摘录 / EXCERPT: Let A be an algebra, and let $\theta\in K_0(\operatorname{proj} A)$. A module $M\in\operatorname{mod} A$ is θ-stable if $\theta(\underline{\dim}M)=0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-014922 | MATHDB-364167

Prasad's conjecture on degenerate Whittaker representations of general linear groups

摘录 / EXCERPT: Let F be a nonarchimedean local field with ring of integers $\mathfrak{o}$, and let $\mathfrak{o}_l=\mathfrak{o}/(\varpi^l)$. Let L and E be unramified extensions of F of de...

conjecture | open | MathDB 页面 | 原文线索

V2-014923 | MATHDB-364240

Ding-Khoroshkin completion conjecture for fused currents

摘录 / EXCERPT: Let $U_q(L\mathfrak{g})$ be the quantum loop group of a finite-type ADE Lie algebra, with negative half $U_q(L\mathfrak{n}^-)$ and a completion $\widehat{U}_q(L\mathfrak{n}^-)$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-014924 | MATHDB-364325

Dade's conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group and let $ u_\ell$ denote the ℓ-adic valuation. For $\chi\in\operatorname{Irr}(G)$, define its ℓ-defect by...

conjecture | open | MathDB 页面 | 原文线索

V2-014925 | MATHDB-364353

The asymmetric growth conjecture

摘录 / EXCERPT: Let S be a non-abelian finite simple group, and let A,B be subsets of S. For $g\in S$, write $A^g=g^{-1}Ag$. Asymmetric growth conjecture. For every $\delta>0$ there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-014926 | MATHDB-364354

The small-set asymmetric growth conjecture

摘录 / EXCERPT: Let S be a non-abelian finite simple group, and let A,B be subsets of S. For $g\in S$, write $A^g=g^{-1}Ag$. Small-set asymmetric growth conjecture. For every $\epsilon>0$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-014927 | MATHDB-364362

Geometric Casselman-Shalika conjecture

摘录 / EXCERPT: Let G be the reductive group, T a maximal torus, and ρ the half-sum of positive roots used to define the affine Grassmannian strata. For a dominant coweight...

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V2-014928 | MATHDB-364364

Zakharevich's perfectness conjecture for scissors automorphism groups

摘录 / EXCERPT: Let E^n be the n-dimensional Euclidean space and let $P\subseteq E^n$ be a polytope. The abelianisation of a group is its quotient by its commutator subgroup. Zakharevich's con...

conjecture | open | MathDB 页面 | 原文线索

V2-014929 | MATHDB-364410

Ballmann-Buyalo periodic rank-one geodesic conjecture

摘录 / EXCERPT: Ballmann-Buyalo's conjecture. If X contains a geodesic of rank 1, then it also contains a G-periodic geodesic of rank 1.

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V2-014930 | MATHDB-364418

Irreducible multiplicity log-concavity conjecture for \(\overline{\mathcal M}_{0,n}\)

摘录 / EXCERPT: Let $\overline{\mathcal M}_{0,n}$ be the moduli space of n-pointed stable rational curves, and let $\mathbb{S}_n$ act by permuting the marked points. For a partition λ of...

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V2-014931 | MATHDB-364452

Normalizer and conjugator description for standard parabolic subgroups of Artin groups

摘录 / EXCERPT: Let A be an Artin group such that every irreducible A_Z with $|Z|>2$ is not of spherical type. For $X\subseteq S$, write ... If A_X has no cyclic irreducible component...

conjecture | open | MathDB 页面 | 原文线索

V2-014932 | MATHDB-364469

The co-context-free group embedding conjecture for Thompson's group V

摘录 / EXCERPT: A finitely generated group is co-context-free if its word problem's complement is context-free. The co-context-free embedding conjecture. A finitely generated group is co-context-f...

conjecture | open | MathDB 页面 | 原文线索

V2-014933 | MATHDB-364470

Brough's conjectures on poly-context-free groups

摘录 / EXCERPT: A group is poly-context-free if its word problem is a finite intersection of context-free languages. A group is virtually a property if it has a finite-index subgroup with that pro...

conjecture | open | MathDB 页面 | 原文线索

V2-014934 | MATHDB-364542

Harada's conjecture II on the integrality of Harada's number

摘录 / EXCERPT: Harada's conjecture II. For every finite group G, $h(G)\in\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-014935 | MATHDB-364543

Harada-Chigira's conjecture on the divisibility of Harada's number

摘录 / EXCERPT: Let G be a finite group, let G' denote its commutator subgroup, and let $h(G)$ be the Harada number ... where $K_1,\ldots,K_s$ are the conjugacy classes of G and...

conjecture | open | MathDB 页面 | 原文线索

V2-014936 | MATHDB-364556

The conjecture on one-element extensions and deletions of difference sets

摘录 / EXCERPT: Let G be a finite group, and let an ADS be obtained from a difference set in G by adding or removing one element. The one-element ADS conjecture. If G is cyclic or has odd or...

conjecture | open | MathDB 页面 | 原文线索

V2-014937 | MATHDB-364570

Kapovich-Millson saturation conjecture for simply laced root systems

摘录 / EXCERPT: Let G be a split reductive group over $\x211a$ and let k_G be the positive integer from the saturation theorem: for dominant integral weights λ, μ, and ν such t...

conjecture | open | MathDB 页面 | 原文线索

V2-014938 | MATHDB-364591

Module-intersection conjecture for Macdonald piece polynomials

摘录 / EXCERPT: Let $S\in\binom{[n]}{k}$ and let λ be a partition inside $R(n,k)$. Write $S(\lambda)$ for the collection of k-subsets defined by the containment conditions in the paper,...

conjecture | open | MathDB 页面 | 原文线索

V2-014939 | MATHDB-364592

The derangement-class product conjecture for finite simple transitive groups

摘录 / EXCERPT: Derangement-class product conjecture. There exist conjugacy classes C and D of derangements such that

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V2-014940 | MATHDB-364632

The mixed tensor product conjecture for Verma modules of \(\mathfrak{gl}(m+1|n)\)

摘录 / EXCERPT: Mixed tensor product conjecture. There is an isomorphism of $U(\mathfrak{gl}(m+1|n))$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-014941 | MATHDB-364655

Moretó's Baby Monster conjecture for groups with two equal-degree characters

摘录 / EXCERPT: Let G be a group, and define ... where $\operatorname{Irr}(G)$ is the set of irreducible characters of G. Let B be the Baby Monster group. Moretó's conjecture. If $m(G)=2$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-014942 | MATHDB-364659

Moore-Tachikawa's TQFT conjecture

摘录 / EXCERPT: Let G be a connected semisimple affine algebraic group with Lie algebra $\mathfrak{g}$. Consider a principal $\mathfrak{sl}_2$-triple $(e,h,f)\in\mathfrak{g}^{\times 3}$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-014943 | MATHDB-364660

Moore-Tachikawa's TQFT conjecture in the algebraic moment-map category

摘录 / EXCERPT: Let G be a connected complex semisimple affine algebraic group with Lie algebra $\mathfrak{g}$. Consider a principal $\mathfrak{sl}_2$-triple $(e,h,f)\in\mathfrak{g}^{\times 3}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-014944 | MATHDB-364710

Categorical Finiteness Conjecture for skein categories

摘录 / EXCERPT: Let G) be a reductive group, let Σ be a closed compact surface, and let $\mathrm{SkCat}_G(\Sigma)$ denote the G-skein category at a generic value of the quantization par...

conjecture | open | MathDB 页面 | 原文线索

V2-014945 | MATHDB-364711

Elliptic cuspidal decomposition conjecture for the SL_N skein category of the torus

摘录 / EXCERPT: Let N be a positive integer, let T^2 be the torus, and let $\mathrm{SkCat}_{\mathrm{SL}_N}(T^2)$ and $\mathrm{SkAlg}_{\mathrm{SL}_{N/d}}$ denote the corresponding skein categor...

conjecture | open | MathDB 页面 | 原文线索

V2-014946 | MATHDB-364821

Liebeck-Nikolov-Shalev conjecture on products of subset conjugates

摘录 / EXCERPT: Let G be a finite simple group and let $A\subseteq G$ be a subset with $|A|\geq 2$. A conjugate of A means a set gAg^{-1} for some $g\in G$. Liebeck-Nikolov-Shalev conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-014947 | MATHDB-364840

The amalgamated-product Tor-vanishing conjecture

摘录 / EXCERPT: Let A, B and C be groups, and let $G=A\ast_C B$ be their amalgamated free product. Let $K\subseteq\mathbb{C}$ be a field closed under complex conjugation, and set ... Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-014948 | MATHDB-364841

The class T Atiyah and universal-localization conjecture

摘录 / EXCERPT: Let G be a group in the class $\mathcal{T}$ with finite $\operatorname{lcm}(G)$, and let $K\subseteq\mathbb{C}$ be any subfield closed under complex conjugation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014949 | MATHDB-364850

Symmetry conjecture for Littlewood-Richardson component counts

摘录 / EXCERPT: Let $\mathsf{k}_{b,c}$ denote the numerical quantity associated in the paper with the relevant Littlewood-Richardson decomposition. Symmetry conjecture. For any positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-014950 | MATHDB-364851

Polynomial formula conjecture for Littlewood-Richardson component counts

摘录 / EXCERPT: Let b and c be positive integers, and let $\mathsf{k}_{b,c}$ be the numerical quantity defined in the paper. Polynomial formula conjecture. Define ... Then ... The formula is motiv...

conjecture | open | MathDB 页面 | 原文线索

V2-014951 | MATHDB-364859

The symmetric-group conjecture for groups generated by horizontal class transpositions

摘录 / EXCERPT: Symmetric-group conjecture. For $k>3$, the group $CT_{(k)}$ is isomorphic to the permutation group S_N.

conjecture | open | MathDB 页面 | 原文线索

V2-014952 | MATHDB-364867

The surjectivity conjecture for Springer-fiber restriction maps

摘录 / EXCERPT: Let $\mathfrak{g}$ be a classical Lie algebra of type B, C, or D, with dual Lie algebra $\mathfrak{g}^\vee$, and let $e^\vee\in\mathfrak{g}^\vee$ be nilpotent with partition $\lambda^\vee$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014953 | MATHDB-364892

Braverman-Kazhdan local ρ-Fourier transform conjecture

摘录 / EXCERPT: Braverman-Kazhdan local ρ-Fourier transform conjecture. There exist a ρ-Schwartz space, a distinguished ρ-basic function, and an invariant distribution satisfying al...

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V2-014954 | MATHDB-364961

The generalised Saxl conjecture for Weyl groups

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra with Weyl group W, Lie group G, nilpotent cone $\mathcal{N}$, and set of nilpotent orbits $G\backslash\mathcal{N}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014955 | MATHDB-364962

The spinorial generalised Saxl conjecture for Weyl groups

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra with Weyl group W, reflection representation V, and Spaltenstein map d on nilpotent orbits. Let Ψ be the map from genuine irre...

conjecture | open | MathDB 页面 | 原文线索

V2-014956 | MATHDB-364963

The minimal-solvable-orbit generalisation of Saxl's conjecture for Weyl groups

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra with Weyl group W. Let $\mathcal{O}$ be the unique minimal solvable nilpotent orbit of $\mathfrak{g}$. For each Lusztig family F, let...

conjecture | open | MathDB 页面 | 原文线索

V2-014957 | MATHDB-364964

The generalised Saxl conjecture for finite Coxeter groups

摘录 / EXCERPT: Let W be any finite Coxeter group. Let $\operatorname{Irr}(W)$ be its irreducible characters, and call a family F of $\operatorname{Irr}(W)$ good if it contains a character occ...

conjecture | open | MathDB 页面 | 原文线索

V2-014958 | MATHDB-364983

Fel'shtyn-Troitsky conjecture on property R_∞ and solvable-by-finite groups

摘录 / EXCERPT: Let G be a finitely generated residually finite group. Fel'shtyn-Troitsky conjecture. The group G either has property R_∞ or is solvable-by-finite. The disjunction is no...

conjecture | open | MathDB 页面 | 原文线索

V2-014959 | MATHDB-364984

The non-amenable residually finite property R_∞ conjecture

摘录 / EXCERPT: Let G be a finitely generated residually finite non-amenable group. Non-amenable property R_∞ conjecture. The group G has property R_∞. This is presented as a wea...

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V2-014960 | MATHDB-365023

Uniqueness conjecture for constant coefficient function theories on Euclidean space

摘录 / EXCERPT: A constant coefficient function theory is a function theory in the sense of F. Sommen whose defining operators have constant coefficients and whose domain is Euclidean space. Uniqu...

conjecture | open | MathDB 页面 | 原文线索

V2-014961 | MATHDB-365044

Bridson-Reid conjecture on profinite rigidity of hyperbolic 3-manifold groups

摘录 / EXCERPT: Let G and H be lattices in $PSL_2(\mathbb{C})$, and let $\widehat{G}$ and $\widehat{H}$ denote their profinite completions. Bridson-Reid conjecture. If ... then ... This is a profi...

conjecture | open | MathDB 页面 | 原文线索

V2-014962 | MATHDB-365082

The average character-degree multiplicity bound for groups without alternating composition factors

摘录 / EXCERPT: Average character-degree multiplicity conjecture. There exists a function $f:\mathbb{Q}\to\mathbb{Q}$ such that, if G is a finite group with no alternating composition factor, th...

conjecture | open | MathDB 页面 | 原文线索

V2-014963 | MATHDB-365083

The average conjugacy-class-size multiplicity bound

摘录 / EXCERPT: Average conjugacy-class-size multiplicity conjecture. There exists a function $f:\mathbb{Q}\to\mathbb{Q}$ such that, if G is a finite group, then

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V2-014964 | MATHDB-365094

Categorical Gelfand-Kazhdan conjecture for cuspidal representations

摘录 / EXCERPT: Let $G=GL_n(\a0\mathcal{K})$ and let P be its mirabolic subgroup, with unipotent subgroup U and character θ as above. Let $\a0\mathcal{C}$ be an irreducible cuspidal cat...

conjecture | open | MathDB 页面 | 原文线索

V2-014965 | MATHDB-365112

Metaplectic-lift conjecture for the Plancherel decomposition of invertible symmetric matrices

摘录 / EXCERPT: Let G be the group acting on the set X of invertible symmetric matrices by $x\cdot g=\,{}^tgxg$, and write the Plancherel decomposition as ... Let G' be a two-fold Kazhdan-Patt...

conjecture | open | MathDB 页面 | 原文线索

V2-014966 | MATHDB-365171

Babai's bounded chromatic number conjecture for semi-minimal Cayley graphs

摘录 / EXCERPT: Let $Gamma$ be a group and let $S={s_1,ldots,s_n}$ be a semi-minimal generating set: S generates $Gamma$, and no s_i is expressible in terms of $s_1,ldots,s_{i-1}$. The graph...

conjecture | open | MathDB 页面 | 原文线索

V2-014967 | MATHDB-365187

Free 2-rig conjecture for the matrix monoid representation 2-rig

摘录 / EXCERPT: Let k be a field and N a positive integer. Let $[?]$ denote the 2-rig of representations of the matrix monoid $[?]$. Free 2-rig conjecture. The 2-rig $[?]$ is the free 2-rig on...

conjecture | open | MathDB 页面 | 原文线索

V2-014968 | MATHDB-365262

The coarse median structures conjecture for mapping class groups

摘录 / EXCERPT: Let S be a non-sporadic finite-type surface, and let $\operatorname{MCG}(S)$ denote its mapping class group. A coarse median structure on a group is a coarse median operation com...

conjecture | open | MathDB 页面 | 原文线索

V2-014969 | MATHDB-365274

Generalized fused Specht polynomial basis conjecture for arbitrary Young diagram shapes

摘录 / EXCERPT: Let λ be a Young diagram, and let Theorem denote the theorem proved in the paper for Young diagrams with two columns, identifying the fused Specht polynomials indexed by...

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V2-014970 | MATHDB-365326

Tadić's independence of unitarizability conjecture

摘录 / EXCERPT: Tadić's independence of unitarizability conjecture.

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V2-014971 | MATHDB-365334

The equivariant Waldhausen A-theory derivative conjecture

摘录 / EXCERPT: Let $\Gmr$ be a finite group, and let ${\bf B}_{\Gmr}{\bf Top}^{(1)}$ denote the first layer of the equivariant Weiss functor ${\bf B}_{\Gmr}{\bf Top}$. Suppose there exists an inc...

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V2-014972 | MATHDB-365335

The equivariant Weiss calculus homotopy-category comparison conjecture

摘录 / EXCERPT: Let $\Gmr$ be a finite group and let $\upalpha$ be an indexing representation. Let $\Ecal_{\Gmr, {\bf 0}}^{\Sigma, {\sf conn}}$ be the category of connected-at-infinity equivariant...

conjecture | open | MathDB 页面 | 原文线索

V2-014973 | MATHDB-365357

Dimension-equality conjecture for centers of cyclotomic Hecke algebras of type $G(r,p,n)$

摘录 / EXCERPT: Let r,p,n be the parameters of the cyclotomic Hecke algebras, let K be the relevant specialization field, let $\mathscr{K}$ be the generic field, and let...

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V2-014974 | MATHDB-365366

Shahidi's strong tempered packet conjecture for quasi-split classical groups

摘录 / EXCERPT: Let F be a local field of characteristic zero, let G be a quasi-split connected reductive F-group, and fix a Whittaker datum $\mathfrak{w}$ for G. Let $\mathfrak{phi}$ be a...

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V2-014975 | MATHDB-365367

Artin-Deligne duality conjecture for Arthur packets

摘录 / EXCERPT: Let G be a quasi-split connected reductive group over a local or global field F. For a parameter $\mathfrak{psi}$, let $mathit{Pi}_{\mathfrak{psi}}(G)^A$ and...

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V2-014976 | MATHDB-365407

Isomorphism versus equivalence by elementary transformation operations

摘录 / EXCERPT: Isomorphism versus equivalence by ETOs. If $T\sim S$, then $T\approx S$.

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V2-014977 | MATHDB-365408

Ground-field dependence and finiteness of the number of isomorphism classes

摘录 / EXCERPT: Ground-field dependence and finiteness conjecture. For $n>3$, the number N_n depends on the ground field. Moreover, N_n is finite if and only if the ground field is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-014978 | MATHDB-365503

de Graaf's characteristic-polynomial factorization conjecture for simple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex simple Lie algebra, let V be an irreducible $\mathfrak{g}$-module with highest weight λ, and let $\Pi(\lambda)$ denote the set of weights...

conjecture | open | MathDB 页面 | 原文线索

V2-014979 | MATHDB-365528

Benjamini's thick embedding conjecture for transitive graphs

摘录 / EXCERPT: Let $d\geq 3$, and let Y be a locally finite, non-planar, transitive graph with separation profile $\simeq n^\alpha$ for some $\alpha \in (0,1]$. A 1-thick embedding of a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-014980 | MATHDB-365599

Uniformly almost flat quotients and virtual nilpotence

摘录 / EXCERPT: Let G be a finitely generated residually finite group. For $d e 0$, write uniformly $(1/d)$-almost flat quotients and uniformly $(1/d)$-almost flat coset spaces for the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-014981 | MATHDB-365600

The induction conjecture for uniformly almost flat quotients

摘录 / EXCERPT: Let $d e 0$, let G be a finitely generated infinite residually finite group with uniformly $(1/d)$-almost flat quotients, and let H be a finite-index subgroup admitting a quoti...

conjecture | open | MathDB 页面 | 原文线索

V2-014982 | MATHDB-365634

Wilson's conjecture for omega-categorical locally nilpotent p-groups

摘录 / EXCERPT: Wilson's conjecture. Every locally nilpotent omega-categorical p-group is nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-014983 | MATHDB-365662

Linear decay conjecture for strongly Ulam stable groups

摘录 / EXCERPT: Let G) be a Strongly Ulam Stable group, and let F_G denote its SUS modulus. Linear decay conjecture. Every Strongly Ulam Stable group G has $O(\varepsilon)$ decay at 0 of t...

conjecture | open | MathDB 页面 | 原文线索

V2-014984 | MATHDB-365709

The conjecture identifying the right-quiver matrix with the toric-chart matrix

摘录 / EXCERPT: Let $U^{\vee}$ and $B_-^{\vee}$ be the groups used in the construction of the matrices, and let u_L, κ_P, $\dot{w}_P$, and $\bar{w}_0$ be the corresponding factors. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-014985 | MATHDB-365725

Infinite-group Erdős-Pósa characterization for allowable A-paths

摘录 / EXCERPT: Infinite-group characterization conjecture. The family $\mathcal{F}$ satisfies the half-integral Erdős-Pósa property, and it satisfies the Erdős-Pósa property if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-014986 | MATHDB-365739

Bridson-Reid's profinite rigidity conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let $\mathcal{M}$ and $\mathcal{N}$ be hyperbolic 3-manifolds. Their profinite completions are the inverse limits of their finite quotient groups. The manifolds are commensurable...

conjecture | open | MathDB 页面 | 原文线索

V2-014987 | MATHDB-365822

The Chevalley polytope Newton-Okounkov body conjecture

摘录 / EXCERPT: Let $\X=\G/\P$ be a homogeneous space with a projective embedding $\X\hookrightarrow\bP(V_\varpi)$, and let $\bs$ be a reduced expression for $\wP$. The Chevalley polytope Newton-O...

conjecture | open | MathDB 页面 | 原文线索

V2-014988 | MATHDB-365843

The non-decreasing stable multiplicities conjecture

摘录 / EXCERPT: Fix a positive integer k and let $\lambda=(a_1,\dots,a_r)$ be a partition of k, so that $k=a_1+\dots+a_r$. For each cohomological degree i, let $d_i(\lambda)$ denote the stab...

conjecture | open | MathDB 页面 | 原文线索

V2-014989 | MATHDB-365844

The first nonzero stable multiplicity bound conjecture

摘录 / EXCERPT: Fix a positive integer k and let $\lambda=(a_1,\dots,a_r)$ be a partition of k, so that $k=a_1+\dots+a_r$. Let $d_0(\lambda),d_1(\lambda),\dots$ be the stable multiplicities as...

conjecture | open | MathDB 页面 | 原文线索

V2-014990 | MATHDB-365872

Almost all permutations are Kostant negative

摘录 / EXCERPT: Let S_n be the symmetric group, and let $\mathbf{p}_n$ denote the number of Kostant positive elements in S_n, where an element is Kostant positive if Kostant's problem f...

conjecture | open | MathDB 页面 | 原文线索

V2-014991 | MATHDB-365873

Almost all involutions are Kostant negative

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathbf{i}_n$ denote the number of involutions in S_n, and let $\mathbf{pi}_n$ denote the number of Kostant positive involutions in S_n....

conjecture | open | MathDB 页面 | 原文线索

V2-014992 | MATHDB-365914

Kleshchev-Martin conjecture on loop-free Ext-quivers of symmetric-group algebras

摘录 / EXCERPT: Let r be a nonnegative integer, let $\Bbbk$ be the underlying field, and let $S_\Bbbk(r,r)$ be the classical Schur algebra. An i-faithful quasi-hereditary cover means a quasi-h...

conjecture | open | MathDB 页面 | 原文线索

V2-014993 | MATHDB-365922

The up-element conjecture for left-orderable closed 3-manifold groups

摘录 / EXCERPT: Let M be a closed 3-manifold. An almost periodic action of a finitely generated group G on $\mathbb R$ is a faithful action in which the generators act uniformly bilipschitzly,...

conjecture | open | MathDB 页面 | 原文线索

V2-014994 | MATHDB-365939

Dey-Kapovich's Steinness conjecture for low-critical-exponent complex hyperbolic quotients

摘录 / EXCERPT: Let $n\geq 1$ and let Γ be a discrete and torsion-free subgroup of $\mathrm{PU}(n,1)$ acting on the complex hyperbolic space $\mathbb{H}^{n}_{\mathbb{C}}$. Define the critic...

conjecture | open | MathDB 页面 | 原文线索

V2-014995 | MATHDB-365950

Real one-orbit Gram graph conjecture

摘录 / EXCERPT: Let V be a finite-dimensional real Hilbert space, and let $G\leq \operatorname{O}(V)$ be finite. For a generic $x\in V$, consider the Gram graph of the orbit Gx. Real one-orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-014996 | MATHDB-365966

Indecomposable-module conjecture for Euler characteristics of crepant resolutions

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let $G\subseteq \mathrm{SL}(n,k)$ be a finite small group of finite representation type, meaning that there ar...

conjecture | open | MathDB 页面 | 原文线索

V2-014997 | MATHDB-366062

Oliver's p-group conjecture

摘录 / EXCERPT: Oliver's p-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-014998 | MATHDB-366081

Levitt-McCammond's conjecture on intrinsically aperiodic flats

摘录 / EXCERPT: Let X be a compact locally CAT(0) triangle-square complex, and let F be an intrinsically aperiodic flat, meaning that F does not admit a cocompact cellular action of...

conjecture | open | MathDB 页面 | 原文线索

V2-014999 | MATHDB-366082

Levitt-McCammond's finiteness conjecture for thoroughly crumpled planes

摘录 / EXCERPT: Let X be a compact locally CAT(0) triangle-square complex. A flat is thoroughly crumpled if it is crumpled and there is a bound on the number of cells in each region. Levitt-McCa...

conjecture | open | MathDB 页面 | 原文线索

V2-015000 | MATHDB-366083

Biautomaticity conjecture under the Levitt-McCammond flat conditions

摘录 / EXCERPT: Let X be a compact locally CAT(0) triangle-square complex such that no intrinsically aperiodic flat embeds in $\widetilde{X}$ and there are only finitely many distinct thoroughly...

conjecture | open | MathDB 页面 | 原文线索

V2-015001 | MATHDB-366146

Profinite rigidity conjecture for cusped hyperbolic 3-manifolds

摘录 / EXCERPT: Profinite rigidity conjecture. Every cusped hyperbolic 3-manifold is profinitely rigid among compact, orientable 3-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-015002 | MATHDB-366172

Stapledon's equivariant volume conjecture

摘录 / EXCERPT: Let P be a G-invariant lattice polytope. For each $g\in G$, define the evaluation $H^*(P;G)1$ of the equivariant volume at g. Stapledon's equivariant volume conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-015003 | MATHDB-366173

Stapledon's positive-coefficient conjecture for equivariant H*-polynomials

摘录 / EXCERPT: Let P be a G-invariant polytope, and suppose that $H^*(P;G)[t]$ is an effective polynomial. Stapledon's positive-coefficient conjecture. If the coefficient of t^m in the ordi...

conjecture | open | MathDB 页面 | 原文线索

V2-015004 | MATHDB-366230

Parabolic branching polynomial positivity conjecture

摘录 / EXCERPT: Parabolic branching polynomial positivity conjecture. The polynomial $b_{\nu,\overline{\lambda}+\gamma}(p)$ has nonnegative integer coefficients at least when...

conjecture | open | MathDB 页面 | 原文线索

V2-015005 | MATHDB-366232

The eigenvalue one conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group, V a finite-dimensional real G-module, and $\rho \colon G \rightarrow \operatorname{GL}(V)$ the representation afforded by V. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015006 | MATHDB-366240

Parker's conjecture on odd orthogonal determinants

摘录 / EXCERPT: Parker's conjecture. If $\mathbb{Q}(\chi)=\mathbb{Q}$, then $\det(\chi)$ is odd.

conjecture | open | MathDB 页面 | 原文线索

V2-015007 | MATHDB-366246

The standard-form conjecture for quasi-isometries of nondegenerate Sol-type groups

摘录 / EXCERPT: Let $G\cong \prod_{i=1}^n \mathbf{N}_i \rtimes \mathbb{R}^k$ be nondegenerate with $n\ge 2$. Let $q:G\to G$ be a quasi-isometry for some, and hence any, left-invariant Riemannian m...

conjecture | open | MathDB 页面 | 原文线索

V2-015008 | MATHDB-366312

Existence of plethystic $\mathfrak{sl}_2$ crystal seeds

摘录 / EXCERPT: Seed-existence conjecture. For all $n\ge 0$, there exists a seed $S(n)$ which is a solution to Problems 1, 2, and 3, and therefore the crystal definition depending on $S(n)$ define...

conjecture | open | MathDB 页面 | 原文线索

V2-015009 | MATHDB-366332

Faithfulness conjecture for the Artin-Tits group action

摘录 / EXCERPT: Faithfulness conjecture. The action of the Artin-Tits group $\mathbb B$ on the triangulated category $\mathcal D$ is faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-015010 | MATHDB-366354

The representation-type characterization of Property PC

摘录 / EXCERPT: Property PC conjecture. For any restricted Lie algebra ${\mathfrak g}$ over k, ${\mathfrak g}$ satisfies Property PC if and only if ${\mathfrak g}$ is of tame or finite represent...

conjecture | open | MathDB 页面 | 原文线索

V2-015011 | MATHDB-366355

The universal Property PB conjecture for restricted Lie algebras

摘录 / EXCERPT: Property PB conjecture. All restricted Lie algebras satisfy Property PB regardless of representation type.

conjecture | open | MathDB 页面 | 原文线索

V2-015012 | MATHDB-366356

The torsion-free cyclic relation-module conjecture

摘录 / EXCERPT: The torsion-free cyclic relation-module conjecture. If G is torsion-free and $N_{\mathbb{Q}}$ is cyclic as a $\mathbb{Q}G$-module, then N is normally generated by a single elem...

conjecture | open | MathDB 页面 | 原文线索

V2-015013 | MATHDB-366401

The Reynolds operator formula for three-column SL_2 invariants

摘录 / EXCERPT: The three-column Reynolds formula. For all nonnegative integers i, j, and k, one has

conjecture | open | MathDB 页面 | 原文线索

V2-015014 | MATHDB-366402

The odd-power vanishing formula for the Reynolds operator

摘录 / EXCERPT: Odd-power vanishing conjecture. For every nonnegative integer n, one has

conjecture | open | MathDB 页面 | 原文线索

V2-015015 | MATHDB-366403

The general linear Reynolds operator formula for a matrix entry

摘录 / EXCERPT: General linear Reynolds formula. For $t=2$ and $n\geq 0$, one has

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V2-015016 | MATHDB-366435

Fusion-lifting conjecture for formal ribbon extensions

摘录 / EXCERPT: Let H be a finite-dimensional quasitriangular Hopf algebra and let $\tilde H$ be its formal ribbon extension. Let M_1,M_2 be indecomposable H-modules, and let...

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V2-015017 | MATHDB-366441

Ren-Xu's conjecture on exceptional Hilbert series of quasi-invariants

摘录 / EXCERPT: Ren-Xu's conjecture. If the Hilbert series of $Q_m(n,\mathbf{F}_p)$ differs from that of $Q_m(n,\mathbf{Q})$, then there exist integers $a\geq 0$ and $k\geq 0$ satisfying the displ...

conjecture | open | MathDB 页面 | 原文线索

V2-015018 | MATHDB-366450

Countable stratification conjecture for equivariant bootstrap categories

摘录 / EXCERPT: Countable stratification conjecture. The category $\Cell{G}$ is countably stratified for every finite group G.

conjecture | open | MathDB 页面 | 原文线索

V2-015019 | MATHDB-366451

Balmer comparison-map conjecture for equivariant cell categories

摘录 / EXCERPT: Balmer's comparison-map conjecture. For every finite group G, the comparison map is a homeomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-015020 | MATHDB-366475

The stability conjecture for discrete series L-packets

摘录 / EXCERPT: Let F be a non-archimedean local field, let G be a connected reductive F-group, and let $\Pi_{\varphi}(G)$ be a discrete series L-packet of irreducible smooth representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-015021 | MATHDB-366503

Hausdorff-dimensional packing conjecture for special-linear symmetries

摘录 / EXCERPT: Let $E\subseteq\mathbb{R}^2$ be a compact set with Hausdorff dimension ... Assume there is a set $R\subseteq\operatorname{SL}_2(\mathbb{R})$ of matrices such that $\theta(E)=E$ for e...

conjecture | open | MathDB 页面 | 原文线索

V2-015022 | MATHDB-366556

The conjecture on endoscopic equivalence without restrictions on ℓ

摘录 / EXCERPT: Endoscopic equivalence conjecture. There exists an endoscopic equivalence as in Theorem $$ without any hypothesis on ℓ.

conjecture | open | MathDB 页面 | 原文线索

V2-015023 | MATHDB-366557

The t-exactness conjecture for the endoscopic equivalence

摘录 / EXCERPT: t-exactness conjecture. The endoscopic equivalence of Theorem $$ is t-exact.

conjecture | open | MathDB 页面 | 原文线索

V2-015024 | MATHDB-366558

The splendid Morita equivalence conjecture for endoscopic equivalence

摘录 / EXCERPT: Splendid Morita equivalence conjecture. The endoscopic equivalence of Theorem $$ induces a splendid Morita equivalence in the sense of Rickard.

conjecture | open | MathDB 页面 | 原文线索

V2-015025 | MATHDB-366573

Reduction-by-stages conjecture for W-algebras

摘录 / EXCERPT: Let $(a_1,\dots,a_s,a_{s+1},\dots,a_r)$ be a partition of n, let f_2 be a nilpotent element of $\mathfrak{sl}_n$ corresponding to this partition, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-015026 | MATHDB-366582

The indecomposability criterion for the n/d-conjecture

摘录 / EXCERPT: The indecomposability criterion for the n/d-conjecture. If, for every nontrivial decomposition $(\mathbb{C}^{n})^{*}=W_{1}\oplus W_{2}$ and every integer $0\leq k\leq d$ satisfyi...

conjecture | open | MathDB 页面 | 原文线索

V2-015027 | MATHDB-366583

Friedl-Löh conjecture on undecidability of the uniform epimorphism problem for virtually abelian groups

摘录 / EXCERPT: Friedl-Löh conjecture. The uniform epimorphism problem onto the class of all finitely generated virtually abelian groups is not decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-015028 | MATHDB-366613

The symmetry-group decomposition conjecture for convex bodies

摘录 / EXCERPT: Let $K\subset\mathbb R^n$, $n\geq 3$, be a convex body, and let $G<\operatorname{O}(\mathbb R^n)$ be its symmetry group. Let π_k and ρ_k denote the subgroup types defined...

conjecture | open | MathDB 页面 | 原文线索

V2-015029 | MATHDB-366620

The semisimplicity criterion for cyclotomic Hecke-Clifford superalgebras

摘录 / EXCERPT: Let $\mathbb{K}$ be the ground field, let q be the parameter, let $\underline{Q}=(Q_1,Q_2,\ldots,Q_m)\in(\mathbb{K}^*)^m$, and let $f=f^{(\bullet)}_{\underline{Q}}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-015030 | MATHDB-366626

Equivariant scissors congruence spectrum conjecture

摘录 / EXCERPT: Let G be a finite group. The equivariant SK^G-groups, as H varies over the subgroups of G, form a Mackey functor, and let $K_G(\underline{\mathbb{Z}})$ denote the genuine...

conjecture | open | MathDB 页面 | 原文线索

V2-015031 | MATHDB-366662

The $\ce1$-conjecture for Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group and let $\chi:A_\Gamma\to\mathbb{R}$ be a character. The living subgraph $\mathrm{Liv}^\chi\subset\Gamma$ is obtained by deleting vertices v with...

conjecture | open | MathDB 页面 | 原文线索

V2-015032 | MATHDB-366663

The converse criterion for $\ce2$-invariants of Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group satisfying the $K(\pi,1)$-conjecture, and let $\chi:A_\Gamma\to\mathbb{R}$ be a character. Write Liv^χ for the living subgraph, and c...

conjecture | open | MathDB 页面 | 原文线索

V2-015033 | MATHDB-366664

Almeida's disconnected living-subgraph conjecture for Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group and let $\chi:A_\Gamma\to\mathbb{R}$ be a character. Let Liv^χ denote the living subgraph associated to χ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015034 | MATHDB-366677

The sign-character formula for the $(1,3)$-bosonic-fermionic coinvariant ring

摘录 / EXCERPT: Let $R_n^{(1,3)}$ be the $(1,3)$-bosonic-fermionic coinvariant ring, let $\operatorname{Frob}(R_n^{(1,3)};q;u,v,w)$ denote its Frobenius series, and let $s_{(1^n)}$ denote the Schu...

conjecture | open | MathDB 页面 | 原文线索

V2-015035 | MATHDB-366690

Unknot linearization conjecture for cyclic actions on the four-sphere

摘录 / EXCERPT: Let $\mathbb Z_n$ act smoothly on S^4, and suppose its fixed-point set is an unknotted two-sphere. Unknot linearization conjecture. The action is smoothly equivalent to a linear...

conjecture | open | MathDB 页面 | 原文线索

V2-015036 | MATHDB-366692

Point-fixing linearization conjecture for cyclic actions on the four-sphere

摘录 / EXCERPT: Let $\mathbb Z_n$ act smoothly on S^4, and suppose the action has exactly two fixed points. Point-fixing linearization conjecture. The action is smoothly equivalent to a linear a...

conjecture | open | MathDB 页面 | 原文线索

V2-015037 | MATHDB-366714

Normalizer-Solubilizer Conjecture

摘录 / EXCERPT: Let G be a finite insoluble group. For $x\in G$, define the solubilizer of x in G by ... Also, let $\mathcal N_G(\langle x\rangle)$ denote the normalizer of the cyclic subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-015038 | MATHDB-366722

Conjecture on the asymptotic normality of nilpotent-subgroup proportions

摘录 / EXCERPT: Asymptotic-normality conjecture. Consider sampling with replacement from the values $\mathfrak{J}(D_2),\mathfrak{J}(D_4),\ldots,\mathfrak{J}(D_n)$ for sufficiently large n. If...

conjecture | open | MathDB 页面 | 原文线索

V2-015039 | MATHDB-366737

Non-realizability of the minimal nilpotent orbit closure of type E_6 as a Hamiltonian reduction

摘录 / EXCERPT: Non-realizability conjecture. The closure $\overline{\mathcal{O}}_{\textrm{min}}^{E_6}$ is not isomorphic to any Hamiltonian reduction of the cotangent space T^*V of a representati...

conjecture | open | MathDB 页面 | 原文线索

V2-015040 | MATHDB-366784

The high-real-rank conjecture for proper surface-group actions

摘录 / EXCERPT: Let G be a linear real simple Lie group and let H be a reductive subgroup. Write (P-surf) for the existence of a proper action of a discrete surface subgroup of genus at least...

conjecture | open | MathDB 页面 | 原文线索

V2-015041 | MATHDB-366788

Natural embedding for distinguished representations and their derivatives

摘录 / EXCERPT: Let G_n be the relevant general linear group, let G_n' be its distinguished subgroup, and let π be an irreducible G_n'-distinguished representation of G_n with depth d...

conjecture | open | MathDB 页面 | 原文线索

V2-015042 | MATHDB-366818

Equality of partial theta series for tensor products involving projective modules

摘录 / EXCERPT: Partial theta series conjecture. The partial theta series associated with the tensor products satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-015043 | MATHDB-366828

Cluster braid group conjecture for Dynkin and Euclidean quivers

摘录 / EXCERPT: Cluster braid group conjecture. There is a canonical isomorphism

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V2-015044 | MATHDB-366829

The conjecture on TF-equivalence classes and cones of g-vectors

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra, let $K_0(\operatorname{proj}\Lambda)_{\mathbb{R}}$ be its real Grothendieck group, and let g be a g-vector. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015045 | MATHDB-366830

The conjecture on density, one-dimensional cones, and finite semistable classes

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra and let g be a g-vector. Write $[g]_{\operatorname{TF}}$ for the TF-equivalence class of g, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015046 | MATHDB-366836

Jacquet's local converse conjecture for general linear groups

摘录 / EXCERPT: Jacquet's local converse conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-015047 | MATHDB-366859

The mixability conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group. The mixability conjecture. G is mixable if and only if it has no nontrivial quotients of odd order. This claims that the obstruction given by nontrivia...

conjecture | open | MathDB 页面 | 原文线索

V2-015048 | MATHDB-366860

The mixability conjecture for finite groups generated by involutions

摘录 / EXCERPT: Let H be a finite group generated by elements of order 2; such a group is called $\{2\}$-generated. The involution-generation mixability conjecture. Every finite $\{2\}$-genera...

conjecture | open | MathDB 页面 | 原文线索

V2-015049 | MATHDB-366861

Unknot Linearization Redux for cyclic actions on the 4-sphere

摘录 / EXCERPT: Let $a4n$ act smoothly on S^4, and let S be its fixed-point set, assumed to be an unknotted 2-sphere. An equivariant bridge trisection is a bridge trisection preserved by the...

conjecture | open | MathDB 页面 | 原文线索

V2-015050 | MATHDB-366862

Homologically trivial linearization conjecture for cyclic actions on the complex projective plane

摘录 / EXCERPT: Let $a4n$ act smoothly on $a4cp^2$. The action is homologically trivial if it acts trivially on the second homology $H_2(a4cp^2)$. A linear action is one induced by the standard li...

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V2-015051 | MATHDB-366863

Trisection reformulation of homologically trivial linearization for cyclic actions on the complex projective plane

摘录 / EXCERPT: Let $a4n$ act smoothly and homologically trivially on $a4cp^2$, and let an equivariant trisection be a trisection preserved by the action. Homologically trivial linearization conje...

conjecture | open | MathDB 页面 | 原文线索

V2-015052 | MATHDB-366865

The rank formula for the even part of the supertrace space of cyclotomic Sergeev algebras

摘录 / EXCERPT: Let ${\rm R}$ be an integral domain in which 2 is invertible. Let $\mathfrak{H}_{\Delta}(n)^{g}$ be the cyclotomic Sergeev algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015053 | MATHDB-366872

Onn's group-algebra isomorphism conjecture for finite-module automorphism groups

摘录 / EXCERPT: Let λ be a partition, and let $\mathfrak{o}$ and $\mathfrak{o}'$ be valuation rings of non-Archimedean local fields with equal residue cardinality q. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015054 | MATHDB-366916

Conjecture on optimal logarithmic diameter and mixing bounds for bounded-rank groups

摘录 / EXCERPT: Let $\underline{G}$ be a simply connected, simple algebraic group over $\mathbb{F}_{p}$, and let $\underline{x}=(x_{1},\ldots,x_{r})$ be a generating set of...

conjecture | open | MathDB 页面 | 原文线索

V2-015055 | MATHDB-366931

Fourier-analytic cuspidal restriction conjecture for Lie algebra orbit closures

摘录 / EXCERPT: Let F be a local or finite field, let $\underline G$ be a split semisimple group with Lie algebra $\mathfrak g$, and let $\Omega\subset\mathfrak g$ be a conjugacy class with clos...

conjecture | open | MathDB 页面 | 原文线索

V2-015056 | MATHDB-366934

Neumann-Praeger covering-subgroup conjecture

摘录 / EXCERPT: Neumann-Praeger conjecture. Is there a function $f:\mathbb{N}\to\mathbb{N}$ such that whenever $U<H\leq A$, where H is a normal subgroup of the finite group A of index n and...

conjecture | open | MathDB 页面 | 原文线索

V2-015057 | MATHDB-366946

Condition keyconj under nonempty root intersection

摘录 / EXCERPT: Let Φ_A and $\Psi_{\lambda_{\mathbf c}}^+$ be the sets defined by ... and ... Suppose that $\Phi_A\cap\Psi_{\lambda_{\mathbf c}}^+\neq\emptyset$. The conjecture. Condition is sati...

conjecture | open | MathDB 页面 | 原文线索

V2-015058 | MATHDB-366984

Tameness conjecture for finite-group representations over finite fields

摘录 / EXCERPT: Let G be a finite group and k a finite field. Write $\mathrm{Rep_{f.d.}}(G,k)$ for the category of finite-dimensional representations of G over k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015059 | MATHDB-366985

The strongly aperiodic SFT conjecture

摘录 / EXCERPT: Let G be a finitely generated group. A strongly aperiodic subshift of finite type (SFT) is an SFT with no periodic configurations. The strongly aperiodic SFT conjecture. G admi...

conjecture | open | MathDB 页面 | 原文线索

V2-015060 | MATHDB-366986

Carroll-Penland conjecture on weakly aperiodic SFTs

摘录 / EXCERPT: Let G be a finitely generated group. A weakly aperiodic subshift of finite type (SFT) is an SFT with no finite-orbit requirement beyond the absence of a fully periodic configurat...

conjecture | open | MathDB 页面 | 原文线索

V2-015061 | MATHDB-366987

Ballier-Stein conjecture on the domino problem

摘录 / EXCERPT: Let G be a finitely generated group. The domino problem asks whether a given finite set of Wang tiles admits a tiling of G. Ballier-Stein conjecture. G has undecidable domino...

conjecture | open | MathDB 页面 | 原文线索

V2-015062 | MATHDB-367088

QF-1 characterisation conjecture for higher Auslander algebras

摘录 / EXCERPT: QF-1 characterisation conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-015063 | MATHDB-367089

QF-1 implication conjecture for higher Auslander algebras

摘录 / EXCERPT: QF-1 implication conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-015064 | MATHDB-367109

Lusztig's symmetry conjecture for dimension polynomials

摘录 / EXCERPT: Let W be a Weyl group, let $\mathcal{J}$ be its set of near involutions, and write $\dim(M_w)=P_w(q)$, where $P_w(t)\in\mathbb{Q}[t]$. For $w\in\mathcal{J}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015065 | MATHDB-367139

Positivity conjecture for Toda eigenfunctions

摘录 / EXCERPT: For $\alpha=\sum_{i=1}^n a_i\alpha_i\in Q^{\geq 0}$, let $\mathfrak{J}_\alpha$ be the Toda eigenfunction and let $(q)_\alpha$ denote the associated q-factorial. Positivity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-015066 | MATHDB-367140

Unimodality conjecture for Toda eigenfunction numerators

摘录 / EXCERPT: For $\alpha\in Q^{\geq 0}$, let $\mathfrak{J}_\alpha$ be the Toda eigenfunction and let $(q)_\alpha$ be the associated q-factorial. A polynomial is unimodal if its coefficient se...

conjecture | open | MathDB 页面 | 原文线索

V2-015067 | MATHDB-367177

Inductive Alperin bound conjecture for blocks

摘录 / EXCERPT: Let $G\unlhd A$ be finite groups, let B be a block of G with defect group D, and let b be its Brauer correspondent in a subgroup M as below. Write $\operatorname{IBr}(B)$...

conjecture | open | MathDB 页面 | 原文线索

V2-015068 | MATHDB-367189

The p-core conjecture for blocks of centraliser algebras

摘录 / EXCERPT: Let l,m,n be as in the definition of the centraliser algebra, and let k be the residue field of the p-modular system $(R,\bbf,k)$. For partitions $\mu_1\setminus\la_1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015069 | MATHDB-367190

Simplicity conjecture for belt modules

摘录 / EXCERPT: Let $\calh_p^k$ be the algebra under consideration, let $\Ga$ be a belt, and let $M_{\Ga}$ denote the associated $\calh_p^k$-module. Simplicity conjecture. The module $M_{\Ga}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-015070 | MATHDB-367196

The flatness-word-problem characterization conjecture

摘录 / EXCERPT: Flatness-word-problem conjecture. G is $\mathbf{C}^\forall$-flat if and only if the word problem of G lies in $\mathbf{C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-015071 | MATHDB-367207

The nilpotent-type skew brace conjecture for finite solvable groups

摘录 / EXCERPT: Nilpotent-type skew brace conjecture. The group G is isomorphic to the multiplicative group of a skew left brace of nilpotent type.

conjecture | open | MathDB 页面 | 原文线索

V2-015072 | MATHDB-367220

Ground-field and parameter independence of dimensions of distinguished subalgebras

摘录 / EXCERPT: Dimension-independence conjecture. For each such subalgebra A, the dimension $\dim_{\mathbb{K}} A$ is independent of the ground field $\mathbb{K}$ and the parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-015073 | MATHDB-367222

Buan-Hanson's conjecture on τ-perpendicular wide subcategories

摘录 / EXCERPT: Let Λ be a finite dimensional algebra. Let $\mathsf{W}\subseteq\operatorname{mod}\Lambda$ be a τ-perpendicular subcategory of $\operatorname{mod}\Lambda$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015074 | MATHDB-367232

Local-to-global conjugacy conjecture for Markov groups

摘录 / EXCERPT: Local-to-global conjugacy conjecture. If $H\leq\operatorname{Aut}(\mathcal{T}_n)$ is locally conjugated into M, then H is globally conjugated into M.

conjecture | open | MathDB 页面 | 原文线索

V2-015075 | MATHDB-367273

The right-angled Artin group conjecture for pure flat braid groups

摘录 / EXCERPT: Let PFB_n denote the pure flat braid group on n strands. Pure flat braid group conjecture. For every $n \geq 3$, PFB_n is a right-angled Artin group. This c...

conjecture | refuted | MathDB 页面 | 原文线索

V2-015076 | MATHDB-367283

Cohomological conjecture for Schur bundles on Debarre-Voisin hyperkähler manifolds

摘录 / EXCERPT: Cohomological conjecture. Among its factors, the vector bundle

conjecture | open | MathDB 页面 | 原文线索

V2-015077 | MATHDB-367310

Classical output-sensitive algorithm conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $\bla$, $\bmu$, and $\bnu$ be partitions such that $|\bla|=|\bmu|+|\bnu|$. Littlewood-Richardson algorithm conjecture. There exists a classical algorithm running in time ... that...

conjecture | open | MathDB 页面 | 原文线索

V2-015078 | MATHDB-367311

Output-sensitive classical algorithm conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $lambda$, mu, and nu be partitions such that $|lambda|=|mu|+|nu|$. Assume $c^{lambda}_{mu,nu}>0$. Output-sensitive Littlewood-Richardson conjecture. There exists a classica...

conjecture | open | MathDB 页面 | 原文线索

V2-015079 | MATHDB-367312

Output-sensitive classical algorithm conjecture for Kronecker coefficients

摘录 / EXCERPT: Let $lambda$, mu, and nu be partitions of n, and suppose that $f^{lambda}\geq f^{mu}\geq f^{nu}$. Kronecker algorithm conjecture. The Kronecker coefficient $g(lambda,mu,nu)$...

conjecture | open | MathDB 页面 | 原文线索

V2-015080 | MATHDB-367337

Degree-three invariant conjecture for generic orbit recovery in symmetric-group representations

摘录 / EXCERPT: Degree-three invariant conjecture. The invariants of degree at most three list resolve the generic orbit in $(\mathbb{C}^n)^d$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-015081 | MATHDB-367342

The spherical-building conjecture for smooth-group actions on reductive groups

摘录 / EXCERPT: Suppose that $(\widetilde G_{k^{\mathrm a}},\Gamma_{k^{\mathrm a}})$ is quasisemisimple. Write G^{red} for the maximal pseudo-reductive quotient of G, which is reducti...

conjecture | open | MathDB 页面 | 原文线索

V2-015082 | MATHDB-367378

Strong circle generator complexity conjecture

摘录 / EXCERPT: Circle generator complexity conjecture (strong version). The average circle generator complexity of class algebras is smaller than the corresponding average for fusion algebras:

conjecture | open | MathDB 页面 | 原文线索

V2-015083 | MATHDB-367379

Limiting average class size ratio conjecture

摘录 / EXCERPT: Average class size conjecture (weak version (b)). The limiting ratio satisfies

open_problem | open | MathDB 页面 | 原文线索

V2-015084 | MATHDB-367418

Kantor's nilpotency conjecture for random subgroups of symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group on n points. A random subgroup of S_n means a subgroup chosen uniformly from the finite set of all subgroups of...

conjecture | open | MathDB 页面 | 原文线索

V2-015085 | MATHDB-367419

Serwene's realizability conjecture for saturated fusion systems

摘录 / EXCERPT: Serwene's conjecture. Let $\mathcal{F}$ be a saturated fusion system. For any finite group G having an $\mathcal{F}$-block b, it follows that $\mathcal{F}$ is realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-015086 | MATHDB-367437

Crystal embedding and realization conjecture for shifted affine quantum groups

摘录 / EXCERPT: Crystal embedding and realization conjecture. (a) The map

conjecture | open | MathDB 页面 | 原文线索

V2-015087 | MATHDB-367557

The mutation-transitivity conjecture for rigid decorated representations

摘录 / EXCERPT: Let $(Q,\mathcal{S})$ be a Jacobi-finite connected quiver with potential. A decorated representation is rigid when it has no nontrivial self-extension, a negative representation is...

conjecture | open | MathDB 页面 | 原文线索

V2-015088 | MATHDB-367558

The regularity conjecture for imaginary coweights

摘录 / EXCERPT: Let $\check{\eta}$ be an indecomposable imaginary coweight, let $M(\check{\eta})$ be the associated representation, and let $\gamma_s(\check{\eta})$ denote its Schur rank. The cowe...

conjecture | open | MathDB 页面 | 原文线索

V2-015089 | MATHDB-367572

Connectivity characterization for Morse boundaries of Coxeter groups

摘录 / EXCERPT: Connectivity conjecture. A Coxeter group has connected, non-empty Morse boundary if and only if it is one-ended and wide-avoidant.

conjecture | open | MathDB 页面 | 原文线索

V2-015090 | MATHDB-367579

Orbit-Method description of the unitary dual for real reductive groups

摘录 / EXCERPT: Let $G_{\mathbb{R}}$ be a real reductive group. Let $\Pi_u(G_{\mathbb{R}})$ be the set of equivalence classes of irreducible unitary $G_{\mathbb{R}}$-representations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015091 | MATHDB-367592

Finite-height subgroups of hyperbolic groups are quasi-convex

摘录 / EXCERPT: Finite-height subgroup conjecture. If H has finite height in G, then H is quasi-convex in G.

conjecture | open | MathDB 页面 | 原文线索

V2-015092 | MATHDB-367607

Shang's character-formula conjecture for free alternative algebras

摘录 / EXCERPT: Shang's character-formula conjecture. In the Grothendieck ring of $GL(V)$,

conjecture | open | MathDB 页面 | 原文线索

V2-015093 | MATHDB-367639

Gaetz-Pierson non-negativity conjecture for increasing-pattern coefficients

摘录 / EXCERPT: Let $v=12\dots k$ be the increasing permutation pattern, let $N_{12\dots k}$ count occurrences of this pattern in permutations, and let $\operatorname{ch}_n(RN_{12\dots k})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-015094 | MATHDB-367707

The torus 3D B-model Fukaya-IndCoh equivalence conjecture

摘录 / EXCERPT: Let T be a torus, let $\mathfrak{t}$ be its Lie algebra, and let $T^{\vee}_{\mathbb{C}}$ be the complex dual torus. Let $\mathrm{BFM}(T^{\vee})$ denote the associated BFN space....

conjecture | open | MathDB 页面 | 原文线索

V2-015095 | MATHDB-367723

The permutational Boone-Higman conjecture

摘录 / EXCERPT: Let a group Γ act faithfully on a set S. The action is of type (A) if Γ is finitely presented, every stabilizer $\operatorname{Stab}_{\Gamma}(s)$ is finitely genera...

conjecture | open | MathDB 页面 | 原文线索

V2-015096 | MATHDB-367737

Prest's conjecture on domestic representation type and Krull-Gabriel dimension

摘录 / EXCERPT: Let K be an algebraically closed field and let A be a finite dimensional algebra over K. The Krull-Gabriel dimension of A, denoted by $\operatorname{KG}(A)$, is the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-015097 | MATHDB-367738

Hayashi's conjecture on finite connected quandles

摘录 / EXCERPT: Let $Q=(Q,\star)$ be a quandle, meaning that $a\star a=a$ for every $a\in Q$ and every left translation $L_a\colon Q\to Q$, $x\mapsto a\star x$, is an automorphism. The left multip...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-015098 | MATHDB-367756

Moore-Tachikawa conjecture for the TQFT associated to a semisimple group

摘录 / EXCERPT: Let G be a connected complex semisimple affine algebraic group with Lie algebra $\mathfrak{g}$. Let $(e,h,f)\in\mathfrak{g}^{\times 3}$ be a principal $\mathfrak{sl}_2$-triple, a...

conjecture | open | MathDB 页面 | 原文线索

V2-015099 | MATHDB-367766

Vdovin's depth conjecture for solvable subgroups

摘录 / EXCERPT: Let G be a finite group with trivial solvable radical, and let $d_G(H)$ denote the depth of a subgroup $H\leqslant G$. A subgroup is solvable if it is a solvable group. Vdovin's...

conjecture | open | MathDB 页面 | 原文线索

V2-015100 | MATHDB-367769

Relative solidity conjecture for biexact groups

摘录 / EXCERPT: Let Γ be a biexact group acting trace-preservingly on a von Neumann algebra N, and let ... Let $P,Q\subset M$ be commuting von Neumann subalgebras. Relative solidity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-015101 | MATHDB-367772

Characterization of non-abelian simple groups by singular-element proportions

摘录 / EXCERPT: Let G be a finite group, and let S be a non-abelian simple group. Write $\mu(G)$ for the set of proportions of r-singular elements of G as r ranges over the primes dividi...

conjecture | open | MathDB 页面 | 原文线索

V2-015102 | MATHDB-367776

BZSV's Plancherel algebra conjecture for hyperspherical Hamiltonian spaces

摘录 / EXCERPT: BZSV's Plancherel algebra conjecture. The following properties hold: $\PL_{X,\hbar}$ is flat over $k[\hbar]$; $\PL_X$ is commutative; and there is an isomorphism of $\Gc$-equivaria...

conjecture | open | MathDB 页面 | 原文线索

V2-015103 | MATHDB-367812

Lorentzian character conjecture for simple highest weight modules

摘录 / EXCERPT: Let $\mathfrak{g} = \oplus_{t=1}^T \mathfrak{sl}_{n_t+1}(\mathbb{C})$ be a direct sum of special linear Lie algebras, with positive roots Δ as above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015104 | MATHDB-367821

Structural conjectures for endomorphism rings of degenerate representations

摘录 / EXCERPT: Let e be a finite sum of primitive idempotents in the center Z_n of $\operatorname{Rep}_{\mathcal{O}}(G_n)$, let $\mathcal{P}_{\lambda}$ be the associated projective object, le...

conjecture | open | MathDB 页面 | 原文线索

V2-015105 | MATHDB-367823

Blumberg-Hill conjecture on tensor products of equivariant commutative operads

摘录 / EXCERPT: Let G be a finite group, and let I and J be indexing categories. Write $\cN_{I\infty}^{\otimes}$ and $\cN_{J\infty}^{\otimes}$ for the corresponding equivariant commutative...

conjecture | open | MathDB 页面 | 原文线索

V2-015106 | MATHDB-367851

The free 2-rig conjecture for representations of the general linear group

摘录 / EXCERPT: Let k be a field of characteristic zero. A 2-rig is the representation-theoretic structure in which the stated universal property is considered; let x be an object, let n be...

conjecture | open | MathDB 页面 | 原文线索

V2-015107 | MATHDB-367852

The free 2-rig conjecture for representations of the special linear group

摘录 / EXCERPT: Let k be a field of characteristic zero, let I be the unit for the tensor product, and let x be an object of dimension n with nth exterior power $\Lambda^n(x)$. The speci...

conjecture | open | MathDB 页面 | 原文线索

V2-015108 | MATHDB-367853

The free 2-rig conjecture for representations of the symplectic group

摘录 / EXCERPT: Let k be a field of characteristic zero, let I be the unit for the tensor product, and let x be a self-dual object of dimension n. Let $\epsilon\colon x\otimes x\to I$ be i...

conjecture | open | MathDB 页面 | 原文线索

V2-015109 | MATHDB-367854

The free 2-rig conjecture for representations of the orthogonal group

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let I be the unit for the tensor product, and let x be a self-dual object of dimension n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015110 | MATHDB-367855

The free 2-rig conjecture for representations of the special orthogonal group

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let I be the unit for the tensor product, and let x be an object of dimension n equipped with an isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-015111 | MATHDB-367861

Characteristic-independence conjecture for IC stalks on the zastava compactification

摘录 / EXCERPT: Let G be a reductive group, let $\overline{\mathrm{Bun}}_{N^-}$ be the compactification used in the paper, and let $\mathrm{IC}_{\overline{\mathrm{Bun}}_{N^-}}$ denote its inters...

conjecture | open | MathDB 页面 | 原文线索

V2-015112 | MATHDB-367895

The eigenvalue one conjecture for odd-dimensional faithful irreducible representations

摘录 / EXCERPT: Let G be a non-trivial finite group and let ... be a representation. Write $\rho^{\mathbb{R}}$ for the representation obtained by extending scalars to $\mathbb{R}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015113 | MATHDB-367913

NIP trace-interpretation conjecture for infinite groups

摘录 / EXCERPT: NIP group conjecture. If $\mathscr{M}$ is $\aleph_0$-categorical and NIP, then $\mathscr{M}$ interprets an infinite group if and only if $\mathscr{M}$ trace defines an infinite gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015114 | MATHDB-368015

Moore-Tachikawa conjecture for the Kostant slice

摘录 / EXCERPT: Moore-Tachikawa conjecture. There exists a two-dimensional TQFT

conjecture | open | MathDB 页面 | 原文线索

V2-015115 | MATHDB-368025

The group-action decomposition conjecture for invariant spaces of continuous functions

摘录 / EXCERPT: Let G be a compact group acting continuously and transitively on a compact Hausdorff space X. A space of continuous functions on X is called minimal if it is closed, invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-015116 | MATHDB-368039

Remeslenikov's conjecture on profinite rigidity of finitely generated free groups

摘录 / EXCERPT: Remeslenikov's conjecture. Finitely generated free groups are profinitely rigid: every finitely generated, residually finite group with the same finite quotients as F is isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-015117 | MATHDB-368054

Quasi-isometry classification conjecture for the groups W_Γ

摘录 / EXCERPT: Quasi-isometry classification conjecture. The groups W_Γ and $W_{\Gamma'}$ are quasi-isometric if and only if they are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-015118 | MATHDB-368093

Infinitesimal-character conjecture for quasi-split groups over finite extensions

摘录 / EXCERPT: Let F be a finite extension of ${\mathbb Q}_p$, and let G be a quasi-split group over F. Infinitesimal-character conjecture for quasi-split groups. Does the infinitesimal-cha...

open_problem | open | MathDB 页面 | 原文线索

V2-015119 | MATHDB-368116

Ciobanu-Evetts conjecture on conjugacy growth series

摘录 / EXCERPT: Let G be a finitely presented group, and let X be a finite generating set. The conjugacy growth series of G with respect to X is the generating function ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-015120 | MATHDB-368117

Geometric comparison of complex and p-adic Arthur-Vogan packets

摘录 / EXCERPT: Let n, r, k_i, and m_i be as above, and let $\psi^\BR$ denote the corresponding parameter on the complex side. On the p-adic side, let ψ^p and the Arthur-Vogan packe...

conjecture | open | MathDB 页面 | 原文线索

V2-015121 | MATHDB-368118

Vogan's geometric form of the local Langlands correspondence

摘录 / EXCERPT: Let $\check G$ be a connected reductive algebraic group over $F=\mathbb{Q}_p$, let G be its Langlands dual group, and let $\Lambda:W_F\to{}^L\check G$ be an infinitesimal charact...

conjecture | open | MathDB 页面 | 原文线索

V2-015122 | MATHDB-368131

Descent of preorderlike topoi under free group actions

摘录 / EXCERPT: Descent conjecture. The descent object $\mathcal{X}/\mathcal{G}$ of $\mathcal{G}$-equivariant sheaves on $\mathcal{X}$ is also preorderlike.

conjecture | open | MathDB 页面 | 原文线索

V2-015123 | MATHDB-368178

Unique representatives conjecture for coadjoint orbits of maximal unipotent subgroups

摘录 / EXCERPT: Unique representatives conjecture. For every coadjoint orbit, there exists a unique linear form $\lambda\in S$ lying on that orbit.

conjecture | open | MathDB 页面 | 原文线索

V2-015124 | MATHDB-368204

The Group Spencer conjecture for regular representations

摘录 / EXCERPT: Let G be a finite group of size n, and let $A_1,\dots,A_n$ be the $n\times n$ matrices corresponding to the regular representation of G. Group Spencer conjecture. The Matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-015125 | MATHDB-368230

Injectivity of the Atiyah-Patodi-Singer invariant for two-dimensional representations

摘录 / EXCERPT: Let $D_{2^{k+1}(2r+1)}$ be the indicated finite group, and let $\varrho_{t,s}$ denote its two-dimensional irreducible representations, with $1\leq t\leq 2r$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015126 | MATHDB-368248

Vanishing Ext conjecture for anti-spherical discrete series

摘录 / EXCERPT: Let $\mathbb H$ be the graded Hecke algebra, let Δ be its set of simple roots, and for $I\subset\Delta$ let $\widetilde{\mathbb H}_I$ be the corresponding graded Hecke algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-015127 | MATHDB-368282

Langlands duality conjecture for homomorphisms from finite subgroups of $SU(2)$

摘录 / EXCERPT: Let Γ be a finite group, and let G be a connected compact simple Lie group. Write $N(\Gamma,G)$ for the number of homomorphisms from Γ to G up to conjugation by...

conjecture | open | MathDB 页面 | 原文线索

V2-015128 | MATHDB-368283

Refined Langlands duality conjecture for representation spaces

摘录 / EXCERPT: Let Γ be a finite subgroup of $SU(2)$. For an Abelian central subgroup $Z\subset G$ of a connected compact simple Lie group G, set $H=G/Z$, with central extension ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015129 | MATHDB-368291

Cannon's quasisymmetric uniformization conjecture

摘录 / EXCERPT: Let G be a Gromov hyperbolic group, and let its boundary at infinity $\partial_\infty G$ be homeomorphic to the 2-sphere $\mathbb S^2$. Cannon's quasisymmetric uniformization con...

conjecture | open | MathDB 页面 | 原文线索

V2-015130 | MATHDB-368301

Symmetry conjecture for projective-injective modules of parabolic category O

摘录 / EXCERPT: Let $\mathfrak{g}$ be an arbitrary quasi-reductive Lie superalgebra and let $\mathfrak p$ be a parabolic subalgebra of $\mathfrak{g}$. Write $\mathcal{PI}^{\mathfrak p}$ for the fu...

conjecture | open | MathDB 页面 | 原文线索

V2-015131 | MATHDB-368329

Keller-Lübeck-Lynd-Semikina conjectures for fusion systems

摘录 / EXCERPT: Let $\mathcal{F}$ be a saturated fusion system on a finite p-group S of order p^e, and let α be a compatible family. Write s for the sectional rank of S, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015132 | MATHDB-368378

Conjecture on support stability and injective transition maps for higher Specht polynomials

摘录 / EXCERPT: Let d and f be positive integers. For a partition $\lambda\vdash n$, let $\operatorname{SSYT}_{d}(\lambda)$ be the relevant set of semistandard Young tableaux, let f_M be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015133 | MATHDB-368395

The group-theoretic conjecture on central translates of conjugacy classes

摘录 / EXCERPT: Let G be a group, let X be a subset of G, and let $H:=\langle X\rangle\leq G$. Assume that X is a conjugacy class of H. Central-translate conjecture. If there exist disti...

conjecture | open | MathDB 页面 | 原文线索

V2-015134 | MATHDB-368418

Cocompact cubulation and CAT(0) conjecture for quandle products

摘录 / EXCERPT: Let $\mathfrak{Q}$ be a quandle product with trivial holonomy of finitely many groups, each of which is cocompactly cubulable or CAT(0). Cocompact cubulation and CAT(0) conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-015135 | MATHDB-368423

Exactness conjecture for the image of the cohomological map

摘录 / EXCERPT: Let T_2 be the torus acting on V_1 with character t_1 and on V^* with character t_2, where $t_1,t_2\in\mathfrak{t}_2^*$ are the projections onto the respective factors. L...

conjecture | open | MathDB 页面 | 原文线索

V2-015136 | MATHDB-368428

Lapid-Prasad conjecture on distinguished representations and invariant L-packets

摘录 / EXCERPT: Lapid-Prasad conjecture. If π is H-distinguished, then the L-packet of π is invariant under the functor

conjecture | open | MathDB 页面 | 原文线索

V2-015137 | MATHDB-368435

Lewis et al.'s conjecture on vanishing conjugacy classes

摘录 / EXCERPT: Let G be a finite group, and let $k(G)$ be the least positive integer k such that there exist k conjugacy classes on which every non-linear irreducible character of G vanis...

conjecture | open | MathDB 页面 | 原文线索

V2-015138 | MATHDB-368445

The Drinfeld coproduct tensor-product equivalence conjecture

摘录 / EXCERPT: Let V and W be $U_q\widehat{\mathfrak{sl}}_2$-modules such that the Drinfeld-coproduct tensor product $\mathrm{V}\otimes_D\mathrm{W}$ is well-defined. The Dri...

conjecture | open | MathDB 页面 | 原文线索

V2-015139 | MATHDB-368451

The $0,2$-word isomorphism classification conjecture

摘录 / EXCERPT: The $\{0,2\}$-word isomorphism classification conjecture. Two words in $W_n^{\{0,2\}}$ are isomorphic as $U_q\widehat{\mathfrak{sl}}_2$-modules if and only if one can be obtained f...

conjecture | open | MathDB 页面 | 原文线索

V2-015140 | MATHDB-368452

The commutativity conjecture for $0^n2^n0^n$

摘录 / EXCERPT: For $n\geq0$, regard the words as tensor products of $U_q\widehat{\mathfrak{sl}}_2$-modules. The commutativity conjecture for $0^n2^n0^n$. ... and, symmetrically, ... The paper verifie...

conjecture | open | MathDB 页面 | 原文线索

V2-015141 | MATHDB-368453

The socle structure conjecture for 0^n2^n

摘录 / EXCERPT: Let 0^n2^n denote the corresponding tensor product module, let Soc_i denote its i-th socle layer, and let $s_2(r)$ be the sum of the binary digits of r. The...

conjecture | open | MathDB 页面 | 原文线索

V2-015142 | MATHDB-368454

The tensor-product strong-graph Cartesian-product conjecture

摘录 / EXCERPT: Let V and W be q-character separated modules. Assume that V, W, and $\mathrm{V}\otimes\mathrm{W}$ are socle-multiplicity free, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015143 | MATHDB-368456

The conf-connected-component tensor-factor conjecture

摘录 / EXCERPT: Let w be a word, and let $w_1,\dots,w_k$ be the subwords corresponding to the conf-connected components of w. Let $H(w)$ be the space of ℓ-singular vectors of weight 0....

conjecture | open | MathDB 页面 | 原文线索

V2-015144 | MATHDB-368467

Adjoint Steinberg conjecture

摘录 / EXCERPT: Let $\tilde{G}$ be the relevant reductive group, let $Z(\tilde{G})$ denote its center, and let $V(\lambda_k)$, for $k=1,\ldots,l$, be the fundamental representations under consider...

conjecture | open | MathDB 页面 | 原文线索

V2-015145 | MATHDB-368468

Steinberg's representation-theoretic conjugacy conjecture

摘录 / EXCERPT: Let G be an algebraic group, let $x,y\in G$, and let ρ range over the irreducible rational representations of G. Steinberg's conjecture. The elements x and y are conju...

conjecture | open | MathDB 页面 | 原文线索

V2-015146 | MATHDB-368469

The conjecture that invariant D-modules factor through the abelianization

摘录 / EXCERPT: Invariant D-module abelianization conjecture. The pull-back induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-015147 | MATHDB-368470

The universal solvability conjecture for unimodular equations

摘录 / EXCERPT: An equation $w(x)=1$ over a group G, where $w(x)\in G\ast\langle x\rangle_\infty$, is unimodular if the exponent sum of x in $w(x)$ equals ±1. It is solvable if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-015148 | MATHDB-368475

Rigidity conjecture for cluster tilting modules over local commutative algebras

摘录 / EXCERPT: Let A be a local commutative algebra with an n-cluster tilting module. Rigidity conjecture. Then $n=1$ and there is an integer $m\geq 1$ such that ... The conjecture is one of th...

conjecture | open | MathDB 页面 | 原文线索

V2-015149 | MATHDB-368514

The incomparability conjecture for multigraded Frobenius series of coinvariant rings

摘录 / EXCERPT: For fixed n and nonnegative integers k,j, let $R_n^{(k,j)}$ be the corresponding multigraded coinvariant ring, and write ... For $0\leq k'\leq k$ and $0\leq j'\leq j$, restrictio...

conjecture | open | MathDB 页面 | 原文线索

V2-015150 | MATHDB-368529

Existence of non-strongly p-artinian groups and non-LFS-realizable fusion systems

摘录 / EXCERPT: Existence conjecture. There are p'-reduced locally finite p-artinian groups that are not locally finite, strongly p-artinian groups, and there are fusion systems that are LF-...

conjecture | open | MathDB 页面 | 原文线索

V2-015151 | MATHDB-368540

Bernstein's unitarizability conjecture for the Aubert-Zelevinsky duality of G_2

摘录 / EXCERPT: Let G_2 be the reductive group under consideration, and let π be an irreducible unitarizable representation of G_2. The Aubert-Zelevinsky duality of π is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-015152 | MATHDB-368545

Losert-Malcev conjecture on quasi-isometric nilpotent Lie groups

摘录 / EXCERPT: Let G and H be simply connected nilpotent Lie groups. Losert-Malcev conjecture. The groups G and H are quasi-isometric if and only if they are isomorphic. This would mean t...

conjecture | open | MathDB 页面 | 原文线索

V2-015153 | MATHDB-368553

The self-extension vanishing conjecture for irreducible symmetric-group representations

摘录 / EXCERPT: Let ${\sf S}_n$ be the symmetric group and let $\mathbb{F}$ be an algebraically closed field of characteristic $p\geq 0$. Let ${\mathscr P}^{\text{\rm p-res}}_n$ denote the set o...

conjecture | open | MathDB 页面 | 原文线索

V2-015154 | MATHDB-368556

The asymptotic-dimension conjecture for the mapping class group

摘录 / EXCERPT: Asymptotic-dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015155 | MATHDB-368577

The profinite rigidity conjecture for automorphic orbits of free-group words

摘录 / EXCERPT: Profinite rigidity conjecture for word automorphic orbits. If there exists $\widehat{f}\in\operatorname{Aut}(\widehat{F}_k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015156 | MATHDB-368578

Simplicity and imaginarity conjecture for successive subquotients

摘录 / EXCERPT: Let $\bs s,\bs s'\in \mathbb L^l$ satisfy $\bs s\rhd_p\bs s'$ for some $0\leq p<l$, and let ... Let $M_0\subseteq M_1\subseteq\cdots\subseteq M_p\subseteq V$ be the chain of highest...

conjecture | open | MathDB 页面 | 原文线索

V2-015157 | MATHDB-368587

Atomic decomposition conjecture for type A_n crystals

摘录 / EXCERPT: Let n and k be as in the construction of the crystal embeddings, let $\mathcal{B}(k\theta)$ be the crystal of highest weight $k\theta$, and let ψ_j^n denote the specified...

conjecture | open | MathDB 页面 | 原文线索

V2-015158 | MATHDB-368591

A depth-bounded local converse conjecture for supercuspidal representations of general linear groups

摘录 / EXCERPT: Let F be the underlying non-Archimedean local field, let $ n$ be a positive integer, and let $ \pi_{1}$ and $ \pi_{2}$ be irreducible supercuspidal representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-015159 | MATHDB-368605

Fields conjecture on the Frobenius characteristic of the superspace coinvariant ring

摘录 / EXCERPT: Let SR_n be the bigraded superspace coinvariant ring with its natural $\mathfrak{S}_n$-action, and let $\operatorname{grFrob}(SR_n;q,z)$ denote its bigraded Frobenius characteris...

conjecture | open | MathDB 页面 | 原文线索

V2-015160 | MATHDB-368606

Equivariant Coxeter-complex surjection conjecture for the superspace coinvariant ring

摘录 / EXCERPT: Let W be a finite reflection group of rank n, let $R^{(1\mid1)}_W$ be its superspace coinvariant-type quotient, and let $\mathcal{OP}_W$ be the Coxeter complex formed by the fa...

conjecture | open | MathDB 页面 | 原文线索

V2-015161 | MATHDB-368637

Separated-subgroup conjecture for discrete torus actions

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and $D>0$. A pointed simply-connected proper geodesic space is a pair (X,p), and $\operatorname{Iso}(X)$ denotes its isometry group. For a discrete subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-015162 | MATHDB-368639

The quantum T-system conjecture for quantum minors

摘录 / EXCERPT: Quantum T-system conjecture. For any i-box $[a,c]$, there exist constants A and B such that

conjecture | open | MathDB 页面 | 原文线索

V2-015163 | MATHDB-368648

The lower-central-series conjecture for generalized class of p-groups

摘录 / EXCERPT: Let G be a finite p-group. Denote by $\gamma_i(G)$ the ith term of the lower central series of G, by $Cl(G)$ the generalized class parameter used in the paper, and by...

conjecture | open | MathDB 页面 | 原文线索

V2-015164 | MATHDB-368653

The two-test hom-counting conjecture for finite groups

摘录 / EXCERPT: Two-test hom-counting conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-015165 | MATHDB-368654

The determinant hom-counting conjecture for finite groups

摘录 / EXCERPT: Determinant hom-counting conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-015166 | MATHDB-368656

Bonatti's splitting conjecture for minimal circle actions

摘录 / EXCERPT: Let $G\leq \operatorname{Homeo}_+(S^1)$ be a subgroup such that every non-trivial element has at most two fixed points, and whose action on S^1 is minimal. Assume that the action...

conjecture | open | MathDB 页面 | 原文线索

V2-015167 | MATHDB-368657

Frankel's conjecture on hyperbolic three-manifold groups and circle actions

摘录 / EXCERPT: A hyperbolic-like action of a group on the circle is an action by orientation-preserving circle homeomorphisms in which every non-trivial element has exactly two fixed points, one...

conjecture | open | MathDB 页面 | 原文线索

V2-015168 | MATHDB-368663

AENY spread-global dimension stabilization conjecture for grid posets

摘录 / EXCERPT: AENY spread-global dimension stabilization conjecture. For every $m\geq 1\in\mathbb{N}$, the function g_m is constant when $k\geq 2+m$.

conjecture | open | MathDB 页面 | 原文线索

V2-015169 | MATHDB-368685

The conjectural Gelfand-Tsetlin theory for b-deformed Jucys-Murphy operators

摘录 / EXCERPT: Let k be a positive integer, let $\mathcal{J}_1,\ldots,\mathcal{J}_k$ be the b-deformed Jucys-Murphy operators, and let $\mathfrak{e}_k$ be the identity pair partition. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-015170 | MATHDB-368702

The regular Kim-Yu type realization conjecture

摘录 / EXCERPT: Assume that $p\neq 2$ is not bad for $\mathbf{G}$, and let $(\mathbf{T},\theta)$ be an unramified Howe-factorizable pair. Let $\vec\phi=(\phi_{-1},\ldots,\phi_d)$ be a Howe factori...

conjecture | open | MathDB 页面 | 原文线索

V2-015171 | MATHDB-368703

The positive-depth Lusztig induction and Kim-Yu type structure conjecture

摘录 / EXCERPT: Assume $\mathbf{G}'$ is a twisted Levi subgroup of $\mathbf{G}$ containing $\mathbf{T}$. Let α be any representation of $\mathbb{G}'_r(\mathbb{F}_q)$ of depth $<r$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015172 | MATHDB-368719

Generalization of the main straight-shape theorem to skew shapes

摘录 / EXCERPT: Skew-shape generalization. Theorem MainStraight holds more generally when $[bc^*]$ and $[bb^*]$ are e-small skew shapes rather than straight shapes.

conjecture | open | MathDB 页面 | 原文线索

V2-015173 | MATHDB-368761

Weissman's local Langlands correspondence for a cover

摘录 / EXCERPT: Let F be the local field and let $\widetilde G$ be the cover under consideration, with associated dual group data $\widetilde H$. Write $\Pi(\widetilde G)$ for the relevant set o...

conjecture | open | MathDB 页面 | 原文线索

V2-015174 | MATHDB-368762

Weissman's local Langlands correspondence for basic covers

摘录 / EXCERPT: Let G be the underlying group, let $\textnormal{\textsf{Basic}}_G$ be the set of basic data, and for each $\beta\in\textnormal{\textsf{Basic}}_G$ let $\widetilde G_\beta$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015175 | MATHDB-368822

The lonely element conjecture for numerical semigroups

摘录 / EXCERPT: Let S be a numerical semigroup, and call a nonzero small element of S lonely if it has the property specified in the source. Lonely element conjecture. If every nonzero small e...

conjecture | open | MathDB 页面 | 原文线索

V2-015176 | MATHDB-368843

Abért-Virág conjecture on Hausdorff dimension of linear groups in W_p

摘录 / EXCERPT: Let R be a commutative pro-p ring and let $G\le \mathrm{GL}_n(R)$ be a linear group over R. The group W_p is the group of p-adic automorphisms, namely a Sylow pro-p sub...

conjecture | open | MathDB 页面 | 原文线索

V2-015177 | MATHDB-368857

Herschend-Iyama-Oppermann conjecture on nonzero regular modules

摘录 / EXCERPT: Herschend-Iyama-Oppermann conjecture. For every n-representation infinite algebra, $\mathscr{R}$ is non-zero.

conjecture | open | MathDB 页面 | 原文线索

V2-015178 | MATHDB-368872

Pseudo \(\ll\)-transitivity for finite semigroups of order at most eight

摘录 / EXCERPT: Pseudo $\ll$-transitivity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-015179 | MATHDB-368881

McDonald-Sahay-Wyman's conjecture on the VC-dimension of Paley graphs

摘录 / EXCERPT: McDonald-Sahay-Wyman's conjecture. As $N\to\infty$ through the primes,

conjecture | open | MathDB 页面 | 原文线索

V2-015180 | MATHDB-369004

The containment conjecture for the global-breadth class

摘录 / EXCERPT: Let $\mathcal{H}$ be the class of finite groups satisfying ... and let Theorem ... . It might also contain simple and nonsimple groups that do not necessarily possess a nontrivial part...

open_problem | open | MathDB 页面 | 原文线索

V2-015181 | MATHDB-369005

The base-two conjecture for nilpotent subgroups of finite simple groups

摘录 / EXCERPT: Vdovin's base-two conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-015182 | MATHDB-369007

Generator conjecture for invariant rings of generalized signatures

摘录 / EXCERPT: Let $n,d\ge 3$, let V be a d-dimensional vector space, and let $\sigma:\mathbb{Z}\to\{1,2,\dots,d-1\}$ be a generalized signature of period n. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-015183 | MATHDB-369008

Moduli-space localization conjecture for admissible generalized signatures

摘录 / EXCERPT: Let σ be a generalized signature, let V be a d-dimensional vector space, and let $\mathfrak{M}(\beta_\sigma)$ be the associated cyclic decorated flag moduli space. If...

conjecture | open | MathDB 页面 | 原文线索

V2-015184 | MATHDB-369009

Natural cluster structure conjecture for d-admissible generalized signatures

摘录 / EXCERPT: Let σ be a d-admissible generalized signature of size n and dimension d, and let $R_\sigma=\mathbb{C}[V^\sigma]^{\operatorname{SL}(V)}$. Natural cluster structure conj...

conjecture | open | MathDB 页面 | 原文线索

V2-015185 | MATHDB-369060

Göpel's birationality conjecture for special trigonal genus-four curves

摘录 / EXCERPT: Göpel's birationality conjecture. The quotient

conjecture | open | MathDB 页面 | 原文线索

V2-015186 | MATHDB-369061

Göpel's orbital degeneracy-locus conjecture for the half-spin representation

摘录 / EXCERPT: Göpel's orbital degeneracy-locus conjecture. Let v be a general element of Δ. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-015187 | MATHDB-369155

The annihilator-map characterization of Verlinde categories

摘录 / EXCERPT: Annihilator-map conjecture. The annihilator map

conjecture | open | MathDB 页面 | 原文线索

V2-015188 | MATHDB-369156

The tensor-subcategory conjecture for injective annihilator maps

摘录 / EXCERPT: Tensor-subcategory conjecture. If $\mathcal{C}$ is Frobenius exact and the annihilator map

conjecture | open | MathDB 页面 | 原文线索

V2-015189 | MATHDB-369173

Asai-Iyama's g-tameness and E-tameness conjecture for Jacobi-finite QPs

摘录 / EXCERPT: Asai-Iyama's conjecture. If (Q,W) is a non-degenerate Jacobi-finite quiver with potential, then

conjecture | open | MathDB 页面 | 原文线索

V2-015190 | MATHDB-369181

Stable germ expansion conjecture for quasi-split reductive groups

摘录 / EXCERPT: Let G be a connected quasi-split reductive group over a non-archimedean local field F. Let $\mathcal{SU}_G$ be the vector space of stable distributions on $G(F)$ with true unip...

conjecture | open | MathDB 页面 | 原文线索

V2-015191 | MATHDB-369201

Foulkes's conjecture for symmetric powers

摘录 / EXCERPT: Let N be a positive integer, and let m,n be integers with $m\leqslant n$. For representations of $\operatorname{SL}_N(\mathbb{C})$, write $U\leqslant V$ when U is isomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-015192 | MATHDB-369202

Generalised Foulkes conjecture

摘录 / EXCERPT: Let $a,b,c,d\in\mathbb{N}$ satisfy $ab=cd$ and $a=\min(a,b,c,d)$. Write $U\leqslant V$ when U is isomorphic to a subrepresentation of V. Generalised Foulkes conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-015193 | MATHDB-369209

Frenkel-Hernandez conjecture on Langlands dual representations

摘录 / EXCERPT: Let $\mathcal{U}_q(\hat{\mathfrak{g}})$ and $\mathcal{U}_q(\leftindex^L{\hat{\mathfrak{g}}})$ be quantum affine algebras of Langlands-dual types, let P' be the sublattice specifi...

conjecture | open | MathDB 页面 | 原文线索

V2-015194 | MATHDB-369211

Irreducibility conjecture for the folding map on quantum affine representations

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type A_{2n-1}, let $\bar\pi$ be the induced homomorphism from the Grothendieck ring of finite-dimensional representations of the non-twisted quantum affi...

conjecture | open | MathDB 页面 | 原文线索

V2-015195 | MATHDB-369223

Conjecture on squares of automorphism cosets in finite simple groups

摘录 / EXCERPT: Let $G=\operatorname{Inn}(G)$ be a non-abelian finite simple group, and let $\alpha\in\operatorname{Aut}(G)$. Coset-square conjecture. There exists $x\in\alpha G$ such that, for th...

conjecture | open | MathDB 页面 | 原文线索

V2-015196 | MATHDB-369323

The pre-Tannakian and semisimplicity conjecture for symplectic interpolation categories

摘录 / EXCERPT: Fix a finite field $\mathbf{F}$ of odd cardinality q. For $n\geq 0$, let $H_n=\operatorname{\mathbf{Sp}}_{2n}(\mathbf{F})$, let k_n be an algebraically closed field, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015197 | MATHDB-369325

Gundlach's multi-invariant counting conjecture for Galois extensions

摘录 / EXCERPT: Let G be a finite group, let K be a number field, and let $\operatorname{inv}_1,\ldots,\operatorname{inv}_n$ denote the multi-invariants used to order G-extensions M/K, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-015198 | MATHDB-369348

Damiani-Martin-Rowell dimension conjecture for the loop Hecke algebra

摘录 / EXCERPT: Let $\mathbb{C}_z$ be the complex numbers $\mathbb{C}$ seen as a left $\mathbb{Z}[t]$-module by evaluating t at $z\in\mathbb{C}$. For $z\neq\pm1$, Damiani-Martin-Rowell's dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-015199 | MATHDB-369377

The bitableaux conjecture for Kronecker coefficients

摘录 / EXCERPT: Bitableaux conjecture. The Kronecker coefficient with the corresponding partitions is counted by lexicographic bitableaux of the given shape and weights whose pair of reading words...

conjecture | open | MathDB 页面 | 原文线索

V2-015200 | MATHDB-369410

Uniqueness conjecture for two-parameter presentations of Q36

摘录 / EXCERPT: Q36 presentation conjecture. If $\mathcal{P}=\mathcal{P}_9(n_1,n_2;m_1,m_2)$ presents Q_{36}, then

conjecture | open | MathDB 页面 | 原文线索

V2-015201 | MATHDB-369416

Commutativity of the truncated category O and KLR Grothendieck-group diagram

摘录 / EXCERPT: Suppose $\mathfrak{g}$ is simply-laced, let $\lambda\in P_+$, let $\mu\in \operatorname{wt}(V(\lambda))$, and choose an integral set of parameters $\mathbf{R}$ of level λ....

conjecture | open | MathDB 页面 | 原文线索

V2-015202 | MATHDB-369432

Gaitsgory's quantum Langlands duality conjecture for affine Hecke categories

摘录 / EXCERPT: Gaitsgory's conjecture. There is an equivalence of monoidal categories

conjecture | open | MathDB 页面 | 原文线索

V2-015203 | MATHDB-369442

The one-ended subgroup characterization for inaccessible finitely generated groups

摘录 / EXCERPT: The one-ended subgroup characterization. The subgroup H of the finitely generated group G is 1-ended in G if and only if for each splitting $A\ast_C B$ of G, where...

conjecture | open | MathDB 页面 | 原文线索

V2-015204 | MATHDB-369443

The Bieri-Stallings connectivity-at-infinity conjecture

摘录 / EXCERPT: The Bieri-Stallings connectivity-at-infinity conjecture. For $n\geq 2$, the group B_n is $(n-2)$-connected at infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-015205 | MATHDB-369444

The splitting conjecture for finitely presented non-semistable groups

摘录 / EXCERPT: The splitting conjecture for finitely presented non-semistable groups. If G does not have semistable fundamental group at infinity, then G splits non-trivially.

conjecture | open | MathDB 页面 | 原文线索

V2-015206 | MATHDB-369487

Shalev's probabilistic identity conjecture for finitely generated pro-p groups

摘录 / EXCERPT: Shalev's probabilistic identity conjecture. If G satisfies some probabilistic identity w, then G satisfies some identity.

conjecture | open | MathDB 页面 | 原文线索

V2-015207 | MATHDB-369496

Generalized Feit conjecture on character conductors and linear constituents

摘录 / EXCERPT: Let G be a finite group, let $\chi\in\operatorname{irr} G$, let $H\leq G$ be a subgroup, and let $\lambda\in\operatorname{irr} H$ be a linear constituent of the restriction χ_H. Generalized Feit conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-015208 | MATHDB-369512

The relative braid group fixed-subgroup conjecture

摘录 / EXCERPT: Let $f_\Gamma:\Sigma\to\underline{\Sigma}=\Sigma/\Gamma$ be the quotient map in the assumptions of Proposition. Write $\operatorname{Br}_\Sigma^\Gamma$ for the subgroup of Br_Σ fixed by...

conjecture | open | MathDB 页面 | 原文线索

V2-015209 | MATHDB-369514

The torsion-freeness conjecture for surface braid groups

摘录 / EXCERPT: Let $\Sigma\in {\bf Surf}$ be a marked surface and let Δ be a triangulation of Σ. Torsion-freeness conjecture. The group Br_Δ is torsion-free. The examples imm...

conjecture | open | MathDB 页面 | 原文线索

V2-015210 | MATHDB-369531

The Grothendieck-semigroup criterion for semiperfectness of localized group algebras

摘录 / EXCERPT: Semiperfectness criterion. The ring $\mathbb{Z}_{(p)}G$ is semiperfect if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-015211 | MATHDB-369600

Hom-finiteness conjecture for tensor products in the category \(\mathscr{X}({\bf G})\)

摘录 / EXCERPT: Let ${\bf G}$ be the group under consideration, let $\Bbbk$ be the coefficient field, and let $\mathscr{X}({\bf G})$ be the abelian category of $\Bbbk {\bf G}$-modules defined by t...

conjecture | open | MathDB 页面 | 原文线索

V2-015212 | MATHDB-369601

Weight-support vanishing conjecture for tensor products in the category \(\mathscr{X}({\bf G})\)

摘录 / EXCERPT: Let ${\bf G}$ be the group under consideration, let $\Bbbk$ be the coefficient field, and let $\mathscr{X}({\bf G})$ be the stated category of $\Bbbk {\bf G}$-modules. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-015213 | MATHDB-369602

Steinberg tensor-product vanishing conjecture for \(\mathscr{X}({\bf G})\)

摘录 / EXCERPT: Let ${\bf G}$ be the group under consideration, let $\Bbbk$ be the coefficient field, let ${\bf T}$ be its torus, and let St denote the Steinberg module. For a cha...

conjecture | open | MathDB 页面 | 原文线索

V2-015214 | MATHDB-369608

Faithfulness of the heart-group functor for finite-dimensional algebras

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra. The group $\operatorname{\mathrm{int}\text{-}\mathrm{heart}}(\operatorname{tors}\Lambda)$ has heart-controlled generators, meaning th...

conjecture | open | MathDB 页面 | 原文线索

V2-015215 | MATHDB-369609

The CAT(0) version of Cannon's conjecture for planar visual boundaries

摘录 / EXCERPT: CAT(0) Cannon conjecture. The group G has a finite-index subgroup that is the fundamental group of an aspherical 3-manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-015216 | MATHDB-369652

The conjecture on almost classically Galois structures of non-abelian simple type

摘录 / EXCERPT: Almost-classical correspondence conjecture. The almost classically Galois Hopf--Galois structures of type N are exactly the Hopf--Galois structures which admit a bijective corres...

conjecture | open | MathDB 页面 | 原文线索

V2-015217 | MATHDB-369656

Vdovin's conjecture on intersections of nilpotent subgroups

摘录 / EXCERPT: Vdovin's conjecture. Then $H \cap H^x = 1$ for some $x \in G$.

conjecture | open | MathDB 页面 | 原文线索

V2-015218 | MATHDB-369657

Lisi-Sabatini conjecture on simultaneous minimal intersections of Sylow subgroups

摘录 / EXCERPT: Lisi-Sabatini conjecture. There exists an element $x \in G$ such that, for each i, $P_i \cap P_i^x$ is inclusion-minimal in the set

conjecture | open | MathDB 页面 | 原文线索

V2-015219 | MATHDB-369658

Lisi-Sabatini conjecture for simple groups

摘录 / EXCERPT: Lisi-Sabatini conjecture for simple groups. There exists an element $x \in G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015220 | MATHDB-369664

He-et al.'s summand-growth conjecture for finite monoids

摘录 / EXCERPT: He-et al.'s conjecture. There is some $n\in\mathbb{N}$ such that $V^{\otimes n}$ contains a direct summand of the form $\operatorname{Ind}(W)$ for a kG-module W.

conjecture | open | MathDB 页面 | 原文线索

V2-015221 | MATHDB-369665

The tensor-power null-summand conjecture for finite semigroups

摘录 / EXCERPT: Null-summand conjecture. There is some $m\ge 1$ such that Z is a direct summand of $V^{\otimes m}$, and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-015222 | MATHDB-369717

Broué-Malle conjecture on Hecke algebras for relative Weyl groups

摘录 / EXCERPT: Let G be a finite reductive group, and let (M,μ) be a Φ_m-cuspidal pair with relative Weyl group $W_{M,\mu}^G$. Let $H_{M,\mu}^G(x)$ be the Broué-Malle algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015223 | MATHDB-369719

Conjecture on derangement proportions in affine symplectic and orthogonal groups

摘录 / EXCERPT: Derangement proportion conjecture. The following formulas hold:

conjecture | open | MathDB 页面 | 原文线索

V2-015224 | MATHDB-369755

Hiraga-Ichino-Ikeda-Langlands conjecture for Plancherel densities

摘录 / EXCERPT: Let G be the reductive group under consideration, let $\operatorname{Temp}(G)$ be its tempered dual, and let $\pi\in\operatorname{Temp}(G)$. Let $\rho_{\pi,\varpi}$ be the repres...

conjecture | open | MathDB 页面 | 原文线索

V2-015225 | MATHDB-369769

Conjectured Wigner 6j-symbol summation identities

摘录 / EXCERPT: For fixed N, let ... let $\lambda_i=i(i+1)$, and let $H_j=1+1/2+\cdots+1/j$ be the j^{th} harmonic number. The conjectured Wigner $6j$ identities. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-015226 | MATHDB-369770

Eventual negativity of averaged Ricci curvature in higher eigenspaces

摘录 / EXCERPT: Let V_ℓ denote the eigenspace indexed by ℓ in the Zeitlin model on $SU(N)$, and let $\widetilde{r}_\ell(N)$ be the averaged Ricci curvature in that eigenspace. Eventual n...

conjecture | open | MathDB 页面 | 原文线索

V2-015227 | MATHDB-369772

The peak-idempotent decomposition conjecture

摘录 / EXCERPT: Peak-idempotent decomposition conjecture. For any odd partition mu with $|mu|\leq n$ one has

conjecture | open | MathDB 页面 | 原文线索

V2-015228 | MATHDB-369786

Higher-rank harmonic map existence conjecture for framed surface-group representations

摘录 / EXCERPT: Let $X \in \mathcal{T}^{\pm}(\mathbb{S},\mathbb{M})$ and let $\hat{X}$ be as before. Write $X_n={\mathrm{PSL}_n(\mathbb{C})}/{\text{PSU}(n)}$ for the symmetric space and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015229 | MATHDB-369795

Davydov-Marcugini-Pambianco orbit-count conjecture for generic lines in $PG(3,q)$

摘录 / EXCERPT: Let q be a power of 3, let $G=PGL_2(q)$ act on the lines of $PG(3,q)$, and let C be the twisted cubic. A line is generic if it neither intersects C nor lies in any osculati...

conjecture | open | MathDB 页面 | 原文线索

V2-015230 | MATHDB-369801

The Second brick-Brauer-Thrall conjecture for finite-dimensional algebras

摘录 / EXCERPT: Second brick-Brauer-Thrall conjecture. If A is brick-infinite (equivalently, τ-tilting infinite), there is a dimension vector $\mathbf{d}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-015231 | MATHDB-369802

Stable-discreteness and brick-discreteness for finite-dimensional algebras

摘录 / EXCERPT: Stable-discreteness conjecture. An algebra A is stably-discrete if and only if it is brick-discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-015232 | MATHDB-369803

The E-finiteness conjecture for finite-dimensional algebras

摘录 / EXCERPT: E-finiteness conjecture. If A is E-finite, then A is brick-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-015233 | MATHDB-369815

The block-kernel characterization of exponential graph growth

摘录 / EXCERPT: Block-kernel characterization. The group L has exponential graph growth if and only if it admits a proper block B such that

conjecture | open | MathDB 页面 | 原文线索

V2-015234 | MATHDB-369816

The asymptotic prevalence conjecture for exponential graph growth

摘录 / EXCERPT: Asymptotic prevalence conjecture. Almost all transitive permutation groups have exponential graph growth, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-015235 | MATHDB-369844

The Ext criterion for irreducibility of parabolic induction

摘录 / EXCERPT: Ext criterion conjecture. The representation $\pi(C)\times\pi(D)$ is irreducible if and only if there exist open subsets $U_C\subseteq C$ and $U_D\subseteq D$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015236 | MATHDB-369847

The pole formula for Zelevinsky representations

摘录 / EXCERPT: Zelevinsky pole conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-015237 | MATHDB-369875

The proper-action characterization of algebraic quantum groups

摘录 / EXCERPT: Proper-action characterization. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-015238 | MATHDB-369876

The support and restriction conjecture for tame categorical local Langlands sheaves

摘录 / EXCERPT: Let $Z^1(W_F,\widehat{G})$ be the moduli of L-parameters over $\overline{\mathbb{Q}}_{\ell}$ for $\mathbf{G}$, let $C_{[\varphi]}$ be the connected components containing a lift of...

conjecture | open | MathDB 页面 | 原文线索

V2-015239 | MATHDB-369878

Determinant conjecture for orbit equivalence relations

摘录 / EXCERPT: Suppose $\Gamma\overset{\alpha}{\curvearrowright}(X,\mu)$ is a probability-measure-preserving action, with associated invariant random subgroup H_α and orbit equivalence rel...

conjecture | open | MathDB 页面 | 原文线索

V2-015240 | MATHDB-369880

The p-length bound for mutually permutable products of p-solvable groups

摘录 / EXCERPT: Let $G=AB$ be a mutually permutable product of two p-solvable subgroups A and B, where $p\in\pi(G)$. Write $l_p(G)$ for the p-length of G, and let $\tau_p(A)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015241 | MATHDB-369897

Benson's conjecture on tensor products of odd-dimensional representations

摘录 / EXCERPT: Benson's conjecture. All indecomposable summands of $V\otimes V^*$ have dimension divisible by 2, except for the single summand isomorphic to the trivial representation $\Bbbk$;...

conjecture | open | MathDB 页面 | 原文线索

V2-015242 | MATHDB-369900

Cannon-Thurston map nonexistence conjecture for hyperbolic normal subgroups of HHGs

摘录 / EXCERPT: Cannon-Thurston map nonexistence conjecture. Such a map does not exist unless either G is hyperbolic or H is hierarchically quasiconvex in some HHG structure on G.

conjecture | open | MathDB 页面 | 原文线索

V2-015243 | MATHDB-369906

Small rational-object equality conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group. Let $f(G)$ denote the number of irreducible characters of G with rational fields of values, and let $h(G)$ denote the number of conjugacy classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-015244 | MATHDB-369942

Gorenstein bimodule conjecture

摘录 / EXCERPT: Gorenstein bimodule conjecture. A is self-injective if and only if A, regarded as an A-bimodule, is Gorenstein-projective.

conjecture | open | MathDB 页面 | 原文线索

V2-015245 | MATHDB-369944

The presentation conjecture for the even subalgebra of the TAMA

摘录 / EXCERPT: Let $U(\mathfrak{so}(n))$ be the universal enveloping algebra, and let $\mathfrak{O}_{\overline{0}}$ denote the even subalgebra of the TAMA. For a sequence $A=(a_1,a_2,a_3,a_4)$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-015246 | MATHDB-369966

Auslander-Reiten conjecture for Auslander-Gorenstein algebras

摘录 / EXCERPT: Let A be an algebra satisfying the Auslander condition, meaning that it is n-Gorenstein for every $n\geq 1$. Auslander-Reiten conjecture. If A satisfies the Auslander conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-015247 | MATHDB-369967

Generalised Nakayama conjecture

摘录 / EXCERPT: Let A be a finite-dimensional K-algebra, and let S be a simple A-module. Write $\operatorname{grade} S$ for the grade of S. Generalised Nakayama conjecture. For every sim...

conjecture | open | MathDB 页面 | 原文线索

V2-015248 | MATHDB-369968

Auslander-Reiten map characterisation of Auslander-Gorenstein algebras

摘录 / EXCERPT: Let A be a finite-dimensional K-algebra over an algebraically closed field K. Say that A has a well-defined Auslander-Reiten map if every indecomposable injective A-modul...

conjecture | open | MathDB 页面 | 原文线索

V2-015249 | MATHDB-370004

Lieb's permanent dominance conjecture for immanants

摘录 / EXCERPT: Let A be a positive semidefinite Hermitian matrix, let $\lambda\vdash n$ be a partition, let χ^{λ} be the irreducible character of $\mathfrak{S}_n$ indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-015250 | MATHDB-370041

Indecomposability conjecture for the modules $\widehat{V}$ and $\widehat{W}$

摘录 / EXCERPT: Let $\widehat{\mathfrak{b}}$ be the algebra whose modules are under consideration, and let $\widehat{V}$ and $\widehat{W}$ be the $\widehat{\mathfrak{b}}$-submodules appearing in t...

conjecture | open | MathDB 页面 | 原文线索

V2-015251 | MATHDB-370050

DWZ-mutation homeomorphism conjecture for Ziegler spectra

摘录 / EXCERPT: DWZ-mutation homeomorphism conjecture. The DWZ-mutation of modules is a homeomorphism between the Ziegler spectra of the Jacobian algebras

conjecture | open | MathDB 页面 | 原文线索

V2-015252 | MATHDB-370097

Twisted conjugacy growth series conjecture

摘录 / EXCERPT: Twisted conjugacy growth series conjecture. The group G is virtually abelian if and only if all its twisted conjugacy growth functions are rational. This is presented as a twiste...

conjecture | open | MathDB 页面 | 原文线索

V2-015253 | MATHDB-370100

Archimedean rationality conjecture for cup products of automorphic forms

摘录 / EXCERPT: Let f and f' be the forms specified in Theorem BS, chosen so that for every $v\in S_\infty$ their local components lie in the indicated coefficient-field-rational subspaces of...

conjecture | open | MathDB 页面 | 原文线索

V2-015254 | MATHDB-370103

The product-or-K3-type conjecture for finite abelian groups acting on rationally connected threefolds

摘录 / EXCERPT: We say that a finite abelian group G is of K3 type if there is an exact sequence ... where H faithfully acts on a K3 surface. In the notation of the threefold classification, gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015255 | MATHDB-370107

Eigenvalue-refined factorization conjecture for unitary groups

摘录 / EXCERPT: Let $\mathrm{GL}^\varepsilon_n$ denote the general linear group when $\varepsilon=1$ and the general unitary group when $\varepsilon=-1$, and let $\mathrm{GL}^\varepsilon_1$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-015256 | MATHDB-370109

Minimum reflection factorization conjecture for symplectic and minus-type orthogonal groups

摘录 / EXCERPT: Let G be either the finite symplectic group $\operatorname{Sp}_{N}(q)$ or the finite minus-type orthogonal group $\operatorname{O}^{-}_{N}(q)$, and let c be a Singer cycle in...

conjecture | open | MathDB 页面 | 原文线索

V2-015257 | MATHDB-370138

The acylindrical hyperbolicity conjecture for Artin groups

摘录 / EXCERPT: Let A_S be an Artin group, and let Z_S denote the centre of A_S. Acylindrical hyperbolicity conjecture. The quotient A_S/Z_S is acylindrically hyperbolic. This would provid...

conjecture | open | MathDB 页面 | 原文线索

V2-015258 | MATHDB-370141

Nonexistence of cross-characteristic composition factors in isospectral groups

摘录 / EXCERPT: Let $L\in\mathcal{L}$, where $\mathcal{L}$ is the specified list of nonabelian simple groups, let S be a finite simple group of Lie type over a field whose characteristic differs...

conjecture | open | MathDB 页面 | 原文线索

V2-015259 | MATHDB-370154

The torsion conjecture for stabilizer abelianizations of groups of finite type

摘录 / EXCERPT: Torsion conjecture. The quotient

conjecture | open | MathDB 页面 | 原文线索

V2-015260 | MATHDB-370174

Internal parametrization conjecture for L-packets over local fields of arbitrary characteristic

摘录 / EXCERPT: Internal parametrization conjecture. There exists a canonical surjection

conjecture | open | MathDB 页面 | 原文线索

V2-015261 | MATHDB-370203

Evaluation correspondence conjecture in the global matrix coefficient case

摘录 / EXCERPT: Let H be a reductive group, let c be its Cartan involution, and let fs_H be a local system with associated Hecke eigensheaf $\LL_{fs_H}$. For a connected component indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-015262 | MATHDB-370240

Kielak's algebraic fibring conjecture for virtually residually torsion-free nilpotent groups

摘录 / EXCERPT: Let G be a virtually residually torsion-free nilpotent group. Its first ℓ^2-Betti number is denoted by $\beta_1^{(2)}(G)$, and G is algebraically fibred if it admits a sur...

conjecture | open | MathDB 页面 | 原文线索

V2-015263 | MATHDB-370258

The depth inequality and equality conjecture for Langlands packets

摘录 / EXCERPT: Let G be a connected reductive group over F that splits over F^t, let φ be a positive-depth Langlands parameter for $G(F)$, and let $\Pi(\varphi)$ be its associated...

conjecture | open | MathDB 页面 | 原文线索

V2-015264 | MATHDB-370259

The depth-zero restricted-parameter conjecture

摘录 / EXCERPT: Let G be a connected reductive group over F that splits over a tame extension. Let π be a depth-zero irreducible representation of $G(F)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015265 | MATHDB-370283

The geometrical Mackey filtration conjecture

摘录 / EXCERPT: Geometrical Mackey filtration conjecture. The restriction of π to P has a filtration whose successive quotients are geometrical Mackey inductions.

conjecture | open | MathDB 页面 | 原文线索

V2-015266 | MATHDB-370284

The closedness conjecture for spectral orbits

摘录 / EXCERPT: Closedness conjecture for spectral orbits. The subset $\mathrm{SO}_U(\pi)$ is closed in $\widehat{U}$.

conjecture | open | MathDB 页面 | 原文线索

V2-015267 | MATHDB-370285

The stable-range theta lift low-rank conjecture

摘录 / EXCERPT: Stable-range theta lift low-rank conjecture. The theta lift $\theta(\tau)$ is a low rank representation of G, meaning that its U-support contains no open L-orbit.

conjecture | open | MathDB 页面 | 原文线索

V2-015268 | MATHDB-370293

The generalized Ramanujan conjecture for cuspidal automorphic representations of general linear groups

摘录 / EXCERPT: Let F be a number field, let $N\in\mathbb{N}$, and let $\pi\in\mathcal{AR}_{cusp}(GL_n)$. For every place v of F, let π_v denote the local component of π, and call it...

conjecture | open | MathDB 页面 | 原文线索

V2-015269 | MATHDB-370294

The parameter exponent bound for local packets

摘录 / EXCERPT: Let $\psi_v\in\Psi_{G_v}$ be a local parameter and let $\pi_v\in\Pi_{\psi_v}$ be a representation in its local packet. Write L_{π_v} and L_{ψ_v} for their respective expon...

conjecture | open | MathDB 页面 | 原文线索

V2-015270 | MATHDB-370369

Tensor-square conjecture for symmetric groups

摘录 / EXCERPT: For each positive integer n, let V_λ denote the irreducible representation of S_n indexed by a partition λ of n. Tensor-square conjecture. There exists a part...

conjecture | open | MathDB 页面 | 原文线索

V2-015271 | MATHDB-370370

Crawley-Boevey's conjectural solution to the multiplicative Deligne-Simpson problem

摘录 / EXCERPT: Let $\mathcal{C}=(C_j)_{1\le j\le k}$ be a tuple of conjugacy classes of GL_n, defining a star-shaped quiver Q, a deformation parameter $\mathbf{q}$, and a dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-015272 | MATHDB-370399

The quotient conjecture for CAT(0) groups with rank-one elements

摘录 / EXCERPT: Let G be a CAT(0) group, meaning a group acting geometrically on a CAT(0) space, and suppose that G contains a rank-one element. Quotient conjecture. G has a proper quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-015273 | MATHDB-370442

The subgroup-fixed-locus compatibility conjecture for Iritani isomorphisms

摘录 / EXCERPT: Subgroup-fixed-locus compatibility conjecture. These inclusions commute with Iritani's partial isomorphisms and with the action of MT. This generalizes the preceding com...

conjecture | open | MathDB 页面 | 原文线索

V2-015274 | MATHDB-370445

The crepant resolution conjecture for quotient singularities

摘录 / EXCERPT: Let V) be a complex vector space and let $\Gamma\subset\operatorname{SU}(m/2)$ be finite. Let (M,ρ) be a crepant resolution of V/Γ, and let the age of a conjugacy cla...

conjecture | open | MathDB 页面 | 原文线索

V2-015275 | MATHDB-370473

Stable primitivity-rank formula for symmetric-group character averages

摘录 / EXCERPT: Let $S_{\bullet}=\{S_N\}_{N\ge 1}$, and let $\chi^{\mu[\bullet]}=\{\chi^{\mu[N]}\}$ denote the stable irreducible character associated with a partition μ. For a word w in the...

conjecture | open | MathDB 页面 | 原文线索

V2-015276 | MATHDB-370476

The compatible normal series conjecture for finite groups

摘录 / EXCERPT: Compatible normal series conjecture. Two groups are compatible if and only if they have compatible normal series.

conjecture | open | MathDB 页面 | 原文线索

V2-015277 | MATHDB-370493

Semisimple reduction conjecture for the Černý conjecture

摘录 / EXCERPT: A semisimple automaton is an automaton in the class considered in the paper for which the semisimple reduction is defined. Semisimple conjecture. If the Černý conjecture is solved...

conjecture | open | MathDB 页面 | 原文线索

V2-015278 | MATHDB-370494

Matrix-algebra conjecture for simple and quasi-simple automata

摘录 / EXCERPT: Let $\mathcal{A}$ be either a simple or quasi-simple automaton with n states, and let $\mathcal{R}(\mathcal{A})$ be its associated synchronized $\mathbb{C}$-algebra. Matrix-algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-015279 | MATHDB-370533

The realization conjecture for strong Markovian actions

摘录 / EXCERPT: Let ρ be a strong Markovian action. A strong Markovian action is an action satisfying the strong Markovian-family conditions defined in the source. Realization conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-015280 | MATHDB-370563

Oliver's Schur multiplier conjecture for saturated fusion systems

摘录 / EXCERPT: Oliver's Schur multiplier conjecture. $\ker(\mu_{\mathcal L})$ is isomorphic to a subgroup of the Schur multiplier of $\mathcal F$.

open_problem | open | MathDB 页面 | 原文线索

V2-015281 | MATHDB-370575

Finite-presentation conjecture for the double dimer group

摘录 / EXCERPT: Finite-presentation conjecture. G has either of the following two equivalent finite presentations:

conjecture | open | MathDB 页面 | 原文线索

V2-015282 | MATHDB-370586

Bernstein's inequality for quantum differential operators at generic parameter

摘录 / EXCERPT: Let q be a complex number that is not a root of unity, and let $G=\operatorname{GL}_2$ or $G=\operatorname{SL}_2$. A $\mathcal{D}_q(G)$-module is called strongly equivariant when...

conjecture | open | MathDB 页面 | 原文线索

V2-015283 | MATHDB-370594

Lam-Postnikov-Pylyavskyy alcoved-polytope Schur-order conjecture

摘录 / EXCERPT: Let ρ,μ,ν,δ be partitions, and let χ_λ denote the corresponding SL_n character. Suppose ... and that μ and ν belong to the minimal a...

conjecture | open | MathDB 页面 | 原文线索

V2-015284 | MATHDB-370603

Hill--Hopkins--Ravenel conjecture on the higher-height Morava module

摘录 / EXCERPT: Hill--Hopkins--Ravenel conjecture. There is a $\mathbb{W}[C_{p^k}]$-module isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-015285 | MATHDB-370606

The freeness conjecture for quasi-flag representation rings

摘录 / EXCERPT: Freeness conjecture. For all $k\in\mathcal{M}(W)$, after possibly inverting $|W|$, the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-015286 | MATHDB-370619

Gillespie-Rhoades higher Specht basis conjecture for Garsia-Procesi modules

摘录 / EXCERPT: Gillespie-Rhoades conjecture. The set $\mathcal{C}'_\mu$ descends to a higher Specht basis of R_μ. This conjecture is attributed in the source to Gillespie and Rhoades; the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-015287 | MATHDB-370666

Polynomial eventual multiplicity conjecture for pure graph braid groups

摘录 / EXCERPT: Let $\mathsf{\Gamma}$ be a graph, let i be a homological degree, and let λ be a partition. Write $\mathcal{H}_i(\mathsf{\Gamma})$ for the relevant representation-valued h...

conjecture | open | MathDB 页面 | 原文线索

V2-015288 | MATHDB-370677

The uniform period bound conjecture for centralizers in spherical Artin-Tits groups

摘录 / EXCERPT: Uniform period bound conjecture. There is a number $n(G)$ such that $p\leq n(G)$ for every $e004gamma i G$. This would provide a uniform bound for the centralizers needed by the ca...

conjecture | open | MathDB 页面 | 原文线索

V2-015289 | MATHDB-370703

The type A_2 shift conjecture for supported edge chains

摘录 / EXCERPT: Let Δ be the Bruhat-Tits building, let $\mathbb{X}$ be the coefficient system, let $X_{0,2}$ be the specified set of edges, and let I be the relevant subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015290 | MATHDB-370720

Atiyah's conjecture for torsion-free groups

摘录 / EXCERPT: Atiyah conjecture. The von Neumann dimension of the kernel of this map is an integer.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-015291 | MATHDB-370721

The Poincaré-duality-group virtual fibring conjecture

摘录 / EXCERPT: Let G be a PD^n-group that is RFRS. Here $\beta_i^{(2)}(G;\mathbb K)$ denotes the ith L^2-Betti number with coefficients in a field $\mathbb K$. Poincaré-duality-g...

conjecture | open | MathDB 页面 | 原文线索

V2-015292 | MATHDB-370725

The RFRS hyperbolic virtual fibring conjecture

摘录 / EXCERPT: A finite-volume hyperbolic manifold is a quotient of real hyperbolic space by a group of isometries acting freely and properly discontinuously. RFRS hyperbolic virtual fibring conj...

conjecture | open | MathDB 页面 | 原文线索

V2-015293 | MATHDB-370739

Kessar-Linckelmann's refined Broué conjecture

摘录 / EXCERPT: Let $\mathcal{O}$ be an arbitrary complete discrete valuation ring, let G be a finite group, and let b be a block idempotent of G over $\mathcal{O}$ with an abelian defect gr...

conjecture | open | MathDB 页面 | 原文线索

V2-015294 | MATHDB-370745

Hilburn's open-embedding conjecture for partial Coulomb-branch resolutions

摘录 / EXCERPT: Let G act on ${\bf N}$, let $\nu\colon\mathbb{C}^{\times}\to\mathsf{A}$ define a partial resolution $\widetilde{\mathcal{M}}_{C,\nu}\to\mathcal{M}_C$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015295 | MATHDB-370746

Normalized character conjecture for the unique Coulomb-branch category-O module

摘录 / EXCERPT: Let $\Phi^{-,\mathrm{re}}_\mu$ be the relevant set of negative real roots, let $\langle\alpha,\mu\rangle$ denote the root-coweight pairing, and let L be the unique irreducible ob...

conjecture | open | MathDB 页面 | 原文线索

V2-015296 | MATHDB-370759

Andruskiewitsch-Schneider generation conjecture for finite Nichols algebras

摘录 / EXCERPT: Let $K=\Bbbk G$ for a finite group G, and let … be a finite-dimensional coradically graded connected Hopf algebra, with R^1 its degree-one component. Andruskiewitsch-Schneider...

conjecture | open | MathDB 页面 | 原文线索

V2-015297 | MATHDB-370761

Feit's conjecture on character conductors and element orders

摘录 / EXCERPT: Let G be a finite group and let n be a positive integer. An irreducible character of G has conductor n when its character field has conductor n. Feit's conjecture. If G...

conjecture | open | MathDB 页面 | 原文线索

V2-015298 | MATHDB-370766

Transitivity conjecture for the braid-group action on $\mathrm{X}^{(2)}_{\gamma,\delta}$

摘录 / EXCERPT: Let B_4 be the braid group on four strands, and let $\mathrm{X}^{(2)}_{\gamma,\delta}$ be the set on which the induced braid-group action is defined. Assume Assumption. Transitiv...

conjecture | open | MathDB 页面 | 原文线索

V2-015299 | MATHDB-370782

Cesi's spectral-gap conjecture for the hyperoctahedral group

摘录 / EXCERPT: Cesi's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015300 | MATHDB-370826

Critical recurrence conjecture for stationary random walks on virtually--Z lamplighter groups

摘录 / EXCERPT: Let G be a virtually-$\mathbb{Z}$ group and let H be a nontrivial finite group. Let λ_c be the critical parameter such that the λ-stationary random walk on...

conjecture | open | MathDB 页面 | 原文线索

V2-015301 | MATHDB-370829

Wiegold's conjecture for T_k-systems of nonabelian simple groups

摘录 / EXCERPT: Let F_k be a free group of rank k, and let G be a group. A subgroup $N\leq F_k$ with $F_k/N\cong G$ is a G-defining subgroup; write $\Sigma_k(G)$ for the set of all such su...

conjecture | open | MathDB 页面 | 原文线索

V2-015302 | MATHDB-370840

Graph localization and G-infinity-operads conjecture

摘录 / EXCERPT: Let G be a finite group, let $(\operatorname{sOp}^{\operatorname{col}})^G$ be the category of G-objects in simplicial coloured operads, and let G-∞-Op...

conjecture | open | MathDB 页面 | 原文线索

V2-015303 | MATHDB-370841

The fundamental local equivalence conjecture for principal equivariant W-algebras

摘录 / EXCERPT: Let G be a reductive group with maximal torus T, coweight lattice $X_{*}(T)$, dominant coweights $X_{*}(T)_{+}$, Langlands dual group $\check{G}$, and principal equivariant aff...

conjecture | open | MathDB 页面 | 原文线索

V2-015304 | MATHDB-370868

Stable-center conjecture for the depth-r image

摘录 / EXCERPT: Depth-r stable-center conjecture. The image of ξ^r is contained in the depth-r stable center:

conjecture | open | MathDB 页面 | 原文线索

V2-015305 | MATHDB-370869

Conjectural tensor-product formulae relative to M

摘录 / EXCERPT: Conjectural tensor-product formulae.

conjecture | open | MathDB 页面 | 原文线索

V2-015306 | MATHDB-370906

Vogan's augmented Kazhdan-Lusztig hypothesis for p-adic groups

摘录 / EXCERPT: Let G be a connected reductive algebraic group over a locally compact non-Archimedean local field F. Let $\lambda:W_F\to{}^LG$ be an infinitesimal parameter. For geometric para...

conjecture | open | MathDB 页面 | 原文线索

V2-015307 | MATHDB-370913

Muić's irreducibility conjecture for big theta lifts

摘录 / EXCERPT: Muić's conjecture. The big theta lift $\Theta(\pi)$ to $H(V_m)$ is irreducible, if non-zero.

conjecture | open | MathDB 页面 | 原文线索

V2-015308 | MATHDB-370929

Isomorphism rigidity conjecture for braided Higman-Thompson groups

摘录 / EXCERPT: Let $A_{n,m}$ be the rooted tree whose root has degree m and whose other vertices have degree $n+1$, and write $\mathrm{br}T_{n,m}:=\mathfrak{mod}(A_{n,m})$ for its braided Higma...

conjecture | open | MathDB 页面 | 原文线索

V2-015309 | MATHDB-370955

Noncongruence conjecture for nonabelian finite simple groups

摘录 / EXCERPT: Noncongruence conjecture. Every nonabelian finite simple group G is noncongruence.

conjecture | open | MathDB 页面 | 原文线索

V2-015310 | MATHDB-371001

Coherence conjecture for finite-type RFRS groups

摘录 / EXCERPT: Let G be a RFRS group of type FP. A group is coherent if every finitely generated subgroup is finitely presented. The algebra $\mathbb{Q}[G]$ is the rational group alg...

conjecture | open | MathDB 页面 | 原文线索

V2-015311 | MATHDB-371007

Functorial braid group action on derived categories of moduli spaces

摘录 / EXCERPT: Let X be the Salvetti complex associated with the discriminantal arrangement, let Θ be its set of vertices, and let $\Pi_1(X,\Theta)$ be the fundamental groupoid. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-015312 | MATHDB-371068

The power-to-enhanced-power cograph conjecture for finite groups

摘录 / EXCERPT: Power-to-enhanced-power cograph conjecture. For every finite group G, if the power graph $\mathcal{P}(G)$ is a cograph, then the enhanced power graph $\mathcal{E}(G)$ is a cograp...

conjecture | open | MathDB 页面 | 原文线索

V2-015313 | MATHDB-371085

Cannon's conjecture on hyperbolic groups with two-sphere boundary

摘录 / EXCERPT: Let G be a Gromov hyperbolic group whose boundary at infinity is homeomorphic to the two-sphere $\mathbb{S}^2$. A group is virtually Kleinian if, up to a finite-index subgroup, i...

conjecture | open | MathDB 页面 | 原文线索

V2-015314 | MATHDB-371094

The $\mathcal{O}$-slice connectivity conjecture for the $RO(G)$-filtered Adams object

摘录 / EXCERPT: Let G be a finite group and let $\mathrm{AN}^{RO}_G(\mathbb S)$ be the $RO(G)$-filtered Adams--Novikov object. Applying the totalization functor gives the filtered G-spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-015315 | MATHDB-371118

Vanishing conjecture for L^2-torsion of subexponentially growing automorphisms

摘录 / EXCERPT: Let G be a $\det$-finite group, meaning that the relevant Fuglede-Kadison determinants are finite, and let $\Phi\colon G\to G$ be an automorphism. Assume that G has subexponent...

conjecture | open | MathDB 页面 | 原文线索

V2-015316 | MATHDB-371130

The complement symmetry conjecture for Varchenko's path sums

摘录 / EXCERPT: Let λ be a partition in an $(n-k)\times k$ rectangle, let λ^c be its complement, and let α be the automorphism of $\mathbb{C}(\hbar,a_1,\ldots,a_n)$ defined b...

conjecture | refuted | MathDB 页面 | 原文线索

V2-015317 | MATHDB-371134

Thomas's multirank conjecture for linearizations

摘录 / EXCERPT: Let $\mathscr{P}$ be a finite poset, and let B_1 and B_2 be $\mathscr{P}$-persistence sets. Write L_{B_1} and L_{B_2} for their linearizations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015318 | MATHDB-371173

Solubility conjecture for finite groups with two composite conjugacy class sizes

摘录 / EXCERPT: Solubility conjecture. Then G is soluble.

conjecture | open | MathDB 页面 | 原文线索

V2-015319 | MATHDB-371174

The generic bicyclic-unit free-group conjecture

摘录 / EXCERPT: Let G be a finite group that is not Dedekind, let R be the coefficient ring, and let $\operatorname{Bic}_R(G)$ be the group generated by all bicyclic units of RG. Fix a bicyc...

conjecture | open | MathDB 页面 | 原文线索

V2-015320 | MATHDB-371175

The shifted bicyclic-unit free-product conjecture

摘录 / EXCERPT: Let G be a finite group, let $H\leq G$ be a finite subgroup such that $\operatorname{Cir}_{\mathbb{Q}}^G(H)$ is non-empty, and let $g\in G$ satisfy $H\cap H^g=1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015321 | MATHDB-371195

The cyclic-character formula for charge connected components

摘录 / EXCERPT: Let $R=(R_1,\ldots,R_k)$ be a sequence of rectangles. Suppose the distinct rectangles R_j occur with multiplicities m_j, and set ... For $i\in[d_R]$, let $\ell_{d_R}^{(i)}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-015322 | MATHDB-371220

Realizability conjecture for compatible pairs of transfer systems

摘录 / EXCERPT: Let G be a finite group. A pair of G-transfer systems is called compatible when it satisfies the compatibility conditions for a pairing of N_{∞}-operads, and it is reali...

conjecture | open | MathDB 页面 | 原文线索

V2-015323 | MATHDB-371227

Tilting conjecture for silting objects over d-representation infinite algebras

摘录 / EXCERPT: Silting mutation conjecture. Then T is tilting and $\operatorname{End}_A(T)$ is a d-representation infinite algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-015324 | MATHDB-371270

Universal enveloping algebra expression for the Burnside transition matrices

摘录 / EXCERPT: Let e,f,h be the basis elements of $\mathfrak{sl}_2$, and define the 2×2 matrices ... For nonnegative integers x,y,z, let $f(x,y,z)$ be the sum over all orders of taking...

conjecture | open | MathDB 页面 | 原文线索

V2-015325 | MATHDB-371271

Multiplicity binomiality conjecture for Burnside-chain eigenvalues

摘录 / EXCERPT: Fix k, and let λ be any nonzero eigenvalue of the Burnside chain on (C_k^n,S_n) for any n. Multiplicity binomiality conjecture. The eigenvalue λ occurs with m...

conjecture | open | MathDB 页面 | 原文线索

V2-015326 | MATHDB-371278

The local twisted Gan-Gross-Prasad conjecture for unitary groups

摘录 / EXCERPT: Let F be a p-adic field, let E and K be quadratic field extensions of F, and let V be an n-dimensional skew-hermitian space relative to E/F. Set $V_K=V\otimes_F K$,...

conjecture | open | MathDB 页面 | 原文线索

V2-015327 | MATHDB-371283

Moretó-Rizo character correspondence conjecture for subnormalisers

摘录 / EXCERPT: Moretó-Rizo's character correspondence conjecture. For any p-element $x\in G$, there exists a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-015328 | MATHDB-371310

Mulmuley's polyhedral conjecture for stretched plethysm coefficients

摘录 / EXCERPT: Mulmuley's conjecture. The generating function $\tilde{\mathsf{A}}_{\mu[\nu]}^\lambda(t)$ is the Ehrhart series of a polytope.

conjecture | open | MathDB 页面 | 原文线索

V2-015329 | MATHDB-371336

The distributivity conjecture for lattices with pure minimal injective coresolutions

摘录 / EXCERPT: Let L be a finite lattice, and let $A=KL$ be its incidence algebra. Assume that A has a pure minimal injective coresolution. Distributivity conjecture. Then L is distributive...

conjecture | open | MathDB 页面 | 原文线索

V2-015330 | MATHDB-371363

Lusztig's character formula conjecture

摘录 / EXCERPT: Lusztig's character formula conjecture. If $p \geq h$ and $w \in W_{\mathrm{aff}}^\varnothing$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-015331 | MATHDB-371364

Finkelberg-Mirković conjecture for general blocks

摘录 / EXCERPT: Finkelberg-Mirković conjecture. There exists an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-015332 | MATHDB-371368

Lusztig-Vogan natural bijection conjecture

摘录 / EXCERPT: Lusztig-Vogan natural bijection conjecture. There is a natural bijection

conjecture | open | MathDB 页面 | 原文线索

V2-015333 | MATHDB-371422

Tărnăuceanu's sum-of-element-orders conjecture for finite abelian groups

摘录 / EXCERPT: Let G and H be finite abelian groups of the same order. Define the sum-of-element-orders function by ... where $o(g)$ denotes the order of g. Tărnăuceanu's conjecture. The grou...

conjecture | open | MathDB 页面 | 原文线索

V2-015334 | MATHDB-371430

Naturality of support for tilting modules under Levi restriction

摘录 / EXCERPT: Naturality conjecture. The conditions in Proposition are satisfied for an arbitrary $\mathfrak{g}$, with the assumptions of Section; equivalently, support of tilting modules is nat...

conjecture | open | MathDB 页面 | 原文线索

V2-015335 | MATHDB-371452

Uniqueness conjecture for subgroups isomorphic to the alternating group in the real space Cremona group

摘录 / EXCERPT: Let $\mathrm{Cr}_3(\mathbb{R})$ denote the Cremona group of birational self-maps of real projective 3-space, and let $\mathfrak{A}_6$ denote the alternating group on six letters. U...

conjecture | open | MathDB 页面 | 原文线索

V2-015336 | MATHDB-371453

Commutativity conjecture for Hecke-algebra shuffle elements

摘录 / EXCERPT: Let ${\mathcal H}_N(q)$ be the Hecke algebra, and let ${\mathfrak s}_k$ denote the sum of all reduced words of length k in ${\mathcal H}_N(q)$. Let ${\mathfrak y}_N(\hat z)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-015337 | MATHDB-371493

Enomoto's semibrick finiteness conjecture for finite-dimensional algebras

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra. A semibrick in $\operatorname{mod} \Lambda$ is a set of bricks whose distinct members have no nonzero morphisms between them. Enomoto...

conjecture | open | MathDB 页面 | 原文线索

V2-015338 | MATHDB-371495

Multiplicity-one conjecture for distinguished representations of \(\operatorname{GL}_{2n}(F)\)

摘录 / EXCERPT: Let E/F be a quadratic extension. Let π be an irreducible representation of $\operatorname{GL}_{2n}(F)$, and let χ be a character of $E^{\times}$. Multiplicity-one conje...

conjecture | open | MathDB 页面 | 原文线索

V2-015339 | MATHDB-371500

Two regular unipotent generation conjecture for quasisimple groups of Lie type

摘录 / EXCERPT: Two regular unipotent generation conjecture. Then G is generated by two regular unipotent elements.

conjecture | open | MathDB 页面 | 原文线索

V2-015340 | MATHDB-371518

Shortest-path bisaturation conjecture for transfer system lattices

摘录 / EXCERPT: Shortest-path bisaturation conjecture. A shortest path between the trivial G-transfer system and the complete G-transfer system on a Hasse diagram will pass through as many bis...

conjecture | open | MathDB 页面 | 原文线索

V2-015341 | MATHDB-371519

Two-component LSP compatibility conjecture for transfer systems

摘录 / EXCERPT: Two-component LSP compatibility conjecture. Then T is not LSP and is compatible with the transfer system generated by adding the edge $e\to G$ to T.

conjecture | open | MathDB 页面 | 原文线索

V2-015342 | MATHDB-371536

The Graded Stanley-Stembridge Conjecture

摘录 / EXCERPT: Let H be a type A Hessenberg space. Let $\mathfrak{ch}(\mathrm{L}_{H})$ be the graded character of the associated left representation, and let Frob denote the Fr...

conjecture | open | MathDB 页面 | 原文线索

V2-015343 | MATHDB-371553

The wavefront set upper bound conjecture for Brylinski-Deligne covers

摘录 / EXCERPT: Let G be a reductive group over a local field F, and let $\overline{G}^{(n)}$ be a Brylinski-Deligne cover of G. Write $\operatorname{Irr}_{\rm gen}(\overline{G}^{(n)})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-015344 | MATHDB-371602

Mathieu's conjecture for finite-type functions on compact connected Lie groups

摘录 / EXCERPT: Let K be a compact connected Lie group, and let f and h be complex-valued finite-type functions on K. Let dk denote Haar measure on K. Mathieu's conjecture. If ... for al...

conjecture | open | MathDB 页面 | 原文线索

V2-015345 | MATHDB-371605

The representation-theoretic Mathieu conjecture for symmetric powers

摘录 / EXCERPT: Let $A=\bigoplus_{n\in {\mathbb{N}}}S^{dn}{\mathbb{C}}^N\otimes S^k({\mathbb{C}}^N)^*$ be a graded $SL(N,{\mathbb{C}})$-algebra. Let P^+ denote the set of dominant highest weight...

conjecture | open | MathDB 页面 | 原文线索

V2-015346 | MATHDB-371657

The finite-dimensional property (T) conjecture for mapping class groups

摘录 / EXCERPT: Let S_g^n be a compact oriented surface of genus g with n boundary components, and let $\mathrm{Mod}(S_g^n)$ denote its mapping class group. For a finite-dimensional unitary...

conjecture | open | MathDB 页面 | 原文线索

V2-015347 | MATHDB-371658

The finite Dehn twist representation cohomology conjecture

摘录 / EXCERPT: Let S_g^n be a compact oriented surface of genus g with n boundary components, and let $\mathrm{Mod}(S_g^n)$ denote its mapping class group. A representation is a finite Dehn...

conjecture | open | MathDB 页面 | 原文线索

V2-015348 | MATHDB-371669

The S-arithmetic polynomial filling conjecture

摘录 / EXCERPT: Let G be an S-arithmetic group in arbitrary characteristic, and let the filling functions and rank be understood as in the cited Lie-group theorem. The S-arithmetic polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-015349 | MATHDB-371677

The charfiniteness conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be an irreducible higher-rank lattice. The charfiniteness conjecture. The group Γ is charfinite, in the sense of the character-rigidity terminology used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-015350 | MATHDB-371705

Kantor's flag-transitive generalized quadrangle conjecture

摘录 / EXCERPT: Let $\mathcal{S}$ be a finite flag-transitive generalized quadrangle. A generalized quadrangle is classical when it is one of the standard generalized quadrangles arising from a cl...

conjecture | open | MathDB 页面 | 原文线索

V2-015351 | MATHDB-371725

The idempotent measure conjecture for automorphism-group actions

摘录 / EXCERPT: Let $\bar m$ enumerate a tuple from a small model $M\preceq\mathfrak{C}$, and let $\mu\in\mathfrak{M}^{\operatorname{def}}_{\bar m}(\mathfrak{C},M)$ be fim over M. By the cited r...

conjecture | open | MathDB 页面 | 原文线索

V2-015352 | MATHDB-371732

Twist-invariance conjecture for multiplicity-free triples

摘录 / EXCERPT: Let $(\mathbf {U},\mathbf{B},\gamma)$ be a multiplicity-free triple, and let $V(\gamma)^\sigma$ denote the twisted module associated with $V(\gamma)$. Twist-invariance conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-015353 | MATHDB-371760

Rationality conjecture for regularized dimensions of infinite Hecke algebra representations

摘录 / EXCERPT: Rationality conjecture. For every partition λ, there exist positive integers a_j and b_i such that

conjecture | open | MathDB 页面 | 原文线索

V2-015354 | MATHDB-371814

Abelianity conjecture for transitive graphs of non-negative Ollivier--Ricci curvature

摘录 / EXCERPT: Let G be a transitive graph with non-negative Ollivier--Ricci curvature. Recall that two metric spaces are quasi-isometric if they are equivalent up to multiplicative and additiv...

conjecture | open | MathDB 页面 | 原文线索

V2-015355 | MATHDB-371878

Demonet's conjecture on tau-tilting finiteness and E-finiteness

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra. It is τ-tilting finite if it has finitely many isomorphism classes of basic 2-term silting complexes in...

conjecture | open | MathDB 页面 | 原文线索

V2-015356 | MATHDB-371879

Brick Brauer-Thrall II' conjecture

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra. A module $M\in\operatorname{mod}\Lambda$ is a brick if $\operatorname{End}_{\Lambda}(M)$ is a division algebra. The algebra Λ...

conjecture | open | MathDB 页面 | 原文线索

V2-015357 | MATHDB-371914

Difference formula for tensor-power multiplicities of covariant representations of $\mathfrak{sl}(m|n)$

摘录 / EXCERPT: Let $\mathfrak{sl}(m|n)$ be the type A Lie superalgebra, let $V_{(2s)}$ be its covariant representation, and write ... where λ ranges over (m,n)-hook Young diagrams. Deco...

conjecture | open | MathDB 页面 | 原文线索

V2-015358 | MATHDB-371932

Classification conjecture for locally complexified-gentle algebras

摘录 / EXCERPT: Let A be a connected $\mathbb{R}$-algebra. An algebra is locally complexified-gentle of uniform type if it is Morita equivalent to an algebra satisfying condition (1) or (2) of T...

conjecture | open | MathDB 页面 | 原文线索

V2-015359 | MATHDB-371935

Stratum-wise constancy conjecture for multiplicities in complex-rank parabolic category O

摘录 / EXCERPT: Let $\scr{X}_n$ be the parameter space with strata W_I, and let $W_I^{\natural}\subset W_I$ consist of points $(\bt,\bs)$ whose coordinates t_i are not integers. A point of...

conjecture | open | MathDB 页面 | 原文线索

V2-015360 | MATHDB-371956

Naor's finite analogue of the Cornulier-Tessera-Valette conjecture

摘录 / EXCERPT: Let $\{G_n\}_n$ be a sequence of finite groups with $\sup_n \operatorname{rank}(G_n)<\infty$. For a generating set S of size r, let d_S be the associated word metric, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015361 | MATHDB-371959

Infinitesimal embedding conjecture for a random profinite action of a free group

摘录 / EXCERPT: Let $(t_n)_{n\geq 0}$ be a rapidly growing sequence of natural numbers, let F_d be the free group, and construct a nested sequence of finite-index subgroups (Λ_n) by taki...

conjecture | open | MathDB 页面 | 原文线索

V2-015362 | MATHDB-371960

Compact-action infinitesimal containment conjecture

摘录 / EXCERPT: Let G be a compact group with Haar probability measure μ, let $\rho\colon\Gamma\to G$ be a homomorphism, and let Γ act on (G,μ) by $\gamma g=\rho(\gamma)g$. For ea...

conjecture | open | MathDB 页面 | 原文线索

V2-015363 | MATHDB-371961

Non-amenable image criterion for infinitesimal containment of compact actions

摘录 / EXCERPT: Let G be a compact group with Haar probability measure μ, let $\rho\colon\Gamma\to G$ be a homomorphism, and let Γ act on (G,μ) by left translation through ρ....

conjecture | open | MathDB 页面 | 原文线索

V2-015364 | MATHDB-371973

Finite-sum conjecture for morphisms between sheet modules

摘录 / EXCERPT: A sheet module is the image of a decorous submodule in a decorous quotient of a representable projective, as defined in the paper. The elementary and multi-elementary morphisms are...

conjecture | open | MathDB 页面 | 原文线索

V2-015365 | MATHDB-371974

Finite-sum conjecture for morphisms between sheets

摘录 / EXCERPT: Let S and S' be sheets, regarded as objects of $\operatorname{Rep}(\Lambda_{\mathbb{I}})$. Elementary morphisms are those constructed in Proposition, and multi-elementary morph...

conjecture | open | MathDB 页面 | 原文线索

V2-015366 | MATHDB-371977

Irreducible minimal-dimension locus conjecture for principal quiver Grassmannians

摘录 / EXCERPT: Irreducible-locus conjecture. There exists a representation M^1 of dimension $\dim P$ such that, for $N\in{\rm Rep}_{\bf d}$, the quiver Grassmannian $\Gr_{\dim P}(N)$ is irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-015367 | MATHDB-371979

Homomorphism-dimension criterion for the minimal-dimension locus

摘录 / EXCERPT: Homomorphism-dimension conjecture. Under either condition, $\dim\Gr_{\dim P}(M)=\langle\dim P,\dim I\rangle$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-015368 | MATHDB-371980

Plücker-dimension characterization of the minimal-dimension locus

摘录 / EXCERPT: Plücker-dimension conjecture. For every $M\in{\rm Rep}_{\bf d}$,

conjecture | open | MathDB 页面 | 原文线索

V2-015369 | MATHDB-371987

Kaimanovich-Le Prince singularity conjecture for hitting measures

摘录 / EXCERPT: Let $\Gamma<\mathrm{SL}_N(\mathbb{R})$ be a discrete subgroup, and let the random walk on Γ have finite support. Its hitting measure is a measure on the boundary at infinity...

conjecture | open | MathDB 页面 | 原文线索

V2-015370 | MATHDB-372041

Tringali-Yan conjecture on automorphisms of reduced power monoids

摘录 / EXCERPT: Tringali-Yan conjecture. If S is a numerical monoid properly contained in $\mathbb{N}$, then the automorphism group of $\mathcal{P}_{0}(S)$ is the trivial group:

conjecture | open | MathDB 页面 | 原文线索

V2-015371 | MATHDB-372062

The finite-group generalization of the random group-equation threshold theorem

摘录 / EXCERPT: Let G be a finite group of order at least two, and let $H_n(\pi,k,m)$ be the random uniquely extendable constraint system arising from the group-equation model described in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-015372 | MATHDB-372065

Existence of supramenable groups without the fixed-point property for cones

摘录 / EXCERPT: A supramenable group is a group G such that every G-set X supports an invariant finitely additive measure normalized on any given non-empty subset $A\subseteq X$. The fixed-p...

conjecture | open | MathDB 页面 | 原文线索

V2-015373 | MATHDB-372092

Polynomial-dimension abelian-structure conjecture for linear approximate groups

摘录 / EXCERPT: Let $\mathbb F$ be a field, let $d\geq1$, and let $A\subset\operatorname{GL}_d(\mathbb F)$ be a K-approximate group for some $K\geq1$. Polynomial-dimension abelian-structure conj...

conjecture | open | MathDB 页面 | 原文线索

V2-015374 | MATHDB-372093

The eigenspace-control conjecture for linear approximate groups

摘录 / EXCERPT: Let $\mathbb F$ be an algebraically closed field, let $d\geq1$, and let $A\subset\operatorname{GL}_d(\mathbb F)$ be a K-approximate group. Suppose every $a\in A^2$ has an eigensp...

conjecture | open | MathDB 页面 | 原文线索

V2-015375 | MATHDB-372094

Subgroups forced by quotients of symmetric groups

摘录 / EXCERPT: For finite groups H_1,H_2, write $H_1\implies H_2$ when every finite group having a quotient isomorphic to H_1 also has a subgroup isomorphic to H_2. The symmetric-group quot...

conjecture | open | MathDB 页面 | 原文线索

V2-015376 | MATHDB-372113

Naive dominance conjecture for polynomial modules of quantum affine superalgebras

摘录 / EXCERPT: Let Ψ be a monomial in $Y_{i,a}$ for $1\leq i\leq M$ and $a\in\mathbb{C}^{\times}$, so that $L(\Psi)$ is a polynomial module. Assume that every ℓ-weight of $L(\Psi)$ othe...

conjecture | open | MathDB 页面 | 原文线索

V2-015377 | MATHDB-372157

The center conjecture for non-semisimple BMW algebras

摘录 / EXCERPT: Center conjecture. The center of $B_n(q,q^{2a})$ is

conjecture | open | MathDB 页面 | 原文线索

V2-015378 | MATHDB-372161

Asymptotic highest-weight and orthogonality conjecture for Cartan subproduct systems

摘录 / EXCERPT: Let $q>0$ and let $\mathbb P_+$; write V_μ for the irreducible $U_q\mathfrak g$-module of highest weight μ, with highest-weight and lowest-weight projections denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-015379 | MATHDB-372193

Schurian-finiteness conjecture for integral type B Hecke algebra blocks

摘录 / EXCERPT: Let e be the quantum characteristic, let $\hhh$ be an integral type B Hecke algebra, and let B be a block of $\hhh$ with defect at least 2. Schurian-finiteness conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-015380 | MATHDB-372194

Schurian-finiteness conjecture for blocks of level 2 cyclotomic q-Schur algebras

摘录 / EXCERPT: Let e be the quantum characteristic, let $\mathcal{S}_n$ be the level 2 cyclotomic q-Schur algebra, and let B be a block of $\mathcal{S}_n$. Schurian-finiteness conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-015381 | MATHDB-372216

The Surface Group Conjecture

摘录 / EXCERPT: Surface Group Conjecture. Must G be isomorphic to either a free group or a surface group?

open_problem | open | MathDB 页面 | 原文线索

V2-015382 | MATHDB-372220

The defining-relations conjecture for truncated shifted iYangians

摘录 / EXCERPT: Let τ be the diagram involution, let $\bar{\mathbf{v}}_j$ denote the associated class of the dimension vector at j, and let $\bm z$ be the central parameters. For a dominant...

conjecture | open | MathDB 页面 | 原文线索

V2-015383 | MATHDB-372233

The rational non-freeness conjecture for two parabolic generators

摘录 / EXCERPT: Let $G_{\alpha}=\left\langle \begin{bmatrix}1&1\\0&1\end{bmatrix},\begin{bmatrix}1&0\\alpha&1\end{bmatrix}\right\rangle<\operatorname{SL}_2(\mathbb{C})$, and call α free whe...

conjecture | open | MathDB 页面 | 原文线索

V2-015384 | MATHDB-372234

Nyberg Brodda's congruence subgroup conjecture for rational parabolic parameters

摘录 / EXCERPT: For $\alpha\in\mathbb{C}$, let ... Let $\frac{q}{p}\in\mathbb{Q}\cap(-4,4)$ with $\gcd(p,q)=1$. Write π for the relevant reduction homomorphism. Nyberg Brodda's congruence subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-015385 | MATHDB-372258

Blockwise spectrum problem for integral group rings

摘录 / EXCERPT: Blockwise Spectrum Problem. For every primitive central idempotent $e\in\operatorname{PCI}({\mathbb Q}G)$,

open_problem | open | MathDB 页面 | 原文线索

V2-015386 | MATHDB-372259

Hertweck's blockwise isomorphism conjecture

摘录 / EXCERPT: Hertweck's blockwise isomorphism conjecture. For every $e\in\operatorname{PCI}({\mathbb Q}G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-015387 | MATHDB-372293

The universal-enveloping-algebra expression conjecture for the hypercube Markov chain

摘录 / EXCERPT: The universal-enveloping-algebra expression conjecture. For every n,

conjecture | open | MathDB 页面 | 原文线索

V2-015388 | MATHDB-372294

The binomial multiplicity conjecture for the Burnside chain

摘录 / EXCERPT: Fix k, and consider the Burnside chain on (C_k^n,S_n) for varying n. Let λ be any nonzero eigenvalue of this chain. The binomial multiplicity conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-015389 | MATHDB-372332

Universal Drinfeld cohomology conjecture for generalized Steinberg representations

摘录 / EXCERPT: Universal Drinfeld cohomology conjecture. For every such splitting s,

conjecture | open | MathDB 页面 | 原文线索

V2-015390 | MATHDB-372350

Chebyshev formulae for Kadar-Yu Gram determinant polynomials

摘录 / EXCERPT: Chebyshev formula conjecture. Formulae for $C^{(l'+2)}(\alpha)$, $C^{(1^{l+2})}(\alpha)$, $P^{(l'+2)}_{l'+4+k}(\alpha)$, and $P^{(1^{l+2})}_{l+4+k}(\alpha)$ hold for all $l\geq -1$...

conjecture | open | MathDB 页面 | 原文线索

V2-015391 | MATHDB-372351

Hook-partition formulae for Kadar-Yu Gram determinant polynomials

摘录 / EXCERPT: Hook-partition formula conjecture. Formulae for $C^{(l'+1,1)}(\alpha)$, $C^{(l+1,1)^T}(\alpha)$, $P^{(l”+1,1)}_{l”+4+k}(\alpha)$, and $P^{(l+1,1)^T}_{l+4+k}(\alpha)$ hold for all...

conjecture | open | MathDB 页面 | 原文线索

V2-015392 | MATHDB-372352

Root-distribution conjecture for Kadar-Yu Chebyshev polynomials

摘录 / EXCERPT: Root-distribution conjecture. For each $k\geq -1$, the polynomial $P^{(\lambda)}_{l+4+k}$ is square free. Furthermore, for sufficiently large k, it has a fixed number of roots ou...

conjecture | open | MathDB 页面 | 原文线索

V2-015393 | MATHDB-372353

Arm-property conjecture for Kadar-Yu Gram determinants

摘录 / EXCERPT: Arm-property conjecture. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-015394 | MATHDB-372354

Soergel's conjecture on the character of indecomposable Soergel bimodules

摘录 / EXCERPT: Let (W,S) be the Coxeter system underlying the Soergel bimodule category, let B_x denote the indecomposable Soergel bimodule indexed by $x\in W$, and let b_x be the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-015395 | MATHDB-372357

Prasad's conjecture on non-vanishing of twisted Jacquet modules

摘录 / EXCERPT: Let F be the underlying local field, let $G_m=GL_m(F)$, and let N and ψ be the subgroup and character defining the twisted Jacquet module. Let π be an irreducible smoo...

conjecture | open | MathDB 页面 | 原文线索

V2-015396 | MATHDB-372366

Range-signature conjecture for same-dimension irreducible representations of symmetric groups

摘录 / EXCERPT: Let G be a symmetric group, and let ρ and τ be irreducible representations of G of the same dimension. Their character range signatures record the multiplicities of t...

conjecture | open | MathDB 页面 | 原文线索

V2-015397 | MATHDB-372367

Strong table-equivalence conjecture for irreducible characters

摘录 / EXCERPT: Let $Gamma$ be a finite group, and let $\textnormal{irr}(Gamma)$ and $\textnormal{cl}(Gamma)$ denote its irreducible characters and conjugacy classes, respectively. Two irreducible c...

conjecture | open | MathDB 页面 | 原文线索

V2-015398 | MATHDB-372368

Similarity conjecture for faithful irreducible representations with identical disjunctive trace identities

摘录 / EXCERPT: Let G and H be finite groups, and let ρ and $sigma$ be faithful irreducible representations. A disjunctive trace identity is an identity obtained from a disjunction of gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015399 | MATHDB-372369

Unitary-similarity conjecture for Lie-group representations with identical Lie-algebra identities

摘录 / EXCERPT: Let $\rho_i:G_i\to\mathbf{U}(H)$ be faithful unirreps of connected simply connected Lie groups in a separable Hilbert space, with corresponding Lie algebras $\mathfrak{g}_i$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015400 | MATHDB-372384

Injectivity conjecture for minimal multisegment derivatives

摘录 / EXCERPT: Generalized injectivity conjecture. Let $\pi\in\mathrm{Irr}_{\rho}$ and let $\mathfrak n\in\mathrm{Mult}_{\rho}$ be minimal to π. Set

conjecture | open | MathDB 页面 | 原文线索

V2-015401 | MATHDB-372385

Uniqueness conjecture for the removal embedding

摘录 / EXCERPT: Unique embedding conjecture. The embedding

conjecture | open | MathDB 页面 | 原文线索

V2-015402 | MATHDB-372386

Factorization conjecture for derivative embeddings

摘录 / EXCERPT: Factorization conjecture. The composite $\iota_2\circ\iota_1$ factors through ι_3.

conjecture | open | MathDB 页面 | 原文线索

V2-015403 | MATHDB-372387

Hernandez's conjecture on specialization of (q,t)-characters

摘录 / EXCERPT: Let $\mathfrak g$ be a general simple Lie algebra, let $\mathcal M_+$ be the set of dominant monomials, and let $\chi_{q,t}(L(m))$ and $\chi_q(L(m))$ denote the (q,t)-character a...

conjecture | open | MathDB 页面 | 原文线索

V2-015404 | MATHDB-372408

The Young-diagram decomposition conjecture for tensor powers of the orthogonal adjoint representation

摘录 / EXCERPT: Let $\mathfrak{so}(N)$ be the orthogonal Lie algebra, and let ad denote its adjoint representation. For a Young diagram λ, write $\lambda\vdash 2m$ when it...

conjecture | open | MathDB 页面 | 原文线索

V2-015405 | MATHDB-372413

The normalized period conjecture in categorical local Langlands

摘录 / EXCERPT: Let G) be a connected quasi-split reductive group over F with a fixed Whittaker datum. Suppose there is a categorical equivalence ... For a dual pair...

conjecture | open | MathDB 页面 | 原文线索

V2-015406 | MATHDB-372414

The functional-equation conjecture for normalized periods

摘录 / EXCERPT: Let G be a reductive group and let $V\in{\operatorname{Rep}}(G)$, with $V^\vee$ its dual representation. Let ${\mathcal{P}}_V^{\operatorname{norm}}$ denote the normalized period...

conjecture | open | MathDB 页面 | 原文线索

V2-015407 | MATHDB-372431

The orientifold LLT polynomial formula for graded decomposition numbers

摘录 / EXCERPT: Let Λ_n index the cell modules $\Delta(\lambda)$ and simple modules $L(\mu)$ of the orientifold quiver Temperley--Lieb algebra, and let $\langle k\rangle$ denote grading sh...

conjecture | solved | MathDB 页面 | 原文线索

V2-015408 | MATHDB-372449

The free-subgroup conjecture for the Manturov group G^2_4

摘录 / EXCERPT: Free-subgroup conjecture. The group G^2_4 has a free subgroup of rank 2.

conjecture | open | MathDB 页面 | 原文线索

V2-015409 | MATHDB-372481

Birational affine Weyl group action for the 3×3 linear q-difference system

摘录 / EXCERPT: Birational affine Weyl group action conjecture. The quotient set of such matrices admits a birational action of the affine Weyl group $W(E_8^{(1)})$, with

conjecture | open | MathDB 页面 | 原文线索

V2-015410 | MATHDB-372484

Shalom's conjecture on uniformly Lipschitz affine actions of hyperbolic groups

摘录 / EXCERPT: Let G be a Gromov hyperbolic group, and consider affine actions of G on a Hilbert space whose linear parts are uniformly bounded representations. An affine action is proper whe...

conjecture | open | MathDB 页面 | 原文线索

V2-015411 | MATHDB-372489

Yun's uniqueness conjecture for minimal reduction types

摘录 / EXCERPT: Let G be the reductive group underlying the affine Grassmannian, let $L^{\heartsuit}{\mathfrak g}$ be the space of elements under consideration, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-015412 | MATHDB-372516

Dynamical McKay correspondence conjecture via gradient-flow limits

摘录 / EXCERPT: Let Γ be a finite subgroup of $SU(2)$, and let X_ζ be the space in the paper's theorem. Let $S_1,\ldots,S_r$ be the components of the S^1-fixed locus of X_ζ ge...

conjecture | open | MathDB 页面 | 原文线索

V2-015413 | MATHDB-372519

James's conjecture for weight-5 blocks of Hecke algebras

摘录 / EXCERPT: James's conjecture. James's conjecture holds for all weight 5 blocks of $\hhh$.

conjecture | open | MathDB 页面 | 原文线索

V2-015414 | MATHDB-372556

Linckelmann's nonvanishing conjecture for Hochschild cohomology of blocks

摘录 / EXCERPT: Let G be a finite group, let k be a field whose characteristic divides the order of G, and let B be a block of the group algebra kG with defect group P. Linckelmann's c...

conjecture | open | MathDB 页面 | 原文线索

V2-015415 | MATHDB-372591

Wei's abelianization conjecture for invariant algebraic D-modules

摘录 / EXCERPT: Let G be a connected algebraic group, let n be a rank, and let $q\colon G\twoheadrightarrow G/[G,G]$ be the quotient map. Write $\sim$ for the relation of D-module isomorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-015416 | MATHDB-372592

Equality conjecture for tensor ideals in tilting modules

摘录 / EXCERPT: Tensor-ideal equality conjecture. For any G and p, all of the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-015417 | MATHDB-372595

Positivity conjecture for subgroup-counting D'Arcais differences

摘录 / EXCERPT: Subgroup-counting positivity conjecture. If $\ell=3$ or $\ell=4$, then for every $n\geq a_{\ell}$ and every real $x\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-015418 | MATHDB-372625

The context-free quasigeodesic characterization of hyperbolic groups

摘录 / EXCERPT: Let G be a finitely generated group. For rational $\lambda\geq 1$, real $\epsilon\geq 0$, and a finite generating set S, consider the language of (λ,ϵ)-quasigeode...

conjecture | open | MathDB 页面 | 原文线索

V2-015419 | MATHDB-372626

The universal quasigeodesic obstruction conjecture

摘录 / EXCERPT: Let G be a non-hyperbolic finitely generated group with generating set S, and let $\Gamma(G,S)$ be its Cayley graph. A word is a (λ,ϵ)-quasigeodesic when it sati...

conjecture | open | MathDB 页面 | 原文线索

V2-015420 | MATHDB-372635

Lehnert's co-context-free group conjecture

摘录 / EXCERPT: A finitely generated group G is co-context-free if its co-word problem is context-free. Thompson's group V is the group of homeomorphisms represented by finite binary tree-pair...

conjecture | open | MathDB 页面 | 原文线索

V2-015421 | MATHDB-372651

Contact-tree conjecture for geometrically finite Bers boundary groups

摘录 / EXCERPT: Let G be a geometrically finite Bers boundary group, and let $\Lambda(G)$ be its limit set. Form the contact tree whose vertices are the components of the domain of discontinuity...

conjecture | open | MathDB 页面 | 原文线索

V2-015422 | MATHDB-372667

Density conjecture for smooth curvature measures

摘录 / EXCERPT: Let n be a positive integer, let $\operatorname{Curv}_k(\mathbb{R}^n)$ be the degree-k space of curvature measures with its topology induced by compact convergence, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015423 | MATHDB-372668

Irreducibility conjecture for zero-total curvature measures

摘录 / EXCERPT: Let n be a positive integer, let $\operatorname{Curv}_k^\pm(\mathbb{R}^n)$ be the even or odd degree-k curvature measures, and define ... This is an invariant closed subspace und...

conjecture | open | MathDB 页面 | 原文线索

V2-015424 | MATHDB-372680

Bader-Kropholler-Vankov's quasi-isometry invariance conjecture for homological filling functions

摘录 / EXCERPT: Let G and H be quasi-isometric groups of type $\FP_n(R)$, where R is an arbitrary coefficient ring. Write $\operatorname{FV}_{R,G}^{n}$ and $\operatorname{FV}_{R,H}^{n}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-015425 | MATHDB-372697

The one-step geometric wavefront set speculation for positive-real representations

摘录 / EXCERPT: One-step geometric wavefront set speculation. Either

conjecture | open | MathDB 页面 | 原文线索

V2-015426 | MATHDB-372698

The conjecture that saturated finite groups are abelian

摘录 / EXCERPT: Abelianness conjecture for saturated groups. If G is a saturated finite group, then G is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-015427 | MATHDB-372780

Hamiltonian reduction conjecture for rank-condition components

摘录 / EXCERPT: Let $\mathcal D^n$ be the rank-condition solution space, with irreducible components related by the birational Weyl group action, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015428 | MATHDB-372781

Vanishing of morphisms from good to exceptional Verma modules

摘录 / EXCERPT: Let a good Verma module mean a Verma module with a weight space decomposition, and let an exceptional Verma module be a module of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-015429 | MATHDB-372782

Finite-dimensional quotient conjecture for exceptional Verma modules

摘录 / EXCERPT: Finite-dimensional quotient conjecture. The module $\mathbb{M}(r+\kappa,r,[\iota;n],\zeta)$ has a finite-dimensional irreducible quotient L if and only if this condition holds. I...

conjecture | open | MathDB 页面 | 原文线索

V2-015430 | MATHDB-372796

The bounded perfect derived subgroup conjecture for pointwise stabilizers

摘录 / EXCERPT: Bounded perfect derived subgroup conjecture. There exists an absolute constant c such that $H^{(c)}$ is perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-015431 | MATHDB-372798

The Indecomposability Conjecture for projective functors in category O

摘录 / EXCERPT: Let $\theta_x^{(n)}$ be the projective functor indexed by $x\in\rm S_n$, and let $L_z^{(n)}$ be the simple highest weight module indexed by $z\in\rm S_n$ in the principal block of...

conjecture | open | MathDB 页面 | 原文线索

V2-015432 | MATHDB-372809

Fixed-point conjecture for Grassmannian braid-group actions

摘录 / EXCERPT: Braid-group fixed-point conjecture. For every $i\in[d-1]$, if (k,n) is $(3,9)$ or $(4,8)$, every unstable fixed point of σ_i in $\mathbb{C}[\operatorname{Gr}(k,n)]$ is a c...

conjecture | open | MathDB 页面 | 原文线索

V2-015433 | MATHDB-372817

Almost all finite Cayley graphs are stable

摘录 / EXCERPT: Stability conjecture. For a group G of order r, the proportion of inverse-closed subsets S of G such that $\operatorname{Cay}(G,S)$ is stable approaches 1 as r tends to...

conjecture | open | MathDB 页面 | 原文线索

V2-015434 | MATHDB-372852

Barker-Gelvin uniform source algebra conjecture for blocks

摘录 / EXCERPT: Let G be a finite group, let p divide $|G|$, and let $\bar{b}$ be a block idempotent of kG with non-trivial defect group. A source algebra of $kG\bar{b}$ is the interior D-...

conjecture | open | MathDB 页面 | 原文线索

V2-015435 | MATHDB-372948

Uniform generation conjecture for minors of finite abelian p-groups

摘录 / EXCERPT: Uniform generation conjecture. There is a $k\in\mathbb{N}$ such that $\mathcal O_{\mathbf{A},\mathbf{R}}$ is uniformly generated by k-ary $(\mathbf{A},\mathbf{R})$-minors.

conjecture | open | MathDB 页面 | 原文线索

V2-015436 | MATHDB-372963

Wise's conjecture on towers and the second L^2-Betti number

摘录 / EXCERPT: Let Y be a 2-complex. Write $b^{(2)}_2(\widetilde Y;\mathcal N(\pi_1(Y)))$ for the second L^2-Betti number of its universal covering, with coefficients in the von Neumann alg...

conjecture | open | MathDB 页面 | 原文线索

V2-015437 | MATHDB-372964

The conjecture that every torsionfree group is a Linnell group

摘录 / EXCERPT: A Linnell group is a group for which, for every field F and crossed product $F\ast G$, a Linnell division ring $\iota\colon F\ast G\to\mathcal D_{F\ast G}$ exists and is unique u...

conjecture | open | MathDB 页面 | 原文线索

V2-015438 | MATHDB-372975

Fargues-Scholze's categorical local Langlands conjecture

摘录 / EXCERPT: Let F be a local field, let G be a quasisplit reductive group over F, and let B be a Borel subgroup with unipotent radical N. Choose a generic continuous character...

conjecture | open | MathDB 页面 | 原文线索

V2-015439 | MATHDB-372978

Conjugacy-class parity conjecture for finite 2-groups

摘录 / EXCERPT: Let π be a finite 2-group and let $N\triangleleft T\triangleleft W\triangleleft\pi$ be normal subgroups such that π/N is cyclic, $T/N\cong\mathbb Z/{2}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-015440 | MATHDB-372988

The lattice-preservation conjecture for convex co-compact complex hyperbolic representations

摘录 / EXCERPT: Let $\mathbb{H}^m_{\mathbb{C}}$ and $\mathbb{H}^M_{\mathbb{C}}$ be complex hyperbolic spaces, and let $\Gamma < \mathsf{Isom}(\mathbb{H}^m_{\mathbb{C}})$ be a uniform lattice. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015441 | MATHDB-373004

Fundamental non-symmetric shift operator conjecture

摘录 / EXCERPT: Let W be a Weyl group, let ρ_k denote the associated weighted half-sum, let l be the shift parameter, and let $S(\mathfrak{h})$ be the symmetric algebra of the Cartan spac...

conjecture | open | MathDB 页面 | 原文线索

V2-015442 | MATHDB-373061

Minimal-polynomial criterion for tree-pair representations of algebraic Bieri-Strebel groups

摘录 / EXCERPT: Minimal-polynomial conjecture. A well-defined tree pair for $\textbf{F}_{\beta}$ can be derived from $P(x)$ only if $P(x)$ is the minimal polynomial having root β among polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-015443 | MATHDB-373065

Naive classification conjecture for class-one representations of $SO(n)$

摘录 / EXCERPT: Let $SO(n)$ be the special orthogonal group, and let $SO(n-1)$ be the subgroup fixing the first coordinate. A class-one representation means an irreducible representation admitting...

conjecture | open | MathDB 页面 | 原文线索

V2-015444 | MATHDB-373094

The Ext-algebra conjecture for categories associated with distinguished nilpotents

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let e be a distinguished nilpotent element, and let $Q=Q(e)$ be the finite centralizer of an associated sl_2-triple. Let $\mathcal{S}_e$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-015445 | MATHDB-373110

Persistent containment conjecture for finitely generated residually finite groups

摘录 / EXCERPT: Let H be a finitely generated residually finite group. A group G persistently contains H if every finite-index subgroup $G'\subset G$ contains a subgroup isomorphic to H. P...

conjecture | open | MathDB 页面 | 原文线索

V2-015446 | MATHDB-373116

The finite-to-one surjective local Langlands correspondence

摘录 / EXCERPT: Let F be a p-adic field, let W_F' be its Weil-Deligne group, and let G be a connected reductive p-adic group. Let $\Pi(G/F)$ be the set of isomorphism classes of irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-015447 | MATHDB-373124

Ben-Zvi-Sakellaridis-Venkatesh conjecture on duals of BZSV quadruples

摘录 / EXCERPT: Ben-Zvi-Sakellaridis-Venkatesh duality conjecture. The dual of Δ is

conjecture | open | MathDB 页面 | 原文线索

V2-015448 | MATHDB-373227

The critical-extremizer characterization of optimal hypercontractive constants on cyclic groups

摘录 / EXCERPT: Critical-extremizer characterization.

conjecture | open | MathDB 页面 | 原文线索

V2-015449 | MATHDB-373232

Property R_∞ for finitely generated special linear groups over integral domains

摘录 / EXCERPT: Let $n\geq 4$ and let $I\subseteq\{1,\dots,n\}$ satisfy the (NG)-condition. For an infinite integral domain R, write $\mathbf{S}^I_n(R)$ for the corresponding group, and assume t...

conjecture | open | MathDB 页面 | 原文线索

V2-015450 | MATHDB-373246

Charney--Davis conjecture for spherical Deligne complexes of braid groups

摘录 / EXCERPT: Let G be a braid group, and let its spherical Deligne complex be the complex associated to the spherical standard parabolic subgroups of G, equipped with its natural piecewise...

conjecture | open | MathDB 页面 | 原文线索

V2-015451 | MATHDB-373264

Deaconescu's solubility conjecture for finite groups with many elements of the same order

摘录 / EXCERPT: Let k be a fixed positive integer and let G be a finite group. Suppose that at least half of the elements of G have order k. Deaconescu's conjecture. Then G is soluble. T...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-015452 | MATHDB-373267

Tringali and Wen's automorphism conjecture for the reduced finitary power monoid of the integers

摘录 / EXCERPT: Let the reduced finitary power monoid of the additive group $(\mathbb{Z},+)$ be the quotient of the finitary power monoid by translation equivalence, and let $X\mapsto -X$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-015453 | MATHDB-373279

Counting conjecture for surface subgroups in finite-volume hyperbolic manifolds

摘录 / EXCERPT: Counting conjecture. There exists a constant $C(M)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015454 | MATHDB-373395

Faithfulness conjecture for the polynomial representation at slope one-half

摘录 / EXCERPT: Faithfulness conjecture. For any $\alpha\in({\mathbb Z}^2)^+$, the polynomial representation $P^{\frac12}_\alpha$ of $A^{\frac12}_\alpha$ is faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-015455 | MATHDB-373419

Hiss-O'Brien conjecture on invariant transversals and commutators

摘录 / EXCERPT: Hiss-O'Brien conjecture. If H admits a G-invariant transversal, then

conjecture | open | MathDB 页面 | 原文线索

V2-015456 | MATHDB-373520

Sibling-pair and sibling-tuple conjecture for groups of orders p^4 and p^5

摘录 / EXCERPT: Sibling-pair and sibling-tuple conjecture. Among the 15 groups of order p^4, there is exactly one sibling-pair. Among the $2p+2\gcd(p-1,3)+\gcd(p-1,4)+61$ groups of order p^5...

conjecture | open | MathDB 页面 | 原文线索

V2-015457 | MATHDB-373573

Character-ratio upper-bound conjecture for irreducible representations of symmetric groups

摘录 / EXCERPT: Let $d\geq10$, and let $\lambda,\mu\vdash d$ satisfy $\lambda\neq(d),(1^d)$. Here $m_i(\mu)$ denotes the multiplicity of the part i in the partition μ, and χ_λ is...

conjecture | open | MathDB 页面 | 原文线索

V2-015458 | MATHDB-373589

The collapse conjecture for finite simple groups

摘录 / EXCERPT: Let G be a finite group. Say that G collapses if every finite-dimensional pointed Hopf algebra H with group of grouplikes $G(H)\simeq G$ is isomorphic to $\Bbbk G$; equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-015459 | MATHDB-373608

The enhanced Shahidi conjecture for generic members of local Arthur packets

摘录 / EXCERPT: Enhanced Shahidi Conjecture. For any quasi-split reductive group G, a local Arthur packet is tempered if and only if it has a generic member.

conjecture | open | MathDB 页面 | 原文线索

V2-015460 | MATHDB-373618

The inner amenability conjecture for strong 1-boundedness

摘录 / EXCERPT: A countable discrete group G is inner amenable if the conjugation action of G on $G\setminus\{1\}$ admits an invariant mean, and it is strongly 1-bounded if its group von Neuma...

conjecture | open | MathDB 页面 | 原文线索

V2-015461 | MATHDB-373659

The Lusztig induction and dual restriction conjecture

摘录 / EXCERPT: Let H be the reductive group and let $\widehat{P}\subseteq\widehat{H}$ be the parabolic data defining the Lusztig induction functor ... On the level of Grothendieck groups, the dua...

conjecture | open | MathDB 页面 | 原文线索

V2-015462 | MATHDB-373660

The Kazhdan-Lusztig conjecture for p-adic representation pairings

摘录 / EXCERPT: Let H be a p-adic reductive group, let $\lambda^{\mathbb{Q}_p}$ be an infinitesimal character, and let $\xi=(C,\mathcal{V})$ and ξ' be parameters indexing the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-015463 | MATHDB-373666

The Baby Remeslennikov Conjecture

摘录 / EXCERPT: Let G be the group and a, d, and P the element, rank, and set of primes occurring in conditions (a), (b), and (c) of the surrounding discussion. Baby Remeslennikov Conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-015464 | MATHDB-373667

The one-ended limit group profinite splitting conjecture

摘录 / EXCERPT: A limit group is a finitely generated subgroup of a group obtained from a free group by finitely many extensions of centralizers, and it is one-ended when it has one end. For a gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015465 | MATHDB-373681

Commensurability invariance of property (LR) for hyperbolic groups

摘录 / EXCERPT: Let G be a hyperbolic group, and suppose that G has property (LR). Let H be a finite index supergroup of G. The commensurability conjecture for hyperbolic groups. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-015466 | MATHDB-373682

Property (LR) for finite index supergroups of limit groups

摘录 / EXCERPT: Let G be a limit group, and let H be a finite index supergroup of G. The limit-group supergroup conjecture. The group H has property (LR). This is motivated by Wilton's the...

conjecture | open | MathDB 页面 | 原文线索

V2-015467 | MATHDB-373691

Rigidity conjecture for skew braces with non-abelian characteristically simple multiplicative group

摘录 / EXCERPT: Rigidity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-015468 | MATHDB-373789

The (weak) Kawamata-Morrison cone conjecture for K-trivial fibrations

摘录 / EXCERPT: Let $f:(X,\Delta)\to S$ be a klt K-trivial fiber space. Define ... and ... A subset is a weak fundamental domain for a group action if its translates cover the space and every transl...

conjecture | open | MathDB 页面 | 原文线索

V2-015469 | MATHDB-373828

Vanishing conjecture for perturbed Dirac cohomology

摘录 / EXCERPT: Let ${\mathfrak g}$ be a Lie superalgebra, ${\mathfrak l}$ a subalgebra, M a ${\mathfrak g}$-supermodule, and $x\in {\mathcal Y}_{{\mathfrak l}}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015470 | MATHDB-373848

Relative Boone--Higman conjecture

摘录 / EXCERPT: Relative Boone--Higman conjecture. Let Γ be a finitely generated group with solvable word problem. Then Γ sharply embeds in a finitely presented relatively simple gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015471 | MATHDB-373849

Relative permutational Boone--Higman conjecture

摘录 / EXCERPT: Relative permutational Boone--Higman conjecture. Let Γ be a finitely generated group with solvable word problem. Then there exists a group G with a type...

conjecture | open | MathDB 页面 | 原文线索

V2-015472 | MATHDB-373861

The three-class vanishing conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group. For a positive integer k, say that G satisfies property $\mathcal{H}_k$ if there is a set $\mathcal{C}$ of at most k conjugacy classes such that ev...

conjecture | open | MathDB 页面 | 原文线索

V2-015473 | MATHDB-373978

The finite-group conjecture for Müger centers of pre-Tannakian categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a finitely tensor-generated pre-Tannakian category of moderate growth. Write $\mathcal{M}\mathcal{Z}(\mathcal{C})$ for its Müger center, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015474 | MATHDB-374001

Existence of beta extensions with full intertwining

摘录 / EXCERPT: Let θ be an m-realization of $\mathfrak{t}$ for $({\rm G},h)$ with parametrization $(({\rm V},h),\varphi,\Lambda,\beta)$. Let $\mathcal{C}$ be a chamber in the building of...

conjecture | open | MathDB 页面 | 原文线索

V2-015475 | MATHDB-374008

Odd-degree Ext dimension conjecture for Auslander-Gorenstein Nakayama algebras

摘录 / EXCERPT: Let A be an Auslander-Gorenstein Nakayama algebra, let S be a simple A-module, and let i be an odd integer. Odd-degree Ext dimension conjecture. One should have ... The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-015476 | MATHDB-374025

The maximal and minimal Arthur-parameter conjecture

摘录 / EXCERPT: Let G be a reductive group and let π be an irreducible admissible representation of G. Define $\Psi(\pi)=\\{\psi\in\Psi^+(G)\mid \pi\in\Pi_\psi\\}$, let ϕ_ψ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015477 | MATHDB-374035

Blind group selection consistency conjecture

摘录 / EXCERPT: Blind group selection consistency conjecture. The spectral-concentration selector is consistent at high signal-to-noise ratio:

conjecture | open | MathDB 页面 | 原文线索

V2-015478 | MATHDB-374050

Enomoto's conjecture on infinite semibricks

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field K. A brick is a module $M\in\operatorname{\mathsf{mod}} A$ whose endomorphism algebra $\operatorname{\mathsf{End}}_A(M)$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-015479 | MATHDB-374051

Demonet's lattice-point conjecture for brick-infinite algebras

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field K. Write $K_0(\operatorname{\mathsf{proj}} A)$ for the Grothendieck group of finitely generated projective A-modules, and l...

open_problem | open | MathDB 页面 | 原文线索

V2-015480 | MATHDB-374082

Relative Serre duality conjecture for parabolic induction

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let $I\subseteq S$ be a subset of simple reflections, and let W_I be the parabolic subgroup generated by I. Let $\mathfrak{h}$ be a realization...

conjecture | open | MathDB 页面 | 原文线索

V2-015481 | MATHDB-374114

The subgroup-count conjecture for groups with few cyclic subgroups

摘录 / EXCERPT: Subgroup-count conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-015482 | MATHDB-374126

Cyclicity conjecture for the periodicity group

摘录 / EXCERPT: Let $\eqp$ be an irreducible ergodic quantum process, and let $\GammaGroup$ denote its periodicity group. Cyclicity conjecture. The group $\GammaGroup$ is cyclic. The group is fini...

conjecture | open | MathDB 页面 | 原文线索

V2-015483 | MATHDB-374155

Betti moduli embedding conjecture for all microsheaves on the affine Springer fiber

摘录 / EXCERPT: Let $B^{\vee,+}$ and $B^{\vee,-}$ be opposite Borel subgroups of $G^{\vee}$ with $B^{\vee,+}\cap B^{\vee,-}=T^{\vee}$. Set $\mathcal{W}=B^{\vee,-}B^{\vee,+}\subset G^{\vee}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015484 | MATHDB-374161

Bardakov-Iskra conjecture on horizontal class transposition groups

摘录 / EXCERPT: Let $n>1$. For each $m\in\{2,3,\ldots,n\}$, let CT_m be the group generated by all horizontal class transpositions $\tau_{r_1(m),r_2(m)}$, and define ... Let...

open_problem | open | MathDB 页面 | 原文线索

V2-015485 | MATHDB-374180

The γ_2-character bridge conjecture for the hyperoctahedral group

摘录 / EXCERPT: Let B_N be the hyperoctahedral group and let $V=V_{((N-1),(1))}$ be its natural representation. Let $T(B_N,V)$ be the tensor product graph, let $I(V)$ denote its identity represe...

conjecture | open | MathDB 页面 | 原文线索

V2-015486 | MATHDB-374220

The folklore polynomial diameter conjecture for transitive permutation groups

摘录 / EXCERPT: Let $G\leq S_n$ be a transitive permutation group. The diameter of G is defined using the maximum over generating sets, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-015487 | MATHDB-374222

Grigorchuk's gap conjecture for finitely generated groups

摘录 / EXCERPT: Let $G=\langle X\rangle$ be a group generated by a finite set X, with growth function ... Write $f\prec g$ for the growth comparison used in the source: there is a constant $C>0$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-015488 | MATHDB-374254

Guo-Tointon conjecture on uniformly controlled coset spaces

摘录 / EXCERPT: Guo-Tointon conjecture. G has the u.c. property if and only if it is virtually nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-015489 | MATHDB-374318

Acyclic Ext-quiver conjectures for fractionally Calabi-Yau and representation-finite algebras

摘录 / EXCERPT: Acyclic Ext-quiver conjectures. 1. If A is twisted fractionally Calabi-Yau, then A has an acyclic Ext-quiver. 2. If A is d-representation-finite, then A has an acyclic Ex...

conjecture | open | MathDB 页面 | 原文线索

V2-015490 | MATHDB-374323

Practical universality conjecture for the Sequential GEVP

摘录 / EXCERPT: Let $\mathbf{R}$ be a Hermitian positive-definite matrix arising from a broad class of physical signal models, and let $\mathcal{B}$ be a basis. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015491 | MATHDB-374326

Non-rigidity conjecture for real projective spaces

摘录 / EXCERPT: Non-rigidity conjecture. For every $n\geqslant 4$ and every finite subgroup $G\subset\mathrm{PGL}_{n+1}(\mathbb{R})$, $\mathbb{P}^n_{\mathbb{R}}$ is not G-birationally rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-015492 | MATHDB-374375

Chai's conjecture on exceptional poles of archimedean Rankin-Selberg L-functions

摘录 / EXCERPT: Chai's conjecture. A complex number $s=s_0$ is an exceptional pole of type 1 with level m for the pair (π_1,π_2) if and only if it is an exceptional pole of type 2 with...

conjecture | open | MathDB 页面 | 原文线索

V2-015493 | MATHDB-374377

The characterization of universally quantified visibly pushdown sets in groups

摘录 / EXCERPT: Let G be a group. A set is universally quantified visibly pushdown, written $4\mathbf{VPL}^\forall4$, if it is defined by the corresponding universal visibly pushdown language co...

conjecture | open | MathDB 页面 | 原文线索

V2-015494 | MATHDB-374378

Uniform exactness conjecture for torsion-free hyperbolic and linear groups

摘录 / EXCERPT: Let Γ be a group, let $\mathcal{U}$ be a free ultrafilter on $\mathbb{N}$, and write $\Gamma^\mathcal{U}$ for its ultrapower. A group is uniformly exact when the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-015495 | MATHDB-374403

Isaacs-Seitz conjecture on solvable-group derived length

摘录 / EXCERPT: For a finite group G, let $\operatorname{Irr}(G)$ be the set of irreducible characters and define the set of irreducible character degrees by ... Let $\operatorname{dl}(G)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-015496 | MATHDB-374405

Brauer height-set conjectures for nonabelian defect groups

摘录 / EXCERPT: Brauer height-set conjectures. The derived length $\operatorname{dl}(D)$ is bounded in terms of e; b is bounded in terms of e; and $\operatorname{dl}(D)$ is bounded in terms...

conjecture | open | MathDB 页面 | 原文线索

V2-015497 | MATHDB-374408

Alperin's Conjecture C

摘录 / EXCERPT: Alperin's Conjecture C. If P is TI, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-015498 | MATHDB-374409

The Subnormalizer Conjecture

摘录 / EXCERPT: Let G be a finite group and let H be a subgroup. Define the subnormalizer ... and the subnormalizer subgroup $\operatorname{Sub}_G(H)=\langle S_G(H)\rangle$. For a cyclic subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-015499 | MATHDB-374415

Zippers-to-flows conjecture for hyperbolic groups

摘录 / EXCERPT: Let G be a hyperbolic group with boundary $\partial_\infty G$ homeomorphic to S^2, and suppose that $Z^\pm\subset\partial_\infty G$ is a minimal G-zipper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015500 | MATHDB-374427

The local Langlands correspondence for disconnected reductive groups

摘录 / EXCERPT: Let G be a connected, quasi-split reductive group over F, let (G',ξ,z) be a rigid inner twist of G, and let $\Phi_\mathrm{e}(G;[z])$ denote the set of $\widehat{G}$-conju...

conjecture | open | MathDB 页面 | 原文线索

V2-015501 | MATHDB-374462

Inner automorphism conjecture for the large power monoids of the integers

摘录 / EXCERPT: Inner automorphism conjecture. The automorphisms of $\mathcal P(\mathbb Z)$ are all inner, and the automorphisms of $\mathcal P_0(\mathbb Z)$ are all inner.

conjecture | open | MathDB 页面 | 原文线索

V2-015502 | MATHDB-374463

Restriction conjecture for automorphisms of large power monoids

摘录 / EXCERPT: Restriction conjecture. Every automorphism of $\mathcal P(H)$ maps $\mathcal P_1(H)$ bijectively onto itself and therefore restricts to an automorphism of $\mathcal P_1(H)$.

conjecture | open | MathDB 页面 | 原文线索

V2-015503 | MATHDB-374543

Characteristic-invariant conjecture for almost periodic actions on the hyperfinite type II_1 factor

摘录 / EXCERPT: Characteristic-invariant conjecture. The action α is classified up to cocycle conjugacy by the characteristic invariant (H,χ).

conjecture | open | MathDB 页面 | 原文线索

V2-015504 | MATHDB-374568

The eventual 8g conjecture for purely non-free actions

摘录 / EXCERPT: Eventual 8g conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015505 | MATHDB-374575

Wilde's conjecture on zeros of characters and orders of elements

摘录 / EXCERPT: Let G be a finite group and let $\chi\in\operatorname{Irr}(G)$. Here $o(g)$ denotes the order of g, and $|G|/\chi(1)$ is the codegree of χ. Wilde's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-015506 | MATHDB-374576

Wilde's nearly simple group conjecture

摘录 / EXCERPT: Let G be a finite group such that $S\lhd G/\mathbf Z(G)\leq\operatorname{Aut}(S)$ for some non-abelian simple group S, and let $\chi\in\operatorname{Irr}(G)$ be faithful. Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-015507 | MATHDB-374577

Wilde's quasi-simple commutator conjecture

摘录 / EXCERPT: Let H be a finite group such that $L=[H,H]$ is quasi-simple and $\mathbf Z(L)=\mathbf Z(H)$, and let $\chi\in\operatorname{Irr}(H)$ be faithful. Wilde's quasi-simple commutator c...

conjecture | open | MathDB 页面 | 原文线索

V2-015508 | MATHDB-374582

Stability conjecture for semisimplicity of twisted cyclotomic quiver Hecke algebras

摘录 / EXCERPT: Assume $p\nmid(e+1)$, let $I=\\{0,1,\ldots,e\\}$, and let $R^{\Lambda}_n(I)$ denote the cyclotomic quiver Hecke algebra associated with a dominant integral weight $\Lambda\in P^+$....

conjecture | open | MathDB 页面 | 原文线索

V2-015509 | MATHDB-374583

Zero-density conjecture for genus numbers in abelian towers

摘录 / EXCERPT: Let G be a finite group of order n, and let $b\mathcal{M}'(G,n,X)d$ denote the subset of $b\mathcal{M}(G,n,X)d$ consisting of number fields K that admit a tower ... in which ea...

conjecture | open | MathDB 页面 | 原文线索

V2-015510 | MATHDB-374628

The multiplicative cohomology conjecture for Burnside Tambara functors

摘录 / EXCERPT: Multiplicative Burnside conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015511 | MATHDB-374688

McGerty's center conjecture for cyclotomic Hecke algebras

摘录 / EXCERPT: Let q be a parameter with $q\neq 1$, and let the cyclotomic Hecke algebra have its Jucys-Murphy center and center. McGerty's center conjecture. The Jucys-Murphy center of the cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-015512 | MATHDB-374692

The abelian-Sylow strong subnormalizer conjecture

摘录 / EXCERPT: Let G be a finite group with abelian Sylow p-subgroups. Let $P\in\operatorname{Syl}_p(G)$ and let $x\in P$ be picky. Let $S_p(x)$ be a set of representatives for the conjugacy...

conjecture | open | MathDB 页面 | 原文线索

V2-015513 | MATHDB-374695

The Hall-picky local-global conjecture

摘录 / EXCERPT: Let G be a finite group, let π be a set of primes, and let H be a nilpotent Hall π-subgroup of G. Let $\mathcal H$ be the set of elements of H that are H-picky, m...

conjecture | open | MathDB 页面 | 原文线索

V2-015514 | MATHDB-374702

Rank-two conjecture for groups even on maximal proper subsets

摘录 / EXCERPT: Rank-two conjecture. If $m\geqslant n\geqslant 2$ and

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-015515 | MATHDB-374705

The virtually-free cogrowth conjecture

摘录 / EXCERPT: Let $\Gamma=\langle S\mid R\rangle$ be a group with finite generating set S, and let ... be its cogrowth series, where ψ_n counts words of length n in the alphabet...

conjecture | open | MathDB 页面 | 原文线索

V2-015516 | MATHDB-374726

The equality of the finite groups \(\mathcal{J}_n^{p,q}\) and \(\mathcal{K}_n^{p,q}\)

摘录 / EXCERPT: Equality conjecture. For all relevant n, p, and q, one has

conjecture | open | MathDB 页面 | 原文线索

V2-015517 | MATHDB-374804

Haiman's positivity and unimodality conjecture for symmetric groups

摘录 / EXCERPT: Haiman's conjecture. For every $z\in S_n$ and $\mu\vdash n$, the nonzero coefficients of α_{μ}^{z} are all positive and form a unimodal sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-015518 | MATHDB-374828

Coarse separability characterization for graph products of virtually nilpotent groups

摘录 / EXCERPT: Coarse separability conjecture. The graph G has a disconnecting clique if and only if the graph product $\Gamma(G)$ is coarsely separable by a family of subexponential growth.

conjecture | open | MathDB 页面 | 原文线索

V2-015519 | MATHDB-374848

Positive-Hausdorff-dimension fixed-point density conjecture for binary tree groups

摘录 / EXCERPT: Let T be the complete infinite rooted binary tree, and let $G\leq\operatorname{Aut}(T)$ be level-transitive if it acts transitively on every level. Define its Hausdorff dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-015520 | MATHDB-374908

BLZ-polynomial counting conjecture for highest-weight states

摘录 / EXCERPT: Let $\beta\in\mathbb C$ and $n\in\mathbb N$. Let $d_n^{(\beta)}$ denote the dimension or counting quantity defined in the source for the degree-n states at momentum β, and...

conjecture | open | MathDB 页面 | 原文线索

V2-015521 | MATHDB-375006

Kostant's tensor-product conjecture for the Weyl vector

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex semisimple Lie algebra with weight lattice notation P^+, and let ρ be the half-sum of its positive roots. For a dominant integral weight...

conjecture | open | MathDB 页面 | 原文线索

V2-015522 | MATHDB-375007

The unequal-parameter Kostant conjecture for multiples of the Weyl vector

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra or an untwisted affine Kac-Moody Lie algebra, let ρ denote the sum of the fundamental weights, and let $V(\gamma)$ be the irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-015523 | MATHDB-375008

The tensor-product embedding conjecture for the dominance order on pairs

摘录 / EXCERPT: Let P^+ be the set of dominant integral weights, and let $V(\gamma)$ denote the irreducible highest weight representation of highest weight γ. For pairs of dominant weight...

conjecture | open | MathDB 页面 | 原文线索

V2-015524 | MATHDB-375009

The affine δ-maximal component conjecture for unequal tensor parameters

摘录 / EXCERPT: Let $\mathfrak{g}$ be an untwisted affine Kac-Moody Lie algebra, let δ be its null root, and let $V(\gamma)$ denote the irreducible highest weight representation of highest...

conjecture | open | MathDB 页面 | 原文线索

V2-015525 | MATHDB-375012

Lewis-Reiner-Stanton parabolic Hilbert-series conjecture

摘录 / EXCERPT: Let $G=GL_n(\mathbb{F}_q)$ act on $S=\mathbb{F}_q[x_1,\ldots,x_n]$, let $Q(m,n)=S/(x_1^{q^m},\ldots,x_n^{q^m})$, and let P_{α} be the parabolic subgroup associated with a co...

conjecture | open | MathDB 页面 | 原文线索

V2-015526 | MATHDB-375014

Non-degeneracy conjecture for Killing forms on real conjugacy classes

摘录 / EXCERPT: Non-degeneracy conjecture. The Killing form on $\mathcal{C}$ is non-degenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-015527 | MATHDB-375022

Three-point higher-order conformal divergence conjecture

摘录 / EXCERPT: Fix $i\in\{1,2,3\}$. Let $\mathbf{s}\in\mathbb{Z}_{\geq 0}^3$, $s_i>0$, $\Delta_j\in\mathbb{C}$ for $j\neq i$, and let the depth satisfy $t_i\in\{0,\ldots,s_i-1\}$. For non-negativ...

conjecture | open | MathDB 页面 | 原文线索

V2-015528 | MATHDB-375029

Kleshchevization's Specht-module conjecture

摘录 / EXCERPT: Let $\boldsymbol{\kappa}=(\kappa_1,\kappa_2)$ be a bicharge, let $\boldsymbol{\lambda}\in\mathcal{MP}^{\boldsymbol{\kappa}}$, and let $\boldsymbol{\lambda}_{\mathrm K}$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-015529 | MATHDB-375044

Arthur's packet conjectures for real reductive groups

摘录 / EXCERPT: Let $\psi\in\Psi({}^{\vee}G^{\Gamma})$ be an Arthur parameter, let φ_ψ be its associated Langlands parameter, and let $\Pi(G)_\varphi$ denote the corresponding L-packe...

conjecture | open | MathDB 页面 | 原文线索

V2-015530 | MATHDB-375045

Conjectural reverse inclusion for microlocal Arthur packets

摘录 / EXCERPT: In the setting of Theorem, let G^{Γ} be an extended real reductive group, let ψ be an Arthur parameter, and let the notation $\Pi^z(G^{\Gamma})^{mic}_{\psi}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015531 | MATHDB-375086

Irreducibility conjecture for imaginary Whittaker modules

摘录 / EXCERPT: Let $\bmod\text{-}\mathcal{H}_q^{I}$ be the subcategory of $\mathcal{H}_q$-modules on which $\gamma=q^I\operatorname{Id}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015532 | MATHDB-375089

Göksel's transitive-subgroup conjugacy conjecture

摘录 / EXCERPT: Let $n\geq 0$, let T_n denote the first n levels of the rooted binary tree, and let $H\leq \operatorname{Aut}(T_n)$ be a group that contains an element acting transitively on...

conjecture | open | MathDB 页面 | 原文线索

V2-015533 | MATHDB-375096

Zero-momentum BRST cohomology conjecture for the bosonic ambitwistor string

摘录 / EXCERPT: Let $\mathsf{H}^n(0)$ denote the absolute BRST cohomology at ghost number n and zero momentum, regarded as a representation of $\operatorname{SO}(25,1)$, and let $\mathbb{V}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-015534 | MATHDB-375097

Middle-degree BRST cohomology splitting conjecture for the bosonic ambitwistor string

摘录 / EXCERPT: Let p be a momentum, let H be its stabiliser, and let $\mathsf{H}^n(p)$ and $\mathsf{H}_{\mathrm{rel}}^n(p)$ denote the absolute and relative BRST cohomology groups at ghost nu...

conjecture | open | MathDB 页面 | 原文线索

V2-015535 | MATHDB-375107

Nuclearity conjecture for the smooth Bernstein-Zelevinsky filtration

摘录 / EXCERPT: Let π be a representation in $\mathrm{Rep}^{\infty,F}(M_{n+1})$, and suppose that π is nuclear. Let π_k denote the terms of the filtration from Theorem 1, whose constru...

conjecture | open | MathDB 页面 | 原文线索

V2-015536 | MATHDB-375113

Conjecture on b[1mb[0mGL(1)\times GL(n-1) distinction over a quaternion division algebra

摘录 / EXCERPT: Main conjecture. An irreducible smooth representation θ of G_n with $n>2$ is $\operatorname{H}_{1,n-1}$-distinguished if and only if θ is either...

conjecture | open | MathDB 页面 | 原文线索

V2-015537 | MATHDB-375125

Fourier positivity conjecture for spherical positive definite functions

摘录 / EXCERPT: Let G be a semisimple linear group over a local field F, let $K<G$ be a maximal compact subgroup, and let $H<G$ be a closed abelian subgroup. Define ... For a continuous bi-K-i...

conjecture | open | MathDB 页面 | 原文线索

V2-015538 | MATHDB-375183

Nonexistence of infinite finitely generated solvable groups with square-free generalized identity length

摘录 / EXCERPT: Let G be an infinite finitely generated solvable group, and let n be a square-free integer. Nonexistence conjecture. There are no such groups G satisfying a generalized ident...

conjecture | open | MathDB 页面 | 原文线索

V2-015539 | MATHDB-375219

Gunningham-Jordan-Vazirani's compact generation conjecture for derived skein categories

摘录 / EXCERPT: Compact generation conjecture. The category $\widehat{\mathcal{S}\mathrm{kCat}_G(\Sigma)}$ admits a compact generator.

conjecture | open | MathDB 页面 | 原文线索

V2-015540 | MATHDB-375247

Matrix Spencer conjecture

摘录 / EXCERPT: Matrix Spencer conjecture. There exists a universal constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-015541 | MATHDB-375253

Casselman's comparison conjecture for Lie algebra homology

摘录 / EXCERPT: Let G be a real reductive group, let P be a parabolic subgroup with nilpotent Lie algebra $\mathfrak{n}$, and let π be a Casselman-Wallach representation of G. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015542 | MATHDB-375254

Socle-filtration conjecture for highest derivatives

摘录 / EXCERPT: Let π be an irreducible representation of GL_n of depth d. Regard its highest derivative π^- as a representation of M_{n-d+1}, and let ... be its increasing so...

conjecture | open | MathDB 页面 | 原文线索

V2-015543 | MATHDB-375255

Homogeneity and indecomposability conjecture for mirabolic restrictions

摘录 / EXCERPT: Let π be an irreducible representation of GL_n, and let M_n be the mirabolic subgroup. Use the paper's definition of a homogeneous representation and write...

conjecture | open | MathDB 页面 | 原文线索

V2-015544 | MATHDB-375274

Strengthened class-breadth conjecture for G-groups

摘录 / EXCERPT: Strengthened class-breadth conjecture. There exists a G-central series C_i, $0\leq i\leq N$, of P such that

conjecture | open | MathDB 页面 | 原文线索

V2-015545 | MATHDB-375280

Stable first-homology representation conjecture for complete graph configuration spaces

摘录 / EXCERPT: Stable first-homology representation conjecture. For $n\geq 5$, there is an isomorphism of S_n-representations

conjecture | open | MathDB 页面 | 原文线索

V2-015546 | MATHDB-375292

Abelianness conjecture for commutators of normal subgroups with two minimal invariant character degrees

摘录 / EXCERPT: Commutator abelianness conjecture. If $\Mcd G N=\{1,f\}$ for some positive integer f, then the commutator subgroup

conjecture | open | MathDB 页面 | 原文线索

V2-015547 | MATHDB-375303

Compact Fourier transform compatibility for pure inner twists

摘录 / EXCERPT: Let G be a connected reductive F-split group. For a unipotent element $u \in G^\text{?}$, a compact pair $(s,h) \in \text{?}$, and a maximal compact open subgroup K of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-015548 | MATHDB-375339

The converse from left-finite semibricks to brick finiteness

摘录 / EXCERPT: Converse conjecture. If every semibrick is left finite, then A is brick finite.

conjecture | open | MathDB 页面 | 原文线索

V2-015549 | MATHDB-375344

Krohn-Rhodes-like decomposition conjecture for Krom programs

摘录 / EXCERPT: Krohn-Rhodes-like decomposition conjecture. Every propositional Krom logic program can be sequentially decomposed into aperiodic and permutation programs.

conjecture | open | MathDB 页面 | 原文线索

V2-015550 | MATHDB-375362

The type A-C correspondence conjecture for homomorphisms between hook Specht modules

摘录 / EXCERPT: Let $R_n(\Gamma,p)$ be the quiver Hecke algebra corresponding to a quiver Γ over a field of characteristic p, and let $\mu,\lambda\vdash n$ be arbitrary partitions. Type A...

conjecture | open | MathDB 页面 | 原文线索

V2-015551 | MATHDB-375394

The eigenvalue conjecture for Radon transforms on flag-quotient functions

摘录 / EXCERPT: Let λ be fixed, let U_{λ} be the subgroup associated with λ, and let the Radon transform $\operatorname{\mathcal{R}}$ act on the complex-valued function spa...

conjecture | open | MathDB 页面 | 原文线索

V2-015552 | MATHDB-375424

The Strong Atiyah Conjecture over a subfield of the complex numbers

摘录 / EXCERPT: Let $\mathbb{K}$ be a subfield of $\mathbb{C}$, let G be a countable group with a bound on the orders of its finite subgroups, and let $\mathrm{lcm}(G)$ denote the least common m...

conjecture | open | MathDB 页面 | 原文线索

V2-015553 | MATHDB-375425

The Weak Atiyah Conjecture over a subfield of the complex numbers

摘录 / EXCERPT: Let $\mathbb{K}$ be a subfield of $\mathbb{C}$, and let G be a countable group with a bound on the orders of its finite subgroups. For a finite matrix A over $\mathbb{K}[G]$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-015554 | MATHDB-375491

Langlands-Shahidi type categorical Langlands conjecture

摘录 / EXCERPT: Fix a quasi-split reductive group G over a local field E, and let W_E be its Weil group. Let $\check{G}$ be the Langlands dual group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015555 | MATHDB-375501

The quadratic-extension algebra-isomorphism conjecture for Hecke measure spaces

摘录 / EXCERPT: Quadratic-extension algebra-isomorphism conjecture. For any quadratic extension E/F, these canonical isomorphisms are algebra isomorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-015556 | MATHDB-375502

The spectral decomposition conjecture for Hecke operators over quadratic extensions

摘录 / EXCERPT: Spectral decomposition conjecture. There exist a set I^E, a function $d^E:I^E\to\mathbb N$, and distinct functions $\lambda_i^E:\mathcal D\to\mathbb C$ for $i\in I^E$, such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-015557 | MATHDB-375505

Mousavand-Paquette conjecture on finite-dimensional bricks and generic bricks

摘录 / EXCERPT: Let A be a finite dimensional algebra over an algebraically closed field. A brick is an A-module whose endomorphism ring is a division ring, and a generic module is an indecomp...

conjecture | open | MathDB 页面 | 原文线索

V2-015558 | MATHDB-375541

Bodirsky-Schneider-Thom conjecture on singular mixed identities in oligomorphic groups

摘录 / EXCERPT: Bodirsky-Schneider-Thom conjecture. If $G\curvearrowright\Omega$ is oligomorphic, then all mixed identities of G are singular.

conjecture | open | MathDB 页面 | 原文线索

V2-015559 | MATHDB-375556

Surjectivity conjecture for the group enumeration function

摘录 / EXCERPT: Let $f(n)$ denote the number of isomorphism classes of finite groups of order n. Surjectivity conjecture. For every positive integer m, there exists an integer n such that .....

conjecture | open | MathDB 页面 | 原文线索

V2-015560 | MATHDB-375567

Irreducible-component positivity conjecture for non-Archimedean Hodge-Riemann relations

摘录 / EXCERPT: Let V have dimension n, let ρ be an irreducible ${GL_n({\cal O})}$-component of $C^\infty(Gr_{n-i,n}(\mathbb F))$, and let ${\cal D}_i$, P_i, and...

conjecture | open | MathDB 页面 | 原文线索

V2-015561 | MATHDB-375574

Polynomial filling conjecture for S-arithmetic lattices

摘录 / EXCERPT: Let k) be a number field, let $\mathbf{G}$ be an absolutely almost simple, k-isotropic k-algebraic group, let S contain all archimedean places of k, and let $\Gamma<G_S$...

conjecture | open | MathDB 页面 | 原文线索

V2-015562 | MATHDB-375624

Conjecture on indecomposable summands of twisted Foulkes modules

摘录 / EXCERPT: Let $H^{(2^m;k)}$ denote the twisted Foulkes module for the symmetric group S_{2m+k}. Let τ be the p-content and θ the odd sequence of a partition of $2m+k$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-015563 | MATHDB-375634

The profinite Bateman-Horn conjecture over S-integers

摘录 / EXCERPT: Profinite Bateman-Horn conjecture. The following limits exist and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-015564 | MATHDB-375647

The de Rham quantum global geometric Langlands conjecture

摘录 / EXCERPT: Let G be a simple reductive group, let $\check{G}$ be its Langlands dual group, let $\check{h}$ be the dual Coxeter number of G, and let r be the maximal multiplicity of arro...

conjecture | open | MathDB 页面 | 原文线索

V2-015565 | MATHDB-375648

The quantum Betti global geometric Langlands conjecture

摘录 / EXCERPT: Let G be a reductive group, let $\check{G}$ be its Langlands dual group, and let $\operatorname{Shv}_{c,\operatorname{Nilp}}(\operatorname{Bun}_G)$ denote the category of c-twi...

conjecture | open | MathDB 页面 | 原文线索

V2-015566 | MATHDB-375649

Gaitsgory-Lurie quantum fundamental local equivalence conjecture

摘录 / EXCERPT: Gaitsgory-Lurie quantum fundamental local equivalence conjecture. The functor FLE is an equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-015567 | MATHDB-375703

The no infinite cocycle crossed-product decomposition conjecture for II$_1$ factors

摘录 / EXCERPT: Let M be a separable II$_1$ factor. Let G be a countably infinite group, let (B,τ) be a tracial von Neumann algebra, and let (σ,c) be a trace-preserving cocycle act...

conjecture | open | MathDB 页面 | 原文线索

V2-015568 | MATHDB-375726

Prasad's adjoint L-function criterion for degenerate Whittaker vanishing

摘录 / EXCERPT: Let F be a non-Archimedean local field, let $G=\operatorname{GL}_{2n}(F)$, and let $P_{n,n}=MN$ be the standard parabolic subgroup with Levi factor...

conjecture | open | MathDB 页面 | 原文线索

V2-015569 | MATHDB-375776

Anagnostopoulou-Merkouri and Burness's unrestricted regularity conjecture

摘录 / EXCERPT: Let G be a finite almost simple group with socle G_0, and let a non-standard subgroup mean a subgroup whose associated primitive action is non-standard in the sense defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-015570 | MATHDB-375802

Conjecture on constructing all inflations by truncation equalities

摘录 / EXCERPT: Fix $J\subseteq I$, and let V_J be an object of $\mathcal{O}_{\nu}^{\lambda}(\mathbf R,\mathfrak g_J)$ for some λ, ν, and $\mathbf R$. An inflation of V_J is a cons...

conjecture | open | MathDB 页面 | 原文线索

V2-015571 | MATHDB-375840

Union restoration conjecture for commutator conductors

摘录 / EXCERPT: Let G be a finite group, let G' denote its commutator subgroup, and for each commutator weight w let $\operatorname{C_{\mathrm{comm}}}(w)$ be the associated commutator conduc...

conjecture | open | MathDB 页面 | 原文线索

V2-015572 | MATHDB-375886

Magnitude conjecture for representation-finite bound path algebras

摘录 / EXCERPT: Magnitude conjecture. 1. We have

conjecture | open | MathDB 页面 | 原文线索

V2-015573 | MATHDB-375888

Generation conjecture for the handlebody Johnson kernel

摘录 / EXCERPT: Let $\mathcal{H}_g$ be the handlebody group, let $\mathcal{K}_g$ be the Johnson kernel of the mapping class group, and define the handlebody Johnson kernel by ... A separating meridi...

conjecture | open | MathDB 页面 | 原文线索

V2-015574 | MATHDB-375905

The maximal product-of-trees embedding conjecture for handlebody groups

摘录 / EXCERPT: Let F_2 be the free group of rank 2, let F_2^n denote its n-fold direct product, and let $\mathcal{H}_g$ be the handlebody group of genus g. A subgroup or map is quasi-is...

conjecture | open | MathDB 页面 | 原文线索

V2-015575 | MATHDB-375983

The cyclicity conjecture for finite groups from skew-plane configurations

摘录 / EXCERPT: Cyclicity conjecture. If $G_{\mathcal L}$ is finite, then $G_{\mathcal L}$ is cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-015576 | MATHDB-376001

Perfect-group square clustering conjecture for Cayley graphs

摘录 / EXCERPT: Let G be a perfect group and let S be a generating set subject to the restrictions required for the square clustering coefficient. Write $\square(\operatorname{Cay}(G,S))$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-015577 | MATHDB-376021

Type B equivariant foam categorification conjecture

摘录 / EXCERPT: Let $\mathbf{B}^n_m$ be the endomorphism category ... and let $(\mathbf{B}^n_m)^\tau$ be the equivariant category associated with the involution τ. The equivariant Grothendieck...

conjecture | open | MathDB 页面 | 原文线索

V2-015578 | MATHDB-376165

Exceptional groups in the triality construction for the symmetric and alternating groups

摘录 / EXCERPT: The construction associates generators ρ_1,ρ_2,ρ_3 to a group G by ... A triality construction conjecture. Apart from two cases, up to coloring symmetries, the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-015579 | MATHDB-376166

Quantitative dynamics conjectures for the relative free factor complex

摘录 / EXCERPT: Quantitative dynamics conjectures. First, if ϕ is fully irreducible with expansion factor $\lambda_\phi>1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-015580 | MATHDB-376208

Hyperbolicity conjecture for one-relator groups without Baumslag-Solitar subgroups

摘录 / EXCERPT: Hyperbolicity conjecture. Every such group G is hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-015581 | MATHDB-376224

King-Tollu-Toumazet positivity conjecture for stretched Littlewood-Richardson coefficients

摘录 / EXCERPT: Let λ,μ,ν be partitions with $|\lambda|+|\mu|=|\nu|$, and let $c^{\nu}_{\lambda\mu}$ be the Littlewood-Richardson coefficient, the multiplicity of the Schur function...

conjecture | open | MathDB 页面 | 原文线索

V2-015582 | MATHDB-376225

BGGH's conjecture on Cannon-Thurston maps for HHG pairs

摘录 / EXCERPT: Let G be a hierarchically hyperbolic group (HHG), and let $H<G$ be an infinite hyperbolic normal subgroup of infinite index. A Cannon-Thurston map is a continuous extension of th...

conjecture | open | MathDB 页面 | 原文线索

V2-015583 | MATHDB-376253

Projectivity conjecture for finitely generated pro-$\mathcal{C}$ subgroups

摘录 / EXCERPT: Projectivity conjecture. H is projective.

conjecture | open | MathDB 页面 | 原文线索

V2-015584 | MATHDB-376254

Adams conjecture for local theta correspondence

摘录 / EXCERPT: Let $m,n\in\mathbb{Z}_{\geq 0}$, let $G_n=\operatorname{Sp}_{2n}(F)$ and $H_m^\pm=\operatorname{O}_{2m}^\pm(F)$ with $m>n$, and set $\alpha=2m-2n-1$. Let $\Pi(G_n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015585 | MATHDB-376264

Weight-space and dimension conjecture for affine nilCoxeter algebra simple modules

摘录 / EXCERPT: Weight-space and dimension conjecture. The weight spaces of simple A-modules are one dimensional. Consequently, the dimension of a simple module is the multinomial coefficient

conjecture | open | MathDB 页面 | 原文线索

V2-015586 | MATHDB-376267

Torsion conjecture for virtual Artin groups

摘录 / EXCERPT: Let Γ be a Coxeter graph, let $\mathrm{VA}[\Gamma]$ be the associated virtual Artin group, and let $\iota_{\mathrm{W}}(\mathrm{W}[\Gamma])$ denote its embedded copy of the a...

conjecture | open | MathDB 页面 | 原文线索

V2-015587 | MATHDB-376268

Minimal classifying-space dimension conjecture for virtual Artin groups

摘录 / EXCERPT: Let Γ be a Coxeter graph, and let $\mathfrak{cl}^f(\Gamma)$ denote the spherical clique number of Γ, namely the number of vertices in a maximal spherical clique. A co...

conjecture | open | MathDB 页面 | 原文线索

V2-015588 | MATHDB-376272

Auslander-Reiten-Smalø conjecture on components of Auslander-Reiten quivers

摘录 / EXCERPT: Auslander-Reiten-Smalø conjecture. If A is of infinite representation type, then Γ_A has infinitely many connected components.

conjecture | open | MathDB 页面 | 原文线索

V2-015589 | MATHDB-376302

Type-II spectral decomposition conjecture for standard quotients

摘录 / EXCERPT: Type-II spectral decomposition conjecture. There exist a topological space $\widehat G_H^L\supset\widehat G_H$, a continuous finite-to-one map

conjecture | open | MathDB 页面 | 原文线索

V2-015590 | MATHDB-376303

Global spectral decomposition conjecture for compact standard quotients

摘录 / EXCERPT: Global spectral decomposition conjecture. There exist measurable Hilbert-space fibers

conjecture | open | MathDB 页面 | 原文线索

V2-015591 | MATHDB-376309

Polyhedral seminorm conjecture for twisted L^2-Betti numbers

摘录 / EXCERPT: Let G be a finitely generated group satisfying the Atiyah conjecture, and let C_* be a finitely generated free L^2-acyclic $\mathbb{Z}G$-chain complex. For a cohomology param...

conjecture | open | MathDB 页面 | 原文线索

V2-015592 | MATHDB-376336

Hall-Paige conjecture on transversals of Cayley tables

摘录 / EXCERPT: For a finite group G, let $L(G)$ be its Cayley table. The Hall-Paige condition is the condition that the sum of the elements of G is the identity in the abelianisation...

conjecture | open | MathDB 页面 | 原文线索

V2-015593 | MATHDB-376355

The symplectic-reduction conjecture for the Lie bracket of simple Lie algebras

摘录 / EXCERPT: The symplectic-reduction conjecture. The symplectic reduction

conjecture | open | MathDB 页面 | 原文线索

V2-015594 | MATHDB-376376

Breitenbchers conjecture on mean equicontinuity and finite-to-one extensions

摘录 / EXCERPT: Let $m\in\mathbb N$, and let a minimal $\mathbb Z$-system mean a minimal dynamical system with a continuous $\mathbb Z$-action. Its maximal equicontinuous factor is the maximal fac...

conjecture | open | MathDB 页面 | 原文线索

V2-015595 | MATHDB-376468

Geometric Langlands reciprocity conjecture for semisimple groups

摘录 / EXCERPT: Let G be a semisimple group, let $G^\lor$ be its Langlands dual, and let X be the underlying curve. Let Λ be the set of isomorphism classes of pairs (P,χ), where...

conjecture | open | MathDB 页面 | 原文线索

V2-015596 | MATHDB-376632

The strong-obstruction conjecture for the permanent versus determinant

摘录 / EXCERPT: Let $g=\operatorname{det}(Y)$, $h=\operatorname{perm}(X)$, and $f=\phi(h)$. A strong obstruction is a representation-theoretic obstruction of the type defined in the paper for the...

conjecture | open | MathDB 页面 | 原文线索

V2-015597 | MATHDB-376633

The strong separability conjecture for the tensor-product subgroup

摘录 / EXCERPT: Let $G=SL_{n^2}(\mathbb{C})$ and let $H=SL_n(\mathbb{C})\times SL_n(\mathbb{C})$ have its natural embedding in $SL(\mathbb{C}^n\otimes\mathbb{C}^n)$. A subgroup is strongly G-sep...

conjecture | open | MathDB 页面 | 原文线索

V2-015598 | MATHDB-376648

The big-group deficiency conjecture

摘录 / EXCERPT: Big-group deficiency conjecture. If Γ is a big group, then

conjecture | open | MathDB 页面 | 原文线索

V2-015599 | MATHDB-376649

The big-group Euler characteristic conjecture for closed 4-manifolds

摘录 / EXCERPT: Big-group Euler characteristic conjecture. If Γ is a big group, then

conjecture | open | MathDB 页面 | 原文线索

V2-015600 | MATHDB-376666

The rationality and integrality conjecture for L^2-Betti numbers

摘录 / EXCERPT: Rationality and integrality conjecture. For every degree p,

conjecture | open | MathDB 页面 | 原文线索

V2-015601 | MATHDB-376671

The correspondence between Green's relations and fine equivalence relations

摘录 / EXCERPT: Correspondence conjecture. The Green's relations on a subsemigroup U of S have as counterparts prolonged images in S, namely the fine equivalence relations on S.

conjecture | open | MathDB 页面 | 原文线索

V2-015602 | MATHDB-376707

Representation-theoretic interpretation of the Standard Model

摘录 / EXCERPT: Representation-theoretic interpretation conjecture. The quantum field theory of the standard model may be understood purely in terms of the representation theory of the automorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-015603 | MATHDB-376731

Kazhdan-Lusztig correspondence conjecture for $(1,p)$ logarithmic models

摘录 / EXCERPT: Kazhdan-Lusztig correspondence conjecture. The category of $\mathcal{W}(p)$-representations is equivalent to the category of finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-015604 | MATHDB-376739

The Kac determinant formula for finite W-algebras

摘录 / EXCERPT: Consider a finite W-algebra associated to an ${\bf sl}_2$ embedding that is principal in $\gf_S\subseteq\gf$. Let $M(\Lambda)$ be a Verma module, let $\bar{\beta}$ be a position in...

conjecture | open | MathDB 页面 | 原文线索

V2-015605 | MATHDB-376740

The Kazhdan-Lusztig theorem for finite W-algebras

摘录 / EXCERPT: Let W_S be the Weyl subgroup associated to $\gf_S$, and let $W_S\setminus W$ be the corresponding coset with its Bruhat ordering. For $\sigma\leq\tau$ in this coset, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015606 | MATHDB-376776

The Cartesian-Klein-Lie approach extends the Erlangen program

摘录 / EXCERPT: Cartesian-Klein-Lie conjecture. The combination of Cartesian and Klein-Lie approaches based on the assumption that coordinates are often a representation space for a group action i...

conjecture | open | MathDB 页面 | 原文线索

V2-015607 | MATHDB-376789

Existence conjecture for monopoles with prescribed magnetic and holomorphic charges

摘录 / EXCERPT: Charge-existence conjecture. For a given μ there are monopoles with any collection of nonnegative magnetic charges $(m_1,\ldots,m_s)$. Given a choice of magnetic charges, there...

conjecture | open | MathDB 页面 | 原文线索

V2-015608 | MATHDB-376836

Admissibility and Weyl-orbit conjecture for soliton contents

摘录 / EXCERPT: Let $B=B^{r_1,l_1}\otimes\cdots\otimes B^{r_L,l_L}$ be a tensor product of crystals of type A_n, and let $m=(m^{(a)}_j)$ be a soliton content. Define ... let $p^{(a)}_j$ be the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-015609 | MATHDB-376838

State-counting formula for invariant soliton sectors

摘录 / EXCERPT: Let $m=(m^{(a)}_j)$ be a content such that ... where ... Let $\Omega(m)$ be the Bethe-ansatz multiplicity, let W be the Weyl group of A_n, and let $\lambda(m)$ be the associated do...

conjecture | open | MathDB 页面 | 原文线索

V2-015610 | MATHDB-376848

Multiplicity-one conjecture for products of Weyl-group orbits

摘录 / EXCERPT: Multiplicity-one conjecture. Under these hypotheses, the multiplicity of $O(w\lambda+\mu)$ in $O(\lambda)\otimes O(\mu)$ is 1.

conjecture | open | MathDB 页面 | 原文线索

V2-015611 | MATHDB-376849

Orbit-product decomposition conjecture with a nontrivial stabilizer

摘录 / EXCERPT: Orbit-product decomposition conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-015612 | MATHDB-376853

Simplicity conjecture for vacuum modules of positive-defect Lie superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be an (almost) simple finite-dimensional Lie superalgebra of positive defect, namely one of $\mathfrak{sl}(m,n)$ with $m,n\geq 1$, $\mathfrak{osp}(m,2n)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-015613 | MATHDB-376864

Deguchi's highest-weight-vector conjecture for irreducible quotient representations

摘录 / EXCERPT: Deguchi's highest-weight-vector conjecture. Every irreducible quotient of a submodule in $U\Omega$ has a highest weight vector of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-015614 | MATHDB-376865

Deguchi's vanishing relation conjecture for highest weight vectors

摘录 / EXCERPT: Let Ω be a highest weight vector, let m_j be the multiplicity associated with the parameter a_j, and let $w_{j^k}(\boldsymbol{\hat a})$ be the operators defined in the a...

conjecture | open | MathDB 页面 | 原文线索

V2-015615 | MATHDB-376886

Injectivity conjecture for braid group actions by generalized Dehn twists

摘录 / EXCERPT: Let (M,β) be a compact symplectic manifold with $\pi_1(M)$ trivial and $c_1(M,\beta)=0$. Let $(S_1,\dots,S_m)$ be an (A_m)-configuration of Lagrangian spheres in M for so...

conjecture | open | MathDB 页面 | 原文线索

V2-015616 | MATHDB-376894

Zel'manov's classification conjecture for just-infinite pro-p groups of finite width

摘录 / EXCERPT: Let G be a just-infinite pro-p-group of finite width, meaning a just-infinite pro-p-group whose finite-index subgroup growth layers have uniformly bounded width. Zel'manov's...

conjecture | open | MathDB 页面 | 原文线索

V2-015617 | MATHDB-376910

Nonexistence of irreducible braid-group representations of corank at least three

摘录 / EXCERPT: Let B_n be the braid group on n strands, and let k denote the corank of an irreducible complex representation of B_n. Nonexistence conjecture. For every $k\geq 3$, for n...

conjecture | open | MathDB 页面 | 原文线索

V2-015618 | MATHDB-376930

Zaslow's crepant-resolution conjecture for stringy Hodge numbers

摘录 / EXCERPT: Let X be a smooth Calabi-Yau manifold with an action of a finite group H preserving the holomorphic volume form, let $Y=X/H$, and let $h_{st}^{p,q}(Y)$ denote its stringy Hodge...

conjecture | open | MathDB 页面 | 原文线索

V2-015619 | MATHDB-376940

The quasi-isometric classification conjecture for polycyclic abelian-by-cyclic groups

摘录 / EXCERPT: Let $M\in\operatorname{SL}(n,\mathbf{Z})$ have no eigenvalues on the unit circle, and let G be a finitely generated group quasi-isometric to the associated group Γ_M. The...

conjecture | open | MathDB 页面 | 原文线索

V2-015620 | MATHDB-376941

The horizontal-preserving conjecture for polycyclic abelian-by-cyclic groups

摘录 / EXCERPT: Horizontal-preserving conjecture. Every quasi-isometry $G_M\to G_N$ is horizontal-respecting.

conjecture | open | MathDB 页面 | 原文线索

V2-015621 | MATHDB-376942

Classification conjecture for polycyclic abelian-by-cyclic groups

摘录 / EXCERPT: Classification conjecture. The groups Γ_M and Γ_N are quasi-isometric if and only if there exist nonzero $a,b\in\mathbb R$ such that M^a and N^b have the same abs...

conjecture | open | MathDB 页面 | 原文线索

V2-015622 | MATHDB-376943

Quasi-isometry group conjecture for Solv geometry

摘录 / EXCERPT: QI-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015623 | MATHDB-376944

Horizontal-respecting conjecture for quasi-isometries of Solv

摘录 / EXCERPT: Horizontal-respecting conjecture. Every quasi-isometry

conjecture | open | MathDB 页面 | 原文线索

V2-015624 | MATHDB-376945

Nonexistence conjecture for upper-half-space quasi-isometric embeddings in Solv

摘录 / EXCERPT: Upper-half-space embedding conjecture. There does not exist a quasi-isometric embedding

conjecture | open | MathDB 页面 | 原文线索

V2-015625 | MATHDB-376951

The contractibility conjecture for the commuting complex

摘录 / EXCERPT: Let G be a group. The commuting complex $C(G)$ is the simplicial complex whose simplices are finite sets of pairwise commuting nonidentity elements of G. The center $Z(G)$ cons...

conjecture | open | MathDB 页面 | 原文线索

V2-015626 | MATHDB-376952

The odd-Euler-characteristic conjecture for non-commuting complexes of simple groups

摘录 / EXCERPT: Let G be a nonabelian simple group. The non-commuting complex $NC(G)$ is the simplicial complex associated to finite sets of pairwise noncommuting nonidentity elements, and...

conjecture | open | MathDB 页面 | 原文线索

V2-015627 | MATHDB-376954

Kervaire's conjecture on balanced presentations of nontrivial groups

摘录 / EXCERPT: Kervaire's conjecture. The resulting group is also non-trivial. The conjecture is an algebraic ingredient in arguments about handle decompositions of 4-manifolds; the source ment...

conjecture | open | MathDB 页面 | 原文线索

V2-015628 | MATHDB-376955

The w.g.s.c. conjecture for fundamental groups of closed aspherical manifolds

摘录 / EXCERPT: Let Γ be the fundamental group of a closed aspherical manifold. A finitely presented infinite group Γ is w.g.s.c. when some compact polyhedron with fundamental group...

conjecture | open | MathDB 页面 | 原文线索

V2-015629 | MATHDB-376970

Geometric reconstruction conjecture for standard Drinfeld-Jimbo quantum groups

摘录 / EXCERPT: Geometric reconstruction conjecture. The tuple T^DJ is a braided tuple, and

conjecture | open | MathDB 页面 | 原文线索

V2-015630 | MATHDB-376971

Flag-tuple parametrization conjecture for standard Drinfeld-Jimbo quantum groups

摘录 / EXCERPT: Flag-tuple parametrization conjecture. The tuple T^DJ is the flag tuple associated to $\OqG$; equivalently, E^DJ parametrizes all flag modules of the shape al...

conjecture | open | MathDB 页面 | 原文线索

V2-015631 | MATHDB-376981

The orbifold elliptic genus conjecture for unramified finite group quotients

摘录 / EXCERPT: Let X be a complex manifold equipped with an effective action of a finite group G. Write X/G for the quotient, and let $\widehat{Ell}(X/G,;y,q)$ denote the normalized ellipti...

conjecture | open | MathDB 页面 | 原文线索

V2-015632 | MATHDB-376984

Katsylo's birationality conjecture for generically free representations

摘录 / EXCERPT: Let G be an algebraic group, and let V and W be generically free linear representations of G. Katsylo's conjecture. If ... then V and W are birationally isomorphic as G...

conjecture | open | MathDB 页面 | 原文线索

V2-015633 | MATHDB-376994

Kirillov-Reshetikhin character conjecture for quantum affine algebras

摘录 / EXCERPT: Let X_n be a finite-dimensional simple Lie algebra of classical or exceptional type, let $W_m^{(a)}$ denote the Kirillov-Reshetikhin module associated with node a and positive...

conjecture | open | MathDB 页面 | 原文线索

V2-015634 | MATHDB-377002

Classification conjecture for non-semisimple homological TQFTs

摘录 / EXCERPT: Let ${\cal V}$ be a non-semisimple TQFT whose mapping class group representations have precisely the Torelli groups as their kernels. Let ${\cal P}^+$ be the class of polynomials d...

conjecture | open | MathDB 页面 | 原文线索

V2-015635 | MATHDB-377012

Horn's conjecture on sums of Hermitian matrices

摘录 / EXCERPT: Horn's conjecture. The relation $\rho\boxplus\sigma\sim_c\tau$ holds if and only if the trace identity holds and the displayed inequality holds whenever i, j, and k are admis...

conjecture | open | MathDB 页面 | 原文线索

V2-015636 | MATHDB-377013

The saturation conjecture for tensor products of unitary-group representations

摘录 / EXCERPT: Saturation conjecture. One can take $N=1$ in the above theorem. Equivalently, for integer λ, μ, and $ u$, the relations $\lambda\boxplus\mu\sim_c\nu$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015637 | MATHDB-377014

Equivalence of nontrivial blocks of Lie superalgebras of type W(n)

摘录 / EXCERPT: Let $W(n)$ be the Lie superalgebra of type $W(n)$, and let its blocks be the blocks of the category of finite-dimensional representations. A block is nontrivial if it does not cont...

conjecture | open | MathDB 页面 | 原文线索

V2-015638 | MATHDB-377025

Nakajima's geometric formula conjecture for the t-analogue of q-characters

摘录 / EXCERPT: Let $M=M_P$ be a standard module, let Q/R be an l-weight of M, and let $M(Q/R)$ be the corresponding l-weight space. Define a filtration on $M(Q/R)$ by ... where ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015639 | MATHDB-377071

Berenstein-Zelevinsky congruence conjecture for quasi-commuting dual canonical basis elements

摘录 / EXCERPT: Let ${\cal M}$ be the set of multisegments. For ${\bf m},{\bf n}\in{\cal M}$, let $G^*({\bf m})$ and $G^*({\bf n})$ be the corresponding dual canonical basis elements, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015640 | MATHDB-377072

Weight-filtration conjecture for spectral-cover eigenspaces

摘录 / EXCERPT: Let $\rho\colon G\to GL(V)$ be an irreducible representation, let λ be a dominant weight, let W_λ be its stabilizer in W, and let m_λ be the multiplicity...

conjecture | open | MathDB 页面 | 原文线索

V2-015641 | MATHDB-377103

The circulator trace conjecture for stable Gelfand-Kazhdan categories

摘录 / EXCERPT: Circulator trace conjecture. In the case of stable Gelfand-Kazhdan categories,

conjecture | open | MathDB 页面 | 原文线索

V2-015642 | MATHDB-377115

Semisimplicity conjecture for the graded pieces gr^N_jW

摘录 / EXCERPT: Let G be the group in the paper, let ${\mathcal I}_G$ be the indicated category of G-modules, and let N denote the filtration used to define gr^N_jW. Semisimplicity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-015643 | MATHDB-377146

Stable deformability conjecture for invariant distributions of geometrically finite groups

摘录 / EXCERPT: Let Γ be a geometrically finite group, let σ be a representation parameter, let $\lambda\in{\mathfrak a}_{\mathbb C}^*$, and let φ be the auxiliary parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-015644 | MATHDB-377183

The Kirillov-Reshetikhin multiplicity formula for Yangian KR modules

摘录 / EXCERPT: Let $\mathfrak{g}$ be a complex simple Lie algebra of rank n. Set $y=(y_a)_{a=1}^n$ with $y_a=e^{-\alpha_a}$ for the simple roots α_a of $\mathfrak{g}$. For the KR module...

conjecture | open | MathDB 页面 | 原文线索

V2-015645 | MATHDB-377184

The canonical-solution conjecture for KR Q-systems

摘录 / EXCERPT: Let $\mathfrak{g}_0$ be the finite-dimensional Lie algebra associated with $U_q(X_N^{(r)})$, let H be the index set of pairs (a,m) for the KR Q-system, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015646 | MATHDB-377185

The original KR character formula for nonexceptional quantum affine algebras

摘录 / EXCERPT: Let $X_N^{(r)}\neq A_{2n}^{(2)}$, let $\Delta_+^{\mathfrak{g}_0}$ be the positive roots of $\mathfrak{g}_0$, and let $\mathcal{Q}^{\nu}(y)$ and $\mathcal{K}^{\nu}_{D,G}(y)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015647 | MATHDB-377186

The $A_{2n}^{(2)}$ KR character formula in type B_n

摘录 / EXCERPT: Let $X_N^{(r)}=A_{2n}^{(2)}$, so that $\mathfrak{g}_0=B_n$, and let $\mathcal{Q}^{\nu}(y)$ and $\mathcal{K}^{\nu}_{D,G}(y)$ be the normalized KR-character product and associated po...

conjecture | open | MathDB 页面 | 原文线索

V2-015648 | MATHDB-377187

The KR character formulas for types $B_n^{(1)}$ and $A_{2n-1}^{(2)}$

摘录 / EXCERPT: Let $\mathcal{Q}^{\nu}(y)$ and $\mathcal{K}^{\nu}_{D,G}(y)$ be the normalized KR-character product and associated power series. Type-B_n and type-D_n KR conjectures. For...

conjecture | open | MathDB 页面 | 原文线索

V2-015649 | MATHDB-377188

Kirillov-Reshetikhin conjecture for normalized characters of KR modules

摘录 / EXCERPT: Kirillov-Reshetikhin conjecture. For every $(a,m)\in H$,

conjecture | open | MathDB 页面 | 原文线索

V2-015650 | MATHDB-377205

Fulton's multiplicity-one conjecture for tensor products of GL(n) representations

摘录 / EXCERPT: Let $\lambda,\mu,\nu \in ({\mathbb Z}^n)^3$ be a triple of dominant weights for $\operatorname{GL}_n({\mathbb C})$, and let V_λ,V_μ,V_ν be the corresponding irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-015651 | MATHDB-377214

Topological Vaught conjecture for Polish group actions

摘录 / EXCERPT: Let G be a Polish group and let X be a Polish G-space, meaning that G acts continuously on X. An orbit is a set of the form $G\cdot x$ for some $x\in X$; two points are e...

conjecture | open | MathDB 页面 | 原文线索

V2-015652 | MATHDB-377230

Natural algebra action on matchings for alternating Kasteleyn matrices

摘录 / EXCERPT: Let A be a nonsingular alternating matrix, and let D be the matrix-algebra deformation of $\mathbb{C}[\operatorname{coker} A]$ defined by the twisted relations in the paper. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-015653 | MATHDB-377233

Uno's conjecture on representation types of Hecke algebras

摘录 / EXCERPT: Let K be a field, let $q\in K$ be an invertible element with $q\ne 1$, and let (W,S) be a finite Weyl group. Let ${\mathcal H}_q(W)$ be the one-parameter Hecke algebra associat...

conjecture | open | MathDB 页面 | 原文线索

V2-015654 | MATHDB-377271

Congruence condition for resolution graphs

摘录 / EXCERPT: Let (X,o) be a ${\mathbb Q}$-Gorenstein surface singularity with ${\mathbb Q}$-homology sphere link, and let its resolution graph be equipped with the associated discriminant-gro...

conjecture | open | MathDB 页面 | 原文线索

V2-015655 | MATHDB-377280

Kazhdan property conjecture for automorphism groups of free groups

摘录 / EXCERPT: Kazhdan property conjecture. For $k\geqslant 4$, $\operatorname{Aut} F_k$ satisfies Kazhdan property (T).

conjecture | open | MathDB 页面 | 原文线索

V2-015656 | MATHDB-377285

Finite quotient detection conjecture for the Andrews--Curtis conjecture

摘录 / EXCERPT: Let $F_2=\langle x,y\rangle$ be the free group of rank two, and let $u(x,y),v(x,y)$ be normal generators of F_2. For a finite factor group G of F_2, write $\bar x,\bar y$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015657 | MATHDB-377292

Khukhro's class-bounded subgroup conjecture for finite p-groups

摘录 / EXCERPT: Khukhro's conjecture. G contains a subgroup of m-bounded nilpotency class and (p,m)-bounded index.

conjecture | open | MathDB 页面 | 原文线索

V2-015658 | MATHDB-377293

Khukhro's derived-length conjecture for finite p-groups

摘录 / EXCERPT: Khukhro's conjecture. G contains a subgroup of m-bounded derived length and (p,n,m)-bounded index.

conjecture | open | MathDB 页面 | 原文线索

V2-015659 | MATHDB-377302

Surjectivity conjecture for the annular skein map onto the centre of $H_{n,p}$

摘录 / EXCERPT: Let $H_{n,p}$ be the Homfly skein algebra with n positive and p negative boundary points, let ${\cal C}$ be the skein of the annulus, and let ... be the algebra homomorphism obta...

conjecture | open | MathDB 页面 | 原文线索

V2-015660 | MATHDB-377312

Existence of contractible relative periodic orbits for Hamiltonian group actions

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold with a Hamiltonian action of a compact connected Lie group ${\rm G}$ and moment map $\mu:M\to\mathfrak g^*$. Assume hypothesis...

conjecture | open | MathDB 页面 | 原文线索

V2-015661 | MATHDB-377330

The cut-number equality conjecture

摘录 / EXCERPT: Let Γ be a group. For each k, let $c_k(\Gamma)$ denote the largest cardinality of a primitive set in $H^1(\Gamma,\mathbb Z)$ for which all k-fold Massey products vanish....

conjecture | open | MathDB 页面 | 原文线索

V2-015662 | MATHDB-377355

Achar's conjecture on equivariant coherent sheaves on the nilpotent cone

摘录 / EXCERPT: Let G be a complex reductive group with nilpotent cone $ ilcone$, let $K_G(ilcone)$ be the Grothendieck group of G-equivariant coherent sheaves on $ ilcone$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015663 | MATHDB-377356

Conjecture on the leading weight of functions on a finite nilpotent-orbit cover

摘录 / EXCERPT: Let ${^{\scriptscriptstyle L\!}}{\mathcal O}$ be a nilpotent orbit in the Lie algebra of the Langlands dual group, let $\mathcal O=d({^{\scriptscriptstyle L\!}}{\mathcal O})$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-015664 | MATHDB-377360

Kostant's canonical extension conjecture for minimal K-types

摘录 / EXCERPT: Let G_0 be a real reductive linear Lie group, and let K_0 be its maximal compact subgroup. Fix an Iwasawa decomposition ... of G_0, where A_0 is abelian and normalizes N_0....

conjecture | open | MathDB 页面 | 原文线索

V2-015665 | MATHDB-377375

Socle-head conjecture for tensor products of quantum affine representations

摘录 / EXCERPT: Let N be a positive integer, and let V and W be finite-dimensional irreducible representations of $U_q(\widehat{\mathfrak{sl}}_N)$. Assume that $V\otimes V$ is irreducible. S...

conjecture | open | MathDB 页面 | 原文线索

V2-015666 | MATHDB-377379

Wildness conjecture for non-quotient surface singularities and related singularities

摘录 / EXCERPT: Wildness conjecture. In each of the following cases, the ring A is Cohen--Macaulay wild:

conjecture | open | MathDB 页面 | 原文线索

V2-015667 | MATHDB-377385

The co-rank conjecture for 3-manifold groups

摘录 / EXCERPT: Co-rank conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015668 | MATHDB-377390

Character map conjecture for tilting modules in \mathcal O_{m|n}

摘录 / EXCERPT: Character map conjecture. If $i:K(\mathcal O_{m|n}^\Delta)\to\mathscr T^{m|n}_{\mathbb Z}$ is the map defined in the cited theorem, then

conjecture | open | MathDB 页面 | 原文线索

V2-015669 | MATHDB-377402

A decomposition multiplicity formula for $S_{\F_2}^{(n-4,2,1^2)}$

摘录 / EXCERPT: Multiplicity formula. The seven asserted formulas in the source hold: for $n\geq9$, $[S_{\mathbb F_2}^{(n-4,2,1^2)}:D_{\mathbb F_2}^{(n-4,4)}]=1$; for $n\geq7$,...

conjecture | open | MathDB 页面 | 原文线索

V2-015670 | MATHDB-377404

Abelianization and commutator conjecture for stabilized fundamental groups

摘录 / EXCERPT: Let X be a simply connected symplectic manifold, let $G_k^0(X,\omega)$ be the reduced stabilized fundamental group, let Λ_k be the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-015671 | MATHDB-377410

The generalized hypergeometric expression conjecture for matrix-valued spherical functions

摘录 / EXCERPT: Generalized hypergeometric expression conjecture. For a given $\ell\geq 0$, the components of the functions $H(t)$ are expressible in terms of generalized hypergeometric functions...

conjecture | open | MathDB 页面 | 原文线索

V2-015672 | MATHDB-377411

The three-term recurrence conjecture for matrix-valued spherical functions

摘录 / EXCERPT: Three-term recurrence conjecture. There exist matrices A_p, B_p, and C_p, independent of t, such that

conjecture | open | MathDB 页面 | 原文线索

V2-015673 | MATHDB-377466

The converse invariant conjecture for complexified Farahat-Higman rings of wreath products

摘录 / EXCERPT: Let ${\Gamma}$ and G be finite groups. For each n, consider the complexified rings $\mathbb C\otimes_{\mathbb Z}\mathcal G_{\Gamma}(n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015674 | MATHDB-377475

Toric Specht module decomposition conjecture

摘录 / EXCERPT: Let $\kappa=\lambda/d/\mu$ be a toric shape, let S^κ be its toric Specht module, and let S^ν denote the irreducible representation of the symmetric group indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-015675 | MATHDB-377476

Higher syzygy degree bounds for finite-group invariants

摘录 / EXCERPT: Let G be a finite group in the nonmodular case acting on a finite-dimensional representation V. Let $\beta_G^i(V)$ denote the degree bound for the i-th syzygies of the invari...

conjecture | open | MathDB 页面 | 原文线索

V2-015676 | MATHDB-377512

The Shapovalov-preserving Weyl group representation conjecture

摘录 / EXCERPT: Shapovalov-preserving Weyl group representation conjecture. There exists a representation of the Weyl group W in U such that each generator s_i is mapped to $\tilde s_i$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-015677 | MATHDB-377515

The classification of exceptional special tuples for the weak Deligne-Simpson problem

摘录 / EXCERPT: Let $\kappa=2$, and consider tuples of conjugacy classes for which conditions i) and ii) of the source's genericity theorem hold, but for which the weak Deligne-Simpson problem is...

open_problem | open | MathDB 页面 | 原文线索

V2-015678 | MATHDB-377516

Quantum character formula for quantum loop modules

摘录 / EXCERPT: Quantum character formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015679 | MATHDB-377518

Quantization conjecture for classical dynamical r-matrices

摘录 / EXCERPT: Let $\mathfrak h$ be the Lie algebra occurring in the classical dynamical Yang-Baxter equation, let V be the representation space, and let a classical dynamical r-matrix be a m...

conjecture | solved | MathDB 页面 | 原文线索

V2-015680 | MATHDB-377520

A-type Kostka polynomial formula

摘录 / EXCERPT: Let $\mathcal{P}$ denote the set of partitions, let $\mu\in\mathcal{P}$ and $\nu\in\mathbb{Z}^n$ satisfy $l(\mu)\leq n$ and $|\mu|=|\nu|$, and let $K_{\mu\nu}$ be the corresponding...

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V2-015681 | MATHDB-377569

Nakayama's conjecture for finite-rank algebras

摘录 / EXCERPT: Let A be a not necessarily commutative algebra of finite rank over a field k, and let A be viewed as a left A-module. Nakayama's conjecture. If A has an injective resolut...

conjecture | open | MathDB 页面 | 原文线索

V2-015682 | MATHDB-377571

Conjecture on maximal Ext-orthogonal sets of irreducible components

摘录 / EXCERPT: Let Q be a Dynkin quiver of type $\mathbb{A}_n$, $\mathbb{D}_n$ or $\mathbb{E}_{6,7,8}$, let Λ be its preprojective algebra, and let R^+ be the set of positive roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-015683 | MATHDB-377585

Character conjecture for dual canonical basis elements and integral Hecke superalgebra modules

摘录 / EXCERPT: Let $r\geqslant 2$, let ${\cal H}_m(t)$ be the relevant Hecke-Clifford superalgebra, and let ${\cal C}_{m,r}$ be its category of integral modules whose eigenvalues lie in...

conjecture | open | MathDB 页面 | 原文线索

V2-015684 | MATHDB-377590

Virtual positive Betti number conjecture for irreducible 3-manifolds

摘录 / EXCERPT: Virtual positive Betti number conjecture. The manifold M has a finite cover N such that

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V2-015685 | MATHDB-377624

The unfoldability conjecture for billiard triangles in Weyl chambers

摘录 / EXCERPT: Unfoldability conjecture. A billiard triangle in Δ with sides $\overline{ox_0}$, $\overline{ox_k}$ and $x_0x_1\ldots x_{k-1}x_k$, and with the special vertices $0,x_0,x_k$,...

conjecture | open | MathDB 页面 | 原文线索

V2-015686 | MATHDB-377628

Positivity conjecture for positive-definite functions on reductive subgroups

摘录 / EXCERPT: Let G be a real reductive group, let K be a maximal compact subgroup of G, and let $\Xi(g)$ be Harish-Chandra's basic spherical function. Let H be a subgroup of G such th...

conjecture | open | MathDB 页面 | 原文线索

V2-015687 | MATHDB-377631

Przebinda's character formula conjecture for theta lifting

摘录 / EXCERPT: Let $R_s(MG_1,MG_2)$ denote the relevant stable-range representation class, let $\theta_s(MG_1,MG_2)(\pi)$ be the stable theta lift, and let ω^c be the contragredient oscill...

conjecture | open | MathDB 页面 | 原文线索

V2-015688 | MATHDB-377632

Barbasch-Vogan characterization of unipotent representations

摘录 / EXCERPT: Let G be either an orthogonal group or a metaplectic group. Let $\mathcal O_{\bold D}$ be the orbit indexing $\Pi_{\mathcal O_{\bold D}}(G)$, let $\mathcal O_{\bold d}$ be a rigi...

conjecture | refuted | MathDB 页面 | 原文线索

V2-015689 | MATHDB-377667

Bogomolov's vanishing conjecture for finite simple groups

摘录 / EXCERPT: Bogomolov's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015690 | MATHDB-377689

Global Weyl module tensor-product conjecture

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra with fundamental representations $V^{(i)}$, $i=1,\dots,r$, and highest weight vectors $v^{(i)}\in V^{(i)}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015691 | MATHDB-377691

Closedness of the derived-wild locus in algebraic families

摘录 / EXCERPT: Closedness conjecture. In the situation of Corollary 44, the set W is always closed in X; equivalently, the set

conjecture | open | MathDB 页面 | 原文线索

V2-015692 | MATHDB-377692

Symplectic resolution conjecture for coverings of nilpotent orbit closures

摘录 / EXCERPT: Let $\bar{p}: X \rightarrow \overline{\0}$ be a covering of a nilpotent orbit in a simple Lie algebra $\mathfrak{g}$ such that $R[1]=0$. Suppose that $\pi: Z\rightarrow X$ is a sym...

conjecture | open | MathDB 页面 | 原文线索

V2-015693 | MATHDB-377695

Sommers's cell-region decomposition conjecture

摘录 / EXCERPT: Sommers's cell-region decomposition conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015694 | MATHDB-377710

The two-random-elements conjecture for full-dimensional closures

摘录 / EXCERPT: Let $\Gamma(p)$ be the profinite automorphism group of the rooted p-ary tree, and let the closure of a subgroup generated by random elements be considered in the profinite topolo...

conjecture | open | MathDB 页面 | 原文线索

V2-015695 | MATHDB-377711

The finitely generated full-dimension conjecture for closed subgroups

摘录 / EXCERPT: Let $\Gamma(p)$ be the profinite automorphism group of the rooted p-ary tree, let G be a closed subgroup, and let Hausdorff dimension be computed in $\Gamma(p)$. A subgroup is...

conjecture | open | MathDB 页面 | 原文线索

V2-015696 | MATHDB-377715

The quantum McKay correspondence for symplectic resolutions

摘录 / EXCERPT: Let M be an algebraic symplectic manifold with a symplectic action of a finite group G, and let $X\to M/G$ be a smooth symplectic resolution. Choose deformation-quantizations...

conjecture | open | MathDB 页面 | 原文线索

V2-015697 | MATHDB-377716

The finiteness conjecture for the Calogero-Moser classifying map

摘录 / EXCERPT: Let V be a symplectic vector space and $G\subset Sp(V)$ a finite group. Let C be the parameter space of G-invariant functions on the symplectic reflections in G, let...

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V2-015698 | MATHDB-377718

The top-dimensional cohomology surjectivity conjecture for elementary abelian actions

摘录 / EXCERPT: Let $G=(\mathbb Z/2)^r$ act freely on ... and let $N=n_1+\dots+n_m$. Write $\pi_G:X/G\to BG$ for the classifying map of the action. Top-dimensional surjectivity conjecture. The induc...

conjecture | open | MathDB 页面 | 原文线索

V2-015699 | MATHDB-377719

The minimal-sphere-product existence problem for finite group actions

摘录 / EXCERPT: Let G be a finite group, and let $r(G)$ denote its rank, namely the maximum rank of an elementary abelian p-subgroup of G over all primes p dividing $|G|$. Minimal-sphere-p...

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V2-015700 | MATHDB-377735

Genericity transfer conjecture for unitary-group representations

摘录 / EXCERPT: Genericity transfer conjecture. If $\pi_1\otimes\pi_2$ is an irreducible subrepresentation of this function space and π_1 is generic, then π_2 is also generic.

open_problem | open | MathDB 页面 | 原文线索

V2-015701 | MATHDB-377742

The Green scalar product formula for symmetric functions

摘录 / EXCERPT: Let n be the fixed positive integer in the construction, let $\mathbf{K}\simeq \Gamma\otimes\mathbb{C}[v,v^{-1}]$, and let $\langle\;,\;\rangle_G$ denote Green's scalar product....

conjecture | open | MathDB 页面 | 原文线索

V2-015702 | MATHDB-377779

Irreducibility conjecture for semi-infinite inductions from cuspidal representations

摘录 / EXCERPT: Let G be the group under consideration, let $\widehat{G}$ denote its associated central extension, and let Π be an irreducible cuspidal representation of G. The functor...

conjecture | open | MathDB 页面 | 原文线索

V2-015703 | MATHDB-377787

The symmetry conjecture for the Solomon homomorphism

摘录 / EXCERPT: Let W be a finite Coxeter group, and let ΣW be the Solomon descent algebra of W. The Solomon homomorphism $[...]$ maps elements of ΣW to class functions on W....

conjecture | open | MathDB 页面 | 原文线索

V2-015704 | MATHDB-377807

The residually finite quasi-Engel group conjecture

摘录 / EXCERPT: Let G be a quasi-Engel group, meaning that for the fixed initial word w there is an integer n such that the identity ${}^w\!u_n(x,y)\equiv 1$ holds in G. A group is residua...

conjecture | open | MathDB 页面 | 原文线索

V2-015705 | MATHDB-377808

The profinite unbounded quasi-Engel group conjecture

摘录 / EXCERPT: Let G be an unbounded quasi-Engel group, meaning that for every $x,y\in G$ there is an integer $n=n(x,y)$ such that ${}^w\!u_n(x,y)=1$, for the fixed initial word w. A group is...

conjecture | open | MathDB 页面 | 原文线索

V2-015706 | MATHDB-377846

The freeness conjecture for Hecke algebras of complex reflection groups

摘录 / EXCERPT: Let W be a finite complex reflection group, let W' be a parabolic subgroup of W, and let ${\mathcal H}(W)$ and ${\mathcal H}(W')$ be their Hecke algebras over...

conjecture | open | MathDB 页面 | 原文线索

V2-015707 | MATHDB-377847

The canonical symmetrizing form conjecture for complex reflection group Hecke algebras

摘录 / EXCERPT: Let W be a finite complex reflection group, let W' be a parabolic subgroup, and let ${\mathcal H}(W)$ be the associated Hecke algebra. A canonical symmetrizing form on...

conjecture | open | MathDB 页面 | 原文线索

V2-015708 | MATHDB-377848

The centre-surjectivity conjecture for Hecke algebras of complex reflection groups

摘录 / EXCERPT: Let $A=\mathbb{Z}[x,x^{-1}]$, let R be a commutative local A-algebra, and let F be its residue field. Let ${\mathcal H}$ be the Hecke algebra of a finite complex reflection g...

conjecture | open | MathDB 页面 | 原文线索

V2-015709 | MATHDB-377861

Weight-string formula for the shape of a tridiagonal pair

摘录 / EXCERPT: Let $w_1,w_2,\ldots,w_d$ be the weights associated with the tridiagonal-pair construction, and let q be the scalar used to define a q-string, namely a set of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-015710 | MATHDB-377863

Irreducibility criterion for the associated quantum affine module

摘录 / EXCERPT: Let $U_q(\widehat{\mathfrak{sl}}_2)$ be the quantum affine algebra and let V be the standard module constructed from the tridiagonal-pair data, with weights $w_1,\ldots,w_d$. Qua...

conjecture | open | MathDB 页面 | 原文线索

V2-015711 | MATHDB-377864

Tridiagonal-pair and type-(1,1) quantum-affine-module correspondence

摘录 / EXCERPT: Let A,A^* be a tridiagonal pair on V satisfying the conditions of Definition 1p1 and the stated eigenvalue-form lemma. Let K,R,L denote the operators in the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-015712 | MATHDB-377865

Serre-pair decomposition and bijectivity problem

摘录 / EXCERPT: Assume K is algebraically closed, $q\in\mathbb K$ is not a root of unity, V is a finite-dimensional nonzero K-vector space, and $R,L:V\to V$ are nilpotent linear maps satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-015713 | MATHDB-377870

Monotonicity conjecture for translation lengths in complex hyperbolic triangle-group deformations

摘录 / EXCERPT: Let I be the critical interval for the family $\rho_t(p,q,r)$, let Γ be the abstract (p,q,r) triangle group, and for an infinite word $W\in\Gamma$ write $W_t=\rho_t(W)$....

conjecture | open | MathDB 页面 | 原文线索

V2-015714 | MATHDB-377871

Polynomial Whitehead-minimal orbit conjecture

摘录 / EXCERPT: Let F_r be a free group of rank r, let F_r^k denote the set of k-tuples of elements of F_r, and let $U_{\min}$ be a minimal representative of the orbit of U under the r...

conjecture | open | MathDB 页面 | 原文线索

V2-015715 | MATHDB-377900

Chevalley-involution criterion for Casimir irreducibility

摘录 / EXCERPT: Let $\mathfrak g$ be a complex simple Lie algebra, and let V be a simple self-dual $\mathfrak g$-module with nonzero zero weight space $V[0]$. Let $\mathcal C_{\mathfrak g}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-015716 | MATHDB-377901

Regular-highest-weight reducibility conjecture for zero weight spaces

摘录 / EXCERPT: Let $\mathfrak g$ be a complex simple Lie algebra not isomorphic to $\mathfrak{sl}_2$ or $\mathfrak{sl}_3$, and let V be a simple self-dual $\mathfrak g$-module with nonzero zero...

conjecture | open | MathDB 页面 | 原文线索

V2-015717 | MATHDB-377911

The two-cyclic-factor conjecture for finite groups realized by 3-nets

摘录 / EXCERPT: Two-cyclic-factor conjecture. If H can be realized by a 3-net in $\mathbb{P}^2$, then H is a direct sum of at most two cyclic groups. If one class of the net is a pencil, then...

conjecture | open | MathDB 页面 | 原文线索

V2-015718 | MATHDB-377955

The generalized projective Euler characteristic conjecture for finite groups

摘录 / EXCERPT: Generalized projective Euler characteristic conjecture. There exists an integer $Q(d)$ such that, whenever G is finite and ${\rm gcd}(\#G,Q(d))=1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-015719 | MATHDB-377973

Kostov's irreducible Deligne-Simpson criterion

摘录 / EXCERPT: Kostov's conjecture. There is an irreducible solution to

conjecture | open | MathDB 页面 | 原文线索

V2-015720 | MATHDB-377974

The direct-link conjecture for indecomposable modules over ad-quivers

摘录 / EXCERPT: Let A be the path algebra of an ad-quiver (Q,σ) without oriented cycles. Let $\beta\in\Delta(Q)^+$ be a positive root, and let $p_F(M)$ and $p_\sigma(\beta)$ denote the pe...

conjecture | open | MathDB 页面 | 原文线索

V2-015721 | MATHDB-378007

King-Tollu-Toumazet stretching conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let λ, μ and $ u$ be partitions, and let $c_{\lambda\mu}^{\nu}$ denote the Littlewood-Richardson coefficient. Assume that $c_{\lambda\mu}^{\nu}>0$. For each positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-015722 | MATHDB-378042

Lunts-Rosenberg conjecture on faithfulness of quantum global sections

摘录 / EXCERPT: Let $\lambda\in\Lambda$ and let $\tilde{\Gamma}_\lambda$ be the left exact global-section functor from the category of modules over the sheaf of twisted differential operators...

conjecture | open | MathDB 页面 | 原文线索

V2-015723 | MATHDB-378043

Lunts-Rosenberg conjecture on quantum differential operator global sections

摘录 / EXCERPT: Let $\lambda\in\Lambda$; let U be the quantized enveloping algebra, let $T(\lambda)$ be the Verma module of highest weight λ, and let J_λ be its annihilator in...

conjecture | open | MathDB 页面 | 原文线索

V2-015724 | MATHDB-378058

The odd-sided polygon representation conjecture

摘录 / EXCERPT: Let $G:=\pi_1({\mathcal P})$ be the fundamental group of a 2-dimensional negatively curved acute polygon ${\mathcal P}$ of finite groups. Theorem asserts that when $n=2k$ is even...

conjecture | open | MathDB 页面 | 原文线索

V2-015725 | MATHDB-378068

Benson's conjecture on the Loewy length of blocks

摘录 / EXCERPT: Let G be a finite group and B a block of kG over a field k of characteristic p, with defect group D. Benson's conjecture. The Loewy length of B should satisfy ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-015726 | MATHDB-378071

Openness of derived tameness in flat families of algebras

摘录 / EXCERPT: Openness conjecture. The set X_{tame} is open in X.

conjecture | open | MathDB 页面 | 原文线索

V2-015727 | MATHDB-378077

Mariño-Vafa formula for two-part Hodge integrals

摘录 / EXCERPT: Let ${\mathcal P}$ be the set of partitions, let μ^+ and μ^- be partitions, and let $G_{\mu^+,\mu^-}(x,y)$, $p^+_{\mu^+}$, and $p^-_{\mu^-}$ denote the corresponding genera...

conjecture | open | MathDB 页面 | 原文线索

V2-015728 | MATHDB-378078

Mariño-Vafa formula in the one-leg specialization

摘录 / EXCERPT: Let ${\mathcal P}_+$ denote the set of nonempty partitions, let μ^+, μ^-, ν^+, and ν^- be partitions, and let $G_{\mu^+,\mu^-}(\tau)$ be the specialization defined...

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V2-015729 | MATHDB-378109

Lusztig's cohomological realization conjecture for reductive groups over local rings

摘录 / EXCERPT: Lusztig's cohomological realization conjecture. Every irreducible representation of G_r^F appears in the virtual representation

conjecture | open | MathDB 页面 | 原文线索

V2-015730 | MATHDB-378140

The inclusion conjecture for the maps $\alpha_{n,s}$

摘录 / EXCERPT: The inclusion conjecture for $\alpha_{n,s}$. The map $\alpha_{n,s}$ is an inclusion.

conjecture | open | MathDB 页面 | 原文线索

V2-015731 | MATHDB-378164

The unequal-parameter cell classification conjecture for type B_n

摘录 / EXCERPT: Let W be of type B_n, with generators $t,s_1,\ldots,s_{n-1}$, where t is not conjugate to the s_i, and let a positive weight function be specified by ... ... Write the weight f...

conjecture | open | MathDB 页面 | 原文线索

V2-015732 | MATHDB-378167

Degenerate Frobenius structure on the dual of a G-Euler Frobenius algebra

摘录 / EXCERPT: Degenerate Frobenius structure conjecture. There is a degenerate G-Frobenius algebra of degree j on the dual of a G-Euler G-Frobenius algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-015733 | MATHDB-378175

Parabolic restriction conjecture for invariant differential-operator quotients

摘录 / EXCERPT: Let V be an n-dimensional vector space over $\Bbbk$, let $G=\operatorname{GL}(V)$ with Lie algebra $\mathfrak g=\mathfrak{gl}(V)$, and let $P\subset G$ be the parabolic subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-015734 | MATHDB-378178

Orbifold chiral-algebra correspondence for DHR representations

摘录 / EXCERPT: Let $\mathcal A$ be a chiral conformal net with a finite gauge-group action by Γ, and let $\mathcal A^\Gamma$ denote its fixed-point orbifold net. Consider the irreducible D...

conjecture | open | MathDB 页面 | 原文线索

V2-015735 | MATHDB-378189

Sela-Kharlampovich-Myasnikov characterization of groups elementarily equivalent to free groups

摘录 / EXCERPT: Characterization conjecture. The group G is elementarily equivalent to a non-abelian free group if and only if G is a non-elementary hyperbolic fully residually free tower.

conjecture | open | MathDB 页面 | 原文线索

V2-015736 | MATHDB-378194

Existence conjecture for towers of algebras with prescribed Cartan invariants

摘录 / EXCERPT: Let $(A_n)_{n\geq 0}$ be a tower of algebras, and let $C^{(n)}$ denote the prescribed matrices associated with the tower's Cartan invariants. Existence conjecture. There exists a t...

conjecture | open | MathDB 页面 | 原文线索

V2-015737 | MATHDB-378195

Realization conjecture for the planar binary tree Hopf algebra

摘录 / EXCERPT: Realization conjecture. There exists a tower of algebras A_n, with a basis $(e_\sigma)_{\sigma\in\mathfrak S_n}$, such that the restriction of $\rho_{m,n}$ to canonical words is...

conjecture | open | MathDB 页面 | 原文线索

V2-015738 | MATHDB-378210

Freeness conjecture for the stalks of standard sheaves

摘录 / EXCERPT: Freeness conjecture. The stalks of $\mathcal I_!(\lambda,\mathbb Z)$ are free.

conjecture | open | MathDB 页面 | 原文线索

V2-015739 | MATHDB-378238

McKay correspondence conjecture for crepant resolutions

摘录 / EXCERPT: Let $G\subset \operatorname{Sl}_n(\mathbb C)$ be a finite group, put $W=\mathbb C^n/G$, and let $X\rightarrow W$ be a crepant resolution. Write $D^b_G(\mathbb C^n)$ for the bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-015740 | MATHDB-378241

Derived exchange-pair conjecture for cluster endomorphism algebras

摘录 / EXCERPT: Derived exchange-pair conjecture. The quotient module categories are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-015741 | MATHDB-378242

Polynomial generation conjecture for the K-homology of the symplectic loop group

摘录 / EXCERPT: Let $Sp(n)$ be the compact symplectic group, let $U(i)$ be the unitary group, and let $K.\Omega Sp(n)$ denote the K-homology ring of its based loop space. For $1\leq i\leq n$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-015742 | MATHDB-378253

Generation of the full Hecke property from the GL_n Hecke property

摘录 / EXCERPT: Full Hecke-property conjecture. Then ${\mathcal S}_E$ satisfies the full Hecke property.

conjecture | open | MathDB 页面 | 原文线索

V2-015743 | MATHDB-378261

The non-linearity conjecture for Kac-Moody groups with hyperbolic Weyl group

摘录 / EXCERPT: Non-linearity conjecture. If W is Gromov-hyperbolic and $q>\!\!>1$, then Λ is not linear.

conjecture | open | MathDB 页面 | 原文线索

V2-015744 | MATHDB-378267

Mukhin-Varchenko conjecture that the norm constant is one

摘录 / EXCERPT: Let $V_{\boldsymbol{\Lambda}}=V_{\boldsymbol{\Lambda}^1}\otimes\dots\otimes V_{\boldsymbol{\Lambda}^k}$ be the tensor product of the relevant $\mathfrak g$-modules. Let z^0 have...

conjecture | open | MathDB 页面 | 原文线索

V2-015745 | MATHDB-378298

Integrality conjecture for type A string polytopes

摘录 / EXCERPT: Let G be of type A_n, let $\underline{w}_0$ be any reduced decomposition of the longest Weyl-group element, and let λ be a weight. Write $Q_{\underline{w}_0}(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-015746 | MATHDB-378303

Conjecture on the minus Drinfeld-Sokolov reduction of category O modules

摘录 / EXCERPT: Minus-reduction conjecture. For any $\ta{ kappa}\in\mathbb C$ and any $V\in\operatorname{Obj}\ta{ O}_{\ta{ kappa}$},

conjecture | open | MathDB 页面 | 原文线索

V2-015747 | MATHDB-378304

Existence of indecomposable graded bimodules realizing the Kazhdan-Lusztig basis

摘录 / EXCERPT: Let V be a reflection-faithful representation of ${\cal W}$ over an infinite field, and let R be the ring of regular functions on V. For $x\in{\cal W}$, let C'_x be the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-015748 | MATHDB-378314

The canonical-basis description of the principal-subspace ideal

摘录 / EXCERPT: Let I_0 be the ideal associated with the principal subspace W_0 in the completed quantum loop-algebra positive part, let $\mathcal{C}_0$ be the class of sheaves...

conjecture | open | MathDB 页面 | 原文线索

V2-015749 | MATHDB-378315

Compatibility of the principal-subspace basis with the Leclerc-Thibon basis

摘录 / EXCERPT: Let L_{Λ_0} be the level-one representation, let W_0 be its principal subspace, let B_0 be the canonical basis induced on W_0, and let B^+ be the Leclerc-Thibon can...

conjecture | open | MathDB 页面 | 原文线索

V2-015750 | MATHDB-378316

The limiting canonical basis conjecture for the level-one representation

摘录 / EXCERPT: Let B_n be the compatible bases of the level-one representation obtained by applying the canonical basis of the completed quantum loop-algebra positive part to the vectors v_n,...

conjecture | open | MathDB 页面 | 原文线索

V2-015751 | MATHDB-378401

The asymptotic index-subgroup count conjecture for braid groups

摘录 / EXCERPT: Let B_n be the braid group on n strands, and let $a_k(B_n)$ denote the number of subgroups of index k in B_n. For n sufficiently large, write $c(k/n)$ for a constant depe...

conjecture | open | MathDB 页面 | 原文线索

V2-015752 | MATHDB-378430

The character identity for roots and the Orlik-Solomon representation

摘录 / EXCERPT: Let W be a finite Coxeter group of rank n, with R its representation on the set of roots. Let G' be the virtual W-module obtained by evaluating at $t=1$ the graded quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-015753 | MATHDB-378456

Conjecture that relatively hyperbolic groups have property R_∞

摘录 / EXCERPT: For a group G, say that it has property R_∞ if every automorphism ϕ of G satisfies $R(\phi)=\infty$, where $R(\phi)$ is the number of ϕ-conjugacy classes. Rela...

conjecture | open | MathDB 页面 | 原文线索

V2-015754 | MATHDB-378466

The graded-module character conjecture for canonical basis elements

摘录 / EXCERPT: Let R be a reduced root system with equal parameters $t_s=t_\ell=:t$. For a weight λ, let C'_{λ} be the canonical basis element and let $P_{v_\mu,v_\lambda}(t)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-015755 | MATHDB-378545

Fibered homotopy K-theory isomorphism conjecture

摘录 / EXCERPT: Fibered homotopy K-theory isomorphism conjecture. The group G satisfies the (fibered) KH-isomorphism conjecture for R if the pair $(G,\mathcal{FIN})$ satisfies the (...

conjecture | open | MathDB 页面 | 原文线索

V2-015756 | MATHDB-378547

The q-multiplicity duality conjecture for root systems of type C

摘录 / EXCERPT: Let $U_{\lambda,\mu}(q)$ be the q-analogue of the multiplicity of the representation $V(\lambda)$ in $\frak{H}^{C_m}(\mu)$, and let $X_{\lambda,\mu}(q)$ be the corresponding crys...

conjecture | open | MathDB 页面 | 原文线索

V2-015757 | MATHDB-378562

The conjectural induction rule for big partitions with two conormal nodes

摘录 / EXCERPT: Induction branching conjecture. There are exactly two such conormal nodes, and

conjecture | open | MathDB 页面 | 原文线索

V2-015758 | MATHDB-378563

The conjectural restriction rule for completely splittable partitions

摘录 / EXCERPT: Restriction branching conjecture. Except for A, the only λ^A-normal node of residue α is B, and

conjecture | open | MathDB 页面 | 原文线索

V2-015759 | MATHDB-378564

The conjectural height-three induction rule

摘录 / EXCERPT: Height-three induction conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-015760 | MATHDB-378569

Enumeration conjectures for odd-order HTSASMs and DSASMs

摘录 / EXCERPT: Enumeration conjectures for odd-order HTSASMs and DSASMs. The number of $(2n+1)\times(2n+1)$ HTSASMs is

conjecture | open | MathDB 页面 | 原文线索

V2-015761 | MATHDB-378577

The tightness characterization of minimal growth rate

摘录 / EXCERPT: Let H be a handlebody with invariant measured lamination $\boldsymbol{\Lambda}$, and let the growth rate refer to the associated irreducible automorphism. A lamination is tight w...

conjecture | open | MathDB 页面 | 原文线索

V2-015762 | MATHDB-378586

The principal-subspace basis conjecture for the Leclerc-Thibon basis

摘录 / EXCERPT: Principal-subspace basis conjecture. The basis

conjecture | open | MathDB 页面 | 原文线索

V2-015763 | MATHDB-378600

The Kac-Moody group conjecture for the monoid of integrable representations

摘录 / EXCERPT: Let $\text{$bf g$}$ be a symmetrizable Kac-Moody algebra, let G be the (minimal) Kac-Moody group, and let M_{drr} be the monoid associated to the category of integrable...

conjecture | open | MathDB 页面 | 原文线索

V2-015764 | MATHDB-378625

Non-triviality of Bethe vectors in the Gaudin model

摘录 / EXCERPT: Let L be a tensor product of irreducible finite-dimensional representations of $sl_{N+1}(\mathbb{C})$, marked by complex parameters $z=(z_1,\ldots,z_n)$. A Bethe vector is a vect...

conjecture | open | MathDB 页面 | 原文线索

V2-015765 | MATHDB-378647

Gaitsgory-Kazhdan irreducibility conjecture for induced representations

摘录 / EXCERPT: Gaitsgory-Kazhdan irreducibility conjecture. The object $\Pi\in \operatorname{Rep}(\BG)$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-015766 | MATHDB-378649

Multiplicity one conjecture for restriction between reductive groups

摘录 / EXCERPT: Multiplicity one conjecture. Multiplicity one holds for restriction of irreducible admissible representations of $\widetilde{{\mathbf{G}}}(k)$ to ${\mathbf{G}}(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-015767 | MATHDB-378719

Pyber's logarithmic lower-bound conjecture for subgroup coverings of finite abelian groups

摘录 / EXCERPT: Pyber's conjecture. There exists a fixed constant $c>1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015768 | MATHDB-378744

The autocorrect-sequence conjecture for products of isomorphic simple groups

摘录 / EXCERPT: Let $\overrightarrow{u}$ be an autocorrect sequence, and let a $\overrightarrow{u}$-Engel automorphism mean an automorphism satisfying the corresponding Engel condition. Consider a...

conjecture | open | MathDB 页面 | 原文线索

V2-015769 | MATHDB-378745

The autocorrect-sequence conjecture for finite simple groups

摘录 / EXCERPT: Let $\overrightarrow{u}$ be an autocorrect sequence, and let a $\overrightarrow{u}$-Engel automorphism mean an automorphism satisfying the corresponding Engel condition. Let G be...

conjecture | open | MathDB 页面 | 原文线索

V2-015770 | MATHDB-378751

Simplicity conjecture for the subgroup of the Burger-Mozes group

摘录 / EXCERPT: Let Λ be the finitely presented $(6,6)$-group defined by the generators and relators in the preceding construction, and let $\Lambda_0<\Lambda$ be the subgroup under consid...

conjecture | open | MathDB 页面 | 原文线索

V2-015771 | MATHDB-378754

Horospherical Cauchy transform conjecture for holomorphic discrete-series parameters

摘录 / EXCERPT: Let ${\Bbb D}(X)$ be the algebra of G-invariant differential operators on X, viewed as holomorphic differential operators on $X_\mathbb{C}$. Let $\mathcal{O}(D_+)_\lambda$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-015772 | MATHDB-378758

Uniqueness conjecture for the spectral parameters of arbitrary-rank quantum orbits

摘录 / EXCERPT: Uniqueness conjecture. At an arbitrary value of p, this solution of the system is unique.

conjecture | open | MathDB 页面 | 原文线索

V2-015773 | MATHDB-378793

Conjecture on homogeneous products of faithful irreducible characters

摘录 / EXCERPT: Let G be a solvable group, and let $\psi,\varphi\in\operatorname{Irr}(G)$ be faithful, where $\operatorname{Irr}(G)$ denotes the irreducible characters of G. Suppose that ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-015774 | MATHDB-378798

Jacquet's distinguishedness conjecture for representations of ${\rm GL}_n(E)$

摘录 / EXCERPT: Let F be a p-adic field, let E/F be a degree-two extension with nontrivial Galois element σ, set $G={\rm GL}_n(E)$ and $H={\rm GL}_n(F)$, and let $\omega_{E/F}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015775 | MATHDB-378799

Flicker-Rallis base-change conjecture for distinguished representations

摘录 / EXCERPT: Let F be a p-adic field, let E/F be a degree-two extension, set $G={\rm GL}_n(E)$ and $H={\rm GL}_n(F)$, and let π be an irreducible admissible representation of G. A r...

conjecture | open | MathDB 页面 | 原文线索

V2-015776 | MATHDB-378800

Jacquet-Ye base-change conjecture for unitary distinguished representations

摘录 / EXCERPT: Let F be a p-adic field, let E/F be a degree-two extension, let ${\rm U}(n)$ be the corresponding unitary group, and let π be a representation of ${\rm GL}_n(E)$. A repre...

conjecture | open | MathDB 页面 | 原文线索

V2-015777 | MATHDB-378826

Irreducibility conjecture for the quotient representations of the double affine Hecke algebra

摘录 / EXCERPT: Let $I^{(k,r)}_m$ be the ideals in the polynomial representation V of the double affine Hecke algebra representation $ \mathcal{H}_n^{(k,r)}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-015778 | MATHDB-378829

Large-rank restriction conjecture for minimal affinizations

摘录 / EXCERPT: Let $\diamondsuit$ be one of the four allowed diagram shapes and let $\{\mathfrak{g}_n\}$ be a nonexceptional family of affine algebras associated with $\diamondsuit$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015779 | MATHDB-378842

Knutson-Tao saturation conjecture

摘录 / EXCERPT: Let $\widehat{G}$ be a connected semisimple complex group, let $\widehat{T}$ be a maximal torus of $\widehat{G}$, and let μ_1,μ_2,λ be weights of $\widehat{T}$. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-015780 | MATHDB-378858

Humphreys's conjecture on simple modules, component-group orbits, and left cells

摘录 / EXCERPT: Fix a regular block of $U_\chi(\mathfrak{g})$, with χ nilpotent, and let $\mathcal{S}$ be a complete set of nonisomorphic simple modules in this block. If χ corresponds t...

conjecture | open | MathDB 页面 | 原文线索

V2-015781 | MATHDB-378863

Feigin-Loktev fusion-product independence conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let $k\in\mathbb{Z}_{>0}$, and let $\lambda_s\in P^+$ for $1\leq s\leq k$. For pairwise distinct complex numbers $a_1,\ldots,a_k$ and pa...

conjecture | open | MathDB 页面 | 原文线索

V2-015782 | MATHDB-378870

The homogeneous-limit conjecture for eigenfunctions and generalized eigenfunctions

摘录 / EXCERPT: Homogeneous-limit conjecture. For $\alpha\in Q_+$, f_α remains finite at $s_1=\cdots=s_n=1$ if and only if $\alpha\in C(\Delta)$. Otherwise it diverges in this homogeneous l...

conjecture | open | MathDB 页面 | 原文线索

V2-015783 | MATHDB-378875

The Lubotzky-Weiss generating-set conjecture for expander Cayley graphs

摘录 / EXCERPT: Let Γ_i be a family of finite groups, and let Σ_i and Σ_i' be generating sets of Γ_i whose sizes are uniformly bounded. Lubotzky-Weiss generating-set co...

conjecture | refuted | MathDB 页面 | 原文线索

V2-015784 | MATHDB-378882

Farb's quasi-isometric rigidity conjecture for mapping class groups

摘录 / EXCERPT: Farb's quasi-isometric rigidity conjecture. There exists a homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-015785 | MATHDB-378896

The quaternion commutator conjecture

摘录 / EXCERPT: Let $Q_{p,l}$ be the set of quaternions under consideration, let $x,y\in Q_{p,l}$, and write $\langle x,y\rangle'$ for the commutator subgroup of the subgroup generated by x and...

conjecture | open | MathDB 页面 | 原文线索

V2-015786 | MATHDB-378917

The dimension conjecture for level-one Demazure and Weyl modules

摘录 / EXCERPT: Let ${\mathfrak g}$ be the simple Lie algebra under consideration, let λ be a dominant weight, and let $D(1,\lambda)$ and $W^1(\lambda)$ denote respectively the level-one D...

conjecture | open | MathDB 页面 | 原文线索

V2-015787 | MATHDB-378918

The fusion-product level-scaling conjecture for simple Lie algebras

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra, let $\lambda^1,\dots,\lambda^n$ be weights, let $V(\lambda^i)$ be the corresponding finite-dimensional ${\mathfrak g}$-modules, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015788 | MATHDB-378919

The fusion conjecture for higher-level Weyl modules

摘录 / EXCERPT: Let $\lambda^1,\dots,\lambda^n$ be weights, let $W^1(\lambda^i)$ be the level-one Weyl modules, and let $W^1(\lambda^i)^{[k]}$ denote their level-k constructions. Let $*$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-015789 | MATHDB-378920

The level-scaling conjecture for two-dimensional Weyl modules

摘录 / EXCERPT: Let ${\mathfrak g}={\mathfrak gl}_r$, let ξ be a partition, let $W^2(\xi)$ be the two-dimensional Weyl module, and let V_ξ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015790 | MATHDB-378921

Connectivity conjecture for the rigid-module and tilting-module graphs

摘录 / EXCERPT: Let Λ be the selfinjective preprojective algebra under consideration, let T be a basic maximal rigid Λ-module, and let $\operatorname{End}_\Lambda(T)$ be its endo...

conjecture | open | MathDB 页面 | 原文线索

V2-015791 | MATHDB-378923

Generalized convergence conjecture for invariant triple products

摘录 / EXCERPT: Generalized convergence conjecture. The same convergence assertion should hold for any semisimple group G with finite center and generic $n_0\in N$. This extension would generali...

conjecture | open | MathDB 页面 | 原文线索

V2-015792 | MATHDB-378949

The covariant representation-ring conjecture for infinite groups

摘录 / EXCERPT: Let G be a group and let F be a field of characteristic zero. Define ... where the colimit uses induction along inclusions of finite subgroups. Covariant representation-ring conj...

conjecture | open | MathDB 页面 | 原文线索

V2-015793 | MATHDB-378952

Flatness, PI degree, and center conjectures for generalized double affine Hecke algebras

摘录 / EXCERPT: Let $H_n(u,t)$ be the generalized double affine Hecke algebra associated with an affine Dynkin diagram $D=\widetilde D_4,\widetilde E_6,\widetilde E_7$, or $\widetilde E_8$, with p...

conjecture | open | MathDB 页面 | 原文线索

V2-015794 | MATHDB-378959

The jump-periodicity-cohomological-dimension equivalence conjecture

摘录 / EXCERPT: Let R be a commutative ring with unit, and let G be a discrete group without R-torsion. The group G has jump cohomology of height k over R if there is an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-015795 | MATHDB-378960

Higher braided symmetric powers conjecture for $U_q(gl_2(\mathbb{C}))$-modules

摘录 / EXCERPT: Let V_ℓ denote the $U_q(gl_2(\mathbb{C}))$-module indexed by $\ell\geq 0$, and let $S_\sigma^n V_\ell$ be its nth braided symmetric power. Higher braided symmetric powers co...

conjecture | open | MathDB 页面 | 原文线索

V2-015796 | MATHDB-378961

The solvable-radical sequence conjecture for finite groups

摘录 / EXCERPT: Solvable-radical sequence conjecture. There exists a sequence $u_n(x,y)$ such that $R(G)$ coincides with the set of elements $y\in G$ having the property that, for every $x\in G$,...

conjecture | open | MathDB 页面 | 原文线索

V2-015797 | MATHDB-378965

Lusztig's relative Kazhdan-Lusztig cell conjecture

摘录 / EXCERPT: Let W be a Coxeter group with weight function L, let $W_I\subseteq W$ be a parabolic subgroup, and let Y_I denote the corresponding set of distinguished coset representatives...

conjecture | open | MathDB 页面 | 原文线索

V2-015798 | MATHDB-379015

Finite generation of subobjects in the geometric Hecke category

摘录 / EXCERPT: Let $\Catgeom$ be the abelian category of objects equipped with compatible Hecke isomorphisms, and call an object finitely generated if it admits a surjection from an object of the...

conjecture | open | MathDB 页面 | 原文线索

V2-015799 | MATHDB-379016

Exactness and full faithfulness of the convolution Hecke functor

摘录 / EXCERPT: Let $\Catgeomdk$ be the geometric category defined in the surrounding text, let $\Pervsmkb$ be the stated target category, and let $\ConvHecke$ be the functor constructed there. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-015800 | MATHDB-379039

Lubotzky's bounded-generator expander conjecture for finite simple groups

摘录 / EXCERPT: Let (G_i) be the family of all non-abelian finite simple groups. A Cayley graph $\mathcal{C}(G_i,S_i)$ is the graph associated with the generating set S_i, and a family of grap...

conjecture | open | MathDB 页面 | 原文线索

V2-015801 | MATHDB-379044

Cluster character conjecture for tilted algebras

摘录 / EXCERPT: Cluster character conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-015802 | MATHDB-379072

Faithfulness conjecture for laminations arising from group actions

摘录 / EXCERPT: Let $\mathcal L$ be a lamination arising from a group action. A fidelity of $\mathcal L$ is a germ map into a path-connected space whose induced map on fundamental germ groupoids i...

conjecture | open | MathDB 页面 | 原文线索

V2-015803 | MATHDB-379139

The multiplicity-free holonomy conjecture for finite groups

摘录 / EXCERPT: Multiplicity-free holonomy conjecture. For every finite group H, there exists a Bieberbach group Γ with $\mathbb{Q}$-multiplicity-free holonomy representation...

conjecture | open | MathDB 页面 | 原文线索

V2-015804 | MATHDB-379144

The equivalence conjecture for the functor $\mathbb B^{\bullet}$

摘录 / EXCERPT: Let k be the base field, let F be the ambient field extension, and let G be its field automorphism group. Let $\mathcal A dm$ denote the category of admissible representation...

conjecture | open | MathDB 页面 | 原文线索

V2-015805 | MATHDB-379145

The irreducible-object conjecture for $\mathcal I_G$

摘录 / EXCERPT: Let F/k be the field extension in the source, let G be its field automorphism group, and let $\mathcal I_G$ be the category of homotopy-invariant smooth G-representations. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-015806 | MATHDB-379150

Local rigidity of generalized quasi-affine actions for higher-rank groups

摘录 / EXCERPT: Let G be a semi-simple Lie group with no compact factors and no simple factors isomorphic to $SO(1,n)$ or $SU(1,n)$, and let $\Gamma<G$ be a lattice. A generalized quasi-affine a...

conjecture | open | MathDB 页面 | 原文线索

V2-015807 | MATHDB-379152

Local rigidity conjecture for weakly irreducible translation actions

摘录 / EXCERPT: Let $G=G_1\times\cdots\times G_k$ be a semisimple Lie group with all factors non-compact, and let $\Lambda<H$ be a cocompact lattice, with $G<H$. A left translation action ρ o...

conjecture | open | MathDB 页面 | 原文线索

V2-015808 | MATHDB-379153

Local deformation conjecture for arithmetic isometric actions of complex hyperbolic lattices

摘录 / EXCERPT: Let ρ be an arithmetic isometric action of a lattice in $SU(1,n)$. Arithmetic-action deformation conjecture. Any sufficiently small perturbation of ρ is of the form obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-015809 | MATHDB-379154

Local rigidity question for boundary actions of complex hyperbolic lattices

摘录 / EXCERPT: Let a lattice in $SU(1,n)$ act on the boundary of complex hyperbolic space. Boundary-action rigidity question. Is the action of any lattice in $SU(1,n)$ on the boundary of complex...

open_problem | open | MathDB 页面 | 原文线索

V2-015810 | MATHDB-379198

The non-indexedness conjecture for the word problem of \(\mathbb Z^2\)

摘录 / EXCERPT: Let $\mathbb Z^2$ be the free abelian group of rank 2, and let its word problem be the language of words over a finite generating set that represent the identity element of...

conjecture | open | MathDB 页面 | 原文线索

V2-015811 | MATHDB-379200

The structure conjecture for tensor product semigroups

摘录 / EXCERPT: The structure conjecture. (1) If R is simply-laced, then

conjecture | open | MathDB 页面 | 原文线索

V2-015812 | MATHDB-379201

The singular-triple criterion for tensor product semigroups

摘录 / EXCERPT: Singular-triple criterion. There exists a triple $\sigma\in {\mathcal P}(G)\cap L^3$ with $Tr(\sigma)\in Q(R)$ and $\sigma\notin Tens(G)$ if and only if there exists a singular tri...

conjecture | open | MathDB 页面 | 原文线索

V2-015813 | MATHDB-379202

The boundary-stratum conjecture for tensor product semigroups

摘录 / EXCERPT: Boundary-stratum conjecture. There exists a subcone $S+\sigma\subset S$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-015814 | MATHDB-379290

Linear-growth Hilbert length conjecture for locally compact groups

摘录 / EXCERPT: Let G be a locally compact, compactly generated group. A Hilbert length function with linear growth on G is a Hilbert length function whose growth is linear. Linear-growth Hilb...

conjecture | open | MathDB 页面 | 原文线索

V2-015815 | MATHDB-379291

Flatness and Koszulity conjecture for preprojective algebras of non-ADET graphs

摘录 / EXCERPT: Let Γ be a connected graph with adjacency matrix A, and let $P({\bf \Gamma})$ be the associated preprojective algebra, constructed from a choice of field k and a modulat...

conjecture | open | MathDB 页面 | 原文线索

V2-015816 | MATHDB-379304

General characterization of arbitrarily distorted diffeomorphisms

摘录 / EXCERPT: General distortion conjecture. The element h is a distortion element in $\operatorname{Diff}^1(M)$ whose distortion function can be chosen to grow faster than any given function...

conjecture | open | MathDB 页面 | 原文线索

V2-015817 | MATHDB-379319

Centrality conjecture for shifted determinants

摘录 / EXCERPT: Let λ be a dominant weight, let V_λ be the corresponding representation space, and let $\Omega_\lambda(u)=\Omega_\lambda+u\otimes \operatorname{id}$. Using a basis...

conjecture | open | MathDB 页面 | 原文线索

V2-015818 | MATHDB-379324

The Isomorphism and Fibered Isomorphism conjectures for equivariant homology theories

摘录 / EXCERPT: Let ${\mathcal H}^?_*$ be an equivariant homology theory with values in R-modules, let G be a group, and let ${\mathcal C}$ be a family of subgroups of G. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015819 | MATHDB-379335

The natural conjecture for subgroups of direct products of limit groups

摘录 / EXCERPT: Direct-product conjecture. S has a subgroup of finite index that is itself a direct product of n or fewer limit groups.

conjecture | open | MathDB 页面 | 原文线索

V2-015820 | MATHDB-379372

The conjectural description of real discrete series for affine Hecke algebras of type B

摘录 / EXCERPT: Let $\mathcal{H}$ be the affine Hecke algebra with labels $q_2=q_1^m$ obtained from the parameters $q_{\check{\alpha}}$ as in the source, with $q_1>1$ and $q_2\geq 1$. Let W_0r b...

conjecture | open | MathDB 页面 | 原文线索

V2-015821 | MATHDB-379387

Geometric categorification conjecture for tensor products of \mathfrak{sl}_2-modules

摘录 / EXCERPT: Let W be the Weyl group, let $S_{\bf d}$ be the subgroup associated with the tensor-product datum ${\bf d}$, and let $H_{\bf d}^i=(B^{\bf d}\otimes C)^{W_i}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015822 | MATHDB-379391

Digne-Michel conjecture on Shintani descents and Fourier matrices

摘录 / EXCERPT: Let G be a finite reductive group with generalized Frobenius map F, let $\boldsymbol{G}$ be the corresponding algebraic group, and let $\boldsymbol{G}^{F^{\delta}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015823 | MATHDB-379404

Automorphic isolation conjecture for cohomological representations

摘录 / EXCERPT: Let G be a semisimple algebraic group over ${\mathbb Q}$, let $\mathfrak{q}$ be a θ-stable parabolic subalgebra, and let $L\subset G$ be its associated Levi subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015824 | MATHDB-379405

Arthur's infinitesimal-character conjecture for congruence spectra

摘录 / EXCERPT: Let G be a semisimple algebraic group over ${\mathbb Q}$, let Γ be a congruence subgroup, and let π be an irreducible representation of G occurring weakly in...

conjecture | open | MathDB 页面 | 原文线索

V2-015825 | MATHDB-379410

Generalized exponent conjecture for restrictions of representations

摘录 / EXCERPT: Let G be a real semisimple group and $H\subset G$ a real semisimple subgroup, with $K^H=H\cap K$ maximal compact in H and $(\mathfrak a^H_0)^+\subset\mathfrak a_0^+$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015826 | MATHDB-379413

Torus case of the Deligne-Lusztig restriction conjecture

摘录 / EXCERPT: Let W be the Weyl group, let $w\in W$, let $\dot w$ be a representative of w, and let $\psi:U_0\to\mathcal{O}^{\times}$ be a regular linear character. Torus restriction conject...

conjecture | open | MathDB 页面 | 原文线索

V2-015827 | MATHDB-379422

Root-system and finite-type conjecture for admissible sets of minimal infinite cluster quivers

摘录 / EXCERPT: Let Q be a minimal infinite cluster quiver, let S be an admissible set of arrows, and let Q_S and Λ_S be the associated quiver and algebra. Let t_S be the correspon...

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V2-015828 | MATHDB-379429

The positivity conjecture for local Braden-MacPherson sections

摘录 / EXCERPT: Let $\mathcal W$ be the Coxeter group with Bruhat order $<$ and length function l. For $x,y\in\mathcal W$ with $y<x$, let $B(x)^{[y]}$ denote the corresponding graded S-module...

conjecture | open | MathDB 页面 | 原文线索

V2-015829 | MATHDB-379430

Soergel's character conjecture for indecomposable projectives

摘录 / EXCERPT: Let $\mathcal W$ be the Coxeter group, let $\mathcal H$ be its Hecke algebra, and let $B(x)$ be the indecomposable projective object associated with $x\in\mathcal W$. For a graded...

conjecture | open | MathDB 页面 | 原文线索

V2-015830 | MATHDB-379471

The zero-intersection characterization of asymptotic shadows in the Thurston boundary

摘录 / EXCERPT: Zero-intersection characterization. For any $F\in\mathcal{PMF}$,

conjecture | open | MathDB 页面 | 原文线索

V2-015831 | MATHDB-379501

Perfect-basis realization for strongly positive parabolic bicrystals

摘录 / EXCERPT: Perfect-basis realization conjecture. Such a based module and linear map exist.

conjecture | open | MathDB 页面 | 原文线索

V2-015832 | MATHDB-379547

Stable one-dimensional sum conjecture for classical affine crystals

摘录 / EXCERPT: Let G be one of the classical groups and let $\mu^{(1)},...,\mu^{(r)}$ and λ index the corresponding q-analogue $\frak{D}_{\mu^{(1)},...,\mu^{(r)}}^{\lambda,G}(q)$ of a...

conjecture | open | MathDB 页面 | 原文线索

V2-015833 | MATHDB-379556

The Kirillov-Reshetikhin conjecture for tensor-product multiplicities

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra with Cartan matrix C, let ${\mathbf R}=\{a_p\omega_{i_p}:1\leq p\leq N\}$ be a collection of dominant integral weights, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015834 | MATHDB-379558

Invariant bilinear form conjecture for finite-dimensional irreducible $\boxtimes_q$-modules

摘录 / EXCERPT: Let V be a finite-dimensional irreducible $\boxtimes_q$-module of type 1, and let ω be the antiautomorphism of $\boxtimes_q$ from Lemma. Invariant-form conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-015835 | MATHDB-379634

Geck-Geck-Hiss parametrization conjecture for unipotent Brauer characters

摘录 / EXCERPT: Geck-Geck-Hiss conjecture. Assume that $cell$ is not “too small” and that the center of G is connected. For every $varphi\throw in \operatorname{UBr}_{cell}(G^F)$, there is a uni...

conjecture | open | MathDB 页面 | 原文线索

V2-015836 | MATHDB-379635

Birget's filling-length characterization of nondeterministic symmetric space

摘录 / EXCERPT: Birget's conjecture. There is a nondeterministic symmetric Turing machine accepting the language of words w representing 1 in G within space

conjecture | open | MathDB 页面 | 原文线索

V2-015837 | MATHDB-379636

Gersten's recursive upper bound for Dehn functions of linear groups

摘录 / EXCERPT: Gersten's conjecture. There is a recursive function $f\colon\mathbb{N}\to\mathbb{N}$, such that every finite presentation of a linear group has

conjecture | open | MathDB 页面 | 原文线索

V2-015838 | MATHDB-379642

Dipper-James conjecture on the centre of the symmetric-group Hecke algebra

摘录 / EXCERPT: Dipper-James conjecture. The centre of ${\mathcal H}$ is the set of symmetric polynomials in the Jucys-Murphy operators $L_1,\ldots,L_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-015839 | MATHDB-379648

Alperin's weight conjecture for blocks

摘录 / EXCERPT: Let ${\mathcal{O}}Gb$ be a block of a finite group G with defect group $D\neq 1$. For a chain ${\mathcal{S}}$ of subgroups ... let $l({\mathcal{S}})=n\geq 1$, let the sum run over...

conjecture | open | MathDB 页面 | 原文线索

V2-015840 | MATHDB-379649

Alperin-Auslander conjecture on stable equivalences

摘录 / EXCERPT: Let A and B be finite-dimensional algebras over an algebraically closed field, and suppose they are stably equivalent, meaning their stable module categories are equivalent. Al...

conjecture | open | MathDB 页面 | 原文线索

V2-015841 | MATHDB-379650

The gluing conjecture for splendid Rickard equivalences

摘录 / EXCERPT: Let G be a finite group, let D be an abelian Sylow ℓ-subgroup, let b and c be the principal block idempotents of kG and kH with $H=N_G(D)$, and form the groups and...

conjecture | open | MathDB 页面 | 原文线索

V2-015842 | MATHDB-379651

Broué's regular-element conjecture for splendid Rickard complexes

摘录 / EXCERPT: Assume the regular-element setting of the source: ${\mathbf L}={\mathbf T}=C_{\mathbf G}(D)$ is a torus, ${\mathcal{O}}Gb$ is the principal block, $\ell\nmid(q-1)$, F acts trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-015843 | MATHDB-379659

The generalized Steinberg invariant norm conjecture

摘录 / EXCERPT: Let G be the group in the paper, let ρ be an irreducible algebraic $\mathbb{Q}_p$-rational representation of G over K, and let π be a generalized Steinberg represent...

conjecture | open | MathDB 页面 | 原文线索

V2-015844 | MATHDB-379660

The non-vanishing conjecture for $B_{\xi,\zeta}$

摘录 / EXCERPT: Non-vanishing conjecture for $B_{\xi,\zeta}$. The Banach space $B_{\xi,\zeta}$ is non-zero. This extends the preceding Satake and unramified Langlands-functoriality construction to...

conjecture | open | MathDB 页面 | 原文线索

V2-015845 | MATHDB-379676

Conjecture on positive definiteness of the invariant-theoretic matrix Xi

摘录 / EXCERPT: For positive integers n and k, let $\operatorname{STab}((k^n))$ be the set of standard tableaux of rectangular shape (k^n). For each $T\in\operatorname{STab}((k^n))$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015846 | MATHDB-379693

Bounded-region conjecture for Whitehead-graph feature vectors

摘录 / EXCERPT: Bounded-region conjecture. The feature vectors of weights of the Whitehead graphs of elements from F are separated into bounded regions in the corresponding space. Each such regi...

conjecture | open | MathDB 页面 | 原文线索

V2-015847 | MATHDB-379705

The twisted Alexander polynomial detection conjecture for fibered 3-manifolds

摘录 / EXCERPT: Twisted Alexander polynomial detection conjecture. If, for every epimorphism $\alpha:\pi_1(N)\to G$ onto a finite group, $\Delta^{\alpha}_{N,\phi}$ is monic and

conjecture | open | MathDB 页面 | 原文线索

V2-015848 | MATHDB-379730

The endomorphism-algebra and tangle-invariant conjecture for Khovanov homology

摘录 / EXCERPT: For each natural number m, let T_{2m} be the full projective tilting module in the relevant parabolic category ${\mathcal O}$ for $\mathfrak{sl}_{2m}$, and let ${\mathcal H}_m$...

conjecture | open | MathDB 页面 | 原文线索

V2-015849 | MATHDB-379741

Alternating-group Beauville structure conjecture for prescribed hyperbolic types

摘录 / EXCERPT: Alternating-group Beauville structure conjecture. Let (r,s,t) and (r',s',t') be two hyperbolic types. Then almost all alternating groups ${\mathcal{A}}_n$ have an unmixed Beauv...

conjecture | open | MathDB 页面 | 原文线索

V2-015850 | MATHDB-379771

Coincidence conjecture for admissible elements in D^4

摘录 / EXCERPT: Coincidence conjecture. For every admissible sequence α, the elements e_α and $\tilde e_\alpha$, respectively $f_{\alpha 0}$ and $\tilde f_{\alpha 0}$, coincide in...

conjecture | open | MathDB 页面 | 原文线索

V2-015851 | MATHDB-379774

The multiplicity-free product conjecture for Hermitian symmetric spaces

摘录 / EXCERPT: Let $G_{{\Bbb R}}/K_{{\Bbb R}}$ be an irreducible Hermitian symmetric space of non-compact type, let $\frak g=\frak k\oplus\frak p^+\oplus\frak p^-$, let K be the complexificatio...

conjecture | open | MathDB 页面 | 原文线索

V2-015852 | MATHDB-379775

The Littlewood-Richardson conjecture for Hermitian symmetric spaces

摘录 / EXCERPT: Let $G_{\Bbb R}/K_{\Bbb R}$ be an irreducible Hermitian symmetric space of rank r, let $X=\frak p^+$, and let Λ be the set of partitions of length at most r. For...

conjecture | open | MathDB 页面 | 原文线索

V2-015853 | MATHDB-379782

Vogan's globalization conjecture for nilradical cohomology

摘录 / EXCERPT: Let $\mathfrak{p}$ be a very nice parabolic subalgebra with associated Levi subgroup L_{0}, stable Levi factor $\mathfrak{l}$, and nilradical $\mathfrak{n}$. Let M be a Harish-...

conjecture | open | MathDB 页面 | 原文线索

V2-015854 | MATHDB-379792

Bimodule decomposition conjecture for the quantum α-determinant cyclic module

摘录 / EXCERPT: Let $\mathcal{V}_{n,q}^{(\alpha)}$ be the $(\mathcal{U}_q(\mathfrak{gl}_n),\mathcal{H}_q(\mathfrak{S}_n))$-cyclic module generated by the quantum α-determinant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015855 | MATHDB-379793

Quantum determinant-permanent reciprocity conjecture

摘录 / EXCERPT: Let $\operatorname{Sing}_{n,q}^{\mathrm{per}}$ and $\operatorname{Sing}_{n,q}$ be the singular sets for the quantum permanent and quantum determinant, respectively. For...

conjecture | open | MathDB 页面 | 原文线索

V2-015856 | MATHDB-379794

Partition-function conjectures for quantum determinant and permanent singularities

摘录 / EXCERPT: Let $\mathcal{P}$ be the set of partitions, and define the partition functions ... Here f^λ is the dimension of the symmetric-group module indexed by λ. A singular po...

conjecture | open | MathDB 页面 | 原文线索

V2-015857 | MATHDB-379819

Conjecture that torsion has density zero in the mapping class group

摘录 / EXCERPT: For a fixed finite generating set of ${\rm Mod}_g$, define the density of a subset S by ... Let $\Tor_g$ denote the set of finite-order elements of ${\rm Mod}_g$. Torsion-density c...

conjecture | open | MathDB 页面 | 原文线索

V2-015858 | MATHDB-379861

Independence of affine fusion products from evaluation parameters

摘录 / EXCERPT: Fusion-product independence conjecture. The corresponding fusion product does not depend on Z as a representation of $\g\otimes\mathbb{C}[u]$.

conjecture | open | MathDB 页面 | 原文线索

V2-015859 | MATHDB-379873

Brady-McCammond-Mühlherr-Neumann twist-equivalence conjecture for Coxeter generating sets

摘录 / EXCERPT: Brady-McCammond-Mühlherr-Neumann conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-015860 | MATHDB-379874

Mühlherr's sharp-angled automorphism conjecture for Coxeter generating sets

摘录 / EXCERPT: Mühlherr's conjecture. There exists a reflection-preserving automorphism α of W such that $\alpha(S)$ is sharp-angled with respect to S'.

conjecture | open | MathDB 页面 | 原文线索

V2-015861 | MATHDB-379888

The positive-power lower-bound conjecture for intermediate growth

摘录 / EXCERPT: Positive-power lower-bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-015862 | MATHDB-379890

The commuting infinite subgroups conjecture for intermediate-growth groups

摘录 / EXCERPT: Commuting infinite subgroups conjecture. The group G contains two infinite subgroups H_1 and H_2 which commute with each other:

conjecture | open | MathDB 页面 | 原文线索

V2-015863 | MATHDB-379898

Lusztig's multiplicity conjecture for baby Verma modules

摘录 / EXCERPT: Let R be a reduced, irreducible, finite root system, X its weight lattice, and k a field whose characteristic is bigger than the Coxeter number of R. Let $\mathfrak g$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-015864 | MATHDB-379899

Braden-MacPherson stalk-rank conjecture

摘录 / EXCERPT: Let R be a reduced, irreducible, finite root system, let $\widehat{\mathcal W}$ be its affine Weyl group, and let k be a field. Form the associated moment graph...

conjecture | open | MathDB 页面 | 原文线索

V2-015865 | MATHDB-379902

Conjecture on homological dimensions and Lusztig's a-function in category O

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semi-simple complex finite-dimensional Lie algebra, let W be its Weyl group with longest element w_0, and let $\mathfrak{t}(w)$ and $\mathfrak{i}(w)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-015866 | MATHDB-379918

The type-B LLT conjecture for the p_1-orbit

摘录 / EXCERPT: Let $I=\{p_1^n;n\in\mathbb Z_{\mathrm{odd}}\}$, assume that none of ±1 and $\pm p_0$ lies in I, and let $\theta(a)=a^{-1}$. Let $\mathfrak g_I$ be the associated Lie algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-015867 | MATHDB-379938

Lam-Postnikov-Pylyavskyy's alcoved-polytope conjecture for sl_n character products

摘录 / EXCERPT: Let $\alpha_{ij}=e_i-e_j$ be the roots of the type A root system. An alcoved polytope is a polytope whose faces lie in hyperplanes $\langle\alpha_{ij},\tau\rangle=m$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-015868 | MATHDB-379948

Self-dual Specht-filtered modules are trivial source modules

摘录 / EXCERPT: Trivial-source conjecture. Indecomposable, self-dual $k\Sigma_d$-modules with Specht filtrations are trivial source modules. This is a proposed weaker replacement for the earlier c...

conjecture | open | MathDB 页面 | 原文线索

V2-015869 | MATHDB-379951

Nonzero-determinant conjecture for the Suzuki-group recognition equations

摘录 / EXCERPT: Let $P,Q\in\mathcal{O}$, let $h\in G$, and let $x\in\operatorname{GL}(4,q)$. Define ... Regard the four equations in as simultaneous linear equations in the variables α^{nt} f...

conjecture | open | MathDB 页面 | 原文线索

V2-015870 | MATHDB-379954

Conjecture on the multiplicative compatibility of the geometric isomorphisms

摘录 / EXCERPT: Multiplicative compatibility conjecture. The isomorphism $\operatorname{End}_{\mathfrak{g}}(P)\cong H^*(\psi(P))$ is a ring homomorphism, and the isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-015871 | MATHDB-379987

The totally geodesic image conjecture for simplicial embeddings of pants graphs

摘录 / EXCERPT: Let Σ_{1} and Σ_{2} be compact, orientable surfaces, and let $\mathcal{P}(\Sigma)$ denote the pants graph of a surface Σ. A subgraph is totally geodesic if eve...

conjecture | open | MathDB 页面 | 原文线索

V2-015872 | MATHDB-379995

The finite-group non-reflection conjecture for the 2-polarization property

摘录 / EXCERPT: Finite-group non-reflection conjecture. Then V does not have the 2-polarization property.

conjecture | open | MathDB 页面 | 原文线索

V2-015873 | MATHDB-380034

Bounded packing conjecture for subgroups of virtually polycyclic groups

摘录 / EXCERPT: Bounded packing conjecture for virtually polycyclic groups. Each subgroup of P has bounded packing in P.

conjecture | open | MathDB 页面 | 原文线索

V2-015874 | MATHDB-380062

Integrality conjecture for functional dimensions

摘录 / EXCERPT: Let G be a connected unipotent algebraic group, let e be a minimal idempotent of ${\mathscr D}_G(G)$, and let its functional dimension be $d_e=(\dim G-n_e)/2$. The integrality...

conjecture | open | MathDB 页面 | 原文线索

V2-015875 | MATHDB-380065

Phillips's tracial Rokhlin property conjecture for outer finite-group actions

摘录 / EXCERPT: Let A be a unital Kirchberg algebra, and let $\alpha \colon G \to \operatorname{Aut}(A)$ be an action of a finite group G on A such that α_g is outer for all...

conjecture | open | MathDB 页面 | 原文线索

V2-015876 | MATHDB-380095

Homological dimension conjecture for virtually cyclic classifying spaces

摘录 / EXCERPT: Let Γ be a discrete subgroup of a Lie group that is not virtually cyclic. Write $vcd(\Gamma)$ for its virtual cohomological dimension, and let $B_{\mathcal{VC}}\Gamma$ denot...

open_problem | open | MathDB 页面 | 原文线索

V2-015877 | MATHDB-380105

W_n-module conjecture for consecutive quotients of the free associative Lie algebra

摘录 / EXCERPT: Wn-module conjecture. Every quotient $A_{n,k}$ for $k>1$ is a W_n-module in the class $\mathcal C$; the Lie bracket on $gr^+(A_n)$ is W_n-equivariant.

conjecture | open | MathDB 页面 | 原文线索

V2-015878 | MATHDB-380110

Lusztig's conjecture for uniform almost characters in type F4

摘录 / EXCERPT: Let $\operatorname{\bf{G}}_0$ be a simple group of type F_4, and let F_0 be the generalized Frobenius map defining a Ree group of type F_4. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015879 | MATHDB-380115

Moore and Russell's character bound for balanced Young diagrams

摘录 / EXCERPT: Let $C>0$. A Young diagram λ with n boxes has at most $C\sqrt{n}$ rows and columns, and let $\pi\in S_n$ be a permutation. Moore and Russell's character bound. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-015880 | MATHDB-380127

Branching component graph product conjecture for horizontally concatenated partitions

摘录 / EXCERPT: Branching component graph product conjecture. The branching component graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-015881 | MATHDB-380139

Okounkov's log-concavity conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $c_{\mu, \nu}^{\lambda}$ denote the Littlewood-Richardson coefficient associated to partitions λ, μ, and ν. For a partition α and a positive integer N,...

conjecture | open | MathDB 页面 | 原文线索

V2-015882 | MATHDB-380140

Log-concavity conjecture for parabolic Kostka numbers

摘录 / EXCERPT: Let λ be a partition and let $R=((m_1^{l_1}), \dots, (m_k^{l_k}))$ be a sequence of rectangular partitions. For an integer $N \geq 1$, define ... Let $K_{\lambda,R}$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-015883 | MATHDB-380151

Three-conjugate characterization of the solvable radical

摘录 / EXCERPT: Let G be a finite group, and let $R(G)$ denote its solvable radical, the largest solvable normal subgroup of G. For $g,a,b,c\in G$, consider the subgroup generated by g and i...

conjecture | open | MathDB 页面 | 原文线索

V2-015884 | MATHDB-380159

The rank conjecture for groups recognised by Cho automata

摘录 / EXCERPT: Rank conjecture. If the word problem of H is accepted by a $2k$-counter or $(2k+1)$-counter Cho automaton, then H is virtually free abelian of rank k.

conjecture | open | MathDB 页面 | 原文线索

V2-015885 | MATHDB-380162

The unique-norm integral Fourier matrix conjecture

摘录 / EXCERPT: Unique-norm integral Fourier matrix conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-015886 | MATHDB-380174

The group-algebra and representation-count polynomiality conjecture for finite O-modules

摘录 / EXCERPT: Let $\mathfrak{o}$ be a discrete valuation ring with maximal ideal $\mathfrak{p}$ and residue-field cardinality $q=|\mathfrak{o}/\mathfrak{p}|$. For a partition $\lambda\in\Lambda$...

conjecture | open | MathDB 页面 | 原文线索

V2-015887 | MATHDB-380175

The degree bound conjecture for primitive representations of rectangular finite O-modules

摘录 / EXCERPT: Let $\mathfrak{o}$ be a discrete valuation ring with maximal ideal $\mathfrak{p}$ and residue-field cardinality $q=|\mathfrak{o}/\mathfrak{p}|$. For positive integers ℓ and...

conjecture | open | MathDB 页面 | 原文线索

V2-015888 | MATHDB-380187

The Hilbertian-projective implies pro-free conjecture

摘录 / EXCERPT: Hilbertian-projective implies pro-free conjecture. If K is Hilbertian and projective, then G_K is profree.

conjecture | open | MathDB 页面 | 原文线索

V2-015889 | MATHDB-380204

The finite-group criterion for holonomicity of quarter-plane walk generating functions

摘录 / EXCERPT: Let ${\mathcal Y}$ be a step set for walks in the quarter plane, and let $Q_{\mathcal Y}$ denote its generating function. Assume that the step set group ${\mathcal Y}$ is not singu...

conjecture | open | MathDB 页面 | 原文线索

V2-015890 | MATHDB-380218

The pattern form conjecture for $(G_{1,1},X)$

摘录 / EXCERPT: Let $(G_{1,1},X)$ be the group-space pair whose patterns are being studied, and let a pattern be a sequence encoded by parenthesized blocks; within a block, $(0,-1)$ or $(1,0)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-015891 | MATHDB-380227

The strict inequality conjecture for primitive detection degrees

摘录 / EXCERPT: Strict inequality conjecture. If G is not p-central, then

conjecture | open | MathDB 页面 | 原文线索

V2-015892 | MATHDB-380228

Lower-bound and minimizer conjecture for Lie algebra representation potentials

摘录 / EXCERPT: Let $\mathfrak{a}$ be a simple Lie algebra, let ${\mathbf d}\geq 1$, and let $\psi_{\mathbf d}$ be the real-valued polynomial obtained by restricting the potential Ψ to the sp...

conjecture | open | MathDB 页面 | 原文线索

V2-015893 | MATHDB-380235

Reduction-lattice conjecture for Weyl modules

摘录 / EXCERPT: Let $\mathbb F$ be an algebraically closed field of positive characteristic, let $\mathbb A$ be the coefficient ring used for reduction modulo the characteristic, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015894 | MATHDB-380252

The analytical zero-divisor conjecture for torsion-free groups

摘录 / EXCERPT: Analytical zero-divisor conjecture. If Γ is torsion-free, then $\lambda_{\Gamma}(f)$ is injective for every non-zero $f\in\mathbf{C}\Gamma$. This is a classical conjecture c...

conjecture | open | MathDB 页面 | 原文线索

V2-015895 | MATHDB-380253

The affine-action dichotomy for amenable groups

摘录 / EXCERPT: Let G be an amenable group, and let λ_G denote its left regular representation. An affine action of G with linear part λ_G has an affine action map whose linear...

conjecture | open | MathDB 页面 | 原文线索

V2-015896 | MATHDB-380278

A linear derived-length bound for products of conjugacy classes

摘录 / EXCERPT: Let G be a finite solvable group and let A be a conjugacy class of G. Write $\eta(AA^{-1})$ for the number of conjugacy classes occurring in the product AA^{-1}, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015897 | MATHDB-380314

The Tame-Wild Dichotomy for coalgebras

摘录 / EXCERPT: Let K be an algebraically closed field and let C be a K-coalgebra. The coalgebra C is of tame comodule type if its finite-dimensional indecomposable comodules are parametri...

conjecture | open | MathDB 页面 | 原文线索

V2-015898 | MATHDB-380318

Absolute rank gradient conjecture for lattices in higher-rank Lie groups

摘录 / EXCERPT: Let Γ be a lattice in a higher-rank Lie group, and let its absolute rank gradient be the infimum of $(d(H)-1)/[\Gamma:H]$ over finite-index subgroups $H\leq\Gamma$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-015899 | MATHDB-380320

Kapovich-Potyagailo-Vinberg's noncoherence conjecture for arithmetic lattices

摘录 / EXCERPT: Let $\Gamma\subset \operatorname{Mob}(\mathbb S^n)$ be an arithmetic lattice with $n\ge 3$. Kapovich-Potyagailo-Vinberg's conjecture. The group Γ is noncoherent: it contains...

conjecture | open | MathDB 页面 | 原文线索

V2-015900 | MATHDB-380323

The cohomological-dimension and critical-exponent conjecture

摘录 / EXCERPT: Let Γ be a Kleinian group in $\operatorname{Mob}(\mathbb S^n)$ without parabolic elements, let $cd(\Gamma)$ denote its cohomological dimension, and let $\delta(\Gamma)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-015901 | MATHDB-380338

Discrete faithful representation conjecture for cycle Artin groups in irreducible symmetric spaces

摘录 / EXCERPT: Let $X=G/K$ be an irreducible symmetric space of rank at least two, let F be a maximal flat in X, and let $p_0\in F$. A geodesic in F through p_0 is singular if it has the...

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V2-015902 | MATHDB-380355

The connectedness problem for cluster tilting graphs of submodule categories

摘录 / EXCERPT: Connectedness problem. For every $w\in W$, the cluster tilting graph of $\operatorname{Sub}\nolimits\Lambda/I_w$ is connected.

open_problem | open | MathDB 页面 | 原文线索

V2-015903 | MATHDB-380359

Simplicity of the neutral-element stabilizer in a simple Moufang loop

摘录 / EXCERPT: Let L be a nonassociative simple Moufang loop and let $H=\operatorname{Mlt}(L)_e$ be the stabilizer of the neutral element in the multiplication group of L. Simplicity conjectu...

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V2-015904 | MATHDB-380360

A polynomial bound for multiplication groups of Moufang loops

摘录 / EXCERPT: Let n be the order of a Moufang loop, and let $\operatorname{Mlt}(L)$ denote its multiplication group. Polynomial-bound conjecture. The function ... solves the problem of finding a...

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V2-015905 | MATHDB-380361

The quarter-distance isomorphism conjecture for Moufang 2-loops

摘录 / EXCERPT: Let $L=(G,\circ)$ and $M=(G,*)$ be Moufang 2-loops on the same set G of order n, and define their distance by ... Quarter-distance isomorphism conjecture. If $d(\circ,*)<n^2/4$...

conjecture | open | MathDB 页面 | 原文线索

V2-015906 | MATHDB-380375

The property (T) rank-gradient conjecture

摘录 / EXCERPT: Let Γ be an infinite group with Kazhdan property (T), and let (Γ_n) be a normal chain in Γ with trivial intersection. Write $\mathrm{RG}(\Gamma,(\Gamma_n))$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-015907 | MATHDB-380380

The Ahlfors Measure Conjecture for mapping class group limit sets

摘录 / EXCERPT: Let S be a surface, let $\operatorname{Mod}(S)$ be its mapping class group, and let Λ_G denote the limit set of a subgroup $G\leq \operatorname{Mod}(S)$ in the projective...

conjecture | open | MathDB 页面 | 原文线索

V2-015908 | MATHDB-380424

The modular decomposition conjecture for principal series representations

摘录 / EXCERPT: Let S be a surface with holes, let S' be obtained by cutting S along a simple loop, and let $\gamma_\pm$ be the two new boundary components. Let ${\bf V}_{G,S;\lambda}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-015909 | MATHDB-380436

Katsylo's birational equivalence conjecture for generically free modules

摘录 / EXCERPT: Katsylo's conjecture. The following properties are equivalent: $\dim V=\dim W$; and there exists a G-equivariant birational map

conjecture | open | MathDB 页面 | 原文线索

V2-015910 | MATHDB-380441

Conjecture on diameters of irreducible spherical actions of large cohomogeneity

摘录 / EXCERPT: Large-cohomogeneity diameter conjecture. For every $\epsilon>0$ and all sufficiently large k with $n>k$,

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V2-015911 | MATHDB-380442

Finiteness conjecture for hyperbolic spaces of irreducible higher-rank lattices

摘录 / EXCERPT: Finiteness conjecture. If Γ is an irreducible lattice in a higher rank Lie group (or in a nontrivial product of locally compact groups) G, there are finitely many Gromov h...

conjecture | open | MathDB 页面 | 原文线索

V2-015912 | MATHDB-380443

Standard-action conjecture for $SL(2)$ over rings of integers

摘录 / EXCERPT: Standard-action conjecture. Every quasi-action by Γ on a Gromov hyperbolic metric space either has an invariant horoball or is standard.

conjecture | open | MathDB 页面 | 原文线索

V2-015913 | MATHDB-380450

Connectedness conjecture for the Hasse diagram of classical tilting modules

摘录 / EXCERPT: Let $B=\operatorname{End}_\Lambda(T_M)$, let ${\mathcal T}_B^{\rm cl}$ denote the relevant class of classical tilting B-modules, and let ${\mathcal F}(\Delta)$ be the category de...

conjecture | open | MathDB 页面 | 原文线索

V2-015914 | MATHDB-380451

Maximality conjecture for the symmetric group in the quantum permutation group

摘录 / EXCERPT: Let $\mathcal Q_n$ be the quantum permutation group on n points, containing the classical symmetric group S_n. Maximality conjecture. The symmetric group S_n is maximal in th...

conjecture | open | MathDB 页面 | 原文线索

V2-015915 | MATHDB-380473

The conjecture identifying diagrammatic and exceptional-point irreducible 2BTL representations

摘录 / EXCERPT: 2BTL representation-identification conjecture. For $n>0$, the irreducible representation $W^{(N,n)}_{\epsilon_1,\epsilon_2}$ is equivalent to $V^{(N,n)}_{\epsilon_1,\epsilon_2}$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-015916 | MATHDB-380474

The categorical McKay correspondence conjecture for finite simply-laced Dynkin diagrams

摘录 / EXCERPT: Categorical McKay correspondence conjecture. There exists such a category and functor satisfying all of the following: $\mathcal{D}$ is 2-periodic, with $T^2=\operatorname{id}$; ev...

conjecture | open | MathDB 页面 | 原文线索

V2-015917 | MATHDB-380483

Stanley's positivity conjecture for Jack character polynomials on multirectangular partitions

摘录 / EXCERPT: Let ${\text{\boldmath p}}=(p_1,\ldots,p_m)$ and ${\text{\boldmath q}}=(q_1,\ldots,q_m)$ be sequences of positive integers with $q_1\geq\cdots\geq q_m$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015918 | MATHDB-380487

Kaplansky's idempotent conjecture for group rings

摘录 / EXCERPT: Let R be an integral domain and let G be a torsionfree group. An idempotent in the group ring RG is an element e satisfying $e^2=e$. Kaplansky's conjecture. Every idempoten...

conjecture | open | MathDB 页面 | 原文线索

V2-015919 | MATHDB-380507

Generation conjecture for invariant operators on symmetric tensor densities

摘录 / EXCERPT: Generation conjecture. The algebra of all invariant operators for the $sp(2n+2)$ action on R_{δ} is generated by the operators X, $i(\alpha)$ and the projectors onto the s...

conjecture | open | MathDB 页面 | 原文线索

V2-015920 | MATHDB-380509

Non-positive-curvature A_2 freeway basis conjecture

摘录 / EXCERPT: Non-positive-curvature A_2 freeway basis conjecture. The set of A_2 freeways of non-positive curvature having these points as endpoints is a basis for

conjecture | open | MathDB 页面 | 原文线索

V2-015921 | MATHDB-380510

Non-positive-curvature C_2 freeway basis conjecture

摘录 / EXCERPT: Non-positive-curvature C_2 freeway basis conjecture. A collection of C_2 freeways of non-positive curvature, with one member in each switching class, constitutes a basis for

conjecture | open | MathDB 页面 | 原文线索

V2-015922 | MATHDB-380528

Cohomological globalization and duality conjecture for Harish-Chandra modules

摘录 / EXCERPT: Let G be a connected semisimple Lie group with finite center, let ${\bf g}$ be its Lie algebra, and let K be a maximal compact subgroup. Let $(\pi,V_{\pi,K})$ be a Harish-Chand...

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V2-015923 | MATHDB-380538

The forcing flexibility conjecture for automorphism-tower heights

摘录 / EXCERPT: Let V be the ground model. For a centreless group $G\in V$, let $\tau(G)$ be the least ordinal α such that the automorphism tower satisfies $G_\beta=G_\alpha$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-015924 | MATHDB-380541

Normalizer Conjecture for closed irreducible 3-manifold groups

摘录 / EXCERPT: Let M be a closed, connected, ${\mathbf{P}^2}$-irreducible 3-manifold with infinite fundamental group. For a subgroup $H\leq\pi_1(M)$, write ... for its normalizer. Normalizer Co...

conjecture | open | MathDB 页面 | 原文线索

V2-015925 | MATHDB-380542

Rational subgroup conjecture for closed irreducible 3-manifold groups

摘录 / EXCERPT: Let M be a closed, connected, ${\mathbf{P}^2}$-irreducible 3-manifold with infinite fundamental group. Rational subgroup conjecture. Every subgroup of $\pi_1(M)$ which embeds i...

conjecture | open | MathDB 页面 | 原文线索

V2-015926 | MATHDB-380578

Conjecture on A_f stratifications for polar representations

摘录 / EXCERPT: Let $G\,|\,V$ be a polar representation, let $f:V\to Q$ be the quotient map, and set ... An A_f stratification is the stratification condition described in Section 2.2.1. The Af st...

conjecture | open | MathDB 页面 | 原文线索

V2-015927 | MATHDB-380605

Monotonicity conjecture for generalized Kostka polynomials

摘录 / EXCERPT: Let $R=R(\eta,\gamma)$ be a dominant sequence of rectangles, and let R' be obtained from R by replacing a subsequence ... by ... where $|\alpha|=|\beta|$ and α^+ dominates...

conjecture | open | MathDB 页面 | 原文线索

V2-015928 | MATHDB-380627

The universal formula for the quantum toroidal central coefficient

摘录 / EXCERPT: Universal formula conjecture. The formula

conjecture | open | MathDB 页面 | 原文线索

V2-015929 | MATHDB-380646

The corrected saturation conjecture for semisimple Lie groups

摘录 / EXCERPT: Saturation conjecture for semisimple Lie groups. If $\lambda+\mu+\nu$ annihilates all elements of T with semisimple centralizer, then

conjecture | open | MathDB 页面 | 原文线索

V2-015930 | MATHDB-380661

The abelian rank conjecture for homologically trivial group actions on 4-manifolds

摘录 / EXCERPT: Abelian rank conjecture. The group G must be abelian of rank at most 2.

conjecture | open | MathDB 页面 | 原文线索

V2-015931 | MATHDB-380666

Displacement-angle characterization of the negative mapping class semigroup

摘录 / EXCERPT: Let $\operatorname{Map}_{g,1}$ be the mapping class group of a genus-g surface with one marked point, and let $\operatorname{Map}_{g,1}(-)\subset\operatorname{Map}_{g,1}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-015932 | MATHDB-380669

The computability conjecture for normalized phase-function integrals

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra with simple roots α_i, let V_1,V_2 be representations of $\mathfrak{g}$ with highest weights λ_1,λ_2, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015933 | MATHDB-380670

The uniqueness conjecture for critical points of Bethe phase functions

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let V_1,V_2 be representations of $\mathfrak{g}$, and let Φ be the corresponding phase function in variables assigned to the simp...

conjecture | open | MathDB 页面 | 原文线索

V2-015934 | MATHDB-380671

The Bethe-vector length conjecture

摘录 / EXCERPT: Let V_1,V_2 be representations of a simple Lie algebra, let X be a Bethe vector obtained from a critical point t^0 of the corresponding phase function Φ, and let B be...

conjecture | open | MathDB 页面 | 原文线索

V2-015935 | MATHDB-380679

The saturation conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let n be a positive integer, let $clambda$, cmu, and cnu be partitions of length at most n, and let T_n consist of the triples for which V_ν occurs in...

conjecture | solved | MathDB 页面 | 原文线索

V2-015936 | MATHDB-380685

The composition-factor multiplicity conjecture for Kac modules of \(\mathfrak{gl}(m/n)\)

摘录 / EXCERPT: The composition-factor multiplicity conjecture. For each of the 2^r integral dominant weights λ_θ, $a_{\lambda_\theta,\mu}=1$, and $a_{\lambda,\mu}=0$ for every othe...

conjecture | open | MathDB 页面 | 原文线索

V2-015937 | MATHDB-380686

The inverse Kazhdan-Lusztig matrix conjecture for \(\mathfrak{gl}(m/n)\)

摘录 / EXCERPT: Let μ be integral dominant and r-fold atypical with respect to the roots $\gamma_1<\gamma_2<\cdots<\gamma_r$ ($\gamma_i\in\Delta_{1,+}$). Let λ_θ, for...

conjecture | open | MathDB 页面 | 原文线索

V2-015938 | MATHDB-380698

Cherlin-Zil'ber conjecture for tame simple K^*-groups

摘录 / EXCERPT: Let G be an infinite simple tame K^*-group, where a K^*-group is a group of finite Morley rank in which every proper definable subgroup is a K-group, and a K-group is one...

conjecture | open | MathDB 页面 | 原文线索

V2-015939 | MATHDB-380709

The fermionic-form identity between M and N

摘录 / EXCERPT: M=N conjecture. The computer experiments suggest

conjecture | open | MathDB 页面 | 原文线索

V2-015940 | MATHDB-380722

Adem's persistence conjecture for torsion in finite-group cohomology

摘录 / EXCERPT: Throughout, let $H^i(S)=H^i(S;\mathbb{Z})$ denote the integral cohomology of a finite group S. For a finite p-group P, let $e(P)$ be the exponent of...

conjecture | open | MathDB 页面 | 原文线索

V2-015941 | MATHDB-380723

Nonexistence conjecture for compact Clifford-Klein forms of SO(2,2m+1)/SU(1,m)

摘录 / EXCERPT: Nonexistence conjecture for $\operatorname{SO}(2,2m+1)/\operatorname{SU}(1,m)$. The homogeneous space

conjecture | open | MathDB 页面 | 原文线索

V2-015942 | MATHDB-380784

Desbois' roof expectation conjecture for locally free groups

摘录 / EXCERPT: Let ${\cal LF}_n^+$ be the locally free semigroup (the heap) and ${\cal LF}_n$ the locally free group (the colored heap). For a random walk word w_N, let $T(w_N)$ denote its roof...

conjecture | open | MathDB 页面 | 原文线索

V2-015943 | MATHDB-380785

The locally free group drift conjecture

摘录 / EXCERPT: Let ${\cal LF}_n$ be a locally free group, and consider its random walk with word length K and drift l, the limiting expected normalized word length. Let $T(w)$ be the roof of...

conjecture | open | MathDB 页面 | 原文线索

V2-015944 | MATHDB-380849

The adiabatic limit conjecture for Hamiltonian vortices

摘录 / EXCERPT: Let $\mu:M\to\mathfrak{g}$ be proper, let 0 be a regular value of μ, assume that $\mu^{-1}(0)$ is nonempty, and suppose that G acts freely on $\mu^{-1}(0)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-015945 | MATHDB-380883

Classification of irreducible modules in a Schur-Weyl category

摘录 / EXCERPT: Let ${\cal SW}(V)$ be the Schur-Weyl category, let V_{λ} denote its distinguished objects, and let ${\bf g}=sl(V_S)$ act on these objects. A linear map between two objects...

conjecture | open | MathDB 页面 | 原文线索

V2-015946 | MATHDB-380892

Abelianization conjecture for the stable symplectic derivation Lie algebra

摘录 / EXCERPT: Let ${\mathfrak h}^{\mathbb Q}_{g,1}$ be the rational graded Lie algebra of symplectic derivations, let $H_{\mathbb Q}$ be the rational symplectic module, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-015947 | MATHDB-380894

Infinite-dimensional bounded cohomology conjecture for mapping class groups

摘录 / EXCERPT: Let ${\cal M}_g$ be the mapping class group and let $H_b^2({\cal M}_g)$ denote its second bounded cohomology. Bounded-cohomology conjecture. For every genus g, the comparison map...

conjecture | open | MathDB 页面 | 原文线索

V2-015948 | MATHDB-380895

Non-uniform-perfectness conjecture for mapping class groups

摘录 / EXCERPT: Let ${\cal M}_g$ be the mapping class group. Recall that a group ${\varGamma}$ is uniformly perfect if there is a natural number N such that every element of ${\varGamma}$ is a p...

conjecture | solved | MathDB 页面 | 原文线索

V2-015949 | MATHDB-380948

Rouquier's graded decomposition-number conjecture for v-Schur algebras

摘录 / EXCERPT: Let ${\cal S}_m$ be the v-Schur algebra, let $W(\lambda)$ be the Weyl module for a partition λ of m, and let $L(\mu)$ be the irreducible module corresponding to a parti...

conjecture | open | MathDB 页面 | 原文线索

V2-015950 | MATHDB-380954

Schoutens's conjecture on the Yangian structure of an affine level-1 module

摘录 / EXCERPT: Let ${\cal L}(\Lambda_k)$ be the level 1 integrable module of $\widehat{sl}_n$, and let the character decomposition in equation (26a) be the decomposition obtained from the irred...

conjecture | open | MathDB 页面 | 原文线索

V2-015951 | MATHDB-380955

Yangian action conjecture for the twisted affine module ${\cal L}(\Lambda_n)$

摘录 / EXCERPT: Let ${\cal L}(\Lambda_n)$ be the level 1 integrable module of $A^{(2)}_{2n}$, and let $Y(B_n)$ be the Yangian of type B_n. Suppose ... where BS is the set of border strips with...

conjecture | open | MathDB 页面 | 原文线索

V2-015952 | MATHDB-380965

Commutation of U_0 with the positive Heisenberg subalgebra

摘录 / EXCERPT: Let H_{+} be the subalgebra generated by $1,B_1,B_2,\dots$, and let U_0 denote the operator acting on the level-1 module under consideration. At the specialization $p=q^{2n}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-015953 | MATHDB-380967

The ordered tensor-product generation and submodule conjecture for fundamental representations

摘录 / EXCERPT: Let I_0 index the fundamental representations, let $i_1,\ldots,i_l\in I_0$, and let $a_1,\ldots,a_l\in\bar k^\times$, where $\bar k=\bigcup_{n>0}\mathbb C((q^{1/n}))$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-015954 | MATHDB-380969

The finite reduction conjecture for fundamental representations

摘录 / EXCERPT: Let I_0 index the fundamental representations, let $\bar k=\bigcup_{n>0}\mathbb C((q^{1/n}))$, and let $\mathfrak m=\bigcup_{n>0}q^{1/n}\mathbb C[[q^{1/n}]]$. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-015955 | MATHDB-380975

The admissibility decomposition conjecture for tensor-product crystals

摘录 / EXCERPT: Admissibility decomposition conjecture. For any $\xi\in(P_{cl}^+)_l$, there exists a disjoint union decomposition, not necessarily unique,

conjecture | open | MathDB 页面 | 原文线索

V2-015956 | MATHDB-380977

The q-Virasoro cohomology conjecture

摘录 / EXCERPT: Let $Q_n^{+[t]}$ be the maps in the complex associated with the screening charge Q_n^+, and let ${\cal H}_{J;n',n}$ denote the proposed irreducible highest weight representation...

conjecture | open | MathDB 页面 | 原文线索

V2-015957 | MATHDB-380990

The finite abelian group bound for phylogenetic complexity

摘录 / EXCERPT: Finite abelian group bound. For any finite abelian group G and any positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-015958 | MATHDB-381004

The character bound conjecture for high-dimensional irreducible representations of symmetric groups

摘录 / EXCERPT: Character bound conjecture. There is a constant A such that, for sufficiently large n, every big λ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-015959 | MATHDB-381064

Gromov's conjecture on hyperbolic groups with spherical boundary

摘录 / EXCERPT: Let G be a hyperbolic group whose boundary ∂G is homeomorphic to S^{n-1}. An aspherical closed manifold is a closed manifold whose universal cover is contractible. Gr...

conjecture | open | MathDB 页面 | 原文线索

V2-015960 | MATHDB-381078

Equality of Kirillov functions and irreducible characters for unitriangular groups

摘录 / EXCERPT: The conjecture. For every $\lambda\in\mathfrak{u}_n(q)^*$,

conjecture | open | MathDB 页面 | 原文线索

V2-015961 | MATHDB-381090

Multiplicity-free restriction conjecture for derived functor modules of symmetric type

摘录 / EXCERPT: Let $(\mathfrak{g},\mathfrak{g}')$ be a reductive symmetric pair, let $\mathfrak{q}$ be a θ-stable parabolic subalgebra of $\mathfrak{g}_{\mathbb{C}}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-015962 | MATHDB-381096

The generic polygonal linkage swap-group conjecture

摘录 / EXCERPT: A polygonal linkage is a closed chain of segments with fixed edge lengths, and the swap group SW_n is generated by the elementary swaps s_i acting on its configurations. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-015963 | MATHDB-381101

Extension-group conjecture for principal series representations of ${\rm SL}_n(k)$

摘录 / EXCERPT: Extension-group conjecture. For $a,b\in Q$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-015964 | MATHDB-381102

Classification conjecture for K-spherical representations of infinite-dimensional general linear groups

摘录 / EXCERPT: Let $\mathbf G=\mathrm{GL}(k\infty,\mathbb{Q}_p)$, and let $\mathbf K$ be its compact subgroup. For each positive integer l and each $r\in\mathrm{Sp}(2kl,\mathbb{Q}_p)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-015965 | MATHDB-381115

Generation of the representation ring by exterior powers for types B and odd D

摘录 / EXCERPT: Let W be a Coxeter group of type B_n or D_{2n+1}. Let V be its reflection representation, and let U be the non-faithful reflection representation of dimension $n-1$ descr...

conjecture | open | MathDB 页面 | 原文线索

V2-015966 | MATHDB-381137

The first and second Brauer-Thrall conjectures for finite-dimensional algebras

摘录 / EXCERPT: Brauer-Thrall conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-015967 | MATHDB-381185

Asymptotic freeness of representation images of the canonical matrix

摘录 / EXCERPT: Let $\lambda^1(N),\lambda^2(N)\in\widehat U(N)$, for $N=1,2,\dots$, be two sequences of signatures satisfying the technical assumption of Definition $\mathrm{regularity}$, and supp...

conjecture | open | MathDB 页面 | 原文线索

V2-015968 | MATHDB-381195

Tokuyama-type character formula for G_2

摘录 / EXCERPT: Let $\Phi=G_2$. For a dominant weight θ, let χ_θ be the character of the irreducible representation V_θ of G_2 of lowest weight $-\theta$, shifted to be a...

conjecture | open | MathDB 页面 | 原文线索

V2-015969 | MATHDB-381202

Mazurov's conjecture on finite groups isospectral to simple groups

摘录 / EXCERPT: Let L be a finite nonabelian simple group, and let G be a finite group. The groups G and L are isospectral when $\omega(G)=\omega(L)$, where $\omega(H)$ denotes the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-015970 | MATHDB-381251

The OD-characterizability conjecture for symmetric groups

摘录 / EXCERPT: Let S_m denote the symmetric group of degree m. A finite group is OD-characterizable if it is uniquely determined, among finite groups, by its order and degree pattern; it is...

conjecture | open | MathDB 页面 | 原文线索

V2-015971 | MATHDB-381264

The maximum-rank-class and majority conjectures for idempotents in Jones monoids

摘录 / EXCERPT: Let $\mathcal{J}_n$ be the Jones monoid, let E_n be its set of idempotents, let $\operatorname{rank}(\alpha)$ denote the rank of $\alpha\in E_n$, let $e_n=|E_n|$, and let d_n^*...

conjecture | open | MathDB 页面 | 原文线索

V2-015972 | MATHDB-381273

The non-zero asymptotic excess conjecture for interval exchanges from partial isometries

摘录 / EXCERPT: Non-zero asymptotic excess conjecture. If Σ_Ψ has non-zero asymptotic excess, then, for every admissible $\mathbf y$, the transformation $T_{\Psi,\mathbf y}$ either is no...

conjecture | open | MathDB 页面 | 原文线索

V2-015973 | MATHDB-381278

Bader-Muchnik irreducibility conjecture for quasi-regular representations

摘录 / EXCERPT: Bader-Muchnik irreducibility conjecture. For every locally compact group G and every spread-out probability measure ... -boundary of G is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-015974 | MATHDB-381303

Simplicity conjecture for multiplicity modules of graded double affine Hecke algebras

摘录 / EXCERPT: For a simple perverse sheaf $\mathcal{S}$ in the block $\mathcal{D}_{G_{\underline0}}(\mathfrak{g}^{nil}_{\underline\eta})_{\xi}$, define ... By the preceding corollary, this is a mo...

conjecture | open | MathDB 页面 | 原文线索

V2-015975 | MATHDB-381317

Affine Macdonald denominator conjecture

摘录 / EXCERPT: Affine denominator conjecture. The affine Macdonald denominator is given by

conjecture | open | MathDB 页面 | 原文线索

V2-015976 | MATHDB-381328

Explicit orthonormal basis conjecture for $SU(2,2)$ coherent states

摘录 / EXCERPT: $SU(2,2)$ coherent-state basis conjecture. The set $\{\psi_{r,s}^{\ell,m}:\ell,m,r,s\in\mathbb{N}\}$ forms an orthonormal basis of $F_{2,2,n}^{U(n)}$. The formulas have been checke...

conjecture | open | MathDB 页面 | 原文线索

V2-015977 | MATHDB-381336

The endo-parameter compatibility conjecture with the local Langlands correspondence

摘录 / EXCERPT: Endo-parameter compatibility conjecture. The diagram relating $\operatorname{lrr}(G)$, $LP_G(W_F)$, $\mathcal{E}_{par}(F)$, and the representations of P_F is commutative.

conjecture | open | MathDB 页面 | 原文线索

V2-015978 | MATHDB-381390

Chen-Xi's derived-equivalence conjecture for Morita algebras

摘录 / EXCERPT: Chen-Xi's conjecture. Suppose two algebras A and B are derived equivalent. If A is a Morita algebra and the dominant dimension of B is at least two, then B is also a Mori...

conjecture | open | MathDB 页面 | 原文线索

V2-015979 | MATHDB-381398

The order-three conjecture for cactus group matrices of \(\mathfrak{sl}_3\)

摘录 / EXCERPT: Let $\lambda=l_1\omega_1+l_2\omega_2\in P^+$, and let $N^{i;\lambda}$, for $i\in\{1,2\}$, be the matrices of the cactus-group generators σ^i with respect to the basis...

conjecture | open | MathDB 页面 | 原文线索

V2-015980 | MATHDB-381406

Subgroup realization conjecture for low energy Ising field theories

摘录 / EXCERPT: Let G be a finite group, let $\mathscr{M}_G$ denote the space of admissible functions on G up to rescaling, and let $Delta_G\subset\mathscr{M}_G$ be the locus of lattice system...

conjecture | open | MathDB 页面 | 原文线索

V2-015981 | MATHDB-381423

Periodicity conjecture for symmetric-power quotients of Q^*

摘录 / EXCERPT: Periodicity conjecture. Under this isomorphism,

conjecture | open | MathDB 页面 | 原文线索

V2-015982 | MATHDB-381428

Discreteness conjecture for commensurators of subgroups of lattices in \(\mathrm{PSL}_2(\mathbb{R})\)

摘录 / EXCERPT: Commensurator discreteness conjecture. The group $\operatorname{Comm}_{\mathrm{PSL}_2(\mathbb{R})}(\Phi(H))$ is discrete provided that $H\neq\Phi(H)$. More generally,

conjecture | open | MathDB 页面 | 原文线索

V2-015983 | MATHDB-381468

Malle-Robinson conjecture on the number of irreducible Brauer characters

摘录 / EXCERPT: Let H be a finite group, let B be an ell-block of H, and let D be a defect group of B. The sectional ell-rank $s(D)$ is the maximum of the ranks of elementary abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-015984 | MATHDB-381471

Exponential dimension conjecture for faithful tropical representations of plactic monoids

摘录 / EXCERPT: Let $\mathbb{P}_n$ denote the plactic monoid on rank n, and let a tropical representation of $\mathbb{P}_n$ be faithful when it distinguishes all elements of $\mathbb{P}_n$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-015985 | MATHDB-381513

The trigonometric shifted quantum affine injectivity conjecture

摘录 / EXCERPT: Let Λ be the coweight lattice, let $\Lambda^+\subset\Lambda$ be the dominant coweights, and let Ψ_D be the homomorphisms defined in for Λ-valued divisors D o...

conjecture | open | MathDB 页面 | 原文线索

V2-015986 | MATHDB-381539

Generalization of the almost-representation theorem to real compact Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a real compact Lie algebra, and replace the operators and relations in Theorem ... holds for all real compact Lie algebras. The theorem for $\mathfrak{su}(2)$ e...

conjecture | open | MathDB 页面 | 原文线索

V2-015987 | MATHDB-381561

Drinfeld-Sokolov reduction conjecture for simple modules of exceptional W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{sp}_4$, let f be a subregular nilpotent element, and let k be an admissible level. Write $L(\xi^{(s)}_{i,j},\chi^{(s)}_{i,j})$ for the simple...

conjecture | open | MathDB 页面 | 原文线索

V2-015988 | MATHDB-381569

The cluster structure conjecture for Grothendieck rings of quantum affine algebras

摘录 / EXCERPT: Let $\mathcal{C}_{\epsilon^\mathbb{Z}}$ be the monoidal category associated to $\mathfrak{g}=\mathfrak{sl}_k$ at a parameter $\epsilon\in\mathbb{C}^*$ such that ϵ^2 is an...

conjecture | open | MathDB 页面 | 原文线索

V2-015989 | MATHDB-381591

Equivalence conjecture for virtually unipotent curves in closed aspherical 3-manifold groups

摘录 / EXCERPT: Let M be a closed aspherical 3-manifold and let $\Gamma=\pi_1(M)$. The paper's statements numbered through concern properties of M and Γ introduced earlier in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-015990 | MATHDB-381665

Elliptic-orbit criterion for discrete-series infinitude on reductive homogeneous spaces

摘录 / EXCERPT: Let G/H be a homogeneous space in the reductive setting, let $\mathfrak g$ and $\mathfrak h$ be the Lie algebras of G and H, and let $\mathfrak h^{\perp}\subset\mathfrak g^*$...

conjecture | open | MathDB 页面 | 原文线索

V2-015991 | MATHDB-381673

Parker's odd orthogonal discriminant conjecture

摘录 / EXCERPT: Let G be a finite group and let χ be an orthogonally stable rational character of G. Its orthogonal discriminant is the square class represented by a unique square-free in...

conjecture | open | MathDB 页面 | 原文线索

V2-015992 | MATHDB-381716

Babai-Godsil-Imrich-Lovász conjecture on almost all Cayley graphs being GRRs

摘录 / EXCERPT: Babai-Godsil-Imrich-Lovász conjecture. The proportion of inverse-closed subsets S of R such that $\operatorname{Cay}(R,S)$ is a GRR of R approaches 1 as r approaches infi...

conjecture | open | MathDB 页面 | 原文线索

V2-015993 | MATHDB-381717

Xu's conjecture for normal Cayley graphs

摘录 / EXCERPT: Xu's conjecture. The proportion of inverse-closed subsets S of R such that $\operatorname{Cay}(R,S)$ is a normal Cayley graph of R approaches 1 as r approaches infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-015994 | MATHDB-381718

Mann's polynomial bound conjecture for Möbius values of open subgroups

摘录 / EXCERPT: Mann's conjecture. The quantity $|\mu(H,G)|$ is bounded by a polynomial function of $|G:H|$, and the number of subgroups H of G of index m satisfying $\mu(H,G)\neq 0$ grows a...

conjecture | open | MathDB 页面 | 原文线索

V2-015995 | MATHDB-381754

Homology-gradient approximation by Hughes-free division rings

摘录 / EXCERPT: Let $\mathbb{F}$ be a skew field. Let G be a torsion-free residually finite group of type $\mathsf{FP}_{n+1}(\mathbb F)$ such that the Hughes-free division ring...

conjecture | open | MathDB 页面 | 原文线索

V2-015996 | MATHDB-381755

Torsion-growth conjecture for Artin kernels

摘录 / EXCERPT: Let L be a flag complex, let $\varphi\colon A_L\to\mathbb{Z}$ be an epimorphism, and let BB_L^φ be the associated Artin kernel. Suppose BB_L^φ is of type...

conjecture | open | MathDB 页面 | 原文线索

V2-015997 | MATHDB-381756

Vanishing minimal volume entropy conjecture for Artin kernels

摘录 / EXCERPT: Let L be a flag complex, let $\mathbb{F}$ be a skew field, and let $\varphi\colon A_L\to\mathbb{Z}$ be an epimorphism. Suppose that BB_L^φ is of type F. Minima...

conjecture | open | MathDB 页面 | 原文线索

V2-015998 | MATHDB-381762

Conjecture on maximal finite subgroups of cactus groups

摘录 / EXCERPT: Let J_n be the cactus group on n strands. A finite group embeds into J_n precisely when it embeds into an iterated permutational wreath product of copies of...

conjecture | open | MathDB 页面 | 原文线索

V2-015999 | MATHDB-381763

Conjecture on free abelian subgroups of cactus groups

摘录 / EXCERPT: Let J_n be the cactus group on n strands, and let k be a nonnegative integer. A free abelian subgroup of rank k is a subgroup isomorphic to $\mathbb{Z}^k$. Free abelian sub...

conjecture | open | MathDB 页面 | 原文线索

V2-016000 | MATHDB-381810

Hyperfocal subalgebra stable unital basis conjecture

摘录 / EXCERPT: Let b be a block of $\mathcal{O}G$ with defect group D, let $A=l\mathcal{O}Gl$ be a source algebra, and let $\mathcal{F}$ be the saturated fusion system of A on D. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016001 | MATHDB-381819

Polynomial-growth conjecture for the dimension of $W_{n,n+K}$

摘录 / EXCERPT: For each $n,K\in\mathbb{N}$, let $W_{n,n+K}$ be the representation considered in the paper, and let $\dim(W_{n,n+K})$ denote its dimension. Dimension-growth conjecture. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-016002 | MATHDB-381820

Early polynomiality conjecture for the dimension of $W_{n,n+K}$

摘录 / EXCERPT: Fix $K\in\mathbb{N}$. Let $W_{n,n+K}$ be the representation considered in the paper, and let $p_K(x)$ and N_K be as in the preceding corollary, so that $p_K(n)=\dim(W_{n,n+K})$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-016003 | MATHDB-381821

The strengthened monotonicity conjecture for multiplicities in $W_{n,n+K}$

摘录 / EXCERPT: Let $K\in\mathbb{N}$, let $\lambda\vdash 4K$, and write $\lambda^{(n-3K)}$ for the partition obtained by adding a first row of length $n-3K$ to λ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016004 | MATHDB-381826

Extension of the main symmetric-group-algebra theorem to Hecke shuffles

摘录 / EXCERPT: Hecke-algebra extension conjecture. Theorem should hold in $\mathcal{H}$ when each t_ℓ is replaced by $t_\ell^{\mathcal{H}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-016005 | MATHDB-381834

The Zassenhaus conjecture for torsion units in integral group rings

摘录 / EXCERPT: Zassenhaus conjecture. If $u\in\mathcal{U}(\mathbb ZG)$ is a unit of finite order, then u is conjugate within $\mathbb QG$ to $\pm g$ for some $g\in G$.

conjecture | open | MathDB 页面 | 原文线索

V2-016006 | MATHDB-381840

Lytchak's Regular Point Conjecture for CAT(0) spaces

摘录 / EXCERPT: Lytchak's Regular Point Conjecture. The Tits boundary $\partial_T X$ contains a regular point.

conjecture | open | MathDB 页面 | 原文线索

V2-016007 | MATHDB-381841

Ballmann's higher rank rigidity conjecture for CAT(0) spaces

摘录 / EXCERPT: Let X be a locally compact CAT(0) space with a geometric group action $Gamma\curvearrowright X$. An n-flat is an isometrically embedded copy of Euclidean n-space, where...

conjecture | open | MathDB 页面 | 原文线索

V2-016008 | MATHDB-381863

Orbit-space classification conjecture for symmetric matrices under the symplectic group

摘录 / EXCERPT: Let $S(2n,\mathbb{R})$ denote the space of real symmetric $2n\times 2n$ matrices, let J be the standard symplectic matrix, and let ρ be the symplectic-group action on...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016009 | MATHDB-381869

The second-smallest-value conjecture for the function f on finite groups

摘录 / EXCERPT: Let f be the function on finite groups studied in the paper, with $f(G)=1$ precisely for cyclic groups. Second-smallest-value conjecture. 2 is the second smallest value of the...

conjecture | open | MathDB 页面 | 原文线索

V2-016010 | MATHDB-381870

Stable-range conjecture for twisted cohomology of \(\operatorname{SL}(n,\mathbb{Q})\)

摘录 / EXCERPT: Let $\ul\la$ be a bipartition and let $n\ge l(\ul\la)$ be an integer. For $i=1,\dots,n$, define ... where μ_i is given in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016011 | MATHDB-381885

CK-admissible groups' Cantor-chain conjecture for visual sublinearly Morse boundaries

摘录 / EXCERPT: Let G be a CK-admissible group acting geometrically on a Hadamard space Y. Let T_{∞} be the Bass-Serre tree associated to the graph-of-groups structure of G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016012 | MATHDB-381921

The upper-bound conjecture for bounds of bounded affine Hecke algebra representations

摘录 / EXCERPT: Let $(\pi_{J,\mathsf{v}},M_{J,\mathsf{v}},\mathsf{B}_{J,\mathsf{v}})$ be a bounded representation, let $\mathbf{a}_{J,\mathsf{v}}$ denote its bound, let J be the relevant subset...

conjecture | open | MathDB 页面 | 原文线索

V2-016013 | MATHDB-381922

The product formula conjecture for bounds of bounded affine Hecke algebra representations

摘录 / EXCERPT: Product formula conjecture. If the representation is bounded, then

conjecture | open | MathDB 页面 | 原文线索

V2-016014 | MATHDB-381923

The cell-recognition conjecture for bounded affine Hecke algebra representations

摘录 / EXCERPT: Let $\Gamma_{J,\mathsf{v}}$ be the cell recognised by the bounded representation $(\pi_{J,\mathsf{v}},M_{J,\mathsf{v}},\mathsf{B}_{J,\mathsf{v}})$, let $\mathbf{a}_{J,\mathsf{v}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-016015 | MATHDB-381953

Stable transfer conjecture for local representations

摘录 / EXCERPT: Let $\Delta=(\Delta_i)_i=(T_i,d_i,\eta_i,\lambda_i)_i$ be a shape of rank N, and let φ_v be a test function on the twisted group $\widetilde{G}(N)$. Stable Transfer conje...

conjecture | open | MathDB 页面 | 原文线索

V2-016016 | MATHDB-381954

Refined exponent conjecture for simple shapes

摘录 / EXCERPT: Refined exponent conjecture. Under all the same conditions, the bound in Theorem may be tightened to

conjecture | open | MathDB 页面 | 原文线索

V2-016017 | MATHDB-381955

Parity-sensitive asymptotic conjecture for simple shapes

摘录 / EXCERPT: Let $\Delta=(T_i,d_i,\lambda_i,\eta_i)_{1\leq i\leq k}\neq\Sigma_{\lambda,\eta}$ satisfy the setup and conditions of the cited theorems, and let $R_0(\Delta)$ be the refined expone...

conjecture | open | MathDB 页面 | 原文线索

V2-016018 | MATHDB-381981

Non-specialized Capparelli-Meurman-Primc-Primc conjecture for C_n^(1)

摘录 / EXCERPT: Let $k_0,\ldots,k_n$ be non-negative integers. Let $\mathbb{Z}_{\mathcal{S}}^+$ be the coloured-part set, let Ω be the set of fictitious parts, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016019 | MATHDB-382015

Finkelberg-Ginzburg's mirabolic monodromy conjecture

摘录 / EXCERPT: Let $H=H^{\mathrm{SL}_n}$ be the maximal torus of $SL_n(\mathbb C)$, let $\vartheta\in\mathfrak h^*/W$, and set $k=c-1$. Let $\nabla^{K}(\vartheta,k)$ be the trigonometric KZ conne...

conjecture | open | MathDB 页面 | 原文线索

V2-016020 | MATHDB-382021

The rigidity conjecture for nonhomogeneous Coxeter connections

摘录 / EXCERPT: Let $\mathcal{F}$ be a $\mathcal{C}$-formal type of slope r/h, and let $\mathscr{O}\subset\operatorname{\mathfrak{g}}$ be a nilpotent orbit. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016021 | MATHDB-382055

Unipotent representation conjecture for graded Langlands parameters

摘录 / EXCERPT: Let G be a connected reductive group, let G_η range over its pure inner forms, and let $\mathbb{L}^u_{\underline q}$ be the graded unipotent Langlands parameter space. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016022 | MATHDB-382091

Ma-Zha-She's quotient-group conjecture for conflict-avoiding codes

摘录 / EXCERPT: Ma-Zha-She's quotient-group conjecture. There is a generator $t\langle -1,2\rangle$ of $\mathbb{Z}_p^{\times}/\langle -1,2\rangle$, for some $t\in\mathbb{Z}_p^{\times}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-016023 | MATHDB-382095

Abdollahi-Akbari-Maimani order conjecture for non-commuting graphs

摘录 / EXCERPT: Let G and H be non-abelian finite groups. For a finite non-abelian group K, let Γ_K be the graph whose vertices are the non-central elements of K, with two vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-016024 | MATHDB-382096

Abdollahi-Akbari-Maimani nilpotency conjecture for non-commuting graphs

摘录 / EXCERPT: Let G be a finite non-abelian nilpotent group, and let H be a group. For a finite non-abelian group K, let Γ_K be the graph whose vertices are the non-central elements...

conjecture | open | MathDB 页面 | 原文线索

V2-016025 | MATHDB-382147

The two-generation conjecture for simple finitely presented groups

摘录 / EXCERPT: A group is simple if it has no nontrivial proper normal subgroups, and finitely presented if it admits a presentation with finitely many generators and finitely many defining relat...

conjecture | open | MathDB 页面 | 原文线索

V2-016026 | MATHDB-382164

Kac-Wakimoto-type conjecture for m-traces and atypicality

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a basic classical Lie superalgebra, and let $L(\lambda)$ be a simple ${\mathfrak{g}}$-supermodule of highest weight λ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016027 | MATHDB-382211

Ballier and Stein's conjecture on the domino problem

摘录 / EXCERPT: Let G be a finitely generated group. Recall that G is virtually free if it has a finite-index free subgroup. Ballier and Stein's conjecture. The group G has decidable domino...

conjecture | open | MathDB 页面 | 原文线索

V2-016028 | MATHDB-382218

Injectivity conjecture for the natural maps associated with the mathbb Z_3-symmetric down-up algebra

摘录 / EXCERPT: Let the maps in Proposition and Theorem be the natural maps associated with the $\mathbb Z_3$-symmetric down-up algebra $\mathbb A=\mathbb A(\alpha,\beta,\gamma)$. Injectivity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-016029 | MATHDB-382227

The sharp upper bound for the Swan-exponent difference in residue characteristic two

摘录 / EXCERPT: Sharp upper-bound conjecture. Assume $p=2$. For any Galois representation ρ, one has

conjecture | open | MathDB 页面 | 原文线索

V2-016030 | MATHDB-382262

Relative asymptotic multiplicity conjecture for simple Lie algebras

摘录 / EXCERPT: Let $\overline{\mathfrak{g}}$ be a finite-dimensional simple Lie algebra, and let $\lambda\in \overline{P}_+\backslash\{0\}$ be a non-zero dominant integral weight. The relative as...

conjecture | open | MathDB 页面 | 原文线索

V2-016031 | MATHDB-382263

The non-factorability conjecture for strongly irreducible shifts on large classes of groups

摘录 / EXCERPT: Non-factorability conjecture. On a large class of groups, not every SI SFT is an SFT factor of a contractible SFT.

conjecture | open | MathDB 页面 | 原文线索

V2-016032 | MATHDB-382275

The primitive-unipotent conjecture for linear groups

摘录 / EXCERPT: Let $G\subseteq GL(n,\mathbb{C})$ be a finitely generated subgroup. An element of G is primitive if it belongs to a generating set obtainable from the fixed generating set by fin...

conjecture | open | MathDB 页面 | 原文线索

V2-016033 | MATHDB-382276

The Schreier Girth Alternative for linear groups

摘录 / EXCERPT: Let G be a finitely generated group, and define its Schreier girth by ... where S is a minimal generating set and $S\sim_{\mathcal{S}}S'$ means that S' can be obtained from S...

conjecture | open | MathDB 页面 | 原文线索

V2-016034 | MATHDB-382288

Uniform boundedness conjecture for fixers of finite primitive permutation groups

摘录 / EXCERPT: Uniform boundedness conjecture. There exists an absolute constant c such that $\rho_0(G)<c$ for all finite primitive permutation groups G. This conjecture asserts that the larg...

conjecture | open | MathDB 页面 | 原文线索

V2-016035 | MATHDB-382289

The homological-dimension-one conjecture for self-dual representations

摘录 / EXCERPT: Homological-dimension-one conjecture. The quotient ${\mathsf V}/\!\!/G$ is either an affine space or a hypersurface; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-016036 | MATHDB-382290

Non-distributivity conjecture for hook Specht modules in characteristic 2

摘录 / EXCERPT: Non-distributivity conjecture. If $r\geq 10$, then S_2^λ is not distributive.

conjecture | open | MathDB 页面 | 原文线索

V2-016037 | MATHDB-382292

Polynomial dimension-and-frequency conjecture for representations of general linear groups over local rings

摘录 / EXCERPT: Polynomial dimension-and-frequency conjecture. There exist $k\geq 1$, polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-016038 | MATHDB-382324

The multiplicity criterion for highest weights in the orthosymplectic cocharacter

摘录 / EXCERPT: Let $(\lambda(1),\emptyset,\lambda(3),\lambda(4))$ be a highest-weight label in the decomposition denoted by $(\lambda(1),\emptyset,\lambda(3),\lambda(4))$, where ... and ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-016039 | MATHDB-382327

Saxl's conjecture on tensor squares of staircase representations

摘录 / EXCERPT: Let k be a positive integer, set $n=k(k+1)/2$, and let $\lambda=(k,k-1,\ldots,2,1)$ be the staircase partition of n. Let σ_λ be the irreducible representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-016040 | MATHDB-382329

Submaximal character degree ratio conjecture for symmetric groups

摘录 / EXCERPT: Let $b_1(n)$ be the largest irreducible character degree of the symmetric group S_n, and for each $m\geq 2$ let $b_m(n)$ be the m-th largest irreducible character degree of...

conjecture | open | MathDB 页面 | 原文线索

V2-016041 | MATHDB-382353

The deformation-space isomorphism conjecture for Kalck-Karmazyn algebras

摘录 / EXCERPT: Let Esubset W be the special fiber consisting of a nodal curve on an algebraic surface, let $\mathop{\operatorname{Def}}_{(E \subset W)}$ be the versal deformation space of the p...

conjecture | open | MathDB 页面 | 原文线索

V2-016042 | MATHDB-382359

Bernstein-Zelevinsky duality conjecture for parabolic Orlik-Strauch representations

摘录 / EXCERPT: Let G be the ambient reductive group, let P be a parabolic subgroup of G containing B, with Lie algebra $\mathfrak{p}$ and opposite parabolic $\overline{P}$. Let M be a...

conjecture | open | MathDB 页面 | 原文线索

V2-016043 | MATHDB-382361

D'Adderio-Iraci-Vanden Wyngaerd Theta conjecture for the $(2,2)$-bosonic-fermionic coinvariant ring

摘录 / EXCERPT: Let $R_n^{(2,2)}$ be the $(2,2)$-bosonic-fermionic coinvariant ring, and let $\operatorname{Frob}(R_n^{(2,2)};q,t;u,v)$ denote its multigraded Frobenius series. Here Θ_{e_k}...

conjecture | open | MathDB 页面 | 原文线索

V2-016044 | MATHDB-382362

D'Adderio-Iraci-Vanden Wyngaerd Theta conjecture at $t=0$ for the $(1,2)$-bosonic-fermionic coinvariant ring

摘录 / EXCERPT: Let $R_n^{(1,2)}$ be the $(1,2)$-bosonic-fermionic coinvariant ring, and let $\operatorname{Frob}(R_n^{(1,2)};q;u,v)$ denote its multigraded Frobenius series. The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-016045 | MATHDB-382417

Translation conjecture for coherent pairs and restriction cohomology

摘录 / EXCERPT: Let (pi,pi') be a coherent pair with Harish-Chandra parameters $lambda$ and $lambda'$ sufficiently far from the walls. Let $(sigma,sigma')$ be an elementary pair, let F and...

conjecture | open | MathDB 页面 | 原文线索

V2-016046 | MATHDB-382422

The cluster monomial conjecture for the shifted category $\mathcal{O}^{sh}$

摘录 / EXCERPT: Let $\mathcal{A}_{\Gamma_\infty'}$ be the cluster algebra associated with the quiver Γ_∞', and let $\mathcal{O}^{sh}$ be the shifted category appearing in the isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-016047 | MATHDB-382426

The Picky Conjecture for finite groups

摘录 / EXCERPT: Picky Conjecture. There exists a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-016048 | MATHDB-382441

The maximum commutative subsemigroup conjecture for full transformation semigroups

摘录 / EXCERPT: Let X be a set, and let $\mathcal{T}(X)$ denote the full transformation semigroup on X. For the notation $N^{X}_{x_1,\ldots,x_t}$ and the functions ξ and α, use the...

conjecture | open | MathDB 页面 | 原文线索

V2-016049 | MATHDB-382456

Universal validity of the Kadar-Yu polynomial formulae

摘录 / EXCERPT: Let $C^{(l+1,1)}(\alpha)$, $C^{(l+1,1)^T}(\alpha)$, $P_{l+4+k}^{(l+1,1)}$, and $P_{l+4+k}^{(l+1,1)^T}$ denote the polynomials and Chebyshev series defined in the preceding formulae...

conjecture | open | MathDB 页面 | 原文线索

V2-016050 | MATHDB-382468

Haines's twisted endoscopic transfer conjecture for the stable Bernstein center

摘录 / EXCERPT: Let G_r be an unramified base-change form of the quasi-split group G, with Hecke algebras $\mathcal{H}(G_r)$ and $\mathcal{H}(G)$, and let $\mathfrak{Z}^{\textnormal{st}}(G_r)$...

conjecture | open | MathDB 页面 | 原文线索

V2-016051 | MATHDB-382492

Generalization of the Amdeberhan-Shareshian-Stanley conjecture for partition division maps

摘录 / EXCERPT: Generalized Amdeberhan-Shareshian-Stanley conjecture. For every partition μ of kn, the symmetric function obtained by applying the partition division map...

conjecture | open | MathDB 页面 | 原文线索

V2-016052 | MATHDB-382575

The PBW filtration conjecture for vacuum representations of affine Kac-Moody algebras

摘录 / EXCERPT: The PBW filtration conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-016053 | MATHDB-382986

Kippenhahn’s conjecture

摘录 / EXCERPT: For Hermitian matrices $A_{1},A_{2}\in M_{n}(\mathbb{C})$, define $p(z_{0},z_{1},z_{2})=\det\!\left(z_{0}I+z_{1}A_{1}+z_{2}A_{2}\right)$. Kippenhahn's conjecture asserts that if th...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016054 | MATHDB-382999

Kostant’s conjecture

摘录 / EXCERPT: For every complex simple Lie algebra $\mathfrak{g}$, with ρ the half-sum of its positive roots, and every dominant integral weight $\lambda\in P_+$, the irreducible representa...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016055 | MATHDB-383135

Kashina’s exponent conjecture

摘录 / EXCERPT: For every finite-dimensional semisimple and cosemisimple Hopf algebra H over a field k, the exponent of H divides its dimension: $\exp(H)\mid\dim_k H$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016056 | MATHDB-383150

Wilson’s conjecture on locally nilpotent countably categorical groups

摘录 / EXCERPT: Every locally nilpotent countably categorical group is nilpotent; equivalently, for every group G, if G is locally nilpotent and countably categorical, then G is nilpotent.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016057 | MATHDB-383244

Quantum Hikita conjecture

摘录 / EXCERPT: For a 3-dimensional $\mathcal{N}=4$ gauge theory with Higgs branch $\mathcal{M}_H$, dual Coulomb branch $\mathcal{M}_C$, and quantized Coulomb-branch algebra $\mathcal{A}_C$, the...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016058 | MATHDB-383298

Thompson’s conjecture

摘录 / EXCERPT: For every finite simple nonabelian group G, there exists a conjugacy class C of G such that $C^{2}=\{xy:x,y\in C\}=G$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016059 | MATHDB-383301

Snashall-Solberg conjecture

摘录 / EXCERPT: For every finite-dimensional associative algebra A over a field K, the quotient of its Hochschild cohomology ring by the ideal of nilpotent elements is a finitely generated K...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016060 | MATHDB-383318

Quotient-submodule equidistribution conjecture

摘录 / EXCERPT: For every finite-dimensional representation-finite algebra A and every integer $i\ge 0$, the number of quotient-closed full additive subcategories...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016061 | MATHDB-383407

Lehnert’s conjecture

摘录 / EXCERPT: For every finitely generated group G, G has co-context-free word problem if and only if G embeds into Thompson's group V. Explicitly, for a finite generating set X of G...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016062 | MATHDB-383497

Aizenbud-Gourevitch regularity conjecture

摘录 / EXCERPT: For every connected complex reductive algebraic group G and every involutive automorphism $\theta:G\to G$, let $H=G^\theta$ and let $\mathfrak{g}=\mathfrak{h}\oplus\mathfrak{s}$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016063 | MATHDB-383683

Miller-Tserunyan orbit-equivalence question

摘录 / EXCERPT: For every $n\in\mathbb{N}\cup\{\infty\}$ and every ergodic treeable probability-measure-preserving equivalence relation $\mathcal{R}$ of cost n on a standard probability space...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016064 | MATHDB-383795

Higher rank abelian group actions (problem 5)

摘录 / EXCERPT: Suppose $\mathbb{R}^{2}$ acts smoothly on a closed manifold X with invariant foliation F. Assume that begin{enumerate} item The set of normally hyperbolic elements is dense in...

open_problem | open | MathDB 页面 | 原文线索

V2-016065 | MATHDB-383807

Property (T) group action

摘录 / EXCERPT: Let Γ be a discrete group with Kazhdan’s property (T). Let Σ be a smooth surface and $\alpha\colon \Gamma\to \textrm{Diff}^{\infty}_{vol}(\Sigma)$ be a smooth volume...

open_problem | open | MathDB 页面 | 原文线索

V2-016066 | MATHDB-384437

Polynomials and Painlev\'e (problem 1)

摘录 / EXCERPT: Is there a relation between the polynomial roots appearing in the Painlev'e equation and representation theory, for example via Young diagrams?

open_problem | open | MathDB 页面 | 原文线索

V2-016067 | MATHDB-384682

Group Actions on Banach Spaces (problem 1)

摘录 / EXCERPT: Does the mapping class group over any closed surface act properly and isometrically on some L_p-space for $p < \infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-016068 | MATHDB-384994

Advanced digital signature functionality from group theory

摘录 / EXCERPT: We would like to use generic group-action tools to build ring signatures, blind signatures, threshold signatures and more. We would also like to apply the "twist" technique present...

open_problem | open | MathDB 页面 | 原文线索

V2-016069 | MATHDB-385066

Group splittings and asymptotic topology (problem 2)

摘录 / EXCERPT: Let G be a finitely generated group which does not split over a finite group. Is it true that (any subset of) a quasi-ray does not separate coarsely the Cayley graph of G?

open_problem | open | MathDB 页面 | 原文线索

V2-016070 | MATHDB-385355

Nonlinear representation theory

摘录 / EXCERPT: Suppose we define an abstract notion of "wild nonabelian Hodge structure", consisting of a hyperkahler manifold with its various algebraic structures, such that it is a wild Hitchi...

open_problem | open | MathDB 页面 | 原文线索

V2-016071 | MATHDB-385533

Group Actions and Submersions (problem 1)

摘录 / EXCERPT: Let M^n be a manifold with $\sec > 0$ or $\sec \geq 0$ or almost non-negative curvature. Does M^n have a positive symmetry degree, i.e. is there an...

open_problem | open | MathDB 页面 | 原文线索

V2-016072 | MATHDB-385534

Group Actions and Submersions (problem 2)

摘录 / EXCERPT: Is there a principal T^2-bundle whose total space admits $\sec > 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-016073 | MATHDB-385537

Group Actions and Submersions (problem 5)

摘录 / EXCERPT: Given a homogeneous space G/H, with G compact, classify all homogeneous metrics with $\sec \geq 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-016074 | MATHDB-385539

Group Actions and Submersions (problem 7)

摘录 / EXCERPT: Given a Riemannian submersion with positively curved total space, is the dimension of the fiber less than the dimension of the base? It is perhaps simpler to decide if there is a b...

open_problem | open | MathDB 页面 | 原文线索

V2-016075 | MATHDB-385541

Group Actions and Submersions (problem 9)

摘录 / EXCERPT: Suppose (M^n,g) is simply-connected and $\sec_M \geq 0$. Is $\operatorname{rank}({Iso}(M^n,g)) \leq \frac{2}{3}n$?

open_problem | open | MathDB 页面 | 原文线索

V2-016076 | MATHDB-385542

Group Actions and Submersions (problem 10)

摘录 / EXCERPT: Suppose we have an isometric group action on M. If one changes the metric on the orbit space, does it lift to an invariant metric on M?

open_problem | open | MathDB 页面 | 原文线索

V2-016077 | MATHDB-385543

Group Actions and Submersions (problem 11)

摘录 / EXCERPT: If a group acts isometrically on a manifold of $\sec \geq 1$ and the fixed point set is a circle, is it's length $\leq 2 \pi$?

open_problem | open | MathDB 页面 | 原文线索

V2-016078 | MATHDB-386381

Questions (problem 12)

摘录 / EXCERPT: Suppose that $f(x_{1},\ldots,x_{n})=f(x_{\sigma(1)},\ldots,x_{\sigma(n)})$ for all $\sigma\in S_{n}$ (the permutation group on n elements). Can we use this group action in a simi...

open_problem | open | MathDB 页面 | 原文线索

V2-016079 | MATHDB-386444

Rational Representations: Induction, Cohomology Vanishing (problem 5)

摘录 / EXCERPT: Let P be a parabolic subgroup of G. Is it true in characteristic p that $R^i\operatorname{Ind}_P^G\operatorname{Sym}(\mathfrak{n}_P^\bullet) = 0$ for all $i > 0$? (Known for...

open_problem | open | MathDB 页面 | 原文线索

V2-016080 | MATHDB-386447

Forced Grading (problem 1)

摘录 / EXCERPT: Is it true for a Schur algebra A, in particular $A = S(5, 5)$ in characteristic 2, that $A \cong \tilde{\operatorname{gr}}A$, the forced graded algebra in the sense of Parshall--...

open_problem | open | MathDB 页面 | 原文线索

V2-016081 | MATHDB-386449

Parity Sheaves and Torsion in IC Sheaves (problem 1)

摘录 / EXCERPT: For which primes p do the IC sheaves on $\operatorname{Perv}_I(Gr, {\\mathbb{Z}}_p)$ have torsion-free stalks? (Related to Problem 10.1 part 2, and to Ext-vanishing between reduc...

open_problem | open | MathDB 页面 | 原文线索

V2-016082 | MATHDB-386451

Parity Sheaves and Torsion in IC Sheaves (problem 3)

摘录 / EXCERPT: Is there a faster algorithm to compute indecomposable parity sheaves on $\operatorname{Perv}_{\mathrm{sph}}(Gr, {\\mathbb{F}}_p)$ than by computing the p-canonical basis? (Motiva...

open_problem | open | MathDB 页面 | 原文线索

V2-016083 | MATHDB-386453

Does this help to describe the center of ... ?

摘录 / EXCERPT: Does this help to describe the center of $\operatorname{Dist}(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-016084 | MATHDB-386461

Steenrod Algebra (problem 1)

摘录 / EXCERPT: begin{enumerate} item What is the representation-theoretic meaning of the Steenrod algebra action on mod-p cohomology? For example, on the cohomology of Gr? item Does the Steenro...

open_problem | open | MathDB 页面 | 原文线索

V2-016085 | MATHDB-386462

Nate Harman's question on lifting FI-modules from $\mathbb{C}$ to $\mathbb{Z}$

摘录 / EXCERPT: Given a map $\phi_{\mathbb{Q}} \colon V_{\mathbb{Q}} \to W_{\mathbb{Q}}$ of finitely generated FI-modules over $\mathbb{Q}$ (or may be over an algebraically closed field...

open_problem | open | MathDB 页面 | 原文线索

V2-016086 | MATHDB-386468

q-tca's

摘录 / EXCERPT: Are modules over this algebra $($as q-tca's$)$ equivalent to modules in the classical case $($as tca's$)$?

open_problem | open | MathDB 页面 | 原文线索

V2-016087 | MATHDB-386469

Quasi-polynomial behavior

摘录 / EXCERPT: begin{enumerate} item Does the same hold if we replace $\mathbb{C}$ by a manifold X $($for some class of manifolds$)$? item For a general finitely generated FI-algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-016088 | MATHDB-386470

Naturally occurring non-trivial FI_2 modules

摘录 / EXCERPT: begin{enumerate}[rm (a)] item What can we say about $\mathrm{H}_i(\mathrm{PConf}(X))$ generally along these lines? item Can we calculate $\mathrm{Ext}(M,N)$ for FI_2-mod...

open_problem | open | MathDB 页面 | 原文线索

V2-016089 | MATHDB-386471

Cohomology of PConf^n

摘录 / EXCERPT: Find the Specht module decomposition for $\mathrm{H}^i(\mathrm{PConf}^n \mathbb{C})$, and its FI-module generators. It might involve a derangement recurrence. Recurrence...

open_problem | open | MathDB 页面 | 原文线索

V2-016090 | MATHDB-386472

S_{∞} structure and configuration spaces

摘录 / EXCERPT: Compute S_{∞} representations of configuration spaces directly.

open_problem | open | MathDB 页面 | 原文线索

V2-016091 | MATHDB-386473

Highly acyclic complexes

摘录 / EXCERPT: Is $R\mathrm{GL}_n(R) \otimes_{R\mathrm{GL}_{n - (\bullet+1)}(R)} R^{n - (\bullet+1)} \to R^n \to 0$ highly acyclic?

open_problem | open | MathDB 页面 | 原文线索

V2-016092 | MATHDB-386476

VI like categories

摘录 / EXCERPT: Does this lead to an interesting analogue of VI-modules?

open_problem | open | MathDB 页面 | 原文线索

V2-016093 | MATHDB-386477

FI like categories

摘录 / EXCERPT: Modify FI to use double cover of symmetric groups.

open_problem | open | MathDB 页面 | 原文线索

V2-016094 | MATHDB-386480

Exact sequences of Hopf algebras

摘录 / EXCERPT: Consider an exact sequence of Hopf algebras $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, where B is finite-dimensional. If A and C satisfy the FGC conjecture,...

open_problem | open | MathDB 页面 | 原文线索

V2-016095 | MATHDB-386481

FGC for Hopf algebras (problem 2)

摘录 / EXCERPT: Let B be a finite-dimensional bialgebra over a field $\mathbf{k}$. What are examples of B for which the FGC conjecture is not true?

open_problem | open | MathDB 页面 | 原文线索

V2-016096 | MATHDB-386482

FGC for Hopf algebras (problem 3)

摘录 / EXCERPT: Let H be a finite-dimensional Hopf algebra. Let $\mathcal{C}$ denote the semisimplification of its representation category. Does $\mathcal{C}$ have nice subcategories...

open_problem | open | MathDB 页面 | 原文线索

V2-016097 | MATHDB-386484

FGC for Hopf algebras (problem 5)

摘录 / EXCERPT: Let H be a finite-dimensional Hopf algebra over a field $\mathbf{k}$. Suppose H acts on a finitely generated algebra A over $\mathbf{k}$. When is $H^0(H, A)$ (i.e.the invaria...

open_problem | open | MathDB 页面 | 原文线索

V2-016098 | MATHDB-386485

FGC for Hopf algebras (problem 6)

摘录 / EXCERPT: There is a list of a few finite-dimensional Nichols algebras $B(X)$ for X an irreducible Yetter-Drinfeld module over a finite non-abelian group G. For these, determine when the...

open_problem | open | MathDB 页面 | 原文线索

V2-016099 | MATHDB-386486

FGC for tensor categories (problem 1)

摘录 / EXCERPT: If $\mathcal{C}$ satisfies the FGC conjecture and $\mathcal{B} \subset \mathcal{Z}(\mathcal{C})$, is it true that the M"uger centralizer...

open_problem | open | MathDB 页面 | 原文线索

V2-016100 | MATHDB-386487

Is there a nice class of categories where the FGC conjecture holds?

摘录 / EXCERPT: Is there a nice class of categories where the FGC conjecture holds? (Something larger than the class of finite tensor categories.)

open_problem | open | MathDB 页面 | 原文线索

V2-016101 | MATHDB-386488

Does a braided finite tensor category have the tensor product property, that is, ... ?

摘录 / EXCERPT: Does a braided finite tensor category have the tensor product property, that is, $V(A \otimes B) = V(A) \cap V(B)$?

open_problem | open | MathDB 页面 | 原文线索

V2-016102 | MATHDB-386489

Construct examples that precisely delineate the boundaries of FGC.

摘录 / EXCERPT: Construct examples that precisely delineate the boundaries of FGC.

open_problem | open | MathDB 页面 | 原文线索

V2-016103 | MATHDB-386490

Balmer spectrum/TTG (problem 1)

摘录 / EXCERPT: Compute the Balmer spectrum of the stable module categories of some pointed Hopf algebras of tame type in characteristic 0 (e.g.$u_q(\mathfrak{sl}(1|2))$).

open_problem | open | MathDB 页面 | 原文线索

V2-016104 | MATHDB-386491

Balmer spectrum/TTG (problem 2)

摘录 / EXCERPT: Let $\mathcal{C}$ be a symmetric FTC. Is it true that $\text{Spc}(\text{St}(\mathcal{C})) = \text{Proj}(\text{Ext}^*_{\mathcal{C}}(\mathbf{1}))$? (This is known for cocommutative H...

open_problem | open | MathDB 页面 | 原文线索

V2-016105 | MATHDB-386492

Balmer spectrum/TTG (problem 3)

摘录 / EXCERPT: Denote the 'particular subalgebra of Ext' from above question as C^*. For pointed Hopf algebras H (set $\mathcal{C} = \text{Rep}(H)$) with abelian grouplikes in characteristic...

open_problem | open | MathDB 页面 | 原文线索

V2-016106 | MATHDB-386493

Take ... to be a tensor triangular category with finite Krull-Schmidt dimension.

摘录 / EXCERPT: Take $\mathbf{K}$ to be a tensor triangular category with finite Krull-Schmidt dimension. Then, is the Balmer spectrum is Noetherian? (See the work of Negron and Stevenson.)

open_problem | open | MathDB 页面 | 原文线索

V2-016107 | MATHDB-386494

Balmer spectrum/TTG (problem 5)

摘录 / EXCERPT: Let $\mathcal{C}$ be an FTC. Is $\text{Spc}(\text{Stmod}(\mathcal{C}))$ independent of the monoidal structure? (e.g.is it true for the quasi-Hopf modifications of small quantum gro...

open_problem | open | MathDB 页面 | 原文线索

V2-016108 | MATHDB-386495

What are the relationships between finite generation, Noetherianity, and topological Noetherianity?

摘录 / EXCERPT: What are the relationships between finite generation, Noetherianity, and topological Noetherianity? (Does this imply the tensor triangular property?)

open_problem | open | MathDB 页面 | 原文线索

V2-016109 | MATHDB-386498

Specific cases (problem 3)

摘录 / EXCERPT: Compute the cohomology of Deligne interpolation categories GL_t for $t \in \mathbf{Z}$.

open_problem | open | MathDB 页面 | 原文线索

V2-016110 | MATHDB-386499

Calculate the Gerstenhaber bracket of the Yoneda algebra in examples.

摘录 / EXCERPT: Calculate the Gerstenhaber bracket of the Yoneda algebra in examples. (e.g.for $\mathcal{C} = \text{Comod}(\mathcal{U}(\mathfrak{g}))$, the Yondea algebra is $\wedge(\mathfrak{g})$...

open_problem | open | MathDB 页面 | 原文线索

V2-016111 | MATHDB-386560

Develop a notion of equivariant trisection; given an action of a finite group on ... which preserves the trisection in an appropriate sense, what does it say about the trisection?

摘录 / EXCERPT: Develop a notion of equivariant trisection; given an action of a finite group on X which preserves the trisection in an appropriate sense, what does it say about the trisection?

open_problem | open | MathDB 页面 | 原文线索

V2-016112 | MATHDB-386777

Is there a symplectic way to understand the entire ... -categorical representation theory framework from which, e.g., KLRW or nil Hecke algebra arises?

摘录 / EXCERPT: Is there a symplectic way to understand the entire 2-categorical representation theory framework from which, e.g., KLRW or nil Hecke algebra arises?

open_problem | open | MathDB 页面 | 原文线索

V2-016113 | MATHDB-386799

Geometric group theory

摘录 / EXCERPT: For the free group F_k on k generators, consider the generating set given by the closure of the standard generating set under conjugation. When considered with the word metric,...

open_problem | open | MathDB 页面 | 原文线索

V2-016114 | MATHDB-391641

Structural Trinity Conjecture in Registration Theory

摘录 / EXCERPT: For the symmetric ACRF-4 registration model, determine whether the canonical permutation-symmetry decomposition $\mathbb{R}^{4}=\operatorname{span}\{\mathbf{1}\}\oplus H_{4}$, wher...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016115 | MATHDB-391899

Isaacs-Navarro-Wolf conjecture

摘录 / EXCERPT: For every finite solvable group G, every element $g\in G$ such that $\chi(g)\neq 0$ for every irreducible complex character $\chi\in\operatorname{Irr}(G)$ belongs to the Fitting...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016116 | MATHDB-392100

Maubach-Willems finite-generation conjecture

摘录 / EXCERPT: For every finite field k of characteristic $p>0$ and every integer $n\ge 3$, the tame polynomial automorphism group $\operatorname{TA}_n(k)$ is not finitely generated; equivalent...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016117 | MATHDB-392125

Bürgisser’s width-two rectangular immanant problem

摘录 / EXCERPT: For a field F of characteristic zero, let $\operatorname{Imm}_{(2^m)}(X)=\sum_{\sigma\in S_{2m}}\chi_{(2^m)}(\sigma)\prod_{i=1}^{2m}x_{i,\sigma(i)}$, where $\chi_{(2^m)}$ is the...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016118 | MATHDB-392274

Takahashi's conjecture on resolving subcategories

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be a commutative Cohen-Macaulay Noetherian local ring, and let $\mathcal X$ be a resolving subcategory of finitely generated R-modules. If $\mathcal X$ ha...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016119 | MATHDB-392298

Gromov’s linear isoperimetric filling conjecture for Hadamard spaces

摘录 / EXCERPT: Let X be a Hadamard space and let $r=\operatorname{asrk}(X)$ be its asymptotic rank. For every integer $k\ge r$, there exists a constant $C=C(X,k)>0$ such that every integral k...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016120 | MATHDB-392299

Osin-Thom conjecture on the first ℓ^2-Betti number and normal rank

摘录 / EXCERPT: For every torsion-free discrete group G, the first ^{2}-Betti number satisfies $\beta^{(2)}_1(G)\leq \operatorname{nrk}(G)-1$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016121 | MATHDB-392457

AGT Conjecture

摘录 / EXCERPT: For generic coupling $\beta=-\epsilon_1/\epsilon_2$, the four-point conformal block for the $SU(2)$ theory with four fundamental hypermultiplets equals, coefficientwise at every in...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016122 | MATHDB-392469

Tringali-Yan conjecture

摘录 / EXCERPT: For all proper numerical monoids S_1 and S_2, if their reduced finitary power monoids are isomorphic, then the underlying monoids are isomorphic:...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016123 | MATHDB-392487

The even-n Auslander-Reiten (n+2)-angle local-finiteness problem

摘录 / EXCERPT: For every even integer n and every $(n+2)$-angulated category $\mathcal C$, if the Auslander-Reiten $(n+2)$-angles of $\mathcal C$ generate the relations in the Grothendieck grou...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016124 | MATHDB-392897

Finite generation for invariants of linear \-dimensional additive-group actions

摘录 / EXCERPT: Is the invariant ring $\mathbb{C}[z_1,\ldots,z_m]^G$ of a linear action of the 2-dimensional additive group $G=\mathbb{G}_a\times\mathbb{G}_a$ finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-016125 | MATHDB-392998

Existence and classification of Beauville surfaces

摘录 / EXCERPT: Determine: (a) which finite groups G can occur in the construction of Beauville surfaces; and (b) classify all possible Beauville surfaces for a given finite group G.

open_problem | open | MathDB 页面 | 原文线索

V2-016126 | MATHDB-393034

Minimum coincidence components for maps into odd-dimensional spherical space forms

摘录 / EXCERPT: Let N be an odd-dimensional spherical space form, i.e. the quotient of S^n by a free action of a finite group. Then, for all maps $f_1,f_2:S^m\to N$, ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-016127 | MATHDB-393288

Local $C^*(G,T)$-Theorem

摘录 / EXCERPT: The report states the following proved-result claim. Let G be a finite group, let p divide $|G|$, and let $T\in\operatorname{Syl}_p(G)$. If G is of characteristic p and...

open_problem | open | MathDB 页面 | 原文线索

V2-016128 | MATHDB-393601

Alperin's Weight Conjecture

摘录 / EXCERPT: Let G be a finite group. A weight is a conjugacy class of pairs (D,P), where D is a p-group in G and P is a projective simple module over...

conjecture | open | MathDB 页面 | 原文线索

V2-016129 | MATHDB-393602

Conjecture 5 on fixed points of subgroups of elementary abelian group actions

摘录 / EXCERPT: Let $G=(\mathbb{Z}/p)^r$ act on a projective $\mathbb{Z}G$-chain complex C with $C_i=0$ for $i<0$ and $H_i(C)=0$ for $i>n$, and let $j:C\to C/G$ be the quotient map. If ... then so...

conjecture | open | MathDB 页面 | 原文线索

V2-016130 | MATHDB-393603

Benson-Carlson homotopy-rank conjecture

摘录 / EXCERPT: For any finite group G, $\operatorname{rk}(G)=h(G)$, where $\operatorname{rk}(G)$ is the maximum p-rank over all primes and $h(G)$ is the least integer k such that G acts f...

conjecture | open | MathDB 页面 | 原文线索

V2-016131 | MATHDB-393605

Decomposition of the symmetric algebra for cyclic group actions

摘录 / EXCERPT: Determine the decomposition of the symmetric algebra $S^*(V)$ into indecomposable kC_{p^n}-modules, where C_{p^n} acts on a polynomial ring $S=k[x_1,\ldots,x_r]$ over a field...

open_problem | open | MathDB 页面 | 原文线索

V2-016132 | MATHDB-393790

Diamond theorem for a finitely generated free profinite group

摘录 / EXCERPT: Let $e\ge 2$ be an integer and let $F=\widehat{F}_e$ be the free profinite group of rank e. Let M_1,M_2 be closed normal subgroups of F, and let M be a closed subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-016133 | MATHDB-393887

Euler characteristics and Gysin sequences for group actions on boundaries

摘录 / EXCERPT: Let G be a torsion-free discrete group and let $G\times\partial X\to\partial X$ be a boundary action, where X is a finite-dimensional simplicial complex with a simplicial actio...

open_problem | open | MathDB 页面 | 原文线索

V2-016134 | MATHDB-393902

Absolute boundedness of first cohomology for faithful absolutely irreducible modules

摘录 / EXCERPT: Is it true that $\dim H^1(G,M)$ is absolutely bounded for M a faithful absolutely irreducible G-module, where G is a finite group?

open_problem | open | MathDB 页面 | 原文线索

V2-016135 | MATHDB-393906

The Navarro-Willems conjecture for blocks of finite groups

摘录 / EXCERPT: Let G be a finite group. For distinct primes p and q, let B_p be a p-block of G and let B_q be a q-block of G. If...

conjecture | open | MathDB 页面 | 原文线索

V2-016136 | MATHDB-393914

Puig equivalence criterion for blocks with cyclic defect groups

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let G be a finite group, and let kG be the group algebra. Suppose that A is a block algebra of kG with def...

open_problem | open | MathDB 页面 | 原文线索

V2-016137 | MATHDB-393919

Minimal-polynomial theorem for semisimple elements of finite groups of Lie type

摘录 / EXCERPT: Let G be a simple algebraic group defined over a field of characteristic r, let $G:=G^F$ for a Frobenius map F, and let $S:=\operatorname{soc}(G/Z(G))$. Let Θ be an ir...

open_problem | open | MathDB 页面 | 原文线索

V2-016138 | MATHDB-393924

MAKRODINU conjecture

摘录 / EXCERPT: Assume $O_\ell(G)=1$. Given an ℓ-block B of a finite group G, one has ... where C runs over chains of ℓ-subgroups modulo G-conjugation, and b runs over ℓ-blo...

conjecture | open | MathDB 页面 | 原文线索

V2-016139 | MATHDB-393973

Closedness of power subgroups in finitely generated profinite groups

摘录 / EXCERPT: For every finitely generated profinite group G and every $q\in\mathbb{N}$, the verbal subgroup G^q, generated algebraically by all qth powers in G, is closed in G.

open_problem | open | MathDB 页面 | 原文线索

V2-016140 | MATHDB-394044

Generating condensed algebras with sufficiently few generators

摘录 / EXCERPT: Let G be a finite group, let F be a field of characteristic $p>0$, and let $e\in FG$ be an idempotent (in the usual application, $e=|K|^{-1}\sum_{k\in K}k$ for a subgroup...

open_problem | open | MathDB 页面 | 原文线索

V2-016141 | MATHDB-394053

Zassenhaus conjecture (first Zassenhaus conjecture)

摘录 / EXCERPT: Let G be a finite group and let $V(\mathbb{Z}G)$ denote the normalized unit group of the integral group ring $\mathbb{Z}G$. Every torsion unit $u\in V(\mathbb{Z}G)$ is conjugate...

conjecture | open | MathDB 页面 | 原文线索

V2-016142 | MATHDB-394326

Existence of a finitely generated non-classifying group

摘录 / EXCERPT: Is there a finitely presented (or at least a finitely generated) infinite group which is not classifying?

open_problem | open | MathDB 页面 | 原文线索

V2-016143 | MATHDB-394501

Character tables of finite groups of Lie type in complete generality

摘录 / EXCERPT: Determine the character table of finite groups of Lie type in complete generality.

open_problem | open | MathDB 页面 | 原文线索

V2-016144 | MATHDB-394534

Bertin obstruction as the sole obstruction to lifting local group actions

摘录 / EXCERPT: Fix a finite group G and an algebraically closed field k of characteristic $p>0$. For local actions $\varphi:G\hookrightarrow\operatorname{Aut}_k(k[[t]])$, the conjecture asser...

open_problem | open | MathDB 页面 | 原文线索

V2-016145 | MATHDB-394653

Cameron’s conjecture on base sizes of non-standard permutation groups

摘录 / EXCERPT: Let $G\leq\operatorname{Sym}(\Omega)$ be a finite non-standard almost simple primitive permutation group, where “non-standard” excludes alternating-group actions on subsets or part...

conjecture | open | MathDB 页面 | 原文线索

V2-016146 | MATHDB-394655

The Palfy-Pudlak question for finite group overgroup lattices

摘录 / EXCERPT: Is each nonempty finite lattice isomorphic to a lattice $\mathcal{O}_G(H)$ of overgroups of some subgroup H in a finite group G?

open_problem | open | MathDB 页面 | 原文线索

V2-016147 | MATHDB-394657

Irreducible restrictions of representations of finite groups with almost simple quotient

摘录 / EXCERPT: Let F be an algebraically closed field of characteristic $\[3m\ell\[0m$. Classify all triples (K,V,H), where K is a finite group with $K/Z(K)$ almost simple, V is an FK-m...

open_problem | open | MathDB 页面 | 原文线索

V2-016148 | MATHDB-394663

Product-free subsets and quasirandomness

摘录 / EXCERPT: Let G be a finite group of order n, and let $k\ge 2$ be the minimal degree of a nontrivial representation of G. Gowers asked whether one can improve the lower bound for the s...

open_problem | open | MathDB 页面 | 原文线索

V2-016149 | MATHDB-394664

Classification of finite primitive flag-transitive generalized hexagons and octagons

摘录 / EXCERPT: Classify all finite thick generalized n-gons, for $n\in\{6,8\}$, admitting a faithful group action that is flag-transitive and primitive on both the point set and the line set.

open_problem | open | MathDB 页面 | 原文线索

V2-016150 | MATHDB-394749

Asphericity of the spaces $B(q,G)$ for finite groups

摘录 / EXCERPT: For a finite group G, are the spaces $B(q,G)$ in the filtration $B(2,G)\subset B(3,G)\subset\cdots\subset B(q,G)\subset\cdots\subset BG$ $K(\pi,1)$-spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-016151 | MATHDB-394753

Classification of tensor-closed localising subcategories of the stable module category

摘录 / EXCERPT: Let G be a finite group and let k be a field of characteristic p. Then the tensor-closed localising subcategories of $\operatorname{StMod}(kG)$ are in one-to-one corresponden...

open_problem | open | MathDB 页面 | 原文线索

V2-016152 | MATHDB-394756

Completion theorem for equivariant stable cohomotopy of infinite groups

摘录 / EXCERPT: Let G be a discrete group admitting a finite G-CW model for EG, and let I_G be the augmentation ideal in the Burnside ring $A(G)$. The report states an isomorphism of pro-...

open_problem | open | MathDB 页面 | 原文线索

V2-016153 | MATHDB-394773

Finiteness of the algebraic fundamental group below the Noether-Reid bound

摘录 / EXCERPT: For a minimal complex surface of general type S, if $K_S^2<4\chi(S)$ and S has no irregular cover, is $\pi_1^{\mathrm{alg}}(S)$ a finite group?

open_problem | open | MathDB 页面 | 原文线索

V2-016154 | MATHDB-394839

First Zassenhaus conjecture (ZC1) for finite groups

摘录 / EXCERPT: For a finite group G, every torsion unit in $\Z G$ is conjugate to an element of $\\pm G$ by a unit of $\Q G$.

conjecture | open | MathDB 页面 | 原文线索

V2-016155 | MATHDB-394841

Sehgal’s Problem on p-adic conjugacy of torsion units

摘录 / EXCERPT: Let G be a finite group with a normal p-subgroup N, and let u be a torsion unit in $\mathbb{Z}G$ which maps to the identity under the map $\mathbb{Z}G \to \mathbb{Z}(G/N)$....

open_problem | open | MathDB 页面 | 原文线索

V2-016156 | MATHDB-394844

Kimmerle’s proposition on Klein four subgroups of normalized units

摘录 / EXCERPT: Let G be a finite group. If the Klein four-group $C_2\times C_2$ is isomorphic to a subgroup of the normalized unit group $V(\mathbb{Z}G)$, then G has a subgroup isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-016157 | MATHDB-394850

Free-product generation by Bass cyclic units

摘录 / EXCERPT: Let G be a finite group, and let $x\in G$ have order n. For $i>0$ with $(i,n)=1$, and for a multiple m of $\varphi(n)$, define the Bass cyclic unit ... Suppose that...

open_problem | open | MathDB 页面 | 原文线索

V2-016158 | MATHDB-394851

Stable isomorphism problem for integral group rings

摘录 / EXCERPT: Let G and H be finite groups. If $M_n(\mathbb{Z}G)$ and $M_n(\mathbb{Z}H)$ are isomorphic rings, does it follow that $\mathbb{Z}G$ and $\mathbb{Z}H$ are isomorphic rings?

open_problem | open | MathDB 页面 | 原文线索

V2-016159 | MATHDB-394855

Simple sections of finite subgroups of normalized units

摘录 / EXCERPT: Let G be a finite group and let $V(\mathbb{Z}G)$ denote the normalized unit group of its integral group ring. Is it true that each simple section of a finite subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-016160 | MATHDB-394856

Structure of endomorphism rings of indecomposable projective modules over p-adic group rings

摘录 / EXCERPT: Let G be a finite group, and let R be the ring of integers in a finite extension field K of the p-adic numbers. Write $RG=\bigoplus_i P_i$ as a direct sum of indecomposable...

open_problem | open | MathDB 页面 | 原文线索

V2-016161 | MATHDB-394857

Kimmerle’s conjugacy problem for torsion units

摘录 / EXCERPT: Let H be a finite group. Is there a finite group G containing H such that each torsion unit in $V(\mathbb{Z}H)$ is conjugate to an element of H by a unit of $\mathbb{Q}G$?...

open_problem | open | MathDB 页面 | 原文线索

V2-016162 | MATHDB-394858

Subgroup separability of unit groups of integral group rings

摘录 / EXCERPT: When is the unit group $U(\mathbb{Z}G)$ of the integral group ring $\mathbb{Z}G$ of a finite group G a subgroup separable group?

open_problem | open | MathDB 页面 | 原文线索

V2-016163 | MATHDB-395022

Perturbative ergodic theorem for random Lie group actions

摘录 / EXCERPT: Let $G\subset GL(L,\mathbb C)$ be a Lie group, let M be a compact, connected, symmetric space given as a quotient of two compact Lie groups, and let G act smoothly and transiti...

open_problem | open | MathDB 页面 | 原文线索

V2-016164 | MATHDB-395037

Non-realizability of DΔ-lattices as finite-group overgroup lattices

摘录 / EXCERPT: Let Λ be a finite $D\Delta$-lattice: writing 0 and ∞ for its least and greatest elements and $\Lambda' = \Lambda \setminus \{0,\infty\}$, regarded as a graph under...

open_problem | open | MathDB 页面 | 原文线索

V2-016165 | MATHDB-395102

Short Presentation Conjecture for Finite Groups

摘录 / EXCERPT: Every finite group G has a presentation whose length is $O\bigl((\log |G|)^3\bigr)$; the report states that this bound is best possible.

conjecture | open | MathDB 页面 | 原文线索

V2-016166 | MATHDB-395103

Cohomology-dimension bounds for absolutely irreducible modules

摘录 / EXCERPT: For each integer $j>0$, there is a constant C_j such that, for every finite group G, every field k, and every absolutely irreducible kG-module M, ...

open_problem | open | MathDB 页面 | 原文线索

V2-016167 | MATHDB-395116

Baer-Suzuki-type characterization of the solvable radical

摘录 / EXCERPT: Let G be a finite group. The solvable radical of G coincides with the collection of all $g\in G$ such that, for any three elements $a,b,c\in G$, the subgroup...

open_problem | open | MathDB 页面 | 原文线索

V2-016168 | MATHDB-395182

Invariant geometric structures for finite group actions on geometric 3-manifolds

摘录 / EXCERPT: Given a smooth finite group action $\rho:G\curvearrowright M$ on a closed geometric 3-manifold M, does there exist a ρ-invariant complete locally homogeneous metric on M?

open_problem | open | MathDB 页面 | 原文线索

V2-016169 | MATHDB-395261

Poisson boundary of simple random walks on wreath products in dimensions 3 and 4

摘录 / EXCERPT: For simple random walk on the wreath product $\mathbb{Z}^d\wr B$, where $d\in\{3,4\}$ and B is a finite group with at least two elements, is the Poisson-Furstenberg boundary equa...

open_problem | open | MathDB 页面 | 原文线索

V2-016170 | MATHDB-395281

Kropholler’s fixed-point question

摘录 / EXCERPT: Does there exist a countably infinite group G for which every finite-dimensional contractible G-CW-complex has a global fixed point?

open_problem | open | MathDB 页面 | 原文线索

V2-016171 | MATHDB-395291

Stable extension of fusion-preserving homomorphisms

摘录 / EXCERPT: Let (S,F,L) and (S',F',L') be p-local finite groups, and suppose that $\rho:S\to S'$ is a fusion-preserving homomorphism. Then there exists an integer $m\ge 0$ and a map ... su...

open_problem | open | MathDB 页面 | 原文线索

V2-016172 | MATHDB-395410

Cofinite universal spaces for proper group actions

摘录 / EXCERPT: For a discrete group Γ, determine when Γ admits a cofinite universal space $E\Gamma$ for proper actions, and find explicit models for such spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-016173 | MATHDB-395418

Dévissage for local deformation functors

摘录 / EXCERPT: Let k be a perfect field of characteristic $p>0$, let G be a finite group, let $N\triangleleft G$, and let $\rho:G\hookrightarrow\operatorname{Aut}_k(k[[t]])$ be a local G-ac...

open_problem | open | MathDB 页面 | 原文线索

V2-016174 | MATHDB-395516

Patching and Schacher’s admissibility conjecture

摘录 / EXCERPT: Let F be the function field of a curve over a complete discretely valued field K whose residue field k is algebraically closed. If a finite group G is admissible over F,...

conjecture | open | MathDB 页面 | 原文线索

V2-016175 | MATHDB-395682

General Version of James’ Conjecture for Modular Principal Series

摘录 / EXCERPT: Let $G=\mathbf G^F$ be a finite group of Lie type over $\mathbb F_q$, with Weyl group W, and let B be an F-stable Borel subgroup. Let k be an algebraically closed field of...

conjecture | open | MathDB 页面 | 原文线索

V2-016176 | MATHDB-395687

Brauer Height Zero Conjecture for 2-blocks of Maximal Defect

摘录 / EXCERPT: Let G be a finite group and let B be a 2-block of G with defect group $P\in\operatorname{Syl}_2(G)$. Then $\chi(1)$ is odd for all $\chi\in\operatorname{Irr}(B)$ if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-016177 | MATHDB-395691

Gluing problem for blocks of finite groups

摘录 / EXCERPT: Let G be a finite group, let k be an algebraically closed field of characteristic p with $p\mid|G|$, and let b be a block of kG. Fix a maximal b-Brauer pair (P,e_P),...

open_problem | open | MathDB 页面 | 原文线索

V2-016178 | MATHDB-396022

Classification of certain product-quotient surfaces with $p_g=0$

摘录 / EXCERPT: Let C_1 and C_2 be compact complex curves of respective genera $g_1,g_2\ge 2$, let G be a finite group acting faithfully on each C_i, let $X:=(C_1\times C_2)/G$ be the quot...

open_problem | open | MathDB 页面 | 原文线索

V2-016179 | MATHDB-396263

Malle-Navarro nilpotent-block question

摘录 / EXCERPT: Let G be a finite group and B a p-block of G. Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-016180 | MATHDB-396306

Geometric proof of the convolution description of affine braid-group actions

摘录 / EXCERPT: Provide a geometric proof of Theorem 1(ii): assuming $p=0$ or that p is greater than the Coxeter number of G, prove that the action of T_w on...

open_problem | open | MathDB 页面 | 原文线索

V2-016181 | MATHDB-396482

Contractibility of the fibres of the fibrations $B(q,G)\to BG(q)$

摘录 / EXCERPT: Let G be a finite group. For every $q\ge 2$, there is a natural fibration $B(q,G)\to BG(q)$ whose fibre is a simply connected finite-dimensional complex K_q. Under what conditi...

open_problem | open | MathDB 页面 | 原文线索

V2-016182 | MATHDB-396486

Blocks with elementary abelian defect groups of order 8

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 2, let G be a finite group, and let b be a block of kG with an elementary abelian defect group P of order 8....

open_problem | open | MathDB 页面 | 原文线索

V2-016183 | MATHDB-396488

Invariant-theory image for a cyclic group action

摘录 / EXCERPT: For an $\Fp[\Z/p]$-module V with no trivial direct summands, describe the image of the composite ... where $S^*(V)$ is the polynomial algebra on V and $V_{\u0005Z/p}$ denotes the...

open_problem | open | MathDB 页面 | 原文线索

V2-016184 | MATHDB-396490

Lifting exotic endotrivial modules from Sylow 2-subgroups

摘录 / EXCERPT: Let G be a finite group over a field k of characteristic 2, and let P be a Sylow 2-subgroup of G that is generalized quaternion or semidihedral. Do the exotic endotrivi...

open_problem | open | MathDB 页面 | 原文线索

V2-016185 | MATHDB-396491

Endotrivial modules with endotrivial Green correspondents

摘录 / EXCERPT: Let G be a finite group with Sylow p-subgroup P, and let k be a field of characteristic p. Which indecomposable endotrivial $kN_G(P)$-modules have an endotrivial kG-Gre...

open_problem | open | MathDB 页面 | 原文线索

V2-016186 | MATHDB-396580

Finite groups with effective M-semi-symplectic actions on Enriques surfaces

摘录 / EXCERPT: Is a finite group embeddable into the symmetric group S_6 of degree 6, if it has an effective M-semi-symplectic action on an Enriques surface?

open_problem | open | MathDB 页面 | 原文线索

V2-016187 | MATHDB-396581

Further restrictions on spectral measures of group-ring operators

摘录 / EXCERPT: What further restrictions do spectral measures of group-ring operators over residually finite groups satisfy, beyond being weak limits of root distributions of monic polynomials wi...

open_problem | open | MathDB 页面 | 原文线索

V2-016188 | MATHDB-396583

Strictness of the first \\2-Betti-number lower bound for cost

摘录 / EXCERPT: Is there some infinite group with a free action such that $\beta_1(\Gamma) < \operatorname{cost}(R_{\Gamma\curvearrowright (X,\mu)}) - 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-016189 | MATHDB-396662

Finite-group quotients of random 2-complex fundamental groups

摘录 / EXCERPT: A result of Meshulam states that, for a finite group Γ, the threshold for the non-existence of an epimorphism $\pi_1(Y(n,n^\alpha))\twoheadrightarrow\Gamma$ occurs at...

open_problem | open | MathDB 页面 | 原文线索

V2-016190 | MATHDB-396936

Quantisation commutes with reduction for proper cocompact noncompact group actions

摘录 / EXCERPT: Let G be a unimodular Lie group, let (M,ω) be a symplectic manifold, and let $G\curvearrowright M$ be a proper strongly Hamiltonian action. Suppose 0 is a regular value...

open_problem | open | MathDB 页面 | 原文线索

V2-016191 | MATHDB-396950

Classification of modules up to Bousfield equivalence

摘录 / EXCERPT: Let G be a finite group and k a field whose characteristic divides $|G|$. In the stable module category $\operatorname{StMod}(kG)$ of all kG-modules, classify modules up to...

open_problem | open | MathDB 页面 | 原文线索

V2-016192 | MATHDB-397106

Undecidability of the 0-divisor problem for a lamplighter-group extension

摘录 / EXCERPT: There exists a finite group H such that the 0-divisor problem for the group $\left(\mathbb{Z}/2 \wr \mathbb{Z}\right)^6 \times H$ is undecidable.

open_problem | open | MathDB 页面 | 原文线索

V2-016193 | MATHDB-397219

Number of conjugacy classes needed to distinguish a finite group from its proper subgroups

摘录 / EXCERPT: In general, how many conjugacy classes should one expect to require before reaching the conclusion that an injective homomorphism $i:\operatorname{Gal}(K/\mathbb{Q})\to G$ is surje...

open_problem | open | MathDB 页面 | 原文线索

V2-016194 | MATHDB-397255

Deciding whether a presentation defines a trivial, finite, or infinite group

摘录 / EXCERPT: Given a presentation, determine whether it defines the trivial group, a finite group, or an infinite group. The report states that this problem is insoluble in general.

open_problem | open | MathDB 页面 | 原文线索

V2-016195 | MATHDB-397352

Decomposition conjecture for gerbes and their dual orbifolds

摘录 / EXCERPT: For a finite group G and a G-gerbe Y over an orbifold B, let $\widehat{Y}$ be the associated dual orbifold and let c be the associated $U(1)$-valued discrete-torsion gerb...

conjecture | open | MathDB 页面 | 原文线索

V2-016196 | MATHDB-397370

Counterexample: localization need not preserve commutative equivariant ring spectra

摘录 / EXCERPT: Let G be a finite group, let P be the family of proper subgroups of G, and let $\widetilde{E}P$ be the cofiber of the natural map $EP_+\to S^0$. This localization of S^0 ca...

open_problem | open | MathDB 页面 | 原文线索

V2-016197 | MATHDB-397385

Asymptotic results for nonfree finite-group actions on curves

摘录 / EXCERPT: Are there asymptotic results for G-actions which are not free?

open_problem | open | MathDB 页面 | 原文线索

V2-016198 | MATHDB-397580

Automorphism-equivariance of Jordan decomposition for \SL_n\

摘录 / EXCERPT: For the finite group of Lie type $G^F=\mathrm{SL}_n(\mathbb F_q)$ arising from the Frobenius endomorphism F in the report, the Jordan decomposition of characters is...

open_problem | open | MathDB 页面 | 原文线索

V2-016199 | MATHDB-397583

The relevant direction of Brauer’s height-zero conjecture

摘录 / EXCERPT: If a p-block of a finite group has abelian defect groups, then the p-parts of the degrees of any two ordinary irreducible characters in the block are equal.

conjecture | open | MathDB 页面 | 原文线索

V2-016200 | MATHDB-397586

A π-version of the McKay counting conjecture for nilpotent Hall subgroups

摘录 / EXCERPT: Let G be a finite group with a nilpotent Hall $\[1m\pi\u001b[0m$-subgroup H. The number of irreducible characters of $\[1m\pi\u001b[0m'$-degree of G is conjectured to equal t...

conjecture | open | MathDB 页面 | 原文线索

V2-016201 | MATHDB-397587

Character-degree data and solvability

摘录 / EXCERPT: Does knowledge of the set of the degrees of a finite group G determine whether G is solvable?

open_problem | open | MathDB 页面 | 原文线索

V2-016202 | MATHDB-397691

Quillen’s conjecture in strong-collapse form

摘录 / EXCERPT: Let G be a finite group, and let $S_p(G)$ denote the poset of nontrivial p-subgroups of G, ordered by containment. Is it true that if the order complex $K(S_p(G))$ is contrac...

conjecture | open | MathDB 页面 | 原文线索

V2-016203 | MATHDB-397942

Oort Conjecture on lifting covers of curves

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let $W(k)$ be its ring of Witt vectors, and let G be a finite group. If $Y\to X$ is a (possibly ramified) G-co...

conjecture | open | MathDB 页面 | 原文线索

V2-016204 | MATHDB-398090

Borel reducibility of isomorphism for nuclear simple separable C*-algebras

摘录 / EXCERPT: Is the isomorphism relation for nuclear simple separable C^*-algebras Borel reducible to the orbit equivalence relation of a Polish group action?

open_problem | open | MathDB 页面 | 原文线索

V2-016205 | MATHDB-398444

Stable component count for Hurwitz spaces over curves of arbitrary genus

摘录 / EXCERPT: For each equipped finite group (G,O), where $O=C_1\sqcup\cdots\sqcup C_m$ and the elements of O generate G, there is a constant T such that, for any projective irreducible...

open_problem | open | MathDB 页面 | 原文线索

V2-016206 | MATHDB-398447

Irreducibility of strata in moduli spaces of curves with symmetry

摘录 / EXCERPT: For a finite group G and a fixed genus g, consider the moduli space $S_g(G)$ of smooth genus-g curves with a G-action. After fixing the genus g' of the quotient curve...

open_problem | open | MathDB 页面 | 原文线索

V2-016207 | MATHDB-398470

Every finite group is involved in a mapping class group

摘录 / EXCERPT: Let Σ_g be a closed orientable surface of genus $g\ge 1$, and let Γ_g be its orientation-preserving mapping class group. A group H is involved in a group G if the...

open_problem | open | MathDB 页面 | 原文线索

V2-016208 | MATHDB-398839

Galois realizations of exceptional finite groups

摘录 / EXCERPT: For sufficiently large primes ℓ, the finite groups $E_8(\mathbb{F}_\ell)$, $F_4(\mathbb{F}_\ell)$, and $G_2(\mathbb{F}_\ell)$ occur as Galois groups over $\mathbb{Q}$ (that is...

open_problem | open | MathDB 页面 | 原文线索

V2-016209 | MATHDB-398849

Counting Galois extensions of \(\mathbb{Q}\) with prescribed Frobenius data

摘录 / EXCERPT: Let G be a nontrivial finite group that is a regular Galois group over $\mathbb{Q}$. For a finite set S of primes and subsets $F_p\subset G$, one for each $p\in S$, where each...

open_problem | open | MathDB 页面 | 原文线索

V2-016210 | MATHDB-398850

Existence of a finite group of Tchebotarev order greater than 1

摘录 / EXCERPT: Does there exist a finite group G such that ${\rm tch}(G)>1$? Here, for a real number $\ell\ge 0$, ${\rm tch}(G)\le \ell$ means that there exist real numbers $m,\delta>0$ such th...

open_problem | open | MathDB 页面 | 原文线索

V2-016211 | MATHDB-399309

An uncompleted Segal-conjecture isomorphism for finite groups

摘录 / EXCERPT: Let G be a finite group with Sylow p-subgroup S, and let $A(\mathcal{F}_p(G))$ be the Burnside ring of G-stable finite S-sets, where G-stability is defined by requiring...

conjecture | open | MathDB 页面 | 原文线索

V2-016212 | MATHDB-399342

Equality of Sullivan entropy and the Poincaré exponent for Kleinian groups

摘录 / EXCERPT: Do the Sullivan entropy $h_S(\Gamma)$ and the Poincaré exponent $\delta(\Gamma)$ agree for classes of Kleinian groups other than the geometrically finite groups considered by Sulli...

open_problem | open | MathDB 页面 | 原文线索

V2-016213 | MATHDB-399347

Finite-index subgroups in finitely generated profinite groups

摘录 / EXCERPT: Let G be a finitely generated profinite group. Then every abstract finite-index subgroup H of G is open in G.

open_problem | open | MathDB 页面 | 原文线索

V2-016214 | MATHDB-399349

Finitely generated infinite homomorphic images of topologically finitely generated profinite groups

摘录 / EXCERPT: Is it possible for a topologically finitely generated profinite group to have a finitely generated infinite homomorphic image? The report states that the answer is no: if G is a...

open_problem | open | MathDB 页面 | 原文线索

V2-016215 | MATHDB-399351

Embedding finitely generated groups into quotients of topologically finitely generated profinite groups

摘录 / EXCERPT: Is it possible to embed every finitely generated group as a subgroup of a quotient of a topologically finitely generated profinite group?

open_problem | open | MathDB 页面 | 原文线索

V2-016216 | MATHDB-399500

Equivariant localization preserving commutative ring spectra

摘录 / EXCERPT: Let G be a finite group, let L_E denote localization with respect to an equivariant spectrum E, and call a spectrum Z E-acyclic if $E\wedge Z$ is contractible. If, for ev...

open_problem | open | MathDB 页面 | 原文线索

V2-016217 | MATHDB-399503

Descent in algebraic K-theory for finite Galois extensions

摘录 / EXCERPT: Assume that G is a finite group and that $A\to B$ is a G-Galois extension in the sense of Rognes satisfying ... Then the canonical map of K-theory spectra ... is an L_n-equivalen...

open_problem | open | MathDB 页面 | 原文线索

V2-016218 | MATHDB-399560

Bertram’s conjecture on the number of conjugacy classes

摘录 / EXCERPT: For every finite group G, one has $\log_{3}|G|<k(G)$, where $k(G)$ denotes the number of conjugacy classes of G.

conjecture | open | MathDB 页面 | 原文线索

V2-016219 | MATHDB-399562

Lower bound for the number of conjugacy classes of a finite group

摘录 / EXCERPT: Does there exist a universal constant $c>0$ such that, for every finite group G, ... where $k(G)$ denotes the number of conjugacy classes of G?

open_problem | open | MathDB 页面 | 原文线索

V2-016220 | MATHDB-399566

Simple modules of double Burnside algebras

摘录 / EXCERPT: For a finite group G and a field k, understand the simple $kB(G,G)$-modules.

open_problem | open | MathDB 页面 | 原文线索

V2-016221 | MATHDB-399567

Jacobson radical of the evaluated biset module

摘录 / EXCERPT: For finite groups G and H and a field k, is the $kB(G,G)$-module $R(G,H)$ the Jacobson radical of $kB(G,H)$ when evaluated at G?

open_problem | open | MathDB 页面 | 原文线索

V2-016222 | MATHDB-399571

Complex group algebras determine the double cover of the alternating group

摘录 / EXCERPT: Let $n\ge 5$, let G be a finite group, and let $\widetilde{A}_n$ be the double cover of the alternating group A_n. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-016223 | MATHDB-399572

Huppert’s conjecture for alternating groups

摘录 / EXCERPT: For every integer $n\ge 5$ and every finite group G such that $\operatorname{cd}(G)=\operatorname{cd}(A_n)$, there exists an abelian group A such that $G\cong A_n\times A$.

conjecture | open | MathDB 页面 | 原文线索

V2-016224 | MATHDB-399629

Finite group actions on spheres with rank-one isotropy

摘录 / EXCERPT: For which finite groups G does there exist a finite G-CW-complex $X \simeq S^n$ with all isotropy subgroups of rank one?

open_problem | open | MathDB 页面 | 原文线索

V2-016225 | MATHDB-399630

Classification of finite groups admitting rank-one-isotropy G-homotopy representations

摘录 / EXCERPT: Determine which finite groups of rank two admit a finite G-homotopy representation X with rank-one isotropy; that is, a finite G-CW-complex such that every fixed-point set...

open_problem | open | MathDB 页面 | 原文线索

V2-016226 | MATHDB-399634

Cellularity of a p-local finite group classifying space

摘录 / EXCERPT: Given a finite p-group P and a p-local finite group $(S,\mathcal F,\mathcal L)$, when is $B\mathcal F\in\mathcal C(BP)$?

open_problem | open | MathDB 页面 | 原文线索

V2-016227 | MATHDB-399639

Restriction criterion for one-dimensional modules and endotrivial modules

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic p, let G be a finite group whose order is divisible by p, and let S be a Sylow p-subgroup of G. For every non...

open_problem | open | MathDB 页面 | 原文线索

V2-016228 | MATHDB-400083

Short laws for finite groups of bounded order

摘录 / EXCERPT: There exists a constant C such that, for every $n\in\mathbb{N}$, there is a law for every finite group of order at most n whose length is at most ...

open_problem | open | MathDB 页面 | 原文线索

V2-016229 | MATHDB-400092

Hochschild cohomology of a smash product in modular characteristic

摘录 / EXCERPT: Fix a field k, an associative unital k-algebra A, a finite group G acting on A, and an AG-bimodule M, where $AG=A\# kG$ is the smash product. How are the algebras...

open_problem | open | MathDB 页面 | 原文线索

V2-016230 | MATHDB-400093

Relation between Hochschild cohomology of a smash product and invariant Hochschild cohomology in modular characteristic

摘录 / EXCERPT: Let k be a field, A an associative unital k-algebra, G a finite group acting on A by algebra automorphisms, and M an $A\# kG$-bimodule. Assume additionally that a map...

open_problem | open | MathDB 页面 | 原文线索

V2-016231 | MATHDB-400262

Finite-automorphism Enriques surfaces in characteristic 2

摘录 / EXCERPT: Does there exist an Enriques surface in characteristic 2 with a finite group of automorphisms whose dual graph of all nonsingular rational curves is of type I, II, $\ldots$,...

open_problem | open | MathDB 页面 | 原文线索

V2-016232 | MATHDB-400341

Descent for quadratic forms invariant under a finite group

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field F, and assume that 2 is a unit in R. Let $\b3$ be a finite group whose order is invertible in R. Let (P,f) and...

open_problem | open | MathDB 页面 | 原文线索

V2-016233 | MATHDB-400434

Structure of the rational double Burnside algebra

摘录 / EXCERPT: For a finite group G, the problem of describing the structure of the rational double Burnside algebra $\mathbb{Q}B(G,G)$ is wide open.

open_problem | open | MathDB 页面 | 原文线索

V2-016234 | MATHDB-400463

Equivariant Kirchberg-Phillips-type absorption for amenable group actions

摘录 / EXCERPT: Does the classical Kirchberg-Phillips absorption theorem have an equivariant analogue, interpreted in the above sense, for outer actions of amenable groups?

open_problem | open | MathDB 页面 | 原文线索

V2-016235 | MATHDB-400468

Classification of lamplighter-group C*-algebras

摘录 / EXCERPT: Classify, up to isomorphism, the C-algebras $C^*(F\wr\mathbb Z)$ as F ranges over finite groups; equivalently, determine for finite groups F_1,F_2 when...

open_problem | open | MathDB 页面 | 原文线索

V2-016236 | MATHDB-400881

Invariant positive semidefinite forms versus invariant sums of squares for finite group actions

摘录 / EXCERPT: Given a finite group G, establish, for forms invariant under the action of G, whether the cone of invariant positive semidefinite forms equals the cone of invariant sums-of-squ...

open_problem | open | MathDB 页面 | 原文线索

V2-016237 | MATHDB-401095

Classification of lattice envelopes

摘录 / EXCERPT: Let Γ be a countable group satisfying $(\dagger)$: (1) Γ is not virtually isomorphic to a product of two infinite groups; (2) Γ has no infinite amenable commen...

open_problem | open | MathDB 页面 | 原文线索

V2-016238 | MATHDB-401096

Determination of finitely generated residually finite groups by finite quotients

摘录 / EXCERPT: Determine when the set of finite quotients of a finitely generated residually finite group determines the group up to isomorphism; equivalently, clarify when isomorphic profinite c...

open_problem | open | MathDB 页面 | 原文线索

V2-016239 | MATHDB-401098

Finite index of profinite closures in finitely generated just-infinite groups

摘录 / EXCERPT: In a finitely generated just-infinite group, the profinite closure of an infinite commensurated subgroup is of finite index.

open_problem | open | MathDB 页面 | 原文线索

V2-016240 | MATHDB-401457

Norm map for a relative finite groupoid

摘录 / EXCERPT: Let $\mathcal C$ be a pre-additive ∞-category admitting limits and colimits indexed by classifying spaces of finite groups. If $f:X\to Y$ is a map of Kan complexes whose hom...

open_problem | open | MathDB 页面 | 原文线索

V2-016241 | MATHDB-401458

Segal’s conjecture for finite groups

摘录 / EXCERPT: For every finite group G, the tautological map $s\colon A(G)\to K(\mathrm{Fin})^0(BG)$ induces an isomorphism ... where $A(G)$ is the Burnside ring and I is the augmentation idea...

conjecture | open | MathDB 页面 | 原文线索

V2-016242 | MATHDB-401473

Parametric extensions for finite groups

摘录 / EXCERPT: The report states that no group G had failed to have a parametric extension over a given number field k: a regular Galois extension $F/k(T)$ that parametrizes, via specializati...

open_problem | open | MathDB 页面 | 原文线索

V2-016243 | MATHDB-401475

Existence of regularly parametric Galois extensions

摘录 / EXCERPT: For a finite group G over a number field k (with $k=\mathbb{Q}$ in the diagram’s examples), does there exist a k-regular Galois extension $F/k(T)$ of group G such that ever...

open_problem | open | MathDB 页面 | 原文线索

V2-016244 | MATHDB-401481

Classification of finite groups admitting finite 1-parametric sets over $\mathbb{Q}$

摘录 / EXCERPT: Determine all finite groups G for which there exists a finite nonempty set S of $\mathbb{Q}$-regular Galois extensions of $\mathbb{Q}(T)$ with group G such that every Galois...

open_problem | open | MathDB 页面 | 原文线索

V2-016245 | MATHDB-401492

Q-admissibility conjecture

摘录 / EXCERPT: The Q-admissibility conjecture states that a finite group is Q-admissible if and only if it is Sylow-metacyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-016246 | MATHDB-401633

Hecke algebra representation theory and determination of the field

摘录 / EXCERPT: Let K be either a number field or a non-archimedean local field of characteristic zero. To what extent does (the representation theory of) the Hecke algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-016247 | MATHDB-401646

Decidable projective profinite groups with undecidable open subgroups

摘录 / EXCERPT: Is there a projective profinite group G which is decidable but which has an open subgroup H which is undecidable? Here, decidable means in the sense of limits of finite quotien...

open_problem | open | MathDB 页面 | 原文线索

V2-016248 | MATHDB-401647

Tame realization with prescribed inertia groups

摘录 / EXCERPT: Given $C_1,\ldots,C_{m(G)+r}$ conjugacy classes of cyclic subgroups generating a finite group G, is there an extension $L/\mathbb{Q}$ which is tame, has Galois group G, and has...

open_problem | open | MathDB 页面 | 原文线索

V2-016249 | MATHDB-401649

Determining fields of definition or moduli of finite covers of the thrice-punctured projective line

摘录 / EXCERPT: Given a finite group G and a surjection $F_2=\langle a,b\rangle\twoheadrightarrow G$, let $Z\to\mathbb{P}^1_{\mathbb{C}}\setminus\{0,1,\infty\}$ be the associated G-cover. Dete...

open_problem | open | MathDB 页面 | 原文线索

V2-016250 | MATHDB-401654

Smoothability of a one-dimensional group action

摘录 / EXCERPT: Let M be a connected 1-manifold and let $G<\operatorname{Homeo}^{+}(M)$ be a group action. Is the original group action smoothable; that is, does there exist a homeomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-016251 | MATHDB-401692

Dual complexes of log Calabi-Yau pairs as finite quotients of spheres

摘录 / EXCERPT: Does the dual complex of a log Calabi-Yau pair of dimension n equal the quotient of a sphere S^k of dimension $k\le n$ by some finite group $G\subset O_{k+1}(\mathbb{R})$?

open_problem | open | MathDB 页面 | 原文线索

V2-016252 | MATHDB-401749

Brylinski’s conjecture for Hochschild homology of compact Lie group actions

摘录 / EXCERPT: Let a compact Lie group H act smoothly on a manifold M, let $H\ltimes M$ be the transformation groupoid, and let A be its smooth convolution algebra. Then the Hochschild homo...

conjecture | open | MathDB 页面 | 原文线索

V2-016253 | MATHDB-401917

Classification of orbit closures for the semigroup action

摘录 / EXCERPT: Can one describe all the orbit closures Γx_0 in X?

open_problem | open | MathDB 页面 | 原文线索

V2-016254 | MATHDB-402013

Description of the kernel of the restriction map for endotrivial modules

摘录 / EXCERPT: For a finite group G, a field k of characteristic p dividing $|G|$, and a Sylow p-subgroup S of G, describe the kernel ...

open_problem | open | MathDB 页面 | 原文线索

V2-016255 | MATHDB-402076

Finiteness of Picard groups of blocks over a discrete valuation ring

摘录 / EXCERPT: Let (K,O,k) be a p-modular system with $k=\overline{k}$, let G be a finite group, and let $B=OGb$ be a block. Is the Picard group $\operatorname{Pic}(B)$ necessarily finite i...

open_problem | open | MathDB 页面 | 原文线索

V2-016256 | MATHDB-402080

Necessity of the one-simple-module hypothesis for simple $HH^1(B)$

摘录 / EXCERPT: For a block B of a finite group algebra kG, is the hypothesis that B has a single isomorphism class of simple modules necessary for Theorem 4.1, which characterizes when the...

conjecture | open | MathDB 页面 | 原文线索

V2-016257 | MATHDB-402083

Spectrum Problem for normalized units in integral group rings

摘录 / EXCERPT: For every finite group G and every positive integer n, does $\mathbb{Z}G$ contain a normalized unit of order n if and only if G contains an element of order n? Here,...

open_problem | open | MathDB 页面 | 原文线索

V2-016258 | MATHDB-402089

Conjecture on the 2-parts of real conjugacy-class sizes

摘录 / EXCERPT: Let G be a finite group. Suppose that all non-central real class sizes of G have the same 2-part; equivalently, $|x^G|_2=2^a$ for all non-central real elements $x\in G$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-016259 | MATHDB-402324

Classes exhibiting the equivariant Gal phenomenon

摘录 / EXCERPT: Let P be a flag simplicial d-dimensional polytope on which a finite group G acts simplicially. The induced graded G-representation on the cohomology of the associated toric...

open_problem | open | MathDB 页面 | 原文线索

V2-016260 | MATHDB-402527

Extension of a residually finite group by a residually finite group is weakly sofic

摘录 / EXCERPT: Let H be a normal subgroup of a group K. If H and $G=K/H$ are residually finite, then K is weakly sofic.

open_problem | open | MathDB 页面 | 原文线索

V2-016261 | MATHDB-402710

Closure of K3 and abelian-surface derived categories under equivariantization

摘录 / EXCERPT: Let S be a smooth complex projective symplectic surface, hence a K3 or abelian surface, and let a finite group G act on $D^b(S)$ such that: (i) every equivalence induced by an...

open_problem | open | MathDB 页面 | 原文线索

V2-016262 | MATHDB-402750

Alperin’s Weight Conjecture for Principal Blocks

摘录 / EXCERPT: If k is an algebraically closed field of characteristic p, G is a finite group, B_0 is the principal block of kG, and $ ... where, for a saturated fusion system $mathcal F...

conjecture | open | MathDB 页面 | 原文线索

V2-016263 | MATHDB-402795

Extension of the braid-group action for products of Dynkin diagrams

摘录 / EXCERPT: Let Δ and Δ' be simply laced Dynkin diagrams with Coxeter numbers k and $n-k$, respectively. Let ρ be the Zamolodchikov transformation of the cluster algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-016264 | MATHDB-403071

Grunwald problem with finitely many prescribed local extensions

摘录 / EXCERPT: Let G be a finite group. It is conjectured that, for every finite set S of primes of $\b Q$ and every choice of G-extensions $L^{(p)}/\Q_p$ for $p\in S$, there exists a G-e...

open_problem | open | MathDB 页面 | 原文线索

V2-016265 | MATHDB-403147

Fredholm criterion for restrictions to isotypical components

摘录 / EXCERPT: Let Γ be a finite group acting by diffeomorphisms on a compact smooth manifold M, and let P be a Γ-invariant classical pseudodifferential operator of order m ac...

open_problem | open | MathDB 页面 | 原文线索

V2-016266 | MATHDB-403252

Statistics of the Galois module structure of Mordell-Weil groups

摘录 / EXCERPT: Fix a number field K, an elliptic curve E/K, a finite group G, and a finitely generated $\[1m\mathbb{Q}[G]\u001b[0m$-module V. Let $\mathcal{L}$ be the finite set of isomor...

open_problem | open | MathDB 页面 | 原文线索

V2-016267 | MATHDB-403270

Distribution of Galois groups of maximal unramified extensions

摘录 / EXCERPT: Fix a finite group Γ. For a totally real (equivalently, split completely at infinity) Γ-extension K of $\mathbb{Q}$ or $\mathbb{F}_q(t)$, let $K^{\#}$ be the maxima...

open_problem | open | MathDB 页面 | 原文线索

V2-016268 | MATHDB-403272

Moment convergence implies measure convergence for finite \Gamma-S-groups

摘录 / EXCERPT: For a finite group Γ and a finite set S of primes not dividing $|\Gamma|$, call a finite group with a Γ-action a Γ-S-group if its order is a product of pow...

open_problem | open | MathDB 页面 | 原文线索

V2-016269 | MATHDB-403302

Schur’s exponent conjecture

摘录 / EXCERPT: For every finite group G, the exponent of the Schur multiplier $M(G)$ divides the exponent of G.

conjecture | open | MathDB 页面 | 原文线索

V2-016270 | MATHDB-403307

Classification of symplectic reflection groups admitting symplectic resolutions

摘录 / EXCERPT: Let V be a finite-dimensional complex symplectic vector space, and let $\Gamma\subset\operatorname{Sp}(V)$ be a finite group generated by symplectic reflections, i.e. elements wh...

open_problem | open | MathDB 页面 | 原文线索

V2-016271 | MATHDB-403310

Embedding extensions with prescribed finite groups

摘录 / EXCERPT: Given finite groups X and G, find all possible finite abelian groups A and actions of G on A such that there is an exact sequence ...

open_problem | open | MathDB 页面 | 原文线索

V2-016272 | MATHDB-403312

Explicit norm relations with optimized or restricted subgroup support

摘录 / EXCERPT: Given a finite group G, find an explicit norm relation ... where $d\in\mathbb Z$, $a_H,b_H\in\mathbb Z[H]$, and $N_H=\sum_{g\in H}g$. In addition, make d as small as possible, or...

open_problem | open | MathDB 页面 | 原文线索

V2-016273 | MATHDB-403389

Grothendieck’s problem on profinite completions

摘录 / EXCERPT: Let $u:\Gamma\to\Lambda$ be a homomorphism of finitely presented, residually finite groups such that the induced homomorphism on profinite completions,...

open_problem | open | MathDB 页面 | 原文线索

V2-016274 | MATHDB-403825

Zeros and roots of unity in finite-group character tables

摘录 / EXCERPT: Conjecture. Let ... and let ... where the minimum is taken over all larger-than-average conjugacy classes K of the finite group G. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-016275 | MATHDB-404038

Equal-product disjoint subsets in finite groups

摘录 / EXCERPT: For every $k\ge 2$, does there exist $c(k)$ such that for all groups G with $|G|=n$ and $S\subseteq G$ with $|S|\ge c(k)\log n$, there are $T_1,\ldots,T_k\subseteq S$ pairwise di...

open_problem | open | MathDB 页面 | 原文线索

V2-016276 | MATHDB-404275

Stratification of rational spectra for profinite groups

摘录 / EXCERPT: Theorem 2. Let G be a profinite group. The category of rational G-spectra $\mathsf{Sp}_{G,\mathbb{Q}}$ is a tensor-triangulated category with Balmer spectrum ... and the followin...

open_problem | open | MathDB 页面 | 原文线索

V2-016277 | MATHDB-404381

Minimal counterexample to the solvable-multiplicative-group assertion for skew braces

摘录 / EXCERPT: Find a minimal counterexample to the assertion that every solvable finite group is the multiplicative group of a skew brace of abelian type, where a skew brace is a triple...

open_problem | open | MathDB 页面 | 原文线索

V2-016278 | MATHDB-404451

Polylogarithmic diameter as a group property

摘录 / EXCERPT: Fix $c\in\mathbb{N}$, and let $G=\langle S\rangle$ be a finite group. Suppose that $|S|\le c$ and $\operatorname{diam}(G,S)\le (\log |G|)^c$. Does there exist $c'(c)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-016279 | MATHDB-404490

Group Spencer conjecture

摘录 / EXCERPT: Let G be a finite group of size n. Conjecture 1 holds in the particular case in which $A_1,\ldots,A_n$ are the $n\times n$ matrices corresponding to the regular representation...

conjecture | open | MathDB 页面 | 原文线索

V2-016280 | MATHDB-404529

Finite-index generation of the outer automorphism group by Dehn twists

摘录 / EXCERPT: Let K be a p-adic field, and let $G_K(p)$ be the maximal pro-p quotient of the absolute Galois group of K. Does the profinite group generated by all Dehn twists have finite...

open_problem | open | MathDB 页面 | 原文线索

V2-016281 | MATHDB-404789

Determining the nonabelian chief-factor lifting case

摘录 / EXCERPT: Let N be a nonabelian minimal normal subgroup of a finite group G, let $d=d(G/N)$, and suppose that $g_1N,\ots,g_dN$ generate G/N. Determine whether case C1 holds, namely whe...

open_problem | open | MathDB 页面 | 原文线索

V2-016282 | MATHDB-404794

Nilpotent decomposition property for nonfaithful cyclic semidirect products

摘录 / EXCERPT: We say that a finite group G has the nilpotent decomposition property if, for every nilpotent element $n\in \mathbb{Z}G$ in the integral group ring of G and every primitive cen...

open_problem | open | MathDB 页面 | 原文线索

V2-016283 | MATHDB-404912

Stable rank one for crossed products of free minimal actions

摘录 / EXCERPT: Does there exist a countably infinite group G, a compact metrizable space X, and a topologically free minimal action $G\curvearrowright X$ with $M_G(X)\ne\varnothing$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-016284 | MATHDB-404999

Noetherian group algebras and polycyclic-by-finite groups

摘录 / EXCERPT: Conjecture 2. If the group algebra of a group G is Noetherian, then G is polycyclic-by-finite.

open_problem | open | MathDB 页面 | 原文线索

V2-016285 | MATHDB-405005

Normal complements for multiplicative groups of finite left braces

摘录 / EXCERPT: Determine classes of finite groups G that are the multiplicative group of a left brace so that G has a normal complement in $U_1(\mathbb Z[G])$.

open_problem | open | MathDB 页面 | 原文线索

V2-016286 | MATHDB-405032

Profinite rigidity of finitely generated free and hyperbolic surface groups

摘录 / EXCERPT: Let Γ be either a finitely generated free group F or a hyperbolic surface group S. Is every finitely generated residually finite group Λ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-016287 | MATHDB-405033

Word-measure rigidity for free-group words

摘录 / EXCERPT: Let F be a non-abelian free group of rank n. For every $w\in F$, every finite group G, and every subset $A\subseteq G$, define the word measure by ... where $w:G^n\to G$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-016288 | MATHDB-405136

Periodicity of points in the permutation-module spectrum

摘录 / EXCERPT: Let G be a (pro-)finite group and $H\leq G$ a (pro-) p-subgroup. Then $P(H,p)$ is periodic if and only if p is. In other words, the periodic locus of...

open_problem | open | MathDB 页面 | 原文线索

V2-016289 | MATHDB-405138

Cusick's conjecture for free elementary abelian 2-group actions on cohomological products of real projective spaces

摘录 / EXCERPT: Let $G=(\mathbb Z/2)^r$, and let X be a finite free G-CW-complex such that ... Must ... where, for each positive integer n, ...

conjecture | open | MathDB 页面 | 原文线索

V2-016290 | MATHDB-405139

Global fixed points for finite-group actions on compact acyclic or contractible CW-complexes

摘录 / EXCERPT: Let G be a finite group acting on a compact CW-complex X, where X is either acyclic over some ring or contractible. Must the fixed-point set X^G be non-empty?

open_problem | open | MathDB 页面 | 原文线索

V2-016291 | MATHDB-405153

Bader-Furman-Gelander-Monod Question 4.1 on property (T_X)

摘录 / EXCERPT: For every $1<p<\infty$, every measure space (Ω,μ), and every real Banach space X that is either a closed subspace of $L_p(\Omega,\mu;\mathbb{R})$ or a quotient of such a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016292 | MATHDB-405212

Question whether the presentation functor preserves tilting objects

摘录 / EXCERPT: ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016293 | MATHDB-405226

Surjectivity conjectures of Dupont and Monod

摘录 / EXCERPT: For every connected semisimple Lie group G with finite centre and every degree $n\ge 0$, the comparison map from bounded continuous cohomology to continuous cohomology,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016294 | MATHDB-405240

Jantzen conjecture for graded affine Hecke algebras

摘录 / EXCERPT: Let $\mathcal H$ be a graded affine Hecke algebra, let M be a standard $\mathcal H$-module, and let $J^0M\supseteq J^1M\supseteq\cdots$ be its Jantzen filtration arising from a o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

B04 代数几何 / Algebraic Geometry | 7,034 条

V2-016295 | MATHDB-53

Reducedness of Hilb^8(S) for surfaces with ADE singularities

摘录 / EXCERPT: If S is a surface with ADE singularities, it is known that $\operatorname{Hilb}^n(S)$ is reduced for $n\leq 7$, while reducedness for arbitrary n remains open. Its description th...

open_problem | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016296 | MATHDB-54

Reduction conjecture for equivariant Gromov-Witten theory of GIT quotients

摘录 / EXCERPT: Let X be a smooth projective variety with an algebraic action of a complex torus $T_{\mathbb C}$, and let $Y=X/\!/T_{\mathbb C}$ be a smooth GIT quotient with no orbifold singula...

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016297 | MATHDB-55

The remaining four-variable Hessian conjecture

摘录 / EXCERPT: Four-variable Hessian conjecture (HC_4). Must φ^L be a polynomial? Equivalently, must every four-variable gradient polynomial map with constant nonzero Jacobian...

conjecture | claimed_progress | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016298 | MATHDB-57

Extra components of Hilbert schemes of points on affine three-space

摘录 / EXCERPT: Open problems. Determine the smallest d for which $H_{d,0}$ is reducible, and the smallest d for which H_d is reducible. The presently known bounds are ... Indeed, both spaces...

open_problem | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016299 | MATHDB-60

Orlov's motivic conjecture for derived-equivalent varieties

摘录 / EXCERPT: Let X and Y be smooth projective varieties. Orlov's motivic conjecture predicts ... where $h(X)_{\mathbb Q}$ denotes the rational Chow motive. A Fourier-Mukai equivalence produce...

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016300 | MATHDB-61

Orlov's conjecture on Rouquier dimension

摘录 / EXCERPT: Conjecture. For every smooth quasi-projective variety X, ... The lower bound $\operatorname{Rdim}D^b(X)\geq\dim X$ is known quite generally. The difficult direction is to construct...

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016301 | MATHDB-62

The Dubrovin-Gamma conjectures for Fano varieties

摘录 / EXCERPT: For a smooth Fano variety X of Hodge-Tate type, the modern form of Dubrovin's conjecture predicts ... The refined Gamma conjecture II also identifies categorical and quantum data....

conjecture | claimed_progress | Vihaan Dheer | MathDB 页面 | 原文线索

V2-016302 | MATHDB-315500

High-dimensional Mean King’s Problem

摘录 / EXCERPT: Can the Mean King’s quantum retrodiction puzzle be implemented beyond qubits with a scalable photonic scheme that retains a clear advantage over the best classical strategy?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-016303 | MATHDB-315670

Improved key-recovery attack on HAWK

摘录 / EXCERPT: Can the best known key-recovery attack against the HAWK signature scheme be made substantially faster?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016304 | MATHDB-315672

Full key-recovery attack on SpoC-128

摘录 / EXCERPT: Is there a practical full-strength key-recovery attack on the unmodified SpoC-128 authenticated-encryption scheme?

open_problem | open | MathDB 页面 | 原文线索

V2-016305 | MATHDB-315677

Bondal-Polishchuk transitivity conjecture for smooth projective varieties

摘录 / EXCERPT: Can the braid-group action on full exceptional collections fail to be transitive for $D^b(X)$ when X is a smooth projective variety?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016306 | MATHDB-315714

Batyrev’s nonnegativity conjecture for stringy Hodge numbers

摘录 / EXCERPT: Nonnegativity conjecture. All stringy Hodge numbers $h^{p,q}_{\rm st}(X)$ are nonnegative.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016307 | MATHDB-315735

Optimal bend-and-break constant for foliations

摘录 / EXCERPT: Let X be a normal projective variety of dimension n, and let $\mathcal{F}$ be a foliation on X of rank r. Let $H_{1},\ldots,H_{n-1},H$ be ample divisors on X, and let C...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016308 | MATHDB-315737

Bounded total Cartier indices in families

摘录 / EXCERPT: Are the total Cartier indices bounded for a bounded family of varieties with rational singularities?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016309 | MATHDB-315748

Unconditional efficient one-bit unclonable encryption

摘录 / EXCERPT: The answer is affirmative. For every $n\geq 1$, there exists a one-time private-key quantum encryption scheme for one-bit classical messages with a $(2n-1)$-bit classical key and a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016310 | MATHDB-315774

Integral local invariant cycles in degree one

摘录 / EXCERPT: Let $f:X\to\Delta$ be a semistable one-parameter family of complex projective varieties over a holomorphic disk, with X smooth and central fiber $X_0=f^{-1}(0)$ a reduced simple...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-016311 | MATHDB-315779

Jacobian conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic zero, let $n\ge 1$ be an integer, and let $f_1,\dots,f_n\in\mathbb{K}[X_1,\dots,X_n]$. Let $F:\mathbb{K}^n\to\mathbb{K}^n$ be the poly...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016312 | MATHDB-315784

Grothendieck’s group-scheme question

摘录 / EXCERPT: Is a finite locally free group scheme killed by its order?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-016313 | MATHDB-315851

Hodge conjecture

摘录 / EXCERPT: Let X be a non-singular complex projective variety of complex dimension n (equivalently, a compact complex manifold admitting a holomorphic embedding into some complex projecti...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016314 | MATHDB-315927

Abundance conjecture

摘录 / EXCERPT: Let k be a field and let X be a projective variety over k with Kawamata log terminal singularities; that is, X is normal, the canonical divisor K_{X} is $\mathbb{Q}$-Cart...

conjecture | open | MathDB 页面 | 原文线索

V2-016315 | MATHDB-315928

Bass conjecture

摘录 / EXCERPT: For a noetherian ring A, let $\mathcal{M}(A)$ be the abelian category of finitely generated A-modules and let $\mathcal{P}(A)$ be the exact category of finitely generated proje...

conjecture | open | MathDB 页面 | 原文线索

V2-016316 | MATHDB-315930

Deligne conjecture

摘录 / EXCERPT: Pierre René, Viscount Deligne is a Belgian mathematician. He is best known for work on the Weil conjectures, leading to a complete proof in 1973. He is the winner of the 1978 Field...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-016317 | MATHDB-315933

Fröberg conjecture

摘录 / EXCERPT: Let $n\ge 1$ and $k\ge 1$ be integers, let $a_1,a_2,\ldots,a_k$ be positive integers, and let $S=\mathbb{C}[x_1,x_2,\ldots,x_n]$ carry its standard grading, with S_a denoting the...

conjecture | open | MathDB 页面 | 原文线索

V2-016318 | MATHDB-315934

Fujita conjecture

摘录 / EXCERPT: Let M be a compact complex manifold of complex dimension n, let K_M denote a canonical line bundle of M, and let L be a positive holomorphic line bundle on M. Then for...

conjecture | open | MathDB 页面 | 原文线索

V2-016319 | MATHDB-315937

Maulik-Nekrasov-Okounkov-Pandharipande conjecture

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau threefold, and let Q^β denote the Novikov monomial associated with an effective curve class $\beta\in H_2(X,\mathbb Z)$. For every n...

conjecture | open | MathDB 页面 | 原文线索

V2-016320 | MATHDB-315938

Nagata's conjecture on curves

摘录 / EXCERPT: Work over an algebraically closed field of characteristic 0, and let $\mathbb{P}^2$ denote the projective plane. Call a tuple of points $(p_1,\dots,p_r) \in (\mathbb{P}^2)^r$ ver...

conjecture | open | MathDB 页面 | 原文线索

V2-016321 | MATHDB-315939

Nagata-Biran conjecture

摘录 / EXCERPT: Let X be a smooth complex projective surface and let L be an ample line bundle on X, and write $d=\sqrt{L^{2}}$ for its degree, where $L^{2}=L\cdot L$ denotes the self-inters...

conjecture | open | MathDB 页面 | 原文线索

V2-016322 | MATHDB-315942

Section conjecture

摘录 / EXCERPT: Let k be a field finitely generated over $\mathbb{Q}$, let $\bar{k}$ be an algebraic closure of k, and set $\operatorname{Gal}(k) = \operatorname{Gal}(\bar{k}/k)$. Let X be a...

conjecture | open | MathDB 页面 | 原文线索

V2-016323 | MATHDB-315943

Standard conjectures

摘录 / EXCERPT: Let k be a field and let H^{*} be a Weil cohomology theory on smooth projective varieties over k with coefficients in a field F of characteristic 0: for each smooth proje...

conjecture | open | MathDB 页面 | 原文线索

V2-016324 | MATHDB-315944

Tate conjecture

摘录 / EXCERPT: Let k be a field that is finitely generated over its prime field, let k_s be a separable closure of k, and let $G=\operatorname{Gal}(k_s/k)$. Let ℓ be a prime number tha...

conjecture | open | MathDB 页面 | 原文线索

V2-016325 | MATHDB-315945

Virasoro conjecture

摘录 / EXCERPT: For every smooth projective complex variety X, let F_g denote its genus-g Gromov-Witten descendent potential and define the total descendent potential ... Let L_n, $n\ge -1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-016326 | MATHDB-316092

Manin conjecture

摘录 / EXCERPT: Let K be a number field and let V be a Fano variety defined over K, i.e. a smooth projective variety over K whose anticanonical divisor $-K_V$ is ample. Let H be a height...

conjecture | open | MathDB 页面 | 原文线索

V2-016327 | MATHDB-316094

Vojta's conjecture

摘录 / EXCERPT: Let F be a number field, let X/F be a non-singular algebraic variety, let D be an effective divisor on X with at worst normal crossings, let H be an ample divisor on X,...

conjecture | open | MathDB 页面 | 原文线索

V2-016328 | MATHDB-316154

Torsion conjecture

摘录 / EXCERPT: For a number field K and an abelian variety A over K, let $A(K)$ denote the group of K-rational points of A and let ... be its torsion subgroup, which is finite since...

conjecture | open | MathDB 页面 | 原文线索

V2-016329 | MATHDB-316171

Weibel's conjecture

摘录 / EXCERPT: Let X be a Noetherian scheme of finite Krull dimension d, and for $i \in \mathbb{Z}$ let $K_i(X)$ denote the i-th nonconnective (Bass-Thomason-Trobaugh) algebraic K-group o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016330 | MATHDB-316179

Lambda g conjecture

摘录 / EXCERPT: In algebraic geometry, the $\lambda _{g}$-conjecture gives a particularly simple formula for certain integrals on the Deligne-Mumford compactification...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016331 | MATHDB-316184

Bogomolov conjecture

摘录 / EXCERPT: Let K be a number field with a fixed algebraic closure $\overline{K}$, let A be an abelian variety over K, let L be a symmetric ample line bundle on A, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016332 | MATHDB-316249

Woo-Yong's conjecture on the Gorenstein locus of Schubert varieties

摘录 / EXCERPT: Let X be a Schubert variety, and let $x\in X$. The Gorenstein-locus conjecture. The point x lies in the Gorenstein locus of X if and only if the generic point of every irredu...

conjecture | open | MathDB 页面 | 原文线索

V2-016333 | MATHDB-316267

Batyrev's stringy Chern-number conjecture

摘录 / EXCERPT: Let $\mathcal{X}$ be a Gorenstein proper Deligne-Mumford stack with projective coarse moduli space X, such that $\mathcal{X}\to X$ is proper and birational and $K_\mathcal{X}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-016334 | MATHDB-316274

Minimal Model Program and abundance conjecture for varieties with Kodaira dimension zero

摘录 / EXCERPT: Minimal Model Program and abundance conjecture. Every smooth projective variety with Kodaira dimension zero admits such a birational model. This is one of the standard conjectures...

conjecture | open | MathDB 页面 | 原文线索

V2-016335 | MATHDB-316275

Abundance conjecture for varieties of Kodaira dimension minus infinity

摘录 / EXCERPT: Let Y be a smooth projective variety. A variety is uniruled if through a general point of Y there passes a rational curve. Abundance conjecture. If $\kappa(Y)=-\infty$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-016336 | MATHDB-316284

Asymptotic irreducibility conjecture for universal Severi varieties

摘录 / EXCERPT: Asymptotic irreducibility conjecture. For every $\varepsilon>0$, if k is sufficiently large with respect to ε, then for every integer h satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-016337 | MATHDB-316309

Berthelot's conjecture D for arithmetic D-modules

摘录 / EXCERPT: Let ${\mathcal X}$ be a smooth formal scheme, let Z be a divisor of its special fibre, and write ${\mathcal U}$ for the complement of Z. Let ${\mathcal E}^\dagger$ be a coheren...

conjecture | open | MathDB 页面 | 原文线索

V2-016338 | MATHDB-316342

The function-field Selmer-module conjecture for abelian varieties

摘录 / EXCERPT: Let F be a function field in one variable over a finite field $\mathbb F_q$, let A/F be an abelian variety with arbitrary reduction, and let $F^{(p)}_\infty/F$ be the constant...

conjecture | open | MathDB 页面 | 原文线索

V2-016339 | MATHDB-316351

Cofinality conjecture for rings of integers of type K(π,1)

摘录 / EXCERPT: Let k be a number field and let p be a prime number. Assume that $p\neq 2$ or that k is totally imaginary. Let S be a finite set of primes of k with...

conjecture | open | MathDB 页面 | 原文线索

V2-016340 | MATHDB-316379

The abstract classification conjecture for hypertoric varieties

摘录 / EXCERPT: A symplectic algebraic variety is a connected algebraic variety equipped with a symplectic structure. Let T^d be an algebraic torus of dimension d acting effectively and hamilt...

conjecture | open | MathDB 页面 | 原文线索

V2-016341 | MATHDB-316404

Cheltsov's conjecture on global log canonical thresholds of quartic and quintic Fano hypersurfaces

摘录 / EXCERPT: Let X be a general quartic threefold or a general quintic fourfold, viewed as a Fano hypersurface in projective space. Cheltsov's conjecture. The global log canonical threshold s...

conjecture | open | MathDB 页面 | 原文线索

V2-016342 | MATHDB-316405

Cheltsov's conjecture on attainment of the global log canonical threshold

摘录 / EXCERPT: Let X be a Fano variety. For each natural number $m\geq 1$, let $\mathrm{lct}_{m}(X)$ denote the corresponding m-th global log canonical threshold, and let $\mathrm{lct}(X)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-016343 | MATHDB-316412

K-theoretic virtual localization conjecture

摘录 / EXCERPT: Let X be an equivariant virtually smooth scheme with fixed-point components X_i, inclusion $\iota:\bigcup X_i\to X$, virtual structure sheaves...

conjecture | open | MathDB 页面 | 原文线索

V2-016344 | MATHDB-316459

Normality conjecture for special pieces

摘录 / EXCERPT: Let P be a special piece in the unipotent variety of a reductive algebraic group. Normality conjecture for special pieces. Every special piece P is normal. This is a necessary...

conjecture | open | MathDB 页面 | 原文线索

V2-016345 | MATHDB-316465

Kanev-Lange isomorphism conjecture for simply ramified B_n-coverings

摘录 / EXCERPT: Let $p:C\overset{\pi}{\longrightarrow}C'\overset{g}{\longrightarrow}\mathbb{P}^1$ be a simply ramified covering of type B_n. Let $P(C,C')$ be its Prym variety, let $P(X,\delta)$...

conjecture | open | MathDB 页面 | 原文线索

V2-016346 | MATHDB-316466

Kanev-Lange polarization-type conjecture for simply ramified B_n-coverings

摘录 / EXCERPT: Let $p:C\overset{\pi}{\longrightarrow}C'\overset{g}{\longrightarrow}\mathbb{P}^1$ be a simply ramified B_n-covering with C irreducible. Let $\mathfrak{D}_s$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-016347 | MATHDB-316473

The vanishing conjecture for intersection numbers on the perfect cone compactification

摘录 / EXCERPT: Vanishing conjecture. The intersection numbers $a_N^{(g)}$ for any N vanish unless $G-N=k(k+1)/2$ for some $k\le g$. The claim extends the vanishing pattern established in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-016348 | MATHDB-316487

Tsuzuki's full-faithfulness conjecture for overconvergent isocrystals

摘录 / EXCERPT: Let k be the perfect field and $\mathcal{S}$ the fixed formal base. Let X be a smooth separated k-scheme of finite type, and let $X\hookrightarrow\overline{X}$ be an open imm...

conjecture | open | MathDB 页面 | 原文线索

V2-016349 | MATHDB-316512

The rationality conjecture for specializations of Hodge classes

摘录 / EXCERPT: Let A be an abelian variety over an algebraic closure of $\mathbb{Q}$ with good reduction to an abelian variety A_{0} over an algebraic closure of a finite field. A Hodge class...

conjecture | open | MathDB 页面 | 原文线索

V2-016350 | MATHDB-316514

Special-points conjecture for Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura p-datum, let $\operatorname{Sh}_{p}(G,X)$ be its integral model, and let $\mathbb{F}$ be the algebraic closure of the residue field. A point of...

conjecture | open | MathDB 页面 | 原文线索

V2-016351 | MATHDB-316515

Zink's theorem for Shimura varieties of Hodge type

摘录 / EXCERPT: Zink's theorem conjecture. Zink's theorem holds for Shimura varieties of Hodge type.

conjecture | open | MathDB 页面 | 原文线索

V2-016352 | MATHDB-316572

The algebraic-closedness criterion for diophantine subfields

摘录 / EXCERPT: Let k be a large field, and let $K_1\subset K_2$ be function fields of k-varieties. Algebraic-closedness criterion. The subfield K_1 is diophantine in K_2 if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-016353 | MATHDB-316578

Zhang's fixed-point growth conjecture for polarized endomorphisms

摘录 / EXCERPT: Let X be a projective variety over $\mathbb{C}$ and let $\varphi:X\to X$ be an endomorphism with $\varphi^*L\cong L^{\otimes q}$ for some ample line bundle L on X. Suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-016354 | MATHDB-316587

n-transitivity conjecture for automorphism groups of rational real algebraic surfaces

摘录 / EXCERPT: n-transitivity conjecture. The group $\operatorname{Aut}(X)$ acts n-transitively on X.

conjecture | open | MathDB 页面 | 原文线索

V2-016355 | MATHDB-316606

Hausel-Letellier-Villegas purity conjecture for character varieties

摘录 / EXCERPT: Let $\M^\mu_\B$ be the smooth character variety associated with generic semisimple conjugacy classes of parabolic type μ, let $\bar{\M}^\mu_\B$ be its corresponding $\PGL_n$ qu...

conjecture | open | MathDB 页面 | 原文线索

V2-016356 | MATHDB-316607

Kac-Moody L^2 Euler characteristic conjecture for parabolic character varieties

摘录 / EXCERPT: Let $\bar{\M}^\mu_\B$ be the character variety quotient associated with a genus-g Riemann surface and parabolic type μ, and let $\chi_{L^2}(\bar{\M}^\mu_\B)$ denote its topol...

conjecture | open | MathDB 页面 | 原文线索

V2-016357 | MATHDB-316650

Donald's Calabi functional conjecture for polarised toric varieties

摘录 / EXCERPT: Let (X,L) be a polarised toric variety with moment polytope P. Let $S(\omega)$ be the scalar curvature of a Kähler metric $\omega\in c_1(L)$, let $\hat{S}$ be its average, and...

conjecture | open | MathDB 页面 | 原文线索

V2-016358 | MATHDB-316652

The toric Harder-Narasimhan subdivision conjecture

摘录 / EXCERPT: Let $(P,d\sigma)$ be the moment polytope of a polarised toric variety with canonical boundary measure $d\sigma$. A pair $(Q,d\sigma)$ is semistable when $\mathcal{L}_A(f)\geq 0$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-016359 | MATHDB-316653

The abundance conjecture for smooth projective varieties

摘录 / EXCERPT: Abundance conjecture. The canonical divisor K_X is abundant, namely

conjecture | open | MathDB 页面 | 原文线索

V2-016360 | MATHDB-316656

Birch-Swinnerton-Dyer-Deligne-Gross conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field K, let F/K be a Galois extension, and let τ be a complex representation of $\operatorname{Gal}(F/K)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016361 | MATHDB-316657

The p-parity conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field K, and let $\mathcal X_p(A/K)$ be the tensor product with $\mathbb Q_p$ of the Pontryagin dual of the p^∞-Selmer group of...

conjecture | open | MathDB 页面 | 原文线索

V2-016362 | MATHDB-316658

The p-parity conjecture for self-dual Artin twists

摘录 / EXCERPT: Let A be an abelian variety over a number field K, let F/K be a Galois extension, let p be a prime, and let τ be a self-dual representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-016363 | MATHDB-316660

Frobenius semisimplicity conjecture for Tate classes

摘录 / EXCERPT: Let X be a smooth projective variety over the algebraic closure $\mathbb{F}$ of $\mathbb{F}_p$, let r be a nonnegative integer, and let $\ell\ne p$. For a Frobenius map π o...

conjecture | open | MathDB 页面 | 原文线索

V2-016364 | MATHDB-316661

The rationality conjecture for CM abelian varieties

摘录 / EXCERPT: Let A be a CM abelian variety over $\mathbb{Q}^{\mathrm{al}}$, with good reduction A_0 over $\mathbb{F}$. A Hodge class on A is an element of $H_{\mathbb{A}}^{2*}(A)(*)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-016365 | MATHDB-316664

The Weyl-group rational smoothness criterion for the Poincaré polynomial

摘录 / EXCERPT: Rational smoothness criterion. The equality $P_w(q)=R_w(q)$ holds if and only if the Schubert variety X_w is rationally smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-016366 | MATHDB-316682

Fulton's conjecture on the cone of curves of the moduli space of stable pointed curves

摘录 / EXCERPT: Let $\overline{\mathbf{M}}_{0,n}$ be the moduli space of stable n-pointed genus-zero curves, and let $\overline{NE}_1$ denote the closure of the real cone of effective curves on...

conjecture | open | MathDB 页面 | 原文线索

V2-016367 | MATHDB-316708

Penkov-Tikhomirov's classification conjecture for vector bundles on ind-Grassmannians

摘录 / EXCERPT: Let the ground field be $\mathbb{C}$, and let $\mathbf E$ be a vector bundle of rank $r\in\mathbb{Z}_{>0}$ on an ind-Grassmannian $\mathbf X=\displaystyle\lim_\to G(k_m;V^{n_m})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-016368 | MATHDB-316727

Isotopy conjecture for analytic continuation of spherical principal series

摘录 / EXCERPT: Let $G=\mathrm{SL}(n,\mathbb R)$, let $\mathrm{Fl}(\mathbb R^n)$ and $\mathrm{Fl}(\mathbb C^n)$ denote the real and complex complete flag varieties, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016369 | MATHDB-316733

Categorified Reshetikhin-Turaev functor conjecture for sl(m)

摘录 / EXCERPT: For a sequence $\beta=(\beta_1,\ldots,\beta_n)$, let Y_β be the corresponding geometric variety, let $K(Y_\beta)=K(D(Y_\beta))$ be its Grothendieck group, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-016370 | MATHDB-316739

Serre's conjecture on the invariant of principally polarized abelian threefolds

摘录 / EXCERPT: Serre's conjecture. There is an invariant $\chi_{18}(A,\lambda)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-016371 | MATHDB-316751

Grothendieck's conjecture on algebraicity of Künneth projectors

摘录 / EXCERPT: Grothendieck's conjecture. The Künneth projectors

conjecture | open | MathDB 页面 | 原文线索

V2-016372 | MATHDB-316759

The base-point-free conjecture for nef line bundles on projective Calabi-Yau varieties

摘录 / EXCERPT: Base-point-free conjecture. If L is nef, then L is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-016373 | MATHDB-316770

Zak's conjecture on cohomological normality of projective varieties

摘录 / EXCERPT: Zak's conjecture. 1. ...

conjecture | open | MathDB 页面 | 原文线索

V2-016374 | MATHDB-316772

Whitney's local holomorphic lamination conjecture for analytic varieties

摘录 / EXCERPT: Let V be an analytic variety in $\mathbb C^n$ and let X be a stratum of an analytic stratification of V. For every point $p\in X$, there should exist a neighborhood U of...

conjecture | open | MathDB 页面 | 原文线索

V2-016375 | MATHDB-316776

Bloch-Kato conjecture on integral motivic cohomology

摘录 / EXCERPT: Let U be a smooth variety over a number field or a local field F, and let $\HM^i(U,n)$ be its motivic cohomology. Let $\HMf^i(U,n)$ be the subgroup whose ℓ-adic Abel-Jacob...

conjecture | open | MathDB 页面 | 原文线索

V2-016376 | MATHDB-316778

Local invariant cycle conjecture

摘录 / EXCERPT: Let $f\colon X\to S$ be proper and flat with regular total space, special fibre Y and geometric generic fibre $\Ubar$. Let I be the inertia subgroup acting on the cohomology of...

conjecture | open | MathDB 页面 | 原文线索

V2-016377 | MATHDB-316780

Standard K-theory conjectures for smooth varieties over finite fields

摘录 / EXCERPT: Let Y be proper and smooth over a finite field k, and let $\bar k$ be an algebraic closure. Tate-Parshin standard conjectures. The following statements hold: - The action of...

conjecture | open | MathDB 页面 | 原文线索

V2-016378 | MATHDB-316798

Erdenberger-Hulek intersection-number vanishing conjecture for principally polarized abelian varieties

摘录 / EXCERPT: Intersection-number vanishing conjecture. The intersection number

conjecture | open | MathDB 页面 | 原文线索

V2-016379 | MATHDB-316799

The indecomposable theta-multiplicity conjecture for principally polarized abelian varieties

摘录 / EXCERPT: Theta-multiplicity conjecture. The maximal multiplicity of the theta function for indecomposable ppavs is

conjecture | open | MathDB 页面 | 原文线索

V2-016380 | MATHDB-316806

The motivic generating-function conjecture for stable-bundle moduli spaces

摘录 / EXCERPT: Let S be the completed and suitably localized Grothendieck ring of motives over $\mathbb{C}$, and let Γ denote the set of characters $\alpha=(n,d)$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-016381 | MATHDB-316807

Nonnegative-coefficient conjecture for nested Hilbert-scheme Euler characteristics

摘录 / EXCERPT: Nonnegative-coefficient conjecture. For every $m\geq 1$, $\mathcal{N}^{(m)}_n(q,t)$ is a polynomial in q and t with nonnegative integer coefficients. The authors further conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-016382 | MATHDB-316841

The adjoint-test ideal correspondence conjecture along a subvariety

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a regular local ring essentially of finite type over a perfect field of prime characteristic p, and let $I \subseteq R$ be a nonzero unmixed ideal. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016383 | MATHDB-316866

Arithmetic mirror ordinarity conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a 3-dimensional Calabi-Yau variety over $\mathbb{Q}$, and assume that X has a mirror over $\mathbb{Q}$. Let the generic family containing X as a member be defined...

conjecture | open | MathDB 页面 | 原文线索

V2-016384 | MATHDB-316867

Arithmetic mirror slope-zeta conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let $\{f,g\}$ be two generic mirror families of a 3-dimensional Calabi-Yau variety over $\mathbb{Q}$, and let $S(f\otimes\mathbb{F}_p,U,T)$ and $S(g\otimes\mathbb{F}_p,U,T)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-016385 | MATHDB-316871

Dimension-codimension formula for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be the group under consideration, let $x\in\widetilde{W}$ be an element of its affine Weyl group, and let $b\in G$. Let I be an Iwahori subgroup, let $X_x(b)$ be the asso...

conjecture | open | MathDB 页面 | 原文线索

V2-016386 | MATHDB-316899

Behrend's vanishing conjecture for canonical parabolic subgroups

摘录 / EXCERPT: Let C be a smooth projective curve and let G be a reductive group scheme over C. Let $P\subset G$ be the canonical parabolic subgroup. Behrend's conjecture. One has ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-016387 | MATHDB-316900

The new t-structure detection conjecture for affine-flag D-modules

摘录 / EXCERPT: Let $\bD_{\mathrm{ren}}(\fD(\Fl^{\operatorname{aff}}_G)_\crit\operatorname{mod})$ carry the old and new t-structures, and let $J_\cla$ be the convolution objects indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-016388 | MATHDB-316904

The Miura-oper description of Iwahori-monodromic critical modules

摘录 / EXCERPT: Let $ MOp$ be the scheme of Miura opers with nilpotent singularities, and let $\bD^b(\hg_\crit\operatorname{mod}_\nilp)^{I^0}$ be the Iwahori-monodromic derived category of critica...

conjecture | open | MathDB 页面 | 原文线索

V2-016389 | MATHDB-316924

Milne's conjecture on integral canonical models of Shimura varieties

摘录 / EXCERPT: Let $(G_1,\mathcal{X}_1)$ be a Shimura pair such that $G_{1,\mathbb{Q}_p}$ is unramified. Let H_1 be a hyperspecial subgroup of $G_1(\mathbb{Q}_p)$, and let v_1 be a prime of...

conjecture | open | MathDB 页面 | 原文线索

V2-016390 | MATHDB-316929

Conjecture on idempotent-generated maximal subgroups of free profinite monoids

摘录 / EXCERPT: Let $\pv H$ be a variety of finite groups closed under extension and containing $\mathbb Z/p\mathbb Z$ for infinitely many primes. Let M be a finitely generated, non-procyclic fr...

conjecture | open | MathDB 页面 | 原文线索

V2-016391 | MATHDB-316959

Suslin's rank conjecture for fractional linear cycles

摘录 / EXCERPT: Let $\mathbbm{k}$ be a field satisfying Suslin's rank conjecture, and let F be a number field as an example. For integers p and $q\geq 0$, consider the cubical higher Chow grou...

conjecture | open | MathDB 页面 | 原文线索

V2-016392 | MATHDB-316978

Finiteness conjecture for wild étale kernels of number fields

摘录 / EXCERPT: Finiteness conjecture. The groups $W_{2i}(F)$ are finite for every number field F.

conjecture | open | MathDB 页面 | 原文线索

V2-016393 | MATHDB-316979

Boundedness conjecture for wild étale kernels in cyclotomic towers

摘录 / EXCERPT: Combined Iwasawa-Greenberg conjecture. The following groups have orders bounded independently of n and i: (i) the ℓ-kernels $W\!K_{4i}(F_n)$; and (ii) the imaginary compon...

conjecture | open | MathDB 页面 | 原文线索

V2-016394 | MATHDB-316984

The Hasse-Weil analytic continuation and functional-equation conjecture for abelian varieties

摘录 / EXCERPT: Let A/K be an abelian variety over a number field. Define its L-series $L(A/K,s)$ and normalized completed function $\Lambda(A/K,s)$ as in the source. Hasse-Weil conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-016395 | MATHDB-316985

The generalized Szpiro conjecture for abelian varieties

摘录 / EXCERPT: Let A/K be an abelian variety of dimension g over a number field K, with conductor ideal $\mathcal{F}_{A/K}$. Generalized Szpiro conjecture. There exist real numbers c_1 an...

conjecture | open | MathDB 页面 | 原文线索

V2-016396 | MATHDB-317001

The higher-codimension Bounded Height Conjecture

摘录 / EXCERPT: Let A be a semi-abelian variety, let V be an irreducible algebraic subvariety of dimension d defined over $\overline{\mathbb{Q}}$, and let Γ be a finite-rank subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-016397 | MATHDB-317010

Griffeth-Ram positivity conjecture for structure sheaf products

摘录 / EXCERPT: Griffeth-Ram positivity conjecture. For all $u,v,w\in W^P$,

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V2-016398 | MATHDB-317026

Equidistribution conjecture for nonseparable and two-stack sortable permutations

摘录 / EXCERPT: Let n be a positive integer. For a permutation, let comp, asc, ldr, and rmax denote the statistics used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-016399 | MATHDB-317060

Generalized Steinberg variety homology conjecture

摘录 / EXCERPT: Generalized Steinberg variety homology conjecture. With the notation above,

conjecture | open | MathDB 页面 | 原文线索

V2-016400 | MATHDB-317062

Equivariant K-theory conjecture for generalized Steinberg varieties

摘录 / EXCERPT: Let ${\mathcal P}$ and ${\mathcal Q}$ be parabolic types, with corresponding longest Weyl-group elements w_P and w_Q. Let $X^{{\mathcal P},{\mathcal Q}}$ be the associated gene...

conjecture | open | MathDB 页面 | 原文线索

V2-016401 | MATHDB-317064

Debarre's conjecture on minimal cohomology classes

摘录 / EXCERPT: Debarre's conjecture. The cycle Y has minimal cohomology class if and only if (A,Θ) is the polarised Jacobian of a curve C of genus n and Y is a translate of...

conjecture | open | MathDB 页面 | 原文线索

V2-016402 | MATHDB-317088

Non-closedness conjecture for symmetric tensors of bounded symmetric rank

摘录 / EXCERPT: Let $\mathcal{Y}_{r}$ be the set of symmetric tensors of symmetric rank at most r, and let $\overline{\mathcal{Y}}_{r}$ denote its closure. Let R_{S} be the maximum sy...

conjecture | open | MathDB 页面 | 原文线索

V2-016403 | MATHDB-317100

T-adic Dwork conjecture for pure slope L-functions

摘录 / EXCERPT: Let Λ be a quasi-projective subvariety of $N(\Delta,\mathbb{F}_p)$ defined over $\mathbb{F}_p$, and let f_Λ be the corresponding family of Laurent polynomials. For...

conjecture | open | MathDB 页面 | 原文线索

V2-016404 | MATHDB-317102

The generation conjecture for quantum cohomology of Hilbert schemes of A_n-resolutions

摘录 / EXCERPT: Let $4\mathcal{A}_{n}4$ be the surface $4A_n4$-resolution, and let $4\mathrm{Hilb}_{m}(\mathcal{A}_{n})4$ be its Hilbert scheme. The operators of quantum multiplication by the divi...

conjecture | open | MathDB 页面 | 原文线索

V2-016405 | MATHDB-317105

The generation conjecture for quantum multiplication on Hilbert schemes

摘录 / EXCERPT: Let M_D and $M_{(1,\omega_i)}$ be the quantum multiplication operators by the divisor classes D and $(1,\omega_i)$, respectively, acting on the quantum cohomology of the Hilber...

conjecture | open | MathDB 页面 | 原文线索

V2-016406 | MATHDB-317106

Generation conjecture for the DT-ring of Hilbert schemes of $\mathcal{A}_n$

摘录 / EXCERPT: Let $m>0$, and let M_D and $M_{(1,\omega_i)}$ be the commuting operators acting on the relative Donaldson-Thomas theory of $\mathrm{Hilb}_m(\mathcal{A}_n)$. A joint eigenspace is...

conjecture | open | MathDB 页面 | 原文线索

V2-016407 | MATHDB-317128

Griffiths-Harris rigidity conjecture for the Segre variety

摘录 / EXCERPT: Let $X=\operatorname{Seg}(\mathbb P^2\times\mathbb P^2)\subset\mathbb P(\mathbb C^3\otimes\mathbb C^3)$. Let $Z^4\subset\mathbb P V$ be a variety, and let $II_{Z,z}$ denote its sec...

conjecture | open | MathDB 页面 | 原文线索

V2-016408 | MATHDB-317149

Grothendieck's conjecture on universally closed morphisms with integral fibers

摘录 / EXCERPT: Grothendieck's conjecture. The morphism $X\to S$ is integral.

conjecture | open | MathDB 页面 | 原文线索

V2-016409 | MATHDB-317171

Otwinowska-Saito monodromy irreducibility conjecture

摘录 / EXCERPT: Let Y be a smooth irreducible complex projective variety of dimension $m+1$, let Z be a closed subscheme of Y, and let $\mathcal I_{Z,Y}(\delta)$ be generated by global sectio...

conjecture | open | MathDB 页面 | 原文线索

V2-016410 | MATHDB-317178

Faber's Gromov-Witten intersection relations

摘录 / EXCERPT: Let X be a smooth projective variety, let $\{T_a\}$ be a basis of $H^*(X,\mathbb Q)$, and write $T^a=g^{ab}T_b$ using the inverse of $g_{ab}=\int_X T_a\cup T_b$. For cohomology cl...

conjecture | open | MathDB 页面 | 原文线索

V2-016411 | MATHDB-317179

Faber's two-point Gromov-Witten relations and Hodge-class identity

摘录 / EXCERPT: Let X be a smooth projective variety and let $\{T_a\}$ be a basis of $H^*(X,\mathbb Q)$, with dual basis $T^a=g^{ab}T_b$ for the Poincaré pairing. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016412 | MATHDB-317194

Geometric McKay correspondence for inverse Fourier-Mukai transforms

摘录 / EXCERPT: Let $G \subset \operatorname{SL}_3(\mathbb{C})$ be a finite abelian group, and let $Y=G\text{-Hilb}(\mathbb{C}^3)$. Write $\Psi:D^G(\mathbb{C}^3)\to D(Y)$ for the inverse Fourier-M...

conjecture | open | MathDB 页面 | 原文线索

V2-016413 | MATHDB-317200

Abhyankar's conjecture on non-rectifiable embeddings of the complex line

摘录 / EXCERPT: A polynomial embedding is an injective polynomial map whose image is isomorphic to the complex line. Two polynomial embeddings $\phi_1,\phi_2:\mathbb{C}\hookrightarrow\mathbb{C}^3$...

conjecture | open | MathDB 页面 | 原文线索

V2-016414 | MATHDB-317207

Griffiths-Green algebraicity conjecture for zero loci of admissible normal functions

摘录 / EXCERPT: Let S be a smooth complex algebraic variety, let $\mathcal H$ be a variation of pure Hodge structure on S, and let ν be an admissible normal function, with zero locus...

conjecture | open | MathDB 页面 | 原文线索

V2-016415 | MATHDB-317217

The toroidalization conjecture for morphisms of nonsingular varieties

摘录 / EXCERPT: Let X and Y be nonsingular varieties over an algebraically closed field of characteristic 0, and let $f:X\longrightarrow Y$ be a dominant morphism. Let D_Y be a simple norm...

conjecture | open | MathDB 页面 | 原文线索

V2-016416 | MATHDB-317229

Aluffi-Mihalcea positivity conjecture for CSM classes of Schubert cells

摘录 / EXCERPT: Let $Gr(k,\mathbb{C}^{n})$ be the Grassmannian, let $P_{k,n-k}$ denote the set of partitions fitting in a $k\times(n-k)$ rectangle, and let X_α° be the Schubert cell in...

conjecture | open | MathDB 页面 | 原文线索

V2-016417 | MATHDB-317246

Natural S^{1}-equivariant lift of the trace for derived categorical sheaves

摘录 / EXCERPT: Natural lift conjecture. The complex $Tr(u)$ has a natural lift

conjecture | solved | MathDB 页面 | 原文线索

V2-016418 | MATHDB-317269

The subclose-family formula for higher weights of Grassmann and Schubert codes

摘录 / EXCERPT: Subclose-family formula. Let r be a positive integer. If $r\leq k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-016419 | MATHDB-317275

The Bloch-type motive decomposition conjecture for finite quotients

摘录 / EXCERPT: Let X be a smooth projective variety over a field k of characteristic 0, and let G be a finite group acting on X. Suppose that ... for all i. Write (X,π) for the moti...

conjecture | open | MathDB 页面 | 原文线索

V2-016420 | MATHDB-317288

The localization conjecture for partial flag variety cluster algebras

摘录 / EXCERPT: Localization conjecture. The localizations of $\widetilde{\cal A}_J$ and $\mathbb{C}[B_K^-d4f G]$ with respect to Σ_J are equal.

conjecture | open | MathDB 页面 | 原文线索

V2-016421 | MATHDB-317290

The total positivity criterion from rigid modules

摘录 / EXCERPT: Let $X=B_K^-d4f G$ be the partial flag variety, let $X_{>0}$ be its totally positive part, and let N_K be the embedded open cell. Let ... be a basic complete rigid Λ-module...

conjecture | open | MathDB 页面 | 原文线索

V2-016422 | MATHDB-317329

The P-alcove conjecture for basic affine Deligne-Lusztig varieties

摘录 / EXCERPT: P-alcove conjecture. $X_x(b)\ne\emptyset$ if and only if, for every semistandard $P=MN$ for which $x\mathbf a$ is a P-alcove, b is σ-conjugate to an element $b'\in M(L)$...

conjecture | open | MathDB 页面 | 原文线索

V2-016423 | MATHDB-317330

Reuman's dimension conjecture for shrunken affine Deligne-Lusztig varieties

摘录 / EXCERPT: Reuman's conjecture. If x lies in the shrunken Weyl chambers, then $X_x(b)\ne\emptyset$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-016424 | MATHDB-317331

The basic-class comparison conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Basic-class comparison conjecture. There exists $N_b\in\mathbb Z_{\ge 0}$ such that, for every $x\in\widetilde W$ with $\ell(x)\ge N_b$,

conjecture | open | MathDB 页面 | 原文线索

V2-016425 | MATHDB-317362

Hitchin's non-discrete image conjecture for branched conformal structures

摘录 / EXCERPT: Let X be a Riemann surface of genus g, let D be an effective divisor of degree d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016426 | MATHDB-317363

The holomorphy conjecture for ideals

摘录 / EXCERPT: Let $\mathcal{I}$ be an ideal defining a subscheme, let π be an embedded resolution of $\mathcal{I}$, and let $Z_{\mathrm{top},\mathcal{I}}^{(d)}$ denote the associated topolog...

conjecture | open | MathDB 页面 | 原文线索

V2-016427 | MATHDB-317385

Morel's conjecture on rational points in the -homotopy category

摘录 / EXCERPT: Let F be the base field, let X be a smooth variety over F, and write ... to X in this category. Define an equivalence relation on $X(F)$ generated by elementary ^{1}-homotopi...

conjecture | open | MathDB 页面 | 原文线索

V2-016428 | MATHDB-317386

Small-rank conjecture for 3-tensors

摘录 / EXCERPT: Assume $3\le m_1\le m_2\le m_3\le (m_1-1)(m_2-1)$ and $(m_1,m_2,m_3)\ne(3,2p+1,2p+1)$ for $p\in\mathbb{N}$, and let $\operatorname{grank}(m_1,m_2,m_3)$ be the generic rank. An inte...

conjecture | open | MathDB 页面 | 原文线索

V2-016429 | MATHDB-317387

Conjecture on linearly normal Legendrian varieties

摘录 / EXCERPT: Conjecture on linearly normal Legendrian varieties. X is one of the subadjoint varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-016430 | MATHDB-317390

Codimension conjecture for the theta-null gradient locus

摘录 / EXCERPT: Let ${\mathcal A}_g$ be the moduli space of principally polarized abelian varieties, let ${\mathcal X}_g$ be the universal principally polarized abelian variety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016431 | MATHDB-317391

Codimension conjecture for the singular theta-divisor locus

摘录 / EXCERPT: Let ${\mathcal X}_g$ be the universal principally polarized abelian variety, and let $\Theta_{\rm sing}\subseteq {\mathcal X}_g$ be the locus of points lying on a theta divisor tog...

conjecture | open | MathDB 页面 | 原文线索

V2-016432 | MATHDB-317392

Codimension conjecture for the loci G_k

摘录 / EXCERPT: Let ${\mathcal A}_g$ be the moduli space of principally polarized abelian varieties, and let G_k denote the loci introduced in the paper in analogy with the Andreotti-Mayer loci....

conjecture | open | MathDB 页面 | 原文线索

V2-016433 | MATHDB-317416

Bounded Height Conjecture for unlikely intersections in abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g, let $V\subset A$ be an irreducible subvariety of dimension d, and let $\Gamma\subset A$ be a subgroup of finite rank. The variety...

conjecture | open | MathDB 页面 | 原文线索

V2-016434 | MATHDB-317418

Functorial Bogomolov-type lower bound for transverse subvarieties

摘录 / EXCERPT: Let $(A,\mathcal{L})$ be a polarized abelian variety of dimension g defined over a number field $\mathbb{K}$, and let $X\subset A$ be a transverse subvariety with finite stabiliz...

conjecture | open | MathDB 页面 | 原文线索

V2-016435 | MATHDB-317424

Kuznetsov's homological projective duality conjecture for Pfaffian varieties

摘录 / EXCERPT: Let W be a vector space of dimension $2n$. For $1\le k\le n-1$, let $\operatorname{Pf}_k(W^*)\subset\mathbb{P}(\Lambda^2W^*)$ be the variety of skew-forms of rank at most $2n-2k$...

conjecture | open | MathDB 页面 | 原文线索

V2-016436 | MATHDB-317425

Semiorthogonal decomposition conjecture for Pfaffian linear sections

摘录 / EXCERPT: Let W be a vector space of dimension $2n$, let $V\subset\Lambda^2W^*$ satisfy $\dim V=2n=\dim W$, and let $V^\perp\subset\Lambda^2W$ be its orthogonal. Define the Pfaffian hypers...

conjecture | open | MathDB 页面 | 原文线索

V2-016437 | MATHDB-317426

Compatibility conjecture for the birational Abel-Jacobi correspondence

摘录 / EXCERPT: For generic $V\subset\Lambda^2W^*$, let X and Y be the associated Calabi-Yau varieties, and let $F(Y)$ and $H(X)$ be the birational moduli spaces described in the source. Since...

conjecture | open | MathDB 页面 | 原文线索

V2-016438 | MATHDB-317436

Fiber regularity conjecture for general linear projections

摘录 / EXCERPT: Let $X\subset \mathbf{P}^r$ be a smooth projective variety of dimension n, and let $\pi:X\to\mathbf{P}^{n+c}$ be a general linear projection. If $Z\subset X\subset\mathbf{P}^r$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-016439 | MATHDB-317437

Comparison conjecture between fiber regularity and the invariant q

摘录 / EXCERPT: Let $X\subset\mathbf{P}^r$ be a smooth projective variety of dimension n, and let $\pi:X\to\mathbf{P}^{n+c}$ be a general linear projection. If $Z\subset X\subset\mathbf{P}^r$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-016440 | MATHDB-317457

The vanishing conjecture for reduced regulators on self-products of very general surfaces

摘录 / EXCERPT: Reduced-regulator vanishing conjecture. The map $\underline{r}_{3,1}$ is zero.

conjecture | open | MathDB 页面 | 原文线索

V2-016441 | MATHDB-317458

The domination conjecture for pairings of rational curves on K3 surfaces

摘录 / EXCERPT: Domination conjecture. The variety ${\mathbb C}^*$ is dominated by one of the components of ${\mathcal M}^\perp\cap {\mathcal N}^\perp$ under the map h.

conjecture | open | MathDB 页面 | 原文线索

V2-016442 | MATHDB-317469

The conjecture on syzygies and regularity of higher secant varieties of curves

摘录 / EXCERPT: The secant-variety syzygy conjecture. Under these hypotheses:

conjecture | open | MathDB 页面 | 原文线索

V2-016443 | MATHDB-317492

Support-variety criterion for wild blocks of Lie superalgebras

摘录 / EXCERPT: Support-variety wildness conjecture. If there exists a simple supermodule S in $\mathcal{B}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-016444 | MATHDB-317494

Quotient structure conjecture for del Pezzo bundles with high-multiplicity fibers

摘录 / EXCERPT: Let $f\colon X\to Z\ni o$ be the germ of a del Pezzo bundle, and let $m_oF_o=f^*(o)$ be a fiber of multiplicity m_o, where $m_o\geq 5$ and F_o is of type $I_{2,3,6}$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-016445 | MATHDB-317504

Murty-Patankar splitting conjecture for absolutely simple abelian varieties

摘录 / EXCERPT: Let X be an absolutely simple abelian variety over a number field K. Say that a prime of K is a splitting prime for X when the reduction of X at that prime is not simple....

conjecture | open | MathDB 页面 | 原文线索

V2-016446 | MATHDB-317505

Korchagin's parity conjecture for degree-9 M-curves with three nests

摘录 / EXCERPT: Let C_9 be an M-curve of degree 9 with real scheme ... Here, the integers α_i count the empty ovals in the three nests. Korchagin's conjecture. At least two of the...

conjecture | open | MathDB 页面 | 原文线索

V2-016447 | MATHDB-317507

Halupczok's tree-level conjecture for schemes over the p-adic integers

摘录 / EXCERPT: Let V be a scheme of finite type over $\mathbb{Z}_p$. Define $\operatorname{T}(V)$ to be the tree whose nodes at depth λ are the image of $V(\mathbb{Z}_p)$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-016448 | MATHDB-317520

Geometric-stratum conjecture for tensegrity fibers

摘录 / EXCERPT: Let G be a graph and let k be a positive integer. The (G,k)-stratum is the set of frameworks $G(P)$ for which the dimension of the fiber $W(G,P)$ is at least k. Such a stra...

conjecture | open | MathDB 页面 | 原文线索

V2-016449 | MATHDB-317523

Pink's conjecture on ordinary reduction of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety defined over $\overline{\mathbb{Q}}$. A model $\mathcal{A}$ of A over the ring of integers $\mathcal{O}_K$ of a number field K satisfies Hypothesi...

conjecture | open | MathDB 页面 | 原文线索

V2-016450 | MATHDB-317535

The Magnus trace map surjectivity conjecture for SL(m,C) character varieties

摘录 / EXCERPT: Magnus trace map surjectivity conjecture. The map $\mathsf{MagTr}_{\mathcal{I}}$ is surjective only for the cases

conjecture | open | MathDB 页面 | 原文线索

V2-016451 | MATHDB-317539

Independence-of-\ell conjecture for compactly supported \ell-adic Betti numbers

摘录 / EXCERPT: Independence-of-l conjecture. The dimension

conjecture | open | MathDB 页面 | 原文线索

V2-016452 | MATHDB-317549

Craw's tilting bundle and moduli conjecture for smooth toric Fano varieties

摘录 / EXCERPT: Let X be a smooth toric Fano variety. A sequence of basepoint-free line bundles is given by ... Assume that this sequence has bound quiver of sections (Q,ϱ), and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016453 | MATHDB-317642

The section conjecture for rational points on curves over number fields

摘录 / EXCERPT: Let k be a number field, let X be a smooth projective curve over k of genus at least 2, and fix a geometric point $\bar x$ of X. Consider the natural exact sequence ... A c...

conjecture | open | MathDB 页面 | 原文线索

V2-016454 | MATHDB-317658

Asymptotic non-rigidity conjecture for curves in projective space

摘录 / EXCERPT: Let $r\geq 5$, and let d and g be the degree and genus of a smooth irreducible and nondegenerate curve in $\mathbb P^{r}$. A curve is rigid if it has no deformations except tho...

conjecture | open | MathDB 页面 | 原文线索

V2-016455 | MATHDB-317661

Uniformization conjecture for parahoric bundle moduli

摘录 / EXCERPT: Assume the setup of a parahoric group scheme $\mathcal G$ over a curve X and let $\mathcal M_{\mathcal G/X}$ denote the moduli stack of $\mathcal G$-torsors. Let $x\in X(k)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-016456 | MATHDB-317665

The rank-three equivariant Poincaré series conjecture for nonorientable surfaces

摘录 / EXCERPT: Rank-three equivariant Poincaré series conjecture. The formulas in equations ... hold for every $\tilde{g}\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-016457 | MATHDB-317668

Factoriality conjecture for nodal complete intersection threefolds

摘录 / EXCERPT: Let X be a nodal complete intersection threefold in $\mathbb{P}^5$ of hypersurfaces F_n and F_k of degrees n and k, respectively, with $n\geq k$, and suppose that F_k i...

conjecture | open | MathDB 页面 | 原文线索

V2-016458 | MATHDB-317684

Very-moving subvarieties have big classes

摘录 / EXCERPT: Let V be smooth and projective, and let $W\subset V$ be a very moving subvariety: through a generic point $y\in V$, and for a generic vector subspace $W'\subset T_{V,y}$ of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-016459 | MATHDB-317685

Effective-cone conjecture for complete-intersection line varieties

摘录 / EXCERPT: Let G be the Grassmannian and let $\mathcal{E}$ and $\mathcal{K}_d$ be the bundles appearing in the construction of the line varieties, with $H=c_1(\mathcal{O}(1))$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-016460 | MATHDB-317717

Continuity conjecture for the radius of convergence of differential equations

摘录 / EXCERPT: Let k be a non-archimedean field, let ${\mathfrak X}$ be a nondegenerate polystable formal scheme over k°, and let $X={\mathfrak X}_\eta$ be its generic fiber. For an obje...

conjecture | open | MathDB 页面 | 原文线索

V2-016461 | MATHDB-317734

Minimality conjecture for the generating sets of $J_{v_{r,s,t,n}}$

摘录 / EXCERPT: Let $v_{r,s,t,n}$ be the bigrassmannian permutation used in type A, and let $J_{v_{r,s,t,n}}$ be its ideal. The two generating sets for this ideal given in Theorem are the genera...

conjecture | open | MathDB 页面 | 原文线索

V2-016462 | MATHDB-317753

The general-position conjecture for critically transverse automorphism orbits

摘录 / EXCERPT: Let $\Bbbk$ be an algebraically closed field of characteristic 0, let $X/\Bbbk$ be a projective variety, let $\sigma \in \operatorname{Aut}_{\Bbbk} X$, and let $Z \subseteq X$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-016463 | MATHDB-317754

Chow-valued GW/DT correspondence for nonsingular projective threefolds

摘录 / EXCERPT: Chow-valued GW/DT conjecture. The equivalence of Corollary TChow, namely the equality of the corresponding Gromov-Witten and Donaldson-Thomas homological push-forwards after...

conjecture | open | MathDB 页面 | 原文线索

V2-016464 | MATHDB-317762

The hyperdeterminantal generation conjecture for principal minors of symmetric matrices

摘录 / EXCERPT: Let Z_n be the variety of principal minors of symmetric $n\times n$ matrices, and let HD denote the hyperdeterminantal module, namely the span of the...

conjecture | open | MathDB 页面 | 原文线索

V2-016465 | MATHDB-317771

Conjecture on syzygies of the first secant variety

摘录 / EXCERPT: Let $X^d\subset\mathbb{P}^n$ be a smooth projective variety such that $H^i(X,\mathcal O(k))=0$ for $i,k\geq1$, and suppose that X satisfies N^{Σ}_{p+d+1}. Conjecture on sy...

conjecture | open | MathDB 页面 | 原文线索

V2-016466 | MATHDB-317782

Conjecture on defective secant varieties of Segre-Veronese varieties embedded by $\mathcal{O}(1,2)$

摘录 / EXCERPT: The defective secant-variety conjecture. The variety $\sigma_s(X_{m,n})$ is defective if and only if (m,n,s) falls into one of the following cases:

conjecture | open | MathDB 页面 | 原文线索

V2-016467 | MATHDB-317802

Higher-dimensional Mostovoy conjecture for algebraic maps between real projective spaces

摘录 / EXCERPT: Let m,n,d be integers with $2\leq m<n$ and $d\geq1$. Write $\operatorname{Alg}_d(\mathbb{R}\operatorname{P}^m,\mathbb{R}\operatorname{P}^n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-016468 | MATHDB-317803

Projection-map conjecture for polynomial representatives of algebraic maps

摘录 / EXCERPT: Let $A_d(m,n)$ and $\widetilde{A}_d(m,n)$ be the spaces of polynomial data representing based and free algebraic maps from $\mathbb{R}\operatorname{P}^m$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-016469 | MATHDB-317804

Quasifibration conjecture for the polynomial-representative projection

摘录 / EXCERPT: Let $A_d(m,n)$ be the polynomial-representative space and let $\Psi_d:A_d(m,n)\to\operatorname{Alg}_d^*(\mathbb{R}\operatorname{P}^m,\mathbb{R}\operatorname{P}^n)$ be the natural p...

conjecture | open | MathDB 页面 | 原文线索

V2-016470 | MATHDB-317810

Namikawa's conjecture on Q-factorial terminalizations of nilpotent orbit closures

摘录 / EXCERPT: Let ${\mathfrak g}$ be a complex simple Lie algebra, let $\O$ be a nilpotent orbit in ${\mathfrak g}$, and let $\tilde{\O}$ be the normalization of its closure $\bar{\O}$. A genera...

conjecture | open | MathDB 页面 | 原文线索

V2-016471 | MATHDB-317814

Birationality and A1-invariance conjecture for A1-connected components of proper schemes

摘录 / EXCERPT: Birationality and ${\mathbb A}^1$-invariance conjecture. For any proper scheme X of finite type over a field k, this canonical morphism is an isomorphism; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-016472 | MATHDB-317826

The nonperiodic dynamical Mordell-Lang conjecture

摘录 / EXCERPT: Let X be a quasiprojective variety defined over ${\mathbb C}$, let $V\subset X$ be a subvariety, let Φ be an endomorphism of X, and let $P\in X({\mathbb C})$. The orbit of...

conjecture | open | MathDB 页面 | 原文线索

V2-016473 | MATHDB-317846

Hausel-Letellier-Villegas conjecture for character varieties

摘录 / EXCERPT: Let ${\bm \mu}=(\mu^1,\ldots,\mu^k)\in{\mathcal{P}_n}^k$ be a tuple of partitions, let ${\mathcal{M}}_{\bm \mu}$ be the corresponding $\operatorname{GL}_n(\mathbb{C})$ character va...

conjecture | open | MathDB 页面 | 原文线索

V2-016474 | MATHDB-317852

Birch and Swinnerton-Dyer special-value formula for modular abelian varieties

摘录 / EXCERPT: Let A_f be the modular abelian variety associated to a weight-2 newform of level N, and suppose that $L_{A_f}(1)\ne 0$. Let Ω_{A_f} be the real volume defined using a...

conjecture | open | MathDB 页面 | 原文线索

V2-016475 | MATHDB-317853

The effective anticanonical divisor conjecture for global log canonical thresholds

摘录 / EXCERPT: Effective anticanonical divisor conjecture. There is an effective $\mathbb{Q}$-divisor D on X such that

conjecture | open | MathDB 页面 | 原文线索

V2-016476 | MATHDB-317870

Landsberg-Tommasi extension conjecture for Fano schemes of lines

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, let $X^{n-1}\subset \mathbb P^n$ be a hypersurface of degree $n-\lambda$, and let $B\subset \mathbb F(X)$ be an irr...

conjecture | open | MathDB 页面 | 原文线索

V2-016477 | MATHDB-317890

Optimality of the volume bound for projective minimal 3-folds of general type

摘录 / EXCERPT: Let X be a projective minimal 3-fold of general type, with canonical volume K_X^3. Optimality conjecture. For all projective minimal 3-folds of general type, the inequality ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-016478 | MATHDB-317946

Stanton's conjecture on multiplicity inequalities in cometric association schemes

摘录 / EXCERPT: Stanton's conjecture. For every integer i with $i<n/2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-016479 | MATHDB-317951

The Maximal Rank Conjecture for general embedded curves

摘录 / EXCERPT: Maximal Rank Conjecture. A general embedded smooth curve in $\mathfrak I_{d,g,r}$ has maximal rank: for every integer $n\geq1$, $\nu_n(C)$ is injective or surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-016480 | MATHDB-317980

Degree-12 equations for the variety of principal minors

摘录 / EXCERPT: Degree-12 equations conjecture. The variety $\mathcal{V}_n$ is cut out by equations of degree 12.

conjecture | open | MathDB 页面 | 原文线索

V2-016481 | MATHDB-317990

The preperiodic-closure conjecture for polarized dynamical systems

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system over an algebraically closed field of characteristic zero. Let Y be an irreducible subvariety of X, and let Y_0 be the...

open_problem | open | MathDB 页面 | 原文线索

V2-016482 | MATHDB-317993

The dynamical Bogomolov conjecture for polarized dynamical systems

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system defined over a number field, and let Y be a subvariety of X. Let the maximal preperiodic subvarieties contained in Y be...

conjecture | open | MathDB 页面 | 原文线索

V2-016483 | MATHDB-317995

Philippon's zero-height conjecture for subvarieties in polarized dynamical systems

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system defined over a number field, and let Y be a subvariety of X with canonical subvariety height $\hat h_{\mathscr L}(Y)$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-016484 | MATHDB-318005

Hartshorne's conjecture on linear normality

摘录 / EXCERPT: Let $X\subset \mathbb P^r$ be a smooth, irreducible, non-degenerate variety of dimension n. Hartshorne's conjecture on linear normality. If $3n>2(r-1)$, then X is linearly norm...

conjecture | open | MathDB 页面 | 原文线索

V2-016485 | MATHDB-318006

Extended Hartshorne's conjecture on generators of ideal sheaves

摘录 / EXCERPT: There is a suitable function $f(r,n,k)$. Let $X\subset \mathbb P^r$ be a k-smooth, irreducible, non-degenerate variety of dimension n. Extended Hartshorne's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-016486 | MATHDB-318013

Bogomolov-Sommese vanishing for log canonical varieties

摘录 / EXCERPT: Let (Z,Δ) be a logarithmic pair, with Z a variety and Δ a boundary divisor, and assume that (Z,Δ) is log canonical. Let $\Omega_Z^{[p]}(\log\Delta)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-016487 | MATHDB-318020

Classification of wonderful varieties by spherical systems

摘录 / EXCERPT: If X is a wonderful G-variety, let $\mathscr S_X=(\Sigma_X,S^p_X,\Delta^a_X)$ be its spherical system, encoding its spherical roots, distinguished subset of simple roots, color...

open_problem | open | MathDB 页面 | 原文线索

V2-016488 | MATHDB-318036

Lang-Silverman conjecture for canonical heights on abelian varieties

摘录 / EXCERPT: Let $g\geq 1$, let k be a number field, let A/k be an abelian variety of dimension g, let $\mathcal{D}\in\operatorname{Div}(A)$ be an ample divisor, and let $P\in A(k)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-016489 | MATHDB-318047

The Dixmier-Moeglin conjecture for twisted homogeneous coordinate rings

摘录 / EXCERPT: Let X be a projective k-scheme of any dimension, let $\sigma:X\to X$ be an automorphism, let $\mathcal{L}$ be a σ-ample invertible sheaf, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016490 | MATHDB-318060

Hwang's algebraicity conjecture for characteristic foliations

摘录 / EXCERPT: Hwang's conjecture. The characteristic foliation on Y cannot have all of its leaves algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-016491 | MATHDB-318070

Uniqueness of algebraic counting by mixed Hodge structures

摘录 / EXCERPT: Mixed Hodge structure conjecture. The algebraic counting c_{alg} specializes to c_{top}, and there are no further algebraic counting invariants specializing to the Hodge--Delig...

conjecture | open | MathDB 页面 | 原文线索

V2-016492 | MATHDB-318071

Linearity of motivic Hirzebruch class transformations

摘录 / EXCERPT: Let X be a complex algebraic variety. Consider a natural transformation ... without the normalization condition on smooth varieties. Let ${{td_y}_*}_i$ be the degree-$2i$ component...

conjecture | open | MathDB 页面 | 原文线索

V2-016493 | MATHDB-318076

The projectively independent configuration criterion for hierarchy adjacency

摘录 / EXCERPT: Let $\operatorname{\mathcal C}$ and $\operatorname{\mathcal D}$ be configurations of k points in $\C^n$. Suppose that the projectivizations of the non-zero vectors in...

conjecture | open | MathDB 页面 | 原文线索

V2-016494 | MATHDB-318133

Conjecture on dense orbits for rational self-maps without invariant fibrations

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, X an irreducible algebraic variety over K, and let $\Phi:X \to X$ be a rational self-map. Suppose there do not...

conjecture | open | MathDB 页面 | 原文线索

V2-016495 | MATHDB-318137

Baur-Draisma-de Graaf non-defectivity conjecture for Grassmannians

摘录 / EXCERPT: Baur-Draisma-de Graaf conjecture. For $k\geq 2$, $\sigma_s(Gr(k,n))$ has the expected dimension except in the following cases:

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V2-016496 | MATHDB-318145

Decomposability conjecture for principally polarized abelian threefolds

摘录 / EXCERPT: Let (A,a) be a principally polarized abelian variety over a field k. Let $\chi_{18}((A,a))$ and $\Sigma_{140}((A,a))$ be the primitive Siegel modular forms introduced in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-016497 | MATHDB-318160

Determinantal equations conjecture for hidden Markov model varieties

摘录 / EXCERPT: Let Σ be the alphabet, let $\mathbf{g}_{n,d}$ denote the parameterization of the relevant hidden Markov model, and let $p=(p(v))_{v\in\Sigma^n}$ be a string function. For wo...

conjecture | open | MathDB 页面 | 原文线索

V2-016498 | MATHDB-318210

Colliot-Thélène's zero-cycle conjecture for geometrically rational varieties

摘录 / EXCERPT: Colliot-Thélène's zero-cycle conjecture. The Hasse principle holds for zero-cycles on X: if

conjecture | open | MathDB 页面 | 原文线索

V2-016499 | MATHDB-318214

Conjecture on a canonical limiting transverse slice for uniqueness procedures

摘录 / EXCERPT: Limiting-slice conjecture. The choice of h alone is enough to define a uniqueness procedure. Moreover, if the varieties $\Lambda(l_j)$ converge to a well-defined limit Λ...

conjecture | open | MathDB 页面 | 原文线索

V2-016500 | MATHDB-318216

Equality of the non-nef locus and the restricted base locus on normal projective varieties

摘录 / EXCERPT: Non-nef locus conjecture. There exists an ample divisor A and an infinite sequence $\tau_k\in H^0(kL+A)$ such that $\tau_k(x)\neq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-016501 | MATHDB-318229

Green-Griffiths algebraicity conjecture for zero loci of normal functions

摘录 / EXCERPT: Let $X\to S$ be a family of complex smooth projective varieties over a quasi-projective base, and let $Z\hookrightarrow X$ be a flat family of cycles of pure codimension i that a...

conjecture | open | MathDB 页面 | 原文线索

V2-016502 | MATHDB-318253

Bannister-Ostergård subdivision conjecture for stack number

摘录 / EXCERPT: Let G be a graph, and let H be a (≤ 1)-subdivision of G, obtained by subdividing each edge of G at most once. Let $\operatorname{sn}(G)$ denote the stack number of G...

conjecture | open | MathDB 页面 | 原文线索

V2-016503 | MATHDB-318266

Local Motivic Monodromy Conjecture for formal schemes

摘录 / EXCERPT: Local Motivic Monodromy Conjecture. There exists a finite subset $\mathcal{S}\subset\mathbb{Z}\times\mathbb{Z}_{>0}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-016504 | MATHDB-318282

The degree-two Newton polygon conjecture for groups of points on abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a finite field, let τ be its Frobenius endomorphism, and let f_A be the characteristic polynomial of τ. Write ... where f_j divides...

conjecture | open | MathDB 页面 | 原文线索

V2-016505 | MATHDB-318318

Ehrenborg's secant-filling conjecture for Grassmannians and Split varieties

摘录 / EXCERPT: Let $\mathbb G(n-1,n+d-1)$ be the Grassmannian of $(n-1)$-dimensional projective subspaces of $\mathbb P^{n+d-1}$, and let $\operatorname{Split}_d(\mathbb P^n)$ be the variety of d...

conjecture | open | MathDB 页面 | 原文线索

V2-016506 | MATHDB-318332

Constancy of quantum walk spectral variety components and projection degrees

摘录 / EXCERPT: Let d and k be fixed, let ${\bf v}^{(1)},\ldots,{\bf v}^{(k)}$ be fixed vectors, and let U vary over unitary matrices. Write ${\mathcal V}$ for the associated variety and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016507 | MATHDB-318351

The finiteness conjecture for fundamental groups with zero Kodaira dimension

摘录 / EXCERPT: Let X be a nonsingular projective variety. Write $\kappa(X)$ for its Kodaira dimension, $q^{\max}(X)$ for its maximal irregularity, and $\pi_1(X)$ for its fundamental group. Fini...

conjecture | open | MathDB 页面 | 原文线索

V2-016508 | MATHDB-318372

Bloch-Kato norm residue conjecture

摘录 / EXCERPT: Bloch-Kato conjecture. This norm residue homomorphism is an isomorphism for any nonnegative integer n.

conjecture | open | MathDB 页面 | 原文线索

V2-016509 | MATHDB-318373

Beilinson-Lichtenbaum conjecture, p-local version

摘录 / EXCERPT: Let k be a field and let $w\geq 0$. The group ... denotes étale hypercohomology with p-local motivic coefficients. Beilinson-Lichtenbaum conjecture. One has ... This is presented a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-016510 | MATHDB-318383

The Green-Griffiths-Brosnan-Fang-Néron-Pełř conjecture for singular normal functions

摘录 / EXCERPT: Let X be a smooth projective variety of dimension $2m$, let L be a very ample line bundle on X, and let $H_{\mathrm{prim}}^{m,m}(X,\mathbb{Z}(m))$ denote the primitive integr...

conjecture | open | MathDB 页面 | 原文线索

V2-016511 | MATHDB-318384

The motivated normal-function zero-locus conjecture

摘录 / EXCERPT: Let $\mathbb F\subset\mathbb C$ be finitely generated over $\overline{\mathbb Q}$, let $S/\mathbb F$ be a smooth quasi-projective variety of dimension D, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016512 | MATHDB-318426

Beilinson's regulator and special-value conjecture for Chow motives

摘录 / EXCERPT: Beilinson's conjecture. (i) The image of the Betti realization map is an E-structure on...

conjecture | open | MathDB 页面 | 原文线索

V2-016513 | MATHDB-318451

Hu-Kriz's conjecture on the motivic homology of $BPGL$

摘录 / EXCERPT: Let $BPGL$ denote the standard summand of the 2-localization of the Thom spectrum MGL of the universal bundle over BGL. Let $\mathcal A$ be the motivic Steenrod algebra over...

conjecture | open | MathDB 页面 | 原文线索

V2-016514 | MATHDB-318456

Delzant's conjecture for multiplicity-free Hamiltonian manifolds

摘录 / EXCERPT: Let K be a connected compact Lie group, let M be a multiplicity-free compact Hamiltonian K-manifold with moment map $\mu:M\rightarrow\mathfrak k$, and fix a maximal torus...

conjecture | open | MathDB 页面 | 原文线索

V2-016515 | MATHDB-318457

The U-invariant algebra conjecture for smooth affine G-varieties

摘录 / EXCERPT: Let G be a reductive group, let B be a Borel subgroup with maximal unipotent subgroup U and maximal torus T, and let X_1 and X_2 be smooth affine G-varieties. Conside...

conjecture | open | MathDB 页面 | 原文线索

V2-016516 | MATHDB-318458

The Luna-Vust birational classification conjecture

摘录 / EXCERPT: Let G be a reductive group and let X be a normal irreducible G-variety. Its Luna-Vust system is the triple...

conjecture | open | MathDB 页面 | 原文线索

V2-016517 | MATHDB-318459

The Luna-Vust automorphism lifting conjecture

摘录 / EXCERPT: Let G be a reductive group and let X be a normal irreducible G-variety with Luna-Vust system $\mathcal{LV}_{G,X}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016518 | MATHDB-318462

Alper's conjecture on étale quotient charts for Artin stacks

摘录 / EXCERPT: Let $\mathcal{X}$ be a finite type Artin stack over $\operatorname{Spec} k$, and let $x\in\mathcal{X}(k)$ have linearly reductive stabilizer G_x. An algebraic space X...

conjecture | open | MathDB 页面 | 原文线索

V2-016519 | MATHDB-318463

Friedlander-Mazur vanishing conjecture for Lawson homology

摘录 / EXCERPT: Let X be a complex smooth projective variety of dimension d, and let $L_qH_n(X)$ denote its Lawson homology. Friedlander-Mazur conjecture. For every q, one has ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-016520 | MATHDB-318475

Nonvanishing conjecture for Hochschild homology of admissible subcategories

摘录 / EXCERPT: Let X be a smooth projective variety, let $\D^b(X)$ denote its bounded derived category of coherent sheaves, and let $\CA\subset \D^b(X)$ be an admissible subcategory. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016521 | MATHDB-318511

Categoricity conjecture for the analytic Zariski structure of generic complex Abelian varieties

摘录 / EXCERPT: Let $\A$ be a generic complex Abelian variety defined over a number field, and let $\Ax(\A(\C)\times\Cm)$ be the analogously defined $\lww$-class. A generic Abelian-variety categ...

conjecture | open | MathDB 页面 | 原文线索

V2-016522 | MATHDB-318520

Chan-Leung's Miyaoka-Yau inequality for Kähler-Einstein toric Fano manifolds

摘录 / EXCERPT: Let X be a Kähler-Einstein toric Fano n-fold, and let H be a nef class on X. Chan-Leung's conjecture. ... Chan and Leung proved this in dimensions $n=2,3,4$ and when every fa...

conjecture | open | MathDB 页面 | 原文线索

V2-016523 | MATHDB-318545

Hilbert-scheme identity for character varieties

摘录 / EXCERPT: Let $\mathcal H_\lambda(z,w)$ be the partition function appearing in the source, let $\phi(0):=0$, and for every nonzero partition λ define ... where the sum is over the boxe...

conjecture | open | MathDB 页面 | 原文线索

V2-016524 | MATHDB-318563

Hard Lefschetz conjecture for Chow groups

摘录 / EXCERPT: Hard Lefschetz conjecture for Chow groups. This map is injective. The conjecture is a Chow-theoretic analogue of the hard Lefschetz theorem on cohomology; surjectivity generally fa...

conjecture | open | MathDB 页面 | 原文线索

V2-016525 | MATHDB-318569

Weak form of Sakellaridis's conjecture on spherical Eisenstein series

摘录 / EXCERPT: Weak form. Under the same assumptions as in the stronger conjecture, for every cusp form ϕ on $G(k)\backslash G(\mathbb A_k)$, the integral

conjecture | open | MathDB 页面 | 原文线索

V2-016526 | MATHDB-318574

Buch-Mihalcea rationality conjecture for Gromov-Witten varieties

摘录 / EXCERPT: Let X be a cominuscule variety, and let $d_{\rm max}$ denote the minimal integer such that a curve of degree $d_{\rm max}$ passes through any two points of X. For...

conjecture | open | MathDB 页面 | 原文线索

V2-016527 | MATHDB-318585

Orbit D_3 stability conjecture under best rank-1 deflation

摘录 / EXCERPT: Let $\underline{\bf X}$ be a tensor in orbit D_3, and let $\underline{\bf Y}$ be a best rank-1 approximation of $\underline{\bf X}$. Orbit D_3 stability conjecture. Almost al...

conjecture | open | MathDB 页面 | 原文线索

V2-016528 | MATHDB-318586

Reid's base point free conjecture for weak log Fano pairs

摘录 / EXCERPT: Reid's base point free conjecture. The linear system

conjecture | open | MathDB 页面 | 原文线索

V2-016529 | MATHDB-318587

Karzhemanov's base point free conjecture for weak log Fano pairs

摘录 / EXCERPT: Karzhemanov's conjecture. Suppose that X is $\mathbb{Q}$-factorial and

conjecture | open | MathDB 页面 | 原文线索

V2-016530 | MATHDB-318657

The primitive-stable domain conjecture for free-group character varieties

摘录 / EXCERPT: Let F_n be the free group of rank n, let ${\mathcal X}(F_n)$ be its ${\operatorname{PSL}_2(\mathbb C)}$ character variety, and let ${\mathcal{PS}}(F_n)$ denote the primitive-st...

conjecture | open | MathDB 页面 | 原文线索

V2-016531 | MATHDB-318660

Degree formula conjecture for Gaussian cycle varieties

摘录 / EXCERPT: Let C_m be the m-cycle and let P_{C_m} be the prime ideal of its Gaussian graphical model. Write $V(P_{C_m})$ for the associated projective variety. Degree formula conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-016532 | MATHDB-318661

Degree growth conjecture for homogeneous Gaussian cycle polynomials

摘录 / EXCERPT: For the m-cycle, let Γ_m denote the homogeneous cycle polynomial defining the relevant algebraic boundary component. Homogeneous cycle-polynomial degree conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-016533 | MATHDB-318693

Kapustin-Rozansky's Hochschild cohomology conjecture for Lagrangian intersections

摘录 / EXCERPT: Let X be a smooth algebraic symplectic variety, and let $Y,Z\subset X$ be smooth Lagrangian submanifolds. Write $H\!H^{\bullet}(\mathscr{C})$ for the Hochschild cohomology of a t...

conjecture | open | MathDB 页面 | 原文线索

V2-016534 | MATHDB-318698

Topological invariants conjecture for maximal representations

摘录 / EXCERPT: Topological invariants conjecture. If G is not locally isomorphic to $\mathrm{Sp}(2n,\mathbf{R})$, then the topological invariants of Theorem distinguish connected components of...

conjecture | open | MathDB 页面 | 原文线索

V2-016535 | MATHDB-318699

Twisted diagonal deformation conjecture for maximal representations

摘录 / EXCERPT: Twisted diagonal deformation conjecture. If G is not locally isomorphic to $\mathrm{Sp}(2n,\mathbf{R})$, then every maximal representation

conjecture | open | MathDB 页面 | 原文线索

V2-016536 | MATHDB-318709

The numerical inequality for irregular compact Kähler manifolds without irregular fibrations

摘录 / EXCERPT: Numerical inequality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-016537 | MATHDB-318723

The finite generation conjecture for log canonical rings

摘录 / EXCERPT: Let X be an n-dimensional smooth projective variety and let B be a boundary $\mathbb Q$-divisor whose support is a simple normal crossing divisor on X. Assume that K_X+B...

conjecture | open | MathDB 页面 | 原文线索

V2-016538 | MATHDB-318724

The relative finite generation conjecture for log canonical rings

摘录 / EXCERPT: Let $f\colon X\to S$ be a proper surjective morphism between normal varieties, and let B be a $\mathbb Q$-divisor on X such that (X,B) is log canonical. Relative finite gener...

conjecture | open | MathDB 页面 | 原文线索

V2-016539 | MATHDB-318733

The Shapiro conjecture for the orthogonal Grassmannian

摘录 / EXCERPT: The Shapiro conjecture for $OG(n)$. If $s_1,\dotsc,s_m$ are distinct real numbers, then

conjecture | open | MathDB 页面 | 原文线索

V2-016540 | MATHDB-318734

The modified Shapiro conjecture for the Lagrangian Grassmannian

摘录 / EXCERPT: Let $LG(n)$ be the Lagrangian Grassmannian, let $F_\bullet(t)$ be the isotropic osculating flag, and let $X_{\boldsymbol{\kappa}}F_\bullet$ be the Schubert variety indexed by a str...

conjecture | open | MathDB 页面 | 原文线索

V2-016541 | MATHDB-318735

The monotone conjecture for flag manifolds

摘录 / EXCERPT: Let $0\leq n_1<\dotsb<n_k<d$, and let $\mathbb F\ell_{{\bf n},d}$ be the flag variety of chains $E_{n_1}\subset\dotsb\subset E_{n_k}\subset\mathbb C_d[t]$ with $\dim E_{n_i}=n_i+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-016542 | MATHDB-318747

The extension conjecture for projective-bundle divisors

摘录 / EXCERPT: Extension conjecture. Then $d\ge b-1$, and $(X,L)\cong (\mathbb{P}(\mathcal{E}),\xi_{\mathbb{P}})$ for an ample vector bundle $\mathcal{E}$ on Z, with p equal to the restrictio...

conjecture | open | MathDB 页面 | 原文线索

V2-016543 | MATHDB-318755

Luna's bijective classification conjecture for wonderful varieties of adjoint groups

摘录 / EXCERPT: Let G be a connected reductive algebraic group of adjoint type, and let X be a wonderful G-variety. Denote by S^p_X, σ_X, and $\mathsf A_X$ the associated combinatoria...

conjecture | open | MathDB 页面 | 原文线索

V2-016544 | MATHDB-318761

The infinitude of rational curves on projective K3 surfaces

摘录 / EXCERPT: Main conjecture. There exist infinitely many rational curves on S.

conjecture | open | MathDB 页面 | 原文线索

V2-016545 | MATHDB-318764

The SHGH conjecture for homogeneous linear systems on Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathcal{L}_n(a,b;m_1,\dots,m_r)$ be a linear system on the Hirzebruch surface $\mathbb{F}_n$, with the notation and -1-systems defined by the intersection pairing and reduc...

conjecture | open | MathDB 页面 | 原文线索

V2-016546 | MATHDB-318765

The SHGH conjecture for planar linear systems with collinear imposed points

摘录 / EXCERPT: Let $L=\mathcal{L}(d;m_1,\dots,m_r,\overline{k_1,\dots,k_s})$ be a planar linear system, where the barred multiplicities are imposed at points on a fixed line, and let $\dim L$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-016547 | MATHDB-318777

Grothendieck's algebraicity conjecture for the Lefschetz operator

摘录 / EXCERPT: Let X be a smooth projective variety of dimension r, let $H^*(X)$ be a Weil cohomology theory, and let L be the Lefschetz operator. Let Λ be the linear operator deter...

conjecture | open | MathDB 页面 | 原文线索

V2-016548 | MATHDB-318778

The homological and numerical equivalence conjecture for algebraic cycles

摘录 / EXCERPT: Let X be a smooth projective variety and let $C^i_{\sim}(X)$ be the group of codimension-i cycles modulo an equivalence relation $\sim$. Write $C^i_{\sim_{\mathrm{hom}}}(X)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-016549 | MATHDB-318788

Strong Mordell conjecture for families of curves

摘录 / EXCERPT: Let k) be a number field. Let T be a smooth connected projective variety, let $U\subseteq T$ be a nonempty Zariski open subset, and let $\pi:X\to U$ be a smooth family of proje...

conjecture | open | MathDB 页面 | 原文线索

V2-016550 | MATHDB-318795

Edoukou's point-count conjecture for projective algebraic sets

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n}(\mathbb{F}_q)$ be a projective algebraic set of degree d and dimension s. Writing $\pi_j=|\mathbb{P}^j(\mathbb{F}_q)|$, Edoukou's point-count conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-016551 | MATHDB-318798

The Cox ring surjectivity conjecture for quiver flag varieties

摘录 / EXCERPT: Let $\mathcal{M}_\vartheta$ be a quiver flag variety, let $\lvert\underline{\det}(\mathscr{W})\rvert$ be the toric variety associated with the sequence of determinant line bundles,...

conjecture | open | MathDB 页面 | 原文线索

V2-016552 | MATHDB-318801

Singular-locus conjecture for character varieties of complex reductive groups

摘录 / EXCERPT: Let G be a complex affine reductive group, and let $\mathfrak{X}_{r}(G)$ be the character variety of the free group of rank r in G. Write $\mathfrak{X}_{r}(G)^{red}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-016553 | MATHDB-318802

Non-Euclidean-neighborhood conjecture for reducible compact character representations

摘录 / EXCERPT: Let K be SU_{n} or U_{n}, let $[\rho]\in\mathfrak{X}_{r}(K)^{red}$, and assume $r,n\geq 2$ with $(r,n)\not=(2,2),(2,3),(3,2)$. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-016554 | MATHDB-318805

The variational Hodge conjecture

摘录 / EXCERPT: Variational Hodge conjecture. If α_b is the restriction of a class $\alpha\in{\operatorname{H}}^{2r}({\mathcal X}^{\operatorname{an}},{\mathbb Q})$, equivalently if it is in...

conjecture | open | MathDB 页面 | 原文线索

V2-016555 | MATHDB-318827

Fano toric variety conjecture for full strongly exceptional collections

摘录 / EXCERPT: Fano toric collection conjecture. X has a full strongly exceptional collection of line bundles.

conjecture | open | MathDB 页面 | 原文线索

V2-016556 | MATHDB-318866

Amoroso-David-Philippon relative obstruction index lower-bound conjecture

摘录 / EXCERPT: Let E be an elliptic curve and let E^N be its N-fold power. Let H be a translate of an abelian subvariety of E^N of dimension n, and let V be transverse in H. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-016557 | MATHDB-318898

Very general hypersurface curve-class divisibility conjecture

摘录 / EXCERPT: Let $X^n\subset \mathbb{P}^{n+1}$ be a very general smooth hypersurface of degree e, with $n\geq 3$, and let $\ell\in H_2(X(\mathbb{C}),\mathbb{Z})$ generate...

conjecture | open | MathDB 页面 | 原文线索

V2-016558 | MATHDB-318914

Conjecture on the optimal Toeplitz feedback scheme for the AWGN channel

摘录 / EXCERPT: Let the system be the AWGN channel with noisy feedback described in the paper, with blocklength N, feedback-noise variance σ^2, signal-to-noise parameter γ, power p...

conjecture | open | MathDB 页面 | 原文线索

V2-016559 | MATHDB-318929

Parabolic maximality conjecture for standard homogeneity of Kazhdan-Lusztig ideals

摘录 / EXCERPT: Let v,w be permutations with $v\leq w$, and let $I_{v,w}$ denote the Kazhdan-Lusztig ideal. Suppose that $I_{v,w}$ is standardly homogeneous. If $ws_i<w$ but $vs_i>v$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-016560 | MATHDB-318933

Function-field Beilinson-Tate conjecture for surfaces

摘录 / EXCERPT: Function-field Beilinson-Tate conjecture. The map $R_{\mathcal D}$ has finite kernel and cokernel; moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-016561 | MATHDB-318941

Attainment and rationality conjecture for equivariant log canonical thresholds

摘录 / EXCERPT: Let X be a variety with a group action by $\bar{G}$, let B_X be a $\bar{G}$-invariant boundary, and suppose that $-(K_X+B_X)$ is ample and $\mathbb{Q}$-Cartier. An effective...

conjecture | open | MathDB 页面 | 原文线索

V2-016562 | MATHDB-318942

Rational ambient characterization conjecture for Du Bois singularities

摘录 / EXCERPT: Let X be a variety, let Y be an ambient variety, and suppose the criterion referred to as the Easy Du Bois Criterion characterizes Du Bois singularities when Y is smooth. A v...

conjecture | open | MathDB 页面 | 原文线索

V2-016563 | MATHDB-318943

Steenbrink's conjecture on Du Bois divisors

摘录 / EXCERPT: Let X be a variety, let $D\subset X$ be a reduced Cartier divisor, and let $x\in D$. Assume that D has only Du Bois singularities in a neighborhood of x. Steenbrink's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-016564 | MATHDB-318944

Rationality conjecture for varieties with Du Bois divisors

摘录 / EXCERPT: Let X be a reduced scheme of finite type over a field of characteristic zero, and let D be a Cartier divisor with Du Bois singularities. Assume that $X\setminus D$ has rational...

conjecture | open | MathDB 页面 | 原文线索

V2-016565 | MATHDB-318978

Colliot-Thélène's period-index conjecture

摘录 / EXCERPT: Period-index conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016566 | MATHDB-318991

Hartshorne conjecture for varieties of lines

摘录 / EXCERPT: Let $X\subset\mathbb{P}^N$ be a smooth irreducible non-degenerate variety of dimension n covered by lines. For a general point $x\in X$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-016567 | MATHDB-318992

Degree bound conjecture for complete intersections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^N$ be a smooth irreducible non-degenerate projective variety of dimension n, and write $\operatorname{degree}(X)$ for its degree. Degree bound conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-016568 | MATHDB-318999

The ell-independence conjecture for motivic zeta polynomials

摘录 / EXCERPT: Let $M\in\mathscr{M}_{k,E}$ be a motive over a number field k. For a finite place v of k, let $P_v(M,T)$ denote its zeta polynomial, defined using the inertia invariants of a...

conjecture | open | MathDB 页面 | 原文线索

V2-016569 | MATHDB-319000

The Hasse-Weil conjecture for motivic L-functions

摘录 / EXCERPT: Let $M\in\mathscr{M}_{k,E}$ be a motive over a number field k, and let $L(M,s)$ be its motivic L-function. Let $M^\vee$ be the dual motive and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016570 | MATHDB-319002

The equivalence conjecture for additive cycle complexes with different modulus conditions

摘录 / EXCERPT: Equivalence conjecture. These natural inclusions are quasi-isomorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-016571 | MATHDB-319030

The assertion for the mixed Brieskorn variety $V_{II,t}$

摘录 / EXCERPT: Let $V_{II,t}$ be the mixed Brieskorn variety considered above, and let the preceding assertion denote the claimed existence of a tangent vector in...

open_problem | open | MathDB 页面 | 原文线索

V2-016572 | MATHDB-319031

Ueno-Kollár conjecture on varieties of Kodaira dimension zero

摘录 / EXCERPT: Ueno-Kollár conjecture. There exists a finite étale covering $\tilde X\to X$ such that $\tilde X$ is birational to the product of an Abelian variety and a simply connected variety...

conjecture | open | MathDB 页面 | 原文线索

V2-016573 | MATHDB-319032

The cone, ampleness, and contraction conjecture for four-dimensional generalized Kummer varieties

摘录 / EXCERPT: Let (X,g) be a polarized variety deformation equivalent to a four-dimensional generalized Kummer variety. Let (R,R) denote the Beauville-Bogomolov form on homology. The general...

conjecture | open | MathDB 页面 | 原文线索

V2-016574 | MATHDB-319033

The complements-are-far-apart conjecture for diagonal links

摘录 / EXCERPT: Let P be a bounded graded poset, let K be its diagonal link, and let x and y be complements in P. Write v_x and v_y for the corresponding vertices of K, and say tha...

conjecture | open | MathDB 页面 | 原文线索

V2-016575 | MATHDB-319034

The modular lattice conjecture on CAT(0) orthoscheme complexes

摘录 / EXCERPT: Let P be a bounded graded modular lattice. A modular lattice is a graded lattice in which, whenever x and y are complements in an interval P_{zw} and x has rank i and c...

conjecture | open | MathDB 页面 | 原文线索

V2-016576 | MATHDB-319036

Metric-completion conjecture for weak Kähler-Ricci flow

摘录 / EXCERPT: Let (X,H,ω) be a triple in which X is a $\mathbf{Q}$-factorial projective variety with log terminal singularities, H is a big semi-ample $\mathbf{Q}$-divisor such that...

conjecture | open | MathDB 页面 | 原文线索

V2-016577 | MATHDB-319039

Generalized Hamilton-Tian conjecture for the Kähler-Ricci flow

摘录 / EXCERPT: Generalized Hamilton-Tian conjecture. As $s\to\infty$, the flow starting from ω converges in the Gromov-Hausdorff sense to a Kähler-Ricci soliton (X_∞,ω_{KR}). T...

conjecture | open | MathDB 页面 | 原文线索

V2-016578 | MATHDB-319052

Babai's conjecture on valencies in primitive association schemes

摘录 / EXCERPT: Babai's conjecture. There exists an integral function f such that, for every primitive association scheme $(X,\{R_0,R_1,\ldots,R_r\})$,

conjecture | open | MathDB 页面 | 原文线索

V2-016579 | MATHDB-319066

The monotonicity conjecture for the saturation sequence of the rational normal curve

摘录 / EXCERPT: Monotonicity conjecture. The saturation sequence is non-increasing:

conjecture | open | MathDB 页面 | 原文线索

V2-016580 | MATHDB-319067

The strict inequality conjecture for the first two saturation indices

摘录 / EXCERPT: For each integer $d\geqslant6$, let α_1 and α_2 denote the first two entries of the saturation sequence of the rational normal curve of degree d. Strict inequality...

conjecture | open | MathDB 页面 | 原文线索

V2-016581 | MATHDB-319068

The eventual terminal-3 conjecture for saturation sequences

摘录 / EXCERPT: For each positive integer d, let $(\alpha_1,\ldots,\alpha_{e_d-1})$ be the saturation sequence of the rational normal curve of degree d. Terminal-3 conjecture. For every positi...

conjecture | open | MathDB 页面 | 原文线索

V2-016582 | MATHDB-319076

Nonderogatory conjecture for quantum cohomology of Hilbert schemes of type-A_r surfaces

摘录 / EXCERPT: Let $\mathcal{A}_r$ be the type-A_r surface, let $\operatorname{Hilb}^n(\mathcal{A}_r)$ be its Hilbert scheme of n points, and let L be the map defined in Section 5. Consider...

conjecture | open | MathDB 页面 | 原文线索

V2-016583 | MATHDB-319083

Regularity conjecture for secant varieties of smooth curves

摘录 / EXCERPT: Let $X \subseteq \mathbb{P}^n$ be a smooth curve of genus g and degree $d \geq 2g+2k+1$. Let Σ_k denote its kth secant variety, and let $\mathcal{I}_{\Sigma_k}$ be the i...

conjecture | open | MathDB 页面 | 原文线索

V2-016584 | MATHDB-319107

Equivariant Chern class formula for flag varieties via arc spaces

摘录 / EXCERPT: Let $G=GL_n$, let $\mathbf{r}=(r_1,\ldots,r_k)$ determine the partial flag variety and the corresponding flag representation $\widetilde{V}_\bullet$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016585 | MATHDB-319111

The determinantal generation conjecture for secant varieties

摘录 / EXCERPT: Let $X \subset \mathbb{P}^r$ be a projective scheme embedded by the complete linear series of a sufficiently ample line bundle, and let $\operatorname{Sec}^k(X)$ denote the Zariski...

conjecture | open | MathDB 页面 | 原文线索

V2-016586 | MATHDB-319127

Lang-Vojta conjecture over number fields

摘录 / EXCERPT: Let K be a number field, let S be a finite set of places of K containing the archimedean places, let X be a variety of log-general type defined over K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016587 | MATHDB-319147

The Geometric Conjecture on general type of Modular Tower components

摘录 / EXCERPT: Let a Modular Tower be associated to a p-perfect centerless group G_0 and the relevant collection of conjugacy classes, and let its level-k spaces have nonsingular projective...

conjecture | open | MathDB 页面 | 原文线索

V2-016588 | MATHDB-319150

The absence of poles conjecture for motivic Donaldson-Thomas invariants

摘录 / EXCERPT: Absence of poles conjecture. For any $3d$ Calabi-Yau category with a stability condition and any strict sector V, the automorphism $x\mapsto A_{V,q}xA_{V,q}^{-1}$ preserves the s...

conjecture | open | MathDB 页面 | 原文线索

V2-016589 | MATHDB-319205

Rosengren's real-zero conjecture for the polynomials p_n

摘录 / EXCERPT: Rosengren's real-zero conjecture. The polynomial p_n has exactly $[(n+1)/2]$ real zeroes.

conjecture | open | MathDB 页面 | 原文线索

V2-016590 | MATHDB-319206

Rosengren's zero-free interval conjecture for the polynomials p_n

摘录 / EXCERPT: Rosengren's zero-free interval conjecture. The polynomial p_n does not vanish for ζ in the interval

conjecture | open | MathDB 页面 | 原文线索

V2-016591 | MATHDB-319222

The formal cell-zeta algebra and framed-motive algebra isomorphism conjecture

摘录 / EXCERPT: The formal cell-zeta isomorphism conjecture. The map m is an isomorphism. This would identify formal cell-zeta values, modulo their motivic relations, with the framed mixed Tate...

conjecture | open | MathDB 页面 | 原文线索

V2-016592 | MATHDB-319266

The presentation conjecture for regular nilpotent Hessenberg variety cohomology

摘录 / EXCERPT: Presentation conjecture. The quotient

conjecture | open | MathDB 页面 | 原文线索

V2-016593 | MATHDB-319269

Kato's homological conjecture for smooth projective varieties

摘录 / EXCERPT: Kato's conjecture. The displayed isomorphism holds. This is a cohomological Hasse-principle prediction for Kato homology of smooth projective varieties over the field k; the supp...

conjecture | open | MathDB 页面 | 原文线索

V2-016594 | MATHDB-319272

Mirror Symmetry conjecture for Hodge structure variations

摘录 / EXCERPT: Let X be a Fano variety. A one-dimensional family is a morphism $Y\to \mathbb{C}$, and its Picard--Fuchs and regularized quantum $\mathcal D$-modules are the associated different...

conjecture | open | MathDB 页面 | 原文线索

V2-016595 | MATHDB-319273

Weak Landau--Ginzburg model existence conjecture for Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano threefold with Picard number 1. A weak Landau--Ginzburg model for X is a Laurent polynomial defining the relevant pencil and satisfying the weak-model...

conjecture | open | MathDB 页面 | 原文线索

V2-016596 | MATHDB-319285

Vermeire's syzygy conjecture for secant varieties of curves

摘录 / EXCERPT: Work over an algebraically closed field k of characteristic zero. Let $C\subset\mathbb{P}^n$ be a smooth linearly normal curve of degree $d\geq2g+2k+1+p$, where $p,k\geq0$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-016597 | MATHDB-319316

Tame reciprocity conjecture for arithmetic Suslin homology

摘录 / EXCERPT: Let X be separated and of finite type over a finite field. Let $H_1^{\operatorname{ar}}(X,\mathbb Z)$ be the first arithmetic homology group, and let $\pi_1^t(X)^{ab}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-016598 | MATHDB-319328

The pseudo-index bound for toric Fano manifolds with a minimal component

摘录 / EXCERPT: Let X^n be a toric Fano manifold of dimension $n\geq 3$. Let ${\mathcal K}$ be a minimal component of degree $p+2$, where p is the associated integer. The pseudo-index bound. I...

conjecture | open | MathDB 页面 | 原文线索

V2-016599 | MATHDB-319329

Versality detected by Sylow subgroups

摘录 / EXCERPT: Let G be a finite group acting faithfully on a variety X. For each prime p, let G_p be a Sylow p-subgroup of G. Sylow versality conjecture. The variety X is G-versa...

conjecture | open | MathDB 页面 | 原文线索

V2-016600 | MATHDB-319330

Bernstein-Lunts conjecture for quasi-minimal characteristic varieties of generic vector fields

摘录 / EXCERPT: Bernstein-Lunts conjecture. The characteristic variety $\operatorname{ch}(\mathcal F_{\xi})$ is a quasi-minimal characteristic variety.

conjecture | open | MathDB 页面 | 原文线索

V2-016601 | MATHDB-319344

The B-strong desingularization conjecture

摘录 / EXCERPT: A B-scheme is a scheme X equipped with a boundary B; here (X,B) is assumed to be qe and generically reduced, and the scheme has characteristic zero. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016602 | MATHDB-319364

Generic linkage preserves local complete intersection log canonical singularities

摘录 / EXCERPT: Let $S=\operatorname{Spec} R$ be a regular affine scheme over k, and let $X\subset S$ be a subscheme defined by $I=(z_1,\ldots,z_t)$ of codimension r. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016603 | MATHDB-319365

Castelnuovo-Mumford regularity bound for log canonical projective schemes

摘录 / EXCERPT: Let $R=k[x_0,\ldots,x_n]$, and let $I=(f_1,\ldots,f_t)$ be a homogeneous ideal generated in degrees $d_1\geq d_2\geq \cdots\geq d_t\geq 1$, of codimension r. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-016604 | MATHDB-319366

The topological-index conjecture for minimal generalized Lagrangian surfaces

摘录 / EXCERPT: Let X be a minimal generalized Lagrangian surface of general type. Thus, for a nonzero $w\in\ker\psi_2\subseteq H^0(X,\Omega_X^2)$ of rank 2, let $V\subset H^0(X,\Omega_X^1)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-016605 | MATHDB-319367

The topological-index conjecture for Lagrangian surfaces

摘录 / EXCERPT: Let X be a Lagrangian surface, meaning a smooth projective surface whose holomorphic two-forms contain a nonzero decomposable element in the relevant kernel defining the Lagrangi...

conjecture | open | MathDB 页面 | 原文线索

V2-016606 | MATHDB-319383

Conjecture on defective two-factor Segre-Veronese varieties

摘录 / EXCERPT: Defective Segre-Veronese conjecture. The variety $X_{\mathbf{n},\mathbf{a}}$ is defective if and only if $(\mathbf{n},\mathbf{a})$ falls into one of the following cases:

conjecture | open | MathDB 页面 | 原文线索

V2-016607 | MATHDB-319384

Conjecture on non-defectivity for degrees at least $(3,3)$

摘录 / EXCERPT: Non-defectivity conjecture for degrees at least $(3,3)$. There are no defective two-factor Segre-Veronese varieties $X_{\mathbf{n},\mathbf{a}}$ for any $\mathbf{n}\in\mathbb{N}^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-016608 | MATHDB-319387

The dimension conjecture for loci of good determinantal schemes

摘录 / EXCERPT: The dimension conjecture for good determinantal schemes. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-016609 | MATHDB-319388

Semistable reduction conjecture

摘录 / EXCERPT: Let R be a henselian discrete valuation ring with fraction field K and residue field k of characteristic $p>0$, and let $S=\operatorname{Spec}(R)$. For a proper morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-016610 | MATHDB-319389

Rigid Clemens-Schmidt conjecture for limiting Frobenius structures

摘录 / EXCERPT: Let $X\to S\subseteq \mathbb A^1_{\mathbb F_q}\setminus\{0\}$ be a morphism of varieties over $\mathbb F_q$ with a good lift around $t=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016611 | MATHDB-319390

Tropical-basis conjecture for minors of a generic matrix

摘录 / EXCERPT: Tropical-basis conjecture. The $(r+1)\times(r+1)$ minors of a $d\times n$ matrix of variables are a tropical basis if and only if $r\leq 2$ or $r\geq \operatorname{min}(d,n)-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-016612 | MATHDB-319415

Hasse-Weil conjecture for regular arithmetic schemes

摘录 / EXCERPT: Let ${\mathfrak X}$ be a regular projective scheme over $\mathbb{Z}$ of dimension d, and let ... be its Hasse-Weil zeta function, with completed function...

conjecture | open | MathDB 页面 | 原文线索

V2-016613 | MATHDB-319416

Deninger's cohomological determinant conjecture for arithmetic schemes

摘录 / EXCERPT: Let ${\mathfrak X}$ be a separated scheme of finite type over $\mathbb{Z}$, of dimension d, and let $\hat{\zeta}_{{\mathfrak X}}(s)$ be its completed Hasse-Weil zeta function. De...

conjecture | open | MathDB 页面 | 原文线索

V2-016614 | MATHDB-319451

The dual-defect conjecture for lattice polytopes of large codegree

摘录 / EXCERPT: The dual-defect conjecture for large codegree. If

conjecture | open | MathDB 页面 | 原文线索

V2-016615 | MATHDB-319455

The semisimplicity conjecture for étale cohomology over number fields

摘录 / EXCERPT: Let K be a number field with algebraic closure $\bar K$, let ℓ be a prime, and let X be a smooth proper variety over K. Consider the representation ... For a finite place...

conjecture | open | MathDB 页面 | 原文线索

V2-016616 | MATHDB-319457

Suslin's conjecture on morphic cohomology and singular cohomology

摘录 / EXCERPT: Suslin's conjecture. The comparison morphism above is a quasi-isomorphism on smooth varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-016617 | MATHDB-319469

Reducedness conjecture for Kazhdan-Lusztig ideal initial schemes

摘录 / EXCERPT: Let $v,w\in S_n$, let $I_{v,w}$ be the associated Kazhdan-Lusztig ideal, and let $\prec_{v,w,\pi}$ be one of the term orders determined by a generalized antidiagonal permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-016618 | MATHDB-319470

Vertex-decomposable degeneration conjecture for Kazhdan-Lusztig ideals

摘录 / EXCERPT: Let $v,w\in S_n$, choose a generalized antidiagonal permutation π satisfying the reducedness conjecture, and let $\Delta_{v,w}$ be the Stanley-Reisner simplicial complex associ...

conjecture | open | MathDB 页面 | 原文线索

V2-016619 | MATHDB-319472

Open Orbit Conjecture for dual canonical and dual semicanonical bases

摘录 / EXCERPT: Let Λ_d be the relevant variety of d-dimensional modules over the preprojective algebra Λ, and let Z be an irreducible component of Λ_d. Let b_Z and...

conjecture | open | MathDB 页面 | 原文线索

V2-016620 | MATHDB-319488

The uniqueness and generation conjecture for the braid arrangement fan

摘录 / EXCERPT: Uniqueness and generation conjecture. (a) $\mathcal A_n$ is a refinement of Σ.

conjecture | open | MathDB 页面 | 原文线索

V2-016621 | MATHDB-319489

Torus-equivariant homological mirror symmetry for toric Fano stacks

摘录 / EXCERPT: Let Δ be a convex lattice polytope in $\bR^n$ containing the origin in its interior. Let X be the associated toric Fano stack, let $\bT$ be the n-dimensional torus actin...

conjecture | open | MathDB 页面 | 原文线索

V2-016622 | MATHDB-319490

Tropical coamoeba realization and equivariant homological mirror symmetry

摘录 / EXCERPT: Let Δ be a convex lattice polytope in $\bR^n$ containing the origin in its interior. Let X be the toric Fano stack associated with Δ, let $W:(\bC^*)^n\to\bC$ be a L...

conjecture | open | MathDB 页面 | 原文线索

V2-016623 | MATHDB-319532

Costa-Miró-Roig's conjecture on exceptional collections for Fano toric varieties

摘录 / EXCERPT: Let X be a smooth complete Fano toric variety. A collection of line bundles on X is full if it generates the bounded derived category $D^b(X)$, and it is strongly exceptional i...

conjecture | open | MathDB 页面 | 原文线索

V2-016624 | MATHDB-319542

Determinantal presentation conjecture for secant varieties of factored line bundles

摘录 / EXCERPT: Let C be a curve of genus g, and let L be a line bundle that factors as ... For an integer k, write $\operatorname{Sec}^k(C)$ for the k-th secant variety of the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-016625 | MATHDB-319602

Contact Fano homogeneity conjecture

摘录 / EXCERPT: Contact Fano homogeneity conjecture. A contact Fano manifold is homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-016626 | MATHDB-319612

Kontsevich-Soibelman motivic integration map conjecture

摘录 / EXCERPT: Kontsevich-Soibelman integration-map conjecture. There exists an L-graded Λ-algebra homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-016627 | MATHDB-319623

General Hodge conjecture

摘录 / EXCERPT: Let X be a smooth projective complex scheme, let $l\ge1$ and $p\ge0$, and define the coniveau filtration by ... Here $F^pH^l(X,\mathbb C)$ is the Hodge filtration. General Hodge co...

conjecture | open | MathDB 页面 | 原文线索

V2-016628 | MATHDB-319624

Milnor K-theory description of zero-cycles on quasi-projective schemes

摘录 / EXCERPT: Let X be a quasi-projective scheme of dimension d over an algebraically closed field k of characteristic zero. Let $CH^d(X)$ denote the Chow group of zero-cycles and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016629 | MATHDB-319625

Roitman-type torsion conjecture for filtered Milnor K-theory

摘录 / EXCERPT: Let X be a projective complex scheme of dimension d. Let $F^dKH^M_0(X)$ be the degree-d part of the Brown filtration on Milnor KH-theory, and let ... be the Chern-class map....

conjecture | open | MathDB 页面 | 原文线索

V2-016630 | MATHDB-319630

The monodromy weight conjecture for ℓ-adic cohomology

摘录 / EXCERPT: Let K be a non-archimedean local field with finite residue field k, let X be a smooth projective variety over K, and let $M_\cdot$ be the monodromy filtration on...

conjecture | open | MathDB 页面 | 原文线索

V2-016631 | MATHDB-319634

The vanishing and global-to-local conjectures over number-field arithmetic schemes

摘录 / EXCERPT: Let K be a number field, let $C=\operatorname{Spec}(\mathcal O_K)$, let $U\subset C$ be an open subscheme on which a prime ℓ is invertible, and let $j:U\hookrightarrow C$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-016632 | MATHDB-319635

The H^2 vanishing conjecture for smooth sheaves over arithmetic curves

摘录 / EXCERPT: Let K be a number field, let $C=\operatorname{Spec}(\mathcal O_K)$, let $U\subseteq C$ be an open subscheme on which a prime ℓ is invertible, and let $j:U\hookrightarrow C$...

conjecture | open | MathDB 页面 | 原文线索

V2-016633 | MATHDB-319642

Connectedness of the approximate-root set

摘录 / EXCERPT: Let A be a regular ring, let $\alpha,\beta\in\operatorname{Sper} A$, and assume that either A is complete or $\sqrt{\langle\alpha,\beta\rangle}=\mathfrak m$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016634 | MATHDB-319644

The KNO conjecture on geometric crystals for fundamental representations

摘录 / EXCERPT: KNO conjecture. For any $i\in I\setminus\{0\}$, there exists a unique variety X endowed with a positive $\mathfrak g$-geometric crystal structure and a rational map...

conjecture | open | MathDB 页面 | 原文线索

V2-016635 | MATHDB-319654

The Friedlander-Suslin tower comparison conjecture

摘录 / EXCERPT: Friedlander-Suslin tower comparison conjecture. For each $X\in{\mathbf{Sm}}/k$ and each quasi-fibrant $E\in{\mathbf{Spt}}_{S^1}(k)$, the map $\varphi_{X,*}$ induces an isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-016636 | MATHDB-319662

Bombieri's conjecture on rational points of varieties of general type

摘录 / EXCERPT: Let X be a smooth variety of general type defined over a number field $K/\mathbb{Q}$. Bombieri's conjecture. The set of rational points $X(K)$ is contained in a proper Zariski cl...

conjecture | open | MathDB 页面 | 原文线索

V2-016637 | MATHDB-319664

The flat unitary structure conjecture for the WZW system

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra, let ${\mathcal W}_\ell(\bm{\lambda})$ be the locally free bundle carrying the WZW system, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016638 | MATHDB-319665

Zak's converse conjecture for Scorza varieties

摘录 / EXCERPT: Let $X \subset \mathbb{P}^N$ be an irreducible, smooth, non-degenerate, projective variety. Write X^* for its projective dual, and let $\operatorname{ord}(X)$ denote its order. S...

conjecture | open | MathDB 页面 | 原文线索

V2-016639 | MATHDB-319682

Ramadoss-Grivaux conjecture on the Euler class of the structure sheaf

摘录 / EXCERPT: Let X be a complex manifold, let TX be its tangent bundle, and let ${\rm eu}(\mathscr{O}_X)$ denote the Euler class of the structure sheaf as defined via the Hochschild-theoret...

conjecture | solved | MathDB 页面 | 原文线索

V2-016640 | MATHDB-319698

Sufficient conditions for extension estimates on line-free plane curves

摘录 / EXCERPT: Given a non-zero polynomial $P(x)\in\mathbb{F}_q[x_1,x_2]$, define the algebraic variety ... Assume that $|V|\sim q$ and that $P(x)$ has no linear factor. Extension-estimate conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-016641 | MATHDB-319752

Cheeger-Goresky-MacPherson conjecture on L^2-cohomology and intersection cohomology

摘录 / EXCERPT: Cheeger-Goresky-MacPherson conjecture. There is a natural isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-016642 | MATHDB-319768

Kontsevich-Soibelman motivic Milnor fiber integral conjecture

摘录 / EXCERPT: Kontsevich-Soibelman motivic Milnor fiber integral conjecture. With these notations and hypotheses, in $\mathscr{M}_{\mathbb{G}_{m,k}}^{\mathbb{G}_{m,k}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-016643 | MATHDB-319770

Helton-Vinnikov determinantal representation conjecture for real zero polynomials

摘录 / EXCERPT: Let $p\in\mathbb{R}[x_1,\ldots,x_n]$ be a real zero polynomial if, for every $x\in\mathbb{R}^n$ and $\mu\in\mathbb{C}$, $p(\mu x)=0$ implies that μ is real. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-016644 | MATHDB-319806

The conjectural abelian variety attached to the sign eigenspaces

摘录 / EXCERPT: Let ℓ be a prime, let K_ℓ be the unramified quadratic extension of $\mathbb{Q}_\ell$, and let T_ϵ/L_ϵ be the rigid analytic quotient associated with the s...

conjecture | open | MathDB 页面 | 原文线索

V2-016645 | MATHDB-319819

Normality conjecture for enhanced nilpotent orbit closures

摘录 / EXCERPT: Let V be a finite-dimensional complex vector space, let $\mathcal{N}\subset\operatorname{End}(V)$ be the nilpotent cone, and let $GL(V)$ act on the enhanced nilpotent cone...

conjecture | open | MathDB 页面 | 原文线索

V2-016646 | MATHDB-319820

Normal complete-intersection conjecture for enhanced quiver varieties

摘录 / EXCERPT: Let $\Lambda_{(r_i),I}$ be the enhanced quiver variety associated with the dimension data (r_i) and index set I, and let $d_{(r_i),I}$ be its expected dimension. Normal complet...

conjecture | open | MathDB 页面 | 原文线索

V2-016647 | MATHDB-319821

Dimension-bound conjecture for the non-open locus of enhanced quiver varieties

摘录 / EXCERPT: Let $\Lambda_{(r_i),I}$ be the enhanced quiver variety, let Λ° denote its specified open locus, and let $d_{(r_i),I}$ be its expected dimension. Dimension-bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-016648 | MATHDB-319822

Stratified dimension conjecture for enhanced quiver varieties

摘录 / EXCERPT: Let Ξ index the strata $\Lambda_{(r_i),I}^{\xi}$ of the enhanced quiver variety $\Lambda_{(r_i),I}$, let $d_{(r_i),I}$ be its expected dimension, and let Λ° be the...

conjecture | open | MathDB 页面 | 原文线索

V2-016649 | MATHDB-319834

The paramodular modularity conjecture for abelian varieties of real multiplication type

摘录 / EXCERPT: Let f be a weight 2 newform for the paramodular group $K(N)$, not in the span of the Gritsenko lifts. Let k_f be the totally real number field generated by the Hecke eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-016650 | MATHDB-319841

Density of rational curves on projective Picard-rank-one K3 surfaces

摘录 / EXCERPT: Density conjecture. The union of rational curves is dense in the Zariski topology on X. Equivalently, X contains infinitely many rational curves. The paper proves density for v...

conjecture | open | MathDB 页面 | 原文线索

V2-016651 | MATHDB-319851

Resolution of divisors by weakly admissible blow-ups

摘录 / EXCERPT: Let $(\Sigma_0,\Delta_0,\xi_0)$ be a coordinated normal crossing scheme over k, and let Φ_0 be any effective divisor of Σ_0. A weakly admissible composition of blowin...

conjecture | open | MathDB 页面 | 原文线索

V2-016652 | MATHDB-319866

The Coates-Sinnott conjecture in characteristic p

摘录 / EXCERPT: Coates-Sinnott conjecture. With this notation, for every prime $\ell\ne p$ and every $n\in\mathbb Z_{\ge 2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-016653 | MATHDB-319915

Strong Carlson-Toledo conjecture for finite étale covers

摘录 / EXCERPT: Let M be a compact Kähler manifold with infinite fundamental group Γ. Strong Carlson-Toledo conjecture. There exists a finite étale cover M' of M such that the Hurewic...

conjecture | open | MathDB 页面 | 原文线索

V2-016654 | MATHDB-319916

Cohomological Carlson-Toledo conjecture for finite étale covers

摘录 / EXCERPT: Cohomological Carlson-Toledo conjecture. This natural map is not injective.

conjecture | open | MathDB 页面 | 原文线索

V2-016655 | MATHDB-319921

Folk conjecture on sections of negative-square line bundles on K3^[2]-type varieties

摘录 / EXCERPT: Folk conjecture. If $(c_1(L),H^2(X;\mathbb Z))_X=\mathbb Z$, then either L or L^{-1} has a nonzero section. If $(c_1(L),H^2(X;\mathbb Z))_X=2\mathbb Z$, then either L^2 or...

conjecture | open | MathDB 页面 | 原文线索

V2-016656 | MATHDB-319931

Griffiths-Harris degree conjecture for ACM bundles on general hypersurfaces

摘录 / EXCERPT: Griffiths-Harris degree conjecture. For every indecomposable ACM vector bundle E on X, the integer d divides $\deg c_2(E)$.

conjecture | open | MathDB 页面 | 原文线索

V2-016657 | MATHDB-319952

Model structure conjecture for algebras over free simplicial operads

摘录 / EXCERPT: Let $\mathcal{O}$ be any simplicial operad such that the action of $Sigma_n$ on $\mathcal{O}(n)$ is free, meaning objectwise and levelwise free away from the basepoint. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016658 | MATHDB-319953

Numerical characterization conjecture for polarized manifolds via sectional geometric and Δ-genera

摘录 / EXCERPT: Let $(X,\mathcal{L})$ be a polarized manifold of dimension $n\geq 3$. For each integer i with $2\leq i\leq n-1$, let K_X denote the canonical divisor, let $g_i(X,\mathcal{L})$...

conjecture | open | MathDB 页面 | 原文线索

V2-016659 | MATHDB-319974

Mossel-Weitz-Wormald hardness conjecture for the hardcore model

摘录 / EXCERPT: Mossel-Weitz-Wormald conjecture. Unless $\mathrm{NP}=\mathrm{RP}$, there does not exist a fully polynomial approximation scheme for the partition function of the hardcore model wit...

conjecture | open | MathDB 页面 | 原文线索

V2-016660 | MATHDB-319982

Mirror-symmetry prediction for BPS invariants of Pfaffian Calabi-Yau threefolds of degrees 5, 7 and 10

摘录 / EXCERPT: Let X_i be the Pfaffian Calabi-Yau threefold for $i\in\{5,7,10\}$, and let $\{n_d^g\}_{d\in\mathbb{N}}$ be the numbers listed in the Appendix. BPS-invariant conjecture. The BPS i...

conjecture | open | MathDB 页面 | 原文线索

V2-016661 | MATHDB-319989

The irreducibility and smoothness conjecture for mathematical instanton bundles

摘录 / EXCERPT: Let $\mathcal{I}(c)$ denote the moduli space of rank 2 mathematical instanton bundles on $\mathbb{C}\mathbb{P}^{3}$, namely holomorphic bundles with vanishing first Chern class a...

conjecture | open | MathDB 页面 | 原文线索

V2-016662 | MATHDB-320005

Witten's conjecture on Fukaya categories of genus-g curves and their moduli spaces

摘录 / EXCERPT: The surface Σ_g is a genus-g curve, and $\EuScript{F}(\Sigma_g)$ denotes its Fukaya category. The space $\EuScript{M}(\Sigma_g)$ is the simply-connected Fano variety of...

conjecture | open | MathDB 页面 | 原文线索

V2-016663 | MATHDB-320009

The essentially large divisor conjecture for big divisors in general position

摘录 / EXCERPT: Let $q\geq 1$ and $r\geq q+2$ be integers. Let $X\subseteq \mathbb{P}^\ell$ be a nonsingular projective variety of dimension q, defined over K. Let $D=\sum_{i=1}^r D_i$ be an e...

conjecture | open | MathDB 页面 | 原文线索

V2-016664 | MATHDB-320027

The nonnegativity and upper semicontinuity conjectures for Schubert local h-polynomials

摘录 / EXCERPT: The h-polynomial conjecture. The polynomial $H_{v,w}(q)$ has nonnegative integral coefficients, and if $v'\leq v\leq w$ in Bruhat order, then

conjecture | open | MathDB 页面 | 原文线索

V2-016665 | MATHDB-320058

Tate's Picard-number conjecture for surfaces over finite fields

摘录 / EXCERPT: Let $k=\mathbb{F}_q$ be a finite field of characteristic p, let X be a geometrically connected, smooth, projective k-surface, and let $P_2(X,t)$ be the characteristic polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-016666 | MATHDB-320090

Generalized Green-Griffiths-Lang conjecture for foliated manifolds

摘录 / EXCERPT: Generalized Green-Griffiths-Lang conjecture. There exists an algebraic subvariety $Y\subsetneq X$ such that any non-degenerate holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-016667 | MATHDB-320096

Conjecture on exceptional collections for smooth complete Fano toric varieties

摘录 / EXCERPT: Let X be a smooth complete Fano toric variety over an algebraically closed field of characteristic zero. A full strongly exceptional collection of line bundles on X is an order...

conjecture | open | MathDB 页面 | 原文线索

V2-016668 | MATHDB-320100

The Chow-group description of the projective generators Ck(X)

摘录 / EXCERPT: Let X be a smooth proper k-variety, and let $C_{k(X)}$ denote the projective-generator sheaf associated with the function field $k(X)$. For a smooth k-variety Y, write...

conjecture | open | MathDB 页面 | 原文线索

V2-016669 | MATHDB-320114

Strassen's rank-preserving pair conjecture for Segre varieties

摘录 / EXCERPT: Let $A_j=A_j'\oplus A_j”$ be decompositions of vector spaces, and set ... Let $X=\operatorname{Seg}(\mathbb P A_1\times\cdots\times\mathbb P A_k)$, and let R_X denote the X-rank....

conjecture | open | MathDB 页面 | 原文线索

V2-016670 | MATHDB-320115

Comon's conjecture on tensor rank versus symmetric rank

摘录 / EXCERPT: Let V be a vector space and let $p\in\mathbb P S^dV\vartriangleleft\mathbb P(V^{\otimes d})$. Let $R_{v_d(\mathbb P V)}(p)$ be the symmetric rank of p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016671 | MATHDB-320117

Power-invariance conjecture for logarithmic Lefschetz classes

摘录 / EXCERPT: Let $f\colon U\to V$ be a finite étale morphism of regular flat separated schemes of finite type over S such that the generic fiber $V_K\to U_K$ is tamely ramified with respect t...

conjecture | open | MathDB 页面 | 原文线索

V2-016672 | MATHDB-320155

Canonical subalgebra conjecture for quantum cohomology of partial flag varieties

摘录 / EXCERPT: Let G/P be a partial flag variety, let $QH^*(G/B)$ be its complete-flag quantum cohomology, and let $\mathcal{F}$ be the filtration on $QH^*(G/B)$ induced by the grading semigrou...

conjecture | open | MathDB 页面 | 原文线索

V2-016673 | MATHDB-320157

Breuil's classification conjecture for p-divisible groups and finite flat group schemes

摘录 / EXCERPT: Let k be a perfect field of characteristic p, let $W(k)$ be its ring of Witt vectors, let $K_0=W(k)[1/p]$, and let K be a finite totally ramified extension of K_0 with ring...

conjecture | open | MathDB 页面 | 原文线索

V2-016674 | MATHDB-320186

Codimension-three conjecture for microlocal perverse sheaves

摘录 / EXCERPT: Let $\La$ be a Lagrangian, and let $\Per_\La(\bZ)$ denote the stack of microlocal perverse sheaves on $\La$ with integral coefficients. Codimension-three conjecture. The codimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-016675 | MATHDB-320220

Winning conjecture for badly approximable vectors on rational quadratic varieties

摘录 / EXCERPT: Winning conjecture. For every $m\in\mathbb{Q}$, the set $BA_{\partial X}^{\psi}(X_m(\mathbb{Z}))$ is a (strong) winning subset of ∂X.

conjecture | open | MathDB 页面 | 原文线索

V2-016676 | MATHDB-320259

Cellular resolution conjecture for consistent toric algebras

摘录 / EXCERPT: Let A be the consistent toric algebra associated to a collection on a Gorenstein affine toric variety X of dimension n. Let $\mathbb{T}^n$ be the quotient torus in which the...

conjecture | open | MathDB 页面 | 原文线索

V2-016677 | MATHDB-320260

Self-rational-map conjecture for very general quintic threefolds

摘录 / EXCERPT: Let X be a very general quintic threefold in $\mathbb{P}^4$, and let $\phi\colon X\dashrightarrow X$ be a rational self-map. Self-rational-map conjecture. The map ϕ cannot h...

conjecture | open | MathDB 页面 | 原文线索

V2-016678 | MATHDB-320266

Bruinier-Yang arithmetic intersection conjecture for Hirzebruch-Zagier divisors

摘录 / EXCERPT: Let F be a real quadratic field, let $K=F(\sqrt{\Delta})$ be a quartic non-biquadratic CM field, and let $\mathcal M$ be the moduli stack of principally polarized abelian surface...

conjecture | open | MathDB 页面 | 原文线索

V2-016679 | MATHDB-320291

Weak approximation for rationally connected varieties

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let B be a curve over k, and let $k(B)$ be its function field. Let X be a rationally connected variety over...

conjecture | open | MathDB 页面 | 原文线索

V2-016680 | MATHDB-320314

Equivariant Suslin's conjecture

摘录 / EXCERPT: Equivariant Suslin's conjecture. This map is an isomorphism for $a\leq 0$ with $a\leq q$ and $b\leq q$, and an injection for $a=1$ with $a\leq q$ and $b\leq q$.

conjecture | open | MathDB 页面 | 原文线索

V2-016681 | MATHDB-320315

The 2-adic Lichtenbaum-Quillen conjecture over the reals

摘录 / EXCERPT: The 2-adic Lichtenbaum-Quillen conjecture. This map is an isomorphism for $n\geq \dim X-1$ and an injection for $n\geq \dim X-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-016682 | MATHDB-320333

The isotriviality conjecture for families over special bases

摘录 / EXCERPT: Let $f:X\to B$ be an algebraic smooth family of canonically polarised manifolds with generically maximal variation, and let B be a special quasi-projective manifold, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-016683 | MATHDB-320342

Pattern characterization of smooth affine Schubert varieties of type A

摘录 / EXCERPT: Let $w\in\widetilde{S}_n$. The Schubert variety X_w is indexed by an affine permutation w, and smoothness is understood in the usual algebraic-geometric sense. Smoothness patte...

conjecture | open | MathDB 页面 | 原文线索

V2-016684 | MATHDB-320343

Bonahon-Wong classification conjecture for irreducible skein-algebra representations

摘录 / EXCERPT: Classification conjecture. Up to isomorphism, there is a unique irreducible representation

conjecture | open | MathDB 页面 | 原文线索

V2-016685 | MATHDB-320352

Debarre's codimension conjecture for Andreotti-Mayer loci

摘录 / EXCERPT: Debarre's codimension conjecture. For any $1\leq k\leq g-3$,

conjecture | open | MathDB 页面 | 原文线索

V2-016686 | MATHDB-320353

The indecomposable theta-multiplicity conjecture

摘录 / EXCERPT: Theta-multiplicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016687 | MATHDB-320354

The Gamma-zero-zero base-locus conjecture for Jacobians

摘录 / EXCERPT: Γ_{00} conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-016688 | MATHDB-320366

Oblomkov-Shende conjecture for the HOMFLY polynomial of a curve singularity

摘录 / EXCERPT: Oblomkov-Shende conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016689 | MATHDB-320431

Affine compactification conjecture for degenerating enhanced projective families

摘录 / EXCERPT: Let T be an affine variety carrying a universal full family of enhanced projective varieties. Let $f:Y\to S$ be a family of projective varieties with fixed prescribed topological...

conjecture | open | MathDB 页面 | 原文线索

V2-016690 | MATHDB-320437

Existence of nonalgebraic quadrature domains in dimensions greater than two

摘录 / EXCERPT: A quadrature domain is understood here in the classical, restricted sense, rather than in the wide sense allowing a compactly supported distribution in the quadrature formula. The...

conjecture | open | MathDB 页面 | 原文线索

V2-016691 | MATHDB-320454

Gamma-function singularity conjecture for beta-ensemble free energies

摘录 / EXCERPT: Let $V_{d+1}(x)$ be any potential of degree $d+1$, and let ${\widetilde{\epsilon}}_i$ be the occupation numbers on the cycles ${\widetilde{{\cal A}}}_i$, for $i=1,\ldots,d$. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-016692 | MATHDB-320458

Conjecture that the global log canonical threshold is attained at finite level

摘录 / EXCERPT: Let X be a Fano variety with at most quotient singularities, and let ... where $\mathrm{lct}_{n}(X)$ is the infimum of the log canonical thresholds of divisors in $|-nK_X|$. Finite...

conjecture | open | MathDB 页面 | 原文线索

V2-016693 | MATHDB-320459

Chen-Wang conjecture on alpha-invariants and Kähler-Einstein metrics

摘录 / EXCERPT: Let X be a Fano variety with at most quotient singularities, and for each $n\in\mathbb{N}$ let $\mathrm{lct}_{n,2}(X)$ be the infimum of the log canonical thresholds of pencils i...

conjecture | open | MathDB 页面 | 原文线索

V2-016694 | MATHDB-320483

The volume bound for Fano manifolds of Picard number one

摘录 / EXCERPT: Volume bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-016695 | MATHDB-320485

Berkovich's henselization conjecture for formal schemes

摘录 / EXCERPT: Let k be a valued field, let $\mathfrak X$ be locally of the form $\operatorname{Spf}(\mathcal A^\circ)$ for k-affinoid algebras $\mathcal A$, so that $\mathfrak X$ is of finit...

conjecture | open | MathDB 页面 | 原文线索

V2-016696 | MATHDB-320488

The secant-syzygy conjecture for Veronese varieties

摘录 / EXCERPT: Let $v_k(\mathbb{P}^d)$ be the k-uple Veronese embedding of $\mathbb{P}^d$, with $d\geq2$ and $k\geq3$, and let Σ be its secant variety. Write $N_{p,q}$ for the indicated...

conjecture | open | MathDB 页面 | 原文线索

V2-016697 | MATHDB-320489

The irreducibility conjecture for loci of general projections

摘录 / EXCERPT: Let $X\subset {\mathbb P}^N$ be a smooth irreducible variety of dimension n. For a general projection of X, denote by X_k the locus in the target where the fiber has degree a...

conjecture | open | MathDB 页面 | 原文线索

V2-016698 | MATHDB-320490

Zak's completeness conjecture for multiple-point loci

摘录 / EXCERPT: Let $X^n\subset {\mathbb P}^N$ be a smooth irreducible variety of dimension n. Consider a general projection $X\to {\mathbb P}^{N-1}$, and let $X_k\subset {\mathbb P}^{N-1}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-016699 | MATHDB-320494

Degree bound for arithmetic supports of finitely generated D-modules

摘录 / EXCERPT: Let X be a smooth affine variety over a characteristic-zero field ${\bf k}$, let M be a finitely generated ${\mathcal D}(X)$-module, and choose a finitely generated ring...

conjecture | open | MathDB 页面 | 原文线索

V2-016700 | MATHDB-320495

Arithmetic support characterization of extended motivic-exponential D-modules

摘录 / EXCERPT: Let ${\bf k}$ be a characteristic-zero field, let X be a smooth algebraic variety over ${\bf k}$, and let M be a holonomic ${\mathcal D}_X$-module. Call M extended motivic-ex...

conjecture | open | MathDB 页面 | 原文线索

V2-016701 | MATHDB-320496

Canonical holonomic D-modules associated with simply connected Lagrangians

摘录 / EXCERPT: Let X be a smooth variety over ${\bf k}={\mathbb C}$, and let $L\subset T^*X$ be a smooth closed connected Lagrangian subvariety satisfying $H_1(L({\mathbb C}),{\mathbb Z})=0$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-016702 | MATHDB-320497

Rigidity implies extended motivic-exponentiality for holonomic D-modules

摘录 / EXCERPT: Let X be a smooth algebraic variety over a characteristic-zero field ${\bf k}$ and let M be a holonomic ${\mathcal D}_X$-module. Rigidity conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-016703 | MATHDB-320498

Quantum integrability criterion from divisibility of characteristic supports

摘录 / EXCERPT: Let X be a smooth affine variety over a characteristic-zero field ${\bf k}$ of dimension $n>1$, and let $P\in{\mathcal D}(X)$ be a differential operator. For a prime p, conside...

conjecture | open | MathDB 页面 | 原文线索

V2-016704 | MATHDB-320500

Atypicality conjecture for support variety dimensions

摘录 / EXCERPT: Atypicality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016705 | MATHDB-320502

Ampleness conjecture for jet differential bundles on generic complete intersections

摘录 / EXCERPT: Let $X\subset \mathbb{P}^N$ be a generic complete intersection variety of dimension n and codimension c. For integers k and m as in the jet-differential construction, let...

conjecture | open | MathDB 页面 | 原文线索

V2-016706 | MATHDB-320511

Formality conjecture for deformation dglas of polarizable principal bundles

摘录 / EXCERPT: Let $X_{{\bullet}}$ be a connected simplicial smooth projective variety satisfying the finite index condition. Let P be a G-principal bundle with λ-connection, or a G...

conjecture | open | MathDB 页面 | 原文线索

V2-016707 | MATHDB-320512

Bruzzo's conjecture on pluri-semistable Higgs bundles

摘录 / EXCERPT: Let X be a smooth proper DM-stack, and let (E,θ) be a Higgs bundle on X. It is pluri-semistable if for every curve Y and map $g:Y\to X$, the pullback Higgs bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-016708 | MATHDB-320516

Algebraic realization conjecture for Hitchin's moduli space

摘录 / EXCERPT: Fix (u,ϑ) and let $I_{FG}(u,\vartheta)$ be the ideal generated by all Fock-Goncharov relations associated with the slabs in the marked BPS polyhedral decomposition. The r...

conjecture | open | MathDB 页面 | 原文线索

V2-016709 | MATHDB-320526

Converse characterization of trivial cocycles

摘录 / EXCERPT: Converse conjecture. If $\psi\in A^p$, then there exists $\varphi\in A^{p-1}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-016710 | MATHDB-320527

BorisovHua conjecture on exceptional collections for nef-Fano toric DM stacks

摘录 / EXCERPT: BorisovHua conjecture. Every smooth nef-Fano toric DM stack possesses a full strong exceptional collection of line bundles.

conjecture | open | MathDB 页面 | 原文线索

V2-016711 | MATHDB-320528

CostaMirRoig conjecture on exceptional collections for Fano toric varieties

摘录 / EXCERPT: CostaMirRoig conjecture. For any smooth, complete Fano toric variety there exists a full, strongly exceptional collection of line bundles.

conjecture | open | MathDB 页面 | 原文线索

V2-016712 | MATHDB-320529

Orlov's conjecture on strong exceptional collections for toric DM stacks

摘录 / EXCERPT: Orlov's conjecture. For any smooth projective toric DM stack $Y,$ the derived category $D^b(Y)$ is generated by a strong exceptional collection.

conjecture | open | MathDB 页面 | 原文线索

V2-016713 | MATHDB-320530

The l-part of the Tamagawa number conjecture

摘录 / EXCERPT: Let U be the open arithmetic base over which the smooth l-adic sheaf V_l^i is defined, let $T_l^i\subseteq V_l^i$ be a Galois-stable $\mathbb Z_l$-lattice, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016714 | MATHDB-320531

Frobenius semisimplicity at the eigenvalue 1

摘录 / EXCERPT: Let V_l^i be the l-adic representation attached to the ith cohomology of the generic fibre, and let $R\Gamma_f(\mathbb Q_p,V_l^i)$ be the corresponding local finite-cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-016715 | MATHDB-320540

The strong maximal rank conjecture for general curves

摘录 / EXCERPT: The strong maximal rank conjecture. For a general $[C]\in\mathcal{M}_g$, the locus $\mathfrak{Quad}_{g,d}^r(C)$ has the expected determinantal dimension

conjecture | open | MathDB 页面 | 原文线索

V2-016716 | MATHDB-320552

Ionescu's effective non-vanishing conjecture for quasi-polarized manifolds

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over the complex numbers, and let L be a nef and big line bundle on X. Thus (X,L) is a quasi-polarized manifold. Ione...

conjecture | open | MathDB 页面 | 原文线索

V2-016717 | MATHDB-320553

Cohomological connectedness conjecture for the mixed elliptic motive algebra

摘录 / EXCERPT: Let E be the elliptic curve and let ${\mathcal A}_{{\frak M}(E)}$ denote the algebra of mixed elliptic motives considered in the paper, graded by i. Cohomological connectedness...

conjecture | open | MathDB 页面 | 原文线索

V2-016718 | MATHDB-320592

The archimedean Tate conjecture for Shimura varieties of quaternionic type

摘录 / EXCERPT: Let X and Y be Shimura varieties of quaternionic type of dimension s. Suppose that they are isospectral for the de Rham Laplacian acting on p-forms for every integer p wi...

conjecture | open | MathDB 页面 | 原文线索

V2-016719 | MATHDB-320635

Geramita's catalecticant ideal equality and inclusion conjecture

摘录 / EXCERPT: Let K be a field, let $n,d,ke\mathbb{Z}$ with $k,n\geq 2$ and $d\geq 2k-2$, and let $Cat(t,d-t;n)$ denote the t-th generic catalecticant. Write $I_k(Cat(t,d-t;n))$ for the idea...

conjecture | open | MathDB 页面 | 原文线索

V2-016720 | MATHDB-320663

Prolongation intersection conjecture for claw-tree varieties

摘录 / EXCERPT: Prolongation intersection conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016721 | MATHDB-320664

Toric intersection conjecture for group-based tree varieties

摘录 / EXCERPT: Toric intersection conjecture. The toric variety $X(T)$ is the toric intersection of all the toric varieties $X(T_i)$.

conjecture | open | MathDB 页面 | 原文线索

V2-016722 | MATHDB-320672

Generalized Mazur conjecture for simple abelian varieties over the algebraic numbers

摘录 / EXCERPT: Let A be a simple abelian variety over $\overline{\mathbb{Q}}$ of dimension g, and let $\Gamma\subset A(\overline{\mathbb{Q}})$ be a subgroup of positive rank. Let C be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-016723 | MATHDB-320673

Lie-algebraic form of the generalized Mazur conjecture

摘录 / EXCERPT: Let A be a simple abelian variety over $\overline{\mathbb{Q}}$, and let $\Gamma\subset A(\overline{\mathbb{Q}})$ be a subgroup of positive rank. Let C be the closure of...

conjecture | open | MathDB 页面 | 原文线索

V2-016724 | MATHDB-320674

Real-model quotient conjecture for commutative group varieties

摘录 / EXCERPT: Let G be a commutative group variety over $\overline{\mathbb{Q}}$, and let C be a connected proper real-analytic submanifold of $G(\mathbb{C})$. Suppose C contains a subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-016725 | MATHDB-320683

Quasi-monomiality conjecture for valuations computing asymptotic Arnold multiplicities

摘录 / EXCERPT: Quasi-monomiality conjecture. The weak version asserts that, for every generic point ξ of an irreducible component of the subscheme defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-016726 | MATHDB-320699

Generating-series conjecture for quasihomogeneous Hilbert schemes of points

摘录 / EXCERPT: Let a,b be positive integers, let $({\mathbb C}^2)^{[n]}_{a,b}$ denote the Hilbert scheme of n points on the plane fixed by the quasihomogeneous action with weights (a,b), le...

conjecture | open | MathDB 页面 | 原文线索

V2-016727 | MATHDB-320726

Stapledon's equivariant mirror symmetry conjecture for orbifold Calabi-Yau varieties

摘录 / EXCERPT: Let P and P^* be d-dimensional reflexive polytopes, with associated mirror Calabi-Yau varieties X and X^*. Let Γ act as in the source, assume that $\det(\rho)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-016728 | MATHDB-320727

Stapledon's orbifold Euler characteristic mirror conjecture

摘录 / EXCERPT: Let P and P^* be d-dimensional reflexive polytopes, with associated mirror Calabi-Yau varieties X and X^*. Assume that $\det(\rho)$ is the trivial representation and that...

conjecture | open | MathDB 页面 | 原文线索

V2-016729 | MATHDB-320737

Hiss-Lübeck-Malle conjecture on Brauer trees of principal blocks

摘录 / EXCERPT: Let $\Gamma^\bullet$ denote the graph obtained from the Brauer tree of the principal ℓ-block by removing the exceptional node and all edges incident to it. For each root of un...

conjecture | open | MathDB 页面 | 原文线索

V2-016730 | MATHDB-320744

Pappas-Rapoport flatness conjecture for the symplectic matrix scheme

摘录 / EXCERPT: Pappas-Rapoport flatness conjecture. The scheme P is flat over

conjecture | open | MathDB 页面 | 原文线索

V2-016731 | MATHDB-320750

Garcia-Stillman-Sturmfels conjecture on equations of secant lines to Segre varieties

摘录 / EXCERPT: Garcia-Stillman-Sturmfels conjecture. The ideal of $\sigma_2(\operatorname{Seg}(\mathbb{P}V_1\times\cdots\times\mathbb{P}V_n))$ is generated by the 3×3 minors of all flatte...

conjecture | open | MathDB 页面 | 原文线索

V2-016732 | MATHDB-320783

Mustaţă's ideal-adic semi-continuity conjecture for minimal log discrepancies of pairs

摘录 / EXCERPT: Let (X,Δ) be a pair, meaning that X is a normal variety and Δ is an effective $\mathbb{R}$-divisor with $K_X+\Delta$ an $\mathbb{R}$-Cartier divisor. Let Z be a c...

conjecture | open | MathDB 页面 | 原文线索

V2-016733 | MATHDB-320794

The semi-orthogonal decomposition conjecture for gaps in Orlov spectra

摘录 / EXCERPT: Let X be a smooth algebraic variety, and let ... be a semi-orthogonal decomposition of $\operatorname{D}^{\operatorname{b}}(\operatorname{coh} X)$. A gap conjecture for semi-orthog...

conjecture | open | MathDB 页面 | 原文线索

V2-016734 | MATHDB-320797

Eisenbud's low-rank quadric syzygy conjecture

摘录 / EXCERPT: Let $\Bbbk$ be an algebraically closed field of characteristic zero, let S be a polynomial ring over $\Bbbk$, and let $I\subseteq S$ be a prime ideal containing no linear form. S...

conjecture | open | MathDB 页面 | 原文线索

V2-016735 | MATHDB-320815

Denis's Mordell-Lang conjecture for Drinfeld modules

摘录 / EXCERPT: Let K be a field extension of ${\mathbb F}_p(t)$, let Φ be a Drinfeld module of generic characteristic over K, let $X\subseteq {\mathbb G}_a^g$ be a subvariety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016736 | MATHDB-320816

The relative Stiefel-Whitney and Hasse-Witt class conjecture

摘录 / EXCERPT: Let S be a normal divisorial scheme over $\mathbb Z[\frac1{2\ell}]$, either separated or noetherian, and let $f:X\to S$ be a proper smooth morphism of relative even dimension n...

conjecture | open | MathDB 页面 | 原文线索

V2-016737 | MATHDB-320832

Geyer-Jarden conjecture on torsion in abelian varieties over large algebraic fields

摘录 / EXCERPT: Let A be an abelian variety of positive dimension defined over a finitely generated field K, let e be a positive integer, and let $\bar K(\sigma)$ denote the fixed field asso...

conjecture | open | MathDB 页面 | 原文线索

V2-016738 | MATHDB-320837

Basic open semialgebraic cone conjecture for graphs (9) and (11)

摘录 / EXCERPT: Let $\mathcal{C}_{\mathcal{G}}$ denote the cone associated with a graph $\mathcal{G}$. A basic open semialgebraic set is an open semialgebraic set defined by finitely many strict p...

conjecture | open | MathDB 页面 | 原文线索

V2-016739 | MATHDB-320841

Conjecture on crystalline points in components of the local character variety

摘录 / EXCERPT: Conjecture sur les points cristallins. Toute composante irréductible de ${\mathfrak X}_d$ contient un point cristallin absolument irréductible.

conjecture | open | MathDB 页面 | 原文线索

V2-016740 | MATHDB-320845

The multiplier-test ideal comparison conjecture

摘录 / EXCERPT: Let Y be a smooth, irreducible variety over an algebraically closed field k of characteristic zero, and let $\mathfrak{a}$ be a nonzero ideal on Y. Given any model Y_A and...

conjecture | open | MathDB 页面 | 原文线索

V2-016741 | MATHDB-320846

The ordinarity conjecture for smooth projective varieties

摘录 / EXCERPT: Let X be a smooth, irreducible n-dimensional projective variety defined over an algebraically closed field k of characteristic zero. If X_A is a model of X defined over a...

conjecture | open | MathDB 页面 | 原文线索

V2-016742 | MATHDB-320856

Conjecture on reduced components of the degree-3 center variety

摘录 / EXCERPT: Component-count conjecture. The number of reduced components of the center variety in degree 3 is 1 in codimension 5, 2 in codimension 6, 4 in codimension 7, 4 in c...

conjecture | open | MathDB 页面 | 原文线索

V2-016743 | MATHDB-320872

Bondal's conjecture for smooth complete toric varieties

摘录 / EXCERPT: Bondal's conjecture. The object $F_{m*}\mathcal O_X(D)$ is a classical generator of $D^b(X)$ for any T-invariant divisor D and a sufficiently large integer m.

conjecture | open | MathDB 页面 | 原文线索

V2-016744 | MATHDB-320877

The expected p-adic rank conjecture for simple Abelian varieties

摘录 / EXCERPT: Let A be a simple Abelian variety over ${\mathbb{Q}}$ of dimension g, let Γ be a subgroup of $A({\mathbb{Q}})$ of rank ℓ over $\mathbb{Z}$, and let p be a prime n...

conjecture | open | MathDB 页面 | 原文线索

V2-016745 | MATHDB-320881

Motivic monodromy conjecture for local motivic zeta functions

摘录 / EXCERPT: Let k be a field of characteristic zero, let X be the relevant k-variety, let $f:X\to\mathbb{A}^1_k$ be the morphism, let x be a closed point with residue field k_x, and...

conjecture | open | MathDB 页面 | 原文线索

V2-016746 | MATHDB-320895

Conjecture on Poincaré duality and ring structures for tropical varieties

摘录 / EXCERPT: Poincaré duality and ring-structure conjecture. The map D_F is surjective for every tropical variety F. If F is regularizable in codimension 1 and irreducible, then its ker...

conjecture | open | MathDB 页面 | 原文线索

V2-016747 | MATHDB-320912

Suciu's Chen-rank formula conjecture from resonance

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a hyperplane arrangement, let $G=G({\mathcal{A}})$ be its arrangement group, and let $R^{1}({\mathcal{A}})$ be its first resonance variety. For each r, let...

conjecture | open | MathDB 页面 | 原文线索

V2-016748 | MATHDB-320935

Drinfeld's conjecture on smooth sheaves on simply connected BunG

摘录 / EXCERPT: Let G^{sc} be the simply connected cover of the derived group of a reductive group G, and let $\operatorname{Bun}_{G^{sc}}$ be the stack of G^{sc}-torsors on X. A smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-016749 | MATHDB-320936

Characteristic-variety conjecture for the minimal automorphic sheaf

摘录 / EXCERPT: Let Bun_H be the stack of H-bundles, let ${\cal K}_H$ be the perverse sheaf constructed in the paper, and let ${\cal C}^0\subset T^*\operatorname{Bun}_H$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-016750 | MATHDB-320944

Uniqueness conjecture for Brauer-Severi varieties of conic-bundle orders

摘录 / EXCERPT: Let X be the Brauer-Severi variety of an order of type F^2_3, F^2_4, or F^{2-}_5, as described in the table in the source. Let Y be another variety birational to X over...

conjecture | open | MathDB 页面 | 原文线索

V2-016751 | MATHDB-320946

The 2-primary hermitian K-theory comparison conjecture

摘录 / EXCERPT: Let S be a scheme with residue fields $k(s)$, let vcd_{2} denote 2-cohomological virtual dimension, and let σ be the comparison map from algebraic to étale herm...

conjecture | open | MathDB 页面 | 原文线索

V2-016752 | MATHDB-320980

Reduced tangent-cone conjecture for Richardson varieties

摘录 / EXCERPT: Let $X_u^v=X_u\cap X^v$ be a Richardson variety in GL_n/B, let $wB\in X_u^v$, and let the tangent cone at wB be defined by the associated graded local ring of X_u^v at that p...

conjecture | open | MathDB 页面 | 原文线索

V2-016753 | MATHDB-320986

The variational Hodge conjecture for de Rham cohomology

摘录 / EXCERPT: Variational Hodge conjecture for de Rham cohomology. For every complex point s of S, the class $\widetilde{\beta}_s$ is the cohomology class of an algebraic cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-016754 | MATHDB-321000

Markman's monodromy conjecture for generalized Kummer varieties

摘录 / EXCERPT: Let X be deformation equivalent to a generalized Kummer variety of dimension $2n$, with $n\geq 2$. Then $H^2(X,\mathbb Z)$ is isometric to...

conjecture | open | MathDB 页面 | 原文线索

V2-016755 | MATHDB-321001

Monodromy-isometry equality for O'Grady's ten-dimensional example

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold deformation equivalent to O'Grady's 10-dimensional exceptional example. Monodromy-isometry conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-016756 | MATHDB-321008

Pink's intersection conjecture for semi-abelian varieties

摘录 / EXCERPT: Pink's conjecture. If X is not contained in a proper algebraic subgroup of G, then

conjecture | open | MathDB 页面 | 原文线索

V2-016757 | MATHDB-321009

Pink's finite-rank intersection conjecture

摘录 / EXCERPT: Pink's finite-rank conjecture. If X is not contained in a translate of a proper algebraic subgroup of G, then

conjecture | open | MathDB 页面 | 原文线索

V2-016758 | MATHDB-321010

Zilber's finite-family conjecture for atypical intersections

摘录 / EXCERPT: Zilber's conjecture. There is a finite family Φ of proper algebraic subgroups of G such that, for every algebraic subgroup H of G, every atypical component of $X\cap H$...

conjecture | open | MathDB 页面 | 原文线索

V2-016759 | MATHDB-321011

David's Lehmer conjecture for tori and abelian varieties

摘录 / EXCERPT: David's Lehmer conjecture. There is a real number $c>0$ such that every point $x\in G(\overline{\mathbf Q})$ not contained in any proper algebraic subgroup of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-016760 | MATHDB-321012

The relative Lehmer conjecture for tori and abelian varieties

摘录 / EXCERPT: Relative Lehmer conjecture. There is a real number $c>0$ such that every point $x\in G(\overline{\mathbf Q})$ not contained in any proper algebraic subgroup of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-016761 | MATHDB-321013

The effective Bogomolov conjecture for tori and abelian varieties

摘录 / EXCERPT: Effective Bogomolov conjecture. There is a real number $c>0$ such that every proper closed irreducible subvariety X of G that is not contained in a translate of a proper algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-016762 | MATHDB-321043

Conjecture on global sections of the Frobenius-pushforward differential operators

摘录 / EXCERPT: Global-sections conjecture. Assume $\ell>h_G$. The natural algebra homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-016763 | MATHDB-321044

Stronger conjecture on global sections for the finite quantized differential operators

摘录 / EXCERPT: Finite global-sections conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-016764 | MATHDB-321045

Conjecture on induction of the quantized module V

摘录 / EXCERPT: Induction conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-016765 | MATHDB-321059

Singularity-geometric/combinatoric duality for Calabi-Yau families

摘录 / EXCERPT: Singularity-geometric/combinatoric duality. The total Milnor number of each arising singular fibre is expressible in terms of the change in the degree of the zeta function when pas...

conjecture | open | MathDB 页面 | 原文线索

V2-016766 | MATHDB-321076

The coupled equations imply alpha-K-polystability for triples

摘录 / EXCERPT: Let (X,L) be a smooth compact polarized variety and let E be a holomorphic vector bundle over X of rank r and degree zero. For positive coupling constants...

conjecture | open | MathDB 页面 | 原文线索

V2-016767 | MATHDB-321094

The motivic multiple zeta value algebra conjecture

摘录 / EXCERPT: Motivic multiple zeta value algebra conjecture. There is an isomorphism of graded algebras

conjecture | open | MathDB 页面 | 原文线索

V2-016768 | MATHDB-321101

Projective-space origin conjecture for the line cycle of Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let X be a smooth $(n-1)$-dimensional Calabi-Yau hypersurface such that its variety of lines $F(X)$ has dimension $n-4$. Let ... be the cycle obtained by taking the union, over lin...

conjecture | open | MathDB 页面 | 原文线索

V2-016769 | MATHDB-321106

The non-Archimedean Hodge-${\mathcal D}$-conjecture for varieties with good reduction

摘录 / EXCERPT: The non-Archimedean Hodge-${\mathcal D}$-conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-016770 | MATHDB-321108

Curious hard Lefschetz conjecture for character varieties

摘录 / EXCERPT: Let $\mathcal{M}_\B$ be the relevant character variety, let $\dim=\dim(\mathcal{M}_\B)$, and let $\alpha\in W_4H^2(\mathcal{M}_\B)$, where $W_\bullet$ is the weight filtration. Cur...

conjecture | open | MathDB 页面 | 原文线索

V2-016771 | MATHDB-321109

P=W conjecture for the Hitchin system

摘录 / EXCERPT: Let $\mathcal{M}_\Dol$ be the Dolbeault moduli space and $\mathcal{M}_\B$ the corresponding Betti character variety. Let $P_\bullet$ be the perverse filtration on...

conjecture | solved | MathDB 页面 | 原文线索

V2-016772 | MATHDB-321124

Conjecture on nodal Enriques surface summands

摘录 / EXCERPT: The paper considers the Artin--Mumford example, the variety V_{10} with 10 singular points, a double covering of a quadric ramified in an octic with 20 nodal singular points,...

conjecture | open | MathDB 页面 | 原文线索

V2-016773 | MATHDB-321125

BFK's gap bound conjecture for the Orlov spectrum

摘录 / EXCERPT: Let X be a smooth variety. Its derived category $D^b(X)$ has an Orlov spectrum whose gaps are intervals between consecutive generation times of strong generators. BFK's gap bound...

conjecture | open | MathDB 页面 | 原文线索

V2-016774 | MATHDB-321126

BFK's birational invariance conjecture for the maximal Orlov-spectrum gap

摘录 / EXCERPT: Let X be a triangulated category arising as D^b of a smooth variety, and consider the maximal gap in its Orlov spectrum. BFK's birational invariance conjecture. The maximal gap...

conjecture | open | MathDB 页面 | 原文线索

V2-016775 | MATHDB-321128

Gap-one conjecture for rational smooth projective threefolds in the Noether--Lefschetz spectrum

摘录 / EXCERPT: Let X be a 3-dimensional smooth projective variety. Let $NL(D^b(X))$ denote the enhanced Noether--Lefschetz spectrum constructed from annihilators of subspaces of Hochschild ho...

conjecture | open | MathDB 页面 | 原文线索

V2-016776 | MATHDB-321142

Pink-Zilber conjecture for curves in semiabelian schemes

摘录 / EXCERPT: Pink-Zilber conjecture. The intersection $V\cap\mathcal{S}^{[2]}$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-016777 | MATHDB-321165

Weak and strong localization conjectures for hypertoric varieties

摘录 / EXCERPT: Let K be a subtorus of a torus T, let $\boldsymbol{\chi}=(\chi_i)$ be the associated characters, and let α be as in the construction of the hypertoric variety. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-016778 | MATHDB-321170

Pirola's nonexistence conjecture for curves in very generic Jacobians

摘录 / EXCERPT: Let J be a very generic Jacobian of dimension $g\geq 4$, and let a curve on J mean the image of a non-constant map from a smooth curve to J. The geometric genus of such a cur...

conjecture | open | MathDB 页面 | 原文线索

V2-016779 | MATHDB-321186

Mirror symmetry of variations of Hodge structures for Fano varieties

摘录 / EXCERPT: Mirror symmetry of variations of Hodge structures. For any Fano variety X there exists a one-parameter family

conjecture | open | MathDB 页面 | 原文线索

V2-016780 | MATHDB-321235

The projective-module conjecture for generic stabilizers of type-L quotients

摘录 / EXCERPT: Let $\mathcal S$ be a spherical system with a quotient $\mathcal S/\Delta'$ of type $(\mathcal L)$ satisfying the stated minimality and color hypotheses. Let M be the Levi factor...

conjecture | open | MathDB 页面 | 原文线索

V2-016781 | MATHDB-321236

The dual-projective embedding conjecture for generic stabilizers

摘录 / EXCERPT: Under the hypotheses and notation of the preceding type-$(\mathcal L)$ quotient construction, let W be the simple $K_{\Delta'}$-module from the projective-module conjecture, let...

conjecture | open | MathDB 页面 | 原文线索

V2-016782 | MATHDB-321237

The higher-defect localization conjecture for spherical systems

摘录 / EXCERPT: Let $\mathcal S$ be a spherical system with spherical roots Σ, let Δ' be a distinguished subset of type $(\mathcal L)$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-016783 | MATHDB-321242

The projective theorem on tangencies for dual varieties

摘录 / EXCERPT: Let $X \subset \mathbb{P}_{\mathbb{C}}^n$ be a non-degenerate, irreducible projective variety, and let $X^* \subset \mathbb{P}_{\mathbb{C}}^N$ be its projective dual. For...

conjecture | open | MathDB 页面 | 原文线索

V2-016784 | MATHDB-321243

Anderson-Klee's principle for objects in a common geometry

摘录 / EXCERPT: Let $X\subset\mathbb{E}^n$ be an object, and let $X\langle r\rangle$ denote the set of r-singular points, where an r-singular point is one whose associated fiber has scheme-the...

conjecture | open | MathDB 页面 | 原文线索

V2-016785 | MATHDB-321244

Anderson-Klee's projective dual singular-locus conjecture

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a smooth, irreducible, non-degenerate projective variety. For each $r\in\{0,\ldots,n\}$, define ... where $p(q^{-1}(H^{\bot}))$ is the contact locus of...

conjecture | open | MathDB 页面 | 原文线索

V2-016786 | MATHDB-321252

Existence of good minimal models for pseudo-effective klt pairs

摘录 / EXCERPT: Let (X,Δ) be a $\mathbb{Q}$-factorial projective kawamata log terminal pair, meaning that X is $\mathbb{Q}$-factorial, projective, and (X,Δ) is kawamata log termina...

conjecture | open | MathDB 页面 | 原文线索

V2-016787 | MATHDB-321254

Genericity conjecture for essential dimension of objects of amenable stacks

摘录 / EXCERPT: Let $\mathcal{X}$ be an amenable stack over k. Let L be an extension of k, and let ξ be an object of $\mathcal{X}(\operatorname{Spec} L)$. Here $\operatorname{ed}_k\xi$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-016788 | MATHDB-321257

Pure-codimension conjecture for the odd two-torsion singularity locus

摘录 / EXCERPT: Let $\mathcal A_g$ be the moduli space of principally polarized abelian varieties of genus g, and let $I^{(g)}\subset\mathcal A_g$ be the locus of pairs (A,Θ) whose theta...

conjecture | open | MathDB 页面 | 原文线索

V2-016789 | MATHDB-321258

Equality of the compactified gradient and odd two-torsion singularity loci

摘录 / EXCERPT: For each genus g, let $I^{(g)}\subset\mathcal A_g$ be the locus of principally polarized abelian varieties whose theta divisor is singular at an odd two-torsion point, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016790 | MATHDB-321259

Codimension conjecture for the simple part of the Andreotti-Mayer locus N_2

摘录 / EXCERPT: Let ${\mathcal A}_g$ be the moduli space of principally polarized abelian varieties of dimension g, and let ... A principally polarized abelian variety is simple if it is not isoge...

conjecture | open | MathDB 页面 | 原文线索

V2-016791 | MATHDB-321260

Kontsevich-Soibelman positivity conjecture for motivic Donaldson-Thomas invariants

摘录 / EXCERPT: Let Q be a symmetric quiver, and let $\boldsymbol{\Omega}_{\boldsymbol{\alpha}}$ denote the Donaldson-Thomas invariant associated with a dimension vector...

conjecture | open | MathDB 页面 | 原文线索

V2-016792 | MATHDB-321263

Motivic zeta-function rationality for curves with finite group actions

摘录 / EXCERPT: Finite-group generalization conjecture. The motivic zeta function $\zeta_{(C,\sigma)}(t)$ is rational.

conjecture | open | MathDB 页面 | 原文线索

V2-016793 | MATHDB-321271

The torsion conjecture for infinitely p-divisible points on abelian varieties

摘录 / EXCERPT: Let K_0 be the function field of a smooth curve over a finite field ${\Bbb F}$ of characteristic p, let $K_0^{\rm s}$ be a separable closure of K_0, and let B be an abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-016794 | MATHDB-321274

The pure-part character formula for quiver varieties

摘录 / EXCERPT: Pure-part character formula.

conjecture | open | MathDB 页面 | 原文线索

V2-016795 | MATHDB-321275

The pure Poincaré polynomial identity for quiver and multiplicative varieties

摘录 / EXCERPT: Quiver-multiplicative Poincaré polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016796 | MATHDB-321276

Chern-Schwartz-MacPherson class conjecture for locally quasi-homogeneous free divisors

摘录 / EXCERPT: Let X be a locally quasi-homogeneous free divisor in a nonsingular variety V. Write $c_{\mathrm{SM}}(1_{V\smallsetminus X})$ for the Chern-Schwartz-MacPherson class of the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-016797 | MATHDB-321282

Coates-Sinnott conjecture in cohomological form

摘录 / EXCERPT: Coates-Sinnott conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-016798 | MATHDB-321292

Tropical Severi irreducibility conjecture for toric surfaces

摘录 / EXCERPT: Let S be a toric surface over an algebraically closed field of characteristic zero, let ${\mathcal L}\in \operatorname{Pic}(S)$ be an effective class, and let $g\geq 0$ be a non-...

conjecture | open | MathDB 页面 | 原文线索

V2-016799 | MATHDB-321304

The closed-web Euler characteristic formula

摘录 / EXCERPT: Let w be a closed web, meaning a web with no boundary, and let D be its dual diskoid. Let $Q(D)$ be the variety of diskoid configurations associated with D, let $\Psi(w)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-016800 | MATHDB-321306

The Satake-basis Euler characteristic expansion conjecture

摘录 / EXCERPT: Let w be a minuscule web with boundary $\vec{\lambda}$ and dual diskoid D. Let $F(\vec{\lambda})$ be the Satake fibre, let $Q(D)$ be the diskoid-configuration variety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016801 | MATHDB-321307

Categorical representability implies rational representability for threefolds

摘录 / EXCERPT: Let X be a smooth projective threefold. Say that X is categorically representable in dimension 1 when its derived category admits the corresponding categorical representation...

conjecture | open | MathDB 页面 | 原文线索

V2-016802 | MATHDB-321326

The p-integrality conjecture for the homological Chern character

摘录 / EXCERPT: The p-integrality conjecture. For every variety X, every positive integer n, and every prime number p, the displayed p-integrality assertion holds. The homological Chern ch...

conjecture | open | MathDB 页面 | 原文线索

V2-016803 | MATHDB-321346

The trisecant and quadrisecant conjecture for the skew quintic

摘录 / EXCERPT: Let $N'_8=\operatorname{Sing}(\mathcal{Q})$ be the singular locus of the skew quintic hypersurface $\mathcal{Q}$, and let $D':\mathcal{Q}\dashrightarrow\mathcal{C}$ be the dual map...

conjecture | open | MathDB 页面 | 原文线索

V2-016804 | MATHDB-321351

The T_n conjecture on compact Kähler manifolds with generically large fundamental group

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n with generically large fundamental group, meaning that the fundamental-group image of every subvariety through a very general...

conjecture | open | MathDB 页面 | 原文线索

V2-016805 | MATHDB-321352

The CZ_n conjecture on torus submersions

摘录 / EXCERPT: The CZ_n conjecture. Up to finite étale cover, X is bimeromorphic to a torus submersion over a variety of general type.

conjecture | open | MathDB 页面 | 原文线索

V2-016806 | MATHDB-321355

Geisser's finite generation conjecture for motivic, Weil-étale and Kato cohomology

摘录 / EXCERPT: Let X be a scheme over a finite field, and let $H_i^c(X,\mathbb Z(n))$, $H_i^F(X,\mathbb Z(n))$, and $H_i^K(X,\mathbb Z(n))$ denote the cycle, Weil-étale, and Kato homology group...

conjecture | open | MathDB 页面 | 原文线索

V2-016807 | MATHDB-321356

Geisser's finite-coefficient finiteness conjecture for the homology sequence

摘录 / EXCERPT: Finite-coefficient finiteness conjecture. Every term in this sequence is finite for all i, n, and X. This is presented as a consequence weaker than the integral finite-genera...

conjecture | open | MathDB 页面 | 原文线索

V2-016808 | MATHDB-321357

Geisser's finite generation conjecture for motivic and Weil-étale cohomology

摘录 / EXCERPT: Let X be smooth and proper over a finite field, and let $H^i_{\mathcal M}(X,\mathbb Z(n))$, $H^i_F(X,\mathbb Z(n))$, $H^{i-1}_K(X,\mathbb Z(n))$, and $H^i_W(X,\mathbb Z(n))$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-016809 | MATHDB-321359

Geisser's finiteness conjecture for the groups $C_i(X)$

摘录 / EXCERPT: Let X be smooth and projective over a finite field, and let $C_i(X)$ be the homology groups of the complex $KC_1^{F\operatorname{et}}(X)$ measuring the difference between Weil-ét...

conjecture | open | MathDB 页面 | 原文线索

V2-016810 | MATHDB-321361

Edixhoven's lower-bound conjecture for Galois orbits of CM points

摘录 / EXCERPT: Let $A_{g,1}$ be the coarse moduli space of principally polarized Abelian varieties. A g-dimensional Abelian variety B is CM if...

conjecture | open | MathDB 页面 | 原文线索

V2-016811 | MATHDB-321362

The Galois-orbit lower-bound conjecture for CM points

摘录 / EXCERPT: Let $A_{g,1}$ be the moduli space of principally polarized complex Abelian varieties of dimension g. Let x be a CM point in $A_{g,1}$ corresponding to a principally polarized A...

conjecture | open | MathDB 页面 | 原文线索

V2-016812 | MATHDB-321373

Perrin-Riou's Euler-system realization conjecture for motivic p-adic L-functions

摘录 / EXCERPT: Let M be the pure motive under consideration, let $\mathcal{T}$ be its p-adic lattice, let $\mathcal{C}$ be the cyclotomic-radial field extension defined in the source, and put...

conjecture | open | MathDB 页面 | 原文线索

V2-016813 | MATHDB-321376

The conjecture that the planar rounding scheme suffices

摘录 / EXCERPT: Let $f_\eta:\mathbb R^2\to\{-1,1\}$ be the family of measurable functions defined in the paper, and let $c(f,g)$ denote the associated Krivine rounding constant. The planar roundin...

conjecture | open | MathDB 页面 | 原文线索

V2-016814 | MATHDB-321408

Asymptotic purity characterization for smooth projective threefolds

摘录 / EXCERPT: Asymptotic purity conjecture. X is AP if and only if the big and ample cones are equal.

conjecture | open | MathDB 页面 | 原文线索

V2-016815 | MATHDB-321437

The semialgebraic universal-cover classification conjecture

摘录 / EXCERPT: Semialgebraic universal-cover classification conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-016816 | MATHDB-321449

Darmon's reciprocity conjecture for generalized Darmon points

摘录 / EXCERPT: Darmon's reciprocity conjecture. The point P_b^{β} lies in $E(K^{\mathrm{ab}})$ and

conjecture | open | MathDB 页面 | 原文线索

V2-016817 | MATHDB-321450

Yoshida's motivic conjecture for quaternionic Shimura varieties

摘录 / EXCERPT: Let E/F be a modular elliptic curve with motive $M=h^1(E)$, let F' be the reflex field, and let M' be the rank 2^r motive over F' conjecturally given by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016818 | MATHDB-321451

Yoshida's multiplicity conjecture for quaternionic Shimura cohomology

摘录 / EXCERPT: Let E/F be a modular elliptic curve with motive $M=h^1(E)$, let F' be the reflex field, and let $m_H(\pi)=\dim \pi_f^H(\varphi)$. Yoshida's multiplicity conjecture. The motive...

conjecture | open | MathDB 页面 | 原文线索

V2-016819 | MATHDB-321454

Half-integral-weight modularity conjecture for refined special cycles

摘录 / EXCERPT: Let $(V,Q)=(B^{\operatorname{Tr}=0},D_0\cdot\operatorname{nr})$, let φ be a Schwartz function, and let $\{Z(t,\varphi;H)\}$ be the refined cohomological class attached to t...

conjecture | open | MathDB 页面 | 原文线索

V2-016820 | MATHDB-321455

Tate's cohomological cycle-class conjecture

摘录 / EXCERPT: Soit ${\mathbb F}$ un corps fini de caractéristique p, soit $\overline{{\mathbb F}}$ une clôture algébrique, et posons $G={\rm Gal}(\overline{{\mathbb F}}/{\mathbb F})$. Soit X...

conjecture | open | MathDB 页面 | 原文线索

V2-016821 | MATHDB-321456

Beilinson's conjecture on numerical equivalence

摘录 / EXCERPT: Soit ${\mathbb F}$ un corps fini et soit X une variété projective et lisse sur ${\mathbb F}$. Pour $i\geq 0$, notons $A^{i}(X)$ le groupe des cycles de codimension i sur X mo...

conjecture | open | MathDB 页面 | 原文线索

V2-016822 | MATHDB-321457

Local-global conjecture for zero-cycles on varieties over finite fields

摘录 / EXCERPT: Soit ${\mathbb F}$ un corps fini, soit $C/{\mathbb F}$ une courbe projective, lisse et géométriquement connexe, et posons $K={\mathbb F}(C)$. Pour chaque point fermé v de C, so...

conjecture | open | MathDB 页面 | 原文线索

V2-016823 | MATHDB-321461

Tan-Wong-Zhang end-invariant conjecture for two-bridge link representations

摘录 / EXCERPT: Let ρ be a $\mathrm{PSL}(2,\mathbb{C})$-representation of $\pi_1(\text{\boldmath$T$})$, and let $\mathcal{E}(\rho)\subset\mathcal{PL}$ denote its set of end invariants. An acc...

conjecture | open | MathDB 页面 | 原文线索

V2-016824 | MATHDB-321463

The Minkowski-sum formula for cluster A-varieties

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster $\mathcal{A}$-variety, and let S_1,S_2 be convex subsets of $\mathcal{A}(\mathbb{A}^t)$. For each $F\in\mathbf{E}(\mathcal{A})$, write $c_{F,S}$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-016825 | MATHDB-321491

The associated-variety equality conjecture for discretely decomposable restrictions

摘录 / EXCERPT: Associated-variety equality conjecture. The inclusion in that theorem is equality:

conjecture | open | MathDB 页面 | 原文线索

V2-016826 | MATHDB-321510

Hausel-Rodriguez-Villegas conjecture for twisted Higgs moduli

摘录 / EXCERPT: Hausel-Rodriguez-Villegas conjecture. The functions $H_n^{(p)}(t)$ are polynomials in t and, for the moduli space $\IM=\IM(L,n,d)$ of twisted Higgs bundles of coprime rank and de...

conjecture | open | MathDB 页面 | 原文线索

V2-016827 | MATHDB-321511

Motivic ADHM recursion formula

摘录 / EXCERPT: Motivic ADHM recursion conjecture. For every $\mu\in\mathbb R$,

conjecture | open | MathDB 页面 | 原文线索

V2-016828 | MATHDB-321514

Takagi's log canonicity and dense sharp F-purity conjecture

摘录 / EXCERPT: Let X be a normal variety over an algebraically closed field K of characteristic zero, and let D be an effective $\mathbb{Q}$-divisor on X such that K_X+D is $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-016829 | MATHDB-321515

Mustață-Srinivas conjecture on Frobenius and top cohomology

摘录 / EXCERPT: Let X be an n-dimensional smooth projective variety over $\overline{\mathbb{Q}}$. Given a model of X over a finitely generated $\mathbb{Z}$-subalgebra B of...

conjecture | open | MathDB 页面 | 原文线索

V2-016830 | MATHDB-321516

Hansen-Johnsen-Ranestad conjecture on higher weights of Grassmann codes

摘录 / EXCERPT: Let $G(l,m)$ be the Grassmann variety of l-dimensional subspaces of an m-dimensional vector space over ${\mathbb F}_q$, embedded in the Plücker projective space...

conjecture | open | MathDB 页面 | 原文线索

V2-016831 | MATHDB-321522

Degeneration conjecture for Chow-weight spectral sequences

摘录 / EXCERPT: Degeneration conjecture. The spectral sequence $T_{w_{Chow}}({\mathcal{H}}^{et}_{{\mathbb{Q}_l}},M)$ degenerates at E_2 for any $M\in \operatorname{Obj}DM_c(S)$.

conjecture | open | MathDB 页面 | 原文线索

V2-016832 | MATHDB-321529

Bronowski's conjecture for Bronowski varieties

摘录 / EXCERPT: Let $X\subset \mathbb{P}^{2n+1}$ be an irreducible, non-degenerate variety of dimension n whose general tangential projection is birational; that is, for a general point $x\in X$...

conjecture | open | MathDB 页面 | 原文线索

V2-016833 | MATHDB-321532

Expected-rank conjecture for finite tensor decompositions

摘录 / EXCERPT: Let $k\ge 3$ and let n_i denote the dimensions of the factors, for $1\le i\le k$. Write $\mathbf{n}=(n_1,\ldots,n_k)$ and define ... A generic tensor of order k and rank r is a...

conjecture | open | MathDB 页面 | 原文线索

V2-016834 | MATHDB-321558

Dihedral-to-cyclotomic Lie coalgebra conjecture at level one

摘录 / EXCERPT: Let ${\cal D}_{\bullet,\bullet}(N)$ be the bigraded dihedral Lie coalgebra associated with N-th roots of unity, and let ${\cal C}_{\bullet,\bullet}(N)$ be the depth-graded cyclot...

conjecture | open | MathDB 页面 | 原文线索

V2-016835 | MATHDB-321585

Product conic-bundle mirror conjecture

摘录 / EXCERPT: Let $X_i \subseteq \mathbb{P}^{n_i}$ be smooth varieties which are either Calabi-Yau or of general type, defined respectively by polynomials f_i of degree d_i. Let ... act on...

conjecture | open | MathDB 页面 | 原文线索

V2-016836 | MATHDB-321594

Conjecture on identifiability of binary Segre products

摘录 / EXCERPT: Identifiability conjecture. For all such m and k, the binary Segre product $({\mathbb{P}}^1)^m$ is k-identifiable.

conjecture | open | MathDB 页面 | 原文线索

V2-016837 | MATHDB-321608

Shub-Smale's polynomial expected-length conjecture for random start systems

摘录 / EXCERPT: Let f be the target polynomial system, let g be a start system chosen from the sphere ${\mathbb S}$ with the uniform probability distribution, and let e_0 be the initial solu...

conjecture | open | MathDB 页面 | 原文线索

V2-016838 | MATHDB-321609

Shub-Smale's optimal-start-pair conjecture

摘录 / EXCERPT: Let $V=\{(h,z)\in{\mathbb S}\times{\mathbb P}({\mathbb C}^{n+1})\mid h(z)=0\}$ be the solution variety, and let (g,e_0) be the initial pair with ... For an initial pair...

conjecture | open | MathDB 页面 | 原文线索

V2-016839 | MATHDB-321616

Motivic rigidity conjecture for Severi-Brauer varieties under division-preserving extensions

摘录 / EXCERPT: Let D be a division algebra over F and let E/F be a field extension such that D_E remains a division algebra. Consider the Severi-Brauer varieties of D and their motivic...

conjecture | open | MathDB 页面 | 原文线索

V2-016840 | MATHDB-321626

Deligne's generalized Thom-Sebastiani conjecture

摘录 / EXCERPT: Let Λ be a noetherian torsion ring such that $m\Lambda=0$ for some integer m relatively prime to p. Let S_1, S_2, and S be henselian traits over...

conjecture | open | MathDB 页面 | 原文线索

V2-016841 | MATHDB-321638

The equations, resolution, and regularity conjecture for Kalman varieties

摘录 / EXCERPT: Kalman-variety resolution conjecture. There is a long exact sequence

conjecture | open | MathDB 页面 | 原文线索

V2-016842 | MATHDB-321685

Larsen's infinite-rank conjecture for abelian varieties over algebraic extensions

摘录 / EXCERPT: Larsen's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-016843 | MATHDB-321689

Algebraic dynamical Manin-Mumford conjecture

摘录 / EXCERPT: Let X be a projective variety over a number field k, let $\psi:X\to X$ be an endomorphism with a polarization, and let Y be a subvariety of X. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016844 | MATHDB-321690

Ghioca-Tucker-Zhang conjecture on tangent-space preperiodicity

摘录 / EXCERPT: Let X be a smooth projective variety over a number field k, let $\psi:X\to X$ be an endomorphism with a polarization, and let Y be a subvariety of X. For a point $x\in Y$,...

conjecture | open | MathDB 页面 | 原文线索

V2-016845 | MATHDB-321761

The product decomposition conjecture for twisted multivector fields

摘录 / EXCERPT: Let X be a projective manifold, let L be a numerically trivial line bundle, and let ... be a non-zero section with $q<\dim X$. A finite étale cover is a finite surjective morphis...

conjecture | open | MathDB 页面 | 原文线索

V2-016846 | MATHDB-321765

Tymoczko's Young-subgroup stabilizer conjecture

摘录 / EXCERPT: Let P be a natural unit interval order on $[n]$. Tymoczko's representation of $\mathfrak S_n$ on the cohomology $H^*({\mathcal H}(P))$ is the representation under consideration....

conjecture | open | MathDB 页面 | 原文线索

V2-016847 | MATHDB-321766

Shareshian-Wachs's h-unimodality conjecture for Hessenberg representations

摘录 / EXCERPT: Let P be a natural unit interval order on $[n]$. For each j, let $\operatorname{ch}H^{2j}({\mathcal H}(P))$ be the Frobenius characteristic of Tymoczko's representation on degr...

conjecture | open | MathDB 页面 | 原文线索

V2-016848 | MATHDB-321862

The knot-state microsupport conjecture

摘录 / EXCERPT: Knot-state microsupport conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016849 | MATHDB-321864

Garoufalidis's geometric AJ conjecture

摘录 / EXCERPT: Geometric AJ conjecture. The deformation variety D_K and the characteristic variety $\operatorname{ch}({\mathcal{A}}^K)$ are M-essentially equal.

conjecture | open | MathDB 页面 | 原文线索

V2-016850 | MATHDB-321870

The Deligne-Lusztig cohomology degree conjecture

摘录 / EXCERPT: Let $G=G(q)$ be a group of Lie type, let $\bar{\Q}_\ell$ be an algebraic closure of the ℓ-adic numbers, and let ℓ be a prime dividing $\Phi_d(q)$. For a unipotent charact...

conjecture | open | MathDB 页面 | 原文线索

V2-016851 | MATHDB-321871

The Deligne-Lusztig cohomology concentration conjecture

摘录 / EXCERPT: Let $G=\b G^F$ be a finite group of Lie type, let B be a unipotent ℓ-block with abelian defect group, let b be its Brauer correspondent, and let Y be the associated Deli...

conjecture | open | MathDB 页面 | 原文线索

V2-016852 | MATHDB-321880

The perverse-Lefschetz conjecture for compactified Jacobians

摘录 / EXCERPT: Let $\mathcal{C}\to\mathcal{B}$ be a family satisfying assumptions (A1)-(A4), and let $\overline{\mathcal{J}}\to\mathcal{B}$ be its family of compactified Jacobians. For every geom...

conjecture | open | MathDB 页面 | 原文线索

V2-016853 | MATHDB-321889

Bloch's generalized conjecture on Chow groups and Hodge coniveau

摘录 / EXCERPT: Bloch's generalized conjecture. For every integer $i\leq c-1$, the cycle map cl is injective. This is presented as a converse to the theorem that injectivity of low-dimensional c...

conjecture | open | MathDB 页面 | 原文线索

V2-016854 | MATHDB-321891

The standard conjecture on supporting algebraic cycles on closed subsets

摘录 / EXCERPT: Supporting-cycle conjecture. There exists a codimension-k cycle Z' on X with rational coefficients, supported on Y, such that $[Z']=[Z]$ in $H^{2k}(X,\mathbb{Q})$. The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-016855 | MATHDB-321892

The alternating projector conjecture for algebraic K3 surfaces

摘录 / EXCERPT: Let S be an algebraic $K3$ surface, let $k\geq2$, and let $\pi_{alt}=\sum_{\sigma\in\mathfrak{S}_k}(-1)^{\epsilon(\sigma)}\sigma$ be the alternating projector on S^k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016856 | MATHDB-321896

The Frobenius bijectivity conjecture for varieties over number fields

摘录 / EXCERPT: Let X be a smooth, geometrically connected, n-dimensional projective variety over ${\mathbf Q}$. For all sufficiently large primes p, let X_p be a smooth reduction of X o...

conjecture | open | MathDB 页面 | 原文线索

V2-016857 | MATHDB-321925

Smith's conjecture on perverse sheaves and the Smith operator

摘录 / EXCERPT: Let X be a complex algebraic variety equipped with a $\boldsymbol{\varpi}$-action, and let P be a $\boldsymbol{\varpi}$-equivariant perverse sheaf of K-vector spaces on X....

conjecture | open | MathDB 页面 | 原文线索

V2-016858 | MATHDB-321933

Viehweg's conjecture on hyperbolicity of bases of families

摘录 / EXCERPT: Let $f^\circ: X^\circ \to Y^\circ$ be a smooth projective family of varieties with semiample canonical bundle over a quasi-projective manifold Y°. The variation of the famil...

conjecture | open | MathDB 页面 | 原文线索

V2-016859 | MATHDB-321934

The MMP-moduli map factorisation conjecture

摘录 / EXCERPT: Let $f^\circ:X^\circ\to Y^\circ$ be a smooth projective family of canonically polarised varieties over a quasi-projective manifold Y°. Let Y be a smooth compactification o...

conjecture | open | MathDB 页面 | 原文线索

V2-016860 | MATHDB-321962

Positivity conjecture for Donaldson-Thomas invariants of symmetric quivers with two disjoint cuts

摘录 / EXCERPT: Let (Q,W) be a symmetric quiver with potential, and suppose that (Q,W) admits two cuts I and I' such that $I\cap I'=\emptyset$. Let $\Om_\al$ denote the Donaldson-Thomas in...

conjecture | open | MathDB 页面 | 原文线索

V2-016861 | MATHDB-321996

The conjecture on intersections of subvarieties with tori

摘录 / EXCERPT: The conjecture on intersections of subvarieties with tori. For every $n\in\mathbb{N}$ and every irreducible subvariety $W\subseteq\mathbb{G}_{\mathrm{m}}^n$, the set...

conjecture | open | MathDB 页面 | 原文线索

V2-016862 | MATHDB-322020

0-coconnectivity conjecture for the loop scheme of a smooth affine scheme

摘录 / EXCERPT: Let Z be an affine smooth scheme of finite type over the ground field, viewed as a 0-coconnective DG scheme, and let $Z\qqart$ denote its loop DG scheme. 0-coconnectivity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-016863 | MATHDB-322035

Rank-one weight semigroup conjecture for affine double coset varieties

摘录 / EXCERPT: Let G be a simple algebraic group and $\operatorname{H} \subset \operatorname{G}$ a connected spherical reductive subgroup. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016864 | MATHDB-322038

Kuznetsov's conjecture on minimal Lefschetz decompositions and categorical resolutions

摘录 / EXCERPT: Let X be a smooth projective variety. A minimal Lefschetz decomposition of $\D^b(X)$ is a Lefschetz decomposition whose first block is minimal under inclusion, and a categorical...

conjecture | open | MathDB 页面 | 原文线索

V2-016865 | MATHDB-322057

Finite lead-coefficient conjecture at Seifert-surface ideal points

摘录 / EXCERPT: Finite lead-coefficient conjecture. The lead coefficient of ${\mathcal{T}_K^{\vphantom{x}}\hspace{-0.5em}\raisebox{0.8ex}{$\scriptstyle X0$}}$ has a finite value at x.

conjecture | open | MathDB 页面 | 原文线索

V2-016866 | MATHDB-322095

Seminormality conjecture for the small nilpotent cone in type E

摘录 / EXCERPT: Let G be a group of type E, and let $\mathcal{N}_{\mathrm{sm}}$ denote the corresponding small nilpotent cone. A variety is seminormal if every bijective map to it is an isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-016867 | MATHDB-322124

Ergodicity and conullity conjectures for redundant representations

摘录 / EXCERPT: Redundant-representation conjecture. (a) The action of $\operatorname{Out}(F_n)$ on R is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-016868 | MATHDB-322175

Simplicity conjecture for the theta-divisor variations

摘录 / EXCERPT: Let (X,Θ) be a principally polarized abelian variety over $\mathbb C$ of dimension $g\geq 2$ with smooth symmetric theta divisor Θ. For $x\in X(\mathbb C)$, set ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-016869 | MATHDB-322176

Simplicity conjecture for the perverse sheaves of a smooth theta divisor

摘录 / EXCERPT: Let (X,Θ) be a principally polarized abelian variety over $\mathbb C$ of dimension $g\geq 2$ with smooth symmetric theta divisor Θ. For the variations...

conjecture | open | MathDB 页面 | 原文线索

V2-016870 | MATHDB-322177

Theta-divisor Tannaka group conjecture

摘录 / EXCERPT: Let (X,Θ) be a principally polarized abelian variety over $\mathbb C$ of dimension $g\geq 2$ with smooth symmetric theta divisor Θ. Let δ_Θ be the pervers...

conjecture | open | MathDB 页面 | 原文线索

V2-016871 | MATHDB-322180

Generic equidistribution conjecture for inverse images of algebraic varieties

摘录 / EXCERPT: Let $f:\mathbb{P}^k\to\mathbb{P}^k$ be a non-invertible holomorphic endomorphism of algebraic degree d, let $1\leq p\leq k$, and let T^p be the Green (p,p)-current of f. If...

conjecture | open | MathDB 页面 | 原文线索

V2-016872 | MATHDB-322186

Harvey-Knapp-Lawson conjecture on pseudoeffective and strongly positive cycles on abelian varieties

摘录 / EXCERPT: Harvey-Knapp-Lawson conjecture. For every $1\leq k\leq m$,

conjecture | open | MathDB 页面 | 原文线索

V2-016873 | MATHDB-322192

The ungraded derived equivalence for perverse coherent sheaves on the nilpotent cone

摘录 / EXCERPT: Let G) be a simply connected semisimple algebraic group over an algebraically closed field of good characteristic, and let $\mathcal{N}$ denote the nilpotent variety in the Lie a...

conjecture | open | MathDB 页面 | 原文线索

V2-016874 | MATHDB-322197

Kawamata's finiteness conjecture for Fourier-Mukai partners

摘录 / EXCERPT: Let a Fourier-Mukai partner of a smooth projective variety X be a smooth projective variety Y such that $\mathrm{D}^{b}(X)\cong\mathrm{D}^{b}(Y)$. Kawamata's conjecture. The nu...

conjecture | open | MathDB 页面 | 原文线索

V2-016875 | MATHDB-322198

The full endofunctor conjecture for derived categories

摘录 / EXCERPT: Let X be a smooth projective variety and let $\mathsf{F}\colon\mathrm{D}^{b}(X)\to\mathrm{D}^{b}(X)$ be an exact endofunctor. Full endofunctor conjecture. If F is full...

open_problem | open | MathDB 页面 | 原文线索

V2-016876 | MATHDB-322199

Kuznetsov's Noetherianity conjecture for admissible subcategories

摘录 / EXCERPT: Let X be a smooth projective variety, and let $\mathbf{A}_1\subseteq\mathbf{A}_2\subseteq\ldots\subseteq\mathbf{A}_i\subseteq\ldots\subseteq\mathrm{D}^{b}(X)$ be a sequence of ad...

conjecture | open | MathDB 页面 | 原文线索

V2-016877 | MATHDB-322200

Kawamata's splitting-functor Fourier-Mukai conjecture

摘录 / EXCERPT: Let X_1 and X_2 be smooth projective varieties. An exact functor $\mathsf{F}\colon\mathrm{D}^{b}(X_1)\to\mathrm{D}^{b}(X_2)$ is called splitting when it is both right and left...

conjecture | open | MathDB 页面 | 原文线索

V2-016878 | MATHDB-322207

The Monotone Secant Conjecture for flag manifolds

摘录 / EXCERPT: Monotone Secant Conjecture. For any Grassmannian Schubert problem $(\sigma_1,\ldots,\sigma_m)$ on $\mathbb{F}\ell(\alpha;n)$ and any disjoint secant flags...

conjecture | open | MathDB 页面 | 原文线索

V2-016879 | MATHDB-322208

The Monotone Osculating Conjecture for flag manifolds

摘录 / EXCERPT: Monotone Osculating Conjecture. For any such Schubert problem and any flags $F_{\bullet}^1,\ldots,F_{\bullet}^m$ osculating γ at points that are monotone with respect to the...

conjecture | open | MathDB 页面 | 原文线索

V2-016880 | MATHDB-322230

The linear-strand Betti-number conjecture for pure ideals

摘录 / EXCERPT: Let E be an exterior algebra and let $J\subset E$ be a pure ideal, meaning an ideal generated by products of linear forms. Suppose that E/J satisfies property (): any two disti...

conjecture | refuted | MathDB 页面 | 原文线索

V2-016881 | MATHDB-322243

Kontsevich/Batyrev-Borisov homological mirror symmetry for toric hypersurfaces

摘录 / EXCERPT: Let $\triangle$ and $\triangle^\vee$ be a dual pair of reflexive polytopes, and let Z and $Z^\vee$ be the corresponding smooth projective Calabi-Yau varieties. Kontsevich/Batyrev...

conjecture | open | MathDB 页面 | 原文线索

V2-016882 | MATHDB-322246

Lehmann's diminished-ideal containment conjecture

摘录 / EXCERPT: Let X be a smooth variety, let L be a pseudo-effective $\mathbb{R}$-divisor on X, and let T_{min} be a current of minimal singularities in the numerical class of L. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016883 | MATHDB-322254

Hochschild homology conjecture for quantizations of symplectic resolutions

摘录 / EXCERPT: Let X be an irreducible affine Poisson variety admitting a symplectic resolution $\widetilde{X}\to X$. Let $A_\hbar$ be a deformation quantization of $\mathcal{O}_X$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-016884 | MATHDB-322274

Non-abelian strong Coates-Sinnott conjecture for higher étale cohomology

摘录 / EXCERPT: Let K/k be a Galois extension of number fields with Galois group G, let p be an odd prime, let $n>1$, and let S and T be finite non-empty disjoint sets of places of k,...

conjecture | open | MathDB 页面 | 原文线索

V2-016885 | MATHDB-322283

The Sylow-versality conjecture for finite group actions

摘录 / EXCERPT: Sylow-versality conjecture. If X is G_p-versal for every prime p, then X is G-versal.

conjecture | open | MathDB 页面 | 原文线索

V2-016886 | MATHDB-322329

Rigidity conjecture for upper motives of generalized Severi-Brauer varieties

摘录 / EXCERPT: Let D be a central division F-algebra, and let $M_{m,D}$ denote its upper motive associated with the generalized Severi-Brauer variety $X(p^m,D)$. Let K/F be a field extensio...

conjecture | open | MathDB 页面 | 原文线索

V2-016887 | MATHDB-322355

Teh's filtration conjecture for real varieties

摘录 / EXCERPT: Teh's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016888 | MATHDB-322362

The cactus-rank conjecture for generic cubic forms

摘录 / EXCERPT: Let $F\in k[x_0,\ldots,x_n]$ be a generic homogeneous cubic form, and let $cr(F)$ denote its cactus rank and $r(F)$ its rank. Cactus-rank conjecture. The cactus rank equals the ran...

conjecture | open | MathDB 页面 | 原文线索

V2-016889 | MATHDB-322367

The three-dimensional G-Hilbert scheme (0,2)-McKay counting conjecture

摘录 / EXCERPT: Let G be a finite group acting on $\mathbb{C}^3$, and let X be its three-dimensional G-Hilbert scheme, a crepant resolution when the construction applies. The $(0,2)$-deform...

conjecture | open | MathDB 页面 | 原文线索

V2-016890 | MATHDB-322368

ACC for lengths of extremal rays of log canonical Fano varieties with Picard number one

摘录 / EXCERPT: ACC for lengths of extremal rays. For every n, the set $\mathcal L_n$ satisfies the ascending chain condition: if X_k is an n-dimensional $\mathbb Q$-factorial log canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-016891 | MATHDB-322369

Geometric Chevalley-Warning conjecture for projective zero loci

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, and let $f_1,\ldots,f_l$ be homogeneous polynomials in $k[x_0,\ldots,x_n]$ satisfying ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016892 | MATHDB-322370

Geometric Chevalley-Warning conjecture in the hypersurface form

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, and let f be a homogeneous polynomial defining a hypersurface of degree d in $\mathbb{P}^n$. A variety is...

conjecture | open | MathDB 页面 | 原文线索

V2-016893 | MATHDB-322372

Logarithmic Kollár vanishing conjecture for semistable reductions in positive characteristic

摘录 / EXCERPT: Let X and Y be proper and smooth S-schemes, let $f:X\rightarrow Y$ be an E-semistable S-morphism, and let D be a simple normal crossing divisor on X containing...

conjecture | open | MathDB 页面 | 原文线索

V2-016894 | MATHDB-322381

The K-theoretic J-function formula for Lagrangian complete flag varieties

摘录 / EXCERPT: The K-theoretic J-function formula. The coefficient $J^{L \operatorname{Fl},K}_d$ is

conjecture | open | MathDB 页面 | 原文线索

V2-016895 | MATHDB-322382

The K-theoretic J-function formula for complete flag varieties of types B and D

摘录 / EXCERPT: The K-theoretic J-function formula. One has

conjecture | open | MathDB 页面 | 原文线索

V2-016896 | MATHDB-322386

Suslin's conjecture for Lawson homology with coefficients

摘录 / EXCERPT: Let A be an abelian group and let X be a smooth quasi-projective variety of dimension n. Let ... be the natural transform from Lawson homology to singular homology. Suslin's co...

conjecture | open | MathDB 页面 | 原文线索

V2-016897 | MATHDB-322396

Irreducibility conjecture for universal Severi varieties of curves on K3 surfaces

摘录 / EXCERPT: Let $\mathcal{B}_g$ be the moduli stack of primitively polarized K3 surfaces of genus $g>2$, parametrizing pairs (S,M) where M is ample and primitive with $(M\cdot M)=2g-2$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-016898 | MATHDB-322432

Generic uniqueness of best rank-one approximations for real tensors

摘录 / EXCERPT: Generic uniqueness conjecture. Every $\mathcal{T}\in\mathbb{R}^{n_1\times\cdots\times n_d}$ has a unique rank-one approximation, unless $\mathcal{T}$ lies on a subvariety; moreover...

conjecture | open | MathDB 页面 | 原文线索

V2-016899 | MATHDB-322465

Hassett-Tschinkel's effective-cone conjecture for irreducible holomorphic symplectic varieties

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic variety with polarization H, and let $N_1(X,\mathbb{Z})$ be its group of curve classes modulo homological equivalence. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-016900 | MATHDB-322466

Hassett-Tschinkel's line-square conjecture for Lagrangian planes

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold of $K3^{[n]}$-type, meaning that it is deformation equivalent to a Hilbert scheme of n points on a $K3$ surface. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016901 | MATHDB-322482

Murty-Patankar's conjecture on simple reductions of abelian varieties

摘录 / EXCERPT: Let A be an absolutely simple abelian variety over a number field K. Let $\mathcal V$ be the set of finite places v of K for which A has good reduction and...

conjecture | open | MathDB 页面 | 原文线索

V2-016902 | MATHDB-322483

Independence-of-ell conjecture for Frobenius conjugacy classes

摘录 / EXCERPT: Let A be an abelian variety over a number field K with $K_A^{\operatorname{conn}}=K$. Let $\mathbf G_A$ be its Mumford-Tate group, let $\rho_{A,\ell}$ be the associated ℓ-a...

conjecture | open | MathDB 页面 | 原文线索

V2-016903 | MATHDB-322498

Geometric categorical action conjecture for simply laced Kac-Moody algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Kac-Moody Lie algebra whose Dynkin diagram is simply laced and contains no loops. A geometric categorical $\mathfrak{g}$ action is the categorical action on...

conjecture | open | MathDB 页面 | 原文线索

V2-016904 | MATHDB-322536

The generalized flop conjecture for derived categories

摘录 / EXCERPT: Let X and X' be birational isomorphic varieties. Suppose that, for a resolution ... of the birational isomorphism $X\stackrel{fl}{\dashrightarrow}X'$, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-016905 | MATHDB-322537

Finiteness conjecture for quasi-projective Fourier-Mukai partners

摘录 / EXCERPT: Finiteness conjecture for Fourier-Mukai partners. If X is a quasi-projective variety, then there are only finitely many quasi-projective Fourier-Mukai partners X'.

conjecture | open | MathDB 页面 | 原文线索

V2-016906 | MATHDB-322556

Affine Grothendieck polynomial conjecture for positroid varieties

摘录 / EXCERPT: Affine Grothendieck polynomial conjecture. The K-theory class of the structure sheaf of Π_f is the image of $\tilde{G}_f$ under this identification:

conjecture | open | MathDB 页面 | 原文线索

V2-016907 | MATHDB-322557

Compatibly split subvariety characterization for the Grassmannian

摘录 / EXCERPT: Compatibly split positroid conjecture. The only compatibly split irreducible subvarieties of $\mathrm{Gr}(k,n)$ are the closed positroid varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-016908 | MATHDB-322559

The Quadratic Zariski Problem for anisotropic quadrics

摘录 / EXCERPT: Let X and Y be anisotropic quadrics of the same dimension over a field k. A rational morphism means a rational map between these varieties, defined on a dense open subset. Qu...

open_problem | open | MathDB 页面 | 原文线索

V2-016909 | MATHDB-322578

D0-brane resolution conjecture for varieties

摘录 / EXCERPT: D0-brane resolution conjecture. For any $r\ge r_0$ in ${\Bbb N}$, the resolution can be factored through an embedding

conjecture | open | MathDB 页面 | 原文线索

V2-016910 | MATHDB-322579

Abundance conjecture for punctual D0-branes

摘录 / EXCERPT: Abundance conjecture. For any $r\ge r_0$ in ${\Bbb N}$, the morphism can be factored through an embedding

conjecture | open | MathDB 页面 | 原文线索

V2-016911 | MATHDB-322580

Bryant-Griffiths-Yang conjecture on the characteristic variety of the isometric embedding system

摘录 / EXCERPT: Bryant-Griffiths-Yang conjecture. The characteristic variety $\Sigma({\bf c})$ is not smooth in $\mathbb R^n\setminus\{0\}$ for every $n\ge 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-016912 | MATHDB-322592

Sikora's conjecture relating the recurrence ideal and the A-ideal

摘录 / EXCERPT: Sikora's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-016913 | MATHDB-322593

Reduced universal character-ring conjecture for knot complements

摘录 / EXCERPT: For a knot K, let X be its complement and let $\mathbb C^{\mathrm{univ}}[\chi(X)]$ denote the universal SL_2-character ring of $\pi_1(X)$, equivalently the skein module at...

conjecture | open | MathDB 页面 | 原文线索

V2-016914 | MATHDB-322597

André-Oort-Pink conjecture for mixed Shimura varieties

摘录 / EXCERPT: André-Oort-Pink conjecture. The set $\Sigma_{\mathrm{max}}(Z)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-016915 | MATHDB-322605

Arithmetic degree conjecture for dominant rational self-maps of varieties

摘录 / EXCERPT: Let X be a nonsingular irreducible algebraic variety defined over $\overline{\mathbb{Q}}$, and let $\varphi:X\dashrightarrow X$ be a dominant rational map defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-016916 | MATHDB-322630

Landsberg-Weyman conjecture on generators of tangential ideals of Segre varieties

摘录 / EXCERPT: Let X be a Segre variety, and let $I(\tau(X))$ denote the homogeneous ideal of its tangential variety. The relevant Schur-module factors are expressed using exterior powers...

conjecture | open | MathDB 页面 | 原文线索

V2-016917 | MATHDB-322635

Oblomkov-Shende conjecture for HOMFLY polynomials of algebraic links

摘录 / EXCERPT: Let C be a curve in $\mathbb{C}^2$ given by $f(x,y)=0$ and passing through the origin. Intersecting C with the boundary of a sufficiently small ball around the origin gives an...

conjecture | open | MathDB 页面 | 原文线索

V2-016918 | MATHDB-322641

Hacon-McKernan's uniformity conjecture for the Iitaka fibration

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n and Kodaira dimension κ, and let $\phi_{|mK_X|}$ denote the rational map defined by the pluricanonical linear system...

conjecture | open | MathDB 页面 | 原文线索

V2-016919 | MATHDB-322686

Erdenberger-Grushevsky-Hulek intersection-number vanishing conjecture

摘录 / EXCERPT: Let g be a positive integer, let ${\cA}_g^{\rm Perf}$ be the perfect-cone toroidal compactification of the moduli space of principally polarized abelian varieties, let D be its...

conjecture | open | MathDB 页面 | 原文线索

V2-016920 | MATHDB-322687

Erdenberger-van der Geer boundary-class conjecture for the Satake compactification

摘录 / EXCERPT: Let $\SatA{g}$ be the Satake compactification, let λ_i be the Chern classes of the Hodge bundle, and for a subset $\alpha\subseteq\{1,2,\ldots,g\}$ define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-016921 | MATHDB-322693

Conjecture B_d on Frobenius bijectivity for varieties with trivial canonical class

摘录 / EXCERPT: Let Z be a d-dimensional projective variety over an algebraically closed field of characteristic zero with only rational singularities and with K_Z linearly trivial. For a mo...

conjecture | open | MathDB 页面 | 原文线索

V2-016922 | MATHDB-322694

Conjecture B_n for models of varieties with rational singularities and trivial canonical class

摘录 / EXCERPT: Let V be an n-dimensional projective variety over an algebraically closed field k of characteristic zero with only rational singularities and with K_V linearly trivial. Giv...

conjecture | open | MathDB 页面 | 原文线索

V2-016923 | MATHDB-322714

The atypicality conjecture for support varieties of basic classical Lie superalgebras

摘录 / EXCERPT: The atypicality conjecture. For every basic classical Lie superalgebra ${\mathfrak{g}}$ and every simple supermodule $L(\lambda)$ in $\mathcal{F}$,

conjecture | open | MathDB 页面 | 原文线索

V2-016924 | MATHDB-322724

Perrin-Riou's Euler characteristic conjecture for dihedral Selmer groups

摘录 / EXCERPT: Let A be a principally polarized abelian variety over F with good ordinary reduction at every prime above p in K, and assume that A satisfies the weak-Heegner hypothesis...

conjecture | open | MathDB 页面 | 原文线索

V2-016925 | MATHDB-322725

Perrin-Riou's nondegeneracy conjecture for the dihedral p-adic height pairing

摘录 / EXCERPT: Let A be an abelian variety over F in the dihedral setting above K, let $G=\operatorname{Gal}(K_{p^\infty}/K)$, and write $\Lambda=\Lambda(G)={\bf Z}_p[[G]]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016926 | MATHDB-322728

The supernoetherianity conjecture for twisted homogeneous coordinate rings of abelian varieties

摘录 / EXCERPT: Let X be an abelian variety and let σ be an automorphism of X that leaves invariant no proper subscheme. Let $\mathcal{L}$ be an ample invertible sheaf on X, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-016927 | MATHDB-322736

Douglas's D-geometry conjecture for D-brane moduli

摘录 / EXCERPT: Let Y be a scheme over $\mathbb{C}$, let $r\in\mathbb{N}$, and let $\mathfrak M^{{0^{A\!z}}^f}_r(Y)$ be the moduli stack of rank-r D0-branes on Y. Write ... for this Azumaya st...

conjecture | open | MathDB 页面 | 原文线索

V2-016928 | MATHDB-322737

The abundance conjecture for punctual D0-brane moduli

摘录 / EXCERPT: Let Y be a reduced quasi-projective variety over $\mathbb{C}$. Let $\mathfrak M^{0^{A\!z^f}_{p}}_r(Y)$ be the moduli stack of punctual D0-branes of rank r on Y. The abundance...

conjecture | open | MathDB 页面 | 原文线索

V2-016929 | MATHDB-322738

The D0-brane resolution conjecture for singularities

摘录 / EXCERPT: Let Y be a reduced quasi-projective variety over $\mathbb{C}$, and let $\rho:Y'\to Y$ be a resolution of the singularities of Y. Let $\mathfrak M^{0^{A\!z^f}_{p}}_r(Y)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-016930 | MATHDB-322765

Piecewise-isomorphism conjecture for birational complements

摘录 / EXCERPT: Piecewise-isomorphism conjecture. The k-schemes

conjecture | open | MathDB 页面 | 原文线索

V2-016931 | MATHDB-322769

Tan-Wong-Zhang's Cantor-set conjecture for end invariants

摘录 / EXCERPT: Let ρ be an $SL(2,\mathbb{C})$-representation of $\pi_1(\text{\boldmath$T$})$, and let $\mathcal{E}(\rho)$ be its set of end invariants in the projective lamination space...

conjecture | open | MathDB 页面 | 原文线索

V2-016932 | MATHDB-322783

The geometric categorical action conjecture for quantum toroidal algebras

摘录 / EXCERPT: Let $\widehat{\mathfrak{g}}$ be the affine Kac-Moody algebra and let the geometric categorical $\widehat{\mathfrak{g}}$ action on...

conjecture | open | MathDB 页面 | 原文线索

V2-016933 | MATHDB-322784

Equivalence conjecture for the two categorifications of the basic toroidal representation

摘录 / EXCERPT: Let $\widehat{\widehat{\mathfrak{g}}}$ denote the double-affine extension associated to the affine Kac-Moody algebra $\widehat{\mathfrak{g}}$. Consider the 2-representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-016934 | MATHDB-322786

Prediction for the moving slope of the moduli space of principally polarized abelian fivefolds

摘录 / EXCERPT: Let $\overline{\mathcal{A}}_5$ be a compactification of the moduli space of principally polarized abelian fivefolds, and let $s'(\overline{\mathcal{A}}_5)$ denote its moving slope....

conjecture | open | MathDB 页面 | 原文线索

V2-016935 | MATHDB-322792

Convergence conjecture for the I(k)-completed slice tower

摘录 / EXCERPT: Let k be a field of finite virtual 2-cohomological dimension and finite p-cohomological dimension for every prime $p>2$. The $I(k)$-completed slice tower is the slice tower a...

conjecture | open | MathDB 页面 | 原文线索

V2-016936 | MATHDB-322798

Definable triangulation conjecture for functions on compact definable sets

摘录 / EXCERPT: Let $f:K\to\mathbb{R}$ be a definable function on a compact definable set $K\subset\mathbb{R}^m$. A definable triangulation of K is a definable triangulation whose open simplices...

conjecture | open | MathDB 页面 | 原文线索

V2-016937 | MATHDB-322818

The log canonical-dense F-pure type conjecture for pairs

摘录 / EXCERPT: Let X be a normal variety over an algebraically closed field of characteristic zero, and let Δ be an effective $\mathbb{Q}$-divisor on X such that $K_X+\Delta$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-016938 | MATHDB-322822

Perfectness conjecture for preorderings of matrix polynomials

摘录 / EXCERPT: Perfectness conjecture. The induced quadratic module satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-016939 | MATHDB-322823

Invariant sums-of-squares induction conjecture

摘录 / EXCERPT: Invariant sums-of-squares induction conjecture. The induced quadratic module satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-016940 | MATHDB-322854

Voisin's conjecture on general type varieties and approximate elliptic connectedness

摘录 / EXCERPT: Let X be a projective variety embedded in projective space. It is approximately elliptically connected in the naïve sense if it can be connected by elliptic curves contained in a...

conjecture | open | MathDB 页面 | 原文线索

V2-016941 | MATHDB-322865

Beltrametti-Sommese equidimensionality conjecture for adjunction-theoretic scrolls

摘录 / EXCERPT: Let (X,L) be a polarised projective manifold of dimension n, and let $\varphi\colon X\rightarrow Y$ be an adjunction-theoretic scroll over a normal variety Y of dimension m...

conjecture | open | MathDB 页面 | 原文线索

V2-016942 | MATHDB-322871

Real-point conjecture for separating extreme rays of nonnegative ternary and quaternary quartics

摘录 / EXCERPT: Real-point conjecture. For $p \in P_{3,6} \setminus \Sigma_{3,6}$ there exist $v_1,\ldots,v_9 \in \mathbb{R}^3$ yielding a separating extreme ray for p. Analogously, for...

conjecture | open | MathDB 页面 | 原文线索

V2-016943 | MATHDB-322875

Tate-Beilinson-Parshin cycle map conjecture

摘录 / EXCERPT: Let k be a finite field, let $\overline{k}$ be a separable closure, and let G_k be its Galois group. Let X be a projective smooth variety over k, let ℓ be a prime diff...

conjecture | open | MathDB 页面 | 原文线索

V2-016944 | MATHDB-322902

Regularity and arithmetically Cohen-Macaulay conjecture for secant varieties of graph curves

摘录 / EXCERPT: Let C_G be a graph curve embedded under the hypotheses of Theorem 1.3, and let kth secant variety denote the union of the linear spans of length-$k+1$ subschemes of C_G. Seca...

conjecture | open | MathDB 页面 | 原文线索

V2-016945 | MATHDB-322904

Secant-plane nonexistence conjecture for graph curves

摘录 / EXCERPT: Let $C_G\subset\mathbb{P}^{g+r}$ be a graph curve of arithmetic genus p_a and degree $d=2g+r$. A $(k+2)$-secant $\mathbb{P}^k$ is a projective k-plane meeting C_G in a subsch...

conjecture | open | MathDB 页面 | 原文线索

V2-016946 | MATHDB-322925

Torsion conjecture for smooth rationally smooth Schubert slices

摘录 / EXCERPT: Torsion conjecture. If U is smooth and N is rationally smooth, then the order of the torsion subgroup of

conjecture | open | MathDB 页面 | 原文线索

V2-016947 | MATHDB-322949

The nilpotent-singular-support form of the geometric Langlands conjecture

摘录 / EXCERPT: Let X be a smooth complete curve, let G be a reductive group, and write $\bun_G$ for the moduli stack of G-bundles on X. Let $\LocSysG$ be the stack of G-local systems on...

conjecture | open | MathDB 页面 | 原文线索

V2-016948 | MATHDB-322953

Weyman's dense orbit conjecture for triangular algebras

摘录 / EXCERPT: Dense orbit conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-016949 | MATHDB-322954

Weyman's multiplicity-free conjecture: Schur-representation-finiteness and MF

摘录 / EXCERPT: Multiplicity-free conjecture. An algebra is Schur-representation-finite if and only if it is MF.

conjecture | open | MathDB 页面 | 原文线索

V2-016950 | MATHDB-322994

Equivariant-morphism conjecture for the Donaldson moduli space

摘录 / EXCERPT: Let $\mathcal{M}_d$ be the Donaldson moduli space of isomorphism classes of rank-two vector bundles $\mathcal{V}$ on $\mathbb{P}^2$ with trivial determinant and second Chern class...

conjecture | open | MathDB 页面 | 原文线索

V2-016951 | MATHDB-322995

Generalized Kollár-Peskine conjecture for infinite Grassmannians

摘录 / EXCERPT: Generalized Kollár-Peskine conjecture. There does not exist any nonconstant morphism

conjecture | open | MathDB 页面 | 原文线索

V2-016952 | MATHDB-323011

Algebraic inequality for invariant formal polynomials on quotient surface singularities

摘录 / EXCERPT: Algebraic inequality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-016953 | MATHDB-323023

LMP conjecture on Frobenius splitting of the diagonal blow-up

摘录 / EXCERPT: Let k be an algebraically closed field of positive characteristic p. Let V be a rational projective homogeneous space, set $X=V\times V$, and let $\widetilde{X}$ be the blow-...

conjecture | open | MathDB 页面 | 原文线索

V2-016954 | MATHDB-323035

Extension of the family-of-representations theorem over locally closed special-fiber subschemes

摘录 / EXCERPT: Extension conjecture. The claim of the theorem and the corollary also holds true if one replaces x_0 by a (locally) closed subscheme of the special fiber over which there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-016955 | MATHDB-323056

Eigenvariety and trianguline deformation space component conjecture

摘录 / EXCERPT: Fix a prime $q\neq p$ splitting in E, choose q' as in the source, and let G be a unitary group in d variables attached to $E/\mathbb{Q}$ satisfying the stated conditions at...

conjecture | open | MathDB 页面 | 原文线索

V2-016956 | MATHDB-323067

Finiteness conjecture on abelian varieties with restricted torsion fields

摘录 / EXCERPT: Finiteness conjecture. Let k be a number field and let $g\geq 0$ be an integer. Then the set $\mathscr{A}(k,g)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-016957 | MATHDB-323084

Tight cover-time conjecture for minimum-degree weighting

摘录 / EXCERPT: Let G_w be a graph equipped with the minimum-degree weighting scheme, let n be its number of vertices, and let d be the parameter used in the locally tree-like analysis. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016958 | MATHDB-323117

Local intersection multiplicity formula for singular T

摘录 / EXCERPT: Let $T\in\Sigma(\alpha)$ be singular, let $Z(T)$ be the associated special cycle, and let $y\in Z(T)(\mathbb F_p^\mathrm{alg})$. With $I_{\mathcal O_{Y_0,y}}$ denoting the local in...

conjecture | open | MathDB 页面 | 原文线索

V2-016959 | MATHDB-323128

The c_2 residue conjecture for primitive log-divergent graphs

摘录 / EXCERPT: The c_2 residue conjecture. If $I_{G_1}=I_{G_2}$ for two primitive log-divergent graphs G_1 and G_2, then

conjecture | open | MathDB 页面 | 原文线索

V2-016960 | MATHDB-323146

Tian's global log canonical threshold conjecture

摘录 / EXCERPT: Let X be a projective Fano variety over an algebraically closed field. Suppose that X is $\mathbb{Q}$-factorial and has at worst log terminal singularities. Write ... for its glo...

conjecture | open | MathDB 页面 | 原文线索

V2-016961 | MATHDB-323199

Artin's supersingularity conjecture for K3 surfaces

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let X be a K3 surface over k. The Picard rank of X is the rank of its Picard group, and X is supersing...

conjecture | open | MathDB 页面 | 原文线索

V2-016962 | MATHDB-323215

Factorization injectivity conjecture for totally nonnegative matrices

摘录 / EXCERPT: Factorization injectivity conjecture. This map should be injective. This conjecture would extend factorization schemes for totally nonnegative upper triangular unipotent matrices t...

conjecture | open | MathDB 页面 | 原文线索

V2-016963 | MATHDB-323225

The no-loss incidence bound for good surfaces in four-dimensional space

摘录 / EXCERPT: No-loss incidence conjecture. The incidence count should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-016964 | MATHDB-323227

The general incidence bound for algebraic surfaces

摘录 / EXCERPT: General surface-incidence conjecture. The incidence count should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-016965 | MATHDB-323231

Minimal generators and Göttingen covariants for powers of binary forms

摘录 / EXCERPT: Assume that r divides d. Let $X=X_{e,d}$ be the variety of binary d-ics that are perfect $\frac{d}{e}$-th powers of binary e-ics, and let I_X be its defining ideal. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-016966 | MATHDB-323275

Harris's dimension conjecture for Hilbert schemes of curves

摘录 / EXCERPT: Let g,n,d be nonnegative integers, and let ... be the Brill-Noether number. Let $\mathcal{H}$ be a component of the Hilbert scheme whose general member corresponds to a smooth, irr...

conjecture | open | MathDB 页面 | 原文线索

V2-016967 | MATHDB-323312

Stable b-infinite transitivity conjecture for unirational varieties

摘录 / EXCERPT: Let X be a unirational algebraic variety over a field $\mathbf{k}$. The field $\mathbf{k}(X)(y_1,\ldots,y_m)$ is said to admit an infinitely transitive model if it has a model th...

conjecture | open | MathDB 页面 | 原文线索

V2-016968 | MATHDB-323329

Bombieri-Masser-Zannier's torsion anomalous openness conjecture

摘录 / EXCERPT: Let G be a semi-abelian variety and let V be a weak-transverse variety in G. Let V^{ta} denote the complement in V of the union of all V-torsion anomalous varieties. Bo...

conjecture | open | MathDB 页面 | 原文线索

V2-016969 | MATHDB-323337

The integral identity conjecture for motivic Milnor fibers

摘录 / EXCERPT: Let f be an element of $\kappa[[x,y,z]]$ invariant under the $\kappa^{\times}$-action of weight $(1,-1,0)$, with $f(0,0,0)=0$. Let $\mathfrak X$ be the formal neighborhood of...

conjecture | open | MathDB 页面 | 原文线索

V2-016970 | MATHDB-323352

Letellier's conjecture on character-variety polynomials

摘录 / EXCERPT: For a partition λ of n, let C_λ be the unipotent conjugacy class of $\operatorname{GL}_n(\mathbb{C})$ whose Jordan-block sizes are given by the dual partition...

conjecture | open | MathDB 页面 | 原文线索

V2-016971 | MATHDB-323353

Folk conjecture on triple points of theta divisors

摘录 / EXCERPT: Folk conjecture on triple points.

conjecture | open | MathDB 页面 | 原文线索

V2-016972 | MATHDB-323354

Salvati Manni-Grushevsky codimension conjecture for triple-point loci

摘录 / EXCERPT: Salvati Manni-Grushevsky codimension conjecture. The loci $F_{3,0}^{(g)}$ and $I^{(g)}$ are purely of codimension g in ${\mathcal A}_g$.

conjecture | open | MathDB 页面 | 原文线索

V2-016973 | MATHDB-323355

Boundary-component conjecture for the extended gradient zero locus

摘录 / EXCERPT: Boundary-component conjecture. The locus

conjecture | open | MathDB 页面 | 原文线索

V2-016974 | MATHDB-323430

Abelian defect group cohomology conjecture for Deligne-Lusztig varieties

摘录 / EXCERPT: Abelian defect group cohomology conjecture. The two $\mathbf G^F$-modules H_c^{odd} and H_c^{even} are disjoint, and F^δ is semisimple on...

conjecture | open | MathDB 页面 | 原文线索

V2-016975 | MATHDB-323450

Green-Lazarsfeld's conjecture on property (N_p) for line bundles on curves

摘录 / EXCERPT: Let $\mathcal L$ be a very ample line bundle on a smooth curve X of genus g, and let $\varphi_{\mathcal L}(X)$ denote its image in projective space. Write $h^1(X,\mathcal L)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-016976 | MATHDB-323501

Generalized determinantal representation conjecture for real-zero polynomials

摘录 / EXCERPT: Let $p \in {\mathbb R}[x_1,\ldots,x_d]$ be an RZ_{x^0} polynomial of degree m with $p(x^0)=1$. Here, RZ_{x^0} means real-zero with respect to x^0. Generalized determinantal...

conjecture | open | MathDB 页面 | 原文线索

V2-016977 | MATHDB-323502

Gurvits's derivative-determinantal representation conjecture

摘录 / EXCERPT: Let $p \in {\mathbb R}[x_1,\ldots,x_d]$ be an RZ_{x^0} polynomial of degree m with $p(x^0)=1$. Here, RZ_{x^0} means real-zero with respect to x^0, and $q^{(k)}_{x^0}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-016978 | MATHDB-323524

Algebraicity conjecture for foliations with a stable algebraic leaf

摘录 / EXCERPT: Let $\mathcal{F}$ be a holomorphic foliation of dimension one with non-degenerate singularities in the complex projective space $\mathbb{C}\mathbb{P}^{3}$, admitting a stable graph...

conjecture | open | MathDB 页面 | 原文线索

V2-016979 | MATHDB-323543

Goldman's volume rigidity conjecture for uniform lattices

摘录 / EXCERPT: Goldman's volume rigidity conjecture. Equality holds if and only if ρ is a discrete, faithful representation of Γ into G. Goldman proved this conjecture for all conne...

conjecture | open | MathDB 页面 | 原文线索

V2-016980 | MATHDB-323567

Tian's partial C^{0}-estimate for Fano manifolds

摘录 / EXCERPT: Tian's partial C^{0}-estimate. Given $t_0\in(0,1]$, there exist $k>0$ and $\delta>0$ such that kL is very ample and, for every $\omega\in\mathcal{K}(X,t_0)$,

conjecture | open | MathDB 页面 | 原文线索

V2-016981 | MATHDB-323609

Absolute irreducibility implies cohomological non-Eisensteinness

摘录 / EXCERPT: Let F be the number field, let G_F be its absolute Galois group, and let $\mathfrak{m}$ be the unique maximal ideal containing $\mathfrak{p}$. The associated residual represent...

conjecture | open | MathDB 页面 | 原文线索

V2-016982 | MATHDB-323621

Stein's conjecture on non-square parts of Tate-Shafarevich groups

摘录 / EXCERPT: Let $A/\mathbb{Q}$ range over all abelian varieties over $\mathbb{Q}$, and suppose that $\sha(A/\mathbb{Q})$ is finite so that its order has a well-defined non-square part. Stein's...

conjecture | open | MathDB 页面 | 原文线索

V2-016983 | MATHDB-323627

Tsukioka's generalized Mukai conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a complex n-dimensional Fano manifold with Picard number ${\rho}_X$. For every extremal ray R of X, define its length by ... Let l_X be the minimum of $l(R)$ over...

conjecture | open | MathDB 页面 | 原文线索

V2-016984 | MATHDB-323645

The sharp asymptotic bounds conjecture for the density of sections on Kähler manifolds

摘录 / EXCERPT: Sharp asymptotic bounds conjecture. For any n,c,V and $\eta<1$, there is a number $k_{0}(n,c,V,\eta)$ such that if $k\geq k_{0}$ then for any X in ${\mathcal K}(n,c,V)$ the dis...

conjecture | open | MathDB 页面 | 原文线索

V2-016985 | MATHDB-323688

The projective normality conjecture for smooth toric varieties

摘录 / EXCERPT: A smooth toric variety is a toric variety that is smooth. The projective normality conjecture. Every smooth toric variety is projectively normal. The conjecture relates smoothness...

conjecture | open | MathDB 页面 | 原文线索

V2-016986 | MATHDB-323700

Numerical characterization of elliptic Calabi-Yau manifolds

摘录 / EXCERPT: Numerical characterization of elliptic Calabi-Yau manifolds. A Calabi-Yau manifold X is elliptic if and only if there is such a $(1,1)$-class L.

conjecture | open | MathDB 页面 | 原文线索

V2-016987 | MATHDB-323703

The higher-loop-space conjecture on iterated integrals over membranes

摘录 / EXCERPT: Higher-loop-space conjecture. (a) The space $Maps((T^\infty,\dot{T}^\infty),(X,\{x_n\}))$, together with all its quotients $Maps((T^n,\dot{T}^n),(X,x_1,\dots,x_n))$, captures the h...

conjecture | open | MathDB 页面 | 原文线索

V2-016988 | MATHDB-323717

Cohomological degree conjecture for Deligne-Lusztig varieties

摘录 / EXCERPT: Let $\ell\neq p$ be primes, let q be a power of p, let d be the multiplicative order of q modulo ℓ, and let $G=G(q)$ be a finite group of Lie type with abelian Sylow...

conjecture | open | MathDB 页面 | 原文线索

V2-016989 | MATHDB-323742

Donagi's Prym map injectivity conjecture

摘录 / EXCERPT: Donagi's conjecture. The restricted Prym map

conjecture | open | MathDB 页面 | 原文线索

V2-016990 | MATHDB-323743

Van Geemen-van der Geer's conjecture on the \Gamma_{00} base locus

摘录 / EXCERPT: Van Geemen-van der Geer's conjecture. If (A,Θ) is the Jacobian of a smooth curve C, then, as a set,

conjecture | open | MathDB 页面 | 原文线索

V2-016991 | MATHDB-323744

Ciliberto-van der Geer's codimension conjecture for Andreotti-Mayer loci

摘录 / EXCERPT: Ciliberto-van der Geer's conjecture. If $1\le \ell\le g-3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-016992 | MATHDB-323769

Geometric description of clasps for the affine Grassmannian of $\mathbb{C}^2$

摘录 / EXCERPT: Let $Y({\underline{i}})$ be the affine-Grassmannian variety associated with ${\underline{i}}=({\0},1^k,{\u2})$ for some $k\in\mathbb N$, and let ... be the projection forgetting the...

conjecture | open | MathDB 页面 | 原文线索

V2-016993 | MATHDB-323782

The existence conjecture for full exceptional collections on projective homogeneous varieties

摘录 / EXCERPT: Exceptional collection conjecture. There exists a full exceptional collection of vector bundles in $\mathsf{D}^{\mathrm{b}}(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-016994 | MATHDB-323812

Donald's conjecture on the greatest Ricci lower bound and conical Kähler-Einstein metrics

摘录 / EXCERPT: Donald's conjecture. There does not exist a conical Kähler-Einstein metric solving

conjecture | open | MathDB 页面 | 原文线索

V2-016995 | MATHDB-323813

Conical Ricci-flow convergence conjecture at the greatest Ricci lower bound

摘录 / EXCERPT: Let X be a Fano manifold and let D be a smooth simple divisor in $|-mK_X|$ for some $m\in\mathbb Z^+$. Consider the conical Ricci flow ... starting from a smooth conical Kähler m...

conjecture | open | MathDB 页面 | 原文线索

V2-016996 | MATHDB-323817

Very free rational curves on nonsingular Fano hypersurfaces

摘录 / EXCERPT: Let X be a nonsingular Fano hypersurface over an algebraically closed field. A very free rational curve on X is a morphism $\varphi:\mathbb{P}^1\to X$ whose pulled-back tangent...

conjecture | open | MathDB 页面 | 原文线索

V2-016997 | MATHDB-323825

Normal distribution conjecture for asymptotic Betti tables

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, let L_d be the very positive line bundles used to define the Betti numbers $k_{p,q}(X;L_d)$, and write r_d for the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-016998 | MATHDB-323842

The irreducibility conjecture for higher-level Deligne-Lusztig cohomology

摘录 / EXCERPT: Let X_h be the closed subscheme of $U^{n,q}_h$ defined over ${\mathbb F}_{q^n}$, with commuting left action of U^1_L/U^h_L and right action of $U^{n,q}_h({\mathbb F}_{q^n})$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-016999 | MATHDB-323843

Convergence conjecture for the modified Calabi flow on relatively K-stable toric varieties

摘录 / EXCERPT: Toric modified-Calabi-flow conjecture. The modified Calabi flow converges exponentially fast to an extremal metric in $c_1(L)$.

conjecture | open | MathDB 页面 | 原文线索

V2-017000 | MATHDB-323855

The Strong Connectedness conjecture for regular rings

摘录 / EXCERPT: Strong Connectedness conjecture. A regular ring Σ has the Strong Connectedness Property at every pair of points $\alpha,\beta\in\operatorname{Sper}\Sigma$ having a common sp...

conjecture | open | MathDB 页面 | 原文线索

V2-017001 | MATHDB-323861

Motivic Galois lifting conjecture for central torus coverings

摘录 / EXCERPT: Let F and E be number fields, and let $H'\twoheadrightarrow H$ be a surjection of linear algebraic groups over E with central torus kernel. Suppose that ... is a homomorphism....

conjecture | open | MathDB 页面 | 原文线索

V2-017002 | MATHDB-323870

Green-Griffiths conjecture on singular normal functions

摘录 / EXCERPT: Let X be a smooth complex projective variety, let $L\to X$ be a very ample line bundle, and let ζ be a non-torsion, primitive Hodge class of type (n,n) on X. For the as...

conjecture | open | MathDB 页面 | 原文线索

V2-017003 | MATHDB-323911

Eisenbud-Harris conjecture on the dimension of restricted Hilbert schemes

摘录 / EXCERPT: Eisenbud-Harris conjecture. The largest irreducible component of $\widetilde{\operatorname{Hilb}}^{b,d}_{\mathbb{P}^n}$ is the one corresponding to the family of degree-d hypersu...

conjecture | open | MathDB 页面 | 原文线索

V2-017004 | MATHDB-323913

Generation conjecture for the universal Kummer variety

摘录 / EXCERPT: Let $\mathcal{I}_3$ be the prime ideal of the universal Kummer variety in the variables $({\bf u},{\bf x})$. The previously constructed equations consist of the Satake polynomial,...

conjecture | open | MathDB 页面 | 原文线索

V2-017005 | MATHDB-323932

Quillen-Lichtenbaum conjecture for global fields

摘录 / EXCERPT: Let F be a global field and let $l\ge 2$. If $l=2$, assume that $\mu_4\subset F$. For $n>1$ and $l\ne\operatorname{char} F$, let K_n^{et} denote étale K-theory. Quil...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-017006 | MATHDB-323944

Motivic vanishing-cycle localization conjecture

摘录 / EXCERPT: Let Z' be a smooth scheme with trivial $\mathbb{G}_m$-action, and let $\mathbb{A}_{\operatorname{\mathbb{C}}}^n$ carry a $\mathbb{G}_m$-action with nonnegative weights. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017007 | MATHDB-323972

McKernan's epsilon-delta conjecture for Mori fiber spaces

摘录 / EXCERPT: McKernan's conjecture. There exists a number $\delta=\delta_{m,n}(\epsilon)>0$ such that, whenever X is a $\mathbb Q$-factorial variety, $f\colon X\to Y$ is a Mori fiber space wi...

conjecture | open | MathDB 页面 | 原文线索

V2-017008 | MATHDB-324015

The open crepant resolution conjecture for toric orbifolds

摘录 / EXCERPT: Let X be a toric variety with at worst Gorenstein quotient singularities. Let $\mathcal{X}$ be the canonical toric orbifold with coarse moduli space X, and let Y be a toric c...

conjecture | open | MathDB 页面 | 原文线索

V2-017009 | MATHDB-324029

Equidistribution of preimages for polarized dynamical systems over nonarchimedean fields

摘录 / EXCERPT: Let K be an algebraically closed complete nonarchimedean field, possibly with trivial absolute value, and let X be an irreducible projective variety over K. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-017010 | MATHDB-324066

The monodromic affine Hecke equivalence for arbitrary tame parameters

摘录 / EXCERPT: Let D_{θ_1,θ_2} be the category of monodromic sheaves on the affine flag variety with generalized monodromy eigenvalues corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-017011 | MATHDB-324078

Hyperbolic Ax-Lindemann conjecture for Shimura varieties

摘录 / EXCERPT: Let S be a Shimura variety and let ${\mathcal X}$ be a complex realisation of a connected component X^+. Let ... be the uniformisation map, and let V be an algebraic subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-017012 | MATHDB-324079

Mumford-Tate and Isogeny conjectures for Shimura-type structures

摘录 / EXCERPT: Let $(V_{\mathbb Z},\mu\circ s,\mu\circ\rho_{\widetilde{x},l})$ be an (∞,l)-integral structure of Shimura type. Let $H_{\mathbb Q_l}$ be the identity component of the Zarisk...

conjecture | open | MathDB 页面 | 原文线索

V2-017013 | MATHDB-324080

Generalised Shafarevich conjecture for Shimura-type integral structures

摘录 / EXCERPT: Let $K=K^lK_l$ be a neat compact open subgroup of $G(\mathbb A_f)$, let $x=[s,\overline{1}]$ be a point of $Sh_K(G,X)(L)$, and fix a lift $\widetilde{x}=[s,1]$ in $Sh_{K^l}(G,X)$....

conjecture | open | MathDB 页面 | 原文线索

V2-017014 | MATHDB-324093

Irreducibility and normality conjecture for nilpotent commuting varieties of type A

摘录 / EXCERPT: Let G be a reductive algebraic group of type A with Lie algebra $\mathfrak{g}$, let B be a Borel subgroup with nilpotent radical $\mathfrak{u}$, and let $\mathcal{N}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-017015 | MATHDB-324096

The safety-bound conjecture for Segre varieties

摘录 / EXCERPT: Safety-bound conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-017016 | MATHDB-324098

Fano version of Hartshorne's complete-intersection conjecture

摘录 / EXCERPT: Let $X\subset {\mathbb P}^N$ be a smooth projective variety of dimension n and codimension c satisfying the hypotheses denoted by $(*)$ in the source. Fano Hartshorne conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-017017 | MATHDB-324099

Local Hartshorne conjecture for varieties covered by lines

摘录 / EXCERPT: Let $X\subset {\mathbb P}^N$ be covered by lines, let $x\in X$ be a general point, and let $\mathcal L_x\subset {\mathbb P}^{n-1}$ be the variety of lines through x. Let T be t...

conjecture | open | MathDB 页面 | 原文线索

V2-017018 | MATHDB-324101

Local-to-global complete-intersection conjecture for varieties covered by lines

摘录 / EXCERPT: Let $X\subset {\mathbb P}^N$ be covered by lines, let $x\in X$ be a general point, and let $\mathcal L_x\subset {\mathbb P}^{n-1}$ be the variety of lines through x. Local-to-glo...

conjecture | open | MathDB 页面 | 原文线索

V2-017019 | MATHDB-324102

The LQEL classification conjecture for cyclic Picard group

摘录 / EXCERPT: The LQEL classification conjecture. X is obtained by taking linear sections and/or isomorphic projections of a rational homogeneous manifold in its natural minimal embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-017020 | MATHDB-324103

The dual-defective-to-LQEL conjecture

摘录 / EXCERPT: The dual-defective-to-LQEL conjecture. X is a local quadratic entry locus manifold (LQELM).

conjecture | open | MathDB 页面 | 原文线索

V2-017021 | MATHDB-324111

The Borisov-Alexeev-Borisov conjecture for quasismooth weighted complete intersections

摘录 / EXCERPT: Let m and ϵ be fixed positive numbers, let α be a negative integer, and consider quasismooth weighted complete intersections of dimension m and amplitude...

conjecture | open | MathDB 页面 | 原文线索

V2-017022 | MATHDB-324118

The conjecture that flat finite-type morphisms of Noetherian schemes are very flat

摘录 / EXCERPT: Very-flatness conjecture. Every flat morphism of finite type between Noetherian schemes is very flat.

conjecture | open | MathDB 页面 | 原文线索

V2-017023 | MATHDB-324134

The Bernstein-Sato specialization equality conjecture

摘录 / EXCERPT: Let $F=(f_1, 3ellots,f_r)$, let $M\in\mathbb{N}^{p\times r}$ be non-zero, let ${\bf m}\in\mathbb{N}^p$, and let F^M and the monomial maps ϕ_M be defined as in the source. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-017024 | MATHDB-324153

The cycle-class form of the Tate conjecture

摘录 / EXCERPT: Let $\mathbf{F}=\mathbf{F}_q$ be a finite field, let X be a smooth projective variety over $\mathbf{F}$, let $\bar X=X\times_{\mathbf{F}}\overline{\mathbf{F}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017025 | MATHDB-324154

Friedlander's conjecture on the K-theoretic Chern character

摘录 / EXCERPT: Let $\mathbf{F}=\mathbf{F}_q$ be a finite field, let X be a smooth variety over $\mathbf{F}$, let $\bar X=X\times_{\mathbf{F}}\overline{\mathbf{F}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017026 | MATHDB-324167

Liu-Sebag cut-and-paste conjecture for the Grothendieck ring of varieties

摘录 / EXCERPT: Let k be a field, and let $K_0(\operatorname{Var}_k)$ be the Grothendieck ring of varieties over k. Liu-Sebag's cut-and-paste conjecture. If varieties X and Y over k sati...

conjecture | open | MathDB 页面 | 原文线索

V2-017027 | MATHDB-324168

Cancellation conjecture for the Lefschetz motive

摘录 / EXCERPT: Let k be a field, let $K_0(\operatorname{Var}_k)$ be the Grothendieck ring of varieties over k, and let $\mathbb{L}=[\mathbb{A}^1_k]$ be the Lefschetz class. Cancellation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-017028 | MATHDB-324169

Aluffi's Euler characteristic conjecture for graph hypersurface complements

摘录 / EXCERPT: Let Γ be a finite graph with graph hypersurface $X_\Gamma\subset\mathbb P^{n-1}$, and write $Y_\Gamma=\mathbb P^{n-1}\smallsetminus X_\Gamma$ for its projective hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-017029 | MATHDB-324170

Beilinson-Goncharov-Schechtman-Varchenko's hyperplane-arrangement generation conjecture

摘录 / EXCERPT: Let k be a number field. Given hyperplanes L_i and M_i, for $i=0,\ldots,n$, in general position in $\mathbb P^n$ and defined over k, consider the mixed Tate motive ... Its re...

conjecture | open | MathDB 页面 | 原文线索

V2-017030 | MATHDB-324178

The codimension-three extension conjecture for microdifferential modules

摘录 / EXCERPT: Let X be a complex manifold, let $Y\subset X$ be a subvariety, and write ... for the open embedding. Let $\mathcal{A}_X$ denote the relevant sheaf of microdifferential operators, a...

conjecture | open | MathDB 页面 | 原文线索

V2-017031 | MATHDB-324180

The codimension-three conjecture for microlocal perverse sheaves with general coefficients

摘录 / EXCERPT: Let X be a complex manifold, let $\Lambda\subset T^*X$ be a conic Lagrangian support, and let $\mathcal{P\!er}_\Lambda(R)$ denote the stack of microlocal perverse sheaves with co...

conjecture | open | MathDB 页面 | 原文线索

V2-017032 | MATHDB-324186

Conjecture on strict separating systems of parameters for explicit varieties

摘录 / EXCERPT: Let $W\subseteq K^m$ be an explicit variety. A strict separating e.s.o.p. is a strict separating homogeneous system of parameters for the coordinate ring $K[W]$; for a weakly expli...

conjecture | open | MathDB 页面 | 原文线索

V2-017033 | MATHDB-324187

Conjecture on explicit defining equations for explicit varieties

摘录 / EXCERPT: Let W be an explicit variety, and distinguish explicit close-to-defining equations from explicit defining equations as in the source. Explicit equations conjecture. (a) Any expli...

conjecture | open | MathDB 页面 | 原文线索

V2-017034 | MATHDB-324198

Beilinson's conjectures on regulators and cycle classes

摘录 / EXCERPT: Let $X\in\mathbf{Reg}_{\mathbb{Z}}$ be proper and flat over $\operatorname{Spec}(\mathbb{Z})$, with potentially good reduction at every prime, and let ... be the regulator. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-017035 | MATHDB-324199

Conjecture on the ML degree of rank-two 3 by n matrices

摘录 / EXCERPT: Let $\mathcal{V}_2$ denote the variety of $3 \times n$ matrices of rank at most 2, with $n \geq 3$. ML-degree conjecture. The ML degree of $\mathcal{V}_2$ equals ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-017036 | MATHDB-324218

The KE moduli-space conjecture for K-semistable Q-Fano varieties

摘录 / EXCERPT: Let h be a Hilbert polynomial. Write $\operatorname{Sch}_{\mathbb C}$ for the category of algebraic schemes over $\mathbb C$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017037 | MATHDB-324223

The Mertens-type conjecture for proper smooth schemes over the integers

摘录 / EXCERPT: Mertens-type conjecture for schemes over $\mathbf Z$.

conjecture | open | MathDB 页面 | 原文线索

V2-017038 | MATHDB-324228

The isotropicity conjecture for the five canonical planes in the Debarre-Voisin tangent space

摘录 / EXCERPT: Isotropicity conjecture. The five vector spaces of dimension 2 canonically defined in the relevant construction are maximal isotropic subspaces for the symplectic form on...

conjecture | open | MathDB 页面 | 原文线索

V2-017039 | MATHDB-324244

The motivic cohomology and K-theory comparison conjecture

摘录 / EXCERPT: For every variety X over ${\mathbb{Q}}$, suppose there is a universal cohomology theory $X\mapsto h^*(X)$ with values in an abelian category ${\mathcal M}{\mathcal M}$ of mixed m...

conjecture | open | MathDB 页面 | 原文线索

V2-017040 | MATHDB-324253

Rational-span conjecture for tangent spaces of regular Hadamard matrices

摘录 / EXCERPT: Rational-span conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017041 | MATHDB-324265

Tate's Birch-Swinnerton-Dyer conjecture for abelian varieties

摘录 / EXCERPT: BSD conjecture for abelian varieties. The rank r equals the analytic rank $\operatorname{ord}_{s=1}L(A,s)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-017042 | MATHDB-324266

The p-adic BSD conjecture for modular abelian varieties

摘录 / EXCERPT: The p-adic BSD conjecture. The Mordell-Weil rank r of A equals $\operatorname{ord}_T(\mathcal{L}_p(A,T))$, and

conjecture | open | MathDB 页面 | 原文线索

V2-017043 | MATHDB-324267

The main conjecture of Iwasawa theory for modular abelian varieties

摘录 / EXCERPT: Main conjecture of Iwasawa theory. There exists an element $u(T)\in\Lambda^{\times}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017044 | MATHDB-324268

Improper marginalization birationality conjecture for translation-invariant matrix product states

摘录 / EXCERPT: Let $N\geq 4$. An improper marginalization maps a translation-invariant N-qubit matrix product state variety $\overline{\operatorname{OB}}(2,2,N)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-017045 | MATHDB-324270

Grothendieck's proper pseudo-coherent direct-image conjecture

摘录 / EXCERPT: Let $f:X\longrightarrow Y$ be a proper and pseudo-coherent morphism of schemes. Write $D_{qcoh}(X)$ and $D_{qcoh}(Y)$ for the derived categories of quasi-coherent complexes on X...

conjecture | open | MathDB 页面 | 原文线索

V2-017046 | MATHDB-324292

Weil rationality for difference zeta functions

摘录 / EXCERPT: Let (X,σ) be a finite-dimensional difference scheme of finite σ-type over a finite difference field (k,φ_0), where φ_0 is the distinguished Frobenius...

conjecture | open | MathDB 页面 | 原文线索

V2-017047 | MATHDB-324298

The centraliser invariant-field conjecture for commuting involutions

摘录 / EXCERPT: Centraliser invariant-field conjecture. For any $e\in{\mathfrak g}_0\cap\mathcal N$,

conjecture | open | MathDB 页面 | 原文线索

V2-017048 | MATHDB-324319

Generalisation of the Lagrangian subvariety proposition to (H,A)-stability

摘录 / EXCERPT: Let Y be a bielliptic surface, let $f\colon X\to Y$ be its canonical covering of degree $n=2$, and let w be a sheaf class on Y. Write $K_{X;f^*H,f^*A}(f^*w)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017049 | MATHDB-324320

Existence of the augmentation polynomial for every knot

摘录 / EXCERPT: Let K be a knot. The augmentation variety V_K is the set of augmentation values in $(\mathbb{C}^*)^3$, and the augmentation polynomial is defined when the maximal-dimensional p...

conjecture | open | MathDB 页面 | 原文线索

V2-017050 | MATHDB-324321

Existence and specialization conjecture for the two-variable augmentation polynomial

摘录 / EXCERPT: Let K be a knot, and let $\operatorname{Aug}_K(\lambda,\mu,U)$ be its three-variable augmentation polynomial. When the $U=1$ slice of the augmentation variety has a maximal-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-017051 | MATHDB-324341

The anticanonical degree conjecture for toric Fano manifolds

摘录 / EXCERPT: Anticanonical degree conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017052 | MATHDB-324349

The Faber-van der Geer conjecture on interior cohomology of \(\mathcal{A}_2\)

摘录 / EXCERPT: Let $\mathcal{A}_2$ be the moduli space of principally polarized abelian surfaces, let $\lambda=\lambda_1\geq\lambda_2\geq0$ be an arbitrary weight, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017053 | MATHDB-324358

The distribution conjecture for Baxter permutations under the stack-sorting map

摘录 / EXCERPT: Let S denote the stack-sorting operator, let $\mathbf{i}$ denote the identity map, and let des, lmax, and comp be the descent,...

conjecture | open | MathDB 页面 | 原文线索

V2-017054 | MATHDB-324366

The log-canonical model conjecture for varieties admitting endomorphisms

摘录 / EXCERPT: Let X be a normal variety, and let $f\colon X\rightarrow X$ be an endomorphism of degree $\deg(f)>1$. Suppose that X admits a log-canonical model ... Let E^{lc}_μ be the sum...

conjecture | open | MathDB 页面 | 原文线索

V2-017055 | MATHDB-324377

ABCH's explicit Bridgeland-Mori wall correspondence conjecture for the Hilbert scheme

摘录 / EXCERPT: Let I_Z be the ideal sheaf of a point $Z\in\mathbb{P}^{2[n]}$. A Bridgeland wall is a wall W_x in the family of walls $W_{I_Z,(r',c',d')}$ along which some ideal sheaf I_Z is...

conjecture | open | MathDB 页面 | 原文线索

V2-017056 | MATHDB-324399

Generalized Gieseker conjecture for regular singular stratified bundles

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, and let X and Y be smooth, connected, separated, finite type k-schemes. Write...

conjecture | solved | MathDB 页面 | 原文线索

V2-017057 | MATHDB-324419

The geometric Bogomolov conjecture for nowhere degenerate abelian varieties

摘录 / EXCERPT: Let A be a nowhere degenerate abelian variety over $\overline{K}$, meaning that its abelian variety structure has no nontrivial degenerate part, and let X be a closed subvariet...

conjecture | open | MathDB 页面 | 原文线索

V2-017058 | MATHDB-324439

Nekrasov-Shatashvili principle for Baxter subalgebras of the Yangian

摘录 / EXCERPT: Nekrasov-Shatashvili principle. The Baxter subalgebras of Y_Q corresponding to $g\in\mathfrak{H}$ are the algebras of modified equivariant quantum multiplication for Nak...

conjecture | open | MathDB 页面 | 原文线索

V2-017059 | MATHDB-324449

Unstable Lichtenbaum-Quillen conjecture for cyclotomic integers

摘录 / EXCERPT: Let ℓ be an odd regular prime and let $KH_{**}(\mathbb Z[\zeta_\ell,1/\ell])\to KH^{et}_{**}(\mathbb Z[\zeta_\ell,1/\ell])$ be the canonical surjection of bi-graded algebras d...

conjecture | open | MathDB 页面 | 原文线索

V2-017060 | MATHDB-324451

Congruence relation for Frobenius on the p-isogeny correspondence

摘录 / EXCERPT: Let Sh_K be the good reduction at p of a PEL Shimura variety, let $p\textrm{-}\mathscr{I}sog$ be its p-isogeny correspondence, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017061 | MATHDB-324494

Vanishing of low-degree Hilbert-Kunz coefficients under the standard conjecture

摘录 / EXCERPT: Vanishing conjecture. If $i\le d/2$, then $\beta_i=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-017062 | MATHDB-324525

Quasi-fractal intersection conjecture for algebraic varieties

摘录 / EXCERPT: Quasi-fractal intersection conjecture. The Zariski closure of $Z(\bar K)\cap Q$ is a union of finitely many points and finitely many components B_i such that, for each i, the i...

conjecture | open | MathDB 页面 | 原文线索

V2-017063 | MATHDB-324542

Algebraic compactification conjecture for KE Fano metrics

摘录 / EXCERPT: Fix a smooth differentiable manifold M^n and a Hilbert polynomial h. Let $\overline{\mathcal{M}}(M^n,h)^{GH}$ denote the Gromov-Hausdorff compactification of the space of isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-017064 | MATHDB-324547

Higher-dimensional smoothing conjecture for nodal cKE Fano varieties

摘录 / EXCERPT: Let $\mathcal{X}\to\Delta_t\subseteq\mathbb{C}$ be a partial smoothing of a nodal constant-scalar-curvature Kähler-Einstein Fano variety (X_0,ω_0), with the specific converg...

conjecture | open | MathDB 页面 | 原文线索

V2-017065 | MATHDB-324562

Stabilizer conjecture for twisted module stacks

摘录 / EXCERPT: Let X be a scheme over an ordinary commutative ring S, let $p:X\rightarrow\operatorname{Spec} S$ be the structure morphism, and let $\alpha\in\operatorname{Br}(S)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017066 | MATHDB-324563

Milnor's product conjecture for fibers of composed polynomial mappings

摘录 / EXCERPT: Let $\psi=(f_1,\ldots,f_k):\mathbb{R}^n\to\mathbb{R}^k$ be a polynomial mapping with an isolated singularity at the origin. Compose ψ with the projection...

conjecture | open | MathDB 页面 | 原文线索

V2-017067 | MATHDB-324576

Reality conjecture for the roots of the polynomial S in Cayley conditions

摘录 / EXCERPT: Let ... where $0<\gamma_{2n-1}<\cdots<\gamma_1$, and let $S(x),P(x)\in\mathbb{R}[x]$ satisfy the relation ... with $P(0)\neq 0$. Reality conjecture for S. The polynomial $S(x)$ has o...

conjecture | open | MathDB 页面 | 原文线索

V2-017068 | MATHDB-324579

The Kummer-coordinate addition relations conjecture

摘录 / EXCERPT: The Kummer-coordinate addition relations conjecture. Suppose that $(P,Q)\in A\times A$ is a good pair. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-017069 | MATHDB-324580

Stability of syzygy bundles in all dimensions

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, let A be the ample line bundle from the preceding setup, and let M_d denote the syzygy bundle associated with the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-017070 | MATHDB-324583

The Zhu algebra variety conjecture for finitely strongly generated simple vertex operator algebras

摘录 / EXCERPT: Let V be a finitely strongly generated simple vertex operator algebra of CFT type. Write ... Here X_V denotes the associated variety of V. The Zhu algebra variety conjecture. W...

conjecture | open | MathDB 页面 | 原文线索

V2-017071 | MATHDB-324592

Hilbert-function conjecture for minimal local Gorenstein schemes computing cubic cactus rank

摘录 / EXCERPT: Hilbert-function conjecture. The Hilbert function of a minimal local Artinian Gorenstein scheme for a general cubic form with local cactus rank l has the displayed form according...

conjecture | open | MathDB 页面 | 原文线索

V2-017072 | MATHDB-324618

Equality of the Terwilliger and orbit centralizer algebras for the conjugacy scheme of the symmetric group

摘录 / EXCERPT: Let $O_1,\dots,O_M$ be the orbits of $\operatorname{Sym}(n)^2$ under $\operatorname{Iso}_1(n)$, and let B_i be the corresponding zero-one matrices. Let $\mathfrak{T}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-017073 | MATHDB-324621

Strictness of the stability inclusions for positive-dimensional arcs

摘录 / EXCERPT: Strictness conjecture. If $\mathfrak{x}$ has positive dimension, each inclusion in this chain is strict. The smooth schemes form the basic class of $\mathfrak{x}$-stable schemes, a...

conjecture | open | MathDB 页面 | 原文线索

V2-017074 | MATHDB-324622

Stability of schemes smooth over fat points

摘录 / EXCERPT: Let $\mathfrak{n}$ be a fat point over κ, and let X be a scheme smooth over $\mathfrak{n}$. Let $\mathfrak{x}$ be an admissible arc in $\operatorname{\bf Arc}_{\kappa}$ co...

conjecture | open | MathDB 页面 | 原文线索

V2-017075 | MATHDB-324623

Piecewise-triviality of truncation maps for affine smooth schemes

摘录 / EXCERPT: Let $X\in\mathbbmss{Sch}_{\kappa}$ be affine. Suppose that $X\to\mathfrak{n}$ and $X\to\mathfrak{m}$ are smooth morphisms, where $\mathfrak{n}\hookrightarrow\mathfrak{m}$ are eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-017076 | MATHDB-324658

The quasi-projectivity conjecture for non-prime 3-manifolds

摘录 / EXCERPT: Let N be a compact 3-manifold with empty or toroidal boundary. A quasi-projective group is the fundamental group of a smooth quasi-projective complex variety. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-017077 | MATHDB-324678

Langlands reciprocity conjecture for Calabi-Yau varieties

摘录 / EXCERPT: Let X be a Calabi-Yau variety defined over $\mathbb Q$. Write $\zeta(X,s)$ for its Hasse-Weil zeta-function. Langlands reciprocity conjecture. The zeta-function $\zeta(X,s)$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-017078 | MATHDB-324688

Collapsing conjectures for projective bundles under the Kähler-Ricci flow

摘录 / EXCERPT: Let $M=\mathbb{P}(E)$ be a projective bundle over a smooth projective variety B, with projection $\pi:M\to B$ and fibers $F\cong\mathbb{P}^{r-1}$. Assume ... for a Kähler metric...

conjecture | open | MathDB 页面 | 原文线索

V2-017079 | MATHDB-324699

Phantom-class conjecture for finite-flat Eisenstein extensions

摘录 / EXCERPT: Let $\mathfrak{q}$ be a prime ideal with $N(\mathfrak{q})=q\not\equiv1\pmod p$. Suppose that the finite-flat extension group $\operatorname{Ext}^1(\mu_p,\mathbf{Z}/p\mathbf{Z})$ ov...

conjecture | open | MathDB 页面 | 原文线索

V2-017080 | MATHDB-324719

The slope inequality conjecture for fibrations with mildly singular fibres

摘录 / EXCERPT: Slope inequality conjecture. Under these assumptions, the fibration satisfies the displayed slope inequality. The inequality is equivalent to the f-positivity of ω_f and f...

conjecture | open | MathDB 页面 | 原文线索

V2-017081 | MATHDB-324728

Hacon-Kovács-Luo-Zhang conjecture on zeros of holomorphic one-forms

摘录 / EXCERPT: Hacon-Kovács-Luo-Zhang conjecture. The zero locus $Z(\omega)$ is nonempty for every global holomorphic one-form $\omega\in H^0(X,\Omega_X^1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-017082 | MATHDB-324729

Luo-Zhang conjecture on zero-free holomorphic one-forms

摘录 / EXCERPT: Luo-Zhang conjecture. The dimension of W can be at most

conjecture | open | MathDB 页面 | 原文线索

V2-017083 | MATHDB-324751

Generic rank-one semi-stable degeneration conjecture for Lagrangian fibrations

摘录 / EXCERPT: Generic rank-one degeneration conjecture. A generic deformation of X preserving the Lagrangian fibration has rank-one semi-stable degenerations as singular fibres in codimension...

conjecture | open | MathDB 页面 | 原文线索

V2-017084 | MATHDB-324753

Matsushita's maximal-or-minimal variation conjecture for Lagrangian fibrations

摘录 / EXCERPT: Matsushita's conjecture. The fibration has either maximal variation or minimal variation.

conjecture | open | MathDB 页面 | 原文线索

V2-017085 | MATHDB-324773

Off-diagonal semi-algebraic Ramsey bound

摘录 / EXCERPT: For integers $k\geq 3$, dimension d, complexity t, and s, let $R_k^{d,t}(s,n)$ be the least integer N such that every sequence of N points in $\mathbb{R}^d$ and every k...

conjecture | open | MathDB 页面 | 原文线索

V2-017086 | MATHDB-324775

Riemann hypothesis for the completed motivic L-function

摘录 / EXCERPT: Fix number fields K,L and a pure motive M of weight w over K with coefficients in L. Let $L(s,M)$ be its motivic L-function, let $L_{\infty}(s,M)$ be its archimedean fa...

conjecture | open | MathDB 页面 | 原文线索

V2-017087 | MATHDB-324845

The pseudo-effective pushforward conjecture for homology classes

摘录 / EXCERPT: Pseudo-effective pushforward conjecture. Any pseudo-effective class $\alpha\in H_{2k}(X,\mathbf{R})$ such that $\varphi_*\alpha=0$ in $H_{2k}(Y,\mathbf{R})$ belongs to the real vec...

conjecture | open | MathDB 页面 | 原文线索

V2-017088 | MATHDB-324846

The pseudo-effective pushforward conjecture for cohomology classes

摘录 / EXCERPT: Cohomological pseudo-effective pushforward conjecture. Any pseudo-effective class $\alpha\in H^{2l}(X,\mathbf{R})$ such that $\varphi_*\alpha=0$ belongs to the real vector space, r...

conjecture | open | MathDB 页面 | 原文线索

V2-017089 | MATHDB-324863

Semistability reduction conjecture for endomorphism bundles

摘录 / EXCERPT: Let X be a smooth complex projective variety, let $\mathcal O_X(1)$ be ample, and let E be a slope-semistable locally free sheaf. Choose a finitely generated model...

conjecture | open | MathDB 页面 | 原文线索

V2-017090 | MATHDB-324865

Non-étale-trivializable reduction conjecture

摘录 / EXCERPT: In the variation setup, let E be a numerically flat vector bundle on a smooth complex projective variety, with model $\mathcal E$ over S. Let Σ^{net} be the set...

conjecture | open | MathDB 页面 | 原文线索

V2-017091 | MATHDB-324866

Aubin's anticanonical volume conjecture for Kähler-Einstein metrics

摘录 / EXCERPT: Aubin's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-017092 | MATHDB-324867

The upper-bound conjecture for the ten Fano potentials

摘录 / EXCERPT: Let W be one of the ten potentials $(W_1,W_2,\dots,W_9,W_Q)$ listed in Galkin and Usnich's work, and let V be the corresponding exchange collection, namely one of...

conjecture | open | MathDB 页面 | 原文线索

V2-017093 | MATHDB-324873

Polyhedral spectrum conjecture for smooth polarized toric varieties

摘录 / EXCERPT: Let P be a lattice polytope corresponding to a polarized smooth toric variety, and let $\operatorname{codeg}_{\mathbb Q}P$ denote its $\mathbb Q$-codegree: if $c>0$ is the maxima...

conjecture | open | MathDB 页面 | 原文线索

V2-017094 | MATHDB-324876

Geiss-Leclerc-Schröer open-orbit conjecture

摘录 / EXCERPT: Let Λ_V be the representation variety under consideration, let G_V act on it, and let Λ_b be an irreducible component indexed by a canonical-basis element b. A...

conjecture | open | MathDB 页面 | 原文线索

V2-017095 | MATHDB-324909

Deformation retraction conjecture for character varieties of right-angled Artin groups

摘录 / EXCERPT: Let G be a complex reductive algebraic group, let K be a maximal compact subgroup, and let Γ be a right-angled Artin group with torsion. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017096 | MATHDB-324945

Radicality conjecture for localized Vandermonde varieties

摘录 / EXCERPT: Let $Vd_{k;I}^{BC}$ be the $\mathcal{BC}_k$-localized Vandermonde variety associated with an integer sequence I, and let $\mathcal I_I^{BC}$ be its defining ideal in the polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-017097 | MATHDB-324947

Structure conjecture for localized Vandermonde varieties of length k+1

摘录 / EXCERPT: Let $I=\{0=i_0<i_1<i_2<\cdots<i_k\}$ be an integer sequence with $\gcd(i_1,\ldots,i_k)=1$, let $Vd_{k;I}^{BC}$ be the localized Vandermonde variety, let $\mathcal R_I^{BC}$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-017098 | MATHDB-324951

Finiteness conjecture for abelian varieties with constrained torsion

摘录 / EXCERPT: Finiteness conjecture. For any K and g, the set $\mathscr{A}(K,g)$ is finite. Equivalently, $\mathscr{A}(K,g,\ell)$ is non-empty for only finitely many primes ℓ.

conjecture | open | MathDB 页面 | 原文线索

V2-017099 | MATHDB-324952

Uniform finiteness conjecture for constrained torsion over number fields

摘录 / EXCERPT: Uniform Version. Let $g>0$ and $n>0$. Then there exists a bound $N=N(g,n)>0$ such that $\mathscr{A}(K,g,\ell)=\varnothing$ for any $K\in\mathscr{F}(\mathbb{Q},n)$ and any prime...

conjecture | open | MathDB 页面 | 原文线索

V2-017100 | MATHDB-324964

Wall-crossing conjecture for moduli stacks of semistable objects

摘录 / EXCERPT: Wall-crossing of moduli stacks. There exist projective schemes S^-, S^0, and S^+, with built-in morphisms

conjecture | open | MathDB 页面 | 原文线索

V2-017101 | MATHDB-324971

Popa's derived-invariance conjecture for cohomological support loci

摘录 / EXCERPT: Let X and Y be smooth projective varieties over the complex numbers, and write ... For each $i\geq 0$, define the cohomological support locus ... and let $V^i(\omega_X)_0$ be the u...

conjecture | open | MathDB 页面 | 原文线索

V2-017102 | MATHDB-324972

Derived invariance of Hodge numbers for smooth projective varieties

摘录 / EXCERPT: Let X and Y be smooth projective varieties over the complex numbers, and write ... For $i\geq 0$, let $h^{0,i}(X)=\dim H^i(X,\mathcal{O}_X)$, and define $h^{0,i}(Y)$ similarly. T...

conjecture | open | MathDB 页面 | 原文线索

V2-017103 | MATHDB-324973

Lombardi's stronger conjecture on derived invariance of canonical cohomology

摘录 / EXCERPT: Let X and Y be smooth projective varieties over the complex numbers, let $\mathbf{D}(X)=\mathbf{D}^{\mathrm{b}}(\operatorname{Coh}(X))$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017104 | MATHDB-324982

The categorical geometric Langlands conjecture for reductive groups

摘录 / EXCERPT: Let G be a connected reductive group, let cG be its Langlands dual, and let $cBun_G$ denote the moduli stack of G-bundles on X. Let $cLocSys_\cG$ be the derived stack of...

conjecture | open | MathDB 页面 | 原文线索

V2-017105 | MATHDB-324983

The extended Whittaker coefficient conjecture

摘录 / EXCERPT: Let G be a connected reductive group, let Bun_G be its moduli stack of bundles, and let $\operatorname{Whit}^{\operatorname{ext}}(G,G)$ be the extended Whittaker...

conjecture | open | MathDB 页面 | 原文线索

V2-017106 | MATHDB-324984

The oper generation conjecture for irreducible local systems

摘录 / EXCERPT: Let $\operatorname{LocSys}_\cG^{\operatorname{irred}}$ be the open substack of irreducible $\cG$-local systems. For every finite set I and map $\lambda^I:I\to\Lambda^+$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017107 | MATHDB-324985

The Ran-space oper contractibility conjecture

摘录 / EXCERPT: Let $\operatorname{LocSys}_\cG$ be the stack of $\cG$-local systems and let ... be the map from global opers with singularities parametrized by the Ran space. Ran-space oper contract...

conjecture | open | MathDB 页面 | 原文线索

V2-017108 | MATHDB-324987

The conjecture on TRI hyperplanes of irreducible hypersurfaces

摘录 / EXCERPT: Let $f(X_0,X_1,\ldots,X_n)$ be a homogeneous irreducible polynomial of degree $d>2$, and let F be the hypersurface in $\operatorname{PG}(n,q)$ defined by $f=0$. A hyperplane is T...

conjecture | open | MathDB 页面 | 原文线索

V2-017109 | MATHDB-324998

Existence of moduli spaces of pointed and unpointed G-covers

摘录 / EXCERPT: Let A be a complete discrete valuation ring with algebraically closed residue field, let $D=\operatorname{Spec}A$, and let G be a finite group. A G-cover of D is the finite...

conjecture | open | MathDB 页面 | 原文线索

V2-017110 | MATHDB-324999

Existence of moduli spaces of G-arcs and G-jets

摘录 / EXCERPT: Let $D=\operatorname{Spec}A$ be as above, let M be a D-variety endowed with an action of a finite group G, and let a G-arc be a G-equivariant D-morphism $E\to M$ from a...

conjecture | open | MathDB 页面 | 原文线索

V2-017111 | MATHDB-325000

Existence of the moduli space of twisted arcs

摘录 / EXCERPT: Let $\mathcal{X}$ be a Deligne-Mumford stack of finite type over D. A stacky formal disk is a connected normal Deligne-Mumford stack birational and finite over D, and a twisted...

conjecture | open | MathDB 页面 | 原文线索

V2-017112 | MATHDB-325001

Existence and truncation properties of twisted jet spaces

摘录 / EXCERPT: Let $\mathcal{X}$ be a Deligne-Mumford stack of finite type over D, and let $\mathcal{J}_{\infty}\mathcal{X}$ be its moduli space of twisted arcs. For $n\in\mathbb{Z}_{\ge0}$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-017113 | MATHDB-325002

The weight-function and change-of-variables conjecture for wild Deligne-Mumford stacks

摘录 / EXCERPT: Let $D=\operatorname{Spec}A$ and let $\mathcal{X}$ be a Deligne-Mumford stack of finite type and pure dimension over D, with twisted-arc space $\mathcal{J}_{\infty}\mathcal{X}$....

conjecture | open | MathDB 页面 | 原文线索

V2-017114 | MATHDB-325003

Factorization of the weight function for quotient stacks

摘录 / EXCERPT: Let $D=\operatorname{Spec}A$, let G be a finite group, and let V be a G-representation over D. Put $\mathcal{X}=[V/G]$, and identify its twisted-arc space with the equivari...

conjecture | open | MathDB 页面 | 原文线索

V2-017115 | MATHDB-325004

Untwisting key lemma for the change-of-variables conjecture

摘录 / EXCERPT: Let $\phi:\mathcal{X}=[M/G]\to X=M/G$ be the proper birational morphism associated with a G-variety M. Let $\gamma:\mathcal{E}\to\mathcal{X}$ be a twisted arc whose generic poi...

conjecture | open | MathDB 页面 | 原文线索

V2-017116 | MATHDB-325006

The basepoint free conjecture for smooth projective threefolds in positive characteristic

摘录 / EXCERPT: Basepoint free conjecture. The linear system $|b(K_X+S+A)|$ is basepoint free for every $b\gg 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-017117 | MATHDB-325007

Trapa's conjecture on connected components of symmetric clan orbits

摘录 / EXCERPT: Trapa's conjecture. For both even and odd n, the orbit Q_γ is disconnected if and only if, for every i such that $c_i\in\mathbb{N}$, one has $c_i\neq c_{n+1-i}$.

conjecture | open | MathDB 页面 | 原文线索

V2-017118 | MATHDB-325008

Type B Richardson-clan conjecture

摘录 / EXCERPT: Type B Richardson-clan conjecture. With these definitions, in cases (1)-(3), X_u^v is one of the two L-orbit closures associated to $\gamma(u,v)$ and $X_{w_0v}^{w_0u}$ is the o...

conjecture | open | MathDB 页面 | 原文线索

V2-017119 | MATHDB-325010

Type D Richardson-clan conjecture

摘录 / EXCERPT: Type D Richardson-clan conjecture. The statement of the type B Richardson-clan conjecture also holds in type D.

conjecture | open | MathDB 页面 | 原文线索

V2-017120 | MATHDB-325018

The multivariate Schmidt-Spitzer conjecture for banded Toeplitz matrices

摘录 / EXCERPT: Main conjecture. For any banded Toeplitz matrix ${\mathcal{A}}$, if $B_{\mathcal{A}}$ and $C_{\mathcal{A}}$ are defined as above, then $B_{\mathcal{A}}=C_{\mathcal{A}}$.

conjecture | solved | MathDB 页面 | 原文线索

V2-017121 | MATHDB-325024

Rationality of the Euler-Chow series and finite generation of Cox rings for rationally connected varieties

摘录 / EXCERPT: Rationality-Cox ring conjecture. $E^1(X)$ is rational if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-017122 | MATHDB-325060

Non-vanishing conjecture for smooth varieties

摘录 / EXCERPT: Let X be a smooth projective variety. Non-vanishing conjecture. If K_X is pseudo-effective, then there exists an effective $\mathbb{Q}$-divisor D such that ... This is the smoo...

conjecture | open | MathDB 页面 | 原文线索

V2-017123 | MATHDB-325099

The Newton-region formula for Segre classes of monomial schemes

摘录 / EXCERPT: Let V be a nonsingular variety, and let $X_1,\dots,X_n$ be nonsingular hypersurfaces meeting with normal crossings in V. For $I=(i_1,\dots,i_n)$, let X^I denote the hypersurf...

conjecture | open | MathDB 页面 | 原文线索

V2-017124 | MATHDB-325111

The Schur-functor conjecture for Tor modules of the orbit variety

摘录 / EXCERPT: Let A be the coordinate ring of the ambient space and let Y be the orbit variety under consideration, with coordinate ring $K[Y]$. The group $\mathbf{GL}(E)\times G$ acts on...

conjecture | open | MathDB 页面 | 原文线索

V2-017125 | MATHDB-325112

The quadratic-generation conjecture for the four-copy restriction

摘录 / EXCERPT: Let Y be the orbit variety in $\operatorname{Hom}(E,V)$, and let E' be a 4-dimensional quotient of E. Then $Y\cap\operatorname{Hom}(E',V)$ is the corresponding restricted v...

conjecture | open | MathDB 页面 | 原文线索

V2-017126 | MATHDB-325136

Finite-basis conjecture for PQS-sortable permutations

摘录 / EXCERPT: A permutation is PQS-sortable if it can be sorted by a pop stack followed by a queue and a stack in series. The class of PQS-sorting permutations consists of 108 permutations, al...

conjecture | open | MathDB 页面 | 原文线索

V2-017127 | MATHDB-325142

Cyclic Douglas-Rachford best approximation conjecture

摘录 / EXCERPT: Cyclic Douglas-Rachford best approximation conjecture. If a best approximation pair relative to (C_1,C_2) exists, then the two-set cyclic Douglas-Rachford scheme converges weakly...

conjecture | refuted | MathDB 页面 | 原文线索

V2-017128 | MATHDB-325181

Kudla's generating-series conjecture for special cycles

摘录 / EXCERPT: Kudla's conjecture. The formal generating series

conjecture | open | MathDB 页面 | 原文线索

V2-017129 | MATHDB-325210

Polynomial defining the maximal-dimensional augmentation component for knots

摘录 / EXCERPT: Let $K\subset\mathbb{R}^3$ be a knot, let $HC_*(K)$ be its contact homology algebra, and let ... Let $\overline{V_K}$ denote the closure of V_K. Polynomial-component conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-017130 | MATHDB-325213

The cohomological product decomposition conjecture for discriminant complements

摘录 / EXCERPT: Let K) be an algebraically closed field of characteristic p, let $\ell\neq p$ be a prime, and let $\underline{d}=(d_1,\ldots,d_k)$. Write $\Pi_{\underline{d},n,K}$ for the spac...

conjecture | open | MathDB 页面 | 原文线索

V2-017131 | MATHDB-325220

Unit-group equality for irreducible polynomials

摘录 / EXCERPT: Let k be the ground field, let f be the polynomial defining the affine variety in the paper, and let T be its coordinate ring. Write $T^\ast$ for the group of units of T. U...

conjecture | open | MathDB 页面 | 原文线索

V2-017132 | MATHDB-325221

Unit-group equality for powers of two

摘录 / EXCERPT: Let k be the ground field, let T be the coordinate ring considered in the paper, let $T^\ast$ be its group of units, and let n be the relevant degree or group order. Let G...

conjecture | open | MathDB 页面 | 原文线索

V2-017133 | MATHDB-325223

Perfectoid approximation conjecture for smooth closed subschemes

摘录 / EXCERPT: Let K be a perfectoid field with tilt $K^\flat$, let $X\subset \mathbb{P}^n_K$ be a smooth closed subscheme, and let $\tilde{X}\subset (\mathbb{P}^n_K)^\mathrm{ad}$ be a small op...

conjecture | open | MathDB 页面 | 原文线索

V2-017134 | MATHDB-325226

Independence-of-ell conjecture for rigid-analytic varieties

摘录 / EXCERPT: Let k be a finite extension of $\mathbb{Q}_p$, let X be a proper smooth rigid-analytic variety over k, and consider $H^i_\mathrm{\acute{e}t}(X_{\bar{k}},\mathbb{Q}_\ell)$. In...

conjecture | open | MathDB 页面 | 原文线索

V2-017135 | MATHDB-325232

The CEQ-rigidity conjecture for plumbing schemes

摘录 / EXCERPT: Let P be the plumbing scheme for a potentially taut dual graph, and suppose that $\gcd(p,n_l)=1$ for all l. Write $\operatorname{CEQ}(P)$ for the set of equivalence classes ass...

conjecture | open | MathDB 页面 | 原文线索

V2-017136 | MATHDB-325323

The maximal quadratic matrix equation conjecture in dimension three

摘录 / EXCERPT: Let A_i, B_j, C_k, D, and F be generic 3×3 matrices, and let $X\in\mathcal{M}_3(\overline{K})$. Define ... The maximal quadratic equation conjecture. The equation...

conjecture | open | MathDB 页面 | 原文线索

V2-017137 | MATHDB-325338

Gordon-Stafford sheaf correspondence conjecture

摘录 / EXCERPT: Let $L_{\frac mn}$ be the unique finite-dimensional irreducible representation of the rational Cherednik algebra with parameter $c=\frac mn$, and let the Gordon-Stafford map send f...

conjecture | open | MathDB 页面 | 原文线索

V2-017138 | MATHDB-325344

Tate's analytic conjecture for abelian varieties

摘录 / EXCERPT: Tate's analytic conjecture. If K is a number field, $L(H_\ell^{2k}(\bar A),s)$ has a meromorphic continuation to $\operatorname{Re}(s)=1+k$ and

conjecture | open | MathDB 页面 | 原文线索

V2-017139 | MATHDB-325355

The length 2 Polydegree Conjecture

摘录 / EXCERPT: Let ${\cal G}$ be the group of polynomial automorphisms of ${\mathbb A}^2_{\mathbb C}$, endowed with its ind-variety Zariski topology, and let ${\cal G}_d$ denote the set of automo...

conjecture | open | MathDB 页面 | 原文线索

V2-017140 | MATHDB-325359

Colombo-Van Geemen nonvanishing conjecture for tautological cycles on Jacobians

摘录 / EXCERPT: Let C be a smooth very general curve of genus $g\geq 2k+1$, and let JC be its Jacobian. Write Z_k for the k-th Beauville component of the cycle associated with C in...

conjecture | open | MathDB 页面 | 原文线索

V2-017141 | MATHDB-325360

Nonvanishing conjecture for tautological cycles on Jacobians of plane curves

摘录 / EXCERPT: Let C be a very general plane curve of degree d, with genus $g=g(C)$ determined by ... Let Z_{d-3} denote the $(d-3)$-rd Beauville component of the cycle associated with C in...

conjecture | open | MathDB 页面 | 原文线索

V2-017142 | MATHDB-325361

Voisin-Pura Bloch-Beilinson filtration conjecture modulo algebraic equivalence

摘录 / EXCERPT: Let Y be a smooth projective variety and let F be a Bloch-Beilinson filtration with rational coefficients on its Chow groups. Assume that F also induces a filtration on Chow...

conjecture | open | MathDB 页面 | 原文线索

V2-017143 | MATHDB-325368

Voisin's injectivity conjecture for the tautological Chow ring of irreducible symplectic varieties

摘录 / EXCERPT: Voisin's injectivity conjecture. The restriction

conjecture | open | MathDB 页面 | 原文线索

V2-017144 | MATHDB-325387

The period conjecture for Hilbert modular abelian varieties

摘录 / EXCERPT: Let $A=A_f$ be the abelian variety associated to a Hilbert modular newform f, and let Ω_A be its real period. For each embedding $\sigma\colon M_f\to\mathbf{C}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017145 | MATHDB-325388

The rational period conjecture for Hilbert modular abelian varieties

摘录 / EXCERPT: Let $A=A_f$ be the abelian variety associated to a Hilbert modular newform f, and let Ω_A be its real period. For each embedding σ of the coefficient field of f,...

conjecture | open | MathDB 页面 | 原文线索

V2-017146 | MATHDB-325389

The quadratic period conjecture for base-changed abelian varieties

摘录 / EXCERPT: Let E/F be the quadratic extension in the paper, let A be the abelian variety associated to a Hilbert modular newform f, and let A_E denote its base change to E. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-017147 | MATHDB-325391

The spinor containment conjecture for codimension-four Gorenstein schemes

摘录 / EXCERPT: Let S be the polynomial ring and let I_R be the ideal associated with a classifying map $\alpha\in\operatorname{SpH}_k(S)$ as in Lemma. For a spinor set $J\cup J^c$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017148 | MATHDB-325411

The Lagrangian-reduction conjecture for link augmentation varieties

摘录 / EXCERPT: Let K be a non-split link, let K_n be its nth component, and let $K-K_n$ be the remaining link. Let $V_K(n)$ and $V_{K_n}(1)$ be the corresponding augmentation varieties, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017149 | MATHDB-325413

Asymptotic existence of nodal curve components with expected moduli

摘录 / EXCERPT: Asymptotic existence conjecture. Fix an integer $r\geq 3$. There is a function $\mathbb{O}_r:\mathbb{N}\to\mathbb{N}$ such that $\lim_{g\to+\infty}\mathbb{O}_r(g)/g=0$ and an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-017150 | MATHDB-325414

Asymptotic existence of nodal curve components with bounded numbers of nodes

摘录 / EXCERPT: Bounded-nodes existence conjecture. Fix integers $r\geq 3$ and $t\geq 0$. There is a function $\mathbb{O}_{r,t}:\mathbb{N}\to\mathbb{N}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-017151 | MATHDB-325424

The test function conjecture for Shimura varieties

摘录 / EXCERPT: Let G be the reductive group in the Shimura datum, let $K_p\subset G(\mathbb Q_p)$ be the prescribed level subgroup, and let $K_{p^r}\subset G(\mathbb Q_{p^r})$ be its base chang...

conjecture | open | MathDB 页面 | 原文线索

V2-017152 | MATHDB-325425

The test function conjecture via nearby cycles

摘录 / EXCERPT: Let Sh_{K_p} be the Shimura variety, let $\mathcal M_{K_p}/\mathcal O_E$ be a natural integral model, and let $R\Psi_x$ denote the nearby-cycles complex at a point x of its spe...

conjecture | open | MathDB 页面 | 原文线索

V2-017153 | MATHDB-325426

The Kottwitz conjecture for parahoric test functions

摘录 / EXCERPT: Assume ${\bf G}_{\mathbb Q_{p^r}}$ is unramified and $K_p\subset{\bf G}(\mathbb Q_p)$ is parahoric. Let $d={\rm dim}(Sh_{K_p})$, and let $z_{-\mu,j}$ be the Bernstein function asso...

conjecture | open | MathDB 页面 | 原文线索

V2-017154 | MATHDB-325443

The minimal model conjecture for nonsingular varieties

摘录 / EXCERPT: Let X be a nonsingular projective variety over Z. Write X/Z for the morphism to the base, and interpret pseudo-effectivity, minimal models, and Mori fiber spaces relatively o...

conjecture | open | MathDB 页面 | 原文线索

V2-017155 | MATHDB-325444

The abundance conjecture for minimal models with terminal singularities

摘录 / EXCERPT: Let X/Z be a normal projective variety with terminal singularities and $\mathbb{Q}$-Cartier canonical divisor. Interpret nefness and semi-ampleness relatively over Z. The abund...

conjecture | open | MathDB 页面 | 原文线索

V2-017156 | MATHDB-325460

Higher-genus quasimap wall-crossing conjecture for Fano index at least two

摘录 / EXCERPT: Let $(W,{\bf G},\theta)$ be a Fano triple of Fano index at least 2, and let $\langle\sigma_1\psi_1^{a_1},\ldots,\sigma_k\psi_k^{a_k}\rangle^\varepsilon_{g,k,\beta}$ denote its de...

conjecture | open | MathDB 页面 | 原文线索

V2-017157 | MATHDB-325473

Weak and strong good-model conjectures for semistable surface families

摘录 / EXCERPT: Let $f:S\to\Delta$ be a semistable family of surfaces, endowed with a line bundle $\mathcal L$, and let δ be a positive integer. A semistable model is δ-absolutely go...

conjecture | open | MathDB 页面 | 原文线索

V2-017158 | MATHDB-325474

Irreducibility conjecture for binodal and cuspidal plane-section curves

摘录 / EXCERPT: Let $S\subset\mathbf{P}^3$ be a general surface of degree $d\geq 4$. Let $V_2(S,\mathcal{O}_S(1))$ be the curve of binodal plane sections of S, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017159 | MATHDB-325495

Cautis-Logvinenko purity conjecture for objects on the G-Hilbert scheme

摘录 / EXCERPT: Let G be a finite, non-abelian subgroup of $\operatorname{SL}(3,\mathbb{C})$, let C be the chamber defining ${G}\operatorname{-Hilb}(\mathbb{C}^3)$, and let S_ρ, L_ρ,...

conjecture | open | MathDB 页面 | 原文线索

V2-017160 | MATHDB-325503

Irreducibility conjecture for the rank varieties of the binary tree model

摘录 / EXCERPT: Let T be a binary tree, let $\mathcal{M}_T$ be its general Markov model, and consider the rank varieties described in (1) and (2): the varieties defined by the rank conditions on...

conjecture | open | MathDB 页面 | 原文线索

V2-017161 | MATHDB-325516

One-parameter Albanese smoothness conjecture for nef-anticanonical varieties

摘录 / EXCERPT: Let M be a normal projective variety with at most canonical Gorenstein singularities. Let $\varphi\colon M\rightarrow C$ be a fibration onto a smooth curve C such that...

open_problem | open | MathDB 页面 | 原文线索

V2-017162 | MATHDB-325549

The completion obstruction for complete subvarieties of moduli-space coverings

摘录 / EXCERPT: Let ${\mathcal M}\to {\mathcal M}_{g,n}$ be a finite orbifold covering of the moduli space of smooth genus-g curves with n marked points, and let X be a smooth complete subva...

conjecture | open | MathDB 页面 | 原文线索

V2-017163 | MATHDB-325573

Classification conjecture for globally generated vector bundles with bounded first Chern class

摘录 / EXCERPT: Let E) be a globally generated vector bundle on $\mathbb{P}^n$, with $n\geq 2$ and $1\leq c_1(E)\leq n-1$, such that ... Write $P(B)$ for the bundle associated with a direct sum...

conjecture | open | MathDB 页面 | 原文线索

V2-017164 | MATHDB-325588

Hyperbolicity implies general type conjecture

摘录 / EXCERPT: Hyperbolicity implies general type conjecture. If X is hyperbolic, then X should be of general type; equivalently, K_X should be big and X should have maximal Kodaira dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-017165 | MATHDB-325594

Faber-van der Geer conjectural formula for strict endoscopic Euler characteristics

摘录 / EXCERPT: Let $\mathcal{A}_2$ be the moduli space of principally polarized abelian surfaces, let $\mathbb{V}(\lambda)$ be the local system of weight $\lambda=(l,m)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017166 | MATHDB-325596

The factoriality conjecture for threefolds with ordinary multiple points

摘录 / EXCERPT: Let $X \subset \mathbb{P}^4$ be a hypersurface of degree d whose singular locus consists of k ordinary multiple points $p_1,\ldots,p_k$ of multiplicities $m_1,\ldots,m_k$. Fact...

conjecture | open | MathDB 页面 | 原文线索

V2-017167 | MATHDB-325597

Hodge-number conjecture for Landau-Ginzburg models

摘录 / EXCERPT: Let X be a smooth Fano variety of dimension $N\geq 3$, and let $LG(X)$ be its N-dimensional Landau-Ginzburg model. Define ... Hodge-number conjecture for Landau-Ginzburg models....

conjecture | open | MathDB 页面 | 原文线索

V2-017168 | MATHDB-325604

Kuznetsov's Fourier-Mukai conjecture for categorical data

摘录 / EXCERPT: Let X and Y be irreducible smooth projective schemes over $\mathbb{C}$, of dimensions d_X and d_Y, respectively, related by the categorical data in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017169 | MATHDB-325605

The verepresentability and Torelli conjecture for Example (E8)

摘录 / EXCERPT: Let X and Y be schemes as in Example (E8), and let $J(X)$ and $J(Y)$ denote their intermediate Jacobians. The E8 verepresentability and Torelli conjecture. The scheme Y is ve...

conjecture | open | MathDB 页面 | 原文线索

V2-017170 | MATHDB-325622

Uniform Rasmussen-Tamagawa conjecture for abelian varieties

摘录 / EXCERPT: Let F be a number field and let $g>0$. For each prime ℓ, let $\mathscr{A}(F,g,\ell)$ be the set of F-isomorphism classes of g-dimensional abelian varieties A/F such th...

conjecture | open | MathDB 页面 | 原文线索

V2-017171 | MATHDB-325652

Unirationality conjecture for norm surfaces defined by sextic polynomials

摘录 / EXCERPT: Let k be a number field and let K/k be a cyclic extension of degree 3. For a polynomial $f\in k[t]$ of degree 6, let $\mathcal{S}_{f}$ denote the variety defined by the nor...

conjecture | open | MathDB 页面 | 原文线索

V2-017172 | MATHDB-325682

Nash's rationality conjecture for compact smooth manifolds

摘录 / EXCERPT: Nash's rationality conjecture. There exists a rational variety X whose real locus is diffeomorphic to M:

conjecture | open | MathDB 页面 | 原文线索

V2-017173 | MATHDB-325683

The geometric Seifert conjecture for orientable real unir ruled threefolds

摘录 / EXCERPT: The geometric Seifert conjecture. The orientable geometric real components of orientable real unir ruled varieties of dimension 3 are exactly the orientable Seifert manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-017174 | MATHDB-325705

Conjecture on secant varieties of Segre products with completely decomposable forms

摘录 / EXCERPT: Segre-split secant conjecture. If $m\geq 1$, then $\sigma_s(\operatorname{Seg}(\mathbb P^m\times\operatorname{Split}_d(\mathbb P^n))$ is nondefective for all cases except those out...

conjecture | open | MathDB 页面 | 原文线索

V2-017175 | MATHDB-325748

Lan-Sheng-Zuo's conjecture on nilpotent semistable Higgs bundles

摘录 / EXCERPT: Let k be a field of positive characteristic, and let a Higgs bundle on a smooth projective variety over k be nilpotent if its Higgs field is nilpotent, with its exponent define...

conjecture | open | MathDB 页面 | 原文线索

V2-017176 | MATHDB-325757

Shokurov's toric characterization conjecture for log canonical pairs

摘录 / EXCERPT: Shokurov's conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-017177 | MATHDB-325758

McKernan's complexity conjecture for log canonical pairs

摘录 / EXCERPT: McKernan's conjecture. For (X,D) as above:

conjecture | open | MathDB 页面 | 原文线索

V2-017178 | MATHDB-325762

Monadicity conjecture for the point functor on eventually coconnective affine DG schemes

摘录 / EXCERPT: Monadicity conjecture. If Z is eventually coconnective, then the functor $\iota^{\operatorname{QCoh},!}$ is monadic. The paper states this as a conjectural assertion in the study...

conjecture | open | MathDB 页面 | 原文线索

V2-017179 | MATHDB-325766

Mori dream space conjecture for log canonical Fano varieties

摘录 / EXCERPT: Log canonical Fano Mori dream space conjecture. Then X is a Mori dream space.

conjecture | open | MathDB 页面 | 原文线索

V2-017180 | MATHDB-325814

Rigidity conjecture for families with decomposable transcendental Hodge structures

摘录 / EXCERPT: Let $\pi: \mathscr{S} \to B$ be a family of surfaces over an analytic space B, and call a point of B very general if it lies outside a countable union of proper analytic subset...

conjecture | open | MathDB 页面 | 原文线索

V2-017181 | MATHDB-325846

Irregular graphical Deligne-Simpson conjecture

摘录 / EXCERPT: Choose a marking of each conjugacy class $\breve{\mathcal{C}}_i\subset \operatorname{GL}(V_i)$ for all $i\in I$. Gluing legs to obtain a supernova graph $\widehat{\Gamma}$ with nod...

conjecture | open | MathDB 页面 | 原文线索

V2-017182 | MATHDB-325847

Weight-zero conjecture for Grothendieck classes

摘录 / EXCERPT: Let X be a smooth quasi-projective variety over $\overline{\mathbb Q}$, and let j and k be integers. Let $H^k_{\rm dRB}(X,\mathbb Q(j))_{\rm Gr}$ denote the Grothendieck clas...

conjecture | open | MathDB 页面 | 原文线索

V2-017183 | MATHDB-325872

The uniform boundedness conjecture for torsion on abelian varieties over finitely generated fields

摘录 / EXCERPT: Let $\Bbbk$ be a finitely generated field over $\mathbb{Q}$, and let d be a positive integer. For an abelian variety A of dimension d over $\Bbbk$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-017184 | MATHDB-325915

The generalized test-ideal and multiplier-ideal reduction conjecture for singular varieties

摘录 / EXCERPT: Let X be a normal variety over a field k of characteristic zero, let Δ be a $\mathbb{Q}$-divisor such that $K_X+\Delta$ is $\mathbb{Q}$-Cartier, and let $\mathfrak{a}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-017185 | MATHDB-325931

Stable independence conjecture for the strata and boundary algebras

摘录 / EXCERPT: Let ... lambda{2i+1}$ and the classes of boundary strata, and no polynomial in the Dm, is identically zero for g sufficiently large. This predicts algebraic independence of the i...

conjecture | open | MathDB 页面 | 原文线索

V2-017186 | MATHDB-325950

Conjecture on crystalline points in local deformation spaces

摘录 / EXCERPT: Fix a prime p and an integer $d\geq 1$, and let X_d be the p-adic character variety of $G=\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ in dimension d. A point...

conjecture | open | MathDB 页面 | 原文线索

V2-017187 | MATHDB-325961

The b.-H. conjecture for projective space

摘录 / EXCERPT: The b.-H. conjecture. Projective space $\mathbb{P}^n$ is b.-H. for any $n\geq 2$. More precisely, there exists a rational fibration

conjecture | open | MathDB 页面 | 原文线索

V2-017188 | MATHDB-325962

Shokurov's toric equality conjecture for log canonical pairs

摘录 / EXCERPT: Let X be a normal $\mathbb{Q}$-factorial algebraic variety with a $\mathbb{Q}$-boundary divisor $D=\sum d_iD_i$, where $0\leq d_i\leq 1$ and the D_i are prime Weil divisors. As...

conjecture | open | MathDB 页面 | 原文线索

V2-017189 | MATHDB-325963

Integrality conjecture for the Chern character

摘录 / EXCERPT: Let p be a prime number and n an integer. Let S be a regular scheme, X a scheme of finite type over S, d an integer, and $x\in K_0'(X)_{(d)}$. For each integer...

conjecture | open | MathDB 页面 | 原文线索

V2-017190 | MATHDB-325967

Generalization of Reichstein-Youssin's fixed-point theorem

摘录 / EXCERPT: Generalized fixed-point conjecture. If $\operatorname{rdim}_k(H_i)\leq d$ for all $i=1,\dots,r$, then there exists a d-dimensional k-variety X with a faithful G-action and...

conjecture | open | MathDB 页面 | 原文线索

V2-017191 | MATHDB-325968

Zilber's conjecture on unlikely intersections in algebraic tori

摘录 / EXCERPT: Let ${\mathbb G}_{\rm m}^N=(\overline{\mathbb Q}^{\times})^N$ be an algebraic torus, and let W be a subvariety of ${\mathbb G}_{\rm m}^N$. For an algebraic subgroup, call it prop...

conjecture | open | MathDB 页面 | 原文线索

V2-017192 | MATHDB-326004

String-space description conjecture for Bott-Samelson Okounkov bodies

摘录 / EXCERPT: Fix a Borel subgroup $B\subset G$ and a sequence $I=(i_1,\ldots,i_\ell)$ of simple roots defining the Bott-Samelson variety R_I. Let $P_v(\mathcal L)$ be the Okounkov body of a l...

conjecture | open | MathDB 页面 | 原文线索

V2-017193 | MATHDB-326009

Caporaso's Brill-Noether generality conjecture for maximally symmetric trivalent graphs

摘录 / EXCERPT: Let $g\geq 2$. A trivalent graph of genus g with largest possible automorphism group means a trivalent graph whose automorphism group has maximal size among trivalent graphs of g...

conjecture | open | MathDB 页面 | 原文线索

V2-017194 | MATHDB-326010

Non-generality conjecture for sufficiently large maximally symmetric trivalent graphs

摘录 / EXCERPT: Let Γ be a trivalent maximally symmetric graph or multigraph of genus g; multigraphs may have loops or may be loopless. A graph is Brill-Noether general if, for every pair...

conjecture | open | MathDB 页面 | 原文线索

V2-017195 | MATHDB-326017

Hausel-Rodriguez-Villegas boundary-complex conjecture for character varieties

摘录 / EXCERPT: Let $\mathcal{R}_{g,\boldsymbol{\mu}}$ be the character variety, let $\overline{\mathcal{R}_{g,\boldsymbol{\mu}}}$ be the compactification constructed from the representation varie...

conjecture | open | MathDB 页面 | 原文线索

V2-017196 | MATHDB-326027

The exponential Betti-number bound for hyperelliptic theta intersections

摘录 / EXCERPT: Exponential Betti-number bound. There exists a universal constant N such that

conjecture | open | MathDB 页面 | 原文线索

V2-017197 | MATHDB-326028

The exponential fiber-cohomology bound for theta correspondences

摘录 / EXCERPT: Exponential fiber-cohomology bound. There exists a universal constant c, independent of g, the field, the curve, a, and b, such that for every $y\in\widetilde{J}^{2g-a-b}$,

conjecture | open | MathDB 页面 | 原文线索

V2-017198 | MATHDB-326029

The uniform exponential bound for translated theta-intersection cohomology

摘录 / EXCERPT: Uniform exponential cohomology bound. There exists a universal constant $N>0$ such that, for every curve of genus $g\gg a,b$ and every such L,

conjecture | open | MathDB 页面 | 原文线索

V2-017199 | MATHDB-326040

Schwede's Du Bois and dense F-injective type conjecture

摘录 / EXCERPT: Let X be a reduced scheme of finite type over an algebraically closed field of characteristic 0. A scheme is of dense F-injective type if it has reductions to characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-017200 | MATHDB-326044

Étale 1-motive realization conjecture in positive characteristic

摘录 / EXCERPT: Étale 1-motive realization conjecture. These functors should satisfy, functorially in X, for every prime $\ell\ne p$,

conjecture | open | MathDB 页面 | 原文线索

V2-017201 | MATHDB-326053

The partial Frobenius conjecture for cohomology of quaternionic Shimura varieties

摘录 / EXCERPT: For each $\mathfrak{p}\in\Sigma_p$, let $n_{\mathfrak{p}}$ be the smallest positive integer such that $\sigma_{\mathfrak{p}}^{n_{\mathfrak{p}}}\mathtt{S}_\infty=\mathtt{S}_\infty$,...

conjecture | open | MathDB 页面 | 原文线索

V2-017202 | MATHDB-326055

Sufficiency of the ampleness conditions for automorphic line bundles

摘录 / EXCERPT: Ampleness sufficiency conjecture. The conditions in Theorem and Corollary are also sufficient for $[\omega^{\underline t}]$ to be ample.

conjecture | open | MathDB 页面 | 原文线索

V2-017203 | MATHDB-326076

Toric polynomial generator conjecture for complex cobordism

摘录 / EXCERPT: Toric polynomial generator conjecture. For each $n\geq 1$, there exists a smooth projective toric variety whose cobordism class can be chosen for the polynomial generator...

conjecture | open | MathDB 页面 | 原文线索

V2-017204 | MATHDB-326077

Number-theoretic conjecture for toric polynomial generators in the remaining even dimensions

摘录 / EXCERPT: Number-theoretic conjecture. There exists an integer

conjecture | open | MathDB 页面 | 原文线索

V2-017205 | MATHDB-326086

The irreducibility implication for ordinary and nilpotent commuting varieties

摘录 / EXCERPT: Irreducibility implication. If $C_r(\mathfrak{gl}_n)$ is irreducible, then $C_r(\mathcal{N}_n)$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-017206 | MATHDB-326087

The parabolic component conjecture for nilpotent commuting varieties

摘录 / EXCERPT: Parabolic component conjecture. The variety V_P is an irreducible component of $C_r(\mathcal{N}_n)$ for all $n,r\ge 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-017207 | MATHDB-326088

The monotonicity conjecture for reducibility of nilpotent commuting varieties

摘录 / EXCERPT: Monotonicity conjecture. If $C_r(\mathcal{N}_n)$ is reducible, then $C_r(\mathcal{N}_{n+1})$ is reducible.

conjecture | open | MathDB 页面 | 原文线索

V2-017208 | MATHDB-326107

Calabi-Yau rational curve conjecture in dimension m

摘录 / EXCERPT: Calabi-Yau rational curve conjecture in dimension m. The variety X should contain a rational curve.

conjecture | open | MathDB 页面 | 原文线索

V2-017209 | MATHDB-326108

The non-algebraic-Lang-hyperbolicity conjecture for absolutely minimal Calabi-Yau varieties

摘录 / EXCERPT: Non-algebraic-Lang-hyperbolicity conjecture. The variety X is not algebraic Lang hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-017210 | MATHDB-326109

Nef-dimension abundance conjecture for minimal varieties

摘录 / EXCERPT: Nef-dimension abundance conjecture. If X has at worst klt singularities, then K_X should be semi-ample, meaning that $|mK_X|$ is base-point free for some $m>0$. If X has at w...

conjecture | open | MathDB 页面 | 原文线索

V2-017211 | MATHDB-326123

The supersimple field conjecture

摘录 / EXCERPT: A supersimple field is a field whose theory is supersimple; a field is PAC if every absolutely irreducible variety over it has a rational point, and its absolute Galois group is sm...

conjecture | open | MathDB 页面 | 原文线索

V2-017212 | MATHDB-326217

The cubic growth conjecture for the Euclidean distance degree of affine multiview varieties

摘录 / EXCERPT: Let $A_1,A_2,\ldots,A_n$ be general projective cameras, and let $X_n\subseteq\mathbb{R}^{2n}$ be the affine multiview variety obtained by dehomogenizing the multiview variety assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-017213 | MATHDB-326252

Concavity conjecture for the variation of cycle classes

摘录 / EXCERPT: Let X be an integral projective variety, and let $N_k(X)_{\mathbb{Z}}$ denote the group of numerical classes of k-cycles. A class is pseudo-effective if it lies in the pseudo-e...

conjecture | open | MathDB 页面 | 原文线索

V2-017214 | MATHDB-326262

Tian's rationality conjecture for the global log canonical threshold

摘录 / EXCERPT: Tian's rationality conjecture. There exists an effective bQ-divisor $D\simq -K_X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017215 | MATHDB-326263

Cat property conjecture for smooth Fano varieties

摘录 / EXCERPT: Cat property conjecture. X satisfies the Cat property if and only if every effective bQ-divisor $D\simq -K_X$ is log canonical in codimension 1.

conjecture | open | MathDB 页面 | 原文线索

V2-017216 | MATHDB-326271

Characteristic-cycle image conjecture for quantized quiver varieties

摘录 / EXCERPT: Let Q be a quiver without loops, let λ be a parameter, and let $\mathcal{A}_\lambda(v)$ be the corresponding quantized quiver-variety algebra. Let CC^λ b...

conjecture | open | MathDB 页面 | 原文线索

V2-017217 | MATHDB-326272

Singular-locus hyperplane conjecture for quantizations of symplectic resolutions

摘录 / EXCERPT: Let X be a symplectic resolution of the affine variety X_0, let $\mathfrak{p}=H^2(X)$, and let $\mathcal{A}_\lambda$ be the algebra of global sections of the quantization speci...

conjecture | open | MathDB 页面 | 原文线索

V2-017218 | MATHDB-326273

Categorical Kac-Moody action conjecture for quantized quiver varieties

摘录 / EXCERPT: Let Q be a quiver, let λ be a parameter, let Π^θ be the system of simple roots for $\mathfrak{a}$ corresponding to the Weyl chamber containing θ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017219 | MATHDB-326274

Categorical Heisenberg wall-crossing conjecture for quantized quiver varieties

摘录 / EXCERPT: Let λ,λ' be parameters with integral difference separated only by the affine wall, and let θ,θ' be the corresponding stability conditions. Categorical Hei...

conjecture | open | MathDB 页面 | 原文线索

V2-017220 | MATHDB-326293

Fock-Goncharov dual bases conjecture for cluster varieties

摘录 / EXCERPT: Fock-Goncharov dual bases conjecture. The set N parameterizes a canonical basis of

conjecture | open | MathDB 页面 | 原文线索

V2-017221 | MATHDB-326300

The J-flow numerical criterion for existence of a solution

摘录 / EXCERPT: Let (M,ω) be a compact Kähler manifold of dimension n, let α be a second Kähler metric on M, and let c be determined by ... For a p-dimensional subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-017222 | MATHDB-326307

b-semi-ampleness conjecture for analytic lc-trivial fibrations

摘录 / EXCERPT: Let X be a normal complex analytic variety in Fujiki's class $\mathcal C$, let Δ be a $\mathbb Q$-divisor such that $K_X+\Delta$ is $\mathbb Q$-Cartier, and let $f:X\to Y$...

conjecture | open | MathDB 页面 | 原文线索

V2-017223 | MATHDB-326311

The conjecture that convex closed semialgebraic sets are spectrahedral shadows

摘录 / EXCERPT: Let a spectrahedral shadow be the affine projection of a spectrahedron, that is, a set of the form ... where $F_k \in {\mathbb S}^m$ for $k=0,1,\ldots,n$ and $G_l \in {\mathbb S}^m$...

conjecture | open | MathDB 页面 | 原文线索

V2-017224 | MATHDB-326352

The Fourier-Mukai formulas for derived duals and adjunction maps

摘录 / EXCERPT: Let Z and X be separated schemes of finite type over k. Let $\mathcal{A}$ and $\mathcal{B}$ be DG-enhancements of $D(Z)$ and $D(X)$, and let $\bar{S}\in D(Z\times X)$ corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-017225 | MATHDB-326380

Compact real BSS separation conjecture

摘录 / EXCERPT: The compact real BSS model has complexity classes $\mathbf{P}_\mathbb{R}^c$ and $\mathbf{NP}_\mathbb{R}^c$. Compact real BSS separation conjecture. ... This is the compact counterpar...

conjecture | open | MathDB 页面 | 原文线索

V2-017226 | MATHDB-326383

The Fourier decomposition conjecture for zero-cycles on hyperkähler varieties

摘录 / EXCERPT: Fourier decomposition conjecture. There exists such an L inducing a canonical splitting

conjecture | open | MathDB 页面 | 原文线索

V2-017227 | MATHDB-326384

The canonical Beauville-Bogomolov cycle conjecture for powers of hyperkähler varieties

摘录 / EXCERPT: Let F be a hyperkähler variety, let $\mathfrak{B}\in\mathrm{H}^4(F\times F,\mathbb{Q})$ be its Beauville-Bogomolov class, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017228 | MATHDB-326389

Noether-Castelnuovo conjecture for irregular Gorenstein 3-folds of Albanese dimension one

摘录 / EXCERPT: Noether-Castelnuovo conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-017229 | MATHDB-326393

Kontsevich's p-support Lagrangian conjecture for holonomic D-modules

摘录 / EXCERPT: Kontsevich's conjecture. There exists a dense open subset U in $\operatorname{Spec} R$ such that for all $(p,v)\in U$, $\operatorname{Supp}_{p,v}(\mathcal{N}_R)$ is Lagrangian.

conjecture | open | MathDB 页面 | 原文线索

V2-017230 | MATHDB-326408

Hausel-Letellier-Villegas' mixed Hodge polynomial formula for character varieties

摘录 / EXCERPT: Let ${\bf C}$ be a generic tuple of conjugacy classes of $\operatorname{GL}_n(\mathbb{C})$, with type ${\bm\omega}\in\overline{\mathbf{T}}^o$, and let $d_{\bf C}$ denote the di...

conjecture | open | MathDB 页面 | 原文线索

V2-017231 | MATHDB-326409

The explicit mixed Hodge polynomial formula for character varieties of fixed type

摘录 / EXCERPT: Let ${\bf C}$ be a generic tuple of conjugacy classes of $\operatorname{GL}_n(\mathbb{C})$ of type ${\bm\omega}\in\overline{\mathbf{T}}^o$. Let $\iota^k({\bm\omega})$ be the ty...

conjecture | open | MathDB 页面 | 原文线索

V2-017232 | MATHDB-326418

Bridgeland stability conjecture for the double tilt of an abelian threefold

摘录 / EXCERPT: Let X be an abelian threefold, let $B\in\operatorname{NS}_{\mathbb{R}}(X)$, and let $\omega\in\operatorname{NS}_{\mathbb{R}}(X)$ be an ample class. With $\mathcal{B}_{\omega,B}$...

conjecture | open | MathDB 页面 | 原文线索

V2-017233 | MATHDB-326420

Log-concavity conjecture for Milnor algebra local cohomology

摘录 / EXCERPT: Let S be the graded polynomial ring defining a projective hypersurface $V(f):f=0$, and let $M(f)=S/J_f$ be its Milnor algebra, where J_f is the Jacobian ideal. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017234 | MATHDB-326421

Mobility log-concavity conjecture

摘录 / EXCERPT: Let X be a projective variety of dimension n. For a class in $N_k(X)$, let mob denote its mobility, and let $\operatorname{\overline{Eff}}_k(X)$ be the pseudo-...

conjecture | open | MathDB 页面 | 原文线索

V2-017235 | MATHDB-326422

Effectivity conjecture for negative parts of Zariski decompositions

摘录 / EXCERPT: Let X be a projective variety, let α be a cycle class admitting a Zariski decomposition, and write ... Here $P(\alpha)$ and $N(\alpha)$ are respectively the positive and neg...

conjecture | open | MathDB 页面 | 原文线索

V2-017236 | MATHDB-326423

Pullback conjecture for basepoint-free classes and movable transforms

摘录 / EXCERPT: Let X be a smooth variety and let $\alpha\in\operatorname{BPF}_k(X)$, where $\operatorname{BPF}_k(X)$ denotes the cone of basepoint-free k-cycle classes. Let $\pi:Y\to X$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-017237 | MATHDB-326450

The generalized André-Oort conjecture for mixed Shimura varieties

摘录 / EXCERPT: A mixed Shimura variety is an algebraic variety arising from a mixed Shimura datum; its special points and special subvarieties are the distinguished points and subvarieties define...

conjecture | open | MathDB 页面 | 原文线索

V2-017238 | MATHDB-326454

Polynomial growth conjecture for component ideal degrees of matrix rigidity varieties

摘录 / EXCERPT: Fix $0<\epsilon<1$ and $0<\delta<1$. Set $r=\epsilon n$ and $s=n^{1+\delta}$. The notation $\hat{\mathcal R}[n,r,s]$ denotes the relevant matrix rigidity variety, and its irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-017239 | MATHDB-326455

Well-presented generators conjecture for joins of secant varieties and coordinate spaces

摘录 / EXCERPT: Let $\mathcal I(\boldsymbol J(\sigma_{r,n},L^S))$ be the ideal of the join of the relevant secant variety $\sigma_{r,n}$ and coordinate linear space L^S. A generator is called we...

conjecture | open | MathDB 页面 | 原文线索

V2-017240 | MATHDB-326461

Grothendieck's infinitesimal Hodge conjecture

摘录 / EXCERPT: Infinitesimal Hodge conjecture. Statement (i) is equivalent to the existence of an element $\xi\in K_0(X)_{\mathbb Q}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017241 | MATHDB-326462

Grothendieck's variational Hodge conjecture

摘录 / EXCERPT: Variational Hodge conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-017242 | MATHDB-326468

Rohlin's maximality conjecture after Shustin

摘录 / EXCERPT: Fix an integer $m\ge 1$, and consider real schemes of degree m, ordered in the lattice of all real schemes. A scheme is of type I when its real curve is dividing. Rohlin's maxima...

conjecture | open | MathDB 页面 | 原文线索

V2-017243 | MATHDB-326470

One-oval rigidity conjecture

摘录 / EXCERPT: Let a smooth real plane curve be of even degree, so that an oval is null-homotopic in $\mathbb RP^2$. Call a scheme rigid when any two curves representing it are rigid-isotopic. On...

conjecture | open | MathDB 页面 | 原文线索

V2-017244 | MATHDB-326485

Modular interpretation conjecture for smooth resolutions of stable-map moduli

摘录 / EXCERPT: Modular interpretation conjecture. The stack $\widetilde{M}_g(\mathbb{P}^n,d)$ inherits a moduli meaning. If $d>2g-2$, its main component is smooth and the entire stack has at wors...

conjecture | open | MathDB 页面 | 原文线索

V2-017245 | MATHDB-326494

Mirror symmetry conjecture for products of general-type hypersurfaces

摘录 / EXCERPT: Let $Z_1,\ldots,Z_t$ be smooth varieties in projective spaces, each either Calabi-Yau or of general type, defined by the polynomials w_j as in the source. Form the grading group...

conjecture | open | MathDB 页面 | 原文线索

V2-017246 | MATHDB-326498

Murre's filtration conjecture for Chow groups

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, and let $\operatorname{CH}^l(X)$ denote its Chow group for $0\leq l\leq n$. A decreasing filtration F^ν for $\nu\leq 0$...

conjecture | open | MathDB 页面 | 原文线索

V2-017247 | MATHDB-326503

The torsion conjecture for admissible subcategories with vanishing Hochschild homology

摘录 / EXCERPT: Let X be a smooth projective variety over a field k of characteristic zero, and let $\mathcal{A}\subset\mathcal{D}^b(X)$ be an admissible subcategory, meaning that its inclusio...

conjecture | open | MathDB 页面 | 原文线索

V2-017248 | MATHDB-326565

Cheltsov-Shramov conjecture on Klein-group Mori fibre spaces over the projective line

摘录 / EXCERPT: Cheltsov-Shramov conjecture. The varieties

conjecture | open | MathDB 页面 | 原文线索

V2-017249 | MATHDB-326574

Amerik-Bogomolov-Rovinsky-Medvedev-Scanlon dense-orbit conjecture

摘录 / EXCERPT: Let X be a quasiprojective variety defined over an algebraically closed field K of characteristic 0, and let $\Phi:X\longrightarrow X$ be an endomorphism defined over K. As...

conjecture | open | MathDB 页面 | 原文线索

V2-017250 | MATHDB-326581

Beilinson-Bloch type conjecture for low-degree Chow groups of complete intersections

摘录 / EXCERPT: Let k be a field and let X be a smooth projective k-subscheme of $\mathbb{P}^n$ that is a complete intersection of multidegree $(d_1,\ldots,d_r)$. Define ... where $[-]$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-017251 | MATHDB-326588

Positive proportion conjecture for locally soluble fibres

摘录 / EXCERPT: Let X satisfy the stated assumptions, let H be an anticanonical height function, and let $\pi:Y\to X$ be a proper surjective almost smooth morphism with geometrically integral...

conjecture | open | MathDB 页面 | 原文线索

V2-017252 | MATHDB-326589

Non-split fibre conjecture for families of varieties

摘录 / EXCERPT: Let F be a number field, let X satisfy the stated assumptions, and let H be an anticanonical height function on X. Let $\pi:Y\to X$ be a non-singular proper faithfully flat...

conjecture | open | MathDB 页面 | 原文线索

V2-017253 | MATHDB-326594

Finiteness conjecture for nonabelian simple subgroups of birational groups

摘录 / EXCERPT: Let X be an irreducible algebraic variety, and let $\operatorname{Bir}(X)$ denote its birational self-maps. A finite group is contained in $\operatorname{Bir}(X)$ when an isomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-017254 | MATHDB-326603

Conjecture on canonical divisors of nonrational Hilbert modular surfaces

摘录 / EXCERPT: Let K be the real quadratic field in the paper, let ρ range over the resolution curves of the cusps of the Hilbert modular surface $Y_{cGamma({\mathscr{O}}_K,\id{a})}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017255 | MATHDB-326645

The quotient modified diagonal conjecture

摘录 / EXCERPT: The quotient modified diagonal conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-017256 | MATHDB-326667

Beltrametti-Sommese-Wisniewski nef-value fibration conjecture

摘录 / EXCERPT: Let $(X,\mathcal L)$ be a nonsingular polarized variety of dimension n, with effective log threshold $\mu(\mathcal L)$ and nef value $\tau(\mathcal L)$. Beltrametti-Sommese-Wisni...

conjecture | open | MathDB 页面 | 原文线索

V2-017257 | MATHDB-326668

Toric nef-value fibration conjecture beyond the smooth Gorenstein case

摘录 / EXCERPT: Let $(X,\mathcal L)$ be an n-dimensional toric polarized variety, not necessarily smooth or Gorenstein, and let $\mu(\mathcal L)$ and $\tau(\mathcal L)$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-017258 | MATHDB-326669

Lamy-Sebag conjecture on extending birational maps to piecewise automorphisms

摘录 / EXCERPT: Let k be an algebraically closed field and let X be a k-variety. A piecewise automorphism of X is an automorphism on each member of a finite partition of X into locally c...

conjecture | open | MathDB 页面 | 原文线索

V2-017259 | MATHDB-326683

The Lie-type conjecture on integrality of Peterson structure constants

摘录 / EXCERPT: Integrality conjecture. In this basis, non-integral structure constants occur only in Lie types D and E.

conjecture | open | MathDB 页面 | 原文线索

V2-017260 | MATHDB-326688

Milne-Ramachandran's special-value conjecture for motivic complexes

摘录 / EXCERPT: Let k be a perfect field and let $\mathsf{DM}(k)$ be a triangulated category of motivic complexes with realization functors, Tate twists, an internal Hom...

conjecture | open | MathDB 页面 | 原文线索

V2-017261 | MATHDB-326689

Milne's rationality conjecture for absolute Hodge classes

摘录 / EXCERPT: Let A be an abelian variety over $\mathbb{Q}^{\mathrm{al}}$ with good reduction A_0 over $\mathbb{F}^{\mathrm{al}}$, let $d=\dim(A)$, and let γ be an absolute Hodge clas...

conjecture | open | MathDB 页面 | 原文线索

V2-017262 | MATHDB-326705

The degree conjecture for the dormant operatic locus

摘录 / EXCERPT: The degree conjecture. The degree of the dormant operatic locus is

conjecture | open | MathDB 页面 | 原文线索

V2-017263 | MATHDB-326706

The finite-fibre conjecture for Frobenius pullback

摘录 / EXCERPT: Finite-fibre conjecture. The quotient scheme $\mathscr{Q}^{r,0}_{V'}$ is finite and reduced. This conjecture is motivated by analogous characteristic-zero results; it is presented...

conjecture | open | MathDB 页面 | 原文线索

V2-017264 | MATHDB-326707

The fibre-description conjecture for Frobenius pullback

摘录 / EXCERPT: Assume that X is an ordinary smooth projective curve and that V' is a general stable bundle of rank r. Let ... be the morphism sending V to the inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-017265 | MATHDB-326719

Purity failure conjecture for non-special semisimple groups

摘录 / EXCERPT: Let k be the ground field, let G be a non-special semisimple k-group scheme, and let X be a smooth affine k-variety. Purity failure conjecture. There exists a smooth affi...

conjecture | open | MathDB 页面 | 原文线索

V2-017266 | MATHDB-326722

The thin-set refinement of Manin's conjecture for Fano varieties

摘录 / EXCERPT: Thin-set refinement of Manin's conjecture. There exists a thin subset $Z \subset X(k)$ such that for every anticanonical height function H on X,

conjecture | open | MathDB 页面 | 原文线索

V2-017267 | MATHDB-326735

Lang-Demailly projective Kobayashi hyperbolicity conjecture

摘录 / EXCERPT: Let X be a compact complex manifold and let ω be a hermitian form on X. Suppose that X is projective algebraic, and write (iii) for the property that every holomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-017268 | MATHDB-326736

Kobayashi's ample canonical bundle conjecture for projective hyperbolic manifolds

摘录 / EXCERPT: Let X be a projective hyperbolic manifold. Kobayashi's conjecture. The canonical bundle K_X is ample. This is a necessary-condition prediction related to the Green-Griffiths-La...

conjecture | open | MathDB 页面 | 原文线索

V2-017269 | MATHDB-326748

The incidence-complex and Hitchin-base sphere correspondence

摘录 / EXCERPT: Let M_B be the character variety with a compactification whose boundary is a normal-crossings divisor, and let M_{Dol} be the Dolbeault moduli space with Hitchin base of comple...

conjecture | open | MathDB 页面 | 原文线索

V2-017270 | MATHDB-326755

PBW filtration conjecture for the character-variety CoHA

摘录 / EXCERPT: Let Σ_g be a genus-g surface, let $\mathcal{H}^{\operatorname{\mathfrak{P}}}_{Q_{\Sigma_g},W_{\Sigma_g}}$ be the associated CoHA, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017271 | MATHDB-326764

The full Mordell-Lang conjecture in positive characteristic

摘录 / EXCERPT: Let L be an algebraically closed field of positive characteristic. Let A be a semiabelian variety over L, let $X\subset A$ be an irreducible subvariety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017272 | MATHDB-326765

The finitely generated-field reduction for full Mordell-Lang

摘录 / EXCERPT: Let L_0 be a field finitely generated over ${\mathbb F}_p$, let L be an algebraic closure of L_0, and let L_0^per be the perfect closure of L_0 in L. Let A be...

conjecture | open | MathDB 页面 | 原文线索

V2-017273 | MATHDB-326768

Rank-two matrix formulation of the 4M-4 conjecture

摘录 / EXCERPT: Let $\phi_1,\ldots,\phi_{4M-5}\in\mathbb{C}^M$, and let L_{Φ} be the Hermitian matrix subspace ... Equivalently, L_{Φ} is the orthogonal complement of the span of...

conjecture | open | MathDB 页面 | 原文线索

V2-017274 | MATHDB-326797

The positive-pair conjecture for SL_n-trace equivalence

摘录 / EXCERPT: Let F_r be a free group, and call two pairs of words (u,v) and (u',v') SL_n-trace equivalent when the corresponding words have equal traces under every repre...

conjecture | open | MathDB 页面 | 原文线索

V2-017275 | MATHDB-326809

Weibel's dimension conjecture for negative K-theory

摘录 / EXCERPT: Let X be a Noetherian scheme of dimension d, and write $K_m(X)$ for its algebraic K-groups. A scheme is K_n-regular when homotopy invariance holds in degree n, equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-017276 | MATHDB-326821

Reduced local-complete-intersection conjecture for lines on the permanental hypersurface

摘录 / EXCERPT: Let $P_{n,n}^n$ be the permanental hypersurface of $n\times n$ matrices, and let $\mathbf{F}_1(P_{n,n}^n)$ be its Fano scheme of lines. Reduced local-complete-intersection conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-017277 | MATHDB-326823

Furter's conjecture on quadratic-to-linear degree-sequence inclusions

摘录 / EXCERPT: Let $a,b,c\ge 2$ be integers, and let $\mathcal{G}_{(c)}$ and $\mathcal{G}_{(a,b)}$ denote the loci of polynomial automorphisms with the indicated degree sequences. Furter's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-017278 | MATHDB-326859

Serre's maximal motive conjecture

摘录 / EXCERPT: Let E be a motive over K, and let $G_{\mathcal{M}(E)}$ be its motivic Galois group. Assume that $G_{\mathcal{M}(E)}$ is connected, and let ... be the adelic Galois representation...

conjecture | open | MathDB 页面 | 原文线索

V2-017279 | MATHDB-326861

Serre's domination by maximal motives conjecture

摘录 / EXCERPT: Let E be a motive over K, and let $G_{\mathcal{M}(E)}$ denote its motivic Galois group. Serre's domination by maximal motives conjecture. After replacing K by a finite extens...

conjecture | open | MathDB 页面 | 原文线索

V2-017280 | MATHDB-326863

Esnault-Levine-Wittenberg conjecture on ELW-indices over Henselian fields

摘录 / EXCERPT: Let K be the quotient field of an excellent, Henselian discrete valuation ring with algebraically closed residue field k, and let X be a proper K-scheme. Esnault-Levine-Wit...

conjecture | open | MathDB 页面 | 原文线索

V2-017281 | MATHDB-326871

Harbourne-Huneke odd alpha conjecture for fat point schemes

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let $\mathbb W$ be a finite set of distinct points in $\mathbb P^n$, and let $\alpha(\mathbb Y)$ denote the least de...

conjecture | open | MathDB 页面 | 原文线索

V2-017282 | MATHDB-326873

Degree conjecture for surfaces of revolution with trigonometric-polynomial support functions

摘录 / EXCERPT: Let $N\geq 2$, and let p be a trigonometric polynomial of degree N with coefficients a_N,b_N satisfying $a_N\ne 0$ or $b_N\ne 0$. Let $f\in\mathbb{R}[x,y,z]$ be the defining...

conjecture | open | MathDB 页面 | 原文线索

V2-017283 | MATHDB-326902

The quiver-variety conjecture for framed sheaves on root stacks

摘录 / EXCERPT: Let X_k be the ALE space, let ${\mathscr X}_k$ be its associated root stack with divisor ${\mathscr D}_\infty$, and let ${\mathcal F}_\infty^{0,\vec w}$ be the framing data index...

conjecture | open | MathDB 页面 | 原文线索

V2-017284 | MATHDB-326905

Cohomological dimension conjecture for the moduli space of principally polarized abelian varieties

摘录 / EXCERPT: Let $\mathcal A_g$ denote the moduli space of principally polarized abelian varieties of dimension g, and let $\operatorname{cd}(\mathcal A_g)$ be its cohomological dimension. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-017285 | MATHDB-326910

Bombieri-Lang conjecture over function fields

摘录 / EXCERPT: Let K_0 be the function field of a smooth variety over an algebraically closed field k_0. Let Z be a variety of general type over K_0. Let Q be the closed reduced subsche...

conjecture | open | MathDB 页面 | 原文线索

V2-017286 | MATHDB-326914

Vakil-Matchett Wood's homological stabilisation conjecture for symmetric complements

摘录 / EXCERPT: Let i be a nonnegative integer, let bb be a partition, and let X be an irreducible smooth complex variety. For each j, let $\bar{w}^c_{1^j\lambda}(X)$ denote the complement...

conjecture | open | MathDB 页面 | 原文线索

V2-017287 | MATHDB-326923

Conjecture F on homological stability for complements of discriminants

摘录 / EXCERPT: Let X be an irreducible smooth complex variety, let $^j\lambda$ denote the partition obtained by adjoining j parts equal to 1 to a partition λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017288 | MATHDB-326927

Homotopy equivalence of polynomial-tuple and algebraic-map approximations

摘录 / EXCERPT: Let $A_D(m,X_{\Sigma};g)$ and $A_D(m,X_{\Sigma})$ be the spaces of tuples of homogeneous polynomials representing, respectively, based and unbased algebraic maps of degree D from...

conjecture | open | MathDB 页面 | 原文线索

V2-017289 | MATHDB-326949

The duality-admissibility conjecture for log-divergent graphs

摘录 / EXCERPT: Let G be a graph with loop number h_G and number of edges n_G, and suppose that G is log-divergent, meaning ... with $h_G,n_G\geq 3$. A graph is duality admissible when it sa...

conjecture | open | MathDB 页面 | 原文线索

V2-017290 | MATHDB-326978

Polynomial-count conjecture for character varieties of split reductive groups

摘录 / EXCERPT: Let G be a split reductive algebraic $\mathbb{C}$-group, and let $\mathfrak{X}_r(G)$ denote the character variety of a free group of rank r in G. Polynomial-count conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-017291 | MATHDB-326990

Parameterized form of Zilber's conjecture on intersections with tori

摘录 / EXCERPT: Let K be an algebraically closed field extending k_0, and let $B=\mathbb{B}(K)$ be a semiabelian variety defined over a field k_0 of characteristic zero. For a subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-017292 | MATHDB-327003

Dedieu's irreducibility conjecture for nodal K3 curve moduli

摘录 / EXCERPT: Let $\mathcal{V}^{n}_{g,k} \subseteq \mathcal{T}^{n}_{g,k}$ be the open substack parametrizing morphisms $[(f:C\to X,L)]$ with $f(C)$ nodal, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-017293 | MATHDB-327020

Simple connectedness conjecture for quasi-log canonical Fano varieties

摘录 / EXCERPT: Let $[X, \omega]$ be a projective quasi-log canonical pair, meaning a projective pair equipped with a quasi-log canonical structure, such that $-\omega$ is ample. Simple connectedn...

conjecture | open | MathDB 页面 | 原文线索

V2-017294 | MATHDB-327021

Simple connectedness conjecture for semi log canonical Fano varieties

摘录 / EXCERPT: Let (X, Δ) be a projective semi log canonical pair, meaning a projective pair with semi log canonical singularities, such that $-(K_X+\Delta)$ is ample; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-017295 | MATHDB-327030

Generalized affine-paving conjecture for cohomologically pure varieties

摘录 / EXCERPT: Generalized affine-paving conjecture. If $\mathfrak{o}$ is a total order such that $P^{\mathfrak{o}}(t)=P_V(t)$, then the decomposition of V according to $\mathfrak{o}$ is a gene...

conjecture | open | MathDB 页面 | 原文线索

V2-017296 | MATHDB-327049

The parahoric dependence conjecture for Rapoport-Zink spaces

摘录 / EXCERPT: Let $\mathcal{D}_{\mathbb{Z}_p}$ be an integral Rapoport-Zink datum such that $\mathcal{G}$ is a parahoric group scheme. The associated rigid space is equipped with an action of...

conjecture | open | MathDB 页面 | 原文线索

V2-017297 | MATHDB-327050

The non-emptiness conjecture for parahoric Rapoport-Zink spaces

摘录 / EXCERPT: Let $\mathcal{D}_{\mathbb{Z}_p}$ be an integral Rapoport-Zink datum such that $[b]\in B(G,\{\mu\})$, and suppose that the group scheme $\mathcal{G}$ is parahoric. Non-emptiness con...

conjecture | open | MathDB 页面 | 原文线索

V2-017298 | MATHDB-327051

Rapoport-Zink duality conjecture for basic local Shimura data

摘录 / EXCERPT: Let $(G,[b],\{\mu\})$ be a basic local Shimura datum, and let $(G^\vee,[b^\vee],\{\mu^\vee\})$ be its dual datum, with $G^\vee=J_b$, $[b^\vee]=[b^{-1}]$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017299 | MATHDB-327052

Harris-Viehmann conjecture for non-basic local Shimura varieties

摘录 / EXCERPT: Assume that G is quasi-split and let $(G,[b],\{\mu\})$ be a non-basic local Shimura datum. Let L be a Levi subgroup, assume that J_b is an inner form of a Levi subgroup conta...

conjecture | open | MathDB 页面 | 原文线索

V2-017300 | MATHDB-327058

Strong liftability conjecture for cyclic covers over smooth projective toric varieties

摘录 / EXCERPT: Strong liftability conjecture. The scheme Y is a smooth projective scheme which is strongly liftable over $W_2(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-017301 | MATHDB-327088

Motivic Serre finiteness

摘录 / EXCERPT: Let k be a field and let $\mathbb{S}_k$ be the motivic sphere spectrum. Motivic Serre finiteness. For every integer $n>0$, ... This is the motivic analogue of the vanishing of posi...

conjecture | open | MathDB 页面 | 原文线索

V2-017302 | MATHDB-327118

Weak Catenary Conjecture for differential varieties

摘录 / EXCERPT: Let $V \subseteq \mathbb A^n$ be a positive-dimensional irreducible differential variety, and let $W \subseteq V$ be a zero-dimensional differential subvariety. Weak Catenary Conje...

conjecture | open | MathDB 页面 | 原文线索

V2-017303 | MATHDB-327153

The nonzero-variety dot-product conjecture

摘录 / EXCERPT: Nonzero-variety dot-product conjecture. If $|E||F|\gg_C q^d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-017304 | MATHDB-327170

Xiao's conjecture on the relative irregularity of fibred surfaces

摘录 / EXCERPT: Let $f:S\rightarrow B$ be a non-trivial fibration from a smooth projective surface S to a smooth projective curve B, with fibres of genus g. Write $q_f=q(S)-g(B)$ for its rel...

conjecture | open | MathDB 页面 | 原文线索

V2-017305 | MATHDB-327187

Generation conjecture for the algebra $W_{0,3}(SL_m(\mathbb{C}))$

摘录 / EXCERPT: Let $W_{0,3}(SL_m(\mathbb{C}))$ be the algebra associated with the three-pointed genus-zero construction, and let extended conformal blocks of level 1 be the generators under con...

conjecture | open | MathDB 页面 | 原文线索

V2-017306 | MATHDB-327194

Highest-weight conjecture for the category O_λ

摘录 / EXCERPT: Assume that X_0 is a singular Poisson variety, X is a smooth symplectic variety, and $\rho:X\rightarrow X_0$ is a Poisson resolution. Suppose that $T_1\times T_2$ acts on X a...

conjecture | open | MathDB 页面 | 原文线索

V2-017307 | MATHDB-327209

The wild character variety and P = W conjecture for plane curve singularities

摘录 / EXCERPT: Let C be a locally planar rational curve with a unique singularity, and let Λ be a rainbow closure of a positive braid such that the topological knot underlying Λ...

conjecture | open | MathDB 页面 | 原文线索

V2-017308 | MATHDB-327273

The secondary Euler characteristic equality conjecture for totally real fields

摘录 / EXCERPT: Let k be a totally real number field, let p be a prime, let E be a finite extension of $mathbb Q_p$, and let chi be an even Artin character of G_k realizable over E. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-017309 | MATHDB-327293

Kobayashi's associated-variety conjecture for discretely decomposable restrictions

摘录 / EXCERPT: Let G be a real reductive group, let G' be a reductive subgroup, and choose a maximal compact subgroup K of G such that $K'=G'\cap K$ is a maximal compact subgroup of G'....

conjecture | open | MathDB 页面 | 原文线索

V2-017310 | MATHDB-327305

Hodge-Witt symmetry conjecture for liftable varieties

摘录 / EXCERPT: Hodge-Witt symmetry conjecture. The variety X satisfies Hodge-Witt symmetry. In particular, Ekedahl's conditions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-017311 | MATHDB-327316

Burns-Flach conjecture at left-of-center special values

摘录 / EXCERPT: Let A be the semisimple coefficient algebra acting on the motive M, let d be its relevant dimension, let i be an integer, and let $n>\frac{i}{2}+1$ be an integer. For an em...

conjecture | open | MathDB 页面 | 原文线索

V2-017312 | MATHDB-327317

Weak equivariant Beilinson-Burns-Flach conjecture

摘录 / EXCERPT: Let A act on the motive M, and let W be an A-submodule of $H^{i+1}_{\mathcal{M}/\mathcal{O}_K}(M,E(n))$. Let r_{B} be the Beilinson regulator, and suppose ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017313 | MATHDB-327347

Tropical basis conjecture for the degree-three del Pezzo surface relations

摘录 / EXCERPT: Let the 270 trinomial relations listed in Proposition be the relations defining the degree-three del Pezzo surface's relevant variety. Tropical basis conjecture. The 270 trinom...

conjecture | open | MathDB 页面 | 原文线索

V2-017314 | MATHDB-327354

Orbit-independence conjecture for elementary abelian subalgebra varieties

摘录 / EXCERPT: Let G be a reductive algebraic group with a Chevalley $\mathbb{Z}$-form $G_{\mathbb{Z}}$, and set ... Let $(\star)$ be the condition on primes used in the paper, and let d be def...

conjecture | open | MathDB 页面 | 原文线索

V2-017315 | MATHDB-327384

The time-invariant time-sharing conjecture for streaming codes

摘录 / EXCERPT: Let d be the block-wise feedback delay. A time-invariant scheme is specified by a vector $\mathbf{x}=[x_1,\ldots,x_d]$ of non-negative integers satisfying ... and transmits x_i l...

conjecture | open | MathDB 页面 | 原文线索

V2-017316 | MATHDB-327390

The cohomology and ring structure conjecture for generic odd linear-tree varieties

摘录 / EXCERPT: Let n be odd, and let $X_n=X_{\mathbb{A}_n}^{\mathtt{generic}}$ be the generic fiber associated with the linear tree $\mathbb{A}_n$. Write WP for the distinguish...

conjecture | open | MathDB 页面 | 原文线索

V2-017317 | MATHDB-327406

Chow stability from tangent-bundle stability for smooth projective varieties

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a complex irreducible smooth variety, and let $T\mathbb{P}^n_{|X}$ denote the restriction to X of the tangent bundle of $\mathbb{P}^n$. Stability is...

conjecture | open | MathDB 页面 | 原文线索

V2-017318 | MATHDB-327407

Polynomially restricted higher-dimensional Bourgain incidence conjecture

摘录 / EXCERPT: Let $n>2$, and let N be a positive parameter. For each r with $1<r<n$, consider N^{n-1} lines in $\mathbb{R}^n$. Higher-dimensional Bourgain conjecture. There exist integers...

conjecture | open | MathDB 页面 | 原文线索

V2-017319 | MATHDB-327413

The $4M-4$ phase-transition conjecture for complex phase retrieval

摘录 / EXCERPT: Let Φ be the intensity-measurement matrix for signals in $\mathbb{C}^M$, with N measurements, and write $\mathrm{Inj}[\Phi;\mathbb{C}^{M\times N}]$ for the property that the...

conjecture | open | MathDB 页面 | 原文线索

V2-017320 | MATHDB-327479

Eichler-Shimura congruence relation for Shimura varieties

摘录 / EXCERPT: Eichler-Shimura congruence relation. The $\operatorname{Gal}(\mathbb Q^{\mathrm{algcl}}/E)$-module V is unramified at all primes $\mathfrak p\mid p$ with $p\notin R$, and

conjecture | open | MathDB 页面 | 原文线索

V2-017321 | MATHDB-327482

Gross-Deligne conjecture on periods of Hodge structures

摘录 / EXCERPT: Let X be a smooth projective variety over $\overline{\mathbb{Q}}$, let $H\subset H^m_B(X)$ be a sub-Hodge structure, let F be an abelian number field of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-017322 | MATHDB-327483

Generalized Yau conjecture for semi-negative holomorphic sectional curvature

摘录 / EXCERPT: Let M be a projective manifold with a Kähler metric of semi-negative holomorphic sectional curvature. Define r_M as the invariant measuring the largest codimension of a maximal...

conjecture | open | MathDB 页面 | 原文线索

V2-017323 | MATHDB-327526

Arithmetic duality conjecture for finite extensions of L_p

摘录 / EXCERPT: Arithmetic duality conjecture. There is a perfect pairing

conjecture | open | MathDB 页面 | 原文线索

V2-017324 | MATHDB-327549

The higher formality conjecture for derived stacks

摘录 / EXCERPT: Let $n\geq 0$, let X be a nice enough derived algebraic stack, and let $\widehat{\mathbb{H}}^{E_{n+1}}(X)$ denote its formal higher Hochschild cohomology, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-017325 | MATHDB-327561

Codimension-one connectedness conjecture for associated graphs

摘录 / EXCERPT: Let X be an irreducible Harish-Chandra $(\mathfrak{g},K)$-module, and let ... be the irreducible decomposition of its associated variety. The graph...

conjecture | open | MathDB 页面 | 原文线索

V2-017326 | MATHDB-327580

The pointed Gromov-Hausdorff convergence conjecture for semi-log canonical degenerations

摘录 / EXCERPT: Consider a flat degeneration whose general fibres are smooth canonical models and whose central fibre is semi-log canonical. The components of the central fibre are regarded with t...

conjecture | open | MathDB 页面 | 原文线索

V2-017327 | MATHDB-327600

Altınok-Brown-Reid conjecture on smoothing Q-Fano threefolds

摘录 / EXCERPT: Let X be a Q-Fano threefold, meaning a three-dimensional Fano variety with $ Q$-Cartier canonical divisor. Let Δ^1 be a unit disc, and let $D\in |{-}K_X|$ when such an ele...

conjecture | open | MathDB 页面 | 原文线索

V2-017328 | MATHDB-327608

The relative Beilinson-Hodge conjecture

摘录 / EXCERPT: Let $k=\overline{k}\subseteq\mathbb C$ be an algebraically closed subfield, and let ... be a smooth proper map of smooth quasi-projective varieties over k. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017329 | MATHDB-327617

Steingrímsson's sorting conjecture for revstack and stack sort

摘录 / EXCERPT: Steingrímsson's sorting conjecture. For all $0\leq t\leq n$, $$

conjecture | open | MathDB 页面 | 原文线索

V2-017330 | MATHDB-327629

Degree-zero Donaldson-Thomas formula for Hilbert schemes of points

摘录 / EXCERPT: Let X be a smooth 3-fold and let $\mathscr{L}_1,\mathscr{L}_2$ be line bundles on X satisfying ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017331 | MATHDB-327652

The minimal categorical resolution conjecture

摘录 / EXCERPT: Let Y be a quasiprojective scheme. A categorical resolution of Y is a smooth triangulated category $\mathcal{T}$ equipped with adjoint functors...

conjecture | open | MathDB 页面 | 原文线索

V2-017332 | MATHDB-327653

The Grassmannian-Pfaffian categorical HP-duality conjecture

摘录 / EXCERPT: Let W be a vector space of dimension m, and let $\operatorname{Pf}(2k,W)$ denote the closure of the locus of bivectors in Λ^2W of rank $2k$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017333 | MATHDB-327656

Completeness conjecture for relations in the tautological algebras of curves

摘录 / EXCERPT: Let $\mathcal{R}^*(\mathcal{M}_{g})$ and $\mathcal{R}^*(\mathcal{M}_{g,*})$ be the tautological algebras of the moduli space of smooth genus-g curves and the moduli space of smoo...

conjecture | open | MathDB 页面 | 原文线索

V2-017334 | MATHDB-327657

The wild McKay correspondence conjecture for linear actions

摘录 / EXCERPT: The wild McKay correspondence conjecture. If the quotient morphism $V\to X$ is étale in codimension one, then

conjecture | open | MathDB 页面 | 原文线索

V2-017335 | MATHDB-327658

The motivic McKay correspondence for non-linear actions

摘录 / EXCERPT: The motivic McKay correspondence for non-linear actions. We have

conjecture | open | MathDB 页面 | 原文线索

V2-017336 | MATHDB-327659

Crepant birational invariance of stringy motifs

摘录 / EXCERPT: Crepant invariance conjecture. If f is proper, birational and crepant, then

conjecture | open | MathDB 页面 | 原文线索

V2-017337 | MATHDB-327660

The conjectural change-of-variables formula for untwisting

摘录 / EXCERPT: Untwisted change-of-variables conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017338 | MATHDB-327661

The conjectural change-of-variables formula for linear wild McKay correspondence

摘录 / EXCERPT: Linear wild change-of-variables conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017339 | MATHDB-327663

The motivic McKay correspondence for linear actions I

摘录 / EXCERPT: The motivic McKay correspondence for linear actions I. We have

conjecture | open | MathDB 页面 | 原文线索

V2-017340 | MATHDB-327664

The motivic McKay correspondence for linear actions II

摘录 / EXCERPT: The motivic McKay correspondence for linear actions II. We have

conjecture | open | MathDB 页面 | 原文线索

V2-017341 | MATHDB-327665

Motivic Krasner formula for discriminant strata

摘录 / EXCERPT: Motivic version of Krasner's formula. In the Grothendieck-type ring $\mathcal{R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-017342 | MATHDB-327675

Existence conjecture for geometric stability conditions

摘录 / EXCERPT: Let X be a smooth projective variety, and let $B+i\omega\in H^2(X,\mathbb{C})$ with ω ample. Define ... where $\operatorname{ch}^B(E)=e^{-B}\operatorname{ch}(E)$. Existence...

conjecture | open | MathDB 页面 | 原文线索

V2-017343 | MATHDB-327677

The three-dimensional connected-generating-function odd qMZV conjecture

摘录 / EXCERPT: Three-dimensional qMZV conjecture. Connected generating functions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-017344 | MATHDB-327701

Bloch-Srinivas conjecture on the stabilisation of relative K-theory filtrations

摘录 / EXCERPT: Let $X'\to X$ be a desingularisation of an integral variety X, with exceptional fibre $E\hookrightarrow X'$. For a positive integer n, let $F^dK_0(X',nE)$ denote the codimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-017345 | MATHDB-327714

Dimca's bound for the Griffiths-Dwork reduction index

摘录 / EXCERPT: Let f be the polynomial defining the rational integral, and let r_f denote the integer for which the reduction operation $[\,]_{r_f}$ has the properties established earlier in...

conjecture | open | MathDB 页面 | 原文线索

V2-017346 | MATHDB-327754

Geometric Langlands duality compatibility with Serre duality

摘录 / EXCERPT: Let G be a reductive group, let X be the underlying smooth projective curve, and let $\bun_G$ be the moduli stack of G-bundles on X. Write $\bG$ for the Langlands dual grou...

conjecture | open | MathDB 页面 | 原文线索

V2-017347 | MATHDB-327773

Distance bound for homogeneous polynomials with roots on the unit circle

摘录 / EXCERPT: Distance-to-discriminant conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017348 | MATHDB-327812

Conjectural isotriviality for fibrations over special orbifold bases

摘录 / EXCERPT: Isotriviality conjecture. Under these assumptions, f is isotrivial.

conjecture | open | MathDB 页面 | 原文线索

V2-017349 | MATHDB-327813

Helmke's conjecture on local global generation of adjoint line bundles

摘录 / EXCERPT: Work over an algebraically closed field k of characteristic zero. Let X be a smooth projective variety of dimension n, let $x\in X$ be a point, and let L be an ample line b...

conjecture | open | MathDB 页面 | 原文线索

V2-017350 | MATHDB-327832

The multivariate Abel-Ruffini conjecture for reduced polynomial systems

摘录 / EXCERPT: Let $A_1,A_2,\ldots,A_n\subset\mathbb{Z}^n$ be a reduced tuple, meaning that the dimension of the convex hull of $\bigcup_{j\in I}A_j$ is greater than $|I|$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-017351 | MATHDB-327876

Hodge-theoretic determination of the Orlov spectrum

摘录 / EXCERPT: Let X be a Fano variety, let $M(\mathbb{P}^1, CY)$ be the zero fiber of the Stability Hodge Structure of $D^b(X)$, and let $\overline{M(\mathbb{P}^1, CY)}$ be its compactificatio...

conjecture | open | MathDB 页面 | 原文线索

V2-017352 | MATHDB-327877

Exclusion of the infinite chain condition for universal covers of surfaces

摘录 / EXCERPT: Let S be a smooth projective surface, and let $\widetilde{S}$ be its universal covering space. Infinite-chain conjecture. The infinite chain condition can be ruled out for...

conjecture | open | MathDB 页面 | 原文线索

V2-017353 | MATHDB-327878

Multiple-filtration conjecture for Landau-Ginzburg Stability Hodge Structures

摘录 / EXCERPT: Let $M(\mathbb{P}^k, CY)$ be the formal scheme occurring as the zero fiber of a Stability Hodge Structure associated with a Fukaya-Seidel category having many potentials. Multiple-...

conjecture | open | MathDB 页面 | 原文线索

V2-017354 | MATHDB-327881

Lachaud's point-count bound for large-dimensional complete intersections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n}$ be a projective algebraic set defined over $\mathbb{F}_{q}$, of dimension $m\geq n/2$ and degree $d\leq q+1$, which is an i.t. complete intersection. W...

conjecture | open | MathDB 页面 | 原文线索

V2-017355 | MATHDB-327883

The p-parity conjecture for abelian varieties and self-dual representations

摘录 / EXCERPT: Let A/K be an abelian variety over a number field, and let p be a prime. Define the p-Selmer vector space ... If F/K is a Galois extension and τ is a self-dual represent...

conjecture | open | MathDB 页面 | 原文线索

V2-017356 | MATHDB-327892

The toric Landau-Ginzburg mirror conjecture for smooth Fano varieties

摘录 / EXCERPT: Toric Landau-Ginzburg mirror conjecture. Every smooth Fano variety has a toric Landau-Ginzburg model.

conjecture | open | MathDB 页面 | 原文线索

V2-017357 | MATHDB-327895

The Landau-Ginzburg moduli-stack conjecture for Fano threefolds

摘录 / EXCERPT: Landau-Ginzburg moduli-stack conjecture. There exists a moduli stack of Landau-Ginzburg models of all three-dimensional Fano varieties, with a cell structure parallel to the basic...

conjecture | open | MathDB 页面 | 原文线索

V2-017358 | MATHDB-327896

The derived-category conjecture for four-dimensional X_{10} varieties

摘录 / EXCERPT: Derived-category conjecture for X_{10}.

conjecture | open | MathDB 页面 | 原文线索

V2-017359 | MATHDB-327897

The birationality conjecture for Landau-Ginzburg moduli of cubic fourfolds and X_{10} varieties

摘录 / EXCERPT: Birationality conjecture. There is an infinite series of such moduli spaces that can be deformed one to another and therefore are birational.

conjecture | open | MathDB 页面 | 原文线索

V2-017360 | MATHDB-327900

The Orlov-spectrum gap conjecture for rational Fano manifolds

摘录 / EXCERPT: Orlov-spectrum gap conjecture. A gap in the Orlov spectrum of $D^b(X)$ is less than or equal to $n-3$.

conjecture | open | MathDB 页面 | 原文线索

V2-017361 | MATHDB-327932

The étale rigid motives equivalence with and without transfers

摘录 / EXCERPT: Let K be a field, let B be a normal variety over K, and let Λ be $\mathbb{Z}$ when $\operatorname{char} K=0$ and $\mathbb{Z}[1/p]$ when $\operatorname{char} K=p$. Den...

conjecture | open | MathDB 页面 | 原文线索

V2-017362 | MATHDB-327939

Finite degeneration conjecture to a weak Kähler-Ricci soliton

摘录 / EXCERPT: Let M be a Fano manifold and consider weak conical Kähler-Ricci solitons on varying pairs (W,Δ), satisfying ... A test-configuration for (M,K_M^{-1}) is a $\mathbf{C}^*$-e...

conjecture | open | MathDB 页面 | 原文线索

V2-017363 | MATHDB-327945

The cancellation conjecture for the Grothendieck ring of varieties

摘录 / EXCERPT: Cancellation conjecture. The element $\mathbb L$ is not a zero divisor in $K_0(Var/k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-017364 | MATHDB-327975

Shokurov's lower semicontinuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Let X be a normal $\mathbb{Q}$-Gorenstein variety, and let Δ be an $\mathbb{R}$-Weil divisor on X such that $K_X+\Delta$ is $\mathbb{R}$-Cartier. For each d, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017365 | MATHDB-327978

The numerical moduli space conjecture for manifolds of Kummer n-type

摘录 / EXCERPT: Numerical moduli space conjecture. The preceding proposition holds for all n: there exists an abelian surface A such that X is the fibre of the Albanese map of a moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-017366 | MATHDB-327987

Manin-Mumford conjecture for abelian varieties

摘录 / EXCERPT: Manin-Mumford conjecture. The Zariski closure of $\{a_i\}_{i\in I}$ is a finite union of torsion subvarieties.

conjecture | open | MathDB 页面 | 原文线索

V2-017367 | MATHDB-328083

Dimension conjecture for quadratic motherbody equations

摘录 / EXCERPT: Dimension conjecture for quadratic motherbody equations. The set $\Omega_{P,Q}$ is a real semi-analytic variety of real dimension n.

conjecture | open | MathDB 页面 | 原文线索

V2-017368 | MATHDB-328088

Polynomial partitioning conjecture for varieties

摘录 / EXCERPT: Let $X\subset \mathbb{C}^d$ be an irreducible variety of dimension e, let $\mathcal{P} \subset X(\mathbb{R})$ be a finite subset, and let ℓ satisfy $\ell \geq c\,\delta(X)$....

conjecture | open | MathDB 页面 | 原文线索

V2-017369 | MATHDB-328089

Point-hypersurface incidence conjecture in higher dimensions

摘录 / EXCERPT: Let $d,h,k\geq1$. Let $\mathcal{P}$ be a finite set of points in $\mathbb{R}^{d}$ and let $\mathcal{H}$ be a finite set of hypersurfaces in $\mathbb{C}^{d}$. Assume that the hypers...

conjecture | open | MathDB 页面 | 原文线索

V2-017370 | MATHDB-328108

Logarithmic subadditivity conjecture for smooth projective pairs

摘录 / EXCERPT: Let $f:X\to Y$ be a surjective morphism between smooth projective varieties with connected fibers. Let D_X and D_Y be simple normal crossing divisors on X and Y, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-017371 | MATHDB-328115

Polynomial partitioning conjecture for irreducible varieties

摘录 / EXCERPT: Let $V\subseteq \mathbb{C}^d$ be an irreducible complex algebraic variety of dimension $k\geq 1$ and degree Δ. Let $Q\subset V\cap\mathbb{R}^d$ be a finite point set, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-017372 | MATHDB-328142

Kuznetsov's categorical McKay conjecture for quotient singularities

摘录 / EXCERPT: Let V be a smooth quasi-projective variety, and let G be a finite subgroup of $\operatorname{Aut}(V)$ such that ω_V is G-equivariantly locally trivial. Let $Z\to V/G$...

conjecture | open | MathDB 页面 | 原文线索

V2-017373 | MATHDB-328172

Greatest-common-divisor conjecture for zeta functions of \Upsilon-schemes

摘录 / EXCERPT: Let an Υ-scheme be given, together with the collection of polynomials associated with its counting data. Greatest-common-divisor conjecture. The zeta function of an...

conjecture | open | MathDB 页面 | 原文线索

V2-017374 | MATHDB-328173

Numerical equivalence implies cohomological triviality

摘录 / EXCERPT: Let X be a variety, and let an algebraic cycle on X be numerically equivalent to zero when its intersection degree with every other algebraic cycle is zero. Numerical-to-cohomo...

conjecture | open | MathDB 页面 | 原文线索

V2-017375 | MATHDB-328174

Canonical decomposition conjecture for homogeneous spaces

摘录 / EXCERPT: Let (G,B,N) be a Tits system, let P be a parabolic subgroup, and let $V(G/P)$ be the corresponding homogeneous variety. Canonical decomposition conjecture. The diagonal or moti...

conjecture | open | MathDB 页面 | 原文线索

V2-017376 | MATHDB-328187

Brion's affine-space conjecture for the moduli scheme of spherical varieties

摘录 / EXCERPT: Let G be a reductive group with Borel subgroup B, maximal torus T in B, and weight lattice Λ; write Λ^+ for the monoid of dominant weights. Let Γ be...

conjecture | open | MathDB 页面 | 原文线索

V2-017377 | MATHDB-328188

Form-scheme affine-space classification conjecture

摘录 / EXCERPT: Let k be the ground field, let $\mathbbmss{Form}_k$ be the category of formal schemes used in the paper, let $[\mathcal{X}]$ denote the class of $\mathcal{X}$ in its Grothendieck...

conjecture | open | MathDB 页面 | 原文线索

V2-017378 | MATHDB-328189

Reduced auto-arc affine-space criterion for smoothness

摘录 / EXCERPT: Let X be a variety over k, let p be a point of X, and let $\mathcal{A}_n(X,p)$ be its auto-arc space at level n. Let $\kappa(p)$ be the residue field at p, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-017379 | MATHDB-328190

Auto-Igusa zeta analytic-isomorphism conjecture

摘录 / EXCERPT: Auto-Igusa zeta analytic-isomorphism conjecture. If $\bar\zeta_{X/Y,p}(t)=0$ for some $q\in Y$, then the pointed germs (X,p) and (Y,q) are analytically isomorphic. The paper mo...

conjecture | open | MathDB 页面 | 原文线索

V2-017380 | MATHDB-328192

One-singularity curve auto-Igusa zeta structure conjecture

摘录 / EXCERPT: Let C be a connected curve over k with only one singular point p, let $f:\bar C\to C$ be its normalization, and let $\bar\zeta_{C,p}(t)$ be the reduced auto-Igusa zeta series...

conjecture | open | MathDB 页面 | 原文线索

V2-017381 | MATHDB-328202

Equality of intrinsic and geometric Beilinson operator rings

摘录 / EXCERPT: Let X be a finite type $\boldsymbol{k}$-scheme, let $\xi=(x_0,\ldots,x_n)$ be a saturated chain of points in X with x_n closed, and put $K:=\boldsymbol{k}(x_0)$. Let K_ξ...

conjecture | open | MathDB 页面 | 原文线索

V2-017382 | MATHDB-328222

Dyn's conjecture on the regularity of generalized Daubechies wavelets

摘录 / EXCERPT: Let $n\ge 2$ and let $\Lambda_n=\{\lambda_0,\ldots,\lambda_{n-1}\}$ be an arbitrary set of real numbers. Let ψ^{Λ_n} be the generalized Daubechies wavelet associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-017383 | MATHDB-328230

The holonomy conjecture for strictly affine representations

摘录 / EXCERPT: A representation is strictly affine when it is nonabelian, is not unitary, and has linear-part angles generating an infinite subgroup of $\mathbb{R}/\mathbb{Z}$. A branched affine...

conjecture | open | MathDB 页面 | 原文线索

V2-017384 | MATHDB-328235

Odaka-Okada conjecture on K-semistability of Picard-rank-one Fano manifolds

摘录 / EXCERPT: Let X be a smooth Fano manifold, meaning a smooth projective variety with ample anticanonical divisor, and suppose that its Picard rank is 1. Odaka-Okada conjecture. The manifo...

conjecture | open | MathDB 页面 | 原文线索

V2-017385 | MATHDB-328236

The Adams differential conjecture for the eta-local motivic sphere

摘录 / EXCERPT: Let A be the mod-2 Steenrod algebra, let h_1 denote the class corresponding to τ_0, and let v_k be the generators in the h_1-localized Ext_A algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-017386 | MATHDB-328237

The conjectural E_∞-page and homotopy of the eta-local motivic sphere

摘录 / EXCERPT: Let A be the mod-2 Steenrod algebra, let h_1 be the class inverted in the h_1-local Adams spectral sequence, and let v_1,v_2 denote the displayed localized generators. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-017387 | MATHDB-328259

Gap-vector conjecture for Veronese embeddings

摘录 / EXCERPT: Let $X=v_d(\mathbb P^n)$ be the dth Veronese embedding of projective n-space, and define ... Here $g_j(X)$ denotes the jth component of the gap vector of X, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017388 | MATHDB-328287

Asymptotic plausibility conjecture for shapes with r=2

摘录 / EXCERPT: A shape (n,κ,r,s) is called plausible when it satisfies the paper's plausibility criterion for detecting whether sufficiently general ideals of that shape are plausibly gene...

conjecture | open | MathDB 页面 | 原文线索

V2-017389 | MATHDB-328305

Kapranov's rigidity conjecture for iterated deformation spaces

摘录 / EXCERPT: Let X be a smooth scheme. Define X_1 to be the moduli space of deformations of X, X_2 the moduli space of deformations of X_1, and so on. Kapranov's rigidity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-017390 | MATHDB-328317

Göttsche-Shende refined Severi degree conjecture

摘录 / EXCERPT: For a projective surface S and line bundle L, let $N^{(S,L),\delta}(y)$ denote the refined Severi degree and let $\widetilde N^{(S,L),\delta}(y)$ denote the refined universal i...

conjecture | open | MathDB 页面 | 原文线索

V2-017391 | MATHDB-328321

Positivity conjecture for refined universal curve-counting invariants

摘录 / EXCERPT: Let S be a smooth projective surface and let L be a δ-very ample line bundle. Let $\widetilde N^{(S,L),\delta}(y)$ be the refined universal invariant. Positivity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-017392 | MATHDB-328323

The higher-dimensional slope inequality for fibred varieties

摘录 / EXCERPT: Higher-dimensional slope inequality. Under these hypotheses, the fibred variety satisfies the slope inequality. This is known for relatively minimal fibred surfaces of genus at lea...

conjecture | open | MathDB 页面 | 原文线索

V2-017393 | MATHDB-328334

Broadhurst-Kreimer Hilbert-series conjecture for depth-graded motivic Lie algebras

摘录 / EXCERPT: Broadhurst-Kreimer Hilbert-series conjecture. There is an identity of formal series in s and t:

conjecture | open | MathDB 页面 | 原文线索

V2-017394 | MATHDB-328342

The higher-dimensional jet ampleness vanishing conjecture for syzygies

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, let $L_d=dA+E$ with A ample and E an arbitrary divisor on X, and let B be a line bundle on X. For the Koszul coho...

conjecture | open | MathDB 页面 | 原文线索

V2-017395 | MATHDB-328361

Kollár's restriction conjecture for local Picard groups of excellent schemes

摘录 / EXCERPT: Let X be an excellent scheme that is S_2 and has pure dimension ≥ 4. Let $x\in X$ be a closed point and let $x\in D\subset X$ be an effective Cartier divisor. Kollár's res...

conjecture | open | MathDB 页面 | 原文线索

V2-017396 | MATHDB-328366

Motivic Broadhurst-Kreimer conjecture for the depth-graded multiple zeta algebra

摘录 / EXCERPT: Motivic Broadhurst-Kreimer conjecture. The depth-and-weight generating series satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-017397 | MATHDB-328377

André-Pink-Zannier conjecture for the universal family of abelian varieties

摘录 / EXCERPT: Let $\mathfrak{A}_g$ be the universal family of principally polarized abelian varieties, and let Y be a subvariety of $\mathfrak{A}_g$. For $s\in\mathfrak{A}_g$, let Σ be...

conjecture | open | MathDB 页面 | 原文线索

V2-017398 | MATHDB-328382

Absolutely minimal Calabi-Yau non-hyperbolicity conjecture

摘录 / EXCERPT: Let X be an absolutely minimal Calabi-Yau variety. Suppose that every birational morphism $X \to Y$ onto a normal projective variety is an isomorphism. Non-hyperbolicity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-017399 | MATHDB-328390

Equivalence of broken-line and seed convexity

摘录 / EXCERPT: Let φ be a tropical function in a cluster-variety setting, and consider convexity along broken lines and convexity with respect to every seed. The convexity-equivalence con...

conjecture | open | MathDB 页面 | 原文线索

V2-017400 | MATHDB-328391

Cluster-complex realization of positive scattering walls

摘录 / EXCERPT: Let φ be a tropical function on the tropicalization of a cluster variety, or of a fiber of a cluster variety, let $\s{Y}$ denote that variety or fiber, and let $\f{w}$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-017401 | MATHDB-328392

Broken-line convexity of tropicalized regular functions

摘录 / EXCERPT: Let Y be a log Calabi-Yau variety and let f be a regular function on Y. Its tropicalization f^{trop} is the associated tropical function. Gross-Hacking-Keel-...

conjecture | open | MathDB 页面 | 原文线索

V2-017402 | MATHDB-328394

The automorphism-group conjecture for rank 2 cluster varieties

摘录 / EXCERPT: Automorphism-group conjecture. For all rank 2 cases,

conjecture | open | MathDB 页面 | 原文线索

V2-017403 | MATHDB-328402

Campana-Peternell width-decreasing conjecture for CP-manifolds

摘录 / EXCERPT: Width-decreasing conjecture. Let X be a CP-manifold which is not a product of positive-dimensional varieties. If $\tau(X)>0$, then there exists a surjective morphism $f:X'\to X$...

conjecture | open | MathDB 页面 | 原文线索

V2-017404 | MATHDB-328426

The Main Question on birational minimal models

摘录 / EXCERPT: The Main Question. Every algebraic variety X is birational to a variety $X^{\rm m}$ that is either of semi-negatively curved or Kodaira--Iitaka type, or of positive fiber type.

conjecture | open | MathDB 页面 | 原文线索

V2-017405 | MATHDB-328461

Billey-Crites conjecture on pattern avoidance for affine Schubert varieties

摘录 / EXCERPT: Let W be an affine Weyl group of type $\tilde{A}_n$, let w be an affine permutation, and let $X^{\emptyset}(w)$ be the corresponding Schubert variety. Billey-Crites conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-017406 | MATHDB-328462

Grassmannian Billey-Postnikov decomposition conjecture for palindromic Schubert polynomials

摘录 / EXCERPT: Let W be a Coxeter group, let J be a subset of its simple generators, and let $w\in W^J$. Let P_w^J denote the corresponding parabolic Kazhdan-Lusztig polynomial, and call it...

conjecture | open | MathDB 页面 | 原文线索

V2-017407 | MATHDB-328474

Snowden's uniform bounded-generation conjecture for GIT quotients

摘录 / EXCERPT: Let V be the vector space underlying the projective variety $\mathbb{P}(V)$, let G be a linearly reductive group, and let X be a G-polarized projective variety. For the inv...

conjecture | open | MathDB 页面 | 原文线索

V2-017408 | MATHDB-328476

Conjecture on smooth representability of low simplicial-degree motivic spheres

摘录 / EXCERPT: Let k be a base ring, and let $i,j\in\mathbb{N}$ satisfy $i<j-1$. The motivic sphere ... means the smash product of the i-fold simplicial sphere with j copies of the multiplica...

conjecture | open | MathDB 页面 | 原文线索

V2-017409 | MATHDB-328478

Jannsen's standard sign conjecture for homological motives

摘录 / EXCERPT: Let k be a field, let H^* be a Weil cohomology theory with coefficients in a characteristic-zero field F, and let $M_{hom}(k)_F$ be the category of homological motiv...

conjecture | open | MathDB 页面 | 原文线索

V2-017410 | MATHDB-328479

Finite-correspondence conjecture for Künneth and sign projectors

摘录 / EXCERPT: Let X/k be a projective smooth variety, set $d=\dim X$, and let ... be the $\mathbb{Q}$-subspace spanned by cohomology classes of codimension-d cycles on $X\times_k X$ that are f...

conjecture | open | MathDB 页面 | 原文线索

V2-017411 | MATHDB-328548

Support-variety analogue of the Borel-Bott-Weil theorem

摘录 / EXCERPT: Let G be semisimple and simply connected, with p good, and let $\lambda\in X(T)$ be arbitrary. Suppose $w\in W$ is such that $w\cdot\lambda\in X(T)_+$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-017412 | MATHDB-328580

Luna's conjecture for spherical subgroups of Kac-Moody groups

摘录 / EXCERPT: Luna's conjecture. Mapping ${\mathcal H}\subseteq{\mathcal G}$ to the homogeneous spherical datum of ${\mathcal G}/{\mathcal H}$ induces a bijection between the set of conjugacy cl...

conjecture | open | MathDB 页面 | 原文线索

V2-017413 | MATHDB-328583

Stabilization conjecture for the expected number of critical tensor approximations

摘录 / EXCERPT: Stabilization conjecture. In the Gaussian setting of Theorem 1, the expected number of critical points of d_v on X does not decrease when n_p is replaced by n_p-1.

conjecture | open | MathDB 页面 | 原文线索

V2-017414 | MATHDB-328596

Batyrev-Tschinkel's modified Manin conjecture for canonical Fano varieties

摘录 / EXCERPT: Let X be a canonical Fano variety over a number field K, with a metrized anti-canonical divisor and associated height function H. For a suitable subset $U\subset X(K)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-017415 | MATHDB-328625

Opposite Schubert localization formula in equivariant K-theory

摘录 / EXCERPT: Consider the Grassmannian $\limfunc{Gr}_{n,N}$ and the ring ... For partitions ${\Greekmath 0115}$ and ${\Greekmath 0116}$ indexing opposite Schubert varieties and fixed points, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017416 | MATHDB-328658

The power-integral Cohen-Macaulay conjecture for affine toric varieties

摘录 / EXCERPT: Power-integral Cohen-Macaulay conjecture. The ring A^{pow} is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-017417 | MATHDB-328672

The completeness conjecture for quasi-compact and quasi-separated algebraic stacks

摘录 / EXCERPT: Let X be a quasi-compact and quasi-separated algebraic stack. A quasi-coherent $\mathcal{O}_X$-module is said to have the completeness property when it is a directed colimit of f...

conjecture | open | MathDB 页面 | 原文线索

V2-017418 | MATHDB-328673

The approximation conjecture for quasi-compact and quasi-separated algebraic stacks

摘录 / EXCERPT: Let X be a quasi-compact and quasi-separated algebraic stack. An approximation of X is a factorization ... where $X\to X_0$ is affine and X_0 is of finite presentation over...

conjecture | open | MathDB 页面 | 原文线索

V2-017419 | MATHDB-328675

Crystalline Variational Tate Conjecture

摘录 / EXCERPT: Let $f:X\to S$ be a smooth, proper morphism of smooth k-varieties, where k is a perfect field of characteristic $p>0$, let $s\in S$ be a closed point, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017420 | MATHDB-328676

Generic generalized Sato-Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension $g\geq 1$ over a number field K. For each prime ideal $\mathfrak{q}$ of good reduction, let $\overline{P}_{\mathfrak{q}}(T)$ be the ren...

conjecture | open | MathDB 页面 | 原文线索

V2-017421 | MATHDB-328694

Nonemptiness conjecture for the nonzero solution set $V(\mathbf{r})^{\times}$

摘录 / EXCERPT: Let $n,k\in\mathbb{N}$ with $n\geq k$, and let $\mathbf{r}\in\Gamma(n,k)$. Let $V(\mathbf{r})^{\times}$ denote the subset of $V(\mathbf{r})$ consisting of the relevant nonzero solu...

conjecture | open | MathDB 页面 | 原文线索

V2-017422 | MATHDB-328705

A linear bound for real critical points of Fermat hypersurfaces

摘录 / EXCERPT: Linear-bound conjecture. The number of real critical points of the critical equation is at most

conjecture | open | MathDB 页面 | 原文线索

V2-017423 | MATHDB-328709

Articulated Zilber-Pink conjecture for mixed Shimura and semi-abelian varieties

摘录 / EXCERPT: Let X be a mixed Shimura variety or a semi-abelian variety defined over $\mathbb{C}$. Let $\langle A\rangle$ denote the smallest special subvariety containing a subvariety A, a...

conjecture | open | MathDB 页面 | 原文线索

V2-017424 | MATHDB-328710

Defect condition for mixed Shimura varieties and weakly special subvarieties

摘录 / EXCERPT: Defect condition. Every mixed Shimura variety, and every weakly special subvariety, has the defect condition. The supplied span does not give the formal definition of the defect co...

conjecture | open | MathDB 页面 | 原文线索

V2-017425 | MATHDB-328712

Weak Complex Ax conjecture, geometric formulation

摘录 / EXCERPT: Let U be a domain in $\mathbb{H}^n$, let U' be a geodesic subvariety of U, and put $X'=\exp U'$. Let A be a complex-analytic component of $W\cap\pi^{-1}(V)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-017426 | MATHDB-328713

Weak Complex Ax conjecture, optimal components formulation

摘录 / EXCERPT: Let $X=Y(1)^n$ with uniformising map $\pi:\mathbb{H}^n\to X(\mathbb{C})$. For a subvariety $V\subseteq X$, a component is an irreducible component of $W\cap\pi^{-1}(V)$ for an alge...

conjecture | open | MathDB 页面 | 原文线索

V2-017427 | MATHDB-328715

Linear Galois-orbit conjecture for abelian varieties and powers of Y(1)

摘录 / EXCERPT: Let K be a field finitely generated over $\mathbb{Q}$. Suppose X is an abelian variety defined over K or $X=Y(1)^n$, and let $V\subseteq X$ be a subvariety defined over K....

conjecture | open | MathDB 页面 | 原文线索

V2-017428 | MATHDB-328716

Distinctness of parabolic bases in ordinary cohomology

摘录 / EXCERPT: Let G be a semisimple algebraic group, let B be a Borel subgroup, and let P and Q be parabolic subgroups distinct from B. Write $\mathcal{B}_P$ and $\mathcal{B}_Q$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-017429 | MATHDB-328724

The globally F-regular type implies log Fano conjecture

摘录 / EXCERPT: Globally F-regular type implies log Fano. A projective globally F-regular type variety of characteristic zero is log Fano.

conjecture | open | MathDB 页面 | 原文线索

V2-017430 | MATHDB-328725

The log Calabi-Yau varieties have Frobenius split type conjecture

摘录 / EXCERPT: Frobenius split type conjecture. If X is a log Calabi-Yau variety of characteristic zero, then X has Frobenius split type.

conjecture | open | MathDB 页面 | 原文线索

V2-017431 | MATHDB-328749

The least-eigenvalue conjecture for the matching derangement graph

摘录 / EXCERPT: Let Γ be the matching derangement graph, whose vertices are the perfect matchings under consideration, and let $\eta_{\min}(\Gamma)$ denote its least eigenvalue. The least-e...

conjecture | open | MathDB 页面 | 原文线索

V2-017432 | MATHDB-328750

The alternating sign property for the matching derangement graph

摘录 / EXCERPT: Let Γ be the matching derangement graph. For a partition $\lambda\vdash n$, define its depth by $d(\lambda)=n-\lambda_1$, and let η_λ be an eigenvalue of Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-017433 | MATHDB-328785

The Du Bois singularities and dense F-injective type conjecture

摘录 / EXCERPT: Let R be a ring essentially of finite type over a field of characteristic zero. Du Bois-F-injective correspondence conjecture. The scheme $\operatorname{Spec} R$ has only Du Bois...

conjecture | open | MathDB 页面 | 原文线索

V2-017434 | MATHDB-328786

The log canonical and dense F-pure type correspondence conjecture

摘录 / EXCERPT: Let (X,Δ) be a normal variety over a field of characteristic zero, with Δ an effective $\mathbb{Q}$-divisor and $K_X+\Delta$ $\mathbb{Q}$-Cartier. Log canonical-dense...

conjecture | open | MathDB 页面 | 原文线索

V2-017435 | MATHDB-328795

Nagata's Kähler packing conjecture for blow-ups of the projective plane

摘录 / EXCERPT: Let $\sigma:X\to\mathbb{CP}^2$ be the blow-up at points $x_1,\ldots,x_k$, with l the Poincaré dual of a line and e_q the Poincaré duals of the exceptional divisors. For...

conjecture | open | MathDB 页面 | 原文线索

V2-017436 | MATHDB-328799

Generalized Grassmannian very weak Landau-Ginzburg model conjecture

摘录 / EXCERPT: Generalized Grassmannian conjecture. The assertion that every smooth Fano complete intersection in a Grassmannian of planes has a very weak Landau-Ginzburg model should hold for co...

conjecture | open | MathDB 页面 | 原文线索

V2-017437 | MATHDB-328800

Rational Landau-Ginzburg model conjecture for smooth Fano varieties

摘录 / EXCERPT: Rational Landau-Ginzburg model conjecture. Every smooth Fano variety has a rational Landau-Ginzburg model of the same dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-017438 | MATHDB-328816

Rational framed motive equivalence conjecture

摘录 / EXCERPT: Let k be a perfect field and let $X\in Sm/k$. Let $M_{fr}(X)$ be the framed motive of X, and let $M_{fr}^{rat}(X)$ be its rational framed motive, obtained from rational framed...

conjecture | open | MathDB 页面 | 原文线索

V2-017439 | MATHDB-328833

Kudla's modularity conjecture for generating series of special cycles

摘录 / EXCERPT: Let (V,Q) be a quadratic space over $\mathbb{Q}$ of signature $(n,2)$, let D^+ be a connected component of the associated orthogonal hermitian symmetric domain, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017440 | MATHDB-328844

Denis-Mordell-Lang conjecture for q-varieties and Drinfeld modules

摘录 / EXCERPT: Denis-Mordell-Lang conjecture. There exists an A-module $G ot i H$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017441 | MATHDB-328845

Generic Denis-Mordell-Lang conjecture for q-varieties and Drinfeld modules

摘录 / EXCERPT: Generic Denis-Mordell-Lang conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017442 | MATHDB-328848

The dg-A_infty module-category equivalence conjecture

摘录 / EXCERPT: Let S be the underlying scheme, and let $\mathcal{B}$ be a sheaf of $\mathbb{Z}$-graded differential-graded $\mathcal{O}_S$-algebras and $\mathcal{A}$ a sheaf of $\mathbb{Z}$-gra...

conjecture | open | MathDB 页面 | 原文线索

V2-017443 | MATHDB-328853

Borisov-Alexeev-Borisov boundedness conjecture for Fano varieties

摘录 / EXCERPT: Borisov-Alexeev-Borisov conjecture. There is a positive number $\delta(n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017444 | MATHDB-328866

Hassett-Tschinkel weak approximation conjecture for rationally connected varieties

摘录 / EXCERPT: Hassett-Tschinkel's weak approximation conjecture. Proper rationally connected varieties defined over F satisfy weak approximation.

conjecture | open | MathDB 页面 | 原文线索

V2-017445 | MATHDB-328872

The defining-equations conjecture for the odeco variety

摘录 / EXCERPT: Let $S^d(\mathbb C^n)$ be the space of symmetric tensors, and let the odeco variety be the Zariski closure of the orthogonally decomposable tensors. For a tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-017446 | MATHDB-328924

Classification conjecture for three-valued Gauss periods in arithmetic progression

摘录 / EXCERPT: Let q be a power of a prime p, let γ be a primitive element of ${\mathbb F}_q$, and let $N>1$ divide $q-1$. The Gauss periods $\psi(\gamma^a C_0^{(N,q)})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-017447 | MATHDB-328927

Split Mukai and generalized Mukai conjectures for Fano manifolds

摘录 / EXCERPT: Split Mukai and generalized Mukai conjectures. The following two assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-017448 | MATHDB-328928

Ando-Mukai-type conjectures for Fano manifolds

摘录 / EXCERPT: Ando-Mukai-type conjectures. The following classifications are conjectured.

conjecture | open | MathDB 页面 | 原文线索

V2-017449 | MATHDB-328952

The uniform effectivity conjecture for Iitaka fibrations

摘录 / EXCERPT: Let W be a smooth projective variety of dimension d with Kodaira dimension $\kappa(W)\ge 0$. An Iitaka fibration of W is the fibration, defined up to birational equivalence,...

conjecture | open | MathDB 页面 | 原文线索

V2-017450 | MATHDB-328963

The higher-dimensional stability-condition conjecture for polarized varieties

摘录 / EXCERPT: Let (X,H) be a smooth polarized n-dimensional variety. Let $\widetilde{\mathfrak P}_n$ denote the universal cover of the parameter space obtained from the rational normal curve...

conjecture | open | MathDB 页面 | 原文线索

V2-017451 | MATHDB-328985

The main cycle-generation conjecture for unitary Shimura varieties

摘录 / EXCERPT: Let $\mathrm{Sh}_{a_{\bullet}}$ and $\mathcal{L}_{\xi}$ be as in the paper. Let λ be a dominant weight occurring in...

conjecture | open | MathDB 页面 | 原文线索

V2-017452 | MATHDB-329005

Campana's almost abelian fundamental group conjecture for special varieties

摘录 / EXCERPT: Let X be a special variety. Write $\pi_1(X)$ for its fundamental group, and define the maximal irregularity by ... Here a group is almost abelian if it contains an abelian subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-017453 | MATHDB-329007

Grothendieck's perfectness conjecture for the Néron component-group pairing

摘录 / EXCERPT: Grothendieck's perfectness conjecture. The pairing above is perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-017454 | MATHDB-329024

Slavov's algebraic Kakeya lower-bound conjecture

摘录 / EXCERPT: Let $L(t_1,t_2),M(t_1,t_2)\in\mathbb{F}_{q_0}[t_1,t_2]$ be arbitrary polynomials. Consider the map ... For each extension $\mathbb{F}_q/\mathbb{F}_{q_0}$, let $E_{\mathbb{F}_q}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-017455 | MATHDB-329046

The BIG condition for Mumford-Tate varieties with infinitely many special curves

摘录 / EXCERPT: Let X be an irreducible Mumford-Tate variety associated to a group G, and suppose that X contains infinitely many special curves. For finitely many compactified special curve...

conjecture | open | MathDB 页面 | 原文线索

V2-017456 | MATHDB-329053

Urbinati's global generation conjecture for Weil divisors

摘录 / EXCERPT: Let X be a complex normal projective variety and D a Weil divisor on X. Global generation conjecture. There is an ample Cartier divisor H on X such that for any...

conjecture | open | MathDB 页面 | 原文线索

V2-017457 | MATHDB-329064

Vakil-Wood's homological stability conjecture for symmetric complements

摘录 / EXCERPT: Let X be an irreducible smooth complex variety. For a partition λ of k, let $W_\lambda(X)$ denote the symmetric complement associated with the stratum indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-017458 | MATHDB-329067

The Landau-Ginzburg/Calabi-Yau correspondence for (Q,G)

摘录 / EXCERPT: Let (Q,G) be an LG pair, let $\mathcal{Z}=\{Q=0\}\subset\mathbb{P}(G)$ be the associated Calabi-Yau variety, and let $\mathscr{L}^{(Q,G)}\subset\mathscr{V}^{(Q,G)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017459 | MATHDB-329094

Zilber's conjecture on intersections with tori

摘录 / EXCERPT: Let $\mathcal{Y}$ be an irreducible subvariety of ${\bf G}_m^n$ defined over ${\bf C}$. For algebraic subgroups $H\subset {\bf G}_m^n$ satisfying ... consider the union of the inters...

conjecture | open | MathDB 页面 | 原文线索

V2-017460 | MATHDB-329095

The maximal Schubert specialization conjecture for Richardson permutations

摘录 / EXCERPT: For a given n, let S_n be the symmetric group and let ${\mathfrak{S}}_w$ denote the Schubert polynomial associated with a permutation $w\in S_n$. Consider the specialization...

conjecture | open | MathDB 页面 | 原文线索

V2-017461 | MATHDB-329110

Equality of the characteristic spans of the restricted and elementary varieties

摘录 / EXCERPT: Let $(M,\mathcal{I})$ be an involutive exterior differential system. Let $\dot{\Xi}$ and Ξ_E be the characteristic varieties defined in the surrounding discussion, viewed as su...

conjecture | open | MathDB 页面 | 原文线索

V2-017462 | MATHDB-329124

Liu's cohomology-class conjecture for diagram varieties

摘录 / EXCERPT: Let D be a finite diagram contained in $[k] \times [n-k]$. Let X_D be the associated diagram variety in $\operatorname{Gr}(k,n)$, with cohomology class...

conjecture | open | MathDB 页面 | 原文线索

V2-017463 | MATHDB-329125

Liu's degree conjecture for diagram varieties

摘录 / EXCERPT: Let D be a diagram contained in $[k] \times [n-k]$, let $D^\vee$ denote its box complement, let X_D be the associated diagram variety in $\operatorname{Gr}(k,n)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017464 | MATHDB-329203

The Harris-Voisin conjecture on special components of the Noether-Lefschetz locus

摘录 / EXCERPT: Let V be the parameter space of smooth complex surfaces of degree d in $\mathbb{P}^{3}$, and let NL_d be the Noether-Lefschetz locus, whose components of codim...

conjecture | open | MathDB 页面 | 原文线索

V2-017465 | MATHDB-329204

Transversality conjecture for the infinitesimal variation of Hodge structures

摘录 / EXCERPT: Let V be a parameter space of smooth projective varieties, let Y/V be the corresponding family, and let the Kodaira-Spencer map be the map referred to as. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017466 | MATHDB-329218

Mozgovoy's degree-independence conjecture for twisted-Higgs Donaldson-Thomas invariants

摘录 / EXCERPT: Mozgovoy's conjecture. The invariants $\Om_D(r,d)$ are independent of d, without requiring r and d to be coprime, and are given by the formula proposed in the cited work. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-017467 | MATHDB-329245

Huh-Sturmfels conjectural inductive formula for maximum likelihood degree

摘录 / EXCERPT: Let X be an irreducible algebraic subvariety of $({\mathbb C}^*)^n$ contained in ... and let $\bar X$ be its closure in $({\mathbb C}^*)^{n-1}\times{\mathbb C}$. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-017468 | MATHDB-329280

The ergodicity conjecture for mapping-class-group actions on punctured-surface character spaces

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a punctured surface, let $\mathcal M(\Sigma_{g,n})$ be its character space, and call a connected component non-extremal when it is not an extremal relative-Eu...

conjecture | open | MathDB 页面 | 原文线索

V2-017469 | MATHDB-329296

The stronger Jacobian Conjecture for restricted polynomial systems

摘录 / EXCERPT: Stronger Jacobian Conjecture. For all $n\geq n'\geq 0$ and all $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-017470 | MATHDB-329307

The no-critical-value conjecture for stable multi-matching percolation

摘录 / EXCERPT: Let D denote the degree of a typical vertex in the stable multi-matching scheme, and let $d_{\star}$ be a proposed critical value of the expected degree. No-critical-value conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-017471 | MATHDB-329308

Weak BAB conjecture for anti-canonical volumes

摘录 / EXCERPT: Weak BAB conjecture. Fix $0<\epsilon<1$ and an integer $n>0$. Then there exists a number $M(n,\epsilon)$ depending only on n and ϵ such that, if X is an n-dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-017472 | MATHDB-329310

Generalized Ambro conjecture for weak log Fano pairs

摘录 / EXCERPT: Generalized Ambro conjecture. Fix $0<\epsilon<1$ and an integer $n>0$. Then there exists a number $\mu(n,\epsilon)>0$ depending only on n and ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017473 | MATHDB-329313

The minimum-distance conjecture for projective nested cartesian codes

摘录 / EXCERPT: Minimum-distance conjecture. The minimum distance of $C_{\mathcal{X}}(d)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-017474 | MATHDB-329333

Automorphic component conjecture for trianguline varieties

摘录 / EXCERPT: Let $\mathfrak{X}^p$ be an irreducible component of $\mathfrak{X}^\square_{\bar{\rho}^p}$. Let X be an irreducible component of $X_{\rm tri}^\square(\bar{\rho}_p)$. By definition...

conjecture | open | MathDB 页面 | 原文线索

V2-017475 | MATHDB-329373

The vanishing of the second homotopy group for real reductive character varieties

摘录 / EXCERPT: Let G be a connected real reductive Lie group, and let $\mathfrak{X}_r(G)$ denote its rank-r character variety. Vanishing conjecture. ... The conjecture extends the vanishing res...

conjecture | open | MathDB 页面 | 原文线索

V2-017476 | MATHDB-329374

The singularity conjecture for reducible character representations

摘录 / EXCERPT: Let G be a reductive group, let r be the rank of the free group, and write $\mathfrak{X}_r(G)^{\mathrm{red}}$ and $\mathfrak{X}_r(G)^{\mathrm{sing}}$ for the reducible and sing...

conjecture | open | MathDB 页面 | 原文线索

V2-017477 | MATHDB-329375

The equivariant symplectomorphism group conjecture for cotangent flag varieties

摘录 / EXCERPT: Let $G_\mathbb{C}$ be a semisimple Lie group over $\mathbb{C}$, let $\mathcal{B}$ be its flag variety, and let $\mathrm{Sympl}_{G_\mathbb{C}}(T^*\mathcal{B})$ denote the group of s...

conjecture | open | MathDB 页面 | 原文线索

V2-017478 | MATHDB-329376

The Steinberg-equivariant symplectomorphism group conjecture for cotangent flag varieties

摘录 / EXCERPT: Let $G_\mathbb{C}$ be a semisimple Lie group over $\mathbb{C}$, let $\mathcal{B}$ be its flag variety, and let $\mathcal{Z}$ be the Steinberg variety, defined as the union of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-017479 | MATHDB-329413

Minimal defining identities conjecture for the variety of code loops

摘录 / EXCERPT: Minimal identities conjecture. These identities form a minimal set of identities defining the variety ${\cal E}$.

conjecture | open | MathDB 页面 | 原文线索

V2-017480 | MATHDB-329415

Dense F-split type for complex Abelian varieties and log Calabi-Yau varieties

摘录 / EXCERPT: Let X be a complex Abelian variety, or more generally a log Calabi-Yau variety. A complex variety has dense F-split type if, for a model over a finitely generated $\bZ$-algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-017481 | MATHDB-329423

The coincidence, density, and dynamical-relation conjecture for marked points

摘录 / EXCERPT: Let V be an irreducible quasiprojective complex algebraic variety, and let $f:V\times\mathbb{P}^1\to\mathbb{P}^1$ be an algebraic family of rational maps of degree $d\geq 2$, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-017482 | MATHDB-329437

Edixhoven's Galois-orbit conjecture for special points

摘录 / EXCERPT: Let (G,X) be a Shimura datum, let K be neat and a product of compact open subgroups K_p of $G(\mathbb Q_p)$, and fix a connected component $S=\Gamma\backslash X^+$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-017483 | MATHDB-329438

Tsimerman's height conjecture for pre-special points

摘录 / EXCERPT: Let $\mathcal F$ be the fixed fundamental set of pre-special points in the chosen connected component, and for a pre-special point $h\in\mathcal F$ retain the associated Mumford-Ta...

conjecture | open | MathDB 页面 | 原文线索

V2-017484 | MATHDB-329491

The generalized Mukai conjecture for complete spherical varieties

摘录 / EXCERPT: Let X be a complete spherical variety. Define ... Here $Delta$ is the set of colors, m_D is the associated coefficient, $rho'(D)$ is the corresponding valuation, Q^* is the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-017485 | MATHDB-329492

The combinatorial generalized Mukai conjecture for spherical skeletons

摘录 / EXCERPT: Let $Rs\coloneqq(\Sigma,S^p,\mathcal{D}^a,\Gamma)$ be a complete spherical skeleton. Define ... and ... Here R is the underlying root system and R_{S^p} is the root subsystem gener...

conjecture | open | MathDB 页面 | 原文线索

V2-017486 | MATHDB-329516

The rational-tail and nonseparating-boundary conjecture for geometric interpretations of type-A conformal blocks

摘录 / EXCERPT: Geometric interpretation conjecture. There are polarized pairs $(\mathcal{X}_x,\mathcal{L}_x)$ such that the isomorphism in Equation holds for every $x\in Z$.

conjecture | open | MathDB 页面 | 原文线索

V2-017487 | MATHDB-329521

Gross-Siebert's period-integral conjecture for toric Calabi-Yau mirrors

摘录 / EXCERPT: Let $\check{X}_t$ be the mirror family of a toric Calabi-Yau variety, let $\check{\Omega}_{\phi(q)}$ be its holomorphic volume form evaluated at the SYZ-map parameter $\phi(q)$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-017488 | MATHDB-329554

Jet Ax-Schanuel conjecture for Shimura varieties

摘录 / EXCERPT: Let $\pi:\mathbb{H}\rightarrow S$ be the uniformizing maps of a Shimura variety S corresponding to a semisimple group G. Let $k\geq 2$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017489 | MATHDB-329566

The pillowcase perturbation conjecture for 2-tangle character varieties

摘录 / EXCERPT: Let (Y,T) be a 2-tangle in the 3-ball or in a homology 3-ball. Write $R_\pi(Y,T)$ for the perturbed traceless character variety and let ... be its restriction map. A pillowcase per...

conjecture | open | MathDB 页面 | 原文线索

V2-017490 | MATHDB-329568

Conjecture on the only generically identifiable perfect tensor formats

摘录 / EXCERPT: Let a tensor have format $(n_1,\dots,n_d)$, and let ... Call the format perfect when $R(n_1,\dots,n_d)$ is an integer, and call a format generically identifiable when a general tenso...

conjecture | open | MathDB 页面 | 原文线索

V2-017491 | MATHDB-329580

Yau's finite-generation conjecture for polynomial-growth holomorphic functions

摘录 / EXCERPT: Yau's finite-generation conjecture. The ring $\mathcal{O}_P(M)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-017492 | MATHDB-329592

Unique ergodicity and uniform density for compact-window model sets

摘录 / EXCERPT: Let $(G\times H,\mathcal{L})$ be a cut and project scheme, let $W\subset H$ be a compact set, and let $\text{\Large $curlywedge$}(W)$ denote the associated model set. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017493 | MATHDB-329604

Finite-dimensionality and uniform bounds for equivariant Bernstein-eigen distributions on spherical pairs

摘录 / EXCERPT: Let G be a reductive group defined over a non-Archimedean field F of characteristic 0, and let $H_1,H_2\subset G$ be its algebraic spherical subgroups. Let χ_i be charac...

conjecture | open | MathDB 页面 | 原文线索

V2-017494 | MATHDB-329613

The dimension and degree conjecture for binomial solution spaces

摘录 / EXCERPT: Let $m,k\in\mathbb{Z}^+$ with $m\ne k$, and let $\mathcal{V}^*(\nabla W_{m,k})$ denote the solution set under consideration. The dimension and degree are conjectured to satisfy the...

conjecture | open | MathDB 页面 | 原文线索

V2-017495 | MATHDB-329628

The Goncharov-Levin comparison conjecture for elliptic curves

摘录 / EXCERPT: Let k be a field, let C be an open elliptic curve over k, and let $\overline{C}$ be its smooth complete elliptic curve. Write $\operatorname{Jac}(\overline{C})$ for its Jacob...

conjecture | open | MathDB 页面 | 原文线索

V2-017496 | MATHDB-329638

Beauville's divisor-generated cycle class conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold, and let $\operatorname{CH}^*(X)$ denote its Chow ring with $\mathbb{Q}$-coefficients. Beauville's conjecture. The cycle class map is...

conjecture | open | MathDB 页面 | 原文线索

V2-017497 | MATHDB-329639

Voisin's opposite-filtration conjecture for hyper-Kähler zero-cycles

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold of dimension $2n$. Let $S_\bullet$ be the filtration on zero-cycles induced by the loci S_iX, and let F'_{BB} denote the even leve...

conjecture | open | MathDB 页面 | 原文线索

V2-017498 | MATHDB-329640

Voisin's coisotropic-generation conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold of dimension $2n$. For $i\leq n$, call a degree-$2i$ cohomology class coisotropic if it satisfies the coisotropic condition defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-017499 | MATHDB-329641

Voisin's injectivity conjecture for classes of constant-cycle subvarieties

摘录 / EXCERPT: Let X be a hyper-Kähler variety of dimension $2n$. For each i, let $C_{2n-i}(X)\subset \operatorname{CH}^i(X)$ be the $\mathbb{Q}$-vector space generated by codimension-i sub...

conjecture | open | MathDB 页面 | 原文线索

V2-017500 | MATHDB-329642

Voisin's zero-cycle opposite-filtration conjecture

摘录 / EXCERPT: Let X be a hyper-Kähler variety of dimension $2n$, and let $S_i\operatorname{CH}_0(X)$ denote the subgroup generated by classes of points whose rational-equivalence orbits have d...

conjecture | open | MathDB 页面 | 原文线索

V2-017501 | MATHDB-329643

Voisin's vanishing conjecture for the Bloch-Beilinson filtration on constant-cycle classes

摘录 / EXCERPT: Let X be a hyper-Kähler variety of dimension $2n$. For an integer i, let $C_i(X)\subset \operatorname{CH}_i(X)=\operatorname{CH}^{2n-i}(X)$ be the $\mathbb{Q}$-vector space gen...

conjecture | open | MathDB 页面 | 原文线索

V2-017502 | MATHDB-329646

Multiplicity-cycle conjecture for nonsingular foliations

摘录 / EXCERPT: Multiplicity-cycle conjecture. There exists a mixed algebraic cycle $\Gamma(P_m;\mathcal F)\subset M$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017503 | MATHDB-329649

Zero-dimensionality conjecture for the systems L_m

摘录 / EXCERPT: Zero-dimensionality conjecture. L_m is zero-dimensional for every even $m\geqslant24$.

conjecture | open | MathDB 页面 | 原文线索

V2-017504 | MATHDB-329655

Weight-monodromy conjecture for smooth proper varieties over local fields

摘录 / EXCERPT: Let X be a proper, smooth, separated scheme of finite type over a non-Archimedean local field K, let l be a prime different from the residue characteristic, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017505 | MATHDB-329676

Secant-obstruction conjecture for linear systems through n+3 points

摘录 / EXCERPT: Let ${\mathcal L}$ be a non-empty linear system in ${\mathbb P}^n$ with $n+3$ base points in general position, and let C be the rational normal curve through those points. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017506 | MATHDB-329679

The arithmetic degree conjectures of Kawaguchi and Silverman

摘录 / EXCERPT: Let K) be an algebraically closed field with a suitable theory of height functions, let X be a smooth projective variety over K, let $\varphi:X\dashrightarrow X$ be a dominan...

conjecture | open | MathDB 页面 | 原文线索

V2-017507 | MATHDB-329692

Existence of bounded uniform stability segments

摘录 / EXCERPT: Let X be a smooth projective variety, and let $\tau\in B(X)_{\mathbb Q}$. Let σ' and σ'' be bounded stability parameters for the fixed topological type τ. Unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-017508 | MATHDB-329738

Toric cell complex and bimodule resolution conjecture for selected toric Fano varieties

摘录 / EXCERPT: Toric cell complex conjecture. The toric cell complex of B exists and is supported on a real five-dimensional torus. Moreover, the cellular resolution exists in the sense of Craw...

conjecture | open | MathDB 页面 | 原文线索

V2-017509 | MATHDB-329798

The main secant-dimension conjecture for varieties of reducible hypersurfaces

摘录 / EXCERPT: Let d be a positive integer, let $\boldsymbol{\rm X}_{n-1,\boldsymbol{\rm }[d_1,\ldots,d_r]}$ denote the variety of reducible degree-d hypersurfaces in $\mathbb P^{n-1}$ with f...

conjecture | open | MathDB 页面 | 原文线索

V2-017510 | MATHDB-329799

The WLP conjecture for joins of tangent-space schemes

摘录 / EXCERPT: Let $I_{(1)},\ldots,I_{(\ell)}$ be the tangent-space ideals at ℓ general points of the variety of reducible hypersurfaces, let B be the coordinate ring of the join of the sc...

conjecture | open | MathDB 页面 | 原文线索

V2-017511 | MATHDB-329801

Abo-Ottaviani-Peterson conjecture on defective Segre varieties

摘录 / EXCERPT: Let $X=\mathbb P^{n_1-1}\times\cdots\times\mathbb P^{n_r-1}$ be a Segre variety, with $2\le n_r\le\cdots\le n_1$, and call it balanced when ... Abo-Ottaviani-Peterson conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-017512 | MATHDB-329815

The Generic Limit conjecture for minimal log discrepancies

摘录 / EXCERPT: Let X be a klt variety over an algebraically closed field of characteristic 0, and let $x\in X$ be a closed point. Given a positive integer r and sequences of nonzero coheren...

conjecture | open | MathDB 页面 | 原文线索

V2-017513 | MATHDB-329845

Surjectivity conjecture for primitive elements of additive-type group schemes

摘录 / EXCERPT: Let A be the base ring, let q be a power of the characteristic prime p, and let $G\in\mathbb{F}_{q}\text{-}\operatorname{Gr}^{+}_{A}$. Suppose that there is a closed embeddin...

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V2-017514 | MATHDB-329846

Primitive-module flatness conjecture for additive-type group schemes

摘录 / EXCERPT: Let A be the base ring and let $G\in\operatorname{Gr}^{+}_{A}$ be a group scheme of additive type. Write $G=\operatorname{Spec}(B_G)$ and let $\operatorname{Prim}(B_G)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-017515 | MATHDB-329927

Conjecture on real rational maps and fields of moduli

摘录 / EXCERPT: Real rational-map conjecture. Every real rational map is definable over its field of moduli.

conjecture | open | MathDB 页面 | 原文线索

V2-017516 | MATHDB-329928

Conjecture on real rational maps over algebraic real fields

摘录 / EXCERPT: Algebraic real-field conjecture. It can be defined over

conjecture | open | MathDB 页面 | 原文线索

V2-017517 | MATHDB-329931

Choice-independence of higher specialization for mixed Tate motives

摘录 / EXCERPT: Let $D(X,J)$ be the higher deformation associated with a simple normal-crossings stratum Y_J and let $D^0(X,J)$ be its open deformation. Successive specialization along the...

conjecture | open | MathDB 页面 | 原文线索

V2-017518 | MATHDB-329932

Affineness conjecture for the geometric motivic group $K_{\mathbb{Z},n}$

摘录 / EXCERPT: Over $\operatorname{Spec}(\mathbb{Z})$, let $G^{\bullet}_{\mathbb{Z},n}$ be the affine derived group scheme associated with the mixed Tate category of $\mathcal{M}_{0,n}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017519 | MATHDB-329933

Boundary-factorization conjecture for motivic groups $K_{\mathbb{Z},n}$

摘录 / EXCERPT: Let $D\simeq\mathcal{M}_{0,n_1}\times\mathcal{M}_{0,n_2}$ be a codimension-one boundary component, and let $K^{\bullet}_{\mathbb{Z},n_1,n_2}$ denote the geometric derived group ass...

conjecture | open | MathDB 页面 | 原文线索

V2-017520 | MATHDB-329934

Motivic realization conjecture for the Grothendieck-Teichmüller group

摘录 / EXCERPT: Let $K_*=(K_n)_n$ be the tower of motivic groups arising from the moduli spaces $\mathcal{M}_{0,n}$, and let $\mathcal{R}_{\mathrm{Betti}}(K_*)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017521 | MATHDB-329946

The pseudovariety decomposition conjecture for partial order-preserving permutations

摘录 / EXCERPT: Let $\mathcal{PODI}_n$ be the monoid of partial order-preserving or order-reversing permutations, let $\mathcal{POI}_n$ be the monoid of partial order-preserving permutations, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017522 | MATHDB-329951

The abundance conjecture for smooth varieties

摘录 / EXCERPT: Let X be a smooth variety with nef canonical divisor. Write $\kappa(X)$ for the Kodaira dimension of X and $\nu(X)$ for the numerical dimension of its canonical divisor. Abunda...

conjecture | open | MathDB 页面 | 原文线索

V2-017523 | MATHDB-329952

Equality or comparability of mobility and the volume of the movable cone

摘录 / EXCERPT: Mobility-volume conjecture. One has

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V2-017524 | MATHDB-329998

The Fano function determines del Pezzo surfaces

摘录 / EXCERPT: Let Y be a smooth projective variety, let S be a del Pezzo surface, and let κ_Y and κ_S denote their Fano functions, where the Fano function counts deformation eq...

conjecture | open | MathDB 页面 | 原文线索

V2-017525 | MATHDB-329999

Algorithmic conjecture for isotypic multiplicities of symmetric semi-algebraic sets

摘录 / EXCERPT: Algorithmic conjecture. There is an algorithm which computes $m_{i,\lambda}(S,\mathbb{Q})$ for every $\lambda\vdash k$ with $m_{i,\lambda}(S,\mathbb{Q})>0$, as well as all the Bett...

conjecture | open | MathDB 页面 | 原文线索

V2-017526 | MATHDB-330002

The cyclotomic trace equivalence conjecture for schemes over finite fields

摘录 / EXCERPT: Let X be a nice scheme over $\mathbf{F}_{p}$, let $K(X)$ denote its algebraic K-theory spectrum, and let $TC(X,p)$ denote its p-typical topological cyclic homology. Cyclotomi...

conjecture | open | MathDB 页面 | 原文线索

V2-017527 | MATHDB-330105

Asymptotic Pfaffian-sum conjecture for general forms

摘录 / EXCERPT: Asymptotic Pfaffian-sum conjecture. A general form of degree k can be expressed as a sum of s pfaffians of such matrices, for

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V2-017528 | MATHDB-330107

Beilinson's conjecture for Grothendieck motives

摘录 / EXCERPT: Let $M=(X,p)$ be a Grothendieck motive over $\mathbb{Q}$ with coefficients in L, where X is smooth projective over $\mathbb{Q}$ and p is a projector in...

conjecture | open | MathDB 页面 | 原文线索

V2-017529 | MATHDB-330108

Weak Beilinson conjecture for Grothendieck motives

摘录 / EXCERPT: Let $M=(X,p)$ be a Grothendieck motive over $\mathbb{Q}$ with coefficients in L, a direct factor of $h^i(X)\otimes L$, and let $M^\vee=(X,{}^tp,i)$ be its dual motive. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017530 | MATHDB-330113

The conjectural endofunctor formula for automorphic D-modules

摘录 / EXCERPT: Let G be the reductive group and Bun_G the moduli stack appearing above, and let $\on{D-mod}(\operatorname{Bun}_G)_{ps-c}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017531 | MATHDB-330116

Finiteness conjecture for equilibrium points of same-sign charges

摘录 / EXCERPT: Same-sign equilibrium finiteness conjecture. For any such set of charges, the set of equilibrium points is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-017532 | MATHDB-330117

Ottaviani-Shapiro conjecture on isolated zeros of sums of squares

摘录 / EXCERPT: Ottaviani-Shapiro conjecture. For any number of variables,

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V2-017533 | MATHDB-330127

The twisted-homogeneous-variety tilting conjecture

摘录 / EXCERPT: Twisted-homogeneous-variety tilting conjecture. The category $D^b\operatorname{Coh}(X)$ admits a tilting object.

conjecture | open | MathDB 页面 | 原文线索

V2-017534 | MATHDB-330143

Saturation conjecture for matrix preorderings on unions of unbounded intervals and a point

摘录 / EXCERPT: Let $K\subseteq \mathbb R$ be either a union of an unbounded interval and a point, or a union of two unbounded intervals and a point. Suppose S is the natural description of K....

conjecture | open | MathDB 页面 | 原文线索

V2-017535 | MATHDB-330156

Semipositivity conjecture for relative pluricanonical direct images

摘录 / EXCERPT: Semipositivity conjecture for relative pluricanonical direct images. The sheaf $f'_*\omega^{\otimes m}_{X'/Y'}$ is nef and locally free for every positive integer m.

conjecture | open | MathDB 页面 | 原文线索

V2-017536 | MATHDB-330187

Restriction isomorphism for relative zero-cycles in characteristic zero

摘录 / EXCERPT: Restriction isomorphism conjecture. For $\operatorname{char}(K)=0$, this map is an isomorphism of pro abelian groups, viewing ${\rm CH}_1(X)_\Lambda$ as a constant pro group.

conjecture | open | MathDB 页面 | 原文线索

V2-017537 | MATHDB-330188

Restriction isomorphism conjecture for higher zero-cycles

摘录 / EXCERPT: Higher zero-cycle restriction conjecture. If $j=d$, corresponding to higher zero-cycles, or if $i-j=d$, corresponding to zero-cycles with coefficients in Milnor K-theory, then th...

conjecture | open | MathDB 页面 | 原文线索

V2-017538 | MATHDB-330204

The weak ordinarity conjecture for projective simple normal crossing varieties

摘录 / EXCERPT: Let V be an n-dimensional projective simple normal crossing variety over an algebraically closed field of characteristic zero, and let A be a finitely generated $\mathbb{Z}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-017539 | MATHDB-330226

The PBW conjecture for the cohomological Hall algebra of character varieties

摘录 / EXCERPT: For $g\geq 0$, define the graded mixed Hodge structures ... ... Here fSym denotes the free symmetric algebra in the relevant graded category. PBW conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-017540 | MATHDB-330227

Schiffmann's Lie algebra and PBW conjecture for character varieties

摘录 / EXCERPT: Let ... and let u be a formal variable. Schiffmann's conjecture. There is a Lie algebra structure on ... and a filtration Y on $\mathcal{H}^{\operatorname{\mathtt{Betti}}}_g$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-017541 | MATHDB-330228

The purity-part isomorphism conjecture for character varieties

摘录 / EXCERPT: Let $\mathcal{H}^{\operatorname{\mathtt{Betti}}}_g$ be the graded mixed Hodge structure associated with the untwisted character stacks, and let...

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V2-017542 | MATHDB-330236

Factoriality conjecture for character varieties of free groups

摘录 / EXCERPT: Let F_g be the free group of rank g, let G be a reductive group, and let DG denote its derived subgroup. Write $\mathcal{X}(F_g,G)$ for the character variety of representat...

conjecture | open | MathDB 页面 | 原文线索

V2-017543 | MATHDB-330249

The distinguished discrete-spectrum decomposition hypothesis for Sp_{2n}

摘录 / EXCERPT: Let $G=Sp_{2n}$ and let $H=Sp_n\times Sp_n$. For a Levi subgroup L of G, write L_H for its distinguished subgroup data, let $\mathfrak{a}_L$ be the corresponding real split c...

conjecture | open | MathDB 页面 | 原文线索

V2-017544 | MATHDB-330277

Lang's divisible-subgroup conjecture for affine subsets of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety defined over $\mathbb{C}$, let U be an open affine subset of A, and let $F\subset \mathbb{C}^n$ be a finitely generated subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017545 | MATHDB-330297

The polynomial formula for symplectic Dellac configuration numbers

摘录 / EXCERPT: Polynomial enumeration conjecture. For any $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-017546 | MATHDB-330306

Regularity formula for Schubert singularities

摘录 / EXCERPT: Let $w=(n-r+1,n-r+2,\dots,n,a_{r+1},\dots,a_{n-1},N)\in\mathcal W_r$, set $a_r=n$ and $a_n=N$, and for $r\leq i\leq n-1$ define $m_i=a_{i+1}-a_i$. Thus $m_i>0$ and $\mathcal U_i$ appe...

conjecture | open | MathDB 页面 | 原文线索

V2-017547 | MATHDB-330307

Generic uniqueness conjecture for representations of the Kauffman skein algebra

摘录 / EXCERPT: Let S be the punctured surface under consideration, let A be a primitive N-root of -1 with N odd, and let $\mathcal S_A(S)$ be its Kauffman skein algebra. For an irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-017548 | MATHDB-330312

The integral Chern-class conjecture for \operatorname{SL}_{\ell^2}/\mu_\ell

摘录 / EXCERPT: Let ℓ be an odd prime, let $\mathrm{G}=\operatorname{SL}_{\ell^2}/\mu_\ell$ over $\mathds{C}$, and let c_2 generate $\operatorname{H}^4(\mathrm{BG},\mathds{Z})$. Integral Ch...

conjecture | open | MathDB 页面 | 原文线索

V2-017549 | MATHDB-330327

Geometric P=W conjecture for character varieties

摘录 / EXCERPT: Let ${\rm M}_{Dol}(S,C_{\cdot})$ be the Dolbeault (Hitchin) moduli space and ${\rm M}_B(S,C_{\cdot})$ the Betti character variety. Let $N^{\ast}_{Dol}$ and $N^{\ast}_B$ be neighbor...

conjecture | open | MathDB 页面 | 原文线索

V2-017550 | MATHDB-330346

The bordering-chamber conjecture for smooth projective toric varieties of Picard number at most four

摘录 / EXCERPT: Let V be a reduced $n\times(n+r)$ F-matrix and let $\gamma\in\mathcal{A}_{\Gamma}(V)$ be a non-singular chamber. A chamber is bordering if it lies on the boundary of the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-017551 | MATHDB-330347

The codimension-two discriminant conjecture for one-dimensional Schubert problems

摘录 / EXCERPT: Let $\mathcal{S}(\lambda_\bullet)\to\mathcal{M}_{0,r}$ be the family of one-dimensional Schubert problems, and let $X\subseteq\mathcal{M}_{0,r}$ be the closure of the locus where t...

conjecture | open | MathDB 页面 | 原文线索

V2-017552 | MATHDB-330348

Smoothness conjecture for real fibers of one-dimensional Schubert problems

摘录 / EXCERPT: Let $\mathcal{S}(\lambda_\bullet)\to\mathcal{M}_{0,r}$ be the family of one-dimensional Schubert problems, let $\mathcal{M}_{0,r}(\mathbb{R})$ denote its real locus, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017553 | MATHDB-330372

Resolution of singularities of the cotangent sheaf

摘录 / EXCERPT: Let X_0 be a reduced algebraic variety over a field of characteristic zero, or a reduced complex- or real-analytic variety, and let $n=\dim X_0$. A resolution of singularities is...

conjecture | open | MathDB 页面 | 原文线索

V2-017554 | MATHDB-330386

The conjectural injectivity of pushforward on Chow groups for very general ample divisors

摘录 / EXCERPT: Let X be a smooth projective variety, let $D\subset X$ be a very general smooth ample divisor of sufficiently large degree, and let $j:D\hookrightarrow X$ be the inclusion. Pushf...

conjecture | open | MathDB 页面 | 原文线索

V2-017555 | MATHDB-330391

The expected-dimension conjecture for higher-dimensional fradeco varieties

摘录 / EXCERPT: Expected-dimension conjecture. The dimension of $\mathcal{T}_{r,n,d}$ equals

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V2-017556 | MATHDB-330440

Injectivity conjecture for the map from graph schemes to finite type invariants

摘录 / EXCERPT: Let M be a closed connected oriented 3-manifold, and let ψ_n be the map defined in the paper from the degree-n graph-scheme space to the associated graded space of finite...

conjecture | open | MathDB 页面 | 原文线索

V2-017557 | MATHDB-330446

The semisimplicity conjecture for motivic Galois representations

摘录 / EXCERPT: Let $M_{\mathbb{Q}}$ be a pure $\mathbb{Q}$-motive over F, let p be a prime number, let M_p be its p-adic étale realization, and let G_F be the absolute Galois group of F...

conjecture | open | MathDB 页面 | 原文线索

V2-017558 | MATHDB-330448

The numerical-homological equivalence conjecture for motives

摘录 / EXCERPT: Let M be a motive for which numerical equivalence and p-adic homological equivalence are defined. Numerical-homological equivalence conjecture. Numerical equivalence is equal t...

conjecture | open | MathDB 页面 | 原文线索

V2-017559 | MATHDB-330464

Langlands reciprocity conjecture for motivic L-functions

摘录 / EXCERPT: Langlands reciprocity conjecture. For each $0\leq i\leq 2n$ there exists an irreducible representation π_i of the group $G(\mathbf{A}_K)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017560 | MATHDB-330468

The derived-category motivic measure conjecture

摘录 / EXCERPT: Let k be a field. Write $K(\mathbf{Var}_{/k})$ for the K-theory spectrum of varieties over k, and let $K(S)$ and $K(k)$ denote the relevant K-theory E_∞-ring spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-017561 | MATHDB-330469

The ring-spectrum motivic measure conjecture

摘录 / EXCERPT: Let k be a field, and let $K(\mathbf{Var}_{/k})$ be the K-theory spectrum of varieties over k. Let $K(S)$ be the E_∞-ring spectrum whose modules contain the K-theory...

conjecture | open | MathDB 页面 | 原文线索

V2-017562 | MATHDB-330470

Compactness or dualizability conjecture for K-theory of smooth schemes

摘录 / EXCERPT: Let X be a smooth scheme, and let $K(X)$ denote its K-theory spectrum, regarded as a module over the $K(S)$ ring spectrum. Compactness or dualizability conjecture. Then $K(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-017563 | MATHDB-330480

Shareshian-Wachs Hessenberg variety representation conjecture

摘录 / EXCERPT: Let ${\bf m}$ be a Hessenberg vector, and let $P({\bf m})$ be the natural unit interval order on $[n]$ defined by $i<_{P({\bf m})}j$ if $j>m_i$. Let ${\mathcal H}({\bf m})$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-017564 | MATHDB-330482

Bogomolov's conjecture for non-isotrivial curves over function fields

摘录 / EXCERPT: Let K be a function field or a number field, let C be a smooth projective curve of genus $g\geq 2$ over $\overline{K}$, let J_C be its Jacobian, and fix a degree-one divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-017565 | MATHDB-330483

Yamaki's geometric Bogomolov conjecture for abelian varieties

摘录 / EXCERPT: Let K be a function field with algebraic closure $\overline{K}$, let A be an abelian variety over $\overline{K}$, and let X be a closed subvariety of A. A closed subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-017566 | MATHDB-330484

The geometric Bogomolov assertion for a closed subvariety

摘录 / EXCERPT: Let A be an abelian variety over $\overline{K}$ and let X be a closed subvariety of A. The geometric Bogomolov assertion. If X has dense small points, then X is a special...

conjecture | open | MathDB 页面 | 原文线索

V2-017567 | MATHDB-330503

Batyrev-Manin's conjecture for big divisors

摘录 / EXCERPT: Let X be a smooth projective variety over a number field K, with Néron-Severi space $\mathrm{NS}(X)_{\mathbb{R}}$ and pseudo-effective cone $\mathrm{PEf}(X)$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-017568 | MATHDB-330544

Fully real eigenconfiguration conjecture near products of linear forms

摘录 / EXCERPT: Let ϕ be any product of d nonzero linear forms in ${\rm Sym}_1(\mathbb{R}^n)$. Every open neighborhood of ϕ in ${\rm Sym}_d(\mathbb{R}^n)$ contains a symmetric tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-017569 | MATHDB-330622

The K-equivalence conjecture for gauged Landau-Ginzburg models

摘录 / EXCERPT: Let $(X_1,\mathcal{O}(\chi),W_1)^G$ and $(X_2,\mathcal{O}(\chi),W_2)^G$ be gauged Landau-Ginzburg models, where G is a group acting on the varieties and W_1,W_2 are χ-semi...

conjecture | open | MathDB 页面 | 原文线索

V2-017570 | MATHDB-330623

Kawamata's converse conjecture for derived equivalence

摘录 / EXCERPT: Let X and Y be smooth projective varieties. They are K-equivalent if they admit a common resolution Z with morphisms $p:Z\to X$ and $q:Z\to Y$ satisfying $p^*K_X=q^*K_Y$. Kaw...

conjecture | refuted | MathDB 页面 | 原文线索

V2-017571 | MATHDB-330627

Kontsevich-Soibelman degeneration conjecture for smooth and compact DG categories

摘录 / EXCERPT: Kontsevich-Soibelman conjecture. This spectral sequence degenerates at the first sheet. The conjecture generalizes the Hodge-to-de Rham degeneration for smooth projective varieties...

conjecture | open | MathDB 页面 | 原文线索

V2-017572 | MATHDB-330633

Combinatorial-geometric facet equality conjecture for Grassmann polytopes

摘录 / EXCERPT: Facet equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017573 | MATHDB-330634

Pole and zero locus conjecture for amplituhedron canonical forms

摘录 / EXCERPT: Pole and zero locus conjecture. The form $\omega_{Z(\Pi_f)}$ has simple poles only along the codimension-one subvarieties $Z(\Pi_g)$, and all its zeroes lie in

conjecture | open | MathDB 页面 | 原文线索

V2-017574 | MATHDB-330664

The auto Igusa-zeta function as a perfect local invariant

摘录 / EXCERPT: Let k be any field, and let X and Y be separated k-schemes of finite type. For points $p\in X$ and $q\in Y$, write $\zeta_{X,p}(t)$ and $\zeta_{Y,q}(t)$ for their auto Igus...

conjecture | open | MathDB 页面 | 原文线索

V2-017575 | MATHDB-330704

André-Pink conjecture for principally polarized abelian varieties

摘录 / EXCERPT: Let $\mathcal{A}_g$ be the moduli space of principally polarized abelian varieties of dimension g over $\mathbb{C}$, let $Z\subset\mathcal{A}_g$ be an irreducible algebraic subva...

conjecture | open | MathDB 页面 | 原文线索

V2-017576 | MATHDB-330705

André-Pink conjecture with unpolarized isogeny classes

摘录 / EXCERPT: Let $\mathcal{A}_g$ be the moduli space of principally polarized abelian varieties of dimension g over $\mathbb{C}$, and let $Z\subset\mathcal{A}_g$ be an irreducible algebraic s...

conjecture | open | MathDB 页面 | 原文线索

V2-017577 | MATHDB-330714

Modified Lee-Zhang conjecture for symplectic and orthogonal Hitchin representations

摘录 / EXCERPT: Modified Lee-Zhang conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017578 | MATHDB-330717

The secant linear virtual dimension formula for Euler characteristics

摘录 / EXCERPT: Let D) be an effective divisor on ${\mathbb P}^n$ blown up at $n+3$ general points, let Y^σ be the corresponding iterated blow-up, and let $\tilde{D}^{\sigma}$ be the stri...

conjecture | open | MathDB 页面 | 原文线索

V2-017579 | MATHDB-330725

Lang-Silverman conjecture for Zariski-dense endomorphism orbits

摘录 / EXCERPT: Lang-Silverman conjecture. There exists a positive quantity $c_{23}=c_{23}(K,g)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017580 | MATHDB-330726

Strong Lang-Silverman conjecture for exceptional points on abelian varieties

摘录 / EXCERPT: Strong Lang-Silverman conjecture. There exist positive quantities $c_{33}=c_{33}(K,g)$ and $c_{34}=c_{34}(K,g)$ such that either there is an abelian subvariety $B\subset A$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-017581 | MATHDB-330738

The algebraic-geometric Novikov conjecture for higher Todd genera

摘录 / EXCERPT: Algebraic geometry Novikov conjecture. The corresponding higher Todd genera agree:

conjecture | open | MathDB 页面 | 原文线索

V2-017582 | MATHDB-330743

Nonrationality conjecture for the varieties X_n

摘录 / EXCERPT: Let X_n be the varieties in the series of G_K-Mori fiber spaces over $\mathbb{P}^1$ whose general fibers are isomorphic to the degree-2 del Pezzo surface S_K. Nonrationalit...

conjecture | open | MathDB 页面 | 原文线索

V2-017583 | MATHDB-330803

Singularity conjecture for higher-degree spectrahedral boundaries

摘录 / EXCERPT: Let $A,B,C \in \operatorname{Sym}_r(\mathbb{R})$ be real symmetric matrices of size r, and set ... Assume that f is a polynomial of degree at least three. Singularity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-017584 | MATHDB-330826

The motivic cycle-class criterion for algebraizability of topological vector bundles

摘录 / EXCERPT: The motivic cycle-class criterion. If this map is an isomorphism, then every topological vector bundle on X^{an} is algebraizable.

conjecture | open | MathDB 页面 | 原文线索

V2-017585 | MATHDB-330827

Nori-Totaro conjecture on Chow groups of very general hypersurfaces

摘录 / EXCERPT: Nori-Totaro conjecture. This restriction map is an isomorphism for every $i<\dim Z$.

conjecture | open | MathDB 页面 | 原文线索

V2-017586 | MATHDB-330832

The stratified Kisin variety determination conjecture for deformation spaces

摘录 / EXCERPT: Let $D^\psi({\bf v},{\rm t},\overline\rho)$ denote the potentially Barsotti-Tate deformation space, and let $\overline{\mathscr{G\!R}}^\psi({\bf v},{\rm t},\overline\rho)$ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-017587 | MATHDB-330833

The product decomposition conjecture for deformation spaces and stratified Kisin factors

摘录 / EXCERPT: Let S be the set of indices at which the projection of the Kisin variety to the corresponding factor of $(\mathbb P^1_{k_E})^f$ is constant, and write $0=i_1<i_2<\cdots<i_r$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-017588 | MATHDB-330869

Humphreys' weight-cell support-variety conjecture

摘录 / EXCERPT: Let G be a semisimple simply connected algebraic group, let W_p^+ be the dominant right-coset representatives, let $c_{[w]}$ be the weight cell associated with the right cell...

conjecture | open | MathDB 页面 | 原文线索

V2-017589 | MATHDB-330870

Cooper's support-variety conjecture for tilting modules

摘录 / EXCERPT: Let G be a semisimple simply connected algebraic group over a field of characteristic p, with Frobenius kernel G_1, weight lattice $X(T)$, and nilpotent cone...

conjecture | open | MathDB 页面 | 原文线索

V2-017590 | MATHDB-330871

The quantum-modular support-variety correspondence conjecture

摘录 / EXCERPT: Let G be a semisimple simply connected algebraic group over a field k of characteristic $p>h$, let G_1 be its Frobenius kernel, and let $\mathbb{U}_{\zeta}(\mathfrak{g})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-017591 | MATHDB-330897

Conjecture on Coxeter-type affine Deligne-Lusztig stratifications

摘录 / EXCERPT: Let G be a reductive group, let mu be a Hodge cocharacter, and let K be a parahoric subgroup. Assume that (G,μ,K) is of Coxeter type, and let the corresponding basic affi...

conjecture | open | MathDB 页面 | 原文线索

V2-017592 | MATHDB-330912

The multi-degree bound conjecture for higher Betti numbers of realizable sign conditions

摘录 / EXCERPT: Let $\mathcal{P}$ and $\mathcal{Q}$ be finite sets of polynomials in $\mathbb{R}[X_1,\ldots,X_k]$, with the same notation and hypotheses as in Theorem B-B; in particular, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017593 | MATHDB-330918

Support conjecture for zero-cycles on complete intersections

摘录 / EXCERPT: Support conjecture. If $A_0X_{\mathbb{Q}}$ is supported on a closed subvariety of dimension ℓ, then Y supports $A_0X_{\mathbb{Q}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-017594 | MATHDB-330919

Hard Lefschetz injectivity conjecture for Abel-Jacobi trivial Chow groups

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, and let L be an ample line bundle. Write $A^j_{AJ}(X)_{\Q}$ for the subgroup of Abel-Jacobi trivial codimension-j cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-017595 | MATHDB-330920

Hard Lefschetz surjectivity conjecture for zero-cycles

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, and let L be an ample line bundle. Let $N^1H^i(X,\mathbb{Q})$ denote the first coniveau piece of cohomology. Hard Lefsche...

conjecture | open | MathDB 页面 | 原文线索

V2-017596 | MATHDB-330948

Morel's exact sequence conjecture for motivic homotopy sheaves

摘录 / EXCERPT: Let k be a base field, let n and i be integers with $n\geq 4$ and $i\geq 0$, and write $S^{(n-1+i)+n\alpha}$ for the corresponding motivic sphere. Let...

conjecture | solved | MathDB 页面 | 原文线索

V2-017597 | MATHDB-330998

The Cohen-Macaulayness conjecture for generalized power-sum varieties

摘录 / EXCERPT: Let $\boldsymbol{\tau}=(\tau_1,\tau_2,\tau_3,\ldots)$ be a partition, and let X_λ denote the variety associated with the partition λ in the algebra of generalized p...

conjecture | open | MathDB 页面 | 原文线索

V2-017598 | MATHDB-331036

The conjecture for the maximal dimension of constant-rank matrix spaces

摘录 / EXCERPT: Constant-rank matrix dimension conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-017599 | MATHDB-331085

Asymptotic enumeration conjecture for deque-sortable and parallel two-stack-sortable permutations

摘录 / EXCERPT: Asymptotic enumeration conjecture. There exist constants μ and γ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017600 | MATHDB-331127

Analytic properties of representation zeta functions of unipotent group schemes

摘录 / EXCERPT: Let $\mathcal{O}$ be the ring of integers of a number field K, and let $\mathbf{G}$ be a unipotent group scheme arising from a finitely generated free torsion-free $\mathcal{O}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-017601 | MATHDB-331128

Topological representation zeta function of infinitesimal Heisenberg groups

摘录 / EXCERPT: Let $\mathbb{Q}[\varepsilon_n]=\mathbb{Q}[x]/(x^n)$, and let $\mathbf{H}[\varepsilon_n]$ denote the group scheme obtained from the Heisenberg Lie algebra tensored with...

conjecture | open | MathDB 页面 | 原文线索

V2-017602 | MATHDB-331153

Tian's alpha-invariant conjecture for projectively normal polarised varieties

摘录 / EXCERPT: Let X be a smooth variety and let L be an ample Cartier divisor on X. For a positive integer n, define ... with $\alpha_n(X,L)=+\infty$ if $|nL|$ is empty, and define ... Assum...

conjecture | open | MathDB 页面 | 原文线索

V2-017603 | MATHDB-331156

SOS conjecture on linear ranks of polynomial mappings

摘录 / EXCERPT: Let $P(z)=(P^1(z),\ldots,P^q(z))$ be a polynomial mapping in $z=(z^1,\ldots,z^n)\in\mathbb C^n$, and let $A(z,\bar z)$ be a Hermitian polynomial satisfying ... If r denotes the lin...

conjecture | open | MathDB 页面 | 原文线索

V2-017604 | MATHDB-331164

Existence of Schurian representations for non-Schurian roots of type one

摘录 / EXCERPT: Schurian representation conjecture. If α is a non-Schurian root of type one, the variety I^{α} contains a Schurian representation.

conjecture | open | MathDB 页面 | 原文线索

V2-017605 | MATHDB-331207

Arrondo-Bernardi nondefectivity conjecture for secant varieties of Chow varieties

摘录 / EXCERPT: Arrondo-Bernardi conjecture. The secant variety $\sigma_s(\operatorname{Split}_d(\mathbb P^n))$ is nondefective unless $d=2$ and $2\leq s\leq\frac{n}{2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-017606 | MATHDB-331208

Upper cluster algebra conjecture for semi-invariant rings of complete flag tuples

摘录 / EXCERPT: Upper cluster algebra conjecture. The quiver S_n^m can be projected from a triple flag quiver, and

conjecture | open | MathDB 页面 | 原文线索

V2-017607 | MATHDB-331211

Multiplicity-one conjecture for the Schwartz space model

摘录 / EXCERPT: Let P be a parabolic subgroup with Levi component M, let X_P be the associated spherical variety, and set $\mathcal{S}:=\mathcal{S}(X_P)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017608 | MATHDB-331230

The anticanonical cylinder conjecture for log del Pezzo surfaces

摘录 / EXCERPT: Let S be a log del Pezzo surface. Let $\mathrm{Amp}(S)$ be its ample cone in $\operatorname{Pic}(S)\otimes\mathbb{R}$, and let $\mathrm{Amp}^{cyl}(S)$ be the set of ample classes...

conjecture | open | MathDB 页面 | 原文线索

V2-017609 | MATHDB-331266

The classical-points characterization of partially de Rham families

摘录 / EXCERPT: Fix a prime $ u mid p$, a nonempty proper subset $J e u$ of the set $ u$-adic embeddings, an integer w, and integers $k_{\tau}2$ with $k_{\tau} w\pmod{2}$ for every $\tau otin J$...

conjecture | open | MathDB 页面 | 原文线索

V2-017610 | MATHDB-331272

Canonical reduction for arithmetic modular heights

摘录 / EXCERPT: Let (X,L) be a normal projective variety over a number field K, and consider integral projective models $(\mathcal{X},\mathcal{L})$ over rings of integers $\mathcal{O}_{K'}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-017611 | MATHDB-331273

Arithmetic MMP with scaling

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{L})$ be a log-canonical arithmetic projective variety whose general fiber satisfies $K_X\equiv aL$ for some $a\in\mathbb{R}$. Arithmetic MMP with scaling...

conjecture | open | MathDB 页面 | 原文线索

V2-017612 | MATHDB-331274

Normalised arithmetic MMP with scaling

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{L})$ be a log-canonical arithmetic projective variety with $K_X\equiv aL$ on the general fiber for some $a\in\mathbb{R}$, and let E denote the divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-017613 | MATHDB-331275

Semistable arithmetic Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Let (X,L) be a smooth projective variety over a number field K. Semistable arithmetic Yau-Tian-Donaldson conjecture. The following conditions are equivalent:...

conjecture | open | MathDB 页面 | 原文线索

V2-017614 | MATHDB-331283

Langlands's motivic-to-automorphic conjecture

摘录 / EXCERPT: Une L-fonction motivique est une fonction provenant de la cohomologie motivique d'une variété arithmétique, tandis qu'une L-fonction automorphe provient de formes modulaires pa...

conjecture | open | MathDB 页面 | 原文线索

V2-017615 | MATHDB-331325

Equivariant Grothendieck-resolution embedding for steep partitions

摘录 / EXCERPT: Embedding conjecture. There exists an SL_2-equivariant embedding

conjecture | open | MathDB 页面 | 原文线索

V2-017616 | MATHDB-331334

Density of mapping class group orbits for non-elementary non-discrete representations

摘录 / EXCERPT: Let Γ_o be the group of the six-point configuration and let X_0 denote the relevant character space of representations into $\mathrm{PSL}_2(\mathbb{R})$. For a representat...

conjecture | open | MathDB 页面 | 原文线索

V2-017617 | MATHDB-331346

Tseng-You's root-stack Gromov-Witten determination conjecture

摘录 / EXCERPT: Root-stack Gromov-Witten determination conjecture. The Gromov-Witten theory of $\mathcal{X}_r$ is determined by the Gromov-Witten theory of X, D, and the restriction map...

conjecture | open | MathDB 页面 | 原文线索

V2-017618 | MATHDB-331347

Tseng-You's projective-bundle and relative Gromov-Witten determination conjecture

摘录 / EXCERPT: Let D be a smooth proper Deligne-Mumford stack with projective coarse moduli space, and let L be a line bundle on D. Put ... Write $\pi:Y\to D$ for the natural projection, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017619 | MATHDB-331403

Deligne's purity and compatible-systems conjecture for lisse sheaves

摘录 / EXCERPT: Let p and ℓ be distinct primes. Let X be a connected normal scheme of finite type over $\mathbb{F}_p$, and let $\mathcal{E}$ be an irreducible lisse...

conjecture | open | MathDB 页面 | 原文线索

V2-017620 | MATHDB-331404

Folklore compatible-systems conjecture for de Rham lisse sheaves

摘录 / EXCERPT: Let ℓ be a rational prime. Let X be an irreducible regular scheme, flat and of finite type over $\mathbb{Z}[\ell^{-1}]$. Let E be a finite extension of $\mathbb{Q}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017621 | MATHDB-331406

Functoriality of the motivic pro-semisimple completion

摘录 / EXCERPT: Let $(X,\tilde X)$ and $(X',\tilde X')$ be pairs as in the paper, with a morphism $(X',\tilde X')\to(X,\tilde X)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017622 | MATHDB-331442

The noetherian property for admissible subcategories of derived categories

摘录 / EXCERPT: Let X be a smooth projective variety, and write $\BD(X)$ for its derived category. Let ... be a decreasing chain of admissible subcategories. Noetherian property. The chain stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-017623 | MATHDB-331443

Nonnegativity conjecture for the dimension of Calabi-Yau subcategories

摘录 / EXCERPT: Let X be a smooth projective variety and let $\CA\subset\BD(X)$ be a Calabi-Yau subcategory. Nonnegativity conjecture. The CY-dimension of $\CA$ is nonnegative. This gives a lowe...

conjecture | open | MathDB 页面 | 原文线索

V2-017624 | MATHDB-331444

Equality case for the CY-dimension bound

摘录 / EXCERPT: Let X be a smooth projective variety and let $\CA\subset\BD(X)$ be an n-Calabi-Yau category with $n=\dim X$. Equality-case conjecture. There is a regular birational morphism .....

conjecture | open | MathDB 页面 | 原文线索

V2-017625 | MATHDB-331454

Birational boundedness conjecture for epsilon-Fano type varieties

摘录 / EXCERPT: Let $n>0$ be an integer and let $0<\epsilon<1$. A normal projective variety is of ϵ-Fano type if there is an effective $\mathbb{Q}$-divisor B such that (X,B) is an...

conjecture | open | MathDB 页面 | 原文线索

V2-017626 | MATHDB-331456

Terminal Fano boundedness and birational boundedness conjectures

摘录 / EXCERPT: Fix an integer n. A terminal Fano variety is Picard-number-one terminal when it is an n-dimensional $\mathbb{Q}$-factorial terminal Fano variety of Picard number one. Terminal...

conjecture | open | MathDB 页面 | 原文线索

V2-017627 | MATHDB-331457

Weak terminal Fano volume and birationality conjectures

摘录 / EXCERPT: Fix an integer n. Weak terminal Fano conjectures. There exists a number $M_0(n)$ depending only on n such that every n-dimensional $\mathbb{Q}$-factorial terminal Fano variet...

conjecture | open | MathDB 页面 | 原文线索

V2-017628 | MATHDB-331458

Fano-type base conjecture for Mori fibrations

摘录 / EXCERPT: Let $n>0$ be an integer and let $0<\epsilon<1$. A normal projective variety is of ϵ-Fano type if there exists an effective $\mathbb{Q}$-divisor B such that (X,B) is an...

conjecture | open | MathDB 页面 | 原文线索

V2-017629 | MATHDB-331470

The boundary Calabi-Yau component conjecture for semiorthogonal decompositions

摘录 / EXCERPT: Let X be a smooth projective variety with a semiorthogonal decomposition ... where $\mathcal{A}$ is a Calabi-Yau category of dimension $n=\dim X$. Boundary Calabi-Yau component con...

conjecture | open | MathDB 页面 | 原文线索

V2-017630 | MATHDB-331488

The split-reduction asymptotic conjecture for abelian surfaces

摘录 / EXCERPT: Let K be a number field and let A/K be a principally-polarizable abelian surface whose absolute endomorphism ring satisfies ... Let $\pi_{\operatorname{split}}(A/K,z)$ count prim...

conjecture | open | MathDB 页面 | 原文线索

V2-017631 | MATHDB-331510

Main local tangent-space dimension conjecture for trianguline varieties

摘录 / EXCERPT: Let $x\in X_{\rm tri}^\square(\overline{r})$ be a crystalline strictly dominant point whose Frobenius eigenvalues $(\varphi_1,\dots,\varphi_n)$ are pairwise distinct. Let w_x be...

conjecture | open | MathDB 页面 | 原文线索

V2-017632 | MATHDB-331511

BHS automorphic trianguline subvariety conjecture

摘录 / EXCERPT: BHS automorphic trianguline subvariety conjecture. The rigid subvariety $X_{\rm tri}^{\mathfrak{X}^p\rm-aut}(\overline{\rho}_p)$ does not depend on $\mathfrak{X}^p$ and is isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-017633 | MATHDB-331516

Ogus' conjecture on de Rham-Tate cycles

摘录 / EXCERPT: Let A be an abelian variety defined over a number field K. Via the comparison isomorphism between Betti and de Rham cohomologies, regard de Rham-Tate cycles as corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-017634 | MATHDB-331517

The relative Frobenius algebraicity conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field K, let L be the coefficient field used for its de Rham cohomology, and let $s\in \operatorname{End}(H^1_{\mathrm{dR}}(A,L))$....

conjecture | open | MathDB 页面 | 原文线索

V2-017635 | MATHDB-331519

Convergence of polylogarithmic Chabauty-Kim loci

摘录 / EXCERPT: Let Z be a totally real integer scheme, and let $\mathfrak{p} in Z$ be a totally split prime. For each n, let $X(\mathcal{O}_\mathfrak{p})_n$ denote the associated polylogarith...

conjecture | open | MathDB 页面 | 原文线索

V2-017636 | MATHDB-331520

Zagier's conjecture on motivic Ext groups and single-valued unipotent polylogarithms

摘录 / EXCERPT: Let K be the field appearing in the motivic category, and for each integer $n\geq 2$ write ... For $t\in K$, let $\operatorname{Li}_n^{U,sv}(t)$ denote the single-valued unipotent...

conjecture | open | MathDB 页面 | 原文线索

V2-017637 | MATHDB-331525

Conjecture on derived invariance of derived Jacobians

摘录 / EXCERPT: Derived-Jacobian invariance conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-017638 | MATHDB-331560

The local and global log-canonical threshold conjecture for diagonal ideals

摘录 / EXCERPT: The local and global log-canonical threshold conjecture. If x lies in the stratum X^n_{λ}, then

conjecture | open | MathDB 页面 | 原文线索

V2-017639 | MATHDB-331561

The canonical and log-canonical singularity classification conjecture for the isospectral Hilbert scheme

摘录 / EXCERPT: Singularity classification conjecture. The singularities of B^n are canonical if and only if $n\leq 7$, log-canonical if $n\leq 8$, and not log-canonical if $n\geq 9$.

conjecture | open | MathDB 页面 | 原文线索

V2-017640 | MATHDB-331562

The log-canonical threshold conjecture for the boundary of the Hilbert scheme

摘录 / EXCERPT: Boundary log-canonical threshold conjecture. The log-canonical threshold of the pair $(X^{[n]},\mathcal{I}_{\partial X^{[n]}})$ is precisely

conjecture | open | MathDB 页面 | 原文线索

V2-017641 | MATHDB-331566

The étale-local quotient stack conjecture

摘录 / EXCERPT: Let X be an algebraic stack locally of finite type over an algebraically closed field k with separated and quasi-compact diagonal, and suppose that X has affine stabilizer gr...

conjecture | open | MathDB 页面 | 原文线索

V2-017642 | MATHDB-331568

Conjectural tropical analog of Descartes' rule of signs

摘录 / EXCERPT: Conjectural tropical analog of Descartes' rule of signs. For any real univariate polynomial $P(x)$, the number of its positive (negative) roots does not exceed the number of positi...

conjecture | open | MathDB 页面 | 原文线索

V2-017643 | MATHDB-331569

The homotopy quotient conjecture for the complex brick operad

摘录 / EXCERPT: Let $\pazocal{B}(n)$ denote the toric brick manifolds associated with Loday's realisations of the associahedra, and let As_{S^1} be the nonsymmetric operad introdu...

conjecture | open | MathDB 页面 | 原文线索

V2-017644 | MATHDB-331571

The infinitude conjecture for rational curves on K3 surfaces

摘录 / EXCERPT: The infinitude conjecture for rational curves on K3 surfaces. Every polarized K3 surface (X,H) over an algebraically closed field contains infinitely many integral rational curve...

conjecture | open | MathDB 页面 | 原文线索

V2-017645 | MATHDB-331583

Closedness of the Calabi-Yau property under deformation

摘录 / EXCERPT: Let X be a smooth projective variety, let $\mathcal{T} \subset \operatorname{D^b}(X)$ be a full admissible subcategory, and let B be a smooth algebraic variety. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017646 | MATHDB-331584

Salamon's relation for Hochschild numbers of hyper-Kähler categories

摘录 / EXCERPT: Let $\mathcal{T}$ be a full admissible subcategory of the derived category of a smooth projective variety. Assume that $\mathcal{T}$ is hyper-Kähler of dimension $2r$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-017647 | MATHDB-331585

Kuznetsov's birationality conjecture for Calabi-Yau subcategories

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, and let $\mathcal{A}\subset \operatorname{D^b}(X)$ be a full admissible subcategory that is Calabi-Yau of dimension n. Ku...

conjecture | open | MathDB 页面 | 原文线索

V2-017648 | MATHDB-331597

The Garcia-Taylor join-primality conjectures for Maltsev conditions

摘录 / EXCERPT: Let a Maltsev condition determine a filter in the lattice of interpretability types of varieties. A filter is join prime when its complement is closed under joins. Garcia-Taylor co...

conjecture | open | MathDB 页面 | 原文线索

V2-017649 | MATHDB-331611

Finite jet detection conjecture for MJ-minimal log discrepancies

摘录 / EXCERPT: Finite jet detection conjecture. There is a number $N_{\delta,d}$ depending only on δ and d such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-017650 | MATHDB-331615

Gorsky-Rasmussen conjecture for twisted projectors

摘录 / EXCERPT: Let n be a positive integer, let $w\in S_n$ be a permutation, and let E be the ring from Definition of the source. Let $x_1,\ldots,x_n$ be formal indeterminates, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017651 | MATHDB-331620

Termination conjecture for the locus generation algorithm

摘录 / EXCERPT: Let the locus generation algorithm operate on a real algebraic locus, using complex detours around ramification points and sufficiently small steps on those detours. Termination co...

conjecture | open | MathDB 页面 | 原文线索

V2-017652 | MATHDB-331625

Yekutieli's conjecture on Beilinson and Yekutieli operator algebras

摘录 / EXCERPT: Let k be a perfect field. Suppose X/k is a finite type k-scheme of pure dimension n, and let ... be a saturated flag of points. Here $E^{\operatorname*{Yek}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017653 | MATHDB-331626

The isotropic loop reductive torsor conjecture

摘录 / EXCERPT: Let $\mathfrak G$ be a loop reductive group scheme over R_n, and suppose that all semisimple quotients of $\mathfrak G$ are isotropic. Isotropic loop reductive torsor conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-017654 | MATHDB-331641

Conjecture on descent of separated envelopes through universally closed birational morphisms

摘录 / EXCERPT: Descent conjecture. Under these hypotheses, the separated envelope of T' descends through p, so that T possesses a separated envelope.

conjecture | open | MathDB 页面 | 原文线索

V2-017655 | MATHDB-331650

MJ-canonicity conjecture for intersections under generic linkage

摘录 / EXCERPT: Let X be a variety in a nonsingular affine space A, and let Y be a generic link of X. Set ... Assume that X is MJ-canonical. MJ-canonicity conjecture. Then Z is also MJ-c...

conjecture | open | MathDB 页面 | 原文线索

V2-017656 | MATHDB-331659

Gorsky-Neguț quantum affine action conjecture for Hilbert-scheme stable bases

摘录 / EXCERPT: Let $K=\bigoplus_{n=0}^{\infty}K_{\mathbb{C}^*\times\mathbb{C}^*}(\operatorname{Hilb}_n)$, let $m=\frac{a}{b}$ with $\gcd(a,b)=1$, and let $\mathcal{A}^{(m)}$ be the slope-m Heis...

conjecture | open | MathDB 页面 | 原文线索

V2-017657 | MATHDB-331666

Compatibility conjecture for the associated graded and block variety functors

摘录 / EXCERPT: Let $\mathcal M^{\rm mix}$ be the graded subcategory from the Hodge-theoretic Koszul duality conjecture. Let κ be the functor from K-equivariant coherent sheaves on...

conjecture | open | MathDB 页面 | 原文线索

V2-017658 | MATHDB-331680

The finite-ramification characterization of postcritically finite morphisms

摘录 / EXCERPT: Let X be a smooth, irreducible, projective variety defined over the field of fractions K of a Dedekind domain R. Let $\varphi:X\to X$ be a morphism defined over K, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017659 | MATHDB-331681

The wild-ramification conjecture for preimage fields

摘录 / EXCERPT: Let X be a smooth, irreducible, projective variety over the field of fractions K of a Dedekind domain R, let $\varphi:X\to X$ be a PCF morphism defined over K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017660 | MATHDB-331684

Kebekus-Kovács conjecture on the Kodaira dimension of bases

摘录 / EXCERPT: Let X° be a smooth quasi-projective variety, and let $f^{\circ}:Y^{\circ}\to X^{\circ}$ be a smooth family of canonically polarized varieties with maximal variation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017661 | MATHDB-331693

Cohomological conjecture for smooth Hessenberg varieties

摘录 / EXCERPT: Let $(G,K)=(SL(N),SO(N))$ with N odd, and let $\operatorname{Hess}(K/P,{\mathfrak g}_1,\Sigma)\to{\mathfrak g}_1$ be a family of Hessenberg varieties associated to a parabolic su...

conjecture | open | MathDB 页面 | 原文线索

V2-017662 | MATHDB-331699

Isolation or a positive-dimensional fixed subvariety in characteristic p

摘录 / EXCERPT: Let F be a field of characteristic p, let φ be an analytic map from the open unit polydisk in n dimensions to itself, and let $x=(0,\ldots,0)$ be a fixed point whose...

conjecture | open | MathDB 页面 | 原文线索

V2-017663 | MATHDB-331703

Integral Brauer-Manin conjecture for log rationally connected varieties

摘录 / EXCERPT: Integral Brauer-Manin conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-017664 | MATHDB-331708

1-affineness conjecture for the de Rham local systems prestack

摘录 / EXCERPT: Let G be an affine algebraic group, and let $\operatorname{LocSys}_G(\overset{\circ}{\mathcal D})$ denote the prestack of de Rham G-local systems on the punctured formal disc...

conjecture | open | MathDB 页面 | 原文线索

V2-017665 | MATHDB-331722

Enhanced Cohen-Lenstra distribution conjecture

摘录 / EXCERPT: Enhanced Cohen-Lenstra conjecture. The $\mathbf{n}$-Cohen-Lenstra distribution $\mathrm{CL}_{\mathbf{n},S}$ accurately models the joint distribution of

conjecture | open | MathDB 页面 | 原文线索

V2-017666 | MATHDB-331732

The four-cusp bound for rational cuspidal curves

摘录 / EXCERPT: A rational cuspidal curve is a plane curve birational to a projective line whose singularities are all cusps. The four-cusp bound. A rational cuspidal curve can have at most four c...

conjecture | open | MathDB 页面 | 原文线索

V2-017667 | MATHDB-331735

The tangent-developable intersection conjecture for cuspidal projections

摘录 / EXCERPT: Let C_n be the rational normal curve in $\mathbb{P}^n$, let T_n be its tangent developable, and let S_n be its secant variety. A cuspidal projection center is a projection ce...

conjecture | open | MathDB 页面 | 原文线索

V2-017668 | MATHDB-331742

Pure-dimensionality conjecture for the critical and asymptotic value set

摘录 / EXCERPT: Let $F:\mathbb{C}^n\to\mathbb{C}^n$ be a dominant polynomial mapping. Write $K_0(F)$ for the set of critical values of F and S_F for its asymptotic set. Pure-dimensionality con...

conjecture | open | MathDB 页面 | 原文线索

V2-017669 | MATHDB-331765

Dimitrov's abelian-variety height lower-bound conjecture

摘录 / EXCERPT: Let A be an abelian variety and let $\lambda:A\dashrightarrow\mathbb{P}^1$ be a rational function, both defined over a number field K. Let h be the standard height on...

conjecture | open | MathDB 页面 | 原文线索

V2-017670 | MATHDB-331846

Explicit reciprocity conjecture for the twisted Perrin-Riou-Stark element

摘录 / EXCERPT: Let A be the CM abelian variety in the source, let $g=\dim A$, let $\mathbb{T}_{\mathfrak{P}}(A)$ be its p-adic representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017671 | MATHDB-331895

Joyce-Song formula for motivic Milnor fibers

摘录 / EXCERPT: Joyce-Song formula. The following identities should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-017672 | MATHDB-331905

Asai motive summand conjecture for the Shimura surface motive

摘录 / EXCERPT: Let (E,A) be an odd-type pair, with associated smooth projective surface X over $\mathbb{Q}$ and abelian variety A. Let $\mathsf{h}^1(A)$ and $\mathsf{h}^2(X)$ denote their C...

conjecture | open | MathDB 页面 | 原文线索

V2-017673 | MATHDB-331912

The local invariant cycle theorem for regular schemes

摘录 / EXCERPT: Let R be a complete discrete valuation ring with fraction field K and finite residue field k of characteristic p, and let X be a proper flat scheme over...

conjecture | open | MathDB 页面 | 原文线索

V2-017674 | MATHDB-331916

Homotopy-limit conjecture for quotient stacks by reductive group schemes

摘录 / EXCERPT: Let G be a reductive group scheme, let X be a space with the quotient simplicial object $[X/G]_{\bullet}$, and let Cpx denote the functor sending a space to the dg-category o...

conjecture | open | MathDB 页面 | 原文线索

V2-017675 | MATHDB-331930

Uniform Rasmussen-Tamagawa finiteness conjecture

摘录 / EXCERPT: Uniform Rasmussen-Tamagawa conjecture. There is a function $B(n,g)$ such that, for every number field K of degree n and every prime $ell>B(n,g)$, the set...

conjecture | open | MathDB 页面 | 原文线索

V2-017676 | MATHDB-331935

Nakajima's equivariant cohomology conjecture for truncated shifted Yangians

摘录 / EXCERPT: Let $B_h(\mathbf R)$ be the B-algebra of the Rees algebra of $Y^\lambda_\mu(\mathbf R)$, with $\mathbf R$ treated as formal variables, and let $\mathcal M(\mathbf m,W)$ be the co...

conjecture | open | MathDB 页面 | 原文线索

V2-017677 | MATHDB-331936

Extended Nakajima conjecture for the B-algebra and quiver-variety cohomology

摘录 / EXCERPT: Let $B(Y^\lambda_\mu(\mathbf R))$ be the B-algebra of the truncated shifted Yangian, let $\mathcal M(\mathbf m,W,\mathbf R)$ be the corresponding specialized quiver variety, and...

conjecture | open | MathDB 页面 | 原文线索

V2-017678 | MATHDB-331953

The hyperelliptic orbital-integral conjecture for classical groups

摘录 / EXCERPT: Orbital integrals for classical groups are expected to be related only to hyperelliptic curves. The precise formulation below concerns the cohomology of Hessenberg varieties. Hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-017679 | MATHDB-331976

The HK conjecture for minimal étale groupoids

摘录 / EXCERPT: HK conjecture. The even and odd homology groups of $\mathcal{G}$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-017680 | MATHDB-331977

The AH conjecture for topological full groups of minimal étale groupoids

摘录 / EXCERPT: AH conjecture. There exists an exact sequence

conjecture | open | MathDB 页面 | 原文线索

V2-017681 | MATHDB-331983

Fourier-Mukai compatibility conjecture for iterated-tilt hearts

摘录 / EXCERPT: Fourier-Mukai compatibility conjecture. For $1\leq k\leq g-1$, there are suitable complexified ample classes Ω_k and Ω'_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-017682 | MATHDB-331984

Fourier-Mukai compatibility conjecture with explicit complexified ample classes

摘录 / EXCERPT: Fourier-Mukai compatibility conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017683 | MATHDB-332043

Point-like limit conjecture for the Zhang-Kawazumi invariant

摘录 / EXCERPT: Point-like limit conjecture. As $t\to0$ in Δ^*, one has

conjecture | open | MathDB 页面 | 原文线索

V2-017684 | MATHDB-332053

The P=W conjecture for non-abelian Hodge moduli spaces

摘录 / EXCERPT: Let X be a smooth complex projective variety and G an algebraic reductive group. Let M_D be the moduli space of Higgs G-bundles, with proper Hitchin map, and let M_B be t...

conjecture | open | MathDB 页面 | 原文线索

V2-017685 | MATHDB-332063

Strong Lang-Silverman-Pazuki height conjecture

摘录 / EXCERPT: Let $g\geq1$ be an integer. For a number field K, an abelian variety A/K of dimension g, an ample symmetric line bundle L on A, and a point $P\in A(K)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017686 | MATHDB-332064

Strong Lang-Silverman-Rémond height conjecture

摘录 / EXCERPT: Let d and g be nonnegative integers. For a number field K of degree d, an abelian variety A/K of dimension g, and a polarization L on A, let $\widehat{h}_{A,L}(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-017687 | MATHDB-332091

Approximation conjecture for algebraic unoriented bordism classes

摘录 / EXCERPT: Let V be a nonsingular real algebraic variety and let M be a compact smooth submanifold of V. Suppose that the unoriented bordism class of the inclusion map...

conjecture | open | MathDB 页面 | 原文线索

V2-017688 | MATHDB-332092

Nice continuous rational approximation conjecture for rational maps into spheres

摘录 / EXCERPT: Let X be a compact nonsingular real algebraic variety, and let p be a positive integer. A continuous rational map $X\to\mathbb{S}^p$ should be approximable by nice continuous r...

conjecture | open | MathDB 页面 | 原文线索

V2-017689 | MATHDB-332097

The nonexistence conjecture for Ulrich Schur bundles on two-step flag varieties

摘录 / EXCERPT: Let $F(k_1,k_2;n)$ be the two-step flag variety parametrizing flags of subspaces of dimensions k_1 and k_2 in an n-dimensional vector space, and let $\mathcal{O}(1)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-017690 | MATHDB-332098

The complete-list conjecture for Ulrich partition types

摘录 / EXCERPT: Let (α,β,γ) be the type of an Ulrich partition, and call two types symmetric when they are identified by the symmetry used in the paper. Here n is a positive inte...

conjecture | open | MathDB 页面 | 原文线索

V2-017691 | MATHDB-332112

The graph-minor generation conjecture for motivic periods

摘录 / EXCERPT: Graph-minor generation conjecture. The motivic periods of G of weight at most k lie in the algebra generated by regularised motivic periods of graph minors of G having at mos...

conjecture | open | MathDB 页面 | 原文线索

V2-017692 | MATHDB-332113

Weight bound conjecture for momentum-space Feynman amplitudes

摘录 / EXCERPT: Weight bound conjecture. The amplitude integrand satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-017693 | MATHDB-332114

Motic-descendant degree conjecture for Feynman amplitudes

摘录 / EXCERPT: Motic-descendant degree conjecture. The amplitude of G is a regularised period of motic descendants of G of degree at most dh.

conjecture | open | MathDB 页面 | 原文线索

V2-017694 | MATHDB-332115

Period conjecture for Galois biconjugates

摘录 / EXCERPT: Let $\xi\in\mathcal{P}^{\mathfrak{m},+}$ be a motivic period, and let $P_{\xi}\subset\mathbb C$ be the $\mathbb Q$-algebra generated by the images under the period homomorphism of...

conjecture | open | MathDB 页面 | 原文线索

V2-017695 | MATHDB-332120

The generating-series conjecture for Hilbert schemes on simple surface singularities

摘录 / EXCERPT: Let $\mathbb{C}^2/G_\Delta$ be a simple singularity associated with an irreducible simply-laced Dynkin diagram Δ with n nodes. Let C_Δ be the Cartan matrix correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-017696 | MATHDB-332133

Affine Deligne-Lusztig induction conjecture for general-position characters of anisotropic tori

摘录 / EXCERPT: Let $G=\operatorname{GL}_n$ and let $b=1$, so that $J_1(F)=G(F)$. For a suitable pair $(f,\underline{w}_f)$, write ... for the affine Deligne-Lusztig induction map, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-017697 | MATHDB-332168

Rank-one factor existence for matrices of psd rank at most k

摘录 / EXCERPT: Rank-one factor abundance conjecture. Every such matrix has a psd factorization in which either at least k matrices A_i and $k-1$ matrices B_j are rank one, or at least $k-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-017698 | MATHDB-332170

Kontsevich-Soibelman motivic integrality conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a category satisfying conditions (7) and (8), let W be its potential, and let ϕ^{mot} denote the motivic vanishing-cycle realization. For each dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-017699 | MATHDB-332171

Kontsevich-Soibelman lambda-conjecture for motivic vanishing cycles

摘录 / EXCERPT: Let $\underline{\operatorname{K}}_0(\operatorname{Sch}^{ft})$ and $\underline{\operatorname{K}}^{\hat{\mu}}(\operatorname{Sch}^{ft})$ be the motivic ring theories over...

conjecture | open | MathDB 页面 | 原文线索

V2-017700 | MATHDB-332180

Deligne's purity conjecture for constructible l-adic sheaves

摘录 / EXCERPT: Let X be a scheme over $\mathbf{F}_p$, and let F be a constructible $\mathbf{Q}_l$-sheaf on X. Purity conjecture. Every constructible $\mathbf{Q}_l$-sheaf on X is mixed. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-017701 | MATHDB-332181

Serre's independence and purity conjecture for bad reduction

摘录 / EXCERPT: Let K be a global field, let X be a smooth projective K-variety of dimension n, and fix $i\in[0,2n]$. For a finite place v and a prime l not dividing the residue charac...

conjecture | open | MathDB 页面 | 原文线索

V2-017702 | MATHDB-332183

Loop-number bound for motivic Feynman amplitudes

摘录 / EXCERPT: Loop-number weight conjecture. The weight of $I_G^{\mathfrak{m}}$ is less than or equal to $4h_G$.

conjecture | open | MathDB 页面 | 原文线索

V2-017703 | MATHDB-332185

Conjecture on the minimizer of the auxiliary matrix bound

摘录 / EXCERPT: Let $L(g)$ be the Kronecker product of g copies of the matrix $M^+(1)$, and let h_0 be the quantity defined by ... Here B is the matrix appearing in the preceding proposition....

conjecture | open | MathDB 页面 | 原文线索

V2-017704 | MATHDB-332192

Motivic Joyce-Song formulas for Behrend function identities

摘录 / EXCERPT: Let E_1, E_2, and $E_1\oplus E_2$ be semi-Schur objects in the derived category of coherent sheaves over Y. Their extension space decomposes as ... Here ${\mathcal S}_0(E)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-017705 | MATHDB-332207

The motivic period conjecture for prime weighted multiple harmonic sums

摘录 / EXCERPT: Let $\mathcal{P}_{\geq 3}$ denote the set of primes at least 3, let $\alpha\in\mathbb{N}^{\ast}$, and let I be one of $\mathcal{P}_{\geq 3}\times\{\alpha\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-017706 | MATHDB-332261

Log-concavity conjecture for the mobility of numerical cycle classes

摘录 / EXCERPT: Let X be an integral projective variety of dimension n, let k be a nonnegative integer, and let mob denote the mobility function on the pseudo-effective cone...

conjecture | open | MathDB 页面 | 原文线索

V2-017707 | MATHDB-332263

Debarre-Jiang-Voisin strong and weak conjectures for contracted cycles

摘录 / EXCERPT: Debarre-Jiang-Voisin conjecture. The class α lies in the vector space generated by k-dimensional subvarieties contracted by π (Weak Conjecture), and in fact lies in th...

conjecture | open | MathDB 页面 | 原文线索

V2-017708 | MATHDB-332274

Conjecture on linear normality of the congruence and regularity of its fundamental scheme

摘录 / EXCERPT: The conjecture. The variety

conjecture | open | MathDB 页面 | 原文线索

V2-017709 | MATHDB-332330

Even-rank associated-variety conjecture for type D affine vertex algebras

摘录 / EXCERPT: Let r be an even integer, let $\mathfrak{g}$ be the type D Lie algebra used in the source, let $V_{2-r}(\mathfrak{g})$ be the simple affine vertex algebra at level $2-r$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017710 | MATHDB-332344

Conjecture on the real rank boundary of ternary quartics

摘录 / EXCERPT: Let f be a ternary quartic, let $C(f)$ be its middle catalecticant, and let $\operatorname{disc}(f)$ and $\operatorname{Bdisc}(f)$ denote respectively the discriminant and the Bl...

conjecture | open | MathDB 页面 | 原文线索

V2-017711 | MATHDB-332345

Hyperdeterminantal boundary conjecture for real six-term decompositions of quartics

摘录 / EXCERPT: Let f be a ternary quartic of real rank 6 and signature $(3,3)$, $(4,2)$, or $(5,1)$. Let $\operatorname{VSP}(f)_\mathbb{R}$ be the real points of the variety of sums of powers...

conjecture | open | MathDB 页面 | 原文线索

V2-017712 | MATHDB-332346

Tangential-secant component conjecture for real rank boundaries

摘录 / EXCERPT: Let $X\subset\mathbb{P}^N$ be a projective variety of dimension d, let r be the generic complex rank, let $\operatorname{VSP}_X(f)$ be the variety of sums of powers, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017713 | MATHDB-332350

Weak subadditivity conjecture for Kodaira dimensions in positive characteristic

摘录 / EXCERPT: Let $f:X\rightarrow Y$ be a fibration between smooth projective varieties over an algebraically closed field k of positive characteristic, with $\dim X=n$ and $\dim Y=m$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-017714 | MATHDB-332351

Canonical-divisor subadditivity conjecture in positive characteristic

摘录 / EXCERPT: Let $f:X\rightarrow Y$ be a fibration between smooth projective varieties over an algebraically closed field k of positive characteristic, with $\dim X=n$ and $\dim Y=m$. Canonic...

conjecture | open | MathDB 页面 | 原文线索

V2-017715 | MATHDB-332363

Extension conjecture for polarized variations of Hodge structure

摘录 / EXCERPT: Let X be a smooth projective variety and let $D\subset X$ be an ample divisor. Let $\mathscr{H}\to D$ be a polarized variation of Hodge structure, and let its numerical invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-017716 | MATHDB-332367

Mirror-symmetry conjecture for elliptic-fibration fibres of K3 surfaces

摘录 / EXCERPT: Let V and W be varieties of dimension 2, let $\pi\colon W\to\mathbb{P}^1$ be an elliptic fibration, and let ρ_p denote the number of irreducible components in the fibre...

conjecture | open | MathDB 页面 | 原文线索

V2-017717 | MATHDB-332368

Categorical-fibration conjecture for Tyurin degenerations

摘录 / EXCERPT: Let V be a Calabi-Yau variety of dimension d admitting a Tyurin degeneration, and let W be a homological mirror of V. For a point $p\in\mathbb{P}^1$, consider the categoric...

conjecture | open | MathDB 页面 | 原文线索

V2-017718 | MATHDB-332383

Frobenius semisimplicity conjecture for proper direct images of intersection complexes

摘录 / EXCERPT: Let k be a finite field, let $f:X\to Y$ be a proper map of varieties over k, and let y be a closed point of Y. Write $\overline{f^{-1}(y)}$ for the geometric fiber and...

conjecture | open | MathDB 页面 | 原文线索

V2-017719 | MATHDB-332390

Sommese's classification conjecture for manifolds containing an ample projective-space bundle

摘录 / EXCERPT: Let X be a smooth projective variety and let $Y\subset X$ be a smooth ample divisor. Suppose that $p:Y\to Z$ exhibits Y as a $\mathbb{P}^d$-bundle over a b-dimensional smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-017720 | MATHDB-332391

Andreatta-Wiśniewski projective-space characterization conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, let $\mathscr{E}$ be an ample vector bundle on X, and let T_X denote the tangent bundle of X. Andreatta-Wiśniewski's conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-017721 | MATHDB-332400

The connected-components conjecture for the stack of G-bundles

摘录 / EXCERPT: Let G be a reductive group, let $\Gamma=\operatorname{Gal}(\overline{E}/E)$, and let $\pi_1(G)_\Gamma$ be the coinvariants of Borovoi's fundamental group. The Kottwitz map gives...

conjecture | open | MathDB 页面 | 原文线索

V2-017722 | MATHDB-332414

The delta-invariant characterization of K-stability for Q-Fano varieties

摘录 / EXCERPT: Main conjecture. The K-stability, respectively K-semistability, of $(X,-K_X)$ is equivalent to

conjecture | open | MathDB 页面 | 原文线索

V2-017723 | MATHDB-332424

The integrality conjecture for character varieties

摘录 / EXCERPT: The integrality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017724 | MATHDB-332435

The non-nef locus and restricted base locus conjecture

摘录 / EXCERPT: Let X be a normal projective variety over an algebraically closed field and let D be an ${\mathbb{R}}$-Cartier ${\mathbb{R}}$-divisor on X. The non-nef locus...

conjecture | open | MathDB 页面 | 原文线索

V2-017725 | MATHDB-332460

The categorical entropy-spectral radius conjecture

摘录 / EXCERPT: Let Y be a smooth proper variety over $\mathbb{C}$. Write ${\mathcal D}^b(Y)$ for its bounded derived category of coherent sheaves, and let ${\rm End}^{FM}({\mathcal D}^b(Y))$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-017726 | MATHDB-332472

Ghioca-Tucker dynamical Mordell-Lang conjecture

摘录 / EXCERPT: Let X be a quasi-projective variety defined over $\mathbb C$, let $f:X\rightarrow X$ be an endomorphism, and let V be any subvariety of X. For a point $x\in X(\mathbb C)$, de...

conjecture | open | MathDB 页面 | 原文线索

V2-017727 | MATHDB-332473

Xie's coherent backward-orbit conjecture

摘录 / EXCERPT: Let X be a quasi-projective variety over $\mathbb C$, let $F:X\to X$ be a finite endomorphism, and let $\{b_i\}_{i\geq 0}$ be points of $X(\mathbb C)$ satisfying ... Let V be a p...

conjecture | open | MathDB 页面 | 原文线索

V2-017728 | MATHDB-332476

Guan's conjecture on equations of secant varieties of Chow varieties

摘录 / EXCERPT: Let $X\in \mathbb{P}W^*$ be an algebraic variety, let $\sigma_r(X)$ denote its r-th secant variety, and let $I_\delta(Y)$ denote the degree-δ component of the ideal of a v...

conjecture | open | MathDB 页面 | 原文线索

V2-017729 | MATHDB-332489

Amitsur's equal-degree conjecture for Brauer-Severi varieties

摘录 / EXCERPT: Amitsur's conjecture. If A and B generate the same cyclic subgroup of $\mathrm{Br}(k)$, then X and Y are birational.

conjecture | open | MathDB 页面 | 原文线索

V2-017730 | MATHDB-332537

Debarre-Pareschi-Popa conjecture on minimal cohomology classes in principally polarized abelian varieties

摘录 / EXCERPT: Debarre-Pareschi-Popa conjecture. The following are equivalent: (1) X is reduced, of pure dimension, and has minimal cohomology class

conjecture | open | MathDB 页面 | 原文线索

V2-017731 | MATHDB-332541

Kimura-O'Sullivan's finite-dimensionality conjecture for motives

摘录 / EXCERPT: Let ${\mathcal M}_k$ be the category of motives over the field k. A motive $M\in{\mathcal M}_k$ is finite-dimensional in the sense of Kimura and O'Sullivan if it admits the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-017732 | MATHDB-332543

Constant-curvature characterization for the filament recursion scheme

摘录 / EXCERPT: Let M be an oriented three-dimensional Riemannian manifold. Define on M the analogues of the differential 2-forms, the filament equation, and the bicycle correspondence whose...

conjecture | open | MathDB 页面 | 原文线索

V2-017733 | MATHDB-332552

The sparse invariant variety intersection conjecture for irreducible reflection groups

摘录 / EXCERPT: Let G be an irreducible and essential reflection group acting on a real vector space, and let $\mathcal{H}_k(G)$ denote the union of the k-dimensional flats of its reflection a...

conjecture | open | MathDB 页面 | 原文线索

V2-017734 | MATHDB-332557

Verschiebung-order lower-bound conjecture for finite unipotent group schemes

摘录 / EXCERPT: Let k be a field of positive characteristic, and let G be a finite commutative unipotent group scheme over k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017735 | MATHDB-332565

Projective-variety conjecture for the Askey scheme and superintegrable systems

摘录 / EXCERPT: The Askey scheme consists of families of hypergeometric orthogonal polynomials, and second order superintegrable systems in dimension two are related to these families through limi...

conjecture | open | MathDB 页面 | 原文线索

V2-017736 | MATHDB-332583

Taylor's modularity conjecture for congruence modular varieties

摘录 / EXCERPT: Let $\var V$ and $\var W$ be varieties, and let $\var V\join\var W$ denote their join in the lattice of interpretability types. A variety is congruence modular when its congruence...

conjecture | open | MathDB 页面 | 原文线索

V2-017737 | MATHDB-332587

Breuil's locally analytic socle conjecture for definite unitary groups

摘录 / EXCERPT: Breuil's conjecture. For a continuous character χ of T_p over E,

conjecture | open | MathDB 页面 | 原文线索

V2-017738 | MATHDB-332588

Breuil's companion-points conjecture for definite unitary eigenvarieties

摘录 / EXCERPT: Let T_p be the relevant diagonal torus, let $\mathcal E(U^p)_{\overline\rho}$ be the eigenvariety attached to the definite unitary group, and retain the notation for...

conjecture | open | MathDB 页面 | 原文线索

V2-017739 | MATHDB-332589

Partial classicality conjecture for companion points

摘录 / EXCERPT: Let ρ, Σ_p, $C(\rho)$, $\Sigma(\rho)$, $\mathcal E(U^p)_{\overline\rho}$, and δ_J^c be as in the preceding setup. Partial classicality conjecture. For every subs...

conjecture | open | MathDB 页面 | 原文线索

V2-017740 | MATHDB-332614

Katz's conjecture on rational curves on quintic threefolds

摘录 / EXCERPT: Let X be a general quintic threefold, and let d be the degree of a smooth rational curve on X. The scheme parametrizing smooth rational curves of degree d on X records th...

conjecture | open | MathDB 页面 | 原文线索

V2-017741 | MATHDB-332622

Green-Griffiths's formal unobstructedness conjecture for tangent spaces to cycle groups

摘录 / EXCERPT: Green-Griffiths's conjecture. For every $\tau\in TZ^{p}(X)$, there are successive lifts $\tau_j\in Z^{M}_{p}(D^{\mathrm{perf}}(X_j))$ for $j=1,2,\ldots$.

conjecture | open | MathDB 页面 | 原文线索

V2-017742 | MATHDB-332623

Green-Griffiths's formal unobstructedness conjecture for rational-equivalence tangent spaces

摘录 / EXCERPT: Green-Griffiths's conjecture. The tangent space $TZ^{p}_{rat}(X)$ is formally unobstructed.

conjecture | open | MathDB 页面 | 原文线索

V2-017743 | MATHDB-332703

Chow-Künneth decomposition conjecture

摘录 / EXCERPT: Chow-Künneth conjecture. There exist correspondences $\pi_{j,X}\in CH^d(X\times X)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-017744 | MATHDB-332707

Necessary and sufficient condition for logarithmic expressions of a hypergeometric value

摘录 / EXCERPT: Let $a,b,q\in\mathbb{Q}$ be non-integers such that none of $q-a$, $q-b$, and $q-a-b$ are integers. Here $B(a,b)$ denotes the beta function, ${}_3F_2$ is the generalized hypergeomet...

conjecture | open | MathDB 页面 | 原文线索

V2-017745 | MATHDB-332709

Rapoport's non-emptiness conjecture for Newton strata of Hodge-type Shimura varieties

摘录 / EXCERPT: Let G be unramified over $\mathbb Q_p$, let $K=K_pK^p$ be hyperspecial at p, and let $\Theta:\mathscr{S}_K(G,X)(\kappa(\wp))\to B(G)$ be the Newton-point map. Define ... The Newt...

conjecture | solved | MathDB 页面 | 原文线索

V2-017746 | MATHDB-332725

Analytic coefficient conjecture for the Hoffman star basis

摘录 / EXCERPT: Analytic coefficient conjecture. The equalities for the real multiple zeta values are satisfied with

conjecture | open | MathDB 页面 | 原文线索

V2-017747 | MATHDB-332726

Motivic Linebarger-Zhao conjecture for multiple zeta star values

摘录 / EXCERPT: Let $a_i,c_i\in\mathbb{N}^{\ast}$ with $c_i\neq 2$, and define ... where $\delta_c=\delta_{c=1}$ equals 1 if $c=1$ and 0 otherwise. Define ... ... and ... Here $\text{\boldmath $1...

conjecture | open | MathDB 页面 | 原文线索

V2-017748 | MATHDB-332738

Iarrobino-Kanev conjecture on Hilbert functions of squared point ideals

摘录 / EXCERPT: Let $n\ge3$, $d\ge2$, and define ... Let Z be a set of s general points in $\mathbb{P}^{n-1}$, where $N_{d-1}\le s<N_d$, and let $I=I(Z)$ be its vanishing ideal. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-017749 | MATHDB-332781

Automorphism group conjecture for Fulton-MacPherson compactifications of varieties of general type

摘录 / EXCERPT: Let X be a smooth projective variety of general type, and let $X[n]$ denote its Fulton-MacPherson compactification. If $n\neq 2$, then the automorphism group conjecture. ... The co...

conjecture | open | MathDB 页面 | 原文线索

V2-017750 | MATHDB-332811

Solvable group action conjecture for the flat irreducible locus

摘录 / EXCERPT: Let $U_{\rm flat,irr}$ be the flat irreducible locus, and let $\pi^{-1}(U_{\rm flat,irr})\to U_{\rm flat,irr}$ be the corresponding family of linear degenerations of flag varieties...

conjecture | open | MathDB 页面 | 原文线索

V2-017751 | MATHDB-332823

Geramita's aCM conjecture for secant varieties of Veronese varieties

摘录 / EXCERPT: Let s,d,n be positive integers, let $\nu_d\mathbb{P}^n$ be the degree-d Veronese re-embedding of projective space, and write $\sigma_s(\nu_d\mathbb{P}^n)$ for its s-secant va...

conjecture | open | MathDB 页面 | 原文线索

V2-017752 | MATHDB-332837

Geometric Bogomolov conjecture for abelian varieties

摘录 / EXCERPT: Geometric Bogomolov conjecture. If X has dense small points, then X should be special.

conjecture | open | MathDB 页面 | 原文线索

V2-017753 | MATHDB-332838

Intermediate-dimensional geometric Bogomolov conjecture for nowhere degenerate abelian varieties

摘录 / EXCERPT: Geometric Bogomolov conjecture in the remaining case. Then X is a torsion subvariety.

open_problem | open | MathDB 页面 | 原文线索

V2-017754 | MATHDB-332840

Mozgovoy's conjecture on stable Higgs bundles over finite fields

摘录 / EXCERPT: Mozgovoy's conjecture. The coefficients $\mathbb{H}_{\bm{\lambda}}(u;-z,w)$ are polynomials in u,z,w with nonnegative integer coefficients. The number of stable Higgs bundles on...

conjecture | open | MathDB 页面 | 原文线索

V2-017755 | MATHDB-332845

Joshi-Rajan's positive-density Hodge-Witt reduction conjecture

摘录 / EXCERPT: Joshi-Rajan's positive-density Hodge-Witt reduction conjecture. The variety X has Hodge-Witt reduction modulo a set of primes of K of positive density.

conjecture | open | MathDB 页面 | 原文线索

V2-017756 | MATHDB-332846

Virtual Kawasaki-Riemann-Roch conjecture for stable-map moduli spaces

摘录 / EXCERPT: Let X be a complex projective manifold, and let $X_{g,n,d}$ denote the moduli stack of genus-g, n-pointed stable maps of degree d to X, equipped with its perfect obstruct...

conjecture | open | MathDB 页面 | 原文线索

V2-017757 | MATHDB-332853

The conjecture that all quasihomogeneous varieties are homogeneous

摘录 / EXCERPT: A quasihomogeneous variety is a variety with a group action having an open orbit, while a homogeneous variety has a transitive group action. Homogeneity conjecture. All quasihomoge...

conjecture | open | MathDB 页面 | 原文线索

V2-017758 | MATHDB-332854

The conjecture on variation-of-GIT boundary-divisor correspondences

摘录 / EXCERPT: Let $MHilb$ be the multiHilbert scheme, and let $\widetilde{E}_{\mathfrak{c},\mathfrak{c}'}$ denote the divisorial locus associated with a wall crossing between chambers...

conjecture | open | MathDB 页面 | 原文线索

V2-017759 | MATHDB-332855

The volume-function conjecture for divisors on the multiHilbert scheme

摘录 / EXCERPT: Volume-function conjecture. The volume function gives the number of intersections of 12 general Weil divisors arising from these (pseudo)sections, away from their base locus. The...

conjecture | open | MathDB 页面 | 原文线索

V2-017760 | MATHDB-332858

Abo-Ottaviani-Peterson conjecture on nondefective Segre varieties

摘录 / EXCERPT: Abo-Ottaviani-Peterson conjecture. $\mathcal{S}_\mathbb{F}$ is nondefective unless one of the following holds: $(n_1,n_2,n_3)=(4,4,3)$; $(n_1,n_2,n_3)=(2k+1,2k+1,3)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-017761 | MATHDB-332859

Chiantini-Ottaviani-Vannieuwenhoven conjecture on generic identifiability of Segre varieties

摘录 / EXCERPT: Chiantini-Ottaviani-Vannieuwenhoven conjecture. $\mathcal{S}_\mathbb{C}$ is generically r-identifiable if

conjecture | open | MathDB 页面 | 原文线索

V2-017762 | MATHDB-332862

K-stability and Kähler-Einstein metrics for birationally rigid Fano varieties

摘录 / EXCERPT: Birational rigidity conjecture. Birationally rigid (orbifold) Fano varieties are K-stable and admit a Kähler-Einstein metric.

conjecture | open | MathDB 页面 | 原文线索

V2-017763 | MATHDB-332864

The asymptotic Poncelet n-gon proportion conjecture for conic pairs

摘录 / EXCERPT: Poncelet n-gon proportion conjecture. The proportion of conic pairs in $\mathbf P^2(\mathbb F_q)$ satisfying the Poncelet n-gon condition is asymptotically equal to

conjecture | open | MathDB 页面 | 原文线索

V2-017764 | MATHDB-332869

Hain's local integrability conjecture for the biextension metric

摘录 / EXCERPT: Let $\boldsymbol{H}$ be an arbitrary polarized variation of Hodge structures of weight -1 on a curve X, with Poincaré bundle $\mathcal{P}$ and polarization λ. Write ... a...

conjecture | open | MathDB 页面 | 原文线索

V2-017765 | MATHDB-332870

Hain's height jump effectiveness conjecture

摘录 / EXCERPT: Let $\overline{\phi}\colon\overline{C}\to\overline{X}$ be a holomorphic test curve whose image is not contained in the boundary divisor D, and let ν be the section defining t...

conjecture | open | MathDB 页面 | 原文线索

V2-017766 | MATHDB-332877

Uniform boundedness conjecture for torsion points on jacobians

摘录 / EXCERPT: Let U be a scheme over ${\mathbb{Q}}$ and C/U a smooth proper curve, with jacobian J. Let $\sigma \in J(U)$ be a section, and let $d\ge 1$ be an integer. Uniform boundedness...

conjecture | open | MathDB 页面 | 原文线索

V2-017767 | MATHDB-332878

The Sato--Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field K, and let $\operatorname{ST}(A)$ be its Sato--Tate group. For each prime $\mathfrak{p}$ of good reduction, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017768 | MATHDB-332879

The Mumford--Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a number field, and fix a prime ℓ. Let $G_\ell^{\rm zar}\subseteq \operatorname{GSp}_{2g}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017769 | MATHDB-332880

The Algebraic Sato--Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a number field, and let $G_\ell^{1,\rm zar}\subseteq \operatorname{Sp}_{2g}$ be the Zariski closure of the determinant-one part...

conjecture | open | MathDB 页面 | 原文线索

V2-017770 | MATHDB-332900

The local model flatness conjecture for parahoric unitary moduli

摘录 / EXCERPT: Let (r,s) be a signature and let $I\subset\{0,\dotsc,m\}$ be nonempty. Let M_I^{naive} be the naive local model, M_I the closed subscheme cut out by the wedge and st...

conjecture | open | MathDB 页面 | 原文线索

V2-017771 | MATHDB-332919

The geometric-origin conjecture for rigid local systems

摘录 / EXCERPT: Let X be a smooth projective variety, and let L be a rigid local system on X. A local system is of geometric origin if, for some Zariski open dense subset $U\subset X$, there...

conjecture | open | MathDB 页面 | 原文线索

V2-017772 | MATHDB-332920

The integrality consequence for rigid local systems

摘录 / EXCERPT: Let X be a smooth projective variety, and let L be a rigid local system on X. A local system is integral when the traces of its monodromy matrices are algebraic integers. The...

conjecture | open | MathDB 页面 | 原文线索

V2-017773 | MATHDB-332923

Twisted parabolic Higgs-moduli perverse Hodge polynomial conjecture

摘录 / EXCERPT: Let C be a complex smooth projective curve of genus g, with divisor $D=p_1+\dots+p_k+rp$, where the points are distinct and p has multiplicity r. Let ${\bm \mu}$ be a k-t...

conjecture | open | MathDB 页面 | 原文线索

V2-017774 | MATHDB-332924

Wild character-variety mixed Hodge polynomial conjecture

摘录 / EXCERPT: Let ${\mathcal{M}}_\B^{{\bm \mu},{\bf r}}$ be the generic wild character variety, let $d_{{\bm \mu},{\bf r}}$ be its dimension, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017775 | MATHDB-332926

Combinatorial identity from a wild-to-tame character-variety correspondence

摘录 / EXCERPT: Let $\mu\in\mathcal{P}_n$, let μ' be the partition obtained by deleting the first part of μ, and let $n'=n-\mu_1$. Set ... Assuming the notation and the wild-to-tame correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-017776 | MATHDB-332948

Fomin-Kirillov nonnegativity conjecture for Schubert polynomials

摘录 / EXCERPT: Let S_n be the symmetric group, let $[ij]$ be the generators of the Fomin-Kirillov algebra, and let θ_i be the Dunkl elements ... Let $\mathfrak{S}_w$ be the Schubert polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-017777 | MATHDB-332949

Aluffi-Mihalcea coefficient positivity conjecture for Schubert CSM classes

摘录 / EXCERPT: Let $X=G/B$ be a flag variety with Weyl group W. For $w\in W$, write ... Let $\leq$ denote the Bruhat order on W. Aluffi-Mihalcea's positivity conjecture. If $v\leq w$, then ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-017778 | MATHDB-332950

Refined positivity conjecture for CSM coefficients and Littlewood-Richardson coefficients

摘录 / EXCERPT: Let W be a Weyl group, let ℓ be its length function, and let $c(w;v)$ be the coefficient of the Schubert class $[X(v)]$ in the CSM class of the Schubert cell $X(w)^\circ$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-017779 | MATHDB-332951

The strong uniform boundedness conjecture for principally polarised abelian varieties

摘录 / EXCERPT: Fix a field K, equal to either $\mathbb{Q}$ or $\mathbb{F}_q(T)$ for a prime power q. Let $g\geq0$, let A/K be a g-dimensional principally polarised abelian variety, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-017780 | MATHDB-332952

The bounded-degree torsion sparsity conjecture for abelian schemes

摘录 / EXCERPT: Bounded-degree torsion sparsity conjecture. The set ${\mathbf{T}}(d)$ is not Zariski dense in S. The paper proves that this conjecture is equivalent to the strong uniform bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-017781 | MATHDB-332954

The sparse small points conjecture for principally polarised abelian schemes

摘录 / EXCERPT: Let S/K be a variety, let A/S be a principally polarised abelian scheme, and let $\sigma\in A(S)$ be a section of infinite order. For an integer $d>0$ and $\epsilon\geq0$, defi...

conjecture | open | MathDB 页面 | 原文线索

V2-017782 | MATHDB-332955

Apéry constants of Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold and let $\gamma\in H^{\:\raisebox{3pt}{\text{\circle*{1.5}}}}(X)$ be a homogeneous coprimitive cohomology class of codimension n with respect to $-K_X$...

conjecture | open | MathDB 页面 | 原文线索

V2-017783 | MATHDB-332956

The converse of quantum minimality and vanishing Apéry constants

摘录 / EXCERPT: Let X be a Fano manifold. Call X quantum minimal when all Apéry constants vanish, equivalently when its Apéry class A is equal to 1. Converse conjecture. If all Apéry const...

conjecture | open | MathDB 页面 | 原文线索

V2-017784 | MATHDB-332993

The compatible-system conjecture for irreducible representations

摘录 / EXCERPT: Compatible-system conjecture. The compatible system comes from algebraic geometry.

conjecture | open | MathDB 页面 | 原文线索

V2-017785 | MATHDB-333009

The cluster variety conjecture for open Richardson varieties

摘录 / EXCERPT: Refined open Richardson conjecture. The open Richardson variety R^u_v is a cluster variety satisfying the Louise property and of full rank.

conjecture | open | MathDB 页面 | 原文线索

V2-017786 | MATHDB-333015

Birationality conjecture for the branching morphism of non-strongly bendable triples

摘录 / EXCERPT: Let (g,d,ℓ) be a triple, and let $\widetilde{\mathcal W}_{g,d,\ell}$ be the corresponding moduli space. A triple is strongly bendable when ... so that the branching morphism ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-017787 | MATHDB-333034

Harari-Wittenberg's strong approximation conjecture for the fibration varieties W

摘录 / EXCERPT: Harari-Wittenberg's conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-017788 | MATHDB-333054

The cohomological Hodge conjecture for singular varieties

摘录 / EXCERPT: Let X be a complex projective variety defined over $\overline{\mathbb Q}$. A weight $2p$ Hodge cycle is a class in $H^{2p}(X,\mathbb Q(p))$ of Hodge type (p,p). Let ... be motivi...

conjecture | open | MathDB 页面 | 原文线索

V2-017789 | MATHDB-333063

Integrality conjecture for logarithmic HLV kernels

摘录 / EXCERPT: Let $\mathbb H_{u_1,u_2,\ldots,u_g}[X_1,X_2,\ldots,X_n;q,t,T]$ be the logarithmic HLV kernel, and let $\mathbb H_{u_1,u_2,\ldots,u_g,\lambda}(q,t)$ denote the coefficient of...

conjecture | open | MathDB 页面 | 原文线索

V2-017790 | MATHDB-333070

The conjecture on étale pro-homology and pro-homotopy satellites

摘录 / EXCERPT: Let X^{et} be the étale site from Example, let A be an abelian group, and let ${\Greekmath 0119}_n^{et}(X)$ and $H_n^{et}(X,A)$ denote étale pro-homotopy and étale pro-homology...

conjecture | open | MathDB 页面 | 原文线索

V2-017791 | MATHDB-333074

The expected-dimension conjecture for rational curves on very general hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a very general hypersurface of degree d, and let $R_e(X)$ denote the space of integral, geometrically rational curves of degree e on X. Expected...

conjecture | open | MathDB 页面 | 原文线索

V2-017792 | MATHDB-333075

Voisin's bounded-degree conjecture for rational curves on general type hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a very general hypersurface of general type, so that its degree satisfies $d>n+1$. A rational curve on X has a degree with respect to the hyperplane...

conjecture | open | MathDB 页面 | 原文线索

V2-017793 | MATHDB-333094

Coleman's finiteness conjecture for endomorphism algebras of abelian varieties

摘录 / EXCERPT: Let F be a totally real field and let $g \in \mathbb Z_{\geq 1}$. For an abelian variety X over F, write...

conjecture | open | MathDB 页面 | 原文线索

V2-017794 | MATHDB-333121

Derived-equivalence conjecture for general Clifford double mirrors

摘录 / EXCERPT: Let $\mathcal{S}=[Y/\overline G]$ be the quotient stack associated with a general Clifford mirror, and let $\mathcal{B}_0$ denote its even Clifford-algebra sheaf. Assume the centra...

conjecture | open | MathDB 页面 | 原文线索

V2-017795 | MATHDB-333147

The gluing conjecture for non-equidimensional eigenvarieties

摘录 / EXCERPT: For each a, let $\mathcal{E}_{a,q}^{q}$, for $q=0,\ldots,a$, be the eigenvarieties constructed over the corresponding weight spaces, and let $\mathcal{W}_a$ be the weight space....

conjecture | open | MathDB 页面 | 原文线索

V2-017796 | MATHDB-333148

Urban's dimension conjecture for eigenvarieties

摘录 / EXCERPT: Let $\mathbb{E}_{\mathcal{H}}$ be the eigenvariety constructed from a Hecke module, let x be a point belonging to exactly one irreducible component of $\mathbb{E}_{\mathcal{H}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-017797 | MATHDB-333150

GW/Hilbert-scheme correspondence for primitive K3 curve classes

摘录 / EXCERPT: Let S be a nonsingular projective K3 surface, let d be a nonnegative integer, and let $\beta\in H_2(S,\mathbb{Z})$ be primitive. For cohomology weighted partitions...

conjecture | open | MathDB 页面 | 原文线索

V2-017798 | MATHDB-333171

The rationality conjecture for Gushel-Mukai fourfolds

摘录 / EXCERPT: Let X be a rational Gushel-Mukai fourfold, and let $\mathcal{A}_X$ denote its GM category. A rationality conjecture. Then ... for some K3 surface S. This is the GM analogue of th...

conjecture | open | MathDB 页面 | 原文线索

V2-017799 | MATHDB-333196

Precise nonvanishing range conjecture for Veronese syzygies

摘录 / EXCERPT: Precise nonvanishing range conjecture. In the situation of that theorem, one has

conjecture | open | MathDB 页面 | 原文线索

V2-017800 | MATHDB-333203

The A-model R-matrix conjecture for mirror Gromov-Witten and FJRW theories

摘录 / EXCERPT: Let $\mathcal{X}$ be a Calabi-Yau variety and let (W,G_{max}) be a pair defining an FJRW theory. Suppose that the Gromov-Witten theory of $\mathcal{X}$ and the FJRW theory of...

conjecture | open | MathDB 页面 | 原文线索

V2-017801 | MATHDB-333224

Weight-detection conjecture for étale homology

摘录 / EXCERPT: Weight-detection conjecture. For every $r,n\ge 0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-017802 | MATHDB-333225

Relative weight-detection conjectures for étale homology

摘录 / EXCERPT: Relative weight-detection conjectures. The conjecture $WD^n(X)$ holds if

conjecture | open | MathDB 页面 | 原文线索

V2-017803 | MATHDB-333244

Novaković's conjecture on exceptional collections on non-split Brauer-Severi varieties

摘录 / EXCERPT: Novaković's conjecture. $D^b(X)$ does not admit a full strongly exceptional collection.

conjecture | open | MathDB 页面 | 原文线索

V2-017804 | MATHDB-333275

Embedded components in powers of non-complete-intersection curves

摘录 / EXCERPT: Let $X\subset\mathbb{P}^3$ be a reduced and unmixed subvariety of codimension 2, and let I_X be its ideal. Suppose there is a point $P\in X$ such that the localization of I_X...

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V2-017805 | MATHDB-333287

The subobject morphism conjecture for overconvergent F-isocrystals

摘录 / EXCERPT: Let X be the fixed variety, and let $\operatorname{\mathbf{F-Isoc}}(X)$ and $\operatorname{\mathbf{F-Isoc}}^\dagger(X)$ denote the categories of convergent and overconvergent F...

conjecture | open | MathDB 页面 | 原文线索

V2-017806 | MATHDB-333288

The cut-by-curves conjecture for overconvergent F-isocrystals

摘录 / EXCERPT: Let X be a variety, let Y be a boundary or compactification datum, and let C denote a curve contained in the indicated variety. Write $\operatorname{\mathbf{F-Isoc}}(X)$ and...

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V2-017807 | MATHDB-333289

The curve restriction conjecture for irreducible overconvergent F-isocrystals

摘录 / EXCERPT: Let X be a variety, let $\operatorname{\mathbf{F-Isoc}}^\dagger(X)$ denote the category of overconvergent F-isocrystals on X, and let $C \subseteq X$ be a curve. Curve restri...

conjecture | open | MathDB 页面 | 原文线索

V2-017808 | MATHDB-333301

Boyarchenko's maximality conjecture for the variety X_h

摘录 / EXCERPT: Let X_h be a finite-type variety over a finite field with geometric Frobenius $\Fr_{q^n}$ acting on its compactly supported cohomology. A variety is maximal in the sense of Boyar...

conjecture | open | MathDB 页面 | 原文线索

V2-017809 | MATHDB-333326

The Prym tropicalization conjecture

摘录 / EXCERPT: Prym tropicalization conjecture. The skeleton of the Prym variety associated with f is the tropical Prym variety associated with ϕ; equivalently, the Prym construction commu...

conjecture | open | MathDB 页面 | 原文线索

V2-017810 | MATHDB-333361

Bombieri-Lang finiteness conjecture for Shimura varieties

摘录 / EXCERPT: Bombieri-Lang conjecture for Shimura varieties. For every finitely generated extension F of $E(\mathbf{G},X)$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-017811 | MATHDB-333365

Galkin-Kontsevich weak zero-divisor conjecture for the Grothendieck ring

摘录 / EXCERPT: Galkin-Kontsevich weak conjecture. If α is represented by a linear combination of varieties of dimension less than or equal to $2(d-2)$ and

conjecture | open | MathDB 页面 | 原文线索

V2-017812 | MATHDB-333367

The conjecture on sum-of-squares representations for varieties of minimal degree

摘录 / EXCERPT: Sum-of-squares representation conjecture. A generic quadratic form nonnegative on X has exactly

conjecture | open | MathDB 页面 | 原文线索

V2-017813 | MATHDB-333391

The rationality-transcendental transcendence dichotomy for affine varieties

摘录 / EXCERPT: Rationality-transcendence conjecture. The power series $F_V(x_1,\ldots,x_d)$ is either rational or transcendentally transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-017814 | MATHDB-333398

Bocklandt-Le Bruyn transitivity conjecture for cyclic quiver varieties

摘录 / EXCERPT: For a quiver Q, let $A_Q:=\mathbb{C}[N_Q]\otimes\mathbb{C}\bar{Q}$, where N_Q is the necklace Lie algebra and $\mathbb{C}\bar{Q}$ is the path algebra of the double of Q. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017815 | MATHDB-333428

Eventual tail conjecture for four-stack Twyst-off positions

摘录 / EXCERPT: Eventual tail conjecture. For every $a>0$, there exist integers $c_0>b_0>a$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017816 | MATHDB-333434

The generalized valuation conjecture for special sections and integral critical points

摘录 / EXCERPT: Let G be a complex reductive group, let B be a Borel subgroup, let w_0 be the longest Weyl-group element, and let $\mathbf{i}$ be a reduced expression of w_0. For a dominan...

conjecture | open | MathDB 页面 | 原文线索

V2-017817 | MATHDB-333435

Harbourne-Hirschowitz conjecture on special linear systems of plane curves

摘录 / EXCERPT: Let $\{p_1,\ldots,p_r\}\subset\mathbb{P}^2$ be general points, let $\nu:\widetilde{\mathbb{P}}^2\to\mathbb{P}^2$ be their blow-up, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017818 | MATHDB-333442

Geometric Lax Conjecture

摘录 / EXCERPT: A hyperbolicity region is the connected component containing the reference point of the set where a real homogeneous polynomial is nonzero. A spectrahedron is a set defined by a li...

conjecture | open | MathDB 页面 | 原文线索

V2-017819 | MATHDB-333443

Helton-Nie Conjecture

摘录 / EXCERPT: A convex semialgebraic set is a convex subset of Euclidean space describable by polynomial equalities and inequalities. A spectrahedral shadow is a linear image of a spectrahedron....

conjecture | open | MathDB 页面 | 原文线索

V2-017820 | MATHDB-333448

Higher-dimensional doubly ruled incidence conjecture for lines

摘录 / EXCERPT: Let $\frak L$ be a set of L lines in $\mathbb{C}^n$, and let $P_2(\frak L)$ denote the set of points incident to at least two lines of $\frak L$. Higher-dimensional doubly ruled...

conjecture | open | MathDB 页面 | 原文线索

V2-017821 | MATHDB-333449

Higher-dimensional doubly ruled incidence conjecture for algebraic curves

摘录 / EXCERPT: Let Γ be a set of L irreducible algebraic curves in $\mathbb{C}^n$, each of degree at most d, and let $P_2(\Gamma)$ denote the set of points incident to at least two cur...

conjecture | open | MathDB 页面 | 原文线索

V2-017822 | MATHDB-333453

Partial-transposition conjecture for pillar entries of Schubert tangent cones

摘录 / EXCERPT: Partial-transposition conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-017823 | MATHDB-333463

Betti geometric Langlands conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, let $G^{\vee}$ be its Langlands dual group, and let $\mathcal Loc_{G^{\vee}}(X)$ be the Betti mo...

conjecture | open | MathDB 页面 | 原文线索

V2-017824 | MATHDB-333469

Affine Langlands-Vogan-Soergel duality conjecture

摘录 / EXCERPT: Let $(X,\alpha)=(\mathbb P^1,\alpha)$ be the real projective line with antipodal conjugation, let (G,θ) be a quasi-split real form with L-group...

conjecture | open | MathDB 页面 | 原文线索

V2-017825 | MATHDB-333490

Li's conjecture on normalized-volume minimizers

摘录 / EXCERPT: Let X be a variety of dimension n, let $x\in X$ be a closed point, and let $\operatorname{Val}_{X,x}$ denote the set of real valuations on X centered at x. For...

conjecture | open | MathDB 页面 | 原文线索

V2-017826 | MATHDB-333506

Conjecture on multiplicity counting for projective schemes over finite fields

摘录 / EXCERPT: Multiplicity-counting conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017827 | MATHDB-333508

1-affineness conjecture for the local space of systems on the punctured disc

摘录 / EXCERPT: Local 1-affineness conjecture. The prestack $\LocSys_\cG(\oD)$ is 1-affine.

conjecture | open | MathDB 页面 | 原文线索

V2-017828 | MATHDB-333541

Manin-Mumford conjecture for special subvarieties

摘录 / EXCERPT: Let X be an algebraic variety over $\mathbb{C}$. Define a special subvariety of X to be a translate by a torsion point of an abelian subvariety when X is an abelian variety,...

conjecture | open | MathDB 页面 | 原文线索

V2-017829 | MATHDB-333587

Thin-set version of Manin's conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over a number field F with ample anticanonical class $-K_X$. Let $\mathcal{L}=(L,\|\cdot\|)$ be a big and nef adelically metrized line bund...

conjecture | open | MathDB 页面 | 原文线索

V2-017830 | MATHDB-333588

Thinness conjecture for geometrically incompatible covers

摘录 / EXCERPT: Let X be a smooth uniruled variety over a number field F and let L be a big and nef $\mathbb Q$-divisor on X. For every F-morphism $f:Y\to X$ from a smooth projective var...

conjecture | open | MathDB 页面 | 原文线索

V2-017831 | MATHDB-333589

Finiteness conjecture for generically finite covers

摘录 / EXCERPT: Let X be a smooth uniruled projective variety over an algebraically closed field of characteristic 0, and let L be a big and nef $\mathbb Q$-divisor on X. For a generically...

conjecture | open | MathDB 页面 | 原文线索

V2-017832 | MATHDB-333615

Mobility conjecture for complete intersection curve classes

摘录 / EXCERPT: Mobility conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017833 | MATHDB-333616

Taylor's abelian-variety conjecture for modular eigenclasses over number fields

摘录 / EXCERPT: Let F be a number field, let ${\mathfrak p}$ be a prime of F, and let $\mathfrak m$ and $\mathfrak d$ be ideals as in the setup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017834 | MATHDB-333666

Orbifold rational connectedness and rational curves conjecture

摘录 / EXCERPT: Orbifold rational curves conjecture. 1. (X,D) is uniruled if and only if K_X+D is not pseudo-effective. 1'. (X,D) is uniruled if and only if it is weakly uniruled. 2. (X,D)...

conjecture | open | MathDB 页面 | 原文线索

V2-017835 | MATHDB-333668

The arithmetically bi-algebraic conjecture for enlarged mixed Shimura varieties

摘录 / EXCERPT: Let S be an algebraic variety over $\overline{\mathbb{Q}}$ whose uniformization $\mathcal{X}$ has a structure of algebraic variety over $\overline{\mathbb{Q}}$. Suppose that Y...

conjecture | open | MathDB 页面 | 原文线索

V2-017836 | MATHDB-333669

The Ax-Schanuel conjecture for enlarged mixed Shimura varieties

摘录 / EXCERPT: Let $S^\natural$ be a connected enlarged mixed Shimura variety associated with $(P,\mathcal{X}^{\natural+})$, with uniformization...

conjecture | open | MathDB 页面 | 原文线索

V2-017837 | MATHDB-333670

The Ax-Schanuel conjecture for mixed Shimura varieties

摘录 / EXCERPT: Let S be a connected mixed Shimura variety associated with $(P,\mathcal{X}^+)$ and let $\mathrm{unif}\colon\mathcal{X}^+\to S$ be its uniformization. Let $\mathbscr{Z}$ be the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-017838 | MATHDB-333671

Mixed André-Oort conjecture

摘录 / EXCERPT: Let S be a connected mixed Shimura variety and let Y be an irreducible subvariety of S. Mixed André-Oort conjecture. If the special points of S contained in Y are Zariski...

conjecture | open | MathDB 页面 | 原文线索

V2-017839 | MATHDB-333692

Toric Landau-Ginzburg component-count conjecture

摘录 / EXCERPT: Let X be a Fano variety of dimension n. Let f_X be its toric Landau-Ginzburg model corresponding to an anticanonical symplectic form on X. Let k_{f_X} be the number of co...

conjecture | open | MathDB 页面 | 原文线索

V2-017840 | MATHDB-333703

Brylinski-Deligne classification conjecture for factorizable line bundles

摘录 / EXCERPT: Let X be a curve and G a constant group scheme over X. Let $(K_2)_{\operatorname{Zar}}$ be the Zariski sheafification of the prestack assigning $K_2(A)$ to an affine scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-017841 | MATHDB-333704

Compatibility of Brylinski-Deligne and étale constructions

摘录 / EXCERPT: Let X be a curve, let G be a constant group scheme over X, and let ℓ be an integer prime to the characteristic. Set $A=\mu_\ell$, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-017842 | MATHDB-333765

Cisinski's converse conjecture for the rational motivic spectrum

摘录 / EXCERPT: Let $SH(\mathbb{F}_q)^c_\mathbb{Q}$ be the compact rational stable motivic homotopy category over $\mathbb{F}_q$, and let Spc denote its tensor-triangular spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-017843 | MATHDB-333769

Irreducibility conjecture for associated varieties with finitely many symplectic leaves

摘录 / EXCERPT: Let V be a simple, finitely strongly generated, positively graded conformal vertex operator algebra with $V_0=\mathbb{C}$, and let $X_V=\operatorname{Specm}(R_V)$ be its associat...

conjecture | open | MathDB 页面 | 原文线索

V2-017844 | MATHDB-333777

The cohomological ell-independence conjectures

摘录 / EXCERPT: Let X/F be a variety, let i be a cohomological degree, and let $H^i_\u0**eell(X)$ and $H^i_{c,\u0**eell}(X)$ denote the corresponding Weil--Deligne representations, with the la...

conjecture | open | MathDB 页面 | 原文线索

V2-017845 | MATHDB-333778

The ell-independence conjecture for unipotent fundamental groups

摘录 / EXCERPT: Let X/F be a geometrically connected variety with a base point $x\in X(F)$, and let $\pi_1^\u0**eell(X,x)$ be its ell-adic unipotent fundamental group viewed as a pro-unipotent r...

conjecture | open | MathDB 页面 | 原文线索

V2-017846 | MATHDB-333779

The finite-level ell-independence conjecture for unipotent fundamental groups

摘录 / EXCERPT: Let X/F be a geometrically connected variety with base point $x\in X(F)$. For each $\u0**eell$, let $\hat{\mathcal{U}}_{\u0**eell,X,x}$ be the completed universal enveloping alge...

conjecture | open | MathDB 页面 | 原文线索

V2-017847 | MATHDB-333809

Mumford's conjecture on rational connectedness

摘录 / EXCERPT: Let X be a complex projective manifold. A manifold is rationally connected if any two general points can be joined by a chain of rational curves. Write Ω^1_X for the cotan...

conjecture | open | MathDB 页面 | 原文线索

V2-017848 | MATHDB-333812

Motivic HyperKähler Resolution Conjecture for global quotients

摘录 / EXCERPT: Let M be a smooth projective holomorphic symplectic variety equipped with a faithful action of a finite group G by symplectic automorphisms of M. If Y is a symplectic resol...

conjecture | open | MathDB 页面 | 原文线索

V2-017849 | MATHDB-333813

Motivic splitting property for holomorphic symplectic varieties

摘录 / EXCERPT: Let X be a holomorphic symplectic variety of dimension $2n$. Then the motivic splitting property. there is a canonical self-dual multiplicative Chow-Künneth decomposition ... of Bl...

conjecture | open | MathDB 页面 | 原文线索

V2-017850 | MATHDB-333842

Fujita-Odaka's delta-invariant criterion for K-stability

摘录 / EXCERPT: Fujita-Odaka's conjecture. X is K-stable, respectively K-semistable, with respect to $-K_X$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-017851 | MATHDB-333852

LZ1's smooth Schubert class conjecture for parabolic Kazhdan-Lusztig classes

摘录 / EXCERPT: Let G/P be a partial flag variety, let J index the parabolic subgroup, and let $v\in W^J$ index the Schubert variety $X(v)$. Write $[X(v)]$ for the topologically defined Schube...

conjecture | open | MathDB 页面 | 原文线索

V2-017852 | MATHDB-333853

Parabolic Kazhdan-Lusztig positivity conjecture

摘录 / EXCERPT: Let G/P_J be a partial flag variety, let $v,w\in W^J$ with $w\le v$, and let C_v^J be the parabolic Kazhdan-Lusztig-Schubert class. Write the coefficient of f_w in the expans...

conjecture | open | MathDB 页面 | 原文线索

V2-017853 | MATHDB-333859

Generalized Sato-Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g defined over $\mathbb{Q}$. For each prime p of good reduction, write the normalized Euler factor as ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-017854 | MATHDB-333879

Endomorphism algebras of categorified projectors and flag Hilbert scheme charts

摘录 / EXCERPT: Let P_T be the categorified Jones--Wenzl projector indexed by a standard Young tableau T of size n, and let $\mathring{\mathrm{FHilb}}_T(\mathbb{C})$ be the corresponding aff...

conjecture | open | MathDB 页面 | 原文线索

V2-017855 | MATHDB-333880

The Behrend-function integral conjecture for locally toric curves

摘录 / EXCERPT: Let Y be any quasi-projective Calabi-Yau threefold, and let $C\subset Y$ be a not necessarily reduced Cohen-Macaulay curve with proper support. Assume that the singularities of...

conjecture | open | MathDB 页面 | 原文线索

V2-017856 | MATHDB-333901

Motivic Gersten's conjecture

摘录 / EXCERPT: Motivic Gersten's conjecture. For any commutative regular local ring R and any integers $1\leq p\leq\dim R$, the inclusion functor...

open_problem | open | MathDB 页面 | 原文线索

V2-017857 | MATHDB-333902

The spherical skeleton conjecture for factorial affine spherical varieties

摘录 / EXCERPT: Spherical skeleton conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-017858 | MATHDB-333903

The spherical skeleton conjecture

摘录 / EXCERPT: Let $\mathscr{R}$ be a complete spherical skeleton, and let $P\subseteq G$ be the parabolic subgroup stabilizing the open B-orbit; the quantity $\dim G/P$ depends only on...

conjecture | open | MathDB 页面 | 原文线索

V2-017859 | MATHDB-333912

Contact-form characterization of smooth isolated singularities

摘录 / EXCERPT: Let $X \subset \mathbb{C}^N$ be a variety of dimension n with a normal isolated singularity at 0. For sufficiently small $\epsilon>0$, let $L_{X,\epsilon}$ be its link, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017860 | MATHDB-333927

The non-vanishing conjecture for smooth projective varieties

摘录 / EXCERPT: Non-vanishing conjecture for smooth varieties. The canonical divisor K_{X} has an effective $\mathbb{Q}$-linearly equivalent representative.

conjecture | open | MathDB 页面 | 原文线索

V2-017861 | MATHDB-333928

Generic subtypical Segre identifiability conjecture

摘录 / EXCERPT: Let $d \ge 3$ and $n_1 \ge \cdots \ge n_d \ge 2$. Let $\mathcal{S}_{\mathbb{F}}=\operatorname{Seg}(\mathbb{P}\mathbb{F}^{n_1}\times\cdots\times\mathbb{P}\mathbb{F}^{n_d})$ be the S...

conjecture | open | MathDB 页面 | 原文线索

V2-017862 | MATHDB-333929

Perfect-rank non-identifiability conjecture for Segre varieties

摘录 / EXCERPT: Let $d\ge3$ and $n_1\ge n_2\ge\cdots\ge n_d\ge2$. Let $\mathcal S=\operatorname{Seg}(\mathbb P\mathbb C^{n_1}\times\cdots\times\mathbb P\mathbb C^{n_d})$ be the Segre variety in...

conjecture | open | MathDB 页面 | 原文线索

V2-017863 | MATHDB-333984

Conjecture on recovering double-point structure from the scheme X_2

摘录 / EXCERPT: The X_2 multiple-line conjecture. For every i, $(X_2)_i$ is curvilinear and $\operatorname{length}(X_2)_i=\delta_{P_i}$. If P_i is an A_{2m_i-1} singularity, then L_i cut...

conjecture | open | MathDB 页面 | 原文线索

V2-017864 | MATHDB-333991

Dynamical-degree degeneration conjecture for families of rational maps

摘录 / EXCERPT: Dynamical-degree degeneration conjecture. For every $\epsilon>0$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-017865 | MATHDB-333998

The direct-sum structure conjecture for étale motivic cohomology

摘录 / EXCERPT: Let X be a scheme in the setting of Lichtenbaum's conjecture, and let $H^i_{\mathrm{et}}(X,\mathbb Z(n))$ denote its étale motivic cohomology groups. Direct-sum structure conject...

conjecture | open | MathDB 页面 | 原文线索

V2-017866 | MATHDB-334005

Geometric Jacquet functor conjecture

摘录 / EXCERPT: Let D_{λ} be a sheaf of twisted differential operators on the complex flag variety, and let ... be the right adjoint of the geometric Bernstein functor. Geometric Jacquet fun...

conjecture | open | MathDB 页面 | 原文线索

V2-017867 | MATHDB-334008

Global stability for sufficiently large invitation parameter

摘录 / EXCERPT: Consider the queueing system with on-demand servers and its fluid limits, and let $gamma$ be the parameter governing the invitation scheme. Large-γ global-stability conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-017868 | MATHDB-334017

Non-vanishing log-one-form conjecture for projective log-smooth pairs

摘录 / EXCERPT: Non-vanishing log-one-form conjecture. For any projective log-smooth pair (X,D) of log-general type, there exists no non-vanishing global holomorphic log-one-form.

conjecture | open | MathDB 页面 | 原文线索

V2-017869 | MATHDB-334051

Fullness conjecture for points of type $S_a(X)$

摘录 / EXCERPT: Let X be a smooth projective variety defined over a number field, of dimension n, and let $S_a(X)$ denote the set of points of type $S_a(X)$. Let $a\geq N$ be a real number. Fu...

conjecture | open | MathDB 页面 | 原文线索

V2-017870 | MATHDB-334055

Maulik-Oblomkov correspondence for Hilbert schemes of Fano surfaces

摘录 / EXCERPT: Maulik-Oblomkov conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017871 | MATHDB-334061

The Lehmer-type height bound for points on group varieties

摘录 / EXCERPT: Let G be a group variety defined over a number field k. Let $b1$ be a point of G, and let B be the irreducible torsion variety of minimal dimension containing $b1$. The fie...

conjecture | open | MathDB 页面 | 原文线索

V2-017872 | MATHDB-334063

The torsion anomalous finiteness conjecture

摘录 / EXCERPT: Let G be a group variety and $V\subset G$ an irreducible variety. A maximal V-torsion anomalous variety is a V-torsion anomalous variety not contained in any V-torsion anom...

conjecture | open | MathDB 页面 | 原文线索

V2-017873 | MATHDB-334064

The Zilber-type non-density consequence for weak-transverse varieties

摘录 / EXCERPT: Let G be a group variety. A subvariety $V\subset G$ is weak-transverse if it is not contained in a proper algebraic subgroup of G. Zilber-type non-density statement. If...

conjecture | open | MathDB 页面 | 原文线索

V2-017874 | MATHDB-334065

The bounded height conjecture for the open anomalous locus

摘录 / EXCERPT: Let G be a group variety, and let $V\subset G$ be irreducible. Let V^{oa} be the set of points of V outside the V-anomalous varieties, and define ... where the union is over...

conjecture | open | MathDB 页面 | 原文线索

V2-017875 | MATHDB-334066

The effective bounded height conjecture for maximal torsion anomalous points

摘录 / EXCERPT: Let G be a group variety and let $V\subset G$ be irreducible. A maximal V-torsion anomalous point is a point that is a maximal V-torsion anomalous variety under the terminolo...

conjecture | open | MathDB 页面 | 原文线索

V2-017876 | MATHDB-334071

Geometric conjecture for character-valued parabolic-bundle polynomials

摘录 / EXCERPT: Geometric conjecture. (i) The quotient exists as an algebraic variety and $X_{\bf A}^{\mathcal{E}}\to\mathcal{X}_{\bf A}^{\mathcal{E}}$ is a principal...

conjecture | open | MathDB 页面 | 原文线索

V2-017877 | MATHDB-334081

Vojta's conjecture for smooth projective schemes

摘录 / EXCERPT: Vojta's conjecture. The stated inequality holds for all such closed points outside Z.

conjecture | open | MathDB 页面 | 原文线索

V2-017878 | MATHDB-334100

Goncharov's five-term relation conjecture for the symbol of $I_{3,1}$

摘录 / EXCERPT: Goncharov's conjecture. There are rational functions $f_j(x,y,z)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017879 | MATHDB-334122

Prasanna-Venkatesh conjecture on motivic classes and rational automorphic cohomology

摘录 / EXCERPT: Prasanna-Venkatesh conjecture. The action of $\wedge^*\mathfrak{a}_G^*$ on $H^*(Y,\mathbf{C})_{\Pi}$ is compatible with rational forms: if an element of $\mathfrak{a}_G^*$ maps to...

conjecture | open | MathDB 页面 | 原文线索

V2-017880 | MATHDB-334167

Strong Chow pushforward conjecture for properly stable smooth Artin stacks

摘录 / EXCERPT: Let $\pi\colon\mathcal {X}\to{\mathbf X}$ be a good moduli space map, where $\mathcal {X}$ is a properly stable smooth Artin stack. For a strong cycle $\mathcal {Z}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-017881 | MATHDB-334168

Strong-cycle formula for the Chow pushforward on smooth Artin stacks

摘录 / EXCERPT: Let $\pi\colon\mathcal {X}\to{\mathbf X}$ be a good moduli space map from a properly stable smooth Artin stack. Let $e_\mathcal {Z}$ be the generic stabilizer order of a strong cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-017882 | MATHDB-334169

Strong Chow ring conjecture for properly stable smooth Artin stacks

摘录 / EXCERPT: Let $\mathcal {X}$ be a properly stable smooth Artin stack with good moduli space ${\mathbf X}$. The strong relative Chow groups $A^k_{st}(\mathcal {X}/{\mathbf X})$ are generated...

conjecture | open | MathDB 页面 | 原文线索

V2-017883 | MATHDB-334170

Injective strong Chow subring conjecture for good moduli spaces

摘录 / EXCERPT: Let $\pi\colon\mathcal {X}\to{\mathbf X}$ be a good moduli space map from a properly stable smooth Artin stack, and let $A^*_{st}(\mathcal {X}/{\mathbf X})_\mathbb{Q}$ denote its s...

conjecture | open | MathDB 页面 | 原文线索

V2-017884 | MATHDB-334198

The Esnault-Srinivas-Viehweg converse for Chow and cohomological products

摘录 / EXCERPT: Let X be a smooth complete variety of dimension n over $\mathbb{C}$, and let $n=n_1+\cdots+n_r$ be a partition with $n_i\in\mathbb{N}_{>0}$. Property (P1) means that there exis...

conjecture | open | MathDB 页面 | 原文线索

V2-017885 | MATHDB-334208

Involutivity and radicality of Pfaffian ideals for Severi strata

摘录 / EXCERPT: Let S be the symplectic manifold with symplectic form Ω, let χ denote the relevant matrix of logarithmic vector fields, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017886 | MATHDB-334209

Cohen-Macaulayness of Severi strata for miniversal deformations of A_{2k}

摘录 / EXCERPT: Let A_{2k} be the indicated singularity, let S be the base of its miniversal deformation, and let $D(\ell)\subset S$ denote the corresponding Severi stratum. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-017887 | MATHDB-334230

Ghioca-Scanlon's dynamical Mordell-Lang conjecture in positive characteristic

摘录 / EXCERPT: Let X be a quasiprojective variety defined over a field K of characteristic p. Let $\alpha\in X(K)$, let $V\subseteq X$ be a subvariety defined over K, and let ... a be an en...

conjecture | open | MathDB 页面 | 原文线索

V2-017888 | MATHDB-334252

The cluster structure conjecture for open positroid varieties

摘录 / EXCERPT: Cluster structure conjecture. The coordinate ring of an open positroid variety has a cluster structure obtained from the quiver associated to any of its reduced plabic graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-017889 | MATHDB-334259

Conjecture on defective weighted projective spaces

摘录 / EXCERPT: Let a weighted projective space be a projective toric variety, and call it defective when its dual variety is not a hypersurface. A cone over a weighted projective space is a weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-017890 | MATHDB-334278

The conjecture that every four-dimensional abelian variety is isogenous to a Jacobian

摘录 / EXCERPT: Let A be an abelian variety of dimension 4 over $\overline{\mathbb{F}}_p$. A Jacobian is the Jacobian variety of a smooth projective curve. The four-dimensional abelian-Jacobia...

conjecture | open | MathDB 页面 | 原文线索

V2-017891 | MATHDB-334279

Geometric conjecture for potentially dense subvarieties of singular varieties of general type

摘录 / EXCERPT: Let X be a $\mathbf{b}{Q}$-Gorenstein variety with a canonical divisor K_{X} big. A closed subvariety $Y\subset X$ is potentially dense if, after a finite extension of the base...

conjecture | open | MathDB 页面 | 原文线索

V2-017892 | MATHDB-334284

Looijenga's affine covering conjecture for the moduli space of curves

摘录 / EXCERPT: Let $\mathcal{M}_g$ be the coarse moduli space of genus g Riemann surfaces, viewed as a quasiprojective complex variety. Looijenga's conjecture. For every $g\geq 2$, the variety...

conjecture | open | MathDB 页面 | 原文线索

V2-017893 | MATHDB-334285

Positivity conjecture for the algebraic stringy Euler number

摘录 / EXCERPT: Let X be a projective algebraic variety with at worst $\mathbb{Q}$-Gorenstein log-terminal singularities. Positivity conjecture. The algebraic stringy Euler number satisfies ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-017894 | MATHDB-334286

Strict monotonicity conjecture for Mori flips

摘录 / EXCERPT: Let $g\colon X\dashrightarrow X^+$ be a birational morphism between $[200~\mathbb{Q}$-Gorenstein varieties, together with birational morphisms $f\colon X\to Z$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017895 | MATHDB-334288

The abundance conjecture for smooth projective minimal varieties of Kodaira dimension zero

摘录 / EXCERPT: Let X be a smooth projective minimal variety of Kodaira dimension 0, and let K_X denote its canonical divisor. Abundance conjecture. The canonical divisor K_X is numericall...

conjecture | open | MathDB 页面 | 原文线索

V2-017896 | MATHDB-334307

The Bloch-Lewis conjecture on algebraically trivial codimension-two cycles

摘录 / EXCERPT: Bloch-Lewis conjecture. Then $CH^2(X)_{\emph{alg}}$ is finite-dimensional, and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-017897 | MATHDB-334324

The nilpotent-freeness conjecture for Kauffman bracket skein algebras

摘录 / EXCERPT: Let M be a compact 3-manifold, let $K M(M;\mathbb{C},-1)$ denote its Kauffman bracket skein algebra at parameter -1, and let $\Phi:K M(M;\mathbb{C},-1)\to\mathcal{R}(M)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-017898 | MATHDB-334364

Periodic approximability of cut-and-project Delone subshifts

摘录 / EXCERPT: Periodic approximability conjecture. The transversal of $\mathcal{D}$ is periodically approximable if $\mathcal{D}$ is defined via a cut-and-project scheme.

conjecture | open | MathDB 页面 | 原文线索

V2-017899 | MATHDB-334365

The local-global equality conjecture for varieties over global function fields

摘录 / EXCERPT: Let K be a global function field over a finite field k of positive characteristic, let S be a finite set of places of K, and let $\Gamma\subset O_S^*$ be a subgroup. Fix a...

conjecture | open | MathDB 页面 | 原文线索

V2-017900 | MATHDB-334378

Sullivant's secant-initial ideal conjecture for Plücker tree ideals

摘录 / EXCERPT: Sullivant's conjecture. For every r, one has

conjecture | open | MathDB 页面 | 原文线索

V2-017901 | MATHDB-334379

The cluster-count criterion for secant and initial ideals

摘录 / EXCERPT: The cluster-count criterion. One has

conjecture | open | MathDB 页面 | 原文线索

V2-017902 | MATHDB-334395

Hosono-Lian-Yau hyperplane conjecture for Calabi-Yau periods

摘录 / EXCERPT: Let $\triangle$ be a reflexive lattice polytope, let $\mathcal T$ be a regular projective triangulation, and let $\widetilde{X^\vee_{\mathcal T}}$ be the associated smooth projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-017903 | MATHDB-334425

Single-exponential mixed complexity conjecture for the semi-algebraic image functor

摘录 / EXCERPT: Let $\opcat{SA}$ be the category whose objects are embedded semi-algebraic sets and whose morphisms are restrictions of polynomial mappings. Let $\cA$ consist of scalar multiplicat...

conjecture | open | MathDB 页面 | 原文线索

V2-017904 | MATHDB-334443

Berest-Samuelson's Brumfiel-Hilden condition for knot meridians and longitudes

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $\pi=\pi_1(S^3\setminus K)$, and let $H[\pi]$ be its Brumfiel-Hilden algebra with commutative subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-017905 | MATHDB-334444

Weak Brumfiel-Hilden conjecture for higher-rank knot representations

摘录 / EXCERPT: Let $\pi=\pi_1(S^3\setminus K)$ be the group of a knot complement, let $H_n[\pi]=\Gamma({\mathrm{Rep}}_n(\pi),M_n(\mathbb{C}))^{\mathrm{GL}_n}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017906 | MATHDB-334445

Weak Brumfiel-Hilden conjecture for the A-polynomial ideal

摘录 / EXCERPT: For a knot $K\subset S^3$, let $J\subset \mathbb{C}[m^{\pm1},l^{\pm1}]$ be the kernel of the algebra map determined by the meridian and longitude, and let $A_K(m,l)$ be the A-pol...

conjecture | open | MathDB 页面 | 原文线索

V2-017907 | MATHDB-334447

Equivalence of the four transitivity conditions for quasiaffine varieties

摘录 / EXCERPT: Let X be an irreducible quasiaffine variety of dimension at least 2. Let conditions (1)-(4) refer to the four conditions in Theorem, whose equivalence is being conjectured. Tra...

conjecture | open | MathDB 页面 | 原文线索

V2-017908 | MATHDB-334456

Limit-measure uniqueness conjecture for decorated étale group schemes

摘录 / EXCERPT: Limit-measure conjecture. The measures mu_g converge to a measure mu on $\mathcal{E}$ such that the expected number of surjections from a mu-random element to every element o...

conjecture | open | MathDB 页面 | 原文线索

V2-017909 | MATHDB-334474

Viehweg's log-general-type conjecture for moduli maps

摘录 / EXCERPT: Let $\mathfrak M$ be the coarse moduli scheme of canonically polarised manifolds with fixed Hilbert polynomial. Let Y° be a smooth quasi-projective variety admitting a gener...

conjecture | open | MathDB 页面 | 原文线索

V2-017910 | MATHDB-334487

Sommers-Tymoczko's Weyl type subset generating-function conjecture

摘录 / EXCERPT: Let $I\subset\Phi^+$ be a lower ideal. A subset $Y\subset I$ is of Weyl type if $\alpha,\beta\in Y$ and $\alpha+\beta\in I$ imply $\alpha+\beta\in Y$, and if...

conjecture | open | MathDB 页面 | 原文线索

V2-017911 | MATHDB-334492

The abundance conjecture for nef divisors on varieties with trivial canonical class

摘录 / EXCERPT: Let X be a projective manifold with $H^1(X,\mathcal O_X)=0$ and $K_X\sim 0$, and let L be a nef divisor. Abundance conjecture for varieties with trivial canonical class. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-017912 | MATHDB-334493

The singular abundance conjecture for klt varieties with numerically trivial canonical class

摘录 / EXCERPT: Let X be a normal projective klt variety with $H^1(X,\mathcal O_X)=0$ and $K_X\sim_{\mathbb Q}0$, and let L be a nef divisor. Singular abundance conjecture. Then L is semiamp...

conjecture | open | MathDB 页面 | 原文线索

V2-017913 | MATHDB-334520

Preservation of MJ-singularities under general residual intersections

摘录 / EXCERPT: Let A be a nonsingular variety, let $X\subset A$ be a closed subscheme, and let Y be a general t-residual intersection of X in the sense of the paper. The singularities MJ-...

conjecture | open | MathDB 页面 | 原文线索

V2-017914 | MATHDB-334522

The h-positivity conjecture for the symmetric function Ψ_{21}

摘录 / EXCERPT: Let $\alpha\in{\bf l}[n]$, and let $\Psi_{21}(\alpha)$ be the symmetric function defined by the coloring scheme whose monochromatic subsequences have standardization...

conjecture | open | MathDB 页面 | 原文线索

V2-017915 | MATHDB-334529

Injectivity of period maps for convergent motivic multiple harmonic values

摘录 / EXCERPT: Let $I\subset\mathcal{P}\times\mathbb{N}$, and let $\widehat{\operatorname{per}}_{\operatorname{har}_I}$ be the period map on weight-adically convergent sums of motivic multiple ha...

conjecture | open | MathDB 页面 | 原文线索

V2-017916 | MATHDB-334578

Fernández de Bobadilla-Luengo Velasco-Melle Hernández-Némethi conjecture for Alexander coefficients

摘录 / EXCERPT: Let $C\subset\mathbb{C}P^2$ be a rational cuspidal curve of degree d, and let K be the connected sum of the links of the singularities of C. Write its Alexander polynomial as...

conjecture | open | MathDB 页面 | 原文线索

V2-017917 | MATHDB-334588

The rough form of the Betti Geometric Langlands correspondence

摘录 / EXCERPT: Let E be a field of characteristic zero. Let G be a reductive group with Langlands dual group $G^{\vee}$, let X be the underlying curve, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017918 | MATHDB-334600

Batyrev-Tschinkel's prediction for constants in the anticanonical counting asymptotic

摘录 / EXCERPT: Let X be a Fano variety over $\mathds{Q}$ with at worst log terminal singularities and a Zariski dense set of rational points. Let $H\colon X(\mathds{Q}) \to \mathds{R}_{>0}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-017919 | MATHDB-334625

Algebraization conjecture for compact Nash sets

摘录 / EXCERPT: A compact Nash set is a compact semialgebraic set endowed with its Nash structure. Two compact Nash sets are strongly asym-Nash cobordant if they are related by the strong form of...

conjecture | open | MathDB 页面 | 原文线索

V2-017920 | MATHDB-334643

Rigidity conjecture for symbolic powers of codimension-two subvarieties

摘录 / EXCERPT: Let $X\factor P^3_k$ be a reduced and unmixed subvariety of codimension 2, with ideal I_X. Suppose there is a point $\mathfrak{p}\in X$ such that the localization of I_X at...

conjecture | open | MathDB 页面 | 原文线索

V2-017921 | MATHDB-334648

Galois-conjugate projective classifying spaces conjecture

摘录 / EXCERPT: Let X be a projective $K(\pi,1)$, meaning a projective variety whose universal cover is contractible, and let $\sigma\in\operatorname{Aut}(\mathbb{C})$. Write X^σ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-017922 | MATHDB-334652

The reducedness conjecture for affine Grassmannian slices

摘录 / EXCERPT: Let G be a complex semisimple group and let $\mu\leq\lambda$ be dominant coweights. Write $\mathcal{G}r_\mu^\lambda=\mathcal{G}r_\mu\cap\overline{\mathcal{G}r^\lambda}$ for the t...

conjecture | open | MathDB 页面 | 原文线索

V2-017923 | MATHDB-334662

The irreducibility conjecture for the ordered-node space of rational curves on polarized K3 surfaces

摘录 / EXCERPT: The ordered-node irreducibility conjecture. Then ${\mathcal W}_{L,0}$ is irreducible. This is identified as the difficult monodromy-related part of the proposed approach to the irr...

conjecture | open | MathDB 页面 | 原文线索

V2-017924 | MATHDB-334704

The intersection-map conjecture for surfaces with $p_g=1$ and $q=2$

摘录 / EXCERPT: Let S be a surface with geometric genus $p_g(S)=1$ and irregularity $q(S)=2$. Let $A^1_{hom}(S)_{\mathbb{Q}}$ be the group of homologically trivial divisor classes with rational...

conjecture | open | MathDB 页面 | 原文线索

V2-017925 | MATHDB-334708

Rapoport-Viehmann's local Shimura variety conjecture

摘录 / EXCERPT: Let $(G,[b],\{\mu\})$ be a local Shimura datum over $\mathbb{Q}_p$, with local reflex field E, flag variety $\mathscr{F}\ell_{G,\mu}$ over $\breve{E}$, weakly admissible locus...

conjecture | open | MathDB 页面 | 原文线索

V2-017926 | MATHDB-334741

Conjecture on quasi-Hamiltonian structures of multiplicative bow varieties

摘录 / EXCERPT: Let V_{i-1} and V_i be the vector spaces associated with adjacent vertices in a bow, and let A_{i-1}, B_{i-1}, B'_i, a_i, and b_{i-1} satisfy conditions (i), (a), (S1...

conjecture | open | MathDB 页面 | 原文线索

V2-017927 | MATHDB-334749

Merel's uniform boundedness conjecture for torsion on abelian varieties

摘录 / EXCERPT: Let d and m be positive integers. An abelian variety of dimension d defined over a number field k of degree m is considered, together with a prime number p. Merel's con...

conjecture | open | MathDB 页面 | 原文线索

V2-017928 | MATHDB-334757

Kawamata's effective non-vanishing conjecture for varieties with trivial first Chern class

摘录 / EXCERPT: Let X be a smooth projective variety with $c_1(X)=0$ in $H^2(X,\mathbb{R})$, and let L be a nef and big line bundle on X. Kawamata's trivial-first-Chern-class conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-017929 | MATHDB-334781

Algebraic identifiability of mixtures of bivariate Gaussians

摘录 / EXCERPT: Let $\mathcal{G}_{n,d}$ denote the Gaussian moment variety of order d in n variables, and let $\operatorname{Sec}_k(\mathcal{G}_{n,d})$ be its k-th secant variety. Algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-017930 | MATHDB-334783

Defect pattern conjecture for fourth-order Gaussian moment varieties

摘录 / EXCERPT: Let $\mathcal{G}_{n,4}$ be the Gaussian moment variety of order 4 in n variables, and let δ_j denote the defect of its j-th secant variety. The model-parameter count i...

conjecture | open | MathDB 页面 | 原文线索

V2-017931 | MATHDB-334786

Palindromicity conjecture for hyperplane sections of high-degree complete intersections

摘录 / EXCERPT: Let $P=\mathbb{P}^N$, and let $X\subset P$ be a general complete intersection of multi-degree $(d_1,d_2,\ldots,d_k)$ with $d_1\leq\cdots\leq d_k$. Assume that $d_1\gg0$. For a hype...

conjecture | open | MathDB 页面 | 原文线索

V2-017932 | MATHDB-334794

The tame-regular singular fundamental group conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of positive characteristic and let X be a smooth variety over k. Write $\pi^t(X)$ for the tame fundamental group and...

conjecture | open | MathDB 页面 | 原文线索

V2-017933 | MATHDB-334817

Reduced Donaldson-Thomas multiple-cover conjecture for abelian threefolds

摘录 / EXCERPT: Let A be an abelian threefold, and let $\beta\in H_2(A,\mathbb Z)$ be a curve class of type (d_1,d_2,d_3). Write $\mathsf{DT}^{\mathrm{red}}_{n,(d_1,d_2,d_3)}$ for the reduced...

conjecture | open | MathDB 页面 | 原文线索

V2-017934 | MATHDB-334845

The Gram locus conjecture for real binary tensors

摘录 / EXCERPT: Gram locus conjecture. The Gram locus $\mathcal{G}(\mathcal{B})$ is given by

conjecture | open | MathDB 页面 | 原文线索

V2-017935 | MATHDB-334849

The smooth-cycle generation conjecture for Chow groups

摘录 / EXCERPT: Smooth-cycle generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017936 | MATHDB-334850

The plci-cycle generation conjecture for Chow rings

摘录 / EXCERPT: Plci-cycle generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-017937 | MATHDB-334853

Base change conjecture for motivic cohomology of a regular model

摘录 / EXCERPT: Let $\mathcal{O}_K$ be an excellent henselian discrete valuation ring with quotient field K and residue field $k=\mathcal{O}_K/\pi\mathcal{O}_K$, assume that $1/n\in k$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017938 | MATHDB-334863

Uniqueness of a tropical variety from its tropical Chow hypersurface

摘录 / EXCERPT: Let $\Sigma_1,\Sigma_2\subset\mathbb{R}^{n+1}/\mathbb{R}$ be pure $(n-k-1)$-dimensional tropical varieties. For a tropical variety Σ, let Z_Σ denote its tropical Chow...

conjecture | refuted | MathDB 页面 | 原文线索

V2-017939 | MATHDB-334869

Stanley-Reisner-type degeneration conjecture for loop-model schemes

摘录 / EXCERPT: Let ρ be a link pattern, let $\mathcal D$ be a diagram of ρ, let L_ρ be the associated scheme, and let $L_{\rho,\pi}$ denote its fixed-connectivity part for a connec...

conjecture | open | MathDB 页面 | 原文线索

V2-017940 | MATHDB-334912

D-equivalence implies L-equivalence for simply connected varieties

摘录 / EXCERPT: Let X and Y be smooth projective simply connected varieties. Write $\mathbb{D}(X)$ for the bounded derived category of coherent sheaves on X, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017941 | MATHDB-334929

The reducibility conjecture for the real rank boundary of quaternary quartics

摘录 / EXCERPT: Reducibility conjecture. The real rank boundary $\partial_{alg}\mathcal{R}_{4,4}$ is reducible. The discriminant of $\Omega(f)$ is one of its components. Further, at least one more...

conjecture | open | MathDB 页面 | 原文线索

V2-017942 | MATHDB-334967

Polyhedrality conjecture for pseudoeffective cones of cycles on k-Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety. For an integer $k\geq 1$, call X k-Fano if its Chern characters $ch_s(X)$ are positive for $1\leq s\leq k$. Let $\Effc kX$ denote the cone of...

conjecture | open | MathDB 页面 | 原文线索

V2-017943 | MATHDB-334974

Mori-Mukai's characterization conjecture for projective space

摘录 / EXCERPT: Let X be a smooth Fano variety of dimension n over an algebraically closed field, and let $C\subseteq X$ be a rational curve. Mori-Mukai's characterization conjecture. If ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-017944 | MATHDB-335031

Kurdyka's extension conjecture for arc-analytic semialgebraic functions

摘录 / EXCERPT: Let X be an $\mathscr{A{\!}R}$-closed set in $\mathbb{R}^n$. Denote by $\mathscr{A}_a(X)$ the ring of arc-analytic semialgebraic functions on X, by...

conjecture | open | MathDB 页面 | 原文线索

V2-017945 | MATHDB-335035

Rational smoothness and smoothness conjecture for symmetric orbit closures

摘录 / EXCERPT: Let K act on the ambient flag variety, let Y_{γ} be the closure of the orbit indexed by γ, and let $\mathcal{O}_{\alpha}$ be the orbit indexed by α with...

conjecture | open | MathDB 页面 | 原文线索

V2-017946 | MATHDB-335046

Existence of real witness sets for special orthogonal groups

摘录 / EXCERPT: Let n be a positive integer, and let a real witness set for an algebraic variety mean a witness set whose points are real. Existence conjecture. For any n, the algebraic group...

conjecture | open | MathDB 页面 | 原文线索

V2-017947 | MATHDB-335048

Logarithmic action conjecture for polynomial components of almost Belyi maps

摘录 / EXCERPT: Let $\varphi(x,w)$ be an almost Belyi map with definition field $K=\mathbb{Q}(w)$. Suppose the vector field ... annihilates $\varphi(x,w)$. A logarithmic action conjecture. This vect...

conjecture | open | MathDB 页面 | 原文线索

V2-017948 | MATHDB-335058

Prediction on singular points of the bitangent congruence

摘录 / EXCERPT: Prediction on singular bitangent lines. If a point of $\operatorname{Bit}(S)$ corresponds to L, then L corresponds to a singular point of $\operatorname{Bit}(S)$.

conjecture | open | MathDB 页面 | 原文线索

V2-017949 | MATHDB-335079

Eigenvalue lower-bound conjecture for stable oscillations in linear PDE schemes

摘录 / EXCERPT: Let a linear partial differential equation be discretized by a symmetric positive difference-scheme matrix. A region of numerically stable oscillations is understood as a parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-017950 | MATHDB-335080

Dominating-eigenvalue conjecture for oscillation-free linear PDE schemes

摘录 / EXCERPT: Let a linear partial differential equation have a well-posed scheme and a consistent scheme matrix. Call an eigenvalue dominating when it supplies the asserted dominant spectral co...

conjecture | open | MathDB 页面 | 原文线索

V2-017951 | MATHDB-335082

Monotonicity conjecture for oscillation-free nonlinear PDE methods

摘录 / EXCERPT: Let a nonlinear partial differential equation have monotonic initial and asymptotic behavior, and let a numerical method be applied to it. Let the method's linearized form be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-017952 | MATHDB-335137

Huang-Katz-Klemm modularity conjecture for elliptic Calabi-Yau varieties

摘录 / EXCERPT: Let X be an elliptically fibered Calabi-Yau variety, and let its Gromov-Witten invariants be the enumerative invariants associated with stable maps to X. Huang-Katz-Klemm's mod...

conjecture | open | MathDB 页面 | 原文线索

V2-017953 | MATHDB-335149

S4-Conjecture for ternary quartics and quadratic forms

摘录 / EXCERPT: Let f be a ternary quartic, that is, a 4-form in $\mathbb{R}[\mathrm{x}_1,\mathrm{x}_2,\mathrm{x}_3]$, and let g be a quadratic form in...

conjecture | open | MathDB 页面 | 原文线索

V2-017954 | MATHDB-335150

Dehomogenized S4-Conjecture for bivariate degree-four and degree-two polynomials

摘录 / EXCERPT: Let f be a polynomial of degree 4 in $\mathbb{R}[\mathrm{x}_1,\mathrm{x}_2]$ and let g be a polynomial of degree 2 in $\mathbb{R}[\mathrm{x}_1,\mathrm{x}_2]$. Suppose there...

conjecture | open | MathDB 页面 | 原文线索

V2-017955 | MATHDB-335168

Zhang's small-height conjecture for simple families of abelian varieties

摘录 / EXCERPT: Zhang's conjecture. The asserted finiteness holds for every such family, line bundle, and non-torsion section.

conjecture | open | MathDB 页面 | 原文线索

V2-017956 | MATHDB-335194

The existence conjecture for toric Landau-Ginzburg models of smooth Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety. A toric Landau-Ginzburg model for X is a Laurent polynomial $f\in{\mathbb T}[x_1,\ldots,x_n]$ satisfying the period, Calabi-Yau, and toric cond...

conjecture | open | MathDB 页面 | 原文线索

V2-017957 | MATHDB-335195

The dominant tower conjecture for projective varieties

摘录 / EXCERPT: Let k be a field and let n be a positive integer. A smooth projective variety X of dimension n is endowed with a tower of smooth fibrations ... where $\dim X_i=n-i$ for each...

conjecture | open | MathDB 页面 | 原文线索

V2-017958 | MATHDB-335197

Higher difference terms and weak difference terms

摘录 / EXCERPT: Let $mathcal V$ be a variety. For each $n i2$, a weak n-difference term is a ternary term $c_n(x,y,z)$ satisfying the corresponding weak n-difference identities, and a variety...

conjecture | open | MathDB 页面 | 原文线索

V2-017959 | MATHDB-335198

Beilinson's conjecture on Chow classes and the Hochschild-Serre filtration

摘录 / EXCERPT: Beilinson's conjecture. With notation as above, if a class in $CH^n(X)_{\mathbb Q}$ vanishes in $H^i\left(k,H^{2n-i}(X_{\bar k},\mathbb Q_\ell(n))\right)$ for $i=0,1$, then it does...

conjecture | open | MathDB 页面 | 原文线索

V2-017960 | MATHDB-335199

Chern character conjecture for Shimura varieties

摘录 / EXCERPT: Chern character conjecture. The map $c_K:\operatorname{Rep}_{\overline{\mathbb Q}}(K_h)\to CH({}_KS(G,X))_{\mathbb Q}$ induces an injective ring homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-017961 | MATHDB-335244

Classicality conjecture for de Rham strictly dominant points

摘录 / EXCERPT: Let $Y(U^p,\overline{\rho})$ be the eigenvariety locus considered in the paper, with points $x=(\rho,\underline{\delta})$ consisting of a Galois representation and a locally...

conjecture | open | MathDB 页面 | 原文线索

V2-017962 | MATHDB-335245

Socle conjecture for refinements of automorphic Galois representations

摘录 / EXCERPT: Let $\mathcal{R}$ be a refinement, let λ be the algebraic weight appearing in the associated locally analytic representation, and let w_0 be the longest Weyl-group elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-017963 | MATHDB-335278

Strict Bott-Samelson desingularization conjecture for Schubert varieties

摘录 / EXCERPT: Strict Bott-Samelson desingularization conjecture. If $w\ngeq v$ for every $v\in\Gamma_n$, then X^w can be desingularized using a sequence of strict Bott-Samelson resolutions.

conjecture | open | MathDB 页面 | 原文线索

V2-017964 | MATHDB-335315

The weight-vector conjecture for toric degenerations of flag varieties

摘录 / EXCERPT: Let $n\geq 3$ be an arbitrary integer, and let $\underline w_0$ be a reduced expression. For each Plücker variable, let ${\bf m}_J$ be the corresponding minimal monomial and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017965 | MATHDB-335331

Beilinson's non-degeneracy conjecture for the height pairing

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over a number field k. Write $\operatorname{CH}^r_{\hom}(X;\mathbb{Q})$ for homologically trivial codimension-r cycles m...

conjecture | open | MathDB 页面 | 原文线索

V2-017966 | MATHDB-335332

Beilinson's Hodge-index conjecture for height pairings

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over a number field k, and let $L_X\in\operatorname{CH}^1(X)$ be the class of a hyperplane section. Assume the hard Lefsch...

conjecture | open | MathDB 页面 | 原文线索

V2-017967 | MATHDB-335339

Simpson's motivicity conjecture for perverse sheaves and constructible complexes

摘录 / EXCERPT: Let X be a complex algebraic variety. Let $\C_X$ be $Perv(X,\mathbb{C})$ or $\mathbf{D}^b_c(X,\mathbb{C})$, and let $S=\{\mathcal{F}\}\subset\operatorname{Iso}(\C_X)$ consist of...

conjecture | open | MathDB 页面 | 原文线索

V2-017968 | MATHDB-335340

Absolute-set decomposition conjecture for local systems

摘录 / EXCERPT: Let X be a smooth complex algebraic variety, and let $\mathcal{M}_B(X)$ be the moduli space of semisimple local systems of all ranks. An absolute K-constructible set is a subse...

conjecture | open | MathDB 页面 | 原文线索

V2-017969 | MATHDB-335341

Special varieties package conjecture for local-system moduli

摘录 / EXCERPT: Let X be a smooth complex algebraic variety and r a natural number. Let $\mathcal{M}_B(X,r)$ be the moduli space of semisimple complex local systems of rank r. An absolute...

conjecture | open | MathDB 页面 | 原文线索

V2-017970 | MATHDB-335355

Slavov's maximal component conjecture for singular hypersurfaces

摘录 / EXCERPT: Let $\mathbb{P}^r$ be projective r-space, and let ℓ and b be integers. Consider the space of hypersurfaces of degree ℓ in $\mathbb{P}^r$ whose singular locus has dime...

conjecture | open | MathDB 页面 | 原文线索

V2-017971 | MATHDB-335358

Igusa-Denef-Loeser motivic monodromy conjecture

摘录 / EXCERPT: Let f define a hypersurface, let x be a point in its domain, and let $Z_{f,x}(T)$ be the local motivic zeta function. Write ${\mathbb L}$ for the Lefschetz motive and let...

conjecture | solved | MathDB 页面 | 原文线索

V2-017972 | MATHDB-335366

Esnault-Langer conjecture on compatible points of Frobenius twists of an abelian variety

摘录 / EXCERPT: Let L be a field finitely generated over an algebraically closed field l_0 of characteristic p, and let C be an abelian variety over L. For each $\ell\geqslant 0$, choose...

conjecture | open | MathDB 页面 | 原文线索

V2-017973 | MATHDB-335405

Pila's modular André-Oort conjecture with derivatives

摘录 / EXCERPT: Let $\mathcal{H}$ be the upper half-plane, and let $V\subseteq\mathbb{C}^{3n}$ be a proper algebraic variety defined over $\overline{\mathbb{Q}}$. Let an...

conjecture | open | MathDB 页面 | 原文线索

V2-017974 | MATHDB-335410

Batyrev's polynomial bound conjecture for components of rational-curve spaces

摘录 / EXCERPT: Batyrev's conjecture. There is a polynomial $P(d) \in \mathbb{Z}[d]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-017975 | MATHDB-335411

Batyrev's polynomial-growth conjecture for rational-curve components on weak Fano varieties

摘录 / EXCERPT: Batyrev's conjecture. Then $h(m\alpha)$, considered as a function of m, is bounded above by a polynomial $P(m)$ whose degree is the largest relative dimension of a map

conjecture | open | MathDB 页面 | 原文线索

V2-017976 | MATHDB-335412

Strong Manin's conjecture on uniqueness of Manin components

摘录 / EXCERPT: Strong Manin's conjecture. For any $\mathbb{Z}$-curve class α contained in the relative interior of $F(X,L)$, there is at most one Manin component parametrizing curves of cl...

conjecture | open | MathDB 页面 | 原文线索

V2-017977 | MATHDB-335413

Existence of dominant rational-curve families in nef classes

摘录 / EXCERPT: Dominant-family existence conjecture. There exists a dominant family of maps from $\mathbb{P}^{1}$ to X whose images have class α.

conjecture | open | MathDB 页面 | 原文线索

V2-017978 | MATHDB-335441

Criterion for quasi-affineness of abstract horospherical stacks

摘录 / EXCERPT: Criterion for quasi-affineness. Under these hypotheses, $\mathcal{X}$ is a quasi-affine scheme.

conjecture | open | MathDB 页面 | 原文线索

V2-017979 | MATHDB-335444

Ambro-Ionescu-Kawamata effective adjunction conjecture

摘录 / EXCERPT: Let (X,L) be a projective variety of dimension n with at most terminal $\mathbb{Q}$-factorial singularities, where L is a nef and big Cartier divisor; such a pair (X,L) is...

conjecture | open | MathDB 页面 | 原文线索

V2-017980 | MATHDB-335445

Effective adjunction threshold conjecture for quasi polarized manifolds

摘录 / EXCERPT: Let (X,L) be a quasi polarized manifold of dimension n, meaning that X is a projective manifold and L is a nef and big Cartier divisor. Let $s>0$. Effective adjunction thre...

conjecture | open | MathDB 页面 | 原文线索

V2-017981 | MATHDB-335450

Batyrev and Dais's conjecture on Hodge numbers of crepant resolutions

摘录 / EXCERPT: Let X be an algebraic variety defined over the complex numbers with at worst Gorenstein canonical singularities. A smooth crepant resolution of X is a smooth resolution preserv...

conjecture | open | MathDB 页面 | 原文线索

V2-017982 | MATHDB-335470

Conjecture on the range of real bitangents of smooth real plane sextics

摘录 / EXCERPT: Let C be a smooth sextic curve in the real projective plane $\mathbb{P}^2_\mathbb{R}$. A bitangent of C is a line tangent to C at two points, and a bitangent is real when it...

conjecture | open | MathDB 页面 | 原文线索

V2-017983 | MATHDB-335477

McKinnon's approximation constant conjecture

摘录 / EXCERPT: McKinnon's conjecture. Il existe une courbe rationnelle C telle que

conjecture | open | MathDB 页面 | 原文线索

V2-017984 | MATHDB-335497

The conjecture that scaled Segre varieties realize every ML degree up to their degree

摘录 / EXCERPT: Let V^c be the scaled Segre variety associated with an $m\times n$ scaling matrix $c\in(\mathbb C^*)^{mn}$, and let $\deg(V)=\binom{m+n-2}{m-1}$. Scaled Segre ML-degree conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-017985 | MATHDB-335498

The self-duality conjecture for no-three-way interaction models

摘录 / EXCERPT: For $i\in\{0,\ldots,m-1\}$ and $j,k\in\{0,1\}$, define the no-three-way interaction model C_m by ... Its associated configuration is denoted A_m, with convex hull...

conjecture | open | MathDB 页面 | 原文线索

V2-017986 | MATHDB-335499

The ML-degree conjecture for no-three-way interaction models

摘录 / EXCERPT: For $i\in\{0,\ldots,m-1\}$ and $j,k\in\{0,1\}$, let C_m be the no-three-way interaction model with one m-ary variable and two binary variables, defined by ... Let A_m be its as...

conjecture | open | MathDB 页面 | 原文线索

V2-017987 | MATHDB-335502

Connected-component bijection for Hodge-Newton irreducible affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be the reductive group, L its relevant local field extension, P the parahoric subgroup, λ a coweight, and $b\in G(L)$. Let $X(\lambda,b)_P$ be the associated cl...

conjecture | open | MathDB 页面 | 原文线索

V2-017988 | MATHDB-335532

Donagi's small Schottky-Jung locus conjecture

摘录 / EXCERPT: Let J_g be the Jacobian locus in $\mathcal A_g$, and let $\mathcal S_g$ be the small Schottky-Jung locus, defined as the intersection of the Schottky-Jung loci associated with al...

conjecture | open | MathDB 页面 | 原文线索

V2-017989 | MATHDB-335533

The totally geodesic subvariety conjecture for Jacobian loci

摘录 / EXCERPT: Let J_g be the Jacobian locus in $\mathcal A_g$, and let a totally geodesic subvariety mean an irreducible subvariety arising as the image of a totally geodesic subvariety of the...

conjecture | open | MathDB 页面 | 原文线索

V2-017990 | MATHDB-335535

The totally geodesic subvariety conjecture for universal Jacobian loci

摘录 / EXCERPT: Let $J_{g,h}$ be the universal Jacobian locus in $\mathcal A_{g,h}$, equipped with the Riemannian metric $ds^2_{g,h;A,B}$. Universal totally geodesic subvariety conjecture. For a s...

conjecture | open | MathDB 页面 | 原文线索

V2-017991 | MATHDB-335558

Wonderful compactification simple connectedness conjecture for exponent canceling groups

摘录 / EXCERPT: Wonderful compactification conjecture. The normalized wonderful compactification of $\mathfrak{X}^0_\Gamma(G)$ should be a simply connected compactification of...

conjecture | open | MathDB 页面 | 原文线索

V2-017992 | MATHDB-335563

The dual Amplituhedron volume conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be an Amplituhedron embedded in an irreducible variety X, and let $\mathcal Y\in X$ lie away from the boundary of ${\mathcal A}$. Dual Amplituhedron volume con...

conjecture | open | MathDB 页面 | 原文线索

V2-017993 | MATHDB-335569

Equality of special-cycle rings on orthogonal Shimura varieties

摘录 / EXCERPT: Let Y_K be the Shimura variety in the paper. Let $\widetilde{{\rm SC}}^\ast(Y_K)$ be the subring of the Chow ring generated by the weighted special cycles, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-017994 | MATHDB-335585

Existence conjecture for real morsifications of plane curve singularities

摘录 / EXCERPT: Existence conjecture. Every real plane curve singularity possesses a real morsification.

conjecture | open | MathDB 页面 | 原文线索

V2-017995 | MATHDB-335609

Local vanishing conjecture for rational singularities

摘录 / EXCERPT: Let Z be a complex variety of dimension $n\geq 2$ with rational singularities, and let $f\colon Y\to Z$ be a resolution of singularities whose reduced exceptional divisor E has...

conjecture | open | MathDB 页面 | 原文线索

V2-017996 | MATHDB-335626

Universal-function numerical conjecture for virtual refined invariants

摘录 / EXCERPT: Universal-function numerical conjecture. The coefficient of $x^{4n-\beta_1-3\beta_4}$ in the s^0-coefficient of the indicated sum of the two universal-function terms equals the c...

conjecture | open | MathDB 页面 | 原文线索

V2-017997 | MATHDB-335631

Ein-Lazarsfeld conjecture on Seshadri constants

摘录 / EXCERPT: Let Y be any projective manifold, and let L be any ample line bundle on Y. The Seshadri constant of L at a point $y\in Y$, denoted $\epsilon(L,y)$, measures the local posit...

conjecture | open | MathDB 页面 | 原文线索

V2-017998 | MATHDB-335633

Effective non-vanishing conjecture for regular pairs

摘录 / EXCERPT: Let $(d;a)=(d_1,\ldots,d_c;a_0,\ldots,a_n) \in \mathbb N^c \times \mathbb N^{n+1}$ be a regular pair such that $a_i \ne 1$ and $d_j \ne a_i$ for any i,j. Assume $c \le n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-017999 | MATHDB-335641

Diverio-Trapani conjecture on ampleness of invariant jet bundles

摘录 / EXCERPT: Let $Z\subset \mathbb{P}^n$ be the complete intersection of c-general hypersurfaces of sufficiently high degree. Let $E_{k,m}T_Z^*$ denote the invariant jet bundle of order k a...

conjecture | open | MathDB 页面 | 原文线索

V2-018000 | MATHDB-335649

Prym-Green conjecture for level paracanonical curves

摘录 / EXCERPT: Prym-Green conjecture. The resolution of a general level ℓ paracanonical curve of genus g is natural.

conjecture | open | MathDB 页面 | 原文线索

V2-018001 | MATHDB-335657

Finite determination conjecture for Mather-Jacobian minimal log discrepancies

摘录 / EXCERPT: Let X be a variety of dimension d, let x be a closed point of X, and define ... where $X_m(x)$ is the local m-jet scheme of X at x. Finite determination conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-018002 | MATHDB-335658

Finite determination conjecture for Mather-Jacobian minimal log discrepancies below a threshold

摘录 / EXCERPT: Let X be a d-dimensional variety, let $x\in X$ be a closed point, and let $s_m(X,x)=(m+1)d-\dim X_m(x)$, where $X_m(x)$ is the local m-jet scheme at x. Let δ be an i...

conjecture | open | MathDB 页面 | 原文线索

V2-018003 | MATHDB-335663

Kirillov-Maeno's Peterson isomorphism conjecture for quantum K-theory of the flag variety

摘录 / EXCERPT: Kirillov-Maeno conjecture. There is a canonical ring isomorphism

conjecture | solved | MathDB 页面 | 原文线索

V2-018004 | MATHDB-335680

The root-stack determination conjecture for Gromov-Witten theory

摘录 / EXCERPT: Let $\mathcal{X}$ be a Deligne-Mumford stack, let $\mathcal{D}\subset\mathcal{X}$ be a smooth irreducible divisor, and let $\mathcal{X}_r$ be the stack of r-th roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-018005 | MATHDB-335686

Schneider's conjecture on étale cohomology

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields, let S be the relevant finite set of places of K, let p be an odd prime, and let $r\ne0$ be an integer. Schneider's co...

conjecture | open | MathDB 页面 | 原文线索

V2-018006 | MATHDB-335687

Leading term conjecture at $s=r$

摘录 / EXCERPT: Let L/K be a finite Galois extension of number fields with Galois group G, let $r>1$ be an integer, and let p be an odd prime. Let ϕ_r, $A_{\phi_r}^S$, ψ_r,...

conjecture | open | MathDB 页面 | 原文线索

V2-018007 | MATHDB-335692

Goto-Watanabe finite generation conjecture for symbolic Rees rings of space monomial curves

摘录 / EXCERPT: Let $R=k[x,y,z]$ be the polynomial ring in three variables over a field k, and let P be the prime ideal defining the space monomial curve (t^a,t^b,t^c) in $\mathbb A^3_k$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018008 | MATHDB-335713

Davison-Meinhardt's conjecture on motivic nearby fibers of weighted homogeneous functions

摘录 / EXCERPT: Let Y be a smooth k-variety with the trivial ${\mathbb G}_{m,k}$-action, and let ${\mathbb G}_{m,k}$ act on ${\mathbb A}^n_k$ with weights $w_1,\ldots,w_n>0$. Let ... be a...

conjecture | open | MathDB 页面 | 原文线索

V2-018009 | MATHDB-335716

Unitary Shimura-subvariety disjointness conjecture for the Torelli locus

摘录 / EXCERPT: Let $M\subset\mathcal{A}_g$ be a Shimura subvariety of the unitary type specified earlier in the source. Unitary disjointness conjecture. The intersection ... is empty. The conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-018010 | MATHDB-335737

The Fourier-Mukai support-dimension invariance conjecture

摘录 / EXCERPT: Fourier-Mukai support-dimension invariance conjecture. If smooth projective varieties X and Y are Fourier-Mukai partners, then

conjecture | open | MathDB 页面 | 原文线索

V2-018011 | MATHDB-335742

Ash-Pollack-Stevens conjecture on non-rigid finite-slope eigensystems for GL_3

摘录 / EXCERPT: Let θ be a finite-slope cuspidal Hecke eigensystem of GL_3. An eigensystem is arithmetically rigid if it does not lie in any irreducible component of an eig...

conjecture | open | MathDB 页面 | 原文线索

V2-018012 | MATHDB-335762

Decomposition conjecture for varieties with stable tangent sheaf and trivial canonical class

摘录 / EXCERPT: Decomposition conjecture. There exists a quasi-étale cover $Y \to X$ such that either Y is an abelian variety, or Y splits as a product of copies of a single Calabi-Yau variety...

conjecture | open | MathDB 页面 | 原文线索

V2-018013 | MATHDB-335768

Hilbert polynomial conjecture for top Kähler differentials of fat point schemes

摘录 / EXCERPT: Let K be the ground field, let $\mathbb W=m_1P_1+\cdots+m_sP_s$ be a fat point scheme in $\mathbb P^n$, and let ... Write $R_{\mathbb W}$ for the homogeneous coordinate ring of...

conjecture | open | MathDB 页面 | 原文线索

V2-018014 | MATHDB-335798

Abuaf's dimension conjecture for higher-dimensional tangency loci

摘录 / EXCERPT: Abuaf's dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018015 | MATHDB-335799

The Euler-characteristic lower-bound conjecture for reflexive toric Calabi-Yau cones

摘录 / EXCERPT: Let ${\cal X}$ be a toric Calabi-Yau n-fold with toric diagram Δ_{n-1}, let Y be its Sasaki-Einstein base, and let $V(b_i^{*};Y)$ denote the minimum volume. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018016 | MATHDB-335800

The dP3-fibration maximum conjecture for reflexive toric Calabi-Yau cones

摘录 / EXCERPT: Let ${\cal X}$ be a reflexive toric Calabi-Yau n-fold with toric diagram Δ_{n-1}, let Y be its Sasaki-Einstein base, let $V(b_i^{*};Y)$ be the minimum volume, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018017 | MATHDB-335802

The geometric Arveson-Douglas conjecture for homogeneous varieties

摘录 / EXCERPT: Geometric Arveson-Douglas conjecture. The quotient module Q is p-essentially normal for every $p>\dim M$.

conjecture | open | MathDB 页面 | 原文线索

V2-018018 | MATHDB-335808

Modified Bloch-Beilinson prediction via the niveau filtration

摘录 / EXCERPT: Let X be an abelian variety of dimension d over $\mathbf{C}$, let $\widetilde{N}^1H^d(X)$ be the first niveau-filtration step, and let $G^dCH_0(X)$ be the deepest piece of the...

conjecture | open | MathDB 页面 | 原文线索

V2-018019 | MATHDB-335812

The naive equivariant ring-spectrum conjecture for motivic modular forms

摘录 / EXCERPT: Let C_2 be the cyclic group of order two, and let tmf_{C_2} be the homotopy pullback spectrum defined by the Tate diagram described above. A C_2-ring spectrum means...

conjecture | open | MathDB 页面 | 原文线索

V2-018020 | MATHDB-335818

Esnault-Kerz conjecture on logarithmic ramification bounds

摘录 / EXCERPT: Esnault-Kerz conjecture. These conditions are equivalent to requiring that, for every birational morphism $f:X'\rightarrow X$ such that X' is smooth over...

conjecture | open | MathDB 页面 | 原文线索

V2-018021 | MATHDB-335819

Esnault-Kerz conjecture on curvewise computation of the Swan divisor

摘录 / EXCERPT: Esnault-Kerz conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018022 | MATHDB-335820

Beilinson's characteristic-cycle direct-image conjecture

摘录 / EXCERPT: Let k be a perfect field, and let $f\colon X\to Y$ be a morphism of smooth schemes over k. Assume that every irreducible component of X has dimension n and every irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-018023 | MATHDB-335861

Monodromy conjecture for tropical Dolbeault cohomology

摘录 / EXCERPT: Let K be an algebraically closed non-Archimedean field whose residue field $\widetilde{K}$ is algebraic over a finite field, and let X be a proper smooth scheme over K. For...

conjecture | refuted | MathDB 页面 | 原文线索

V2-018024 | MATHDB-335890

The rank conjecture for the period matrix of linear algebraic cycles

摘录 / EXCERPT: Let n be even, let $d_1,\cldots,d_{\frac{n}{2}+1}$ be the cardinalities of subsets $B_1,B_2,\ldots,B_{\frac{n}{2}+1}$ of the set of complex roots of $\zeta^d+1=0$, and let ${\sf p}_i$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-018025 | MATHDB-335897

The weight-generating-function conjecture for Poisson-de Rham homology

摘录 / EXCERPT: Let X be a conical Poisson variety with a symplectic resolution, and let d be the degree of its generic symplectic form. Let $\widetilde X$ be the resolution and, for each i,...

conjecture | open | MathDB 页面 | 原文线索

V2-018026 | MATHDB-335909

The Gromov-Yomdin type conjecture for categorical entropy

摘录 / EXCERPT: Let X be a smooth projective variety over $\a0\mathbb{C}$, and let $\a0\Phi \in \operatorname{Aut}(D^b(X))$ be an autoequivalence. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018027 | MATHDB-335969

Curve-counting conjecture for rationality of $X_{4,6}$

摘录 / EXCERPT: Curve-counting conjecture for $X_{4,6}$. For a generic point $p\in E^4$, one has $\#(p)=1$. Moreover, for a generic point $p\in D_{0,1,2,1}$, one has $\#(p)=1$. This conjecture imp...

conjecture | open | MathDB 页面 | 原文线索

V2-018028 | MATHDB-335970

Curve-counting conjecture for rationality of $X_{5,6}$

摘录 / EXCERPT: Curve-counting conjecture for rationality of $X_{5,6}$. For a generic point $p\in D_{1,0,\zeta,0,-1}$, one has $\#(p)=1$. Together with rationality of $X_{4,6}$, the source states...

conjecture | open | MathDB 页面 | 原文线索

V2-018029 | MATHDB-335974

Gross's modularity conjecture for abelian varieties over the rationals

摘录 / EXCERPT: Gross-Langlands conjecture. There exists a globally generic cuspidal automorphic representation Π on $\operatorname{GSpin}_{2g+1}({\mathbf A}_{\mathbf Q})$ of weight 2 and pa...

conjecture | open | MathDB 页面 | 原文线索

V2-018030 | MATHDB-335975

Gross's modularity conjecture for abelian varieties over number fields

摘录 / EXCERPT: Gross-Langlands conjecture. There exists a globally generic cuspidal automorphic representation Π on $\operatorname{GSpin}_{2g+1}({\mathbf A}_E)$ of weight 2 and paramodular...

conjecture | open | MathDB 页面 | 原文线索

V2-018031 | MATHDB-335995

Conjecture on Fano manifolds with a specified family of lines

摘录 / EXCERPT: Let $X \subset \mathbb{P}^N$ be a Fano manifold of Picard number 1 which is covered by lines. For a general point $x \in X$, suppose that a family of lines on X through x, em...

conjecture | open | MathDB 页面 | 原文线索

V2-018032 | MATHDB-336013

Agreement of naive and Deligne Hodge filtrations for character stacks

摘录 / EXCERPT: Let G be a complex reductive group, and let $\mathcal{Loc}_G(S)$ denote the character stack of a surface S. Its Borel-Moore homology carries both the naive Hodge filtration ari...

conjecture | open | MathDB 页面 | 原文线索

V2-018033 | MATHDB-336014

Grunewald-O'Halloran conjecture for nilpotent Leibniz algebras

摘录 / EXCERPT: Let A be an n-dimensional nilpotent Leibniz algebra with $n\ge 2$. A degeneration $B\to A$ means that the n-dimensional non-nilpotent Leibniz algebra B degenerates to A....

conjecture | open | MathDB 页面 | 原文线索

V2-018034 | MATHDB-336015

Vergne-Grunewald-O'Halloran conjecture for Leibniz varieties

摘录 / EXCERPT: Let $n\ge 2$, and consider an irreducible component of the variety of n-dimensional Leibniz algebras. Vergne-Grunewald-O'Halloran conjecture. Every such irreducible component con...

conjecture | open | MathDB 页面 | 原文线索

V2-018035 | MATHDB-336021

Conjecture on Hodge monodromy and E-polynomial differences for SL and PGL character varieties

摘录 / EXCERPT: Let $G=SL(n,\mathbb{C})$ or $G=PGL(n,\mathbb{C})$, and let $\overline{X}_{C}(G)$ be the representation space corresponding to the moduli space $\mathcal{M}_{C}(G)$. For a regular s...

conjecture | open | MathDB 页面 | 原文线索

V2-018036 | MATHDB-336044

Resolution conjecture for the singular loci and osculating projections of Grassmannians

摘录 / EXCERPT: Let R_i be the relevant subvariety of $\mathbb G(r,n)$, and for $1\leq i\leq r$ let α_j be the successive blow-ups along the smooth centers introduced in the construction....

conjecture | open | MathDB 页面 | 原文线索

V2-018037 | MATHDB-336117

The positive Yangian-cohomological Hall algebra isomorphism conjecture

摘录 / EXCERPT: Let Q be a quiver, let $\mathbb{Y}_+$ be the positive half of the Maulik-Okounkov Yangian, let K be the scalar extension used in the paper, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-018038 | MATHDB-336194

Huneke-Harbourne containment conjecture for fat point ideals

摘录 / EXCERPT: Let $Z\subseteq\mathbb P^N$ be a fat point subscheme, let $R=K[x_0,\ldots,x_N]$ be the homogeneous coordinate ring, and let $M=(x_0,\ldots,x_N)$. Huneke-Harbourne containment conje...

conjecture | open | MathDB 页面 | 原文线索

V2-018039 | MATHDB-336203

The maximal-rank conjecture for space curves in the range $g\leq Kd^{3/2}$

摘录 / EXCERPT: Maximal-rank conjecture. There exists a constant K such that, for all natural numbers d,g satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-018040 | MATHDB-336204

The general Frobenius decomposition conjecture for flag varieties

摘录 / EXCERPT: Let $n\in\mathbb N$. Define subcategories of $\mathop{\mathrm D\kern0pt}\nolimits^b(X_n)$ by ... with $\mathcal A_0=\langle\mathcal O_{X_n}\rangle$, and for $k<0$ let $\mathcal A_k$...

conjecture | open | MathDB 页面 | 原文线索

V2-018041 | MATHDB-336232

F-regularity conjecture for the irreducible polynomial of a generic matrix

摘录 / EXCERPT: F-regularity conjecture for $\mathcal{P}(X)$. The quotient ring

conjecture | open | MathDB 页面 | 原文线索

V2-018042 | MATHDB-336255

The real \SU(2) character curve conjecture for canonical components

摘录 / EXCERPT: Real character curve conjecture. There is a real curve of $\operatorname{\mathrm{SU}}(2)$ characters on the canonical component of the $\operatorname{\mathrm{SL}}_2(\mathbb{C})$ ch...

conjecture | open | MathDB 页面 | 原文线索

V2-018043 | MATHDB-336280

Non-separation and Steiner systems in Johnson schemes

摘录 / EXCERPT: Non-separation conjecture. There is a function G such that, if $n\geqslant G(k)$ and the Johnson scheme $J(n,k)$ is non-separating, then there exists a Steiner system $S(t,k,n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-018044 | MATHDB-336281

Divisibility characterization of non-separation in Johnson schemes

摘录 / EXCERPT: Divisibility conjecture. There is a function H such that, if $n\geqslant H(k)$, then the Johnson scheme is non-separating if and only if the divisibility conditions for...

conjecture | open | MathDB 页面 | 原文线索

V2-018045 | MATHDB-336282

Synchronization and large sets of Steiner systems in Johnson schemes

摘录 / EXCERPT: Large-set conjecture. There is a function L such that, if $n\geqslant L(k)$, then the Johnson scheme $J(n,k)$ is non-synchronizing if and only if there exists a large set of Stei...

conjecture | open | MathDB 页面 | 原文线索

V2-018046 | MATHDB-336283

Maximum cocliques in the projective-plane Johnson graph

摘录 / EXCERPT: Projective-plane coclique conjecture. For $q>2$, a coclique of maximum size in the graph $\Gamma_q(q^2+q+1,q+1)$ must consist of all the $(q+1)$-sets containing two given points; s...

conjecture | open | MathDB 页面 | 原文线索

V2-018047 | MATHDB-336302

The base-locus characterization of Jacobians via the Gauss map

摘录 / EXCERPT: Let (A,Θ) be a principally polarized abelian variety, and for a fixed $p\in\mathbb P^{g-1}$ let $Z_p\subset |2\Theta|$ be the linear system generated by the divisors ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-018048 | MATHDB-336304

The generalized Ginzburg-Kazhdan conjecture for Coulomb branches

摘录 / EXCERPT: Generalized Ginzburg-Kazhdan conjecture. The variety $W^{g,b}$ is isomorphic to the Coulomb branch of the gauge theory associated with the quiver for a Riemann surface with b leg...

conjecture | open | MathDB 页面 | 原文线索

V2-018049 | MATHDB-336311

Maximal Rank Conjecture for the restriction maps

摘录 / EXCERPT: Let the restriction maps be those in Theorem, and let $d\geqslant 1$. The paper reports that no counterexamples were found in the tested range $d<d_0(m)$. Maximal Rank Conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-018050 | MATHDB-336317

Nondegeneracy and compatibility of coisotropic intersections

摘录 / EXCERPT: Let X be a derived Artin stack, and let $L_1\longrightarrow X$ and $L_2\longrightarrow X$ be morphisms equipped with n-shifted coisotropic structures. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018051 | MATHDB-336328

The refined square-zero upper-triangular matrix conjecture

摘录 / EXCERPT: Let k be an algebraically closed field, let r be a positive integer, let $N=2n$ be an even positive integer, and let $d=(d_1,\ldots,d_N)$ be an N-tuple of nonincreasing integ...

conjecture | open | MathDB 页面 | 原文线索

V2-018052 | MATHDB-336352

Generic rank equality for odd-order Hankel tensors

摘录 / EXCERPT: Let $m\geq 5$ be odd, and let $\mathcal{H}\in\operatorname{H}^m(\mathbb{C}^n)$ be a generic Hankel tensor. The generic Hankel rank conjecture. The answer to whether ... is yes. The e...

conjecture | open | MathDB 页面 | 原文线索

V2-018053 | MATHDB-336410

Debarre-Manivel's conjecture for Fano varieties of lines on cubics

摘录 / EXCERPT: Debarre-Manivel's conjecture. The inclusion $F\hookrightarrow G$ induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-018054 | MATHDB-336424

The des-Wilf equivalence for two sets of West-2-stack-sortable patterns

摘录 / EXCERPT: For a permutation pattern τ, let $F_n^{\operatorname{des}}(\tau;q)$ denote the descent generating polynomial over permutations in $\operatorname{Av}_n(\tau)$, and for sets of...

conjecture | open | MathDB 页面 | 原文线索

V2-018055 | MATHDB-336426

Affineness conjecture for moduli spaces of instanton bundles on the flag variety

摘录 / EXCERPT: Let $F=F(0,1,2)$ be the flag variety, and let $MI_F(k)$ denote the moduli space of instanton bundles on F with charge k. Affineness conjecture. $MI_F(k)$ is affine for any k....

conjecture | open | MathDB 页面 | 原文线索

V2-018056 | MATHDB-336430

Reid's geography conjecture for surfaces of general type

摘录 / EXCERPT: Let $g=2,3,\ldots$. A smooth surface X of general type has canonical self-intersection K_X^2 and holomorphic Euler characteristic $\chi(X,\mathcal{O}_X)$. A pencil of curves of...

conjecture | open | MathDB 页面 | 原文线索

V2-018057 | MATHDB-336437

Grothendieck's standard conjectures

摘录 / EXCERPT: Grothendieck's standard conjectures. The following assertions hold: (1) there exist correspondences $\Lambda_j\in CH^j(Y\times Y)$ such that $\Lambda_{j*}:H^{2d_Y-j}(Y)\to H^j(Y)$...

conjecture | open | MathDB 页面 | 原文线索

V2-018058 | MATHDB-336503

Deformation conjecture for dessins d'enfants

摘录 / EXCERPT: A dessin d'enfant is a combinatorial object associated with a trigonal curve; it is maximal when it has one singular fiber in each region and is connected. Deformation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-018059 | MATHDB-336504

Relation between Bottacin's Poisson structure and shifted Poisson structures on anticanonical divisors

摘录 / EXCERPT: Let S be a smooth projective Poisson surface and let $C\subset S$ be a smooth anticanonical divisor. Consider the moduli stack of semistable torsion-free sheaves on S and the m...

conjecture | open | MathDB 页面 | 原文线索

V2-018060 | MATHDB-336507

Elliptic-fibration quasimodularity conjecture

摘录 / EXCERPT: Let X and B be nonsingular projective varieties, let $\pi:X\to B$ be an elliptic fibration with integral fibers and a section $\sigma:B\to X$, and let N_{ι} be the normal...

conjecture | open | MathDB 页面 | 原文线索

V2-018061 | MATHDB-336523

Conjecture on canonical central leaves in Shimura varieties of abelian type

摘录 / EXCERPT: Canonical central-leaf conjecture. Canonical central leaves are non-empty, and every canonical central leaf is a union of connected components of the adjoint central leaf containin...

conjecture | open | MathDB 页面 | 原文线索

V2-018062 | MATHDB-336552

Fukaya-Morse-Zinger conjecture on deformation equivalence of NC symplectic smoothings

摘录 / EXCERPT: Let $(\mathcal{Z},\omega_{\mathcal{Z}},\pi)$ be a one-parameter family of smoothings of a compact NC symplectic variety $(X_{\emptyset},\omega_{\emptyset})$. Deformation-equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-018063 | MATHDB-336568

Lewis-Sheng dimension bound for Γ-equivalent points on complete intersections

摘录 / EXCERPT: Lewis-Sheng dimension conjecture. The locus of points Γ-equivalent to p satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-018064 | MATHDB-336588

Carvalho-Lopez-Neumann-López minimum-distance conjecture for projective nested cartesian codes

摘录 / EXCERPT: Let $K=\mathbb{F}_q$ be a finite field, let $A_1,\ldots,A_s$ be subsets of K defining a projective nested cartesian set ... and write $d_i=|A_i|$ for $i=1,\ldots,s$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018065 | MATHDB-336605

Walton's conjecture on simple representations of 3-dimensional Sklyanin algebras

摘录 / EXCERPT: Let A be a 3-dimensional Sklyanin algebra, and let τ be a torsion point of order n on its associated elliptic curve, with $(n,3)=3$. Consider the simple finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-018066 | MATHDB-336623

Conjecture on powers of characteristic-polynomial coefficients for twisted Carlitz modules

摘录 / EXCERPT: Powers-of-coefficients conjecture. For any i,j,m,n, there exists $\gamma\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018067 | MATHDB-336644

Ambient Lipschitz equivalence conjecture for normally embedded real surface germs

摘录 / EXCERPT: Let $(S_0,0)$ and $(S_1,0)$ be two normally embedded real semialgebraic surface germs. They are normally embedded when their inner and outer metrics are bi-Lipschitz equivalent; fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018068 | MATHDB-336663

Chen-Zhu's conjecture on irreducible components of affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be the reductive group in the local Shimura-variety setting, let mu be the relevant dominant cocharacter, and let $b\in B(G)$ be a $sigma$-conjugacy class. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018069 | MATHDB-336664

The Satake compatibility conjecture for the spectral action

摘录 / EXCERPT: Let $\boldsymbol{J}$ be the algebra acting on the compactly supported cohomology of the Shimura variety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018070 | MATHDB-336665

The Hecke compatibility conjecture for the spectral action on Shimura cohomology

摘录 / EXCERPT: Let $\boldsymbol{J}$ be the algebra of regular functions on the stack $[\hat G\sigma_p/\hat G]$, acting on $\operatorname{H}_c(\operatorname{Sh}_{\mu,\overline{k}_v})$, and identif...

conjecture | open | MathDB 页面 | 原文线索

V2-018071 | MATHDB-336668

The operational-to-primarily-strong Chow isomorphism conjecture

摘录 / EXCERPT: Let $\mathcal{X}$ be a smooth connected properly stable Artin stack with good moduli space $\pi\colon\mathcal{X}\to X$. Let $A^*_{\operatorname{op}}(X)_\mathbb Q$ be the operationa...

conjecture | open | MathDB 页面 | 原文线索

V2-018072 | MATHDB-336682

The Fano compactification conjecture for Euler-symmetric projective varieties

摘录 / EXCERPT: Fano compactification conjecture. X can be realized as an Euler-symmetric projective variety under a suitable projective embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-018073 | MATHDB-336683

The nonsingular VMRT conjecture for Fano compactifications of vector groups

摘录 / EXCERPT: Nonsingular VMRT conjecture. For some choice of ${\mathcal K}$, the VMRT ${\mathcal C}_x$ at a general point $x\in X$ is nonsingular.

conjecture | open | MathDB 页面 | 原文线索

V2-018074 | MATHDB-336689

Multiplicity estimate for singular loci of projective subschemes

摘录 / EXCERPT: Let K be a number field, and let $\delta\geqslant1$, $d\geqslant1$, and $s\geqslant0$ be integers. For a reduced pure-dimensional closed subscheme X of $\mathbb P^n_K$ of dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-018075 | MATHDB-336693

The covering gonality conjecture for very general hypersurfaces

摘录 / EXCERPT: Let $X\subset \mathbb{P}^{n+1}$ be a very general hypersurface of degree $d\geqslant 2n$. Here $\operatorname{cov.gon}(X)$ denotes the covering gonality of X, namely the least go...

conjecture | open | MathDB 页面 | 原文线索

V2-018076 | MATHDB-336694

The Seshadri criterion for the strict interior of the nef cone

摘录 / EXCERPT: Let X be a projective variety and let $\alpha\in\operatorname{Nef}^k(X)$. Suppose that there exists $\varepsilon>0$ such that ... for all $x\in X$. Seshadri criterion. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-018077 | MATHDB-336699

Relative Humphreys conjecture for antispherical cells

摘录 / EXCERPT: Let G, G_1, W, ${}^{\mathrm{f}}W^{\mathrm{f}}$, $\mathcal{C}_p$, $\iota_{\Bbbk}$, and the corresponding nilpotent orbit $\mathscr{O}^{\mathbb{C}}_{\mathbf{c}(w)}$ be as in th...

conjecture | open | MathDB 页面 | 原文线索

V2-018078 | MATHDB-336713

The cosmall-root characterization conjecture for simply laced flag varieties

摘录 / EXCERPT: Cosmall-root characterization conjecture. Assume that R is simply laced and let $\alpha\in R^+\setminus R^+_P$. Then α is P-cosmall if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-018079 | MATHDB-336714

André-Oort-type conjecture for integral subvarieties of the foliation on $S_{\sharp}$

摘录 / EXCERPT: Let $S_{\sharp}$ be the ambient moduli space, let $\mathcal{T}S^{+}$ be the foliation on it, and let S_{fol} denote the distinguished integral subvariety introduced in t...

conjecture | open | MathDB 页面 | 原文线索

V2-018080 | MATHDB-336715

Conjectural generating series for kernels of the matrices $C_k^{(j)}$

摘录 / EXCERPT: Kernel generating-series conjecture. The following identities are expected:

conjecture | open | MathDB 页面 | 原文线索

V2-018081 | MATHDB-336716

Conjecture that the $(r-1)$-st almost totally odd part spans the full space

摘录 / EXCERPT: For integers $k>r\ge3$, let $\mathcal H^{al}_{k,r}$ be the $\mathbb Q$-vector space defined from products of depth-graded motivic multiple zeta values with one even index, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018082 | MATHDB-336728

The forcing-algebra component intersection conjecture

摘录 / EXCERPT: Let $R=k[x_1,\ldots,x_r]$ be a polynomial ring in the homogeneous case, and suppose ... where the P_i are the minimal primes for $i=1,\ldots,s$. Let $V(P_i)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018083 | MATHDB-336731

Kuznetsov-Schinder's derived equivalence implies L-equivalence conjecture

摘录 / EXCERPT: Let X and Y be smooth projective simply connected varieties. They are L-equivalent if $[X]-[Y]$ vanishes in the localization...

conjecture | open | MathDB 页面 | 原文线索

V2-018084 | MATHDB-336746

Tameness conjecture for the Lagrangians in the twisted character variety

摘录 / EXCERPT: Let $\Sigma^{\#}$ be the punctured Heegaard surface and let $X_{\operatorname{tw}}(\Sigma^{\#})$ be its twisted character variety. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018085 | MATHDB-336755

Ng's conjecture relating abelian contact homology to the branched-cover character ring

摘录 / EXCERPT: Let K be a knot in bS^3, let $\mathbb{Sigmad}_2K$ be its 2-fold branched covering, and let $X(\mathbb{Sigmad}_2K)$ be the $bSL_2(bC)$ character variety of $bpi_1(\mathbb{Sigmad}_2K)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018086 | MATHDB-336765

Chow-Künneth projector conjecture for the $(2n-2)$-nd motive

摘录 / EXCERPT: Let X be an n-dimensional connected scheme smooth and projective over ${\bf C}$. Let $M_2(X)^*$ and $i_{2n-2}:M_2(X)^*\otimes {\bf L}^{n-2}\to M(X)$ be as obtained from the dec...

conjecture | open | MathDB 页面 | 原文线索

V2-018087 | MATHDB-336766

Jacobian-kernel conjecture for codimension-two cycles

摘录 / EXCERPT: Let X be an n-dimensional connected scheme smooth and projective over ${\bf C}$, with $n\geq 2$. Write $CH_{Jac,\mathbf Q}^2(X)$ for the kernel of the Abel-Jacobi map on homolo...

conjecture | open | MathDB 页面 | 原文线索

V2-018088 | MATHDB-336767

Motivic description conjecture for the slice filtration

摘录 / EXCERPT: Let X be an n-dimensional connected scheme smooth and projective over ${\bf C}$, and suppose that ... is a conjectural Chow-Künneth decomposition. Let ${\bf L}$ denote the Lefsch...

conjecture | open | MathDB 页面 | 原文线索

V2-018089 | MATHDB-336768

Homotopy classification conjecture for vector bundles on rigid analytic varieties

摘录 / EXCERPT: Let X be a smooth rigid analytic quasi-Stein variety over a complete, non-archimedean, non-trivially valued field k, and let n be a positive integer. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018090 | MATHDB-336776

The class-number formula conjecture for local factors of Weil polynomials

摘录 / EXCERPT: Class-number formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018091 | MATHDB-336815

Demailly's jet-curvature characterization conjecture for Kobayashi hyperbolicity

摘录 / EXCERPT: Demailly's conjecture. The variety X is Kobayashi hyperbolic if and only if there exists $k\in\mathbb{N}$ such that (X,T_X) has negative k-jet curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-018092 | MATHDB-336818

The integer-valued Iitaka dimension conjecture for numerical cycle classes

摘录 / EXCERPT: Let X be a projective variety over $\mathbb{C}$, let $N_k(X)_{\mathbb{Z}}$ denote the group of k-cycle classes up to numerical equivalence, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018093 | MATHDB-336819

Lehmann's conjecture on the Iitaka dimension of contracted cycles

摘录 / EXCERPT: Let X be a projective variety of dimension n, let $\pi:X\to Z$ be a surjective morphism of projective varieties of relative dimension e, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018094 | MATHDB-336828

Trace compatibility for obstruction classes of determinants

摘录 / EXCERPT: Let $A\to A_0$ be a square-zero extension with kernel I, and let $X\to\operatorname{Spec} A$ be a flat Artin stack such that ... Write $X_0=X\otimes_A A_0$. Suppose that P_0 is a...

conjecture | open | MathDB 页面 | 原文线索

V2-018095 | MATHDB-336846

Real-lift conjecture for tropical bitangent classes of quartics

摘录 / EXCERPT: Real-lift conjecture. A tropical bitangent class of a tropicalization of a generic smooth quartic defined over the real Puiseux series has either 0 or 4 real lifts.

conjecture | open | MathDB 页面 | 原文线索

V2-018096 | MATHDB-336850

Embedded resolution of singularities for quasi-excellent regular schemes

摘录 / EXCERPT: Let $X\subset Z$ be a reduced closed subscheme of a quasi-excellent regular Noetherian scheme Z. Write X' for the strict transform of X under a proper birational morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-018097 | MATHDB-336862

Uniqueness conjecture for minimal surfaces of maximal representations

摘录 / EXCERPT: Let S be a closed oriented surface of genus at least two, let $\Gamma=\pi_1(S)$, and let $\mathsf G$ be a real Lie group of Hermitian type. For a representation...

conjecture | open | MathDB 页面 | 原文线索

V2-018098 | MATHDB-336869

Kimura's finite-dimensionality conjecture for smooth projective varieties

摘录 / EXCERPT: Kimura's conjecture. All smooth projective varieties have finite-dimensional motive.

conjecture | open | MathDB 页面 | 原文线索

V2-018099 | MATHDB-336870

Nilpotence conjecture for homologically trivial correspondences of Chow motives

摘录 / EXCERPT: Nilpotence conjecture. If $\Gamma\in \operatorname{End}_{\mathcal M}_{k,R}}(M)$ is homologically trivial, then there exists an integer $n\gg0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018100 | MATHDB-336871

Torsion nilpotence conjecture for algebraic correspondences

摘录 / EXCERPT: Torsion nilpotence conjecture. Every cohomologically trivial n-torsion correspondence Γ is nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-018101 | MATHDB-336872

Universal-series correspondence for Hilbert schemes and strange duality

摘录 / EXCERPT: Universal-series correspondence. The series satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-018102 | MATHDB-336873

Finite Quot scheme and tautological Chern-class conjecture

摘录 / EXCERPT: Finite Quot tautological Chern-class conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018103 | MATHDB-336881

KLMW conjecture on equations for the affine Grassmannian of SL_n

摘录 / EXCERPT: KLMW conjecture. The square is Cartesian. Equivalently, the shuffle equations cut out $\mathcal{G} r_{SL_n}$ inside $\mathcal{S}\mathcal{G} r$. The paper proves this conjecture ove...

conjecture | open | MathDB 页面 | 原文线索

V2-018104 | MATHDB-336882

Nonreducedness conjecture for invariant subspaces of nilpotent operators

摘录 / EXCERPT: Nonreducedness conjecture. The scheme $\mathcal{G}^T$ is not reduced unless $T=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-018105 | MATHDB-336883

Finite-dimensional KLMW conjecture for nilpotent invariant-subspace schemes

摘录 / EXCERPT: Finite-dimensional KLMW conjecture. The diagram

conjecture | open | MathDB 页面 | 原文线索

V2-018106 | MATHDB-336895

The uniruledness conjecture via Kodaira dimension and RC-positivity

摘录 / EXCERPT: Let X be a projective manifold, let K_X be its canonical bundle, let K_X^{-1} be its anticanonical bundle, and let $\kappa(X)$ denote its Kodaira dimension. Uniruledness conj...

conjecture | open | MathDB 页面 | 原文线索

V2-018107 | MATHDB-336907

Artin vanishing conjecture for Zariski-constructible sheaves on affinoids

摘录 / EXCERPT: Let X be an affinoid rigid space over a complete algebraically closed nonarchimedean field C. Let n be prime to the residue characteristic of C, and let $\mathscr{G}$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-018108 | MATHDB-336908

Six-functor stability conjecture for Zariski-constructible complexes

摘录 / EXCERPT: Let X be a finite-dimensional rigid space over a complete algebraically closed field C. Set $\Lambda=\mathbf{Z}/n\mathbf{Z}$, where n is prime to the residue characteristic,...

conjecture | open | MathDB 页面 | 原文线索

V2-018109 | MATHDB-336917

Carlini-Catalisano-Geramita conjecture on generic disjoint unions of linear spaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a generic union of linear spaces whose components are pairwise non-intersecting. Recall that X has good postulation if, for every degree $d\geq 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-018110 | MATHDB-336923

Positivity conjecture for the K-theory stable-basis transition coefficients

摘录 / EXCERPT: Let G/B be the flag variety for a semisimple group of rank n, let W be its Weyl group, and let τ_w^+ be the coefficients in the expansion of the positive stable basis in...

conjecture | open | MathDB 页面 | 原文线索

V2-018111 | MATHDB-336925

Vanishing property conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over a field k of characteristic zero. For a positive integer r, the r-th vanishing property means that the relevant motivic Hom groups vanish un...

conjecture | open | MathDB 页面 | 原文线索

V2-018112 | MATHDB-336926

Generalized Beilinson-Soulé vanishing conjecture for an elliptic curve

摘录 / EXCERPT: Let E be an elliptic curve over a field k. In the non-CM case, consider the motivic Hom groups ... In the CM case, write $M_1(E)=M\oplus\bar{M}$ over $\mathbb{K}$ and consider th...

conjecture | open | MathDB 页面 | 原文线索

V2-018113 | MATHDB-336927

Marian-Oprea-Pandharipande conjecture relating higher-rank Segre series and Verlinde series

摘录 / EXCERPT: Marian-Oprea-Pandharipande conjecture. The universal series are

conjecture | open | MathDB 页面 | 原文线索

V2-018114 | MATHDB-336928

Closed formulas for the first nontrivial Verlinde series at r=±2

摘录 / EXCERPT: The $r=\pm2$ Verlinde-series conjecture. The first nontrivial series satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-018115 | MATHDB-336933

The conjecture on the two de Rham Chern-class morphisms

摘录 / EXCERPT: Let X be the smooth proper rigid analytic space considered above, let $ u yields A$ be the morphism to the abeloid variety associated with Pic^0_X, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018116 | MATHDB-336945

Kontsevich's conjecture on transformations of period integrals

摘录 / EXCERPT: A period is a number admitting an integral representation whose functions and domains of integration are algebraic with coefficients in $\bar{\mathbb{Q}}$. The relevant transformat...

conjecture | open | MathDB 页面 | 原文线索

V2-018117 | MATHDB-336961

The Yau-Tian-Donaldson conjecture for K-stable maps

摘录 / EXCERPT: The Yau-Tian-Donaldson conjecture for maps. The class $c_1(L)$ admits a $p^*\alpha$-twisted cscK metric if and only if the map $p:(X,L)\to(Y,T)$ is uniformly K-stable. This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-018118 | MATHDB-336982

The Verschiebung lower-bound conjecture for finite unipotent group schemes

摘录 / EXCERPT: Verschiebung lower-bound conjecture. The essential dimension of G over k satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-018119 | MATHDB-337007

The finite-determinant conjecture for the associated F-isocrystal

摘录 / EXCERPT: Finite-determinant conjecture. There exist elements $u\in\mathbb{Z}_{q'}^*$ such that $(\det V,\det\nabla,u\det\Phi)$ corresponds to a finite character of

conjecture | open | MathDB 页面 | 原文线索

V2-018120 | MATHDB-337009

Ruan's cohomological crepant resolution conjecture

摘录 / EXCERPT: Let M be a smooth projective variety endowed with a faithful action of a finite group G, and let $X:=M/G$ be its quotient. Assume that X is Gorenstein, and let $Y\to X$ be an...

conjecture | open | MathDB 页面 | 原文线索

V2-018121 | MATHDB-337010

Motivic hyper-Kähler resolution conjecture

摘录 / EXCERPT: Let M be a smooth projective holomorphic symplectic variety endowed with a faithful symplectic action of a finite group G. Let $X:=M/G$ be the quotient, and suppose that X ha...

conjecture | open | MathDB 页面 | 原文线索

V2-018122 | MATHDB-337041

Bondal's Frobenius generation conjecture for smooth proper toric varieties

摘录 / EXCERPT: Let X be a smooth proper toric variety. For each positive integer ℓ, the ℓth toric Frobenius morphism $F_{\ell}:X\to X$ is defined by extending...

conjecture | open | MathDB 页面 | 原文线索

V2-018123 | MATHDB-337073

Semi-ampleness conjecture for nef line bundles on Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a projective manifold with torsion canonical bundle, and let L be a nef line bundle on X. Semi-ampleness conjecture. The line bundle L is numerically equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-018124 | MATHDB-337075

Radicality conjecture for patches of semisimple Hessenberg varieties

摘录 / EXCERPT: Radicality conjecture. The ideal $I_{w,\mathcal{B}(S,H)}$ is radical for any semisimple Hessenberg variety $\mathcal{B}(S,H)$, and is therefore the patch ideal of...

conjecture | open | MathDB 页面 | 原文线索

V2-018125 | MATHDB-337078

The multiplicative self-dual Chow--Künneth conjecture for holomorphic symplectic varieties

摘录 / EXCERPT: Let X be a holomorphic symplectic variety, let $\operatorname{CH}(X)_{(0)}$ denote the degree-zero part of the Chow grading associated with a Chow--Künneth decomposition, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018126 | MATHDB-337079

The Section Property for holomorphic symplectic varieties

摘录 / EXCERPT: Let X be a smooth projective holomorphic symplectic variety. Write $\operatorname{CH}(X)$ for its Chow ring with rational coefficients and $\overline{\operatorname{CH}}(X)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-018127 | MATHDB-337080

The distinguished marking conjecture for holomorphic symplectic varieties

摘录 / EXCERPT: Let X be a smooth projective holomorphic symplectic variety. A marking of X is required to satisfy Condition $(\star)$, namely that the small diagonal is distinguished with res...

conjecture | open | MathDB 页面 | 原文线索

V2-018128 | MATHDB-337081

The Section Property for Calabi--Yau and Fano Fermat hypersurfaces

摘录 / EXCERPT: Let X be a Fermat hypersurface of degree d and dimension parameter r, in the notation of the paper. Let $(\star)$ denote the distinguished-marking condition defined there. Fe...

conjecture | open | MathDB 页面 | 原文线索

V2-018129 | MATHDB-337082

Modified Xiao conjecture for the relative irregularity

摘录 / EXCERPT: Let $f\colon S\to B$ be a fibred surface, meaning a fibration from a smooth projective surface S to a smooth projective curve B, with general fibre of genus $g\geq 2$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018130 | MATHDB-337083

Unitary-rank analogue of the modified Xiao conjecture

摘录 / EXCERPT: Let $f\colon S\to B$ be a fibred surface, meaning a fibration from a smooth projective surface S to a smooth projective curve B, with general fibre of genus $g\geq 2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018131 | MATHDB-337086

Finite-dimensionality conjecture for the module generated by closed surfaces

摘录 / EXCERPT: Let G be a complex algebraic group, and let $\mathrm{K}\operatorname{M}_{G}$ denote the ambient module of motivic data associated with G. Define ... the submodule generated by th...

conjecture | open | MathDB 页面 | 原文线索

V2-018132 | MATHDB-337091

Two-vertex conjecture for minimal geometrically irreducible algebras

摘录 / EXCERPT: Let $A=KQ/{\mathcal I}$ be a finite-dimensional algebra presented by a quiver with relations. Call A minimal if ${\mathcal I}=0$, or if every generating set of relations satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-018133 | MATHDB-337113

The conjectured maximal-sink-set formula for Hessenberg cohomology coefficients

摘录 / EXCERPT: Let $h:[n]\to[n]$ be a Hessenberg function, and let Γ_h be its associated graph. Let $m=m(\Gamma_h)$ be the number of parts of the partition $\lambda\vdash n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018134 | MATHDB-337115

The isogeny conjecture for automorphic-period tori

摘录 / EXCERPT: Let R_π be the coefficient ring acting on the objects in the construction, let X and Y be the period lattices, and let A_Π be the abelian variety attached to Π. For...

conjecture | open | MathDB 页面 | 原文线索

V2-018135 | MATHDB-337128

The Kapranov potential conjecture for smooth symplectic schemes

摘录 / EXCERPT: Let (X,ω) be a smooth symplectic scheme. Kapranov's cyclic L_∞ structure on $\mathrm{T}_X[-1]$ has a potential in...

conjecture | open | MathDB 页面 | 原文线索

V2-018136 | MATHDB-337129

The nil-isomorphic symplectic realization conjecture

摘录 / EXCERPT: Let L be a shifted Poisson stack. A symplectic realization of L is an n-shifted Lagrangian morphism $L\to X$ lifting the given $(n-1)$-shifted Poisson structure along the for...

conjecture | open | MathDB 页面 | 原文线索

V2-018137 | MATHDB-337152

Seventeen-point conjecture for generic cubic-quartic relations

摘录 / EXCERPT: Let A be a connected graded algebra with two degree-one generators, and let Γ_5 denote its truncated point scheme of length 6. Suppose the defining ideal of A is gener...

conjecture | open | MathDB 页面 | 原文线索

V2-018138 | MATHDB-337156

The finite-generation conjecture for gauge modules

摘录 / EXCERPT: Let X be an affine variety with coordinate algebra A_X, let $A\mathcal{V}_X$ denote the corresponding algebra of functions and vector fields, and let a gauge module mean an...

conjecture | open | MathDB 页面 | 原文线索

V2-018139 | MATHDB-337158

Van Dam-Sotirov's smallest-eigenvalue conjecture for Hamming graphs

摘录 / EXCERPT: Van Dam-Sotirov's conjecture. If $j\ge d-\frac{d-1}{q}$, with j even when $q=2$, then the smallest eigenvalue of $H(d,q,j)$ is $K_j(1)$. The conjecture was proved in the paper: t...

conjecture | open | MathDB 页面 | 原文线索

V2-018140 | MATHDB-337159

Karloff's smallest-eigenvalue conjecture for distance graphs of Johnson graphs

摘录 / EXCERPT: Karloff's conjecture. If $n=2d$ and $j>d/2$, then the smallest eigenvalue of $J(n,d,j)$ is $E_j(1)$. The surrounding text says that this conjecture is settled in the paper, so the...

conjecture | open | MathDB 页面 | 原文线索

V2-018141 | MATHDB-337160

Unimodality conjecture for dual polar scheme eigenvalues

摘录 / EXCERPT: Unimodality conjecture for dual polar schemes. The sequence $|C_j(i)|$ is unimodal except when $(q,e)=(2,1)$ or $(q,e,j)=(2,2,d-4)$ with $8\le d\le12$. If its minimum is at i_0,...

conjecture | open | MathDB 页面 | 原文线索

V2-018142 | MATHDB-337161

Smallest-eigenvalue conjecture for bilinear forms graphs

摘录 / EXCERPT: Let $B_j(i)$ denote the eigenvalues of the distance-j graph of the bilinear forms association scheme, with parameters q,d,e and $1\le j\le d$. Smallest-eigenvalue conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018143 | MATHDB-337180

Fourier-Mukai equivalence of conjectural stability hearts on abelian varieties

摘录 / EXCERPT: Fourier-Mukai heart-equivalence conjecture. The Fourier-Mukai transform gives an equivalence of the conjecturally constructed stability-condition hearts:

conjecture | open | MathDB 页面 | 原文线索

V2-018144 | MATHDB-337183

Ressayre's flag-variety structure-constant conjecture for the Belkale-Kumar product

摘录 / EXCERPT: Ressayre's conjecture. For G/B and any $u,v,w\in W$, the structure constant $\widetilde c_{uv}^w$ is equal to 1 if

conjecture | open | MathDB 页面 | 原文线索

V2-018145 | MATHDB-337186

Ressayre's root-partition conjecture for complete flag varieties

摘录 / EXCERPT: Ressayre's root-partition conjecture. If $u,v,w\in W$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-018146 | MATHDB-337198

Coskun-Harris-Starr conjecture on rational curves on general Fano hypersurfaces

摘录 / EXCERPT: Let $X\subset \mathbb{P}^n$ be a general hypersurface of degree $d\leq n$ and dimension at least 3. Let $R_e(X)$ be the open locus in the Hilbert scheme of X parameterizing smo...

conjecture | open | MathDB 页面 | 原文线索

V2-018147 | MATHDB-337210

The folklore conjecture on first-order model checking in interpretations of bounded-expansion classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph class with bounded expansion, let I be a simple first-order graph interpretation scheme, and let φ be a first-order property to be tested. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-018148 | MATHDB-337218

Polishchuk-Vandenberghe conjecture on semiorthogonal decompositions of equivariant derived categories

摘录 / EXCERPT: Let X be a smooth variety over an algebraically closed field of characteristic zero, and let a finite group G act effectively on X. For $\lambda\in G/{\sim}$, let X_λ...

conjecture | open | MathDB 页面 | 原文线索

V2-018149 | MATHDB-337224

Claborn-Fossum conjecture on Chow groups of regular local rings

摘录 / EXCERPT: Let $(A,\mathfrak{m}_A,L)$ be a regular local ring, and let $\operatorname{CH}^p(A)$ denote its codimension-p Chow group. Claborn-Fossum Conjecture. ... The conjecture extends the...

conjecture | open | MathDB 页面 | 原文线索

V2-018150 | MATHDB-337226

Beilinson-Parshin conjecture on torsion in the K-theory of finite-field varieties

摘录 / EXCERPT: Let X be a smooth, projective variety over $\mathbb{F}_p$. Beilinson-Parshin Conjecture. ... This conjecture concerns torsion in the algebraic K-theory of smooth projective varie...

conjecture | open | MathDB 页面 | 原文线索

V2-018151 | MATHDB-337243

Zariski-density conjecture for intermediate components of SO(n,n+1)-character varieties

摘录 / EXCERPT: Let X be a closed Riemann surface of genus g, let $n\geq 3$, and let ${\mathcal X}_d({\mathsf {SO}}(n,n+1))$ be the component of the ${\mathsf {SO}}(n,n+1)$-character variety d...

conjecture | open | MathDB 页面 | 原文线索

V2-018152 | MATHDB-337255

Calabi-Yau property conjecture for the moduli stack of one-dimensional sheaves

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau threefold, let $\beta\in N_1(X)$, and let $\mathcal{M}_X(\beta)$ be the moduli stack of one-dimensional sheaves with class β. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018153 | MATHDB-337281

Subregular coset ideal parametrization conjecture

摘录 / EXCERPT: For $n\geq4$, let K_n be the ideal defining the subregular coset quotient and let $V(K_n)\subseteq\mathbb C^2$ be its affine variety in coordinates (c,λ). Subregular cose...

conjecture | open | MathDB 页面 | 原文线索

V2-018154 | MATHDB-337283

Property $B(M)^*$ for motivic Lefschetz projectors

摘录 / EXCERPT: Let M be a smooth projective variety, let $L\in\operatorname{Corr}^{1}(M,M)$ be the Lefschetz correspondence, and let λ be the inverse Lefschetz operator on cohomology. P...

conjecture | open | MathDB 页面 | 原文线索

V2-018155 | MATHDB-337295

Comparison conjecture for the Maulik-Okounkov and cuspidal quiver Lie algebras

摘录 / EXCERPT: Comparison conjecture. The $\mathbb{Z}$-graded Lie algebras $\mathbf{g}_Q$ and $\widetilde{\mathfrak{g}}_Q$ are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-018156 | MATHDB-337331

Galois-orbit lower bound for Hecke translates

摘录 / EXCERPT: Let $(\mathbf G,X)$ be a Shimura datum and let S be an associated Shimura variety component. Let $(\mathbf H,X_{\mathbf H})\subset(\mathbf G,X)$ be a sub-Shimura datum, and for e...

conjecture | open | MathDB 页面 | 原文线索

V2-018157 | MATHDB-337332

Polynomial field-of-definition bound for Hecke translates

摘录 / EXCERPT: In the Shimura-datum setup, let $S_{\mathbf H,\gamma}$ be the special subvariety associated with $\gamma\in\Omega\subset\mathbf G(\mathbb Q)_+$ and let $N(\gamma)$ be the complexit...

conjecture | open | MathDB 页面 | 原文线索

V2-018158 | MATHDB-337333

Finiteness conjecture for non-toric Fano combinatorial divisorial polytopes

摘录 / EXCERPT: Finiteness conjecture. In any fixed dimension d, there are only finitely many equivalence classes of non-toric Fano CDPs.

conjecture | open | MathDB 页面 | 原文线索

V2-018159 | MATHDB-337340

The volume formula for slices of extended infinitesimal Okounkov bodies

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, let $\pi:{\rm Bl}_{r}(X)\rightarrow X$ be the blow-up at r general points $x_{1},\dots,x_{r}$, and let D be a big...

conjecture | open | MathDB 页面 | 原文线索

V2-018160 | MATHDB-337349

Manin-Mumford conjecture for curves in their Jacobians

摘录 / EXCERPT: Let C be an algebraic curve of genus $g \geq 2$, defined over a number field and embedded in its Jacobian J. Manin-Mumford conjecture. The set of torsion points of J which li...

conjecture | open | MathDB 页面 | 原文线索

V2-018161 | MATHDB-337350

Coleman's conjecture on the ramification of torsion points of curves

摘录 / EXCERPT: Let C be a curve of genus $g \geq 2$ embedded in its Jacobian J, all defined over a number field K. Let J_{tors} be the torsion subgroup of J and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018162 | MATHDB-337379

Ample canonical height zero-set conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over $\overline{\mathbb Q}$, let f be an endomorphism of X with $\delta_f>1$, and let K be a number field. Write $Z_f(K)$ for the set o...

conjecture | open | MathDB 页面 | 原文线索

V2-018163 | MATHDB-337381

Generalized Lefschetz standard conjecture for the coniveau filtration

摘录 / EXCERPT: Generalized Lefschetz standard conjecture. For every such p,q,k, the homomorphism u^{q-p}_a is an isomorphism. The assertion is presented as a strengthened or generalized form...

conjecture | open | MathDB 页面 | 原文线索

V2-018164 | MATHDB-337392

Motivic rationality conjecture for general special cubic fourfolds

摘录 / EXCERPT: Let $\mathcal{C}_d$ be the special cubic-fourfold divisor whose discriminant d satisfies ... For a general cubic fourfold $X\in\mathcal{C}_d$, say that its motive is associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-018165 | MATHDB-337395

Broadhurst-Kreimer-Brown conjecture on the depth-graded motivic multiple zeta values

摘录 / EXCERPT: Broadhurst-Kreimer-Brown conjecture. The generating series satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-018166 | MATHDB-337396

The exact-sequence conjecture for totally odd motivic multiple zeta values

摘录 / EXCERPT: For each $r\geq 3$, let $gr_r^{\mathfrak{D}}\mathcal{A}^{odd}$ denote the depth-graded totally odd part modulo $\zeta^{\mathfrak{m}}(2)$, let $\mathbb{P}^{\vee}$ be the weight-dual...

conjecture | open | MathDB 页面 | 原文线索

V2-018167 | MATHDB-337397

The long exact-sequence conjecture for depth-graded motivic multiple zeta values

摘录 / EXCERPT: Let $\mathcal{A}=\mathcal{H}/(\zeta^{\mathfrak{m}}(2))$, with depth-graded pieces $gr_r^{\mathfrak{D}}\mathcal{A}$, and let $\mathbb{P}^{\vee}$ be the weight-dual of the restricted...

conjecture | open | MathDB 页面 | 原文线索

V2-018168 | MATHDB-337398

The vanishing conjecture for quasi-uneven motivic Lie elements

摘录 / EXCERPT: Let $\mathfrak{d}\mathfrak{g}_1$ be the depth-one part of the depth-graded motivic Lie algebra, let $\mathcal{U}\mathfrak{g}$ be its enveloping algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018169 | MATHDB-337399

The surjective conjecture for the quasi-uneven quotient

摘录 / EXCERPT: Let $\mathfrak{d}\mathfrak{g}_1$ be the depth-one part of the depth-graded motivic Lie algebra, let $\mathcal{U}\mathfrak{g}$ be its enveloping algebra, and let γ be the nat...

conjecture | open | MathDB 页面 | 原文线索

V2-018170 | MATHDB-337412

Motivic Peterson conjecture

摘录 / EXCERPT: Let M_n be the mod 2 motivic cohomology algebra of the classifying space of the elementary abelian 2-group of rank n, and let $QM_n^{d,*}$ denote its degree-$(d,*)$ quotien...

conjecture | open | MathDB 页面 | 原文线索

V2-018171 | MATHDB-337420

The Beilinson-Bloch regular-model conjecture

摘录 / EXCERPT: Beilinson-Bloch's regular-model conjecture. There exists a regular proper flat model ${\mathcal{X}}$ of $X over $\operatorname{Spec} O_E$.

conjecture | open | MathDB 页面 | 原文线索

V2-018172 | MATHDB-337430

The DK hypothesis for K-equivalent varieties

摘录 / EXCERPT: DK hypothesis. If $X\sim_K Y$, then

conjecture | open | MathDB 页面 | 原文线索

V2-018173 | MATHDB-337508

Marion's finiteness conjecture for rigid triangle generation

摘录 / EXCERPT: Let G be a simple algebraic group over an algebraically closed field of characteristic $p>0$. For a positive integer m, let $j_m(G)$ be the dimension of the subvariety of eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-018174 | MATHDB-337542

Toric pullback irreducibility conjecture for finite maps

摘录 / EXCERPT: Toric pullback irreducibility conjecture. There is a finite union $\mathcal{E}$ of proper subtori of $\mathbb{G}_{\rm m}^n$ and a finite set Λ of isogenies of...

conjecture | open | MathDB 页面 | 原文线索

V2-018175 | MATHDB-337555

Poonen's smoothness conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field k, let S_A be the set of primes of good reduction for A, and let $u\in(0,1)$. Here an integer is B^u-smooth if every prime...

conjecture | open | MathDB 页面 | 原文线索

V2-018176 | MATHDB-337559

Topological and algebraic perfection conjecture for the formal punctured neighborhood of a coherent derived category

摘录 / EXCERPT: Let X be a proper scheme over a perfect field $\mathrm k$. Write $\operatorname{Perf}_{\mathrm{top}}(\widehat{\mathfrak D}^b_{coh}(X)_{\infty})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018177 | MATHDB-337600

Chin-Lung Wang's integral Chow-motive conjecture for K-equivalent varieties

摘录 / EXCERPT: Chin-Lung Wang's conjecture. If X and Y are K-equivalent smooth projective complex varieties, then they have isomorphic integral Chow motives.

conjecture | open | MathDB 页面 | 原文线索

V2-018178 | MATHDB-337601

Motivic t-structure conjecture for rational geometric motives

摘录 / EXCERPT: Motivic t-structure conjecture. There is a motivic t-structure on $\textup{DM}_{\textup{gm}}(k;\mathbb{Q})$ whose heart has semisimple part $\textup{Num}(k;\mathbb{Q})$, every...

conjecture | open | MathDB 页面 | 原文线索

V2-018179 | MATHDB-337612

The local weak equivalence-hypercover conjecture

摘录 / EXCERPT: Let $(\mathcal{C},\tau)$ be a site, let $\mathbb{A}$ be an augmentation category, and let $f\colon\mathcal{F}\to\mathcal{F}'$ be a map in $\mathbb{A}\textbf{-Pr}(\mathcal{C})$. A l...

conjecture | open | MathDB 页面 | 原文线索

V2-018180 | MATHDB-337613

The augmentation-category derived adjunction conjecture

摘录 / EXCERPT: Let $\mathbb{A}$ be an augmentation category, and let $\mathcal{G}_n^{\mathrm{sm}}(\textbf{dAff})$ and $\mathbb{A}\text{-}\mathcal{G}_n^{\mathrm{sm}}(\textbf{dAff})$ denote the homotop...

conjecture | open | MathDB 页面 | 原文线索

V2-018181 | MATHDB-337648

Nori's conjecture on algebraic correspondences for the transcendental motive

摘录 / EXCERPT: Let T be the smooth projective threefold considered under Hypothesis, and let $t_3(T)$ denote its transcendental degree-three motive. Write $\mathcal{M}(\mathbb{L},t_3(T))$ for m...

conjecture | open | MathDB 页面 | 原文线索

V2-018182 | MATHDB-337652

Green-Lazarsfeld's secant conjecture for property N_p

摘录 / EXCERPT: Secant Conjecture. The pair (C,L) fails property (N_p) if and only if L is not $p+1$-very ample.

conjecture | open | MathDB 页面 | 原文线索

V2-018183 | MATHDB-337660

The versal-family pullback question for p-divisible groups

摘录 / EXCERPT: Versal-family pullback conjecture. Does there exist a variety Y and a height-2, dimension-1 p-divisible group $\mathscr{H}$ on Y, everywhere versally deformed, together w...

open_problem | open | MathDB 页面 | 原文线索

V2-018184 | MATHDB-337662

Hindry-Ratazzi's torsion-growth conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety isogenous to a product ... where the A_i are simple abelian varieties that are pairwise non-isogenous. Let $\gamma(A)$ be the optimal exponent governi...

conjecture | open | MathDB 页面 | 原文线索

V2-018185 | MATHDB-337693

Isogonality growth conjecture for arithmetic covers

摘录 / EXCERPT: Let k be a finitely generated field of characteristic $p>0$, let X be a smooth geometrically connected k-variety equipped with a GLP $[?\rho$ and associated groups and covers...

conjecture | open | MathDB 页面 | 原文线索

V2-018186 | MATHDB-337694

Beauville's injectivity conjecture for the divisor-generated Chow subalgebra

摘录 / EXCERPT: Beauville's conjecture. The cycle class map is injective on the subalgebra of $A^{*}(X)$ generated by divisors.

conjecture | open | MathDB 页面 | 原文线索

V2-018187 | MATHDB-337711

Mordell-Weil rank and rational-point conjecture for a genus-6 plane curve

摘录 / EXCERPT: Let C be the genus 6 affine plane curve defined by ... Its Jacobian is an abelian variety, and its Mordell-Weil rank refers to the rank of its rational points over $\mathbb{Q}$....

conjecture | open | MathDB 页面 | 原文线索

V2-018188 | MATHDB-337774

Main conjecture on atypical loci as finite unions of special subvarieties

摘录 / EXCERPT: Let S be an irreducible smooth quasi-projective variety endowed with a variation of mixed Hodge structures $\mathbb{V} \longrightarrow S$. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-018189 | MATHDB-337775

Classical André-Oort conjecture for connected mixed Shimura varieties

摘录 / EXCERPT: Let $Y\subset {\rm Sh}_K^0({\mathbf P},X)$ be a closed irreducible algebraic subvariety of a connected mixed Shimura variety. Classical André-Oort conjecture. If Y contains a Zar...

conjecture | open | MathDB 页面 | 原文线索

V2-018190 | MATHDB-337776

Ax-Schanuel conjecture for variations of mixed integral Hodge structures

摘录 / EXCERPT: Let $(S,\mathbb{V})$ be a $\mathbb{Z}$VMHS, let $U\subset\widetilde S\times S$ be an algebraic subvariety, and let W be an irreducible component of $U\cap\Delta$, where Δ...

conjecture | open | MathDB 页面 | 原文线索

V2-018191 | MATHDB-337777

Ax-Lindemann conjecture for variations of mixed integral Hodge structures

摘录 / EXCERPT: Let $(S,\mathbb{V})$ be a $\mathbb{Z}$VMHS and let $Y\subset\widetilde S$ be an algebraic subvariety. Ax-Lindemann for $\mathbb{Z}$VMHS. The Zariski closure ... is a bi-algebraic sub...

conjecture | open | MathDB 页面 | 原文线索

V2-018192 | MATHDB-337778

Rational tower conjecture for unirational varieties

摘录 / EXCERPT: Let X be a unirational variety over a field k. A rational tower is a sequence of dominant rational maps ... over k, with rational source X_n and geometrically rational irredu...

conjecture | open | MathDB 页面 | 原文线索

V2-018193 | MATHDB-337779

Finiteness conjecture for unramified cohomology of unirational varieties

摘录 / EXCERPT: Let X be a unirational variety over an algebraically closed field of characteristic zero, and let H^i_{nr} denote unramified cohomology. Unramified-cohomology finiteness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-018194 | MATHDB-337781

Rational quotient conjecture for finite-group actions

摘录 / EXCERPT: Let G be a finite group. Rational-quotient conjecture. There exists a k-rational G-variety X with a generically free G-action such that X/G is k-rational. This is a n...

conjecture | open | MathDB 页面 | 原文线索

V2-018195 | MATHDB-337798

The parity conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a global field $\mathcal{K}$. Write $\operatorname{rk} A/\mathcal{K}$ for the rank of its Mordell--Weil group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018196 | MATHDB-337799

The twisted parity conjecture for abelian varieties

摘录 / EXCERPT: Let $\mathcal{F}/\mathcal{K}$ be a finite Galois extension, let A be an abelian variety over the global field $\mathcal{K}$, and let $A(\mathcal{F})_{\mathbb{C}}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-018197 | MATHDB-337833

The motivic formula conjecture for the moduli stack of vector bundles

摘录 / EXCERPT: The motivic formula conjecture. Suppose that $C(k)\neq\emptyset$. Then in $\mathrm{DM}(k,R)$,

conjecture | open | MathDB 页面 | 原文线索

V2-018198 | MATHDB-337834

The transition-map conjecture for motivic Białynicki-Birula decompositions

摘录 / EXCERPT: Let C be a curve and let $D'\geq D$ be effective divisors on C. For the motivic Białynicki-Birula decompositions, write $C^{(\underline m)}$ for the relevant product of symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-018199 | MATHDB-337873

Irreducibility conjecture for associated varieties of quasi-lisse conical vertex algebras

摘录 / EXCERPT: Irreducibility conjecture. The associated variety of a quasi-lisse conical vertex algebra is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-018200 | MATHDB-337880

Algebraicity conjecture for the universal Segre series A_3 and A_4

摘录 / EXCERPT: Algebraicity conjecture. For all ranks, the functions A_3 and A_4 are algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-018201 | MATHDB-337885

The linking-class surjectivity conjecture for regular sections

摘录 / EXCERPT: Linking-class surjectivity conjecture. The homomorphism O^* is surjective. In particular, for any smooth projective G-variety X, the Division Phenomena holds for all but fini...

conjecture | open | MathDB 页面 | 原文线索

V2-018202 | MATHDB-337890

GIT quotient conjecture for semistable syzygy points of canonical curves

摘录 / EXCERPT: Let g and p be integers, and let $\operatorname{Hilb}(\mathbb{P}^{g-1})_{p,2}^{\operatorname{ss}}$ denote the locus in the Hilbert scheme of genus g canonically embedded curv...

conjecture | open | MathDB 页面 | 原文线索

V2-018203 | MATHDB-337899

Fujino's simple connectedness conjecture for projective quasi-log canonical pairs

摘录 / EXCERPT: Fujino's simple connectedness conjecture. X is simply connected.

conjecture | open | MathDB 页面 | 原文线索

V2-018204 | MATHDB-337913

Li-Hansen Hodge symmetry conjecture for proper smooth rigid spaces

摘录 / EXCERPT: Let C be the algebraically closed p-adic field, let X be a proper smooth rigid-analytic variety over C, and suppose that X admits a formal model $\mathfrak X$ whose speci...

conjecture | open | MathDB 页面 | 原文线索

V2-018205 | MATHDB-337915

Global Kottwitz cohomology conjecture

摘录 / EXCERPT: Let Kt_F be Kottwitz's F-linear tensor category for a local or global field F, and let there be natural functors ... For a variety X over...

conjecture | open | MathDB 页面 | 原文线索

V2-018206 | MATHDB-337919

Vishik's syzygies conjecture for algebraic cobordism

摘录 / EXCERPT: Let X be a smooth variety of dimension d. A free ${\mathbb{L}}$-resolution of $\Omega^*(X)$ is one whose j-th term has generators concentrated in codimensions between j and...

conjecture | open | MathDB 页面 | 原文线索

V2-018207 | MATHDB-337921

Maulik-Pandharipande's conjecture on Gromov-Witten invariants of blow-ups

摘录 / EXCERPT: Let X be a smooth variety, let $Z\subset X$ be a complete intersection of l divisors $W_1,\ldots,W_l$ of X, and let $\widetilde X$ be the blow-up of X along Z. Maulik-Pan...

conjecture | open | MathDB 页面 | 原文线索

V2-018208 | MATHDB-337924

David's conjecture on lower bounds for Néron-Tate heights over torsion fields

摘录 / EXCERPT: Let A be an abelian variety defined over a number field K, equipped with a Néron-Tate height coming from a symmetric and ample line bundle. Write A_{tor} for the tor...

conjecture | open | MathDB 页面 | 原文线索

V2-018209 | MATHDB-337944

The fourfold divisor Hodge-number correction conjecture

摘录 / EXCERPT: Let V be a simplicial toric variety of dimension $n+1$, let X be a smooth Calabi-Yau n-fold hypersurface in V, and set $N_n=h^{1,1}(X)+n+1$. Let ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-018210 | MATHDB-337950

The cohomological spectrum conjecture for infinitesimal unipotent supergroup schemes

摘录 / EXCERPT: Let k be algebraically closed, and let G be an infinitesimal unipotent k-supergroup scheme of height $\leq r$. Let $H(G,k)$ denote the cohomology algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018211 | MATHDB-337951

The support-variety detection conjecture for height-1 infinitesimal unipotent supergroups

摘录 / EXCERPT: Suppose k is algebraically closed. Let G be a height-1 infinitesimal unipotent k-supergroup scheme, let M be a finite-dimensional rational G-supermodule, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018212 | MATHDB-337966

Bloch's conjecture for hyperkähler varieties with non-symplectic automorphisms

摘录 / EXCERPT: Bloch's conjecture. The sum of the actions of the powers of σ annihilates homologically trivial zero-cycles. This prediction follows in the source from the correspondence fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018213 | MATHDB-337967

Optimistic conjecture for quotient EPW sextics

摘录 / EXCERPT: Let $\mathcal C$ be the class of varieties whose Chow rings admit the multiplicative bigrading described in Beauville's weak splitting problem. Let X be an EPW sextic, meaning a...

conjecture | open | MathDB 页面 | 原文线索

V2-018214 | MATHDB-337969

Cycle-class injectivity for powers of quotient EPW sextics

摘录 / EXCERPT: For a projective quotient variety X, let $E^\ast(X^r)\subset A^\ast(X^r)$ be the $\mathbb{Q}$-subalgebra generated by pullbacks of elements of $A^1(X)$ and $A^2(X)$, pullbacks of...

conjecture | open | MathDB 页面 | 原文线索

V2-018215 | MATHDB-338001

Strong reciprocity conjecture for the infinitesimal Chow dilogarithm

摘录 / EXCERPT: Strong reciprocity conjecture. The residue map $res_{|C|}:\Gamma(k(C),3)\to\Gamma(k,2)$ is homotopic to zero. More precisely, there exists a map h fitting into a commutative diag...

conjecture | open | MathDB 页面 | 原文线索

V2-018216 | MATHDB-338037

The vanishing conjecture for the connecting map in local duality

摘录 / EXCERPT: Vanishing conjecture. If $i\notin\{n+1,\ldots,n+d\}$, or if X has good reduction and $i\neq 2n$, then $\delta=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-018217 | MATHDB-338045

Cylinder conjecture for low-degree del Pezzo surfaces

摘录 / EXCERPT: Let S be a del Pezzo surface of anticanonical degree at most 2 with at worst DuVal singularities. The affine Fano variety $\mathbb{P}\setminus S$ is the complement of S in th...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-018218 | MATHDB-338052

The logarithmic Chow-Künneth decomposition conjecture

摘录 / EXCERPT: Logarithmic Chow-Künneth conjecture. The motive $h(X)$ should admit a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-018219 | MATHDB-338053

The numerical triviality conjecture for log motives

摘录 / EXCERPT: Numerical triviality conjecture. In $\mathrm{LM}(S)$, $f\sim g$ implies $f=g$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-018220 | MATHDB-338081

McKernan's conjecture on singularities of Mori fibre space bases

摘录 / EXCERPT: McKernan's conjecture. There is $\delta\in\mathbb R^{>0}$ such that if $f\colon X\to Z$ is a Mori fibre space, where X is an ϵ-lc $\mathbb Q$-factorial variety of dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-018221 | MATHDB-338116

The formal expansion kernel conjecture for logarithmic crystalline cohomology

摘录 / EXCERPT: Let X be a smooth toric variety of dimension n, let Y be a smooth hypersurface, and consider logarithmic crystalline cohomology $H^n_{\operatorname{cris}}(X_0,Y_0)$ together...

conjecture | open | MathDB 页面 | 原文线索

V2-018222 | MATHDB-338117

The period-integral description of tautological-system solutions for complete intersections

摘录 / EXCERPT: Let $X=G/P$ be a homogeneous variety, let $E=L_1\oplus\cdots\oplus L_r$ be a direct sum of homogeneous line bundles, and let Y_i be the zero locus of the L_i-component of a sec...

conjecture | open | MathDB 页面 | 原文线索

V2-018223 | MATHDB-338125

Ottaviani's conjecture on the generic k-rank of forms

摘录 / EXCERPT: Let S be the polynomial ring in n variables over the complex numbers. For positive integers k,d,n, let $\operatorname{rk}_k^\circ(kd,n)$ denote the k-rank of a general form...

conjecture | open | MathDB 页面 | 原文线索

V2-018224 | MATHDB-338130

The maximal zero-set conjecture for sums of squares

摘录 / EXCERPT: For even n and even degree m, let $P_{n,m}$ be the cone of non-negative real homogeneous forms in n variables of degree m, and let $\Sigma_{n,m}$ be the subset consisting o...

conjecture | open | MathDB 页面 | 原文线索

V2-018225 | MATHDB-338148

The non-degenerated conjecture for the depth-graded motivic Lie algebra

摘录 / EXCERPT: Let $\mathfrak{dg}_1$ be the depth-one component of the depth-graded motivic Lie algebra, let $\mathbb{P}$ be the space of restricted even period polynomials, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018226 | MATHDB-338155

Saito-Sato conjecture on Kato homology over a henselian discrete valuation ring

摘录 / EXCERPT: Let R be an excellent henselian discrete valuation ring with residue field k, and let $B=\operatorname{Spec}R$. Let $\mathcal Y$ be a regular scheme, flat and proper over B,...

conjecture | open | MathDB 页面 | 原文线索

V2-018227 | MATHDB-338156

Beilinson's Albanese isomorphism conjecture for zero-cycles

摘录 / EXCERPT: Let K be a global field with separable closure K^s, and let X^s be a smooth projective variety over K^s. Let $\operatorname{CH}_0(X^s)^0$ denote the degree-zero Chow group...

conjecture | open | MathDB 页面 | 原文线索

V2-018228 | MATHDB-338172

The Bell-Ghioca-Tucker conjecture for intersections of two étale orbits

摘录 / EXCERPT: Bell-Ghioca-Tucker conjecture. The set $S_{f,g}(P,Q)$ is a union of finitely many sets of the form

conjecture | open | MathDB 页面 | 原文线索

V2-018229 | MATHDB-338175

Aizenbud-Avni convolution powers conjecture for algebraic groups

摘录 / EXCERPT: Let K be a field of characteristic zero. Let X be a smooth, absolutely irreducible K-variety, let G be a K-algebraic group, and let $\varphi:X\to G$ be a morphism such th...

conjecture | open | MathDB 页面 | 原文线索

V2-018230 | MATHDB-338185

Brunella's algebraic separatrix conjecture for foliations on projective spaces

摘录 / EXCERPT: Let $d4d5$ be a codimension one foliation on $d4d9^n$, with $n d45e 3$. An invariant algebraic hypersurface is an algebraic hypersurface preserved by $d4d5$; $d4d5$ is everywhere t...

conjecture | open | MathDB 页面 | 原文线索

V2-018231 | MATHDB-338216

Central ascent enumeration conjecture for two-pop-stack-sortable permutations

摘录 / EXCERPT: Let $a(n,k)$ denote the number of two-pop-stack-sortable permutations of length n with k ascents. Central ascent enumeration conjecture. The sequence $\{a(2n+1,n)\}$ is enumera...

conjecture | open | MathDB 页面 | 原文线索

V2-018232 | MATHDB-338223

Closed-form intersection conjecture for moving-line Mallavibarrena-Sols loci

摘录 / EXCERPT: Let H be the locus under consideration, and let $(0,0,\mathbf{m})$ be a general Mallavibarrena-Sols locus along a moving line, with $\mathbf{m}_i$ points on that line. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018233 | MATHDB-338229

Landman's duality defect conjecture

摘录 / EXCERPT: Landman's duality defect conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-018234 | MATHDB-338235

The eventual classification conjecture for Boolean degree 1 functions on Grassmann graphs

摘录 / EXCERPT: Eventual Grassmann graph classification conjecture. Let q be a prime power. Then there exists a constant n_q such that a Boolean degree 1 function on $J_q(n,k)$ is trivial fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018235 | MATHDB-338236

The eventual classification conjecture for Boolean degree 1 functions on polar graphs

摘录 / EXCERPT: Polar graph classification conjecture. Let $k\geq 2$. Then there exists $n_{q,k}$ such that every Boolean degree 1 function f on $C_q(n,k,e)$ with $n\geq n_{q,k}$ is trivial, t...

conjecture | open | MathDB 页面 | 原文线索

V2-018236 | MATHDB-338237

The Hermitian forms graph classification conjecture

摘录 / EXCERPT: Hermitian forms graph classification conjecture. Let q be a prime power. Let f be a Boolean degree 1 function on $H_q(\ell,k)$, where $k+\ell$ is sufficiently large depending...

conjecture | open | MathDB 页面 | 原文线索

V2-018237 | MATHDB-338238

Homomorphism conjecture for fixed-point Lie algebras and quiver varieties

摘录 / EXCERPT: Let Γ be a Dynkin diagram, let θ be the involution determined by $w_0(\alpha_i)=-\alpha_{\theta(i)}$, and let $\mathfrak g_{\Gamma}$ be the associated simple Lie alge...

conjecture | open | MathDB 页面 | 原文线索

V2-018238 | MATHDB-338270

Uniform boundedness for Brauer groups in families

摘录 / EXCERPT: Let k be a finitely generated field of characteristic zero, let ℓ be a prime, let S be a quasi-projective variety over k, and let $\pi:X\rightarrow S$ be a smooth projec...

conjecture | open | MathDB 页面 | 原文线索

V2-018239 | MATHDB-338274

The representation-variety conjecture detecting the trefoil

摘录 / EXCERPT: Given a knot K in the 3-sphere, let ... where μ is a chosen meridian and ... For a trefoil T, its representation variety is $\mathscr{R}(T,\mathbf{i})\cong\{*\}\sqcup S^1$, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-018240 | MATHDB-338288

Radicality conjecture for off-diagonal entries of Gram matrices

摘录 / EXCERPT: Radicality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018241 | MATHDB-338300

The categorical intersection conjecture for the varieties X_w

摘录 / EXCERPT: Let Z be the variety introduced earlier in the construction, let X_0 be the variety associated with the zero face, and for each chamber indexed by w let X_w be the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-018242 | MATHDB-338301

The homotopy-colimit conjecture for the varieties X_C

摘录 / EXCERPT: Let $\mathcal{C}$ be the poset of faces and let X_C be the associated varieties, with the derived categories $D^b(X_C)$ forming the corresponding diagram. Homotopy-colimit conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-018243 | MATHDB-338313

Structure conjecture for varieties with semi-positive holomorphic sectional curvature

摘录 / EXCERPT: Structure conjecture. There exists a smooth morphism $X\to Y$ such that a fiber is rationally connected and Y admits a finite étale cover $A\to Y$ by an abelian variety A. More...

conjecture | open | MathDB 页面 | 原文线索

V2-018244 | MATHDB-338318

Kottwitz's Frobenius-Hecke trace conjecture for intersection cohomology

摘录 / EXCERPT: Let (G,X) be a Shimura datum with reflex field E, let $K=K_pK^p\subset G(\mathbb A_f)$ be neat with K_p hyperspecial at a prime p coprime to λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018245 | MATHDB-338319

The K3-type classification conjecture for Fano complete intersections in Grassmannians

摘录 / EXCERPT: Let $X=X_{d_1,\ldots,d_c}\subset\operatorname{Gr}(k,n)$ be a smooth Fano complete intersection of even dimension, and assume that X is not a cubic fourfold. The variety X is of...

conjecture | open | MathDB 页面 | 原文线索

V2-018246 | MATHDB-338338

MNOP rationality conjecture for stable-pairs partition functions

摘录 / EXCERPT: Let X be a smooth projective threefold, let $\operatorname{Chow}(X)$ be the Chow variety of 1-dimensional cycles in X, and let ... be the stable-pairs partition function pushed...

conjecture | open | MathDB 页面 | 原文线索

V2-018247 | MATHDB-338339

GW/DT correspondence for stable-pairs and Gromov-Witten partition functions

摘录 / EXCERPT: Let X be a smooth projective threefold, let $\operatorname{Chow}(X)$ be the Chow variety of 1-dimensional cycles, and write $\operatorname{vir}\, dim}=c_1(X)\cdot\textup{cycle}$...

conjecture | open | MathDB 页面 | 原文线索

V2-018248 | MATHDB-338340

Relative GW/DT correspondence for Nakajima creation operators

摘录 / EXCERPT: Let D be a divisor and consider the relative correspondence map ... where $\mathbb{F}$ denotes the functions of u and z, and let $\gamma\in H^{\bullet}(D)$. Relative correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-018249 | MATHDB-338352

Harris-Viehmann conjecture for cohomology of unramified Rapoport-Zink spaces

摘录 / EXCERPT: Let G be a connected, quasisplit reductive group over $4p$, let $4mu$ be a dominant cocharacter, and let $b\in \mathbf{B}(G,\mu)$. Let $M_b\subset M_S\subset G$ be standard ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-018250 | MATHDB-338372

Combinatorial invariance of the TNS property for projective toric varieties

摘录 / EXCERPT: Let X and Y be nonsingular projective toric varieties of complex dimension n whose moment polytopes are combinatorially equivalent. Combinatorial invariance conjecture. X i...

conjecture | open | MathDB 页面 | 原文线索

V2-018251 | MATHDB-338373

The flag-polytope characterization of TNS toric manifolds

摘录 / EXCERPT: Let M^{2n} be a nonsingular projective toric variety of complex dimension n, and let P^n be its moment polytope. Flag-polytope characterization conjecture. M^{2n} is a TNS-...

conjecture | open | MathDB 页面 | 原文线索

V2-018252 | MATHDB-338376

Kontsevich-Soibelman integral identity conjecture for regular functions

摘录 / EXCERPT: Let (x,y,z) be the standard coordinates of the vector space $k^d=k^{d_1}\times k^{d_2}\times k^{d_3}$. Let $f\in k[x,y,z]$ satisfy $f(0,0,0)=0$ and ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-018253 | MATHDB-338384

The proximality conjecture for Θ-positive elements

摘录 / EXCERPT: Let G be a semisimple Lie group with a Θ-positive structure, let G/P_Θ be its associated flag variety, and let $G_\Theta^{>0}$ denote the Θ-positive semigrou...

conjecture | open | MathDB 页面 | 原文线索

V2-018254 | MATHDB-338393

Equality conjecture for the Betti-map rank of a section

摘录 / EXCERPT: Betti-map rank equality conjecture. If $\mathbb{Z}\xi$ is Zariski-dense in A, and A has no fixed part over any finite étale covering of S, then the preceding inequality is an...

conjecture | open | MathDB 页面 | 原文线索

V2-018255 | MATHDB-338413

Local divisibility conjecture for Albanese kernels over number fields

摘录 / EXCERPT: Let X be a smooth projective geometrically connected variety over a number field K with a K-rational point. For a finite place v of K, let ... be the base change to the com...

conjecture | open | MathDB 页面 | 原文线索

V2-018256 | MATHDB-338488

Bloch-type conjecture on intersections of homologically trivial divisors

摘录 / EXCERPT: Bloch-type conjecture. If this cup product map is zero, then the intersection product map

conjecture | open | MathDB 页面 | 原文线索

V2-018257 | MATHDB-338493

Dimension bound for normalized orbits in very general abelian varieties

摘录 / EXCERPT: Normalized-orbit dimension conjecture. If A is a very general abelian variety, then

conjecture | open | MathDB 页面 | 原文线索

V2-018258 | MATHDB-338515

Dubrovin's residual-category conjecture for Fano varieties

摘录 / EXCERPT: Let X be a complex Fano variety of Picard rank one, let H be the ample generator of $\operatorname{Pic}(X)$, and write $-K_X=mH$. Let $\operatorname{QH}(X)$ denote the small qu...

conjecture | open | MathDB 页面 | 原文线索

V2-018259 | MATHDB-338517

Existence and uniqueness of optimal degenerations for modified K-stability

摘录 / EXCERPT: Let X be a $ Q$-Fano variety. An $ R$-degeneration is a possibly non-equivariant degeneration of X over which the Donaldson-Futaki invariant is evaluated; a T-equivariant deg...

conjecture | open | MathDB 页面 | 原文线索

V2-018260 | MATHDB-338545

Guth's polynomial Wolff axiom conjecture

摘录 / EXCERPT: Guth's polynomial Wolff axiom conjecture. There is a constant $C(n,D,b5)>0$ such that the number of these $b4$-tubes contained in the $b4$-neighborhood of Z is at most

conjecture | open | MathDB 页面 | 原文线索

V2-018261 | MATHDB-338549

Connectedness conjecture for regular rings

摘录 / EXCERPT: Let Σ be a ring and let $\alpha,\beta\in\operatorname{Sper}\Sigma$. The connectedness property at α and β requires that, for every finite collection...

conjecture | open | MathDB 页面 | 原文线索

V2-018262 | MATHDB-338559

Conjecture (E) for zero-cycles on rationally connected varieties

摘录 / EXCERPT: Let X be a smooth proper variety over a number field k. Let Ω_f be the set of finite places of k and Ω_∞ the set of infinite places. For each infinite plac...

conjecture | open | MathDB 页面 | 原文线索

V2-018263 | MATHDB-338562

Geometric realization conjecture for absolutely cuspidal quiver polynomials

摘录 / EXCERPT: Let Q be a quiver, let $\mathbf{d}$ be a dimension vector, let $C_{Q,\mathbf{d}}^{\mathrm{abs}}(t)$ be the absolutely cuspidal polynomial, and let $\mathbf{k}$ be a finite field....

conjecture | open | MathDB 页面 | 原文线索

V2-018264 | MATHDB-338595

Strong generalized Hodge conjecture via algebraic correspondences

摘录 / EXCERPT: Let X be a complex smooth projective variety of dimension d, and let H be a sub-Hodge structure of $\mathrm{H}^k(X,\mathbb{Q})$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-018265 | MATHDB-338596

Voisin's antisymmetrization conjecture for powers of K3 surfaces

摘录 / EXCERPT: Voisin's conjecture. For all $l<N$, the anti-symmetrization projector $p_{\wedge^{N+1}}$ acts as zero on $\ker(\operatorname{pr}_*)\otimes\operatorname{CH}_0(S)_{\mathrm{num}}$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-018266 | MATHDB-338623

Precise inversion of adjunction for minimal log discrepancies

摘录 / EXCERPT: Let $P\in X$ be the germ of a normal variety, and let $S+B$ be an effective $\mathbf{R}$-divisor on X such that S is a normal prime divisor not appearing in B. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-018267 | MATHDB-338649

Koblitz's conjecture for abelian varieties

摘录 / EXCERPT: Let $A/\mathbb{Q}$ be an abelian variety satisfying the hypothesis (Triv_A), and suppose that $C_A\neq 0$. Koblitz's conjecture for abelian varieties. ... In particu...

conjecture | open | MathDB 页面 | 原文线索

V2-018268 | MATHDB-338654

Barja-Stoppino slope conjecture for fibered varieties

摘录 / EXCERPT: Barja-Stoppino slope conjecture. Under these hypotheses, f satisfies the slope inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-018269 | MATHDB-338659

The associated-form invariant lifting conjecture

摘录 / EXCERPT: Let X_n^d be the variety of degree-d forms in ${\mathbb C}[z_1,\dots,z_n]_d$ with nonzero discriminant, and let ... be the associated-form morphism. A function is called regular...

conjecture | open | MathDB 页面 | 原文线索

V2-018270 | MATHDB-338666

Ben-Zvi-Nadler's Betti geometric Langlands conjecture for nodal curves

摘录 / EXCERPT: Let Σ be a compact Riemann surface, let G be a reductive group, and let $\check{G}$ be its Langlands dual group. Write $\operatorname{Bun}_G(\Sigma)$ for the moduli stack...

conjecture | open | MathDB 页面 | 原文线索

V2-018271 | MATHDB-338678

André-Oort-Manin-Mumford conjecture

摘录 / EXCERPT: Let $T=S\times A$ be the product of a Shimura variety S and an abelian variety A. A subvariety of T is weakly special generic if it is not contained in any smaller weakly spe...

conjecture | open | MathDB 页面 | 原文线索

V2-018272 | MATHDB-338679

André-Pink-Mordell-Lang conjecture

摘录 / EXCERPT: Let $T=S\times A$ be the product of a Shimura variety S and an abelian variety A, and let $X\varsubsetneq T$ be a weakly special generic closed irreducible subvariety. Fix a su...

conjecture | open | MathDB 页面 | 原文线索

V2-018273 | MATHDB-338682

Quasi-projective K(π,1) conjecture for Artin groups

摘录 / EXCERPT: Quasi-projective $K(\pi,1)$ conjecture. There exists a smooth, connected, quasi-projective Eilenberg-MacLane space X such that

conjecture | open | MathDB 页面 | 原文线索

V2-018274 | MATHDB-338687

Sphericality conjecture for smooth Schubert varieties

摘录 / EXCERPT: Let G be a connected complex semisimple group, let Q be a parabolic subgroup of G, and let X_{wQ}, for $w\in W^Q$, be a smooth Schubert variety in G/Q. Let L be a Levi...

conjecture | open | MathDB 页面 | 原文线索

V2-018275 | MATHDB-338695

Zilber's conjecture on intersection with tori

摘录 / EXCERPT: Let $V\subseteq (\mathbb{C}^{\times})^n$ be an algebraic variety. Atypical subvarieties are the components of intersections with algebraic subtori having larger-than-expected dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-018276 | MATHDB-338696

Weak modular Zilber-Pink with derivatives for parametric families

摘录 / EXCERPT: Let C be an algebraically closed field of characteristic zero. Let $S\subseteq C^{3n}$ be an upper triangular D-special variety associated with a j-special variety...

conjecture | open | MathDB 页面 | 原文线索

V2-018277 | MATHDB-338697

Weak modular Zilber-Pink with derivatives for differential-field points

摘录 / EXCERPT: Let $(F;+,\cdot,\delta)$ be a differential field with algebraically closed field of constants C. Let $S\subseteq C^{3n}$ be an upper triangular D-special variety defined over C...

conjecture | open | MathDB 页面 | 原文线索

V2-018278 | MATHDB-338698

Modular André-Oort with derivatives

摘录 / EXCERPT: Let $V\subsetneq\mathbb{C}^{3n}$ be an algebraic variety. A J-special subvariety is an irreducible component of the Zariski closure of the image under J of an $\mathbb{H}$-spec...

conjecture | open | MathDB 页面 | 原文线索

V2-018279 | MATHDB-338699

Cluster-singularity conjecture for the width-7 symplectic 2-frieze

摘录 / EXCERPT: A symplectic 2-frieze is a frieze satisfying the symplectic 2-frieze rules, and a frieze is cluster-singular when it is singular in the associated cluster variety. Let the width-7...

conjecture | open | MathDB 页面 | 原文线索

V2-018280 | MATHDB-338724

Pink's conjecture on Hodge generic points of Shimura varieties

摘录 / EXCERPT: Let S be the Shimura variety under consideration, and let $z\in S(\mathbb{C})$. A point z is Hodge generic if it is not contained in any proper special subvariety of S, and i...

conjecture | open | MathDB 页面 | 原文线索

V2-018281 | MATHDB-338726

Guichard-Wienhard equivalence for higher Teichmüller spaces

摘录 / EXCERPT: Guichard-Wienhard equivalence. There are connected components of $\operatorname{Hom}(\pi_1(S),G)/G$ which are not distinguished by characteristic invariants if and only if there ex...

conjecture | open | MathDB 页面 | 原文线索

V2-018282 | MATHDB-338727

Guichard-Wienhard geometric-structure conjecture for higher Teichmüller representations

摘录 / EXCERPT: Guichard-Wienhard conjecture. There exists a generalized flag variety X and a compact fiber bundle $M\to S$ such that $\overline{\rho}$ is the holonomy of a locally homogeneous...

conjecture | open | MathDB 页面 | 原文线索

V2-018283 | MATHDB-338729

Guichard-Labourie-Wienhard conjecture on positive representations

摘录 / EXCERPT: Guichard-Labourie-Wienhard conjecture. The set of Θ-positive representations $\rho:\pi_1(\Sigma_g)\to G$ is open and closed in $\operatorname{Hom}(\pi_1(S),G)/G$. In particu...

conjecture | open | MathDB 页面 | 原文线索

V2-018284 | MATHDB-338760

Ulrich wildness conjecture for general linear standard determinantal schemes

摘录 / EXCERPT: Let $X\subset \mathbb{P}^n$ be a general linear standard determinantal scheme of codimension $c\ge 2$ defined by the maximal minors of a $t\times (t+c-1)$ matrix with linear entrie...

conjecture | open | MathDB 页面 | 原文线索

V2-018285 | MATHDB-338766

Countably many isogeny classes for every attainable Picard number

摘录 / EXCERPT: Let $R_g^{(p)}$ denote the set of Picard numbers of g-dimensional abelian varieties in characteristic $p>0$. Let $g \geq 1$ and let $\rho \in R_g^{(p)}$. Countable-isogeny-class...

conjecture | open | MathDB 页面 | 原文线索

V2-018286 | MATHDB-338768

The multiplier-ideal locus conjecture for polarized abelian varieties

摘录 / EXCERPT: Let X be a g-dimensional abelian variety and let B be an ample rational divisor on X. The cosupport of a multiplier ideal is the locus where that ideal is nontrivial. Multi...

conjecture | open | MathDB 页面 | 原文线索

V2-018287 | MATHDB-338782

Motivic McKay correspondence conjecture for the group scheme α_p

摘录 / EXCERPT: Let k be a perfect field of characteristic $p>0$, let $G=\alpha_p$, and let V be a finite-dimensional representation of G with quotient V/G. Let $H=\mathbb{Z}/p\mathbb{Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-018288 | MATHDB-338800

Finite-generation conjecture for infinite transitivity on generically flexible varieties

摘录 / EXCERPT: A family $\mathfrak{F}$ of $\mathbb{G}_a$-subgroups of $\operatorname{Aut}(X)$ is called saturated if it contains every replica of each of its members and is closed under conjugati...

conjecture | open | MathDB 页面 | 原文线索

V2-018289 | MATHDB-338801

Lie-algebraic closure conjecture for finitely generated additive subgroups

摘录 / EXCERPT: Let X be an affine variety and let $A=\mathcal{O}_X(X)$ be its structure algebra. For locally nilpotent derivations $\partial_1,\ldots,\partial_k\in\operatorname{LND}(A)$, set .....

conjecture | open | MathDB 页面 | 原文线索

V2-018290 | MATHDB-338802

The general-type and rational-point conjecture for high modular-tower levels

摘录 / EXCERPT: High-level modular-tower conjecture. At sufficiently high levels, there are no K-points on a modular tower. Moreover, sufficiently high levels are algebraic varieties of general...

conjecture | open | MathDB 页面 | 原文线索

V2-018291 | MATHDB-338803

The strong-torsion bound for regular realizations

摘录 / EXCERPT: Strong-torsion bound conjecture. Some expression in r and m, independent of the prime of good reduction v, bounds p^u.

conjecture | open | MathDB 页面 | 原文线索

V2-018292 | MATHDB-338805

Direct recurrence conjecture for the coefficients of Hermite subdivision schemes

摘录 / EXCERPT: Direct recurrence conjecture. The coefficients can be computed directly by

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-018293 | MATHDB-338860

Local-to-global decomposition conjecture for deformations of Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a Calabi-Yau variety with $\mathbb{Q}$-factorial terminal singularities and isolated singularities P. The Kuranishi space parametrizes complex deformations of X, and...

conjecture | open | MathDB 页面 | 原文线索

V2-018294 | MATHDB-338864

Discrete-torsion conjecture for multiple fibers of Calabi-Yau genus-one fibrations

摘录 / EXCERPT: Let a Calabi-Yau variety admit a relatively minimal genus-one fibration with multiple fibers, and let a multiple fiber have multiplicity m. Discrete-torsion conjecture. The multi...

conjecture | open | MathDB 页面 | 原文线索

V2-018295 | MATHDB-338870

The compatibility conjecture for the K-theoretic and motivic parameterizations

摘录 / EXCERPT: the compatibility conjecture. The canonical map induces the identity map on the homotopy groups. This asserts compatibility between the K-theoretic parameterization and the mod-...

conjecture | open | MathDB 页面 | 原文线索

V2-018296 | MATHDB-338871

The factorization conjecture for the motivic parameterization map

摘录 / EXCERPT: The factorization conjecture. The canonical map factors through this fully faithful map, and the first arrow in the factorization

conjecture | open | MathDB 页面 | 原文线索

V2-018297 | MATHDB-338872

The motivic compatibility conjecture for factorizable line bundles and gerbes

摘录 / EXCERPT: Let X be the smooth scheme considered above, let G be the reductive group with maximal torus T, and let ℓ be the chosen prime. Consider the diagram ... The horizontal and...

conjecture | open | MathDB 页面 | 原文线索

V2-018298 | MATHDB-338890

Pfister's u-invariant conjecture for real fields

摘录 / EXCERPT: Pfister's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018299 | MATHDB-338909

Demailly's Brody-algebraic hyperbolicity conjecture

摘录 / EXCERPT: A complex projective variety X is algebraically hyperbolic if there exists an $\epsilon>0$ such that every curve $C\subset X$ of geometric genus $g(C)$ satisfies ... It is Brody hy...

conjecture | open | MathDB 页面 | 原文线索

V2-018300 | MATHDB-338913

The $\mathbb{F}_1$-module structure conjecture for the integral Moonshine module

摘录 / EXCERPT: Let $V^\natural_{\mathbb{Z}}$ be a self-dual integral form of $V^\natural$ with Monster symmetry, and let $W_\infty M$ denote the big Witt vectors of a graded module M....

conjecture | open | MathDB 页面 | 原文线索

V2-018301 | MATHDB-338929

The local model conjecture for parahoric local Shimura pairs

摘录 / EXCERPT: The local model conjecture. There exists a projective and flat scheme ... over ... which supports an action of ${\mathcal G}\otimes_{\mathbb Z_p}{\mathcal O}_E$ and is such that: (i) t...

conjecture | open | MathDB 页面 | 原文线索

V2-018302 | MATHDB-338930

The existence conjecture for integral models of Shimura varieties of abelian type

摘录 / EXCERPT: The integral-model conjecture. There is a scheme ${\mathscr S}_{K_p}({\mathbb G},X)$ over ... with a right action of ${\mathbb G}({\mathbb A}_f^p)$ satisfying: (a) sufficiently small...

conjecture | open | MathDB 页面 | 原文线索

V2-018303 | MATHDB-338931

The Langlands-Rapoport conjecture for parahoric integral models

摘录 / EXCERPT: Langlands-Rapoport conjecture. There is a $\langle\Phi_E\rangle\times{\mathbb G}({\mathbb A}_f^p)$-equivariant bijection

conjecture | open | MathDB 页面 | 原文线索

V2-018304 | MATHDB-338939

Converse characterization of abelian duality spaces via the Albanese map

摘录 / EXCERPT: Let X be an n-dimensional smooth complex quasi-projective variety with proper Albanese map ... An abelian duality space of dimension n is a space whose cohomology with coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-018305 | MATHDB-338940

Suslin's conjecture on the image of the Hurewicz map

摘录 / EXCERPT: Suslin's conjecture. The image of Suslin's Hurewicz map coincides with

conjecture | open | MathDB 页面 | 原文线索

V2-018306 | MATHDB-338970

Separability conjecture for key polynomials of maximally real elliptic curves

摘录 / EXCERPT: Separability conjecture. If f has three distinct real roots, then P_n is separable for every $n\geq1$.

conjecture | open | MathDB 页面 | 原文线索

V2-018307 | MATHDB-338971

Discriminant-degree conjecture for key polynomials in elliptic curve families

摘录 / EXCERPT: Discriminant-degree conjecture. For the Legendre family $f=x(x-1)(x-\lambda)$, the degree of the x-discriminant of P_n is $3n^2-3n+2$; for the family...

conjecture | open | MathDB 页面 | 原文线索

V2-018308 | MATHDB-338972

Topological-profile conjecture for the real key-polynomial curves

摘录 / EXCERPT: For each integer $n\geq2$, let $\mathcal{C}(n)$ be the affine plane curve defined by the key polynomial $P_n(x,\lambda)=0$, and let $\mathcal{C}(n)(\mathbb{R})$ denote its real loc...

conjecture | open | MathDB 页面 | 原文线索

V2-018309 | MATHDB-338973

Discriminant-root conjecture for key polynomials

摘录 / EXCERPT: Let $P_n=P_n(x,\lambda)$ be the key polynomial and let $\Delta_n=\Delta_n(x)$ be its discriminant with respect to λ. Discriminant-root conjecture. For every $n\geq2$, the s...

conjecture | open | MathDB 页面 | 原文线索

V2-018310 | MATHDB-338994

Grothendieck's Hom conjecture for hyperbolic curves

摘录 / EXCERPT: Let k be finitely generated over $\mathbb{Q}$. Let T/k be a smooth variety, and let X/k be a smooth, proper, geometrically connected hyperbolic curve. Grothendieck's Hom conj...

conjecture | open | MathDB 页面 | 原文线索

V2-018311 | MATHDB-339014

Equality conjecture for signature variety stabilization bounds

摘录 / EXCERPT: Let $\lambda_{d,k}$ and the bounds in ... for all integers $d,m\geq 2$. If true, this determines the thresholds at which the signature varieties stabilize and yields the formula for...

conjecture | open | MathDB 页面 | 原文线索

V2-018312 | MATHDB-339015

Rational identifiability conjecture for signature varieties below stabilization

摘录 / EXCERPT: Let $\mathcal{P}_{d,k,m}$ and $\mathcal{L}_{d,k,m}$ denote the polynomial and piecewise-linear signature varieties, and let M be the stabilization constant from the preceding cor...

conjecture | open | MathDB 页面 | 原文线索

V2-018313 | MATHDB-339040

Scholze's local-model conjecture

摘录 / EXCERPT: Let $(G,\{\mu\},K)$ be a local model triple, with G split over a tame extension, and let ${\mathbb M}^{\mathrm{loc}}_K(G,\{\mu\})$ be the associated flat projective local model o...

conjecture | open | MathDB 页面 | 原文线索

V2-018314 | MATHDB-339063

Unimodality conjecture for Poincaré polynomials of degenerate flag varieties

摘录 / EXCERPT: For each of the five types considered-type A, odd symplectic, even symplectic, odd orthogonal, and even orthogonal-let $P(q)$ denote the corresponding Poincaré polynomial. Unimod...

conjecture | open | MathDB 页面 | 原文线索

V2-018315 | MATHDB-339110

The universal amplituhedron form triangulation-independence conjecture

摘录 / EXCERPT: Let $\mathcal{T}$ be a degree-one (n,k,m)-triangulation. The universal amplituhedron form $\omega^{(\mathcal{T})}_{\mathcal{A}_{n,k,\geq m}}$ is the signed sum of the pushforward...

conjecture | open | MathDB 页面 | 原文线索

V2-018316 | MATHDB-339111

Positive Laurent formula conjecture for the stacked twist map

摘录 / EXCERPT: Let V and W be matrices whose stacked matrix is $U:=\operatorname{stack}(V,W)\in\operatorname{Mat}^{\circ}(k+\ell,n)$, and let the stacked twist map satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-018317 | MATHDB-339118

The Picard-generation conjecture for Hurwitz spaces

摘录 / EXCERPT: Work over an algebraically closed field of characteristic zero. Let $\widetilde H_{d,g}$ be the Hurwitz space of degree-d covers of smooth genus-0 curves by irreducible genus-...

conjecture | open | MathDB 页面 | 原文线索

V2-018318 | MATHDB-339121

Shafarevich morphism conjecture for toroidal compactifications of arithmetic complex hyperbolic lattices

摘录 / EXCERPT: Let $\Gamma<PU(n,1)$ be a non-uniform arithmetic lattice, and let $\bar X_{\Gamma}$ be its toroidal compactification. Assume that Γ is torsion-free and torsion-free at infin...

conjecture | open | MathDB 页面 | 原文线索

V2-018319 | MATHDB-339123

DT/PT/GW correspondence for Hilbert schemes in K-theory

摘录 / EXCERPT: Let $KDT(\mathbb P^1\times\mathbb C^2)$ denote the K-theoretic Donaldson-Thomas invariants of $\mathbb P^1\times\mathbb C^2$, and let $QK_{\mathrm T}(\operatorname{Hilb})$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-018320 | MATHDB-339124

Complex Chern slope accumulation conjecture for line arrangements

摘录 / EXCERPT: A line arrangement is a finite collection of lines in the projective plane, and its Chern slope is the quotient of its logarithmic Chern numbers, denoted by $\bar{c}_1^2/\bar{c}_2$...

conjecture | open | MathDB 页面 | 原文线索

V2-018321 | MATHDB-339125

Complex H-constant accumulation conjecture for line arrangements

摘录 / EXCERPT: For a line arrangement $\mathcal{A}$, let H_L denote its associated H-constant. Consider all infinite families of line arrangements over $\mathbb{C}$ with the number of lines t...

conjecture | open | MathDB 页面 | 原文线索

V2-018322 | MATHDB-339146

Viro's conjecture on Betti numbers of real projective surfaces

摘录 / EXCERPT: Let V be a non-singular real projective surface, and let ${\mathbb R}V$ and ${\mathbb C}V$ denote its real and complex points. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018323 | MATHDB-339147

Itenberg's primitive patchworking conjecture for Betti numbers

摘录 / EXCERPT: Let V be a real hypersurface in ${\mathbb P}^{n+1}$ obtained by a primitive patchworking, where $n=\dim V$. For $0\leq q\leq n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-018324 | MATHDB-339148

Second-page degeneration conjecture for tropical hypersurface spectral sequences

摘录 / EXCERPT: Let $(X,\mathcal E)$ be a compact non-singular real tropical hypersurface. Let ${\mathcal K}_1,\dots,{\mathcal K}_n$ and ${\mathcal S}_{\mathcal E}$ be the coefficient systems appe...

conjecture | open | MathDB 页面 | 原文线索

V2-018325 | MATHDB-339152

Invertibility conjecture for the depth-graded sum odd multiple zeta matrix

摘录 / EXCERPT: Invertibility conjecture. For any $r\geq3$, $N\geq r+2$, and $N-r\equiv0\pmod 2$, the order $|T_{N,r}|$ matrix

conjecture | open | MathDB 页面 | 原文线索

V2-018326 | MATHDB-339157

K-theory structure constants for minuscule Kac-Moody flag varieties

摘录 / EXCERPT: Let $X=G/P$ be a Kac-Moody flag variety, let $m \in W^P$ be Λ-minuscule, and let $\mathcal{P}_m$ be the associated d-complete poset. For $u,v,w \leq m$ in Bruhat order, l...

conjecture | open | MathDB 页面 | 原文线索

V2-018327 | MATHDB-339172

Power-saving refinement of the Manin-Peyre conjecture

摘录 / EXCERPT: Refinement of the Manin-Peyre conjectures. There exist a Zariski open subset U of V, a polynomial $P_{U,H}$ of degree ρ, and $\delta\in(0,1)$ such that, for $B\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-018328 | MATHDB-339191

Nori's Chow-theoretic Lefschetz conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over the complex numbers, and let $D\subset X$ be a very general smooth ample divisor of sufficiently large degree. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018329 | MATHDB-339192

Nori's push-forward conjecture for Chow groups

摘录 / EXCERPT: Let X be a smooth projective variety over the complex numbers, let $D\subset X$ be a very general smooth ample divisor of sufficiently large degree, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018330 | MATHDB-339193

Bloch-Beilinson Abel-Jacobi injectivity conjecture

摘录 / EXCERPT: Let W be a smooth projective variety over $\overline{\mathbb{Q}}$, and let $r\geq 0$ be an integer. Let $J(H^{2r-1}(W(\mathbb{C}),\mathbb{Q}(r)))$ denote the relevant intermediat...

conjecture | open | MathDB 页面 | 原文线索

V2-018331 | MATHDB-339194

Bloch-Beilinson Abel-Jacobi injectivity conjecture for higher Chow groups

摘录 / EXCERPT: Let W be a smooth projective variety over $\overline{\mathbb{Q}}$, and let r and m be integers for which the displayed higher Chow group is defined. Let ... be the generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-018332 | MATHDB-339196

The refined Birch and Swinnerton-Dyer conjecture for abelian varieties over function fields

摘录 / EXCERPT: Let A be an abelian variety over a function field K in one variable over a finite field of characteristic p. Let L/K be a finite Galois extension with group G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018333 | MATHDB-339226

The higher-dimensional zero-dimensional count formula in dimensions divisible by four

摘录 / EXCERPT: Let $d\geq4$ with $d\equiv0\pmod4$, let $\ell\in\mathbb{Z}$, and let $L=\mathcal{O}_{\mathbb{C}^d}\otimes t_d^{-\ell}$ be the trivial equivariant line bundle. Let T be the Calabi...

conjecture | open | MathDB 页面 | 原文线索

V2-018334 | MATHDB-339227

The orientation uniqueness conjecture for higher-dimensional counts

摘录 / EXCERPT: Let $d\geq4$ with $d\equiv0\pmod4$, and consider the specialized generating series for the trivial equivariant line bundle $L=\mathcal{O}_{\mathbb{C}^d}\otimes t_d^{-\ell}$ and the...

conjecture | open | MathDB 页面 | 原文线索

V2-018335 | MATHDB-339286

Anosov combination conjecture

摘录 / EXCERPT: Anosov combination conjecture. If each Γ_i is $P_{{\tau_{\mathrm{mod}}}}$-Anosov and, for every $i=1,\ldots,n$ and every $\gamma\in\Gamma_i-\{1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-018336 | MATHDB-339363

Honda's conjecture on linear rank growth of abelian varieties

摘录 / EXCERPT: Honda's conjecture. There is $c=c(A,k)>0$ such that for every finite extension L/k,

conjecture | open | MathDB 页面 | 原文线索

V2-018337 | MATHDB-339381

Voisin's moving rational curve conjecture for projective IHS varieties

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic variety of dimension $2n$. A primitive rational curve means a rational curve whose cohomology class is primitive. Voisin'...

conjecture | open | MathDB 页面 | 原文线索

V2-018338 | MATHDB-339384

Beauville's filtration-compatibility conjecture for the Grothendieck group

摘录 / EXCERPT: Let A be an abelian variety of dimension g over an algebraically closed field of characteristic zero, and let $K_0(A)\otimes\mathbf Q$ carry its usual gamma filtration...

conjecture | open | MathDB 页面 | 原文线索

V2-018339 | MATHDB-339391

The Brauer-Manin conjecture for zero-cycles of degree one

摘录 / EXCERPT: Let k be a number field, and let $\{X_\omega\}_\omega$ be the family of all smooth, proper, geometrically integral varieties over k. The Brauer-Manin obstruction to weak approx...

conjecture | open | MathDB 页面 | 原文线索

V2-018340 | MATHDB-339443

Ruan's big quantum cohomology conjecture for birational Calabi-Yau manifolds

摘录 / EXCERPT: Let X and Y be birational Calabi-Yau manifolds, meaning smooth projective varieties with trivial first Chern class. Their quantum cohomology includes the big quantum cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-018341 | MATHDB-339469

Grothendieck's standard conjecture of Künneth type

摘录 / EXCERPT: Standard conjecture of Künneth type. There exists a unique decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-018342 | MATHDB-339534

Lazarsfeld's conjecture on Seshadri constants of ample line bundles

摘录 / EXCERPT: Let X be a projective manifold and let L be an ample line bundle on X. For a very general point $x\in X$, write ... the maximal value of the Seshadri constant of L on X. La...

conjecture | open | MathDB 页面 | 原文线索

V2-018343 | MATHDB-339585

Moduli and CM polarization conjecture for K-semistable klt Fano varieties

摘录 / EXCERPT: Moduli and CM polarization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018344 | MATHDB-339595

Surjective endomorphism conjecture for Fano manifolds of Picard number one

摘录 / EXCERPT: Surjective endomorphism conjecture. Every surjective endomorphism $f:X\to X$ must be bijective.

conjecture | open | MathDB 页面 | 原文线索

V2-018345 | MATHDB-339600

Capacity conjecture for symmetric PIR from MDS-coded storage with adversarial servers

摘录 / EXCERPT: Let an $[n,k]$ MDS storage code store a database accessed by a Symmetric Private Information Retrieval scheme, with t-colluding servers, b Byzantine servers, and r unresponsi...

conjecture | open | MathDB 页面 | 原文线索

V2-018346 | MATHDB-339630

Constructibility and openness of F-rational fibers over higher-dimensional bases

摘录 / EXCERPT: F-rational fiber-locus conjecture. The locus of $t\in\mathfrak{mSpec}D$ for which X_t is F-rational is constructible. If the family is flat and proper, or sufficiently local, t...

conjecture | open | MathDB 页面 | 原文线索

V2-018347 | MATHDB-339646

Morel's conjecture on the Witt-valued Euler characteristic of split reductive groups

摘录 / EXCERPT: Let G be a split reductive group-scheme over a perfect field k, let T be a maximal split torus of G, and let N_T be the normalizer of T in G. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018348 | MATHDB-339658

Farb's conjecture on sections of configuration-space bundles over cubic plane curves

摘录 / EXCERPT: Farb's conjecture. The smallest value of n for which ξ_n admits a continuous section is

conjecture | open | MathDB 页面 | 原文线索

V2-018349 | MATHDB-339736

The asymptotic abundance conjecture for sparse paving matroid stack delta-matroids

摘录 / EXCERPT: A matroid stack delta-matroid is a delta-matroid whose proper set systems in its stack are matroids. It is sparse paving when every matroid in its stack is sparse paving, meaning t...

conjecture | open | MathDB 页面 | 原文线索

V2-018350 | MATHDB-339749

Pareschi-Popa conjecture on summands of theta divisors

摘录 / EXCERPT: Pareschi-Popa conjecture. If $\Theta=X+Y$ decomposes nontrivially as a sum, then either (A,Θ) is the Jacobian of a smooth projective curve, or (A,Θ) is the intermedia...

conjecture | open | MathDB 页面 | 原文线索

V2-018351 | MATHDB-339750

Pareschi-Popa adjoint-module conjecture for indecomposable ppav's

摘录 / EXCERPT: Pareschi-Popa adjoint-module conjecture. Every summand $\mathscr M\hookrightarrow\mathscr{A}\hspace*{-0.15em}d_\Theta$ is either a skyscraper sheaf or has support A.

conjecture | open | MathDB 页面 | 原文线索

V2-018352 | MATHDB-339785

Green--Griffiths--Lang conjecture for projective varieties

摘录 / EXCERPT: Green--Griffiths--Lang conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-018353 | MATHDB-339786

Fibration property for algebraically hyperbolic varieties

摘录 / EXCERPT: Fibration property. If Y is algebraically hyperbolic over k and, for all $y\in Y(k)$, the projective scheme X_y is algebraically hyperbolic over k, then X is algebraicall...

conjecture | open | MathDB 页面 | 原文线索

V2-018354 | MATHDB-339794

Tame purity conjecture for open de Rham spaces

摘录 / EXCERPT: Purity conjecture. The map ν_a^* preserves mixed Hodge structures and is an isomorphism on the pure parts. This predicts that the pure cohomological information of the tame cha...

conjecture | open | MathDB 页面 | 原文线索

V2-018355 | MATHDB-339795

Irregular purity conjecture for wild character varieties

摘录 / EXCERPT: Irregular purity conjecture. The map ν induces

conjecture | open | MathDB 页面 | 原文线索

V2-018356 | MATHDB-339826

The rational Lagrangian fibration conjecture for irreducible symplectic varieties

摘录 / EXCERPT: Rational Lagrangian fibration conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018357 | MATHDB-339871

Relative twist formula for étale characteristic classes

摘录 / EXCERPT: Let S be a regular Noetherian scheme over $\mathbb Z[1/\ell]$ and let $f\colon X\to S$ be a smooth proper morphism purely of relative dimension n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018358 | MATHDB-339872

Relative push-forward conjecture for characteristic classes

摘录 / EXCERPT: Let S be a smooth connected scheme over a perfect field k of characteristic p. Let $f\colon X\to S$ and $g\colon Y\to S$ be smooth morphisms of relative dimensions n and...

conjecture | open | MathDB 页面 | 原文线索

V2-018359 | MATHDB-339873

Relative characteristic class cycle-class conjecture

摘录 / EXCERPT: Let S be a smooth connected scheme over a perfect field k of characteristic p. Let $f\colon X\to S$ be a smooth morphism purely of relative dimension n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018360 | MATHDB-339882

The signed Selmer torsion conjecture

摘录 / EXCERPT: Let F be a number field, let p be an odd prime, let F_∞/F be the $\mathbf{Z}_p$-cyclotomic extension, and let A be an abelian variety in the setting of the signed Selm...

conjecture | open | MathDB 页面 | 原文线索

V2-018361 | MATHDB-339890

Exceptional-set description in Manin's conjecture for Fano varieties

摘录 / EXCERPT: Let X be a smooth projective geometrically integral Fano variety over a number field, and set $L=-K_X$. For a thin map $f:Y\to X$ from a smooth projective geometrically integral...

conjecture | open | MathDB 页面 | 原文线索

V2-018362 | MATHDB-339924

Cotangent-stack invariance conjecture for Coulomb and Higgs categories

摘录 / EXCERPT: Let $\calY$ and $\calY'$ be stacks such that their cotangent dg-stacks are isomorphic as symplectic dg-stacks. Write $\calC_C(\calY)$ and $\calC_H(\calY)$ for the corresponding Cou...

conjecture | open | MathDB 页面 | 原文线索

V2-018363 | MATHDB-339925

Small-loops comparison conjecture for mapping dg-stacks

摘录 / EXCERPT: For a dg-stack $\calY$, let $L\calY=\calY\underset{\calY\times\calY}{\times}\calY$ be its derived loop stack, with both maps given by the diagonal, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018364 | MATHDB-339928

Ind-coherent Hecke equivalence conjecture for the $\GL(n)$ theory

摘录 / EXCERPT: Let ... be the stack specified in the source. Let $\IndCoh$ denote ind-coherent sheaves and let $D(\Gr_{\GL(n)})^{\operatorname{Hecke}}$ be the Hecke-eigenmodule category on the affi...

conjecture | open | MathDB 页面 | 原文线索

V2-018365 | MATHDB-339930

The motivic Grothendieck-Teichmüller criterion for Kashiwara-Vergne groups

摘录 / EXCERPT: Motivic Grothendieck-Teichmüller criterion. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-018366 | MATHDB-339938

The low-point Mori dream space conjecture for Hirzebruch-surface Hilbert schemes

摘录 / EXCERPT: Low-point Mori dream space conjecture. The Hilbert scheme $\mathbb{F}_r^{[r-k]}$ is a Mori dream space, and the decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-018367 | MATHDB-339948

The nef-canonical-bundle conjecture on semi-orthogonal decompositions

摘录 / EXCERPT: Let X be a smooth projective variety. Let K_X be its canonical bundle and let $h^0(K_X)$ denote the dimension of its global sections. A line bundle is nef if it has nonnegative...

conjecture | open | MathDB 页面 | 原文线索

V2-018368 | MATHDB-339960

Hasse-Witt bijectivity and unobstructed Witt-vector deformations

摘录 / EXCERPT: Let X be a smooth projective variety with trivial canonical class over an algebraically closed field k of characteristic $p>0$. Let ... be the Hasse-Witt operation induced by the...

conjecture | open | MathDB 页面 | 原文线索

V2-018369 | MATHDB-339977

The threefold numerical-effectivity conjecture for nef divisors

摘录 / EXCERPT: Threefold numerical-effectivity conjecture. Then L is numerically equivalent to an effective divisor.

conjecture | open | MathDB 页面 | 原文线索

V2-018370 | MATHDB-340000

Lang's conjecture on rationally swept-out projective manifolds

摘录 / EXCERPT: Lang's conjecture. X is rationally swept out by abelian varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-018371 | MATHDB-340010

The strong maximal rank conjecture for linear series on general curves

摘录 / EXCERPT: Let g, r, and d be positive integers such that $g-d+r\geq 0$ and $0\leq \rho(g,r,d)\leq r-2$. For a general curve X of genus g, let $W^r_d(X)$ denote the variety of divis...

conjecture | open | MathDB 页面 | 原文线索

V2-018372 | MATHDB-340029

Keel-McKernan's logarithmic bend-and-break conjecture

摘录 / EXCERPT: Logarithmic bend-and-break conjecture. Through a general point of C there is a rational curve meeting D at most once.

conjecture | open | MathDB 页面 | 原文线索

V2-018373 | MATHDB-340032

Weak-BAB conjecture on boundedness of anti-canonical volumes

摘录 / EXCERPT: Fix a positive integer n and a real number $\varepsilon>0$. Let $\mathfrak{D}$ be the set of all log pairs (X,Δ) such that X is a projective variety of dimension n,...

conjecture | solved | MathDB 页面 | 原文线索

V2-018374 | MATHDB-340039

Kivinen's conjectural description of Hilbert-scheme homology for rational-component curves

摘录 / EXCERPT: Let C be the unique compactification with rational components and no other singularities of the curve $\{x^n=y^{dn}\}$. Let ... The algebra A_n is the algebra acting on this homo...

conjecture | open | MathDB 页面 | 原文线索

V2-018375 | MATHDB-340040

The Procesi-bundle conjecture for affine Springer fiber homology

摘录 / EXCERPT: Let $\mathcal{P}$ be the Procesi bundle on $\operatorname{Hilb}^n(\mathbb{C}^2)$, let $\mathcal{P}^*$ denote its dual, and let $\mathcal{O}(d)$ be the indicated determinant-line-bu...

conjecture | open | MathDB 页面 | 原文线索

V2-018376 | MATHDB-340064

Gross-Deligne-type conjecture for abelian schemes with complex multiplication

摘录 / EXCERPT: Let K be a finite Galois extension of $\mathbb Q$ and let c be a central element of ${\rm Gal}(K/\mathbb Q)$ inducing complex conjugation under every embedding. Assume that all...

conjecture | open | MathDB 页面 | 原文线索

V2-018377 | MATHDB-340065

Vanishing conjecture for eigenspaces of abelian-scheme cohomology

摘录 / EXCERPT: Assume the hypotheses of the Gross-Deligne-type conjecture for the abelian scheme $f:{\cal A}\to Y$, including the $\mathcal O_K$-action, the parity condition on χ and l, an...

conjecture | open | MathDB 页面 | 原文线索

V2-018378 | MATHDB-340134

Section property conjecture for symplectic varieties

摘录 / EXCERPT: Section property conjecture. This algebra epimorphism admits a multiplicative section whose image contains all Chern classes of the tangent bundle of X.

conjecture | open | MathDB 页面 | 原文线索

V2-018379 | MATHDB-340137

Nash blow-up conjecture for covexillary Schubert varieties

摘录 / EXCERPT: Nash blow-up conjecture for covexillary Schubert varieties. Let w be a covexillary permutation with parabolic P defined as above. The Nash blow-up of X_w^P is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-018380 | MATHDB-340157

Global arithmetic intersection conjecture

摘录 / EXCERPT: Let $\mathcal M$ be the regular, flat, proper Deligne-Mumford stack over $\operatorname{Spec}O_F$ in the global setup, let $\widehat z$ be the arithmetic diagonal cycle with a Gree...

conjecture | open | MathDB 页面 | 原文线索

V2-018381 | MATHDB-340204

Regularity of rational actions with regular image

摘录 / EXCERPT: Let G be an algebraic group, let X be a variety, and let $\rho\colon G\to\operatorname{Bir}(X)$ be a rational action, meaning a group homomorphism represented by a rational act...

conjecture | open | MathDB 页面 | 原文线索

V2-018382 | MATHDB-340208

Finite-value conjecture for stability thresholds in Q-Fano families

摘录 / EXCERPT: Let $\pi:X\to T$ be a projective family of varieties. Assume that T is normal, every fiber X_t is klt, and $-K_{X/T}$ is $\mathbb{Q}$-Cartier and ample. For a geometric point...

conjecture | open | MathDB 页面 | 原文线索

V2-018383 | MATHDB-340218

Flatness conjecture for high-rank polynomial pullback maps

摘录 / EXCERPT: Let k be a field, let $\bb V$ and $\bb W$ be k-vector spaces with $\dim(\bb W)\leq m$, and let $P:\bb V\to k$ be a polynomial of degree d. For an affine map...

conjecture | open | MathDB 页面 | 原文线索

V2-018384 | MATHDB-340236

The non-ordinary base-change classification conjecture

摘录 / EXCERPT: Let $\mathcal E_{\mathrm{par}}$ be the parallel-weight eigenvariety, let $\mathcal C_M$ denote the relevant Coleman-Mazur eigencurve at level parameter M, and let $[\varphi]$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-018385 | MATHDB-340238

Conjecture on intersections of Lagrangian subvarieties in the Chow ring

摘录 / EXCERPT: Let X be a hyperkähler variety of dimension $2m$, and let $A^i(X):=CH^i(X)_{\mathbb{Q}}$ denote its codimension-i Chow group with rational coefficients. Assume the Chow ring ha...

conjecture | open | MathDB 页面 | 原文线索

V2-018386 | MATHDB-340241

Non-archimedean grouplessness-purity conjecture

摘录 / EXCERPT: Let K be the field under consideration, and let X be a projective variety over K. Recall that X is groupless if every morphism from a finite type connected group scheme ove...

conjecture | open | MathDB 页面 | 原文线索

V2-018387 | MATHDB-340251

Batyrev-Manin Conjecture C' for Fano varieties

摘录 / EXCERPT: Batyrev-Manin Conjecture C'. If U is sufficiently small, then

conjecture | open | MathDB 页面 | 原文线索

V2-018388 | MATHDB-340280

Boucksom-Broustet-Pacienza conjecture on restricted base loci and non-nef loci

摘录 / EXCERPT: Let X be a normal projective variety, and let D be a pseudoeffective $d53d$-Cartier divisor on X. The Boucksom-Broustet-Pacienza conjecture. The restricted base locus and the...

conjecture | open | MathDB 页面 | 原文线索

V2-018389 | MATHDB-340302

McKinnon's rational-curve conjecture for best approximation constants

摘录 / EXCERPT: McKinnon's conjecture. There exists a rational curve C on X passing through Q achieving the best approximation constant at Q with respect to L.

conjecture | open | MathDB 页面 | 原文线索

V2-018390 | MATHDB-340304

Conjecture R2 on rank 2 local systems coming from abelian varieties

摘录 / EXCERPT: Let $X/\mathbb{F}_{q}$ be a smooth, geometrically connected, quasi-projective variety, let $l\neq p$ be a prime, and let L be a lisse $\overline{\mathbb{Q}}_{l}$-sheaf of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-018391 | MATHDB-340305

The companions conjecture for irreducible coefficient objects

摘录 / EXCERPT: Let $X/\mathbb{F}_{q}$ be a smooth variety, and let $\mathcal{F}$ be an irreducible coefficient object on X with algebraic determinant. Companions conjecture. There exists a numb...

conjecture | open | MathDB 页面 | 原文线索

V2-018392 | MATHDB-340325

Generalised Nonvanishing Conjecture

摘录 / EXCERPT: Generalised Nonvanishing Conjecture. The numerical class of $K_X+\Delta+tL$ belongs to the effective cone for every $t\geq0$.

conjecture | open | MathDB 页面 | 原文线索

V2-018393 | MATHDB-340326

Weak Generalised Nonvanishing Conjecture

摘录 / EXCERPT: Weak Generalised Nonvanishing Conjecture. Under these hypotheses, the numerical class of $K_X+\Delta+tL$ belongs to the effective cone for every $t\geq0$.

conjecture | open | MathDB 页面 | 原文线索

V2-018394 | MATHDB-340340

Colmez's product formula for CM abelian varieties

摘录 / EXCERPT: Let X be a CM abelian variety over a number field K, let E be its CM algebra with CM-type Φ, let $H_E=\operatorname{Hom}_Q(E,Q^{\rm alg})$, and assume the notation...

conjecture | open | MathDB 页面 | 原文线索

V2-018395 | MATHDB-340344

A binomial identity for valid compositions

摘录 / EXCERPT: Let n and k be nonnegative integers, and let C_q and D_q be the quantities defined in Theorem 8 for compositions q. Write $\operatorname{Comp}_{k+1}(n-k)$ for the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-018396 | MATHDB-340345

Unimodality conjecture for stack-sorting preimages of increasing-pattern classes

摘录 / EXCERPT: For an integer $m\geq 2$ and $1\leq n\leq 2^{m-1}-1$, let $\operatorname{Av}_n(123\cdots m)$ be the set of permutations in S_n avoiding the increasing pattern $123\cdots m$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-018397 | MATHDB-340356

Chen-Zhu conjecture on irreducible components of affine Deligne-Lusztig varieties

摘录 / EXCERPT: Chen-Zhu conjecture. There exists a natural bijection

conjecture | open | MathDB 页面 | 原文线索

V2-018398 | MATHDB-340357

Hyper-Kähler resolution conjectures for orbifold invariants

摘录 / EXCERPT: Let $\mathcal{X}$ be a smooth algebraic orbifold with coarse moduli space $|\mathcal{X}|$, and suppose there is a crepant resolution $Y\to |\mathcal{X}|$ such that Y is hyper-Käh...

conjecture | open | MathDB 页面 | 原文线索

V2-018399 | MATHDB-340379

Infertility numbers infinitude conjecture

摘录 / EXCERPT: Infertility numbers infinitude conjecture. There are infinitely many infertility numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-018400 | MATHDB-340380

Second exceptional fertility number conjecture

摘录 / EXCERPT: Second exceptional fertility number conjecture. The smallest fertility number that is congruent to 3 modulo 4 and is greater than 27 is 95.

conjecture | open | MathDB 页面 | 原文线索

V2-018401 | MATHDB-340381

Sublinear minimal permutation size conjecture for fertility numbers

摘录 / EXCERPT: Sublinear minimal permutation size conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-018402 | MATHDB-340386

Conjecture on the ordering of weights in determinantal codes

摘录 / EXCERPT: Let $ell leq m$ be positive integers and let t be an integer satisfying $1<t<ell$. For the determinantal code, write $hat{w}_r(t;ell,m)$ for the weight indexed by r. Weight-ord...

conjecture | open | MathDB 页面 | 原文线索

V2-018403 | MATHDB-340400

The conjecture on intersection probabilities with irreducible varieties

摘录 / EXCERPT: Let X be a geometrically irreducible variety of dimension m and degree d in $cmathbb{P}^n(\mathbb{F}_q)$, where q is a prime power, and suppose that X has the simple tang...

conjecture | open | MathDB 页面 | 原文线索

V2-018404 | MATHDB-340409

Nonnegativity conjecture for the BCOV coefficient of maximally unipotent Calabi-Yau threefold degenerations

摘录 / EXCERPT: Let $f \colon X \to \mathbb D$ be a projective degeneration of 3-dimensional Calabi-Yau varieties with maximally unipotent monodromy, and let κ_f denote its BCOV logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-018405 | MATHDB-340415

Kebekus-Kovács hyperbolicity conjecture I

摘录 / EXCERPT: Let $f_U:U\to V$ be a smooth projective family whose general fiber admits a good minimal model. Write $\operatorname{Var}(f_U)$ for the variation of the family. Kebekus-Kovács hype...

conjecture | solved | MathDB 页面 | 原文线索

V2-018406 | MATHDB-340416

Kebekus-Kovács hyperbolicity conjecture II

摘录 / EXCERPT: Let $f_U:U\to V$ be a smooth family of projective varieties, and let (X,D) be a smooth compactification of V with $V\cong X\smallsetminus D$. Assume that the geometric general...

conjecture | open | MathDB 页面 | 原文线索

V2-018407 | MATHDB-340417

Campana's isotriviality conjecture for polarized families with semi-ample canonical bundle

摘录 / EXCERPT: Let $f_U:U\to V$ be a smooth projective family, and suppose that V has a morphism $\mu:V\to P_h$, where P_h is the coarse moduli scheme associated with the moduli functor of po...

conjecture | open | MathDB 页面 | 原文线索

V2-018408 | MATHDB-340422

The Hodge-$\mathcal D$-Conjecture for the constructed regulator cycles

摘录 / EXCERPT: Let d be the degree parameter of the family considered in the source, let X_t be the corresponding fiber, and let $\mathscr{I}_0$, $\mathscr{I}_{\infty}$, and $\mathscr{D}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-018409 | MATHDB-340446

The quasi-finite fundamental group scheme conjecture for affine space

摘录 / EXCERPT: Let R be the base ring, let K be its fraction field, and let $\mathbb{A}^n_R$ be n-dimensional affine space with origin 0. Write $pi^{\text{qf}}(\mathbb{A}^n_R,0)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-018410 | MATHDB-340457

The Persistence Conjecture for arithmetically hyperbolic schemes

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. A finite type separated scheme X over k is arithmetically hyperbolic if, for every $\mathbb{Z}$-finitely genera...

conjecture | open | MathDB 页面 | 原文线索

V2-018411 | MATHDB-340458

The mild boundedness conjecture for arithmetically hyperbolic varieties

摘录 / EXCERPT: Throughout, k is an algebraically closed field of characteristic zero, and a variety over k is a finite type separated integral k-scheme. A variety X over k is arithmetic...

conjecture | open | MathDB 页面 | 原文线索

V2-018412 | MATHDB-340499

The minimal-degeneration conjecture for absolutely special vertices

摘录 / EXCERPT: Let G be a reductive group over the relevant field, let x be an absolutely special vertex, and let $\tGr_{G,x}^{\leq \bar{\mu}}$ be the corresponding normalized Schubert variet...

conjecture | open | MathDB 页面 | 原文线索

V2-018413 | MATHDB-340516

The hare-tortoise fertility gap conjecture

摘录 / EXCERPT: Let w be a word of length m, and let $\langle w\rangle_{\operatorname{\mathsf{hare}}}$ and $\langle w\rangle_{\operatorname{\mathsf{tortoise}}}$ denote its hare- and tortoise-f...

conjecture | open | MathDB 页面 | 原文线索

V2-018414 | MATHDB-340517

The hare-tortoise fertility comparison conjecture

摘录 / EXCERPT: Let w be a word, and let $\langle w\rangle_{\operatorname{\mathsf{hare}}}$ and $\langle w\rangle_{\operatorname{\mathsf{tortoise}}}$ denote its hare- and tortoise-fertilities. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-018415 | MATHDB-340518

The monotonicity conjecture for exceptional-word ratios

摘录 / EXCERPT: For each positive integer m, let $\mathcal E_m$ be the set of exceptional normalized words of length m, and let $\mathcal{NW}_m$ be the set of normalized words of length m. H...

conjecture | open | MathDB 页面 | 原文线索

V2-018416 | MATHDB-340522

Strong mirror symmetry conjecture for Fano manifolds and toric Landau-Ginzburg models

摘录 / EXCERPT: Strong mirror symmetry conjecture. Every pair consisting of a smooth Fano manifold and a divisor on it has a toric Landau-Ginzburg model.

conjecture | open | MathDB 页面 | 原文线索

V2-018417 | MATHDB-340523

Mirror Hodge-number conjecture for Fano manifolds and Landau-Ginzburg models

摘录 / EXCERPT: Mirror Hodge-number conjecture. For every p,q, one has

conjecture | open | MathDB 页面 | 原文线索

V2-018418 | MATHDB-340524

Nef-partition conjecture for smooth well-formed weighted complete intersections

摘录 / EXCERPT: Nef-partition conjecture. The variety X has a good nef-partition and a toric Landau-Ginzburg model.

conjecture | open | MathDB 页面 | 原文线索

V2-018419 | MATHDB-340543

The semi-abelian characterization conjecture for non-Jordan birational automorphism groups

摘录 / EXCERPT: Semi-abelian characterization conjecture. The group $\mathrm{Bir}(X)$ is not Jordan if and only if X is birational to some variety having a faithful action of a non-trivial semi-...

conjecture | open | MathDB 页面 | 原文线索

V2-018420 | MATHDB-340547

The Polydegree Conjecture for plane polynomial automorphisms

摘录 / EXCERPT: Polydegree Conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-018421 | MATHDB-340582

The quasi-coherent description conjecture for D-modules on the affine Grassmannian

摘录 / EXCERPT: Let $\mathcal{W}_n^{\mathrm{triv}}$ be the dg-stack defined from chains of local systems on $\mathcal{D}^*=\operatorname{Spec}(\mathbb{C}((z)))$, with the rank-$n$ local system ide...

conjecture | open | MathDB 页面 | 原文线索

V2-018422 | MATHDB-340595

Gottsche-Kool monopole contribution conjecture for prime-rank refined Vafa-Witten invariants

摘录 / EXCERPT: Let S and c_1 satisfy the hypotheses of Theorem 1 in the source, and let r be prime. Write $\mathrm{VW}_{r,c_1,c_2}^{\mathrm{monopole}}(S,y)$ for the monopole contribution to...

conjecture | open | MathDB 页面 | 原文线索

V2-018423 | MATHDB-340622

Kerz's filtration conjecture for Chow groups of one-cycles

摘录 / EXCERPT: Let A be an excellent henselian discrete valuation ring with uniformising parameter π and residue field k, let X be a smooth projective scheme over...

conjecture | open | MathDB 页面 | 原文线索

V2-018424 | MATHDB-340630

Geometric Broué-Malle conjecture for Deligne-Lusztig varieties

摘录 / EXCERPT: Let $(K,\mathcal{O},k)$ be an ℓ-modular system, let B be a unipotent block with defect group D, and let d be the order of q modulo ℓ. Let C be the reduction mod...

conjecture | open | MathDB 页面 | 原文线索

V2-018425 | MATHDB-340631

Craven-Rouquier perversity conjecture for Broué's equivalence

摘录 / EXCERPT: Let C be the cohomology complex associated with the relevant Deligne-Lusztig variety, and let $\mathcal{S}_{B_0(H)}$ denote the simple-module indexing set for the local principal...

conjecture | open | MathDB 页面 | 原文线索

V2-018426 | MATHDB-340636

The irreducible-component conjecture for the family $\nu_n(\alpha)$

摘录 / EXCERPT: Let $\mathbf{k}$ be the ground field, let $n\geq 1$, and let $\nu_n(\alpha)$ be the n-dimensional terminal algebra ... For an algebra A, write $O(A)$ for its orbit under the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-018427 | MATHDB-340668

Abramovich-Karu's semistable reduction conjecture

摘录 / EXCERPT: Let $f\colon X\rightarrow B$ be a dominant morphism of finite type of integral qe schemes. A morphism is semistable when, in the characteristic-zero setting, its source and target...

conjecture | open | MathDB 页面 | 原文线索

V2-018428 | MATHDB-340675

Derived invariance of semistable degeneration and good reduction

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field K. Let X and Y be derived equivalent smooth projective varieties over K. They have semistable degeneration over R...

conjecture | open | MathDB 页面 | 原文线索

V2-018429 | MATHDB-340687

The conjectured formula for the first minimum zero eigenvalue multiplicity

摘录 / EXCERPT: Conjectured identity. For $w\geq 5$ and $n\geq 2w+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-018430 | MATHDB-340688

Boundary fixed-point conjecture for Coulomb branches

摘录 / EXCERPT: Boundary fixed-point conjecture. The fixed-point set $\mathcal M(\lambda,\mu)^{\chi_i}$ is either empty or isomorphic to a Coulomb branch $\mathcal M_{A_1}(\lambda',\mu')$ of an...

conjecture | open | MathDB 页面 | 原文线索

V2-018431 | MATHDB-340703

Q-torus characterization for maximal projective-amenable action

摘录 / EXCERPT: Let X be a compact Kähler manifold such that $\kappa(X)=0$ and $b_1(X)\neq 0$. Let $G\leq\operatorname{Aut}(X)$ be a subgroup of zero-entropy automorphisms. Say that G has MPA...

conjecture | open | MathDB 页面 | 原文线索

V2-018432 | MATHDB-340706

Kac's cellular decomposition conjecture for indecomposable quiver representations

摘录 / EXCERPT: Kac's cellular decomposition conjecture. This set admits a cellular decomposition by locally closed subvarieties isomorphic to affine spaces, with c_i equal to the number of cell...

conjecture | open | MathDB 页面 | 原文线索

V2-018433 | MATHDB-340715

Finite generation conjecture for the automorphic-form algebra on the stack of G-zips

摘录 / EXCERPT: Let $R_{\rm zip}$ be the graded algebra ... where $\mathscr V(\lambda)$ is the automorphic vector bundle associated with λ. Finite generation conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-018434 | MATHDB-340716

Geometric automorphic forms conjecture for the zip cone

摘录 / EXCERPT: Let C_{GS} denote the cone of weights arising from the geometric automorphic forms considered in the source, and let C_{zip} be the cone of charac...

conjecture | open | MathDB 页面 | 原文线索

V2-018435 | MATHDB-340725

Gusein-Zade, Luengo, and Melle-Hernández's product formula for diagonal $[?]$-equivariant Hilbert schemes

摘录 / EXCERPT: Let the cyclic group $\mathbb{Z}_3$ act diagonally on $\mathbb{C}^2$, and let $b_k(-)$ denote the kth topological Betti number. The equivariant Hilbert scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-018436 | MATHDB-340726

Quantum D-module conjecture for primitive extremal transitions

摘录 / EXCERPT: Let X and Y be smooth projective varieties related by a primitive extremal transition. Denote by $\mathcal{H}(X)$ the ambient part of the quantum D-module of X. After analy...

conjecture | open | MathDB 页面 | 原文线索

V2-018437 | MATHDB-340742

Fullness conjecture for line bundles on smooth toric Fano threefolds and fourfolds

摘录 / EXCERPT: Let X be a smooth toric Fano 3-fold or 4-fold, and consider an exceptional collection of line bundles on X whose length is maximal. Toric Fano fullness conjecture. Every su...

conjecture | open | MathDB 页面 | 原文线索

V2-018438 | MATHDB-340764

Fujita-number subadditivity conjecture for smooth fibrations

摘录 / EXCERPT: Let X and Y be smooth projective varieties, let $\pi\colon X\to Y$ be a smooth fibration, and let f_X, f_Y, and f_π denote the corresponding Fujita numbers. Fujita-numb...

conjecture | open | MathDB 页面 | 原文线索

V2-018439 | MATHDB-340775

Generic rank-one certifying representations for taut genus-two handlebodies

摘录 / EXCERPT: Let M be a taut sutured genus-two handlebody. A representation $\alpha:\pi_1(M)\to\operatorname{GL}_1(\mathbb C)$ is certifying when the inclusion-induced maps ... are isomorphisms....

conjecture | open | MathDB 页面 | 原文线索

V2-018440 | MATHDB-340776

The conjecture on the Satake action of the algebra \(\mathbf J\)

摘录 / EXCERPT: The Satake-action conjecture. Under the Satake isomorphism, the action of $\mathbf J$ coincides with the usual Hecke algebra action on

conjecture | open | MathDB 页面 | 原文线索

V2-018441 | MATHDB-340780

Green-Griffiths conjecture on algebraicity of normal-function zero loci

摘录 / EXCERPT: Let $\mathsf B$ be a complex algebraic manifold, let $\mathcal H$ be a variation of pure negative weight integral Hodge structures over $\mathsf B$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018442 | MATHDB-340781

Charles's arithmetic conjecture for motivated normal-function zero loci

摘录 / EXCERPT: Let $F\subseteq\mathbb C$ be a subfield, and let ν be an algebraically F-motivated (respectively, F-motivated) normal function on a complex algebraic manifold, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-018443 | MATHDB-340794

The exceptional collection conjecture for semisimple homogeneous varieties

摘录 / EXCERPT: Exceptional collection conjecture. There is a full exceptional collection of vector bundles on G/P.

conjecture | open | MathDB 页面 | 原文线索

V2-018444 | MATHDB-340814

Colliot-Thélène-Karpenko-Merkurjev conjecture on essential dimension of projective modules

摘录 / EXCERPT: Let k be a field, let $d\geq 1$, and let D be a central division algebra of degree d over k. For a positive rational number r, write $\operatorname{Mod}_{D,r}$ for the fu...

conjecture | open | MathDB 页面 | 原文线索

V2-018445 | MATHDB-340827

Strassen's additivity conjecture for tensor rank

摘录 / EXCERPT: Let $\mathcal{S}^{\underline{n}}$ be the Segre variety for triple tensors, and let $\mathcal{S}^{(a_1,b_1,c_1)}$ and $\mathcal{S}^{(a_2,b_2,c_2)}$ be the Segre varieties associated...

conjecture | open | MathDB 页面 | 原文线索

V2-018446 | MATHDB-340849

Existence of motivic modular forms spectra over smooth bases

摘录 / EXCERPT: Let S be a smooth base, and let mmf denote a motivic modular forms spectrum, intended to have cohomology given by the motivic analogue of $A/\kern -0.2em/ A(2)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-018447 | MATHDB-340914

The cotangent-bundle isomorphism conjecture for smooth affine varieties

摘录 / EXCERPT: Let X and Y be smooth affine varieties over $\mathbb{C}$, and let $T^*(X)$ and $T^*(Y)$ denote their cotangent bundles. The cotangent-bundle isomorphism conjecture. If ... then .....

conjecture | open | MathDB 页面 | 原文线索

V2-018448 | MATHDB-340915

The companion conjecture for algebraic coefficient objects

摘录 / EXCERPT: Companion conjecture. Any algebraic ℓ-adic coefficient object on X admits an ℓ'-adic companion.

conjecture | open | MathDB 页面 | 原文线索

V2-018449 | MATHDB-340927

Oort's amalgamated-sum conjecture for Newton polygons on the Torelli locus

摘录 / EXCERPT: For $i=1,2$, let $g_i\in\mathbb{Z}_{\ge 1}$, and let ν_i be a symmetric Newton polygon appearing on the open Torelli locus $\mathcal{T}_{g_i}^{\circ}$. Write ... Let ν be the...

conjecture | open | MathDB 页面 | 原文线索

V2-018450 | MATHDB-340944

Finite generation of log canonical rings for projective lc pairs

摘录 / EXCERPT: Finite-generation conjecture. The ring $R(X,\Delta)$ is a finitely generated $\mathbb C$-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-018451 | MATHDB-340950

Equivalence conjecture for effective hearts under ${\mathbb{G}_m}$-stabilization

摘录 / EXCERPT: Work over a perfect field k. For $n\geqslant 1$, let $\mathcal{SH}^{S^1}\!(k)(n)^\heartsuit$ be the heart of the indicated effective filtration on motivic S^1-spectra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018452 | MATHDB-340951

Equivalence conjecture for the effective heart and shifted S^1-stable heart

摘录 / EXCERPT: Let k be a perfect field. For $n\geq 1$, let $\mathcal{SH}(k)^{{\text{eff}}\heartsuit}$ be the effective heart and $\mathcal{SH}^{S^1}\!(k)(n)^\heartsuit$ the heart of the n-th...

conjecture | open | MathDB 页面 | 原文线索

V2-018453 | MATHDB-340954

The Calabi-Yau type conjecture for varieties with polarized endomorphisms

摘录 / EXCERPT: Calabi-Yau type conjecture. Then X is of Calabi-Yau type.

conjecture | open | MathDB 页面 | 原文线索

V2-018454 | MATHDB-340972

The point-count conjecture for the double cover Q_2

摘录 / EXCERPT: Let Q_2 be the double cover considered in the paper, and let $[Q_2]_p$ denote its number of points over $\mathbb F_p$. Let b_p be the Hecke eigenvalue of the newform of weight...

conjecture | open | MathDB 页面 | 原文线索

V2-018455 | MATHDB-340973

The point-count conjecture for the double cover Q_3

摘录 / EXCERPT: Let Q_3 and R_3 be the quotient varieties associated with the involution ι_3, and let $[Q_3]_p$ and $[R_3]_p$ denote their numbers of points over $\mathbb F_p$. Let a_p...

conjecture | open | MathDB 页面 | 原文线索

V2-018456 | MATHDB-340974

The cohomological modularity conjecture for Q_3

摘录 / EXCERPT: Let Q_3 be the quotient variety associated with the involution ι_3. A strongly rigid resolution is a resolution of singularities with the rigidity property intended in the...

conjecture | open | MathDB 页面 | 原文线索

V2-018457 | MATHDB-340975

The point-count conjecture for the quotients of V_{32} by α_1 and α_2

摘录 / EXCERPT: Let V_{32} be the double cover considered in the paper, and let α_1 and α_2 be the two involutions acting on it. Write $[V_{32}/\alpha_i]_p$ for the number of point...

conjecture | open | MathDB 页面 | 原文线索

V2-018458 | MATHDB-340976

The point-count and Calabi-Yau resolution conjecture for V_{32}/G_4

摘录 / EXCERPT: Let V_{32} be the double cover considered in the paper, and let $G_4=\langle\alpha_1,\alpha_2\rangle$. Write $[V_{32}/G_4]_p$ for the number of points of the quotient over...

conjecture | open | MathDB 页面 | 原文线索

V2-018459 | MATHDB-340990

Berest-Le Bruyn transitivity conjecture for quiver varieties

摘录 / EXCERPT: Let $Q=(I,E)$ be a quiver, let $\bar{Q}$ be its double quiver, and let $\mathfrak{M}^{\tau}(Q,\beta)$ be the Nakajima quiver variety associated to parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-018460 | MATHDB-341007

The chiral D-affineness conjecture for quotient stacks

摘录 / EXCERPT: Chiral D-affineness conjecture. The stack quotient $(R\oplus\Pi V)/G$ is chiral D-affine.

conjecture | open | MathDB 页面 | 原文线索

V2-018461 | MATHDB-341033

Weak abelian Schanuel conjecture

摘录 / EXCERPT: Weak abelian Schanuel conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018462 | MATHDB-341035

Strong convergence conjecture for the stochastic Allen-Cahn equation with infinite-dimensional noise

摘录 / EXCERPT: Consider the stochastic Allen-Cahn equation driven by a multiplicative infinite-dimensional ${\bf Q}$-Wiener process, with the drift-implicit Euler-Galerkin scheme and numerical ap...

conjecture | open | MathDB 页面 | 原文线索

V2-018463 | MATHDB-341045

The quadratic-zero conjecture for positive semidefinite quartic forms

摘录 / EXCERPT: Let $P_{n,4}$ denote the space of homogeneous quartic forms on $\mathbb{R}^n$, and let $G\in P_{n,4}$ be positive semidefinite if $G(x)\geq 0$ for all $x\in\mathbb{R}^n$. Suppose t...

conjecture | refuted | MathDB 页面 | 原文线索

V2-018464 | MATHDB-341064

Flattening conjecture for Gorsky-Negut-Rasmussen's tautological complex

摘录 / EXCERPT: Flattening conjecture. The flattening of ${\mathcal{V}}$ is a complex ${\mathcal{E}}$ satisfying the requirements of Gorsky-Negut-Rasmussen's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018465 | MATHDB-341067

Kim's conjecture for motivically integral points on hyperbolic curves

摘录 / EXCERPT: Let $Z\subset\operatorname{Spec}\mathbb{Z}$ be open, let Z_p be the complete local scheme at a prime p of Z, and let X be a hyperbolic curve over Z. In the commuting squa...

conjecture | open | MathDB 页面 | 原文线索

V2-018466 | MATHDB-341103

The SYZ conjecture for mirror Calabi-Yau varieties

摘录 / EXCERPT: SYZ conjecture. There is an affine manifold with singularities B and maps $\phi:X\rightarrow B$ and $\check{\phi}:\check{X}\rightarrow B$ which are dual special Lagrangian fibrat...

conjecture | open | MathDB 页面 | 原文线索

V2-018467 | MATHDB-341113

Conjectural formulas for the Voronoi degree of generic hypersurfaces

摘录 / EXCERPT: Voronoi degree conjecture. The Voronoi degree of a generic hypersurface of degree d in $\mathbb{R}^n$ equals

conjecture | open | MathDB 页面 | 原文线索

V2-018468 | MATHDB-341124

Algebraic quadruple conjecture for basepoint-free pencils of binary quartics

摘录 / EXCERPT: Let $f=(f_1,f_2)$ be a basepoint-free pencil of binary quartics, viewed as a morphism $f\colon\mathbb{P}^1\to\mathbb{P}^1$. Let (E,V,ϕ,ξ) denote an algebraic quadruple, and...

conjecture | open | MathDB 页面 | 原文线索

V2-018469 | MATHDB-341190

Integral Prasanna-Venkatesh regulator relation for base change

摘录 / EXCERPT: Let f be a normalized newform, F the number field in the paper, and f_F its base change. Let $\widetilde{u}_2(f_F)$ be the normalized degree-two period quantity, Ω_f^+...

conjecture | open | MathDB 页面 | 原文线索

V2-018470 | MATHDB-341194

Strict log-concavity conjecture for coefficients of non-equivariant motivic Chern classes

摘录 / EXCERPT: Strict log-concavity conjecture. The coefficients in the $\mathbf{[w]}$-expansion of the non-equivariant $\mC[p]$ classes are strictly log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-018471 | MATHDB-341195

Sign conjecture for motivic Chern class coefficients in new positivity variables

摘录 / EXCERPT: New-variable sign conjecture. The coefficients of the s_i,δ-monomials in the $\mathbf{[w]}$-expansion of the $\mC[p]$ classes have sign $(-1)^{\ell(w)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-018472 | MATHDB-341212

The asymptotic unlabeled compression conjecture for powers of C_5

摘录 / EXCERPT: Let C_5 be the family defined earlier in the source, let $\mathcal F^{*n}$ denote its n-fold join, and let $\mathrm{UCS}(\mathcal F)$ be the minimum size of an unlabeled compre...

conjecture | open | MathDB 页面 | 原文线索

V2-018473 | MATHDB-341226

Conjectural Sato-Tate group formula for even-genus trinomial hyperelliptic curves

摘录 / EXCERPT: Let C_1 be the hyperelliptic curve ... where $g=2k$ is its genus and c is the constant appearing in the defining equation. Let $\operatorname{ST}^0(C_1)$ denote the identity comp...

conjecture | open | MathDB 页面 | 原文线索

V2-018474 | MATHDB-341247

Mazur's First Conjecture on rational points of varieties

摘录 / EXCERPT: Let V be any variety over $\mathbb{Q}$, and let $V(\mathbb{Q})$ and $V(\mathbb{R})$ denote its rational and real points, respectively. Consider the topological closure of...

conjecture | open | MathDB 页面 | 原文线索

V2-018475 | MATHDB-341248

The generalized Mazur conjecture for rings of localized rationals

摘录 / EXCERPT: Let P be the set of all primes and let $W_Q e P$ be a subset of P. Define ... For a variety V over $\mathbb{Q}$, write $V(O_{Q,W_Q})$ for its points over this ring. Generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-018476 | MATHDB-341258

Compliciality conjecture for perfect complexes on quasi-compact quasi-separated schemes

摘录 / EXCERPT: Compliciality conjecture. $\mathcal{P}\mathrm{erf}(X)_{\geqslant 0}$ is n-complicial for some n.

conjecture | open | MathDB 页面 | 原文线索

V2-018477 | MATHDB-341265

King's conjecture on exceptional collections of line bundles

摘录 / EXCERPT: King's conjecture. Every smooth toric variety possesses a full strong exceptional collection of line bundles.

conjecture | open | MathDB 页面 | 原文线索

V2-018478 | MATHDB-341283

Extension of the parabolic Calogero-Moser results to all compositions

摘录 / EXCERPT: Extension conjecture. Theorem, Theorem, and Theorem hold for all compositions α.

conjecture | open | MathDB 页面 | 原文线索

V2-018479 | MATHDB-341302

Kontsevich's holonomic-module and Lagrangian-subvariety conjecture

摘录 / EXCERPT: Let W_n be the n-th Weyl algebra, let irreducible holonomic $\mathcal D$-modules mean irreducible holonomic modules over W_n, and let Lagrangian subvarieties be taken in the...

conjecture | open | MathDB 页面 | 原文线索

V2-018480 | MATHDB-341306

Yau's finiteness conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: A Calabi-Yau threefold is a Calabi-Yau threefold in the usual sense, and its topological data are recorded by the data referred to as ... . This conjecture concerns whether the compa...

conjecture | open | MathDB 页面 | 原文线索

V2-018481 | MATHDB-341308

Classification question for families of differential operators with infinite-dimensional algebraic kernel

摘录 / EXCERPT: Let $\mathbb K$ be the ground field, let $\underline{x}=(x_1,\ldots,x_n)$, and let ... be the universal parameter space of linear partial differential operators on...

open_problem | open | MathDB 页面 | 原文线索

V2-018482 | MATHDB-341314

Weak Manin's conjecture for rational point counting

摘录 / EXCERPT: Let X be a geometrically uniruled smooth projective variety defined over a number field k, let L be a big and nef divisor on X, and let ... be a height function. For a suitab...

conjecture | open | MathDB 页面 | 原文线索

V2-018483 | MATHDB-341315

The rationally connected inequality for the delta-invariant

摘录 / EXCERPT: Let X be a geometrically rationally connected smooth projective variety of dimension n, let L be a big and nef $\mathbb Q$-divisor on X, let ... and let $\delta(X,L)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-018484 | MATHDB-341319

Divisorial computation and dreaminess conjecture for the delta-invariant

摘录 / EXCERPT: Divisorial computation and dreaminess conjecture. If $\delta(X,\Delta)\leq 1$, then $\delta(X,\Delta)$ is computed by a divisor over X, and every divisor over X that computes i...

conjecture | open | MathDB 页面 | 原文线索

V2-018485 | MATHDB-341327

Sakellaridis-Venkatesh conjecture on Plancherel support

摘录 / EXCERPT: Let F be a non-archimedean local field, let $\mathbf{G}$ be a reductive group over F, and let $\mathbf{X}$ be a $\mathbf{G}$-spherical F-variety. Write G for the group of...

conjecture | open | MathDB 页面 | 原文线索

V2-018486 | MATHDB-341336

Fujita's log spectrum conjecture on iterated accumulation points

摘录 / EXCERPT: Work over the complex numbers. For all triples (X,Δ,M), where (X,Δ) is an n-dimensional log smooth variety with Δ a reduced divisor and M an ample Cartier...

conjecture | open | MathDB 页面 | 原文线索

V2-018487 | MATHDB-341338

Schubert degree formula via symplectic mitosis

摘录 / EXCERPT: Schubert degree formula via symplectic mitosis. The degree of the Schubert variety X_w with respect to the weight parameter λ should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-018488 | MATHDB-341351

Semialgebraic structure conjecture for equal-volume Dehn fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be a 1-cusped hyperbolic 3-manifold with volume function $\Theta_{\mathcal{M}}(x,y)$, and let $\mathcal{Z}$ be the analytic variety defined by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018489 | MATHDB-341362

Beilinson's injectivity conjecture for Deligne cycle classes

摘录 / EXCERPT: Beilinson's conjecture. The composition

conjecture | open | MathDB 页面 | 原文线索

V2-018490 | MATHDB-341363

The F^{2}-vanishing form of Beilinson's conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over a number field k, let p be a positive integer, and let $F^{\bullet}CH^{p}(X_{\mathbb{C}})_{\mathbb{Q}}$ be the Bloch-Beilinson filtr...

conjecture | open | MathDB 页面 | 原文线索

V2-018491 | MATHDB-341368

The abelian multiplicative geometric Langlands conjecture

摘录 / EXCERPT: Let E be an elliptic curve, and let $\operatorname{Diff}_q(X)$ denote the category of q-difference modules on a variety or stack X with automorphism q. The abelian multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-018492 | MATHDB-341371

The p-adic period conjecture for motivic iterated integrals

摘录 / EXCERPT: The p-adic period conjecture. For every open integer scheme Z, the period map $\operatorname{per}_{\pf}\colon A(Z)\to\mathbb{Q}_p$ is injective. This is presented as a p-adic a...

conjecture | open | MathDB 页面 | 原文线索

V2-018493 | MATHDB-341379

Colliot-Thélène-Sansuc conjecture on the Hilbert property of unirational varieties

摘录 / EXCERPT: Let X be a unirational variety over a number field. The Colliot-Thélène-Sansuc conjecture. X has the Hilbert property. This conjecture connects the abundance of rational points...

conjecture | refuted | MathDB 页面 | 原文线索

V2-018494 | MATHDB-341396

Eisenbud-Green-Harris Cayley-Bacharach conjecture

摘录 / EXCERPT: Let Γ be any subscheme of a zero-dimensional complete intersection of hypersurfaces of degrees $d_1\leq\dots\leq d_c$ in a projective space P^c. Say that Γ fails to...

conjecture | open | MathDB 页面 | 原文线索

V2-018495 | MATHDB-341399

Refined Manin conjecture with the canonical exceptional set

摘录 / EXCERPT: Let X be a geometrically uniruled, geometrically integral, smooth projective variety over a number field F, and let L be a big and nef $\mathbb Q$-divisor on X. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-018496 | MATHDB-341406

The symplectic-singularity conjecture for multiplicative quiver varieties

摘录 / EXCERPT: Let $\mathcal{M}_{q,\theta}(Q,\alpha)$ denote a multiplicative quiver variety. A symplectic singularity is a normal singular variety whose smooth locus carries a symplectic form ex...

conjecture | open | MathDB 页面 | 原文线索

V2-018497 | MATHDB-341407

Simpson's isosingularity conjecture for filtered local systems and parabolic Higgs bundles

摘录 / EXCERPT: Let X be a compact Riemann surface, let $D\subset X$ be a finite set of marked points, and fix corresponding parameters for filtered local systems and parabolic Higgs bundles of...

conjecture | open | MathDB 页面 | 原文线索

V2-018498 | MATHDB-341408

The symplectic-resolution correspondence for parabolic Higgs moduli spaces

摘录 / EXCERPT: Let X be a compact Riemann surface with marked points and fixed parabolic parameters, and let the corresponding character variety parametrize representations of the fundamental g...

conjecture | open | MathDB 页面 | 原文线索

V2-018499 | MATHDB-341419

Containment and reduction for cherry-unresolved reticulated-cherry stack CPNs

摘录 / EXCERPT: Let N and N' be CPNs on the same leaf-set in the $(,)$ class. Here, a cherry-picking sequence (CPS) of N is a sequence used to reduce N', and N' is contained in N acco...

conjecture | open | MathDB 页面 | 原文线索

V2-018500 | MATHDB-341445

Homma-Kim's conjectural bound for rational points on even-dimensional hypersurfaces

摘录 / EXCERPT: Homma-Kim's conjecture. The number of rational points satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-018501 | MATHDB-341473

Numerical $P=W$ conjecture for the Hitchin fibration

摘录 / EXCERPT: Numerical $P=W$ conjecture. The perverse numbers of the Hitchin fibration and the Hodge numbers of the corresponding character variety should agree:

conjecture | open | MathDB 页面 | 原文线索

V2-018502 | MATHDB-341482

Chai-Faltings congruence conjecture for PEL-type Shimura varieties

摘录 / EXCERPT: Let Sh_K be a Kottwitz PEL-type Shimura variety attached to (G,X), with $K=K_pK^p$, where G is nonramified at p and K_p is hyperspecial. Let E be the reflex field, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018503 | MATHDB-341492

Chai's formal-torus conjecture for ordinary points on mod p Shimura varieties

摘录 / EXCERPT: Chai's conjecture. Then X is a Shimura subvariety.

conjecture | open | MathDB 页面 | 原文线索

V2-018504 | MATHDB-341515

Conjecture on principally polarized varieties in almost ordinary isogeny classes

摘录 / EXCERPT: Principally polarized isogeny-class size conjecture. For a positive proportion of integers n, their number is

conjecture | open | MathDB 页面 | 原文线索

V2-018505 | MATHDB-341522

Classification conjecture for the Exists Right-Factor Problem and bounded f-cores

摘录 / EXCERPT: Classification conjecture. The Exists Right-Factor Problem with fixed algebra Z is in polynomial time for a given variety if and only if the variety has bounded f-cores.

conjecture | open | MathDB 页面 | 原文线索

V2-018506 | MATHDB-341529

Compatibility conjecture for the Betti realization of perverse motives

摘录 / EXCERPT: Let X be a smooth k-variety. Let $d4b3(X)$ be the category of perverse motives constructed by Ivorra, and let ... be the triangulated functor constructed by Ivorra. Betti-realiza...

conjecture | open | MathDB 页面 | 原文线索

V2-018507 | MATHDB-341543

Absolute equivariant DT/PT correspondence conjecture for renormalized descendents

摘录 / EXCERPT: Let X be a toric variety with torus action T, let $\gamma\in H^*_T(X)$, and let $\widehat{\mathrm{ch}}z$ be the renormalized descendent generating function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018508 | MATHDB-341556

Birkar's bounded complement conjecture for log canonical Fano pairs

摘录 / EXCERPT: Let d be a natural number and let $\mathfrak{R}\subset [0,1]$ be a finite set of rational numbers. For a finite set $\mathfrak{R}$, write ... An n complement of K_X+B is a divi...

conjecture | open | MathDB 页面 | 原文线索

V2-018509 | MATHDB-341557

Bounded canonical index conjecture for semi-log canonical Calabi-Yau pairs

摘录 / EXCERPT: Let d be a natural number and let $\mathfrak{R}\subset [0,1]$ be a finite set of rational numbers. For a finite set $\mathfrak{R}$, write ... A pair (X,B) is semi-log canonical i...

conjecture | open | MathDB 页面 | 原文线索

V2-018510 | MATHDB-341558

Uniform admissible-section conjecture for disconnected klt Calabi-Yau pairs

摘录 / EXCERPT: For a positive integer n and a natural number d, let $A(X,D)$ denote the space of admissible sections associated with the divisor D. Uniform admissible-section conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-018511 | MATHDB-341581

Voisin's alternating zero-cycle conjecture for surfaces

摘录 / EXCERPT: Let S be a smooth projective surface, and let $A^2_{AJ}(S)_{\mathbb{Z}}$ denote the group of Abel-Jacobi-trivial 0-cycles on S. For an integer $m>p_g(S)$, where $p_g(S)$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-018512 | MATHDB-341592

Conjecture on special manifolds and lifted entire curves

摘录 / EXCERPT: Let X be a projective manifold, let $X_1:=\mathbb P(T_X)$ be the projectivized tangent bundle, and for every entire curve $f:\mathbb C\to X$ let $f_{[1]}:\mathbb C\to X_1$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-018513 | MATHDB-341653

Finite-rank Lehmer conjecture for products of tori and abelian varieties

摘录 / EXCERPT: Finite-rank Lehmer conjecture for products. There exists $c_\Gamma>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018514 | MATHDB-341660

Augmentation-polynomial conjecture for monotone Lagrangian two-tori

摘录 / EXCERPT: Let $L\subset(\mathbb{C}P^2,\omega_{\operatorname{FS}})$ be an embedded monotone Lagrangian two-torus, and let Λ be the Legendrian lift of its canonical threefold Bohr-Somm...

conjecture | open | MathDB 页面 | 原文线索

V2-018515 | MATHDB-341662

Surjectivity conjecture for products of sections of odd theta characteristics

摘录 / EXCERPT: Surjectivity conjecture. The map is surjective when D and $D+v$ range over all effective odd theta characteristic divisors. The conjecture would ensure that the conditional step...

conjecture | open | MathDB 页面 | 原文线索

V2-018516 | MATHDB-341701

Diagonal dimension conjecture for irrational curves

摘录 / EXCERPT: Diagonal dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018517 | MATHDB-341708

Primality conjecture for odd-size jet pfaffian ideals

摘录 / EXCERPT: Let r and k be positive integers, and let $I^{2r+1,k}_r$ denote the jet pfaffian ideal associated with the ideal of size-$2r$ pfaffians of a generic $(2r+1)\times(2r+1)$ skew-s...

conjecture | open | MathDB 页面 | 原文线索

V2-018518 | MATHDB-341723

The deformation-limit conjecture over a complex variety

摘录 / EXCERPT: Let $\pi:\mathcal{X}\rightarrow Y$ be a holomorphic family of compact complex manifolds over a complex variety Y, let $V\subset Y$ be a proper subvariety, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-018519 | MATHDB-341762

Conjecture on smooth and oversized Hodge loci for cubic hypersurfaces

摘录 / EXCERPT: Let X_0 be a cubic hypersurface in the full family of cubic hypersurfaces, and let $\mathbb{P}^{\frac{n}{2}}$ and $\check{\mathbb{P}}^{\frac{n}{2}}$ be linear cycles in X_0. Fo...

conjecture | solved | MathDB 页面 | 原文线索

V2-018520 | MATHDB-341768

Positive-characteristic structure conjecture for Suslin homology

摘录 / EXCERPT: Positive-characteristic structure conjecture. There are a uniquely divisible group V, integers $s_{i,\ell}$ and r_i, and a torsion group...

conjecture | open | MathDB 页面 | 原文线索

V2-018521 | MATHDB-341776

Voisin's gonality conjecture for very general abelian varieties

摘录 / EXCERPT: Let $k\in\mathbb{Z}_{>0}$, and let A be a very general abelian variety. Its gonality is the least degree of a nonconstant map from a smooth projective curve to A. Voisin's gona...

conjecture | open | MathDB 页面 | 原文线索

V2-018522 | MATHDB-341777

Voisin's strong orbit conjecture for very general abelian varieties

摘录 / EXCERPT: Let $k\in\mathbb{Z}_{>0}$, let A be a very general abelian variety, and let $\mathscr{G}(k)$ be the least dimension such that a very general abelian variety of that dimension has...

conjecture | open | MathDB 页面 | 原文线索

V2-018523 | MATHDB-341778

Modified André-Pink-Zannier conjecture over a curve

摘录 / EXCERPT: Modified André-Pink-Zannier conjecture. If $\mathcal{A}_{\Gamma}\cap\mathcal{V}$ is Zariski dense in $\mathcal{V}$, then one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-018524 | MATHDB-341786

The link and double-point image criterion for bi-Lipschitz equivalence of map germs

摘录 / EXCERPT: Link and double-point image criterion. The map germs f and g are bi-Lipschitz equivalent if and only if $link(f)$ and $link(g)$ are bi-Lipschitz equivalent and the images...

conjecture | open | MathDB 页面 | 原文线索

V2-018525 | MATHDB-341820

Real Jacobian conjecture

摘录 / EXCERPT: Let $F:\mathbb{R}^n\to\mathbb{R}^n$ be a polynomial map. Real Jacobian conjecture. If ... then F is injective. The source explains that this real analogue is false: Pinchuk constru...

conjecture | open | MathDB 页面 | 原文线索

V2-018526 | MATHDB-341836

Polynomial formulas for signed pancake stacks requiring five to nine flips

摘录 / EXCERPT: Let $R_k^B(n)$ denote the number of signed permutations of size n whose burnt-pancake distance from the sorted stack is k. Polynomial formulas conjecture. If $n\geq1$, then ... .

conjecture | open | MathDB 页面 | 原文线索

V2-018527 | MATHDB-341837

Gregory-Newton summation identity for signed pancake distances

摘录 / EXCERPT: Let $R_k^B(n)$ denote the number of signed permutations of size n whose burnt-pancake distance from the sorted stack is k. Summation identity conjecture. If $k,n\geq1$, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-018528 | MATHDB-341874

Koszulness conjecture for De Concini-Procesi wonderful models

摘录 / EXCERPT: Wonderful-model Koszulness conjecture. The ring

conjecture | open | MathDB 页面 | 原文线索

V2-018529 | MATHDB-341876

Yoshihara's conjecture on complements of irreducible plane curves

摘录 / EXCERPT: Yoshihara's conjecture. Then C and D are projectively equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-018530 | MATHDB-341886

The rational Schubert Galois group conjecture

摘录 / EXCERPT: Let ${\boldsymbol{\lambda}}$ be a Schubert problem, and let $\text{\rm Gal}_{{\boldsymbol{\lambda}}}$ be its geometric Schubert Galois group and...

conjecture | open | MathDB 页面 | 原文线索

V2-018531 | MATHDB-341948

Generalized Sturm separation inequality for non-globally convex curves

摘录 / EXCERPT: Let $n\geq2$, and let $\gamma:\mathbb S^1\to\mathbb RP^{n-1}$, or $\gamma:I\to\mathbb RP^{n-1}$, be locally but not globally convex. For a complete flag f, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018532 | MATHDB-342006

Codogni-Dervan's Loewy filtration conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold with automorphism group $\operatorname{Aut}(X)$. A Loewy filtration is the canonical filtration of the section ring constructed from the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-018533 | MATHDB-342011

Brady-McCammond's CAT(0) conjecture for the orthoscheme complex of non-crossing partitions

摘录 / EXCERPT: Let $\operatorname{NC}(n)$ be the lattice of non-crossing partitions, and let its orthoscheme complex be its simplicial realisation equipped with the orthoscheme metric obtained by...

conjecture | open | MathDB 页面 | 原文线索

V2-018534 | MATHDB-342034

Adaptive-radius homology computation conjecture for intersections of balls

摘录 / EXCERPT: Adaptive-radius homology computation conjecture. One can compute the homology of

conjecture | open | MathDB 页面 | 原文线索

V2-018535 | MATHDB-342037

de Jong-Debarre conjecture on lines in smooth hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a smooth hypersurface of degree d, and let $F_1(X)\subset\mathbb{G}(1,n)$ denote the Hilbert scheme of lines contained in X. The expected dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-018536 | MATHDB-342065

Real separability conjecture for elliptic inflection polynomials

摘录 / EXCERPT: Let $f=x(x-1)(x-\lambda)$ be the Legendre cubic with three distinct real roots, so $\lambda\in\mathbb{R}\setminus\{0,1\}$. For each pair of indices μ,k considered in the paper,...

conjecture | open | MathDB 页面 | 原文线索

V2-018537 | MATHDB-342066

Real-root count conjecture for elliptic inflection polynomials

摘录 / EXCERPT: Let $\mu\geq 1$ and $k\geq\mu+1$ be nonnegative integers, let f be associated with a real Legendre parameter λ, and fix $\lambda\neq 0,1$. Let $P_{\mu,k}$ be the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-018538 | MATHDB-342086

The obstruction conjecture for the expected dimension of path signature varieties

摘录 / EXCERPT: Obstruction conjecture. The variety $\mathcal{A}_{\nu,k}$ is defective if and only if there is a subsequence $\nu'=(\nu_i,\dots,\nu_{i+r})$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018539 | MATHDB-342091

The extension conjecture for varieties containing affine subspaces

摘录 / EXCERPT: The extension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018540 | MATHDB-342111

Effective-bound conjecture for incidence strata

摘录 / EXCERPT: Effective-bound conjecture. These three quantities equal 2^k-1 for $k\geq 1$. This conjecture gives the expected sharp value of the embedding dimension bound for the incidence st...

conjecture | open | MathDB 页面 | 原文线索

V2-018541 | MATHDB-342112

Reduced-fiber stratification conjecture for incidence strata

摘录 / EXCERPT: Let X be the variety occurring in the family $\Delta^k(X)$ over $(\mathbb{C}^k)^{\circ}$. For $(m_1,\ldots,m_k),(n_1,\ldots,n_k)\in(\mathbb{C}^k)^{\circ}$, define an equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-018542 | MATHDB-342116

Kato-Saito local-to-global conjecture for zero-cycles

摘录 / EXCERPT: Let K be a number field, let X be a smooth projective geometrically integral variety over K, and let P_K be the set of places of K. Write X_{K_v} for the base change to...

conjecture | open | MathDB 页面 | 原文线索

V2-018543 | MATHDB-342117

Exactness conjecture for the higher Chow cycle-class complex

摘录 / EXCERPT: Let K be a global field, let X/K be a smooth projective geometrically integral variety of dimension d, let n be prime to the characteristic of K, and let $i>0$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-018544 | MATHDB-342123

Gamma II conjecture for Fano varieties

摘录 / EXCERPT: Let X be a Fano variety with semisimple quantum cohomology at a point ${\boldsymbol\tau}\in H^2(X;\mathbb{C})$. Let $\theta\in\mathbb{R}$ be an admissible phase, meaning that the...

conjecture | open | MathDB 页面 | 原文线索

V2-018545 | MATHDB-342152

The P=W identity for diffeomorphic cluster varieties and Lefschetz torus fibrations

摘录 / EXCERPT: P=W conjecture. For any diffeomorphism $f:X\to Y$, the $P=W$ identity holds under f^*.

conjecture | open | MathDB 页面 | 原文线索

V2-018546 | MATHDB-342154

The Cat-HA version of the non-abelian Hodge correspondence

摘录 / EXCERPT: Let X be a smooth projective complex curve. The derived stack $\mathbf{Coh}_{/\mathbb A^1}^\ast(X_\mathsf{Del})$ parametrizes Deligne's λ-connections, interpolating betwe...

conjecture | open | MathDB 页面 | 原文线索

V2-018547 | MATHDB-342168

The weak Gelfand-type classification conjecture for the Cayley plane and Freudenthal variety

摘录 / EXCERPT: Let $\mathcal{O}_q(\mathbb{OP}^2)$ be the quantum Cayley plane and $\mathcal{O}_q(\mathbf{F})$ the quantum Freudenthal variety, each equipped with its Heckenberger--Kolb calculus.…

conjecture | open | MathDB 页面 | 原文线索

V2-018548 | MATHDB-342172

Zaidenberg's torus characterization of affine varieties without additive actions

摘录 / EXCERPT: Let X be an affine variety over an algebraically closed field ${\mathbb K}$ of characteristic zero. Let ${\mathbb G}_a$ denote the additive group of ${\mathbb K}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018549 | MATHDB-342173

Perepechko-Regeta conjecture on additive actions and nested automorphism groups

摘录 / EXCERPT: Let X be an affine variety over an algebraically closed field ${\mathbb K}$ of characteristic zero. Let ${\mathcal U}(X)\subset \operatorname{Aut}(X)$ be the subgroup generated b...

conjecture | open | MathDB 页面 | 原文线索

V2-018550 | MATHDB-342184

Suspension conjecture for topologically contractible smooth complex varieties

摘录 / EXCERPT: Let X be a topologically contractible smooth complex variety with a chosen base-point $x\in X({\mathbb C})$. Write $\Sigma^n_{{\mathbb P}^1}(X,x)$ for the n-fold...

conjecture | open | MathDB 页面 | 原文线索

V2-018551 | MATHDB-342185

A1-contractibility conjecture for topologically contractible surfaces

摘录 / EXCERPT: Let X be a smooth topologically contractible surface. Surface contractibility conjecture. The variety X is ${\mathbb A}^1$-contractible if and only if it is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-018552 | MATHDB-342186

Suspension conjecture for topologically contractible smooth complex affine varieties

摘录 / EXCERPT: Let X be a topologically contractible smooth complex affine variety, and let $x\in X({\mathbb C})$ be a chosen base-point. Write $\Sigma^n_{{\mathbb P}^1}(X,x)$ for the n-fold...

conjecture | open | MathDB 页面 | 原文线索

V2-018553 | MATHDB-342189

Additive realization conjecture for multiplicative hypertoric varieties

摘录 / EXCERPT: Additive realization conjecture. Every multiplicative hypertoric variety admits an additive realization: there exists an additive hypertoric variety $\mathfrak{M}$, a divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-018554 | MATHDB-342202

Gillet-Soulé's non-negativity conjecture for arithmetic-surface cohomology

摘录 / EXCERPT: Let $\pi: X\rightarrow B=\operatorname{Spec}(\mathcal{O}_{K})$ be an integral, flat, projective scheme of dimension 2, and let L be a metrized line bundle on X. Suppose there...

conjecture | open | MathDB 页面 | 原文线索

V2-018555 | MATHDB-342205

Bóna's upper-bound conjecture for t-stack-sortable permutations

摘录 / EXCERPT: Let $W_t(n)$ denote the number of t-stack-sortable permutations of length n, for integers $n,t\geq 1$. Bóna's upper-bound conjecture. For all $n,t\geq 1$, we have ... This is one...

conjecture | open | MathDB 页面 | 原文线索

V2-018556 | MATHDB-342206

Bóna's parity conjecture for iterated stack-sortable permutations

摘录 / EXCERPT: Let $W_t(n)$ denote the number of t-stack-sortable permutations of length n. Bóna's parity conjecture. If t is even, then $W_t(n)$ is frequently odd. If t is odd, then...

conjecture | open | MathDB 页面 | 原文线索

V2-018557 | MATHDB-342207

Growth-rate conjecture for 3-stack-sortable permutations

摘录 / EXCERPT: Let $W_3(n)$ denote the number of 3-stack-sortable permutations of length n. Growth-rate conjecture. The exponential growth rate satisfies ... The bounds were proposed from numeric...

conjecture | open | MathDB 页面 | 原文线索

V2-018558 | MATHDB-342229

Semistable families are related by admissible blow-ups

摘录 / EXCERPT: Let X and Y be two semistable families over S with an isomorphism ... of generic fibers. A semistable-family modification conjecture. There is a sequence of rational maps ... suc...

conjecture | open | MathDB 页面 | 原文线索

V2-018559 | MATHDB-342247

Unitary representations of the four-generator relation group approximate complex curves

摘录 / EXCERPT: Unitary representation conjecture. Representations of this group approximate, as $N\to+\infty$, complex curves in the abelian variety $(\mathbb{C}/(\mathbb{Z}+i\mathbb{Z}))^2$. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-018560 | MATHDB-342263

Regularity conjecture for zero-dimensional subschemes in toric varieties

摘录 / EXCERPT: Regularity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-018561 | MATHDB-342264

The Frobenius structure conjecture: weak Gromov-Witten version

摘录 / EXCERPT: Weak Gromov-Witten Frobenius structure conjecture. There exists a product $*$ on $\operatorname{QH}_{\log}^0(Y,D)$ making it a commutative associative...

conjecture | open | MathDB 页面 | 原文线索

V2-018562 | MATHDB-342268

The DT/PT correspondence for Calabi-Yau 4-folds

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau 4-fold, let $I_n(X,\beta)$ be the Hilbert scheme of one-dimensional subschemes with Chern character $(0,0,0,\beta,n)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018563 | MATHDB-342285

Finite-file PIR rate upper-bound conjecture for coded colluding servers

摘录 / EXCERPT: Finite-file PIR rate conjecture. Any such PIR scheme has rate at most

conjecture | open | MathDB 页面 | 原文线索

V2-018564 | MATHDB-342290

The rank-two motivic extension conjecture for multiple Lambda-values

摘录 / EXCERPT: Let $\mathcal{E}$ be an extension of $\mathbb{Q}$ by a rank-two motive M in a suitable category of realisations of motives over $\mathbb{Q}$, with weight $m\leq -2$: ... Assume tha...

conjecture | open | MathDB 页面 | 原文线索

V2-018565 | MATHDB-342310

Massey's same-cluster query lower-bound conjecture

摘录 / EXCERPT: Let $k=2$, let $\bfX\in\{0,1\}^n$ be the label vector, and let a querying scheme use m same cluster' queries, each involving two elements ($\Delta=2$). A scheme is $(1-\delta)$-g...

conjecture | open | MathDB 页面 | 原文线索

V2-018566 | MATHDB-342341

The directed-graph containment conjecture for Landau-Ginzburg model moduli

摘录 / EXCERPT: Let the directed graphs of Fano varieties be those from Theorem, and let the associated Landau-Ginzburg models be as defined in DKK13. The Landau-Ginzburg moduli containment conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-018567 | MATHDB-342342

The directed-graph containment conjecture for stability-condition moduli

摘录 / EXCERPT: Let the directed graphs of Fano varieties be those from Theorem, and consider the associated categories and their Bridgeland stability conditions. The stability-condition moduli co...

conjecture | open | MathDB 页面 | 原文线索

V2-018568 | MATHDB-342343

The strong mirror-symmetry conjecture for toric Landau-Ginzburg models

摘录 / EXCERPT: Let X be a smooth Fano variety. A toric Landau-Ginzburg model for X is a toric Landau-Ginzburg model associated with X as in the source. Strong version of Mirror Symmetry for...

conjecture | open | MathDB 页面 | 原文线索

V2-018569 | MATHDB-342356

Schneider's conjecture on Tate-Shafarevich groups

摘录 / EXCERPT: Let L/K be the number-field extension and S the finite set of places fixed above, and let $\Sha^1(\mathcal{O}_{L,S},\mathbb{Z}_p(r))$ denote the Tate-Shafarevich group, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-018570 | MATHDB-342392

Hilbert-scheme model for a wrapped unknot invariant

摘录 / EXCERPT: Let $\operatorname{Hilb}^n(\mathbb{C}^2)$ be the Hilbert scheme of n points, let $\mathcal{I}$ be its tautological ideal sheaf, and let X,Y be its two commuting endomorphisms....

conjecture | open | MathDB 页面 | 原文线索

V2-018571 | MATHDB-342409

Big tautological classes conjecture for Hilbert schemes of points on K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface and let $X={\operatorname{Hilb}}_n(S)$ be the Hilbert scheme of n points on S. Let π_1 and π_2 be the projections from $X\times S$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018572 | MATHDB-342415

The conjecture on FK3 complete intersections in Grassmannians

摘录 / EXCERPT: Let $X=X_{d_1,\ldots,d_c}\subset\operatorname{Gr}(k,n)$ be a smooth Fano complete intersection of even dimension. A Fano variety is of K3 type if its cohomology contains a K3-type...

conjecture | open | MathDB 页面 | 原文线索

V2-018573 | MATHDB-342430

Gupta's three-set polynomial Sylvester-Gallai conjecture

摘录 / EXCERPT: Gupta's three-set polynomial Sylvester-Gallai conjecture. There is a function $\lambda(r)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018574 | MATHDB-342456

Closedness conjecture for uniform matrix product state varieties

摘录 / EXCERPT: Closedness conjecture. Except for the trivial cases in Proposition, $\operatorname{uMPS}(D,d,N)$ is only closed if it fills the ambient space $\operatorname{Cyc}^N(\mathbb{C}^d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-018575 | MATHDB-342514

The curve semistability conjecture for Higgs bundles

摘录 / EXCERPT: Curve semistability conjecture. A curve semistable Higgs bundle ${\mathfrak E}$ has vanishing discriminant, $\Delta(E)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-018576 | MATHDB-342542

Fine-Panov conjecture on six-dimensional monotone Hamiltonian circle manifolds

摘录 / EXCERPT: Let (M,ω) be a six dimensional closed monotone symplectic manifold, meaning that $\langle c_1(TM),[\Sigma]\rangle>0$ for every symplectic surface Σ, equipped with an...

conjecture | open | MathDB 页面 | 原文线索

V2-018577 | MATHDB-342548

Lagrangianity conjecture for the attracting scheme

摘录 / EXCERPT: Lagrangianity conjecture. The attracting scheme $\mathfrak Z$ is a Lagrangian subscheme of $\mathfrak M_1$.

conjecture | refuted | MathDB 页面 | 原文线索

V2-018578 | MATHDB-342588

Brambilla-Dumitrescu-Postinghel conjecture for divisors on blow-ups at $n+3$ points

摘录 / EXCERPT: Let $X_{n,s}$ be the blow-up of projective n-space at s points, and let D be a divisor. For $I\subset\{1,\ldots,n+3\}$ and a secant parameter t, define ... and ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-018579 | MATHDB-342592

Bondal's degeneracy-locus conjecture for Poisson structures on Fano varieties

摘录 / EXCERPT: Let X be a connected Fano variety of dimension n and let π be a Poisson structure on it. For an integer $k\geq 0$ satisfying $2k<n$, define the degeneracy locus...

conjecture | open | MathDB 页面 | 原文线索

V2-018580 | MATHDB-342594

The absolute index conjecture for klt Calabi-Yau varieties

摘录 / EXCERPT: The absolute index conjecture. There exists a positive integer n, depending only on d, such that

conjecture | open | MathDB 页面 | 原文线索

V2-018581 | MATHDB-342638

Conjecture on rational points of terminal Fano varieties over C_1 fields

摘录 / EXCERPT: Let k be a field of characteristic 0 that is C_1, meaning that every hypersurface of degree at most n in $\mathbb{P}^n_k$ has a k-rational point. Let X be a terminal...

conjecture | open | MathDB 页面 | 原文线索

V2-018582 | MATHDB-342647

Riemann-Hilbert correspondence for discrete Painlevé equations

摘录 / EXCERPT: The generalised Halphen surface and the corresponding Looijenga pair are the geometric objects associated with the discrete Painlevé equation under consideration. Riemann-Hilbert c...

conjecture | open | MathDB 页面 | 原文线索

V2-018583 | MATHDB-342657

Duality conjecture for the genus-five Prym fibration

摘录 / EXCERPT: Let S' be a K3 surface that is a branched double cover of a degree-one del Pezzo surface T', and let D' be a genus-two curve on T' covered by a genus-five curve C' on...

conjecture | open | MathDB 页面 | 原文线索

V2-018584 | MATHDB-342658

Stringy Hodge-number conjecture for the Debarre dual fibration

摘录 / EXCERPT: Let $X\rightarrow\mathbb{P}^g$ be a Debarre system and let $\widehat{X}$ be its dual fibration, with fibres related by dual exact sequences of Jacobians and abelian varieties. The...

conjecture | open | MathDB 页面 | 原文线索

V2-018585 | MATHDB-342669

The motivic analogue of the Tate conjecture

摘录 / EXCERPT: Assume that the field k is finitely generated, and let M be a motive over k. Write $\mathop{\mathrm{G}}_{\ell,k}(M)$ for the ℓ-adic monodromy group of M, and...

conjecture | open | MathDB 页面 | 原文线索

V2-018586 | MATHDB-342683

Grothendieck's Standard Conjecture D

摘录 / EXCERPT: Grothendieck's Standard Conjecture D. For any field k, Weil cohomology theory H^*, and smooth, projective variety X over k, if two cycles on X are numerically equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-018587 | MATHDB-342690

Chevalley restriction conjecture for the commuting quotient

摘录 / EXCERPT: Chevalley restriction conjecture. This morphism is an isomorphism. Equivalently, the categorical quotient ${\mathfrak C}^d_G\sslash G$ is reduced and normal. The source records thi...

conjecture | open | MathDB 页面 | 原文线索

V2-018588 | MATHDB-342691

Universal spectral data morphism conjecture

摘录 / EXCERPT: Let G be a reductive group with Lie algebra ${\mathfrak g}$, let ${\mathfrak t}$ be a Cartan subalgebra, let W be the Weyl group, and let ${\mathfrak C}^d_G$ be the scheme of c...

conjecture | open | MathDB 页面 | 原文线索

V2-018589 | MATHDB-342692

Surjectivity conjecture for the Hitchin morphism onto the postulated image

摘录 / EXCERPT: Let X be a proper smooth algebraic variety over an algebraically closed field k of characteristic zero, let $\mathcal{M}_X$ be the moduli stack of Higgs bundles, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018590 | MATHDB-342720

The semicontinuity conjecture for minimal log discrepancies

摘录 / EXCERPT: The semicontinuity conjecture for minimal log discrepancies. The function

conjecture | open | MathDB 页面 | 原文线索

V2-018591 | MATHDB-342731

The interval-fibre conjecture for Minkowski sums of moment curves

摘录 / EXCERPT: For integers k,n with $k>3$, let $\mathcal{A}_{k,n}$ denote the Minkowski sum of n copies of the moment-curve segment $\{(t,t^2,\ldots,t^k)\mid -1\leq t\leq 1\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018592 | MATHDB-342738

Klingler's conjecture on the distribution of CM points

摘录 / EXCERPT: Let S be a smooth connected quasi-projective variety over $\mathbb{C}$ and let $f:\mathcal{X}\rightarrow S$ be a smooth projective family of relative dimension n. The family ca...

conjecture | open | MathDB 页面 | 原文线索

V2-018593 | MATHDB-342743

Zhang's Zariski dense orbit conjecture for polarized endomorphisms

摘录 / EXCERPT: Zhang's conjecture. There exists a point $x\in X({\mathbf{k}})$ whose orbit $O_f(x)$ is Zariski dense in X. This is a polarized special case of the Zariski dense orbit problem an...

conjecture | open | MathDB 页面 | 原文线索

V2-018594 | MATHDB-342744

Ghioca-Tucker's dynamical Mordell-Lang conjecture for rational self-maps

摘录 / EXCERPT: Let ${\mathbf{k}}$ be an algebraically closed field of characteristic 0, let X be a variety defined over ${\mathbf{k}}$, and let $f:X\dashrightarrow X$ be a rational self-map....

conjecture | open | MathDB 页面 | 原文线索

V2-018595 | MATHDB-342745

Nonemptiness conjecture for the set of regular nilpotent orbits

摘录 / EXCERPT: Let (G,H) be a spherical pair obtained by parabolic induction from (G_0,H_0,ξ), and let ${\mathcal{N}}(G,H,\xi)$ be the subset of $Nil_{reg}({\mathfrak{g}}(F))$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-018596 | MATHDB-342746

Euler-Poincaré multiplicity formula conjecture for spherical varieties

摘录 / EXCERPT: Let $G(F)$ be a reductive group and $H(F)$ a spherical subgroup with character data $\omega\otimes\xi$. For a finite length smooth representation π of $G(F)$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-018597 | MATHDB-342747

Conjectural trace formula for spherical varieties

摘录 / EXCERPT: Let $I(f)$ be the trace-formula distribution for a test function f, and let $I_{geom}(f)=m_{geom}(\theta_f)$ be its geometric expansion. Define $I_{spec}(f)$ by the cuspidal spec...

conjecture | open | MathDB 页面 | 原文线索

V2-018598 | MATHDB-342766

Ayoub's contraction conjecture for n-motives

摘录 / EXCERPT: Let $\boldsymbol{k}$ be a perfect field, let R be a $\mathbb{Q}$-algebra, and let $\mathrm{DM}^{\mathrm{eff}}_{\acute et}(\boldsymbol{k},R)$ be the effective étale motivic catego...

conjecture | open | MathDB 页面 | 原文线索

V2-018599 | MATHDB-342773

Dodgson-intersection model conjecture for nonzero c_2-invariants

摘录 / EXCERPT: A c_2-invariant has a minimal model if it is represented by a projective variety of smallest possible dimension, and a Dodgson intersection is the variety defined by an intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-018600 | MATHDB-342774

Quadratic-reduction dimension bound for c_2-invariants

摘录 / EXCERPT: Let G be a primitive graph with first Betti number $h_1(G)$, and suppose its quadratic n-invariant ${}^n\Psi_G^2$ exists. Quadratic-reduction dimension conjecture. Then ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-018601 | MATHDB-342775

Universal dimension bound for primitive graph c_2-invariants

摘录 / EXCERPT: Let G be a primitive graph with first Betti number $h_1(G)$, and let $\dim(c_2(G))$ denote the dimension of a minimal model of its c_2-invariant. Universal dimension-bound conj...

conjecture | open | MathDB 页面 | 原文线索

V2-018602 | MATHDB-342787

The mirror P=W conjecture for log Calabi-Yau mirror pairs

摘录 / EXCERPT: Let U be a log Calabi-Yau variety and assume that its homological mirror $U^\vee$ is also a log Calabi-Yau variety with the same dimension. Set $d=\dim U=\dim U^\vee$. The perver...

conjecture | open | MathDB 页面 | 原文线索

V2-018603 | MATHDB-342788

Auroux's boundary conjecture for special Lagrangian torus fibrations

摘录 / EXCERPT: Let X be a variety with a smooth anticanonical divisor D, set $U=X\setminus D$, and let $\pi:U\to B$ be a special Lagrangian torus fibration, meaning that its smooth fibers are...

conjecture | open | MathDB 页面 | 原文线索

V2-018604 | MATHDB-342847

Motivic Telescope Conjecture in telescopic form

摘录 / EXCERPT: Let X be a finite motivic spectrum of type (m,n) with a non-nilpotent self-map $v:\Sigma^{-r,-s}X\to X$ of type (m,n). Write $\langle E\rangle$ for the Bousfield class of a m...

conjecture | open | MathDB 页面 | 原文线索

V2-018605 | MATHDB-342848

Motivic Telescope Conjecture in localization form

摘录 / EXCERPT: Motivic Telescope Conjecture. The natural transformation

conjecture | open | MathDB 页面 | 原文线索

V2-018606 | MATHDB-342849

Motivic Smashing Conjecture

摘录 / EXCERPT: For each pair (m,n), let L_{mn} denote the motivic chromatic Bousfield localization defined in the source. A localization functor is smashing when its value on X is obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-018607 | MATHDB-342852

Non-commutative crepant resolution conjecture for extended Dynkin quivers

摘录 / EXCERPT: Non-commutative crepant resolution conjecture. The algebra $\Lambda^1(Q)$ is a 2-dimensional non-commutative crepant resolution of its center, the ring of functions on...

conjecture | open | MathDB 页面 | 原文线索

V2-018608 | MATHDB-342858

Conjecture on unimodality of minimal filling widths

摘录 / EXCERPT: Fix an activation degree r, a depth h, and input and output widths d_0 and d_h. Let $\bm d=(d_0,d_1,\ldots,d_h)$ be a minimal filling architecture. Unimodality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-018609 | MATHDB-342859

Conjecture on uniform high-degree bounds for powers of homogeneous polynomials

摘录 / EXCERPT: Let d, k, and s be integers, and let $p_1,\ldots,p_k\in\mathbb{R}[x_1,\ldots,x_d]$ be homogeneous polynomials of the same degree s, with no two linearly dependent. Proposit...

conjecture | open | MathDB 页面 | 原文线索

V2-018610 | MATHDB-342887

The virtual projective dimension bound for smooth toric varieties

摘录 / EXCERPT: Virtual projective dimension bound. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-018611 | MATHDB-342892

The semisimple Langlands duality conjecture for the *-generated category

摘录 / EXCERPT: Let X be a smooth projective curve, let G be a reductive group, let $\Gch$ be its Langlands dual group, and let $\LSGch$ be the derived stack of $\Gch$-local systems on X. De...

conjecture | open | MathDB 页面 | 原文线索

V2-018612 | MATHDB-342893

Existence and profile covariance of the modified scalar scattering matrix and interacting fields

摘录 / EXCERPT: Let $\boldsymbol{\eta}$ be a profile, let $C\in\mathcal{F}$ be a polynomial in fields and their derivatives, and let $\Psi,\Psi'\in\mathcal{D}_2$. A renormalization scheme for time...

conjecture | open | MathDB 页面 | 原文线索

V2-018613 | MATHDB-342895

The conjectured subquadratic bound for parent-identifying schemes

摘录 / EXCERPT: Let $t\ge 4$ be an integer and define ... For a t-parent-identifying scheme with block size w on a v-point ground set, let $I_t(w,v)$ be its maximum number of blocks. Subquadra...

conjecture | open | MathDB 页面 | 原文线索

V2-018614 | MATHDB-342916

Schur-Siegel-Smyth trace conjecture

摘录 / EXCERPT: Let TP be the set of totally positive algebraic integers, and for an algebraic integer α with Galois orbit $\alpha_1,\ldots,\alpha_n$, let ... The Schur-Siegel-Smyt...

conjecture | open | MathDB 页面 | 原文线索

V2-018615 | MATHDB-342932

Negative-a-invariant conjecture for square-free initial ideals

摘录 / EXCERPT: Let $I\subset \mathbb{K}[x_1,\ldots,x_n]$ be a homogeneous prime ideal defining a nonsingular projective variety. Suppose that $\operatorname{in}(I)$ is square-free for some monomi...

conjecture | open | MathDB 页面 | 原文线索

V2-018616 | MATHDB-342948

Strict identity-variety growth conjecture for upper triangular tropical matrices

摘录 / EXCERPT: Strict identity-variety growth conjecture. For every positive integer n, there is a semigroup identity satisfied by $UT_n(b$\mathbb{T})$ but not by $UT{n+1}(b$\mathbb{T})$.

conjecture | open | MathDB 页面 | 原文线索

V2-018617 | MATHDB-342954

Non-Cohen-Macaulay invariant rings and normality of the dual nilpotent cone

摘录 / EXCERPT: Non-normality question. Is the dual nilpotent cone $\mathcal{N}^*$ not a normal variety?

open_problem | open | MathDB 页面 | 原文线索

V2-018618 | MATHDB-342979

Projectivity conjecture for connected components of the moduli space of polarized Calabi-Yau pairs

摘录 / EXCERPT: Let a polarized Calabi-Yau pair consist of a semi-log-canonical pair (X,Δ) with $K_X+\Delta$ ${\mathbb Q}$-linearly equivalent to 0, together with an ample ${\mathbb Q}$-C...

conjecture | open | MathDB 页面 | 原文线索

V2-018619 | MATHDB-342987

Yakimov's containment conjecture for quantum partial flag varieties

摘录 / EXCERPT: Let G be a split, simply connected, semisimple algebraic group over a field K of characteristic zero, let P_I be the standard parabolic subgroup associated with a set of simp...

conjecture | open | MathDB 页面 | 原文线索

V2-018620 | MATHDB-342991

Beauville's conjecture on the Chow decomposition of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over $\mathbb C$ of dimension g. Its Beauville decomposition is ... where ... Here n^* is induced by multiplication by n on A. Beauville's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-018621 | MATHDB-343010

Birational abelian-variety fibration conjecture for varieties with nowhere-vanishing one-forms

摘录 / EXCERPT: Let X be a smooth projective variety admitting a holomorphic one-form without zeros. Birational abelian-fibration conjecture. Then X is birational to a smooth projective variet...

conjecture | open | MathDB 页面 | 原文线索

V2-018622 | MATHDB-343034

Uniform Dynamical Mordell-Lang Conjecture

摘录 / EXCERPT: Let X and Y be quasi-projective varieties over a field K of characteristic 0. Let $f\colon X\longrightarrow Y$ be a morphism defined over K, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018623 | MATHDB-343035

Height Gap Conjecture

摘录 / EXCERPT: Let X be a quasi-projective variety defined over $\overline{\mathbb{Q}}$, let $\Phi\colon X\dashrightarrow X$ be a rational self-map, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018624 | MATHDB-343036

Dense-orbit liminf height gap conjecture

摘录 / EXCERPT: Let X be an irreducible quasi-projective variety over $\overline{\mathbb{Q}}$, let $\Phi\colon X\dashrightarrow X$ be a rational self-map, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018625 | MATHDB-343037

Zariski's multivariable Poincaré-series conjecture

摘录 / EXCERPT: Zariski's multivariable Poincaré-series conjecture. The series $M_{X,D,D_1,D_2,\ldots,D_l}(t_1,t_2,\ldots,t_l)$ is rational: there exist $f,g\in{\mathbb Z}[t_1,t_2,\ldots,t_l]$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-018626 | MATHDB-343038

Rationality conjecture for cohomology series of numerically trivial line bundles

摘录 / EXCERPT: Let X be a smooth projective surface over ${\mathbb C}$, and let C be a closed subscheme of X of pure dimension 1, supported on a divisor with simple normal crossings on...

conjecture | open | MathDB 页面 | 原文线索

V2-018627 | MATHDB-343079

Existence and choice-independence conjecture for higher arithmetic degrees

摘录 / EXCERPT: Let X be a smooth projective variety, let $f\colon X\dashrightarrow X$ be a dominant rational map, and let $V\subset X$ be a subvariety of dimension k. Let $\lambda_k(f;V)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-018628 | MATHDB-343080

Higher Kawaguchi-Silverman conjecture for subvariety arithmetic degrees

摘录 / EXCERPT: Let X be a smooth projective variety, let $f\colon X\dashrightarrow X$ be a dominant rational map, and let $V\subset X$ be a subvariety of dimension k. Let $\alpha_{k+1}(f;V)$...

conjecture | open | MathDB 页面 | 原文线索

V2-018629 | MATHDB-343081

Product formula conjecture for higher arithmetic and dynamical degrees

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d, and let $f\colon X\dashrightarrow X$ be a dominant rational self-map. For each integer $1\leqslant k\leqslant d$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018630 | MATHDB-343129

Conjecture on arithmetic hyperbolicity for divisors in a nef cone

摘录 / EXCERPT: Let X be a projective variety defined over a number field k, of dimension n. Let $E_1,\ldots,E_r$ be nef Cartier divisors on X such that $\sum_{j=1}^rE_j$ is ample. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018631 | MATHDB-343132

Conjecture on maximal-length strong exceptional collections for Fano toric Deligne-Mumford stacks

摘录 / EXCERPT: Maximal-length fullness conjecture. Every strong exceptional collection of line bundles of maximal length on a Fano toric Deligne-Mumford stack is a full strong exceptional collect...

conjecture | open | MathDB 页面 | 原文线索

V2-018632 | MATHDB-343158

Cheltsov-Rubinstein conjecture on small-angle Kähler-Einstein edge metrics

摘录 / EXCERPT: Let X be a smooth projective variety and let D be a smooth divisor such that (X,D) is asymptotically log Fano. For $\beta\in(0,1]$, write $(X,(1-\beta)D)$ for the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-018633 | MATHDB-343159

Finiteness conjecture for optimal subvarieties in mixed Shimura varieties

摘录 / EXCERPT: Let M be a mixed Shimura variety, let $Z\subset M$ be a subvariety, and let $\delta(A)=\dim\langle A\rangle-\dim A$ denote the defect of a subvariety A, where...

conjecture | open | MathDB 页面 | 原文线索

V2-018634 | MATHDB-343179

Tradeoff between repair bandwidth and sub-packetization for Reed-Solomon codes

摘录 / EXCERPT: Let $[n,k]$ denote a Reed-Solomon code with an ϵ-MSR repair scheme, meaning that the repair bandwidth for every single failed node is at most $(1+\epsilon)$ times the cut-...

conjecture | open | MathDB 页面 | 原文线索

V2-018635 | MATHDB-343189

Conjecture on the asymptotic proportion of simple abelian-variety isogeny classes

摘录 / EXCERPT: Asymptotic simplicity conjecture. The proportion of isogeny classes that are simple should converge to one:

conjecture | open | MathDB 页面 | 原文线索

V2-018636 | MATHDB-343196

Finite Quot schemes and strange duality

摘录 / EXCERPT: Assume a general sheaf in $M(e)$ is locally free, let V have Chern character e^*+f, and assume $\chi(e\cdot f)=0$. Let $\operatorname{Quot}(V,f)$ parametrize quotients...

conjecture | open | MathDB 页面 | 原文线索

V2-018637 | MATHDB-343197

Johnson's Quot-scheme integral conjecture

摘录 / EXCERPT: Let X be a smooth projective surface with $\chi(\mathcal{O}_X)=1$, let f be the invariants of an ideal sheaf of k points, let V be a vector bundle such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018638 | MATHDB-343199

Budur's hyperplane-arrangement Bernstein-Sato variety conjecture

摘录 / EXCERPT: Budur's conjecture. The affine hyperplane

conjecture | open | MathDB 页面 | 原文线索

V2-018639 | MATHDB-343204

The Optimal Destabilization Conjecture for log Fano pairs

摘录 / EXCERPT: Optimal Destabilization Conjecture. If (X,Δ) is a log Fano pair that is not uniformly K-stable, then there exists a prime divisor F over X computing the stability thresh...

conjecture | open | MathDB 页面 | 原文线索

V2-018640 | MATHDB-343205

The anticanonical boundary conjecture for klt Fano varieties

摘录 / EXCERPT: Anticanonical Boundary Conjecture. Suppose X is a klt Fano variety of dimension n with $\delta(X)<1$. Then there is a general $D\in |-K_X|_{\mathbb Q}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018641 | MATHDB-343241

The Dirac-Motzkin conjecture for real line arrangements

摘录 / EXCERPT: Let t_2 denote the number of crossing points of multiplicity 2 in a real line arrangement, and let s be the number of its lines. Dirac-Motzkin conjecture. The inequality ... ho...

conjecture | open | MathDB 页面 | 原文线索

V2-018642 | MATHDB-343242

The simple-crossing conjecture for complex supersolvable line arrangements

摘录 / EXCERPT: Let t_2 denote the number of crossing points of multiplicity 2 in a complex line arrangement. An arrangement is nontrivial if it is neither a pencil nor a near pencil, and it i...

conjecture | open | MathDB 页面 | 原文线索

V2-018643 | MATHDB-343243

Anzis-Tohăneanu's simple-crossing conjecture for complex supersolvable arrangements

摘录 / EXCERPT: Let t_2 denote the number of crossing points of multiplicity 2 in a complex line arrangement, and let s be the number of its lines. A non-pencil complex supersolvable line ar...

conjecture | open | MathDB 页面 | 原文线索

V2-018644 | MATHDB-343259

Geometric rationality and affine stratifications for varieties with finite-dimensional DG models

摘录 / EXCERPT: Let Z be a projective scheme, let ${\Bbbk}$ be the base field, and let $\mathscr R$ be a finitely-dimensional DG algebra. Suppose that the DG category...

conjecture | open | MathDB 页面 | 原文线索

V2-018645 | MATHDB-343288

The gradedness conjecture for mixed Q-P flag posets

摘录 / EXCERPT: Let $n>0$, let $k\in[n]$, and let $Q,P\in\operatorname{POS}(n)$. Consider the poset $(Q^P_B(\mathbb{C}),\subseteq)$. The gradedness conjecture. The poset...

conjecture | open | MathDB 页面 | 原文线索

V2-018646 | MATHDB-343290

Periodic cyclic homology conjecture for categorical DT theory of critical loci

摘录 / EXCERPT: Let $\mathfrak{M}$ be a quasi-smooth QCA derived stack, let $\mathcal{N}=t_0(\Omega_{\mathfrak{M}}[-1])$ be its associated d-critical locus, and let $\mathcal{N}_{\mathrm{st}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-018647 | MATHDB-343299

The trichotomy conjecture for cyclic dilations of affine varieties

摘录 / EXCERPT: Let M be an n-dimensional smooth affine variety over a field $\mathbb{K}$ of characteristic zero. Let $\kappa(M)$ denote its relevant Kodaira dimension. A cyclic dilation is un...

conjecture | open | MathDB 页面 | 原文线索

V2-018648 | MATHDB-343300

The Gutt-Hutchings capacity conjecture for affine varieties of negative Kodaira dimension

摘录 / EXCERPT: Let M be a smooth affine variety with $\kappa(M)=-\infty$, and let $c_1^{\mathit{GH}}(M)$ denote its first Gutt-Hutchings capacity. Gutt-Hutchings capacity conjecture. ... The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-018649 | MATHDB-343317

Guichard-Wienhard conjecture for extra components of $SO(p+q,q)$-Higgs-bundle moduli

摘录 / EXCERPT: Guichard-Wienhard conjecture. The extra components in the moduli space of $SO(p+q,q)$-Higgs bundles conjectured to contain positive representations are those connected components c...

conjecture | open | MathDB 页面 | 原文线索

V2-018650 | MATHDB-343328

The p-power base point free conjecture for klt pairs

摘录 / EXCERPT: Let k be a field of characteristic $p>0$. Let (X,Δ) be a klt pair over k, and let $f:X\to Z$ be a projective k-morphism to a quasi-projective k-scheme Z. Let L b...

conjecture | open | MathDB 页面 | 原文线索

V2-018651 | MATHDB-343335

The hyper-Kähler derived-equivalence conjecture for Chow motives

摘录 / EXCERPT: Hyper-Kähler motivic conjecture. If $\operatorname{D}^b(X)$ and $\operatorname{D}^b(Y)$ are exactly equivalent, then $\mathfrak{h}(X)$ and $\mathfrak{h}(Y)$ are isomorphic as Frobe...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-018652 | MATHDB-343336

The motivic global Torelli conjecture

摘录 / EXCERPT: Motivic global Torelli conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-018653 | MATHDB-343341

Mild boundedness conjecture for projective varieties without rational curves

摘录 / EXCERPT: Mild boundedness conjecture. If $X_{\overline{k}}$ has no rational curves, then X is mildly bounded over k.

conjecture | open | MathDB 页面 | 原文线索

V2-018654 | MATHDB-343343

Vojta's general-type and pseudo-algebraic-hyperbolicity conjecture

摘录 / EXCERPT: Vojta's conjecture. The projective variety X is of general type over k if and only if it is pseudo-algebraically hyperbolic over k.

conjecture | open | MathDB 页面 | 原文线索

V2-018655 | MATHDB-343344

Mild boundedness and grouplessness equivalence conjecture

摘录 / EXCERPT: Mild boundedness and grouplessness conjecture. In this situation, mild boundedness is equivalent to having no rational curves, and the condition that...

conjecture | open | MathDB 页面 | 原文线索

V2-018656 | MATHDB-343371

The supersolvable line arrangement double-point conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be a supersolvable line arrangement consisting of $d=|{\mathcal A}|$ lines, and suppose that ${\mathcal A}$ is not a pencil. Denote by n_2 the number of inters...

conjecture | open | MathDB 页面 | 原文线索

V2-018657 | MATHDB-343401

Relative Fujita-type freeness conjecture for mixed pluricanonical sheaves

摘录 / EXCERPT: Let $f:X\to Y$ be a surjective morphism between smooth projective varieties with $\dim Y=n$, let $\mathcal L$ be any ample invertible sheaf on Y, and let k be a positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-018658 | MATHDB-343402

Nakayama's numerical-dimension growth conjecture

摘录 / EXCERPT: Let X be a smooth projective variety and let D be a pseudo-effective $\mathbb R$-divisor on X. Nakayama's numerical-dimension growth conjecture. There exist a positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-018659 | MATHDB-343419

Shibata's refined multiplicity conjecture for log canonical complete intersection singularities

摘录 / EXCERPT: Let X be an n-dimensional variety that is locally a complete intersection with log canonical singularities. For a closed point x of X, let $\mathrm{emb}(X,x)$ be the embedd...

conjecture | open | MathDB 页面 | 原文线索

V2-018660 | MATHDB-343432

Ricci-flow conjecture for the trigonometric deformation of the flag-variety sigma model

摘录 / EXCERPT: Let G be a compact group with complexification $G_{\mathbb C}$, let B be a Borel subgroup, and let s be the trigonometric deformation parameter. The resulting family of metri...

conjecture | open | MathDB 页面 | 原文线索

V2-018661 | MATHDB-343580

Beilinson-Bloch conjecture on the rank of homologically trivial cycles

摘录 / EXCERPT: Beilinson-Bloch conjecture. Assume that k is a number field. Then $\operatorname{CH}^m_{\mathrm{hom}}(X)$ is finitely generated, and its rank equals the order of zero at $s=m$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-018662 | MATHDB-343581

Beilinson-Bloch injectivity conjecture for the higher Abel-Jacobi map

摘录 / EXCERPT: Beilinson-Bloch conjecture. The m-th higher Abel-Jacobi map AJ^m is injective. This is presented as another part of the Beilinson-Bloch conjecture for smooth projective varieti...

conjecture | open | MathDB 页面 | 原文线索

V2-018663 | MATHDB-343589

Simpson's nestedness conjecture for Dolbeault and de Rham stratifications

摘录 / EXCERPT: Let M be a (quasi-)projective variety with a stratification by locally closed subsets ... The stratification is nested if there is a partial order $(\Lambda,\leq)$ such that ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-018664 | MATHDB-343593

Deng's isotriviality conjecture for families with semi-ample canonical bundle

摘录 / EXCERPT: Let $f:X\to Y$ be a smooth projective family between complex quasi-projective manifolds. Assume that each fiber of f has semi-ample canonical bundle, and that the Kobayashi pseud...

conjecture | open | MathDB 页面 | 原文线索

V2-018665 | MATHDB-343599

Genus formula conjecture for the spectral curves

摘录 / EXCERPT: Let l be a positive integer, and let Γ_l be the irreducible l-th spectral curve. Its geometric genus is the genus of the normalization of Γ_l. Genus formula conje...

conjecture | open | MathDB 页面 | 原文线索

V2-018666 | MATHDB-343602

Absence of real critical points for the normalized spectral-curve projection

摘录 / EXCERPT: Real-critical-point conjecture. This projection has no real critical points (μ,ω).

conjecture | open | MathDB 页面 | 原文线索

V2-018667 | MATHDB-343637

Sensible Caveman conjecture

摘录 / EXCERPT: Let a Q-polynomial association scheme have 3 or more classes, and suppose that it is primitive, meaning that it has no nontrivial equivalence relation arising from a union of r...

conjecture | open | MathDB 页面 | 原文线索

V2-018668 | MATHDB-343645

A Manin-type conjecture for Campana Fano orbifolds

摘录 / EXCERPT: Let (X,D_ϵ) be a Campana orbifold over a number field F such that $-(K_X+D_\epsilon)$ is ample, and let $(\mathcal X,\mathcal D_\epsilon)$ be a good integral model over...

conjecture | open | MathDB 页面 | 原文线索

V2-018669 | MATHDB-343650

The intersection-cohomology Galois subquotient conjecture for Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura datum with reflex field E, let K be a neat open compact subgroup, and let $W_p(\pi^\infty)$ be the finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-018670 | MATHDB-343669

Functorial monoid realization conjecture for affine $G^\vee$-varieties

摘录 / EXCERPT: Functorial monoid realization conjecture. For any complex affine $G^\vee$ variety $X^\vee$, there is a monoid $A_{X^\vee}$ in $\operatorname{\mathbf{Mult}}_G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018671 | MATHDB-343682

Tanisaki's conjecture on associated varieties of orbital varieties

摘录 / EXCERPT: Let $\mathcal{O}$ be a special nilpotent orbit, let $\mathscr{C}$ be the corresponding double cell, and let $\mathscr{C}/ {\small \stackrel{R}{\sim}}$ be the set of right cells con...

conjecture | open | MathDB 页面 | 原文线索

V2-018672 | MATHDB-343688

Habegger-Pila optimal-subvariety conjecture for abelian varieties

摘录 / EXCERPT: Habegger-Pila optimality conjecture. The subvariety V contains at most finitely many optimal subvarieties of defect at most d.

conjecture | open | MathDB 页面 | 原文线索

V2-018673 | MATHDB-343692

Generalized Bloch conjecture on support of zero-cycles

摘录 / EXCERPT: Let X be a smooth projective complex variety and let m be a nonnegative integer. Say that $\operatorname{CH}_0(X)$ is supported on an m-dimensional algebraic subset...

conjecture | open | MathDB 页面 | 原文线索

V2-018674 | MATHDB-343758

Burns-Macias Castillo equivariant Birch and Swinnerton-Dyer conjecture

摘录 / EXCERPT: Let A be an abelian variety, let F/k be a finite Galois extension with group G, let S be the specified finite set of places, and let $\widehat G$ be the set of irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-018675 | MATHDB-343759

Darmon-style norm and Mazur-Tate height conjecture for Rubin lattices

摘录 / EXCERPT: Let A be an abelian variety, let F/k be a finite Galois extension with group G, let $J\leq G$ with $E=F^J$, and let p be a prime. Let A^t be the dual abelian variety, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018676 | MATHDB-343786

Cao-Maulik-Toda Katz/GV conjecture for one-dimensional sheaves on Calabi-Yau fourfolds

摘录 / EXCERPT: Let X be a Calabi-Yau 4-fold and let $M_1(X,\beta)$ be the moduli scheme of one-dimensional stable sheaves F with $[F]=\beta\in H_2(X,\mathbb Z)$ and $\chi(F)=1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018677 | MATHDB-343802

Conjecture on the components of the rank-two locus of the cubic discriminant

摘录 / EXCERPT: Let V_k be the locus of cubic surfaces for which the matrix M has rank at most $20-k$, and let N_2 denote the locus of binodal cubic surfaces. The 19×19 minors of M...

conjecture | open | MathDB 页面 | 原文线索

V2-018678 | MATHDB-343803

Conjecture on minors of the cubic discriminant matrix over singularity loci

摘录 / EXCERPT: Let M be the matrix whose determinantal rank loci are denoted by V_k, and let N_k denote the locus of cubic surfaces with k singularities. For each k, consider the minors...

conjecture | open | MathDB 页面 | 原文线索

V2-018679 | MATHDB-343809

Conjecture on the signed count of real bitangents to a plane quartic

摘录 / EXCERPT: Let Q be a smooth plane quartic defined over $\mathbb{R}$, and let $L_\infty \subset \mathbb{P}^2_{\mathbb{R}}$ be a line defined over $\mathbb{R}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018680 | MATHDB-343818

Divisibility of the Hurwitz form by the eigendiscriminant

摘录 / EXCERPT: Divisibility conjecture. Restricted to the 3-planes coming from eigenschemes, the eigendiscriminant divides the Hurwitz form.

conjecture | open | MathDB 页面 | 原文线索

V2-018681 | MATHDB-343828

Semi-positivity conjecture for divisors with unitary-flat normal bundle

摘录 / EXCERPT: Let X be a projective manifold and let $Y\subset X$ be a reduced subvariety of codimension 1. Let $L=[Y]$ be the associated line bundle, and write ... for the inclusion, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-018682 | MATHDB-343830

Fujino's relative Fujita-type conjecture for direct images of relative pluricanonical bundles

摘录 / EXCERPT: Let $f:X\to Y$ be a surjective morphism of smooth projective varieties over an algebraically closed field of characteristic zero, with Y of dimension n, and let $\mathcal L$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-018683 | MATHDB-343838

Flach-Morin's local conjecture ${\bf D}_p(\mathcal{X},n)$

摘录 / EXCERPT: Fix a prime number p and let $\mathcal X$ be regular, proper and flat over $\operatorname{Spec}(\mathbb{Z})$. Write $R\Gamma_{dR}(\mathcal{X}_{\mathbb{Q}_p}/\mathbb{Q}_p)/F^n$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018684 | MATHDB-343839

Flach-Morin's Artin-Verdier duality conjecture

摘录 / EXCERPT: Let $\mathcal X$ be the regular proper arithmetic scheme under consideration, of dimension d, and work on its étale site $\mathcal X_{et}$. For any integer n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018685 | MATHDB-343840

Bogomolov-Mumford conjecture on rational curves on K3 surfaces

摘录 / EXCERPT: Bogomolov-Mumford conjecture. X contains infinitely many rational curves.

conjecture | open | MathDB 页面 | 原文线索

V2-018686 | MATHDB-343853

Kleppe-Ellia conjecture on cubic-surface space-curve families

摘录 / EXCERPT: Kleppe-Ellia conjecture. The family W is an irreducible component of $(H(d,g)^{sc})_{\operatorname{red}}$, and $H(d,g)^{sc}$ is generically non-reduced along W.

conjecture | open | MathDB 页面 | 原文线索

V2-018687 | MATHDB-343895

Non-emptiness criterion for the $e_0^t\mathbf{g}^k$ family

摘录 / EXCERPT: For non-negative integers t and k, let $e_0^t\mathbf{g}^k$ denote the set defined by the corresponding Ext comparison with $e_0^t g^k$. Non-emptiness conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-018688 | MATHDB-343896

Non-emptiness criterion for the $e_0\mathbf{g}^k$ family

摘录 / EXCERPT: For a non-negative integer k, let $e_0\mathbf{g}^k$ denote the corresponding subset of Ext. Non-emptiness conjecture. The set $e_0\mathbf{g}^k$ is non-empty if and only if ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-018689 | MATHDB-343897

Non-emptiness criterion for the $\Delta h_1e_0\mathbf{g}^k$ family

摘录 / EXCERPT: For integers $t\geq 1$ and $k\geq 0$, let $\Delta h_1e_0^t\mathbf{g}^k$ denote the corresponding subset of Ext; the paper notes that the case $t=0$ is not considered because...

conjecture | open | MathDB 页面 | 原文线索

V2-018690 | MATHDB-343905

The no-real-roots conjecture for the Stern polynomials B_{h_n}

摘录 / EXCERPT: For $n\in\mathbb{N}$, define ... Let $B_{h_n}(t)$ denote the associated Stern polynomial. No-real-roots conjecture. For each $n\in\mathbb{N}$, the polynomial $B_{h_n}(t)$ is complete...

conjecture | open | MathDB 页面 | 原文线索

V2-018691 | MATHDB-343915

Mirabolic Iwahori-equivariant Satake conjecture

摘录 / EXCERPT: Let V_1 and V_2 be N-dimensional vector spaces, let $\bI$ be an Iwahori subgroup, and let $\bFfl_i$ be the variety of complete flags in V_i. Define the dg-scheme with zero...

conjecture | open | MathDB 页面 | 原文线索

V2-018692 | MATHDB-343925

Lang-Vojta's pseudo-1-boundedness characterization

摘录 / EXCERPT: Lang-Vojta's pseudo-1-boundedness conjecture. A projective integral variety X over k is of general type if and only if X is pseudo-1-bounded over k.

conjecture | open | MathDB 页面 | 原文线索

V2-018693 | MATHDB-343926

Lang-Vojta's pseudo-algebraic-hyperbolicity conjecture

摘录 / EXCERPT: Lang-Vojta's pseudo-algebraic-hyperbolicity conjecture. A projective integral variety X over k is of general type if and only if X is pseudo-algebraically hyperbolic over k...

conjecture | open | MathDB 页面 | 原文线索

V2-018694 | MATHDB-343928

Truong's equality conjecture for cohomological and numerical dynamical degrees

摘录 / EXCERPT: Let X be a smooth projective variety defined over an algebraically closed field $\mathbf{k}$, and let $f\colon X \vdash X$ be a dominant self-correspondence. For a prime ℓ d...

conjecture | open | MathDB 页面 | 原文线索

V2-018695 | MATHDB-343932

Versality conjecture for vanishing flags on a general twice-marked curve

摘录 / EXCERPT: Let C be a curve of genus g with marked points p and q, and fix $N\geq 2g-1$. Over $\operatorname{Pic}^N(C)$, let $\mathcal{H}$ be the vector bundle whose fiber at $[L]$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-018696 | MATHDB-343968

Duality conjecture for ideal access structures

摘录 / EXCERPT: Let $\mathcal A$ be an access structure. It is ideal if it can be realized by an ideal entropic polymatroid, equivalently by an ideal distribution scheme. Let $\mathcal A^\bot$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-018697 | MATHDB-344005

Equivalence of formal K-flatness and K-flatness

摘录 / EXCERPT: Formal K-flatness conjecture. Formal K-flatness is equivalent to K-flatness.

conjecture | open | MathDB 页面 | 原文线索

V2-018698 | MATHDB-344026

The Hodge conjecture for complex projective manifolds

摘录 / EXCERPT: The Hodge conjecture. Every element of $H^{p,p}(X;\Q)$ is representable by some algebraic $(n-p)$-cycles with $\Q$-coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-018699 | MATHDB-344060

The recursive formula for critical spheres over complex configurations

摘录 / EXCERPT: Let $\mathcal H_{m,n}$ denote the variety associated with critical $(n-1)$-spheres for a generic configuration of m points in $\mathbb C^n$. Here $EDdegree(\mathcal H_{m,n})$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-018700 | MATHDB-344061

The multiplicative formula for the Euclidean distance degree of hypersurface classes

摘录 / EXCERPT: Let $\mathcal H_{m,n}$ be the variety parametrizing the relevant hypersurface class, let $C_{m,n}$ denote the corresponding parameter variety, and let π be a generically finite...

conjecture | open | MathDB 页面 | 原文线索

V2-018701 | MATHDB-344062

The recursive formula for critical spheres over real configurations

摘录 / EXCERPT: Let Z_m denote the variety of critical $(n-1)$-spheres associated with a generic configuration of m points in $\mathbb R^n$, and let $EDdegree(Z_m)$ be its Euclidean distance d...

conjecture | open | MathDB 页面 | 原文线索

V2-018702 | MATHDB-344063

The recursive formula for the Euclidean distance degree of plane circles

摘录 / EXCERPT: Let C_m denote the variety associated with critical circles through a generic configuration of m points in the plane, and let $EDD(C_m)$ denote its Euclidean distance degree. D...

conjecture | open | MathDB 页面 | 原文线索

V2-018703 | MATHDB-344064

The multiplicative Euclidean distance degree formula for parameterized hypersurface classes

摘录 / EXCERPT: Let $X_{m,n}^d$ be a variety parameterizing a set of hypersurfaces with the same Euclidean distance degree, let H be a point of $X_{m,n}^d$, and let k be the number of paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-018704 | MATHDB-344066

Uniform-in-time convergence conjecture for the nondiffusive advection scheme

摘录 / EXCERPT: Let u^0 be a function with total bounded variation, and associate to it the initialization ... Suppose that the ratio $\lambda=\frac{\Delta t}{V\Delta x}$ is fixed and belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-018705 | MATHDB-344069

Strong approximation conjecture for varieties associated to pairs of polynomials and field extensions

摘录 / EXCERPT: Let k be a number field, let $P_1(t),\cdots, P_n(t)$ be pairwise distinct irreducible polynomials in $k[t]$. Set $k_i=k[t]/(P_i(t))$, let a_i be the class of t in k_i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018706 | MATHDB-344073

Integral Hecke stability conjecture for coherent cohomology lattices

摘录 / EXCERPT: Let κ be a dominant weight for $M_\mu/Z_s(G)$, let V_κ be the corresponding Weyl representation over $\mathcal{O}_{E',\lambda'}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018707 | MATHDB-344075

Kedlaya's minimal slope conjecture for overconvergent F-isocrystals

摘录 / EXCERPT: Let X be a smooth connected scheme separated of finite type over $\operatorname{Spec} k$. Let $\mathcal M^\dagger$ and $\mathcal N^\dagger$ be irreducible overconvergent F-isoc...

conjecture | open | MathDB 页面 | 原文线索

V2-018708 | MATHDB-344098

Derived invariance of the Hasse-Weil zeta function

摘录 / EXCERPT: Derived zeta-invariance conjecture. Derived equivalent nice varieties defined over finite fields have the same Hasse-Weil zeta function.

conjecture | open | MathDB 页面 | 原文线索

V2-018709 | MATHDB-344099

Derived invariance of Hodge numbers

摘录 / EXCERPT: Derived Hodge-invariance conjecture. Derived equivalent nice varieties defined over the complex numbers have the same Hodge numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-018710 | MATHDB-344100

Derived invariance of rational point counts modulo the characteristic

摘录 / EXCERPT: Mod-p point-count conjecture. Derived equivalent nice varieties defined over finite fields of characteristic p have the same number, modulo p, of rational points.

conjecture | open | MathDB 页面 | 原文线索

V2-018711 | MATHDB-344104

The general Hadamard convergence conjecture for smooth sections

摘录 / EXCERPT: Let $X/\mathbb{F}_q$ be a smooth projective variety equipped with an ample line bundle $\mathcal{L}$, and let $U_d \subset \Gamma(X,\mathcal{L}^d)$ be the open subvariety of smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-018712 | MATHDB-344128

The Brieskorn variety dilation conjecture

摘录 / EXCERPT: Brieskorn dilation conjecture. The variety $V_{a_0,\ldots,a_n}$ admits a k-dilation if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-018713 | MATHDB-344149

Sreekantan regulator factorization conjecture

摘录 / EXCERPT: Sreekantan regulator factorization conjecture. This composition factors via the Sreekantan regulator.

conjecture | open | MathDB 页面 | 原文线索

V2-018714 | MATHDB-344170

Briançon-Iarrobino lexsegment conjecture on maximal tangent spaces

摘录 / EXCERPT: Let $\mathbf{k}$ be a field, let $S=\mathbf{k}[x_1,\ldots,x_n]$, and for $d\in\mathbf{N}$ let $E(d)\subseteq S$ be the unique lexsegment ideal such that ... and, for some r,...

conjecture | open | MathDB 页面 | 原文线索

V2-018715 | MATHDB-344188

The Chow-resolution conjecture for generalized punctual Hilbert schemes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, and let $\operatorname{Hilb}(\mathfrak{g})$ be its generalized Hilbert scheme with Chow morphism ch to $\mathfrak{h}^2/W$. Chow-re...

conjecture | open | MathDB 页面 | 原文线索

V2-018716 | MATHDB-344189

The generalized GIT quotient conjecture for generalized punctual Hilbert schemes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, and let $\operatorname{Hilb}(\mathfrak{g})$ denote its generalized Hilbert scheme. Generalized GIT quotient conjecture. There is a g...

conjecture | open | MathDB 页面 | 原文线索

V2-018717 | MATHDB-344190

The nilpotent-orbit chart conjecture for generalized punctual Hilbert schemes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, and suppose a smooth version of $\operatorname{Hilb}(\mathfrak{g})$ exists. For classical $\mathfrak{g}$, let m be the dimension o...

conjecture | open | MathDB 页面 | 原文线索

V2-018718 | MATHDB-344191

The nilpotent centralizer conjecture for generalized punctual Hilbert schemes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra of rank at least 3, let $A\in\mathfrak{g}$ be nilpotent, and write $Z(A)$ for its centralizer and $\operatorname{rk}\mathfrak{g}$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-018719 | MATHDB-344192

The Hitchin-component conjecture for generalized complex-structure moduli

摘录 / EXCERPT: Let Σ be the underlying surface, let $\mathfrak{g}$ be a real split Lie algebra, let G be the real split Lie group associated to $\mathfrak{g}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018720 | MATHDB-344193

The symplectic zero-fiber conjecture for generalized punctual Hilbert schemes

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $\operatorname{Hilb}(\mathfrak{g})$ be its generalized Hilbert scheme, and let $\operatorname{Hilb}_0(\mathfrak{g})$ be the zero-fiber of t...

conjecture | open | MathDB 页面 | 原文线索

V2-018721 | MATHDB-344209

Absolute rigidity conjecture for divisorially canonical Fano varieties

摘录 / EXCERPT: Let V) be a divisorially canonical Fano variety, and let X be a rationally connected variety with $\dim X=\dim V$. A rational dominant map is a rational map...

conjecture | open | MathDB 页面 | 原文线索

V2-018722 | MATHDB-344210

Non-abelian Galois-cover generalization of the rational-cover rigidity theorem

摘录 / EXCERPT: Let V be a divisorially canonical Fano variety satisfying the hypotheses of Theorem 1, and let $X\dashrightarrow V$ be a rationally connected Galois rational cover. A Galois grou...

conjecture | open | MathDB 页面 | 原文线索

V2-018723 | MATHDB-344216

Characterization of rank-one subshifts by repetition schemes

摘录 / EXCERPT: Let (X,T) be a subshift. Call (X,T) irreducible if, for every pair of finite words u,v occurring in points of X, some finite word w occurs in a point of X such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018724 | MATHDB-344258

Bombieri-Masser-Zannier torsion openness conjecture for anomalous intersections

摘录 / EXCERPT: Let $\mathcal{X}$ be an algebraic variety, and let $\{H_i\}_{i\in\mathcal{I}}$ be the algebraic subgroups for which $\mathcal{X}\cap H_i$ is an anomalous subvariety of...

conjecture | open | MathDB 页面 | 原文线索

V2-018725 | MATHDB-344265

Rémond's generalized Lehmer conjecture

摘录 / EXCERPT: Rémond's generalized Lehmer conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-018726 | MATHDB-344285

Absolute Hodge conjecture for isomorphic Hodge structures

摘录 / EXCERPT: Let X and Y be smooth projective complex varieties, and let i be an integer. An absolute Hodge class is a Hodge class whose image under every automorphism of $\mathbb{C}$ rem...

conjecture | open | MathDB 页面 | 原文线索

V2-018727 | MATHDB-344322

Geometric realization conjecture for double-affine Bruhat intervals

摘录 / EXCERPT: Geometric realization conjecture. For every $\pi^\mu w,\pi^\nu v\in W_{\mathcal{T}}$, there exists an affine algebraic variety defined over $\Bbbk$, with a contracting...

conjecture | open | MathDB 页面 | 原文线索

V2-018728 | MATHDB-344341

Uniform finiteness conjecture for isogeny classes in one-parameter families

摘录 / EXCERPT: Let K be a number field, S a set of finite places of K, and $g\geq 2$ an integer. Let $\mathcal{F}$ be a one-parameter family of complex abelian varieties A of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-018729 | MATHDB-344376

Ax--Schanuel conjecture for variations of mixed Hodge structure

摘录 / EXCERPT: Let X be a smooth connected algebraic variety over $\mathbb{C}$, with base point $b \in X(\mathbb{C})$. Let $\mathscr{H}_{\mathbb{Z}}$ be a $\mathbb{Z}$-variation of mixed Hodge...

conjecture | open | MathDB 页面 | 原文线索

V2-018730 | MATHDB-344381

Li's conjecture on a homomorphism from the sigma-fixed Lie algebra envelope

摘录 / EXCERPT: Let $b[?mU(b[0m\mathfrak g_{\Gamma}^{\sigma})$ be the enveloping algebra of the σ-fixed subalgebra of the Lie algebra $\mathfrak g_{\Gamma}$ associated to the underlying Dyn...

conjecture | open | MathDB 页面 | 原文线索

V2-018731 | MATHDB-344390

Dense globally F-split type conjecture for varieties with int-amplified endomorphisms

摘录 / EXCERPT: Let X be a normal projective variety over an algebraically closed field k of characteristic zero. An endomorphism $f\colon X\to X$ is int-amplified if there exists an ample Car...

conjecture | open | MathDB 页面 | 原文线索

V2-018732 | MATHDB-344397

Asok-Fasel conjecture on the th -homotopy sheaf of punctured affine space

摘录 / EXCERPT: Let k be a perfect field with ${\rm char}(k)\ne 2$, and let d be the dimension of a k-scheme X. Write $\mathbf{K}^{\mathrm{M}}_{d+2}/24$ for the quotient of the Milnor K-...

conjecture | open | MathDB 页面 | 原文线索

V2-018733 | MATHDB-344398

Tateness conjecture for motivic classes of translation quiver representation spaces

摘录 / EXCERPT: Let Q be a quiver with an automorphism $\ta$, let $\bv\in\bN^{(Q_0)}$ be a dimension vector, and let $\cR^\ta(Q,\bv)$ denote the space of pairs $(M,\vi)$ with $M\in\cR(Q,\bv)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-018734 | MATHDB-344402

Scheme-theoretic multiplicity formula for pullbacks of higher associated varieties

摘录 / EXCERPT: Let $\lambda=[\lambda_1,\ldots,\lambda_n]$ be a partition, let $m_1(\lambda)$ denote the multiplicity of its parts equal to 1, and let $\mathcal{F}_j(\lambda)$ be the family of d...

conjecture | open | MathDB 页面 | 原文线索

V2-018735 | MATHDB-344406

Hopkins's Wilson space hypothesis for algebraic cobordism

摘录 / EXCERPT: Let k be a field, let $\bMGL$ denote the motivic algebraic cobordism spectrum, let $\bT$ be the motivic suspension coordinate, and let $\bDM(k)$ be the category of motives over...

conjecture | open | MathDB 页面 | 原文线索

V2-018736 | MATHDB-344418

Medvedev-Scanlon conjecture on dense orbits of rational difference varieties

摘录 / EXCERPT: Let k be an algebraically closed field, let V be a variety over k, and let (V,ϕ) be a σ-variety over k, with ϕ the self-map defining its difference structu...

conjecture | open | MathDB 页面 | 原文线索

V2-018737 | MATHDB-344443

Geiss-Leclerc-Schröer conjecture on square-irreducibility

摘录 / EXCERPT: Let $\mathfrak{m}$ be a multisegment, let V be the graded vector space determined by its supercuspidal support, and let $\mathfrak{C}_{\mathfrak{m}}$ be the corresponding irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-018738 | MATHDB-344444

Geometric criterion for subrepresentation and irreducibility of parabolic induction

摘录 / EXCERPT: Let $\mathfrak{m}$ and $\mathfrak{m}'$ be multisegments satisfying the condition referred to in the source as ... , (1) $\operatorname{SA}(\mathfrak{m},\mathfrak{m}')$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018739 | MATHDB-344460

Friedlander-Mazur conjecture on the topological, geometric and Hodge filtrations

摘录 / EXCERPT: Friedlander-Mazur conjecture. For any smooth projective variety X and $k\geq 2p\geq 0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-018740 | MATHDB-344461

The virtual Hodge and Betti number conjecture for Chow varieties of projective space

摘录 / EXCERPT: For integers $n\geq p\geq 0$ and $d\geq 0$, let $C_{p,d}({\mathbb{P}}^n)$ denote the Chow variety of effective p-dimensional algebraic cycles of degree d in projective space, l...

conjecture | open | MathDB 页面 | 原文线索

V2-018741 | MATHDB-344462

The Chow group and Lawson homology conjecture for Chow varieties of projective space

摘录 / EXCERPT: Let $C_{p,d}({\mathbb{P}}^n)$ denote the Chow variety of effective p-dimensional algebraic cycles of degree d in projective space. Write Ch_q for the codimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-018742 | MATHDB-344463

Carrell's rationality conjecture for varieties with isolated vector-field zeroes

摘录 / EXCERPT: Let X be a smooth (or nonsingular) irreducible projective algebraic variety admitting a holomorphic vector field whose zero scheme is nonempty and consists of isolated zeroes. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-018743 | MATHDB-344475

Modified Morel stable connectivity conjecture

摘录 / EXCERPT: Modified Morel stable connectivity conjecture. If $\mathcal F$ is 0-connective, then

conjecture | open | MathDB 页面 | 原文线索

V2-018744 | MATHDB-344478

Rational homotopy fixed points compute motivic localization

摘录 / EXCERPT: Let k be a perfect field, let $G=\operatorname{Gal}(\bar{k}/k)$, and let $\mathbf{H}^{\mathbb{A}^1}\mathbb{Q}$ denote rational $\mathbb{A}^1$-homology. Let $\mathcal{X}$ be an...

conjecture | open | MathDB 页面 | 原文线索

V2-018745 | MATHDB-344480

The associator scheme conjecture for Ran-space fiber functors

摘录 / EXCERPT: Let $\mathcal M^{B}_{un}(Ran(\mathbb A^1))$ and $\mathcal M^{dR}_{un}(Ran(\mathbb A^1))$ be the Betti and de Rham categories equipped with their natural tensor structures and fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-018746 | MATHDB-344503

Even-dimensional stacky Euler-class formula with Catalan correction factor

摘录 / EXCERPT: Let $n\geq 2$ be even, let π^*V be the pullback of the vector bundle to the stack considered in the paper, and let $\mathbb{H}=\langle 1\rangle+\langle -1\rangle$ denote a hype...

conjecture | open | MathDB 页面 | 原文线索

V2-018747 | MATHDB-344514

Motivic generation conjecture for singular and open moduli spaces on K3 or abelian surfaces

摘录 / EXCERPT: Let S be a projective K3 or abelian surface, let α be a Brauer class, let $\bm{\mathrm v}$ be a not necessarily primitive Mukai vector, and let σ be a not necessari...

conjecture | open | MathDB 页面 | 原文线索

V2-018748 | MATHDB-344515

Noncommutative motivic generation conjecture for moduli spaces

摘录 / EXCERPT: Let Y be a smooth projective variety and let $\mathcal{A}\subset D^b(Y)$ be an admissible triangulated subcategory that is 2-Calabi--Yau. Let $\bm{\mathrm v}$ be an element of th...

conjecture | open | MathDB 页面 | 原文线索

V2-018749 | MATHDB-344518

Shen-Vial's multiplicative Chow-Künneth conjecture for hyper-Kähler varieties

摘录 / EXCERPT: A hyper-Kähler variety is a smooth projective variety with the usual hyper-Kähler structure; a multiplicative Chow-Künneth decomposition is a Chow-Künneth decomposition compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-018750 | MATHDB-344519

Murre's conjecture on Chow-Künneth decompositions and the Bloch-Beilinson filtration

摘录 / EXCERPT: Murre's conjecture. There exists a Chow-Künneth decomposition $\{\pi^0,\ldots,\pi^{2d}\}$, and the induced filtration satisfies $F^0\operatorname{CH}^i(X)=\operatorname{CH}^i(X)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-018751 | MATHDB-344585

Christol's geometric origin conjecture for differential operators

摘录 / EXCERPT: Let L be a differential operator. Say that L has a strong Frobenius structure for a prime p if it has the corresponding p-adic Frobenius structure, and say that L comes f...

conjecture | open | MathDB 页面 | 原文线索

V2-018752 | MATHDB-344601

Sommese's extendability conjecture for ample subvarieties

摘录 / EXCERPT: Sommese's extendability conjecture. Any such morphism extends uniquely to a morphism

conjecture | open | MathDB 页面 | 原文线索

V2-018753 | MATHDB-344637

The fundamental-group conjecture for open Richardson varieties

摘录 / EXCERPT: Fundamental-group conjecture. For every $v\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-018754 | MATHDB-344647

Higher-dimensional moment asymptotics for volumes of Kostlan zero sets

摘录 / EXCERPT: Let M be a real algebraic manifold of dimension n, let s_d be a random section in the complex Fubini--Study model in dimension n and codimension $r\in\{1,\dots,n\}$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-018755 | MATHDB-344672

The quiver flag variety reconstruction conjecture

摘录 / EXCERPT: Quiver flag variety reconstruction conjecture. The morphism

conjecture | open | MathDB 页面 | 原文线索

V2-018756 | MATHDB-344674

The Fano-fibration degeneration conjecture

摘录 / EXCERPT: Let a Fano fibration be a fibration whose fibres are Fano varieties, possibly with singular fibres in a degeneration. Fano-fibration degeneration conjecture. A Fano fibration is po...

conjecture | open | MathDB 页面 | 原文线索

V2-018757 | MATHDB-344677

The critical-exponent gap conjecture for compact complex subvarieties

摘录 / EXCERPT: Let $\Gamma<\operatorname{PU}(n,1)$ be a discrete, torsion-free subgroup, let δ_Γ be its critical exponent, and let k denote the dimension of a compact complex subvar...

conjecture | open | MathDB 页面 | 原文线索

V2-018758 | MATHDB-344681

The high-dimensional non-reflection lattice conjecture

摘录 / EXCERPT: There exists an integer N such that for every $n\ge N$ and every lattice $\Gamma<\operatorname{PU}(n,1)$, the following hold: Γ is not a reflection subgroup, and the under...

conjecture | open | MathDB 页面 | 原文线索

V2-018759 | MATHDB-344714

The Bogomolov property of the torsion field of an abelian variety

摘录 / EXCERPT: Soit A une variété abélienne définie sur un corps de nombres K. Pour un sous-corps $L\subset\overline{K}$, on dit que L a la propriété de Bogomolov relativement à A si, pou...

conjecture | open | MathDB 页面 | 原文线索

V2-018760 | MATHDB-344729

Genus stability conjecture for algebraic-geometric \(\mathbb{Z}_p\)-towers

摘录 / EXCERPT: Let $(C_n)_{n\geq 0}$ be a $\mathbb{Z}_p$-tower of curves over a finite field, with genus $g_n=g(C_n)$. Assume the tower comes from algebraic geometry. Genus stability conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-018761 | MATHDB-344731

Main stability conjecture for zeta functions of algebraic-geometric \(\mathbb{Z}_p\)-towers

摘录 / EXCERPT: Let $(C_n)_{n\geq 0}$ be a $\mathbb{Z}_p$-tower of curves from algebraic geometry. Let g_n be the genus of C_n, let $d_\alpha(n)$ be the slope-counting quantity defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-018762 | MATHDB-344733

Voevodsky's slice-filtration conjecture for infinite loop spaces

摘录 / EXCERPT: Let k be a perfect field. Write $\mathcal{SH}(k)$ for the motivic stable homotopy category and $\mathcal{SH}^{S^1}(k)$ for the S^1-stable motivic homotopy category. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018763 | MATHDB-344745

The Eudave-Muñoz surgery characterization for Seifert graph manifolds

摘录 / EXCERPT: Let $T_{a,b}$ and $T_{c,d}$ be nontrivial torus knots, and let $Y=Y(T_{a,b},T_{c,d})$. A surgery has half-integral toroidal slope when its slope is half-integral and the resulting...

conjecture | open | MathDB 页面 | 原文线索

V2-018764 | MATHDB-344746

Conjectural Iwahori-equivariant orthosymplectic Satake equivalence

摘录 / EXCERPT: Let V_0 and V_1 be respectively orthogonal and symplectic vector spaces, let $\Ffl_1$ be the variety of complete self-orthogonal flags in V_1, and let $\Ffl_0$ be a chosen co...

conjecture | open | MathDB 页面 | 原文线索

V2-018765 | MATHDB-344748

Unique nearest-point conjecture for the stochastic phylogenetic variety

摘录 / EXCERPT: Unique nearest-point conjecture. The distance from P_{0} to $\mathcal{V}_{T}^{+}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-018766 | MATHDB-344768

Polynomial canonical-height bound for integral points on abelian varieties

摘录 / EXCERPT: Let B be a smooth projective curve, let A/K be an abelian variety with a model $f\colon\mathcal{A}\to B$, where $\mathcal{A}_K=A$. Let $D\subset A$ be a reduced effective ample...

conjecture | open | MathDB 页面 | 原文线索

V2-018767 | MATHDB-344770

Noncommutative K3 conjecture for the Debarre-Voisin hyperplane section

摘录 / EXCERPT: Let $Y\subset\operatorname{Gr}(3,V_{10})$ be a very general hyperplane section, and let A be the K3 category in ... where the E_i are exceptional objects. Noncommutative...

conjecture | open | MathDB 页面 | 原文线索

V2-018768 | MATHDB-344771

K3-component decompositions for nested Debarre-Voisin varieties

摘录 / EXCERPT: Let $Y\subset\operatorname{Gr}(3,V_{10})$ be a very general hyperplane section, and let A be the K3 subcategory of $\operatorname{D}^b(Y)$ obtained as the semiorthogonal...

conjecture | open | MathDB 页面 | 原文线索

V2-018769 | MATHDB-344772

Lefschetz decomposition conjecture for the congruence of lines and Peskine variety

摘录 / EXCERPT: Let A be the K3 category of the Debarre-Voisin hyperplane section Y, and let T and P be the associated congruence of lines and Peskine variety. Let $\mathcal{U}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-018770 | MATHDB-344773

Crepant categorical resolution conjecture for the Coble cubic

摘录 / EXCERPT: Let C be the singular Coble cubic, let W be the associated smooth variety, let q^*W be its indicated $\mathbb{P}^1$-bundle, and let Γ be the associated genus-2 curve...

conjecture | open | MathDB 页面 | 原文线索

V2-018771 | MATHDB-344803

Holonomic D-module-Lagrangian correspondence conjecture

摘录 / EXCERPT: Let X be a smooth variety, let $\mathcal{D}(X)$ denote its ring of differential operators, and let T^*X be its cotangent bundle. A holonomic $\mathcal{D}(X)$-module is a...

conjecture | open | MathDB 页面 | 原文线索

V2-018772 | MATHDB-344812

The unobstructed-deformation conjecture for polarizations of Artinian monomial ideals

摘录 / EXCERPT: Let I be an Artinian monomial ideal and let J be a polarization of I. Consider a first-order deformation of J, not necessarily homogeneous for the standard grading, and sup...

conjecture | open | MathDB 页面 | 原文线索

V2-018773 | MATHDB-344827

Kummini-Lakshmibai-Sastry-Seshadri regularity conjecture for Grassmannian Schubert patches

摘录 / EXCERPT: Let $B\subseteq GL_n(\mathbb{C})$ be the Borel subgroup of upper triangular matrices, and let $w\in S_n$ be a Grassmannian permutation with descent at position k. Let Y be the...

conjecture | open | MathDB 页面 | 原文线索

V2-018774 | MATHDB-344828

Degree conjecture for Grothendieck polynomials of Grassmannian Schubert patches

摘录 / EXCERPT: Let $w\in S_n$ be a Grassmannian permutation with descent at position k, let Y be the associated standard open patch, and let J_w define the corresponding patch Y_w. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018775 | MATHDB-344876

Langlands duality conjecture for character varieties of free groups

摘录 / EXCERPT: Let $\Gamma=F_r$ be the free group of rank r, and let G and G^L be complex reductive Langlands dual groups. Write $\mathcal{X}_{rG}$ and $\mathcal{X}_{rG^L}$ for their character...

conjecture | open | MathDB 页面 | 原文线索

V2-018776 | MATHDB-344878

The monodromy conjecture for ideals

摘录 / EXCERPT: Let $\mathcal I=(f_1,\ldots,f_r)$ be an ideal in $\mathbb C[x_0,\ldots,x_n]$ whose associated subscheme $Y=V(\mathcal I)$ in $\mathbb C^{n+1}$ contains the origin. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018777 | MATHDB-344900

Frieze-Lovász generic rank conjecture for three-way tensors

摘录 / EXCERPT: Frieze-Lovász generic rank conjecture. Equality

conjecture | open | MathDB 页面 | 原文线索

V2-018778 | MATHDB-344919

Recursive definability of curves and Jacobians in perfect fields

摘录 / EXCERPT: Recursive definability conjecture. There are formulas $\phi_{n,g}(\vec{x})$, $\psi_{n,g}(\vec{x},\vec{y})$, and $\rho_{n,g}(\vec{x},\vec{z})$, depending recursively on n and g,...

conjecture | open | MathDB 页面 | 原文线索

V2-018779 | MATHDB-344920

The Weil conjecture for the counting Euler characteristic

摘录 / EXCERPT: Weil conjecture. The ℓ-adic component of $\chi(V(K_1))$ is

conjecture | open | MathDB 页面 | 原文线索

V2-018780 | MATHDB-344945

Sakellaridis-Venkatesh Plancherel conjecture for spherical varieties

摘录 / EXCERPT: Let F be a p-adic field, let G be a split reductive group over F, and let $X=H\backslash G$ be a homogeneous spherical variety admitting an invariant measure. Let G_X be...

conjecture | open | MathDB 页面 | 原文线索

V2-018781 | MATHDB-344979

Kasprzyk-Tveiten's mirror conjecture for Fano varieties

摘录 / EXCERPT: Let a rigid maximally mutable Laurent polynomial be a Laurent polynomial with the rigidity and maximal-mutation properties used in the source, and identify Laurent polynomials up t...

conjecture | open | MathDB 页面 | 原文线索

V2-018782 | MATHDB-344984

The antisymmetric motivic v-adic multiple zeta conjecture

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$, let v be a prime, and let $\mathbf{s}=(s_i)_{i=1}^n$. Assume that the s_i are distinct, q-even, and that n is odd. Let S_n be the symmetric group...

conjecture | open | MathDB 页面 | 原文线索

V2-018783 | MATHDB-344985

The characteristic-two motivic relation conjecture for repeated q-even exponents

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$, let v be a prime, and assume $\operatorname{char}(\mathbb{F}_q)=2$. Let $n\geq1$, let $k_i\geq1$, and suppose that the s_i and the $2s_i$ occurring whe...

conjecture | open | MathDB 页面 | 原文线索

V2-018784 | MATHDB-344994

General Mordell-Lang conjecture for function fields

摘录 / EXCERPT: Let K/K_0 be an extension of algebraically closed fields, let A/K be an abelian variety, let $X\subseteq A$ be a subvariety, and let $\Lambda\subseteq A(K)$ be a subgroup of fi...

conjecture | open | MathDB 页面 | 原文线索

V2-018785 | MATHDB-345022

Extension of Poincaré-dual independence to types E7 and E8

摘录 / EXCERPT: Let h be a Hessenberg function and let $\operatorname{Hess}(N,h)$ be the associated regular nilpotent Hessenberg variety. For a Hessenberg function $h'\subset h$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-018786 | MATHDB-345027

Arithmetic Gan-Gross-Prasad conjecture in the Selmer rank one case

摘录 / EXCERPT: Assume the setup of the paper: Π_0 and Π_1 are the relevant representations under consideration, F is the CM field, E is their coefficient field, and the notation...

conjecture | open | MathDB 页面 | 原文线索

V2-018787 | MATHDB-345029

Counting conjecture for real critical points of cyclotomic polynomials

摘录 / EXCERPT: Let n be an odd square-free integer with k distinct prime factors, and let N_n denote the number of real roots of $\Phi_n^{\prime}$. For primes p not dividing n, write...

conjecture | open | MathDB 页面 | 原文线索

V2-018788 | MATHDB-345038

Bijection conjecture for irreducible components of Mustafin varieties under the natural projection

摘录 / EXCERPT: Let $\boldsymbol{\mathcal{M}}_{\mathbb{P}}\left(\overline{\bigwedge^k\Gamma^{\mathrm{st}}}\right)_{\kappa}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018789 | MATHDB-345047

Uniform rational points conjecture for K-trivial varieties over finite fields

摘录 / EXCERPT: Let d be a positive integer, and let q_d be an integer depending only on d. For a smooth projective variety X of dimension d over $\mathbb{F}_q$ with trivial canonical di...

conjecture | open | MathDB 页面 | 原文线索

V2-018790 | MATHDB-345055

K-polystability implies reduced uniform K-stability

摘录 / EXCERPT: Let (X,Δ) be a log Fano pair. K-polystability conjecture. If (X,Δ) is K-polystable, then it is reduced uniformly K-stable. This is the K-polystable analogue of the co...

conjecture | open | MathDB 页面 | 原文线索

V2-018791 | MATHDB-345066

Tautologicality conjecture for Lagrangian constant-cycle subvarieties

摘录 / EXCERPT: Let X be a K3 surface and ${\mathsf M}$ a moduli space of stable sheaves on X with special zero-cycle $c_{\mathsf M}$. A subvariety $V\subset {\mathsf M}$ is Lagrangian constan...

conjecture | open | MathDB 页面 | 原文线索

V2-018792 | MATHDB-345068

Constant-cycle subvariety vanishing conjecture for moduli spaces of sheaves on K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface and ${\mathsf M}$ a moduli space of stable sheaves on X of dimension $2n$. Let $V\subset {\mathsf M}$ be a constant-cycle subvariety, meaning that all poi...

conjecture | open | MathDB 页面 | 原文线索

V2-018793 | MATHDB-345082

Ishii's bounded blow-up conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $n\in\mathbb N$. For a singularity (X,x) of dimension n embedded into a smooth variety A with $\dim A=\mathrm{emb}(X,x)$, let I_X be the defining ideal of X, set...

conjecture | open | MathDB 页面 | 原文线索

V2-018794 | MATHDB-345083

Ishii's bounded jet-level conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $n\in\mathbb N$, let $I\subset(x_1,\dots,x_{n+c})$ define the n-dimensional variety $X=\operatorname{Spec}k[x_1,\dots,x_{n+c}]/I$, and let $s_m(0;X)$ be the jet-scheme invari...

conjecture | open | MathDB 页面 | 原文线索

V2-018795 | MATHDB-345101

Finiteness conjecture for Calabi-Yau threefold topological types

摘录 / EXCERPT: A Calabi-Yau threefold is a compact Calabi-Yau threefold, and its topological type is described by its topological invariants. Yau's finiteness conjecture. The number of topologica...

conjecture | open | MathDB 页面 | 原文线索

V2-018796 | MATHDB-345116

Duality conjecture for planar scaffold equations

摘录 / EXCERPT: Duality conjecture. For every association scheme with $d\geq\delta$ classes, the dual scaffold equations

conjecture | open | MathDB 页面 | 原文线索

V2-018797 | MATHDB-345139

The conjecture on meromorphic continuation and nonvanishing of motivic L-functions

摘录 / EXCERPT: Let M be a pure motive of weight w, with L-function $L(M,s)$ defined by its Euler product. The meromorphic continuation conjecture. The function $L(M,s)$ has a meromorphic ex...

conjecture | open | MathDB 页面 | 原文线索

V2-018798 | MATHDB-345140

The conjectural functional equation for motivic L-functions

摘录 / EXCERPT: Let M be a pure motive of weight w, let $M^{\vee}$ denote its dual, and define the completed L-function by $\Lambda(M,s)=L(M,s)L_{\infty}(M,s)$. The functional-equation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-018799 | MATHDB-345143

The Harris-Hassett rationality conjecture for Gushel-Mukai fourfolds

摘录 / EXCERPT: Let X be a GM fourfold. An associated K3 surface of X is a K3 surface arising from a period point of X lying on a hypersurface $\mathcal{D}_d$, with a nef algebraic class h...

conjecture | open | MathDB 页面 | 原文线索

V2-018800 | MATHDB-345154

The higher-dimensional subvariety exclusion conjecture for complex-hyperbolic quotients

摘录 / EXCERPT: Let $\Gamma<\operatorname{Aut}(\mathbf B^n)$ be a discrete, torsion-free subgroup, and let $M_\Gamma=\mathbf B^n/\Gamma$. For a positive integer k, suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-018801 | MATHDB-345159

Kuznetsov-Smirnov conjecture on residual categories and quantum spectra

摘录 / EXCERPT: Let X be a Fano variety of index m over an algebraically closed field of characteristic zero, and assume that its big quantum cohomology $\operatorname{BQH}(X)$ is generically...

conjecture | open | MathDB 页面 | 原文线索

V2-018802 | MATHDB-345160

Conjecture on rectangular Lefschetz collections for adjoint varieties

摘录 / EXCERPT: Let X be the adjoint variety of a simple algebraic group G over an algebraically closed field of characteristic zero. Adjoint-variety conjecture. If the Dynkin diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-018803 | MATHDB-345161

Conjecture on residual categories of coadjoint varieties

摘录 / EXCERPT: Let X be the coadjoint variety of a simple algebraic group G over an algebraically closed field of characteristic zero. Coadjoint-variety conjecture. The category...

conjecture | open | MathDB 页面 | 原文线索

V2-018804 | MATHDB-345182

Smoothness conjecture for the determinantal curve associated with nodal elliptic curves

摘录 / EXCERPT: Let X_0 be a general hyperelliptic K3 surface, and let Z_{2r} be the determinantal curve constructed to dominate the curve Γ_{2r} parametrizing divisors in...

conjecture | open | MathDB 页面 | 原文线索

V2-018805 | MATHDB-345200

Modern formulation of Manin's conjecture

摘录 / EXCERPT: Manin's conjecture. There exists a finite set of thin maps $f_i:Y_i\to X_i$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018806 | MATHDB-345307

Flag Hilbert scheme conjecture for the Borel moment-map quotients

摘录 / EXCERPT: Flag Hilbert scheme conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-018807 | MATHDB-345308

Compatibility conjecture for Borel and general linear moment-map quotients

摘录 / EXCERPT: Compatibility conjecture. The following diagram commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-018808 | MATHDB-345313

The Weak Specialness Conjecture for rational points

摘录 / EXCERPT: Let X be a projective variety defined over a number field. A variety is weakly special if some (equivalently, any) desingularization admits no dominant rational map to a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-018809 | MATHDB-345314

The Weak Specialness Conjecture for entire curves and function fields

摘录 / EXCERPT: Let X be a projective variety defined over ${\mathbb C}$. An entire curve on X is a holomorphic map from ${\mathbb C}$ to X. Let Y_{exc} be a proper closed subset,...

conjecture | open | MathDB 页面 | 原文线索

V2-018810 | MATHDB-345319

The arithmetic-Brody-general-type hyperbolicity conjecture

摘录 / EXCERPT: Hyperbolicity equivalence conjecture. The following are equivalent: X is arithmetically hyperbolic; $X_{\mathbb C}$ is Brody hyperbolic; and every integral subvariety of X is o...

conjecture | open | MathDB 页面 | 原文线索

V2-018811 | MATHDB-345320

Vojta's height conjecture for log-general-type pairs over function fields

摘录 / EXCERPT: Vojta's function-field conjecture. For every $\varepsilon>0$ there exist a constant C and a proper closed subvariety Z such that, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-018812 | MATHDB-345354

Twisted wave-front containment and orbit-dimension conjecture

摘录 / EXCERPT: Let F be a non-Archimedean local field, let $\bf G$ be an algebraic reductive group over F, and let $G=\bf G(F)$. Let $\bf H\subset G$ be a spherical subgroup defined over F,...

conjecture | open | MathDB 页面 | 原文线索

V2-018813 | MATHDB-345402

Beauville's conjecture on polynomial relations among divisors on hyper-Kähler manifolds

摘录 / EXCERPT: Beauville's conjecture. Any polynomial cohomological relation among divisor classes on M that holds in $H^*(M,\mathbb{Q})$ already holds in ${\rm CH}(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-018814 | MATHDB-345468

Debarre's lower-bound conjecture for Seshadri constants of indecomposable polarized abelian varieties

摘录 / EXCERPT: Let (A,L) be a g-dimensional indecomposable polarized abelian variety, meaning that it is not a product of polarized abelian varieties. Let (J_C,Θ_C) denote the Jacobian...

conjecture | open | MathDB 页面 | 原文线索

V2-018815 | MATHDB-345470

Debarre's conjecture on small Seshadri constants of principally polarized abelian varieties

摘录 / EXCERPT: Debarre's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-018816 | MATHDB-345471

Hodge's conjecture on algebraic cycles and Hodge classes

摘录 / EXCERPT: Let X be a smooth projective variety over $\operatorname{\mathbb{C}}$. For each i, let $\operatorname{CH}^i(X)$ be the codimension-i Chow group, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018817 | MATHDB-345472

Tate's conjecture on algebraic cycles and Galois-fixed étale classes

摘录 / EXCERPT: Let X be a smooth projective variety over a field k, let l be invertible in k, and write $\bar{k}$ for an algebraic closure of k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018818 | MATHDB-345473

The p-adic Tate conjecture for crystalline cohomology

摘录 / EXCERPT: Let X be a smooth projective variety over the finite field $k=\operatorname{\mathbb{F}}_q$ of characteristic p, with $q=p^m$. Let $W(k)$ be the ring of p-typical Witt vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-018819 | MATHDB-345476

Thomason's Tate conjecture via K-theory localization

摘录 / EXCERPT: Let X be a smooth projective variety over $\operatorname{\mathbb{F}}_q$, let l be a prime, and for each positive integer n consider...

conjecture | open | MathDB 页面 | 原文线索

V2-018820 | MATHDB-345477

Reichstein-Rogalski-Zhang conjecture on wild automorphisms

摘录 / EXCERPT: Reichstein-Rogalski-Zhang conjecture. If X admits a wild automorphism, then X is isomorphic to an abelian variety.

conjecture | open | MathDB 页面 | 原文线索

V2-018821 | MATHDB-345478

Zero-entropy conjecture for wild automorphisms

摘录 / EXCERPT: Zero-entropy conjecture. Every wild automorphism σ of a projective variety X has zero entropy.

conjecture | open | MathDB 页面 | 原文线索

V2-018822 | MATHDB-345479

Weak abundance conjecture for uniruled varieties

摘录 / EXCERPT: Weak abundance conjecture. X is uniruled if and only if it has negative Kodaira dimension:

conjecture | open | MathDB 页面 | 原文线索

V2-018823 | MATHDB-345485

Debarre-de Jong conjecture on lines on Fano hypersurfaces

摘录 / EXCERPT: Let $X\subset \mathbb{P}^n$ be a smooth hypersurface of degree d, and let $\mathcal{F}(X)$ denote its Fano variety of lines. A smooth hypersurface is Fano when $d\leq n$. The exp...

conjecture | open | MathDB 页面 | 原文线索

V2-018824 | MATHDB-345486

Generalized Debarre-de Jong conjecture for Fano complete intersections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a smooth Fano complete intersection of s hypersurfaces of degrees $d_1,\dots,d_s$, and write $d=\sum d_i$. The expected dimension of its Fano variet...

conjecture | open | MathDB 页面 | 原文线索

V2-018825 | MATHDB-345499

The partition characterization of relaxed SLMFs

摘录 / EXCERPT: The partition characterization of relaxed SLMFs. Suppose $\#\Omega=r(m+n-r)$ and every vertex of G_Ω has degree at least $r+1$. Then Ω is a relaxed (r,r,m)-SLMF if...

conjecture | open | MathDB 页面 | 原文线索

V2-018826 | MATHDB-345514

K-semistability and the entropy equality conjecture for Fano manifolds

摘录 / EXCERPT: K-semistability and entropy equality conjecture. A Fano manifold is K-semistable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-018827 | MATHDB-345522

Chen's conjecture on the essential minimum and asymptotic maximal slope

摘录 / EXCERPT: Let X be a projective variety over a global field, let D be a divisor on X, and let $\overline{D}$ be a metrized divisor. For each integer $n\geq 1$, set $V_n=H^0(X,nD)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018828 | MATHDB-345545

The growth constant for the stack-sorting map

摘录 / EXCERPT: Stack-sorting growth conjecture. The value of this limit lies in the interval $(1.68,1.73)$.

conjecture | open | MathDB 页面 | 原文线索

V2-018829 | MATHDB-345561

The maximal-set realization conjecture for Borel-fixed ideals

摘录 / EXCERPT: Maximal-set realization conjecture. For every set of ideals $J_1,\ldots,J_s\in\textnormal{\textsf{S}}_{p(t)}^{n}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-018830 | MATHDB-345623

sAND Conjecture on small arithmetic degree loci

摘录 / EXCERPT: Let $f:X\to X$ be a surjective endomorphism of a projective variety X defined over a number field K. For $d>0$, define ... where $\alpha_f(x)$ is the arithmetic degree and...

conjecture | open | MathDB 页面 | 原文线索

V2-018831 | MATHDB-345625

Relative sAND Conjecture for fiberwise endomorphisms

摘录 / EXCERPT: Let X and S be quasi-projective varieties over a number field K, and let $\pi:X\to S$ be a projective morphism. Let $f:X\to X$ be a surjective morphism satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-018832 | MATHDB-345626

Finite-union conjecture for small arithmetic degree loci

摘录 / EXCERPT: Let $f:X\to X$ be a surjective endomorphism of a projective variety with first dynamical degree $\delta_f>1$. Let $\{Z_\lambda\}_{\lambda\in\Lambda}$ be the collection of all f-p...

conjecture | open | MathDB 页面 | 原文线索

V2-018833 | MATHDB-345627

Conjectural Chern-class compatibility for toroidal embeddings

摘录 / EXCERPT: Let $L\subset\mathbb{C}$ be a number field with ring of integers $\mathcal{O}_L$, let X/L be a smooth projective variety equipped with a toroidal embedding $Z\subset X$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018834 | MATHDB-345629

The conjectural extension of the module-variety reconstruction theorem

摘录 / EXCERPT: Conjecture. Theorem 1 would remain valid without any assumptions on the algebras A and B other than their being isomorphic as graded S-modules; equivalently, the conditions i...

conjecture | open | MathDB 页面 | 原文线索

V2-018835 | MATHDB-345640

Lang's affine hyperbolicity conjecture for hyperbolically embeddable varieties

摘录 / EXCERPT: Lang's affine hyperbolicity conjecture. If $X_{0,\mathbb{C}}$ is hyperbolically embeddable, then X is absolutely arithmetically hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-018836 | MATHDB-345645

Weak Lang-Vojta conjecture I

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. A variety X over k is strongly-Brody hyperbolic over k if, for every subfield $k_0\subset k$, every model...

conjecture | open | MathDB 页面 | 原文线索

V2-018837 | MATHDB-345646

Strong Lang-Vojta conjecture II

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. For a projective variety X over k, consider pseudo-Mordellicity, pseudo-arithmetic hyperbolicity, pseudo-groupl...

conjecture | open | MathDB 页面 | 原文线索

V2-018838 | MATHDB-345647

Strong Lang-Vojta conjecture III

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. A variety over k is strongly-pseudo-Brody hyperbolic (respectively, strongly-pseudo-Kobayashi hyperbolic) if ever...

conjecture | open | MathDB 页面 | 原文线索

V2-018839 | MATHDB-345648

Strong Lang-Vojta conjecture IV

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. For a projective variety X over k, consider strong pseudo-Brody and pseudo-Kobayashi hyperbolicity, pseudo-Mord...

conjecture | open | MathDB 页面 | 原文线索

V2-018840 | MATHDB-345649

Strongest Lang-Vojta exceptional-locus conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero and let X be a projective variety over k. Let Δ_X^{gr}, Δ_X^{Mor},...

conjecture | open | MathDB 页面 | 原文线索

V2-018841 | MATHDB-345650

Pseudo-Noguchi finiteness conjecture I

摘录 / EXCERPT: Let X be a projective integral variety over $\mathbb{C}$. A pseudo-Brody hyperbolic variety is a variety that is Brody hyperbolic outside a proper closed subset. Pseudo-Noguchi c...

conjecture | open | MathDB 页面 | 原文线索

V2-018842 | MATHDB-345651

Non-archimedean Lang-Vojta conjecture

摘录 / EXCERPT: Let K be an algebraically closed complete non-archimedean valued field of characteristic zero. A variety X over K is K-analytically Brody hyperbolic if every morphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-018843 | MATHDB-345673

Strengthened independence of intersection points conjecture

摘录 / EXCERPT: Strengthened independence conjecture. For “most” sections, the following maps are injections:

conjecture | open | MathDB 页面 | 原文线索

V2-018844 | MATHDB-345674

Few-zeros conjecture for ample line bundles on curves

摘录 / EXCERPT: Few-zeros conjecture. For “most” ample line bundles L, every section of L^m has at least $g(C)$ zeros for every $m\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-018845 | MATHDB-345675

Abelian-variety few-zeros conjecture

摘录 / EXCERPT: Abelian-variety few-zeros conjecture. For “most” $p\in A(K)$, the equation

conjecture | open | MathDB 页面 | 原文线索

V2-018846 | MATHDB-345677

Adjoint Reidemeister torsion vanishing conjecture

摘录 / EXCERPT: Let M be an oriented compact 3-manifold with torus boundary, and let $X^\mathrm{irr}(M)$ be the character variety of irreducible $\mathrm{SL}_{2}(\mathbb{C})$-representations....

conjecture | open | MathDB 页面 | 原文线索

V2-018847 | MATHDB-345709

The limsup Height Gap Conjecture

摘录 / EXCERPT: Let X be a quasi-projective variety, let $\Phi\colon X\dashrightarrow X$ be a rational self-map, and let $f\colon X\dashrightarrow \mathbb{P}^1$ be a non-constant rational functi...

conjecture | open | MathDB 页面 | 原文线索

V2-018848 | MATHDB-345759

Simpson's integrality conjecture for infinitesimally rigid representations

摘录 / EXCERPT: Simpson's conjecture. Every such representation has integral image. This connects rigidity of local systems on projective varieties with arithmeticity; the source presents it as Si...

conjecture | open | MathDB 页面 | 原文线索

V2-018849 | MATHDB-345765

Generalized Abhyankar conjecture for the affine line

摘录 / EXCERPT: Let k be a field of characteristic $p>5$, and let $\mathbb{A}^1_k$ be the affine line over k. Let $\pi^{\rm loc}_A(\mathbb{A}^1_k)$ denote the set of finite local k-group sch...

open_problem | open | MathDB 页面 | 原文线索

V2-018850 | MATHDB-345801

Chan-Ivanov conjecture on representability of Coxeter Deligne-Lusztig spaces

摘录 / EXCERPT: Let G be the reductive group, W its Weyl group, b the relevant element, and $X_w(b)$ the associated p-adic Deligne-Lusztig space for $w obreak\relax\in W$. An element w i...

conjecture | open | MathDB 页面 | 原文线索

V2-018851 | MATHDB-345809

The smoothness conjecture for the EKOR zip stack

摘录 / EXCERPT: Smoothness conjecture. The morphism

conjecture | open | MathDB 页面 | 原文线索

V2-018852 | MATHDB-345813

The cost classification conjecture for skew-layered permutations

摘录 / EXCERPT: Let π be a skew-layered permutation of size n, and let $\mathsf{cost}(\pi)$ denote its pop-stack sorting cost. Exclude id and $-\mathsf{id}$. Skew-layered cost cla...

conjecture | open | MathDB 页面 | 原文线索

V2-018853 | MATHDB-345822

Ahmadi-Shparlinski's maximal angle rank conjecture for Jacobians

摘录 / EXCERPT: Ahmadi-Shparlinski's conjecture. Every ordinary, geometrically simple Jacobian over a finite field has maximal angle rank.

conjecture | open | MathDB 页面 | 原文线索

V2-018854 | MATHDB-345823

The CM Galois group realization conjecture for Weil numbers

摘录 / EXCERPT: Let $W_{2g}=C_2^g\rtimes S_g$ be the wreath product acting on the $2g$ conjugates in g complex-conjugate pairs, and let G be a transitive subgroup of W_{2g} containing comple...

conjecture | open | MathDB 页面 | 原文线索

V2-018855 | MATHDB-345824

Shankar-Tsimerman's Jacobian isogeny conjecture for abelian fourfolds

摘录 / EXCERPT: Let p be a prime, let A be a 4-dimensional abelian variety over $\overline{\mathbf{F}}_p$, and let a stable curve be allowed to be possibly reducible. Shankar-Tsimerman's con...

conjecture | open | MathDB 页面 | 原文线索

V2-018856 | MATHDB-345837

Strengthened West conjecture for average stack-sorting depth

摘录 / EXCERPT: Let $\mathcal D_n$ be the average number of iterations of the stack-sorting map needed to sort a permutation in S_n into the identity permutation. Strengthened West conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-018857 | MATHDB-345838

Revstack fertility monotonicity conjecture

摘录 / EXCERPT: Let $\operatorname{revstack}=s\circ\operatorname{rev}$, where s is the stack-sorting map and rev reverses a permutation. For a permutation $\sigma\in S_n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-018858 | MATHDB-345839

Conjectured limiting average depth for pop-stack-sorting

摘录 / EXCERPT: Let Pop be the pop-stack-sorting map, which reverses every maximal decreasing run of a permutation, and let $\mathcal D_n^{\mathsf{Pop}}$ be the average number of iterat...

conjecture | open | MathDB 页面 | 原文线索

V2-018859 | MATHDB-345849

The toric Fano ch_2-positivity conjecture

摘录 / EXCERPT: Let X be a smooth toric Fano d-fold. A smooth toric Fano variety is ch_2-positive when its second Chern character has positive intersection with every relevant surfa...

conjecture | open | MathDB 页面 | 原文线索

V2-018860 | MATHDB-345850

The toric Fano surface-intersection conjecture

摘录 / EXCERPT: Let X be a smooth toric Fano d-fold, and let V^d denote the exceptional toric Fano variety used in the paper. A torus invariant subsurface is a torus-invariant surface...

conjecture | open | MathDB 页面 | 原文线索

V2-018861 | MATHDB-345892

Veronese secant-index conjecture

摘录 / EXCERPT: Let $v_d^{(n)}(\mathbb{P}^n)$ be the degree-d Veronese variety, let $\mathfrak{L}(X)$ be its sequence of secant indices, and let ${\mathfrak{R}\mathfrak{L}}^G(X)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-018862 | MATHDB-345893

Planar Veronese secant-index formula

摘录 / EXCERPT: Planar Veronese secant-index conjecture. The maximal number of points in the finite reduced intersection of r linearly independent degree-d curves is b_{r-1}. The statement p...

conjecture | open | MathDB 页面 | 原文线索

V2-018863 | MATHDB-345894

Segre secant-index conjecture

摘录 / EXCERPT: Let $X=\sigma_{n,m}(\mathbb{P}^n\times\mathbb{P}^m)$ be the Segre variety, and let ${\mathfrak{R}\mathfrak{L}}^G(X)$ and $\mathfrak{L}(X)$ denote its greedy and secant-index sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-018864 | MATHDB-345895

Segre final-gap formula

摘录 / EXCERPT: Let $X=\sigma_{n,m}(\mathbb{P}^n\times\mathbb{P}^m)$, so that the relevant codimension is nm. Let $\mathfrak{L}_j(X)$ denote the jth secant index. Segre final-gap conjecture. O...

conjecture | open | MathDB 页面 | 原文线索

V2-018865 | MATHDB-345897

The topological mirror partner conjecture for toric degenerations

摘录 / EXCERPT: Let Y be a smooth complete algebraic variety that is the generic fiber of a flat family with special fiber Y_0, where Y_0 is a complete intersection in a complete toric varie...

conjecture | open | MathDB 页面 | 原文线索

V2-018866 | MATHDB-345898

The reverse-transition conjecture for f-mirror partners

摘录 / EXCERPT: Let Y be a smooth complete variety with toric degeneration Y_0, let $Y_0^\vee$ be its f-mirror partner, and suppose there is a geometric transition ... Let k_Z and m_Z deno...

conjecture | open | MathDB 页面 | 原文线索

V2-018867 | MATHDB-345937

The matroid rank conjecture for Hadamard products of linear spaces

摘录 / EXCERPT: Let $L_1,\dots,L_d \subseteq \mathbb{R}^E$ be finite-dimensional linear spaces, and let $M_1,\dots,M_d$ be their algebraic matroids. Write r_{M_i} for the rank function of M_i,...

conjecture | open | MathDB 页面 | 原文线索

V2-018868 | MATHDB-345939

Beilinson's conjecture for higher K-groups of smooth projective varieties

摘录 / EXCERPT: Let $X_{{\mathbb Q}}$ be a smooth projective variety over ${{\mathbb Q}}$, and let $X_{{\mathbb R}}:=X_{{\mathbb Q}}\times_{{\mathbb Q}}{{\mathbb R}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018869 | MATHDB-345951

The $cell$-adic monodromy-weight conjecture for $cell\neq p$

摘录 / EXCERPT: Let K be a complete discrete valuation field of characteristic zero with finite residue field k of characteristic p, let W_K be its Weil group with valuation map v, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-018870 | MATHDB-345981

Fontaine-Messing's p-adic variational Hodge conjecture

摘录 / EXCERPT: Fix a complete discretely valued field K of mixed characteristic $(0,p)$, with ring of integers $\mathcal{O}_K$ and perfect residue field k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018871 | MATHDB-345985

The conjecture that the feasible region for classical patterns is full-dimensional

摘录 / EXCERPT: Full-dimensionality conjecture. The feasible region clP_k is full-dimensional inside a manifold of dimension $|\mathcal{L}_k|$.

conjecture | open | MathDB 页面 | 原文线索

V2-018872 | MATHDB-345996

Polynomiality conjecture for Schur classes of tautological line bundles

摘录 / EXCERPT: Let S) be a smooth connected surface with $h^1(\mathcal{O}_S)=0$ whose Picard group is generated by line bundles $\mathcal{L}_1,\cdots,\mathcal{L}_m$. Consider a line bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-018873 | MATHDB-346008

Compatibility of global and critical obstruction theories for local Quot schemes

摘录 / EXCERPT: Compatibility conjecture. The restriction $\iota_{n,\alpha}^*{\mathbb E}$ agrees, as a symmetric perfect obstruction theory, with ${\mathbb E}_{\operatorname{crit}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-018874 | MATHDB-346011

Nearly-overconvergent universal eigenform conjecture

摘录 / EXCERPT: Nearly-overconvergent universal eigenform conjecture. There is an affinoid neighbourhood

conjecture | open | MathDB 页面 | 原文线索

V2-018875 | MATHDB-346115

The noncommutative variational Hodge conjecture

摘录 / EXCERPT: Let $f\colon X\to S$ be a smooth proper morphism of complex varieties, and let $\mathcal{C}\subset\operatorname{D}_{\mathrm{perf}}(X)$ be an S-linear admissible subcategory. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018876 | MATHDB-346120

Lower semi-continuity conjecture for log plurigenera of proper SNCL schemes

摘录 / EXCERPT: Let κ be a field, let s be the log point of κ, and let X/s be a proper SNCL scheme of pure dimension d. For $n,r\in\mathbb Z_{\geq 1}$, define the log geometric...

conjecture | open | MathDB 页面 | 原文线索

V2-018877 | MATHDB-346129

Parametrization conjecture for monomial ideals by bounded Hilbert-Burch matrices

摘录 / EXCERPT: Parametrization conjecture. For a monomial ideal E, the set $\mathcal{N}(E)_{<\underline{d}}$ parametrizes $V(E)$.

conjecture | open | MathDB 页面 | 原文线索

V2-018878 | MATHDB-346166

Motivic factorisation of the derived Hecke action

摘录 / EXCERPT: Let $Y(K)$ be the arithmetic manifold, let E be the coefficient field, and let M_{coad} be the coadjoint motive with its integral model $(M_{\mathrm{coad}})_{\mathbb Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-018879 | MATHDB-346199

Separate balance and poles conjecture for elliptic stable envelopes

摘录 / EXCERPT: Let X be a smooth symplectic variety with finite X^{T}, and suppose that the elliptic stable envelope exists. For $4\lambda,\mu\in X^{\mathsf{T}}$, define ... Here a a...

conjecture | open | MathDB 页面 | 原文线索

V2-018880 | MATHDB-346210

Embedding conjecture for canonical maps of general hypersurfaces in Abelian varieties

摘录 / EXCERPT: Let A be an Abelian variety of dimension g, and let $X\subset A$ be a smooth ample hypersurface of dimension $n=g-1$. Assume that (A,X) is a general pair giving a polarizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-018881 | MATHDB-346232

Kato-Saito and Colliot-Thélène's exactness conjecture for zero-cycles

摘录 / EXCERPT: Let k be a number field and let V be a proper smooth geometrically integral variety over k. Consider the complex ... where the modified Chow groups are the usual Chow groups at...

conjecture | open | MathDB 页面 | 原文线索

V2-018882 | MATHDB-346263

Central limit conjecture for descents after stack-sorting

摘录 / EXCERPT: Central limit conjecture. The sequence $(Y_n)_{n\geq 1}$ converges in distribution to a random variable Y such that

conjecture | open | MathDB 页面 | 原文线索

V2-018883 | MATHDB-346272

Logarithmic rigidity conjecture for étale motives

摘录 / EXCERPT: Logarithmic rigidity conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-018884 | MATHDB-346273

Comparison conjecture for étale and logarithmic étale motives

摘录 / EXCERPT: Étale comparison conjecture. There is an equivalence of triangulated categories

conjecture | open | MathDB 页面 | 原文线索

V2-018885 | MATHDB-346274

Coefficient comparison conjecture for logarithmic and ordinary étale motives

摘录 / EXCERPT: Coefficient comparison conjecture. There are equivalences of triangulated categories

conjecture | open | MathDB 页面 | 原文线索

V2-018886 | MATHDB-346278

The finite-completion characterization for low-rank matrix patterns

摘录 / EXCERPT: Finite-completion conjecture. Without loss of generality, let the number of 1's at every row and column of Ω be at least $r+1$. Then a generic rank-r matrix X observed...

conjecture | open | MathDB 页面 | 原文线索

V2-018887 | MATHDB-346283

Coleman-Mazur-Buzzard-Kilford halo conjecture for eigencurves

摘录 / EXCERPT: Let p be an odd prime, let $\mathscr{W}$ be the rigid analytic space parametrizing continuous characters of $\mathbb{Z}_p^\times$, and write $T(w)=w(\exp(p))-1$ for the T-coord...

conjecture | open | MathDB 页面 | 原文线索

V2-018888 | MATHDB-346298

Compatibility conjecture for the critical and framed-sheaf obstruction theories

摘录 / EXCERPT: The compatibility conjecture. The isomorphism η induces an isomorphism of perfect obstruction theories

conjecture | open | MathDB 页面 | 原文线索

V2-018889 | MATHDB-346352

The anticanonical component-count conjecture for log Calabi-Yau compactifications

摘录 / EXCERPT: Let X be a smooth Fano variety, and let (Z,f) be a log Calabi-Yau compactification of its toric Landau-Ginzburg model. The anticanonical component-count conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-018890 | MATHDB-346361

The homological projective duality conjecture for the Cayley plane

摘录 / EXCERPT: Homological projective duality conjecture. The homological projective dual variety of the Cayley plane is a non-commutative resolution of the Cartan cubic hypersurface in...

conjecture | open | MathDB 页面 | 原文线索

V2-018891 | MATHDB-346417

The Γ-André-Oort conjecture

摘录 / EXCERPT: Let S_Γ be the variety under consideration, and let a Γ-special point and a Γ-special subvariety be defined as in the paper. The Γ-André-Oort conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-018892 | MATHDB-346418

The $\mathbb{Z}$-special-point characterization conjecture

摘录 / EXCERPT: Let (G,X,Γ) be a real Shimura datum and let $W\subset S_\Gamma$ be an algebraic subvariety. The $\mathbb{Z}$-special-point characterization conjecture. The subvariety W co...

conjecture | open | MathDB 页面 | 原文线索

V2-018893 | MATHDB-346419

The CM-point characterization conjecture

摘录 / EXCERPT: Let (G,X,Γ) be a real Shimura datum and let $W\subset S_\Gamma$ be an irreducible algebraic subvariety. A CM-point is a point of S_Γ at which the Mumford-Tate group o...

conjecture | open | MathDB 页面 | 原文线索

V2-018894 | MATHDB-346420

The bi-arithmetic-point characterization conjecture

摘录 / EXCERPT: Let S_Γ be the variety under consideration, and let bi-arithmetic points and bi-arithmetic subvarieties be defined as in the paper. The bi-arithmetic-point characterization...

conjecture | open | MathDB 页面 | 原文线索

V2-018895 | MATHDB-346425

Teissier's conjecture on Arnold exponents and polar invariants

摘录 / EXCERPT: Let X be a smooth complex n-dimensional algebraic variety, let $f\in\operatorname{\mathcal O}_X(X)$ be nonzero, and let $P\in X$ be a point in the zero-locus of f such that...

conjecture | open | MathDB 页面 | 原文线索

V2-018896 | MATHDB-346442

Degree formulas for the Euclidean joint signature of a generic plane curve

摘录 / EXCERPT: Let $\mathcal{J}^{\mathcal{E}}_d$ denote the Euclidean joint signature for a generic plane curve of degree d. The multiprojective witness-set dimensions are...

conjecture | open | MathDB 页面 | 原文线索

V2-018897 | MATHDB-346445

Functional equation conjecture for zeta functions of arithmetic schemes

摘录 / EXCERPT: Functional equation conjecture. The function $\zeta(\mathcal X,s)$ should have a meromorphic continuation to all $s\in\mathbb C$ and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-018898 | MATHDB-346453

Biregularity conjecture for non-natural involutions on Hilbert schemes of points on K3 surfaces

摘录 / EXCERPT: Biregularity conjecture. If $k\geq 3$, then the non-natural non-symplectic involution which generates $\operatorname{Bir}(S^{[n]})$ is biregular.

conjecture | open | MathDB 页面 | 原文线索

V2-018899 | MATHDB-346456

Lower-bound conjecture for the Waldschmidt constant of fat point schemes

摘录 / EXCERPT: Let $Z\subset {\mathbb{P}}^N$ be a fat point scheme. For a homogeneous ideal J, write $\alpha(J)$ for its least nonzero degree, and write $\widehat{\alpha}(I(Z))$ for the Waldsch...

conjecture | open | MathDB 页面 | 原文线索

V2-018900 | MATHDB-346469

Uniformizability conjecture for moduli stacks of polarized varieties

摘录 / EXCERPT: Uniformizability conjecture. The stack $\mathcal{M}_h$ is uniformizable by a quasi-projective scheme over $\mathbb{Q}$. Uniformizability would allow finiteness results proved for s...

conjecture | open | MathDB 页面 | 原文线索

V2-018901 | MATHDB-346470

The persistence conjecture for arithmetic hyperbolicity

摘录 / EXCERPT: Persistence conjecture. If X is arithmetically hyperbolic over k, then its base change X_L is arithmetically hyperbolic over L. This asserts that finiteness of integral poi...

conjecture | open | MathDB 页面 | 原文线索

V2-018902 | MATHDB-346487

The Fermat mirror-map conjecture

摘录 / EXCERPT: For each positive integer n, let $\mathscr{F}_n$ be the Fermat-type Calabi-Yau n-fold and let t denote the mirror map for its Fermat pencil, with Fermat point $\psi=0$. Mirro...

conjecture | open | MathDB 页面 | 原文线索

V2-018903 | MATHDB-346488

The Fermat Deligne-period quotient conjecture

摘录 / EXCERPT: Assume the sub-Hodge-structure splitting conjecture, and let $\mathbf{H}^n_{a,j}$ be a two-dimensional direct summand in the resulting split, defined over the smallest real algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-018904 | MATHDB-346498

Generalized affine Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Let Y be a normal affine variety which is $\mathbb{Q}$-Gorenstein and endowed with a good action by a compact torus T. Let $\mathbb{g}$ be the Lie algebra of T, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018905 | MATHDB-346532

General strength equals general slice rank conjecture

摘录 / EXCERPT: Let f be a general form of degree $d\geq2$ in $n+1$ variables. Write $\mathrm{str}(f)$ for its strength and $\mathrm{sl.rk}_{n,d}^\circ$ for the general slice rank in degree d...

conjecture | open | MathDB 页面 | 原文线索

V2-018906 | MATHDB-346542

Algebraic PDE and PDI characterization conjecture for polynomial neural-network spaces

摘录 / EXCERPT: Let $\mathcal{N}$ be a feedforward neural network with inputs labeled by $x_1,\dots,x_n$, and let $\mathbf{F}^{\rm act}_d(\mathcal{N})$ and $\mathbf{F}_d(\mathcal{N})$ denote the p...

conjecture | open | MathDB 页面 | 原文线索

V2-018907 | MATHDB-346562

Inversion of stability conjecture for divisors computing minimal log discrepancies

摘录 / EXCERPT: Let $\Gamma\subseteq[0,1]$ be a finite set, let X be a normal quasi-projective variety, and let $x\in X$ be a closed point. For an $\mathbb{R}$-divisor $B=\sum_i b_iB_i$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-018908 | MATHDB-346569

The disk-potential deformation conjecture for knot augmentations

摘录 / EXCERPT: Disk-potential deformation conjecture. For any knot K, there is a collection of functions $\epsilon=\{\alpha_{ij}(t,p)\}$ such that $U_{K}^{0,\epsilon}(p,t)$ is invariant under d...

conjecture | open | MathDB 页面 | 原文线索

V2-018909 | MATHDB-346571

Kuznetsov's conjecture on maximal exceptional collections of line bundles

摘录 / EXCERPT: Let X be a smooth projective variety, and let $D(X)$ denote its bounded derived category of coherent sheaves. An exceptional collection of line bundles is a sequence of line bund...

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V2-018910 | MATHDB-346572

Lunts-Yan-Zhang conjecture for smooth toric Fano threefolds and fourfolds

摘录 / EXCERPT: Let X be a smooth toric Fano threefold or fourfold. An exceptional collection of line bundles of maximal length is an exceptional collection of line bundles whose length is maxim...

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V2-018911 | MATHDB-346573

The Noether inequality for minimal threefolds of general type with small geometric genus

摘录 / EXCERPT: Noether inequality. Then

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V2-018912 | MATHDB-346605

The consistent-sign-pattern conjecture for cluster configuration spaces

摘录 / EXCERPT: Consistent-sign-pattern conjecture. The number of connected components of ${\mathcal M}_D(\mathbb{R})$ equals the number of sign patterns of the u-variables consistent with the e...

conjecture | open | MathDB 页面 | 原文线索

V2-018913 | MATHDB-346637

The non-Archimedean Green-Griffiths-Lang conjecture

摘录 / EXCERPT: Let K be an algebraically closed, complete, non-Archimedean valued field of characteristic zero, and let X/K be a projective variety. Say that X is groupless over K when it...

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V2-018914 | MATHDB-346638

Nondefectivity conjecture for secant varieties of Chow-Veronese varieties

摘录 / EXCERPT: Let $\boldsymbol d$ be a composition of the degree data, let n be a nonnegative integer, and let $\operatorname{CV}_{\boldsymbol d}(\mathbb P^n)$ denote the corresponding Chow-Ve...

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V2-018915 | MATHDB-346649

The strong density conjecture for arithmetic representations

摘录 / EXCERPT: Let X_0 be a smooth geometrically connected variety over a finite field $k={\mathbb F}_q$ of characteristic $p\neq\ell$, let $X=X_0\otimes_k\bar k$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-018916 | MATHDB-346656

Small arithmetic non-density conjecture

摘录 / EXCERPT: Let K be a number field, let X be a projective variety over K, and let $f:X\to X$ be a surjective endomorphism. For a positive constant d, let $X(d)$ denote the degree-boun...

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V2-018917 | MATHDB-346675

Cohomological rigidity conjecture for toric Fano manifolds

摘录 / EXCERPT: Let X and Y be toric Fano manifolds, and let $H^*(X)$ and $H^*(Y)$ denote their cohomology rings. A cohomology ring isomorphism preserves their first Chern classes when it maps...

conjecture | open | MathDB 页面 | 原文线索

V2-018918 | MATHDB-346686

Spectral halo conjecture for Hilbert modular eigenvarieties

摘录 / EXCERPT: Let I index the relevant embeddings, let $\rX_D^{>r}$ be the boundary region of the eigenvariety, let $\rW^{>r}$ be the corresponding weight-space region, let w be the weight m...

conjecture | open | MathDB 页面 | 原文线索

V2-018919 | MATHDB-346693

Hyperkähler structure conjecture for the cotangent bundle of higher Teichmüller space

摘录 / EXCERPT: The cotangent bundle $T^*\bm\hat{\mathcal{T}}^n$ is obtained by Hamiltonian reduction from the hyperkähler punctual Hilbert scheme, with its given symplectic structure near the zer...

conjecture | open | MathDB 页面 | 原文线索

V2-018920 | MATHDB-346694

Character-variety conjecture for the double Hamiltonian reduction

摘录 / EXCERPT: Let $\mathcal{A}//\mathcal{P}$ denote the space of parabolic connections with its infinitesimal higher-diffeomorphism action, and let Symp_0 denote the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-018921 | MATHDB-346749

Sikora's strong AJ conjecture for knots

摘录 / EXCERPT: Strong AJ conjecture. Suppose K is a knot in S^3. Then

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V2-018922 | MATHDB-346775

The geometric properties conjecture for Weil-Deligne parameter stacks

摘录 / EXCERPT: Let F be a non-archimedean local field, let $\check G$ be the dual group of a split reductive group, and let $X_{\check G}^{\rm WD}$ be the scheme of $\check G$-valued Weil-Delig...

conjecture | open | MathDB 页面 | 原文线索

V2-018923 | MATHDB-346804

Combinatorial irreducibility and irreducibility of the drawing variety

摘录 / EXCERPT: Let ${\mathcal{T}}_{\pentagon}$ be a constrained triangulation with an arbitrary boundary polygon. Combinatorial irreducibility conjecture. The triangulation...

conjecture | open | MathDB 页面 | 原文线索

V2-018924 | MATHDB-346805

Toroidal irreducibility and irreducibility of the drawing variety

摘录 / EXCERPT: Let ${\mathcal{T}}$ be a constrained triangulation of a parallelogram, and call it toroidally irreducible if it is combinatorially irreducible and there do not exist constraints...

conjecture | open | MathDB 页面 | 原文线索

V2-018925 | MATHDB-346806

Hyperness conjecture for generalized dissections

摘录 / EXCERPT: Let ${\mathcal{D}}$ be a generalized dissection, let ${\mathcal{T}}({\mathcal{D}})$ be its associated constrained triangulation, and let $V({\mathcal{D}})$ and $Y({\mathcal{D}})$ d...

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V2-018926 | MATHDB-346818

Conjecture on irreducibility of the Bloch variety for periodic Schrödinger operators

摘录 / EXCERPT: Let V be a periodic potential and let $B(V)$ denote its Bloch variety. An analytic set is irreducible if it cannot be represented as the union of two nonempty proper analytic sub...

conjecture | open | MathDB 页面 | 原文线索

V2-018927 | MATHDB-346819

Conjecture on irreducibility of Fermi varieties for periodic Schrödinger operators

摘录 / EXCERPT: Let $d\geq 2$, let V be a periodic potential, and let $F_{\lambda}(V)$ denote the Fermi variety at energy λ. An analytic set is irreducible if it cannot be represented as...

conjecture | open | MathDB 页面 | 原文线索

V2-018928 | MATHDB-346821

Positivity conjecture for Mather classes and local Euler obstructions

摘录 / EXCERPT: Let $X=G/P$ be a cominuscule space, and let $v,w\in W^P$. Write the Mather class of the Schubert variety X_w^P as ... The local Euler obstruction coefficients $e_{w,v}$ are non-neg...

conjecture | open | MathDB 页面 | 原文线索

V2-018929 | MATHDB-346822

Unimodality and log-concavity conjecture for Mather polynomials

摘录 / EXCERPT: Unimodality and log-concavity conjecture for Mather polynomials. The polynomial M_w has strictly positive coefficients and is unimodal. If in addition...

conjecture | open | MathDB 页面 | 原文线索

V2-018930 | MATHDB-346825

Reflexive differential nonvanishing conjecture

摘录 / EXCERPT: Reflexive differential nonvanishing conjecture. The sheaf $\Omega^{[q]}_X$ is pseudoeffective if and only if there exists a positive integer m such that

conjecture | open | MathDB 页面 | 原文线索

V2-018931 | MATHDB-346826

MRC-fibration conjecture for pseudoeffective differential forms

摘录 / EXCERPT: MRC-fibration pseudoeffectivity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-018932 | MATHDB-346830

The recursive lower-bound conjecture for pluricanonical birationality thresholds

摘录 / EXCERPT: For each positive integer n, let r_n be an integer such that the pluricanonical maps of smooth projective varieties of general type and dimension n are birational for all...

conjecture | open | MathDB 页面 | 原文线索

V2-018933 | MATHDB-346831

The variance bound for higher-degree defining equations

摘录 / EXCERPT: Let $X \subset \mathbb{P}$ be a smooth well formed Fano weighted complete intersection of dimension at least 2 which is not an intersection with a linear cone. Let $s_2(X)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-018934 | MATHDB-346843

Equivariant Beilinson-Bloch conjecture for unitary Shimura varieties

摘录 / EXCERPT: Let E/F be a CM extension of number fields, with complex conjugation $\mathtt{c}$, and let $n=2r$ be an even positive integer. Let $W_r=E^n$ with the specified skew-hermitian for...

conjecture | open | MathDB 页面 | 原文线索

V2-018935 | MATHDB-346845

Derived equivalence conjecture for Calabi-Yau fibrations from homogeneous roof bundles

摘录 / EXCERPT: Let G/P be a homogeneous roof, and $\mathcal{Z}$ a homogeneous roof bundle of type G/P with projective bundle structures ... Given a general section...

conjecture | open | MathDB 页面 | 原文线索

V2-018936 | MATHDB-346860

The Hecke orbit conjecture for PEL type Shimura varieties

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic p, and let S be a PEL type Shimura variety over k. For a point $x\in S$, let its prime-to-p Hecke orbit be the or...

conjecture | open | MathDB 页面 | 原文线索

V2-018937 | MATHDB-346878

The main conjecture on decompositions of Mustafin varieties

摘录 / EXCERPT: Fix $d\in\mathbb{Z}_{>0}$, a tuple $\underline{n}=(n_1,\dots,n_{d-1})\in\mathbb{Z}_{>0}^{d-1}$ with $n_1<\dots<n_{d-1}$, a positive integer n, and matrices M_I and lattices...

conjecture | open | MathDB 页面 | 原文线索

V2-018938 | MATHDB-346884

P=W conjecture for resolutions of singular character varieties

摘录 / EXCERPT: Let X be a compact Riemann surface, let G be a complex reductive algebraic group, and choose resolutions of singularities ... Let...

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V2-018939 | MATHDB-346891

van Garens-Ranganathan log-local conjecture for normal crossing divisors

摘录 / EXCERPT: Let X be a smooth projective variety and let $D=D_1+\cdots+D_n$ be an effective reduced normal crossing divisor whose components D_i are smooth, irreducible, and nef. For a cur...

conjecture | open | MathDB 页面 | 原文线索

V2-018940 | MATHDB-346892

Generalized log-local conjecture for normal crossing divisors

摘录 / EXCERPT: Let X be a smooth projective variety and let $D=D_1+\cdots+D_n$ be an effective reduced normal crossing divisor with each D_i smooth, irreducible, and nef. Let $I_1,\ldots,I_m$...

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V2-018941 | MATHDB-346893

Generalized local-orbifold conjecture for normal crossing divisors

摘录 / EXCERPT: Let X be a smooth projective variety, let $D=D_1+\cdots+D_n$ be an effective reduced normal crossing divisor with smooth, irreducible, nef components, and let $I_1,\ldots,I_m$ pa...

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V2-018942 | MATHDB-346902

Nef cotangent conjecture for projective manifolds with Stein universal cover

摘录 / EXCERPT: Let Y be a projective manifold, and let $\widetilde{Y}$ be its universal cover. Nef cotangent conjecture. If $\widetilde{Y}$ is a Stein manifold, then the cotangent bundle...

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V2-018943 | MATHDB-346915

The optimal labeling conjecture for bounded twin-width classes

摘录 / EXCERPT: Optimal labeling conjecture. Every bounded twin-width class has a $(1+o(1))\log n$-bits labeling scheme.

conjecture | open | MathDB 页面 | 原文线索

V2-018944 | MATHDB-346917

Deligne's companions conjecture for normal varieties

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field of characteristic p, let X_0 be a connected normal variety over $\mathbb{F}_q$, let $\ell\ne p$ be a prime, and let $\mathcal{V}_0$ be an i...

conjecture | open | MathDB 页面 | 原文线索

V2-018945 | MATHDB-346918

Conjecture that morphisms to normal varieties are lambda-uniform

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, and let Y_0 and Z_0 be varieties over $\mathbb{F}_q$. A morphism $f_0:Z_0\to Y_0$ is λ-uniform if, for every untwisted compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-018946 | MATHDB-346920

The polarization-independence conjecture for refined Vafa-Witten generating functions

摘录 / EXCERPT: Let S be the surface under consideration, let $c_1(S)$ denote its first Chern class, and let $v_0(S)$ be the null vector specified in the source. Write...

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V2-018947 | MATHDB-346925

Fraser-Sherman-Bennett conjecture on quasi-equivalence of relabeled plabic cluster structures

摘录 / EXCERPT: The source concerns open positroid varieties and the cluster structures determined by relabeled plabic graphs. For a relabeled graph G^ρ and an ordinary plabic graph H with...

conjecture | open | MathDB 页面 | 原文线索

V2-018948 | MATHDB-346927

Conjecture on units of open positroid coordinate rings

摘录 / EXCERPT: Let $\widetilde \Pi^{\circ}_\pi$ be the relevant open positroid variety, and let $\mathbb{P}_\pi$ be the free abelian group of Laurent monomials in the target frozen variables. The...

conjecture | open | MathDB 页面 | 原文线索

V2-018949 | MATHDB-346931

The Cartier base-point-free cone conjecture for b-classes

摘录 / EXCERPT: Let $\mathcal{X}$ be the space of smooth birational models of a smooth projective variety, and let $\operatorname{c-N}^k(\mathcal{X})$ and $\operatorname{w-N}^k(\mathcal{X})$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-018950 | MATHDB-346946

Serre's Sato-Tate equidistribution conjecture for generic abelian varieties

摘录 / EXCERPT: Let A be a generic abelian g-fold over $\mathbb{Q}$, meaning that the image of its adelic Galois representation is open in $\operatorname{GSp}_{2g}(\hat{\mathbb{Z}})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018951 | MATHDB-346947

Lang-Trotter conjecture for generic abelian varieties

摘录 / EXCERPT: Let A be a generic abelian g-fold over $\mathbb{Q}$, and let $\pi_{A,T}(x)$ count primes $p\leq x$ of good reduction for which $a_p(A)=T$. Write H_A for the adelic Galois ima...

conjecture | open | MathDB 页面 | 原文线索

V2-018952 | MATHDB-346948

Multiplicative Chow-Künneth decomposition for Fano varieties of K3 type

摘录 / EXCERPT: The multiplicative Chow-Künneth conjecture. The variety X has a multiplicative Chow-Künneth decomposition. This is presented as a partial answer to the problem of describing vari...

conjecture | open | MathDB 页面 | 原文线索

V2-018953 | MATHDB-346961

Factorization of the rational orbit action through the restricted Weyl group

摘录 / EXCERPT: Let k be a field with $\operatorname{char} k\neq 2$, let W_k be the restricted Weyl group, and let $F(W_k)$ be the group generated by involutions $\widetilde{s}_\alpha$ indexed...

conjecture | open | MathDB 页面 | 原文线索

V2-018954 | MATHDB-346963

Finite abelian descent VSA conjecture for all number fields

摘录 / EXCERPT: Let k be a number field. A variety satisfies VSA for $(\fab,f,k)$ if the corresponding finite abelian descent obstruction has the stated property of the paper. Finite abelian des...

conjecture | open | MathDB 页面 | 原文线索

V2-018955 | MATHDB-346965

Ramified étale-Brauer obstruction conjecture for stratified varieties

摘录 / EXCERPT: Let k be a number field and let X/k be a variety with $X(k)=\emptyset$. A stratification of X is a decomposition ... For each stratum, let $\cl{X_i}$ be its closure and let...

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V2-018956 | MATHDB-346994

The algebraic Weyl law

摘录 / EXCERPT: Algebraic Weyl law.

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V2-018957 | MATHDB-347126

Manin-Mumford conjecture for formal groups

摘录 / EXCERPT: Let G be a semi-abelian variety. Define a special subvariety to be a translate of a semi-abelian subvariety of G by a torsion point, and call the zero-dimensional special subva...

conjecture | open | MathDB 页面 | 原文线索

V2-018958 | MATHDB-347145

The nef-cone conjecture for codimension-two classes on the compactification of A_3

摘录 / EXCERPT: Nef-cone conjecture. The cone of nef codimension-two classes is

conjecture | open | MathDB 页面 | 原文线索

V2-018959 | MATHDB-347159

Free-algebra conjecture for the Betti character-stack Hall algebra

摘录 / EXCERPT: Let $g\geq 2$, let Σ_g be a genus-g surface, and let $p\colon \mathfrak{M}_{g,r}^{\operatorname{Betti}}\to \mathcal{M}_{g,r}^{\operatorname{Betti}}$ be the semisimplificat...

conjecture | open | MathDB 页面 | 原文线索

V2-018960 | MATHDB-347208

Motivic decomposition conjecture for moduli spaces of vector bundles

摘录 / EXCERPT: Let C be a smooth projective curve over an algebraically closed field of characteristic zero, let L be a line bundle on C of degree coprime to r, and let $M(r,L)$ be the mo...

conjecture | open | MathDB 页面 | 原文线索

V2-018961 | MATHDB-347212

Existence of Kähler-Einstein metrics on the manifolds \(\mathcal M_{s,p,n}\)

摘录 / EXCERPT: Let $\mathcal M_{s,p,n}$ be the manifold considered above, and let s,p,n be its defining parameters. A line bundle L on a projective variety X is called big when its highest...

conjecture | open | MathDB 页面 | 原文线索

V2-018962 | MATHDB-347223

Kisin's connectedness conjecture for Kisin varieties

摘录 / EXCERPT: Let K be a finite extension of $\Q_p$ with $p>2$, let $\mathbb F$ be a finite field, and let ... be an n-dimensional continuous representation of the absolute Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-018963 | MATHDB-347238

Nonexistence of a universal regular homomorphism for a specific fourfold

摘录 / EXCERPT: Let A be the generic abelian 3-fold described in the preceding context, with field of constants k_0 finitely generated over $\mathbb{Q}$. Choose an elliptic curve $E/k_0(t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-018964 | MATHDB-347249

Brosnan's decomposition and Weyl-group monodromy conjecture

摘录 / EXCERPT: Let G be a connected reductive algebraic group over $\mathbb{C}$, let M be a Hessenberg subspace, and let $\pi_M:G\times^B M\to\mathfrak{g}$ be the Hessenberg map. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-018965 | MATHDB-347250

The converse to the projective support characterization for quasi-reductive supergroups

摘录 / EXCERPT: Let G be a quasi-reductive supergroup, and let $M\in\operatorname{Rep}(G)$. Write $\mathfrak{g}_{\mathrm{neat}}$ for the neat locus in the odd part of the Lie superalgebra, and...

conjecture | open | MathDB 页面 | 原文线索

V2-018966 | MATHDB-347258

Rational contractibility conjecture for logarithmic de Rham-Witt sheaves

摘录 / EXCERPT: Let $W_r\Omega^n_{\log}$ denote the logarithmic de Rham-Witt sheaf. Rational contractibility conjecture. Each logarithmic de Rham-Witt sheaf $W_r\Omega^n_{\log}$ is rationally cont...

conjecture | open | MathDB 页面 | 原文线索

V2-018967 | MATHDB-347266

The sphericality characterization for Schubert varieties

摘录 / EXCERPT: Let G be a complex reductive group with Borel subgroup B, Weyl group W, and let $X_w=\overline{BwB/B}$ be the Schubert variety associated with $w\in W$. For $I\subseteq J(w)$...

conjecture | open | MathDB 页面 | 原文线索

V2-018968 | MATHDB-347267

The asymptotic rarity conjecture for maximally spherical permutations

摘录 / EXCERPT: For each n, let ${\mathfrak S}_n$ be the symmetric group, let $J(w)$ be the descent set used to define maximal sphericality, and call w maximally spherical when it is $J(w)$-sp...

conjecture | open | MathDB 页面 | 原文线索

V2-018969 | MATHDB-347268

The pattern-avoidance conjecture for maximally spherical permutations

摘录 / EXCERPT: For $u\in{\mathfrak S}_n$ and $v\in{\mathfrak S}_N$, write $u\hookrightarrow v$ when there are indices $\phi_1<\cdots<\phi_n$ such that $v(\phi_1),\ldots,v(\phi_n)$ have the same r...

conjecture | open | MathDB 页面 | 原文线索

V2-018970 | MATHDB-347269

The staircase-weight multiplicity-free converse conjecture

摘录 / EXCERPT: Suppose $w\in{\mathfrak S}_n$ and $I\subseteq J(w)$. Let $D=[n-1]\setminus I$, and let $\lambda^{\mathrm{staircase}}=(n,n-1,\ldots,3,2,1)$. Write κ_β for the relevant ch...

conjecture | open | MathDB 页面 | 原文线索

V2-018971 | MATHDB-347270

The distinct-weight multiplicity obstruction conjecture

摘录 / EXCERPT: Suppose $w\in{\mathfrak S}_n$ and $I\subseteq J(w)$, let $D=[n-1]\setminus I$, and let $\lambda^{\mathrm{distinct}}=(\lambda_1>\lambda_2>\cdots>\lambda_n)$. Write κ_β fo...

conjecture | open | MathDB 页面 | 原文线索

V2-018972 | MATHDB-347325

The dimension conjecture for determinantal loci

摘录 / EXCERPT: The dimension conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-018973 | MATHDB-347326

The fixed-degree matrix dimension conjecture

摘录 / EXCERPT: Fix integers $r\ge1$, $c>2-r$, $t\ge2$, and $d\ge1$. Let $\mathcal{A}$ be a homogeneous $t\times(t+c-1)$ matrix with entries of degree e, and set $A=R/I_{t-r+1}(\mathcal{A})$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-018974 | MATHDB-347327

The generic smoothness and matrix-deformation conjecture for determinantal schemes

摘录 / EXCERPT: Let $r\ge1$, $c>2-r$, and let $\mathcal{A}$ be a homogeneous $t\times(t+c-1)$ matrix with entries of degree a_j-b_i. Set $A=R/I_{t-r+1}(\mathcal{A})$ and suppose...

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V2-018975 | MATHDB-347346

Cerveau-Lins Neto conjecture on foliations with infinite transverse action

摘录 / EXCERPT: Cerveau-Lins Neto conjecture. If the transverse action of $f\in\operatorname{End}(X,\mathcal F)$ is infinite, then $\mathcal F$ is virtually transversely additive.

conjecture | open | MathDB 页面 | 原文线索

V2-018976 | MATHDB-347349

Conjecture on determinantal equations of eigenschemes of symmetric tensors

摘录 / EXCERPT: Let $n\geq 2$. For a homogeneous polynomial $f\in\mathbb{C}[x_0,\dots,x_n]_d$, let $E(f)$ denote its eigenscheme. Consider a set...

conjecture | open | MathDB 页面 | 原文线索

V2-018977 | MATHDB-347380

Geyer-Jarden's torsion conjecture for abelian varieties

摘录 / EXCERPT: Let K be a finitely generated field over its prime field, let e be a positive integer, and let σ range over $\mathrm{Gal}(K^\mathrm{sep}/K)^e$. For almost all such...

conjecture | open | MathDB 页面 | 原文线索

V2-018978 | MATHDB-347381

The MRC-foliation characterization for varieties with nef anti-canonical bundle

摘录 / EXCERPT: Let X be a smooth n-dimensional projective variety with nef anti-canonical bundle, and let $\mathcal{F}$ be a foliation on X. For some ample divisors $H_1,\ldots,H_{n-1}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-018979 | MATHDB-347394

The finite-subgroup criterion for solidity of toric Fano varieties

摘录 / EXCERPT: Let X be a toric Fano variety with at most terminal singularities. Let G_X be the normalizer of a maximal torus $\mathbb{T}$ in $\operatorname{Aut}(X)$, and let...

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V2-018980 | MATHDB-347397

Gizatullin's automorphism conjecture for complements of smooth cubic surfaces

摘录 / EXCERPT: Let S be a smooth cubic surface in $\mathbb P^3$. Gizatullin's automorphism conjecture. Every automorphism of the affine variety $\mathbb P^3\setminus S$ is induced by an automor...

conjecture | open | MathDB 页面 | 原文线索

V2-018981 | MATHDB-347400

The solid Fano hypersurface conjecture for index at least two

摘录 / EXCERPT: Let X be a quasi-smooth Fano hypersurface, and let ι_X denote its Fano index. A Fano variety is solid if it admits no birational map to a Mori fibre space over a positive-d...

conjecture | open | MathDB 页面 | 原文线索

V2-018982 | MATHDB-347420

Maximal rank conjecture for varieties with a hypersurface secant variety

摘录 / EXCERPT: Let $X \subset \mathbb{P}^N$ be a smooth non-degenerate projective variety, and let g be the generic X-rank. Denote by r_{max} the maximum X-rank and by...

conjecture | open | MathDB 页面 | 原文线索

V2-018983 | MATHDB-347430

Morel's functoriality conjecture for geometric transfers

摘录 / EXCERPT: Let k be the base field, let $M\in\mathbf{HI}(k)$ be a homotopy sheaf, and let F/E be a finite field extension generated by a system $(x_1,\ldots,x_r)$. Write ... for the composi...

conjecture | open | MathDB 页面 | 原文线索

V2-018984 | MATHDB-347431

Bachmann-Yakerson conservativity conjecture for motivic stabilization

摘录 / EXCERPT: For a natural number $d\geq 1$, let $\mathbf{SH}^{S^1}(k)(d)$ be the localizing subcategory generated by $\Sigma^d_{\mathbb{G}_m}\Sigma^\infty_{S^1}X_+$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-018985 | MATHDB-347466

Mumford's conjecture that varieties of Kodaira dimension minus infinity are uniruled

摘录 / EXCERPT: Let X be a smooth projective variety, and let $\kappa^h(X)$ denote its holomorphic Kodaira dimension. Mumford's conjecture. If ... then X is uniruled. This is the converse to the...

conjecture | open | MathDB 页面 | 原文线索

V2-018986 | MATHDB-347483

The periodic-point conjecture for the stack-sorting maps $s_{\{132,213\}}$ and $s_{\{231,213\}}$

摘录 / EXCERPT: Let S_n be the set of permutations of length n, and let $s_{\{132,213\}}$ and $s_{\{231,213\}}$ be the corresponding stack-sorting maps. A permutation is periodic if some posit...

conjecture | open | MathDB 页面 | 原文线索

V2-018987 | MATHDB-347498

The general-type conjecture for high modular-tower levels

摘录 / EXCERPT: Let $\mathbb H$ be a modular tower, with levels parametrized by k, and let $\mathbb Q$ denote the rational field. A variety is of general type when some multiple of its canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-018988 | MATHDB-347500

The bounded--power conjecture for regular realizations

摘录 / EXCERPT: Let G be a finite group with a regular realization over $\mathbb Q$, let P_ℓ be an ℓ-Sylow subgroup of G, let $m=[G:P_\ell]$, and let r be the branch-point number o...

conjecture | open | MathDB 页面 | 原文线索

V2-018989 | MATHDB-347503

Hamaker-Pechenik-Weigandt's CDG Gröbner basis pattern-avoidance conjecture

摘录 / EXCERPT: Hamaker-Pechenik-Weigandt's conjecture. The CDG generators are a diagonal Gröbner basis for I_w if and only if w avoids all eight patterns

conjecture | open | MathDB 页面 | 原文线索

V2-018990 | MATHDB-347504

Hamaker-Pechenik-Weigandt's multiplicity-free Schubert polynomial conjecture

摘录 / EXCERPT: Hamaker-Pechenik-Weigandt's conjecture. If the Schubert polynomial of w is a multiplicity-free sum of monomials, then the CDG generators of I_w are a diagonal Gröbner basis.

conjecture | open | MathDB 页面 | 原文线索

V2-018991 | MATHDB-347639

The Higgs-bundle curve-semistability characterization

摘录 / EXCERPT: Curve-semistability characterization. The following are equivalent:

conjecture | solved | MathDB 页面 | 原文线索

V2-018992 | MATHDB-347640

Freitag's freeness conjecture for modular varieties that are weighted projective spaces

摘录 / EXCERPT: Freitag's freeness conjecture. If the modular variety $(\mathcal D / \Gamma)^*$ is a weighted projective space, then there exists a weight k_0 and a multiplier system v_0 of we...

conjecture | open | MathDB 页面 | 原文线索

V2-018993 | MATHDB-347642

Eigenvalue sign and maximum conjecture for the perfect matching association scheme

摘录 / EXCERPT: Eigenvalue sign and maximum conjecture. The greatest eigenvalue in the row indexed by $[2k-2\ell,2\ell]$ occurs in the class $[2k-2\ell,2\ell]$, and every eigenvalue in that row be...

conjecture | open | MathDB 页面 | 原文线索

V2-018994 | MATHDB-347673

Euler-symmetric embedding conjecture for Picard-rank-one equivariant compactifications

摘录 / EXCERPT: Let X be a smooth Fano variety of Picard number one that is an equivariant compactification of a vector group. Euler-symmetric embedding conjecture. There is a suitable projectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-018995 | MATHDB-347691

The projective-space conjecture for smooth Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety of Picard number one, meaning that $-K_X$ is ample and $\rho(X)=1$. Suppose that X admits a non-isomorphic surjective endomorphism f. The proje...

conjecture | open | MathDB 页面 | 原文线索

V2-018996 | MATHDB-347692

The linear-subspace conjecture for inverse-periodic subvarieties of projective space

摘录 / EXCERPT: Let $X:=\mathbb{P}^n$, let $f:X\to X$ be an endomorphism with $\deg(f)\geq 2$, and let Y be a closed subvariety of X. The linear-subspace conjecture. If Y is f^{-1}-invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-018997 | MATHDB-347709

Gorsky-Neguț-Rasmussen conjecture on coherent computation of Khovanov-Rozansky homology

摘录 / EXCERPT: Gorsky-Neguț-Rasmussen conjecture. There is an isomorphism of bigraded vector spaces

conjecture | open | MathDB 页面 | 原文线索

V2-018998 | MATHDB-347724

Ito's torsion analogue of the weight-monodromy conjecture

摘录 / EXCERPT: Torsion analogue of the weight-monodromy conjecture. For every integer i, there exists a non-zero monic polynomial $P_i(T)\in\mathbb Z[T]$ such that its roots have complex absolu...

conjecture | open | MathDB 页面 | 原文线索

V2-018999 | MATHDB-347726

Conjecture B on the radius of Hilbert schemes

摘录 / EXCERPT: Let $Q_d(t)$ be a polynomial and let n_d be a positive integer for each $d\in\mathbf{N}$. For a Hilbert scheme $\operatorname{Hilb}^{Q_d(t)}\mathbf{P}^{n_d}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-019000 | MATHDB-347736

The periodic-point avoidance conjecture for consecutive-pattern-avoiding stack-sorting maps

摘录 / EXCERPT: Periodic-point avoidance conjecture. The periodic points of SC_σ are precisely the permutations in

conjecture | open | MathDB 页面 | 原文线索

V2-019001 | MATHDB-347737

The stabilization-time conjecture for the map \operatorname{SC}_{231}

摘录 / EXCERPT: Stabilization-time conjecture. For every $\pi\in S_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-019002 | MATHDB-347738

The universal fertility conjecture for consecutive-pattern-avoiding stack-sorting maps

摘录 / EXCERPT: Universal fertility conjecture. For every $\sigma\in S_3$ and every positive integer f, there exists a permutation π such that

conjecture | open | MathDB 页面 | 原文线索

V2-019003 | MATHDB-347739

The \operatorname{SC}_{132} image conjecture for maximally nonperiodic permutations

摘录 / EXCERPT: operatorname{SC}{132} image conjecture. If $\tau\in S_n$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-019004 | MATHDB-347746

Chow's basis conjecture for the equivariant cohomology of the permutohedral variety

摘录 / EXCERPT: Erasing marks conjecture. The collection

conjecture | open | MathDB 页面 | 原文线索

V2-019005 | MATHDB-347748

Poisson structure conjecture for the stack of semistable Higgs bundles over a stacky curve

摘录 / EXCERPT: Let $\mathcal{X}$ be a stacky curve, let α be a fixed parabolic structure, and let $\widetilde{\mathcal{M}}^{ss}_H(\mathcal{X},\alpha)$ denote the moduli stack of semistable...

conjecture | open | MathDB 页面 | 原文线索

V2-019006 | MATHDB-347784

The analytic continuation conjecture for motivic L-functions

摘录 / EXCERPT: Analytic continuation conjecture. La función $L(M,-)$ tiene una continuación meromorfa a todo $\mathbb{C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-019007 | MATHDB-347792

Stable pairs Virasoro constraints for simply connected smooth projective 3-folds

摘录 / EXCERPT: Let X be a simply-connected smooth projective 3-fold. Let $\mathbb D_{\textup{PT}}^{X}$ be the algebra generated by stable-pairs descendents $\textup{ch}_i(\gamma)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019008 | MATHDB-347801

Kontsevich-Zagier's period conjecture

摘录 / EXCERPT: A period is a complex number represented by an integral of an algebraic differential form over a domain defined by algebraic equations and inequalities, with algebraic coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-019009 | MATHDB-347802

The period conjecture for motivic periods

摘录 / EXCERPT: The period conjecture. The period map is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-019010 | MATHDB-347803

Panzer-Schnetz coaction conjecture for ϕ^4-periods

摘录 / EXCERPT: Panzer-Schnetz coaction conjecture. The Galois coaction closes on ϕ^4-periods:

conjecture | open | MathDB 页面 | 原文线索

V2-019011 | MATHDB-347804

Brown's coaction conjecture for generalised scalar Feynman integrals

摘录 / EXCERPT: Brown's coaction conjecture. The space $\mathcal{P}_{g}^{\rm m}$ is stable under the Galois coaction:

conjecture | open | MathDB 页面 | 原文线索

V2-019012 | MATHDB-347805

Brown's small graph principle for Feynman periods

摘录 / EXCERPT: Let G be a primitive log-divergent Feynman graph, and let $[H_G,\omega_G,\sigma]^{\rm m}$ denote the motivic realisation of its Feynman integral $I_G^{\rm m}$. Let Δ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-019013 | MATHDB-347809

Homologically trivial cycles are admissible cycles

摘录 / EXCERPT: Let X be a smooth projective variety over the finitely generated field K, and let $CH^i_l(X)$ denote the subgroup of cycles homologically equivalent to zero for l-adic cohomo...

conjecture | open | MathDB 页面 | 原文线索

V2-019014 | MATHDB-347810

Finite-kernel specialization for Lang-Néron groups

摘录 / EXCERPT: Let B be a smooth variety over a field k, let $K=k(B)$, and let A be an abelian K-variety. Assume that A has semistable reduction at every point of $B^{(1)}$. For any pro...

conjecture | open | MathDB 页面 | 原文线索

V2-019015 | MATHDB-347820

Chen-Sun-Wang's algebraic uniqueness conjecture for Kähler-Ricci flow limits

摘录 / EXCERPT: Let X be a smooth Fano manifold. A metric degeneration from X to a Gromov-Hausdorff limit X_∞ induces a finitely generated filtration $\mathcal{F}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-019016 | MATHDB-347821

Uniqueness and specialness conjecture for the minimizing valuation

摘录 / EXCERPT: Let v_* be a minimizer of the valuation functional $\tilde{\beta}$ for a ${\mathbb{Q}}$-Fano variety X, and let $\mathcal{F}_{v_*}$ be the filtration induced by v_*. Minimizi...

conjecture | open | MathDB 页面 | 原文线索

V2-019017 | MATHDB-347826

Strictly nef anticanonical bundles under smooth morphisms

摘录 / EXCERPT: Direct-image conjecture for strict nefness. If K_X^{-1} is strictly nef, then K_Y^{-1} is strictly nef.

conjecture | open | MathDB 页面 | 原文线索

V2-019018 | MATHDB-347834

The motivic origin conjecture for Beilinson's geometric height pairing

摘录 / EXCERPT: Let B be the base and X the smooth proper variety over the relevant function field, with $d=\operatorname{dim}X$ and $p+q=d+1$. Write $CH^p_{\rm hom}(X)_{\bf Q}$ for the group...

conjecture | open | MathDB 页面 | 原文线索

V2-019019 | MATHDB-347835

Beilinson's Arakelov height-pairing conjecture for arithmetic varieties

摘录 / EXCERPT: Let R be a regular arithmetic ring, let B_0 be a regular integral scheme flat and of finite type over R, and let X_0 be a smooth projective integral variety of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-019020 | MATHDB-347870

Compatibility conjecture for orbit embeddings and Steinberg maps

摘录 / EXCERPT: Compatibility conjecture. The diagram commutes; equivalently, the orbit embedding ι is compatible with the Steinberg maps:

conjecture | open | MathDB 页面 | 原文线索

V2-019021 | MATHDB-347891

The crystal-basis conjecture for spherical varieties

摘录 / EXCERPT: Crystal-basis conjecture. The crystal $\mathcal B_{X^\bullet}$ is isomorphic to the unique crystal basis of a finite-dimensional $\check G$-module V_X.

conjecture | open | MathDB 页面 | 原文线索

V2-019022 | MATHDB-347892

The crystal-basis conjecture for general spherical varieties

摘录 / EXCERPT: Crystal-basis conjecture. The crystal $\mathcal B_X$ is isomorphic to the unique crystal basis of a finite-dimensional $\check G$-module ρ_X.

conjecture | open | MathDB 页面 | 原文线索

V2-019023 | MATHDB-347893

The wrapped Fukaya interpretation of the universal deformation for hypertoric varieties

摘录 / EXCERPT: Let $\mathcal{V}$ be a polarized real arrangement, let $\mathcal{X}_{\mathcal{V}}$ be its complexified hyperplane complement, and let $\widetilde{B}(\mathcal{V})$ be the universal...

conjecture | open | MathDB 页面 | 原文线索

V2-019024 | MATHDB-347911

Real critical points conjecture for reciprocal likelihoods of d-pencils

摘录 / EXCERPT: Real critical points conjecture. There exists $s=(s_1,\ldots,s_n)\in\mathbb{R}^n$ such that $\widetilde{\ell}_S$ has $2r-3$ distinct real critical points.

conjecture | open | MathDB 页面 | 原文线索

V2-019025 | MATHDB-347916

The special-stabilizer lifting conjecture for arcs of Artin stacks

摘录 / EXCERPT: Special-stabilizer lifting conjecture. If all stabilizers of $\mathcal{X}$ are special groups, then $\operatorname{sep}_\pi^{-1}(0) \subset \mathscr{L}(X)$ is measurable and

conjecture | open | MathDB 页面 | 原文线索

V2-019026 | MATHDB-347933

Base point freeness conjecture for nef and big Kähler classes

摘录 / EXCERPT: Let (X,B) be a klt pair, where X is a normal Q-factorial compact Kähler variety. Let $\alpha\in H^{1,1}_{\rm BC}(X)$ be a nef class on X. If $\alpha-(K_X+B)$ is nef and big...

conjecture | open | MathDB 页面 | 原文线索

V2-019027 | MATHDB-347948

Geometric connectedness conjecture for Langlands-parameter summands

摘录 / EXCERPT: Connectedness conjecture. For any pair (ϕ,α), the $\mathcal{O}_{K_e}[1/p]$-scheme ${\underline Z}^{1}(W_F^0,{\hat G})_{\phi,\alpha}$ is connected and remains connected af...

conjecture | open | MathDB 页面 | 原文线索

V2-019028 | MATHDB-347955

Standard conjecture for nearby and vanishing cohomology

摘录 / EXCERPT: Let $X\to S$ be the projective family in the paper, let X_s be its special fiber, let X_η be its generic fiber, and let $H^*_{\psi}(X_s)$ and $H^*_{\phi}(X_s)$ denote the ne...

conjecture | open | MathDB 页面 | 原文线索

V2-019029 | MATHDB-347966

The motivic -homotopy contraction conjecture for torus actions

摘录 / EXCERPT: The motivic $\mathbf{A}^1$-homotopy contraction conjecture. The inclusion $Y\to U$ is an $\mathbf{A}^1$-homotopy equivalence, that is, an isomorphism in the $\mathbf{A}^1$-homotopy...

conjecture | open | MathDB 页面 | 原文线索

V2-019030 | MATHDB-347967

The Frobenius compatibility conjecture for abelian varieties

摘录 / EXCERPT: Let A be as above, put $L:=K_A^{\operatorname{conn}}$, and let $\operatorname{MT}_A^\sharp$ denote the set of conjugacy classes in MT_A. For a nonzero prime idea...

conjecture | open | MathDB 页面 | 原文线索

V2-019031 | MATHDB-347970

Dynamically weakly special grand-orbit intersection conjecture

摘录 / EXCERPT: Let $\mathcal{X}_1,\dots,\mathcal{X}_n$ be projective varieties, let f_i be endomorphisms of $\mathcal{X}_i$ respectively, and let $f=(f_1,\dots,f_n)$ have degree at least 2. S...

conjecture | open | MathDB 页面 | 原文线索

V2-019032 | MATHDB-347971

Conjecture comparing the elements $\xi_1(a,b)$ and Eisenstein motivic elements

摘录 / EXCERPT: For an integer $N\geq 1$ and $a,b\in\mathbb{Z}/N\mathbb{Z}$, let $\xi_1(a,b)$ and $\mathrm{Eis}^{0,0,1}(a,b)$ be the motivic elements introduced in the paper in...

conjecture | open | MathDB 页面 | 原文线索

V2-019033 | MATHDB-347974

The degree-3 covering map conjecture for elliptic surfaces

摘录 / EXCERPT: Let π_2 be the elliptic fibration constructed from a point of ${\mathcal M}'_{K,1}$, let σ_0 be its zero section, and let ϕ be the map associated with the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-019034 | MATHDB-348014

General infinitesimal Terracini Lemma

摘录 / EXCERPT: General infinitesimal Terracini Lemma. X is m-defective if and only if, given a general zero-dimensional curvilinear scheme γ of length $m+1$ contained in X, the dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-019035 | MATHDB-348029

The unirationality conjecture for varieties with full étale-exceptional collections

摘录 / EXCERPT: Unirationality conjecture. Then X is k-unirational.

conjecture | open | MathDB 页面 | 原文线索

V2-019036 | MATHDB-348050

The conjectural picture for supersingular symplectic varieties

摘录 / EXCERPT: Supersingularity conjecture. If X is second-Artin supersingular, then:

conjecture | open | MathDB 页面 | 原文线索

V2-019037 | MATHDB-348061

The non-viability criterion for the (110)-space

摘录 / EXCERPT: Non-viability criterion. If

open_problem | open | MathDB 页面 | 原文线索

V2-019038 | MATHDB-348072

Finkelberg-Mirković reducedness conjecture for spherical Schubert varieties

摘录 / EXCERPT: Let G be the group under consideration and let λ be a dominant coweight. Let $\overline{\mathcal{Y}^\lambda}\subset\operatorname{Gr}_G$ be the moduli scheme of principal...

conjecture | open | MathDB 页面 | 原文线索

V2-019039 | MATHDB-348084

Moore's attractor conjecture for Calabi-Yau varieties

摘录 / EXCERPT: Moore's attractor conjecture. If X is an attractor variety for a non-zero class $\gamma\in H^d(X,\mathbb{Q})$ such that $\gamma^{d,0}\ne 0$, then it has a model over...

conjecture | open | MathDB 页面 | 原文线索

V2-019040 | MATHDB-348110

Odaka's asymptotics conjecture for complete Ricci-flat Kähler metrics

摘录 / EXCERPT: Let X^{o} be a non-proper separated complex variety with at most Kawamata log terminal singularities, carrying a complete Ricci-flat weak Kähler metric g and a base point p....

conjecture | open | MathDB 页面 | 原文线索

V2-019041 | MATHDB-348113

Odaka's conjecture on minimally non-collapsing limits

摘录 / EXCERPT: In a polarized flat punctured holomorphic family $\pi^*:(\mathcal X^*,\mathcal L^*)\to\Delta^*$ of n-dimensional polarized klt projective varieties with continuous Kähler metrics...

conjecture | open | MathDB 页面 | 原文线索

V2-019042 | MATHDB-348122

The rationality conjecture for the archimedean motivic action

摘录 / EXCERPT: Archimedean rationality conjecture. The action of $U_f^\vee\otimes E\subseteq U_f^\vee\otimes_\iota\mathbb{C}$ on $H^\ast(X_\mathbb{C},\mathcal{E}_{\underline 1,1})_f$ preserves th...

conjecture | open | MathDB 页面 | 原文线索

V2-019043 | MATHDB-348136

Voisin's surface decomposability conjecture for hyper-Kähler varieties

摘录 / EXCERPT: A hyper-Kähler variety X of dimension $2n$ is called surface decomposable if there exist a smooth projective variety Γ, smooth projective surfaces $S_1,\ldots,S_n$, and ge...

conjecture | open | MathDB 页面 | 原文线索

V2-019044 | MATHDB-348137

Co-multiplicative birational Chow-Künneth decomposition conjecture for hyper-Kähler varieties

摘录 / EXCERPT: A birational Chow-Künneth decomposition of a smooth projective variety is co-multiplicative when the induced grading on its zero-cycles is compatible with the diagonal coproduct. C...

conjecture | open | MathDB 页面 | 原文线索

V2-019045 | MATHDB-348138

Bloch-Beilinson co-generation conjecture for birational motives

摘录 / EXCERPT: Let X be a smooth projective variety and let i be a positive integer. Assume that $\mathrm{H}^{*}_{\mathrm{tr}}(X)$ is generated by $\mathrm{H}^i_{\mathrm{tr}}(X)$. Bloch-Beili...

conjecture | open | MathDB 页面 | 原文线索

V2-019046 | MATHDB-348139

Strict grading conjecture for birational motives of hyper-Kähler varieties

摘录 / EXCERPT: Strict grading conjecture. This unital grading is a strict grading. More strongly, the graded co-algebra morphism co-induced by the graded split surjection...

conjecture | open | MathDB 页面 | 原文线索

V2-019047 | MATHDB-348213

The field-of-definition conjecture for special subvarieties of variations of Hodge structure

摘录 / EXCERPT: Let $\mathbb V$ be a $\mathbb Z$-variation of Hodge structure defined over a number field $L\subset\mathbb C$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019048 | MATHDB-348232

The bounding-cochain conjecture for pillowcase tangle invariants

摘录 / EXCERPT: Let T be a 2-tangle, let $R_\pi(T)\looparrowright P^*$ be its holonomy-perturbed traceless character variety, and let $b\in \mathit{CF}(R_\pi(T),R_\pi(T))$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019049 | MATHDB-348247

Voisin's integral Hodge/Tate conjecture for one cycles on rationally connected varieties

摘录 / EXCERPT: Let X be a smooth projective separably rationally connected variety of dimension d over an algebraically closed field. The groups ... and, when X is defined over the complex nu...

conjecture | open | MathDB 页面 | 原文线索

V2-019050 | MATHDB-348307

The injectivity and birationality criterion for positroid cells

摘录 / EXCERPT: Let S_M be a km-dimensional positroid cell of ${\rm Gr}_+(k,n)$. Let π_+ be the positive amplituhedron map and let π denote its complexified rational map on the closure...

conjecture | open | MathDB 页面 | 原文线索

V2-019051 | MATHDB-348351

Whitney's fibering conjecture

摘录 / EXCERPT: Whitney's fibering conjecture. Every analytic subvariety $V\subset U$ has a stratification such that each point $p_0\in V$ has a neighborhood U_0 with a semi-analytic fibration.

conjecture | open | MathDB 页面 | 原文线索

V2-019052 | MATHDB-348368

Klingler's characterization conjecture for variations of Hodge structures with dense CM points

摘录 / EXCERPT: Let ${\mathbb V}$ be a ${\mathbb Z}$-variation of Hodge structure on a smooth irreducible complex quasi-projective variety S, with generic Hodge datum $({\pmb G},{\mathcal D})$....

conjecture | open | MathDB 页面 | 原文线索

V2-019053 | MATHDB-348369

Klingler's Shimura-type conjecture for variations of Hodge structures

摘录 / EXCERPT: Let ${\mathbb V}$ be a ${\mathbb Z}$-variation of Hodge structure on a smooth irreducible complex quasi-projective variety S. A point of S is a CM point for $(S,{\mathbb V})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-019054 | MATHDB-348370

Klingler's geometric André-Oort conjecture for variations of Hodge structures

摘录 / EXCERPT: Let ${\mathbb V}$ be a ${\mathbb Z}$-variation of Hodge structure on a smooth irreducible complex quasi-projective variety S. A special subvariety for $(S,{\mathbb V})$ is positi...

conjecture | open | MathDB 页面 | 原文线索

V2-019055 | MATHDB-348371

The conical kinematics amplitude conjecture

摘录 / EXCERPT: Let $\mathcal{K}(k,n)$ be the kinematic space, let $J\in\binom{[n]}{k}^{\mathrm{nf}}$, and define ... where $(i,j)\in[1,k-1]\times[1,n-k+1]$. For $\alpha\in\mathbb{R}^{(k-1)(n-k+1)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-019056 | MATHDB-348381

Ito-Lozovanu conjecture on syzygies of polarized abelian varieties

摘录 / EXCERPT: Ito-Lozovanu conjecture. Under these assumptions, L satisfies property (N_p), meaning that the first p steps of the minimal graded free resolution of its section algebra are...

conjecture | open | MathDB 页面 | 原文线索

V2-019057 | MATHDB-348382

Caucci's conjecture on the basepoint-freeness threshold and syzygies

摘录 / EXCERPT: Let (A,L) be a polarized abelian variety of dimension g, and let p be a positive integer. Define the $\mathbb Q$-ample divisor ... Assume that for every abelian subvariety B...

conjecture | open | MathDB 页面 | 原文线索

V2-019058 | MATHDB-348413

The conjecture that Fano contact manifolds with cyclic Picard group are homogeneous

摘录 / EXCERPT: Let X be a Fano manifold with a contact structure $D \subset T_X$, and let $L=T_X/D$ be the quotient line bundle. Assume that $\operatorname{Pic} X \cong \mathbb{Z}\cdot L$. Fano...

conjecture | open | MathDB 页面 | 原文线索

V2-019059 | MATHDB-348427

Conjecture on divisors parameterized by stable objects on singular cubic surfaces

摘录 / EXCERPT: Let Y be a smooth cubic threefold, let S be a singular hyperplane section, and let D be a Weil divisor on S such that $\mathcal{O}_S(D)$ is a stable object parameterized by...

conjecture | open | MathDB 页面 | 原文线索

V2-019060 | MATHDB-348436

The semiampleness conjecture for K-trivial varieties

摘录 / EXCERPT: Semiampleness conjecture. Then D is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-019061 | MATHDB-348464

The Chow-ring multiplicativity conjecture for the Lefschetz eigenspace decomposition

摘录 / EXCERPT: Multiplicativity conjecture. The intersection product preserves the eigenspace grading: products of classes in ${\rm CH}^i(X)_s$ and ${\rm CH}^j(X)_t$ lie in...

conjecture | open | MathDB 页面 | 原文线索

V2-019062 | MATHDB-348501

André's generalized period conjecture

摘录 / EXCERPT: Let M be a pure or mixed motive defined over a subfield K of $\mathbb{C}$, and let $\mathcal{G}_{\mathrm{mot}}(M)$ denote its motivic Galois group. Generalized period conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-019063 | MATHDB-348502

Relative semi-abelian Schanuel conjecture

摘录 / EXCERPT: Let A be an abelian variety defined over $\overline{\mathbb{Q}}$, and let G be an extension of A by the torus $\mathbb{G}_m^s$ parametrized by...

conjecture | open | MathDB 页面 | 原文线索

V2-019064 | MATHDB-348503

Explicit relative semi-abelian conjecture

摘录 / EXCERPT: Let G be as in the relative semi-abelian conjecture, with notation c_i and t defined there. Explicit relative semi-abelian conjecture. ... This is stated as a corollary of the...

conjecture | open | MathDB 页面 | 原文线索

V2-019065 | MATHDB-348508

Span of critical points and irreducibility of structured low-rank varieties

摘录 / EXCERPT: Let $S \varnothing$ be a zero pattern, let $U\in\mathbb{R}^{m\times n}$, and let $r\in[m-1]$. Denote by $Z_{U,r}^{S}$ the critical-point variety and by H_U^{S} the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-019066 | MATHDB-348510

Euclidean distance degree for diagonal-zero rank-two varieties

摘录 / EXCERPT: Let $\mathcal{L}_2^S\subset\mathbb{C}^{m\times n}$ be the variety of matrices of rank at most 2 with diagonal zero pattern $S=\{(1,1),\ldots,(s,s)\}$, where $s\in[4]$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-019067 | MATHDB-348511

Quadratic ED-degree growth for square diagonal-zero rank-two varieties

摘录 / EXCERPT: Let $\mathcal{L}_2^S\subset\mathbb{C}^{m\times m}$ be the variety of square matrices of rank at most 2 with zero pattern $S=\{(1,1),\ldots,(s,s)\}$ for some $s\ge2$. Quadratic-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-019068 | MATHDB-348512

Euclidean distance degree for symmetric diagonal-zero rank-two varieties

摘录 / EXCERPT: Let $\mathcal{L}_2^S\subset\operatorname{Sym}_n(\mathbb{C})$ be the variety of symmetric matrices of rank at most 2 with diagonal zero pattern $S=\{(1,1),\ldots,(s,s)\}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-019069 | MATHDB-348519

The integrality conjecture for special points on Shimura subvarieties

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k including all the archimedean places, and let X be a special subvariety of a Shimura variety, defined over k...

conjecture | open | MathDB 页面 | 原文线索

V2-019070 | MATHDB-348543

The Gamma Conjecture of local mirror symmetry

摘录 / EXCERPT: Let X be a noncompact toric Calabi-Yau variety and let Y be its Hori-Vafa mirror. For mirror objects ... write $Z_t(E)$ for the central charge of E and $C_t(L)$ for the central...

conjecture | open | MathDB 页面 | 原文线索

V2-019071 | MATHDB-348544

The Gamma Conjecture for local mirror symmetry

摘录 / EXCERPT: Let X be the noncompact toric variety considered in the paper, with K-groups $K(X)$ and $K^c(X)$ paired by ... Choose bases $\{F_1,\ldots,F_d\}$ of $K(X)$ and $\{S_1,\ldots,S_d\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-019072 | MATHDB-348551

Optimal destabilization conjecture for K-unstable log Fano pairs

摘录 / EXCERPT: Optimal destabilization conjecture. If (X,D) is a K-unstable log Fano pair, then there exists a divisor E over X computing the infimum:

conjecture | open | MathDB 页面 | 原文线索

V2-019073 | MATHDB-348592

Factorization criterion for nongeneric Betti numbers of bihomogeneous ideals

摘录 / EXCERPT: Let W be a three-dimensional space of bihomogeneous forms defining the ideal I_W in the setting of the paper, let n be the relevant degree parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019074 | MATHDB-348599

The periodic de Rham conjecture for irreducible bundles over curves

摘录 / EXCERPT: Periodic de Rham conjecture. Any irreducible periodic de Rham bundle over U is motivic.

conjecture | open | MathDB 页面 | 原文线索

V2-019075 | MATHDB-348600

André's nilpotency conjecture for algebraic connections

摘录 / EXCERPT: André's nilpotency conjecture. Then (V,∇) is motivic.

conjecture | open | MathDB 页面 | 原文线索

V2-019076 | MATHDB-348604

Periodic Higgs conjecture

摘录 / EXCERPT: Let C and D be as in the source's theorem on periodicity for motivic Higgs bundles, and let C_{log} denote the corresponding logarithmic curve. A periodic parabolic...

conjecture | open | MathDB 页面 | 原文线索

V2-019077 | MATHDB-348609

Periodic Higgs conjecture for motivic parabolic Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve, and let $\mathrm{MHG}(C)$ and $\mathrm{PHG}(C)$ denote respectively the motivic and periodic parabolic Higgs categories from the source. Let .....

conjecture | open | MathDB 页面 | 原文线索

V2-019078 | MATHDB-348627

Bannister-Oudot subdivision conjecture for stack-number

摘录 / EXCERPT: Let G be a graph, let k be an integer, and let G' be any $(\leqslant k)$-subdivision of G, obtained by replacing each edge of G by an internally disjoint path with at mos...

conjecture | open | MathDB 页面 | 原文线索

V2-019079 | MATHDB-348637

Finiteness conjecture for F^N-projective structures

摘录 / EXCERPT: Finiteness conjecture. For every smooth projective variety X over k and every positive integer N, the set

conjecture | open | MathDB 页面 | 原文线索

V2-019080 | MATHDB-348656

Compatibility conjecture for the shifted symplectic structures on de Rham mapping stacks

摘录 / EXCERPT: Compatibility conjecture. This isomorphism is compatible with the $(n-2)$-shifted symplectic structures on both sides. The mapping stack $\mathrm{Map}(C_{\mathrm{dR}},X)$ is known...

conjecture | open | MathDB 页面 | 原文线索

V2-019081 | MATHDB-348658

Zannier's conjecture on torsion points in the universal abelian family

摘录 / EXCERPT: Let $\pi: \mathfrak{A}_{g,l} \to A_{g,l}$ be the universal family of complex principally polarized abelian varieties of dimension g with symplectic level l-structure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019082 | MATHDB-348669

Approximation conjecture for maps from spheres into real algebraic homogeneous spaces

摘录 / EXCERPT: Approximation conjecture. For every positive integer n, the set of regular maps

conjecture | open | MathDB 页面 | 原文线索

V2-019083 | MATHDB-348670

Odd-dimensional sphere approximation conjecture for real algebraic varieties

摘录 / EXCERPT: Odd-dimensional sphere approximation conjecture. The set of regular maps $\mathcal{R}(X,\mathbb{S}^k)$ is dense in the space of smooth maps $\mathcal{C}^{\infty}(X,\mathbb{S}^k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-019084 | MATHDB-348671

Oriented-boundary conjecture for algebraic models of 4k-manifolds

摘录 / EXCERPT: Oriented-boundary conjecture. The disjoint union of two copies of M is an oriented boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-019085 | MATHDB-348682

Sharifi's Eisenstein conjecture for the cyclotomic K_2 map

摘录 / EXCERPT: Sharifi's conjecture. The map Π_N factors through a map

conjecture | open | MathDB 页面 | 原文线索

V2-019086 | MATHDB-348684

Birational superrigidity implies K-stability

摘录 / EXCERPT: Birational superrigidity conjecture. A birationally superrigid Fano variety is K-stable.

conjecture | open | MathDB 页面 | 原文线索

V2-019087 | MATHDB-348696

Pappas-Rapoport reducedness conjecture for a symplectic nilpotent scheme

摘录 / EXCERPT: Pappas-Rapoport's conjecture. The scheme W is reduced.

conjecture | open | MathDB 页面 | 原文线索

V2-019088 | MATHDB-348705

The Quasi-diagonal Conjecture for invariant subvarieties

摘录 / EXCERPT: Fix $n>1$. For each $i\in\{1,\ldots,n\}$, let $\mathcal{S}_i$ be a component of a stratum of Abelian or quadratic differentials, and let $\mathcal{M}\subset\prod_i\mathcal{S}_i$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-019089 | MATHDB-348712

Chai-Oort Hecke orbit conjecture for Hodge type Shimura varieties

摘录 / EXCERPT: Let $\mathcal{S}$ be the canonical integral model of a Shimura variety of Hodge type with hyperspecial level, and let $\mathcal{S}_{\overline{\mathbb{F}}_p}$ be its special fiber....

conjecture | open | MathDB 页面 | 原文线索

V2-019090 | MATHDB-348729

Fuentes García's projective normality conjecture for polarized abelian varieties

摘录 / EXCERPT: Fuentes García's conjecture. L is projectively normal if

conjecture | open | MathDB 页面 | 原文线索

V2-019091 | MATHDB-348741

The finite-generation conjecture for valuations computing the delta invariant

摘录 / EXCERPT: Finite-generation conjecture. The graded ring ${\rm gr}_vR$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-019092 | MATHDB-348743

The properness conjecture for K-moduli spaces of K-polystable Fano varieties

摘录 / EXCERPT: Properness conjecture. The good moduli space $X^{\rm Kps}_{n,V}$ is proper.

conjecture | open | MathDB 页面 | 原文线索

V2-019093 | MATHDB-348745

Sommese's extension conjecture for morphisms from ample subvarieties

摘录 / EXCERPT: Let X be a smooth complex variety and $Y\subset X$ a smooth subvariety of codimension r defined by a regular section of an ample vector bundle on X. Let Z be a variety and...

conjecture | open | MathDB 页面 | 原文线索

V2-019094 | MATHDB-348746

Sommese's extension conjecture for ample subvarieties

摘录 / EXCERPT: Let X be a smooth complex projective variety and $Y\subset X$ a smooth ample subvariety of codimension r. Let Z be a variety and let $\pi\colon Y\to Z$ be a morphism such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-019095 | MATHDB-348756

Geiss-Leclerc-Schröer conjecture on characteristic cycles and semi-canonical bases

摘录 / EXCERPT: Let (I,Q) be a type A quiver with orientation $i\rightarrow i+1$, let V be an I-graded vector space, and let S and S' be G_V-orbits in the representation variety...

conjecture | open | MathDB 页面 | 原文线索

V2-019096 | MATHDB-348759

The Alexander-Hirschowitz property for ample line bundles on smooth projective varieties

摘录 / EXCERPT: Alexander-Hirschowitz conjecture. Suppose that X is a smooth projective variety of any dimension and $\mathcal{L}$ is ample. Then $(X,\mathcal{L})$ satisfies AH.

conjecture | open | MathDB 页面 | 原文线索

V2-019097 | MATHDB-348768

Flach-Morin special value conjecture for arithmetic schemes

摘录 / EXCERPT: Let ${\mathcal X}$ be a regular scheme of pure dimension d proper over $\operatorname{Spec}(\mathbb Z)$, and let $n\in\mathbb Z$. Assume that ${\mathcal X}$ satisfies Assumptions...

conjecture | open | MathDB 页面 | 原文线索

V2-019098 | MATHDB-348780

Positivity conjecture for characteristic classes of determinantal varieties

摘录 / EXCERPT: Let $\tau_{n,k}^{\circ}$, $\tau_{n,k}^{\wedge \circ}$, and $\tau_{n,k}^{S \circ}$ denote the corresponding generic determinantal, skew-symmetric determinantal, and symmetric determ...

conjecture | open | MathDB 页面 | 原文线索

V2-019099 | MATHDB-348782

Equivalence conjecture for Picard and analytic hyperbolicity

摘录 / EXCERPT: Picard-analytic equivalence conjecture. The manifold X is $b[31m\ellb[0m$-Picard hyperbolic if and only if it is $b[31m\ellb[0m$-analytically hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-019100 | MATHDB-348783

The Weil-divisor conjecture for rational polytopes

摘录 / EXCERPT: Weil-divisor conjecture. The divisor associated to a full-dimensional rational convex polytope is a Weil divisor on the toric variety associated to the polytope if and only if the...

conjecture | open | MathDB 页面 | 原文线索

V2-019101 | MATHDB-348784

Weak K-moduli folklore conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ be a moduli algebraic stack of polarized log-terminal Calabi-Yau varieties with quasi-projective coarse moduli variety M. A weak K-moduli stack is a proper Deli...

conjecture | open | MathDB 页面 | 原文线索

V2-019102 | MATHDB-348785

Satake-Baily-Borel type compactification conjecture for Calabi-Yau moduli

摘录 / EXCERPT: Let M be a moduli space of polarized log-terminal Calabi-Yau varieties satisfying the assumptions specified in the source. Satake-Baily-Borel compactification conjecture. There s...

conjecture | open | MathDB 页面 | 原文线索

V2-019103 | MATHDB-348786

Minimal log canonical center parametrization conjecture

摘录 / EXCERPT: Let $M\subset\overline{M}^{\mathrm{SBB}}$ be one of the conjectural compactifications, and define ... where $\sim$ is generated by log crepant birational maps. Minimal log canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-019104 | MATHDB-348791

Campana-Corvaja-Zannier conjecture on the weak-Hilbert property

摘录 / EXCERPT: Let X be a smooth proper geometrically connected variety over a finitely generated field k of characteristic zero. Campana-Corvaja-Zannier conjecture. The following are equival...

conjecture | open | MathDB 页面 | 原文线索

V2-019105 | MATHDB-348802

Projective-equivalence conjecture for the sixth Terracini locus of a triple Segre variety

摘录 / EXCERPT: Projective-equivalence conjecture. The subsets $S\in X$ of cardinality 6 such that $\pi_i(S)$ and $\pi_j(S)$ are projectively equivalent, for some fixed $i\neq j$, fill a dense o...

conjecture | open | MathDB 页面 | 原文线索

V2-019106 | MATHDB-348824

The Batyrev-Manin conjecture for rational points of bounded height

摘录 / EXCERPT: Batyrev-Manin conjecture. There exists an open subvariety $U \subseteq V$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019107 | MATHDB-348838

Fock-Goncharov's duality conjecture for ${\rm SL}_3$-${\rm PGL}_3$

摘录 / EXCERPT: Let $\frak{S}$ be a generalized marked surface. The set $\mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^{T})$ consists of tropical-integer points of the $\mathscr{A}$-moduli space, a...

conjecture | open | MathDB 页面 | 原文线索

V2-019108 | MATHDB-348839

Geometric model conjecture for tropical ${\rm SL}_3$ laminations

摘录 / EXCERPT: Let $\mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^{T})$ be the set of tropical-integer points of the ${\rm SL}_3$ cluster $\mathscr{A}$-space of a generalized marked surface...

conjecture | open | MathDB 页面 | 原文线索

V2-019109 | MATHDB-348841

Laurent coefficient positivity conjecture for the ${\rm SL}_3$-${\rm PGL}_3$ duality map

摘录 / EXCERPT: Let $\mathbb{I}$ be the classical duality map and let $\mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^{T})$ index its basic functions. Laurent coefficient positivity conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-019110 | MATHDB-348842

${A}_2$-bracelets basis conjecture for ${\rm SL}_3$ character varieties

摘录 / EXCERPT: Let $\mathscr{A}_{\rm L}(\frak{S};\mathbb{Z})$ be the set of ${\rm SL}_3$ laminations and let $\mathbb{I}^{\rm br}_{{\rm SL}_3}$ be the map obtained by replacing loop components wi...

conjecture | open | MathDB 页面 | 原文线索

V2-019111 | MATHDB-348843

${A}_2$-bracelets basis conjecture for ${\rm PGL}_3$ character varieties

摘录 / EXCERPT: Let $\mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^{T})$ be the tropical-integer lamination set, and let $\mathbb{I}^{\rm br}$ be the map obtained by applying the paper's constructi...

conjecture | open | MathDB 页面 | 原文线索

V2-019112 | MATHDB-348844

Laurent positivity conjecture for the ${A}_2$-bracelets basis

摘录 / EXCERPT: Let $\mathbb{I}^{\rm br}$ be the conjectural ${A}_2$-bracelets basis map for $\mathscr{O}(\mathscr{X}_{{\rm PGL}_3,\frak{S}})$, and let $\mathbb{I}$ denote the paper's correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-019113 | MATHDB-348845

Structure-constant positivity conjecture for the ${A}_2$-bracelets basis

摘录 / EXCERPT: Let $\mathbb{I}^{\rm br}$ and $\mathbb{I}$ denote the conjectural bracelets and bangles basis maps, respectively, and let their structure constants be defined by multiplication in...

conjecture | open | MathDB 页面 | 原文线索

V2-019114 | MATHDB-348846

GHKK compatibility conjecture for the ${\rm SL}_3$ duality map

摘录 / EXCERPT: Let $\mathbb{I}^{\rm GHKK}$ be the Gross-Hacking-Keel-Kontsevich duality map for the cluster varieties associated with the mutation class of the 3-triangulation quiver, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019115 | MATHDB-348856

The Selmer classes conjecture for GSp(4) × GL(2) × GL(2)

摘录 / EXCERPT: Let $(\Pi,\underline{\mathcal G})$ satisfy the stated Galois-representation, ordinarity, residual non-Eisenstein, and p-distinguished hypotheses. For a classical specialisation...

conjecture | open | MathDB 页面 | 原文线索

V2-019116 | MATHDB-348857

The family Selmer rank and algebraic cycles conjecture

摘录 / EXCERPT: Let $(\Pi,\underline{\mathcal G})$ be a family as in the stated Galois-representation setting, and let $\widetilde H^1(\mathbf Q,\Pi\otimes\underline{\mathcal G};\spadesuit)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-019117 | MATHDB-348864

Symmetry conjecture for the genus of the central curve

摘录 / EXCERPT: Let m be the size of the symmetric matrices and let d denote the dimension parameter for the central curve. Write $\mathrm{genus}(m,d)$ for its genus. The complementary paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-019118 | MATHDB-348869

Generalized abundance conjecture for Calabi-Yau varieties

摘录 / EXCERPT: Let X be a Calabi-Yau variety, meaning a normal projective variety with only ${\mathbb Q}$-factorial terminal singularities, $h^1(X,{\mathscr O}_X)=0$, and numerically trivial ca...

conjecture | open | MathDB 页面 | 原文线索

V2-019119 | MATHDB-348877

Torsion conjecture for abelian varieties over number fields

摘录 / EXCERPT: Torsion conjecture. For every $d\in\mathbb{N}_{\geq 1}$ and every $g\in\mathbb{N}_{\geq 1}$, there exists $c(g,d)\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019120 | MATHDB-348891

Birch-Swinnerton-Dyer conjecture for the Picard variety

摘录 / EXCERPT: Let X be a smooth projective geometrically connected variety over a finitely generated field K, with a smooth projective model $f:{\mathcal X}\to{\mathcal Y}$ as above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019121 | MATHDB-348892

Tate's pole-order conjecture for algebraic cycles

摘录 / EXCERPT: Let X be a smooth projective geometrically connected variety over a finitely generated field K, let $d=\dim X$, and let $A^i(X)$ be the group of codimension-i algebraic cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-019122 | MATHDB-348904

The polygraph module conjecture for the Procesi bundle

摘录 / EXCERPT: The polygraph module conjecture. As a module over $\mathbb{C}[{\mathbf{x}},{\mathbf{y}}] \rtimes S_n$, M is the image of P under this isomorphism; the higher derived functors...

conjecture | open | MathDB 页面 | 原文线索

V2-019123 | MATHDB-348972

The equality of planarity and uniruledness invariants for affine varieties

摘录 / EXCERPT: The affine uniruledness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019124 | MATHDB-348977

Dynamical local and global height ratio conjecture

摘录 / EXCERPT: Let K be a number field, let X be a smooth projective geometrically irreducible variety over K, and let $f\colon X\longrightarrow X$ be a surjective morphism. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019125 | MATHDB-348978

Height growth conjecture for subvarieties under iteration

摘录 / EXCERPT: Let K be a number field and let X be a smooth projective geometrically irreducible variety over K. Let $f\colon X\longrightarrow X$ be a surjective morphism, let $Y\subset X$...

conjecture | open | MathDB 页面 | 原文线索

V2-019126 | MATHDB-348989

Convergence conjecture for non-negative dyadic subdivision schemes

摘录 / EXCERPT: Let $I=\{i:c_i>0\}$ be the support of the mask of a dyadic subdivision scheme with non-negative coefficients. Convergence conjecture. The subdivision scheme converges unless the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-019127 | MATHDB-348990

Göttsche-Shende universality conjecture for the stable-pair generating series

摘录 / EXCERPT: Let S be a smooth projective surface and let $\mathcal{L}$ be a line bundle on S. Let $g(\mathcal{L})$ denote the genus associated with $\mathcal{L}$, and define Q and $w(Q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-019128 | MATHDB-348991

Srinivas-de Fernex local class group generation conjecture

摘录 / EXCERPT: Srinivas-de Fernex conjecture. There exist a normal variety Y, a closed point $y\in Y$, and divisors $\{D^Y_i:i\in I\}$ on Y such that

conjecture | open | MathDB 页面 | 原文线索

V2-019129 | MATHDB-349017

The generalized Bloch conjecture for smooth hypersurfaces

摘录 / EXCERPT: Let $Y\subset\mathbb{P}^{n}$ be a smooth hypersurface of degree d. Write $A_i(Y):=CH_i(Y)_{\mathbb{Q}}$ for the Chow group of i-dimensional cycles with rational coefficients mo...

conjecture | open | MathDB 页面 | 原文线索

V2-019130 | MATHDB-349018

The Chow-group conjecture for Plücker hyperplane sections

摘录 / EXCERPT: Let $\operatorname{Gr}(k,n)$ denote the Grassmannian of k-dimensional subspaces of an n-dimensional vector space, embedded by the Plücker embedding. Let … be a smooth hyperplan...

conjecture | open | MathDB 页面 | 原文线索

V2-019131 | MATHDB-349023

Index conjecture for canonical Calabi-Yau varieties

摘录 / EXCERPT: Index conjecture. There exists a positive integer $I=I(d)$ such that every complete variety or space X of dimension d with only canonical singularities and $K\equiv 0$ has inde...

conjecture | open | MathDB 页面 | 原文线索

V2-019132 | MATHDB-349041

The monodromy conjecture for motivic zeta functions of Calabi-Yau varieties

摘录 / EXCERPT: Let K be a complete discretely valued field with algebraically closed residue field k, and let $X\to\operatorname{Spec}K$ be a smooth Calabi-Yau variety with volume form...

conjecture | open | MathDB 页面 | 原文线索

V2-019133 | MATHDB-349078

Lang's intermediate pseudo-conjectures

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let X be a projective variety over k, and let $pgeq 1$ be an integer. The notions of general type, pseudo-Morde...

conjecture | open | MathDB 页面 | 原文线索

V2-019134 | MATHDB-349079

Lang's intermediate conjecture for exceptional loci

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, let X be a projective variety over k, let $\Delta\subset X$ be a closed subset, and let p be a positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-019135 | MATHDB-349080

The conjectural cohomological stable-envelope formula for bow varieties

摘录 / EXCERPT: Let $\DD$ be a brane diagram and let $f\in\Ch(\DD)^{\T}$ be a fixed point. For each tangent-weight term $\alpha_{x,y,k}\frac{x}{y}\mathbf h^k$, call it f-small when its restricti...

conjecture | open | MathDB 页面 | 原文线索

V2-019136 | MATHDB-349083

Lejmi-Szekelyhidi's numerical criterion for the J-equation

摘录 / EXCERPT: Lejmi-Szekelyhidi's conjecture. Under this normalization, the J-equation admits a unique smooth solution ω if and only if, for every m-dimensional analytic subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-019137 | MATHDB-349084

Singular J-equation conjecture for normal Kähler varieties

摘录 / EXCERPT: Singular J-equation conjecture. If (α,β) is J-positive, then this equation admits a unique solution ω as a Kähler current with bounded local potentials.

conjecture | open | MathDB 页面 | 原文线索

V2-019138 | MATHDB-349094

Fullness conjecture for logarithmic and logarithmic transfer-invariant motives

摘录 / EXCERPT: Let $\omega_{\operatorname{\mathbf{CI}}}^{\log}$ and $\omega_{\operatorname{\mathbf{CI}}}^{\rm ltr}$ be the functors from logarithmic and logarithmic transfer-invariant motives to...

conjecture | open | MathDB 页面 | 原文线索

V2-019139 | MATHDB-349104

The decomposition conjecture for F-liftable varieties

摘录 / EXCERPT: Decomposition conjecture. There exists a finite Galois cover

conjecture | open | MathDB 页面 | 原文线索

V2-019140 | MATHDB-349122

Subvariety-corrected stability criterion for Z-critical connections

摘录 / EXCERPT: Subvariety stability conjecture. E admits a Z-critical connection if and only if for all coherent sheaves F supported on a subvariety $V\subset X$ with $F \subset E|_{V}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019141 | MATHDB-349123

Kurokawa's absolute Euler product conjecture for arithmetic schemes

摘录 / EXCERPT: Let X be an arithmetic scheme. Its counting function $N(t)$ satisfies $N(1)=\chi_{\mathrm{abs}}(X)$, the absolute Euler characteristic. An absolute Euler product is an infinite p...

conjecture | open | MathDB 页面 | 原文线索

V2-019142 | MATHDB-349124

Mukai's projective-isomorphism conjecture for K3 curve sections

摘录 / EXCERPT: Let $g_1=7,8,9,10$, let M_{g_1} be the Mukai variety associated with genus g_1, let G_{g_1} act on M_{g_1}, and set $m=\dim(M_{g_1})$. Let $V\subseteq\mathbf{P}^6$ be the M...

conjecture | open | MathDB 页面 | 原文线索

V2-019143 | MATHDB-349142

Finite generation conjecture for negative-weight motivic cohomology

摘录 / EXCERPT: Let X be an arithmetic scheme, let $n<0$ be an integer, and let $H^i(X_{\text{\it \'{e}t}},\mathbb{Z}^c(n))$ denote the cohomology of the negative-weight motivic complex. Finite...

conjecture | open | MathDB 页面 | 原文线索

V2-019144 | MATHDB-349143

Flach-Morin finite generation conjecture for negative-weight motivic cohomology

摘录 / EXCERPT: Let X be a proper regular arithmetic scheme of dimension d, let $n<0$, and let $H^i(X_{\text{\it \'{e}t}},\mathbb{Z}^c(n))$ denote negative-weight motivic cohomology. Flach-Mor...

conjecture | open | MathDB 页面 | 原文线索

V2-019145 | MATHDB-349144

Finite generation conjecture for all negative weights

摘录 / EXCERPT: Let X be an arithmetic scheme, let $n<0$ be an integer, and let $H^i(X_{\text{\it \'{e}t}},\mathbb{Z}^c(n))$ denote its negative-weight motivic cohomology. Finite generation conj...

conjecture | open | MathDB 页面 | 原文线索

V2-019146 | MATHDB-349154

Pseudoeffective threshold characterization of projective spaces and hyperquadrics

摘录 / EXCERPT: Let X be a smooth projective variety and A an ample divisor on X. Let ... where $\xi=c_1(\mathcal{O}_{\mathbb{P}(T_X)}(1))$ is the tautological class on $\mathbb{P}(T_X)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019147 | MATHDB-349186

Variational filtered log p-adic hard Lefschetz conjecture

摘录 / EXCERPT: Let X/S_0 be a proper SNCL scheme with structural morphism $f\colon X\longrightarrow S_0\overset{\subset}{\longrightarrow} S$, and suppose that the relative dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-019148 | MATHDB-349190

The tropical Hodge conjecture

摘录 / EXCERPT: Let $\mathfrak X$ be a smooth projective tropical variety. For an integer p, consider the tropical cohomology groups ... and let N be the tropical monodromy operator between them...

conjecture | open | MathDB 页面 | 原文线索

V2-019149 | MATHDB-349203

The relative Fontaine-Mazur conjecture for local systems

摘录 / EXCERPT: Let F be a number field, let X be a smooth variety over F, and let ${\mathbb L}$ be an irreducible $\overline{\mathbb{Q}}_p$-local system on X. Assume that there is a finit...

conjecture | open | MathDB 页面 | 原文线索

V2-019150 | MATHDB-349213

Conjecture on eventual crystalline growth and unique minimizing movement

摘录 / EXCERPT: Let φ be an anisotropy and let α be a suitable parameter. For the discrete solutions $\{E^k\}$ of scheme ... . Conjecture on crystalline growth. Under suitable assumpt...

conjecture | open | MathDB 页面 | 原文线索

V2-019151 | MATHDB-349223

The Diophantine approximation conjecture for global fields

摘录 / EXCERPT: Let K be a global field and let v be a place of K. Let X and Y be positive dimensional irreducible projective varieties over K, let $f:X\dashrightarrow Y$ be a dominant...

conjecture | open | MathDB 页面 | 原文线索

V2-019152 | MATHDB-349244

The Hadamard meta-conjecture for natural sequences of zeta functions

摘录 / EXCERPT: Let q be a prime power, let $a_n\in\mathcal{M}_{\mathbb{F}_q}$ be a natural sequence of classes, and let $\mathcal{H}_1$ denote the set of Hadamard functions f satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-019153 | MATHDB-349245

Hadamard convergence conjecture for smooth hypersurfaces in projective space

摘录 / EXCERPT: Let $V_d:=\Gamma(\mathbb{P}^n,\mathcal{O}(d))$ and let $U_d\subset V_d$ be the space of smooth hypersurfaces of degree d in $\mathbb{P}^n$. Projective-space Hadamard convergence...

conjecture | open | MathDB 页面 | 原文线索

V2-019154 | MATHDB-349275

Density conjecture for geometric-origin representations

摘录 / EXCERPT: Density conjecture. The set of G-representations of geometric origin is Zariski dense in

conjecture | open | MathDB 页面 | 原文线索

V2-019155 | MATHDB-349276

Density conjecture for geometric-origin representations on special subschemes

摘录 / EXCERPT: Special-locus density conjecture. The set of complex points of Z corresponding to representations of geometric origin is dense in Z. In particular, the special points are dense...

conjecture | open | MathDB 页面 | 原文线索

V2-019156 | MATHDB-349346

Dimension conjecture for constant-bond-dimension-two tensor network varieties

摘录 / EXCERPT: Dimension conjecture. For all such d, $\mathbf{m}$, and $\mathbf{n}$, the displayed dimension formula holds outside precisely the listed exceptional cases.

conjecture | open | MathDB 页面 | 原文线索

V2-019157 | MATHDB-349347

The permutation-basis conjecture for regular semisimple Hessenberg cohomology

摘录 / EXCERPT: Let $h:[n]\to[n]$ be a Hessenberg function, meaning a nondecreasing function with $h(i)\geq i$ for all i. Let $\mathcal{H}ess(\mathsf{S},h)$ be the regular semisimple Hessenberg...

conjecture | open | MathDB 页面 | 原文线索

V2-019158 | MATHDB-349375

P=W conjecture for twisted character varieties

摘录 / EXCERPT: P=W conjecture for twisted character varieties. For every k,

conjecture | open | MathDB 页面 | 原文线索

V2-019159 | MATHDB-349376

PI=WI conjecture for character varieties

摘录 / EXCERPT: PI=WI conjecture. For every k,

conjecture | open | MathDB 页面 | 原文线索

V2-019160 | MATHDB-349377

Intersection curious hard Lefschetz conjecture for character varieties

摘录 / EXCERPT: Intersection curious hard Lefschetz conjecture. There exists a class $\alpha\in H^2(M_{\mathrm{B}}(C,G))$ such that, for every k, cup product induces isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-019161 | MATHDB-349378

P=W conjecture for untwisted character varieties

摘录 / EXCERPT: P=W conjecture for untwisted character varieties. For every k,

conjecture | open | MathDB 页面 | 原文线索

V2-019162 | MATHDB-349379

P=W for resolutions of character varieties

摘录 / EXCERPT: P=W for resolution. The lift satisfies, for every k,

conjecture | open | MathDB 页面 | 原文线索

V2-019163 | MATHDB-349380

Topological mirror symmetry in degree zero for rank two character varieties

摘录 / EXCERPT: Topological mirror symmetry in degree zero. For every $\kappa\in\hat{\Gamma}$, with $\gamma=w(\kappa)$,

conjecture | open | MathDB 页面 | 原文线索

V2-019164 | MATHDB-349381

Perverse topological mirror symmetry for rank two character varieties

摘录 / EXCERPT: Perverse topological mirror symmetry.

conjecture | open | MathDB 页面 | 原文线索

V2-019165 | MATHDB-349384

Conjectural formula for the function phi of prime-power degrees

摘录 / EXCERPT: Let $\phi:\mathbb{Z}_+\to\mathbb{Z}_+$ be defined from the indecomposable elements of M_m by ... Here p denotes an odd prime and k,l are positive integers subject to the condit...

conjecture | open | MathDB 页面 | 原文线索

V2-019166 | MATHDB-349389

Wang-Zhou-Krug generating-series conjecture for tautological exterior powers

摘录 / EXCERPT: Let X be a smooth projective scheme over $\boldsymbol{k}$ of dimension $d\geq 2$, and let $K,L\in\operatorname{Pic}(X)$. For a line bundle M on X, write $M^{[n]}$ for its tau...

conjecture | open | MathDB 页面 | 原文线索

V2-019167 | MATHDB-349392

Wang-Zhou-Krug tautological Euler-characteristic conjecture

摘录 / EXCERPT: Wang-Zhou-Krug conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019168 | MATHDB-349393

Equivariant local-structure conjecture for non-Borel extra-dimension-six ideals

摘录 / EXCERPT: Let I_{λ} be a non-Borel monomial ideal of $\Bbbk[X_1,X_2,X_3]$ with extra dimension 6. Let $\widehat{G}(2,6)$ be the affine cone over the Grassmannian $G(2,6)$, and call...

conjecture | open | MathDB 页面 | 原文线索

V2-019169 | MATHDB-349394

Critical-locus conjecture for Borel Hilbert-scheme neighborhoods

摘录 / EXCERPT: Critical-locus conjecture. The Haiman neighborhood of every such Borel monomial ideal is a critical locus in the stated tangent space.

conjecture | open | MathDB 页面 | 原文线索

V2-019170 | MATHDB-349395

Minimal extra-dimension-six conjecture for Hilbert schemes of points

摘录 / EXCERPT: For $n\geq4$, consider points of $\operatorname{Hilb}^n(\mathbb{A}^3)$ and their extra dimension, namely the excess of the embedded dimension over the expected dimension $3n$. Mini...

conjecture | open | MathDB 页面 | 原文线索

V2-019171 | MATHDB-349396

Borel extra-dimension-six ideals are tripod ideals

摘录 / EXCERPT: A monomial ideal of $\Bbbk[X_1,X_2,X_3]$ is a tripod ideal if it has minimal generators of the form ... A Borel ideal is a monomial ideal fixed in the relevant lexicographic order. T...

conjecture | open | MathDB 页面 | 原文线索

V2-019172 | MATHDB-349397

Extra-dimension criterion for Borel tripod ideals

摘录 / EXCERPT: Extra-dimension criterion. The extra dimension of I is 6 if and only if at least one of b and d is equal to 1 or $a-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-019173 | MATHDB-349398

Uniform local-structure conjecture for extra-dimension-six points

摘录 / EXCERPT: Let z be a point on $\operatorname{Hilb}^n(\mathbb{A}^3)$. If the embedded dimension at z is $3n+6$, let $\widehat{G}(2,6)$ denote the affine cone over $G(2,6)$. Uniform local-...

conjecture | open | MathDB 页面 | 原文线索

V2-019174 | MATHDB-349399

Same-singularity-type conjecture for similarly shaped partitions

摘录 / EXCERPT: Same-singularity-type conjecture. Then the germs

conjecture | open | MathDB 页面 | 原文线索

V2-019175 | MATHDB-349400

Global equivariant Hilbert-function conjecture for Haiman neighborhoods

摘录 / EXCERPT: Let $\mathscr{P}_r$ be the set of r-dimensional partitions, including the empty partition. For $\lambda\in\mathscr{P}_r$, let A_{λ} be the coordinate ring of the Haiman n...

conjecture | open | MathDB 页面 | 原文线索

V2-019176 | MATHDB-349401

K-theoretic McKay pushforward conjecture for the Hilbert-Chow morphism

摘录 / EXCERPT: Let X be a smooth projective scheme, let $X^{(n)}=X^n/\mathfrak{S}_n$ be its nth symmetric product, and let ... be the Hilbert-Chow morphism. K-theoretic McKay conjecture. The K-...

conjecture | open | MathDB 页面 | 原文线索

V2-019177 | MATHDB-349402

K-theoretic McKay conjecture for tautological exterior powers

摘录 / EXCERPT: Let X be a scheme of dimension at least 2, let K,L be line bundles on X, and let $p_i:X^n\to X$ be the projections. Set ... Let $\rho:X^{[n]}\to X^{(n)}$ be the Hilbert-Chow...

conjecture | open | MathDB 页面 | 原文线索

V2-019178 | MATHDB-349429

Strong Birch-Swinnerton-Dyer conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a number field F, let A^t be its dual abelian variety, and let $L^*(A,1)$ denote the leading coefficient of $L(A,s)$ at...

conjecture | open | MathDB 页面 | 原文线索

V2-019179 | MATHDB-349430

Strong Birch-Swinnerton-Dyer formula in the everywhere-good-reduction case

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a number field F, assume that A has good reduction everywhere, and let $P_{A,\infty}(\eta)$ and $\mathfrak a_\eta$ be the ar...

conjecture | open | MathDB 页面 | 原文线索

V2-019180 | MATHDB-349437

The \mathrm{SL}_n-quantum trace map

摘录 / EXCERPT: Fix a n^2-root $\omega=q^{1/n^2}\in\mathbb{C}-\{0\}$. For each ideal triangulation λ of the punctured surface $\mathfrak{S}$, let $\mathscr{S}^q_n(\mathfrak{S})$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-019181 | MATHDB-349442

Harbourne-Huneke containment conjecture for fat point schemes

摘录 / EXCERPT: Let $Z a derive\subseteq \mathbb{P}^n$ be a fat point scheme, let $I:=I_Z$ be its defining ideal, and let $\mathcal{M}=(x_0,\dots,x_n)$ be the graded maximal ideal. Harbourne-Hunek...

conjecture | open | MathDB 页面 | 原文线索

V2-019182 | MATHDB-349484

Thapa's conjecture on ICC and fractional-partitioned clique-cover broadcast rates

摘录 / EXCERPT: Let $\mathcal{I}=\{(i\mid A_i),i\in[m]\}$ be an index coding instance. The ICC and fractional-partitioned clique-cover coding schemes have broadcast rates...

conjecture | open | MathDB 页面 | 原文线索

V2-019183 | MATHDB-349487

K-stability conjecture for smooth Fano hypersurfaces

摘录 / EXCERPT: Let $X\subseteq\mathbb{P}^{n+1}$ be a smooth Fano hypersurface of degree $d\ge 3$. K-stability conjecture. The variety X is K-stable. This folklore conjecture concerns the relati...

conjecture | open | MathDB 页面 | 原文线索

V2-019184 | MATHDB-349519

Comparison of simplicial and E-infinity formal-group Hochschild homology

摘录 / EXCERPT: Let $\formalgroup$ be a formal group over a perfect field k. For a simplicial commutative k-algebra A, let $\theta(A)$ be its underlying E_∞-algebra, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-019185 | MATHDB-349530

The generalized upper-bound conjecture for (b,k)-hashing

摘录 / EXCERPT: Let $R_{(b,k)}$ denote the rate of a (b,k)-hashing scheme, with integers $b\geq k>3$. For j in the range $2\leq j\leq k-2$, let $b^{\underline{j+1}}$ denote the falling factori...

conjecture | open | MathDB 页面 | 原文线索

V2-019186 | MATHDB-349531

Algebraicity conjecture for deformations of varieties of general type

摘录 / EXCERPT: Algebraicity conjecture for deformations of varieties of general type. The space $\operatorname{Def}(X_0)$ has a representative ${\mathbf X}\to S$ where S is a scheme of finite t...

conjecture | open | MathDB 页面 | 原文线索

V2-019187 | MATHDB-349532

Deformation invariance of plurigenera for canonical varieties

摘录 / EXCERPT: Deformation invariance of plurigenera. The plurigenera $h^0(X_s,\omega_{X_s}^{[r]})$ are independent of $s\in S$ for every r, in some neighborhood of $0\in S$.

conjecture | open | MathDB 页面 | 原文线索

V2-019188 | MATHDB-349534

Archimedean/non-archimedean μ-entropy equality conjecture

摘录 / EXCERPT: Let (X,L) be a polarized variety. Denote by $\mathcal{H}^{\mathbb{R}}_{\mathrm{NA}}(X,L)$ the space of non-archimedean metrics and by $\mathcal{H}(X,L)$ the space of Kähler metri...

conjecture | open | MathDB 页面 | 原文线索

V2-019189 | MATHDB-349544

Refined Dubrovin conjecture and Gamma-conjecture II for Fano varieties

摘录 / EXCERPT: Let X be an n-dimensional smooth Fano variety satisfying $h^{p,q}(X)=0$ for every $p\ne q$. Let $\widetilde\mu=\dim_{\mathbb C}\bigoplus_{p\in\mathbb Z}H^p(X;\mathbb C)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-019190 | MATHDB-349558

The dimension growth conjecture for rational points on projective varieties

摘录 / EXCERPT: Let $X\subseteq \mathbb{P}_{\mathbb{Q}}^{n}$ be an integral projective variety of degree $d\geq 2$. Let H be the absolute projective multiplicative height. Dimension growth conje...

conjecture | open | MathDB 页面 | 原文线索

V2-019191 | MATHDB-349590

Cao-Kool's MacMahon formula for tautological insertions on Calabi-Yau fourfolds

摘录 / EXCERPT: Let X) be a projective Calabi-Yau 4-fold and let L be a line bundle on X. Define ... Here $M(q)=\prod_{i=1}^{\infty}(1-q^i)^{-i}$ is the MacMahon function. Cao-Kool's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-019192 | MATHDB-349593

The prime-to-p Hecke orbit conjecture for K3 moduli strata

摘录 / EXCERPT: Let $\shS^i$ be the height and Artin-invariant strata in the special fiber of the orthogonal Shimura variety parametrizing degree-$2d$ polarized K3 surfaces, and let the prime-to-...

conjecture | open | MathDB 页面 | 原文线索

V2-019193 | MATHDB-349597

Przyjalkowski-Shinder-type conjecture on reducible fibers and primitive Hodge numbers

摘录 / EXCERPT: Let X be a smooth Fano variety of dimension n, and let (Y,w) be a Calabi-Yau compactification of a toric Landau-Ginzburg model for X. Define ... Here the primitive Hodge numb...

conjecture | open | MathDB 页面 | 原文线索

V2-019194 | MATHDB-349602

The virtual Euler class conjecture for quasi-smooth derived schemes

摘录 / EXCERPT: Let $\boldsymbol{Y}$ be a quasi-smooth derived scheme, write $Y=t_0(\boldsymbol{Y})$, and let $\widetilde{Y}=t_0(\mathbf{T}^*[-1]\boldsymbol{Y})$. The class...

conjecture | open | MathDB 页面 | 原文线索

V2-019195 | MATHDB-349621

The local Moishezon morphism criterion

摘录 / EXCERPT: The local Moishezon morphism criterion. The morphism g is locally Moishezon if and only if every irreducible component of every fiber X_s is Moishezon.

conjecture | open | MathDB 页面 | 原文线索

V2-019196 | MATHDB-349639

Conjecture on coisotropic subvarieties of ample hypersurfaces

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold of dimension $2n$, and let Y be a smooth ample hypersurface in X. A subvariety $Z\subset Y$ is coisotropic when its ta...

conjecture | open | MathDB 页面 | 原文线索

V2-019197 | MATHDB-349651

Independence of Selmer coranks from the prime and localization

摘录 / EXCERPT: Let p be a prime, let Y/L be an abelian variety, let $B\subset\operatorname{End}_{L}(Y)$ be an integral domain, and let $\mathfrak{q}$ and $\mathfrak{q}'$ be primes of B abov...

conjecture | open | MathDB 页面 | 原文线索

V2-019198 | MATHDB-349654

Projective-dual linear-section Hodge conjecture

摘录 / EXCERPT: Let $\mathsf{X}\subset\mathbb{P}(\mathsf{V})$ be a smooth projective variety whose Hodge conjecture is true, and let $\mathsf{Y}\subset\mathbb{P}(\mathsf{V}^{\ast})$ be its project...

conjecture | open | MathDB 页面 | 原文线索

V2-019199 | MATHDB-349665

Ichino-Ikeda type conjecture for strongly tempered spherical varieties

摘录 / EXCERPT: Let (G,H) be one of the models in Table 1 other than the first, and let D/k be a quaternion algebra, possibly split (or $D\in H^1(H/Z_{G,H},k)$ for Model 2). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019200 | MATHDB-349666

Ichino-Ikeda type conjecture for the general linear quaternionic model

摘录 / EXCERPT: Consider the first model in Table 1, $(G,H)=({\mathrm{GL}}_4\times{\mathrm{GL}}_2,{\mathrm{GL}}_2\times{\mathrm{GL}}_2)$, for which $Z_H/Z_{G,H}\cong {\mathrm{GL}}_1$. Under the no...

conjecture | open | MathDB 页面 | 原文线索

V2-019201 | MATHDB-349667

Weak global Ichino-Ikeda conjecture for strongly tempered spherical varieties

摘录 / EXCERPT: Let $\Pi_\phi=\bigcup_D\Pi_\phi(G_D)$ be the global tempered cuspidal L-packet and retain the notation for the models (G_D,H_D), the embeddings ν, and the period integrals...

conjecture | open | MathDB 页面 | 原文线索

V2-019202 | MATHDB-349682

The dynamical degree comparison conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$, and let f be a dynamical correspondence of X. Write $\chi_{2k}(f)_{\iota}$ for the relevant cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-019203 | MATHDB-349683

The generalized Weil's Riemann hypothesis for polarized endomorphisms

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$, and let f be a self-morphism of X. Suppose that f is polarized, meaning that ... for an ample divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-019204 | MATHDB-349684

The generalized semisimplicity conjecture for polarized endomorphisms

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$, and let f be a polarized self-morphism as in the generalized Weil's Riemann hypothesis:...

conjecture | open | MathDB 页面 | 原文线索

V2-019205 | MATHDB-349685

The norm comparison conjecture for dynamical correspondences

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$. Let $\|\cdot\|_{\iota}$ denote the chosen operator norm on ℓ-adic étale cohomology and $\|\cdot\|$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-019206 | MATHDB-349686

The generalized Lefschetz standard conjecture for homological correspondences

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$, and for each $r\in\mathbf{Q}_{>0}$ let γ_r be the homological correspondence acting on $H^i(X)$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-019207 | MATHDB-349687

The standard conjecture of Künneth type

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\mathbf{K}$, and write the Künneth decomposition of the diagonal class as ... with $\pi_i\in H^i(X)\otimes H^{2n-i}(X)$....

conjecture | open | MathDB 页面 | 原文线索

V2-019208 | MATHDB-349691

Nondivisibility of leak terms in singular-locus equations

摘录 / EXCERPT: Let $\widetilde{\mathcal{M}}=(G,In,Out,Leak)$ be a strongly connected linear compartmental model with at least one input and at least one leak. A singular-locus equation f is the...

conjecture | open | MathDB 页面 | 原文线索

V2-019209 | MATHDB-349701

The rigid polystable reduction conjecture

摘录 / EXCERPT: Rigid polystable reduction conjecture. There is a rig-isomorphism $\mathcal{X}'\to\mathcal{X}$ such that $\mathcal{X}'$ is polystable.

conjecture | open | MathDB 页面 | 原文线索

V2-019210 | MATHDB-349724

Brosnan's p-incompressibility conjecture for multiplication maps on abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a field L of characteristic zero, and let $[p]:A\to A$ denote multiplication by a prime p. Brosnan's conjecture. For every prime p, the map...

conjecture | open | MathDB 页面 | 原文线索

V2-019211 | MATHDB-349726

Leal's ramification bound conjecture for nearby cohomology

摘录 / EXCERPT: Let $\mathcal S$ be a henselian trait with perfect residue field of characteristic $p>0$, with closed point s, generic point η, and geometric generic point $\overline\eta$....

conjecture | open | MathDB 页面 | 原文线索

V2-019212 | MATHDB-349775

Discriminant-semi-orthogonal decomposition multiplicity conjecture

摘录 / EXCERPT: Let Z be the toric variety associated to a wall-crossing, and suppose its derived category has a semi-orthogonal decomposition ... where each $D^b(Z_i)$ occurs some number of times...

conjecture | open | MathDB 页面 | 原文线索

V2-019213 | MATHDB-349786

The Kikuta-Takahashi categorical Gromov-Yomdin conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over $\mathbb{C}$ and let $D^b(X)$ denote its bounded derived category of coherent sheaves. Let $\mathcal{N}(X)$ be the numerical Grothendiec...

conjecture | open | MathDB 页面 | 原文线索

V2-019214 | MATHDB-349806

Chi Li's Fujita approximation conjecture with asymptotic \mathfrak r_1 convergence

摘录 / EXCERPT: Let ${\mathcal X}$ be a smooth projective manifold and let ${\mathcal L}$ be a big line bundle on ${\mathcal X}$. A Fujita approximation consists of a birational morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-019215 | MATHDB-349816

Grassmann graph eigenvalue monotonicity and exceptional minimum conjecture

摘录 / EXCERPT: Grassmann eigenvalue conjecture. (i) If $(n,q) eq (2d,2)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-019216 | MATHDB-349817

Bilinear forms graph smallest eigenvalue conjecture

摘录 / EXCERPT: Bilinear forms eigenvalue conjecture. For $q\geq 3$, or $q=2$ and $d\neq e$, $B_j(d-j+1)$ is the smallest eigenvalue in the distance-j graph for $1\leq j\leq d$.

conjecture | open | MathDB 页面 | 原文线索

V2-019217 | MATHDB-349818

Hermitian forms graph smallest eigenvalue conjectures

摘录 / EXCERPT: Hermitian forms eigenvalue conjectures. (i) If j is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-019218 | MATHDB-349819

Hermitian forms graph absolute eigenvalue conjecture

摘录 / EXCERPT: Hermitian absolute eigenvalue conjecture. Let $d\geq 3$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-019219 | MATHDB-349820

Hamming graph distinct-eigenvalue conjecture

摘录 / EXCERPT: Hamming distinct-eigenvalue conjecture. If $H(d,q,j)$ is connected, it has more than $d/2$ distinct eigenvalues.

conjecture | open | MathDB 页面 | 原文线索

V2-019220 | MATHDB-349835

The character-variety finiteness conjecture for profinite distinguishability of Dehn fillings

摘录 / EXCERPT: Let M be a one-cusped, finite-volume, hyperbolic 3-manifold, and let $M_{m/n}$ be a hyperbolic Dehn filling with surgery coefficient m/n. Write $\Gamma=\pi_1(M_{m/n})$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-019221 | MATHDB-349836

Profinite recoverability conjecture for character varieties of 3-manifold groups

摘录 / EXCERPT: Let Γ be a 3-manifold group, let $\widehat{\Gamma}$ be its profinite completion, and let $\chi^\mathbb{C}(\Gamma)$ denote its character variety. Profinite character-vari...

conjecture | open | MathDB 页面 | 原文线索

V2-019222 | MATHDB-349837

Profinite detection conjecture for zero-dimensional character-variety components

摘录 / EXCERPT: Let Γ be a 3-manifold group, let $\widehat{\Gamma}$ be its profinite completion, and let $\chi^l_\mathbb{C}(\Gamma)$ denote its character variety. Zero-dimensional compone...

conjecture | open | MathDB 页面 | 原文线索

V2-019223 | MATHDB-349845

Descent conjecture for perfect and finite projective modules in the noetherian setting

摘录 / EXCERPT: Work in either of the two settings considered in the subsection: the derived analytic or derived algebraic geometric setting with the associated period and Robba rings. Consider pe...

conjecture | open | MathDB 页面 | 原文线索

V2-019224 | MATHDB-349846

Self-duality conjecture for the Hilbert scheme under 3D-mirror symmetry

摘录 / EXCERPT: Let $\mathsf{X}=\operatorname{Hilb}^{n}(\mathbb{C}^{2})$, and let $\mathsf{X}^{!}$ denote its 3D-mirror dual. Let $\mathbf{U}(a,z)$ and $\mathbf{L}(a,z)$ be the matrices of twisted...

conjecture | open | MathDB 页面 | 原文线索

V2-019225 | MATHDB-349847

Local structure conjecture for localized intrinsic étale morphisms

摘录 / EXCERPT: Let $T_1\to T_2$ be a localized intrinsic étale morphism of strongly sheafy adic spaces. Local structure conjecture. Any localized intrinsic étale morphism of strongly sheafy adic...

conjecture | open | MathDB 页面 | 原文线索

V2-019226 | MATHDB-349867

Secant-dimension conjecture for varieties of reducible forms

摘录 / EXCERPT: Secant-dimension conjecture. For each integer $r\geq 1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-019227 | MATHDB-349870

K3^[n]-type multiple-cover conjecture

摘录 / EXCERPT: Let X be a variety of $K3^{[n]}$ type and let $\beta\in H_2(X,\mathbb{Z})$ be effective. For each divisor $k\mid\beta$, choose a variety X_k of $K3^{[n]}$ type and a parallel t...

conjecture | open | MathDB 页面 | 原文线索

V2-019228 | MATHDB-349877

Real-rootedness conjecture for Narayana polynomials over binary nestohedra

摘录 / EXCERPT: For a positive integer n, define the Narayana polynomial ... Given a composition $\mathbf q=(q_0,\ldots,q_k)\in\mathbb Z_{>0}^{k+1}$, define ... An integral binary nestohedron...

conjecture | open | MathDB 页面 | 原文线索

V2-019229 | MATHDB-349880

Conjecture on quasi-log schemes and short affine-line curves

摘录 / EXCERPT: Let $[X,\omega]$ be a quasi-log scheme and let $\pi\colon X\to S$ be a projective morphism between schemes such that $-\omega$ is π-ample and the induced morphism from the non-ql...

conjecture | open | MathDB 页面 | 原文线索

V2-019230 | MATHDB-349882

Flip termination conjecture for non-pseudo-effective klt pairs

摘录 / EXCERPT: Let (X,Δ) be a $\mathbb{Q}$-factorial klt pair, and let $\pi\colon X\to S$ be a projective surjective morphism between normal quasi-projective varieties such that $K_X+\Delta$ is not...

conjecture | open | MathDB 页面 | 原文线索

V2-019231 | MATHDB-349883

Lichtenbaum's finiteness conjecture for étale motivic cohomology

摘录 / EXCERPT: Let X be an arithmetic scheme, meaning a separated scheme of finite type over $\operatorname{Spec}\mathbb{Z}$, and let $n<0$. Write $X_{\text{ét}}$ for its étale site and...

conjecture | open | MathDB 页面 | 原文线索

V2-019232 | MATHDB-349884

Beshenov's regulator conjecture for Weil-étale motivic cohomology

摘录 / EXCERPT: Let X be an arithmetic scheme such that $X_{\mathbb{C}}$ is smooth and quasi-projective, and let $n<0$. Let $G_{\mathbb{R}}=\operatorname{Gal}(\mathbb{C}/\mathbb{R})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-019233 | MATHDB-349885

Beshenov's vanishing-order conjecture for arithmetic schemes

摘录 / EXCERPT: Let X be an arithmetic scheme and let $n<0$ be an integer. Assume that $\zeta(X,s)$ has a meromorphic continuation around $s=n$, and let $H^i_{\mathrm{W,c}}(X,\mathbb{Z}(n))$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-019234 | MATHDB-349886

Beshenov's special-value conjecture for Weil-étale cohomology

摘录 / EXCERPT: Beshenov's special-value conjecture. The special value $\zeta^*(X,n)$ is determined up to sign by

conjecture | open | MathDB 页面 | 原文线索

V2-019235 | MATHDB-349922

Homeomorphism conjecture for the topological classification of G-bundles

摘录 / EXCERPT: Let G be a reductive group over E, let Bun_G be the moduli stack of G-bundles on the relevant Fargues-Fontaine curve, and let $B(G)$ be the set of Kottwitz c...

conjecture | open | MathDB 页面 | 原文线索

V2-019236 | MATHDB-349934

The semisimplicity conjecture for three-variable invertible Calabi-Yau LG pairs

摘录 / EXCERPT: Let W be an invertible Calabi-Yau polynomial in three variables, and let (W,G) be an admissible Landau-Ginzburg pair. Semisimplicity conjecture for three-variable Calabi-Yau LG...

conjecture | open | MathDB 页面 | 原文线索

V2-019237 | MATHDB-349935

The seven-model classification conjecture for three-variable Calabi-Yau LG theories

摘录 / EXCERPT: Let (W,G) be an admissible LG pair, where W is an invertible Calabi-Yau polynomial of three variables from the classified list. Seven-model classification conjecture. The LG A-...

conjecture | open | MathDB 页面 | 原文线索

V2-019238 | MATHDB-349967

Levine's hat problem conjecture

摘录 / EXCERPT: Consider t players, each with a stack of n independently random black or white hats, and let $p(t,n)$ be the maximal probability that all players select black hats under a join...

conjecture | open | MathDB 页面 | 原文线索

V2-019239 | MATHDB-349999

Cheeger-Goresky-MacPherson conjecture on intersection cohomology and L^2-cohomology

摘录 / EXCERPT: Let $X\subset \mathbb{P}^N$ be a projective variety, let $X_{\rm reg}$ be its regular locus, and let $ds^2_{\rm FS}$ be the Fubini-Study metric on $X_{\rm reg}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019240 | MATHDB-350032

The p-adic local invariant cycle conjecture

摘录 / EXCERPT: The p-adic local invariant cycle conjecture. If the residue field of R is finite, there is a surjective natural map

conjecture | open | MathDB 页面 | 原文线索

V2-019241 | MATHDB-350072

Plessisiso's Bogomolov property conjecture for torsion extensions of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety defined over a number field K, let $\mathcal{L}$ be a symmetric ample line bundle on A/K, and let L/K be a finite extension. For the Néron-Tate...

conjecture | open | MathDB 页面 | 原文线索

V2-019242 | MATHDB-350073

Bogomolov property conjecture from bounded local degree

摘录 / EXCERPT: Let K be a number field, let v be a finite place of K, and let L/K be an algebraic extension. For an extension w of v to L, write L_w for the corresponding completi...

conjecture | open | MathDB 页面 | 原文线索

V2-019243 | MATHDB-350075

Zilber's Zilber-Pink conjecture for distinguished categories

摘录 / EXCERPT: Let K be an algebraically closed field and let $\mathfrak{C}$ be a distinguished category over K. A distinguished variety is an object of $\mathfrak{C}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-019244 | MATHDB-350099

Weyl-cycle conjecture for joins over secant varieties

摘录 / EXCERPT: Let X^n_{n+3} be the blow-up of $\mathbb{P}^n$ at $n+3$ points. Let L_I denote the linear space spanned by the relevant subset of base points, and let σ_t denote the t...

open_problem | open | MathDB 页面 | 原文线索

V2-019245 | MATHDB-350127

Containment of the open-orbit Schwartz space

摘录 / EXCERPT: Let F be the field in the paper, let P be the relevant parabolic subgroup, and let $X_P^{\circ}(F) \subset X_P(F)$ be the open subset appearing in the construction of the Schwa...

conjecture | open | MathDB 页面 | 原文线索

V2-019246 | MATHDB-350204

Generalized Cayley-Bacharach conjecture for quadrics

摘录 / EXCERPT: Let Γ be a complete intersection of n quadrics in $\mathbb{P}^n$. Let $X\subset\mathbb{P}^n$ be a hypersurface of degree d containing a subscheme $\Omega\subset\Gamma$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-019247 | MATHDB-350223

Log-resolution conjecture for schemes over local quasi-excellent rings

摘录 / EXCERPT: Log-resolution conjecture. There exists such a morphism $f:\tilde{X}\to X$ for which $\tilde{X}$ is a regular scheme and $\tilde{X}_s$ is a normal crossings divisor.

conjecture | open | MathDB 页面 | 原文线索

V2-019248 | MATHDB-350245

Quadratic generation conjecture for Richardson variety initial ideals

摘录 / EXCERPT: Let X_w^v be a Richardson variety in the Grassmannian, let $G_{k,n,\ell}|_w^v$ be the corresponding restricted set of relations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019249 | MATHDB-350256

The full Mordell-Lang conjecture for abelian varieties over function fields

摘录 / EXCERPT: Let p be a prime, let k be a finite field of characteristic p, let K/k be a regular extension with algebraic closure $K\subseteq\overline K$, and let A be a K-abelian v...

conjecture | open | MathDB 页面 | 原文线索

V2-019250 | MATHDB-350264

Miyanishi's conjecture on endomorphisms injective outside a closed subvariety

摘录 / EXCERPT: Let X be an algebraic variety over an algebraically closed field of characteristic zero, and let $\phi:X \longrightarrow X$ be an endomorphism. Let Y be a proper closed subvari...

conjecture | open | MathDB 页面 | 原文线索

V2-019251 | MATHDB-350267

Budur-Wang special varieties package conjecture for absolute sets

摘录 / EXCERPT: Budur-Wang's special varieties package conjecture. The following conditions hold in $M_B(X,r)$: (A1) absolute $\overline{\mathbb{Q}}$-constructible subsets are generated from absol...

conjecture | open | MathDB 页面 | 原文线索

V2-019252 | MATHDB-350271

The hypertoric intersection-cohomology conjecture for type A

摘录 / EXCERPT: The hypertoric intersection-cohomology conjecture. For all n, there exists an isomorphism of graded $\mathfrak{S}_n$-representations

conjecture | open | MathDB 页面 | 原文线索

V2-019253 | MATHDB-350290

Geometric Lang's conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of general type. Consider the irreducible positive-dimensional subvarieties of X that are not of general type. Geometric Lang's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-019254 | MATHDB-350321

Parity conjecture for type-B stack-sorting fibers

摘录 / EXCERPT: Let B_n be the type-B Coxeter group and let $\operatorname{\mathtt{s}}_B:B_n\to B_n$ be the type-B stack-sorting map. For $v\in B_n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-019255 | MATHDB-350322

Asymptotic average-sorting-time conjecture for type-B stack-sorting

摘录 / EXCERPT: Let $\mathcal D_n$ be the average number of iterations of the ordinary stack-sorting map $\operatorname{\mathtt{s}}$ needed to send an element of S_n to the identity permutation,...

conjecture | open | MathDB 页面 | 原文线索

V2-019256 | MATHDB-350325

Hodges and Yong's maximal sphericity conjecture for Schubert varieties

摘录 / EXCERPT: Let G/B be a flag variety, let $w\in S_n$ index the Schubert variety $X_w\subseteq G/B$, and let $J(w)$ be the left descent set of w. Write $P_{J(w)}$ for the corresponding par...

conjecture | open | MathDB 页面 | 原文线索

V2-019257 | MATHDB-350326

Hodges and Yong's pattern-avoidance conjecture for spherical permutations

摘录 / EXCERPT: Let w be a permutation, and let ... A permutation is said to avoid the patterns in P when none of these twenty-one patterns occurs as a pattern in it. Hodges and Yong's conjectur...

conjecture | solved | MathDB 页面 | 原文线索

V2-019258 | MATHDB-350327

Long-time convergence conjecture for generalized Kähler-Ricci flow on Fano varieties

摘录 / EXCERPT: Suppose (g, I, J) is a symplectic type generalized Kähler structure on a Fano manifold (M, J), with $F \in 2\pi c_1(M, J)$. Let (g_t,I_t,J) be the solution of the normalized...

conjecture | open | MathDB 页面 | 原文线索

V2-019259 | MATHDB-350332

Rationality conjecture for sortable elements of m-Tamari lattices

摘录 / EXCERPT: For integers $m,t\geq 1$, let $h_t(m,n)$ denote the number of t-$\operatorname{\mathsf{Pop}}$-sortable m-ballot paths in the m-Tamari lattice $\operatorname{Tam}_n(m)$. Ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-019260 | MATHDB-350347

Fargues' conjecture for irreducible representations of the Weil group of GL_n

摘录 / EXCERPT: Let E be the local field underlying the Weil group W_E, let $G=\mathrm{GL}_n$, and let $B(G)$ denote its Kottwitz set. Write Bun_n for the moduli stack of G-bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-019261 | MATHDB-350352

Kawamata's relative Fujita conjecture

摘录 / EXCERPT: Let $f:Y\to X$ be a proper morphism between smooth projective algebraic varieties. Assume that the degenerate locus of f is contained in a normal crossing divisor $D\subset X$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-019262 | MATHDB-350375

Discriminant-negligibility of heights of special points on Shimura varieties

摘录 / EXCERPT: Discriminant-negligibility conjecture. There exists a Weil height function $h\colon S(\overline{\mathbb Q})\to\mathbb R$ such that, for every special point $p\in S$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-019263 | MATHDB-350387

Antieau-Williams conjecture on failure of purity for non-special semisimple groups

摘录 / EXCERPT: Let k be a field, let G be a connected reductive group over k, and let X be an irreducible smooth k-scheme with function field K. Purity for G-torsors over X means...

conjecture | open | MathDB 页面 | 原文线索

V2-019264 | MATHDB-350388

The universal quantum Laurentness conjecture for quantum duality elements

摘录 / EXCERPT: Let $\frak{S}$ be a triangulable punctured surface. A congruent ${\rm SL}_3$-lamination is an element $\ell \in \mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^t)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-019265 | MATHDB-350389

The quantum duality basis conjecture for the full cluster algebra

摘录 / EXCERPT: Let $\frak{S}$ be a triangulable punctured surface, and let $\mathscr{A}_{{\rm SL}_3,\frak{S}}(\mathbb{Z}^t)$ denote the set indexing the quantum duality elements...

conjecture | open | MathDB 页面 | 原文线索

V2-019266 | MATHDB-350437

Gang's vanishing conjecture for adjoint Reidemeister torsion

摘录 / EXCERPT: Let M be a compact oriented 3-manifold with torus boundary, and let $\mathcal{X}(M)$ be the character variety of irreducible representations...

conjecture | open | MathDB 页面 | 原文线索

V2-019267 | MATHDB-350448

Arakelov equality characterization for semistable families of varieties

摘录 / EXCERPT: Arakelov equality characterization. The general fiber has Kodaira dimension zero, and the family is Shimura in the following sense: Y parameterizes a compactified universal famil...

conjecture | open | MathDB 页面 | 原文线索

V2-019268 | MATHDB-350474

Dilation conjecture for affine conic bundles

摘录 / EXCERPT: Let M be an affine conic bundle of the form described in the paper, and suppose that the polynomial $p(z_3,\ldots,z_{n+1})$ is weighted homogeneous and invertible. Dilation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-019269 | MATHDB-350571

Motivic Gerasimov-Shatashvili amplitude conjecture

摘录 / EXCERPT: Let $A_{\mathbb{Q}_p}^{(N)}(\boldsymbol{s})$ be the p-adic Koba-Nielsen amplitudes, and let $A_{\mathbb{C((}T\mathbb{))}}^{(N)}(\boldsymbol{L})$ denote the motivic string amplitu...

conjecture | open | MathDB 页面 | 原文线索

V2-019270 | MATHDB-350610

The Doran-Harder-Thompson conjecture on mirror fibrations

摘录 / EXCERPT: Let $\frX \to \AA^1$ be a Tyurin degeneration to $\frZ_0 \cup_{\frD} \frZ_1$, with mirror Calabi-Yau $\widecheck{\frX_{gen}}$ and mirror divisor $\frD$. A Tyurin degeneration is a...

conjecture | open | MathDB 页面 | 原文线索

V2-019271 | MATHDB-350611

The Doran-Harder-Thompson reconstruction conjecture via gluing

摘录 / EXCERPT: Let $\widecheck{\frZ}_0 \to \AA^1$ and $\widecheck{\frZ}_1 \to \AA^1$ be Landau-Ginzburg models, and suppose that the fibres of $\widecheck{\frZ}_0 \to \AA^1$ have one dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-019272 | MATHDB-350639

The monoid conjecture for stratified embedded Kisin varieties

摘录 / EXCERPT: Let SEKV be the monoid of shape-stratified closed subvarieties of $(\mathbb{P}^1)^n$, for varying n, with multiplication given by products. For a tame inertial type...

conjecture | open | MathDB 页面 | 原文线索

V2-019273 | MATHDB-350640

The tensor-product conjecture for deformation rings of stratified Kisin varieties

摘录 / EXCERPT: Let $\mathbb{X}$ be a fixed nondegenerate gene, and write its stratified embedded Kisin variety as ... inside $(\mathbb{P}^1)^f$. Let $R^{\psi}({\rm t},\overline\rho)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019274 | MATHDB-350644

Non-Archimedean Campana's specialness conjecture

摘录 / EXCERPT: Let K be an algebraically closed, complete, non-Archimedean valued field of characteristic zero, and let X/K be a smooth projective variety. Write X^{an} for i...

conjecture | open | MathDB 页面 | 原文线索

V2-019275 | MATHDB-350652

Griffiths-Green singularity conjecture for normal functions

摘录 / EXCERPT: Griffiths-Green conjecture. For every $\alpha\in Hg^p(X)$, the associated admissible normal function ν_α is singular on $\bar{S}=|L^{\otimes d}|$ for some $d>0$. This conj...

conjecture | open | MathDB 页面 | 原文线索

V2-019276 | MATHDB-350669

Hodge-proper stack conjecture on rational étale cohomology

摘录 / EXCERPT: Let K be a finite extension of $\mathbb Q_p$, let $\mathcal O_K$ be its ring of integers, and let $\mathpzc X$ be a smooth Hodge-proper stack over $\mathcal O_K$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019277 | MATHDB-350670

Formally proper stack conjecture on mod-p étale cohomology

摘录 / EXCERPT: Let K be a finite extension of $\mathbb Q_p$, let $\mathcal O_K$ be its ring of integers, and let $\mathpzc X$ be a smooth formally proper stack over $\mathcal O_K$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019278 | MATHDB-350671

Formally proper Artin stack conjecture on integral étale cohomology

摘录 / EXCERPT: Let K be a finite extension of $\mathbb Q_p$, let $\mathcal O_K$ be its ring of integers, and let $\mathpzc X$ be a smooth formally proper Artin stack over $\mathcal O_K$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019279 | MATHDB-350684

The orbit-invariance conjecture for horospherical varieties

摘录 / EXCERPT: Let X be a horospherical complexity-zero variety, let τ be a face of its cone $\sigma^\vee$, and let O_τ be the corresponding G-orbit. Let $p_1,\ldots,p_k$ be the pri...

conjecture | open | MathDB 页面 | 原文线索

V2-019280 | MATHDB-350689

The lattice-theoretic exact sequence for the component group of a unipotent group

摘录 / EXCERPT: Let C be a curve, let G be the generic fiber of the relevant group scheme, and let $\mathbf{U}$ be its Néron model. Write $D(\operatorname{NS}(X))$ for the discriminant group o...

conjecture | open | MathDB 页面 | 原文线索

V2-019281 | MATHDB-350697

The Chow-ring intersection conjecture for Lagrangian fibrations

摘录 / EXCERPT: Let X be a hyperkähler variety of dimension 4 admitting a Lagrangian fibration, and let A be a general fibre. Write $A^i(X):=CH^i(X)_{\mathbb{Q}}$ for the Chow groups with ra...

conjecture | open | MathDB 页面 | 原文线索

V2-019282 | MATHDB-350701

The ampleness conjecture for strictly nef divisors on K-trivial manifolds

摘录 / EXCERPT: Ampleness conjecture. Any strictly nef divisor on X is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-019283 | MATHDB-350711

Gross-Hacking-Keel theta-function conjecture for positive log Calabi-Yau varieties

摘录 / EXCERPT: Let U be a positive log Calabi-Yau variety with maximal boundary, and let $N_U(\mathbb{Z})$ denote its integral tropical points. A mirror positive log Calabi-Yau variety is a pos...

conjecture | open | MathDB 页面 | 原文线索

V2-019284 | MATHDB-350712

Batyrev-Borisov-type mirror conjecture for log Calabi-Yau varieties

摘录 / EXCERPT: Let U and $V=U^\star$ be mirror log Calabi-Yau varieties with theta-function bases, and let $P\subset N_U$ and $Q=P^\star\subset M_U$ be dual reflexive polygons. The polytope P...

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V2-019285 | MATHDB-350731

The asymptotic polynomial-count conjecture for complex varieties

摘录 / EXCERPT: A complex variety X is a variety over $\mathbb{C}$. It is asymptotically polynomial count when its point counts over finite fields agree with a polynomial in the field size for a...

conjecture | open | MathDB 页面 | 原文线索

V2-019286 | MATHDB-350732

The strong polynomial-count conjecture for complex varieties

摘录 / EXCERPT: Let $\mathcal{X}$ be a reduced $\mathbb{Z}$-scheme and let $X=\mathcal{X}(\mathbb{C})$ be the associated complex algebraic variety. Say that X is strongly polynomial count when i...

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V2-019287 | MATHDB-350759

The seven-Pfaffian conjecture for the rank-one catalecticant image

摘录 / EXCERPT: Let C_1 denote the rank-one locus of the relevant catalecticant variety, let A be a point on C_1, and let $\phi(A)$ be its image under the map ϕ. Seven-Pfaffian conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-019288 | MATHDB-350760

The 27-Pfaffian conjecture for the reciprocal catalecticant variety

摘录 / EXCERPT: Let $\mathrm{Cat}(2,3)$ be the catalecticant variety, let $\mathbb{P}\\,\mathrm{Cat}(2,3)^{-1}$ denote its reciprocal variety, and let a cubic Pfaffian mean the Pfaffian of a...

conjecture | open | MathDB 页面 | 原文线索

V2-019289 | MATHDB-350782

Weighted blow-up bound for minimal log discrepancies of smooth varieties

摘录 / EXCERPT: Weighted blow-up bound. Assume $N\geq3$. For every prime divisor E over A with the center at 0, there exists a prime divisor F obtained by at most $N-1$ weighted blow-ups s...

conjecture | open | MathDB 页面 | 原文线索

V2-019290 | MATHDB-350797

Pro-étale uniformisation conjecture for curves

摘录 / EXCERPT: Let K be the base field, and let C be a connected smooth projective curve over K of genus $g\geq 1$. In the diamantine setting, one expects a universal pro-étale perfectoid c...

conjecture | open | MathDB 页面 | 原文线索

V2-019291 | MATHDB-350799

Zilber's Exponential Algebraic Closedness conjecture for semiabelian varieties

摘录 / EXCERPT: Let S be a complex semiabelian variety of dimension n, and write $\exp_S: \mathbb{C}^n \to S$ for its exponential map. Let $V \subseteq \mathbb{C}^n \times S$ be a free and rot...

conjecture | open | MathDB 页面 | 原文线索

V2-019292 | MATHDB-350801

Zilber-Pink finiteness conjecture for optimal subvarieties

摘录 / EXCERPT: Let S be a Shimura variety and let V be a subvariety of S. For an irreducible subvariety W of S, let $\langle W\rangle$ be the smallest special subvariety of S containi...

conjecture | open | MathDB 页面 | 原文线索

V2-019293 | MATHDB-350832

Finiteness conjecture for fundamental groups of klt Calabi-Yau varieties

摘录 / EXCERPT: Let Y be a projective klt variety with numerically trivial canonical divisor, meaning that K_Y is numerically equivalent to zero. Its augmented irregularity is the supremum of...

conjecture | open | MathDB 页面 | 原文线索

V2-019294 | MATHDB-350833

Maximal loss of quadrics under projection of Veronese varieties

摘录 / EXCERPT: Maximal-loss conjecture. In each projection step, one loses the maximal number of quadrics, namely the codimension of the projected variety. Equivalently, successive general-point...

conjecture | open | MathDB 页面 | 原文线索

V2-019295 | MATHDB-350841

The improved threshold conjecture for rational points on hypersurfaces

摘录 / EXCERPT: Let $g_1(d)$ be an effectively computable function satisfying ... Let $X\subset\mathbb{A}^n_{\mathbb{F}_q}$ be a geometrically irreducible hypersurface of degree d. Rational-point...

conjecture | open | MathDB 页面 | 原文线索

V2-019296 | MATHDB-350845

Desargues configuration criterion for five-tuples of matrices

摘录 / EXCERPT: Let $A_1,\dots,A_5$ be real matrices whose degeneracy locus ${\mathcal X}:={\mathcal X}(A_1,\dots,A_5)$ is formed from the associated ten lines in ${\mathbb P}^3$. Desargues config...

conjecture | open | MathDB 页面 | 原文线索

V2-019297 | MATHDB-350846

Generalized Desargues configuration criterion for matrix degeneracy loci

摘录 / EXCERPT: Let $A_1,\dots,A_{m+1}$ be linear operators on ${\mathbb R}^m$, and let ${\mathcal X}$ be their degeneracy locus in ${\mathbb P}^{m-1}({\mathbb R})$. The locus has expected dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-019298 | MATHDB-350903

Conjecture that Arakelov and theta densities coincide for homogeneous subsets

摘录 / EXCERPT: Let $\mathcal{X}$ be a regular projective arithmetic variety of dimension n, and let $\overline{\mathcal{L}}$ be an ample line bundle on $\mathcal{X}$. Let ... be a subset such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-019299 | MATHDB-350941

The conjectural product rule for spherical functions of type A

摘录 / EXCERPT: The type-A spherical product conjecture. For $\lambda,\mu,\nu\in\Lambda_X^+$,

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V2-019300 | MATHDB-350942

The conjectural product rule for symmetric spherical functions of type A

摘录 / EXCERPT: The symmetric type-A product conjecture. For $\lambda,\mu,\nu\in\Lambda_X^+$,

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V2-019301 | MATHDB-350944

Bounded Gotsman-Linial conjecture for semi-algebraic sets

摘录 / EXCERPT: Let $S\subseteq\mathbb{R}^d$ be a semi-algebraic set of description complexity r, and let $a_1,\dots,a_n\in\mathbb{R}^d$. Define $f:\\{-1,1\\}^n\to\\{0,1\\}$ by letting...

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V2-019302 | MATHDB-350965

The Ulrich bundle existence conjecture for polarized varieties

摘录 / EXCERPT: Let (X,L) be a polarized pair, meaning a projective variety X equipped with a polarization L, and let an Ulrich bundle mean a vector bundle satisfying the Ulrich conditions w...

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V2-019303 | MATHDB-350990

Curious Lefschetz conjecture for unipotent and Steinberg varieties

摘录 / EXCERPT: Let W be a finite Coxeter group, let $\beta\in\mathit{Br}_W^+$, and let $w\in W$ be its image. Let $\cal{U}(\beta)$ and $\cal{Z}(\beta)$ be the associated varieties, and let...

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V2-019304 | MATHDB-351005

The global Shimura variety local-model diagram conjecture

摘录 / EXCERPT: Let ${\mathscr S}_{\sf K}$ be an integral model of a Shimura variety equipped with its v-sheaf theoretic local model diagram. A scheme-theoretic local model diagram is a diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-019305 | MATHDB-351038

Srinivas-Takagi F-nilpotent type and Hodge filtration conjecture

摘录 / EXCERPT: Let X be an n-dimensional scheme over $\mathbb C$ and let ${\mathfrak x}$ be a normal isolated singularity of X. Write X^{an} for the analytification and...

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V2-019306 | MATHDB-351047

Geometric duality conjecture for Whittaker sheaves

摘录 / EXCERPT: Fix a regular weight $ \lambda\in\mathfrak{h}^*$ and let X be the flag variety of $ \mathfrak{g}$. Let Q be an open B-orbit in a P_η-orbit on X, let $ \mathcal{O}_Q$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-019307 | MATHDB-351052

Generic cohomology vanishing conjecture for unitary Shimura varieties

摘录 / EXCERPT: Generic cohomology vanishing conjecture. If $H^i(S_K,\overline{\mathbf F}_{\ell})_{\mathfrak m_p}\neq0$, then $i\geq d$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-019308 | MATHDB-351058

A periodicity-one self-map conjecture for the real-motivic spectrum $\mathcal{Z}_\mathbb{R}$

摘录 / EXCERPT: Periodicity-one self-map conjecture. The spectrum $\mathcal{Z}_\mathbb{R}$ is of type $(2,2)$ and admits a $v_{(2,\mathrm{nil})}$-self-map

conjecture | open | MathDB 页面 | 原文线索

V2-019309 | MATHDB-351072

The cancellation conjecture for surjective self-morphisms of projective varieties

摘录 / EXCERPT: Let X be a projective variety defined over a number field K, and let $f\colon X\longrightarrow X$ be a surjective self-morphism. Cancellation conjecture. There exists a non-neg...

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V2-019310 | MATHDB-351092

Lang-Vojta conjecture for varieties of general type

摘录 / EXCERPT: Lang-Vojta conjecture. The stated degree inequality holds uniformly for all such curves and morphisms outside the proper exceptional subset Z.

conjecture | open | MathDB 页面 | 原文线索

V2-019311 | MATHDB-351095

Blasius--Rogawski conjecture on Eichler--Shimura relations for Shimura varieties

摘录 / EXCERPT: Let $\mathbf{G}$ be a connected reductive group over $\mathds{Q}$ with Shimura datum $(\mathbf{G},\mathcal{X})$, let $K=K^pK_p$ be sufficiently small with K_p hyperspecial at a p...

conjecture | open | MathDB 页面 | 原文线索

V2-019312 | MATHDB-351096

Frobenius-cycle conjecture for the Blasius--Rogawski Hecke polynomial

摘录 / EXCERPT: Assume that $\operatorname{Sh}_K(\mathbf{G},\mathcal{X})$ is of Hodge type, let $\mathscr{S}_K$ be its integral model over $\mathcal{O}_{E_{\mathfrak{p}}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019313 | MATHDB-351117

Generic injectivity of recovery from coprime power sums

摘录 / EXCERPT: Generic-injectivity conjecture. The recovery of a set of n complex numbers from $n+1$ power sums with coprime powers is unique: for generic $z\in\mathbb{C}^n$, the fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-019314 | MATHDB-351118

Weighted generic injectivity from power sums

摘录 / EXCERPT: Let $\tau=(\tau_1,\ldots,\tau_n)\in\mathbb{R}^n_{>0}$, and define ... Let $\operatorname{Stab}(\tau)$ be the subgroup of S_n consisting of coordinate permutations that fix τ....

conjecture | open | MathDB 页面 | 原文线索

V2-019315 | MATHDB-351126

The completion conjecture for plus test ideals

摘录 / EXCERPT: Let X be a normal integral scheme of finite type over a complete local ring $(R,\mathfrak{m})$ such that $R/\mathfrak{m}$ has positive characteristic. Let $B\geq 0$ be a...

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V2-019316 | MATHDB-351127

The perturbation invariance conjecture for plus test modules

摘录 / EXCERPT: Let X be a normal integral scheme projective over a complete local ring $(R,\mathfrak{m})$ with residual characteristic $p>0$, and let B be a $\mathbb{Q}$-divisor on X. Pertu...

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V2-019317 | MATHDB-351128

The subadditivity conjecture for plus test ideals

摘录 / EXCERPT: Let X be a regular scheme. Subadditivity conjecture. For divisors Δ_1 and Δ_2 on X, one has ... This is the mixed-characteristic analogue of subadditivity for multi...

conjecture | open | MathDB 页面 | 原文线索

V2-019318 | MATHDB-351182

Homeomorphism and codimension conjecture for Calabi-Yau collapse limits

摘录 / EXCERPT: Let M^m be a projective Calabi-Yau manifold with $K_M\cong\mathcal{O}_M$, equipped with a fiber space structure $f:M\to N$, where N^n is a normal projective variety and $0<n<m$...

conjecture | open | MathDB 页面 | 原文线索

V2-019319 | MATHDB-351197

Hu-Ye's geometric-versus-algebraic nullity conjecture for hypermatrices

摘录 / EXCERPT: Hu-Ye's conjecture. For any order-k hypermatrix M,

conjecture | open | MathDB 页面 | 原文线索

V2-019320 | MATHDB-351203

Generic simplicity conjecture for the billiard polynomial \(\mathcal{B}^n_{a,b}\)

摘录 / EXCERPT: For each integer $n\geq 3$, let $\mathcal{B}^n_{a,b}$ be the polynomial defined in the manuscript for the parameters a and b. Generic simplicity conjecture. For all $n\geq 3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-019321 | MATHDB-351216

Bloch's conjecture on Chern classes of flat vector bundles

摘录 / EXCERPT: Let X be a smooth complex projective variety. Let (V,∇) be a flat vector bundle on X, and let ... be the conjectural functorial Bloch-Beilinson filtration. Bloch's conject...

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V2-019322 | MATHDB-351217

Bloch's weak conjecture for Chern classes of flat vector bundles

摘录 / EXCERPT: Let X be a smooth variety, let $B^i(X)$ denote codimension-i algebraic cycles modulo algebraic equivalence, and let (V,∇) be a flat vector bundle on X. Bloch's weak co...

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V2-019323 | MATHDB-351218

Torsion conjecture for Chern classes of properly rigid representations

摘录 / EXCERPT: Let X be a smooth complex quasi-projective variety with base point x, and let ... be a properly rigid representation with quasi-unipotent monodromy at infinity. Let...

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V2-019324 | MATHDB-351219

Torsion conjecture for Chern classes of flat bundles of geometric origin

摘录 / EXCERPT: Let X be a smooth complex quasi-projective variety and let $\mathbb V$ be a local system of geometric origin on X. Let (V,∇) be the associated algebraic flat vector bund...

conjecture | open | MathDB 页面 | 原文线索

V2-019325 | MATHDB-351220

Chow-theoretic Chern class conjecture for Lie-irreducible representations

摘录 / EXCERPT: Let X be a smooth complex quasi-projective variety with base point x, and let ... be Lie irreducible with quasi-unipotent monodromy at infinity. Let (V_ρ,∇_ρ) be the...

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V2-019326 | MATHDB-351221

Chow-theoretic Chern class conjecture for stable Higgs bundles

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over an algebraically closed field k, let H be an ample divisor on X, and let (E,θ) be a slope H-stable Higgs...

conjecture | open | MathDB 页面 | 原文线索

V2-019327 | MATHDB-351222

Weak Bloch conjecture for semistable Higgs bundles

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over an algebraically closed field k of characteristic zero, let H be an ample divisor, and let (E,θ) be a slope...

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V2-019328 | MATHDB-351226

The reduced orbifold associated-variety conjecture

摘录 / EXCERPT: Let ${\mathcal V}$ be a simple, strongly finitely generated VOA that is $\mathbb N$-graded by conformal weight, with ${\mathcal V}[0]\cong\mathbb C$. Let G be a finite group of a...

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V2-019329 | MATHDB-351232

The local model conjecture for tubular neighborhoods of p-adic shtuka spaces

摘录 / EXCERPT: Local model conjecture. For every closed point $x\in |({X_{\mathscr{G}}^{\leq\mu}(b)})_{{k_F}}|$ there exists a closed point $y\in |({{\mathcal{A}}_{\mathscr{G},\mu}})_{k_F}|$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-019330 | MATHDB-351257

Zhou's connected-components conjecture for Hodge-Newton irreducible affine Deligne-Lusztig varieties

摘录 / EXCERPT: Zhou's conjecture. If (λ,b) is Hodge-Newton irreducible, then η_G induces a bijection

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V2-019331 | MATHDB-351258

Bannai-Ito conjecture on metric and cometric association schemes

摘录 / EXCERPT: Bannai-Ito conjecture. For sufficiently large diameter, a primitive association scheme is metric if and only if it is cometric.

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V2-019332 | MATHDB-351271

Shadow-boundary conjecture for strongly generic polynomials

摘录 / EXCERPT: Shadow-boundary conjecture. (i) $\x1d4fe_P\subset\x0043\x006f\x006e\x0076_P$ is a concave domain. (ii) The boundary of $\x1d4fe_P$ is contained in the union, over...

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V2-019333 | MATHDB-351281

Sakellaridis-Venkatesh weak local conjecture for spherical varieties

摘录 / EXCERPT: Let $Y=H\backslash G$ be a strongly tempered spherical variety over a local field F. Let $G_Y^\vee$ be the dual group of Y, let...

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V2-019334 | MATHDB-351282

Rationality conjecture for canonical local periods

摘录 / EXCERPT: Let E be a field, let G be a reductive group over a local field F, and let $Y=H\backslash G$ be a spherical variety. For a smooth admissible $G(F)$-representation π over...

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V2-019335 | MATHDB-351296

Algebraicity conjecture for real symplectic del Pezzo surfaces of degree 1

摘录 / EXCERPT: Let a real symplectic del Pezzo surface of degree 1 be equipped with a J-holomorphic structure satisfying the Bertini condition. Algebraicity conjecture. Each deformation class...

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V2-019336 | MATHDB-351320

Equivalent strong Zilber-Pink conditions for the atypical Hodge locus

摘录 / EXCERPT: Strong Zilber-Pink equivalence. The following conditions are equivalent: (a) the atypical Hodge locus is a finite union of maximal atypical special subvarieties; (b) it is a strict...

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V2-019337 | MATHDB-351321

Quantitative semi-algebraic basic lemma conjecture

摘录 / EXCERPT: Let $X \subset \mathbb{R}^k$ be a semi-algebraic set defined by a closed formula of complexity bounded by C, and let $\dim X=n$. A nested semi-algebraic filtration is a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-019338 | MATHDB-351338

The abundance conjecture for canonical sheaves

摘录 / EXCERPT: Let X be a smooth projective variety, and let ω_X denote its canonical sheaf. A line bundle is semi-ample if some positive tensor power is globally generated; in the formu...

conjecture | open | MathDB 页面 | 原文线索

V2-019339 | MATHDB-351339

The non-vanishing conjecture for canonical sheaves

摘录 / EXCERPT: Let X be a smooth projective variety. A $\mathbb{Q}$-divisor is effective if it is a nonnegative $\mathbb{Q}$-linear combination of prime divisors, and two $\mathbb{Q}$-divisors...

conjecture | open | MathDB 页面 | 原文线索

V2-019340 | MATHDB-351345

The minimality conjecture for indecomposable derived categories

摘录 / EXCERPT: Let X be a smooth projective variety. Assume that the canonical bundle K_X is nef and effective. A minimality conjecture for indecomposable derived categories asserts that...

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V2-019341 | MATHDB-351346

Kawatani's converse conjecture on minimality and derived indecomposability

摘录 / EXCERPT: Let X be a smooth projective variety. A variety is derived-indecomposable if $D(X)$ has no nontrivial semi-orthogonal decompositions. Kawatani's converse conjecture. If $D(X)$ ha...

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V2-019342 | MATHDB-351347

Refined indecomposability conjecture for varieties with positive irregularity

摘录 / EXCERPT: Let X be a smooth projective variety, let K_X be nef and effective, and define the irregularity by $q(X)=h^1(\mathcal{O}_X)$. Refined indecomposability conjecture. If...

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V2-019343 | MATHDB-351350

The threaded-digon complex conjecture for the flag Hilbert scheme

摘录 / EXCERPT: Let $a_1,\ldots,a_m,b\geq0$ and set $\bm{a}=(a_1,\ldots,a_m)$. A one-sided twisted complex of threaded digons is a chain of complexes obtained by increasing the number of crossings...

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V2-019344 | MATHDB-351352

The colored full-twist dg-algebra conjecture

摘录 / EXCERPT: For ${\bm{b}}=(b_1,\ldots,b_m)$, let $E_{\bm{b}}$ be the colored polynomial algebra and let $I_{\bm{b}}\vartriangleleft E_{\bm{b}}$ be the colored full-twist ideal. Let...

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V2-019345 | MATHDB-351354

The Lyndon-permutation dimension conjecture for feasible regions

摘录 / EXCERPT: Lyndon-permutation dimension conjecture. The feasible region clP_k contains an open ball of dimension $|\mathcal{L}_k|$.

conjecture | open | MathDB 页面 | 原文线索

V2-019346 | MATHDB-351386

The broad and Hodge-generic conjecture for quotient maps of Shimura domains

摘录 / EXCERPT: Let $q: Omegato S$ be the quotient map associated with the Shimura setting, and let $EqsubsetOmegatimes S$ denote its graph. Let $Vsubsetmathbb C^{N}times S$ be an irreducible alge...

conjecture | open | MathDB 页面 | 原文线索

V2-019347 | MATHDB-351387

The special-point non-density conjecture for Shimura quotient graphs

摘录 / EXCERPT: Let $\Gamma\subset G(\mathbb Q)$ be an arithmetic subgroup, let $S=\Gamma\backslash\Omega$, and let $V\subset\mathbb C^{N}\times S$ be an irreducible algebraic variety. Let...

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V2-019348 | MATHDB-351417

Algebraization conjecture for the formal group H_Σ

摘录 / EXCERPT: Let H_Σ be the formal group over the stack Σ introduced above. An algebraization of H_Σ is an isomorphism class of a smooth affine group scheme over Σ w...

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V2-019349 | MATHDB-351441

Vitoria's converse to Zhang's equivalence conjecture for skew projective spaces

摘录 / EXCERPT: Let S_{α} and $S_{\alpha'}$ be standard graded skew polynomial algebras. Their noncommutative projective schemes are the pairs ... where...

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V2-019350 | MATHDB-351489

The converse vanishing theorem in Verdier quotients of stable categories

摘录 / EXCERPT: Assume $(A,\mathfrak m)$ is a complete local complete intersection of dimension d and codimension c, with algebraically closed residue field $k=A/\mathfrak m$. Let X be an al...

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V2-019351 | MATHDB-351506

Bi-Lipschitz invariance of degree at infinity for complex algebraic sets

摘录 / EXCERPT: Degree-at-infinity conjecture. If X and Y are bi-Lipschitz homeomorphic at infinity, then

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V2-019352 | MATHDB-351550

Debarre-Iliev-Manivel conjecture on the period fiber of ordinary Gushel-Mukai threefolds

摘录 / EXCERPT: Let $\mathcal{X}_{10}$ be the moduli space of ordinary Gushel-Mukai threefolds, let $\mathcal{A}_{10}$ be the moduli space of 10-dimensional principally polarised abelian varieti...

conjecture | open | MathDB 页面 | 原文线索

V2-019353 | MATHDB-351552

Foliated Fujita conjectures for adjoint line bundles

摘录 / EXCERPT: Let $\mathcal{F}$ be a regular foliation of rank p on a smooth projective variety X, and let L be an ample line bundle on X. Foliated Fujita conjectures. The line bundle .....

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V2-019354 | MATHDB-351581

Stacky Batyrev-Manin-Malle conjecture of Ellenberg-Satriano-Zureick-Brown

摘录 / EXCERPT: Let $\mathfrak{X}$ be a “nice” algebraic stack defined over a number field K, and let $\mathcal{E}$ be a “nice” vector bundle on $\mathfrak{X}$. Let...

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V2-019355 | MATHDB-351592

Extension of the equivalence lemma for quasi-Hermitian varieties to odd dimensions

摘录 / EXCERPT: Let r be an odd integer with $r\geq 3$, and let α and β be the coefficients appearing in the construction of the quasi-Hermitian varieties. The relevant equivalence...

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V2-019356 | MATHDB-351598

The Hodge filtration criterion for rational singularities of local complete intersections

摘录 / EXCERPT: Let X be a smooth variety and let $Z\subseteq X$ be a local complete intersection of pure codimension r. Write $F_1\mathcal{H}_Z^r(\mathscr{O}_X)$ for the first Hodge-filtratio...

conjecture | open | MathDB 页面 | 原文线索

V2-019357 | MATHDB-351600

The depth criterion for logarithmic de Rham vanishing

摘录 / EXCERPT: Let Z be a reduced closed subscheme of a smooth variety X of dimension n, let $f:Y\to X$ be a log resolution, and let E be the reduced exceptional divisor over Z. Depth-v...

conjecture | open | MathDB 页面 | 原文线索

V2-019358 | MATHDB-351634

The classical Zariski dense orbit conjecture

摘录 / EXCERPT: Zariski dense orbit conjecture. Either there exists $\alpha\in X(K)$ whose orbit under Φ is well-defined and Zariski dense in X, or there exists a non-constant rational func...

conjecture | open | MathDB 页面 | 原文线索

V2-019359 | MATHDB-351653

Inverse Galois conjecture for Schubert calculus

摘录 / EXCERPT: Let $1\leq a_1<\dotsb<a_r<b$ be positive integers, write $a_\bullet$ for the sequence $a_1<\dotsb<a_r$, and let $S\binom{[b]}{a_\bullet}$ denote the permutation group induced by...

conjecture | open | MathDB 页面 | 原文线索

V2-019360 | MATHDB-351686

Hida-Urban non-abelian Leopoldt conjecture for eigenvariety components

摘录 / EXCERPT: Let $\tilde\pi$ be a non-critical cuspidal representation, let $x_{\tilde\pi}\in\cE_{\cU,h}^\bullet$ be the associated point, and let $\cW$ be the weight space. Write ℓ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-019361 | MATHDB-351695

Local Prym parity product formula for genus 2 and 3 curves

摘录 / EXCERPT: Let C be a curve of genus 2 or 3 over a number field K, with an unramified double cover ... and associated Prym variety $\operatorname{Prym}(D/C)$ over K. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019362 | MATHDB-351705

Kollár's quotient-singularity conjecture for finite covers

摘录 / EXCERPT: Let $f\colon X\xymatrix@1@=15pt{\ar[r]&} Y$ be a finite and dominant morphism from a smooth variety X onto a normal variety Y. Kollár's conjecture. The variety Y has quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-019363 | MATHDB-351712

Shende's conjecture relating braid varieties and affine Springer fibers

摘录 / EXCERPT: Let C be an irreducible plane curve singularity, let L be the corresponding algebraic knot, and let β be the corresponding braid on n strands. Let $X(\beta\Delta;w_0)$...

conjecture | open | MathDB 页面 | 原文线索

V2-019364 | MATHDB-351713

Gorsky-Negut-Rasmussen conjecture for sheaves on Hilbert schemes

摘录 / EXCERPT: Let β be a braid on n strands. Let $\operatorname{Hilb}^n(\mathbb{C}^2,\mathbb{C})$ be the indicated Hilbert scheme, let $\mathcal{T}$ be its tautological rank-n bundle,...

conjecture | open | MathDB 页面 | 原文线索

V2-019365 | MATHDB-351730

The Abelian/non-Abelian correspondence for Givental cones

摘录 / EXCERPT: Let G be a reductive group acting on a smooth projective variety A, let T be a maximal torus of G, and let W be the Weyl group. Write $A /\!/ G$ and $A /\!/ T$ for the co...

conjecture | open | MathDB 页面 | 原文线索

V2-019366 | MATHDB-351731

The Abelian/non-Abelian correspondence with bundles

摘录 / EXCERPT: Let G be a reductive group acting on a smooth projective variety A, let T be a maximal torus of G, and let W be the Weyl group. A representation of G induces vector bun...

conjecture | open | MathDB 页面 | 原文线索

V2-019367 | MATHDB-351751

Unlikely Intersection Conjecture for semiabelian varieties

摘录 / EXCERPT: Let G be a semiabelian variety defined over $\overline{\mathbb{Q}}$ and let $X\subseteq G$ be an algebraic subvariety. For an algebraic subgroup $H\subseteq G$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-019368 | MATHDB-351783

Kollár's topological reconstruction conjecture for varieties

摘录 / EXCERPT: Let X and X' be seminormal, geometrically irreducible varieties over fields k and k', respectively, with $\noperatorname{char} k=0$ and $ \dim X\geq 2$. A homeomorphism ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-019369 | MATHDB-351799

Generic isomorphism conjecture for the motivic multiple zeta value quotient

摘录 / EXCERPT: Let $C_{u,1-u}$ be the zinbiel sub-algebra parametrized by u, and let $\Pi:C_{u,1-u}\to\mathsf{A}_{\mathsf{MZV}}$ be the corresponding morphism of commutative algebras. Generic i...

conjecture | open | MathDB 页面 | 原文线索

V2-019370 | MATHDB-351808

Nonexistence conjecture for diameter perfect codes in Johnson spaces

摘录 / EXCERPT: Diameter-perfect-code conjecture. There are no nontrivial diameter perfect codes in $J(n,w)$, except for Steiner systems and their complements.

conjecture | open | MathDB 页面 | 原文线索

V2-019371 | MATHDB-351829

The affine-bundle conjecture for p-adic Coxeter orbits

摘录 / EXCERPT: Let $X_{c,b}^{\caG_{\bfx}^{\rm ad}}$ be the affine Deligne-Lusztig variety associated with the Coxeter element c, the element b, and the adjoint parahoric group scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-019372 | MATHDB-351837

Surjectivity conjecture for the motivic polylogarithm map

摘录 / EXCERPT: Soit $\psi_n:\ker(\delta_n)\rightarrow\ker(\delta:\mathcal{C}_n(F)\rightarrow(\Lambda^2\mathcal{C}(F))_n)$ le morphisme construit à partir des polylogarithmes motiviques, où F es...

conjecture | open | MathDB 页面 | 原文线索

V2-019373 | MATHDB-351838

Goncharov's freeness conjecture for the motivic Lie coalgebra

摘录 / EXCERPT: Pour $n\geq 1$, soit $\iota^\bullet(n):\mathcal{B}^\bullet(F,n)\rightarrow\mathrm{CE}^\bullet(\mathcal{C}(F))_n$ le morphisme de complexes induced by motivic polylogarithms. Conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-019374 | MATHDB-351839

Goncharov's universality conjecture for motivic multiple polylogarithms

摘录 / EXCERPT: Pour un corps F, notons $\mathcal{C}(F)$ la cogèbre de Lie motivique et $\operatorname{Li}^{\mathcal{C}}_{n_1,\ldots,n_r}(x_1,\ldots,x_r)$ les polylogarithmes multiples motivique...

conjecture | open | MathDB 页面 | 原文线索

V2-019375 | MATHDB-351840

Goncharov's depth conjecture for motivic multiple polylogarithms

摘录 / EXCERPT: Pour un corps F, soit $D_k\mathcal{C}(F)$ la filtration croissante par la profondeur, définie par $D_{-1}\mathcal{C}(F)=0$, $D_0\mathcal{C}(F)=\mathcal{C}_1(F)$ et la condition s...

conjecture | open | MathDB 页面 | 原文线索

V2-019376 | MATHDB-351841

Goncharov-Rudenko conjecture on cluster polylogarithms

摘录 / EXCERPT: Pour $n\geq 3$, soit $\mathcal{A}_n$ l’associaèdre du polygone Π_n, $C_\bullet(\mathcal{A}_n)$ son complexe cellulaire, et $\mathcal{C}(F_n)$ la cogèbre de Lie motivique of the...

conjecture | open | MathDB 页面 | 原文线索

V2-019377 | MATHDB-351934

Griffiths-Harris conjecture on degrees of curves on very general hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^4$ be a very general hypersurface of degree $d\ge6$, and let $C\subset X$ be a curve. Griffiths-Harris conjecture. The degree of C is divisible by d. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-019378 | MATHDB-351935

Generalized Griffiths-Harris degree-divisibility conjecture for hypersurfaces

摘录 / EXCERPT: Let $n\ge1$ be an integer, let $X\subset\mathbb{P}^{n+1}$ be a very general hypersurface of degree $d\ge2n$, and let $Z\subset X$ be a positive-dimensional closed subvariety. Gener...

conjecture | open | MathDB 页面 | 原文线索

V2-019379 | MATHDB-351937

Dimensional reduction conjecture for -2-shifted symplectic derived schemes

摘录 / EXCERPT: Let $\boldsymbol{Y}$ be a -2-shifted symplectic derived scheme over $\operatorname{Spec}{\mathbb C}$, equipped with an orientation, and set $Y=t_0(\boldsymbol{Y})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019380 | MATHDB-351938

DT4 square-root Euler class conjecture for -2-shifted symplectic derived schemes

摘录 / EXCERPT: Let $\boldsymbol{Y}$ be an oriented -2-shifted symplectic derived scheme, set $Y=t_0(\boldsymbol{Y})$, and suppose the dimensional reduction conjecture holds. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019381 | MATHDB-351944

The socle conjecture for locally analytic representations

摘录 / EXCERPT: Let $v\in S_p$. For each v, let $\widetilde v$ be the chosen place above v, let $\rho_{\widetilde v}$ be the associated generic potentially crystalline Galois representation, l...

conjecture | open | MathDB 页面 | 原文线索

V2-019382 | MATHDB-351945

The local-to-global companion-point criterion

摘录 / EXCERPT: Let y be a point of the relevant patched, global, or local Bernstein variety, and let ϱ be its associated generic potentially crystalline Galois representation with disti...

conjecture | open | MathDB 页面 | 原文线索

V2-019383 | MATHDB-351947

The Hilbert-Samuel cycle formula for Bernstein eigenvarieties

摘录 / EXCERPT: Let L be the relevant local field, let $x=(\rho,\underline{x},\chi)$ be a point of $X_{\Omega,\mathbf h}(\overline\rho)$ with ρ de Rham and $(\underline{x},\chi)$ generic, a...

conjecture | open | MathDB 页面 | 原文线索

V2-019384 | MATHDB-351952

Gang-Kim-Yoon's vanishing conjecture for adjoint Reidemeister torsions

摘录 / EXCERPT: Let M be a compact 3-manifold with a torus boundary whose interior admits a hyperbolic structure. Suppose that the character variety of irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-019385 | MATHDB-351953

Gang-Kim-Yoon's norm-curve vanishing conjecture for adjoint Reidemeister torsions

摘录 / EXCERPT: Let M be an oriented compact hyperbolic 3-manifold with a torus boundary. A norm curve is a one-dimensional component of the character variety on which every trace function is...

conjecture | open | MathDB 页面 | 原文线索

V2-019386 | MATHDB-352032

Bloch's semi-regularity conjecture for embedded deformations

摘录 / EXCERPT: Bloch's semi-regularity conjecture. The semi-regularity map annihilates every obstruction to embedded deformations of Y in X. The conjecture concerns the obstruction theory of...

conjecture | open | MathDB 页面 | 原文线索

V2-019387 | MATHDB-352033

Semi-regularity conjecture via the universal obstruction space

摘录 / EXCERPT: Let X be a smooth projective variety over a field k of characteristic zero, and let $Y\subset X$ be a closed locally complete intersection subvariety of codimension p. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019388 | MATHDB-352035

The infinitesimal Hodge conjecture for algebraic K-theory classes

摘录 / EXCERPT: Let $\mathcal{X}/S$ be a smooth projective scheme, where $S=\operatorname{Spec}(k[[t]])$ for a field k of characteristic zero. Set $S_n=\operatorname{Spec}(k[t]/t^{n+1})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019389 | MATHDB-352077

Minimal-volume canonical Calabi-Yau hypersurface conjecture

摘录 / EXCERPT: Let $s_0=2$ and $s_m=s_{m-1}(s_{m-1}-1)+1$ for $m\geq 1$ be Sylvester's sequence. For a positive integer n, set ... A general hypersurface X of degree d in the weighted project...

conjecture | open | MathDB 页面 | 原文线索

V2-019390 | MATHDB-352078

Minimal anticanonical-volume terminal Fano conjecture

摘录 / EXCERPT: Let $s_0=2$ and $s_m=s_{m-1}(s_{m-1}-1)+1$ for $m\geq 1$ be Sylvester's sequence. For each integer $n\geq 2$, set ... A general hypersurface X of degree d in ... is a terminal Fano...

conjecture | open | MathDB 页面 | 原文线索

V2-019391 | MATHDB-352079

Largest-vanishing-range conjecture for canonical Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let $s_0=2$ and $s_m=s_{m-1}(s_{m-1}-1)+1$ for $m\geq 1$ be Sylvester's sequence. For an integer $n\geq 2$, set ... Let X be a general hypersurface of degree d in ... It is a quasi...

conjecture | open | MathDB 页面 | 原文线索

V2-019392 | MATHDB-352080

Minimal-volume canonical Calabi-Yau divisor conjecture

摘录 / EXCERPT: Let $s_0=2$ and $s_m=s_{m-1}(s_{m-1}-1)+1$ for $m\geq 1$ be Sylvester's sequence. For a positive integer n, let X be the n-fold given by the canonical Calabi-Yau hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-019393 | MATHDB-352081

Largest-vanishing-range conjecture for quasi-smooth Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let $s_0=2$ and $s_m=s_{m-1}(s_{m-1}-1)+1$ for $m\geq 1$ be Sylvester's sequence. For a positive integer n, let X be the n-fold from the quasi-smooth Calabi-Yau hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-019394 | MATHDB-352100

Neguț's stable-basis realization conjecture for the quantum affine quiver algebra

摘录 / EXCERPT: Let $\mathcal A$ be the algebra defined by the action-product map from the shuffle-algebra construction, and let $U_q(\widehat{\mathfrak g}^Q)$ be the Hopf algebra obtained from th...

conjecture | open | MathDB 页面 | 原文线索

V2-019395 | MATHDB-352102

Recognizability conjecture for canonical constant cycle polarizing divisors on K3 surfaces

摘录 / EXCERPT: Recognizability conjecture. If R is a constant cycle curve, then it is recognizable.

conjecture | open | MathDB 页面 | 原文线索

V2-019396 | MATHDB-352124

The Bruhat-cell boundary reconstruction conjecture

摘录 / EXCERPT: Let $\mathcal{F}\ell_n$ denote the flag variety, whose Bruhat decomposition has cells indexed by elements of the Weyl group. For a Bruhat cell, its boundary is the collection of ce...

conjecture | open | MathDB 页面 | 原文线索

V2-019397 | MATHDB-352125

Jelonek's conjecture on polynomial mappings with non-properness set of codimension at least two

摘录 / EXCERPT: Let $F=(f_1,\ldots,f_n):\mathbb R^n\to\mathbb R^n$ be a polynomial mapping. Write $J(F)$ for its Jacobian determinant and S_F for its non-properness set, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019398 | MATHDB-352127

The surjectivity conjecture for Jacobian operators of polynomial mappings

摘录 / EXCERPT: Surjectivity conjecture. If $J(F)$ is nowhere vanishing and $\operatorname{codim} S_F\geq2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-019399 | MATHDB-352129

The equivariant infinity-category construction conjecture

摘录 / EXCERPT: Let G be a smooth algebraic group and let X be a left G-variety over $\operatorname{Spec} K$. Let $\mathsf{Ner}(\operatorname{\mathbf{SfResl}}_G(X)^{\operatorname{op}})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-019400 | MATHDB-352140

Least-eigenvalue conjecture for set-wise 3-intersecting perfect matchings

摘录 / EXCERPT: Least-eigenvalue conjecture. The row sum and least eigenvalue of $\hat{B}_{3}(2k)$ are, respectively,

conjecture | open | MathDB 页面 | 原文线索

V2-019401 | MATHDB-352141

Pattern conjecture for eigenvalues in the perfect matchings association scheme

摘录 / EXCERPT: Eigenvalue pattern conjecture. The eigenvalues of the class $[2k]$ corresponding to these modules are, respectively,

conjecture | open | MathDB 页面 | 原文线索

V2-019402 | MATHDB-352191

Langlands-Rapoport-tau conjecture

摘录 / EXCERPT: Let $(G,X,p,\mathcal G)$ be an unramified Shimura datum. For each admissible morphism ϕ, let I_ϕ, $X(\phi)$, and $\mathcal H(\phi)$ be the associated objects, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019403 | MATHDB-352204

Mixed-structure compactification conjecture for rank-two Hitchin and maximal components

摘录 / EXCERPT: Let S be a surface and let G be a real Lie group of rank 2. A $(\overline{\mathfrak{a}^{+}},k)$-mixed structure on S is the structure whose dual $(\mathfrak{a},W)$-complex...

conjecture | open | MathDB 页面 | 原文线索

V2-019404 | MATHDB-352219

Movable Bend and Break conjecture

摘录 / EXCERPT: Movable Bend and Break conjecture. There exists a positive integer $c(n)$ such that, if M is a component of $$ generically parametrizing free rational curves of anticanonical deg...

conjecture | open | MathDB 页面 | 原文线索

V2-019405 | MATHDB-352228

The nef-and-effective canonical bundle conjecture for semi-orthogonal indecomposability

摘录 / EXCERPT: Nef-and-effective canonical bundle conjecture. If ω_X is nef and effective, then ${\mathbf{D}}^b(X)$ has no non-trivial semi-orthogonal decompositions.

conjecture | open | MathDB 页面 | 原文线索

V2-019406 | MATHDB-352235

Koolen's conjecture on Terwilliger algebras of finite connected simple graphs

摘录 / EXCERPT: Koolen's conjecture. For almost all finite connected simple graphs, the Terwilliger algebras coincide with the full matrix algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-019407 | MATHDB-352238

Artin-Hochster conjecture on the Cohen-Macaulayness of the commuting scheme

摘录 / EXCERPT: Artin-Hochster conjecture. The scheme $\mathfrak C_n$ is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-019408 | MATHDB-352244

Kudla-Rapoport arithmetic Siegel-Weil conjecture for nonsingular intersections

摘录 / EXCERPT: Let $\mathcal{X}$ be a suitable regular integral model of the relevant orthogonal or unitary Shimura variety, let $n=\dim\mathcal{X}$, and let $\mathcal{Z}(m_i)$ be integral models...

conjecture | open | MathDB 页面 | 原文线索

V2-019409 | MATHDB-352246

The partial Dyck path Tor vanishing conjecture

摘录 / EXCERPT: Let U_k be the open locus defined in the paper, and let $U_{\pi,l}$ be the corresponding smooth open locus associated with a partial Dyck path (π,l). The partial Dyck path To...

conjecture | open | MathDB 页面 | 原文线索

V2-019410 | MATHDB-352250

Zhang's finiteness conjecture for small specializations of sections

摘录 / EXCERPT: Let $A\to T^0$ be a non-isotrivial family of abelian varieties over a number field K, with generic fiber A_{η} simple of dimension at least two. A section $P:T^0\to A$ is no...

conjecture | open | MathDB 页面 | 原文线索

V2-019411 | MATHDB-352265

Motion conjecture for primitive coherent configurations

摘录 / EXCERPT: Motion conjecture for primitive coherent configurations. If $\mathfrak X$ is not a Cameron scheme, then there is a positive constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-019412 | MATHDB-352276

Strongly exact mirror identification conjecture

摘录 / EXCERPT: Let $\mathcal{R}$ be a simple eigenray diagram, let $M_{\mathcal{R}}$ be the associated symplectic manifold, let $\mathcal{Y}_{\mathcal{R}}$ be the analytification of the scheme co...

conjecture | open | MathDB 页面 | 原文线索

V2-019413 | MATHDB-352292

Touzet's conjecture on regular foliations of rationally connected manifolds

摘录 / EXCERPT: Let X be a projective rationally connected manifold and let $\mathcal{F}$ be a regular foliation on X. Touzet's conjecture. The foliation $\mathcal{F}$ is induced by a smooth m...

conjecture | open | MathDB 页面 | 原文线索

V2-019414 | MATHDB-352311

Integral cohomology and non-isomorphism conjecture for generalized roofs

摘录 / EXCERPT: Integral cohomology and non-isomorphism conjecture. One has an isomorphism of integral cohomology groups

conjecture | open | MathDB 页面 | 原文线索

V2-019415 | MATHDB-352351

Comparison conjecture for the weight and stable rank filtrations

摘录 / EXCERPT: Weight-stable-rank comparison conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019416 | MATHDB-352352

Rational comparison conjecture for motivic and stable rank spectral sequences

摘录 / EXCERPT: Motivic-stable-rank comparison conjecture. The motivic spectral sequence and the stable rank spectral sequence are rationally isomorphic starting from their E^2-terms:

conjecture | open | MathDB 页面 | 原文线索

V2-019417 | MATHDB-352353

Ein-Erman-Lazarsfeld's normal distribution conjecture for syzygy Betti numbers

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, let B be a line bundle, and for sufficiently large d set $L_d:=\mathscr{O}_X(dA+P)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019418 | MATHDB-352362

The R = T conjecture for Klingen-ordinary deformations

摘录 / EXCERPT: Let $\bar{\rho}$ be the residual Galois representation attached to a Klingen-ordinary automorphic representation, let $\mathbf{T}^{\mathrm{wKl}}_{\bar{\rho}}$ be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-019419 | MATHDB-352368

Generalization conjecture for admissible sets and compatible orders of smooth elements

摘录 / EXCERPT: Generalization conjecture. The set A is admissible, a compatible order on $A_{\mathcal{T}}$ exists, and multiplying the elements of $A_{\mathcal{T}}$ in any compatible order give...

conjecture | open | MathDB 页面 | 原文线索

V2-019420 | MATHDB-352406

Conjecture on the defining ideal of orthogonal tensor trains

摘录 / EXCERPT: Defining-ideal conjecture. The Zariski closure of $SOT_{2,n}$ in $\mathbb{C}^{n^4}$ is cut out by the vanishing of $\mathcal P_n$, $\mathcal Q_n$, and h_n. The ideal defined by t...

conjecture | open | MathDB 页面 | 原文线索

V2-019421 | MATHDB-352416

Martin's scaffold duality conjecture for association schemes

摘录 / EXCERPT: A standard planar scaffold is a weakly connected planar scaffold whose edge weights are among the adjacency matrices $A_0,\dots,A_c$ of an association scheme. For sequences of such...

conjecture | open | MathDB 页面 | 原文线索

V2-019422 | MATHDB-352418

Varshavsky-author conjecture on the cocenter summand and Langlands-dual data

摘录 / EXCERPT: Let Z_e^{red} be the quotient of Z_e by its unipotent radical. Let ${\bf Z}_e\subset (Z_e^{red})^2$ be the algebraic variety of pairs of commuting elements in Z_e^{red}. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019423 | MATHDB-352425

The dR-absolute specialness conjecture

摘录 / EXCERPT: Let $(\mathbb{V}^{\sigma})_{\sigma}$ be an absolute variation of Hodge structure on a smooth irreducible quasi-projective complex algebraic variety S. For a closed irreducible su...

conjecture | open | MathDB 页面 | 原文线索

V2-019424 | MATHDB-352426

The Deligne absolute Hodge cycle conjecture

摘录 / EXCERPT: Let $Z\subset S$ be a closed irreducible subvariety of a variation of Hodge structure. Write G_Z for its generic Mumford-Tate group and G_Z^{AH} for its generic dR-absolute Mum...

conjecture | open | MathDB 页面 | 原文线索

V2-019425 | MATHDB-352427

The bialgebraic characterization of dR-absolutely special subvarieties

摘录 / EXCERPT: Let S be defined over $\bar{\mathbb{Q}}$ and let $(\mathbb{V}^{\sigma})_{\sigma}$ be a $\bar{\mathbb{Q}}$-absolute variation on S. A closed irreducible subvariety is...

conjecture | open | MathDB 页面 | 原文线索

V2-019426 | MATHDB-352433

Aprodu-Farkas strong maximal rank conjecture

摘录 / EXCERPT: Aprodu-Farkas strong maximal rank conjecture. The determinantal variety

conjecture | open | MathDB 页面 | 原文线索

V2-019427 | MATHDB-352436

Semiorthogonal decomposition and Hochschild homology conjecture for Fano varieties

摘录 / EXCERPT: Let X be a complex Fano variety such that the condition ... bm{kappa}X(QS(X)) cap ({mathbb{A}}^1 setminus {0}) = {z1,dots,zN} subset {mathbb{A}}^1 setminus {0} ... holds. Semiorthogo...

conjecture | open | MathDB 页面 | 原文线索

V2-019428 | MATHDB-352438

Coadjoint varieties residual-category conjecture

摘录 / EXCERPT: Let X be the coadjoint variety of a simple complex algebraic group G. Let $D_s(G)$ denote the short-roots subdiagram of the Dynkin diagram of G, which is the entire Dynkin di...

conjecture | open | MathDB 页面 | 原文线索

V2-019429 | MATHDB-352463

The higher-dimensional continuous Fermi variety irreducibility conjecture

摘录 / EXCERPT: Let $d\geq 3$, let $V_c:\mathbb{T}^d\to\mathbb{C}^d$ be a periodic function, and let $-\Delta_c+V_c$ be the corresponding continuous periodic Schrödinger operator, where Δ_c...

conjecture | open | MathDB 页面 | 原文线索

V2-019430 | MATHDB-352466

The absolute Mumford-Tate conjecture for smooth projective varieties

摘录 / EXCERPT: Let X be a smooth projective variety over K, let G_X^{AH} denote its absolute Mumford-Tate group, and let $G_{\ell,X}$ denote its ℓ-adic algebraic Galois group. Absolute...

conjecture | open | MathDB 页面 | 原文线索

V2-019431 | MATHDB-352467

The absolute Hodge conjecture for generic Mumford-Tate groups

摘录 / EXCERPT: Let S be the base of the family considered in the paper, and for a subvariety $Z\subset S$ let G_Z and G_Z^{AH} be its generic Mumford-Tate and generic absolute Mumford-Tate...

conjecture | open | MathDB 页面 | 原文线索

V2-019432 | MATHDB-352468

Special subvarieties are absolutely special

摘录 / EXCERPT: Let S be the base of the family considered in the paper. A special subvariety is a closed irreducible complex subvariety maximal among those having a fixed generic Mumford-Tate g...

conjecture | open | MathDB 页面 | 原文线索

V2-019433 | MATHDB-352469

\ell-Galois special subvarieties are absolutely special

摘录 / EXCERPT: Let S be the base of the family considered in the paper. An ℓ-Galois special subvariety is a closed irreducible complex subvariety maximal among those having a fixed generic...

conjecture | open | MathDB 页面 | 原文线索

V2-019434 | MATHDB-352479

Irreducibility conjecture for the exposed-point hypersurface of a generic discotope

摘录 / EXCERPT: Let $\mathcal{D}$ be a generic discotope of type $(0,N_2,\dots,N_d)$, and let $\mathcal{S}$ denote the hypersurface associated with the exposed points of $\mathcal{D}$. Irreducibil...

conjecture | open | MathDB 页面 | 原文线索

V2-019435 | MATHDB-352488

The XOP2 characterization of binary disc2,3-error

摘录 / EXCERPT: Let $\mathcal{L}={R(x,y,z)}$ be the language of ternary 3-graph relations, and let XOP_2 be a proposed definition given by an infinite scheme of existential sentences. X...

conjecture | open | MathDB 页面 | 原文线索

V2-019436 | MATHDB-352507

Surjectivity conjecture for the Lawson homology cycle class map on threefolds

摘录 / EXCERPT: Let X be a complex projective variety of dimension 3, and let ... be the Friedlander-Mazur cycle class map, with rational coefficients. Surjectivity conjecture. The map...

conjecture | open | MathDB 页面 | 原文线索

V2-019437 | MATHDB-352515

The relative local Langlands conjecture for spherical varieties

摘录 / EXCERPT: Let X be a spherical variety over a local field F, let $\bar X$ denote X together with its pure inner forms, and let $ u_{{^LG_X}}$ be the natural measure on the set of tempe...

conjecture | open | MathDB 页面 | 原文线索

V2-019438 | MATHDB-352516

The global relative trace formula conjecture for spherical varieties

摘录 / EXCERPT: Let $\mathfrak X$ be the global spherical period space associated with X, let $\mathcal L_k$ be the global Langlands group, and let ${^LG_X}\to {^LG}$ be the dual-group map. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019439 | MATHDB-352517

The local Plancherel density conjecture for the basic function

摘录 / EXCERPT: Let F be a split non-Archimedean local field with ring of integers $\mathfrak o$, and let X be an affine spherical variety with a good model over $\mathfrak o$, satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-019440 | MATHDB-352518

The derived Satake equivalence conjecture for spherical varieties

摘录 / EXCERPT: Let $F=\mathbb F_q((t))$, let L and L^+ denote formal loop and arc spaces, and let X and G be defined over $\mathbb F_q$, with X smooth. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019441 | MATHDB-352520

Suslin's factorial conjecture for the sheaf $\mathbf{S}_n$

摘录 / EXCERPT: For an integer $n\geq 2$, let $\mathbf{S}_n$ be the strictly $\mathbb{A}^1$-invariant sheaf occurring in the exact sequence for $\bm{\pi}^{\mathbb{A}^1}_n(\mathrm{BGL}_n)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019442 | MATHDB-352521

Vanishing conjecture for the unstable sheaf $\mathbf{U}_{n+1}$

摘录 / EXCERPT: Vanishing conjecture. For $n\geq 4$, the sheaf $\mathbf{U}_{n+1}$ is zero. The preceding theorem establishes the corresponding vanishing at $n=3$; the supplied text does not indica...

conjecture | open | MathDB 页面 | 原文线索

V2-019443 | MATHDB-352522

Suslin's cancellation conjecture in corank one

摘录 / EXCERPT: Let k be an algebraically closed field, and let X be a smooth affine k-scheme of dimension $d\geq2$. Suslin's cancellation conjecture. If $\mathcal{E}$ and $\mathcal{E}'$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-019444 | MATHDB-352523

Algebraic realization conjecture for vector bundles on cellular complex varieties

摘录 / EXCERPT: Let X be a smooth cellular $\mathbb{C}$-variety, let Gr_r be the Grassmannian classifying rank-r algebraic bundles, and let $\mathrm{Vect}^{top}(X)$ denote topological vector...

conjecture | open | MathDB 页面 | 原文线索

V2-019445 | MATHDB-352527

Goodman's conjecture on the normal bundle of an affine divisor complement

摘录 / EXCERPT: Let X be a compact algebraic manifold and let $Y\subset X$ be a prime divisor. The normal bundle $N_{Y/X}$ is the bundle of normal directions to Y in X. Goodman's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-019446 | MATHDB-352529

The Fano criterion for affine canonical extensions with nef tangent bundle

摘录 / EXCERPT: Let M be a compact Kähler manifold with nef tangent bundle T_M, and let Z_M be its canonical extension associated with a Kähler class. Fano criterion conjecture. The canonica...

conjecture | open | MathDB 页面 | 原文线索

V2-019447 | MATHDB-352575

Geiss-Schröer rigidity conjecture for irreducibility of induced representations

摘录 / EXCERPT: Let $C_i\in\operatorname{Comp}(\bm{d}_i)$ for $i=1,2$, and suppose that at least one of C_1,C_2 is rigid, meaning that it contains an open orbit. Let $\pi(C_i)$ be the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-019448 | MATHDB-352576

Mellit's conormal-bundle and Schubert-smoothness conjecture

摘录 / EXCERPT: Let $x,w\in S_n$ with $Y_w\subseteq X_x$, equivalently $w\leq x$ in the Bruhat order, where Y_w is the Bruhat cell and X_x its Schubert-variety closure. Suppose that X_x is s...

conjecture | open | MathDB 页面 | 原文线索

V2-019449 | MATHDB-352577

Lapid-Mínguez conjecture on Jordan-Hölder constituents of parabolic induction

摘录 / EXCERPT: Let $C_i\in\operatorname{Comp}(\bm{d}_i)$ for $i=1,\dots,k$, let $V=V^1\oplus\cdots\oplus V^k$, and let $C\in\operatorname{Comp}(V)$. Let $\pi(C)$ denote the irreducible representa...

conjecture | open | MathDB 页面 | 原文线索

V2-019450 | MATHDB-352580

Benedetti-Shiota conjecture on real algebraic links

摘录 / EXCERPT: Let $f:\mathbb{R}^4\to\mathbb{R}^2$ be a polynomial map with an isolated singularity at the origin, meaning that $f(0)=(0,0)$, its first derivatives vanish at the origin, and its J...

conjecture | open | MathDB 页面 | 原文线索

V2-019451 | MATHDB-352594

Uniform multiplicative lower bound for degrees of curves in very general complete intersections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+f}$ be a very general complete intersection variety of dimension n and type $(d_1,d_2,\ldots,d_f)$. Let D be a positive constant depending only on...

conjecture | open | MathDB 页面 | 原文线索

V2-019452 | MATHDB-352615

Tits-type alternative for groups generated by additive subgroups of affine varieties

摘录 / EXCERPT: Let X be an affine algebraic variety defined over an algebraically closed field, and let G be a subgroup of $\operatorname{Aut}(X)$ generated by a finite collection of...

conjecture | open | MathDB 页面 | 原文线索

V2-019453 | MATHDB-352617

Real eigenvectors of regular simplex tensors

摘录 / EXCERPT: Let $\mathcal{T}$ be a regular simplex tensor generated by a regular simplex frame. An eigenvector is a nonzero vector $\mathbf{x}$ satisfying ... for some scalar λ, with eig...

conjecture | open | MathDB 页面 | 原文线索

V2-019454 | MATHDB-352638

The higher complex structures conjecture for the Hitchin component

摘录 / EXCERPT: Let Σ be a closed oriented surface, let $\mathcal{T}^n$ be the moduli space of higher complex structures on Σ, and let ... be the formal map whose image is an open subs...

conjecture | open | MathDB 页面 | 原文线索

V2-019455 | MATHDB-352651

Mumford-Tate conjecture for étale cohomology

摘录 / EXCERPT: Let X be a smooth projective variety defined over a number field k, let ℓ be a prime, and set ... Let G_ℓ be the generic fiber of the Zariski closure of the image of th...

conjecture | open | MathDB 页面 | 原文线索

V2-019456 | MATHDB-352691

Beilinson's arithmetic Hodge index conjecture for intermediate Jacobians

摘录 / EXCERPT: Let X be the fourfold considered in the paper, let $J^2(X)$ be its intermediate Jacobian, and let ${\mathsf L}:J^2(X)\to J^3(X)$ be induced by intersection with an ample cycle. D...

conjecture | open | MathDB 页面 | 原文线索

V2-019457 | MATHDB-352700

The additivity conjecture for Kodaira dimension in smooth families

摘录 / EXCERPT: Let $f\colon X\to Y$ be a smooth algebraic fiber space between smooth projective varieties, and let F be its general fiber. The additivity conjecture. One should have ... This foll...

conjecture | open | MathDB 页面 | 原文线索

V2-019458 | MATHDB-352701

The additivity conjecture for smooth projective families over quasi-projective bases

摘录 / EXCERPT: Let $f\colon U\to V$ be a smooth projective algebraic fiber space between smooth quasi-projective varieties, and let F be its general fiber. The quasi-projective additivity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-019459 | MATHDB-352702

The logarithmic additivity conjecture for algebraic fiber spaces

摘录 / EXCERPT: Let $f\colon X\to Y$ be an algebraic fiber space between smooth projective varieties. Let E be a simple normal crossing divisor on X and D a simple normal crossing divisor on...

conjecture | open | MathDB 页面 | 原文线索

V2-019460 | MATHDB-352707

Oblomkov-Pandharipande multiple cover conjecture for Hilbert schemes of K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $S^{[n]}$ be its Hilbert scheme of points, and let $\beta eq 0$ be a curve class on S. For classes $\beta+mA$ on $S^{[n]}$, classes...

conjecture | open | MathDB 页面 | 原文线索

V2-019461 | MATHDB-352714

Igusa's conjecture on uniform exponential-sum bounds

摘录 / EXCERPT: Let $f\in\operatorname{O}_K[x_1,\ldots,x_n]$ be a nonconstant polynomial, let Z be a subscheme of $\mathbb A^n_{\operatorname{O}_K}$, and let L be a non-Archimedean local field...

conjecture | open | MathDB 页面 | 原文线索

V2-019462 | MATHDB-352717

The ergodicity criterion for nilpotent character varieties

摘录 / EXCERPT: Let Γ be a finitely generated nilpotent group, and let G be a connected compact Lie group. Write $\mathsf{Out}(\Gamma)$ for the outer automorphism group of Γ, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-019463 | MATHDB-352737

The large Galois orbits conjecture for endomorphism loci in principally polarised abelian varieties

摘录 / EXCERPT: Large Galois orbits conjecture. There exist positive constants C_1 and C_2, depending only on g, L, and C, such that for every $s\in C\cap\Sigma$,

conjecture | open | MathDB 页面 | 原文线索

V2-019464 | MATHDB-352738

The Zilber--Pink conjecture for Shimura varieties

摘录 / EXCERPT: Zilber--Pink conjecture. If V is an irreducible Hodge generic subvariety of S, then the intersection of V with the special subvarieties of S having codimension greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-019465 | MATHDB-352739

The large Galois orbits conjecture for endomorphism-generic special subvarieties

摘录 / EXCERPT: Large Galois orbits conjecture. There exist positive constants C_1 and C_2 such that, for every $s\in C\cap\Sigma$,

conjecture | open | MathDB 页面 | 原文线索

V2-019466 | MATHDB-352748

Yong's Gröbner basis conjecture for Schubert patch ideals

摘录 / EXCERPT: Let $G=GL_n(\mathbb{K})$ be the general linear group, let B_{-} and B_{+} be the lower and upper triangular subgroups, and let $B_{-}\backslash G$ be the complete flag variety....

conjecture | open | MathDB 页面 | 原文线索

V2-019467 | MATHDB-352754

Ueno's Conjecture K strengthening for algebraic fiber spaces over abelian varieties

摘录 / EXCERPT: Ueno's Conjecture K strengthening. There exists an isogeny $A' \to A$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019468 | MATHDB-352771

The Levi-spherical Schubert variety classification conjecture

摘录 / EXCERPT: Let W be a Coxeter group with simple generators, let $w\in W$, let $J(w)$ denote the support of w, and fix $I\subseteq J(w)$. Let X_w be the Schubert variety indexed by w,...

conjecture | open | MathDB 页面 | 原文线索

V2-019469 | MATHDB-352773

Toric-stratum conjecture for hypercube decompositions

摘录 / EXCERPT: Let $S_{x,y}$ be the standard slice through xB/B inside the Schubert variety associated with y, embedded in its tangent representation. Fix a hypercube decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-019470 | MATHDB-352788

The wall-crossing conjecture for iterated Hilbert schemes

摘录 / EXCERPT: Wall-crossing conjecture. There is a path $\gamma\colon[0,1]\to\Theta(Q_G,\ub{d})$ passing through walls $\mathfrak{w}_1,\dots,\mathfrak{w}_m$ satisfying the conditions in the wall...

conjecture | open | MathDB 页面 | 原文线索

V2-019471 | MATHDB-352813

Positivity of Chern numbers for irreducible hyper-Kähler manifolds

摘录 / EXCERPT: The known irreducible hyper-Kähler manifolds include Hilbert schemes of points on K3 surfaces, generalized Kummer varieties, and two further examples in dimensions 6 and 10. A...

conjecture | open | MathDB 页面 | 原文线索

V2-019472 | MATHDB-352841

The degree bound for components of theta divisors from reducible curves

摘录 / EXCERPT: Let C be a reducible curve of arithmetic genus g, and let the associated theta divisor be constructed as in the paper. Consider an irreducible component of this theta divisor....

conjecture | open | MathDB 页面 | 原文线索

V2-019473 | MATHDB-352873

The Gibbs stability conjecture for Kähler-Einstein Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold. A Kähler-Einstein metric on X is denoted by ω_{KE}, and let δ_N be the empirical measure of the corresponding canonical point process. T...

conjecture | open | MathDB 页面 | 原文线索

V2-019474 | MATHDB-352874

The K-stability-Gibbs stability equivalence conjecture

摘录 / EXCERPT: Let X be a Fano manifold. The K-stability-Gibbs stability conjecture. X is (uniformly) K-stable if and only if X is (uniformly) Gibbs stable. Here uniform Gibbs stability mea...

conjecture | open | MathDB 页面 | 原文线索

V2-019475 | MATHDB-352875

The arithmetic intersection conjecture for Gibbs partition functions

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{L})$ be an arithmetic variety as above, and let X be its corresponding Fano manifold. Assume that X admits a unique Kähler-Einstein metric, with volu...

conjecture | open | MathDB 页面 | 原文线索

V2-019476 | MATHDB-352920

Stable motivic connectivity and negative homotopy sheaves of the sphere spectrum

摘录 / EXCERPT: Particular case of the framed-motive conjecture. For $i<j$, the stable motivic homotopy group sheaves are trivial, and

conjecture | open | MathDB 页面 | 原文线索

V2-019477 | MATHDB-352921

The conjecture that the quasivariety of odd antilattices is not a variety

摘录 / EXCERPT: Odd-antlattice quasivariety conjecture. The quasivariety of odd antilattices is proper; that is, it is not a variety.

conjecture | open | MathDB 页面 | 原文线索

V2-019478 | MATHDB-352931

The geometric description of the Braverman-Kazhdan-Polishchuk incidence scheme

摘录 / EXCERPT: Let $\mathcal C$ be a smooth complete irreducible curve of genus $g\geq 2$, and let $\mathcal E$ be a stable bundle on $\mathcal C$ of degree $2g-1$. Define $F_{\mathcal E}$ to be...

conjecture | open | MathDB 页面 | 原文线索

V2-019479 | MATHDB-352932

The Hecke eigenbasis conjecture for automorphic half-forms

摘录 / EXCERPT: Let F be a non-archimedean or archimedean local field, let $\mathcal C$ be a smooth complete curve over F, and let Bun be the moduli stack of principal G-bun...

conjecture | open | MathDB 页面 | 原文线索

V2-019480 | MATHDB-352935

The Schwartz-space and multiplicity-one conjecture for automorphic half-forms

摘录 / EXCERPT: Let $\mathcal C$ be a smooth projective curve over $\mathbb C$, let G be a split connected semi-simple group, and let Bun be the moduli stack of principal G-bu...

conjecture | open | MathDB 页面 | 原文线索

V2-019481 | MATHDB-352938

The L-system parametrization conjecture for real curves

摘录 / EXCERPT: Let G be a connected complex semi-simple group with real structure, let $\mathcal C$ be a compact complex Riemann surface with antiholomorphic involution τ and no real point...

conjecture | open | MathDB 页面 | 原文线索

V2-019482 | MATHDB-352940

The Torelli conjecture for mirror Landau-Ginzburg models

摘录 / EXCERPT: Let X_1 and X_2 be Fano varieties with mirror Landau-Ginzburg models $(X_1^\vee,W_1)$ and $(X_2^\vee,W_2)$. Write $\pi_{W_i}(t)$ for the classical period of the model, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019483 | MATHDB-352962

The irreducibility conjecture for universal Severi varieties of K3 surfaces

摘录 / EXCERPT: Universal Severi irreducibility conjecture. For every $0\leq \delta\leq g$, the universal Severi variety $\mathcal{V}_{g,\delta}$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-019484 | MATHDB-352990

Unique real critical point conjecture for the aED problem

摘录 / EXCERPT: Let U be a nonnegative tensor whose entries sum to 1, and let $\mathcal{M}_\Sigma$ be a marginal independence model contained in the probability simplex Δ_{D-1}. Let the...

conjecture | open | MathDB 页面 | 原文线索

V2-019485 | MATHDB-352991

Refined Kuznetsov conjecture for geometric Calabi-Yau components

摘录 / EXCERPT: Refined Kuznetsov conjecture. One can find a Calabi-Yau variety Y' such that X is a certain blow-up of Y', and

conjecture | open | MathDB 页面 | 原文线索

V2-019486 | MATHDB-353038

Non-vanishing conjecture for strictly nef anti-canonical divisors

摘录 / EXCERPT: Non-vanishing conjecture for strictly nef anti-canonical divisors. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-019487 | MATHDB-353042

Kuznetsov-Smirnov conjecture for coadjoint varieties

摘录 / EXCERPT: Let X be the coadjoint variety of a simple algebraic group G over an algebraically closed field of characteristic zero. Let $\operatorname{Aut}(X)$ denote its automorp...

conjecture | open | MathDB 页面 | 原文线索

V2-019488 | MATHDB-353045

Pro-étale Poincaré lemma for topological derived de Rham complexes

摘录 / EXCERPT: Pro-étale Poincaré conjecture. The construction in this section should extend to the pro-étale sites, giving two strictly exact long exact sequences of topological derived de Rham...

conjecture | open | MathDB 页面 | 原文线索

V2-019489 | MATHDB-353046

Logarithmic Kummer pro-étale Poincaré lemma

摘录 / EXCERPT: Logarithmic Kummer Poincaré conjecture. The logarithmic versions of the strictly exact long exact sequences in the Poincaré lemma for the rigid-analytic situation should hold for t...

conjecture | open | MathDB 页面 | 原文线索

V2-019490 | MATHDB-353047

Kummer pro-étale extension for logarithmic topological constructions

摘录 / EXCERPT: Kummer pro-étale extension conjecture. The construction developed in the section should extend to suitable Kummer pro-étale sites with logarithmic perfectoid subdomains forming a b...

conjecture | open | MathDB 页面 | 原文线索

V2-019491 | MATHDB-353048

Logarithmic Kummer pro-étale Poincaré sequences for differential crystals

摘录 / EXCERPT: Let X be a smooth space in the setting of the paper, and consider the projective maps ... Let M be a corresponding Z-projective differential crystal spectrum. Logarithmic cryst...

conjecture | open | MathDB 页面 | 原文线索

V2-019492 | MATHDB-353054

The homological standard conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over an algebraically closed field k, and let ℓ be a prime distinct from the characteristic of k. Numerical equival...

conjecture | open | MathDB 页面 | 原文线索

V2-019493 | MATHDB-353056

Higher-rank Landau-Ginzburg compactification conjecture

摘录 / EXCERPT: Let X be a quasi-Fano variety with a smooth anticanonical divisor D, and let D_1+D_2 be a simple normal crossings anticanonical divisor of X. Let $(X^\beth,h)$ denote the r...

conjecture | open | MathDB 页面 | 原文线索

V2-019494 | MATHDB-353057

Maximal-unipotent-monodromy fibration conjecture for Calabi-Yau mirrors

摘录 / EXCERPT: Let X be a Calabi-Yau variety, or let $X\setminus D$ be a log Calabi-Yau variety. Suppose that X (or $X\setminus D$) admits a semi-stable degeneration connected to a point of m...

conjecture | open | MathDB 页面 | 原文线索

V2-019495 | MATHDB-353058

Higher-rank Landau-Ginzburg gluing conjecture for simple normal-crossings pairs

摘录 / EXCERPT: Let X be a quasi-Fano variety with a simple normal crossings anticanonical divisor D, and suppose that X admits a semi-stable degeneration into $X_1\cup_{D_0}X_2$. The mirror...

conjecture | open | MathDB 页面 | 原文线索

V2-019496 | MATHDB-353062

Morel's -invariance conjecture for -connected components

摘录 / EXCERPT: Let k be a field, let $\mathcal{X}$ be a space, and write $\pi_0^{\mathbb{A}^1}(\mathcal{X})$ for its sheaf of $\mathbb{A}^1$-connected components. Morel's conjecture. The sheaf...

conjecture | open | MathDB 页面 | 原文线索

V2-019497 | MATHDB-353095

Deligne's absolute Hodge-tensor conjecture

摘录 / EXCERPT: Let $({\mathbb V}^{\sigma})_{\sigma}$ be a de Rham motivic variation of Hodge structure on a smooth connected complex quasi-projective variety S. Let ${\mathbf G}$ be the group d...

conjecture | open | MathDB 页面 | 原文线索

V2-019498 | MATHDB-353129

Tate's strong conjecture on poles of zeta functions

摘录 / EXCERPT: Let X be a smooth projective variety over a finitely generated field F, let ${\mathrm{Ch}}^r_{\hom}(X)$ be the quotient of the Chow group of codimension-r cycles by ℓ-ad...

conjecture | open | MathDB 页面 | 原文线索

V2-019499 | MATHDB-353134

The toric Landau-Ginzburg model existence conjecture

摘录 / EXCERPT: Let X be a smooth Fano variety. A toric Landau-Ginzburg model existence conjecture asserts that X has a Laurent polynomial satisfying the period, Calabi-Yau, and toric conditio...

conjecture | open | MathDB 页面 | 原文线索

V2-019500 | MATHDB-353135

The mutation connectivity conjecture for mirror partners

摘录 / EXCERPT: Let X be a Fano variety, and let f and g be Laurent-polynomial mirror partners to X. A mutation connectivity conjecture asserts that f and g are connected by a finite s...

conjecture | open | MathDB 页面 | 原文线索

V2-019501 | MATHDB-353137

The primitive Hodge number conjecture for toric Landau-Ginzburg models

摘录 / EXCERPT: Let X be a smooth Fano variety of dimension n, and let Y be a Calabi-Yau compactification of its toric Landau-Ginzburg model. Let k_Y be the number of irreducible component...

conjecture | open | MathDB 页面 | 原文线索

V2-019502 | MATHDB-353140

Parity-homogeneity conjecture for relations among alternating multiple t-values

摘录 / EXCERPT: Parity-homogeneity conjecture. Every relation between alternating multiple t-values is homogeneous in r modulo 2; equivalently, terms involving an even number of negative sig...

conjecture | open | MathDB 页面 | 原文线索

V2-019503 | MATHDB-353187

Pseudo-rationality conjecture for p-adic local models

摘录 / EXCERPT: Let $\calM_{\calG,\mu}^{\on{sch}}$ be the scheme-theoretic local model attached to a parahoric group scheme $\calG$ and a minuscule cocharacter μ. Pseudo-rationality conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-019504 | MATHDB-353224

Isotropic Chow groups conjecture over flexible fields

摘录 / EXCERPT: Let k be a field, and let $\operatorname{Ch}^*_{k/k}$ denote the isotropic Chow groups with $\mathbb{F}_p$-coefficients, obtained from Chow groups by quotienting by classes comin...

conjecture | open | MathDB 页面 | 原文线索

V2-019505 | MATHDB-353225

Isotropic Chow groups conjecture with finite coefficients

摘录 / EXCERPT: Let k be a flexible field and let X be a smooth projective variety over k. Write $\operatorname{CH}_{k/k}(X,\mathbb{Z}/p^m)$ for the isotropic Chow group with...

conjecture | open | MathDB 页面 | 原文线索

V2-019506 | MATHDB-353240

The modified generalized Bogomolov-Gieseker conjecture

摘录 / EXCERPT: Let (X,H) be a smooth complex projective polarized variety. Let $\gamma=\gamma_{X,H}$ be a class as in the source's definition of γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019507 | MATHDB-353247

Birational categorical Torelli conjecture for Gushel-Mukai varieties

摘录 / EXCERPT: Let X_1 and X_2 be Gushel-Mukai varieties of the same dimension, and let $\mathcal{K}\!u(X_i)$ denote their Kuznetsov components. Birational categorical Torelli conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-019508 | MATHDB-353265

Uniform Sobolev inequality for blowups of compact Kähler varieties

摘录 / EXCERPT: Uniform Sobolev inequality conjecture. The metrics (Y,ω_ϵ) have a uniform Sobolev constant for $0<\epsilon\leqslant1$: there exists a constant C_S, independent of...

conjecture | open | MathDB 页面 | 原文线索

V2-019509 | MATHDB-353269

The fundamental conjecture for overconvergent isocrystals on partial Frobenius stacks

摘录 / EXCERPT: Let I be an index set, let X_i be the varieties equipped with partial Frobenius actions, and let X/Φ be the associated partial Frobenius stack. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019510 | MATHDB-353271

Weil's product formula for the moduli stack of G-bundles

摘录 / EXCERPT: Let X be a smooth proper curve over $\mathbb{F}_p$, let $G\to X$ be a smooth affine group scheme with semisimple simply connected generic fiber and connected fibers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019511 | MATHDB-353276

Expected codimension and irreducibility of Spohn CI varieties for Bayesian networks

摘录 / EXCERPT: Expected-codimension and irreducibility conjecture. The variety $\mathcal{V}_{X,\mathcal{C}}$ has expected codimension n inside $\mathcal{M}_\mathcal{C}$. It is positive-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-019512 | MATHDB-353289

Cohomological dimension conjecture for the Hodge-Tate stack of a regular ring

摘录 / EXCERPT: Let R be a p-complete noetherian regular local ring with perfect residue field. Let G be the group scheme identified in the source with the Hodge-Tate stack presentation...

conjecture | open | MathDB 页面 | 原文线索

V2-019513 | MATHDB-353290

Regularity criterion via the Hodge-Tate stack

摘录 / EXCERPT: Let X be a noetherian excellent p-adic formal scheme, and write $\operatorname{WCart}_X^{\mathrm{HT}}$ for its derived Hodge-Tate stack, namely the Hodge-Tate locus in...

conjecture | open | MathDB 页面 | 原文线索

V2-019514 | MATHDB-353346

Letellier's twisted mixed-Hodge polynomial conjecture for Weyl group actions

摘录 / EXCERPT: Let $\mathcal{C}_{\boldsymbol{\mu},\boldsymbol{\sigma}}$ be a generic k-uple of conjugacy classes, let...

conjecture | open | MathDB 页面 | 原文线索

V2-019515 | MATHDB-353360

Scholl's vanishing and non-negative weight conjecture for mixed motives

摘录 / EXCERPT: Scholl's conjecture. We expect that:

conjecture | open | MathDB 页面 | 原文线索

V2-019516 | MATHDB-353361

Scholl's integral-part conjecture for motivic extensions

摘录 / EXCERPT: Integral-part conjecture. For any prime ℓ not lying below $\mathfrak p$,

conjecture | open | MathDB 页面 | 原文线索

V2-019517 | MATHDB-353362

Regulated local good-reduction conjecture for A-motives

摘录 / EXCERPT: Regulated local good-reduction conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019518 | MATHDB-353364

Gopakumar-Vafa integrality conjecture for Gromov-Witten invariants

摘录 / EXCERPT: Let X be the smooth projective variety under consideration, let $N^{\operatorname{GW}}_{g,\beta}(X)$ denote its Gromov-Witten invariants, let u be a formal variable, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019519 | MATHDB-353365

Toda's product formula conjecture for stable pair invariants

摘录 / EXCERPT: Let X be the smooth projective variety under consideration, let q and t be formal variables, let t^β record effective curve classes β, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-019520 | MATHDB-353398

Mirabolic Bezrukavnikov equivalence for the Iwahori version

摘录 / EXCERPT: For $M<N$, let V_1 be an M-dimensional vector space and V_2 an N-dimensional vector space, and let $\Ffl_i$ denote the variety of complete flags in V_i. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019521 | MATHDB-353404

Minimal generating set conjecture for the integral Chow ring of the moduli stack of maps

摘录 / EXCERPT: Let r be a positive integer and d a positive odd number. The classes $4c_1,c_24$ generate the polynomial ring describing the presentation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019522 | MATHDB-353408

Integrality conjecture for Gopakumar-Vafa invariants of holomorphic symplectic 4-folds

摘录 / EXCERPT: Let X be a holomorphic symplectic 4-fold, let $\beta\beta\in H_2(X,\mathbb{Z})$ be a primitive curve class, and let $n_{0,\beta}(\gamma_1,\ldots,\gamma_n)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-019523 | MATHDB-353409

DT4/Gopakumar-Vafa correspondence for holomorphic symplectic 4-folds

摘录 / EXCERPT: Let X be a holomorphic symplectic 4-fold and let M_β be the moduli scheme of one-dimensional stable sheaves F on X with $\operatorname{ch}_3(F)=\beta$ and $\chi(F)=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-019524 | MATHDB-353436

The Prym-image conjecture for the divisor $D(\mu;3)$

摘录 / EXCERPT: Let $\mathcal{R}_6$ be the moduli space of Prym curves of genus 6, let $\mathcal{A}_5$ be the moduli space of principally polarized abelian varieties of dimension 5, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019525 | MATHDB-353473

A relative Bogomolov-Gieseker-type conjecture for fibred threefolds

摘录 / EXCERPT: Let $\mathcal{X}$ be the three-dimensional fibred variety considered above, with polarization data H and F, and let $\operatorname{Coh}_C^{\beta H}(\mathcal{X})$ be the tilted...

conjecture | open | MathDB 页面 | 原文线索

V2-019526 | MATHDB-353490

K-stability criterion for solutions of the complexified equation

摘录 / EXCERPT: Let (X,L) be a polarized complex variety, and let $\eta\in H^1(X,\mathcal{E}_L)$ be a deformation class of (X,L). A test configuration is compatible with η if its first-or...

conjecture | open | MathDB 页面 | 原文线索

V2-019527 | MATHDB-353509

Strong conjecture on the Picard rank of toric Fano varieties

摘录 / EXCERPT: Strong conjecture. The Picard rank satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-019528 | MATHDB-353510

Weak conjecture on bounded Picard rank at fixed codegree

摘录 / EXCERPT: Weak conjecture. There exists a constant A_q, depending only on q, such that

conjecture | open | MathDB 页面 | 原文线索

V2-019529 | MATHDB-353521

Infinite transitivity criterion for affine toric varieties

摘录 / EXCERPT: Infinite transitivity conjecture for affine toric varieties. The group G acts on the open orbit of X infinitely transitively if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-019530 | MATHDB-353548

Multiple Cover Conjecture for Gromov-Witten invariants of Hilbert schemes of points on K3 surfaces

摘录 / EXCERPT: Multiple Cover Conjecture. For all $\ell>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-019531 | MATHDB-353549

Quasi-Jacobi form property for Gromov-Witten series of Hilbert schemes of points on K3 surfaces

摘录 / EXCERPT: Quasi-Jacobi form property. For all $\ell>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-019532 | MATHDB-353551

Multiple cover formula for Gromov-Witten invariants of Hilbert schemes of points on K3 surfaces

摘录 / EXCERPT: Multiple cover formula.

conjecture | open | MathDB 页面 | 原文线索

V2-019533 | MATHDB-353568

Conjecture O for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold. Write $H(X)=H^{even}(X)$, let ρ be its Fano index, and let $\boldsymbol{c_1(X)\star_0}$ denote the linear operator on the unital commutative ring...

conjecture | open | MathDB 页面 | 原文线索

V2-019534 | MATHDB-353569

Galkin-Golyshev-Iritani's Gamma conjecture I

摘录 / EXCERPT: Gamma conjecture I. There exists a constant $C\in\mathbb{C}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019535 | MATHDB-353573

The characteristic variety conjecture for solution sheaves

摘录 / EXCERPT: Let $\cM$ be a module over the sheaf of convergent-exponential differential operators $\cD^{\mathrm{ce}, \Lambda_0}$, with characteristic variety $\mathrm{Char}(\cM)$ and solution...

conjecture | open | MathDB 页面 | 原文线索

V2-019536 | MATHDB-353574

The endless continuability conjecture for Borel-transformed WKB solutions

摘录 / EXCERPT: Let $\cQ$ be an $\hbar$-differential operator, lifted to an $\hbar$-connection after fixing a spin structure, and assume WKB-regularity. Let L be the characteristic variety of...

conjecture | open | MathDB 页面 | 原文线索

V2-019537 | MATHDB-353585

Bannister-Ojakian's shallow topological-minor conjecture for stack-number

摘录 / EXCERPT: Let G and H be graphs, and let $k\geqslant 0$ be a half-integer. A graph H is a k-shallow topological minor of G if a subgraph of G is isomorphic to a subdivision of...

conjecture | open | MathDB 页面 | 原文线索

V2-019538 | MATHDB-353606

Ax-Schanuel conjecture for strata of abelian differentials

摘录 / EXCERPT: Let $\textnormal{S}_{\alpha}$ be a stratum of abelian differentials up to scaling, let $\widetilde{\textnormal{S}_{\alpha}^{\mathrm{an}}}$ be the universal cover of its analytifica...

conjecture | open | MathDB 页面 | 原文线索

V2-019539 | MATHDB-353607

Polynomial-time approximate-root conjecture for circuit-system Gale duals

摘录 / EXCERPT: Polynomial-time approximate-root conjecture. Following this notation, one can find approximate roots for all the real roots of g in time polynomial in $\log^n(BH)$.

conjecture | open | MathDB 页面 | 原文线索

V2-019540 | MATHDB-353610

Regularity equals the degree of the Hilbert numerator for Schubert varieties

摘录 / EXCERPT: Regularity conjecture. The Castelnuovo-Mumford regularity satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-019541 | MATHDB-353611

Semicontinuity of regularity along Bruhat order

摘录 / EXCERPT: Let GL_n/B be the complete flag variety, with Schubert varieties X_w indexed by permutations $w\in\mathfrak S_n$. For $u\leq v\leq w$ in Bruhat order, let $R_{u,w}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019542 | MATHDB-353612

Cohen-Macaulayness of Schubert tangent-cone rings

摘录 / EXCERPT: Let GL_n/B be the complete flag variety, with Schubert varieties X_w indexed by permutations $w\in\mathfrak S_n$. For $v\leq w$ in Bruhat order, let $R_{v,w}$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-019543 | MATHDB-353613

Coefficientwise semicontinuity of Schubert Hilbert numerators

摘录 / EXCERPT: Let GL_n/B be the complete flag variety, with Schubert varieties X_w indexed by permutations $w\in\mathfrak S_n$. For $u\leq v\leq w$ in Bruhat order, let $H_{u,w}(q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019544 | MATHDB-353614

Quadratic growth of maximal Schubert regularity

摘录 / EXCERPT: For each n, define ... where $R_{v,w}$ is the associated graded ring of the local ring of the Schubert variety X_w at the torus-fixed point e_v. Maximal-regularity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-019545 | MATHDB-353615

The strengthened Zariski dense orbit conjecture for birational maps of dynamical degree one

摘录 / EXCERPT: Let X be a smooth projective variety defined over an algebraically closed field K of characteristic 0, and let $\phi:X\dashrightarrow X$ be a birational self-map. Its first d...

conjecture | open | MathDB 页面 | 原文线索

V2-019546 | MATHDB-353616

The dynamical-degree-one alternative for invariant subvarieties

摘录 / EXCERPT: Let X be a smooth projective variety over an algebraically closed field K of characteristic 0, and let $\phi:X\dashrightarrow X$ be a dominant rational self-map. Assume that...

conjecture | refuted | MathDB 页面 | 原文线索

V2-019547 | MATHDB-353619

Bounded Weil index of K-semistable Fano varieties

摘录 / EXCERPT: Weil-index boundedness conjecture. There exists a constant $N>0$ depending only on n such that the Weil index of every n-dimensional K-semistable Fano variety is at most N.

conjecture | open | MathDB 页面 | 原文线索

V2-019548 | MATHDB-353620

Hausel-Letellier-Rodriguez-Villegas conjecture for branched character varieties

摘录 / EXCERPT: Let $\mathcal{C}$ be a generic tuple of conjugacy classes, let $\bs\mu$ encode the eigenvalue multiplicities, and let $\mathcal{M}_{\mathcal{C}}$ be the associated character variet...

conjecture | open | MathDB 页面 | 原文线索

V2-019549 | MATHDB-353621

Mixed-Hodge-polynomial conjecture for wreath character varieties

摘录 / EXCERPT: Let $\mathcal{C}=(C_j)_{1\le j\le 2k}$ be a strongly generic tuple of semisimple conjugacy classes in $\operatorname{GL}_n\langle\sigma\rangle$ satisfying the condition referred to...

conjecture | open | MathDB 页面 | 原文线索

V2-019550 | MATHDB-353622

Connected-components conjecture for nongeneric wreath character varieties

摘录 / EXCERPT: Let $\mathcal{C}$ be a tuple of semisimple conjugacy classes in $\operatorname{GL}_n\langle\sigma\rangle$, not necessarily generic, and suppose that the character variety...

conjecture | open | MathDB 页面 | 原文线索

V2-019551 | MATHDB-353660

Complete intersection classification conjecture for reduced subvarieties of Veronese surfaces

摘录 / EXCERPT: Complete intersection classification conjecture. Such a subvariety exists if and only if it is one of the following:

conjecture | open | MathDB 页面 | 原文线索

V2-019552 | MATHDB-353684

Equivariant Hikita-Nakajima kernel conjecture for general quivers

摘录 / EXCERPT: Let $I=(I_0,I_1)$ be a finite quiver with dimension vector ${\bf v}=(v_i)_{i\in I_0}$ and framing ${\bf w}=(w_i)_{i\in I_0}$. Let A act on the Nakajima quiver variety...

conjecture | open | MathDB 页面 | 原文线索

V2-019553 | MATHDB-353685

Flatness conjecture for fixed-point algebras of general quiver Coulomb branches

摘录 / EXCERPT: Let $I=(I_0,I_1)$ be a finite quiver with dimension vector ${\bf v}$ and framing ${\bf w}$, let A be the natural torus acting on $\mathfrak{M}({\bf v},{\bf w})$, let $G_{\bf v}$...

conjecture | open | MathDB 页面 | 原文线索

V2-019554 | MATHDB-353691

Stabilization conjecture for cohomology rings of affine group schemes

摘录 / EXCERPT: Let p) be a prime number, and let $\mathbb{G}(-)$ be an affine group scheme over the p-adic integers $\mathbb{Z}_p$. For each $n\geq 1$, let K be a field of characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-019555 | MATHDB-353703

The Stein universal-cover conjecture for nef cotangent bundles

摘录 / EXCERPT: Stein universal-cover conjecture. If the universal cover of Y is a Stein manifold, then the cotangent bundle of Y is nef.

conjecture | open | MathDB 页面 | 原文线索

V2-019556 | MATHDB-353708

Ambro's nonvanishing conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\a0\mathbb{C}$, and let K_X be its canonical divisor. Ambro's conjecture. If D is an ample divisor on X and...

conjecture | open | MathDB 页面 | 原文线索

V2-019557 | MATHDB-353709

Adjunction termination implies uniruledness

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\a0\mathbb{C}$, with canonical divisor K_X, and let L be an effective divisor. Say that adjunction terminates in t...

conjecture | open | MathDB 页面 | 原文线索

V2-019558 | MATHDB-353710

Generalized Ambro nonvanishing conjecture for 1-big divisors

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over $\a0\mathbb{C}$, and let K_X be its canonical divisor. A nef divisor is 1-big when $L^{n-1}\cdot H>0$ for some ample...

conjecture | open | MathDB 页面 | 原文线索

V2-019559 | MATHDB-353721

The real symmetric substitution conjecture

摘录 / EXCERPT: Let I be an index set, let d be a positive integer, let $\mathcal{C}\in(\mathbb{R}^{I})^{\otimes d}$ be a symmetric tensor, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019560 | MATHDB-353732

Minimax conjecture for μ-entropy

摘录 / EXCERPT: Let (X,L) be a polarized normal variety with only klt singularities. For $\lambda\le 0$, the extended non-archimedean and Perelman μ-entropies should satisfy ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-019561 | MATHDB-353735

Li's slope formula for entropy

摘录 / EXCERPT: Let (X,L) be a polarized variety and let $\varphi\in\mathcal{E}_{\mathrm{NA}}^2(X,L)$ subordinate to its maximal geodesic ray $\{\ell_{\varphi,t}\}_{t\in[0,\infty)}$. The entropy...

conjecture | open | MathDB 页面 | 原文线索

V2-019562 | MATHDB-353737

Regularization conjecture for non-archimedean entropy

摘录 / EXCERPT: Let (X,L) be a polarized variety. For any $\varphi\in\mathcal{E}_{\mathrm{NA}}^1(X,L)$, there exists a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-019563 | MATHDB-353738

Weak-to-strong convergence conjecture for finite-energy metrics

摘录 / EXCERPT: Let (X,L) be a polarized variety. Every d_2-bounded weakly convergent sequence in $\mathcal{E}_{\mathrm{NA}}^2(X,L)$ should be d_1-convergent. This conjecture would provide t...

conjecture | open | MathDB 页面 | 原文线索

V2-019564 | MATHDB-353742

Polyhedrality conjecture for cones of codimension-2 special cycles

摘录 / EXCERPT: Let X be an orthogonal Shimura variety arising from an even unimodular lattice of signature $(b,2)$, where $b>2$. The cone of special cycles of codimension 2 on X is the cone...

conjecture | open | MathDB 页面 | 原文线索

V2-019565 | MATHDB-353772

The weak Schottky problem for Hirota varieties of rational nodal curves

摘录 / EXCERPT: Let g be a genus, and let $\mathcal{H}_\mathcal{C}$ be the Hirota variety associated with the rational nodal curve data, with main component $\mathcal{H}_\mathcal{C}^{M}$ and the...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-019566 | MATHDB-353773

The strong Schottky problem for Hirota varieties

摘录 / EXCERPT: Let $\mathcal{H}_\mathcal{C}^{M}$ be the main component of the Hirota variety $\mathcal{H}_\mathcal{C}$, and let $\mathcal{H}_\mathcal{C}^{I}$ denote its distinguished irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-019567 | MATHDB-353779

Rank-variety detection of projectivity for quasireductive Lie superalgebras

摘录 / EXCERPT: Rank-variety projectivity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-019568 | MATHDB-353780

Campana's conjecture over function fields

摘录 / EXCERPT: Let X be a smooth projective variety over $\a9\mathbb{C}$ and let D be a strict normal crossings divisor on X with irreducible components $D_1,\ldots,D_q$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-019569 | MATHDB-353789

Subloop freeness conjecture for free Bol loops of exponent two

摘录 / EXCERPT: A free Bol loop of exponent two is a Bol loop freely generated in the variety of Bol loops satisfying $x^2=1$. Subloop freeness conjecture. Every subloop of a free Bol loop of expo...

conjecture | open | MathDB 页面 | 原文线索

V2-019570 | MATHDB-353791

The conjectural mixed Hodge polynomial formula for unbranched character varieties

摘录 / EXCERPT: The conjectural mixed Hodge polynomial formula. (i) The rational function $\mathbb{H}_{\bs\mu}(z,w)$ is a polynomial of degree d in each variable, every monomial has even degree,...

conjecture | open | MathDB 页面 | 原文线索

V2-019571 | MATHDB-353793

Existence of a twisted scheme for residual categories of quadric surface bundles

摘录 / EXCERPT: Let $p\colon \mathcal{Q}\to S$ be a quadric surface bundle. Write S_2 and S_3 for the loci where the fibers have corank at least 2 and 3, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019572 | MATHDB-353811

Non-semi-algebraicity conjecture for logarithmic Voronoi cells of the elliptope

摘录 / EXCERPT: For $m=3$, the unrestricted correlation model ... is the elliptope. Its logarithmic Voronoi cells are unbounded 3-dimensional convex sets.
Non-semi-algebraicity conjecture. The log...

conjecture | open | MathDB 页面 | 原文线索

V2-019573 | MATHDB-353816

The weakened Roberts conjecture for cubic Hurwitz stacks

摘录 / EXCERPT: Let k be a field of characteristic not dividing 6, let g be the genus, and let $\mathrm{Hur}_{3,g,k}$ be the degree-3 Hurwitz stack. Write $\mathbb L$ for the class of the...

conjecture | open | MathDB 页面 | 原文线索

V2-019574 | MATHDB-353830

Support-recovery incidence-variety conjecture for sparse phase retrieval

摘录 / EXCERPT: Support-recovery conjecture. If $|S-S|=|S'-S'|\geq K$, then $I_{S,S'}$ has dimension strictly less than K. This would imply that a generic vector supported on S cannot share it...

conjecture | open | MathDB 页面 | 原文线索

V2-019575 | MATHDB-353831

Known-support incidence-variety conjecture for sparse phase retrieval

摘录 / EXCERPT: Known-support recovery conjecture. If $|S-S|>|S|$, then I_S has dimension $|S|$ and degree $2|D_S|$. This is the signal-recovery step for generic signals whose support is known;...

conjecture | open | MathDB 页面 | 原文线索

V2-019576 | MATHDB-353832

Sharpness conjecture for Newton-Okounkov bodies of Hilbert schemes on toric surfaces

摘录 / EXCERPT: Sharpness conjecture. If X is $\mathbb P^2$, $\mathbb P^1\times\mathbb P^1$, or a Hirzebruch surface, then

conjecture | open | MathDB 页面 | 原文线索

V2-019577 | MATHDB-353841

Campana's numerical cotangent bound for special manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold that is special, and let $p>0$. For every rank-one coherent subsheaf $L\subset\Omega_X^p$, the numerical cotangent bound conjecture. One has .....

conjecture | open | MathDB 页面 | 原文线索

V2-019578 | MATHDB-353842

Campana's strengthening for pseudoeffective and uniruled varieties

摘录 / EXCERPT: Let X be a smooth complex projective manifold. If K_X is pseudoeffective, write $\nu_1(X):=\nu(X,\Omega_X^1)$. If X is uniruled, let $r:X\to R_X$ be its maximal rationally co...

conjecture | open | MathDB 页面 | 原文线索

V2-019579 | MATHDB-353869

Kuznetsov-Perry's birational categorical Torelli conjecture for GM varieties

摘录 / EXCERPT: Let X and X' be GM varieties of the same dimension. The Kuznetsov-Perry conjecture. If there exists an equivalence ... then X and X' are birationally equivalent. This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-019580 | MATHDB-353881

Asymptotic trace conjecture for quantum representations of pseudo-Anosov mapping tori

摘录 / EXCERPT: Let S be a punctured surface, let $\varphi\colon S\to S$ be a pseudo-Anosov diffeomorphism, let $[r]\in\mathcal X_{\mathrm{SL}_2(\mathbb C)}(S)$ be a smooth character fixed by...

conjecture | open | MathDB 页面 | 原文线索

V2-019581 | MATHDB-353890

Cluster algebra structure on the homogeneous coordinate ring of a partial flag variety

摘录 / EXCERPT: Cluster-structure conjecture. The homogeneous coordinate ring equals the lifted cluster algebra:

conjecture | open | MathDB 页面 | 原文线索

V2-019582 | MATHDB-353936

The conjectural formula for the U_3 sum

摘录 / EXCERPT: The U_3 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019583 | MATHDB-353937

The vanishing criterion for the U_1 sum

摘录 / EXCERPT: The U_1 vanishing conjecture. For $n\geq|\mathbf{d}|$,

conjecture | open | MathDB 页面 | 原文线索

V2-019584 | MATHDB-353938

The conjectural formulas for the V_1, V_2, and V_3 sums

摘录 / EXCERPT: The V_1-V_3 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019585 | MATHDB-353939

Asymptotic enumerativity conjecture for Tevelev degrees of Fano varieties

摘录 / EXCERPT: Asymptotic enumerativity conjecture. There exists a constant $C_{X,g}$ depending only on X and g such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-019586 | MATHDB-353954

Semitoric compactification conjecture for weak K-moduli Artin stacks

摘录 / EXCERPT: A weak K-moduli space is an Artin stack when continuous isotropy is allowed. Consider such a stack parametrizing polarized irreducible holomorphic symplectic varieties, polarized a...

conjecture | open | MathDB 页面 | 原文线索

V2-019587 | MATHDB-353981

Singular Singer-Hopf conjecture for subvarieties

摘录 / EXCERPT: Let X be an aspherical complex projective manifold and let $Z\subseteq X$ be a closed irreducible subvariety. Let Eu_Z be MacPherson's local Euler obstruction function, let...

conjecture | open | MathDB 页面 | 原文线索

V2-019588 | MATHDB-353984

Tensor and lavish support conjecture for arbitrary almost-simple groups

摘录 / EXCERPT: Standard conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-019589 | MATHDB-353985

E2-formality conjecture for the quantum-group endomorphism scheme

摘录 / EXCERPT: Formality conjecture. The E_3-scheme $\mathscr{X}$ is E_2-formal, so that there is a monoidal equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-019590 | MATHDB-354007

The J^H-semistability conjecture

摘录 / EXCERPT: Let (X,L) be an n-dimensional polarized normal variety, meaning that X is normal and L is ample, and let H be a pseudoeffective line bundle. J^H-semistability conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-019591 | MATHDB-354019

Spectrality conjecture for the Kolmogorov quotient of infinite affine orbit space

摘录 / EXCERPT: Let K be a field, let n be a positive integer, and let G act on the infinite affine space ${\mathbb A}^n_{K,\infty}$. Write $\textnormal{KQ}({\mathbb A}^n_{K,\infty}/G)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-019592 | MATHDB-354029

Hacon-Lamarche-Schwede subadditivity conjecture for $+$-test ideals

摘录 / EXCERPT: Hacon-Lamarche-Schwede subadditivity conjecture. Given effective ${\mathbb Q}$-divisors D and E on ${\mathscr X}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-019593 | MATHDB-354032

Generalized geometric Satake conjecture for quiver varieties

摘录 / EXCERPT: Generalized geometric Satake conjecture. The vector space

conjecture | open | MathDB 页面 | 原文线索

V2-019594 | MATHDB-354033

Characteristic-cycle isomorphism conjecture for top category O

摘录 / EXCERPT: Characteristic-cycle conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-019595 | MATHDB-354040

O'Grady's Lagrangian covering family conjecture

摘录 / EXCERPT: Let M be a projective hyperkähler variety. A Lagrangian covering family for M is a closed subscheme $\mathcal{U}\subset M\times B$, pure of dimension $\dim(B)+n$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-019596 | MATHDB-354055

Kontsevich-Soibelman's conjecture on affine structures for maximally degenerating Calabi-Yau varieties

摘录 / EXCERPT: Kontsevich-Soibelman's conjecture. For maximal degenerating polarized algebraic Calabi-Yau varieties, the IAMS structure induced by the Collapse Picture coincides with the IAMS str...

conjecture | open | MathDB 页面 | 原文线索

V2-019597 | MATHDB-354104

The unlikely intersections conjecture for generically ordinary subvarieties of simple Shimura varieties

摘录 / EXCERPT: Let $\mathcal S$ be a simple Shimura variety over some base, and let $V,W\subseteq\mathcal S$ be generically ordinary subvarieties of complementary dimension. Unlikely intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-019598 | MATHDB-354135

Affine Springer-Hilbert-scheme sheaf correspondence

摘录 / EXCERPT: For $G=GL_n$, let $gamma\bin g derbrace{\llbracket t\rrbracket}_{\mathcal O}$ be regular semisimple, let $cbeta=\beta(gamma)$ be the associated braid, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019599 | MATHDB-354138

The conjecture that the u-field equals the trace field

摘录 / EXCERPT: Equality conjecture. The u-field equals the trace field:

conjecture | open | MathDB 页面 | 原文线索

V2-019600 | MATHDB-354139

The conjecture on nontrivial knots and parabolic representations

摘录 / EXCERPT: Parabolic-representation existence conjecture. A knot K is nontrivial if and only if $8par(K)$ is nonempty. Moreover, $8par(K)$ has a zero-dimensional component.

conjecture | open | MathDB 页面 | 原文线索

V2-019601 | MATHDB-354146

The Euler-characteristic discrepancy conjecture for degeneracy-locus Hilbert squares

摘录 / EXCERPT: Let $m\geq1$, let $s\in\{m+3,\ldots,2m+3\}$, and set $n=2s-2m-3$. Let $\mathsf{M}_{m,s}$ be the degeneracy locus considered in the paper and let $Z_{n,s,m}$ be the associated varie...

conjecture | open | MathDB 页面 | 原文线索

V2-019602 | MATHDB-354148

Rational Hodge isometries conjecture for hyper-Kähler varieties of K3[n]-type

摘录 / EXCERPT: Let ${\mathcal G}^{[n]}_{an}$ be the groupoid of the relevant cohomological morphisms between projective irreducible holomorphic symplectic manifolds of $K3^{[n]}$-type, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019603 | MATHDB-354151

The Kisin-variety and gene conjecture for potentially crystalline deformation rings

摘录 / EXCERPT: Let $K/{\mathbb Q}_p$ be unramified, let $\overline\rho$ be absolutely irreducible, let $\lambda=(0,0)$ for each embedding, and let t be a tame inertial type of level f. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019604 | MATHDB-354192

Scholze's representability conjecture for integral local Shimura varieties

摘录 / EXCERPT: Let (G,b,μ) be a local Shimura datum, with G a reductive group over $\mathbb{Q}_p$, $b\in G(\breve{\mathbb{Q}}_p)$, and μ a minuscule conjugacy class of cocharacters, and...

conjecture | open | MathDB 页面 | 原文线索

V2-019605 | MATHDB-354193

The discrete-to-finite general representation type conjecture

摘录 / EXCERPT: Let A be a finite-dimensional algebra. Say that A has discrete general representation type when every irreducible component of each representation variety for A has a dense o...

conjecture | open | MathDB 页面 | 原文线索

V2-019606 | MATHDB-354220

Zhang's dynamical Manin-Mumford conjecture

摘录 / EXCERPT: Let X be a variety and let $\Phi:X\to X$ be a dominant map. A point of X is preperiodic for Φ if its forward orbit under Φ is finite, and a subvariety Y is preperio...

conjecture | open | MathDB 页面 | 原文线索

V2-019607 | MATHDB-354235

Algebraization conjecture for 6-webs with cubic-threefold properties

摘录 / EXCERPT: Let $\boldsymbol{\mathcal W}$ be a 6-web satisfying the five listed properties: it is linearizable and skew, has maximal 2-rank equal to 10, has 1-rank at least 5, and has a 5-...

conjecture | open | MathDB 页面 | 原文线索

V2-019608 | MATHDB-354236

McKernan-Prokhorov boundedness conjecture for rationally connected log Calabi-Yau pairs

摘录 / EXCERPT: McKernan-Prokhorov conjecture. The set of such varieties X is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-019609 | MATHDB-354263

Leclerc's irreducible-factor conjecture for Richardson varieties

摘录 / EXCERPT: Let G be a semisimple algebraic group, let N be its maximal unipotent subgroup, and let $\Rich_v^w$ be the open Richardson variety associated to $v\leq w$ in the Weyl group. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-019610 | MATHDB-354264

Leclerc-Ing quiver conjecture for Richardson varieties

摘录 / EXCERPT: Let $G=\operatorname{SL}_n(\mathbb{C})$, let $v\leq w$ be elements of the Weyl group W, and let Q_{Lec} and Q_{Ing} be the quivers constructed f...

conjecture | open | MathDB 页面 | 原文线索

V2-019611 | MATHDB-354265

Fixed-point conjecture for the twist on the longest Richardson variety

摘录 / EXCERPT: Let $G=\operatorname{SL}_n(\mathbb{C})$, let $v=\operatorname{id}$ and $w=w_0$, and let $\twist_v^w$ be the twist automorphism of the open Richardson variety, preserving its totall...

conjecture | open | MathDB 页面 | 原文线索

V2-019612 | MATHDB-354299

The Epsilon Dichotomy Conjecture for strongly tempered spherical models

摘录 / EXCERPT: Fix a local field F, a model (G,H) in the paper's table, a tempered L-parameter ϕ, its component group S_ϕ, and the representation ρ_X attached to the spherica...

conjecture | open | MathDB 页面 | 原文线索

V2-019613 | MATHDB-354305

Infinite transitivity conjecture for homogeneous quasi-affine varieties

摘录 / EXCERPT: Let X be an irreducible homogeneous quasi-affine variety with ... and with $\mathbb{K}[X]^{\times}=\mathbb{K}^{\times}$. Here $\theta(X)$ denotes the transitivity degree of the aut...

conjecture | open | MathDB 页面 | 原文线索

V2-019614 | MATHDB-354308

Generic 2:2 relation for natural parameters of degenerate tetrahedra

摘录 / EXCERPT: Let $\alpha,\beta,\gamma,\delta,\varsigma\in\mathbb R$ satisfy ... and let $\varrho_+(\alpha,\beta,\gamma,\delta)$ denote the unique positive root of the discriminant quadratic descr...

conjecture | open | MathDB 页面 | 原文线索

V2-019615 | MATHDB-354313

Serre's algebraic Sato-Tate conjecture

摘录 / EXCERPT: Let A be an abelian variety with rational symplectic representation (V,ψ), and let K be a field over which A is defined. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019616 | MATHDB-354314

Serre's conjecture on the component-group image of algebraic monodromy

摘录 / EXCERPT: Let A be an abelian variety over K, let $\operatorname{G}_{\ell,K,1}^{\operatorname{alg}}$ be its normalized algebraic ℓ-adic monodromy group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019617 | MATHDB-354358

Quasi-F-splitting of general fibers with trivial relative canonical divisor

摘录 / EXCERPT: General-fiber quasi-F-splitting conjecture. If the relative canonical divisor is trivial and X is quasi-F-split, then a general fiber of π is quasi-F-split.

conjecture | open | MathDB 页面 | 原文线索

V2-019618 | MATHDB-354365

Pattern-avoidance criterion for rational smoothness in types AI and AII

摘录 / EXCERPT: Pattern-avoidance equivalence conjecture. In types AI and AII, an involution π avoids all bad patterns in the sense of this section if and only if it avoids all bad pattern...

conjecture | open | MathDB 页面 | 原文线索

V2-019619 | MATHDB-354367

Mirror symmetry conjecture for Calabi-Yau fractional complete intersections

摘录 / EXCERPT: Let $X\to\mathbf{P}_{\Delta}$ and $X^{\vee}\to\mathbf{P}_{\nabla}$ be smooth MPCP desingularizations admitting the stated nef-partitions, and let Y and $Y^{\vee}$ be the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-019620 | MATHDB-354412

Campana-Corvaja-Zannier conjecture for special varieties

摘录 / EXCERPT: Let X be a special smooth projective geometrically connected variety over a number field k. Campana-Corvaja-Zannier conjecture. There is a number field L/k such that X_L sa...

conjecture | open | MathDB 页面 | 原文线索

V2-019621 | MATHDB-354422

Lockhart-Rosen-Silverman partial generalization of Szpiro's conjecture

摘录 / EXCERPT: Lockhart-Rosen-Silverman's partial generalization of Szpiro's conjecture. There exist constants $C_1(F,d)$ and $C_2(F,d)$, depending only on F and d, such that

conjecture | open | MathDB 页面 | 原文线索

V2-019622 | MATHDB-354426

Holomorphic arithmetic modularity of the arithmetic special-divisor generating series

摘录 / EXCERPT: Modularity conjecture. Under the stated assumption, the arithmetic generating series $z(g,\phi)_{{\mathfrak e},{\mathfrak a}}^{\overline{\mathcal L}}$ is holomorphic and modular, w...

conjecture | open | MathDB 页面 | 原文线索

V2-019623 | MATHDB-354427

Weak Rost Nilpotence Principle

摘录 / EXCERPT: Let k be a perfect field, let X be a smooth projective scheme over k, and let $M(X)$ be its Chow motive in the category ${\mathcal M}_{k}$ of Chow motives. For a field extens...

conjecture | open | MathDB 页面 | 原文线索

V2-019624 | MATHDB-354444

The arithmetic Fujita height inequality for K-semistable arithmetic Fano varieties

摘录 / EXCERPT: Let $\mathcal{X}$ be an arithmetic Fano variety of relative dimension n over $\mathbb{Z}$, and let X be its complexification. Assume that X is K-semistable. Assume also that...

conjecture | open | MathDB 页面 | 原文线索

V2-019625 | MATHDB-354445

Odaka's modular height conjecture for Fano varieties

摘录 / EXCERPT: Let $X_{\mathbb{Q}}$ be a Fano variety defined over $\mathbb{Q}$, and let $h(X_{\mathbb{Q}},-K_{X_{\mathbb{Q}}})$ denote Odaka’s modular invariant with the volume-normalized metric...

conjecture | open | MathDB 页面 | 原文线索

V2-019626 | MATHDB-354446

The toric K-semistable Fano volume gap hypothesis

摘录 / EXCERPT: Let X be an n-dimensional toric K-semistable Fano manifold. Gap hypothesis. If $X\neq\mathbb{P}^n$, then ... This gap would extend the proof of the paper’s toric volume theorem t...

conjecture | open | MathDB 页面 | 原文线索

V2-019627 | MATHDB-354447

The arithmetic Ding invariant conjecture over number fields

摘录 / EXCERPT: Let X_F be a K-semistable Fano variety defined over a number field F. Define $\mathcal{D}(X_F)$ as the infimum of the normalized arithmetic Ding functional over integral models...

conjecture | open | MathDB 页面 | 原文线索

V2-019628 | MATHDB-354478

Folklore conjecture on liftings of globally F-split varieties

摘录 / EXCERPT: Let X be a globally F-split normal projective variety over an algebraically closed field k of characteristic $p>0$. The variety X is globally F-split if its Frobenius mor...

conjecture | open | MathDB 页面 | 原文线索

V2-019629 | MATHDB-354494

Equal-degree and equal-genus conjecture for components of coupler curves

摘录 / EXCERPT: Equal-degree and equal-genus conjecture. For almost all edge length assignments, for all $1\leq i,j\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-019630 | MATHDB-354538

Corvaja-Zannier's conjecture on the Hilbert Property

摘录 / EXCERPT: Let K be a number field and let X be a variety over K. Assume that $X(K)$ is Zariski dense and that X has a smooth, projective model over K which is algebraically simply...

conjecture | open | MathDB 页面 | 原文线索

V2-019631 | MATHDB-354541

Cohomological concentration conjecture for 0-toral characters

摘录 / EXCERPT: Let $\mathbb{T}_r$ and $\mathbb{U}_r$ be the groups appearing in the cohomology varieties $Y_{\mathbb{T}_r,\mathbb{U}_r}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019632 | MATHDB-354573

Implicitness conjecture for symmetric resonance-based schemes

摘录 / EXCERPT: Implicitness conjecture. Every symmetric resonance-based scheme is necessarily implicit.

conjecture | open | MathDB 页面 | 原文线索

V2-019633 | MATHDB-354588

Definite symmetric determinantal representation from Rayleigh differences

摘录 / EXCERPT: Let $f\in\mathbb{R}[x_1,\ldots,x_m]$ be multiaffine in $x_1,\ldots,x_n$, with nonzero coefficient of $x_1\cdots x_n$. For indices i,j, let $\Delta_{ij}(f)$ denote the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-019634 | MATHDB-354589

The orbit classification conjecture for horospherical varieties

摘录 / EXCERPT: The orbit classification conjecture. Two G-orbits $\mathcal{O}$ and $\mathcal{O}'$ lie in the same $\mathrm{Aut}(X)^0$-orbit if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-019635 | MATHDB-354602

Silverman's conjecture on geometric divisibility sequences

摘录 / EXCERPT: Let $\mathcal{G}/\mathbb{Z}$ be a group scheme, let $\sigma_P\in\mathcal{G}(\mathbb{Z})$, and write ... for its generic fiber. Assume that G is an irreducible commutative algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-019636 | MATHDB-354618

Tate's numerical-cycles conjecture for varieties over finite fields

摘录 / EXCERPT: Let V be a smooth projective variety over $\mathbb{F}_q$, let $r\geq 0$, and let $A^r(V)$ be the quotient of the space of algebraic cycles of codimension r on V by the subspa...

conjecture | open | MathDB 页面 | 原文线索

V2-019637 | MATHDB-354619

The l-Fano characterization conjecture for projective space

摘录 / EXCERPT: Let X be an n-dimensional smooth l-Fano variety, meaning that X is smooth and Fano and its Chern characters $\mathrm{ch}_i(X)$ are positive for every...

conjecture | open | MathDB 页面 | 原文线索

V2-019638 | MATHDB-354678

Equivariant degree-interval conjecture for quantum K-theory products

摘录 / EXCERPT: Equivariant degree-interval conjecture. The power q^d occurs in

conjecture | open | MathDB 页面 | 原文线索

V2-019639 | MATHDB-354679

Alternating-sign and leading-term conjecture for exceptional-degree pushforwards

摘录 / EXCERPT: Exceptional-degree pushforward conjecture. Under these assumptions:

conjecture | open | MathDB 页面 | 原文线索

V2-019640 | MATHDB-354705

Chai-Oort Lie algebra characterization of strongly Tate-linear deformation spaces

摘录 / EXCERPT: Let Y be the object used to define the formally smooth deformation space $\mathbf{Def}_{\mathrm{sus}}(Y)$, and let $\mathfrak{a}_Y$ be its ambient Lie algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019641 | MATHDB-354732

The primitive Hodge conjecture

摘录 / EXCERPT: Primitive Hodge conjecture. Every primitive Hodge class of degree $2n$ is algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-019642 | MATHDB-354802

Rational smoothness conjecture for regular generalized Hessenberg varieties

摘录 / EXCERPT: Let X be an element of the group under consideration, let $\underline{s}$ be a reduced word, and let $\mathcal{Y}_{\underline{s}}(X)$ be the associated variety. A variety is rati...

conjecture | open | MathDB 页面 | 原文线索

V2-019643 | MATHDB-354803

Intersection-cohomology restriction conjecture for generalized Hessenberg varieties

摘录 / EXCERPT: Let G be the group, let $X\in G$ be regular, let $w\in W$, and let $\mathcal{Y}_w(X)$ be the corresponding variety. Let Ω_w be the Schubert variety and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019644 | MATHDB-354804

Cohomology-basis conjecture for smooth generalized Hessenberg varieties

摘录 / EXCERPT: Let $w\in S_n$ be smooth, and let $X\in SL(n)$ be regular. For $u\leq w$, let $[\mathcal{Y}_u(X)]\in H^*(\mathcal{Y}_w(X))$ be the corresponding geometric class. Cohomology-basis c...

conjecture | open | MathDB 页面 | 原文线索

V2-019645 | MATHDB-354836

The Hessenberg variety homeomorphism conjecture for smooth permutations

摘录 / EXCERPT: Hessenberg variety homeomorphism conjecture. There exists a codominant permutation w' such that

conjecture | open | MathDB 页面 | 原文线索

V2-019646 | MATHDB-354845

Single-survivor conjecture for competing urns on hypercubes and discrete tori

摘录 / EXCERPT: Let $K\ge3$, let the underlying graph be either a hypercube or a discrete torus, and let the initial configuration be any nonzero configuration of the K-type competing urn scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-019647 | MATHDB-354846

Unequal-rate single-survivor conjecture for competing urns on cycles

摘录 / EXCERPT: Let C_N be a cycle, let $K\ge3$, and consider the K-type competing urn scheme with a nonzero initial configuration. For each type j, let balls of type j have independent Po...

conjecture | open | MathDB 页面 | 原文线索

V2-019648 | MATHDB-354849

The tangent-sheaf cohomology dimension claim for boundary s.l.c. sextic surfaces

摘录 / EXCERPT: Let S be a semi-log-canonical (s.l.c.) sextic surface lying in any boundary divisor of the closure $\overline{M}^{\operatorname{sextic}}$ of the Gieseker moduli stack of sextic s...

conjecture | open | MathDB 页面 | 原文线索

V2-019649 | MATHDB-354854

The folklore conjecture on the lower bound of Seshadri constants

摘录 / EXCERPT: Seshadri lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019650 | MATHDB-354862

Motivic jet-space integral formula for formal power series

摘录 / EXCERPT: Motivic jet-space integral conjecture. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-019651 | MATHDB-354863

Motivic Milnor fiber comparison with the topological Milnor fiber

摘录 / EXCERPT: Let f be a complex analytic function in d variables vanishing at the origin O. Let $F_{f,O}$ be its topological Milnor fiber, let $\mathscr S_{f,O}$ be the motivic Milnor fib...

conjecture | open | MathDB 页面 | 原文线索

V2-019652 | MATHDB-354909

Charlton's generalized cyclic insertion conjecture

摘录 / EXCERPT: Let $I^{\mathfrak{m}}(0,a_1,\ldots,a_k,1)$ denote the motivic iterated integral along the specified tangential path from 0 to 1. For an odd index $(l_1,\ldots,l_d)$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-019653 | MATHDB-354911

Exceptional motivic block-shuffle relations

摘录 / EXCERPT: Let $p_{a,b,c}$ and $q_{a,b,c}$ be the polynomials and let $r_{a,b,c}$ and $r'_{a,b,c}$ be the integers defined in the statement. Let $I_{\mathrm{bl}}^{\mathfrak{m}}$ denote the mo...

conjecture | open | MathDB 页面 | 原文线索

V2-019654 | MATHDB-354920

Iarrobino-Kanev conjecture on the small tangent space condition for apolar algebras

摘录 / EXCERPT: Let $S=k[\alpha_1,\ldots,\alpha_n]$ and $P=k[x_1,\ldots,x_n]$ be polynomial rings over a field k, with S acting on P by contraction. For a homogeneous polynomial $f\in P$, wr...

conjecture | open | MathDB 页面 | 原文线索

V2-019655 | MATHDB-354952

Orlov's graded Manin-invariant conjecture

摘录 / EXCERPT: Orlov's conjecture. Each graded piece $\mathcal{F}^i$ is a derived invariant for smooth projective varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-019656 | MATHDB-354954

Modified Batyrev-Kreuzer torsion mirror-symmetry conjecture

摘录 / EXCERPT: Modified Batyrev-Kreuzer conjecture. If m is relatively prime to each of 2, 3, 5, and 7, then the m-torsion subgroups

conjecture | open | MathDB 页面 | 原文线索

V2-019657 | MATHDB-354958

Pluricanonical birationality and embedding conjectures for varieties of general type

摘录 / EXCERPT: Let X be a minimal variety of general type of dimension n with at worst $\mathbb{Q}$-factorial terminal singularities, and suppose that its canonical divisor K_X is Cartier....

conjecture | open | MathDB 页面 | 原文线索

V2-019658 | MATHDB-354965

Uniform logarithmic plurigenus characterization of quasi-abelian varieties

摘录 / EXCERPT: Uniform plurigenus conjecture. There exists a positive integer k, independent of the dimension of V, such that

conjecture | open | MathDB 页面 | 原文线索

V2-019659 | MATHDB-354971

Biholomorphic Geometric Arveson-Douglas conjecture for quotient submodules

摘录 / EXCERPT: Let $V\subset\mathbb{B}^n$ be an affine homogeneous algebraic variety, meaning that there exist finitely many holomorphic homogeneous polynomials $q_1,\ldots,q_m$ such that ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019660 | MATHDB-354985

Eisenbud-Goto regularity conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb P^r$ be a nondegenerate projective variety. Its Castelnuovo-Mumford regularity is denoted by $\operatorname{reg} X$, its degree by $\deg X$, and its codimens...

conjecture | open | MathDB 页面 | 原文线索

V2-019661 | MATHDB-355006

Smooth Bruhat intervals imply dual zircons in symmetric groups

摘录 / EXCERPT: Let $[u,w]$ be a Bruhat interval in the symmetric group of type A_n, and let $[u,w]^\ast$ denote its dual interval. A Bruhat interval is smooth when its associated Schubert varie...

conjecture | open | MathDB 页面 | 原文线索

V2-019662 | MATHDB-355007

Smooth-interval descent-set conjecture in symmetric groups

摘录 / EXCERPT: Let $[u,w]$ be a Bruhat interval in the symmetric group of type A_n. Write $D_L(u)$ for the left descent set of u, and let S be the set of simple reflections. A Bruhat interv...

conjecture | open | MathDB 页面 | 原文线索

V2-019663 | MATHDB-355017

Algebraic-geometric genericity conjecture for the combinatorial derived matroid

摘录 / EXCERPT: Fix a field $\mathbb{F}$ with algebraic closure $\overline{\mathbb{F}}$. Let M be representable over $\mathbb{F}$, with $|E(M)|=n$ and $r(M)=k$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019664 | MATHDB-355033

Uniqueness of sum-of-squares decompositions up to orthogonal equivalence

摘录 / EXCERPT: Let $f\in \operatorname{Sym}^{2d}V$ be generic of SOS-rank $k\leq n$, and let ... For a decomposition of f into k squares, write $A\in\mathbb C^k\otimes\mathbb C^N$ for the coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-019665 | MATHDB-355063

Strict upper bound for real umbilical points of a general algebraic surface

摘录 / EXCERPT: Real-umbilic bound. The surface X has strictly fewer than

conjecture | open | MathDB 页面 | 原文线索

V2-019666 | MATHDB-355064

Isolated critical curvature points for general hypersurfaces

摘录 / EXCERPT: Isolation conjecture. The hypersurface X has isolated complex critical curvature points.

conjecture | open | MathDB 页面 | 原文线索

V2-019667 | MATHDB-355067

Mirror P=W conjecture for mirror log Calabi-Yau varieties

摘录 / EXCERPT: Let U and $U^\vee$ be mirror n-dimensional log Calabi-Yau varieties. For a nonsingular quasi-projective variety V, let ... be its perverse-mixed Hodge polynomial. Mirror P=W co...

conjecture | open | MathDB 页面 | 原文线索

V2-019668 | MATHDB-355091

Song's class formula conjecture for Lagrangian planes of $K3^{[n]}$-type

摘录 / EXCERPT: Song's conjecture. The cohomology class of P is

conjecture | open | MathDB 页面 | 原文线索

V2-019669 | MATHDB-355104

The coaction and weight conjectures for six-dimensional phi-three periods

摘录 / EXCERPT: Let $\mathcal{P}_{\phi^3}^{\mathfrak{m}}$ denote the space of motivic periods of six-dimensional ϕ^3 theory, and let $\mathcal{P}^{\mathfrak{dr}}$ denote the de Rham period sp...

conjecture | open | MathDB 页面 | 原文线索

V2-019670 | MATHDB-355108

Bounded-complement conjecture for Fano type varieties of bounded coregularity

摘录 / EXCERPT: Let c be a nonnegative integer, and let $s(c)$ be the c-th Sylvester's number. Let X be a Fano type variety of coregularity at most c. Bounded-complement conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-019671 | MATHDB-355109

Sphere conjecture for coregularity-zero Fano type varieties

摘录 / EXCERPT: Let X be an n-dimensional Fano type variety of coregularity zero. A 2-complement (X,B) is a complement with (X,B) log canonical and $2(K_X+B)\sim 0$. Let (Y,B_Y) be the...

conjecture | open | MathDB 页面 | 原文线索

V2-019672 | MATHDB-355110

Boundedness conjecture for minimal dlt centers

摘录 / EXCERPT: Let c and N be positive integers. A minimal dlt center of a log Calabi-Yau pair is a minimal log canonical center on a dlt modification of that pair. Boundedness conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-019673 | MATHDB-355111

Exceptional quotient singularity existence conjecture

摘录 / EXCERPT: An exceptional quotient singularity is a quotient singularity satisfying the source's exceptional condition. Exceptional quotient singularity existence conjecture. For every dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-019674 | MATHDB-355112

Finiteness conjecture for regional fundamental groups of dlt rationally connected pairs

摘录 / EXCERPT: Let X be a rationally connected variety, and let (X,Δ) be a dlt pair. Write $\pi_1^{\rm reg}(X)$ for the regional fundamental group. Finiteness conjecture for regional fun...

conjecture | open | MathDB 页面 | 原文线索

V2-019675 | MATHDB-355113

Coregularity structure conjecture for regional fundamental groups

摘录 / EXCERPT: Let X be a rationally connected variety of dimension n, regularity r, and coregularity c. Assume that $G:=\pi_1^{\rm reg}(X)$ is finite. Coregularity structure conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-019676 | MATHDB-355126

The small-resolution conjecture for maximally edge-connected graph potentials

摘录 / EXCERPT: Small-resolution conjecture. The graph potential toric variety $X_{P_{mathord{\gamma},c},M}$ admits a small resolution of singularities. Equivalently in the stated combinatorial fo...

conjecture | open | MathDB 页面 | 原文线索

V2-019677 | MATHDB-355150

Bültel-Pappas representability conjecture for adjoint nilpotent displays

摘录 / EXCERPT: Bültel-Pappas conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-019678 | MATHDB-355151

Conjecture that the display-to-local-Shimura comparison morphism is an isomorphism

摘录 / EXCERPT: Comparison-isomorphism conjecture. The morphism Υ^{sm} is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-019679 | MATHDB-355169

Mutation equivalence conjecture for matching field polytopes of partial flag varieties

摘录 / EXCERPT: Let B_c^σ be the matching fields considered in the paper, with $\sigma \in S_n$ and $c \in [p-1]$. Suppose that the matching field polytope associated to B_c^σ gives...

conjecture | open | MathDB 页面 | 原文线索

V2-019680 | MATHDB-355180

Fargues's étale cohomology comparison conjecture for the Fargues-Fontaine curve

摘录 / EXCERPT: Let E be a p-adic field, let F be the coefficient field defining the Fargues-Fontaine curve, and let $X^{\operatorname{alg}}_{E,F}$ and $X^{\operatorname{ad}}_{E,F}$ denote i...

conjecture | open | MathDB 页面 | 原文线索

V2-019681 | MATHDB-355181

Fargues's cohomological dimension conjecture for constructible sheaves on the Fargues-Fontaine curve

摘录 / EXCERPT: Let E be a p-adic field, let F be the coefficient field defining the Fargues-Fontaine curve, and let $X^{\operatorname{alg}}_{E,F}$ denote its algebraic version. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019682 | MATHDB-355185

The SYZ conjecture on special Lagrangian fibrations

摘录 / EXCERPT: Let $f\colon\mathcal{X}^*\to\Delta^*=\{t\in\mathbb{C}\mid 0<|t|<1\}$ be a flat proper family that is a maximal degeneration, meaning that its monodromy has maximal unipotency index...

conjecture | open | MathDB 页面 | 原文线索

V2-019683 | MATHDB-355223

The p-adic weight-monodromy conjecture for Hyodo-Kato cohomology

摘录 / EXCERPT: Let Y be a proper smooth algebraic variety over K, and let $H^i_{\mathrm{HK}}(Y_{\bar{K}})$ denote its Hyodo-Kato cohomology, equipped with Frobenius, monodromy, and the monodr...

conjecture | open | MathDB 页面 | 原文线索

V2-019684 | MATHDB-355224

The p-adic weight-monodromy conjecture for proper smooth varieties over p-adic fields

摘录 / EXCERPT: Let X be a proper smooth algebraic variety over $\mathbb{C}_p$ defined over $\bar{\mathbb{Q}}_p$. Its Hyodo-Kato cohomology $H^i_{\mathrm{HK}}(X^{\mathrm{an}})$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-019685 | MATHDB-355242

Grothendieck's finite generation conjecture for punctured-spectrum fundamental groups

摘录 / EXCERPT: Grothendieck's conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-019686 | MATHDB-355248

Quantum cohomology semisimplicity conjecture for linear sections of (co)adjoint varieties

摘录 / EXCERPT: Let X be an adjoint or quasi-minuscule variety as above, and let $Y\subset X$ be a general linear section. Write ${\rm BQH}(Y)$ and ${\rm QH}(Y)$ for its small and big quantum co...

conjecture | open | MathDB 页面 | 原文线索

V2-019687 | MATHDB-355249

Canonical quantum cohomology conjecture for hyperplane sections of type A_n adjoint varieties

摘录 / EXCERPT: Let X be an adjoint variety of type A_n, and let $Y\subset X$ be a general hyperplane section. Let ${\rm QH}(X)_{\rm can}$ and ${\rm QH}(Y)_{\rm can}$ denote the restrictions o...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-019688 | MATHDB-355250

Kuznetsov-Smirnov Lefschetz collection conjecture for adjoint varieties

摘录 / EXCERPT: Let X be an adjoint variety that is not quasi-minuscule, let $Y\subset X$ be a general hyperplane section, let $c_1(X)$ and $c_1(Y)$ denote their first Chern indices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019689 | MATHDB-355264

Non-Archimedean Fujita approximation conjecture

摘录 / EXCERPT: Let $\overline{\mathcal{X}}$ be a smooth $(n+1)$-dimensional smooth projective variety, and let $\overline{\mathcal{L}}$ be a big line bundle over $\overline{\mathcal{X}}$. Fujita...

conjecture | open | MathDB 页面 | 原文线索

V2-019690 | MATHDB-355266

Semiampleness conjecture on klt Calabi--Yau pairs

摘录 / EXCERPT: Semiampleness conjecture on klt Calabi--Yau pairs. The divisor L is num-semiample; that is, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L' on X such that

conjecture | open | MathDB 页面 | 原文线索

V2-019691 | MATHDB-355276

The strong Green-Griffiths-Lang conjecture for special loci

摘录 / EXCERPT: Let X be a projective complex algebraic variety. Define $\,\operatorname{Sp}(X)$ as the union of the positive-dimensional integral closed subvarieties of X that are not of gene...

conjecture | open | MathDB 页面 | 原文线索

V2-019692 | MATHDB-355278

The equivariant Beilinson conjecture for abelian extensions

摘录 / EXCERPT: Equivariant Beilinson conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-019693 | MATHDB-355279

The motivic Lerch element conjecture for totally noncritical characters

摘录 / EXCERPT: Motivic Lerch element conjecture. The image of this element belongs to

conjecture | open | MathDB 页面 | 原文线索

V2-019694 | MATHDB-355285

Secure-torsor counting conjecture for finite étale group schemes

摘录 / EXCERPT: Secure-torsor counting conjecture. If F is a number field, there exists $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019695 | MATHDB-355286

The stacky Batyrev--Manin conjecture I

摘录 / EXCERPT: Let $\mathcal{X}$ be a nice stack over F, and let $(\mathcal{L},c)$ be a raised line bundle that is big and adequate. Suppose that $\mathcal{X}\langle F\rangle$ is Zariski dense....

conjecture | open | MathDB 页面 | 原文线索

V2-019696 | MATHDB-355304

Vakil-Wood motivic stabilization conjecture

摘录 / EXCERPT: Let $\mathcal U$ be a geometrically irreducible ungraded variety, and let $\mathcal X_n$ denote its space of effective zero-cycles of degree n. Write $\mathbb L$ for the Lefschet...

conjecture | open | MathDB 页面 | 原文线索

V2-019697 | MATHDB-355305

Farb-Wolfson-Wood motivic stabilization conjecture

摘录 / EXCERPT: Let $\mathcal U$ be a geometrically connected and smooth ungraded variety. For $\in\mathbb Z_{\geq 0}^{r}$, let $\mathcal Z_{,n}$ be the locus of ordered r-tuples of effective ze...

conjecture | open | MathDB 页面 | 原文线索

V2-019698 | MATHDB-355306

Generic invertibility conjecture for multilateration matrices

摘录 / EXCERPT: Let $m=5$ sensors have pairwise distinct, noncoplanar positions $\mathbf{a}_1,\ldots,\mathbf{a}_5\in\mathbb{R}^3$. For emission times determined by the multilateration model, let...

conjecture | refuted | MathDB 页面 | 原文线索

V2-019699 | MATHDB-355315

The motivic-weight-22 classification conjecture for geometric Galois representations

摘录 / EXCERPT: Let V_f be the geometric Galois representation associated with a Hecke eigenform, and let $\phi_1,\ldots,\phi_{11}$ be the representations associated with the 11 automorphic re...

conjecture | open | MathDB 页面 | 原文线索

V2-019700 | MATHDB-355319

Birational isotriviality conjecture for smooth families over special bases

摘录 / EXCERPT: Let $f:X\to Y$ be a smooth projective morphism with connected fibres between two quasi-projective manifolds. Assume that K_{X_y} is pseudo-effective for general $y\in Y$, and tha...

conjecture | open | MathDB 页面 | 原文线索

V2-019701 | MATHDB-355346

McLeman's Massey-product criterion for finiteness of p-class field towers

摘录 / EXCERPT: Let $p>3$ be a prime, and let K be an imaginary quadratic field whose class group has p-rank two. Set ... Choose a basis x,y of $H^1(G,\mathbb{Z}/p\mathbb{Z})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019702 | MATHDB-355373

Popa's conjecture on cohomological support loci and variation

摘录 / EXCERPT: Let $f\colon X\to A$ be a fibration from a smooth projective variety X to an abelian variety A. For a coherent sheaf $\mathcal{F}$ on A, write ... Let $\operatorname{Var}(f)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-019703 | MATHDB-355378

Existence conjecture for a smooth integral model of the orthogonal Shimura variety

摘录 / EXCERPT: Existence conjecture. There exists a flat, formally smooth and universally closed $\mathbb Z_p\otimes\mathcal O_K$-scheme $\mathcal M$ with $H(\mathbb A^{\infty,p}\otimes K)$-actio...

conjecture | open | MathDB 页面 | 原文线索

V2-019704 | MATHDB-355379

Conjectural invariant six-dimensional subvariety of the special fiber

摘录 / EXCERPT: Invariant-subvariety conjecture. The special fiber of $\mathfrak M_{\mathcal R,\Omega}$ contains a nonempty $Z(\mathbb A^{\infty,p})\times G(\mathbb A^{\infty,p}\otimes L^+)$-invar...

conjecture | open | MathDB 页面 | 原文线索

V2-019705 | MATHDB-355391

The homogeneous-space p-adic Grothendieck period conjecture

摘录 / EXCERPT: Homogeneous-space p-adic Grothendieck period conjecture. The variety $\mathcal H_M$ is irreducible and

conjecture | open | MathDB 页面 | 原文线索

V2-019706 | MATHDB-355396

Polynomiality of the stringy E-function for varieties with NCCRs

摘录 / EXCERPT: Stringy E-function polynomiality conjecture. The stringy E-function $E_{st}(X,u,v)$ is a polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-019707 | MATHDB-355442

Hermitianity conjecture for degree-$\sqrt{q}+1$ Frobenius nonclassical hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a Frobenius nonclassical hypersurface over $\mathbb{F}_q$, where $n\geq 2$, of degree $\sqrt{q}+1$. Assume that X is reduced and smooth at its...

conjecture | open | MathDB 页面 | 原文线索

V2-019708 | MATHDB-355529

The tame representation-type moduli product conjecture

摘录 / EXCERPT: Let (Q,I) be a bound quiver, and let $A=kQ/I$ be its bound quiver algebra. For a dimension vector $\mathbf{d}$, let $\operatorname{rep}_Q(I,\mathbf{d})$ denote the variety of rep...

conjecture | open | MathDB 页面 | 原文线索

V2-019709 | MATHDB-355532

Berkesch-Erman-Smith's virtual resolution length conjecture

摘录 / EXCERPT: Berkesch-Erman-Smith's conjecture. The module M admits a virtual resolution of length at most

conjecture | open | MathDB 页面 | 原文线索

V2-019710 | MATHDB-355537

Good-reduction conjecture for polarized morphisms

摘录 / EXCERPT: Let k be any complete non-Archimedean field, and let X be a projective k-variety. A morphism $f\colon X\to X$ is polarized if there exists an ample line bundle $L\to X$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-019711 | MATHDB-355538

Invariance of topological entropy under complete metrized field extension

摘录 / EXCERPT: Let k be a complete metrized field, let X be a projective variety defined over k, and let $f\colon X\dashrightarrow X$ be a dominant rational map. For any complete metrized f...

open_problem | open | MathDB 页面 | 原文线索

V2-019712 | MATHDB-355554

Dolgachev's lattice-polarized mirror conjecture for toric K3 hypersurfaces

摘录 / EXCERPT: Let P be a three-dimensional reflexive polytope, and let S_P and $S_{P^\turnstile}$ be generic anticanonical $K3$ hypersurfaces in the toric varieties associated with P and i...

conjecture | open | MathDB 页面 | 原文线索

V2-019713 | MATHDB-355585

Medvedev-Scanlon Zariski dense orbit conjecture

摘录 / EXCERPT: Let X be a projective variety over an algebraically closed field k of characteristic zero, and let $f:X\dasharrow X$ be a dominant rational self-map. Write $k(X)^f$ for the fie...

conjecture | open | MathDB 页面 | 原文线索

V2-019714 | MATHDB-355593

Alternating-sign expansion conjecture for CSM classes of Richardson cells

摘录 / EXCERPT: Alternating-sign conjecture. For every $u,v,w\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-019715 | MATHDB-355611

Kontsevich-Soibelman motivic integral identity conjecture

摘录 / EXCERPT: Let k be the coefficient field, let d_1,d_2,d_3 be nonnegative integers, and write $x=(x_1,\ldots,x_{d_1})$, $y=(y_1,\ldots,y_{d_2})$, and $z=(z_1,\ldots,z_{d_3})$. For a tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-019716 | MATHDB-355631

Yau's compactification conjecture for complete Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a complete Calabi-Yau manifold with Euclidean volume growth. Yau's compactification conjecture. The manifold X is naturally a quasi-projective variety. This is a speci...

conjecture | open | MathDB 页面 | 原文线索

V2-019717 | MATHDB-355633

Uniform equivalence conjecture for compatible Calabi-Yau metrics

摘录 / EXCERPT: Let X be a normal affine variety, let v be a fixed polystable negative valuation on X, and let $\mathcal M_v$ be the space of compatible Calabi-Yau metrics ω on X sa...

conjecture | open | MathDB 页面 | 原文线索

V2-019718 | MATHDB-355644

The simplex-like compactification conjecture for cluster varieties

摘录 / EXCERPT: Let Y be a cluster variety that is locally acyclic and of full rank. Write $Y_{>0}$ for its positive part and $Omega(Y_{>0})$ for its natural top form. The simplex-like compactif...

conjecture | open | MathDB 页面 | 原文线索

V2-019719 | MATHDB-355657

The hyperkähler twistor conjecture for higher complex structures

摘录 / EXCERPT: Let Σ be a smooth surface, let n be a positive integer, and let $\mathcal{T}^n$ be the moduli space of higher complex structures of order n on Σ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019720 | MATHDB-355665

Continuity conjecture for variations of Newton-Okounkov bodies

摘录 / EXCERPT: Let X be a projective variety of dimension d, let L be a big line bundle on X, and let D be a simple normal crossing divisor whose dual cone complex $\Sigma(D)$ is pure o...

conjecture | open | MathDB 页面 | 原文线索

V2-019721 | MATHDB-355666

Continuity conjecture on higher rank skeleta

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over an algebraically closed field κ, and let L be a big line bundle on X. For each valuation ν in the higher...

conjecture | open | MathDB 页面 | 原文线索

V2-019722 | MATHDB-355683

The improved bound for big and nef tautological bundles on Enriques surfaces

摘录 / EXCERPT: Let F be a $(k-1)$-very ample bundle of rank r on an Enriques surface. Write ... where $v=v(F)=\operatorname{ch}(F)\sqrt{\operatorname{Td}(X)}$ is the Mukai vector. Improved boun...

conjecture | open | MathDB 页面 | 原文线索

V2-019723 | MATHDB-355708

He’s conjecture on connected components of affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be the reductive group, let $b\in G(\breve{F})$ and let μ be a cocharacter, with associated affine Deligne-Lusztig variety $X^{\mathcal{K}_p}_\mu(b)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019724 | MATHDB-355715

Algebraic Green-Griffiths-Lang conjecture for log varieties

摘录 / EXCERPT: Let X be a smooth projective variety and D a simple normal crossings divisor such that K_X+D is big. Let L be an ample line bundle on X, and let $g(C)$ denote the geometr...

conjecture | open | MathDB 页面 | 原文线索

V2-019725 | MATHDB-355734

Borcherds-lift conjecture for motivic cycles on the twisted moduli space

摘录 / EXCERPT: Let X_Δ be the twisted moduli space, let Ξ_{Δ_1} be the sum of the motivic cycles $\Xi_{\Delta_1}^c$, and let ρ_L be the Weil representation attached to the tr...

conjecture | open | MathDB 页面 | 原文线索

V2-019726 | MATHDB-355775

The orientation conjecture for stable-pair moduli on Calabi-Yau fourfolds

摘录 / EXCERPT: Let $f:\mathcal{X}\to\mathcal{B}$ be a smooth projective morphism with connected fibres of relative dimension 4 to a smooth connected affine scheme $\mathcal{B}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-019727 | MATHDB-355780

The rigidity conjecture for automorphism groups of affine varieties

摘录 / EXCERPT: Let $\mathbb K$ be an algebraically closed field of characteristic zero. Let Y be an affine algebraic variety. Call Y rigid if it admits no effective $\mathbb G_{\rm a}$-action...

conjecture | open | MathDB 页面 | 原文线索

V2-019728 | MATHDB-355781

Discreteness conjecture for automorphism groups without additive and multiplicative actions

摘录 / EXCERPT: Let Y be an affine variety over an algebraically closed field of characteristic zero. Assume that Y admits no effective action of the additive group $\mathbb G_{\rm a}$ and no...

conjecture | open | MathDB 页面 | 原文线索

V2-019729 | MATHDB-355787

Nekovář-Scholl's plectic conjecture for Shimura varieties

摘录 / EXCERPT: Consider a Shimura datum (G,X) with $G=\operatorname{Res}_{F/\mathbb{Q}}H$, where H is a connected reductive group over a totally real field F. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019730 | MATHDB-355791

The polynomial decomposition criterion for holomorphic forms

摘录 / EXCERPT: Let X be a smooth projective variety. A degree l polynomial decomposition of the diagonal means a decomposition of the diagonal whose non-supported part lies in the subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-019731 | MATHDB-355792

Nori's conjecture on restrictions of cycles to subvarieties

摘录 / EXCERPT: Let X be a smooth projective variety, let i be an integer, and let $w\in\operatorname{CH}^i(X)$ be a cycle. Nori's conjecture. If $w|_T=0\in\operatorname{CH}_0(T)$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-019732 | MATHDB-355793

Voisin's polynomial decomposition conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X be a smooth projective hyper-Kähler $2n$-fold, let h_X be a polarization, and let $Z_{\mathrm{lef}}\in\operatorname{CH}^2(X\times X)$ be a cycle such that its cohomologic...

conjecture | open | MathDB 页面 | 原文线索

V2-019733 | MATHDB-355801

Bounded multiplicities for irreducible fibers of Fano fibrations

摘录 / EXCERPT: Let d be a positive integer and ϵ a positive real number. Let X be a normal variety of dimension d and let $\pi:X\to C$ be a contraction to a curve C. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-019734 | MATHDB-355803

Non-accumulation conjecture for minimal log discrepancies of rationally connected Calabi-Yau varieties

摘录 / EXCERPT: Let d be a positive integer. Consider the minimal log discrepancies of all normal projective Calabi-Yau rationally connected varieties of dimension d. Minimal-log-discrepancy n...

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V2-019735 | MATHDB-355804

Bounded index conjecture for rationally connected klt Calabi-Yau varieties

摘录 / EXCERPT: Let d be a positive integer. Consider any d-dimensional klt Calabi-Yau rationally connected projective variety X. Rationally connected Calabi-Yau index conjecture. There exis...

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V2-019736 | MATHDB-355805

Boundedness modulo flops for rationally connected klt Calabi-Yau varieties

摘录 / EXCERPT: Let d be a positive integer. Consider all d-dimensional klt Calabi-Yau rationally connected projective varieties. Rationally connected Calabi-Yau boundedness conjecture modulo...

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V2-019737 | MATHDB-355818

Component-count conjecture for positive definite X_9 singularities

摘录 / EXCERPT: $+X_9$ component conjecture. Each of the seven virtual components is represented by exactly one connected component of the complement of the discriminant variety of any versal defo...

conjecture | open | MathDB 页面 | 原文线索

V2-019738 | MATHDB-355819

Component-count conjecture for X_9^2 singularities

摘录 / EXCERPT: X_9^2 component conjecture. These representatives exhaust the components of the complement of the discriminant variety; in total there are 52 such components.

conjecture | open | MathDB 页面 | 原文线索

V2-019739 | MATHDB-355820

Component-count conjecture for J_{10}^1 singularities

摘录 / EXCERPT: J_{10}^1 component conjecture. The complement of the discriminant variety has exactly 11 components, represented by these six original and five transformed topological types.

conjecture | open | MathDB 页面 | 原文线索

V2-019740 | MATHDB-355821

Component-count conjecture for J_{10}^3 singularities

摘录 / EXCERPT: J_{10}^3 component conjecture. The complement of the discriminant variety has exactly 33 distinct components.

conjecture | open | MathDB 页面 | 原文线索

V2-019741 | MATHDB-355844

Rational-singularities conjecture for mixed-characteristic local models

摘录 / EXCERPT: Let $\underline{\mathcal{G}}$ be the relevant parahoric group scheme and let μ be the specified cocharacter datum, and write $\widetilde{M}_{\underline{\mathcal{G}},\mu}$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-019742 | MATHDB-355855

The coefficient formula conjecture for generalized Hessenberg varieties

摘录 / EXCERPT: Coefficient formula conjecture. For every partition $\lambda=(n-j,j)$, the coefficient c_λ in the equivariant Poincaré polynomial of X_{h_2} is given by the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-019743 | MATHDB-355870

Electrical canonical basis stratification conjecture for electroid varieties

摘录 / EXCERPT: Let $\mathcal{X}_n$ be the space of electrical networks, let $\mathcal{X}_\tau$ be the electroid variety indexed by a matching τ on $2n$ points, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019744 | MATHDB-355895

The filtration-singular-support conjecture for nilpotent subschemes

摘录 / EXCERPT: The filtration-singular-support conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019745 | MATHDB-355951

Dense-set equality of multiplier ideals and test ideals

摘录 / EXCERPT: Let X be a smooth, irreducible complex algebraic variety and let $f\in\mathcal{O}_X(X)$ be nonzero. Choose a model (X_A,f_A) over a finitely generated ${\mathbf Z}$-subalgebra...

open_problem | open | MathDB 页面 | 原文线索

V2-019746 | MATHDB-355952

Stapledon's equivalence conjecture for equivariant H*-series

摘录 / EXCERPT: Let G be a finite group acting on a lattice, and let P be a G-invariant lattice polytope. Let Y be the toric variety with ample line bundle L associated to P, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019747 | MATHDB-355988

Conjectural t-deformation of Shalika germs via the Hilbert-scheme sheaf

摘录 / EXCERPT: t-deformation conjecture. The coefficients $\widehat{\Gamma}_\lambda(\gamma)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-019748 | MATHDB-355994

Divisibility conjecture for the tame reciprocity map

摘录 / EXCERPT: Let X be a smooth quasi-projective variety over the local field k. Write ... for the tame reciprocity map, where $E^{\rm t}(X)$ and $J^{\rm t}(X)$ are the tame class group and ta...

conjecture | open | MathDB 页面 | 原文线索

V2-019749 | MATHDB-356036

Liu's minimal mld conjecture for standard-coefficient Calabi-Yau pairs

摘录 / EXCERPT: Liu's minimal mld conjecture. The number $1/(s_{n+1}-1)$ is the smallest possible minimal log discrepancy among all n-dimensional klt Calabi-Yau pairs with standard coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-019750 | MATHDB-356037

The extremal orbifold Betti-number and Euler-characteristic conjecture

摘录 / EXCERPT: Let n be a positive integer, and consider projective n-dimensional varieties with quotient singularities and trivial canonical class. Write H for the sum of their orbifold Be...

conjecture | open | MathDB 页面 | 原文线索

V2-019751 | MATHDB-356038

The maximal-index conjecture for klt Calabi-Yau varieties

摘录 / EXCERPT: Let $n\geq2$, let X be the weighted Calabi-Yau hypersurface constructed in the source, and let G be the cyclic subgroup acting freely in codimension one. Let $m=|G|$. Maximal-i...

conjecture | open | MathDB 页面 | 原文线索

V2-019752 | MATHDB-356047

Expected-codimension conjecture for Weddle schemes of general points

摘录 / EXCERPT: Let $d\geq 3$ and let Z be a set of $\binom{d+2}{2}-1$ general points in $\mathbb{P}^3$. The d-Weddle scheme of Z is the scheme associated to the loci of points from which th...

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V2-019753 | MATHDB-356054

Formal Ax-Lindemann conjecture for ordinary Siegel moduli schemes

摘录 / EXCERPT: Formal Ax-Lindemann conjecture. Assume that $\mathfrak{V}$ is a closed formal subscheme of an open formal subscheme $\mathfrak{U}$ of $\mathfrak{S}$, that $x\in\mathfrak{V}(k)$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-019754 | MATHDB-356055

Joint unlikely almost-intersections conjecture for ordinary Siegel moduli schemes

摘录 / EXCERPT: Joint unlikely almost-intersections conjecture. If $Z\subset\mathfrak{V}_k$ is a closed subscheme contained in that Zariski closure for all $\epsilon>0$, then $\mathfrak{V}$ contai...

conjecture | open | MathDB 页面 | 原文线索

V2-019755 | MATHDB-356056

Emerton-Gee-Hellmann overconvergence conjecture for étale (φ,Γ)-modules

摘录 / EXCERPT: Emerton-Gee-Hellmann overconvergence conjecture. Every étale (φ,Γ)-module over $\mathrm{{\bf A}}_{K,A}$ canonically descends to an étale (φ,Γ)-module over...

conjecture | open | MathDB 页面 | 原文线索

V2-019756 | MATHDB-356087

The induction formula for fundamental classes in equivariant Lagrangian homology

摘录 / EXCERPT: Let X and X' be the varieties in the induction diagram, with a map $g\relationships W\to X'$ and the induced map ... where ... Let...

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V2-019757 | MATHDB-356088

The conjectural dotted-arrow identification for seminilpotent quiver varieties

摘录 / EXCERPT: Let Q be the quiver, let $\mathbf{d}\in\mathbb{N}^{Q_0}$, and let $\mathfrak{N}_{\Pi_Q,\mathbf{d}}^{\mathcal{SSN}}$ be the seminilpotent stack, with irreducible components...

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V2-019758 | MATHDB-356092

Marian-Oprea-Pandharipande's rank-zero Segre integral conjecture

摘录 / EXCERPT: Let S be a smooth projective surface over $\mathbb{C}$, let $S^{[n]}$ be the Hilbert scheme of n points on S, and let...

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V2-019759 | MATHDB-356120

Mordell-Lang conjecture for Shimura varieties

摘录 / EXCERPT: Let S be a Shimura variety. For a subset $\boldsymbol{\Xi}\to S$, let $\boldsymbol{\bar{\Xi}}$ be the union in S of the generalised Hecke orbits of...

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V2-019760 | MATHDB-356121

Mordell-Lang conjecture for Shimura varieties: Zariski-closure formulation

摘录 / EXCERPT: Let S be a Shimura variety, let $\boldsymbol{\Sigma}\to S$ be a structure of finite rank, and call a weakly special subvariety $\boldsymbol{\Sigma}$-s...

conjecture | open | MathDB 页面 | 原文线索

V2-019761 | MATHDB-356122

André-Pink-Zannier conjecture for Shimura varieties

摘录 / EXCERPT: Let S be a Shimura variety and let $\boldsymbol{\Xi}\to S$ be a subset of the generalised Hecke orbit of a point in S. Write...

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V2-019762 | MATHDB-356132

Endoscopic averaging formula for local Shimura cohomology

摘录 / EXCERPT: Let G be a reductive group, let μ be a cocharacter with reflex field E_{μ}, and let $(H,\mathcal{H},s,\eta)$ be an endoscopic datum such that an L-parameter ϕ fact...

conjecture | open | MathDB 页面 | 原文线索

V2-019763 | MATHDB-356134

Signature independence of pure supergroup gauge-theory partition functions

摘录 / EXCERPT: Signature-independence conjecture. The partition function Z does not depend on the signatures of the NS5-branes surrounding the stack of D4-branes (or D5-branes), nor on the orde...

conjecture | open | MathDB 页面 | 原文线索

V2-019764 | MATHDB-356159

The moduli conjecture for uniformly stable varieties with big anticanonical class

摘录 / EXCERPT: Consider birational equivalence classes of varieties (X,L) with L big, in the situation where some representative has $L=-K_X$. A variety is uniformly K-stable or uniformly Din...

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V2-019765 | MATHDB-356171

Finiteness conjecture for vanishing modified diagonals of Hurwitz curves

摘录 / EXCERPT: Let X be a smooth projective Hurwitz curve over $\mathbb{C}$, meaning that its automorphism group has order $84(g-1)$ when X has genus g. Let ξ be the distinguished degre...

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V2-019766 | MATHDB-356177

Demailly-Green-Griffiths-Javanpeykar-Kamenova-Lang hyperbolicity equivalence conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over L. Consider the four properties: algebraic hyperbolicity over L, boundedness over L, grouplessness over L, and hyperbolicity ove...

conjecture | open | MathDB 页面 | 原文线索

V2-019767 | MATHDB-356189

Saito's conjecture on cohomological characteristic classes

摘录 / EXCERPT: Let k be a perfect field, let X be a closed subscheme of a smooth scheme over k, and let Λ be the finite local coefficient ring in the setup. Let $\mathcal F$ be a co...

conjecture | open | MathDB 页面 | 原文线索

V2-019768 | MATHDB-356190

Milnor-type formula for the non-acyclicity class

摘录 / EXCERPT: Let k be a perfect field, let X be a smooth scheme over k, let $\mathcal F\in D_{\rm ctf}(X,\Lambda)$, and let $SS(\mathcal F)$ be its singular support. Let $Z\subseteq X$ be...

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V2-019769 | MATHDB-356216

The strong existential closedness conjecture for the modular j-function

摘录 / EXCERPT: Let $n\geq 1$ and let $V\subseteq\mathbb{C}^n\times\mathrm{Y}(1)^n$ be an algebraic variety. A point of V is generic over a field K if its transcendence degree over K equals...

conjecture | open | MathDB 页面 | 原文线索

V2-019770 | MATHDB-356217

The existential and strong existential closedness conjectures for the J-function

摘录 / EXCERPT: Let $n\geq 1$, let $\mathrm{Y}_2(1)$ be the target variety for the tuple-valued function J, and let E_J^n be its graph over $\mathbb{H}^n$. A variety...

conjecture | open | MathDB 页面 | 原文线索

V2-019771 | MATHDB-356234

The MCK decomposition conjecture for Fano varieties of K3 type

摘录 / EXCERPT: MCK decomposition conjecture. Every Fano variety X of K3 type has an MCK decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-019772 | MATHDB-356283

Conjecture on instanton corrections and the local mirror map

摘录 / EXCERPT: Let X be a smooth variety with a smooth nef anticanonical divisor D, and consider the local Calabi-Yau manifold $\mathcal O_X(-D)$. The instanton corrections are the correction...

conjecture | open | MathDB 页面 | 原文线索

V2-019773 | MATHDB-356314

Conjectural Verlinde-Segre correspondence for Hilbert schemes

摘录 / EXCERPT: Verlinde-Segre correspondence. The universal series satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-019774 | MATHDB-356316

The modified Hodge conjecture for smooth projective manifolds

摘录 / EXCERPT: Modified Hodge conjecture. The group

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V2-019775 | MATHDB-356324

The tropical Poincaré-Prym formula

摘录 / EXCERPT: The tropical Poincaré-Prym formula. The cycle class of $\widetilde{Y}_d$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-019776 | MATHDB-356335

Popa's superadditivity conjecture for logarithmic Kodaira dimension

摘录 / EXCERPT: Let $f:X\to Y$ be a smooth projective morphism with connected fibers between smooth quasi-projective varieties. Write $\bar\kappa$ for logarithmic Kodaira dimension, and let F be...

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V2-019777 | MATHDB-356336

The two-singular-fiber uniruledness conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, and let $f:X\to\mathbb{P}^1$ be a surjective morphism with at most 2 singular fibers. Two-singular-fiber uniruledness conjecture. Then X...

conjecture | open | MathDB 页面 | 原文线索

V2-019778 | MATHDB-356337

Higher-dimensional uniruledness conjecture for fibrations over projective space

摘录 / EXCERPT: Let X be a smooth projective variety, and let $f:X\to\mathbb{P}^n$ be a surjective morphism. Let $\Delta(f)$ denote the discriminant locus of f. Higher-dimensional uniruledness...

conjecture | open | MathDB 页面 | 原文线索

V2-019779 | MATHDB-356340

Kuga-Satake Hodge conjecture for projective hyper-Kähler varieties

摘录 / EXCERPT: Let X be a projective hyper-Kähler variety, and let $H^2_{\mathrm{tr}}(X)$ denote its transcendental second cohomology. Let $\mathrm{KS}(X)$ be the Kuga-Satake variety associated...

conjecture | open | MathDB 页面 | 原文线索

V2-019780 | MATHDB-356342

Viehweg hyperbolicity conjecture for KSB-stable Whitney equisingular families

摘录 / EXCERPT: Viehweg hyperbolicity conjecture for KSB-stable families. If f has maximal variation, then V is of log general type.

conjecture | open | MathDB 页面 | 原文线索

V2-019781 | MATHDB-356353

Kontsevich-Zagier conjecture on linear relations between periods

摘录 / EXCERPT: Conjecture de Kontsevich-Zagier. Toutes les relations linéaires entre périodes découlent de l'additivité, du changement de variables et de la formule de Stokes.

conjecture | open | MathDB 页面 | 原文线索

V2-019782 | MATHDB-356354

Ayoub's precise Kontsevich-Zagier conjecture for algebraic holomorphic functions

摘录 / EXCERPT: Conjecture précisée de Kontsevich-Zagier. Le noyau de cette application est le $\mathbb{Q}$-sous-espace vectoriel engendré par les éléments

conjecture | open | MathDB 页面 | 原文线索

V2-019783 | MATHDB-356355

Grothendieck's period conjecture for motivic Galois groups

摘录 / EXCERPT: Conjecture des périodes de Grothendieck. Le degré de transcendance de ce sous-anneau coïncide avec la dimension du groupe de Galois motivique G_p^X.

conjecture | open | MathDB 页面 | 原文线索

V2-019784 | MATHDB-356383

Popa-Schnell's conjecture on direct images of pluricanonical bundles

摘录 / EXCERPT: Let the base field be an algebraically closed field of characteristic zero. Let $f:X\to Y$ be a morphism from a smooth projective variety X to a smooth projective variety Y of...

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V2-019785 | MATHDB-356386

The Virasoro constraint for moduli of sheaves

摘录 / EXCERPT: Let X be a smooth projective variety and let M be a moduli space of sheaves on X with virtual fundamental class $[M]^{\mathrm{vir}}$. Let $\mathbb{D}^X$ be the descendent alg...

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V2-019786 | MATHDB-356387

The Virasoro conjecture for moduli of pairs

摘录 / EXCERPT: Let X be a smooth projective variety, let V be a fixed sheaf on X, and let P be a moduli space of pairs consisting of a sheaf F and a map $V\to F$, with universal pair...

conjecture | open | MathDB 页面 | 原文线索

V2-019787 | MATHDB-356396

The stratification conjecture for proper morphisms to abelian varieties

摘录 / EXCERPT: Stratification conjecture. There is a stratification $\{S_\alpha\}$ of X by locally closed smooth subvarieties such that, for each α, there is a positive-dimensional abeli...

conjecture | open | MathDB 页面 | 原文线索

V2-019788 | MATHDB-356414

The trace conjecture for perverse schobers

摘录 / EXCERPT: Let X be an algebraic variety or stack, and let $\mathbb{L}\subset T^*X$ be a conic Lagrangian. Write $\mathfrak{L}X$ for the loop space, and let...

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V2-019789 | MATHDB-356424

Weak and strong Eisenbud-Goto conjectures for projective toric varieties

摘录 / EXCERPT: Let $X\subseteq\mathbb{P}^r$ be a projective toric variety. Its maximal generator degree is denoted by $\operatorname{maxdeg} X$, and its regularity, degree, and codimension by...

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V2-019790 | MATHDB-356456

Finite N holographic conjecture for local operators of six-dimensional superconformal theories

摘录 / EXCERPT: Let N be a positive integer, and let $Obs_N(0)$ denote the local operators of the theory on a stack of N fivebranes wrapping C^3 in $R\times C^5$. Let $\mathcal G_N$ be the l...

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V2-019791 | MATHDB-356474

Asymptotic Riemann-Roch conjecture for the relative second Chern class

摘录 / EXCERPT: Assume that the base field k has characteristic zero. Let $f:\tilde X\to X$ be the morphism under consideration, and let $\mathcal F$ be a rank r vector bundle on $\tilde X$. W...

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V2-019792 | MATHDB-356479

Trinh's unipotent-variety twisting conjecture

摘录 / EXCERPT: Let G be a complex, connected, reductive algebraic group, let B_+ and B_- be opposed Borel subgroups, and let U_+ and U_- be their unipotent radicals. Let...

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V2-019793 | MATHDB-356480

Twisted braid homology-unipotent variety conjecture

摘录 / EXCERPT: Let G be a split reductive group with Weyl group W, root lattice Φ, and $V=\mathbf{Z}\Phi\otimes_{\mathbf{Z}}\mathbf{Q}$. Let Br_W^+ denote the positive braid m...

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V2-019794 | MATHDB-356549

Conjectured minimal volume for exceptional klt Fano varieties

摘录 / EXCERPT: Minimal-volume conjecture for exceptional Fano varieties. For every integer $n\geq 2$, X_n has the minimum anticanonical volume among all exceptional klt Fano n-folds.

conjecture | open | MathDB 页面 | 原文线索

V2-019795 | MATHDB-356558

Poincaré-type cscK metric and relative K-polystability conjecture

摘录 / EXCERPT: Let X be a polarized variety with polarization L_X, and let D be a divisor. A metric on $X\setminus D$ is said to be of Poincaré type when it has the prescribed Poincaré asym...

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V2-019796 | MATHDB-356559

Log Yau-Tian-Donaldson conjecture for cscK cone metrics

摘录 / EXCERPT: Let X be a polarized variety with polarization L_X, let D be a divisor, and fix a cone angle along D. A cscK cone metric is a Kähler cone metric on (X,D) with constant sc...

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V2-019797 | MATHDB-356583

Leclerc's conjecture on the cluster algebra of Richardson varieties

摘录 / EXCERPT: Let $v\leq w$ be elements indexing an open Richardson variety $\mathring{\mathcal{R}}_{v,w}$, and let $\mathbf{w}$ be the chosen word defining Leclerc's seed...

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V2-019798 | MATHDB-356584

Fraser's quasi-cluster transformation conjecture for Richardson twists

摘录 / EXCERPT: Let $v\leq w$ and let $\tau_{v,w}:\mathring{\mathcal{R}}_{v,w}\to\mathring{\mathcal{R}}_{v,w}$ be the right twist map. A quasi-cluster transformation means a sequence of cluster mu...

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V2-019799 | MATHDB-356587

Bronowski's conjecture on identifiability and tangential projections

摘录 / EXCERPT: Bronowski's conjecture. The variety X is h-identifiable if and only if a general $(h-1)$-tangential projection

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V2-019800 | MATHDB-356588

Modified Bronowski conjecture on identifiability and secant-variety degree

摘录 / EXCERPT: Modified Bronowski conjecture. If a general $(h-1)$-tangential projection of X is birational and its image is a variety of minimal degree, then X is h-identifiable and

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V2-019801 | MATHDB-356589

Bronowski conjecture for varieties with non-degenerate Gauss map

摘录 / EXCERPT: Non-degenerate Gauss-map conjecture. The variety X is h-identifiable if and only if a general $(h-1)$-tangential projection

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V2-019802 | MATHDB-356617

The divisible-plus-finite conjecture for the Albanese kernel of a smooth projective variety

摘录 / EXCERPT: Divisible-plus-finite conjecture. The group $F^2(X)$ is the direct sum of a divisible group and a finite group.

conjecture | open | MathDB 页面 | 原文线索

V2-019803 | MATHDB-356626

The cubic hypersurface limit conjecture

摘录 / EXCERPT: Cubic hypersurface limit conjecture. K-polystable limits of cubic hypersurfaces are cubic hypersurfaces.

conjecture | open | MathDB 页面 | 原文线索

V2-019804 | MATHDB-356646

The associated-variety conjecture for the simple vertex algebra $L_{-2}(G_2)$

摘录 / EXCERPT: Let $L_{-2}(G_2)$ be the simple vertex algebra at level -2, and let $X_{L_{-2}(G_2)}$ denote its associated variety. Let $\mathbb{O}_{\mathrm{subreg}}$ be the subregular nilpoten...

conjecture | open | MathDB 页面 | 原文线索

V2-019805 | MATHDB-356648

Bloch-Beilinson conjecture on Abel-Jacobi maps

摘录 / EXCERPT: Let $V/\overline{\mathbb Q}$ be a smooth quasiprojective variety. Let ... be the Abel-Jacobi map. Bloch-Beilinson conjecture on Abel-Jacobi maps. The map $\Phi_{k,\mathbb Q}$ is inje...

conjecture | open | MathDB 页面 | 原文线索

V2-019806 | MATHDB-356654

The mixed-motive properad conjecture for framed curves

摘录 / EXCERPT: Let $\mathbfcal M_{g,k}$ denote the logarithmic stack of formal curves with framings, and let $\textsf N^{fr}_{\mathrm{LMM}}$, $\textsf N^{fr}_{\mathrm{MHS}}$, and...

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V2-019807 | MATHDB-356701

The tensor-induction local-global compatibility conjecture

摘录 / EXCERPT: Let $K=F_v$ be unramified, let π be an automorphic admissible smooth representation of ${\rm GL}_2(K)$ over $\mathbb{F}$, and let $\overline{r}_v$ be its corresponding local Ga...

conjecture | open | MathDB 页面 | 原文线索

V2-019808 | MATHDB-356723

Birationality conjecture for pluricanonical systems of general type varieties with many two-forms

摘录 / EXCERPT: Let $n\geq 5$. For a smooth projective n-fold X of general type, write $h^{2,0}(X)$ for the dimension of its space of global holomorphic two-forms, and let r_{n-2} denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-019809 | MATHDB-356730

The ideal-generation conjecture for restricted secant varieties of Grassmannians

摘录 / EXCERPT: Let V be an n-dimensional vector space and let $X=\sigma_s^r(\operatorname{Gr}(k,n))$ be a non-trivial restricted secant variety. For $w\in\sigma_s^r(\operatorname{Gr}(k,V))$,...

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V2-019810 | MATHDB-356768

Arithmetic Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Let F be a number field, (X_{η},L_{η}) a normal polarized projective variety over F, and let (X,L,h_L) range over all metrized polarized normal models over...

conjecture | open | MathDB 页面 | 原文线索

V2-019811 | MATHDB-356779

Flat/sharp Iwasawa main conjecture for modular abelian varieties over an anticyclotomic extension

摘录 / EXCERPT: Let f be the modular form and K_∞ the anticyclotomic extension considered in the paper, let A_f be the associated p-primary representation, and let Λ be the re...

conjecture | open | MathDB 页面 | 原文线索

V2-019812 | MATHDB-356782

The crystalline companion conjecture for irreducible local systems

摘录 / EXCERPT: Let S be the base scheme, let X be the relevant smooth variety over S with boundary divisors D_i, and let T_i prescribe the monodromy at infinity. Suppose there is one, o...

conjecture | open | MathDB 页面 | 原文线索

V2-019813 | MATHDB-356784

Exterior-power Euler characteristic conjecture for higher-rank Quot schemes

摘录 / EXCERPT: Exterior-power conjecture. For every $k\leq n+r(a+1)$,

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V2-019814 | MATHDB-356785

Symmetric-power Euler characteristic conjecture for higher-rank Quot schemes

摘录 / EXCERPT: Symmetric-power conjecture. For every $k\leq n+r(a+1)$,

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V2-019815 | MATHDB-356786

The simple-factor decomposition conjecture for Jacobians of Fermat-type curves

摘录 / EXCERPT: Let m be a positive odd integer, and let J_m denote the Jacobian associated with the curve $C_m\colon y^2=x^m-1$. Write $\phi(d)$ for Euler's totient function. The simple-facto...

conjecture | open | MathDB 页面 | 原文线索

V2-019816 | MATHDB-356787

The Mumford-Tate projection conjecture for J_{p^3}

摘录 / EXCERPT: Let p be an odd prime, let J_{p^3} be the Jacobian of $C_{p^3}\colon y^2=x^{p^3}-1$, and write $J_{p^3}\simeq X_2\times X_1\times J_p$, where X_2 is the absolutely simple fac...

conjecture | open | MathDB 页面 | 原文线索

V2-019817 | MATHDB-356797

Bloch-Beilinson's filtration conjecture for higher Chow groups

摘录 / EXCERPT: Bloch-Beilinson filtration conjecture. There exists a filtration

conjecture | open | MathDB 页面 | 原文线索

V2-019818 | MATHDB-356798

Kollár-Szabó's finiteness conjecture for zero-cycles over local fields

摘录 / EXCERPT: Kollár-Szabó conjecture. The group

conjecture | open | MathDB 页面 | 原文线索

V2-019819 | MATHDB-356805

Heath, Pemmaraju and Trenk's bounded stack-number conjecture for outerplanar DAGs

摘录 / EXCERPT: Let G be a directed acyclic outerplanar graph, and let $\operatorname{sn}(G)$ denote its stack number. Heath, Pemmaraju and Trenk's conjecture. The stack number of the class of d...

conjecture | open | MathDB 页面 | 原文线索

V2-019820 | MATHDB-356808

Finiteness conjecture for motivic cohomology of modular motives

摘录 / EXCERPT: Let K be a number field, let X be the Kuga-Sato variety defining the modular motive $\mathcal{M}$, and let $H^1_\operatorname{mot}(K,\mathcal{M})$ be its first motivic cohomolo...

conjecture | open | MathDB 页面 | 原文线索

V2-019821 | MATHDB-356809

Finite generation conjecture for Chow groups over number fields

摘录 / EXCERPT: Let Y be a smooth projective variety defined over a number field L, and let $\operatorname{CH}^n(Y)$ denote its Chow group of codimension-n algebraic cycles. Chow-group finit...

conjecture | open | MathDB 页面 | 原文线索

V2-019822 | MATHDB-356810

Non-degeneracy conjecture for the Gillet-Soulé height pairing

摘录 / EXCERPT: Height non-degeneracy conjecture. The pairing $\langle\cdot,\cdot\rangle_{\operatorname{GS},\infty}$ is non-degenerate. The conjecture is motivated by arithmetic analogues of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-019823 | MATHDB-356814

Integral p-adic regulator conjecture for modular motives

摘录 / EXCERPT: Integral regulator conjecture. The map $\mathtt{reg}_{L,p}$ is an isomorphism of ${\mathcal O}_p$-modules for all but finitely many primes p at which f is p-isolated. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-019824 | MATHDB-356841

The parity conjecture for ranks of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field K. Write $\operatorname{rk} A$ for its Mordell-Weil rank, and let $w(A/K_v)$ denote the local root number at each place v of...

conjecture | open | MathDB 页面 | 原文线索

V2-019825 | MATHDB-356848

Non-equivariant positivity conjecture for CSM classes of Schubert cells

摘录 / EXCERPT: Let G/B be the flag variety of a reductive group G over $\mathbb{C}$, with B a fixed Borel subgroup, and let W be its Weyl group. For $w\in W$, write $Y(w)^\circ=B^-wB/B$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-019826 | MATHDB-356849

Kumar's positivity conjecture for CSM classes of Richardson cells

摘录 / EXCERPT: Let G/B be the flag variety of a reductive group G over $\mathbb{C}$, with Weyl group W. For $u,v\in W$, let $X(u)^\circ$ and $Y(v)^\circ$ be the Schubert and opposite Schube...

conjecture | open | MathDB 页面 | 原文线索

V2-019827 | MATHDB-356850

Mihalcea's alternating positivity conjecture for SSM classes

摘录 / EXCERPT: Let G/B be the flag variety with Weyl group W, and let $Y(u)^\circ$ denote the opposite Schubert cell. Write s_{SM} for the Segre-Schwartz-MacPherson class and...

conjecture | open | MathDB 页面 | 原文线索

V2-019828 | MATHDB-356858

Homogeneous-fibre conjecture for stationary measures in Case 2.3

摘录 / EXCERPT: Let $G=\operatorname{PGL}_n(\mathbb{R})$. Let X and the groups Q, R_0, R, $S\simeq Q/R_0$, $\Lambda\simeq R/R_0$, and H be as defined above. Suppose that we are in Case 2...

conjecture | open | MathDB 页面 | 原文线索

V2-019829 | MATHDB-356871

Analytification conjecture for the closed string mirror

摘录 / EXCERPT: Let $\mathcal{R}$ be an eigenray diagram, let $Y_\mathcal{R}$ be the non-proper scheme over the Novikov field obtained from the associated toric variety by the stated Kaliman modif...

conjecture | open | MathDB 页面 | 原文线索

V2-019830 | MATHDB-356878

Smoothness criterion for morphisms to simple abelian varieties

摘录 / EXCERPT: Smoothness criterion. The morphism f is smooth if and only if there is a holomorphic 1-form $\omega\in H^0(A,\Omega_A^1)$ such that $f^*\omega$ has no zero.

conjecture | open | MathDB 页面 | 原文线索

V2-019831 | MATHDB-356882

The analytic-rank-zero Birch and Swinnerton-Dyer conjecture for abelian varieties

摘录 / EXCERPT: Let $A/\mathbb{Q}$ be an abelian variety with L-function $L(A,s)$, and let $\Omega(A)$ denote its real period. For each prime p, let $c_p(A)$ be the Tamagawa number of...

conjecture | open | MathDB 页面 | 原文线索

V2-019832 | MATHDB-356883

Agashe-Stein divisibility conjecture for optimal modular abelian varieties

摘录 / EXCERPT: Let $A/\mathbb{Q}$ be an abelian variety with $L(A,1)\neq 0$ that is an optimal quotient of $J_0(N)/\mathbb{Q}$ attached to a newform. Let $c_{\infty}(A)$ be the number of connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-019833 | MATHDB-356915

Fubini-Study Chern number conjecture for singular K3 surfaces

摘录 / EXCERPT: Let $X\subseteq\mathbb{P}_\mathbb{C}^3$ be a possibly singular K3 surface, and let X_s be its smooth locus with curvature form J induced by the Fubini-Study metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-019834 | MATHDB-356916

Global finite quotient conjecture for the singular K3 family

摘录 / EXCERPT: Let $X_{\lambda^\sharp}$ be a member of the singular algebraic-surface family defined in the paper, with finite parameter $\lambda^\sharp<\infty$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019835 | MATHDB-356918

Reformulated Severi dimensionality conjecture for unicuspidal rational curves

摘录 / EXCERPT: Let ${\rm S}$ be a numerical semigroup with conductor c, and let ${\bf k}=(k_1,\dots,k_n)$ be a ramification profile. Let B be the set of Betti elements of the ramification ord...

conjecture | open | MathDB 页面 | 原文线索

V2-019836 | MATHDB-356925

Quadratic-form conjecture for the CCSD working equations

摘录 / EXCERPT: Let n_s and n_d denote the numbers of single and double excitations, respectively. For each doubly excited determinant Φ_{ij}^{ab}, let $k=k(i,j,a,b)$ be its canonically or...

conjecture | open | MathDB 页面 | 原文线索

V2-019837 | MATHDB-356942

Algebraic correspondence conjecture for the universal Calabi-Yau threefold construction

摘录 / EXCERPT: Let F be an algebraically closed field, let $\E\to\mathbf{P}^1_F$ be a rational elliptic fibration with multiplicative fibers at 0 and 1 and an additive fiber at ∞, an...

conjecture | open | MathDB 页面 | 原文线索

V2-019838 | MATHDB-356945

Boundedness conjecture for slc stable minimal models

摘录 / EXCERPT: Boundedness conjecture for slc stable minimal models. The family

conjecture | open | MathDB 页面 | 原文线索

V2-019839 | MATHDB-356946

The Hodge-locus dimension and algebraicity conjecture for cubic sixfolds and eightfolds

摘录 / EXCERPT: Let ${\sf T}$ be the local deformation space of smooth hypersurfaces $X_t\subset\mathbb{P}^{n+1}$ of even dimension n, and let $V_Z\subset V_{[Z]}$ be respectively the locus wher...

conjecture | open | MathDB 页面 | 原文线索

V2-019840 | MATHDB-356947

The non-smoothness conjecture for Hodge loci of intersecting linear cycles

摘录 / EXCERPT: Let $V_{[Z]}$ be the Hodge locus associated with the class of $Z=r\mathbb{P}^{\frac n2}+\check r\check{\mathbb{P}}^{\frac n2}$ on a smooth even-dimensional hypersurface, where the...

conjecture | open | MathDB 页面 | 原文线索

V2-019841 | MATHDB-356948

The non-smoothness conjecture for cubic Hodge loci

摘录 / EXCERPT: Let $V_{[Z]}$ be the Hodge locus associated with $Z=r\mathbb{P}^{\frac n2}+\check r\check{\mathbb{P}}^{\frac n2}$ on a smooth cubic hypersurface, where the two linear cycles meet i...

conjecture | open | MathDB 页面 | 原文线索

V2-019842 | MATHDB-356949

The generalized cubic-scroll description conjecture for a cubic Hodge locus

摘录 / EXCERPT: Let $V_{[Z]}$ be the Hodge locus associated with $Z=r\mathbb{P}^{\frac n2}+\check r\check{\mathbb{P}}^{\frac n2}$, where the two linear cycles meet in $\mathbb{P}^m$. Generalized c...

conjecture | open | MathDB 页面 | 原文线索

V2-019843 | MATHDB-356987

Expected class identity for symmetric powers of twisted projective spaces

摘录 / EXCERPT: Let $\mathcal{E}$ be an α-twisted locally free sheaf on $\operatorname{Spec}K$ of rank n, and let $\mathbb{P}\mathcal{E}$ denote its associated twisted projective space. T...

conjecture | open | MathDB 页面 | 原文线索

V2-019844 | MATHDB-357015

The topological quotient conjecture for quantized coordinate rings

摘录 / EXCERPT: Let A be a quantized coordinate ring of an affine variety V. Topological quotient conjecture. The spaces $\operatorname{Spec} A$ and $\operatorname{Prim} A$ are compatible topo...

conjecture | open | MathDB 页面 | 原文线索

V2-019845 | MATHDB-357016

The secondary semiclassical limit conjecture

摘录 / EXCERPT: Assume that $\operatorname{char} k=0$. Let A and A' be suitably generic members of some flat family of quantized coordinate rings for an affine variety V with semiclassical l...

conjecture | open | MathDB 页面 | 原文线索

V2-019846 | MATHDB-357017

The stratification closure-maps conjecture for quantized coordinate rings

摘录 / EXCERPT: Let A and H be as in Theorem \textup{(c)}. For $J\subset J'$ in $H\text{-}\operatorname{Spec} A$, let $\phi^p_{JJ'}$ and $\phi^s_{JJ'}$ be the maps between closed subsets of the...

conjecture | open | MathDB 页面 | 原文线索

V2-019847 | MATHDB-357032

The smooth-normalization conjecture for varieties

摘录 / EXCERPT: Let X be an algebraic or analytic variety. A smooth blowing-up is a blowing-up with smooth centre, and it is admissible when it satisfies the local normal-crossings and Hilbert-S...

conjecture | open | MathDB 页面 | 原文线索

V2-019848 | MATHDB-357033

The minimal-singularities conjecture for partial desingularization

摘录 / EXCERPT: Let X be an algebraic or analytic variety, and let X^{nc} denote its normal-crossings locus. A smooth blowing-up is a blowing-up with smooth centre, and it is admissib...

conjecture | open | MathDB 页面 | 原文线索

V2-019849 | MATHDB-357040

The integral Tate conjecture for separably rationally connected varieties in codimension one

摘录 / EXCERPT: Let X be a smooth projective variety over a finite field, and let $d=\dim X$. Say that X is separably rationally connected in codimension 1 if it admits a morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-019850 | MATHDB-357044

Griffiths-Schmid conjecture on the saturated cone of automorphic vector bundles

摘录 / EXCERPT: Griffiths-Schmid conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-019851 | MATHDB-357045

Goldring-Koskivirta conjecture on global sections over G-zips

摘录 / EXCERPT: Let G be a reductive group over a finite field $\mathbb{F}_q$, let $k=\overline{\mathbb{F}}_q$, and let $\mu\colon\mathbb{G}_{\mathrm{m},k}\to G_k$ be a cocharacter. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019852 | MATHDB-357046

Goldring-Koskivirta conjecture for flag spaces over G-zips

摘录 / EXCERPT: Let k be a field, let G be a reductive group over a finite field, let $\mu\colon\mathbb{G}_{\mathrm{m},k}\to G_k$ be a cocharacter, and let (X,ζ) consist of a k-scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-019853 | MATHDB-357048

Amerik's boundedness conjecture for finite morphisms

摘录 / EXCERPT: Let $f:Y\to X$ be a finite morphism between smooth projective varieties of dimension $n\ge 2$, each having second Betti number 1. Suppose that $X\not\cong\mathbb{P}^n$. Amerik's...

conjecture | open | MathDB 页面 | 原文线索

V2-019854 | MATHDB-357049

The projective-space conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold of Picard number 1, meaning that $-K_X$ is ample and the Picard number of X is 1. Projective-space conjecture. If X admits a non-isomorphic surje...

conjecture | open | MathDB 页面 | 原文线索

V2-019855 | MATHDB-357058

Perepechko-Zaidenberg conjecture on automorphisms of rigid affine varieties

摘录 / EXCERPT: Let Y be a rigid affine algebraic variety over ${\mathbb K}$. The connected component $\operatorname{Aut}^0(Y)$ is an algebraic torus of rank at most $\dim Y$. Perepechko-Zaidenb...

conjecture | open | MathDB 页面 | 原文线索

V2-019856 | MATHDB-357059

Finiteness question for automorphism groups of toral varieties

摘录 / EXCERPT: Let Y be a toral variety, and suppose that its maximal torus in $\operatorname{Aut}(Y)$ is trivial. Finiteness question. Is $\operatorname{Aut}(Y)$ a finite group? This is posed...

open_problem | open | MathDB 页面 | 原文线索

V2-019857 | MATHDB-357067

The cone conjecture for automorphic forms on Hodge-type Shimura varieties

摘录 / EXCERPT: The cone conjecture. For any Hodge-type Shimura variety, one has

conjecture | open | MathDB 页面 | 原文线索

V2-019858 | MATHDB-357068

The cone conjecture for automorphic forms on schemes over a G-zip

摘录 / EXCERPT: The cone conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-019859 | MATHDB-357069

The saturated zip-cone conjecture for Hodge-type Shimura varieties

摘录 / EXCERPT: The saturated zip-cone conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-019860 | MATHDB-357070

The orbitwise weight inequality conjecture for automorphic forms

摘录 / EXCERPT: The orbitwise weight inequality conjecture. If $f\in H^0(S_K,{\mathcal V}_I(\lambda))$ is a nonzero automorphic form of weight $\lambda\in X^*(\mathbf{T})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-019861 | MATHDB-357071

Hasse-regularity conjecture for flag strata of Hodge-type Shimura varieties

摘录 / EXCERPT: Let (G,μ) be a cocharacter datum over $\mathbb{F}_q$, and let (X,ζ) satisfy the paper's stated assumptions. Let Y be the flag space of X, stratified by flag strata...

conjecture | open | MathDB 页面 | 原文线索

V2-019862 | MATHDB-357086

Fulger-Murayama's characterization of projective space by tangent-sheaf Seshadri constants

摘录 / EXCERPT: Let X be a smooth projective variety over an algebraically closed field. For a coherent sheaf E on X and a point $p\in X$, write $\varepsilon(E,p)$ for its Seshadri constant...

open_problem | open | MathDB 页面 | 原文线索

V2-019863 | MATHDB-357088

Archimedean motivic action conjecture for Shimura varieties

摘录 / EXCERPT: Let $\Pi=\Pi_f\otimes\Pi_\infty$ satisfy the paper's Assumption, with Π_∞ a non-discrete-series representation that is not a discrete series. Set ... Regard both as...

conjecture | open | MathDB 页面 | 原文线索

V2-019864 | MATHDB-357091

Motivic action conjecture for Hilbert modular varieties

摘录 / EXCERPT: Let f be a parallel weight-one form, let $U_f^\vee$ be the dual space appearing in the cited construction, and for $u\in U_f^\vee$ let u_i denote its σ_i-component. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019865 | MATHDB-357117

The mixed random fewnomial square-root bound conjecture

摘录 / EXCERPT: For $i=1,\ldots,n$, let $A_i\subseteq\mathbb{R}^n$ be finite nonempty sets of cardinality t_i, with convex hulls P_i. Let V_0 be the number of vertices of the Minkowski sum...

conjecture | open | MathDB 页面 | 原文线索

V2-019866 | MATHDB-357129

The conjecture that projective spaces are the only toric 2-Fano manifolds

摘录 / EXCERPT: Toric 2-Fano uniqueness conjecture. The only toric 2-Fano manifolds are projective spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-019867 | MATHDB-357130

The cohomological singular Hodge conjecture

摘录 / EXCERPT: The cohomological singular Hodge conjecture. The image of cl_p coincides with $H^{2p}_{\mathrm{Hdg}}(X)$. This extends the classical Hodge conjecture to singular varieti...

conjecture | open | MathDB 页面 | 原文线索

V2-019868 | MATHDB-357131

The algebraic cohomology exactness conjecture for resolutions

摘录 / EXCERPT: Let X be a singular projective variety, let $\phi:\widetilde{X}\to X$ be any resolution of singularities, and let E be the exceptional divisor. Let p be an integer such that...

conjecture | open | MathDB 页面 | 原文线索

V2-019869 | MATHDB-357143

Batyrev--Manin--Peyre--Tschinkel conjecture for rational points

摘录 / EXCERPT: Let κ be a number field and let X be a smooth Fano variety over κ. Let $\mathcal{L}=(L,\{\|\cdot\|_v\})$ be an adelically metrized big and nef $\mathbb{Q}$-divisor,...

conjecture | open | MathDB 页面 | 原文线索

V2-019870 | MATHDB-357147

Motivic or homological stability conjecture for Manin components

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a Fano fibration and let M range over the Manin components of $\operatorname{Sec}(\mathcal{X}/B)$, with degree tending to infinity. Stability conjec...

open_problem | open | MathDB 页面 | 原文线索

V2-019871 | MATHDB-357149

Generalized weight-monodromy conjecture for rigid-analytic varieties with projective reduction

摘录 / EXCERPT: Let K be a p-adic local field with residue field of cardinality q, let X be a smooth proper rigid-analytic K-variety, let $\ell\neq p$ be a prime number, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-019872 | MATHDB-357150

Local weight-monodromy conjecture for nearby cycles

摘录 / EXCERPT: Let K be a local field with residue characteristic p, let $\ell\neq p$ be a prime number, and let $\mathcal X$ be an admissible formal $\mathcal O_K$-scheme with smooth generic...

conjecture | open | MathDB 页面 | 原文线索

V2-019873 | MATHDB-357151

Weights conjecture for intersection cohomology of rigid-analytic varieties

摘录 / EXCERPT: Let K be a p-adic local field with residue field of cardinality q, let X be a quasi-compact and quasi-separated rigid-analytic K-variety of dimension d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019874 | MATHDB-357157

Smale's sixth problem for generic masses

摘录 / EXCERPT: Let n bodies have masses $m_1,dots,m_n$ and pairwise distances r_{ij} satisfying the Albouy-Chenciner equations and the Cayley-Menger equations for planar configurations. A nor...

conjecture | open | MathDB 页面 | 原文线索

V2-019875 | MATHDB-357158

Chow-Künneth conjecture for algebraic motives

摘录 / EXCERPT: Let k be a base field, let H^* be a Weil cohomology, and let X be a smooth projective variety of dimension d_X. Write $\mathfrak{h}(X)$ for its motive in the...

conjecture | open | MathDB 页面 | 原文线索

V2-019876 | MATHDB-357161

Positivity conjecture for algebraic classes of motives

摘录 / EXCERPT: Let the base field k have characteristic p. Let M be a homological motive over k, and let ... be a morphism in the category of homological motives. Suppose that q is the re...

conjecture | open | MathDB 页面 | 原文线索

V2-019877 | MATHDB-357162

Homological equivalence equals numerical equivalence conjecture

摘录 / EXCERPT: Let k be a base field and let π' be the quotient functor from homological motives to numerical motives in the diagram of motive categories. Homological-equals-numerical conje...

conjecture | open | MathDB 页面 | 原文线索

V2-019878 | MATHDB-357163

Hodge standard conjecture

摘录 / EXCERPT: Hodge standard conjecture. The quadratic form q_Z is positive definite. This is Grothendieck's Hodge-type standard conjecture; it is known in characteristic zero through Hodge-Ri...

conjecture | open | MathDB 页面 | 原文线索

V2-019879 | MATHDB-357164

Finite-dimensionality conjecture for Chow motives

摘录 / EXCERPT: Let M be a Chow motive. Finite-dimensionality conjecture. There exists a non-unique decomposition ... for which $\Lambda^N M_+=0$ and $\operatorname{Sym}^N M_-=0$ for some sufficie...

conjecture | open | MathDB 页面 | 原文线索

V2-019880 | MATHDB-357165

Weak p-adic Grothendieck period conjecture

摘录 / EXCERPT: Let X be a variety over $\overline{\mathbb{Q}}$ with good reduction X_p at a prime p. Let Z_0 be the $\mathbb{Q}$-vector space of algebraic classes on X, let Z_p be the...

open_problem | open | MathDB 页面 | 原文线索

V2-019881 | MATHDB-357188

Letellier's conjecture on twisted mixed Hodge polynomials of partial character varieties

摘录 / EXCERPT: Let $\mathcal{M}_{\mu,\nu}$ be the partial resolution of the character variety defined in the source, and let w be an n-cycle in the Weyl group relative to the third puncture....

conjecture | open | MathDB 页面 | 原文线索

V2-019882 | MATHDB-357194

Drinfeld's Grothendieck-Teichmüller conjecture for the motivic Galois group

摘录 / EXCERPT: Let $\mathcal{MTM}(\mathbb{Z})$ be the Tannakian category of mixed Tate motives over $\mathbb{Z}$, and let $\mathcal{G}_{1}=\mathbb{G}_{m}\ltimes\mathcal{U}_{1}$ be its motivic Gal...

conjecture | open | MathDB 页面 | 原文线索

V2-019883 | MATHDB-357212

The essential skeleton conjecture for log canonical pairs and tropicalization

摘录 / EXCERPT: Let (X,Δ) be a log canonical pair such that $U:= X \mathbin{\rule[0.2em]{0.67em}{0.12em}} \Delta$ is a subvariety of a torus. Let ... be the tropicalization map, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019884 | MATHDB-357213

Effective Iitaka fibration conjecture

摘录 / EXCERPT: Effective Iitaka fibration conjecture. There exists a positive integer m_d depending only on d, such that for every smooth projective variety X of dimension d with non-nega...

conjecture | open | MathDB 页面 | 原文线索

V2-019885 | MATHDB-357249

Expectations for derived locally compact modules over arithmetic schemes

摘录 / EXCERPT: Let X be a regular scheme that is flat, of finite type, and separated over a ring of S-integers $\mathcal{O}_{S}$ of a number field. Suppose that C_{X} is a category...

conjecture | open | MathDB 页面 | 原文线索

V2-019886 | MATHDB-357258

The local discriminant-complement component count for the real singularity X_{10}^3

摘录 / EXCERPT: Let X_{10}^3 be the real singularity under consideration, and let its discriminant variety be the locus of singular members in the corresponding deformation space. Component-coun...

conjecture | open | MathDB 页面 | 原文线索

V2-019887 | MATHDB-357299

Integer-program dimension conjecture for secant varieties of mixture varieties

摘录 / EXCERPT: Let $\mathcal{M}_{n,d}$ be the mixture variety, let $\sigma_r(\mathcal{M}_{n,d})$ denote its rth secant variety, and let the integer linear program in the source provide a bound...

conjecture | open | MathDB 页面 | 原文线索

V2-019888 | MATHDB-357301

The log uniformization conjecture for quasi-excellent schemes

摘录 / EXCERPT: Log uniformization conjecture. Every valuation on a quasi-excellent integral scheme X is log uniformizable.

conjecture | open | MathDB 页面 | 原文线索

V2-019889 | MATHDB-357304

The log-smooth modification conjecture for rig-smooth formal schemes

摘录 / EXCERPT: Log-smooth modification conjecture. There exists a finite extension l/k and an admissible blowing up

conjecture | open | MathDB 页面 | 原文线索

V2-019890 | MATHDB-357313

Linear bound for Lie algebras of linearly reductive height-one group schemes

摘录 / EXCERPT: Let $p\geq5$, let ${\rm Ver}_p$ be the Verlinde category, and let G be a linearly reductive height-1 subgroup scheme of $GL(Y)$ for an object $Y\in {\rm Ver}_p$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-019891 | MATHDB-357314

Uniform bound for simple linearly reductive Lie algebras in Verlinde categories

摘录 / EXCERPT: Let $p\geq5$, let ${\rm Ver}_p$ be the Verlinde category, and let G be a simple linearly reductive height-1 group scheme in ${\rm Ver}_p$, with Lie algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-019892 | MATHDB-357338

Arithmetic signed count conjecture for 2-torsion points on principally polarized abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a field k with $\operatorname{char}(k)\neq 2$, and let $\lambda\colon A\to A^\vee$ be a principal polarization. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-019893 | MATHDB-357339

The 1-loop polynomial conjecture for \(\mathbb C^2\)-torsion

摘录 / EXCERPT: The 1-loop polynomial conjecture. The two polynomials agree up to multiplication by signs and monomials in t:

conjecture | open | MathDB 页面 | 原文线索

V2-019894 | MATHDB-357352

The pseudo-Abelian variety conjecture for coframed projective manifolds

摘录 / EXCERPT: Let X be a k-coframed projective manifold, let $q=h^0(\Omega^1_X)$, and suppose that $q=k$. Assume also that the canonical divisor K_X is nef. Pseudo-Abelian variety conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-019895 | MATHDB-357381

Existence of Bridgeland stability conditions on smooth projective varieties

摘录 / EXCERPT: Let X be a smooth projective variety. A stability condition means a Bridgeland stability condition on $\operatorname{D^b}(X)$, defined with respect to the numerical lattice...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-019896 | MATHDB-357409

Ampleness conjecture for log canonical varieties

摘录 / EXCERPT: Let X be a projective log canonical variety with numerically trivial canonical divisor, so that $K_X\equiv 0$, and let L be a strictly nef $\mathbb Q$-Cartier divisor on X. A...

conjecture | open | MathDB 页面 | 原文线索

V2-019897 | MATHDB-357410

Log version of Campana--Peternell's conjecture

摘录 / EXCERPT: Let X be a projective log canonical variety, and suppose that $-K_X$ is a strictly nef divisor. Log version of Campana--Peternell's conjecture. The anti-canonical divisor $-K_X$...

conjecture | open | MathDB 页面 | 原文线索

V2-019898 | MATHDB-357411

The point-choosing conjecture for smooth hypersurfaces

摘录 / EXCERPT: You can't choose a point. For $d\geq 3$ and $N\geq 2$, there is no holomorphic, or even continuous, section of ${\mathcal E}_{d,N}\to{\mathcal C}_{d,N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-019899 | MATHDB-357434

Finite-generation conjecture for modules over the completed algebra of vector fields

摘录 / EXCERPT: Let X be a quasi-projective variety, let U be an affine open subset, and write $A={\mathcal A}(U)$ and $AV={\mathcal {AV}}(U)$. Let M be an AV-module that is finitely gener...

conjecture | open | MathDB 页面 | 原文线索

V2-019900 | MATHDB-357464

Positivity conjecture for Welschinger invariants of $(\mathbb{R}P^1)^3$

摘录 / EXCERPT: Let $(\mathbb{R}P^1)^3$ be the product of three real projective lines with the standard real structure, and let $\mathfrak{s}_{(\mathbb{R}P^1)^3}$ be the $Spin_3$-structure over...

conjecture | open | MathDB 页面 | 原文线索

V2-019901 | MATHDB-357480

Converses characterizing positivity for partial tropical flag varieties

摘录 / EXCERPT: Positivity characterization conjecture. 1. ${\bm{\mu}}$ is totally non-negative if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-019902 | MATHDB-357493

The Amerik-Rovinsky-Van de Ven conjecture on endomorphisms of Fano varieties

摘录 / EXCERPT: Amerik-Rovinsky-Van de Ven conjecture. Then X is isomorphic to projective space.

conjecture | open | MathDB 页面 | 原文线索

V2-019903 | MATHDB-357503

The root-count conjecture for p_k

摘录 / EXCERPT: Let $p_k(x)$ be the polynomial defined in the surrounding argument for an integer $k\geq2$. Root-count conjecture for p_k. The polynomial $p_k(x)$ has at most two real roots in...

conjecture | open | MathDB 页面 | 原文线索

V2-019904 | MATHDB-357527

Serre's conjecture relating the Mumford-Tate and motivic Mumford-Tate groups

摘录 / EXCERPT: Let M be a motive in the relevant motivic category, with rational polarized Hodge realization (V,ψ). Let $\operatorname{MMT}_{\sim}(M)$ be its motivic Mumford-Tate group. Se...

conjecture | open | MathDB 页面 | 原文线索

V2-019905 | MATHDB-357528

The motivic Mumford-Tate conjecture

摘录 / EXCERPT: Let M be a motive in the chosen motivic category, let $G_{l,K}^{\operatorname{alg}}$ be the algebraic group attached to its l-adic realization, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-019906 | MATHDB-357529

The motivic Sato-Tate conjecture

摘录 / EXCERPT: Let M be a motive and let $\operatorname{MS}_{\sim}(M)$ be its motivic Serre group. Motivic Sato-Tate conjecture. For any prime number l, ... The paper obtains this formulation a...

conjecture | open | MathDB 页面 | 原文线索

V2-019907 | MATHDB-357551

Codimension formula for Severi varieties of unicuspidal rational curves

摘录 / EXCERPT: Let ${\bf r}=(r_1,\ldots,r_n)\in \mathbb{N}_{>0}^n$, and let $\mathcal{V}_{\bf r}\subset M^n_{d,g;\mathrm{S}}$ be the subvariety of maps $f:\mathbb{P}^1\rightarrow\mathbb{P}^n$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-019908 | MATHDB-357552

Unirationality conjecture for Severi varieties of unicuspidal rational curves

摘录 / EXCERPT: Let $\mathcal{V}_{\bf r}$ be the Severi variety of unicuspidal rational curves of type $({\rm S},{\bf r})$, under the hypotheses and notation of the codimension conjecture above. U...

conjecture | open | MathDB 页面 | 原文线索

V2-019909 | MATHDB-357553

Covering gonality bound for Gorenstein unicuspidal rational curves

摘录 / EXCERPT: Let C be a Gorenstein unicuspidal rational curve of genus g, and let $\gon_F(C)$ denote its covering gonality, namely the smallest degree of a map from C to $\mathbb{P}^1$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-019910 | MATHDB-357595

Conjecture on the non-lci locus of Schubert varieties

摘录 / EXCERPT: Let X_w be a Schubert variety, and let $({\mathfrak S},\prec_I)$ be the poset of permutation intervals governing its local singularity types. The ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-019911 | MATHDB-357596

Conjecture on the non-Gorenstein locus of Schubert varieties

摘录 / EXCERPT: Let X_w be a Schubert variety, and let $({\mathfrak S},\prec_I)$ be the poset of permutation intervals. The order ideal of intervals corresponding to points where Schubert variet...

conjecture | open | MathDB 页面 | 原文线索

V2-019912 | MATHDB-357597

Cohen-Macaulay tangent-cone conjecture for Kazhdan-Lusztig varieties

摘录 / EXCERPT: Let $R={\mathbb C}[{\bf z}^{(v)}]$, let $I_{v,w}$ be the Kazhdan-Lusztig ideal defining the local chart of X_w at $E^{(v)}_\bullet$, and let $\overline I_{v,w}$ be its ideal of l...

conjecture | open | MathDB 页面 | 原文线索

V2-019913 | MATHDB-357598

Lusztig's interval conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: For permutations a,b,v,w, let $P_{a,b}(q)$ and $P_{v,w}(q)$ denote Kazhdan-Lusztig polynomials, and let $[a,b]$ and $[v,w]$ be the corresponding Bruhat order intervals. Lusztig's...

conjecture | open | MathDB 页面 | 原文线索

V2-019914 | MATHDB-357627

Logarithmic K-theory comparison conjecture

摘录 / EXCERPT: Let S be a scheme. Write $\Kthlog(S)$ for its logarithmic K-theory spectrum and $\Kth(S)$ for its algebraic K-theory spectrum. Logarithmic K-theory comparison conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-019915 | MATHDB-357639

The integral motive conjecture for moduli spaces of sheaves on Poisson surfaces

摘录 / EXCERPT: Integral motive conjecture. The integral Chow motive of $M_H(S,v)$ is in the tensor category generated by the motives of the Hilbert schemes $\{S^{[n]}\}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-019916 | MATHDB-357647

The linear-obstruction characterization of Frobenius generators for smooth toric varieties

摘录 / EXCERPT: Linear-obstruction conjecture. $(F_\ell)_*\mathcal O_X(D)$ is a classical generator of $D^b\operatorname{Coh}(X)$ for some $\ell\in\mathbb N$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-019917 | MATHDB-357648

The zonotope criterion for generating line bundles on smooth toric varieties

摘录 / EXCERPT: Zonotope generation conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-019918 | MATHDB-357650

Galkin-Shinder's rationality conjecture for cubic fourfolds

摘录 / EXCERPT: Let X be a smooth cubic fourfold, let $F(X)$ be its Fano variety of lines, and let $S^{[2]}$ be the Hilbert scheme of length-two subschemes of a K3 surface S. Galkin-Shinder's...

conjecture | open | MathDB 页面 | 原文线索

V2-019919 | MATHDB-357685

The degeneracy-locus conjecture for torsion-dense subvarieties

摘录 / EXCERPT: Let $g\in\mathbb{N}$, and let X be an irreducible closed subvariety of the universal family $\mathfrak{A}_g$ defined over $\overline{\mathbb{Q}}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019920 | MATHDB-357704

HKL conjecture for degree-8 del Pezzo pairs

摘录 / EXCERPT: Let ${\mathcal F}^\ast$ be the locally symmetric variety, let λ be its Hodge line bundle, and for $s\in[0,1]\cap{\mathbb Q}$ define ... Let $P^\ast$ denote the relevant open...

conjecture | open | MathDB 页面 | 原文线索

V2-019921 | MATHDB-357708

Grothendieck's p-curvature conjecture for algebraic connections

摘录 / EXCERPT: Let X be a smooth quasi-projective variety over $\mathbb C$, and let (E,∇) be an integrable algebraic connection. Choose a finite-type scheme S over $\mathbb Z$ over whi...

conjecture | open | MathDB 页面 | 原文线索

V2-019922 | MATHDB-357709

André's generalized Grothendieck p-curvature conjecture for geometric local systems

摘录 / EXCERPT: Let $\mathbb L$ be a geometric local system on a smooth variety X, with underlying vector bundle with connection (E,∇). For all sufficiently large primes p, require that...

conjecture | open | MathDB 页面 | 原文线索

V2-019923 | MATHDB-357710

de Jong's conjecture on isocrystals over simply connected varieties

摘录 / EXCERPT: Let X be a smooth projective variety over an algebraically closed field k of characteristic $p>0$, and let $\pi_1(X)$ denote its étale fundamental group. An isocrystal on X i...

conjecture | open | MathDB 页面 | 原文线索

V2-019924 | MATHDB-357714

Breuil-Hellmann-Schraen conjecture for the two-flag variety

摘录 / EXCERPT: Let G be a connected split reductive group over k, with Weyl group W and Lie algebra $\gfrak$. Define ... Let $\pi\colon X\to G/B\times G/B$ be the natural projection, let...

conjecture | open | MathDB 页面 | 原文线索

V2-019925 | MATHDB-357718

Strong deformation retract conjecture for twisted Hopf link character varieties

摘录 / EXCERPT: Let H_n be a twisted Hopf link, and let $X(H_n,G)$ denote its G-character variety. For an integer $r\geq 1$, consider the character varieties for $G=\operatorname{SU}(r)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019926 | MATHDB-357733

The de Rham-Betti conjecture for smooth projective varieties

摘录 / EXCERPT: Let k be a number field, let Y be a smooth projective k-variety, and fix an embedding $k\hookrightarrow\mathbf{C}$. For a codimension-$n$ de Rham-Betti cycle, take a pair...

conjecture | open | MathDB 页面 | 原文线索

V2-019927 | MATHDB-357734

The Ogus conjecture for smooth projective varieties

摘录 / EXCERPT: Let k be a number field and let Y be a smooth projective k-variety. For a finite unramified place v of k where Y has good reduction, use the Berthelot isomorphism ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-019928 | MATHDB-357739

The generic étaleness conjecture for the Joyce-structure parameter map

摘录 / EXCERPT: Let $\mathcal O=\mathcal O(g,m)$ be the space of data (C,Q_0,Q_1,Q_2) satisfying the prescribed pole, apparent-singularity, and residue conditions, and let $X^\#$ be the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-019929 | MATHDB-357767

The motivic Higgs-bundle locus and arithmetic periodicity

摘录 / EXCERPT: Let $X-D$ be a smooth complex quasi-projective variety, and consider the moduli space of semistable Higgs bundles with trivial Chern classes on $X-D$. Suppose there is a family of...

conjecture | open | MathDB 页面 | 原文线索

V2-019930 | MATHDB-357769

The bounded-preperiodic-length criterion for motivic Higgs bundles

摘录 / EXCERPT: Let $\mathcal X$ be a smooth scheme over the ring of integers of some number field, and let (E,θ) be a semistable Higgs bundle on $\mathcal X$ with trivial Chern classes. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-019931 | MATHDB-357772

The three-dimensional restriction conjecture for quadratic monomial projection algebras

摘录 / EXCERPT: The three-dimensional restriction conjecture. The algebra $\operatorname{A}(X_{n,d}^G)$ is quadratic if and only if, for all $i<j<k$, the algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-019932 | MATHDB-357810

Totaro's cone conjecture for klt Calabi-Yau pairs

摘录 / EXCERPT: Totaro's cone conjecture. There exists a rational polyhedral cone Π that is a fundamental domain for the action of $\operatorname{Aut}(X)$ on $\operatorname{Nef^e}(X)$, namely

conjecture | open | MathDB 页面 | 原文线索

V2-019933 | MATHDB-357836

Equality of the universal variety and the universal Grassmannian image

摘录 / EXCERPT: Let $\mathcal V_{d,h}$ denote the universal variety, let $\mathfrak A_d$ be the relevant algebra, and let $\hat{\mathcal G}^{h}(\mathfrak A_d)$ denote the h-fold universal Grassm...

conjecture | open | MathDB 页面 | 原文线索

V2-019934 | MATHDB-357840

The geometric AJ conjecture for SO(3)-Berezin-Toeplitz operators

摘录 / EXCERPT: Let K be a knot in S^3, let $A(p,q)$ denote the A-polynomial expressed in the variables used for the torus character variety, let $T_{r+\frac{1}{2}}(A)$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-019935 | MATHDB-357844

Billey and Postnikov's bound on the minimal Kazhdan-Lusztig degree

摘录 / EXCERPT: Billey and Postnikov's conjecture. If $X_w\subset G/B$ is singular, then

conjecture | open | MathDB 页面 | 原文线索

V2-019936 | MATHDB-357845

Positivity conjecture for the signed Euler characteristic of three Schubert cells

摘录 / EXCERPT: Positivity conjecture for the signed Euler characteristic. One has

conjecture | open | MathDB 页面 | 原文线索

V2-019937 | MATHDB-357851

Finite differential inequality characterization for semialgebraic systems

摘录 / EXCERPT: Fix $m\ge 0$. Let $(A_{ij}(x))_{1\le i\le N,1\le j\le M}$ be a matrix of semialgebraic functions on $\mathbb{R}^n$, and let the system ... L{mu,nu}f(x)=sum{i=1}^Nsum{|alpha|le bar m}...

conjecture | open | MathDB 页面 | 原文线索

V2-019938 | MATHDB-357852

Strong Segre symmetry for equivariant Quot schemes of the plane

摘录 / EXCERPT: Let $r,N>0$, let $S=\mathbb{C}^2$, and set $E=\bigoplus_{i=1}^N\mathcal{O}_S\langle y_i\rangle$ and $V=\bigoplus_{i=1}^r\mathcal{O}_S\langle v_i\rangle$. Strong Segre symmetry. The...

conjecture | open | MathDB 页面 | 原文线索

V2-019939 | MATHDB-357853

Degree-zero Segre and Verlinde coefficient formulas for rank one

摘录 / EXCERPT: Let $A_0(q),A_1(q),B_0(q),B_1(q)$ be the series introduced from the degree-zero reduced Segre and Verlinde series, with $N=1$. Degree-zero coefficient formulas. For...

conjecture | open | MathDB 页面 | 原文线索

V2-019940 | MATHDB-357854

Weak Segre-Verlinde correspondence and symmetry for four-dimensional affine space

摘录 / EXCERPT: Let $X=\mathbb{C}^4$, let $E=\bigoplus_{i=1}^{N}\mathcal{O}_X\langle y_i\rangle$ and $V=\bigoplus_{i=1}^{r}\mathcal{O}_X\langle v_i\rangle$, and let $\alpha\in K_{\mathsf{T}}(X)$....

conjecture | open | MathDB 页面 | 原文线索

V2-019941 | MATHDB-357855

Nekrasov's conjecture for Quot schemes of affine space

摘录 / EXCERPT: Let $Y=\mathbb{C}^d$, let $E=\bigoplus_{i=1}^{r+1}\mathcal{O}_Y\langle y_i\rangle$ and $V=\bigoplus_{i=1}^{r}\mathcal{O}_Y\langle v_i\rangle$. Nekrasov's conjecture for Quot scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-019942 | MATHDB-357856

Nekrasov-Piazzalunga's explicit formula for the rank-equal Nekrasov genus

摘录 / EXCERPT: Let X be the affine four-space occurring in the definition of the Nekrasov genus, let $E=\bigoplus_{i=1}^{N}\mathcal{O}_X\langle y_i\rangle$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-019943 | MATHDB-357861

Chern-Segre positivity conjecture for tautological integrals

摘录 / EXCERPT: Let X be a smooth complex variety, let F be the bundle appearing in the tautological integral of Theorem, and write that integral as a polynomial in the Chern classes of F an...

conjecture | open | MathDB 页面 | 原文线索

V2-019944 | MATHDB-357862

Regularity conjecture for monomial geometric subsets

摘录 / EXCERPT: Let $(A_1,\ldots,A_s)$ be a tuple of monomial algebras, and let $\mathrm{Hilb}^{A_1,\ldots,A_s}(X)$ be the associated monomial geometric subset. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-019945 | MATHDB-357882

The conjecture that compact rationally connected Kähler manifolds are Oka-1

摘录 / EXCERPT: A complex manifold X is rationally connected if every pair of points in X can be connected by a rational curve $\mathbb{CP}^1\to X$. A compact Kähler manifold is a compact comp...

conjecture | open | MathDB 页面 | 原文线索

V2-019946 | MATHDB-357884

Ayoub's conjecture on the heart of the perverse homotopy t-structure

摘录 / EXCERPT: Let S be a noetherian excellent finite dimensional base scheme of characteristic 0. The Ayoub conjecture. The heart of the perverse homotopy t-structure on $\mathrm{DM}(S)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-019947 | MATHDB-357892

Baldi-Klingler-Ullmo's density conjecture for typical Hodge loci

摘录 / EXCERPT: Let $\mathbb V$ be a polarizable $\mathbb Z$-VHS on a smooth, irreducible and quasi-projective variety S over $\mathbb C$. The typical Hodge-locus density conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-019948 | MATHDB-357894

Density threshold for typical Hodge loci in the Torelli locus

摘录 / EXCERPT: Let $g\geqslant 4$ be an integer, let $S\subset\mathcal A_g$ be a closed Hodge generic subvariety of dimension q, and let $\mathbb V$ be the induced polarizable $\mathbb Z$-VHS on...

conjecture | open | MathDB 页面 | 原文线索

V2-019949 | MATHDB-357911

Motivic filtration conjecture for relative topological Hochschild homology

摘录 / EXCERPT: Motivic filtration conjecture. There is a filtration $\mathrm{F}^\star_{\mathrm{mot}}\mathrm{THH}(R/J(p))$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-019950 | MATHDB-357912

Motivic filtration conjecture for relative topological periodic homology

摘录 / EXCERPT: Motivic filtration conjecture. The spectrum $\mathrm{TP}(R/X(p))$ admits a motivic filtration satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-019951 | MATHDB-357941

Defectivity conjecture for Taylor varieties

摘录 / EXCERPT: Let $\mathcal{T}^n_{d,e,m}$ denote the Taylor variety, and call it defective when its dimension is smaller than the expected dimension. Defectivity conjecture. The following statem...

conjecture | open | MathDB 页面 | 原文线索

V2-019952 | MATHDB-357946

SYZ conjecture for potentially Lagrangian classes

摘录 / EXCERPT: Let Y be a hyper-Kähler variety, let q be its Beauville-Bogomolov quadratic form on $H^2(Y)$, and let l be a divisor class on Y. The class l is potentially Lagrangian if...

conjecture | open | MathDB 页面 | 原文线索

V2-019953 | MATHDB-357955

Segre-Harbourne-Gimigliano-Hirschowitz conjecture for plane blow-ups

摘录 / EXCERPT: Segre-Harbourne-Gimigliano-Hirschowitz conjecture. Either

conjecture | open | MathDB 页面 | 原文线索

V2-019954 | MATHDB-357965

Irreducibility conjecture for periodic Schrödinger Bloch varieties

摘录 / EXCERPT: Let $L=-\Delta+V(x)$ be a periodic Schrödinger operator, or more generally a second-order elliptic periodic operator, with Bloch variety $\mathcal{B}_L$. Bloch-variety irreducibili...

conjecture | open | MathDB 页面 | 原文线索

V2-019955 | MATHDB-357974

The differential-generation conjecture for the homological slice spectral sequence of $BPGL\langle m\rangle$

摘录 / EXCERPT: Let $BPGL\langle m\rangle$ be the truncated motivic Brown-Peterson spectrum, and let its homological slice spectral sequence (HSSS) have differentials d_r. The differentials disp...

conjecture | open | MathDB 页面 | 原文线索

V2-019956 | MATHDB-357978

Görtz-Haines-Kottwitz-Reuman conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let $G=Sp_4(L)$, where L is as in the paper. For $\in G(L)$, let ... be the σ-centralizer of b, and let $\operatorname{def}(b)$ be the difference between the F-rank of...

conjecture | open | MathDB 页面 | 原文线索

V2-019957 | MATHDB-357979

Collins-Jacob-Yau conjecture for the deformed Hermitian Yang-Mills equation

摘录 / EXCERPT: Let X be a rational homogeneous variety of complex dimension n, let $[\psi]$ be a class with phase angle $\hat{\Theta}\in \big((n-2)\frac{\pi}{2},n\frac{\pi}{2}\big)$, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-019958 | MATHDB-357987

Elliot-Pestun's brane duality conjecture for multiplicative Higgs bundles

摘录 / EXCERPT: Let X be a Calabi-Yau variety, let G be a Langlands self-dual group, and let $\theta\in\operatorname{Aut}_2(G)$ be an involution. The group $\check{G}_{G/G^\theta}$ is the dual...

conjecture | open | MathDB 页面 | 原文线索

V2-019959 | MATHDB-357990

The Jacobi conjecture for polynomial maps

摘录 / EXCERPT: Let $f:{\mathbb{C}}^n\rightarrow{\mathbb{C}}^n$ be a polynomial map, meaning that each component $f_i:{\mathbb{C}}^n\rightarrow{\mathbb{C}}$ is polynomial. Jacobi conjecture. If th...

conjecture | open | MathDB 页面 | 原文线索

V2-019960 | MATHDB-357999

Generic-rank conjecture for simultaneous Waring decompositions

摘录 / EXCERPT: Generic simultaneous Waring rank conjecture. The complex simultaneous Waring rank of a general polynomial vector is

conjecture | open | MathDB 页面 | 原文线索

V2-019961 | MATHDB-358012

Corti's cluster-variety conjecture for toric specializations of Fano varieties

摘录 / EXCERPT: Let X be a generic Fano variety with klt singularities. A polarized cluster variety is a cluster variety equipped with a polarization; write it as $U=Y\setminus D$ with polarizat...

conjecture | open | MathDB 页面 | 原文线索

V2-019962 | MATHDB-358067

The amended Bronowski conjecture on identifiability and tangential projections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{N}$ be an irreducible and non-degenerate variety with non-degenerate Gauss map. The variety X is h-identifiable if a general fiber of its $(h+1)$-secan...

conjecture | open | MathDB 页面 | 原文线索

V2-019963 | MATHDB-358076

Su-ion Ih's S-integrality conjecture for abelian varieties

摘录 / EXCERPT: Su-ion Ih's S-integrality conjecture. The set of torsion points of $A(\overline{k})$ whose closures in $\mathcal{A}_S$ are disjoint from $\overline{D}$ is not Zariski dense in...

conjecture | open | MathDB 页面 | 原文线索

V2-019964 | MATHDB-358079

Bound on the index of a multisection for Calabi-Yau elliptic fibrations

摘录 / EXCERPT: Multisection-index conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-019965 | MATHDB-358091

Colliot-Thélène-Kahn conjecture on unramified cohomology of uniruled threefolds

摘录 / EXCERPT: Let $\mathbb{F}$ be the finite field under consideration, let ℓ be a prime, and let X be a smooth projective geometrically uniruled $\mathbb{F}$-variety of dimension 3. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-019966 | MATHDB-358095

Existence and connectedness of B-minimal resolutions

摘录 / EXCERPT: B-minimal resolution conjecture. For every projective variety X over the complex numbers:

conjecture | open | MathDB 页面 | 原文线索

V2-019967 | MATHDB-358131

Interlacing roots conjecture for the dual polynomial

摘录 / EXCERPT: Interlacing-roots conjecture. The polynomial $p(x)$ has a unique root in each interval (r_s,r_{s+1}) for $1\leq s<k$.

conjecture | open | MathDB 页面 | 原文线索

V2-019968 | MATHDB-358140

Generic meridian-trace irreducibility conjecture for nonabelian links

摘录 / EXCERPT: Generic meridian-trace conjecture. Given generic $t_1,\ldots,t_k$, there exists an irreducible representation

open_problem | open | MathDB 页面 | 原文线索

V2-019969 | MATHDB-358141

Motivic symplectic cohomology detects smoothly but not symplectically isotopic symplectomorphisms

摘录 / EXCERPT: Let X be a symplectic manifold, and let $g_0,g_1:X\rightarrow X$ be symplectomorphisms. The maps $(g_0)_*$ and $(g_1)_*$ are induced on the groups $SH_{\bullet}^M(Y,X)$ associate...

conjecture | open | MathDB 页面 | 原文线索

V2-019970 | MATHDB-358142

Motivic symplectic cohomology distinguishes symplectic forms in a connected deformation class

摘录 / EXCERPT: Let $X_1=(X,\omega_1)$ and $X_2=(X,\omega_2)$ be symplectic manifolds on the same underlying manifold, where $ \omega_1$ and ω_2 are connected by a path of symplectic forms....

conjecture | open | MathDB 页面 | 原文线索

V2-019971 | MATHDB-358143

e-positivity conjecture for the symmetric functions g_k of Hessenberg graphs

摘录 / EXCERPT: Let $\mathbf{m}\colon[n]\to[n]$ be a Hessenberg function, let $G_\mathbf{m}$ be its indifference graph, and let $g_k(G_\mathbf{m};x,q)$ be the symmetric functions defined in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-019972 | MATHDB-358146

Karpenko's injectivity conjecture for generic flag varieties

摘录 / EXCERPT: Karpenko's injectivity conjecture. The morphism φ is injective for the generic flag variety $X=E/P$.

conjecture | open | MathDB 页面 | 原文线索

V2-019973 | MATHDB-358152

The A-twisted Higgs branch conjecture for boundary vertex algebras

摘录 / EXCERPT: Let $\mathcal{V}$ be the boundary vertex algebra of an A-twisted gauge theory with purely free fermion boundary degrees of freedom. Its associated variety is ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-019974 | MATHDB-358154

K-stability conjecture for birationally rigid Fano varieties

摘录 / EXCERPT: Let X be a Fano variety with terminal singularities such that $\operatorname{rk}\operatorname{Cl}(X)=1$. Suppose that X is birationally rigid, meaning that X is not birationa...

conjecture | open | MathDB 页面 | 原文线索

V2-019975 | MATHDB-358155

The optimality conjecture for crepant resolutions of log-terminal singularities

摘录 / EXCERPT: Let Y be an irreducible $\mathbb{Q}$-Gorenstein variety, and let ... be a crepant resolution such that $\mathcal{X}$ has affine diagonal. Optimality conjecture for crepant resoluti...

conjecture | open | MathDB 页面 | 原文线索

V2-019976 | MATHDB-358168

Drinfeld's extension conjecture for Lau's group scheme

摘录 / EXCERPT: Drinfeld's extension conjecture. The isomorphism between these restrictions extends to an isomorphism over the whole Disp_1^G.

conjecture | open | MathDB 页面 | 原文线索

V2-019977 | MATHDB-358170

Drinfeld's algebraicity conjecture for Shimurian Barsotti-Tate stacks

摘录 / EXCERPT: Drinfeld's algebraicity conjecture. This restriction is a quasicompact smooth algebraic stack over $\mathbb Z/p^m\mathbb Z$ with affine diagonal.

conjecture | open | MathDB 页面 | 原文线索

V2-019978 | MATHDB-358207

Minimal mld conjecture for the constructed klt Calabi-Yau varieties

摘录 / EXCERPT: For each integer $n\geq2$, let V/G be the quotient of the hypersurface constructed in the paper by the cyclic group action described there; it is a klt Calabi-Yau variety of dime...

conjecture | open | MathDB 页面 | 原文线索

V2-019979 | MATHDB-358209

Geometric Breuil--Mézard conjecture for the Emerton--Gee stack of \(\mathrm{GSp}_4\)

摘录 / EXCERPT: Let ${\mathcal X}_{{\mathrm{Sym}}},K,\mathrm{red}}}$ be the reduced Emerton--Gee stack, and let ${\mathcal Z}_{\lambda,\tau}$ be the $4f$-dimensional cycle attached to a regular Hodg...

conjecture | open | MathDB 页面 | 原文线索

V2-019980 | MATHDB-358249

Simpson's algebraicity conjecture for the non-abelian Hodge locus

摘录 / EXCERPT: Simpson's conjecture. The non-abelian Hodge locus $\mathrm{NHL}({\mathcal Y}/{\mathcal S},\operatorname{GL}_n)$ is an algebraic variety, and its inclusions into...

conjecture | open | MathDB 页面 | 原文线索

V2-019981 | MATHDB-358257

Plectic Jacobian descent conjecture

摘录 / EXCERPT: Let F be a totally real number field of narrow class number one, let B/F be a non-split quaternion algebra split at the Archimedean places $ u_1,\dots,\nu_r$, and let X_B be...

conjecture | open | MathDB 页面 | 原文线索

V2-019982 | MATHDB-358258

Plectic parametrization conjecture

摘录 / EXCERPT: Let F, B, X_B, E, and Σ be as above, and let φ_ν be the morphism of abelian varieties obtained from the plectic Oda conjecture and the computation of the p...

conjecture | open | MathDB 页面 | 原文线索

V2-019983 | MATHDB-358259

Algebraicity conjecture for plectic points

摘录 / EXCERPT: Let F be totally real, let E/F be a quadratic CM extension, let $E_\mathfrak c/E$ be the anticyclotomic extension of conductor $\mathfrak c$, and let ψ be an optimal embed...

conjecture | open | MathDB 页面 | 原文线索

V2-019984 | MATHDB-358273

The weak Landau-Ginzburg model formula for cyclic covers

摘录 / EXCERPT: Let Y be a smooth Fano variety of index r and $D_Y\thinspace\subseteq\thinspace Y$ a smooth anti-canonical divisor. Let X be the r-fold cyclic covering of Y branched alon...

conjecture | open | MathDB 页面 | 原文线索

V2-019985 | MATHDB-358274

The unconditional weak Landau-Ginzburg model formula for cyclic covers

摘录 / EXCERPT: Let Y be a smooth Fano variety of index r, let $D\subseteq Y$ be a smooth anti-canonical divisor, and let X be the corresponding cyclic cover. A graded exact Lagrangian torus...

conjecture | open | MathDB 页面 | 原文线索

V2-019986 | MATHDB-358315

Irreducibility conjecture for mixed-critical preperiodic loci

摘录 / EXCERPT: Mixed-critical irreducibility conjecture. This family is an irreducible quasi-affine subvariety of $\mathcal{M}_{d}$.

conjecture | open | MathDB 页面 | 原文线索

V2-019987 | MATHDB-358369

The Cohen-Lenstra rationality conjecture for motivic curve zeta series

摘录 / EXCERPT: Let X be a reduced curve over $\mathbb{F}_q$, and let $\widetilde X$ be its normalization. Let $\#_q$ denote the point-count homomorphism from the completed Grothendieck ring to...

conjecture | open | MathDB 页面 | 原文线索

V2-019988 | MATHDB-358370

The exact motivic formula conjecture for the cusp singularity

摘录 / EXCERPT: The exact motivic formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-019989 | MATHDB-358371

The motivic polynomiality conjecture for cusp punctual quotient schemes

摘录 / EXCERPT: Let (X,p) be the cusp singularity $x^2=y^3$, and let $\operatorname{Quot}_{d,n}(X,p)$ be its punctual quotient scheme. The motivic polynomiality conjecture. The motive of...

conjecture | open | MathDB 页面 | 原文线索

V2-019990 | MATHDB-358372

The polynomiality conjecture for pure-K commuting matrix varieties

摘录 / EXCERPT: Let $\alpha=(K(a_1),\dots,K(a_d))$ be a pure-K leading term datum, and let k be any field. Consider the affine variety ... defined by ... and by the vanishing conditions $X_{ij}=0$...

conjecture | open | MathDB 页面 | 原文线索

V2-019991 | MATHDB-358373

The conjectural recurrence and functional equation for cusp Hilbert series

摘录 / EXCERPT: For each $d\geq 0$, let $\mathit{NH}_d(t)$ be the normalized cusp Hilbert series, let ${d\brack r}_q$ be the q-binomial coefficient, and let $(t;q)_m$ be the finite q-Pochhamme...

conjecture | open | MathDB 页面 | 原文线索

V2-019992 | MATHDB-358374

The explicit normalized Hilbert-series polynomial conjecture for the cusp

摘录 / EXCERPT: Let A be a commutative ring with 1, let $q\in A$, and let $\mathit{NH}_d(t)\in A[t]$. Let ${d\brack r}_q$ be the q-binomial coefficient, $(t;q)_m$ the finite q-Pochhammer s...

conjecture | open | MathDB 页面 | 原文线索

V2-019993 | MATHDB-358375

The irreducibility conjecture for normalized cusp polynomials

摘录 / EXCERPT: Define ... Let $\Phi_r(q)$ denote the cyclotomic polynomial of order r. The normalized cusp-polynomial irreducibility conjecture. The polynomial $P_d(t;q)$ lies in...

conjecture | open | MathDB 页面 | 原文线索

V2-019994 | MATHDB-358381

The extended Dubrovin conjecture for Fano varieties

摘录 / EXCERPT: Extended Dubrovin conjecture. If $\mathrm{BQH}(X)$ is generically semisimple and $[K_X]$ is invertible in $\mathrm{QH}(X)_{can}$, then there is an exceptional collection...

conjecture | open | MathDB 页面 | 原文线索

V2-019995 | MATHDB-358404

Vanishing identity for adjoint Reidemeister torsions of closed 3-manifolds

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra of a semisimple complex Lie group G, let M be a connected closed oriented 3-manifold, and let $R^{\rm irr}_G(M)$ be the set of conjugacy...

conjecture | open | MathDB 页面 | 原文线索

V2-019996 | MATHDB-358415

The relative monodromy conjecture for abelian schemes

摘录 / EXCERPT: Let $\pi:\mathcal{A}\rightarrow B$ be an abelian scheme of relative dimension g with no fixed part, and let $\sigma:B\rightarrow\mathcal{A}$ be a section. Let $B^*\rightarrow B$...

conjecture | open | MathDB 页面 | 原文线索

V2-019997 | MATHDB-358420

The integral Hodge conjecture

摘录 / EXCERPT: Let X be a smooth projective complex variety. Write ... for the integral cycle class map, and let ... be the group of integral Hodge classes. Integral Hodge conjecture. For every suc...

conjecture | open | MathDB 页面 | 原文线索

V2-019998 | MATHDB-358421

Schoen's integral Tate conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over an algebraically closed field k, let $\ell\neq\operatorname{char}(k)$ be a prime, and choose a subfield $k_0\subseteq k$ over which...

conjecture | open | MathDB 页面 | 原文线索

V2-019999 | MATHDB-358432

Beauville-Voisin conjecture for birational automorphisms of hyper-Kähler varieties

摘录 / EXCERPT: Let Y be a smooth projective hyper-Kähler variety of dimension $2n$. Define ... where $\mathrm{S}_{s}{\rm CH}_0(Y)$ is the Beauville-Voisin filtration piece generated by points who...

conjecture | open | MathDB 页面 | 原文线索

V2-020000 | MATHDB-358444

The coincidence conjecture for the two-torsion line bundles on the Hessian surface

摘录 / EXCERPT: Let $[f]\in U$ be general, and let Y be the associated surface. The surface Y has two 2-torsion line bundles: $[?]$ the line bundle $[?]$ defined by $K_Y=3H|_Y+\eta$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-020001 | MATHDB-358477

Bloch-Beilinson regulator conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, let $L(E,s)$ be its L-function, and let $H^2_{\mathscr{M}}(E,\mathbb{Q}(2))_{\mathbb{Z}}$ denote the integral part of its motivic...

conjecture | open | MathDB 页面 | 原文线索

V2-020002 | MATHDB-358478

Mirror-symmetry compactification conjecture for genus-zero relative character varieties

摘录 / EXCERPT: Mirror-symmetry compactification conjecture. For every $n>3$ and generic $\boldsymbol{t}\in\mathbb{C}^n$, there exists a smooth compactification...

conjecture | open | MathDB 页面 | 原文线索

V2-020003 | MATHDB-358513

Commutativity conjecture for Hecke correspondences on Rapoport-Zink spaces

摘录 / EXCERPT: Let $t\neq t'$ and let A be any closed formal subscheme of the Rapoport-Zink space $\mathcal{N}_n$. The correspondence $\mathcal{T}_n^{\leq t}$ is obtained by composing the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-020004 | MATHDB-358516

The general-type non-density conjecture over function fields

摘录 / EXCERPT: Let K be a finitely generated field over a field k of characteristic 0, with k algebraically closed in K, and let X be a projective variety over K. General-type non-d...

conjecture | open | MathDB 页面 | 原文线索

V2-020005 | MATHDB-358517

The enhanced algebraic special-set conjecture

摘录 / EXCERPT: Let K be a finitely generated field over an algebraically closed field k of characteristic 0, and let X be a projective variety over K. Enhanced special-set conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-020006 | MATHDB-358539

Rigid-character lower-bound conjecture for skein modules

摘录 / EXCERPT: For a closed 3-manifold M, let $S(M,\mathbb{Q}(A))$ be its Kauffman bracket skein module over $\mathbb{Q}(A)$. Call an isolated, reduced character in the character variety...

conjecture | open | MathDB 页面 | 原文线索

V2-020007 | MATHDB-358559

Hilbert-scheme degeneration conjecture for quartic double solids

摘录 / EXCERPT: Let $Y\to\mathbb{P}^3$ be a quartic double solid whose ramification divisor contains no lines, and let $\cX/B$ be the family of prime Fano threefolds of genus 6 constructed in Th...

conjecture | open | MathDB 页面 | 原文线索

V2-020008 | MATHDB-358571

The canonical-degree linear bound conjecture for curves on smooth projective surfaces

摘录 / EXCERPT: Canonical-degree bound conjecture. There exist constants A and B, depending only on X, such that for every integral curve C on X,

conjecture | open | MathDB 页面 | 原文线索

V2-020009 | MATHDB-358578

Okounkov-Pandharipande parity conjecture for Hilbert schemes of points on threefolds

摘录 / EXCERPT: Okounkov-Pandharipande parity conjecture. For any $[Z]\in\operatorname{Hilb}^d X$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-020010 | MATHDB-358604

General form of averaged Igusa conjecture for exponential sums

摘录 / EXCERPT: Let K be a number field, let $\mathcal{I}\subset\mathcal{O}_K[x_1,\dots,x_n]$ be a non-zero ideal with $\mathcal{I}_K\neq(1)$, and let $Z\subset\mathbb{A}^n_{\mathcal{O}_K}$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-020011 | MATHDB-358605

Degree-sensitive averaged exponential-sum conjecture

摘录 / EXCERPT: Let $J\subset\mathbb{N}_{\geq1}$ be finite and non-empty. For each $i\in J$, let r_i be positive, let $f_{ij}\in\mathcal{O}_K[x_1,\dots,x_n]$ be non-constant with...

conjecture | open | MathDB 页面 | 原文线索

V2-020012 | MATHDB-358606

Uniform degree-sensitive averaged exponential-sum conjecture

摘录 / EXCERPT: Let J, $(r_i)_{i\in J}$, the polynomials f_{ij}, the ideal $\mathcal{I}$, r, and $\sigma_0((f_{ij}))$ be as in the degree-sensitive averaged exponential-sum conjecture. Unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-020013 | MATHDB-358607

Higher rational and Du Bois singularities via motivic oscillation

摘录 / EXCERPT: Let e be a positive integer, and let Z be a locally complete intersection variety of pure dimension over a number field K. Define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-020014 | MATHDB-358608

Raf Cluckers' conjecture on parametric suprema of constructible functions

摘录 / EXCERPT: Let $X\subset(0,\infty)\times\mathbb{R}^k$ be a subanalytic set, and let $F\colon X\to\mathbb{R}$ be a constructible, non-negative function, written $F(\varepsilon,\bm{x})$. A func...

conjecture | open | MathDB 页面 | 原文线索

V2-020015 | MATHDB-358611

The multiview conjecture for the Euclidean distance degree

摘录 / EXCERPT: Let X_m be the affine multiview variety of $m\geq 3$ cameras in general position, and let $\operatorname{EDdeg}(X_m)$ denote its Euclidean distance degree. Multiview conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-020016 | MATHDB-358625

Shiota's conjecture on Nash images of semialgebraic sets

摘录 / EXCERPT: Shiota's conjecture. ${\EuScript S}$ is a Nash image of ${\mathbb R}^d$ if and only if ${\EuScript S}$ is pure dimensional and there exists an analytic path

conjecture | open | MathDB 页面 | 原文线索

V2-020017 | MATHDB-358632

The compatibility conjecture for the motivic group structure on projective-line endomorphisms

摘录 / EXCERPT: Let $\mathcal{J}$ be the Jouanolou device of $\mathbb{P}^1$, and let $[\mathcal{J},\mathbb{P}^1]^\mathrm{N}$ and $[\mathbb{P}^1,\mathbb{P}^1]^{\mathbb{A}^1}$ denote the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-020018 | MATHDB-358633

The equality conjecture for the explicit subgroup of naive projective-line maps

摘录 / EXCERPT: Let $\mathcal{J}$ be the Jouanolou device of $\mathbb{P}^1$, and let $\mathbf{G}_0\subseteq\mathbf{G}$ be the subgroups introduced from the explicit generators, with...

conjecture | open | MathDB 页面 | 原文线索

V2-020019 | MATHDB-358634

Equidimensionality conjecture for polarized diagonal initial ideals

摘录 / EXCERPT: Let I_w be the relevant matrix Schubert ideal, let $\prec'$ be the lex-diagonal term order, and let $\widetilde{J}$ be the ideal obtained by polarizing...

conjecture | open | MathDB 页面 | 原文线索

V2-020020 | MATHDB-358635

Multiplicity-free bumpless pipe dream conjecture for the tablet

摘录 / EXCERPT: Let $\mathfrak X_w$ denote the relevant combinatorial object, let $\prec'$ be the chosen term order, and let $\mathcal T(\mathfrak X_w,\prec')$ be its tablet; let the support of a...

conjecture | open | MathDB 页面 | 原文线索

V2-020021 | MATHDB-358645

Viehweg's conjecture on the strengthened subadditivity of Kodaira dimension

摘录 / EXCERPT: Let $f\colon X \to Y$ be a fibration between smooth projective varieties with general fiber F and $\kappa(Y)\ge0$. Define $\operatorname{Var}(f)$ to be the variation of the fibra...

conjecture | open | MathDB 页面 | 原文线索

V2-020022 | MATHDB-358646

The determinant equality conjecture for Kodaira dimensions of fibrations

摘录 / EXCERPT: Let $f\colon X \to Y$ be a fibration between smooth projective varieties with general fiber F and $\kappa(Y)\ge0$. For a sufficiently divisible positive integer m such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020023 | MATHDB-358666

Generalized Bloch conjecture for correspondences on surfaces

摘录 / EXCERPT: Generalized Bloch conjecture. If $[Z]^*$ vanishes on $H^0(X,\Omega^2_X)$, then Z_* vanishes on $F^2CH_0(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-020024 | MATHDB-358673

Reducibility conjecture for Severi varieties on rational elliptic surfaces

摘录 / EXCERPT: Let R be a rational elliptic surface, and let $V(R,L)$ denote the genus 0 Severi variety associated to a line bundle L on R. Reducibility conjecture. There exist line bundl...

conjecture | open | MathDB 页面 | 原文线索

V2-020025 | MATHDB-358700

Exact optimality of k-closest encoding for rotationally symmetric codebooks

摘录 / EXCERPT: Let $U^M=(U_1,\ldots,U_M)$ be a rotationally symmetric codebook, and let k-closest encoding denote the proposed encoding scheme that selects the k closest codewords to the inpu...

conjecture | open | MathDB 页面 | 原文线索

V2-020026 | MATHDB-358713

Integral Milnor-Witt motivic decomposition conjecture for Stiefel varieties

摘录 / EXCERPT: Let S be the base scheme, let $V_k(\mathbb{A}^n)$ denote the Stiefel variety of k-frames in affine n-space, and let $\widetilde{\mathrm{M}}$ denote its Milnor-Witt motive. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-020027 | MATHDB-358721

The refined Gopakumar-Vafa/Pandharipande-Thomas correspondence

摘录 / EXCERPT: Refined Gopakumar-Vafa/Pandharipande-Thomas correspondence. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-020028 | MATHDB-358730

The shtuka cohomology conjecture for proper rigid-analytic varieties

摘录 / EXCERPT: Let C be the completed algebraic closure of a p-adic field, let $\mathcal{Y}_{\operatorname{FF}}=\operatorname{Spa}(A_{\inf},A_{\inf})\setminus V([p^{\flat}])$, and let A be...

conjecture | open | MathDB 页面 | 原文线索

V2-020029 | MATHDB-358739

The shrunken Weyl chamber conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be the reductive group in the setup, let $x=w\varepsilon^\mu\in\widetilde W$ lie in a shrunken Weyl chamber, so that $\operatorname{LP}(x)=\{v\}$ for a uniquely determined...

conjecture | open | MathDB 页面 | 原文线索

V2-020030 | MATHDB-358740

The integral basic-element conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let $x\in\widetilde W$ and let $[b]\in B(G)$ be integral, meaning that $\operatorname{defect}(b)=0$. For each $v\in\operatorname{LP}(x)$ and $u\in W$, define the multiset ... Set...

conjecture | open | MathDB 页面 | 原文线索

V2-020031 | MATHDB-358741

The Borisov-Alexeev-Borisov conjecture for Fano varieties

摘录 / EXCERPT: Let d be a positive integer and let $\boldsymbol{\mathcal{X}_{d,\varepsilon}}$ denote the class of ε-klt Fano varieties of dimension d, where $\varepsilon>0$ is rat...

conjecture | open | MathDB 页面 | 原文线索

V2-020032 | MATHDB-358762

Okounkov-Pandharipande parity conjecture for the Hilbert scheme of points on affine three-space

摘录 / EXCERPT: Fix $n\in\mathbb{Z}_{\geq 0}$, let $\mathrm{H}_n=\operatorname{Hilb}^n(\mathbb{A}^3)$ be the Hilbert scheme of ideals of colength n in $\mathbb{C}[x,y,z]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020033 | MATHDB-358771

Ambro--Kawamata effective non-vanishing conjecture

摘录 / EXCERPT: Ambro--Kawamata effective non-vanishing conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-020034 | MATHDB-358822

The two Igusa zeta function conjectures for degree five

摘录 / EXCERPT: The degree-five Igusa zeta function conjectures. The following two assertions are conjectured, with (b) stronger than (a): (a) $I_5(s)$ is a power series in q and t with positi...

conjecture | open | MathDB 页面 | 原文线索

V2-020035 | MATHDB-358842

Monomial-ideal characterization conjecture for geometric subsets of Hilbert schemes

摘录 / EXCERPT: Let $A_1,\ldots,A_s$ be finite-dimensional quotient algebras of $\mathbb{C}[x_1,\ldots,x_n]$, with $\dim_{\mathbb{C}}(A_i)=k_i$ and $k=k_1+\cdots+k_s$. Define the geometric subset...

conjecture | open | MathDB 页面 | 原文线索

V2-020036 | MATHDB-358846

The higher arithmetic-degree Kawaguchi-Silverman conjecture for subvarieties

摘录 / EXCERPT: Let X be a projective variety over a number field K, let $f:X\dashrightarrow X$ be a dominant rational self-map, and let V be an irreducible subvariety of dimension k. Supp...

conjecture | open | MathDB 页面 | 原文线索

V2-020037 | MATHDB-358901

The folklore conjecture on algebraic quantizations of Feigin-Odesskii varieties

摘录 / EXCERPT: Let X be a Calabi-Yau curve and let ξ be a simple vector bundle on X. Let $N(\xi)$ be the coarse moduli scheme of the moduli stack of non-splitting complexes $O_X\to V$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-020038 | MATHDB-358918

Smooth fibration conjecture for varieties with maximal PLI one-forms

摘录 / EXCERPT: Let X be a smooth minimal model, and let g be the maximal number of pointwise linearly independent (PLI) holomorphic one-forms that X carries. Smooth fibration conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-020039 | MATHDB-358989

Finite-depth conjecture for étale sheaves

摘录 / EXCERPT: Let X be a smooth variety over an algebraically closed field k of characteristic $p\neq 2$, and let $\mathcal{F}\in D(X)$. Write $SS\mathcal{F}$ for its singular support and le...

conjecture | open | MathDB 页面 | 原文线索

V2-020040 | MATHDB-358990

Uniform stability conjecture for vanishing cycles

摘录 / EXCERPT: Let X be a smooth variety over an algebraically closed field k of characteristic $p\neq 2$, let $\mathcal{F}\in D(X)$, and let (x,ξ) be a smooth point of $SS\mathcal{F}$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-020041 | MATHDB-359014

Asok-Østvær's stable \mathbb{A}^1-contractibility conjecture

摘录 / EXCERPT: Let X be a topologically contractible smooth affine complex variety, and let x be a chosen closed basepoint. Asok-Østvær's conjecture. The pointed variety (X,x) is stably...

conjecture | open | MathDB 页面 | 原文线索

V2-020042 | MATHDB-359032

The mirror conjecture for gauge-fixed double-cover families

摘录 / EXCERPT: Let Y and $Y^{\vee}$ be fibers of the gauge-fixed double-cover families associated with dual nef-partitions on toric varieties X and $X^{\vee}$, respectively. Double-cover mirr...

conjecture | open | MathDB 页面 | 原文线索

V2-020043 | MATHDB-359057

Generic injectivity conjecture for three-dimensional stiffness tensors

摘录 / EXCERPT: Let $E(3)$ be the space of three-dimensional stiffness tensors, let F map a stiffness tensor to its slowness polynomial, and write the target as $\mathbb{R}[p_1,p_2,p_3]$. The sl...

conjecture | open | MathDB 页面 | 原文线索

V2-020044 | MATHDB-359076

Xiao-Reid's moduli-stack Morsification conjecture

摘录 / EXCERPT: Xiao-Reid's Morsification conjecture. There exists a moduli stack $\mathcal{M}^{\star}_{g}$ of curves which contains all good smoothable curves of genus g as geometric points suc...

conjecture | open | MathDB 页面 | 原文线索

V2-020045 | MATHDB-359112

The geometric form of the dynamical Mordell-Lang conjecture

摘录 / EXCERPT: Let ${\mathbf{k}}$ be an algebraically closed field, let X be a quasi-projective variety over ${\mathbf{k}}$, let $f:X\dashrightarrow X$ be a dominant rational self-map, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020046 | MATHDB-359119

The single-relation conjecture for Calogero-Moser spaces and invariant commuting varieties

摘录 / EXCERPT: Single-relation conjecture. The coordinate rings satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-020047 | MATHDB-359122

The canonical isomorphism conjecture for generalized Barsotti-Tate quotient 2-stacks

摘录 / EXCERPT: Let G be a smooth affine group scheme over ${\mathbb{Z}}/p^n{\mathbb{Z}}$ and let $\mu:{\mathbb{G}}_m\to G$ be 1-bounded. Let $\operatorname{BT}_n^{G,\mu,?}$ be the quotient 2-st...

conjecture | open | MathDB 页面 | 原文线索

V2-020048 | MATHDB-359141

Conjecture on regular maps between homogeneous projective varieties

摘录 / EXCERPT: Let $X=G/P$ and $X'=G'/P'$ be homogeneous projective varieties, where G and G' are connected simple algebraic groups and P and P' are parabolic subgroups. For a realization...

conjecture | open | MathDB 页面 | 原文线索

V2-020049 | MATHDB-359160

The homological Broadhurst-Kreimer conjecture for the depth graded motivic Lie algebra

摘录 / EXCERPT: Let ${\mathfrak{d}}{\mathfrak{g}}$ be the depth graded motivic Lie algebra, and let S be the space of cusp forms for $\mathrm{SL}_2(\mathbb{Z})$. Homological Broadhurst-...

conjecture | open | MathDB 页面 | 原文线索

V2-020050 | MATHDB-359161

Brown's uneven Broadhurst-Kreimer conjecture for totally odd motivic multiple zeta values

摘录 / EXCERPT: Uneven Broadhurst-Kreimer conjecture. The bigraded dimensions of the totally odd part are given by

conjecture | open | MathDB 页面 | 原文线索

V2-020051 | MATHDB-359162

Brown's even-part conjecture for the depth graded motivic Lie algebra

摘录 / EXCERPT: Let ${\mathfrak{d}}{\mathfrak{g}}$ be the depth graded motivic Lie algebra, let ${\mathfrak{s}}{\mathfrak{d}}{\mathfrak{g}}$ be the subalgebra generated by the non-exceptional elem...

conjecture | open | MathDB 页面 | 原文线索

V2-020052 | MATHDB-359163

The totally odd spanning conjecture for motivic multiple zeta values

摘录 / EXCERPT: Totally odd spanning conjecture. Totally odd zeta values of depth r span this vector space. The claim is a reformulation of the proposed identification between the totally odd pa...

conjecture | open | MathDB 页面 | 原文线索

V2-020053 | MATHDB-359164

The depth-to-even-block isomorphism conjecture

摘录 / EXCERPT: Depth-to-even-block isomorphism conjecture. The map ${\mathfrak{s}}{\mathfrak{d}}{\mathfrak{g}}\to{\mathfrak{e}}{\mathfrak{b}}{\mathfrak{g}}$ is an isomorphism. It is stated to hol...

conjecture | open | MathDB 页面 | 原文线索

V2-020054 | MATHDB-359165

The lower-depth totally odd spanning conjecture

摘录 / EXCERPT: Let ${\mathcal{D}}_n$ and ${\mathcal{B}}_{n-1}$ denote the depth and block filtrations, and work modulo products. Lower-depth totally odd spanning conjecture. The vector space ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-020055 | MATHDB-359166

The depth-strengthening conjecture for block-to-depth relations

摘录 / EXCERPT: Let ξ_B and ξ_D be the block-graded and depth-graded functionals associated with the relations in the preceding block-to-depth reduction theorem, and let σ be an ele...

conjecture | open | MathDB 页面 | 原文线索

V2-020056 | MATHDB-359178

Pullback functors are strong tangent morphisms for formal schemes

摘录 / EXCERPT: Let $\operatorname{\mathfrak{X}}$ and $\operatorname{\mathfrak{Y}}$ be formal schemes over a base scheme S, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020057 | MATHDB-359179

Closed immersions are preserved by the formal tangent scheme

摘录 / EXCERPT: Let $\operatorname{\mathfrak{X}}=(X_i)_{i\in I}$ be a formal scheme over a base scheme S, presented as an inductive system of schemes. Suppose that every morphism appearing in th...

conjecture | open | MathDB 页面 | 原文线索

V2-020058 | MATHDB-359195

Genus-one STE generation conjecture for degree-15 cohomology

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable curves, and let $H^{11}(\overline{\mathcal{M}}_{1,11})$ and $H^{15}(\overline{\mathcal{M}}_{g,n})$ denote their coh...

conjecture | open | MathDB 页面 | 原文线索

V2-020059 | MATHDB-359214

Schiffmann's positivity conjecture for Kac polynomials of smooth projective curves

摘录 / EXCERPT: Let $(g,r,d)\in\mathbf{N}\times\mathbf{Z}_{\geqslant 1}\times\mathbf{Z}$ be a triple consisting of a genus g, a rank r, and a degree d. The polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-020060 | MATHDB-359217

Chi-independence of the module structure for Betti stacks

摘录 / EXCERPT: Let $r,d\in\mathbf{Z}$ with $r>0$, and let $\mathscr{A}_{\mathbf{Z}_{\geqslant1}\cdot(r,d)}^{\operatorname{Betti}}$ be the indicated Betti CoHA subalgebra object. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020061 | MATHDB-359222

Asymptotic minimal-model invariants conjecture for root covers

摘录 / EXCERPT: Let $X_n \to Y_n \to (Z,D)$ be a terminal good partial resolution of singularities of the n-th root cover construction. Assume that K_{Y_n} is nef, and let X'_n be a minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-020062 | MATHDB-359223

Jonsson-Mustață quasi-monomiality conjecture for log canonical thresholds

摘录 / EXCERPT: Let X be a klt variety and let $\mathfrak{a}_\bullet$ be a graded sequence of ideals on X with finite log canonical threshold $\operatorname{lct}(\mathfrak{a}_\bullet)$. A valu...

conjecture | open | MathDB 页面 | 原文线索

V2-020063 | MATHDB-359250

Orbit propagation principles for rational points

摘录 / EXCERPT: Let K) be a number field, let X/K be a smooth projective variety, and let $f:X\to X$ be an endomorphism defined over K. For $P\in X$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-020064 | MATHDB-359251

Silverman's arithmetic-order trichotomy conjecture

摘录 / EXCERPT: Let X/K be a smooth projective variety over a number field, let H be a Weil height associated to an ample divisor, and define the arithmetic order $\mathsf{A}(X)$ using the ite...

conjecture | open | MathDB 页面 | 原文线索

V2-020065 | MATHDB-359252

Finite-orbit-union conjecture for endomorphisms of dynamical degree greater than one

摘录 / EXCERPT: Let K be a number field, let X/K be a smooth projective variety, and let $f:X\to X$ be an endomorphism defined over K with dynamical degree $\delta(f)>1$. Finite-orbit-union...

conjecture | open | MathDB 页面 | 原文线索

V2-020066 | MATHDB-359287

Smoothness conjecture for Hessenberg Schubert varieties

摘录 / EXCERPT: Smoothness conjecture. The following statements are equivalent: $\Omega_{w,h}$ is smooth; $\Gamma_{w,h}$ is a regular graph; and w avoids all the patterns $\overline{2143}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-020067 | MATHDB-359318

Finiteness conjecture for Homs in isotropic motivic categories

摘录 / EXCERPT: Finiteness conjecture for isotropic motivic Homs. Homs between compact objects of $DM(k/k;\mathbb{F}_p)$ are finite groups.

conjecture | open | MathDB 页面 | 原文线索

V2-020068 | MATHDB-359336

The Gamma-conjecture III on exponential-type quantum connections

摘录 / EXCERPT: Let X be a smooth projective variety, let $\tau\in H^*(X)$ be a point where the quantum connection is of exponential type, and let C index the exponential factors. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020069 | MATHDB-359337

The crepant transformation conjecture for quantum D-modules

摘录 / EXCERPT: Let X_1 and X_2 be smooth projective varieties or smooth Deligne-Mumford stacks with projective coarse moduli spaces, and let $\varphi\colon X_1\dashrightarrow X_2$ be a crepan...

conjecture | open | MathDB 页面 | 原文线索

V2-020070 | MATHDB-359345

Kontsevich's sphere conjecture for dual boundary complexes of log Calabi-Yau varieties

摘录 / EXCERPT: Let X be a log Calabi-Yau variety, and let $\mathbb{D}\partial X$ denote its dual boundary complex. Kontsevich's conjecture. The dual boundary complex $\mathbb{D}\partial X$ is h...

conjecture | open | MathDB 页面 | 原文线索

V2-020071 | MATHDB-359352

The irreducible hypersurface Pythagoras-number conjecture

摘录 / EXCERPT: Irreducible hypersurface Pythagoras-number conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-020072 | MATHDB-359354

Kumar's conjecture on morphisms between homogeneous projective varieties

摘录 / EXCERPT: Let X and X' be connected indecomposable homogeneous projective varieties. For such a variety, write $\operatorname{minss\ rank} X$ and $\operatorname{maxss\ rank} X$ for its m...

conjecture | open | MathDB 页面 | 原文线索

V2-020073 | MATHDB-359358

Monodromy quantum group isomorphism conjecture

摘录 / EXCERPT: Monodromy quantum group isomorphism conjecture. There are isomorphisms of algebras

conjecture | open | MathDB 页面 | 原文线索

V2-020074 | MATHDB-359393

Han's eigenvariety dimension conjecture for G

摘录 / EXCERPT: Let G be the reductive group considered in the paper, let $K=K^p\mathrm{Iw}$ be Iwahori level at p, and let $\sE_K^G$ be its eigenvariety. Let $x\in\sE_K^G$ be a non-critical c...

conjecture | open | MathDB 页面 | 原文线索

V2-020075 | MATHDB-359394

The transfer conjecture for symplectic families of refined RASCARs

摘录 / EXCERPT: Let $\tilde\pi$ be a p-refined RASCAR of GL_{2n}, and let $P_{\tilde\pi}$ be the associated parabolic. A symplectic family is a family through $\tilde\pi$ in the eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-020076 | MATHDB-359395

The classical-family transfer conjecture for refined RASCARs on GL_4

摘录 / EXCERPT: Let $\tilde\pi$ be a p-refined RASCAR of GL_4, and let $P_{\tilde\pi}$ be its associated parabolic. A classical family is a family through $\tilde\pi$ consisting of cl...

conjecture | open | MathDB 页面 | 原文线索

V2-020077 | MATHDB-359429

Characteristic-two Nim-polynomial formula for divided-power cohomology

摘录 / EXCERPT: Let $\boldsymbol{k}$ be a field of characteristic 2, let $d and $egeq d-1$, and let $mathcal Nm denote the Nim symmetric polynomial. For each $q=2^r$ with $r\geq1$ and each...

conjecture | open | MathDB 页面 | 原文线索

V2-020078 | MATHDB-359457

Defant and Zheng's maximal-time conjecture for the consecutive-pattern-avoiding stack-sorting map

摘录 / EXCERPT: Let S_n be the set of permutations of length n, let $SC_{231}:S_n\to S_n$ be the stack-sorting map whose stack avoids consecutive occurrences of 231, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-020079 | MATHDB-359472

Largest anticanonical volume conjecture for Gorenstein terminal Fano varieties

摘录 / EXCERPT: For each odd integer $n\geq5$, let s_i be the sequence used in the paper, and let X be the n-fold in the theorem constructing the Gorenstein terminal weighted projective spac...

conjecture | open | MathDB 页面 | 原文线索

V2-020080 | MATHDB-359489

The Tate-linear conjecture for products of GSpin Shimura varieties

摘录 / EXCERPT: Let $f:X_0\to\mathscr{S}_{\mathbf{I},F_{\infty}}^{\operatorname{ord}}$ be a locally closed immersion, so that X is a subvariety of...

conjecture | open | MathDB 页面 | 原文线索

V2-020081 | MATHDB-359490

The characteristic p Mumford-Tate conjecture for products of GSpin Shimura varieties

摘录 / EXCERPT: Let (X,x), X_0, and f be as in the source, let $u\in\operatorname{fpl}(\Q)$, and let $G(\L_{\mathbf{I},u,X_0},x)$ and $\operatorname{Lg}(f)_{\Q_u}$ be the groups defined ther...

conjecture | open | MathDB 页面 | 原文线索

V2-020082 | MATHDB-359491

The characteristic p André-Oort conjecture for products of GSpin Shimura varieties

摘录 / EXCERPT: Let $f:X_0\to\mathscr{S}_{\mathbf{I},F_{\infty}}^{\operatorname{ord}}$ be a locally closed immersion, and let X be its image. Suppose that X possesses a Zariski-dense collectio...

conjecture | open | MathDB 页面 | 原文线索

V2-020083 | MATHDB-359492

The zip-cone conjecture for Hodge-type Shimura varieties

摘录 / EXCERPT: Let G be the reductive group over $\mathbb{F}_p$ attached to a Hodge-type Shimura variety, let μ be its Hodge cocharacter, and let $\mathop{\mathrm{Zip}}\nolimits_G^\mu$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-020084 | MATHDB-359493

The orbit-inequality conjecture for automorphic forms on Hodge-type Shimura varieties

摘录 / EXCERPT: Let S_K be the special fiber of a Hodge-type Shimura variety at a prime p of good reduction which splits in the reflex field $\mathbf E$. Let G be the attached reductive grou...

conjecture | open | MathDB 页面 | 原文线索

V2-020085 | MATHDB-359494

The cactus flower moduli-space homeomorphism conjecture

摘录 / EXCERPT: Let $\breve D_n$ be the cube complex whose cubes are indexed by $[n]$-labelled cyclic forests, and let $\overline M_{n+2}^\sigma(\mathbb{R})$ be the indicated real locus of the mod...

conjecture | open | MathDB 页面 | 原文线索

V2-020086 | MATHDB-359512

Geometric Verdier duality for pro-étale cohomology of Stein rigid analytic varieties

摘录 / EXCERPT: Let C be the base field, let X be a smooth Stein rigid analytic variety over C, connected and of dimension d, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-020087 | MATHDB-359513

The sectional-degree formula for the Euclidean distance degree

摘录 / EXCERPT: Let $X \subseteq \mathbb{C}^n$ be an irreducible affine variety, and let $\overline{X} \subset \mathbb{P}^n$ be its projective closure. Let Q_{∞} denote the quadric at infin...

conjecture | open | MathDB 页面 | 原文线索

V2-020088 | MATHDB-359519

Esser's coprimality conjecture for the large-index Calabi-Yau construction

摘录 / EXCERPT: For each integer $n\geq 2$, let m'_n and E_n be the explicitly defined integers associated with the weighted hypersurface construction. Two integers are relatively prime if the...

conjecture | open | MathDB 页面 | 原文线索

V2-020089 | MATHDB-359561

The logarithmic Hochschild-Kostant-Rosenberg conjecture

摘录 / EXCERPT: Let X be a log scheme for which the constructions in the paper are defined, let $i:X\to B$ be the local complete intersection morphism used to define logarithmic Hochschild cohom...

conjecture | open | MathDB 页面 | 原文线索

V2-020090 | MATHDB-359596

Fine-Panov conjecture on monotone Hamiltonian circle actions

摘录 / EXCERPT: Fine-Panov conjecture. M is diffeomorphic to a smooth Fano threefold.

conjecture | open | MathDB 页面 | 原文线索

V2-020091 | MATHDB-359603

Existence of rational points on varieties with nef anti-canonical class over finite fields

摘录 / EXCERPT: Rational-point conjecture. For every positive integer d there exists an integer q_d such that, whenever $-K_X$ is nef and $q\geq q_d$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-020092 | MATHDB-359604

Characteristic polynomial fiber-dimension conjecture for partially symmetric tensors

摘录 / EXCERPT: Characteristic polynomial fiber-dimension conjecture. The generic fiber of Φ is equidimensional of dimension n.

conjecture | open | MathDB 页面 | 原文线索

V2-020093 | MATHDB-359610

Linear dependence conjecture for Norton algebra vectors

摘录 / EXCERPT: Let $\Gamma=(X,\mathcal R)$ be a distance-regular graph that is Q-polynomial with respect to a primitive idempotent E. For distinct vertices $x,y\in X$, let $C(x,y)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020094 | MATHDB-359611

Conjecture on the size of isogeny classes of abelian varieties over finite fields

摘录 / EXCERPT: Conjecture on isogeny class sizes. We have

conjecture | open | MathDB 页面 | 原文线索

V2-020095 | MATHDB-359617

The anomaly-free symplectic representation conjecture for strongly tempered spherical varieties

摘录 / EXCERPT: Let (G,H) be a strongly tempered spherical pair that is the Whittaker induction of (G_0,H_0,ξ), with G admitting a quasi-split pure inner form G_{qs}. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-020096 | MATHDB-359618

The well-defined epsilon-character conjecture for strongly tempered spherical varieties

摘录 / EXCERPT: The epsilon-character conjecture. The function ω_{ϕ',ρ_X} is well defined, independent of the extended endoscopic triple and the lifting, descends to a function on...

conjecture | open | MathDB 页面 | 原文线索

V2-020097 | MATHDB-359619

The multiplicity conjecture for strongly tempered spherical varieties

摘录 / EXCERPT: Multiplicity conjecture. There exists a choice of Whittaker datum of G_{qs}, depending only on (G,H,ξ) and not on π, such that $m(\pi)$ equals the number of irreducible r...

conjecture | open | MathDB 页面 | 原文线索

V2-020098 | MATHDB-359642

Representability conjecture for truncated affine Grassmannian double quotients

摘录 / EXCERPT: Representability conjecture. The fpqc sheaf $\prescript{}{1}{\mathcal{C}_1^{\leq\mu}}$ is represented by a normal k-scheme. This predicts that the indicated truncated double quot...

conjecture | open | MathDB 页面 | 原文线索

V2-020099 | MATHDB-359645

Elliptic dynamical quantum supergroup actions for type A quiver varieties

摘录 / EXCERPT: Type A quiver-variety action conjecture. There exists an action of the elliptic dynamical quantum supergroup $E_{\hbar,\tau}(\mathfrak g_Q)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-020100 | MATHDB-359651

Gowers's intersection conjecture for approximate quadratic varieties

摘录 / EXCERPT: Gowers's conjecture. If ε is sufficiently small in terms of δ and c_0, then there exists a quadratic variety Q such that

conjecture | open | MathDB 页面 | 原文线索

V2-020101 | MATHDB-359674

Secant-variety extremality conjecture for blow ups along rational normal curves

摘录 / EXCERPT: Let $C\subset\mathbb{P}^n$ be a rational normal curve, and let $X_n=\operatorname{Bl}_C\mathbb{P}^n$. For an integer d, write $\operatorname{Sec}_d(C)$ for the d-secant variety...

conjecture | open | MathDB 页面 | 原文线索

V2-020102 | MATHDB-359708

Combinatorial-conformal-field-theoretic mirror conjecture for Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let Δ and Δ^* be dual reflexive polyhedra, and let MIR be the involution sending the family ${\cal F}(\Delta)$ to ${\cal F}(\Delta^*)$. Let ${\cal MIR}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-020103 | MATHDB-359709

Toroidal boundary singularities conjecture for Calabi-Yau mirror moduli

摘录 / EXCERPT: Let ${\cal M}$ be a moduli space of complex structures on an n-dimensional Calabi-Yau variety W, and let ${\cal M}'$ be a moduli space for a mirror Calabi-Yau variety W'. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020104 | MATHDB-359726

Clemens's single-vanishing-cycle tube mapping conjecture

摘录 / EXCERPT: Let $X\subseteq \mathbb P^N$ be a smooth projective variety of dimension n, let $Y=X\cap H$ be a smooth hyperplane section, let $U\subseteq (\mathbb P^N)^*$ be the open set of sm...

conjecture | open | MathDB 页面 | 原文线索

V2-020105 | MATHDB-359769

Lazić-Peternell's generalized abundance conjecture

摘录 / EXCERPT: Let X be a projective manifold with K_X pseudoeffective, and let $L\to X$ be a nef line bundle such that ... is nef for some $m\geqslant 1$. Lazić-Peternell's generalized abundan...

conjecture | open | MathDB 页面 | 原文线索

V2-020106 | MATHDB-359787

Beltrametti-Sommese conjecture on effective strictly nef divisors

摘录 / EXCERPT: Beltrametti-Sommese conjecture. If $D-K_X$ is nef, then D is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-020107 | MATHDB-359796

Coherent completeness and formal functions for geometrically reductive quotient stacks

摘录 / EXCERPT: Coherent completeness and formal functions conjecture. The stack $[X/G]$ is cohomologically proper over A^G, and the pair $([X/G],\mathcal{G}_x)$ is coherently complete and satis...

conjecture | open | MathDB 页面 | 原文线索

V2-020108 | MATHDB-359828

The determinantal presentation conjecture for the invariant ring

摘录 / EXCERPT: Let Q be the quiver and I the relations defining the representation space in the preceding construction, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020109 | MATHDB-359832

Comparison conjecture for quantum and elliptic cuspidal modules

摘录 / EXCERPT: Let G be a connected reductive group, let E be an elliptic curve, and let $\operatorname{Bun}_G^{ss,0}(E)$ denote the moduli stack of semistable degree-zero G-bundles on E....

conjecture | open | MathDB 页面 | 原文线索

V2-020110 | MATHDB-359858

The Euclidean distance degree formula for multiview camera resectioning

摘录 / EXCERPT: Let $X_{\bar{\mathbf{q}}, n}$ be the affine variety of image configurations arising from a camera and world points $\bar{\mathbf{q}} \in (\mathbf P^3)^n$. For generic data, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020111 | MATHDB-359884

The supercritical collection conjecture for hard Lefschetz classes

摘录 / EXCERPT: Supercritical collection conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020112 | MATHDB-359885

The arbitrary-collection conjecture for hard Lefschetz kernels

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n, and let $\mathfrak{L}=(L_1,\ldots,L_{n-2})$ be a collection of nef classes on X. Define ... and let ... where the span is ove...

conjecture | open | MathDB 页面 | 原文线索

V2-020113 | MATHDB-359907

The corrected twisted Hodge-number generating-series conjecture for Hilbert schemes

摘录 / EXCERPT: Let S be a smooth projective surface, let $L\in \operatorname{Pic} S$, and let L_n denote the induced line bundle on Hilb^nS. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020114 | MATHDB-359908

The twisted Hodge McKay correspondence conjecture for Hilbert schemes

摘录 / EXCERPT: Let S be a smooth projective surface, let $L\in \operatorname{Pic} S$, and let L_n and $L^{\{n\}}$ denote the induced coefficients on Hilb^nS and...

conjecture | open | MathDB 页面 | 原文线索

V2-020115 | MATHDB-359909

Beilinson's conjecture on the Albanese kernel of varieties over the algebraic closure of the rationals

摘录 / EXCERPT: Let X be a smooth projective variety over $\overline{\mathbb{Q}}$. The filtration on its zero-cycle Chow group is ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-020116 | MATHDB-359920

The alternating Euler-sum permutation conjecture for unramified motivic sums

摘录 / EXCERPT: Let $a\in\mathbb{N}_0$, $\ell\in\mathbb{N}$, and let $u_j,v_j\in X:=\{1,\bar2,3,\bar4,\dots\}$ for $j=1,\dots,\ell$, with ... For $\mathfrak S_\ell$ the symmetric group on ℓ let...

conjecture | open | MathDB 页面 | 原文线索

V2-020117 | MATHDB-359921

Dream Building Blocks conjecture for surjective endomorphisms

摘录 / EXCERPT: Let X be a normal projective variety and let $f:X\to X$ be a non-isomorphic surjective endomorphism. The properties weakly δ-imprimitive, strongly imprimitive, quasi-abeli...

conjecture | open | MathDB 页面 | 原文线索

V2-020118 | MATHDB-359932

The flexible affine variety conjecture for RR-spaces

摘录 / EXCERPT: Flexible affine variety conjecture. Every smooth flexible complex affine variety is an RR-space.

conjecture | open | MathDB 页面 | 原文线索

V2-020119 | MATHDB-359943

tt-separatedness conjecture for abelian varieties

摘录 / EXCERPT: Let X be an abelian variety. A smooth projective variety is tt-separated when the relevant Fourier-Mukai copies in its triangular spectrum are pairwise disjoint. Abelian tt-separ...

conjecture | open | MathDB 页面 | 原文线索

V2-020120 | MATHDB-359944

Kawamata's DK hypothesis for K-equivalent varieties

摘录 / EXCERPT: Let X and Y be smooth projective varieties. They are K-equivalent if there is a crepant birational map between them, and they are Fourier-Mukai partners if their perfect deri...

conjecture | open | MathDB 页面 | 原文线索

V2-020121 | MATHDB-359945

Proper Fourier-Mukai locus conjecture for triangulated categories

摘录 / EXCERPT: Let $\mathcal T$ be a triangulated category with $X\in\operatorname{FM}\mathcal T$. The proper Fourier-Mukai locus $\operatorname{Spec}^{\mathsf{pFM}}\mathcal T$ and the Serre-inva...

conjecture | open | MathDB 页面 | 原文线索

V2-020122 | MATHDB-359946

Topological structure conjecture for the Serre-invariant locus

摘录 / EXCERPT: Let $\mathcal T$ be a triangulated category with $X\in\operatorname{FM}\mathcal T$. A topological space is locally spectral if it has an open cover by spectral spaces, and locally...

conjecture | open | MathDB 页面 | 原文线索

V2-020123 | MATHDB-359954

Quasi-BPS equivalence conjecture for zero-dimensional sheaves on the plane

摘录 / EXCERPT: Let $d\geqslant 1$, let $w\in\mathbb{Z}$, and let $\mathscr{C}(d)^{\rm{red}}$ be the reduced derived moduli stack of length-d zero-dimensional sheaves on $\mathbb{C}^2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020124 | MATHDB-359960

Generic torsion vanishing conjecture for Shimura varieties

摘录 / EXCERPT: Let $(\mathbf{G},X)$ be a Shimura datum such that $G=\mathbf{G}_{\mathbb{Q}_{p}}$ is unramified, and let $K=K_{p}K^{p}$ be a sufficiently small level with...

conjecture | open | MathDB 页面 | 原文线索

V2-020125 | MATHDB-359961

Weak Langlands-Shahidi torsion vanishing conjecture for Shimura varieties

摘录 / EXCERPT: Let ϕ be a semi-simple L-parameter of weakly Langlands-Shahidi type with cuspidal support (M,ϕ_{M}). Let $\mathcal{S}(\mathbf{G},X)_{K^{p},C}$ be the corresponding Shim...

conjecture | open | MathDB 页面 | 原文线索

V2-020126 | MATHDB-359975

Beauville's generic vanishing conjecture for the Albanese morphism

摘录 / EXCERPT: Beauville's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-020127 | MATHDB-359992

The extension conjecture for non-toric staged tree models

摘录 / EXCERPT: Let $\mathcal T$ be a staged tree with non-toric $V(\ker \varphi_{\mathcal T})$ that uses color set S. Let $\mathcal T'$ be a staged tree such that its restriction to color set...

conjecture | open | MathDB 页面 | 原文线索

V2-020128 | MATHDB-359999

DK conjecture for simple flips of homogeneous type

摘录 / EXCERPT: Let X_{α} and X_{β} be the varieties associated with the simple flip in Proposition, and let $D(X_{\alpha})$ and $D(X_{\beta})$ denote their triangulated derived categ...

conjecture | open | MathDB 页面 | 原文线索

V2-020129 | MATHDB-360023

The motivic t-structure conjecture for Voevodsky motives

摘录 / EXCERPT: Let k be the base field, let $\mathcal{DM}_c(k)$ denote the triangulated category of constructible Voevodsky motives over k, and let the ℓ-adic realisation be the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-020130 | MATHDB-360026

André-Baldassarri equivalence conjecture for arithmetic flat bundles

摘录 / EXCERPT: Let X be a smooth algebraic variety over $\overline{\mathbb{Q}}$ and let $\mathscr{F}$ be a flat bundle on X. A G-connection is a flat bundle satisfying the arithmetic condit...

conjecture | open | MathDB 页面 | 原文线索

V2-020131 | MATHDB-360029

Rational points conjecture for generic marked hypersurfaces

摘录 / EXCERPT: Rational points conjecture. If $d\geq4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-020132 | MATHDB-360043

K-polystability conjecture for Casagrande-Druel varieties

摘录 / EXCERPT: A Casagrande-Druel variety X is constructed from a Fano double cover with base space B. K-polystability conjecture. If V and B are K-polystable Fano varieties, then X is...

conjecture | open | MathDB 页面 | 原文线索

V2-020133 | MATHDB-360068

Polishchuk's localization conjecture for Kazhdan-Laumon categories

摘录 / EXCERPT: Let G be a semisimple algebraic group split over a finite field $\mathbb{F}_q$, let $\mathcal{A}_{\mathbb{F}_q}=\mathcal{A}_{e,\mathbb{F}_q}$ be the category of Weil sheaves in t...

conjecture | open | MathDB 页面 | 原文线索

V2-020134 | MATHDB-360072

Regularizability conjecture for bimeromorphic maps of polarized abelian families

摘录 / EXCERPT: Let $f\colon \mathcal{X}\dashrightarrow \mathcal{X}$ be a bimeromorphic map of a family of polarized abelian varieties that fixes the 0-section. A regularizability conjecture ass...

conjecture | open | MathDB 页面 | 原文线索

V2-020135 | MATHDB-360073

Degree formula for realization spaces of matrix multiplication varieties

摘录 / EXCERPT: Realization-space degree conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020136 | MATHDB-360102

The non-degeneracy conjecture for the realizable generic Enriques surface Y_{158}

摘录 / EXCERPT: Let Y_{158} be the realizable $(\tau,\overline{\tau})$-generic Enriques surface associated with the family indexed by 158, and let $\mathrm{nd}(Y_{158})$ denote its non-degener...

conjecture | open | MathDB 页面 | 原文线索

V2-020137 | MATHDB-360108

De Cataldo-Hausel-Migliorini's P=W conjecture

摘录 / EXCERPT: Let $\mathcal{M}_{B}(r,n)$ and $\mathcal{M}_{Dol}(r,n)$ be the Betti and Dolbeault moduli spaces, let h be the Hitchin map, and let $W_\bullet$ and $P_\bullet$ denote the weight...

conjecture | open | MathDB 页面 | 原文线索

V2-020138 | MATHDB-360109

The PI=WI conjecture for singular character varieties

摘录 / EXCERPT: Let r,n be integers with $r>0$, not necessarily coprime, and let $\mathcal{M}_{B}(r,n)$ and $\mathcal{M}_{Dol}(r,n)$ be the Betti and Dolbeault moduli spaces. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020139 | MATHDB-360111

Filtered synthetic deformation models motivic realizations

摘录 / EXCERPT: Let $\mathcal{C}$ be the realization target, let L be its Tate object, and let $\nu(L^{\otimes *})$ be the filtered object induced by the slice tower. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020140 | MATHDB-360112

Frobenius minimal ED-degree conjecture for Segre-Veronese varieties

摘录 / EXCERPT: Frobenius minimal ED-degree conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020141 | MATHDB-360128

The intermediate-variety dimension conjecture for orbit closures

摘录 / EXCERPT: Let ${\cal K}$ and ${\cal H}_{\overline{y_e}}$ be the stabilizer-related groups associated with the orbit-closure data, and let $\overline{O(z)}\subseteq\overline{O(y)}$ be orbit c...

conjecture | open | MathDB 页面 | 原文线索

V2-020142 | MATHDB-360133

Simpson's geometric origin conjecture for rigid local systems

摘录 / EXCERPT: Let X be a smooth projective complex variety, and let $\mathcal{L}$ be an irreducible rigid local system on X, meaning that it cannot be deformed continuously or algebraically...

conjecture | open | MathDB 页面 | 原文线索

V2-020143 | MATHDB-360148

Debarre's strong Schottky conjecture for theta divisors

摘录 / EXCERPT: Debarre's strong Schottky conjecture. If (A,Θ) is not a hyperelliptic Jacobian or the intermediate Jacobian of a smooth cubic threefold, then

conjecture | open | MathDB 页面 | 原文线索

V2-020144 | MATHDB-360149

Casalaina-Martin's singular-locus conjecture for theta divisors

摘录 / EXCERPT: Casalaina-Martin's conjecture. If (A,Θ) is an indecomposable principally polarized abelian variety, then

conjecture | open | MathDB 页面 | 原文线索

V2-020145 | MATHDB-360158

Nil-projection conjecture for double flag varieties

摘录 / EXCERPT: Let $\mathfrak{X}$ be a double flag variety of finite type, with projection ... Let $\mathcal{N}_{\mathfrak{s}}$ denote the nilpotent variety in $\mathfrak{s}$. Nil-projection conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-020146 | MATHDB-360191

Chai's additivity conjecture for base change conductors

摘录 / EXCERPT: Chai's additivity conjecture. Assume that T is a torus and that A is an Abelian variety. Then

conjecture | open | MathDB 页面 | 原文线索

V2-020147 | MATHDB-360192

Hausel-Letellier-Rodriguez-Villegas conjecture for generic character stacks

摘录 / EXCERPT: Let $\mathcal{C}$ be a generic k-tuple of semisimple conjugacy classes, let $\alpha_{\mathcal{C}}$ be its associated dimension vector, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020148 | MATHDB-360193

Mixed-Poincare conjecture for non-generic character stacks

摘录 / EXCERPT: Let $\mathcal{C}$ be a k-tuple of semisimple orbits, let $\alpha_{\mathcal{C}}$ be its associated dimension vector, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020149 | MATHDB-360194

Dual Log-compatibility conjecture for multiplicative quiver-stack point counts

摘录 / EXCERPT: Let $Q=(I,\Omega)$ be a quiver, let $\mathcal{M}_{s,\alpha}$ be the associated multiplicative quiver stack over $\mathbb{F}_q$, and define the class function...

conjecture | open | MathDB 页面 | 原文线索

V2-020150 | MATHDB-360202

Broadness and freeness conjecture for intersections with the Gamma graph

摘录 / EXCERPT: Let n be a positive integer. For coordinate projections $\mathrm{Pr}_{\mathbf{i}}$ onto any selected ℓ pairs of coordinates, call an algebraic set...

conjecture | open | MathDB 页面 | 原文线索

V2-020151 | MATHDB-360232

The c_1-cohomological rigidity conjecture for smooth toric Fano varieties

摘录 / EXCERPT: c_1-cohomological rigidity conjecture. If there exists a c_1-preserving graded ring isomorphism between the integral cohomology rings of X and Y, then X and Y are isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-020152 | MATHDB-360240

Quantum K Whitney relations for partial flag varieties

摘录 / EXCERPT: Quantum K Whitney relations. For $j=1,\dots,k$, the following relations hold in $\operatorname{QK}_T(X)$:

conjecture | open | MathDB 页面 | 原文线索

V2-020153 | MATHDB-360295

Beilinson's conjecture for motivic cohomology of genus-one curves

摘录 / EXCERPT: Let C be a smooth projective curve over $\mathbb Q$ of genus 1, let E be its Jacobian, and let α be a nontrivial element of $H^2_\mathcal M(C,\mathbb Q(3))$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020154 | MATHDB-360305

Fano reduced-invariant conjecture

摘录 / EXCERPT: Let X be a Fano variety for which both the reduced and standard Gromov-Witten invariants are defined. Fano reduced-invariant conjecture. The reduced and standard Gromov-Witten in...

conjecture | open | MathDB 页面 | 原文线索

V2-020155 | MATHDB-360307

The extended Lang-Vojta conjecture for quasi-projective varieties

摘录 / EXCERPT: Extended Lang-Vojta conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-020156 | MATHDB-360315

The tame Breuil-Mézard conjecture for unramified groups

摘录 / EXCERPT: Let G be an unramified group over $\mathbf{Q}_p$ satisfying Hypothesis, and let $d=\dim\check{\mathcal{B}}$ be the dimension of the flag variety of the dual group. For each Serre...

conjecture | open | MathDB 页面 | 原文线索

V2-020157 | MATHDB-360316

Smoothness conjecture for Fano schemes of bounded-rank symmetric matrices

摘录 / EXCERPT: Let $\mathrm{\textbf{F}}_{k}(\mathrm{SD}_n^r)$ denote the Fano scheme of k-planes contained in the space of symmetric $n\times n$ matrices of rank at most r. Let $\kappa(s)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-020158 | MATHDB-360328

André-Oort conjecture for meromorphic strata in genus zero

摘录 / EXCERPT: André-Oort conjecture in genus zero. The variety S is $\overline{\mathbb Q}$-bi-algebraic. Furthermore, if not all residues are identically zero on S, then the fibers of the re...

conjecture | open | MathDB 页面 | 原文线索

V2-020159 | MATHDB-360332

Quasi-projective soliton limit conjecture for Ricci-vertex blow-ups

摘录 / EXCERPT: Let $\mathcal{X}$ be a metric-flow limit of Type I blow-ups of a Kähler-Ricci flow, with time slices $(\mathcal{X}_t,d_t)$ as in Theorem 2main1. A quasi-projective normal variety i...

conjecture | open | MathDB 页面 | 原文线索

V2-020160 | MATHDB-360378

Finite-type LPA and positivity conjecture for the Freudenthal variety

摘录 / EXCERPT: Finite-type LPA and positivity conjecture. There is a finite-type LPA structure of rank 10 on

conjecture | open | MathDB 页面 | 原文线索

V2-020161 | MATHDB-360394

Uniform geometric torsion conjecture for abelian varieties over function fields

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, and let C/k be a smooth, projective, and geometrically connected curve of genus g with function field $K=k(C)$....

conjecture | open | MathDB 页面 | 原文线索

V2-020162 | MATHDB-360444

Cellular-decomposition conjecture for Fano manifolds with nef tangent bundle

摘录 / EXCERPT: Let X be a Fano manifold, so K_X^{-1} is ample, and suppose that its tangent bundle TX is nef. An algebraic cellular decomposition is a chain of Zariski closed subsets whose...

conjecture | open | MathDB 页面 | 原文线索

V2-020163 | MATHDB-360445

Diagonal Hodge-number conjecture for Fano manifolds with nef tangent bundle

摘录 / EXCERPT: Let X be a Fano manifold with nef tangent bundle TX, and let $h^{p,q}(X)$ denote its Hodge numbers. Diagonal Hodge-number conjecture. ... Equivalently, the Hodge numbers of X a...

conjecture | open | MathDB 页面 | 原文线索

V2-020164 | MATHDB-360446

Odd-Betti-number conjecture for Fano manifolds with nef tangent bundle

摘录 / EXCERPT: Let X be a Fano manifold with nef tangent bundle TX, and let $b_i(X)$ denote its i-th Betti number. Odd-Betti-number conjecture. ... for every integer k for which the index i...

conjecture | open | MathDB 页面 | 原文线索

V2-020165 | MATHDB-360454

Polo's Schubert filtration conjecture for tensor products of Demazure modules

摘录 / EXCERPT: Let G be a semisimple algebraic group with Borel subgroup B, Weyl group W, and dominant weights $\lambda,\mu\in P^+$. For $v,w\in W$, let $V_v(\lambda)$ and $V_w(\mu)$ be Dem...

conjecture | open | MathDB 页面 | 原文线索

V2-020166 | MATHDB-360476

Kaneko-Tsumura's conjecture on unramified multiple T-values

摘录 / EXCERPT: Kaneko-Tsumura's conjecture. For even weights, the values $T(k,l,m)$ with k,r odd and at least 3 and q even, together with their duals, are in MZV. Together with the single...

conjecture | open | MathDB 页面 | 原文线索

V2-020167 | MATHDB-360477

Conjecture on unramified motivic multiple t-values with repeated even components

摘录 / EXCERPT: Unramified MtV conjecture. For $m,n,e\text{\in}N$ and with all components of q at least two, the MtV

conjecture | open | MathDB 页面 | 原文线索

V2-020168 | MATHDB-360478

Kottwitz's conjecture on the cohomology of Shimura varieties

摘录 / EXCERPT: Kottwitz's conjecture. The displayed equality holds as an equality of virtual representations of $\operatorname{Gal}(\overline{F}/F)$, and all integers i for which...

conjecture | open | MathDB 页面 | 原文线索

V2-020169 | MATHDB-360535

The log Campana-Peternell conjecture

摘录 / EXCERPT: Let (X,Δ) be a klt pair, where X is a normal projective variety and Δ is an effective $\mathbb{Q}$-divisor such that $K_X+\Delta$ is $\mathbb{Q}$-Cartier. A divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-020170 | MATHDB-360536

Kovács's exterior-power characterization conjecture

摘录 / EXCERPT: Let X be a smooth complex projective n-dimensional variety, let $\mathcal{F}$ be an ample vector bundle on X that is a subsheaf of $\wedge^pT_X$, and let $p>0$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-020171 | MATHDB-360544

The Hilb-vs-Quot conjecture for plane curve germs

摘录 / EXCERPT: Hilb-vs-Quot conjecture. For any plane curve germ,

conjecture | open | MathDB 页面 | 原文线索

V2-020172 | MATHDB-360545

The parabolic Hilb-vs-Quot conjecture

摘录 / EXCERPT: Assume $R=\mathbb{C}[[x]][y]/(f)$ defines a generically separable degree-n cover of the x-axis, fully ramified at the origin. For an integer composition ν of n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020173 | MATHDB-360546

The KhR-vs-Quot conjecture

摘录 / EXCERPT: Let $f(x,y)=0$ be a plane curve germ as above, let $\bar{X}_f(a,q,t)$ be the normalized Khovanov--Rozansky homology polynomial, and let Ψ be the map from the relevant symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-020174 | MATHDB-360547

The y-ified KhR-vs-Quot conjecture

摘录 / EXCERPT: Let L_f be the link of the plane curve germ $f=0$. Let $\widetilde{\operatorname{Quot}}^{\vec{x},\vec{y}}$ be the bigraded $\mathbb{C}[\vec{x},\vec{y}]$-module assembled from the...

conjecture | open | MathDB 页面 | 原文线索

V2-020175 | MATHDB-360548

The enhanced y-ified KhR-vs-Quot conjecture

摘录 / EXCERPT: Let $\bar{Y}_f$ be the enhanced y-ified Khovanov--Rozansky module of the link L_f. Let $\widetilde{Q}_{S}^{\vec{x},\vec{y}}$ be the bigraded $\mathbb{C}[\vec{x},\vec{y}]$-modul...

conjecture | open | MathDB 页面 | 原文线索

V2-020176 | MATHDB-360551

Qiu-Zhang conjecture for isotypic motivic components

摘录 / EXCERPT: Let $A/{\mathbb C}$ be a simple abelian variety. For a curve $X/{\mathbb C}$, let $h^A(X)$ denote the A-isotypic component of $h^1(X)$, and let $[\Delta]_{h^A(X)}$ denote the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-020177 | MATHDB-360552

Beilinson-Bloch conjecture for projected modified diagonal cycles

摘录 / EXCERPT: Let F be a number field and let X/F be a curve. Let A_1,A_2,A_3 be three simple isogeny factors of Jac_X. Assume that the L-function ... has holomorphic cont...

conjecture | open | MathDB 页面 | 原文线索

V2-020178 | MATHDB-360557

Weyl-group module isomorphism for dual hyperspherical varieties

摘录 / EXCERPT: Let G be a complex reductive group with Lie algebra $\fg$, let $\fg^\vee$ be the Langlands-dual Lie algebra, and let $Z_\fg=T^*\CB\times_{\CN_\fg}T^*\CB$ be the Steinberg variety...

conjecture | open | MathDB 页面 | 原文线索

V2-020179 | MATHDB-360558

Koszul duality conjecture for quantizations of dual hyperspherical varieties

摘录 / EXCERPT: Let $\scrX$ be a hyperspherical G-variety and $\scrX^\vee$ its dual hyperspherical $G^\vee$-variety, with quantizations $Q\scrX$ and $Q\scrX^\vee$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020180 | MATHDB-360566

Formality conjecture for de-equivariantized Ext algebras in relative Langlands duality

摘录 / EXCERPT: Formality conjecture. The dg-algebra A_X is formal: it is quasi-isomorphic to the graded algebra

conjecture | open | MathDB 页面 | 原文线索

V2-020181 | MATHDB-360576

Finiteness conjecture for congruence Fuchsian groups with vanishing modified diagonal cycles

摘录 / EXCERPT: Finiteness conjecture. There exist only finitely many congruence Fuchsian groups $Gamma$ up to conjugation such that $Gamma\backslash \mathbb H^*$ has vanishing modified diagonal c...

conjecture | open | MathDB 页面 | 原文线索

V2-020182 | MATHDB-360592

The Hodge-Zucker conjecture for intersection and L^2-cohomology

摘录 / EXCERPT: Hodge-Zucker conjecture. The isomorphism in Theorem 2.8 is an isomorphism of $\mathbb{R}$-Hodge structures. The question refines Zucker's conjecture by asking whether its cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-020183 | MATHDB-360595

Commutative-hyperplane derived-equivalence conjecture

摘录 / EXCERPT: Let $\mathscr C=[E/\Gamma]$ be the orbifold curve, let $S_{com}\subset S$ be the subscheme corresponding to the hyperplane of deformations perpendicular to the D-module direction,...

conjecture | open | MathDB 页面 | 原文线索

V2-020184 | MATHDB-360598

Wall-correspondence conjecture for orbifold Hilbert-scheme moduli

摘录 / EXCERPT: Let $s\in HH_0([E/\Gamma])$, let S be the deformation base, and let $M_{\underline\sigma}(v;\mathcal O_{D_\infty})$ be the relative moduli spaces associated to geometric relative...

conjecture | open | MathDB 页面 | 原文线索

V2-020185 | MATHDB-360622

The conjecture on minimal monomial Togliatti systems and Laplace equations

摘录 / EXCERPT: The minimal monomial Togliatti conjecture. Every minimal monomial Togliatti system of degree d satisfies precisely one Laplace equation of order $d-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-020186 | MATHDB-360646

Uniform boundedness conjecture for torsion points on abelian varieties

摘录 / EXCERPT: Uniform boundedness conjecture. There exists $C \geq 1$, depending only on g and d, such that

conjecture | open | MathDB 页面 | 原文线索

V2-020187 | MATHDB-360656

Ein-Lazarsfeld syzygy conjecture for Veronese varieties

摘录 / EXCERPT: Let $\mathbf P^n$ be projective n-space, embedded by $\mathcal{O}_{\mathbf P^n}(d)$, and let $\mathrm{K}_{p,q}(\mathbf P^n,\mathcal{O}_{\mathbf P^n}(d))$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-020188 | MATHDB-360657

Ein-Lazarsfeld conjecture for linear syzygies

摘录 / EXCERPT: Let $\mathbf P^n$ be projective n-space, embedded by $\mathcal{O}_{\mathbf P^n}(d)$, and let $\mathrm{K}_{p,1}(\mathbf P^n,\mathcal{O}_{\mathbf P^n}(d))$ denote the Koszul cohomo...

conjecture | open | MathDB 页面 | 原文线索

V2-020189 | MATHDB-360685

Conjecture on the necessary ambient-dimension dependence of covering numbers

摘录 / EXCERPT: Let V be a (K,n) regular set in $\mathcal{B}^N$, and let $\mathcal{N}(V,\eps)$ denote its covering number at scale $\eps$. Covering-number dependence conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-020190 | MATHDB-360730

Lang's rational-point conjecture for separably rationally connected varieties over C_1 fields

摘录 / EXCERPT: Let k be a C_{1} field, meaning that every hypersurface of degree at most n in $\mathbb{P}^{n}_{k}$ has a k-rational point. A variety is separably rationally connected if t...

conjecture | open | MathDB 页面 | 原文线索

V2-020191 | MATHDB-360740

Incomplete Datta-Ghorpade conjecture for maximum projective point counts

摘录 / EXCERPT: Let $e_r(d,m)$ denote the maximal number of $\mathbb{F}_q$-rational points of a projective algebraic set defined by r linearly independent homogeneous degree-d polynomials in...

conjecture | open | MathDB 页面 | 原文线索

V2-020192 | MATHDB-360741

Complete Beelen-Datta-Ghorpade conjecture for projective point counts

摘录 / EXCERPT: Let $e_r(d,m)$ be the maximal number of $\mathbb{F}_q$-rational points of a projective algebraic set defined by r linearly independent homogeneous degree-d polynomials in $m+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-020193 | MATHDB-360743

Global refinement conjecture for tempered and equivariant elliptic cohomology

摘录 / EXCERPT: Let ${\mathscr{F}}$ be a geometric oriented abelian sheaf over a derived stack S. Write $\operatorname{Sp}_{\mathrm{fin}-\mathrm{gl}}$ for the symmetric monoidal...

conjecture | open | MathDB 页面 | 原文线索

V2-020194 | MATHDB-360752

The monodromy conjecture for Bernstein-Sato polynomials of ideals

摘录 / EXCERPT: Let X be a smooth affine variety, let Z be a closed subscheme defined by an ideal $I\subset\mathcal O_X$, and let $x\in Z$. Write $b_I(s)$ and $b_{I,x}(s)$ for the global and l...

conjecture | open | MathDB 页面 | 原文线索

V2-020195 | MATHDB-360757

The affine-space parametrization conjecture for complements of codimension-two subvarieties

摘录 / EXCERPT: Conjectural approach. There exist such N, hypersurfaces W_i, $\mathbb{G}_a$-subgroups G_i, a point p, and a subvariety H for which the restricted map $\psi|_H$ is surject...

conjecture | open | MathDB 页面 | 原文线索

V2-020196 | MATHDB-360762

The entropy-variety divergence conjecture for unexpectedness

摘录 / EXCERPT: Let U denote unexpectedness in Simplicity Theory, with causal or generative complexity describing the world and description complexity describing the observer. Let generative ent...

conjecture | open | MathDB 页面 | 原文线索

V2-020197 | MATHDB-360779

Hosono's central charge formula

摘录 / EXCERPT: Let $\hat X_0$ and X_p be a mirror pair of Calabi-Yau threefolds that are complete intersections inside toric varieties, and let X_b be the corresponding B-model fiber. Fix a b...

conjecture | open | MathDB 页面 | 原文线索

V2-020198 | MATHDB-360785

Generalization of geometric delineability without the relative-dimension-one condition

摘录 / EXCERPT: Let R be the real closed field under consideration, let X and Y be affine R-varieties, and let $pi:X\to Y$ be a morphism such that $pi_*\mathcal{O}_X$ is a free...

conjecture | open | MathDB 页面 | 原文线索

V2-020199 | MATHDB-360788

The correspondence conjecture for strictly Morse quartic polynomials

摘录 / EXCERPT: Correspondence conjecture. For any rigid isotopy class, the isotopy classes of strictly Morse polynomials contained in it are in a natural one-to-one correspondence with the extens...

conjecture | open | MathDB 页面 | 原文线索

V2-020200 | MATHDB-360793

Divisorial instability conjecture for blowups of Fano hypersurfaces along linear subspaces

摘录 / EXCERPT: Let $\Gamma \simeq \mathbb P^k \subset X$ be a k-dimensional linear subspace in a smooth Fano hypersurface $X \subset \mathbb P^{n+1}$ of degree d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020201 | MATHDB-360834

K-polystability and polar cylinders for higher secant varieties of rational normal curves

摘录 / EXCERPT: Let $C\subseteq \mathbf{P}^d$ be a rational normal curve of degree d, and let ... be its k-th secant variety, defined as the closure of the union of $(k+1)$-secant k-planes to...

conjecture | open | MathDB 页面 | 原文线索

V2-020202 | MATHDB-360861

Voevodsky's smash-nilpotence conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over a field k, and let a cycle on X be smash-nilpotent if some positive external power of it is rationally equivalent to zero on a corre...

conjecture | open | MathDB 页面 | 原文线索

V2-020203 | MATHDB-360887

Maximal-seed formula for the Fock-Goncharov pairing

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster $\mathcal{A}$-variety and let $\mathcal{X}^\vee$ be its Langlands-dual cluster $\mathcal{X}$-variety. For $l\in\mathcal{A}(\mathbb{Z}^t)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020204 | MATHDB-360937

Strong twisted Hodge D conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, and let $\underline{CH}^{k}(X,m)$ be the relevant higher Chow group, $\underline{r}_{k,m}$ its regulator, and $D(k,m)$ the indicated denomin...

conjecture | open | MathDB 页面 | 原文线索

V2-020205 | MATHDB-360952

One-dimensionality conjecture for the path variety of a finite union of algebraic curves

摘录 / EXCERPT: Let $M\subset\mathbb{R}^d$ be a finite union of algebraic curves, and let $P_{\in M}$ denote the associated path variety. Path-variety dimension conjecture. Then $P_{\in M}$ is one...

conjecture | open | MathDB 页面 | 原文线索

V2-020206 | MATHDB-360962

Goldring-Koskivirta Cone Conjecture for Hodge-type Shimura varieties

摘录 / EXCERPT: Let S_K be a Hodge-type Shimura variety, let ${\mathcal C}_K$ be the saturation of the set of characters whose associated automorphic vector bundles on S_K have nonzero degree-...

conjecture | solved | MathDB 页面 | 原文线索

V2-020207 | MATHDB-361016

Boolean degree 1 classification conjecture for families of lines in polar spaces

摘录 / EXCERPT: Let ${\mathcal P}$ be a rank d polar space, and let Y be a family of lines with characteristic vector $\chi_Y \in \langle \bm{j}\rangle + V_{10} + V_{11}$. Consider the example...

conjecture | open | MathDB 页面 | 原文线索

V2-020208 | MATHDB-361045

Extremal image-size conjecture for pop-stack operators on type A Cambrian lattices

摘录 / EXCERPT: Extremal image-size conjecture. For every Coxeter element c of A_n, we have

conjecture | open | MathDB 页面 | 原文线索

V2-020209 | MATHDB-361049

Ueno's Conjecture K on varieties of Kodaira dimension zero

摘录 / EXCERPT: Let X be a smooth projective variety with $\kappa(X)=0$, and let $\alpha:X\rightarrow A$ be its Albanese map, with general fibre F. Ueno Conjecture K. The map α is surje...

conjecture | open | MathDB 页面 | 原文线索

V2-020210 | MATHDB-361050

Lazić-Peternell's generalized nonvanishing conjecture

摘录 / EXCERPT: Let (X,Δ) be a klt pair on a normal projective variety X such that $K_X+\Delta$ is pseudo-effective. Let L be a nef $\mathbb{Q}$-divisor on X. Generalized Nonvanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-020211 | MATHDB-361060

Combinatorial criteria for maximal tangent spaces of Borel-fixed ideals

摘录 / EXCERPT: Let $N\geq 3$. Let I be a zero-dimensional Borel-fixed ideal in $\mathbb{C}[x_1,\ldots,x_N]$ whose convex hull is spanned by its monomial generators. Suppose that ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-020212 | MATHDB-361061

Briançon-Iarrobino conjecture on maximal tangent spaces of punctual Hilbert schemes

摘录 / EXCERPT: Let $N\geq 1$ and let $\mathfrak{m}=(x_1,\ldots,x_N)\subset\mathbb{C}[x_1,\ldots,x_N]$. Briançon-Iarrobino conjecture. The ideal $\mathfrak{m}^k$ has the maximum dimension tangent...

conjecture | open | MathDB 页面 | 原文线索

V2-020213 | MATHDB-361062

Partial classification conjecture for maximal tangent spaces in three variables

摘录 / EXCERPT: Let $k\geq 1$ and let $n=\binom{k+2}{3}+i$, where $i\in\{0,1,2,3,4,k+1,2k+1,\binom{k+3}{3}-\binom{k+2}{3}-1\}$ and $n<\binom{k+3}{3}$. Partial-classification conjecture. For each s...

conjecture | open | MathDB 页面 | 原文线索

V2-020214 | MATHDB-361069

Bloch-Kato dimension formula for étale cohomology Selmer groups

摘录 / EXCERPT: Let X be as above, let p be a prime of good reduction, and let $m<2r-1$. Write $H^m_{\operatorname{\acute{e}t}}(X_{\overline{\mathbb{Q}}},\mathbb{Q}_p(r))$ for the p-adic éta...

conjecture | open | MathDB 页面 | 原文线索

V2-020215 | MATHDB-361071

The logarithmic arithmetic Fujita conjecture for K-semistable Fano varieties

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{D})$ be an arithmetic log Fano variety, with generic fibre (X,Δ) of relative dimension n. Let $-(K_X+\Delta)$ carry a volume-normalized continuo...

conjecture | open | MathDB 页面 | 原文线索

V2-020216 | MATHDB-361090

Hilbert Nullstellensatz conjecture for consistent M-sets

摘录 / EXCERPT: Let $\mathbb{F}_{p}$ be the finite field with p elements, let $M\colon\mathbb{F}_{p}^{d}\to\mathbb{F}_{p}$ be a non-degenerate quadratic form, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020217 | MATHDB-361112

Maxwell's equilibrium-point conjecture for electric charges

摘录 / EXCERPT: Consider a configuration of N fixed electric charges in $\mathbb{R}^3$ having finitely many points of equilibrium. Maxwell's conjecture. The number of equilibrium points is at mo...

conjecture | open | MathDB 页面 | 原文线索

V2-020218 | MATHDB-361113

Finiteness conjecture for equilibrium points of positive electric charges

摘录 / EXCERPT: Let N positive electric charges be fixed in $\mathbb{R}^d$. Finiteness conjecture. The configuration has only finitely many points of equilibrium. The source calls this a folklor...

conjecture | open | MathDB 页面 | 原文线索

V2-020219 | MATHDB-361126

Polynomial-kind uniformity conjecture for factorized A_2-Leonard pairs

摘录 / EXCERPT: A factorized A_2-Leonard pair consists of compatible parameter arrays ... and ... where the arrays are of type II when their eigenvalues are quadratic in the index. A parameter array...

conjecture | open | MathDB 页面 | 原文线索

V2-020220 | MATHDB-361129

Type A characterization by LND images and stable points

摘录 / EXCERPT: Let Y be a variety, let ${\mathbb K}[Y]$ be its coordinate ring, and let I be the ideal of ${\mathbb K}[Y]$ generated by the images of all locally nilpotent derivations on Y....

conjecture | open | MathDB 页面 | 原文线索

V2-020221 | MATHDB-361130

Explicit modified Derksen invariant for type B varieties

摘录 / EXCERPT: Let Y be a variety, let ${\mathbb K}[Y]$ be its coordinate ring, let I be the ideal generated by the images of all locally nilpotent derivations on Y, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020222 | MATHDB-361161

Smooth manifold structure conjecture for hyperbolic cone metrics

摘录 / EXCERPT: Let S be the marked surface, let $\mathcal{P}$ be its marked points, let n be the relevant parameter, and let $\alpha\in(0,2\pi)^{\mathcal{P}}$ be the tuple of angle defects. F...

conjecture | open | MathDB 页面 | 原文线索

V2-020223 | MATHDB-361168

Quadratic motivic Donaldson-Thomas conjecture for real threefolds

摘录 / EXCERPT: Let X be a smooth projective threefold over $\mathbb{R}$, equipped with an isomorphism ... for some line bundle L on X. For $n\geq 0$, let $\tilde{I}_n$ be the quadratic degree...

conjecture | open | MathDB 页面 | 原文线索

V2-020224 | MATHDB-361175

Principal A-determinant stratification for the triple Segre product

摘录 / EXCERPT: Let w and w' be scalings for the Segre embedding of $\mathbb P^1\times\mathbb P^1\times\mathbb P^{n}$, and let P_n denote the relevant polytope. For every face...

conjecture | open | MathDB 页面 | 原文线索

V2-020225 | MATHDB-361183

Non-defectiveness conjecture for inverse Gaussian and gamma moment varieties

摘录 / EXCERPT: Let $\mathcal{M}^\mathrm{IG}_d$ and $\mathcal{M}^{\Gamma}_d$ denote the moment varieties associated with the inverse Gaussian and gamma distributions, respectively. For $k\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-020226 | MATHDB-361186

The conjecture on the shape of Beloch's surface

摘录 / EXCERPT: Shape conjecture. The shape of $z=F(x,y)$ is determined as follows: if $4p+q^2<0$, the surface has a unique local maximum at $(p,q,0)$ and a unique saddle point; if $4p+q^2=0$, it...

conjecture | open | MathDB 页面 | 原文线索

V2-020227 | MATHDB-361211

Full faithfulness conjecture for motivic realization on compact motives

摘录 / EXCERPT: Full faithfulness conjecture. The motivic realization functor

conjecture | open | MathDB 页面 | 原文线索

V2-020228 | MATHDB-361221

Movasati's codimension conjecture for Hodge loci of two intersecting planes

摘录 / EXCERPT: Let $k\geq 2$ be an integer. Let X be a hypersurface of degree 3 in $\mathbf{P}^{2k+1}$ containing two k-planes Π_1 and Π_2 whose intersection has dimension $k-3$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-020229 | MATHDB-361222

Global functional equation conjecture for dualizing sheaves on singular curves

摘录 / EXCERPT: Global functional equation conjecture. As rational functions in t,

conjecture | open | MathDB 页面 | 原文线索

V2-020230 | MATHDB-361223

Local functional equation conjecture for dualizing modules of singularities

摘录 / EXCERPT: Local functional equation conjecture. As rational functions in t,

conjecture | open | MathDB 页面 | 原文线索

V2-020231 | MATHDB-361225

Positivity conjecture for even plane curve singularities

摘录 / EXCERPT: For $m\geq 1$ and $d\geq 0$, let $R=R^{(2,2m)}$. Let $\mathit{NZ}_{R^d}(t)$ be the motivic numerator and $\widehat{\mathit{NZ}}_R(t)$ its completed version. Positivity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-020232 | MATHDB-361228

Huybrechts' conjecture on curves of elliptic nodes on polarized K3 surfaces

摘录 / EXCERPT: Let (X,L) be a polarized K3 surface of genus $g\geqslant 3$. Let C_{en} denote the curve of elliptic nodes of (X,L), namely the curve swept out by the singular points of the...

conjecture | open | MathDB 页面 | 原文线索

V2-020233 | MATHDB-361230

The Beilinson-Bloch conjecture for the μ6-invariant Jacobian motive

摘录 / EXCERPT: Beilinson-Bloch conjecture. The Beilinson-Bloch conjecture for the motive $\mathfrak h^3(J)^{\mu_6}$ predicts

conjecture | open | MathDB 页面 | 原文线索

V2-020234 | MATHDB-361244

C2-algebra isomorphism conjecture for the chiral differential operators on quiver varieties

摘录 / EXCERPT: C2-algebra isomorphism conjecture. The map above is an isomorphism. This would identify the C_2 algebra of the chiral differential operator vertex algebra with the function algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-020235 | MATHDB-361245

Kac-Roan-Wakimoto embedding conjecture for T-sigma quiver vertex algebras

摘录 / EXCERPT: Let $(Q,\mathbf v,\mathbf w)$ be a $T^{\sigma}_{[1^n]}[\SU(n)]$ quiver. Let $\CW^{-n+1}(\mathfrak{gl}_n,\sigma)$ be the corresponding affine $\mathcal W$-algebra, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020236 | MATHDB-361246

Namikawa-Weyl orbifold conjecture for T-sigma quiver chiral algebras

摘录 / EXCERPT: Let $(Q,\mathbf v,\mathbf w)$ be a $T^{\sigma}_{[1^n]}[\SU(n)]$ quiver, and let $\mathbb W=\mathfrak S_n$ be its Namikawa-Weyl group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020237 | MATHDB-361248

Global and local conjectures on Fano morphisms and klt singularities in positive characteristic

摘录 / EXCERPT: Global and local conjectures. The following statements should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-020238 | MATHDB-361253

Algebraic-variety anticoncentration conjecture for Rademacher sums

摘录 / EXCERPT: Let $0\le d<r$ and q be integers. Let $\mathcal{Z}\subseteq\mathbb{R}^{r}$ be an algebraic variety of dimension d and degree at most q. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020239 | MATHDB-361276

Prokhorov-Shokurov's effective base-point-freeness conjecture

摘录 / EXCERPT: Let d be a positive integer and let Γ_0 be a finite set of rational numbers. An lc-trivial fibration is a fibration $f:(X,B)\rightarrow Z$ with the usual lc-triviality con...

conjecture | open | MathDB 页面 | 原文线索

V2-020240 | MATHDB-361277

Li's Γ-effective adjunction conjecture

摘录 / EXCERPT: Let d be a positive integer and let $\Gamma\subset [0,1]$ be a DCC set of real numbers. A Γ-base-point-free b-divisor $\mathbf{D}$ on a normal projective variety X is...

conjecture | open | MathDB 页面 | 原文线索

V2-020241 | MATHDB-361281

Murthy's splitting conjecture

摘录 / EXCERPT: Let R be a smooth affine algebra of Krull dimension $d\geq 3$ over an algebraically closed field. Let P be a projective R-module of rank $d-1$. Murthy's splitting conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-020242 | MATHDB-361282

Asok--Fasel's exactness conjecture for the \mathbb{A}^1-homotopy sheaf sequence

摘录 / EXCERPT: Asok--Fasel's exactness conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-020243 | MATHDB-361286

Index conjecture for terminal Calabi-Yau varieties

摘录 / EXCERPT: For $n\ge1$, let $I_{\mathrm{klt}}(n-1,\Phi_{\mathrm{st}})$ denote the set of indices of klt Calabi-Yau pairs of dimension $n-1$ whose coefficients lie in the set...

conjecture | open | MathDB 页面 | 原文线索

V2-020244 | MATHDB-361312

The modified Kudla-Rapoport conjecture for maximal parahoric Rapoport-Zink spaces

摘录 / EXCERPT: Let $L\subset \mathbb{V}$ be an O_F-lattice. The modified derived local density is ... where the coefficients $c_{n,i}\in\mathbb{Q}$ are chosen so that...

conjecture | open | MathDB 页面 | 原文线索

V2-020245 | MATHDB-361317

Kononov-Pi-Shen's P=C conjecture for moduli spaces of sheaves on the plane

摘录 / EXCERPT: Kononov-Pi-Shen's P=C conjecture. For $d\geq 3$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-020246 | MATHDB-361320

The Kac-Moody extension conjecture for motivic Chern class Chevalley formulae

摘录 / EXCERPT: Kac-Moody extension conjecture. The analogues of the equations giving the Chevalley formulae for λ and $-\lambda$ hold for W.

conjecture | open | MathDB 页面 | 原文线索

V2-020247 | MATHDB-361328

Algorithmic conjecture for detecting zero-divisor ACR

摘录 / EXCERPT: Let (G,κ^*) be a mass-action system with n species and rational-number rate constants, and let I be its steady-state ideal. For each species X_i, Algorithm computes a...

conjecture | open | MathDB 页面 | 原文线索

V2-020248 | MATHDB-361398

Modularity conjecture for abelian varieties of GL(2)-type over totally real fields

摘录 / EXCERPT: Let A be an abelian variety of $GL(2,E)$-type over a number field K, with conductor $\mathfrak{n}$, and let $\rho_{A,\lambda}$ denote its Galois representation for a prime...

conjecture | open | MathDB 页面 | 原文线索

V2-020249 | MATHDB-361399

Beilinson's motivic spectral sequence conjecture for algebraic K-theory

摘录 / EXCERPT: Let X be a reasonable scheme. Write $K_*(X)$ for the non-connective algebraic K-theory of X, and let $H^i_{\mathcal{M}}(X,\mathbb{Z}(n))$ denote the motivic cohomology of X...

conjecture | open | MathDB 页面 | 原文线索

V2-020250 | MATHDB-361401

Lusztig's microlocal characterization conjecture for perverse sheaves

摘录 / EXCERPT: Let Q be a finite quiver without loops, let E be a variety of representations of Q, and let $\mathcal{F}$ be an equivariant simple perverse sheaf on E. Let the singular sup...

conjecture | open | MathDB 页面 | 原文线索

V2-020251 | MATHDB-361412

Generalised André-Pink-Zannier conjecture

摘录 / EXCERPT: Let $\Gamma\backslash X^+$ be a Shimura variety, let V be an algebraic subvariety of it, and let $x\in X^+$. Write ${\mathcal{H}}_{\Gamma\backslash X^+}(x)$ for the generalised H...

conjecture | open | MathDB 页面 | 原文线索

V2-020252 | MATHDB-361413

André-Oort conjecture for arithmetic varieties

摘录 / EXCERPT: Let $\Gamma\backslash X^+$ be an arithmetic variety, let V be an algebraic subvariety, and let Σ be the set of special points of $\Gamma\backslash X^+$ that belong to V....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-020253 | MATHDB-361414

Density formulation of the André-Oort conjecture

摘录 / EXCERPT: Let $\Gamma\backslash X^+$ be an arithmetic variety, let V be an algebraic subvariety, and let $\{Z_1;\ldots\}$ be the special subvarieties $Z_i\subseteq V$. Density formulation...

conjecture | open | MathDB 页面 | 原文线索

V2-020254 | MATHDB-361415

Hybrid André-Oort-André-Pink-Zannier conjecture

摘录 / EXCERPT: Let $\Gamma\backslash X^+$ be an arithmetic variety, let V be an irreducible algebraic subvariety, let (M,X_M) be a Shimura datum, and let $x\in X_M$ be Hodge generic. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020255 | MATHDB-361416

Finite-maximal-elements formulation of the hybrid conjecture

摘录 / EXCERPT: Let $\Gamma\backslash X^+$ be an arithmetic variety, let V be an irreducible algebraic subvariety, let (M,X_M) be a Shimura datum, and let $x\in X_M$ be Hodge generic. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020256 | MATHDB-361417

Variant of Zilber-Pink for weakly special subvarieties

摘录 / EXCERPT: Let $V'\subsetneq\Gamma\backslash X^+$ be an irreducible subvariety, and let $ZP_{V',W}$ denote the union of the atypical intersection components defined in the paper using...

conjecture | open | MathDB 页面 | 原文线索

V2-020257 | MATHDB-361420

Quartic generation conjecture for squared Grassmannians

摘录 / EXCERPT: Let ${\rm sGr}(d,n)$ denote the squared Grassmannian associated with the Grassmannian ${\rm Gr}(d,n)$, and let $\mathcal{I}({\rm sGr}(d,n))$ be its prime ideal. Quartic generation...

conjecture | open | MathDB 页面 | 原文线索

V2-020258 | MATHDB-361421

Degree conjecture for projection Grassmannians

摘录 / EXCERPT: Let ${\rm pGr}(2,n)$ be the projection Grassmannian, an embedded variety of dimension $2n-4$. Its degree is the number of complex intersection points with a generic affine-linear s...

conjecture | open | MathDB 页面 | 原文线索

V2-020259 | MATHDB-361455

The arithmetic weak Hilbert property conjecture

摘录 / EXCERPT: Let R be an arithmetic ring with fraction field k, and let $\mathcal{X}$ be a regular quasi-projective arithmetic scheme over R. Write $\mathcal{X}(R)^{(1)}$ for the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-020260 | MATHDB-361483

The algebraic Sato-Tate conjecture

摘录 / EXCERPT: Let A be a nonsingular projective g-dimensional variety over $\mathbb C$. For each prime ℓ, let $G_{A,\ell}$ be the Zariski closure of the image of the ℓ-adic represe...

conjecture | open | MathDB 页面 | 原文线索

V2-020261 | MATHDB-361487

Shan's associated-variety conjecture for the critical simple affine vertex algebra

摘录 / EXCERPT: Assume that Z_e is finite. Let $\mathfrak{g}$ be a simple Lie algebra, let $\mathcal{N}_{\mathfrak{g}}$ and $\mathcal{N}_{\mathfrak{g}^\vee}$ be the nilpotent cones, and let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-020262 | MATHDB-361492

Lusztig's conjecture on generic positive-depth parabolic induction

摘录 / EXCERPT: Let G be a connected reductive group over the maximal unramified extension of a non-archimedean local field, let G_r be the truncated parahoric group scheme associated to a poi...

conjecture | open | MathDB 页面 | 原文线索

V2-020263 | MATHDB-361528

Vanishing of the regular semisimple obstruction for symmetric pairs

摘录 / EXCERPT: Let $(\operatorname{G},\mathrm{H})$ be a symmetric pair over k such that G is quasi-split over k and $\operatorname{G}_{\mathrm{der}}$ is simply connected. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-020264 | MATHDB-361555

The analytic subspace conjecture for Shimura varieties

摘录 / EXCERPT: Let S be a connected Shimura variety with uniformization $u\colon X\to S^{\mathrm{an}}$, let $[o]\in S(\overline{\mathbb{Q}})$ be a special point, and choose $o\in u^{-1}([o])$....

conjecture | open | MathDB 页面 | 原文线索

V2-020265 | MATHDB-361574

The real-intersection conjecture for Chebyshev T-curves

摘录 / EXCERPT: Real-intersection conjecture. Any hyperplane through the origin intersects ${\cal X}_{A,T}$ in at least

conjecture | open | MathDB 页面 | 原文线索

V2-020266 | MATHDB-361600

Flatness conjecture for the local-model weight morphisms

摘录 / EXCERPT: Let X_L be the local model with closed subschemes X_w, let $\widehat{\mathcal X}^{\flat}_{L,\widehat y}$ and $\widehat{\mathcal X}^{\flat}_{L,w,\widehat y}$ be the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-020267 | MATHDB-361654

Gabber-Abe conjecture on Ψ-factorizability of Ψ-good pairs

摘录 / EXCERPT: Gabber-Abe conjecture. If (f,K) is Ψ-good, then $(f,R(\Psi_f)_u^tK)$ is Ψ-good and (f,K) is Ψ-factorizable.

conjecture | open | MathDB 页面 | 原文线索

V2-020268 | MATHDB-361668

Stability of motivic p-completion under connective covers

摘录 / EXCERPT: Connective-cover p-completion conjecture. If X is p-complete, then $\tau_{\geq 1}X$ is also p-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-020269 | MATHDB-361677

Geometric Langlands duality for the spectral period of the determinantal cone

摘录 / EXCERPT: Let C be the underlying curve and let $G=\operatorname{GL}_2\times\operatorname{GL}_2$, with Langlands dual group $\check{G}$. Define ... whose R-points are triples (F_1,F_2,s)...

conjecture | open | MathDB 页面 | 原文线索

V2-020270 | MATHDB-361680

Euclidean distance degree conjecture for point resectioning varieties

摘录 / EXCERPT: For a generic configuration of n points, let $\mathcal{R}^{N,h}_n$ denote the corresponding point resectioning variety, and let EDD denote its Euclidean distance degre...

conjecture | open | MathDB 页面 | 原文线索

V2-020271 | MATHDB-361681

Euclidean distance degree conjectures for point multiview varieties

摘录 / EXCERPT: For generic arrangements of cameras, let $\mathcal{M}_n^{N,h}$ denote the point multiview variety with n cameras, ambient dimension parameters (N,h), and let EDD den...

conjecture | open | MathDB 页面 | 原文线索

V2-020272 | MATHDB-361682

Euclidean distance degree conjectures for anchored line multiview varieties

摘录 / EXCERPT: For generic arrangements of cameras, let $\mathcal{L}_n^{L,3,3}$, $\mathcal{L}_n^{L,3,2}$, $\mathcal{L}_n^{L^2,3,3}$, $\mathcal{L}_n^{L^2,3,2}$, $\mathcal{L}_n^{L^3,3,3}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-020273 | MATHDB-361683

Euclidean distance degree conjectures for resectioning varieties

摘录 / EXCERPT: For generic configurations of points, let $\mathcal{R}_n^{N,h}$ denote the corresponding resectioning variety, and let EDD denote Euclidean distance degree. Resectioning...

conjecture | open | MathDB 页面 | 原文线索

V2-020274 | MATHDB-361686

Asymptotic expected-dimension conjecture for polynomial neural network neurovarieties

摘录 / EXCERPT: Asymptotic expected-dimension conjecture. There exists a threshold $d53fdd53f(d53d)$ such that, whenever $r>d53f$, the neurovariety $d4b1_{d53d,r}$ attains the expected dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-020275 | MATHDB-361687

Expected-dimension conjecture for non-increasing-width polynomial neural networks

摘录 / EXCERPT: Non-increasing-width expected-dimension conjecture. For every r, the neurovariety $d4b1_{d53d,r}$ attains the expected dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-020276 | MATHDB-361688

Unimodality conjecture for minimal filling neural-network architectures

摘录 / EXCERPT: Unimodality conjecture. Every minimal width vector whose architecture is filling is unimodal: there exists $i\in\{0,1,d55fd53dots,L\}$ such that $(d_0,d55fd53dots,d_i)$ is weakly i...

conjecture | open | MathDB 页面 | 原文线索

V2-020277 | MATHDB-361733

Density conjecture for the typical Hodge locus

摘录 / EXCERPT: Let $\mathbb V$ be a polarizable $\mathbb Z$-variation of Hodge structure on an irreducible smooth quasi-projective variety S, with period map...

conjecture | open | MathDB 页面 | 原文线索

V2-020278 | MATHDB-361750

Equality of finite abelian descent, Brauer, and closure sets for smooth subvarieties of abelian varieties

摘录 / EXCERPT: Let k be a global field, let A be an abelian variety over k, and let $X\subset A$ be a smooth closed subvariety. Write $X(\mathbb{A}_k)_\bullet^{\operatorname{Br}}$ for the B...

conjecture | open | MathDB 页面 | 原文线索

V2-020279 | MATHDB-361802

The DT-PT$_0$ topological vertex correspondence

摘录 / EXCERPT: Fix plane partitions $\boldsymbol{\mu}=\{\mu_a\}_{a=1}^{4}$ corresponding to $(\mathbb C^*)^3$-fixed subschemes of $\mathbb C^3$ of dimension at most one, viewed as asymptotic cros...

conjecture | open | MathDB 页面 | 原文线索

V2-020280 | MATHDB-361853

Hansen's rigid-analytic Artin-Grothendieck vanishing conjecture

摘录 / EXCERPT: Let C be an algebraically closed non-archimedean field, let A be a C-affinoid algebra, and let n be an integer invertible in $\mathcal{O}_C$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020281 | MATHDB-361856

Motivic $\mathfrak{sl}_2$-triple for Lagrangian compactified Jacobian fibrations

摘录 / EXCERPT: Let $\pi:\overline{J}_C\to B$ be as in the motivic Beauville decomposition conjecture, and let $\operatorname{Corr}^k_B(\overline{J}_C,\overline{J}_C)$ denote the degree-k relati...

conjecture | open | MathDB 页面 | 原文线索

V2-020282 | MATHDB-361857

Chow-theoretic perversity conjecture for Chern classes of compactified Jacobian fibrations

摘录 / EXCERPT: Let $\pi:\overline{J}_C\to B$ be a Lagrangian compactified Jacobian fibration as in the motivic Beauville decomposition conjecture, with motivic summands $h_{2i}(\overline{J}_C/B)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020283 | MATHDB-361874

Finite generation conjecture for the ring of automorphic forms on the stack of G-zip flags

摘录 / EXCERPT: Finite generation conjecture. The ring R_{zip} is a finitely generated k-algebra; equivalently, the stack $\mathop{\text{$G$-\tt ZipFlag}}\nolimits^\mu$ is a Mori drea...

conjecture | open | MathDB 页面 | 原文线索

V2-020284 | MATHDB-361879

The finite-quotient sphere conjecture for dual complexes

摘录 / EXCERPT: Let (X,B) be a log Calabi-Yau pair, meaning that X is a proper variety, B is an effective $\mathbb{R}$-divisor, (X,B) is log canonical, and K_X+B is $\mathbb{R}$-linearly...

conjecture | open | MathDB 页面 | 原文线索

V2-020285 | MATHDB-361880

Algebro-geometric Poincaré conjecture for dual complexes

摘录 / EXCERPT: Let (X,B) be an n-dimensional log Calabi-Yau pair, meaning that X is a proper variety, B is an effective $\mathbb{R}$-divisor, (X,B) is log canonical, and K_X+B is...

conjecture | open | MathDB 页面 | 原文线索

V2-020286 | MATHDB-361914

The real spectral-curve component conjecture

摘录 / EXCERPT: Real spectral-curve component conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-020287 | MATHDB-361918

Quantum K-theory positivity conjecture for homogeneous spaces

摘录 / EXCERPT: Let $X=G/P_X$ be a rational projective homogeneous space. Schubert varieties in X are indexed by W^X, and for $u\in W^X$ let X^u be the Schubert variety with...

conjecture | open | MathDB 页面 | 原文线索

V2-020288 | MATHDB-361935

Raskind-Spiess conjecture on the Albanese kernel of varieties over p-adic fields

摘录 / EXCERPT: Raskind-Spiess conjecture. The group $T(Y)$ is the direct sum of its maximal divisible subgroup and a finite group.

conjecture | open | MathDB 页面 | 原文线索

V2-020289 | MATHDB-361940

Unconditional equivalence between prismatic higher displays and integral local Shimura varieties

摘录 / EXCERPT: Unconditional local Shimura conjecture. The natural transformation

conjecture | open | MathDB 页面 | 原文线索

V2-020290 | MATHDB-361956

Codimension conjecture for Spohn CI varieties of binary-choice games

摘录 / EXCERPT: Let G be the undirected graphical model modelling a generic n-player game X with binary choices in normal form. Let $\mathcal{C}=\operatorname{global}(G)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020291 | MATHDB-362024

Finite-exception conjecture for the basic pattern condition of quad equations

摘录 / EXCERPT: Finite-exception conjecture. For any quad equation, there exist at most finitely many values in $\mathbb{P}^1(\mathbb{C})$ for which the basic pattern condition fails.

conjecture | open | MathDB 页面 | 原文线索

V2-020292 | MATHDB-362044

The no mysterious points conjecture for skew-symmetric locally acyclic cluster algebras

摘录 / EXCERPT: Let A be a skew-symmetric, locally acyclic cluster algebra in which all frozen variables are invertible, and let $\mathcal{A}=\operatorname{Spec}A$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020293 | MATHDB-362046

The no mysterious points conjecture for locally acyclic cluster algebras

摘录 / EXCERPT: Let A be a locally acyclic cluster algebra and let $\mathcal{A}$ be its corresponding cluster variety. Let $\mathcal{D}(\mathcal{A})$ denote the deep locus and...

conjecture | open | MathDB 页面 | 原文线索

V2-020294 | MATHDB-362047

The deep-locus conjecture for positive braid Bott-Samelson varieties

摘录 / EXCERPT: Let $\beta\in\mathrm{Br}^{+}_{n}$ be a positive braid word, let $\mathrm{BS}(\beta)$ be the associated double Bott-Samelson variety, and let $\mathcal{D}(\mathrm{BS}(\beta))$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-020295 | MATHDB-362048

The torus-generation conjecture for Fukaya categories of homogeneous varieties

摘录 / EXCERPT: Let G be a simple simply connected complex algebraic group, let $P\subset G$ be a parabolic subgroup, and let $V^\vee\subset G^\vee/P^\vee$ be the top projected open Richardson s...

conjecture | open | MathDB 页面 | 原文线索

V2-020296 | MATHDB-362058

Nonexistence of discrete-normalizer sections for X-state varieties

摘录 / EXCERPT: Let $n>2$, and let $\mathscr X_n$ be the variety of X-states equipped with the action of G. A section is a subvariety meeting the relevant general G-orbits with normalizer in...

conjecture | open | MathDB 页面 | 原文线索

V2-020297 | MATHDB-362068

Algebraic-cycle conjecture for simple geodesics on arithmetic curves

摘录 / EXCERPT: Let $F\subset \operatorname{PSL}_2(\mathbb{R})$ be a lattice such that the complex algebraic curve $S=F\backslash\mathbb{H}$, or equivalently its Jacobian, is defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-020298 | MATHDB-362070

The complete-intersection conjecture for maximal permanental minors

摘录 / EXCERPT: Let M be a generic $k\times(k+1)$ matrix with $k\geq 2$. For a matrix M, let $P_{k,k+1}$ denote the locus where all $k\times k$ permanents of M vanish. Complete-intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-020299 | MATHDB-362071

The radical-decomposition conjecture for singular permanental loci

摘录 / EXCERPT: Let M be a generic $k\times k$ matrix, let $P=\{\mathrm{perm}(M)=0\}$ be its permanental hypersurface, and let $\mathrm{Sing}(P)$ be its singular locus. For a subset R of rows,...

conjecture | open | MathDB 页面 | 原文线索

V2-020300 | MATHDB-362095

Bondal-Orlov localization conjecture

摘录 / EXCERPT: Let X be a singular variety and let ... be a resolution of rational singularities. Write $\mathbf{D}^{b}(X)$ and $\mathbf{D}^{b}(\widetilde{X})$ for their bounded derived categorie...

conjecture | open | MathDB 页面 | 原文线索

V2-020301 | MATHDB-362101

Irreducibility conjecture for Santaló varieties of generic polytopes

摘录 / EXCERPT: Irreducibility conjecture. For generic matrices $A \in \mathbb{R}^{d \times n}_{\geq 0}$ and each cell $C \in \mathcal{C}_A$, there exists a dense open subset...

conjecture | open | MathDB 页面 | 原文线索

V2-020302 | MATHDB-362131

Morel's rational motivic cohomology identification conjecture

摘录 / EXCERPT: Let S be a derived scheme. Write $\mathrm{H}\mathbb Q$ for the rational motivic cohomology spectrum over S, and let $\mathbb{1}_{\mathbb Q}^+$ denote the positive eigenspace i...

conjecture | open | MathDB 页面 | 原文线索

V2-020303 | MATHDB-362147

McKinnon's best-approximation curve conjecture

摘录 / EXCERPT: Let X be a smooth projective variety defined over a number field k, let $P\in X(k)$, and let A be an ample $\mathbb{Q}$-Cartier divisor on X. For an algebraic point P, wr...

conjecture | open | MathDB 页面 | 原文线索

V2-020304 | MATHDB-362148

Non-Zariski-dense best-approximation conjecture

摘录 / EXCERPT: Let X be a smooth, projective variety defined over a number field k, let $P\in X(k)$, and let A be an ample line bundle on X. Let $\alpha(P,A)$ denote the infimum of approx...

conjecture | open | MathDB 页面 | 原文线索

V2-020305 | MATHDB-362153

The BAB conjecture for log Fano varieties

摘录 / EXCERPT: Fix an algebraically closed field k, an integer $d>0$, and a rational number $\varepsilon>0$. Let X be a d-dimensional normal projective variety over k, and let B be a...

conjecture | open | MathDB 页面 | 原文线索

V2-020306 | MATHDB-362157

Parity conjecture for Quot schemes of points on smooth threefolds

摘录 / EXCERPT: Let $r,d\begin{in}\mathbb{N}\end{in}$ be two positive integers and let X be an irreducible smooth threefold. For a point $[\mathcal F]$ of the Quot scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-020307 | MATHDB-362184

Arakawa-Moreau's irreducibility conjecture for associated varieties of quasi-lisse vertex algebras

摘录 / EXCERPT: Let V be a simple quasi-lisse conformal vertex algebra, and let X_V denote its associated variety. Arakawa-Moreau's conjecture. ... The paper explains that the theorem proved the...

conjecture | open | MathDB 页面 | 原文线索

V2-020308 | MATHDB-362186

Arakawa-van Ekeren-Moreau's finite-extension dominance conjecture

摘录 / EXCERPT: Let V and W be vertex algebras such that W is a finite extension of V. Let X_W and X_V be their associated Poisson algebraic varieties, and let ... be the corresponding m...

conjecture | open | MathDB 页面 | 原文线索

V2-020309 | MATHDB-362223

Glick-type algebraicity conjecture for generalized pentagram-map collapse points

摘录 / EXCERPT: Generalized Glick conjecture. Let $k\geq 2$. If $n=3k+1$, the point $\widehat P_{\infty}$ is a fixed point for the projective action of $G_{P,k,k}$. If $n=3k+2$, the point...

conjecture | open | MathDB 页面 | 原文线索

V2-020310 | MATHDB-362241

Chung--Moon's Euler characteristic conjecture for line bundles on moduli spaces

摘录 / EXCERPT: Let $M_{d,\chi}$ be the moduli space of one-dimensional sheaves on $\mathbb{P}^2$ with degree d and Euler characteristic χ, let $c_0(2)$ denote the normalized tautological c...

conjecture | open | MathDB 页面 | 原文线索

V2-020311 | MATHDB-362290

Radicality conjecture for the Hasse-Witt ideal of genus-4 double covers

摘录 / EXCERPT: Radicality conjecture. The ideal I_M is radical for every prime $p\geq 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-020312 | MATHDB-362303

The bounded-chamber conjecture for signed reduced A-discriminants of bivariate 5-nomials

摘录 / EXCERPT: Bounded-chamber conjecture. Whenever the parametrization map has two critical points, the complement of the signed reduced A-discriminant contains a bounded connected component.

conjecture | open | MathDB 页面 | 原文线索

V2-020313 | MATHDB-362304

The horospherical spherical amoeba convergence conjecture

摘录 / EXCERPT: Let G/H be a spherical homogeneous space, let $Y\subseteq G/H$ be a subvariety, and let $\operatorname{sLog}_{\Gamma,t}(Y)$ be its spherical amoeba and $\operatorname{strop}(Y)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020314 | MATHDB-362305

Batyrev's conjecture on K-orbits and the valuation cone

摘录 / EXCERPT: Let G/H be a spherical homogeneous space, let K be a maximal compact subgroup of G, and let $\mathcal{V}_{G/H}$ be the valuation cone. For each face σ of...

conjecture | open | MathDB 页面 | 原文线索

V2-020315 | MATHDB-362306

The horospherical characterization by the K-orbit space

摘录 / EXCERPT: Let G/H be a spherical homogeneous space and let K be a maximal compact subgroup of G. Horospherical characterization conjecture. The homogeneous space G/H is horospherical...

conjecture | open | MathDB 页面 | 原文线索

V2-020316 | MATHDB-362318

Lang-Vojta finiteness conjecture for rational curves in plane complements

摘录 / EXCERPT: Let $B\subset\mathbb{P}^2$ be a reduced curve with at most simple normal crossing singularities, that is, nodes. For a curve $C\subset\mathbb{P}^2$, let $\nu_C:C^\nu\to C$ be its n...

conjecture | open | MathDB 页面 | 原文线索

V2-020317 | MATHDB-362320

Semigroup conjecture for matrix Schubert coefficient supports

摘录 / EXCERPT: Fix a permutation w, and let ... Here $\underline\lambda$ and $\underline\mu$ are partition-tuples and $c^w_{\underline\lambda|\underline\mu}$ is the associated coefficient. Semigr...

conjecture | open | MathDB 页面 | 原文线索

V2-020318 | MATHDB-362321

SNP conjecture for matrix Schubert coefficient supports

摘录 / EXCERPT: Fix a permutation w. Identify each partition-tuple $(\underline\lambda|\underline\mu)$ with a point in ${\mathbb R}^{m+n}$, and let ... A lattice point means a point of the ambient...

conjecture | open | MathDB 页面 | 原文线索

V2-020319 | MATHDB-362325

Rost nilpotence principle for smooth projective schemes

摘录 / EXCERPT: Rost nilpotence principle. For every $\Gamma\in \operatorname{End}_k(X)$ such that $\Gamma_E=0$ for some field extension E/k, Γ is nilpotent as a correspondence.

conjecture | open | MathDB 页面 | 原文线索

V2-020320 | MATHDB-362359

The finite-subset criterion for nested ind-groups

摘录 / EXCERPT: Let $\mathfrak G$ be an ind-group, and let X be a variety with automorphism group $\operatorname{Aut}(X)$. The finite-subset criterion. If every finite subset of $\mathfrak G$ (r...

conjecture | open | MathDB 页面 | 原文线索

V2-020321 | MATHDB-362360

The closed-conjugacy-class conjecture for Cartan subalgebras of nested ind-groups

摘录 / EXCERPT: Let X be a variety, let $\mathfrak G=\bigcup_i G_i$ be a connected nested ind-subgroup of $\operatorname{Aut}(X)$, and let $T\subseteq\mathfrak G$ be a maximal torus. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020322 | MATHDB-362361

The finite algebraic-generation dichotomy for automorphism groups

摘录 / EXCERPT: Let X be a variety, and let $G_1,\ldots,G_n$ be algebraic subgroups of $\operatorname{Aut}(X)$. Set $G=\langle G_1,\ldots,G_n\rangle$. The finite algebraic-generation dichotomy....

conjecture | open | MathDB 页面 | 原文线索

V2-020323 | MATHDB-362362

The two-element algebraic-generation dichotomy

摘录 / EXCERPT: Let X be a variety, and let $g,h\in\operatorname{Aut}(X)$ be algebraic elements, meaning that each is contained in an algebraic subgroup. The two-element algebraic-generation dic...

conjecture | open | MathDB 页面 | 原文线索

V2-020324 | MATHDB-362363

The non-torsion automorphism conjecture for infinite-dimensional automorphism groups

摘录 / EXCERPT: Let X be a variety. The non-torsion automorphism conjecture. If $\operatorname{Aut}(X)$ is infinite dimensional (respectively, has a nontrivial connected component), then...

conjecture | open | MathDB 页面 | 原文线索

V2-020325 | MATHDB-362366

The finite-filtration generation conjecture for centers and derived subgroups

摘录 / EXCERPT: Let $\operatorname{Aut}^0(X)=\lim A_n$ be an admissible filtration by irreducible affine subvarieties. Let $G\subseteq\operatorname{Aut}^0(X)$ be nilpotent or solvable and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-020326 | MATHDB-362367

The rigid-affine trivial-automorphism conjecture

摘录 / EXCERPT: Let X be an affine variety admitting no effective $\mathbb G_{\mathrm a}$-action and no effective $\mathbb G_{\mathrm m}$-action. The rigid-affine trivial-automorphism conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-020327 | MATHDB-362398

The braided fusion and real weight functor conjecture for symmetric varieties

摘录 / EXCERPT: Let $K\subset G$ be the symmetric subgroup of a real form $G_{\mathbb R}\subset G$. Write $X({\mathcal K})$ for the corresponding symmetric variety, $G({\mathcal O})$ for the posit...

conjecture | open | MathDB 页面 | 原文线索

V2-020328 | MATHDB-362399

Mukai's conjecture on the index and Picard number of Fano varieties

摘录 / EXCERPT: Let X be an n-dimensional smooth Fano variety, with index ... Let ρ_X be the Picard number of X. Mukai's conjecture. One has ... with equality if and only if X is isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-020329 | MATHDB-362400

Generalized Mukai conjecture on the pseudoindex and Picard number

摘录 / EXCERPT: Let X be an n-dimensional smooth Fano variety, and let its pseudoindex be ... Let ρ_X be the Picard number of X. Generalized Mukai conjecture. One has ... with equality if a...

conjecture | open | MathDB 页面 | 原文线索

V2-020330 | MATHDB-362429

The CDK mirror conjecture for arrow pencils on Grassmannians

摘录 / EXCERPT: Let $G(r,n)$ be the Grassmannian of r-dimensional subspaces of an n-dimensional vector space, let $H_{n,r}$ be the finite group acting effectively on $G(r,n)$ as defined above,...

conjecture | open | MathDB 页面 | 原文线索

V2-020331 | MATHDB-362433

The cone conjecture for global sections on G-zip period spaces

摘录 / EXCERPT: Cone conjecture. Under these assumptions, one has

conjecture | open | MathDB 页面 | 原文线索

V2-020332 | MATHDB-362492

Kawamata's stacky DK conjecture for varieties with finite quotient singularities

摘录 / EXCERPT: Let X and Y be projective varieties with only finite quotient singularities, and let $\mathcal{X}(X)$ and $\mathcal{X}(Y)$ be their associated stacks. The varieties X and Y...

conjecture | open | MathDB 页面 | 原文线索

V2-020333 | MATHDB-362583

The regularity conjecture for the quasi-canonical intersection space

摘录 / EXCERPT: For $0\leq r\leq n$, let $\widetilde{\mathcal{N}}^{[r]}_n$ be the formal scheme constructed from the relevant Rapoport--Zink spaces, and let p be the residue characteristic. Regu...

conjecture | open | MathDB 页面 | 原文线索

V2-020334 | MATHDB-362596

Steinberg tensor product conjecture for affine group schemes in the Verlinde category

摘录 / EXCERPT: Steinberg tensor product conjecture. Any irreducible representation of G is uniquely a tensor product of a one-irreducible representation and one inflated from G_0.

conjecture | open | MathDB 页面 | 原文线索

V2-020335 | MATHDB-362600

Euclidean distance degree conjecture for two-camera conic multiview varieties

摘录 / EXCERPT: Let $\textit{\textbf{C}}$ be an arrangement of two generic cameras, and let $\mathcal C_{\textit{\textbf{C}},2}$ denote the corresponding conic multiview variety. Euclidean distanc...

conjecture | open | MathDB 页面 | 原文线索

V2-020336 | MATHDB-362613

Zagier's polylogarithm conjecture for motivic Lie coalgebras

摘录 / EXCERPT: Let F be a number field. For $x\in F\setminus\{0,1\}$ and $n\geq 1$, let $\operatorname{Li}_n^{\mathcal{C}}(x)\in\mathcal{C}_n(F)$ be the motivic polylogarithm, where...

conjecture | open | MathDB 页面 | 原文线索

V2-020337 | MATHDB-362614

Goncharov's universality conjecture for motivic iterated integrals

摘录 / EXCERPT: Let F be a number field. For elements $a_0,\ldots,a_{n+1}\in F$, let ... be the motivic iterated integral associated with the corresponding motivic fundamental groupoid, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020338 | MATHDB-362659

Geometric properties conjecture for degenerate flag varieties on boundary faces

摘录 / EXCERPT: Geometric properties conjecture. For every such $\mathbf{e}$, $\mathcal{F}^{\mathbf{e}}(\lambda)$ is normal, Cohen-Macaulay, and has rational singularities. For parameters in...

conjecture | open | MathDB 页面 | 原文线索

V2-020339 | MATHDB-362668

Ogus' principle for Hodge-type Shimura varieties

摘录 / EXCERPT: Ogus' principle. Assume that the conjugacy class of μ is defined over $\mathbf{F}_p$. For every $x\in S_K(k)$,

conjecture | open | MathDB 页面 | 原文线索

V2-020340 | MATHDB-362675

Affine boundedness conjecture for log Calabi-Yau pairs of birational complexity zero

摘录 / EXCERPT: For a positive integer n, let $\mathcal{F}_n$ be the family of n-dimensional projective varieties X such that there exists a log Calabi--Yau pair (X,B) with ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-020341 | MATHDB-362682

The ideal-adic semi-continuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Let X be a klt variety, let $x \in X$ be a closed point, and let $\mathfrak{m}_x$ denote the maximal ideal corresponding to x. Ideal-adic semi-continuity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-020342 | MATHDB-362686

Goncharov's motivic polylogarithmic complex conjecture

摘录 / EXCERPT: Goncharov's conjecture. For any field $\mathbb K$ there is a functorial isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-020343 | MATHDB-362687

Quasi-isomorphism conjecture for the map from the cycle complex to the Lambda complex

摘录 / EXCERPT: Let $\mathbb K$ be a field of characteristic zero. Let $\mathcal N(\mathbb K,m)$ be the cycle complex introduced in the paper, let $\Lambda(\mathbb K,m)$ be the complex built from...

conjecture | open | MathDB 页面 | 原文线索

V2-020344 | MATHDB-362693

Gersten's injectivity conjecture for functors of motivic type

摘录 / EXCERPT: Gersten's injectivity conjecture. A functor $\mathscr{F}$ of “motivic type” satisfies Gersten's injectivity for a subcategory $\mathcal{S}$ of “non-singular” rings.

conjecture | open | MathDB 页面 | 原文线索

V2-020345 | MATHDB-362711

Arthur-type uniqueness conjecture for smooth orbit closures

摘录 / EXCERPT: Let G be a classical group over a p-adic field F, let λ be an infinitesimal parameter, and let C be an orbit in the Vogan variety V_λ whose closure...

conjecture | open | MathDB 页面 | 原文线索

V2-020346 | MATHDB-362744

The Norton-balanced kite-function conjecture for Q-polynomial distance-regular graphs

摘录 / EXCERPT: Let $\Gamma=(X,\mathcal R)$ be a Q-polynomial distance-regular graph with diameter $D\geq 3$, and let E be a Q-polynomial primitive idempotent of Γ. Assume that the se...

conjecture | open | MathDB 页面 | 原文线索

V2-020347 | MATHDB-362754

Beauville's weak splitting conjecture for hyper-Kähler manifolds

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold. Consider the cycle class map from the Chow ring $CH^*(X)$ to cohomology. Beauville's weak splitting conjecture. The cycle class map i...

conjecture | open | MathDB 页面 | 原文线索

V2-020348 | MATHDB-362755

Voisin map action conjecture on zero-cycles of higher Fano varieties of planes

摘录 / EXCERPT: Let Y be a cubic hypersurface and let $X=F_r(Y)$ be the variety of r-planes in Y, equipped with the Voisin rational self-map $\Psi:X\dashrightarrow X$. For $r\geq 2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020349 | MATHDB-362756

Constant-cycle fixed locus conjecture for the Voisin map

摘录 / EXCERPT: Let $X=F_r(Y)$ be as above, with Voisin rational self-map $\Psi:X\dashrightarrow X$, and let $F\subset X$ be the closure of its fixed locus. A closed subvariety is constant-cycle w...

conjecture | open | MathDB 页面 | 原文线索

V2-020350 | MATHDB-362784

Generic semistability of projected push-forwards for cubic and Gushel-Mukai fourfolds

摘录 / EXCERPT: Generic projected-push-forward conjecture. For every $[E]\in U$, the object $\operatorname{pr}_X(j_*E)$ is σ_X-semistable.

conjecture | open | MathDB 页面 | 原文线索

V2-020351 | MATHDB-362823

Conjecture on mixed differentials in the \a0-motivic Bockstein spectral sequence

摘录 / EXCERPT: Mixed-differential conjecture. There exists a τ^{2^n}-periodic differential

conjecture | open | MathDB 页面 | 原文线索

V2-020352 | MATHDB-362833

Brück-Santos Rego-Sroka vanishing conjecture for Euclidean Chevalley groups

摘录 / EXCERPT: Let ${\mathfrak{O}}$ be a Euclidean number ring and let ${\mathcal{G}}$ be a Chevalley-Demazure group scheme. Brück-Santos Rego-Sroka's conjecture. ... This conjecture extends known...

conjecture | open | MathDB 页面 | 原文线索

V2-020353 | MATHDB-362870

The Puncturing Conjecture for potential density of integral points

摘录 / EXCERPT: Let K be a number field, let X be a smooth (quasi-)projective variety over K, and let Z be a subvariety of codimension at least 2. Assume that the rational points on X...

conjecture | open | MathDB 页面 | 原文线索

V2-020354 | MATHDB-362938

Oort's conjecture on automorphisms of supersingular abelian varieties

摘录 / EXCERPT: Let k be a field of characteristic $p>0$, let E be a supersingular elliptic curve over k, and for $g\geq 2$ let $\mathcal{A}_g$ denote the moduli space of principally polariz...

conjecture | open | MathDB 页面 | 原文线索

V2-020355 | MATHDB-362947

Periodic-point conjecture for three generalized stack-sorting maps

摘录 / EXCERPT: Let S_n be the set of permutations of length n. For a pair of patterns (σ,τ), let $s_{\sigma,\tau}:S_n\to S_n$ denote the corresponding generalized stack-sorting map,...

conjecture | open | MathDB 页面 | 原文线索

V2-020356 | MATHDB-362948

Enumerative conjecture for minimally-sorted permutations

摘录 / EXCERPT: Let $\mathfrak{M}_n$ be the set of minimally-sorted permutations under $s_{123,132}$. Enumerative conjecture. For every positive integer n, ... This gives a recurrence between the...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-020357 | MATHDB-362949

Enumerative recurrence conjecture for generalized stack-sortable permutations

摘录 / EXCERPT: For positive integers t and n, let $\operatorname{Sort}_{t,n}(123,132)$ be the set of length-n permutations that are t-stack-sortable under $s_{123,132}$. Enumerative recur...

conjecture | open | MathDB 页面 | 原文线索

V2-020358 | MATHDB-362968

The isogeny-class size estimate for abelian varieties of Lubin-Tate type

摘录 / EXCERPT: Let $A\in \mathcal{A}_{g,1}({\mathbb F}_{q})$ be a geometrically simple principally polarised abelian variety with Weil q-number α_A, and suppose that its p-divisible gr...

conjecture | open | MathDB 页面 | 原文线索

V2-020359 | MATHDB-362976

The unramified period-index conjecture for smooth projective varieties

摘录 / EXCERPT: Unramified period-index conjecture. For every $\alpha \in \operatorname{Br}(X)$,

conjecture | open | MathDB 页面 | 原文线索

V2-020360 | MATHDB-362977

The noncommutative variational integral Hodge conjecture

摘录 / EXCERPT: Noncommutative variational integral Hodge conjecture. If there is a point $0\in S(\mathbf{C})$ such that $v_0\in\mathrm{K}_{0}^{\mathrm{top}}(\mathscr{C}_0)$ is algebraic, then...

conjecture | open | MathDB 页面 | 原文线索

V2-020361 | MATHDB-362978

The generalized Bogomolov-Gieseker stability conjecture for twisted varieties

摘录 / EXCERPT: Twisted stability-condition conjecture. There exists a bounded t-structure on $\mathrm{D}_{\mathrm{perf}}(X,\alpha)$ with heart $\mathcal{A}_{\omega,D}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020362 | MATHDB-363014

The abundance conjecture for compact Kähler manifolds

摘录 / EXCERPT: Abundance conjecture. K_X is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-020363 | MATHDB-363026

Degree formula for the Grassmannian of 2-planes

摘录 / EXCERPT: Let $\operatorname{Gr}(2,\mathbb{R}^n)$ be the Grassmannian of 2-dimensional subspaces of $\mathbb{R}^n$, viewed in its projection model. Degree formula. Its degree is ... The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-020364 | MATHDB-363041

Serrano's conjecture on strictly nef divisors

摘录 / EXCERPT: Let X be a projective variety of dimension n, and let D be a strictly nef divisor on X, meaning that $D\cdot C>0$ for every irreducible curve C in X. Serrano's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-020365 | MATHDB-363057

Finite-type conjecture for étale sites over fields

摘录 / EXCERPT: Let k be a field, and let $({\rm Sm}/k)_{\acute{\mathrm{e}t}}$ denote the étale site of smooth schemes of finite type over k. A field extension L/k is finite when it has fini...

conjecture | open | MathDB 页面 | 原文线索

V2-020366 | MATHDB-363097

Relative compactification conjecture for semiabelian Néron models

摘录 / EXCERPT: Relative compactification conjecture. For some $n_0\in{\mathbf N}$, P_{n_0} is a relative compactification of ${\mathcal G}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-020367 | MATHDB-363111

Completeness of component orbits in fixed-point Springer fibers

摘录 / EXCERPT: Let e be a nilpotent element in a classical Lie algebra, let Fix denote the fixed-point variety associated with the Springer fiber, and let S_e be the specifie...

conjecture | open | MathDB 页面 | 原文线索

V2-020368 | MATHDB-363131

The tensor-algebra injectivity conjecture for the weight-zero cohomology of the moduli of abelian varieties

摘录 / EXCERPT: Tensor-algebra injectivity conjecture. This induced map is injective. The later sections provide various kinds of evidence for this conjecture, while the equivalent formulation in...

conjecture | open | MathDB 页面 | 原文线索

V2-020369 | MATHDB-363132

The zeta-extension conjecture for compactly supported cohomology of the moduli of abelian varieties

摘录 / EXCERPT: Zeta-extension conjecture. The mixed Hodge structure on $H^{2g}_c(\mathcal{A}_g)$ has a subquotient isomorphic to such an extension whose class is given by a nonzero rational multi...

conjecture | open | MathDB 页面 | 原文线索

V2-020370 | MATHDB-363137

Categorical Singer-Hopf conjecture for projective manifolds with large fundamental group

摘录 / EXCERPT: Let X be a complex projective manifold with large fundamental group, meaning that for every irreducible subvariety $Z\subset X$, the image of ... is infinite, where...

conjecture | open | MathDB 页面 | 原文线索

V2-020371 | MATHDB-363155

Hartshorne's unstable rank-two bundle conjecture

摘录 / EXCERPT: Hartshorne's unstable rank-two bundle conjecture. There exists $n_0'\geq 5$ such that any unstable rank 2 vector bundle E on$\mathbb{P}^n$, with $n\geq n_0'$, is decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-020372 | MATHDB-363185

Higher Nielsen formulae for motivic multiple polylogarithms

摘录 / EXCERPT: Let F be the field under consideration, let k be a positive integer, and let $(I_i)\in\{0,1\}^k$ specify which arguments are specialized to 1. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020373 | MATHDB-363186

Higher Gangl formula for motivic multiple polylogarithms

摘录 / EXCERPT: Let F be the field under consideration, let k be a positive integer, and let $\mathcal{G}_k$ be the quotient of $\operatorname{gr}_k^\mathcal{D}\mathcal{L}_{2k}(F)$ by the subs...

conjecture | solved | MathDB 页面 | 原文线索

V2-020374 | MATHDB-363194

The deformation conjecture for Frobenius pushforwards on smooth projective varieties

摘录 / EXCERPT: Let X be a smooth projective variety with nontrivial first-order deformations, meaning $h^1(T_X)>0$, where T_X is the tangent sheaf. The absolute Frobenius pushforward is denot...

conjecture | open | MathDB 页面 | 原文线索

V2-020375 | MATHDB-363195

The D-affinity conjecture for Frobenius pushforwards

摘录 / EXCERPT: Let X be a globally F-split variety. Write $D_X^{(e)}$ for the sheaf of $O_X^{p^e}$-linear endomorphisms of $\mathcal O_X$, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-020376 | MATHDB-363210

Uniform rationality conjecture for totally negative quiver jet-counts

摘录 / EXCERPT: Let Q be a quiver and let $\mathbf{d}\in\mathbb{Z}_{\geq0}^{Q_{0}}\setminus\{0\}$ such that $(Q,\mathbf{d})$ has property (P). Let $\mu_{Q,\mathbf{d}}$ be the corresponding momen...

conjecture | open | MathDB 页面 | 原文线索

V2-020377 | MATHDB-363211

Bialgebra categorification conjecture for the multiplicity-α preprojective stack

摘录 / EXCERPT: Fix $\alpha\geq1$, and let $\mathcal{A}_{\Pi_{(A_{2},\alpha)}}$ be the graded algebra associated with the preprojective construction for the quiver of type A_{2} with multiplicit...

conjecture | open | MathDB 页面 | 原文线索

V2-020378 | MATHDB-363247

Ito's conjecture on the Fourier-Mukai and Serre spectra

摘录 / EXCERPT: Ito's conjecture. The Fourier-Mukai spectrum of $\mathcal T$ should coincide with its Serre spectrum:

conjecture | open | MathDB 页面 | 原文线索

V2-020379 | MATHDB-363261

Positive geometry conjecture for the nonnegative ABCT variety

摘录 / EXCERPT: Positive geometry conjecture. The closure

conjecture | open | MathDB 页面 | 原文线索

V2-020380 | MATHDB-363279

Motivic Langlands generalization using the group $M_{K,2}$

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((z))$ with $\ell\neq p$ and $p>2$. The source considers motivic Schur-irreducible sheaves in the derived categories...

conjecture | open | MathDB 页面 | 原文线索

V2-020381 | MATHDB-363287

K-polystability conjecture for regular semisimple Hessenberg varieties and pairs

摘录 / EXCERPT: Let G be a reductive group with Lie algebra $\mathfrak{g}$, let $\mathfrak{g}^{\mathrm{rs}}$ denote its regular semisimple locus, and let $X(s)$ and $Y(s)$ be the associated Fano...

conjecture | open | MathDB 页面 | 原文线索

V2-020382 | MATHDB-363288

Type B, D, E, F, and G cohomology conjecture for parabolic bundles

摘录 / EXCERPT: Let G be a simple group of type different from type A and type C, let $\mathcal{B}$ be its flag variety, let $\mathfrak{n}_P$ be the relevant nilradical, and let θ den...

conjecture | open | MathDB 页面 | 原文线索

V2-020383 | MATHDB-363292

The skein-module finiteness conjecture for closed 3-manifolds

摘录 / EXCERPT: The skein-module finiteness conjecture. For any closed 3-manifold M, the following are equivalent: M contains no 2-sided essential surface (that is, M is irreducible and no...

conjecture | open | MathDB 页面 | 原文线索

V2-020384 | MATHDB-363308

Product-metric convergence conjecture for the real locus of CICY2

摘录 / EXCERPT: Product-metric convergence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020385 | MATHDB-363339

Geography of log models conjecture

摘录 / EXCERPT: Fix a reduced divisor $F=\sum_iD_i$ on a normal variety X/Z of dimension d, where Z is normal and affine. Let ... and let N_F be the set of divisors $B\in B_F$ such that the...

conjecture | open | MathDB 页面 | 原文线索

V2-020386 | MATHDB-363340

Strong geography of log models conjecture

摘录 / EXCERPT: Fix a reduced divisor $F=\sum_iD_i$ on a normal variety X/Z of dimension d, where Z is normal and affine. Let ... and let N_F be the set of divisors $B\in B_F$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020387 | MATHDB-363356

Lang's conjecture on rational points of hyperbolic varieties

摘录 / EXCERPT: Let Y be a projective variety defined over a number field k. Write $Y_\mathbb{C}$ for its base change to $\mathbb{C}$, and let L be a number field extending k. A complex pr...

conjecture | open | MathDB 页面 | 原文线索

V2-020388 | MATHDB-363359

Abelian/non-Abelian correspondence for twisted quantum K-rings

摘录 / EXCERPT: Let G be a reductive group acting on a vector space, let T be a maximal torus with Weyl group W, and write $Y=V//T$ and $X=V//G$. Let L_r be the bundle on Y associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-020389 | MATHDB-363385

Higher Θ-stratification by h-invariant of Fano varieties

摘录 / EXCERPT: Let S be a finite type $\mathbbm{k}$-scheme and let $\pi\colon\mathcal{X}\to S$ be a $\mathbb{Q}$-Gorenstein family whose h-invariants and volumes are constant on closed fibers...

conjecture | open | MathDB 页面 | 原文线索

V2-020390 | MATHDB-363412

Smallest minimal log discrepancy conjecture for exceptional Fano varieties

摘录 / EXCERPT: Smallest minimal log discrepancy conjecture. For any even (respectively, odd) integer $n\geq 2$, the smallest minimal log discrepancy of an exceptional Fano variety of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-020391 | MATHDB-363433

Strong Campana uniruledness conjecture for Fano orbifolds

摘录 / EXCERPT: Let (X,Δ_ϵ) be a klt Campana orbifold over a field of characteristic 0. It is strongly Campana uniruled when there exists a free Campana curve $f:C\to X$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020392 | MATHDB-363439

The geometric generation conjecture for commuting stacks

摘录 / EXCERPT: Let $n\in\mathbb{N}$, let $(m,n)\in\mathbb{Z}\times\mathbb{N}$ be coprime, and let $d\geq 1$. For the equivariant derived category...

conjecture | open | MathDB 页面 | 原文线索

V2-020393 | MATHDB-363441

The eccentric commuting-stack object conjecture

摘录 / EXCERPT: Let $(m,n)\in\mathbb{Z}\times\mathbb{N}$ be coprime. Let $\mathcal{H}_{m,n}$ and $\mathcal{H}'_{m,n}$ be the derived objects obtained from the ordinary and eccentric flag commuting...

conjecture | open | MathDB 页面 | 原文线索

V2-020394 | MATHDB-363457

Koszulity conjecture for cohomology rings of graph modular compactifications

摘录 / EXCERPT: Let $\operatorname{\Gamma}$ be a connected graph, and let $H^{\bullet}(\operatorname{\overline{\mathcal{M}}}(\operatorname{\Gamma}))$ denote the cohomology ring of its graph modula...

conjecture | open | MathDB 页面 | 原文线索

V2-020395 | MATHDB-363493

Non-vanishing conjecture for cotangent bundles

摘录 / EXCERPT: Let X be a smooth projective variety, and let $1 \leq q \leq \dim X$. A vector bundle $\mathcal{E}$ on X is pseudoeffective when the tautological line bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-020396 | MATHDB-363497

Edoukou's conjecture on intersections with Hermitian threefolds

摘录 / EXCERPT: Let V_3 be a non-degenerate Hermitian threefold in $\mathbb P^4(\mathbb F_{q^2})$, and let $F\in\mathbb F_{q^2}[x_0,\dots,x_4]$ be homogeneous of degree d. Edoukou's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-020397 | MATHDB-363514

The Strong Monodromy Conjecture for local motivic zeta functions

摘录 / EXCERPT: Let $f: X \longrightarrow \mathbb{C}$ be a non-constant regular function and let $x \in X$. Denote by $Z_{\textnormal{mot}, x}(f; s)$ the local motivic zeta function and by...

conjecture | open | MathDB 页面 | 原文线索

V2-020398 | MATHDB-363571

All-dimensional Hessian irreducibility conjecture for smooth cubic hypersurfaces

摘录 / EXCERPT: Let $f\in\bK[x_0,\dots,x_n]$ define a smooth cubic hypersurface $X=V(f)\subset\bP^n$, and let $\cH_f$ be its Hessian hypersurface. Say that f is of Thom-Sebastiani type if, after...

conjecture | open | MathDB 页面 | 原文线索

V2-020399 | MATHDB-363603

The kernel conjecture for formal multiple zeta values

摘录 / EXCERPT: Kernel conjecture. For every $N\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-020400 | MATHDB-363620

The exact-degree conjecture for the motivic polynomial of punctual Hilbert schemes

摘录 / EXCERPT: For each positive integer d, let $\mathsf P_d(t)$ be the polynomial appearing in the motivic formula for Hilbert schemes of d points on smooth varieties. Exact-degree conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-020401 | MATHDB-363621

The half-data conjecture for motivic Hilbert-scheme polynomials

摘录 / EXCERPT: Let $\mathsf P_d(t)$ denote the polynomial in the motivic formula for Hilbert schemes of d points, and suppose that $\mathsf P_e(t)$ is known for every integer e with...

conjecture | open | MathDB 页面 | 原文线索

V2-020402 | MATHDB-363622

The polynomial-error conjecture for higher-dimensional partitions

摘录 / EXCERPT: For positive integers n and d, let Ω^n_d be the effective classes defined by ... and let χ denote the Euler characteristic. Polynomial-error conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-020403 | MATHDB-363623

The generating-function conjecture for motivic Omega-classes

摘录 / EXCERPT: For each positive integer d, define ... Let $\zeta_{{\mathbb P}^{d-1}}(t)$ denote the zeta function of projective $(d-1)$-space. Omega-class generating-function conjecture. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-020404 | MATHDB-363624

The rational-generating-function conjecture for motivic partition errors

摘录 / EXCERPT: For nonnegative integers k and d, let $e_{k,d}$ denote the numerical discrepancy associated with the partition/Hilbert-scheme comparison, and let $\mathsf M_k(t)$ be a polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-020405 | MATHDB-363625

Beilinson's conjecture for higher Chow groups

摘录 / EXCERPT: Beilinson's conjecture for higher Chow groups. This map should be injective.

conjecture | open | MathDB 页面 | 原文线索

V2-020406 | MATHDB-363630

Galaxy-model conjecture for minimal non-collapsing bubbles

摘录 / EXCERPT: Let $(\mathcal{X},\mathcal{L})\to\Delta$ be a non-log-terminal semi-log-canonical, locally stable degeneration of polarized log-terminal projective Calabi--Yau varieties, and enhan...

conjecture | open | MathDB 页面 | 原文线索

V2-020407 | MATHDB-363646

The characteristic-polynomial conjecture for the Hermitian Killing form of a harmonic polynomial

摘录 / EXCERPT: Let n be a positive integer and let $a,b\in\mathbb{C}$ satisfy $|a|\neq 1$. Set $h(z,w)=z^n+aw^n+b$ and $h^{\ast}$ for the polynomial obtained by conjugating the coefficients and...

conjecture | open | MathDB 页面 | 原文线索

V2-020408 | MATHDB-363652

Homological mirror symmetry for the non-Calabi-Yau suspension

摘录 / EXCERPT: Let $\bfW$ be an invertible polynomial not of Calabi--Yau type, let U be the associated quotient stack, let $\Uv$ be the corresponding Milnor fiber, and let $\scoh U$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020409 | MATHDB-363657

Hilbert's conjectured resolvent degrees for polynomials of degrees six, seven, and eight

摘录 / EXCERPT: Hilbert's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020410 | MATHDB-363672

Breuil-Schneider style functor for prismatic geometrization

摘录 / EXCERPT: Let $\mathrm{Stack}_{\mathrm{Bun},G,X,n}$ be the stack of G-bundles used in the paper, and let ... and ... be the indicated endomorphism-line and local subcategories. Breuil-Schneide...

conjecture | open | MathDB 页面 | 原文线索

V2-020411 | MATHDB-363681

The component-count conjecture for the lifted scattering correspondence

摘录 / EXCERPT: Component-count conjecture. The lifted scattering correspondence $\widetilde{\rm C}(3,n,1)$ decomposes into five irreducible components, with exactly one component whose map onto...

conjecture | open | MathDB 页面 | 原文线索

V2-020412 | MATHDB-363708

Weight-aspect sup-norm conjecture for Hecke eigenforms on Siegel modular varieties

摘录 / EXCERPT: Weight-aspect sup-norm conjecture. If F is an eigenfunction of all Hecke operators, then as $k\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-020413 | MATHDB-363713

Bejleri-McKean conjecture on symmetric powers of null motivic Euler characteristic

摘录 / EXCERPT: Let k be a field of characteristic not 2. Let $K_0(\operatorname{Var}_k)$ be the Grothendieck ring of k-varieties, let $\operatorname{GW}(k)$ be the Grothendieck-Witt ring, a...

conjecture | open | MathDB 页面 | 原文线索

V2-020414 | MATHDB-363737

The hybrid conjecture for Shimura varieties

摘录 / EXCERPT: Let (M,X_M) and (G,X) be Shimura data, let $x\bin X_M$ be Hodge generic, and let $\bSigma$ be the image in $Sh_K(G,X)$ of the hybrid orbit of (M,X_M,x). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020415 | MATHDB-363738

Large Galois orbits conjecture for hybrid orbits

摘录 / EXCERPT: Let S be a Shimura variety, let F be a field of finite type over the reflex field of S, and let $s=[x,1]\in S(F)$. Let $\Sigma^g(x)$ denote the geometric hybrid orbit of x,...

conjecture | open | MathDB 页面 | 原文线索

V2-020416 | MATHDB-363739

Large Galois orbits lower-bound conjecture for hybrid orbits

摘录 / EXCERPT: Let S be a Shimura variety, let F/E be a finite type extension of the reflex field of S, let $s=[x,1]\in S(F)$, and let $\Sigma^g(s)$ be a geometric hybrid orbit. Let $w(x')$...

conjecture | open | MathDB 页面 | 原文线索

V2-020417 | MATHDB-363750

The Q-polynomial triple-intersection vanishing conjecture

摘录 / EXCERPT: Consider the Q-polynomial distance-regular graph $\Gamma=(X,\mathcal R)$, with standard module V, diameter D, and Krein parameters $q^h_{i,j}$. Let Λ be the fundament...

conjecture | open | MathDB 页面 | 原文线索

V2-020418 | MATHDB-363751

The conjectured bases for distinguished subspaces of the fundamental module

摘录 / EXCERPT: Consider the Q-polynomial distance-regular graph $\Gamma=(X,\mathcal R)$ of diameter D, its standard module V, and the fundamental $\mathbb T$-submodule...

conjecture | open | MathDB 页面 | 原文线索

V2-020419 | MATHDB-363764

Height inequality conjecture for Beilinson-Bloch heights

摘录 / EXCERPT: Let S be a quasi-projective variety over a number field K, let $f\colon X\to S$ be a smooth projective family with relatively ample line bundle N, and let Z be the family o...

conjecture | open | MathDB 页面 | 原文线索

V2-020420 | MATHDB-363765

Nef adelic extension conjecture for the Beilinson-Bloch biextension

摘录 / EXCERPT: Let S be a variety over a number field K, and let L be the biextension line bundle on S whose adelic fiberwise structures have degrees equal to the Beilinson-Bloch heights...

conjecture | open | MathDB 页面 | 原文线索

V2-020421 | MATHDB-363766

Enough torsion-free objects conjecture for universal Nori motives

摘录 / EXCERPT: Let X be an algebraic variety over $\mathbb{Q}$. An object M of the abelian category $\mathcal{M}_\mathrm{univ}(X,\mathbb{Z})$ is torsion-free if multiplication by every nonzer...

conjecture | open | MathDB 页面 | 原文线索

V2-020422 | MATHDB-363767

Qin's quasi-modularity conjecture for Hilbert schemes of points

摘录 / EXCERPT: Let X be a smooth projective complex surface, and let $X^{[n]}$ be the Hilbert scheme of n points on X. For line bundles $L_1,\ldots,L_N$ on X, let ${\rm ch}_k(L_i^{[n]})$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-020423 | MATHDB-363768

Okounkov's multiple q-zeta value conjecture for Hilbert schemes of points

摘录 / EXCERPT: Let X be a smooth projective surface, let $X^{[n]}$ be the Hilbert scheme of n points on X, and let L be a line bundle on X. For the reduced generating series...

conjecture | open | MathDB 页面 | 原文线索

V2-020424 | MATHDB-363793

The toric integral-model Hilbert property conjecture

摘录 / EXCERPT: Toric integral-model Hilbert property conjecture. If $N_M=N$, then $(\mathcal{X},\mathcal{M})$ satisfies the $\mathcal{M}$-Hilbert property over every open $B\subsetneq C$. If furt...

conjecture | open | MathDB 页面 | 原文线索

V2-020425 | MATHDB-363795

Serre's dimension growth conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb{P}_{\mathbb{Q}}^{M-1}$ be an irreducible projective variety of degree at least two defined over $\mathbb{Q}$. Let $N_X(B)$ be the number of rational points o...

conjecture | open | MathDB 页面 | 原文线索

V2-020426 | MATHDB-363825

The multiplicative motivic perverse filtration conjecture for the Hitchin system

摘录 / EXCERPT: Multiplicative motivic perverse filtration conjecture. The motivic perverse filtration is multiplicative with respect to cup-product:

conjecture | open | MathDB 页面 | 原文线索

V2-020427 | MATHDB-363853

Libgober's Hodge-Tate conjecture for rationally elliptic quasi-projective varieties

摘录 / EXCERPT: Libgober's conjecture. If X is rationally elliptic, then its mixed Hodge structure is of Hodge-Tate type; equivalently, every non-zero mixed Hodge number $h^{k,p,q}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-020428 | MATHDB-363855

Poincaré-polynomial classification conjecture for rationally elliptic complex varieties

摘录 / EXCERPT: Poincaré-polynomial conjecture. There exist integers k,s,n_i,m_j such that

conjecture | open | MathDB 页面 | 原文线索

V2-020429 | MATHDB-363873

The strong exceptional collection conjecture for generalized Grassmannians

摘录 / EXCERPT: Let $\mathbf{G}$ be a simple algebraic group over an algebraically closed field k of characteristic 0, and let $\mathbf{P}\subset \mathbf{G}$ be a maximal parabolic su...

conjecture | open | MathDB 页面 | 原文线索

V2-020430 | MATHDB-363875

The full exceptional collection conjecture for odd symplectic flag varieties

摘录 / EXCERPT: Let V be a vector space of dimension $2n+1$ equipped with a skew-symmetric form of maximal rank $2n$, and let odd symplectic flag varieties be defined analogously to the odd isot...

conjecture | open | MathDB 页面 | 原文线索

V2-020431 | MATHDB-363890

The 1-invariance conjecture for tame cohomology

摘录 / EXCERPT: Let X be a scheme over a base scheme S, and let $pr:\mathbb A^1_X\to X$ be the natural projection. For a torsion sheaf $F\in \operatorname{Sh}_t(X/S)$, write $H^q_t(X/S,F)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-020432 | MATHDB-363892

The Gersten conjecture for tame logarithmic de Rham-Witt sheaves

摘录 / EXCERPT: Let k be a perfect field of characteristic $p>0$, and let X be a smooth d-dimensional k-scheme. Write $X^{(a)}$ for the points of codimension a, $i_{*,x}$ for extension b...

conjecture | open | MathDB 页面 | 原文线索

V2-020433 | MATHDB-363893

The Gersten conjecture for p-adic tame Tate twists

摘录 / EXCERPT: Let X be a smooth scheme over S of dimension d. Write $X^{(a)}$ for the points of codimension a, $i_{*,x}$ for extension by zero from x, ϵ for the relevant morph...

conjecture | open | MathDB 页面 | 原文线索

V2-020434 | MATHDB-363899

Weyl-Heisenberg boundedness conjecture

摘录 / EXCERPT: For each $d\geq1$, except $d=3$, let $\WHSIC{d}$ denote the set of Weyl-Heisenberg SICs and identify scalar-equivalent SICs when they share a fiducial vector up to a nonzero scalar...

conjecture | open | MathDB 页面 | 原文线索

V2-020435 | MATHDB-363900

Weyl-Heisenberg algebraicity conjecture

摘录 / EXCERPT: Let $d\geq1$, $d\neq3$, and let a Weyl-Heisenberg line-SIC consist of projective-algebraic lines in $\C^d$ when its vectors can be represented by algebraic coordinates. Weyl-Heisen...

conjecture | open | MathDB 页面 | 原文线索

V2-020436 | MATHDB-363924

Berkovich skeleton complement conjecture for virtual open disks

摘录 / EXCERPT: Let K be a complete discrete valued field with equal characteristic 0, and let $\mathcal{X}$ be a smooth proper integral variety of dimension n over K. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020437 | MATHDB-363925

Non-Archimedean uniruledness conjecture for analytifications

摘录 / EXCERPT: Let K be a complete discrete valued field with equal characteristic 0, and let $\mathcal{X}$ be a uniruled variety over K. Say that a proper geometrically integral smooth K...

conjecture | open | MathDB 页面 | 原文线索

V2-020438 | MATHDB-363931

Generalized abundance conjecture for projective smooth pairs

摘录 / EXCERPT: Let X be a smooth projective variety and let E be a simple normal crossing divisor on X. Let κ_σ denote Nakayama's numerical dimension. Generalized abundance conj...

conjecture | open | MathDB 页面 | 原文线索

V2-020439 | MATHDB-363941

Grothendieck's anabelian reconstruction conjecture

摘录 / EXCERPT: Grothendieck's conjecture. The variety U may be reconstructed group-theoretically from $\pi^{\acute{e}\mathrm{t}}_1(U,\ast)$ and this projection. The conjecture for hyperbolic cu...

conjecture | open | MathDB 页面 | 原文线索

V2-020440 | MATHDB-363942

The geometrically m-step solvable Grothendieck conjecture

摘录 / EXCERPT: Let k be a field of characteristic $p\geq 0$, finitely generated over the prime field, with absolute Galois group G_k. Let U be an anabelian variety over k with a geometric...

conjecture | open | MathDB 页面 | 原文线索

V2-020441 | MATHDB-363945

The strong bi-anabelian m-step solvable Grothendieck conjecture

摘录 / EXCERPT: Let k be a field with absolute Galois group G_k, let U_1,U_2 be the curves under consideration, and let m,n be non-negative integers. For each r, write...

conjecture | solved | MathDB 页面 | 原文线索

V2-020442 | MATHDB-363957

Geometric Bombieri-Lang conjecture

摘录 / EXCERPT: Let $K/\mathbb{C}$ be a finitely generated field extension of transcendence degree one, and let X be a variety over K. Say that X is groupless if it admits no non-constant mo...

conjecture | open | MathDB 页面 | 原文线索

V2-020443 | MATHDB-363961

Geometric Dynamical Bogomolov Conjecture for families of morphisms

摘录 / EXCERPT: Let S be a complex variety, let $K=\a0\mathbb{C}(S)$, and let $\Phi:S\times\mathbb{P}^N\to S\times\mathbb{P}^N$ be an algebraic family of morphisms of degree >1. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020444 | MATHDB-363962

Nonexistence of singular -harmonic 1-forms when the representation variety is zero-dimensional

摘录 / EXCERPT: Nonexistence conjecture. There exist no $\mathbb Z_2$-harmonic 1-forms on Y with $\mathcal Z\neq\emptyset$ with respect to any metric. In particular, there exist no...

conjecture | open | MathDB 页面 | 原文线索

V2-020445 | MATHDB-363979

Derived-stack universal vector bundle conjecture

摘录 / EXCERPT: Derived-stack universal vector bundle conjecture. These choices satisfy the assumptions introduced for the moduli stack of vector bundles and its universal vector bundle.

conjecture | open | MathDB 页面 | 原文线索

V2-020446 | MATHDB-363981

Category-theoretic and quiver-bundle moduli stack equivalence conjecture

摘录 / EXCERPT: Category-theoretic and quiver-bundle moduli stack equivalence conjecture. There is an equivalence of stacks

conjecture | open | MathDB 页面 | 原文线索

V2-020447 | MATHDB-363986

Semistable reduction conjecture for abelian varieties and dynamical systems

摘录 / EXCERPT: Let K be a complete discretely valued field. An extension L/K is weakly totally ramified if it is separable and has the relevant ramification property while preserving the resi...

conjecture | open | MathDB 页面 | 原文线索

V2-020448 | MATHDB-364004

The left dual exceptional collection conjecture for the \(\mathcal X_v\)'s

摘录 / EXCERPT: Let $\mathbf G/\mathbf B$ be the generalized flag variety, let W be its Weyl group with longest element w_0 and length function ℓ, and let $\mathcal X_v$ be the objects in...

conjecture | open | MathDB 页面 | 原文线索

V2-020449 | MATHDB-364015

Basic semialgebraicity conjecture for symmetric nonnegative realizable spectra

摘录 / EXCERPT: Basic semialgebraicity conjecture. The set of spectra realizable by symmetric nonnegative matrices forms a basic semi-algebraic set.

conjecture | open | MathDB 页面 | 原文线索

V2-020450 | MATHDB-364016

Calegari-Emerton vanishing conjecture for completed cohomology

摘录 / EXCERPT: Let G be a reductive group over $\mathbb{Q}$ admitting a Shimura datum (G,X), and let $(\operatorname{Sh}_{K,\mathbb{C}})_K$ be the associated inverse system of Shimura varieti...

conjecture | open | MathDB 页面 | 原文线索

V2-020451 | MATHDB-364021

Hausel's equivariant K-theory conjecture for fixed point schemes

摘录 / EXCERPT: Let $\mathrm G$ be a principally paired group acting regularly on a smooth projective variety X. Define the fixed point scheme $\operatorname{Fix}_{\mathrm G}(X)$ by the conditio...

conjecture | open | MathDB 页面 | 原文线索

V2-020452 | MATHDB-364030

Sign conjecture for fundamental identities of generalised spherical minors

摘录 / EXCERPT: Sign conjecture. For every $b\in\mathcal B$,

conjecture | open | MathDB 页面 | 原文线索

V2-020453 | MATHDB-364058

Semi-ampleness conjecture for nef integral divisors on Calabi-Yau threefolds

摘录 / EXCERPT: Semi-ampleness conjecture. Any nef integral divisor on a SCY3 is semi-ample.

conjecture | open | MathDB 页面 | 原文线索

V2-020454 | MATHDB-364068

Cattani-Cox-Dickenstein's Codimension One conjecture for toric varieties

摘录 / EXCERPT: Cattani-Cox-Dickenstein's Codimension One conjecture. The critical-degree component of the quotient by the F_i should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-020455 | MATHDB-364107

Voisin's standard conjecture on algebraic cycles supported on a closed subset

摘录 / EXCERPT: Let X be a smooth complex projective variety and let $Y\subset X$ be a closed algebraic subset. Suppose $Z\subset X$ is a codimension-k algebraic cycle whose class...

conjecture | open | MathDB 页面 | 原文线索

V2-020456 | MATHDB-364108

The generalized Bloch conjecture for zero-cycles

摘录 / EXCERPT: Let X and Y be smooth projective varieties of dimension n, and let $Z\in CH^n(X\times Y)_{\mathbb Q}$ be a correspondence. Let $F^\bullet$ be the Bloch-Beilinson filtration a...

conjecture | open | MathDB 页面 | 原文线索

V2-020457 | MATHDB-364109

The generalized Bloch conjecture for varieties with vanishing low off-diagonal Hodge numbers

摘录 / EXCERPT: Let X be a smooth projective variety of dimension m. Assume that $H^{p,q}(X)=0$ for $p\ne q$ and $p<c$ (or $q<c$). Generalized Bloch conjecture. The cycle class map ... is inject...

conjecture | open | MathDB 页面 | 原文线索

V2-020458 | MATHDB-364110

Bloch's conjecture for strict Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a strict Calabi-Yau manifold of dimension n, and let $\omega_X\in H^0(X,K_X)$ be a nowhere-zero top-degree holomorphic form. Let $Z\in CH^n(X\times X)_{\mathbb Q}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-020459 | MATHDB-364113

Voisin's constant-cycle subvariety conjecture

摘录 / EXCERPT: Let X be a hyper-Kähler manifold of dimension $2n$, and let $L,L'\subset X$ be n-dimensional constant-cycle subvarieties, meaning that all their points are rationally equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-020460 | MATHDB-364114

The fixed-locus constant-cycle conjecture for Voisin maps

摘录 / EXCERPT: Let $X=F_r(Y)$ with $r\geq2$, let $\Psi:X\dashrightarrow X$ be the Voisin map, and let $F\subset X$ be the closure of its fixed locus. Fixed-locus constant-cycle conjecture. The va...

conjecture | open | MathDB 页面 | 原文线索

V2-020461 | MATHDB-364129

Valuations detecting rational connectedness over tame valued fields

摘录 / EXCERPT: Let K be a tame valued field with divisible value group and residue field k. Let V be a separably rationally connected variety over K, and set $F=K(V)$. Valuation conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-020462 | MATHDB-364130

Rational points transfer from residue fields to henselian defectless fields

摘录 / EXCERPT: Let K be a henselian defectless valued field with divisible value group and residue field k. A field has the separably rationally connected rational-point property if every sep...

open_problem | open | MathDB 页面 | 原文线索

V2-020463 | MATHDB-364139

The equitable semialgebraic regularity conjecture

摘录 / EXCERPT: Equitable regularity conjecture. The equitable semialgebraic regularity lemma holds with partitions into

conjecture | open | MathDB 页面 | 原文线索

V2-020464 | MATHDB-364144

Pythagoras number conjecture for two-dimensional regular local rings

摘录 / EXCERPT: Let R be a regular local ring of dimension 2, and let K be its field of fractions. The Pythagoras number conjecture for regular local rings asserts that ... Here $p(R)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020465 | MATHDB-364145

Descent of uniform rationality from an algebraic closure

摘录 / EXCERPT: Let K be a field, let $\bar{K}$ be an algebraic closure of K, and let $X_K\subset K^n$ be a nonsingular K-rational algebraic variety. Assume that...

open_problem | open | MathDB 页面 | 原文线索

V2-020466 | MATHDB-364146

Pythagoras numbers of regular functions on rational surfaces

摘录 / EXCERPT: Let K be a formally real field and let X be a nonsingular surface over K, with the parenthetical K-rationality assumption as in the source. The surface Pythagoras number co...

conjecture | open | MathDB 页面 | 原文线索

V2-020467 | MATHDB-364180

The delta-nine conjecture for ternary sextics

摘录 / EXCERPT: Let P be a nonnegative ternary sextic, let $\Delta_{3,6}$ denote the cone of such forms that are not sums of squares, and let $\delta(P)$ denote its delta invariant. A form is st...

conjecture | open | MathDB 页面 | 原文线索

V2-020468 | MATHDB-364208

Igusa-stack conjecture for Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura datum satisfying $\operatorname{rank}_{\mathbb Q}(\mathsf{Z}^\circ)=\operatorname{rank}_{\mathbb R}(\mathsf{Z}^\circ)$, let Ξ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-020469 | MATHDB-364216

Disjointness conjecture for components of the embedded Nash space

摘录 / EXCERPT: Let X be an integral separated surface, C an effective Cartier divisor in X, and o a closed point in C. Assume that X is smooth away from C. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020470 | MATHDB-364219

Call-Silverman conjecture on variation of dynamical degrees

摘录 / EXCERPT: Let T be an irreducible variety, let X/T be a family of smooth irreducible projective varieties, and let $f:X/T\dashrightarrow X/T$ be a family of dominant rational maps. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020471 | MATHDB-364220

Shibata's conjecture on ample canonical heights

摘录 / EXCERPT: Let $X/\overline{\mathbb{Q}}$ be a smooth projective variety, let $f:X\to X$ be a dominant morphism with $\delta(f)>1$, and define ... with the infimum of the empty set defined as...

conjecture | open | MathDB 页面 | 原文线索

V2-020472 | MATHDB-364244

Resolution of singularities for quasi-excellent schemes

摘录 / EXCERPT: Resolution conjecture. There exists such a morphism π.

conjecture | open | MathDB 页面 | 原文线索

V2-020473 | MATHDB-364245

Embedded resolution

摘录 / EXCERPT: Embedded resolution. There exists a projective, surjective morphism

conjecture | open | MathDB 页面 | 原文线索

V2-020474 | MATHDB-364250

The motivic PT generating-series conjecture for Enriques fiber classes

摘录 / EXCERPT: Motivic PT generating-series conjecture. The generating series of the fiber-class invariants is given by the displayed infinite-product and plethystic-exponential formula.

conjecture | open | MathDB 页面 | 原文线索

V2-020475 | MATHDB-364263

Lunts's conjecture on Fourier-Mukai functors

摘录 / EXCERPT: Lunts's conjecture. The functor f is Fourier-Mukai.

conjecture | open | MathDB 页面 | 原文线索

V2-020476 | MATHDB-364286

Global conjecture on the decomposition of Beilinson-Bloch heights

摘录 / EXCERPT: Let f be the family considered above, and let Z_1 and Z_2 be cycles supported on a single common fiber of f. Denote their global Beilinson-Bloch height pairing by...

conjecture | open | MathDB 页面 | 原文线索

V2-020477 | MATHDB-364327

The proposed growth-control conjecture for motivic cohomology of fields

摘录 / EXCERPT: The paper considers motivic cohomology groups $\operatorname H_{\mathrm M}^{n,i}(K)$ of a field K, with $\delta(K)$ denoting its Kronecker dimension. Growth-control conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-020478 | MATHDB-364332

Fujita's freeness and very ampleness conjectures

摘录 / EXCERPT: Fujita's conjectures. The linear system

conjecture | open | MathDB 页面 | 原文线索

V2-020479 | MATHDB-364334

The authors' uniform moving-Seshadri-constant conjecture

摘录 / EXCERPT: Uniform moving-Seshadri-constant conjecture. There exist integer-valued functions $c(n)$ and $d(n)$ and a real-valued function $e(n)$, all depending only on n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-020480 | MATHDB-364338

Polynomial decomposition conjecture for universally defined cycles on powers

摘录 / EXCERPT: Let X be a smooth variety of dimension d, and let X^k be its k-th power. A cycle $\mathcal{Z}$ is universally defined if it is assigned compatibly to smooth families as in...

conjecture | open | MathDB 页面 | 原文线索

V2-020481 | MATHDB-364352

Equivariant Stolz-Teichner conjecture for field theories and equivariant TMF

摘录 / EXCERPT: Let G be a compact Lie group acting on a manifold X, let $X{/\!/}^{\nabla}G$ denote the quotient stack encoding the action with connection data, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020482 | MATHDB-364377

Noether-Lefschetz tautological ring socle and vanishing conjecture for abelian varieties

摘录 / EXCERPT: Let $\mathcal{A}_g$ be the moduli space of principally polarized abelian varieties of dimension g. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020483 | MATHDB-364383

Katzarkov-Kontsevich-Pantev conjecture for Fano varieties and Landau-Ginzburg models

摘录 / EXCERPT: Katzarkov-Kontsevich-Pantev conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-020484 | MATHDB-364400

Nonabelian mirror-brane conjecture for Hamiltonian group actions

摘录 / EXCERPT: Let Y be a Hamiltonian G-manifold, with maximal torus T and Weyl group W, and suppose the T-action on Y extends to a Hamiltonian G-action. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020485 | MATHDB-364433

Poisson boundary conjecture for totally strongly irreducible random walks

摘录 / EXCERPT: Let E be a Euclidean space of dimension $d\geq 3$, and let ν be a totally strongly irreducible probability distribution supported on a discrete subgroup of $\mathrm{SL}(E)$....

conjecture | open | MathDB 页面 | 原文线索

V2-020486 | MATHDB-364434

Specialness conjecture for bi-Hom schemes

摘录 / EXCERPT: Specialness conjecture. For any maximal non-rigid family $H_1\times H_2\to M$, both H_i are weakly special subvarieties with respect to the $\mathbb Z$-PVHS of the universal fami...

conjecture | open | MathDB 页面 | 原文线索

V2-020487 | MATHDB-364435

Finiteness conjecture for structurally atypical intersections

摘录 / EXCERPT: Finiteness conjecture. There exists a finite set $\Sigma_{\mathrm{p}}\cup\Sigma_M$, where Σ_{p} consists of weakly special product subvarieties and Σ_M consi...

conjecture | open | MathDB 页面 | 原文线索

V2-020488 | MATHDB-364437

BKU atypical Hodge locus conjecture

摘录 / EXCERPT: BKU atypical Hodge locus conjecture. The following conditions are equivalent: $\mathrm{HL}(S,\mathbb V)_{\mathrm{atyp}}$ is a finite union of maximal atypical special subvarieties;...

conjecture | open | MathDB 页面 | 原文线索

V2-020489 | MATHDB-364439

Structural decomposition conjecture for the non-rigid locus

摘录 / EXCERPT: Structural decomposition conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020490 | MATHDB-364440

Specialness conjecture for maximally non-rigid morphisms

摘录 / EXCERPT: Specialness conjecture. Both H_i are weakly special subvarieties with respect to the $\mathbb Z$-PVHS associated with the universal family.

conjecture | open | MathDB 页面 | 原文线索

V2-020491 | MATHDB-364443

Conjecture on groups of product type or K3 type for rationally connected threefolds

摘录 / EXCERPT: Let X be a rationally connected variety of dimension 3, and let $G\subset \mathrm{Bir}(X)$ be a finite abelian group. By the third type, we mean the case in which G faithfull...

conjecture | open | MathDB 页面 | 原文线索

V2-020492 | MATHDB-364447

Oeding's symmetric border-rank conjecture for general monomials

摘录 / EXCERPT: Let $\alpha=(0<\alpha_0\leq\cdots\leq\alpha_{d-1})\in\mathbb{N}^{d+1}$. For the monomial $x_0^{\alpha_0}\cdots x_{d-1}^{\alpha_{d-1}}$, its symmetric border rank is the least numb...

conjecture | open | MathDB 页面 | 原文线索

V2-020493 | MATHDB-364448

The generic tensor-rank conjecture for multipartite quantum states

摘录 / EXCERPT: Let $\mathcal{H}=\mathbb{C}^{d_1}\otimes\cdots\otimes\mathbb{C}^{d_n}$ be the Hilbert space of multipartite quantum states. The expected tensor rank is the smallest k for which...

conjecture | open | MathDB 页面 | 原文线索

V2-020494 | MATHDB-364496

McKernan's ACC conjecture for interpolated thresholds of nested foliations

摘录 / EXCERPT: Let X be a normal variety of dimension d, and let $\mathcal{F}\subset\mathcal{G}\subset T_X$ be foliations. McKernan's nested-foliation ACC conjecture. For every positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-020495 | MATHDB-364503

The D-equivalence conjecture for birational Calabi-Yau varieties

摘录 / EXCERPT: Let X and X' be nonsingular projective birational Calabi-Yau varieties. D-equivalence conjecture. There is an equivalence of bounded derived categories ... This conjecture predic...

conjecture | open | MathDB 页面 | 原文线索

V2-020496 | MATHDB-364535

Goncharov's spanning conjecture for motivic iterated integrals over number fields

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be a number field. The ring $\mathcal{H}(K)$ is the ring of effective motivic periods of the category of mixed Tate motives over K, and motivic iterated...

conjecture | open | MathDB 页面 | 原文线索

V2-020497 | MATHDB-364555

The residue-class description of order-two kinematic varieties

摘录 / EXCERPT: The residue-class description. For $d \equiv 0,1,6,7 \pmod{8}$, the kinematic variety $\mathcal{K}^{(2)}_{d,n}$ consists of all symmetric $n\times n$ matrices with zero diagonal an...

conjecture | open | MathDB 页面 | 原文线索

V2-020498 | MATHDB-364565

Kontsevich-Kollár-Xu conjecture on dual boundary complexes of log Calabi-Yau varieties

摘录 / EXCERPT: A log Calabi-Yau variety is a smooth complex quasi-projective variety admitting a log compactification whose boundary is a simple normal crossing divisor and whose logarithmic cano...

conjecture | open | MathDB 页面 | 原文线索

V2-020499 | MATHDB-364569

The main conjecture on exceptional points in the Teichmüller stack

摘录 / EXCERPT: Main conjecture on exceptional points. I. If X_0 is Kähler, then X_0 is neither exceptional in $\mathscr{T}(M)$ nor $\mathbb{Z}$-exceptional in $\mathscr{T}^{\mathbb{Z}}(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-020500 | MATHDB-364643

The modularity conjecture for higher theta series

摘录 / EXCERPT: Let $n\geq 1$ and $n\geq m\geq 1$. For a triple $(\mathcal{G},h,\mathcal{E})$ in the moduli stack of skew-Hermitian bundles with a Lagrangian sub-bundle, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020501 | MATHDB-364644

Beilinson-Bloch-Kato conjecture for Venkatesh's motivic cohomology

摘录 / EXCERPT: Let π be a cohomological cuspidal automorphic representation and let M_π be the motive hypothetically attached to its dual adjoint representation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020502 | MATHDB-364645

Prasanna-Venkatesh rationality conjecture for automorphic cohomology

摘录 / EXCERPT: Let G be a reductive group over $\mathbf{Q}$, let π_f be a finite part of a cohomological cuspidal automorphic representation, and let $E(\pi_f)$ be a number field over which...

conjecture | open | MathDB 页面 | 原文线索

V2-020503 | MATHDB-364646

Venkatesh's p-adic motivic rationality conjecture

摘录 / EXCERPT: Let π be as in the p-adic cohomological setting, let $\text{\dn{v}}_{\pi,p}$ be the p-adic motivic cohomology space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020504 | MATHDB-364647

The Trace Conjecture for derived Hitchin stacks

摘录 / EXCERPT: Let $\mathcal M$ be a derived Hitchin stack, let $\operatorname{Sht}_{\mathcal M}$ be its associated shtuka stack, and let c_1 be the correspondence map whose derived dimension i...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-020505 | MATHDB-364663

Conjecture on the weight variety of the maximal ideal at admissible affine sl_3 levels

摘录 / EXCERPT: Let $k=-3+\frac{2}{2m+1}$ with $m\geqslant 0$, let $V^k(sl_3)$ be the universal affine vertex operator algebra, and let $\mathcal I_k$ be its maximal ideal. Let $\mathcal V(I_k)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-020506 | MATHDB-364664

Dubrovin's refined conjecture for Fano varieties

摘录 / EXCERPT: Let X be a Fano variety of complex dimension D with odd-vanishing cohomology, and let $F_0(\textbf{v})$ be a power series with convergence domain Ω. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020507 | MATHDB-364673

Stabilization conjecture for bow-variety generating-series coefficients

摘录 / EXCERPT: For $e\in \mathbb{Z}^n$ and $f\in \mathbb{Z}^m$, let ... Here $P_s(t)$ is a polynomial, and N_s is a sequence of integers with $N_s\to\infty$. Stabilization conjecture. The polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-020508 | MATHDB-364674

Divisibility conjecture for quotient generating series of bow varieties

摘录 / EXCERPT: Let n,m be positive integers and let $e\in\mathbb{Z}^n$ and $f\in\mathbb{Z}^m$ be arbitrary. Write $Z_{\mathrm{quotient}}^c(q,t)$ for the quotient generating series, and interpre...

conjecture | open | MathDB 页面 | 原文线索

V2-020509 | MATHDB-364688

The Sibirsky ideal component conjecture

摘录 / EXCERPT: Consider the $1:-1$ resonant polynomial system ... Let I_S be its Sibirsky ideal and let B be the Bautin ideal, with ${\bf V}(I_S)$ and ${\bf V}(B)$ their varieties. Sibirsky ide...

conjecture | open | MathDB 页面 | 原文线索

V2-020510 | MATHDB-364689

The reversibility conjecture for restricted integrability equations

摘录 / EXCERPT: Let ${\bf W}$ be a component of the integrability variety ${\bf V}(B)$ of the system under consideration. Let $\mathcal A|_{\bf W}$ denote the restriction to ${\bf W}$ of the opera...

conjecture | open | MathDB 页面 | 原文线索

V2-020511 | MATHDB-364690

The Bautin-variety invariance conjecture for polynomial vector fields

摘录 / EXCERPT: Consider the polynomial system denoted by (gs), its Bautin ideal B, and the associated parameter-space vector field denoted by (Vvfsh). Bautin-variety invariance conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-020512 | MATHDB-364693

Kashiwara-Kim-Oh-Park's monoidal categorification conjecture for open Richardson varieties

摘录 / EXCERPT: Let W be the Weyl group of a Dynkin diagram, and let $w,v,u\in W$ satisfy ... Let $\mathscr{C}_{w,v}$ be the subcategory associated with w and v, let $K_q(\mathscr{C}_{w,v})$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-020513 | MATHDB-364694

Purity conjecture for additive character varieties

摘录 / EXCERPT: Let Y be the additive character variety associated to a reductive group G, a genus g, and a tuple of G-orbits in the Lie algebra, under the additive genericity assumptions....

conjecture | open | MathDB 页面 | 原文线索

V2-020514 | MATHDB-364695

Bogomolov-Karzhemanov-Kuyumzhiyan conjecture on birational stable flexibility

摘录 / EXCERPT: Bogomolov-Karzhemanov-Kuyumzhiyan conjecture. Every unirational algebraic variety X is birationally stably flexible.

conjecture | open | MathDB 页面 | 原文线索

V2-020515 | MATHDB-364729

S-duality of bow varieties

摘录 / EXCERPT: Let V_i and V_j be finite-dimensional vector spaces, and let $\operatorname{GL}(V_i)$ and $\operatorname{GL}(V_j)$ act on the Hamiltonian spaces $\mathbf M_{\medcirc}(V_i,V_j)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020516 | MATHDB-364739

Symplectic-duality conjecture for associated varieties of boundary VOAs

摘录 / EXCERPT: Let boundary VOAs arise from topologically twisted $\mathcal{N}=4$ SQFTs, and let their associated varieties be the corresponding Poisson-geometric spaces. Let Slodowy varieties an...

conjecture | open | MathDB 页面 | 原文线索

V2-020517 | MATHDB-364750

The toric section conjecture for hyperbolic curves over p-adic fields

摘录 / EXCERPT: Toric p-adic section conjecture. This map is bijective. The statement is supported by the paper's proof of the corresponding result for torsors under abelian varieties over finit...

conjecture | open | MathDB 页面 | 原文线索

V2-020518 | MATHDB-364825

Equivariant Seidel multiplication conjecture in quantum K-theory

摘录 / EXCERPT: Let $X=G/P$ be a flag variety over $\mathbb C$, let W^{comin} be the subgroup of point representatives of cominuscule flag varieties together with the identity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020519 | MATHDB-364836

The arbitrary-level extension of the signature matrix dimension theorem

摘录 / EXCERPT: Let k denote the level in the signature construction, and let the theorem referred to as the main result assert the stated dimension result for signature matrices of membranes. A...

conjecture | open | MathDB 页面 | 原文线索

V2-020520 | MATHDB-364844

Beauville's conjecture on the numerical characterization of Kähler classes

摘录 / EXCERPT: Let X be a projective variety, and let $\{\alpha\}\in H^{1,1}(X,\mathbb{R})$ be a real $(1,1)$-class. For every reduced irreducible subvariety $Y\subset X$ of dimension d, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-020521 | MATHDB-364849

Smoothness conjecture for Kuranishi spaces of Schur functor bundles

摘录 / EXCERPT: Let $X\subset\operatorname{Gr}(2,V_6)$ be the Fano variety of lines of a smooth cubic fourfold. For a partition $\lambda=(\lambda_1,\lambda_2,\lambda_3,\lambda_4)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020522 | MATHDB-364853

The Exponential Closedness conjecture

摘录 / EXCERPT: Let $V \subseteq \mathbb{C}^n \times (\mathbb{C}^{\times})^n$ be an algebraic subvariety. An exponential point of V is a point of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-020523 | MATHDB-364855

Associated-variety conjecture for boundary VOAs of abelian A-models

摘录 / EXCERPT: Let a three-dimensional abelian gauge theory be equipped with the A-twist and additional boundary free fermions, and let V^A be the resulting boundary VOA. Write X_{V^A} for it...

conjecture | open | MathDB 页面 | 原文线索

V2-020524 | MATHDB-364860

The orbit conjecture for apolar schemes to powers of quadrics

摘录 / EXCERPT: Let q be the quadric and let $N(n,r)=\binom{n+r}{n}$ be the cactus rank of q^r. Denote by $\operatorname{VAPS}_G(q^r,N(n,r))$ the variety of smoothable apolar schemes of length...

conjecture | open | MathDB 页面 | 原文线索

V2-020525 | MATHDB-364863

Generalized complete-intersection conjecture for Jordan-type loci

摘录 / EXCERPT: Let Q be a stable partition of length ℓ, let J_Q be a Jordan matrix of type Q, and let $\mathcal N_{J_Q}$ be the variety of nilpotent matrices commuting with J_Q. For...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-020526 | MATHDB-364864

Bosch-Lütkebohmert-Raynaud's conjecture on Néron models of smooth algebraic groups

摘录 / EXCERPT: Let S be a global excellent Dedekind scheme with function field K, and let G be a smooth algebraic group over K. A Néron lft-model of G is a smooth separated S-group al...

conjecture | open | MathDB 页面 | 原文线索

V2-020527 | MATHDB-364880

Cooper's conjectural description of tautological rings of Shimura varieties

摘录 / EXCERPT: Let G be a reductive group over a field k, let P be a parabolic subgroup, and let Δ be the set of simple roots. For each $J\subset\Delta$, let L_J be the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-020528 | MATHDB-364887

Kato's conjecture for the Kato complex of 0-cycles

摘录 / EXCERPT: Let S be the spectrum of a finite field or a henselian discrete valuation ring with finite residue field, and let X be a smooth projective scheme over S. For an integer n i...

conjecture | open | MathDB 页面 | 原文线索

V2-020529 | MATHDB-364888

The conjecture on specialization of 1-cycles

摘录 / EXCERPT: Let X be as in the cited theorem of Kollár--Tian, with generic fiber X_K and integer n as in that theorem. Write $C\!H_2(X)$ and $C\!H_1(X_K)$ for the relevant Chow groups, a...

conjecture | open | MathDB 页面 | 原文线索

V2-020530 | MATHDB-364917

The Whitney presentation conjecture for quantum K-theory of partial flag varieties

摘录 / EXCERPT: Let $X=Fl(v_1,\dots,v_k;N)$ be the partial flag variety of flags $V_1\subset\dots\subset V_k\subset\mathbb{C}^N$ with $\dim(V_i)=v_i$. Let $\mathcal{S}_i$ be the tautological subbu...

conjecture | open | MathDB 页面 | 原文线索

V2-020531 | MATHDB-364918

Rimányi-Tarasov-Varchenko's Wronskian presentation conjecture for quantum K-theory of partial flags

摘录 / EXCERPT: Let $Fl=Fl(v_1,\dots,v_k;N)$ be a partial flag variety, let W_Q be the discrete Wronskian matrix appearing in the Bethe-algebra description, and let $QK(Fl)$ denote its quantum...

conjecture | open | MathDB 页面 | 原文线索

V2-020532 | MATHDB-364919

The quasimap-stable-map quantum K-theory conjecture for partial flags

摘录 / EXCERPT: Let Fl be a partial flag variety. Denote by $QK(Fl)$ its stable-map quantum K-ring and by $QK^{QM}(Fl)$ its quasimap quantum K-ring. The quasimap-stable-map conjecture. In th...

conjecture | open | MathDB 页面 | 原文线索

V2-020533 | MATHDB-364921

Rimányi-Tarasov-Varchenko's explicit Wronskian presentation conjecture for partial flags

摘录 / EXCERPT: Let $Fl=Fl(v_1,\dots,v_k;N)$ be a partial flag variety. Introduce variables γ^i_j for $i=0,\dots,n$ and $j=1,\dots,\dim(\mathcal{R}_i)$, and let $Sym(\gamma)$ be the invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-020534 | MATHDB-364923

Regular-quotient gluing conjecture for spherical varieties

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety with abelian regular centralizers and no type N roots. Let $\mathfrak c=(\mathfrak h^\perp){/\!/}H$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020535 | MATHDB-364924

Symplectic Pfaffian conjecture for spherical dual representations

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety with abelian regular centralizers and no type N roots. Let S_X be the corresponding symplectic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-020536 | MATHDB-364925

Matching-divisors conjecture for regular quotients and symplectic Pfaffians

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety with abelian regular centralizers and no type N roots. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020537 | MATHDB-364926

Polarized relative Dolbeault geometric Langlands conjecture

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety with abelian regular centralizers and no type N roots, and let S_X be a polarizable symplectic dual representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-020538 | MATHDB-364927

Equivalence of abelian regular centralizers and Borel parabolic conjecture

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety without type N roots, and let $P(X)$ be the associated parabolic subgroup. Abelian-centralizer equivalence conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-020539 | MATHDB-364928

Regular-centralizer determination of the spherical root system

摘录 / EXCERPT: Let $X=G/H$ be an affine homogeneous spherical variety with abelian regular centralizers and no type N roots. Let J_A be the associated group scheme and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020540 | MATHDB-364929

Independence of optimal degenerations from the weighting function

摘录 / EXCERPT: Let (X,Δ) be a log Fano pair and let $g,\bar{g}$ be functions satisfying the conditions in the source. Let (Y,Δ_Y,ξ_0) and...

conjecture | open | MathDB 页面 | 原文线索

V2-020541 | MATHDB-364935

Gunningham's conjecture relating instanton Floer homology and generic skein modules

摘录 / EXCERPT: Let G be a reductive group and M a closed, connected, oriented 3-manifold. Let $\Loc^{\ff}_{G}(M)$ be the representation variety of G-local systems, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020542 | MATHDB-364952

Broken Gorenstein characterization conjecture for smoothable points in the affine threefold Hilbert scheme

摘录 / EXCERPT: Let $S={\Bbbk}[x,y,z]$, let d be a positive integer, and let $[S/I]\in\mathrm{Hilb}^d(\mathbb{A}^3)$ be a smoothable point. Broken Gorenstein characterization conjecture. The fol...

conjecture | open | MathDB 页面 | 原文线索

V2-020543 | MATHDB-364953

Hu's tangent-space lower-bound conjecture for singular smoothable points

摘录 / EXCERPT: Let $S={\Bbbk}[x,y,z]$, let d be a positive integer, and let $[S/I]\in\mathrm{Hilb}^d(\mathbb{A}^3)$ be smoothable and singular. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020544 | MATHDB-364954

Hu's Grassmannian-cone local-model conjecture for minimal tangent dimension

摘录 / EXCERPT: Let $S={\Bbbk}[x,y,z]$, let d be a positive integer, and let $[S/I]\in\mathrm{Hilb}^d(\mathbb{A}^3)$ be smoothable with $\dim_{\Bbbk}T(I)=3d+6$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-020545 | MATHDB-364959

The categorical crepant resolution conjecture for Pfaffian varieties

摘录 / EXCERPT: Categorical crepant resolution conjecture. The categorical crepant resolution $\widetilde{D}^b({\mathrm {Pf}}(2k,V^{\vee}))$ has the non-rectangular Lefschetz decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-020546 | MATHDB-364980

Batyrev-Manin-Peyre counting conjecture for strongly saturated varieties

摘录 / EXCERPT: Let F be a number field, and let V be a smooth projective variety over F that is strongly $\mathcal{L}$-saturated for a metrized line bundle $\mathcal{L}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020547 | MATHDB-364996

The D5 singular-type converse for celestial surfaces

摘录 / EXCERPT: The D5 converse. If

conjecture | open | MathDB 页面 | 原文线索

V2-020548 | MATHDB-364997

The excluded D4 visible types conjecture for celestial surfaces

摘录 / EXCERPT: Excluded D4 visible types conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020549 | MATHDB-365013

Schnell's algebraic fiber-space non-vanishing conjecture

摘录 / EXCERPT: Let $f:X\to Y$ be an algebraic fiber space, meaning that X and Y are smooth projective varieties and f is surjective with connected fibers. Let F be a generic fiber satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-020550 | MATHDB-365014

Campana-Peternell reformulation for algebraic fiber spaces with Kodaira dimension zero fibers

摘录 / EXCERPT: Let $f:X\to Y$ be an algebraic fiber space between smooth projective varieties, and suppose that the general fiber F satisfies $\kappa(F)=0$. Let H be an ample divisor on Y....

conjecture | open | MathDB 页面 | 原文线索

V2-020551 | MATHDB-365043

The weak Albanese conjecture for compact Kähler C-pairs

摘录 / EXCERPT: Let (X,D) be a C-pair, meaning a pair consisting of a complex space X and a boundary divisor D with Campana's orbifold multiplicities, and suppose that X is compact Kähler....

conjecture | open | MathDB 页面 | 原文线索

V2-020552 | MATHDB-365049

Conjecture on irreducible components of varieties with symmetric compound matrices

摘录 / EXCERPT: Let N be the matrix-size parameter, let $[N+1]=\{1,\ldots,N+1\}$, and let $S=S^2\mathbb{C}^{N+1}$ be the subspace of symmetric $(N+1)\times(N+1)$ matrices. For $r\in[N+1]$, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-020553 | MATHDB-365057

Serre's ordinariness conjecture for smooth varieties

摘录 / EXCERPT: Serre's ordinariness conjecture. There exists an infinite set of primes $\mathfrak{p}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-020554 | MATHDB-365060

The motivic Newton-Hodge inequality and ordinariness conjectures

摘录 / EXCERPT: The motivic Newton-Hodge inequality and ordinariness conjectures. For every integer k: (b) the representations R_{ℓ} form a strictly compatible system; (c) there exists a fi...

conjecture | open | MathDB 页面 | 原文线索

V2-020555 | MATHDB-365078

Projective-space quotient conjecture for bases of Lagrangian fibrations

摘录 / EXCERPT: Lagrangian-fibration base conjecture. The base B is isomorphic to a quotient of

conjecture | open | MathDB 页面 | 原文线索

V2-020556 | MATHDB-365086

The power-of-p conjecture for Hochschild-Kostant-Rosenberg differentials

摘录 / EXCERPT: Let A be an $mathbb F_p$-algebra and let X/A be a scheme. Consider the Hochschild-Kostant-Rosenberg spectral sequence for X/A, with differentials denoted by d_r. Power-of-...

conjecture | open | MathDB 页面 | 原文线索

V2-020557 | MATHDB-365121

Primitive-point conjecture for the unit sphere over finite fields

摘录 / EXCERPT: Let q be a prime power with $q>25$, and let ... be the unit sphere over ${\mathbb F}_q$. An ${\mathbb F}_q$-primitive point is a point on this sphere whose coordinates generate the...

conjecture | open | MathDB 页面 | 原文线索

V2-020558 | MATHDB-365169

The quantum rim-hook rule for cominuscule homogeneous spaces

摘录 / EXCERPT: Let $X=G/P$ be a cominuscule homogeneous space, let $\ideal\subideal\minposet$ be a generalized Young diagram, and let $\sigma_\ideal$ denote the corresponding Schubert class. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020559 | MATHDB-365204

The rational-curves conjecture for torsion-canonical manifolds

摘录 / EXCERPT: Rational-curves conjecture for torsion-canonical manifolds. The manifold X is an étale quotient of an Abelian variety.

conjecture | open | MathDB 页面 | 原文线索

V2-020560 | MATHDB-365206

Lang's conjecture on entire curves and weakly special manifolds

摘录 / EXCERPT: Lang's entire-curve conjecture. (i) A projective manifold of general type does not contain a Zariski dense entire curve. Equivalently, (ii) a projective manifold containing a Zaris...

conjecture | open | MathDB 页面 | 原文线索

V2-020561 | MATHDB-365209

Tevelev's log Fano conjecture for the blow-up of the moduli space

摘录 / EXCERPT: Tevelev's conjecture. M_1 is log Fano.

conjecture | open | MathDB 页面 | 原文线索

V2-020562 | MATHDB-365210

Bhatt-Lurie’s obstruction-class conjecture for Frobenius liftings and the Hodge-Tate gerbe

摘录 / EXCERPT: Let X be a smooth scheme over a field of characteristic p. Write X_{fpqc} for its fpqc site, let $T_{X/k}$ be its tangent bundle, and let $T_{X/k}^{\sharp}$ be the f...

conjecture | open | MathDB 页面 | 原文线索

V2-020563 | MATHDB-365211

The Weakly Special Conjecture for potential density

摘录 / EXCERPT: Let X be a smooth projective variety over a number field K. A variety is weakly special if no finite étale cover of $X_{\overline{K}}$ rationally dominates a positive-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-020564 | MATHDB-365213

The equivalence conjecture for analytic specialness

摘录 / EXCERPT: Let (X,Δ) be a smooth proper orbifold pair. Call it Kobayashi-special when its Kobayashi pseudometric vanishes identically, Picard-special when it admits the specified dense...

conjecture | open | MathDB 页面 | 原文线索

V2-020565 | MATHDB-365287

Simpson's standard motivicity conjecture for connections

摘录 / EXCERPT: Simpson's standard conjecture. By spreading out, the $\overline{\mathbb{Q}}$-points in the intersection

conjecture | open | MathDB 页面 | 原文线索

V2-020566 | MATHDB-365288

Simpson's motivicity conjecture for rigid integrable connections

摘录 / EXCERPT: Simpson's motivicity conjecture. Rigid integrable connections are motivic.

conjecture | open | MathDB 页面 | 原文线索

V2-020567 | MATHDB-365297

Chow-class coefficient conjecture for Schubert varieties of subspace arrangements

摘录 / EXCERPT: Let $Y_V \subset \mathbb{P}^{\mathbf n}$ be a Schubert variety of a subspace arrangement, and write its Chow class as ... For a basis $\mathbf b=(b_1,\ldots,b_N)$, Chow-class coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-020568 | MATHDB-365315

Relative integral identity for equivariant functions on a vector bundle

摘录 / EXCERPT: Relative integral identity. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-020569 | MATHDB-365336

Geometric Stanley-Stembridge conjecture

摘录 / EXCERPT: For a proper smooth variety $\mathscr X$ equipped with a $G[\![z]\!]$-action and a $G[\![z]\!]$-equivariant map ... assume that ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-020570 | MATHDB-365355

Codimension conjecture for the bad locus of super Riemann surfaces

摘录 / EXCERPT: Let M be the relevant moduli space and let B be its bad locus. For a component arising from the stratum indexed by an integer $r\geq 2$, the surrounding argument considers comp...

conjecture | solved | MathDB 页面 | 原文线索

V2-020571 | MATHDB-365389

The motivic-DAHA coincidence conjecture for colored algebraic links

摘录 / EXCERPT: Let $\mathcal{R}$ be a plane-curve-singularity ring with branches indexed by $1\le i\le\kappa$, and let $\sigma=(c_1\ge\cdots\ge c_\kappa>0)$. Define the motivic superpolynomial ……

conjecture | open | MathDB 页面 | 原文线索

V2-020572 | MATHDB-365390

The colored-link motivic-DAHA embedding conjecture

摘录 / EXCERPT: Let $\mathcal{R}=\mathbf{F}_q[[x,y]]\subset\Omega=\mathbf{F}_q[[\epsilon_j,\zeta_j]]$, and let $\widetilde{\mathcal R}=\mathbf{F}_q[[\widetilde x,\widetilde y]]$ be the sufficientl...

conjecture | open | MathDB 页面 | 原文线索

V2-020573 | MATHDB-365413

The renormalized mixed-Hodge-polynomial correspondence

摘录 / EXCERPT: Let $C_{g,n}$ be the character variety of the $SU(2)/\mathbb{Z}_2$ Hitchin system on the punctured surface, and let $PH_{g,n}(q)$ be its pure mixed Hodge polynomial. Define the ren...

conjecture | open | MathDB 页面 | 原文线索

V2-020574 | MATHDB-365425

The improved Adams-filtration bound for higher $c4^n$-torsion

摘录 / EXCERPT: Improved Adams-filtration bound. Then x has Adams filtration at least $3n$.

conjecture | open | MathDB 页面 | 原文线索

V2-020575 | MATHDB-365430

Refined real cycle class conjecture for Chow-Witt groups

摘录 / EXCERPT: Refined real cycle class conjecture. The image of $\widetilde{\gamma}^c$ contains

conjecture | open | MathDB 页面 | 原文线索

V2-020576 | MATHDB-365431

Refined I-Hodge conjecture

摘录 / EXCERPT: Refined $\mathbf I$-Hodge conjecture. Every connected smooth $\mathbb R$-scheme satisfies $P(X,\mathcal L,c)$ for every line bundle $\mathcal L$ and every $c\geqslant0$. Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-020577 | MATHDB-365432

Baldi-Klingler-Ullmo's typical and atypical Hodge locus conjecture

摘录 / EXCERPT: Let $(\mathbb{V},\mathscr{F}^{\bullet})$ be an integral polarized variation of Hodge structures on a smooth connected quasi-projective variety S. The Baldi-Klingler-Ullmo conject...

conjecture | open | MathDB 页面 | 原文线索

V2-020578 | MATHDB-365434

The quotient-motive conjecture for irreducible torus-knot representations

摘录 / EXCERPT: Let n,m be coprime positive integers, and let … For a positive integer r, write … for the irreducible representation locus, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020579 | MATHDB-365436

Equality of the sorting classes for the 312-type stack-sorting maps

摘录 / EXCERPT: Let SC_{312}, $\mathrm{SC}_{\underline{31}2}$, and $\mathrm{SC}_{3\underline{12}}$ be the vincular-pattern-avoiding stack-sorting maps, and let $\mathrm{Sort}_n(f)$ denote the p...

conjecture | open | MathDB 页面 | 原文线索

V2-020580 | MATHDB-365437

Second-largest preimage conjecture for the $1\underline{23}$ stack-sorting map

摘录 / EXCERPT: Let $\mathfrak S_n$ be the set of permutations of length n, and let $\mathrm{SC}_{1\underline{23}}$ be the vincular-pattern-avoiding stack-sorting map. Second-largest-preimage conj...

conjecture | open | MathDB 页面 | 原文线索

V2-020581 | MATHDB-365438

Maximum preimage conjecture for the $1\underline{23}$ and $3\underline{21}$ stack-sorting maps

摘录 / EXCERPT: Let $\mathfrak S_n$ be the set of permutations of length n, and let $\mathrm{SC}_{1\underline{23}}$ and $\mathrm{SC}_{3\underline{21}}$ be the corresponding vincular-pattern-avoiding...

conjecture | open | MathDB 页面 | 原文线索

V2-020582 | MATHDB-365440

Schröder enumeration conjecture for the $\underline{23}1$ sorting class

摘录 / EXCERPT: Let $\mathrm{SC}_{\underline{23}1}$ be the vincular-pattern-avoiding stack-sorting map, let $\mathrm{Sort}_n(\mathrm{SC}_{\underline{23}1})$ denote its sorting class in $\mathfrak S_n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-020583 | MATHDB-365455

The Relative Manin-Mumford conjecture for semi-abelian schemes

摘录 / EXCERPT: Relative Manin-Mumford conjecture. The stated alternative holds.

conjecture | open | MathDB 页面 | 原文线索

V2-020584 | MATHDB-365456

Pink-Mordell-Lang conjecture for semi-abelian varieties

摘录 / EXCERPT: Pink-Mordell-Lang conjecture. If Z is not contained in any translation of a proper algebraic subgroup of G, then $Z\cap(\Gamma+G^{[>d]})$ is not Zariski dense in Z.

conjecture | open | MathDB 页面 | 原文线索

V2-020585 | MATHDB-365457

Pink's original relative Manin-Mumford conjecture for semi-abelian schemes

摘录 / EXCERPT: Pink's original relative Manin-Mumford conjecture. The intersection $Z\cap\mathcal{G}^{[>d]}$ is not Zariski dense in Z.

conjecture | open | MathDB 页面 | 原文线索

V2-020586 | MATHDB-365458

The revised relative Manin-Mumford conjecture for semi-abelian schemes

摘录 / EXCERPT: Revised relative unlikely-intersection conjecture. The intersection $Z\cap\mathcal{G}^{[>d]}$ is not Zariski dense in Z.

conjecture | open | MathDB 页面 | 原文线索

V2-020587 | MATHDB-365459

The revised relative Manin-Mumford conjecture in torsion-point form

摘录 / EXCERPT: Revised relative Manin-Mumford conjecture. If $Z(\mathbb{C})\cap\mathcal{G}_{\mathrm{tor}}$ is Zariski dense in Z, then either Z is contained in a special subvariety of...

conjecture | open | MathDB 页面 | 原文线索

V2-020588 | MATHDB-365477

The Rasmussen-Tamagawa finiteness conjecture for heavenly abelian varieties

摘录 / EXCERPT: Rasmussen-Tamagawa conjecture. The set $\mathcal{H}(K,g)$ is finite for any choice of K and g. This strengthens the known finiteness of $\mathcal{H}(K,g,\ell)$ when the prime...

conjecture | open | MathDB 页面 | 原文线索

V2-020589 | MATHDB-365480

Strongly tempered inverse Satake conjecture

摘录 / EXCERPT: Let X be a strongly tempered spherical variety, so that its dual group satisfies $\check{G}_X=\check{G}$. Let $\mathbf{A}=\mathbf{A}_X$, let δ denote the relevant modular...

conjecture | open | MathDB 页面 | 原文线索

V2-020590 | MATHDB-365482

Universal volume bounds for reflexive toric Calabi-Yau cones

摘录 / EXCERPT: Let $X(\Delta_{n-1})$ be the toric variety associated with a reflexive toric diagram Δ_{n-1}, and let $\widetilde{X(\Delta_{n-1})}$ denote the relevant resolution. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020591 | MATHDB-365507

The Brauer-Manin sufficiency conjecture for rationally connected surfaces over finite-field function fields

摘录 / EXCERPT: Colliot-Thélène-Sansuc analogue. The surface X verifies (BM).

conjecture | open | MathDB 页面 | 原文线索

V2-020592 | MATHDB-365517

Bondal-Orlov-Kawamata DK conjecture

摘录 / EXCERPT: Let $\mu:X_1\longrightarrow X_2$ be a birational map of smooth varieties, resolved by morphisms ... such that $f_1^*K_{X_1}$ and $f_2^*K_{X_2}$ are linearly equivalent; this is a K-e...

conjecture | open | MathDB 页面 | 原文线索

V2-020593 | MATHDB-365522

Hu-Ye's eigenvariety bound for tensor eigenvalue multiplicity

摘录 / EXCERPT: Let $\mathcal{A}$ be a k-th order, n-dimensional tensor, let $lambda$ be an eigenvalue, and let ... be its eigenvariety. Suppose that this variety has irreducible components...

conjecture | open | MathDB 页面 | 原文线索

V2-020594 | MATHDB-365523

Multiplicity-eigenvector-variety conjecture for connected uniform hypergraphs

摘录 / EXCERPT: Let H be a connected k-uniform hypergraph, let $\mathcal{A}(H)$ be its adjacency tensor, let rho be its spectral radius, and let $\mathbb{V}_{rho}(\mathcal{A}(H))$ be the pro...

conjecture | open | MathDB 页面 | 原文线索

V2-020595 | MATHDB-365529

Uniform boundedness conjecture for integral Tate obstructions in families

摘录 / EXCERPT: Uniform boundedness conjecture. For every integer $d\geq 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-020596 | MATHDB-365561

Deery-Geramita conjecture on catalecticant minors

摘录 / EXCERPT: Let V) be a finite-dimensional vector space, and let $\Upsilon^{i,d-i}$ denote the corresponding catalecticant map; write $\operatorname{Minors}_r(\Upsilon^{i,d-i})$ for the idea...

conjecture | open | MathDB 页面 | 原文线索

V2-020597 | MATHDB-365562

Conjecture that saturation preserves linear ideal sheaves

摘录 / EXCERPT: Let Q be a base scheme, let W be a finite-dimensional vector space with homogeneous coordinate algebra ... and let ${\mathcal{I}}\subset{}_{Q}{\mathcal{S}}$ be a linear ideal she...

conjecture | open | MathDB 页面 | 原文线索

V2-020598 | MATHDB-365575

Birational Fano varieties conjecture for cubic fourfolds

摘录 / EXCERPT: Let X and X' be smooth cubic fourfolds, and let $F(X)$ and $F(X')$ be their Fano varieties of lines. Birational Fano varieties conjecture. If ... then X and X' are birational...

conjecture | open | MathDB 页面 | 原文线索

V2-020599 | MATHDB-365580

Crepant transformation correspondence for simple Grassmannian flops

摘录 / EXCERPT: Let X and X' be the two varieties related by the simple Grassmannian flop under discussion, and let the isomorphism in Proposition $\mathrm{prop_coh_isom}$ be the cohomologic...

conjecture | open | MathDB 页面 | 原文线索

V2-020600 | MATHDB-365616

The rank-three refined Poincaré polynomial formula

摘录 / EXCERPT: Let $N_{3,1}$ be the moduli space of stable rank-three bundles with fixed degree one on a smooth projective curve of genus g, and let $\Omega(N_{3,1},q,t)$ denote its refined Poi...

conjecture | open | MathDB 页面 | 原文线索

V2-020601 | MATHDB-365617

The general-rank specialization formula for refined Poincaré polynomials

摘录 / EXCERPT: Let $N_{r,d}$ be the moduli space of stable rank-r bundles of degree d on a smooth projective curve of genus g, and let $\Omega(N_{r,d},q,t)$ denote its refined Poincaré poly...

conjecture | open | MathDB 页面 | 原文线索

V2-020602 | MATHDB-365638

Modularity conjecture for d-elliptic Noether-Lefschetz cycles

摘录 / EXCERPT: For integers $d,g\ge 1$, let $\widetilde{\mathsf{NL}}_{g,d}$ be the moduli space of morphisms of abelian varieties $h:E\to A$, where E is an elliptic curve, A is a principally...

conjecture | open | MathDB 页面 | 原文线索

V2-020603 | MATHDB-365639

Modularity conjecture for compactified d-elliptic cycles

摘录 / EXCERPT: Let $\mathcal{A}_1^*=\overline{\mathcal{M}}_{1,1}$ and let $[\widetilde{\mathsf{NL}}_{g,d}]^+\in \operatorname{CH}^{g}(\mathcal{A}_1^*\times\mathcal{A}_g)$ be the class of the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-020604 | MATHDB-365647

Generalized Denef-Jacobs-Veys vanishing conjecture

摘录 / EXCERPT: Let X be a smooth irreducible complex projective variety, let $(E_i)_{i\in S}$ be a finite set of mutually distinct prime divisors on X, and let $\bm a=(a_i)_{i\in S}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-020605 | MATHDB-365691

Double mirror conjecture for affine log Calabi-Yau varieties

摘录 / EXCERPT: Let U be an affine log Calabi-Yau variety and let V be a general fibre of its mirror family; assume the mirror construction extends to the canonical-singularity setting conside...

conjecture | open | MathDB 页面 | 原文线索

V2-020606 | MATHDB-365692

Symmetry conjecture for tropical theta pairings

摘录 / EXCERPT: Let U and V be affine log Calabi-Yau varieties with maximal boundary, and let $\operatorname{Sk}(U)$ and $\operatorname{Sk}(V)$ be their skeleta. For $u\in\operatorname{Sk}(U)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020607 | MATHDB-365693

Tropical independence conjecture for theta functions

摘录 / EXCERPT: Let V be an affine log Calabi-Yau variety with maximal boundary, let $v_1,\dots,v_k,w\in\operatorname{Sk}(V)$, and let $\alpha_1,\dots,\alpha_k\in k$ be non-zero constants. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020608 | MATHDB-365696

Cox ring conjecture for universal torsors of log Calabi-Yau pairs

摘录 / EXCERPT: Let $U'\subset Y'$ be an snc minimal model of an affine log Calabi-Yau variety with maximal boundary, let $Y\to Y'$ be the universal torsor, and let $U\subset Y$ be its restriction...

conjecture | open | MathDB 页面 | 原文线索

V2-020609 | MATHDB-365721

Structural decomposition conjecture for local systems with restricted variation

摘录 / EXCERPT: Structural decomposition conjecture. The stack $\operatorname{LS}_G^{\mathrm{restr}}(D^{\circ})$ is a disjoint union of formal completions of a collection of pairwise non-intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-020610 | MATHDB-365751

Huang-Jiang's conjectural evaluation for the even torus-link singularity

摘录 / EXCERPT: Let $R^{(2,2k)}$ be the germ of the plane curve singularity $y(y-x^k)=0$, and let $\widehat{N\!Z}_{R^{(2,2k)}}(t)$ denote its motivic Cohen-Lenstra zeta function. Here k is a pos...

conjecture | open | MathDB 页面 | 原文线索

V2-020611 | MATHDB-365752

Huang-Jiang's nonnegativity conjecture for torus-link zeta functions

摘录 / EXCERPT: Let $R^{(2,2k)}$ be the germ of the plane curve singularity $y(y-x^k)=0$, let $R^{(2,2k)}\oplus N$ denote the direct sum of N copies of this singularity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020612 | MATHDB-365753

Torsion finiteness conjecture for smooth affinoid rigid spaces

摘录 / EXCERPT: Torsion finiteness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020613 | MATHDB-365758

Conjecture on extensions of covariant structures for full quantum flag manifolds

摘录 / EXCERPT: Let $\mathcal{O}_q(\mathrm{F}_{n+1})$ be a full quantum flag manifold, endowed with the first-order differential calculus $\Omega^1_q(\mathrm{F}_{n+1})$, and let Π be the set o...

conjecture | open | MathDB 页面 | 原文线索

V2-020614 | MATHDB-365763

The period-lattice conjecture for Hilbert modular abelian varieties

摘录 / EXCERPT: Let f be a Hilbert newform of parallel weight 2 over the totally real field F, with Hecke field K_f of degree g. Let $s=s_i$ be a sign vector corresponding to the real em...

conjecture | open | MathDB 页面 | 原文线索

V2-020615 | MATHDB-365795

Dimension conjecture for the conformal two-point variety

摘录 / EXCERPT: Dimension conjecture. The dimension of the two-point variety is

conjecture | open | MathDB 页面 | 原文线索

V2-020616 | MATHDB-365834

The tropical Brill-Noether cycle conjecture

摘录 / EXCERPT: Tropical Brill-Noether cycle conjecture. Assume $\rho\geq 0$. Then there exists a canonical tropical subvariety $Z^r_d\subseteq W^r_d$ of pure dimension ρ such that

conjecture | open | MathDB 页面 | 原文线索

V2-020617 | MATHDB-365867

Quadratic period relations and Shimura subvarieties for CM abelian varieties

摘录 / EXCERPT: Let A be a CM abelian variety of dimension $g\geq 1$, let $[o]\in\mathbb{A}_g(\overline{\mathbb{Q}})$ parametrize A, and set $[\mathbf{o}]:=([o],\ldots,[o])$. An elementary qua...

conjecture | open | MathDB 页面 | 原文线索

V2-020618 | MATHDB-365923

Dimension bound and equality characterization for binary graphical descriptions

摘录 / EXCERPT: Fix a tuple of graphs $(\mathcal{G}_1,\dots,\mathcal{G}_d)$ on vertex set $[r]$, and let $W \subset (\mathbb{R}^2)^{\otimes d}$ be the set described by this tuple. Assume that the...

conjecture | open | MathDB 页面 | 原文线索

V2-020619 | MATHDB-365928

Bounded-height conjecture for nondegenerate points in fiberwise small group subschemes

摘录 / EXCERPT: Let $\mathcal{A}\rightarrow S$ be the abelian scheme in the paper, and let $\mathcal{X}\subseteq\mathcal{A}$ be a subvariety. For $t\in\mathbb{N}$, let $\mathcal{A}_{(\leq t)}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-020620 | MATHDB-365936

Equality of quantum and classical learning coefficients on affine open subsets

摘录 / EXCERPT: Quantum-classical learning-coefficient conjecture. The functions K and K^Q behave similarly under the fundamental conditions; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-020621 | MATHDB-365937

Koszulity conjecture for isotropic Milnor K-rings

摘录 / EXCERPT: Let k be a flexible field and let F/k be a finitely generated field extension. The isotropic Milnor K-ring of F/k, denoted $\operatorname{\mathrm{k}}^M_*(F/k)$, is the quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-020622 | MATHDB-365942

Ash-Pollack-Stevens conjecture on classical points of higher-rank eigenvarieties

摘录 / EXCERPT: Let ${\mathcal E}(\mathcal N^\bullet)$ be the eigenvariety constructed from the cohomological complex $\mathcal N^\bullet$ for $\operatorname{GL}_n/\mathbb Q$, and let a cuspidal c...

conjecture | open | MathDB 页面 | 原文线索

V2-020623 | MATHDB-365963

The restricted two-stack sorting characterization for first-stack avoidance of 1324

摘录 / EXCERPT: Let a permutation be sortable by two stacks in series under the restriction that the permutation on the first stack avoids the classical pattern 1324. Let $\operatorname{Av}(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020624 | MATHDB-365967

Guarantee of LADC for non-trivial radial updates

摘录 / EXCERPT: Let X be a radial Riemannian manifold with stationary probability distribution $p(x)\propto{\mathrm{e}^{-V(x)}}$. Suppose an update scheme changes the radial component in a non-t...

conjecture | open | MathDB 页面 | 原文线索

V2-020625 | MATHDB-365987

The weight-monodromy conjecture for smooth projective varieties

摘录 / EXCERPT: Weight-monodromy conjecture. The two filtrations agree up to the shift specified by

conjecture | open | MathDB 页面 | 原文线索

V2-020626 | MATHDB-366024

Beilinson's L-value conjecture for singular K3 surfaces with Picard rank 20

摘录 / EXCERPT: Let X be a singular $K3$ surface with Picard rank 20 over $\mathbb{Q}$, let $H^3_\mathcal{M}(X,\mathbb{Q}(4))_{\operatorname{tr}}$ be the transcendental motivic cohomology, and...

conjecture | open | MathDB 页面 | 原文线索

V2-020627 | MATHDB-366067

Mabuchi-Nakagawa conjecture for canonical cones of Kähler-Ricci solitons

摘录 / EXCERPT: Let X be a smooth Fano manifold admitting a Kähler-Ricci soliton, and let $Y:=K_X^{\times}$ be the complement of the zero section in its canonical bundle. A Calabi-Yau cone struc...

conjecture | open | MathDB 页面 | 原文线索

V2-020628 | MATHDB-366074

Tian's stabilization conjecture for alpha-invariants

摘录 / EXCERPT: Tian's stabilization conjecture. There exists $k_0\in\mathbb{N}_+$ such that, for every $k\geq k_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-020629 | MATHDB-366076

Tian's strong stabilization conjecture for alpha-invariants

摘录 / EXCERPT: Tian's strong stabilization conjecture. If $R(X,L)$ is generated by

conjecture | open | MathDB 页面 | 原文线索

V2-020630 | MATHDB-366077

Recovered-component conjecture for Chow-Lam loci

摘录 / EXCERPT: Recovered-component conjecture. If $i\geq k$, then all recovered components of Ω_1^{ki} are of the form described in Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-020631 | MATHDB-366078

Generic Grassmannian Chow-Lam recovery conjecture

摘录 / EXCERPT: Generic Grassmannian recovery conjecture. The algebraic sets $W_{\mathcal V}$ and $\mathcal V$ coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-020632 | MATHDB-366102

Refined inverse Galois problem for abelian varieties

摘录 / EXCERPT: Fix a prime number p and let $G\subset W_{2d}$ be a transitive subgroup containing the complex conjugation element. The weighted permutation representation associated to an abeli...

conjecture | open | MathDB 页面 | 原文线索

V2-020633 | MATHDB-366116

Arithmetic level raising conjecture for unitary Shimura varieties

摘录 / EXCERPT: Let $\mathfrak{m}$ be a maximal ideal of the Hecke algebra, and assume Hypotheses and. The relevant special fibre is denoted by ${{\Sh}_{1,n-1}^{{\rm{ss}}}}$, its ambient Shimura v...

conjecture | open | MathDB 页面 | 原文线索

V2-020634 | MATHDB-366147

Additional hyperplanes conjecture for directed-cycle model singular loci

摘录 / EXCERPT: Let $n \geq 3$, and let $\mathcal{M} = (G, \textit{In}, \textit{Out}, \textit{Leak})$ be an n-compartment directed-cycle model with $\textit{In}=\{1\}$ and $\textit{Out}=\{p\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-020635 | MATHDB-366148

The universal Psi constraint for Hodge integrals with lambda insertions

摘录 / EXCERPT: Let X be a smooth projective variety, and let $\Psi_{g,n}$ denote the universal expression for Hodge integrals with one λ_{g-1} or λ_g insertion defined in the so...

conjecture | open | MathDB 页面 | 原文线索

V2-020636 | MATHDB-366152

The rational basis-function conjecture for non-hyperelliptic Jacobians

摘录 / EXCERPT: Let $\mathcal{C}$ be a non-hyperelliptic curve, let $\operatorname{Jac}(\mathcal{C})$ be its Jacobian, let Σ denote the relevant theta divisor, and let the basis $\wp$-funct...

conjecture | open | MathDB 页面 | 原文线索

V2-020637 | MATHDB-366165

Cazassus-Herald-Kirk-Kotelskiy conjecture on bounding cochains

摘录 / EXCERPT: Let T be a 2-tangle, let $R_\pi(T)$ be its holonomy-perturbed traceless character variety, and let P^* be the pillowcase with its singular points removed. A bounding cochain is...

conjecture | open | MathDB 页面 | 原文线索

V2-020638 | MATHDB-366171

Stapledon's effectiveness conjecture for equivariant Ehrhart series

摘录 / EXCERPT: Let P be a G-invariant lattice polytope, and let Y be the corresponding toric variety with torus-invariant line bundle L. The equivariant H^*-series is the series...

conjecture | open | MathDB 页面 | 原文线索

V2-020639 | MATHDB-366176

The Monodromy Conjecture for motivic zeta functions

摘录 / EXCERPT: Monodromy Conjecture. If s_0 is a pole of $Z_{f,W}^{\mathrm{mot}}(s)$, then $(s-s_0)\mid b_f(s)$, and $e^{s_0\cdot 2\pi \sqrt{-1}}$ is an eigenvalue of a local monodromy at some...

conjecture | open | MathDB 页面 | 原文线索

V2-020640 | MATHDB-366181

Torelli conjecture for deformation-equivalent log Calabi-Yau pairs

摘录 / EXCERPT: Let (Y,D) and (Y',D') be n-dimensional, deformation-equivalent log Calabi-Yau pairs, where a log Calabi-Yau pair is a smooth projective variety together with an anticanonical...

conjecture | open | MathDB 页面 | 原文线索

V2-020641 | MATHDB-366182

Conjecture on the maximum number of conic-line curves in large degree

摘录 / EXCERPT: Let P be a pencil of degree-d curves, and let m denote the number of conic-line curves in the pencil. Large-degree conic-line pencil conjecture. There does not exist a pencil...

conjecture | open | MathDB 页面 | 原文线索

V2-020642 | MATHDB-366221

Pappas-Rapoport's conformal-blocks conjecture for parahoric bundle stacks

摘录 / EXCERPT: Let X be a smooth projective curve, let $S\subseteq X$ be a nonempty set containing the bad points $\mathcal{R}$ of a parahoric Bruhat-Tits group scheme $\mathcal{G}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020643 | MATHDB-366222

The ordinarity conjecture for smooth projective schemes

摘录 / EXCERPT: Let X be a smooth projective scheme defined over a field of characteristic zero. After spreading out X, for a prime p, write $X_{p=0}$ for its reduction modulo p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020644 | MATHDB-366226

The coherent-subset classification conjecture for weak localizing subcategories

摘录 / EXCERPT: Let X be a noetherian scheme. A coherent subset of X is a subset of the underlying topological space of the kind appearing in Krause's classification for commutative noetherian...

conjecture | open | MathDB 页面 | 原文线索

V2-020645 | MATHDB-366229

Cheltsov-Park-Won conjecture on polarized cylinders in log del Pezzo surfaces

摘录 / EXCERPT: Let S be a log del Pezzo surface, meaning a del Pezzo surface with at worst klt singularities. Let ... be its cylindrical ample set, where $\mathrm{Amp}(S)$ is the ample cone. Chel...

conjecture | open | MathDB 页面 | 原文线索

V2-020646 | MATHDB-366271

The forbidden-pattern characterization conjecture for one-stack one-queue layouts

摘录 / EXCERPT: Forbidden-pattern characterization conjecture. The ordered matching admits

conjecture | open | MathDB 页面 | 原文线索

V2-020647 | MATHDB-366288

Thomsen's comparison conjecture for the MVdK and KLT splittings

摘录 / EXCERPT: Let $G=\operatorname{GL}_n$, let P be a parabolic subgroup containing the Borel subgroup B, and let $\mathfrak{u}_P$ be the nilpotent radical of the Lie algebra of P. Conside...

conjecture | open | MathDB 页面 | 原文线索

V2-020648 | MATHDB-366297

Finiteness conjecture for characteristic numbers of complex line arrangements

摘录 / EXCERPT: Let a line arrangement be a finite collection of lines in the complex projective plane, and let $\gamma(\mathcal{L})$ denote its characteristic number. For a real number...

conjecture | open | MathDB 页面 | 原文线索

V2-020649 | MATHDB-366298

Finiteness conjecture for characteristic numbers of conic-line arrangements

摘录 / EXCERPT: Let a conic-line arrangement be an arrangement $\mathcal{CL}$ of lines and smooth conics in the complex projective plane with ordinary singularities, and let $\gamma(\mathcal{CL})$...

conjecture | open | MathDB 页面 | 原文线索

V2-020650 | MATHDB-366334

Dubrovin's conjecture on Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety. The Dubrovin conjecture. The derived category $\mathrm{D}^b(X)$ admits a full exceptional collection if and only if the big quantum cohomology rin...

conjecture | open | MathDB 页面 | 原文线索

V2-020651 | MATHDB-366342

Limit-height conjecture for translated complete intersections in toric varieties

摘录 / EXCERPT: Let $\mathbb{K}$ be a number field with set of places $\mathfrak{M}$, let $\mathbb{T}$ be a split algebraic torus over $\mathbb{K}$ of dimension n with character lattice M, and...

conjecture | open | MathDB 页面 | 原文线索

V2-020652 | MATHDB-366353

Piecewise-polynomial top-term conjecture for Euler characteristics of isoresidual fibers

摘录 / EXCERPT: Piecewise-polynomial top-term conjecture. The top term of the Euler characteristic of $\overline{\mathcal F}_{\lambda}$ is a piecewise homogeneous polynomial of degree $p-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-020653 | MATHDB-366360

The DK conjecture on K-equivalent smooth varieties

摘录 / EXCERPT: Let X_1 and X_2 be smooth varieties, and let $\mu:X_1\longrightarrow X_2$ be a birational map resolved by morphisms ... such that $f_1^*K_{X_1}$ and $f_2^*K_{X_2}$ are linearly e...

conjecture | open | MathDB 页面 | 原文线索

V2-020654 | MATHDB-366363

Transition-system conjecture for symmetric matrix Schubert varieties

摘录 / EXCERPT: Let n be a positive integer, let $\boldsymbol{\rm Mat}^{\mathrm{sym}}_{n\times n}$ be the space of symmetric $n\times n$ matrices, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-020655 | MATHDB-366365

Groebner-basis conjecture for symmetric matrix Schubert ideals

摘录 / EXCERPT: Let n be a positive integer, let $\boldsymbol{\rm Mat}^{\mathrm{sym}}_{n\times n}$ be the space of symmetric $n\times n$ matrices, and let I^{sym}_w and...

conjecture | open | MathDB 页面 | 原文线索

V2-020656 | MATHDB-366366

Primality conjecture for symmetric matrix Schubert ideals

摘录 / EXCERPT: Let n be a positive integer, let $\boldsymbol{\rm Mat}^{\mathrm{sym}}_{n\times n}$ be the space of symmetric $n\times n$ matrices, and for each symmetric matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-020657 | MATHDB-366367

Involution pipe-dream decomposition conjecture for symmetric initial ideals

摘录 / EXCERPT: Involution pipe-dream conjecture. If $y\in I_\infty^{(n)}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-020658 | MATHDB-366404

Iwasawa's main conjecture for definite and indefinite Rankin-Selberg eigenvarieties

摘录 / EXCERPT: Let $\mathbf{V}$ be a datum with $\mathfrak{m}=\emptyset$, let $\xi=(\xi^n,\xi^{n+1})$ be a pair of $\Sigma^+_p\setminus\lozenge^+$-trivial dominant hermitian weights, let γ...

conjecture | open | MathDB 页面 | 原文线索

V2-020659 | MATHDB-366421

Schnell's conjecture for algebraic fiber spaces

摘录 / EXCERPT: Let $f\colon X\to Y$ be an algebraic fiber space between smooth projective varieties, and let F be its general fiber with $\kappa(F)=0$. Suppose that there are an ample divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-020660 | MATHDB-366470

Mozgovoy's motivic ADHM formula for twisted Higgs-bundle moduli

摘录 / EXCERPT: Mozgovoy's motivic ADHM conjecture. For every $r\geq 1$, $H_r(t)$ is a polynomial in t and

conjecture | open | MathDB 页面 | 原文线索

V2-020661 | MATHDB-366509

The algebraicity conjecture for the Kuga-Satake correspondence

摘录 / EXCERPT: Algebraicity conjecture for the Kuga-Satake correspondence. The morphism μ' is induced by an algebraic cycle on $S\times \mathrm{KS}(S)^2$. This asserts that the Hodge-theoreti...

conjecture | open | MathDB 页面 | 原文线索

V2-020662 | MATHDB-366519

Flatness conjecture for matrix Hessenberg degenerations over arbitrary sheets

摘录 / EXCERPT: Let $h:[n]\to[n]$ be a Hessenberg function, let $\mathsf{x}\in\mathfrak{gl}_n(\mathbb{C})$, and let $\varphi_{\mathsf{x},h}$ be the one-parameter family constructed from the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-020663 | MATHDB-366548

Strong real Jacobian conjecture

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be a real polynomial map, and let $f'(x)=\det\operatorname{Jac}_x f$. Strong real Jacobian conjecture. If ... for every $x\in\mathbb{R}^n$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-020664 | MATHDB-366549

Real topological Jacobian conjecture

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be a real polynomial map with Jacobian determinant $f'(x)>0$ for every $x\in\mathbb{R}^n$, and let S_f denote its non-properness set. Real top...

conjecture | open | MathDB 页面 | 原文线索

V2-020665 | MATHDB-366555

Canonical-map perfectoidness conjecture for Albanese towers

摘录 / EXCERPT: Let X/C be a smooth proper connected variety of pure dimension d with globally generated 1-forms, and fix a point $x\in X(C)$. Let ... be a closed subgroup. The canonical map s...

conjecture | open | MathDB 页面 | 原文线索

V2-020666 | MATHDB-366569

Bonavero-Casagrande-Debarre-Druel generalized Mukai conjecture

摘录 / EXCERPT: Let X be a smooth Fano variety. Let ρ_X denote its Picard number and let ι_X denote its pseudo-index, defined by ... Here $-K_X$ is the anticanonical divisor of X. The...

conjecture | open | MathDB 页面 | 原文线索

V2-020667 | MATHDB-366570

Gagliardi-Hofscheier spherical-skeleton conjecture

摘录 / EXCERPT: Let $\mathcal R=(\Delta,S^p,\Sigma,\Gamma)$ be a complete spherical skeleton, where R is the underlying root system, R^+ is its set of positive roots, and $R^+_{S^p}$ is the se...

conjecture | open | MathDB 页面 | 原文线索

V2-020668 | MATHDB-366586

Gabrielov-Novikov-Shapiro line-restriction conjecture for generalized electrostatic potentials

摘录 / EXCERPT: Gabrielov-Novikov-Shapiro conjecture. For all $p\geq 1$, the number of equilibria of the restriction of V_p to any straight line in $\mathbb{R}^3$ is at most $2n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-020669 | MATHDB-366600

Harder-Narasimhan recursion conjecture for higher asymptotic ranges

摘录 / EXCERPT: For each d, let $P_d(y)$ be the Poincaré polynomial of the corresponding moduli space, let $\widetilde P_{d,\chi}(y)$ be the Poincaré series of the stack of semistable sheaves, a...

conjecture | open | MathDB 页面 | 原文线索

V2-020670 | MATHDB-366642

Krämer-Weissauer's Tannakian Schottky conjecture

摘录 / EXCERPT: Let $\mathcal{J}_g$ denote the locus of g-dimensional Jacobians in the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020671 | MATHDB-366709

Classification conjecture for tight symmetric inner distribution codes in Johnson schemes

摘录 / EXCERPT: Let r,s be positive integers, let $n:=r+s$, and consider the Johnson scheme $J(2n,n)$. A tight symmetric inner distribution code is a code for which equality holds in the relevan...

conjecture | open | MathDB 页面 | 原文线索

V2-020672 | MATHDB-366717

Mirror existence conjecture for Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let X be a space with canonical bundle cK_X and let f be an anticanonical section defining a Calabi-Yau hypersurface $Z_f\subset X[c_1]$. A corresponding space Y has an ant...

conjecture | open | MathDB 页面 | 原文线索

V2-020673 | MATHDB-366718

Complete-intersection realization conjecture for toric varieties

摘录 / EXCERPT: A toric variety is an algebraic variety equipped with the torus action arising from a toric construction; a Calabi-Yau hypersurface is a hypersurface with trivial canonical class....

conjecture | open | MathDB 页面 | 原文线索

V2-020674 | MATHDB-366719

Laurent smoothing conjecture for Tyurin-degenerate Calabi-Yau manifolds

摘录 / EXCERPT: Let a Tyurin-degenerate Calabi-Yau n-fold be a degenerate Calabi-Yau space whose smoothing may require Laurent deformations, and let ε denote the deformation paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-020675 | MATHDB-366721

Multiple-mirror conjecture for transpolar VEX multitopes

摘录 / EXCERPT: Let $(\mathcal{D}^{\star}_X,\mathcal{D}_X)$ be a transpolar pair of VEX multitopes. A center-enclosing sub-simplex reduction is a pair of lattice simplices...

conjecture | open | MathDB 页面 | 原文线索

V2-020676 | MATHDB-366727

Polynomial surfaces of revolution parametrization conjecture

摘录 / EXCERPT: Let S be a polynomial surface of revolution, and let $p(t)$ be its associated polynomial. A proper real polynomial parametrization is a real polynomial map that parametrizes S...

conjecture | open | MathDB 页面 | 原文线索

V2-020677 | MATHDB-366729

Nakajima's canonical tensor-product isomorphism conjecture for quiver varieties

摘录 / EXCERPT: Let $\tilde{\frac?}$ be the tensor product variety associated with dominant weights λ_1,λ_2, and let ω^1,ω^2 be the corresponding framing data, with...

conjecture | open | MathDB 页面 | 原文线索

V2-020678 | MATHDB-366754

Algebraicity and integrality for formal leaves of foliations

摘录 / EXCERPT: Let $R\subset\mathbb{C}$ be an integral domain finitely generated over $\mathbb{Z}$, let $(X,\mathscr{F})$ be a smooth scheme with a foliation over R, and let $x\in X(R)$. The le...

conjecture | open | MathDB 页面 | 原文线索

V2-020679 | MATHDB-366755

André-Christol conjecture for formal flat sections

摘录 / EXCERPT: Let $R\subset\mathbb{C}$ be a finitely generated $\mathbb{Z}$-algebra, let X be a smooth R-scheme, let $(\mathscr{E},\nabla)$ be a flat vector bundle on X/R, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020680 | MATHDB-366757

Variational motivic conjecture for local systems

摘录 / EXCERPT: Let $\overline{k}$ be a separable algebraic closure of a finitely generated field k, let $f:X\to S$ be a smooth proper morphism of smooth $\overline{k}$-schemes, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020681 | MATHDB-366760

Formal meta-conjecture for Picard-Fuchs equations

摘录 / EXCERPT: Let $R\subset\mathbb{C}$ be a finitely generated $\mathbb{Z}$-algebra, let $f:X\to S$ be a smooth proper morphism of smooth R-schemes, fix $s\in S(R)$, and let $\widehat{S}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020682 | MATHDB-366780

Purity, splitness and cellularity conjectures for mixed motives

摘录 / EXCERPT: Purity, splitness and cellularity conjectures. The following properties should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-020683 | MATHDB-366781

Ben-Zvi-Nadler's local monodromic Arkhipov-Bezrukavnikov equivalence

摘录 / EXCERPT: Let G be the Langlands dual group, $\mathsf{B}\subset\mathsf{G}$ a Borel subgroup, and Fl the flag variety with Iwahori subgroup I. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020684 | MATHDB-366786

The deformation conjecture for cocompact hyperbolic lattices

摘录 / EXCERPT: For $n\geqslant 2$, let Γ be a cocompact lattice of $\operatorname{O}(n,1)$, and let $N\geqslant n$. Consider the standard embedding ... A representation is strictly dominated...

conjecture | open | MathDB 页面 | 原文线索

V2-020685 | MATHDB-366809

Somoza's endomorphism-ring conjecture for a cyclic plane quintic

摘录 / EXCERPT: Let C be the smooth plane curve ... Let J be its Jacobian, let $K=\mathbb{Q}(\zeta_9)^+(\zeta_5)$, and let $\mathcal{O}_K$ denote the ring of integers of K. Somoza's endomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-020686 | MATHDB-366817

Conjectural splitting of the virtual class for twisted maps to the universal root stack

摘录 / EXCERPT: Let X be a smooth pair with boundary divisor D, let r be a positive integer, and let Λ denote the discrete data of the twisted maps. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020687 | MATHDB-366847

The Fermat conjecture inside projective varieties

摘录 / EXCERPT: Fermat conjecture. Fermat holds inside V if and only if $(V\cap Z)(\mathbb Q)$ is trivial. This gives a proposed characterization of the projective varieties for which the genera...

conjecture | open | MathDB 页面 | 原文线索

V2-020688 | MATHDB-366853

Equivalence of the matrix factorization categories for the six-line hourglass

摘录 / EXCERPT: Let $\mathrm{X}_{1.9}$ and $\mathrm{X}'_{1.9}$ be the hourglass varieties, and for each i let $\mathcal{MF}_{i}(\mathrm{X}_{1.9})$ denote the left orthogonal component of the exc...

conjecture | open | MathDB 页面 | 原文线索

V2-020689 | MATHDB-366879

Reduction conjecture for equivariant and quotient quantum D-modules

摘录 / EXCERPT: Let X be a smooth projective variety equipped with an algebraic $T_\mathbb{C}$-action, and let Y be a smooth GIT quotient of X without orbifold singularities. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020690 | MATHDB-366941

The quasi-lisse conjecture for the vertex algebras \(\mathcal{C}_p\)

摘录 / EXCERPT: Let $\mathcal{C}_p$ be the vertex operator algebra considered in the paper. A vertex operator algebra is quasi-lisse when its associated variety has finitely many symplectic leaves...

conjecture | open | MathDB 页面 | 原文线索

V2-020691 | MATHDB-366947

Conjecture that spinor modifications of conic bundles are hyperbolically equivalent

摘录 / EXCERPT: Spinor-modification hyperbolic-equivalence conjecture. If X'/S is a spinor modification of X/S, then X'/S is hyperbolic equivalent to X/S.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-020692 | MATHDB-366950

Markman-Mukai lattice conjecture for twisted derived categories of hyper-Kähler varieties

摘录 / EXCERPT: Let X and Y be hyper-Kähler varieties of $K3^{[n]}$-type. Let $L(X)$ and $L(Y)$ be their Markman-Mukai lattices, with primitive generators v and w of the rank-one orthogona...

conjecture | open | MathDB 页面 | 原文线索

V2-020693 | MATHDB-366977

Lehmer's conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety defined over a number field K, and let $\mathcal{L}$ be a symmetric ample line bundle on A. For a point $P\in A(\overline{K})$, let $D=[K(P):K]$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-020694 | MATHDB-366981

Conjectural deformation of secant sheaves on abelian sixfolds

摘录 / EXCERPT: Let X be an abelian threefold, let $\hat X$ be its dual, and let ${{{{{{{{\mathcal G}}}}}}}}_{F_1,F_2}$ be the sheaf constructed from non-zero coherent sheaves F_1,F_2 whose Ch...

conjecture | open | MathDB 页面 | 原文线索

V2-020695 | MATHDB-366982

Semiregularity conjecture for twisted coherent sheaves

摘录 / EXCERPT: Let $\pi:{{{{{{{\mathcal M}}}}}}}\rightarrow S$ be a deformation of a smooth complex projective variety M_0 over a smooth germ $(S,0)$, and write $M_s=\pi^{-1}(s)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020696 | MATHDB-367005

Motivic-cycle realization conjecture for regularized theta lifts

摘录 / EXCERPT: Let A^j_{η} be the j-fold power of the universal elliptic-curve family over the generic point η, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020697 | MATHDB-367025

Equality of the images of the fans associated to L_1 and L_{d-1}

摘录 / EXCERPT: Let L_1 and L_{d-1} be the fans associated to the corresponding loci in the tropical parameter space, and let M^{tr}_2 be the tropical moduli space of principally polarized a...

conjecture | open | MathDB 页面 | 原文线索

V2-020698 | MATHDB-367043

Coleman-Oort conjecture, arithmetic version

摘录 / EXCERPT: Let $\mathcal{M}_g$ denote the moduli space of smooth curves of genus g. A curve in $\mathcal{M}_g$ has a CM Jacobian when its Jacobian is a CM abelian variety. Coleman-Oort conj...

conjecture | open | MathDB 页面 | 原文线索

V2-020699 | MATHDB-367044

Coleman-Oort conjecture, geometric version

摘录 / EXCERPT: Let $\mathcal{M}_g\subset\mathcal{A}_g$ be the Torelli locus, viewed inside the moduli space of principally polarised abelian varieties. A sub-Shimura variety means a special subva...

conjecture | open | MathDB 页面 | 原文线索

V2-020700 | MATHDB-367045

Zilber-Pink conjecture for real multiplication loci in the Torelli locus

摘录 / EXCERPT: Real multiplication finiteness conjecture. Fix $g\geq 7$. Each $E_{g,K}$ is contained in a finite union of Hilbert modular varieties. Moreover, $E_{g,K}$ is empty for all but finit...

conjecture | open | MathDB 页面 | 原文线索

V2-020701 | MATHDB-367048

Klingler-Lerer conjecture on arithmetic points of strata

摘录 / EXCERPT: Let $\Omega\mathcal{M}_g(\kappa)$ be a stratum of abelian differentials, and let $S\subset\Omega\mathcal{M}_g(\kappa)$ be an irreducible algebraic subvariety. Call a subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-020702 | MATHDB-367049

Symplecticity conjecture for parabolic partial implosions

摘录 / EXCERPT: Let G be a connected semisimple group, let P be a parabolic subgroup of G, and let U_P be its unipotent radical. Let B be a Borel subgroup and set ... as a homogeneous vect...

conjecture | open | MathDB 页面 | 原文线索

V2-020703 | MATHDB-367054

Xu's index conjecture for klt Calabi-Yau pairs

摘录 / EXCERPT: Xu's index conjecture. There is a positive integer $c(n,I)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-020704 | MATHDB-367055

Smooth maximal-index conjecture for Calabi-Yau varieties

摘录 / EXCERPT: Smooth maximal-index conjecture. The largest possible index of any smooth, projective Calabi-Yau n-fold is

conjecture | open | MathDB 页面 | 原文线索

V2-020705 | MATHDB-367056

Wang's maximal-Betti-number conjecture for smooth Calabi-Yau varieties

摘录 / EXCERPT: Wang's maximal-Betti-number conjecture. The variety $W^{(n)}$ has the largest sum of Betti numbers among smooth, projective Calabi-Yau n-folds. When n is odd, it also has the s...

conjecture | open | MathDB 页面 | 原文线索

V2-020706 | MATHDB-367077

The intrinsic-limit conjecture for Laurent-deformed Calabi-Yau hypersurfaces

摘录 / EXCERPT: Intrinsic-limit conjecture. Laurent-deformed Calabi-Yau hypersurfaces closed or completed by the intrinsic limit are not algebraic varieties; they are toric spaces equipped with a...

conjecture | open | MathDB 页面 | 原文线索

V2-020707 | MATHDB-367078

The deformation-limit conjecture for rational sections smoothing Tyurin degenerations

摘录 / EXCERPT: Deformation-limit conjecture. The rational sections $\vec\d q_0$ that smooth Z_{q_0} are deformation limits, in the rightward direction, of regular (not rational-monomial) sectio...

conjecture | open | MathDB 页面 | 原文线索

V2-020708 | MATHDB-367079

The flip-folded transposition-mirror conjecture for non-Fano toric hypersurfaces

摘录 / EXCERPT: Flip-folded transposition-mirror conjecture. (1) The transposition mirror may be found as the transposed hypersurface $Z_{f^{\mathrm T}}$ in $\widetilde X$. (2) The toric space...

conjecture | open | MathDB 页面 | 原文线索

V2-020709 | MATHDB-367081

The regularizing fractional-transformation conjecture for VEX-multitope Calabi-Yau hypersurfaces

摘录 / EXCERPT: Regularizing fractional-transformation conjecture. All such Calabi-Yau Laurent hypersurfaces are also describable, via suitable fractional transformations, in terms of general-type...

conjecture | open | MathDB 页面 | 原文线索

V2-020710 | MATHDB-367101

The unbounded-degree ruled-divisors conjecture for projective IHS manifolds

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic manifold, and let a polarization on X determine degrees of curves. The unbounded-degree ruled-divisors conjecture. X...

conjecture | open | MathDB 页面 | 原文线索

V2-020711 | MATHDB-367102

Javanpeykar-Rousseau's characterization of Campana-special varieties

摘录 / EXCERPT: Javanpeykar-Rousseau's conjecture. X is geometrically special if and only if X is (Campana-)special.

conjecture | open | MathDB 页面 | 原文线索

V2-020712 | MATHDB-367110

The exceptional-collection conjecture for permutation representations of partial flag varieties

摘录 / EXCERPT: Let G be an algebraic group, let $P\supset B$ be a parabolic subgroup, and let $\mathsf{k}[(G/P)(\mathbb{F}_q)]$ be the permutation representation over k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020713 | MATHDB-367134

Weak non-Archimedean Green-Griffiths-Lang-Vojta conjecture

摘录 / EXCERPT: Let X be a K-variety. Weak non-Archimedean Green-Griffiths-Lang-Vojta conjecture. The following are equivalent: ... This is the weak form of the non-Archimedean Green-Griffiths-L...

conjecture | open | MathDB 页面 | 原文线索

V2-020714 | MATHDB-367138

Regular-sequence conjecture for Zastava spaces

摘录 / EXCERPT: Let $Q^{\geq 0}$ be the nonnegative cone in the root lattice, let $\alpha=\sum_{i=1}^n a_i\alpha_i\in Q^{\geq 0}$, and let Z^α_0 be the central fiber of the corresponding Za...

conjecture | open | MathDB 页面 | 原文线索

V2-020715 | MATHDB-367151

Hosono's central charge formula conjecture

摘录 / EXCERPT: Let X be a Calabi-Yau variety, let J denote its polarization class, and let $w_{0}^{T_{o}}(x,J/(2i))$ be the cohomology-valued hypergeometric series on $U_{T_{o}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020716 | MATHDB-367156

The conjecture on Morgan-Shalen accumulation of holonomy intersections

摘录 / EXCERPT: Let S be the underlying surface, let $\mathcal{T}$ be its Teichmüller space, let $\mathcal{T}^\ast$ denote the corresponding space with the opposite orientation, and let χ b...

conjecture | open | MathDB 页面 | 原文线索

V2-020717 | MATHDB-367162

Coherence conjecture for holonomic arithmetic \(\mathcal{D}\)-modules over the generic fibre

摘录 / EXCERPT: Let S be the base formal scheme, let η denote its generic point, and let $\mathbb{D}_S$ and $\mathbb{D}_{\mathfrak{S}}$ be the corresponding arithmetic differential-operator...

conjecture | open | MathDB 页面 | 原文线索

V2-020718 | MATHDB-367163

Comparison conjecture for arithmetic \(\mathcal{D}\)-module and rigid cohomology

摘录 / EXCERPT: Let X_η be the generic fibre of the strictly semi-stable scheme considered in the paper, let $\mathcal{E}^{\dagger}_K$ be the bounded Robba-ring coefficient field, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-020719 | MATHDB-367185

Multiplicity bounds for \d835\dcb-Gorenstein toric varieties of rank r

摘录 / EXCERPT: Let X be an n-dimensional $\delta 835\dcb$-Gorenstein toric variety of rank r and Gorenstein index $k\ge 2$. Let Q° be the polar weight matrix defined in the source, a...

conjecture | open | MathDB 页面 | 原文线索

V2-020720 | MATHDB-367235

Higher-dimensional unobstructedness conjecture for rigid tropical curves

摘录 / EXCERPT: Let Q be a tropical abelian variety, and let $V\subset Q$ be a rigid tropical curve admitting a pair-of-pants decomposition. Higher-dimensional unobstructedness conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-020721 | MATHDB-367236

B-realizability conjecture for rigid tropical curves

摘录 / EXCERPT: Let Q be a tropical abelian variety and let $V\subset Q$ be a rigid tropical curve admitting a pair-of-pants decomposition, as in the higher-dimensional unobstructedness conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-020722 | MATHDB-367250

The vanishing conjecture for fixed log étale cohomology

摘录 / EXCERPT: Let $s=\operatorname{Spec}(k)$ be the standard log point, let X be a projective vertical log smooth fs log scheme over s, let ℓ be invertible in k, and write ... The fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-020723 | MATHDB-367251

The log Tate conjecture in degree twice the codimension

摘录 / EXCERPT: Let $s=\operatorname{Spec}(k)$ be the standard log point, let X be a projective vertical log smooth fs log scheme over s, let ℓ be invertible in k, and define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-020724 | MATHDB-367253

The log Tate conjecture for a log Tate curve

摘录 / EXCERPT: Let s be the standard log point over a finite field k, let X be a projective vertical log smooth fs log scheme over s, let $m\geq 2r$, and let E be a log Tate curve over...

conjecture | open | MathDB 页面 | 原文线索

V2-020725 | MATHDB-367254

The logarithmic K-theoretic Tate conjecture

摘录 / EXCERPT: Let s be the standard log point over a finite field k, let X be a projective vertical log smooth fs log scheme over s, and define $K_0^{\log}(X)$ as the quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-020726 | MATHDB-367255

The finer logarithmic Tate conjecture

摘录 / EXCERPT: Let s be the standard log point over a finite field k, let X be a projective vertical log smooth fs log scheme over s, and let E be a log Tate curve. Define $K(X,m,r)$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-020727 | MATHDB-367256

Bloch's pole-order conjecture in logarithmic K-theory

摘录 / EXCERPT: Let V be a variety with strictly semistable reduction X over a local field with residue field k, assume that k is finite, and define ... where φ is geometric Frobeniu...

conjecture | open | MathDB 页面 | 原文线索

V2-020728 | MATHDB-367258

Existence conjecture for the monodromy cycle

摘录 / EXCERPT: Let s be the standard log point over a finite field, let X be a projective vertical log smooth fs log scheme over s, let $d=\dim X$, and let q be a section of the log struc...

conjecture | open | MathDB 页面 | 原文线索

V2-020729 | MATHDB-367297

The weak-order chain-length conjecture for alternating sign matrices

摘录 / EXCERPT: Weak-order chain-length conjecture. The weak-order saturated chains from the identity to A are all of the same length if and only if X_B is Cohen--Macaulay for every...

conjecture | open | MathDB 页面 | 原文线索

V2-020730 | MATHDB-367299

The cohomological Hall algebra conditional surjectivity conjecture

摘录 / EXCERPT: Let X be a smooth surface, let Z_1 and Z_2 be curves with $Z=Z_1\cup Z_2$, and let $\mathbf{HA}^T_{X,Z}$ denote the corresponding T-equivariant cohomological Hall algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-020731 | MATHDB-367301

The stratified cohomological Hall algebra amalgamation conjecture

摘录 / EXCERPT: Let $Z=Z_1\cup\cdots\cup Z_s$ be a union of curves, let $\langle Z\rangle$ be the relevant semigroup of classes, and let $\overline{\mathfrak a}_\gamma$ be the amalgamation map in...

conjecture | open | MathDB 页面 | 原文线索

V2-020732 | MATHDB-367326

The generalised Mukai conjecture for smooth Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety. Denote by ρ_X its Picard number, by ι_X its pseudo-index, and by $\dim X$ its dimension. The generalised Mukai conjecture. ... with equal...

conjecture | open | MathDB 页面 | 原文线索

V2-020733 | MATHDB-367327

Gongyo's Mukai-type conjecture for smooth Fano varieties

摘录 / EXCERPT: Gongyo's Mukai-type conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020734 | MATHDB-367361

Voevodsky's conjecture that algebraic cobordism is logarithmic

摘录 / EXCERPT: Let $S\in\mathrm{Rg}$ be a base ring, and let $\mathrm{SH}(S)$ denote the stable motivic homotopy category over S. Let $\mathbf{MGL}\in\mathrm{SH}(S)$ be the algebraic cobordism...

conjecture | open | MathDB 页面 | 原文线索

V2-020735 | MATHDB-367362

The conjecture that every algebraic cobordism module is logarithmic

摘录 / EXCERPT: Let $S\in\mathrm{Rg}$ be a base ring, let $\mathrm{MS}_{\mathrm{Nis}}(S)$ denote the Nisnevich motivic stable category over S, and let $\mathbf{MGL}$ be the algebraic cobordism s...

conjecture | open | MathDB 页面 | 原文线索

V2-020736 | MATHDB-367367

The re-embedding conjecture for non-simplicial MaxDeg planar order ideals

摘录 / EXCERPT: Let ${\mathcal{O}}$ be a non-simplicial MaxDeg planar order ideal, and let $\mathbb{B}_{\mathcal{O}}$ denote its border basis scheme. A Z-separating re-embedding means a re-embed...

conjecture | open | MathDB 页面 | 原文线索

V2-020737 | MATHDB-367369

$varphi$-rationality of perfect Schubert schemes

摘录 / EXCERPT: Let $\Fl_{\calG}$ be an affine flag variety, and let $\Fl_{\calG,\leq w}$ be its Schubert subvariety indexed by a double coset w of the Iwahori--Weyl group. Its perfection is den...

conjecture | open | MathDB 页面 | 原文线索

V2-020738 | MATHDB-367386

Stable homology conjecture for Hurwitz spaces of higher-genus curves

摘录 / EXCERPT: Let $g\geq2$ be an integer and let G be a finite group. Let CHur_g^G be the algebraic stack over $\mathbb C$ whose objects are geometrically connected finite éta...

conjecture | open | MathDB 页面 | 原文线索

V2-020739 | MATHDB-367408

Torsion-vanishing conjecture for weak Langlands-Shahidi parameters

摘录 / EXCERPT: Let $\bb G$ be a connected reductive group with Shimura datum $(\bb G,\bb X)$, let $\msf G$ be its relevant local base change, let $\mdc K^p$ be the prime-to-p level, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020740 | MATHDB-367431

Joyce-Safronov conjecture on matrix factorizations for shifted Lagrangians

摘录 / EXCERPT: Let U be a smooth $\mathbb{C}$-scheme and let $\phi\colon U\to\mathbb{C}$ be a regular function. Let $\mathbf{M}\to\mathbf{Crit}(\phi)$ be a (-1)-shifted Lagrangian such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020741 | MATHDB-367443

Liu-Xu's triangulation conjecture for special valuation spaces

摘录 / EXCERPT: Liu-Xu's triangulation conjecture. There is a rational triangulation of $\mathcal{D}^{KV}(X,D+\Delta)$ such that, in the interior of $\mathcal{D}^{KV}(X,D+\Delta)$, the triangulati...

conjecture | open | MathDB 页面 | 原文线索

V2-020742 | MATHDB-367471

Kronenthal-Lazebnik's uniqueness conjecture for generalized quadrangles over algebraically closed fields

摘录 / EXCERPT: Let $\mathbb F$ be an algebraically closed field of characteristic zero, and let $f_2,f_3\in\mathbb F[p_1,l_1]$. Kronenthal-Lazebnik's conjecture. Every graph...

conjecture | open | MathDB 页面 | 原文线索

V2-020743 | MATHDB-367485

RCD compactness and homeomorphism conjecture for singular Kähler spaces

摘录 / EXCERPT: Let X be an n-dimensional projective variety with log terminal singularities, and let θ_X be a smooth Kähler metric on X. Let $(X,\omega)\in\mathcal{RK}(n)$, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-020744 | MATHDB-367565

Faithfulness conjecture for higher-rank character varieties

摘录 / EXCERPT: Let $M_g^{n,d}$ be the moduli space of flat connections on a $U(n)$-bundle over $\a0\Sigma_g$ with fixed determinant connection and $c_1(P)[\Sigma_g]=d$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020745 | MATHDB-367566

Faithfulness conjecture for mapping class group actions on punctured character varieties

摘录 / EXCERPT: Let g be a non-negative integer, let k be an odd positive integer, let π be a set of k points in Σ_g, and define ... where μ runs over all small meridional loops...

conjecture | open | MathDB 页面 | 原文线索

V2-020746 | MATHDB-367582

Fock-Goncharov duality conjecture for the $\mathcal{A}_{SL_2,\Sigma_{g,p}}$ cluster variety

摘录 / EXCERPT: Let $\Sigma=\Sigma_{g,p}$, let $Sk^{ta}(\Sigma)$ be the tagged skein algebra, let $U(\Sigma)$ be the upper cluster algebra, and let $\nu:Sk^{ta}(\Sigma)\to U(\Sigma)$ be the natura...

conjecture | open | MathDB 页面 | 原文线索

V2-020747 | MATHDB-367583

Full Fock-Goncharov conjecture for a cluster variety

摘录 / EXCERPT: Let V be a cluster variety, let $V^{\vee}(\mathbb{Z}^t)$ be the tropical integral points on its dual cluster variety, let Θ be the proposed theta-function index set, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-020748 | MATHDB-367603

Sheaf quantization conjecture for the Givental-Hori-Vafa mirror

摘录 / EXCERPT: Let X_Σ be the toric variety with Givental-Hori-Vafa mirror W_Σ, and let $\widehat{\Lambda}_{\geq}$ denote the Novikov ring. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020749 | MATHDB-367604

HMS of log Calabi-Yau varieties over the Novikov ring

摘录 / EXCERPT: Let (X,D) be a log Calabi-Yau manifold, with D an anticanonical divisor of X, and let $W_X\colon \check X\to\mathbb C$ be a mirror of (X,D). Let $\widehat{\Lambda}_{\geq}$...

conjecture | open | MathDB 页面 | 原文线索

V2-020750 | MATHDB-367608

Finiteness conjecture for elementary wild character varieties of fixed even dimension

摘录 / EXCERPT: Let n be an even integer. An elementary wild character variety is a wild character variety arising from an elementary connection, considered up to isomorphism. Finiteness conject...

conjecture | open | MathDB 页面 | 原文线索

V2-020751 | MATHDB-367632

The octonion multiplication tensor's real subrank

摘录 / EXCERPT: Let $\mathbb O$ be the octonion algebra, viewed as an 8-dimensional real algebra, and let $T_{\mathbb O}\in\mathbb R^8\otimes\mathbb R^8\otimes\mathbb R^8$ be its multiplication...

conjecture | open | MathDB 页面 | 原文线索

V2-020752 | MATHDB-367644

The Galois-width conjecture for determinantal maximum-likelihood covers

摘录 / EXCERPT: Galois-width conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020753 | MATHDB-367684

Griffiths' conjecture on Nevanlinna's second theorem

摘录 / EXCERPT: Griffiths' conjecture. (1) For every holomorphic curve $f:\mathbb{C}\to X$ with Zariski-dense image,

conjecture | open | MathDB 页面 | 原文线索

V2-020754 | MATHDB-367699

DT4 correspondence conjectures for Calabi-Yau fourfold invariants

摘录 / EXCERPT: Let X be a projective Calabi-Yau 4-fold. Write $\operatorname{Hilb}^n(X)$ for the Hilbert scheme of n points on X, let $L\to X$ be a line bundle, and distinguish Gromov-Wit...

conjecture | open | MathDB 页面 | 原文线索

V2-020755 | MATHDB-367729

Pattern-avoidance characterization of restricted-stack-sortable permutations

摘录 / EXCERPT: Let $t\geq 1$, $n\geq 0$, and $p\in\{213,321\}$. A permutation is t-stack-sortable if it can be sorted using t applications of the stack-sorting operator, and it avoids a patte...

conjecture | open | MathDB 页面 | 原文线索

V2-020756 | MATHDB-367751

The Hikita conjecture for symplectically dual resolutions

摘录 / EXCERPT: Let X and $X^!$ be symplectically dual, and let Y be the affinization of X. Let Y^T denote the scheme-theoretic fixed locus of Y under the T-action, with grading induce...

conjecture | open | MathDB 页面 | 原文线索

V2-020757 | MATHDB-367760

Bayer-Macrì-Toda stability conjecture for threefolds

摘录 / EXCERPT: Bayer-Macrì-Toda stability conjecture. The pair $(\mathcal{A}_{H}^{\alpha,\beta},Z^{3}_{\alpha,\beta})$ is a stability condition on $D^{b}(X)$. This conjecture is equivalent to the...

conjecture | open | MathDB 页面 | 原文线索

V2-020758 | MATHDB-367805

Polishchuk-Van den Bergh conjecture on equivariant derived categories

摘录 / EXCERPT: Let G be a finite group acting effectively on a smooth quasiprojective variety X. For $g\in G$, let X^g denote the invariant subvariety and let $C(g)$ be the centralizer of...

conjecture | open | MathDB 页面 | 原文线索

V2-020759 | MATHDB-367806

The dimension conjecture for unreduced evaluation schemes with 2^m multiplications

摘录 / EXCERPT: Let $m>2$ and let $\Pi^*_{2^m}$ denote the polynomial set associated with evaluation schemes using 2^m matrix-matrix multiplications. Numerical computations suggest that generic...

conjecture | open | MathDB 页面 | 原文线索

V2-020760 | MATHDB-367824

Higher unramified cohomology extension conjecture

摘录 / EXCERPT: Let k be the base field, let $\mathcal{X}=(X,D)$ be an object of $\category{mSm}$, and let $q\geq 0$. Write $\mathcal{X}^{\circ}$ for the complement of the boundary,...

conjecture | open | MathDB 页面 | 原文线索

V2-020761 | MATHDB-367845

Smith-Thom inequality for real Deligne-Mumford stacks

摘录 / EXCERPT: Real stack Smith-Thom conjecture. The dimension of the mod-2 cohomology of the real locus should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-020762 | MATHDB-367860

The semi-infinite Finkelberg-Mirković conjecture

摘录 / EXCERPT: Let $\ell\geq h$. The category $\mathrm{Perv}_{I_\mathrm{u}}(\mathcal{F}l^{\frac{\infty}{2}})$ has simple objects $\mathrm{IC}_x^{\frac{\infty}{2}}$, standard objects...

conjecture | open | MathDB 页面 | 原文线索

V2-020763 | MATHDB-367877

The genericity conjecture for L-packets and open Vogan orbits

摘录 / EXCERPT: Assume the p-adic Kazhdan-Lusztig hypothesis. Let ϕ be a Langlands parameter, let λ_ϕ be its infinitesimal parameter, and let C_ϕ be the orbit associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-020764 | MATHDB-367878

Vogan's conjecture identifying Arthur and ABV-packets

摘录 / EXCERPT: Let G be a connected reductive p-adic group and let $\psi:W_F”\to{}^LG$ be an Arthur parameter. Write ϕ_ψ for the associated Langlands parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020765 | MATHDB-367903

Coba's conjecture on curves of best approximation

摘录 / EXCERPT: Let X be a smooth variety defined over a number field k, let $P\in X(k)$, and let A be an ample line bundle on X. For a curve $C\subseteq X$ through which approximation is...

conjecture | open | MathDB 页面 | 原文线索

V2-020766 | MATHDB-367944

Prokhorov-Shokurov conjecture on the b-semi-ampleness of moduli divisors

摘录 / EXCERPT: Let X be a compact complex manifold with an lc-trivial contraction $\sigma\colon (X,R)\to Y$ over a smooth projective variety Y, where R is a $\mathbb{Q}$-divisor on X. The...

conjecture | open | MathDB 页面 | 原文线索

V2-020767 | MATHDB-367955

The global geometric Langlands conjecture

摘录 / EXCERPT: Let X be a smooth projective curve over $F={\mathbb C}$. Let G be a reductive group, let $\Bun_G$ be the moduli stack of principal G-bundles on X, and let $\Loc_{\hat{G}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-020768 | MATHDB-367958

The Shimura-variety local Langlands action conjecture

摘录 / EXCERPT: Let (G,X) and (G',X') be Shimura data with G' a prime-to-p trivialized inner form of G, and let $K'_p\subset G'({\mathbb Q}_p)$ be open compact. Let $\Loc_{\widehat G,p}$...

conjecture | open | MathDB 页面 | 原文线索

V2-020769 | MATHDB-368065

Smooth Schubert enumeration conjecture for signed permutations

摘录 / EXCERPT: Smooth Schubert enumeration conjecture. The number of signed permutations in S_n^B that simultaneously index smooth Schubert varieties in types B and C is equal to the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-020770 | MATHDB-368074

The motivic control conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Motivic control conjecture. The variety X should be motivated by $\mathrm{KS}(X)$, and the motives of X and Y should be isomorphic. These assertions express the expectation t...

conjecture | open | MathDB 页面 | 原文线索

V2-020771 | MATHDB-368109

Conjecture on degenerations of rationally connected varieties

摘录 / EXCERPT: Let R be a DVR with fraction field K and residue field k. Let X be a flat, projective R-scheme such that the generic fiber X_K is smooth and separably rationally connec...

conjecture | open | MathDB 页面 | 原文线索

V2-020772 | MATHDB-368120

Abreu-Nigro's homeomorphism conjecture for regular semisimple Lusztig varieties

摘录 / EXCERPT: Let $G=\operatorname{GL}_n(\mathbb{C})$, let $\mathbf{s}\in G^{\mathrm{rs}}$ be regular semisimple, and let $w,w'\in S_n$ be smooth permutations associated, via the natural combina...

conjecture | open | MathDB 页面 | 原文线索

V2-020773 | MATHDB-368143

Tautological generation conjecture for Higgs-bundle stacks on smooth curves

摘录 / EXCERPT: Tautological generation conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-020774 | MATHDB-368156

BZSV global numerical conjecture for periods on spherical varieties

摘录 / EXCERPT: Let G be a reductive group, let H be a subgroup equipped with a homomorphism $\boldsymbol{\rho}_H$ as in the BZSV setting, and let ι determine the dual homomorphism ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-020775 | MATHDB-368175

Local-to-global multiplicativity conjecture for perverse filtrations

摘录 / EXCERPT: Local-to-global multiplicativity conjecture. The perverse filtration associated with f is multiplicative if and only if the local perverse filtrations on all fibers are multiplic...

conjecture | open | MathDB 页面 | 原文线索

V2-020776 | MATHDB-368233

Poonen-Rains conjecture for abelian varieties with real multiplication

摘录 / EXCERPT: Poonen-Rains conjecture for real multiplication. If p is odd, then, as (A,ι,λ) varies in $\mathrm{RM}_{\mathcal {O}}(F)$ ordered by conductors,

conjecture | open | MathDB 页面 | 原文线索

V2-020777 | MATHDB-368234

Quadrics base locus conjecture for exceptional dimensions

摘录 / EXCERPT: Let $n\in\{8,16\}$, and let $L\subset\mathbb{R}_{\mathrm{sym}}^{n\times n}$ be a subspace satisfying ... and such that $\boldsymbol{\mathrm{P}}(L^\vee)\cap\boldsymbol{\mathrm{Z}}$ exists and...

conjecture | open | MathDB 页面 | 原文线索

V2-020778 | MATHDB-368235

Quadrics base locus conjecture for normed-division-algebra dimensions

摘录 / EXCERPT: For $n\in\{2,4,8,16\}$, let $W_{n,\operatorname{sym}}$ be a linear subspace of $\mathbb{R}_{\operatorname{sym}}^{n\times n}$ with ... and suppose every nonzero element of...

conjecture | open | MathDB 页面 | 原文线索

V2-020779 | MATHDB-368249

LRS asymptotic formula for rational points on multiple conics

摘录 / EXCERPT: Let X be a smooth, projective, geometrically integral, weak Fano variety over $\mathbb{Q}$. Let $U \subseteq X$ be an open subset and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020780 | MATHDB-368259

Relative affinization conjecture for the filtered nodal curve

摘录 / EXCERPT: Let R be a commutative ring, and let ... be the map classifying the filtered nodal curve over R. Denote the corresponding filtered nodal curve by $\mathcal{N}_{\mathrm{fil}}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-020781 | MATHDB-368272

Beilinson-Bloch conjecture for smooth projective varieties over global fields

摘录 / EXCERPT: Let Y be a smooth projective variety over a global field, and let $i\in\mathbb{Z}$. Write $\operatorname{CH}^i(Y)_0$ for the kernel of the cycle class map on...

conjecture | open | MathDB 页面 | 原文线索

V2-020782 | MATHDB-368273

Jannsen's generalized Tate conjecture for arbitrary varieties

摘录 / EXCERPT: Jannsen's generalized Tate conjecture. This cycle class map is surjective. It extends the smooth-variety formulation of the Tate conjecture to arbitrary varieties using étale homol...

conjecture | open | MathDB 页面 | 原文线索

V2-020783 | MATHDB-368274

Jannsen's generalized Beilinson and semisimplicity conjecture

摘录 / EXCERPT: Generalized Beilinson and semisimplicity conjecture. The cycle class map induces an isomorphism, and the eigenvalue 1 of Frob_q acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-020784 | MATHDB-368275

Beilinson-Bloch conjecture over global fields

摘录 / EXCERPT: Let K be a global field, let Y be a smooth projective variety over K, and let $i\in\mathbb{Z}$. Write $\operatorname{CH}^i(Y)_0$ for the kernel of the cycle class map and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020785 | MATHDB-368288

The rationality conjecture for Mumford-Tate-valued Weil-Deligne representations

摘录 / EXCERPT: Let X be a proper, smooth algebraic variety over a number field ${\mathrm{E}}\subset\mathbb C$, and let ${\mathbf{G}}$ be the Mumford-Tate group of the Hodge structure...

conjecture | open | MathDB 页面 | 原文线索

V2-020786 | MATHDB-368295

The conjecture that landscape signature matrices fill the loop signature variety

摘录 / EXCERPT: Let $\mathcal{M}_{d,2}^{\mathrm{LS}}$ be the Zariski closure of the set of second signature matrices of landscape paths in $\mathbb R^d$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020787 | MATHDB-368319

Hamann-Lee generic vanishing conjecture for Shimura varieties of abelian type

摘录 / EXCERPT: Hamann-Lee's generic vanishing conjecture. Assume that ξ is generic. Then

conjecture | open | MathDB 页面 | 原文线索

V2-020788 | MATHDB-368353

Colmez's original height formula conjecture

摘录 / EXCERPT: Let ${\mathcal{CM}}^0$ be the space of class functions associated with CM-types, and let ht and $Z(\mathord\cdot,s)$ be the linear height and logarithmic-derivative functionals d...

conjecture | open | MathDB 页面 | 原文线索

V2-020789 | MATHDB-368367

Ciliberto-Di Gennaro conjecture for nodal hypersurfaces

摘录 / EXCERPT: Let $X\subset\operatorname{\mathbb{P}}^4$ be a nodal hypersurface, meaning a hypersurface of degree d whose singularities are all ordinary double points. Assume that X has at m...

conjecture | open | MathDB 页面 | 原文线索

V2-020790 | MATHDB-368369

Motivic stability of commuting tuples in general linear groups

摘录 / EXCERPT: For a positive integer n, let $C_n(\operatorname{GL}_r(k))$ denote the variety of commuting n-tuples in $\operatorname{GL}_r(k)$. Motivic stability conjecture. The limit ... exis...

conjecture | open | MathDB 页面 | 原文线索

V2-020791 | MATHDB-368370

Motivic limit conjecture for representation varieties of surface groups

摘录 / EXCERPT: Let G be a connected linear algebraic group over a field k, and let M_g be a smooth compact surface of genus g. Write $\operatorname{Rep}_G(M_g)$ for the G-representation...

conjecture | open | MathDB 页面 | 原文线索

V2-020792 | MATHDB-368389

Dominance conjecture for higher Albanese maps of special varieties

摘录 / EXCERPT: Dominance conjecture for higher Albanese maps. The map

conjecture | open | MathDB 页面 | 原文线索

V2-020793 | MATHDB-368405

Realizability conjecture for universally pseudoeffective surface classes

摘录 / EXCERPT: Let X be a smooth complex projective variety, and set ... Let $\mathrm{B}_2(X)$ denote the cone of universally pseudoeffective 2-dimensional classes, and let a class be realizabl...

conjecture | open | MathDB 页面 | 原文线索

V2-020794 | MATHDB-368431

Iitaka fibration conjecture for integral points on polarized log Fano varieties

摘录 / EXCERPT: Let $\mathcal{X}=(\mathfrak{X},D,L)$ be a polarized log Fano variety, with integral-point scheme $\mathfrak{U}=\mathfrak{X}\setminus\overline{D}$, height function H_L, and counti...

conjecture | open | MathDB 页面 | 原文线索

V2-020795 | MATHDB-368462

Mazur's algebraic equidistribution conjecture for Hecke orbits

摘录 / EXCERPT: Let $S_K(G,X)$ be the Shimura variety under consideration, and let $Y\subset S_K(G,X)$ be a positive-dimensional subvariety intersecting the ordinary locus. The Hecke orbit of Y...

conjecture | open | MathDB 页面 | 原文线索

V2-020796 | MATHDB-368466

Normality conjecture for toric orbit closures in generalized flag varieties

摘录 / EXCERPT: Let G be a semisimple algebraic group, let P_{λ} be the parabolic subgroup associated with a dominant weight λ, and let G/P_{λ} be embedded by the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-020797 | MATHDB-368478

Positivity conjecture for fixed-domain Gromov-Witten invariants of Fano varieties

摘录 / EXCERPT: Let X be a smooth Fano variety of complex dimension r, let g be a genus, and let A be a curve class. The fixed-domain Gromov-Witten invariants are the virtual counts associ...

conjecture | open | MathDB 页面 | 原文线索

V2-020798 | MATHDB-368479

The conjectural structure of the hypergeometric discriminant for type B_n

摘录 / EXCERPT: The conjectural structure of the hypergeometric discriminant for type B_n. The hypergeometric discriminant of $\texttt{B}_n$ is given by the formal sum

conjecture | open | MathDB 页面 | 原文线索

V2-020799 | MATHDB-368522

Strong form of Mazur's conjecture for abelian varieties

摘录 / EXCERPT: Let A be a simple abelian variety over $\mathbb{Q}$ of dimension at least 1, and let $P\in A(\mathbb{Q})$ be non-torsion. Strong Mazur conjecture. The subgroup generated by P...

conjecture | open | MathDB 页面 | 原文线索

V2-020800 | MATHDB-368523

Nori's structural-rank conjecture for closures of algebraic-group subgroups

摘录 / EXCERPT: Let $\mathcal{A}$ be a commutative algebraic group over $\mathbb{Q}$ that is either a torus or an abelian variety, with Lie algebra $\mathfrak{a}$ over $\mathbb{Q}$ and exponential...

conjecture | open | MathDB 页面 | 原文线索

V2-020801 | MATHDB-368526

Buch-Mihalcea's quantum K-theoretic divisor axiom for flag manifolds

摘录 / EXCERPT: Let G be a connected, simply-connected, simple linear algebraic group over $\a0\mathbb{C}$ of Lie type A, let P be a parabolic subgroup containing a maximal torus T and a B...

conjecture | open | MathDB 页面 | 原文线索

V2-020802 | MATHDB-368548

Manin-Peyre conjecture for smooth weak Fano varieties

摘录 / EXCERPT: Let V be a smooth weak Fano variety over a number field F such that $V(F)$ is Zariski dense, and let H_F be a relative adelic height on the anticanonical line bundle. Manin-P...

conjecture | open | MathDB 页面 | 原文线索

V2-020803 | MATHDB-368549

Peyre's leading-constant formula for weak Fano varieties

摘录 / EXCERPT: Let V be a smooth weak Fano variety over a number field F, let Γ_F be the absolute Galois group, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-020804 | MATHDB-368555

Shenfeld-van Handel's hard Lefschetz kernel conjecture for supercritical collections

摘录 / EXCERPT: Shenfeld-van Handel's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020805 | MATHDB-368588

Virtual dimension formula for the moduli of stable supermaps

摘录 / EXCERPT: Let ${\mathcal Y}$ be a projective smooth superscheme of dimension $r|s$, with bosonic reduction Y, and let $\beta_0\in A_1(Y)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020806 | MATHDB-368621

Birational equivalence conjecture for products of Severi-Brauer varieties

摘录 / EXCERPT: Product Severi-Brauer conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-020807 | MATHDB-368628

The BAB conjecture for epsilon-lc Fano varieties

摘录 / EXCERPT: BAB conjecture. The set of ϵ-lc Fano varieties of dimension d forms a bounded family.

conjecture | open | MathDB 页面 | 原文线索

V2-020808 | MATHDB-368651

Achinger-Zdanowicz conjecture on Witt-vector liftings of Frobenius-split varieties

摘录 / EXCERPT: Achinger-Zdanowicz conjecture. There is a flat deformation of X over the Witt vectors $W(k)$, not only over $W_2(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-020809 | MATHDB-368652

Achinger-Witaszek-Zdanowicz conjecture on Frobenius-liftable varieties

摘录 / EXCERPT: Achinger-Witaszek-Zdanowicz conjecture. There exists a finite étale Galois cover $f:Y\to X$ such that the Albanese morphism

conjecture | open | MathDB 页面 | 原文线索

V2-020810 | MATHDB-368658

The motivic action conjecture for Hilbert modular forms

摘录 / EXCERPT: The motivic action conjecture. The classes

conjecture | open | MathDB 页面 | 原文线索

V2-020811 | MATHDB-368675

Strong Zilber-Pink conjecture II for optimal subvarieties

摘录 / EXCERPT: Let S be a Shimura variety and let V be an irreducible subvariety of S. An irreducible subvariety W of V is called optimal if, for every irreducible subvariety Y of V...

conjecture | open | MathDB 页面 | 原文线索

V2-020812 | MATHDB-368676

André-Pink-Zannier conjecture for Hecke orbits

摘录 / EXCERPT: Let S be a Shimura variety and let V be an irreducible subvariety of S. André-Pink-Zannier conjecture. Suppose that the intersection of V with a single Hecke orbit in S i...

conjecture | open | MathDB 页面 | 原文线索

V2-020813 | MATHDB-368677

Arithmetic Zilber-Pink conjecture for optimal points

摘录 / EXCERPT: Let S be a Shimura variety and let V be an irreducible subvariety of S. An optimal point is an optimal subvariety of V of dimension zero, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-020814 | MATHDB-368715

The Steenrod-span conjecture for motivic cohomology operations

摘录 / EXCERPT: Let F be a field. Write $\mathcal{A}^{\star,\star}_{F}$ for the algebra of motivic Steenrod operations and $H\mathbb{F}_{p,F}^{\star,\star}H\mathbb{F}_{p,F}$ for the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-020815 | MATHDB-368716

The Hopkins-Morel isomorphism aka the motivic Quillen theorem

摘录 / EXCERPT: Let F be a field, and let SH_F denote the motivic stable homotopy category over F. Let MGL be the algebraic cobordism spectrum, let a_i be generators of...

conjecture | solved | MathDB 页面 | 原文线索

V2-020816 | MATHDB-368717

Motivic absolute purity conjecture

摘录 / EXCERPT: Let $H\mathbb{Z}^{\mathbb{A}^1}$ denote the motivic spectrum representing motivic cohomology in the $\mathbb{A}^1$-homotopy setting. A motivic spectrum is absolutely pure when it s...

conjecture | open | MathDB 页面 | 原文线索

V2-020817 | MATHDB-368735

Oort's Hecke orbit conjecture for principally polarized abelian varieties

摘录 / EXCERPT: Let $p>0$, let g be a positive integer, and let $x=\left[\left(A_x,\lambda_x\right)\right]$ be a $\overline{\mathbb{F}}_p$-point of the moduli space $\mathcal{A}_g$ of g-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-020818 | MATHDB-368736

The p-adic nowhere density conjecture for Hecke orbits of positive-dimensional subvarieties

摘录 / EXCERPT: Let $\mathcal{S}h$ be a Shimura variety and let $Z\subseteq\mathcal{S}h$ be a positive-dimensional subvariety. Denote by the Hecke orbit of Z the union of its images under the He...

conjecture | open | MathDB 页面 | 原文线索

V2-020819 | MATHDB-368768

Degree bounds for points on universal complete intersections

摘录 / EXCERPT: Let $\underline{d}=(d_1,\ldots,d_e)$ be a tuple of positive integers, and let $X_{n,\underline{d}}$ denote the universal complete intersection of type $\underline{d}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-020820 | MATHDB-368792

Belmans-Smirnov's non-Hochschild-globality conjecture for isotropic Grassmannians

摘录 / EXCERPT: Belmans-Smirnov's conjecture. If X is neither (co)minuscule nor (co)adjoint, then X is not Hochschild global; equivalently, for some $i>0$ and some j,

conjecture | solved | MathDB 页面 | 原文线索

V2-020821 | MATHDB-368796

Virasoro conjecture for Hodge integrals

摘录 / EXCERPT: Virasoro conjecture for Hodge integrals. For any smooth projective variety,

conjecture | open | MathDB 页面 | 原文线索

V2-020822 | MATHDB-368809

The realizability criterion for complete intersection rings

摘录 / EXCERPT: Assume that a local ring R has minimal regular presentation Q/I, let k be its residue field, and fix minimal generators $f_1,\ldots,f_n$ of I. Let ... and let ... where each...

conjecture | open | MathDB 页面 | 原文线索

V2-020823 | MATHDB-368862

Ito's proper Fourier-Mukai locus conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over an algebraically closed field k of characteristic zero. Write $\mathrm{Spec}_{\triangle}^{\mathrm{pFM}}\mathrm{Perf}(X)$ for the prope...

conjecture | open | MathDB 页面 | 原文线索

V2-020824 | MATHDB-368867

Schwede-Smith conjecture on globally F-regular type varieties

摘录 / EXCERPT: Schwede-Smith conjecture. X is of Fano type.

conjecture | open | MathDB 页面 | 原文线索

V2-020825 | MATHDB-368887

Sato's conjecture for smooth Fano varieties of Picard number one

摘录 / EXCERPT: Sato's conjecture. X is a projective space.

conjecture | open | MathDB 页面 | 原文线索

V2-020826 | MATHDB-368897

General existence conjecture for higher Theta-stratifications of Fano-fibration moduli

摘录 / EXCERPT: Let $\mathcal{M}^o$ be a finite-type algebraic $\mathbbm{k}$-stack underlying a $\mathbb{Q}$-Gorenstein faithfully flat affine family of $\mathbb{Q}$-Fano fibrations, equipped with...

conjecture | open | MathDB 页面 | 原文线索

V2-020827 | MATHDB-368901

The modular-flat vanishing conjecture for hyperplane-arrangement forms

摘录 / EXCERPT: Let $\mathcal{K}$ be a hyperplane arrangement, let F be a flat, let J be a set of hyperplanes, let ι be as above, and let $\omega_{\mathcal{K},\iota,F,J}$ denote the asso...

conjecture | open | MathDB 页面 | 原文线索

V2-020828 | MATHDB-368905

Equality conjecture for the dimension of cone-volume fibers

摘录 / EXCERPT: Let $U \in \mathcal{U}(n,m)$, and let ... be its unique partition into irreducible sets. Let $\gamma \in C_{\operatorname{cv}}(U) \cap \mathbb{R}_{>0}^m$, and define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-020829 | MATHDB-368917

Kawaguchi-Silverman conjecture for surjective endomorphisms

摘录 / EXCERPT: Kawaguchi-Silverman conjecture. The arithmetic degree at x equals the dynamical degree of f:

conjecture | open | MathDB 页面 | 原文线索

V2-020830 | MATHDB-368920

Occhetta-Wiśniewski's toric image conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, let Z be a complete toric variety, and let ... be a surjective map. Occhetta-Wiśniewski's conjecture. Then X is a toric variety. The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-020831 | MATHDB-368930

Indecomposability conjecture for varieties with nef effective canonical bundle

摘录 / EXCERPT: Indecomposability conjecture. If the canonical bundle K_X is nef and effective, then X admits no non-trivial semiorthogonal decompositions.

conjecture | open | MathDB 页面 | 原文线索

V2-020832 | MATHDB-368954

The cardinality conjecture for the Yang-Baxter variety associated with A_3

摘录 / EXCERPT: Let $[? Let $[?

conjecture | open | MathDB 页面 | 原文线索

V2-020833 | MATHDB-368959

The indecomposable-root conjecture for simply-laced Weyl groups

摘录 / EXCERPT: Indecomposable-root conjecture. The root $\check\eta_w(k)$ is indecomposable in $\check R^+(w)$. In particular, the set

conjecture | open | MathDB 页面 | 原文线索

V2-020834 | MATHDB-368965

Rémond's Lehmer conjecture for tori and abelian varieties

摘录 / EXCERPT: Let K be a number field, and let G be either $\mathbb{G}_m^t$ for some $t\in\mathbb{Z}_{\geq 0}$ or an abelian variety over K. Let $\Gamma\subset G(\overline{K})$ be a subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-020835 | MATHDB-368966

Rémond's Dobrowolski conjecture for finite-rank subgroups

摘录 / EXCERPT: Let K be a number field, let G be a semiabelian variety over K, and let $\Gamma\subset G(\overline{K})$ be a subgroup of finite rank. For a subvariety $V\subset G$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-020836 | MATHDB-368988

Waldschmidt's density conjecture for rational points on abelian varieties

摘录 / EXCERPT: Let $\mathcal{A}$ be a simple abelian variety of dimension g over a number field K embedded in $\mathbb{R}$. Let $\mathcal{A}(\mathbb{R})^0$ be the identity component, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020837 | MATHDB-368989

Diophantine exponent conjecture for abelian varieties

摘录 / EXCERPT: The Diophantine exponent conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020838 | MATHDB-369006

Touzet's fibration conjecture for regular foliations on rationally connected manifolds

摘录 / EXCERPT: Touzet's fibration conjecture. The foliation $\mathcal{F}$ is induced by a fibration $X\rightarrow Y$ onto a projective manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-020839 | MATHDB-369011

Stable-pair integrality and symmetry conjecture for logarithmic Calabi-Yau 3-folds

摘录 / EXCERPT: Let $\bold X$ be a logarithmic Calabi-Yau 3-fold, let $D^{[\beta]}$ be the product of the relevant Hilbert schemes of points on the interiors of the boundary divisors, and let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-020840 | MATHDB-369031

Woo-Billey-Weed conjecture on low-degree Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let $w\in S_n$, let $P_{\mathrm{id},w}(q)$ be the Kazhdan-Lusztig polynomial, and let $\operatorname{ms}(w)$ denote the set of permutations indexing the irreducible components of t...

conjecture | open | MathDB 页面 | 原文线索

V2-020841 | MATHDB-369034

Conjecture on representability of integral local Shimura varieties

摘录 / EXCERPT: Let (G,μ,b) be an integral local Shimura datum. Denote by $\mathcal{M}^{\textup{int}}_{(G,\mu,b)}$ the integral model for the local Shimura variety at hyperspecial level. Repre...

conjecture | open | MathDB 页面 | 原文线索

V2-020842 | MATHDB-369035

Conjecture on perfectoidness of Shimura varieties at infinite p-level

摘录 / EXCERPT: Global perfectoidness conjecture. This inverse limit is representable by a perfectoid space. Perfectoidness at infinite p-level is a central representability question for Shimura...

conjecture | open | MathDB 页面 | 原文线索

V2-020843 | MATHDB-369054

The standard-slice conjecture for Bridgeland stability conditions

摘录 / EXCERPT: Let (X,H) be an n-dimensional irreducible smooth polarized variety over $\mathbb{C}$. Let Λ_H be the H-polarized lattice, and let $\mathfrak U_n$ be the specified ope...

conjecture | open | MathDB 页面 | 原文线索

V2-020844 | MATHDB-369055

The reduced-stability conjecture

摘录 / EXCERPT: Let (X,H) be a smooth polarized variety over $\mathbb{C}$, and let Λ_H be the H-polarized lattice. Let $\mathfrak B_n$ and $\pm\mathfrak B_n$ be the spaces of reduced c...

conjecture | open | MathDB 页面 | 原文线索

V2-020845 | MATHDB-369056

The Stab^d conjecture for truncated stability-condition spaces

摘录 / EXCERPT: Let (X,H) be a smooth polarized variety over $\mathbb{C}$ and let $d\geq 0$. Let $\mathfrak U_n^{>d}$ be the subspace of central charges defined by the interlacing conditions and...

conjecture | open | MathDB 页面 | 原文线索

V2-020846 | MATHDB-369057

The reduced Sb^d conjecture

摘录 / EXCERPT: Let (X,H) be a smooth polarized variety over $\mathbb{C}$ and let $d\geq 0$. Let $\mathfrak B_n^{>d}$ and $\pm\mathfrak B_n^{>d}$ denote the reduced-central-charge spaces satisfyi...

conjecture | open | MathDB 页面 | 原文线索

V2-020847 | MATHDB-369065

Very efficient format conjecture for moment matrix extension

摘录 / EXCERPT: Very efficient format conjecture. 1. The format $(n,R_c(n))$ is very efficient if c is such that $|E_1|\geq |Y|$. 2. The format (n,r) is efficient if there exists c with...

conjecture | open | MathDB 页面 | 原文线索

V2-020848 | MATHDB-369071

Conjectural unstable Poincaré-Hopf formula at non-rational points

摘录 / EXCERPT: Let $f/g:\mathbb{P}^1\to\mathbb{P}^1$ be a pointed rational function with vanishing locus $D=\{r_1,\ldots,r_n\}\subset\mathbb{A}^1_k$. Let $m_i(x)\in k[x]$ denote the monic minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-020849 | MATHDB-369072

Shellability and Hilbert-series conjecture for principal components of determinantal jet schemes

摘录 / EXCERPT: Shellability and Hilbert-series conjecture. The abstract simplicial complex corresponding to $\tilde{I^{m,n}_{r,k}}$ is shellable, and the Hilbert series of the principal component...

conjecture | open | MathDB 页面 | 原文线索

V2-020850 | MATHDB-369095

The mirror-triviality characterization of the physical geometric window

摘录 / EXCERPT: Mirror-triviality conjecture. A level ℓ is in the physical geometric window if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-020851 | MATHDB-369097

The flag-manifold Coulomb-branch relations conjecture

摘录 / EXCERPT: Flag-manifold relations conjecture. For ℓ in the geometric window, the $\prod_i S_{k_i}$-invariant combinations of the Coulomb-branch equations generate the ideal of relations...

conjecture | open | MathDB 页面 | 原文线索

V2-020852 | MATHDB-369100

Potential density of integral points in relative character varieties

摘录 / EXCERPT: Potential-density conjecture. Integral points are potentially Zariski-dense in the K-scheme $X_{G,\underline{C}}(Y)_K$: there exists a finite extension L/K such that the Zarisk...

conjecture | open | MathDB 页面 | 原文线索

V2-020853 | MATHDB-369112

Identification conjecture for the pre-quotient Calabi-Yau complete intersection

摘录 / EXCERPT: Let X be the smooth Calabi-Yau complete intersection whose existence is conjectured to match the A-periods and GKZ system of Y_{nc}, and let X_{pre} be th...

conjecture | open | MathDB 页面 | 原文线索

V2-020854 | MATHDB-369117

Generalized abundance for symplectic varieties

摘录 / EXCERPT: Generalized abundance for symplectic varieties. Then L is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-020855 | MATHDB-369120

Finiteness of the topological fundamental group of an irreducible symplectic regular locus

摘录 / EXCERPT: Finiteness conjecture. The group $\pi_1(X^{\mathrm{reg}})$ is finite:

conjecture | open | MathDB 页面 | 原文线索

V2-020856 | MATHDB-369158

The Newton non-degenerate homaloidal hypersurface conjecture

摘录 / EXCERPT: Let f be a Newton non-degenerate homogeneous polynomial of degree $d\geq 3$ defining a homaloidal hypersurface. Let $D_0,\dots,D_n$ be the polytopes associated with f in Propos...

conjecture | open | MathDB 页面 | 原文线索

V2-020857 | MATHDB-369177

The secant non-containment conjecture for Brill-Noether loci of rank two

摘录 / EXCERPT: Secant non-containment conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-020858 | MATHDB-369178

The general secant non-containment conjecture for Brill-Noether loci

摘录 / EXCERPT: General secant non-containment conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-020859 | MATHDB-369183

Prasanna--Venkatesh rationality conjecture for automorphic cohomology periods

摘录 / EXCERPT: Prasanna--Venkatesh conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020860 | MATHDB-369185

Beilinson's determinant formula for the adjoint motive of an elliptic curve

摘录 / EXCERPT: Beilinson's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020861 | MATHDB-369191

Beelen-Datta-Ghorpade conjecture for generalized Hamming weights

摘录 / EXCERPT: Let $e_r(d,m;q)$ denote the maximum number of $\mathbb{F}_q$-rational points in the common zero set of an r-dimensional subspace of degree-d homogeneous polynomials in...

conjecture | open | MathDB 页面 | 原文线索

V2-020862 | MATHDB-369193

Jet closure formula for weighted homogeneous polynomials

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let $R=k[[x_{1},...,x_{n}]]$ with maximal ideal $\mathbf{m}$, and let $f\in R$ be a weighted homogeneous polynomial....

conjecture | open | MathDB 页面 | 原文线索

V2-020863 | MATHDB-369194

Equality of the jet Tjurina index and nilpotency index

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let $f\in k[[x_{1},...,x_{n}]]$ define an isolated singularity at 0, and let $J(f)$ be the Jacobian ideal of f....

conjecture | open | MathDB 页面 | 原文线索

V2-020864 | MATHDB-369208

Voronoi decomposition conjecture for tropical Prym second moments

摘录 / EXCERPT: Let $R_{\widetilde{G}/G}$ be the cone associated with a double cover $\widetilde{G}\to G$. As a double cover of tropical curves $\widetilde{\Gamma}\to\Gamma$ varies over...

conjecture | open | MathDB 页面 | 原文线索

V2-020865 | MATHDB-369239

The equivalent modularity criteria for simple abelian varieties

摘录 / EXCERPT: Equivalent modularity criteria. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-020866 | MATHDB-369245

Zhang-Kitaev equinumerosity conjecture for 213- and 321-avoiding permutations

摘录 / EXCERPT: Let $\mathfrak{S}_n^t$ be the set of permutations of length n that become the identity after t applications of the stack-sorting operation, and let $\mathfrak{S}_n^(p)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-020867 | MATHDB-369250

Asymptotic comparison conjecture for Gaussian periods and cyclovarieties

摘录 / EXCERPT: For odd integers n, let α_n be the Gaussian-period algebraic integer and let $C_{\kappa(n)}$ be the associated cyclovariety, with Mahler measures denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-020868 | MATHDB-369272

Beilinson-Bloch conjecture on the Albanese kernel

摘录 / EXCERPT: Let K be a number field and let Z/K be a smooth projective irreducible variety of dimension d. Define the Albanese kernel by ... Here $\mathrm{CH}^d(Z)_0$ is the group of homol...

conjecture | open | MathDB 页面 | 原文线索

V2-020869 | MATHDB-369326

Quantum K-theoretic Seidel operator and Pieri rule for complete flag varieties

摘录 / EXCERPT: Let $F\ell_n$ be the complete flag variety, let S_n be its Weyl group with simple reflections $s_1,\ldots,s_{n-1}$, and let $QK(F\ell_n)$ denote its quantum K-theory with quant...

conjecture | open | MathDB 页面 | 原文线索

V2-020870 | MATHDB-369331

The Hausel partition-function conjecture for mixed Hodge polynomials of character varieties

摘录 / EXCERPT: Fix the number k of marked points and the genus g. For partitions $\mu_1,\ldots,\mu_k$ and the corresponding generic character variety $X_{g;J_{\mu_1},\ldots,J_{\mu_k}}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-020871 | MATHDB-369332

The local-cohomology interpretation of the character-variety partition function

摘录 / EXCERPT: Let $L\subset\mathrm{Hilb}_n(\mathbb C^2)$ be the Lagrangian cut out by $\sum_{i=1}^n x_i^r=0$ for $r>0$, let P be the Procesi bundle, let Ω^* be the bundle of differentia...

conjecture | open | MathDB 页面 | 原文线索

V2-020872 | MATHDB-369333

The primitive local-cohomology conjecture for character-variety cohomology

摘录 / EXCERPT: Let $M_{n,k}=H_L^n(O(k))^\vee$, and let $M_{n,k}^{\mathrm{shuffles}}\subset M_{n,k}$ be spanned by shuffle products $f*g$ with $f\in M_{n',k}$, $g\in M_{n-n',k}$, and $0<n'<n$. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-020873 | MATHDB-369334

The generators conjecture for primitive local cohomology

摘录 / EXCERPT: Generators conjecture. The space $M_{n,k}^{\mathrm{prim}}$ is spanned by products of $1/\Delta_x^{k-1}$ and polynomials of the forms $\psi_s(\mathrm{pt})$ and $\psi_s(1)$. The clai...

conjecture | open | MathDB 页面 | 原文线索

V2-020874 | MATHDB-369335

The ring-structure conjecture for primitive local cohomology

摘录 / EXCERPT: Let $M_{n,k}^{\mathrm{prim}}$ be the primitive quotient and let the preceding primitive vector-space conjecture identify it with $H^*(X)[\sum_{i=1}^nx_i,\sum_{i=1}^ny_i]$. Ring-str...

conjecture | open | MathDB 页面 | 原文线索

V2-020875 | MATHDB-369356

Kudla's boundary-correction conjecture for compactified Shimura varieties

摘录 / EXCERPT: Let X_Γ^Σ be a toroidal compactification of an orthogonal type Shimura variety, or of a unitary type Shimura variety of signature $(n+1,1)$. The special cycles of codim...

conjecture | open | MathDB 页面 | 原文线索

V2-020876 | MATHDB-369357

All-codimension cohomological Kudla conjecture for unitary Shimura varieties

摘录 / EXCERPT: Let $\mathrm{U}(n+1,1)$ be the unitary group considered in the paper, and let g denote the codimension of the special cycles. The paper's main completion and explicit-form statem...

conjecture | open | MathDB 页面 | 原文线索

V2-020877 | MATHDB-369369

Nagai's conjecture on monodromy nilpotency for hyper-Kähler varieties

摘录 / EXCERPT: Nagai's conjecture. The monodromy operators satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-020878 | MATHDB-369370

The p-adic analogue of Nagai's conjecture

摘录 / EXCERPT: Fix a prime p, let K be a complete discretely valued field of mixed characteristic $(0,p)$ with perfect residue field, and let X be a hyper-Kähler variety over K of dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-020879 | MATHDB-369394

Degenerational realizability of rational Hodge classes

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to\Delta$ be a semistable degeneration, with smooth projective fibers X_t for $t\neq 0$, singular central fiber X_0, smooth total space, and monodromy loga...

conjecture | open | MathDB 页面 | 原文线索

V2-020880 | MATHDB-369428

The improved Milnor K-theory conjecture for local rings

摘录 / EXCERPT: Let A be a local ring, let $i\geq 0$ be an integer, let $\widehat{\text{K}}^{\text{M}}_i(A)$ denote the improved Milnor K-group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020881 | MATHDB-369431

F-alpha positivity conjecture for Q-Fano varieties

摘录 / EXCERPT: Let X be a d-dimensional $\mathbb{Q}$-Fano variety over $\mathbb{C}$. Q-Fano F-alpha conjecture. ... This is a weaker global version of the positivity conjecture for log Fano pai...

conjecture | open | MathDB 页面 | 原文线索

V2-020882 | MATHDB-369434

The Large Galois Orbits conjecture for endomorphism-generic PEL subvarieties

摘录 / EXCERPT: Large Galois Orbits conjecture. There exist positive constants C_{mult} and C_{exp} such that, for all $s\in C\cap\Sigma$,

conjecture | open | MathDB 页面 | 原文线索

V2-020883 | MATHDB-369452

The Breuil-Kisin torsion monotonicity conjecture

摘录 / EXCERPT: Let $X/\mathcal{O}_K$ be the fixed scheme, let $\mathfrak{S}$ be the Breuil-Kisin ring, and let E be its Eisenstein polynomial. For the Breuil-Kisin cohomology module ... write...

conjecture | open | MathDB 页面 | 原文线索

V2-020884 | MATHDB-369453

Nagata's conjecture for plane curves through very general points

摘录 / EXCERPT: Let $p_1,\dots,p_r$ be r very general points in $\mathbb P^2$. For an integral curve $C\subset\mathbb P^2$, write $\operatorname{mult}_{p_i}(C)$ for its multiplicity at p_i. Eq...

conjecture | open | MathDB 页面 | 原文线索

V2-020885 | MATHDB-369463

Representable-functor conjecture for cohomology of integral nilpotent group schemes

摘录 / EXCERPT: Let G be a nilpotent affine group scheme over the integers, and let A be a representable functor of commutative graded $\mathbb{Z}$-algebras. Integral group-scheme representabi...

conjecture | open | MathDB 页面 | 原文线索

V2-020886 | MATHDB-369473

Lagrangianity conjecture for singular support on rigid analytic varieties

摘录 / EXCERPT: Lagrangianity conjecture. $SS(\mathcal{F})$ is an analytic Lagrangian subspace of T^*X, meaning that it is an analytic closed subset and is Lagrangian on its regular locus. In pa...

conjecture | open | MathDB 页面 | 原文线索

V2-020887 | MATHDB-369535

The domain conjecture for irreducible Artin groups modulo their centers

摘录 / EXCERPT: Domain conjecture. Every quotient $A/Z(A)$ is a domain.

conjecture | open | MathDB 页面 | 原文线索

V2-020888 | MATHDB-369538

Component correspondence for Grassmann cactus varieties and Hilbert schemes

摘录 / EXCERPT: Component correspondence conjecture. The irreducible components of $\mathfrak{K}_{r,k}(X)$ are in bijection with the irreducible components of $\operatorname{Hilb}_r(X)$ whose gene...

conjecture | open | MathDB 页面 | 原文线索

V2-020889 | MATHDB-369549

Conjecture F on algebraic integrability of foliations

摘录 / EXCERPT: Let X be a normal complex quasi-projective variety, and let $\mathcal{F}\subset T_X$ be a coherent subsheaf closed under the Lie bracket. The distribution $\mathcal{F}$ is algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-020890 | MATHDB-369677

Density conjecture for pleated surfaces in the complex character variety

摘录 / EXCERPT: Let S be a closed surface with fundamental group Γ, let $d\geq 2$, and let λ be a maximal geodesic lamination on S. Denote by $\mathfrak R(\lambda,d)$ the space...

conjecture | open | MathDB 页面 | 原文线索

V2-020891 | MATHDB-369689

Green-Griffiths-Lang conjecture for Mordellic varieties

摘录 / EXCERPT: Let k be a number field and let k^{alg} be an algebraic closure of k. Let X be a smooth projective geometrically irreducible variety over k. The properties of...

conjecture | open | MathDB 页面 | 原文线索

V2-020892 | MATHDB-369690

The largest xeric open set conjecture

摘录 / EXCERPT: Let k be a number field and let X be a smooth projective geometrically irreducible variety over k. Let $Z_{rc}(X)\subseteq X$ be the Zariski closure of the union of all possi...

conjecture | open | MathDB 页面 | 原文线索

V2-020893 | MATHDB-369691

Kodaira-dimension conjecture for dense p-adic analytic maps

摘录 / EXCERPT: Let X be a smooth projective variety over $\mathbb{C}_p$, and let X^{an} denote its analytification. Kodaira-dimension conjecture. If there is a p-adic analytic map...

conjecture | open | MathDB 页面 | 原文线索

V2-020894 | MATHDB-369734

Orbifold correction conjecture for quantum Singleton bounds

摘录 / EXCERPT: Let X) be a weighted projective variety with orbifold singularities, and let C be a quantum stabilizer code constructed from self-orthogonal algebraic geometry codes on X. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-020895 | MATHDB-369735

Refined Singleton-bound conjecture for quotient-projective-line quantum codes

摘录 / EXCERPT: Refined Singleton-bound conjecture. The minimum distance satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-020896 | MATHDB-369743

Beilinson extension conjecture for geometric height pairings

摘录 / EXCERPT: Beilinson extension conjecture. For each $a_{1}\in CH^{p}_{\operatorname{hom}}(X)_{\mathbb{Q}}$, there exists a Beilinson extension...

conjecture | open | MathDB 页面 | 原文线索

V2-020897 | MATHDB-369744

Beilinson extension conjecture for vanishing étale cycle classes

摘录 / EXCERPT: Beilinson extension conjecture. For every $\alpha^{p}\in CH^{p}_{\operatorname{hom}}(X)_{\mathbb{Q}}$, there exists an extension z^{p}_{B} on $\mathcal{X}$ such that the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-020898 | MATHDB-369763

Faithfulness of tensoring by a quasi-coherent sheaf on Deligne-Mumford stacks

摘录 / EXCERPT: Let X be a quasi-separated Deligne-Mumford stack with generically trivial stabilizer. Let $\mathcal{M}\in \operatorname{QCoh}(X)$ and let $\alpha\in \mathrm{Br}(X)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-020899 | MATHDB-369776

The rationality conjecture for absolute Hodge classes on abelian varieties

摘录 / EXCERPT: Let w be a prime of $\Q^{\mathrm{al}}$ lying over a prime p, let $\mathbb{F}$ be the residue field at p, and let A be an abelian variety over $\mathbb{Q}^{\mathrm{al}}$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-020900 | MATHDB-369777

The weak rationality conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over $\mathbb{Q}^{\mathrm{al}}$ with good reduction at w, and let γ be a Hodge class on A. Call γ w-Lefschetz if its specializatio...

conjecture | open | MathDB 页面 | 原文线索

V2-020901 | MATHDB-369778

The CM-lift conjecture for Hodge classes on abelian varieties

摘录 / EXCERPT: Let w be a prime of $\mathbb{Q}^{\mathrm{al}}$ and let A be an abelian variety over $\mathbb{Q}^{\mathrm{al}}$ with good reduction at w. Let γ be a Hodge class on A....

conjecture | open | MathDB 页面 | 原文线索

V2-020902 | MATHDB-369820

The LP optimum conjecture for odd-dimensional generators in D_n

摘录 / EXCERPT: LP optimum conjecture for D_n. If n and d are odd integers with $1\leq d\leq n$, then

conjecture | open | MathDB 页面 | 原文线索

V2-020903 | MATHDB-369825

Motivic mirror symmetry for character stacks

摘录 / EXCERPT: Let G be a connected, simply connected, and semisimple complex Lie group. Let F be a finite subgroup of its center $Z(G)$, and let ${}^L\tilde{G}$ be the universal covering of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-020904 | MATHDB-369829

K-stable-fibre conjecture for smooth uniruled varieties

摘录 / EXCERPT: Let X be a smooth uniruled variety. K-stable-fibre conjecture. The variety X is birational to a Mori fibre space whose general fibre is K-stable. This conjecture links biration...

conjecture | open | MathDB 页面 | 原文线索

V2-020905 | MATHDB-369843

The rigid-component conjecture for parabolic induction

摘录 / EXCERPT: Rigid-component conjecture. For $C\in\mathbf{Comp}(\mathbf{d})$, the induced representation $\pi(C)\times\pi(C)$ is irreducible if and only if C contains an open $G_{\mathbf{d}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-020906 | MATHDB-369845

The generic-extension subrepresentation conjecture of Lapid and Mínguez

摘录 / EXCERPT: Generic-extension subrepresentation conjecture. The representation $\pi(C*D)$ is a subrepresentation of

conjecture | open | MathDB 页面 | 原文线索

V2-020907 | MATHDB-369846

The component formula for intertwining-operator poles

摘录 / EXCERPT: Intertwining-pole conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-020908 | MATHDB-369848

Elmer's separating-set conjecture for quiver representations

摘录 / EXCERPT: Let Q be a quiver, and for each dimension vector α let $\mathcal{V}_{\alpha}$ be its representation variety, with $G=\prod_i \operatorname{SL}_{\alpha(x_i)}(\mathbb{C})$ ac...

conjecture | open | MathDB 页面 | 原文线索

V2-020909 | MATHDB-369849

Dynamical Manin-Mumford conjecture for polarized endomorphisms

摘录 / EXCERPT: Let $f:X\to X$ be a polarized endomorphism of a smooth projective variety over a field of characteristic zero, and let $Z\subset X$ be a subvariety containing a Zariski dense set o...

conjecture | open | MathDB 页面 | 原文线索

V2-020910 | MATHDB-369850

DeMarco-Mavraki relative Dynamical Manin-Mumford conjecture

摘录 / EXCERPT: Let S be a smooth and irreducible quasi-projective variety over $\mathbb C$, let N be a positive integer, and let $\Phi:S\times\mathbb P^N\to S\times\mathbb P^N$ be an algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-020911 | MATHDB-369851

Relative degree-stabilization conjecture for families of endomorphisms

摘录 / EXCERPT: Let S be a smooth irreducible quasi-projective variety over $\mathbb C$, let K be a positive integer, and let $\Phi:S\times\mathbb P^K\to S\times\mathbb P^K$ be a family given...

conjecture | open | MathDB 页面 | 原文线索

V2-020912 | MATHDB-369913

Characteristic-two Newton-slope criterion for hyperelliptic Jacobians

摘录 / EXCERPT: Let A be a 3-dimensional abelian variety over a finite field of characteristic 2, with Weil polynomial having Newton polygon whose first slope is $\frac{1}{3}$; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-020913 | MATHDB-369914

Asymptotic proportions of non-hyperelliptic isogeny classes and obstruction-rule polynomials

摘录 / EXCERPT: Let $\Pi_3(q)$ be the proportion of isogeny classes of 3-dimensional abelian varieties over $\mathbb{F}_q$ that do not contain the Jacobian of a genus-3 hyperelliptic curve. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-020914 | MATHDB-369921

The symmetric-power compatibility conjecture for the graded Grothendieck-ring involution

摘录 / EXCERPT: Let $K_0(\operatorname{Var}_k^{\dim})$ be the graded Grothendieck ring of varieties, equipped with the symmetric-power operations Sym^m and the involution...

conjecture | open | MathDB 页面 | 原文线索

V2-020915 | MATHDB-369922

The symmetric powers conjecture for $\mathbb{D}$-singularities

摘录 / EXCERPT: Let X be a smooth variety over k, and let $m\geq 1$. The symmetric power $\operatorname{Sym}^m X=X^m/S_m$ is the quotient by the symmetric group. A variety has $\mathbb{D}$-sin...

conjecture | open | MathDB 页面 | 原文线索

V2-020916 | MATHDB-369928

Higher-dimensional Hodge-number conjecture for flops and flips

摘录 / EXCERPT: Let X be a projective variety with $\mathbb{Q}$-factorial terminal singularities of dimension $n\geq 4$, and let X^+ be the variety associated with X in the setting of the ci...

conjecture | open | MathDB 页面 | 原文线索

V2-020917 | MATHDB-369954

Dominance conjecture for associated varieties of finite vertex-algebra extensions

摘录 / EXCERPT: Let W be a vertex algebra that is a finite extension of a vertex algebra L, and let X_W and X_L denote their associated Poisson varieties. Dominance conjecture. The morphis...

conjecture | open | MathDB 页面 | 原文线索

V2-020918 | MATHDB-369955

Associated-variety conjecture for the level $-12+9/7$ affine E_6 vertex algebra

摘录 / EXCERPT: Let X_V denote the associated variety of a vertex algebra V, and let $\mathbb{O}_{E_6(a_1)}$ be the nilpotent orbit of Bala-Carter type $E_6(a_1)$ in E_6. Associated-variety...

conjecture | open | MathDB 页面 | 原文线索

V2-020919 | MATHDB-369956

Collapsing and associated-variety conjecture for level -6 affine F_4 W-algebras

摘录 / EXCERPT: Let X_V denote the associated variety of a vertex algebra V, and let $\mathbb{O}_{F_4(a_3)}$ be the nilpotent orbit of Bala-Carter type $F_4(a_3)$ in F_4. Level -6 F_4 co...

conjecture | open | MathDB 页面 | 原文线索

V2-020920 | MATHDB-369957

Associated-variety conjecture for the level $-18+14/5$ affine E_7 vertex algebra

摘录 / EXCERPT: Let X_V denote the associated variety of a vertex algebra V, and let $\mathbb{O}_{E_7(a_5)}$ be the nilpotent orbit of Bala-Carter type $E_7(a_5)$ in E_7. Associated-variety...

conjecture | open | MathDB 页面 | 原文线索

V2-020921 | MATHDB-369963

Universal parameterized LMO invariant for flat connections on 3-manifolds

摘录 / EXCERPT: Let W be a 3-manifold, let $\mathrm{Char}(W)$ be its character variety, and let $\mathcal{L}\in\mathrm{Char}(W)$ be a flat connection. Existence of a universal parameterized inva...

conjecture | open | MathDB 页面 | 原文线索

V2-020922 | MATHDB-369981

Filtered D-module degeneration to the limit category

摘录 / EXCERPT: Let $\mathcal{X}$ be a smooth QCA stack, and let $\operatorname{L}(\Omega_{\mathcal{X}})_{1/2}$ be the half-twisted limit category associated with its cotangent stack. For a cohere...

conjecture | open | MathDB 页面 | 原文线索

V2-020923 | MATHDB-369982

Extension by zero and restriction for magic categories

摘录 / EXCERPT: Let $\mathcal{Y}$ be a smooth stack, let $\mathcal{S}\subset\mathcal{Y}$ be the relevant closed stratum, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020924 | MATHDB-369983

D-module and nilpotent limit categories for genus-zero vector bundles

摘录 / EXCERPT: Let $C=\mathbb{P}^1$ and let $\operatorname{Bun}_{\operatorname{GL}_r}(0)$ be the degree-zero component of the moduli stack of rank-r vector bundles. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020925 | MATHDB-369997

Stoll's finite-subscheme conjecture for Chabauty-Coleman loci

摘录 / EXCERPT: Let X) be a smooth projective curve of genus $g\geq2$ whose Jacobian has Mordell-Weil rank at most $g-2$. Let U_1^p denote the abelianisation of the fundamental group of...

conjecture | open | MathDB 页面 | 原文线索

V2-020926 | MATHDB-370038

The variety-invariance conjecture for Bol-Moufang homology

摘录 / EXCERPT: Let Vij and $Wk\ell$ be identities defining the same variety of quasigroups, let X be any quasigroup in that variety, and let (t,s) be a substitution. Write $H_2(X;Vij,t,s)$...

conjecture | open | MathDB 页面 | 原文线索

V2-020927 | MATHDB-370063

Schreyer's geometric syzygy conjecture for canonical curves

摘录 / EXCERPT: Geometric Syzygy Conjecture. All linear syzygy spaces $\mathrm{K}_{i,1}(C,\omega_C)$ are spanned by syzygies of minimal rank $i+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-020928 | MATHDB-370074

Injectivity conjecture for primitive quantum algebras of Fano hypersurfaces

摘录 / EXCERPT: Injectivity conjecture. With the same notation as in the quantum Lefschetz proposition, there is a natural injective morphism of algebras

conjecture | open | MathDB 页面 | 原文线索

V2-020929 | MATHDB-370081

Künneth conjecture for p-power torsion Azumaya algebras

摘录 / EXCERPT: Let X be a quasi-compact and quasi-separated scheme in which p is invertible, and let A and A' be p-power torsion Azumaya algebras over X. The Künneth map is ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-020930 | MATHDB-370085

Nonsmooth subvariety conjecture for higher cylindrical singular strata

摘录 / EXCERPT: Let $n\geq 3$ and $k\geq 2$. For a mean curvature flow $\mathbf{M}$ in $\mathbb{R}^{n+1}$, let $\mathcal{S}_{k}(\mathbf{M})_+$ denote its positive part of the k-th cylindrical si...

conjecture | open | MathDB 页面 | 原文线索

V2-020931 | MATHDB-370101

Weighted analogue of the Batyrev-Manin conjecture

摘录 / EXCERPT: Let W be a smooth weighted variety in $\mathbb{WP}^n_{\mathbf{q}}(k)$ with weights $\mathbf{q}=(q_0,\ldots,q_n)$, let $q=\operatorname{lcm}(q_0,\ldots,q_n)$, and let A be an am...

conjecture | open | MathDB 页面 | 原文线索

V2-020932 | MATHDB-370114

The BPGL\langle 2 \rangle motivic Adams spectral sequence collapse claim

摘录 / EXCERPT: Let F be a field of low cohomological dimension. Consider the motivic Adams spectral sequence based on $\operatorname{BPGL}\langle 2 \rangle$, and let n be a positive integer....

conjecture | open | MathDB 页面 | 原文线索

V2-020933 | MATHDB-370136

Modular Zilber-Pink conjecture, optimal version

摘录 / EXCERPT: Let n be a positive integer and let $V\subseteq\mathbb{A}_{\mathbb{C}}^{n}$ be an algebraic variety. For a subvariety $X\subseteq V$, define its defect by ... The subvariety X is...

conjecture | open | MathDB 页面 | 原文线索

V2-020934 | MATHDB-370155

The integral approximation conjecture for weakly log Fano schemes

摘录 / EXCERPT: Let $(\mathfrak{X},\mathfrak{D})$ be a weakly log Fano scheme over the ring of integers $\mathfrak{o}$, with generic fiber (X,D), and write...

conjecture | open | MathDB 页面 | 原文线索

V2-020935 | MATHDB-370169

The converse from Kollár-hyperbolicity to V-hyperbolicity

摘录 / EXCERPT: Converse conjecture. If X is Kollár-hyperbolic with respect to H, then X is V-hyperbolic with respect to H.

conjecture | open | MathDB 页面 | 原文线索

V2-020936 | MATHDB-370205

The p-adic monodromy-weight conjecture for projective log smooth schemes

摘录 / EXCERPT: Let $\overset{\circ}{S}$ be a p-adic formal scheme and let $\overset{\circ}{X}\to\overset{\circ}{S}$ be projective. For a log smooth scheme Y/S, let ... be the monodromy operator...

conjecture | open | MathDB 页面 | 原文线索

V2-020937 | MATHDB-370206

The relative p-adic monodromy-weight conjecture for a proper SNCL scheme with a relative SNCD

摘录 / EXCERPT: Let S be a p-adic scheme, let X/S_0 be a proper SNCL scheme with relative SNCD D, and let $f\colon X\to S$ be the structural morphism. Let P and P^D be the weight filtr...

conjecture | open | MathDB 页面 | 原文线索

V2-020938 | MATHDB-370225

Beilinson-Bloch conjecture for decomposable triple product diagonal cycles

摘录 / EXCERPT: Let F be the totally real field, let X_U be the Shimura curve, and let π_1,π_2,π_3 be the automorphic representations and $[\Delta_3]_{f_1\otimes f_2\otimes f_3}$ the c...

conjecture | open | MathDB 页面 | 原文线索

V2-020939 | MATHDB-370226

Vanishing of the diagonal cycle for equal self-dual representations

摘录 / EXCERPT: Let π be a self-dual automorphic representation, let $\pi_1=\pi_2=\pi_3=\pi$, and let f range over $\boxtimes_{i=1}^3\pi_i^U$. Vanishing conjecture for equal self-dual repres...

conjecture | open | MathDB 页面 | 原文线索

V2-020940 | MATHDB-370227

The K-theoretic Hikita conjecture for quiver gauge theories

摘录 / EXCERPT: K-theoretic Hikita conjecture. There is an isomorphism of algebras over $\mathbb C[(T_{\bf{v}} / S_\bv) \times F] \otimes \mathbb C[q^{\pm 1}]$

conjecture | open | MathDB 页面 | 原文线索

V2-020941 | MATHDB-370228

The homological Hikita conjecture for quiver gauge theories

摘录 / EXCERPT: Homological Hikita conjecture. There is an isomorphism of graded $\mathbb C[(\mathfrak t_\bv/S_\bv)\times\ff]\otimes\mathbb C[\hbar]$-algebras

conjecture | open | MathDB 页面 | 原文线索

V2-020942 | MATHDB-370264

General order convergence conjecture for subdivision schemes

摘录 / EXCERPT: Consider $d53d^n$ endowed with the metric $d(x,y)=\lVert x-y\rVert$, where the norm is any norm on $d53d^n$. Let S_2 be a subdivision scheme on $d53d^n$ satisfying, for...

conjecture | open | MathDB 页面 | 原文线索

V2-020943 | MATHDB-370298

Quadratic generation of the Parke-Taylor toric kernel

摘录 / EXCERPT: Quadratic kernel-generation conjecture. For $n\geq 7$, the set $\calB_n$ generates $\ker_\mathbb{Z}(A_n)$. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-020944 | MATHDB-370299

Quadratic generation of the Parke-Taylor ideal

摘录 / EXCERPT: Quadratic Parke-Taylor generation conjecture. The ideal $\tilde{\mathcal{I}}(\mathrm{PT}_n^\circ)\subseteq\mathbb{C}[z_\sigma^{\pm}]$ is generated by the quadratics in $\calF$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-020945 | MATHDB-370322

Finiteness conjecture for the fundamental group of the regular locus of ICY and IHS varieties

摘录 / EXCERPT: Fundamental-group finiteness conjecture. The fundamental group

conjecture | open | MathDB 页面 | 原文线索

V2-020946 | MATHDB-370362

Bondal-Orlov-Kawamata crepant-resolution derived-equivalence conjecture

摘录 / EXCERPT: Let X be a normal algebraic variety with Gorenstein singularities, and let ... be two crepant resolutions by schemes or Deligne-Mumford stacks. An equivalence of triangulated categ...

conjecture | open | MathDB 页面 | 原文线索

V2-020947 | MATHDB-370363

Van den Bergh's conjecture on derived equivalence of crepant resolutions

摘录 / EXCERPT: Let X be a variety admitting crepant resolutions, including commutative resolutions and non-commutative crepant resolutions. Van den Bergh's conjecture. All crepant resolutions o...

conjecture | open | MathDB 页面 | 原文线索

V2-020948 | MATHDB-370364

Partial-tilting extension conjecture for toric vector bundles

摘录 / EXCERPT: Let Σ be a complete, simplicial fan such that the toric Deligne-Mumford stack $\mathcal{X}_\Sigma$ has a partial tilting complex $\mathcal{T}$ with finite global dimension e...

conjecture | open | MathDB 页面 | 原文线索

V2-020949 | MATHDB-370368

Griffiths-Lang conjecture for holomorphic curves and simple normal crossing divisors

摘录 / EXCERPT: Griffiths-Lang conjecture. There exists a proper Zariski closed subset $Z_D\subsetneq X$ such that, for all holomorphic curves $\mathbf{x}:\mathbb{C}\to X$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-020950 | MATHDB-370379

Restriction conjecture for cubic forms of slice rank at most r

摘录 / EXCERPT: Let V be a finite-dimensional vector space, let $f\in S^3V$, and let $E\subseteq V^*$ be a linear subspace. Write $f|_E$ for the restriction of f to E, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020951 | MATHDB-370381

Picard-groupoid refinement conjecture for derived Looijenga line bundles

摘录 / EXCERPT: Let $\operatorname{Bil}\colon \operatorname{Lat}^{\operatorname{op}}\to\operatorname{AbGroup}\subset\mathrm{PicGrpds}$ be the functor of bilinear forms, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020952 | MATHDB-370384

Algebraic refinement conjecture for the multivariable theta function

摘录 / EXCERPT: Let b be a symmetric bilinear form and let θ_b be the holomorphic section of the complex Looijenga line bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-020953 | MATHDB-370437

Kato's localization conjecture for Kato homology over global fields

摘录 / EXCERPT: Let F be a global field of characteristic $p\geq 0$, let $l\neq p$ be a prime, and for each place v let F_v be the completion of F at v. Let X be the scheme under consi...

conjecture | open | MathDB 页面 | 原文线索

V2-020954 | MATHDB-370440

The quotient-stack blowup F-bundle isomorphism conjecture

摘录 / EXCERPT: Let X be a smooth projective variety with an action of a group G, and let $Z\subset X$ be a G-invariant submanifold of codimension $c\geq 2$. Write ... with $c-1$ copies of Z...

conjecture | open | MathDB 页面 | 原文线索

V2-020955 | MATHDB-370441

The quotient-stack and ordinary blowup isomorphism compatibility conjecture

摘录 / EXCERPT: Let X be a smooth projective variety with a G-action, let $Z\subset X$ be a G-invariant submanifold of codimension $c\geq 2$, and set $\widetilde X=\operatorname{Bl}_Z(X)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-020956 | MATHDB-370468

Arithmetic level raising, self-dual case

摘录 / EXCERPT: In the indefinite case, let $\mathcal{S}_0^\circ$ be the relevant Shimura variety, let $\widetilde{\mathcal{S}}_0^\circ$ be its special fiber, let S'_0 parameterize the...

conjecture | open | MathDB 页面 | 原文线索

V2-020957 | MATHDB-370469

Ihara lemma, indefinite case

摘录 / EXCERPT: Assume the indefinite Shimura-variety setup of the source. Let $\mathcal{S}_0^\circ$ and $\mathcal{S}_0^\bullet$ be the relevant Shimura varieties, let Ω_λ be the loca...

conjecture | open | MathDB 页面 | 原文线索

V2-020958 | MATHDB-370471

Coherent-pair conjecture for non-trivial maps of coherent cohomology

摘录 / EXCERPT: Let (π,π') be a coherent pair linked by the chain of diagrams constructed in the paper to an elementary pair (σ,σ'). Coherent-pair cohomology conjecture. This cha...

conjecture | open | MathDB 页面 | 原文线索

V2-020959 | MATHDB-370488

Broustet-Gongyo's Calabi-Yau type conjecture for polarized endomorphisms

摘录 / EXCERPT: Broustet-Gongyo's conjecture. Every normal projective variety admitting a polarized endomorphism is of Calabi-Yau type. This has been proved for surfaces, smooth threefolds, and ra...

conjecture | open | MathDB 页面 | 原文线索

V2-020960 | MATHDB-370490

Generalized log-Calabi-Yau conjecture for polarized endomorphisms

摘录 / EXCERPT: Let X be a normal projective variety, let $f\colon X\to X$ be a q-polarized endomorphism, and let Δ be an effective $\mathbb{Q}$-divisor such that (X,Δ) is an f...

conjecture | open | MathDB 页面 | 原文线索

V2-020961 | MATHDB-370501

Simpson's geometricity conjecture for quasi-geometric flat bundles

摘录 / EXCERPT: Let Y be the complement of a normal crossing divisor in a smooth proper Deligne-Mumford stack. A flat bundle on Y is quasi-geometric if it extends as an algebraic regular-singu...

conjecture | open | MathDB 页面 | 原文线索

V2-020962 | MATHDB-370557

Purity conjecture for generalized affine Springer fibers

摘录 / EXCERPT: A generalized affine Springer fiber is the variety associated with a reductive group, a finite-dimensional representation, and an element of the corresponding loop-space representa...

conjecture | open | MathDB 页面 | 原文线索

V2-020963 | MATHDB-370558

Cherednik-Danilenko-Philipp conjecture for motivic superpolynomials

摘录 / EXCERPT: Let $(C,0)$ be a planar curve singularity. Let $L(q,t,a)$ be its Hilbert L-function and let $\mathrm{H}^{\mathrm{mot}}(q,t,a)$ be its normalized motivic superpolynomial. Cheredni...

conjecture | open | MathDB 页面 | 原文线索

V2-020964 | MATHDB-370559

Cherednik's Riemann-hypothesis conjecture for the Hilbert L-function

摘录 / EXCERPT: Let $(C,0)$ be a planar curve singularity, and let $L(q,t,0)$ be the specialization at $a=0$ of its Hilbert L-function. A t-zero is a complex number ξ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-020965 | MATHDB-370562

The twisted-diagonal-cycle factorisation conjecture for the a-region

摘录 / EXCERPT: Let $\underline{\Pi}$ be a family on GSp_4 with root number -1 and no endoscopic specialisations, let $\underline{\sigma}$ be the accompanying GL_2 fa...

conjecture | open | MathDB 页面 | 原文线索

V2-020966 | MATHDB-370577

Multiplicity bound conjecture in terms of minimal log discrepancies

摘录 / EXCERPT: Let $x\in X$ be a point on a variety with log canonical singularities. Denote $e:=\dim m_x/m_x^2$ and $d:=\dim X$ at x. The minimal log discrepancy at x is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-020967 | MATHDB-370588

The cornering and GIT-specialisation diagram for Kleinian quiver varieties

摘录 / EXCERPT: Let $0 \in J \subseteq J' \subseteq I$, set $K \coloneqq I \setminus J$ and $K' \coloneqq I \setminus J'$, and let $\sigma_{K,K'}$ be the cone defined by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020968 | MATHDB-370589

The Hilbert-Chow and symmetric-power diagram for Kleinian quiver varieties

摘录 / EXCERPT: Let $0 \in J \subseteq I$ and set $K \coloneqq I \setminus J$. Let X_J be the corresponding partial resolution of the Kleinian singularity, let $g:X_J\to X_0$ be the natural morp...

conjecture | open | MathDB 页面 | 原文线索

V2-020969 | MATHDB-370604

The Toposic Hom Conjecture

摘录 / EXCERPT: Let K be an absolutely finitely generated field, meaning that it is finitely generated over its prime field. Let ... be the localization of the category of finite type K-schemes...

conjecture | open | MathDB 页面 | 原文线索

V2-020970 | MATHDB-370614

Holonomy Chen ranks conjecture for simple matroids

摘录 / EXCERPT: Let M be a simple matroid. For each sub-matroid $\mathsf{M}'\subset \mathsf{M}$, let $n_k(\mathsf{M}')$ be the number of k-multinets supported by M', and le...

conjecture | open | MathDB 页面 | 原文线索

V2-020971 | MATHDB-370615

Yuzvinsky's uniqueness conjecture for $(4,d)$-multinets

摘录 / EXCERPT: Let a hyperplane arrangement be given, and let a $(4,d)$-multinet be a multinet with four classes and degree d. Yuzvinsky's uniqueness conjecture. The only $(4,d)$-multinet on a...

conjecture | open | MathDB 页面 | 原文线索

V2-020972 | MATHDB-370626

Restrictedness conjecture for Hecke-lisse sheaves with level structure

摘录 / EXCERPT: Let X be a smooth complete curve over k, let $\ul{x}$ be a finite set of k-points, and let $\operatorname{Bun}_G^{\operatorname{level}_{\ul{x}}}$ be the moduli stack of G-b...

conjecture | open | MathDB 页面 | 原文线索

V2-020973 | MATHDB-370628

Ramified trace conjecture for enhanced automorphic sheaves

摘录 / EXCERPT: Let X be a smooth complete curve, let G be reductive, let $\ul{x}$ be a finite set of points, and let Isoc_G be the stack of G-isocrystals. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-020974 | MATHDB-370660

The Coleman-Oort conjecture for special subvarieties in the Torelli locus

摘录 / EXCERPT: Let $\mathcal{A}_g$ be the moduli space of principally polarized complex abelian varieties of dimension g, let T_g be the Torelli locus, and let T_g^o be its open Torelli loc...

conjecture | open | MathDB 页面 | 原文线索

V2-020975 | MATHDB-370669

Conjecture on the image of the motivic birational invariant for projective four-space

摘录 / EXCERPT: Let c be the motivic birational invariant considered in the paper, and let S and S' be D-equivalent K-nef surfaces. The corresponding elements are the differences...

conjecture | open | MathDB 页面 | 原文线索

V2-020976 | MATHDB-370670

Conjecture on the motivic birational invariant for projective three-space over arbitrary fields

摘录 / EXCERPT: Let $\mathbb k$ be a field, and let C and C' be smooth projective curves over $\mathbb k$ that are D-equivalent and whose geometric irreducible components have genus 1. The a...

conjecture | open | MathDB 页面 | 原文线索

V2-020977 | MATHDB-370674

The Tian-Yau-Donaldson conjecture on K-stability and Kähler-Einstein metrics

摘录 / EXCERPT: Let M be a Fano manifold, meaning a compact complex manifold with positive first Chern class. A Fano manifold is K-stable when it satisfies the algebro-geometric K-stability cond...

conjecture | open | MathDB 页面 | 原文线索

V2-020978 | MATHDB-370680

The infinite-rank conjecture for abelian varieties over finitely generated Galois extensions

摘录 / EXCERPT: The infinite-rank conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-020979 | MATHDB-370681

The infinite-rank conjecture for abelian varieties over fields with finitely generated Galois group

摘录 / EXCERPT: The infinite-rank conjecture. If G_K is finitely generated, then A has infinite rank over K.

conjecture | open | MathDB 页面 | 原文线索

V2-020980 | MATHDB-370683

The SYZ conjecture for primitive symplectic varieties

摘录 / EXCERPT: A primitive symplectic variety is a normal compact Kähler variety with symplectic singularities whose smooth locus carries a symplectic form and whose reflexive powers of the canon...

conjecture | open | MathDB 页面 | 原文线索

V2-020981 | MATHDB-370704

Tamagawa number conjecture for $(M,\mathcal{O}_F)$ in analytic rank zero

摘录 / EXCERPT: Let M be a critical motive over K with coefficient field $\mathcal{F}$, let $F=\mathcal{F}_\lambda$, and let T be a stable $\mathcal{O}_F$-lattice in the associated represent...

conjecture | open | MathDB 页面 | 原文线索

V2-020982 | MATHDB-370705

Motivic cohomology conjecture for M

摘录 / EXCERPT: Let $(M,\mathcal{F})$ be a motive defined over K with coefficient field $\mathcal{F}$, and let $V_\lambda(M)$ be its λ-adic realization for $\lambda\in S_p(\mathcal{F})$....

conjecture | open | MathDB 页面 | 原文线索

V2-020983 | MATHDB-370706

Height-pairing conjecture for critical motives

摘录 / EXCERPT: Let M be a motive over K with coefficient field $\mathcal{F}$, and let $M^*(1)$ denote its dual Tate twist. Let $H_f^1(K,M)$ and $H_f^1(K,M^*(1))$ be the motivic cohomology gro...

conjecture | open | MathDB 页面 | 原文线索

V2-020984 | MATHDB-370709

Tamagawa number conjecture for $(M,\mathcal{O}_F)$

摘录 / EXCERPT: Let M be a motive over K, let $F=\mathcal{F}_\lambda$, and let T be a stable $\mathcal{O}_F$-lattice in its realization. Assume the local hypotheses and the motivic cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-020985 | MATHDB-370727

Eisenbud-Koh-Stillman conjecture on determinantal presentations of secant varieties of curves

摘录 / EXCERPT: Let C be a smooth projective curve of genus g. For line bundles A and B on C of sufficiently large degree with respect to g and k, set $L=A\otimes B$. The k-th seca...

conjecture | open | MathDB 页面 | 原文线索

V2-020986 | MATHDB-370728

Sidman-Smith conjecture on determinantal ideals of cactus varieties

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n. Let $\kappa_k(X,L)$ be the k-th cactus variety, namely the closure of the union of the k-planes spanned by finite subsc...

conjecture | open | MathDB 页面 | 原文线索

V2-020987 | MATHDB-370729

Han-Lee-Moon-Park conjecture on rank-three quadrics of sufficiently ample embeddings

摘录 / EXCERPT: Let X be a projective scheme and let L be a sufficiently ample line bundle on X. A quadric of rank 3 means a degree-two equation whose associated quadratic form has rank...

conjecture | open | MathDB 页面 | 原文线索

V2-020988 | MATHDB-370736

Projective normality and quadratic generation for polarized Hilbert squares of K3 surfaces

摘录 / EXCERPT: Projective-normality and quadratic-generation conjecture. If $g\geq8$ (equivalently, $q(L_2-2\delta)\geq6$), then this embedding is projectively normal; and if $g\geq10$ (equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-020989 | MATHDB-370741

Hosono-Lee-Lian-Yau Hodge number duality conjecture for singular Calabi-Yau mirror pairs

摘录 / EXCERPT: Let $(\widetilde{T}_\Delta,\widetilde{T}_{\check{\Delta}})$ be the pair of d-dimensional singular Calabi-Yau varieties obtained as branched double covers of the toric varieties a...

conjecture | open | MathDB 页面 | 原文线索

V2-020990 | MATHDB-370743

Zero-dimensional quiver-variety vertex-function factorization conjecture

摘录 / EXCERPT: Let Q be a quiver without loops, let $\mathcal{M}=\mathcal{M}_{Q,\theta}(\mathsf{v},\mathsf{w})$ be a zero-dimensional quiver variety with $\theta=(1,1,\ldots,1)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-020991 | MATHDB-370744

Tangent-character conjecture for duals of zero-dimensional quiver varieties

摘录 / EXCERPT: Let $\mathcal{M}^{!}=\overline{\mathcal{W}}^{\lambda}_{\mu}$ be the 3d mirror dual of a zero-dimensional quiver variety, with unique torus fixed point $p^{!}$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-020992 | MATHDB-370753

Hasse-Weil conjecture for smooth projective varieties

摘录 / EXCERPT: Hasse-Weil conjecture. The function $\zeta_X(s)$ extends to a meromorphic function of $\mathbf C$. There exists a positive real number $A\in\mathbf R^{>0}$, non-zero rational funct...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-020993 | MATHDB-370755

Conjectured formula for the first matching number of paired constructions

摘录 / EXCERPT: Let n,k,t be non-negative integers with $k\leq n$ and $t\leq\min(k,n-k)$. Let $m_1(n,k,t)$ denote the first matching number associated with the paired construction for Boolean fu...

conjecture | open | MathDB 页面 | 原文线索

V2-020994 | MATHDB-370756

Touzé's conjecture on cohomological finite generation for reductive group schemes

摘录 / EXCERPT: Let G be a reductive algebraic group scheme over a commutative Noetherian ring k. The Touzé conjecture. G satisfies the cohomological finite generation property. Over a field...

conjecture | open | MathDB 页面 | 原文线索

V2-020995 | MATHDB-370803

Autoduality conjecture for the ring stack ${}^{s}W$

摘录 / EXCERPT: Let Q be the quotient of the Witt vector ring stack W by its formal subgroup $\hat W$, let $T_F(Q)$ denote the associated Tate module, and let ${}^{s}W$ be the extension of W...

conjecture | open | MathDB 页面 | 原文线索

V2-020996 | MATHDB-370806

Hadamard non-defectivity conjecture for secant varieties of Segre-Veronese varieties

摘录 / EXCERPT: Let $SV_{\mathbf{d},\mathbf{n}}$ be a Segre-Veronese variety, let $\sigma_{\mathbf{r}}(SV_{\mathbf{d},\mathbf{n}})$ denote its secant variety, and let a Hadamard product mean the c...

conjecture | open | MathDB 页面 | 原文线索

V2-020997 | MATHDB-370807

Cox category tensor-product conjecture for semiprojective toric varieties

摘录 / EXCERPT: Let $\Xfr$ be a smooth semiprojective toric variety. The category $\Sh_{\Lambda_\mathrm{Cox}}(M_\RR / M)$ is closed under $\otimes_k$. Cox category tensor-product conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-020998 | MATHDB-370810

Mirror-pair conjecture for gauge-fixed double covers from dual nef-partitions

摘录 / EXCERPT: Let $(\Delta,\{\Delta_i\}_{i=1}^{r})$ and $(\nabla,\{\nabla_i\}_{i=1}^{r})$ be dual nef-partitions, and let X and $X^{\vee}$ be smooth maximal projective crepant partial desingul...

conjecture | open | MathDB 页面 | 原文线索

V2-020999 | MATHDB-370856

Potent relative Langlands duality

摘录 / EXCERPT: Let $G\circlearrowright M$ be a polarized hyperspherical G-variety with $M=T^*X$, and let $\check G\circlearrowright\check M$ be its Langlands-dual polarized hyperspherical varie...

conjecture | open | MathDB 页面 | 原文线索

V2-021000 | MATHDB-370894

Quantum Hikita conjecture for Higgs-Coulomb duality

摘录 / EXCERPT: Let X and $X^!$ be symplectically dual varieties arising respectively from Coulomb-branch and Higgs-branch constructions for the gauge theory (G,N). Let R be the coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-021001 | MATHDB-370895

K-theoretic quantum Hikita conjecture at roots of unity

摘录 / EXCERPT: Let X and $X^!$ be symplectically dual varieties arising from Coulomb-branch and resolved Higgs-branch constructions for the gauge theory (G,N). Let ζ be the relevant roo...

conjecture | open | MathDB 页面 | 原文线索

V2-021002 | MATHDB-370901

Refined cTC^- conjecture for the prismatization of the sphere

摘录 / EXCERPT: Let $\bbS^{\widehat{\calN}}$ be the stack of one-dimensional commutative formal groups with a section, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021003 | MATHDB-370902

The ring-structure reconstruction conjecture for the prismatized sphere

摘录 / EXCERPT: Let $\bbS^\mathrm{Syn}$ be the syntomic stack and let $\bbM_{\mathrm{m}\ifthenelse{\equal{\bbS}{}}{}{,\bbS}}^\mathrm{Syn}$ be its associated monoid stack. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021004 | MATHDB-370903

Full-faithfulness conjecture for the syntomic sphere and its monoid stack

摘录 / EXCERPT: Let $\bbS^\mathrm{Syn}$ be the syntomic stack, and let $\bbM_{\mathrm{m}\ifthenelse{\equal{\bbS}{}}{}{,\bbS}}^\mathrm{Syn}$ be the associated monoid stack. Consider the map from...

conjecture | open | MathDB 页面 | 原文线索

V2-021005 | MATHDB-370915

The non-Hodge-maximal adelic Mumford-Tate conjecture

摘录 / EXCERPT: Let $M=H^w(X)$ arise from a smooth projective variety X over a number field F, and fix an embedding $\sigma:F\hookrightarrow\mathbb C$. Let G_{MT} be the Mumford-Tate group o...

conjecture | open | MathDB 页面 | 原文线索

V2-021006 | MATHDB-370916

The absolute Hodge conjecture for Hodge classes

摘录 / EXCERPT: Absolute Hodge conjecture. If this pair comes from a Hodge class in $(H^w_B(X(\mathbb C),\mathbb Q))^{\otimes}$ for one embedding $\sigma:k\hookrightarrow\mathbb C$, then for every...

conjecture | open | MathDB 页面 | 原文线索

V2-021007 | MATHDB-370917

The compatibility of absolute Hodge classes with the étale-de Rham comparison

摘录 / EXCERPT: Let X be a smooth projective variety over a number field F. Let $(s_{\mathbb A_f},s_{\mathrm{dR}})$ be an absolute Hodge class in ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021008 | MATHDB-370927

Hara-Watanabe conjecture on strongly F-regular type couples

摘录 / EXCERPT: Hara-Watanabe conjecture. (X,Δ) is of strongly F-regular type if and only if (X,Δ) is of klt type.

conjecture | open | MathDB 页面 | 原文线索

V2-021009 | MATHDB-370930

The holomorphy conjecture for twisted topological zeta functions

摘录 / EXCERPT: Let $f\in \mathbb C[x_{1},\dots,x_{n}]\setminus \mathbb C$ be a non-constant polynomial, let $d\in\mathbb Z_{\geq 1}$, and let $Z^{\mathrm{top},(d)}_{(f)}(s)$ denote the twisted to...

conjecture | open | MathDB 页面 | 原文线索

V2-021010 | MATHDB-370944

Plane containment conjecture for low-degree curves on general hypersurfaces

摘录 / EXCERPT: Low-degree curve plane-containment conjecture. Every curve C of degree $\delta \leq d$ must be contained in a 2-plane. This extends the stated minimality theorem for plane curv...

conjecture | open | MathDB 页面 | 原文线索

V2-021011 | MATHDB-370961

Ben-Zvi-Sakellaridis-Venkatesh period formula for spherical varieties

摘录 / EXCERPT: Assume that $\check{M}=T^*\check{X}$, where $\check{X}$ is a $\check{G}$-spherical variety. Let π be a tempered, everywhere unramified automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-021012 | MATHDB-370966

Arinkin's tamely ramified geometric Langlands conjecture for parabolic connections

摘录 / EXCERPT: Let $\mathcal{M}$ be the moduli stack of parabolic connections, let P be the corresponding coarse moduli space of indecomposable quasi-parabolic $\mathfrak{sl}_2$-bundles, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-021013 | MATHDB-370968

Derived Langlands equivalence for parabolic connections via the partially compactified Radon transform

摘录 / EXCERPT: Let $\mathcal{M}$ be the moduli stack of parabolic connections, let $P^{\vee}$ be the dual projective space, and let $D_{\boldsymbol\nu}$ be the corresponding twisted differential-...

conjecture | open | MathDB 页面 | 原文线索

V2-021014 | MATHDB-370971

Reducedness conjecture for the Poisson ideal defining the symmetric quotient Schubert variety

摘录 / EXCERPT: Let $K_0\backslash\mathcal{G}_0$ be the quotient under consideration, let λ be a dominant coweight, let ω_i be the fundamental coweights for $1\leq i\leq n-1$, and s...

conjecture | open | MathDB 页面 | 原文线索

V2-021015 | MATHDB-370972

Erman-Martinova sharpness conjecture for Veronese Betti bounds

摘录 / EXCERPT: Let M be the coordinate ring of the Veronese, under the hypotheses of Theorem, and let $\beta_{i,i+1}$ denote the corresponding graded Betti numbers. Erman-Martinova conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-021016 | MATHDB-370998

Jacobian decomposition conjecture for elementary abelian actions

摘录 / EXCERPT: Let $p\geqslant 2$ be a prime number and let (S,N) be a $\mathbb{Z}_p^m$-action of signature $(0;p^{n+1})$, where $2\leqslant m\leqslant n-1$. For each subgroup $L\leqslant N$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-021017 | MATHDB-371005

Conjectural resolution counts for Type I four-dimensional quiver varieties

摘录 / EXCERPT: Let $N(\I(\D[4],n))$, $N(\I(\D[m],1))$, $N(\I(\D[m],n)')$, $N(\I(\D[m],0)')$, $N(\I(\E[6,7],n))$, and $N(\I(m))$ denote the numbers of projective crepant resolutions for the indica...

conjecture | open | MathDB 页面 | 原文线索

V2-021018 | MATHDB-371006

Unique projective crepant resolution for Type II quiver varieties

摘录 / EXCERPT: Let $\root\in\Sig$ be a dimension vector of Type $\II$, and let the associated quiver variety be the variety determined by this dimension vector. Type II uniqueness conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-021019 | MATHDB-371016

The monadic description of derived D-modules for derived affine schemes

摘录 / EXCERPT: Derived D-module equivalence. There exists a symmetric monoidal equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-021020 | MATHDB-371039

Smallest spectral-gap conjecture for perfect matching association scheme graphs

摘录 / EXCERPT: Smallest spectral-gap conjecture. Among the adjacency matrices

conjecture | open | MathDB 页面 | 原文线索

V2-021021 | MATHDB-371040

Largest-diameter conjecture for perfect matching association scheme graphs

摘录 / EXCERPT: Largest-diameter conjecture. The diameter of $X_{[2,1^{n-2}]}$ is the largest diameter among all graphs in the perfect matching association scheme.

conjecture | open | MathDB 页面 | 原文线索

V2-021022 | MATHDB-371043

Hausel's K-theoretic conjecture for fixed-point schemes

摘录 / EXCERPT: Hausel's K-theoretic conjecture. The ring of invariant functions is, as an algebra over $\mathbb C[\mathrm G]^{\mathrm G}\cong K^0_{\mathrm G}(\operatorname{pt})\otimes\mathbb C$,...

conjecture | open | MathDB 页面 | 原文线索

V2-021023 | MATHDB-371049

Morel's conjecture on the motivic homotopy sheaves of punctured affine space

摘录 / EXCERPT: Morel's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021024 | MATHDB-371089

Losev's simplicity conjecture for canonical quantizations

摘录 / EXCERPT: Let X be a Poisson variety and let U be its canonical quantization, the preferred quantization associated with Losev's classification. A quantization is simple if it has no non...

conjecture | open | MathDB 页面 | 原文线索

V2-021025 | MATHDB-371119

Mixed linear independence conjecture for partial Dirichlet L-values

摘录 / EXCERPT: Mixed linear independence conjecture. As ${\mathbb Q}$-subspaces of ${\mathbb R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-021026 | MATHDB-371120

Canonical splitting conjecture for the regulator-value space

摘录 / EXCERPT: Canonical splitting conjecture. There is a canonical direct-sum decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-021027 | MATHDB-371121

Weak mixed linear independence conjecture for regulator-value spaces

摘录 / EXCERPT: Weak mixed linear independence conjecture. As ${\mathbb Q}$-subspaces of ${\mathbb R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-021028 | MATHDB-371126

Motivic standard conjecture of Hodge type

摘录 / EXCERPT: Let $\mathcal X$ be a smooth projective scheme of relative dimension d over a field or over the ring of integers of a p-adic field, satisfying the Künneth standard conjecture,...

conjecture | open | MathDB 页面 | 原文线索

V2-021029 | MATHDB-371131

Existence of non-equivariant stable-envelope integrals for type A quiver varieties

摘录 / EXCERPT: Let X be a type A Nakajima quiver variety with its natural torus action, let p be a T-fixed point, and let $\boldsymbol{a}\to 0$ denote the non-equivariant limit. Qu...

conjecture | open | MathDB 页面 | 原文线索

V2-021030 | MATHDB-371132

Uniqueness conjecture for relative motivic spheres in dimension 2

摘录 / EXCERPT: Uniqueness conjecture for relative motivic spheres. If X is $\mathbb{A}^1$-homotopic to $\mathbb{A}^2_S\backslash\{0\}$, then X is isomorphic to $\mathbb{A}^2_S\backslash\{0\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-021031 | MATHDB-371151

The elliptic quantum group and geometric representation conjectures

摘录 / EXCERPT: Elliptic quantum group and geometric representation conjectures. (1) Vertex operators of $U_{q,p}(\widehat{\mathfrak g})$ or $U_{t_1,t_2,p}(\mathfrak g_{tor})$ defined as intertwin...

conjecture | open | MathDB 页面 | 原文线索

V2-021032 | MATHDB-371152

The level-$(0,-1)$ geometric action conjecture for affine type A

摘录 / EXCERPT: Level-$(0,-1)$ geometric action conjecture. The representation $\mathcal F^{(0,-1)(k)}_u$ of $U_{t_1,t_2,p}(\mathfrak{gl}_{N,tor})$ is equivalent to an expected level-$(0,-1)$ geom...

conjecture | open | MathDB 页面 | 原文线索

V2-021033 | MATHDB-371153

The level-$(0,-|\boldsymbol w|)$ geometric action conjecture

摘录 / EXCERPT: Level-$(0,-|\boldsymbol w|)$ geometric action conjecture. The tensor product representation

conjecture | open | MathDB 页面 | 原文线索

V2-021034 | MATHDB-371164

Batyrev--Borisov homological mirror symmetry conjecture

摘录 / EXCERPT: Let $(Z,\check{Z})$ be a Batyrev--Borisov mirror pair, with Z and $\check{Z}$ smooth subvarieties of Fano toric varieties X_{Σ} and $X_{\check{\Sigma}}$ associated to dual...

conjecture | open | MathDB 页面 | 原文线索

V2-021035 | MATHDB-371192

Integral weak Hilbert property conjecture over arithmetic schemes

摘录 / EXCERPT: Integral weak Hilbert property conjecture. If $\mathcal{Y}(S)^{(1)}$ is Zariski dense in Y, then $\mathcal{Y}$ has WHP over S potentially.

conjecture | open | MathDB 页面 | 原文线索

V2-021036 | MATHDB-371206

The unramified relative local Langlands conjecture for polarized hyperspherical varieties

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth affine spherical G-variety with connected stabilizers in the open Borel orbit, and set $M=T^*(2)(X)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021037 | MATHDB-371218

Adelic Mordell-Lang conjecture for Brauer-Manin components

摘录 / EXCERPT: Let k be a global field, let A be an abelian variety over k, and let $X\subset A$ be a closed subvariety. Assume that $A_{\overline{k}}$ has no positive-dimensional isotrivia...

conjecture | open | MathDB 页面 | 原文线索

V2-021038 | MATHDB-371219

Affine paving conjecture for punctual noncommutative Hilbert schemes

摘录 / EXCERPT: Affine paving conjecture. There exist affine pavings of ${}^0\operatorname{Hilb}^{(m)}(V)$ and $Z^{(m)}(V)$ by the parts $S_T^{\rm new}$ and $S_{\widehat{T}}$, respectively, such t...

conjecture | open | MathDB 页面 | 原文线索

V2-021039 | MATHDB-371236

Debarre-Voisin realization conjecture for Hilbert squares of genus 16 K3 surfaces

摘录 / EXCERPT: Let (S,h) be a general polarized K3 surface of genus 16, let S' be its unique nontrivial Fourier-Mukai partner, and let V_{10} and V_{10}' be the rank 10 vector spaces...

conjecture | open | MathDB 页面 | 原文线索

V2-021040 | MATHDB-371245

Free rational curves in the smooth locus of klt Fano varieties

摘录 / EXCERPT: Let (X,Δ) be a klt Fano pair over an algebraically closed field of characteristic 0. A rational curve in X is free when its pullback tangent bundle is nef. Free-curve co...

conjecture | open | MathDB 页面 | 原文线索

V2-021041 | MATHDB-371259

Uniform torsion-freeness conjecture for smooth projective varieties

摘录 / EXCERPT: Uniform torsion-freeness conjecture. There exist polynomials $\psi(x),\phi(x)\in\mathbb{Z}[x]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-021042 | MATHDB-371262

The coherent cohomological dimension conjecture for Shimura varieties

摘录 / EXCERPT: The coherent cohomological dimension conjecture. One conjectures that

conjecture | open | MathDB 页面 | 原文线索

V2-021043 | MATHDB-371267

Grothendieck's de Rham-Betti cycle conjecture

摘录 / EXCERPT: Grothendieck's de Rham-Betti cycle conjecture. Every de Rham-Betti class in V_{B} comes from a $\mathbb{Q}$-coefficient algebraic cycle on X for $0\leq p\leq \dim(X)$....

conjecture | open | MathDB 页面 | 原文线索

V2-021044 | MATHDB-371268

The Grothendieck period conjecture for de Rham-Betti classes

摘录 / EXCERPT: Let X be a smooth projective variety defined over $\overline{\mathbb{Q}}$, and let $0\leq j\leq\dim(X)$ be an integer. Consider the de Rham-Betti structure ... If...

conjecture | open | MathDB 页面 | 原文线索

V2-021045 | MATHDB-371273

The monodromy conjecture for coherent ideal sheaves

摘录 / EXCERPT: Let X be a smooth irreducible complex variety and let $\mathcal I$ be a coherent ideal sheaf of X with a non-empty zero locus. Let $Z_{\mathcal I}^{\mathrm{mot}}(T)$ be its mot...

conjecture | open | MathDB 页面 | 原文线索

V2-021046 | MATHDB-371274

The holomorphy conjecture for generalized topological zeta functions

摘录 / EXCERPT: Let X be a smooth irreducible complex variety and let $\mathcal I$ be a coherent ideal sheaf of X with a non-empty zero locus. Let $E(\mathcal I)$ be the set of Verdier monodro...

conjecture | open | MathDB 页面 | 原文线索

V2-021047 | MATHDB-371279

The motivic multiple multiplicativity defect bound

摘录 / EXCERPT: Let X be a smooth connected projective variety of dimension n equipped with a CK decomposition. For an integer $m\geq 2$, an m-fold r-multiplicative CK decomposition is one...

conjecture | open | MathDB 页面 | 原文线索

V2-021048 | MATHDB-371280

Bloch-Beilinson compatibility conjecture for multiple intersection products

摘录 / EXCERPT: Let X be a smooth connected projective variety of dimension n with CK projectors $\{\pi_i^X\}_{0\leq i\leq 2n}$, and let $\Delta_I^X\in\operatorname{CH}^{mn}(X^{m+1})$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-021049 | MATHDB-371281

Franchetta conjecture for families with motivic multiplicativity

摘录 / EXCERPT: Let X and B be smooth connected varieties, and let $f:X\to B$ be a smooth projective morphism such that a very general fiber of f admits motivic 0-multiplicativity. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-021050 | MATHDB-371284

Vojta's conjecture for bounded-degree points

摘录 / EXCERPT: Vojta's conjecture. The stated inequality holds for all such data and points outside the exceptional subvariety Z. The conjecture is motivated by Nevanlinna theory and, as noted...

conjecture | open | MathDB 页面 | 原文线索

V2-021051 | MATHDB-371322

Ewald's bounded-coordinate conjecture for smooth Fano polytopes

摘录 / EXCERPT: Let $P \subset N_{\operatorname{\mathbb{Q}}}$ be a smooth Fano polytope of dimension n. A smooth Fano polytope is a lattice polytope whose vertices are primitive lattice points a...

conjecture | open | MathDB 页面 | 原文线索

V2-021052 | MATHDB-371330

Covering conjecture for generalized connected subsets over real closed fields

摘录 / EXCERPT: Let R be a real closed field. For a smooth algebraic variety X over R, write X^{h} for its associated space of real points. A generalized connected open subs...

conjecture | open | MathDB 页面 | 原文线索

V2-021053 | MATHDB-371355

The generalized Green-Griffiths-Lang conjecture for smooth quasi-projective varieties

摘录 / EXCERPT: Let X be a smooth quasi-projective variety. Define the special loci ... as follows: $\mathrm{Sp}_{\mathrm{sab}}(X)$ is the Zariski closure of the union of images of non-constant ra...

conjecture | open | MathDB 页面 | 原文线索

V2-021054 | MATHDB-371365

Traditional Humphreys conjecture on support varieties of tilting modules

摘录 / EXCERPT: Traditional Humphreys conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021055 | MATHDB-371366

Relative Humphreys conjecture on support varieties

摘录 / EXCERPT: Relative Humphreys conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021056 | MATHDB-371367

Scheme-theoretic traditional Humphreys conjecture

摘录 / EXCERPT: Scheme-theoretic traditional Humphreys conjecture. For every $w\in W_{\mathrm{ext}}^\varnothing$, the coherent sheaf

conjecture | open | MathDB 页面 | 原文线索

V2-021057 | MATHDB-371407

The constrained crossing-uncrossing closure conjecture for Go-diagrams

摘录 / EXCERPT: Let D' be a Go-diagram in a partition λ, and let $c\prec c'$ be a crossing and uncrossing pair in D' with $D'(c)=\circ$. Let D be obtained by replacing the stones at...

conjecture | open | MathDB 页面 | 原文线索

V2-021058 | MATHDB-371409

Lang's weak and strong conjectures for varieties of general type

摘录 / EXCERPT: Lang's conjecture. (a) Weak form: The set $X(k)$ of k-rational points is not Zariski dense in X. (b) Strong form: There exists a proper Zariski-closed subset $Z\subset X$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-021059 | MATHDB-371427

Characteristic class version of the Hodge index theorem

摘录 / EXCERPT: Let X be a compact pure-dimensional complex algebraic variety. The Goresky-MacPherson homology L-class of X is denoted by $L_*(X)$, and its intersection cohomology Hirzebruch...

conjecture | open | MathDB 页面 | 原文线索

V2-021060 | MATHDB-371442

Seiberg duality conjecture for Gromov-Witten invariants of quiver critical loci

摘录 / EXCERPT: Let $(\mathbf{Q},\mathbf{W},\mathbf{v})$ be a quiver with potential and dimension vector, with associated gauge group G, representation space N, stability con...

conjecture | open | MathDB 页面 | 原文线索

V2-021061 | MATHDB-371443

Cluster algebra conjecture for quantum cohomology of quiver critical loci

摘录 / EXCERPT: Let $(\mathbf{Q},\mathbf{W},\mathbf{v})$ be a quiver with potential and dimension vector, with associated quiver variety X and superpotential, and let Z be its critical locus;...

conjecture | open | MathDB 页面 | 原文线索

V2-021062 | MATHDB-371454

Generalized Bloch conductor formula

摘录 / EXCERPT: Generalized Bloch conductor formula. If $p:X\to S$ is an arithmetic S-scheme with S-proper singular locus Z, then

conjecture | open | MathDB 页面 | 原文线索

V2-021063 | MATHDB-371462

The cutoff conjectures for multi-stack and restricted random-to-random shuffles

摘录 / EXCERPT: Multi-stack and restricted shuffle cutoff conjecture. Both chains exhibit cutoff, with the multi-stack chain around time

conjecture | open | MathDB 页面 | 原文线索

V2-021064 | MATHDB-371481

Grothendieck's special case of the generalised Hodge conjecture

摘录 / EXCERPT: Grothendieck's conjecture. There exists a dense open $U\subseteq Y$ such that this restriction map vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-021065 | MATHDB-371499

The Oka-1 conjecture for rationally connected projective manifolds

摘录 / EXCERPT: Let X be a projective manifold that is rationally connected, meaning that any two points of X can be joined by a rational curve. Let Oka-1 denote the property that, for every o...

conjecture | open | MathDB 页面 | 原文线索

V2-021066 | MATHDB-371540

Non-commutative correspondence conjecture for derived Γ-schemes

摘录 / EXCERPT: Let X_{Γ}^{nc} be a compact non-commutative derived Γ-scheme, and let $\mathcal C_\Gamma^{\mathrm{dg}}$ be the dg-category of Γ-representations. Non-c...

conjecture | open | MathDB 页面 | 原文线索

V2-021067 | MATHDB-371541

Tangent-cotangent equivalence conjecture for derived Γ-schemes

摘录 / EXCERPT: Let X be a Γ-scheme, let $\mathbb T_X^\Gamma$ denote its Γ-tangent ∞-category, and let $\mathbb L_X^\Gamma$ denote its cotangent complex. Tangent-cotangent e...

conjecture | open | MathDB 页面 | 原文线索

V2-021068 | MATHDB-371550

Virtual nilpotence of orbifold fundamental groups of log canonical Calabi-Yau pairs

摘录 / EXCERPT: Let (X,D_X) be a log canonical pair with standard coefficients, where ... with each D_i a prime divisor and $m_i\in\mathbb{Z}_{>0}\cup\{\infty\}$. The orbifold fundamental group...

conjecture | open | MathDB 页面 | 原文线索

V2-021069 | MATHDB-371557

Differentiability conjecture for holonomic modules over the completed algebra of vector fields

摘录 / EXCERPT: Let X be the underlying smooth variety, and let $\mathcal{M}$ be a holonomic $\widehat{\mathcal{A}\mathcal{V}}$-module. Here holonomicity is the finiteness condition used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-021070 | MATHDB-371574

Smoothness and rationality conjecture for triangularization varieties

摘录 / EXCERPT: Let k be a number field, let $M\in \mathrm{M}_n(\mathcal{O}_k)$, and suppose that the triangularization variety X_M has a rational point, $X_M(k)\neq\varnothing$. An irreducibl...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-021071 | MATHDB-371603

The Jacobian Conjecture over algebraically closed fields of characteristic zero

摘录 / EXCERPT: Let n be a positive integer, let k be an algebraically closed field of characteristic zero, and let $f:k^n\rightarrow k^n$ be a polynomial map, meaning that each component...

conjecture | open | MathDB 页面 | 原文线索

V2-021072 | MATHDB-371620

Tate's conjecture on polarized endomorphisms and étale cohomology

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d over the algebraically closed field $\mathbb{k}$, and let f be a (q-)polarized endomorphism of X: there is an ample d...

conjecture | open | MathDB 页面 | 原文线索

V2-021073 | MATHDB-371631

The weak weak approximation conjecture for smooth unirational varieties

摘录 / EXCERPT: Let K be a number field. A K-variety X satisfies weak weak approximation (WWA) if there is a finite set S_0 of places of K such that, for every finite set S_1 of places...

conjecture | open | MathDB 页面 | 原文线索

V2-021074 | MATHDB-371632

The potential weak Hilbert property conjecture for special varieties

摘录 / EXCERPT: Let K be a finitely generated field and let X be a smooth projective variety over K. The variety X has potential WHP if X_L has the weak Hilbert property for some finite...

conjecture | open | MathDB 页面 | 原文线索

V2-021075 | MATHDB-371646

Rigidity conjecture for polarized abelian varieties from Bergman densities

摘录 / EXCERPT: Rigidity conjecture. Then Φ is either a biholomorphism or an anti-biholomorphism, and moreover

conjecture | open | MathDB 页面 | 原文线索

V2-021076 | MATHDB-371661

Peternell's algebraic approximation conjecture for minimal Kähler varieties

摘录 / EXCERPT: Peternell's conjecture. Any minimal Kähler variety has an algebraic approximation.

conjecture | open | MathDB 页面 | 原文线索

V2-021077 | MATHDB-371691

Hyperbolicity conjecture for adjoint linear series

摘录 / EXCERPT: Hyperbolicity conjecture. The linear system $|K_X+mL|$ is hyperbolic if $m\geq 3n+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-021078 | MATHDB-371711

Agreement of orientations for motivic Donaldson-Thomas invariants

摘录 / EXCERPT: Let the orientation constructed by Joyce and Upmeier be the orientation used to define the motivic Donaldson-Thomas invariants in this paper, and let the orientation used by Behren...

conjecture | open | MathDB 页面 | 原文线索

V2-021079 | MATHDB-371740

Chai's Tate-linear conjecture

摘录 / EXCERPT: Let $f:X\rightarrow\mathscr{A}_{g,\mathbb{F}}^{\mathrm{ord}}$ be a map from a smooth connected variety, let $x\in X(\mathbb{F})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021080 | MATHDB-371742

Char p geometric André-Oort conjecture

摘录 / EXCERPT: Let $f:X\rightarrow\mathscr{A}_{g,\mathbb{F}}$ be a locally closed immersion with image in the ordinary locus. A subvariety is quasi-weakly special when its ordinary locus is conta...

conjecture | open | MathDB 页面 | 原文线索

V2-021081 | MATHDB-371743

Mod p geometric André-Oort conjecture for factor-separating subvarieties

摘录 / EXCERPT: Let $\mathbf{I}$ be a finite index set, let $f:X\rightarrow\mathscr{A}_{g,\mathbb{F}}$ separate factors over $\mathscr{A}_{g_{\mathbf I},\mathbb{F}}$, and for each $i\in\mathbf I$...

conjecture | open | MathDB 页面 | 原文线索

V2-021082 | MATHDB-371746

Finite-slope universal Rankin-Selberg p-adic L-function conjecture

摘录 / EXCERPT: Let U be the affinoid neighbourhood of a finite-slope classical point x_0 on the Rankin-Selberg eigenvariety, with U smooth, the weight map finite étale, and classical cuspid...

conjecture | open | MathDB 页面 | 原文线索

V2-021083 | MATHDB-371752

Cheltsov's cylindrical ample cone conjecture for log del Pezzo surfaces

摘录 / EXCERPT: Let S be a log del Pezzo surface. The cylindrical ample cone conjecture. If the cylindrical ample cone ... contains the anticanonical class, then ... Equivalently, the cylindrical am...

conjecture | open | MathDB 页面 | 原文线索

V2-021084 | MATHDB-371771

The conjectural analytic continuation and functional equation for motivic L-functions

摘录 / EXCERPT: Let M be an object of the category of premotivic structures over K that is L-admissible at every prime. Let $L(M,s)$ be its $K\otimes\mathbb{C}$-valued L-function, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021085 | MATHDB-371772

Fontaine-Perrin-Riou's order-of-vanishing conjecture for motivic L-functions

摘录 / EXCERPT: Let M be an L-admissible premotivic structure over K, let $\tau:K\to\mathbb{C}$ be an embedding, and let λ be any finite prime of K. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-021086 | MATHDB-371773

The Bloch-Kato conjecture for critical premotivic structures

摘录 / EXCERPT: Let M be a critical object of $\operatorname{{\bf PM}}_K$, let $b(M)$ be the basis supplied by Deligne's period conjecture, let λ be a place of K for which the stated v...

conjecture | open | MathDB 页面 | 原文线索

V2-021087 | MATHDB-371778

Jiao's bounded complement conjecture for Calabi-Yau type varieties

摘录 / EXCERPT: Jiao's conjecture. There exists a positive integer m with the following property: there is an effective $\mathbb{Q}$-Weil divisor Δ on X such that (X,Δ) is klt an...

conjecture | open | MathDB 页面 | 原文线索

V2-021088 | MATHDB-371802

Vexillary partial permutation minimality conjecture for real matrix Schubert varieties

摘录 / EXCERPT: Let ω be a partial permutation, and let $X_\omega\subset\mathfrak{M}_{m,n}(\mathbb{R})$ denote the corresponding open dense regular part of the real matrix Schubert variety....

conjecture | open | MathDB 页面 | 原文线索

V2-021089 | MATHDB-371810

Shokurov's minimal discrepancy conjecture

摘录 / EXCERPT: Let V be a normal affine variety of complex dimension n with an isolated singularity at 0 that is numerically $\mathbb{Q}$-Gorenstein. Write $\operatorname{min.disc}(V,0)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-021090 | MATHDB-371820

The Lefschetz standard conjecture

摘录 / EXCERPT: The Lefschetz standard conjecture. For each $k\leq n$, there exists an algebraic self-correspondence

conjecture | open | MathDB 页面 | 原文线索

V2-021091 | MATHDB-371830

Grimm-Monnée complexity conjecture for atypical special subvarieties

摘录 / EXCERPT: Let $(\mathbb{V},Q)$ be a variation of polarized integral Hodge structures of even weight $2k$ on a smooth quasi-projective variety S. Let $\mathcal{A}_{0}(1,q)$ denote the degre...

conjecture | open | MathDB 页面 | 原文线索

V2-021092 | MATHDB-371835

Classical eigensystem decomposition for the Fontaine-kernel cohomology

摘录 / EXCERPT: Let $\mathscr{F}\!\ell$ be the flag variety, let N_0 be the Fontaine operator, let d be the relevant cohomological degree, and let $\mathfrak{m}$ be the specified Hecke ideal....

conjecture | open | MathDB 页面 | 原文线索

V2-021093 | MATHDB-371836

Classicality conjecture for geometric de Rham complex cohomology

摘录 / EXCERPT: Let $\mathscr{F}\!\ell$ be the flag variety, let GDR^{la}_0 be the geometric locally analytic de Rham complex, let d be the relevant cohomological degree, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021094 | MATHDB-371847

Kontsevich's canonical semi-orthogonal decomposition conjecture

摘录 / EXCERPT: Let X be a smooth projective variety and let $D^b\operatorname{Coh}(X)$ denote its derived category of coherent sheaves. A semi-orthogonal decomposition is a decomposition of thi...

conjecture | open | MathDB 页面 | 原文线索

V2-021095 | MATHDB-371868

Amitsur's birationality conjecture for Severi-Brauer varieties

摘录 / EXCERPT: Let F be a field, and let A and B be central simple algebras over F. Write $SB(A)$ and $SB(B)$ for their associated Severi-Brauer varieties, and let $[A]$ and $[B]$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-021096 | MATHDB-371869

Hartshorne's Chow ring conjecture for smooth hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a smooth hypersurface, and let $h\in A^1(X)$ denote the hyperplane class. Hartshorne's conjecture. ... Apart from some easy results when...

conjecture | open | MathDB 页面 | 原文线索

V2-021097 | MATHDB-371870

Hartshorne's product conjecture for Chow groups of hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a smooth hypersurface, and let $h\in A^1(X)$ denote the hyperplane class. Hartshorne's product conjecture. ... This is presented as a co...

conjecture | open | MathDB 页面 | 原文线索

V2-021098 | MATHDB-371872

Voisin's ambientness conjecture for the triple-cycle Gamma

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a general hypersurface of degree $n+2$, and let $\Gamma\in A^{2n}(X\times X\times X)$ be the cycle defined in the source. Voisin's con...

conjecture | open | MathDB 页面 | 原文线索

V2-021099 | MATHDB-371873

Triple-intersection conjecture for Chow groups of smooth hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a smooth hypersurface, let $h\in A^1(X)$ denote the hyperplane class, and let $i_1,i_2,i_3>0$. Triple-intersection conjecture. ... The s...

conjecture | open | MathDB 页面 | 原文线索

V2-021100 | MATHDB-371884

The combinatorial Hodge conjecture for projective toric varieties

摘录 / EXCERPT: Let Σ be a rational fan defining a projective toric variety X_Σ. The combinatorial intersection cohomology is denoted $IH_{\mathrm{comb}}^*(\Sigma,\mathbb{Q})$, and f...

conjecture | open | MathDB 页面 | 原文线索

V2-021101 | MATHDB-371895

Algebraization conjecture for coherent sheaves on quantum elliptic curves

摘录 / EXCERPT: Algebraization conjecture. (i) If $\lvert q\rvert=1$ and q is not a root of 1, then F is full and faithful. (ii) If, in addition, there exists $L>0$ such that for every...

conjecture | open | MathDB 页面 | 原文线索

V2-021102 | MATHDB-371896

Sturmfels's determinantal ideal conjecture for tensor train varieties

摘录 / EXCERPT: Let $\mathbf{k}=(k_0,\dots,k_n)$ and $\mathbf{r}=(r_1,\dots,r_n)$, let ψ be a $k_0\times\cdots\times k_n$ tensor of indeterminates, and let ψ^i denote its flattenings. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-021103 | MATHDB-371897

Binary tensor train Gröbner basis conjecture

摘录 / EXCERPT: Let $\mathbf{k}=(2)_{n+1}$ and $\mathbf{r}=(1,r_2,\dots,r_{n-1},1)$. For each i, let $G_i^{(r_i+1)}$ denote the set of (r_i+1)-minors of the flattening ψ^{i-1}, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021104 | MATHDB-371898

Sum-of-squares conjecture for isotropic RR discriminant factors

摘录 / EXCERPT: Let V be the variety under consideration, let its RR discriminant be defined by a polynomial, and call a factor isotropic when it belongs to the isotropic part of that discrimina...

conjecture | open | MathDB 页面 | 原文线索

V2-021105 | MATHDB-371899

RR discriminant degree conjectures for products of projective spaces

摘录 / EXCERPT: For the products $\mathbb P^1\times\mathbb P^{n-1}$ and $\mathbb P^2\times\mathbb P^{n-1}$, consider the corresponding RR discriminants and their nonisotropic parts. Product RR dis...

conjecture | open | MathDB 页面 | 原文线索

V2-021106 | MATHDB-371906

Flexibility conjecture for affine spherical varieties

摘录 / EXCERPT: Flexibility conjecture. X is flexible.

conjecture | open | MathDB 页面 | 原文线索

V2-021107 | MATHDB-371928

Gammage-McBreen-Webster conjecture on multiplicative hypertoric varieties

摘录 / EXCERPT: Gammage-McBreen-Webster conjecture. There exists an additive variety $\mathfrak{M}$, a divisor $\mathfrak{D} \in \mathfrak{M}$, and an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-021108 | MATHDB-371930

Billey-Fan-Losonczy's Lehmer-code conjecture for rationally smooth elements

摘录 / EXCERPT: Billey-Fan-Losonczy's conjecture. The interval $[e,w]$ admits a Lehmer code.

conjecture | solved | MathDB 页面 | 原文线索

V2-021109 | MATHDB-371931

Richmond-Slofstra's Grassmannian BP-decomposition conjecture

摘录 / EXCERPT: Richmond-Slofstra's conjecture. The element w has a Grassmannian BP decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-021110 | MATHDB-371938

Bonnafé's fixed-point conjecture for two-sided cells

摘录 / EXCERPT: Let G be a complex reductive group with Cartan subalgebra $\mathfrak{t}$, Weyl group W, and $A=\mathbb{C}[\mathfrak{t}\oplus\mathfrak{t}^*]^{\mathrm{sgn}}$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-021111 | MATHDB-371939

Kivinen's symbolic-Losev variety isomorphism conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra. Choose a well-chosen $\mathbb{Q}$-factorial terminalization ... and let $X_{\mathfrak{g},\mathrm{Losev}}$ be the intermediate symplecti...

conjecture | open | MathDB 页面 | 原文线索

V2-021112 | MATHDB-371940

Kivinen's fixed-point and equivariant K-theory conjecture for symbolic varieties

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra with Weyl group W, let $\mathfrak{C}_W$ be the set of two-sided cells in W, and let $\mathbf{a}:\mathfrak{C}_W\to\mathbb{N}$ be Lu...

conjecture | open | MathDB 页面 | 原文线索

V2-021113 | MATHDB-371976

Reduced path-relation scheme conjecture for principal quiver Grassmannians

摘录 / EXCERPT: Reduced scheme conjecture. For every principal quiver Grassmannian, the scheme structure defined by the quadratic relations corresponding to all paths in Q is reduced.

conjecture | open | MathDB 页面 | 原文线索

V2-021114 | MATHDB-371981

Bass's finite generation conjecture for algebraic K-theory

摘录 / EXCERPT: Let X be a scheme of finite type over $\operatorname{Spec}\mathbf{Z}$, and let $n\geq 0$. The K'-theory (also called G-theory) group $K'_n(X)$ is the abelian group associated...

conjecture | open | MathDB 页面 | 原文线索

V2-021115 | MATHDB-371982

Arithmetic finiteness conjecture for standard Milnor-Witt modules

摘录 / EXCERPT: Let $S=\operatorname{Spec}\mathbf{Z}$. For an R-linear MW-module M over S, write $\big(\mathrm{FT}^{R}\big)$ for the property that, for every S-scheme X of finite type, al...

conjecture | open | MathDB 页面 | 原文线索

V2-021116 | MATHDB-371992

Moraga-Yeong algebraic hyperbolicity conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of dimension $n\geq 2$, let K_X be its canonical line bundle, and let L be an ample line bundle on X. Moraga-Yeong's conjecture. The co...

conjecture | open | MathDB 页面 | 原文线索

V2-021117 | MATHDB-371999

Morel's stable $\mathbb{A}^1$-connectivity conjecture

摘录 / EXCERPT: Morel's stable $\mathbb{A}^1$-connectivity conjecture. Under these hypotheses, $L_{\operatorname{mot}}E$ is n-connective.

conjecture | open | MathDB 页面 | 原文线索

V2-021118 | MATHDB-372017

Piatetski-Shapiro-Shafarevich conjecture on fields of definition of varieties of CM type

摘录 / EXCERPT: A variety is said to be of CM type when it has the complex multiplication property described in the surrounding theory. Piatetski-Shapiro-Shafarevich conjecture. Varieties of CM ty...

conjecture | open | MathDB 页面 | 原文线索

V2-021119 | MATHDB-372024

Conjecture on the decreasing proportion of toric Fano varieties with exceptional collections

摘录 / EXCERPT: Monotonicity and vanishing conjecture. The function $f(d)$ is strictly monotone decreasing for $d>1$, and

conjecture | open | MathDB 页面 | 原文线索

V2-021120 | MATHDB-372056

The maximal transcendence-degree conjecture for double zeta values of equal parity

摘录 / EXCERPT: Maximal transcendence-degree conjecture. The transcendence degree of the $\mathbb Q$-extension field generated by this set is maximal.

conjecture | open | MathDB 页面 | 原文线索

V2-021121 | MATHDB-372074

O'Grady's covering-family conjecture for Lagrangian subvarieties

摘录 / EXCERPT: O'Grady's conjecture. There is a positive integer $a\geq 1$ and a family of Lagrangian subvarieties $\pi:\mathcal{Z}_S\to X_n$ over a base S such that $\mathcal{Z}_S$ covers...

conjecture | open | MathDB 页面 | 原文线索

V2-021122 | MATHDB-372166

Darmon’s large residual image conjecture for GL_2-type abelian varieties

摘录 / EXCERPT: Let K be a totally real field and L a number field. Let A/L be an abelian variety of GL_2-type such that ... For a prime $\mathfrak{p}$ of K above a rational prime...

conjecture | open | MathDB 页面 | 原文线索

V2-021123 | MATHDB-372192

Bass-Quillen conjecture on projective modules over polynomial rings

摘录 / EXCERPT: Let A be a regular ring of finite Krull dimension, and let $n\ge 1$. A finitely generated projective module over $A[x_1,\ldots,x_n]$ is extended from A if it is obtained from a...

conjecture | open | MathDB 页面 | 原文线索

V2-021124 | MATHDB-372217

Kawamata-Morrison cone conjecture for K-trivial fiber spaces

摘录 / EXCERPT: Let $f:X\to S$ be a K-trivial fiber space, meaning a normal $\mathbb Q$-factorial klt pair (X,Δ) endowed with a proper surjective morphism with connected fibres such that...

conjecture | open | MathDB 页面 | 原文线索

V2-021125 | MATHDB-372227

The virtually nilpotent fundamental-group conjecture for special and h-special varieties

摘录 / EXCERPT: Virtually nilpotent fundamental-group conjecture. The fundamental group $\pi_1(X)$ is virtually nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-021126 | MATHDB-372228

The virtually nilpotent fundamental-group conjecture for h-special and special varieties

摘录 / EXCERPT: Nilpotency conjecture. The fundamental group $\pi_1(X)$ is virtually nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-021127 | MATHDB-372241

Generic projection invariance of Euclidean distance degree for arbitrary varieties

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^n$ be an affine variety, let $K\subseteq\mathbb{R}^n$ be a generic linear subspace with $\dim K=k<n-d$, where $d=\dim X$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021128 | MATHDB-372273

Geometric height pairing and Lear extension conjecture

摘录 / EXCERPT: Geometric height pairing and Lear extension conjecture. There is a canonical isomorphism of $\mathbb{Q}$-line bundles over $\overline{S}$:

conjecture | open | MathDB 页面 | 原文线索

V2-021129 | MATHDB-372274

Beilinson--Bloch positivity conjectures

摘录 / EXCERPT: Beilinson--Bloch positivity conjectures. For $p\leq \frac{d+1}{2}$, the map

conjecture | open | MathDB 页面 | 原文线索

V2-021130 | MATHDB-372275

Griffiths's conjecture on incidentally trivial cycles

摘录 / EXCERPT: Griffiths's conjecture on incidentally trivial cycles. For every $p\in\{1,\ldots,d\}$ and every $z\in I^p(X)$, there is a positive integer N such that

conjecture | open | MathDB 页面 | 原文线索

V2-021131 | MATHDB-372286

Godsil's edge-connectivity conjecture for colour classes in association schemes

摘录 / EXCERPT: Godsil's conjecture. If G is a connected graph which is a colour class in an association scheme, then

conjecture | open | MathDB 页面 | 原文线索

V2-021132 | MATHDB-372341

Shafarevich conjecture for varieties with large fundamental group

摘录 / EXCERPT: Let X be a smooth complex projective variety. Its fundamental group is large if, for every closed irreducible positive-dimensional subvariety $Z\subset X$, the image ... is infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-021133 | MATHDB-372342

De Oliveira-Katzarkov-Ramachandran deformation conjecture

摘录 / EXCERPT: Let $f:\mathscr{X}\to\mathbb{D}$ be a smooth projective family over the unit disk, and write $X_t=f^{-1}(t)$. A fiber has big fundamental group when the image of the fundamental gr...

conjecture | open | MathDB 页面 | 原文线索

V2-021134 | MATHDB-372343

Generalized Green-Griffiths-Lang conjecture

摘录 / EXCERPT: Let X be a smooth quasi-projective variety. A variety is pseudo-Picard hyperbolic if it satisfies the punctured-disk extension property outside a proper Zariski closed subset, an...

conjecture | open | MathDB 页面 | 原文线索

V2-021135 | MATHDB-372345

Semisimple big-quotient conjecture for strong hyperbolicity

摘录 / EXCERPT: Let X be a quasi-projective normal variety. A finitely generated group is semisimple if every abelian normal subgroup is finite. A quotient $\varrho:\pi_1(X)\twoheadrightarrow G$...

conjecture | open | MathDB 页面 | 原文线索

V2-021136 | MATHDB-372347

Revised Campana conjecture on virtually nilpotent fundamental groups

摘录 / EXCERPT: Let X be a smooth quasi-projective variety. It is special or h-special according to the notions defined in the source, and a group is virtually nilpotent if it has a nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-021137 | MATHDB-372348

Claudon-Höring-Kollár conjecture on projective varieties with quasi-projective universal cover

摘录 / EXCERPT: Let X be a complex projective manifold with infinite fundamental group, and let $\widetilde{X}$ be its universal cover. Suppose that $\widetilde{X}$ is quasi-projective. Claudon-...

conjecture | open | MathDB 页面 | 原文线索

V2-021138 | MATHDB-372349

Kollár's structure conjecture for varieties with big fundamental group

摘录 / EXCERPT: Let X be a smooth projective variety with big fundamental group and Kodaira dimension satisfying ... A fundamental group is big in the sense defined in the source. Kollár's structu...

conjecture | open | MathDB 页面 | 原文线索

V2-021139 | MATHDB-372362

Equality of the configuration varieties for finite representation type algebras

摘录 / EXCERPT: Let Λ be a finite representation type algebra, and let $\widetilde{\mathcal{U}}_\Lambda$ and $\widetilde{\mathcal{M}}_\Lambda$ be the affine varieties associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-021140 | MATHDB-372389

Rational integrability criterion for diagonal hypersurface reciprocals

摘录 / EXCERPT: Rational integrability conjecture. The function f is rationally integrable with respect to $x_1,\ldots,x_m$ if and only if $n\neq m$. This extends the discussion of rational inte...

conjecture | open | MathDB 页面 | 原文线索

V2-021141 | MATHDB-372393

Monotonicity conjecture for the three-variable violation sets

摘录 / EXCERPT: Monotonicity conjecture. For any $n\in\mathbf{N}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-021142 | MATHDB-372443

Algebraic homomorphism conjecture for the tautological projection on the moduli of abelian varieties

摘录 / EXCERPT: Algebraic homomorphism conjecture. The homomorphism property holds for all classes $d6fc,d6fd\in\mathsf{CH}^*(\mathcal{A}_g)$ defined over $d53e$.

conjecture | open | MathDB 页面 | 原文线索

V2-021143 | MATHDB-372469

Exceptional-curve conjecture for cyclic quotient groups

摘录 / EXCERPT: Exceptional-curve conjecture. $\operatorname{EC}(a+b+c)$ corresponds to the exceptional curves of $G\text{-}\operatorname{Hilb}(\mathbb{C}^{3})$.

conjecture | open | MathDB 页面 | 原文线索

V2-021144 | MATHDB-372495

The cohomological comparison conjecture for algebraic double loop spaces of flag varieties

摘录 / EXCERPT: Let Fl_{n+1} be the flag variety, let β be a strictly monotonic class, and write $\Omega^{2}_{\beta}(\operatorname{Fl}_{n+1})$ for the corresponding algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-021145 | MATHDB-372497

Scholze's fiber product conjecture for Igusa stacks

摘录 / EXCERPT: Let (G,X) be a Shimura datum with associated $G(\mathbb{C})$-conjugacy class of minuscule cocharacters $[\mu^{-1}]$ and reflex field E_0. Fix a prime p, let E be the comple...

conjecture | open | MathDB 页面 | 原文线索

V2-021146 | MATHDB-372505

Ekedahl-Shepherd-Barron-Taylor-Bost algebraicity conjecture for foliations

摘录 / EXCERPT: Let $R\subset\mathbb{C}$ be a finitely-generated $\mathbb{Z}$-algebra, and let M be a smooth R-scheme. A foliation $\mathscr{F}\subset T_{M/R}$ is a sub-bundle of Lie algebras....

conjecture | open | MathDB 页面 | 原文线索

V2-021147 | MATHDB-372506

Integral-points conjecture for Betti moduli spaces

摘录 / EXCERPT: Let X be a smooth projective complex variety, and let $M_B(X,r)$ denote its Betti moduli space of rank r representations as defined in the paper. An integral point is a point d...

conjecture | open | MathDB 页面 | 原文线索

V2-021148 | MATHDB-372507

Theta-vanishing conjecture for isomonodromy actions

摘录 / EXCERPT: Let $f:X\to S$ be a smooth projective morphism of smooth complex varieties, and fix $s\in S$. Let $\Theta_{X/S}$ be the quantity on the Dolbeault moduli space appearing in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-021149 | MATHDB-372512

Minets' critical CoHA conjecture for zero-dimensional sheaves

摘录 / EXCERPT: Let X be a quasiprojective 3-Calabi-Yau variety. Let $\operatorname{W}^+(X)$ be the algebra generated by symbols $T_m^+(\lambda)$, with $m\in\mathbb{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021150 | MATHDB-372514

The Betti Langlands duality conjecture for BPS cohomology

摘录 / EXCERPT: Let M be a three-manifold and let G and $G^{\vee}$ be Langlands dual groups. Write $\mathfrak{M}_G(M)$ and $\mathfrak{M}_{G^{\vee}}(M)$ for the corresponding stacks of local sy...

conjecture | open | MathDB 页面 | 原文线索

V2-021151 | MATHDB-372590

Chevalley restriction conjecture for commuting schemes

摘录 / EXCERPT: Chevalley restriction conjecture for commuting schemes. The morphism ϑ is an isomorphism. This asserts that invariant functions on commuting n-tuples are completely det...

conjecture | open | MathDB 页面 | 原文线索

V2-021152 | MATHDB-372598

Goldring-Koskivirta's cone conjecture for Hodge type Shimura varieties

摘录 / EXCERPT: Goldring-Koskivirta's cone conjecture. The two cones coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-021153 | MATHDB-372599

The de Rham weight part of Serre's conjecture for Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura datum of Hodge type. Let $\mathbb{T}$ be the spherical Hecke algebra and let $\mathfrak{m}\subseteq\mathbb{T}$ be a maximal ideal. For an algebraic represe...

conjecture | open | MathDB 页面 | 原文线索

V2-021154 | MATHDB-372600

Generic de Rham cohomology concentration conjecture

摘录 / EXCERPT: Let $\overline{\operatorname{Sh}}$ be the Siegel threefold or a unitary Shimura variety, and let $\mathfrak{m}\subseteq\mathbb{T}$ be generic and non-Eisenstein. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021155 | MATHDB-372605

Rationality conjecture for Betti strata of normal functions

摘录 / EXCERPT: Let S be a quasi-projective complex variety, let $f\colon X\to S$ be a smooth family of projective varieties of relative dimension d, and let N be a relatively ample line bun...

conjecture | open | MathDB 页面 | 原文线索

V2-021156 | MATHDB-372623

The generalized Lax conjecture for rigidly convex sets

摘录 / EXCERPT: Let $p\in\mathbb{R}[\mathbf{x}]$ be a real-zero (RZ) polynomial with $p(\mathbf{0})=1$. For a polynomial f, write $\operatorname{rcs}(f)$ for its rigidly convex set. Generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-021157 | MATHDB-372637

Characterization of $(\mathcal N,0)$-ats semialgebraic sets

摘录 / EXCERPT: A semialgebraic set $\EuScript T\subset\mathbb R^n$ is locally connected by analytic paths if, for every $x\in\EuScript T$ and every open semialgebraic neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-021158 | MATHDB-372642

The field-independence conjecture for volume polynomials

摘录 / EXCERPT: Let k be a field, and let $\mathbb{V}^d_n(\mathbb{R},k)$ denote the set of volume polynomials over k. The field-independence conjecture. The set of volume polynomials over k...

conjecture | open | MathDB 页面 | 原文线索

V2-021159 | MATHDB-372643

The minor-approximation conjecture for volume polynomials

摘录 / EXCERPT: Let k be a field. A minor of a volume polynomial is obtained by applying deletions and contractions of variables, and limits are taken in the relevant coefficient space. The mino...

conjecture | open | MathDB 页面 | 原文线索

V2-021160 | MATHDB-372654

Vergne's conjecture on rigid complex nilpotent Lie algebras

摘录 / EXCERPT: Vergne's conjecture. There are no rigid complex nilpotent Lie algebras in L_n.

conjecture | open | MathDB 页面 | 原文线索

V2-021161 | MATHDB-372679

Cadoret-Tamagawa genus-growth conjecture for torsion multisections

摘录 / EXCERPT: Cadoret-Tamagawa conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-021162 | MATHDB-372688

Non-rationality conjecture for Mordell-Weil and Néron-Severi height specializations

摘录 / EXCERPT: Non-rationality conjecture. The specializations are not rational in t with coefficients in $K_0(\operatorname{Stck}_{\mathbb{C}})[\mathbb{L}^{-1}][v]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021163 | MATHDB-372691

Berman's Gibbs stability criterion for Fano varieties

摘录 / EXCERPT: Let X be a Fano variety, and let $b3(X)$ denote the Gibbs stability invariant. Recall that K-stability and K-semistability are algebraic stability conditions for Fano varieties....

conjecture | open | MathDB 页面 | 原文线索

V2-021164 | MATHDB-372730

Duality for boundary pro-étale cohomology

摘录 / EXCERPT: Let Y be a smooth partially proper rigid analytic variety over K of dimension d, let ∂Y denote its boundary, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-021165 | MATHDB-372735

Characterization of generically reduced elementary components for two-step nestings

摘录 / EXCERPT: The nesting conjecture. The variety $V_{\underline{\boldsymbol{h}}}$ is a generically reduced elementary component if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-021166 | MATHDB-372738

Self-reciprocity conjecture for the partial Hasse polynomials of the double cover of W_{17}

摘录 / EXCERPT: Let Π_{17} be the double cover of the Teichmüller curve $W_{17}\hookrightarrow\mathbb{H}^2/\mathrm{SL}(\mathcal{O}_{17}\oplus\mathcal{O}^{\vee}_{17})$. For $j\in\{1,2\}$ and ev...

conjecture | open | MathDB 页面 | 原文线索

V2-021167 | MATHDB-372752

Rigidity conjecture for Calabi-Yau families with isolated singularities

摘录 / EXCERPT: Let $f\colon (\mathcal{X},\mathcal{L})\to S$ be a flat family of n-dimensional polarized smooth Calabi-Yau manifolds over a quasiprojective curve S. Let $\overline{S}$ be a pro...

conjecture | open | MathDB 页面 | 原文线索

V2-021168 | MATHDB-372768

Odaka's K-moduli conjecture for K-polystable polarized varieties

摘录 / EXCERPT: Let a polarized variety mean a projective variety equipped with a polarization, and consider the moduli space of K-polystable polarized varieties. K-moduli conjecture. A set of K-p...

conjecture | open | MathDB 页面 | 原文线索

V2-021169 | MATHDB-372774

Extremal-contraction conjecture for slope-unstable Fano tangent bundles

摘录 / EXCERPT: Let X be a Fano manifold, let $n=\dim X$, and define the $(-K_X)$-slope of a subsheaf $\mathcal{E}\subseteq T_X$ by ... The tangent bundle has $(-K_X)$-slope $\mu(T_X)=(-K_X)^n/n$....

conjecture | open | MathDB 页面 | 原文线索

V2-021170 | MATHDB-372779

Weyl-invariant square-root cluster functions conjecture

摘录 / EXCERPT: Let $\mathcal X_{|\mathcal A_n|}$ be the cluster variety associated with the $\mathcal A_n$-quiver, let $O(\sqrt{\mathcal X_{|\mathcal A_n|}})$ be the ring of universal Laurent pol...

conjecture | open | MathDB 页面 | 原文线索

V2-021171 | MATHDB-372819

Fu-Vial's multiplicative Orlov conjecture

摘录 / EXCERPT: Let X and X' be derived equivalent hyper-Kähler varieties. The Chow motive $h(X)$ is equipped with its cup-product structure, written $(h(X),\cup)$. Fu-Vial's multiplicative Or...

conjecture | open | MathDB 页面 | 原文线索

V2-021172 | MATHDB-372820

The homological multiplicative Orlov conjecture

摘录 / EXCERPT: Let X and X' be derived equivalent hyper-Kähler varieties, and let $h_{\mathrm{hom}}(X)$ denote the homological motive of X, equipped with its cup-product. The homological mu...

conjecture | open | MathDB 页面 | 原文线索

V2-021173 | MATHDB-372821

The Fourier vanishing conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X and Y be hyper-Kähler varieties of dimension $2n$, and let $\mathfrak{F}$ and $\mathfrak{F}^{-1}$ be the Fourier transforms and their inverses considered in the source, w...

conjecture | open | MathDB 页面 | 原文线索

V2-021174 | MATHDB-372838

Halpern-Leistner's spanning conjecture for Fano varieties

摘录 / EXCERPT: Halpern-Leistner's spanning conjecture. For any Fano variety X, there exist $\tau\in\mathrm{H}^2(X)$ and a sector $\mathscr{S}\subset\mathbf{C}^*$ such that, for every τ-adm...

conjecture | open | MathDB 页面 | 原文线索

V2-021175 | MATHDB-372839

The Kuznetsov decomposition conjecture for Fano complete intersections

摘录 / EXCERPT: The Kuznetsov decomposition conjecture. For a Fano complete intersection $X\subset\mathbf{P}^n$, there exist $\tau\in\mathrm{H}^\bullet(X)$ and a sector $\mathscr{S}$ such that the...

conjecture | open | MathDB 页面 | 原文线索

V2-021176 | MATHDB-372840

Ballini-Lombardo-Verzobio's L^1 Katz-Sarnak conjecture for $\mathcal A_g$

摘录 / EXCERPT: Let $g\ge 2$ be an integer. For each prime l, let $v_l(t)$ denote the local factor defined from the trace distribution in $\operatorname{GSp}_{2g}(\mathbb{Z}_l)$, with $v_p(t)=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-021177 | MATHDB-372842

Power-saving approximation of local factors

摘录 / EXCERPT: Let $v_{l,k}(f)$ be the finite-level local factor and let $v_l(f)$ be its limit. Power-saving local-factor approximation. There is a constant $\delta>0$ such that ... This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-021178 | MATHDB-372859

Hasse-Weil conjecture for motivic pieces of curves

摘录 / EXCERPT: Let C be a curve over a number field K with an action by a finite group G, let τ be an irreducible representation of G, and let C^τ be the corresponding motivic p...

conjecture | open | MathDB 页面 | 原文线索

V2-021179 | MATHDB-372861

Deligne's period conjecture for superelliptic curve motivic pieces

摘录 / EXCERPT: Let $C/\mathbb{Q}(\zeta_m)$ be a superelliptic curve $y^m=f(x)$ as in the paper, let τ be a character of C_m, and fix an embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-021180 | MATHDB-372873

Česnavičius's compactification conjecture for isotrivial reductive groups

摘录 / EXCERPT: Let S be a Noetherian scheme and let G be an isotrivial reductive group over S, meaning that G becomes split after a finite étale cover of S. A G-equivariant S-fiberw...

conjecture | open | MathDB 页面 | 原文线索

V2-021181 | MATHDB-372874

Generation-degree conjecture for the square variety of skew-symmetric matrices

摘录 / EXCERPT: Let X be a real skew-symmetric $n\times n$ matrix, and let $\mathcal{V}_{X^2}$ denote the variety of symmetric matrices arising as squares X^2. Generation-degree conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-021182 | MATHDB-372875

Generic ED and GD degree formulas for surfaces and threefolds in ${\rm Gr}(2,4)$

摘录 / EXCERPT: Let a surface in ${\rm Gr}(2,4)$ be defined by generic polynomials of degrees d_1,d_2, and let a threefold in ${\rm Gr}(2,4)$ be defined by a generic polynomial f of degree d...

conjecture | open | MathDB 页面 | 原文线索

V2-021183 | MATHDB-372876

Optimal-line intersection conjecture for Chow threefolds

摘录 / EXCERPT: Let $\mathcal{C}\subset\mathbb{P}^3$ be a curve, and consider the associated Chow threefold, secant surface, and tangent curve of lines, together with a data line and an optimal li...

conjecture | refuted | MathDB 页面 | 原文线索

V2-021184 | MATHDB-372877

SD-degree conjecture for Chow threefolds and secant surfaces

摘录 / EXCERPT: Let $\mathcal{C}\subset\mathbb{P}^3$ be a curve, and let the Chow threefold and secant surface associated with $\mathcal{C}$ have Grassmannian-distance degree ${\rm GDdegree}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021185 | MATHDB-372878

Explicit-generator conjecture for squared Schubert varieties

摘录 / EXCERPT: For $1\leq i<j\leq n$, let $\mathcal{S}_{ij}$ be the Schubert variety in ${\rm Gr}(2,n)$, let $\mathcal{S}_{ij}^2$ be its square image, let P be the associated symmetric matrix,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-021186 | MATHDB-372879

Degree recursion conjecture for squared Schubert varieties

摘录 / EXCERPT: Let $1\leq i<j\leq n$ with $j<n$, and let $\mathcal{S}_{ij}^2$ be the square image of the Schubert variety $\mathcal{S}_{ij}$. Degree recursion conjecture. The degrees satisfy ... an...

conjecture | open | MathDB 页面 | 原文线索

V2-021187 | MATHDB-372880

ED-degree formula conjecture for Schubert varieties

摘录 / EXCERPT: Let $1\leq i<j\leq n$, and let $\mathcal{S}_{ij}$ be the Schubert variety in ${\rm Gr}(2,n)$. ED-degree conjecture. If $i<j-1$, then ... The formula was verified in the paper for...

conjecture | open | MathDB 页面 | 原文线索

V2-021188 | MATHDB-372881

GD-degree-two conjecture for non-extreme Schubert varieties

摘录 / EXCERPT: Let $\mathcal{S}_{ij}$ be a Schubert variety in ${\rm Gr}(2,n)$. GD-degree conjecture. If $1<j-i<n-1$, then the Schubert variety $\mathcal{S}_{ij}$ has GD degree 2. The paper rep...

conjecture | open | MathDB 页面 | 原文线索

V2-021189 | MATHDB-372882

Distinguished representations and the moment map conjecture

摘录 / EXCERPT: Distinguished-representation moment map conjecture. If $Z(\pi)^\vee$ is X-distinguished, then

conjecture | open | MathDB 页面 | 原文线索

V2-021190 | MATHDB-372902

Face-relative-interior conjecture for stubborn forms

摘录 / EXCERPT: Let $X\subset \mathbb{P}^n$ be a totally real variety, let P_X denote its cone of nonnegative forms, and let $\mathcal{F}$ be a face of P_X. Face-relative-interior conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-021191 | MATHDB-372909

Automorphic and geometric equivalence for commutative associative linear algebras

摘录 / EXCERPT: Automorphic-geometric equivalence conjecture. Despite the fact that

conjecture | open | MathDB 页面 | 原文线索

V2-021192 | MATHDB-372910

Automorphic and geometric equivalence in subvarieties of groups

摘录 / EXCERPT: Automorphic-geometric equivalence conjecture. In every subvariety Θ of the variety of all groups, automorphic equivalence coincides with geometric equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-021193 | MATHDB-372922

Existence of a totally real codimension-two linear slice of SO-star(5)

摘录 / EXCERPT: Let ${\textrm{SO}}^\star(5)$ denote the variety of orthogonal matrices with the relevant zero pattern, and let a totally real linear slice mean a codimension-two linear section who...

conjecture | open | MathDB 页面 | 原文线索

V2-021194 | MATHDB-372951

Deformation invariance of pseudo-effectivity and uniruledness in Kähler families

摘录 / EXCERPT: Let $f:X\to S$ be a family from a Kähler variety X onto a smooth, connected, and relatively compact curve S. Assume that every fiber X_t has canonical singularities. Deformat...

conjecture | open | MathDB 页面 | 原文线索

V2-021195 | MATHDB-372953

Das-Hacon's transcendental base point free conjecture

摘录 / EXCERPT: Let $(X,B+\boldsymbol{\beta})$ be a generalized klt Kähler pair, where X is a normal compact Kähler variety, and let $\alpha\in H_{\mathrm{BC}}^{1,1}(X,\mathbb{R})$ be a nef clas...

conjecture | open | MathDB 页面 | 原文线索

V2-021196 | MATHDB-372971

Rigidity and absolute irreducibility of reductions of automorphic Galois representations

摘录 / EXCERPT: Let F be the number field in the setup, let Π be a $\mathfrak d$-REASDC representation of $\operatorname{GL}_{2m}(\mathbb A_F)$, let E be a strong coefficient field of...

conjecture | open | MathDB 页面 | 原文线索

V2-021197 | MATHDB-372976

Zhu's unramified categorical local Langlands conjecture

摘录 / EXCERPT: Let u be a closed point of U, let $\LS_{\prescript{L}{}{G},F_u}$ be the local Langlands parameter stack, and let $\LS_{\prescript{L}{}{G},\cO_u}$ denote its unramified locus. L...

conjecture | open | MathDB 页面 | 原文线索

V2-021198 | MATHDB-372981

Mustață-Popa's local cohomological vanishing conjecture

摘录 / EXCERPT: Suppose X is embedded in a smooth variety Y of dimension d. Choose a log resolution ... of the pair (Y,X) that is an isomorphism away from X, and let ... Thus E is a simple...

conjecture | open | MathDB 页面 | 原文线索

V2-021199 | MATHDB-372983

Closure conjecture for positive tilted Richardson cells

摘录 / EXCERPT: Positive-part closure conjecture. The closure of the positive part equals the nonnegative part:

conjecture | open | MathDB 页面 | 原文线索

V2-021200 | MATHDB-372984

Regularity conjecture for the CW-complex of tilted Richardson cells

摘录 / EXCERPT: Regularity conjecture for tilted Richardson cells. The CW-complex in the preceding stratification is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-021201 | MATHDB-373018

Semiorthogonal decomposition conjecture for Fano schemes of lines on intersections of two quadrics

摘录 / EXCERPT: Let $X=Q_1\cap Q_2\subset\mathbb{P}^{n+2}$ be a smooth complete intersection of two quadrics in an odd-dimensional projective space. Let C be the hyperelliptic curve associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-021202 | MATHDB-373021

Kollár's almost abelianity conjecture for projective manifolds with Kodaira dimension zero

摘录 / EXCERPT: Kollár's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-021203 | MATHDB-373022

RCD homeomorphism conjecture for singular Kähler varieties

摘录 / EXCERPT: RCD homeomorphism conjecture. The metric completion of $(X^\circ,\omega,\omega^n)$ is a non-collapsed RCD space homeomorphic to X.

conjecture | open | MathDB 页面 | 原文线索

V2-021204 | MATHDB-373029

Oort's direct-sum conjecture for Newton polygons

摘录 / EXCERPT: Let p be a prime, and let ξ_1 and ξ_2 be Newton polygons that are realized in characteristic p. Oort's direct-sum conjecture. Then $\xi_1 \oplus \xi_2$ is realized in c...

conjecture | open | MathDB 页面 | 原文线索

V2-021205 | MATHDB-373033

Karpenko's conjecture on Chow rings and topological filtrations

摘录 / EXCERPT: Let G be a split semisimple linear algebraic group over a field F, let $B\subset G$ be a Borel subgroup, and let E be a generic G-torsor. Write $X=E/B$ and let ... be the can...

conjecture | open | MathDB 页面 | 原文线索

V2-021206 | MATHDB-373041

Multiplicity refinement for specialized spectral-invariant varieties

摘录 / EXCERPT: Let $V(I')$ be the affine variety defined by the specialized ideal I' associated with the specialized spectral-invariant system, and let q denote the period. A point is regular...

conjecture | open | MathDB 页面 | 原文线索

V2-021207 | MATHDB-373073

Canonical-basis characterization conjecture for totally nonnegative double flag varieties

摘录 / EXCERPT: Canonical-basis characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021208 | MATHDB-373104

Base-change criterion for fundamental group schemes under purely inseparable extensions

摘录 / EXCERPT: Let K/k be a purely inseparable extension of degree p, let X be a connected scheme proper over k, and let $x_K\in X_K(K)$ lie over $x\in X(k)$. Let $\mathcal{C}_X$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021209 | MATHDB-373120

The classical-limit conjecture for the augmentation ideal

摘录 / EXCERPT: Let L be a link, let $I_q(L)$ be the ideal of quantum relations annihilating the distinguished class $[L]$, let $I_q'(L)$ be the annihilator of its evaluation in antisymmetric co...

conjecture | open | MathDB 页面 | 原文线索

V2-021210 | MATHDB-373135

Tangency flattening conjecture for Hilbert-scheme tangent spaces

摘录 / EXCERPT: Tangency flattening conjecture. The displayed equality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-021211 | MATHDB-373136

Minor flattening conjecture for Hilbert-scheme tangent spaces

摘录 / EXCERPT: Minor flattening conjecture. The displayed equality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-021212 | MATHDB-373140

The local monodromy conjecture for motivic and topological zeta functions

摘录 / EXCERPT: Let $f \in \mathbb{C}[x_1,\dots,x_n] \setminus \mathbb{C}$ and assume $f(o)=0$. The local motivic zeta function $Z_{\operatorname{mot},o}(f;s)$ and the local topological zeta funct...

conjecture | open | MathDB 页面 | 原文线索

V2-021213 | MATHDB-373141

The monodromy conjecture for remarkable numbers

摘录 / EXCERPT: Let X be an n-dimensional smooth complex irreducible variety and let $f:X\to\mathbb A^1$ be a regular function with non-empty zero locus. Let α be a remarkable number of...

conjecture | open | MathDB 页面 | 原文线索

V2-021214 | MATHDB-373148

Flat-locus dimension conjecture for linear degenerations of Schubert varieties

摘录 / EXCERPT: Flat-locus dimension conjecture. A tuple $f\in R^{\iota}_{\bf d}$ is in the flat locus of π if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-021215 | MATHDB-373155

Wan-Zhang weak global conjecture for the spherical variety $\mathbf{X}$

摘录 / EXCERPT: Let F be a number field and let $G=\mathbf{GL}_4\times\mathbf{GL}_2$. Embed $\mathbf{GL}_2\times\mathbf{GL}_2$ into G by $(g_1,g_2)\mapsto(\operatorname{diag}(g_1,g_2),g_2)$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-021216 | MATHDB-373164

Support conjecture for mirror duals of Procesi bundles

摘录 / EXCERPT: Let S be the surface in the Hilbert-scheme setup, let $\mathcal{M}_n$ be the corresponding moduli space, and let $I\in\mathcal{M}_n^{\mathbb{T}}$ be a fixed ideal. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021217 | MATHDB-373178

Galashin-Lam's deformation-retract and filtration conjecture for open positroid varieties

摘录 / EXCERPT: Galashin-Lam's conjecture. There exists a non-algebraic deformation retract

conjecture | open | MathDB 页面 | 原文线索

V2-021218 | MATHDB-373181

Integrability conjecture for split tangent bundles on rationally connected manifolds

摘录 / EXCERPT: Let X be a projective manifold with a splitting ... Assume that X is rationally connected. Integrability conjecture. The subbundles V_1 and V_2 are integrable. The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-021219 | MATHDB-373191

Miyanishi's conjecture on endomorphisms of varieties

摘录 / EXCERPT: Miyanishi's conjecture. Then ϕ is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-021220 | MATHDB-373202

Jet-point counting conjecture for varieties over finite fields

摘录 / EXCERPT: Jet-point counting conjecture. Then there exists a constant D such that

conjecture | open | MathDB 页面 | 原文线索

V2-021221 | MATHDB-373203

The saturation conjecture for toric varieties

摘录 / EXCERPT: Let X_Σ be the toric variety and let $f_1,\ldots,f_k$ satisfy Assumption. Let $Y\subseteq\mathbb{C}^n$ be the affine variety constructed from these polynomials as above. Its...

conjecture | open | MathDB 页面 | 原文线索

V2-021222 | MATHDB-373211

Generic quadratic graph degree conjecture

摘录 / EXCERPT: Let m and s be nonnegative integers, and consider the rational map ... whose coordinates on $\mathbb P^m\setminus \operatorname{V}(z_0)$ are given by ... where the $\gamma_i(z)$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-021223 | MATHDB-373212

CC degree formula conjecture for four-electron truncation varieties

摘录 / EXCERPT: CC degree formula conjecture. For $d=4$ and $n\leq 12$, the CC degree of $V_{\{2\}}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-021224 | MATHDB-373218

The log Fano type conjecture for strongly globally +-regular varieties

摘录 / EXCERPT: Let (X,Δ) be a strongly globally $+$-regular variety, projective over an open subset of $\operatorname{Spec}(\mathbb{Z})$. The log Fano type conjecture asserts that...

conjecture | open | MathDB 页面 | 原文线索

V2-021225 | MATHDB-373219

The vanishing criterion for non-pseudo-effectivity of the canonical divisor

摘录 / EXCERPT: Let X be a klt projective variety over $\C$. Suppose that ... for $i>0$ and every semi-ample line bundle L on X. The vanishing criterion conjecture asserts that K_X is not ps...

conjecture | open | MathDB 页面 | 原文线索

V2-021226 | MATHDB-373262

Corti-Hanamura's motivic decomposition conjecture

摘录 / EXCERPT: Corti-Hanamura's motivic decomposition conjecture. Every such morphism admits a motivic decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-021227 | MATHDB-373271

The visibility-dimension claim for the abelian surface 9603.2.a.o

摘录 / EXCERPT: Visibility-dimension claim. This element is not visible in any abelian threefold; in particular, its visibility dimension is 4.

conjecture | open | MathDB 页面 | 原文线索

V2-021228 | MATHDB-373272

The abelian motive conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X be a hyper-Kähler variety, and let $\mathfrak{h}(X)$ denote its motive. For each degree i and prime ell, let $\mathbf{G}_{\ell,i}(X)$ be the ell-adic algebraic monodr...

conjecture | open | MathDB 页面 | 原文线索

V2-021229 | MATHDB-373273

Larsen's Galois maximality conjecture for compatible systems

摘录 / EXCERPT: Let K be a finitely generated field, let X be a smooth projective variety over K, and consider a $\mathds{Q}$-compatible system of Galois representations ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021230 | MATHDB-373281

Göttsche-Kool universal-function conjecture for rank-two virtual Segre numbers

摘录 / EXCERPT: Let S be the surface and M the moduli space defined above, let $s\in\mathbb Z$, and let $\alpha\in K(S)$ have $\operatorname{rk}(\alpha)=s$. The universal functions...

conjecture | open | MathDB 页面 | 原文线索

V2-021231 | MATHDB-373284

The G-noncommutative minimal model program conjecture

摘录 / EXCERPT: Let G be an algebraic group, let X be a smooth projective variety with a G-action, and let $\pi:X\to Y$ be a G-contraction. Let $\mathrm{Stab}(X_G)$ denote the space of sta...

conjecture | open | MathDB 页面 | 原文线索

V2-021232 | MATHDB-373285

Conjecture on higher-dimensional toric Fano manifolds with coupled but no ordinary Kähler-Einstein metrics

摘录 / EXCERPT: Higher-dimensional coupled Kähler-Einstein conjecture. For every integer $n\geq 4$, there exists a toric Fano manifold of dimension n that admits a two-coupled Kähler-Einstein me...

conjecture | open | MathDB 页面 | 原文线索

V2-021233 | MATHDB-373293

The global and local Strong Monodromy Conjecture

摘录 / EXCERPT: Let $f \in \operatorname{\mathbb{C}}[x_1,\ldots,x_n] \setminus \mathbb{C}$ with $f(0)=0$. Write $Z_f^{\operatorname{mot}}(s)$ and $Z_{f,0}^{\operatorname{mot}}(s)$ for the global a...

conjecture | open | MathDB 页面 | 原文线索

V2-021234 | MATHDB-373298

Conjecture on rational torsion orders for GL2-type abelian varieties

摘录 / EXCERPT: Torsion-order conjecture. The possible orders $|\operatorname{Tors}(A)(\mathbb Q)|$ are precisely the integers listed below:

conjecture | open | MathDB 页面 | 原文线索

V2-021235 | MATHDB-373299

Conjecture on prime divisors of rational torsion for GL2-type abelian varieties

摘录 / EXCERPT: Prime-divisor conjecture. If $g=2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-021236 | MATHDB-373340

The global integral-model conjecture for $\Gamma_1(p)$-level Shimura varieties

摘录 / EXCERPT: Assume that $({\mathcal G},\{\mu\})$ is strictly convex, that the divisor conjecture holds, and that an O_E-integral model ${\mathcal S}_{K_0}$ of ${\rm Sh}_{K_0}(G,X)_E$ with it...

conjecture | open | MathDB 页面 | 原文线索

V2-021237 | MATHDB-373341

The torus-torsor extension conjecture for $\Gamma_1(p)$-level covers

摘录 / EXCERPT: Assume the divisor conjecture and let $\pi:{\rm Sh}_{K_1}(G,X)_E% \ifbool{@display}{\longrightarrow}{\rightarrow}% {\rm Sh}_{K_0}(G,X)_E$ be the generic-fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-021238 | MATHDB-373342

The local integral-model conjecture for $\Gamma_1(p)$-level local Shimura varieties

摘录 / EXCERPT: Let $(G,\{\mu\},b)$ be a local Shimura datum, let ${\mathcal G}$ be parahoric, set $K_0={\mathcal G}(\mathbb Z_p)$, and let K_1 be the kernel of the reduction map to the reductiv...

conjecture | open | MathDB 页面 | 原文线索

V2-021239 | MATHDB-373345

Ducat's log rationality conjecture for three-dimensional Calabi-Yau pairs

摘录 / EXCERPT: Ducat's conjecture. Then the pair (X,B) is log rational. The conjecture extends the demonstrated surface and projective three-space examples, where Cremona transformations relate...

conjecture | open | MathDB 页面 | 原文线索

V2-021240 | MATHDB-373346

Geometric Jordan conjecture for the Cremona group

摘录 / EXCERPT: Let n be a positive integer, let X be an n-dimensional rational variety, and let G be a finite group acting on X. Geometric Jordan conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-021241 | MATHDB-373354

Sublinear projective Schwartz-Zippel conjecture

摘录 / EXCERPT: Let X be a projective variety of degree d and pure dimension k in ${\mathbb P}_{\mathbb Q}^n$, and let $X({\mathbb Q},B)$ denote its rational points of height at most B. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-021242 | MATHDB-373380

Clausen's étale-local light profiniteness conjecture for smooth p-adic analytic Artin stacks

摘录 / EXCERPT: Let X be a smooth p-adic analytic Artin stack, and let $L(X)$ denote its associated light condensed anima. Clausen's conjecture. Étale-locally, $L(X)$ is a light profinite anim...

conjecture | open | MathDB 页面 | 原文线索

V2-021243 | MATHDB-373390

Greer-Lian's modularity conjecture for special cycles on Siegel modular varieties

摘录 / EXCERPT: Greer-Lian's modularity conjecture. The series $\Theta_{\mathcal A_n}(q)$ is a holomorphic modular form of weight $2n$ for $\operatorname{SL}_2(\mathbb Z)$, valued in...

conjecture | open | MathDB 页面 | 原文线索

V2-021244 | MATHDB-373391

Virtual modularity conjecture for higher-codimension elliptic-factor cycles

摘录 / EXCERPT: Virtual modularity conjecture. The series $\Theta^{2,\mathrm{vir}}_{\mathcal A_n}$ is a holomorphic Siegel modular form of weight $2n$ for $\operatorname{Sp}_4(\mathbb Z)$, valued...

conjecture | open | MathDB 页面 | 原文线索

V2-021245 | MATHDB-373392

Looijenga-Saper-Stern conjecture for modularity in L2-cohomology

摘录 / EXCERPT: Let X be a Shimura variety with special cycles $\mathcal Z(m)$ whose generating series is modular in cohomology, and let X^* be its Satake-Baily-Borel compactification. Looijen...

conjecture | open | MathDB 页面 | 原文线索

V2-021246 | MATHDB-373393

Local constancy of the automorphic category for families of curves

摘录 / EXCERPT: Let $\pi:\mathfrak{X}\to S$ be the family of curves in the paper, let $\operatorname{Bun}_G(\pi_U)$ be the moduli stack of G-bundles on the base-changed family over a contractibl...

conjecture | open | MathDB 页面 | 原文线索

V2-021247 | MATHDB-373407

Hu's integral cycle class conjecture for Chow varieties

摘录 / EXCERPT: Hu's integral cycle class conjecture. For all $d\geq 0$ and $0\leq p\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-021248 | MATHDB-373408

Hu's rational cycle class conjecture for Chow varieties

摘录 / EXCERPT: Hu's rational cycle class conjecture. For all $d\geq 0$ and $0\leq p\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-021249 | MATHDB-373410

Generalised Franchetta conjecture for powers of hyperkähler varieties

摘录 / EXCERPT: Let $\mathcal{F}$ be the moduli stack of a locally complete family of polarised hyperkähler varieties, and let $\mathcal{X}\to\mathcal{F}$ be its universal family. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-021250 | MATHDB-373426

The Hasse-Witt invariant and mirror-map expansion conjecture

摘录 / EXCERPT: Let X_z be a family of Calabi-Yau n-folds defined over a perfect field of characteristic p, let α_z be a holomorphic n-form, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021251 | MATHDB-373427

The Calabi-Yau modular-form Hasse-Witt polynomial conjecture

摘录 / EXCERPT: Let ${\sf S}$ be the affine moduli scheme of pairs (X,α) with universal family ${\sf X}\to{\sf S}$ and universal holomorphic form...

conjecture | open | MathDB 页面 | 原文线索

V2-021252 | MATHDB-373458

Dual EL-shellability conjecture for Coxeter-type admissible sets

摘录 / EXCERPT: Dual EL-shellability conjecture. The poset $\widehat{\operatorname{Cox}(\mu)^K}$ is dual EL-shellable.

conjecture | open | MathDB 页面 | 原文线索

V2-021253 | MATHDB-373459

The Hodge-theoretic anabelian conjecture for varieties embedded in products of hyperbolic curves

摘录 / EXCERPT: Let X and Y be smooth, geometrically connected varieties over ... mathbb{C}$. Here $X($\mathbb{C})^{\sim}$ and $Y($mathbb{C})^{sim}$ denote their schematic homotopy types, and $m...

conjecture | open | MathDB 页面 | 原文线索

V2-021254 | MATHDB-373519

Zero-dimensionality conjecture for Weyl-Heisenberg-covariant SIC varieties

摘录 / EXCERPT: Let $d>3$, and consider the polynomial equations defining a SIC-POVM that is covariant under the action of the Weyl-Heisenberg group. Their solution set is an algebraic variety. Ze...

conjecture | open | MathDB 页面 | 原文线索

V2-021255 | MATHDB-373567

Koszul-dual and intersection-theoretic formula for the non-linear D-Fredholm index

摘录 / EXCERPT: Let X be a smooth scheme. Consider the non-linear $\mathcal{D}$-Fredholm index, its symbol complex, and the cycle defined by the associated graded symbol morphism in the total sp...

conjecture | open | MathDB 页面 | 原文线索

V2-021256 | MATHDB-373569

Zhang's moduli-space conjecture for hyperkähler varieties of K3^[n]-type

摘录 / EXCERPT: Let X be a hyperkähler variety of K3$^{[n]}$-type with $n\geq 2$. A K3 category is a category of K3 type, and let $\mathcal A_X^{[n]}$ denote its Hilbert-scheme-type construction...

conjecture | open | MathDB 页面 | 原文线索

V2-021257 | MATHDB-373571

The anti-canonical polar cylinder criterion for K-polystability of Fano varieties

摘录 / EXCERPT: Let X be a Fano variety with at most Kawamata log terminal singularities. An anti-canonical polar cylinder means a cylinder polarized by $-K_X$. The anti-canonical cylinder conje...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-021258 | MATHDB-373620

The virtual 2-step nilpotency conjecture for fundamental groups of special varieties

摘录 / EXCERPT: 2-step nilpotency conjecture. The group $\pi_1(X)$ is virtually nilpotent of class at most 2.

conjecture | open | MathDB 页面 | 原文线索

V2-021259 | MATHDB-373653

Rational Lawson homology conjecture for Chow varieties

摘录 / EXCERPT: Let $C_{p,d}(\mathbb{P}^n)$ denote the Chow variety of effective algebraic p-cycles of degree d in complex projective space, let $L_qH_k(-)_{\mathbb{Q}}$ denote Lawson homology...

conjecture | open | MathDB 页面 | 原文线索

V2-021260 | MATHDB-373693

The symmetric-part conjecture for lattice varieties

摘录 / EXCERPT: Symmetric-part conjecture. Each lattice variety $\mathcal{V}$ equals its symmetric part:

conjecture | open | MathDB 页面 | 原文线索

V2-021261 | MATHDB-373694

The bounded-arity symmetric-part conjecture for lattice varieties

摘录 / EXCERPT: Bounded-arity symmetric-part conjecture. For any lattice variety axiomatized by identities of arity at most k, one has

conjecture | open | MathDB 页面 | 原文线索

V2-021262 | MATHDB-373696

The Jacobi Bound Conjecture for differential algebraic varieties

摘录 / EXCERPT: Jacobi Bound Conjecture. The absolute dimension of Σ_1 satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-021263 | MATHDB-373714

Wachspress' conjecture on adjoint curves of regular real polycons

摘录 / EXCERPT: Let ${\mathcal P}$ be a regular real polycon, and let A be its adjoint curve. Write ${\mathcal P}_{\geq 0}$ for the closed real region defining the polycon, and $A(\mathbb R)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-021264 | MATHDB-373715

Wachspress' conjecture for regular polycons stable under line replacement

摘录 / EXCERPT: Let ${\mathcal P}$ be a regular real polycon. Assume that whenever one boundary conic of ${\mathcal P}$, or two adjacent boundary lines, is replaced by a line, the resulting polyco...

conjecture | open | MathDB 页面 | 原文线索

V2-021265 | MATHDB-373728

The non-abelian Leopoldt conjecture for eigenvariety components

摘录 / EXCERPT: Let G be a reductive group, let $K^p\subset G(\mathbb{A}_f^p)$ be a tame level, let $\operatorname{Iw}\subset G(\mathbb{Q}_p)$ be an Iwahori subgroup, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-021266 | MATHDB-373732

Euler stratification conjecture for the principal A-determinant

摘录 / EXCERPT: Let W and W' be coefficient points for the varieties $Y_{W,n}$ and $Y_{W',n}$, respectively, and let E_A denote the principal A-determinant, whose factors are the discrimin...

conjecture | open | MathDB 页面 | 原文线索

V2-021267 | MATHDB-373754

Structural rank conjecture for the p-adic logarithm matrix

摘录 / EXCERPT: Fix an embedding $\iota_p:\overline{\mathbb{Q}}\hookrightarrow\overline{\mathbb{Q}}_p$, let A be an abelian variety over $\overline{\mathbb{Q}}$ of dimension d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021268 | MATHDB-373760

Parshin's torsion conjecture for algebraic K-groups over finite fields

摘录 / EXCERPT: Let k be a finite field of characteristic $p>0$, and let X be a smooth projective variety over k. For each integer $n\geq 1$, let $K_n(X)$ denote the algebraic K-group of...

conjecture | open | MathDB 页面 | 原文线索

V2-021269 | MATHDB-373764

The geometry and categorical moduli space conjecture for the stack of all curves

摘录 / EXCERPT: The stack of all curves conjecture. 1. $\mathcal{M}_g^{\text{all}}$ satisfies Vakil's Murphy's law. 2. $\operatorname{Pic}(\mathcal{M}_g^{\text{all}})=0$. 3. The structure morphism

conjecture | open | MathDB 页面 | 原文线索

V2-021270 | MATHDB-373766

The finite generation conjecture for motivic cohomology of arithmetic schemes

摘录 / EXCERPT: Finite generation conjecture. For every $n\in\mathbb{Z}$ and every $i\leq 2n+1$, the group

conjecture | open | MathDB 页面 | 原文线索

V2-021271 | MATHDB-373776

The Hodge realization conjecture for Nori motivic local systems

摘录 / EXCERPT: Hodge realization conjecture. The functor $\iota_{X,\sigma}^{\textup{Hdg}}$ is fully faithful, with essential image stable under subquotients. This is a Hodge-type conjecture for N...

conjecture | open | MathDB 页面 | 原文线索

V2-021272 | MATHDB-373793

Potential good reduction conjecture for basic admissible pairs

摘录 / EXCERPT: Let $L/\mathbb{Q}_p$ be a p-adic field and let X/L be a smooth rigid analytic variety. Let $\mathrm{AdmPair}^{\mathrm{basic},\mathrm{pgr}}(X^\diamond)$ denote basic admissible...

conjecture | open | MathDB 页面 | 原文线索

V2-021273 | MATHDB-373801

Finiteness conjecture for the fundamental group of the regular locus

摘录 / EXCERPT: Finiteness conjecture. The fundamental group $\pi_1(M_\mathrm{reg})$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-021274 | MATHDB-373803

Beckmann's conjecture on simple atomic objects

摘录 / EXCERPT: Beckmann's conjecture. The object E is 1-obstructed.

conjecture | open | MathDB 页面 | 原文线索

V2-021275 | MATHDB-373804

Beckmann's hard Lefschetz conjecture for stable atomic sheaves

摘录 / EXCERPT: Beckmann's conjecture. The element $\operatorname{obs}_E(\eta)$ has the hard Lefschetz property for the Yoneda product on $\operatorname{Ext}^{\bullet}(E,E)$.

conjecture | open | MathDB 页面 | 原文线索

V2-021276 | MATHDB-373814

Projective Dimension Growth Conjecture

摘录 / EXCERPT: Let $V\subset\mathbb{P}^N$ be an irreducible projective variety of degree $d\geq 2$ defined over $\mathbb{Q}$, and let $N_{\mathrm{proj}}(V,B)$ denote the number of rational points...

conjecture | open | MathDB 页面 | 原文线索

V2-021277 | MATHDB-373815

Uniform projective Dimension Growth Conjecture

摘录 / EXCERPT: Let $V\subset\mathbb{P}^N$ be an irreducible projective variety, and let $C(V)$ and, where relevant, $\gamma(V)$ be the constants in the Projective Dimension Growth Conjecture. Uni...

conjecture | open | MathDB 页面 | 原文线索

V2-021278 | MATHDB-373822

Hindry-Pacheco conjecture on the positive-characteristic Arakelov inequality

摘录 / EXCERPT: Let $K = \overline{\mathbb{F}}_p(B)$ be the function field of a curve B over $\overline{\mathbb{F}}_p$ and let A/K be a g-dimensional abelian variety with semistable reductio...

conjecture | open | MathDB 页面 | 原文线索

V2-021279 | MATHDB-373834

Van den Bergh's NCCR existence conjecture for affine Gorenstein toric varieties

摘录 / EXCERPT: Van den Bergh's NCCR existence conjecture. Every affine Gorenstein toric variety has an NCCR.

conjecture | open | MathDB 页面 | 原文线索

V2-021280 | MATHDB-373837

Geometric Lang-Vojta conjecture for integral points on log-general-type pairs

摘录 / EXCERPT: Let k be an algebraically closed field, let B be a nonsingular irreducible algebraic curve over k with function field $K=k(B)$, and let (X,D) be a pair of log general type...

conjecture | open | MathDB 页面 | 原文线索

V2-021281 | MATHDB-373845

The Mordell-Weil rank bound via Mori fiber spaces

摘录 / EXCERPT: Let $X\to B$ be as in Theorem MMP, with X a Calabi-Yau variety. Suppose that B is birational to a Mori fiber space $B'\to U$ with fiber F, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021282 | MATHDB-373852

The integral-model conjecture for Igusa stacks and Shimura varieties

摘录 / EXCERPT: Integral-model conjecture. The composition

conjecture | open | MathDB 页面 | 原文线索

V2-021283 | MATHDB-373865

Factorization conjecture for LS discriminants of rational Landau diagrams

摘录 / EXCERPT: Let $\mathcal{L}$ be a rational Landau diagram, and let V_G be the associated variety with irreducible components $V_{G,\sigma}$. Let $\Delta_{\mathcal{L}}$ denote its LS discrim...

conjecture | open | MathDB 页面 | 原文线索

V2-021284 | MATHDB-373866

Reality conjecture for positive planar Landau diagrams

摘录 / EXCERPT: Let G be an outerplanar graph with cyclically labeled vertices in $[\ell]$, and let $\mathcal{L}=G_u\cup H_u$ be any planar Landau diagram. For a positive tuple $\mathbf{M}$ of e...

conjecture | open | MathDB 页面 | 原文线索

V2-021285 | MATHDB-373875

The generic spectrum conjecture for SL(2,R)-representations of closed surfaces

摘录 / EXCERPT: Generic spectrum conjecture. For a generic $\rho: \Sigma_g \longrightarrow \mathrm{SL}(2,\mathbb{R})$ which is not simple Fuchsian, the spectrum of ρ is closed and has empty i...

conjecture | open | MathDB 页面 | 原文线索

V2-021286 | MATHDB-373930

Algebraic-rank criterion and exterior-power description of plectic Heegner classes

摘录 / EXCERPT: Let A be the abelian variety, E the number field, $L_{\mathfrak c}$ the anticyclotomic extension, and let χ be a character. Let r be the number of primes in...

conjecture | open | MathDB 页面 | 原文线索

V2-021287 | MATHDB-373936

Colmez's conjecture on Faltings heights of CM abelian varieties

摘录 / EXCERPT: Colmez's conjecture. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-021288 | MATHDB-373937

The refined Colmez conjecture for individual CM periods

摘录 / EXCERPT: Refined Colmez conjecture. There exists an integer $m\geq1$ and an element $\beta=\beta(A,m,\phi)\in\omega_{\cal A}^{\otimes m}$ such that (a) β generates...

conjecture | open | MathDB 页面 | 原文线索

V2-021289 | MATHDB-373939

Lang-Vojta-Chevalley-Weil weakly special conjecture

摘录 / EXCERPT: Lang-Vojta-Chevalley-Weil conjecture. If X satisfies potential density of integral points, then X is weakly special.

conjecture | open | MathDB 页面 | 原文线索

V2-021290 | MATHDB-373940

Weakly special potential-density conjecture

摘录 / EXCERPT: Weakly Special Conjecture. If X is a weakly special variety over a number field K, then X has a potentially dense set of integral points.

conjecture | open | MathDB 页面 | 原文线索

V2-021291 | MATHDB-373941

Campana's specialness and potential-density conjecture

摘录 / EXCERPT: Campana's conjecture. X is special if and only if the set of integral points on X is potentially dense.

conjecture | open | MathDB 页面 | 原文线索

V2-021292 | MATHDB-374029

The Oblomkov-Rasmussen-Shende conjecture for Hilbert schemes of points on singular curves

摘录 / EXCERPT: Let $C\subset\mathbb{C}^2$ be a singular curve with a planar singularity at the origin, and let L be the algebraic link obtained by intersecting C with a sufficiently small 3...

conjecture | open | MathDB 页面 | 原文线索

V2-021293 | MATHDB-374030

Chen-Ngô surjectivity conjecture for the Hitchin morphism

摘录 / EXCERPT: Chen-Ngô's conjecture. For every point $s\in\mathscr{B}^r_X$, the fiber

conjecture | open | MathDB 页面 | 原文线索

V2-021294 | MATHDB-374041

The algebraic reduction conjecture for compact Kähler varieties

摘录 / EXCERPT: Let X be a compact Kähler variety. The algebraic reduction map of X is a meromorphic fibration associated with the maximal algebraic subfield of the field of meromorphic functi...

conjecture | open | MathDB 页面 | 原文线索

V2-021295 | MATHDB-374062

Zhang's relative dynamical Manin-Mumford conjecture for families

摘录 / EXCERPT: Let S be a smooth irreducible quasi-projective variety over ${\mathbb C}$, let K be a positive integer, and let ... be a family of endomorphisms with $\Phi(s,p)=(s,F_s(p))$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-021296 | MATHDB-374077

Ein-Lazarsfeld-Mustopa syzygy bundle stability conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, and let A and P be line bundles on X. Assume that A is very ample, and set $L_d:=dA+P$ for every positive integer d. Ein-Lazarsfel...

conjecture | open | MathDB 页面 | 原文线索

V2-021297 | MATHDB-374091

Tate conjecture in characteristic p for integral Shimura models

摘录 / EXCERPT: Let $k(v)$ be the residue field at v, let κ be a perfect field over $k(v)$, and let $x\in \mathscr{S}_K(\kappa)$. For each prime ℓ, write $I_{\ell,x}$ for the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-021298 | MATHDB-374116

Colliot-Thélène's conjecture on rationally connected varieties

摘录 / EXCERPT: Colliot-Thélène's conjecture. Let X be a rationally connected smooth variety over k. Then $X(k)$ is dense in $X(k_{\Omega})^{\operatorname{Br}_{\mathrm{nr}}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-021299 | MATHDB-374117

Wittenberg's descent conjecture for torsors under linear algebraic groups

摘录 / EXCERPT: Wittenberg's descent conjecture. Let X be a smooth k-variety and let G be a linear algebraic k-group. Let $f:Y\to X$ be a G-torsor with Y rationally connected. Assume t...

conjecture | open | MathDB 页面 | 原文线索

V2-021300 | MATHDB-374131

The local leakage-resilience conjecture for Massey schemes

摘录 / EXCERPT: Let C be an MDS code over a prime field $\mathbb{F}_p$, and let the Massey scheme be the secret-sharing scheme defined by C. A scheme is one-bit local leakage resilient if it r...

conjecture | open | MathDB 页面 | 原文线索

V2-021301 | MATHDB-374196

Bezrukavnikov-Gaitsgory-... relative Langlands duality conjecture for spherical varieties

摘录 / EXCERPT: Let G be a connected reductive group over $\mathbb C$, let $\fY$ be a smooth affine spherical G-variety with connected stabilizer in a Borel subgroup B of a general point, an...

conjecture | open | MathDB 页面 | 原文线索

V2-021302 | MATHDB-374197

Finkelberg-Gaitsgory-Travkin Koszul duality conjecture for spherical varieties

摘录 / EXCERPT: Let G be a connected reductive group, let $\fY$ be the spherical variety above, and let $B\subset G$ and $B^{\vee}\subset G^{\vee}$ be Borel subgroups. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021303 | MATHDB-374198

Finkelberg-Gaitsgory-Travkin $\bZ/2$-graded Koszul duality conjecture

摘录 / EXCERPT: Let $B\subset G$ and $B^{\vee}\subset G^{\vee}$ be Borel subgroups, and let $D^{\mathbb Z/2}$ denote the $\mathbb Z/2$-graded version of a graded category. $\mathbb Z/2$-graded Kos...

conjecture | open | MathDB 页面 | 原文线索

V2-021304 | MATHDB-374199

General-monodromy Koszul duality conjecture for spherical varieties

摘录 / EXCERPT: Let $\sT\subset B$ be a Cartan subgroup, let λ be a regular coweight of G, and let $\sT_{\lambda}$ be the subgroup of $\sT$ generated by $\exp(2\pi i\lambda)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021305 | MATHDB-374200

The Lagrangian correspondence conjecture for the Dolbeault geometric Langlands correspondence

摘录 / EXCERPT: Let $\mathbb L_H(d)$ be the Lagrangian correspondence constructed between the relevant Hitchin moduli space and the Hilbert scheme of points on T^*C, and let the Dolbeault geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-021306 | MATHDB-374258

Simpson-type completeness conjecture for quiver-variety descending manifolds

摘录 / EXCERPT: Let $\mathfrak M_{(0,-2i\zeta_{\mathbb{R}})}$ be the indicated quiver-variety moduli space, let p_0 be a fixed point of the $\mathbb{C}^{\star}$-action, and let $W^1(p_0)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-021307 | MATHDB-374259

Trace conjecture for non-nef Hodge bundles

摘录 / EXCERPT: Let $\mathcal{X}/C$ be an abelian scheme and suppose it does not admit an isotrivial abelian subscheme. If ... then for some finite étale cover $C'\twoheadrightarrow C$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-021308 | MATHDB-374260

Non-liftability conjecture for maps into isogeny leaves

摘录 / EXCERPT: Let Z be a smooth irreducible variety over k, and let $\varphi:Z\to I(x)$ be a nontrivial map, where $I(x)$ is an isogeny leaf. Non-liftability conjecture for maps into isogeny...

conjecture | open | MathDB 页面 | 原文线索

V2-021309 | MATHDB-374283

Goncharov's kernel conjecture for polylogarithmic groups

摘录 / EXCERPT: Let F be a field, let $B_n(F)$ be Goncharov's polylogarithmic group, and let ... be the associated map. Write $K^{(n)}_{2n-1}(F)_\mathbb Q$ for the n-th graded quotient of the...

conjecture | open | MathDB 页面 | 原文线索

V2-021310 | MATHDB-374285

Minimal-degree conjecture for non-degenerate integral curves in weighted projective three-space

摘录 / EXCERPT: Let C be a non-degenerate integral curve in the weighted projective space $\mathbf{P}(1,1,m,n)$, where m and n are positive integers. Write $n\bmod m$ for the remainder on di...

conjecture | open | MathDB 页面 | 原文线索

V2-021311 | MATHDB-374299

Classification conjecture for toric 2-Fano manifolds

摘录 / EXCERPT: Toric 2-Fano classification conjecture. The only toric 2-Fano manifolds are projective spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-021312 | MATHDB-374316

Necessary bitangency condition for full-dimensional medial-axis components

摘录 / EXCERPT: Let X be a codimension-one variety and let $\mathcal{X}_B=\bigcup_i X_i$ be its stratification with respect to a given polyhedral norm, assuming that this stratification exists....

conjecture | open | MathDB 页面 | 原文线索

V2-021313 | MATHDB-374337

The localisation injectivity and vanishing conjecture for symmetric powers

摘录 / EXCERPT: Localisation injectivity and vanishing conjecture. For all $0\leq i\leq n$, this map is injective, and

conjecture | open | MathDB 页面 | 原文线索

V2-021314 | MATHDB-374351

The lower-bound conjecture for polynomial volume growth

摘录 / EXCERPT: Let X be a normal projective variety of dimension $d\ge 2$ over $\mathbf{k}$, and let f be an automorphism of X such that $\deg_1(f^n) \asymp n^{k}$ as $n\to\infty$. Lower-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-021315 | MATHDB-374361

The rational symmetric equilibrium conjecture

摘录 / EXCERPT: Let $r\geq 1$, let $x\in\mathbb{C}$, and seek nonzero pairwise distinct complex numbers $\alpha_1,\dots,\alpha_r$ with no relation $\alpha_\mu=-\alpha_\nu$ for distinct indices. Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-021316 | MATHDB-374363

Grothendieck's standard conjecture of Hodge type

摘录 / EXCERPT: Grothendieck's standard conjecture of Hodge type. For any $i \le d/2$, the symmetric bilinear form

conjecture | open | MathDB 页面 | 原文线索

V2-021317 | MATHDB-374371

Conjecture on curve components of the log-free variety for even-degree divisors

摘录 / EXCERPT: Let $L = \sum_{i = 1}^{N} \ell_i p_i$ be a divisor with degree $\ell = \deg L$ even, and let V denote the log-free variety associated with the polynomial system for the equation...

conjecture | open | MathDB 页面 | 原文线索

V2-021318 | MATHDB-374388

Mixed hybrid conjecture for universal abelian schemes

摘录 / EXCERPT: Mixed hybrid conjecture. (1) The subvariety $S\subseteq {\mathbb A}_\Gamma$ is weakly special. (2) If, for some $s\in S$, one has $V\cap (A_s)_{\operatorname{tors}}\neq\varnothing$...

conjecture | open | MathDB 页面 | 原文线索

V2-021319 | MATHDB-374417

Dual defectivity and connectedness of the matroid

摘录 / EXCERPT: Dual-defectivity conjecture. The reciprocal linear space L^{-1} is dual defective, meaning

conjecture | open | MathDB 页面 | 原文线索

V2-021320 | MATHDB-374418

Degree conjecture for the dual variety of a reciprocal linear space

摘录 / EXCERPT: Degree conjecture. If the matroid $M(L)$ is connected, then $\nabla(L^{-1})$ is a hypersurface of degree

conjecture | open | MathDB 页面 | 原文线索

V2-021321 | MATHDB-374422

Oeding's Cohen-Macaulayness conjecture for secant varieties of Segre products

摘录 / EXCERPT: Oeding's conjecture. All secant varieties of Segre products of projective spaces are arithmetically Cohen-Macaulay; equivalently, $\widehat{\sigma}_r$ is Cohen-Macaulay for every...

conjecture | open | MathDB 页面 | 原文线索

V2-021322 | MATHDB-374424

Reduced resonance scheme conjecture for two-step nilpotent Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional nilpotent Lie algebra over $\mathbb{k}$, and set $A=\operatorname{CE}(\mathfrak{g})$. Let S be the symmetric algebra governing the Kosz...

conjecture | open | MathDB 页面 | 原文线索

V2-021323 | MATHDB-374425

Mukai's conjecture for smooth Fano varieties

摘录 / EXCERPT: Mukai's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-021324 | MATHDB-374458

Golyshev-Kerr-Sasaki Arithmetic Mirror Symmetry conjecture for Fano varieties

摘录 / EXCERPT: Arithmetic Mirror Symmetry conjecture. For every such X, the stated geometric construction produces its Apéry numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-021325 | MATHDB-374459

Landau-Ginzburg compactification conjecture for Fano varieties

摘录 / EXCERPT: Landau-Ginzburg compactification conjecture. X admits a Landau-Ginzburg model with a smooth log Calabi-Yau compactification.

conjecture | open | MathDB 页面 | 原文线索

V2-021326 | MATHDB-374460

Extremality conjecture for the variable quantum local system of Fano varieties

摘录 / EXCERPT: Extremality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021327 | MATHDB-374467

Weak ordinarity conjecture for smooth proper Calabi-Yau families

摘录 / EXCERPT: Let S be a scheme smooth and dominant over $\operatorname{Spec} \mathbb{Z}$, and let $Y_S\to S$ be a smooth proper morphism whose fibers are Calabi-Yau varieties. Weak ordinarity...

conjecture | open | MathDB 页面 | 原文线索

V2-021328 | MATHDB-374468

Perfectoid splitting conjecture for smooth proper Calabi-Yau families

摘录 / EXCERPT: Let S be a scheme smooth and dominant over $\operatorname{Spec} \mathbb{Z}$, and let $Y_S\to S$ be a smooth proper morphism whose fibers are Calabi-Yau varieties. For a closed po...

conjecture | open | MathDB 页面 | 原文线索

V2-021329 | MATHDB-374473

Semi-topological Quillen-Lichtenbaum conjecture for smooth complex schemes

摘录 / EXCERPT: Let X be a smooth separated scheme over $\mathbb{C}$ of finite type and let A be an abelian group. The semi-topological Quillen-Lichtenbaum conjecture. The canonical map ... is a...

conjecture | open | MathDB 页面 | 原文线索

V2-021330 | MATHDB-374474

Semi-topological Quillen-Lichtenbaum conjecture for coherent sheaves

摘录 / EXCERPT: Let X be a separated scheme over $\mathbb{C}$ of finite type and let A be an abelian group. The semi-topological Quillen-Lichtenbaum conjecture for coherent sheaves. The canoni...

conjecture | open | MathDB 页面 | 原文线索

V2-021331 | MATHDB-374479

Toric homological mirror symmetry monodromy conjecture

摘录 / EXCERPT: Let $\check X$ be a toric variety with d rays in its fan, and let T be an algebraic torus of rank $\dim(\check X)$. Let $V(E_A)$ be the hypersurface cut out by the principal...

conjecture | open | MathDB 页面 | 原文线索

V2-021332 | MATHDB-374496

Codimension-two transverse torsion principle

摘录 / EXCERPT: Let X be a complex variety with a codimension-two stratum ... whose transverse singularity is a normal surface singularity with local discriminant package (E,q). Then the higher-...

conjecture | open | MathDB 页面 | 原文线索

V2-021333 | MATHDB-374503

The Zilber-Pink conjecture for curves in powers of the modular curve

摘录 / EXCERPT: Let $Y(1)$ denote the modular curve, and let $S\subset Y(1)^n$ be an irreducible curve that is not contained in a proper special subvariety of $Y(1)^n$. For a special subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-021334 | MATHDB-374550

Kontsevich's homological mirror symmetry conjecture

摘录 / EXCERPT: Let X and Y be mirror varieties, let $\mathcal{F}uk(X)$ denote the Fukaya category of X, and let $D^b\operatorname{Coh}(Y)$ denote the bounded derived category of coherent sh...

conjecture | open | MathDB 页面 | 原文线索

V2-021335 | MATHDB-374554

Néron mapping property for logarithmic abelian varieties

摘录 / EXCERPT: Let $A\to S$ be a family of log abelian varieties over a log smooth base S. Let $U\subseteq S$ be the dense open subset of S on which the log structure is trivial. For a log sm...

conjecture | open | MathDB 页面 | 原文线索

V2-021336 | MATHDB-374614

One-dimensional de Rham-Betti structure conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over $\overline{\mathbb{Q}}$. Write $\langle\mathrm{H}^1_{\mathrm{dRB}}(A,\mathbb{Q})\rangle^{\otimes}$ for the tensor category generated by its first...

conjecture | open | MathDB 页面 | 原文线索

V2-021337 | MATHDB-374632

Iitaka's conjecture on the Kodaira dimension of algebraic fiber spaces

摘录 / EXCERPT: Let (X,Δ) be a Klt pair, and let $f:X\to Y$ be an algebraic fiber space, where X and Y are smooth projective varieties of dimensions n and m, respectively. Let F b...

conjecture | open | MathDB 页面 | 原文线索

V2-021338 | MATHDB-374651

Manin's conjecture for rational points on smooth Fano varieties

摘录 / EXCERPT: Let k be a number field and let X be a smooth Fano variety over k, so that $-K_X$ is ample. Fix an adelic metrization of the anticanonical bundle $\mathcal O(-K_X)$, with as...

conjecture | open | MathDB 页面 | 原文线索

V2-021339 | MATHDB-374689

The cocenter conjecture for the Gieseker variety and cyclotomic Hecke algebra

摘录 / EXCERPT: Let $\mathcal{L}(n,r)$ be the specified closed projective subvariety of the Gieseker space, let $K_{\mathcal{T}}(\mathcal{L}(n,r))$ denote its equivariant K-theory, and let the c...

conjecture | open | MathDB 页面 | 原文线索

V2-021340 | MATHDB-374690

Integral cocenter isomorphism conjecture for the Gieseker variety

摘录 / EXCERPT: Let $\mathcal{R}$ be the coefficient ring, let $P_{\mathrm{loc}}=P\otimes_{\mathcal{R}}\operatorname{Frac}\mathcal{R}$ for every $\mathcal{R}$-module P, and let ... be the unique i...

conjecture | open | MathDB 页面 | 原文线索

V2-021341 | MATHDB-374718

The singular Bayer-Macrì-Toda conjecture

摘录 / EXCERPT: Bayer-Macrì-Toda conjecture. There exists $546d53953c(X)_{4d9}$ with $546.Hd5390$ such that, for every $w>\frac{1}{2}b^2+534$ and every $539_{b,w}$-semistable object...

conjecture | open | MathDB 页面 | 原文线索

V2-021342 | MATHDB-374739

The no-connected-hypersurfaces conjecture for Jacobi spectral curves

摘录 / EXCERPT: No-connected-hypersurfaces conjecture. For every $n\ge4$, every codimension-one component of the reducible locus is one of the cutting hyperplanes $b_i=0$. Equivalently, the reduci...

conjecture | open | MathDB 页面 | 原文线索

V2-021343 | MATHDB-374740

The primitive codimension-growth conjecture for Jacobi spectral curves

摘录 / EXCERPT: Primitive codimension-growth conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-021344 | MATHDB-374784

IsoPrymian characterization by the two-dimensional Schrödinger equation

摘录 / EXCERPT: Let X be a principally polarized Abelian variety with theta function θ. Let $\widehat{\theta}_1$ denote the Prym theta function associated with the first involution, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021345 | MATHDB-374799

The modified log Manin conjecture for geometrically \(\mathbb A^1\)-connected varieties

摘录 / EXCERPT: Let $(\underline{X},\Delta)$ be a log pair and set $\underline{U}=\underline{X}\setminus\operatorname{Supp}(\Delta)$. Assume that $\underline{U}$ is geometrically $\mathbb A^1$-con...

conjecture | open | MathDB 页面 | 原文线索

V2-021346 | MATHDB-374800

The bubble-tree blow-up and comparison conjecture for KSBA moduli

摘录 / EXCERPT: Let M_N be the KSBA moduli stack of index one covers, let M_N^{Bub} be the moduli stack of bubble-tree index one cover DM stacks, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021347 | MATHDB-374801

Gerbe-independence conjecture for virtual fundamental classes

摘录 / EXCERPT: Let W be the coarse projective variety in the KSBA moduli problem, let ${\mathfrak W}\to W$ be a μ_N-gerbe, and let M be the associated KSBA moduli space. Suppose that the...

conjecture | open | MathDB 页面 | 原文线索

V2-021348 | MATHDB-374803

The Q-smoothing conjecture for terminal Fano threefolds

摘录 / EXCERPT: Let X be a Fano threefold with terminal singularities. A Q-smoothing of X is a Q-Gorenstein family $\mathcal{X}\rightarrow C\ni 0$ such that ... and $\mathcal{X}_t$ has quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-021349 | MATHDB-374808

Jordan-type conjecture for affine growth diagrams

摘录 / EXCERPT: Jordan-type conjecture. There is an open set $\mathcal{N}\subseteq\mathfrak{n}_F\cap\mathfrak{n}_{F'}$ such that, for every $er\in\mathcal{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-021350 | MATHDB-374811

The semisimplicity conjecture for Galois actions on cohomology

摘录 / EXCERPT: Semisimplicity conjecture. The Galois action on

conjecture | open | MathDB 页面 | 原文线索

V2-021351 | MATHDB-374812

The generalized Tate conjecture

摘录 / EXCERPT: Generalized Tate conjecture. There is a closed algebraic subset $Z\subset X$ of codimension k such that

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V2-021352 | MATHDB-374871

Maximal-rank conjecture for the cohomological map alpha_T

摘录 / EXCERPT: Let $V=\mathbb{C}^{n_1+1}\otimes\dots\otimes\mathbb{C}^{n_k+1}\otimes\mathbb{C}^{n+2}$, where $n=\sum_{j=1}^k n_j$, and let $T\in V$ be a general tensor. Let α_T be the coho...

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V2-021353 | MATHDB-374872

Higher-factor defective-span conjecture beyond boundary format

摘录 / EXCERPT: Let $n=\sum_{j=1}^k n_j$ and consider a $(k+1)$-tensor space V of format $(n_1+1,\dots,n_k+1,n+2)$, with $T\in V$ a general tensor. Higher-factor defective-span conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-021354 | MATHDB-374886

Bochnak-Kucharz approximation conjecture for odd-dimensional real algebraic sets

摘录 / EXCERPT: Let n be a positive odd integer, and let X be a nonsingular compact connected real algebraic set of dimension n. A regular map means a map given by regular functions on X....

conjecture | open | MathDB 页面 | 原文线索

V2-021355 | MATHDB-374887

Degree-parity conjecture for regular maps from products of spheres

摘录 / EXCERPT: Let n and m be positive integers. A continuous map is considered from the product of spheres $\mathbb{S}^n\times\mathbb{S}^m$ to $\mathbb{S}^{n+m}$, and $\binom{n+m}{n}$ denote...

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V2-021356 | MATHDB-374927

Quasi-projectivity conjecture for singular Kähler-Ricci shrinkers

摘录 / EXCERPT: Let (X,d_X) be the singular Kähler-Ricci shrinker arising as the metric soliton modeled on the $\mathbb{F}$-limit described in the paper, with regular part...

conjecture | open | MathDB 页面 | 原文线索

V2-021357 | MATHDB-374969

Weak Harris-Morrison slope conjecture

摘录 / EXCERPT: Weak Harris-Morrison slope conjecture. For every g,

conjecture | open | MathDB 页面 | 原文线索

V2-021358 | MATHDB-375011

Boucksom-Jonsson regularization conjecture

摘录 / EXCERPT: Let (X,L) be a smooth polarized variety, and let $\mathbb{G}\subset \operatorname{Aut}^\circ(X,L)$ be a reductive complex Lie group. Let $(\mathcal{X},\mathcal{L})$ be a dominati...

conjecture | open | MathDB 页面 | 原文线索

V2-021359 | MATHDB-375038

Geometric Manin's conjecture on the number of Manin components

摘录 / EXCERPT: Let X be a smooth Fano variety over $\mathbb C$ with Brauer group $Br(X)$. Let C be a smooth connected genus g curve. A component of $\operatorname{Mor}(C,X,\alpha)$ is under...

conjecture | open | MathDB 页面 | 原文线索

V2-021360 | MATHDB-375066

Semi-algebraicity conjecture for the maximal honest open in ReLU-network parameter space

摘录 / EXCERPT: For a fixed ReLU-network architecture, let $\mathcal{P}$ denote the parameter space obtained from the space of weights by taking into account the trivial scaling and permutation sy...

conjecture | open | MathDB 页面 | 原文线索

V2-021361 | MATHDB-375081

Essential normality conjecture for one-dimensional varieties in the infinite-variable Drury-Arveson module

摘录 / EXCERPT: Let V be a one-dimensional smooth holomorphic zero variety of an open neighborhood of the closed unit ball $\overline{\mathbb B}$, transversal to the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-021362 | MATHDB-375094

Bhatt-Lurie’s cohomological dimension conjecture for the Hodge-Tate locus

摘录 / EXCERPT: Bhatt-Lurie’s cohomological dimension conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-021363 | MATHDB-375098

Bena et al.'s tadpole conjecture

摘录 / EXCERPT: Tadpole conjecture. The tadpole contribution satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-021364 | MATHDB-375109

Multi-oval generic non-algebraicity conjecture for exterior Cauchy transforms

摘录 / EXCERPT: Multi-oval generic non-algebraicity conjecture. For a generic affine Euclidean realization, the exterior Cauchy transform of the domain bounded by any one of these ovals is non-alg...

conjecture | open | MathDB 页面 | 原文线索

V2-021365 | MATHDB-375165

Shao-Yin-Zhao's conjecture for associated varieties at non-admissible integer levels

摘录 / EXCERPT: Let $\fg$ be a simple simply-laced Lie algebra, let $k\in\BQ$ satisfy $k+\check{\operatorname{\mathbbm{h}}}=m\geq1$, and let $L_k(\fg)$ be the simple affine vertex algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021366 | MATHDB-375166

The covering-duality conjecture for associated varieties of affine vertex algebras

摘录 / EXCERPT: Let $\fg$ be simple of simply-laced type and let $k\in\BQ$ satisfy $k+\check{\operatorname{\mathbbm{h}}}=m/u$ in lowest terms with $m\geq1$. Let $\check{\fg}^{(u)}$ be the metaplec...

conjecture | open | MathDB 页面 | 原文线索

V2-021367 | MATHDB-375167

The sheet formulation of the covering-duality conjecture

摘录 / EXCERPT: Let $\fg$ be simple of simply-laced type, let $k\in\BQ$ satisfy $k+\check{\operatorname{\mathbbm{h}}}=m/u$ in lowest terms with $m\geq1$ and $\operatorname{gcd}(m,u)=1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021368 | MATHDB-375172

Hyperelliptic characterization by the Seshadri constant

摘录 / EXCERPT: Let (A,θ) be an indecomposable principally polarized abelian variety of dimension $g\geq 4$. The hyperelliptic characterization conjecture. If $\varepsilon(A,\theta)<2$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-021369 | MATHDB-375173

Gonality-genus bound for the Seshadri constant of a polarized abelian variety

摘录 / EXCERPT: Let (A,θ) be a polarized abelian variety, and let $C\subset A$ be a smooth curve of genus h. Write c for the gonality of C and ... for its θ-degree. Gonality-genu...

conjecture | open | MathDB 页面 | 原文线索

V2-021370 | MATHDB-375224

Contact Joyce conjecture for oriented Legendrians

摘录 / EXCERPT: Let $(X,\mathcal{L},\alpha)$ be an oriented -1-shifted contact derived stack over $\mathbb{K}$, and let $\varphi\colon L\to X$ be a proper oriented Legendrian. Write ... for the re...

conjecture | open | MathDB 页面 | 原文线索

V2-021371 | MATHDB-375244

Hassett-Kuznetsov rationality conjecture for cubic fourfolds

摘录 / EXCERPT: Let X be a smooth cubic fourfold, and let $\mathcal{C}_d$ denote the Hassett divisor parametrizing special cubic fourfolds with a labelling of discriminant d. The numerical con...

conjecture | open | MathDB 页面 | 原文线索

V2-021372 | MATHDB-375275

Kawakami-Totaro conjecture on endomorphisms of Fano varieties of Picard number 1

摘录 / EXCERPT: Let X be a smooth Fano variety of Picard number $\rho(X)=1$ over an algebraically closed field of characteristic zero. An endomorphism of degree greater than 1 is a self-map of...

conjecture | open | MathDB 页面 | 原文线索

V2-021373 | MATHDB-375327

Compositionality conjecture for partial flag supervarieties

摘录 / EXCERPT: Let $X = \operatorname{\mathbf{Flag}}(p_1|q_1, \dots, p_t|q_t;\;V)$ be a flag supervariety of type τ, and let Y be the variety associated to X by the construction in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-021374 | MATHDB-375334

Lusztig's support conjecture for semicanonical basis elements

摘录 / EXCERPT: Let Z be an irreducible component in the relevant module variety, and let f_Z denote the corresponding semicanonical basis element. Lusztig's support conjecture. The semicanoni...

conjecture | open | MathDB 页面 | 原文线索

V2-021375 | MATHDB-375335

Gorenstein-injective realization conjecture for positroid categories

摘录 / EXCERPT: Let $v \otin\quotmax$ and $w\geq v$, let $b1a\sigma=v^{-1}w$ be the resulting permutation decorated by declaring a fixed index j black if $j\in v^{-1}[k]$, and set $I=v^{-1}[k]$....

conjecture | open | MathDB 页面 | 原文线索

V2-021376 | MATHDB-375356

Toric Property N_p conjecture inspired by Mukai's conjecture

摘录 / EXCERPT: Let X be a projective toric variety of dimension n, let T be its algebraic torus, and let L be an ample line bundle on X. Let p be a nonnegative integer, and let Proper...

conjecture | open | MathDB 页面 | 原文线索

V2-021377 | MATHDB-375375

The losing-hook stack conjecture

摘录 / EXCERPT: Let λ be a losing partition and let $\Gamma_{r,c}$ be a losing hook. Write ... when the stack operation $\lhd$ is defined. Losing-hook stack conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-021378 | MATHDB-375376

Mirror line bundle conjecture for parabolic very stable upward flows

摘录 / EXCERPT: Let G be a reductive group with Langlands dual $G^\vee$, let C be the underlying curve, and let $D=\{c_i\}$ be the parabolic divisor. Let (E,φ,Q) be a smooth fixed poin...

conjecture | open | MathDB 页面 | 原文线索

V2-021379 | MATHDB-375378

Partial flag Toeplitz limit conjecture

摘录 / EXCERPT: Partial flag Toeplitz limit conjecture. There exists a sequence $u^{(n)}d4d9\text{in}X_{P_{d4c3}^{(n)}}(\mathbb R_{>0})$ converging to u^{d4c4}; conversely, every infinite totall...

conjecture | open | MathDB 页面 | 原文线索

V2-021380 | MATHDB-375379

Partial flag Schubert polynomial limit conjecture

摘录 / EXCERPT: Let $(\mathbf n,\mathbf m)$ be as above, and let $\boldsymbol\alpha_{\mathbf n}$ and $\boldsymbol\beta_{\mathbf m}$ be positive sequences satisfying the stated conditions and summa...

conjecture | open | MathDB 页面 | 原文线索

V2-021381 | MATHDB-375382

Chen-Ngô's spectral-base image conjecture for the Hitchin morphism

摘录 / EXCERPT: Let $\mathbb{K}$ be an algebraically closed field of characteristic 0, let X be an irreducible smooth projective variety over $\mathbb{K}$, and let G be a reductive group ove...

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V2-021382 | MATHDB-375399

Cofibration conjecture for closed subspaces of reasonable and PA-spaces

摘录 / EXCERPT: Let X be a reasonable topological space or a PA-space, and let $Y\subset X$ be a closed subspace. A map is a cofibration if it has the homotopy extension property. Cofibration co...

conjecture | open | MathDB 页面 | 原文线索

V2-021383 | MATHDB-375400

Differential-operator conjecture for PA-forms

摘录 / EXCERPT: Let X be a PA-space, let f be a PA-form on X, let $\Omega(X)$ denote the space of smooth differential forms on X, and let P be a differential operator. Differential-opera...

conjecture | open | MathDB 页面 | 原文线索

V2-021384 | MATHDB-375412

Value-variety decomposition conjecture for multisymmetric isotypic Hilbert series

摘录 / EXCERPT: Let $K[X^1,\dots,X^k]$ be the polynomial ring in k sets of countably many variables, with $\operatorname{Sym}(\infty)$ permuting the indices. Let $I\subset K[X^1,\dots,X^k]$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-021385 | MATHDB-375447

The generic-fiber conjecture for two-body eigenvector varieties

摘录 / EXCERPT: Generic-fiber conjecture. Equality holds in the dimension bound

conjecture | open | MathDB 页面 | 原文线索

V2-021386 | MATHDB-375448

Lyubeznik's étale cohomological dimension conjecture

摘录 / EXCERPT: Lyubeznik's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021387 | MATHDB-375455

Exact formula conjecture for the minimal total degree of flag-variety regularity

摘录 / EXCERPT: Let ${\mathcal{F}}I(\mathbb C^n)$ be the complete flag variety, and let $\operatorname{reg}(\iota_*{\mathcal{O}}_{{\mathcal{F}}I(\mathbb C^n)})$ be its multigraded regularity regio...

conjecture | open | MathDB 页面 | 原文线索

V2-021388 | MATHDB-375461

The motivic cohomology decomposition conjecture for \(\overline{M_{0,n}}\)

摘录 / EXCERPT: Let k be any field of characteristic not 2, let $n\geq 3$ be a natural number, and let $\mathcal{L}$ be a line bundle on $\overline{M_{0,n}}$. Write $K^{MW}_q(\mathcal{L})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-021389 | MATHDB-375474

Voisin's strong Beauville-Voisin conjecture for hyper-Kähler varieties

摘录 / EXCERPT: Let X be a complex hyper-Kähler variety, and let $A^*(X)=\bigoplus_i A^i(X)$ be its Chow ring with $\mathbb{Q}$-coefficients. A subvariety $Z\subset X$ is a constant cycle subvar...

conjecture | open | MathDB 页面 | 原文线索

V2-021390 | MATHDB-375482

Orbit characterization of the filtration on generalized Kummer varieties

摘录 / EXCERPT: Let X be an abelian surface, let $K_n(X)$ be the generalized Kummer variety, and let $z=\{z_0,\ldots,z_n\}\in K_n(X)$ with $z_m\in X$. For a point z, let O_z denote its ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-021391 | MATHDB-375490

Ogus's canonical F-isocrystal conjecture for relative rigid spaces

摘录 / EXCERPT: Let X be a regular p-adic formal scheme, and let $Y_\eta\to X_\eta$ be a proper smooth rigid space with analytically good reduction over X_η. For each $i\in\mathbb{N}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-021392 | MATHDB-375492

Splitting conjecture for the Langlands-Shahidi derived category

摘录 / EXCERPT: Let G be a quasi-split reductive group over E, let Bun_G be its moduli stack of G-bundles, and let Bun_G^b denote the stratum indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-021393 | MATHDB-375523

Conjecture on monotonicity of successive average stack-sorting depths

摘录 / EXCERPT: Monotonicity conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-021394 | MATHDB-375533

The dynamical Ax-Lindemann-Weierstrass conjecture for Böttcher coordinates

摘录 / EXCERPT: Let f be a non-exceptional polynomial of degree $d\ge 2$. Let $V_1\subset \mathbb{C}^n$ be an irreducible subvariety with non-empty intersection with $\mathbb{D}_R^n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021395 | MATHDB-375550

Reducedness and irreducibility conjecture for Weddle schemes

摘录 / EXCERPT: Let Σ be the d-Weddle scheme associated with a general set of $\binom{d+n}{n}$ points in $\mathbb{P}^{n+1}$, as in the preceding theorem. Reducedness and irreducibility co...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-021396 | MATHDB-375594

Harris's conjecture on Noether-Lefschetz loci for octic surfaces

摘录 / EXCERPT: Let $X_0: x_0^8+x_1^8+x_2^8+x_3^8=0$ be the Fermat octic, and let C_1 and C_2 be respectively a line, namely a complete intersection of type $(1,1)$, and a complete intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-021397 | MATHDB-375604

S. David's Bogomolov property conjecture for torsion fields of abelian varieties

摘录 / EXCERPT: Let A be an abelian variety defined over a number field, and let A_{tors} denote its torsion subgroup. S. David's conjecture. The field $\mathbb{Q}(A_{\mathrm{tors}})$...

conjecture | open | MathDB 页面 | 原文线索

V2-021398 | MATHDB-375658

Song-Zhu conjecture on Property N_p for projective toric varieties

摘录 / EXCERPT: Let X be a projective toric variety of dimension $n\geq 2$, let L be a line bundle on X, and let $p\geq 0$. Property N_p is the syzygy property defined by the minimal grade...

conjecture | open | MathDB 页面 | 原文线索

V2-021399 | MATHDB-375659

The Thomsen stratification intersection-number conjecture for smooth proper toric varieties

摘录 / EXCERPT: Let X be a smooth proper toric variety, and consider its Thomsen stratification and the associated intersection numbers. Thomsen stratification intersection-number conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-021400 | MATHDB-375675

The irreducible-component conjecture for Kottwitz-Viehmann varieties

摘录 / EXCERPT: Let G be the reductive group defining the Kottwitz-Viehmann variety, let F be its base field, let T be a maximal torus, and let $G(F)^{\mathrm{rs}}$ denote the regular semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-021401 | MATHDB-375679

Finiteness conjecture for geometric representations over finite fields

摘录 / EXCERPT: Let p be a prime number, and let k be a finite field of characteristic different from p. Let X be a normal geometrically connected variety over k, let $\overline X$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-021402 | MATHDB-375709

The discreteness conjecture for affine varieties without additive or multiplicative actions

摘录 / EXCERPT: Work over an algebraically closed field $\mathbb K$ of characteristic zero. Let Y be an affine variety, and let $\mathbb{G}_{\rm a}$ and $\mathbb{G}_{\rm m}$ denote the additive...

conjecture | open | MathDB 页面 | 原文线索

V2-021403 | MATHDB-375718

Prym-Brill-Noether dimension conjecture for general k-gonal covers

摘录 / EXCERPT: Let $r \geq 1$ and $k \geq 3$. For a general étale double cover $f \colon \widetilde{C} \to C$ in the k-gonal locus in $\mathcal{R}_g$, the Prym-Brill-Noether variety $V^r(C,f)$...

conjecture | open | MathDB 页面 | 原文线索

V2-021404 | MATHDB-375732

Geometric Bombieri--Lang's constant-family conjecture

摘录 / EXCERPT: Let K/k be a finitely generated extension of characteristic zero, and let X be a projective variety of general type over K. Geometric Bombieri--Lang's conjecture. After remov...

conjecture | open | MathDB 页面 | 原文线索

V2-021405 | MATHDB-375734

Geometric Bombieri--Lang's generic-sequence height conjecture

摘录 / EXCERPT: Let $K=k(B)$, where B is a smooth projective curve, and let X be a projective variety of general type over K. A sequence of K-rational points in X is generic if every inf...

conjecture | open | MathDB 页面 | 原文线索

V2-021406 | MATHDB-375735

Xie--Yuan's non-degeneracy conjecture for partial heights

摘录 / EXCERPT: Let $K=\mathbb C(B)$, where B is a smooth projective curve, and let X be a projective variety over K containing no rational curve. Let h_L be a Weil height attached to an a...

conjecture | open | MathDB 页面 | 原文线索

V2-021407 | MATHDB-375738

The reductive-group triangulation uniqueness conjecture

摘录 / EXCERPT: Reductive-group triangulation uniqueness conjecture. Then D has at most one triangulation of parameter δ.

conjecture | open | MathDB 页面 | 原文线索

V2-021408 | MATHDB-375760

Subdivision conjecture for moment polyhedra of Kac-Moody flag varieties

摘录 / EXCERPT: Subdivision conjecture. The moment polyhedron $\Phi_T(G/B)$ has a locally finite subdivision into Bruhat interval polytopes

conjecture | open | MathDB 页面 | 原文线索

V2-021409 | MATHDB-375773

Refined Stanley-Stembridge conjecture for Hessenberg graphs

摘录 / EXCERPT: Let k be a positive integer, let $h\colon [k]\rightarrow [k]$ be a Hessenberg function, and let G_h be the graph associated with h. For a proper coloring κ of G_h, w...

conjecture | open | MathDB 页面 | 原文线索

V2-021410 | MATHDB-375774

Permutation-module decomposition conjecture for Hessenberg-variety cohomology

摘录 / EXCERPT: Let k be a positive integer, let $h\colon [k]\rightarrow [k]$ be a Hessenberg function, let S be a regular semisimple operator on $\mathbb C^k$, and let $\mathfrak S_k$ act on...

conjecture | open | MathDB 页面 | 原文线索

V2-021411 | MATHDB-375775

Cho-Hong-Lee permutation-module generator conjecture for Hessenberg cohomology

摘录 / EXCERPT: Let $h\colon [k]\rightarrow [k]$ be a Hessenberg function, let S be a regular semisimple operator on $\mathbb C^k$, and let $\mathfrak S_k$ act on $H^*(\operatorname{Hess}(S,h))$...

conjecture | open | MathDB 页面 | 原文线索

V2-021412 | MATHDB-375791

Jiang's second-smallest alpha invariant conjecture for K-semistable Fano manifolds

摘录 / EXCERPT: Jiang's conjecture. $\alpha(X)<\frac{1}{n}$ if and only if $X\cong \mathbb{P}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-021413 | MATHDB-375808

Classification conjecture for compact Kähler manifolds without analytic subvarieties

摘录 / EXCERPT: Classification conjecture. The manifold X should be either a complex torus or an irreducible hyperkähler manifold. In particular, K_X should be trivial, and the two cases shoul...

conjecture | open | MathDB 页面 | 原文线索

V2-021414 | MATHDB-375828

Bloch's conjecture on zero-cycles and vanishing holomorphic forms

摘录 / EXCERPT: Let X be a smooth, projective variety over $\mathbb{C}$, and write $h^{p,0}(X)$ for the Hodge number measuring global holomorphic p-forms. Bloch's conjecture. If ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-021415 | MATHDB-375835

Cohen-Macaulayness conjecture for closed electroid varieties

摘录 / EXCERPT: Let χ be the electroid space equipped with the Frobenius splitting under consideration, and let closed electroid varieties be its closed electroid subvarieties. Cohen-Macaulay...

conjecture | open | MathDB 页面 | 原文线索

V2-021416 | MATHDB-375836

Lam's positive-geometry conjecture for electroid spaces

摘录 / EXCERPT: Let χ_n be the electroid space, let closed electroid varieties be its closed electroid subvarieties, and let the electroids be partially ordered by their poset of electroids.…

conjecture | open | MathDB 页面 | 原文线索

V2-021417 | MATHDB-375872

Pirola-Farkas conjecture on rational sections of Kummer fibrations

摘录 / EXCERPT: Let $\mathcal{M}_g$ be the moduli space of curves of genus g, let ${J}\rightarrow \mathcal{M}_g$ be the associated Jacobian fibration, and let ... be the Kummer fibration. A ration...

conjecture | open | MathDB 页面 | 原文线索

V2-021418 | MATHDB-375875

Arc-descent conjecture for rational motivic cohomology presheaves

摘录 / EXCERPT: For an algebraic space X, let $\Cmot(X;\Q(r)):=\Gamma(X;\Q(r))$ denote the rational motivic cohomology spectrum with Tate twist r. For every integer $r\in\Z$, consider the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-021419 | MATHDB-375880

Extension conjecture for locally analytic functions on the bc-ordinary Igusa variety

摘录 / EXCERPT: Let ... where the locally convex inductive limit is over open neighborhoods $U\subset N\backslash P^{\operatorname{an}}_{\operatorname{dR}}$ containing the closure of the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-021420 | MATHDB-375885

The Mumford-Tate conjecture for smooth projective varieties

摘录 / EXCERPT: Let X be a smooth projective variety defined over a finitely generated subfield $F\subset\mathbb{C}$. Under the Artin comparison identification ... let...

conjecture | open | MathDB 页面 | 原文线索

V2-021421 | MATHDB-375917

Collins-Jacob-Yau hypercritical-phase conjecture

摘录 / EXCERPT: Collins-Jacob-Yau hypercritical-phase conjecture. The set $\mathcal{H}_{\Omega}$ is non-empty and Ω has hypercritical phase if and only if, for every irreducible subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-021422 | MATHDB-375952

The global numerical conjecture for polarized hyperspherical Hamiltonian varieties

摘录 / EXCERPT: Let $M=T^*X$ and $\check{M}=T^*\check{X}$ be a pair of dual polarized hyperspherical Hamiltonian varieties, where X is a G-spherical variety and $\check{X}$ is the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-021423 | MATHDB-376026

Corti's bounded Sarkisov link conjecture for terminal Fano threefolds

摘录 / EXCERPT: Let X and X' be birationally equivalent terminal Fano varieties of dimension 3 and Picard rank one over $\mathbb{C}$. A birational map $f\colon X\dashrightarrow X'$ is said t...

conjecture | open | MathDB 页面 | 原文线索

V2-021424 | MATHDB-376065

Classification of two-dimensional leaves in quiver Coulomb branches

摘录 / EXCERPT: Let $\mathbf{v}$ be a dimension vector, let $\GL_{\mathbf{v}}$ be the corresponding product of general linear groups, and let $\mathbf{N}_{\mathbf{v}}$ be the associated representa...

conjecture | open | MathDB 页面 | 原文线索

V2-021425 | MATHDB-376072

Lusztig's quotient conjecture for special pieces

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra of an exceptional simple algebraic group, let $\0$ be a special nilpotent orbit, and let ${\mathcal P}(\0)$ be its special piece. Let H be t...

conjecture | open | MathDB 页面 | 原文线索

V2-021426 | MATHDB-376084

The unlikely intersection bound with multiplicities

摘录 / EXCERPT: Unlikely intersection bound. For every $\varepsilon>0$, there exists $c=c(n,\mathcal{V},\varepsilon,K)\in\mathbb{R}$ such that, for every flat subgroup scheme...

conjecture | open | MathDB 页面 | 原文线索

V2-021427 | MATHDB-376113

Existence of MDS LCD codes on the maximal hyperelliptic curve

摘录 / EXCERPT: Let q be a prime power with $q\geq 3$, and let $\mathcal{X}$ be the maximal hyperelliptic curve over $\mathbb{F}_{q^2}$ given by ... Let $u_B\in\mathbb{F}_{q^2}[x]$ be an irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-021428 | MATHDB-376116

Asymptotic half-density conjecture for the $(2,2,2)$ neuromanifold over finite fields

摘录 / EXCERPT: Asymptotic half-density conjecture. The arithmetic expressive capacity approaches $\frac{1}{2}$ as $p\to\infty$:

conjecture | open | MathDB 页面 | 原文线索

V2-021429 | MATHDB-376154

The Brauer-Manin zero-cycle conjecture for smooth projective varieties

摘录 / EXCERPT: Brauer-Manin zero-cycle conjecture. If $X({\bf A}_k)^{\operatorname{Br}}$ is non-empty, then $I(X)=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-021430 | MATHDB-376157

Campana's conjecture on virtually nilpotent fundamental groups

摘录 / EXCERPT: Let X be a normal complex algebraic variety. Suppose that $\pi_1(X)$ is virtually nilpotent. Campana's conjecture. Then $\pi_1(X)$ is virtually at most 2-step nilpotent. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-021431 | MATHDB-376167

DeMarco-Mavraki dynamical unlikely-intersection conjecture

摘录 / EXCERPT: Let S be a smooth irreducible quasi-projective variety over $\mathbb{C}$, let $\Phi:S\times\mathbb{P}^N\to S\times\mathbb{P}^N$ be an algebraic family of endomorphisms of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-021432 | MATHDB-376186

Effective characterization of semi-abelian varieties

摘录 / EXCERPT: Effective semi-abelian characterization conjecture. There exists an integer $k = k(d) \geq 1$ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-021433 | MATHDB-376204

Algebraic versus span multiplicity conjecture for tensor eigenvalues

摘录 / EXCERPT: Let n,d be positive integers, let $T,\mathbf{t}\in\mathbb{C}^n\otimes((\mathbb{C}^n)^*)^{\otimes d}$, and assume that $\mathbf{t}$ is non-singular. For $\lambda_0\in\mathbb{C}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-021434 | MATHDB-376220

Keller's Jacobian conjecture

摘录 / EXCERPT: Keller's Jacobian conjecture. If F has $\operatorname{Jac} F\in\Bbbk^{\times}$, then F is invertible, with polynomial inverse.

conjecture | open | MathDB 页面 | 原文线索

V2-021435 | MATHDB-376242

Tate-Voloch conjecture on torsion points and subvarieties

摘录 / EXCERPT: Let K be a field complete with respect to a non-archimedean absolute value, let A/K be a semiabelian variety, and let $X \subset A$ be a closed subvariety. Tate-Voloch conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-021436 | MATHDB-376245

A sharp Euler-sum identity for height-one motivic multiple mixed values

摘录 / EXCERPT: Let d be a positive integer with $d\ge 3$. Write 1_{d-3} for a string of $d-3$ entries equal to 1, let $\zeta^\sharp$ denote the Euler sharp sum, and let $\zeta(m,n)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-021437 | MATHDB-376246

Unramified MTV classification by width and dual depth

摘录 / EXCERPT: Let $\bfs$ be an index, let $T(\bfs)$ be a multiple t-value (MTV), and let $\bfs^*$ be its dual. Define the width $\operatorname{wth}(\bfs)$ as the largest component of $\bfs$; cal...

conjecture | open | MathDB 页面 | 原文线索

V2-021438 | MATHDB-376247

Depth-four classification conjecture for unramified multiple mixed values

摘录 / EXCERPT: Let $M(\bs;\varepsilon)$ denote a multiple mixed value of depth 4, let $T(\bs)$ denote a multiple t-value, and classify indices as wide, slender, or narrow by whether their large...

conjecture | open | MathDB 页面 | 原文线索

V2-021439 | MATHDB-376248

Depth-five classification conjecture for unramified multiple mixed values

摘录 / EXCERPT: Let $M(\bs;\varepsilon)$ denote a multiple mixed value and $T(\bs)$ a multiple t-value. Classify indices as wide, slender, or narrow according to whether their largest component is...

conjecture | open | MathDB 页面 | 原文线索

V2-021440 | MATHDB-376260

Cotangent-case Borel-orbit duality conjecture

摘录 / EXCERPT: Let G be a reductive group with Langlands dual group $G^\vee$, and let W be the Weyl group. Let X be a smooth spherical G-variety such that T^*X is hyperspherical and has...

conjecture | open | MathDB 页面 | 原文线索

V2-021441 | MATHDB-376274

The Dolbeault-Betti symplectic resolution conjecture for G-Higgs moduli spaces

摘录 / EXCERPT: Let G be a complex connected reductive group, and let $M_{Dol}^{vc}(G)$ be the moduli space of G-Higgs bundles with vanishing Chern classes over a compact Riemann surface of ge...

conjecture | open | MathDB 页面 | 原文线索

V2-021442 | MATHDB-376287

Duncan's Sylow-subgroup criterion for versal group actions

摘录 / EXCERPT: Let k be a field, let G be a finite group, and let X be a k-variety with a faithful G-action. For every prime p, let G_p be a Sylow p-subgroup of G. Duncan's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-021443 | MATHDB-376291

The Miyaoka-Yau inequality and equality criterion for arbitrary weights

摘录 / EXCERPT: Assume Setup, with X a Fano manifold and $\omega \in c_1(X)$. Let $v \colon P \to \mathbb{R}_{>0}$ be a weight, let $\mathscr{V}$ be a canonical extension sheaf with extension cl...

conjecture | open | MathDB 页面 | 原文线索

V2-021444 | MATHDB-376320

The Bernstein-Sato divisibility conjecture for regular sequences

摘录 / EXCERPT: Let X be the smooth variety and let $f_1,\dots,f_r\in \mathscr{O}_X(X)$ be a regular sequence. Set $g=f_1\cdots f_r$ and let U denote the open subset on which the relevant rest...

conjecture | open | MathDB 页面 | 原文线索

V2-021445 | MATHDB-376321

Blasius-Rogawski conjecture on Eichler-Shimura relations

摘录 / EXCERPT: Let (G,X) be a Shimura datum with reflex field E, let G be its associated reductive group, and let μ be the Hodge cocharacter. Let (r_μ,V) be the re...

conjecture | open | MathDB 页面 | 原文线索

V2-021446 | MATHDB-376325

Cautis-Logvinenko conjecture for the three-dimensional McKay correspondence

摘录 / EXCERPT: Let $G\subset \operatorname{SL}(3,\mathbb{C})$ be a finite subgroup, let $Y={G}\operatorname{-Hilb}(\mathbb{C}^3)$ be the G-Hilbert scheme, and let ... be the inverse of the Bridge...

conjecture | open | MathDB 页面 | 原文线索

V2-021447 | MATHDB-376359

The cluster-variety P=W and PI=WI conjectures

摘录 / EXCERPT: P=W and PI=WI conjectures. There exists a real Lagrangian fibration h such that

conjecture | open | MathDB 页面 | 原文线索

V2-021448 | MATHDB-376363

Weak Morrison-Kawamata cone conjecture

摘录 / EXCERPT: Let $(X/Z,\Delta)$ be a klt log Calabi-Yau pair over Z, where X is $\mathbb{Q}$-factorial. Write $\mathcal{A}^e(X/Z)$ for the effective nef cone and $\mathcal{M}^e(X/Z)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-021449 | MATHDB-376364

Morrison-Kawamata cone conjecture

摘录 / EXCERPT: Let $(X/Z,\Delta)$ be a klt log Calabi-Yau pair over Z, where X is $\mathbb{Q}$-factorial and Δ is an $\mathbb{R}$-boundary. Let $\mathcal{A}^e(X/Z)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021450 | MATHDB-376394

Quadratic-generation conjecture for canonical-form and log canonical compactifications

摘录 / EXCERPT: Let $\mathcal{X}_{\mathrm{cf}}(3,6)$ be the closure of the variety parametrized by the canonical forms of the real regions of $\mathcal{X}(3,6)$, and let the log canonical compacti...

conjecture | open | MathDB 页面 | 原文线索

V2-021451 | MATHDB-376433

Nori's conjecture for Chow groups of a general hypersurface

摘录 / EXCERPT: Nori's conjecture. If n is sufficiently large, then the natural map

conjecture | open | MathDB 页面 | 原文线索

V2-021452 | MATHDB-376434

The mirror-family conjecture for Calabi-Yau complete intersections in projective space

摘录 / EXCERPT: Mirror-family conjecture. The one-parameter family of d-dimensional varieties V' yields the mirror family for V.

conjecture | open | MathDB 页面 | 原文线索

V2-021453 | MATHDB-376438

The toric compactification conjecture for mirrors of Calabi-Yau complete intersections

摘录 / EXCERPT: Toric compactification conjecture. The mirror Calabi-Yau varieties with respect to V are Calabi-Yau compactifications of the complete intersection in ${\bf T}$ defined by

conjecture | open | MathDB 页面 | 原文线索

V2-021454 | MATHDB-376445

Simpson's motivic conjecture for rigid semisimple representations

摘录 / EXCERPT: Simpson's motivic conjecture. The representation $\varrho\otimes k$ is a direct summand over k of the monodromy representation of a motive over X, that is, it comes from geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-021455 | MATHDB-376449

Resolution-independence conjecture for string-theoretic Hodge numbers

摘录 / EXCERPT: Let X be a variety with mild Gorenstein singularities, and let $\widehat X_1$ and $\widehat X_2$ be smooth crepant resolutions of X, when such resolutions exist. Resolution-ind...

conjecture | open | MathDB 页面 | 原文线索

V2-021456 | MATHDB-376450

Coherence conjecture for locally coherent functors

摘录 / EXCERPT: Coherence conjecture for locally coherent functors. Every locally coherent X-functor is coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-021457 | MATHDB-376451

Rank-condition conjecture for configuration varieties of diagrams

摘录 / EXCERPT: Rank-condition conjecture. For an arbitrary diagram D, $\mathcal F_D$ is the set of configurations satisfying these conditions for every list of columns $C,C',\ldots$ of D. Equ...

conjecture | open | MathDB 页面 | 原文线索

V2-021458 | MATHDB-376452

Compatible-splitting conjecture for diagram configuration varieties

摘录 / EXCERPT: Compatible-splitting conjecture. For any diagram D and any Weyl group elements $u_1,\ldots,u_r$, the pairs

conjecture | open | MathDB 页面 | 原文线索

V2-021459 | MATHDB-376455

Iitaka-Severi finiteness conjecture for varieties of general type

摘录 / EXCERPT: Let X be an algebraic variety over a field of characteristic zero. Let $R(X,Y)$ denote the set of dominant rational maps $f:X\rightarrow Y$, and let ${\cal F}={\cal F}(X)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-021460 | MATHDB-376457

Uniform pluricanonical birationality conjecture for smooth complex threefolds

摘录 / EXCERPT: Fix a complex projective variety X, and let ${\cal G}_m$ be the set of nonsingular complex projective varieties Y for which the m-th pluricanonical map...

conjecture | open | MathDB 页面 | 原文线索

V2-021461 | MATHDB-376458

Caporaso-Harris-Mazur's Conjecture H for fibered powers

摘录 / EXCERPT: Let $X\to B$ be a morphism of varieties in characteristic 0, with general fiber a variety of general type. Denote by X^n_B the n-th fibered power of X over B. Caporaso-Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-021462 | MATHDB-376459

The deformation bound conjecture for the index of grad f

摘录 / EXCERPT: Let f be a real polynomial, let f^s be a deformation of f, and let $p\in {\bf L}”$. For $c\in {\bf R}$, write $i_{p,c}$ for the index associated with f at p and level c...

conjecture | open | MathDB 页面 | 原文线索

V2-021463 | MATHDB-376464

Caporaso-Harris-Mazur correlation conjecture for varieties of general type

摘录 / EXCERPT: Let $f:X \longrightarrow B$ be a proper morphism of integral varieties whose general fiber is an integral variety of general type. For a sufficiently large integer n, consider th...

conjecture | open | MathDB 页面 | 原文线索

V2-021464 | MATHDB-376465

Lang's geometric conjecture for varieties of general type

摘录 / EXCERPT: Let W be a variety of general type. Let $\Xi_W \subset W$ denote the locus formed by the union of all irreducible positive-dimensional subvarieties of W that are not of general...

conjecture | open | MathDB 页面 | 原文线索

V2-021465 | MATHDB-376466

The motivic quasi-isomorphism conjecture for the elliptic Bloch complex

摘录 / EXCERPT: Let E be an elliptic curve over a field k, let $B(E,3)$ denote the elliptic Bloch complex, and let ${\cal M}{\cal M}_k$ be the hypothetical abelian category of mixed motivic sh...

conjecture | open | MathDB 页面 | 原文线索

V2-021466 | MATHDB-376470

Harris's weak conjecture on rational points and Lang dimension

摘录 / EXCERPT: Let X be a variety defined over a number field K. Define the Lang dimension of X to be the maximal dimension of a variety of general type dominated by X. Harris's weak conj...

conjecture | open | MathDB 页面 | 原文线索

V2-021467 | MATHDB-376471

The uniruledness and Iitaka-fibration conjecture

摘录 / EXCERPT: Let X be a variety in characteristic 0. The uniruledness and Iitaka-fibration conjecture. Either X is uniruled, or $\operatorname{Kod}(X)\geq 0$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-021468 | MATHDB-376475

The nonabelian Hodge conjecture for motivic representations

摘录 / EXCERPT: Nonabelian Hodge conjecture. The points of $NL(X_s,GL(n))$, that is, the $\mathbb Z$-variations of Hodge structure, are motivic representations on X_s.

conjecture | open | MathDB 页面 | 原文线索

V2-021469 | MATHDB-376480

Gillet-Soulé's arithmetic analogues of the standard conjectures

摘录 / EXCERPT: Let X be an arithmetic variety, meaning a regular scheme projective and flat over $\mathbb Z$ whose generic fibre $X_{\mathbb Q}$ is smooth over $\mathbb Q$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021470 | MATHDB-376485

Very presentable targets have locally geometric mapping stacks

摘录 / EXCERPT: Let T be a geometric n-stack that is very presentable in the sense that its fundamental groups over artinian bases are affine and its higher homotopy groups are vector sheaves....

conjecture | open | MathDB 页面 | 原文线索

V2-021471 | MATHDB-376486

Locally geometricity conjecture for $K({\bf G}_m,2)$ and finite-type group schemes

摘录 / EXCERPT: Let S be a base scheme, let $X\to S$ be a flat projective morphism, and let $K({\bf G}_m,2)$ denote the Eilenberg-Mac Lane stack. For any group scheme G of finite type, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021472 | MATHDB-376487

Higher-vector-homotopy mapping stacks are locally geometric

摘录 / EXCERPT: Let T be a geometric n-stack whose homotopy sheaves π_i are vector sheaves for every $i\geq 3$, and let $X\to S$ be the morphism under consideration. The higher-vector-homo...

conjecture | open | MathDB 页面 | 原文线索

V2-021473 | MATHDB-376488

Flat projective mapping stacks are slightly geometric

摘录 / EXCERPT: Let T be any geometric, locally geometric, or slightly geometric n-stack, and let $X\to S$ be a flat projective morphism. The slightly geometric mapping-stack conjecture. The m...

conjecture | open | MathDB 页面 | 原文线索

V2-021474 | MATHDB-376489

de Rham mapping stacks of smooth projective varieties are geometric

摘录 / EXCERPT: Let T be a very presentable geometric n-stack, and let X be a smooth projective variety. Write X_{DR} for its de Rham formal stack. The de Rham mapping-stack conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-021475 | MATHDB-376503

van Geemen-van der Geer-Donagi Schottky characterization via second-order theta divisors

摘录 / EXCERPT: Let (P,Ξ) be a principally polarized abelian variety of dimension $p\geq 4$, with Ξ a symmetric theta divisor. Let $\Gamma=H^0(P,2\Xi)$, and let $\Gamma_{00}\subset\Gamma$...

conjecture | open | MathDB 页面 | 原文线索

V2-021476 | MATHDB-376504

Beauville-Debarre infinitesimal Schottky conjecture for Prym and Jacobian theta divisors

摘录 / EXCERPT: Let (P,Ξ) be a principally polarized abelian variety of dimension $p\geq 4$, and let $V_{inf,00}$ be the base locus of the linear system of quartic tangent cones at the origin...

conjecture | open | MathDB 页面 | 原文线索

V2-021477 | MATHDB-376505

Izadi's base-locus conjecture for Prym second-order theta divisors

摘录 / EXCERPT: Let (P,Ξ) be a Prym variety associated with a double cover $\pi:\widetilde C\to C$, and let Σ be the relevant Prym-theta subvariety. Let Γ_{00}' consist of sectio...

conjecture | open | MathDB 页面 | 原文线索

V2-021478 | MATHDB-376507

Schapira-Schneiders conjecture on the Euler class of a holonomic complex

摘录 / EXCERPT: Let X be a complex manifold of dimension d, let ${\cal M}^\bullet$ be an object of $\operatorname{D}^b_{good}({\cal D}^{op}_X)$, and let Λ be a conic subvariety of...

conjecture | open | MathDB 页面 | 原文线索

V2-021479 | MATHDB-376509

B. Shapiro and M. Shapiro's conjecture on real osculating Schubert calculus

摘录 / EXCERPT: Let $s_1,\ldots,s_n$ be distinct real numbers, and let $\gamma(s)$ be a parameterization of a rational normal curve in $\mathbf{R}^{m+p}$ of degree $m+p-1$. For integers k_i, def...

conjecture | open | MathDB 页面 | 原文线索

V2-021480 | MATHDB-376510

Semistable reduction for families of complex projective varieties

摘录 / EXCERPT: Semistable reduction conjecture. There is a projective alteration $B_1\to B$ and a projective modification $Y\to X\times_B B_1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-021481 | MATHDB-376517

The effectiveness conjecture for homotopy projectors on n-stacks

摘录 / EXCERPT: An n-stack is a functor on a site satisfying descent with respect to sieves, and $nSTACK/{\cal X}$ denotes the resulting higher category of n-stacks. Let U' be an n-stack w...

conjecture | open | MathDB 页面 | 原文线索

V2-021482 | MATHDB-376520

The mixed twistor structure conjecture for relative higher Malcev completion

摘录 / EXCERPT: Let X be a projective variety, and let ρ be a reductive representation of its fundamental group, assumed reductive after restriction to the fundamental group of the normaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-021483 | MATHDB-376521

The nef-cone conjecture for the moduli space of principally polarized abelian varieties

摘录 / EXCERPT: Nef-cone conjecture. For any $g\geq2$, the nef cone on ${\cal A}_g^*$ is given by the divisors $aL-bD$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-021484 | MATHDB-376523

Chain conjecture for closures of cells in \mathbb{G}_T

摘录 / EXCERPT: Let $\mathbb{G}_T$ be the parameter space of graded quotients with Hilbert function T, and let U_c and U_e be cells in $\mathbb{G}_T$ of dimensions c and e, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-021485 | MATHDB-376540

The rationality conjecture for the polarized index

摘录 / EXCERPT: Rationality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-021486 | MATHDB-376541

The boundedness-of-index conjecture for canonical Q-Fano varieties

摘录 / EXCERPT: Boundedness-of-index conjecture. The set of possible values of $r(W)$ for canonical ${\bf Q}$-Fano varieties W of dimension n is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-021487 | MATHDB-376542

The canonical Q-Fano contraction conjecture for primitive varieties

摘录 / EXCERPT: Canonical ${\bf Q}$-Fano contraction conjecture. There exist a resolution of singularities

conjecture | open | MathDB 页面 | 原文线索

V2-021488 | MATHDB-376543

The vanishing conjecture for primitive varieties

摘录 / EXCERPT: Vanishing conjecture. For every $i>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-021489 | MATHDB-376544

The polyhedrality conjecture for the effective cone of primitive varieties

摘录 / EXCERPT: Polyhedrality conjecture. The cone $\Lambda_{\rm eff}(V,{\cal L})$ is a rational finitely generated polyhedral cone.

conjecture | open | MathDB 页面 | 原文线索

V2-021490 | MATHDB-376546

The strong saturation conjecture for smooth quasi-projective varieties

摘录 / EXCERPT: Strong saturation conjecture. The complex analytic variety $V({\bf C})$ is ${\cal L}$-primitive.

conjecture | open | MathDB 页面 | 原文线索

V2-021491 | MATHDB-376631

The second fundamental theorem conjecture for the determinant orbit closure

摘录 / EXCERPT: Let v be the determinant function and let $Delta_V[v]$ be the class variety associated with NC, namely the orbit closure of the determinant. Let $X(Pi_v)$ be the variety and sc...

conjecture | open | MathDB 页面 | 原文线索

V2-021492 | MATHDB-376696

The elliptic genus formula for generalized Kummer varieties

摘录 / EXCERPT: Let A be an abelian surface over $\mathbb{C}$, let $A^{[\ell]}$ be its Hilbert scheme of length-ℓ subschemes, and let $A^{\langle\ell-1\rangle}$ be the fiber over 0 of the...

conjecture | open | MathDB 页面 | 原文线索

V2-021493 | MATHDB-376713

The geometric McKay correspondence

摘录 / EXCERPT: Let Γ be a finite subgroup of $SU(n)$, let $M=\mathbb{C}^n/\Gamma$, and let $\widetilde{M}$ be the crepant resolution introduced above. The relevant objects are the cohomolo...

conjecture | open | MathDB 页面 | 原文线索

V2-021494 | MATHDB-376726

Badness-level conjecture for Calabi-Yau modular forms

摘录 / EXCERPT: Let X be a d-dimensional Calabi-Yau variety whose associated modular form has weight $d+1$ on some $\Gamma_0(N)$, and write the level as ... where the product is over primes of b...

conjecture | open | MathDB 页面 | 原文线索

V2-021495 | MATHDB-376741

Global Plücker formula for W-surfaces associated with simple Lie algebras

摘录 / EXCERPT: Global Plücker formula. For each k, one has

conjecture | open | MathDB 页面 | 原文线索

V2-021496 | MATHDB-376748

Universal recursive-polynomial conjecture for Fano hypersurfaces

摘录 / EXCERPT: Universal recursion conjecture. There are universal recursive polynomial laws expressing

conjecture | open | MathDB 页面 | 原文线索

V2-021497 | MATHDB-376770

The eigenvalue-support conjecture for rational KP solitons

摘录 / EXCERPT: Eigenvalue-support conjecture. The support of the distributions annihilating $z^n\psi_M$ is the set of eigenvalues of the matrices Y and Z, with the highest derivative taken at...

conjecture | open | MathDB 页面 | 原文线索

V2-021498 | MATHDB-376781

Conjecture on the semicanonical functions for semistable diagonals

摘录 / EXCERPT: Let $I'=I\setminus\{p\}$ index the irreducible components Y_k of the exceptional variety $\mathfrak{L}=\mathfrak{L}_m$, and let $\widetilde E_k(m)$ be the characteristic function...

conjecture | open | MathDB 页面 | 原文线索

V2-021499 | MATHDB-376881

Generic-condition conjecture for periodic Schrödinger potentials

摘录 / EXCERPT: Let $H_0=-\Delta+q(x)$ be a Schrödinger operator with periodic potential q, let $F_{\,\lambda}(q)$ be its Fermi variety, and let the Condition mean that, for every λ in t...

conjecture | open | MathDB 页面 | 原文线索

V2-021500 | MATHDB-376884

Conjecture on PG-shell Lefschetz chains for projective embeddings

摘录 / EXCERPT: Let X be a complex projective manifold of dimension $n>0$, let $j:X\hookrightarrow P=\mathbb{P}^{N}(\mathbb{C})$ be an arithmetically normal embedding, and let $\{W_p\}_{p=0}^n$...

conjecture | open | MathDB 页面 | 原文线索

V2-021501 | MATHDB-376906

Conjecture on osculating Schubert families for classical flag manifolds

摘录 / EXCERPT: Let V be a vector space with a bilinear form, and let $G:=\operatorname{Aut}(V,\langle\cdot,\cdot\rangle)$ be respectively a general linear, orthogonal, or symplectic group accor...

conjecture | open | MathDB 页面 | 原文线索

V2-021502 | MATHDB-376907

Classification of smooth Mori fibrations birational to the double cone variety

摘录 / EXCERPT: Classification conjecture. Any smooth Mori fibration birational to U is biregular to either U or V_l for some curve l.

conjecture | open | MathDB 页面 | 原文线索

V2-021503 | MATHDB-376911

Expected intersection-point conjecture for two plane curves

摘录 / EXCERPT: Expected intersection-point conjecture. For $d>m$ and $d\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-021504 | MATHDB-376913

Goncharov's regulator image conjecture

摘录 / EXCERPT: Goncharov's regulator image conjecture. The image of this regulator map in Deligne cohomology coincides with the image of Beilinson's regulator map.

conjecture | open | MathDB 页面 | 原文线索

V2-021505 | MATHDB-376917

Nakajima's zero-weight-space conjecture for quiver varieties

摘录 / EXCERPT: Let $M_{0,\lambda}(d,v)$ be the quiver variety in question, and set ... Let $(L_d)_0$ denote the 0-weight space of the Kac-Moody algebra associated to the quiver, for the represent...

conjecture | open | MathDB 页面 | 原文线索

V2-021506 | MATHDB-376918

Connectedness and normality conjecture for finite-type quiver varieties

摘录 / EXCERPT: Fix a quiver of finite type, take $m=(1,\dots,1)$ and $\lambda=0$, and let d,v be fixed. Write $M_{m,0}(d,v)$ and $M_{0,0}(d,v)$ for the corresponding quiver varieties. Connected...

conjecture | open | MathDB 页面 | 原文线索

V2-021507 | MATHDB-376956

The local cohomology conjecture for additive Okamoto-Painlevé pairs

摘录 / EXCERPT: Local cohomology conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-021508 | MATHDB-376963

Elliptic analogue of the Beilinson-Soulé conjecture

摘录 / EXCERPT: Let ${\cal E} \to B$ be an elliptic curve over a regular noetherian base. For integers $k \ge 0$ and j, let $H^i_{\cal M}({\cal E}^k,j)_{-,\ldots,-}$ denote the indicated minus e...

conjecture | open | MathDB 页面 | 原文线索

V2-021509 | MATHDB-376965

Universal nonabelian mixed Hodge structure conjecture

摘录 / EXCERPT: Let X be a simply connected smooth projective variety. Universal nonabelian mixed Hodge structure conjecture. There is a universal morphism to a simply connected nonabelian mixed...

conjecture | open | MathDB 页面 | 原文线索

V2-021510 | MATHDB-376966

Formality criterion for simply connected finite CW complexes

摘录 / EXCERPT: Let X be a simply connected finite CW complex, and let X_B denote the constant ∞-stack whose values are the Poincaré ∞-groupoid of X. Formality criterion conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-021511 | MATHDB-376967

Relative sections conjecture for variations of nonabelian mixed Hodge structures

摘录 / EXCERPT: Let $f:X\to S$ be a projective morphism of smooth varieties, and let $\underline{{\cal U}}$ be a (linearized) pre-vnamhs over X. Relative sections conjecture. There is a (lineari...

conjecture | open | MathDB 页面 | 原文线索

V2-021512 | MATHDB-376969

Reducedness conjecture for the nilpotent local-model scheme

摘录 / EXCERPT: Reducedness conjecture. The scheme $\overline N$ is reduced. This reducedness assertion is used to identify the special fiber of the canonical flat model and to establish its norma...

conjecture | open | MathDB 页面 | 原文线索

V2-021513 | MATHDB-376996

The fewnomial bound for non-degenerate positive roots

摘录 / EXCERPT: Fewnomial root-count conjecture. The maximum number of non-degenerate roots of F in the positive orthant is

conjecture | open | MathDB 页面 | 原文线索

V2-021514 | MATHDB-377008

Polar de Rham theorem for smooth projective manifolds

摘录 / EXCERPT: Polar de Rham theorem. The mapping ρ is an isomorphism. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-021515 | MATHDB-377040

Resonance formula for Chen groups of arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a complex hyperplane arrangement with arrangement group $G=G(\mathcal{A})$. Write the resonance variety as $R_1(\mathcal{A})=\bigcup_i L_i$, where the L_i ar...

conjecture | open | MathDB 页面 | 原文线索

V2-021516 | MATHDB-377041

Resonance formula for lower central series ranks of arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a complex hyperplane arrangement with arrangement group $G=G(\mathcal{A})$. Let $R_1(\mathcal{A})=\bigcup_i L_i$ be its resonance variety, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-021517 | MATHDB-377046

Goncharov's freeness conjecture for the motivic Galois Lie algebra

摘录 / EXCERPT: Goncharov's conjecture. The Lie algebra $\mathfrak g$ is a prounipotent Lie algebra freely generated by elements $z_3,z_5,z_7,\dots$, where z_m has weight $-2m$.

conjecture | solved | MathDB 页面 | 原文线索

V2-021518 | MATHDB-377055

The $n!$ conjecture for derivative spaces of lattice determinants

摘录 / EXCERPT: $n!$ conjecture. The dimension of D_{μ} is equal to $n!$.

conjecture | open | MathDB 页面 | 原文线索

V2-021519 | MATHDB-377056

The graded character conjecture for derivative spaces D_{μ}

摘录 / EXCERPT: Graded character conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-021520 | MATHDB-377057

Normality and Cohen-Macaulayness of higher-dimensional isospectral Hilbert schemes

摘录 / EXCERPT: Normality and Cohen-Macaulayness conjecture. For all d and n, $X_n(\mathbb{C}^d)$ is normal and Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-021521 | MATHDB-377058

Higher-dimensional polygraph freeness conjecture

摘录 / EXCERPT: Higher-dimensional polygraph conjecture. The coordinate ring of $Z(n,l)$ over $E=\mathbb{A}^d$ is a free $k[\mathbf{z}]$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-021522 | MATHDB-377064

Questions on negative indices of rigidity in the Deligne-Simpson problem

摘录 / EXCERPT: Questions on negative indices of rigidity. 1) Is the dimension of the variety of $(p+1)$-tuples with non-trivial centralizers always smaller than the expected dimension of the vari...

open_problem | open | MathDB 页面 | 原文线索

V2-021523 | MATHDB-377074

Batyrev's boundedness conjecture for the string-theoretic index

摘录 / EXCERPT: Let X be an r-dimensional normal complex variety having at most Gorenstein canonical singularities. Its string-theoretic index is the least positive integer l such that...

conjecture | open | MathDB 页面 | 原文线索

V2-021524 | MATHDB-377076

Goncharov's Grassmannian Lie coalgebra and mixed Tate motives conjecture

摘录 / EXCERPT: Let F be a field and let ... The maps ... are required to make the induced sequence $G_\bullet(F)\to\Lambda^2G_\bullet(F)\to\Lambda^3G_\bullet(F)\to\cdots$ a complex. Goncharov's mot...

conjecture | open | MathDB 页面 | 原文线索

V2-021525 | MATHDB-377077

Goncharov's Aomoto-Grassmannian comparison conjecture

摘录 / EXCERPT: Let $\widetilde A_n$ be the free abelian group generated by admissible pairs of simplices, let $\overline C_{2n}$ be the Grassmannian configuration group, and suppose the maps...

conjecture | open | MathDB 页面 | 原文线索

V2-021526 | MATHDB-377078

Goncharov's weight-four Grassmannian-to-polylogarithmic complex conjecture

摘录 / EXCERPT: Let $G_4(F)$ be the weight-four Grassmannian group, let $\mathcal B_4(F)$ be the corresponding polylogarithmic group, and let ... be the canonical cobracket described in the construc...

conjecture | open | MathDB 页面 | 原文线索

V2-021527 | MATHDB-377083

Relative local Fujita freeness conjecture for higher direct images

摘录 / EXCERPT: Let $f:Y\to X$ be a surjective morphism from a smooth projective variety to a smooth projective variety of dimension n, and suppose that f is smooth over $X_0=X\setminus B$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-021528 | MATHDB-377091

The irreducible-smooth characteristic scheme conjecture

摘录 / EXCERPT: Let $\mathcal{C}=(\mathbb{Z}/2)\langle a_1,\ldots,a_n\rangle/I$ be the characteristic algebra, and let X be the affine scheme in $\mathbb{A}^n$ over $\mathbb{Z}/2$ obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-021529 | MATHDB-377105

Stanley's reformulation of Kontsevich's polynomial-counting conjecture

摘录 / EXCERPT: Let G be a finite graph with edge set $E(G)$, and let ... where the sum runs over spanning trees T of G. Let $V(Q_G)$ be the zero scheme of Q_G in $\mathbf A^{E(G)}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021530 | MATHDB-377106

Stembridge's polynomial-counting conjecture for nondegenerate sparse symmetric matrices

摘录 / EXCERPT: Let G be a simple graph with vertex set $V(G)=\{v_1,\ldots,v_n\}$. Let Z^o_G be the scheme of nondegenerate symmetric $n\times n$ matrices M satisfying ... Equivalently, for a...

conjecture | open | MathDB 页面 | 原文线索

V2-021531 | MATHDB-377107

Affineness and affinization of the Eilenberg-Mac Lane stack of the unipotent completion

摘录 / EXCERPT: For an integer n, let $\bbH$ be the affine group scheme of formal series $P(X)\in 1+XA[[X]]$ satisfying $P(X)P(Y)=P(X+Y+XY)$. The assignment $m\mapsto(1+X)^m$ defines a morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-021532 | MATHDB-377116

Chow-group identification conjecture for smooth proper varieties

摘录 / EXCERPT: Let $k\subset F$ be the fields in the paper, let $n=\infty$, and let X be a smooth irreducible proper variety over k. Write $CH_0(X_F)_{\mathbb Q}$ for the rationalized group o...

conjecture | open | MathDB 页面 | 原文线索

V2-021533 | MATHDB-377117

Strong Fujita freeness conjecture

摘录 / EXCERPT: Strong Fujita freeness conjecture. The linear system

conjecture | open | MathDB 页面 | 原文线索

V2-021534 | MATHDB-377135

Uniqueness conjecture for the tensor-product isomorphism of quiver-variety homology modules

摘录 / EXCERPT: Let $\mathfrak g$ be the Lie algebra used in the quiver-variety construction, let $\mathbf w^1$ and $\mathbf w^2$ be the two framing data, let $\mathfrak L(\mathbf w^i)$ be the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-021535 | MATHDB-377149

Grothendieck-Hodge conjecture for coniveau

摘录 / EXCERPT: Let X be a proper smooth scheme over $\mathbb C$. For integers i,j, define the coniveau filtration by ... The inclusion ... identifies the left-hand side with a subspace of rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-021536 | MATHDB-377150

Grothendieck-Hodge conjecture for the weight-graded cohomology of singular schemes

摘录 / EXCERPT: Let X be a proper integral scheme over $\mathbb C$, and let $\nu^{i,l}$ be the natural map ... where $(N^lH^t)^\bullet$ is induced by the coniveau filtration on the components of a...

conjecture | open | MathDB 页面 | 原文线索

V2-021537 | MATHDB-377151

Grothendieck-Hodge conjecture for algebraic 1-motives

摘录 / EXCERPT: Let X be smooth and proper over $\mathbb C$. Let $\operatorname{gr}^0_{F_m}CH^p(X)$ denote the degree-zero graded piece of the motivic filtration on codimension-p Chow groups,...

conjecture | open | MathDB 页面 | 原文线索

V2-021538 | MATHDB-377152

The Hodge 1-motive conjecture for proper complex schemes

摘录 / EXCERPT: Let X be a proper scheme over $\mathbb C$, let $\pi:X_\bullet\to X$ be a smooth proper hypercovering, and let $H^{2p+i}(X)$ be Deligne's mixed $\mathbb Q$-Hodge structure on...

conjecture | open | MathDB 页面 | 原文线索

V2-021539 | MATHDB-377159

The nilpotence conjecture for higher K-theory of monoid rings

摘录 / EXCERPT: Let M be an arbitrary commutative, cancellative, torsion-free monoid without non-trivial units, let R be a regular ring, and let $i>0$. For every sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-021540 | MATHDB-377175

Goncharov's motivic higher Chow group conjecture

摘录 / EXCERPT: Let F be a field. For $m\ge 3$, let ${\mathcal G}_m(F)$ be Goncharov's motivic complex group in degree one, let ${\delta}_m$ be its differential, and define ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-021541 | MATHDB-377176

Bloch group conjecture for $CH^3(F,5)$

摘录 / EXCERPT: Let F be a field. Let $B_2(F)$ be the Bloch group, let $R_3(F)$ be the subgroup generated by the stated relations in ${\mathbb Z}[{\mathbb P}^1_F]$, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021542 | MATHDB-377178

Köhler-Rössler's equivariant arithmetic Riemann-Roch conjecture for principally polarized abelian schemes

摘录 / EXCERPT: Köhler-Rössler's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021543 | MATHDB-377211

Beilinson-Lichtenbaum generalized Hilbert 90 conjecture for motivic cohomology

摘录 / EXCERPT: Let k be a field and let $n\ge 0$. The motivic complex ${\bf Z}(n)$ is evaluated on $\operatorname{Spec}(k)$ in the étale topology. Beilinson-Lichtenbaum's generalized Hilbert 90...

conjecture | open | MathDB 页面 | 原文线索

V2-021544 | MATHDB-377212

Beilinson-Lichtenbaum comparison conjecture for motivic cohomology

摘录 / EXCERPT: Beilinson-Lichtenbaum comparison conjecture. This map is an isomorphism for $p\le q+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-021545 | MATHDB-377213

Beilinson-Lichtenbaum generalized Hilbert 90 conjecture in Lichtenbaum motivic cohomology

摘录 / EXCERPT: Beilinson-Lichtenbaum generalized Hilbert 90 conjecture. For every $n\ge 0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-021546 | MATHDB-377216

Kouchnirenko's fewnomial conjecture for positive solutions

摘录 / EXCERPT: Consider a system of n polynomials in n variables, where the ith polynomial has m_i monomials and all solutions are simple. Kouchnirenko's conjecture. The system has at mos...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-021547 | MATHDB-377217

Full reality conjecture for Grassmannian Schubert data on partial flag manifolds

摘录 / EXCERPT: Let ${\bf a}=1<a_1<a_2<\cdots<a_r<a_{r+1}=n$, and let $(\alpha^1,i_1),\ldots,(\alpha^s,i_s)$ be Grassmannian Schubert data for the partial flag manifold $\mathbb{F}\ell_{\bf a}$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-021548 | MATHDB-377225

Multiplicity conjecture for Schubert varieties at arbitrary points

摘录 / EXCERPT: Let $w,\tau\in W/W_{P_d}$. Let R^+ and $R_{P_d}^+$ denote the positive roots of the ambient root system and of the parabolic subsystem, respectively. Define $S'_{w,\tau}$ to be t...

conjecture | open | MathDB 页面 | 原文线索

V2-021549 | MATHDB-377228

Uniqueness conjecture for canonical parallel translation

摘录 / EXCERPT: Let X_K be a smooth connected scheme over K, let $x_0,x_1\in X(K)$, and let F_{x_i} denote the corresponding fibre functors on the category under consideration. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-021550 | MATHDB-377229

Monodromy conjecture for crystalline cohomology

摘录 / EXCERPT: Let X_K be a smooth scheme of finite type over K. Let $H^{n}_{pst}(X_K)$ be the finite-dimensional $\mathcal{H}^{ur}_1$-module equipped with its increasing weight filtration...

conjecture | open | MathDB 页面 | 原文线索

V2-021551 | MATHDB-377250

Generalized Carlson-Clemens-Morgan conjecture for crossover mappings

摘录 / EXCERPT: Let X be a complex algebraic manifold and let d be a positive integer. Write $QH^k(X)$ for the quotient cohomology groups appearing in the exact sequence associated with iterat...

conjecture | open | MathDB 页面 | 原文线索

V2-021552 | MATHDB-377251

The differential-form criterion for rational connectedness

摘录 / EXCERPT: Rational connectedness criterion. X is rationally connected if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-021553 | MATHDB-377253

The M-regularity index conjecture for syzygies of abelian varieties

摘录 / EXCERPT: Let X be an abelian variety and let A be an ample line bundle on X. The M-regularity index $m(A)$ is the largest integer l such that ... is M-regular for all distinct point...

conjecture | open | MathDB 页面 | 原文线索

V2-021554 | MATHDB-377260

Rational quotient conjecture for special fibrations

摘录 / EXCERPT: Let $f:X\dashrightarrow Y$ be a special fibration with $\kappa(Y,f)=-\infty$. Let $\Delta(f)$ be its base orbifold, and let κ-rationally generated mean that the general orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-021555 | MATHDB-377261

Deformation conjecture for the specialness invariant

摘录 / EXCERPT: Let $X\in\mathcal C$ be smooth, and let $\operatorname{ess}(X)$ denote the specialness invariant introduced in the paper. Deformation conjecture. The invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-021556 | MATHDB-377262

Conjecture III_H: special manifolds and Kobayashi speciality

摘录 / EXCERPT: Let $X\in\mathcal C$, and call X H-special when its Kobayashi pseudometric satisfies $d_X\equiv 0$. Conjecture IIIH. The manifold X is special if and only if it is H-specia...

conjecture | open | MathDB 页面 | 原文线索

V2-021557 | MATHDB-377268

Positive-density conjecture for Hodge-Witt reduction

摘录 / EXCERPT: Let K be a number field and let X/K be a smooth projective variety. Say that X has Hodge-Witt reduction at a prime of K when its smooth reduction at that prime is Hodge-Wit...

conjecture | open | MathDB 页面 | 原文线索

V2-021558 | MATHDB-377270

Toën's full-faithfulness conjecture for the functor from derived schemes to geometric stacks

摘录 / EXCERPT: Full-faithfulness conjecture. The functor Θ is fully faithful. This would embed the homotopy theory of DG-schemes into the homotopy theory of geometric stacks over the mod...

conjecture | open | MathDB 页面 | 原文线索

V2-021559 | MATHDB-377298

The birational parametrization conjecture for highly movable rational curves

摘录 / EXCERPT: Let W be an arbitrary projective variety of dimension n. Assume that W contains a rational curve and that every rational curve of W moves in a family of dimension at least...

conjecture | open | MathDB 页面 | 原文线索

V2-021560 | MATHDB-377299

The normality conjecture for surface fibers of symplectic contractions

摘录 / EXCERPT: Let F be the generic fiber of a birational contraction from a smooth projective symplectic variety. Normality conjecture for surface fibers. If F is a surface, then F is norm...

conjecture | open | MathDB 页面 | 原文线索

V2-021561 | MATHDB-377300

The projective-space exceptional-locus conjecture for symplectic contractions

摘录 / EXCERPT: Let X be a smooth projective symplectic variety of dimension $2n$, and let ... be a birational contraction whose degenerate locus B is contracted to isolated points. Projective-s...

conjecture | open | MathDB 页面 | 原文线索

V2-021562 | MATHDB-377301

The strict semismallness conjecture for symplectic contractions

摘录 / EXCERPT: Let X be a smooth projective symplectic variety and let ... be a birational contraction. A contraction is strictly IH-semismall when its relevant strata satisfy the strict semismal...

conjecture | open | MathDB 页面 | 原文线索

V2-021563 | MATHDB-377321

The linear Picard-number bound for toric Fano varieties

摘录 / EXCERPT: Let X be an n-dimensional toric Fano variety, and let ρ denote its Picard number. The preceding discussion suggests that varieties with maximal Picard number are S_3-bun...

conjecture | open | MathDB 页面 | 原文线索

V2-021564 | MATHDB-377322

Stratification conjecture for quasiderivation fiber dimensions

摘录 / EXCERPT: Let G/P be a homogeneous space, let M_λ be the maximal proper P-submodule associated with a parameter λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021565 | MATHDB-377350

The principal nilpotent-pair conjecture for non-exceptional pairs

摘录 / EXCERPT: Let $\mathfrak g$ be a simple Lie algebra with Cartan subalgebra $\mathfrak h$, let $\mathfrak g^d=\mathfrak g\oplus\mathfrak g$, and let $\mathbf e=(e_1,e_2)\in\mathfrak g^d$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-021566 | MATHDB-377366

The homeomorphism conjecture for the Springer variety and the crossingless-matching space

摘录 / EXCERPT: Let $S=\mathbb{S}^2$ and let B be the set of crossingless matchings on $2n$ points. For each $a\text{ in }B$, let $S_a\text{ be the submanifold of }S^{\mathord{\times}2n}$ consis...

conjecture | open | MathDB 页面 | 原文线索

V2-021567 | MATHDB-377368

The Springer cohomology conjecture for parabolic category $\mathcal{O}^{\kappa}$

摘录 / EXCERPT: Let $\kappa=(k_1,\dots,k_m)$ be a decomposition of n, let $\mathcal{O}^{\kappa}$ be the corresponding parabolic highest weight category for $\mathfrak{sl}_n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021568 | MATHDB-377373

Geometric syzygy conjecture for general canonical curves

摘录 / EXCERPT: Let $C \subset \mathbb{P}^{g-1}$ be a general canonical curve of genus g. A scrollar syzygy is a kth syzygy of rank $k+2$, and the corresponding schemes of such syzygies are ca...

conjecture | open | MathDB 页面 | 原文线索

V2-021569 | MATHDB-377376

Goncharov's isomorphism conjecture for the dihedral and motivic Lie coalgebras

摘录 / EXCERPT: Goncharov's isomorphism conjecture. The map $\nu_{\bullet,\bullet}^{\cal M}(\mu_N)$ is an isomorphism for either $N=1$, or $N=p$ is a prime and $w=m$.

conjecture | open | MathDB 页面 | 原文线索

V2-021570 | MATHDB-377377

Goncharov's scissor-congruence realization conjecture for mixed Tate motives

摘录 / EXCERPT: Goncharov's scissor-congruence realization conjecture. The map constructed in theorem 11.18.01.5(c) is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-021571 | MATHDB-377378

Goncharov's polynomiality conjecture for cyclotomic motivic dimensions

摘录 / EXCERPT: Goncharov's polynomiality conjecture. If p is a prime, then $d_w(p)$ and $d_{w,m}(p)$ are polynomials in p.

conjecture | open | MathDB 页面 | 原文线索

V2-021572 | MATHDB-377395

Hertling's asymptotic conjecture for CDV-structures on singularity unfoldings

摘录 / EXCERPT: Let M be the base space of an unfolding F of a singularity with a CDV-structure, let $R\subset M$ be the real analytic exceptional subvariety, let $E+\overline E$ be the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-021573 | MATHDB-377399

Generalized Amitsur conjecture for generalized Severi-Brauer varieties

摘录 / EXCERPT: Let F be an infinite field, and let A and B be central simple algebras over F of the same degree. For each $r<\deg(A)$, write $F_r(A)$ and $F_r(B)$ for the function fields...

conjecture | open | MathDB 页面 | 原文线索

V2-021574 | MATHDB-377423

The Broué-Malle-Michel cohomological conjecture for good elements

摘录 / EXCERPT: Let $\mathbf G$ be an algebraic group over $\bar{\mathbb F}_q$, let F be a Frobenius map, let $G=\mathbf G^F$, and let W be the F-split Weyl group. Let $w\in W$ be a good ele...

conjecture | open | MathDB 页面 | 原文线索

V2-021575 | MATHDB-377437

The mu-class closure conjecture for rational parametrizations

摘录 / EXCERPT: Mu-class closure conjecture. For every $\mu\leq\lfloor n/2\rfloor$,

conjecture | open | MathDB 页面 | 原文线索

V2-021576 | MATHDB-377444

The iterated-projective-line-bundle conjecture for intersections of two-row Springer-fiber components

摘录 / EXCERPT: Let Springer fibers be associated with two-row Young shapes, and let their irreducible components be described by the corresponding tableau and cup-diagram data. An iterated...

conjecture | open | MathDB 页面 | 原文线索

V2-021577 | MATHDB-377445

Vanishing of the moving Seshadri constant on the effective-cone boundary

摘录 / EXCERPT: Let X be the variety under consideration, let $D\in \operatorname{Div}(X)\otimes \mathbb{Q}$ be a divisor class on the boundary of the effective cone, and let $\eta\in X$ be a ge...

conjecture | open | MathDB 页面 | 原文线索

V2-021578 | MATHDB-377449

Mazur's conjecture for abelian varieties

摘录 / EXCERPT: Mazur's conjecture for abelian varieties. The preceding assertion should hold for every such A and G.

conjecture | open | MathDB 页面 | 原文线索

V2-021579 | MATHDB-377452

Projectivity conjecture for noncommutative Hilbert schemes

摘录 / EXCERPT: Projectivity conjecture. This union of schemes is actually a projective scheme.

conjecture | open | MathDB 页面 | 原文线索

V2-021580 | MATHDB-377467

The Constant Conjecture for cohomology rings of Hilbert schemes

摘录 / EXCERPT: Let X be a quasi-projective surface. For each n, let $H^*(X^{[n]})$ be expressed in the Heisenberg monomial basis $(\mathfrak h_\mu(n))$. Constant Conjecture. The structure con...

conjecture | open | MathDB 页面 | 原文线索

V2-021581 | MATHDB-377477

The first-n-coefficients conjecture for rational smoothness in simply laced root systems

摘录 / EXCERPT: First-n-coefficients conjecture. If (Φ_+,w) is not a rationally smooth pair, then comparing the first n coefficients and the last n coefficients of $P_w(t)$ suffices to...

conjecture | open | MathDB 页面 | 原文线索

V2-021582 | MATHDB-377479

Motivic fundamental lemma for orthogonal Lie algebras

摘录 / EXCERPT: Let n, k, and r be the parameters indexing the relevant classical groups and equal-valuation strips. Let $\Theta^{G,+}_{n,k,r}$ and $\Theta^{G,-}_{n,k,r}$ be the Chow motives...

conjecture | open | MathDB 页面 | 原文线索

V2-021583 | MATHDB-377481

Andreatta-Wiśniewski fiber normality conjecture

摘录 / EXCERPT: Let $\varphi:X\rightarrow Z$ be a Fano-Mori contraction of a manifold of dimension at most 4. Andreatta-Wiśniewski's conjecture. All fibers are normal, with the exception describ...

conjecture | open | MathDB 页面 | 原文线索

V2-021584 | MATHDB-377483

The termination conjecture for flips

摘录 / EXCERPT: Let X be a terminal $\mathbb Q$-factorial variety which is not minimal. Termination conjecture. After finitely many flips, there is an extremal ray whose exceptional locus has co...

conjecture | open | MathDB 页面 | 原文线索

V2-021585 | MATHDB-377484

The minimal model conjecture for non-uniruled varieties

摘录 / EXCERPT: Let X be a projective variety with at most terminal $\mathbb Q$-factorial singularities. Minimal model conjecture. There exists a minimal model X' birational to X if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-021586 | MATHDB-377485

The minimal model conjecture in terms of Kodaira dimension

摘录 / EXCERPT: Let X be a projective variety with at most terminal singularities. Minimal model conjecture. There is a minimal model X' birational to X if and only if ... The source presents...

conjecture | open | MathDB 页面 | 原文线索

V2-021587 | MATHDB-377486

Reid's general elephant conjecture for Q-Fano threefolds

摘录 / EXCERPT: Let X be a $\mathbb Q$-Fano threefold, and let $H\in|-K_X|$ be a generic section of the anticanonical divisor. Reid's general elephant conjecture. The surface H has at worst ca...

conjecture | open | MathDB 页面 | 原文线索

V2-021588 | MATHDB-377507

Functorial cyclic-homology decomposition for quotient schemes

摘录 / EXCERPT: Let X be a quasiprojective scheme over a field k, let G be a finite group acting on X, and assume that $\operatorname{char} k$ does not divide $|G|$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021589 | MATHDB-377508

Ruan's orbifold product conjecture for periodic cyclic homology

摘录 / EXCERPT: Let $\mathcal{X}$ be a smooth Deligne--Mumford stack with an étale cover $\pi:U\to\mathcal{X}$ by a scheme, and set $Z=U\times_{\mathcal{X}}U$. The correspondence $Z\times_UZ\to Z$...

conjecture | open | MathDB 页面 | 原文线索

V2-021590 | MATHDB-377510

Shapiro-Shapiro conjecture for maximally inflected rational curves

摘录 / EXCERPT: Let $d>r$ be integers and let $\calpha^1,\ldots,\alpha^n$ be ramification data for degree d rational curves in $\mathbb{P}^r$. Let $s_1,\ldots,s_n$ be distinct points of...

conjecture | open | MathDB 页面 | 原文线索

V2-021591 | MATHDB-377511

Deformation invariance of solitary points on maximally inflected plane quintics

摘录 / EXCERPT: A maximally inflected plane quintic is a real rational curve of degree 5 in $\mathbb{P}^2$ whose ramification occurs only at real points; a solitary point is an isolated real nod...

conjecture | open | MathDB 页面 | 原文线索

V2-021592 | MATHDB-377525

Motivic variation conjecture for the Grassmannian n-logarithm

摘录 / EXCERPT: Let $\widehat G_{n-1}^n$ be the configuration space used to define the Grassmannian n-logarithm, and let a_i,b_j be the maps on this space appearing in its functional equations...

conjecture | open | MathDB 页面 | 原文线索

V2-021593 | MATHDB-377526

Homotopy conjecture for the residue map on motivic complexes

摘录 / EXCERPT: Let X be a projective regular curve over an algebraically closed field k, and let $F=k(X)$. Let $\Gamma(F;n)$ and $\Gamma(k;n-1)$ be the weight-n and weight-$(n-1)$ motivic c...

conjecture | open | MathDB 页面 | 原文线索

V2-021594 | MATHDB-377527

Kato's conjecture on the algebraic cycle representing the monodromy operator

摘录 / EXCERPT: Let $S^{[3]}$ and $S^{[4]}$ be the strata appearing in the semistable degeneration, and let $[N]$ denote the cohomology class of the monodromy operator. The relevant restriction ma...

conjecture | open | MathDB 页面 | 原文线索

V2-021595 | MATHDB-377547

The equality-case conjecture for Hansen's theorem

摘录 / EXCERPT: Let d be the parameter in Hansen's theorem and let its lower estimate be the stated inequality involving d. The equality-case conjecture. Hansen's theorem continues to hold whe...

conjecture | open | MathDB 页面 | 原文线索

V2-021596 | MATHDB-377549

Atiyah's linear independence conjecture for configurations of points in three-space

摘录 / EXCERPT: Atiyah's conjecture. The polynomials $p_1,\ldots,p_N$ are linearly independent. Equivalently, their $N\times N$ coefficient matrix is nonsingular. Atiyah's conjecture is known for...

conjecture | open | MathDB 页面 | 原文线索

V2-021597 | MATHDB-377564

The (q,t) Catalan character conjecture for rational Cherednik algebras

摘录 / EXCERPT: The (q,t) Catalan character conjecture. In particular, $\chi_{\mathbf e\cdot L(\mathsf{triv})}(q,t)=C_n^{(k)}(q,t)$, the (q,t) Catalan number. This conjecture refines the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-021598 | MATHDB-377565

Goncharov's motivic iterated-integral isomorphism conjecture

摘录 / EXCERPT: The motivic Hopf algebra ${\cal A}_{\bullet}(F)$ is built from motivic iterated integrals, and the map in the source identifies the corresponding motivic iterated-integral construc...

conjecture | open | MathDB 页面 | 原文线索

V2-021599 | MATHDB-377566

Goncharov's period-surjectivity conjecture for mixed Tate motives

摘录 / EXCERPT: Let F be a number field, let ${\cal O}_{F,S}$ be a ring of S-integers, and let ${\cal A}_{\bullet}({\cal O}_{F,S})$ be its motivic Hopf algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021600 | MATHDB-377567

Hansen's parameter estimate for toric codes

摘录 / EXCERPT: Hansen's conjecture. If q is sufficiently large, then every $f\in L(G)=H^0(X,\operatorname{\mathcal O}(G))$ has at most n zeros in $X(\mathbb{F}_q)$, and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-021601 | MATHDB-377568

Toric-code estimate from the degree of the evaluation set

摘录 / EXCERPT: Toric-code degree conjecture. If q is sufficiently large, then every $f\in L(G)=H^0(X,\operatorname{\mathcal O}(G))$ has at most $n=\operatorname{deg}(P)$ zeros in...

conjecture | refuted | MathDB 页面 | 原文线索

V2-021602 | MATHDB-377576

Equal-degree dimension conjecture for good determinantal schemes

摘录 / EXCERPT: Let $W(\underline{0};\underline{d})$ be the locus of good determinantal schemes in $\mathbb{P}^{n+c}$ of codimension c given by the maximal minors of a $t\times(t+c-1)$ matrix wh...

conjecture | open | MathDB 页面 | 原文线索

V2-021603 | MATHDB-377604

Ambro's codimension inequality for minimal log discrepancies

摘录 / EXCERPT: Let X be a normal $\mathbb{Q}$-Gorenstein variety and let $Y=\sum_i q_i\cdot Y_i$, where $q_i\in\mathbb{R}_+$ and each $Y_i\subset X$ is a proper closed subscheme. For irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-021604 | MATHDB-377608

Existence of an algebraically stable ramified model for maps with large dynamical degree

摘录 / EXCERPT: Let $\f: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ be a rational map of topological degree e, and let $\lambda_1=\lim_{n\to\infty}\operatorname{deg}(\f^n)^{1/n}$ be its first dy...

conjecture | open | MathDB 页面 | 原文线索

V2-021605 | MATHDB-377616

The multiple monomial scheme conjecture

摘录 / EXCERPT: Let $X=\mathbb{P}^N\subset\mathbb{P}^{N+M}$ be a linear subspace of dimension $N\geq 6$. Let W be a Cohen-Macaulay multiple monomial scheme with reduced subscheme...

conjecture | open | MathDB 页面 | 原文线索

V2-021606 | MATHDB-377619

The additive polylogarithm cycle boundary conjecture

摘录 / EXCERPT: Let $T\mathcal M(n)$ denote the additive part of the degree-n component of the indecomposable space, and let $Z_n(a)$ be a codimension-n cycle on ... For $a\in\mathbb A^1-\{0\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-021607 | MATHDB-377636

Conjectural derived moduli stack of rank-n vector bundles

摘录 / EXCERPT: Let X be a fixed smooth projective variety and let n be a positive integer. For a commutative differential graded algebra A, let $\mathbb{R}\underline{Vect}_{n}(X)(A)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-021608 | MATHDB-377637

Automorphism-group PD-set conjecture for one-point AG codes

摘录 / EXCERPT: Let C be a one-point algebraic-geometric code associated to the curve ... over $GF(p^2)$, with length $n=2p(p-1)+p$, and let X be the corresponding curve over a field F. Arrang...

conjecture | open | MathDB 页面 | 原文线索

V2-021609 | MATHDB-377661

Stability conjecture for holomorphic maps with multiple degree

摘录 / EXCERPT: Let X be a nilpotent complex algebraic variety such that $\pi_2(X)$ is a free abelian group of rank r. Assign to each holomorphic map $f:\mathbb{C}\mathbb{P}^1\to X$ a multiple...

conjecture | open | MathDB 页面 | 原文线索

V2-021610 | MATHDB-377671

The torsion-translated subtori conjecture for characteristic varieties

摘录 / EXCERPT: Torsion-translation conjecture. The characteristic variety is a union of translated subtori of

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V2-021611 | MATHDB-377693

Canonical Kazhdan-Lusztig cell conjecture on completely prime primitive ideals

摘录 / EXCERPT: Canonical cell conjecture. The element of minimal length is unique, and $J(\lambda)$ is a completely prime maximal ideal of ${\mathbf U}({^L \mathfrak{g}})$ whose associated variet...

conjecture | open | MathDB 页面 | 原文线索

V2-021612 | MATHDB-377698

Kobayashi's converse conjecture for the meromorphic intrinsic pseudovolume form

摘录 / EXCERPT: Kobayashi's converse conjecture. If X is not of general type, then

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V2-021613 | MATHDB-377699

Equivalence conjecture for the intrinsic forms Φ_X and Ψ_X

摘录 / EXCERPT: Equivalence conjecture for Φ_X and Ψ_X. There exists a nonzero constant α depending on X such that

conjecture | open | MathDB 页面 | 原文线索

V2-021614 | MATHDB-377700

Self-K-isocorrespondence conjecture for Kodaira dimension

摘录 / EXCERPT: Self-K-isocorrespondence conjecture. If there exists a self-K-isocorrespondence $\Sigma\subset X\times X$ passing through an arbitrary pair (x,y), then

open_problem | open | MathDB 页面 | 原文线索

V2-021615 | MATHDB-377701

Discreteness conjecture for abelian varieties over maximal abelian extensions

摘录 / EXCERPT: Let K be a number field, and let A/K be an abelian variety. The discreteness conjecture for abelian varieties. 1. $A(K^{\rm ab})$ has the discreteness property. 2. If A does...

conjecture | open | MathDB 页面 | 原文线索

V2-021616 | MATHDB-377712

Conjecture on bijective actions on dense sets of rational points

摘录 / EXCERPT: Let X be a proper algebraic variety over a finitely generated field K of characteristic zero, and let $f:X\to X$ be a morphism. Suppose there is a subset $S\subseteq X(K)$ that...

conjecture | open | MathDB 页面 | 原文线索

V2-021617 | MATHDB-377713

Motivic zeta-function rationality conjecture

摘录 / EXCERPT: Let k be a field, let X be a variety over k, and let ${\mathcal M}_k$ denote the relevant localization of the Grothendieck ring of varieties. Let $Z_{\rm mot}$ be the motivic...

conjecture | open | MathDB 页面 | 原文线索

V2-021618 | MATHDB-377747

Boundary-intersection conjecture for affine flag fibers

摘录 / EXCERPT: Let $\fg$ be the Lie algebra under consideration, with index set I, and let $\fF^\lambda_{\fg,\fb}$ be the corresponding fiber space with open stratum...

conjecture | open | MathDB 页面 | 原文线索

V2-021619 | MATHDB-377749

Batyrev's index boundedness conjecture for log terminal Fano varieties

摘录 / EXCERPT: Index boundedness conjecture. The family of n-dimensional log terminal Fano varieties of index I is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-021620 | MATHDB-377750

Hard Lefschetz conjecture for algebraic cycle cohomology

摘录 / EXCERPT: Hard Lefschetz conjecture. For all k, the Lefschetz operator induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-021621 | MATHDB-377758

Cubic generation conjecture for binary naive Bayes models

摘录 / EXCERPT: Cubic generation conjecture. The prime ideal Q_G is generated by the 3×3-subdeterminants of any two-dimensional table obtained by flattening the n-dimensional table...

conjecture | open | MathDB 页面 | 原文线索

V2-021622 | MATHDB-377759

The logarithmic Batyrev-Manin conjecture for integral points

摘录 / EXCERPT: Let X be a variety over a number field F, let D be a divisor on X, and let L be a big divisor. For a finite set of places S and a model $(\mathcal U,\mathcal L)$ of a d...

conjecture | open | MathDB 页面 | 原文线索

V2-021623 | MATHDB-377761

The abundance conjecture for minimal models

摘录 / EXCERPT: Let X be a minimal model, meaning a complex projective variety with at most terminal singularities and nef canonical divisor K. For a nef divisor K, its nef dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-021624 | MATHDB-377774

Morita's conjecture on potentially good reduction of abelian varieties

摘录 / EXCERPT: Let E) be a number field, let A/E be an abelian variety, and let $\mathrm{MT}(A)$ denote its Mumford-Tate group. A point of an algebraic group is unipotent if it is unipotent i...

conjecture | open | MathDB 页面 | 原文线索

V2-021625 | MATHDB-377780

Inverse-shift conjecture for almost regular rectangular Birkhoff schemes

摘录 / EXCERPT: Let (A,S) be a Birkhoff interpolation scheme, with S a lower set, and fix non-negative integers p and q. An inverse shift moves elements of A downwards and to the left wi...

conjecture | open | MathDB 页面 | 原文线索

V2-021626 | MATHDB-377809

Beilinson's conjectural filtration on Chow groups

摘录 / EXCERPT: Beilinson's conjecture. For every $X\in\mathcal V_k$ there exists a decreasing filtration $F^\bullet$ on $A^i(X)$ such that: (a) $F^0A^j(X)=A^j(X)$ and $F^1A^j(X)=(A^j(X))_{hom}$;...

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V2-021627 | MATHDB-377810

Bloch's correspondence conjecture for the Albanese filtration

摘录 / EXCERPT: Bloch's conjecture. For every $\gamma\in A^2(Y\times X)$, the induced action $Gr^*_F\gamma$ depends only on $cl(\gamma)$.

conjecture | open | MathDB 页面 | 原文线索

V2-021628 | MATHDB-377829

Character-variety conjecture for the complexified cord ring

摘录 / EXCERPT: Let K be a knot, let $\Sigma_2(K)$ be its double branched cover, and let $HC_0^{\operatorname{ab}}(K)$ denote the abelianized degree-zero contact homology, identified with the co...

conjecture | open | MathDB 页面 | 原文线索

V2-021629 | MATHDB-377837

David-Philippon's height lower-bound conjecture for subvarieties of abelian varieties

摘录 / EXCERPT: David-Philippon's conjecture. There is a constant $c(A,L)$ depending only on A and L such that

conjecture | open | MathDB 页面 | 原文线索

V2-021630 | MATHDB-377843

Beilinson's Tate conjecture for regulator maps

摘录 / EXCERPT: Beilinson's Tate conjecture. In case k is finitely generated over $\mathbb Q$, $\operatorname{reg}_{U,\mathrm{et}}^q$ is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-021631 | MATHDB-377849

Nagata-Szemberg conjecture on maximal Seshadri constants

摘录 / EXCERPT: Let X be a smooth variety of dimension n and let L be an ample divisor on X. For very general points $p_1,\dots,p_r\in X$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-021632 | MATHDB-377879

Birational torus conjecture for symplectic reduction of a conjugation and dressing orbit

摘录 / EXCERPT: Birational torus conjecture. There is a birational correspondence between the variety $({\mathcal O}\times {\mathcal O}_-)///B_+$ described above and a symplectic torus of the same...

conjecture | open | MathDB 页面 | 原文线索

V2-021633 | MATHDB-377880

Birational torus conjecture for the triple-orbit symplectic reduction

摘录 / EXCERPT: Birational torus conjecture. There is a birational equivalence between the symplectic variety $({\mathcal O}_1\times{\mathcal O}_2\times\bar{{\mathcal O}}_3)///G$ and a complex tor...

conjecture | open | MathDB 页面 | 原文线索

V2-021634 | MATHDB-377886

A-stability conjecture for the third-order biased IMEX scheme

摘录 / EXCERPT: Let the biased IMEX scheme be ... Here λ is the explicit eigenvalue, μ is the implicit eigenvalue, and $\mathcal{S}$ is the explicit stability domain. A-stability conject...

conjecture | open | MathDB 页面 | 原文线索

V2-021635 | MATHDB-377887

A(alpha)-stability conjecture for the fourth-order biased IMEX scheme

摘录 / EXCERPT: Let the biased IMEX scheme be ... Here λ is the explicit eigenvalue, μ is the implicit eigenvalue, and $\mathcal{S}$ is the explicit stability domain. The notation...

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V2-021636 | MATHDB-377898

Injectivity of the cohomological monodromy restriction for Hilbert-scheme-type manifolds

摘录 / EXCERPT: Monodromy restriction conjecture. The restriction homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-021637 | MATHDB-377904

Projective-join characterization of defect polytopes

摘录 / EXCERPT: Let P be an integral polytope. A polytope is called a defect polytope when its associated projective toric variety has degenerate dual variety, equivalently when the invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-021638 | MATHDB-377912

Inversion of adjunction for arbitrary-codimension subvarieties

摘录 / EXCERPT: Let X be a nonsingular variety over a field of characteristic zero, and let $Y=\sum_{i=1}^k t_iY_i$ be a formal combination where $t_i>0$ are real numbers and each...

conjecture | open | MathDB 页面 | 原文线索

V2-021639 | MATHDB-377924

The log canonical ampleness conjecture for numerically positive divisors

摘录 / EXCERPT: Let X be an n-dimensional nonsingular projective algebraic variety over the complex numbers, and let $\Delta=\sum_{i\in I}\Delta_i$ be a reduced simple normal crossing divisor,...

conjecture | open | MathDB 页面 | 原文线索

V2-021640 | MATHDB-377938

Conjecture on the classification of varieties with full ramification locus

摘录 / EXCERPT: Let X be a smooth complex variety of dimension $n\ge 2$. Let L be an ample line bundle on X, spanned by a subspace $V\subseteq H^0(L)$ with $\dim(|V|)\ge n+1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021641 | MATHDB-377945

The characteristic equals deformation variety conjecture for knots

摘录 / EXCERPT: Let K be a knot in S^3. Its deformation variety $D(K)$ is the image of the character variety of $\mathrm{SL}_2(\mathbb C)$ representations of the complement...

conjecture | open | MathDB 页面 | 原文线索

V2-021642 | MATHDB-377947

Denef-Loeser's motivic monodromy conjecture

摘录 / EXCERPT: Let k be a field of characteristic zero, let $X=\mathbb A^d_{\mathbb C}$, let $h\in\mathbb C[x_1,\ldots,x_d]$ satisfy $h(\mathbf 0)=0$, and let $Z_{DL}(h,T)$ be the local Denef-L...

conjecture | open | MathDB 页面 | 原文线索

V2-021643 | MATHDB-377956

Higher-twisting Verlinde index formula

摘录 / EXCERPT: Let G be a compact Lie group, let $\mathfrak{M}$ be the moduli stack of G-bundles over a genus-g surface, let τ_0 and σ be the twistings appearing in the index co...

conjecture | open | MathDB 页面 | 原文线索

V2-021644 | MATHDB-377957

Twisted Gysin formula for the index over the moduli stack

摘录 / EXCERPT: Let G be a compact Lie group, let Σ be a closed surface of genus g, and let $\mathfrak{M}$ be the moduli stack of G-bundles on Σ. Let $\Pi:G^{2g}\to G$ be the p...

conjecture | open | MathDB 页面 | 原文线索

V2-021645 | MATHDB-377958

Naive localization formula for the index on the moduli stack

摘录 / EXCERPT: Let G be a rank-one compact Lie group with maximal torus T, let $\mathfrak{M}$ be the moduli stack of G-bundles, and let $\mathfrak{M}_T$ be the corresponding moduli stack of...

conjecture | open | MathDB 页面 | 原文线索

V2-021646 | MATHDB-377961

Knutson's puzzle conjecture for two-step flag varieties

摘录 / EXCERPT: Knutson's conjecture. The integral

conjecture | open | MathDB 页面 | 原文线索

V2-021647 | MATHDB-377971

Existence of coarse moduli spaces for semistable derived objects

摘录 / EXCERPT: Let X be a nonsingular projective variety, let $\sigma\in\operatorname{Stab}(X)$ be a stability condition, and let $\alpha\in\mathcal{N}(X)$ be a numerical equivalence class. Mod...

conjecture | open | MathDB 页面 | 原文线索

V2-021648 | MATHDB-377978

Pliability of Mori fibre spaces is an algebraic variety

摘录 / EXCERPT: Let $X\to S$ be a Mori fibre space, and let $\mathcal{P}(X/S)$ denote its pliability. Pliability algebraicity conjecture. The pliability $\mathcal{P}(X/S)$ is, in a natural way, an...

conjecture | open | MathDB 页面 | 原文线索

V2-021649 | MATHDB-377999

Equivalence of stacks of twisted differential-module categories

摘录 / EXCERPT: Equivalence conjecture. Any stack of $\mathbb{C}_X$-twisted $\mathcal{D}_X$-modules is $\mathbb{C}$-equivalent to a stack of the form

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V2-021650 | MATHDB-378011

The weight-monodromy conjecture for proper smooth varieties

摘录 / EXCERPT: Let K be the local field and F its residue field, let X_K be a proper smooth variety over K, and let ... Let M be the monodromy filtration on V and W the weight filtrat...

conjecture | open | MathDB 页面 | 原文线索

V2-021651 | MATHDB-378013

Finiteness conjecture for Cox rings of log Fano varieties

摘录 / EXCERPT: Finiteness of Cox ring. The ring

conjecture | open | MathDB 页面 | 原文线索

V2-021652 | MATHDB-378027

The dynamics conjecture for the once-punctured Klein bottle Fricke space

摘录 / EXCERPT: Dynamics conjecture. The action of Γ on Ω^K is wandering, and for each $c>2$ its action on

conjecture | open | MathDB 页面 | 原文线索

V2-021653 | MATHDB-378028

Segre-like conjecture for linear systems on generic K3 surfaces

摘录 / EXCERPT: Let S) be a generic $K3$ surface with ${\rm Pic}(S)=\mathbb{Z}H$, where $H^2=2g-2\geq 2$. Let $p_1,\ldots,p_r$ be points in general position on S, with prescribed multiplicitie...

conjecture | open | MathDB 页面 | 原文线索

V2-021654 | MATHDB-378029

Structure conjecture for linear systems on generic K3 surfaces

摘录 / EXCERPT: Let S and $\mathcal L$ be as above: S is a generic $K3$ surface with ${\rm Pic}(S)=\mathbb{Z}H$, $H^2=2g-2\geq 2$, and $\mathcal L=\mathcal L^n(d,m_1,\ldots,m_r)$ is the system...

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V2-021655 | MATHDB-378032

Quantum K-theory Chevalley formula for flag varieties

摘录 / EXCERPT: Quantum K-theory Chevalley conjecture. For every $w\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-021656 | MATHDB-378044

Lunts-Rosenberg conjecture on the quantum Beilinson-Bernstein equivalence

摘录 / EXCERPT: Let $\lambda\in\Lambda^+$, let $\tilde{\mathcal D}_{\mathcal B_q,\lambda}$ be the sheaf of twisted differential operators on the quantum flag variety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021657 | MATHDB-378053

Power-residue density conjecture for motivic Galois representations

摘录 / EXCERPT: Let $\rho:\operatorname{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})\to\operatorname{GL}_n\mathbf{Q}_\ell$ be a motivic Galois representation, and let $\delta_m(\rho)$ be the density rat...

conjecture | open | MathDB 页面 | 原文线索

V2-021658 | MATHDB-378054

Cebatorev-set power-residue density conjecture

摘录 / EXCERPT: Let ρ be a motivic Galois representation with open image as above, let $m\geq 2$, and let $\mathcal{P}$ be a Cebatorev set, meaning that, up to finitely many primes, it is spe...

conjecture | open | MathDB 页面 | 原文线索

V2-021659 | MATHDB-378059

(-1)-speciality conjecture for linear systems on rational surfaces

摘录 / EXCERPT: Let S be a rational surface, let D be an ample, non-special divisor on S, and let $p_1,\ldots,p_r$ be r points on S. Write ... for the corresponding linear system, and call...

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V2-021660 | MATHDB-378064

Quantum Dunkl-element isomorphism conjecture for crystallographic Coxeter systems

摘录 / EXCERPT: Let (W,S) be a crystallographic finite Coxeter system, let $qBE(W,S)$ be the associated quantum algebra, and let ${\bf R}[q_s][\tilde{\theta}_s\mid s\in S]$ denote the subalgebra...

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V2-021661 | MATHDB-378069

Boden-Hu conjecture on small resolutions of parabolic bundle moduli spaces

摘录 / EXCERPT: Let N and s be integers with $0<s<N$, and let ... For $\alpha\in W(N,s)^{\circ}$, let $M(\underline{\mathbf{1}})^\alpha$ be the moduli scheme of quasiparabolic bundles with multi...

conjecture | open | MathDB 页面 | 原文线索

V2-021662 | MATHDB-378075

Rational curves on Calabi-Yau varieties over finite fields

摘录 / EXCERPT: Let X be a smooth projective variety over a finite field k. Assume that X has trivial canonical class and that $X_{\bar{k}}$ has trivial algebraic fundamental group. An algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-021663 | MATHDB-378115

The motivic fundamental lemma

摘录 / EXCERPT: Motivic fundamental lemma. For every such pair and every $r\in\mathbb{Q}$, one has the identity in $\mathbb{K}$

conjecture | open | MathDB 页面 | 原文线索

V2-021664 | MATHDB-378117

Friedlander's generalized isomorphism conjecture

摘录 / EXCERPT: Let G be an algebraic group over an algebraically closed field k, and let n be a positive integer such that the characteristic of k does not divide n. The simplicial sche...

conjecture | open | MathDB 页面 | 原文线索

V2-021665 | MATHDB-378124

Generalized Tauberian Nagao conjecture

摘录 / EXCERPT: Let $f:\mathcal{X}\to\mathcal{Y}$ be a fibration in smooth projective varieties over a number field k. Let R be the finite set of excluded prime ideals, let $q_{\mathfrak p}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-021666 | MATHDB-378125

Analogue of Tate's conjecture for fibrations in varieties

摘录 / EXCERPT: Let $f:\mathcal{X}\to\mathcal{Y}$ be a fibration in varieties over a number field K. Let R be the finite set of excluded prime ideals, let $q_{\mathfrak p}$ denote the residue-...

conjecture | open | MathDB 页面 | 原文线索

V2-021667 | MATHDB-378128

Green-Griffiths conjecture on pseudo-hyperbolicity

摘录 / EXCERPT: Green-Griffiths conjecture. If V is of general type, then V is pseudo-hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-021668 | MATHDB-378129

Lang's strong Diophantine conjecture

摘录 / EXCERPT: Lang's strong Diophantine conjecture. There is a proper, closed subset V_0 of V such that, for any number field K, all but finitely many K-rational points of V lie in...

conjecture | open | MathDB 页面 | 原文线索

V2-021669 | MATHDB-378130

Harris-Tschinkel potential-density conjecture for varieties with negative canonical bundle

摘录 / EXCERPT: Harris-Tschinkel conjecture. Rational points are potentially dense on

conjecture | open | MathDB 页面 | 原文线索

V2-021670 | MATHDB-378146

Steenbrink's spectrum conjecture for curve singular loci

摘录 / EXCERPT: Let $f:X\to\mathbf A^1$ be a function on a smooth complex algebraic variety, and let x be a closed point of $f^{-1}(0)$. Suppose the singular locus of f is a curve with local c...

conjecture | open | MathDB 页面 | 原文线索

V2-021671 | MATHDB-378153

Toën's characterization conjecture for Tannakian Segal categories

摘录 / EXCERPT: Let A be a commutative ring spectrum, let T be an A-tensor Segal category, and let $FIB_A(T)$ be its stack of fiber functors. For an A-Tannakian Segal category, the source...

conjecture | open | MathDB 页面 | 原文线索

V2-021672 | MATHDB-378159

Algebraic-cycle conjecture for Springer fibres

摘录 / EXCERPT: Let $X=T^*(G/B)$ be the Springer resolution of the nilpotent cone $Y={\mathcal N}\subset{\mathfrak g}^*$ in the coadjoint representation of a semisimple algebraic group G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021673 | MATHDB-378160

Vishik's dimension conjecture for anisotropic quadratic forms in powers of the fundamental ideal

摘录 / EXCERPT: Vishik's conjecture. There is some $i\in\{1,2,\dots,n+1\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-021674 | MATHDB-378165

Generalized Lehmer conjecture for abelian varieties

摘录 / EXCERPT: Generalized Lehmer conjecture. There exists a positive constant $c=c(A,K,\mathcal L)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-021675 | MATHDB-378196

Mukhin-Varchenko population-flag variety conjecture with Bruhat-cell decomposition

摘录 / EXCERPT: Let $\mathfrak g$ be a Kac-Moody algebra, let $\Lambda_1,\dots,\Lambda_n$ be dominant integral weights, let $z_1,\dots,z_n$ be points, and let $\boldsymbol b$ be symmetric paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-021676 | MATHDB-378230

Invariance of log Lefschetz classes under prime-to-order powers

摘录 / EXCERPT: Let $f:V\to U$ be a finite étale morphism of connected separated and smooth schemes of finite type purely of dimension d over a perfect field F, and let σ be a non-trivi...

conjecture | open | MathDB 页面 | 原文线索

V2-021677 | MATHDB-378231

Integrality and pullback conjecture for Swan classes

摘录 / EXCERPT: Let U be a smooth connected scheme of dimension d over a perfect field F, and let $\mathcal F$ be a smooth $\overline{\mathbb F}_\ell$-sheaf on U. Swan class conjecture. (1...

conjecture | open | MathDB 页面 | 原文线索

V2-021678 | MATHDB-378232

Swan trace formula for tensor products over finite fields

摘录 / EXCERPT: Swan trace formula conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-021679 | MATHDB-378233

Kato's rank-one Swan class comparison and existence conjecture

摘录 / EXCERPT: Let X be a separated scheme of finite type over a perfect field F, let $U\subset X$ be dense, connected, smooth, and purely d-dimensional, let ℓ be invertible in F, an...

conjecture | open | MathDB 页面 | 原文线索

V2-021680 | MATHDB-378258

David's conjecture on Lehmer-type lower bounds for abelian varieties

摘录 / EXCERPT: David's conjecture. For every $\varepsilon>0$, there exists a constant $c(A/K,\mathcal{L})>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-021681 | MATHDB-378273

Conjecture that the intermediate variety is the Narasimhan-Ramanan desingularization

摘录 / EXCERPT: Let K_{σ} be the intermediate variety in the factorization of the morphism from the Kirwan desingularization K to the desingularization S. Narasimhan-Ramanan desingulari...

conjecture | open | MathDB 页面 | 原文线索

V2-021682 | MATHDB-378280

Mustăța's Minimal Resolution Conjecture for points on projective varieties

摘录 / EXCERPT: Let $X\subset\mathbb P^n$ be a projective variety of dimension $d\geq 1$, with $\operatorname{reg}(X)=m$ and Hilbert polynomial P_X. Let δ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-021683 | MATHDB-378286

The algebraicity conjecture for higher cycles in universal hypersurface families

摘录 / EXCERPT: The algebraicity conjecture. For all integers $b\in\mathbb N^*$ and $p\in\mathbb N$ there is an integer D depending only on b, p, Y and ${\mathcal O}(1)$, such that for all...

conjecture | open | MathDB 页面 | 原文线索

V2-021684 | MATHDB-378292

Dubrovin's reduced quantum cohomology conjecture

摘录 / EXCERPT: Dubrovin's reduced quantum cohomology conjecture. The variety X has generically semisimple reduced quantum cohomology, that is, M' is generically semisimple, if and only if the...

conjecture | open | MathDB 页面 | 原文线索

V2-021685 | MATHDB-378307

O'Grady's L conjecture for degree-2 polarized deformations of Hilbert schemes of K3 surfaces

摘录 / EXCERPT: Let $n\geq 2$, and let (X,H) be a degree-2 polarized irreducible symplectic variety, meaning that X is an irreducible symplectic manifold deformation equivalent to...

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V2-021686 | MATHDB-378308

O'Grady's I conjecture for birational involutions on deformations of Hilbert schemes of K3 surfaces

摘录 / EXCERPT: Let X be an irreducible symplectic manifold deformation equivalent to $(K3)^{[n]}$, let $(\,\ )$ be its Beauville form, and let $h\in H^{1,1}_{\mathbb Z}(X)$ satisfy $(h,h)=2$. F...

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V2-021687 | MATHDB-378339

Schneider's rank conjecture for Selmer groups over p-adic Lie extensions

摘录 / EXCERPT: Let A be an abelian variety over a finite extension F of $\mathbb Q$, let $p>2\dim(A)+1$ be prime, let $F_\infty=F(A_{p^\infty})$, and put...

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V2-021688 | MATHDB-378345

Jannsen's vanishing conjecture for étale cohomology

摘录 / EXCERPT: Let X be a smooth, projective scheme over ${\cal O}_K[1/S]$, and let $\bar{X}:=X\times_{{\cal O}_K[1/S]}\bar{K}$ be its base change to an algebraic closure. For integers i and...

conjecture | open | MathDB 页面 | 原文线索

V2-021689 | MATHDB-378347

Tautological Chow conjecture for stable maps to general flag varieties

摘录 / EXCERPT: Let G be a semisimple algebraic group, let P be a parabolic subgroup, and let $X=G/P$ be a general flag variety. For nonnegative integers n and curve class β, write...

conjecture | open | MathDB 页面 | 原文线索

V2-021690 | MATHDB-378348

Degree-independence conjecture for Chow rings of open stable-map spaces

摘录 / EXCERPT: Let X be any flag variety, and let $\mathcal M_{0,n}(X,\beta)$ be the open part of the moduli space of stable maps, parametrizing maps without boundary degenerations. Let the ind...

conjecture | open | MathDB 页面 | 原文线索

V2-021691 | MATHDB-378360

Translated-subgroup support conjecture for homotopy characteristic varieties

摘录 / EXCERPT: Translated-subgroup support conjecture. The support of $\pi_n(\partial B_{\epsilon}-D)$ is a union of translated subgroups of $\operatorname{Char}\pi_1(\partial B_{\epsilon}-D)$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-021692 | MATHDB-378361

Quasi-adjunction faces and characteristic-variety components conjecture

摘录 / EXCERPT: Quasi-adjunction characteristic-variety conjecture. The Zariski closure of $\exp(\mathcal F)\subset\operatorname{Char}H_1(U,\mathbb Z)$ is a component of the characteristic variety...

conjecture | open | MathDB 页面 | 原文线索

V2-021693 | MATHDB-378366

The K3 Hilbert-scheme conjecture for Jacobian abelian fibrations

摘录 / EXCERPT: Let X^{2n} be an irreducible holomorphic symplectic manifold equipped with an abelian fibration whose fibres are isomorphic to Jacobians of curves. A multisection is a subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-021694 | MATHDB-378367

The generalized Kummer conjecture for polarized abelian fibrations

摘录 / EXCERPT: Let X^{2n} be an irreducible holomorphic symplectic manifold equipped with an abelian fibration whose fibres have polarization type $(1,\ldots,1,n+1)$. Assume the fibration admit...

conjecture | open | MathDB 页面 | 原文线索

V2-021695 | MATHDB-378376

Kahn's integral comparison conjecture for Weil motivic cohomology

摘录 / EXCERPT: Kahn's conjecture. For every prime l and any i, the map c induces an isomorphism

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V2-021696 | MATHDB-378377

Tate-Beilinson conjecture on cycles and numerical equivalence over finite fields

摘录 / EXCERPT: Tate-Beilinson conjecture. For every prime l: (1) the cycle map above is surjective; (2) the $\hat G$-module $H^{2n}_{cont}(\bar X,\mathbb{Q}_l(n))$ is semisimple at the eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-021697 | MATHDB-378385

Generalized Lang conjecture for pre-periodic points

摘录 / EXCERPT: Generalized Lang conjecture. The set of pre-periodic points contained in Y is not Zariski-dense in Y.

conjecture | open | MathDB 页面 | 原文线索

V2-021698 | MATHDB-378399

Campana's abundance-type conjecture for decompositions of the canonical divisor

摘录 / EXCERPT: Let X be a projective manifold. Suppose that, for some positive integer N, ... where A is an effective divisor and B is a pseudo-effective line bundle. Campana's abundance-ty...

conjecture | open | MathDB 页面 | 原文线索

V2-021699 | MATHDB-378406

The abelian-variety entire-curve and Kobayashi-distance conjecture

摘录 / EXCERPT: Abelian-variety entire-curve conjecture. Every non-constant holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-021700 | MATHDB-378407

The node-count conjecture for factoriality of nodal threefolds

摘录 / EXCERPT: Node-count conjecture. The inequalities

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V2-021701 | MATHDB-378426

Fano-Iskovskikh degree bound conjecture for canonical Gorenstein Fano threefolds

摘录 / EXCERPT: Fano-Iskovskikh conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021702 | MATHDB-378440

Conjecture on the maximal number of vertices of a reflexive polytope

摘录 / EXCERPT: Vertex bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021703 | MATHDB-378441

Conjecture on the maximal number of vertices of a simplicial reflexive polytope

摘录 / EXCERPT: Simplicial vertex bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021704 | MATHDB-378486

Normal generation conjecture for polarized abelian varieties of type $(1,3,\ldots,3,6)$

摘录 / EXCERPT: Let (A,L) be a polarized abelian variety of dimension g. A polarization is of type $(d_1,\ldots,d_g)$ when its elementary divisors are $d_1\mid\cdots\mid d_g$. Normal generatio...

conjecture | open | MathDB 页面 | 原文线索

V2-021705 | MATHDB-378502

Spanning conjecture for TD systems

摘录 / EXCERPT: Let ${\cal F}$ be a field, let V be a finite-dimensional nonzero vector space over ${\cal F}$, and let Φ be a TD system on V with diameter d. Let R and L be the maps...

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V2-021706 | MATHDB-378505

Kontsevich-Soibelman conjecture on the non-archimedean collapse map

摘录 / EXCERPT: Let ${X}^{an}$ be the smooth analytic space associated with the Calabi-Yau variety over $K={{\bf C}}((t))$, and let π_{mer} map meromorphic points to their limiting points in t...

conjecture | open | MathDB 页面 | 原文线索

V2-021707 | MATHDB-378513

Harbourne-Hirschowitz conjecture for plane linear systems

摘录 / EXCERPT: Let $m_1,\ldots,m_r$ be positive integers corresponding to general points $p_1,\ldots,p_r\in\mathbb{P}^2$. Let $\mathcal{L}=\mathcal{L}_d(m_1,\ldots,m_r)$ be the linear system of d...

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V2-021708 | MATHDB-378515

Existence of the augmentation polynomial for all knots

摘录 / EXCERPT: Let K be a knot in S^3. Its augmentation variety is the locus in $(\mathbb{C}^*)^2$ of pairs (λ_0,μ_0) for which the framed knot DGA of K, specialized at...

conjecture | open | MathDB 页面 | 原文线索

V2-021709 | MATHDB-378524

Monotonicity and positivity conjecture for Welschinger invariants on toric Del Pezzo surfaces

摘录 / EXCERPT: Positivity and monotonicity conjecture. The Welschinger invariants $W_{D,m}$ are positive if $m<[(c_1(\Sigma)\cdot D-1)/2]$, non-negative if $m=[(c_1(\Sigma)\cdot D-1)/2]$, and sat...

conjecture | open | MathDB 页面 | 原文线索

V2-021710 | MATHDB-378532

Desingularization conjecture for orbifold Euler characteristics of quotients

摘录 / EXCERPT: Let G be a finite group, or more generally a linear algebraic group, acting on a variety X, and let $X\to Y:=X/G$ be the quotient. A G-variety is a variety with a G-action,...

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V2-021711 | MATHDB-378534

The minimal-class characterization by strong theta-regularity

摘录 / EXCERPT: The minimal-class characterization by strong theta-regularity. The subvariety Y represents a minimal class if and only if its ideal sheaf is strongly 3-Θ-regular. This p...

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V2-021712 | MATHDB-378535

The Prym characterization by the minimal number of conditions

摘录 / EXCERPT: The Prym characterization by the minimal number of conditions. Then $n\geq 2g$, and equality characterizes Prym varieties. This is proposed as a higher Castelnuovo-Schottky problem...

conjecture | open | MathDB 页面 | 原文线索

V2-021713 | MATHDB-378538

Geometric divisibility sequence constancy conjecture for higher-dimensional semiabelian groups

摘录 / EXCERPT: Geometric divisibility sequence conjecture. The geometric divisibility sequence $(D_{n\mathcal{P}})_{n\geq 1}$ corresponding to $\mathcal{P}$ satisfies

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V2-021714 | MATHDB-378559

Virtual-class compatibility under the Kobayashi-Hitchin isomorphism

摘录 / EXCERPT: Let (V,g) be a surface with a Kähler metric g, let $m\in H^2(V,\mathbb{Z})$, and let β be a real closed 2-form of type $(1,1)$. Let τ be a $Spin^c(4)$-structure re...

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V2-021715 | MATHDB-378560

Canonical-divisor formula for Hilbert-scheme virtual classes

摘录 / EXCERPT: Let V be a surface with $p_g(V)>0$, let K be an effective canonical divisor, and fix $m\in H^2(V,\mathbb{Z})$. Let ... be the zero-cycle on...

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V2-021716 | MATHDB-378572

The Gopakumar-Vafa invariant formula via relative Hilbert schemes

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, fix $g\ge0$, and suppose $I_{1-g}(X,\beta)$ parametrizes ideals of local complete intersection curves of arithmetic genus g. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021717 | MATHDB-378576

The adelic amoeba half-space conjecture for lower-dimensional varieties

摘录 / EXCERPT: Let $X\subset\mathbb{G}_{\text{m}}^d$ be an irreducible variety of dimension r such that its adelic amoeba $\mathscr{A}_{\mathbb{A}}(X)$ is not contained in any hyperplane. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021718 | MATHDB-378585

Higher-dimensional failure of semigroup decompositions for semi-algebraic sets

摘录 / EXCERPT: For each dimension d, let $\mathfrak{A}_d$ be the class of sets admitting the semigroup decomposition considered in the paper, and call a set semi-algebraic if it is defined by p...

conjecture | open | MathDB 页面 | 原文线索

V2-021719 | MATHDB-378598

The weakly special closure conjecture for canonical-lift torsion loci

摘录 / EXCERPT: Let ${\mathcal X}_{\mathbb C}$ be the complex universal abelian variety considered in the paper, let $Y\subseteq {\mathcal X}_{\mathbb C}$ be an irreducible subvariety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021720 | MATHDB-378599

Uniform p-adic proximity conjecture for canonical-lift torsion points

摘录 / EXCERPT: Let ${\mathbb C}_p$ be the completion of the algebraic closure of ${\mathbb Q}_p$, with v the extension of the p-adic valuation, and let ... be its ring of integers. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021721 | MATHDB-378621

Alexeev-Borisov boundedness conjecture for epsilon-lc weak log Fano varieties

摘录 / EXCERPT: AlexeevbBorisov boundedness conjecture. For every $b4>0$ and $91bb[0,1]$, the varieties X for which such a pair (X,B) exists are elements of an algebraic family.

conjecture | open | MathDB 页面 | 原文线索

V2-021722 | MATHDB-378622

Explicit Littelmann-path formula for duality of homogeneous coordinate rings

摘录 / EXCERPT: Path-reversal conjecture. In these Littelmann path models, $\widetilde{\Theta}$ is given by reversing Lakshmibai-Seshadri paths and translating the initial points of the reversed p...

conjecture | open | MathDB 页面 | 原文线索

V2-021723 | MATHDB-378624

The Schubert calculus conjecture on transversality of osculating Schubert intersections

摘录 / EXCERPT: The Schubert calculus conjecture. For generic $z=(z_1,\ldots,z_n)$, the intersection of Schubert varieties $\mathcal{I}_{\{{\bf w}\}}(z)$ is transversal. Here generic means that...

conjecture | open | MathDB 页面 | 原文线索

V2-021724 | MATHDB-378639

The orbital variety closure conjecture for $\mathfrak{sl}_n$

摘录 / EXCERPT: Let ${\mathfrak g}={\mathfrak{sl}}_n$ and let ${\mathcal V}$ be an orbital variety, that is, an irreducible component of ${\mathcal O}\cap{\mathfrak n}$ for a nilpotent orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-021725 | MATHDB-378641

Rapoport's surjectivity conjecture for Newton strata of parahoric Shimura varieties

摘录 / EXCERPT: Let Sh_{K_p} be a PEL Shimura variety with standard parahoric level subgroup K_p, and let ... be the map sending a point to the $c3$-conjugacy class of its associated isocrystal....

conjecture | open | MathDB 页面 | 原文线索

V2-021726 | MATHDB-378642

The maximal non-Gorenstein locus characterization for Schubert varieties

摘录 / EXCERPT: Let X_w be a Schubert variety, and let X_v range over the Schubert varieties contained in its maximal singular locus. The maximal non-Gorenstein locus characterization proposes...

open_problem | open | MathDB 页面 | 原文线索

V2-021727 | MATHDB-378657

Browning-Heath-Brown-Salberger's rational-point conjecture for integral varieties

摘录 / EXCERPT: Browning-Heath-Brown-Salberger's conjecture. For every $\varepsilon>0$,

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V2-021728 | MATHDB-378674

Knutson's scheme-theoretic equations conjecture for upper-upper schemes

摘录 / EXCERPT: Knutson's conjecture. The variety $\overline{E}_\pi$ is defined as a scheme by the following three sets of equations and rank conditions: XY and YX are upper triangular;

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V2-021729 | MATHDB-378675

Brauer-loop groundstate and upper-upper scheme degree correspondence conjecture

摘录 / EXCERPT: Brauer-loop degree correspondence conjecture. The groundstate element of the Brauer Hamiltonian for $L=2n$ corresponding to $\pi\in S_n$ equals the degree of $\overline{E}_\pi$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-021730 | MATHDB-378676

Extremal metric-relative K-stability conjecture

摘录 / EXCERPT: Let (X,L) be a polarised variety, let T be a maximal torus of automorphisms of X, and let $F_\chi(\alpha)$ denote the modified Futaki invariant associated with the extremal v...

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V2-021731 | MATHDB-378677

F-conjugacy invariance of Deligne-Lusztig cohomology

摘录 / EXCERPT: Let ${\bf t}\in B^+$, and let F act on B^+ as in the definition of F-conjugacy. Two elements are F-conjugate if ${\bf t}'={\bf w}^{-1}{\bf t}F({\bf w})$ for some...

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V2-021732 | MATHDB-378678

Periodicity conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Let $N=l(w_0)$, let $\rho\in{\mathcal E}({\bf G}^F,1)$, and let A_ρ and a_ρ be respectively the degree and valuation of the polynomial in q giving $\rho(1)$. Let...

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V2-021733 | MATHDB-378679

Duality periodicity conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Duality periodicity conjecture. If ${\bf w}\in B^+$ satisfies ${\bf w}\preccurlyeq\boldsymbol\pi^n$, then

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V2-021734 | MATHDB-378680

Ennola duality conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Let $N=l(w_0)$, let $\rho\in{\mathcal E}({\bf G}^F,1)$, and let A_ρ and a_ρ be respectively the degree and valuation of the polynomial in q giving $\rho(1)$. Let E be...

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V2-021735 | MATHDB-378681

The A_2 character-formula conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Assume that ${\bf G}$ is split of type A_2. Let $\varphi:B^+\to\mathbb Z$ be the class function defined in the source by its values on the listed representatives and by...

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V2-021736 | MATHDB-378691

Generalized potential-density conjecture for varieties with non-nef anticanonical divisor

摘录 / EXCERPT: Generalized potential-density conjecture. Under these assumptions, rational points on V are potentially dense.

conjecture | open | MathDB 页面 | 原文线索

V2-021737 | MATHDB-378692

Weak Lang conjecture for varieties of general type

摘录 / EXCERPT: Weak Lang conjecture. Rational points on V are not potentially dense.

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V2-021738 | MATHDB-378702

The conjecture that general-type varieties have no nowhere-vanishing holomorphic one-forms

摘录 / EXCERPT: Let X be a smooth projective variety of general type. A nowhere vanishing holomorphic one-form on X is a holomorphic section of the cotangent bundle $4\Omega_X^14$ with no zero...

conjecture | open | MathDB 页面 | 原文线索

V2-021739 | MATHDB-378724

Numerical special effect conjecture for plane linear systems

摘录 / EXCERPT: Numerical special effect conjecture. The system ${\mathcal{L}}_{2,d}(-\sum_{i=1}^hm_iP_i)$ is special if and only if it is numerically special.

conjecture | open | MathDB 页面 | 原文线索

V2-021740 | MATHDB-378732

Intersection-pairing conjecture for the ring of a unimodular hypertoric variety

摘录 / EXCERPT: Let $\mathcal{A}$ be a unimodular arrangement, let $\mathfrak{M}(\mathcal{A})$ be the associated hypertoric variety, and let $R(\mathcal{A})$ be the ring generated by the elements...

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V2-021741 | MATHDB-378736

Sturmfels-Zworski conjecture on cyclic resultants of reciprocal polynomials

摘录 / EXCERPT: Let K be the coefficient field, and let $f(x)\in K[x]$ be a reciprocal monic polynomial of even degree d, meaning that $f(1/x)=x^d f(x)$. Its cyclic resultants are the associat...

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V2-021742 | MATHDB-378740

Minimal Hilbert function for double schemes supported on linear configurations

摘录 / EXCERPT: Minimal-Hilbert-function conjecture. Among these double schemes, there is a minimal Hilbert function, and it occurs when the support is $C_{t,r}$.

conjecture | open | MathDB 页面 | 原文线索

V2-021743 | MATHDB-378746

The slope mirror conjecture over the integers

摘录 / EXCERPT: Let X and Y be schemes of finite type over $\mathbb{Z}$ whose generic fibres $X\otimes\mathbb{Q}$ and $Y\otimes\mathbb{Q}$ form a usual weak mirror pair of d-dimensional Cala...

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V2-021744 | MATHDB-378747

The generically ordinary conjecture for maximally generic Calabi-Yau mirror pairs

摘录 / EXCERPT: Let $d\leq 3$, and let X and Y be a maximally generic mirror pair of d-dimensional smooth projective Calabi-Yau schemes over the Witt ring $W(\mathbb{F}_q)$. A reduction is g...

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V2-021745 | MATHDB-378777

Kollár's realization and exclusion conjectures for real components of uniruled varieties

摘录 / EXCERPT: Kollár's conjectures. 1. If M is an orientable Seifert fibered manifold, then there is a uniruled algebraic variety X such that M is diffeomorphic to a connected component of...

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V2-021746 | MATHDB-378782

Finiteness conjecture for diffeomorphism types of real forms

摘录 / EXCERPT: Let $M(\mathbb C)$ be a smooth projective manifold. A real form of $M(\mathbb C)$ is a real algebraic manifold whose complexification is $M(\mathbb C)$. Finiteness conjecture. The...

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V2-021747 | MATHDB-378783

Conjecture that simple complex manifolds have simple real forms

摘录 / EXCERPT: Let a simple complex manifold mean a complex manifold satisfying the source’s informal notion of simplicity, and likewise let a simple real form mean a real form satisfying the cor...

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V2-021748 | MATHDB-378790

The l-independence conjecture for traces of correspondences on smooth proper varieties

摘录 / EXCERPT: Let K be a nonarchimedean local field with residue characteristic p, let X/K be a proper smooth variety purely of dimension n, let $\sigma \in W_K$, and let...

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V2-021749 | MATHDB-378804

The generalized volume bounds for Gorenstein Fano varieties with canonical singularities

摘录 / EXCERPT: Let X be a Gorenstein Fano variety with canonical singularities. The source's Theorem A' gives, for $d=2$, $(-K_X)^2\leq 9$ with equality if and only if $X\cong\mathbb{P}^2$; for...

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V2-021750 | MATHDB-378806

Infinitesimal mirror symmetry for Calabi-Yau varieties

摘录 / EXCERPT: Let Y be a Calabi-Yau variety of dimension n. Infinitesimal mirror symmetry conjecture. There exists a Calabi-Yau variety Y° and isomorphisms of complex vector spaces...

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V2-021751 | MATHDB-378810

Logarithmic asymptotics conjecture for Welschinger invariants of blown-up planes

摘录 / EXCERPT: Assume that $\mathbb{P}^2_k$ is obtained from $\mathbb{P}^2$ by blowing up k generic real points and is equipped with its natural real structure. Let $D\subset \mathbb{P}^2_k$ be...

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V2-021752 | MATHDB-378813

The Kleiman-Mori cone conjecture for complete toric varieties

摘录 / EXCERPT: Kleiman-Mori cone conjecture.

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V2-021753 | MATHDB-378821

Pseudo-effectivity conjecture for the second Chern class of varieties with nef anticanonical divisor

摘录 / EXCERPT: Pseudo-effectivity conjecture for the second Chern class.

conjecture | open | MathDB 页面 | 原文线索

V2-021754 | MATHDB-378855

The transversality conjecture for monotone Grassmannian Schubert problems

摘录 / EXCERPT: Let $a=(a_1<\dotsb<a_k)$ and n be positive integers with $a_k<n$. Let W^a index Schubert varieties on $\mathbb F\ell(a;n)$, and let $w_1,\dotsc,w_m\in W^a$ be data for a Schube...

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V2-021755 | MATHDB-378859

Alexeev-Borisov boundedness conjecture for delta-log-canonical weak log Fano varieties

摘录 / EXCERPT: Let $\delta>0$ be a real number. A delta-log-canonical weak log Fano pair is a pair $(X/\operatorname{pt.},B)$ of dimension d that is δ-log canonical and weak log Fano, wh...

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V2-021756 | MATHDB-378861

Fulton-Lazarsfeld's connectivity conjecture for symmetric degeneracy loci

摘录 / EXCERPT: Fulton-Lazarsfeld's connectivity conjecture. If

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V2-021757 | MATHDB-378864

The abelian Lehmer problem

摘录 / EXCERPT: Abelian Lehmer problem. There is a constant $c(A/K,\mathcal{L})>0$ such that, for every point $P\in A(\overline{K})$ of infinite order modulo every proper abelian subvariety of A...

open_problem | open | MathDB 页面 | 原文线索

V2-021758 | MATHDB-378865

David-Hindry's multihomogeneous abelian Lehmer conjecture

摘录 / EXCERPT: David-Hindry's conjecture. For every integer $n\in\mathbb{N}$, there is a constant $c(A/K,\mathcal{L},n)>0$ such that, for every n-tuple $(P_1,\ldots,P_n)$ of points of infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-021759 | MATHDB-378877

Weibel's vanishing and regularity conjecture for negative K-theory

摘录 / EXCERPT: Let X be a Noetherian scheme of dimension d. For a contravariant functor F on schemes, call X F-regular if, for every $r\geq 0$, the map $F(X)\to F(X\times\mathbb{A}^r)$...

conjecture | open | MathDB 页面 | 原文线索

V2-021760 | MATHDB-378889

Hu-Yau conjecture on strict semismallness of symplectic contractions

摘录 / EXCERPT: Let X be a smooth projective symplectic variety and let $\pi:X\to Z$ be a small contraction. A small contraction is perverse semi-small if there is a stratification...

conjecture | open | MathDB 页面 | 原文线索

V2-021761 | MATHDB-378890

Eriksson-Linusson's realizability conjecture for permutation arrays

摘录 / EXCERPT: A permutation array is an element P of $[n]^d$ satisfying the permutation-array conditions, and X^o_P denotes the corresponding permutation array variety, consisting of tuples...

conjecture | open | MathDB 页面 | 原文线索

V2-021762 | MATHDB-378892

The invariance criterion for tautological equations

摘录 / EXCERPT: Invariance criterion. If

conjecture | open | MathDB 页面 | 原文线索

V2-021763 | MATHDB-378895

Lichtenbaum-Milne conjecture on special values of zeta functions

摘录 / EXCERPT: Lichtenbaum-Milne conjecture. For every relevant integer r, as $t\to q^{-r}$,

conjecture | open | MathDB 页面 | 原文线索

V2-021764 | MATHDB-378901

Dimension conjecture for the deformed ring of a type A cluster quiver

摘录 / EXCERPT: Let n be a positive integer. Define the commutative ring $\mathbf{M}^{\text{def}}(n)$ generated by variables S_i for $i\in[n]$ and S_α for...

conjecture | open | MathDB 页面 | 原文线索

V2-021765 | MATHDB-378905

The Schubert-union conjecture for higher weights of Grassmann codes

摘录 / EXCERPT: Let $G(l,m)$ be a Grassmann variety and let its associated Grassmann code have higher weights. A Schubert union is a union of Schubert varieties in the Grassmann variety. Schubert-...

conjecture | open | MathDB 页面 | 原文线索

V2-021766 | MATHDB-378938

Conjecture on Fano complex contact manifolds with second Betti number one

摘录 / EXCERPT: Let Y be a Fano complex contact manifold with second Betti number $b_2=1$. For a simple Lie group G, let $\mathfrak{g}$ denote its Lie algebra, and let G act on the projectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-021767 | MATHDB-378939

Hollmann's pseudocyclicity conjecture for the Frobenius fusion scheme

摘录 / EXCERPT: Hollmann's conjecture. If m is an odd prime, then $({\cal E},\{\Delta_a\})$ is pseudocyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-021768 | MATHDB-378946

Conjecture on restriction functors and full faithfulness of specialization

摘录 / EXCERPT: Let Y_0 be a k-scheme, let X_0 be a smooth ambient scheme when specified, and let $\rho_{Y_0,X_0}:\mathrm{Isoc}^{\dag}(Y_0/K)\to\mathrm{Isoc}^{\dag}(Y_0,X_0/K)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021769 | MATHDB-378951

Tate's correspondence conjecture for isomorphic two-dimensional Galois representations

摘录 / EXCERPT: Let X_1 and X_2 be varieties defined over $\mathbb{Q}$, and let ρ_1 and ρ_2 be isomorphic two-dimensional Galois representations occurring in their étale cohomology....

conjecture | open | MathDB 页面 | 原文线索

V2-021770 | MATHDB-378953

The isomorphism conjecture for the representation variety map

摘录 / EXCERPT: Isomorphism conjecture. The map Φ is an isomorphism of algebraic varieties. This would identify the moduli of the relevant irreducible representations with the explicitly defi...

conjecture | open | MathDB 页面 | 原文线索

V2-021771 | MATHDB-378966

Positivity on movable curves implies bigness

摘录 / EXCERPT: Let A be a projective manifold, and let ${\mathcal L} \subset \bigwedge^r \Omega^j_A$ be a line bundle. A curve C on A is movable if its deformations fill up A. The positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-021772 | MATHDB-378967

The positivity conjecture for Grothendieck polynomials in type A

摘录 / EXCERPT: Let S_n be the symmetric group, let ${\cal G}_w(x_1,\ldots,x_n)$ be the Grothendieck polynomial associated with $w\in S_n$, and let $G_i^{A_{n-1}}$ be the elements defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-021773 | MATHDB-378969

McKinnon's rational-curve conjecture for best approximations

摘录 / EXCERPT: Let V be an algebraic variety defined over a field k, let D be an ample divisor on V, and let $P\in V(k)$ be a k-rational point. Assume that there is a rational curve C...

conjecture | open | MathDB 页面 | 原文线索

V2-021774 | MATHDB-378971

Chen-Chen-Zhang nonvanishing conjecture for adjoint Q-divisors

摘录 / EXCERPT: Let X be a nonsingular projective variety, and let L be a ${\bf Q}$-divisor on X. Assume that L is nef and big, that K_X+L is nef, and that either L is a Cartier integr...

conjecture | open | MathDB 页面 | 原文线索

V2-021775 | MATHDB-378979

Rapoport's dimension conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be a connected reductive group over F, let $b\in G(L)$, and let μ be a dominant coweight such that the affine Deligne-Lusztig variety $X_\mu(b)$ is non-empty. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-021776 | MATHDB-378980

Eventual translation-shift conjecture for affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let G be one of the groups SL_2, SL_3, or Sp_4. Let $b=\epsilon^\nu$, and let $\ell(b)$ denote the length of the translation in $\widetilde W$ determined by ν. Eventual...

conjecture | open | MathDB 页面 | 原文线索

V2-021777 | MATHDB-378989

Gordon-Hanamura-Murre's Motivic Decomposition Conjecture

摘录 / EXCERPT: Let $\overline{A}$ and $\overline{X}$ be quasi-projective varieties over $\mathbb C$, with $\overline{A}$ smooth, and let $\overline{f}:\overline{A}\to\overline{X}$ be a projective...

conjecture | open | MathDB 页面 | 原文线索

V2-021778 | MATHDB-378990

Real analyticity conjecture for the volume function

摘录 / EXCERPT: Let X be a smooth projective variety of dimension d, and let $\textnormal{Big}(X)$ denote its big cone. An open subset $U\subseteq\textnormal{Big}(X)$ is called “large” in the...

conjecture | open | MathDB 页面 | 原文线索

V2-021779 | MATHDB-379010

Conjecture that non-singular hypersurfaces are not unions of maximal sub-boundary planes

摘录 / EXCERPT: Let $n\geq 5$, let $X\subset\mathbb{P}^n$ be a non-singular hypersurface of degree $d\geq 3$, and set ... An m-plane means a linear projective subspace of dimension m contained i...

conjecture | open | MathDB 页面 | 原文线索

V2-021780 | MATHDB-379020

Vatne's conjecture on Cohen-Macaulay triple structures

摘录 / EXCERPT: Vatne's conjecture. There is a degree-two Cohen-Macaulay subscheme Z such that

conjecture | open | MathDB 页面 | 原文线索

V2-021781 | MATHDB-379034

The component-count conjecture for cubic Poincare systems

摘录 / EXCERPT: Component-count conjecture. The variety X_{∞} has exactly 1 component of codimension 5, exactly 2 components of codimension 6, exactly 4 components of codimension...

conjecture | open | MathDB 页面 | 原文线索

V2-021782 | MATHDB-379051

Unobstructedness conjecture for quantized symplectic quotient algebras

摘录 / EXCERPT: Let M be a symplectic C^∞-manifold or an affine complex algebraic variety, let G be a finite group acting on M by symplectic transformations, and let B be a G-equi...

conjecture | open | MathDB 页面 | 原文线索

V2-021783 | MATHDB-379092

Radicality of determinantal ideals for unions of splitting-type strata

摘录 / EXCERPT: Radicality conjecture. The ideal above is radical. This stronger assertion includes cases involving certain unions of splitting-type strata. The source records the case $r=2$ as as...

conjecture | open | MathDB 页面 | 原文线索

V2-021784 | MATHDB-379112

Strong Green-Griffiths conjecture on a uniform exceptional subvariety

摘录 / EXCERPT: Strong Green-Griffiths conjecture. There exists a closed subvariety $E\subsetneq X$ such that E contains the image $f(\mathbf{C})$ for every non-constant holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-021785 | MATHDB-379119

Equality of associated varieties for irreducible highest-weight modules

摘录 / EXCERPT: Associated-variety conjecture. For any integral dominant $\lambda\in\frak h^*$ with degree of atypicality k,

conjecture | open | MathDB 页面 | 原文线索

V2-021786 | MATHDB-379120

The upper-bound conjecture for critical points of real line arrangements

摘录 / EXCERPT: Let $F(x,y)$ be a polynomial whose zero set is a plane curve consisting of the product of d real lines. Consider the critical points of F having the same non-zero critical valu...

conjecture | open | MathDB 页面 | 原文线索

V2-021787 | MATHDB-379123

Deligne-Beilinson conjecture on the order of vanishing of motivic L-functions

摘录 / EXCERPT: Deligne-Beilinson conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021788 | MATHDB-379124

Deligne-Beilinson rationality conjecture for motivic L-values

摘录 / EXCERPT: Let M be a K-motive, let $\Delta_K(M)$ be its fundamental line, and let $\vartheta_\infty:K_{\mathbb R}\otimes_K\Delta_K(M)\cong\mathbf{1}_{K_{\mathbb R}}$ be the period-regula...

conjecture | open | MathDB 页面 | 原文线索

V2-021789 | MATHDB-379128

Orlov's kernel functor conjecture for splitting functors

摘录 / EXCERPT: Let X and Y be smooth projective varieties, and let ... be a splitting functor. A kernel functor is a functor represented by a kernel $K\in D^b\operatorname{Coh}(X\times Y)$. Ker...

conjecture | open | MathDB 页面 | 原文线索

V2-021790 | MATHDB-379138

Smoothness-frequency conjecture for abelian varieties

摘录 / EXCERPT: Let k be a number field, let A be an abelian variety over k, and let S_A be the set of places of good reduction for A. For $u\in(0,1)$, call an integer B^u-smooth if al...

conjecture | open | MathDB 页面 | 原文线索

V2-021791 | MATHDB-379141

The discriminant-sign conjecture for monotone Grassmannian Schubert problems

摘录 / EXCERPT: Let $(w_1,\dotsc,w_m)$ be a Grassmannian Schubert problem for $\mathbb{F}\ell(\alpha;n)$, let $\delta(w_i)$ be the unique descent of w_i, and let $\Delta_w(t_1,\dotsc,t_m)$ be it...

conjecture | open | MathDB 页面 | 原文线索

V2-021792 | MATHDB-379146

The structural conjectures for the category $\mathcal A$

摘录 / EXCERPT: Let F/k be the field extension in the source, let $\mathfrak m$ be the object and $\mathcal C$ the category appearing in the source, let $\mathcal A$ be the category of admissibl...

conjecture | open | MathDB 页面 | 原文线索

V2-021793 | MATHDB-379149

Kollár-Shokurov inversion of adjunction conjecture

摘录 / EXCERPT: Let (X,Y) be a pair, where Y is a formal rational or real linear combination of closed subvarieties of X. For a log resolution $f:X'\xrightarrow{\ \ } X$, write ... and define...

conjecture | open | MathDB 页面 | 原文线索

V2-021794 | MATHDB-379158

The Vogan-chain characterization conjecture for orbital-variety closures

摘录 / EXCERPT: Let ${\bf T}_n$ be the set of standard Young tableaux with n boxes, let I_T be the primitive ideal attached to T, let ${\mathcal V}_T$ be its orbital variety, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021795 | MATHDB-379166

The fixed-point criterion equivalent to the Jacobian conjecture

摘录 / EXCERPT: Let $P:\mathbb C^2\rightarrow\mathbb C^2$ satisfy $J(P)\equiv 1$. Suppose there exists an algebraic function defined on some open set ${\cal U}$ such that $P\circ Q=P$, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-021796 | MATHDB-379172

Conjecture on the p-reductivity threshold for Bergman quotient modules

摘录 / EXCERPT: Let $I\subseteq\mathbb{C}[\mathbf{z}]$ be an ideal, let $\mathcal{Z}$ be its zero variety, and let $\mathcal{S}$ be the submodule obtained by taking the closure of I in...

conjecture | open | MathDB 页面 | 原文线索

V2-021797 | MATHDB-379187

Uniqueness conjecture for nice forms of homogeneous polynomials

摘录 / EXCERPT: Uniqueness conjecture. Every nonsingular homogeneous polynomial is nice. For degree greater than 2, there is precisely one choice of coordinates in which a nice polynomial reache...

conjecture | open | MathDB 页面 | 原文线索

V2-021798 | MATHDB-379191

Qin's Euler partition-function conjecture for moduli spaces of curves

摘录 / EXCERPT: Let X) be a smooth projective complex variety. Let $\mathfrak I_n(X,\beta)$ be the moduli space of 1-dimensional closed subschemes Z of X satisfying the paper's fixed numer...

conjecture | open | MathDB 页面 | 原文线索

V2-021799 | MATHDB-379203

Global semistable reduction conjecture for partially overconvergent F-isocrystals

摘录 / EXCERPT: Let X be a smooth irreducible k-variety, let $\overline{X}$ be a partial compactification of X, and let $\mathcal{E}$ be an F-isocrystal on X overconvergent along...

conjecture | open | MathDB 页面 | 原文线索

V2-021800 | MATHDB-379205

Reduced real Lawson homology filtration conjecture

摘录 / EXCERPT: Let X be a smooth real projective variety. Let $RT_tH_n(X)$ denote the topological filtration on $H_n(ReX;\mathbb Z_2)$, given by the image of $\Phi_{t,n}$, and let $RG_tH_n(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-021801 | MATHDB-379222

Tautness conjecture for pseudo primitive idempotents

摘录 / EXCERPT: Let Γ be a bipartite distance-regular graph, and let T denote its subconstituent algebra. Assume that, up to isomorphism, there exist exactly two irreducible T-modules w...

conjecture | open | MathDB 页面 | 原文线索

V2-021802 | MATHDB-379223

Q-polynomial quotient conjecture for taut distance-regular graphs

摘录 / EXCERPT: Let Γ be a taut distance-regular graph with odd diameter $D\geq 5$. By the source's stated theorem, Γ is an antipodal 2-cover; hence its antipodal quotient is defin...

conjecture | open | MathDB 页面 | 原文线索

V2-021803 | MATHDB-379224

The discriminant-complement identification with the two-component link complement

摘录 / EXCERPT: Let V_2 be the two-dimensional complex Cartan subalgebra for the Lie algebra of type G_2, and let $V_2^{\rm reg}$ be the complement of the six root-kernel lines. Let W_{G_2}...

conjecture | open | MathDB 页面 | 原文线索

V2-021804 | MATHDB-379232

The endomorphism conjecture for F-roots of the full twist

摘录 / EXCERPT: Let B be the braid group, let B^+ be its positive braid monoid, let F act on B by the Frobenius diagram automorphism, and let ${\boldsymbol\pi}$ be the positive generator o...

conjecture | open | MathDB 页面 | 原文线索

V2-021805 | MATHDB-379234

The cyclotomic Hecke factorization conjecture for regular Deligne-Lusztig varieties

摘录 / EXCERPT: Let ${\bf w}$ be an F-root of ${\boldsymbol\pi}$, let $w=\beta({\bf w})$, let $B(w)$ be the braid group of $C_W(wF)$, and let ${\mathcal H}(w)$ be a cyclotomic Hecke algebra for...

conjecture | open | MathDB 页面 | 原文线索

V2-021806 | MATHDB-379235

The character and specialness conjecture for Deligne-Lusztig cohomology

摘录 / EXCERPT: Let ${\bf w}$ be an F-root of ${\boldsymbol\pi}$, let ${\mathcal H}_q(w)$ be the specialization at $x^{1/a}\mapsto q^{1/a}$ of the cyclotomic Hecke algebra ${\mathcal H}(w)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-021807 | MATHDB-379236

The disjointness conjecture for Deligne-Lusztig cohomology degrees

摘录 / EXCERPT: Let ${\bf w}$ be an F-root of ${\boldsymbol\pi}$ and let ${\bf X}({\bf w})$ be the associated Deligne-Lusztig variety. The compactly supported cohomology groups...

conjecture | open | MathDB 页面 | 原文线索

V2-021808 | MATHDB-379237

The virtual cohomology decomposition conjecture for the full twist

摘录 / EXCERPT: Let W be the Weyl group, let F be the Frobenius automorphism, and let ${\boldsymbol\pi}$ be the positive generator of the center of the pure braid group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021809 | MATHDB-379240

The Cantor-set conjecture for end invariants of one-holed torus characters

摘录 / EXCERPT: Let ρ be an $\mathrm{SL}(2,\mathbb C)$ character, let $\mathcal E(\rho)$ be its set of end invariants, and let $\mathscr{PL}$ denote the projective lamination space. Cantor-se...

conjecture | open | MathDB 页面 | 原文线索

V2-021810 | MATHDB-379245

Goldman's invariant-forms conjecture for compact-group character varieties

摘录 / EXCERPT: Invariant-forms conjecture. The symplectic structures $\omega_\mathbb{B}$ generate the subalgebra of $\Omega^*(\mathsf{Hom}(\pi,G)/G)$ consisting of Mod_{Σ}-invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-021811 | MATHDB-379246

Andersen's scalar differential-operator conjecture for compact-group character varieties

摘录 / EXCERPT: Andersen's scalar-operator conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-021812 | MATHDB-379247

Ergodicity conjecture for outer automorphisms of free groups

摘录 / EXCERPT: Free-group ergodicity conjecture. If $r\geq 3$, the action of $\mathsf{Out}(\pi)$ on $\mathsf{Hom}(\pi,G)$ is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-021813 | MATHDB-379248

Ergodicity conjecture for non-extreme Euler-class components

摘录 / EXCERPT: Euler-component ergodicity conjecture. For each integer $1\leq k\leq 2g+b-2$, the Mod_{Σ}-action on the component $e^{-1}(2-2g+b+k)$ of $\mathsf{Hom}(\pi,G)$ is erg...

conjecture | open | MathDB 页面 | 原文线索

V2-021814 | MATHDB-379249

Neumann's surjectivity conjecture for the Euler-class parametrization

摘录 / EXCERPT: Neumann's surjectivity conjecture. If $k=1$, then $\mathbb{U}$ is onto. In general, every $\mathsf{PSL}(2,\mathbb{R})$-representation with dense image lies in...

conjecture | open | MathDB 页面 | 原文线索

V2-021815 | MATHDB-379274

Radical-generation conjecture for the three-factor model

摘录 / EXCERPT: Three-factor radical-generation conjecture. The radical of $J_{p,3}$ is the prime ideal $I_{p,3}$:

conjecture | open | MathDB 页面 | 原文线索

V2-021816 | MATHDB-379284

Hindry's integral torsion-point conjecture for abelian varieties

摘录 / EXCERPT: Let k) be a number field, let S be a finite set of places of k containing the archimedean places, and let ${\mathcal O}_{k,S}$ be the ring of S-integers. Let A/k be an ab...

conjecture | open | MathDB 页面 | 原文线索

V2-021817 | MATHDB-379288

The higher Todd genus birational-invariance conjecture

摘录 / EXCERPT: Let V be a nonsingular complex projective variety of complex dimension n, viewed as a complex manifold, with fundamental class $[V]\in H_{2n}(V,\mathbb Z)$ and total Todd class...

conjecture | open | MathDB 页面 | 原文线索

V2-021818 | MATHDB-379289

The blow-up formulation of higher Todd genus birational invariance

摘录 / EXCERPT: Let $\phi\colon\widehat V\to V$ be a blowing down between nonsingular complex projective varieties, with $\widehat V$ obtained by blowing up V along a nonsingular irreducible cen...

conjecture | open | MathDB 页面 | 原文线索

V2-021819 | MATHDB-379306

Rationality conjecture for Weil-Petersson characteristic integrals

摘录 / EXCERPT: Let M be a Weil-Petersson variety, and let X be a Weil-Petersson subvariety of dimension q in M. Let $c_k(\omega_{WP})$ denote the k-th elementary polynomial of the curva...

conjecture | open | MathDB 页面 | 原文线索

V2-021820 | MATHDB-379312

Finite-order translation conjecture for characteristic varieties

摘录 / EXCERPT: Finite-order translation conjecture. The characteristic variety is a union of translated subtori, with each translation given by a point of finite order.

conjecture | open | MathDB 页面 | 原文线索

V2-021821 | MATHDB-379320

The toric M-variety conjecture for homology with closed support

摘录 / EXCERPT: A real algebraic variety is a complex variety X defined over the real numbers, with real part $X(\mathbb{R})$ equal to the fixed-point set of complex conjugation. It is an M-vari...

conjecture | open | MathDB 页面 | 原文线索

V2-021822 | MATHDB-379321

The G^1-degeneration conjecture for real toric varieties

摘录 / EXCERPT: The G^1-degeneration conjecture. The spectral sequence $G^k_{p,q}$ for $H_*(X(\mathbb{R}))$ degenerates at the G^1 level.

conjecture | open | MathDB 页面 | 原文线索

V2-021823 | MATHDB-379325

Vanishing of the polyhedral Euler-complex relations

摘录 / EXCERPT: Let K be the number field, ${\cal P}$ a prime ideal in its ring of integers, and let $\gamma {\Bbb B}_d$ be a polyhedron in the Bianchi decomposition of the orbifold...

conjecture | open | MathDB 页面 | 原文线索

V2-021824 | MATHDB-379326

Voevodsky's S^1-connectedness conjecture

摘录 / EXCERPT: Let k be the base field and let ${\operatorname{{\mathcal S}{\mathcal H}}}_{S^1}(k)$ be the Morel-Voevodsky homotopy category of S^1-spectra. For an integer d, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021825 | MATHDB-379327

The motivic connectivity conjecture for $\mathbb{P}^1$-spectra

摘录 / EXCERPT: Let k be the base field, let ${\operatorname{{\mathcal S}{\mathcal H}}}_{S^1}(k)$ be the homotopy category of S^1-spectra, and let $\Sigma^\infty_{{\mathbb P}^1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021826 | MATHDB-379328

The multiplicative lifting conjecture for the slice filtration

摘录 / EXCERPT: Let ${\mathbf{Spt}}^\Sigma_{{\mathbb P}^1}(k)$ be the category of symmetric $\mathbb{P}^1$-spectra over k, and let ${\operatorname{{\mathcal S}{\mathcal H}}}(k)$ be its homotopy...

conjecture | open | MathDB 页面 | 原文线索

V2-021827 | MATHDB-379330

Finiteness conjecture for symplectic resolutions

摘录 / EXCERPT: Finiteness conjecture. The variety W has at most finitely many non-isomorphic symplectic resolutions.

conjecture | solved | MathDB 页面 | 原文线索

V2-021828 | MATHDB-379333

Hu-Yau's conjecture on symplectic resolutions and Mukai flops

摘录 / EXCERPT: Let W be a symplectic variety, and let $Z\to W$ and $Z^+\to W$ be projective symplectic resolutions. A Mukai flop in codimension 2 is a birational transformation that becomes a...

conjecture | open | MathDB 页面 | 原文线索

V2-021829 | MATHDB-379343

Kim's Selmer-dimension conjecture for hyperbolic curves

摘录 / EXCERPT: Let $F=\mathbb{Q}$ and let X be a hyperbolic curve. For the étale unipotent fundamental group U^{et} and its level-n quotient U_n^{et}, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021830 | MATHDB-379347

General validity of the Gromov-Witten formula for transversal ADE singularities

摘录 / EXCERPT: Let Y be a variety with transversal A_n-singularities, $n\geq 2$, and let $\rho:Z\to Y$ be its crepant resolution. Let K be the line bundle on the singular locus appearing in...

conjecture | open | MathDB 页面 | 原文线索

V2-021831 | MATHDB-379360

Serrano's strict nefness conjecture for projective manifolds

摘录 / EXCERPT: Let X be a projective manifold of dimension n, and let L be a strictly nef line bundle, meaning that $L\cdot C>0$ for every irreducible curve $C\subset X$. Serrano's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-021832 | MATHDB-379364

The mirror conjecture for quantum differential equations of G/P

摘录 / EXCERPT: Let G/P be a partial flag variety, let $Z_P^{\mathfrak h^\vee}$ be the pullback of the mirror family to $(\mathfrak h^\vee)^{W_P}$, and let ω and $\mathcal F_P$ be respect...

conjecture | open | MathDB 页面 | 原文线索

V2-021833 | MATHDB-379365

The equivariant mirror conjecture for quantum differential equations of G/P

摘录 / EXCERPT: Equivariant mirror conjecture. A full set of solutions to the T-equivariant quantum differential equations of G/P is given by these integrals. This is the T-equivariant exten...

conjecture | open | MathDB 页面 | 原文线索

V2-021834 | MATHDB-379374

The normality Gröbner-basis conjecture for projectively normal toric varieties

摘录 / EXCERPT: Let I be a toric ideal defining a projectively normal r-dimensional toric variety. The normality Gröbner-basis conjecture. The ideal I has a Gröbner basis consisting of binom...

conjecture | open | MathDB 页面 | 原文线索

V2-021835 | MATHDB-379379

Higher-dimensional exceptional cover conjecture

摘录 / EXCERPT: Higher-dimensional exceptional cover conjecture. There exists a constant C, depending only on n, the dimensions of X and Y, and $b_1,\dots,b_{2r-1}$, such that if $q>C$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021836 | MATHDB-379380

Birational invariance of higher Todd genera

摘录 / EXCERPT: Let M be a smooth projective variety, let π be a group, and let $f:M\to B\pi$ be a continuous map. For $\alpha\in H^*(B\pi;\mathbb{Q})$, let $\operatorname{Td}(M)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-021837 | MATHDB-379383

Viehweg's conjecture on maximal variation and log general type

摘录 / EXCERPT: Let $f^\circ:X^\circ\to S^\circ$ be a smooth family of canonically polarized varieties. The family has maximal variation when its variation equals the dimension of the base. Viehwe...

conjecture | open | MathDB 页面 | 原文线索

V2-021838 | MATHDB-379390

The factoriality conjecture for nodal hypersurfaces in projective four-space

摘录 / EXCERPT: Let $V\subset\mathbb{P}^{4}$ be a hypersurface of degree d with at most isolated ordinary double points, and let $\mathrm{Sing}(V)$ denote its singular locus. Factoriality conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-021839 | MATHDB-379393

The dendrite conjecture for end invariants of one-holed torus characters

摘录 / EXCERPT: Let ρ be an $\mathrm{SL}(2,\mathbb C)$ character and let ${\mathcal E}(\rho)$ denote its set of end invariants, viewed as a subset of the projective lamination space...

conjecture | open | MathDB 页面 | 原文线索

V2-021840 | MATHDB-379394

The ending invariant rigidity conjecture for one-holed torus characters

摘录 / EXCERPT: Let ${\mathcal X}_{\kappa}$ be the character variety with parameter κ, and let ${{\mathcal E}(\rho)}$ be the set of end invariants of a character ρ. The ending invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-021841 | MATHDB-379395

Sharp modular-degree conjecture for families of n-gonal curves

摘录 / EXCERPT: Let V be an irreducible variety of dimension $2g+2n-5$, with $4\leq 2n-2<g$, and let $\mathcal{F}\rightarrow V$ be a family of smooth n-gonal curves of genus g with maximal v...

conjecture | open | MathDB 页面 | 原文线索

V2-021842 | MATHDB-379403

Projection conjecture for point sets in projective four-space

摘录 / EXCERPT: Projection conjecture. At most $k(n-1)$ points of $\phi_4(\Sigma)$ can be contained in a curve of degree k in $\mathbb{P}^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-021843 | MATHDB-379414

Stratified cohomology conjecture for Deligne-Lusztig varieties

摘录 / EXCERPT: Let W be the Weyl group, let $w\in W$, and let $Y_x(\dot w)$ be the stratum defined by ... For a G-regular linear character $\psi:U_0\to\mathcal{O}^{\times}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-021844 | MATHDB-379424

Constructible bijectivity conjecture for filtered Weyl and Poisson automorphism schemes

摘录 / EXCERPT: For each $n,N\geq 1$, let ${\underline{\operatorname{Aut}}}^{\leq N}(A_{n,\mathbb Q})$ and ${\underline{\operatorname{Aut}}}^{\leq N}(P_{n,\mathbb Q})$ denote the filtered automorp...

conjecture | open | MathDB 页面 | 原文线索

V2-021845 | MATHDB-379431

Yasuda's smoothness conjecture for higher Nash blowups

摘录 / EXCERPT: Let k be a field of characteristic zero, let X be a variety of dimension d, and let $J\subseteq X$ be the Jacobian subscheme. Denote by $J^{(d-1)}$ its $(d-1)$-th neighborhoo...

conjecture | open | MathDB 页面 | 原文线索

V2-021846 | MATHDB-379435

The morphic conjecture for even and odd morphic cohomology

摘录 / EXCERPT: Let X be a smooth projective variety of dimension m, let ${\mathbb F}$ be the coefficient field used in the source, and let $\widetilde{L}^tH^k(X;{\mathbb F})$ be the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-021847 | MATHDB-379436

Cohomological Friedlander-Mazur conjecture

摘录 / EXCERPT: Let X be a smooth projective variety, and let $T^pH^l(X;\Q)$ be the topological filtration defined using the natural transformation from morphic cohomology to singular cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-021848 | MATHDB-379438

Joyner's conjecture 4.3 on parameters of toric codes

摘录 / EXCERPT: Let $\mathcal{C}(E,D,X)$ be the toric code associated to the 1-cycle E, the T-invariant Cartier divisor D, and the toric variety X. Assume that X is a nonsingular toric...

conjecture | open | MathDB 页面 | 原文线索

V2-021849 | MATHDB-379457

Bogomolov conjecture for Drinfeld modules

摘录 / EXCERPT: Bogomolov conjecture. If X_n is Zariski dense in X for every $n\geq 1$, then X is a torsion subvariety of $\mathbb{G}_a^g$. This is presented as a consequence of the proposed...

conjecture | open | MathDB 页面 | 原文线索

V2-021850 | MATHDB-379467

The negative-curve conjecture for very general blow-ups of the plane

摘录 / EXCERPT: Negative-curve conjecture. The curve C is a (-1)-curve of Y, meaning that it is a smooth rational curve with self-intersection -1.

conjecture | open | MathDB 页面 | 原文线索

V2-021851 | MATHDB-379468

The cone-of-curves formulation of the negative-curve conjecture

摘录 / EXCERPT: Cone-of-curves formulation. The extremal rays of $\mathcal{B}_1$ that do not lie on $\operatorname{Nul}(Y)$ are spanned by classes of (-1)-curves, and

conjecture | open | MathDB 页面 | 原文线索

V2-021852 | MATHDB-379472

The motivic Tamagawa conjecture for simply connected groups

摘录 / EXCERPT: Let C be a smooth projective geometrically connected curve over k, of genus g, and let G be a split semisimple connected algebraic group over k. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-021853 | MATHDB-379473

Pareschi-Popa syzygy conjecture for ample line bundles on abelian varieties

摘录 / EXCERPT: Let X be an abelian variety, let A be an ample line bundle on X, and let $m(A)$ denote its index of M-regularity. For non-negative integers $p\ge m$, Pareschi-Popa's syzygy...

conjecture | open | MathDB 页面 | 原文线索

V2-021854 | MATHDB-379474

The minimal-class conjecture for subvarieties of principally polarized abelian varieties

摘录 / EXCERPT: Let (X,Θ) be an irreducible principally polarized abelian variety, let $Y\subset X$ be a non-degenerate subvariety of dimension m, and write $\theta=c_1(\Theta)$. Say that...

conjecture | open | MathDB 页面 | 原文线索

V2-021855 | MATHDB-379479

The spectral gap conjecture for the \operatorname{Out}(F_n) representation

摘录 / EXCERPT: Let F_n be the free group of rank n, let K be the compact group appearing in the paper, and let $\operatorname{Hom}(F_n,K)/K$ denote the quotient of the representation variet...

conjecture | open | MathDB 页面 | 原文线索

V2-021856 | MATHDB-379503

Perfect bases for Schubert cells

摘录 / EXCERPT: Schubert-cell perfect-basis conjecture. For every $w\in W$, the coordinate algebra $\mathbb{C}[X_w^\vee]$ has a perfect basis.

conjecture | open | MathDB 页面 | 原文线索

V2-021857 | MATHDB-379504

Crystal realization of Schubert cells

摘录 / EXCERPT: Schubert-cell crystal conjecture. For every $w\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-021858 | MATHDB-379506

The p-adic generalized Hodge conjecture for third cohomology

摘录 / EXCERPT: Let X be a variety with totally degenerate reduction, and let $N_{1,L}$ and $N'_{1,L}$ be the enriched monodromy operators defined in Section 4.3. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-021859 | MATHDB-379527

The strong factorization conjecture for smooth complete varieties

摘录 / EXCERPT: Let X and Y be smooth complete varieties over a field of characteristic zero, and let $f:X\dashrightarrow Y$ be a birational map. A factorization of f consists of a successio...

conjecture | open | MathDB 页面 | 原文线索

V2-021860 | MATHDB-379533

The Separation conjecture for real spectra

摘录 / EXCERPT: Separation conjecture. If $g\text{$in$ \hspace{-.8em}/}\langle\alpha,\beta\rangle$, then α and β lie in the same connected component of

conjecture | open | MathDB 页面 | 原文线索

V2-021861 | MATHDB-379534

Converse characterization of motivic weight bounds by singular cohomology

摘录 / EXCERPT: Converse weight-bound conjecture. If, for all l, the weights of $H^l\otimes\mathbb{Q}(M)$ lie between $l+a$ and $l+b$, then

conjecture | open | MathDB 页面 | 原文线索

V2-021862 | MATHDB-379544

The PSL(2,C) Casson invariant decomposition conjecture

摘录 / EXCERPT: Let Σ be a closed, orientable 3-manifold. Write $\lambda_{\operatorname{PSL}_2(\mathbb C)}(\Sigma)$ and $\lambda_{\operatorname{SL}_2(\mathbb C)}(\Sigma)$ for the respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-021863 | MATHDB-379549

Looijenga's affine stratification conjecture for moduli spaces of pointed curves

摘录 / EXCERPT: Looijenga's conjecture. There is such a stratification in which every locally closed stratum S_i is affine.

conjecture | open | MathDB 页面 | 原文线索

V2-021864 | MATHDB-379550

Roth-Vakil affine stratification conjecture for partial moduli spaces of curves

摘录 / EXCERPT: Roth-Vakil's conjecture. The coarse space of $\mathcal{M}^{\leq k}_{g,n}$ has an affine stratification with $g+k$ strata; the coarse spaces of $\mathcal{M}^{rt}_{g,n}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-021865 | MATHDB-379580

Conjecture on the finiteness of the characteristic cycle for a unique critical slope

摘录 / EXCERPT: Let ${\mathscr F}$ be a constructible étale sheaf of flat Λ-modules on η, and let $r>1$ be a critical slope of ${\mathscr F}$. Assume that ... Thus r is the unique cri...

conjecture | open | MathDB 页面 | 原文线索

V2-021866 | MATHDB-379583

R=T and étaleness conjectures for refined deformation rings

摘录 / EXCERPT: Let $R_{\rho,\mathcal F}$ be the ring pro-representing the refined deformation functor, let $\mathbb T$ be the completed local Hecke algebra at the corresponding eigenvariety point...

conjecture | open | MathDB 页面 | 原文线索

V2-021867 | MATHDB-379587

Finite-pole conjecture for the Frobenius matrix

摘录 / EXCERPT: The Frobenius matrix $F(\Gamma)$ is the matrix arising from the Frobenius action in the recursive computation of the zeta function, with parameter Γ. At infinity, a function...

conjecture | open | MathDB 页面 | 原文线索

V2-021868 | MATHDB-379597

The injectivity conjecture for polynomial cycles on powers of a K3 surface

摘录 / EXCERPT: Polynomial-cycle injectivity conjecture. If the cohomology class of P vanishes, then P vanishes in the Chow group:

conjecture | open | MathDB 页面 | 原文线索

V2-021869 | MATHDB-379663

The gamma-polynomial conjecture for the matroid invariant

摘录 / EXCERPT: Gamma-polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021870 | MATHDB-379682

The monodromy-weight conjecture for smooth proper varieties

摘录 / EXCERPT: Let K be the local field, let X be a proper smooth variety over K, and let $H^j(X^{\mathrm{ca}},\overline{\mathbb{Q}}_l)$ carry its monodromy filtration $M_\bullet$ and weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-021871 | MATHDB-379684

Irrationality conjecture for the cusp translation parameter of hyperbolic knot representations

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3, let R_0 be the relevant component of the $SL_2(\mathbb{C})$-representation variety, and let μ and λ denote the meridian and lon...

conjecture | open | MathDB 页面 | 原文线索

V2-021872 | MATHDB-379701

Hyperdeterminantal module generation conjecture for the symmetric principal-minor ideal

摘录 / EXCERPT: Let P_n be the prime ideal of the algebraic variety that is the image of the principal-minor map for complex symmetric $n\times n$-matrices. Let the hyperdeterminantal module be...

conjecture | open | MathDB 页面 | 原文线索

V2-021873 | MATHDB-379709

MNOP dimension-zero Donaldson-Thomas conjecture for smooth projective threefolds

摘录 / EXCERPT: MNOP dimension-zero Donaldson-Thomas conjecture. For any smooth projective threefold X, the dimension-zero Donaldson-Thomas series has the form

conjecture | open | MathDB 页面 | 原文线索

V2-021874 | MATHDB-379710

The conjecture on infinitely many nondifferentiable points and semi-algebraicity

摘录 / EXCERPT: Nondifferentiability conjecture. If f is non-differentiable at infinitely many points, then C is not semi-algebraic.

conjecture | open | MathDB 页面 | 原文线索

V2-021875 | MATHDB-379712

Eisenbud-Harris conjecture on curves of large genus in projective space

摘录 / EXCERPT: Let C be a reduced, irreducible, nondegenerate curve of genus g and degree d in ${\bf P}^n$, and let $\pi_\alpha(d,n)$ denote the Eisenbud-Harris bound, with α in the...

conjecture | open | MathDB 页面 | 原文线索

V2-021876 | MATHDB-379717

Pukhlikov's birational rigidity conjecture for Fano varieties of index one

摘录 / EXCERPT: Let V be a smooth Fano variety of dimension $\dim V\geq4$ whose Picard group is generated by the canonical class K_V. Pukhlikov's conjecture. If $\dim V\geq4$, then V is bira...

conjecture | open | MathDB 页面 | 原文线索

V2-021877 | MATHDB-379719

The surjectivity conjecture for cohomology of smoothings

摘录 / EXCERPT: Let X_0 be a normal crossing variety with smoothing X_t, and let X denote the associated total space. Write $H^k(-,\mathbb Z)_f$ for the torsion-free quotient of integral coh...

conjecture | open | MathDB 页面 | 原文线索

V2-021878 | MATHDB-379720

Shokurov's ascending chain conjecture for log canonical thresholds

摘录 / EXCERPT: Let X be a variety of dimension n and let $Y\subset X$ be a subvariety. The log canonical threshold of the pair (X,Y) is denoted by $\operatorname{lc}(X,Y)$. Shokurov's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-021879 | MATHDB-379728

The Schubert-calculus form of the rational total reality conjecture

摘录 / EXCERPT: Let $0\le d<n$, let $t_1<t_2<\dots<t_{(n+1)(d-n)}$ be real numbers, and let $\mathbf r:\mathbb C\to\mathbb C^{d+1}$ be a rational normal curve with coordinates $r_i(t)=t^i$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-021880 | MATHDB-379732

Caro's surholonomicity conjecture for arithmetic D-modules of overconvergent F-isocrystals

摘录 / EXCERPT: Let k be a field, let Y be a separated smooth k-scheme, let E be an overconvergent F-isocrystal on Y, and let $\mathrm{sp}_{Y+}(E)$ denote its specialization. An arithm...

conjecture | open | MathDB 页面 | 原文线索

V2-021881 | MATHDB-379734

Generalized Kashiwara conjecture for proper pushforwards

摘录 / EXCERPT: Let $f:Y'\to Y”$ be a proper map between schemes over an algebraically closed field, and let $\CF$ be an irreducible ℓ-adic perverse sheaf on Y' obtained by the six operatio...

conjecture | open | MathDB 页面 | 原文线索

V2-021882 | MATHDB-379735

Bloch-Beilinson-type conjecture for zero-cycles over p-adic fields

摘录 / EXCERPT: Let V be a smooth projective geometrically integral variety over a p-adic field k. Let $A_0(V)$ be the degree-zero subgroup of the Chow group of zero-cycles, and let ... be the...

conjecture | open | MathDB 页面 | 原文线索

V2-021883 | MATHDB-379795

The geometric endomorphism conjecture for the unramified Hecke module

摘录 / EXCERPT: Let $\mathbf{G}$ be a reductive group over the nonarchimedean field k, let $\mathbf{X}$ be a spherical $\mathbf{G}$-variety, let K be a maximal compact subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-021884 | MATHDB-379816

The odd-weight invertible motive conjecture

摘录 / EXCERPT: Let k be a field, let $\mathcal{M}(k)$ be the category of motives over k for a fixed adequate equivalence relation, and let $L\in\mathcal{M}(k)$ be an invertible motive. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-021885 | MATHDB-379817

The Picard-group generation conjecture for motives

摘录 / EXCERPT: Let k be a field and let $\mathcal{M}_\sim(k)$ be the category of motives over k for an adequate equivalence relation $\sim$. Let $\operatorname{Pic}(\mathcal{M}_\sim(k))$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-021886 | MATHDB-379836

Donagi-Pantev conjecture on derived categories of complementary elliptic fibrations

摘录 / EXCERPT: Let X be a complex manifold elliptically fibered with at worst I_{1} fibers over a normal analytic variety B such that ... Let $\alpha,\beta\in\mathop{\mathcal{S}h}_{an}(X)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-021887 | MATHDB-379848

Mazur's principle for U(2,1) Shimura varieties

摘录 / EXCERPT: Mazur's principle. Suppose that m is not pseudo-Eisenstein and that $N^r=0$ on a representation space for $\overline{\rho}_m$. Then there is a level U' containing U, isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-021888 | MATHDB-379849

Traverso's truncation conjecture for p-divisible groups

摘录 / EXCERPT: Traverso's truncation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-021889 | MATHDB-379862

The isomorphism conjecture for the Fomin-Kirillov algebra and the Nichols-Woronowicz algebra

摘录 / EXCERPT: The isomorphism conjecture. The algebra homomorphism η is an isomorphism. The statement appears as a conjecture in the source, but the supplied material gives no resolution ev...

conjecture | open | MathDB 页面 | 原文线索

V2-021890 | MATHDB-379863

The quantum K-theory realization conjecture for generalized flag varieties

摘录 / EXCERPT: Let G be a simple Lie group with Borel subgroup B, let G/B be its flag variety, let P be the weight lattice, and let R be the coefficient ring used in the quantum deforma...

conjecture | solved | MathDB 页面 | 原文线索

V2-021891 | MATHDB-379869

Soulé's zeta-function order-of-vanishing conjecture

摘录 / EXCERPT: Let X be a scheme of finite type over $\mathbb{Z}$, let $d=\dim X$, and assume that higher Chow groups $CH^r(X,i)$ are defined for such schemes, have finite ranks, and satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-021892 | MATHDB-379891

The higher-stack equivalence conjecture for Morita linear and abelian categories

摘录 / EXCERPT: For a commutative ring A, let $A\textrm{-}\mathcal{CAT}^{Mor}$ denote the Morita localization of linear categories and let $A\textrm{-}\mathcal{AB}^{proj}$ be the subcategory of...

conjecture | open | MathDB 页面 | 原文线索

V2-021893 | MATHDB-379910

Filtration conjecture for étale homotopy sheaves

摘录 / EXCERPT: Assume that the characteristic exponent p is inverted and take the étale topology. For every smooth k-variety X, write $h_0^{\text{\scriptsize{\rm {\'e}t}}}(X)$ for its zerot...

conjecture | open | MathDB 页面 | 原文线索

V2-021894 | MATHDB-379911

Positivity conjecture for Chern-Schwartz-MacPherson classes of Schubert cells

摘录 / EXCERPT: Positivity conjecture. For all ${\underline\alpha}$, $c_{\text{SM}}({\mathbb{S}}({\underline\alpha})^\circ)\in A_*({\mathbb{S}}({\underline\alpha}))$ is represented by an effective...

conjecture | open | MathDB 页面 | 原文线索

V2-021895 | MATHDB-379919

The converse to Karpenko's non-ruledness theorem for anisotropic quadratic forms

摘录 / EXCERPT: Converse to Karpenko's theorem. If the first Witt index of q is greater than 1, then q is ruled over k.

conjecture | open | MathDB 页面 | 原文线索

V2-021896 | MATHDB-379941

Demailly-Campana-Peternell conjecture on cotangent bundles with affine contraction

摘录 / EXCERPT: Let M be a smooth algebraic variety, set $X=T^*M$, and let ... be the natural map, assumed to be projective and birational. Demailly-Campana-Peternell conjecture. Then $M=G/P$, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-021897 | MATHDB-379963

Ruan-Wang conjecture on quantum cohomology of K-equivalent varieties

摘录 / EXCERPT: Two smooth varieties X and X' are K-equivalent if there exist birational morphisms ... from a smooth variety Y such that ... Let their quantum cohomology rings be considered over...

conjecture | open | MathDB 页面 | 原文线索

V2-021898 | MATHDB-379967

Homological projective duality conjecture for the Pfaffian-Grassmannian pair

摘录 / EXCERPT: Let $\mathbf{G}$ be the Grassmannian in its Plücker embedding, let $\mathbf{Pf}$ be its projective-dual Pfaffian variety, and write $\mathbf{P}^*$ for the dual projective space. A...

conjecture | open | MathDB 页面 | 原文线索

V2-021899 | MATHDB-379971

Kahn's conjecture on Somekawa K-groups of semi-abelian varieties

摘录 / EXCERPT: Kahn's conjecture. If $r=s+1$, then the Somekawa K-group $K(k;G_1,\ldots,G_r)$ is torsion. If $r>s+1$ and k does not admit an ordered field structure, then

conjecture | open | MathDB 页面 | 原文线索

V2-021900 | MATHDB-379978

Multiplicity-one conjecture for truncated wedges of reduced monomial hypersurfaces

摘录 / EXCERPT: Let k be a field and let ... be a reduced monomial hypersurface. For a nonnegative integer m, let $\mathcal{W}_{m}(X)$ denote its truncated wedge scheme. An irreducible component...

conjecture | open | MathDB 页面 | 原文线索

V2-021901 | MATHDB-379981

Combinatorial description of slopes of projectivized A-hypergeometric systems

摘录 / EXCERPT: Let A be an integer matrix, let $\mathcal{M}_A(\beta)$ be the projectivized A-hypergeometric system with parameter β, and let Y be a coordinate variety. For a filtratio...

conjecture | open | MathDB 页面 | 原文线索

V2-021902 | MATHDB-379998

The localization conjecture for partial flag coordinate rings

摘录 / EXCERPT: Let G be a simple simply connected complex algebraic group of simply-laced type A, D, or E, let J be a subset of the simple roots, and let K be its complementary subset...

conjecture | open | MathDB 页面 | 原文线索

V2-021903 | MATHDB-380004

Minimality conjecture for categorical resolutions

摘录 / EXCERPT: Let Y be a singular variety, and let a categorical resolution of Y correspond to a Lefschetz decomposition of $4D^b(\mathit{Z})4D$, where Z is the exceptional diviso...

conjecture | open | MathDB 页面 | 原文线索

V2-021904 | MATHDB-380005

Indecomposable crepant categorical resolutions conjecture

摘录 / EXCERPT: Let Y be a singular variety admitting crepant categorical resolutions, and call a categorical resolution indecomposable when it has no nontrivial decomposition in the relevant ca...

conjecture | open | MathDB 页面 | 原文线索

V2-021905 | MATHDB-380006

Existence conjecture for minimal Lefschetz decompositions

摘录 / EXCERPT: Let X be a smooth projective variety and let L be an ample line bundle on X. A Lefschetz decomposition of $\mathit{D}^b(X)$ is minimal when it is strict and minimal with resp...

conjecture | open | MathDB 页面 | 原文线索

V2-021906 | MATHDB-380039

The mirror conjecture for projective bundles

摘录 / EXCERPT: Let X be a smooth projective variety and let $V=\bigoplus_{i=0}^n L_i$ be a direct sum of line bundles, where L_i, $i=1,\ldots,n$, are nef and $-K_X-c_1(V)$ is ample. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-021907 | MATHDB-380043

Automorphism conjecture for indecomposable Legendrian subvarieties

摘录 / EXCERPT: Let $X\subset \P^{2n-1}$ be an irreducible indecomposable Legendrian subvariety, and let $G<\P\operatorname{GL}_{2n}$ be a connected subgroup preserving X. Automorphism conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-021908 | MATHDB-380044

Nilpotent one-parameter subgroup conjecture for Legendrian subvarieties

摘录 / EXCERPT: Let V be a symplectic vector space, let $X'\subset \P(V)$ be an irreducible Legendrian subvariety, let $g\in\mathfrak{sp}(V)$ be a nilpotent endomorphism, and let m be an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-021909 | MATHDB-380073

Existence of a motivic Galois pro-NGLA realizing the Hom conjecture

摘录 / EXCERPT: Let K be a number field and let X and X' be smooth hyperbolic curves defined over K. Denote by $\pi_1^{mot}(\bar X)$ the motivic fundamental group over $\bar K$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021910 | MATHDB-380074

The resultant-square conjecture for the discriminant of a trigonal genus-three curve

摘录 / EXCERPT: Let C be the trigonal genus-three curve defined by $f(x,y)=0$, with coefficients $\mu_1,\mu_2,\mu_3,\mu_4,\mu_5,\mu_6,\mu_8,\mu_9,\mu_{12}$. For variables x and y, write...

conjecture | open | MathDB 页面 | 原文线索

V2-021911 | MATHDB-380077

The secant-variety regularity and projective-normality conjecture for embedded curves

摘录 / EXCERPT: Work over an algebraically closed field of characteristic 0. Let $C\subset\mathbb{P}^n$ be a linearly normal embedding of a smooth curve of genus g by a line bundle L, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-021912 | MATHDB-380088

The abelian/nonabelian correspondence for equivariant Frobenius structures

摘录 / EXCERPT: Equivariant abelian/nonabelian correspondence. Let $\varphi:P\longrightarrow N$ be the isomorphism of formal schemes over $N(X//{\bf G})\otimes{\mathbb C}[\lambda]$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-021913 | MATHDB-380102

Amoroso-David essential minimum conjecture over the rationals

摘录 / EXCERPT: Let $X\subset \mathbf{G}_m^N(\overline{\mathbf{Q}})$ be a variety, and let $U_X^{\mathbf{Q}}$ be the smallest union of torsion varieties containing X that is defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-021914 | MATHDB-380103

Amoroso-David geometric essential minimum conjecture for torus translates

摘录 / EXCERPT: Let $X\subset \mathbf{G}_m^N(\overline{\mathbf{Q}})$ be a variety, and let V_X be the smallest translate of a subtorus containing X. Write $\widehat{\mu}^{\mathrm{ess}}(X)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-021915 | MATHDB-380109

The étale and Zariski motivic complex comparison conjectures for p-adic étale Tate twists

摘录 / EXCERPT: Comparison conjecture. (1) There is an isomorphism in $D^b(X_{{\mathrm{\acute{e}t}}},{\Bbb Z}/p^r{\Bbb Z})$

conjecture | open | MathDB 页面 | 原文线索

V2-021916 | MATHDB-380112

Knutson's K-theoretic geometric Littlewood-Richardson conjecture

摘录 / EXCERPT: Let X_1 and X_2 be the two components arising when a puzzle variety breaks in the geometric Littlewood-Richardson degeneration, and let X_3 be the puzzle variety correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-021917 | MATHDB-380113

Knutson's two-step Littlewood-Richardson puzzle conjecture

摘录 / EXCERPT: Let α, β, and γ be strings indexing Schubert classes in the two-step flag variety, and let $c^{\gamma}_{\alpha,\beta}$ denote the corresponding two-step Littlew...

conjecture | open | MathDB 页面 | 原文线索

V2-021918 | MATHDB-380114

Vakil's geometric Littlewood-Richardson conjecture for partial flag varieties

摘录 / EXCERPT: Vakil's geometric Littlewood-Richardson conjecture. There exists a subset $U\subseteq S\times S\times\{d_0,\dots,d_{\binom n2}\}$ such that: $(\alpha,\beta,d_0)\in U$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-021919 | MATHDB-380124

Generalized stringy Grothendieck standard conjecture

摘录 / EXCERPT: Let X be a d-dimensional normal irreducible projective variety with at worst log-terminal singularities, and let ϕ be a family of bounded motivic invariants of type...

conjecture | open | MathDB 页面 | 原文线索

V2-021920 | MATHDB-380132

The graded homology conjecture for real loci of complex-conjugation-stable building sets

摘录 / EXCERPT: Let ${\cal G}$ be an $\mathbb R$-rational building set. A real subspace $C\in{\cal C}_{\cal G}$ is purely complex when it decomposes as ... Write $\mathbb R({\cal G})$ for the real s...

conjecture | open | MathDB 页面 | 原文线索

V2-021921 | MATHDB-380133

The irrationality conjecture for peripheral translation parameters of hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3, let R_0 be the relevant irreducible component of its $SL_2(\mathbb{C})$-representation variety, and let μ and λ denote the meri...

conjecture | open | MathDB 页面 | 原文线索

V2-021922 | MATHDB-380142

The complex-subalgebra characterization conjecture for idempotent varieties

摘录 / EXCERPT: Complex-subalgebra characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021923 | MATHDB-380144

Mohan Kumar's affineness conjecture for algebraic manifolds

摘录 / EXCERPT: Let Y be an algebraic manifold of dimension n, let X be a completion of Y, and let D be an effective boundary divisor with support $X-Y$. For integers $i,j\geq 0$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-021924 | MATHDB-380149

Homological projective duality conjecture for higher-dimensional Pfaffian varieties

摘录 / EXCERPT: Higher-dimensional Pfaffian HPD conjecture. There exists a noncommutative resolution of singularities $(Y,\CR)$ of Y which is Homologically Projectively Dual to X.

conjecture | open | MathDB 页面 | 原文线索

V2-021925 | MATHDB-380150

Homological projective duality conjecture for generalized Pfaffian varieties

摘录 / EXCERPT: Generalized Pfaffian HPD conjecture. For every $0\leq t\leq \lfloor n/2\rfloor$, there exist noncommutative resolutions of singularities of X and Y which are Homologically Proj...

conjecture | open | MathDB 页面 | 原文线索

V2-021926 | MATHDB-380171

The spherical weight conjecture for the associated subgroup

摘录 / EXCERPT: Spherical weight conjecture. For every $\theta\in \cowts_{A^0}\cap \CV(G/S)$, the irreducible object $'\IC^\theta\in \catp_\CH({}'Z)$ belongs to $\catq(Z)$.

conjecture | open | MathDB 页面 | 原文线索

V2-021927 | MATHDB-380184

Range-extension conjecture for the algebraic degree of semidefinite programming

摘录 / EXCERPT: Let $\delta(m,n,r)$ denote the algebraic degree of semidefinite programming, and let the formula in Theorem be the formula established there under a stated restriction on the range...

conjecture | open | MathDB 页面 | 原文线索

V2-021928 | MATHDB-380192

The hyperelliptic Jacobian torsion conjecture

摘录 / EXCERPT: Hyperelliptic Jacobian torsion conjecture. For sufficiently large k, there is no such C for which G_K acts on C through its cyclotomic action on...

conjecture | open | MathDB 页面 | 原文线索

V2-021929 | MATHDB-380215

Disconnectedness conjecture for finite abelian G-Hilbert schemes

摘录 / EXCERPT: Let G be a finite abelian subgroup of ${\operatorname{GL}}(n,\Bbbk)$, and let ${G}\operatorname{-Hilb}$ denote the associated G-Hilbert scheme. Disconnectedness conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-021930 | MATHDB-380217

Borisov-Alexeev boundedness conjecture for log Fano varieties

摘录 / EXCERPT: Borisov-Alexeev boundedness conjecture. Log varieties X with $-K_X$ ample, $a(X)\geq \epsilon$, and $\dim(X)=d$ form a bounded family.

conjecture | open | MathDB 页面 | 原文线索

V2-021931 | MATHDB-380225

Beilinson-Lichtenbaum conjectures for motivic complexes on arithmetic schemes

摘录 / EXCERPT: Let ${\mathscr X}$ be the arithmetic scheme in the paper, with Zariski and small étale sites ${\mathscr X}_{{\mathrm{Zar}}}$ and ${\mathscr X}_{{\mathrm{\acute{e}t}}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021932 | MATHDB-380243

Existence conjecture for hypersurfaces with prescribed isolated singularities

摘录 / EXCERPT: Let X be a projective algebraic variety of dimension n, let W be a very ample linear system on X, and let $S_1,\dots,S_r$ be singularity types with Milnor numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-021933 | MATHDB-380268

Conjecture on integrability of split tangent factors on rationally connected manifolds

摘录 / EXCERPT: Let X be a projective manifold with split tangent bundle ... A subbundle $V\subset T_X$ is integrable if it is closed under the Lie bracket of local sections. Integrability conject...

conjecture | open | MathDB 页面 | 原文线索

V2-021934 | MATHDB-380271

The genus-one partition identity for mixed Hodge polynomials

摘录 / EXCERPT: Genus-one partition identity. The following combinatorial identity holds:

conjecture | open | MathDB 页面 | 原文线索

V2-021935 | MATHDB-380272

Hausel's main mixed-Hodge polynomial conjecture for character varieties

摘录 / EXCERPT: Hausel's mixed-Hodge polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-021936 | MATHDB-380273

The polynomiality and positivity conjectures for the character-variety coefficients

摘录 / EXCERPT: Polynomiality and positivity conjectures. The following two assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-021937 | MATHDB-380274

The genus-one product identity for the character-variety generating function

摘录 / EXCERPT: Genus-one product identity.

conjecture | open | MathDB 页面 | 原文线索

V2-021938 | MATHDB-380291

Abel-Jacobi family formulation of the general Hodge conjecture

摘录 / EXCERPT: Let X be a smooth projective variety of dimension g, and let $V\subset H^m(X,\mathbb{Q})$ be a $\mathbb{Q}$-Hodge substructure of level at most $m-2p$. A nonsingular projective...

conjecture | open | MathDB 页面 | 原文线索

V2-021939 | MATHDB-380301

The $(-\infty)$-conjecture

摘录 / EXCERPT: Let X be a smooth projective variety. A variety is uniruled if there are a variety B of dimension $\dim X-1$ and a dominant rational map...

conjecture | open | MathDB 页面 | 原文线索

V2-021940 | MATHDB-380302

Lang's geometric conjecture

摘录 / EXCERPT: Let X be a smooth complex projective variety of general type. Lang's geometric conjecture. There is a proper Zariski-closed subset $S(X)\subset X$ whose irreducible components ar...

conjecture | open | MathDB 页面 | 原文线索

V2-021941 | MATHDB-380304

Truncated Vojta conjecture

摘录 / EXCERPT: Let X be a smooth proper variety over a number field k, let D be a normal crossings divisor, and let A be an ample line bundle on X. Let r be a positive integer and let...

conjecture | open | MathDB 页面 | 原文线索

V2-021942 | MATHDB-380309

Irreducibility conjecture for codimension-one intersections of Springer fiber components

摘录 / EXCERPT: Irreducibility conjecture. The intersection in codimension one of two irreducible components of a Springer fiber of type A is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-021943 | MATHDB-380312

The Liu-Xu universal vanishing conjecture for Gromov-Witten invariants

摘录 / EXCERPT: Let X be a smooth projective variety. Given a basis $\{\gamma_a\}$ for $H^*(X,\mathbb Q)$, let $x_i,y_i\in H^*(X)$ and let $k\geq 2g-3+r+s$. Using Gathmann's conventions ... and .....

conjecture | open | MathDB 页面 | 原文线索

V2-021944 | MATHDB-380313

The two-point vanishing conjecture for Gromov-Witten invariants

摘录 / EXCERPT: Let X be a smooth projective variety, let $\{\gamma_a\}$ and $\{\gamma^a\}$ be dual bases of $H^*(X,\mathbb Q)$, and let g and k be integers with $k>g$. Two-point vanishing c...

conjecture | open | MathDB 页面 | 原文线索

V2-021945 | MATHDB-380316

Andreotti-Mayer codimension conjecture for general-endomorphism components

摘录 / EXCERPT: Let g and k be integers with $1\leq k\leq g-3$, and let N be an irreducible component of the Andreotti-Mayer locus $N_{g,k}$ whose general point corresponds to a principally...

conjecture | open | MathDB 页面 | 原文线索

V2-021946 | MATHDB-380317

Boundary Andreotti-Mayer conjecture for tangentially degenerate loci

摘录 / EXCERPT: Let (B,Ξ) be a simple principally polarized abelian variety of dimension g, and suppose that $(B,\Xi)\notin N_{g,i}$ for every $i\geq k\geq1$. Here $N_k(B,\Xi)\subset B$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-021947 | MATHDB-380325

The virtual adjoint-cohomology conjecture for hyperbolic lattices

摘录 / EXCERPT: Let $\Gamma\subset\operatorname{Isom}(\mathbb H^n)$ be a lattice, let $\Gamma'\subset\Gamma$ be a finite-index subgroup, and let ρ denote the relevant natural representation....

conjecture | open | MathDB 页面 | 原文线索

V2-021948 | MATHDB-380328

Algebraic diffeomorphism transitivity conjecture for rational surfaces

摘录 / EXCERPT: Let X be a smooth projective rational surface, and let n be a natural integer. The group of algebraic diffeomorphisms $\operatorname{Diff}_{\mathrm{alg}}(X(\mathbb{R}))$ consis...

conjecture | open | MathDB 页面 | 原文线索

V2-021949 | MATHDB-380335

Q-factoriality conjecture for nodal hypersurfaces in projective four-space

摘录 / EXCERPT: Let Y be a nodal hypersurface of degree d in $\mathbb{P}^4$ with s singular points. Q-factoriality conjecture. If $s<(d-1)^2$, then Y is $\mathbb{Q}$-factorial; if Y cont...

conjecture | open | MathDB 页面 | 原文线索

V2-021950 | MATHDB-380347

The association-scheme conjecture for the meet-table graphs on 12 points

摘录 / EXCERPT: Association-scheme conjecture for 12 points. The graphs ... form an association scheme on $\mathcal{U}^{12}_3$.

conjecture | open | MathDB 页面 | 原文线索

V2-021951 | MATHDB-380349

The association-scheme conjecture for the meet-table graphs on 15 points

摘录 / EXCERPT: Association-scheme conjecture for 15 points. The graphs G_i, for $i=2,\dots,13$, form an association scheme on $\mathcal{U}^{15}_3$.

conjecture | open | MathDB 页面 | 原文线索

V2-021952 | MATHDB-380393

The dynamical model conjecture for motivic local systems on curves

摘录 / EXCERPT: Dynamical model conjecture. There exists an endomorphism Φ_C such that:

conjecture | open | MathDB 页面 | 原文线索

V2-021953 | MATHDB-380396

The algebra conjecture for parameterizing motivic local systems

摘录 / EXCERPT: Algebra conjecture for motivic local systems. There exists a commutative associative unital algebra $A=A_{Y,D,N}$ in $\mathcal C_{\mathbb F_q}$ such that, for every $n\ge 1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-021954 | MATHDB-380399

Colliot-Thélène's Hasse principle conjecture for zero-cycles

摘录 / EXCERPT: Let K be a global field, and let V/K be a smooth projective variety such that its base change $V_{\overline K}=V\operatorname*{\otimes}\overline K$ is rational and has Picard g...

conjecture | open | MathDB 页面 | 原文线索

V2-021955 | MATHDB-380404

Pappas's spin local model flatness conjecture for ramified unitary groups

摘录 / EXCERPT: Pappas's spin flatness conjecture. The scheme M_I is flat over $\mathcal O_E$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-021956 | MATHDB-380405

Pappas's corrected local model flatness conjecture for orthogonal groups

摘录 / EXCERPT: Corrected local model flatness conjecture. The scheme M is flat over

conjecture | open | MathDB 页面 | 原文线索

V2-021957 | MATHDB-380423

The canonical isomorphism and quantum deformation conjecture for the double

摘录 / EXCERPT: Let ${\cal D}$ be the cluster double, with Langlands dual ${\cal D}^{\vee}$, and let ${\mathbb Z}_+\{S\}$ denote the semigroup of finitely supported nonnegative-integer-valued func...

conjecture | open | MathDB 页面 | 原文线索

V2-021958 | MATHDB-380428

Katz's local decomposition conjecture for arithmetic families on curves

摘录 / EXCERPT: Let Y be a scheme over $\mathbb Z$, and let $\{(Y_p,{\mathscr G}_p)\}$ be a family arising from a fixed constructible complex and morphisms as in the setup. Suppose Y is an alg...

conjecture | open | MathDB 页面 | 原文线索

V2-021959 | MATHDB-380449

Existence and termination of flips

摘录 / EXCERPT: Flip conjecture. There exists a unique birational map $\phi:X\dashrightarrow X^+$ to a $\mathbb{Q}$-factorial projective variety X^+ with only terminal singularities, together wi...

conjecture | open | MathDB 页面 | 原文线索

V2-021960 | MATHDB-380452

Moduli conjecture for A^1-contractible smooth varieties

摘录 / EXCERPT: Moduli conjecture. For every $m\geq 3$ and every $n\geq 0$, there exists such an S and f for which the fibers over the k-points of S are all non-isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-021961 | MATHDB-380457

Existence of a divisor computing the global log canonical threshold of a Fano variety

摘录 / EXCERPT: Global threshold conjecture. For some such divisor D, one has

conjecture | open | MathDB 页面 | 原文线索

V2-021962 | MATHDB-380463

The irreducibility conjecture for fibers of conical Hamiltonian varieties

摘录 / EXCERPT: Irreducibility conjecture. Every conical irreducible Hamiltonian G-variety X satisfies (Irr).

conjecture | open | MathDB 页面 | 原文线索

V2-021963 | MATHDB-380464

The nilpotent-model irreducibility conjecture for Hamiltonian fibers

摘录 / EXCERPT: Nilpotent-model irreducibility conjecture. If $X=M_G(H,\eta,V)$, where η is nilpotent, then X satisfies (Irr).

conjecture | open | MathDB 页面 | 原文线索

V2-021964 | MATHDB-380466

The moment-map fiber dimension conjecture

摘录 / EXCERPT: Moment-map fiber dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-021965 | MATHDB-380475

p-independence of parity for abelian-variety Selmer ranks

摘录 / EXCERPT: Let K be a number field, let A/K be an abelian variety, and for each prime p let $\selrk{p}(A/K)$ denote the p-Selmer rank. p-independence conjecture. The parity of...

conjecture | open | MathDB 页面 | 原文线索

V2-021966 | MATHDB-380482

Derived equivalence conjecture for mirror Calabi-Yau complete intersections

摘录 / EXCERPT: Let $\{\Delta_1',\ldots,\Delta_r'\}$ and $\{\Delta_1”,\ldots,\Delta_r”\}$ be the translated nef-partitions described in the preceding discussion, with corresponding mirror Calabi-Y...

conjecture | open | MathDB 页面 | 原文线索

V2-021967 | MATHDB-380501

Jannsen's vanishing conjecture for global Galois cohomology

摘录 / EXCERPT: Let F be a number field with algebraic closure $\overline{F}$, let X be a smooth projective variety over F, let p be a prime, and let S contain the places above ∞...

conjecture | open | MathDB 页面 | 原文线索

V2-021968 | MATHDB-380532

Invariance of plurigenera under smooth projective deformations

摘录 / EXCERPT: Invariance of plurigenera conjecture. The plurigenera is invariant under smooth projective deformations; equivalently, for every integer $m\geq 1$, $P_m(X_s)$ is independent of...

conjecture | open | MathDB 页面 | 原文线索

V2-021969 | MATHDB-380537

Shapiro-Shapiro conjecture for real pole placement

摘录 / EXCERPT: Let m,p,n be positive integers with $n=mp$. Let $(\bar{A},\bar{B},\bar{C})$ be a minimal realization of the system represented through a coprime factorization $D^{-1}(s)N(s)$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-021970 | MATHDB-380552

The etale realization conjecture for motivic complexes over finite fields

摘录 / EXCERPT: Etale realization conjecture. This morphism is an isomorphism. Equivalently, after tensoring with ${\Bbb Q}$, the induced morphism is an isomorphism. The mod-l reduction is alrea...

conjecture | open | MathDB 页面 | 原文线索

V2-021971 | MATHDB-380553

Zariski realization and torsion-freeness conjecture for etale motivic complexes

摘录 / EXCERPT: Zariski realization conjecture. (a) This morphism is an isomorphism. (b) The Zariski sheaf $R^{n+1}\alpha_*{\Bbb Z}_l(n)^c$ is torsion-free. Under resolution of singularities and t...

conjecture | open | MathDB 页面 | 原文线索

V2-021972 | MATHDB-380554

Finiteness conjecture for Zariski cohomology of etale cohomology sheaves

摘录 / EXCERPT: Finiteness conjecture. These groups are finite. This conjecture is introduced to obtain finite-generation consequences for motivic cohomology when combined with the etale realizati...

conjecture | open | MathDB 页面 | 原文线索

V2-021973 | MATHDB-380555

Schneider's extension of the characteristic-zero comparison theorem to negative twists

摘录 / EXCERPT: Let S be the open arithmetic scheme and let $A(n)$ denote the real Tate-twist object used in the source. Theorem 12.1(a) gives a comparison isomorphism for positive twists betwee...

conjecture | open | MathDB 页面 | 原文线索

V2-021974 | MATHDB-380556

Leopoldt's conjectural description of the zero Tate twist over an arithmetic base

摘录 / EXCERPT: Let S be the arithmetic scheme in the source, let j be the relevant open immersion, let e be the indicated cup-product class, and let $A(1)$ be the real Tate-twist object. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-021975 | MATHDB-380557

Beilinson-Lichtenbaum-type comparison over the rationals

摘录 / EXCERPT: Let α^* be the inverse-image functor from etale to Zariski motivic complexes, let $j^*{\Bbb Z}_l(n)_S$ be the indicated arithmetic etale Tate-twist complex, and let $n\ge0$....

conjecture | open | MathDB 页面 | 原文线索

V2-021976 | MATHDB-380601

Nakamura's G-Hilbert space conjecture in dimension three

摘录 / EXCERPT: Let G be any finite subgroup of $\operatorname{SL}(3,\mathbb{C})$, and let $\mathcal{H}^G(\mathbb{C}^3)$ denote the G-Hilbert scheme. Nakamura's conjecture. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-021977 | MATHDB-380603

Hirschowitz's conjecture on special plane linear systems

摘录 / EXCERPT: Let $\mathcal{L}=\mathcal{L}(d;m_0,m^n)$ be a linear system of plane curves with one base point of multiplicity m_0 and n base points of equal multiplicity m. A system is...

conjecture | open | MathDB 页面 | 原文线索

V2-021978 | MATHDB-380604

Broer's positivity conjecture for generalized Kostka polynomials

摘录 / EXCERPT: Let G/B be the flag variety, let $\mathcal{T}'$ be the sheaf used in the paper's Euler-characteristic definition, and let $\mathcal{L}_\gamma$ be the line bundle associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-021979 | MATHDB-380623

Generic Green's conjecture for canonical curves

摘录 / EXCERPT: Generic Green's conjecture. The generic curve C of genus g satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-021980 | MATHDB-380653

Poonen's Mordell-Lang-Bogomolov conjecture for semiabelian varieties

摘录 / EXCERPT: Poonen's Mordell-Lang-Bogomolov conjecture. For $i \geq 1$, suppose

conjecture | open | MathDB 页面 | 原文线索

V2-021981 | MATHDB-380654

Positive-integral-point equidistribution conjecture for real genus-zero curves

摘录 / EXCERPT: Let $C\subset\mathbb{C}^2$ be a curve defined over $\mathbb{Z}$ and irreducible over $\mathbb{C}$. Let $C_{\mathbb{R}}$ be an irreducible component of $C\cap\mathbb{R}^2$ whose int...

conjecture | open | MathDB 页面 | 原文线索

V2-021982 | MATHDB-380656

The converse to the Weak Lang Conjecture for varieties with negative canonical bundle

摘录 / EXCERPT: The negative-canonical-bundle converse. If the canonical bundle K_X of X is negative, then for some finite extension K' of K the set $X(K')$ of K'-rational points of X...

conjecture | open | MathDB 页面 | 原文线索

V2-021983 | MATHDB-380658

The Strong Lang Conjecture

摘录 / EXCERPT: The Strong Lang Conjecture. If X is of general type, then there exists a proper subvariety $\Sigma\subsetneqq X$ such that for any finite extension K' of K,

conjecture | open | MathDB 页面 | 原文线索

V2-021984 | MATHDB-380659

Correlation conjecture for stably integral points on quasi-abelian subvarieties

摘录 / EXCERPT: Correlation conjecture. The intersection ${\mathcal P}\cap H$ is correlated with respect to the morphism $H\to B$.

conjecture | open | MathDB 页面 | 原文线索

V2-021985 | MATHDB-380663

The MSV sheaf conjecture for Gorenstein toroidal varieties

摘录 / EXCERPT: MSV sheaf conjecture. For every variety X with only Gorenstein toroidal singularities, there exists a local sheaf $\mathcal{MSV}(X)$ locally isomorphic to the product of...

conjecture | open | MathDB 页面 | 原文线索

V2-021986 | MATHDB-380664

Multiplicity inequality for simple transversal modules on a non-commutative smooth surface

摘录 / EXCERPT: Multiplicity inequality. The following inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-021987 | MATHDB-380672

The product decomposition conjecture for the kappa-zero case

摘录 / EXCERPT: Let X satisfy the assumptions that its cotangent bundle contains a nef locally free subsheaf $L^*\subset\Omega_X^1$ proportional to K_X. If $\kappa(X)=0$, then the product deco...

conjecture | open | MathDB 页面 | 原文线索

V2-021988 | MATHDB-380673

Ueno's Conjecture K on Albanese maps

摘录 / EXCERPT: Let X be a nonsingular projective variety with $\kappa(X)=0$. Ueno's Conjecture K asserts that the Albanese map is birational to an étale fiber bundle over...

conjecture | open | MathDB 页面 | 原文线索

V2-021989 | MATHDB-380712

Ginzburg's hyperkähler desingularization conjecture for reflection-group quotients

摘录 / EXCERPT: Ginzburg's conjecture. The quotient variety X can be naturally desingularized, and this desingularization is holomorphically symplectic and admits a hyperkähler structure. The co...

conjecture | open | MathDB 页面 | 原文线索

V2-021990 | MATHDB-380724

The L^2-Stokes conjecture for complex projective varieties

摘录 / EXCERPT: Let V be a complex projective variety, let $M=V\setminus\operatorname{sing} V$ be its regular part, and equip M with the metric induced by a smooth Kähler metric on projective...

conjecture | open | MathDB 页面 | 原文线索

V2-021991 | MATHDB-380725

Deformation-theoretic conjectures for geometric n-stacks

摘录 / EXCERPT: Let A be a geometric n-stack, and let $D=\operatorname{Spec}(k[\epsilon]/(\epsilon^2))$. Let $TDef_A$ denote the $(n+1)$-category of deformations of A parametrized by D. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-021992 | MATHDB-380746

The lower semicontinuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Let (X,B) be a log variety, and let $a(x)=a(x;X,B)$ denote the minimal log discrepancy at a closed point $x\in X$. Lower semicontinuity conjecture. For every closed point...

conjecture | open | MathDB 页面 | 原文线索

V2-021993 | MATHDB-380748

The moduli-part freeness conjecture

摘录 / EXCERPT: Positivity conjecture. Then K_Y+B_Y is $\mathbb{Q}$-Cartier and the moduli part M_Y is $\mathbb{Q}$-free: for some $\nu\in\mathbb{N}$, $\mathcal O_Y(\nu M_Y)$ is generated by g...

conjecture | open | MathDB 页面 | 原文线索

V2-021994 | MATHDB-380749

The optimal bound for effective building of singularities

摘录 / EXCERPT: Let (X,B) be a log variety, let $x\in X\setminus LCS(X,B)$ be a closed point, and let $H\in\operatorname{Div}(X)\otimes\mathbb{Q}$ be ample and normalized at x. Effective-build...

conjecture | open | MathDB 页面 | 原文线索

V2-021995 | MATHDB-380762

The mixed Tate motivic Galois group conjecture over the integers

摘录 / EXCERPT: Let $P_{\mathbb{Z},\mathrm{Tate}}$ be the subalgebra of periods generated by $(2\pi i)^{\pm1}$ and periods of mixed Tate motives unramified over $\mathsf{Spec}(\mathbb{Z})$. Let th...

conjecture | open | MathDB 页面 | 原文线索

V2-021996 | MATHDB-380766

The motivic Galois action conjecture for free massless field theories

摘录 / EXCERPT: Let $\mathfrak{g}^*$ be the homotopy Lie algebra associated with the free massless theory in dimension d, and let $G_{M,\mathrm{Betti}}$ be the motivic Galois group in the Betti...

conjecture | open | MathDB 页面 | 原文线索

V2-021997 | MATHDB-380769

Higher-rank Gauß-Manin determinant formula for irregular connections on curves

摘录 / EXCERPT: Higher-rank Gauß-Manin determinant formula. Assuming these conditions, one has

conjecture | open | MathDB 页面 | 原文线索

V2-021998 | MATHDB-380776

Shapiro-Shapiro conjecture for real Schubert intersections

摘录 / EXCERPT: Let $m,p\geq 1$, let γ be the real rational normal curve, and let $w_1,\ldots,w_n$ be Schubert conditions on p-planes in $(m+p)$-space. For points...

conjecture | open | MathDB 页面 | 原文线索

V2-021999 | MATHDB-380777

Reducedness conjecture for real Schubert intersections

摘录 / EXCERPT: Let $m,p\geq 1$, let γ be the real rational normal curve, and let $w_1,\ldots,w_n$ be Schubert conditions satisfying ... For distinct real points $s_1,\ldots,s_n\in\gamma$, co...

conjecture | open | MathDB 页面 | 原文线索

V2-022000 | MATHDB-380802

Slope conjecture for the moduli space of curves

摘录 / EXCERPT: For $g\geq 3$, define the slope of the moduli space of stable curves by ... where $\delta=\delta_0+\delta_1+\cdots+\delta_{\lfloor g/2\rfloor}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-022001 | MATHDB-380811

Ampleness conjecture for trace-zero bundles over Picard-rank-one Fano manifolds

摘录 / EXCERPT: Fano ampleness conjecture. If Y is Fano with $\rho(Y)=1$, then ${\cal E}$ is ample, and is spanned in most cases. This is presented as the guideline for the paper’s section on co...

conjecture | open | MathDB 页面 | 原文线索

V2-022002 | MATHDB-380829

Termination conjecture for the motivic filtration on higher Chow groups

摘录 / EXCERPT: Termination conjecture for the motivic filtration. The filtration terminates, meaning

conjecture | open | MathDB 页面 | 原文线索

V2-022003 | MATHDB-380830

Beilinson's Abel-Jacobi and Galois-invariance conjecture

摘录 / EXCERPT: Let $r\geq 2$ be an integer. For a nonsingular projective variety X over $\overline{\bold Q}$, let ... be the Abel-Jacobi map. If $X_{\bold C}=X\otimes_{\overline{\bold Q}}\bold C$...

conjecture | open | MathDB 页面 | 原文线索

V2-022004 | MATHDB-380835

Lang's conjecture for intersections with finite-rank subgroups

摘录 / EXCERPT: Let A be a complex abelian variety, let Γ be a subgroup of finite rank in $A(\mathbb{C})$, and let X be a subvariety of A. Here $\overline{X(\mathbb{C}) \cap \Gamma}$...

conjecture | open | MathDB 页面 | 原文线索

V2-022005 | MATHDB-380836

Injectivity conjecture for the second Albanese map

摘录 / EXCERPT: Let X be a projective nonsingular surface over ${\bold C}$, and let $T(X)$ be the kernel of the Albanese map on degree-zero zero-cycles. Let $\underline{{\mathrm{M}}}({\bold C})$...

conjecture | open | MathDB 页面 | 原文线索

V2-022006 | MATHDB-380843

The stronger factor-sequence conjecture for rectangle diagrams

摘录 / EXCERPT: The stronger factor-sequence conjecture. Under these hypotheses, $(\dots,X',Y',\dots)$ is also a factor sequence for γ.

conjecture | open | MathDB 页面 | 原文线索

V2-022007 | MATHDB-380856

Huybrechts' projectivity and bigness conjecture for holomorphic symplectic manifolds

摘录 / EXCERPT: Huybrechts' projectivity and bigness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-022008 | MATHDB-380858

Effective divisors conjecture for holomorphic symplectic fourfolds

摘录 / EXCERPT: Let (F,g) be a polarized irreducible holomorphic symplectic fourfold deformation equivalent to the Hilbert scheme of length-two subschemes of a K3 surface. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-022009 | MATHDB-380864

The pseudo-symmetry or F-equivalence conjecture for nonsingular toric Fano varieties

摘录 / EXCERPT: Pseudo-symmetry or F-equivalence conjecture. Any nonsingular toric Fano d-fold is either pseudo-symmetric or F-equivalent to the d-dimensional projective space $\mathbf P^{d}$.

conjecture | open | MathDB 页面 | 原文线索

V2-022010 | MATHDB-380865

The equivariant blow-up decomposition conjecture for nonsingular toric Fano varieties

摘录 / EXCERPT: Let X and Y be nonsingular toric Fano d-folds, and let $\varphi:Y\longrightarrow X$ be an equivariant morphism. An equivariant blow-up is the blow-up along a T_N-invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-022011 | MATHDB-380866

The weak F-equivalence conjecture for nonsingular toric weak Fano varieties

摘录 / EXCERPT: Let X be a nonsingular toric weak Fano d-fold. Two such varieties are weakly-F-equivalent if they are connected by a sequence of equivariant blow-ups, blow-downs, and flops thr...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022012 | MATHDB-380872

Eisenbud-Koh-Stillman secant-variety determinantal conjecture for embedded curves

摘录 / EXCERPT: Let X be a smooth curve and let L be a very ample line bundle defining an embedding of X. For each integer k, let Sec^kX denote the k-th secant variety. Eisenbud-Koh-St...

conjecture | open | MathDB 页面 | 原文线索

V2-022013 | MATHDB-380873

Asymptotic equations and syzygies conjecture for secant varieties

摘录 / EXCERPT: Let X be a smooth variety, let L be an ample line bundle on X, and fix an integer $k\geq 1$. For sufficiently large n, the line bundle L^n defines an embedding of X; wr...

conjecture | open | MathDB 页面 | 原文线索

V2-022014 | MATHDB-380874

Degree bounds for equations and syzygies of secant varieties of curves

摘录 / EXCERPT: Let X be a smooth curve embedded by a line bundle L, and let $Sec^{k-1}X$ denote its $(k-1)$-st secant variety. Secant-variety degree-bound conjecture. If $\deg(L)\geq 2g+2k$,...

conjecture | open | MathDB 页面 | 原文线索

V2-022015 | MATHDB-380875

Linear growth conjecture for character varieties of Montesinos knots

摘录 / EXCERPT: Let $\mathfrak{m}=\mathfrak{m}(a_1/b_1, a_2/b_2,\ldots,a_t/b_t)$ be a Montesinos knot with t tangles. Consider its $\operatorname{PSL}_2(\mathbb C)$- and...

conjecture | open | MathDB 页面 | 原文线索

V2-022016 | MATHDB-380891

Faber's tautological-algebra conjecture for the moduli space of curves

摘录 / EXCERPT: Let ${\mathbf M}_g$ be the moduli space of smooth genus-g curves, and let ${\cal R}^*({\mathbf M}_g)$ be its tautological algebra, with tautological classes κ_i. Faber's c...

conjecture | solved | MathDB 页面 | 原文线索

V2-022017 | MATHDB-380901

The wreath-product Hilbert quotient morphism conjecture

摘录 / EXCERPT: Let Γ be a finite subgroup of $SL_2(\mathbb{C})$, let $\Gamma_n=\Gamma^n\rtimes S_n$, and let $X_{\Gamma,n}$ be the union of connected components of the Γ-fixed locus...

conjecture | open | MathDB 页面 | 原文线索

V2-022018 | MATHDB-380909

General semisimple-group conjecture for totally positive varieties

摘录 / EXCERPT: General semisimple-group conjecture. Theorems on the contractions and product structure-and consequently the orientability theorem and the relative-homology corollary-hold for any...

conjecture | open | MathDB 页面 | 原文线索

V2-022019 | MATHDB-380921

Boundedness conjecture for log Fano varieties

摘录 / EXCERPT: Boundedness conjecture for log Fano varieties. Then $(X,\operatorname{Supp}(D))$ belongs to a finite number of algebraic families. This conjecture is known to be true when...

conjecture | open | MathDB 页面 | 原文线索

V2-022020 | MATHDB-380950

Kazhdan-Lusztig singular-support conjecture for Schubert varieties

摘录 / EXCERPT: Kazhdan-Lusztig singular-support conjecture. The singular support of $\sideset{^{\pi}}{_{X_w}}{\Bbb C}$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-022021 | MATHDB-380988

The phylogenetic invariants conjecture for binary trees

摘录 / EXCERPT: Let τ be a binary tree on binary random variables, and let I_f be the ideal of phylogenetic invariants of its general Markov model. Each edge of τ induces a split of th...

conjecture | open | MathDB 页面 | 原文线索

V2-022022 | MATHDB-381011

The exp-injectivity criterion for fat point schemes in the plane

摘录 / EXCERPT: Let $Z=\sum_i m_iP_i$ be a fat point scheme in $\mathbb{P}^2$ for general points P_i, and suppose that $h^1(\mathcal I_Z(k))=0$. Let $\mu_k(Z)$ denote the multiplication map in d...

conjecture | open | MathDB 页面 | 原文线索

V2-022023 | MATHDB-381024

The weight-monodromy conjecture

摘录 / EXCERPT: Let K be a discretely valued field with separable closure K^{sep}, let X be a variety over K with semistable model, and let $E_r^{p,q}$ be its weight spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-022024 | MATHDB-381040

Conjectural properties of the Schwartz space on an affine spherical variety

摘录 / EXCERPT: Let X be an affine spherical variety over k with trivial arithmetic multiplicity. Let $\mathcal S(X(\mathbb A_k))$ be a Schwartz space in the sense described in the source, let...

conjecture | open | MathDB 页面 | 原文线索

V2-022025 | MATHDB-381048

The toric fibration conjecture for full strongly exceptional collections

摘录 / EXCERPT: Let X be a smooth complete toric variety. Suppose that X is an F-fibration over Y, where Y and F are smooth toric varieties, and both Y and F have a full strongly e...

conjecture | open | MathDB 页面 | 原文线索

V2-022026 | MATHDB-381049

Q-abelianity conjecture for non-uniruled varieties with polarized endomorphisms

摘录 / EXCERPT: Let X be a normal projective variety admitting a non-isomorphic polarized endomorphism. A normal projective variety is non-uniruled if it is not covered by rational curves, and i...

conjecture | open | MathDB 页面 | 原文线索

V2-022027 | MATHDB-381051

The cell-zeta motivic isomorphism conjecture

摘录 / EXCERPT: Let ${\cal FC}$ be the algebra of formal cell-zeta numbers, let ζ_2 be the element whose period is $\zeta(2)$, and let $\widehat{\mathcal{M}}(\mathbb Z)$ denote the completed...

conjecture | open | MathDB 页面 | 原文线索

V2-022028 | MATHDB-381062

The definable connectedness conjecture for regular rings

摘录 / EXCERPT: Let A be a regular ring, let $\alpha,\beta\in\operatorname{Sper} A$, and let $g_1,\dots,g_s$ be a finite collection of elements of A not belonging to...

conjecture | open | MathDB 页面 | 原文线索

V2-022029 | MATHDB-381084

Dolbeault topological mirror symmetry conjecture for character varieties

摘录 / EXCERPT: Let n be a positive integer, let d,e be integers satisfying $(d,n)=(e,n)=1$, and let $\mathcal{M}_{\Dol}^d$ and $\hat{\mathcal{M}}_{\Dol}^e$ denote the corresponding Dolbeault...

conjecture | open | MathDB 页面 | 原文线索

V2-022030 | MATHDB-381085

Betti topological mirror symmetry conjecture for character varieties

摘录 / EXCERPT: Let $\mathcal{M}_{\B}^d$ be the GL_n character variety with central monodromy $\zeta_n^d I_n$, and let $\hat{\mathcal{M}}_{\B}^e$ be the corresponding...

conjecture | solved | MathDB 页面 | 原文线索

V2-022031 | MATHDB-381142

Urbano's conjecture on uniformly generated divisorial sheaves

摘录 / EXCERPT: Urbano's conjecture. There exists a very ample divisor A such that $\mathcal{O}_X(mD)\otimes \mathcal{O}_X(A)^{\otimes m}$ is globally generated for every $m\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-022032 | MATHDB-381149

Extension-class conjecture for general linear supergroup schemes

摘录 / EXCERPT: Assume that k is a perfect field of characteristic $p>2$, let $m,n,r\in\mathbb{N}$, and write $GL(m|n)_1$ for the first Frobenius kernel of the general linear supergroup scheme....

conjecture | open | MathDB 页面 | 原文线索

V2-022033 | MATHDB-381153

The si-full-rank conjecture for Deligne-Lusztig cohomology maps

摘录 / EXCERPT: Let $w=sw's$ be an element of the Weyl group with $ll(w)=ll(w')+2$, and let $Z=X(w)\cup X(sw')$. For $i\geq 0$, write ... where ... and ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-022034 | MATHDB-381154

The strong Pick set conjecture for rational inner functions on the bidisc

摘录 / EXCERPT: Let f be a rational inner function on $\mathbb D^2$, and let $V=Z_p$ be an irreducible inner variety. Write $\deg_i(f)$ and $\deg_i(p)$ for the degrees in the variable z_i. Str...

conjecture | open | MathDB 页面 | 原文线索

V2-022035 | MATHDB-381207

The classification conjecture for smooth fibre-like polytopes in odd prime dimension

摘录 / EXCERPT: Let Δ be a smooth fibre-like polytope of dimension d, and let $X(\Delta)$ denote the associated smooth toric Fano variety. Assume that d is an odd prime number. Classifi...

conjecture | open | MathDB 页面 | 原文线索

V2-022036 | MATHDB-381214

Bóna's real-rootedness conjecture for descent polynomials of stack-sortable permutations

摘录 / EXCERPT: Bóna's real-rootedness conjecture. The descent polynomial $W_{n,t}(x)$ has only real zeros for any integer $1 \le t \le n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-022037 | MATHDB-381219

Dimension conjecture for Kontsevich spaces on general Fano hypersurfaces

摘录 / EXCERPT: Let X be a general hypersurface of degree d in $\mathbb{P}^n$. Let $\overline{\mathcal{M}}_{0,0}(X,e)$ denote the Kontsevich space of degree e stable rational maps to X. Di...

conjecture | open | MathDB 页面 | 原文线索

V2-022038 | MATHDB-381229

Huh-Sturmfels conjecture on the maximum likelihood degree

摘录 / EXCERPT: Let X be a closed irreducible subvariety of the hyperplane $H=\{(p_1,\ldots,p_n)\in({\mathbb{C}}^*)^n:p_1+\cdots+p_n=1\}$, and let $d=\dim X$. The Huh-Sturmfels conjecture. One h...

conjecture | open | MathDB 页面 | 原文线索

V2-022039 | MATHDB-381234

Weak ordinarity conjecture

摘录 / EXCERPT: Let X be a smooth projective variety over $\bC$, let $A=\operatorname{Spec}R$ where R is a finitely generated $\bZ$-algebra containing $\bZ$, and let $X_A\to A$ be a spreading...

conjecture | open | MathDB 页面 | 原文线索

V2-022040 | MATHDB-381239

The Loewy filtration destabilisation conjecture for polarised varieties

摘录 / EXCERPT: Loewy filtration destabilisation conjecture. If (X,L) has non-reductive automorphism group, then the Loewy filtration destabilises (X,L).

conjecture | open | MathDB 页面 | 原文线索

V2-022041 | MATHDB-381246

Local finiteness conjecture for slopes of holonomic D-modules

摘录 / EXCERPT: Let X be a complex variety, let $\mathcal{M}$ be a holonomic $\mathcal{D}$-module on X, and let $S(\mathcal{M})$ be the union, over germs of hypersurfaces Z of X, of the se...

conjecture | open | MathDB 页面 | 原文线索

V2-022042 | MATHDB-381259

The completely reducible matrix-polynomial conjecture for canonical compactified Jacobians

摘录 / EXCERPT: Completely reducible matrix-polynomial conjecture. This moduli space, consisting of completely reducible $n\times n$ degree m matrix polynomials with spectral curve C, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-022043 | MATHDB-381262

Lehn's conjecture on Segre-number generating series for tautological bundles

摘录 / EXCERPT: Let S be a smooth projective surface and let L be a line bundle on it. Define ... Here H is the divisor corresponding to L, K is the canonical divisor, and ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-022044 | MATHDB-381263

Humphreys' support variety conjecture for tilting modules

摘录 / EXCERPT: Let G be the reductive algebraic group, G_1 its first Frobenius kernel, h the Coxeter number, and W_p^+ the set of minimum-length right coset representatives in...

conjecture | open | MathDB 页面 | 原文线索

V2-022045 | MATHDB-381287

Colliot-Thélène-Kato-Saito conjecture on zero-cycles of degree one

摘录 / EXCERPT: Let X be a smooth, proper, geometrically irreducible variety over a number field k. A zero-cycle of degree one on X is a formal integer linear combination of closed points wh...

conjecture | open | MathDB 页面 | 原文线索

V2-022046 | MATHDB-381290

The generalized partnership and duality conjecture for Gushel-Mukai categories

摘录 / EXCERPT: Let X_1 and X_2 be Gushel-Mukai varieties, and let $ \mathsf{A}(X_1)$ and $ \mathsf{A}(X_2)$ be their associated Lagrangian subspaces. Assume that neither subspace contains dec...

conjecture | open | MathDB 页面 | 原文线索

V2-022047 | MATHDB-381307

The 58-line bound conjecture for K3 quartic surfaces in characteristic 3

摘录 / EXCERPT: Let X be a K3 quartic surface defined over a field of characteristic 3. The 58-line bound conjecture. The surface X contains 112 lines or at most 58 lines. The theorem pr...

conjecture | open | MathDB 页面 | 原文线索

V2-022048 | MATHDB-381318

Baur-Draisma-de Graaf conjecture on secant defectivity of Grassmannians

摘录 / EXCERPT: Let $\mathbb G(r,n)$ denote the Grassmannian of r-planes in projective n-space, and let h be a positive integer. A projective variety is called h-defective when its h-sec...

conjecture | open | MathDB 页面 | 原文线索

V2-022049 | MATHDB-381329

Normality and Cohen-Macaulayness of conormal Schubert varieties

摘录 / EXCERPT: Let I be a k-subset of ${1,\ldots,n}$, and let CS_I denote the conormal Schubert variety, equivalently the unique component of $\mu^{-1}(\mathcal O_{k,n})$ satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-022050 | MATHDB-381334

The fixed-point formula conjecture for the motivic generating series of points on affine three-space

摘录 / EXCERPT: Fixed-point formula conjecture. The motivic generating series satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-022051 | MATHDB-381365

Algebraic specification conjecture for quasi-admissibility in the Belgian Chocolate Problem

摘录 / EXCERPT: Algebraic specification conjecture. For all n, $[?]$ is quasi-admissible by some x,y,z that are formed via algebraic specification.

conjecture | open | MathDB 页面 | 原文线索

V2-022052 | MATHDB-381377

Torus-knot conjecture for nontrivial SU(2)-averse knots

摘录 / EXCERPT: Torus-knot conjecture. If K is a nontrivial $SU(2)$-averse knot, then K is a torus knot.

conjecture | open | MathDB 页面 | 原文线索

V2-022053 | MATHDB-381382

Injectivity conjecture I(k;R) for geometric and slice motives

摘录 / EXCERPT: Let k be a field and R a coefficient ring. Write $\operatorname{DM}_{\operatorname{gm}}^{\operatorname{eff}}(k;R)$ for geometric effective motives and...

conjecture | open | MathDB 页面 | 原文线索

V2-022054 | MATHDB-381389

Geometric representation conjecture for symmetric pairs via sigma-fixed Nakajima varieties

摘录 / EXCERPT: Let $(\mathfrak g,\mathfrak k)$ be a symmetric pair listed in Table 1 in Section. Let the relevant Nakajima varieties carry an involution σ, and let the associated Steinberg...

conjecture | open | MathDB 页面 | 原文线索

V2-022055 | MATHDB-381426

Irreducibility conjecture for the resultant factor of a generic form

摘录 / EXCERPT: Let F be a generic homogeneous form, and let $\mathcal{R}_F$ denote the factor of the resultant appearing in Theorem. Irreducibility conjecture. The polynomial $\mathcal{R}_F$ is...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022056 | MATHDB-381436

Sign conjecture for coefficients of orbit-stratification characteristic classes

摘录 / EXCERPT: Sign conjecture. The coefficient of $[w]$ in the expansion of $\mathcal{C}[p]$ is a Laurent polynomial in $\tau_1,\ldots,\tau_n,y$ whose terms have sign $(-1)^{\ell(p)-\ell(w)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-022057 | MATHDB-381497

Equal matroids conjecture for CFN join varieties of a pair of tree pairs

摘录 / EXCERPT: Equal matroids conjecture. The two join varieties have the same algebraic matroid:

conjecture | open | MathDB 页面 | 原文线索

V2-022058 | MATHDB-381500

Nodal singular locus conjecture for projected complete intersection curves

摘录 / EXCERPT: Let $A\in\mathbb Z^{n+2}$ be a finite set of full dimension containing $0\in\mathbb Z^{n+2}$ and satisfying Assumption. For generic polynomials $f_1,\ldots,f_{n+1}\in\mathbb C^{A}$...

conjecture | open | MathDB 页面 | 原文线索

V2-022059 | MATHDB-381555

Unirationality conjecture for moduli components of smooth polarized Calabi-Yau pairs

摘录 / EXCERPT: Let Q be a connected component of the coarse moduli space of triples $(X,E=E_1+\dots+E_n,\Theta)$ such that X is a connected smooth projective complex variety,...

conjecture | open | MathDB 页面 | 原文线索

V2-022060 | MATHDB-381556

Iterated orthogonal extension conjecture from log-canonical centres

摘录 / EXCERPT: Let X be the space and let $\vphi_L$, ψ, ℓ, $\spH$, $\spE$, σ, $\sigma_{\operatorname{mlc}}$, $\lcc$, $\multidl_{}\paren{\vphi_L}$, and $\RTF|\cdot|(t)[\sigma]$...

conjecture | open | MathDB 页面 | 原文线索

V2-022061 | MATHDB-381575

André's motivic formulation of the Grothendieck period conjecture

摘录 / EXCERPT: Let X be a pure or mixed motive defined over $\overline{\mathbb{Q}}$. Write $\mathrm{periods}(X)$ for its periods and $\mathcal{G}_{\mathrm{mot}}(X)$ for its motivic Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-022062 | MATHDB-381581

Polynomial codegrees and dual defects of general determinantal loci

摘录 / EXCERPT: Let $D^{s,n}_{n^2-m}$ denote the determinantal locus of general $n\times n$ matrices of the indicated rank parameter. For any fixed $m>0$, the general determinantal-locus conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-022063 | MATHDB-381582

Polynomial codegrees and dual defects of skew-symmetric determinantal loci

摘录 / EXCERPT: Let $AD^{2s,2n}_{\binom{2n}{2}-m}$ denote the skew-symmetric determinantal locus in even dimension with the indicated rank parameter. For any fixed $m>0$, the even skew-symmetric d...

conjecture | open | MathDB 页面 | 原文线索

V2-022064 | MATHDB-381595

Conjecture on semiampleness and ampleness of the extended Hodge line bundle

摘录 / EXCERPT: Let B be the base of a period map $\Phi:B\to\Gamma\backslash D$, let $\overline{B}$ be a completion with boundary divisors Z_i, and let Λ_e denote the extended...

conjecture | open | MathDB 页面 | 原文线索

V2-022065 | MATHDB-381607

Cluster algebra structure conjecture for braid varieties

摘录 / EXCERPT: Let η be a braid word and let $X(\eta)$ denote the corresponding braid variety. A cluster algebra structure conjecture asserts that the coordinate ring of any braid variety...

conjecture | open | MathDB 页面 | 原文线索

V2-022066 | MATHDB-381613

Castelnuovo's conjecture for triple-point linear systems on rational threefolds

摘录 / EXCERPT: Castelnuovo's conjecture. The linear system $\mathcal{L}_{\bullet}$ has dimension $r-10$, and one of the following conditions occurs: (A) a general element...

conjecture | open | MathDB 页面 | 原文线索

V2-022067 | MATHDB-381623

Potential density under int-amplified endomorphisms

摘录 / EXCERPT: Potential density under int-amplified endomorphisms. If X admits an int-amplified endomorphism, then X satisfies potential density.

conjecture | open | MathDB 页面 | 原文线索

V2-022068 | MATHDB-381629

Weak Manin's conjecture for geometrically uniruled varieties

摘录 / EXCERPT: Weak Manin's conjecture. There exists a non-empty Zariski open subset $U\subset X$ such that, for every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-022069 | MATHDB-381638

The binary-operation conjecture for representations of GL_n

摘录 / EXCERPT: Binary-operation conjecture. $\pi(C_1*C_2)$ is a subrepresentation of $\pi(C_1)\times\pi(C_2)$. In particular, if $\pi(C_1)$ or $\pi(C_2)$ is $\square$-irreducible, then

conjecture | open | MathDB 页面 | 原文线索

V2-022070 | MATHDB-381647

Finite-dimensionality conjecture for integral motivic extension spaces

摘录 / EXCERPT: Let F be a number field, let X be a smooth projective variety over F, and let $M=h^{i-1}(X)(n)$. Write $の\operatorname{Ext}^1_{\mathcal{O}_F}(\mathbb{1},M)$ for the subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-022071 | MATHDB-381661

Polynomial growth conjecture for Specht multiplicities of symmetric real algebraic sets

摘录 / EXCERPT: Polynomial growth conjecture. For any fixed $p\geq 0$, $m_{p,\{\lambda\}_n}(V_n(I))$ is eventually given by a polynomial in n.

conjecture | open | MathDB 页面 | 原文线索

V2-022072 | MATHDB-381664

Multiplicity conjecture for the model $({\mathrm{GU}}_4\times {\mathrm{GU}}_2,({\mathrm{GU}}_2\times {\mathrm{GU}}_2)^0)$

摘录 / EXCERPT: Multiplicity conjecture. The unique distinguished element belongs to $\Pi_\phi(G)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-022073 | MATHDB-381691

Optimal index conjecture for terminal and canonical Calabi-Yau varieties

摘录 / EXCERPT: Optimal index conjecture. The index m of X is the largest possible index among all terminal Calabi-Yau varieties of dimension n, and also among all canonical Calabi-Yau varie...

conjecture | open | MathDB 页面 | 原文线索

V2-022074 | MATHDB-381700

Hamilton-Tian conjecture for the Kähler-Ricci flow on Fano manifolds

摘录 / EXCERPT: Hamilton-Tian conjecture. For any global solution $\omega(t)$ as above, any sequence $(X,\omega(t))$ along the Kähler-Ricci flow contains a subsequence converging to a $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-022075 | MATHDB-381705

Maximality conjecture for real circles tangent to three conics

摘录 / EXCERPT: Let $Q_1,Q_2,Q_3\subseteq\mathbb{R}^2$ be three general real conics. A real circle is a circle defined over $\mathbb{R}$ that is tangent to each of Q_1,Q_2,Q_3. Maximality conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-022076 | MATHDB-381708

Kuznetsov-Smirnov's quantum-spectrum conjecture for Fano varieties

摘录 / EXCERPT: Let X be a Fano variety of index m over an algebraically closed field $\boldsymbol{\Bbbk}$ of characteristic zero, and assume that the big quantum cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-022077 | MATHDB-381709

Kuznetsov-Smirnov's residual-category conjecture for coadjoint varieties

摘录 / EXCERPT: Let X be the coadjoint variety of a simple algebraic group $\mathrm G$ over an algebraically closed field of characteristic zero. Let $\mathrm T(\mathrm G)$ be the Dynkin diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-022078 | MATHDB-381719

The Medvedev-Scanlon conjecture on dense forward orbits

摘录 / EXCERPT: Let X be an irreducible variety defined over an algebraically closed field F of characteristic 0, and let $\phi\colon X\dashrightarrow X$ be a dominant rational map. A domina...

conjecture | open | MathDB 页面 | 原文线索

V2-022079 | MATHDB-381728

Shokurov's toric-model conjecture for maximal log Calabi-Yau threefolds

摘录 / EXCERPT: Shokurov's toric-model conjecture. Every maximal log Calabi-Yau pair (X,Δ_X) whose underlying variety X is a rational 3-fold has a toric model.

conjecture | open | MathDB 页面 | 原文线索

V2-022080 | MATHDB-381748

The class-module rank and Fitting-ideal conjecture for Carlitz twists

摘录 / EXCERPT: Let $Z(x,n)$ be written uniquely as ... in $A[\![x-1]\!]$, with $h_n\geq 0$ and $\zeta_C^*(n)\in A$. Let $\operatorname{Cl}(\underline{A}(n))$ denote the class module of the Carlitz...

conjecture | open | MathDB 页面 | 原文线索

V2-022081 | MATHDB-381780

Parity conjecture for epsilon-torsion in the 2-complete real motivic stable homotopy ring

摘录 / EXCERPT: Parity conjecture. If $(1-\epsilon)\alpha$ is non-zero, then w is even.

conjecture | open | MathDB 页面 | 原文线索

V2-022082 | MATHDB-381817

Colliot-Thélène's exactness conjecture for zero-cycles

摘录 / EXCERPT: Let k be a global field and let X be a smooth, projective, geometrically integral variety over k. Write $CH_0(X)$ for the Chow group of zero-cycles modulo rational equivalenc...

conjecture | open | MathDB 页面 | 原文线索

V2-022083 | MATHDB-381849

Chern-Simons reciprocity for closed 3-manifolds with finite irreducible character variety

摘录 / EXCERPT: Let M be an oriented closed 3-manifold, let G be a simply connected complex simple Lie group, and let $\mathrm{R}^{\rm irr}_G(M)$ be the character variety of irreducible repr...

conjecture | open | MathDB 页面 | 原文线索

V2-022084 | MATHDB-381850

Chern-Simons reciprocity for boundary-parabolic representations

摘录 / EXCERPT: Let M be an oriented compact 3-manifold with a torus boundary, and let $\mathrm{R}^{\rm para}_{\mathrm{SL}_2(\mathbb{C})}(M)$ be the set of conjugacy classes of boundary-parabo...

conjecture | open | MathDB 页面 | 原文线索

V2-022085 | MATHDB-381880

The inhomogeneous AJ conjecture for the nonabelian A-polynomial factor

摘录 / EXCERPT: Let K be a knot. Let $P_K^0(q,E,Q)$ be the factor of the A_q-polynomial corresponding to the inhomogeneous recursive relation, let $A_K(I,\alpha)$ be the A-polynomial, let...

conjecture | open | MathDB 页面 | 原文线索

V2-022086 | MATHDB-381882

Defant's linear-time conjecture for Pop-Stack Sorting on random permutations

摘录 / EXCERPT: Let $D70E_n$ be a uniformly random permutation in the symmetric group $D70E_n$, and let $D43E\operatorname{Pop}$ denote one step of Pop-Stack Sorting, which reverses every maximal...

conjecture | open | MathDB 页面 | 原文线索

V2-022087 | MATHDB-381892

The Hamiltonian-reduction implication for Condition (GIT)

摘录 / EXCERPT: Let V be a linear representation of a complex reductive algebraic group G and let ... be the associated moment map. Set $V:=\mu^{-1}(0)$. For a generic...

conjecture | open | MathDB 页面 | 原文线索

V2-022088 | MATHDB-381898

The bridge-graph quantum max-flow conjecture in the symmetric regions

摘录 / EXCERPT: Let w be a positive integer and let (a,b) and (a',b') lie in the union of the regions $\mathbf{U}_w$, $\mathbf{V}_w$, and $\mathbf{W}_w$. Consider the bridge graph with bond...

conjecture | open | MathDB 页面 | 原文线索

V2-022089 | MATHDB-381899

The bridge-graph quantum max-flow conjecture below the cut threshold

摘录 / EXCERPT: Let w be a positive integer and let (a,b) and (a',b') lie in $\mathbf{U}_w\cup\mathbf{V}_w\cup\mathbf{W}_w$, with $ab'\leq a'b$. Consider the corresponding bridge graph with...

conjecture | open | MathDB 页面 | 原文线索

V2-022090 | MATHDB-381900

The three-bond-dimension reduction for the bridge-graph conjecture

摘录 / EXCERPT: Let (a,b) and (a',b') lie in $\mathbf{U}_3\cup\mathbf{V}_3\cup\mathbf{W}_3$, and consider the bridge graph with bond dimension 3 and boundary dimensions a,b,b',a'. Three-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-022091 | MATHDB-381927

Monodromy conjecture for log mixed objects

摘录 / EXCERPT: Let k be the base field, let ${\mathcal {P}}$ be the class of schemes considered in the source, and let $X\in {\mathcal {P}}$. The monodromy operator is ... The associated objects...

conjecture | open | MathDB 页面 | 原文线索

V2-022092 | MATHDB-381957

Strong formality conjecture for the half-quantum flag variety

摘录 / EXCERPT: Let G be an almost-simple algebraic group with dual group $\check{G}$, let q be a chosen complex root of unity, let X_q be the half-quantum flag variety, let...

conjecture | open | MathDB 页面 | 原文线索

V2-022093 | MATHDB-381959

Strong formality conjecture for the unit-generated subcategories

摘录 / EXCERPT: Let X_q be the half-quantum flag variety, let $\widetilde{\mathcal{N}}$ be the Springer resolution, let $\mathcal{N}$ be the nilpotent cone, and use...

conjecture | open | MathDB 页面 | 原文线索

V2-022094 | MATHDB-381967

Strong positivity conjecture for CSM classes of Schubert cells

摘录 / EXCERPT: Strong positivity conjecture. One has $c_{v,w}>0$ for every $v\le w$. The weaker non-negativity assertion is known, but strict positivity remains the stronger conjectural statement...

conjecture | open | MathDB 页面 | 原文线索

V2-022095 | MATHDB-381969

Positivity conjecture for equivariant motivic Chern classes of Schubert cells

摘录 / EXCERPT: Positivity conjecture for motivic Chern classes. For every $u\leq w\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-022096 | MATHDB-381988

ℓ-independent Lefschetz standard conjecture in degree i

摘录 / EXCERPT: ℓ-independent Lefschetz standard conjecture in degree i. For X, h, and i as above, there exists a codimension i cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-022097 | MATHDB-381989

ℓ-adic integral Tate conjecture in codimension c

摘录 / EXCERPT: ℓ-adic integral Tate conjecture in codimension c. The cycle class map

conjecture | open | MathDB 页面 | 原文线索

V2-022098 | MATHDB-381990

The period-index conjecture for unramified Brauer classes

摘录 / EXCERPT: Unramified period-index conjecture. There exists a positive integer e such that, for all $b1 \in \operatorname{Br}(X)$,

conjecture | open | MathDB 页面 | 原文线索

V2-022099 | MATHDB-382003

Permutation-representation conjecture for local systems on parabolic Lusztig varieties

摘录 / EXCERPT: Permutation-representation conjecture. The local systems $L_{z,w}^J$ are induced by permutation representations of $(S_n)_{J_z}^z$. This would strengthen the preceding theorem, whi...

conjecture | open | MathDB 页面 | 原文线索

V2-022100 | MATHDB-382004

Parabolic Haiman-Stanley-Stembridge h-positivity conjecture

摘录 / EXCERPT: Parabolic h-positivity conjecture. The Frobenius character

conjecture | open | MathDB 页面 | 原文线索

V2-022101 | MATHDB-382005

Codominant-support conjecture for parabolic Lusztig decomposition

摘录 / EXCERPT: Codominant-support conjecture. The local system $L_{z,w}^{J',\emptyset}$ is zero for every non-codominant permutation z. The conjecture would restrict the decomposition-theorem s...

conjecture | open | MathDB 页面 | 原文线索

V2-022102 | MATHDB-382008

Lie-algebra closure conjecture for finitely generated unipotent subgroups

摘录 / EXCERPT: Let X be an irreducible affine variety. For locally nilpotent derivations $\partial_1,\ldots,\partial_k$ of ${\mathbb K}[X]$, let ... A Lie-algebra closure conjecture asserts that...

conjecture | open | MathDB 页面 | 原文线索

V2-022103 | MATHDB-382009

Tits-alternative conjecture for finitely generated unipotent subgroups

摘录 / EXCERPT: Let X be an affine variety of dimension at least 2, and let ... be generated by finitely many ${\mathbb G}_a$-subgroups. Suppose that G acts 2-transitively on one of its orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-022104 | MATHDB-382048

The equality conjecture for the volume of the even-dimensional Eisenstein orbifold

摘录 / EXCERPT: Let $n \in 2\mathbb{Z}_{\geq 1}$. Write $\Gamma^n_{\mathrm{eis}} \setminus \mathbb{R} H^n$ for the non-arithmetic hyperbolic orbifold $M({\mathscr{L}}_{\mathrm{eis}}^n,\alpha_0)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-022105 | MATHDB-382050

The period-index conjecture for Brauer groups of complex varieties

摘录 / EXCERPT: Period-index conjecture. For every $alpha in \operatorname{Br}(X)$,

conjecture | open | MathDB 页面 | 原文线索

V2-022106 | MATHDB-382061

Ooguri-Vafa integrality product formula for framed geometric-seed wavefunctions

摘录 / EXCERPT: Ooguri-Vafa integrality conjecture. The wavefunction has the product expansion

conjecture | open | MathDB 页面 | 原文线索

V2-022107 | MATHDB-382073

Generalized relative Manin-Mumford conjecture

摘录 / EXCERPT: Generalized relative Manin-Mumford conjecture. Under these hypotheses, Y is contained in a proper torsion coset.

conjecture | open | MathDB 页面 | 原文线索

V2-022108 | MATHDB-382106

Mazur's finite generation conjecture over cyclotomic extensions

摘录 / EXCERPT: Let ℓ be a rational prime, let K be a number field, and let $K_\infty=K\cdot\mathbb{Q}_{\infty,\ell}$ be the cyclotomic $\mathbb{Z}_\ell$-extension of K. Let A/K_∞...

conjecture | open | MathDB 页面 | 原文线索

V2-022109 | MATHDB-382110

Compactness of increasing nets of non-Archimedean psh metrics

摘录 / EXCERPT: Let X be a unibranch variety, let L be a line bundle on X, and consider the space of non-Archimedean plurisubharmonic metrics $\mathrm{PSH}(L^{\mathrm{an}})$. An increasing n...

conjecture | open | MathDB 页面 | 原文线索

V2-022110 | MATHDB-382146

Fujita's Spectrum Conjecture for smooth polarized varieties

摘录 / EXCERPT: Fujita's Spectrum Conjecture. For every $\varepsilon>0$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-022111 | MATHDB-382156

Determinant vanishing conjecture for finite semigroups in ECOM

摘录 / EXCERPT: Let S be the finite semigroup under consideration, let $u,u'\in S$, let ${u^{+}_r}$ and ${u'}^{+}_l$ denote their associated right and left idempotents, let $\boldsymbol{*}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-022112 | MATHDB-382157

Naive dimension conjecture for secant loci of scrolls over curves

摘录 / EXCERPT: Let C be a nonhyperelliptic curve of genus $g\geq 9$. Let $E\to C$ be a vector bundle of rank r and slope at most $g-2$ such that the tautological line bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-022113 | MATHDB-382179

Singular-locus conjecture for secant varieties of spinor varieties

摘录 / EXCERPT: Let $N\geq 7$, let $\mathbb S_N^+$ be the Spinor variety, let $\sigma_2(\mathbb S_N^+)$ be its secant variety of lines, and let $\Sigma_{2,N}$ be the distance--2 orbit. The singu...

conjecture | open | MathDB 页面 | 原文线索

V2-022114 | MATHDB-382193

Deligne's weight-monodromy conjecture

摘录 / EXCERPT: Let X be a proper smooth variety over a local field k, and let V be its ith ℓ-adic étale cohomology group. Write N for the monodromy operator, let $W_\bullet$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-022115 | MATHDB-382210

Coates-Corti-Iritani-Tseng's quantum Lefschetz conjecture

摘录 / EXCERPT: Let X be a smooth proper Deligne-Mumford stack with projective coarse moduli space, and let Y be a smooth complete intersection in X. Let I^{tw} be a point on the twisted L...

conjecture | refuted | MathDB 页面 | 原文线索

V2-022116 | MATHDB-382213

Non-degeneracy conjecture for partial heights

摘录 / EXCERPT: Let $K=\mathbb{C}(B)$ be the function field of a smooth projective curve B over $\mathbb{C}$. Let X be a projective variety over K that does not contain any possibly singular...

conjecture | open | MathDB 页面 | 原文线索

V2-022117 | MATHDB-382241

Conjecture on irreducible components of relative tangency varieties

摘录 / EXCERPT: Conjecture on the components of $\mathcal{SC}_r$. For all $r\in[N+1]$:

conjecture | open | MathDB 页面 | 原文线索

V2-022118 | MATHDB-382251

Boguslavsky-Tsfasman conjecture for projective point counts over finite fields

摘录 / EXCERPT: Let $S(m)=\mathbb{F}_q[x_0,\dots,x_m]$, and let $S_d(m)$ be its degree-d graded component. For $m,d\geq 1$ and $1\leq r\leq \binom{m+d}{d}$, let $e_r(d,m)$ be the maximum number...

conjecture | open | MathDB 页面 | 原文线索

V2-022119 | MATHDB-382253

The main conjecture on rigid isotopy classes of quartic Morse polynomials

摘录 / EXCERPT: Let $f\colon\mathbb{R}^2\to\mathbb{R}$ be a Morse polynomial of degree 4, and consider its principal part. The principal part may be non-discriminant positive definite, vanish on...

conjecture | open | MathDB 页面 | 原文线索

V2-022120 | MATHDB-382258

Generalized analytic delineability conjecture for iterated resultants

摘录 / EXCERPT: Let $f_1,\ldots,f_t$ be relatively prime real polynomials of positive degrees with respect to $x_{f+1},\ldots,x_r$, and let $x=(x_1,\ldots,x_f)$. Let $D(x)$ be a generalized discri...

conjecture | open | MathDB 页面 | 原文线索

V2-022121 | MATHDB-382299

The determinant criterion for finiteness of rational points

摘录 / EXCERPT: Let $V(k)$) be the set of k-rational points of a projective variety V over a number field k, and let A^1_{∞} be the infinite matrix associated with V in the paper. D...

conjecture | open | MathDB 页面 | 原文线索

V2-022122 | MATHDB-382309

Non-triviality of graded Satake diagrams

摘录 / EXCERPT: A graded Satake diagram is a decorated Dynkin diagram arising in the classification above, with the specified type I or type II $\mathbb{Z}_2$-graded Satake-diagram structure. Non-...

conjecture | open | MathDB 页面 | 原文线索

V2-022123 | MATHDB-382310

Voisin map conjecture for homologically trivial zero-cycles

摘录 / EXCERPT: Let Y be the variety in the preceding construction and let $X=F_r(Y)$, with $r\geq 2$, where $\Psi:X\dashrightarrow X$ is the Voisin rational map. For $z\in CH_0(X)_{hom}$, the s...

conjecture | open | MathDB 页面 | 原文线索

V2-022124 | MATHDB-382320

Conjecture on the top codimension of the product-extended tautological ring

摘录 / EXCERPT: Top-codimension conjecture. For all $g\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-022125 | MATHDB-382328

Tautologicality of the Tor class of the product locus

摘录 / EXCERPT: Let $\mathcal M_g^{\leq 1}$ be the moduli space of stable curves of genus g with degenerations of torus rank at most 1, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-022126 | MATHDB-382334

The small p-adic Simpson comparison conjecture for semi-stable formal schemes

摘录 / EXCERPT: Small p-adic Simpson comparison conjecture. The canonical morphism induces a quasi-isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-022127 | MATHDB-382338

Existence of supersingular curves of genus five via unramified double covers

摘录 / EXCERPT: Let p be an odd prime, and let X be a smooth curve of genus 3 over $\overline{\mathbf F}_p$. An unramified double cover of X is a degree-two étale morphism $\pi:Y\to X$; by...

conjecture | open | MathDB 页面 | 原文线索

V2-022128 | MATHDB-382339

Mustață-Popa depth conjecture for the Du Bois complex

摘录 / EXCERPT: Let X be a variety, let i be a nonnegative integer, and write $\underline{\Omega}^{i}_{X}$ for the i-th graded piece of the Du Bois complex of X. For a sheaf or complex, le...

conjecture | open | MathDB 页面 | 原文线索

V2-022129 | MATHDB-382346

Detection of all $S_4(q,r)$ surfaces by the two-dimensional character-variety component

摘录 / EXCERPT: Let $\mathbb{F}$ be an algebraically closed field of characteristic p. Consider N_Φ a Seifert fibered space with base orbifold $S^2(2,2,2,3)$ and gluing matrix Φ $=$ .....

conjecture | open | MathDB 页面 | 原文线索

V2-022130 | MATHDB-382355

The semi-algebraicity conjecture for K-semistable domains

摘录 / EXCERPT: Let X be a normal projective variety, and let D_j for $1\leq j\leq k$ be effective Weil divisors on X. The notation $\operatorname{Kss}(X,\sum_{j=1}^k D_j)$ denotes the K-sem...

conjecture | open | MathDB 页面 | 原文线索

V2-022131 | MATHDB-382368

Existence and uniqueness of Frobenius-fixed paths in mixed motives

摘录 / EXCERPT: Let Z be as in paragraph A16, let X be the corresponding variety, and suppose that x,y are Z-points of X. Let $s_\mathfrak{p}^*$ denote the specialization functor to the...

conjecture | open | MathDB 页面 | 原文线索

V2-022132 | MATHDB-382386

Cluster-theoretic conjecture for splicing braid varieties

摘录 / EXCERPT: Let $\beta=\beta^1\beta^2$ be the braid and let r_1 and w be as in the construction of the open subset $\mathcal{U}_{r_1,w}(\beta)\subseteq X(\beta)$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-022133 | MATHDB-382395

Conjecture on tangent-space inclusion for singular real algebraic hypersurfaces

摘录 / EXCERPT: Let $\mathbb{R}\langle \xi\rangle$ be the real closed field of algebraic Puiseux series, let $\mathcal{P}=\{P_1,\ldots,P_s\}\subset \mathbb{R}[X_1,\ldots,X_n]$, and set ... For a bou...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022134 | MATHDB-382397

Zero-dimensional intersection conjecture for degree-\d hypersurfaces

摘录 / EXCERPT: Zero-dimensional intersection conjecture. Given $d,m\geq 1$ and $m\leq r\leq \binom{d+m}{m}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-022135 | MATHDB-382411

The asymptotic proportion conjecture for hyperelliptic Jacobians in dimension three

摘录 / EXCERPT: Let q range over powers of 2, and consider isogeny classes of 3-dimensional abelian varieties over the finite field of characteristic 2 with q elements. The classes under...

conjecture | open | MathDB 页面 | 原文线索

V2-022136 | MATHDB-382420

The local companion points conjecture

摘录 / EXCERPT: The local companion points conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-022137 | MATHDB-382447

Char p Mumford-Tate conjecture for ordinary abelian schemes

摘录 / EXCERPT: Let X_0 be a geometric connected smooth variety over $\mathbb{F}_q$, let $f_0:X_0\rightarrow \mathscr{A}_{g,\mathbb{F}_q}^{{\mathrm{ord}}}$ be a morphism as above, and let A be...

conjecture | open | MathDB 页面 | 原文线索

V2-022138 | MATHDB-382463

Noncommutative minimal model conjecture for Fano varieties

摘录 / EXCERPT: Noncommutative minimal model conjecture. For any smooth Fano variety X, there exist $\tau\in {\mathrm H}^\bullet(X)$, a sector $\mathscr{S}\subset\mathbf{C}^*$, $\epsilon>0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-022139 | MATHDB-382475

Gieseker-Knörrer-Trubowitz conjecture on irreducibility of Fermi varieties

摘录 / EXCERPT: Let q_1 and q_2 be two distinct odd primes, and let V be a nonconstant real-valued $q_1\be Z\binom{}{}\oplus q_2\be Z$-periodic potential. For $[?]$

conjecture | open | MathDB 页面 | 原文线索

V2-022140 | MATHDB-382499

Braverman-Makarychev-Makarychev-Naor conjecture on an explicit Krivine rounding scheme

摘录 / EXCERPT: Braverman-Makarychev-Makarychev-Naor conjecture. There exists $\eta>0$ such that f_η and g_η form a Krivine rounding scheme and

conjecture | open | MathDB 页面 | 原文线索

V2-022141 | MATHDB-382521

Bounded denominator conjecture for stringy Euler numbers

摘录 / EXCERPT: Let X be a d-dimensional algebraic variety with at worst Gorenstein canonical singularities. The bounded denominator conjecture asserts that the denominator of the rational num...

conjecture | open | MathDB 页面 | 原文线索

V2-022142 | MATHDB-382539

Connectedness of closures of varieties of matrix tuples without irreducible tuples

摘录 / EXCERPT: Connectedness conjecture. The closures of ${\cal V}$ and ${\cal W}$ should be connected without the assumption that there are irreducible tuples.

open_problem | open | MathDB 页面 | 原文线索

V2-022143 | MATHDB-382545

Shafarevich conjecture for families of polarized Calabi-Yau varieties

摘录 / EXCERPT: Let C be a smooth projective curve, let $S\subset C$ be a finite subset, and fix an integer $q\geq 2$. Let $\mathbb T$ denote the parameterizing scheme for non-isotrivial familie...

conjecture | open | MathDB 页面 | 原文线索

V2-022144 | MATHDB-382552

Terwilliger systems' subdegree bound conjecture

摘录 / EXCERPT: Terwilliger systems' subdegree bound conjecture. The inequality above holds for every TD system.

conjecture | open | MathDB 页面 | 原文线索

V2-022145 | MATHDB-382553

Terwilliger systems' rank-generating polynomial factorization conjecture

摘录 / EXCERPT: Terwilliger systems' rank-generating polynomial factorization conjecture. There exist positive integers $d_1,d_2,\ldots,d_n$ whose sum is d such that

conjecture | open | MathDB 页面 | 原文线索

V2-022146 | MATHDB-382564

The degree doubling conjecture for braid monodromy factorizations

摘录 / EXCERPT: Let X be a complex algebraic manifold, and let k be a positive integer for which the degree doubling formula is defined. A braid factorization is considered up to Hurwitz equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-022147 | MATHDB-382567

The smooth birational model conjecture for rational maps and entropy

摘录 / EXCERPT: Let Y be a smooth projective variety and $f:Y\dashrightarrow Y$ be a rational dominating map. A smooth birational model is a smooth projective variety $\hat Y$ together with a bi...

conjecture | open | MathDB 页面 | 原文线索

V2-022148 | MATHDB-382572

Orthogonality conjecture for the four split decompositions

摘录 / EXCERPT: Under Assumption, let D be the diameter and let ${\tilde V}^{\downarrow \downarrow}_{ij}$, ${\tilde V}^{\uparrow \uparrow}_{rs}$, ${\tilde V}^{\downarrow \uparrow}_{ij}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-022149 | MATHDB-382573

Atypicality and rank support varieties for stable polar Lie superalgebras

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be a simple basic classical Lie superalgebra, and let $L(\lambda)$ be a finite-dimensional simple ${\mathfrak{g}}$-supermodule. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-022150 | MATHDB-382578

Deligne's algebraic definition conjecture for one-motives

摘录 / EXCERPT: Let V be an arbitrary variety over an arbitrary field k, let n be an integer, and let $\bar{k}$ be an algebraic closure of k. Write T_{ℓ} and T_{DR} for the ℓ-a...

conjecture | open | MathDB 页面 | 原文线索

V2-022151 | MATHDB-382583

Generic Semisimplicity of the Big Quantum Cohomology of Flag Varieties

摘录 / EXCERPT: The big quantum cohomology of G/P is generically semisimple.

conjecture | partial | Chi Zhang | MathDB 页面 | 原文线索

V2-022152 | MATHDB-383000

Quartic Hessian conjecture in dimension four

摘录 / EXCERPT: For every polynomial $f\in\mathbb{C}[x_1,x_2,x_3,x_4]$ satisfying $\det\bigl(\operatorname{Hess}(f)\bigr)\in\mathbb{C}^{\times}$, the gradient map...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022153 | MATHDB-383136

Atiyah’s Minkowski Space Conjecture

摘录 / EXCERPT: For every integer $n\ge 2$ and every admissible marked configuration of n complete pairwise disjoint timelike affine worldlines in Minkowski space, let $F_1,\ldots,F_n$ be the as...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022154 | MATHDB-383225

Mistretta-Stoppino conjecture

摘录 / EXCERPT: Let C be a smooth projective curve and let (L,V) be a generated linear series with $\deg L=d$ and $\dim V=r+1$, where $r\ge 2$. Let $M_{L,V}$ be its syzygy bundle, defined by t...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022155 | MATHDB-383292

Sato’s weak F-equivalence conjecture

摘录 / EXCERPT: For every integer $d\geq 2$, any two nonsingular projective toric d-dimensional weak Fano varieties are connected by a finite zigzag of projective toric birational morphisms, eac...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022156 | MATHDB-383348

Oort’s conjecture on supersingular abelian varieties

摘录 / EXCERPT: For every integer $g\ge 2$ and every prime p, let $\mathcal{S}_g$ denote the supersingular locus in the moduli space of principally polarized abelian varieties of dimension g o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022157 | MATHDB-383449

Linear secret-sharing scheme size problem

摘录 / EXCERPT: For an n-party monotone access structure $\Gamma\subseteq 2^{[n]}$, let ℓ_{Γ} be the minimum, over all linear secret-sharing schemes for one-bit secrets realizing...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-022158 | MATHDB-383505

Generators of the automorphism group of the Fermat quartic surface

摘录 / EXCERPT: Let $X_4\subset\mathbb{P}^3_{\mathbb{C}}$ be the Fermat quartic surface $X_4=\{[x_0:x_1:x_2:x_3]\mid x_0^4+x_1^4+x_2^4+x_3^4=0\}$. Find an explicit finite collection of automorphis...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-022159 | MATHDB-383557

Hodge conjecture for Fermat fourfolds

摘录 / EXCERPT: For every integer $m\ge 1$, let $X_m^4=\{ x_0^m+x_1^m+x_2^m+x_3^m+x_4^m+x_5^m=0\}\subset\mathbb{P}^5$ be the Fermat fourfold. The Hodge conjecture asserts that every rational Hodge...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-022160 | MATHDB-383612

Hessian conjecture in Lorentzian signature

摘录 / EXCERPT: Does every polynomial $\phi\in\mathbb{R}[x_1,x_2,x_3,x_4]$ satisfying $\det\bigl(\operatorname{Hess}\phi(x)\bigr)=-1$ and having Lorentzian Hessian, namely...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022161 | MATHDB-383718

Yau-Tian-Donaldson conjecture for constant scalar curvature metrics

摘录 / EXCERPT: For every polarized smooth projective complex manifold (X,L), the polarized manifold (X,L) is K-polystable if and only if the Kähler class $c_1(L)$ admits a constant scalar c...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022162 | MATHDB-383823

Describe the singularities of the smoothable component of ... .

摘录 / EXCERPT: Describe the singularities of the smoothable component of ${\rm Hilb}^d(\mathbb{A}^n)$. To be more specific: How large can the dimension of the Zariski tangent space to this compon...

open_problem | open | MathDB 页面 | 原文线索

V2-022163 | MATHDB-383824

Can you describe the Zariski tangent space to the smoothable component of ... ?

摘录 / EXCERPT: Can you describe the Zariski tangent space to the smoothable component of ${\rm Hilb}^d(\mathbb{A}^n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022164 | MATHDB-383825

Hilbert schemes of points (problem 3)

摘录 / EXCERPT: What is the smallest d such that ${\rm Hilb}^d(\mathbb{A}^3)$ is reducible?

open_problem | open | MathDB 页面 | 原文线索

V2-022165 | MATHDB-383826

Hilbert schemes of points (problem 4)

摘录 / EXCERPT: Is there a nonreduced component of ${\rm Hilb}^d(\mathbb{A}^n)$? If so, find it.

open_problem | open | MathDB 页面 | 原文线索

V2-022166 | MATHDB-383827

Hilbert schemes of points (problem 5)

摘录 / EXCERPT: Is the Gr"obner fan a discrete invariant that distinguishes the components of ${\rm Hilb}^d(\mathbb{A}^n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022167 | MATHDB-383828

Hilbert schemes of points (problem 6)

摘录 / EXCERPT: Does there exist a nonrational component of ${\rm Hilb}^d(\mathbb{A}^n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022168 | MATHDB-383829

If a component of the Hilbert scheme contains a smooth Borel fixed point, does the component have to be rational?

摘录 / EXCERPT: If a component of the Hilbert scheme contains a smooth Borel fixed point, does the component have to be rational?

open_problem | open | MathDB 页面 | 原文线索

V2-022169 | MATHDB-383830

Hilbert schemes of points (problem 8)

摘录 / EXCERPT: Can ${\rm Hilb}^d(C)$ have a component of dimension less than d?

open_problem | open | MathDB 页面 | 原文线索

V2-022170 | MATHDB-383834

Does the set of monomial ideals contained in a component of ... determine that component?

摘录 / EXCERPT: Does the set of monomial ideals contained in a component of ${\rm Hilb}^d(\mathbb{A}^n)$ determine that component? (Or more generally for ${\rm Hilb}^P(\mathbb{P}^n)$ where the Hil...

open_problem | open | MathDB 页面 | 原文线索

V2-022171 | MATHDB-383835

What can we say about the set ... ?

摘录 / EXCERPT: What can we say about the set Z^e? Can we construct exotic components (i.e. components that only exist in characteristic p) using this stratification? Study the action by the G...

open_problem | open | MathDB 页面 | 原文线索

V2-022172 | MATHDB-383836

Is the Hilbert scheme of local Cohen-Macaulay curves in ... connected?

摘录 / EXCERPT: Is the Hilbert scheme of local Cohen-Macaulay curves in $\mathbb{P}^3$ connected?

open_problem | open | MathDB 页面 | 原文线索

V2-022173 | MATHDB-383837

Is ... always a complete intersection, assuming that the characteristic is 0?

摘录 / EXCERPT: Is C_0 always a complete intersection, assuming that the characteristic is 0?

open_problem | open | MathDB 页面 | 原文线索

V2-022174 | MATHDB-383838

Hilbert schemes of curves (problem 4)

摘录 / EXCERPT: There exists $n_0\in\mathbb{Z}$ such that for any smooth curve $C\subset S$ of degree $n\ge n_0$, let H_C be the irreducible component of the Hilbert scheme containing C, then...

open_problem | open | MathDB 页面 | 原文线索

V2-022175 | MATHDB-383839

Hilbert schemes of curves (problem 5)

摘录 / EXCERPT: Fix d,g,n. Find a good/sharp lower bound for the dimension of components of ${\rm Hilb}_{d,g}(\mathbb{P}^n)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022176 | MATHDB-383840

Hilbert schemes of curves (problem 6)

摘录 / EXCERPT: (1) Let C be of bidegree $(3,7)$ on a nonsingular quadric surface in $\mathbb{P}^3$. Can C be connected to an extremal curve in ${\rm Hilb}_{10,12}(\mathbb{P}^3)$? (2) Given...

open_problem | open | MathDB 页面 | 原文线索

V2-022177 | MATHDB-383841

What is the largest number of points in general position you can make these curves pass through?

摘录 / EXCERPT: What is the largest number of points in general position you can make these curves pass through?

open_problem | open | MathDB 页面 | 原文线索

V2-022178 | MATHDB-383842

Hilbert schemes of curves (problem 8)

摘录 / EXCERPT: What is the best pair (d,g) (in the vague sense that d is as small as possible and g is close to 0 as possible) such that there is a component of...

open_problem | open | MathDB 页面 | 原文线索

V2-022179 | MATHDB-383843

Hartshorne-Rao modules

摘录 / EXCERPT: Let $\rho=(p,q,r,0,0,\dots)$ and define $E_{p,q,r}:=E_\rho$. For which p, q, r is $E_{p,q,r}$ irreducible?

open_problem | open | MathDB 页面 | 原文线索

V2-022180 | MATHDB-383844

Hilbert schemes of curves (problem 10)

摘录 / EXCERPT: Describe the irreducible component of E_ρ.

open_problem | open | MathDB 页面 | 原文线索

V2-022181 | MATHDB-383845

What do properties of the Rao modules imply about ... ?

摘录 / EXCERPT: What do properties of the Rao modules imply about C? For example, if M_C is Gorenstein or annihilated by a linear form, does C have any nice properties?

open_problem | open | MathDB 页面 | 原文线索

V2-022182 | MATHDB-383846

What is the geography of locally Cohen-Macaulay surfaces in ... ?

摘录 / EXCERPT: What is the geography of locally Cohen-Macaulay surfaces in $\mathbb{P}^4$? To be more precise, what numerical invariants (degree, sectional genus, etc.) occur?

open_problem | open | MathDB 页面 | 原文线索

V2-022183 | MATHDB-383847

Is there a multigraded Hilbert scheme with a connected component isomorphic to a fat point?

摘录 / EXCERPT: Is there a multigraded Hilbert scheme with a connected component isomorphic to a fat point?

open_problem | open | MathDB 页面 | 原文线索

V2-022184 | MATHDB-383848

Is every multigraded Hilbert scheme connected if the polynomial ring is ... ?

摘录 / EXCERPT: Is every multigraded Hilbert scheme connected if the polynomial ring is $R=k[x_1,x_2,x_3]$?

open_problem | open | MathDB 页面 | 原文线索

V2-022185 | MATHDB-383849

Does there exist a nondegenerate rigid curve in ... other than the rational normal curve?

摘录 / EXCERPT: Does there exist a nondegenerate rigid curve in $\mathbb{P}^n$ other than the rational normal curve?

open_problem | open | MathDB 页面 | 原文线索

V2-022186 | MATHDB-383850

Are there necessary or sufficient conditions on Borel fixed monomial ideals such that they are the generic initial ideals of local Cohen-Macaulay curves?

摘录 / EXCERPT: Are there necessary or sufficient conditions on Borel fixed monomial ideals such that they are the generic initial ideals of local Cohen-Macaulay curves?

open_problem | open | MathDB 页面 | 原文线索

V2-022187 | MATHDB-383852

Is there a rigid local Artinian algebra besides ... ?

摘录 / EXCERPT: Is there a rigid local Artinian algebra besides k^n?

open_problem | open | MathDB 页面 | 原文线索

V2-022188 | MATHDB-383853

What is the generic point of the component it lies on?

摘录 / EXCERPT: What is the generic point of the component it lies on?

open_problem | open | MathDB 页面 | 原文线索

V2-022189 | MATHDB-383855

The u-invariant problem (problem 2)

摘录 / EXCERPT: Find interesting classes $\mathcal F$ of fields such that for $F \in \mathcal F$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-022190 | MATHDB-383856

The u-invariant problem (problem 3)

摘录 / EXCERPT: When $F = {\\mathbb{R}}$, we have $u(F) = \infty$ but $\tau_F < \infty$. Is there an example where this holds for F not formally real?

open_problem | open | MathDB 页面 | 原文线索

V2-022191 | MATHDB-383857

The u-invariant problem (problem 4)

摘录 / EXCERPT: Can we compute $u_k(F)$? How about $u_3(F)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022192 | MATHDB-383859

The u-invariant problem (problem 6)

摘录 / EXCERPT: If $\nu(F) < \infty$, is it true that $\nu(F') < \nu(F) +1$ for all F'/F finite?

open_problem | open | MathDB 页面 | 原文线索

V2-022193 | MATHDB-383861

The Brauer group (problem 2)

摘录 / EXCERPT: Higher cohomology $H^d(F, \pmb{\mu}_n^{\otimes d})$, where $d = \dim (F)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022194 | MATHDB-383862

The Brauer group (problem 3)

摘录 / EXCERPT: When is $\operatorname{Br}(X) = \operatorname{Br'}(X)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022195 | MATHDB-383864

Characteristic $p>0$ invariants (problem 2)

摘录 / EXCERPT: Realize effective computations of numerical F-invariants.

open_problem | open | MathDB 页面 | 原文线索

V2-022196 | MATHDB-383866

Characteristic $p>0$ invariants (problem 4)

摘录 / EXCERPT: For a graded sequence of ideals I_m of a ring R, does $\lim \limits_{m \to \infty} m \cdot \operatorname{fpt}(I_m) $ exist?

open_problem | open | MathDB 页面 | 原文线索

V2-022197 | MATHDB-383868

Characteristic $p>0$ invariants (problem 6)

摘录 / EXCERPT: Investigate the discreteness and rationality of F-jumping numbers in the non-${\\mathbb{Q}}$-Gorenstein case.

open_problem | open | MathDB 页面 | 原文线索

V2-022198 | MATHDB-383877

Generalize the Hara-Yoshida Theorem on restriction of (Hacon-de Fernex) multiplier ideals to test ideals to the non- ... -Gorenstein case.

摘录 / EXCERPT: Generalize the Hara-Yoshida Theorem on restriction of (Hacon-de Fernex) multiplier ideals to test ideals to the non-${\\mathbb{Q}}$-Gorenstein case. Can this be done in the numeric...

open_problem | open | MathDB 页面 | 原文线索

V2-022199 | MATHDB-383880

Investigate lifting sections (for cohomology) using test ideals.

摘录 / EXCERPT: Investigate lifting sections (for cohomology) using test ideals.

open_problem | open | MathDB 页面 | 原文线索

V2-022200 | MATHDB-383882

Other and Related Problems (problem 6)

摘录 / EXCERPT: Realize effective computations of test ideals.

open_problem | open | MathDB 页面 | 原文线索

V2-022201 | MATHDB-383883

Other and Related Problems (problem 8)

摘录 / EXCERPT: Extend tight closure to mixed characteristic.

open_problem | open | MathDB 页面 | 原文线索

V2-022202 | MATHDB-383886

Do determinantal rings have finite ... -representation type?

摘录 / EXCERPT: Do determinantal rings have finite F-representation type?

open_problem | open | MathDB 页面 | 原文线索

V2-022203 | MATHDB-383888

Other and Related Problems (problem 13)

摘录 / EXCERPT: Suppose that R is a local ring and $f \in R$ is a regular element such that $R/(f)$ is F-injective. Does this imply that R is F-injective?

open_problem | open | MathDB 页面 | 原文线索

V2-022204 | MATHDB-383889

Other and Related Problems (problem 14)

摘录 / EXCERPT: Given a log pair (X,Δ), we know that there exists a finite map $\phi: Y \to X$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022205 | MATHDB-383891

Investigate possible Bertini theorems for ... -singularities.

摘录 / EXCERPT: Investigate possible Bertini theorems for F-singularities.

open_problem | open | MathDB 页面 | 原文线索

V2-022206 | MATHDB-383906

Is there an intrinsic description of the relation between these twisted K3 surfaces?

摘录 / EXCERPT: Is there an intrinsic description of the relation between these twisted K3 surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-022207 | MATHDB-383907

K3 surfaces (problem 3)

摘录 / EXCERPT: Is Y unique? How many such Y are there? If X is defined over $\mathbb{Q}$ and Y is defined over some nontrivial extension of $\mathbb{Q}$, is it possible that α is d...

open_problem | open | MathDB 页面 | 原文线索

V2-022208 | MATHDB-383917

If we replace ... by another regular surface, is the condition of not having phantom subcategories equivalent to rationality?

摘录 / EXCERPT: If we replace $\mathbb{P}^n$ by another regular surface, is the condition of not having phantom subcategories equivalent to rationality? What if we take $K_0(X) \otimes \mathbb{Q}$...

open_problem | open | MathDB 页面 | 原文线索

V2-022209 | MATHDB-383921

Other (problem 5)

摘录 / EXCERPT: Let X be a smooth projective complex threefold and a class $\gamma \in H^2(X,{\\mathbb{Z}}/2)$. Consider the integral Bockstein of its square,...

open_problem | open | MathDB 页面 | 原文线索

V2-022210 | MATHDB-383922

Other (problem 6)

摘录 / EXCERPT: Fix a variety X and a Brauer class α. Is the set of β such that the $D^b(X,\alpha)$ are equivalent finite?

open_problem | open | MathDB 页面 | 原文线索

V2-022211 | MATHDB-383924

Connectedness

摘录 / EXCERPT: Is there a positive-characteristic analog of the connectedness lemma?

open_problem | open | MathDB 页面 | 原文线索

V2-022212 | MATHDB-383925

Terminal and Cohen-Macaulay

摘录 / EXCERPT: Are (log) terminal singularities Cohen-Macaulay? Rational?

open_problem | open | MathDB 页面 | 原文线索

V2-022213 | MATHDB-383926

Normality of plt pairs

摘录 / EXCERPT: Suppose that $(X,D = S+B)$ is a plt pair. Must S be normal?

open_problem | open | MathDB 页面 | 原文线索

V2-022214 | MATHDB-383927

Grauert-Riemenschneider for F-regular varieties

摘录 / EXCERPT: Is there Grauert-Riemenschneider vanishing for F-regular varieties which admit a resolution?

open_problem | open | MathDB 页面 | 原文线索

V2-022215 | MATHDB-383928

Global F-regularity and rational chain connectedness

摘录 / EXCERPT: Does globally F-regular imply rationally chain connected?

open_problem | open | MathDB 页面 | 原文线索

V2-022216 | MATHDB-383929

Global F-regularity and Fano type

摘录 / EXCERPT: If X is of globally F-regular type, does it follow that it is log Fano (in the sense that there exists a boundary divisor Δ with (X,Δ) klt and $-(K_X+\Delta)$ amp...

open_problem | open | MathDB 页面 | 原文线索

V2-022217 | MATHDB-383930

Construction of F-pure centers

摘录 / EXCERPT: Suppose that (X,Δ) is a strongly F-regular pair. Can we find a boundary $D \geq 0$ such that $(X,\Delta+D)$ is F-pure, with $p \nmid \text{index}(K_X+\Delta+D)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022218 | MATHDB-383932

Invariance of plurigenera

摘录 / EXCERPT: What is the status of invariance of plurigenera?

open_problem | open | MathDB 页面 | 原文线索

V2-022219 | MATHDB-383933

Sections and section rings (problem 3)

摘录 / EXCERPT: What if we look at $\text{dim } S^0(mK_X+A)$ instead?

open_problem | open | MathDB 页面 | 原文线索

V2-022220 | MATHDB-383936

Effective Fujita vanishing

摘录 / EXCERPT: Is there an effective Fujita vanishing result in flat families? Suppose that $f: \mathcal X \to S$ is flat, $\mathcal F$ is a coherent sheaf on $\mathcal X$, and $\mathcal L$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-022221 | MATHDB-383939

Pulling back forms to a resolution

摘录 / EXCERPT: Suppose that X is log canonical, and that there exists a log resolution $f: \tilde{X} \to X$ which is an isomorphism over the smooth locus. Let ω be an $(n-1)$-form on X...

open_problem | open | MathDB 页面 | 原文线索

V2-022222 | MATHDB-383940

Universal lower bounds for Seshadri constants

摘录 / EXCERPT: Suppose that X is a smooth variety over an uncountable field k of positive characteristic. Does there exist a constant $c = c(n)$ such that $\epsilon(L,x) \geq c$ for any ample...

open_problem | open | MathDB 页面 | 原文线索

V2-022223 | MATHDB-383942

Singularities and point-counting

摘录 / EXCERPT: Suppose that X is a variety over a finite field. Is there any relation between the singularities of X and the number of points over $\mathbb F_q$? For example, suppose that X...

open_problem | open | MathDB 页面 | 原文线索

V2-022224 | MATHDB-383943

Of the ten possibilities, how many of the solutions can be real?

摘录 / EXCERPT: Of the ten possibilities, how many of the solutions can be real?

open_problem | open | MathDB 页面 | 原文线索

V2-022225 | MATHDB-383945

How do the 24 linear constraints lead a minimal problem?

摘录 / EXCERPT: How do the 24 linear constraints lead a minimal problem? How many complex solutions are there?

open_problem | open | MathDB 页面 | 原文线索

V2-022226 | MATHDB-383956

What is the functor of points of multiview geometry?

摘录 / EXCERPT: What is the functor of points of multiview geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-022227 | MATHDB-383957

What is the functor of points in the calibrated case?

摘录 / EXCERPT: What is the functor of points in the calibrated case?

open_problem | open | MathDB 页面 | 原文线索

V2-022228 | MATHDB-383960

What is the signature curve of a general canonical curve?

摘录 / EXCERPT: What is the signature curve of a general canonical curve?

open_problem | open | MathDB 页面 | 原文线索

V2-022229 | MATHDB-383961

What can we say about the signature of the signature of a curve?

摘录 / EXCERPT: What can we say about the signature of the signature of a curve?

open_problem | open | MathDB 页面 | 原文线索

V2-022230 | MATHDB-383962

Which curves occur as signatures of another curve?

摘录 / EXCERPT: Which curves occur as signatures of another curve?

open_problem | open | MathDB 页面 | 原文线索

V2-022231 | MATHDB-383963

Can we use differential (integral, other...) invariants to construct a moduli space?

摘录 / EXCERPT: Can we use differential (integral, other...) invariants to construct a moduli space?

open_problem | open | MathDB 页面 | 原文线索

V2-022232 | MATHDB-383964

How can we use these signatures for object recognition?

摘录 / EXCERPT: How can we use these signatures for object recognition?

open_problem | open | MathDB 页面 | 原文线索

V2-022233 | MATHDB-383967

Can we find equations of visual event surfaces?

摘录 / EXCERPT: Can we find equations of visual event surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-022234 | MATHDB-383970

Which curves with cusps come from rims of algebraic surfaces?

摘录 / EXCERPT: Which curves with cusps come from rims of algebraic surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-022235 | MATHDB-383972

Is there a Carlsson-Weinshallduality for point and line correspondence?

摘录 / EXCERPT: Is there a Carlsson-Weinshallduality for point and line correspondence?

open_problem | open | MathDB 页面 | 原文线索

V2-022236 | MATHDB-383977

Motivic Homotopy Theory (problem 1)

摘录 / EXCERPT: What is the motivic analogue of an N_{∞}-operad or in other words, what are the analogues of G-commutative ring spectra in motivic homotopy theory?

open_problem | open | MathDB 页面 | 原文线索

V2-022237 | MATHDB-383980

Long Term

摘录 / EXCERPT: Do we have a motivic description akin to adding transfers?

open_problem | open | MathDB 页面 | 原文线索

V2-022238 | MATHDB-383981

Can we construct/prove the above notions in the context of motivic homotopy theory?

摘录 / EXCERPT: Can we construct/prove the above notions in the context of motivic homotopy theory?

open_problem | open | MathDB 页面 | 原文线索

V2-022239 | MATHDB-383982

What more can we say about the tensor triangular geometry of the motivic stable homotopy category?

摘录 / EXCERPT: What more can we say about the tensor triangular geometry of the motivic stable homotopy category? (See for a list of questions.)

open_problem | open | MathDB 页面 | 原文线索

V2-022240 | MATHDB-383984

What is the description of ... using the above model?

摘录 / EXCERPT: What is the description of $Comm_G(Sp^G)$ using the above model?

open_problem | open | MathDB 页面 | 原文线索

V2-022241 | MATHDB-383985

Equivariant Stable Homotopy Theory (problem 2)

摘录 / EXCERPT: If $\underline{R}$ is a commutative Green functor, what is a Witt-vector model for $\pi_0(N^{S^1}_{C_p} H\underline{R})^{C_{p^n}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-022242 | MATHDB-383986

Equivariant Stable Homotopy Theory (problem 3)

摘录 / EXCERPT: Can we describe such a C_3-commutative ring spectrum $MU_{{\\mathbb{R}}}$ so that its underlying Bousfield type is MU and...

open_problem | open | MathDB 页面 | 原文线索

V2-022243 | MATHDB-383987

What should the correct notion of a Real spectrum be?

摘录 / EXCERPT: What should the correct notion of a Real spectrum be? Is it “more" than a genuine C_2-spectrum? (see Sec B12)

open_problem | open | MathDB 页面 | 原文线索

V2-022244 | MATHDB-383989

Equivariant Stable Homotopy Theory (problem 6)

摘录 / EXCERPT: Is a similar statement true equivariantly? That is, is there an equivalence between G-symmetric monoidal categories and the homotopy category of connective G-spectra?

open_problem | open | MathDB 页面 | 原文线索

V2-022245 | MATHDB-383990

Long Term

摘录 / EXCERPT: Can we construct a decompleted version of the Tate spectrum?

open_problem | open | MathDB 页面 | 原文线索

V2-022246 | MATHDB-383991

Can we get a better understanding of the tensor triangular geometry of ... ?

摘录 / EXCERPT: Can we get a better understanding of the tensor triangular geometry of Sp^G? (See Problem 1.6)

open_problem | open | MathDB 页面 | 原文线索

V2-022247 | MATHDB-383992

Can we prove a similar result in the setting of equivariant spectra?

摘录 / EXCERPT: Can we prove a similar result in the setting of equivariant spectra? This would aid in understanding the tensor triangular geometry of Sp^G.

open_problem | open | MathDB 页面 | 原文线索

V2-022248 | MATHDB-383994

Connecting Equivariant Notions and Derived Algebraic Geometry (problem 2)

摘录 / EXCERPT: What are the explicit generators of $Pic(Sp^G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022249 | MATHDB-383995

Connecting Equivariant Notions and Derived Algebraic Geometry (problem 3)

摘录 / EXCERPT: (1) Suppose we are given a linear representation of a finite group G on ${\\mathbb{C}}^n$ and a G-equivariant polynomial function...

open_problem | open | MathDB 页面 | 原文线索

V2-022250 | MATHDB-383996

Long Term

摘录 / EXCERPT: How best should we be making sense of equivariant derived algebraic geometry? (e.g. What is Spec? What site to use? What sheaf? etc.)

open_problem | open | MathDB 页面 | 原文线索

V2-022251 | MATHDB-383997

Extensions to Profinite, Compact Lie Groups, etc. (problem 1)

摘录 / EXCERPT: If G is a profinite absolute Galois group of K, is there a theory of G-spectra$ such that $Sp_{G} \rightarrow Sp^{Mot}_K$ is “rich”?

open_problem | open | MathDB 页面 | 原文线索

V2-022252 | MATHDB-383999

Are there analogues of multiplicative norms or multiplicative transfers for compact Lie groups?

摘录 / EXCERPT: Are there analogues of multiplicative norms or multiplicative transfers for compact Lie groups?

open_problem | open | MathDB 页面 | 原文线索

V2-022253 | MATHDB-384000

Extensions to Profinite, Compact Lie Groups, etc. (problem 4)

摘录 / EXCERPT: For any finite or compact Lie group G, is it also the case that $\pi^G_*(MU)$ is even?

open_problem | open | MathDB 页面 | 原文线索

V2-022254 | MATHDB-384001

Long Term

摘录 / EXCERPT: How should we think about equivariant stable homotopy theory in the context of infinite compact Lie groups?

open_problem | open | MathDB 页面 | 原文线索

V2-022255 | MATHDB-384003

Chromatic Homotopy Theory (problem 1)

摘录 / EXCERPT: Can this construction be extended to commutative genuine G-spectra or G-commutative genuine G-spectra?

open_problem | open | MathDB 页面 | 原文线索

V2-022256 | MATHDB-384006

Chromatic Homotopy Theory (problem 4)

摘录 / EXCERPT: Why should taking the Tate spectrum bring an object down in chromatic height (i.e. why do we see a blue shift)? More generally, we'd like to look at "chromatic primes" and "equivar...

open_problem | open | MathDB 页面 | 原文线索

V2-022257 | MATHDB-384009

Long Term

摘录 / EXCERPT: Can we describe the moduli stack of formal groups in the derived setting?

open_problem | open | MathDB 页面 | 原文线索

V2-022258 | MATHDB-384011

What can we say about modules over ... ?

摘录 / EXCERPT: What can we say about modules over $S[BU_+]$?

open_problem | open | MathDB 页面 | 原文线索

V2-022259 | MATHDB-384012

Effective cones of projective bundles

摘录 / EXCERPT: What is the effective cone of surfaces $\text{Eff}_2(X)$ where X is a $\mathbb{P}^2$-bundle over $\mathbb{P}^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-022260 | MATHDB-384013

Computing higher codimension effective and nef cones in explicit examples (problem 2)

摘录 / EXCERPT: What is the effective cone of surfaces $\text{Eff}^2(X)$ where X is a $\mathbb{P}^2$-bundle over a K3 surface?

open_problem | open | MathDB 页面 | 原文线索

V2-022261 | MATHDB-384014

Computing higher codimension effective and nef cones in explicit examples (problem 3)

摘录 / EXCERPT: If $\mathcal{E}\cong \mathcal{L}_1 \oplus... \oplus \mathcal{L}_r$ is a split vector bundle on a smooth projective surface S, what are the effective cones of k-cycles...

open_problem | open | MathDB 页面 | 原文线索

V2-022262 | MATHDB-384015

Effective cones of hypersurfaces

摘录 / EXCERPT: If $X\subset Gr(m,n)$ is a very general hypersurface, what is the effective cone of k-cycles $\text{Eff}_k(K)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022263 | MATHDB-384017

Computing higher codimension effective and nef cones in explicit examples (problem 6)

摘录 / EXCERPT: If $X\subset Y$ is a very general ample divisor of sufficiently high degree, does the pushforward map $i_*: N_k(X) \rightarrow N_k(Y)$ induce an isomorphism of effective cones for...

open_problem | open | MathDB 页面 | 原文线索

V2-022264 | MATHDB-384019

Computing higher codimension effective and nef cones in explicit examples (problem 8)

摘录 / EXCERPT: What is the effective cone of surfaces $\text{Eff}_2(\overline{\mathcal{M}}_{0,7}^{\mathfrak{S}_7})$?

open_problem | open | MathDB 页面 | 原文线索

V2-022265 | MATHDB-384020

Effective cones of blowups

摘录 / EXCERPT: Let $Z\subset \mathbb{P}^n$ be a smooth subvariety with $1\le \text{dim}Z \le n-2$. What is the effective cone of k-cycles $\text{Eff}_k(X)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022266 | MATHDB-384021

Mobility and mobility counts

摘录 / EXCERPT: Let $[\ell]\in N_1(\mathbb{P}^3)_\mathbb{Z}$ be the class of a line. What is $\text{mob}([\alpha])$?

open_problem | open | MathDB 页面 | 原文线索

V2-022267 | MATHDB-384022

Measuring positivity of cycles (problem 2)

摘录 / EXCERPT: What is $K(\alpha)$ for $\alpha \in N_k(\text{Gr}(k,n))$ a Schubert cycle?

open_problem | open | MathDB 页面 | 原文线索

V2-022268 | MATHDB-384024

Other measures of positivity

摘录 / EXCERPT: Is α in the interior of the effective cone?

open_problem | open | MathDB 页面 | 原文线索

V2-022269 | MATHDB-384025

Obstructions to rationality (problem 1)

摘录 / EXCERPT: Let X be a separably rationally connected variety. Does $H^{i}(X,\mathcal{O}_{X})=0$ for $i>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-022270 | MATHDB-384026

Look for new obstruction to diagonal decompositions / rationality.

摘录 / EXCERPT: Look for new obstruction to diagonal decompositions / rationality.

open_problem | open | MathDB 页面 | 原文线索

V2-022271 | MATHDB-384034

4th day

摘录 / EXCERPT: Do there exist a cubic 4-fold defined over number fields which is conjecured to be irrational but many reductions are conjectured to be ratioal?

open_problem | open | MathDB 页面 | 原文线索

V2-022272 | MATHDB-384035

Can irrational varieties over ... fields specialize to rational varieties in char.

摘录 / EXCERPT: Can irrational varieties over $\#$ fields specialize to rational varieties in char. $p>0$? (Colliot-th'elene-Ojanguren's examples)

open_problem | open | MathDB 页面 | 原文线索

V2-022273 | MATHDB-384037

Rational subvarieties of the moduli space of rational curves and of abelian varieties (problem 2)

摘录 / EXCERPT: Find rational curves in $\mathcal{A}_{g}$ in char. p, that do not lift to char. 0.

open_problem | open | MathDB 页面 | 原文线索

V2-022274 | MATHDB-384038

Are complete rational curves in ... dense in char.

摘录 / EXCERPT: Are complete rational curves in $\mathcal{A}_{g}$ dense in char. p?

open_problem | open | MathDB 页面 | 原文线索

V2-022275 | MATHDB-384039

The moduli space of rational curves of hypersurfaces (problem 1)

摘录 / EXCERPT: In char. p if $n\geq d+2$, are the moduli spaces of rational curves on a general hypersurface of degree d in $\mathbb{P}^{n}$ irreducible of the expected dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-022276 | MATHDB-384040

The moduli space of rational curves of hypersurfaces (problem 2)

摘录 / EXCERPT: What is the dimension of rational curves in hypersurfaces of degree d in $\mathbb{P}^{n}$ in char. p? Does char. 0 proof go through?

open_problem | open | MathDB 页面 | 原文线索

V2-022277 | MATHDB-384041

New applications of Kollar's and Totaro's techniques (problem 1)

摘录 / EXCERPT: Is the very general hyperusrfae of degree $d\geq 2 \lceil(n+3)/3\rceil$ not rational / stably rational / ruled in char $p>0$ for $p>>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-022278 | MATHDB-384044

New applications of Kollar's and Totaro's techniques (problem 4)

摘录 / EXCERPT: Look for new applications of Koll'ar / Totaro's techique.

open_problem | open | MathDB 页面 | 原文线索

V2-022279 | MATHDB-384046

Unirational parameterizations of cubic hypersurfaces (problem 1)

摘录 / EXCERPT: Do cubic 3-folds have odd degree unirational prametrizations in char p?

open_problem | open | MathDB 页面 | 原文线索

V2-022280 | MATHDB-384047

Fano varieties, characterization of projective spaces (problem 1)

摘录 / EXCERPT: Does every Fano variety in char. p contain a rational, unirational and rationally-connected surface?

open_problem | open | MathDB 页面 | 原文线索

V2-022281 | MATHDB-384048

Is every smooth Fano hypersurface / variety separably rationally connected?

摘录 / EXCERPT: Is every smooth Fano hypersurface / variety separably rationally connected?

open_problem | open | MathDB 页面 | 原文线索

V2-022282 | MATHDB-384049

Does every smooth projective Fano lift to char.

摘录 / EXCERPT: Does every smooth projective Fano lift to char. 0? Are they all simply-connected? Does boundedness hold in char. p?

open_problem | open | MathDB 页面 | 原文线索

V2-022283 | MATHDB-384050

What extra hypothesis need to be added to make characterizations of ... in terms of ampleness of a subbundle of the tangent bundle hold in char.

摘录 / EXCERPT: What extra hypothesis need to be added to make characterizations of $\mathbb{P}^{n}$ in terms of ampleness of a subbundle of the tangent bundle hold in char. p?

open_problem | open | MathDB 页面 | 原文线索

V2-022284 | MATHDB-384051

Fano varieties, characterization of projective spaces (problem 5)

摘录 / EXCERPT: If there exist $\phi\colon E\longrightarrow T_{X}$ with E ample, X smooth projective irreducible, does it follow $X\cong\mathbb{P}^{n}$?

open_problem | open | MathDB 页面 | 原文线索

V2-022285 | MATHDB-384052

Can we find a separably rationally connected Fano hypersurface of every degree and dimension over ... ?

摘录 / EXCERPT: Can we find a separably rationally connected Fano hypersurface of every degree and dimension over $\mathbb{F}_{p}$?

open_problem | open | MathDB 页面 | 原文线索

V2-022286 | MATHDB-384056

Is there a supersingular hypersurface of every degree, dimension and char.

摘录 / EXCERPT: Is there a supersingular hypersurface of every degree, dimension and char. p?

open_problem | open | MathDB 页面 | 原文线索

V2-022287 | MATHDB-384059

Can a surface with ... contain a rational curve in char.

摘录 / EXCERPT: Can a surface with $c_{1}^{2}=3c_{2}$ contain a rational curve in char. p?

open_problem | open | MathDB 页面 | 原文线索

V2-022288 | MATHDB-384062

Other problems (problem 10)

摘录 / EXCERPT: For every p, d with $d\geq n+1$, can we find hypersurface over $\mathbb{F}_{p}$ of degree d in $\mathbb{P}^{n}$ such that either $\# X \not\equiv 1 \text{mod} p$ or X is not...

open_problem | open | MathDB 页面 | 原文线索

V2-022289 | MATHDB-384065

Other problems (problem 13)

摘录 / EXCERPT: Can we get alterations of coprime degrees?

open_problem | open | MathDB 页面 | 原文线索

V2-022290 | MATHDB-384067

Other problems (problem 15)

摘录 / EXCERPT: If X is separably rationally connected, there exists $N(\text{dim} X, \text{deg} X, m)$ such that, for any $p_{1},\cdots, p_{m}\in X(\mathbb{F}_{p})$, there exists a rational cur...

open_problem | open | MathDB 页面 | 原文线索

V2-022291 | MATHDB-384069

K-stability and KSBA stability (problem 1)

摘录 / EXCERPT: Is there a modification of K-stability/KSBA stability such that curves with singularities “worse” than nodes (e.g. cusps, tacnodes) become stable?

open_problem | open | MathDB 页面 | 原文线索

V2-022292 | MATHDB-384070

K-stability and KSBA stability (problem 2)

摘录 / EXCERPT: (Coskun) If (X,L) is K-stable, what can one say about the ϵ for which the perturbation $(X, L-\epsilon D)$ remains K-stable? (Dervan) If X is a smooth projective v...

open_problem | open | MathDB 页面 | 原文线索

V2-022293 | MATHDB-384071

K-stability and KSBA stability (problem 3)

摘录 / EXCERPT: What is twisted K-stability/KSBA stability for the universal family $\pi:\mathcal{C}=\overline{\mathcal{M}}_{g,1}\to\overline{\mathcal{M}}_g$, $\mathcal{L}=\omega_\pi$,...

open_problem | open | MathDB 页面 | 原文线索

V2-022294 | MATHDB-384072

What are semistable limits of non-reduced quintic surfaces?

摘录 / EXCERPT: What are semistable limits of non-reduced quintic surfaces? For instance, $\lim_{t\to 0} (tF_5+F_3\cdot F_1^2)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022295 | MATHDB-384073

K-stability and KSBA stability (problem 5)

摘录 / EXCERPT: When is the set of K-semistable (singular) Fano varieties of dimension n and volume $\geq c (>0)$ bounded? This is known for surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-022296 | MATHDB-384074

Bridgeland stability (problem 1)

摘录 / EXCERPT: (Macri) Construct Bridgeland stability conditions on degenerate varieties, for example, the hypersurfaces with equations $x_0x_1\dots x_n=0$ in $\mathbb{P}^n$, especially for...

open_problem | open | MathDB 页面 | 原文线索

V2-022297 | MATHDB-384075

Bridgeland stability (problem 2)

摘录 / EXCERPT: Let X be a smooth projective surface. Let $L\in\mathrm{Pic}(X)$ be a globally generated line bundle. Consider the Lazarsfeld-Mukai vector bundle...

open_problem | open | MathDB 页面 | 原文线索

V2-022298 | MATHDB-384076

Bridgeland stability (problem 3)

摘录 / EXCERPT: In the same setting as problem 2.2, can one prove property (N_p) for K_X+nL in terms of Bridgeland stability? For $X=C$ a curve, and $\deg L\geq 2g+p+1$, Green in 1984 proved t...

open_problem | open | MathDB 页面 | 原文线索

V2-022299 | MATHDB-384077

Bridgeland stability (problem 4)

摘录 / EXCERPT: Let $C \subset \mathbb{P}^3$ be a smooth curve. Is there a characterization for walls in Bridgeland or tilt stability for its ideal sheaf $\mathcal{I}_{C/\mathbb{P}^3}$? It is know...

open_problem | open | MathDB 页面 | 原文线索

V2-022300 | MATHDB-384078

Bridgeland stability (problem 5)

摘录 / EXCERPT: Given one of the known exceptional collections of stable objects on $\overline{\mathcal{M}}_{0,n}$ (e.g. or forthcoming work of Castravet-Tevelev), can one use the corresponding Br...

open_problem | open | MathDB 页面 | 原文线索

V2-022301 | MATHDB-384079

Do moduli spaces of Bridgeland stable objects have a projective/good coarse moduli space?

摘录 / EXCERPT: Do moduli spaces of Bridgeland stable objects have a projective/good coarse moduli space? The Bayer-Macri divisor is nef. When is it ample? Try the blow-up of $\mathbb{P}^2$ at two...

open_problem | open | MathDB 页面 | 原文线索

V2-022302 | MATHDB-384080

Bridgeland stability (problem 7)

摘录 / EXCERPT: Let X be a smooth surface with $\rho(X)\geq 2$. Vary the polarzation not only along a line but along a plane and describe the structure of walls for tilt stability. A first case...

open_problem | open | MathDB 页面 | 原文线索

V2-022303 | MATHDB-384081

GIT stability (problem 1)

摘录 / EXCERPT: begin{enumerate} item (Question by Wang) Does the moduli space of K-stable del Pezzo surfaces have an asymptotic GIT description? item (Conjecture by Laza-Odaka) Asymptotic GIT for...

open_problem | open | MathDB 页面 | 原文线索

V2-022304 | MATHDB-384082

Can one exploit the convexity properties of the estimates in for asymptotic stability for nodal curves to get more effective and sharper stability estimates, e.g.

摘录 / EXCERPT: Can one exploit the convexity properties of the estimates in for asymptotic stability for nodal curves to get more effective and sharper stability estimates, e.g. 5-canonical cur...

open_problem | open | MathDB 页面 | 原文线索

V2-022305 | MATHDB-384083

Describe the quotients ... for suitable linearizations, characterizing the stable points.

摘录 / EXCERPT: Describe the quotients $\overline{\mathcal{M}}_g(\mathbb{P}^1,d)\mathbin{/\mkern-6mu/}PGL(2,\mathbb{C})$ for suitable linearizations, characterizing the stable points. Do any of t...

open_problem | open | MathDB 页面 | 原文线索

V2-022306 | MATHDB-384084

GIT stability (problem 4)

摘录 / EXCERPT: Let X be a smooth subvariety of $\mathbb{P}^n$ of dimension d. Let λ be a one-parameter subgroup of $SL(n+1,\mathbb{C})$. Does the following inequality ... hold? Here...

open_problem | open | MathDB 页面 | 原文线索

V2-022307 | MATHDB-384085

Clemens conjecture

摘录 / EXCERPT: A smooth quintic threefold has only finitely many rational curves of each degree.

open_problem | open | MathDB 页面 | 原文线索

V2-022308 | MATHDB-384087

Moduli space of metric graphs

摘录 / EXCERPT: Does tropical geometry give us a reason why the moduli space of metric graphs should be a formal space?

open_problem | open | MathDB 页面 | 原文线索

V2-022309 | MATHDB-384088

Realization problem for tropical curves

摘录 / EXCERPT: Can tropical scheme structure distinguish between realizable and unrealizable tropical curves (in $\mathbb{R}^3$, e.g.).

open_problem | open | MathDB 页面 | 原文线索

V2-022310 | MATHDB-384089

Hilbert schemes of points

摘录 / EXCERPT: Construct and compare $\text{Trop}(\text{Hilb}^n(X))$ and $\text{Hilb}^n(\text{Trop}(X))$.

open_problem | open | MathDB 页面 | 原文线索

V2-022311 | MATHDB-384090

Applications and long-term goals (problem 7)

摘录 / EXCERPT: Given 2 monomial ideals, when is there a connected emph{smooth} curve $(\mathbb{P}^1 \hookrightarrow \text{Hilb}^n(\mathbb{C}^2))$ joining them?

open_problem | open | MathDB 页面 | 原文线索

V2-022312 | MATHDB-384091

Applications and long-term goals (problem 8)

摘录 / EXCERPT: $\text{Hilb}^n(\mathbb{C}^3)$ is generically reduced.

open_problem | open | MathDB 页面 | 原文线索

V2-022313 | MATHDB-384092

More generally, when are Hilbert schemes of points on threefolds reducible?

摘录 / EXCERPT: More generally, when are Hilbert schemes of points on threefolds reducible?

open_problem | open | MathDB 页面 | 原文线索

V2-022314 | MATHDB-384093

Brill-Noether theory

摘录 / EXCERPT: Develop a tropical version of limit linear series.

open_problem | open | MathDB 页面 | 原文线索

V2-022315 | MATHDB-384094

Tropical Euler characteristic

摘录 / EXCERPT: For a tropical scheme Y, define $\chi(\mathcal{O}_Y)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022316 | MATHDB-384095

Abelian varieties

摘录 / EXCERPT: What is a tropical abelian variety? How do we tropicalize subschemes of abelian varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022317 | MATHDB-384096

Extend the results of Giansiracusa-Giansiracusa's first paper to maximally degenerate abelian varieties.

摘录 / EXCERPT: Extend the results of Giansiracusa-Giansiracusa's first paper to maximally degenerate abelian varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-022318 | MATHDB-384097

Combinatorics

摘录 / EXCERPT: Which questions in combinatorics are amenable to tropical scheme theory?

open_problem | open | MathDB 页面 | 原文线索

V2-022319 | MATHDB-384098

Use tropical scheme theory to approach MacPherson's conjecture that the “space of oriented matroids” is homotopy equivalent to the real oriented Grassmannian.

摘录 / EXCERPT: Use tropical scheme theory to approach MacPherson's conjecture that the “space of oriented matroids” is homotopy equivalent to the real oriented Grassmannian.

open_problem | open | MathDB 页面 | 原文线索

V2-022320 | MATHDB-384099

Cohomology and intersection theory

摘录 / EXCERPT: Develop a theory of cohomology for tropical schemes.

open_problem | open | MathDB 页面 | 原文线索

V2-022321 | MATHDB-384100

Does tropical cohomology satisfy Hard Lefschetz?

摘录 / EXCERPT: Does tropical cohomology satisfy Hard Lefschetz? Can we use it to approach problems in combinatorial algebraic geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-022322 | MATHDB-384101

Applications and long-term goals (problem 18)

摘录 / EXCERPT: Develop scheme-theoretic tropical intersection theory.

open_problem | open | MathDB 页面 | 原文线索

V2-022323 | MATHDB-384102

Prove a tropical Riemann-Roch theorem by cohomological methods.

摘录 / EXCERPT: Prove a tropical Riemann-Roch theorem by cohomological methods.

open_problem | open | MathDB 页面 | 原文线索

V2-022324 | MATHDB-384103

Commutative algebra aspects (problem 1)

摘录 / EXCERPT: What is a good definition of textbf{prime} tropical ideals?

open_problem | open | MathDB 页面 | 原文线索

V2-022325 | MATHDB-384104

Produce examples of tropical ideals whose varieties are non-realizable.

摘录 / EXCERPT: Produce examples of tropical ideals whose varieties are non-realizable.

open_problem | open | MathDB 页面 | 原文线索

V2-022326 | MATHDB-384105

Commutative algebra aspects (problem 3)

摘录 / EXCERPT: How do we intersect tropical ideals? How do we add tropical ideals?

open_problem | open | MathDB 页面 | 原文线索

V2-022327 | MATHDB-384106

How can we describe a tropical ideal with a finite amount of information?

摘录 / EXCERPT: How can we describe a tropical ideal with a finite amount of information?

open_problem | open | MathDB 页面 | 原文线索

V2-022328 | MATHDB-384107

Is there a primary decomposition for tropical ideals?

摘录 / EXCERPT: Is there a primary decomposition for tropical ideals?

open_problem | open | MathDB 页面 | 原文线索

V2-022329 | MATHDB-384108

Commutative algebra aspects (problem 6)

摘录 / EXCERPT: Develop descent theory for modules over a $\mathbb{T}$-algebra A. This will likely require finding an appropriate subcategory of the category of A-modules.

open_problem | open | MathDB 页面 | 原文线索

V2-022330 | MATHDB-384109

Commutative algebra aspects (problem 7)

摘录 / EXCERPT: Are there syzygies in tropical geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-022331 | MATHDB-384110

How does the matroidal condition defining tropical ideals interact with Berkovich skeleta?

摘录 / EXCERPT: How does the matroidal condition defining tropical ideals interact with Berkovich skeleta?

open_problem | open | MathDB 页面 | 原文线索

V2-022332 | MATHDB-384111

End of workshop problems

摘录 / EXCERPT: Generate (more) examples of interesting tropical ideals, especially non-realizable tropical ideals.

open_problem | open | MathDB 页面 | 原文线索

V2-022333 | MATHDB-384112

Find a new example of a tropical ideal which is prime in the naive sense.

摘录 / EXCERPT: Find a new example of a tropical ideal which is prime in the naive sense.

open_problem | open | MathDB 页面 | 原文线索

V2-022334 | MATHDB-384113

Commutative algebra aspects (problem 11)

摘录 / EXCERPT: For a given tropical ideal I, describe the set of tropical ideals which agree with I in low degree.

open_problem | open | MathDB 页面 | 原文线索

V2-022335 | MATHDB-384114

Commutative algebra aspects (problem 12)

摘录 / EXCERPT: Develop a notion of prime ideals such that under the reduction map $\mathbb{T}_+[x_1, \ldots, x_n] \to \mathbb{B}[x_1, \ldots, x_n]$ (here...

open_problem | open | MathDB 页面 | 原文线索

V2-022336 | MATHDB-384115

What is the semiring analogue of a DVR (or valuation ring more generally)?

摘录 / EXCERPT: What is the semiring analogue of a DVR (or valuation ring more generally)?

open_problem | open | MathDB 页面 | 原文线索

V2-022337 | MATHDB-384116

Geometric aspects (problem 1)

摘录 / EXCERPT: What is a morphism of tropical schemes? What is the right definition of flat, proper, smooth, étale?

open_problem | open | MathDB 页面 | 原文线索

V2-022338 | MATHDB-384117

Geometric aspects (problem 2)

摘录 / EXCERPT: What is a smooth subscheme of a tropical scheme? What is a smooth point?

open_problem | open | MathDB 页面 | 原文线索

V2-022339 | MATHDB-384118

What is the multiplicity of a point on a tropical scheme?

摘录 / EXCERPT: What is the multiplicity of a point on a tropical scheme?

open_problem | open | MathDB 页面 | 原文线索

V2-022340 | MATHDB-384119

What are differential forms on tropical schemes?

摘录 / EXCERPT: What are differential forms on tropical schemes?

open_problem | open | MathDB 页面 | 原文线索

V2-022341 | MATHDB-384120

Define the tangent space of a tropical scheme.

摘录 / EXCERPT: Define the tangent space of a tropical scheme.

open_problem | open | MathDB 页面 | 原文线索

V2-022342 | MATHDB-384121

Geometric aspects (problem 6)

摘录 / EXCERPT: Does the balancing condition hold for arbitrary (i.e. non-realizable) tropical schemes?

open_problem | open | MathDB 页面 | 原文线索

V2-022343 | MATHDB-384122

Is there projective duality for tropical varieties?

摘录 / EXCERPT: Is there projective duality for tropical varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022344 | MATHDB-384123

Are there Chow forms in tropical geometry?

摘录 / EXCERPT: Are there Chow forms in tropical geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-022345 | MATHDB-384124

How can the tropical scheme structure change under tropical modifications?

摘录 / EXCERPT: How can the tropical scheme structure change under tropical modifications?

open_problem | open | MathDB 页面 | 原文线索

V2-022346 | MATHDB-384125

Can we see faithful tropicalizations on the level of schemes?

摘录 / EXCERPT: Can we see faithful tropicalizations on the level of schemes?

open_problem | open | MathDB 页面 | 原文线索

V2-022347 | MATHDB-384126

What are Deligne-Mumford stacks over tropical schemes?

摘录 / EXCERPT: What are Deligne-Mumford stacks over tropical schemes?

open_problem | open | MathDB 页面 | 原文线索

V2-022348 | MATHDB-384127

Geometric aspects (problem 12)

摘录 / EXCERPT: Prove that (a) implies (b). If this fails to hold in general, does it hold for some subclass of $\text{CPA}(\Delta)$-modules?

open_problem | open | MathDB 页面 | 原文线索

V2-022349 | MATHDB-384128

Tropicalization in other settings

摘录 / EXCERPT: Endow the skeleton of a Berkovich curve with a tropical scheme structure.

open_problem | open | MathDB 页面 | 原文线索

V2-022350 | MATHDB-384129

Develop a way to tropicalize subschemes of toric vector bundles.

摘录 / EXCERPT: Develop a way to tropicalize subschemes of toric vector bundles.

open_problem | open | MathDB 页面 | 原文线索

V2-022351 | MATHDB-384130

Describe the relationship between tropicalization and perfect obstruction theories, using the tropicalization of the moduli space of stable maps as a guiding example.

摘录 / EXCERPT: Describe the relationship between tropicalization and perfect obstruction theories, using the tropicalization of the moduli space of stable maps as a guiding example.

open_problem | open | MathDB 页面 | 原文线索

V2-022352 | MATHDB-384132

Geometric aspects (problem 17)

摘录 / EXCERPT: Given an n-marked genus g tropical curve Γ, what is the smallest dimension of a toric variety X such that a curve $C \hookrightarrow X$ gives a faithful tropicalizatio...

open_problem | open | MathDB 页面 | 原文线索

V2-022353 | MATHDB-384133

Find a modular interpretation of the tropical Dressian and tropical Grassmannian.

摘录 / EXCERPT: Find a modular interpretation of the tropical Dressian and tropical Grassmannian.

open_problem | open | MathDB 页面 | 原文线索

V2-022354 | MATHDB-384134

Geometric aspects (problem 19)

摘录 / EXCERPT: Compute more examples of tropical Hilbert schemes (e.g. $\text{Hilb}^3(\mathbb{A}^1), \text{Hilb}^2(\mathbb{A}^2)$, ldots). Describe the Hilbert-Chow morphism.

open_problem | open | MathDB 页面 | 原文线索

V2-022355 | MATHDB-384135

Find a way to recover/detect properties (reducedness, smoothness, irreducibility, number of components) of a scheme from its tropicalization.

摘录 / EXCERPT: Find a way to recover/detect properties (reducedness, smoothness, irreducibility, number of components) of a scheme from its tropicalization.

open_problem | open | MathDB 页面 | 原文线索

V2-022356 | MATHDB-384136

What is the étale topology for tropical schemes?

摘录 / EXCERPT: What is the étale topology for tropical schemes? Étale maps?

open_problem | open | MathDB 页面 | 原文线索

V2-022357 | MATHDB-384137

Develop tropical scheme theory over the tropical hyperfield.

摘录 / EXCERPT: Develop tropical scheme theory over the tropical hyperfield. Is this a better setting for thinking about prime ideals? Does the hyperfield topology help with localization?

open_problem | open | MathDB 页面 | 原文线索

V2-022358 | MATHDB-384138

Geometric aspects (problem 23)

摘录 / EXCERPT: Give a geometric interpretation of Jo'o-Mincheva's tropical Krull dimension.

open_problem | open | MathDB 页面 | 原文线索

V2-022359 | MATHDB-384139

Construct an analog of generalised parabolic Hitchin pairs, the corresponding moduli spaces, and the Hitchin maps, for other complex algebraic groups ... .

摘录 / EXCERPT: Construct an analog of generalised parabolic Hitchin pairs, the corresponding moduli spaces, and the Hitchin maps, for other complex algebraic groups $G_{\mathbb{C}}$. (I.e. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-022360 | MATHDB-384140

Generalised parabolic Higgs bundles (problem 2)

摘录 / EXCERPT: Compare generalised parabolic Higgs bundles on nodal curves emph{a la} Bhosle to the work of (a) Bezrukovnikov and Kapranov (microlocal sheaves) and (b) Crawley-Boevey (or). Also,...

open_problem | open | MathDB 页面 | 原文线索

V2-022361 | MATHDB-384141

Give a geometric description of the holonomy group scheme.

摘录 / EXCERPT: Give a geometric description of the holonomy group scheme.

open_problem | open | MathDB 页面 | 原文线索

V2-022362 | MATHDB-384142

Establish Morse theoretic approach to the study of the topology of Hitchin moduli spaces in high rank.

摘录 / EXCERPT: Establish Morse theoretic approach to the study of the topology of Hitchin moduli spaces in high rank. In particular, find a geometric reason for the independence of the Poincar'e...

open_problem | open | MathDB 页面 | 原文线索

V2-022363 | MATHDB-384143

Formulate a non-abelian Hodge theorem and mirror symmetry for the moduli spaces of ... integrable Hitchin systems.

摘录 / EXCERPT: Formulate a non-abelian Hodge theorem and mirror symmetry for the moduli spaces of $2k$ integrable Hitchin systems.

open_problem | open | MathDB 页面 | 原文线索

V2-022364 | MATHDB-384144

2k integrable Hitchin equations (problem 2)

摘录 / EXCERPT: Classify exact solutions to these $2k$ integrable Hitchin equations, and interpret them in terms of 4d $\mathcal{N}=1$ SUSY theories using marked points on the Riemann surface C.

open_problem | open | MathDB 页面 | 原文线索

V2-022365 | MATHDB-384145

Give a geometric realisation of a meromorphic Hitchin system of type A or D using a non-compact Calabi-Yau.

摘录 / EXCERPT: Give a geometric realisation of a meromorphic Hitchin system of type A or D using a non-compact Calabi-Yau.

open_problem | open | MathDB 页面 | 原文线索

V2-022366 | MATHDB-384146

Extend this realisation over the elliptic part of the singular locus.

摘录 / EXCERPT: Extend this realisation over the elliptic part of the singular locus.

open_problem | open | MathDB 页面 | 原文线索

V2-022367 | MATHDB-384147

Meromorphic Hitchin systems and Calabi-Yau integrable systems (problem 3)

摘录 / EXCERPT: For local $\mathbb{P}^2$/quintic/others we have the Donagi-Markman system (for local $\mathbb{P}^2$ this is on the mirror curve). Can it be realised as a Hitchin integrable system?

open_problem | open | MathDB 页面 | 原文线索

V2-022368 | MATHDB-384148

Compute the Fukaya category of the total space of the Hitchin fibration, and compare with representations of the double affine Hecke algebra (DAHA).

摘录 / EXCERPT: Compute the Fukaya category of the total space of the Hitchin fibration, and compare with representations of the double affine Hecke algebra (DAHA).

open_problem | open | MathDB 页面 | 原文线索

V2-022369 | MATHDB-384149

Characterize situation where the central fibre of the Hitchin fibration has known, smooth components, intersecting transversely.

摘录 / EXCERPT: Characterize situation where the central fibre of the Hitchin fibration has known, smooth components, intersecting transversely.

open_problem | open | MathDB 页面 | 原文线索

V2-022370 | MATHDB-384150

Four-dimensional $\mathcal{N}=3$ theories (problem 1)

摘录 / EXCERPT: Obtain mathematical constructions of $d=4$ $\mathcal{N}=3$ theories.

open_problem | open | MathDB 页面 | 原文线索

V2-022371 | MATHDB-384151

Construct explicit examples of how gauge groups and representations appear from elliptic fibrations.

摘录 / EXCERPT: Construct explicit examples of how gauge groups and representations appear from elliptic fibrations.

open_problem | open | MathDB 页面 | 原文线索

V2-022372 | MATHDB-384152

Is there an intrinsic characterisation of the gauge group arising from the singularities of an elliptic fibration that doesn't require passing through Dynkin diagrams?

摘录 / EXCERPT: Is there an intrinsic characterisation of the gauge group arising from the singularities of an elliptic fibration that doesn't require passing through Dynkin diagrams?

open_problem | open | MathDB 页面 | 原文线索

V2-022373 | MATHDB-384153

What singularities can elliptic fibrations have in higher codimension?

摘录 / EXCERPT: What singularities can elliptic fibrations have in higher codimension?

open_problem | open | MathDB 页面 | 原文线索

V2-022374 | MATHDB-384154

Intermediate Jacobian and Hitchin System.

摘录 / EXCERPT: Give a geometric description of how the intermediate Jacobian systems of and are related to the Hitchin system.

open_problem | open | MathDB 页面 | 原文线索

V2-022375 | MATHDB-384155

Calabi-Yau integrable systems (problem 2)

摘录 / EXCERPT: Given a non-compact Calabi-Yau integrable system, recognize whether it is a Nakajima quiver variety.

open_problem | open | MathDB 页面 | 原文线索

V2-022376 | MATHDB-384156

Simplify the fibres of the Hodge bundle.

摘录 / EXCERPT: Seek notion of genericity for complex structures simplifying the stratification of the singular locus of the Hitchin fibration (for $SL_2(\mathbb{C})$).

open_problem | open | MathDB 页面 | 原文线索

V2-022377 | MATHDB-384157

Relation to Hilbert schemes of points.

摘录 / EXCERPT: Better understand the relationship between the moduli of Higgs bundles on C and the Hilbert scheme of points $\text{Hilb}^{\text{pt}}(T^*C)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022378 | MATHDB-384158

Deformations from Hochschild homology.

摘录 / EXCERPT: Give a geometric realisation of objects given by deforming a Hitchin pair by a non-zero class in $HH_0(C)$ (c.f. Keller).

open_problem | open | MathDB 页面 | 原文线索

V2-022379 | MATHDB-384159

Branes in the moduli of Higgs bundles.

摘录 / EXCERPT: Review techniques for constructing Lagrangians in hyperkahler spaces, see which can be applied to the moduli space of Higgs bundles, and explicitly translate/apply those techniques...

open_problem | open | MathDB 页面 | 原文线索

V2-022380 | MATHDB-384160

tt^*-equations and Higgs bundles.

摘录 / EXCERPT: Make explicit the relationship between the tt^*-equations and Higgs bundles in a specific example.

open_problem | open | MathDB 页面 | 原文线索

V2-022381 | MATHDB-384164

To what extend can computers compute HHH ... directly from the definition?

摘录 / EXCERPT: To what extend can computers compute HHH$^\text{geom}$ directly from the definition?

open_problem | open | MathDB 页面 | 原文线索

V2-022382 | MATHDB-384168

Parity (problem 2)

摘录 / EXCERPT: Are algebraic braids parity? If so find a (recursive) formula for HHH of an algebraic braid.

open_problem | open | MathDB 页面 | 原文线索

V2-022383 | MATHDB-384178

Is there an equivalence of categories between DAHA representations and a suitable extension of the Fukaya category of the character variety of a single punctured torus?

摘录 / EXCERPT: Is there an equivalence of categories between DAHA representations and a suitable extension of the Fukaya category of the character variety of a single punctured torus?

open_problem | open | MathDB 页面 | 原文线索

V2-022384 | MATHDB-384181

How does one describe the Procesi bundle in the other settings, like K ... ?

摘录 / EXCERPT: How does one describe the Procesi bundle in the other settings, like K$^b(\mathbb{S}\text{Bim}_{\text{fin}})$?

open_problem | open | MathDB 页面 | 原文线索

V2-022385 | MATHDB-384184

Does unirationality specialize in smooth projective families?

摘录 / EXCERPT: Does unirationality specialize in smooth projective families?

open_problem | open | MathDB 页面 | 原文线索

V2-022386 | MATHDB-384195

Which birational transformations preserve the following property?

摘录 / EXCERPT: Which birational transformations preserve the following property? ... has a semiorthogonal decomposition with $\mathscr A_i \hookrightarrow D(Y_i)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022387 | MATHDB-384196

Is there a rationally connected irrational variety ... with property ()?

摘录 / EXCERPT: Is there a rationally connected irrational variety X with property ()? Recall Property () from a previous question was defined as follows. ... has a semiorthogonal decomposition wi...

open_problem | open | MathDB 页面 | 原文线索

V2-022388 | MATHDB-384203

Show that smooth quartic double 5-folds are irrational.

摘录 / EXCERPT: Show that smooth quartic double 5-folds are irrational.

open_problem | open | MathDB 页面 | 原文线索

V2-022389 | MATHDB-384208

K-stability of smooth Fano 3-folds (problem 1)

摘录 / EXCERPT: (C. Xu) Do we expect all Picard number $\rho=1$ ones to be K-semistable?

open_problem | open | MathDB 页面 | 原文线索

V2-022390 | MATHDB-384209

K-stability of smooth Fano 3-folds (problem 2)

摘录 / EXCERPT: (Z. Patakfalvi) Find out the K-polystable moduli compactifications. (C. Xu) Especially $X_4 \subset \mathbb{P}^4$, since in this case, K-stability $\neq$ GIT stability. (A. Kaloghi...

open_problem | open | MathDB 页面 | 原文线索

V2-022391 | MATHDB-384210

Distribution of invariants (problem 1)

摘录 / EXCERPT: (C. Li) Give an algebraic proof of stronger version of K.Fujita's result that for smooth Fano X, if $\alpha(X)=\frac{n}{n+1}$, then X is uniform K-stable.

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V2-022392 | MATHDB-384211

Distribution of invariants (problem 2)

摘录 / EXCERPT: (I. Cheltsov, Y. Liu, Z. Zhuang) Find as large α invariant as possible for K-unstable smooth Fanos.

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V2-022393 | MATHDB-384212

Distribution of invariants (problem 3)

摘录 / EXCERPT: (C. Jiang) For K-semistable klt Fano varieties, in a fixed dimension, does the set of all α-invariants sastisfy the DCC? Is there a gap above $\frac{1}{n+1}$? If so, how lar...

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V2-022394 | MATHDB-384213

Distribution of invariants (problem 4)

摘录 / EXCERPT: (L. Qi) Does the set of α-invariants of K-semistable Fanos satisfy the ACC? What about δ-invariant?

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V2-022395 | MATHDB-384214

Distribution of invariants (problem 5)

摘录 / EXCERPT: (C. Jiang) For smooth Fano varieties, is it true that $\alpha \leq 1$? Conjecture of Kawamata that $|-K_X| \neq \varnothing$ implies this one.

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V2-022396 | MATHDB-384215

Distribution of invariants (problem 6)

摘录 / EXCERPT: (C. Li) How large is the gap of global volumes of K-semistable Fano varieties from $(n+1)^n$? (C. Xu) Is the second largest volume achieved by quadratic?

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V2-022397 | MATHDB-384216

Distribution of invariants (problem 7)

摘录 / EXCERPT: (C. Jiang) Find volume upper bound of birational superrigid Fano varieties.

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V2-022398 | MATHDB-384217

Distribution of invariants (problem 8)

摘录 / EXCERPT: (C. Xu) What about similar questions for $\widehat{\mathrm{vol}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-022399 | MATHDB-384218

Special degeneration (problem 1)

摘录 / EXCERPT: (C. Xu) Conj: Given a klt Fano variety X, there is a bounded family of special degeneration of X. The local (global?) version of Higher Rank Finite Generation Conjecture (HRFG)...

open_problem | open | MathDB 页面 | 原文线索

V2-022400 | MATHDB-384219

Special degeneration (problem 2)

摘录 / EXCERPT: (I. Cheltsov) What about Manetti surfaces? $\mathbb{P}^2 \rightsquigarrow \mathbb{P}^2(a^2,b^2,c^2), a^2+b^2+c^2=3abc$.

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V2-022401 | MATHDB-384220

Special degeneration (problem 3)

摘录 / EXCERPT: (Z. Zhuang) Can you get the finite wall-chamber structure from global HRFG Conjecture?

open_problem | open | MathDB 页面 | 原文线索

V2-022402 | MATHDB-384221

Special degeneration (problem 4)

摘录 / EXCERPT: (Y. Liu) There exists special degeneration $\mathbb{P}^2 \rightsquigarrow \mathbb{P}(1,1,4), \mathbb{P}(1,4,25), X_{26} \subset \mathbb{P}(1,2,13,25)$ (Ascher-DeVleming-Liu). Are t...

open_problem | open | MathDB 页面 | 原文线索

V2-022403 | MATHDB-384222

Special degeneration (problem 5)

摘录 / EXCERPT: (I. Cheltsov) Given a del Pezzo surface filtrations over a curve (Mori fiber space), assume all fibers are irreducible, what are the singularities of the central fiber? Conj: bound...

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V2-022404 | MATHDB-384223

Special degeneration (problem 6)

摘录 / EXCERPT: (C. Li) Find explicit description of isotrivial degeneration $\mathbb{P}^2 \rightsquigarrow \mathbb{P}(a^2,b^2,c^2)$.

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V2-022405 | MATHDB-384224

Anti-hyperbolicity of K-moduli (problem 1)

摘录 / EXCERPT: (Z. Patakfalvi) Assume deformation is unobstructed for some K-moduli space M, is it true that K_M should not be more positive than $c \cdot \lambda_{CM}$ (i.e....

open_problem | open | MathDB 页面 | 原文线索

V2-022406 | MATHDB-384225

Normalized volume under MMP (problem 1)

摘录 / EXCERPT: (C. Xu) How does $\widehat{\mathrm{vol}}$ behave under MMP? Can we define a motivic integration of $\widehat{\mathrm{vol}}$ such that it behaves nicely?

open_problem | open | MathDB 页面 | 原文线索

V2-022407 | MATHDB-384226

$\mathbb{G}_a$-action on the section ring (problem 1)

摘录 / EXCERPT: (I. Cheltsov) Let X be a K-stable Fano variety, $R=R(X,-K_X)$. Does R admits an action of $\mathbb{G}_a$? (For $dP_{1,2,3},$ No. For $dP_{4,5}, $ Yes). Any relation between the...

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V2-022408 | MATHDB-384227

K-semistability of fibration (problem 1)

摘录 / EXCERPT: (K. Fujita) Let $X \to Y$ be a smooth morphism between smooth Fano varieties. If X is K-stable or K-semistable, does it hold for Y?

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V2-022409 | MATHDB-384228

K-semistability of fibration (problem 2)

摘录 / EXCERPT: (Z. Patakfalvi) Under the same assumption, is the general fiber K-semistable?

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V2-022410 | MATHDB-384229

K-stability in the general polarization case (problem 1)

摘录 / EXCERPT: (M. Jonsson) Define the Futaki invariant for general filtrations in the Fano case, so that it behaves well. In the general polarized case, is the definition of uniform K-stability...

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V2-022411 | MATHDB-384230

K-stability in the general polarization case (problem 2)

摘录 / EXCERPT: (I. Cheltsov) Develop δ-criterion for K-stability in the general polarization case, see Dervan for α as an example.

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V2-022412 | MATHDB-384231

Reduced uniform K-stability (problem 1)

摘录 / EXCERPT: (C. Li) How to effectively test reduced uniform K-stability using similar divisors to basis type divisors?

open_problem | open | MathDB 页面 | 原文线索

V2-022413 | MATHDB-384232

Berman-Gibbs invariant (problem 1)

摘录 / EXCERPT: (H. Blum) Does the Berman-Gibbs invariant γ equal to δ? (Fujita-Odaka: $\gamma \leq \delta$)

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V2-022414 | MATHDB-384233

Berman-Gibbs invariant (problem 2)

摘录 / EXCERPT: (Z. Zhuang) γ is tested by $G=\mathrm{Aut}(X)$-invariant divisors. So maybe $\gamma = \delta^{G}$ when one of them $\leq$ 1?

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V2-022415 | MATHDB-384234

Other problems (problem 1)

摘录 / EXCERPT: (Z. Patakfalvi) Consider $f: \mathfrak{X} \to T$(smooth projective curve), assume the general fiber is unifrom K-stable. Let $c_{nef}:=$ minimum of c such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022416 | MATHDB-384235

Other problems (problem 2)

摘录 / EXCERPT: (I. Cheltsov, A. Castravet, S. Donaldson) Let X be a smooth Fano variety, $\mathrm{Pic}(X)=\mathbb{Z}(-K_X)$. Is X always K-semistable? (S. Donaldson) Is the moduli spcae of ve...

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V2-022417 | MATHDB-384236

Other problems (problem 3)

摘录 / EXCERPT: (C. Araujo) Let X be a smooth Fano variety. Let V denote the variety of minimal degree rational curves through a general point. Does the K-(semi)stability of X implies specia...

open_problem | open | MathDB 页面 | 原文线索

V2-022418 | MATHDB-384237

Other problems (problem 4)

摘录 / EXCERPT: (Y. Liu) Explicitly describe the K-moduli compactification of $(\mathbb{P}^1 \times \mathbb{P}^1, c \cdot C)$, where $C \in |\mathscr{O}(3,3)|$, $0 < c < 2/3$. $\cdot$ Birational mod...

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V2-022419 | MATHDB-384239

Can we see the mirror to this fact explicitly in case e.g.

摘录 / EXCERPT: Can we see the mirror to this fact explicitly in case e.g. of mirrors to $X=\{xy=0\}\subset \mathbb{A}_{\mathbb{F}_p}$? More generally, what is the symplectic mirror to Frobenius f...

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V2-022420 | MATHDB-384240

Mirror constructions (problem 3)

摘录 / EXCERPT: Probe what is know about symp. mirrors to algebraic geometric operations (not necessary toric) such as blow-ups?

open_problem | open | MathDB 页面 | 原文线索

V2-022421 | MATHDB-384242

Does every smooth toric variety admits an exceptional collection of sheaves ...

摘录 / EXCERPT: Does every smooth toric variety admits an exceptional collection of sheaves begin{itemize} item $\bullet$ if replace sheaves (good objects of $D^b(X)$, answer is yes) item...

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V2-022422 | MATHDB-384244

Compare Δ-resolutions

摘录 / EXCERPT: Can we directly compare the Hanlon-Hicks-Lazarev and Brown-Erman construction of resolutions? $\bullet$ Potential mechanism: (starting in degree 0) {vertices/points in stratificati...

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V2-022423 | MATHDB-384247

Resolutions (problem 4)

摘录 / EXCERPT: Given a cellular resolution, is there a meaningful way to algebraically interpret topology of the cell complex beyond homology e.g. π_1?

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V2-022424 | MATHDB-384249

Is there a well-behaved moduli theory of virtual resolutions equipped with extra structure?

摘录 / EXCERPT: Is there a well-behaved moduli theory of virtual resolutions equipped with extra structure? e.g. condition under which there exists a finite set? or space/stacks/...? (c.f. [Boij-...

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V2-022425 | MATHDB-384251

Modules over the Cox ring (problem 1)

摘录 / EXCERPT: Given M an S-module, $S=Cox(X)$, begin{enumerate} item can we always find another S-module M' satisfying begin{itemize} item $\bullet$ $\tilde{M}=\tilde{M}'$ item $\bullet$...

open_problem | open | MathDB 页面 | 原文线索

V2-022426 | MATHDB-384252

What can be said about the Orlov spectrum of a toric variety (or any ...)?

摘录 / EXCERPT: What can be said about the Orlov spectrum of a toric variety (or any X)? (e.g. about gaps, sequences, etc.) Known: begin{itemize} item $OrSpec(\mathbb{P}^1)=\{1,2\}$ item On a to...

open_problem | open | MathDB 页面 | 原文线索

V2-022427 | MATHDB-384253

Rdim of $D^b Sing$

摘录 / EXCERPT: What about Rouquier dimension or orlov spectrum of $D^b(Sing(X))=D^b(Coh(X))/D^b(Perf(X))$?

open_problem | open | MathDB 页面 | 原文线索

V2-022428 | MATHDB-384254

Orlov spectrum and Rouquier dimension (problem 3)

摘录 / EXCERPT: Can (1) be used to bound (2) or vice-versa?

open_problem | open | MathDB 页面 | 原文线索

V2-022429 | MATHDB-384255

What can be said about the Rouquier dimension of non-toric varieties and/for their symplectic mirrors?

摘录 / EXCERPT: What can be said about the Rouquier dimension of non-toric varieties and/for their symplectic mirrors? begin{itemize} item Toric varieties of dim n $\longleftrightarrow$(mirror) (...

open_problem | open | MathDB 页面 | 原文线索

V2-022430 | MATHDB-384256

What operations on Fukaya categories have meaningful commutative algebra translations?

摘录 / EXCERPT: What operations on Fukaya categories have meaningful commutative algebra translations? e.g. maybe a spherical twist (via mirror symmetry Dehn twist) around a spherical object suppo...

open_problem | open | MathDB 页面 | 原文线索

V2-022431 | MATHDB-384257

Line bundles over toric stacks (problem 1)

摘录 / EXCERPT: begin{enumerate} item Understand symplectic mirror of these criteria. item Understand consequences of generalizations of these critera (e.g. if there exists a toric subvariety whos...

open_problem | open | MathDB 页面 | 原文线索

V2-022432 | MATHDB-384258

Parellelization

摘录 / EXCERPT: How to parellize Macaulay2? Do we need a new garbage collector to do this? Which algorithms are inherently parallelizable and which need more thought?

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V2-022433 | MATHDB-384259

What features at the user level should be parallelized?

摘录 / EXCERPT: What features at the user level should be parallelized? For which computations do we want to run multiple algorithms simultaneously and take whichever finishes first? Can we do oth...

open_problem | open | MathDB 页面 | 原文线索

V2-022434 | MATHDB-384260

Benchmarks

摘录 / EXCERPT: What "benchmark" calculations should be tracked in M2? What is the best way to keep track of these benchmarks (e.g., wiki, on the website, etc.)?

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V2-022435 | MATHDB-384261

Other internal improvements

摘录 / EXCERPT: How can we implement a faster basis calculation without breaking anything?

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V2-022436 | MATHDB-384262

What commands should we add "DegreeLimit" to, and how can we better document "DegreeLimit" and its expected behavior in these functions?

摘录 / EXCERPT: What commands should we add "DegreeLimit" to, and how can we better document "DegreeLimit" and its expected behavior in these functions?

open_problem | open | MathDB 页面 | 原文线索

V2-022437 | MATHDB-384264

Local cohomology in characteristic p

摘录 / EXCERPT: Given an F-finite F-module, how can one efficiently compute a unit F-module? For example, given the map $\mathrm{Ext}^i(R/I,R)\to \mathrm{Ext}^i(R/I^{[p]},R)$, how can one find a r...

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V2-022438 | MATHDB-384265

Applying differential operators

摘录 / EXCERPT: Can we add a convenient way to let the differential operators on a ring R act on elements of R? Can we have them act on elements of the fraction field?

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V2-022439 | MATHDB-384266

What functionality should be added to the DGAlgebras package?

摘录 / EXCERPT: What functionality should be added to the DGAlgebras package? Can we handle DG ideals and modules? What about semifree resolutions? Can it be better incorporated with the AInfinity...

open_problem | open | MathDB 页面 | 原文线索

V2-022440 | MATHDB-384268

DG Algebras and Resolutions (problem 3)

摘录 / EXCERPT: Can we add simplicial resolutions to Macaulay2?

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V2-022441 | MATHDB-384269

What is the best way to implement a "morphism of sheaves" data type?

摘录 / EXCERPT: What is the best way to implement a "morphism of sheaves" data type? What's the best way to calculate the maps on sheaf cohomology induced by such a morphism on a projective variet...

open_problem | open | MathDB 页面 | 原文线索

V2-022442 | MATHDB-384270

Can we compute all the degree-0 (or degree-≤ e) homomorphisms of a graded module M without computing the entirety of End M?

摘录 / EXCERPT: Can we compute all the degree-0 (or degree-≤ e) homomorphisms of a graded module M without computing the entirety of End M? What are other ways to speed up the decomposition of mod...

open_problem | open | MathDB 页面 | 原文线索

V2-022443 | MATHDB-384271

Convenience functions and user experience (problem 1)

摘录 / EXCERPT: What quality-of-life code should be added to M2?

open_problem | open | MathDB 页面 | 原文线索

V2-022444 | MATHDB-384272

What error or warning messages are currently missing that should be added?

摘录 / EXCERPT: What error or warning messages are currently missing that should be added?

open_problem | open | MathDB 页面 | 原文线索

V2-022445 | MATHDB-384273

Topology of dual complexes (problem 1)

摘录 / EXCERPT: What are the possible fundamental groups $\pi_1(\mathcal{D}(X,B))$ for a log Calabi--Yau 4-fold (X,B) of coregularity zero? Does $\mathbb{Z}/7\mathbb{Z}$ appear as such a group?

open_problem | open | MathDB 页面 | 原文线索

V2-022446 | MATHDB-384274

Does the Poincare homology ... arise as the dual complex of a log Calabi--Yau pair?

摘录 / EXCERPT: Does the Poincare homology $\mathbb{S}^3$ arise as the dual complex of a log Calabi--Yau pair?

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V2-022447 | MATHDB-384275

Does the Wu manifold appear as the dual complex of a log Calabi-Yau pair?

摘录 / EXCERPT: Does the Wu manifold appear as the dual complex of a log Calabi-Yau pair?

open_problem | open | MathDB 页面 | 原文线索

V2-022448 | MATHDB-384276

Topology of dual complexes (problem 4)

摘录 / EXCERPT: Does there exist a log smooth simple normal crossing Calabi--Yau variety X for which $\mathcal{D}(X)$ is homotopy equivalent to the Wu manifold?

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V2-022449 | MATHDB-384277

Log rational and cluster type pairs (problem 1)

摘录 / EXCERPT: Let T be a toric 3-fold and $B\sim |-K_T|$ be an slc hypersurface of coregularity zero. Show that (T,B) is log rational.

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V2-022450 | MATHDB-384278

Log rational and cluster type pairs (problem 2)

摘录 / EXCERPT: Let (X,B) be a log Calabi--Yau pair of dimension n. Let $(Y,B_Y)\rightarrow (X,B)$ be a dlt modification. Assume that every strata of (Y,B_Y), inlcuding Y itself, is a rati...

open_problem | open | MathDB 页面 | 原文线索

V2-022451 | MATHDB-384279

Classify log rational smooth Fano ... -folds.

摘录 / EXCERPT: Classify log rational smooth Fano 3-folds.

open_problem | open | MathDB 页面 | 原文线索

V2-022452 | MATHDB-384280

Log rational and cluster type pairs (problem 4)

摘录 / EXCERPT: Let (X,B) be a log Calabi--Yau pair of dimension n, index one, and Picard rank ρ. Assume that the number of components of B equals $n+\rho-1$. Show that (X,B) is of cl...

open_problem | open | MathDB 页面 | 原文线索

V2-022453 | MATHDB-384281

Classify which smooth Fano blow-ups of ... are of cluster type.

摘录 / EXCERPT: Classify which smooth Fano blow-ups of $\mathbb{P}^3$ are of cluster type.

open_problem | open | MathDB 页面 | 原文线索

V2-022454 | MATHDB-384282

Log rational and cluster type pairs (problem 6)

摘录 / EXCERPT: Let X be a cluster type variety defined over an algebraically closed field k of characteristic $p>1$. Show that X is liftable over $\mathbb{W}_2(k)$.

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V2-022455 | MATHDB-384283

Mukai conjecture and related problems (problem 1)

摘录 / EXCERPT: Let X be a klt Fano variety. Write $-K_X\equiv \sum_{i=1}^k a_i M_i$ where each $a_i \in \mathbb{R}_{>0}$ and each M_i is a nef Cartier divisor which is not numerically trivial...

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V2-022456 | MATHDB-384284

Mukai conjecture and related problems (problem 2)

摘录 / EXCERPT: Let X be a klt Fano variety of dimension n. Write $-K_X\equiv \sum_{i=1}^k a_i M_i$ where each $a_i\in \mathbb{R}_{>0}$ and each M_i is a nef Cartier divisor which is not num...

open_problem | open | MathDB 页面 | 原文线索

V2-022457 | MATHDB-384285

Mukai conjecture and related problems (problem 3)

摘录 / EXCERPT: Let $(X,B,\mathbf{M})$ be a generalized log Calabi--Yau pair. Let $B=\sum_{i=1}^k a_i B_i$ be its decomposition into prime components. Let X' be the model where the b-nef divisor...

open_problem | open | MathDB 页面 | 原文线索

V2-022458 | MATHDB-384286

Mukai conjecture and related problems (problem 4)

摘录 / EXCERPT: Let X be a smooth Fano variety all whose extremal contractions are fibrations. Prove that ${\rm Nef}(X)$ is a simplicial cone.

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V2-022459 | MATHDB-384287

Symplectic and log symplectic pairs (problem 1)

摘录 / EXCERPT: Let $\mathcal{X}\longrightarrow\Delta$ be a dlt degeneration of Hyperkähler varieties with $K_{\mathcal{X}/\Delta}\simeq\mathcal{O_X}$. Is ... Given...

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V2-022460 | MATHDB-384288

Symplectic and log symplectic pairs (problem 2)

摘录 / EXCERPT: Can $\text{K3}^{[5]}$ and OG_{10} be distinguished via combinatorics of $\Gamma(\mathcal{X}_0)$ for degeneration $\mathcal{X}\longrightarrow\Delta$? "Combinatorics": integ...

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V2-022461 | MATHDB-384289

Symplectic and log symplectic pairs (problem 3)

摘录 / EXCERPT: Do 1-parameter Hyperk"ahler degenerations admit K-trivial, SNC models, after a finite base change? Note: minimal dlt exist, due to Fujino, Lai. This is true in Type I (KLSV).

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V2-022462 | MATHDB-384290

Symplectic and log symplectic pairs (problem 4)

摘录 / EXCERPT: If (X,D) is log sympletic dlt (or plt) pair and $X\backslash X^{\text{SNC}}$ has codimension ≥3 in X, are the singularities rigid (in the sense of definitions of log symple...

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V2-022463 | MATHDB-384291

Are Type III log sympletic pairs rational, log rational, or cluster type?

摘录 / EXCERPT: Are Type III log sympletic pairs rational, log rational, or cluster type?

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V2-022464 | MATHDB-384292

Local deformation theory of dlt log sympletic singularities?

摘录 / EXCERPT: Local deformation theory of dlt log sympletic singularities?

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V2-022465 | MATHDB-384293

Symplectic and log symplectic pairs (problem 7)

摘录 / EXCERPT: Examples of dual complexes which are not Hilb^n or Kum^n of a K3 degeneration? degeneration of cubic 4-fold $X\rightsquigarrow V(x_0x_1x_2)$....

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V2-022466 | MATHDB-384294

Boundedness and moduli theory (problem 1)

摘录 / EXCERPT: Let $\mathcal{M}$ be a moduli space of smooth Calabi--Yau varieties. Does there exist a compactification...

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V2-022467 | MATHDB-384295

Boundedness and moduli theory (problem 2)

摘录 / EXCERPT: Near a cusp of $\mathcal{M}^*$ prove that there is a mirror X^{mir} of the general fibres $\mathcal{X}\longrightarrow\mathcal{M}$ such that ... where σ is the fan mod...

open_problem | open | MathDB 页面 | 原文线索

V2-022468 | MATHDB-384296

Boundedness and moduli theory (problem 3)

摘录 / EXCERPT: If we degenerate a log CY pair (X,D) coming from a Fano to the normal cone $\text{Cone}_N(D)$, does the deformation theory of the normal cone recover the moduli space? More preci...

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V2-022469 | MATHDB-384297

Boundedness and moduli theory (problem 4)

摘录 / EXCERPT: Let (X,D) be a log Calabi--Yau 3-fold. Is the period map ... injective?

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V2-022470 | MATHDB-384298

Boundedness and moduli theory (problem 5)

摘录 / EXCERPT: Consider $(X,L)\in F_{2d}$, the universal family $\mathcal{X}\overset{\pi}{\longrightarrow}F_{2d}$, and $\mathbb{P}(\pi_*nL)\simeq\mathbb{P}^g$-bundle over F_{2d} with a section...

open_problem | open | MathDB 页面 | 原文线索

V2-022471 | MATHDB-384299

Boundedness and moduli theory (problem 6)

摘录 / EXCERPT: Let X be a Calabi--Yau variety, and let L be a semi-ample Cartier divisor on X. Consider the universal family $(\mathcal X,\mathcal L) \to \mathrm{Def}(X,L)$ of deformations...

open_problem | open | MathDB 页面 | 原文线索

V2-022472 | MATHDB-384300

Is the relatie Albanese variety of a Calabi--Yau variety itself Calabi--Yau?

摘录 / EXCERPT: Is the relatie Albanese variety of a Calabi--Yau variety itself Calabi--Yau? How many abelian fibered Calabi--Yau varieties admit the same relative Albanese fibration?

open_problem | open | MathDB 页面 | 原文线索

V2-022473 | MATHDB-384301

What characterizes a variety Z with ... reflexive for all (some) k?

摘录 / EXCERPT: What characterizes a variety Z with Ω_Z^k reflexive for all (some) k? Does Ω_Z^k being reflexive for all (some) k implies X is smooth or lci?

open_problem | open | MathDB 页面 | 原文线索

V2-022474 | MATHDB-384302

Properties of K\"ahler differentials (problem 2)

摘录 / EXCERPT: What characterizes a variety $\Omega_Z^k/\text{torsion}$ being reflexive or S_2?

open_problem | open | MathDB 页面 | 原文线索

V2-022475 | MATHDB-384303

Positive/mixed characteristic (problem 1)

摘录 / EXCERPT: Does there exist a map $X \to \mathbb{Z}_p$ such that (X,X_p) is log canonical (where X_p is the closed fiber) and X_p is irreducible but not S_2?

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V2-022476 | MATHDB-384304

Positive/mixed characteristic (problem 2)

摘录 / EXCERPT: Is there an analogue of $\underline{\Omega}_X^\bullet$ in positive characteristic which makes the Hodge-to-de Rham spectral sequence degenerate at E_1?

open_problem | open | MathDB 页面 | 原文线索

V2-022477 | MATHDB-384305

Is there an analog of the minimal exponent in positive characteristics?

摘录 / EXCERPT: Is there an analog of the minimal exponent in positive characteristics?

open_problem | open | MathDB 页面 | 原文线索

V2-022478 | MATHDB-384306

Positive/mixed characteristic (problem 4)

摘录 / EXCERPT: Is there a notion of pseudo p-rationality? That is, $H^d_m(R) \rightarrow H^d_m(\mathbb{R} \pi_*\mathcal{O}_Y)$ is injective, where $\pi: Y \rightarrow X$ is a proper birational ma...

open_problem | open | MathDB 页面 | 原文线索

V2-022479 | MATHDB-384307

Positive/mixed characteristic (problem 5)

摘录 / EXCERPT: If X is rational, this means something like $\mathcal{O}_X \rightarrow B$ where B is Cohen-Macaulay algebra, then $H^i_x(\mathcal{O}_X) \rightarrow H^i_x(B)$ is injective. begin{it...

open_problem | open | MathDB 页面 | 原文线索

V2-022480 | MATHDB-384308

Understand KSBA moduli spaces in char p (Hodge bundles, local freeness + commutes with base change, E1 degeneration etc).

摘录 / EXCERPT: Understand KSBA moduli spaces in char p (Hodge bundles, local freeness + commutes with base change, E1 degeneration etc).

open_problem | open | MathDB 页面 | 原文线索

V2-022481 | MATHDB-384309

Properties of higher singularities and minimal exponents (problem 1)

摘录 / EXCERPT: Let X be a variety, and H a Cartier divisor on X. Suppose H is m-Du Bois. Is X m-Du Bois around H?

open_problem | open | MathDB 页面 | 原文线索

V2-022482 | MATHDB-384310

Properties of higher singularities and minimal exponents (problem 2)

摘录 / EXCERPT: What can we say about higher singularities for pure morphisms (e.g. direct summands, linearly reductive group quotients)?

open_problem | open | MathDB 页面 | 原文线索

V2-022483 | MATHDB-384311

What is the relation between non-smoothability and 1-Du Bois singularities?

摘录 / EXCERPT: What is the relation between non-smoothability and 1-Du Bois singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-022484 | MATHDB-384312

Properties of higher singularities and minimal exponents (problem 4)

摘录 / EXCERPT: Let X be a normal variety of dimension n. X is pre-$\lceil \frac{n}{2} \rceil$-Du Bois. Then is X pre-n-Du Bois? X is pre-$\lfloor \frac{n}{2} \rfloor$-rational. Then is X pre-n-ra...

open_problem | open | MathDB 页面 | 原文线索

V2-022485 | MATHDB-384313

Are there universal bounds on the multiplicity of a p-DB and p-rational singularities?

摘录 / EXCERPT: Are there universal bounds on the multiplicity of a p-DB and p-rational singularities? For p=0, dim X = n, embedding dimension of x=e, then $mult_x \le \binom{e}{n}$ when X is Du B...

open_problem | open | MathDB 页面 | 原文线索

V2-022486 | MATHDB-384314

Properties of higher singularities and minimal exponents (problem 6)

摘录 / EXCERPT: Let H = Z(f), $p\in H$. Is the minimal exponent $\tilde{\alpha}_p(H) \le lct(\mathfrak{m}_p \cdot Jac(f))$? This is related to Tessier's conjecture and also to the work of Dano Kim...

open_problem | open | MathDB 页面 | 原文线索

V2-022487 | MATHDB-384315

Properties of higher singularities and minimal exponents (problem 7)

摘录 / EXCERPT: Consider a family of hypersurfaces with non-isolated singularities. Is there a condition for the minimal exponent to be constant in the family? (When the hypersurfaces have isolate...

open_problem | open | MathDB 页面 | 原文线索

V2-022488 | MATHDB-384316

How do we decide if a hypersurface H is m-Du Bois or m-rational in the non-isolated case?

摘录 / EXCERPT: How do we decide if a hypersurface H is m-Du Bois or m-rational in the non-isolated case?

open_problem | open | MathDB 页面 | 原文线索

V2-022489 | MATHDB-384317

Properties of higher singularities and minimal exponents (problem 9)

摘录 / EXCERPT: What are the $(n+1)\times (n+1)$ tables with entries either 0 or 1, depending on whether $\mathcal(H)^q \underline{\Omega}_X^p$ is zero or not? Most simple case: Is there a surface...

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V2-022490 | MATHDB-384318

Properties of higher singularities and minimal exponents (problem 10)

摘录 / EXCERPT: Maxim-Yang prove a result characterizing m-Du Bois and m-rational hypersurface singularities in terms of characteristic classes. Can we extend this to the lci case?

open_problem | open | MathDB 页面 | 原文线索

V2-022491 | MATHDB-384319

Properties of higher singularities and minimal exponents (problem 11)

摘录 / EXCERPT: Can we prove that a variety with enough' endomorphisms is pre-m-Du Bois for all m? This would support a conjecture which predicts that such varieties have to be toric.

open_problem | open | MathDB 页面 | 原文线索

V2-022492 | MATHDB-384320

Study the global behavior of period map on ... -Du Bois loci.

摘录 / EXCERPT: Study the global behavior of period map on m-Du Bois loci.

open_problem | open | MathDB 页面 | 原文线索

V2-022493 | MATHDB-384321

Understand the higher Du Bois singularities in the case of cubics and see what are some examples of Hodge structures in the case of cubics.

摘录 / EXCERPT: Understand the higher Du Bois singularities in the case of cubics and see what are some examples of Hodge structures in the case of cubics.

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-022494 | MATHDB-384322

Is there an analytic characterization of the minimal exponent in the lci case?

摘录 / EXCERPT: Is there an analytic characterization of the minimal exponent in the lci case?

open_problem | open | MathDB 页面 | 原文线索

V2-022495 | MATHDB-384323

What is the correct notion of higher DB singularities in the symplectic case?

摘录 / EXCERPT: What is the correct notion of higher DB singularities in the symplectic case?

open_problem | open | MathDB 页面 | 原文线索

V2-022496 | MATHDB-384324

What is the a definition of m-Du Bois singularities satisfying: ...

摘录 / EXCERPT: What is the a definition of m-Du Bois singularities satisfying: begin{enumerate} item 0-Du Bois = Du Bois; item m-Du Bois $\Rightarrow$ (m-1)-Du Bois; item $R^i f_* \Omega_{X/S}^m$...

open_problem | open | MathDB 页面 | 原文线索

V2-022497 | MATHDB-384325

Is there a notion of minimal exponent in the non-lci case related to higher singularities?

摘录 / EXCERPT: Is there a notion of minimal exponent in the non-lci case related to higher singularities? This requires p-Du Bois $\Rightarrow$ (p-1)-rational. Is there an invariant characterizin...

open_problem | open | MathDB 页面 | 原文线索

V2-022498 | MATHDB-384326

Is there a good notion of a relative Du Bois complex?

摘录 / EXCERPT: Is there a good notion of a relative Du Bois complex?

open_problem | open | MathDB 页面 | 原文线索

V2-022499 | MATHDB-384327

Exploit wall-crossing techniques to prove the DT/PT correspondence for Calabi-Yau 3-folds with tautological insertions.

摘录 / EXCERPT: Exploit wall-crossing techniques to prove the DT/PT correspondence for Calabi-Yau 3-folds with tautological insertions.

open_problem | open | MathDB 页面 | 原文线索

V2-022500 | MATHDB-384328

Exploit wall-crossing to prove the rationality and symmetry of the PT invariants for Calabi-Yau 3-folds.

摘录 / EXCERPT: Exploit wall-crossing to prove the rationality and symmetry of the PT invariants for Calabi-Yau 3-folds.

open_problem | open | MathDB 页面 | 原文线索

V2-022501 | MATHDB-384329

Wall-crossing for Calabi-Yau varieties (problem 3)

摘录 / EXCERPT: Let X be a Fano 3-fold, and K_X a local Calabi-Yau 4-fold, obtained as the canonical bundle of X. Prove DT/PT type correspondences on X via wall-crossing for the Calabi-Yau...

open_problem | open | MathDB 页面 | 原文线索

V2-022502 | MATHDB-384330

Wall-crossing for Calabi-Yau varieties (problem 4)

摘录 / EXCERPT: Let X be a Fano 3-fold, and $p:Y\to X$ an elliptically fibered Calabi-Yau 4-fold. Compute invariants, or prove correspondences on X by performing wall-crossing on Y, with res...

open_problem | open | MathDB 页面 | 原文线索

V2-022503 | MATHDB-384331

Wall-crossing for Calabi-Yau varieties (problem 5)

摘录 / EXCERPT: Let X be a Calabi-Yau 4-fold. Use wall-crossing to formulate new (conjectural) relations on DT and PT type invariants on X, extending the known (conjectural) DT/PT correspondenc...

open_problem | open | MathDB 页面 | 原文线索

V2-022504 | MATHDB-384332

Rank-reduction wall-crossing (problem 1)

摘录 / EXCERPT: Let S be a surface of general type. Express rank r Vafa-Witten invariants on S in terms of rank 1 Vafa-Witten invariants of S via wall-crossing.

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V2-022505 | MATHDB-384333

Rank-reduction wall-crossing (problem 2)

摘录 / EXCERPT: Let S be a surface of general type. Use wall-crossing to find an explicit (possibly conjectural) formula for rank $r=4$ Vafa-Witten invariants of S.

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V2-022506 | MATHDB-384334

Rank-reduction wall-crossing (problem 3)

摘录 / EXCERPT: Extend the rank reduction techniques of Feyzbakhsh-Thomas to include refined invariants (e.g. motivic invariants).

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V2-022507 | MATHDB-384335

Can we extend Joyce’s wallcrossing techniques to non-linear settings?

摘录 / EXCERPT: Can we extend Joyce’s wallcrossing techniques to non-linear settings?

open_problem | open | MathDB 页面 | 原文线索

V2-022508 | MATHDB-384336

Properly formulate wall-crossing formulas for triangulated categories.

摘录 / EXCERPT: Properly formulate wall-crossing formulas for triangulated categories.

open_problem | open | MathDB 页面 | 原文线索

V2-022509 | MATHDB-384337

Extend wall-crossing to moduli spaces of sheaves on Calabi-Yau orbifolds.

摘录 / EXCERPT: Extend wall-crossing to moduli spaces of sheaves on Calabi-Yau orbifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-022510 | MATHDB-384338

Find some formulation of Virasoro constraints for moduli spaces of sheaves on orbifolds, and prove them using wall-crossing techniques as in Bojko-Lim-Moreira.

摘录 / EXCERPT: Find some formulation of Virasoro constraints for moduli spaces of sheaves on orbifolds, and prove them using wall-crossing techniques as in Bojko-Lim-Moreira.

open_problem | open | MathDB 页面 | 原文线索

V2-022511 | MATHDB-384443

Other (problem 2)

摘录 / EXCERPT: For general β, can one prove that the polynomials related to the coefficients of ... have zeros on the unit circle?

open_problem | open | MathDB 页面 | 原文线索

V2-022512 | MATHDB-384559

Splitting schemes for general advection-reaction-diffusion equations

摘录 / EXCERPT: In, the authors provide a splitting scheme to get an existence result for a general advection-reaction-diffusion equation, i.e....

open_problem | open | MathDB 页面 | 原文线索

V2-022513 | MATHDB-384820

Projective sampling scheme for sparse graphs

摘录 / EXCERPT: For each n, let $\mathcal{D}_n$ be a distribution over graphs of n vertices. A projective sampling scheme is as follows. For any $k < n$ we have a procedure to, given a sample...

open_problem | open | MathDB 页面 | 原文线索

V2-022514 | MATHDB-384896

What are combinatorially and geometrically interesting tropical varieties, in particular as they arise from root systems and/or matroids?

摘录 / EXCERPT: What are combinatorially and geometrically interesting tropical varieties, in particular as they arise from root systems and/or matroids?

open_problem | open | MathDB 页面 | 原文线索

V2-022515 | MATHDB-385126

Other problems (problem 5)

摘录 / EXCERPT: Is there a f.p. infinite torsion group? textbf{Conjecture:}(Grigorchuk) no.

open_problem | open | MathDB 页面 | 原文线索

V2-022516 | MATHDB-385334

Open problems (problem 9)

摘录 / EXCERPT: Questions: i) $H(\mathcal{L}_X)$ is a subvariety of B of half dimension. What is the intersection of $\mathcal{L}$ and fibres of the Hitchin map, i.e. the connected components of...

open_problem | open | MathDB 页面 | 原文线索

V2-022517 | MATHDB-385340

Langlands duality for meromorphic Higgs bundles

摘录 / EXCERPT: The existence of the hyperkahler metrics on the wild Hitchin spaces means the moduli spaces of meromorphic connection on curves (and the corresponding wild character varieties) hav...

open_problem | open | MathDB 页面 | 原文线索

V2-022518 | MATHDB-385344

Hilbert schemes of points on $K2$-surfaces.

摘录 / EXCERPT: Is each of these $4n$ dimensional hyperkahler manifolds diffeomorphic to (or a deformation of) the Hilbert scheme of n-points on the underlying $K2$ surface?

open_problem | open | MathDB 页面 | 原文线索

V2-022519 | MATHDB-385434

Complex geometry (problem 7)

摘录 / EXCERPT: (Conjecture of Lejmi-Sz'ekelyhidi) There exists a solution if and only if ... for all proper subvarieties $V \subset M$, where $p = \mathrm{dim}(V)$

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V2-022520 | MATHDB-385435

Complex geometry (problem 8)

摘录 / EXCERPT: Can one establish regularity properties of geodesics in the space of K"ahler potentials for singular K"ahler varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022521 | MATHDB-385583

The Problems (problem 17)

摘录 / EXCERPT: Consider the Gaussian measure of sections of the cube B^n_∞. It is known that if $n \geq 3$, ... Is this the best upper bound? If we introduce a dilation factor $r>0$, for whi...

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V2-022522 | MATHDB-385585

The Problems (problem 19)

摘录 / EXCERPT: Given $n+1$ points $x_1,\ldots, x_{n+1}$ on S^{n-1}, find the configuration such that the convex hull of $x_1,\ldots, x_{n+1}$ has the largest mean width. Is it maximal for the r...

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V2-022523 | MATHDB-385590

The Problems (problem 24)

摘录 / EXCERPT: begin{enumerate}[(a)] item (V. Milman) Is the minimum of $V(a,t)$ for fixed $t>0$ and $a \in S^{n-1}$ (variable) always attained for a standard vector $a=f_i$? item If...

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V2-022524 | MATHDB-385606

Stratifications of braid varieties (problem 1)

摘录 / EXCERPT: If we consider instead a braid variety, ie. a Legendrian associated to a (-1)-closure, does this still hold?

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V2-022525 | MATHDB-385607

For more general Legendrians, is the augmentation variety cluster?

摘录 / EXCERPT: For more general Legendrians, is the augmentation variety cluster?

open_problem | open | MathDB 页面 | 原文线索

V2-022526 | MATHDB-385617

What are the tropical points of the braid variety?

摘录 / EXCERPT: What are the tropical points of the braid variety? What are the Θ-functions on the braid variety?

open_problem | open | MathDB 页面 | 原文线索

V2-022527 | MATHDB-385620

What relationship is there between braid varieties and affine Springer fibers in the affine Grassmannian?

摘录 / EXCERPT: What relationship is there between braid varieties and affine Springer fibers in the affine Grassmannian?

open_problem | open | MathDB 页面 | 原文线索

V2-022528 | MATHDB-385622

Is there a similar story for other cluster varieties or braid varieties?

摘录 / EXCERPT: Is there a similar story for other cluster varieties or braid varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022529 | MATHDB-385624

Combinatorics of cluster structures on braid varieties (problem 1)

摘录 / EXCERPT: Investigate: begin{enumerate} item cluster variables, item quivers, item positive parametrizations, item twist maps and Donaldson-Thomas transformations, item and the relationship...

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V2-022530 | MATHDB-385625

For each positroid stratum of a Grassmanian, there is a Legendrian link whose augmentation variety is that stratum.

摘录 / EXCERPT: For each positroid stratum of a Grassmanian, there is a Legendrian link whose augmentation variety is that stratum. Positroid strata are also ordered with respect to closure. Is th...

open_problem | open | MathDB 页面 | 原文线索

V2-022531 | MATHDB-385633

How can we define totally non negative braid varieties in a way which generalizes the totally non negative part of ... ?

摘录 / EXCERPT: How can we define totally non negative braid varieties in a way which generalizes the totally non negative part of $G(k,n)/\Pi_f$?

open_problem | open | MathDB 页面 | 原文线索

V2-022532 | MATHDB-385676

When do resolutions of toric subvarieties from the work of Hanlon-Hicks-Lazarev (and their minimizations) have nice properties?

摘录 / EXCERPT: When do resolutions of toric subvarieties from the work of Hanlon-Hicks-Lazarev (and their minimizations) have nice properties? More specifically, when is the resolution of the tor...

open_problem | open | MathDB 页面 | 原文线索

V2-022533 | MATHDB-385677

Computational and structural aspects of resolutions (problem 2)

摘录 / EXCERPT: Given a coherent sheaf F on a toric variety X (or a complex of sheaves), can we implement a code in M2 that computes its free monad in Bondal-Thompsen collection?

open_problem | open | MathDB 页面 | 原文线索

V2-022534 | MATHDB-385695

Are there quiver-theoretic descriptions of certain moduli of sheaves (e.g ... on toric varieties) via resolutions by the Bondal-Thompsen collection?

摘录 / EXCERPT: Are there quiver-theoretic descriptions of certain moduli of sheaves (e.g $Hilb_{n}$ on toric varieties) via resolutions by the Bondal-Thompsen collection?

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V2-022535 | MATHDB-385735

The Problems (problem 1)

摘录 / EXCERPT: We know how to construct points on an elliptic curve E over ${\\mathbb{F}}_q(T)$ when the analytic rank is 1 by using (Drinfel'd) Heegner points. What if X is a hyperelliptic c...

open_problem | open | MathDB 页面 | 原文线索

V2-022536 | MATHDB-385737

The Problems (problem 3)

摘录 / EXCERPT: Let X be a Hilbert modular variety. At some level (say, that of cohomology), these look like a product of modular curves. Under what circumstances does X have Tate classes not...

open_problem | open | MathDB 页面 | 原文线索

V2-022537 | MATHDB-385739

The Problems (problem 5)

摘录 / EXCERPT: Consider the curve $C_{d,a}: y^d=x(x-1)(x-a)$ defined over ${\\mathbb{C}}$ (though $\bar{\\mathbb{Q}}$ or ${\\mathbb{F}}_q$ will also work). For which values of a does it happen...

open_problem | open | MathDB 页面 | 原文线索

V2-022538 | MATHDB-385740

The Problems (problem 6)

摘录 / EXCERPT: Let X be a Hilbert modular surface over ${\\mathbb{Q}}$. All of the Tate classes that arise from modular curves on X are defined over ${\\mathbb{Q}}^{ab}$; however, there also...

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V2-022539 | MATHDB-385742

A Hilbert modular surface in characteristic ... has interesting cycles parametrizing Hilbert-Blumenthal abelian varieties with specified Newton polygon behavior.

摘录 / EXCERPT: A Hilbert modular surface in characteristic p has interesting cycles parametrizing Hilbert-Blumenthal abelian varieties with specified Newton polygon behavior. Are these containe...

open_problem | open | MathDB 页面 | 原文线索

V2-022540 | MATHDB-385745

The Problems (problem 11)

摘录 / EXCERPT: Let A be CM abelian variety over $\bar{\\mathbb{Q}}$ and let c be an absolute Hodge class. Reduce A to A_0 over $\overline{{\\mathbb{F}}_p}$ and take a Lefschetz class x...

open_problem | open | MathDB 页面 | 原文线索

V2-022541 | MATHDB-385747

The Problems (problem 13)

摘录 / EXCERPT: C. Simpson proved that if X is a variety and if $\rho_X:\pi_1^{geom}(X)\to GL_2({\\mathbb{Q}}_\ell)$ is a representation whose algebraic enveloping contains SL_2, then ρ_X...

open_problem | open | MathDB 页面 | 原文线索

V2-022542 | MATHDB-385748

The Problems (problem 14)

摘录 / EXCERPT: Take an abelian variety over a finite extension K of ${\\mathbb{Q}}_p$ with multiplicative reduction. Show the isomorphism ... Comments: 1. This implies the analogous statement for...

open_problem | open | MathDB 页面 | 原文线索

V2-022543 | MATHDB-385749

The Problems (problem 15)

摘录 / EXCERPT: According to the Tate Conjecture, if one takes the L-function of a variety X over K at the edge of the critical strip, and then increases K, the order of the pole should st...

open_problem | open | MathDB 页面 | 原文线索

V2-022544 | MATHDB-385750

The Problems (problem 16)

摘录 / EXCERPT: Let $X(\ell)$ be a modular curve. What can we say about the splitting field of the ℓ-torsion of the Jacobian $J(X(\ell))$? Comments: 1. This is needed to speak about the L-f...

open_problem | open | MathDB 页面 | 原文线索

V2-022545 | MATHDB-385751

Can we find algebraic cycles (living in characteristic ...), which account for such Tate classes?

摘录 / EXCERPT: Can we find algebraic cycles (living in characteristic p), which account for such Tate classes?

open_problem | open | MathDB 页面 | 原文线索

V2-022546 | MATHDB-385754

The Problems (problem 20)

摘录 / EXCERPT: Take a principally polarized abelian variety A and assume $\text{rank}(NS(A))=1$. What is the index of $C^i(A)$ in $H^{2i}(A,{\\mathbb{Z}}(i))$? What is the image? What if we jus...

open_problem | open | MathDB 页面 | 原文线索

V2-022547 | MATHDB-385757

The Problems (problem 23)

摘录 / EXCERPT: Using the current definition of the Weil-'{e}tale topology, define the Weil-'{e}tale topos for an arbitrary arithmetic scheme (i.e., a scheme of finite type over...

open_problem | open | MathDB 页面 | 原文线索

V2-022548 | MATHDB-385765

ABV for covering groups

摘录 / EXCERPT: There is a canonical isomorphism as K-groups ... where X is a $\mathbb C$-algebraic variety and H is a $\mathbb C$-algebraic group (for the trivial cover, $H={^\vee}G$).

open_problem | open | MathDB 页面 | 原文线索

V2-022549 | MATHDB-385895

Provide a public key encryption scheme using only a one-way function given as a circuit.

摘录 / EXCERPT: Provide a public key encryption scheme using only a one-way function given as a circuit.

open_problem | open | MathDB 页面 | 原文线索

V2-022550 | MATHDB-385896

Construct a PKE scheme from hardness of discrete-log problem in a specific group.

摘录 / EXCERPT: Construct a PKE scheme from hardness of discrete-log problem in a specific group.

open_problem | open | MathDB 页面 | 原文线索

V2-022551 | MATHDB-385903

When can you find algebraically defined multilinear maps involving group schemes of p-power order?

摘录 / EXCERPT: When can you find algebraically defined multilinear maps involving group schemes of p-power order?

open_problem | open | MathDB 页面 | 原文线索

V2-022552 | MATHDB-385947

Hilbert's Tenth Problem for Subrings of $\mathbb Q$ (problem 2)

摘录 / EXCERPT: Let $X/\mathbb Q$ be a variety, and let $\{Y_a\}_{a\in A}$ be a set of uniformly definable subsets of X. begin{enumerate} item Is $\overline{Y_a(\mathbb Q)} = Y_a(\mathbb R)$? it...

open_problem | open | MathDB 页面 | 原文线索

V2-022553 | MATHDB-385953

Can the ideas of Xiao-Zhu and Liu-Tian-Xiao-Zhang-Zhu apply to “higher level” affine Deligne-Lusztig varieties or loop Deligne-Lusztig varieties?

摘录 / EXCERPT: Can the ideas of Xiao-Zhu and Liu-Tian-Xiao-Zhang-Zhu apply to “higher level” affine Deligne-Lusztig varieties or loop Deligne-Lusztig varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022554 | MATHDB-385956

Can one prove that the image of the Ichino-Prasanna Tate class in the special fiber comes from an algebraic cycle?

摘录 / EXCERPT: Can one prove that the image of the Ichino-Prasanna Tate class in the special fiber comes from an algebraic cycle?

open_problem | open | MathDB 页面 | 原文线索

V2-022555 | MATHDB-385957

Problems on geometric realizations of functoriality (problem 1)

摘录 / EXCERPT: Can one construct cycles that realize other instances of functoriality (i.e. other than Jacquet-Langlands) in special fibers of Shimura varieties? For example: the Saito-Kurokawa l...

open_problem | open | MathDB 页面 | 原文线索

V2-022556 | MATHDB-385959

Can one use cycles (Hodge or Tate) to produce correspondences between eigenvarieties of different inner forms?

摘录 / EXCERPT: Can one use cycles (Hodge or Tate) to produce correspondences between eigenvarieties of different inner forms?

open_problem | open | MathDB 页面 | 原文线索

V2-022557 | MATHDB-385963

Can one write down an explicit correspondence on a Shimura variety whose induced endomorphism on cohomology doesn't come from a Hecke correspondence?

摘录 / EXCERPT: Can one write down an explicit correspondence on a Shimura variety whose induced endomorphism on cohomology doesn't come from a Hecke correspondence?

open_problem | open | MathDB 页面 | 原文线索

V2-022558 | MATHDB-385965

Can you construct classes in motivic cohomology or higher Chow groups of the generic fiber of Shimura varieties?

摘录 / EXCERPT: Can you construct classes in motivic cohomology or higher Chow groups of the generic fiber of Shimura varieties? e.g do Rong Zhou's classes on the special fiber of Shimura varietie...

open_problem | open | MathDB 页面 | 原文线索

V2-022559 | MATHDB-385966

To what extent are the integral Hodge conjecture and integral Tate conjectures true for (... -isotypic components of) Shimura varieties?

摘录 / EXCERPT: To what extent are the integral Hodge conjecture and integral Tate conjectures true for (π-isotypic components of) Shimura varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022560 | MATHDB-385968

What are the irreducible components of affine Deligne-Lusztig varieties outside the fully Hodge-Newton decomposable setting?

摘录 / EXCERPT: What are the irreducible components of affine Deligne-Lusztig varieties outside the fully Hodge-Newton decomposable setting? e.g. for $G = U(3,2)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022561 | MATHDB-386004

Geometry and Topology of Moduli Spaces (problem 2)

摘录 / EXCERPT: Is $\mathcal{M}_d^N[n]$ (the moduli space of maps with marked points of period n) a variety of general type for $n\geq C(d,N)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022562 | MATHDB-386006

Geometry and Topology of Moduli Spaces (problem 4)

摘录 / EXCERPT: Is the class of $\mathcal{M}_d^1$ the same as that of $\Bbb{A}^{2d-2}$ in the Grothendieck ring of varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022563 | MATHDB-386015

Generalization of Complex Dynamical Results to Higher Dimensions (problem 2)

摘录 / EXCERPT: Is a dynamically stable algebraic family of degree d maps $\Bbb{CP}^N\rightarrow\Bbb{CP}^N$ (i.e. a subvariety of ${\rm{End}}^N_d$) which is not Lattès or isotrivial necessarily...

open_problem | open | MathDB 页面 | 原文线索

V2-022564 | MATHDB-386038

Is it possible to define a generating series of special cycles over Shimura varieties related to automorphic representations of higher weight?

摘录 / EXCERPT: Is it possible to define a generating series of special cycles over Shimura varieties related to automorphic representations of higher weight? Do we have similar geometric/arithmet...

open_problem | open | MathDB 页面 | 原文线索

V2-022565 | MATHDB-386054

Central derivative of Rankin--Selberg L-functions on U(n)

摘录 / EXCERPT: Prove an analogue of Problem 1.2 for algebraic cycles.

open_problem | open | MathDB 页面 | 原文线索

V2-022566 | MATHDB-386073

Initial Problem Session (problem 5)

摘录 / EXCERPT: Let X/k be a nice variety, and let $\delta(X/k)=\{d:$ degree d points on X are Zariski dense in $X\}$ be its density degree set. How different can $\min\delta(X/k)$ be from...

open_problem | open | MathDB 页面 | 原文线索

V2-022567 | MATHDB-386089

Brauer Group (problem 1)

摘录 / EXCERPT: What is the (relative to E_n) Brauer group of the category of $K(n)$-local spectra?

open_problem | open | MathDB 页面 | 原文线索

V2-022568 | MATHDB-386093

Pro-etale cohomology of Lubin-Tate space (problem 1)

摘录 / EXCERPT: What is $H^*_\text{proet}(\text{LT}_n, \mathscr{O}^+)$ (e.g. when $n = 1,2$)? What is $H^*_\text{proet}([\text{LT}_n/\mathbf{G}_n], \mathscr{O}^+)$? Is it of finite cohomological d...

open_problem | open | MathDB 页面 | 原文线索

V2-022569 | MATHDB-386102

Torelli group orbits

摘录 / EXCERPT: Classify the finite orbits of $T_{g,n}$ on the character varieties $M(\Sigma_{g,n}, r)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022570 | MATHDB-386105

Fixed points of special elements of the mapping class group

摘录 / EXCERPT: Fix $\gamma \in \mathrm{Mod}(\Sigma_g)$ can we understand the set of fixed points $M(\Sigma_g, 2)(\overline{\mathbb{Z}})^{\gamma}$ of its action on the rank 2 character variety?

open_problem | open | MathDB 页面 | 原文线索

V2-022571 | MATHDB-386106

Examples of MCG-finite irreducible representations of surface groups not arising from abelian varieties

摘录 / EXCERPT: Can we find an examples of an irreducible representation $\rho: \pi_1(\Sigma_{g,n}) \to {\\textrm{GL}}_r(\mathbb{C})$ with finite orbit under $\mathrm{Mod}(\Sigma_{g,n})$ that do n...

open_problem | open | MathDB 页面 | 原文线索

V2-022572 | MATHDB-386109

Stable cohomology of character varieties

摘录 / EXCERPT: Prove (or disprove) that the cohomology of $\mathcal{U}_{G,g}$ stabilizes with respect to g and compute its stable value. A conjecture for the value of the putative stable cohomo...

open_problem | open | MathDB 页面 | 原文线索

V2-022573 | MATHDB-386110

Log-Calabi-Yau compactifications of character varieties

摘录 / EXCERPT: Do there exist “canonical” algebraic compactifications of the character variety $M(\Sigma_{g,n}, r)$ of a punctured surface? It is a conjecture (of Simpson?) that...

open_problem | open | MathDB 页面 | 原文线索

V2-022574 | MATHDB-386111

Predictions of the Relative Fontaine-Mazur and Simpson conjectures (problem 1)

摘录 / EXCERPT: Suppose a flat bundle $(\mathcal{E}, \nabla)$ on a smooth projective variety $X / \overline{\mathbb{Q}}$ is defined over $\overline{\mathbb{{\\mathbb{Q}}}}$ and its associated Higg...

open_problem | open | MathDB 页面 | 原文线索

V2-022575 | MATHDB-386112

Decidability of arithmeticity and motivicty

摘录 / EXCERPT: Let $X / \mathbb{C}$ be a smooth projective variety. Is there an algorithm which given an integral point of the character variety $x \in M_B(X, r)(\mathcal{O}_K)$, where K is a n...

open_problem | open | MathDB 页面 | 原文线索

V2-022576 | MATHDB-386113

Predictions of the Relative Fontaine-Mazur and Simpson conjectures (problem 3)

摘录 / EXCERPT: Let $X / \mathbb{F}_q$ be a smooth projective variety over a finite field. Given a representation...

open_problem | open | MathDB 页面 | 原文线索

V2-022577 | MATHDB-386114

Isolated singularities of the character variety

摘录 / EXCERPT: Let $X / \mathbb{C}$ be a smooth quasi-projective variety and $x \in M_B(X, r)$ an isolated singularity or isolated point of a cohomology jump loci. Is the corresponding local syst...

open_problem | open | MathDB 页面 | 原文线索

V2-022578 | MATHDB-386115

Geometric proof of a theorem of McMullen--Imayoshi-Shiga

摘录 / EXCERPT: Is there a proof of this theorem using primarily algebraic geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-022579 | MATHDB-386312

Problems (problem 4)

摘录 / EXCERPT: Let $X \sim MVN(\theta, I_d)$, where I_d denotes the d-dimensional identity matrix. Consider the problem of estimating θ using estimators of the form $X + h(X)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-022580 | MATHDB-386315

Problems (problem 7)

摘录 / EXCERPT: Consider a Markov queueing system where there are N servers who serve at rate 1, and new customers arrive into the system at a constant rate of λ. When a new customer arr...

open_problem | open | MathDB 页面 | 原文线索

V2-022581 | MATHDB-386330

What tropical objects characterize all tropical RBMs?

摘录 / EXCERPT: What tropical objects (varieties) characterize all tropical RBMs?

open_problem | open | MathDB 页面 | 原文线索

V2-022582 | MATHDB-386356

Modify the above scheme by adding a new mutator' ball to each of player A's urns.

摘录 / EXCERPT: Modify the above scheme by adding a new mutator' ball to each of player A's urns. When this ball is selected, player A invents a new signal and sends that. Player B responds by gue...

open_problem | open | MathDB 页面 | 原文线索

V2-022583 | MATHDB-386382

Can we use mathematical analysis to design better optimization/training schemes?

摘录 / EXCERPT: Can we use mathematical analysis to design better optimization/training schemes?

open_problem | open | MathDB 页面 | 原文线索

V2-022584 | MATHDB-386413

#9 Other (Metropolized) chains for small perimeter

摘录 / EXCERPT: Can we find other Metropolis-Hasting schemes besides UST RevReCom to sample partitions with small perimeters, for instance under the isoparametric ratio or Polsby-Popper score? Is...

open_problem | open | MathDB 页面 | 原文线索

V2-022585 | MATHDB-386439

29 Phase trasition for exponential weighting schemes

摘录 / EXCERPT: If you are exponentially down-weighting partitions based on boundary length, at what value do you see a phase transition between compact and non-compact partitions?

open_problem | open | MathDB 页面 | 原文线索

V2-022586 | MATHDB-386483

Does there exist a scheme parameterizing types of Hopf algebras?

摘录 / EXCERPT: Does there exist a scheme parameterizing types of Hopf algebras? (Nichols algebras would be one of the types.)

open_problem | open | MathDB 页面 | 原文线索

V2-022587 | MATHDB-386496

Specific cases (problem 1)

摘录 / EXCERPT: Take the Verlinde category Ver_p and consider $GL(X)$ for $X \in Ver_p$. Compute the cohomology of this group scheme or other finite group schemes in Ver_p.

open_problem | open | MathDB 页面 | 原文线索

V2-022588 | MATHDB-386497

Specific cases (problem 2)

摘录 / EXCERPT: Let $\mathbf{k}$ be a Noetherian ring and G a reductive group scheme over $\mathbf{k}$. Suppose that G acts on a finitely generated $\mathbf{k}$-algebra A. Is $H^*(G, A)$ fin...

open_problem | open | MathDB 页面 | 原文线索

V2-022589 | MATHDB-386501

Does a rank variety theory exist for quantum complete intersections (e.g.~Nichols algebras under certain conditions)?

摘录 / EXCERPT: Does a rank variety theory exist for quantum complete intersections (e.g.Nichols algebras under certain conditions)? What plays the role of $\mathbf{Z}/p$?

open_problem | open | MathDB 页面 | 原文线索

V2-022590 | MATHDB-386578

Find Casson type invariants (based on character variety representations) of knotted surfaces.

摘录 / EXCERPT: Find Casson type invariants (based on character variety representations) of knotted surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-022591 | MATHDB-386627

Embedding problems and configurations of curves (problem 4)

摘录 / EXCERPT: [HF criterion, Bezout, involutive HF, constraint from algebraic geometry]

open_problem | open | MathDB 页面 | 原文线索

V2-022592 | MATHDB-386748

Lagrangian Floer theory of symmetric products and Hilbert schemes (problem 1)

摘录 / EXCERPT: For a symplectic manifold (X, ω), what is the relationship between $\mathcal{F}^{S_n}(X^n)$ and $\mathcal{F}(\text{Sym}^n(X))$? Here $\mathcal{F}^{S_n}(X^n)$ is (some sort o...

open_problem | open | MathDB 页面 | 原文线索

V2-022593 | MATHDB-386749

Lagrangian Floer theory of symmetric products and Hilbert schemes (problem 2)

摘录 / EXCERPT: Continuing the previous problem, if X is a Weinstein manifold of dimension at least 4, are there (canonical) resolutions of $\text{Sym}^n(X)$ that have well-behaved Floer theor...

open_problem | open | MathDB 页面 | 原文线索

V2-022594 | MATHDB-387005

Isomorphism problems in a variety (problem 1)

摘录 / EXCERPT: Is every Σ^0_2 equivalence relation on ω which is a quotient of $=^{ce}$ actually Δ^0_2? Fix a variety. Consider the equivalence on the c.e. sets by isomorphis...

open_problem | open | MathDB 页面 | 原文线索

V2-022595 | MATHDB-387040

Khovanov homology (problem 3)

摘录 / EXCERPT: For Khovanov homology develop analogues of begin{enumerate}[label=alph)] item The Oblomkov-Rasmussen-Shende conjecture. item Braid varieties. item Hilb$^n(\mathbb{C}^2)$ should be...

open_problem | open | MathDB 页面 | 原文线索

V2-022596 | MATHDB-387044

Hilbert schemes (problem 4)

摘录 / EXCERPT: Relate Hilb$^n(x^2=0)$ to HH of the arc algebra.

open_problem | open | MathDB 页面 | 原文线索

V2-022597 | MATHDB-387053

What is the relation between braid varieties and singular braid varieties?

摘录 / EXCERPT: What is the relation between braid varieties and singular braid varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022598 | MATHDB-387054

Braid varieties (problem 4)

摘录 / EXCERPT: Compute $H^*_T(X(\sigma_1\dots\sigma_{n-1})^m\Delta)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022599 | MATHDB-387055

Are there "braid variety analogues" of projectors?

摘录 / EXCERPT: Are there "braid variety analogues" of projectors?

open_problem | open | MathDB 页面 | 原文线索

V2-022600 | MATHDB-387059

Relate 3-row and more general Springer fibers to bow varieties and link homology.

摘录 / EXCERPT: Relate 3-row and more general Springer fibers to bow varieties and link homology.

open_problem | open | MathDB 页面 | 原文线索

V2-022601 | MATHDB-389239

Conjecture 4.18

摘录 / EXCERPT: Let A be a simple abelian variety over $\mathbb{Q}$, $\exp_A:\mathbb{R}^g\rightarrow A(\mathbb{R})^0$ the exponential map of the Lie group $A(\mathbb{R})^0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-022602 | MATHDB-390463

Chudnovsky’s conjecture and Demailly’s conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let $Z\subset\mathbb{P}^n_k$ be a finite set of points, and let $I=I(Z)$ be its homogeneous radical ideal. For a non...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022603 | MATHDB-390598

Suslin’s cancellation conjecture

摘录 / EXCERPT: For every smooth affine variety X over $\mathbb{R}$ of dimension $d\ge 3$, and every pair of vector bundles E and F on X of rank d, if...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022604 | MATHDB-391727

Does every separated Noetherian scheme have the resolution property?

摘录 / EXCERPT: Open problem. Does every separated Noetherian scheme have the resolution property? Equivalently, can one construct such an X and a coherent sheaf $\mathcal F$ that is not a quoti...

open_problem | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-022605 | MATHDB-391901

Guedj-Trusiani conjecture on local alpha invariants

摘录 / EXCERPT: Let (X,x) be an n-dimensional isolated log terminal singularity. If $\alpha_x^{\mathrm{an}}(X)$ and $\alpha_x^{\mathrm{alg}}(X)$ denote the two local alpha invariants introduce...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022606 | MATHDB-391902

Gross-Joyce-Tanaka wall-crossing conjecture

摘录 / EXCERPT: The conjecture asserts that the Gross-Joyce-Tanaka wall-crossing formula holds for equivariant Donaldson-Thomas 4 invariants of Calabi-Yau 4 categories equipped with framing fu...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022607 | MATHDB-391904

Frey’s height conjecture

摘录 / EXCERPT: For every number field K and every integer $g\ge 1$, there exist constants $C_{K,g},C'_{K,g}>0$ such that every g-dimensional abelian variety A/K satisfies...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022608 | MATHDB-391914

Yuan’s Conjecture 5.4.1 on equidistribution

摘录 / EXCERPT: Let K be a finitely generated field, let X be a quasi-projective variety over K, and let $\overline{L}$ be an adelically metrized line bundle satisfying the hypotheses of Yua...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022609 | MATHDB-391933

Iarrobino-Fröberg Conjecture; conjecture on Weak Lefschetz Property failure for powers of general linear forms

摘录 / EXCERPT: Let $R=\mathbb{K}[x_1,\ldots,x_n]$ be a polynomial ring over a field of characteristic 0, and let $\ell_1,\ldots,\ell_{n+k}\in R_1$ be general linear forms. For integers $d\ge 1$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022610 | MATHDB-392097

Fröberg’s conjecture

摘录 / EXCERPT: For every field k of characteristic zero and all positive integers n,r,d, let $S=k[x_1,\ldots,x_n]$ and let $f_1,\ldots,f_r\in S_d$ be generic homogeneous forms of degree d....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022611 | MATHDB-392106

Zilber-Pink conjecture for curves in abelian varieties

摘录 / EXCERPT: Let A be an abelian variety over a number field, let $V\subseteq A$ be an irreducible algebraic curve not contained in any proper algebraic subgroup of A, and define...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022612 | MATHDB-392276

Multiplicative Thom-Sebastiani property for Bernstein-Sato polynomials of ideals

摘录 / EXCERPT: Let $\mathfrak a\subseteq \mathbb C[x_1,\ldots,x_m]$ and $\mathfrak b\subseteq \mathbb C[y_1,\ldots,y_n]$ be nonzero ideals, and regard them as ideals in...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-022613 | MATHDB-392279

Oprea-Pandharipande question on Segre pushforwards and Lehn’s conjecture

摘录 / EXCERPT: Let $p\colon C\to B$ be a smooth proper family of curves, let L be a line bundle on C, and let $\pi_n\colon C^{[n]}\to B$ be the relative Hilbert scheme of length-n subscheme...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-022614 | MATHDB-392281

V. Lafforgue's constant-sheaf conjecture in geometric Langlands

摘录 / EXCERPT: Let X be a smooth projective curve over an algebraically closed field of characteristic zero, let G be a reductive group, and let $\check G$ be its Langlands dual group. Under...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022615 | MATHDB-392283

Troncoso-Urzúa Chern-slope density conjecture

摘录 / EXCERPT: Let G be the topological fundamental group of a nonsingular complex projective surface. Then the set...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022616 | MATHDB-392302

Generalised Cone Conjecture

摘录 / EXCERPT: The Generalised Cone Conjecture asserts that the cone-conjecture theorem for Calabi--Yau surfaces extends to a broader class of algebraic surfaces beyond the Calabi--Yau case. The...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022617 | MATHDB-392303

Arcara-Miles conjecture

摘录 / EXCERPT: Let X be a connected smooth complex projective surface, let L be a line bundle on X, and let σ be a divisorial Bridgeland stability condition. If L, or the appropria...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-022618 | MATHDB-392322

Multivariable Strong Monodromy Conjecture

摘录 / EXCERPT: For every tuple $F=(f_1,\ldots,f_r)$ of nonzero holomorphic germs on a smooth complex germ $(X,0)$, every actual polar hyperplane H of the local multivariable topological zeta fu...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022619 | MATHDB-392485

Ekedahl-Shepherd-Barron-Taylor conjecture

摘录 / EXCERPT: Let X be a smooth projective complex variety and let $\mathcal{F}$ be a foliation on X. Assume that, for almost all primes p, the reduction of $\mathcal{F}$ modulo p is p...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-022620 | MATHDB-392718

Second-order convergence of collocation schemes for retarded potential integral equations

摘录 / EXCERPT: For the stable collocation schemes for the single-layer retarded potential equation described in the report, using piecewise-linear spatial approximation and piecewise-linear or pi...

open_problem | open | MathDB 页面 | 原文线索

V2-022621 | MATHDB-392727

2-torsion obstructs the decomposition theorem for a type D_4 Schubert variety

摘录 / EXCERPT: Let G be a semisimple complex algebraic group of type D_4, and order the simple reflections so that s_1,s_2,s_3 commute. Set $w=s_1s_2s_3s_4s_3s_2s_1$ and $x=s_1s_2s_3$. For...

open_problem | open | MathDB 页面 | 原文线索

V2-022622 | MATHDB-392729

Alternating signs of K-theoretic quiver coefficients

摘录 / EXCERPT: Let $E_0\to E_1\to\cdots\to E_n$ be a sequence of vector bundles and bundle maps over a nonsingular variety X. Let $r=\{r_{ij}\}$ be occurring rank conditions, with...

open_problem | open | MathDB 页面 | 原文线索

V2-022623 | MATHDB-392751

Claimed Schur-nilpotency theorem for Chow motives

摘录 / EXCERPT: Let X be a smooth projective variety over a field, and suppose that its Chow motive $M(X)$ is Schur-finite. Then every numerically trivial correspondence of degree zero from X...

open_problem | open | MathDB 页面 | 原文线索

V2-022624 | MATHDB-392752

Sign Conjecture for algebraic cycles

摘录 / EXCERPT: For every smooth projective variety X over a field, the diagonal class $ ... where $[3mpiX^+$ and $[3mpiX^-$ are orthogonal projectors, with $[3mpiX^+$ carrying the even part and $...

conjecture | open | MathDB 页面 | 原文线索

V2-022625 | MATHDB-392754

Existence and uniqueness of the p-adic étale Tate twist

摘录 / EXCERPT: Let X be a regular scheme flat over an algebraic integer ring O, let p be a prime, let $n\ge 0$, and let $r>0$. If X is smooth or semistable over $\operatorname{Spec}(O)$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-022626 | MATHDB-392764

Entropy stability of a semi-discrete conservative scheme

摘录 / EXCERPT: Let $u_ u(t)$ be a semi-discrete conservative approximation ... to the system ... where $f=(f_1,\ldots,f_N)^\mathsf{T}$ is a smooth flux and (U,F) is an entropy pair for the system....

open_problem | open | MathDB 页面 | 原文线索

V2-022627 | MATHDB-392779

Extension of Yamanoi's truncated Second Main Theorem to semi-abelian varieties

摘录 / EXCERPT: Extend (1.3) to the case of semi-abelian varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-022628 | MATHDB-392781

Geometric reasons for strict inequality between numerical and Kodaira dimensions

摘录 / EXCERPT: For a nef line bundle L on a smooth projective variety X, are there geometric reasons for the strict inequality $\kappa(X,L)<\nu(X,L)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022629 | MATHDB-392785

Fujita's k-very ampleness conjecture

摘录 / EXCERPT: Let H be an ample divisor on a smooth projective variety X of dimension n. Then the adjoint line bundle $\omega_X(mH)$ is k-very ample if $m>k+n$, or if $m=k+n$ and $H^n>1$...

conjecture | open | MathDB 页面 | 原文线索

V2-022630 | MATHDB-392789

Finiteness of rational points on complex two-ball quotients

摘录 / EXCERPT: Let M be an algebraic variety over a number field k such that the underlying complex variety is a smooth complex two-ball quotient. Then the cardinality of $M(k)$ is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-022631 | MATHDB-392815

Quantum modification of the congruent zeta function for mirror pairs

摘录 / EXCERPT: Can a "quantum modification" of the congruent zeta function be defined so that the zeta functions of mirror pairs of Calabi-Yau varieties are inverses of one another?

open_problem | open | MathDB 页面 | 原文线索

V2-022632 | MATHDB-392818

Nakamura's conjecture on the three-dimensional G-Hilbert scheme

摘录 / EXCERPT: For every finite subgroup $G\subset SL(3,\mathbb{C})$, the G-Hilbert scheme $\operatorname{Hilb}_G(\mathbb{C}^3)$ is a crepant resolution of the quotient $\mathbb{C}^3/G$. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-022633 | MATHDB-392850

PTAS claim for the ε-dense Group Steiner Tree Problem

摘录 / EXCERPT: For every fixed $\\varepsilon>0$, the $\\varepsilon$-dense Group Steiner Tree Problem on unweighted graphs admits a polynomial-time approximation scheme. Here an instance consists...

open_problem | open | MathDB 页面 | 原文线索

V2-022634 | MATHDB-392884

Correspondences between minimal-degree varieties in projective and abelian geometry

摘录 / EXCERPT: Is there a geometric correspondence relating rational normal scrolls to W_d's and Veronese surfaces to Fano surfaces of lines on cubic threefolds? What are the similar properties...

open_problem | open | MathDB 页面 | 原文线索

V2-022635 | MATHDB-392889

D-equivalence versus K-equivalence

摘录 / EXCERPT: Let $f: Z \to X$ and $g: Z \to Y$ be birational morphisms between smooth projective varieties such that the pull-backs of canonical divisors are equal, $f^*K_X = g^*K_Y$. Then the...

open_problem | open | MathDB 页面 | 原文线索

V2-022636 | MATHDB-392890

Log K-equivalence and derived-category equivalence for KLT pairs

摘录 / EXCERPT: Let (X,B) and (Y,C) be pairs of projective varieties with $\mathbb{Q}$-divisors which are KLT. Assume that there are birational morphisms $f:Z\to X$ and $g:Z\to Y$ from a smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-022637 | MATHDB-392892

Littlewood-Richardson rules for other types of Grassmannians and flag varieties

摘录 / EXCERPT: Provide an LR rule for other types of Grassmannians/flag varieties (i.e. of types B,C,D,E,F and G).

open_problem | open | MathDB 页面 | 原文线索

V2-022638 | MATHDB-392893

Quantum Littlewood-Richardson rule for flag varieties

摘录 / EXCERPT: Provide a quantum LR rule for flag varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-022639 | MATHDB-392915

Property (N1) for ample line bundles on smooth toric varieties

摘录 / EXCERPT: Does every ample line bundle on a smooth toric variety satisfy property $(N1)$?

open_problem | open | MathDB 页面 | 原文线索

V2-022640 | MATHDB-392929

Uniqueness of a Hasse-derivation structure on a projective variety

摘录 / EXCERPT: If D is a Hasse derivation, then there is at most one D-structure, up to D-isomorphism, on a projective variety.

open_problem | open | MathDB 页面 | 原文线索

V2-022641 | MATHDB-392958

Purity for the Brauer group of a punctured strictly henselian regular local scheme

摘录 / EXCERPT: Let R be a strictly henselian regular local ring, let $U_R=\operatorname{Spec}(R)\setminus\{\mathfrak m\}$ be its punctured spectrum, and let $\operatorname{Br}'(X)$ denote the t...

open_problem | open | MathDB 页面 | 原文线索

V2-022642 | MATHDB-392961

Period-index conjecture for function fields of varieties

摘录 / EXCERPT: Conjecture. If K is the function field of an n-dimensional variety over an algebraically closed field, then for any $\b1\in \operatorname{Br}(K)$ one has...

conjecture | open | MathDB 页面 | 原文线索

V2-022643 | MATHDB-392965

Guiding principle for rational sections over surfaces

摘录 / EXCERPT: Suppose that $f:Y\to S$ is a dominant projective morphism of varieties, where S is a surface and the generic fibre $Y_{k(S)}$ is rationally 1-connected. The guiding principle p...

open_problem | open | MathDB 页面 | 原文线索

V2-022644 | MATHDB-392966

Fano varieties with negative higher-order Fano condition

摘录 / EXCERPT: Let X be a smooth projective variety over $\mathbb{C}$ with $\operatorname{Pic}(X)=\mathbb{Z}\mathcal{O}_X(1)$. Suppose that X is Fano and that...

open_problem | open | MathDB 页面 | 原文线索

V2-022645 | MATHDB-392967

Generalized André-Pink-Zannier conjecture for mixed Shimura varieties

摘录 / EXCERPT: Let S be a mixed Shimura variety over $ C$ and $ Lambda \subset S$ the generalized Hecke orbit of a point $s\in S$. Let $Z\subset S$ be an irreducible closed algebraic subvariety...

conjecture | open | MathDB 页面 | 原文线索

V2-022646 | MATHDB-392968

Motivic degeneration theorem for a boundary stratum of a Shimura variety

摘录 / EXCERPT: There is a canonical isomorphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-022647 | MATHDB-392973

André-Oort conjecture for special subvarieties

摘录 / EXCERPT: Let S be a Shimura variety and let X be a subvariety of S. There exists a finite set of special subvarieties $(Z_1,\ldots,Z_r)$ of S, each contained in X, such that every...

conjecture | open | MathDB 页面 | 原文线索

V2-022648 | MATHDB-392985

Classification of local Todd-class coefficient functions

摘录 / EXCERPT: Classify all functions μ that give a local formula for the Todd class, meaning that for every fan Σ they satisfy ... Here X_Σ is the toric variety associated with...

open_problem | open | MathDB 页面 | 原文线索

V2-022649 | MATHDB-393002

Uniform bound for lines through arbitrary points of a projective variety

摘录 / EXCERPT: Let X be an irreducible projective variety of dimension n in a projective space. Is the number of lines lying on X through any given point of X either infinite or bounded b...

open_problem | open | MathDB 页面 | 原文线索

V2-022650 | MATHDB-393090

Weak version of Zagier’s conjecture for elliptic curves

摘录 / EXCERPT: Let K be a number field and let B be a smooth, quasi-projective, connected scheme over K or $\d4aa_K$. Let $\d4c0:E\to B$ be an elliptic curve. For $k\ge 2$, set ... where P...

conjecture | open | MathDB 页面 | 原文线索

V2-022651 | MATHDB-393160

Existence and uniqueness for the reflection-transmission problem under an abstract flux scheme

摘录 / EXCERPT: For the abstract wave-propagation system $w'(z)=A(z;\omega)w$, with $w\in\mathbb{C}^{2m}$ and real parameter ω, suppose there is a Hermitian matrix I such that $F=w^*Iw$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-022652 | MATHDB-393217

Error estimate for the fully discrete mixed finite-element scheme for Richards’ equation

摘录 / EXCERPT: Assuming, in addition, that the flux satisfies $q\in H^1(\Omega)^d$ rather than only $q\in H(\operatorname{div},\Omega)$, the fully discrete mixed finite-element scheme (4)-(5) sat...

open_problem | open | MathDB 页面 | 原文线索

V2-022653 | MATHDB-393233

Full strong exceptional sequences of line bundles on smooth toric varieties

摘录 / EXCERPT: Does every smooth toric variety admit a full strong exceptional sequence of line bundles, similar to the sequence on $\mathbb{P}^n$? The report states that this was even open for t...

open_problem | open | MathDB 页面 | 原文线索

V2-022654 | MATHDB-393234

Classification of spherical varieties

摘录 / EXCERPT: The classification of spherical varieties is a long-standing open question.

open_problem | open | MathDB 页面 | 原文线索

V2-022655 | MATHDB-393237

Lang-Manin congruence for Fano varieties over finite fields

摘录 / EXCERPT: Let X be a Fano variety defined over $\\mathbb{F}_q$. Then $\lvert X(\mathbb{F}_q)\rvert \equiv 1 \pmod q$.

open_problem | open | MathDB 页面 | 原文线索

V2-022656 | MATHDB-393240

Inversion of adjunction for minimal log discrepancies on normal \Q-Gorenstein varieties

摘录 / EXCERPT: Conjecture: Let X be a normal $\Q$-Gorenstein variety, let Y be a proper closed subscheme of X, and let $\lambda>0$. If $D\subset X$ is a normal effective Cartier divisor wit...

open_problem | open | MathDB 页面 | 原文线索

V2-022657 | MATHDB-393243

Heier’s de Franchis problem for curves

摘录 / EXCERPT: Let X and Y be smooth complex projective varieties of general type of dimension n. Define $M(X,Y)=\{f:X\dashrightarrow Y\mid f\text{ is a rational dominant map}\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-022658 | MATHDB-393244

Holomorphic one-forms on varieties of general type

摘录 / EXCERPT: Let X be a smooth projective variety of general type. Then X does not admit a nowhere vanishing holomorphic one-form.

open_problem | open | MathDB 页面 | 原文线索

V2-022659 | MATHDB-393248

Modularity of smooth varieties and Calabi-Yau varieties

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n_{\mathbb{Z}}$ be a smooth variety defined over the integers, and let n be its complex dimension. For every prime p of good reduction, let...

open_problem | open | MathDB 页面 | 原文线索

V2-022660 | MATHDB-393255

Unconditional instability of an explicit partitioned fluid-structure scheme

摘录 / EXCERPT: For the axisymmetric potential-flow/thin-elastic-tube model on $\Omega=\{(z,r)\in\mathbb{R}^2:0<z<L,\ 0<r<R\}$, with interface $\Sigma=\overline\Omega\cap\{r=R\}$, the explicit par...

open_problem | open | MathDB 页面 | 原文线索

V2-022661 | MATHDB-393300

Decidability of MSO theories for level-2 recursion-scheme trees without safety

摘录 / EXCERPT: The monadic second-order (MSO) theory of every infinite tree generated by a level-2 grammar is decidable; the previously imposed “safety” condition can be dropped.

open_problem | open | MathDB 页面 | 原文线索

V2-022662 | MATHDB-393313

Generation of the motivic Steenrod algebra by Voevodsky’s reduced power operations

摘录 / EXCERPT: For a prime p and a field k with $p\nmid\operatorname{char}(k)$, do Voevodsky’s stable bigraded operations ... generate, over $H^{*,*}_{\mathrm{mot}}(k,\mathbb Z/p)$, the bigrade...

open_problem | open | MathDB 页面 | 原文线索

V2-022663 | MATHDB-393314

Construction of functorial pull-backs in algebraic cobordism

摘录 / EXCERPT: Construct functorial pull-back maps for algebraic cobordism for morphisms between smooth schemes, reducing to the construction for a strict normal-crossing divisor...

open_problem | open | MathDB 页面 | 原文线索

V2-022664 | MATHDB-393394

Complexity reduction for the Hilbert polynomial of smooth equidimensional complex projective varieties

摘录 / EXCERPT: The problem $\mathrm{Hilbert}_{\mathrm{sm}}$ of computing the Hilbert polynomial of a smooth equidimensional complex projective variety can be reduced in polynomial time to...

open_problem | open | MathDB 页面 | 原文线索

V2-022665 | MATHDB-393413

Monodromy-weight map for a p-adically uniformized variety

摘录 / EXCERPT: In the setup of pages 10-11, let X be the smooth projective variety over a finite extension K of $\mathbb{Q}_p$ associated with a quotient of Drinfeld’s p-adic symmetric doma...

open_problem | open | MathDB 页面 | 原文线索

V2-022666 | MATHDB-393414

Finite generation of arithmetic cohomology groups

摘录 / EXCERPT: For a variety X over a finite field $\mathbb F_q$, with coefficients in the motivic complex $\mathbb Z(n)$, the cohomology groups $H_c^i(X_{\overline{\mathbb F}_q},\mathbb Z(n))$...

open_problem | open | MathDB 页面 | 原文线索

V2-022667 | MATHDB-393415

Torsion of higher Chow groups of smooth projective varieties over finite fields

摘录 / EXCERPT: For a fixed integer d, is $CH_0(S,i)$ a torsion group for every smooth and projective variety S over a finite field, of dimension at most d, and every integer $i>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-022668 | MATHDB-393416

Parshin’s conjecture for smooth projective schemes over finite fields

摘录 / EXCERPT: For every smooth and projective scheme X over a finite field $\F_q$, and every integer $i>0$, the algebraic K-group $K_i(X)$ is torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-022669 | MATHDB-393419

Relative semisimple monodromy representation for a Frobenius connection

摘录 / EXCERPT: Let K be a complete discretely valued field with algebraically closed residue field, let P be a smooth formal scheme over $\o_K$, and let $\mathcal Q$ be a formal disc bundle o...

open_problem | open | MathDB 页面 | 原文线索

V2-022670 | MATHDB-393479

Determining the free rank of a Mordell-Weil group

摘录 / EXCERPT: Problem: Determine the free rank of the Mordell-Weil group $J(K)$ of an abelian variety J over a number field K.

open_problem | open | MathDB 页面 | 原文线索

V2-022671 | MATHDB-393607

p-adic equidistribution of points of small height

摘录 / EXCERPT: Let F be a number field, let X be a projective variety over F, and let $\bar L$ be an ample line bundle on X endowed with a semipositive adelic metric. Fix a place v of...

open_problem | open | MathDB 页面 | 原文线索

V2-022672 | MATHDB-393608

Arithmetic intersection theory on regular Deligne-Mumford stacks over \(\operatorname{Spec}(\mathbb{Z})\)

摘录 / EXCERPT: For every regular Deligne-Mumford stack X flat and proper over $\operatorname{Spec}(\mathbb{Z})$, with rational coefficients, there is a canonical isomorphism ... where the inverse...

open_problem | open | MathDB 页面 | 原文线索

V2-022673 | MATHDB-393671

Switching/shunting problem

摘录 / EXCERPT: Given a set of stacks of a given maximum depth (tracks); an ordered list of items (cars), one for each stack; a set of sets of items, with each set identified by its color (request...

open_problem | open | MathDB 页面 | 原文线索

V2-022674 | MATHDB-393694

Kawamata’s conjecture: birational varieties with trivial canonical class are D-equivalent

摘录 / EXCERPT: Kawamata’s conjecture, in the form stated in the report, asserts that birational smooth projective varieties with trivial canonical class should be D-equivalent; equivalently, for...

conjecture | open | MathDB 页面 | 原文线索

V2-022675 | MATHDB-393699

Presentation-independence of string theories on stacks under renormalization-group flow

摘录 / EXCERPT: We conjecture that, after applying the renormalization group, the different physical theories associated to different presentations of a single stack all evolve to the same physica...

open_problem | open | MathDB 页面 | 原文线索

V2-022676 | MATHDB-393777

A nonnormal G-Hilbert scheme

摘录 / EXCERPT: There is a subgroup $G\subseteq \mathrm{GL}(6,\mathbb{C})$ isomorphic to $(\mathbb{Z}/5\mathbb{Z})^4$ for which $G\text{-Hilb}$ is not normal.

open_problem | open | MathDB 页面 | 原文线索

V2-022677 | MATHDB-393782

Quadratic generation of defining ideals of smooth projective toric varieties

摘录 / EXCERPT: Let P be a lattice polytope whose corresponding projective toric variety is smooth. Is the defining ideal I_P generated by quadrics?

open_problem | open | MathDB 页面 | 原文线索

V2-022678 | MATHDB-393786

King’s conjecture for smooth compact toric varieties

摘录 / EXCERPT: For every smooth compact toric variety X, there exists a tilting bundle T that is a direct sum of line bundles, $T=\bigoplus_{i=1}^{n}L_i$, where n is the rank of the Grothen...

conjecture | open | MathDB 页面 | 原文线索

V2-022679 | MATHDB-393787

Versal deformations of tents

摘录 / EXCERPT: We would like to know what the versal deformation space of a tent is-in the example above, is there a larger variety still containing all four toric deformations?

open_problem | open | MathDB 页面 | 原文线索

V2-022680 | MATHDB-393795

Strong Torsion Conjecture

摘录 / EXCERPT: Given integers $g,d\ge 1$, there exists an integer $n(d,g)\ge 1$ such that there is no abelian variety of dimension at most g, defined over a number field of degree at most d,...

conjecture | open | MathDB 页面 | 原文线索

V2-022681 | MATHDB-393807

Rank of abelian varieties over maximal abelian extensions

摘录 / EXCERPT: Is $\operatorname{rk}(A(K^{\mathrm{ab}}))=\infty$ for any non-zero abelian variety A over a number field K? Here K^{ab} denotes the maximal abelian extension of K.

open_problem | open | MathDB 页面 | 原文线索

V2-022682 | MATHDB-393808

Realizing abelian varieties as quotients of Jacobians of abelian Galois covers

摘录 / EXCERPT: Does every abelian variety arise as a quotient of the Jacobian J_T of an abelian Galois cover $T\to\mathbb{P}^1$?

open_problem | open | MathDB 页面 | 原文线索

V2-022683 | MATHDB-393809

Ranks of quadratic twists of abelian varieties over \(\mathbb{Q}\)

摘录 / EXCERPT: Let $A/\mathbb{Q}$ be a non-zero abelian variety. For $D\in\mathbb{Q}^{\times}/2$, denote by $A_D/\mathbb{Q}$ the twist of A by the extension $\mathbb{Q}(\sqrt D)/\mathbb{Q}$. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-022684 | MATHDB-393812

Simply connected subvarieties of the moduli space of curves in positive characteristic

摘录 / EXCERPT: Does the moduli space (or stack) $\mathcal{M}_g$ of smooth, projective, geometrically connected curves of genus $g\ge 2$ contain simply connected subvarieties in positive character...

open_problem | open | MathDB 页面 | 原文线索

V2-022685 | MATHDB-393842

Realizability of algebraic cones as complete-intersection support varieties

摘录 / EXCERPT: Let R be the complete local complete-intersection ring with algebraically closed residue field k and codimension c fixed in the report, and let $V_R^*(M)\subseteq k^c$ denote...

open_problem | open | MathDB 页面 | 原文线索

V2-022686 | MATHDB-393852

Full strongly exceptional sequences of line bundles on smooth toric varieties

摘录 / EXCERPT: Every smooth toric variety admits a full strongly exceptional sequence of line bundles. Here a sequence $\varepsilon=(L_1,\ldots,L_t)$ is strongly exceptional when it is exceptiona...

open_problem | open | MathDB 页面 | 原文线索

V2-022687 | MATHDB-393913

Variety criterion for objects in the homotopy category of injective kG-complexes

摘录 / EXCERPT: The report states the following theorem. Let G be a finite p-group, let X be an object of $K(\\operatorname{Inj} kG)$, and let $\mathfrak p$ be a homogeneous prime ideal of...

open_problem | open | MathDB 页面 | 原文线索

V2-022688 | MATHDB-393962

Existence of full exceptional sequences of line bundles on smooth projective toric varieties

摘录 / EXCERPT: For every smooth projective toric variety X, there exists a sequence of line bundles $\epsilon=(L_1,\ldots,L_n)$, with $n=\operatorname{rank}K_0(X)$, such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022689 | MATHDB-393976

Leading-term interpolation formula for a motive attached to an abelian variety

摘录 / EXCERPT: Theorem (as reported). Let $M=h^1(A)(1)$ be the motive of an abelian variety A over $\boldsymbol{Q}$ with good ordinary reduction at a fixed prime $p\ne 2$. Assume that the archi...

open_problem | open | MathDB 页面 | 原文线索

V2-022690 | MATHDB-394012

Termination conjecture for flips of klt pairs

摘录 / EXCERPT: There does not exist an infinite sequence of flips ... projective over a fixed variety S, starting from a klt pair (X,B) whose boundary B is an $\mathbb{R}$-divisor.

conjecture | open | MathDB 页面 | 原文线索

V2-022691 | MATHDB-394014

Codimension of higher-multiplicity loci of indecomposable theta divisors

摘录 / EXCERPT: For an indecomposable principally polarized abelian variety (A,Θ) of dimension g over the complex numbers, is it true that ... for every integer $k\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-022692 | MATHDB-394016

Generic smoothness of Hom schemes to projective homogeneous spaces

摘录 / EXCERPT: Let G/P be a projective homogeneous space, e.g., a Grassmann variety, and let X be a curve with general modulus. Is every component of $\operatorname{Hom}(X,G/P)$ generically s...

open_problem | open | MathDB 页面 | 原文线索

V2-022693 | MATHDB-394019

Relative and degenerate stable maps for normal-crossings divisors

摘录 / EXCERPT: The problem set out to solve is: (1) define relative stable maps to (Y,D), where D is a normal-crossings divisor on a smooth variety Y; (2) define degenerate stable maps to a...

open_problem | open | MathDB 页面 | 原文线索

V2-022694 | MATHDB-394023

Viehweg’s hyperbolicity conjecture

摘录 / EXCERPT: Let $f^{\circ}: X^{\circ} \to S^{\circ}$ be a smooth family of canonically polarized varieties. If f° is of maximal variation, then S° is of log general type.

conjecture | open | MathDB 页面 | 原文线索

V2-022695 | MATHDB-394031

Decomposition of Chow motives into indecomposable objects

摘录 / EXCERPT: For a smooth projective variety X over a field F, how can its Chow motive $M(X)$ be expressed as a direct sum of indecomposable objects?

open_problem | open | MathDB 页面 | 原文线索

V2-022696 | MATHDB-394048

Computing generators of invariant rings

摘录 / EXCERPT: Let G be a linear algebraic group acting on an affine variety X by a morphism $G\times X\to X$. Assuming that the invariant ring $K[X]^G$ is finitely generated as a K-algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-022697 | MATHDB-394067

Weak Bloch-Beilinson conjecture for zero-cycles over p-adic fields

摘录 / EXCERPT: Conjecture 1. Let V be a smooth projective variety over a p-adic field k. Then $\operatorname{Ker}(\phi_V)$ is the direct sum of a finite group and a divisible group.

conjecture | open | MathDB 页面 | 原文线索

V2-022698 | MATHDB-394073

Lichtenbaum’s DBS determinant conjecture for special values of motivic L-functions

摘录 / EXCERPT: For a motive M over $\Q$ with L-function $L_M(s)$, let $H_f^0(M)$, $H_f^1(M)$, $H_c^1(M)$, $H_c^2(M)$, $H_B(M)^+$, and t_M be the finite-dimensional $\Q$-vector spaces associ…

conjecture | open | MathDB 页面 | 原文线索

V2-022699 | MATHDB-394141

Viehweg’s conjecture for two-dimensional bases

摘录 / EXCERPT: In the two-dimensional-base case, let $f^{\circ}:X^{\circ}\to S^{\circ}$ be a smooth family of canonically polarized varieties. If f° is of maximal variation, then…

conjecture | open | MathDB 页面 | 原文线索

V2-022700 | MATHDB-394142

Quotient singularities of the variety $S(X)$

摘录 / EXCERPT: Is it true that $S(X)$ has only quotient singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-022701 | MATHDB-394146

Pluricanonical nonvanishing or rational curves

摘录 / EXCERPT: Let X be a smooth projective variety. Then either (1) $H^0(X, \mathcal{O}_X(mK_X))\neq 0$ for some $m>0$, or (2) X is covered by rational curves.

open_problem | open | MathDB 页面 | 原文线索

V2-022702 | MATHDB-394151

Viehweg’s conjecture for families over quasi-projective surfaces

摘录 / EXCERPT: Let S° be a smooth quasi-projective complex surface and $f^\circ:X^\circ\to S^\circ$ a smooth non-isotrivial family of canonically polarized complex varieties. Then: 1. If…

conjecture | open | MathDB 页面 | 原文线索

V2-022703 | MATHDB-394152

Generalized Viehweg conjecture for smooth families of canonically polarized varieties

摘录 / EXCERPT: Let $f^{\circ}:X^{\circ}\to S^{\circ}$ be a smooth family of canonically polarized varieties. Then either $\kappa(S^{\circ})=-\infty$ and…

conjecture | open | MathDB 页面 | 原文线索

V2-022704 | MATHDB-394154

Generalized Mukai conjecture for smooth Fano varieties

摘录 / EXCERPT: For every smooth Fano variety X of dimension n, with Picard number $\rho(X)$ and pseudo-index $\\iota_X$, one has … with equality if and only if …

conjecture | open | MathDB 页面 | 原文线索

V2-022705 | MATHDB-394169

Cohomology classes of singularity and multisingularity loci for families of maps of curves

摘录 / EXCERPT: Given a diagram of complex nonsingular varieties … with $\dim X=\dim Y=\dim B+1$, each fiber of p a nodal complex curve, each fiber of q smooth, f proper, and p restricted…

open_problem | open | MathDB 页面 | 原文线索

V2-022706 | MATHDB-394201

Weak two-probe conjecture for static set membership

摘录 / EXCERPT: Let m be sufficiently large, let $n=\lfloor m^{1/3}\rfloor$, and let $\varphi:([m]^{\le n})\to\{0,1\}^s$ be a scheme representing subsets of $[m]$ as bit strings. The report conj…

conjecture | open | MathDB 页面 | 原文线索

V2-022707 | MATHDB-394285

Splitting off a free summand when the top Chern class vanishes

摘录 / EXCERPT: Let $X=\operatorname{Spec}A$ be a smooth affine variety of dimension $n\ge 2$ over a field k, and let P be a projective A-module of rank n. Does $C_n(P)=0$ imply that P h…

open_problem | open | MathDB 页面 | 原文线索

V2-022708 | MATHDB-394287

Splitting of top-rank projective modules over smooth real affine varieties

摘录 / EXCERPT: Let $X=\operatorname{Spec}A$ be a smooth affine variety of dimension $n\ge 2$ over $\mathbb{R}$, and let P be a projective A-module of rank n. Under what further restrictions…

open_problem | open | MathDB 页面 | 原文线索

V2-022709 | MATHDB-394288

Splitting of top-rank projective modules in odd characteristic-zero dimension

摘录 / EXCERPT: Let $X=\operatorname{Spec} A$ be a smooth affine variety of odd dimension $n\ge 2$ over a field k of characteristic 0. Let P be a projective A-module of rank n such that…

open_problem | open | MathDB 页面 | 原文线索

V2-022710 | MATHDB-394302

Zaidenberg’s question on exotic affine algebraic varieties

摘录 / EXCERPT: Is there a complex affine algebraic variety biholomorphic to C^n but not isomorphic to C^n?

open_problem | open | MathDB 页面 | 原文线索

V2-022711 | MATHDB-394307

Extension of the fiberwise-isomorphism proposition to arbitrary fields

摘录 / EXCERPT: Let k be an algebraically closed field of infinite transcendence degree over its prime field. Let $p:S\to X$ and $q:T\to X$ be morphisms of k-varieties. Assume that for every…

open_problem | open | MathDB 页面 | 原文线索

V2-022712 | MATHDB-394311

Equivariant Jacobian Conjecture

摘录 / EXCERPT: Let X be a normal affine variety defined over $\bb C$ with an effective algebraic action of an algebraic group G, and let $\cvarphi:X\to X$ be an unramified endomorphism which…

conjecture | open | MathDB 页面 | 原文线索

V2-022713 | MATHDB-394485

Bounded Height Conjecture for Subvarieties of a Multiplicative Torus

摘录 / EXCERPT: Let $X\subset \mathbb{G}_m^n$ be an irreducible closed subvariety defined over $\mathbb{Q}$. The set $X^{\mathrm{oa}}(\mathbb{Q})\cap (\mathbb{G}_m^n)^{[\dim X]}$ has uniformly bou...

conjecture | open | MathDB 页面 | 原文线索

V2-022714 | MATHDB-394498

Integral points on finite covers of semi-abelian varieties

摘录 / EXCERPT: Conjecture. Let G be a semi-abelian variety defined over a number field k; let V be a smooth variety, and let $[U+007F]\pi: V \to G$ be a finite morphism, all defined over k. If t...

open_problem | open | MathDB 页面 | 原文线索

V2-022715 | MATHDB-394500

Luna’s spherical-system classification conjecture

摘录 / EXCERPT: Conjecture. Spherical systems classify wonderful varieties.

conjecture | open | MathDB 页面 | 原文线索

V2-022716 | MATHDB-394574

Levin’s currents and the topological realization of the polylogarithm

摘录 / EXCERPT: For a complex abelian scheme, the currents constructed by Levin provide an explicit description of the polylogarithm at the topological level.

open_problem | open | MathDB 页面 | 原文线索

V2-022717 | MATHDB-394575

Birch-Swinnerton-Dyer rank conjecture for an abelian variety

摘录 / EXCERPT: Let F be a finite extension of $\d53d$ and let A be an abelian variety of dimension g defined over F. The L-function $L(A/F,s)$ of A has an analytic continuation to...

conjecture | open | MathDB 页面 | 原文线索

V2-022718 | MATHDB-394578

Non-emptiness and dimension of affine Deligne-Lusztig varieties in the shrunken Weyl chambers

摘录 / EXCERPT: Let b be basic, for example $b=1$, and let $x\in\widetilde W$ lie in the shrunken Weyl chambers. Then $X_x(b)$ is non-empty if and only if $\eta_2(x)^{-1}\eta_1(x)\eta_2(x)$ is n...

open_problem | open | MathDB 页面 | 原文线索

V2-022719 | MATHDB-394581

Statistically hiding commitments from one-way functions

摘录 / EXCERPT: Assuming that one-way functions exist, there exists a statistically-hiding and computationally-binding commitment scheme.

open_problem | open | MathDB 页面 | 原文线索

V2-022720 | MATHDB-394623

Chabauty in higher dimension for varieties with injective or birational Albanese map

摘录 / EXCERPT: For a variety X of dimension at least 2, if its Albanese map $j:X\to\operatorname{Alb}(X)$ is injective, or if $j(X)$ is merely birational to X, can Chabauty’s method enable...

open_problem | open | MathDB 页面 | 原文线索

V2-022721 | MATHDB-394646

Lipschitz stability of median interpolating subdivision

摘录 / EXCERPT: Determine whether the median interpolating subdivision scheme is Lipschitz stable, in the sense that there exists a finite constant C such that, for sufficiently close initial da...

open_problem | open | MathDB 页面 | 原文线索

V2-022722 | MATHDB-394693

Refuted resonance-critical-variety codimension conjecture

摘录 / EXCERPT: For a hyperplane arrangement $\mathcal A$ with complement M, and a one-form $\omega_\lambda\in A^1(\mathcal A)$, let $\Sigma_\lambda=V(\omega_\lambda)\subset M$. The conjecture w...

conjecture | open | MathDB 页面 | 原文线索

V2-022723 | MATHDB-394698

Conditions for an equality involving the BNS invariant and first resonance variety

摘录 / EXCERPT: Give conditions on a finitely generated group G such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-022724 | MATHDB-394700

Dimensions of translated components of characteristic varieties of line arrangements

摘录 / EXCERPT: Let $\mathcal A$ be a line arrangement in $\mathbb P^2$, and let $G=\pi_1!\left(\mathbb P^2\setminus\bigcup_{\ell\in\mathcal A}\ell\right)$. Prove or provide a counterexample to t...

open_problem | open | MathDB 页面 | 原文线索

V2-022725 | MATHDB-394734

Quasi-isomorphism between periodic cyclic chains and global twisted equivariant cohomology

摘录 / EXCERPT: For a quotient stack $[M/G]$, where G is a compact Lie group and $\b1 \b2 H^3_G(M,\mathbb{Z})$, the explicitly constructed chain map from the periodic cyclic chain complex of...

open_problem | open | MathDB 页面 | 原文线索

V2-022726 | MATHDB-394766

Existence of minimal models or Mori fibre spaces

摘录 / EXCERPT: Let X be a smooth projective variety. Then there exists a K_X-negative birational map $f\colon X\dashrightarrow Y$, whose inverse does not contract any divisors, such that Y...

open_problem | open | MathDB 页面 | 原文线索

V2-022727 | MATHDB-394781

A Gromov-bubbling formulation of the Bloch principle

摘录 / EXCERPT: Suppose that for a quasi-projective variety X with boundary ∂ there is a Zariski subset Z of X such that every entire map $f:\mathbb{C}\to X$ factors through Z. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-022728 | MATHDB-394816

Paramodular Conjecture for abelian varieties with real multiplication

摘录 / EXCERPT: Let $N\in\mathbb{N}$ be odd and squarefree. Let K be a totally real number field of degree d over $\mathbb{Q}$, and let $\mathcal{O}$ be its ring of integers. There is a biject...

conjecture | open | MathDB 页面 | 原文线索

V2-022729 | MATHDB-394864

Quantum message-passing decoding on non-trees

摘录 / EXCERPT: Can a quantum version of this algorithm-i.e. a scheme that passes qudits instead of the original distributions along the edges of the graph-be proven to work correctly for non-tree...

open_problem | open | MathDB 页面 | 原文线索

V2-022730 | MATHDB-394869

Shellability of the space of tropically collinear points

摘录 / EXCERPT: Develin conjectured that the tropical variety $T_{d,n}$ of the determinantal ideal generated by the 3×3 minors of a $d\times n$ matrix is shellable for all d and n. The...

open_problem | open | MathDB 页面 | 原文线索

V2-022731 | MATHDB-394870

Secant dimensions of low-dimensional homogeneous varieties

摘录 / EXCERPT: The secant varieties of $(\mathbb{P}^1)^i$ for $i=1,2,3$, $\mathbb{P}^2$, and $\mathbb{P}^1\times\mathbb{P}^2$, in all equivariant embeddings, are as expected, with an explicit lis...

open_problem | open | MathDB 页面 | 原文线索

V2-022732 | MATHDB-394933

Finiteness properties of positive-characteristic S-arithmetic groups

摘录 / EXCERPT: In the setup where G is a non-commutative, absolutely almost simple group scheme over the global function field K, S is a non-empty set of places, $\mathcal O_S$ is the ring...

open_problem | open | MathDB 页面 | 原文线索

V2-022733 | MATHDB-394934

Nonemptiness and dimension of affine Deligne-Lusztig varieties in shrunken Weyl chambers

摘录 / EXCERPT: In the setup of a split semisimple simply connected group with affine Weyl group W_a, finite Weyl group W, finite simple-reflection set S, and base alcove a, define...

open_problem | open | MathDB 页面 | 原文线索

V2-022734 | MATHDB-394944

Affineness of the two exceptional Deligne-Lusztig varieties of type G_2

摘录 / EXCERPT: Let G_0 be split of type G_2 over $\\mathbf F_q$, and let $X(w)$ denote the Deligne-Lusztig variety associated with w. In the exceptional case $q=2$, the report expects the t...

open_problem | open | MathDB 页面 | 原文线索

V2-022735 | MATHDB-395055

Reconstruction of character-variety images from Witten-Reshetikhin-Turaev invariants

摘录 / EXCERPT: For a semisimple Lie group G, a positive integer r, a closed orientable surface F, and a 3-manifold M with boundary F, the Witten-Reshetikhin-Turaev quantum invariants...

open_problem | open | MathDB 页面 | 原文线索

V2-022736 | MATHDB-395065

Sturmfels' quadratic-generation conjecture for smooth toric varieties

摘录 / EXCERPT: If X is smooth and the section ring $R(X,L)$ is normal, then the ideal of relations $I(X,L)$ is generated in degree two.

conjecture | open | MathDB 页面 | 原文线索

V2-022737 | MATHDB-395067

Bigness of classes of very moving subvarieties

摘录 / EXCERPT: Let Y be a smooth projective variety and let k be an integer. Let $A^{2k}(Y)\subset H^{2k}(Y,\mathbb{R})$ be the real vector space generated by classes of codimension-k algeb...

open_problem | open | MathDB 页面 | 原文线索

V2-022738 | MATHDB-395070

Regularity of fibers of general projections

摘录 / EXCERPT: Work over an algebraically closed field of characteristic zero. Let $X\subset \mathbf P^r$ be a smooth projective variety of dimension n, and let $\pi:X\to\mathbf P^{n+c}$ be a g...

open_problem | open | MathDB 页面 | 原文线索

V2-022739 | MATHDB-395071

Lower bound for the fiber invariant in terms of regularity

摘录 / EXCERPT: Let $X\subset\mathbf P^r$ be a smooth projective variety of dimension n over an algebraically closed field of characteristic 0, and let $\pi:X\to\mathbf P^{n+c}$ be a general p...

open_problem | open | MathDB 页面 | 原文线索

V2-022740 | MATHDB-395076

Sturmfels conjecture on quadratic generation of toric section-ring relations

摘录 / EXCERPT: Let X be a projective toric variety with an ample line bundle L, and let $R(X,L)=\bigoplus_{m\ge 0}H^0(X,L^m)$. If X is smooth and $R(X,L)$ is normal, then the ideal of relat...

conjecture | open | MathDB 页面 | 原文线索

V2-022741 | MATHDB-395077

Oda's multiplication-map problem for smooth projective toric varieties

摘录 / EXCERPT: Let X be a projective toric variety with an ample line bundle L and a nef line bundle L'. If X is smooth, is the natural multiplication map ... necessarily surjective?

open_problem | open | MathDB 页面 | 原文线索

V2-022742 | MATHDB-395078

Bøgvad's conjecture on section rings of smooth projective toric varieties

摘录 / EXCERPT: Let X be a projective toric variety and let L be an ample line bundle on X. If X is smooth, is the section ring ... necessarily normal and Koszul?

conjecture | open | MathDB 页面 | 原文线索

V2-022743 | MATHDB-395079

A Schubert variety in the G_2 flag variety is not diagonally split

摘录 / EXCERPT: There is a Schubert variety in the G_2 flag variety that is not diagonally split.

open_problem | open | MathDB 页面 | 原文线索

V2-022744 | MATHDB-395082

Ample cone and extremal contractions for manifolds deformation equivalent to Hilbert schemes of K3 surfaces

摘录 / EXCERPT: Conjecture 3. Let (F,g) be a polarized holomorphic symplectic manifold deformation equivalent to $S^{[n]}$, where S is a K3 surface. A divisor class h is ample if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-022745 | MATHDB-395113

Polynomiality of the point count of the commuting variety

摘录 / EXCERPT: Is $|V(\mathbb{F}_q)|$ a polynomial function in q? Here V is the affine algebraic variety over $\mathbb{Z}$ described in the report by the vanishing of the Lie bracket on the w...

open_problem | open | MathDB 页面 | 原文线索

V2-022746 | MATHDB-395184

Finiteness of class numbers for affine algebraic groups in characteristic 2

摘录 / EXCERPT: Conjecture: Let k be a global function field with $\\operatorname{char}(k)=2$. Then every affine algebraic k-group scheme has finite class numbers; that is, for every finite se...

open_problem | open | MathDB 页面 | 原文线索

V2-022747 | MATHDB-395185

Bogomolov conjecture for totally degenerate abelian varieties

摘录 / EXCERPT: The Bogomolov conjecture holds for abelian varieties over function fields which are totally degenerate with respect to some place.

conjecture | open | MathDB 页面 | 原文线索

V2-022748 | MATHDB-395188

Syntomic realization of the motivic Eisenstein class via p-adic Eisenstein series

摘录 / EXCERPT: The report presents an explicit theorem, not an unresolved problem. Let $\pi:E\to M$ be the universal elliptic curve over the moduli scheme with full level-N structure, where...

open_problem | open | MathDB 页面 | 原文线索

V2-022749 | MATHDB-395189

Fujiwara’s theorem for stacks: eventual fixed-point structure

摘录 / EXCERPT: Let $X_0/\\mathbb{F}_q$ be an algebraic stack of finite type, let $c=(c_1,c_2):C_0\to X_0\times X_0$ be a correspondence over $\mathbb{F}_q$ with c_2 representable and quasi-fini...

open_problem | open | MathDB 页面 | 原文线索

V2-022750 | MATHDB-395191

Adelic version of a conjecture of Tate for Shimura varieties

摘录 / EXCERPT: For every point $y:\operatorname{Spec}(F)\to N$, each tensor t_α ($\alpha\in J$) is the crystalline realization of an algebraic cycle on the abelian variety A associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-022751 | MATHDB-395192

Langlands-Rapoport conjecture for the stated Shimura varieties

摘录 / EXCERPT: Under the hypotheses of the Main Theorem-namely, that (G,X) has compact factors and every simple factor of $(G^{\mathrm{ad}},X^{\mathrm{ad}})$ is of type A_n, B_n, C_n, or...

conjecture | open | MathDB 页面 | 原文线索

V2-022752 | MATHDB-395193

Conjecture 2 for basic points of Shimura varieties of type B_n, C_n, or D_{\mathbb R}^n

摘录 / EXCERPT: Let (G_0,H_0) be an arbitrary simple, adjoint Shimura pair of type A_n, B_n, C_n, or $D_{\mathbb R}^n$, and assume that $G_{0,\mathbb Q_p}$ is unramified. For the embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-022753 | MATHDB-395225

Topological restrictions on real loci with Kodaira dimension $-\inf$

摘录 / EXCERPT: For a smooth real projective variety W, to what extent does the condition $\kappa(W)=-\infty$ impose strong restrictions on the topology of a connected component...

open_problem | open | MathDB 页面 | 原文线索

V2-022754 | MATHDB-395231

A categorical replacement between the derived and perfect categories of a singular variety

摘录 / EXCERPT: Find something between $D^b(\operatorname{Coh}(X))$ and $\operatorname{Perf}(X)$ for a singular variety X that satisfies the three properties listed in the report: (1) it is of f...

open_problem | open | MathDB 页面 | 原文线索

V2-022755 | MATHDB-395232

Viehweg’s hyperbolicity conjecture for maximal variation

摘录 / EXCERPT: Let Y° be a quasi-projective manifold admitting a morphism $\mu:Y^{\circ}\to\mathfrak{M}$ to the moduli stack of canonically polarized varieties. If μ is generically f...

conjecture | open | MathDB 页面 | 原文线索

V2-022756 | MATHDB-395233

Viehweg conjecture for families over threefolds

摘录 / EXCERPT: Let $f^{\circ}:X^{\circ}\to Y^{\circ}$ be a smooth projective family of varieties with semi-ample canonical bundle, over a quasi-projective manifold Y° with...

conjecture | open | MathDB 页面 | 原文线索

V2-022757 | MATHDB-395234

Refined Viehweg conjecture for families over threefolds

摘录 / EXCERPT: Let $f^\bullet: X^\bullet \to Y^\bullet$ be a smooth projective family of canonically polarized varieties over a quasi-projective manifold $Y^\bullet$ of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-022758 | MATHDB-395237

Campana’s special-variety characterization by the Kobayashi pseudodistance

摘录 / EXCERPT: Conjecture (Campana): A manifold X is special if and only if its Kobayashi pseudodistance satisfies $d_X \equiv 0$ on $X \times X$.

open_problem | open | MathDB 页面 | 原文线索

V2-022759 | MATHDB-395333

André-Oort conjecture for subvarieties of Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura datum, let $K\subset G(\mathbb{A}_f)$ be a compact open subgroup, and let $Z\subset \operatorname{Sh}_K(G,X)_\mathbb{C}$ be a closed algebraic subvariety t...

conjecture | open | MathDB 页面 | 原文线索

V2-022760 | MATHDB-395334

Geometric criterion for constructing a higher-dimensional special subvariety

摘录 / EXCERPT: Theorem. Let (G,X) be a Shimura datum and let $K=\prod_p K_p\subset G(\mathbb A_f)$ be an open compact subgroup. Assume that there is a prime p_0 such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022761 | MATHDB-395403

Combinatorial description of the Bruhat order on symmetric-variety orbit sets

摘录 / EXCERPT: A full combinatorial description of the Bruhat order on the sets V_A (or V_k) of orbits of minimal parabolic k-subgroups acting on symmetric k-varieties remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-022762 | MATHDB-395404

Realizing convex contractible subcomplexes via Kempf-Rousseau instability

摘录 / EXCERPT: Given a convex contractible subcomplex Y of X, can we find a G-action on a variety V and a point $v\in V$ such that $C_0(v)$ is contained in Y and...

open_problem | open | MathDB 页面 | 原文线索

V2-022763 | MATHDB-395419

Classification of wild quotient singularities of surfaces

摘录 / EXCERPT: Let X be a regular scheme of dimension 2 with an action of a finite group G, let $q:X\to Y=X/G$ be the quotient, and assume that $\operatorname{Sing}Y$ is finite of codimensi...

open_problem | open | MathDB 页面 | 原文线索

V2-022764 | MATHDB-395423

Representability of the schematic image of a finite flat group action

摘录 / EXCERPT: Let X be an R-scheme locally of finite type, separated, flat and pure, of type (FA). Let G be a finite flat R-group scheme acting on X, faithfully on the generic fibre....

open_problem | open | MathDB 页面 | 原文线索

V2-022765 | MATHDB-395424

Effective models and strong extension of torsors

摘录 / EXCERPT: Let R be a discrete valuation ring of unequal characteristic with fraction field K, let G be an abstract finite group, let X be a normal faithfully flat scheme over R, an...

open_problem | open | MathDB 页面 | 原文线索

V2-022766 | MATHDB-395465

Classification of normal complexity-one T-varieties by hyperfans

摘录 / EXCERPT: The normal T-varieties X of complexity one, with a fixed birational equivariant isomorphism $X\approx T\times Z$, are in a one-to-one correspondence with the hyperfans...

open_problem | open | MathDB 页面 | 原文线索

V2-022767 | MATHDB-395466

Classification of effective Hamiltonian torus actions

摘录 / EXCERPT: Describe the effective Hamiltonian actions of compact $(n-1)$-dimensional tori on $2n$-dimensional symplectic varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-022768 | MATHDB-395469

Explicit ampleness bounds for determinantal equations of secant varieties

摘录 / EXCERPT: For an embedding $X\subset \mathbb{P}^r$ given by the complete linear series of a sufficiently ample line bundle, the report suggests that the homogeneous ideal of the kth secant...

open_problem | open | MathDB 页面 | 原文线索

V2-022769 | MATHDB-395472

Self-duality of the Hilbert scheme of points on \(\mathbb{C}^2\)

摘录 / EXCERPT: The Hilbert scheme of n points in $\mathbb{C}^2$ should be self-dual under symplectic duality.

open_problem | open | MathDB 页面 | 原文线索

V2-022770 | MATHDB-395525

Infinite rank of abelian varieties over ample fields

摘录 / EXCERPT: Let F be an ample field which is not algebraic over a finite field. Then every non-zero abelian variety A/F satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-022771 | MATHDB-395529

Adelic closure and rational points for subvarieties of abelian varieties

摘录 / EXCERPT: Let K be a number field or a function field in one variable over a finite field, let A be an abelian variety over K, and let X be a closed K-subscheme of A. Then ... wher...

open_problem | open | MathDB 页面 | 原文线索

V2-022772 | MATHDB-395600

Support varieties and rank varieties for quantum complete intersections

摘录 / EXCERPT: Let k be algebraically closed, let $c\ge 1$ and $a\ge 2$, define ... and let $q\in k$ be a primitive bth root of unity. Set ... Let...

open_problem | open | MathDB 页面 | 原文线索

V2-022773 | MATHDB-395603

Finite generation of endotrivial modules for general group schemes

摘录 / EXCERPT: For general group schemes, is the group of endotrivial modules finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-022774 | MATHDB-395604

A good definition of rank varieties

摘录 / EXCERPT: Find a good definition of “rank varieties”.

open_problem | open | MathDB 页面 | 原文线索

V2-022775 | MATHDB-395609

Principal ideal of the rank variety for the Specht module S^{(p^p)}

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, where p is odd, let $E_p\leq S_{p^2}$ be the elementary abelian subgroup generated by the p disjoint p-cycle...

open_problem | open | MathDB 页面 | 原文线索

V2-022776 | MATHDB-395610

Jantzen’s support-variety conjecture for induced modules

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let $G=GL_n(k)$, and let $\mathfrak g=\mathfrak{gl}_n(k)$. For each dominant weight λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-022777 | MATHDB-395611

Atypicality and support-variety dimension for simple classical Lie superalgebra modules

摘录 / EXCERPT: For a classical Lie superalgebra $\mathfrak g=\mathfrak g_{\bar 0}\oplus\mathfrak g_{\bar 1}$ over $\mathbb C$, and a simple $\mathfrak g$-module $L(\lambda)$ in the category...

open_problem | open | MathDB 页面 | 原文线索

V2-022778 | MATHDB-395612

Module varieties measuring complexity for classical Lie superalgebras

摘录 / EXCERPT: Develop a theory of module varieties for modules in the category $\mathcal F$ of finite-dimensional modules over a classical Lie superalgebra that measures the rate of growth of th...

open_problem | open | MathDB 页面 | 原文线索

V2-022779 | MATHDB-395644

Regular CW-complex and ball conjecture for the totally nonnegative part of a partial flag variety

摘录 / EXCERPT: The cell decomposition of $(G/P)_{\geq 0}$ is a regular CW complex homeomorphic to a ball.

conjecture | open | MathDB 页面 | 原文线索

V2-022780 | MATHDB-395648

Fiebig’s moment-graph rank conjecture

摘录 / EXCERPT: Let $G_{\\leq w}$ be the graph associated to a finite or affine Schubert variety corresponding to an element w of the associated Weyl group, and let $B_k(w)$ be its Braden-MacPhe...

conjecture | open | MathDB 页面 | 原文线索

V2-022781 | MATHDB-395649

Equality of the two characteristic-cycle constructions

摘录 / EXCERPT: Assume that $n>2$ and that $c\in\mathbb{C}\setminus\mathbb{Q}_{<0}$. Then, for every finitely generated U_c-module M, one has an equality of algebraic cycles ...

open_problem | open | MathDB 页面 | 原文线索

V2-022782 | MATHDB-395652

Luna’s classification conjecture for wonderful varieties

摘录 / EXCERPT: Luna conjectured that there corresponds a unique wonderful variety to any spherical system.

conjecture | open | MathDB 页面 | 原文线索

V2-022783 | MATHDB-395661

Soergel-Lunts conjecture for equivariant constructible derived categories

摘录 / EXCERPT: Let a complex reductive group G act on a projective variety X with finitely many G-orbits. Let $D^b_{G,c}(X)$ be the bounded G-equivariant constructible derived category of...

conjecture | open | MathDB 页面 | 原文线索

V2-022784 | MATHDB-395689

Theorem A: equivalences between fusion systems of finite groups of Lie type

摘录 / EXCERPT: Fix a prime p, a connected reductive integral group scheme $\mathbb G$, and a pair of prime powers q and q' both prime to p. Then: (a) If...

open_problem | open | MathDB 页面 | 原文线索

V2-022785 | MATHDB-395713

Shokurov’s ACC conjecture for log canonical thresholds

摘录 / EXCERPT: For every fixed dimension, the set of all log canonical thresholds of proper ideal sheaves on log canonical varieties over an algebraically closed field of characteristic zero sati...

conjecture | open | MathDB 页面 | 原文线索

V2-022786 | MATHDB-395726

Connectedness of the regular-sequence locus in a fixed-Hilbert-function Hilbert scheme

摘录 / EXCERPT: Let $S=k[x_1,\ldots,x_n]$, let $F=(f_1,\ldots,f_r)\subset S$ be a regular sequence of homogeneous elements with degrees $\deg f_i=e_i$ for $2\le e_1\le\cdots\le e_r$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-022787 | MATHDB-395740

Toroidalization in higher dimensions

摘录 / EXCERPT: Suppose that $\Phi:X\to Y$ is a dominant morphism of varieties over a field of characteristic zero. Construct a commutative diagram ... such that the vertical arrows are products of...

open_problem | open | MathDB 页面 | 原文线索

V2-022788 | MATHDB-395741

Sufficiency of bigness of the anticanonical divisor

摘录 / EXCERPT: Let X be a normal projective variety over k, and define $A(X)=\{a(R)\mid R\text{ is quasi-Gorenstein and }\operatorname{Proj}(R)=X\}$. Is the necessary condition that $-K_X$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-022789 | MATHDB-395779

Grothendieck-Serre conjecture under Panin’s isotropy condition

摘录 / EXCERPT: Let R be a regular semi-local domain containing an infinite perfect field, or let R be the semi-local ring of finitely many points on a smooth irreducible k-variety X over...

conjecture | open | MathDB 页面 | 原文线索

V2-022790 | MATHDB-395781

Description of indecomposable motivic summands of products of projective homogeneous varieties

摘录 / EXCERPT: Let F be a field, let G be a semisimple affine algebraic group over F, and fix a prime p. Let X_G be the class of all finite direct products of projective homogeneous var...

open_problem | open | MathDB 页面 | 原文线索

V2-022791 | MATHDB-395840

Faltings height pairing of arithmetic special divisors and CM cycles

摘录 / EXCERPT: Assume there is a regular scheme $X_K\to\operatorname{Spec}(\mathbb Z)$, projective and flat over $\mathbb Z$, whose associated complex variety is a smooth compactification of...

open_problem | open | MathDB 页面 | 原文线索

V2-022792 | MATHDB-395848

Wiesend’s idele class group for an arithmetic scheme

摘录 / EXCERPT: Let X be a regular, connected scheme that is flat, separated, and of finite type over $\b\mathbb{Z}\u0007d$. How can one define an idele class group $C(X)$ generalizing the class...

open_problem | open | MathDB 页面 | 原文线索

V2-022793 | MATHDB-395852

Kato-type vanishing conjecture for Suslin-Kato homology

摘录 / EXCERPT: Fix a finite field $\mathbb{F}_q$. For every smooth and connected scheme X over $\mathbb{F}_q$, one has ...

conjecture | open | MathDB 页面 | 原文线索

V2-022794 | MATHDB-395860

Gubeladze’s conjecture for the K-theory of toric varieties

摘录 / EXCERPT: Let A be a finitely generated, cancellative, torsion-free, normal abelian monoid, let k be a field, and let θ_c be the endomorphism induced by $a\mapsto a^c$ for a posit...

conjecture | open | MathDB 页面 | 原文线索

V2-022795 | MATHDB-395862

Motivic slices over a perfect field are motivic-cohomology modules

摘录 / EXCERPT: Let $q\in\mathbb{Z}$ and let E be an arbitrary symmetric T-spectrum in $\operatorname{Spt}(\mathcal{M}_X)$. If the base scheme X is a perfect field k, then the q-slice...

open_problem | open | MathDB 页面 | 原文线索

V2-022796 | MATHDB-395864

Existence of left adjoints to the inclusion of n-motivic sheaves

摘录 / EXCERPT: Fix a field k of characteristic 0 and work with rational coefficients. For a fixed integer $n\in\mathbb{N}$, let $\mathrm{HI}(k)$ be the category of homotopy-invariant sheaves...

open_problem | open | MathDB 页面 | 原文线索

V2-022797 | MATHDB-395865

Exactness of Kato’s niveau complex over a finite field

摘录 / EXCERPT: Let k be a finite field, let $l\ne\operatorname{char}(k)$ be a prime, and set $\Lambda=\mathbb{Q}_l/\mathbb{Z}_l$. For a smooth projective variety X/k of dimension d, Kato’s...

open_problem | open | MathDB 页面 | 原文线索

V2-022798 | MATHDB-395866

Possible motivic decomposition types of quadrics

摘录 / EXCERPT: Let k be a field of characteristic different from 2, and let Q be a smooth projective quadric over k. Writing $M(Q)$ for its motive in the category of Chow motives, over an...

open_problem | open | MathDB 页面 | 原文线索

V2-022799 | MATHDB-395881

Evaluation of torsors under tamely split tori

摘录 / EXCERPT: Let K be a finite extension of a p-adic field, with ring of integers O and residue field k. Let X be a smooth variety over K, let $\mathcal X/O$ be a smooth (not necess...

open_problem | open | MathDB 页面 | 原文线索

V2-022800 | MATHDB-395885

Decision problem for rational points over number fields

摘录 / EXCERPT: For a fixed number field k, does there exist an algorithm that takes as input a k-variety X and outputs YES or NO according to whether X has a k-rational point?

open_problem | open | MathDB 页面 | 原文线索

V2-022801 | MATHDB-395887

Generalizing the crude heuristic to predict the Bombieri-Lang conjecture

摘录 / EXCERPT: Can one generalize the crude heuristic so that it predicts the Bombieri-Lang conjecture for all varieties of general type?

conjecture | open | MathDB 页面 | 原文线索

V2-022802 | MATHDB-395888

Mazur’s conjecture on the topology of rational points

摘录 / EXCERPT: Conjecture 9 (Mazur). Let X be a $\a4$-variety. Then the closure of $X(\a4)$ in $X(\a1)$ with respect to the usual analytic topology has at most finitely many connected component...

conjecture | open | MathDB 页面 | 原文线索

V2-022803 | MATHDB-395893

Algebraicity of the modular correspondence \\Pi^{?} for generalized Heegner cycles

摘录 / EXCERPT: For the fixed integer $r\ge 0$ and the varieties W_r and elliptic curve A in the setting of Assumption 2, there exists an algebraic cycle class ... such that ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-022804 | MATHDB-396003

Global jet-scheme isomorphism problem

摘录 / EXCERPT: Let k be an algebraically closed field of arbitrary characteristic. For a morphism $f:X\to Y$ of k-schemes, let $f_m:X_m\to Y_m$ be the induced morphism of m-jet schemes. If...

open_problem | open | MathDB 页面 | 原文线索

V2-022805 | MATHDB-396009

Existence of weak Zariski decompositions

摘录 / EXCERPT: Work over an algebraically closed field of characteristic zero, in the relative setting where $X\to Z$ is a projective morphism of normal quasi-projective varieties. A weak Zariski...

open_problem | open | MathDB 页面 | 原文线索

V2-022806 | MATHDB-396011

Shokurov's ACC conjecture for log canonical thresholds on smooth varieties

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. For every n, the set ... of log canonical thresholds on smooth varieties of dimension n satisfies the ascending c...

conjecture | open | MathDB 页面 | 原文线索

V2-022807 | MATHDB-396012

Nonexistence of nontrivial stratified bundles on simply connected projective varieties in characteristic $p>0$

摘录 / EXCERPT: Let X be a smooth projective variety over a perfect field k of characteristic $p>0$. If ... then there are no nontrivial $\mathcal O_X$-coherent $\mathcal D_X$-modules.

open_problem | open | MathDB 页面 | 原文线索

V2-022808 | MATHDB-396013

Limit of the Fourier-Mukai transform under a rank-one degeneration

摘录 / EXCERPT: Let $X\to S$ be a degeneration over a discrete valuation ring whose generic fiber is a principally polarized abelian variety and whose special fiber is a rank-one semi-abelian vari...

open_problem | open | MathDB 页面 | 原文线索

V2-022809 | MATHDB-396020

Properness of the Mabuchi energy along degenerations

摘录 / EXCERPT: Let $X^n\hookrightarrow\mathbb{P}^N$ be a smooth, linearly normal complex projective variety of degree $d\ge 2$, embedded by a very ample complete linear system. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-022810 | MATHDB-396021

Boundedness of the Mabuchi energy along all degenerations

摘录 / EXCERPT: Let $X^n\hookrightarrow\mathbb{P}^N$ be a smooth complex projective variety of degree $d\ge 2$, embedded by a very ample complete linear system. Let $G=\mathrm{SL}(N+1,\mathbb{C})$...

open_problem | open | MathDB 页面 | 原文线索

V2-022811 | MATHDB-396023

Constancy of the movable-cone chamber decomposition under deformations

摘录 / EXCERPT: Does the chamber decomposition of the movable cone remain constant under deformations of a $\[0mQ$-factorial terminal Fano variety X_0? In particular, would the nef cone remain c...

open_problem | open | MathDB 页面 | 原文线索

V2-022812 | MATHDB-396024

Volume criterion for ampleness on Q-factorial terminal Fano varieties

摘录 / EXCERPT: Does the "volume criterion for ampleness" hold for Q-factorial terminal Fano varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022813 | MATHDB-396148

Finite CW-complex homotopy type of smooth projective Berkovich analytifications

摘录 / EXCERPT: Let k be the complete non-Archimedean valued field under discussion, and let V be a smooth projective variety over k. Berkovich conjectured that V^{an} has the hom...

open_problem | open | MathDB 页面 | 原文线索

V2-022814 | MATHDB-396159

Extension theorem for differential forms on log canonical pairs

摘录 / EXCERPT: Let X be a normal complex algebraic variety of dimension n and let $\Delta\subset X$ be a $\mathbb{Q}$-divisor with coefficients in $[0,1]\cap\mathbb{Q}$. Assume that...

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V2-022815 | MATHDB-396160

Birationality of the genus-16 K3 universal-family map

摘录 / EXCERPT: Conjecture 3. The induced rational map $\mathbb P^{36}//\mathrm{PGL}(4)\dashrightarrow \mathcal S_{16}$ between 21-dimensional varieties is birational.

open_problem | open | MathDB 页面 | 原文线索

V2-022816 | MATHDB-396302

Singular loci of affine Schubert varieties

摘录 / EXCERPT: For an affine Schubert variety $X(w)$ in the affine Grassmannian $X=G(F)/G(A)$, with $G=\mathrm{SL}_n$, determine the singular locus of $X(w)$; the singular locus is itself a union...

open_problem | open | MathDB 页面 | 原文线索

V2-022817 | MATHDB-396304

Recognizing versal actions

摘录 / EXCERPT: Let k be a field of characteristic 0, let G be a linear algebraic group, and let X be an algebraic k-variety with a generically free G-action. Here generically free mea...

open_problem | open | MathDB 页面 | 原文线索

V2-022818 | MATHDB-396366

Parshin’s conjecture for smooth proper varieties over finite fields

摘录 / EXCERPT: Parshin’s conjecture: the rational higher K-theory of a smooth and proper variety over a finite field is torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-022819 | MATHDB-396367

Equivalence of finiteness properties for Suslin homology

摘录 / EXCERPT: Under resolution of singularities, the following statements are equivalent: 1. Conjecture P_0. 2. For every scheme X of finite type over a finite field, the groups...

open_problem | open | MathDB 页面 | 原文线索

V2-022820 | MATHDB-396368

Beilinson-Parshin cup-product perfectness conjecture

摘录 / EXCERPT: Conjecture BP(n). For all smooth and projective schemes X over the finite field $\mathbb{F}_q$, the cup-product pairing ... is perfect, where $d=\dim X$.

conjecture | open | MathDB 页面 | 原文线索

V2-022821 | MATHDB-396369

Conjecture P_m for motivic cohomology with compact support

摘录 / EXCERPT: For all smooth and projective schemes X over the finite field $ \mathbb{F}_q$, and all $i\ne 2m$, the group $H_c^i(X,\mathbb{Z}(m))$ is torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-022822 | MATHDB-396370

Torsion in the gamma filtration and Chow groups of Borel varieties

摘录 / EXCERPT: Let X be a variety of Borel subgroups of a strongly inner simple linear algebraic group G over a field k. Then the torsion part of the second quotient of the Grothendieck...

open_problem | open | MathDB 页面 | 原文线索

V2-022823 | MATHDB-396396

Voisin’s conjecture on very moving subvarieties

摘录 / EXCERPT: For a smooth complex projective variety X, if $Y\subset X$ is “very moving”-that is, through a general point $x\in X$ there is a deformation of Y passing through x with gener...

conjecture | open | MathDB 页面 | 原文线索

V2-022824 | MATHDB-396401

Lifting tropical intersections

摘录 / EXCERPT: Let X and X' be subvarieties of a torus over a nonarchimedean field. If $\operatorname{Trop}(X)$ and $\operatorname{Trop}(X')$ meet properly at w, then...

open_problem | open | MathDB 页面 | 原文线索

V2-022825 | MATHDB-396402

Characterization of sextic covers arising from the Gulliksen-Negard complex

摘录 / EXCERPT: For a scheme T, determine precisely which sextic covers $\pi:X\to T$-that is, finite, flat covers of degree 6-arise from a Gulliksen-Negard complex, equivalently belong to the...

open_problem | open | MathDB 页面 | 原文线索

V2-022826 | MATHDB-396403

Irreducibility of the Enriques 10-tuple cover

摘录 / EXCERPT: The authors conjecture that the Galois 2^9-cover H'_{En} of the Hilbert scheme H_{En} of Fano models of Enriques surfaces, corresponding to a choice of ten...

open_problem | open | MathDB 页面 | 原文线索

V2-022827 | MATHDB-396420

Quantitative lifting of integral points modulo ideals

摘录 / EXCERPT: Let X be an affine variety defined over F and equipped with a transitive F-action of a connected F-simple algebraic group. Then there exists a finite subset S_0 of V_F,...

open_problem | open | MathDB 页面 | 原文线索

V2-022828 | MATHDB-396560

Standard conjectures for irreducible symplectic manifolds of K3 Hilbert-scheme deformation type

摘录 / EXCERPT: The standard conjectures hold for irreducible symplectic manifolds deformation equivalent to Hilbert schemes of points on K3 surfaces.

conjecture | open | MathDB 页面 | 原文线索

V2-022829 | MATHDB-396564

Lefschetz standard conjecture for varieties of K3^[n]-type

摘录 / EXCERPT: For every smooth projective variety of K3$^{[n]}$-type, the Lefschetz standard conjecture holds. Here a variety is of K3$^{[n]}$-type if it is deformation equivalent to the Hilbert...

conjecture | open | MathDB 页面 | 原文线索

V2-022830 | MATHDB-396565

Standard conjectures for varieties of K3^[n]-type

摘录 / EXCERPT: The standard conjectures hold for any smooth projective variety of $K3^{[n]}$-type.

conjecture | open | MathDB 页面 | 原文线索

V2-022831 | MATHDB-396566

Whether the multiplier-ideal subscheme is a divisor

摘录 / EXCERPT: Let X be an n-dimensional Fano manifold without a Kähler-Einstein metric, let G be a compact subgroup of $\\operatorname{Aut}(X)$, and let $\\varphi_\infty$ be a weak limit o...

open_problem | open | MathDB 页面 | 原文线索

V2-022832 | MATHDB-396568

Embedded deformations of the Segre embedding

摘录 / EXCERPT: Let $\{C_t\subset \mathbb{P}^{2n-5}\}_{t\in\Delta}$ be a family of smooth projective varieties such that, for $t\ne 0$, $C_t\subset\mathbb{P}^{2n-5}$ is the Segre embedding of...

open_problem | open | MathDB 页面 | 原文线索

V2-022833 | MATHDB-396569

Uniqueness of a seven-dimensional Fano manifold with Hirzebruch-surface VMRT

摘录 / EXCERPT: There exists, up to biholomorphism, at most one 7-dimensional Fano manifold of Picard number 1 whose variety of minimal rational tangents at a general point is isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-022834 | MATHDB-396571

Intersection theory of analytic graphs

摘录 / EXCERPT: Let F consist of one one-variable function f. For an algebraic subvariety $W\subseteq\mathbb{C}^{2m}$ of dimension m, written in variables $x_1,\ldots,x_m,y_1,\ldots,y_m$, de...

open_problem | open | MathDB 页面 | 原文线索

V2-022835 | MATHDB-396575

Cone conjecture for abelian varieties

摘录 / EXCERPT: Theorem 5. Let X be an abelian variety over an algebraically closed field. Then the cone conjecture holds for X. For an abelian variety, the nef and movable cones coincide, so...

conjecture | open | MathDB 页面 | 原文线索

V2-022836 | MATHDB-396578

Class of the closure of the gradient-theta locus in the perfect cone compactification

摘录 / EXCERPT: For $g\le 5$, let $G(g)\subset\mathcal A_g$ be the locus of principally polarized abelian varieties for which there exists an odd two-torsion point $m\in A[2]$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022837 | MATHDB-396579

Pure codimension of the odd triple-point theta locus

摘录 / EXCERPT: For every genus g, let ... where A is the principally polarized abelian variety represented by τ. Is $G(g)$ purely of codimension g in $\mathcal A_g$?

open_problem | open | MathDB 页面 | 原文线索

V2-022838 | MATHDB-396629

p-local equivalence of finite groups of Lie type

摘录 / EXCERPT: Fix a connected, reductive, integral group scheme G and a prime p. For any pair of prime powers q and q' such that $p\nmid qq'$ and $\langle q\rangle=\langle q'\rangle$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-022839 | MATHDB-396640

Initial-degree inequality for fat point schemes

摘录 / EXCERPT: Let X be a finite set of points in projective n-space, let r be any positive integer, and let Y be the fat point scheme supported on X with each point having multiplicity...

open_problem | open | MathDB 页面 | 原文线索

V2-022840 | MATHDB-396646

Jet-spannedness of toric vector bundles from Seshadri-type positivity

摘录 / EXCERPT: Let X be a smooth, complete d-dimensional toric variety and let E be a toric vector bundle on X. For each invariant curve C through a fixed point, write ... and define...

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V2-022841 | MATHDB-396647

Determining the Mordell-Weil rank from a Weierstrass equation

摘录 / EXCERPT: Let cmathit{V}omathrmcto B be a non-trivial elliptic fibration with section, where cmathit{V}omathrmand B are smooth projective varieties and cmathrmcdim B1. Let cmathcal Lomathrmc...

open_problem | open | MathDB 页面 | 原文线索

V2-022842 | MATHDB-396745

Precise asymptotics of the coarse Schottky density constants

摘录 / EXCERPT: For every genus g, let A_g be the moduli space of principally polarized abelian varieties with its invariant metric d, and let $J(M_g)\subset A_g$ be the Jacobian locus. The...

open_problem | open | MathDB 页面 | 原文线索

V2-022843 | MATHDB-396778

Linking 1-motives with unipotent part to the full motivic picture

摘录 / EXCERPT: This way, the 1-motive is no longer A^1-invariant and a precise way to link this to the full motivic picture is to be understood.

open_problem | open | MathDB 页面 | 原文线索

V2-022844 | MATHDB-396779

Sharp cohomology for proper possibly singular varieties

摘录 / EXCERPT: For a proper, possibly singular variety X, and its associated 1-motive M (for example, via generalized Picard varieties), construct a sharp cohomology theory of X whose fir...

open_problem | open | MathDB 页面 | 原文线索

V2-022845 | MATHDB-396782

Manin obstruction for torsors under semi-abelian varieties

摘录 / EXCERPT: Let k be a number field, let G be a semi-abelian variety over k, and let X be a torsor under G. Assume that the Tate-Shafarevich group of the abelian quotient of G is f...

open_problem | open | MathDB 页面 | 原文线索

V2-022846 | MATHDB-396948

Nonnegativity of cohomological quiver coefficients

摘录 / EXCERPT: Let Q be a quiver without oriented cycles, let $V=\bigoplus_{i\to j}\operatorname{Hom}(E_i,E_j)$, and let $\Omega\subset V$ be a G-stable closed irreducible subvariety (a quive...

open_problem | open | MathDB 页面 | 原文线索

V2-022847 | MATHDB-396972

Sakellaridis-Venkatesh local conjecture for spherical varieties

摘录 / EXCERPT: Let X be a spherical variety over a local field. Then ... where ψ varies over X-distinguished Arthur parameters modulo conjugacy, and μ_X belongs to the class of Lebesgu...

conjecture | open | MathDB 页面 | 原文线索

V2-022848 | MATHDB-396973

Sakellaridis-Venkatesh local Plancherel decomposition for spherical varieties

摘录 / EXCERPT: Under the report’s standing assumption that the reductive group G is split, let $X=H\backslash G$ be a spherical variety over a local field k (with $H=H(k)$ and $G=G(k)$). The...

open_problem | open | MathDB 页面 | 原文线索

V2-022849 | MATHDB-396984

Long-time preservation of numerical actions for the one-dimensional cubic NLS

摘录 / EXCERPT: For the one-dimensional cubic nonlinear Schrödinger equation and the fully discrete Fourier time-splitting scheme described in the report, let...

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V2-022850 | MATHDB-397026

Extension of Lelong-class plurisubharmonic functions without additional growth

摘录 / EXCERPT: For an algebraic subvariety $X\subset\mathbb{C}^n$, is every plurisubharmonic function $\eta\in\mathcal{L}(X)$ the restriction of a plurisubharmonic function in...

open_problem | open | MathDB 页面 | 原文线索

V2-022851 | MATHDB-397027

Extension of \c9-plurisubharmonic functions from subvarieties

摘录 / EXCERPT: Let X be a subvariety of a projective manifold V equipped with a Hodge form $\c9$, meaning a Kahler form whose cohomology class is integral. Then every $\c9|_X$-plurisubharmoni...

open_problem | open | MathDB 页面 | 原文线索

V2-022852 | MATHDB-397028

Cheeger-Goresky-MacPherson conjecture for projective varieties

摘录 / EXCERPT: Let $Y\subset\mathbb{CP}^{N}$ be a projective variety. Then there is a natural isomorphism ... Here the L^{2}-de Rham cohomology is taken on the regular part with respect to the re...

conjecture | open | MathDB 页面 | 原文线索

V2-022853 | MATHDB-397099

Finiteness of Shimura subvarieties in the Torelli locus

摘录 / EXCERPT: It is conjectured that, as g varies, there are only finitely many Shimura subvarieties of $\mathcal A_g$ that are contained in $t(\mathcal M_g)$ and meet $t(\mathcal M_g)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-022854 | MATHDB-397100

Finiteness of Shimura subvarieties in the Schottky locus

摘录 / EXCERPT: It is conjectured that, as the genus g ranges over all genera, there are only finitely many Shimura subvarieties of $ \mathcal A_g$ that are contained in the closure of the Schot...

open_problem | open | MathDB 页面 | 原文线索

V2-022855 | MATHDB-397101

Hilbert modular varieties and the Schottky locus

摘录 / EXCERPT: Conjecture: For every sufficiently large genus g, all but finitely many Hilbert modular varieties in $\mathcal A_g$ are disjoint from the Schottky locus $t(\mathcal M_g)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022856 | MATHDB-397123

Vanishing theorem for negative K-groups and K-regularity

摘录 / EXCERPT: Let X be a scheme of finite type over a field F of characteristic 0, and suppose that the dimension of X is d. Then $NK_i(X)=0$ for all $d\le -i$, and, in fact, X is...

open_problem | open | MathDB 页面 | 原文线索

V2-022857 | MATHDB-397129

Parshin’s conjecture for higher algebraic K-theory over finite fields

摘录 / EXCERPT: For every smooth and proper scheme X over a finite field, the algebraic K-group $K_i(X)$ is torsion for every integer $i>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-022858 | MATHDB-397130

Beilinson’s non-degeneracy conjecture for Chow groups

摘录 / EXCERPT: For all smooth and proper schemes X over a finite field, the intersection pairing ... is non-degenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-022859 | MATHDB-397173

Weight-monodromy for smooth complete intersections in proper toric varieties

摘录 / EXCERPT: Let $X \subset T$ be a smooth complete intersection in a proper toric variety, over a p-adic field. Then, for $\ell \ne p$, the ℓ-adic cohomology groups of X satisfy the w...

open_problem | open | MathDB 页面 | 原文线索

V2-022860 | MATHDB-397175

p-adic variational Hodge conjecture for algebraic cycles

摘录 / EXCERPT: Let k be a perfect field of characteristic $p>0$, let W be the ring of Witt vectors of k, and let $X\to W$ be a smooth projective scheme. Write $X_1=X\otimes_W W/p$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-022861 | MATHDB-397221

General Sato-Tate equidistribution conjecture

摘录 / EXCERPT: Let X_i ($i\in I$) be a finite family of smooth projective varieties over $\mathbb{Q}$, and let n_i be positive integers. Choose a finite set S of primes such that the X_i...

conjecture | open | MathDB 页面 | 原文线索

V2-022862 | MATHDB-397222

Generic principal Sato-Tate distribution for genus-2 curves with non-split Jacobians

摘录 / EXCERPT: For genus-2 curves with non-split Jacobians, after restricting to the principal coset and averaging over fibers over a base scheme, the expected (conjectural) generic Sato-Tate dis...

open_problem | open | MathDB 页面 | 原文线索

V2-022863 | MATHDB-397223

Numerical Sato-Tate conjecture for abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of dimension g over a number field k. For each prime ideal $\p$ of k at which A has good reduction, let $L_{\p}(T)$ be the Frobenius character...

conjecture | open | MathDB 页面 | 原文线索

V2-022864 | MATHDB-397224

Classification of exceptional Sato-Tate cases in dimension greater than one

摘录 / EXCERPT: For abelian varieties over number fields of dimension $g>1$, classify the exceptional cases to the generic situation $\operatorname{ST}(A)=\operatorname{SU}(2g)$, particularly the...

open_problem | open | MathDB 页面 | 原文线索

V2-022865 | MATHDB-397383

Numerical characterization of Du Bois singularities

摘录 / EXCERPT: Let X be a proper variety over $\mathbb{C}$ with a fixed basepoint-free linear system Δ. Then X has only Du Bois singularities if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-022866 | MATHDB-397390

Hochschild homology of a variety and the cohomology of its mirror

摘录 / EXCERPT: Given a mirror pair $S, (\check{S},F_{\check{S}})$ of the construction, conjecture that $\mathrm{HH}_{k}(S)\cong H^{k}(\check{S},\mathbb{C})$.

open_problem | open | MathDB 页面 | 原文线索

V2-022867 | MATHDB-397394

Birational characterization under maximal real-rank automorphism action

摘录 / EXCERPT: In Theorem 3, if H is of real rank $n-1$ (with n not assumed to be 3), is X birational to a quotient of an abelian variety of dimension n?

open_problem | open | MathDB 页面 | 原文线索

V2-022868 | MATHDB-397455

Minimal class of singularities preserving the normal-crossings locus

摘录 / EXCERPT: For reduced algebraic or analytic varieties X over a field of characteristic zero, can one find the smallest class of singularities S such that: 1. S includes all normal-cros...

open_problem | open | MathDB 页面 | 原文线索

V2-022869 | MATHDB-397551

Coherent cohomology rings of group schemes over discrete valuation rings

摘录 / EXCERPT: Describe the coherent cohomology ring of group schemes over (say) discrete valuation rings.

open_problem | open | MathDB 页面 | 原文线索

V2-022870 | MATHDB-397560

Cohen-Macaulay and Gorenstein properties of the normalized isospectral commuting variety

摘录 / EXCERPT: Theorem 0.6. X^{norm} is a Cohen-Macaulay and Gorenstein variety with trivial canonical sheaf.

open_problem | open | MathDB 页面 | 原文线索

V2-022871 | MATHDB-397563

Covering relations in spherical systems have positive height at most two

摘录 / EXCERPT: Let M be a wonderful variety with spherical system (Σ,Δ,c). For $E,F\in\mathbb N[\Delta]$, write $F<_{\Sigma}E$ when $E-F\in\mathbb N[\Sigma]$. If F is maximal with...

open_problem | open | MathDB 页面 | 原文线索

V2-022872 | MATHDB-397566

Action of the Feigin-Frenkel center on induced critical-level modules

摘录 / EXCERPT: Conjecture 1. If the center $Z(\mathfrak g_n)$ acts on $V\in \mathfrak g_n\text{-mod}$ through the character $\Lambda\in(\mathfrak h^*/W)_n$, then ... is scheme-theoretically support...

open_problem | open | MathDB 页面 | 原文线索

V2-022873 | MATHDB-397567

Koszul categories and mixed Hodge modules

摘录 / EXCERPT: Let X be a smooth complex algebraic variety, and let $ \mathcal S$ be a stratification of X by affine spaces. Can one generate a Koszul category from the category of mixed Hodg...

open_problem | open | MathDB 页面 | 原文线索

V2-022874 | MATHDB-397568

AGT conjecture for the one-vertex, one-loop quiver variety

摘录 / EXCERPT: Let Γ be the quiver with one vertex and one loop, let $r\ge 0$, and let M_r be the corresponding quiver variety, namely the moduli space of torsion-free sheaves of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-022875 | MATHDB-397576

Representative-volume size in numerical homogenization

摘录 / EXCERPT: Definition of representative volume size in the context of numerical homogenization schemes.

open_problem | open | MathDB 页面 | 原文线索

V2-022876 | MATHDB-397585

First non-trivial cohomology degree for finite Chevalley groups

摘录 / EXCERPT: Let G be a reductive algebraic group scheme defined over $\mathbb{F}_p$, let $q=p^r$, and let $G(\mathbb{F}_q)$ be the corresponding finite Chevalley group. For k an algebraica...

open_problem | open | MathDB 页面 | 原文线索

V2-022877 | MATHDB-397611

Exact number of real-form orbits on the real part of a wonderful variety

摘录 / EXCERPT: Given a connected reductive algebraic group G over $\mathbb{C}$, the involution σ defining its split real form, and a wonderful G-variety X equipped with its canonical...

open_problem | open | MathDB 页面 | 原文线索

V2-022878 | MATHDB-397641

Losev’s theorem proving the former Knop conjecture

摘录 / EXCERPT: Let G be a connected reductive complex algebraic group and let X be a smooth affine spherical G-variety. Then X is uniquely determined by the set of highest weights occurri...

conjecture | open | MathDB 页面 | 原文线索

V2-022879 | MATHDB-397642

Finite termination of iterated Semple-Nash modifications

摘录 / EXCERPT: Let X be an algebraic variety over a field of characteristic zero, and let $\nu_X\colon N X\to X$ denote its Semple-Nash modification. Does iterating this operation eventually le...

open_problem | open | MathDB 页面 | 原文线索

V2-022880 | MATHDB-397645

Classification of toric degenerations of low-degree smooth Fano threefolds

摘录 / EXCERPT: All degenerations of smooth Fano threefolds of degree ≤ 12 in their anticanonical embeddings to Gorenstein toric Fano varieties are completely classified.

open_problem | open | MathDB 页面 | 原文线索

V2-022881 | MATHDB-397801

Dense algebraic orbit for the residual-line self-map

摘录 / EXCERPT: For the variety $X=F(V)$ of lines on a smooth cubic fourfold $V\subset\mathbb{P}^5$ equipped with the rational self-map f sending a general line to its residual line in the tange...

open_problem | open | MathDB 页面 | 原文线索

V2-022882 | MATHDB-397802

Zhang’s conjecture on dense orbits of polarized self-maps

摘录 / EXCERPT: Let S be a smooth projective variety over a number field, and let $f:S\to S$ be a regular polarized self-map: there is an ample line bundle L on S such that...

conjecture | open | MathDB 页面 | 原文线索

V2-022883 | MATHDB-397808

The Tate conjecture for smooth projective varieties over finite fields

摘录 / EXCERPT: Let k be a finite field, let X be a smooth projective variety over k, let i be an integer, and let $\ll$ be a prime number invertible in k. The image of the cycle class m...

conjecture | open | MathDB 页面 | 原文线索

V2-022884 | MATHDB-397809

Decomposition of the tangent sheaf for varieties with numerically trivial canonical divisor

摘录 / EXCERPT: Let X be a normal projective variety with at worst canonical singularities, defined over the complex numbers, and assume that $K_X\equiv 0$. Then there exist an Abelian variety...

open_problem | open | MathDB 页面 | 原文线索

V2-022885 | MATHDB-397810

Effective cone of the Hilbert scheme of points on \(\mathbf P^2\)

摘录 / EXCERPT: Determine the effective cone of H_n, where H_n is the Hilbert scheme of zero-dimensional subschemes of $\mathbf P^2$ of degree n, for every n.

open_problem | open | MathDB 页面 | 原文线索

V2-022886 | MATHDB-397816

Smoothness and geometric-point equivalence of the truncated display functor

摘录 / EXCERPT: For truncated p-divisible groups of level n over schemes of characteristic p, the induced morphism of algebraic stacks ... is smooth and is an equivalence on geometric points.

open_problem | open | MathDB 页面 | 原文线索

V2-022887 | MATHDB-397820

Connected components of minuscule affine Deligne-Lusztig varieties

摘录 / EXCERPT: Let $F=k((t))$ or $F=W(k)[1/p]$, let L be respectively $k((t))$ or $W(k)[1/p]$, and let G be a classical connected unramified group over $\mathcal O_F$. Let $b\in G(L)$ and let...

open_problem | open | MathDB 页面 | 原文线索

V2-022888 | MATHDB-397823

Classicality of overconvergent Hilbert modular forms of parallel weight one

摘录 / EXCERPT: Let F be a totally real number field and let p be a prime with $p\mathcal O_F=\prod_{i=1}^r\pi_i^{e_i}$. Let $X_0(\pi)$ be the Hilbert modular variety described in the report,...

open_problem | open | MathDB 页面 | 原文线索

V2-022889 | MATHDB-397935

Raynaud-Gruson flatification theorem

摘录 / EXCERPT: Let S be a scheme, let $U \subset S$ be an open subscheme, and let $f:X\to S$ be a morphism of finite type such that $f|_U$ is flat and of finite presentation. Then there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-022890 | MATHDB-397938

Derived Cartier isomorphism (Theorem 1.2)

摘录 / EXCERPT: Let $f:X\to S$ be a map of $\mathbf F_p$-schemes. Then $dR_{X/S}$ admits an increasing, bounded-below, exhaustive filtration $\operatorname{Fil}^{\bullet}_{\mathrm{conj}}$ with gra...

open_problem | open | MathDB 页面 | 原文线索

V2-022891 | MATHDB-398017

Lagrangian tori and fibrations on compact hyperkähler manifolds

摘录 / EXCERPT: Let $A\subset M$ be a Lagrangian torus in a compact hyperkähler manifold. Then there exists a meromorphic map $f\colon M\dashrightarrow B$ dominant over a projective variety B, s...

open_problem | open | MathDB 页面 | 原文线索

V2-022892 | MATHDB-398041

Homological Hilbert-scheme formula for the HOMFLY-PT homology of an algebraic link

摘录 / EXCERPT: Let C be a planar curve singularity at $(0,0)$, let $L_{C,(0,0)}$ be its link, and let … be the Milnor number. The conjecture is … where P^{vir}_t denotes the virtual…

open_problem | open | MathDB 页面 | 原文线索

V2-022893 | MATHDB-398044

Correct hypotheses for comparing invariant algebra deformations with abstract scheme deformations

摘录 / EXCERPT: Let $X=X'/G$ be a scheme presented as a good quotient of a quasi-affine scheme X' by a linearly reductive group G, with X' an open invariant subscheme of an affine G-scheme…

open_problem | open | MathDB 页面 | 原文线索

V2-022894 | MATHDB-398051

Derive the classification of finite-root diagonal braidings from Cartan-scheme classification

摘录 / EXCERPT: Deduce the classification in [9] from the classification of Cartan schemes [7].

open_problem | open | MathDB 页面 | 原文线索

V2-022895 | MATHDB-398052

Relation between Cartan-scheme and arithmetic-root-system classifications

摘录 / EXCERPT: Deduce the classification in [9] from the classification of Cartan schemes [7]. There are some objects in the list in [7] that do not appear in the list in [9]. Are they related to…

open_problem | open | MathDB 页面 | 原文线索

V2-022896 | MATHDB-398060

Topological data encoded in the motivic Milnor fibre

摘录 / EXCERPT: What kind of topological data is structurally encoded in the motivic Milnor fibre?

open_problem | open | MathDB 页面 | 原文线索

V2-022897 | MATHDB-398061

Real motivic Milnor fibre and resolution data

摘录 / EXCERPT: Assuming that one can establish a real substitute for the complex motivic Milnor fibre and prove a real version of the Denef-Loeser formula, is it still true that the Euler-Poincar…

open_problem | open | MathDB 页面 | 原文线索

V2-022898 | MATHDB-398066

Chen-rank formula for arrangement groups

摘录 / EXCERPT: If $G=G(\mathcal A)$ is an arrangement group, then, for $k\gg 0$, … where h_r is the number of irreducible components of dimension r in the first resonance variety $R^1(G)$.

open_problem | open | MathDB 页面 | 原文线索

V2-022899 | MATHDB-398075

Relating $E_\[4m\[1m\infty\[0m$-coalgebra invariants to realization spaces

摘录 / EXCERPT: Every realization space-the set of arrangements with a given combinatorial structure-is an algebraic variety defined over $\mathbb{Z}$. In the previous example, the defining equati…

open_problem | open | MathDB 页面 | 原文线索

V2-022900 | MATHDB-398120

Berkovich analytic tubular descent for coherent sheaves

摘录 / EXCERPT: Let k be a field with the trivial valuation, let X be a k-variety, let $Z\hookrightarrow X$ be a closed subvariety, and let $U=X\setminus Z$. Write $\widehat{X}_Z$ for the fo…

open_problem | open | MathDB 页面 | 原文线索

V2-022901 | MATHDB-398126

Comparison of the stable completion with the Berkovich analytification

摘录 / EXCERPT: Let $F=F^{\mathrm{alg}}$ be an $\mathbb{R}$-valued field, and let V be an algebraic variety over F. For the base structure F^c constructed using the field sort F, the value…

open_problem | open | MathDB 页面 | 原文线索

V2-022902 | MATHDB-398129

Fully homomorphic encryption without circular security assumptions

摘录 / EXCERPT: Devise a fully homomorphic encryption scheme for all functions without making a circular security assumption.

open_problem | open | MathDB 页面 | 原文线索

V2-022903 | MATHDB-398250

Local quotient structure conjecture for algebraic stacks

摘录 / EXCERPT: Conjecture. Let $\mathcal{X}$ be an algebraic stack with separated and quasi-compact diagonal of finite type over an algebraically closed field k. Suppose that all points of…

conjecture | open | MathDB 页面 | 原文线索

V2-022904 | MATHDB-398251

Local quotient structure of the stack of semistable curves

摘录 / EXCERPT: Does the algebraic stack $\mathcal{M}^{ss}_g$ of Deligne-Mumford semistable curves, allowing rational components to meet the rest of the curve in only two points, satisfy Conjectur…

open_problem | open | MathDB 页面 | 原文线索

V2-022905 | MATHDB-398255

Class of the closure of the zero section in the partial universal semiabelian family

摘录 / EXCERPT: Let d538g' be Mumford7s partial toroidal compactification of d538g parameterizing semiabelian varieties of torus rank one, and let d539g'tou0001d538g' be the corresponding partial…

open_problem | open | MathDB 页面 | 原文线索

V2-022906 | MATHDB-398326

Recursive formula for curve neighborhoods of Schubert varieties

摘录 / EXCERPT: Let $X=G/P$ be a generalized flag variety, with Weyl group W, parabolic Weyl group W_P, and Hecke product $[U+007F]$. For a positive root $\b1\in R^+$ with $s_\b1\notin W_P$, let…

open_problem | open | MathDB 页面 | 原文线索

V2-022907 | MATHDB-398328

Positivity in T-equivariant K-theory of flag varieties associated to Kac-Moody groups

摘录 / EXCERPT: Let G be a symmetrizable Kac-Moody group, let W be its Weyl group, and let $[U+007F]\alpha_1,\ldots,\alpha_r$ be the simple roots. If $p^w_{u,v}\in R(T)$ are defined by the product...

open_problem | open | MathDB 页面 | 原文线索

V2-022908 | MATHDB-398329

Dependence of degenerate flag varieties on the support of the highest weight

摘录 / EXCERPT: Let G be a semisimple connected algebraic group over an algebraically closed field of characteristic zero, and let λ,μ be dominant integral weights. If...

open_problem | open | MathDB 页面 | 原文线索

V2-022909 | MATHDB-398334

Realization of affine varieties as character varieties

摘录 / EXCERPT: Let S be an affine algebraic variety defined over $\mathbb{Q}$. There exist a finitely generated group Γ having Kazhdan's property (T) and an integer $n\geq 1$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022910 | MATHDB-398336

Smoothness of the principal component in the three-dimensional classical-group cases

摘录 / EXCERPT: For the invariant Hilbert schemes H_W considered for $G=\mathrm{GL}(V)$, $\mathrm{O}(V)$, or $\mathrm{SO}(V)$ with $\dim(V)=3$, is the principal component H_W^{prin} s...

open_problem | open | MathDB 页面 | 原文线索

V2-022911 | MATHDB-398338

Resolution of the symplectic invariant Hilbert scheme in dimension at least three

摘录 / EXCERPT: For a classical group $G\subset GL(V)$ acting on $W=V^{\oplus d}\oplus V^{*\oplus d}$, with moment map $\mu:W\to\mathfrak g^*$, determine whether the restriction of the Hilbert-Cho...

open_problem | open | MathDB 页面 | 原文线索

V2-022912 | MATHDB-398417

Luna conjecture on the classification of wonderful varieties

摘录 / EXCERPT: The wonderful varieties are classified by the abstract spherical systems.

conjecture | open | MathDB 页面 | 原文线索

V2-022913 | MATHDB-398419

Plancherel decomposition for spherical varieties via the dual group

摘录 / EXCERPT: Let X be a spherical variety for a reductive group G over a local field k, assuming that $X(k)$ carries a $G(k)$-invariant measure. The conjecture predicts a direct-integral...

open_problem | open | MathDB 页面 | 原文线索

V2-022914 | MATHDB-398420

Simple roots of the dual group of a spherical variety

摘录 / EXCERPT: The conjecture is that, for a spherical variety X, the set $\Sigma_{\mathrm{SV}}(X)$ of normalized simple spherical roots is the set of simple roots of the "dual group" of X de...

open_problem | open | MathDB 页面 | 原文线索

V2-022915 | MATHDB-398422

Affine-space conjecture for orbit closures in the invariant Hilbert moduli scheme

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0, let G be a connected reductive algebraic group, and let Γ be a finitely generated submonoid of the dominant...

conjecture | open | MathDB 页面 | 原文线索

V2-022916 | MATHDB-398423

Computing global extension modules on a smooth projective toric variety

摘录 / EXCERPT: Let X be a smooth projective toric variety over a computable field k. Identify $\operatorname{Pic}(X)$ with $\mathbb{Z}^r$, let $S=k[x_1,\ldots,x_n]$ be the $\mathbb{Z}^r$-grad...

open_problem | open | MathDB 页面 | 原文线索

V2-022917 | MATHDB-398430

Characterization of support filtrations restricting to t-structures on coherent complexes

摘录 / EXCERPT: For a noetherian semiseparated scheme X, characterize the decreasing filtrations by supports $\varphi:\cZ\to\mathcal P(X)$ for which the aisle...

open_problem | open | MathDB 页面 | 原文线索

V2-022918 | MATHDB-398437

Classification of abelian-subvariety-stable Brill-Noether subvarieties

摘录 / EXCERPT: In [1] the authors posed the problem of studying, and possibly classifying, situations like the following: C is a smooth, projective, complex curve of genus g, Z is an irredu...

open_problem | open | MathDB 页面 | 原文线索

V2-022919 | MATHDB-398438

Classification of abelian-stable subvarieties in Brill-Noether loci

摘录 / EXCERPT: Let C be a smooth projective complex curve of genus g, let Z be an irreducible r-dimensional subvariety of a proper Brill-Noether locus $W_d^s(C)\subsetneq J_d(C)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-022920 | MATHDB-398441

Converse to the Fano-type implication for globally F-regular type

摘录 / EXCERPT: Let X be a projective variety over a field of characteristic zero. If X is of globally F-regular type, must X be of Fano type?

open_problem | open | MathDB 页面 | 原文线索

V2-022921 | MATHDB-398442

Lipman-Zariski conjecture for klt spaces

摘录 / EXCERPT: If X is a complex variety with Kawamata log terminal (klt) singularities and its tangent sheaf T_X is locally free, then X is smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-022922 | MATHDB-398450

Simply connected conjugate varieties that are non-homeomorphic

摘录 / EXCERPT: Do there exist simply connected conjugate varieties X, X^{σ} which are non-homeomorphic?

open_problem | open | MathDB 页面 | 原文线索

V2-022923 | MATHDB-398451

Conjugate varieties with prescribed fundamental group and non-isomorphic real cohomology algebras

摘录 / EXCERPT: Corollary 2. Let G be the fundamental group of a smooth complex projective variety. Then there exist conjugate smooth complex projective varieties with fundamental group G, but...

open_problem | open | MathDB 页面 | 原文线索

V2-022924 | MATHDB-398455

The wild McKay correspondence

摘录 / EXCERPT: Let K be a local field with residue field $\a0\kappa$, let Γ be a finite group, and let $X=\mathbb{A}^{n}_{\mathcal{O}_{K}}/\Gamma$ be the quotient scheme associated to a...

open_problem | open | MathDB 页面 | 原文线索

V2-022925 | MATHDB-398484

Almost independence after adjoining all roots of unity in positive characteristic

摘录 / EXCERPT: Let K be a finitely generated field of characteristic $p>0$, let $L'=L\setminus\{p\}$, let X be a separated algebraic K-scheme, and let $q\in\mathbb N$. Then the restricted f...

open_problem | open | MathDB 页面 | 原文线索

V2-022926 | MATHDB-398486

Jarden’s conjecture on Hilbertianity of torsion fields of abelian varieties

摘录 / EXCERPT: Let K be a Hilbertian field and let A/K be an abelian variety. If $K(A_{\rm tor})$ denotes the extension obtained by adjoining all torsion points of A, then...

conjecture | open | MathDB 页面 | 原文线索

V2-022927 | MATHDB-398503

Grothendieck-Serre conjecture for reductive group schemes over regular semi-local domains containing an infinite field

摘录 / EXCERPT: Let R be a regular semi-local domain containing an infinite field, let K be its field of fractions, and let G be a reductive group scheme over R. Then the map ... induced by...

conjecture | open | MathDB 页面 | 原文线索

V2-022928 | MATHDB-398504

A modification theorem for principal bundles on the relative projective line

摘录 / EXCERPT: Let R be the semi-local ring of finitely many closed points on an irreducible smooth affine variety over an infinite field k, and set $U=\operatorname{Spec}R$. Let G be a sim...

open_problem | open | MathDB 页面 | 原文线索

V2-022929 | MATHDB-398505

Grothendieck-Serre conjecture for reductive group schemes

摘录 / EXCERPT: Let R be a regular local ring, let K be its field of fractions, and let G be a reductive group scheme over R. Then the map...

conjecture | open | MathDB 页面 | 原文线索

V2-022930 | MATHDB-398512

ldh descent exactness without resolution of singularities

摘录 / EXCERPT: Suppose k is a perfect field of exponential characteristic p, and let F be a presheaf of $\mathbb Z(l)$-modules with transfers on the category of separated schemes of finite...

open_problem | open | MathDB 页面 | 原文线索

V2-022931 | MATHDB-398514

Soulé’s conjecture on the order of vanishing of scheme zeta-functions

摘录 / EXCERPT: Assume that the scheme zeta-function $\zeta(X,s)$ admits a meromorphic continuation to the complex plane. For every connected scheme X of finite type over...

conjecture | open | MathDB 页面 | 原文线索

V2-022932 | MATHDB-398516

Property $(\ast)$ under reduction of CM-type abelian varieties

摘录 / EXCERPT: Let A be an abelian variety of CM-type over $\mathbb{Q}^{\mathrm{alg}}$. Property $(\ast)$ for A means that all Hodge classes are Lefschetz on all powers of A; for a reductio...

open_problem | open | MathDB 页面 | 原文线索

V2-022933 | MATHDB-398518

Factorization of motivic Hopf maps through Milnor-Witt K-theory modulo 24

摘录 / EXCERPT: For any integer $n\ge 3$, the morphism $(\nu_n)_*$ factors through a morphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-022934 | MATHDB-398521

Suslin-Weil-Suslin homology comparison conjecture

摘录 / EXCERPT: Let F be a finite field and let X be a smooth separated scheme of finite type over F. The natural homomorphism ... is conjectured to be an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-022935 | MATHDB-398522

Slices of co-operations for KGL

摘录 / EXCERPT: If S is smooth over a perfect field, then there is an isomorphism of cosimplicial motivic spectra ...

open_problem | open | MathDB 页面 | 原文线索

V2-022936 | MATHDB-398523

Slices of co-operations for algebraic K-theory

摘录 / EXCERPT: Theorem. Suppose that S is a finite-dimensional noetherian scheme. If, in addition, $s_0(1)\cong s_0(\mathrm{KGL})$ and $[s_0(1),s_0(1)]$ is torsion-free, then the maps in part (...

open_problem | open | MathDB 页面 | 原文线索

V2-022937 | MATHDB-398587

Tolerance choice for the inexact Rothe scheme for an SPDE

摘录 / EXCERPT: Under the stated assumptions that $A:D(A)\subset U\to U$ is linear, densely defined, strictly negative definite, self-adjoint, has 0 in its resolvent set, has compact inverse...

open_problem | open | MathDB 页面 | 原文线索

V2-022938 | MATHDB-398643

Autoequivalences arising from variation of GIT quotient

摘录 / EXCERPT: For a balanced wall crossing of a smooth quasiprojective variety X by a reductive group G, under a hypothesis on ω_X, the equivalence...

open_problem | open | MathDB 页面 | 原文线索

V2-022939 | MATHDB-398722

Uniqueness of the contact structure from rational curves

摘录 / EXCERPT: Let X be a complex-projective contact manifold with $b_{2}(X)=1$, in the case $X\not\cong\mathbb{P}^{\dim X}$. At a general point $x\in X$, the variety of tangent directions to c...

open_problem | open | MathDB 页面 | 原文线索

V2-022940 | MATHDB-398754

Generators of the cohomology ring for type A cluster varieties

摘录 / EXCERPT: For a linear tree (Dynkin diagram of type A) with n odd, one free coefficient variable α, and a generic fixed value of α, is the cohomology ring generated by the...

open_problem | open | MathDB 页面 | 原文线索

V2-022941 | MATHDB-398824

Vanishing condition for homogeneous bundles on generalized flag varieties

摘录 / EXCERPT: For a generalized flag variety $X=G/P$ and a simple P-module of highest weight $\bc$, with $E_\mu=G\times_P E_\mu$ and cotangent projection $\pi:(T^{1,0})^*\to X$, the vanishing...

open_problem | open | MathDB 页面 | 原文线索

V2-022942 | MATHDB-398827

Finite orbit criterion for quotients by the derived unipotent subgroup

摘录 / EXCERPT: Let G be a connected reductive complex algebraic group, let B be a Borel subgroup with unipotent radical U, and let X be an affine spherical G-variety. Then the categoric...

open_problem | open | MathDB 页面 | 原文线索

V2-022943 | MATHDB-398840

Coleman’s conjecture on multiplication fields of GL2-type abelian varieties

摘录 / EXCERPT: For each fixed dimension d, only finitely many number fields F occur as multiplication fields of GL$_2$-type abelian varieties of dimension d over $\Q$. Equivalently, among w...

conjecture | open | MathDB 页面 | 原文线索

V2-022944 | MATHDB-398845

Minimal conditions for non-Zariski-density of exceptional specializations

摘录 / EXCERPT: Let k be a finitely generated field of characteristic $p\ge 0$, let X be a smooth, separated, geometrically connected k-scheme, and let L be an infinite set of primes with...

open_problem | open | MathDB 页面 | 原文线索

V2-022945 | MATHDB-398880

A domain of discontinuity for the four-holed sphere character variety

摘录 / EXCERPT: Let S be the four-holed sphere, viewed as having fundamental group $F_3=\langle\alpha,\beta,\gamma,\delta:\alpha\beta\gamma\delta=I\rangle$, and let ... The mapping class group...

open_problem | open | MathDB 页面 | 原文线索

V2-022946 | MATHDB-398890

Separability criterion for the saturation morphism

摘录 / EXCERPT: Let $H\leq G$ be smooth group schemes over K, let Y be a G-scheme, and let $X\subset Y$ be an irreducible H-stable subscheme. For $x\in X(K)$, the orbit map $H\to X$,...

open_problem | open | MathDB 页面 | 原文线索

V2-022947 | MATHDB-398896

Cohomological realization of Kac polynomials via extended-quiver Nakajima varieties

摘录 / EXCERPT: Let Q be a quiver with dimension vector $v=(v_i)$. Extend Q by attaching to each vertex i a leg whose successive added dimensions are $v_i-1,v_i-2,\2601,1$, obtaining an indi...

open_problem | open | MathDB 页面 | 原文线索

V2-022948 | MATHDB-398902

Counterexample to the Lunts-Orlov extension without full faithfulness

摘录 / EXCERPT: Assume that k is an algebraically closed field of characteristic 0. Let X and Y be smooth projective k-schemes. Let $i_\eta:\eta\to X$ be the generic point of X, let...

open_problem | open | MathDB 页面 | 原文线索

V2-022949 | MATHDB-398924

Structure of the augmented-base-locus decomposition of the big cone

摘录 / EXCERPT: For a smooth projective variety X, the big cone naturally decomposes into subsets on which the augmented base locus $B^+(D)$ is constant. How do these subsets look like? Here...

open_problem | open | MathDB 页面 | 原文线索

V2-022950 | MATHDB-398925

Higher-cohomology vanishing for globally generated toric vector bundles

摘录 / EXCERPT: Do globally generated toric vector bundles on smooth toric varieties necessarily have vanishing higher cohomology?

open_problem | open | MathDB 页面 | 原文线索

V2-022951 | MATHDB-398926

Ample toric vector bundles need not be globally generated

摘录 / EXCERPT: There are ample but not globally generated toric vector bundles on smooth toric varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-022952 | MATHDB-398928

Higher-rank interpolation problem for zero-dimensional schemes

摘录 / EXCERPT: Fix a zero-dimensional scheme $Z\subset\mathbb{P}^2$. Determine the set of rational numbers $\mu\in\mathbb{Q}$ which are the slopes of vector bundles E such that...

open_problem | open | MathDB 页面 | 原文线索

V2-022953 | MATHDB-398929

Convergence in probability of normalized log-eigenvalues

摘录 / EXCERPT: Let X be a projective smooth variety over a field K that is either the complex numbers or a non-archimedean field, let L be a big line bundle on X equipped with continuous...

open_problem | open | MathDB 页面 | 原文线索

V2-022954 | MATHDB-398930

Kudla’s intersection-product conjecture for special-cycle generating series

摘录 / EXCERPT: In the Shimura-variety case, let $\Theta_{\mathrm{CH}}(\tau)$ be the generating series of special-cycle classes with values in the Chow group of a compactification. Kudla conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-022955 | MATHDB-398985

Oort conjecture on Shimura subvarieties in the Jacobian locus

摘录 / EXCERPT: For sufficiently large genus g, there do not exist non-zero-dimensional Shimura subvarieties generically contained in the locus of Jacobians of smooth curves.

conjecture | open | MathDB 页面 | 原文线索

V2-022956 | MATHDB-399014

Irreducible components of varieties associated with ROG cones

摘录 / EXCERPT: For a ROG cone $K\subset S_+^n$, the set $\{x\in\mathbb{R}^n\mid xx^T\in K\}$ defines a projective algebraic variety. Which irreducible components can occur?

open_problem | open | MathDB 页面 | 原文线索

V2-022957 | MATHDB-399044

Multiplier biholomorphisms of irreducible varieties and multiplier-algebra isomorphisms

摘录 / EXCERPT: Let $V,W\subset \mathbb{B}_d$ be irreducible varieties with $d<\infty$ that are multiplier biholomorphic. Are $\mathcal{M}_V$ and $\mathcal{M}_W$ isomorphic?

open_problem | open | MathDB 页面 | 原文线索

V2-022958 | MATHDB-399047

Geometric Arveson-Douglas conjecture

摘录 / EXCERPT: Let V be a homogeneous variety in $\mathbb{C}^{d}$, and let ... If $S_{j}=P_{\mathcal{M}^{\perp}}M_{z_{j}}|_{\mathcal{M}^{\perp}}$ for $j=1,\ldots,d$, then ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-022959 | MATHDB-399049

Geometric Arveson-Douglas conjecture for smooth homogeneous varieties

摘录 / EXCERPT: Let V be a homogeneous variety in $\\mathbb{C}^d$ such that $V\setminus\{0\}$ is a complex submanifold of $\mathbb{C}^d\setminus\{0\}$ of dimension n, let $\alpha\in\mathbb{R}$...

conjecture | open | MathDB 页面 | 原文线索

V2-022960 | MATHDB-399067

Kudla’s modularity conjecture

摘录 / EXCERPT: For an orthogonal Shimura variety X, the generating series $f_X(\tau)=\sum_t Z(t)\exp\!\bigl(2\pi i\,\operatorname{tr}(t\tau)\bigr)$, whose coefficients are the special cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-022961 | MATHDB-399071

Deligne’s conjecture on algebraicity of critical values of motivic L-functions

摘录 / EXCERPT: Let M be a motive over $\\mathbb{Q}$ with coefficients in an algebraic number field E. If $n\in\mathbb{Z}$ is a critical point of M, then ... where the sign $\pm$ is the same a...

conjecture | open | MathDB 页面 | 原文线索

V2-022962 | MATHDB-399081

Codimension-three extension theorem for holonomic DQ-modules

摘录 / EXCERPT: Let X be a complex manifold endowed with a DQ-algebroid stack $\cal A_X$ such that the associated Poisson structure is symplectic. Let $\cal Lambda$ be a closed Lagrangian analyt...

open_problem | open | MathDB 页面 | 原文线索

V2-022963 | MATHDB-399082

Coherence of extensions of reflexive formal power-series modules

摘录 / EXCERPT: Let X be a complex manifold, let Y be a subvariety of X, and let $j:X\setminus Y\to X$ be the open embedding. If N is a coherent reflexive $\O_X\setminus Y[[\hbar]]$-module...

open_problem | open | MathDB 页面 | 原文线索

V2-022964 | MATHDB-399099

Positivity of Chern classes of Schubert cells in other flag varieties

摘录 / EXCERPT: Are the corresponding numbers nonnegative for Schubert cells in other flag varieties? Equivalently, is the Chern class of a B-orbit in G/P effective, where B is a Borel subgr...

open_problem | open | MathDB 页面 | 原文线索

V2-022965 | MATHDB-399102

Conjecture 19: a generalized Mordell-Lang finiteness statement

摘录 / EXCERPT: Let $A\subset (\mathbb{C}^{\times})^n$ be an algebraic variety of codimension $>k$, and let A' be a subvariety of anomalous points of A. Conjecture 19. Let L be a k-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-022966 | MATHDB-399103

Effective cone of a Bott-Samelson variety

摘录 / EXCERPT: Let G be a reductive (or Kac-Moody) group with Borel subgroup B, maximal torus T, and Weyl group W. For a sequence of simple roots...

open_problem | open | MathDB 页面 | 原文线索

V2-022967 | MATHDB-399110

Analyticity of the singularity formation set for the Kähler-Ricci flow

摘录 / EXCERPT: The singularity formation set Σ is an analytic subvariety of X. The report states that this decade-old conjecture of Feldman-Ilmanen-Knopf (2003) was proved by Valentino T...

open_problem | open | MathDB 页面 | 原文线索

V2-022968 | MATHDB-399127

Gross-Hacking-Keel canonical-basis and mirror-duality conjecture

摘录 / EXCERPT: Conjecture 1 (GHK 2011). Let U be an affine log Calabi-Yau variety with maximal boundary. Let A be the complex vector space with basis $U^{\trop}(\mathbb{Z})$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-022969 | MATHDB-399135

Numerical versus rational triviality for polynomials in symmetric divisor classes

摘录 / EXCERPT: Let A be an abelian variety. If $P\in \operatorname{CH}^*(A)$ is a polynomial with rational coefficients in the first Chern classes of symmetric line bundles on A, then P is...

open_problem | open | MathDB 页面 | 原文线索

V2-022970 | MATHDB-399136

Are all tautological relations motivic?

摘录 / EXCERPT: For the tautological ring $T(J_{g,1})$ of the universal Jacobian, are all relations of motivic nature?

open_problem | open | MathDB 页面 | 原文线索

V2-022971 | MATHDB-399137

Conjecture 0.14 for Hilbert schemes of K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface and let $X=S^{[m]}$ be the m-th Hilbert scheme of S. With the decomposition ... Conjecture 0.14 asserts that ... Moreover, the cycle class map ... is injectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-022972 | MATHDB-399138

Motivic nature of relations in tautological rings

摘录 / EXCERPT: For the tautological rings $T(J_{g,1})$ and $R(C^{[n]}_{g,1})$, are all relations of motivic nature?

open_problem | open | MathDB 页面 | 原文线索

V2-022973 | MATHDB-399147

Explicit description of Eisenstein cohomology classes for Hilbert-Blumenthal varieties

摘录 / EXCERPT: For a totally real field F of degree g, with $S_K=K K_\∞\backslash \mathrm{GL}_2(\mathbb A_F)/\mathrm{GL}_2(F)$, boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-022974 | MATHDB-399151

Comparison between rigid cohomology with coefficients and overconvergent de Rham-Witt cohomology

摘录 / EXCERPT: Let X be a smooth quasi-projective scheme over k, and let $E\in \operatorname{Isoc}^{\dagger}(X\subset \overline{X}/W(k))$ be a locally free isocrystal. Then there is a natural...

open_problem | open | MathDB 页面 | 原文线索

V2-022975 | MATHDB-399155

Colliot-Thélène-Kato-Saito exactness conjecture for zero-cycles

摘录 / EXCERPT: For every smooth, proper, irreducible variety X over a number field k, the complex ... is exact. Here $\widehat{M}=\varprojlim_{n\geq 1}M/nM$,...

conjecture | open | MathDB 页面 | 原文线索

V2-022976 | MATHDB-399156

Colliot-Thélène-Kato-Saito conjecture for zero-cycles

摘录 / EXCERPT: Let k be a number field, let X be a smooth, proper, irreducible variety over k, and write $\Omega=\Omega_f\sqcup\Omega_\infty$ for the set of places of k. For an abelian gr...

conjecture | open | MathDB 页面 | 原文线索

V2-022977 | MATHDB-399247

Yau-Tian-Donaldson converse for Kähler-Ricci solitons

摘录 / EXCERPT: Let c(X,V) be a singular Fano variety together with a holomorphic vector field, with c(V) understood to be the Tian-Zhu extremal vector field in the intended formulation. If c((X,V...

open_problem | open | MathDB 页面 | 原文线索

V2-022978 | MATHDB-399248

Proper coarse moduli space of modified K-polystable Fano varieties

摘录 / EXCERPT: Conjecture 6. Consider the moduli functor of families $\pi:\mathcal{X}\to B$ of Fano varieties with log-terminal singularities such that there exists a holomorphic vector field V...

open_problem | open | MathDB 页面 | 原文线索

V2-022979 | MATHDB-399252

Rationally connected half-dimensional subschemes on projective hyper-Kähler manifolds

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold of dimension $2n$. Does there exist a rationally connected subscheme Y of X of pure dimension n?

open_problem | open | MathDB 页面 | 原文线索

V2-022980 | MATHDB-399260

Yau-Tian-Donaldson conjecture for arbitrary polarizations

摘录 / EXCERPT: Let (X,L) be a smooth projective complex variety endowed with an ample line bundle, and assume that the automorphism group of (X,L) is finite. The Yau-Tian-Donaldson conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-022981 | MATHDB-399316

Coefficients ensuring geometric convergence of the iterative scheme

摘录 / EXCERPT: Determine suitable coefficients b[?]$\b[?]\gamma,\gamma_c$b[?] for which the scheme has geometric convergence.

open_problem | open | MathDB 页面 | 原文线索

V2-022982 | MATHDB-399323

Existence via PDE techniques on more general Fano varieties

摘录 / EXCERPT: Can PDE techniques yield existence on more general Fano varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-022983 | MATHDB-399365

Question 1 for Cluckers-Loeser motivic integration

摘录 / EXCERPT: For Cluckers-Loeser motivic integration, if two integrals satisfy $\int_{X_1} f_1=\int_{X_2} f_2$, does it follow that one integral can be transformed into the other using the stan...

open_problem | open | MathDB 页面 | 原文线索

V2-022984 | MATHDB-399406

The Paris-Wilkie problem of the consistency of no collection and no exponentiation

摘录 / EXCERPT: Does $I\Delta_0 + \neg\mathrm{Exp}$ prove $B\Sigma_1$? Here $B\Sigma_1$ is the theory extending $I\Delta_0$ by all instances of the Σ_1 collection scheme ... where φ r...

open_problem | open | MathDB 页面 | 原文线索

V2-022985 | MATHDB-399506

Relative Cartier criterion

摘录 / EXCERPT: Let C be a smooth curve and $f:X\to C$ a proper, flat family of normal varieties of dimension n. Let D be a Weil divisor on X such that its restriction D_c is an ample Ca...

open_problem | open | MathDB 页面 | 原文线索

V2-022986 | MATHDB-399507

Grothendieck-Lefschetz theorem for the local Picard group

摘录 / EXCERPT: Let $(x\in X)$ be a local scheme that is essentially of finite type over a field, and let $x\in D\subset X$ be a Cartier divisor. Set $U:=X\setminus\{x\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-022987 | MATHDB-399525

The Nash problem on families of arcs

摘录 / EXCERPT: Describe the Nash valuations over a complex algebraic variety X in terms of resolutions of singularities.

open_problem | open | MathDB 页面 | 原文线索

V2-022988 | MATHDB-399553

Detection of projectivity by π-cosupport

摘录 / EXCERPT: Let k be a field of characteristic $p>0$, let G be a finite group scheme, and let M be a kG-module. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-022989 | MATHDB-399591

Entropy-conservative numerical fluxes for conservative schemes

摘录 / EXCERPT: Given an entropy pair (η,F) and the entropy variables $v=\eta'(u)$, consider scheme (2),...

open_problem | open | MathDB 页面 | 原文线索

V2-022990 | MATHDB-399609

Equality of Hodge numbers for Pfaffian double mirrors

摘录 / EXCERPT: For any odd $n\ge 5$, the associated varieties satisfy ...

open_problem | open | MathDB 页面 | 原文线索

V2-022991 | MATHDB-399610

Mirror-symmetry isomorphism for homogeneous spaces in the cominuscule case

摘录 / EXCERPT: Let $X=G/P$ be a homogeneous space with P cominuscule, and let $(\check X,W)$ be the mirror Landau-Ginzburg model obtained by projecting the Richardson variety to...

open_problem | open | MathDB 页面 | 原文线索

V2-022992 | MATHDB-399625

Donkin’s \(p,r)-Filtration Conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic $p>0$, let G be a simple, simply connected algebraic group scheme, and let $r\ge 1$. For a finite-dimensional rational...

conjecture | open | MathDB 页面 | 原文线索

V2-022993 | MATHDB-399651

Equidimensionality of nilpotent commuting varieties for parabolic subgroups

摘录 / EXCERPT: Let $P\subseteq \mathrm{GL}_n(\mathbb{C})$ be a standard parabolic subgroup, let $\mathfrak p$ be its Lie algebra, and let $\mathcal N_{\mathfrak p}=\mathcal N\cap\mathfrak p$, whe...

open_problem | open | MathDB 页面 | 原文线索

V2-022994 | MATHDB-399652

Exponential prime torsion in Schubert varieties for \(\mathrm{SL}_N\)

摘录 / EXCERPT: Let p-torsion in a variety X mean p-torsion in a stalk or costalk of the intersection cohomology complex $IC(X;\mathbb Z)$. There exists a constant $c>1$ such that, for all s...

open_problem | open | MathDB 页面 | 原文线索

V2-022995 | MATHDB-399654

Forgetful functor for mixed modular derived categories

摘录 / EXCERPT: Let F be a field, and let X be a complex algebraic variety equipped with an algebraic stratification $X=\bigsqcup_{s\in S}X_s$ whose strata are affine spaces. With $D^b_S(X,F)$...

open_problem | open | MathDB 页面 | 原文线索

V2-022996 | MATHDB-399664

Limit of the segmented linearisation scheme for quantum time-correlation functions

摘录 / EXCERPT: Determine the precise nature of the limit of the approximate symmetrised time-correlation function obtained by concatenating linearised short-time propagators as the number of segm...

open_problem | open | MathDB 页面 | 原文线索

V2-022997 | MATHDB-399839

Character attached to the Hodge bundle and the case $p=2$

摘录 / EXCERPT: Conjecturally, for Hodge-type Shimura varieties, the character attached to the Hodge bundle ω is always a multiple of a minuscule character; consequently, the result is conj...

open_problem | open | MathDB 页面 | 原文线索

V2-022998 | MATHDB-399841

Coincidence with the Bruhat-Tits stratification in all Görtz-He cases

摘录 / EXCERPT: For every case of affine Deligne-Lusztig varieties studied by Görtz and He, the stratification defined by the invariant f_g should coincide with the corresponding Bruhat-Tits str...

open_problem | open | MathDB 页面 | 原文线索

V2-022999 | MATHDB-399844

Generic Tate cycles on a unitary Shimura variety

摘录 / EXCERPT: Let F be a real quadratic field, let E be an imaginary quadratic extension of F, and let p be inert in F and split in E. For the PEL-type Shimura variety...

open_problem | open | MathDB 页面 | 原文线索

V2-023000 | MATHDB-399845

Extra algebraic cycles when Frobenius eigenvalues are not distinct

摘录 / EXCERPT: If the assumption in Theorem 1(ii) that the eigenvalues of $\rho_{\pi_v}(\operatorname{Frob}_{p^2})$ are distinct is not satisfied, construct additional algebraic cycles sufficient...

open_problem | open | MathDB 页面 | 原文线索

V2-023001 | MATHDB-399847

Green-Griffiths conjecture

摘录 / EXCERPT: If X is a projective manifold of general type, then there exists an algebraic proper subvariety of X which contains the image of all holomorphic curves from the complex plane t...

conjecture | open | MathDB 页面 | 原文线索

V2-023002 | MATHDB-399848

Extension theorem for affine toric varieties

摘录 / EXCERPT: Let Y be an irreducible, not necessarily normal affine toric variety. Let S be a factorial subvariety of a reduced Stein space X such that the maps...

open_problem | open | MathDB 页面 | 原文线索

V2-023003 | MATHDB-399849

Extension dichotomy for holomorphic maps into affine toric varieties

摘录 / EXCERPT: Let Y be an irreducible, not necessarily normal affine toric variety of dimension d in $\\mathbb{C}^n$, with $0\in Y$. Suppose that the normalisation of Y is $\mathbb{C}^d$.n...

open_problem | open | MathDB 页面 | 原文线索

V2-023004 | MATHDB-399850

Approximation of holomorphic maps into affine toric varieties

摘录 / EXCERPT: Proposition 5. Let Ω be a Runge domain of finite embedding dimension in a factorial Stein space X, such that $H^p(\Omega,\mathbb{Z})=0$ for $p=1,2$. Let Y be an irreduci...

open_problem | open | MathDB 页面 | 原文线索

V2-023005 | MATHDB-399851

Bogomolov’s transverse-embedding realization question

摘录 / EXCERPT: For any compact complex manifold (X,J), does there exist a triple (Z,D,α) formed by a smooth complex projective variety Z, an algebraic foliation D on Z, and an isot...

open_problem | open | MathDB 页面 | 原文线索

V2-023006 | MATHDB-399872

Higher-order energy-stable multirate partitioned Runge-Kutta schemes for mechanical FSI

摘录 / EXCERPT: Can higher-order energy-stable partitioned schemes be found within the class of multirate partitioned Runge-Kutta methods for mechanical fluid-structure interaction?

open_problem | open | MathDB 页面 | 原文线索

V2-023007 | MATHDB-399919

Quantitative Hartshorne connectedness for arithmetically Gorenstein schemes

摘录 / EXCERPT: Let $X \subset \mathbb{P}^n$ be an arithmetically Gorenstein projective scheme of Castelnuovo-Mumford regularity $r+1$. (A) If every irreducible component of X has regularity at...

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V2-023008 | MATHDB-399962

Generic finiteness or birationality of the moduli map on Severi components

摘录 / EXCERPT: Let K_p be the moduli space of primitively polarized K3 surfaces (X,L) with $L^2=2p-2>0$, and let $V_{p,n,\delta}$ be the universal Severi variety of irreducible curves in...

open_problem | open | MathDB 页面 | 原文线索

V2-023009 | MATHDB-399964

Maximal rank of the moduli map for components of Severi varieties on K3 surfaces

摘录 / EXCERPT: Let $p\ge 3$, $n\ge 1$, and $0\le \delta\le \pi_a(p,n):=n^2(p-1)+1$, and let $\psi_{p,n,\delta}:V_{p,n,\delta}\to M_g$ be the moduli map, where $g=\pi_a(p,n)-\delta$. For an irredu...

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V2-023010 | MATHDB-399996

Efficient one-round delegation for deterministic computations

摘录 / EXCERPT: Do there exist efficient 1-round delegation schemes for all deterministic computations?

open_problem | open | MathDB 页面 | 原文线索

V2-023011 | MATHDB-399997

The conjectured \Omega(n^{1/3}) lower bound for 2-server PIR

摘录 / EXCERPT: Every 2-server private information retrieval scheme for an n-bit database has communication cost $\Omega(n^{1/3})$.

conjecture | open | MathDB 页面 | 原文线索

V2-023012 | MATHDB-400026

Desingularization in positive characteristic

摘录 / EXCERPT: Does every integral algebraic variety X over a field of positive characteristic admit a proper birational morphism $f:X'\to X$ such that X' is regular?

open_problem | open | MathDB 页面 | 原文线索

V2-023013 | MATHDB-400029

Hyperbolic Ax-Lindemann theorem for the modular function

摘录 / EXCERPT: Let $\mathbb{H}$ denote the complex upper half-plane and let $j:\mathbb{H}\to\mathbb{C}$ be the elliptic modular function. An algebraic subvariety of $\mathbb{H}^n$ means the trace...

open_problem | open | MathDB 页面 | 原文线索

V2-023014 | MATHDB-400032

Refined tropical multiplicity and motivic curve-counting invariant

摘录 / EXCERPT: In the setting where $C\to B$ is the universal curve over the locus of reduced and irreducible curves in a general δ-dimensional linear series, and where Γ is a tropi...

open_problem | open | MathDB 页面 | 原文线索

V2-023015 | MATHDB-400067

Nondefectivity of balanced Segre varieties

摘录 / EXCERPT: Let $X=u(\d53f\bigl(P^{n_1}\d53f\times\cdots\times P^{n_t}\d53f\bigr)\subset P^N$, where $n_1\ge\cdots\ge n_t\ge2$, $N=(n_1+1)\cdots(n_t+1)-1$, and X is the Segre variety. Call...

open_problem | open | MathDB 页面 | 原文线索

V2-023016 | MATHDB-400068

Secant varieties of varieties of reducible hypersurfaces

摘录 / EXCERPT: Let $\lambda=[d_1,d_2,\ldots,d_r]$ be a partition of d, with $r\ge 2$, $d_1\ge\cdots\ge d_r\ge 1$, and $d=d_1+\cdots+d_r$. For $n\ge 3$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-023017 | MATHDB-400077

Elekes-Szabó theorem for algebraic varieties

摘录 / EXCERPT: Let A,B,C be irreducible k-dimensional complex algebraic varieties, and let $F\subset A\times B\times C$ be an irreducible subvariety whose projection to each of $A\times B$,...

open_problem | open | MathDB 页面 | 原文线索

V2-023018 | MATHDB-400079

Proper intersections for purely non-modular relations

摘录 / EXCERPT: Let X,Y be L-definable sets (say varieties over a pseudo-finite field F), and let $U\leq X\times Y$ be purely non-modular. Then there exists an L-definable finite partition...

open_problem | open | MathDB 页面 | 原文线索

V2-023019 | MATHDB-400082

Exponential Ax-Lindemann theorem

摘录 / EXCERPT: Let $\exp:\mathbb{C}^n\to(\mathbb{C}^{\times})^n$ be the morphism $(z_1,\ldots,z_n)\mapsto(\exp(z_1),\ldots,\exp(z_n))$. Let V be an irreducible algebraic subvariety of...

open_problem | open | MathDB 页面 | 原文线索

V2-023020 | MATHDB-400120

Postulation of generic linear subspaces and one fat linear subspace

摘录 / EXCERPT: Let $X \subset \mathbb{P}^{2n+1}$ be a subscheme consisting of $s \ge 1$ generic reduced linear subspaces of dimension n and a single generic linear subspace of dimension $n-1$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-023021 | MATHDB-400132

Criterion for unexpected curves

摘录 / EXCERPT: Let k be an infinite field and let $Z\subset \mathbb{P}^2_k$ be a reduced zero-dimensional scheme. For a general point $P\in\mathbb{P}^2_k$, define ... and ... Then Z has an unexpe...

open_problem | open | MathDB 页面 | 原文线索

V2-023022 | MATHDB-400161

Entropy monotonicity of the semi-Lagrangian discontinuous Galerkin scheme

摘录 / EXCERPT: Determine whether the semi-Lagrangian discontinuous Galerkin scheme for the Vlasov-Poisson equation satisfies the second law of thermodynamics, in the sense that it does not decrea...

open_problem | open | MathDB 页面 | 原文线索

V2-023023 | MATHDB-400163

Correspondence between semipositive toric metrics and concave functions

摘录 / EXCERPT: Let K be a non-archimedean field as above, let X_Σ be the toric variety associated with the fan Σ, and let D be a base-point-free toric Cartier divisor on...

open_problem | open | MathDB 页面 | 原文线索

V2-023024 | MATHDB-400165

Global finite abelian quotients of varieties with tame abelian quotient singularities

摘录 / EXCERPT: If X is a variety over an algebraically closed field with tame abelian quotient singularities, then is it of the form U/G with U a smooth variety and G a finite abelian gro...

open_problem | open | MathDB 页面 | 原文线索

V2-023025 | MATHDB-400166

Fulton’s global quotient question for varieties with quotient singularities

摘录 / EXCERPT: Let k be an algebraically closed field. If X is a variety over k with quotient singularities-meaning that there is an étale cover $\{X_i\to X\}_i$ with each $X_i=U_i/G_i$, wh...

open_problem | open | MathDB 页面 | 原文线索

V2-023026 | MATHDB-400167

Classification of additive actions on toric varieties

摘录 / EXCERPT: Classify additive actions on toric varieties; namely, for an irreducible algebraic toric variety X of dimension n over an algebraically closed field K of characteristic zero,...

open_problem | open | MathDB 页面 | 原文线索

V2-023027 | MATHDB-400168

Weak approximability of tropical currents associated to loopless matroids

摘录 / EXCERPT: Let X be the toric variety of the permutohedron, and let T_C be the tropical current associated to a loopless matroid C. Is T_C a weak limit of effective algebraic cycles o...

open_problem | open | MathDB 页面 | 原文线索

V2-023028 | MATHDB-400169

Gorenstein property of the affine cone over a five-factor Segre secant variety

摘录 / EXCERPT: Is the affine cone over $\sigma_2(\mathbb{P}^1\times\mathbb{P}^1\times\mathbb{P}^1\times\mathbb{P}^1\times\mathbb{P}^1)$ Gorenstein?

open_problem | open | MathDB 页面 | 原文线索

V2-023029 | MATHDB-400170

Positivity of toric vector bundles detected on invariant curves

摘录 / EXCERPT: Let X_Σ be a smooth toric variety and let E be a toric vector bundle of rank r. For every invariant curve $C\subset X_\Sigma$, write ... Then E is nef if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-023030 | MATHDB-400206

Kawaguchi-Silverman conjecture for dominant self-morphisms

摘录 / EXCERPT: Let X be a smooth projective variety, let $f:X\to X$ be a dominant self-morphism, and let $P\in X(\overline{K})$, with X, f, and an ample divisor H defined over a number fi...

conjecture | open | MathDB 页面 | 原文线索

V2-023031 | MATHDB-400207

Arithmetic degree of a Zariski-dense orbit on an abelian variety

摘录 / EXCERPT: Let X be an abelian variety over a number field, and let $f:X\to X$ be a dominant self-morphism. For $P\in X(\overline{K})$, if the orbit $O_f(P)=\{f^n(P):n\ge 0\}$ is Zariski de...

open_problem | open | MathDB 页面 | 原文线索

V2-023032 | MATHDB-400208

Explicit lower bound for the Néron-Tate height

摘录 / EXCERPT: Let k be a number field of degree $D=[k:\mathbb{Q}]$, let (A,L) be a polarized abelian variety over k of dimension g, and let $p\in A(k)\setminus A_{\mathrm{tors}}$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-023033 | MATHDB-400210

Lang-Silverman conjecture for simple abelian varieties

摘录 / EXCERPT: Let $g\ge 1$ be an integer. For every number field K, there exists a positive constant $c_2=c_2(K,g)$ such that, for every simple abelian variety A/K of dimension g, every am...

conjecture | open | MathDB 页面 | 原文线索

V2-023034 | MATHDB-400213

André’s logarithmic Ax-Schanuel theorem for an abelian variety

摘录 / EXCERPT: Let $K=\mathbb{C}(S)$ be the function field of an algebraic curve over $\mathbb{C}$, let A/K be an abelian variety with $K/\mathbb{C}$-trace A_0, and let $y\in A(K)$. Let B b...

open_problem | open | MathDB 页面 | 原文线索

V2-023035 | MATHDB-400214

Pink’s relative Manin-Mumford conjecture for sections

摘录 / EXCERPT: Conjecture 1. If the relative dimension of A over C is at least two and $s(C)$ is not contained in a proper algebraic subgroup scheme, then there exist only finitely many torsi...

conjecture | open | MathDB 页面 | 原文线索

V2-023036 | MATHDB-400215

Weak Ax-Schanuel conjecture for the universal vectorial extension

摘录 / EXCERPT: Let $\widetilde{Z}^{\natural}$ be a complex analytic irreducible subvariety of $\mathbb{C}^{2g}\times\mathcal{H}_{g}^{+}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023037 | MATHDB-400226

The Bounded Height Conjecture for semiabelian varieties

摘录 / EXCERPT: Let G be a semiabelian variety endowed with a G-linearized ample line bundle L, and let X be a closed subvariety, with G, L, and X defined over $\Q$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023038 | MATHDB-400227

Brauer-Siegel asymptotics for abelian varieties

摘录 / EXCERPT: Fix a global field K and an integer g, and let A/K range over abelian varieties of dimension g for which the Shafarevich-Tate group $X(A/K)$ is finite. With ... does the clas...

open_problem | open | MathDB 页面 | 原文线索

V2-023039 | MATHDB-400323

Uniform boundedness of full level-p^r structures, conditional on Lang’s conjecture

摘录 / EXCERPT: Assume Lang’s conjecture. Fix a number field K, a dimension g, and a prime p. Then there is a number N such that, for every r with $p^r>N$, no abelian variety of dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-023040 | MATHDB-400324

Motivic hyper-Kähler resolution conjecture for generalized Kummer varieties

摘录 / EXCERPT: For an abelian surface A and a positive integer n, let $A^{n+1}_0:=\ker(A^{n+1}\to A)$ be the kernel of the summation map, with its natural action by $\mathfrak S_{n+1}$. The g...

conjecture | open | MathDB 页面 | 原文线索

V2-023041 | MATHDB-400330

Deformation invariance of rationality for smooth projective threefolds

摘录 / EXCERPT: Let B be an irreducible complex variety and let $\d719:X\to B$ be a smooth projective morphism with relative dimension $\dim(X/B)=3$. Is rationality deformation invariant; equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-023042 | MATHDB-400353

Field of definition of the intermediate Jacobian

摘录 / EXCERPT: Is the field of definition of the intermediate Jacobian (assuming this to be algebraic, or its algebraic part if it is not) of a smooth projective complex variety X the same as t...

open_problem | open | MathDB 页面 | 原文线索

V2-023043 | MATHDB-400355

Kato-Saito’s conjecture for relative Milnor K-theory

摘录 / EXCERPT: Let X be a smooth d-dimensional variety over a finite field of characteristic p, and let $Y\hookrightarrow X$ be a normal crossing divisor. Writing Y_s for the sth infini...

conjecture | open | MathDB 页面 | 原文线索

V2-023044 | MATHDB-400356

Grothendieck’s standard conjecture of type D

摘录 / EXCERPT: Let k be a base field of characteristic zero and let X be a smooth projective k-scheme. Writing $Z^*(X)_{\bQ\}$ for the $\bQ\$-vector space of algebraic cycles on X, Grothe...

conjecture | open | MathDB 页面 | 原文线索

V2-023045 | MATHDB-400358

Homological projective duality invariance for standard conjectures D and V

摘录 / EXCERPT: Let k be a field of characteristic zero, let V be a finite-dimensional k-vector space, and let X be a smooth projective k-scheme equipped with a map $f:X\to\mathbb{P}(V)$...

conjecture | open | MathDB 页面 | 原文线索

V2-023046 | MATHDB-400360

Theorem 4, bilinear-divisor stack case of Grothendieck’s standard conjecture

摘录 / EXCERPT: Let k be a field of characteristic zero, let W be a finite-dimensional k-vector space of odd dimension, and let $X=(\mathbb{P}(W)\times\mathbb{P}(W))/\mu_2$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-023047 | MATHDB-400361

Grothendieck’s standard conjecture of Hodge type for abelian varieties

摘录 / EXCERPT: Let A be a polarized abelian variety over a field k of dimension g. For $2i\leq g$, let $V^i(A)$ be the quotient, modulo numerical equivalence, of the algebraic cycles of cod...

conjecture | open | MathDB 页面 | 原文线索

V2-023048 | MATHDB-400362

Standard conjecture of Hodge type

摘录 / EXCERPT: Let k be a base field and let (X,L) be a polarized, smooth, projective, geometrically connected variety over k of dimension g. For each integer i with $2i\le g$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-023049 | MATHDB-400364

Motivic description of a curve with non-reduced modulus after applying \(\omega_!\)

摘录 / EXCERPT: Let k be a perfect field, let C be a smooth proper connected curve over k having a k-rational point, and let D be a non-reduced effective divisor on C. Is there an isom...

open_problem | open | MathDB 页面 | 原文线索

V2-023050 | MATHDB-400365

Conjectural Weil-Arakelov and Weil-étale cohomology satisfying properties (1)-(9)

摘录 / EXCERPT: For arithmetic schemes X (separated schemes of finite type over $\mathbb{Z}$), all $n\in\mathbb{Z}$, and coefficients $A\in\mathrm{LCA}$ of finite ranks, do there exist Weil-Arak...

open_problem | open | MathDB 页面 | 原文线索

V2-023051 | MATHDB-400366

Conjecture B(X,n): Perfectness of the regulator pairing

摘录 / EXCERPT: Let $X/\mathbb{Z}$ be a proper regular arithmetic scheme of pure dimension d, and let $n\in\mathbb{Z}$. Conjecture B(X,n) asserts that the pairing ... is a perfect pairing of fin...

conjecture | open | MathDB 页面 | 原文线索

V2-023052 | MATHDB-400367

Deninger-style long exact sequence and semisimplicity

摘录 / EXCERPT: For an arithmetic scheme X and $n\in\mathbb Z$, one expects a long exact sequence ... such that the map...

open_problem | open | MathDB 页面 | 原文线索

V2-023053 | MATHDB-400368

Vanishing order of the zeta function at integer points

摘录 / EXCERPT: For a proper regular arithmetic scheme $X/\mathbb{Z}$ of pure dimension d and $n\in\mathbb{Z}$, the vanishing order of its zeta function at $s=n$ is conjectured to satisfy ...

open_problem | open | MathDB 页面 | 原文线索

V2-023054 | MATHDB-400371

Descent of the algebraic intermediate Jacobian

摘录 / EXCERPT: Given a smooth projective variety X defined over a subfield $K\subseteq\mathbb{C}$, does the algebraic intermediate Jacobian $J^{2n+1}_a(X_{\mathbb{C}})$ admit a model J over...

open_problem | open | MathDB 页面 | 原文线索

V2-023055 | MATHDB-400375

Canonical isomorphism for motives of tubes and nearby motivic sheaves

摘录 / EXCERPT: Let k be a field of characteristic zero, $R=k[[t]]$, and $K=k((t))$. Let $f:X\to\operatorname{Spec}(R)$ be a separated finite-type R-scheme, with generic fibre X_η smooth...

open_problem | open | MathDB 页面 | 原文线索

V2-023056 | MATHDB-400387

Multivariate characterization of convergence for nonnegative subdivision schemes

摘录 / EXCERPT: Find a simple characterization, analogous to the univariate greatest-common-divisor criterion, for uniform convergence of multivariate stationary subdivision schemes with finitely...

open_problem | open | MathDB 页面 | 原文线索

V2-023057 | MATHDB-400442

Conjecture 6 on Frobenius eigenvectors in convergent completed group algebras

摘录 / EXCERPT: Let X/k be any variety satisfying condition $(*)$ above, namely that I^n/I^{n+1} is torsion-free for every n, where I is the augmentation ideal in...

conjecture | open | MathDB 页面 | 原文线索

V2-023058 | MATHDB-400445

Finiteness of the atypical locus for variations of mixed Hodge structures

摘录 / EXCERPT: Let S be an irreducible smooth quasi-projective variety over $\a0\mathbb{C}$, and let $\mathbb{V}\to S$ be a variation of mixed $\mathbb{Q}$-Hodge structures. For an irreducible...

open_problem | open | MathDB 页面 | 原文线索

V2-023059 | MATHDB-400447

André-Oort conjecture for variations of mixed Hodge structures

摘录 / EXCERPT: Let $V\to S$ be any variation of mixed Hodge structure over a smooth irreducible complex quasi-projective variety S. Suppose that the union of CM-points for (S,V) is Zariski-de...

conjecture | open | MathDB 页面 | 原文线索

V2-023060 | MATHDB-400448

Atypical-locus conjecture, main conjecture form 2

摘录 / EXCERPT: For any irreducible smooth quasi-projective variety S endowed with a variation of mixed $\mathbb{Q}$-Hodge structures $V \to S$, the subset $S_{\mathrm{atyp}}(V)$ is a strict alg...

conjecture | open | MathDB 页面 | 原文线索

V2-023061 | MATHDB-400449

D-module reductions and isotriviality

摘录 / EXCERPT: Let X be a smooth projective geometrically connected variety over a number field K, and let $M=(E,\nabla)$ be a vector bundle with an integrable connection on X. After choosi...

open_problem | open | MathDB 页面 | 原文线索

V2-023062 | MATHDB-400499

Convergence of adaptive refinement driven by auxiliary-subspace error estimators

摘录 / EXCERPT: Even for basic problems like the Poisson problem in 2D, there is no proof that typical refinement schemes will lead to convergent adaptive algorithms when this class of estimator i...

open_problem | open | MathDB 页面 | 原文线索

V2-023063 | MATHDB-400516

Kaledin’s conjecture on conical structures for symplectic singularities

摘录 / EXCERPT: Let a0$0\in V$ be a germ of a normal variety over $\mathbf C$ whose smooth locus V_{reg} admits a holomorphic symplectic 2-form ω, extending to a holomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-023064 | MATHDB-400522

Oblomkov-Rasmussen-Shende conjecture for HOMFLY-PT homology

摘录 / EXCERPT: Let $(X,0)$ be a germ of a unibranch plane curve singularity with Milnor number $\bc$, let $K\subset S^3$ be its link, and let $X^{[n]}$ denote the nth Hilbert scheme of X. If...

conjecture | open | MathDB 页面 | 原文线索

V2-023065 | MATHDB-400523

Conjectural adjoint functors relating Soergel bimodules and flag Hilbert schemes

摘录 / EXCERPT: Conjecture 3 asserts that there are adjoint functors ... with the additional properties that ι_* is monoidal and ... where FT_k is the full twist on the first k stra...

open_problem | open | MathDB 页面 | 原文线索

V2-023066 | MATHDB-400524

Adjoint functors from Soergel bimodules to the flag Hilbert scheme

摘录 / EXCERPT: Conjecture 3 asserts that there are adjoint functors ... with ι^* monoidal and satisfying ... where FT_k is the full twist on the first k strands. Assuming these fun...

open_problem | open | MathDB 页面 | 原文线索

V2-023067 | MATHDB-400530

Positivity of the algebraic stringy Euler number

摘录 / EXCERPT: For every projective algebraic variety X over $\mathbb{C}$ with at worst $\mathbb{Q}$-Gorenstein log-terminal singularities, the algebraic stringy Euler number satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-023068 | MATHDB-400552

Geyer-Jarden conjecture on torsion of abelian varieties over large algebraic fields

摘录 / EXCERPT: Let K be a finitely generated field, let A be a non-trivial abelian variety over K, and let e be a positive integer. Then, for almost all...

conjecture | open | MathDB 页面 | 原文线索

V2-023069 | MATHDB-400553

Positive-characteristic part (a) of the Geyer-Jarden torsion conjecture

摘录 / EXCERPT: Let K be a finitely generated field of positive characteristic, let A be a non-trivial abelian variety over K, and let K^s be a separable closure of K. For almost all...

conjecture | open | MathDB 页面 | 原文线索

V2-023070 | MATHDB-400554

Mazur’s question on whether the integers are diophantine over \(\mathbb Q\)

摘录 / EXCERPT: Does there exist a $\mathbb Q$-variety X and a $\mathbb Q$-morphism $f:X\to\mathbb A^1_{\mathbb Q}$ such that $f(X(\mathbb Q))=\mathbb Z$?

open_problem | open | MathDB 页面 | 原文线索

V2-023071 | MATHDB-400555

Colliot-Thélène-Sansuc conjecture on rational points of geometrically rationally connected varieties

摘录 / EXCERPT: Let k be a number field and let X be a projective and smooth variety over k. If X is geometrically rationally connected, then $X(k)$ is dense in $X(\[Ak]^{\mathrm{Br}}$).

conjecture | open | MathDB 页面 | 原文线索

V2-023072 | MATHDB-400578

Mazur’s non-Archimedean topological conjecture

摘录 / EXCERPT: Let K be a number field, $v\in M_K$, and X/K a projective variety. For $x\in X(K_v)$, let $Z_x\subseteq X$ be the limit of the Zariski closure of $X(K)\cap U$ in X, as U va...

conjecture | open | MathDB 页面 | 原文线索

V2-023073 | MATHDB-400579

Pasten’s Conjecture 4 on proximity functions and heights

摘录 / EXCERPT: Let K be a global field and let S be a finite set of places of K. Let X,Y be projective varieties over K. Let $D\in \mathrm{CD}_{+}(X/K)$ and let $f:X\to Y$ be a K-morp...

conjecture | open | MathDB 页面 | 原文线索

V2-023074 | MATHDB-400582

Completeness of lists of \-integral curves and hypersurfaces

摘录 / EXCERPT: Given a smooth projective variety $X/\mathbb{C}$ of dimension greater than two, an invertible sheaf L on X, an integer r, and a section...

open_problem | open | MathDB 页面 | 原文线索

V2-023075 | MATHDB-400601

Shareshian-Wachs conjecture for regular semisimple Hessenberg varieties

摘录 / EXCERPT: Let $m=(m_1,\dots,m_n)$ be a Hessenberg function, meaning $m_1\le \dots\le m_n$ and $i\le m_i\le n$, and let $y\in\mathfrak g_{\mathrm{rs}}$ be regular semisimple. For the graph...

conjecture | open | MathDB 页面 | 原文线索

V2-023076 | MATHDB-400604

Fixed points of finite abelian p-group actions

摘录 / EXCERPT: Let p be a prime number different from the characteristic of k. Let G be a finite abelian p-group acting on a smooth projective k-variety X without zero-dimensional com...

open_problem | open | MathDB 页面 | 原文线索

V2-023077 | MATHDB-400607

Conservativity of the motivic functor

摘录 / EXCERPT: What extra assumptions are necessary in order for the natural functor $M:\mathrm{SH}(k)\to\mathrm{DM}(k)$, associating to a spectrum its motive, to be conservative?

open_problem | open | MathDB 页面 | 原文线索

V2-023078 | MATHDB-400638

Height of rational points on random Fano varieties

摘录 / EXCERPT: Fix integers $n\ge 3$, $r\ge 1$, and $d_1,\ldots,d_r\ge 2$, put $\mathbf d=(d_1,\ldots,d_r)$ and $d=d_1+\cdots+d_r$, and assume $n+1-d>0$. Let $\mathcal V_{\mathbf d,n}$ be the par...

open_problem | open | MathDB 页面 | 原文线索

V2-023079 | MATHDB-400667

Looijenga’s affine-open-covering conjecture for the moduli space of curves

摘录 / EXCERPT: For every integer $g\ge 2$, the quasiprojective variety $\mathcal M_g$ of smooth genus-g curves can be covered by $(g-1)$ open affine subsets.

conjecture | open | MathDB 页面 | 原文线索

V2-023080 | MATHDB-400678

Complete complex surfaces in \(\mathcal{M}_4\)

摘录 / EXCERPT: Does $\mathcal{M}_4$ contain a complete complex 2-dimensional subvariety?

open_problem | open | MathDB 页面 | 原文线索

V2-023081 | MATHDB-400681

Maximal dimension of a complete subvariety of A_4

摘录 / EXCERPT: What is the maximal dimension of a complete algebraic subvariety of A_4, the moduli space of principally polarized abelian fourfolds, over the field of complex numbers?

open_problem | open | MathDB 页面 | 原文线索

V2-023082 | MATHDB-400687

Resolution of singularities in dimension at least four

摘录 / EXCERPT: Let X be a reduced, separated, quasi-excellent, Noetherian scheme of dimension at least 4. Does there exist a proper birational morphism $\pi:X'\to X$ such that: (i) X' is ev...

open_problem | open | MathDB 页面 | 原文线索

V2-023083 | MATHDB-400844

Is the morphism Ψ an isomorphism for all tame algebras?

摘录 / EXCERPT: Let A be a finite-dimensional algebra and let $C\subseteq\operatorname{rep}(A,d)$ be a $\mathrm{GL}(d)$-invariant, irreducible, closed subvariety with θ-stable decompositi...

open_problem | open | MathDB 页面 | 原文线索

V2-023084 | MATHDB-400853

Minimum quantile problem for rank-based rewards

摘录 / EXCERPT: Given a reward budget R_0 and a number $\alpha\in(0,1)$, what rank-based reward scheme minimizes the α-quantile of the equilibrium completion times?

open_problem | open | MathDB 页面 | 原文线索

V2-023085 | MATHDB-400874

Hyperbolic variety with multiple real-projective components

摘录 / EXCERPT: Find a smooth irreducible hyperbolic variety $X\subseteq \mathbb{P}^n_{\mathbb{R}}$ of dimension at least two such that more than one connected component of $X(\mathbb{R})$ is home...

open_problem | open | MathDB 页面 | 原文线索

V2-023086 | MATHDB-400888

Number of variety drops in a convergent extension sequence

摘录 / EXCERPT: Let $M_d\to M_{d+1}\to\cdots\to M_{d+k}\to M_{d+k+1}$ be a convergent extension sequence of positive moment matrices, with $k\ge 0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-023087 | MATHDB-400971

Eventual vanishing of higher extensions between \(\mathbb G_a\) and \(\mathbb G_m\)

摘录 / EXCERPT: Let $\mathcal C$ be the abelian category of commutative group schemes of finite type over a fixed field k. Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-023088 | MATHDB-400972

Zariski cohomology of twisted loop reductive group schemes over Laurent polynomial rings

摘录 / EXCERPT: For an algebraically closed field k and $n\ge 3$, determine the Zariski cohomology pointed set $H^1_{\mathrm{Zar}}(R_n,E_G)$ for each loop G-torsor E, where...

open_problem | open | MathDB 页面 | 原文线索

V2-023089 | MATHDB-400977

Chow ring of generic flag varieties

摘录 / EXCERPT: Let k be a field, let G be a split semisimple affine algebraic group over k, and let $B\subset G$ be a Borel subgroup. For an embedding $G\hookrightarrow \mathrm{GL}(n)$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-023090 | MATHDB-400981

Characterization of anisotropy over perfect fields by closed orbits

摘录 / EXCERPT: Suppose that k is perfect. Then G is k-anisotropic if and only if, for every k-defined affine G-variety W and every $w\in W(k)$, the orbit $G\cdot w$ is closed in W.

open_problem | open | MathDB 页面 | 原文线索

V2-023091 | MATHDB-400991

Manin’s conjecture for Fano varieties over number fields

摘录 / EXCERPT: Let X be a Fano variety over a number field k, and let $H:X(k)\to\mathbb{R}_{\ge 0}$ be the height function defined by the anticanonical bundle. Then there exists an open subse...

conjecture | open | MathDB 页面 | 原文线索

V2-023092 | MATHDB-401011

Finiteness of optimal points in a subvariety of a mixed Shimura variety

摘录 / EXCERPT: Let S be a mixed Shimura variety and let V be a subvariety of S. Then V contains only finitely many points P such that $[U+007F]P[U+007F]$ is optimal in V: for every subvariety Y o...

open_problem | open | MathDB 页面 | 原文线索

V2-023093 | MATHDB-401012

Weak hyperbolic Ax-Schanuel conjecture

摘录 / EXCERPT: For a fixed subvariety V of S, let A be a Zariski-optimal intersection component of $\b5^{-1}(V)$. Then the Zariski closure $\b5cA\d5_{\mathrm{Zar}}$ is pre-weakly special.

conjecture | open | MathDB 页面 | 原文线索

V2-023094 | MATHDB-401028

Classification of level-1 cuspidal algebraic automorphic representations of motivic weight at most 22

摘录 / EXCERPT: Let $n\ge 1$ and let π be a cuspidal algebraic automorphic representation of GL_n over $\mathbb{Q}$ with motivic weight $w(\pi)\le 22$, such that π_p is unramifi...

open_problem | open | MathDB 页面 | 原文线索

V2-023095 | MATHDB-401029

Finiteness of level-1 algebraic cuspidal automorphic representations of bounded motivic weight

摘录 / EXCERPT: For every integer $N\ge 1$, there are only finitely many cuspidal automorphic representations π of GL_n over $\mathbb{Q}$, with n varying, whose Artin conductor is...

open_problem | open | MathDB 页面 | 原文线索

V2-023096 | MATHDB-401041

Deligne’s conjecture for critical values of motivic L-functions

摘录 / EXCERPT: For a motive M, the critical values at $s=m\in\mathbb{Z}$ of its L-function satisfy ...

conjecture | open | MathDB 页面 | 原文线索

V2-023097 | MATHDB-401052

Kottwitz’s stabilization conjecture for the intersection cohomology of Shimura varieties

摘录 / EXCERPT: Let (G,X) be a Shimura datum, let $K=K_pK^p\subset G(\mathbb A_f)$ be neat, and let Sh_K be the associated Shimura variety over its reflex field E. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023098 | MATHDB-401053

Kottwitz’s conjectural local Euler-factor formula for orthogonal Shimura varieties

摘录 / EXCERPT: Conjecture A (Kottwitz). Let (G,X) be a Shimura datum, let $K\subset G(\mathbb A_f)$ be neat, and let $\operatorname{Sh}_K=\operatorname{Sh}_K(G,X)$ be the associated Shimura var...

open_problem | open | MathDB 页面 | 原文线索

V2-023099 | MATHDB-401054

Kottwitz’s Conjecture B for orthogonal Shimura varieties

摘录 / EXCERPT: Let (G,X) be the Shimura datum with $G=\mathrm{SO}(V)$, where V is a quadratic space over $\mathbb{Q}$ of signature $(n,2)$ with $n\ge 5$, and where X is the space of oriente...

conjecture | open | MathDB 页面 | 原文线索

V2-023100 | MATHDB-401194

Extension of the quaternion-algebra assignment to the canonical component

摘录 / EXCERPT: For a hyperbolic knot K, with canonical component $C(K)$ of the $SL_2(\mathbb C)$ character variety and smooth projective model $\widetilde C$, determine whether the assignment...

open_problem | open | MathDB 页面 | 原文线索

V2-023101 | MATHDB-401205

Borel hyperbolicity of the moduli stack of smooth canonically polarized varieties

摘录 / EXCERPT: Is the moduli stack of smooth canonically polarized varieties Borel hyperbolic?

open_problem | open | MathDB 页面 | 原文线索

V2-023102 | MATHDB-401207

Holonomy of klt varieties with numerically trivial canonical divisor

摘录 / EXCERPT: Let X be a klt variety with numerically trivial canonical divisor, and fix a point $x\in X_{\mathrm{reg}}$. Then: - The holonomy group Hol_x of the Eyssidieux-Gu...

open_problem | open | MathDB 页面 | 原文线索

V2-023103 | MATHDB-401211

Deformation of a generically finite map to a hypersurface embedding

摘录 / EXCERPT: Consider a one-dimensional family of smooth projective varieties of dimension n, namely a smooth projective holomorphic map $p:W\to T$ with T a smooth holomorphic curve, togeth...

open_problem | open | MathDB 页面 | 原文线索

V2-023104 | MATHDB-401213

K-stability criterion for Fano spherical varieties

摘录 / EXCERPT: Let $X \hookleftarrow G$ be a Fano spherical manifold, with moment polytope Δ^+, valuation cone V_-, and weighted barycenter $\overline{\mathrm{DH}}(\Delta^+)$. Then X a...

open_problem | open | MathDB 页面 | 原文线索

V2-023105 | MATHDB-401256

Hartshorne’s Conjecture A

摘录 / EXCERPT: If Y is a smooth subvariety with ample normal bundle of a smooth projective variety X, then $n[Y]$ moves in a large algebraic family for n large.

conjecture | open | MathDB 页面 | 原文线索

V2-023106 | MATHDB-401258

Equality of the movable and weakly movable cones

摘录 / EXCERPT: For a projective variety X of dimension n and $1\le d\le n-1$, is $\operatorname{Mov}_d(X)=\operatorname{WMov}_d(X)$ in general?

open_problem | open | MathDB 页面 | 原文线索

V2-023107 | MATHDB-401261

Finite generation of value semigroups for ample divisors

摘录 / EXCERPT: For which varieties is it possible to find a flag such that the value semigroup of an ample divisor is finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-023108 | MATHDB-401290

Kuznetsov-Shinder conjecture on D-equivalence and L-equivalence

摘录 / EXCERPT: Let k be a field of characteristic 0, and let X and Y be smooth projective simply connected varieties over k. If their bounded derived categories of coherent sheaves are...

conjecture | open | MathDB 页面 | 原文线索

V2-023109 | MATHDB-401293

Finite generation of the geometric component group of an automorphism group

摘录 / EXCERPT: Suppose that X is a projective variety over a field K. Writing … for the group of geometric components, must $\pi_0(\operatorname{Aut}(X))$ be finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-023110 | MATHDB-401294

The Tate conjecture for smooth projective schemes over finitely generated fields of characteristic zero

摘录 / EXCERPT: For every field K finitely generated over $\a0\mathbb{Q}$, every smooth projective scheme X over K, every integer $i\ge 0$, and every prime number ℓ, the cycle class map...

conjecture | open | MathDB 页面 | 原文线索

V2-023111 | MATHDB-401337

Indistinguishability obfuscation from quasi-polynomially secure private-key functional encryption

摘录 / EXCERPT: Assuming a quasi-polynomially secure private-key functional encryption scheme for polynomial-size circuits, there exists an indistinguishability obfuscator for the class of circuit...

open_problem | open | MathDB 页面 | 原文线索

V2-023112 | MATHDB-401338

Efficiency of one-way-function-based FSS for point functions with at least three parties

摘录 / EXCERPT: What is the efficiency of one-way-function-based constructions of FSS schemes for the class of point functions in the case of $m\ge 3$ parties?

open_problem | open | MathDB 页面 | 原文线索

V2-023113 | MATHDB-401340

Adaptive security of Yao’s circuit garbling scheme

摘录 / EXCERPT: Can Yao’s circuit garbling scheme be extended to guarantee security in the face of an adversary that corrupts both parties, adaptively, as the computation proceeds?

open_problem | open | MathDB 页面 | 原文线索

V2-023114 | MATHDB-401402

Connections between graph Allen-Cahn, MBO, and mean-curvature flows

摘录 / EXCERPT: Are the graph Allen-Cahn equation, graph MBO scheme, and graph mean-curvature flow as intimately connected as their continuum counterparts?

open_problem | open | MathDB 页面 | 原文线索

V2-023115 | MATHDB-401431

Adaptive layer selection in stochastic optimization for continuous-depth CNNs

摘录 / EXCERPT: How can an adaptive scheme for choosing layers, such as shallow-to-deep training, be incorporated into stochastic optimization methods?

open_problem | open | MathDB 页面 | 原文线索

V2-023116 | MATHDB-401472

Colliot-Thélène’s Brauer-Manin obstruction conjecture

摘录 / EXCERPT: For varieties over a number field that are geometrically rationally connected (in particular, unirational varieties), is the Brauer-Manin obstruction the only obstruction to the ex...

conjecture | open | MathDB 页面 | 原文线索

V2-023117 | MATHDB-401477

Arithmetic operad of genus-zero moduli spaces from rational tangential structures

摘录 / EXCERPT: Conjecture. The set of $\Q$-tangential structures $\\{\vec{s}\u0005\}$ over $\\mathcal{M}_{0,m}\$ defines an operad ... in $\\mathbb{Q}$-proschemes whose geometric étale homotopy typ...

open_problem | open | MathDB 页面 | 原文线索

V2-023118 | MATHDB-401478

Arithmetic Galois action on the operad of genus-zero moduli spaces

摘录 / EXCERPT: Does the set of $\[Q$-tangential structures $\[\{\vec{s}\}\]$ over $M_{0,m}$ define an operad ... in $\mathbf{Q}$-proschemes whose geometric étale homotopy type is endowed with a...

open_problem | open | MathDB 页面 | 原文线索

V2-023119 | MATHDB-401483

Colliot-Thélène conjecture on the Brauer-Manin obstruction

摘录 / EXCERPT: The Colliot-Thélène conjecture predicts that, for a geometrically rationally connected variety X, the Brauer-Manin obstruction to weak approximation is the only obstruction.

conjecture | open | MathDB 页面 | 原文线索

V2-023120 | MATHDB-401486

Functorial retraction from étale homotopy-type isomorphisms

摘录 / EXCERPT: Let $k/\mathbb{Q}$ be a finitely generated field. Let X and Y be smooth geometrically connected varieties over k such that each admits a locally closed embedding into a produ...

open_problem | open | MathDB 页面 | 原文线索

V2-023121 | MATHDB-401489

Rational points in families of rationally connected varieties

摘录 / EXCERPT: Let k be a number field. Let $p:Z\to B$ be a dominant morphism between smooth and proper varieties over k. Assume that B and the generic fibre of p are rationally connected...

open_problem | open | MathDB 页面 | 原文线索

V2-023122 | MATHDB-401490

Uniform nontriviality modulo a power of ℓ for arithmetic representations

摘录 / EXCERPT: Let X be a normal, geometrically connected variety over a finitely generated field k of characteristic zero, and let ℓ be a prime. Then there exists an integer...

open_problem | open | MathDB 页面 | 原文线索

V2-023123 | MATHDB-401491

Nonabelian generalization of the tamely ramified torsor criterion

摘录 / EXCERPT: Does the theorem above remain valid for a finite group scheme G/k that is not necessarily abelian?

open_problem | open | MathDB 页面 | 原文线索

V2-023124 | MATHDB-401562

Grothendieck-Serre conjecture for reductive group schemes over semi-local Dedekind domains

摘录 / EXCERPT: Let R be a semi-local Dedekind domain, let K be the field of fractions of R, and let G be a reductive R-group scheme. Is the kernel of the map ... induced by the inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-023125 | MATHDB-401573

Can the splitting ind-variety be chosen to be a variety?

摘录 / EXCERPT: Let k be an infinite field, let $n\ge 3$, let M be a finite étale group scheme over k, and let $s\in H^n(k,M)$. Does there exist a smooth, geometrically integral k-variety...

open_problem | open | MathDB 页面 | 原文线索

V2-023126 | MATHDB-401618

Special properties of Fourier-Mukai functors for symmetric powers and Hilbert schemes

摘录 / EXCERPT: What are the special properties of the Fourier-Mukai functor from C (respectively, S) to $\[${operatorname{Sym}}^i C$ (respectively, ${operatorname{Hilb}}^n S$)?

open_problem | open | MathDB 页面 | 原文线索

V2-023127 | MATHDB-401631

Non-birational reconstruction of a variety from function-field data

摘录 / EXCERPT: The birational equivalence class of a normal proper geometrically integral variety X of dimension ≥ 2 over a perfect field k is described by inductive systems of the groups...

open_problem | open | MathDB 页面 | 原文线索

V2-023128 | MATHDB-401632

Conjecture of Geyer-Jarden on torsion over large algebraic extensions

摘录 / EXCERPT: Let K be a finitely generated field over $\mathbb{Q}$, let A be a non-zero abelian variety over K, and let e be a positive integer. Fix an algebraic closure $\widetilde K$...

conjecture | open | MathDB 页面 | 原文线索

V2-023129 | MATHDB-401637

Colliot-Thélène’s conjecture on the Brauer-Manin obstruction and weak weak approximation

摘录 / EXCERPT: Colliot-Thélène’s conjecture says that the Brauer-Manin obstruction is the only obstruction to weak weak approximation for rationally connected varieties, and in particular for var...

conjecture | open | MathDB 页面 | 原文线索

V2-023130 | MATHDB-401687

Theorem 3: homomorphisms from the Mori birational groupoid

摘录 / EXCERPT: Let $n\ge 3$. There is an integer $d\ge 2$ such that, for every conic bundle X/B, where X is a rationally connected variety of dimension n, there is a groupoid homomorphism .....

open_problem | open | MathDB 页面 | 原文线索

V2-023131 | MATHDB-401689

Smooth rational real affine varieties with infinitely many real forms

摘录 / EXCERPT: For every $d\ge 4$, there exist smooth rational real affine varieties of dimension d which have at least countably infinitely many pairwise non-isomorphic real forms.

open_problem | open | MathDB 页面 | 原文线索

V2-023132 | MATHDB-401695

Katzarkov-Kontsevich-Pantev Conjecture 2 for Landau-Ginzburg models

摘录 / EXCERPT: If (Y,w) is a Landau-Ginzburg model for a Fano variety X of dimension n, then $h^{p,q}(X)=f^{n-p,q}(Y,w)$.

conjecture | open | MathDB 页面 | 原文线索

V2-023133 | MATHDB-401698

Alternative proof of the Jordan property for rationally connected varieties

摘录 / EXCERPT: Can one give an alternative proof of the Jordan property for rationally connected varieties over a field of characteristic zero, following the approach of [9] instead of [6]?

open_problem | open | MathDB 页面 | 原文线索

V2-023134 | MATHDB-401710

Compatibility of the p-adic and classical Riemann-Hilbert correspondences

摘录 / EXCERPT: Let X be a smooth algebraic variety over a number field E. Fix an isomorphism $\iota:\overline{\mathbb{Q}}_p\xrightarrow{\sim}\mathbb{C}$ and a field homomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-023135 | MATHDB-401711

Compatibility of the p-adic and classical Riemann-Hilbert functors

摘录 / EXCERPT: Let X be a smooth algebraic variety over a number field E. Fix an isomorphism $\iota:\mathbb{Q}_p\xrightarrow{\sim}\mathbb{C}$ and a field homomorphism $\sigma:E\to\mathbb{C}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-023136 | MATHDB-401714

Resolvent degree of the universal p-torsion cover of the moduli space of principally polarized abelian varieties

摘录 / EXCERPT: For a prime p, if $(g,p)\ne(2,2)$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-023137 | MATHDB-401715

Resolvent-degree lower bound for alternating-group quotients of the level-2 abelian-variety cover

摘录 / EXCERPT: For every integer $n\ge 6$, Conjecture 12 asserts ... where $A_n\hookrightarrow \operatorname{Sp}_{2g}(\mathbb F_2)$ is the embedding from Proposition 11,...

open_problem | open | MathDB 页面 | 原文线索

V2-023138 | MATHDB-401718

Positive examples of regular orbits of positive-dimensional subvarieties

摘录 / EXCERPT: Let $L\subset k$ be fields with k algebraically closed, let G be a connected affine algebraic group of finite type over L, write $G=G(k)$, and let X be a projective algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-023139 | MATHDB-401719

Algebraic cycle realizing the CM motive comparison

摘录 / EXCERPT: Let $A/\mathbf{Q}$ be an elliptic curve with complex multiplication by an imaginary quadratic field K, and let ψ be its Hecke character of infinity type $(1,0)$. If W is...

open_problem | open | MathDB 页面 | 原文线索

V2-023140 | MATHDB-401724

Grothendieck’s period conjecture for a motive

摘录 / EXCERPT: Let M be a motive over $\mathbb{Q}$. Grothendieck’s period conjecture asserts that the scheme ... is connected and that its dimension equals the transcendence degree ... of the field...

conjecture | open | MathDB 页面 | 原文线索

V2-023141 | MATHDB-401754

Debarre’s minimal class conjecture

摘录 / EXCERPT: Let (A,Θ) be an indecomposable principally polarized abelian variety of dimension g. Let X be a closed subscheme of A of dimension $1\le d\le g-2$. The following are e...

conjecture | open | MathDB 页面 | 原文线索

V2-023142 | MATHDB-401755

Generic vanishing and minimal cohomology classes on principally polarized abelian varieties

摘录 / EXCERPT: Let (A,Θ) be an indecomposable principally polarized abelian variety of dimension g, and let $X\subseteq A$ be a closed subscheme of dimension $1\leq d\leq g-2$. Conjectur...

open_problem | open | MathDB 页面 | 原文线索

V2-023143 | MATHDB-401756

Multiplicity bound for points on theta divisors

摘录 / EXCERPT: Let (A,Θ) be an indecomposable principally polarized abelian variety of dimension g. Is it true that, for every $x\in\Theta$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-023144 | MATHDB-401760

Further examples of minifolds

摘录 / EXCERPT: Find more examples of minifolds beyond those listed in the table. Here a minifold is a smooth Fano variety X of dimension n satisfying at least one of: (1)...

open_problem | open | MathDB 页面 | 原文线索

V2-023145 | MATHDB-401761

Voisin’s conjecture on special subvarieties of holomorphic symplectic varieties

摘录 / EXCERPT: Let X be holomorphic symplectic of dimension $2n$. For all $0 \le i \le n$, there exist special subvarieties $Z \subset X$ in codimension i.

conjecture | open | MathDB 页面 | 原文线索

V2-023146 | MATHDB-401762

Stable cohomology of complements of discriminants

摘录 / EXCERPT: Let M be a nonsingular projective variety and let L be a very ample line bundle on M. For $d\geq 1$, let $X(M,L),d$ denote the complement of the discriminant in...

open_problem | open | MathDB 页面 | 原文线索

V2-023147 | MATHDB-401763

Manin-Kollár conjecture on rational points over real curve function fields

摘录 / EXCERPT: If C is an integral real curve with $C(\mathbb{R})=\varnothing$, every rationally connected variety X over $\mathbb{R}(C)$ has an $\mathbb{R}(C)$-point.

conjecture | open | MathDB 页面 | 原文线索

V2-023148 | MATHDB-401767

Geometry controlling the d-primary Brauer-Manin obstruction

摘录 / EXCERPT: Determine how the geometry of a smooth projective variety controls the sets of positive integers d for which BM_d, $BM_d^{\perp}$, or $\widehat{BM}_d$ holds; equivalently, dete...

open_problem | open | MathDB 页面 | 原文线索

V2-023149 | MATHDB-401970

Relation between folding and the modulated-graph construction

摘录 / EXCERPT: Taking the fixed points of the composition of a diagram automorphism and a Galois automorphism of the same order, acting on the symmetric quiver variety, gives an instance of the m...

open_problem | open | MathDB 页面 | 原文线索

V2-023150 | MATHDB-401972

Supports of the linear degeneration family of flag varieties

摘录 / EXCERPT: For $n\ge 1$, let V be an $(n+1)$-dimensional complex vector space and let $\mathcal F\to U$ be the flat family of linear degenerations of the complete flag variety, with fibers...

open_problem | open | MathDB 页面 | 原文线索

V2-023151 | MATHDB-402147

Equality of the saturated G-zip and Shimura-variety cones

摘录 / EXCERPT: For the Hodge-type Shimura datum, compact open level K, and prime p of good reduction described in the report, the conjecture is ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-023152 | MATHDB-402148

Classification of level-one algebraic cusp forms of motivic weight 23 with repeated top weight

摘录 / EXCERPT: Classify the level-one cuspidal algebraic automorphic representations π of PGL_m over $\mathbb{Q}$, with m varying, having motivic weight 23 and for which the we...

open_problem | open | MathDB 页面 | 原文线索

V2-023153 | MATHDB-402152

Classification of the zero-dimensional extremal case of an affine Deligne-Lusztig variety

摘录 / EXCERPT: Assume that G is adjoint and simple over F, that $\tau\in\breve G$ is the basic element associated to ${\mu}$, and that ${\mu}$ is non-central in every simple factor of G. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-023154 | MATHDB-402170

Existence criterion for good moduli spaces of algebraic stacks

摘录 / EXCERPT: Let M be an algebraic stack locally of finite type with affine diagonal over a quasi-separated and locally noetherian algebraic space. Then M admits a good moduli space if and...

open_problem | open | MathDB 页面 | 原文线索

V2-023155 | MATHDB-402171

Structure of coarse moduli spaces for refined Higgs-bundle strata

摘录 / EXCERPT: Let Σ,L be a compact Riemann surface with a line bundle L of positive degree, and let $mathcal H_{r,d}(\Sigma,L)$ be the moduli stack of Higgs bundles of coprime rank r...

open_problem | open | MathDB 页面 | 原文线索

V2-023156 | MATHDB-402177

Implicit entropy-stable asymptotic-preserving scheme for the two-speed relaxation solver

摘录 / EXCERPT: Can one derive an implicit numerical scheme for the two-speed relaxation solver that retains both the asymptotic-preserving property in the low-Mach limit and a discrete entropy in...

open_problem | open | MathDB 页面 | 原文线索

V2-023157 | MATHDB-402195

Ring structure on symplectic cohomology and theta functions

摘录 / EXCERPT: Let (M,D) be a positive maximally degenerate pair, meaning that M is a smooth projective variety, D is a simple normal-crossings anticanonical divisor with a zero-dimensional...

open_problem | open | MathDB 页面 | 原文线索

V2-023158 | MATHDB-402202

Six operations for integral étale motives with p-torsion

摘录 / EXCERPT: For any morphism of any varieties $X\to Y$ over k, there should be a six operations formalism, including Verdier duality, between the hypothetical categories...

open_problem | open | MathDB 页面 | 原文线索

V2-023159 | MATHDB-402204

Motivic origin of the \-adic height pairing

摘录 / EXCERPT: In the setup where F is a finite field, B is a smooth proper integral F-scheme of finite type with function field $K=F(B)$, and X is a smooth projective K-variety of dime...

open_problem | open | MathDB 页面 | 原文线索

V2-023160 | MATHDB-402207

Dugger’s conjecture on the first motivic stable homotopy group

摘录 / EXCERPT: Dugger conjectured that the elements $\nu\in\pi_{1,2}\mathbf{1}$ and $\eta_{\mathrm{top}}\in\pi_{1,0}\mathbf{1}$ generate $\pi_1\mathbf{1}$ as a K^{MW}-module. The repor...

conjecture | open | MathDB 页面 | 原文线索

V2-023161 | MATHDB-402214

Closedness of the set of Θ-positive Anosov representations

摘录 / EXCERPT: For a pair (G_R,P_Θ) admitting a Θ-positive structure, the set of Θ-positive Anosov representations is open. Is this set also closed in the character variety $X(S,G_R)$?

open_problem | open | MathDB 页面 | 原文线索

V2-023162 | MATHDB-402219

GIT stability and log K-polystability for plurianticanonical divisors

摘录 / EXCERPT: Is it true that, for any K-polystable (KE Fano) variety X and for β big enough, log K-polystability (existence of log KE metrics) for $(X,(1-\beta)D)$, with D correspondi...

open_problem | open | MathDB 页面 | 原文线索

V2-023163 | MATHDB-402262

Conservativity of algebraic cobordism for normed motivic spectra

摘录 / EXCERPT: Let $E \in \mathrm{NAlg}(\mathrm{SH}(S))$. If $E \wedge \mathrm{MGL} \simeq 0$, is it then also true that $E \simeq 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-023164 | MATHDB-402304

Realisability of weighted polyhedral fans by tropical ideals

摘录 / EXCERPT: Which weighted polyhedral fans are the variety of some tropical ideal?

open_problem | open | MathDB 页面 | 原文线索

V2-023165 | MATHDB-402373

Multiplicity formula for general spherical varieties

摘录 / EXCERPT: Let F be a local field of characteristic 0, let G be a connected reductive group over F, and let H be a closed (not necessarily reductive) spherical subgroup of G. For...

open_problem | open | MathDB 页面 | 原文线索

V2-023166 | MATHDB-402396

Geometry of the irreducible components of the maximal flat degeneration

摘录 / EXCERPT: Let V) be a complex vector space of dimension $n+1$, and let $\mathrm{Fl}_{\mathrm{mf}}(V)$ be the maximal flat linear degeneration of the complete flag variety, with irreducible...

open_problem | open | MathDB 页面 | 原文线索

V2-023167 | MATHDB-402397

Problem 6: Study the stratification of the universal family of linear flag degenerations

摘录 / EXCERPT: Let V be a complex vector space of dimension $n+1$, and let $\mathcal U$ be the variety of tuples $(f_1,\ldots,f_{n-1})\in\operatorname{End}(V)^{n-1}$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-023168 | MATHDB-402398

Orbit structure of the automorphism action on the maximally flat degenerate flag variety

摘录 / EXCERPT: Let V be a complex vector space of dimension $n+1$, and let ... where $\operatorname{pr}_{i,i+1}$ is the coordinate projection associated with the fixed basis that annihilates...

open_problem | open | MathDB 页面 | 原文线索

V2-023169 | MATHDB-402400

Does the FFLV-polytope arise from a cluster seed?

摘录 / EXCERPT: For the full flag variety $F\f\relax\require{HTML} \f\relax\ell_n$ of type A with $n\ge 5$, let $A_n=\mathbb{C}[p_J\mid J\subset [n]]/I_n$ be its multihomogeneous coordinate ring...

open_problem | open | MathDB 页面 | 原文线索

V2-023170 | MATHDB-402490

Rationality of Seshadri constants

摘录 / EXCERPT: Let X be a projective variety of dimension n over the complex numbers, let $x\in X$ be a smooth point, and let L be a big line bundle that is ample at x. Is the Seshadri co...

open_problem | open | MathDB 页面 | 原文线索

V2-023171 | MATHDB-402491

Rationality of local positivity invariants

摘录 / EXCERPT: Let X be a projective variety over $\a0\mathbb{C}$, let $x\in X$ be a smooth point, and let L be a big line bundle that is ample at x. With $\pi:Y\to X$ the blow-up of X at...

open_problem | open | MathDB 页面 | 原文线索

V2-023172 | MATHDB-402493

Fujita’s conjecture

摘录 / EXCERPT: Let k be an algebraically closed field, and let X be a smooth projective variety over k of dimension n. If L is an ample line bundle on X, then...

conjecture | open | MathDB 页面 | 原文线索

V2-023173 | MATHDB-402529

Finite presentation of the étale fundamental group of an arithmetic open scheme

摘录 / EXCERPT: Is the étale fundamental group $\pi_1^{\acute{\mathrm{e}t}}\!\left(\operatorname{Spec}(\mathbb{Z})\setminus\{p_1,\ldots,p_n\}\right)$ finitely presented?

open_problem | open | MathDB 页面 | 原文线索

V2-023174 | MATHDB-402565

Resolution of singularities for quasi-excellent reduced noetherian schemes

摘录 / EXCERPT: It is conjectured that any quasi-excellent reduced noetherian scheme has a resolution of singularities.

open_problem | open | MathDB 页面 | 原文线索

V2-023175 | MATHDB-402572

Hartshorne’s conjecture on smooth subvarieties of projective space

摘录 / EXCERPT: Every smooth subvariety X of $[U+007F]\mathbf{P}^n$ of codimension $<n/3$ is a complete intersection.

conjecture | open | MathDB 页面 | 原文线索

V2-023176 | MATHDB-402580

Irreducibility of the arc space after a proper birational modification

摘录 / EXCERPT: Given a variety X, does there exist a birational and proper morphism $Y \to X$ such that Y_∞ is irreducible?

open_problem | open | MathDB 页面 | 原文线索

V2-023177 | MATHDB-402636

Landau-Ginzburg potentials from cluster structures on Fano compactifications

摘录 / EXCERPT: Given a Fano variety Y and an anticanonical divisor $D \subset Y$ such that $U=Y\setminus D$ has a cluster structure, can the cluster structure on U be used to construct a Land…

open_problem | open | MathDB 页面 | 原文线索

V2-023178 | MATHDB-402637

Landau-Ginzburg potential from theta functions for a Fano compactification

摘录 / EXCERPT: Let Y be a Fano variety and let $D\subset Y$ be an anticanonical divisor such that $U=Y\setminus D$ has a cluster structure given by a cluster variety V, with a birational map…

open_problem | open | MathDB 页面 | 原文线索

V2-023179 | MATHDB-402638

Fock-Goncharov canonical-basis conjecture for cluster varieties

摘录 / EXCERPT: For a cluster variety V, the Fock-Goncharov conjecture asserts that the algebra of regular functions on V has a canonical vector-space basis parameterized by the tropical space…

conjecture | open | MathDB 页面 | 原文线索

V2-023180 | MATHDB-402708

Dubrovin’s conjecture on semisimplicity and full exceptional collections

摘录 / EXCERPT: For a variety X, $\operatorname{BQH}^{\bullet}(X)$ is generically semisimple if and only if $\mathbf{D}^{\mathrm b}(X)$ admits a full exceptional collection.

conjecture | open | MathDB 页面 | 原文线索

V2-023181 | MATHDB-402724

Specialization order on the stack of G-bundles on the Fargues-Fontaine curve

摘录 / EXCERPT: Fix a prime p and a finite extension $F/\mathbb{Q}_p$. Let C be an algebraically closed complete extension of $\mathbb{Q}_p$, let $C^\flat$ be its tilt, and let X be the Farg…

open_problem | open | MathDB 页面 | 原文线索

V2-023182 | MATHDB-402725

Density of arithmetic representations of function fields

摘录 / EXCERPT: Let X be a smooth connected scheme of finite type over $\overline{\mathbb{F}}_p$, and let $\ell\neq p$ be a prime. Fix a semisimple continuous residual representation…

open_problem | open | MathDB 页面 | 原文线索

V2-023183 | MATHDB-402731

Fargues-Scholze categorical equivalence for \[1mGL_n\[0m

摘录 / EXCERPT: Let $X_{\widehat G}$ be the Artin stack of n-dimensional ℓ-adic representations of $W_{\mathbb Q_p}$, i.e. homomorphisms $W_{\mathbb Q_p}\to\widehat G(\mathbb Q_\ell)$ taken…

open_problem | open | MathDB 页面 | 原文线索

V2-023184 | MATHDB-402733

Classification of normal Schubert varieties in the bad-characteristic case

摘录 / EXCERPT: Classify the normal Schubert varieties in the partial affine flag variety $\mathrm{Fl}_{G^\flat,\mathfrak f^\flat}$ over $\overline{\mathbb F}_p$ when G is absolutely almost simp…

open_problem | open | MathDB 页面 | 原文线索

V2-023185 | MATHDB-402734

Representability of minuscule mixed-characteristic affine Grassmannian Schubert varieties

摘录 / EXCERPT: If bc is minuscule, then $\operatorname{Gr}_{G,\mu}$ is representable by a unique normal, flat, projective O_E-scheme $M_{G,\mu}$ with reduced special fiber.

open_problem | open | MathDB 页面 | 原文线索

V2-023186 | MATHDB-402749

High-order, unconditionally gradient-stable multistage variational schemes

摘录 / EXCERPT: For the multistage minimizing-movement scheme … can the parameters $\gamma_{m,i}$ be chosen so that the unconditional gradient-stability condition $E(u^{n+1})\le E(u^n)$ holds whil…

open_problem | open | MathDB 页面 | 原文线索

V2-023187 | MATHDB-402757

Finiteness of the regional fundamental group of a klt singularity

摘录 / EXCERPT: Let X be an algebraic variety defined over $\b[?]mathbb{C}$ and let $x\in X$ be a klt singularity. Then the regional fundamental group $\pi_1^{\mathrm{reg}}(X,x)$ is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-023188 | MATHDB-402763

Fields of definition of special subvarieties for ZVHS

摘录 / EXCERPT: Let S be a smooth quasi-projective complex variety and let $\mathbb V$ be a polarizable variation of $\mathbb Z$-Hodge structure on S, defined over a number field...

open_problem | open | MathDB 页面 | 原文线索

V2-023189 | MATHDB-402768

Properness of the K-moduli space of K-polystable Fano varieties

摘录 / EXCERPT: The moduli good space $X^{Kps}_{n,V}$ is proper.

open_problem | open | MathDB 页面 | 原文线索

V2-023190 | MATHDB-402769

Projectivity of the toroidal completion of a period-map image

摘录 / EXCERPT: Given a triple $(B,Z;\Phi)$ consisting of a smooth projective variety B, a reduced normal-crossing divisor $Z\subset B$, and a period map $\Phi:B\setminus Z\to\Gamma\backslash D$...

open_problem | open | MathDB 页面 | 原文线索

V2-023191 | MATHDB-402772

Analytic-variety structure on the SBB-type completion of a period map

摘录 / EXCERPT: The compact Hausdorff topological space $\mathcal{P}_S$ admits the structure of an analytic variety.

open_problem | open | MathDB 页面 | 原文线索

V2-023192 | MATHDB-402803

Multiplicity coefficients in the decomposition theorem for quiver Grassmannians

摘录 / EXCERPT: For the proper map $p_2:\operatorname{Gr}^{Q}_{\mathbf e}(\mathbf d)\to R_{\mathbf d}(Q)$ associated with the incidence variety of quiver subrepresentations, the decomposition theo...

open_problem | open | MathDB 页面 | 原文线索

V2-023193 | MATHDB-402851

Classification of point configurations admitting unexpected curves

摘录 / EXCERPT: Let $Z=P_1+\\\\\\ P_s$ be a reduced scheme of mutually distinct points in $\\\\\\\mathbb{P}^2_{\mathbb{C}}$. Classify all such configurations Z that admit an unexpected curve: na...

open_problem | open | MathDB 页面 | 原文线索

V2-023194 | MATHDB-403047

Convergence of fully discrete approximations for general hyperbolic systems

摘录 / EXCERPT: “A major remaining open problem is the convergence of fully discrete approximations, such as the Lax-Friedrichs or the Godunov scheme” for general hyperbolic systems.

open_problem | open | MathDB 页面 | 原文线索

V2-023195 | MATHDB-403048

Stable convergent numerical schemes for the full Rosensweig system

摘录 / EXCERPT: Does there exist a stable numerical scheme whose approximations converge to solutions of the Rosensweig system (1)?

open_problem | open | MathDB 页面 | 原文线索

V2-023196 | MATHDB-403061

Simpson’s motivic conjecture for rigid local systems

摘录 / EXCERPT: Let X be a connected smooth quasi-projective variety over $\mathbb{C}$, and let $j:X\hookrightarrow \overline{X}$ be an open immersion into a smooth projective variety such that...

conjecture | open | MathDB 页面 | 原文线索

V2-023197 | MATHDB-403062

Simpson’s conjecture on the geometric origin of rigid local systems

摘录 / EXCERPT: Let X be a connected smooth quasi-projective variety over $\a0\mathbb{C}$, and let $j:X\hookrightarrow \overline{X}$ be an open immersion into a smooth projective variety such th...

conjecture | open | MathDB 页面 | 原文线索

V2-023198 | MATHDB-403063

Motivic/geometricity conjecture for rigid local systems with reductive structure group

摘录 / EXCERPT: Let X be a connected smooth quasi-projective variety over $\a0\mathbb{C}$, let $j:X\hookrightarrow\overline{X}$ be a smooth projective compactification with strict normal-crossin...

conjecture | open | MathDB 页面 | 原文线索

V2-023199 | MATHDB-403067

Section conjecture for localisations of curves under two conditions

摘录 / EXCERPT: Let X/k be a smooth, proper curve and let $S\subseteq X_{\mathrm{cl}}$ be a set of closed points. Let X_S denote the localisation of X at S, defined by ... in the scheme-theo...

conjecture | open | MathDB 页面 | 原文线索

V2-023200 | MATHDB-403076

Tannakian Chebotarev density theorem

摘录 / EXCERPT: Let X be a smooth, geometrically connected variety over a finite field k of characteristic $p>0$, and let $S\subset |X|$ have upper density $\delta_u(S)=1$. For a local system...

open_problem | open | MathDB 页面 | 原文线索

V2-023201 | MATHDB-403077

Tannakian Čebotarev density for non-\[?]-adic coefficient systems

摘录 / EXCERPT: Let X be a smooth, geometrically connected variety over a finite field k of characteristic $p>0$, and let F be one of the following coefficient objects on X: a $\[?]$-local...

open_problem | open | MathDB 页面 | 原文线索

V2-023202 | MATHDB-403078

Existence of the motivic Tannakian group and Frobenius conjugacy classes

摘录 / EXCERPT: For a semisimple motivic $\mathbb{Q}$-local system F on X, there should exist a reductive group $G(F^{\mathrm{mot}})$ over $\mathbb{Q}$, together with a faithful finite-dimensi...

open_problem | open | MathDB 页面 | 原文线索

V2-023203 | MATHDB-403079

Existence of the conjectural motivic Tannakian group and Frobenius conjugacy classes

摘录 / EXCERPT: Let X be smooth and geometrically connected over a finite field k of characteristic $p>0$, and let F be a semisimple motivic Q-local system, where Q is one of...

open_problem | open | MathDB 页面 | 原文线索

V2-023204 | MATHDB-403101

Convergence of the Guermond-Popov invariant-domain-preserving scheme to a dissipative measure-valued solution

摘录 / EXCERPT: Prove convergence of the invariant-domain-preserving finite-element scheme proposed by J.-L. Guermond and B. Popov to a dissipative measure-valued solution of the complete compress...

open_problem | open | MathDB 页面 | 原文线索

V2-023205 | MATHDB-403113

R-equivalence over complete equicharacteristic regular local rings

摘录 / EXCERPT: Let A be a complete equicharacteristic regular local ring, with residue field k and fraction field K. Then, for every reductive A-group scheme G, there are canonical isom...

open_problem | open | MathDB 页面 | 原文线索

V2-023206 | MATHDB-403114

Specialization of R-equivalence for reductive group schemes over henselian local domains

摘录 / EXCERPT: Let A be a henselian local domain with residue field k and fraction field K, and let G be a reductive group scheme over A. Does there exist a specialization homomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-023207 | MATHDB-403116

Finite-type representability of homomorphism schemes

摘录 / EXCERPT: When is the representing scheme M of the functor ${\rm Hom}_{\rm gp}(G,H)$ of finite type?

open_problem | open | MathDB 页面 | 原文线索

V2-023208 | MATHDB-403166

Strong coniveau of integral cohomology of rationally connected varieties

摘录 / EXCERPT: Let X be a rationally connected smooth projective variety. Is $H^i(X,\mathbb{Z})=\widetilde N^1H^i(X,\mathbb{Z})$ for $i>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-023209 | MATHDB-403168

Geometric interpretation or proof of the exceptional Hirzebruch-Riemann-Roch-type formula

摘录 / EXCERPT: For the permutohedral variety X_E, Theorem 3 asserts that there is a ring isomorphism $\zeta_{X_E}:K_0(X_E)\cong A^\bullet(X_E)$ satisfying ... for every $[\mathcal E]\in K_0(X_E)$...

open_problem | open | MathDB 页面 | 原文线索

V2-023210 | MATHDB-403177

Equivariant linearizability of finite-group actions

摘录 / EXCERPT: Let X be a smooth projective variety of dimension n over an algebraically closed field k of characteristic zero. Assume that X is rational, i.e. birational to...

open_problem | open | MathDB 页面 | 原文线索

V2-023211 | MATHDB-403178

Cohomology groups of line bundles on complete flag varieties

摘录 / EXCERPT: Determine the cohomology groups of line bundles on $\operatorname{Flag}(V)$.

open_problem | open | MathDB 页面 | 原文线索

V2-023212 | MATHDB-403179

Cohomology of line bundles on complete flag varieties

摘录 / EXCERPT: Let V be a vector space of dimension $n\ge 3$ over a field k, and let $\operatorname{Flag}(V)$ parametrize complete flags in V. Determine the cohomology groups of all line bu...

open_problem | open | MathDB 页面 | 原文线索

V2-023213 | MATHDB-403180

Cohomology of the specified line bundles on complete flag varieties

摘录 / EXCERPT: Let V be a vector space of dimension $n\ge 3$ over a field k. Let $\operatorname{Flag}(V)$ denote the variety of complete flags in V. Suppose that...

open_problem | open | MathDB 页面 | 原文线索

V2-023214 | MATHDB-403181

Borel-Haefliger question on generation of Chow groups by smooth subvarieties

摘录 / EXCERPT: Fix two integers $c,d\ge 1$. Let X be a smooth projective variety of dimension $c+d$ over $\mathbb{C}$. Is the Chow group $\mathrm{CH}_d(X)=\mathrm{CH}^c(X)$ generated by classes...

open_problem | open | MathDB 页面 | 原文线索

V2-023215 | MATHDB-403182

Generation of Chow groups by subvarieties smooth along their real loci

摘录 / EXCERPT: Fix integers $c,d\geq 1$. Let X be a smooth projective variety of dimension $c+d$ over $\mathbb{R}$. Is the Chow group $\mathrm{CH}_d(X)$ generated by classes of subvarieties...

open_problem | open | MathDB 页面 | 原文线索

V2-023216 | MATHDB-403183

Generation of the kernel of the real cycle class map

摘录 / EXCERPT: Fix integers $c,d\ge 1$. Let X be a smooth projective variety of dimension $c+d$ over $\mathbb{R}$. Is ... generated by classes of subvarieties $Y\subset X$ whose real locus...

open_problem | open | MathDB 页面 | 原文线索

V2-023217 | MATHDB-403184

Optimal lower bounds for volumes of smooth varieties of general type

摘录 / EXCERPT: For each positive integer n, determine the optimal constant v_n such that every smooth complex projective n-fold X of general type satisfies $\operatorname{vol}(X)\ge v_n$;...

open_problem | open | MathDB 页面 | 原文线索

V2-023218 | MATHDB-403268

Infinitely many simple abelian varieties over \(\mathbb F_2\) of every prescribed positive order

摘录 / EXCERPT: For every positive integer m, there exist infinitely many simple abelian varieties over $\mathbb F_2$, of various dimensions, whose groups of $\mathbb F_2$-rational points have o...

open_problem | open | MathDB 页面 | 原文线索

V2-023219 | MATHDB-403329

Integral models of Shimura varieties and their local shtuka descriptions

摘录 / EXCERPT: Conjecture 2.1. Assume that G is a parahoric group scheme. There exists a pro-system of normal and flat integral models $\mathcal{S}_K$ over $\mathcal{O}_E$, with generic fiber...

open_problem | open | MathDB 页面 | 原文线索

V2-023220 | MATHDB-403331

Beilinson-Drinfeld geometric Langlands conjecture

摘录 / EXCERPT: Let G be a split, connected reductive group, and let X be a smooth projective curve over a field k of characteristic zero. With $[U+007F]operatorname{Bun}_G$ the moduli stack of G...

conjecture | open | MathDB 页面 | 原文线索

V2-023221 | MATHDB-403333

Connected components of the arithmetic local-systems stack

摘录 / EXCERPT: What are the connected components of $\operatorname{LocSys}^{\mathrm{arthm}}_{\check G}$?

open_problem | open | MathDB 页面 | 原文线索

V2-023222 | MATHDB-403443

Brasselet-Schürmann-Yokura conjecture for rational homology manifolds

摘录 / EXCERPT: Let X be a compact complex algebraic variety that is a rational homology manifold. Then ... where $T_{1,*}$ is the specialization at $y=1$ of the Hirzebruch homology-class transfor...

conjecture | open | MathDB 页面 | 原文线索

V2-023223 | MATHDB-403446

3d mirror symmetry for elliptic characteristic classes of singularities

摘录 / EXCERPT: For a pair of 3d mirror-dual bow varieties X and X', the corresponding torus fixed points $p\leftrightarrow p'$ are conjectured to satisfy ... where $\big|_q$ denotes restriction...

open_problem | open | MathDB 页面 | 原文线索

V2-023224 | MATHDB-403634

Smoothness and normality of K-moduli spaces of Fano varieties

摘录 / EXCERPT: The assertion that K-moduli stacks of Fano varieties are always smooth and their good moduli spaces are always normal is false. More precisely, for every integer $n\ge 3$, there ex...

open_problem | open | MathDB 页面 | 原文线索

V2-023225 | MATHDB-403639

Unobstructedness of affine toric Gorenstein varieties in the mutation subspace

摘录 / EXCERPT: Conjecture. An affine toric Gorenstein variety X is unobstructed in $T(f)$, i.e. there exists a flat family over $\mathbb{C}[[T(f)]]\subset\mathbb{C}[[T^1X]]$ with the fiber over...

open_problem | open | MathDB 页面 | 原文线索

V2-023226 | MATHDB-403640

Motivic change-of-variables formula for Artin stacks

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 0. Let X be a smooth irreducible finite-type Artin stack over k with affine geometric stabilizers and separated dia...

open_problem | open | MathDB 页面 | 原文线索

V2-023227 | MATHDB-403641

Nonnegativity of Batyrev’s stringy Hodge numbers

摘录 / EXCERPT: For a log-terminal scheme Y with $\cmathbb{Q}$-Gorenstein singularities, Batyrev conjectured that its stringy Hodge numbers satisfy $h_{\cmathrm{st}}^{p,q}(Y)\ge 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-023228 | MATHDB-403643

Relations among toric degenerations from maximal tropical cones

摘录 / EXCERPT: For an arbitrary projective variety X, what is the relationship between the different toric degenerations arising from different maximal cones in $\operatorname{trop}(X)$?

open_problem | open | MathDB 页面 | 原文线索

V2-023229 | MATHDB-403666

Relative Bogomolov conjecture

摘录 / EXCERPT: Let $\pi:\mathcal A\to S$ be a family of abelian varieties over an irreducible $\mathbb Q$-variety S, and let $X\subset\mathcal A$ be irreducible with $\pi(X)=S$. Choose a global...

conjecture | open | MathDB 页面 | 原文线索

V2-023230 | MATHDB-403668

Relative Manin-Mumford conjecture for abelian schemes

摘录 / EXCERPT: Let S be an irreducible quasi-projective variety defined over $\overline{\mathbb{Q}}$, and let $\pi:\mathcal{A}\to S$ be an abelian scheme of relative dimension $g\ge 1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023231 | MATHDB-403672

Vojta’s conjecture for truncated proximity to a normal-crossing divisor

摘录 / EXCERPT: Let X be a smooth projective variety, D a reduced normal-crossing divisor, K a canonical divisor, and A an ample divisor, all defined over a number field k. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-023232 | MATHDB-403706

Agreement between the even and motivic filtrations on algebraic K-theory

摘录 / EXCERPT: Does the even filtration of algebraic K-theory of a scheme agree with the motivic filtration thereof in the cases where the latter is defined?

open_problem | open | MathDB 页面 | 原文线索

V2-023233 | MATHDB-403707

p-adic weight-monodromy conjecture for complete intersections in toric varieties

摘录 / EXCERPT: The Fontaine-Jannsen p-adic version of Deligne’s monodromy-weight conjecture holds for a smooth complete intersection in a projective toric variety over a complete discrete valuati...

conjecture | open | MathDB 页面 | 原文线索

V2-023234 | MATHDB-403711

Special-value conjecture for proper regular arithmetic schemes

摘录 / EXCERPT: Let $X/\mathbb{Z}$ be a proper regular arithmetic scheme and let $n\in\mathbb{Z}$. Under the assumptions that the relevant perfect complexes exist and Beilinson’s regulator behaves...

conjecture | open | MathDB 页面 | 原文线索

V2-023235 | MATHDB-403712

Existence of higher K-theory classes of varieties

摘录 / EXCERPT: Are there any classes in $K_n(\operatorname{Var}_k)$ for $n\ge 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-023236 | MATHDB-403713

A motivic Drinfeld lemma for two smooth projective curves

摘录 / EXCERPT: Let X_1,X_2 be smooth projective curves over $\\mathbb{F}_p$. Is there an equivalence ...

open_problem | open | MathDB 页面 | 原文线索

V2-023237 | MATHDB-403714

Weibel’s conjecture for non-Noetherian schemes using valuative dimension

摘录 / EXCERPT: Do the assertions of Weibel’s conjecture continue to hold for non-Noetherian schemes when Krull dimension is replaced by valuative dimension; in particular, for a scheme X of fin...

conjecture | open | MathDB 页面 | 原文线索

V2-023238 | MATHDB-403717

Hopkins-Morel conjecture

摘录 / EXCERPT: For every scheme S, the canonical map ... is an equivalence, where MGL is Voevodsky’s algebraic cobordism spectrum, $H\mathbb{Z}$ is Spitzweck’s motivic cohomology spectrum, and...

conjecture | open | MathDB 页面 | 原文线索

V2-023239 | MATHDB-403792

Additivity of Kodaira dimension for smooth projective algebraic fiber spaces

摘录 / EXCERPT: Let $f:U\to V$ be a smooth projective algebraic fiber space, with U and V smooth quasi-projective varieties, and with general fiber F. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-023240 | MATHDB-403799

Strominger-Yau-Zaslow conjecture for maximally degenerate Calabi-Yau families

摘录 / EXCERPT: Consider a projective family $(X_t)_t$ of Calabi-Yau varieties of dimension n over a punctured disk, such that the family is maximally degenerate, i.e. the monodromy operator on...

conjecture | open | MathDB 页面 | 原文线索

V2-023241 | MATHDB-403800

Deduction of the Gushel-Mukai Tate conjecture from the hyper-Kähler case

摘录 / EXCERPT: Can the authors’ results proving the Tate conjecture for even-dimensional Gushel-Mukai varieties over finitely generated fields of characteristic $p\ge 5$ be deduced from the Tate...

conjecture | open | MathDB 页面 | 原文线索

V2-023242 | MATHDB-403801

Precise structure of the local Nori fundamental group scheme

摘录 / EXCERPT: Determine the precise structure of the local Nori fundamental group scheme $\pi_N^{\mathrm{loc}}(X)$ associated with an isolated singularity X over an algebraically closed field...

open_problem | open | MathDB 页面 | 原文线索

V2-023243 | MATHDB-403803

Frobenius-ampleness criterion for projective space

摘录 / EXCERPT: Let X/k be a d-dimensional smooth projective variety such that the locally free sheaf B_X^1 is anti-ample. Is X isomorphic to $\boldsymbol{P}_k^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-023244 | MATHDB-403805

Uniform exponent bounds for torsion in higher Chow groups

摘录 / EXCERPT: Let $\pi:X\to S$ be a family of smooth projective varieties over a quasi-projective scheme over $\mathbb{Q}$, and let d be a positive integer. Does there exist a positive integer...

open_problem | open | MathDB 页面 | 原文线索

V2-023245 | MATHDB-403863

Rigid analytic moduli stacks of (φ,Γ)-modules

摘录 / EXCERPT: The stack $\mathcal{X}_d$ is a rigid analytic Artin stack of dimension $d^2[F:\mathbf{Q}_p]$ (equal to $\dim \operatorname{Res}_{F/\mathbf{Q}_p}\mathrm{GL}_d$) and a local complete...

open_problem | open | MathDB 页面 | 原文线索

V2-023246 | MATHDB-403868

Essential image of the p-adic Simpson correspondence

摘录 / EXCERPT: Let X be a connected smooth projective variety over $\\mathbb{C}_p$. Determine which Higgs bundles on X lie in the essential image of the category of continuous representations...

open_problem | open | MathDB 页面 | 原文线索

V2-023247 | MATHDB-403929

Configurations of tensor eigenschemes in projective space

摘录 / EXCERPT: What configurations of points in $\mathbb{P}^n$ are eigenschemes $E(T)$ of a tensor?

open_problem | open | MathDB 页面 | 原文线索

V2-023248 | MATHDB-403937

Classification of discrete statistical models with ML degree one

摘录 / EXCERPT: It is an open problem to classify, in statistical terms, all discrete statistical models with ML degree 1, even when restricted to toric varieties.

open_problem | open | MathDB 页面 | 原文线索

V2-023249 | MATHDB-403959

Enumeration of the West-stack-sorting preimage of \(\operatorname{Av}(321)\)

摘录 / EXCERPT: Determine the generating function of the preimage of $\operatorname{Av}(321)$ under the West-stack-sorting operation, namely the class $\operatorname{Av}(34251,35241,45231)$; in pa...

open_problem | open | MathDB 页面 | 原文线索

V2-023250 | MATHDB-403978

Polynomial interpolation on arbitrary varieties

摘录 / EXCERPT: Let $V_1,\ldots,V_n$ be subvarieties of $\mathbb{C}^d$ and let $p_1,\ldots,p_n$ be polynomials in $\mathbb{C}[x]=\mathbb{C}[x_1,\ldots,x_d]$. When does there exist a polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-023251 | MATHDB-403979

Sufficient condition for polynomial interpolation on varieties

摘录 / EXCERPT: Let $V_1,\ldots,V_n$ be varieties in $\mathbb{C}^d$, let $J_i=J(V_i)$ for $i=1,\ldots,n$, and let $p_1,\ldots,p_n\in\mathbb{C}[x_1,\ldots,x_d]$ be polynomial data on the varieties...

open_problem | open | MathDB 页面 | 原文线索

V2-023252 | MATHDB-403986

Homogeneous equations for mean-value coordinate varieties

摘录 / EXCERPT: Given a polygon with n sides in the plane, let $\b2_v$ be its mean-value barycentric coordinate for each vertex v. These coordinates define a map from the polygon to the affine...

open_problem | open | MathDB 页面 | 原文线索

V2-023253 | MATHDB-403999

Optimality of the Euler-scheme rate for fractional-noise SDEs

摘录 / EXCERPT: For the SDE in (3), ... is it possible, in the case $H=1/2$, to show that the Euler-scheme convergence rate $(1+\alpha)/2$ cannot be improved for some $b\in C^\alpha$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-023254 | MATHDB-404001

Removal of the logarithmic loss in a stochastic quadrature estimate

摘录 / EXCERPT: Under the setting and assumptions for the tamed Euler-Maruyama scheme described in the report, can the logarithmic factor in the bound for the weighted quadrature error be removed,...

open_problem | open | MathDB 页面 | 原文线索

V2-023255 | MATHDB-404090

Campana’s abelianity conjecture for special or Brody special quasi-projective varieties

摘录 / EXCERPT: A smooth quasi-projective variety X that is special or Brody special has a virtually abelian fundamental group.

conjecture | open | MathDB 页面 | 原文线索

V2-023256 | MATHDB-404094

Beilinson Conjecture for smooth projective varieties over finite fields

摘录 / EXCERPT: Let F be a finite field, let ℓ be a prime invertible in F, and let X be a smooth projective variety over F. For every codimension i, the cycle class map ... is injecti...

conjecture | open | MathDB 页面 | 原文线索

V2-023257 | MATHDB-404124

Toroidal Hodge embedding for integral models of Shimura varieties

摘录 / EXCERPT: Theorem 5 (Xu). Let (G,X) be a Shimura datum of Hodge type. For each $K\subset G(\mathbb{A}_f)$ sufficiently small, there exist collections Σ and Σ' of cone decompo...

open_problem | open | MathDB 页面 | 原文线索

V2-023258 | MATHDB-404128

Arithmetic duality conjecture for smooth Stein varieties

摘录 / EXCERPT: Conjecture 1. Let X be a smooth, geometrically irreducible, Stein variety of dimension d over K. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-023259 | MATHDB-404143

Openness of Kähler-Einstein metrics in families of Fano varieties

摘录 / EXCERPT: Let X be an $(n+1)$-dimensional $\Q$-Gorenstein variety and let $\\pi:X\to D$ be a proper holomorphic surjective map with connected fibres. Suppose that $-K_{X/D}$ is relatively...

open_problem | open | MathDB 页面 | 原文线索

V2-023260 | MATHDB-404278

Spectral moduli-theoretic interpretations of the spectral schemes

摘录 / EXCERPT: Do the spectral schemes appearing in the equivalences above admit natural spectral moduli-theoretic interpretations?

open_problem | open | MathDB 页面 | 原文线索

V2-023261 | MATHDB-404283

Classification of tensor-compatible compactly generated t-structures on a noetherian scheme

摘录 / EXCERPT: Let X be a noetherian scheme and let $D=D(\operatorname{Qcoh}(X))$. Denote by $(D^{\leq 0}_{\mathrm{can}},D^{\geq 0}_{\mathrm{can}})$ the canonical t-structure, and call a full...

open_problem | open | MathDB 页面 | 原文线索

V2-023262 | MATHDB-404300

Finiteness of the Tate-Shafarevich set of an abelian variety over a number field

摘录 / EXCERPT: Let k be a number field, let $\bar{k}$ be a separable closure, and let A be an abelian variety over k. Conjecturally, the Tate-Shafarevich group ... is finite, where Ω i...

open_problem | open | MathDB 页面 | 原文线索

V2-023263 | MATHDB-404302

Cadoret-Stix Conjecture 1 on the degeneracy locus of an ℓ-adic local system

摘录 / EXCERPT: Let k be a number field, let X be a smooth, separated, geometrically connected variety over k, and let V_ℓ be a $\mathbb{Q}_\ell$-local system on X. Let G_ℓ and...

conjecture | open | MathDB 页面 | 原文线索

V2-023264 | MATHDB-404303

Cadoret-Stix conjecture on the toric locus of an \bb-adic local system

摘录 / EXCERPT: Let k be a number field, let X be a smooth, separated, geometrically connected variety over k, and let $f:Y\to X$ be a smooth projective morphism. Fix a prime ℓ and put...

conjecture | open | MathDB 页面 | 原文线索

V2-023265 | MATHDB-404304

Non-density of the toric locus of an \-adic local system

摘录 / EXCERPT: Let k be a number field, let X be a smooth, separated, geometrically connected variety over k, and let $f:Y\to X$ be a smooth projective morphism. Fix a prime ℓ and set...

open_problem | open | MathDB 页面 | 原文线索

V2-023266 | MATHDB-404316

Other varieties with the finite-covering and centralizer properties

摘录 / EXCERPT: Find other nontrivial varieties, beyond hyperbolic curves and semi-abelian varieties, whose associated étale homotopy exact sequences have both properties used in the report: (1) t...

open_problem | open | MathDB 页面 | 原文线索

V2-023267 | MATHDB-404362

Realization of conical subvarieties as cohomological support varieties

摘录 / EXCERPT: Fix a local ring R. Given a conical subvariety V of $\4\mathbb{A}^n\u00024$, is there $M\in D^b(R)$ such that $V_R(M)=V$?

open_problem | open | MathDB 页面 | 原文线索

V2-023268 | MATHDB-404369

Normality of orbit closures for Dynkin quivers

摘录 / EXCERPT: Even for Dynkin quivers, are orbit closures always normal varieties?

open_problem | open | MathDB 页面 | 原文线索

V2-023269 | MATHDB-404447

Colliot-Thélène’s conjecture on the Hilbert Property

摘录 / EXCERPT: Every unirational variety over $\mathbb{Q}$ has HP.

conjecture | open | MathDB 页面 | 原文线索

V2-023270 | MATHDB-404511

Affine Weyl-group classification of marked noncommutative del Pezzo surfaces

摘录 / EXCERPT: Conjecture 3. A pair of geometric points on the moduli stack of marked noncommutative del Pezzo surfaces give an isomorphic pair of noncommutative del Pezzo surfaces if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-023271 | MATHDB-404519

Polishchuk-Van den Bergh semi-orthogonal decomposition conjecture

摘录 / EXCERPT: Suppose that a finite group G acts effectively on a smooth variety X and that, for all $g\in G$, the geometric quotient $\bar X_g=X^g/C(g)$ is smooth. Then there is a semi-orth...

conjecture | open | MathDB 页面 | 原文线索

V2-023272 | MATHDB-404522

Generalised Section Conjecture for étale homotopy types

摘录 / EXCERPT: Let k be a field with separable closure $\bar{k}\supset k$, and let X be a geometrically connected quasi-compact and quasi-separated scheme over k. Define ... The Generalised S...

conjecture | open | MathDB 页面 | 原文线索

V2-023273 | MATHDB-404523

Equivariant triangulability of separated finite-type real schemes

摘录 / EXCERPT: Any separated scheme of finite type $X/\mathbb{R}$ is equivariantly triangulable; that is, $X(\mathbb{C})$, with the action of $\operatorname{Gal}_{\mathbb{R}}$ induced by complex...

open_problem | open | MathDB 页面 | 原文线索

V2-023274 | MATHDB-404525

Conjecture 3 on algebraic leaves of integrable foliations

摘录 / EXCERPT: Let $\pi:M\to S$ be a smooth morphism of smooth schemes, and let $F\subset T_M$ be an integrable foliation on M/S, i.e. a subsheaf closed under the Lie bracket such that the natu...

conjecture | open | MathDB 页面 | 原文线索

V2-023275 | MATHDB-404526

Theorem B: Algebraicity of flat sections for symmetric powers of genus-one Gauss-Manin connections

摘录 / EXCERPT: Let $f:X\to S$ be a smooth proper morphism of smooth $\mathbb{C}$-schemes, with X/S a relative genus-one curve. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-023276 | MATHDB-404527

Conjecture 1 on the index of pullbacks under multiplication by p^n

摘录 / EXCERPT: Let X be a smooth projective curve of genus $\a0\2$ embedded in an abelian variety A over a field k finite over $\a0\Q_p$. Assume that $X\subset A$ contains no torsion points...

conjecture | open | MathDB 页面 | 原文线索

V2-023277 | MATHDB-404537

Existence of a smooth abelian subvariety with Tannaka group of type E_7

摘录 / EXCERPT: Do there exist smooth subvarieties $X\subset A$ whose Tannaka group $G_{X,\omega}$ is of type E_7?

open_problem | open | MathDB 页面 | 原文线索

V2-023278 | MATHDB-404668

Finite-element convergence for a triple-junction curvature-flow scheme

摘录 / EXCERPT: Theorem 1 (h-convergence, ε fixed). Assume that there is a unique sufficiently regular solution $\Gamma=(u^{(1)},u^{(2)},u^{(3)})$ with $0<c_0\leq |u_x^{(i)}|\leq 1/c_0$. For all s...

open_problem | open | MathDB 页面 | 原文线索

V2-023279 | MATHDB-404750

Inverse Galois problem for connected algebraic groups

摘录 / EXCERPT: If G is a smooth connected algebraic group defined over a field k, can G be realized as the automorphism group of a smooth projective variety?

open_problem | open | MathDB 页面 | 原文线索

V2-023280 | MATHDB-404752

Automorphism groups determine connected affine varieties

摘录 / EXCERPT: Let X be a connected affine variety over a field k of arbitrary characteristic. If $\\operatorname{Aut}(X)$ is isomorphic to $\operatorname{Aut}(\mathbb{A}^n)$ as an abstract g...

open_problem | open | MathDB 页面 | 原文线索

V2-023281 | MATHDB-404795

Odd-degree cohomology of the moduli space of principally polarized abelian varieties

摘录 / EXCERPT: Does the moduli space $\mathcal A_g$ of principally polarized abelian varieties of dimension g have any cohomology class in odd degree?

open_problem | open | MathDB 页面 | 原文线索

V2-023282 | MATHDB-404796

D-equivalence for hyper-Kähler varieties of K3^[n]-type

摘录 / EXCERPT: If X and X' are birational hyper-Kähler varieties of K3$^{[n]}$-type, then there is an equivalence of bounded derived categories ...

open_problem | open | MathDB 页面 | 原文线索

V2-023283 | MATHDB-404797

Virtual structure-sheaf pushforward for the Iarrobino scheme of points on $\mathbb A^2$

摘录 / EXCERPT: For $X=\mathbb A^2$, conjecturally, $R\pi_*\mathcal O^{\mathrm{vir}}_{\operatorname{Iar}_d(\mathbb A^2)}\simeq \mathcal O_{\operatorname{Hilb}^d(\mathbb A^2)}$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-023284 | MATHDB-404799

D-equivalence for birational hyper-Kähler varieties of K3[n]-type

摘录 / EXCERPT: Let $X\dashrightarrow X'$ be a birational map of hyper-Kähler varieties of $K3^{[n]}$-type inducing an identification $\operatorname{Br}(X)=\operatorname{Br}(X')$. Then, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-023285 | MATHDB-404800

Huybrechts' period-index conjecture for hyper-Kähler varieties

摘录 / EXCERPT: For every hyper-Kähler variety X and every Brauer class $\alpha\in\operatorname{Br}(X)$, one has ...

conjecture | open | MathDB 页面 | 原文线索

V2-023286 | MATHDB-404807

Relations among generators of twisted Chow groups

摘录 / EXCERPT: For a locally constant étale sheaf E on a smooth k-scheme X, describe the relations among the generators of $CH^i(X,E)$ supplied by Theorem 2, analogously to the known cokern...

open_problem | open | MathDB 页面 | 原文线索

V2-023287 | MATHDB-404808

Question 1: Beauville-Bogomolov decomposition for log canonical varieties

摘录 / EXCERPT: Given a log canonical variety (or pair) X with $K_X\sim_{\mathbb{Q}}0$, does the universal cover Y of X_{reg} admit a decomposition ... where F, Y_i, and Z_j are...

open_problem | open | MathDB 页面 | 原文线索

V2-023288 | MATHDB-404814

Nekovář-Scholl global plectic conjecture for Shimura varieties

摘录 / EXCERPT: Let F be a totally real field of degree d, let H/F be a reductive group, and let $G:=\operatorname{Res}_{F/\mathbb{Q}}H$ be part of a Shimura datum (G,X) with Hodge cochara...

conjecture | open | MathDB 页面 | 原文线索

V2-023289 | MATHDB-404816

Special case of Grothendieck's generalized Hodge conjecture

摘录 / EXCERPT: Let X be an irreducible smooth projective complex algebraic variety. If $H^{0}(X,\Omega_X^{i})=0$, then there exists a dense open subset $U\subset X$ such that the restriction ho...

conjecture | open | MathDB 页面 | 原文线索

V2-023290 | MATHDB-404817

Generic-fibre vanishing predicted from dense mod-\c0 fibrewise vanishing

摘录 / EXCERPT: Fix an integer i. Let S be a smooth affine scheme over $\mathbb{Z}$ and let $X_S\to S$ be smooth and proper. If there is a dense set of closed points $s\in S$ such that, for ea...

open_problem | open | MathDB 页面 | 原文线索

V2-023291 | MATHDB-404818

The p-GGHS problem

摘录 / EXCERPT: Let X be a smooth proper scheme over $\bb{Z}_p$. If $H^0(X_1,\Omega^i_{X_1})=0$, is there an open dense $U\subset X$, smooth over $\bb{Z}_p$, with $U_1\subset X_1$ dense, such th...

open_problem | open | MathDB 页面 | 原文线索

V2-023292 | MATHDB-404825

Fully homomorphic encryption from mathematical objects other than integer lattices

摘录 / EXCERPT: Whether one can construct a fully homomorphic encryption scheme from mathematical objects other than integer lattices remains a tantalizing open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-023293 | MATHDB-404917

Equivalence of motivic and logarithmic motivic spectra

摘录 / EXCERPT: For any derived scheme S, one has an equality of symmetric monoidal categories ...

open_problem | open | MathDB 页面 | 原文线索

V2-023294 | MATHDB-404918

Colliot-Thélène’s conjecture on the Albanese kernel over a p-adic field

摘录 / EXCERPT: Let X be a smooth projective geometrically connected variety over a finite extension k of the p-adic field $\mathbb{Q}_p$. Let $F^1(X)=\ker(\deg:CH_0(X)\to\mathbb{Z})$ and le...

conjecture | open | MathDB 页面 | 原文线索

V2-023295 | MATHDB-404923

Towards a categorical Künneth formula for motives

摘录 / EXCERPT: Let X and Y be algebraic varieties over a field k with finite Galois cohomological dimension. Let $ ...

open_problem | open | MathDB 页面 | 原文线索

V2-023296 | MATHDB-404925

Categorical Künneth formula for motives generated by smooth proper schemes

摘录 / EXCERPT: Let X and Y be algebraic varieties over a field k of finite Galois cohomological dimension, and let $\\operatorname{DM}(X)^{\prime\prime}$ denote the full stable presentable...

open_problem | open | MathDB 页面 | 原文线索

V2-023297 | MATHDB-404976

Polishchuk-Van den Bergh semiorthogonal decomposition conjecture

摘录 / EXCERPT: Suppose that G is a finite group acting effectively on a smooth variety X and that, for all $g\in G$, the geometric quotient $\bar X_g=X^g/C(g)$ is smooth. Then there is a semi...

conjecture | open | MathDB 页面 | 原文线索

V2-023298 | MATHDB-404983

Corti-R. conjecture on log crepant resolutions and smoothings

摘录 / EXCERPT: Log schemes obtained as in (b) of the Proposition permit a log crepant log resolution and also a smoothing.

conjecture | open | MathDB 页面 | 原文线索

V2-023299 | MATHDB-404986

Computing the discriminant-weighted motivic generating series

摘录 / EXCERPT: For a smooth curve f (and eventually for any reduced plane curve), compute ... where s is a formal parameter, $X_f=\operatorname{Spec}(\mathbb{C}[[t]][x,y]/(f-t))$, X_f^0 is th...

open_problem | open | MathDB 页面 | 原文线索

V2-023300 | MATHDB-405027

Potential weak Hilbert property versus potential density

摘录 / EXCERPT: Let k be a number field and let X be a smooth projective k-variety. Is it true that X has potential weak Hilbert property if and only if there exists a finite extension...

open_problem | open | MathDB 页面 | 原文线索

V2-023301 | MATHDB-405028

Torsion Anomalous Conjecture (TAC)

摘录 / EXCERPT: Let G be a torus or an abelian variety defined over $\mathbb{Q}$, and let $V\subsetneq G$ be a proper irreducible subvariety. A torsion variety is a finite union of translates of...

conjecture | open | MathDB 页面 | 原文线索

V2-023302 | MATHDB-405030

Recovery of the topological fundamental group from the algebraic fundamental group

摘录 / EXCERPT: Let X be a smooth complex projective variety. Can we recover $\[1m\pi_1(X)\u001b[0m$ up to isomorphism from its algebraic fundamental group $\[1m\pi_1^{\mathrm{alg}}(X)\u001b[0m$...

open_problem | open | MathDB 页面 | 原文线索

V2-023303 | MATHDB-405098

Bialgebraic varieties of the Gamma function

摘录 / EXCERPT: Let $W\subseteq\mathbb{C}^n$ be an algebraic variety. If W is bialgebraic for the coordinatewise Gamma function, meaning that $\dim W=\dim \Gamma(W)^{\mathrm{Zar}}$, then W is...

open_problem | open | MathDB 页面 | 原文线索

V2-023304 | MATHDB-405099

Algebraicity of the non-abelian Hodge loci

摘录 / EXCERPT: Let $X\to S$ be a smooth projective morphism with S a smooth complex quasi-projective variety, and let $M_{\mathrm{dR}}(X/S)$ and $M_{\mathrm{Dol}}(X/S)$ be the relative de Rham...

open_problem | open | MathDB 页面 | 原文线索

V2-023305 | MATHDB-405100

A bound of 3 for the degree of nonminimality of generic difference equations

摘录 / EXCERPT: Let $K\subseteq U\models\mathrm{ACFA}$ be an embedding of difference fields with distinguished automorphism σ. Let V be a variety over K and let...

open_problem | open | MathDB 页面 | 原文线索

V2-023306 | MATHDB-405116

Equality relating the Poincaré polynomial of a quiver variety to the Kac polynomial

摘录 / EXCERPT: Let Q be a quiver, let $d\in\mathbb{N}^{Q_0}$ be indivisible, and let $\zeta\in\mathbb{Q}^{Q_0}$ be generic. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-023307 | MATHDB-405146

Fujita’s freeness conjecture

摘录 / EXCERPT: For every smooth complex projective variety X of dimension n and every ample Cartier divisor L on X, the adjoint divisor $K_X+(n+1)L$ is globally generated.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023308 | MATHDB-405149

Maximum number of circles tangent to three conics

摘录 / EXCERPT: Determine the maximum number of distinct real circles tangent to each of three general real conics in the Euclidean plane. The proposed conjecture was that every such triple has at...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023309 | MATHDB-405159

Jiang-Tseng Virasoro conjecture

摘录 / EXCERPT: For every smooth projective effective orbifold curve X, the total descendant Gromov-Witten partition function $\mathcal{Z}_X$ satisfies the Jiang-Tseng Virasoro constraints...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023310 | MATHDB-405162

Conjectures on algebraic, geometric, and span multiplicities of tensor eigenvalues

摘录 / EXCERPT: For every tensor T and every tensor eigenvalue λ, the algebraic multiplicity dominates the geometric and span multiplicities:...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023311 | MATHDB-405168

Kim’s conjecture for S-integral points on the thrice-punctured line

摘录 / EXCERPT: Let K be a number field, let S be a finite set of finite places of K, and let $X=\mathbb{P}^{1}\smallsetminus\{0,1,\infty\}$. Kim's conjecture asserts that the motivic Chabau...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023312 | MATHDB-405175

Generalized Grothendieck-Katz p-curvature conjecture

摘录 / EXCERPT: Let X be a smooth irreducible variety over a field of characteristic zero, and let $\mathcal{F}\subseteq T_X$ be an algebraic foliation of rank r admitting reductions modulo al...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023313 | MATHDB-405181

N. Alon’s question on zeros of sums of square roots; Maxwell’s conjecture-related zero bound

摘录 / EXCERPT: For every $n\ge 1$, real coefficients $c_1,\ldots,c_n$, and polynomials $P_1,\ldots,P_n\in\mathbb{R}[x]$ satisfying $\deg P_k\le 2$ and $P_k(x)>0$ for all $x\in\mathbb{R}$, if...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023314 | MATHDB-405183

Brochard’s duality conjecture

摘录 / EXCERPT: Let S be a base scheme and let G be a proper, flat, finitely presented commutative group stack over S with finite flat inertia. Define its dual by...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023315 | MATHDB-405188

Eisenbud-Schreyer-Weyman Ulrich Existence Problem

摘录 / EXCERPT: For every projective variety $X\subseteq\mathbb{P}^N$ over a field, equipped with its given embedding, does there exist a nonzero coherent sheaf $\mathcal{E}$ on X that is Ulrich...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023316 | MATHDB-405193

Relative non-nef loci conjecture

摘录 / EXCERPT: For a projective morphism $f\colon X\to S$ and a relatively pseudo-effective $\mathbb{R}$-Cartier divisor D on X, the relative non-nef locus equals the relative restricted base...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023317 | MATHDB-405198

Prescribed Lelong Numbers for One-Pole Green Functions on Complex Projective Space

摘录 / EXCERPT: For every integer $n\geq 2$, every point $x\in\mathbf P^n$, and every $\lambda\in[0,1]$, does there exist a function $u\in DMA(\mathbf P^n,\omega_{\mathrm{FS}})$ such that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023318 | MATHDB-405207

Borovik-Cherlin conjecture

摘录 / EXCERPT: Let $n>0$, and let a connected group G of finite Morley rank act faithfully and transitively on a definable set X of Morley rank n. If the action is generically $(n+2)$-trans...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023319 | MATHDB-405208

Existence of Neumaier graphs of coherent rank five

摘录 / EXCERPT: Does there exist a Neumaier graph whose coherent rank is exactly 5? Equivalently, determine whether there exists a finite simple graph G that is a Neumaier graph and whose cohe...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023320 | MATHDB-405213

Goldman’s conjecture for type-preserving representations

摘录 / EXCERPT: For every punctured surface $\Sigma_{g,p}$ with $g\geq 2$, the mapping class group $\operatorname{Mod}(\Sigma_{g,p})$ acts ergodically on every non-Teichmüller component of the rel...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023321 | MATHDB-405218

Oort’s conjecture for split unitary Shimura varieties

摘录 / EXCERPT: Let $\mathcal{S}$ be a split unitary Shimura variety of signature (g,g) at a split prime p, and let $\mathcal{S}_{\mathrm{basic}}$ denote its basic locus. Outside the degenerat...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023322 | MATHDB-405221

Briançon-Iarrobino Conjecture and Conjecture B on maximal singularity of threefold Hilbert schemes

摘录 / EXCERPT: For a smooth threefold X and an integer $\ell\geq 1$, characterize the points $[Z]\in \operatorname{Hilb}^{\ell}(X)$ having maximal singularity, namely those satisfying...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023323 | MATHDB-405222

Dan Abramovich’s question on contractions defined by differential or divisor data

摘录 / EXCERPT: Given a proper morphism $f:\mathcal{X}\to\mathcal{Z}$ with connected fibers between smooth separated Deligne--Mumford stacks, and a smooth effective Cartier divisor...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-023324 | MATHDB-405225

Morse-Cairns finiteness conjecture for equilibria of Coulomb fields

摘录 / EXCERPT: For every integer $N\ge 1$, every collection of distinct points $\mathbf a_1,\ldots,\mathbf a_N\in\mathbb R^3$, and every collection of nonzero charges $q_1,\ldots,q_N$ having the...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023325 | MATHDB-405229

Shelton-Yuzvinsky conjecture on Koszul Orlik-Solomon algebras

摘录 / EXCERPT: For every complex hyperplane arrangement $\mathcal{A}$, if its Orlik-Solomon algebra $\operatorname{OS}(\mathcal{A})$ is Koszul, then the intersection lattice $L(\mathcal{A})$ is s...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023326 | MATHDB-405237

Rational noncommutative Hodge conjecture

摘录 / EXCERPT: For every smooth proper differential graded category $\mathcal{A}$, the rational Chern character is surjective onto the rational noncommutative Hodge classes:...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023327 | MATHDB-405252

The b-conjecture and Airyβ soft-edge limit for the Jack-Plancherel process

摘录 / EXCERPT: For every $\beta>0$, the standard edge-scaled Jack-Plancherel process converges at the soft edge, in its multi-time finite-dimensional statistics (in particular, in moments), to th...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023328 | MATHDB-405259

The telescope conjecture for derived Artin motives over local fields

摘录 / EXCERPT: Let F be a nonarchimedean local field and let k be a field of characteristic $p>0$. Does the tensor-triangulated category $\operatorname{DAM}(F;k)$ of derived Artin motives sat...

open_problem | claimed_progress | MathDB 页面 | 原文线索

B05 几何 / Geometry | 6,599 条

V2-023329 | MATHDB-59

The DK hypothesis: K-equivalence implies derived equivalence

摘录 / EXCERPT: DK hypothesis. K-equivalence should imply an equivalence ... More generally, if $p^*K_X\geq q^*K_Y$, one expects a fully faithful embedding from $D^b\operatorname{Coh}(Y)$ into...

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-023330 | MATHDB-315725

Rank-two Poisson Conjecture

摘录 / EXCERPT: Must every polynomial endomorphism of a rank-n symplectic Poisson algebra that preserves the canonical bracket be an automorphism?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023331 | MATHDB-315957

Carathéodory conjecture

摘录 / EXCERPT: Let $\alpha\in(0,1)$ and let $S\subset\mathbb{R}^{3}$ be a compact, connected, embedded surface without boundary of class $C^{3,\alpha}$ which is convex, i.e. S bounds a convex b...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023332 | MATHDB-315962

Hopf conjecture

摘录 / EXCERPT: Let $M=\mathbb{S}^{2}\times\mathbb{S}^{2}$, the product of two 2-spheres, regarded as a closed smooth 4-manifold. For a Riemannian metric g on M and a point $p\in M$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-023333 | MATHDB-315963

Osserman conjecture

摘录 / EXCERPT: Let $(M^{n},\langle\cdot,\cdot\rangle)$ be a connected Riemannian manifold of dimension n with Riemann curvature tensor R. For $p\in M^{n}$ and a unit vector $X\in T_{p}M^{n}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-023334 | MATHDB-315964

Yau's conjecture on the first eigenvalue

摘录 / EXCERPT: Let $n\ge 2$ and let $S^{n+1}=\{x\in\mathbb{R}^{n+2}:|x|=1\}$ carry its standard round metric of constant sectional curvature 1. Let $M^{n}\subset S^{n+1}$ be a smooth, closed (c...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023335 | MATHDB-315976

Atiyah conjecture on configurations

摘录 / EXCERPT: For $v=(v_1,v_2,v_3)\in\mathbb{R}^3\setminus\{0\}$ with $|v|=(v_1^2+v_2^2+v_3^2)^{1/2}$, let $h(v)\in\mathbb{CP}^1$ be the point ... which is well defined since...

conjecture | open | MathDB 页面 | 原文线索

V2-023336 | MATHDB-315981

Falconer's conjecture

摘录 / EXCERPT: Let $d \ge 2$ be an integer. For a nonempty set $S \subset \mathbb{R}^d$ define its distance set ... where $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^d$, and let $\dim_H$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-023337 | MATHDB-315984

Lebesgue's universal covering problem

摘录 / EXCERPT: For a nonempty set $S \subset \mathbb{R}^2$ let ... where $|\cdot|$ denotes the Euclidean norm. Say that a set $C \subset \mathbb{R}^2$ covers S if there is an isometry φ o...

open_problem | open | MathDB 页面 | 原文线索

V2-023338 | MATHDB-315985

Mahler's conjecture

摘录 / EXCERPT: Let $n\ge 1$. A convex body in $\mathbb{R}^n$ is a compact convex set with non-empty interior, and $|\cdot|$ denotes Lebesgue measure on $\mathbb{R}^n$. For a convex body...

conjecture | open | MathDB 页面 | 原文线索

V2-023339 | MATHDB-315990

Hilbert's third problem

摘录 / EXCERPT: Let X be either the round 3-sphere S^{3} (spherical geometry) or hyperbolic 3-space H^{3}, in each case with its standard metric of constant curvature, and let G be the...

open_problem | open | MathDB 页面 | 原文线索

V2-023340 | MATHDB-316170

Einstein problem

摘录 / EXCERPT: In plane discrete geometry, the einstein problem asks about the existence of a single prototile that by itself forms an aperiodic set of prototiles; that is, a shape that can tesse...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023341 | MATHDB-316172

Yau's conjecture

摘录 / EXCERPT: In differential geometry, Yau's conjecture is a mathematical conjecture which states that any closed Riemannian 3-manifold has infinitely many smooth closed immersed minimal surfac...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023342 | MATHDB-316177

Ending lamination theorem

摘录 / EXCERPT: In hyperbolic geometry, the ending lamination theorem, originally conjectured by William Thurston as the eleventh problem out of his twenty-four questions, states that hyperbolic 3...

open_problem | open | MathDB 页面 | 原文线索

V2-023343 | MATHDB-316178

Carpenter's rule problem

摘录 / EXCERPT: The carpenter's rule problem is a discrete geometry problem, which can be stated in the following manner: Can a simple planar polygon be moved continuously to a position where all...

open_problem | open | MathDB 页面 | 原文线索

V2-023344 | MATHDB-316186

Dodecahedral conjecture

摘录 / EXCERPT: The dodecahedral conjecture in geometry is intimately related to sphere packing.

conjecture | open | MathDB 页面 | 原文线索

V2-023345 | MATHDB-316231

Surface subgroup conjecture

摘录 / EXCERPT: Let ${\cal M}$ be a closed hyperbolic 3-manifold. A surface subgroup is the image of the fundamental group of a closed surface of genus at least 2 under a map inducing an injec...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023346 | MATHDB-316238

Berry-Robbins problem

摘录 / EXCERPT: In mathematics, the Berry-Robbins problem asks whether there is a continuous map from configurations of n points in R3 to the flag manifold U(n)/Tn that is compatible with the acti...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-023347 | MATHDB-316250

Baum-Connes vanishing conjecture for the fibrewise L^2-rho form

摘录 / EXCERPT: Let $\pi\colon M\to B$ be a smooth fibre bundle of compact manifolds, let $p\colon \widetilde M\to M$ be a normal covering with torsion-free deck-transformation group Γ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-023348 | MATHDB-316273

The reduced Abel-Jacobi map conjecture for tropical Jacobians

摘录 / EXCERPT: Let Γ be a compact tropical curve of genus g, let $J(\Gamma)=\Omega(\Gamma)^\ast/H_1(\Gamma,\mathbb{Z})$ be its tropical Jacobian, and let $\mathcal{D}^g(\Gamma)$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-023349 | MATHDB-316320

The bounded h-star coefficient conjecture for lattice pyramids

摘录 / EXCERPT: Lattice-pyramid conjecture. There is a function f_i depending only on c and d such that any n-dimensional lattice polytope P with $h^*_i=c$ and degree $\deg(P)=d$ is a la...

conjecture | open | MathDB 页面 | 原文线索

V2-023350 | MATHDB-316334

Conjectured surface-area inequality for arbitrary lattices

摘录 / EXCERPT: Let $\Lambda\subset\mathbb R^n$ be a lattice with determinant $\det\Lambda>0$, and let K be a convex body with $\dim(K\cap\Lambda)=n$. Write $\mathrm{F}(K)$ for the surface area...

conjecture | open | MathDB 页面 | 原文线索

V2-023351 | MATHDB-316337

The moduli-space structure conjecture for stable smooth exploded curves

摘录 / EXCERPT: The moduli-space structure conjecture. The moduli space

conjecture | open | MathDB 页面 | 原文线索

V2-023352 | MATHDB-316338

The exploded Dolbeault-bundle conjecture for stable smooth curves

摘录 / EXCERPT: The exploded Dolbeault-bundle conjecture. The space $\mathfrak E_{g,n}(\mathfrak B,\omega)$ has the structure of a Fréchet orbifold exploded fibration and is a vector bundle over...

conjecture | open | MathDB 页面 | 原文线索

V2-023353 | MATHDB-316356

Abouzaid-Seidel conjecture on the wrapped Floer algebra of a cotangent fibre

摘录 / EXCERPT: Let Z be a closed oriented manifold, let $M=T^*Z$, and let $x\in Z$. Write $L=T_x^*Z$ for the cotangent fibre, $CW^*(L,L)$ for its wrapped Floer cochain algebra, and Ω_x f...

conjecture | open | MathDB 页面 | 原文线索

V2-023354 | MATHDB-316364

The $\frac{3\pi}{2}$-open spanning path conjecture

摘录 / EXCERPT: $\frac{3\pi}{2}$-open spanning path conjecture. Every finite point set in general position in the plane has a $\frac{3\pi}{2}$-open spanning path.

conjecture | open | MathDB 页面 | 原文线索

V2-023355 | MATHDB-316365

The Vaisman conjecture on locally conformal Kähler manifolds

摘录 / EXCERPT: Let M be a locally conformal Kähler manifold, meaning that it is a Hermitian manifold whose metric is locally conformally equivalent to Kähler metrics. Suppose that M has the “...

conjecture | open | MathDB 页面 | 原文线索

V2-023356 | MATHDB-316366

The flat normal bundle conjecture for Ricci-limit spaces

摘录 / EXCERPT: The flat normal bundle conjecture. If X is a limit space as above, then either X has the loops to infinity property or it has a split double cover and it is the flat normal bun...

conjecture | open | MathDB 页面 | 原文线索

V2-023357 | MATHDB-316367

The cut-off covering spectrum theorem for Ricci-limit spaces

摘录 / EXCERPT: The cut-off covering spectrum conjecture. Theorem 7, namely the corresponding cut-off covering spectrum conclusion established earlier in the paper, holds for such limit spaces X...

conjecture | open | MathDB 页面 | 原文线索

V2-023358 | MATHDB-316368

The Margulis-type cut-off covering spectrum conjecture

摘录 / EXCERPT: The Margulis-type cut-off covering spectrum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023359 | MATHDB-316388

The projective-space conjecture for Fano manifolds of Picard number one

摘录 / EXCERPT: Let X be a Fano manifold with Picard number $\rho(X)=1$, and let $f:X\to X$ be an endomorphism. The projective-space conjecture. If $\deg f>1$, then $X\simeq\mathbb P_n$. The sou…

conjecture | open | MathDB 页面 | 原文线索

V2-023360 | MATHDB-316426

The log minimal model conjecture for lc pairs

摘录 / EXCERPT: Minimal model conjecture. Any lc pair $(X/Z,B)$ has a log minimal model or a Mori fibre space.

conjecture | open | MathDB 页面 | 原文线索

V2-023361 | MATHDB-316427

Abundance conjecture for minimal lc pairs

摘录 / EXCERPT: Abundance conjecture. There is an $m\in\mathbb N$ such that $m(K_Y+B)$ is base point free. In particular, there are a contraction $h\colon Y\to S$, called the Iitaka fibration, and…

conjecture | open | MathDB 页面 | 原文线索

V2-023362 | MATHDB-316432

The SHGH conjecture for negative curves on blowups of the plane

摘录 / EXCERPT: SHGH negative-curve formulation. Every reduced irreducible curve $C\subset X$ with $C^2<0$ is an exceptional curve, and for every nef divisor F either

conjecture | open | MathDB 页面 | 原文线索

V2-023363 | MATHDB-316445

The naive mirror construction conjecture for anticanonical complements

摘录 / EXCERPT: Let (X,ω,J) be a compact Kähler manifold, let D be an anticanonical divisor in X, and let Ω be a holomorphic volume form on $X\setminus D$. A complexified moduli…

conjecture | open | MathDB 页面 | 原文线索

V2-023364 | MATHDB-316447

The renormalized mirror limit conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold, or more generally a manifold for which $-K_X$ is nef, with Kähler form ω. For a family of Kähler forms ω_k, write $W_{(\omega_k)}$ for the…

conjecture | open | MathDB 页面 | 原文线索

V2-023365 | MATHDB-316448

The boundary structure conjecture for special Lagrangian moduli spaces

摘录 / EXCERPT: Let X be a compact Kähler manifold, let D be an anticanonical divisor, let U be a neighborhood of D, and let M be the moduli space of pairs (L,∇) consisting of a s…

conjecture | open | MathDB 页面 | 原文线索

V2-023366 | MATHDB-316449

The SYZ mirror identification conjecture for an anticanonical divisor

摘录 / EXCERPT: The SYZ mirror identification conjecture. The subset M_D is the Strominger-Yau-Zaslow mirror of D. This identifies the mirror of the anticanonical divisor with the locus where…

conjecture | open | MathDB 页面 | 原文线索

V2-023367 | MATHDB-316455

Completeness of fiber products of complete transverse exploded morphisms

摘录 / EXCERPT: Let f and g be complete and transverse smooth exploded morphisms … The fiber product $\mathfrak A{}_{\hspace{3pt}f\hspace{-2pt}}\times_{g}\mathfrak B$ exists, and its natural m…

conjecture | open | MathDB 页面 | 原文线索

V2-023368 | MATHDB-316464

Gursky's hyperbolic metric conjecture for compact 3-manifolds

摘录 / EXCERPT: Gursky's conjecture. If $Scal_g<0$ and $\sigma_2(C_g)>0$ on M, then M carries a hyperbolic metric. The conjecture weakens the expected characterization of hyperbolic metrics by…

conjecture | open | MathDB 页面 | 原文线索

V2-023369 | MATHDB-316518

Polygonal generalization of the equilateral-triangle eigenvalue inequalities

摘录 / EXCERPT: Let $P(n)$ denote a polygon with n sides and let $R(n)$ denote a regular polygon with n sides. Write A for area, R for inradius, and $\lambda_1<\lambda_2$ for the first two…

conjecture | open | MathDB 页面 | 原文线索

V2-023370 | MATHDB-316529

Virtual fibering conjecture for taut sutured manifolds

摘录 / EXCERPT: Let (M,γ) be a taut sutured manifold, meaning a sutured manifold satisfying the tautness condition, and suppose that $\chi(M)<0$. A finite-sheeted cover is a covering … with…

conjecture | open | MathDB 页面 | 原文线索

V2-023371 | MATHDB-316536

Local and global harmonic morphism conjecture for irreducible Riemannian symmetric spaces

摘录 / EXCERPT: Let (M^m,g) be an irreducible Riemannian symmetric space of dimension $m\ge 2$. For each point $p\in M$, consider complex-valued harmonic morphisms defined on an open neighbourho…

conjecture | open | MathDB 页面 | 原文线索

V2-023372 | MATHDB-316546

Symmetrization conjecture for the PBW algebra and Kontsevich star-algebra

摘录 / EXCERPT: Let $\bbar{V}$ be a finite-dimensional vector space, let $\bbar{S}(V)\bbar{\bmathbb{C}}[[\hbar]]$ carry the Kontsevich star product associated with a polynomial Poisson bivector...

conjecture | open | MathDB 页面 | 原文线索

V2-023373 | MATHDB-316560

The DDVV conjecture for submanifolds of space forms

摘录 / EXCERPT: Let M^n be an immersed submanifold of the space form $N^{n+m}(c)$, with scalar curvature ρ, normal scalar curvature $\rho^{\perp}$, and mean curvature tensor...

conjecture | solved | MathDB 页面 | 原文线索

V2-023374 | MATHDB-316561

The DDVV matrix inequality for symmetric matrices

摘录 / EXCERPT: Let $A_1,\ldots,A_m$ be symmetric $n\times n$ matrices, with Frobenius norm $\lVert A_r\rVert$ and commutator $[A_r,A_s]=A_rA_s-A_sA_r$. DDVV matrix conjecture. One has ... This is t...

conjecture | solved | MathDB 页面 | 原文线索

V2-023375 | MATHDB-316563

The pinching conjecture for totally geodesic and Veronese submanifolds

摘录 / EXCERPT: Let M be the submanifold under consideration, let σ denote its second fundamental form, and let n be its dimension. Pinching conjecture. There is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-023376 | MATHDB-316568

Conjecture on embedded turnovers in orbifolds containing immersed $(2,4,6)$ turnovers

摘录 / EXCERPT: Let Q be a hyperbolic 3-orbifold. A $(2,4,6)$ turnover is a turnover with cone-point orders 2, 4, and 6; it is immersed, non-embedded if its immersion into Q is not an...

conjecture | open | MathDB 页面 | 原文线索

V2-023377 | MATHDB-316581

The 27-line conjecture for smooth tropical cubic surfaces

摘录 / EXCERPT: A smooth tropical cubic surface is a smooth tropical surface of degree 3. The 27-line conjecture. A general smooth tropical cubic surface contains exactly 27 tropical lines. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-023378 | MATHDB-316582

The 27-line count conjecture for smooth tropical surfaces

摘录 / EXCERPT: Let X be a smooth tropical surface. Let f be the number of two-point families on X, and let i be the number of tropical lines on X that are not part of any two-point fami...

conjecture | open | MathDB 页面 | 原文线索

V2-023379 | MATHDB-316619

Convergence of Ricci iterations on Kähler classes

摘录 / EXCERPT: Let M be a Fano manifold, let $L=-K_M$, and fix a sufficiently large integer r such that M is embedded by the holomorphic sections of L^r. Let $h_{\omega_{l,r}}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-023380 | MATHDB-316651

Donald's toric scalar-curvature existence conjecture

摘录 / EXCERPT: Donald's toric scalar-curvature existence conjecture. If $\mathcal{L}_A(f)>0$ for all non-affine-linear convex functions $f\in\mathcal{C}_1$, then there exists a symplectic potenti...

conjecture | open | MathDB 页面 | 原文线索

V2-023381 | MATHDB-316654

The conjecture on asymptotic expansion of Bergman kernels

摘录 / EXCERPT: Bergman-kernel asymptotics conjecture. The upper and lower local volumes coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-023382 | MATHDB-316676

Nonsingularity conjecture for torus-fixed stable-pairs loci

摘录 / EXCERPT: Nonsingularity conjecture. The loci $P_n(X,\beta)^{\mathbf T_0}$ are nonsingular. This is proved in the local toric surface case in the source, while the general case is reduced to...

conjecture | open | MathDB 页面 | 原文线索

V2-023383 | MATHDB-316702

The effective curves conjecture for holomorphic symplectic fourfolds

摘录 / EXCERPT: Let (F,g) be a polarized irreducible holomorphic symplectic fourfold deformation equivalent to $S^{[2]}$, where S is a K3 surface. Let $\mathrm{N}^1_+(F,g)$ be the positive hal...

conjecture | open | MathDB 页面 | 原文线索

V2-023384 | MATHDB-316703

The nodal classes conjecture for holomorphic symplectic fourfolds

摘录 / EXCERPT: Let (F,g) be a polarized irreducible holomorphic symplectic fourfold deformation equivalent to $S^{[2]}$, and let E^*_{nod} be the classes corresponding to the extrema...

conjecture | open | MathDB 页面 | 原文线索

V2-023385 | MATHDB-316715

Vanishing conjecture for coefficients in equivariant heat trace expansions

摘录 / EXCERPT: Equivariant heat coefficient vanishing conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-023386 | MATHDB-316728

Beauville's symplectic strange-duality conjecture

摘录 / EXCERPT: Beauville's symplectic strange-duality conjecture. The morphism D is an isomorphism. Beauville established equality of the dimensions of the two spaces of sections, but...

conjecture | open | MathDB 页面 | 原文线索

V2-023387 | MATHDB-316737

Smallest-eigenvalue weights for even-dimensional spin Grassmannians

摘录 / EXCERPT: Let $G_{2k,2\ell}=SO(2k+2\ell)/(SO(2k)\times SO(2\ell))$ be the even-dimensional spin Grassmannian, and let the spinor-bundle decomposition be the one described in the preceding de...

conjecture | open | MathDB 页面 | 原文线索

V2-023388 | MATHDB-316738

Smallest Dirac-eigenvalue contribution for even-dimensional spin Grassmannians

摘录 / EXCERPT: Let $G_{2k,2\ell}=SO(2(k+\ell))/(SO(2k)\times SO(2\ell))$ be the even-dimensional spin Grassmannian, and let λ denote a weight occurring in the spinor-bundle decomposition....

conjecture | open | MathDB 页面 | 原文线索

V2-023389 | MATHDB-316740

Extension of Theorem GML to rigid surfaces with modified vertical components

摘录 / EXCERPT: Let S be a rigid surface admitting a morphism $b:S\to\mathbb{P}^1$. Suppose that the divisor at infinity constructed as in Lemma has a graph differing from the graph in that lemm...

conjecture | open | MathDB 页面 | 原文线索

V2-023390 | MATHDB-316757

Completeness conjecture for constant-curvature holomorphic Riemannian manifolds

摘录 / EXCERPT: Let M be a compact complex manifold endowed with a holomorphic Riemannian metric of constant non-vanishing curvature. Completeness conjecture. The metric on M is complete; in p...

conjecture | open | MathDB 页面 | 原文线索

V2-023391 | MATHDB-316768

Kollár's accumulation conjecture for log canonical thresholds

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero. For each integer $n\geq 0$, let $\mathcal{T}_n(k)$ be the set of log canonical thresholds of pairs (X,Y) where...

conjecture | open | MathDB 页面 | 原文线索

V2-023392 | MATHDB-316769

Tian's uniform estimate conjecture for degenerate complex Monge-Ampère equations

摘录 / EXCERPT: Let (X,ω_X) be a polarized compact connected Kähler manifold of complex dimension n, let (Y,ω_Y) be a compact irreducible Kähler space of complex dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-023393 | MATHDB-316800

Compatibility conjecture for algebraic and analytic index maps under strict quantization

摘录 / EXCERPT: Let M be a Poisson manifold with a strict C^*-algebraic deformation quantization, and let ${\cal G}$ be a Lie groupoid with tangent-groupoid deformation quantization. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-023394 | MATHDB-316837

Infinitesimal rigidity conjecture for weakly convex decomposable polyhedra

摘录 / EXCERPT: A polyhedron $P\subset {\mathbb R}^3$ is weakly convex if its vertices are in convex position in ${\mathbb R}^3$, and it is decomposable if it can be triangulated without adding ne...

conjecture | open | MathDB 页面 | 原文线索

V2-023395 | MATHDB-316855

The generalized-quadratic criterion for sets good for double recurrence

摘录 / EXCERPT: Generalized-quadratic criterion. If $(p(r_n))_{n\in\mathbb{N}}$ is uniformly distributed in $\mathbb{T}$ for every irrational generalized quadratic p, and if

conjecture | open | MathDB 页面 | 原文线索

V2-023396 | MATHDB-316861

Unimodality conjecture for stable weight multiplicity polynomials of \(\mathfrak{spo}(2n,M)\)

摘录 / EXCERPT: Let $P^+(n,m)$ be the set of dominant weights, and let $K_{\lambda,\mu}^{\mathrm{stab}}(q)$ denote the stable q-analog of the weight multiplicity for the orthosymplectic Lie supe...

conjecture | open | MathDB 页面 | 原文线索

V2-023397 | MATHDB-316864

Adolphson-Sperber congruence criterion for generic ordinarity

摘录 / EXCERPT: Let $\triangle$ be a polytope, let $I(\triangle)$ and $I_{\infty}(\triangle)$ be the exponents defined by ... ... and let $\operatorname{disc}_{\triangle}$, $GNP(\triangle,p)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-023398 | MATHDB-316882

Wang's Penrose inequality conjecture for asymptotically hyperbolic 3-manifolds

摘录 / EXCERPT: Let (M,g) be an asymptotically hyperbolic 3-manifold with scalar curvature $R\geq -6$ and mass M. Let Σ_0 be an outermost sphere with mean curvature $H(\Sigma_0)=2$. W...

conjecture | open | MathDB 页面 | 原文线索

V2-023399 | MATHDB-316891

Gromov's conjecture on rational curves and mean dimension

摘录 / EXCERPT: Let $X\subset \mathbb{C} P^N$ be a projective manifold, and let $\mathcal{M}(X)$ be the space of Brody curves in X. Gromov's conjecture. ... The conjecture relates positivity of th...

conjecture | open | MathDB 页面 | 原文线索

V2-023400 | MATHDB-316912

The characterization of amenable Hilbert geometries as G_n-polygons

摘录 / EXCERPT: Amenability characterization. A Hilbert geometry is amenable if and only if it is a G_n-polygon.

open_problem | open | MathDB 页面 | 原文线索

V2-023401 | MATHDB-316934

The $P(\mathfrak g,e)$ conjecture for split semisimple Lie algebras

摘录 / EXCERPT: Let $\mathfrak g$ be a split semisimple real Lie algebra and let $e\in\mathfrak g$ be nilpotent. Denote by $P(\mathfrak g,e)$ the statement that, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-023402 | MATHDB-316983

The simplicial half-net conjecture for point sets

摘录 / EXCERPT: Let S be a finite point set in $cmathbb{R}^d$. A simplicial half-net is a family of simplices determined by points of S such that every halfspace containing at least half the p...

conjecture | open | MathDB 页面 | 原文线索

V2-023403 | MATHDB-316996

The symplectic cone polytope conjecture

摘录 / EXCERPT: Symplectic cone polytope conjecture. There exists such a polytope C satisfying:

conjecture | open | MathDB 页面 | 原文线索

V2-023404 | MATHDB-317002

Second conformal eigenvalue conjecture for round spheres

摘录 / EXCERPT: Let $n\geq 2$, let $({\mathbb S}^n,[g_0])$ be the n-sphere with its conformal class, and define its k-th conformal eigenvalue by ... where ω_n is the volume of the unit ro...

conjecture | open | MathDB 页面 | 原文线索

V2-023405 | MATHDB-317005

Gromov-Fraser conjecture on fundamental groups of manifolds with positive isotropic curvature

摘录 / EXCERPT: Gromov-Fraser conjecture. $\pi_1(M)$ is virtually free, i.e., it is a finite extension of a free group.

conjecture | open | MathDB 页面 | 原文线索

V2-023406 | MATHDB-317017

The symplectic-leaf identification conjecture for Poisson Lie groups

摘录 / EXCERPT: Let K be the compact Lie group under consideration, with Poisson Lie group structure π_K, and let W be its Weyl group. Fix $w\in W$, choose a representative $\mathbf w$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-023407 | MATHDB-317035

Unbounded amplitude, non-graph behavior, and curvature of iterated bicycle tracks

摘录 / EXCERPT: Iterated bicycle-track conjectures. Unless γ is a segment, the amplitude of ${\cal T}$ is unbounded; equivalently, ${\cal T}$ is not contained in any horizontal strip. Unles...

conjecture | open | MathDB 页面 | 原文线索

V2-023408 | MATHDB-317038

Gordon's conjecture on stabilization of connected-sum Heegaard splittings

摘录 / EXCERPT: Gordon's conjecture. The connected-sum Heegaard splitting $\mathcal{V}\cup_S\mathcal{W}$ is stabilized if and only if at least one of the summand splittings

conjecture | open | MathDB 页面 | 原文线索

V2-023409 | MATHDB-317075

Completeness from complete vector fields for coframings

摘录 / EXCERPT: Let M be an n-dimensional manifold with a coframing, and let its canonical metric be the Riemannian metric induced by the coframing. Consider the Lie algebra of vector fields g...

conjecture | open | MathDB 页面 | 原文线索

V2-023410 | MATHDB-317076

Projective Blaschke conjecture

摘录 / EXCERPT: Let M be a manifold with a normal projective connection, and suppose that its geodesics are embedded curves. Projective Blaschke conjecture. If all geodesics are closed embedded...

conjecture | open | MathDB 页面 | 原文线索

V2-023411 | MATHDB-317077

Tameness conjecture for \(\mathcal{O}(n)\)-geometries

摘录 / EXCERPT: Let an $\mathcal{O}(n)$-geometry be a Cartan geometry modelled on the corresponding model, and call it tame according to the source's definition. Tameness conjecture. An...

conjecture | open | MathDB 页面 | 原文线索

V2-023412 | MATHDB-317078

Tameness conjecture for the rolling-surface \(G_2/P_2\)-geometry

摘录 / EXCERPT: Let S and S' be Riemannian surfaces whose Gauss curvatures differ at every pair of points, and let M be the space of linear isometries between tangent planes of S and S'....

conjecture | open | MathDB 页面 | 原文线索

V2-023413 | MATHDB-317079

Positive-Ricci-curvature conjecture for parabolic geometries

摘录 / EXCERPT: Let G be the split real form of a semisimple Lie group with parabolic subgroup P, and let $P\to E\to M$ be a G/P-geometry. Suppose that the Lie algebra $\mathfrak{g}$ has its...

conjecture | open | MathDB 页面 | 原文线索

V2-023414 | MATHDB-317082

Representative conjecture for generalized stationary space-times

摘录 / EXCERPT: Let $(\mathcal M,g)$ be a generalized stationary space-time, and let $(g_\varepsilon)_\varepsilon$ be a symmetric representative of its metric. The representative conjecture. On re...

conjecture | open | MathDB 页面 | 原文线索

V2-023415 | MATHDB-317083

Representative-independence conjecture for the causal hypersurface condition

摘录 / EXCERPT: Let $(\mathcal M,g)$ be a generalized space-time with a symmetric representative $(g_\varepsilon)_\varepsilon$, and let Σ be a past-compact space-like hypersurface. For suff...

conjecture | open | MathDB 页面 | 原文线索

V2-023416 | MATHDB-317087

Convolution exhaustion conjecture for mirabolic character sheaves

摘录 / EXCERPT: Let D be a vector space, and let V and W be vector spaces with $\dim(V)+\dim(W)=\dim(D)$. Let $\CG\in\calC(\operatorname{GL}(V))$ and $\CG'\in\calC(\operatorname{GL}(W))$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-023417 | MATHDB-317112

The Gromov-Hausdorff limit conjecture for maximally unipotent Calabi-Yau degenerations

摘录 / EXCERPT: The Gromov-Hausdorff limit conjecture. A convergent subsequence of $(\mathcal{X}_{t_i},g_i)$ converges to a metric space (X_{∞},d_{∞}), where X_{∞} is homeomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-023418 | MATHDB-317113

The toric degeneration conjecture for large complex structure limits

摘录 / EXCERPT: Toric degeneration conjecture. The degeneration f is birationally equivalent to a toric degeneration

conjecture | open | MathDB 页面 | 原文线索

V2-023419 | MATHDB-317119

Lower bound conjecture for connected non-simply connected triangulated manifolds

摘录 / EXCERPT: Let $d\geq 3$, and let X be a connected, non-simply connected triangulated d-manifold. Lower bound conjecture for non-simply connected manifolds. Its face numbers satisfy ... Equ...

conjecture | open | MathDB 页面 | 原文线索

V2-023420 | MATHDB-317126

The homogeneity and pole-order conjecture for the pairing on the Gauss-Manin system

摘录 / EXCERPT: Let f be the linear free divisor considered in the paper, let $\mathbf{G}$ and $\overline{\mathbf{G}}$ be the corresponding Gauss-Manin systems, and let G and $\overline{G}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-023421 | MATHDB-317127

The quaternionic Calabi conjecture for compact HKT-manifolds

摘录 / EXCERPT: Quaternionic Calabi conjecture. If the necessary condition

conjecture | open | MathDB 页面 | 原文线索

V2-023422 | MATHDB-317130

Coincidence of the cotangent-bundle and diffeomorphism-group symplectic structures

摘录 / EXCERPT: Let $M=S^2$ be a two-dimensional sphere, let $\mathcal{D}_\mu$ be the group of area-preserving diffeomorphisms, and identify $T^*\mathcal{D}_\mu$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-023423 | MATHDB-317142

Plumbing open books realizes contact connected sum

摘录 / EXCERPT: Plumbing conjecture. There are deformations $\tilde\psi_i$ of ψ_i, isotopic to ψ_i as symplectomorphisms, such that the open book

conjecture | open | MathDB 页面 | 原文线索

V2-023424 | MATHDB-317144

The normal scalar curvature conjecture for submanifolds

摘录 / EXCERPT: Let M^n be an n-dimensional manifold isometrically immersed in the space form $N^{n+m}(c)$ of constant sectional curvature c. Let h be the second fundamental form, let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-023425 | MATHDB-317146

The pinching gap conjecture for minimal submanifolds in spheres

摘录 / EXCERPT: Let M be an n-dimensional minimal submanifold in S^{n+m}. Let λ_2 denote the quantity used in the paper, and suppose that $||\sigma||^2+\lambda_2$ is constant. Pinchi...

conjecture | open | MathDB 页面 | 原文线索

V2-023426 | MATHDB-317161

Conjecture on the number of segments of each depth

摘录 / EXCERPT: Let $P\subset\mathbb{R}^3$ be a set of n points in convex position. The depth of a segment is the minimum number of points that must be removed from P so that the segment is no...

conjecture | open | MathDB 页面 | 原文线索

V2-023427 | MATHDB-317162

Conjectural abundance of pairs with large circle depth

摘录 / EXCERPT: Let P be a set of n points in the plane in general position, meaning that no three are collinear and no four are cocircular. A pair of points has the required circle depth when...

conjecture | open | MathDB 页面 | 原文线索

V2-023428 | MATHDB-317188

The special Lagrangian torus fibration conjecture for the projective plane

摘录 / EXCERPT: Let $D\subset\mathbb{CP}^2$ be a smooth elliptic curve, and equip $\mathbb{CP}^2\setminus D$ with the holomorphic volume form having poles along D. Special Lagrangian torus fibra...

conjecture | open | MathDB 页面 | 原文线索

V2-023429 | MATHDB-317189

The special Lagrangian fibration conjecture for rational elliptic surfaces

摘录 / EXCERPT: Let X be a rational elliptic surface obtained by blowing up $\mathbb{CP}^2$ at the nine base points of a pencil of cubics, and let $\hat D$ be a smooth elliptic fiber. Rational e...

conjecture | open | MathDB 页面 | 原文线索

V2-023430 | MATHDB-317190

The special Lagrangian fibration conjecture for Calabi-Yau double covers

摘录 / EXCERPT: Let (X,ω,J) be a compact Kähler manifold, let H represent twice the anticanonical class, and let Y be the Calabi-Yau double cover of X branched along H. Let $\bar B$...

conjecture | open | MathDB 页面 | 原文线索

V2-023431 | MATHDB-317191

The rational elliptic surface fibration deformation conjecture

摘录 / EXCERPT: Let X be the rational elliptic surface and let $H=D_+\cup D_-$ be the union of two smooth fibers of its elliptic fibration. Rational elliptic surface deformation conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-023432 | MATHDB-317224

Asymptotic volume conjecture for cusped orientable hyperbolic 3-manifolds

摘录 / EXCERPT: Asymptotic volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023433 | MATHDB-317225

Non-orientable minimal limit volume conjecture

摘录 / EXCERPT: Non-orientable minimal limit volume conjecture. The minimal limit volume of non-orientable hyperbolic 3-manifolds is V_8.

conjecture | open | MathDB 页面 | 原文线索

V2-023434 | MATHDB-317244

Moussu-Pelletier-Sullivan Godbillon-Vey growth conjecture

摘录 / EXCERPT: Let ${\mathcal F}$ be a codimension-one, C^2-foliation of a closed manifold M, and let ... be its Godbillon-Vey class. Moussu-Pelletier-Sullivan conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-023435 | MATHDB-317248

The rational integral-distance embedding conjecture

摘录 / EXCERPT: Let $\mathcal{P}\subset\mathbb{Q}^m$ be a finite set of points such that the distances between any two points of $\mathcal{P}$ are integers. Embedding conjecture. There is a Euclid...

conjecture | open | MathDB 页面 | 原文线索

V2-023436 | MATHDB-317249

The divisibility and scaling conjecture for integral point sets

摘录 / EXCERPT: Let $\mathcal{P}\subset\mathbb{Z}^m$ be a finite set of points such that the distances between any two points of $\mathcal{P}$ are integers divisible by an integer k. Scaling con...

conjecture | open | MathDB 页面 | 原文线索

V2-023437 | MATHDB-317250

The maximality conjecture for decompose crabs

摘录 / EXCERPT: Let h be an integer, and let $\mathcal{P}=\operatorname{decompose}(h)$ be the plane integral point set constructed as a crab of the order specified in the source. Decompose-crab...

conjecture | open | MathDB 页面 | 原文线索

V2-023438 | MATHDB-317251

The maximality conjecture for semi-crabs

摘录 / EXCERPT: Let gh and g be integers for which the preceding semi-crab construction defines the plane integral point set $\mathcal{P}=\operatorname{decompose}(gh,g)$. Semi-crab maximality...

conjecture | open | MathDB 页面 | 原文线索

V2-023439 | MATHDB-317257

Exact value of the normalized integral-point-set number when -1 is a square

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $\square_q$ denote the set of squares in $\mathbb{F}_q$, and let $\overline{\mathcal{I}}(\mathbb{F}_q,2)$ denote the normalized maximum si...

conjecture | open | MathDB 页面 | 原文线索

V2-023440 | MATHDB-317261

Conjecture on the tightness of the two lower bounds for integral point sets over $\mathbb{Z}_n^2$

摘录 / EXCERPT: Let $\mathcal{I}(n,2)$ denote the maximum number of points in an integral point set over $\mathbb{Z}_n^2$. The paper gives two lower bounds for this quantity, one in Lemma 1 and on...

conjecture | open | MathDB 页面 | 原文线索

V2-023441 | MATHDB-317263

The collinear-subset conjecture for minimum-diameter plane integral point sets

摘录 / EXCERPT: Collinear-subset conjecture. For $n\ge 9$, a plane integral point set with minimum diameter contains a subset of $n-1$ collinear points.

conjecture | open | MathDB 页面 | 原文线索

V2-023442 | MATHDB-317264

The sharpness conjecture for upper bounds from circular plane integral point sets

摘录 / EXCERPT: Sharpness conjecture. The upper bounds from Table $$ are sharp.

conjecture | open | MathDB 页面 | 原文线索

V2-023443 | MATHDB-317271

Aubin's conjecture for the Yamabe invariant

摘录 / EXCERPT: Let (M,g) be a smooth compact Riemannian manifold of dimension $n\geq 3$. For nonzero $\psi\in H_1(M)$, define the Yamabe functional ... and set ... Here R_g is the scalar curvatur...

conjecture | open | MathDB 页面 | 原文线索

V2-023444 | MATHDB-317283

The equality-case conjecture for maximal-volume clean lattice tetrahedra

摘录 / EXCERPT: Let $\mathscr{S}_k^3$ denote the family of clean lattice tetrahedra in $\mathbb{R}^3$ with k interior lattice points, let $\operatorname{Vol}(T)$ denote the normalized volume, an...

conjecture | open | MathDB 页面 | 原文线索

V2-023445 | MATHDB-317312

Kusner's conjecture on balanced configurations and minimal surfaces

摘录 / EXCERPT: Let $\{v_1,\ldots,v_n\}$ be a balanced configuration of n vectors, meaning that their vector sum is zero, and suppose that for all i and j, $v_i\neq r\cdot v_j$ for every pos...

conjecture | open | MathDB 页面 | 原文线索

V2-023446 | MATHDB-317316

Kobayashi's logarithmic conjecture for non-log-general-type pairs

摘录 / EXCERPT: Let (X,D) be a pair consisting of a projective manifold X and a normal crossing Weil divisor $D\subset X$, whose positive part is reduced. The pair is of log-general type when...

conjecture | open | MathDB 页面 | 原文线索

V2-023447 | MATHDB-317352

The symplectic product conjecture for closed 3-manifolds

摘录 / EXCERPT: Let N be a closed 3--manifold. The symplectic product conjecture. If $S^1 \times N$ is symplectic, then there exists a $\phi \in H^1(N;\mathbb{Z})$ such that (N,ϕ) fibers...

open_problem | open | MathDB 页面 | 原文线索

V2-023448 | MATHDB-317360

Bishop volume comparison conjecture for locally tessellating planar graphs

摘录 / EXCERPT: Let $p,q\ge 3$ with $1/p+1/q\le 1/2$ be given. Let ${\cal G}=({\cal V},{\cal E},{\cal F})$ be a locally tessellating planar graph without cut locus, meaning that it has no cut locu...

conjecture | open | MathDB 页面 | 原文线索

V2-023449 | MATHDB-317371

Representability conjecture for triangulordinary filtrations

摘录 / EXCERPT: Let $\mathscr{X}/E$ be a reduced, separated rigid analytic space, and let $\mathscr{D}$ be a locally free sheaf of (φ,Γ_K)-modules over...

conjecture | open | MathDB 页面 | 原文线索

V2-023450 | MATHDB-317383

Shokurov's effective log adjunction conjecture

摘录 / EXCERPT: Let (X,Δ) be a log klt pair and let $f\colon X\to Y$ be a surjective morphism such that the Kodaira dimension of $K_X+\Delta$ restricted to the general fibre is zero. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-023451 | MATHDB-317393

Lück's conjecture on L^2-invariants of aspherical manifolds with zero simplicial volume

摘录 / EXCERPT: Let M be a closed aspherical orientable manifold with vanishing simplicial volume. Lück's conjecture. The L^2-Betti numbers $b^{(2)}_n({\tilde M})$ vanish, the Novikov-Shubin i...

conjecture | open | MathDB 页面 | 原文线索

V2-023452 | MATHDB-317394

Morrison-Kawamata finiteness conjecture for minimal models

摘录 / EXCERPT: Let F be a simply-connected Calabi-Yau manifold. A minimal model of F is a minimal model in its birational class. Morrison-Kawamata finiteness conjecture. There are finitely ma...

conjecture | open | MathDB 页面 | 原文线索

V2-023453 | MATHDB-317399

Dumitrescu-Zeghib completeness conjecture for constant-curvature holomorphic Riemannian manifolds

摘录 / EXCERPT: Let M be a compact complex manifold endowed with a holomorphic Riemannian metric of constant non-vanishing curvature. The dimension of M is the complex dimension. Dumitrescu-Ze...

conjecture | open | MathDB 页面 | 原文线索

V2-023454 | MATHDB-317430

The rank-period conjecture for complex multiplication elliptic curves

摘录 / EXCERPT: Let E_{CM} be an elliptic curve with complex multiplication, and let $K={\mathbb Q}(j(E_{CM}))$. The group of K-rational points $E_{CM}(K)$ has infinite part of rank $R\geq 0$....

conjecture | open | MathDB 页面 | 原文线索

V2-023455 | MATHDB-317459

LeBrun-Mason type correspondence conjecture for space-like Zoll Einstein-Weyl structures

摘录 / EXCERPT: LeBrun-Mason type correspondence conjecture. There is a natural one-to-one correspondence, at least in a neighborhood of the standard objects, between equivalence classes of space-...

conjecture | open | MathDB 页面 | 原文线索

V2-023456 | MATHDB-317481

Reid's conjecture on conjugate generators of knot groups

摘录 / EXCERPT: Let K be a knot, and let $\pi_1(S^3\smallsetminus K)$ denote its knot group. Suppose this group is generated by two conjugate elements. Reid's conjecture. The elements are periph...

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V2-023457 | MATHDB-317482

Simon's conjecture on surjections between knot groups

摘录 / EXCERPT: A knot group is the fundamental group of the complement of a knot in S^3. Simon's conjecture. A knot group can surject onto only finitely many other knot groups. This is an attri...

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V2-023458 | MATHDB-317486

The security conjecture for Riemannian tori

摘录 / EXCERPT: Let (M,g) be a Riemannian torus, meaning a manifold diffeomorphic to a torus equipped with a Riemannian metric. A finite set $B\subset M\setminus\{x,y\}$ is a blocking set for po...

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V2-023459 | MATHDB-317491

Conjecture on Ricci-flat Kähler metrics on crepant resolutions of Kähler cones

摘录 / EXCERPT: Let $X=C(S)$ be an isolated singularity whose cone admits a Ricci-flat Kähler cone metric, and let ... be a crepant resolution. Let (S,g) be the Sasaki-Einstein manifold associated...

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V2-023460 | MATHDB-317537

The nef anticanonical Albanese submersion conjecture

摘录 / EXCERPT: Albanese submersion conjecture. The Albanese map of X is a surjective submersion.

conjecture | open | MathDB 页面 | 原文线索

V2-023461 | MATHDB-317538

The generically nef tangent bundle conjecture for nef anticanonical manifolds

摘录 / EXCERPT: Generically nef tangent bundle conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023462 | MATHDB-317563

The finite blocking number conjecture for compact Riemannian manifolds

摘录 / EXCERPT: Finite blocking number conjecture. A compact Riemannian manifold has finite blocking number if and only if it is flat.

conjecture | open | MathDB 页面 | 原文线索

V2-023463 | MATHDB-317569

Kaledin-Lehn-Sorger conjecture on smoothness of simple quotients

摘录 / EXCERPT: Let G be a semisimple complex linear algebraic group, let V be a finite-dimensional representation of G, and let $V\oplus V^*$ carry its natural symplectic form with moment m...

conjecture | open | MathDB 页面 | 原文线索

V2-023464 | MATHDB-317576

Converse of (*) for small and rigid geometric transitions

摘录 / EXCERPT: A geometric transition is abbreviated g.t.; a geometric transition is small and rigid when it has those properties. A factorization of a geometric transition T is an expression ……

conjecture | open | MathDB 页面 | 原文线索

V2-023465 | MATHDB-317578

Direct connectedness conjecture for the Calabi-Yau web

摘录 / EXCERPT: Direct connectedness conjecture. If $\mathcal{M}_1$ and $\mathcal{M}_2$ are nodes of the Calabi-Yau web with distinct Picard numbers $\rho_1>\rho_2$, then there exists a unique arr...

conjecture | open | MathDB 页面 | 原文线索

V2-023466 | MATHDB-317581

Laugesen-Morpurgo conjecture for the Neumann heat kernel of the unit ball

摘录 / EXCERPT: Let $p_{\mathbb{U}}(t,x,y)$ denote the heat kernel for the Laplacian with Neumann boundary conditions on the unit ball ... in $\mathbb{R}^{n}$, where $n\in\mathbb{N}^{\ast}$. The fun...

conjecture | open | MathDB 页面 | 原文线索

V2-023467 | MATHDB-317666

Conjecture on the number of projection cones defining maximal cells in equidistant tree space

摘录 / EXCERPT: Projection-cone bound conjecture. Every maximal cell in $\mathcal{Q}_n$ is obtained as the intersection of at most $(n-1)!$ projection cones.

conjecture | open | MathDB 页面 | 原文线索

V2-023468 | MATHDB-317678

Fukaya's Witten-complex conjecture for Lagrangian disc instantons

摘录 / EXCERPT: Let M be a manifold, let f_i be functions on M, and let $L_i={\rm graph\ }df_i\subset T^*M$ for $i=1,2,3$. For intersection points $p_{ij}\in L_i\cap L_j$, let ψ_{ij} be...

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V2-023469 | MATHDB-317711

Batyrev's projection conjecture for lattice polytopes without interior lattice points

摘录 / EXCERPT: Let P be an n-dimensional lattice polytope without interior lattice points. A lattice polytope is exceptional if it is among the finite exceptional cases excluded from the asse...

conjecture | open | MathDB 页面 | 原文线索

V2-023470 | MATHDB-317713

The natural measure conjecture for strongly regular Poisson manifolds

摘录 / EXCERPT: Let P be a strongly regular Poisson manifold whose simply connected symplectic leaves are the fibres of a smooth fibration $P\to M$. Assume that the lattice of differentials of p...

conjecture | open | MathDB 页面 | 原文线索

V2-023471 | MATHDB-317719

Conjecture on gaps between generalized Hadwiger numbers

摘录 / EXCERPT: Let $F,B\subset\mathbb{R}^d$ be convex domains, and let $N(F,B)$ denote the set of generalized Hadwiger numbers associated with F and B. For consecutive elements of $N(F,B)$, c...

conjecture | open | MathDB 页面 | 原文线索

V2-023472 | MATHDB-317720

Conjecture on second-order asymptotics of generalized Hadwiger numbers

摘录 / EXCERPT: Let F,B be the domains under consideration, let $N_{c(B)/r}(F,B)$ be the generalized Hadwiger number at scale $c(B)/r$, and write $\{r\}=r-[r]$ for the fractional part of r. Se...

conjecture | open | MathDB 页面 | 原文线索

V2-023473 | MATHDB-317725

Finite generation of the canonical algebra near rational surface sections

摘录 / EXCERPT: Finite-generation conjecture. The sheaf $\bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ is a finitely generated sheaf of $\mathcal{O}_X$-algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-023474 | MATHDB-317746

Geodesic rigidity conjecture for projectively equivalent metrics

摘录 / EXCERPT: Let M^n be a closed connected manifold, and let g and $\bar g$ be Riemannian metrics on M^n that are projectively equivalent and nonproportional. Geodesic rigidity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-023475 | MATHDB-317765

Strong deformation retraction for moduli spaces of metric graphs of arbitrary genus

摘录 / EXCERPT: Let $MG_{g,n}^{\mathfrak{v}}$ be the moduli space of metric graphs of genus g with n marked points on vertices, and let $TM_{g,n}$ be the corresponding tropical moduli space. S...

conjecture | open | MathDB 页面 | 原文线索

V2-023476 | MATHDB-317774

Equidistribution of dilated polynomial curves in nilmanifolds

摘录 / EXCERPT: Equidistribution conjecture. The canonical projections onto X of the dilations of p equidistribute if and only if their projections onto the horizontal torus of X equidistrib...

conjecture | open | MathDB 页面 | 原文线索

V2-023477 | MATHDB-317778

Local mirror symmetry prediction for the Gromov-Witten prepotential

摘录 / EXCERPT: Let $\mathbb{P}_{\Delta^{\circ}}$ be the toric surface in the preceding construction, let $W_{j}^{(1)}$ be the logarithmic period coordinates, let $Q_{j}=\exp(2\pi iW_{j}^{(1)})$,...

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V2-023478 | MATHDB-317780

Reeh-Schlieder conjecture for the free Dirac field

摘录 / EXCERPT: Let M be a globally hyperbolic spin spacetime and let $O\subset\mathcal{M}$ be a bounded cc-region with non-empty causal complement. Let ${\mathbf{F}}^0$ and ${\mathbf{R}}^0$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-023479 | MATHDB-317793

The measurable-to-smooth invariant-metric conjecture

摘录 / EXCERPT: Let G be a semisimple Lie group with property (T), or let Γ be a lattice in such a group, acting smoothly on a compact manifold M as in the source. The measurable-to-s...

conjecture | open | MathDB 页面 | 原文线索

V2-023480 | MATHDB-317795

The uniformly partially hyperbolic quasi-affine conjecture

摘录 / EXCERPT: Let G be a semisimple Lie group all of whose simple factors have property (T), and let $\Gamma<G$ be a lattice. Assume that G or Γ acts smoothly on a compact manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-023481 | MATHDB-317801

Accessibility-class regularity conjecture for partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Accessibility-class regularity conjecture. Every accessibility class for f is an injectively immersed, C^r submanifold of M.

conjecture | open | MathDB 页面 | 原文线索

V2-023482 | MATHDB-317805

The two negative eigenvalues conjecture for Ricci operators of non-unimodular solvable Lie algebras

摘录 / EXCERPT: Let $\mathfrak{s}$ be a non-unimodular solvable Lie algebra of arbitrary dimension, and let Q be an inner product on $\mathfrak{s}$. The two negative eigenvalues conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-023483 | MATHDB-317821

The Harris-Morrison slope conjecture for the moduli space of curves

摘录 / EXCERPT: Let $g\beq 4$, let $\overline{\mathcal{M}}_g$ be the moduli space of stable curves of genus g, and let $s(\overline{\mathcal{M}}_g)$ denote its slope, defined as the infimum of t...

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V2-023484 | MATHDB-317827

The maximal entropy conjecture for Hilbert geometries

摘录 / EXCERPT: Let K be an n-dimensional convex set equipped with its Hilbert geometry. Fix a volume and, for a metric ball $B(o,r)$, define the upper volume entropy by ... The hyperbolic space...

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V2-023485 | MATHDB-317828

Hsiang-Kleiner diffeomorphism conjecture for positively curved 4-manifolds

摘录 / EXCERPT: Let M be an orientable closed positively curved 4-manifold with an effective isometric S^1-action. Hsiang-Kleiner's conjecture. M is diffeomorphic to S^4 or ${\bf CP}^2$....

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V2-023486 | MATHDB-317829

Hsiang-Kleiner conjecture for nonnegatively curved 4-manifolds

摘录 / EXCERPT: Let M be an orientable closed simply connected nonnegatively curved 4-manifold with an effective isometric S^1-action. Hsiang-Kleiner's conjecture. M is diffeomorphic to ei...

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V2-023487 | MATHDB-317832

Surjective sphere-preserving maps of the extended Euclidean space

摘录 / EXCERPT: Let n and r be the dimensions under consideration, and let $f:\hat{\mathbb{R}}^n\to\hat{\mathbb{R}}^n$ be a map that sends every r-dimensional sphere into an r-dimensional...

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V2-023488 | MATHDB-317833

Surjective hyperplane-preserving maps of hyperbolic space

摘录 / EXCERPT: Let n and r be the dimensions under consideration, and let $f:\mathbb{H}^n\to\mathbb{H}^n$ be a map that sends every r-dimensional hyperplane into an r-dimensional hyperpla...

conjecture | open | MathDB 页面 | 原文线索

V2-023489 | MATHDB-317886

Green-Griffiths conjecture for holomorphic maps from complex Euclidean spaces

摘录 / EXCERPT: Let $f:{\mathbb C}^{p}\rightarrow X$ be a holomorphic map into a compact complex manifold X of general type, meaning that its canonical bundle K_X is big. Green-Griffiths conje...

conjecture | open | MathDB 页面 | 原文线索

V2-023490 | MATHDB-317887

The fundamental gap conjecture for triangular domains

摘录 / EXCERPT: Triangular fundamental-gap conjecture.

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V2-023491 | MATHDB-317888

The iterated-circuit characterization of dual defectiveness

摘录 / EXCERPT: Let $A\thinspace\subset\mathbb Z^k$ be a finite set whose convex hull is k-dimensional. An iterated circuit is a finite configuration obtained recursively by the iterated-circuit...

conjecture | open | MathDB 页面 | 原文线索

V2-023492 | MATHDB-317889

The B-nondegeneracy conjecture for collections of supports

摘录 / EXCERPT: Let $A_0,\ldots,A_l$ be finite collections of exponent vectors, and let $S_{A_0,\ldots,A_l}$ be the bifurcation set: the minimal closed subset of the coefficient space outside whic...

conjecture | open | MathDB 页面 | 原文线索

V2-023493 | MATHDB-317907

The virtual circle-action conjecture for lattices in SO(1,n)

摘录 / EXCERPT: Let Γ be any lattice in $\mathop{\mathrm{SO}}\nolimits(1,n)$. Virtual circle-action conjecture. Some finite-index subgroup of Γ has a C^∞ action on S^1 with...

conjecture | open | MathDB 页面 | 原文线索

V2-023494 | MATHDB-317912

The symplectic product fibering conjecture for closed oriented 3-manifolds

摘录 / EXCERPT: Let M be a closed oriented 3-manifold, and let $S^1\times M$ denote its product with the circle. Suppose that $S^1\times M$ admits a symplectic structure. The symplectic produc...

conjecture | open | MathDB 页面 | 原文线索

V2-023495 | MATHDB-317913

Baldridge's fibering conjecture for free circle actions

摘录 / EXCERPT: Let X be a closed symplectic 4-manifold admitting a free S^1-action, and let M be the orbit space of this action. Baldridge's fibering conjecture. Then the quotient manifol...

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V2-023496 | MATHDB-317947

The K-stability criterion for Kähler-Einstein Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold whose anticanonical polarization $-K_X$ has discrete automorphism group. For sufficiently large powers of $-K_X$, consider K-stability in the sense of th...

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V2-023497 | MATHDB-317953

Uniform Hessian bound for the fermionic free energy

摘录 / EXCERPT: Uniform Hessian bound. There is a constant C, independent of k, such that

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V2-023498 | MATHDB-318007

The convex-polygon realization conjecture for maximum rectilinear crossing numbers

摘录 / EXCERPT: Convex-polygon realization conjecture. The maximum rectilinear crossing number of G can be realized in a convex-polygon drawing.

conjecture | open | MathDB 页面 | 原文线索

V2-023499 | MATHDB-318022

Extension of holomorphic principal bundles characterized by vanishing Atiyah class

摘录 / EXCERPT: Let M be a domain in a Stein manifold, let $\hat{M}$ be its envelope of holomorphy, and let $E \to M$ be a holomorphic principal bundle. The extension conjecture. The bundle E...

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V2-023500 | MATHDB-318023

Extension conjecture for holomorphic first order structures with trivial kernel

摘录 / EXCERPT: Let M be a domain in a Stein manifold, let $\hat{M}$ be its envelope of holomorphy, and let a holomorphic first order structure on M have an underlying holomorphic principal bu...

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V2-023501 | MATHDB-318024

Extension conjecture for holomorphic Engel plane fields

摘录 / EXCERPT: Let M be a complex manifold and let $S \subset M$ be a subset of complex codimension at least two. A holomorphic Engel 2-plane field is a holomorphic 2-plane field satisfying...

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V2-023502 | MATHDB-318025

Hartogs extension conjecture for Kähler holomorphic fibrations

摘录 / EXCERPT: Let $E \to M \to B$ be a holomorphic fibration whose total space is Kähler. A complex manifold is a Hartogs extension target for local biholomorphisms when local biholomorphisms in...

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V2-023503 | MATHDB-318026

Equivalence of extension for torsion relations and invariant relations

摘录 / EXCERPT: Let $G \to \operatorname{GL}(n,\mathbb{C})$ be a representation with discrete kernel and closed image. Let R be a torsion relation for G, and let R' be the induced invariant...

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V2-023504 | MATHDB-318029

The high-dimensional balanced non-group-balanced configuration conjecture

摘录 / EXCERPT: High-dimensional balanced non-group-balanced configuration conjecture. If n is sufficiently large, then there exists a discrete subset of $\mathbb{R}^n$ that is balanced but not...

conjecture | open | MathDB 页面 | 原文线索

V2-023505 | MATHDB-318038

Existence and uniqueness conjecture for constant mean curvature spheres in Sol_3

摘录 / EXCERPT: Let Sol_3 be the three-dimensional Sol geometry, and let $H>0$. A constant mean curvature (CMC) H sphere is an immersed sphere in Sol_3 with constant mean c...

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V2-023506 | MATHDB-318057

Tyurin's birationality conjecture for Fano threefolds of degree 10

摘录 / EXCERPT: Let X_{10} be a general Fano threefold of degree 10, and let a quartic double solid be a smooth double cover of $\mathbb{P}^3$ branched over a quartic surface. Tyurin's conject...

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V2-023507 | MATHDB-318061

The holomorphic Weinstein conjecture for characteristic foliations

摘录 / EXCERPT: Holomorphic Weinstein conjecture. The generic leaf of the characteristic foliation on Y is a rational curve. In particular, if Y is smooth, then it is a $\mathbb{P}^1$-bundle o...

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V2-023508 | MATHDB-318097

The unique sum-point conjecture for complete arcs in $PG(2,q)$

摘录 / EXCERPT: Let q be even, and let a complete arc in the projective plane $PG(2,q)$ mean an arc that cannot be enlarged by adding another point. Two arcs are projectively equivalent when a p...

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V2-023509 | MATHDB-318105

The convex-fillability characterization of contact-type embeddability

摘录 / EXCERPT: Let $\Xi(3)$ denote the collection of closed 3--manifolds with positive, cooriented contact structures. Let $\Xi_{\text{\textsf{embed}}}(3)$ consist of those $(M,\xi)\in\Xi(3)$ t...

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V2-023510 | MATHDB-318115

Donaldson's Ricci-form conjecture for almost-Hermitian four-manifolds

摘录 / EXCERPT: Let (M,Ω) be a compact symplectic four-manifold with $b^+(M)=1$, equipped with an almost complex structure J tamed by Ω. Let g_{Ω} be the metric determined b...

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V2-023511 | MATHDB-318116

The epsilon-regularity conjecture for the almost complex Calabi-Yau equation

摘录 / EXCERPT: Epsilon-regularity conjecture. There exist constants $\varepsilon,C,r_0>0$, depending only on Ω, J, and σ, such that if

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V2-023512 | MATHDB-318134

Palais-Terng conjecture on integrable orbit-normal distributions

摘录 / EXCERPT: Let a compact Lie group G act isometrically on a Riemannian manifold M, and suppose that the distribution of normal spaces to the regular G-orbits is integrable. Palais-Terng...

conjecture | open | MathDB 页面 | 原文线索

V2-023513 | MATHDB-318135

Bolsinov-Jovanović conjecture for invariant polynomials on homogeneous spaces

摘录 / EXCERPT: Let G/H be a homogeneous space of a compact Lie group G. Choose an Ad_G-invariant positive-definite scalar product on $\mathfrak g=\operatorname{Lie}(G)$, let...

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V2-023514 | MATHDB-318139

Strong Gohberg-Markus-Boltyanski-Hadwiger conjecture

摘录 / EXCERPT: Let $d\geq 1$, and let K be a convex body in $\mathbb R^d$, meaning a compact convex set with non-empty interior. A translate of λK, with $0<\lambda<1$, is a smaller pos...

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V2-023515 | MATHDB-318183

Index conjecture for constant mean curvature hypersurfaces in spheres

摘录 / EXCERPT: CMC index conjecture. Either

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V2-023516 | MATHDB-318225

The boundary-behavior conjecture for special Lagrangian fibrations

摘录 / EXCERPT: Let X be a Kähler manifold, let $D\subset X$ be a smooth Calabi-Yau hypersurface in the anticanonical linear system, and let $\pi:X\setminus D\to B$ be a reasonable special Lagra...

conjecture | open | MathDB 页面 | 原文线索

V2-023517 | MATHDB-318230

Log canonical model conjecture for Chow quotient compactifications of hyperplane moduli spaces

摘录 / EXCERPT: Let $X(r,n)$ be the moduli space of ordered n hyperplanes in general position in $\mathbb P^{r-1}$, and let $\overline{X}(3,n)$ denote its Chow quotient compactification. Log can...

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V2-023518 | MATHDB-318231

Supertropical Bézout conjecture

摘录 / EXCERPT: Supertropical Bézout conjecture. The intersection of the sets of tangible roots of f and g has at most mn tangible connected components.

conjecture | open | MathDB 页面 | 原文线索

V2-023519 | MATHDB-318234

Mirror-symmetry conjecture for the Landau-Ginzburg Givental tau-function

摘录 / EXCERPT: Let ${\mathscr D}_{W,G}$ be the Landau-Ginzburg A-model tau-function for a suitable symmetry group G, and let ${\mathscr D}_{Giv,\check{W}}$ be the Givental tau-function associat...

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V2-023520 | MATHDB-318240

The conjecture on L^2-torsion for aspherical manifolds

摘录 / EXCERPT: Let M be a closed aspherical manifold of odd dimension and let $\widetilde{M}$ be its universal covering. L^2-torsion conjecture. The space $\widetilde{M}$ is determinant-L^2...

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V2-023521 | MATHDB-318241

The conjecture relating simplicial volume and L^2-invariants

摘录 / EXCERPT: Let M be an aspherical closed orientable manifold, let $\widetilde{M}$ be its universal covering, let $\lVert M\rVert$ denote its simplicial volume, and let $b_p^{(2)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-023522 | MATHDB-318242

The Zero-in-the-Spectrum conjecture

摘录 / EXCERPT: Let $\widetilde{M}$ be a complete Riemannian manifold and suppose it is the universal covering of an aspherical closed Riemannian manifold M, with the metric induced from M. Le...

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V2-023523 | MATHDB-318252

The higher-dimensional volume comparison conjecture

摘录 / EXCERPT: Higher-dimensional volume comparison conjecture. There exists a positive $\epsilon_0(n)<1$ such that, whenever

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V2-023524 | MATHDB-318267

Bounded curvature conjecture for the Hodge metric on Calabi-Yau threefold moduli

摘录 / EXCERPT: Bounded-curvature conjecture. The curvature of the Hodge metric ω_H is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-023525 | MATHDB-318268

Analytic torsion formula for cones over odd-dimensional spheres

摘录 / EXCERPT: Let $C_\alpha S^{2p-1}_{l\sin\alpha}$ be the cone of angle α and length $l>0$ over the odd-dimensional sphere S^{2p-1}, equipped with the standard metric induced by the im...

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V2-023526 | MATHDB-318271

Conjecture on conformally covariant operators on odd-dimensional spheres

摘录 / EXCERPT: Conjecture. For some, and hence all, such k_0,

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V2-023527 | MATHDB-318279

Equivariant classification conjecture for positively curved 4-manifolds with circle actions

摘录 / EXCERPT: Let M be a simply connected positively curved 4-manifold equipped with an isometric circle action. A linear circle action is a circle action induced by a linear action on either...

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V2-023528 | MATHDB-318287

Monopole classes of connected sums of 3-manifolds

摘录 / EXCERPT: Connected-sum monopole-class conjecture. The rational part of a monopole class of $N_1\#\cdots\#N_n$ is expressed as

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V2-023529 | MATHDB-318308

The tropical Grassmannian containment conjecture for weighted trees

摘录 / EXCERPT: Weighted-tree tropical Grassmannian conjecture. For $m\geq 5$, one has

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V2-023530 | MATHDB-318316

Alekseevsky's conjecture on homogeneous Einstein manifolds of negative Ricci curvature

摘录 / EXCERPT: Alekseevsky's conjecture. Every homogeneous Einstein manifold $\mathcal{M}$ of negative Ricci curvature is normal, namely there exists a normal metric solvable Lie algebra...

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V2-023531 | MATHDB-318339

The finite-orbit characterization of countable end limit sets

摘录 / EXCERPT: Finite-orbit characterization. The end limit set associated with E is countable if and only if it consists of finitely many orbits of parabolic fixed points.

conjecture | open | MathDB 页面 | 原文线索

V2-023532 | MATHDB-318369

Minimal-energy conjecture for endomorphisms of noncommutative tori

摘录 / EXCERPT: Let A_θ be a noncommutative torus with smooth subalgebra A_θ^∞, and let φ_A be the $*$-automorphism defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-023533 | MATHDB-318377

Alexander's plank-cutting conjecture

摘录 / EXCERPT: Let ${\bf C}$ be a convex body in ${\mathbb E}^d$. Slicing ${\bf C}$ by $n-1$ hyperplane cuts means making successive cuts while keeping each resulting piece intact after each cut....

conjecture | open | MathDB 页面 | 原文线索

V2-023534 | MATHDB-318378

Bezdek-Bezdek relative-width conjecture

摘录 / EXCERPT: Bezdek-Bezdek relative-width conjecture. If ${\bf K}$ is covered by planks ${\bf P}_1,\dots,{\bf P}_n$ in ${\mathbb E}^d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-023535 | MATHDB-318379

Local support conjecture for the Funk transform on compact symmetric spaces

摘录 / EXCERPT: Let M be a compact symmetric space that is not a sphere, let $d(M)$ denote the relevant upper radius bound, let $p\in M$, and let $\bar B_r(p)$ be the closed ball of radius r a...

conjecture | open | MathDB 页面 | 原文线索

V2-023536 | MATHDB-318402

K-semistability characterization by the greatest Ricci lower bound

摘录 / EXCERPT: Let M be a Fano manifold, and let $R(M)$ denote its greatest Ricci lower bound, as defined in the source. The manifold M is K-semistable if its Futaki invariant satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-023537 | MATHDB-318409

Waldhausen's Heegaard genus conjecture in terms of rank

摘录 / EXCERPT: Let M be a closed 3-manifold, let $\pi_1(M)$ be its fundamental group, and let $g(M)$ denote its Heegaard genus. Waldhausen's conjecture. For every k there is a number g su...

conjecture | open | MathDB 页面 | 原文线索

V2-023538 | MATHDB-318449

The pyramid conjecture for Gorenstein polytopes of degree d

摘录 / EXCERPT: Let P be an n-dimensional Gorenstein polytope of degree d. A polytope is a pyramid if it is the lattice pyramid over another lattice polytope. Pyramid conjecture. Every $3d$-...

conjecture | open | MathDB 页面 | 原文线索

V2-023539 | MATHDB-318473

Minimum-dimension degree vectors near centers of nonempty determinantal boxes

摘录 / EXCERPT: Minimum-dimension box-center conjecture. If the data $\boldsymbol l,\boldsymbol d,\boldsymbol s$ are determinantal, then determinantal degree vectors of minimum matrix dimension li...

conjecture | open | MathDB 页面 | 原文线索

V2-023540 | MATHDB-318501

Partial boundary distance rigidity for conformal simple surfaces

摘录 / EXCERPT: Let (X,g_1) and (X,g_2) be simple compact Riemannian manifolds of dimension 2 with boundary, and let d_1 and d_2 denote their respective Riemannian distances. Let Y be...

conjecture | open | MathDB 页面 | 原文线索

V2-023541 | MATHDB-318504

Tian's birational Fano extinction conjecture for the Kähler-Ricci flow with surgery

摘录 / EXCERPT: Let the unnormalized Kähler-Ricci flow be ... started from an initial projective manifold, and allow surgeries along the flow. The flow is said to become extinct in finite time when...

conjecture | solved | MathDB 页面 | 原文线索

V2-023542 | MATHDB-318505

The Mori MMP termination and Mori fibre space conjecture

摘录 / EXCERPT: Let (Z,Φ) be a Kawamata log terminal pair. Run the $(K_Z+\Phi)$-minimal model program via a birational map $f\colon Z\dashrightarrow X$, and write $\Delta=f_*\Phi$. Mori MMP c...

conjecture | open | MathDB 页面 | 原文线索

V2-023543 | MATHDB-318521

Chan-Leung's facet-point conjecture for reflexive polytopes

摘录 / EXCERPT: Chan-Leung's facet-point conjecture. If $b_P=0$, then for every facet F there exists a point $x_F\in\operatorname{aff}(F)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-023544 | MATHDB-318529

Finite generation of canonical rings for projective log canonical pairs

摘录 / EXCERPT: Canonical ring finite-generation conjecture. The canonical ring $R(X,K_X+\Delta)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-023545 | MATHDB-318530

Finite generation of multigraded adjoint rings

摘录 / EXCERPT: Adjoint-ring finite-generation conjecture. The adjoint ring $R(X;D_1,\dots,D_\ell)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-023546 | MATHDB-318557

Universal recursive formula for GJMS-operators in the locally conformally flat case

摘录 / EXCERPT: Let (M,g) be a locally conformally flat Riemannian manifold of dimension $n\geq3$. For $N\geq1$, let ${\mathcal M}_{2N}$ denote the operator defined from the recursive constructi...

conjecture | open | MathDB 页面 | 原文线索

V2-023547 | MATHDB-318558

Universal formula for the eighth-order GJMS-operator and Q-curvature

摘录 / EXCERPT: Eighth-order recursive formula.

conjecture | open | MathDB 页面 | 原文线索

V2-023548 | MATHDB-318560

Generating-function square-root conjecture for Q-curvature defects

摘录 / EXCERPT: Let ... where $\Lambda_{2N}=Q_{2N}-{\mathcal Q}_{2N}$. Let $v(r)$ be the generating function defined by the volume coefficients. Square-root conjecture. ... This reformulates the recur...

conjecture | open | MathDB 页面 | 原文线索

V2-023549 | MATHDB-318639

The dual Tverberg conjecture for families with the d-intersection property

摘录 / EXCERPT: The dual Tverberg conjecture. There exist a point $xinmathbb R^d$ and r pairwise disjoint nonempty subfamilies $mathcal F_1,ldots,mathcal F_rsubseteqmathcal F$ such that, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-023550 | MATHDB-318640

The dual Tverberg transversal conjecture for families with the $d-m$-intersection property

摘录 / EXCERPT: The dual Tverberg transversal conjecture. There exist an m-flat L and, for every $i=0,ldots,m$, r_i pairwise disjoint nonempty subfamilies...

conjecture | open | MathDB 页面 | 原文线索

V2-023551 | MATHDB-318652

The Thurston-Bennequin inequality conjecture for convex hypersurfaces in standard spheres

摘录 / EXCERPT: Let S^{2n+1}, with $n>1$, carry its standard contact structure. Consider any convex hypersurface in S^{2n+1} with contact-type boundary. The standard-sphere Thurston-Bennequin...

conjecture | open | MathDB 页面 | 原文线索

V2-023552 | MATHDB-318667

Finiteness conjecture for essential compact hyperbolic Coxeter polytopes

摘录 / EXCERPT: Finiteness conjecture. The number of essential compact hyperbolic Coxeter polytopes of dimension at least four is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-023553 | MATHDB-318673

The fill-radius conjecture for positive isotropic curvature

摘录 / EXCERPT: Let (M,g) be a complete Riemannian n-dimensional manifold whose positive isotropic curvature is bounded below by α, where $\alpha>0$: for every point $p\in M$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-023554 | MATHDB-318678

Minimal mixed volume implies standard-simplex containment

摘录 / EXCERPT: Let $\Sigma_1,\ldots,\Sigma_m$ be an essential collection of integer polyhedra in $\mathbb{R}^m$, and let their mixed volume be $1/m!$, the minimal possible value. Minimal mixed-vo...

conjecture | open | MathDB 页面 | 原文线索

V2-023555 | MATHDB-318719

Continuous Hirsch conjecture

摘录 / EXCERPT: For a polytope P of dimension d defined by n inequalities and a linear objective function c, let $\lambda_c(P)$ be the total curvature of its central path. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023556 | MATHDB-318762

Nakayama's splitting conjecture for elliptic fibrations

摘录 / EXCERPT: Nakayama's splitting conjecture. Under these hypotheses, $X\simeq Y\times E$ after a base change.

conjecture | open | MathDB 页面 | 原文线索

V2-023557 | MATHDB-318784

The weak nonvanishing conjecture for dlt pairs

摘录 / EXCERPT: Weak nonvanishing conjecture. If K_X+B is pseudo-effective over Z, then it is effective over Z.

conjecture | open | MathDB 页面 | 原文线索

V2-023558 | MATHDB-318807

The quasi-line foliation conjecture for projective space

摘录 / EXCERPT: Quasi-line foliation conjecture. There exists a birational map

conjecture | open | MathDB 页面 | 原文线索

V2-023559 | MATHDB-318814

Restriction functor conjecture for Liouville subdomains

摘录 / EXCERPT: Let $M^{in} \subset M$ be an inclusion of Liouville subdomains, and let $\EuScript{W}(M)$ denote the wrapped Fukaya category of M, with $\operatorname{Tw}(\EuScript{W}(M^{in}))$...

conjecture | open | MathDB 页面 | 原文线索

V2-023560 | MATHDB-318815

Bi-rigidity conjecture for standard degree-three del Pezzo fibrations

摘录 / EXCERPT: Let X be a standard degree-three del Pezzo fibration, meaning a standard del Pezzo fibration whose generic fibre is a cubic surface. Bi-rigidity conjecture. The threefold X is...

conjecture | open | MathDB 页面 | 原文线索

V2-023561 | MATHDB-318817

Conjecture on tensor multiplication of generalized tensor distributions

摘录 / EXCERPT: Let M be the underlying manifold, and let $\tilde D'^{a}_{b E A}(M)$ and $\tilde D'^{m}_{n E A}(M)$ denote the indicated classes of generalized tensor distributions, while...

conjecture | open | MathDB 页面 | 原文线索

V2-023562 | MATHDB-318818

Conjecture on associated fields and tensorial transformation

摘录 / EXCERPT: Let M be the underlying manifold, and let $D'^{m}_{n (A' o)}(M)$ be the indicated class of tensor distributions. An associated field means the field associated to such an object...

conjecture | open | MathDB 页面 | 原文线索

V2-023563 | MATHDB-318819

Conjecture on association commuting with tensor products

摘录 / EXCERPT: Let M be the underlying manifold. Let $\Gamma^{m}_{(A' o)}(M)$ and $\Gamma^{n}_{S}(M)$ be the indicated classes, and let A_s denote the association map. Association-tensor-prod...

conjecture | open | MathDB 页面 | 原文线索

V2-023564 | MATHDB-318820

Conjecture on preservation of equivalence under tensor multiplication

摘录 / EXCERPT: Let M be the underlying manifold, and let $C,D,F,B\in\Gamma^{m}_{A}(M)$ belong to the same indicated classes $\Gamma^{m}_{A(At(\tilde A))}(M)$. Assume that, for every relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-023565 | MATHDB-318822

Conjecture on equivalence and iterated covariant derivatives

摘录 / EXCERPT: Let M be the underlying manifold, let U^i be a piecewise smooth tensor field, let the connection belong to $\Gamma^{3}_{S}(M)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-023566 | MATHDB-318826

Functoriality of Weil prolongation over the Cahiers site

摘录 / EXCERPT: Let $\mathbf{FS}_{\mathbf{WE}}$ be the full subcategory of Weil-exponentiable Frölicher spaces, let $\mathbf{D}$ be the full subcategory of $\mathcal{C}^{\infty}$-rings whose objec...

conjecture | open | MathDB 页面 | 原文线索

V2-023567 | MATHDB-318832

Fletcher's codimension conjecture for weighted threefold complete intersections

摘录 / EXCERPT: Fletcher's codimension conjecture. There are no canonically embedded threefold complete intersections in WPS of codimension >5, and there are no anticanonically embedded...

conjecture | open | MathDB 页面 | 原文线索

V2-023568 | MATHDB-318833

Fletcher's completeness conjecture for lists of threefold weighted complete intersections

摘录 / EXCERPT: Fletcher's completeness conjecture. The lists of threefold weighted complete intersections are complete without any degree constraints.

conjecture | open | MathDB 页面 | 原文线索

V2-023569 | MATHDB-318834

Weak asymptotic magnitude conjecture for compact Euclidean subsets

摘录 / EXCERPT: Weak magnitude conjecture. There is a unique function $P\colon\bar{\mathcal{C}}\to\mathbb R$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-023570 | MATHDB-318835

Strong asymptotic magnitude conjecture for all compact Euclidean subsets

摘录 / EXCERPT: Strong magnitude conjecture. There are unique functions $P,q\colon\mathcal{C}\to\mathbb R$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-023571 | MATHDB-318850

Homogeneous classification conjecture for affine surface areas

摘录 / EXCERPT: Let ${\cal K}_0^n$ be the space of convex bodies in $\mathbb R^n$ containing the origin in their interiors. A valuation $\Psi:{\cal K}_0^n\to(-\infty,\infty]$ is homogeneous of deg...

open_problem | open | MathDB 页面 | 原文线索

V2-023572 | MATHDB-318861

Sturmfels's unit-ideal conjecture for constant coefficients of Laurent polynomial powers

摘录 / EXCERPT: Let m,n be positive integers and consider the Laurent polynomial ... in the variables $x_{-m+1},\dotsc,x_{n-1}$. For a positive integer i, let $\llbracket f^i\rrbracket$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-023573 | MATHDB-318864

The square knot does not have Property 2R

摘录 / EXCERPT: Let $Q\subset S^3$ be the square knot, and let Property $2R$ mean that whenever Q is a component of a 2-component link whose integral surgery yields ... a sequence of handle slid...

conjecture | open | MathDB 页面 | 原文线索

V2-023574 | MATHDB-318907

Expected dimension conjecture for the tropical restricted Boltzmann machine

摘录 / EXCERPT: Let TM_n^k denote the tropical restricted Boltzmann machine model, a polyhedral fan in the tropical projective space $\mathbb{TP}^{2^n-1}$. Tropical RBM expected dimension conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-023575 | MATHDB-318908

Zaidenberg-Lin conjecture on singular points of affine plane curves

摘录 / EXCERPT: Let $C\subset\mathbb{C}^2$ be a reduced, irreducible, algebraic plane curve with first Betti number b_1. The singular points at finite distance are the singular points of C lyi...

conjecture | open | MathDB 页面 | 原文线索

V2-023576 | MATHDB-318938

Existence of exceptional quotient singularities in every dimension

摘录 / EXCERPT: An exceptional quotient singularity is a quotient singularity of the form $(V\ni O)=\mathbb{C}^{n+1}\!/G$ that is exceptional in the sense of the paper. Existence conjecture. Excep...

conjecture | open | MathDB 页面 | 原文线索

V2-023577 | MATHDB-318949

Restriction-determination conjecture for zero-series eigenfunctions on the Vicsek set

摘录 / EXCERPT: Let V_m be the level-m discrete approximation of the Vicsek set, let D_m be its intersection with one arm of the diagonal, and let Z_m be the span of the zero-series eigenf...

conjecture | open | MathDB 页面 | 原文线索

V2-023578 | MATHDB-318967

Tubular-neighborhood conjecture for Poincaré duality angles

摘录 / EXCERPT: Let M^m be a closed, smooth, oriented Riemannian manifold and let N^n be a closed submanifold of codimension $m-n \geq 2$. Define the compact Riemannian manifold ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-023579 | MATHDB-318977

Generalized Nitsche conjecture for properly embedded minimal annuli

摘录 / EXCERPT: Generalized Nitsche conjecture. Then E has finite total curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-023580 | MATHDB-318987

The unique-arc characterization of lion-winning Euclidean metric spaces

摘录 / EXCERPT: Unique-arc conjecture. The lion has a winning strategy in the lion and man game on X from all starting positions if and only if any two points of X are joined by a unique arc.

conjecture | open | MathDB 页面 | 原文线索

V2-023581 | MATHDB-318989

Fixed-point conjecture for convex subsets of spherical buildings

摘录 / EXCERPT: Let B be a spherical building and let $C\subseteq B$ be a closed convex subset. Fixed-point conjecture. Either C is a subbuilding, or the action ... has a fixed point. This gener...

conjecture | open | MathDB 页面 | 原文线索

V2-023582 | MATHDB-318990

Radius-gap conjecture for convex subsets of spherical buildings

摘录 / EXCERPT: Let B be a spherical building and let $C\subseteq B$ be a closed convex subset. Write $rad_C(C)$ for the intrinsic radius of C, namely the infimum of the radii of balls centere...

conjecture | open | MathDB 页面 | 原文线索

V2-023583 | MATHDB-319012

Binding number conjecture for Giroux torsion

摘录 / EXCERPT: Let (Y,ξ) be a contact 3-manifold. The binding number $bn(\xi)$ is the minimal number of binding components among open books supporting ξ whose pages have minimal genus. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-023584 | MATHDB-319048

Two-point convex-hull conjecture for the secrecy-channel curve

摘录 / EXCERPT: Two-point convex-hull conjecture. Any point in the convex hull of $\mathscr{C}$ can be expressed as the convex combination of only two points belonging to $\mathscr{C}$. This geome...

conjecture | open | MathDB 页面 | 原文线索

V2-023585 | MATHDB-319071

Lichnerowicz's conjecture on canonical metrics and conformal isometries

摘录 / EXCERPT: Lichnerowicz's conjecture. There exists a metric $\tilde g$ conformal to g, with constant scalar curvature $R_{\tilde g}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-023586 | MATHDB-319125

The projective space derangement graph conjecture for PGL(n+1,q)

摘录 / EXCERPT: Let $\mathbb{P}_q^n$ be projective n-space, with $n\geq 1$, and let $\Gamma_{\operatorname{PGL}(n+1,q)}$ be the derangement graph of $\operatorname{PGL}(n+1,q)$ acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-023587 | MATHDB-319143

Coset-fibration conjecture for totally geodesic foliations in compact Lie groups

摘录 / EXCERPT: Let G be a simple compact Lie group with a bi-invariant metric, and let $\pi:G\rightarrow B$ be a Riemannian submersion with connected totally geodesic fibers. Coset-fibration co...

conjecture | open | MathDB 页面 | 原文线索

V2-023588 | MATHDB-319163

Grothendieck-Katz conjecture for iterative q-difference modules

摘录 / EXCERPT: Grothendieck-Katz conjecture for iterative q-difference modules. The iterative q-difference module $\mathcal{M}$ is isotrivial, meaning that it becomes trivial after a finite bas...

conjecture | open | MathDB 页面 | 原文线索

V2-023589 | MATHDB-319167

Strong Gordon conjecture for the figure-8 knot complement

摘录 / EXCERPT: A non-hyperbolic Dehn filling is a Dehn filling of a 1-cusped hyperbolic 3-manifold that does not remain hyperbolic. Strong Gordon conjecture. The figure-8 knot complement is...

conjecture | open | MathDB 页面 | 原文线索

V2-023590 | MATHDB-319168

Milnor's conjecture on rational-angle Lobachevsky relations

摘录 / EXCERPT: For an angle θ, let $\operatorname{L}(\theta)$ denote the Lobachevsky function. Consider angles θ that are rational multiples of π, and rational coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-023591 | MATHDB-319198

Sign-change invariance conjecture for holonomy angles

摘录 / EXCERPT: Let Γ be a lattice with trace field K, where K is the field generated by the traces of elements of Γ. Let h_K and h_K^+ denote the class number and narrow cla...

conjecture | open | MathDB 页面 | 原文线索

V2-023592 | MATHDB-319199

Surface-polynomial invariance for swapped block and hole polyhedra

摘录 / EXCERPT: Let $\mathcal{P}$ be a generically isostatic block and hole polyhedron, and let $\overline{\mathcal{P}}$ denote its swapped block and hole polyhedron. Write $T(\mathcal{P})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-023593 | MATHDB-319209

Isotropic-orbit F-structure conjecture for collapsed Kähler manifolds

摘录 / EXCERPT: Let $n\in \mathbb{N}$. For a closed Kähler n-manifold (M,ω,J,g) with $[\omega]\in H^{2}(M,\mathbb{Z})$, define ... An F-structure is understood in the sense of Cheeger-Gromo...

conjecture | open | MathDB 页面 | 原文线索

V2-023594 | MATHDB-319240

Regularity bound for fibers of general projections

摘录 / EXCERPT: Regularity bound for fibers of general projections. Every fiber satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-023595 | MATHDB-319255

Faber's lower-bound conjecture for Kakeya sets in odd-order projective planes

摘录 / EXCERPT: Let π_q be a projective plane of odd order q, and let K be a Kakeya set in π_q, meaning that K contains at least one point on each line of a fixed line pencil. Faber'...

conjecture | open | MathDB 页面 | 原文线索

V2-023596 | MATHDB-319270

Buser's asymptotic conjecture for Bers' constants

摘录 / EXCERPT: Let g and n denote the genus and number of cusps of a hyperbolic surface, respectively. Its Bers' constant is the smallest number such that the surface admits a pants decomposi...

conjecture | open | MathDB 页面 | 原文线索

V2-023597 | MATHDB-319281

Zariski-density conjecture for modular points

摘录 / EXCERPT: Let ${\bar\rho}$ be a Galois representation of type $\mathrm{U}(n)$, let ${\mathfrak X}({\bar\rho})$ be its deformation space, and let ${\mathfrak X}^{\rm mod}$ denote the set of m...

conjecture | open | MathDB 页面 | 原文线索

V2-023598 | MATHDB-319282

Genericity conjecture for global Galois representations

摘录 / EXCERPT: Let n be an odd integer, let ${\bar\rho}$ be modular, let ${\mathfrak X}({\bar\rho})$ be its deformation space, and let $\rho_{y,\widetilde v}$ denote the local representation as...

conjecture | open | MathDB 页面 | 原文线索

V2-023599 | MATHDB-319302

Functoriality of Weil prolongation over the category of smooth algebras

摘录 / EXCERPT: Functoriality conjecture. The right-hand side of this definition is functorial in C, so that there is a bifunctor

conjecture | open | MathDB 页面 | 原文线索

V2-023600 | MATHDB-319306

The Big-Line-Big-Chromatic-Number Conjecture for planar visibility graphs

摘录 / EXCERPT: Let P be a finite set of points in the plane, and let $\mathcal{V}(P)$ be its visibility graph. A $(k-1)$-colouring assigns one of $k-1$ colours to each point. Big-Line-Big-Chrom...

conjecture | open | MathDB 页面 | 原文线索

V2-023601 | MATHDB-319307

The superlinear blocking conjecture for general-position planar point sets

摘录 / EXCERPT: For a finite set P of points in the plane, a set B of points disjoint from P blocks P if every segment joining two distinct points of P contains a point of B. If P is...

conjecture | open | MathDB 页面 | 原文线索

V2-023602 | MATHDB-319312

The superlinear blocker conjecture with bounded collinearities

摘录 / EXCERPT: For fixed ℓ, let $b_\ell(n)$ be the minimum integer such that every set of n points in the plane with no ℓ collinear points can be blocked by a set of $b_\ell(n)$ point...

conjecture | open | MathDB 页面 | 原文线索

V2-023603 | MATHDB-319324

Symplectic invariant-tensor character conjecture

摘录 / EXCERPT: Let $n\ge 1$, let $\mathfrak{g}=\mathfrak{sp}(2n)$, and let V be the vector representation. For a partition μ, write $\overline{2\mu}$ for the partition obtained by doubling...

conjecture | open | MathDB 页面 | 原文线索

V2-023604 | MATHDB-319326

Schmutz Schaller's multiplicity-six conjecture for once-punctured tori

摘录 / EXCERPT: Schmutz Schaller's conjecture. The multiplicity is bounded above by 6.

conjecture | open | MathDB 页面 | 原文线索

V2-023605 | MATHDB-319370

Deser-Schwimmer conjecture on global conformal invariants

摘录 / EXCERPT: Let M^n be a Riemannian manifold with n even, and let $P(g)$ be a Riemannian invariant of weight $-n$ such that ... is unchanged under conformal rescalings $g\mapsto e^{2\phi}g$...

conjecture | open | MathDB 页面 | 原文线索

V2-023606 | MATHDB-319378

Uniqueness of the k-sector for disjoint closed sets

摘录 / EXCERPT: Let P and Q be two disjoint nonempty closed sets in Euclidean space, and let a k-sector be the geometric object associated with P and Q that separates their distance-base...

conjecture | open | MathDB 页面 | 原文线索

V2-023607 | MATHDB-319380

Shioda's unirationality conjecture for K3 surfaces

摘录 / EXCERPT: Shioda's conjecture. A K3 surface is unirational if and only if it is Shioda-supersingular.

conjecture | open | MathDB 页面 | 原文线索

V2-023608 | MATHDB-319381

Subcritical heat kernels decay faster than critical heat kernels

摘录 / EXCERPT: Let M be a connected noncompact Riemannian manifold, and let $k_P^M(x,y,t)$ denote the positive minimal heat kernel of a second-order elliptic operator P on M. An operator is...

conjecture | open | MathDB 页面 | 原文线索

V2-023609 | MATHDB-319382

Equivalence conjecture for heat kernels under compact perturbations

摘录 / EXCERPT: Let M be a Riemannian manifold, and let P_1 and P_2 be subcritical elliptic operators of the form specified in the paper. Two heat kernels are equivalent when they are compar...

conjecture | open | MathDB 页面 | 原文线索

V2-023610 | MATHDB-319386

Existence conjecture for genus-three minimal surfaces with Scherk-type ends

摘录 / EXCERPT: Existence conjecture. There exists a one-parameter family of genus-three, embedded minimal surfaces M_k^{+-} with $4k$ Scherk-type ends, for all $k\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-023611 | MATHDB-319391

The symplectic automorphism conjecture for Chow groups of K3 surfaces

摘录 / EXCERPT: Symplectic automorphism conjecture. If f is a symplectomorphism, then

conjecture | open | MathDB 页面 | 原文线索

V2-023612 | MATHDB-319404

Baldridge's symplectic circle action conjecture for closed 4-manifolds

摘录 / EXCERPT: Let M be a closed 4-manifold that admits a symplectic form and a smooth circle action. Baldridge's conjecture. M also admits a symplectic circle action, possibly with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-023613 | MATHDB-319410

The local-to-global principle for integral Apollonian circle packings

摘录 / EXCERPT: Local-to-global principle for Apollonian circle packings. There exists $X_P\in\mathbb Z$ such that every integer $x>X_P$ whose residue modulo 24 lies in P_{24} is the curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-023614 | MATHDB-319428

Bromberg's local-connectivity conjecture for surface-group deformation spaces

摘录 / EXCERPT: Bromberg's surface-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023615 | MATHDB-319429

Bromberg's conjecture on local connectivity of Bers-slice closures

摘录 / EXCERPT: Bromberg's Bers-slice conjecture. The closure of any Bers slice in $QF(S)$ is not locally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-023616 | MATHDB-319430

The non-local-connectivity conjecture for hyperbolizable 3-manifold deformation spaces

摘录 / EXCERPT: Non-local-connectivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023617 | MATHDB-319434

The exotic sphere pinching conjecture

摘录 / EXCERPT: An exotic sphere is a smooth manifold homeomorphic but not diffeomorphic to a standard sphere. A Riemannian metric is strictly $1/4$-pinched in the global sense when its sectional...

conjecture | open | MathDB 页面 | 原文线索

V2-023618 | MATHDB-319448

Baker-Norine lifting conjecture for divisors on tropical curves

摘录 / EXCERPT: Lifting conjecture. Every divisor of degree d and rank r on Γ lifts to a divisor of degree d and rank r on X. Furthermore, if ρ is zero, then this lift is uni...

conjecture | open | MathDB 页面 | 原文线索

V2-023619 | MATHDB-319450

Di Rocco's non-negativity conjecture for the face-volume invariant

摘录 / EXCERPT: Di Rocco's non-negativity conjecture. For every n-dimensional lattice polytope P,

conjecture | open | MathDB 页面 | 原文线索

V2-023620 | MATHDB-319500

Generic stratification conjecture for Pareto critical sets

摘录 / EXCERPT: Let W be a compact manifold of dimension $n\geq m$, let $u\in C^\infty(W,\mathbb{R}^m)$, and let θ be its Pareto critical set, with θ_S denoting the singular part o...

conjecture | open | MathDB 页面 | 原文线索

V2-023621 | MATHDB-319503

Quadratic-size conjecture for k-blocked point sets

摘录 / EXCERPT: A finite point set in the plane is k-blocked if its points are assigned one of k colours such that two distinct points have the same colour exactly when some other point of the...

conjecture | open | MathDB 页面 | 原文线索

V2-023622 | MATHDB-319504

Colour-class-size conjecture for k-blocked point sets

摘录 / EXCERPT: Let P be a k-blocked point set, with its points assigned one of k colours so that two distinct points have the same colour exactly when some other point of P blocks them. C...

conjecture | open | MathDB 页面 | 原文线索

V2-023623 | MATHDB-319509

The characterization conjecture for curvature-invariant spacetimes

摘录 / EXCERPT: Characterization conjecture. If a spacetime is characterised by its invariants, weakly or strongly, then it is either $\mathcal{I}$-non-degenerate or of type D to all orders; that...

conjecture | open | MathDB 页面 | 原文线索

V2-023624 | MATHDB-319575

Monodromy conjecture for compact LCK manifolds with automorphic potential

摘录 / EXCERPT: Let M be a compact locally conformally Kähler (LCK) manifold with automorphic potential, meaning that M admits a Kähler covering with an automorphic potential. Monodromy conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-023625 | MATHDB-319576

Structure conjecture for compact LCK nilmanifolds

摘录 / EXCERPT: Let M be a compact nilmanifold admitting a locally conformally Kähler (LCK) structure, and let $H(n)$ be the generalized Heisenberg group. Structure conjecture. Up to covering,...

conjecture | open | MathDB 页面 | 原文线索

V2-023626 | MATHDB-319586

Makeev's inscribed quadrangle conjecture for plane curves

摘录 / EXCERPT: Let C be a smooth simple closed curve in $\mathbb R^2$, and let Q be four points on a single circle. Makeev's inscribed quadrangle conjecture. There is a similarity transform...

conjecture | open | MathDB 页面 | 原文线索

V2-023627 | MATHDB-319599

Finiteness conjecture for hyperkähler manifolds up to deformation

摘录 / EXCERPT: Finiteness conjecture. There are finitely many hyperkähler manifolds up to deformation in each complex dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-023628 | MATHDB-319600

Lagrangian fibration conjecture for hyperkähler manifolds

摘录 / EXCERPT: Lagrangian fibration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023629 | MATHDB-319613

Gross-Wilson-Kontsevich-Soibelman large complex structure limit conjecture

摘录 / EXCERPT: Large complex structure limit conjecture. There is a subsequence $(X_{s_{i_j}},g_{i_j})$ converging in the Gromov-Hausdorff sense to a metric space (X_∞,d_∞) such that...

conjecture | open | MathDB 页面 | 原文线索

V2-023630 | MATHDB-319615

Theta-functional general-solution conjecture for the generalized ultradiscrete Toda lattice

摘录 / EXCERPT: Let Γ be the tropical spectral curve, let Θ be its tropical theta function, and let $\vec{L}$ and $\vec{\lambda}$ be the vectors defined by the integrals between the...

conjecture | open | MathDB 页面 | 原文线索

V2-023631 | MATHDB-319616

Jacobian parametrization conjecture for the generalized ultradiscrete Toda isolevel set

摘录 / EXCERPT: Let Γ be the tropical spectral curve associated with $\operatorname{\mathcal{T}}(M,N)$, let $J(\Gamma)$ denote its Jacobian, and let $\Phi^{-1}(\xi)$ denote the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-023632 | MATHDB-319617

The vanishing conjecture for Wodzicki-Chern classes of pseudodifferential operator bundles

摘录 / EXCERPT: Let $\Psi{\rm DO}_0^*$ be the group of invertible zeroth-order classical pseudodifferential operators, and let $\mathcal E$ be a $\Psi{\rm DO}_0^*$-bundle over a base manifold admi...

conjecture | open | MathDB 页面 | 原文线索

V2-023633 | MATHDB-319620

Nonexistence of an irreducible radial minimal cubic in dimension 17

摘录 / EXCERPT: Dimension-17 nonexistence claim. There is no irreducible radial minimal cubic in $\mathbb{R}^{17}$.

conjecture | open | MathDB 页面 | 原文线索

V2-023634 | MATHDB-319621

Radiality conjecture for irreducible minimal cubics

摘录 / EXCERPT: Radiality conjecture. Any minimal irreducible cubic is radial.

conjecture | open | MathDB 页面 | 原文线索

V2-023635 | MATHDB-319646

Pointwise convergence of the vortex metric to the holomorphic-map metric

摘录 / EXCERPT: Let Σ be a Riemann surface, let $n>1$, and let $\mathcal M^o$ be the vortex moduli space identified with the space $\mathcal H$ of holomorphic maps...

conjecture | open | MathDB 页面 | 原文线索

V2-023636 | MATHDB-319647

Riley-Wielenberg conjecture on commensurable hyperbolic knots

摘录 / EXCERPT: Riley-Wielenberg conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-023637 | MATHDB-319674

The sub-exponential volume growth conjecture for the essential spectrum

摘录 / EXCERPT: Sub-exponential volume growth conjecture. The L^p essential spectrum of M is $[0,+\infty)$ for any $p\in[1,+\infty]$.

conjecture | open | MathDB 页面 | 原文线索

V2-023638 | MATHDB-319676

Urbański's boundary dimension conjecture for non-polygonal self-affine tilings

摘录 / EXCERPT: Let $\mathcal T$ be a d-dimensional non-polygonal self-affine tiling, and let each tile be $T=(A,i)$. Urbański's boundary dimension conjecture. Each tile satisfies ... This folklor...

conjecture | open | MathDB 页面 | 原文线索

V2-023639 | MATHDB-319683

Complex-holomorphic-data conjecture for Willmore surfaces

摘录 / EXCERPT: Complex-holomorphic-data conjecture. Then f is given by complex holomorphic data.

conjecture | open | MathDB 页面 | 原文线索

V2-023640 | MATHDB-319688

The punctured class C Barlet-space properness conjecture

摘录 / EXCERPT: Let Δ be a disc, let $\Delta^{\star}:=\Delta\setminus\{0\}$, and let ... be a complex analytic family of compact complex manifolds, with fibres $X_t:=\pi^{-1}(t)$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-023641 | MATHDB-319700

The torus level-set length bound in dimension two

摘录 / EXCERPT: Let g be a metric on the two-dimensional torus T^2, and let $f:T^2\rightarrow\mathbb{R}$. Naive conjecture. There is a function f such that, for every $y\in\mathbb{R}$, ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-023642 | MATHDB-319701

The torus level-set area bound in dimension three

摘录 / EXCERPT: Let g be a metric on the three-dimensional torus T^3, and let $f:T^3\rightarrow\mathbb{R}$. Naive conjecture. There is a function f such that, for every $y\in\mathbb{R}$, ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-023643 | MATHDB-319702

Geroch's conjecture on positive scalar curvature of the torus

摘录 / EXCERPT: The n-torus is the manifold T^n. A metric has positive scalar curvature when its scalar-curvature function is positive everywhere. Geroch's conjecture. The n-torus does not a...

conjecture | open | MathDB 页面 | 原文线索

V2-023644 | MATHDB-319703

Gromov's generalized Geroch conjecture

摘录 / EXCERPT: Fix a scale $r>0$. For a Riemannian metric on a closed manifold, let $Scal_r(p)$ denote the macroscopic scalar curvature at scale r at p, defined by comparing the volume of the...

conjecture | open | MathDB 页面 | 原文线索

V2-023645 | MATHDB-319704

The three-torus fiber-length bound

摘录 / EXCERPT: Let g be a Riemannian metric on the three-dimensional torus T^3, and let $f:T^3\rightarrow\mathbb{R}^2$. Fiber-length conjecture. There is a function f such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-023646 | MATHDB-319705

Open Gromov-Witten sum-formula conjecture for Lagrangian real projective 3-spaces

摘录 / EXCERPT: Let X be an almost Calabi-Yau 3-fold and let $L\cong \mathbb{R}P^3\subset X$ be an embedded Lagrangian. For odd classes $\beta\in\Pi_2(X,L)$, the open Gromov-Witten invariants...

conjecture | open | MathDB 页面 | 原文线索

V2-023647 | MATHDB-319720

Positivity conjecture for volumes of Hermitian potentials

摘录 / EXCERPT: Positivity conjecture. For any compact Hermitian manifold (X,ω), one has

conjecture | open | MathDB 页面 | 原文线索

V2-023648 | MATHDB-319724

The knot-and-link generation conjecture for exotic smooth structures on $E(2)$

摘录 / EXCERPT: Knot-and-link generation conjecture. All distinct exotic smoothness structures of the compact 4-manifold $M=E(2)$ can be generated by knots and links.

conjecture | open | MathDB 页面 | 原文线索

V2-023649 | MATHDB-319730

The symmetric-group Killing-form signature conjecture

摘录 / EXCERPT: Let S_n be the symmetric group with $n>4$, and let K be the Killing form associated with a conjugacy class. Symmetric-group signature conjecture. Every conjugacy class has nond...

conjecture | open | MathDB 页面 | 原文线索

V2-023650 | MATHDB-319762

The split-closed Fukaya-to-Landau-Ginzburg branes embedding conjecture

摘录 / EXCERPT: Let M be one of the hypersurfaces considered in the source, with Fukaya category $\mathcal{F}(M)$, and let $\hat W$ be the associated Landau-Ginzburg potential. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023651 | MATHDB-319763

Kontsevich-Soibelman conjecture for global torus fibrations

摘录 / EXCERPT: Let B be a closed $\mathbb{Z}$-affine manifold, let $M=T^*B/T^*_{\mathbb{Z}}B$ be the associated symplectic torus fibration, and let $\mathcal{O}_B$ be its natural sheaf of...

conjecture | open | MathDB 页面 | 原文线索

V2-023652 | MATHDB-319764

The Fukaya category and signed tropical-graph sheaves embedding conjecture

摘录 / EXCERPT: Let C be a trivalent graph, possibly with semi-infinite edges, equipped with a signed tropical structure, and let M be the symplectic surface obtained by gluing the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-023653 | MATHDB-319767

Burger-Li-Sarnak purity conjecture for congruence hyperbolic manifolds

摘录 / EXCERPT: Let G be a non-trialitarian semisimple $mathbf Q$-group obtained by restriction of scalars from a special orthogonal group over a totally real number field, with $G(\mathbf R)$ e...

conjecture | open | MathDB 页面 | 原文线索

V2-023654 | MATHDB-319769

Gromov-Witten duality conjecture for étale gerbes

摘录 / EXCERPT: Let $cmathcal{Y}$ be a G-gerbe over a compact symplectic orbifold $cmathcal{B}$, and let $(cmathcal{\widehat{Y}},c)$ be its dual pair, where c is the flat $U(1)$-gerbe defining...

conjecture | open | MathDB 页面 | 原文线索

V2-023655 | MATHDB-319778

Conjecture on CR maps preserving the Tanaka-Webster Ricci tensor in generalized ellipsoids

摘录 / EXCERPT: Let the model be the generalized ellipsoid described in Section 2, equipped with the contact form ϑ defined in equation (polo). A CR map is said to preserve the Tanaka-We...

conjecture | open | MathDB 页面 | 原文线索

V2-023656 | MATHDB-319789

The flexibility conjecture for 3-ribbon surfaces

摘录 / EXCERPT: Let f be a 3-ribbon surface, let $\mathcal D_\gamma$ be the operator defined for the surface, and let χ be its characteristic function. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-023657 | MATHDB-319791

The Fano volume inequality conjecture

摘录 / EXCERPT: Let Z be a smooth Fano manifold of complex dimension $n-1$, with Fano index $I(Z)mathrel{\in}\mathbb{Z}_{>0}$. Fano volume inequality conjecture. ... Equality should hold if and on...

conjecture | open | MathDB 页面 | 原文线索

V2-023658 | MATHDB-319797

Spectrum conjecture for complete arcs in projective planes

摘录 / EXCERPT: Let $PG(2,q)$ be the projective plane of order q, let $\overline{t}_{2}(2,q)$ be the smallest known size of a complete arc, and let M_q be the upper endpoint defined in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-023659 | MATHDB-319803

The infinite exotic smooth structures conjecture for simply connected 4-manifolds

摘录 / EXCERPT: Infinite exotic smooth structures conjecture. There are infinitely many mutually nondiffeomorphic 4-manifolds homeomorphic to X.

conjecture | open | MathDB 页面 | 原文线索

V2-023660 | MATHDB-319809

The toric Sasaki polystability conjecture

摘录 / EXCERPT: Let M be a compact co-oriented toric contact manifold of Reeb type. Let $\mathcal{C}$ be its strictly convex good cone, let $\mathcal{C}_+^*$ denote the Reeb cone, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-023661 | MATHDB-319814

The security-uniform security-flatness conjecture for closed Riemannian manifolds

摘录 / EXCERPT: Let (M,g) be a closed C^{∞} Riemannian manifold. Here, (M,g) is secure if every pair of points has the finite blocking property, and uniformly secure if there is a unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-023662 | MATHDB-319842

Buzzard-Lu conjecture on holomorphic domination of K3 surfaces

摘录 / EXCERPT: Let X be a complex $K3$ surface. It is holomorphically dominable by ${\mathbb C}^2$ if there exists a holomorphic map ... whose image is dense, equivalently a holomorphic map domin...

conjecture | open | MathDB 页面 | 原文线索

V2-023663 | MATHDB-319869

Elashvili's index conjecture for centralizers of reductive Lie algebras

摘录 / EXCERPT: Let ${\mathfrak g}$ be a reductive Lie algebra, and let ${\mathfrak g}^x$ denote the centralizer of $x\in{\mathfrak g}^*$. Write ${\rm ind}\,{\mathfrak h}$ for the index of a Lie a...

conjecture | open | MathDB 页面 | 原文线索

V2-023664 | MATHDB-319886

Bounded-length-distortion embedding conjecture for compact length metric spaces

摘录 / EXCERPT: Let X be a compact length metric space of finite Hausdorff dimension. A map from X into a Euclidean space has bounded length distortion if it distorts lengths by a controlled r...

conjecture | open | MathDB 页面 | 原文线索

V2-023665 | MATHDB-319911

The abundance conjecture for projective log canonical pairs

摘录 / EXCERPT: Throughout, work over the complex number field. Let (X,Δ) be a projective log canonical pair, where $K_X+\Delta$ is the log canonical divisor, and let $\nu(K_X+\Delta)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-023666 | MATHDB-319913

Finiteness conjecture for B-pluricanonical representations

摘录 / EXCERPT: Finiteness conjecture for B-pluricanonical representations. If $K_X+\Delta$ is nef, then $\rho_m(\operatorname{Bir}(X,\Delta))$ is finite for a sufficiently large and divisible p...

conjecture | open | MathDB 页面 | 原文线索

V2-023667 | MATHDB-319927

The quotient monotonicity conjecture for totally skew embeddings

摘录 / EXCERPT: Quotient monotonicity conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-023668 | MATHDB-319928

The product lower-bound conjecture for totally skew embeddings

摘录 / EXCERPT: Product lower-bound conjecture. The invariant should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-023669 | MATHDB-319929

Existence of hyperbolic surface bundles with trace fields having real places

摘录 / EXCERPT: Let $S=S(g,p)$ be an orientable, connected surface of genus g with p punctures, and let M be a hyperbolic surface bundle over S^1 with fiber of type $(-\chi,p)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-023670 | MATHDB-319981

Polynomiality conjecture for the leading coefficients of node polynomials

摘录 / EXCERPT: Let $C_\tau(\delta)$ denote the coefficient vector associated with a type τ in the computation of the node polynomial $N_\delta(d)$. Leading-coefficient polynomiality conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-023671 | MATHDB-319984

The lcK existence conjecture for closed smooth four-manifolds

摘录 / EXCERPT: Let X be a closed smooth four-manifold. The lcK existence conjecture. If X admits a complex structure, then X also admits a locally conformally Kähler (lcK) structure, not ne...

conjecture | open | MathDB 页面 | 原文线索

V2-023672 | MATHDB-319995

Generalized mirror-transformation conjecture for K_{F_0} and the Calabi-Yau hypersurface in WP_1

摘录 / EXCERPT: Generalized mirror-transformation conjecture. For K_{F_0}, respectively for the Calabi-Yau hypersurface in WP_1, equation (intromirr) gives the mirror map, and after substituti...

conjecture | open | MathDB 页面 | 原文线索

V2-023673 | MATHDB-319996

Conjecture relating weighted-projective correlation functions to w_d

摘录 / EXCERPT: Weighted-projective conjecture. In this example,

conjecture | open | MathDB 页面 | 原文线索

V2-023674 | MATHDB-320002

The existence of an infinite M-set in a strictly convex smooth four-dimensional Minkowski space

摘录 / EXCERPT: Infinite M-set conjecture. There exists a strictly convex, smooth 4-dimensional Minkowski space admitting an infinite M-set.

conjecture | open | MathDB 页面 | 原文线索

V2-023675 | MATHDB-320003

The lower-bound conjecture for M-sets in finite-dimensional Minkowski spaces

摘录 / EXCERPT: M-set lower-bound conjecture. Each d-dimensional Minkowski space has an M-set of at least $d+1$ points.

conjecture | open | MathDB 页面 | 原文线索

V2-023676 | MATHDB-320025

The perfect-power conjecture for reptile simplices

摘录 / EXCERPT: For integers $d\geq 3$ and $k\geq 2$, a d-dimensional k-reptile simplex is a d-dimensional simplex that can be tiled without overlaps by k smaller simplices congruent and s...

conjecture | open | MathDB 页面 | 原文线索

V2-023677 | MATHDB-320033

Continuous Hirsch Conjecture for central-path curvature

摘录 / EXCERPT: Continuous Hirsch Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023678 | MATHDB-320034

Continuous d-step Conjecture for central-path curvature

摘录 / EXCERPT: Continuous d-step Conjecture. The function $\Lambda(2d,d)$ grows linearly in its input; that is,

conjecture | open | MathDB 页面 | 原文线索

V2-023679 | MATHDB-320050

Gillet-Soulé's boundedness conjecture for Dolbeault determinants

摘录 / EXCERPT: Gillet-Soulé's conjecture. The determinant of $\Delta_{\bar{\partial}}$, viewed as a functional on the space of all smooth Hermitian metrics on L, is bounded from above.

conjecture | open | MathDB 页面 | 原文线索

V2-023680 | MATHDB-320052

Existence of analytic torsion critical points on curves

摘录 / EXCERPT: Curve critical-point conjecture. For every k, the functional $\mathcal{F}_{k}$ admits a critical point.

conjecture | open | MathDB 页面 | 原文线索

V2-023681 | MATHDB-320053

Coercivity conjecture for the analytic torsion functional

摘录 / EXCERPT: Analytic torsion coercivity conjecture. Under these assumptions, $\mathcal{F}_{\omega_{0}}$ is coercive.

conjecture | open | MathDB 页面 | 原文线索

V2-023682 | MATHDB-320064

Canonical flat coordinates conjecture for the instanton-corrected toric Calabi-Yau mirror

摘录 / EXCERPT: Let X be a toric Calabi-Yau manifold with complexified Kähler moduli space $\mathcal{M}_K(X)$, and let $\check{X}$ be its Hori-Iqbal-Vafa mirror family with complex moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-023683 | MATHDB-320066

Modified weak Purdy conjecture on incident hyperplanes

摘录 / EXCERPT: Let S be a set of n points in Euclidean d-space. Say that S is r-degenerate if it can be covered by a set of nonzero-dimensional flats whose dimensions sum to less than...

conjecture | open | MathDB 页面 | 原文线索

V2-023684 | MATHDB-320067

Beck-Erdős-type conjecture for nondegenerate point sets

摘录 / EXCERPT: Let S be a set of n points in Euclidean d-space, and let k be such that no d-degenerate subset of S has more than $n-k$ points. Here S is d-degenerate when it can b...

conjecture | open | MathDB 页面 | 原文线索

V2-023685 | MATHDB-320068

Reversible projective spherically symmetric Finsler metrics are Riemannian

摘录 / EXCERPT: Let $F=F(x,y)$ be a projective spherically symmetric Finsler metric with constant flag curvature on a convex domain $\Omega\subseteq\mathbb{R}^n$. A Finsler metric is reversible wh...

conjecture | open | MathDB 页面 | 原文线索

V2-023686 | MATHDB-320088

The noncommutative Weyl formula

摘录 / EXCERPT: Let ${\mathcal F}_N$ be the lifted foliation on the conormal bundle, let $\operatorname{tr}_{{\mathcal F}_N}$ be its canonical transverse trace, and let Δ_{ε} be t...

conjecture | open | MathDB 页面 | 原文线索

V2-023687 | MATHDB-320094

Integrality conjecture for open-closed mirror maps

摘录 / EXCERPT: Integrality conjecture. The Taylor series of $q_0,q_1,\dots,q_N$ in $x_0,x_1,\dots,x_N$ have integer coefficients, and the inverse Taylor series likewise have integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-023688 | MATHDB-320095

Product-expansion conjecture for two-variable open-closed mirror maps

摘录 / EXCERPT: Product-expansion conjecture. There are integers $\alpha^{(i)}_{m_0,m_1}$ and $\beta^{(i)}_{m_0,m_1}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-023689 | MATHDB-320097

The DDVV conjecture for submanifolds of real space forms

摘录 / EXCERPT: DDVV conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023690 | MATHDB-320128

The modified openness conjecture for plurisubharmonic functions

摘录 / EXCERPT: Modified openness conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-023691 | MATHDB-320162

Abhyankar-Sathaye embedding conjecture

摘录 / EXCERPT: Let A be a commutative ring with one, let $A^{[n]}$ denote a polynomial ring in n variables over A, and let $f\in A^{[n]}$. A polynomial f is a coordinate if it can be comp...

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V2-023692 | MATHDB-320163

Stable coordinate conjecture

摘录 / EXCERPT: Let A be a commutative ring with one, let $A^{[n]}$ denote a polynomial ring in n variables over A, and let $f\in A^{[n]}$. A polynomial f is a stable coordinate if it is a...

conjecture | open | MathDB 页面 | 原文线索

V2-023693 | MATHDB-320164

Coordinate equivalence conjecture for transformed polynomials

摘录 / EXCERPT: Let A be a commutative ring with one, let $A^{[n]}$ be a polynomial ring over A, let $a,b\in A^{[n]}$, and let $g\in A^{[n]}[y]$. Coordinate equivalence conjecture. If A is a...

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V2-023694 | MATHDB-320167

Liu's LBQ 67 inequality for triangle geometry

摘录 / EXCERPT: Liu's LBQ 67 conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-023695 | MATHDB-320169

The finite-codimension image conjecture for the quantization map

摘录 / EXCERPT: Let $P_n({\mathbb Z})$ be the polynomial Poisson algebra in $2n$ variables over ${\mathbb Z}$, let $\operatorname{Aut}(P_n({\mathbb Z}))$ be its Poisson automorphism group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-023696 | MATHDB-320241

Disk minimality conjecture for sums of Dirichlet eigenvalues

摘录 / EXCERPT: Let Ω be a bounded plane domain of diameter D, and let $0<\lambda_1\leq\lambda_2\leq\cdots$ be the Dirichlet eigenvalues of its Laplacian. Disk eigenvalue-sum conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-023697 | MATHDB-320257

Relative minimal singularity metric extension conjecture

摘录 / EXCERPT: Let $f:X\to S$ be a surjective proper Kähler morphism with connected fibers between connected complex manifolds, let S° be the complement of the discriminant locus, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-023698 | MATHDB-320290

Uniform stability conjecture for the toric Abreu equation

摘录 / EXCERPT: Let Δ be a Delzant polytope, let $K\in C^\infty(\overline{\Delta})$, and let $\mathcal L_K$ be the linear functional defining relative K-stability. The pair (Δ,K) is...

conjecture | open | MathDB 页面 | 原文线索

V2-023699 | MATHDB-320317

The slicing conjecture for isotropic constants

摘录 / EXCERPT: For a convex body $K\subseteq\mathbb{R}^d$, let L_K denote its isotropic constant. The slicing conjecture. There exists a universal constant $c_3>0$ such that for every dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-023700 | MATHDB-320320

The asymptotic Faber-Krahn shape conjecture for subgraphs of the integer lattice

摘录 / EXCERPT: Asymptotic Faber-Krahn shape conjecture. After possibly translating the G_n, the Hausdorff distance between G_n^* and D converges to 0 as $n\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-023701 | MATHDB-320329

Campana's conjecture on the cotangent Kodaira dimension

摘录 / EXCERPT: Let X be a complex projective connected n-fold. For a rank-one coherent subsheaf $L\subset\Omega_X^p$, define ... where $\bar h^0(X,mL)$ is the dimension of the space of sections...

conjecture | open | MathDB 页面 | 原文线索

V2-023702 | MATHDB-320331

The orbifold cotangent-dimension conjecture under C_{n,m}^{orb}

摘录 / EXCERPT: Assume the orbifold $C_{n,m}^{orb}$ conjecture, so that the fibration $r^*:(X\vert\Delta)\dasharrow R^*$ is defined. Let $(R^*\vert\Delta_{(r^*\vert\Delta)})$ be the orbifold base...

conjecture | open | MathDB 页面 | 原文线索

V2-023703 | MATHDB-320332

The log-canonical abundance conjecture

摘录 / EXCERPT: Let $(X\vert\Delta)$ be a log-canonical pair. Define the numerical dimension $\nu(X,D)$ for a rational divisor D by asymptotic growth of sections after adding an ample divisor. A...

conjecture | open | MathDB 页面 | 原文线索

V2-023704 | MATHDB-320355

The polynomial-field conjecture on real Grassmannians

摘录 / EXCERPT: The polynomial-field conjecture. There exists $n(d,k)$ such that, for every section of $\Sigma^d(\gamma_n^k)$ over G_n^k with $n\geq n(d,k)$, there exists $V\in G_n^k$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-023705 | MATHDB-320357

Three-halves logarithmic dimension conjecture for Furstenberg sets

摘录 / EXCERPT: Three-halves logarithmic dimension conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-023706 | MATHDB-320367

Thomas-Yau conjecture for flow-stable Lagrangians

摘录 / EXCERPT: Let (M^{2n},ω,J,Ω) be a Calabi-Yau manifold, and let $L\subseteq M$ be a zero-Maslov-class, almost-calibrated Lagrangian. Suppose that L is flow-stable, meaning that,...

conjecture | open | MathDB 页面 | 原文线索

V2-023707 | MATHDB-320368

Simplified Thomas-Yau conjecture for zero-Maslov-class Lagrangians

摘录 / EXCERPT: Let M be a Calabi-Yau manifold and let Σ be a compact embedded Lagrangian submanifold with zero Maslov class. Simplified Thomas-Yau conjecture. The mean curvature flow of...

conjecture | open | MathDB 页面 | 原文线索

V2-023708 | MATHDB-320402

Tiling conjecture for minimal diskbusting lists of words

摘录 / EXCERPT: Let F_n be a free group, and let a list of cyclically reduced words in F_n be minimal and diskbusting, meaning that it cannot be decomposed into a proper sublist with the same...

conjecture | open | MathDB 页面 | 原文线索

V2-023709 | MATHDB-320422

The Euclidean conjecture on polynomial independence for point configurations

摘录 / EXCERPT: Euclidean conjecture. For every such configuration, the polynomials $\beta_1(z),\dotsc,\beta_n(z)$ are linearly independent over $\mathbb{C}$. This is the first and simplest versio...

conjecture | open | MathDB 页面 | 原文线索

V2-023710 | MATHDB-320423

The hyperbolic conjecture on polynomial independence for point configurations

摘录 / EXCERPT: Hyperbolic conjecture. For every such configuration, the polynomials $\beta_1(z),\dotsc,\beta_n(z)$ are linearly independent over $\mathbb{C}$. This is presented as a stronger vers...

conjecture | open | MathDB 页面 | 原文线索

V2-023711 | MATHDB-320442

The convex-domain bounds conjecture for the Dirichlet fundamental tone

摘录 / EXCERPT: Let Ω be a bounded convex plane domain, with area A, moment of inertia I, and first Dirichlet Laplace eigenvalue λ_1. Convex-domain bounds conjecture. ... Equality...

conjecture | open | MathDB 页面 | 原文线索

V2-023712 | MATHDB-320443

The disk maximizer conjecture for Neumann eigenvalue sums

摘录 / EXCERPT: Let Ω be a bounded convex plane domain of area A and moment of inertia I, and let $\mu_2,\ldots,\mu_n$ be its positive Neumann Laplace eigenvalues. Neumann sum conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-023713 | MATHDB-320461

Kirby's geometric 1-connectedness conjecture for smooth 4-manifolds

摘录 / EXCERPT: Kirby's conjecture. Every simply connected closed smooth 4-manifold is geometrically 1-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-023714 | MATHDB-320462

LC triangulation conjecture for Mazur's 4-manifold

摘录 / EXCERPT: Mazur-manifold LC triangulation conjecture. Mazur's 4-manifold admits an LC triangulation.

conjecture | open | MathDB 页面 | 原文线索

V2-023715 | MATHDB-320463

Heegaard genus versus discrete Morse critical edges

摘录 / EXCERPT: Heegaard-discrete Morse conjecture. Some closed 3-manifold with Heegaard genus g admits discrete Morse functions with less than g critical edges.

conjecture | open | MathDB 页面 | 原文线索

V2-023716 | MATHDB-320470

The k-harmonic submanifold conjecture in Euclidean spaces

摘录 / EXCERPT: The k-harmonic submanifold conjecture. The only k-harmonic submanifolds in Euclidean spaces are the minimal ones.

conjecture | open | MathDB 页面 | 原文线索

V2-023717 | MATHDB-320501

Higher-dimensional Shafarevich conjecture for canonically polarized manifolds

摘录 / EXCERPT: Higher-dimensional Shafarevich conjecture. (B) Families of canonically polarized manifolds over U with Hilbert polynomial h fall into finitely many deformation equivalence clas...

conjecture | open | MathDB 页面 | 原文线索

V2-023718 | MATHDB-320515

Gaiotto-Moore-Neitzke-Hitchin metric coincidence conjecture

摘录 / EXCERPT: Let $\mathcal{M}(R)$ be Hitchin's moduli space with sufficiently large parameter R. The Gaiotto-Moore-Neitzke construction gives an instanton-corrected hyperkähler metric, while...

conjecture | open | MathDB 页面 | 原文线索

V2-023719 | MATHDB-320517

Mirror enumerative correspondence for Hitchin moduli spaces

摘录 / EXCERPT: Let $\mathcal{M}$ be an $SU(2)$ Hitchin moduli space with prescribed singularities, and let $\hat{\mathcal{M}}$ be its SYZ mirror, an $PGL(2)$ Hitchin moduli space with prescribed...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-023720 | MATHDB-320545

Limiting Riemannian Penrose inequality conjecture for constrained maximizers

摘录 / EXCERPT: Limiting Riemannian Penrose inequality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-023721 | MATHDB-320591

Kitaoka's spectral commensurability conjecture for flat tori

摘录 / EXCERPT: Kitaoka's conjecture. If the flat tori $\mathbb{R}^n/L_1$ and $\mathbb{R}^n/L_2$ are spectrally commensurable, then, up to an isometry of $\mathbb{R}^n$, the lattices L_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-023722 | MATHDB-320615

The solid-torus realization conjecture for lens space surgeries

摘录 / EXCERPT: Let $S^1\times D^2$ be the solid torus and let $S^1\times S^2$ be the product of the circle with the 2-sphere. An integer lens space surgery is integer surgery on a knot in...

conjecture | open | MathDB 页面 | 原文线索

V2-023723 | MATHDB-320617

Triangulation conjecture for clean intersections

摘录 / EXCERPT: Let M be a smooth compact manifold and let $(M_i)_{i \in I}$ be a finite family of closed subspaces of M that intersects cleanly. A family intersects cleanly when, locally, its...

conjecture | open | MathDB 页面 | 原文线索

V2-023724 | MATHDB-320622

Classification of surfaces enveloped by two-parameter families of cones of revolution

摘录 / EXCERPT: Let S be a surface, and suppose there is a 2-parametric family of cones of revolution touching S along certain curves distinct from the directrices. Classification conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-023725 | MATHDB-320636

The strict minimal filling conjecture for simple manifolds

摘录 / EXCERPT: Let M be a compact Riemannian manifold with boundary. It is a strict minimal filling if, for every compact Riemannian manifold M' with $\partial M'=\partial M$, the inequalitie...

conjecture | open | MathDB 页面 | 原文线索

V2-023726 | MATHDB-320652

The anisotropic Calderón conjecture for Riemannian manifolds

摘录 / EXCERPT: Let (M,g_1) and (M,g_2) be two compact oriented Riemannian manifolds with smooth boundary and dimension $n\geq 3$. For a metric g, let ... be the Cauchy data set of harmonic fu...

conjecture | open | MathDB 页面 | 原文线索

V2-023727 | MATHDB-320653

Fang-Zhang-Zhang conjecture for normalized Ricci flow

摘录 / EXCERPT: Let X be a closed oriented smooth Riemannian 4-manifold. Write $\lVert X\rVert$ for its Gromov simplicial volume and $\bar{\lambda}(X)$ for Perelman's invariant. A solution to...

conjecture | open | MathDB 页面 | 原文线索

V2-023728 | MATHDB-320654

Volume-entropy generalization of the Fang-Zhang-Zhang conjecture

摘录 / EXCERPT: Let X be a closed oriented smooth Riemannian 4-manifold, and let $\mu(X)$ denote its volume entropy. Assume that ... and that X admits a quasi-non-singular solution to the norm...

conjecture | open | MathDB 页面 | 原文线索

V2-023729 | MATHDB-320661

Generic Hausdorff dimension conjecture for Takagi-function abscissa level sets

摘录 / EXCERPT: Abscissa generic level sets. A full Lebesgue measure set of abscissa points $x\in[0,1]$ have level sets $L(\tau(x))$ that are uncountable and have Hausdorff dimension 0.

conjecture | open | MathDB 页面 | 原文线索

V2-023730 | MATHDB-320670

The completeness conjecture for Constructions A, B and C

摘录 / EXCERPT: Let $PG(2,q)$ be the projective plane of order q. Constructions A, B, and C produce families of complete k-arcs in the size regions stated in the paper's theorem: Construction...

conjecture | open | MathDB 页面 | 原文线索

V2-023731 | MATHDB-320720

Rigidity conjecture for smooth structures on even-dimensional manifold bundles

摘录 / EXCERPT: Let $M\to B$ and $M'\to B$ be tangentially homeomorphic smooth manifold bundles with common fiber a closed oriented even-dimensional manifold, and let $\Theta(M',M)$ denote their s...

conjecture | open | MathDB 页面 | 原文线索

V2-023732 | MATHDB-320728

Conjecture on free relations for symplectic matrix invariants in characteristic two

摘录 / EXCERPT: Let p be the characteristic of the ground field, let $R^{S\!p(n)}$ be the invariant ring for $n\times n$ generic matrices under the symplectic group, and let K_{∞} denote...

conjecture | open | MathDB 页面 | 原文线索

V2-023733 | MATHDB-320763

Higher-dimensional Goldberg-Sachs conjecture for complex Riemannian manifolds

摘录 / EXCERPT: Let $(\mathcal{M},\bm{g})$ be a $(2m+\epsilon)$-dimensional complex Riemannian manifold, and let $\mathcal{N}$ be a maximal totally null subbundle of its tangent bundle, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-023734 | MATHDB-320764

The essential spectral gap conjecture for convex co-compact groups

摘录 / EXCERPT: Essential spectral gap conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-023735 | MATHDB-320785

The plane radial-density conjecture for circles, undularies, and pinched circles

摘录 / EXCERPT: Plane radial-density conjecture. For $k\in[0,\infty)$, the following curves are isoperimetric:

conjecture | open | MathDB 页面 | 原文线索

V2-023736 | MATHDB-320786

The radial-density hypersphere conjecture in Euclidean space

摘录 / EXCERPT: Radial-density hypersphere conjecture. In $\mathbb{R}^{n}$ with density r^{p}, hyperspheres through the origin are uniquely isoperimetric.

conjecture | open | MathDB 页面 | 原文线索

V2-023737 | MATHDB-320789

The good minimal model conjecture for klt pairs

摘录 / EXCERPT: Good minimal model conjecture. If $K_X+\Delta$ is pseudo-effective, then (X,Δ) has a good minimal model.

conjecture | open | MathDB 页面 | 原文线索

V2-023738 | MATHDB-320790

Genus-like expansion conjecture for the D3-brane path integral

摘录 / EXCERPT: Let Y be a target space-time and let $Z_{D3}(Y)$ denote the path integral of a perturbative D3-brane theory, with contributions summed over fundamental closed orientable 4-mani...

conjecture | open | MathDB 页面 | 原文线索

V2-023739 | MATHDB-320800

The higher-dimensional filling volume accumulation conjecture

摘录 / EXCERPT: Let M be a finite-volume hyperbolic n-manifold with torus cusps, let V be its complete hyperbolic interior, and let $M(T_1,\ldots,T_m)$ denote the manifold obtained by fillin...

conjecture | open | MathDB 页面 | 原文线索

V2-023740 | MATHDB-320842

Subpolynomial restriction norm conjecture for geodesic circles

摘录 / EXCERPT: Subpolynomial restriction norm conjecture. For any fixed $\varepsilon>0$ and every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-023741 | MATHDB-320850

The converse geodesic conjecture for zero-dimensional cusps of the Kähler moduli space

摘录 / EXCERPT: Converse geodesic conjecture. Every geodesic in $KM(X)$ converging to $[v]$ is a linear degeneration.

conjecture | open | MathDB 页面 | 原文线索

V2-023742 | MATHDB-320883

Redundancy of the diagonal transitivity condition for finite-dimensional unit balls

摘录 / EXCERPT: Let $\mathcal{S}$ be a finite-dimensional compact convex set with $\operatorname{Aff}(\mathcal{S})=\mathbb{R}^n$ for some finite n, and let G be the group of bijective affine t...

conjecture | open | MathDB 页面 | 原文线索

V2-023743 | MATHDB-320898

Magnitude's conjecture for intrinsic volumes of convex Euclidean sets

摘录 / EXCERPT: Let A be a compact convex subspace of ℓ_2^N. For $0\leq i\leq N$, let $V_i(A)$ denote its i-th intrinsic volume, and let ω_i be the volume of the Euclidean unit bal...

conjecture | open | MathDB 页面 | 原文线索

V2-023744 | MATHDB-320899

Magnitude's conjecture for intrinsic volumes of compact ℓ_1-convex sets

摘录 / EXCERPT: Let A be a compact ℓ_1-convex subspace of ℓ_1^N, and let $V'_i(A)$ denote its i-th ℓ_1-intrinsic volume. The ℓ_1 magnitude-intrinsic-volume conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-023745 | MATHDB-320905

Minimal fillings in general position are non-degenerate binary trees

摘录 / EXCERPT: General-position conjecture. Every minimal filling of a finite metric space in general position is a non-degenerate binary tree.

conjecture | open | MathDB 页面 | 原文线索

V2-023746 | MATHDB-320906

Minimal fillings possess exact tours

摘录 / EXCERPT: Exact-tour conjecture. Every minimal filling possesses an exact tour.

conjecture | open | MathDB 页面 | 原文线索

V2-023747 | MATHDB-320907

The maximal-tour bound for minimal filling weight

摘录 / EXCERPT: Maximal-tour conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-023748 | MATHDB-320908

The tour formula for minimal filling weight

摘录 / EXCERPT: Minimal-filling tour formula. For an arbitrary pseudo-metric space ${\cal M}=(M,\rho)$,

conjecture | open | MathDB 页面 | 原文线索

V2-023749 | MATHDB-320911

Polar-dual Neumann eigenvalue sum conjecture for convex domains

摘录 / EXCERPT: Polar-dual Neumann eigenvalue sum conjecture. The scale-invariant quantity

conjecture | open | MathDB 页面 | 原文线索

V2-023750 | MATHDB-320914

Turaev-Viro commensurability conjecture for geometric 3-manifolds

摘录 / EXCERPT: Let M and N be closed irreducible geometric 3-manifolds. Their abelian and $SU(2)$ Turaev-Viro invariants are the collections of invariants associated to abelian and $SU(2)$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-023751 | MATHDB-320915

Finiteness and unboundedness for Turaev-Viro classes of hyperbolic fibered 3-manifolds

摘录 / EXCERPT: For a closed 3-manifold M, let ${\mathcal X}^{TV}(M)$ denote the set of homeomorphism classes of closed orientable 3-manifolds having the same Turaev-Viro invariants as M. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-023752 | MATHDB-320919

Rogalski-Stafford GK-dimension-4 non-noetherianity conjecture

摘录 / EXCERPT: Let S be a connected graded domain of GK-dimension 4, generated in degree 1, and birational to a commutative surface. Rogalski-Stafford's conjecture. Then S is not noetheri...

conjecture | refuted | MathDB 页面 | 原文线索

V2-023753 | MATHDB-320933

The supermanifold Hodge numbers and singularity-category gaps conjecture

摘录 / EXCERPT: Consider supermanifolds, their Hodge numbers, and the associated categories of singularities. A gap is a missing interval in the generation-time spectrum of such a category, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-023754 | MATHDB-320943

Moduli conjecture for conic bundles associated with F^{2+}_5

摘录 / EXCERPT: Let C be the canonically embedded complete-intersection curve of degree 8 and genus 5 arising as the base locus of a net of quadrics in $\mathbb{P}^4$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-023755 | MATHDB-320992

Chudnovsky's conjecture on Waldschmidt constants

摘录 / EXCERPT: Let $S=\{p_1,\dots,p_r\}$ be distinct points in $\mathbb P^N$ over an algebraically closed field, and let $a(S,m)$ and $e(S)$ be defined by ... Chudnovsky's conjecture, proved for...

conjecture | open | MathDB 页面 | 原文线索

V2-023756 | MATHDB-321020

The non-occurrence conjecture for the \mathbb U_n configuration in symplectic contractions

摘录 / EXCERPT: Let $\mathbb U_n$ be the intersection-matrix type arising from the folded root system $\mathbb A_{2n}$. A symplectic contraction is a contraction $\pi:X\rightarrow Y$ from a symple...

conjecture | open | MathDB 页面 | 原文线索

V2-023757 | MATHDB-321021

The root-and-weight lattice conjecture for symplectic contractions

摘录 / EXCERPT: Suppose the preceding non-occurrence conjecture holds, and let $\pi:X\rightarrow Y$ be a local symplectic contraction satisfying the purely divisorial conditions. Then there are a...

conjecture | open | MathDB 页面 | 原文线索

V2-023758 | MATHDB-321050

Integrability conjecture for the periodic Full Kostant-Toda system

摘录 / EXCERPT: Let $\mathcal{T}^{(k)}_{\lambda}$ be the family of phase spaces for the periodic Full Kostant-Toda system, let $\widetilde{\mathcal{F}}^{(k)}_{\lambda}$ be the indicated family of...

conjecture | open | MathDB 页面 | 原文线索

V2-023759 | MATHDB-321051

Disk minimization conjecture for convex-domain Neumann eigenvalue sums

摘录 / EXCERPT: Let Ω be a convex domain of given diameter D, and let $0=\mu_1(\Omega)<\mu_2(\Omega)\leq\mu_3(\Omega)\leq\cdots$ be the Neumann Laplacian eigenvalues. Disk minimization co...

conjecture | open | MathDB 页面 | 原文线索

V2-023760 | MATHDB-321057

The polynomial bounded relative K-theory conjecture for a closed solvmanifold group

摘录 / EXCERPT: Let M^3 be the closed 3-dimensional solvmanifold described in the source, let $\pi=\pi_1M^3$, and let $\mathcal P$ be the bounding class of polynomial functions. Let $E\pi$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-023761 | MATHDB-321069

The polyhedral symplectic cone conjecture for symplectic 4-manifolds

摘录 / EXCERPT: Polyhedral symplectic cone conjecture. There exists a polyhedral symplectic cone C with the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-023762 | MATHDB-321070

Generalized lower bound conjecture for triangulated manifolds

摘录 / EXCERPT: Let M be a triangulation of a connected closed d-manifold. Write $g_j(M)$ for its g-numbers, $\beta_i(M;\mathbb F)$ for Betti numbers over a field $\mathbb F$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-023763 | MATHDB-321083

Weak Calabi flow convergence conjecture for big line bundles

摘录 / EXCERPT: Weak Calabi flow convergence conjecture. The Ω-Kähler flow exists in a weak sense and converges to a solution of the Calabi problem

conjecture | open | MathDB 页面 | 原文线索

V2-023764 | MATHDB-321086

The cone-angle existence conjecture for Fano manifolds

摘录 / EXCERPT: The cone-angle existence conjecture. There is a cone-singularity solution ω_β to (27) for every parameter $\beta\in(0,R(X))$. If $R(X)<1$, there is no solution for param...

conjecture | open | MathDB 页面 | 原文线索

V2-023765 | MATHDB-321087

The degeneration conjecture at the critical cone angle

摘录 / EXCERPT: Let X be a Fano manifold and D a smooth divisor in $-K_X$. Let $0<\beta\leq 1$, and for a pair (Y,Δ) with an S^1 action define the log Futaki invariant by ... where H...

conjecture | open | MathDB 页面 | 原文线索

V2-023766 | MATHDB-321088

Spherical cap contact-density conjecture

摘录 / EXCERPT: Let $\mathbf{c}_i+\mathbf{B}\in {\cal P}$ be tangent to $\mathbf{c}_j+\mathbf{B}\in {\cal P}$ for all $j\in T_i$, where $T_i\subset\{1,2,\dots,n\}$ is the set of indices satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-023767 | MATHDB-321093

The conjecture on the asymptotic growth of one-dimensional geodesics

摘录 / EXCERPT: Asymptotic growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023768 | MATHDB-321103

Campana's closedness conjecture for the class C property

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to\Delta$ be a holomorphic family of compact complex manifolds, with fibres X_t. A compact complex manifold is said to have the class $\mathcal{C}$ property...

conjecture | open | MathDB 页面 | 原文线索

V2-023769 | MATHDB-321104

Demailly's closedness conjecture for strongly Gauduchon manifolds

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to\Delta$ be a holomorphic family of compact complex manifolds, with fibres X_t. A compact complex manifold is strongly Gauduchon (sG) if it has the strongly...

conjecture | open | MathDB 页面 | 原文线索

V2-023770 | MATHDB-321105

Closedness conjecture for balanced manifolds

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to\Delta$ be a holomorphic family of compact complex manifolds, with fibres X_t. A compact complex manifold is balanced if it has the balanced property. Bala...

conjecture | open | MathDB 页面 | 原文线索

V2-023771 | MATHDB-321119

Götsche's polynomiality conjecture for nodal curves on surfaces

摘录 / EXCERPT: Let S be a smooth, irreducible, complex projective surface, let $4\mathscr{L}4$ be a line bundle on S, and write $Y=|\mathscr{L}|=\mathbb{P}(H^0(S,\mathscr{L}))$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-023772 | MATHDB-321121

Bell-polynomial formula for multiple-point corrections

摘录 / EXCERPT: Let S be the surface, $\mathscr{L}$ the line bundle, and $Y=|\mathscr{L}|$ of dimension N. Let f be the relevant map, let $M_i(S,\mathscr{L})$ be the classes defined by ... let...

conjecture | open | MathDB 页面 | 原文线索

V2-023773 | MATHDB-321122

Polynomiality conjecture for the residual Segre class

摘录 / EXCERPT: Let S be a smooth projective surface, let $\mathscr{L}$ be a line bundle, and let $F=S\times Y$. Let $L=c_1(\mathscr{L})$, $K=c_1(\mathscr{K}_S)$, and let $x=c_2(S)$; regard thes...

conjecture | open | MathDB 页面 | 原文线索

V2-023774 | MATHDB-321136

A quantitative Busemann-Petty conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: Quantitative Busemann-Petty conjecture. There exists a universal constant $c>0$ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-023775 | MATHDB-321156

Hyperbolic torsion approximation conjecture for finite-volume 3-manifolds

摘录 / EXCERPT: Let Y be a finite-volume hyperbolic 3-manifold, and let $Y_n_{n\in \mathbb{N}}$ be a nested collection of finite covers such that ... is trivial. Here $\tau(Y_n,\alpha_{can})$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-023776 | MATHDB-321168

Rank-gradient versus Heegaard-gradient conjecture

摘录 / EXCERPT: Let M be a finite-volume hyperbolic 3-manifold and let $\{M_i\longrightarrow M\}$ be a family of finite-sheeted covers. Write $\operatorname{rgr}(M,\{M_i\})$ for the rank gradi...

conjecture | open | MathDB 页面 | 原文线索

V2-023777 | MATHDB-321182

Atiyah's non-vanishing conjecture for the Atiyah determinant

摘录 / EXCERPT: Let $x_1,dots,x_n$ be distinct points in $\mathbb{R}^3$, and let $D(x_1,dots,x_n)$ denote the Atiyah determinant constructed from them. Atiyah's non-vanishing conjecture. ... The con...

conjecture | open | MathDB 页面 | 原文线索

V2-023778 | MATHDB-321183

Atiyah-Sutcliffe's multiplicative conjecture for the Atiyah determinant

摘录 / EXCERPT: Let $x_1,dots,x_n$ be distinct points in $\mathbb{R}^3$, and let $D(x_1,dots,x_n)$ be the Atiyah determinant; for each i, write $D(x_1,dots,x_{i-1},x_{i+1},dots,x_n)$ for the det...

conjecture | open | MathDB 页面 | 原文线索

V2-023779 | MATHDB-321184

The four-point inequality conjecture for Atiyah-Sutcliffe angular quantities

摘录 / EXCERPT: For four distinct points x_1,x_2,x_3,x_4 in $\mathbb{R}^3$, let the inequality denoted by ... holds for every such four-point configuration. The paper notes that this inequality ex...

conjecture | open | MathDB 页面 | 原文线索

V2-023780 | MATHDB-321185

The four-point comparison inequality conjecture for Atiyah-Sutcliffe quantities

摘录 / EXCERPT: For four distinct points x_1,x_2,x_3,x_4 in $\mathbb{R}^3$, let the expressions denoted by ... in the source be the corresponding angular quantities. Four-point comparison inequali...

conjecture | open | MathDB 页面 | 原文线索

V2-023781 | MATHDB-321198

Quadratic curvature blow-up conjecture for Ricci-flat manifolds

摘录 / EXCERPT: Let M^n be a Ricci-flat manifold, and let $alpha$ be the exponent appearing in Corollary. Quadratic curvature blow-up conjecture. $alpha$ in Corollary can be chosen to be 1. A...

conjecture | open | MathDB 页面 | 原文线索

V2-023782 | MATHDB-321205

Niederreiter's splitting subspace conjecture

摘录 / EXCERPT: Let $\alpha\in{\mathbb F}_{q^{mn}}$ satisfy ... An m-dimensional α-splitting subspace is an m-dimensional ${\mathbb F}_q$-linear subspace $W\subseteq{\mathbb F}_{q^{mn}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-023783 | MATHDB-321206

Pointed splitting subspace conjecture

摘录 / EXCERPT: Let $\alpha\in{\mathbb F}_{q^{mn}}$ satisfy ... and let $x\in{\mathbb F}_{q^{mn}}^*$. An m-dimensional pointed α-splitting subspace with base point x is a pair (W,x) whe...

conjecture | open | MathDB 页面 | 原文线索

V2-023784 | MATHDB-321245

Frobenius manifold correspondence for orbifold projective lines, quivers, and simple elliptic singularities

摘录 / EXCERPT: For positive integers a_1,a_2,a_3, let $M_{{\mathbb P}^1_{a_1,a_2,a_3}}$ be the Frobenius manifold constructed from the Gromov--Witten theory of the orbifold projective line, let...

conjecture | open | MathDB 页面 | 原文线索

V2-023785 | MATHDB-321246

FJRW and singularity Frobenius manifold correspondence for simple elliptic singularities

摘录 / EXCERPT: Let T_{a_1,a_2,a_3} define a simple elliptic singularity of type $\widetilde{E}_6$, $\widetilde{E}_7$, or $\widetilde{E}_8$. Let $d=6,7,8$ for these three types, respectively. De...

conjecture | open | MathDB 页面 | 原文线索

V2-023786 | MATHDB-321288

The explicit leading constant for optimal spherical \(\mathbb{L}_2\)-discrepancy on \(\mathbb{S}^2\)

摘录 / EXCERPT: The spherical discrepancy constant conjecture. If both the fundamental optimal Riesz-energy conjecture and the Kuijlaars--Saff conjecture hold, then

conjecture | open | MathDB 页面 | 原文线索

V2-023787 | MATHDB-321338

Five-link characterization of the Reinhardt minimizer

摘录 / EXCERPT: Let K be a hexameral domain with multi-curve σ around its boundary. Let t_0,t_1 be the parameters defining the non-repeating boundary interval. Suppose that the entire n...

conjecture | open | MathDB 页面 | 原文线索

V2-023788 | MATHDB-321344

Non-existence conjecture for endomorphisms satisfying weak Condition dagger

摘录 / EXCERPT: Let M be the closed manifold under consideration, let $Z^1(M)$ denote the space of closed 1-forms on M, and let μ be an endomorphism of the relevant space of 1-forms. C...

conjecture | open | MathDB 页面 | 原文线索

V2-023789 | MATHDB-321349

Iitaka's uniformisation conjecture for compact Kähler manifolds

摘录 / EXCERPT: Iitaka's conjecture. Then X is a torus, up to finite étale cover.

conjecture | open | MathDB 页面 | 原文线索

V2-023790 | MATHDB-321350

Non-vanishing conjecture for klt pairs

摘录 / EXCERPT: Non-vanishing conjecture for klt pairs. Then

conjecture | open | MathDB 页面 | 原文线索

V2-023791 | MATHDB-321360

The Fano Kähler-Ricci soliton conjecture for Sasaki-Einstein circle bundles

摘录 / EXCERPT: Sasaki-Einstein circle-bundle conjecture. If there exists a Kähler-Ricci soliton on M, then the S^1-bundle associated to K_M admits a Sasaki-Einstein metric with a suitable c...

conjecture | open | MathDB 页面 | 原文线索

V2-023792 | MATHDB-321367

Symplectic Seifert fibration conjecture for four-manifolds

摘录 / EXCERPT: Let X be a symplectic 4-manifold and let $\pi:X\rightarrow Y$ be a Seifert fibration. A fibered 3-manifold is a 3-manifold admitting a fibration over ${\mathbb S}^1$. Sympl...

conjecture | open | MathDB 页面 | 原文线索

V2-023793 | MATHDB-321368

The fat-points symbolic-power evolution conjecture

摘录 / EXCERPT: Let $K[\mathbb{P}^{N}]$ be the homogeneous coordinate ring, let M be its homogeneous maximal ideal, and let ... be any fat-points ideal. Fat-points evolution conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-023794 | MATHDB-321371

Smooth lattice-polytope volume conjecture

摘录 / EXCERPT: Smooth lattice-polytope volume conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-023795 | MATHDB-321377

The structure conjecture for the algebras \(\mathcal V^{\mathbb{C},\lambda,n}\)

摘录 / EXCERPT: The structure conjecture. For every $i>n$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-023796 | MATHDB-321384

Keith's dimension conjecture for the passant-internal incidence null space

摘录 / EXCERPT: Keith's conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-023797 | MATHDB-321390

Tian-Yau-Donaldson conjecture for constant scalar curvature Kähler metrics

摘录 / EXCERPT: Let (X,J) be a polarized manifold, with L an ample line bundle on (X,J). A polarization (X,J,L) is K-stable in the sense of algebraic K-stability, and $c_1(L)$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-023798 | MATHDB-321391

Donaldson's cone-singularity conjecture along an anticanonical divisor

摘录 / EXCERPT: Donaldson's conjecture. There is a cone-singularity solution ω_β to this equation for every parameter $\beta\in(0,R(X))$. If $R(X)<1$, there is no solution for any param...

conjecture | open | MathDB 页面 | 原文线索

V2-023799 | MATHDB-321392

Logarithmic Tian-Yau-Donaldson conjecture for conic Kähler metrics

摘录 / EXCERPT: Let (X,Y) be a pair consisting of a compact Kähler manifold X and a divisor Y, and let (X,Y) be log-K-stable in the logarithmic sense. A constant scalar curvature conic Käh...

conjecture | open | MathDB 页面 | 原文线索

V2-023800 | MATHDB-321444

Stability of the positive mass theorem for special-surface tubular neighborhoods

摘录 / EXCERPT: Let $\mathcal{M}$ be a subclass of asymptotically flat three-dimensional Riemannian manifolds with nonnegative scalar curvature and no interior closed minimal surfaces, with either...

conjecture | open | MathDB 页面 | 原文线索

V2-023801 | MATHDB-321445

Uniform lower bound conjecture for the Ricci potential before finite-time singularities

摘录 / EXCERPT: Uniform lower bound conjecture. There is a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-023802 | MATHDB-321468

Väisälä's uniqueness, prolongation and convexity conjectures for quasihyperbolic balls

摘录 / EXCERPT: Väisälä's uniqueness, prolongation and convexity conjectures. There are universal constants $c_u,c_p,c_c>0$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-023803 | MATHDB-321471

Minimal-dilatation asymptotics for closed surfaces from the magic manifold

摘录 / EXCERPT: Magic-manifold minimal-dilatation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023804 | MATHDB-321472

Asymptotic minimal dilatation for fibrations on the magic manifold filling N(1)

摘录 / EXCERPT: N(1)-fibration asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023805 | MATHDB-321498

Generalized log minimal model and termination conjecture for lc pairs

摘录 / EXCERPT: Generalized log minimal model conjecture. The pair $(X/Z,B)$ has either a Mori fibre space or a log minimal model $(Y/Z,B_Y)$; if K_Y+B_Y is nef$/Z$, then it is semi-ample$/Z$; a...

conjecture | open | MathDB 页面 | 原文线索

V2-023806 | MATHDB-321499

Relative semi-ampleness conjecture for dlt pairs

摘录 / EXCERPT: Let $(X/Z,B)$ be a $\mathbb Q$-factorial dlt pair, and let ... Assume that B is a $\mathbb Q$-divisor and Z is affine. Suppose that K_X+B is nef$/Z$, that $(K_X+B)|_S$ is semi-...

conjecture | open | MathDB 页面 | 原文线索

V2-023807 | MATHDB-321540

Boavida de Brito-Weiss non-discretized operadic model for Taylor towers

摘录 / EXCERPT: Let M be a manifold, let $\mathcal{O}^s_\infty(M)$ and $\widetilde{\mathcal{O}}^s_k(\mathbb{R}^m)$ be the source categories of the continuous context-free functors under consider...

conjecture | open | MathDB 页面 | 原文线索

V2-023808 | MATHDB-321546

Chebyshev and interval-based fractal characterization of gapless spectra

摘录 / EXCERPT: Let K be the finitely ramified fractal, let $R(z)$ be the associated rational map, and let $\mathcal{J}_R$ denote its Julia set. Under the conditions of Theorems and, call the s...

conjecture | open | MathDB 页面 | 原文线索

V2-023809 | MATHDB-321547

Complex-dimension conjecture for non-interval fractals

摘录 / EXCERPT: Let K be the fractal under consideration, and assume the conditions of Theorems and. A complex dimension is called nontrivial when it is one of the complex dimensions considered...

conjecture | open | MathDB 页面 | 原文线索

V2-023810 | MATHDB-321559

Bibikov's hyperbolic Feuerbach conjecture

摘录 / EXCERPT: Let ABC be a triangle with sides a, b, and c. Let M_a, M_b, and M_c be points on the sides a, b, and c, respectively, such that the cevians AM_a, BM_b, and...

conjecture | open | MathDB 页面 | 原文线索

V2-023811 | MATHDB-321573

Lackenby's Heegaard gradient conjecture

摘录 / EXCERPT: Lackenby's Heegaard gradient conjecture. If $\chi^h_{\mathcal{L}}(M)=0$ for a family of covers, then M virtually fibres over a circle. The source motivates this by noting that vi...

conjecture | open | MathDB 页面 | 原文线索

V2-023812 | MATHDB-321575

Dickenstein-Nill codegree conjecture for lattice polytopes

摘录 / EXCERPT: Dickenstein-Nill conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-023813 | MATHDB-321576

Dickenstein-Nill rational codegree conjecture for lattice polytopes

摘录 / EXCERPT: Dickenstein-Nill rational codegree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-023814 | MATHDB-321581

The lower-bound claim for Hermite's constant of symplectic lattices in dimension a power of two

摘录 / EXCERPT: Lower-bound claim.

conjecture | open | MathDB 页面 | 原文线索

V2-023815 | MATHDB-321611

Prokhorov-Shokurov conjectures on the moduli part of the canonical bundle formula

摘录 / EXCERPT: Prokhorov-Shokurov conjectures. The following properties are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-023816 | MATHDB-321645

Pandharipande's direct BPS-count conjecture for unramified Gromov-Witten invariants

摘录 / EXCERPT: Pandharipande's direct BPS-count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023817 | MATHDB-321660

The spherical generalized upper bound conjecture for random hyperplane splits

摘录 / EXCERPT: Let $n\geq d$ points in general position in $\mathbb{R}^d$ be given, and let N be the number of points on the larger side of a random hyperplane split. Let N^* be the analogous...

conjecture | open | MathDB 页面 | 原文线索

V2-023818 | MATHDB-321696

The generic nonvanishing conjecture for conformally invariant nonlinear Dirac equations

摘录 / EXCERPT: Let M be a connected Riemannian spin manifold, and set ... Let ϕ be a nonzero spinor satisfying ... for some Riemannian metric g_0 and $c\in\mathbb{R}$. Generic nonvanishing c...

conjecture | open | MathDB 页面 | 原文线索

V2-023819 | MATHDB-321723

Global Chen conjecture for complete biharmonic submanifolds

摘录 / EXCERPT: Global Chen conjecture. Any complete biharmonic immersed submanifold in $\mathbb{E}^N$ is minimal.

conjecture | open | MathDB 页面 | 原文线索

V2-023820 | MATHDB-321725

Bijection conjecture for gradient staircases under metric degeneration

摘录 / EXCERPT: Gradient-staircase bijection conjecture. There exists $\delta>0$ such that, for every s with $0<s\leq\delta$, there is a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-023821 | MATHDB-321740

Sharpness conjecture for tropical gonality bounds of lattice polygons

摘录 / EXCERPT: Let $\Delta\subset\mathbb{R}^2$ be a two-dimensional lattice polygon. A regular subdivision $\Delta_1,\ldots,\Delta_r$ determines a metric graph $\Gamma(\Delta_1,\ldots,\Delta_r)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-023822 | MATHDB-321759

The conjecture on foliations from non-generically ample cotangent bundles

摘录 / EXCERPT: Let X be a non-uniruled terminal n-fold. Suppose that $ Omega^1_X$ is not generically ample and $K_X \not\equiv 0$. A finite cover $X' \to X$ is étale in codimension 1 if it is...

conjecture | open | MathDB 页面 | 原文线索

V2-023823 | MATHDB-321760

The conjecture on zeros of twisted multivector fields

摘录 / EXCERPT: Let X be a projective manifold, let L be a numerically trivial line bundle, and let $v \in H^0(X,\bigwedge^q T_X \otimes L)$ be a non-trivial section. A zero of v is a point...

conjecture | open | MathDB 页面 | 原文线索

V2-023824 | MATHDB-321762

The Albanese submersion conjecture for nef anticanonical manifolds

摘录 / EXCERPT: Let X be a projective manifold. The anticanonical bundle is nef when $-K_X$ has nonnegative degree on every curve. The Albanese map is the canonical morphism from X to its Alba...

conjecture | open | MathDB 页面 | 原文线索

V2-023825 | MATHDB-321763

Sufficient nefness conjecture for cotangent bundles of non-uniruled manifolds

摘录 / EXCERPT: Let X be a non-uniruled projective manifold. A vector bundle E on X is sufficiently nef if, for every $x \in X$, there is a family of curves through x covering X such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-023826 | MATHDB-321772

The non-amenability criterion for dimension gaps on hyperbolic surfaces

摘录 / EXCERPT: Let S be a hyperbolic surface given by some Fuchsian group. Assume S is of infinite type with strongly bounded geometry, and let $\mathcal{G}$ be the graph associated to some u...

conjecture | open | MathDB 页面 | 原文线索

V2-023827 | MATHDB-321775

The algebraic-dimension conjecture for submanifolds with ample normal bundle

摘录 / EXCERPT: Algebraic-dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-023828 | MATHDB-321776

The interior-of-the-Mori-cone conjecture for curves with ample normal bundle

摘录 / EXCERPT: Interior-of-the-Mori-cone conjecture. The class $[C]$ lies in the interior of

conjecture | open | MathDB 页面 | 原文线索

V2-023829 | MATHDB-321782

Converse gluing conjecture for Alexandrov spaces

摘录 / EXCERPT: Let $X,Y\in \text{Alex}^n(\kappa)$ be Alexandrov spaces with non-empty boundary, and let $f:\partial X\to\partial Y$ be a homeomorphism. If the gluing $X\cup_fY$ is an Alexandrov s...

conjecture | open | MathDB 页面 | 原文线索

V2-023830 | MATHDB-321783

Equivariant Gopakumar-Vafa correspondence for local \(\mathbb{P}^1\)

摘录 / EXCERPT: Equivariant Gopakumar-Vafa correspondence. The Gopakumar-Vafa formula holds:

conjecture | open | MathDB 页面 | 原文线索

V2-023831 | MATHDB-321810

Smurov's plank covering conjecture for the unit ball

摘录 / EXCERPT: Smurov's conjecture. There exists a constant C_d, possibly depending on d, such that any collection of planks in $\mathbb R^d$ with total width at least C_d can be translated...

conjecture | open | MathDB 页面 | 原文线索

V2-023832 | MATHDB-321811

Smurov's plank covering conjecture for planar convex bodies

摘录 / EXCERPT: Smurov's conjecture. The planks can be translated so that their translates cover B.

conjecture | open | MathDB 页面 | 原文线索

V2-023833 | MATHDB-321834

The Abelianity Conjecture for special compact Kähler manifolds

摘录 / EXCERPT: Abelianity Conjecture. Every special compact Kähler manifold X has an almost abelian fundamental group; that is, $\pi_1(X)$ has an abelian subgroup of finite index.

conjecture | solved | MathDB 页面 | 原文线索

V2-023834 | MATHDB-321841

Equality of the geometric and analytic Grauert tube radii

摘录 / EXCERPT: Equality conjecture. The two radii should be equal:

conjecture | open | MathDB 页面 | 原文线索

V2-023835 | MATHDB-321845

Gronwall's conjecture on uniqueness of linearization for non-hexagonal 3-webs

摘录 / EXCERPT: Let $\mathcal W$ and $\mathcal W'$ be germs of linear 3-webs on $(\mathbb C^2,0)$ with non-vanishing curvature, and let $\varphi:(\mathbb C^2,0)\to(\mathbb C^2,0)$ be a germ of b...

conjecture | open | MathDB 页面 | 原文线索

V2-023836 | MATHDB-321900

Nonexistence of positive sectional curvature on $\mathbb{S}^{2}\times\mathbb{S}^{2}$

摘录 / EXCERPT: Let $\mathbb{S}^{2}\times\mathbb{S}^{2}$ be the product of two 2-spheres. Hopf's product conjecture. There is no Riemannian metric on $\mathbb{S}^{2}\times\mathbb{S}^{2}$ with posi...

conjecture | open | MathDB 页面 | 原文线索

V2-023837 | MATHDB-321904

Camere's conjecture on fixed loci of symplectic involutions on K3^[2]-type manifolds

摘录 / EXCERPT: Let X be a hyperkähler manifold deformation equivalent to the Hilbert square of a $K3$ surface, and let φ be an involution of X preserving the holomorphic symplectic fo...

conjecture | open | MathDB 页面 | 原文线索

V2-023838 | MATHDB-321926

K-semistability conjecture for Picard-rank-one Fano manifolds

摘录 / EXCERPT: Let X be an arbitrary Fano manifold of Picard rank 1, so that $\rho(X)=1$, and let $-K_X$ be its anticanonical polarization. K-semistability conjecture. The polarized Fano mani...

conjecture | open | MathDB 页面 | 原文线索

V2-023839 | MATHDB-321927

Global maximizer conjecture for Perelman's entropy on a Fano manifold

摘录 / EXCERPT: Let M be a Fano manifold of complex dimension n, let X be the holomorphic vector field used to define the modified Mabuchi K-energy, let V denote the volume normalization a...

conjecture | open | MathDB 页面 | 原文线索

V2-023840 | MATHDB-321940

The Log-Convex Density Conjecture for radial densities

摘录 / EXCERPT: Log-Convex Density Conjecture. In $\mathbb{R}^{n}$ with radial log-convex simple density, balls about the origin are isoperimetric for every volume.

conjecture | open | MathDB 页面 | 原文线索

V2-023841 | MATHDB-321966

Volume determination conjecture for stable scissors congruence classes

摘录 / EXCERPT: Let M be a hyperbolic manifold, and let its stable scissors congruence class mean its scissors congruence class after adjoining a common hyperbolic polyhedron. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023842 | MATHDB-321977

Diffeomorphism conjecture for Luttinger surgeries on K3 and Enriques surfaces

摘录 / EXCERPT: Let X be a K3 surface or an Enriques surface, and perform a Luttinger surgery along any embedded Lagrangian torus in X, obtaining $\tilde X$. Diffeomorphism conjecture. The man...

conjecture | open | MathDB 页面 | 原文线索

V2-023843 | MATHDB-321978

Luttinger-surgery generation conjecture for torus bundles over the torus

摘录 / EXCERPT: Let X be any smooth oriented torus bundle over the torus. Then there exists a symplectic structure ω on X such that (X,ω) can be obtained by applying Luttinger su...

conjecture | open | MathDB 页面 | 原文线索

V2-023844 | MATHDB-322009

Lower bound for the minimum distance of complete intersections

摘录 / EXCERPT: Complete-intersection distance conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-023845 | MATHDB-322066

Convergence of differential forms on fat triangulations of almost Riemannian manifolds

摘录 / EXCERPT: Let $M^n \subset \mathbb{R}^N$ be a $C^{1,1}$ compact manifold with boundary, and let $\{P_m\}_{m \in \mathbb{N}}$ be a sequence of fat PL (polyhedral) manifolds with boundary cl...

conjecture | open | MathDB 页面 | 原文线索

V2-023846 | MATHDB-322069

The exact Lagrangian cobordism generation conjecture for Legendrian knots

摘录 / EXCERPT: Let $cLambda$ be a Legendrian knot in $cmathbb{R}^{3}$, and let $L_{cLambda}$ be an exact Lagrangian filling with $cemptysetcprec^{ex}_{L_{cLambda}}cLambda$. The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-023847 | MATHDB-322076

Boundary and bulk scaling-limit conjecture for the lambda-self-avoiding walk

摘录 / EXCERPT: Boundary and bulk lambda-self-avoiding-walk scaling conjecture. The partition function and rescaled measures satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-023848 | MATHDB-322080

The MMA existence and derived equivalence conjecture for Gorenstein threefolds

摘录 / EXCERPT: Let R be a normal Gorenstein threefold over $\mathbb{C}$ with only canonical, equivalently rational, singularities. An MMA is a maximal modification algebra of R, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-023849 | MATHDB-322083

Generalized mirror transformation for open Gromov-Witten invariants

摘录 / EXCERPT: Let N and k be the parameters of the projective hypersurface, let P_f be the set of partitions $\sigma_f=(f_1,\ldots,f_{l(\sigma_f)})$ of f, and let ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-023850 | MATHDB-322110

Birational diversity conjecture for the constructed symplectic Calabi-Yau manifolds

摘录 / EXCERPT: Birational diversity conjecture. No symplectic Calabi-Yau manifold arising in the proof of the main theorem is symplectically birational to an algebraic Calabi-Yau manifold. Moreov...

conjecture | open | MathDB 页面 | 原文线索

V2-023851 | MATHDB-322137

s-injectivity conjecture for tensor tomography on nontrapping surfaces

摘录 / EXCERPT: Let (M,g) be a compact nontrapping surface with strictly convex boundary. For a smooth symmetric m-tensor field f on M, let $I_m f$ denote its ray transform. A tensor field...

conjecture | open | MathDB 页面 | 原文线索

V2-023852 | MATHDB-322139

Isometry overlap bound for finite planar point sets

摘录 / EXCERPT: Let ${\mathcal P}\subset\mathbb R^2$ be a finite point set. An isometry is a distance-preserving map $\varphi:\mathbb R^2\to\mathbb R^2$. The isometry overlap conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-023853 | MATHDB-322140

Affine overlap conjecture for proper planar point sets

摘录 / EXCERPT: Let ${\mathcal P}\subset\mathbb R^2$ be any finite point set, let $3\le k\le|{\mathcal P}|$, and let $C>0$. An intersection is proper 2-dimensional up to C in the sense used by t...

conjecture | open | MathDB 页面 | 原文线索

V2-023854 | MATHDB-322141

Higher-dimensional affine overlap conjecture

摘录 / EXCERPT: Let G be the group of non-degenerate, equivalently bijective, affine mappings of $\mathbb R^d$. Let $f_G^d(N,k)$ denote the corresponding maximum number of maps having at least...

conjecture | open | MathDB 页面 | 原文线索

V2-023855 | MATHDB-322142

Incidence bound for rich affine subspaces in finite point sets

摘录 / EXCERPT: Let $C>0$ and let ${\mathcal P}\subset\mathbb R^D$ be a finite point set that is proper D-dimensional up to C. For $r<D$, let $S^{(r)}\subset\mathbb R^D$ range over affine subs...

conjecture | open | MathDB 页面 | 原文线索

V2-023856 | MATHDB-322150

The consistency and self-duality conjecture for weighted-game chains

摘录 / EXCERPT: Let J_n be the poset of linear games on n voters, let $\mathcal{W}_n$ be its weighted-games subposet, and let $\mathcal{C}_n$ denote the weighted voting polygon. A saturated ch...

conjecture | open | MathDB 页面 | 原文线索

V2-023857 | MATHDB-322172

The refined symbolic-power containment conjecture with one maximal-ideal factor

摘录 / EXCERPT: Let K be an algebraically closed field, let $M=(x_0,\ldots,x_N)$ be the maximal homogeneous ideal of $K[\mathbb{P}^{N}]$, and let I be the radical ideal of a finite set of poin...

conjecture | open | MathDB 页面 | 原文线索

V2-023858 | MATHDB-322173

The two-part refined symbolic-power containment conjecture

摘录 / EXCERPT: Let K be an algebraically closed field, let I be the radical ideal of a finite set of points in $\mathbb{P}^{N}$, and let $M=(x_0,\ldots,x_N)$ be the maximal homogeneous ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-023859 | MATHDB-322179

Folklore conjecture on totally invariant algebraic subsets of projective space

摘录 / EXCERPT: Let $f:\mathbb{P}^k\to\mathbb{P}^k$ be a non-invertible holomorphic endomorphism, and let $E\subset\mathbb{P}^k$ be an algebraic subset satisfying $f^{-1}(E)=E=f(E)$. Folklore conj...

conjecture | open | MathDB 页面 | 原文线索

V2-023860 | MATHDB-322185

The order-three conjecture for noncommutative matrix involutions

摘录 / EXCERPT: The order-three conjecture. The transformation $(I_1\circ I_2\circ I_3)^3$ is equal to the identity modulo the...

conjecture | open | MathDB 页面 | 原文线索

V2-023861 | MATHDB-322187

Wu-Zheng's splitting conjecture for complete Kähler manifolds

摘录 / EXCERPT: Wu-Zheng's splitting conjecture. The manifold $\widetilde M$ is holomorphically isometric to

conjecture | open | MathDB 页面 | 原文线索

V2-023862 | MATHDB-322234

Sawon's conjecture on Lagrangian fibrations by Jacobians

摘录 / EXCERPT: Let $X\rightarrow \mathbb{P}^n$ be a holomorphic symplectic manifold of dimension $2n$ with a proper surjective morphism whose fibres are Lagrangian and whose generic fibre is an...

conjecture | open | MathDB 页面 | 原文线索

V2-023863 | MATHDB-322235

Strict convexity of the logarithmic gap function for triangles

摘录 / EXCERPT: Let a triangle be parametrized by grid coordinates (ν,τ), with vertices $(0,0)$, $(1,0)$), and (x,y) where ... The equilateral triangle corresponds to $\nu=\tau=1$. Strict-c...

conjecture | open | MathDB 页面 | 原文线索

V2-023864 | MATHDB-322236

Regular simplex conjecture for the spectral gap

摘录 / EXCERPT: Let $\mathfrak{M}_n$ be the moduli space of all n-simplices with unit diameter. For $n\geq 2$, let $p_0,p_1,\ldots,p_n\in\mathbb{R}^n$ satisfy ... so that these points define a reg...

conjecture | open | MathDB 页面 | 原文线索

V2-023865 | MATHDB-322237

Strict fundamental-gap inequality for higher-dimensional convex domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a convex domain, and assume $n>1$. Write $\xi(\Omega)$ for the gap function, namely the difference between the first two Dirichlet eigenvalues of...

conjecture | open | MathDB 页面 | 原文线索

V2-023866 | MATHDB-322261

Nadler-Tanaka's conjecture relating Lagrangian cobordisms to wrapped Fukaya categories

摘录 / EXCERPT: Nadler-Tanaka's wrapped-Fukaya conjecture. The continuation-map functor induces an equivalence of stable ∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-023867 | MATHDB-322286

Growth conjecture for heat trace coefficients on non-polynomial-Plancherel locally symmetric spaces

摘录 / EXCERPT: Let $\mathcal{M}$ be modeled on an irreducible simply connected symmetric space G/K of non-compact type. Let $\mathcal{A}_n(\mathcal{M})$ denote its heat trace coefficients, and...

conjecture | open | MathDB 页面 | 原文线索

V2-023868 | MATHDB-322292

The generalized Alekseevsky conjecture for expanding homogeneous Ricci solitons

摘录 / EXCERPT: Let $M=G/H$ be a homogeneous Ricci soliton with $c<0$, where G is a transitive Lie group and H is the isotropy subgroup. A homogeneous Ricci soliton is a Riemannian homogeneous...

conjecture | open | MathDB 页面 | 原文线索

V2-023869 | MATHDB-322308

Farb and Weinberger's characterization of locally symmetric manifolds

摘录 / EXCERPT: Farb and Weinberger's conjecture. 1. M is aspherical, smoothly irreducible, $\pi_1(M)$ has no non-trivial normal abelian subgroup, and

conjecture | open | MathDB 页面 | 原文线索

V2-023870 | MATHDB-322328

Infinitely many folds on the upper solution branch

摘录 / EXCERPT: Upper-branch fold conjecture. The upper solution branch undergoes infinitely many folds, and λ tends to a finite value bounded away from zero.

conjecture | open | MathDB 页面 | 原文线索

V2-023871 | MATHDB-322349

The covering-map conjecture for nonsurjective developing maps

摘录 / EXCERPT: Let M be a compact conformally flat Lorentzian manifold with developing map $\operatorname{dev}:\widetilde M\to \mathbb{R}^{2n+2}$, and suppose that dev is not s...

conjecture | open | MathDB 页面 | 原文线索

V2-023872 | MATHDB-322357

The Brieskorn sphere pseudoconvex embedding conjecture

摘录 / EXCERPT: Brieskorn embedding conjecture. No Brieskorn sphere, with either orientation, admits a pseudoconvex embedding in $\mathbb C^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-023873 | MATHDB-322359

The virtual specialness conjecture for aspherical compact 3-manifolds

摘录 / EXCERPT: Let M be an aspherical compact 3-manifold. The virtual specialness conjecture. The manifold M is nonpositively curved if and only if its fundamental group virtually embeds in...

conjecture | open | MathDB 页面 | 原文线索

V2-023874 | MATHDB-322398

Cheeger's measurable non-degeneracy conjecture for coordinate charts

摘录 / EXCERPT: Cheeger's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-023875 | MATHDB-322401

Maehara's strengthened common-point conjecture for unit spheres

摘录 / EXCERPT: Maehara's conjecture. All the spheres in $\mathfrak{F}$ have a point in common.

conjecture | open | MathDB 页面 | 原文线索

V2-023876 | MATHDB-322402

Concavity conjecture for maximal-area disk-polygons

摘录 / EXCERPT: Concavity conjecture for maximal-area disk-polygons. For all $n\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-023877 | MATHDB-322406

The SYZ mirror-map inversion conjecture for toric Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a toric Calabi-Yau manifold with mirror family $\check{X}_y$, and let ... be the SYZ map, defined by ... where $1+\delta_i(q)$ are generating functions of the genus 0 open...

conjecture | open | MathDB 页面 | 原文线索

V2-023878 | MATHDB-322407

Székelyhidi's conjecture for asymptotically hyperbolic Kähler metrics

摘录 / EXCERPT: Let X be a complex manifold, D a smooth divisor in X, and $L\to X$ a polarization. The triple (X,D,L) is K-stable when it satisfies the relevant stability condition for the...

conjecture | open | MathDB 页面 | 原文线索

V2-023879 | MATHDB-322441

Direct-downward arcs conjecture for triangulation graphs

摘录 / EXCERPT: For a given 3-manifold $\mathcal{M}$, let $\phi_n(\mathcal{M})$ denote the fraction of nodes at level n of $\mathscr{P}_1(\mathcal{M})$ that have an arc leading directly down to...

conjecture | open | MathDB 页面 | 原文线索

V2-023880 | MATHDB-322442

Sublinear simplification-length conjecture for 3-sphere triangulations

摘录 / EXCERPT: Let $L_n(S^3)$ denote the worst-case simplification path length for nodes at level n in the graph $\mathscr{P}_1(S^3)$. Sublinear simplification-length conjecture. From any node...

conjecture | open | MathDB 页面 | 原文线索

V2-023881 | MATHDB-322473

The partition-model conjecture for optimally approximating exponential families

摘录 / EXCERPT: Let N be the size of the finite sample space and let k be the dimension of an exponential family. For a class $\mathcal{H}$ of exponential families, define ... In particular, let...

conjecture | open | MathDB 页面 | 原文线索

V2-023882 | MATHDB-322484

The type D stratified Mukai flop equivalence conjecture

摘录 / EXCERPT: Let W be a rank $2N$ vector bundle equipped with a fibrewise non-degenerate symmetric bilinear form, and let $\mathbb{IG}(N,2N)^-$ and $\mathbb{IG}(N,2N)^+$ be the two components...

conjecture | open | MathDB 页面 | 原文线索

V2-023883 | MATHDB-322496

The p-adic section conjecture for curves

摘录 / EXCERPT: Let $k/\mathbb Q_p$ be a finite extension, and let X/k be a smooth, projective and geometrically connected curve of genus at least 2. A rational point $a\in X(k)$ determines a...

conjecture | open | MathDB 页面 | 原文线索

V2-023884 | MATHDB-322497

The valuative section conjecture for curves

摘录 / EXCERPT: Let k be a number field or a finite extension of $\mathbb Q_p$, and let X/k be a smooth, projective and geometrically connected curve of genus at least 2, with function field...

conjecture | open | MathDB 页面 | 原文线索

V2-023885 | MATHDB-322513

Conjectural composition factors of the third lower-central quotient of the symplectic free associative algebra

摘录 / EXCERPT: Composition-factor conjecture. The vectors $v_{k,0}$ and $z_{k,0}$ are non-zero in $B_3(A'_{2n})$, and the H_{2n}-module composition factors of $B_3(A'_{2n})$ are

conjecture | open | MathDB 页面 | 原文线索

V2-023886 | MATHDB-322533

Neumann's strong conjecture on Bloch groups of number fields

摘录 / EXCERPT: Let $F\subset\mathbb C$ be a concrete number field which is not contained in $\mathbb R$. Let $\mathcal B(F)$ be the Bloch group of F, and call a hyperbolic 3-manifold admissib...

conjecture | open | MathDB 页面 | 原文线索

V2-023887 | MATHDB-322547

The logarithmic upper-bound conjecture for complete arcs in projective planes

摘录 / EXCERPT: Let $PG(2,q)$ be the projective plane of order q, and let $t_{2}(2,q)$ denote the smallest size of a complete arc in $PG(2,q)$. The logarithmic upper-bound conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-023888 | MATHDB-322548

The finite-range upper-bound conjecture for complete arcs in projective planes

摘录 / EXCERPT: Let $PG(2,q)$ be the projective plane of order q, and let $t_{2}(2,q)$ denote the smallest size of a complete arc in $PG(2,q)$. The finite-range upper-bound conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-023889 | MATHDB-322554

Schmutz Schaller's systole-length conjecture for hyperbolic surfaces

摘录 / EXCERPT: Let $\mathcal M_g$ denote the moduli space of closed hyperbolic surfaces of genus g, and let $\operatorname{sys}(S)$ be the systole length of S. Schmutz Schaller's systole-leng...

conjecture | open | MathDB 页面 | 原文线索

V2-023890 | MATHDB-322560

Totaro's ruledness conjecture for anisotropic quadrics

摘录 / EXCERPT: Let X be an anisotropic quadric over a field k. Write $\mathfrak{i}_1(X)$ for its first Witt index, and say that X is ruled if it is birational to $Y\times\mathbb{P}_k^1$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-023891 | MATHDB-322561

Totaro's birational product conjecture for anisotropic quadrics

摘录 / EXCERPT: Let X be an anisotropic quadric over a field k. Write $\mathfrak{i}_1(X)$ for its first Witt index. A subquadric $X'\subset X$ is a quadric contained in X. Totaro's birationa...

conjecture | open | MathDB 页面 | 原文线索

V2-023892 | MATHDB-322586

Instability conjecture for the Chen-LeBrun-Weber Ricci-flat cone

摘录 / EXCERPT: Let $C(\mathbb{C}P^2\sharp 2\overline{\mathbb{C}P^2})$ denote the Ricci-flat cone over $\mathbb{C}P^2\sharp 2\overline{\mathbb{C}P^2}$ equipped with the Chen-LeBrun-Weber metri...

conjecture | open | MathDB 页面 | 原文线索

V2-023893 | MATHDB-322587

Instability conjecture for the Chen-LeBrun-Weber Ricci shrinker

摘录 / EXCERPT: Let $\mathbb{C} P^2\sharp 2\overline{\mathbb{C} P^2}$ carry the Chen-LeBrun-Weber metric, viewed as a Ricci shrinker. Instability conjecture. The manifold ... with the Chen-LeBrun-We...

conjecture | open | MathDB 页面 | 原文线索

V2-023894 | MATHDB-322589

Donaldson's definite-triple conjecture for compact 4-manifolds

摘录 / EXCERPT: Let X be a compact 4-manifold admitting symplectic forms $\omega_1,\omega_2,\omega_3$ that span a definite 3-plane in Λ^2 at each point. Donaldson's conjecture. Then X...

conjecture | open | MathDB 页面 | 原文线索

V2-023895 | MATHDB-322620

Algebraic torsion conjecture for contact manifolds containing glued Giroux domains

摘录 / EXCERPT: Algebraic torsion conjecture. Then (V,ξ) has algebraic 1-torsion, and it is also algebraically overtwisted if N contains any blown down boundary components.

conjecture | open | MathDB 页面 | 原文线索

V2-023896 | MATHDB-322623

Classification conjecture for proper biharmonic hypersurfaces in spheres

摘录 / EXCERPT: Classification conjecture. The only proper biharmonic hypersurfaces in $\mathbb{S}^{m+1}$ are the open parts of hyperspheres $\mathbb{S}^m(1/\sqrt{2})$ or of the standard products...

conjecture | open | MathDB 页面 | 原文线索

V2-023897 | MATHDB-322632

The arbitrary-weights character conjecture for homogeneous components of sheaf moduli spaces

摘录 / EXCERPT: Let ${\overline w}\in{\mathbb Z}^r$, let $\alpha,\beta\geq 1$ satisfy $0\leq w_i<\alpha+\beta$ and $\gcd(\alpha,\beta)=1$, and let $a_i=\sharp\{j\mid w_j=i\}$. Let α' be the...

conjecture | open | MathDB 页面 | 原文线索

V2-023898 | MATHDB-322633

The rank-two Betti-number generating-function conjecture for two torus actions

摘录 / EXCERPT: Let $P_q(X)=\sum_{i\geq0}\dim H_i(X)q^{i/2}$ be the Poincare polynomial, and consider the rank-two moduli spaces ${\mathcal M}(2,n)$ with the two fixed components corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-023899 | MATHDB-322655

The equality of Picard-Fuchs and Lagrangian Floer superpotentials

摘录 / EXCERPT: Let X be a toric manifold with $-K_X$ numerically effective. Let $W^{\textrm{PF}}$ and $W^{\textrm{LF}}$ be the Picard-Fuchs and Lagrangian Floer superpotentials in the mirror, r...

conjecture | open | MathDB 页面 | 原文线索

V2-023900 | MATHDB-322656

The open mirror map conjecture for semi-Fano toric manifolds

摘录 / EXCERPT: Let X be a semi-Fano toric manifold, and let m denote the number of its toric divisors. For $i=1,\ldots,m$, let $\delta_i(q)$ be the generating-function correction determined b...

conjecture | open | MathDB 页面 | 原文线索

V2-023901 | MATHDB-322669

Yau's structure conjecture for compact Kähler manifolds with nonpositive bisectional curvature

摘录 / EXCERPT: Yau's structure conjecture. There exists a finite cover M' of M that is a holomorphic and metric fibre bundle over a compact Kähler manifold N with nonpositive bisectional cu...

conjecture | open | MathDB 页面 | 原文线索

V2-023902 | MATHDB-322689

No faithful p-adic action conjecture

摘录 / EXCERPT: Let $\mathbb Z_p$ denote the p-adic integers, and let M be a connected n-manifold. A faithful action means a continuous injection into $\operatorname{Homeo}(M)$ with its comp...

conjecture | open | MathDB 页面 | 原文线索

V2-023903 | MATHDB-322690

Small-homeomorphism compact-subgroup conjecture

摘录 / EXCERPT: Small-homeomorphism compact-subgroup conjecture. There exists $\epsilon>0$ such that this subset of $\operatorname{Homeo}(M)$ contains no nontrivial compact subgroup. For any speci...

conjecture | open | MathDB 页面 | 原文线索

V2-023904 | MATHDB-322691

Almost periodic homeomorphism flow conjecture

摘录 / EXCERPT: Let M be a connected n-manifold. A homeomorphism f of M is almost periodic when the subgroup of $\operatorname{Homeo}(M)$ that it generates has compact closure. Almost peri...

conjecture | open | MathDB 页面 | 原文线索

V2-023905 | MATHDB-322699

Excess bound conjecture for the positive semidefinite cone

摘录 / EXCERPT: Let Sym^k_+ denote the cone of positive semidefinite matrices, and let $e(C)$ be the excess of a closed convex cone C over the Lorentz cone. Positive-semidefinit...

conjecture | open | MathDB 页面 | 原文线索

V2-023906 | MATHDB-322707

Honda's classification conjecture for Moishezon twistor spaces on $4\bar{\mathbb{CP}}^2$

摘录 / EXCERPT: Honda's classification conjecture. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-023907 | MATHDB-322750

Truncated binomial polynomial conjecture for boundary points of Steiner cones

摘录 / EXCERPT: Truncated binomial polynomial conjecture. There exist a truncated binomial polynomial $P_{j,k}^n(z)$ with $0<j<k<n$ and a real number $\lambda>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-023908 | MATHDB-322753

Invariance of pedal unfolding and Legendrian duality under A-equivalence

摘录 / EXCERPT: Let $\varphi_1,\varphi_2:(\mathbb{R}^2,0)\to(\mathbb{R}^3,0)$ be two C^∞ map-germs of pedal unfolding type. Let $\Phi_1,\Phi_2:(\mathbb{R}^2,0)\to(\mathbb{R}^3,0)$ be two no...

conjecture | open | MathDB 页面 | 原文线索

V2-023909 | MATHDB-322771

End-invariant exclusion conjecture for hyperbolic cone manifolds

摘录 / EXCERPT: Let r be a parameter for the hyperbolic cone manifold $M(r;\theta^-,\theta^+)$, with $0\leq\theta^-\leq\pi$ and $0\leq\theta^+\leq\pi$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023910 | MATHDB-322794

Generalized Chern Conjecture for closed oriented aspherical manifolds

摘录 / EXCERPT: Let M be a closed oriented aspherical manifold. Its tangent bundle is denoted by TM, and a flat structure on TM means that it is induced by a representation of the fundamenta...

conjecture | open | MathDB 页面 | 原文线索

V2-023911 | MATHDB-322795

Loewner's bound for aspherical surfaces

摘录 / EXCERPT: Loewner's bound. Every aspherical surface satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-023912 | MATHDB-322813

The conjecture that beta(r; Q_p) equals 4r

摘录 / EXCERPT: For a field K, let $\beta(r;K)$ be the least integer n such that every system of r quadratic forms in more than n variables over K has a non-trivial common zero. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-023913 | MATHDB-322814

Monotone convergence of Riemannian gradient descent for the center of mass

摘录 / EXCERPT: Let M be a Riemannian manifold, let $\rho\leq r_{\operatorname{cx}}$, and let $\{x_i\}_{i=1}^N\subset B(o,\rho)\subset M$. Set $\bar{x}_2$ to be their L^2 center of mass, and d...

conjecture | open | MathDB 页面 | 原文线索

V2-023914 | MATHDB-322817

The non-vanishing conjecture for log canonical pairs

摘录 / EXCERPT: Non-vanishing conjecture. There exists an effective $\mathbb{K}$-divisor D such that

conjecture | open | MathDB 页面 | 原文线索

V2-023915 | MATHDB-322820

Main existence conjecture for the L^2 curvature flow

摘录 / EXCERPT: Let (M^n,g) be a compact Riemannian manifold. The L^2 curvature flow is the gradient flow of the functional $\mathcal F$, and Rm denotes the Riemann curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-023916 | MATHDB-322836

The approximation-radius gap conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^n$ be a centrally symmetric convex body, let $R(X,\sigma)$ denote its minimax risk, and let $z_{c,k}(X)$ denote the approximation-radius quantity defined...

conjecture | open | MathDB 页面 | 原文线索

V2-023917 | MATHDB-322838

The simplex-inverted simplex conjecture for volume ratios

摘录 / EXCERPT: Let n be a positive integer, and let a simplex and its inverted simplex be given in n-dimensional space. Consider their Minkowski interpolations and define the volume ratio as...

conjecture | open | MathDB 页面 | 原文线索

V2-023918 | MATHDB-322846

The complete Chen conjecture for biharmonic submanifolds in Euclidean space

摘录 / EXCERPT: Complete Chen conjecture. Any complete biharmonic submanifold in $\mathbb{E}^N$ is minimal.

conjecture | open | MathDB 页面 | 原文线索

V2-023919 | MATHDB-322848

The CH-rigidity characterization for compact submanifolds

摘录 / EXCERPT: CH-rigidity characterization. N is CH-rigid if and only if N is coisotropic.

conjecture | open | MathDB 页面 | 原文线索

V2-023920 | MATHDB-322866

Equidimensionality conjecture for maximal-length fibre-type contractions

摘录 / EXCERPT: Maximal-length equidimensionality conjecture. If $n\geq 2m-1$, then φ is equidimensional. The paper presents this as a more general setting in which the Beltrametti-Sommese...

conjecture | open | MathDB 页面 | 原文线索

V2-023921 | MATHDB-322874

Finiteness conjecture for dihedral knot surgeries

摘录 / EXCERPT: Let m and n be positive integers, and consider the Seifert-fibered dihedral manifolds ... A knot surgery is a Dehn surgery on a knot in the 3-sphere, and the family is consider...

conjecture | open | MathDB 页面 | 原文线索

V2-023922 | MATHDB-322885

Curvature bound and tangent-cone convergence for conifold contractions

摘录 / EXCERPT: Curvature and rescaling conjecture. There exists $C>0$ such that, for all $t\in(0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-023923 | MATHDB-322940

Infinite exotic structures conjecture for the manifolds Y and Z

摘录 / EXCERPT: Let Y_{2m} and Z_{2m} denote the manifolds obtained from Y and Z, respectively, by generalized cork twists of exponent $2m$. Infinite exotic structures conjecture. The fami...

conjecture | open | MathDB 页面 | 原文线索

V2-023924 | MATHDB-322942

The birational compatibility conjecture for Enriques moduli maps

摘录 / EXCERPT: Let $\mathcal{M}_{{\text{En}}}(2)^{{\text{ndeg}}}$ be the moduli space of non-degenerate degree-two Enriques surfaces, let $\mathcal{X}_4$ be the moduli space of pointed genus-4...

conjecture | open | MathDB 页面 | 原文线索

V2-023925 | MATHDB-322944

Smoothness of the normalization of a free divisor with radical Jacobian ideal

摘录 / EXCERPT: Let $D \subseteq S$ be a divisor in a complex manifold S that is locally at a point p given by $h=0$, and let $\pi:\widetilde D \rightarrow D$ be its normalization. Suppose tha...

conjecture | open | MathDB 页面 | 原文线索

V2-023926 | MATHDB-322945

Freeness of divisors with equidimensional radical Jacobian ideal of depth two

摘录 / EXCERPT: Let $D \subseteq S$ be a divisor in a complex manifold S, defined at a point p by $h \in \mathcal{O}_{S,p}$. Assume that its Jacobian ideal J_h is radical, equidimensional, a...

conjecture | open | MathDB 页面 | 原文线索

V2-023927 | MATHDB-322946

Complete-intersection conjecture for equidimensional radical Jacobian ideals of depth at least three

摘录 / EXCERPT: Let D be a divisor in a complex manifold S, defined locally at a point p by a reduced $h \in \mathcal{O}_{S,p}$. Suppose that the Jacobian ideal J_h is radical, equidimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-023928 | MATHDB-323038

Generalized lattice-point bound for families of convex bodies

摘录 / EXCERPT: Generalized lattice-point bound conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-023929 | MATHDB-323052

Iarrobino-Evain conjecture for generic points in projective space

摘录 / EXCERPT: Let $I\subset k[\mathbb{P}^N]$ be the ideal of $s\gg0$ generic points of $\mathbb{P}^N$. The quantity $\gamma(I)$ denotes the asymptotic initial degree of I. Iarrobino-Evain conj...

conjecture | open | MathDB 页面 | 原文线索

V2-023930 | MATHDB-323053

Asymptotic resurgence conjecture for sufficiently general lines in projective 3-space

摘录 / EXCERPT: Let I be the ideal of s sufficiently general lines in $\mathbb{P}^3$, with $s\geq q$ for some integer q. Let g be the largest real root of ... The quantity $\gamma(I)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-023931 | MATHDB-323060

Li-Wu-Zheng classification conjectures for Kähler C-spaces with nonnegative quadratic bisectional curvature

摘录 / EXCERPT: Li-Wu-Zheng's conjectures. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-023932 | MATHDB-323074

Equivariant Quillen determinant line bundle conjecture

摘录 / EXCERPT: Equivariant Quillen determinant conjecture. The Quillen determinant line bundle satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-023933 | MATHDB-323118

Legendrian Gronwall conjecture for Legendrian 3-webs

摘录 / EXCERPT: Legendrian Gronwall conjecture. Every Legendrian 3-web admits at most one distinct local linearization in $\mathbb{P}^3$, except when it is locally equivalent to the dual 3-web of...

conjecture | open | MathDB 页面 | 原文线索

V2-023934 | MATHDB-323131

Checkerboard Conjecture for energy minimisation on toric grids

摘录 / EXCERPT: Let a d-dimensional toric grid have side lengths $n_1,\ldots,n_d$, total size $n=n_1\cdots n_d$, and Lee distance given by shortest-path distance in the grid graph. Assume that a...

conjecture | open | MathDB 页面 | 原文线索

V2-023935 | MATHDB-323144

Blow-up conjecture for cscK metrics

摘录 / EXCERPT: Let X be a compact Kähler manifold. A finite sequence of blow-ups is a finite succession of blow-up constructions starting from X. Blow-up conjecture for cscK metrics. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-023936 | MATHDB-323185

Calabi-Yau threefold symplectic Monge-Ampère conjecture

摘录 / EXCERPT: Calabi-Yau threefold symplectic Monge-Ampère conjecture. The equation has a unique solution $\varphi:M\to\mathbb{R}$, modulo adding a constant, such that...

conjecture | open | MathDB 页面 | 原文线索

V2-023937 | MATHDB-323186

Mirror Kähler potential conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Mirror Kähler potential conjecture. For every $\tilde{\Omega}\sim\Omega$, there exists a function $\varphi:M\to\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-023938 | MATHDB-323201

Cone duality conjecture for compact balanced manifolds

摘录 / EXCERPT: Cone duality conjecture. The dual cone of $\mathcal E$ is the closure of the balanced cone:

open_problem | open | MathDB 页面 | 原文线索

V2-023939 | MATHDB-323226

The higher-dimensional incidence bound for equal-dimensional surfaces

摘录 / EXCERPT: Higher-dimensional incidence conjecture. The incidence count should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-023940 | MATHDB-323247

Colin-Honda exponential-growth conjecture for Reeb orbits

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold with a universally tight contact structure ξ, and let α be a non-degenerate contact form defining ξ. Denote by…

conjecture | open | MathDB 页面 | 原文线索

V2-023941 | MATHDB-323260

Local freeness conjecture for families of (φ,Γ_K)-modules

摘录 / EXCERPT: Local freeness conjecture. For every (φ,Γ_K)-module M over $\mathcal{R}_A(\pi_K)$, there exists a finite admissible cover $\{\operatorname{Max}(A_i)\}_{i=1,\dots,n}$…

conjecture | open | MathDB 页面 | 原文线索

V2-023942 | MATHDB-323269

CR Obata-type conjecture for pseudohermitian manifolds

摘录 / EXCERPT: Let (M,θ) be a closed pseudohermitian $(2n+1)$-manifold with $n\ge 2$. Assume that the Paneitz operator is nonnegative if $n=1$. Suppose there is a positive constant k_0 s…

conjecture | open | MathDB 页面 | 原文线索

V2-023943 | MATHDB-323295

Xu-Zhao diffeomorphic sphere conjecture for submanifolds

摘录 / EXCERPT: Let M be a smooth closed n-dimensional submanifold, with $n\geq2$, in the space form $\mathbb{F}^{n+d}(c)$, where $c\geq0$. Let $\mathring{A}$ denote its traceless second funda…

conjecture | open | MathDB 页面 | 原文线索

V2-023944 | MATHDB-323301

Unique smooth minimum-degree toric surface of maximal Picard rank in an m×m box

摘录 / EXCERPT: Fix $m\in\mathbb{Z}_{>0}$. Consider toric surfaces S with ample anticanonical divisor $-K_S$, without imposing conditions on their singularities, such that the anticanonical poly…

conjecture | open | MathDB 页面 | 原文线索

V2-023945 | MATHDB-323302

Box-dimension bound for real projective Sierpinski gaskets

摘录 / EXCERPT: Let ${\boldsymbol{A}}\subset SL^\pm_n(\mathbb{R})$ be a real projective Sierpinski gasket, and let $R_{\boldsymbol{A}}$ be its residual set. Write $s_{\boldsymbol{A}}$ for the expo…

conjecture | open | MathDB 页面 | 原文线索

V2-023946 | MATHDB-323306

The quartic growth conjecture for radial orderings

摘录 / EXCERPT: Let $g(n)$ denote the minimum number of distinct radial orderings determined by a set of n points in strong general position in the plane. Quartic growth conjecture. … The paper…

conjecture | open | MathDB 页面 | 原文线索

V2-023947 | MATHDB-323307

The general-position quartic lower-bound conjecture for radial orderings

摘录 / EXCERPT: Let S be a set of n points in general position in the plane, meaning that no three points are collinear. General-position quartic lower-bound conjecture. S has at least…

conjecture | open | MathDB 页面 | 原文线索

V2-023948 | MATHDB-323309

The conjectural presentation of the valuation algebra of complex space forms

摘录 / EXCERPT: The presentation conjecture. For the valuation algebra $\mathcal{V}^n_\lambda$,

conjecture | open | MathDB 页面 | 原文线索

V2-023949 | MATHDB-323310

The equality conjecture for angular valuations

摘录 / EXCERPT: Angular-valuation equality conjecture. The algebra of angular valuations on M equals $LK(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-023950 | MATHDB-323324

Coloured Tverberg's theorem

摘录 / EXCERPT: Let $F_1, F_2, \ldots, F_{d+1}$ be colour classes, each consisting of k points in $\mathbb{R}^d$. A colourful k-partition is a partition of their union into k pairwise disjoi…

conjecture | open | MathDB 页面 | 原文线索

V2-023951 | MATHDB-323334

Stoll metric bounded-distance conjecture

摘录 / EXCERPT: Let G be a simply connected nilpotent Lie group and let Γ be a discrete co-compact subgroup. For a finite symmetric generating set $S\subset\Gamma$, let ρ_S be the as…

conjecture | open | MathDB 页面 | 原文线索

V2-023952 | MATHDB-323335

Stoll's bounded-distance conjecture for word and subFinsler metrics

摘录 / EXCERPT: Let G be a simply connected nilpotent Lie group, let Γ be a discrete co-compact subgroup, and let S be a finite symmetric generating set containing the identity. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023953 | MATHDB-323382

Existence of metrics with harmonic spinors on closed spin manifolds

摘录 / EXCERPT: Harmonic-spinor existence conjecture. There exists a Riemannian metric g on M such that

conjecture | open | MathDB 页面 | 原文线索

V2-023954 | MATHDB-323408

Lens space surgery conjecture for the links A_{m,n}

摘录 / EXCERPT: Lens space surgery conjecture for $A_{m,n}$. M is a lens space if and only if either

conjecture | open | MathDB 页面 | 原文线索

V2-023955 | MATHDB-323439

Equality conjecture for the Brunn-Minkowski inequality for the Bernoulli constant

摘录 / EXCERPT: Brunn-Minkowski equality conjecture. Equality holds if and only if Ω_0 and Ω_1 are homothetic. This is the equality case left open for the Brunn-Minkowski inequality...

conjecture | open | MathDB 页面 | 原文线索

V2-023956 | MATHDB-323451

The LERF conjecture for non-Seifert-separated 3-manifolds

摘录 / EXCERPT: Let N be a compact, orientable, irreducible 3-manifold with empty or toroidal boundary. Suppose no torus of the JSJ decomposition bounds a Seifert fibered 3-manifold on both...

conjecture | open | MathDB 页面 | 原文线索

V2-023957 | MATHDB-323452

The fully relatively quasi-convex virtual-retract conjecture

摘录 / EXCERPT: Let N be an orientable, compact, non-positively curved 3-manifold with empty or toroidal boundary, and write $\pi=\pi_1(N)$. Equip π with its natural relatively hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-023958 | MATHDB-323453

The Zero Divisor Conjecture for aspherical 3-manifold groups

摘录 / EXCERPT: Let N be an aspherical 3-manifold with empty or toroidal boundary. Zero Divisor Conjecture. The group ring $\mathbb{Z}[\pi_1(N)]$ has no non-trivial zero divisors. The source s...

conjecture | open | MathDB 页面 | 原文线索

V2-023959 | MATHDB-323460

Metric collar conjecture for compact boundaries of Alexandrov spaces

摘录 / EXCERPT: Metric collar conjecture. There exists a positive number ϵ such that $X^n-X_{\epsilon}^n$ is a collar neighborhood of ∂X^n.

conjecture | open | MathDB 页面 | 原文线索

V2-023960 | MATHDB-323462

Soul neighborhood conjecture for Alexandrov manifolds

摘录 / EXCERPT: Soul neighborhood conjecture. The space X is homeomorphic to a small neighborhood of S that has the structure of an open disk bundle over S.

conjecture | open | MathDB 页面 | 原文线索

V2-023961 | MATHDB-323463

The maximal essential covering conjecture for nonnegatively curved Alexandrov spaces

摘录 / EXCERPT: Let M be an n-dimensional compact Alexandrov space with nonnegative curvature. The maximal essential covering conjecture asserts that ... Here, $\tau_n(M)$ denotes the relevant c...

conjecture | open | MathDB 页面 | 原文线索

V2-023962 | MATHDB-323469

Wang's generalized Ehrhart conjecture for convex bodies

摘录 / EXCERPT: Wang's generalized Ehrhart conjecture. If K contains only the origin as an interior lattice point, then

conjecture | open | MathDB 页面 | 原文线索

V2-023963 | MATHDB-323474

Link-volume slope finiteness conjecture for one-cusped hyperbolic manifolds

摘录 / EXCERPT: Link-volume slope finiteness conjecture. For any $V>0$, there exists a finite set of slopes $\mathcal{F}$ on ∂X, so that if $\lv[X(\alpha)] < V$, then α intersects...

conjecture | open | MathDB 页面 | 原文线索

V2-023964 | MATHDB-323475

Existence of bounded-volume, arbitrarily large-link-volume 3-manifolds

摘录 / EXCERPT: Bounded-volume, arbitrarily large-link-volume conjecture. For every $V>0$ there exists a manifold M so that

conjecture | open | MathDB 页面 | 原文线索

V2-023965 | MATHDB-323478

Compactifiable universal cover conjecture for compact Kähler manifolds

摘录 / EXCERPT: Compactifiable universal cover conjecture. After passing to a finite étale cover, there exists a locally trivial fibration $X\rightarrow A$ with simply connected fibre F onto a c...

conjecture | open | MathDB 页面 | 原文线索

V2-023966 | MATHDB-323479

Campana's special-manifold conjecture S

摘录 / EXCERPT: Conjecture S. Some finite étale cover of X is bimeromorphic to a complex torus.

conjecture | open | MathDB 页面 | 原文线索

V2-023967 | MATHDB-323498

The volume conjecture for hyperbolic knots

摘录 / EXCERPT: Let $\mathfrak{K}$ be a hyperbolic knot in $\mathbb{S}_3$, let $J_N(\mathfrak{K},q)$ be its colored Jones polynomial, and let $\mathsf{Vol}(\mathbb{S}_3\setminus\mathfrak{K})$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-023968 | MATHDB-323526

Flow-closedness conjecture for the norm-square of a complex moment map

摘录 / EXCERPT: Flow-closedness conjecture. If $\mu_\mathbb{C}$ is a complex moment map associated to a linear action by a compact group K on a vector space, then $|\mu_\mathbb{C}|^2$ is flow-cl...

conjecture | open | MathDB 页面 | 原文线索

V2-023969 | MATHDB-323547

Finite-genus classification conjecture for planar exceptional domains

摘录 / EXCERPT: Let an exceptional domain in $\mathbb{R}^2$ be a domain supporting the overdetermined potential-theoretic problem considered in the paper, and let its genus be finite. The three ex...

conjecture | open | MathDB 页面 | 原文线索

V2-023970 | MATHDB-323561

Genus and auxiliary-term formula conjecture for ADE orbifolds

摘录 / EXCERPT: Let $\mathbb{P}^1$ be an orbifold of ADE type, with orbifold-point multiplicities p, q, and r, and let t_n denote the distinguished coordinate appearing in the genus-zero F...

conjecture | open | MathDB 页面 | 原文线索

V2-023971 | MATHDB-323562

Degree-jumping conic containment conjecture

摘录 / EXCERPT: Degree-jumping conic containment conjecture. Then Z is contained in a single conic.

conjecture | open | MathDB 页面 | 原文线索

V2-023972 | MATHDB-323569

Diagonalisation conjecture for integrable Killing tensors on spheres

摘录 / EXCERPT: Diagonalisation conjecture. Under the action of the isometry group, any integrable Killing tensor on S^n is equivalent to one with a diagonal algebraic curvature tensor.

conjecture | open | MathDB 页面 | 原文线索

V2-023973 | MATHDB-323571

Conjecture on degeneration of singular AC metrics to smooth AC metrics

摘录 / EXCERPT: Metric contraction conjecture. The singular AC metrics on the r-th root of $K_{\mathbb{P}(E)}$ contract along one-parameter families to the smooth AC metrics on $E\otimes L$ by s...

conjecture | open | MathDB 页面 | 原文线索

V2-023974 | MATHDB-323593

The generalised Taubes conjecture for singular loci of circle-action quotients

摘录 / EXCERPT: Generalised Taubes conjecture. The singular locus L is a meridionally fibered link.

conjecture | open | MathDB 页面 | 原文线索

V2-023975 | MATHDB-323595

Thurston-Waldhausen virtually positive Betti number conjecture

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold. Property (FAb) means that every finite-index subgroup has finite abelianization. Thurston-Waldhausen's VPBN conjecture. The manifold M has...

conjecture | open | MathDB 页面 | 原文线索

V2-023976 | MATHDB-323596

Lubotzky-Sarnak conjecture on property tau for hyperbolic 3-manifold groups

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold, and let $\pi_1(M)$ be its fundamental group. Lubotzky-Sarnak conjecture. The group $\pi_1(M)$ does not have property (τ). This is present...

conjecture | open | MathDB 页面 | 原文线索

V2-023977 | MATHDB-323599

Conformal-class characterization of negatively infinite Yamabe constant

摘录 / EXCERPT: Let M be a nonempty manifold of dimension $n\geq 3$, and let $g\in\operatorname{Metr}(M)$. For a metric $\overline{g}$ conformal to g, write...

conjecture | open | MathDB 页面 | 原文线索

V2-023978 | MATHDB-323611

The unbounded-face conjecture for Dirichlet domains in the bidisk

摘录 / EXCERPT: Unbounded-face conjecture. Every $\gamma=(g_1,g_2)$ where both g_i's are hyperbolic admits a Dirichlet domain with more than two faces.

conjecture | open | MathDB 页面 | 原文线索

V2-023979 | MATHDB-323619

Nonexistence conjecture for proper biharmonic semi-Riemannian submersions in nonpositive curvature

摘录 / EXCERPT: Let M be a $(2n+1)$-dimensional semi-Riemannian space form $M_{1}^{2n+1}(c)$ of index 1, and let B be a $2n$-dimensional Riemannian manifold. A proper biharmonic semi-Riemann...

conjecture | open | MathDB 页面 | 原文线索

V2-023980 | MATHDB-323631

d'Hildebrandt's Hölder continuity conjecture for conformally invariant variational problems

摘录 / EXCERPT: Let E be a coercive conformally invariant Lagrangian with quadratic growth, and let u be a critical point of E with finite energy. d'Hildebrandt's conjecture. The map u is...

conjecture | open | MathDB 页面 | 原文线索

V2-023981 | MATHDB-323632

Willmore's conjecture for immersed tori

摘录 / EXCERPT: Let $\vec{\Phi}:T^2\to\mathbb{R}^m$ be an immersion, and let $\vec{H}_{\vec{\Phi}}$ and $d\operatorname{vol}_{\vec{\Phi}^*g_{\mathbb{R}^m}}$ denote its mean curvature vector and in...

conjecture | open | MathDB 页面 | 原文线索

V2-023982 | MATHDB-323684

Stein's equidissection conjecture for balanced polygons

摘录 / EXCERPT: Stein's conjecture. A balanced polygon cannot be cut into an odd number of triangles of equal areas.

conjecture | open | MathDB 页面 | 原文线索

V2-023983 | MATHDB-323701

Strong abundance for Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a Calabi-Yau manifold and let $L\in H^2(X,\mathbb{Q})$ be a $(1,1)$-class such that ... for every algebraic curve $C\subset X$. Strong abundance for Calabi-Yau manifolds....

conjecture | open | MathDB 页面 | 原文线索

V2-023984 | MATHDB-323713

The signed-curvature conjecture for closed knotted curves of constant torsion

摘录 / EXCERPT: Let a closed knotted curve be a closed curve in three-dimensional space that is knotted, and suppose its torsion is constant. Signed-curvature conjecture. Every closed, knotted cur...

conjecture | open | MathDB 页面 | 原文线索

V2-023985 | MATHDB-323714

The ovaloid constant-torsion conjecture

摘录 / EXCERPT: Let an ovaloid be the boundary of a strictly convex body in three-dimensional space. Consider a closed curve lying on an ovaloid and having nonzero constant torsion. Ovaloid consta...

conjecture | open | MathDB 页面 | 原文线索

V2-023986 | MATHDB-323734

Generic absence of rigid eigenspaces for conformally covariant operators

摘录 / EXCERPT: Generic absence of rigid eigenspaces. The operator P_g has no rigid eigenspaces for a dense set of metrics in $\mathcal M$. This conjecture would remove the non-rigidity assumpti...

conjecture | open | MathDB 页面 | 原文线索

V2-023987 | MATHDB-323794

The core-resolution conjecture for positively curved G-manifolds

摘录 / EXCERPT: Let M be a positively curved G-manifold with principal isotropy group $H\ne\{1\}$. Write ${}^r\!M$ for the core resolution of M. Core-resolution conjecture. Either ... or ... In...

conjecture | open | MathDB 页面 | 原文线索

V2-023988 | MATHDB-323847

Positivity and numerator growth for tropical cluster recurrences

摘录 / EXCERPT: Let (d_n) be a sequence satisfying ... with initial conditions $d_1=-1$ and $d_2=\cdots=d_N=0$ up to shifting the index, and suppose that (d_n) is not periodic. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-023989 | MATHDB-323874

The weak monotonicity conjecture for Minkowski endomorphisms

摘录 / EXCERPT: Let n be the dimension, and let $\mathcal{K}^n$ denote the space of convex bodies in $\mathbb{R}^n$. A Minkowski endomorphism is a continuous, rotation-intertwining, Minkowski-ad...

conjecture | open | MathDB 页面 | 原文线索

V2-023990 | MATHDB-323875

The invariant-range conjecture for Blaschke-Minkowski homomorphisms

摘录 / EXCERPT: Let $\mathcal{K}^n$ be the space of convex bodies in $\mathbb{R}^n$. A Blaschke-Minkowski homomorphism is a map from convex bodies, equipped with Blaschke addition, to convex bodie...

conjecture | open | MathDB 页面 | 原文线索

V2-023991 | MATHDB-323876

Hyperbolicity conjecture for log canonical divisors

摘录 / EXCERPT: Hyperbolicity conjecture for log canonical divisors. Let (X,D) be a log smooth projective Mori hyperbolic (respectively, Brody hyperbolic) pair. Then the log canonical divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-023992 | MATHDB-323887

The angle formula for geodesic hinges in the space of metric measure spaces

摘录 / EXCERPT: Let $(\boldsymbol{X}_\bullet,\boldsymbol{X}'_\bullet)$ be a geodesic hinge with endpoint distances R and R', and let ${\sf f}_0,{\sf f}_1,{\sf f}'_1$ be the associated function...

conjecture | open | MathDB 页面 | 原文线索

V2-023993 | MATHDB-323893

Ghorpade-Ram splitting subspace conjecture

摘录 / EXCERPT: Ghorpade-Ram conjecture. The number of σ-splitting subspaces is

conjecture | open | MathDB 页面 | 原文线索

V2-023994 | MATHDB-323903

F-functional geodesic stability conjecture for Kähler-Einstein metrics

摘录 / EXCERPT: F-functional geodesic stability conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-023995 | MATHDB-323904

Modified F-functional geodesic stability conjecture for Kähler-Ricci solitons

摘录 / EXCERPT: Modified F-functional geodesic stability conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-023996 | MATHDB-323916

Bounded ratio conjecture for negatively and positively curved triangulations

摘录 / EXCERPT: Let M be a closed 3-manifold, let K denote the number of 3-simplices in a triangulation, and let $\mathcal{N}^-_{K,M}$ and $\mathcal{N}^+_{K,M}$ denote the numbers of trian...

conjecture | open | MathDB 页面 | 原文线索

V2-023997 | MATHDB-323921

Existence and uniqueness conjecture for AdS manifolds with prescribed convex boundary metrics

摘录 / EXCERPT: AdS convex-boundary conjecture. There exists a unique globally hyperbolic $\mathbb{A}\mathbbm{d}\mathbb{S}^3$ manifold homeomorphic to $S\times[-1,1]$, with smooth, space-like and...

conjecture | open | MathDB 页面 | 原文线索

V2-023998 | MATHDB-323922

Mess's uniqueness conjecture for maximal globally hyperbolic AdS manifolds

摘录 / EXCERPT: Mess's conjecture. There is a unique maximal globally hyperbolic $\mathbb{A}\mathbbm{d}\mathbb{S}^3$ manifold N such that the induced metric on the boundary of the convex core of...

conjecture | open | MathDB 页面 | 原文线索

V2-023999 | MATHDB-323943

Point-separation conjecture for type A conformal blocks divisors

摘录 / EXCERPT: Point-separation conjecture. Every non-trivial conformal blocks divisor of type A with strictly positive weights separates all points on $\operatorname{M}_{0,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-024000 | MATHDB-323973

Shokurov's effective moduli-part conjecture for Fano type fibrations

摘录 / EXCERPT: Shokurov's conjecture. If (X,Δ) is ϵ-log terminal, then there exists $\delta=\delta_{m,n}(\epsilon)>0$ and an effective moduli part B such that $(Y,R+B)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-024001 | MATHDB-323976

Gaiotto-Moore-Neitzke metric identification conjecture for Hitchin moduli spaces

摘录 / EXCERPT: Let $\mathcal{M}$ be an $SU(2)$ Hitchin moduli space with only regular singularities and regular semisimple residues, equipped with the hyperkähler metric constructed from Fock-Gon...

conjecture | open | MathDB 页面 | 原文线索

V2-024002 | MATHDB-323979

Local SYZ metric identification conjecture for Ooguri-Vafa models

摘录 / EXCERPT: Let $\mathcal{M}^{0}$ be the local Ooguri-Vafa model, let Q be the quotient group defining its SYZ mirror $\bar{\mathcal{M}}^{0}=\mathcal{M}^{0}/Q$, and fix sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-024003 | MATHDB-323980

Ooguri-Vafa local model conjecture for singular Hitchin fibers

摘录 / EXCERPT: Let $\bar{\mathcal{M}}^{0}=\mathcal{M}^{0}/Q$ be the SYZ mirror of the local model, and let $\Omega(\xi)$ be the holomorphic symplectic form of its hyperkähler metric. Let u_0 be...

conjecture | open | MathDB 页面 | 原文线索

V2-024004 | MATHDB-323996

Symplectic rational connectedness conjecture

摘录 / EXCERPT: Symplectic rational connectedness conjecture. There is a non-zero Gromov-Witten invariant of this form. This conjecture proposes a symplectic analogue of rational connectedness, an...

conjecture | open | MathDB 页面 | 原文线索

V2-024005 | MATHDB-324017

The closed crepant resolution conjecture for orbifold Gromov-Witten theory

摘录 / EXCERPT: Let $\mathcal{X}$ be a toric orbifold and Y a crepant resolution, with Givental symplectic spaces $\mathcal{H}_{\mathcal{X}}$ and $\mathcal{H}_Y$ and Lagrangian cones...

conjecture | open | MathDB 页面 | 原文线索

V2-024006 | MATHDB-324018

The square-root grid general position subset conjecture

摘录 / EXCERPT: Let $f(n,\ell)$ denote the largest guaranteed size of a subset in general position among every set of n points in the plane with at most ℓ collinear. The...

conjecture | open | MathDB 页面 | 原文线索

V2-024007 | MATHDB-324019

The general-position colouring conjecture for square-root collinearities

摘录 / EXCERPT: Let P be a set of n points in the plane, with at most $\sqrt{n}$ collinear. A colouring of P has each colour class in general position when no colour class contains three col...

conjecture | open | MathDB 页面 | 原文线索

V2-024008 | MATHDB-324020

The fixed-collinearity general position subset conjecture

摘录 / EXCERPT: Let $f(n,\ell)$ be the largest guaranteed size of a subset in general position among every set of n points in the plane with at most ℓ collinear, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024009 | MATHDB-324021

The fixed-collinearity general-position colouring conjecture

摘录 / EXCERPT: Let P be a set of n points in the plane with at most ℓ collinear. A colouring of P has each colour class in general position when no colour class contains three collinea...

conjecture | open | MathDB 页面 | 原文线索

V2-024010 | MATHDB-324025

Hensley's dimension-denominator conjecture for continued-fraction alphabets

摘录 / EXCERPT: Hensley's conjecture. The denominators contain every sufficiently large natural number if and only if the Hausdorff dimension exceeds one half:

conjecture | open | MathDB 页面 | 原文线索

V2-024011 | MATHDB-324027

Hertz-Hertz-Ures Anosov-torus conjecture

摘录 / EXCERPT: Let M be a 3-manifold and let $f:M\to M$ be a partially hyperbolic diffeomorphism with invariant bundles E^u, E^c, and E^s. An embedded torus is periodic if $f^k(S)=S$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-024012 | MATHDB-324031

The one-local-maximum-per-way conjecture for polygon fillings

摘录 / EXCERPT: One-local-maximum-per-way conjecture. There is at most one local maximum per way.

conjecture | open | MathDB 页面 | 原文线索

V2-024013 | MATHDB-324032

The nearby-optimal-way conjecture for polygon fillings

摘录 / EXCERPT: Nearby-optimal-way conjecture. Given the optimal way of distributing $N-1$ discs, $\{n'_i\}_1^{K}$, the optimal way of distributing N discs is nearby, where nearby means

conjecture | open | MathDB 页面 | 原文线索

V2-024014 | MATHDB-324033

The eventual junction-occupation conjecture for optimal polygon fillings

摘录 / EXCERPT: Eventual junction-occupation conjecture. For a given G and $M(G)$, there is an N' such that for $N\geq N'$, the junction points are always occupied in the optimal filling solut...

conjecture | open | MathDB 页面 | 原文线索

V2-024015 | MATHDB-324034

The eventual junction-occupation conjecture for all local-maximal polygon fillings

摘录 / EXCERPT: Eventual all-local-maxima junction-occupation conjecture. For a given G and $M(G)$, there is an N' such that for $N\geq N'$, the junction points are always occupied in all fill...

conjecture | open | MathDB 页面 | 原文线索

V2-024016 | MATHDB-324044

The reconstruction conjecture for holonomic systems

摘录 / EXCERPT: Let X be a complex analytic manifold and let $\boldsymbol{P}$ denote the complex projective line. Consider the contravariant functors ... obtained from tempered holomorphic solutio...

conjecture | open | MathDB 页面 | 原文线索

V2-024017 | MATHDB-324055

Kodaira's conjecture on the Kähler condition for compact complex surfaces

摘录 / EXCERPT: Kodaira's conjecture. A compact complex surface is Kählerian if and only if its first Betti number is even.

conjecture | open | MathDB 页面 | 原文线索

V2-024018 | MATHDB-324059

The ideal conjecture for affine-Grassmannian slices at the base point

摘录 / EXCERPT: Let $\Gr^{\overline{\lambda}}_0$ be the slice at $\mu=0$ inside $\Gr_0=G[[t^{-1}]]_1$, and let J^λ_0 be the Poisson ideal of $\mathcal O(\Gr_0)$ generated by the coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-024019 | MATHDB-324060

The ideal conjecture for general affine-Grassmannian slices

摘录 / EXCERPT: Let $\Gr^{\overline{\lambda}}_\mu$ be the affine-Grassmannian slice in $\Gr_\mu$, and let J^λ_μ be the Poisson ideal of $\mathcal O(\Gr_\mu)$ generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-024020 | MATHDB-324061

The universality conjecture for the quantization of affine-Grassmannian slices

摘录 / EXCERPT: Let $\tilde{Y}^\lambda_\mu$ be the parameterized algebra obtained from $\tilde{Y}_\mu$ by base change along the map from the $r_i^{(s)}$ to the $c_i^{(s)}$ and imposing...

conjecture | open | MathDB 页面 | 原文线索

V2-024021 | MATHDB-324072

Davies' strong ratio limit conjecture for heat kernels

摘录 / EXCERPT: Let $Lu=u_t+P(x,\partial_x)u$ be a parabolic operator defined on a noncompact Riemannian manifold M, with $\lambda_0(P,M)\geq 0$. Fix a reference point $x_0\in M$. Davies' conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-024022 | MATHDB-324073

Critical-subcritical heat-kernel comparison conjecture

摘录 / EXCERPT: Let M be a Riemannian manifold, and let P_+ and P_0 be respectively subcritical and critical operators in M. Critical-subcritical decay conjecture. The ratio ... locally unif...

conjecture | open | MathDB 页面 | 原文线索

V2-024023 | MATHDB-324075

Bagchi-Datta generalized lower bound conjecture for triangulated manifolds

摘录 / EXCERPT: Let Δ be a connected triangulated $(d-1)$-manifold without boundary, and let $h_r(\Delta)$ denote its h-numbers and $\beta_{j-1}(\Delta)$ its Betti numbers. Bagchi-Datta's...

conjecture | open | MathDB 页面 | 原文线索

V2-024024 | MATHDB-324084

The slope inequality for genus-g Lefschetz fibrations over S^2

摘录 / EXCERPT: Slope inequality conjecture. For every genus-g Lefschetz fibration $f:X\rightarrow S^2$, the slope inequality

conjecture | open | MathDB 页面 | 原文线索

V2-024025 | MATHDB-324086

The non-smooth curvature-dimension conjecture for warped products

摘录 / EXCERPT: Non-smooth warped-product conjecture. Then the N-warped product $B\times^N_f F$ satisfies $CD((N+d-1)K,N+d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-024026 | MATHDB-324100

Small-degree complete-intersection conjecture for quadratic manifolds

摘录 / EXCERPT: Let $X\subset {\mathbb P}^N$ be a quadratic manifold of dimension n. Small-degree complete-intersection conjecture. If ... then X is a complete intersection, unless it is project...

conjecture | open | MathDB 页面 | 原文线索

V2-024027 | MATHDB-324147

The variance conjecture for isotropic log-concave vectors

摘录 / EXCERPT: Let X be an isotropic log-concave random vector in $\mathbb R^n$, and let $|X|$ denote its Euclidean norm. The variance conjecture. There exists an absolute constant C such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-024028 | MATHDB-324161

Boundary Colin de Verdière conjecture for the first Steklov eigenvalue

摘录 / EXCERPT: Let M be a compact surface with boundary, and let $\sigma_1(M)$ denote its first nonzero Steklov eigenvalue. Define $\mathrm{Chr}(M,\partial M)$ to be the number of vertices of t...

conjecture | open | MathDB 页面 | 原文线索

V2-024029 | MATHDB-324179

Equivalence conjecture for regular holonomic microdifferential modules

摘录 / EXCERPT: Let X be the underlying complex manifold, and let $\mathcal{E}_X$ and $\widehat{\mathcal{E}}_X$ denote the two sheaves of microdifferential operators appearing in the paper. Cons...

conjecture | open | MathDB 页面 | 原文线索

V2-024030 | MATHDB-324205

Phantom components in Fukaya categories of exotic Del Pezzo manifolds

摘录 / EXCERPT: Let M be a symplectic manifold homeomorphic to a Del Pezzo surface, and let its Fukaya category be considered. A phantom orthogonal component is an orthogonal component of the Fu...

conjecture | open | MathDB 页面 | 原文线索

V2-024031 | MATHDB-324206

Nonrationality conjecture for conic bundles with phantom decompositions

摘录 / EXCERPT: Let X be a conic bundle of dimension 3. Let $D^b(X)$ denote its bounded derived category, and suppose ... is a semiorthogonal decomposition into a phantom category $\mathcal A$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-024032 | MATHDB-324217

Corti's completion conjecture for families of Del Pezzo surfaces

摘录 / EXCERPT: Let C be a smooth curve, let $p\in C$, and let $f\colon \mathcal{X}\to C\setminus\{p\}$ be a smooth family of Del Pezzo surfaces. Corti's conjecture. The family can be completed...

conjecture | open | MathDB 页面 | 原文线索

V2-024033 | MATHDB-324250

Beloshapka's automorphism-dimension conjecture for real-analytic CR hypersurfaces

摘录 / EXCERPT: For every $n\ge 2$, let $\mathfrak C_n$ be the class of all connected real-analytic CR-hypersurfaces M of dimension $2n-1$ such that $\dim\operatorname{hol}(M,p)<\infty$ for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-024034 | MATHDB-324255

The minimal model conjecture for log canonical pairs

摘录 / EXCERPT: Let $(X/Z,B)$ be a log canonical (lc) pair, meaning that $(X/Z,B)$ is a pair with log canonical singularities over Z. A log minimal model is a birational model on which K_Y+B_Y...

conjecture | open | MathDB 页面 | 原文线索

V2-024035 | MATHDB-324256

The abundance conjecture for log canonical pairs

摘录 / EXCERPT: Let $(Y/Z,B_Y)$ be a log canonical (lc) pair, and suppose that K_Y+B_Y is nef over Z. A divisor is semi-ample over Z if it defines a contraction over Z after a suitable pos...

conjecture | open | MathDB 页面 | 原文线索

V2-024036 | MATHDB-324257

The nonvanishing conjecture for log canonical pairs

摘录 / EXCERPT: Let $(X/Z,B)$ be a log canonical (lc) pair, and suppose that K_X+B is pseudo-effective over Z. A divisor M is effective, written $M\ge 0$, when all its coefficients are nonne...

conjecture | open | MathDB 页面 | 原文线索

V2-024037 | MATHDB-324258

The finite generation conjecture for log canonical algebras

摘录 / EXCERPT: Let $(X/Z,B)$ be a log canonical (lc) pair such that K_X+B is a $\mathbb Q$-divisor, and let $f\colon X\to Z$ be the given morphism. Define the log canonical algebra by ... Finite...

conjecture | open | MathDB 页面 | 原文线索

V2-024038 | MATHDB-324273

Margulis' short-geodesic counting conjecture for arithmetic lattices

摘录 / EXCERPT: Let G be the ambient semisimple Lie group, let X be its symmetric space, and let Γ range over arithmetic lattices with $\operatorname{vol}(\Gamma\backslash X)\leq V$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-024039 | MATHDB-324325

Mirzakhani's covering conjecture for affine invariant submanifolds

摘录 / EXCERPT: Mirzakhani's conjecture. If an affine invariant submanifold $\cM$ is defined over $\bQ$, and $\cM$ is not a connected component of a stratum, then every translation surface in...

conjecture | open | MathDB 页面 | 原文线索

V2-024040 | MATHDB-324326

Mirzakhani's dimension conjecture for non-rational affine invariant submanifolds

摘录 / EXCERPT: Mirzakhani's conjecture. If an affine invariant submanifold $\cM$ is not defined over $\bQ$, then $p(T(\cM))$ has dimension 2.

conjecture | open | MathDB 页面 | 原文线索

V2-024041 | MATHDB-324331

The Laface-Ugaglia conjecture for Cremona-reduced linear systems in \(\mathbb P^3\)

摘录 / EXCERPT: Let $\mathcal L=\mathcal L_{3,d}(m_1,\dots,m_s)$ be a linear system in $\mathbb P^3$ with general base points $p_1,\dots,p_s$. It is Cremona reduced when ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-024042 | MATHDB-324336

Pull-apart obstruction conjecture for non-repeating Whitney towers

摘录 / EXCERPT: Let $A:\Sigma\to X$ be a collection of immersed 2-spheres in a 4-manifold, and suppose that $\lambda_n(A):=\lambda_n(\mathcal{W})$ is defined using an order n non-repeating W...

conjecture | open | MathDB 页面 | 原文线索

V2-024043 | MATHDB-324337

Second-order obstruction conjecture for pulling apart four 2-spheres

摘录 / EXCERPT: Let X be a 4-manifold and let $A:\amalg^4 S^2\to X$ be a quadruple of immersed 2-spheres. Suppose that A admits an order 2 non-repeating Whitney tower $\mathcal{W}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024044 | MATHDB-324352

The cluster-intersection criterion for Poisson primitive ideals

摘录 / EXCERPT: Let G be a semisimple group, let ${\bf x}_{{\bf w}_0,{\bf w}_0'}$ be a cluster as defined in the paper, and let ... be the subring of $\mathbb{C}[G]$ generated by the cluster varia...

conjecture | open | MathDB 页面 | 原文线索

V2-024045 | MATHDB-324362

Ivanov-Tuzhilin half-perimeter formula for minimal fillings

摘录 / EXCERPT: Let ${\mathcal M}=(M,\rho)$ be a finite metric space. For a tree G connecting M, let ${\mathcal O}(M,G)$ be the set of tours of M with respect to G, and for each...

conjecture | open | MathDB 页面 | 原文线索

V2-024046 | MATHDB-324387

The superlinear triangulation conjecture in higher dimensions

摘录 / EXCERPT: Let S be a finite point set in $\mathbb{R}^d$, and let a triangulation of S be a subdivision of its convex hull into simplices whose vertices belong to S and whose interiors...

conjecture | open | MathDB 页面 | 原文线索

V2-024047 | MATHDB-324391

The h-principle and almost abelianity of the fundamental group

摘录 / EXCERPT: The h-principle fundamental-group conjecture. Every projective manifold satisfying the h-principle should have an almost abelian fundamental group.

conjecture | open | MathDB 页面 | 原文线索

V2-024048 | MATHDB-324392

The core-map decomposition conjecture

摘录 / EXCERPT: Core-map decomposition conjecture. For any such X,

conjecture | open | MathDB 页面 | 原文线索

V2-024049 | MATHDB-324393

The projective h-principle fundamental-group conjecture

摘录 / EXCERPT: Projective h-principle fundamental-group conjecture. If X is a complex projective manifold satisfying the h-principle, then $\pi_1(X)$ is almost abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-024050 | MATHDB-324402

Fractal Weyl upper bound for resonances on geometrically finite hyperbolic manifolds

摘录 / EXCERPT: Let Γ be a geometrically finite discrete group of isometries of $\mathbb{H}^{n+1}$ such that $X=\Gamma\backslash\mathbb{H}^{n+1}$ is a smooth noncompact manifold. Let $R(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-024051 | MATHDB-324411

Strong-coupling convergence conjecture for abelian vortex moduli metrics

摘录 / EXCERPT: Let M be a Kähler manifold, let X be a toric target, and let $\mathcal{H}_\xi$ be the space of holomorphic maps $M\rightarrow X$ associated with parameter ξ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024052 | MATHDB-324412

Volume and total scalar-curvature limit conjecture for vortex moduli spaces

摘录 / EXCERPT: Let $\mathcal{H}_\xi$ be the space of holomorphic maps associated with parameter ξ, and let $\mathcal{M}_\xi$ be the corresponding vortex moduli space. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-024053 | MATHDB-324436

The Uryson width conjecture for metrics on spheres

摘录 / EXCERPT: Let m and k be as in the claim, let $1\leq c\leq m$, and let g_0 denote the unit sphere metric on S^m. For a metric g on S^m, write $\Lambda^k g\leq\Lambda^k g_0$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-024054 | MATHDB-324471

Donaldson's conjecture on conical Kähler-Einstein metrics and the greatest Ricci lower bound

摘录 / EXCERPT: Donald's conjecture. For all $0<\beta<R(M)$ there exists a cone-singularity solution to this equation, and there is no solution for $R(M)<\beta<1$. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-024055 | MATHDB-324475

The distance-sum characterization of regular simplices

摘录 / EXCERPT: Regular simplex conjecture. Then $A_1\ldots A_{n+1}$ are the vertices of a regular simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-024056 | MATHDB-324476

The distance-sum characterization of cross-polytopes

摘录 / EXCERPT: Cross-polytope conjecture. Then $A_1,\ldots,A_{2n}$ are the vertices of an n-dimensional cross-polytope.

conjecture | open | MathDB 页面 | 原文线索

V2-024057 | MATHDB-324477

The polar-duality extremum conjecture for regular polytopes

摘录 / EXCERPT: Polar-duality conjecture. If the first sum attains its maximum at a point N, then the second sum attains its minimum at N, and conversely. If the first sum is independent of th...

conjecture | open | MathDB 页面 | 原文线索

V2-024058 | MATHDB-324486

The conjecture that the essential spectrum is a half-line under a Ricci lower bound

摘录 / EXCERPT: Let M be a complete noncompact Riemannian manifold with Ricci curvature bounded below. Denote by the L^2 essential spectrum of the Laplacian on functions the spectral set assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-024059 | MATHDB-324502

Virtual Betti number conjecture for mapping tori with reducible fibers

摘录 / EXCERPT: Virtual Betti number conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024060 | MATHDB-324503

Symplectic mapping-torus conjecture for reducible fibers

摘录 / EXCERPT: Symplectic reducible-fiber conjecture. Then $Y=S^2\times S^1$ and

conjecture | open | MathDB 页面 | 原文线索

V2-024061 | MATHDB-324557

Neuwirth's conjecture for non-split links

摘录 / EXCERPT: Let L be a non-trivial, non-split link, and let $E(L)$ denote its exterior. A closed surface F containing L has each component of L non-separating in F when no component...

conjecture | open | MathDB 页面 | 原文线索

V2-024062 | MATHDB-324566

Rigidity conjecture for the bottom of the spectrum of regular Hilbert geometries

摘录 / EXCERPT: Let $(\mathcal{C},d_{\mathcal{C}})$ be a regular Hilbert geometry, and let $\lambda_1(\mathcal{C})$ denote the bottom of the spectrum of its Finsler Laplacian. Rigidity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-024063 | MATHDB-324585

The Standard Simplex Conjecture for Gaussian noise stability

摘录 / EXCERPT: Let $n\geq2$, let $\rho\in[-1,1]$, and let $3\leq k\leq n+1$. Let $\{A_i\}_{i=1}^k$ be a measurable partition of $\mathbb{R}^n$, meaning that $\bigcup_{i=1}^k A_i=\mathbb{R}^n$ and...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-024064 | MATHDB-324589

The secure closed Riemannian manifold conjecture

摘录 / EXCERPT: Let M be a closed Riemannian manifold. It is secure if every pair of points has a finite blocking set intersecting every geodesic segment joining them, and it is flat if its Riem...

conjecture | open | MathDB 页面 | 原文线索

V2-024065 | MATHDB-324597

The Short Curve conjecture for Weil-Petersson geodesics

摘录 / EXCERPT: Short Curve conjecture. If g is a Weil-Petersson geodesic with end invariant (ν^{-},ν^{+}), then:

conjecture | open | MathDB 页面 | 原文线索

V2-024066 | MATHDB-324598

Metric comparison conjecture for k-metrics on ℓ^s- and ℓ^t-type domains

摘录 / EXCERPT: For $0<s<t$, let G_s and G_t be the planar domains ... For points $z_1,z_2\in G_s$, write $k_{G_s}(z_1,z_2)$ and $k_{G_t}(z_1,z_2)$ for their quasihyperbolic distances in the res...

conjecture | open | MathDB 页面 | 原文线索

V2-024067 | MATHDB-324609

The optimal packing conjecture for nine congruent circles on a square flat torus

摘录 / EXCERPT: Nine-point packing conjecture. There is one unique, up to isometry, optimal arrangement of 9 points in ${\mathbb T}^2$, shown in the paper's Figure 9, with

conjecture | open | MathDB 页面 | 原文线索

V2-024068 | MATHDB-324610

Numerical exceptional-spectrum conjecture for certain complex hyperbolic arithmetic quotients

摘录 / EXCERPT: Let Q be the complex-hermitian form $\operatorname{diag}(I_n,-a)$ over $\mathbb{Z}[\sqrt{-3}]$, with $n\in\{3,4,5\}$ and $1\leq a\leq 15$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024069 | MATHDB-324613

Kontsevich's convergence conjecture for quadratic Poisson structures

摘录 / EXCERPT: Let cpi be a quadratic Poisson bivector on $cmathbf{A}^n$, so that the bracket $cbrace-,-cbrace$ is homogeneous, and let $ck=cmathbf{C}$. Kontsevich's deformation quantization gi...

conjecture | open | MathDB 页面 | 原文线索

V2-024070 | MATHDB-324626

The conjecture on Neumann heat content asymptotics with singular data

摘录 / EXCERPT: Let M be a compact Riemannian manifold with boundary, let D be the Laplace-type operator, let $\beta(\rho,\phi,D,\mathcal{B}_{\mathcal{D}/\mathcal{R}})(t)$ denote the heat cont...

conjecture | open | MathDB 页面 | 原文线索

V2-024071 | MATHDB-324639

Compact-by-Lie conjecture for groups quasi-isometric to negatively curved homogeneous manifolds

摘录 / EXCERPT: A locally compact group is compact-by-Lie if it has a compact normal subgroup such that the quotient is a Lie group. Let G be a compactly generated locally compact group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024072 | MATHDB-324640

The Poincaré conjecture for three-dimensional Alexandrov spaces with nonnegative curvature

摘录 / EXCERPT: Let A and A' be nonnegatively curved Alexandrov spaces with isometric boundaries, and let $A\cup_\partial A'$ denote their gluing along a boundary isometry. Let D^3,...

conjecture | open | MathDB 页面 | 原文线索

V2-024073 | MATHDB-324643

Infinite generation of invariant Griffiths groups for the cubic sevenfold and weighted quartic families

摘录 / EXCERPT: Let X_3 and X_4 be general members of the families of cubic sevenfolds and smooth hypersurfaces in $\mathbb{P}(1^6;2)$ described above, respectively. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-024074 | MATHDB-324661

Microlocalization conjecture for hyperbolic restriction

摘录 / EXCERPT: Microlocalization conjecture. (a) $\eta^*\bmu(\operatorname{IC}(X))$ is constant in any chamber $\fC$. (b) There is a canonical isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-024075 | MATHDB-324707

Gross-Wilson and Kontsevich-Soibelman metric SYZ conjecture

摘录 / EXCERPT: Let $\mathcal{X}\rightarrow D$ be a maximally unipotent degeneration of simply-connected Calabi-Yau manifolds with full $SU(n)$ holonomy. Let $t_i\in D$ satisfy $t_i\rightarrow 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-024076 | MATHDB-324708

Limiting Strominger-Yau-Zaslow conjecture

摘录 / EXCERPT: Let $(\mathcal{X}_{t_i},g_i)$ converge to (X_{∞},g_{∞}). Let (B_i,d_i) be non-empty for large i and converge to (B_{∞},d_{∞}), where B_i are compactif...

conjecture | open | MathDB 页面 | 原文线索

V2-024077 | MATHDB-324709

Symplectic compactification conjecture for polarized toric degenerations

摘录 / EXCERPT: Let $\mathcal{X}\rightarrow D$ be a polarized toric degeneration with intersection complex $(\check B,\check{\mathcal P})$. Let ω_t be a Kähler form on $\mathcal{X}_t$ repre...

conjecture | open | MathDB 页面 | 原文线索

V2-024078 | MATHDB-324710

Gevrey solution conjecture for holonomic \mathscr{D}-modules on the subanalytic site

摘录 / EXCERPT: Let X be a complex manifold, let $\rho_{\rm sa}$ denote the morphism from the subanalytic site of X to X, and let $\rho_{\rm sa!}$, $\rho_{\rm sa}^{-1}$, and $\rho_{\rm sa*}$...

conjecture | open | MathDB 页面 | 原文线索

V2-024079 | MATHDB-324733

Small-resolution conjecture for the mirror transition between $\mathcal{X}^*_C$ and $\mathcal{X}^*_{BB}$

摘录 / EXCERPT: Small-resolution conjecture. For a generic member $X \in \mathcal{X}^*_{BB}$, the map

conjecture | open | MathDB 页面 | 原文线索

V2-024080 | MATHDB-324735

The hyperbolic linear programming bound for sphere packing density

摘录 / EXCERPT: Let $f \colon [0,\infty) \to \mathbb{R}$ be continuous and integrable on $\mathbb{H}^n$, and suppose ... while $\widehat{f}(\lambda) \ge 0$ for all $\lambda>0$ and $\widehat{f}(0)>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-024081 | MATHDB-324791

Yamaguchi's tangent-splitting conjecture for noncompact Alexandrov spaces

摘录 / EXCERPT: Let $A\in \operatorname{Alex}^n(0)$ be a noncompact Alexandrov space with a soul S. Let C_i be the subsets associated with the soul as in 1.1, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024082 | MATHDB-324816

Geodesic strengthenings of the Norine and Feder-Subi conjectures

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, with edge set $E(Q_n)$, and let a geodesic be a shortest path in Q_n. For a 2-colouring of $E(Q_n)$, call it antipodal if no two ant...

conjecture | open | MathDB 页面 | 原文线索

V2-024083 | MATHDB-324817

The two-sided limit shadowing characterization of Anosov diffeomorphisms

摘录 / EXCERPT: Let M be the compact manifold under consideration, let $f\in\operatorname{Diff}^1(M)$, and let $\mathcal{TLS}$ denote the set of diffeomorphisms with the two-sided limit shadowin...

conjecture | open | MathDB 页面 | 原文线索

V2-024084 | MATHDB-324827

Molino's conjecture on closures of leaves of singular Riemannian foliations

摘录 / EXCERPT: Molino's conjecture. The partition $\overline{\mathcal{F}}$ given by the closures of the leaves of $\mathcal{F}$ is again a singular Riemannian foliation.

conjecture | open | MathDB 页面 | 原文线索

V2-024085 | MATHDB-324828

Conjecture on simultaneous resolution of related tropical conifold nodes

摘录 / EXCERPT: Related-nodes conjecture. A set of nodes $p_1,\ldots,p_k$ in $(B,\mathcal P,\phi)$ can be simultaneously resolved by the above process if and only if they are related.

conjecture | open | MathDB 页面 | 原文线索

V2-024086 | MATHDB-324831

The vanishing-cycle conjecture for stable framed objects in 3CY categories

摘录 / EXCERPT: Let $\mathcal C$ be a $3CY$ triangulated category and let $\mathcal M^{sfr}$ be the moduli space of stable framed objects. Vanishing-cycle conjecture. There is a formal manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-024087 | MATHDB-324834

Uniform heat-content conjecture for nodal sets

摘录 / EXCERPT: Let (M,g) be a compact smooth manifold without boundary. Let u be a Laplacian eigenfunction with eigenvalue λ, and let $p_t(x)$ be globally defined with respect to its...

conjecture | open | MathDB 页面 | 原文线索

V2-024088 | MATHDB-324839

Bound on the negative self-intersection of divisible embedded surfaces

摘录 / EXCERPT: Let X be a simply-connected 4-manifold, and let B be an embedded surface in X of genus g_B with self-intersection ... Assume that the class $[B]$ is divisible by 2. Embed...

conjecture | open | MathDB 页面 | 原文线索

V2-024089 | MATHDB-324849

The planar robustness conjecture for regular polygons

摘录 / EXCERPT: For $n>2$, let ρ_n denote the maximal robustness among planar convex solids in the equilibrium class with n points, and let $\rho(\text{regular n-gon})$ denote the robustnes...

conjecture | open | MathDB 页面 | 原文线索

V2-024090 | MATHDB-324850

The robustness equality for the equilibrium classes $(2,n)$ and $(n,2)$

摘录 / EXCERPT: For the three-dimensional equilibrium classes indexed by pairs $(2,n)$ and $(n,2)$, let $\rho_{2,n}$ and $\rho_{n,2}$ denote their maximal full robustness, and let ρ_n denote...

conjecture | open | MathDB 页面 | 原文线索

V2-024091 | MATHDB-324851

Monotonicity of robustness across equilibrium classes

摘录 / EXCERPT: Let $\rho_{i,j}$ denote the maximal robustness of the three-dimensional equilibrium class indexed by (i,j). Equilibrium-class monotonicity conjecture. If $i\geq k$ and $j\geq l$,...

conjecture | open | MathDB 页面 | 原文线索

V2-024092 | MATHDB-324910

Batyrev-Ciocan-Fontanine-Kim-van Straten conjecture for the mirror conifold family

摘录 / EXCERPT: Batyrev-Ciocan-Fontanine-Kim-van Straten's conjecture. The closure Y_0^* in a maximal projective crepant partial resolution $\hat{\mathbb{P}}_{\nabla}$ has p nodes and satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-024093 | MATHDB-324918

Formality conjecture for derived self-intersections

摘录 / EXCERPT: Let X and Y be a pair of complex analytic spaces for which the derived self-intersection is defined, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024094 | MATHDB-324931

Kawamata's K-dominance conjecture

摘录 / EXCERPT: Kawamata's conjecture. If X K-dominates X', then there is a fully faithful functor

conjecture | open | MathDB 页面 | 原文线索

V2-024095 | MATHDB-324981

Guillemin's diffeomorphism-type conjecture for three-dimensional Zollfrei manifolds

摘录 / EXCERPT: Guillemin's conjecture. Every Zollfrei manifold in dimension three has the same diffeomorphism type as one of the standard examples.

conjecture | open | MathDB 页面 | 原文线索

V2-024096 | MATHDB-324995

Sarnak-Xue lattice-point counting conjecture

摘录 / EXCERPT: Let $G=\operatorname{SO}(n,1)^\circ$, let $\rho=\tfrac{n-1}{2}$, and let $\Gamma<G(\mathbb{Z})$ satisfy $[G(\mathbb{Z}):\Gamma]<\infty$. Fix $o\in\mathbb H^n$, let d be hyperboli...

conjecture | open | MathDB 页面 | 原文线索

V2-024097 | MATHDB-324997

Geometrically finite lattice-point counting conjecture for congruence covers

摘录 / EXCERPT: Let $G=\operatorname{SO}(n,1)^\circ$, let $\rho=\tfrac{n-1}{2}$, and let $\Gamma<G(\mathbb{Z})$ be geometrically finite and Zariski dense, with critical exponent δ. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-024098 | MATHDB-325014

Core-tunnel geodesic conjecture for genus-two compression bodies

摘录 / EXCERPT: Let C be a compression body with $\partial_{-}C$ a torus and $\partial_{+}C$ a genus-two surface. A core tunnel is the arc in C corresponding to the single tunnel in the deform...

conjecture | open | MathDB 页面 | 原文线索

V2-024099 | MATHDB-325015

Dual-Ford geodesic conjecture for compression-body core tunnels

摘录 / EXCERPT: Let C be a $(1;2)$--compression body, meaning a compression body whose negative boundary is a torus and whose positive boundary has genus two. Equip C with a geometrically fini...

conjecture | open | MathDB 页面 | 原文线索

V2-024100 | MATHDB-325036

Nussbaum-Vandehey conjecture on Hilbert nonexpansive maps

摘录 / EXCERPT: Let C be a bounded convex domain in $\mathbb{R\mathrm{^{N}}}$ equipped with Hilbert's metric d, and let $f:C\rightarrow C$ be a 1-Lipschitz map. Nussbaum-Vandehey conjecture. E...

conjecture | open | MathDB 页面 | 原文线索

V2-024101 | MATHDB-325052

Bellows conjecture for flexible polyhedra

摘录 / EXCERPT: Let P be a flexible polyhedron in Euclidean space, and let its generalized volume be the volume invariant associated with its possibly self-intersecting realization. A bending is...

conjecture | open | MathDB 页面 | 原文线索

V2-024102 | MATHDB-325055

Pair-correlation density formula for hyperbolic lattice angles

摘录 / EXCERPT: Pair-correlation density conjecture. The pair-correlation measure $R_2(\xi)$ exists on $[0,\infty)$ and is given by a C^1 function expressed as a series of three-dimensional volu...

conjecture | open | MathDB 页面 | 原文线索

V2-024103 | MATHDB-325058

Finite generation conjecture for purely log terminal pairs of log general type

摘录 / EXCERPT: Finite generation conjecture. The log canonical ring $R(X,\Delta)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-024104 | MATHDB-325069

The triple-line conjecture for point sets

摘录 / EXCERPT: Triple-line conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-024105 | MATHDB-325076

Eventual concavity conjecture for parallel volume

摘录 / EXCERPT: Eventual concavity conjecture. There exists t_0 such that $V_{A,B}(t)$ is $\frac{1}{n}$-concave on $[t_0,+\infty)$. This is proposed as a weaker form of the Costa-Cover conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-024106 | MATHDB-325078

Ball equality conjecture for intermediate convex-hull constants

摘录 / EXCERPT: Intermediate constant conjecture. For every $K\in\mathcal{K}_n$,

open_problem | open | MathDB 页面 | 原文线索

V2-024107 | MATHDB-325079

Ellipsoid characterization from the translative constant volume property

摘录 / EXCERPT: Translative constant volume conjecture. Then K is an ellipsoid.

conjecture | open | MathDB 页面 | 原文线索

V2-024108 | MATHDB-325107

The stable-region splitting conjecture for higher-dimensional hyperplanes

摘录 / EXCERPT: Let $\Sigma=\mathbb{R}^n\subset\mathbb{R}^{n+1}$ be a self-shrinker with $n\geq 3$. Let r_1 be the dimension-dependent radius from the cited remark. Then there should exist a num...

conjecture | open | MathDB 页面 | 原文线索

V2-024109 | MATHDB-325109

Rigidity conjecture for Lipschitz equivalence of self-similar sets

摘录 / EXCERPT: Let ${\boldsymbol{\rho}}=(\rho_1,\ldots,\rho_m)$ be a contraction vector such that $\operatorname{rank}\langle {\boldsymbol{\rho}}\rangle=m$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024110 | MATHDB-325117

The Lagrangian conjecture for hyperkähler manifolds

摘录 / EXCERPT: Let X be a hyperkähler manifold, and let L be a nontrivial nef line bundle on X. Denote by $q(L)$ its Beauville-Bogomolov square. Lagrangian conjecture. If ... then some power...

conjecture | open | MathDB 页面 | 原文线索

V2-024111 | MATHDB-325120

Givental-space formulation of the Crepant Resolution Conjecture

摘录 / EXCERPT: Let $\mathcal{X}$ be a toric Gorenstein orbifold and Y a crepant resolution, with Givental spaces ... for $\mathcal{Z}=\mathcal{X},Y$, equipped with the symplectic form...

conjecture | open | MathDB 页面 | 原文线索

V2-024112 | MATHDB-325160

Contact conjecture for odd-dimensional moment-angle manifolds

摘录 / EXCERPT: Contact conjecture for odd-dimensional moment-angle manifolds. For every admissible configuration $\pmb{\Lambda}\subset\mathbb{R}^{2m}$, the manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-024113 | MATHDB-325164

Kaledin-Lehn-Sorger conjecture on symplectic reductions

摘录 / EXCERPT: Let V be a finite-dimensional vector space, let $W=(V\oplus V^*)^{\oplus m}$, and let $G\subset GL(V)$ be a reductive algebraic subgroup acting naturally on W. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024114 | MATHDB-325190

Nonpositivity conjecture for the logarithmic tangent sheaf of cuspidal curves on Hirzebruch surfaces

摘录 / EXCERPT: Let C be a rational cuspidal curve with four or more cusps on a Hirzebruch surface $\mathbb{F}_e$. Let (V,D) be the minimal embedded resolution of C, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024115 | MATHDB-325224

Locality conjecture for perfectoid algebras

摘录 / EXCERPT: Let K be a perfectoid field, let (A,A^+) be a complete affinoid K-algebra, and set $X=\operatorname{Spa}(A,A^+)$. Assume that X has a cover by rational subsets...

conjecture | open | MathDB 页面 | 原文线索

V2-024116 | MATHDB-325231

Conjecture on cusp sections arising from cubulations

摘录 / EXCERPT: Cusp-section conjecture. The cusp section of a hyperbolic 4-manifold with a single cusp arising from a cubulation is a flat 3-torus.

conjecture | open | MathDB 页面 | 原文线索

V2-024117 | MATHDB-325264

Analytic torsion convergence conjecture for Benjamini-Schramm covers

摘录 / EXCERPT: Let M be a fixed closed hyperbolic 3-manifold, and let M_n be covers of M that Benjamini-Schramm converge to hyperbolic 3-space $\mathbb{H}^3$. The analytic torsion conve...

conjecture | open | MathDB 页面 | 原文线索

V2-024118 | MATHDB-325275

Boundary length decrease conjecture for higher-dimensional free boundary minimal submanifolds

摘录 / EXCERPT: Let B^n be the unit ball with boundary S^{n-1}. Let Σ be a k-dimensional minimal submanifold in B^n with boundary $\partial\Sigma\subset S^{n-1}$, and suppose that t...

conjecture | open | MathDB 页面 | 原文线索

V2-024119 | MATHDB-325284

Orevkov's four-cusp conjecture for plane rational cuspidal curves

摘录 / EXCERPT: Let $C\subset \mathbb{P}^2$ be a plane rational cuspidal curve, meaning a rational plane curve whose singularities are cusps. Orevkov's four-cusp conjecture. The curve C can not...

conjecture | open | MathDB 页面 | 原文线索

V2-024120 | MATHDB-325294

The odd Alexandrov-Fenchel quermassintegral inequality for horospherical convex hypersurfaces

摘录 / EXCERPT: Let n and k satisfy ... Let $\Sigma\subset\mathbb{H}^n$ be a horospherical convex hypersurface, let $|\Sigma|$ denote its area, let ω_{n-1} denote the area of the unit sph...

conjecture | open | MathDB 页面 | 原文线索

V2-024121 | MATHDB-325326

Song's metric uniformization conjecture for smooth flips

摘录 / EXCERPT: Let $\phi_-:X^-\to Y\leftarrow X^+:\phi_+$ be a smooth flip of two smooth projective manifolds, and let T be a finite singular time. Assume that $g(t)$ is a smooth Kähler-Ricci f...

conjecture | open | MathDB 页面 | 原文线索

V2-024122 | MATHDB-325327

Song's metric uniformization conjecture for singularities of smooth flips

摘录 / EXCERPT: Let $\phi_-:X^-\to Y\leftarrow X^+:\phi_+$ be a smooth flip of two smooth projective manifolds. For each point y in the singular set of Y, choose $p^-\in\phi_-^{-1}(y)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-024123 | MATHDB-325371

The chromatic bound conjecture for the first Steklov eigenvalue

摘录 / EXCERPT: Let Σ be a compact surface with boundary. Write $m_1(\Sigma)$ for the maximal multiplicity of the first Steklov eigenvalue $\sigma_1(\Sigma,g,\rho,\gamma)$ as the metric g...

conjecture | open | MathDB 页面 | 原文线索

V2-024124 | MATHDB-325450

Finiteness conjecture for essential projective transformations

摘录 / EXCERPT: Finiteness conjecture. Unless (M,g) is a finite quotient of the standard Riemannian sphere,

conjecture | open | MathDB 页面 | 原文线索

V2-024125 | MATHDB-325457

Tensor telescope conjecture

摘录 / EXCERPT: Let $\mathscr{T}$ be a compactly generated $\otimes$-triangulated category and let $\mathscr{U}$ be a compactly generated triangulated category, both with all coproducts. Suppose .....

conjecture | open | MathDB 页面 | 原文线索

V2-024126 | MATHDB-325482

The natural atlas conjecture for nonsingular arcs

摘录 / EXCERPT: Let X be a complex space, and let $\operatorname{Arc}^*(X)\subset \operatorname{Arc}(X)$ be the space of arcs that are not contained in $\operatorname{Sing} X$. A finite type fam...

conjecture | open | MathDB 页面 | 原文线索

V2-024127 | MATHDB-325483

The curve selection conjecture for irreducible components of arc spaces

摘录 / EXCERPT: Let X be a complex space, let W be an irreducible component of $\operatorname{Arc}^{\circ}(X)$ or $\operatorname{ShArc}(X)$, and let $[\phi_1],\dots,[\phi_m]$ be arcs in W. C...

conjecture | open | MathDB 页面 | 原文线索

V2-024128 | MATHDB-325484

The neighborhood-independence conjecture for arc spaces of isolated singularities

摘录 / EXCERPT: Let $(0\in X)$ be an isolated singularity, and let $0\in U\subset X$ be a contractible, Stein neighborhood. Neighborhood-independence conjecture. The homeomorphism types of...

conjecture | open | MathDB 页面 | 原文线索

V2-024129 | MATHDB-325488

Campana's classification conjecture for simple compact Kähler manifolds

摘录 / EXCERPT: Let X be a simple compact Kähler manifold of dimension n, and write $q=h^1(X,\mathcal O_X)$ and $\kappa(X)$ for its irregularity and Kodaira dimension. A holomorphic 2-form...

conjecture | open | MathDB 页面 | 原文线索

V2-024130 | MATHDB-325515

Genus two G-function vanishing conjecture for ADE Frobenius manifolds

摘录 / EXCERPT: Genus two G-function vanishing conjecture. For every such M,

conjecture | open | MathDB 页面 | 原文线索

V2-024131 | MATHDB-325528

Almost-sure constancy of the dimension of uniformly random self-affine sets

摘录 / EXCERPT: A uniformly random self-affine set is obtained as the attractor of a random iterated function system of affine contractions, with the randomness specified by the model under consid...

conjecture | open | MathDB 页面 | 原文线索

V2-024132 | MATHDB-325557

Politeness of coadjoint actions with locally closed orbits

摘录 / EXCERPT: Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let G act on $\mathfrak{g}^{\ast}$ by the coadjoint action. Assume that every coadjoint orbit is locally closed in...

conjecture | open | MathDB 页面 | 原文线索

V2-024133 | MATHDB-325571

Universal polynomial formula for tropical fans of higher-ary strata

摘录 / EXCERPT: Universal polynomial conjecture. There exists a universal function P_m of pairwise disjoint sets $B_1,\ldots,B_m\subset\mathbb Z^n$, with values in...

conjecture | open | MathDB 页面 | 原文线索

V2-024134 | MATHDB-325578

The Cairo-Prismatic conjecture for perimeter-minimizing pentagonal tilings

摘录 / EXCERPT: Let a planar tiling consist of unit-area polygons with at most five sides, and measure its perimeter by the perimeter per tile. The Cairo and Prismatic tiles are the two specified...

conjecture | open | MathDB 页面 | 原文线索

V2-024135 | MATHDB-325598

Kawamata's nonnegativity conjecture for the second Chern class

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n with nef anticanonical bundle, and let $\omega_1,\ldots,\omega_{n-2}$ be nef classes. Kawamata's second-Chern-class conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-024136 | MATHDB-325620

Gritsenko-Nikulin's completeness conjecture for rank-3 hyperbolic generalized Cartan matrices

摘录 / EXCERPT: Let A be a hyperbolic generalized Cartan matrix of rank 3 and elliptic type, with a lattice Weyl vector, whose geometric realization is twisted to a symmetric generalized Carta...

conjecture | open | MathDB 页面 | 原文线索

V2-024137 | MATHDB-325624

Chernov-Kauffman conjecture on virtual Legendrian isotopy

摘录 / EXCERPT: Let F be either S^2 or $\mathbb{R}^2$, and let two Legendrian knots lie in the unit cotangent bundle ST^*F. Suppose they are isotopic as virtual Legendrian knots. Chernov-Kau...

conjecture | open | MathDB 页面 | 原文线索

V2-024138 | MATHDB-325625

Minimal-genus virtual Legendrian isotopy conjecture

摘录 / EXCERPT: Let F be a surface, and let K_1 and K_2 be Legendrian knots in ST^*F that are isotopic as virtual Legendrian knots. Suppose that F is the surface of smallest genus realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-024139 | MATHDB-325639

Cuspless sub-Riemannian exponential-map conjecture in $SE(3)$

摘录 / EXCERPT: Let $\widetilde{Exp}_e$ be the reduced exponential map, let $\mathcal{D}$ be its domain, let $\mathcal{R}$ be its range, and let $\mathcal{C}$ and s_{max} be as defined in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-024140 | MATHDB-325667

Necessity of condition (A4) for translation ovals

摘录 / EXCERPT: The paper considers sets ${\cal C}$ and their corresponding spreads in a finite even order plane. Condition (A4) is one of the structural conditions imposed on these objects. Neces...

open_problem | open | MathDB 页面 | 原文线索

V2-024141 | MATHDB-325668

Classification of spreads yielding projectively inequivalent hyperovals

摘录 / EXCERPT: Let ${\cal S}$ be a spread containing t_N and t_∞, whose other lines each meet exactly one ${\cal C}$-line. Classification of inequivalent hyperovals. Which regular spread...

open_problem | open | MathDB 页面 | 原文线索

V2-024142 | MATHDB-325671

Akutagawa-Maeta conjecture for complete biharmonic Euclidean submanifolds

摘录 / EXCERPT: Let M be a complete submanifold of Euclidean space $\mathbb{E}^n$. An isometric immersion $\phi:(M,g)\rightarrow\mathbb{E}^n$ is biharmonic when its bitension field satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-024143 | MATHDB-325678

Exotic monotone torus conjecture for \(\mathbb{CP}^1\times\mathbb{CP}^1\)

摘录 / EXCERPT: Let $T^c_{r,0}$ be a Lagrangian torus in $\mathbb{CP}^1\times\mathbb{CP}^1$ of type $T(1,2,9)$, and let a family of Maslov-index-2 holomorphic discs mean a family of such discs w...

conjecture | open | MathDB 页面 | 原文线索

V2-024144 | MATHDB-325713

Young's weighted geodesic L^2 conjecture for Hecke cusp forms

摘录 / EXCERPT: Let $\Gamma=PSL_2(\mathbb{Z})$, let $E(z,s)$ denote the Eisenstein series, let u_j be an even Hecke-Maass cusp form, and let f be a holomorphic Hecke cusp form of weight k. F...

conjecture | open | MathDB 页面 | 原文线索

V2-024145 | MATHDB-325715

Young's logarithmic geodesic L^2 conjecture for Maass forms

摘录 / EXCERPT: Let u_j be an even Hecke-Maass cusp form with spectral parameter t_j, so its Laplace eigenvalue is $1/4+t_j^2$. Young's logarithmic geodesic L^2 conjecture. ... as...

conjecture | open | MathDB 页面 | 原文线索

V2-024146 | MATHDB-325727

The characterization of real valuations with linear growth

摘录 / EXCERPT: Linear-growth characterization. A real valuation v has linear growth if and only if there exists a divisorial valuation v' such that $v\leqslant v'$ at the center of v on som...

conjecture | open | MathDB 页面 | 原文线索

V2-024147 | MATHDB-325732

The upper-envelope switching-point conjecture for polynomials

摘录 / EXCERPT: Upper-envelope switching-point conjecture. The function g has at most $d(n-1)$ points at which it switches between a pair of the polynomials f_i.

conjecture | open | MathDB 页面 | 原文线索

V2-024148 | MATHDB-325746

Castelnuovo bound conjecture for vertices of tropical curves

摘录 / EXCERPT: Castelnuovo bound conjecture. The number of vertices of G is bounded by

conjecture | open | MathDB 页面 | 原文线索

V2-024149 | MATHDB-325747

Characterization of conformal factors for flat hexagonal triangulations

摘录 / EXCERPT: Characterization conjecture. A hexagonally triangulated flat plane is conformal to the regular one if and only if the PL conformal factor w is linear.

conjecture | open | MathDB 页面 | 原文线索

V2-024150 | MATHDB-325753

Connectedness conjecture for the source fibers of presymplectic groupoids

摘录 / EXCERPT: Connectedness conjecture. Theorem 1 holds also when the s-fibers are not connected.

conjecture | open | MathDB 页面 | 原文线索

V2-024151 | MATHDB-325759

The tropical Lagrangian torus conjecture for Fano hypersurfaces

摘录 / EXCERPT: Tropical Lagrangian torus conjecture. The torus satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024152 | MATHDB-325761

Densest hyperball packing conjecture for two 5-dimensional Coxeter tilings

摘录 / EXCERPT: Densest hyperball packing conjecture. The above described optimal hyperball packings to Coxeter tilings $[5,3,3,3,3]$ and $[5,3,3,3,3^{1,1}]$ provide the densest hyperball packings...

conjecture | open | MathDB 页面 | 原文线索

V2-024153 | MATHDB-325763

Effective basepoint-free conjecture for nef log canonical divisors

摘录 / EXCERPT: Let $(X,Delta)$ be a projective klt pair of dimension n, and let k be a positive integer such that $k(K_X+Delta)$ is Cartier. Assume that $K_X+Delta$ is nef. Basepoint-free con...

conjecture | open | MathDB 页面 | 原文线索

V2-024154 | MATHDB-325798

Fat CAT conjecture for spherical triangulations

摘录 / EXCERPT: Let T be a geodesic triangulation of $\mathbb{S}^2$, with triangles $R_i=A_iB_iC_i$ for $i=1,2,\ldots,m$. A spherical triangle is fatter than another when each of its correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-024155 | MATHDB-325808

Pugh's conjecture on one-dimensional stable manifolds in Pesin theory

摘录 / EXCERPT: Let $f\colon M\to M$ be a C^1-diffeomorphism, and let $Orb(p)$ be an orbit with well-defined non-vanishing Lyapunov exponents. In the two-dimensional case, assume that ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-024156 | MATHDB-325816

Costa-Cover conjecture on the concavity of parallel volume

摘录 / EXCERPT: Let A be a bounded measurable set in $cmathbb{R}^n$, let B_2^n denote the Euclidean closed unit ball, and let $|\cdot|$ denote Lebesgue measure. Costa-Cover conjecture. The fun...

conjecture | open | MathDB 页面 | 原文线索

V2-024157 | MATHDB-325820

Gramination conjecture for the stretch locus

摘录 / EXCERPT: Let $G=\mathrm{PO}(n,1)$, let Γ_0 be the group in the representation space under consideration, and let $(j,\rho)\in\operatorname{Hom}(\Gamma_0,G)^2$. For $n=2$, assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-024158 | MATHDB-325839

Landau-Ginzburg/Calabi-Yau correspondence

摘录 / EXCERPT: Consider the Lagrangian cones $\mathcal L_{\operatorname{FJRW}}$ and $\mathcal L_{\operatorname{GW}}$ in the symplectic spaces $\mathcal V_{\operatorname{FJRW}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-024159 | MATHDB-325851

Basak's generation conjecture for the Allcock orbifold fundamental group

摘录 / EXCERPT: Let I_{26} be the incidence diagram of the 13 points and 13 lines of $\mathbb{P}^2(3)$, let $A(I_{26})$ be its Artin group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024160 | MATHDB-325852

The mirror-avoidance conjecture for the Allcock 13-polytope

摘录 / EXCERPT: Let $\mathbb{B}(V)$ be the real hyperbolic ball of dimension 13 through the Weyl point w_0, containing the 26 geodesics from w_0 to the nearest mirrors. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024161 | MATHDB-325853

The mirror-avoidance conjecture for the Allcock 26-cell

摘录 / EXCERPT: Let V be the Lorentzian real vector space of dimension 14 associated with the incidence diagram I_{26}, let $\mathbb{B}(V)$ be its real hyperbolic ball, and let ... be the 26...

open_problem | open | MathDB 页面 | 原文线索

V2-024162 | MATHDB-325883

Kobayashi's pseudo-Riemannian space-form conjecture

摘录 / EXCERPT: Let a pseudo-Riemannian space form of signature (p,q) be a pseudo-Riemannian manifold with constant sectional curvature κ, where $p,q\neq 0$. Kobayashi's pseudo-Riemannian...

conjecture | open | MathDB 页面 | 原文线索

V2-024163 | MATHDB-325900

Minkowski-plane bead-gap tightness conjecture for Steiner minimum paths

摘录 / EXCERPT: Let N be a set of n terminals in a Minkowski plane M with unit ball B, and let T_S and T_{opt} be a d_M-SMT and a d_M-MSPT on N, respectively. Minkowski-...

conjecture | open | MathDB 页面 | 原文线索

V2-024164 | MATHDB-325901

Full Euclidean Steiner tree degeneracy conjecture for minimum Steiner point trees

摘录 / EXCERPT: Let N be a set of terminals embedded in the Euclidean plane, and let T_S be a full Steiner minimum tree on N. Full-tree degeneracy conjecture. There exists a minimum Steiner...

conjecture | open | MathDB 页面 | 原文线索

V2-024165 | MATHDB-325905

Markus conjecture on completeness of compact affine manifolds

摘录 / EXCERPT: Soit M une variété compacte munie d'une structure affine, c'est-à-dire d'une structure modelée sur $(\operatorname{Aff}(\mathbb{R}^n),\mathbb{R}^n)$. Une forme volume parallèle e...

conjecture | open | MathDB 页面 | 原文线索

V2-024166 | MATHDB-325910

Smooth density conjecture for inextensible centerlines with nonvanishing curvature

摘录 / EXCERPT: Smooth density conjecture. Smooth functions are strongly dense in $\Xi^{\kappa_m,p}$. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-024167 | MATHDB-325916

Henrot's reverse Faber-Krahn conjecture for the Steklov-type eigenvalue

摘录 / EXCERPT: Henrot's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024168 | MATHDB-325917

Reverse Bossel-Daners conjecture for negative Robin parameter

摘录 / EXCERPT: Reverse Bossel-Daners conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024169 | MATHDB-325918

Eigenvalue ratio conjecture for nonnegatively curved Alexandrov spaces

摘录 / EXCERPT: Eigenvalue ratio conjecture. For every natural number k, there exists a positive constant C_k depending only on k such that

conjecture | open | MathDB 页面 | 原文线索

V2-024170 | MATHDB-325948

Makeev's universal covering conjecture for convex bodies

摘录 / EXCERPT: Makeev's conjecture. Every convex body K in $\mathbb{R}^d$ of diameter ≤ 1 can be covered by a translate of R_Δ for some non-degenerate simplex Δ of diameter...

conjecture | open | MathDB 页面 | 原文线索

V2-024171 | MATHDB-326058

Rovenski's normalized partial Ricci flow convergence conjecture

摘录 / EXCERPT: Let (M^{n+p}, g) be a compact Riemannian manifold that is the total space of a smooth fiber bundle ... with totally geodesic p-dimensional fibers. Suppose that all mixed sectiona...

conjecture | open | MathDB 页面 | 原文线索

V2-024172 | MATHDB-326069

Schoen's harmonic extension conjecture for hyperbolic spaces

摘录 / EXCERPT: Schoen's conjecture. There exists a unique harmonic and quasi-isometric map

conjecture | open | MathDB 页面 | 原文线索

V2-024173 | MATHDB-326081

Artin's classification conjecture for function skewfields of noncommutative surfaces

摘录 / EXCERPT: Let A be a noetherian, connected graded domain over ${\Bbbk}$ with $\operatorname{GKdim} A=3$. Its graded quotient ring has the form $Q_{gr}(A)\cong D[t,t^{-1};\alpha]$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-024174 | MATHDB-326083

Duality pairing conjecture for better-behaved GKZ systems

摘录 / EXCERPT: Let C be a rational polyhedral cone in a lattice N, let C° denote the associated dual data, and let $\bbGKZ(C,0)$ and $\bbGKZ(C^\circ,0)$ be the corresponding better-behav...

conjecture | open | MathDB 页面 | 原文线索

V2-024175 | MATHDB-326102

Boucksom-Demailly-Păun-Peternell conjecture on transcendental holomorphic Morse inequalities

摘录 / EXCERPT: Let X be an n-dimensional compact complex manifold. For a real d-closed $(1,1)$-form α, let $X(\alpha,\leq 1)$ be the set where α has at most one negative eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-024176 | MATHDB-326106

The Mori/Brody hyperbolicity conjecture for log canonical divisors

摘录 / EXCERPT: Mori/Brody hyperbolicity conjecture. If (X,D) is Mori hyperbolic, then K_X+D is nef; if (X,D) is Brody hyperbolic, then K_X+D is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-024177 | MATHDB-326110

Finitely generated pants-like COL2 groups and hyperbolic 3-manifold groups

摘录 / EXCERPT: Let G be a finitely generated torsion-free discrete subgroup of $\operatorname{Homeo}_+(S^1)$. A pants-like COL_2 group has a pants-like collection of two invari...

conjecture | open | MathDB 页面 | 原文线索

V2-024178 | MATHDB-326111

The Cannon-boundary-action conjecture for virtual COL2 groups

摘录 / EXCERPT: Let G be a word-hyperbolic group whose ideal boundary is homeomorphic to a 2-sphere, and suppose that G acts faithfully on its boundary. Virtual COL2 conjecture. G is virtu...

conjecture | open | MathDB 页面 | 原文线索

V2-024179 | MATHDB-326113

The exceptional set conjecture for the 3-adic Cantor set

摘录 / EXCERPT: Exceptional set conjecture. The 3-adic exceptional set has Hausdorff dimension zero:

conjecture | open | MathDB 页面 | 原文线索

V2-024180 | MATHDB-326114

Existence of asymptotically many tight simplices in quaternionic projective space

摘录 / EXCERPT: Existence conjecture. As $d\to\infty$, there exist tight N-point simplices in ${\mathbb H}{\mathbb P}^{d-1}$ for every N satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-024181 | MATHDB-326115

Local dimension conjecture for 12- and 13-point quaternionic simplices

摘录 / EXCERPT: Local dimension conjecture. There exists a 12-point, respectively 13-point, tight simplex in ${\mathbb H}{\mathbb P}^2$ such that, in a neighborhood of it, the space of tight s...

conjecture | open | MathDB 页面 | 原文线索

V2-024182 | MATHDB-326116

Local dimension conjecture for 24- and 25-point octonionic simplices

摘录 / EXCERPT: Local dimension conjecture. There exists a 24-point, respectively 25-point, tight simplex in ${\mathbb O}{\mathbb P}^2$ such that, in a neighborhood of it, the space of tight s...

conjecture | open | MathDB 页面 | 原文线索

V2-024183 | MATHDB-326117

Local dimension of tight 39-point codes in octonionic projective 2-space

摘录 / EXCERPT: Local dimension conjecture. In a neighborhood of the explicitly constructed code, the space of tight 39-point codes modulo the action of F_4 is a manifold of dimension 24.

conjecture | open | MathDB 页面 | 原文线索

V2-024184 | MATHDB-326135

Conjecture on constant-mean-curvature hypersurfaces in the four-sphere

摘录 / EXCERPT: Let M be a closed hypersurface with constant mean curvature H in $\mathbb{S}^{4}$, and let S denote the squared length of its second fundamental form. Write $\beta(3,H)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-024185 | MATHDB-326153

The Hurwitz ball quotient automorphism conjecture

摘录 / EXCERPT: Hurwitz ball quotient automorphism conjecture. Theorems concerning the bound $|\operatorname{Aut}(S)| \leq 288 e(S)$ and its equality case should hold without the arithmetic assump...

conjecture | open | MathDB 页面 | 原文线索

V2-024186 | MATHDB-326168

Perverse stable-pair partition-function conjecture for nodal K3 fibrations

摘录 / EXCERPT: Perverse stable-pair partition-function conjecture. The partition functions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-024187 | MATHDB-326174

The three lens space fillings conjecture

摘录 / EXCERPT: Three lens space fillings conjecture. If M admits three lens space fillings along distinct slopes, then either M is the exterior of a torus knot in a lens space or M belongs...

conjecture | open | MathDB 页面 | 原文线索

V2-024188 | MATHDB-326175

The longitudinal surgery duals conjecture

摘录 / EXCERPT: Longitudinal surgery duals conjecture. If $K_1\subset L(p_1,q_1)$ and $K_2\subset L(p_2,q_2)$ are longitudinal surgery duals, then, up to reindexing, K_2 is a $(1,1)$--knot and...

conjecture | open | MathDB 页面 | 原文线索

V2-024189 | MATHDB-326176

Kawamata's finiteness conjecture for projective klt wlc models

摘录 / EXCERPT: Let (X,B) be a pair with boundary B. A projective klt wlc model of (X,B) is a model occurring in the log birational class of (X,B), considered up to log isomorphism. Kawama...

conjecture | open | MathDB 页面 | 原文线索

V2-024190 | MATHDB-326180

Categorical equivalence between finite real spectral triples and spectral C*-categories

摘录 / EXCERPT: Categorification conjecture. The category of finite even S^o-real spectral triples over $\mathbb{C}$ and spectral triple morphisms is equivalent to the category of finite-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-024191 | MATHDB-326181

The nef Cartier b-divisor determination conjecture

摘录 / EXCERPT: Nef Cartier b-divisor determination conjecture. If $D\in\operatorname{\mathbf{CDiv}}_{\mathbb{Q}}X$ is nef, then there exists a determination of D with center equal to the Cartie...

conjecture | open | MathDB 页面 | 原文线索

V2-024192 | MATHDB-326182

The functorial determination conjecture for nef integral Cartier b-divisors

摘录 / EXCERPT: Functorial determination conjecture. There exist a covariant functor $Y:\mathcal{C}'\to\mathcal{C}$ and a natural transformation $F:Y\to\iota$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-024193 | MATHDB-326190

The projective-space systolic conjecture for reversible optical hypersurfaces

摘录 / EXCERPT: Let Σ be a reversible optical hypersurface in the cotangent bundle $T^*\mathbb{R}P^n$ of real projective n-space, enclosing a volume V. A periodic characteristic on...

conjecture | open | MathDB 页面 | 原文线索

V2-024194 | MATHDB-326191

The Lagrangian evolution-relation conjecture for wave equations

摘录 / EXCERPT: Let (M,g) be a compact oriented pseudo-Riemannian manifold with boundary, isometrically embedded into some Euclidean space $\mathbb{R}^N$ equipped with a constant metric. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024195 | MATHDB-326192

The no-heavy-curves conjecture for few-distance point sets

摘录 / EXCERPT: Let $\mathcal P$ be a set of n points in $\mathbb R^2$, and let $D(\mathcal P)$ denote the number of distinct distances determined by $\mathcal P$. No-heavy-curves conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-024196 | MATHDB-326195

Mirzakhani's conjecture on higher-rank affine invariant submanifolds

摘录 / EXCERPT: Let $\mathcal{N}$ be a higher-rank affine invariant submanifold of a stratum of translation surfaces. A translation surface is not primitive if it covers a quadratic differential o...

conjecture | open | MathDB 页面 | 原文线索

V2-024197 | MATHDB-326231

Classification conjecture for normalized 3-Buchsbaum rank 2 bundles on projective 3-space

摘录 / EXCERPT: Classification conjecture. Every normalized, 3-Buchsbaum rank 2 bundle on ${\mathbb P}^3$ is either an instanton bundle of charge 3, 4, or 5 when $c_1=0$, or the cohomolo...

conjecture | open | MathDB 页面 | 原文线索

V2-024198 | MATHDB-326245

Wills' conjecture on Ehrhart coefficients of centrally symmetric lattice polytopes

摘录 / EXCERPT: Let $\mathcal{P}^n$ denote the family of lattice polytopes in $\mathbb{R}^n$, and let $C_n=[-1,1]^n$. For $P\in\mathcal{P}^n$, write ... for its Ehrhart polynomial. Suppose that P...

conjecture | open | MathDB 页面 | 原文线索

V2-024199 | MATHDB-326253

The K3 Halphen pencil conjecture for birationally rigid Fano threefold hypersurfaces

摘录 / EXCERPT: Let X be the birationally rigid Fano threefold hypersurface considered above, and let a Halphen pencil on X be a pencil whose general members are irreducible surfaces of Kodair...

conjecture | open | MathDB 页面 | 原文线索

V2-024200 | MATHDB-326257

Gromov's zero-in-the-spectrum conjecture

摘录 / EXCERPT: Gromov's zero-in-the-spectrum conjecture. There is $p \geq 0$ such that zero belongs to the spectrum of $(\Delta_p)_{\operatorname{min}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-024201 | MATHDB-326290

The finite-field Furstenberg set lower-bound conjecture

摘录 / EXCERPT: Finite-field Furstenberg set conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-024202 | MATHDB-326294

Fock-Goncharov finite-generation and codimension-two affineness conjecture

摘录 / EXCERPT: Fock-Goncharov affineness consequence. If the Fock-Goncharov conjecture holds, then $H^0(\mathcal X,\mathcal O_{\mathcal X})$ is finitely generated, and the canonical morphism abov...

open_problem | open | MathDB 页面 | 原文线索

V2-024203 | MATHDB-326301

The inverse σ_k-flow numerical criterion for existence

摘录 / EXCERPT: Let M be an n-dimensional compact Kähler manifold, let α be a fixed Kähler metric, and let ω vary in a fixed Kähler class. For $k=1,\ldots,n$, define the inverse...

conjecture | refuted | MathDB 页面 | 原文线索

V2-024204 | MATHDB-326308

Finite-generation conjecture for log canonical rings in Fujiki's class

摘录 / EXCERPT: Fujiki-class finite-generation conjecture. The ring $R(X,\Delta)$ is a finitely generated $\mathbb C$-algebra. The source states that this conjecture is equivalent, after taking a...

conjecture | open | MathDB 页面 | 原文线索

V2-024205 | MATHDB-326323

Projective-flatness conjecture for vector bundles of radial Kupka components

摘录 / EXCERPT: Let M be an algebraic manifold with $\dim_{\mathbb{C}}M\geq3$, let L be a very ample line bundle on M, and let $\omega\in K(M,L,x\,dy-y\,dx)$ be a foliation with normal bundl...

conjecture | open | MathDB 页面 | 原文线索

V2-024206 | MATHDB-326329

The KP decomposition conjecture for tropical amplitudes

摘录 / EXCERPT: KP decomposition conjecture. There exists such a KP decomposition for which, as $\alpha'\to0$, the following hold: (i) integrating over the outer domain gives only a subleading con...

conjecture | open | MathDB 页面 | 原文线索

V2-024207 | MATHDB-326353

Harbourne-Huneke degree-bound conjecture for symbolic powers

摘录 / EXCERPT: Harbourne-Huneke degree-bound conjecture. For every $r>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-024208 | MATHDB-326354

Harbourne-Huneke refined Chudnovsky question

摘录 / EXCERPT: Refined Chudnovsky question. For every $r>0$,

open_problem | open | MathDB 页面 | 原文线索

V2-024209 | MATHDB-326355

Bocci-Cooper-Harbourne generalized symbolic-power containment conjecture

摘录 / EXCERPT: Bocci-Cooper-Harbourne conjecture. For all $m\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-024210 | MATHDB-326392

Symmetries of linear finite-type Frobenius systems

摘录 / EXCERPT: Symmetry conjecture. The pair $(\hat{Y}^{(n)},\mu^*\hat{\rm I}^{(2n+2)})$ is a Frobenius system. Moreover, each Jacobi field $\textnormal{A}^k$, $k\leq n$, has an associated locall...

conjecture | open | MathDB 页面 | 原文线索

V2-024211 | MATHDB-326403

Conjecture on the order of gaps between consecutive Dirichlet eigenvalues

摘录 / EXCERPT: Let Ω be a bounded domain in an n-dimensional complete Riemannian manifold M, and let ... be the Dirichlet eigenvalues of the Laplacian on Ω. Conjecture on consecut...

conjecture | open | MathDB 页面 | 原文线索

V2-024212 | MATHDB-326428

The degree bound conjecture for Calabi-Yau threefolds in $\mathbb{P}^6$

摘录 / EXCERPT: Let $X\subset\mathbb{P}^6$ be a smooth Calabi-Yau threefold, and let $d=\deg(X)$. Degree bound conjecture. There are no Calabi-Yau threefolds of degree $d\geq 19$ in $\mathbb{P}^6$...

conjecture | open | MathDB 页面 | 原文线索

V2-024213 | MATHDB-326429

The degree bound conjecture for canonical surfaces in $\mathbb{P}^5$

摘录 / EXCERPT: Let $S\subset\mathbb{P}^5$ be a canonical surface, and let $d=\deg(S)$. Degree bound conjecture. There are no canonical surfaces of degree $d\geq 19$ in $\mathbb{P}^5$. This is pre...

conjecture | open | MathDB 页面 | 原文线索

V2-024214 | MATHDB-326432

Hermitian Einstein characterization of absolute Weyl-functional minimizers

摘录 / EXCERPT: Hermitian Einstein minimizer conjecture. The conformal class $[g]$ is an absolute minimizer of $\mathscr{W}$. Moreover, for the given M, every absolute minimizer arises from a me...

conjecture | open | MathDB 页面 | 原文线索

V2-024215 | MATHDB-326443

Lower-bound conjecture for the moduli group of positive scalar curvature metrics

摘录 / EXCERPT: Moduli-group rank conjecture. The rank of $\widetilde P(M)$ is at least $N_{\mathrm{fin}}(\Gamma)$.

conjecture | open | MathDB 页面 | 原文线索

V2-024216 | MATHDB-326446

The variance conjecture for the Euclidean norm

摘录 / EXCERPT: Variance conjecture-bis. There exists a constant C such that for every isotropic log-concave random vector X,

conjecture | open | MathDB 页面 | 原文线索

V2-024217 | MATHDB-326459

The facet orthogonality conjecture for the metric space of polyhedra

摘录 / EXCERPT: Let V be the vector configuration defining the metric space $M_{\mathcal{K}}(V)$, and let $M_{\mathcal{K}}^C$ be the subsets of its boundary defined using dihedral angles. A face...

conjecture | open | MathDB 页面 | 原文线索

V2-024218 | MATHDB-326460

The nonlocality conjecture for dihedral angles of polyhedral metrics

摘录 / EXCERPT: Let $C_1,C_2\subset[n]$ be disjoint circuits of a vector configuration V, and let Δ be a polytopal fan containing $\operatorname{pos}(V_{C_1})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-024219 | MATHDB-326475

Pesin's nonzero center Lyapunov exponents conjecture

摘录 / EXCERPT: Let M be a compact smooth Riemannian manifold, and let $\operatorname{Diff}^r_m(M)$ denote the C^r diffeomorphisms preserving volume m. For $f\in\operatorname{Diff}^r_m(M)$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-024220 | MATHDB-326513

Nagata-Iarrobino conjecture for generic points in affine space

摘录 / EXCERPT: Nagata-Iarrobino conjecture. If I is the ideal of $r\gg0$ generic points of $\mathbb{A}^{n}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024221 | MATHDB-326520

Subdifferential condition for minimizing trajectories

摘录 / EXCERPT: Subdifferential condition. For almost every $t\in(0,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-024222 | MATHDB-326566

Pliability conjecture for terminal factorial quartic hypersurfaces

摘录 / EXCERPT: Pliability conjecture. The set $\mathcal P(X)$ is finite, and

conjecture | open | MathDB 页面 | 原文线索

V2-024223 | MATHDB-326577

Hamilton-Tian conjecture for normalized Kähler-Ricci flows on Fano manifolds

摘录 / EXCERPT: Hamilton-Tian conjecture. The pair $(M,g(t))$ converges, at least along a subsequence, to a shrinking Kähler-Ricci soliton with mild singularities.

conjecture | solved | MathDB 页面 | 原文线索

V2-024224 | MATHDB-326592

Tosatti-Weinkove conjecture for the modified Hermitian Monge-Ampère equation

摘录 / EXCERPT: Tosatti-Weinkove's conjecture. There exists a unique pair (u,b), with u a smooth function on M and b a constant, such that

conjecture | open | MathDB 页面 | 原文线索

V2-024225 | MATHDB-326600

Gromov's conjecture on convex hypersurfaces in Alexandrov spaces

摘录 / EXCERPT: Let $X\in\operatorname{Alex}^n(\kappa)$ have no boundary, and let a convex hypersurface in X be equipped with its intrinsic metric. Gromov's conjecture. The hypersurface is an Al...

conjecture | open | MathDB 页面 | 原文线索

V2-024226 | MATHDB-326611

Hamilton-Jacobi conjecture for logarithmic Husimi limits

摘录 / EXCERPT: Hamilton-Jacobi conjecture. Any such limit G solves the Hamilton-Jacobi equation

conjecture | open | MathDB 页面 | 原文线索

V2-024227 | MATHDB-326612

Weighted Levi-Hadwiger covering conjecture

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^{n}$ be a convex body, and let $N_{\omega}(K,T)$ denote its weighted covering number. Weighted Levi-Hadwiger covering conjecture. ... Moreover, equality hol...

conjecture | open | MathDB 页面 | 原文线索

V2-024228 | MATHDB-326617

Spherical correlation conjecture

摘录 / EXCERPT: Let $\mathbb{S}^n$ be the canonical Riemannian sphere, let o be the center of a fixed hemisphere, and write the hemisphere as $B(o,\pi/2)\subset\mathbb{S}^n$. Let K_1,K_2 be ge...

conjecture | open | MathDB 页面 | 原文线索

V2-024229 | MATHDB-326618

Spherical Bishop-Gromov correlation conjecture

摘录 / EXCERPT: Let $\mathbb{S}^n$ be the sphere with its canonical Riemannian geometry, and let $o\in\mathbb{S}^n$. Let $X\subset\mathbb{S}^n$ be convex with o in its interior, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024230 | MATHDB-326621

Modified virtual localization for complete intersections in algebraic GKM manifolds

摘录 / EXCERPT: Modified localization conjecture. An analog of the theorem asserting that modified virtual localization computes the genus-one Gromov-Witten invariants of a Calabi-Yau hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-024231 | MATHDB-326644

Finite-order automorphism conjecture for surfaces with vanishing irregularity

摘录 / EXCERPT: Finite-order automorphism conjecture. The induced map

conjecture | open | MathDB 页面 | 原文线索

V2-024232 | MATHDB-326649

The b-semiampleness conjecture for klt-trivial fibrations

摘录 / EXCERPT: Let $f\colon(X,B)\rightarrow Z$ be a klt-trivial fibration, meaning that (X,B) is Kawamata log terminal at the generic point of Z and $K_X+B\sim_{\mathbb{Q}f}0$. The canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-024233 | MATHDB-326659

Bertram's GIT quotient conjecture for the last birational model of the blow-up

摘录 / EXCERPT: Bertram's conjecture. The last birational model of $\operatorname{Bl}_{X}E_0^-$ is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-024234 | MATHDB-326666

Batyrev-Nill codegree conjecture for Cayley decompositions

摘录 / EXCERPT: For a lattice polytope P, define its codegree by ... For example, the unimodular simplex satisfies $\operatorname{codeg}(\Delta_n)=n+1$. Batyrev-Nill conjecture. There is a functio...

conjecture | open | MathDB 页面 | 原文线索

V2-024235 | MATHDB-326676

Generic infinite-energy conjecture for wave solutions on Lorentzian manifolds

摘录 / EXCERPT: Let Σ_0 and Σ_1 be the initial and final spacelike hypersurfaces, let n_{Σ_1} denote the future-directed unit normal to Σ_1, and consider the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-024236 | MATHDB-326686

Stolz's fundamental-class conjecture for twisted spin cobordism

摘录 / EXCERPT: Let M be a connected, closed, smooth manifold whose universal cover is spin, let $\pi=\pi_1(M)$, and let $w^\nu(M):B\pi\to K$ be the twisting determined by its stable normal bund...

conjecture | open | MathDB 页面 | 原文线索

V2-024237 | MATHDB-326699

The conjecture on biholomorphicity of non-autonomous basins

摘录 / EXCERPT: Non-autonomous basin conjecture. The basin Ω is always biholomorphic to $\mathbb C^m$. This stronger conjecture is used to approach Bedford's stable-manifold conjecture; the...

conjecture | open | MathDB 页面 | 原文线索

V2-024238 | MATHDB-326711

The unconditional log-Brunn-Minkowski conjecture for symmetric convex bodies

摘录 / EXCERPT: The unconditional log-Brunn-Minkowski conjecture. Inequality $V(\lambda\cdot K+_o(1-\lambda)\cdot L)\geq V(K)^\lambda V(L)^{1-\lambda}$ is true for all $\lambda\in[0,1]$ and all sy...

conjecture | open | MathDB 页面 | 原文线索

V2-024239 | MATHDB-326712

The symmetric L^p Brunn-Minkowski conjecture

摘录 / EXCERPT: Let K and L be symmetric convex bodies in $\mathbb{R}^n$, let V denote volume, and let $\lambda\cdot K+_p(1-\lambda)\cdot L$ denote their L^p-Minkowski combination. The sym...

conjecture | open | MathDB 页面 | 原文线索

V2-024240 | MATHDB-326713

The symmetric L^p Minkowski conjecture

摘录 / EXCERPT: Let K and L be symmetric convex bodies in $\mathbb{R}^n$, let V denote volume, let h_K and h_L be their support functions, and let S_L be the surface area measure of...

conjecture | open | MathDB 页面 | 原文线索

V2-024241 | MATHDB-326737

Kashiwara-Schapira's R-constructibility conjecture for tempered solutions

摘录 / EXCERPT: Let X be a complex manifold, let $\mathcal{D}_X$ be its sheaf of differential operators, and let $\mathcal{M}\in D^b_h(\mathcal{D}_X)$ be a bounded complex with holonomic cohomol...

conjecture | open | MathDB 页面 | 原文线索

V2-024242 | MATHDB-326787

Gromov's waist comparison conjecture for positively curved Riemannian manifolds

摘录 / EXCERPT: Let X be an n-dimensional Riemannian manifold with sectional curvature $K\geq 1$, and let $f:X\to\mathbb{R}^k$ be a continuous map. For a subset $Y\subset X$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024243 | MATHDB-326791

Pants decomposition filling-volume conjecture for embedded surfaces

摘录 / EXCERPT: Let K) be a surface embedded in ${\mathbb{R}}^N$, of genus g, and let $\gamma_1,\dots,\gamma_{3g-3}$ be the boundary curves of a pants decomposition of K. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-024244 | MATHDB-326792

Quadratic pants decomposition conjecture for bilipschitz embedded surfaces

摘录 / EXCERPT: Let K be a surface of genus g embedded in ${\mathbb{R}}^N$ by a bilipschitz map, and let $\gamma_1,\dots,\gamma_{3g-3}$ be the boundary curves of a pants decomposition of K....

conjecture | open | MathDB 页面 | 原文线索

V2-024245 | MATHDB-326816

Least dense hyperball covering conjecture for the Coxeter tiling $[5,3,3,3,3]$

摘录 / EXCERPT: Let $\mathbf{H}^5$ be the 5-dimensional hyperbolic space, and consider the Coxeter tiling $[5,3,3,3,3]$ and the hyperball covering described in the paper. Least dense hyperball cov...

conjecture | open | MathDB 页面 | 原文线索

V2-024246 | MATHDB-326828

Pitts-Rubinstein conjecture on the Morse index of min-max limits

摘录 / EXCERPT: Let M^3 be a Riemannian 3-manifold, and let a min-max limit arising from a Heegaard splitting have components Γ_i. Call a component unstable if it is unstable as a minim...

conjecture | open | MathDB 页面 | 原文线索

V2-024247 | MATHDB-326833

The Riemannian positive-definite curvature sphere conjecture

摘录 / EXCERPT: Riemannian positive-definite curvature sphere conjecture. S^4 and $\overline{\mathbb C\mathbb P}^2$ are the only compact 4-manifolds admitting Riemannian metrics satisfying this...

conjecture | open | MathDB 页面 | 原文线索

V2-024248 | MATHDB-326858

Constant scalar curvature Sasakian metrics and K-polystability conjecture

摘录 / EXCERPT: Let (M,ξ) be a Sasakian manifold polarized by the Reeb vector field ξ, with polarized affine cone (Y,ξ), where $Y=C(M)$ is the fixed complex cone. A constant scalar cur...

conjecture | open | MathDB 页面 | 原文线索

V2-024249 | MATHDB-326867

Noncommutative Kato distinguished-triangle conjecture

摘录 / EXCERPT: Noncommutative distinguished-triangle conjecture. There is a natural distinguished triangle

conjecture | open | MathDB 页面 | 原文线索

V2-024250 | MATHDB-326869

Knieper's rigidity conjecture for complete Riemannian planes

摘录 / EXCERPT: Let $(\mathbb{R}^2,g)$ be a complete Riemannian plane. For a geodesic γ, say that a geodesic through a point $x\notin\gamma$ is disjoint from γ if it does not interse...

conjecture | open | MathDB 页面 | 原文线索

V2-024251 | MATHDB-326870

Higher-dimensional Riemannian foliation rigidity conjecture

摘录 / EXCERPT: Higher-dimensional rigidity conjecture. If every geodesic is a line and $S_{\gamma,\gamma}$ foliates $\mathbb{R}^n$ for every line γ, then g is flat. This is proposed as a...

conjecture | open | MathDB 页面 | 原文线索

V2-024252 | MATHDB-326900

Nagata-type conjecture for very general quasimonomial valuations

摘录 / EXCERPT: Let t be a real parameter, and let $\widehat\mu(t)$ denote the invariant for a very general quasimonomial valuation introduced in the paper. Nagata-type conjecture for quasimonom...

conjecture | open | MathDB 页面 | 原文线索

V2-024253 | MATHDB-326952

Norm convergence conjecture for relaxed Douglas-Rachford methods on Hilbert lattice cones

摘录 / EXCERPT: Let K be a Hilbert lattice cone and let A be a finite-codimension closed affine manifold. For parameters $0\leq a,b,c<1$, consider the relaxed Douglas-Rachford iteration define...

conjecture | open | MathDB 页面 | 原文线索

V2-024254 | MATHDB-326955

The Euclidean-space conjecture for expanding algebraic solitons

摘录 / EXCERPT: An expanding algebraic soliton is a homogeneous Ricci soliton whose soliton structure is algebraic. Euclidean-space conjecture. Any expanding algebraic soliton is diffeomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-024255 | MATHDB-326987

Riemannianity conjecture for restrictively CW-homogeneous left invariant (α_1,α_2)-metrics

摘录 / EXCERPT: Let F be a left invariant (α_1,α_2)-metric on a compact connected simple Lie group G, with a decomposition $\mathfrak{g}=\mathbf{V}_1+\mathbf{V}_2$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-024256 | MATHDB-326988

Dimension-decomposition conjecture for restrictively CW-homogeneous left invariant (α_1,α_2)-metrics

摘录 / EXCERPT: Let F be a left invariant (α_1,α_2)-metric on a compact connected simple Lie group G with a dimension decomposition (n_1,n_2), where $n_1\geq n_2>1$. Dimension-de...

conjecture | open | MathDB 页面 | 原文线索

V2-024257 | MATHDB-327017

Semistability-modified birational rigidity conjecture for cubic del Pezzo fibrations

摘录 / EXCERPT: Semistability-modified conjecture. If X satisfies the K-condition, then X is birationally rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-024258 | MATHDB-327044

Finiteness conjecture for connected 3-CS-transitive 2-level posets

摘录 / EXCERPT: Finiteness conjecture. The poset M is finite if and only if the interval I is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-024259 | MATHDB-327056

The omnioriented quasitoric representative conjecture for unitary equivariant bordism

摘录 / EXCERPT: Let $\mathcal{Z}_{2n}^{U}(T^n)$ be the group of T^n-equivariant unitary bordism classes of $2n$-dimensional unitary toric manifolds, let $\mathcal{Q}_*$ be the graded noncommutat...

conjecture | open | MathDB 页面 | 原文线索

V2-024260 | MATHDB-327057

The sharp fixed-point bound conjecture for nonbounding unitary toric manifolds

摘录 / EXCERPT: Let M^{2n} be a nonbounding unitary toric manifold, meaning that it does not bound equivariantly, and let $\lceil n/2\rceil$ denote the least integer greater than or equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-024261 | MATHDB-327070

The rank conjecture for injective map germs from surfaces to three-space

摘录 / EXCERPT: Let $g:(\mathbb{C}^2,0)\to(\mathbb{C}^3,0)$ be an injective map germ, and let $\operatorname{grad} g$ denote its gradient. The rank conjecture. The rank of the gradient at the ori...

conjecture | open | MathDB 页面 | 原文线索

V2-024262 | MATHDB-327075

Brunn-Minkowski conjecture for the alpha-Riesz capacity

摘录 / EXCERPT: Let $N\geq 2$ and let $\alpha\in(0,N)$. For a convex body $K\subset\mathbb{R}^N$, write ${\rm Cap}_{\alpha}(K)$ for its α-Riesz capacity. For convex bodies K_0,K_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-024263 | MATHDB-327076

Isoperimetric conjecture for the alpha-Riesz energy

摘录 / EXCERPT: Let $N\geq 2$, let $\alpha\in(0,N)$, and let $I_{\alpha}(K)$ denote the α-Riesz energy of a convex body $K\subset\mathbb{R}^N$. For $r>0$, consider convex bodies with perime...

conjecture | open | MathDB 页面 | 原文线索

V2-024264 | MATHDB-327093

The balanced Calabi-Yau metric conjecture

摘录 / EXCERPT: The balanced Calabi-Yau metric conjecture. There is a balanced metric $\tilde{\omega}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-024265 | MATHDB-327094

Diagonalizability of adjoint-abelian operators in smooth Minkowski spaces

摘录 / EXCERPT: Let a smooth Minkowski space be a finite-dimensional real normed space whose norm is smooth away from the origin, and let an adjoint-abelian operator be a linear operator A satis...

conjecture | open | MathDB 页面 | 原文线索

V2-024266 | MATHDB-327106

Logarithmic injectivity-radius conjecture for random mapping tori

摘录 / EXCERPT: Let S be a surface, and let N denote the length of a random monodromy word defining a mapping torus of S. The injectivity radius is the radius of the largest embedded metric...

conjecture | open | MathDB 页面 | 原文线索

V2-024267 | MATHDB-327107

Asymptotic eigenvalue conjecture for random mapping tori

摘录 / EXCERPT: Let N denote the length of a random monodromy word and let N^3 be the associated mapping torus. Write $\lambda_1(N)$ for the bottom eigenvalue of the Laplacian. Asymptotic eige...

conjecture | open | MathDB 页面 | 原文线索

V2-024268 | MATHDB-327133

Tian's uniform Bergman kernel lower-bound conjecture for Kähler-Einstein manifolds

摘录 / EXCERPT: Let $\{(M_i,g^i)\}$ be a sequence of n-dimensional Kähler-Einstein manifolds with constant scalar curvature n. For each M_i, let $\rho_l(M_i,g^i)$ denote the Bergman kernel o...

conjecture | open | MathDB 页面 | 原文线索

V2-024269 | MATHDB-327152

Koh-Shen even-dimensional finite-field distance conjecture

摘录 / EXCERPT: Koh-Shen conjecture. If $|E||F|\gg_C q^d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024270 | MATHDB-327162

Equivalence conjecture for regular relational symplectic groupoids

摘录 / EXCERPT: Equivalence conjecture. Any two regular relational symplectic groupoids integrating M are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-024271 | MATHDB-327163

Categoroid equivalence between Poisson manifolds and regular relational symplectic groupoids

摘录 / EXCERPT: Categoroid equivalence conjecture. There is an equivalence of categoroids between $\textbf{Poiss}^{Ext}$ and $\textbf{RRSGpd}$:

conjecture | open | MathDB 页面 | 原文线索

V2-024272 | MATHDB-327210

Bezdek-Bezdek's successive inradius plank conjecture

摘录 / EXCERPT: Let $\mathbf{K}$ and $\mathbf{C}$ be convex bodies in $\mathbb{E}^d$, with $d\geq 2$, and let m be a positive integer. For each body, let $r_{\mathbf{C}}(\cdot,m)$ denote its m...

conjecture | open | MathDB 页面 | 原文线索

V2-024273 | MATHDB-327280

Gukov-Murakami generalized volume conjecture

摘录 / EXCERPT: Gukov-Murakami generalized volume conjecture. The limit $L(u)$ exists for small u, and the volume $\operatorname{Vol}(S^3\setminus K;u)$ of the hyperbolic structure associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-024274 | MATHDB-327281

The asymptotic colored-Jones formula with twisted Reidemeister torsion

摘录 / EXCERPT: Let K be a hyperbolic knot, let $u\ne0$ be small, and let $S(u)$ be the asymptotic function defined by ... Let $\mathbb{T}^{K}_{\mu}(u)$ be the twisted Reidemeister torsion of the...

conjecture | open | MathDB 页面 | 原文线索

V2-024275 | MATHDB-327319

The strong Gromov conjecture for classifying maps

摘录 / EXCERPT: Strong Gromov conjecture. The classifying map can be deformed into the $(n-2)$-skeleton $B\pi^{(n-2)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-024276 | MATHDB-327338

Belolipetsky-Emery conjecture on compact and noncompact hyperbolic manifolds

摘录 / EXCERPT: Let $cmathcal{M}$ be a compact hyperbolic manifold of dimension $n eq 3$. Belolipetsky-Emery conjecture. There exists a noncompact hyperbolic n-manifold $cmathcal{M}_1$ whose vol...

conjecture | open | MathDB 页面 | 原文线索

V2-024277 | MATHDB-327348

The minimal-inflection conjecture for closed plane curves

摘录 / EXCERPT: Let γ be a closed plane curve with n double points. A plane diffeomorphism can draw γ with a number of inflection points no greater than $n+1$. Minimal-inflection c...

conjecture | open | MathDB 页面 | 原文线索

V2-024278 | MATHDB-327399

O'Grady's deformation conjecture for IHS fourfolds numerically equivalent to Hilbert squares

摘录 / EXCERPT: Let X be an IHS fourfold, and let S be a $K3$ surface. Say that X is numerically equivalent to $Hilb^2(S)$ when $c=3$ and $(H^2(X,\mathbb{Z}),q)$ is isometric to...

conjecture | open | MathDB 页面 | 原文线索

V2-024279 | MATHDB-327409

The base-locus curve conjecture for almost maximal Hilbert growth

摘录 / EXCERPT: Let Z be the point set, let I_Z be its ideal, and let J be the Artinian reduction considered in this section. Assume that $[J]_n$ has a base locus consisting of a zero-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-024280 | MATHDB-327411

Schürmann's nine-dimensional bound conjecture for lattice basis constants

摘录 / EXCERPT: Let Λ be a lattice of dimension $n=9$. Write $Q_b(\Lambda)$ for its basis constant, let C_9 be the unique binary $[9,2,6]$ code, and let a centred cubic lattice be a latt...

conjecture | open | MathDB 页面 | 原文线索

V2-024281 | MATHDB-327412

The well-roundedness conjecture for lattice basis and generator constants

摘录 / EXCERPT: Let Λ be a lattice, and let $Q_b(\Lambda)$ and $Q_g(\Lambda)$ denote its basis and generator constants. A lattice is well-rounded if its minimal vectors span the ambient Eu...

conjecture | open | MathDB 页面 | 原文线索

V2-024282 | MATHDB-327415

The space-like limit-direction conjecture for Lorentzian Coxeter systems

摘录 / EXCERPT: Let W_∞ denote the set of infinite-order elements, and let $w\in W_\infty$. A sequence of the form $(w^k)_{k\in\mathbb{N}}$ is obtained by taking the successive powers of...

conjecture | open | MathDB 页面 | 原文线索

V2-024283 | MATHDB-327428

Theorem of Completeness for holomorphic Poisson structures

摘录 / EXCERPT: Theorem of Completeness for holomorphic Poisson structures. If φ_0 is surjective, then the Poisson analytic family $(\mathcal{X},\Lambda,B,\pi)$ is complete at $0\in B$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-024284 | MATHDB-327429

Kuranishi-type completeness conjecture for holomorphic Poisson manifolds

摘录 / EXCERPT: Kuranishi-type completeness conjecture. A complete family of deformations of (X,Λ_0) whose Poisson Kodaira-Spencer map is an isomorphism exists without assuming...

conjecture | open | MathDB 页面 | 原文线索

V2-024285 | MATHDB-327462

Ni's maximal volume growth conjecture for Kähler manifolds

摘录 / EXCERPT: Let M^n be a complete noncompact Kähler manifold with nonnegative bisectional curvature. Suppose that M admits a nonconstant holomorphic function with polynomial growth and tha...

conjecture | open | MathDB 页面 | 原文线索

V2-024286 | MATHDB-327473

Elekes-Rónyai-type conjecture for bilinear forms on algebraic curves

摘录 / EXCERPT: Let $C\subset \mathbb C^2$ be an algebraic curve of degree d_C, and let $F:C\times C\to \mathbb C$ be a polynomial of degree d_F. For a finite set $S\subset C$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024287 | MATHDB-327502

Hyperplane classification conjecture for Veronese spaces over polar spaces

摘录 / EXCERPT: Let ${\goth M}:=\VerSpace(k,\PolarSpace(\varpi))$, and let ${\mathscr H}$ be a hyperplane of $\VerSpace(k,\text{\boldmath$goth P$})$ determined by a polarity $\varkappa$. The inter...

conjecture | open | MathDB 页面 | 原文线索

V2-024288 | MATHDB-327504

The classification conjecture for irreducible polar actions on compact nonnegatively curved manifolds

摘录 / EXCERPT: Let a compact Lie group act smoothly and isometrically on a simply connected compact manifold of nonnegative curvature. The action is irreducible when its associated reflection gro...

conjecture | open | MathDB 页面 | 原文线索

V2-024289 | MATHDB-327519

Dolgachev's conjecture for toric K3 hypersurfaces

摘录 / EXCERPT: Dolgachev's conjecture. The lattice $M_\Delta^\bot$ contains a 1-admissible isotropic vector; there exists a primitive embedding

conjecture | open | MathDB 页面 | 原文线索

V2-024290 | MATHDB-327520

The rank-five K-theory conjecture for the algebraic noncommutative toroidal orbifold

摘录 / EXCERPT: Let $\mathcal A_\theta^{alg} \rtimes \mathbb Z_2$ be the algebraic noncommutative toroidal orbifold, and let $1,p^\theta,q_0^\theta,q_1^\theta,r^\theta$ be the five projections con...

conjecture | open | MathDB 页面 | 原文线索

V2-024291 | MATHDB-327524

Dušek's $k=0$ conjecture for null homogeneous geodesics on g.o. spaces

摘录 / EXCERPT: Dušek's $k=0$ conjecture. For all null homogeneous geodesics in G/H, one has $k=0$ in the geodesic lemma.

conjecture | open | MathDB 页面 | 原文线索

V2-024292 | MATHDB-327529

Uniqueness conjecture for lattices associated with symplectic K3 degenerations

摘录 / EXCERPT: Uniqueness conjecture. For a fixed abstract group type of H (equivalently, a fixed n) and a fixed Dynkin-diagram type of degeneration, the corresponding lattice

conjecture | open | MathDB 页面 | 原文线索

V2-024293 | MATHDB-327582

All-order upper-bound conjecture for complete arcs

摘录 / EXCERPT: Let $t_{2}(2,q)$ be the smallest size of a complete arc in $\mathrm{PG}(2,q)$, and let the bounds denoted in the source by ... be the explicit upper bounds obtained from the computed...

conjecture | open | MathDB 页面 | 原文线索

V2-024294 | MATHDB-327593

Displacement-energy subadditivity for plank coverings

摘录 / EXCERPT: Let V be a finite-dimensional real vector space, let $K\subset V$ be a convex body, and let $B\subset V^*$ be the unit ball of the dual norm. For the canonical symplectic structu...

conjecture | open | MathDB 页面 | 原文线索

V2-024295 | MATHDB-327594

Subadditivity of Hofer-Zehnder capacity for symplectic cylinders

摘录 / EXCERPT: For a convex body $S\subset\mathbb C$, a symplectic cylinder is a set $S\times\mathbb C^{n-1}\subset\mathbb C^n$ or any of its images under symplectic transformations. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024296 | MATHDB-327595

Euclidean cylinder covering conjecture

摘录 / EXCERPT: Let B^n be the Euclidean unit ball, and let $Z_1,\ldots,Z_N$ be m-dimensional Euclidean cylinders covering B^n. Write $\sigma_{n-m}(Z_i)$ for the $(n-m)$-dimensional volume o...

open_problem | open | MathDB 页面 | 原文线索

V2-024297 | MATHDB-327597

Coisotropic non-squeezing conjecture for symplectic homeomorphisms

摘录 / EXCERPT: Let h be a symplectic homeomorphism that sends a coisotropic submanifold C to a smooth, hence coisotropic, submanifold, and let $\hat h$ be its reduction. Let k be the codime...

conjecture | open | MathDB 页面 | 原文线索

V2-024298 | MATHDB-327598

Isotropic-disc flexibility conjecture in $\mathbb{C}^3$

摘录 / EXCERPT: A symplectic disc is a smoothly embedded two-dimensional disc in the standard symplectic manifold $\mathbb C^3$, while a smooth isotropic disc is a smoothly embedded disc whose tan...

conjecture | open | MathDB 页面 | 原文线索

V2-024299 | MATHDB-327605

Herbig-Seaton's symplecticity conjecture for Hilbert series of symplectic quotients

摘录 / EXCERPT: Herbig-Seaton's conjecture. The Hilbert series $\operatorname{Hilb}_{\mathbb{R}[M_0]}(t)$ is symplectic at $t=1$ of order

conjecture | open | MathDB 页面 | 原文线索

V2-024300 | MATHDB-327610

The noncommutative pencil conjecture for anticanonical Lefschetz pencils

摘录 / EXCERPT: Noncommutative pencil conjecture. The A_∞-algebra $\EuScript A$ carries a canonical structure of a noncommutative pencil satisfying $\EuScript P=\EuScript A^\vee[-n]$, whose...

conjecture | open | MathDB 页面 | 原文线索

V2-024301 | MATHDB-327611

The open-closed string map isomorphism conjecture for Lefschetz fibrations

摘录 / EXCERPT: Open-closed string map conjecture. For any automorphism ϕ and $\delta\gg0$, the open-closed string map is an isomorphism. Such an isomorphism would relate the Fukaya-theoretic...

conjecture | open | MathDB 页面 | 原文线索

V2-024302 | MATHDB-327615

The reduction conjecture for Wintgen ideal submanifolds

摘录 / EXCERPT: Let $x:M^m\longrightarrow \mathbb{R}^{m+p}$ be a Wintgen ideal submanifold without umbilic points. Let $\mathbb{D}_2=\operatorname{Span}\{E_1,E_2\}$ be its canonical distribution,...

conjecture | open | MathDB 页面 | 原文线索

V2-024303 | MATHDB-327616

The closed invariant characterization conjecture for irreducible Wintgen ideal submanifolds

摘录 / EXCERPT: Let M^k be an irreducible Wintgen ideal submanifold of dimension $k\ge 3$, meaning that the only integrable distribution containing the canonical distribution $\mathbb{D}_2$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-024304 | MATHDB-327627

Steklov spectral rigidity of disk metrics up to boundary-fixed conformal equivalence

摘录 / EXCERPT: Let g and g' be Riemannian metrics on the unit disk ${\mathbb D}$, and let Λ_g and $\Lambda_{g'}$ be their Dirichlet-to-Neumann operators. Disk-metric Steklov rigidity...

conjecture | solved | MathDB 页面 | 原文线索

V2-024305 | MATHDB-327646

Generic classification conjecture for generalized distance-squared mappings

摘录 / EXCERPT: Generic classification conjecture. There exists a subset $\Sigma\subset(\mathbb{R}^{n+1})^{n+1}$ of Lebesgue measure zero such that, for every $(p_0,p_1,\ldots,p_n)\notin\Sigma$: (...

conjecture | open | MathDB 页面 | 原文线索

V2-024306 | MATHDB-327668

Heisenberg-Virasoro extension relating the KW and Hodge tau-functions

摘录 / EXCERPT: Heisenberg-Virasoro extension. The KW and Hodge tau-functions are related by

conjecture | open | MathDB 页面 | 原文线索

V2-024307 | MATHDB-327669

The homogeneous Landsberg-Berwald conjecture

摘录 / EXCERPT: Homogeneous Landsberg-Berwald conjecture. Every homogeneous Landsberg space must be a Berwald space.

conjecture | open | MathDB 页面 | 原文线索

V2-024308 | MATHDB-327685

The ellipsoid full-packing conjecture for closed symplectic four-manifolds

摘录 / EXCERPT: Let $(X,omega)$ be a closed 4-dimensional symplectic manifold. A symplectic ellipsoid is a domain $E(a,b)\subset\mathbb{C}^2$ defined by ... It fully packs X if there is a symple...

conjecture | open | MathDB 页面 | 原文线索

V2-024309 | MATHDB-327692

Line-completion conjecture for affinizations of products of projective spaces

摘录 / EXCERPT: Let $\mathfrak{A}$ be the complement of a spiky hyperplane in a product of projective spaces. Line-completion conjecture. The family of lines of $\mathfrak{A}$ cannot be completed...

conjecture | open | MathDB 页面 | 原文线索

V2-024310 | MATHDB-327702

Tropical wall-crossing conjecture for reduced open Gromov-Witten invariants

摘录 / EXCERPT: Let γ be a primitive charge. Suppose that u crosses a wall consisting of relative classes γ_i, for $i=1,\ldots,n$. For each i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-024311 | MATHDB-327706

Bocci-Harbourne containment conjecture for symbolic powers

摘录 / EXCERPT: Let I be a homogeneous ideal in the coordinate ring $\mathbb K[\mathbf P^N]$ of projective space $\mathbf P^N$ over a field $\mathbb K$, and let r and N be positive integers....

conjecture | open | MathDB 页面 | 原文线索

V2-024312 | MATHDB-327742

The isogeny conjecture for mixed-signature Darmon points

摘录 / EXCERPT: Isogeny conjecture. There exists an isogeny

conjecture | open | MathDB 页面 | 原文线索

V2-024313 | MATHDB-327746

The non-archimedean globality and reciprocity conjecture for Darmon points

摘录 / EXCERPT: Non-archimedean Darmon point conjecture. The local point P_ψ is rational over $H_{\mathcal O}^+$, and, for every $\alpha\in\operatorname{Pic}^+({\mathcal O})$,

conjecture | open | MathDB 页面 | 原文线索

V2-024314 | MATHDB-327753

Stable-pairs rationality conjecture for vertical classes on K3 fibrations

摘录 / EXCERPT: Let V be a nonsingular projective threefold that is the total space of a K3 fibration $\pi:V\to C$ over a nonsingular projective curve, and let $\beta\in H_2(V,\mathbb{Z})^\pi$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-024315 | MATHDB-327758

Conway-sphere conjecture for L-space knots

摘录 / EXCERPT: Let K be an L-space knot, meaning a knot for which $S^3_n(K)$ is an L-space for some $n>0$. An essential Conway sphere is an essential 4-punctured sphere in the knot complement...

conjecture | open | MathDB 页面 | 原文线索

V2-024316 | MATHDB-327760

Packing measure formula for regular-polygon self-similar sets

摘录 / EXCERPT: Regular-polygon packing measure formula. The conjectured packing measure satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024317 | MATHDB-327762

Coincidence of symplectic capacities on convex domains

摘录 / EXCERPT: Let $\Sigma\subset{\mathbb R}^{2n}$ be a convex domain. Denote by $\underline c(\Sigma)$ its Gromov radius and by $\overline c(\Sigma)$ its cylindrical capacity. Coincidence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-024318 | MATHDB-327775

Kepler's sphere-packing conjecture

摘录 / EXCERPT: In three-dimensional space, consider packings of congruent non-overlapping spheres, and let their density be the proportion of space occupied by the spheres, interpreted as the asy...

conjecture | open | MathDB 页面 | 原文线索

V2-024319 | MATHDB-327784

Calabi's conjecture on Kähler-Einstein metrics with nonpositive first Chern class

摘录 / EXCERPT: Calabi's conjecture. Then (M,J) admits a Kähler-Einstein metric, unique up to homothety when $c_1(M,J)<0$, and unique in each Kähler class when $c_1(M,J)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-024320 | MATHDB-327785

Tian's conjecture on conical Kähler-Einstein metrics

摘录 / EXCERPT: Tian's conjecture. Then (M,J) admits a conical Kähler-Einstein metric with angle $2\pi\beta_j$ along D_j, unique in the negative case and unique in each Kähler class in the zer...

conjecture | open | MathDB 页面 | 原文线索

V2-024321 | MATHDB-327804

Bonatti-Díaz hyperbolicity conjecture

摘录 / EXCERPT: Let M be a compact manifold and let $\operatorname{Diff}^1(M)$ be the space of C^1 diffeomorphisms. A diffeomorphism is hyperbolic if its nonwandering dynamics has a hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-024322 | MATHDB-327805

Bonatti's finiteness conjecture for tame diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold, let Tang denote the set of diffeomorphisms exhibiting a homoclinic tangency, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024323 | MATHDB-327807

Conjecture I on tameness of infinitely punctured planes

摘录 / EXCERPT: Conjecture I. If there exists an automorphism of infinite order $h\in\operatorname{Aut}(\mathbb{C}\setminus A)$ such that the quotient space...

conjecture | open | MathDB 页面 | 原文线索

V2-024324 | MATHDB-327857

Funano and Shioya's dimension-free eigenvalue ratio conjecture for Alexandrov spaces

摘录 / EXCERPT: Funano and Shioya's conjecture. For every natural number k, there exists a positive constant C_k depending only on k such that

conjecture | open | MathDB 页面 | 原文线索

V2-024325 | MATHDB-327858

Farb-Weinberger's quantitative symmetry-gap conjecture

摘录 / EXCERPT: Let M be a smooth closed, aspherical, smoothly irreducible manifold such that $pi_1(M)$ contains no nontrivial normal abelian subgroups. Then there exists $C>1$ depending only on...

conjecture | open | MathDB 页面 | 原文线索

V2-024326 | MATHDB-327867

Butruille's conjecture on compact nearly Kähler manifolds

摘录 / EXCERPT: Butruille's conjecture. Every compact nearly Kähler manifold is a 3-symmetric space.

conjecture | open | MathDB 页面 | 原文线索

V2-024327 | MATHDB-327879

Landau-Ginzburg families conjecture for Sarkisov links

摘录 / EXCERPT: A Sarkisov link is a birational map, or birational cobordism, connecting two Mori fibrations. Landau-Ginzburg families conjecture. One-parameter families of Landau-Ginzburg models...

conjecture | open | MathDB 页面 | 原文线索

V2-024328 | MATHDB-327899

The moduli-deformation conjecture for cubic, X_{10}, and Kuechle fourfolds

摘录 / EXCERPT: Moduli-deformation conjecture. There is an infinite series of associated moduli spaces that can be deformed one to another.

conjecture | open | MathDB 页面 | 原文线索

V2-024329 | MATHDB-327905

The moving-set conjecture for local categorical invariants

摘录 / EXCERPT: Moving-set conjecture. The geometry of the moving set determines the number d/k and the gap of the spectrum of the corresponding category.

conjecture | open | MathDB 页面 | 原文线索

V2-024330 | MATHDB-327925

The constant-length second fundamental form conjecture for self-shrinkers

摘录 / EXCERPT: Let $\Sigma^n\subset\mathbb{R}^{n+1}$ be a smooth, complete, embedded self-shrinker, meaning that its mean curvature satisfies ... Assume that Σ has polynomial volume growth a...

conjecture | open | MathDB 页面 | 原文线索

V2-024331 | MATHDB-327952

The weak rigidity conjecture for rational cuspidal curves

摘录 / EXCERPT: Let $\bar E\subseteq\mathbb{P}^2$ be a rational cuspidal curve of log general type, and let $\pi:(X,D)\to(\mathbb{P}^2,\bar E)$ be the minimal log resolution of singularities. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024332 | MATHDB-328004

Uniqueness of tangent spaces for Calabi-Yau conifolds

摘录 / EXCERPT: Let $X\in\widetilde{\mathscr{KS}}(n,\kappa)$ be a Calabi-Yau conifold and let $x_0\in X$. The tangent space at x_0 is the limiting tangent object associated with the conifold str...

conjecture | open | MathDB 页面 | 原文线索

V2-024333 | MATHDB-328008

Gravin-Robins-Shiryaev lattice-enumeration conjecture for multiple tilings

摘录 / EXCERPT: Let P be a convex polytope. Suppose that P k-tiles $\mathbb R^d$ with a discrete multiset Λ. Gravin-Robins-Shiryaev's lattice-enumeration conjecture. Then P m-til...

conjecture | open | MathDB 页面 | 原文线索

V2-024334 | MATHDB-328053

Higher-dimensional Alexandrov isodiametric conjecture

摘录 / EXCERPT: Higher-dimensional Alexandrov conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-024335 | MATHDB-328054

Cahn-Levi-Kálmán minimal-genus conjecture for virtual Legendrian knots

摘录 / EXCERPT: Let K_1 and K_2 be Legendrian knots in the spherical cotangent bundle ST^*M of a surface M. Suppose that they are isotopic as virtual Legendrian knots and that M has the...

conjecture | open | MathDB 页面 | 原文线索

V2-024336 | MATHDB-328055

The spherical-cotangent-bundle conjecture for virtual Legendrian knots

摘录 / EXCERPT: Let S^2 be the two-sphere, and let ST^*S^2 denote its spherical cotangent bundle. Consider two Legendrian knots in ST^*S^2 that are isotopic as virtual Legendrian knots. Sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-024337 | MATHDB-328058

The $1/4$ conjecture for the first eigenvalue of closed hyperbolic surfaces

摘录 / EXCERPT: Let $g\geq 2$ be an integer, and let $\lambda_1(S)$ denote the first positive Laplace eigenvalue of a closed hyperbolic surface S. The $1/4$ eigenvalue conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-024338 | MATHDB-328070

Scaling-limit conjecture for affine preferential attachment looptrees

摘录 / EXCERPT: Scaling-limit conjecture. There exists a random compact metric space $\mathcal{L}^{(S)}_{\delta}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024339 | MATHDB-328151

Non-geometric 3-manifold virtual embedding conjecture

摘录 / EXCERPT: Virtual embedding conjecture. Every non-geometric closed aspherical 3-manifold contains a closed essentially immersed subsurface which is not virtually embedded.

conjecture | open | MathDB 页面 | 原文线索

V2-024340 | MATHDB-328152

Immersion and embedding bounds for manifolds over properly colored polytopes

摘录 / EXCERPT: Let P^n be a simple convex n-dimensional polytope that admits a proper coloring with n colors. Let $\alpha(n)$ denote the number of 1's in the binary expansion of n, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024341 | MATHDB-328193

Generalized abundance conjecture

摘录 / EXCERPT: Generalized abundance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-024342 | MATHDB-328206

Linearity for translation generalized quadrangles

摘录 / EXCERPT: Let Γ^x be an infinite translation generalized quadrangle with translation group T. Let z be an affine point, meaning a point not collinear with x, and let $\{U\}_z$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-024343 | MATHDB-328218

The boundary-generic realization conjecture for traversing fields

摘录 / EXCERPT: Boundary-generic realization conjecture. The constraints

conjecture | open | MathDB 页面 | 原文线索

V2-024344 | MATHDB-328219

The generic-path conjecture for traversing vector fields

摘录 / EXCERPT: Generic-path conjecture. Given v_0 and v_1, there is a one-parameter family of traversing fields $\{v_t\}_{t\in[0,1]}$ connecting v_0 to v_1 such that...

conjecture | open | MathDB 页面 | 原文线索

V2-024345 | MATHDB-328220

Schapira's reflexive extension conjecture for DQ-modules

摘录 / EXCERPT: Schapira's conjecture. If $\mathcal{M}$ is d-reflexive, then $j_*\mathcal{M}$ is a coherent $\mathcal{A}_X$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-024346 | MATHDB-328258

Leung's weak pinching conjecture for minimal submanifolds

摘录 / EXCERPT: Leung's weak pinching conjecture. If n is odd and $\sigma(W)\leq\frac{n}{n-1}$, then W is homeomorphic to S^n.

conjecture | open | MathDB 页面 | 原文线索

V2-024347 | MATHDB-328264

The positive mass conjecture for asymptotically flat manifolds with boundary

摘录 / EXCERPT: The simplest model is the closed half-space $\mathbb R^n_+$ with its standard flat metric δ. Let (M,g) be an asymptotically flat manifold with decay rate $\tau>(n-2)/2$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-024348 | MATHDB-328265

Generic super-rigidity conjecture for index-zero pseudoholomorphic curves

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold of real dimension at least four, and let ${\mathcal J}_{\operatorname{reg}}$ denote a subset of the space of smooth ω-tame a...

conjecture | open | MathDB 页面 | 原文线索

V2-024349 | MATHDB-328269

Fine's positive definite connection conjecture

摘录 / EXCERPT: Let M be a closed 4-manifold. A connection on an associated sphere bundle is called positive definite when its induced symplectic structure has fibre normal degree $+2$. Fine's...

conjecture | open | MathDB 页面 | 原文线索

V2-024350 | MATHDB-328278

The linear instability conjecture for the Einstein CLW metric

摘录 / EXCERPT: The Einstein CLW metric is a distinguished Einstein metric arising in the context above. A conformal perturbation means a variation of the metric obtained by multiplying it infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-024351 | MATHDB-328318

Smooth toric refined Severi degree conjecture

摘录 / EXCERPT: Let Δ be a convex lattice polygon, let $S=X(\Delta)$ be the associated smooth toric surface, and let $L=L(\Delta)$ be a δ-very ample line bundle. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024352 | MATHDB-328319

Refined Severi degrees on classical toric del Pezzo surfaces

摘录 / EXCERPT: Let S be a classical toric del Pezzo surface and let L be a line bundle. Assume that the loci in $|L|$ of non-reduced curves and of curves having a (-1)-curve as a component...

conjecture | open | MathDB 页面 | 原文线索

V2-024353 | MATHDB-328349

Embedding conjecture for Hitchin harmonic maps

摘录 / EXCERPT: Embedding conjecture. The map f_{ρ} is an embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-024354 | MATHDB-328371

The convex hypersurface conjecture for infinitesimal strip deformations

摘录 / EXCERPT: For each arc $\alpha\in\mathscr{A}$, let $\underline{\alpha}$ be a minimally intersecting geodesic representative and let p_{α} be a waist; with widths m_{α}, define...

conjecture | open | MathDB 页面 | 原文线索

V2-024355 | MATHDB-328372

The cap-induced factorization conjecture for even-dimensional projective spaces

摘录 / EXCERPT: An M-cap in $\mathrm{PG}(n,q)$ is a set of M points with no three collinear. By Ebert's theorem, if $n\geq 2$ is even, then $\mathrm{PG}(2n-1,q)$ can be partitioned into disjoi...

conjecture | open | MathDB 页面 | 原文线索

V2-024356 | MATHDB-328373

The elliptic-quadric factorization conjecture in projective 3-space

摘录 / EXCERPT: An elliptic quadric is a cap in $\mathrm{PG}(3,q)$, and $\mathrm{PG}(3,q)$ can be partitioned into $q+1$ disjoint elliptic quadrics. The set of elliptic quadrics induces factors on...

conjecture | open | MathDB 页面 | 原文线索

V2-024357 | MATHDB-328374

The subspace-induced factorization conjecture in finite projective spaces

摘录 / EXCERPT: Let i and n be such that $(i+1)\mid(n+1)$, and let the i-spaces of $\mathrm{PG}(n,q)$ be the relevant subspaces. Subspace-induced factorization conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-024358 | MATHDB-328376

Leydold's conjecture on maximal nodal domains of spherical harmonics

摘录 / EXCERPT: Let $\cal H_\ell$ be the eigenspace of spherical harmonics of degree ℓ on the sphere, and let $\mu(u)$ denote the number of nodal domains of a spherical harmonic u. Leydold'...

conjecture | open | MathDB 页面 | 原文线索

V2-024359 | MATHDB-328378

The maximal affine-set direction conjecture

摘录 / EXCERPT: Maximal affine-set direction conjecture. If $\mathcal{U}$ is maximal, then

conjecture | open | MathDB 页面 | 原文线索

V2-024360 | MATHDB-328379

The maximal affine-set linearity conjecture

摘录 / EXCERPT: Maximal affine-set linearity conjecture. If $t=s>2$, then $\mathcal{U}$ is an affine $\operatorname{GF}(s)$-linear set.

conjecture | open | MathDB 页面 | 原文线索

V2-024361 | MATHDB-328389

The regular simplex maximization conjecture for minimum projective distance

摘录 / EXCERPT: Let $\mathbf{u}_0,\dots,\mathbf{u}_n$ be unit vectors representing the vertices of a simplex in complex projective space, in general position, and let ... Let $d_{\min}$ denote the m...

conjecture | open | MathDB 页面 | 原文线索

V2-024362 | MATHDB-328395

Asymptotic codimension-one Cheeger constant conjecture for finite buildings

摘录 / EXCERPT: Let $A_{n+1}({\mathbb F}_q)$ be the n-dimensional building associated with the projective geometry over the finite field ${\mathbb F}_q$, and let h_{n-1} denote its normalized...

conjecture | open | MathDB 页面 | 原文线索

V2-024363 | MATHDB-328399

The Sylvester-form conjecture for the Rees ideal of a linearly presented ideal

摘录 / EXCERPT: Let $k[\mathbf{x}]=k[x_1,x_2,x_3]$ and $k[\mathbf{y}]=k[y_1,y_2,y_3,y_4]$. Consider the matrix $\varphi:=\varphi_{3,r}$ as in the paper's template, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024364 | MATHDB-328413

Rigidity of flat labels modulo Möbius transformations

摘录 / EXCERPT: Let $f\in C_{\hat{\alpha},\hat{\eta}}$ be a flat label, where $C_{\hat{\alpha},\hat{\eta}}$ denotes the space of labels with the prescribed data $\hat{\alpha}$ and $\hat{\eta}$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-024365 | MATHDB-328433

The F-conjecture for nef divisors on the moduli space of pointed rational curves

摘录 / EXCERPT: Let $\overline{M}_{0,n}$ be the Deligne-Mumford moduli space of stable n-pointed rational curves. A line bundle on $\overline{M}_{0,n}$ is F-nef if it has nonnegative intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-024366 | MATHDB-328444

The non-homaloidality conjecture for the generic Hankel determinant

摘录 / EXCERPT: Let R be the polynomial ring containing the entries of the generic $m\times m$ Hankel matrix $\mathcal{H}_m$. Hankel non-homaloidality conjecture. The determinant of...

conjecture | open | MathDB 页面 | 原文线索

V2-024367 | MATHDB-328446

The non-trivial self-cover conjecture for infra-nilmanifolds

摘录 / EXCERPT: The non-trivial self-cover conjecture. The infra-nilmanifold $\Gamma \backslash G$ admits a non-trivial self-cover if and only if the nilmanifold $N \backslash G$ admits an expandi...

conjecture | open | MathDB 页面 | 原文线索

V2-024368 | MATHDB-328449

Gromov's minimal volume conjecture for finite-volume hyperbolic manifolds

摘录 / EXCERPT: Let (M^n,g_0) be a complete hyperbolic manifold with finite volume, where g_0 is a hyperbolic metric on M^n. The minimal volume of M is denoted by ... Gromov's conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-024369 | MATHDB-328454

The determinant inequality conjecture for configurations of points

摘录 / EXCERPT: Let C_n be the configuration space under consideration, and let $D(\boldsymbol{x})$ denote the associated determinant for $\boldsymbol{x}\in C_n$. Determinant inequality conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-024370 | MATHDB-328467

Finsler soul conjecture for Berwaldian manifolds with large volume growth

摘录 / EXCERPT: Finsler soul conjecture. M is diffeomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-024371 | MATHDB-328480

Fornaess-Stensønes basin conjecture for uniformly attracting sequences

摘录 / EXCERPT: Fornaess-Stensønes basin conjecture. The basin Ω is always biholomorphic to $\mathbb C^m$. A positive answer would imply a positive answer to Bedford's stable-manifold conje...

conjecture | open | MathDB 页面 | 原文线索

V2-024372 | MATHDB-328569

The derived formal-neighborhood conjecture for Lie pairs

摘录 / EXCERPT: Let (L,A) be a Lie pair over $\mathbb{R}$, and let $(\mathscr{L},\mathscr{A})$ be a pair of simply source-connected local Lie groupoids integrating it, with $\mathscr{A}$ a close...

conjecture | open | MathDB 页面 | 原文线索

V2-024373 | MATHDB-328571

Torsion conjecture for three-dimensional closed CR Yamabe solitons

摘录 / EXCERPT: Let (M^3,J,θ,f,μ) be a three-dimensional closed CR Yamabe soliton, where P_0 is the CR Paneitz operator. Assume that $P_0f=0$ and that f is nowhere vanishing. Torsion...

conjecture | open | MathDB 页面 | 原文线索

V2-024374 | MATHDB-328619

The two-tripod conjecture for smooth closed convex curves in spherical and hyperbolic geometry

摘录 / EXCERPT: Two-tripod conjecture. Every smooth closed convex curve in the spherical or hyperbolic geometry has at least two tripod configurations.

conjecture | open | MathDB 页面 | 原文线索

V2-024375 | MATHDB-328629

Bogomolov-Pop's mod-ℓ isomorphism conjecture

摘录 / EXCERPT: Let $K|k$ and $L|l$ be function fields over algebraically closed fields. Let $\mathcal{G}_K^a$ and $\mathcal{G}_L^a$ denote the mod-ℓ abelianized Galois groups, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024376 | MATHDB-328643

The multiplicity version of the joints problem

摘录 / EXCERPT: Multiplicity joints conjecture. The number of joints counted with multiplicities satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024377 | MATHDB-328653

Conjecture on lower-order controls and generalized Kähler-Einstein limits for Kähler-Ricci flow

摘录 / EXCERPT: Lower-control and limit conjecture. Uniformly on $X\times[0,\infty)$, there is a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-024378 | MATHDB-328656

The F-integral Cohen-Macaulay conjecture for complete toric domains

摘录 / EXCERPT: F-integral Cohen-Macaulay conjecture. The ring $\mathbf{F}_{\!}^{\operatorname{int}}(R)$ is Cohen-Macaulay, and hence is a small MCM algebra over R.

conjecture | open | MathDB 页面 | 原文线索

V2-024379 | MATHDB-328657

The small MCM conjecture for tame bi-partite toric rings

摘录 / EXCERPT: Small MCM conjecture for tame bi-partite toric rings. For some sufficiently high power q of p, R admits the small MCM

conjecture | open | MathDB 页面 | 原文线索

V2-024380 | MATHDB-328714

Modular Ax-Schanuel conjecture with derivatives

摘录 / EXCERPT: Let A be as in the complex Modular Ax-Schanuel setting, with geodesically independent coordinates $z_1,\ldots,z_n$ and the modular function j. Modular Ax-Schanuel conjecture wi...

conjecture | open | MathDB 页面 | 原文线索

V2-024381 | MATHDB-328718

Finiteness conjecture for globally hyperbolic Lorentzian manifolds

摘录 / EXCERPT: Let M be a non-spacelike geodesically complete globally hyperbolic Lorentzian manifold with a compact Cauchy hypersurface of dimension $m \geqslant 3$. Suppose there is a positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-024382 | MATHDB-328737

Yamabe's conjecture

摘录 / EXCERPT: Yamabe's conjecture. There exists a metric g on X conformally equivalent to g_0 and having constant scalar curvature.

conjecture | solved | MathDB 页面 | 原文线索

V2-024383 | MATHDB-328738

Hamilton's Yamabe flow convergence conjecture

摘录 / EXCERPT: Hamilton's conjecture. The metric $g(t)$ converges to a metric of constant scalar curvature as $t\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-024384 | MATHDB-328772

Rigidity conjecture for maximal representations of complex hyperbolic lattices

摘录 / EXCERPT: Rigidity conjecture. If $\rho\colon\Gamma\to G$ is maximal, then $G=\operatorname{SU}(p,q)$ with $p\geq nq$, and ρ is a trivial deformation of ρ_{std}, namely

conjecture | open | MathDB 页面 | 原文线索

V2-024385 | MATHDB-328779

The solvmanifold conjecture for expanding homogeneous Ricci solitons

摘录 / EXCERPT: An expanding homogeneous Ricci soliton is a homogeneous Ricci soliton with soliton constant $\lambda<0$. A solvmanifold is a homogeneous space associated with a solvable Lie group;...

conjecture | open | MathDB 页面 | 原文线索

V2-024386 | MATHDB-328809

Tightness of geodesic deviation in the hyperbolic Poisson-Voronoi graph

摘录 / EXCERPT: Let $\mathscr{V}^\lambda$ be the hyperbolic Poisson-Voronoi graph. For $r>0$, consider a geodesic in $\mathscr{V}^\lambda$ connecting the Voronoi cells containing $-r$ and r, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024387 | MATHDB-328811

The B-conjecture for even log-concave measures

摘录 / EXCERPT: B-conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-024388 | MATHDB-328812

Saroglou's upper-bound conjecture for the second projection body of zonoids

摘录 / EXCERPT: Let K be a zonoid in $\mathbb{R}^d$, let $V(K)$ be its volume, and let $\Pi^2K:=\Pi(\Pi K)$ be its second projection body. Saroglou's upper-bound conjecture. ... with equality if a...

conjecture | open | MathDB 页面 | 原文线索

V2-024389 | MATHDB-328813

Saroglou's three-dimensional ellipsoid conjecture for a section functional

摘录 / EXCERPT: Let K be a symmetric convex body in $\mathbb{R}^3$, let h_K be its support function, let $V(K)$ be its volume, and let V_2 denote planar area. For $x\in S^2$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024390 | MATHDB-328814

Saroglou's lower-bound conjecture for the second projection body

摘录 / EXCERPT: Let K be a d-dimensional convex body containing the origin in its interior. Define ... For $x\in S^{d-1}$, let $K\cap(sx+x^\bot)$ be the $(d-1)$-dimensional section and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024391 | MATHDB-328815

Saroglou's lower-order projection-body minimizer conjecture

摘录 / EXCERPT: Saroglou's lower-order projection-body conjecture. For $i=1,\dots,d-2$, the functionals $P_i(\cdot)$ and $Q_i(\cdot)$ are minimized precisely for Euclidean balls. The source notes...

conjecture | open | MathDB 页面 | 原文线索

V2-024392 | MATHDB-328818

Cobordism rationality and functional equation conjecture for stable-pair partition functions

摘录 / EXCERPT: Cobordism rationality and functional equation conjecture. The function $Z_{\Omega_\ast,\beta}^{X}(q)$ is a rational function in q; more precisely, every $f_I(q)$ is the Laurent e...

conjecture | open | MathDB 页面 | 原文线索

V2-024393 | MATHDB-328819

Common-poles conjecture for stable-pair Chern-number partition functions

摘录 / EXCERPT: Common-poles conjecture. All these partition functions are Laurent expansions of rational functions having the same poles.

conjecture | open | MathDB 页面 | 原文线索

V2-024394 | MATHDB-328837

Optimality conjecture for stabilized Pell ellipsoid embeddings

摘录 / EXCERPT: Let P_n and H_n be the Pell numbers and half companion Pell numbers defined by ... and ... Define ... Let $E(1,a_2,\dots,a_n)$ denote the symplectic ellipsoid and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024395 | MATHDB-328847

Extreme-point conjecture for Nikol'skii constants

摘录 / EXCERPT: Let $D\subset\mathbb{R}^d$ be a convex compact set, and let $C(\mathcal{P}_N,D,\boldsymbol{x})$ denote the Nikol'skii constant at $\boldsymbol{x}\in D$ for the polynomial space...

conjecture | open | MathDB 页面 | 原文线索

V2-024396 | MATHDB-328854

Rational-curve conjecture for non-pseudo-effective adjoint divisors

摘录 / EXCERPT: Rational-curve conjecture. If K_X+L is not pseudo-effective, then there is a rational curve C deforming to cover X such that

conjecture | open | MathDB 页面 | 原文线索

V2-024397 | MATHDB-328855

The discriminant-form surjectivity conjecture for invariant lattices

摘录 / EXCERPT: Let L_G be a lattice of type (a) or (b), let L^G be the associated invariant lattice, let A_{L^G} be its discriminant group, and let $\overline{O}(L^G)$ denote the image of i...

conjecture | open | MathDB 页面 | 原文线索

V2-024398 | MATHDB-328863

Grigoriev's effective tropical dual Nullstellensatz conjecture

摘录 / EXCERPT: Let $F=\{f_1,\ldots,f_k\}$ be a system of tropical polynomials in n variables, and let M_N be the finite Macaulay matrix obtained from the monomial multiples of the f_i of de...

conjecture | open | MathDB 页面 | 原文线索

V2-024399 | MATHDB-328888

Sogge-Smith conjecture for nodal extrema of Dirichlet eigenfunctions

摘录 / EXCERPT: Let (M^n,g) be a manifold with boundary, and let u_λ be a Dirichlet eigenfunction associated to the eigenvalue parameter λ. For each nodal domain A_i of...

conjecture | open | MathDB 页面 | 原文线索

V2-024400 | MATHDB-328893

Invariant Reeb orbit conjecture for contactomorphisms

摘录 / EXCERPT: Let (Y,ξ) be a closed contact manifold equipped with a selected contact form α such that $\ker\alpha=\xi$. Let ϕ be a contactomorphism of Y that preserves...

conjecture | open | MathDB 页面 | 原文线索

V2-024401 | MATHDB-328894

Assarf-Joswig-Paffenholz product decomposition conjecture for toric Fano manifolds

摘录 / EXCERPT: Assarf-Joswig-Paffenholz conjecture. If $d>3k$, then X is isomorphic to $(S_3)^m\times Y$, where m is a positive integer and Y is a toric Fano manifold of dimension at most...

conjecture | open | MathDB 页面 | 原文线索

V2-024402 | MATHDB-328974

Monotonicity conjecture for the turning angle of positive--parameter self-shrinkers

摘录 / EXCERPT: Let ... denote the energy, and let ... -self-shrinker solution. Monotonicity conjecture. When >0, ... This extends the monotonicity known for the self-shrinker case $=0$, attributed...

conjecture | open | MathDB 页面 | 原文线索

V2-024403 | MATHDB-329009

Finite-surgery minimal-model conjecture for Kähler-Ricci flow

摘录 / EXCERPT: Finite-surgery minimal-model conjecture. There are only finitely many surgery times

conjecture | open | MathDB 页面 | 原文线索

V2-024404 | MATHDB-329017

Conjecture on deformations of two-component degenerate Poisson structures

摘录 / EXCERPT: A two-component Poisson structure of hydrodynamic type with degenerate metric has deformation parameters depending on the single variable u^2. Deformation-characterisation conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-024405 | MATHDB-329018

Conjecture on deformations of degenerate hydrodynamic-type Poisson structures

摘录 / EXCERPT: Let P be an n-component Poisson structure of the form $P^{ij}=g^{ij}(u)\frac{d}{dx}+b^{ij}_k(u)u^k_x$, where the matrix $g=(g^{ij})$ has rank $m<n$. The variables...

conjecture | open | MathDB 页面 | 原文线索

V2-024406 | MATHDB-329021

The indecomposable locally homogeneous pp-wave conjecture

摘录 / EXCERPT: Let $({\cal M},\mathrm{g})$ be an indecomposable locally homogeneous pp-wave of dimension greater than 3. A locally homogeneous pp-wave is a pp-wave whose local Killing vector fi...

conjecture | open | MathDB 页面 | 原文线索

V2-024407 | MATHDB-329026

Engliš-Peetre conjecture on higher Laplacians for Hermitian symmetric spaces

摘录 / EXCERPT: Let $X=G/K$ be a simple Hermitian symmetric space, where K is a compact subgroup of G. For each $m\in\mathbb{N}$, let L_m denote the higher Laplacian of order $2m$ on X, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-024408 | MATHDB-329031

Nodal rational curves conjecture for very general polarized K3 surfaces

摘录 / EXCERPT: Let (X,L) be a very general polarized $K3$ surface. A rational curve in $|L|$ is a curve of geometric genus 0. Nodal rational curves conjecture. All rational curves in $|L|$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-024409 | MATHDB-329033

Non-negativity conjecture for the inverse matrix in the effective case

摘录 / EXCERPT: In the effective case $|G_0|=1$, let $g_1=i_0$ and consider the matrix ... The non-negativity conjecture. The entries of the inverse of the above matrix are non-negative integers. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-024410 | MATHDB-329056

The one-dimensionality conjecture for analytic centers of simplicial polytopes

摘录 / EXCERPT: Let a center assign a point to every simplicial polytope in ${\mathbb R}^n$. Assume that the center depends analytically on the polytope, commutes with dilations, and satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-024411 | MATHDB-329060

The critical-number monotonicity conjecture for convex bodies

摘录 / EXCERPT: Let K be a convex body. For each integer $k\geq 1$, partition $[0,1]$ into the intervals $p_i=[i/k,(i+1)/k)$ for $i=0,1,\dots,k-1$, and define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-024412 | MATHDB-329087

The long-edge conjecture for the integer decomposition property

摘录 / EXCERPT: Long-edge conjecture. Simple lattice polytopes with sufficiently long edges have the integer decomposition property, where the required meaning of “sufficiently long” is given by a...

conjecture | open | MathDB 页面 | 原文线索

V2-024413 | MATHDB-329091

Divergence of nodal domains along infinite eigenvalue sequences in the square

摘录 / EXCERPT: Consider the Neumann problem in the square and a sequence of eigenfunctions associated with an infinite sequence of eigenvalues. Nodal-domain divergence conjecture. The number of n...

conjecture | open | MathDB 页面 | 原文线索

V2-024414 | MATHDB-329108

Involutivity and ordinariness of the elementary integral manifold

摘录 / EXCERPT: Let $(M,\mathcal{I})$ be an analytic involutive exterior differential system, considered over $\mathbb{C}$. Write $\operatorname{elem}(\mathcal{I})$ for its elementary system, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024415 | MATHDB-329109

Equality of the characteristic sheaves of E and the restricted tableau

摘录 / EXCERPT: Let $(M,\mathcal{I})$ be an involutive exterior differential system. Let E and $\dot{A}$ denote the tableaux introduced in the surrounding discussion, and let their characteristi...

conjecture | open | MathDB 页面 | 原文线索

V2-024416 | MATHDB-329128

Glick's conjecture on the point of collapse of axis-aligned polygons

摘录 / EXCERPT: Let $P\subset\mathbb{P}^2$ be an axis-aligned $2n$-gon. Let T denote the pentagram map, and let ... be its point of collapse. Let $\mathscr C(P)$ denote the center of mass of P....

conjecture | open | MathDB 页面 | 原文线索

V2-024417 | MATHDB-329131

The uniform coarse Baum-Connes isomorphism conjecture for Rips complexes

摘录 / EXCERPT: Uniform coarse Baum-Connes conjecture. The map μ_u is an isomorphism. This is the discrete Rips-complex formulation of the uniform coarse Baum-Connes conjecture, connecting uni...

conjecture | open | MathDB 页面 | 原文线索

V2-024418 | MATHDB-329134

Non-leaf conjecture for open 4-manifolds with isolated end sums of Euclidean spaces

摘录 / EXCERPT: Let M be an open 4-manifold with an isolated end diffeomorphic to $\natural^n\mathbf{R}$, where $1\leq n\leq\infty$. A $C^{1,0}$ non-leaf conjecture. No such M is diffeomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-024419 | MATHDB-329150

Affine-trap density conjecture for the real two-map parameter space

摘录 / EXCERPT: Let Ω_2 be the real parameter space for the two-map iterated function system, and call a parameter an affine trap when it satisfies the affine-trap condition described by th...

conjecture | open | MathDB 页面 | 原文线索

V2-024420 | MATHDB-329151

Goulden-Jackson polynomiality conjecture for single Hurwitz numbers

摘录 / EXCERPT: Let g be fixed and let $\eta=(\eta_1,\ldots,\eta_n)$ be a ramification profile with $n=\ell(\eta)$. Set $r=2g-2+d+n$, where $d=\sum_i\eta_i$, and let $H_g(\eta)$ denote the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-024421 | MATHDB-329164

Stable-manifold and asymptotic-stability conjecture for cubic NLS on hyperbolic spaces

摘录 / EXCERPT: Stable-manifold and asymptotic-stability conjecture. In ${\mathbb H}^2$ and ${\mathbb H}^3$, there is a codimension-one manifold of stable perturbations of a soliton for the cubic...

conjecture | open | MathDB 页面 | 原文线索

V2-024422 | MATHDB-329169

The definite intersection-form conjecture for positively curved 4-manifolds

摘录 / EXCERPT: Let M be a compact oriented 4-manifold with positive sectional curvature and nonvanishing second Betti number. Its intersection form is the bilinear form on $H^2(M;\mathbb{R})$...

conjecture | open | MathDB 页面 | 原文线索

V2-024423 | MATHDB-329183

Bounded regularity subdivision conjecture for simplicial complexes

摘录 / EXCERPT: Bounded regularity subdivision conjecture. There exist M_ϵ and M_ϵ' such that

conjecture | open | MathDB 页面 | 原文线索

V2-024424 | MATHDB-329197

Sandon's small oscillation conjecture for translated points

摘录 / EXCERPT: Let (N,ξ) be a contact manifold with a global contact form α, and let ψ be a contactomorphism. Say that ψ has small oscillation energy when its oscillation ene...

conjecture | open | MathDB 页面 | 原文线索

V2-024425 | MATHDB-329206

Projective-equivalence conjecture for isometric Hilbert geometries

摘录 / EXCERPT: Let (Ω,d_Ω) and $(\Omega',d_{\Omega'})$ be Hilbert geometries, where Ω and Ω' are convex domains equipped with their Hilbert metrics. They are projectivel...

conjecture | open | MathDB 页面 | 原文线索

V2-024426 | MATHDB-329276

The Gromoll-Meyer conjecture for leaf-wise intersections

摘录 / EXCERPT: Let (Σ,α) be a Liouville-fillable contact manifold with Liouville filling (M_1,λ_1), and let M be the symplectic completion obtained by attaching...

conjecture | open | MathDB 页面 | 原文线索

V2-024427 | MATHDB-329288

Asymptotic stability conjecture for the discrete NLS soliton manifold

摘录 / EXCERPT: Consider the discrete nonlinear Schrödinger evolution and its soliton family $e^{-i(-\mu t+\nu)}\alpha_\mu$, parametrized by (μ,ν). The associated soliton manifold is the set...

conjecture | open | MathDB 页面 | 原文线索

V2-024428 | MATHDB-329309

Weak BAB conjecture for Mori fiber spaces

摘录 / EXCERPT: Weak BAB conjecture for Mori fiber spaces. Fix $0<\epsilon<1$ and an integer $n>0$. Then there exists a number $M(n,\epsilon)$ depending only on n and ϵ such that, if...

conjecture | open | MathDB 页面 | 原文线索

V2-024429 | MATHDB-329311

Fenchel-Nielsen parameter conjecture for Zariski-dense free subgroups of ${\rm SU}(3,1)$

摘录 / EXCERPT: Fenchel-Nielsen parameter conjecture. The subgroup $\langle A,B\rangle$ is determined up to conjugacy by

conjecture | open | MathDB 页面 | 原文线索

V2-024430 | MATHDB-329323

Incidence correspondence between ball intersections and ball-hull boundaries

摘录 / EXCERPT: Let $K=\{p_1,p_2,\dots,p_n\}$ be a finite set in a normed plane $\mathbb{M}^2$, and let $\lambda\geq\lambda_K$. The ball intersection $\operatorname{bi}(K,\lambda)$ and the ball hu...

conjecture | open | MathDB 页面 | 原文线索

V2-024431 | MATHDB-329325

Craw-Ishii conjecture for minimal models and quiver GIT

摘录 / EXCERPT: Let S be an arbitrary complete local normal Gorenstein 3-fold with rational singularities. Let R be a ring appearing in the setup, and let N be an R-module such that...

conjecture | open | MathDB 页面 | 原文线索

V2-024432 | MATHDB-329336

Conjecture on control of higher eigenvalues of anticanonical powers

摘录 / EXCERPT: Higher-eigenvalue control conjecture. For each natural number k such that $2\leq k\leq n$, the k-th eigenvalue of K_M^{-m} is controlled.

conjecture | open | MathDB 页面 | 原文线索

V2-024433 | MATHDB-329357

The regular polygon conjecture for the determinant

摘录 / EXCERPT: Amongst all convex n-gons of fixed area, let the determinant refer to the determinant of the associated Laplacian. Regular polygon conjecture. Amongst all convex n-gons of fixe...

conjecture | open | MathDB 页面 | 原文线索

V2-024434 | MATHDB-329379

The minimum-norm Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Minimum-norm Yau-Tian-Donaldson conjecture. A polarised manifold (X,L) with discrete automorphism group admits a cscK metric if and only if it is uniformly K-stable with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-024435 | MATHDB-329380

Mabuchi-Tian coercivity conjecture

摘录 / EXCERPT: Mabuchi-Tian coercivity conjecture. Suppose X has discrete automorphism group. Then there exists a cscK metric in $c_1(L)$ if and only if the Mabuchi functional is coercive.

conjecture | open | MathDB 页面 | 原文线索

V2-024436 | MATHDB-329382

Twisted Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Twisted Yau-Tian-Donaldson conjecture. A smooth polarised manifold (X,L) together with an ample line bundle T admits a twisted cscK metric if and only if it is uniformly twiste...

conjecture | open | MathDB 页面 | 原文线索

V2-024437 | MATHDB-329386

Universal tightness conjecture for concave neighborhoods of circular spherical divisors

摘录 / EXCERPT: Let (X,ω) be a closed symplectic 4-manifold obtained as a symplectic blowup of $\mathbb{CP}^2$ with the standard Kähler form. Let $D=C_1\cup\cdots\cup C_n\subset X$ be a c...

conjecture | open | MathDB 页面 | 原文线索

V2-024438 | MATHDB-329392

Batyrev-Nill's derived equivalence conjecture for mirror Calabi-Yau complete intersections

摘录 / EXCERPT: Let Δ be a reflexive polytope with unique interior point 0, and let Δ_i and Δ_i' be two translated nef-partitions of Δ whose associated dual nef-parti...

conjecture | open | MathDB 页面 | 原文线索

V2-024439 | MATHDB-329393

Tian's coercivity conjecture for cscK metrics

摘录 / EXCERPT: Let (X,ω) be a Kähler manifold, and let the Mabuchi functional be the functional on the space of Kähler metrics in the fixed Kähler class represented by ω. Tian's con...

conjecture | open | MathDB 页面 | 原文线索

V2-024440 | MATHDB-329408

Ultrametric approximation conjecture for compact semimetric spaces

摘录 / EXCERPT: Let X be a compact nonempty semimetric space and let $\varepsilon>0$. A compact semimetric approximation conjecture asserts that there exist continuous mappings ... such that Y i...

conjecture | open | MathDB 页面 | 原文线索

V2-024441 | MATHDB-329420

Non-existence of non-isometric Dirac-isospectral spin lens spaces in dimensions congruent to 1 modulo 4

摘录 / EXCERPT: Non-existence conjecture. Two Dirac isospectral spin lens spaces of dimension $n\equiv 1\pmod4$ are necessarily isometric.

conjecture | open | MathDB 页面 | 原文线索

V2-024442 | MATHDB-329436

Wendl's uniqueness conjecture for fillings of unit cotangent bundles of surfaces

摘录 / EXCERPT: Wendl's conjecture. The diffeomorphism type of a Stein or exact filling of (Y,ξ_{std}) is unique and is given by the disk cotangent bundle for every g.

conjecture | open | MathDB 页面 | 原文线索

V2-024443 | MATHDB-329444

Existence of Frobenius structures for non-general multiplets

摘录 / EXCERPT: Existence conjecture. For each non-general multiplet A, there exists a Frobenius structure satisfying conditions (i), (ii), (iii), (v), and (vi) of Theorem 1, but not condition (...

conjecture | open | MathDB 页面 | 原文线索

V2-024444 | MATHDB-329450

Isoperimetric inequality for fundamental tones of free plates

摘录 / EXCERPT: Free-plate isoperimetric conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024445 | MATHDB-329451

Simple angular dependence conjecture for fundamental free-plate modes

摘录 / EXCERPT: Simple angular dependence conjecture. The fundamental modes of $\mathbb{B}(R)$ can be written as linear combinations of $u_1(r,\hat{\theta})$.

conjecture | open | MathDB 页面 | 原文线索

V2-024446 | MATHDB-329470

Transitivity conjecture for weakly proper pairs of analytic stacks

摘录 / EXCERPT: Transitivity conjecture. If $X \Subset_S Y$ and $Y \Subset_S Z$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024447 | MATHDB-329473

Yau's ampleness conjecture for manifolds with negative holomorphic sectional curvature

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold with strictly negative holomorphic sectional curvature, and let K_X denote its canonical line bundle. Yau's ampleness conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-024448 | MATHDB-329519

The open mirror theorem for semi-Fano toric manifolds

摘录 / EXCERPT: Let X be a semi-Fano toric manifold, meaning that its anti-canonical divisor $-K_X$ is nef. Let W^{HV} and W^{LF} denote its Hori-Vafa and Lagrangian Floer supe...

conjecture | open | MathDB 页面 | 原文线索

V2-024449 | MATHDB-329520

The SYZ map as the inverse toric mirror map

摘录 / EXCERPT: The SYZ-map conjecture. The SYZ map ϕ is inverse to the mirror map ψ. This is an equivalent formulation of the equality between the Hori-Vafa and Lagrangian Floer superpo...

conjecture | open | MathDB 页面 | 原文线索

V2-024450 | MATHDB-329529

Conjecture on maximal Fourier coefficients of automorphic packets for symplectic groups

摘录 / EXCERPT: Conjecture 4.2. For every $\psi\in\widetilde{\Psi}_2(\operatorname{Sp}_{2n})$:

conjecture | open | MathDB 页面 | 原文线索

V2-024451 | MATHDB-329569

Nonexistence of first-type regulators for singular polarizations with non-Lagrangian isotropic fibers

摘录 / EXCERPT: Nonexistence conjecture. There do not exist regulators of the first type for singular polarizations ${\mathcal P}$ such that there exist points $x\in M$ for which ${\mathcal P}_x$...

conjecture | open | MathDB 页面 | 原文线索

V2-024452 | MATHDB-329581

Ni's equivalence conjecture for polynomial-growth functions and curvature

摘录 / EXCERPT: Ni's equivalence conjecture. The following conditions are equivalent: (1) $\mathcal{O}_P(M)\neq\mathbb{C}$; (2) M has maximal volume growth; and (3) there is a constant C...

conjecture | open | MathDB 页面 | 原文线索

V2-024453 | MATHDB-329582

The polynomial-growth harmonic-function conjecture for Ricci-nonnegative manifolds

摘录 / EXCERPT: Polynomial-growth harmonic-function conjecture. There exists a nonconstant harmonic function on M of polynomial growth.

open_problem | open | MathDB 页面 | 原文线索

V2-024454 | MATHDB-329586

The interval conjecture for region counts of circles with pairwise intersections

摘录 / EXCERPT: Let C_n be the set of numbers of regions formed by n circles in the plane, in an arrangement not in general position, such that every two circles have at least one common point...

conjecture | open | MathDB 页面 | 原文线索

V2-024455 | MATHDB-329591

Pujals' transitive classification conjecture for three-dimensional partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let f be a partially hyperbolic diffeomorphism of a 3-manifold. Pujals' classification conjecture. If f is transitive, then f is finitely covered by one of the following: a p...

conjecture | open | MathDB 页面 | 原文线索

V2-024456 | MATHDB-329622

The divergence conjecture for CR Q-curvature

摘录 / EXCERPT: Divergence conjecture. On CR manifolds, the Q-curvature is a divergence: there exist form-valued local invariants T_a and $T_{\overline{a}}$ of θ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024457 | MATHDB-329623

Geometric representation conjecture for the von Neumann continuous geometry

摘录 / EXCERPT: Geometric representation conjecture. The von Neumann example of a continuous geometry admits a geometric representation over an elementary extension of the reals.

conjecture | open | MathDB 页面 | 原文线索

V2-024458 | MATHDB-329654

The coarse assembly map conjecture for warped cones

摘录 / EXCERPT: The warped-cone assembly conjecture. The coarse assembly map is not surjective for the warped cone $X=\mathcal{O}_G M$. This would provide a geometric obstruction to the coarse Bau...

conjecture | open | MathDB 页面 | 原文线索

V2-024459 | MATHDB-329670

Nonrealizability conjecture for supersolvable arrangements with a triple-point line

摘录 / EXCERPT: Let $\mathcal A\subset\mathbb P^2$ be a full-rank supersolvable arrangement of n lines with a singular modular point P of maximum multiplicity $m\geq 3$. A line has only triple...

conjecture | open | MathDB 页面 | 原文线索

V2-024460 | MATHDB-329699

Conjecture on minimizers and maximizers of the generalized eigenvalue quotient

摘录 / EXCERPT: Generalized eigenvalue quotient conjecture. If $q<p$, then functionals of the form $\mathcal{R}_{p,q}(\Omega)$ admit a minimizer and are bounded above. Any sequence of fixed-volume...

conjecture | open | MathDB 页面 | 原文线索

V2-024461 | MATHDB-329745

The BDD conjecture for minimal log discrepancies

摘录 / EXCERPT: BDD conjecture. For fixed $d \in \mathbb{Z}_{>0}$, there exists a real number $a(d)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024462 | MATHDB-329782

The mixed eigenvalue ordering conjecture for triangles

摘录 / EXCERPT: Mixed eigenvalue ordering conjecture. For an arbitrary triangle,

conjecture | open | MathDB 页面 | 原文线索

V2-024463 | MATHDB-329789

The standard-manifold classification conjecture for nonnegatively curved simply connected 4-manifolds

摘录 / EXCERPT: Standard-manifold classification conjecture. The manifold N is diffeomorphic to one of the 5 standard manifolds

conjecture | open | MathDB 页面 | 原文线索

V2-024464 | MATHDB-329790

Classification of complete cohomogeneity-one nearly Kähler structures on four six-manifolds

摘录 / EXCERPT: Classification conjecture. The Main Theorem yields all inhomogeneous complete cohomogeneity-one nearly Kähler structures. In particular, $S^2\times S^4$ admits no cohomogeneity-one...

conjecture | open | MathDB 页面 | 原文线索

V2-024465 | MATHDB-329791

Monotonicity of maximal-orbit volume for the nearly Kähler families

摘录 / EXCERPT: Monotonicity conjecture. The functions $V(\Psi_a)$ and $V(\Psi_b)$ are strictly increasing in a and b, respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-024466 | MATHDB-329792

Monotonicity and bounds for the zero count before maximal volume

摘录 / EXCERPT: Zero-count conjecture. The count satisfies: 1. $\mathcal{C}(a)$ and $\mathcal{C}(b)$ are decreasing in a and b, respectively. 2. For $a>0$ sufficiently small,...

conjecture | open | MathDB 页面 | 原文线索

V2-024467 | MATHDB-329819

Oriented Akbulut-Kirby conjecture on concordance from 0-surgery

摘录 / EXCERPT: Oriented Akbulut-Kirby conjecture. If 0-surgeries on two knots give the same 3-manifold, then the knots with relevant orientations are concordant.

conjecture | open | MathDB 页面 | 原文线索

V2-024468 | MATHDB-329835

Conjecture on the extended constrained KP hierarchy

摘录 / EXCERPT: Consider the constrained KP hierarchy and its extension obtained by adjoining the additional set of flows associated with the logarithm of its Lax operator. Let the associated Frob...

conjecture | open | MathDB 页面 | 原文线索

V2-024469 | MATHDB-329871

Uniform boundedness conjecture for the non-zero Hodge Laplace spectrum

摘录 / EXCERPT: Let X be a compact manifold of dimension $n\geq 3$. Write $\Delta_{(X,g)}^\bullet$ for the Hodge Laplacian acting on all differential-form degrees for a metric g on X, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-024470 | MATHDB-329873

The discrete Lichnerowicz conjecture for Delaunay spherical triangulations

摘录 / EXCERPT: Let a triangulation of ${\mathbb S}^{n-1}$ be equipped with a spherical cone metric, and let M be its associated discrete Laplacian matrix. Assume that all cone angles are less t...

conjecture | open | MathDB 页面 | 原文线索

V2-024471 | MATHDB-329883

The local Gan-Gross-Prasad conjecture for the symplectic-metaplectic case

摘录 / EXCERPT: Let W be a symplectic space over F, let ψ be a non-trivial additive character of F, and let ω_ψ be the Weil representation of $\widetilde{\mathrm{Sp}}(W)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024472 | MATHDB-329894

Pseudoeffectivity and uniruledness conjecture for compact Kähler manifolds

摘录 / EXCERPT: Pseudoeffectivity and uniruledness conjecture. The canonical class K_X is pseudoeffective if and only if X is not uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-024473 | MATHDB-329906

Relative Donaldson-Thomas 4 virtual cycle conjecture for simple degenerations

摘录 / EXCERPT: D-orbifold conjecture. Under these hypotheses, there exists a D-orbifold associated with $\mathfrak{I}^{\delta}_{\mathfrak{X}_{0}^{\dag}/\mathfrak{C}_{0}^{\dag}}$. If the canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-024474 | MATHDB-329929

The binomial vanishing ideal conjecture for projective sets over finite fields

摘录 / EXCERPT: Let $K=\mathbb{F}_q$ be a finite field, let $\mathbb{Y}$ be a subset of $\mathbb{P}^{s-1}$, and let $I(\mathbb{Y})$ denote its vanishing ideal. A projective set is parameterized by...

conjecture | open | MathDB 页面 | 原文线索

V2-024475 | MATHDB-329989

Bedford's stable-manifold conjecture

摘录 / EXCERPT: Let X be a complex manifold endowed with a Riemannian metric, and let $f\colon X\to X$ be an automorphism acting hyperbolically on an invariant compact subset $K\subset X$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-024476 | MATHDB-329990

Universal characteristic-zero quasi-isomorphism conjecture for plumbing DG-algebras

摘录 / EXCERPT: Let Γ be any tree, let $\mathscr{B}_\Gamma$ be the DG-algebra arising from the plumbing construction, and let $\mathscr{G}_\Gamma$ be the corresponding Ginzburg DG-algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-024477 | MATHDB-329997

Differentiability of the volume function for Kähler classes

摘录 / EXCERPT: Let X be a Kähler manifold of dimension n. Let α be a big class and let β be a pseudo-effective class. The notation $\langle \alpha^{n-1}\rangle$ denotes the posit...

conjecture | open | MathDB 页面 | 原文线索

V2-024478 | MATHDB-330011

Conjecture on Gaussian estimates and Riesz transform boundedness under Kato curvature conditions

摘录 / EXCERPT: Gaussian estimates and Riesz transform conjecture. The heat kernel of the Hodge Laplacian should have Gaussian estimates, and the Riesz transform should be bounded on L^p for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-024479 | MATHDB-330060

Density and finite-to-one conjecture for horizontal Y-mesh configurations

摘录 / EXCERPT: Density and finite-to-one conjecture. The map π_1 has dense image. Restricted to twisted configurations, the map π_1 is finite-to-one. This predicts both generic realizabil...

conjecture | open | MathDB 页面 | 原文线索

V2-024480 | MATHDB-330061

Existence conjecture for generic Y-meshes

摘录 / EXCERPT: Let S be a Y-pin with maximal admissible dimension $D(S)$, and let D be a positive integer satisfying $D\leq D(S)$. A Y-mesh is D-generic when its D-fractals have the g...

conjecture | open | MathDB 页面 | 原文线索

V2-024481 | MATHDB-330066

Boundedness of analytic slopes of holonomic D-modules

摘录 / EXCERPT: Let X be a complex manifold and let f be a holonomic f-module on X. For each hypersurface Z of X, consider the analytic slopes of f along Z, namely the real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-024482 | MATHDB-330083

Constant-mean-curvature conjecture for non-minimal biharmonic submanifolds of spheres

摘录 / EXCERPT: Let M be a non-minimal biharmonic submanifold of the unit sphere $S^n(1)$. Constant-mean-curvature conjecture. The mean curvature of M is constant. The claim is a weaker statem...

conjecture | open | MathDB 页面 | 原文线索

V2-024483 | MATHDB-330086

Gross-Matessi intersection conjecture for Lagrangian spheres

摘录 / EXCERPT: Let C be a bounded connected component of $N_{\mathbb R}-\Delta$, let L_ℓ be a Lagrangian sphere of type L_ℓ over C, and let $w_\ell:M\to\mathbb Z$ be its winding fun...

conjecture | open | MathDB 页面 | 原文线索

V2-024484 | MATHDB-330087

Morrison's conjecture on geometric transitions and mirror symmetry

摘录 / EXCERPT: Let $\widehat{X}$ and $\widetilde{X}$ be Calabi--Yau manifolds, and suppose they are related by a geometric transition ... where $\widehat{X} \dashrightarrow X$ is a birational contr...

conjecture | open | MathDB 页面 | 原文线索

V2-024485 | MATHDB-330088

Log abundance conjecture for klt threefolds in characteristic greater than 5

摘录 / EXCERPT: Log abundance conjecture. If K_X+B is nef, then it is semi-ample.

conjecture | open | MathDB 页面 | 原文线索

V2-024486 | MATHDB-330089

Minimal simplicial volume conjecture for hyperbolic 3-manifolds with geodesic boundary

摘录 / EXCERPT: For each $g\geq 2$, let $\mathcal{M}_g$ be the set of 3-manifolds with boundary that admit an ideal triangulation by g tetrahedra and have Euler characteristic $\chi(M)=1-g$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-024487 | MATHDB-330119

Forsgård-Shapiro tropical bound for real zeros

摘录 / EXCERPT: Forsgård-Shapiro tropical conjecture. The number of real zeros of f does not exceed the number of corners of the continuous piecewise-linear function f_{trop} on...

conjecture | open | MathDB 页面 | 原文线索

V2-024488 | MATHDB-330124

The class-number conjecture for pseudo-Anosov maps with fixed dilatation

摘录 / EXCERPT: Let $(\Lambda,[I],K)$ be the triple corresponding to a pseudo-Anosov map $\phi\in Mod~(X)$, and let λ_ϕ be its dilatation. The integral order Λ has class number...

conjecture | open | MathDB 页面 | 原文线索

V2-024489 | MATHDB-330125

The Langlands conjecture for noncommutative tori

摘录 / EXCERPT: Let E be a finite extension of $\mathbb Q$ with Galois group $Gal~(E|\mathbb Q)$, and let $\sigma_{n+1}:Gal~(E|\mathbb Q)\to GL_{n+1}(\mathbb C)$ be an irreducible representation...

conjecture | open | MathDB 页面 | 原文线索

V2-024490 | MATHDB-330148

Canonical L_{∞} morphism for stable generalized complex structures

摘录 / EXCERPT: Canonical L_{∞} morphism conjecture. There is a canonical L_{∞} morphism from $\Omega^{\bullet}_{\mathbb J,H}$ to the total complex of, extending the given quasi-isom...

conjecture | open | MathDB 页面 | 原文线索

V2-024491 | MATHDB-330155

Fujino's logarithmic injectivity conjecture

摘录 / EXCERPT: Fujino's logarithmic injectivity conjecture. The multiplication homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-024492 | MATHDB-330181

CR analogue of Yau's first eigenvalue conjecture for p-minimal surfaces

摘录 / EXCERPT: CR analogue of Yau's conjecture. The first eigenvalue of L_{0} on any closed embedded p-minimal surface of genus at most 1 in the standard CR 3-sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-024493 | MATHDB-330182

CR analogue of Lawson's conjecture for p-minimal tori

摘录 / EXCERPT: CR analogue of Lawson's conjecture. Any closed embedded p-minimal torus with $\alpha=0$ in the standard CR 3-sphere $\mathbf{S}^{3}$ is the Clifford torus.

conjecture | open | MathDB 页面 | 原文线索

V2-024494 | MATHDB-330185

Conjecture on the space of Alexandrov embedded Lawson symmetric CMC surfaces

摘录 / EXCERPT: A Lawson symmetric CMC surface is a compact constant-mean-curvature surface in $\mathbb{S}^3$ of genus $g\geq 1$ with a cyclic symmetry of order $g+1$ having four fixed points. For...

conjecture | open | MathDB 页面 | 原文线索

V2-024495 | MATHDB-330233

Support rigidity conjecture for Liouville currents of flat metrics

摘录 / EXCERPT: Let S be the underlying surface and let $\mathrm{Flat}(S)$ denote the space of flat metrics on S. For $\varphi\in\mathrm{Flat}(S)$, let L_{φ} be its Liouville geodesic...

conjecture | open | MathDB 页面 | 原文线索

V2-024496 | MATHDB-330242

Optimal error-term conjecture for hyperbolic conjugacy classes

摘录 / EXCERPT: Let Γ be a cocompact or cofinite discrete group, let $\mathcal{H}$ be a hyperbolic conjugacy class of Γ, and let $E(\mathcal{H},X;z)$ denote the corresponding error t...

conjecture | open | MathDB 页面 | 原文线索

V2-024497 | MATHDB-330244

Planar-surface horoball-packing conjecture

摘录 / EXCERPT: Planar-surface horoball-packing conjecture. For any horoball packing of F, there is at least one horoball of area less than

conjecture | open | MathDB 页面 | 原文线索

V2-024498 | MATHDB-330271

The conjecture that three-dimensional ε-isospectral lens spaces are ε-spin-isometric

摘录 / EXCERPT: Let two three-dimensional lens spaces be ε-isospectral if their Dirac spectra agree up to the equivalence encoded by ε, and let them be ε-spin-isometric if there is an ε-compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-024499 | MATHDB-330320

Fatabbi-Lorenzini's generalized Segre bound for fat points

摘录 / EXCERPT: Fatabbi-Lorenzini's generalized Segre bound. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024500 | MATHDB-330321

Modified Segre conjecture for fat points in arbitrary position

摘录 / EXCERPT: Fix integers $d\ge 2$ and $n>2$. Let $S\subseteq {\mathbb P}^n$ be a finite collection of s points, fix integers $m_1,\dots,m_s$, and set ... Let $\operatorname{w}(Z)$ be the total...

conjecture | open | MathDB 页面 | 原文线索

V2-024501 | MATHDB-330332

The Knot Complement Conjecture

摘录 / EXCERPT: Let K be a knot in a closed connected orientable 3--manifold Y, with exterior $M=Y\setminus\operatorname{int}N(K)$ irreducible and not homeomorphic to the solid torus. A slop...

conjecture | open | MathDB 页面 | 原文线索

V2-024502 | MATHDB-330338

The Gromov-Witten determination conjecture for the canonical natural transformation

摘录 / EXCERPT: Let a Lefschetz fibration arise from an anticanonical Lefschetz pencil, and choose the bulk term specified by the corresponding theorem. The Fukaya category has a canonical natural...

conjecture | open | MathDB 页面 | 原文线索

V2-024503 | MATHDB-330339

The connection conjecture for deformed Serre natural transformations

摘录 / EXCERPT: Let $\bar b$ be the restriction of a bulk term $ b$ satisfying the paper's main assumption, and suppose the deformation $\EuScript A_{q,\bar b}$ is trivial. Let $\rho_{q,\bar b}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-024504 | MATHDB-330340

The noncommutative divisor deformation conjecture for the fibrewise compactification

摘录 / EXCERPT: Let $\EuScript A$ be the Fukaya A_∞-algebra of the Lefschetz fibration and $\EuScript B_{q,\bar b|M}$ the category associated to the fibrewise compactification. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-024505 | MATHDB-330412

The conjecture on uniform continuity and weakly Euclidean points

摘录 / EXCERPT: The conjecture on uniform continuity and weakly Euclidean points. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-024506 | MATHDB-330413

Conjecture on symplectic leaves of Calogero-Moser spaces for finite Coxeter groups

摘录 / EXCERPT: Symplectic-leaf conjecture. Each conjugacy class of parabolic subgroups (W') labels at most one symplectic leaf, and the geometric ordering on leaves equals the algebraic orderin...

conjecture | open | MathDB 页面 | 原文线索

V2-024507 | MATHDB-330437

Modified Weyl-Berry conjecture for Minkowski measurable fractal strings

摘录 / EXCERPT: Modified Weyl-Berry conjecture. There is a positive constant c_D depending only on D such that

conjecture | open | MathDB 页面 | 原文线索

V2-024508 | MATHDB-330439

The CPE conjecture on critical metrics being Einstein

摘录 / EXCERPT: A CPE metric is a 3-tuple (M^n,g,f), where (M^n,g) is a compact oriented Riemannian manifold of dimension $n\geq 3$ with constant scalar curvature, and f is a smooth function...

conjecture | open | MathDB 页面 | 原文线索

V2-024509 | MATHDB-330441

Bridgeland-type stability conjecture for symplectic autoequivalences of cubic K3 categories

摘录 / EXCERPT: Let $X\subset\mathbb P^5$ be a smooth cubic fourfold, let $\mathcal A_X$ be its K3 category, and let $\operatorname{Aut}_{\rm s}(\mathcal A_X)$ denote the group of symplectic autoe...

conjecture | open | MathDB 页面 | 原文线索

V2-024510 | MATHDB-330442

First-candidate conjecture for type I singularities in Lagrangian mean curvature flow

摘录 / EXCERPT: Let $T_1,T_2,\ldots$ be the candidate singular times associated with a Lagrangian mean curvature flow as in Theorem C. First-candidate conjecture. A type I singularity can occur on...

conjecture | open | MathDB 页面 | 原文线索

V2-024511 | MATHDB-330443

Type II singularity conjecture for loss of embeddedness in Lagrangian mean curvature flow

摘录 / EXCERPT: Let a Lagrangian mean curvature flow start from embedded initial data, and suppose that it becomes immersed before its singular time. Embeddedness-loss conjecture. The flow must ha...

conjecture | open | MathDB 页面 | 原文线索

V2-024512 | MATHDB-330458

The one-log-zone conjecture for resonance-free regions

摘录 / EXCERPT: Let M be a cusp manifold and let $\mathcal{G}(M)$ denote the space of C^∞ metrics on M for which (M,g) is a cusp manifold with negative sectional curvature. The theore...

conjecture | open | MathDB 页面 | 原文线索

V2-024513 | MATHDB-330459

The generic abundance conjecture for resonances of negatively curved cusp manifolds

摘录 / EXCERPT: Generic resonance abundance conjecture. For an open and dense set of $g\in\mathcal{G}(M)$, there is an infinite number of resonances outside any such strip. The conjecture asserts...

conjecture | open | MathDB 页面 | 原文线索

V2-024514 | MATHDB-330466

Local-global conjecture for reduced curvatures in K-Apollonian packings

摘录 / EXCERPT: Local-global conjecture for K-Apollonian packings. There exists an M dividing 24 such that all sufficiently large integers whose residues modulo M lie in S_M occur as red...

conjecture | open | MathDB 页面 | 原文线索

V2-024515 | MATHDB-330486

Maximal covering density of the lattices LD_n among covering maxima

摘录 / EXCERPT: For each $n\geq 6$, let LD_n be the lattice defined in Dutour (2005), and let a covering maximum be a lattice whose covering density decreases under perturbation. The covering-de...

conjecture | open | MathDB 页面 | 原文线索

V2-024516 | MATHDB-330523

Weak Borsuk conjecture for polytopes

摘录 / EXCERPT: Let $P\subseteq\mathbb R^d$ be a polytope with m facets. A covering by smaller-diameter sets is a cover of P by sets whose diameters are strictly smaller than the diameter of...

conjecture | open | MathDB 页面 | 原文线索

V2-024517 | MATHDB-330530

Convergence of discrete A-flow to smooth Ricci flow

摘录 / EXCERPT: Fix a smooth Riemannian surface (M,g). Let $(M_n,\mathcal{T}_n,\Phi_n,r_n)$ be a sequence of weighted triangulations of M with initial circle packing metrics r_n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024518 | MATHDB-330535

Isoperimetric inequality for a spherical simplex

摘录 / EXCERPT: Isoperimetric inequality for a spherical simplex. Among $(d-1)$-dimensional spherical simplices with fixed volume, the regular simplex has minimal surface $(d-1)$-volume.

conjecture | open | MathDB 页面 | 原文线索

V2-024519 | MATHDB-330536

Generalized minimal solid-angle conjecture for faces of simplices

摘录 / EXCERPT: Generalized minimal solid-angle conjecture. For every $k<d-1$, T has a k-face whose corresponding $(d-k-1)$-dimensional solid angle is not greater than the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-024520 | MATHDB-330560

Klee-Novik's balanced lower bound conjecture

摘录 / EXCERPT: Let Δ be a balanced normal pseudomanifold of dimension $d-1\geq 3$. Klee-Novik's balanced lower bound conjecture. ... This extends the balanced Lower Bound Theorem by includin...

conjecture | open | MathDB 页面 | 原文线索

V2-024521 | MATHDB-330566

The centered convex-body lattice-point inequality

摘录 / EXCERPT: The centered convex-body lattice-point inequality. For every $K \in \mathcal{K}_{c}^{d}$,

conjecture | open | MathDB 页面 | 原文线索

V2-024522 | MATHDB-330578

Han's conjecture on complete p-biharmonic submanifolds

摘录 / EXCERPT: Let M be a complete p-biharmonic submanifold of a nonpositively curved Riemannian manifold. Han's conjecture. Every complete p-biharmonic submanifold in a nonpositively curve...

conjecture | open | MathDB 页面 | 原文线索

V2-024523 | MATHDB-330607

Simpliciality conjecture for monotone and nonnegative lattice-invariant valuations

摘录 / EXCERPT: Let d be a positive integer, let $\mathcal{P}(\mathbb{Z}^d)$ denote the collection of lattice polytopes in $\mathbb{R}^d$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024524 | MATHDB-330612

Markus conjecture on completeness and parallel volume

摘录 / EXCERPT: Let B be a connected affine manifold, with affine structure group $\mathrm{GL}_n(\mathbb{Z})\ltimes H^n$, and let completeness mean that all geodesics on B are parametrised by...

conjecture | open | MathDB 页面 | 原文线索

V2-024525 | MATHDB-330644

Optimal curved Brunn-Minkowski inequality for the symmetric group

摘录 / EXCERPT: Let $S(n)$ be the symmetric group equipped with the metric d, and let $A,B\subseteq S(n)$ be nonempty. Write M for their midpoint set, and let $\mathrm{d}(A,B)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-024526 | MATHDB-330661

Uniform projection dimension conjecture for planar Brownian level sets

摘录 / EXCERPT: Let L^d be the Brownian level set and let $\mathcal{D}$ be the exceptional set defined in the source. For each direction $\theta\in[-\frac{\pi}{2},\frac{\pi}{2})$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024527 | MATHDB-330682

Fractal ringlet conjecture for the Asteroid Belt

摘录 / EXCERPT: Consider the Asteroid Belt of the Solar System and the corresponding variational problem. A fractal solution is an exact solution with fractal structure, and ringlets are isolated...

conjecture | open | MathDB 页面 | 原文线索

V2-024528 | MATHDB-330702

Relative Gromov-Lawson-Rosenberg conjecture

摘录 / EXCERPT: Relative Gromov-Lawson-Rosenberg conjecture. If the relative higher index of D is zero, then N admits a metric of positive scalar curvature that is collared near ∂N....

conjecture | open | MathDB 页面 | 原文线索

V2-024529 | MATHDB-330707

Nonexistence of aperiodic totally separable sphere packings

摘录 / EXCERPT: A totally separable sphere packing is a packing of congruent spheres in Euclidean space such that every pair of touching spheres admits a hyperplane separating their interiors from...

conjecture | open | MathDB 页面 | 原文线索

V2-024530 | MATHDB-330737

The Gromov-Lawson conjecture on higher Â-genera

摘录 / EXCERPT: Gromov-Lawson conjecture. The higher $\widehat A$-genus vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-024531 | MATHDB-330760

Okun-Schreve conjecture on L2-Betti numbers of aspherical manifolds with boundary

摘录 / EXCERPT: Let $(M,\partial M)$ be a compact aspherical n-manifold-with-boundary, and write $b^{(2)}_{>n/2}(M)=0$ to mean that all L^2-Betti numbers in degrees greater than $n/2$ vanish....

conjecture | open | MathDB 页面 | 原文线索

V2-024532 | MATHDB-330761

Liu-Xu-Ye-Zhao convergence conjecture for pinched submanifolds in spheres

摘录 / EXCERPT: Let M_{0} be an n-dimensional complete submanifold in the sphere $\mathbb{S}^{n+q}\left(1/\sqrt{c}\right)$. Let h be its second fundamental form, H its mean curvature vecto...

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V2-024533 | MATHDB-330762

Lawson-Simons conjecture for submanifolds in spheres

摘录 / EXCERPT: Let $F_{0}:M^{n}\to\mathbb{S}^{n+q}$ be an n-dimensional closed submanifold satisfying ... where h is the second fundamental form. Lawson-Simons conjecture. The mean curvature fl...

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V2-024534 | MATHDB-330767

Donaldson's twisted cscK conjecture

摘录 / EXCERPT: Let M be a compact Kähler manifold with Kähler class $[\omega]$, let χ be a closed positive $(1,1)$-form, and let $t\in[0,1]$. A metric solving the corresponding twisted con...

conjecture | open | MathDB 页面 | 原文线索

V2-024535 | MATHDB-330769

Chen's global existence conjecture for the twisted Calabi flow

摘录 / EXCERPT: Let φ be a smooth initial Kähler potential, and let the twisted Calabi flow be the evolution equation introduced in the paper for the corresponding Kähler metrics and twist...

conjecture | open | MathDB 页面 | 原文线索

V2-024536 | MATHDB-330770

Donaldson's asymptotic conjecture for the twisted Calabi flow

摘录 / EXCERPT: Let (M,ω,χ,J) be the initial twisted Kähler data, and suppose that the twisted Calabi flows have global existence. The transformed data along the flow may be written...

conjecture | open | MathDB 页面 | 原文线索

V2-024537 | MATHDB-330771

Donaldson's coercivity conjecture for the K-energy

摘录 / EXCERPT: Let $(M,[\omega])$ be a polarized Kähler manifold, and let E denote the K-energy functional. The maximal invariant, totally geodesic subspace induced by the automorphism group is...

conjecture | open | MathDB 页面 | 原文线索

V2-024538 | MATHDB-330772

Twisted K-stability and solvability conjecture

摘录 / EXCERPT: Let (Ω,χ,t) be twisted Kähler data with $t<1$, and let the corresponding twisted continuity-path equation be the equation whose solvability at parameter t defines a twi...

conjecture | open | MathDB 页面 | 原文线索

V2-024539 | MATHDB-330773

Twisted cscK existence and properness conjecture for generalized twisting forms

摘录 / EXCERPT: Let μ_k be the closed positive twisting data used to define the functional J_{μ_k}, and let ... be the corresponding twisted K-energy functional. A twisted cscK metric is a s...

conjecture | open | MathDB 页面 | 原文线索

V2-024540 | MATHDB-330796

Hexagonal conjecture for asymptotic minimal partition energies

摘录 / EXCERPT: Let Ω be a bounded planar domain, and let $\mathfrak L_k(\Omega)$ be the energy of a minimal k-partition of Ω. Let $\hexagon$ be the regular hexagon of area 1, an...

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V2-024541 | MATHDB-330798

Tropical Noether formula

摘录 / EXCERPT: Tropical Noether formula.

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V2-024542 | MATHDB-330799

Tropical Hirzebruch signature conjecture for tropical surfaces

摘录 / EXCERPT: Tropical Hirzebruch signature conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-024543 | MATHDB-330822

Degeneracy conjecture for isolated complex tangents of 3-manifold embeddings

摘录 / EXCERPT: Let M be a smooth 3-manifold smoothly embedded in $\mathbb{C}^3$, and let $p\in M$ be an isolated complex tangent. The degeneracy conjecture asserts that the Bishop invariant at...

conjecture | open | MathDB 页面 | 原文线索

V2-024544 | MATHDB-330844

The analytic structure conjecture for the left adjoint to the underlying algebra functor

摘录 / EXCERPT: Analytic structure conjecture. For every such $\mathcal O$, the unit $\mathcal O \to \overline{\Psi}(\mathcal O)^{\mathrm{alg}}$ has this universal analytification property. This g...

conjecture | open | MathDB 页面 | 原文线索

V2-024545 | MATHDB-330846

The Lagrangian-image and quasifiniteness conjecture for projector moduli

摘录 / EXCERPT: Let $\Psi:{\cal M}_B\to{\cal M}_A$ be the morphism induced by the homomorphism between the algebras associated with the projector systems, and let k and n denote the numbers of...

conjecture | open | MathDB 页面 | 原文线索

V2-024546 | MATHDB-330914

Cannon-Thurston conjecture for surface Kleinian groups

摘录 / EXCERPT: Cannon-Thurston conjecture. The inclusion $\tilde{i}:\widetilde{S^h}\rightarrow\mathbb{H}^3$ extends continuously to the boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-024547 | MATHDB-330945

Intrinsicness conjecture for weighted tangent cones and their filtrations

摘录 / EXCERPT: Intrinsicness conjecture. The filtration

conjecture | open | MathDB 页面 | 原文线索

V2-024548 | MATHDB-330964

The affine-surface higher-cancellation conjecture

摘录 / EXCERPT: Let X be a smooth affine surface. Say that X has negative logarithmic Kodaira dimension when its logarithmic Kodaira dimension is negative. Say that X fails the...

conjecture | open | MathDB 页面 | 原文线索

V2-024549 | MATHDB-330978

Majorization conjecture for hyperbolic-cell angle data

摘录 / EXCERPT: Let $\mathcal{B}$ and $\mathcal{B}^*$ be the decreasing rearrangements of the collections of angles $\{\alpha_{j,k}\}$ and $\{\alpha_{j,k}^*\}$, respectively, as defined before. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-024550 | MATHDB-330981

Palais' slice conjecture for the Chern-Ricci geodesic distribution

摘录 / EXCERPT: Let (X,J) be a Fano manifold that is complex deformation unobstructed, and let $\omega\in 2\pi c_1(X)$ be a Kähler form. For $(g,\Omega)\in\mathcal{S}_{\omega}$, set...

conjecture | open | MathDB 页面 | 原文线索

V2-024551 | MATHDB-330983

The trisection classification conjecture for the standard four-sphere

摘录 / EXCERPT: Let $\mathcal{T}$ be a trisection of the standard smooth four-sphere $\mathcal{S}^4$. Let $\mathcal{S}^0$ denote its standard $(0,0)$-trisection, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024552 | MATHDB-330988

Conjecture on homogeneous convex divisible domains in flag manifolds

摘录 / EXCERPT: Let G be a connected non-compact simple Lie group with trivial center, and let $P \leq G$ be a parabolic subgroup. A convex divisible domain in G/P is a bounded open convex sub...

conjecture | open | MathDB 页面 | 原文线索

V2-024553 | MATHDB-331011

Alexandrov-space bi-Lipschitz embedding conjecture

摘录 / EXCERPT: Let $D>0$. Let A be a bounded subset with $\operatorname{diam}(A)\leq D$ of an n-dimensional complete Alexandrov space X^n with curvature $K\geq -1$. Alexandrov-space embeddi...

conjecture | open | MathDB 页面 | 原文线索

V2-024554 | MATHDB-331012

Noncollapsed Ricci-curvature bi-Lipschitz embedding conjecture

摘录 / EXCERPT: Let A be a subset with $\operatorname{diam}(A)\leq D$ of an n-dimensional complete Riemannian manifold (M^n,g) with Ricci curvature ... Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-024555 | MATHDB-331018

The cyclic fundamental group conjecture for unknotted surfaces

摘录 / EXCERPT: Let $\mathcal{K}$ be a knotted surface in S^4, and let $S^4\setminus\mathcal{K}$ be its exterior. The fundamental group of the exterior is $\pi_1(S^4\setminus\mathcal{K})$. Cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-024556 | MATHDB-331019

The standardness conjecture for $(b;b-1,c_2,c_3)$-bridge trisections

摘录 / EXCERPT: Let $\mathcal{K}$ be a knotted surface in S^4 admitting a $(b;b-1,c_2,c_3)$--bridge trisection. Such a bridge trisection is a surface of the indicated type, with b the bridge p...

conjecture | open | MathDB 页面 | 原文线索

V2-024557 | MATHDB-331020

Finite-time collapsing conjecture for the Kähler-Ricci flow

摘录 / EXCERPT: Finite-time collapsing conjecture. There exists a Kähler metric ω_0 such that the Kähler-Ricci flow has finite-time collapsing if and only if X admits a Fano fibration...

conjecture | open | MathDB 页面 | 原文线索

V2-024558 | MATHDB-331021

Diameter-collapse conjecture for finite-time Kähler-Ricci flow

摘录 / EXCERPT: Diameter-collapse conjecture. As $t\to T$,

conjecture | open | MathDB 页面 | 原文线索

V2-024559 | MATHDB-331035

The homogeneous uniform bundle conjecture on projective space

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, and let $\boldsymbol{P}^n$ denote projective n-space. A vector bundle on $\boldsymbol{P}^n$ is uniform if its res...

conjecture | open | MathDB 页面 | 原文线索

V2-024560 | MATHDB-331043

Spectral comparison conjecture for positively curved Euclidean weighted manifolds

摘录 / EXCERPT: Let $(\mathbb{R}^n,g,\mu)$ be a connected weighted manifold satisfying $CD(\rho,\infty)$ with $\rho>0$, and let $\lambda_k(\mathbb{R}^n,g,\mu)$ denote its ordered eigenvalues. Spec...

conjecture | open | MathDB 页面 | 原文线索

V2-024561 | MATHDB-331044

Contraction conjecture for positively curved Euclidean weighted manifolds

摘录 / EXCERPT: Let $(\mathbb{R}^n,g,\mu)$ satisfy $CD(\rho,\infty)$ with $\rho>0$, and let γ^n_ρ be the n-dimensional Gaussian measure. A map pushes forward γ^n_ρ onto μ...

conjecture | open | MathDB 页面 | 原文线索

V2-024562 | MATHDB-331045

Contraction conjecture for positively curved spheres

摘录 / EXCERPT: Let $(S^n,g,\operatorname{vol}_g)$ satisfy $\operatorname{Ric}_g\geq\rho g$ with $\rho>0$, and let g^ρ_{can} be the canonical metric on S^n rescaled so that...

conjecture | open | MathDB 页面 | 原文线索

V2-024563 | MATHDB-331046

Graded curvature-dimension version of the contraction and spectral conjectures

摘录 / EXCERPT: Let a weighted manifold be diffeomorphic to $\mathbb{R}^n$ and satisfy the separate curvature bounds ... Here Ric_g is the Riemannian Ricci tensor and ∇_g^2V...

conjecture | open | MathDB 页面 | 原文线索

V2-024564 | MATHDB-331093

The C^k-map versus ring-homomorphism conjecture

摘录 / EXCERPT: The C^k-map versus ring-homomorphism conjecture. The following statements are equivalent: (1) $\varphi^{\sharp}$ is a ring-homomorphism over ${\Bbb R}\hookrightarrow{\Bbb C}$; (2...

conjecture | open | MathDB 页面 | 原文线索

V2-024565 | MATHDB-331094

The C^k-map to Euclidean space conjecture

摘录 / EXCERPT: Let X be a C^k-manifold and E a complex C^k vector bundle of rank r on X. Let $(y^1,\ldots,y^n)$ be a global coordinate system on $\Bbb R^n$, and let ... be an assignment...

conjecture | open | MathDB 页面 | 原文线索

V2-024566 | MATHDB-331101

Inscribability conjecture for 2-neighborly simplicial polytopes

摘录 / EXCERPT: Inscribability conjecture for 2-neighborly simplicial polytopes. All 2-neighborly simplicial polytopes are inscribable.

conjecture | open | MathDB 页面 | 原文线索

V2-024567 | MATHDB-331102

Equality conjecture for Welschinger invariants of real projective three-space

摘录 / EXCERPT: Let d be a positive integer, and let $W_{\mathbb{R}P^3}(d,l)$ denote the Welschinger invariant of degree d with the parameter l specifying the real-point constraint. Equality...

conjecture | open | MathDB 页面 | 原文线索

V2-024568 | MATHDB-331106

Donaldson-Tian-Yau conjecture for constant-scalar-curvature Kähler metrics

摘录 / EXCERPT: Donaldson-Tian-Yau conjecture. (X,L) admits a cscK metric in $c_1(L)$ if and only if it is K-polystable.

conjecture | open | MathDB 页面 | 原文线索

V2-024569 | MATHDB-331129

Existence of non-$(0,d-1)$-scribable polytopes

摘录 / EXCERPT: A d-polytope is a polytope of dimension d, and a polytope is $(0,d-1)$-scribable if it has a realization whose vertices lie on a sphere and whose facets are tangent to a concen...

conjecture | open | MathDB 页面 | 原文线索

V2-024570 | MATHDB-331130

Non-$(1,d-1)$-scribable polytopes in odd dimensions

摘录 / EXCERPT: A d-polytope is a polytope of dimension d, and a polytope is $(1,d-1)$-scribable if it has a realization with a sphere separating its vertices from its facets in the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-024571 | MATHDB-331131

Circumscribability conjecture for neighborly polytopes

摘录 / EXCERPT: A polytope is neighborly if every set of at most half the dimension's vertices forms a face, and a polytope is circumscribable if it has a realization whose facets are tangent to a...

conjecture | open | MathDB 页面 | 原文线索

V2-024572 | MATHDB-331135

The ten-triangle conjecture for acute spherical diangles

摘录 / EXCERPT: Ten-triangle conjecture. Any dissection of an acute spherical diangle into acute spherical triangles requires at least ten triangles.

conjecture | open | MathDB 页面 | 原文线索

V2-024573 | MATHDB-331149

Malkoun's linear independence conjecture for the symplectic-group construction

摘录 / EXCERPT: Malkoun's conjecture 1 for $Sp(n)$. For every such configuration, the polynomials $p_1,q_1,\ldots,p_n,q_n$ are linearly independent over $\mathbb{C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-024574 | MATHDB-331150

Malkoun's determinant conjecture for the symplectic-group construction

摘录 / EXCERPT: Malkoun's conjecture 2 for $Sp(n)$. For every such configuration,

conjecture | open | MathDB 页面 | 原文线索

V2-024575 | MATHDB-331155

Huang-Ji-Yin Gap Conjecture for CR mappings between spheres

摘录 / EXCERPT: For $n\geq 2$, let κ_0 be the largest integer such that ... and define ... Let $f\colon\mathbb S^n\to\mathbb S^N$ be a sufficiently smooth CR mapping. Huang-Ji-Yin Gap Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-024576 | MATHDB-331158

Uniqueness of the limiting geometric objects for K-unstable Fano manifolds

摘录 / EXCERPT: Uniqueness conjecture. The geometric objects $\overline X$, $\mathcal F$, and X_∞ are uniquely determined by X; equivalently, they are independent of the choice of the ini...

conjecture | open | MathDB 页面 | 原文线索

V2-024577 | MATHDB-331162

The lecture-hall Sperner triangulation conjecture

摘录 / EXCERPT: Lecture-hall Sperner conjecture. There exists a regular, flag, unimodular triangulation of R_n that admits a shelling order such that the maximal simplices of the triangulation a...

conjecture | open | MathDB 页面 | 原文线索

V2-024578 | MATHDB-331170

The singularity formation set conjecture for finite-time Kähler-Ricci flow singularities

摘录 / EXCERPT: Let $(X,\omega(t))$ be a solution of the Kähler-Ricci flow developing a finite-time singularity, and let Σ denote its singularity formation set. Singularity formation set co...

conjecture | open | MathDB 页面 | 原文线索

V2-024579 | MATHDB-331171

Zhang's uniform lower-bound conjecture for the Kähler-Ricci potential

摘录 / EXCERPT: Let X be a compact Kähler manifold, let $\varphi(t)$ solve the parabolic complex Monge-Ampère equation associated with the Kähler-Ricci flow, and suppose the solution exists on...

conjecture | open | MathDB 页面 | 原文线索

V2-024580 | MATHDB-331172

The bounded-potential conjecture for nef classes on compact Kähler manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[\alpha]$ be a nef $(1,1)$ class such that $[\alpha]+\lambda c_1(X)$ is a Kähler class for some $\lambda>0$. A closed positive current...

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V2-024581 | MATHDB-331173

The uniruledness conjecture for null-locus components

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[\alpha]$ be a nef and big $(1,1)$ class which is not Kähler, with $[\alpha]+\lambda c_1(X)$ a Kähler class for some $\lambda>0$. The...

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V2-024582 | MATHDB-331174

The Kähler contraction conjecture for null loci

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[\alpha]$ be a nef and big $(1,1)$ class which is not Kähler, with $[\alpha]+\lambda c_1(X)$ a Kähler class for some $\lambda>0$. Let...

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V2-024583 | MATHDB-331175

Tian's diameter-collapse conjecture for finite-time Kähler-Ricci flow

摘录 / EXCERPT: Let (X^n,ω_0) be a compact Kähler manifold, and let $\omega(t)$ be the Kähler-Ricci flow solution on its maximal interval $[0,T)$, where $T<\infty$. Tian's diameter-collapse...

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V2-024584 | MATHDB-331176

The finite-time diameter bound conjecture for Kähler-Ricci flow

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $\omega(t)$ be a solution of the Kähler-Ricci flow on a maximal interval $[0,T)$ with $T<\infty$. Finite-time diameter bound conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-024585 | MATHDB-331177

The infinite-time rescaled diameter bound conjecture

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $\omega(t)$ be a solution of the Kähler-Ricci flow defined for all $t\geqslant0$. Rescaled diameter bound conjecture. There is a consta...

conjecture | open | MathDB 页面 | 原文线索

V2-024586 | MATHDB-331178

The volume-growth conjecture for infinite-time Kähler-Ricci flow

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $\omega(t)$ be a solution of the Kähler-Ricci flow defined on $[0,\infty)$. Volume-growth conjecture. The Kodaira dimension satisfies...

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V2-024587 | MATHDB-331179

The initial-metric independence conjecture for singularity type at infinity

摘录 / EXCERPT: Let X be a compact Kähler manifold with K_X nef, so every solution of the Kähler-Ricci flow exists for all positive time. Initial-metric independence conjecture. The singularit...

conjecture | open | MathDB 页面 | 原文线索

V2-024588 | MATHDB-331196

Mirzakhani's arithmeticity conjecture for higher-rank affine invariant submanifolds

摘录 / EXCERPT: Mirzakhani's arithmeticity conjecture. Higher-rank affine invariant submanifolds are arithmetic.

conjecture | open | MathDB 页面 | 原文线索

V2-024589 | MATHDB-331197

Mirzakhani's covering conjecture for higher-rank arithmetic affine invariant submanifolds

摘录 / EXCERPT: Mirzakhani's covering conjecture. Higher-rank arithmetic affine invariant submanifolds are either connected components of strata or branched covering constructions.

conjecture | open | MathDB 页面 | 原文线索

V2-024590 | MATHDB-331252

Conjecture on boundary signs of Neumann eigenfunctions on Platonic solids

摘录 / EXCERPT: Let u be a Neumann eigenfunction on a Platonic solid. Boundary-sign conjecture. Neumann eigenfunctions that are nonnegative on the boundary of a tetrahedron or octahedron do not...

conjecture | open | MathDB 页面 | 原文线索

V2-024591 | MATHDB-331253

Conjecture on nodal lines of second Neumann eigenfunctions on triangles

摘录 / EXCERPT: Let D be a non-equilateral triangle, and let u be a second Neumann eigenfunction on D. Call D superequilateral if it is isosceles with aperture angle greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-024592 | MATHDB-331263

Rationality conjecture for $[200~\mathbb{Q}$-factorial nodal quartic double solids

摘录 / EXCERPT: Rationality conjecture for $\mathbb{Q}$-factorial nodal quartic double solids. Every $\mathbb{Q}$-factorial nodal quartic double solid is irrational.

conjecture | open | MathDB 页面 | 原文线索

V2-024593 | MATHDB-331264

Shokurov's irrationality conjecture for standard conic bundles

摘录 / EXCERPT: Shokurov's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-024594 | MATHDB-331326

The coincidence conjecture for symplectic capacities on convex bodies

摘录 / EXCERPT: The convex-body coincidence conjecture. For any convex body K in ${\mathbb R}^{2n}$,

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V2-024595 | MATHDB-331353

The minimal-generation conjecture for rigid multiview ideals

摘录 / EXCERPT: Minimal-generation conjecture. The rigid multiview ideal J_A is minimally generated by

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V2-024596 | MATHDB-331358

Cartwright's Picard-group conjecture for products of graphs

摘录 / EXCERPT: Cartwright's Picard-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-024597 | MATHDB-331372

Conjecture on finite-covolume actions of discrete groups on hyperbolic isometry groups

摘录 / EXCERPT: Let $n\geq 2$. Let Γ be a discrete group and let $j\times\rho$ be a faithful representation of Γ into $\operatorname{SO}_0(n,1)\times\operatorname{SO}_0(n,1)$. Suppos...

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V2-024598 | MATHDB-331381

The classification conjecture for immersions with the central cross-cut property

摘录 / EXCERPT: Classification conjecture. Every complete immersion $F\colon M^{n}\to\mathbf{R}^{n+1}$ with the central cross-cut property must be either a central cylinder or a tubular quadric.

conjecture | open | MathDB 页面 | 原文线索

V2-024599 | MATHDB-331397

Higher-order tangency bound for algebraic curves

摘录 / EXCERPT: Let k be a field, let $1\leq s<\operatorname{char}(k)$ (with only $s\geq 1$ required when $\operatorname{char}(k)=0$), and let $\mathcal{L}$ be a set of irreducible algebraic cur...

conjecture | open | MathDB 页面 | 原文线索

V2-024600 | MATHDB-331398

Higher-dimensional tangency bound for hypersurfaces

摘录 / EXCERPT: Let k be a field and let $\mathcal{S}$ be a set of irreducible hypersurfaces in k^d. A directed point of tangency is a pair (p,H), where $p\in k^d$ and H is a hyperplane co...

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V2-024601 | MATHDB-331401

The norm conjecture for entropy of projections of symmetric log-concave vectors

摘录 / EXCERPT: The norm conjecture. The function N_X defines a norm on $\mathbb{R}^n$. This would give a geometric formulation of entropy inequalities for one-dimensional projections of symmetr...

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V2-024602 | MATHDB-331413

Böröczky-Lutwak-Yang-Zhang log-Brunn-Minkowski inequality

摘录 / EXCERPT: Let K,L be symmetric convex bodies in $\mathbb R^n$. For $\lambda\theta \in [0,1]$, define their logarithmic Minkowski combination by ... where h_K and h_L are support function...

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V2-024603 | MATHDB-331414

Measure-valued extension of the log-Brunn-Minkowski inequality

摘录 / EXCERPT: Let $p \in [0,1]$. Let μ be a symmetric measure in $\mathbb R^n$ with an α-concave density function, where $\alpha \geq -\frac{p}{n}$. Let $\oplus_p$ denote the p-Fire...

conjecture | open | MathDB 页面 | 原文线索

V2-024604 | MATHDB-331418

Normalized determinant bound for observer- and star-based maps

摘录 / EXCERPT: Normalized determinant conjecture. For any configuration $\mathbf{x}\in C_n(X)$,

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V2-024605 | MATHDB-331453

Gross-Popescu conjecture on generators of ideals of (1,d)-polarized abelian surfaces

摘录 / EXCERPT: Let X be an abelian surface with a polarization of type $(1,d)$, embedded by the associated polarization, and let its homogeneous ideal be the ideal defining this embedded surfac...

conjecture | open | MathDB 页面 | 原文线索

V2-024606 | MATHDB-331462

Intermediate intersection-volume inequality for convex bodies

摘录 / EXCERPT: Intermediate-section conjecture. For every $K\in\mathcal{K}_d$,

open_problem | open | MathDB 页面 | 原文线索

V2-024607 | MATHDB-331463

Translative equal-volume property and Euclidean constant width

摘录 / EXCERPT: Translative constant-width conjecture. If $d\geq3$ and some $K\in\mathcal{K}_d$ satisfies the translative equal volume property, then K is a convex body of constant width in a Eu...

conjecture | open | MathDB 页面 | 原文线索

V2-024608 | MATHDB-331466

Schmutz Schaller's systolic disk area conjecture

摘录 / EXCERPT: Let S_g be a closed hyperbolic surface of genus g, and let $D_{\operatorname{inj}(S_g)}$ denote an embedded metric disk of radius equal to the injectivity radius of S_g. Schm...

conjecture | open | MathDB 页面 | 原文线索

V2-024609 | MATHDB-331482

Square-root dimension dependence conjecture for discrete edge-isoperimetric stability

摘录 / EXCERPT: Let $S\subset\mathbb{Z}^d$ be finite and suppose ... Then the edge-isoperimetric stability conjecture. There exists a cube $C\subset\mathbb{Z}^d$ such that ... where c' is an absolut...

conjecture | open | MathDB 页面 | 原文线索

V2-024610 | MATHDB-331512

Donaldson's long-time asymptotics conjecture for the Calabi flow

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold, let $\mathcal H_\omega$ denote the space of Kähler potentials in the class $[\omega]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024611 | MATHDB-331518

Formality-independence of the quasi-classical limit of a DQ-algebroid

摘录 / EXCERPT: Formality-independence claim. The quasi-classical limit $\widetilde{\operatorname{gr}^\times}\mathcal{C}$ is equivalent to $\operatorname{gr}^\times\mathcal{C}$ equipped with the...

conjecture | open | MathDB 页面 | 原文线索

V2-024612 | MATHDB-331526

Conjecture on derived invariance of derived Jacobians as derived Jacobians

摘录 / EXCERPT: Derived-Jacobian isogeny conjecture. If X and Y are derived equivalent, then $Pic^0(X)_{dg}$ and $Pic^0(Y)_{dg}$ are isogenous as derived Jacobians.

conjecture | open | MathDB 页面 | 原文线索

V2-024613 | MATHDB-331546

The equality of the mixed and skew-symmetric curvature-tensor bounds

摘录 / EXCERPT: Let R be a canonical algebraic curvature tensor, and let $\mu(R)$ denote the minimum number of canonical curvature tensors arising from either symmetric or anti-symmetric operato...

conjecture | open | MathDB 页面 | 原文线索

V2-024614 | MATHDB-331564

The polar form of the Beta Conjecture

摘录 / EXCERPT: Let $f:(\mathcal{U},\mathbf{0})\to(\mathbb{C},0)$ be a complex analytic function with one-dimensional critical locus at the origin, choose a generic linear coordinate z_0, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-024615 | MATHDB-331590

Soliton resolution for equivariant wave maps from hyperbolic space

摘录 / EXCERPT: Soliton resolution conjecture. The Cauchy problem is globally well-posed, and its solution scatters to P_λ as $t\to\pm\infty$. The harmonic maps P_λ minimize the en...

conjecture | open | MathDB 页面 | 原文线索

V2-024616 | MATHDB-331591

Global asymptotic stability of explicit hyperbolic solitons

摘录 / EXCERPT: Non-equivariant soliton resolution conjecture. The unique wave-map evolution associated with (u_0,u_1) is globally regular and scatters to $(\mathcal{P}_\lambda,0)$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-024617 | MATHDB-331603

Lu-Zhou conjecture on the L^p-essential spectrum

摘录 / EXCERPT: Let M be a complete, non-compact Riemannian manifold with Ricci curvature bounded below and uniformly sub-exponential volume growth. For $p\geq 1$, the L^p-essential spectrum i...

conjecture | open | MathDB 页面 | 原文线索

V2-024618 | MATHDB-331631

Equality of geodesic dimension and curvature exponent for Carnot groups

摘录 / EXCERPT: Let (X,d,μ) be a Carnot group with negligible cut loci. The geodesic-dimension conjecture. The geodesic dimension coincides with the curvature exponent. The equality would clar...

conjecture | open | MathDB 页面 | 原文线索

V2-024619 | MATHDB-331657

The uniruledness conjecture for compact Kähler manifolds

摘录 / EXCERPT: Uniruledness conjecture. X is uniruled, meaning covered by rational curves, if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-024620 | MATHDB-331658

The rational connectedness conjecture for compact Kähler manifolds

摘录 / EXCERPT: Rational connectedness conjecture. X is rationally connected, meaning that any two generic points are joined by some rational curve, if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-024621 | MATHDB-331695

Riemann-Roch conjecture for tropical surfaces

摘录 / EXCERPT: Let Δ be a tropical surface, let D be a Cartier divisor on Δ, and let K_Δ be the divisor obtained by summing $(\operatorname{deg} r-2)[r]$ over the ridges...

conjecture | open | MathDB 页面 | 原文线索

V2-024622 | MATHDB-331704

Small-radius convexity conjecture for Cassinian balls in the upper half-plane

摘录 / EXCERPT: Let $\mathbb{H}^2=\{z\in\mathbb{C}:\operatorname{Im}z>0\}$, let $\delta_{\mathbb{H}^2}(x)$ denote the Euclidean distance from x to the boundary, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024623 | MATHDB-331705

Starlikeness conjecture for Cassinian balls in starlike domains

摘录 / EXCERPT: Let $G\subset\mathbb{R}^n$ be a domain that is starlike with respect to $x\in G$, and let $B_{c_G}(x,r)$ denote the Cassinian ball of radius r. Starlikeness conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-024624 | MATHDB-331706

Sharp Cassinian-to-j-metric ball inclusion conjecture

摘录 / EXCERPT: Let $D\subsetneq\mathbb{R}^n$ be a proper subdomain, let $x\in D$, and let $\delta_D(x)$ be the Euclidean distance from x to ∂D. For a metric m, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024625 | MATHDB-331707

Sharp Cassinian-to-quasihyperbolic ball inclusion conjecture in a punctured space

摘录 / EXCERPT: Let $a\in\mathbb{R}^n$, let $D=\mathbb{R}^n\setminus\{a\}$, let $x\in D$, and let $B_k(x,s)$ and $B_c(x,s)$ denote quasihyperbolic and Cassinian metric balls, respectively. Sharp C...

conjecture | open | MathDB 页面 | 原文线索

V2-024626 | MATHDB-331733

The Coolidge-Nagata conjecture for rational cuspidal curves

摘录 / EXCERPT: A rational cuspidal curve is a rational plane curve all of whose singularities are cusps. A curve is rectifiable if it can be transformed into a line by successive Cremona transfor...

conjecture | open | MathDB 页面 | 原文线索

V2-024627 | MATHDB-331738

Beauville's finiteness conjecture for deformation types of irreducible hyperkähler manifolds

摘录 / EXCERPT: Beauville's finiteness conjecture. The number of deformation types of compact irreducible hyperkähler manifolds is finite in any dimension, at least for a fixed value of b_2.

conjecture | open | MathDB 页面 | 原文线索

V2-024628 | MATHDB-331755

Rank conjecture for the lifted inclusion matrices of arcs

摘录 / EXCERPT: Let $\mathbb{F}_{q}$ be a finite field with characteristic p, let $G\subset\mathbb{F}_{q}^{k}$ be an arc, and let $M_{G}^{\uparrow n}$ denote the matrix defined in the paper from...

conjecture | open | MathDB 页面 | 原文线索

V2-024629 | MATHDB-331756

Uniqueness conjecture for the normal rational curve

摘录 / EXCERPT: Let $\mathbb{F}_{q}$ be a finite field with characteristic p. For an arc $G\subset\mathbb{F}_{q}^{k}$ of size $2k$, let $H_{G}^{\uparrow 2}$ be the matrix defined in the paper, a...

conjecture | open | MathDB 页面 | 原文线索

V2-024630 | MATHDB-331757

Linear coloring conjecture for touching sets of strings

摘录 / EXCERPT: A touching set of strings is a finite family of strings in the plane in which no pair crosses, and it is k-touching if every point of the plane belongs to at most k strings. Li...

conjecture | open | MathDB 页面 | 原文线索

V2-024631 | MATHDB-331758

Additive coloring conjecture for 1-intersecting touching strings

摘录 / EXCERPT: A 1-intersecting set of strings is a set of strings in which any two strings intersect in at most one point, and it is k-touching if every point of the plane belongs to at most...

conjecture | open | MathDB 页面 | 原文线索

V2-024632 | MATHDB-331769

Generalized abundance conjecture for klt pairs

摘录 / EXCERPT: Generalized abundance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-024633 | MATHDB-331770

Extension conjecture for dlt pairs

摘录 / EXCERPT: Extension conjecture for dlt pairs. The restriction map

conjecture | open | MathDB 页面 | 原文线索

V2-024634 | MATHDB-331793

The Cremona equivalence conjecture for almost-grid-free hypersurfaces

摘录 / EXCERPT: Let H be an irreducible hypersurface in $\mathbb{P}^s\times\mathbb{P}^s$. A set $V\subset\mathbb{P}^s\times\mathbb{P}^s$ is almost-(s,t)-grid-free if there are nonempty Zariski...

conjecture | open | MathDB 页面 | 原文线索

V2-024635 | MATHDB-331794

The biregular equivalence conjecture for grid-free hypersurface restrictions

摘录 / EXCERPT: Let H be an irreducible hypersurface in $\mathbb{P}^s\times\mathbb{P}^s$, and let X,Y be nonempty Zariski-open subsets of $\mathbb{P}^s$. A hypersurface restriction is (s,t)-...

conjecture | open | MathDB 页面 | 原文线索

V2-024636 | MATHDB-331795

The affine automorphism conjecture for grid-free hypersurfaces

摘录 / EXCERPT: Let H be an irreducible affine hypersurface in $\mathbb{A}^s\times\mathbb{A}^s$. It is (s,t)-grid-free when the associated bipartite algebraic graph contains no complete bipart...

conjecture | open | MathDB 页面 | 原文线索

V2-024637 | MATHDB-331803

Symplectic surface representability conjecture for symplectic 4-manifolds of Kodaira dimension minus infinity

摘录 / EXCERPT: Symplectic surface representability conjecture. The class A is represented by a connected $\delta0omega$-symplectic surface if and only if $[\delta0omega]\delta0cdot A>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-024638 | MATHDB-331804

Symplectic Kodaira dimension zero classification conjecture

摘录 / EXCERPT: Let $(M,\domega)$ be a minimal symplectic 4-manifold with $\delta0kappa^s(M,\domega)=0$. Symplectic Kodaira dimension zero classification conjecture. The manifold M is diffeomo...

conjecture | open | MathDB 页面 | 原文线索

V2-024639 | MATHDB-331805

Bogomolov-Miyaoka-Yau conjecture for symplectic Kodaira dimension two

摘录 / EXCERPT: Let $(M,\domega)$ be a minimal symplectic 4-manifold with symplectic Kodaira dimension $\delta0kappa^s(M)=2$. Bogomolov-Miyaoka-Yau conjecture. The symplectic canonical class sat...

conjecture | open | MathDB 页面 | 原文线索

V2-024640 | MATHDB-331853

Chen's conjecture on global existence of the Calabi flow

摘录 / EXCERPT: Chen's conjecture. The Calabi flow exists for all time.

conjecture | open | MathDB 页面 | 原文线索

V2-024641 | MATHDB-331854

Donaldson's convergence conjecture for the Calabi flow

摘录 / EXCERPT: Donaldson's conjecture. If the Calabi flow exists for all time and there exists a cscK metric in the Kähler class, then the Calabi flow converges to a cscK metric.

conjecture | open | MathDB 页面 | 原文线索

V2-024642 | MATHDB-331855

Uniqueness conjecture for TEP-structures over generically semisimple F-manifolds

摘录 / EXCERPT: Let $((M,0),\circ,e,E)$ be an irreducible germ of a generically semisimple F-manifold, meaning that the multiplication is semisimple at generic points, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-024643 | MATHDB-331887

The connectedness conjecture for the essential spectrum of the Laplacian

摘录 / EXCERPT: Connectedness conjecture. The essential spectrum of $-\triangle_g$ on functions is a connected subset of the positive real line $[a,\infty)$.

conjecture | open | MathDB 页面 | 原文线索

V2-024644 | MATHDB-331921

Existence of XY-manifolds with non-constant holomorphic sectional curvature

摘录 / EXCERPT: Existence conjecture. There exist XY-manifolds with non-constant holomorphic sectional curvature. The source explains that manifolds with constant holomorphic sectional curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-024645 | MATHDB-331934

Universal quantization conjecture for smooth symplectic resolutions of Lusztig slices

摘录 / EXCERPT: Let $\widetilde{\mathop{\mathrm{Gr}}}{}^\lambda_\mu$ be a smooth symplectic resolution equipped with a conic $\mathbb C^*$-action, and let $Y^\lambda_\mu(\mathbf R)$ be the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-024646 | MATHDB-331942

Fibration conjecture for compact Clifford-Klein forms

摘录 / EXCERPT: Fibration conjecture. Under these hypotheses, such a manifold B and fibration exist. This generalizes a theorem of Guérithaud-Kassel for certain quotients of...

conjecture | open | MathDB 页面 | 原文线索

V2-024647 | MATHDB-332000

Discrete bilinear-form values conjecture

摘录 / EXCERPT: Let F be the real or complex field, let ω be a non-degenerate symmetric or skew-symmetric bilinear form on F^2, and let $P\subset F^2\setminus\{0\}$ be an N-element po...

conjecture | open | MathDB 页面 | 原文线索

V2-024648 | MATHDB-332067

Existence of flips for pl-flipping contractions with universally homeomorphic boundary normalization

摘录 / EXCERPT: Flip existence conjecture. A flip of f exists.

conjecture | open | MathDB 页面 | 原文线索

V2-024649 | MATHDB-332069

Riemannian-curvature spectral bound conjecture for the complex Laplacian

摘录 / EXCERPT: Let (M^n,h) be a compact complex manifold, and let $\square$ denote its complex Laplacian. The Riemannian-curvature spectral bound conjecture asserts that there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-024650 | MATHDB-332070

Riemannian-curvature bound conjecture for Hermitian torsion

摘录 / EXCERPT: Let (M^n,h) be a compact complex manifold, let η be Gauduchon's torsion one-form, and let $\|\eta\|$ denote its pointwise norm. The Riemannian-curvature torsion-bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-024651 | MATHDB-332072

Rigidity conjecture for nonnegative Q-curvature on closed flat manifolds

摘录 / EXCERPT: Let $n\geq 3$, and let $(M^n,\bar{g})$ be a connected closed flat Riemannian manifold. A metric has pointwise positive Q-curvature if its Q-curvature is positive at every point. Ri...

conjecture | open | MathDB 页面 | 原文线索

V2-024652 | MATHDB-332080

Dubrovin-Zhang identification with the universal Whitham hierarchy

摘录 / EXCERPT: Let M be a semisimple Frobenius manifold, let its Dubrovin-Zhang integrable hierarchy be the hierarchy associated with M, and let the spectral curve of M be constructed using...

conjecture | open | MathDB 页面 | 原文线索

V2-024653 | MATHDB-332081

Whitham tau-function conjecture for toric Calabi-Yau brane partition functions

摘录 / EXCERPT: Let the space be a toric Calabi-Yau 3-manifold or 3-orbifold with Aganagic-Vafa outer D-branes, let its partition function be the corresponding brane partition function, and let it...

conjecture | open | MathDB 页面 | 原文线索

V2-024654 | MATHDB-332083

Fermionic reformulation of the Dubrovin-Zhang hierarchy

摘录 / EXCERPT: Let M be a semisimple Frobenius manifold, and let its Dubrovin-Zhang integrable hierarchy be the associated hierarchy. Dubrovin-Zhang fermionic reformulation conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-024655 | MATHDB-332086

Complex polynomial partitioning conjecture

摘录 / EXCERPT: Let $\mathcal{P}\subset\mathbb{C}^d$ be a set of n points, and let $\iota\colon\mathbb{C}^d\to\mathbb{R}^{2d}$ be the usual identification. For a complex polynomial P, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024656 | MATHDB-332110

Donaldson's K-polystability conjecture for conformally Kähler Einstein-Maxwell metrics

摘录 / EXCERPT: Donaldson's conjecture. There exists a solution of the modified Abreu equation in $\mathcal{S}(\Delta, {\bf L})$ if and only if $(\Delta, {\bf L}, f)$ is K-polystable.

conjecture | open | MathDB 页面 | 原文线索

V2-024657 | MATHDB-332121

Tohaneanu-Van Tuyl degree-one minimum-distance conjecture for complete intersections

摘录 / EXCERPT: Let $\mathbb{X}$ be a finite set of points in $\mathbb{P}^{s-1}$, and let $I(\mathbb{X})$ be a complete intersection generated by $f_1,\ldots,f_{s-1}$. Set $e_i=\deg(f_i)$ and assu...

conjecture | open | MathDB 页面 | 原文线索

V2-024658 | MATHDB-332130

Infinitely many families of Sasaki-Einstein metrics on odd-dimensional spheres

摘录 / EXCERPT: Let S^{2n+1} denote the $(2n+1)$-dimensional sphere, with n a positive integer. Sasaki-Einstein metric abundance conjecture. There are infinitely many families of Sasaki-Einste...

conjecture | open | MathDB 页面 | 原文线索

V2-024659 | MATHDB-332131

Conjecture on the covering radius of random points on the sphere

摘录 / EXCERPT: Covering-radius conjecture. The expected covering radius satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024660 | MATHDB-332153

The weighted orthogonality Poincaré inequality conjecture

摘录 / EXCERPT: Weighted orthogonality Poincaré conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024661 | MATHDB-332165

Classification conjecture for rigid two-geodesic square-torus packings

摘录 / EXCERPT: Let $N\geq 6$ and consider a rigid packing of N equal disks in a square torus. Assume that its packing graph consists of two linear geodesics, and let M be its Markov constant....

conjecture | open | MathDB 页面 | 原文线索

V2-024662 | MATHDB-332175

Costability conjecture for deformations of holomorphic Poisson maps

摘录 / EXCERPT: Let (X,Λ_0) and (Y,Π_0) be compact holomorphic Poisson manifolds, let $f:(X,\Lambda_0)\to(Y,\Pi_0)$ be a holomorphic Poisson map, and let ... be a Poisson analytic family...

conjecture | open | MathDB 页面 | 原文线索

V2-024663 | MATHDB-332193

Hyperbolic analogue of the Bezdek-Connelly-Kertész average-degree conjecture

摘录 / EXCERPT: Let ${\mathcal P}_r$ be a packing of disks of radius $r>0$ in the hyperbolic plane $\mathbb{H}^{2}$, and let ${\rm deg}_{\rm avr}({\mathcal P}_r)$ denote the average degree of its...

conjecture | open | MathDB 页面 | 原文线索

V2-024664 | MATHDB-332201

Crepant categorical-resolution conjecture for Clifford double mirrors

摘录 / EXCERPT: Let the Clifford double mirror of Definition be the Clifford construction associated with the two quartics $\mathbb{Q}_+$ and $\mathbb{Q}_-$ in $\mathbb{C}\mathbb{P}^3$. The corres...

conjecture | open | MathDB 页面 | 原文线索

V2-024665 | MATHDB-332202

Clifford-limit equivalence conjecture for toric complete intersections

摘录 / EXCERPT: Let $(\mathcal{S},\mathcal{B}_0)$ be the Clifford-limit object defined using the central even Clifford algebra sheaf, and let $D_B(K,c,\Sigma)$ denote the category associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-024666 | MATHDB-332210

The abundance conjecture for nef line bundles on Calabi-Yau manifolds

摘录 / EXCERPT: Abundance conjecture. $mathcal L$ is semiample, i.e. there exists a positive integer m such that $mathcal L^{\otimes m}$ is generated by global sections.

conjecture | open | MathDB 页面 | 原文线索

V2-024667 | MATHDB-332225

RRP codes exceed the LMRD bound outside the exceptional packet lengths

摘录 / EXCERPT: RRP bound conjecture. For any packet length $v\geq 7$ with $v\notin\{8,10\}$, the largest subspace codes that can be obtained by solving the RRP exceed the LMRD code bound and thus...

conjecture | open | MathDB 页面 | 原文线索

V2-024668 | MATHDB-332226

Shahidi's generic cuspidal representation conjecture for symplectic Arthur packets

摘录 / EXCERPT: Let $\phi=\boxplus_{i=1}^r(\tau_i,1)\in\widetilde{\Phi}_2({\mathrm{Sp}}_{2n})$ be a generic global Arthur parameter. Here $\widetilde{\Pi}_{\phi}({\mathrm{Sp}}_{2n})$ denotes its g...

conjecture | open | MathDB 页面 | 原文线索

V2-024669 | MATHDB-332233

Product decomposition conjecture for densities on the torus

摘录 / EXCERPT: Product decomposition conjecture. Given $\mu\in\mathrm{Dens}({\mathbb T}^n)$, there exist $\eta\in\mathrm{Diff}_{\mathrm{d}x}({\mathbb T}^n)$ and functions...

conjecture | open | MathDB 页面 | 原文线索

V2-024670 | MATHDB-332239

Tropical Martens' conjecture for metric graphs

摘录 / EXCERPT: Let Γ be a metric graph of genus g. For integers d and r satisfying ... let $w^r_d(\Gamma)$ denote the Brill-Noether rank, namely the largest integer ρ such that ev...

conjecture | open | MathDB 页面 | 原文线索

V2-024671 | MATHDB-332240

L. Ni's existence conjecture for polynomial-growth harmonic functions

摘录 / EXCERPT: Let (M^n,g) be a complete Riemannian manifold of positive sectional curvature. A manifold has maximum volume growth when its volume growth is maximal in the relevant comparison s...

conjecture | open | MathDB 页面 | 原文线索

V2-024672 | MATHDB-332249

The general flop formula for Donaldson-Thomas invariants

摘录 / EXCERPT: Let f be a flop between nonsingular projective 3-folds, and let ... denote the corresponding flop identities for the Donaldson-Thomas invariants and their contributions from the...

conjecture | open | MathDB 页面 | 原文线索

V2-024673 | MATHDB-332250

The MNOP primary Gromov-Witten/Donaldson-Thomas correspondence

摘录 / EXCERPT: Let X be a nonsingular projective 3-fold, let $\gamma_1,\ldots,\gamma_m\in H^*(X,\mathbb{Q})$ with $m\geqslant0$, and let $\beta\in H_2(X,\mathbb{Z})$. Write Z'_{GW} and...

conjecture | open | MathDB 页面 | 原文线索

V2-024674 | MATHDB-332251

The MNOP point-descendant Gromov-Witten/Donaldson-Thomas correspondence

摘录 / EXCERPT: Let X be a nonsingular projective 3-fold, let P be the class of a point in X, let $\gamma_1,\ldots,\gamma_m\in H^{>0}(X,\mathbb{Q})$ with $m\geqslant0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-024675 | MATHDB-332275

Conjecture on the classification of Cohen-Macaulay order-one congruences

摘录 / EXCERPT: The classification conjecture. The congruences listed in the theorem and the proposition form the exhaustive list of Cohen-Macaulay, order 1, irreducible congruences with smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-024676 | MATHDB-332308

Unipotent-orbit conjecture for theta representations of metaplectic symplectic groups

摘录 / EXCERPT: Let r be odd and let $r<2n$. Write $2n=a(n,r)r+b(n,r)$ with $a(n,r)$ and $b(n,r)$ nonnegative and $0\leq b(n,r)\leq r-1$. For a partition of $2n$, let its Sp collapse be the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-024677 | MATHDB-332309

Equality conjecture for the residual and theta representations of metaplectic symplectic groups

摘录 / EXCERPT: Let r be odd with $r<2n$, and let $\sigma_{2n-r+1}^{(2r)}$ and $\Theta_{2n-r+1}^{(2r)}$ denote the representations constructed in the paper on the metaplectic symplectic group of...

conjecture | open | MathDB 页面 | 原文线索

V2-024678 | MATHDB-332316

Perfect gauge fixing by superconformal selection rules and unitarity

摘录 / EXCERPT: A hyperkähler cone is a space whose deformation quantization is constrained by superconformal selection rules and unitarity conditions. Perfect gauge-fixing conjecture. Whenever th...

conjecture | open | MathDB 页面 | 原文线索

V2-024679 | MATHDB-332336

The expansion conjecture for polynomials on curves

摘录 / EXCERPT: Let $C\subset\mathbb C^2$ be an algebraic curve of degree at most d, let $F:C\times C\to\mathbb C$ be a polynomial of degree at most d, and let $P\subset C$. The curve expansio...

conjecture | open | MathDB 页面 | 原文线索

V2-024680 | MATHDB-332337

Elekes's conjecture on collinear triples

摘录 / EXCERPT: Let $P\subset\mathbb R^2$ be a finite point set of size n. Elekes's collinear-triples conjecture. If P determines $\Omega(n^2)$ collinear triples, then at least ten points of...

conjecture | open | MathDB 页面 | 原文线索

V2-024681 | MATHDB-332338

The higher-dimensional Elekes-Szabó conjecture

摘录 / EXCERPT: Let $D\geq 1$, let $F\in\mathbb C[x_1,\ldots,x_D]$ be irreducible of degree d, and assume each partial derivative $F_{x_1},\ldots,F_{x_D}$ is not identically zero. The higher-dim...

conjecture | open | MathDB 页面 | 原文线索

V2-024682 | MATHDB-332360

Universal eigenvalue ratio conjecture for weighted compact Alexandrov spaces

摘录 / EXCERPT: Let (X,μ) be a weighted compact finite-dimensional Alexandrov space of $\operatorname{CD}(0,\infty)$, and let k be a natural number. Eigenvalue ratio conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-024683 | MATHDB-332361

Tohăneanu-Van Tuyl conjecture on the minimum distance of complete intersections

摘录 / EXCERPT: Let $\mathbb{X}$ be a finite set of points in $\mathbb{P}^{s-1}$, and let $I(\mathbb{X})$ be its vanishing ideal. Suppose that $I(\mathbb{X})$ is a complete intersection generated...

conjecture | open | MathDB 页面 | 原文线索

V2-024684 | MATHDB-332404

Strong Wilhelm's conjecture for positively curved Riemannian foliations

摘录 / EXCERPT: Let M be a compact manifold with positive curvature, and let $\cal F$ be a Riemannian foliation on M. A horizontal vector is a vector tangent to the horizontal distribution of...

conjecture | open | MathDB 页面 | 原文线索

V2-024685 | MATHDB-332410

Weak same-level reducibility conjecture for involutions

摘录 / EXCERPT: Let $m\geq 2$ be even, let $R\in SO(p)$ be an involution, and suppose that R has level m. A sign-change matrix is a matrix in $\mathcal I_p^+$, and R is sign-change reducible...

conjecture | open | MathDB 页面 | 原文线索

V2-024686 | MATHDB-332411

The Kähler-like and G-Kähler-like curvature conjecture

摘录 / EXCERPT: Let (M^n,g) be a Hermitian manifold. The metric g is Kähler-like if its holomorphic curvature tensor R^h has the curvature symmetries of a Kähler metric, and G-Kähler-like if...

conjecture | open | MathDB 页面 | 原文线索

V2-024687 | MATHDB-332443

Regularity conjecture for minimizers of the extended K-energy

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold, let $(\mathcal E^1,d_1)$ be the metric completion of the space of Kähler potentials, and let the extended Mabuchi K-energy be a map...

conjecture | open | MathDB 页面 | 原文线索

V2-024688 | MATHDB-332444

Cyclic pursuit convergence conjecture on compact Riemannian manifolds

摘录 / EXCERPT: Let M be a compact Riemannian manifold. For cyclic pursuit curves $b_i(t)$, let β^t be the piecewise geodesic loop obtained by concatenating the shortest geodesics between...

conjecture | open | MathDB 页面 | 原文线索

V2-024689 | MATHDB-332445

Angle convergence conjecture for cyclic pursuit on compact manifolds

摘录 / EXCERPT: Let M be a compact manifold, let $\{b_i(t)\}_{i\in \mathbb{Z}/n}$ be pursuit curves, and let $l_i(t)$ and $\theta_i(t)$ denote the associated edge lengths and angles. Angle conve...

conjecture | open | MathDB 页面 | 原文线索

V2-024690 | MATHDB-332447

Convex-hull conjecture for almost toric fibrations

摘录 / EXCERPT: Let M be a symplectic manifold with an almost toric fibration (ATF), and let an almost toric base diagram (ATBD) describe it. Assume the cuts of the ATBD lie inside the eigenline...

conjecture | open | MathDB 页面 | 原文线索

V2-024691 | MATHDB-332448

Distinct orbifold limits give nonsymplectomorphic monotone Lagrangian fibers

摘录 / EXCERPT: Let two monotone Lagrangian fibers of almost toric fibrations lie in the same symplectic manifold, and suppose their almost toric base diagrams have different orbifold limits. The...

conjecture | open | MathDB 页面 | 原文线索

V2-024692 | MATHDB-332449

Equality of the tight hyperbolic and hyperbolic minimal discrepancy densities

摘录 / EXCERPT: Let $\rho_{\mathbb{H}}$ be the minimal discrepancy density for hyperbolic zero packing, and let $\rho_{\mathbb{H}}^\ast$ be the minimal discrepancy density for tight hyperbolic zer...

conjecture | open | MathDB 页面 | 原文线索

V2-024693 | MATHDB-332451

Strong-field convergence of hyperbolic to planar beta-exponent packing density

摘录 / EXCERPT: For positive real parameters α and β, let $\rho_{\alpha,\beta}(\mathbb{H})$ be the minimal discrepancy density for hyperbolic zero packing with field strength α...

conjecture | open | MathDB 页面 | 原文线索

V2-024694 | MATHDB-332459

Torus-relative stability conjecture for constant scalar curvature polarised manifolds

摘录 / EXCERPT: Torus-relative stability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-024695 | MATHDB-332466

Boucksom-Demailly-P5un-Peternell duality conjecture for compact K4hler manifolds

摘录 / EXCERPT: Let X be a compact K4hler manifold of complex dimension n. The cone $\mathcal{E}$ of pseudoeffective classes consists of classes in $H^{1,1}_{\mathbb{R}}(X)$ containing a close...

conjecture | open | MathDB 页面 | 原文线索

V2-024696 | MATHDB-332486

Degree 3 dilation conjecture for the infinite square lattice

摘录 / EXCERPT: Let Λ be the infinite square lattice, and let $\delta_0(\Lambda,3)$ denote its minimum stretch factor among degree-3 spanners. Degree 3 dilation conjecture. ... The paper h...

conjecture | open | MathDB 页面 | 原文线索

V2-024697 | MATHDB-332505

Atiyah-Floer type conjectures for cyclic Cerf decompositions

摘录 / EXCERPT: Let $[Y]=[Y_{01}]\circ\cdots\circ[Y_{(k-1)k}]$ be a Cerf decomposition of a closed 3-manifold Y into simple cobordisms $Y_{i(i+1)}$, and let $L_{Y_{i(i+1)}}$ denote the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-024698 | MATHDB-332528

The separation conjecture for convex bodies

摘录 / EXCERPT: Let $\mathbf{K}$ be a convex body in $\mathbb{E}^d$, $d\geq 3$, and let $\mathbf{o}$ be an interior point. A face is the intersection of $\mathbf{K}$ with a supporting hyperplane....

conjecture | open | MathDB 页面 | 原文线索

V2-024699 | MATHDB-332529

The constant-width illumination conjecture in three dimensions

摘录 / EXCERPT: Let $\mathbf{K}$ be a three-dimensional convex body of constant width. Constant-width illumination conjecture. The illumination number satisfies $I(\mathbf{K})=4$. The known genera...

conjecture | open | MathDB 页面 | 原文线索

V2-024700 | MATHDB-332531

Boltyanski's Helly-dimension illumination bound

摘录 / EXCERPT: Let $\mathbf{K}$ be a convex body in $\mathbb{E}^d$, $d\geq 3$, and let $\operatorname{him}(\mathbf{K})$ be its Helly dimension, defined as the least h for which the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-024701 | MATHDB-332532

Bezdek-Zamfirescu's X-ray conjecture

摘录 / EXCERPT: Let $X(\mathbf{K})$ be the smallest number of lines such that every boundary point of a convex body $\mathbf{K}$ is X-rayed along at least one of them. X-ray conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-024702 | MATHDB-332533

Zong's blocking-number conjecture

摘录 / EXCERPT: Let $\beta(\mathbf{K})$ denote the blocking number of a d-dimensional convex body $\mathbf{K}$. Zong's blocking-number conjecture. For every d-dimensional convex body...

conjecture | open | MathDB 页面 | 原文线索

V2-024703 | MATHDB-332535

The ellipsoid conjecture for cylindrical covering parameters

摘录 / EXCERPT: Let $\mathbf{K}$ be an ellipsoid in $\mathbb{E}^d$, and let $\operatorname{cyl}_k(\mathbf{K})$ denote its k-th cylindrical covering parameter. Ellipsoid conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-024704 | MATHDB-332598

Minkowski's twin-cube conjecture for lattice cube tilings

摘录 / EXCERPT: Let a lattice tiling of $\mathbb{R}^{n}$ by unit cubes be a tiling whose translation set is a lattice. Two cubes are twins when they share a whole $(n-1)$-dimensional face. Minkows...

conjecture | open | MathDB 页面 | 原文线索

V2-024705 | MATHDB-332608

Bounds for derivatives of hyperbolic type distances

摘录 / EXCERPT: Let G be a starlike domain containing the origin, let $d_G(0)$ denote the Euclidean distance from the origin to the boundary of G, and let m be a hyperbolic type distance wit...

conjecture | open | MathDB 页面 | 原文线索

V2-024706 | MATHDB-332618

Polynomial stabilization conjecture for iterative convex-hull construction

摘录 / EXCERPT: Polynomial stabilization conjecture. The iterative construction stabilizes after a polynomially bounded number of iterations:

conjecture | open | MathDB 页面 | 原文线索

V2-024707 | MATHDB-332650

Reduction of singularities to weakly reduced foliations

摘录 / EXCERPT: Let X be a Kähler manifold equipped with a foliation $\mathcal{F}$ by curves. A birational model of $(X,\mathcal{F})$ is obtained by replacing it through a birational modificatio...

conjecture | open | MathDB 页面 | 原文线索

V2-024708 | MATHDB-332651

Dyadic minimum conjecture for recurrence coefficients of Cantor-type measures

摘录 / EXCERPT: Let $\gamma=(\gamma_k)_{k=1}^\infty$ be the parameter sequence defining $K(\gamma)$, and let $\mu_{K(\gamma)}$ be its associated measure with recurrence coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-024709 | MATHDB-332652

Positive-measure and recurrence-coefficient conjecture for Cantor-type sets

摘录 / EXCERPT: Let $\gamma=(\gamma_k)_{k=1}^\infty$ be the parameter sequence defining $K(\gamma)$, set $\varepsilon_k:=1-4\gamma_k$ for $k\in\mathbb{N}$, and let $(a_n)_{n=1}^\infty$ be the recu...

conjecture | open | MathDB 页面 | 原文线索

V2-024710 | MATHDB-332654

Widom-factor characterization of Parreau-Widom Cantor-type sets

摘录 / EXCERPT: Let $K(\gamma)$ be the Cantor-type compact set associated with γ, let $\mu_{K(\gamma)}$ be its associated measure, and let $W_n^2(\mu_{K(\gamma)})$ and $W_n(K(\gamma))$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-024711 | MATHDB-332656

Zero-spacing conjecture for orthogonal polynomials on Cantor-type sets

摘录 / EXCERPT: Zero-spacing conjecture. For each $\gamma=(\gamma_k)_{k=1}^\infty$ with $\inf_k\gamma_k>0$, $(A_{2^k})_{k=1}^\infty$ is unbounded. If $s=2^k$ for some $k\in\mathbb{N}$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-024712 | MATHDB-332659

Conjecture on the Volume Conjecture for hyperbolic polyhedra

摘录 / EXCERPT: Volume conjecture for Kirillov-Reshetikhin invariants. Then

conjecture | open | MathDB 页面 | 原文线索

V2-024713 | MATHDB-332666

Baer-subplane covering conjecture

摘录 / EXCERPT: Baer-subplane covering conjecture. If $\mathcal{L}$ covers every point of Π_q outside Π', then $\mathcal{L}$ is a covering set; in particular, it also covers every point of...

conjecture | open | MathDB 页面 | 原文线索

V2-024714 | MATHDB-332669

The curvature-improved multilinear restriction conjecture

摘录 / EXCERPT: Let $S_i\subset\mathbb{R}^{n+1}$ be hypersurfaces, and let $\mathcal R^*(2\times\cdots\times 2\rightarrow p)$ denote the k-linear restriction estimate associated with the surface...

conjecture | open | MathDB 页面 | 原文线索

V2-024715 | MATHDB-332732

The positive mass conjecture for asymptotically Euclidean manifolds

摘录 / EXCERPT: Positive mass conjecture. Then $m(g)\geq 0$, with equality if and only if $(M,g)=(\mathbb{R}^n,g_E)$.

conjecture | open | MathDB 页面 | 原文线索

V2-024716 | MATHDB-332743

Finiteness conjecture for refined Hofer-Zehnder capacities of inessential manifolds

摘录 / EXCERPT: Finiteness conjecture. For every such Γ, the capacity $C_{HZ}^{\Gamma}(DM)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-024717 | MATHDB-332744

Systolic lower-bound conjecture for Hofer-Zehnder capacity

摘录 / EXCERPT: Systolic capacity conjecture. There is a constant D_n depending only on n such that

conjecture | open | MathDB 页面 | 原文线索

V2-024718 | MATHDB-332745

The weight-one vector conjecture for inclusion matrices of arcs

摘录 / EXCERPT: Weight-one vector conjecture. The matrix M_{n} has a vector of weight one in its column space.

conjecture | open | MathDB 页面 | 原文线索

V2-024719 | MATHDB-332760

Projective affine-analogue conjecture for homogeneous polynomial zeros

摘录 / EXCERPT: Let $m\geq1$ and $d\geq1$, let S_d be the vector space of homogeneous degree-d polynomials in ${\mathbb F}_q[x_0,\ldots,x_m]$, and let $H_r(d-1,m)$ denote the affine maximum de...

conjecture | open | MathDB 页面 | 原文线索

V2-024720 | MATHDB-332770

Positive mass theorem for the fractional Yamabe problem

摘录 / EXCERPT: Let $\gamma \in (0,1)$, $n > 2\gamma$, and let (X^{n+1},g^+) be Poincaré-Einstein. Suppose that $\Lambda^{\gamma}(M,[\hat{h}])>0$ and that either $(M^n,[\hat{h}])$ is locally con...

conjecture | open | MathDB 页面 | 原文线索

V2-024721 | MATHDB-332787

The super-neatness conjecture for finite projective planes

摘录 / EXCERPT: A finite projective plane is a 2-design whose parameters satisfy the projective-plane axioms; its incidence graph is the bipartite graph joining each point to the blocks containi...

conjecture | open | MathDB 页面 | 原文线索

V2-024722 | MATHDB-332793

Berry cancellation conjecture for Lipschitz-Killing curvatures

摘录 / EXCERPT: Let T be a Gaussian eigenfunction on a compact manifold $\mathbb M$. For $u\in\mathbb R$, define the excursion set ... For each Lipschitz-Killing curvature of $A_u(T,\mathbb M)$, c...

conjecture | open | MathDB 页面 | 原文线索

V2-024723 | MATHDB-332804

Conjecture on projective dimension of the exceptional line in an elliptic blowup

摘录 / EXCERPT: Exceptional-line projective-dimension conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-024724 | MATHDB-332808

Conjecture on the Smith classification of the pencil $f_{\alpha,\beta}$

摘录 / EXCERPT: Let $f_{\alpha,\beta}$ be the holomorphic Lefschetz pencil constructed from the elements α and β, and let $X_{\alpha,\beta}$ be its total space. Smith classification c...

conjecture | open | MathDB 页面 | 原文线索

V2-024725 | MATHDB-332810

Adiabatic limit conjecture for co-associative Kovalev-Lefschetz fibrations

摘录 / EXCERPT: Adiabatic limit conjecture. If there is a positive section h of ${\cal H}_{\chi}$ which avoids excess -2 classes, then for sufficiently large R there is a closed positive 3...

conjecture | open | MathDB 页面 | 原文线索

V2-024726 | MATHDB-332815

Infinite-dimensionality conjecture for Besse metrics on spherical 2-orbifolds

摘录 / EXCERPT: The infinite-dimensionality conjecture. If $-1\notin G$, then the moduli space of Besse metrics on S^2/G is infinite-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-024727 | MATHDB-332817

The genomic Q-tableaux upper-bound conjecture for symplectic Schubert constants

摘录 / EXCERPT: Let $Z={\rm LG}(n,2n)$ be the Lagrangian Grassmannian, and let $c_{\lambda,\mu}^{\nu}$ denote its Schubert structure constants for strict partitions λ, μ, and ν. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-024728 | MATHDB-332818

The genomic Q-tableaux dagger lower-bound conjecture for symplectic Schubert constants

摘录 / EXCERPT: Let $Z={\rm LG}(n,2n)$ be the Lagrangian Grassmannian, and let $c_{\lambda,\mu}^{\nu}$ denote its Schubert structure constants. Let $\mathtt{QBallot}_{\mu}(\nu/\lambda)$ be the set...

conjecture | open | MathDB 页面 | 原文线索

V2-024729 | MATHDB-332822

The Brill-Noether dimension conjecture for general k-gonal curves

摘录 / EXCERPT: Let g,k,r,d be nonnegative integers with $g-d+r>0$ and $2\leq k\leq \frac{g+3}{2}$. Let C be a general k-gonal curve of genus g over an algebraically closed field, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024730 | MATHDB-332839

Pandharipande-Farkas conjecture for principal rubber 1-log-canonical divisors

摘录 / EXCERPT: Fix a triplet $(g,n,\vec{\mathbf{c}})$ and write $\mathbf{m}_i:=\mathbf{c}_i-1$. In the strictly meromorphic case, let $\mathcal{D}^{*(1)}_{\mathrm{pr}}(g,n,\vec{\mathbf{c}})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-024731 | MATHDB-332851

The Gaussian minimal-energy asymptotics conjecture

摘录 / EXCERPT: Let $f(t)=e^{-\alpha t^2}$, with $0<\alpha<\pi e$. For configurations of density ρ in $\mathbb{R}^n$, let the minimal f-energy be the infimum of the corresponding Gaussian p...

conjecture | open | MathDB 页面 | 原文线索

V2-024732 | MATHDB-332868

Double Gaussian half-plane conjecture in the plane

摘录 / EXCERPT: Let $f(x,y)=e^{\psi(x,y)}$ be the normalized sum of two Gaussian densities with the same variance and different centers. The line connecting the two centers determines a perpendicu...

conjecture | open | MathDB 页面 | 原文线索

V2-024733 | MATHDB-332887

Holographic renormalization conjecture for bulk Laplacian eigenvalue variation

摘录 / EXCERPT: Let (X,G) be the bulk with boundary M, and let $\tilde{D}=-\Delta_G$ and $D=-\Delta_g$ be the bulk and boundary Laplace--Beltrami operators, respectively. Here λ is an...

conjecture | open | MathDB 页面 | 原文线索

V2-024734 | MATHDB-332888

The distinct perpendicular bisectors conjecture

摘录 / EXCERPT: Let P be a set of n points in the Euclidean plane, and let $\mathcal B$ be the set of distinct perpendicular bisectors determined by pairs of points of P. Distinct perpendicu...

conjecture | open | MathDB 页面 | 原文线索

V2-024735 | MATHDB-332889

The point-circle incidence bound conjecture

摘录 / EXCERPT: Let P be a set of n points in the Euclidean plane, and let c_k denote the maximum number of circles containing at least k points of P. Point-circle incidence bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-024736 | MATHDB-332890

The refined bisector energy bound conjecture

摘录 / EXCERPT: Refined bisector energy bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-024737 | MATHDB-332903

Characterization of the $CD(0,m)$-condition by monotonicity of W_m-entropy

摘录 / EXCERPT: $CD(0,m)$ characterization conjecture. Then the $CD(0,m)$-condition holds, that is,

conjecture | open | MathDB 页面 | 原文线索

V2-024738 | MATHDB-332905

Generalized Jørgensen inequality for purely loxodromic free Kleinian groups

摘录 / EXCERPT: Generalized Jørgensen conjecture. If α_n is the only real root of $p_n(x)$ greater than $(2n-1)^2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024739 | MATHDB-332906

Generalized Jørgensen trace inequality under displacement hypotheses

摘录 / EXCERPT: Let $\Gamma=\langle\xi_1,\xi_2,\dots,\xi_n\rangle$, and let α_n be as described in the generalized Jørgensen conjecture. For $i\ne1$, define ... Let z_1 and z_2 be the mid...

conjecture | open | MathDB 页面 | 原文线索

V2-024740 | MATHDB-332936

The super-polynomial lower-bound conjecture for type A1 semisimple Lie groups

摘录 / EXCERPT: Let H be a semisimple Lie group of type A_1 with more than one simple factor, and let $\mathrm{IL}_{H}(x)$ denote the maximal size of a family of pairwise isospectral,...

conjecture | open | MathDB 页面 | 原文线索

V2-024741 | MATHDB-332941

Ball-Nayar-Tkocz Busemann norm conjecture for Shannon entropy

摘录 / EXCERPT: Ball-Nayar-Tkocz's conjecture. The function M_1^X defines a norm on $\mathbb{R}^d$. The conjecture is the Shannon-entropy case of a Busemann-type theorem. The source records a la...

conjecture | open | MathDB 页面 | 原文线索

V2-024742 | MATHDB-332943

Busemann-type norm conjecture for symmetric κ-concave measures

摘录 / EXCERPT: Fix $\kappa\in[-\infty,\frac1d]$. Let (U,V) be the coordinates of a symmetric κ-concave random vector in $\mathbb{R}^2$. Symmetric κ-concave Busemann conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-024743 | MATHDB-332947

Conjecture on the optimal dimension dependence in quantitative isoperimetric inequalities

摘录 / EXCERPT: Let $n\geq 2$, and let $C(K,n)$ and $C(n)$ denote the optimal constants in the quantitative anisotropic isoperimetric inequality and quantitative Brunn-Minkowski inequality, respec...

conjecture | open | MathDB 页面 | 原文线索

V2-024744 | MATHDB-332963

Finiteness conjecture for birational symmetries of simple foliations

摘录 / EXCERPT: Finiteness conjecture. If D is simple, then

conjecture | open | MathDB 页面 | 原文线索

V2-024745 | MATHDB-332973

The balanced metric conjecture for compact complex manifolds satisfying the $\partial\overline{\partial}$-lemma

摘录 / EXCERPT: Balanced metric conjecture. Each compact complex manifold X satisfying the $\partial\overline{\partial}$-lemma admits a balanced metric.

conjecture | open | MathDB 页面 | 原文线索

V2-024746 | MATHDB-332977

Supporting conjecture on quadrics containing a general hyperplane section

摘录 / EXCERPT: Let $\Gamma\subset\mathbf{P}^r$ be a general hyperplane section of a nondegenerate reduced irreducible curve of degree n. Assume that Γ imposes $2r+1+\alpha$ conditions on...

conjecture | open | MathDB 页面 | 原文线索

V2-024747 | MATHDB-332982

Stability conjecture for the edge points of linear steps

摘录 / EXCERPT: Let $a,b\geqslant 1$ be real numbers, and for $N\geqslant 3$ let $c_b^N(a)$ denote the stabilized symplectic embedding capacity, while $c_b(a)$ denotes the corresponding four-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-024748 | MATHDB-332983

Exceptional-class conjecture for symplectic embeddings into integral polydiscs

摘录 / EXCERPT: Let $c_b(a)$ be the infimum of the $\lambda>0$ for which the ellipsoid $E(1,a)$ symplectically embeds into the polydisc $P(\lambda,\lambda b)$, and let ... ... The associated exceptiona...

conjecture | open | MathDB 页面 | 原文线索

V2-024749 | MATHDB-332984

Gross-Popescu's ideal-generation conjecture for polarized abelian surfaces

摘录 / EXCERPT: Gross-Popescu's conjecture. The homogeneous ideal of A in this embedding is generated by quadrics and cubics.

conjecture | open | MathDB 页面 | 原文线索

V2-024750 | MATHDB-333047

Waring-locus additivity conjecture for disjoint-variable forms

摘录 / EXCERPT: Waring-locus additivity conjecture. If $F=\sum_{i=1}^sF_i\in S$ is a degree $d\geq3$ form such that $F_i\in S^{[i]}$ for all $i=1,\ldots,s$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024751 | MATHDB-333050

Makai Jr.'s covering-minimum conjecture

摘录 / EXCERPT: Let $K\in\mathcal{K}^n$ be a convex body and let $\Lambda\in\mathcal{L}^n$ be a lattice. Let $\mu_1(K,\Lambda)$ denote the first covering minimum. Makai Jr.'s conjecture. One has ……

conjecture | open | MathDB 页面 | 原文线索

V2-024752 | MATHDB-333051

The covering-product lower-bound conjecture for convex bodies

摘录 / EXCERPT: For a convex body $K\in\mathcal{K}^n$, let $\mu_i(K)$ denote its ith covering minimum. Covering-product lower-bound conjecture. For every $K\in\mathcal{K}^n$, … with equality, fo…

conjecture | open | MathDB 页面 | 原文线索

V2-024753 | MATHDB-333059

Logarithmic K-semistability conjecture for polarized manifolds with smooth divisors

摘录 / EXCERPT: Logarithmic K-semistability conjecture. If $\sigma\leq 0$, then $(X,D,L,0)$ is logarithmically K-semistable.

conjecture | open | MathDB 页面 | 原文线索

V2-024754 | MATHDB-333064

Tian's weak-compactness conjecture for Kähler Ricci flow

摘录 / EXCERPT: Let (M,ω_t) be a Kähler Ricci flow on a Fano manifold, and let $(M_{\infty},\omega_{\infty})$ denote a subsequential limit. A subset of real Hausdorff codimension at least…

conjecture | open | MathDB 页面 | 原文线索

V2-024755 | MATHDB-333113

Harmonic-series conjecture for rotationally invariant geodesic balls

摘录 / EXCERPT: Let $\mathbb{M}_{h}^{m}$ be a rotationally invariant model manifold, let $B(o,r)\subset \mathbb{M}_{h}^{m}$ be a geodesic ball, and let $\lambda_{l,i}(B(o,r))$ denote the eigenvalu…

conjecture | open | MathDB 页面 | 原文线索

V2-024756 | MATHDB-333122

K-theory rank conjecture for noncommutative torus orbifold crossed products

摘录 / EXCERPT: Let $\mathcal A_\theta^{alg}$ be the algebraic noncommutative torus and let Γ be one of the cyclic groups $\mathbb Z_3$, $\mathbb Z_4$, or $\mathbb Z_6$. The group…

conjecture | open | MathDB 页面 | 原文线索

V2-024757 | MATHDB-333154

Cauchy-BTZ extension maximality conjecture for globally hyperbolic singular manifolds

摘录 / EXCERPT: Cauchy-BTZ extension maximality conjecture. The manifold M_3 is both Cauchy-maximal and BTZ-maximal.

conjecture | open | MathDB 页面 | 原文线索

V2-024758 | MATHDB-333181

John ellipsoid half-volume conjecture

摘录 / EXCERPT: John ellipsoid half-volume conjecture. The ellipsoid obtained by scaling B_K by a factor of $O(\sqrt{n})$ about its center contains at least half of the volume of K.

conjecture | open | MathDB 页面 | 原文线索

V2-024759 | MATHDB-333197

Gaussian asymptotics conjecture for Betti numbers of large-degree embeddings

摘录 / EXCERPT: Gaussian Betti-number conjecture. Fix $q\in[1,n]$. There is a normalizing function $F_q(d)$, depending on X and geometric data, such that for every sequence of integers p_d sat…

conjecture | open | MathDB 页面 | 原文线索

V2-024760 | MATHDB-333200

The arrangement (a) conjecture for the square punctured torus Hopf differential

摘录 / EXCERPT: Arrangement (a) conjecture. The Hopf differential Φ^h of Corollary $ has divisor data described by arrangement (a).

conjecture | open | MathDB 页面 | 原文线索

V2-024761 | MATHDB-333207

The perimeter-diameter lower bound for curves covering a planar convex shape

摘录 / EXCERPT: Perimeter-diameter conjecture. The length of $ satisfies

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V2-024762 | MATHDB-333211

Ball-to-shell transition conjecture for the first Robin eigenvalue

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a domain of fixed volume, and let $\lambda_1^\alpha(\Omega)$ be the first eigenvalue of the Robin problem … where $\alpha<0$. Ball-to-shell trans…

conjecture | open | MathDB 页面 | 原文线索

V2-024763 | MATHDB-333212

Convex-domain Robin maximisation conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a convex domain of fixed volume, and let $\lambda_1^\alpha(\Omega)$ be the first eigenvalue of the Robin problem with $\alpha<0$. Convex-domain R…

conjecture | open | MathDB 页面 | 原文线索

V2-024764 | MATHDB-333213

Surface-area Robin maximisation conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a domain with prescribed surface area, and let $\lambda_1^\alpha(\Omega)$ be the first eigenvalue of the Robin problem with $\alpha<0$. Surface-a…

conjecture | open | MathDB 页面 | 原文线索

V2-024765 | MATHDB-333217

Hyperbolic event-domain conjecture for Thiagarajan and Badouel-Darondeau-Raoult

摘录 / EXCERPT: Let an event domain be the domain of an event structure, and call it hyperbolic when its underlying graph is hyperbolic. Hyperbolic event-domain conjecture. Conjectures Thiagarajan…

conjecture | open | MathDB 页面 | 原文线索

V2-024766 | MATHDB-333218

Hyperbolic strongly regular event structure labeling conjecture

摘录 / EXCERPT: Let $\mathcal E$ be a strongly regular event structure, and let its domain be hyperbolic. A regular nice labeling is a finite nice labeling satisfying the regularity condition for...

conjecture | open | MathDB 页面 | 原文线索

V2-024767 | MATHDB-333226

The Bochner conjecture for harmonic maps into NPC metric spaces

摘录 / EXCERPT: Let M be a Riemannian manifold with non-negative Ricci curvature, and let X be an NPC metric space. A harmonic map is a map $u:M\rightarrow X$ satisfying the harmonic-map condi...

conjecture | open | MathDB 页面 | 原文线索

V2-024768 | MATHDB-333230

The limiting density conjecture for rays of random toric surfaces

摘录 / EXCERPT: The limiting density conjecture. For $k>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-024769 | MATHDB-333241

Invariant-measure universality conjecture for rescaled curvature flow

摘录 / EXCERPT: Let $\hat{\Omega}$ be curvature flow on embedded graphs in $\mathbb{R}^n$, rescaled so that a characteristic length scale remains equal to one, and let $\mathcal{G}^n$ be the class...

conjecture | open | MathDB 页面 | 原文线索

V2-024770 | MATHDB-333246

Self-intersection conjecture for flexible cross-polytopes in Euclidean and Lobachevsky spaces

摘录 / EXCERPT: Let $n\ge 3$. A flexible cross-polytope is a flexible polyhedral cross-polytope in the indicated ambient space. Self-intersection conjecture. All flexible cross-polytopes in...

conjecture | open | MathDB 页面 | 原文线索

V2-024771 | MATHDB-333248

The generalized bellows conjecture for flexible polyhedra in Euclidean space

摘录 / EXCERPT: Let $P\colon K\to\mathbb{R}^n$ be a flexible, not necessarily embedded polyhedron, and let $V_K(P)$ denote its generalized oriented volume, defined by integrating its characteristi...

conjecture | open | MathDB 页面 | 原文线索

V2-024772 | MATHDB-333249

The bellows conjecture for non-Euclidean spaces

摘录 / EXCERPT: Let $n\ge 3$, and let a flexible polyhedron lie either in Lobachevsky space Λ^n or in the open hemisphere S^n_+. Its generalized oriented volume is defined in the relevan...

conjecture | open | MathDB 页面 | 原文线索

V2-024773 | MATHDB-333257

Negative curvature conjecture for minimal immersions from Hitchin representations

摘录 / EXCERPT: Negative curvature conjecture. The minimal immersion is never tangential to any flat inside the symmetric space. Consequently, the sectional curvature of the immersed minimal surfa...

conjecture | open | MathDB 页面 | 原文线索

V2-024774 | MATHDB-333260

The magnitude formula for compact ell_1-convex sets

摘录 / EXCERPT: Let A be a compact ℓ_1-convex set, meaning a geodesic subset of ℓ_1^n, and let $V'_i(A)$ denote its ith ℓ_1-intrinsic volume. These volumes are defined by the St...

conjecture | open | MathDB 页面 | 原文线索

V2-024775 | MATHDB-333295

Non-expansion conjecture for bounded-degree graphs with nonnegative Bakry-Émery curvature

摘录 / EXCERPT: Let $d\in{\mathbb N}$. Non-expansion conjecture. No infinite family of finite, increasing, d-regular graphs satisfying $CD(0,\infty)$ can be a family of expander graphs. This is...

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V2-024776 | MATHDB-333296

$CD(2,\infty)$ conjecture for strongly regular graphs of girth three

摘录 / EXCERPT: Let $G=(V,E)$ be a strongly regular graph. Its girth is the length of its shortest cycle; in particular, girth 3 means that G contains a triangle. Strongly regular girth-three...

conjecture | open | MathDB 页面 | 原文线索

V2-024777 | MATHDB-333317

The cone conjecture for the poset of rational cones

摘录 / EXCERPT: Cone conjecture. For every d, the order in $\operatorname{Cones}(d)$ is the inclusion order.

conjecture | open | MathDB 页面 | 原文线索

V2-024778 | MATHDB-333319

The hypercellular complex characterization conjecture

摘录 / EXCERPT: The hypercellular complex characterization conjecture. For a graph G, the following conditions are equivalent:

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V2-024779 | MATHDB-333325

The BLMPR conjecture on effective theta characteristics of tropical plane quartics

摘录 / EXCERPT: BLMPR conjecture. Each of the seven odd theta characteristics of Γ is the specialization of four effective theta characteristics of X, counted with multiplicity.

conjecture | open | MathDB 页面 | 原文线索

V2-024780 | MATHDB-333333

The first shape-coefficient conjecture for consecutive Dirichlet eigenvalue gaps

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{n}$ be a bounded domain with piecewise smooth boundary $\partial\Omega$, and let λ_i be the i-th Dirichlet eigenvalue. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024781 | MATHDB-333341

Generic lines and one fat linear space have good postulation

摘录 / EXCERPT: Let $n,d,r\in\mathbb{N}$ with $n\geq r+2\geq 3$. Let $\Pi\cong\mathbb{P}^r\subset\mathbb{P}^n$ and let $X\subset\mathbb{P}^n$ consist of $s\geq 1$ generic lines and one m-multipl...

conjecture | open | MathDB 页面 | 原文线索

V2-024782 | MATHDB-333342

Extension of birational rigidity to mildly singular del Pezzo fibrations

摘录 / EXCERPT: A del Pezzo fibration is birationally rigid if it has only one Mori fiber space structure in its birational class. Let X be a del Pezzo fibration satisfying the hypotheses of The...

conjecture | open | MathDB 页面 | 原文线索

V2-024783 | MATHDB-333388

Tropical soliton-resolution conjecture for affine-finite quivers

摘录 / EXCERPT: Let Q be any quiver in the affine-finite classification, with affine component graph Γ and finite component graph Δ, and let $\mathfrak{t}^{\lambda}_v(t)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-024784 | MATHDB-333389

Tropical speed-conservation conjecture under Dynkin involution

摘录 / EXCERPT: Assume the tropical T-system has tensor-product form $\Lambda'\otimes\widehat{\Lambda}$, with Λ' of type A_m, D_{2m+1}, or E_6, and let η be the canonical Dynk...

conjecture | open | MathDB 页面 | 原文线索

V2-024785 | MATHDB-333390

Tropical speed-conservation conjecture for general affine slices

摘录 / EXCERPT: Let the tropical T-system be in any other classified case, meaning either it belongs to a different family or it is a tensor product whose finite factor is not of type A_m,...

conjecture | open | MathDB 页面 | 原文线索

V2-024786 | MATHDB-333411

Smoothness and positivity conjecture for band-limited nodal-area distributions

摘录 / EXCERPT: Let $f=f_{\alpha;T}$ be a band-limited Gaussian function on an n-dimensional compact manifold, with $n\geq 2$ and $\alpha\leq 1$, and let $\Psi_{n;\alpha}$ be the limiting distri...

conjecture | open | MathDB 页面 | 原文线索

V2-024787 | MATHDB-333424

Chen-Zheng-Yang gap conjecture for drifting Laplacian eigenvalues

摘录 / EXCERPT: Let (M^{n},g,f) be a complete smooth metric measure space, and let λ_i be the i-th eigenvalue, for $i=1,2,\ldots,k$, of the drifting-Laplacian eigenvalue problem conside...

conjecture | open | MathDB 页面 | 原文线索

V2-024788 | MATHDB-333436

The main conjecture on local density in stable lattices

摘录 / EXCERPT: Let $\mathcal{L}\subset\mathbb{R}^n$ be an n-dimensional lattice such that every sublattice of $\mathcal{L}$ has determinant at least 1. For $r>0$, let $N_{\mathcal{L}}(r)$ den...

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V2-024789 | MATHDB-333491

Hawking Mass Compactness conjecture

摘录 / EXCERPT: Let M_j^3 be a sequence of three-dimensional oriented manifolds satisfying ... Assume also ... and either that there are no closed interior minimal surfaces or that...

conjecture | open | MathDB 页面 | 原文线索

V2-024790 | MATHDB-333492

Almost rigidity of the Positive Mass Theorem for special surfaces

摘录 / EXCERPT: Fix $D>0$ and $r_0>0$. Let $\mathcal M$ be the class of asymptotically flat three-dimensional Riemannian manifolds with nonnegative scalar curvature and no interior closed minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-024791 | MATHDB-333494

Almost rigidity of the Scalar Torus Theorem

摘录 / EXCERPT: Almost-rigidity conjecture. A subsequence should converge intrinsically flatly to a flat torus:

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V2-024792 | MATHDB-333495

Generalized nonnegative scalar curvature under intrinsic-flat convergence

摘录 / EXCERPT: Generalized scalar-curvature conjecture. Then: (a) M_∞ is a nonzero integral current space, with no cancellation without collapse; (b) M_∞ is geodesic; (c) angles bet...

conjecture | open | MathDB 页面 | 原文线索

V2-024793 | MATHDB-333509

Ekstrand-Heluani-Källén-Zabzine's Shatashvili-Vafa algebra conjecture for G_{2}-manifolds

摘录 / EXCERPT: Shatashvili-Vafa algebra conjecture. The section pairs $\{\Phi_{+},K_{+}\}$ and $\{\Phi_{-},K_{-}\}$ generate two commuting copies of the Shatashvili-Vafa G_{2} superconformal al...

conjecture | open | MathDB 页面 | 原文线索

V2-024794 | MATHDB-333531

The additivity conjecture for filtered operads

摘录 / EXCERPT: Additivity conjecture. For every $k\geq 1$, the additivity isomorphism $E_{k+1}\simeq E_1\otimes^{\mathbf{L}}_{BV}E_k$ induces the stated map...

conjecture | open | MathDB 页面 | 原文线索

V2-024795 | MATHDB-333563

Jordan-algebra representation conjecture for convex state spaces

摘录 / EXCERPT: Jordan-algebra representation conjecture. If C has a local regret function and a transitive symmetry group, then C can be represented as the positive elements of a Jordan algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-024796 | MATHDB-333570

Donaldson's a priori estimates conjecture for the Calabi-Yau equation

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold of real dimension 4, equipped with an almost complex structure J tamed by ω. Let $\tilde{\omega}$ be another symplectic...

conjecture | open | MathDB 页面 | 原文线索

V2-024797 | MATHDB-333597

Even-lacuna conjecture for isolated plane critical points

摘录 / EXCERPT: Even-lacuna conjecture. The deformations of f have no even local lacunas unless f has an extremum at the origin. If f has a minimum at the origin, then all critical values of...

conjecture | open | MathDB 页面 | 原文线索

V2-024798 | MATHDB-333603

Conjecture on densest prism-group ball arrangements in Nil geometry

摘录 / EXCERPT: Let $\boldsymbol{Nil}$ denote Nil geometry, and let $Gamma_p(q)=\mathbf{pq2_1}$ be the associated isometry group for parameters $(p,q)=(4,4),(3,6),(6,3)$. Let $\mathcal{B}_p(q)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-024799 | MATHDB-333623

Conjecture on global analytic lifts of weakly analytic maps

摘录 / EXCERPT: Let Y be a k-affinoid space, let X be a basic tube, and let $f:X\to Y$ be a weakly analytic map, meaning locally a map of the form $\pi_{K/k}\circ F\circ\sigma_{K/k}$ for a c...

conjecture | open | MathDB 页面 | 原文线索

V2-024800 | MATHDB-333625

Beloshapka's maximal symmetry conjecture for holomorphically nondegenerate CR-hypersurfaces

摘录 / EXCERPT: Let M be a real-analytic connected holomorphically nondegenerate CR-hypersurface of CR-dimension n, and let $\mathfrak{s}(M)$ denote its symmetry algebra. Beloshapka's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-024801 | MATHDB-333650

Salis's conjecture on projectively induced rotation invariant Kähler-Einstein manifolds

摘录 / EXCERPT: Let (M,g) be a Kähler-Einstein manifold admitting a Kähler immersion into $(\mathds{C}\mathrm{P}^N,g_{FS})$ with $N<\infty$, where g_{FS} is the Fubini-Study metric. Salis's co...

conjecture | open | MathDB 页面 | 原文线索

V2-024802 | MATHDB-333663

The conjecture that all Hirzebruch-property arrangements are complex reflection arrangements

摘录 / EXCERPT: Hirzebruch-property reflection-arrangement conjecture. All arrangements satisfying the Hirzebruch property are complex reflection arrangements.

conjecture | open | MathDB 页面 | 原文线索

V2-024803 | MATHDB-333667

Orbifold fundamental group finiteness conjecture for slope rationally connected pairs

摘录 / EXCERPT: Orbifold fundamental group conjecture. The group $\pi_1(X,D)$ is almost abelian, finite if (X,D) is klt, and trivial if (X,D) is purely logarithmic or if...

conjecture | open | MathDB 页面 | 原文线索

V2-024804 | MATHDB-333691

Landau-Ginzburg mirror symmetry for Hodge numbers

摘录 / EXCERPT: Let (Y,w) be a Landau-Ginzburg model of Fano type arising, together with a tame compactification, as a mirror of a projective Fano manifold X of dimension $n=\dim Y$. Let...

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V2-024805 | MATHDB-333708

The zero-area hitting measure conjecture for random planar curves

摘录 / EXCERPT: Let γ be an unbounded one-sided curve in the Euclidean plane, and let Ω be a simply connected, open, bounded planar domain. Reroot γ at a root chosen in...

conjecture | open | MathDB 页面 | 原文线索

V2-024806 | MATHDB-333748

Linearity conjecture for translation generalized quadrangles

摘录 / EXCERPT: Linearity conjecture. Every translation generalized quadrangle has a projective representation; equivalently, the kernel k is a division ring.

conjecture | open | MathDB 页面 | 原文线索

V2-024807 | MATHDB-333770

The conjectured lower bound for ordinary planes

摘录 / EXCERPT: Let $e_3(n)$ denote the minimum number of ordinary planes determined by a set of n points in the relevant geometric setting. Lower-bound conjecture for $e_3(n)$. For all n, ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-024808 | MATHDB-333771

The higher-dimensional lower-bound conjecture for ordinary hyperplanes

摘录 / EXCERPT: For $d\geqslant4$, let $e_d(n)$ denote the minimum number of ordinary hyperplanes determined by a set of n points in the relevant d-dimensional geometric setting. Higher-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-024809 | MATHDB-333810

Mumford's nonvanishing conjecture for pseudoeffective canonical bundles

摘录 / EXCERPT: Let X be a complex projective manifold, and let K_X denote its canonical bundle. A line bundle is pseudoeffective when its numerical class lies in the closure of the cone of ef...

conjecture | open | MathDB 页面 | 原文线索

V2-024810 | MATHDB-333843

The submersion version of generalized Chen's conjecture

摘录 / EXCERPT: Let $M^m(c)$ be an m-dimensional space form of constant sectional curvature c, let N^{m-1} be a manifold, and let $\varphi:M^m(c)\to N^{m-1}$ be a Riemannian submersion. Subm...

open_problem | open | MathDB 页面 | 原文线索

V2-024811 | MATHDB-333846

Brianchon's theorem in descriptive proximity geometry

摘录 / EXCERPT: Descriptive proximity form of Brianchon's theorem. Theorem Brianchon's theorem has descriptive proximity form.

conjecture | open | MathDB 页面 | 原文线索

V2-024812 | MATHDB-333847

Descriptive proximity duality for spinning and massless spin particles

摘录 / EXCERPT: Spinning-particle duality conjecture. The duality asserting that the dual of a spinning particle is a massless spin particle has descriptive proximity form:

conjecture | open | MathDB 页面 | 原文线索

V2-024813 | MATHDB-333871

Non-symmetric L_p zonoid simplex volume conjecture

摘录 / EXCERPT: Let S^{n-1} be the unit sphere in $\mathbb{R}^n$. A measure on S^{n-1} is isotropic if its second-moment matrix is the identity, and it is centred if ... For a centred isotropic...

conjecture | open | MathDB 页面 | 原文线索

V2-024814 | MATHDB-333873

The null-locus characterization of the non-Kähler locus

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[b1]$ be a pseudoeffective $(1,1)$ class. Let $E_{nK}(b1)$ denote its non-Kähler locus and let ... where the union is over irreducible a...

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V2-024815 | MATHDB-333892

Critical-threshold conjecture for equivariant wave-map blowup

摘录 / EXCERPT: Critical-threshold conjecture. For $d\in\{3,4,5,6\}$, the threshold of blowup is given by the codimension-one stable manifold of the self-similar solution f_1.

conjecture | open | MathDB 页面 | 原文线索

V2-024816 | MATHDB-333906

Casimir-count conjecture for the invariant Kontsevich bracket

摘录 / EXCERPT: Let $\mathcal A^+=\mathbb C\langle u,v\rangle$ be the free associative algebra on two generators, let $c=uvu^{-1}v^{-1}$, and let φ denote the representation map to invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-024817 | MATHDB-333907

Conjectural modified double Poisson brackets on the free algebra with three generators

摘录 / EXCERPT: Let $Free_3=\mathbb C\langle x_1,x_2,x_3\rangle$, and let brackets I and II be the biderivations defined by the displayed formulas in the source, with all omitted brackets of gener...

conjecture | open | MathDB 页面 | 原文线索

V2-024818 | MATHDB-333932

Erdős's conjecture on the first unresolved triangle count

摘录 / EXCERPT: Erdős's conjecture. Any set of seven points in the plane determines at least four distinct triangles; consequently,

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V2-024819 | MATHDB-333942

Generalized Lichnerowicz conjecture for compact Lorentzian manifolds

摘录 / EXCERPT: Generalized Lichnerowicz conjecture. If a compact Lorentzian manifold has an essential conformal group, then it is conformally flat.

conjecture | open | MathDB 页面 | 原文线索

V2-024820 | MATHDB-333957

The weak union-of-lines conjecture

摘录 / EXCERPT: Let L be a set of lines in $\mathbb{F}_q^3$, and define the union of lines by ... Weak union-of-lines conjecture. If ... and no plane contains $\omega(q)$ lines of L, then ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-024821 | MATHDB-333964

The coherent-constructible correspondence for smooth complete fans

摘录 / EXCERPT: Let M be a free abelian group of rank n, let N be its dual, and let $Sigma$ be a smooth complete fan in $N_\bR:=N\otimes_\bZ\bR$. Let X_Σ be the associated smooth tori...

conjecture | open | MathDB 页面 | 原文线索

V2-024822 | MATHDB-333975

The maximal rank conjecture for general algebraic curves

摘录 / EXCERPT: The maximal rank conjecture. The map μ_m is either injective or surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-024823 | MATHDB-333986

The covering-radius conjecture for lattice zonotopes in general linear position

摘录 / EXCERPT: Let $S=\{\boldsymbol{z}_1,\dotsc,\boldsymbol{z}_m\}\subseteq\mathbb{Z}^n$ and let Z be the lattice zonotope generated by S. The set S is in LGP (general linear position) when...

conjecture | open | MathDB 页面 | 原文线索

V2-024824 | MATHDB-333988

The zonotopal reformulation of the Lonely Runner Conjecture

摘录 / EXCERPT: Let n be a positive integer. Let $Z\subseteq\mathbb{R}^n$ be a zonotope generated by $n+1$ vectors of $\mathbb{Z}^n$ in LGP (general linear position), and let $\boldsymbol{x}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-024825 | MATHDB-333999

Uniqueness conjecture for the index-4 free boundary minimal surface

摘录 / EXCERPT: Let Σ be a free boundary minimal surface in the unit ball $\mathbb{B}^3$ with index 4. Uniqueness conjecture. Then Σ is isometric to the critical catenoid. The crit...

conjecture | open | MathDB 页面 | 原文线索

V2-024826 | MATHDB-334000

Betti-number conjecture for lattice polygons and toric surfaces

摘录 / EXCERPT: Let $\Delta\subseteq\mathbb{R}^2$ be a lattice polygon whose interior polygon $\Delta^{(1)}$ is two-dimensional and contains $g\geq 4$ lattice points. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-024827 | MATHDB-334009

Yau's existence conjecture for the Strominger system

摘录 / EXCERPT: Let (X,Ω) be a compact Calabi-Yau threefold endowed with a balanced class τ. Let $\mathcal{E}$ be a holomorphic vector bundle over X satisfying $c_1(\mathcal{E})=0$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-024828 | MATHDB-334031

The hypoelliptic damping resonance conjecture for Anosov geodesic flows

摘录 / EXCERPT: Hypoelliptic damping conjecture. The second part of the theorem for the elliptically perturbed operator P_ϵ holds with $\widetilde P_\epsilon$ in place of P_ϵ. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-024829 | MATHDB-334095

The determinantal representation conjecture for Calabi-Yau potentials

摘录 / EXCERPT: Determinantal representation conjecture. The unique smooth normalized solution φ may be represented as a limit in $L^{1}(X)$:

conjecture | open | MathDB 页面 | 原文线索

V2-024830 | MATHDB-334096

The negative-temperature equivalence conjecture for Gibbs measures

摘录 / EXCERPT: Negative-temperature equivalence conjecture. For every $\beta>\beta_0$, the following are equivalent: the partition functions $Z_{N_k,\beta}$ are finite for all sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-024831 | MATHDB-334102

Demailly-Paun conjecture on generic Kähler stability in deformations

摘录 / EXCERPT: Demailly-Paun conjecture. There exists a countable union $S'\subsetneq S$ of analytic subsets in the base such that X_t is Kähler for $t\in S\setminus S'$. Moreover, S can be c...

conjecture | open | MathDB 页面 | 原文线索

V2-024832 | MATHDB-334166

The affine-embedding conjecture for self-similar sets

摘录 / EXCERPT: Let $\Phi=\{\phi_i(x)=\alpha_i x+a_i\}_{i=1}^m$ and $\Psi=\{\psi_j(x)=\beta_jx+b_j\}_{j=1}^{\ell}$ be self-similar iterated function systems on $\mathbb R$, with attractors...

conjecture | open | MathDB 页面 | 原文线索

V2-024833 | MATHDB-334179

Dutour-Sikirić's local-maxima conjecture for covering radii

摘录 / EXCERPT: For each integer $n \geq 6$, let $[L_n,Q_n]$ be the lattice family defined in the cited work, and consider the covering radius function on lattices in dimension n. A local maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-024834 | MATHDB-334191

Neumann's Bloch-group generation conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let $F \subset \mathbb{C}$ be a number field not contained in $\mathbb{R}$, and let $\mathcal{M}$ be the set of manifolds with invariant trace fields contained in F, ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024835 | MATHDB-334192

Neumann's conjecture on invariant trace fields of hyperbolic manifolds

摘录 / EXCERPT: A non-real concrete number field is a non-real number field considered with its specified embedding into the complex numbers. Neumann's weaker conjecture. Every non-real concrete n...

conjecture | open | MathDB 页面 | 原文线索

V2-024836 | MATHDB-334193

Neumann's conjecture for volumes of hyperbolic 3-manifolds

摘录 / EXCERPT: Let $F \subset \mathbb{C}$ be a number field not contained in $\mathbb{R}$. Let ... Let r_2 be the number of complex places of F. The volume-basis conjecture. The lattice generat...

conjecture | open | MathDB 页面 | 原文线索

V2-024837 | MATHDB-334194

Neumann's abstract-field conjecture for hyperbolic manifolds

摘录 / EXCERPT: A non-real abstract number field is a field presented in the form $\mathbb{Q}[x]/\langle p \rangle$. Neumann's abstract-field conjecture. Every non-real abstract number field...

conjecture | open | MathDB 页面 | 原文线索

V2-024838 | MATHDB-334202

Holomorphic Seifert-Weinstein conjecture for locally conformally symplectic manifolds

摘录 / EXCERPT: Holomorphic Seifert-Weinstein conjecture. For any l.c.s. pair (ω,J) on $S^{2k+1}\times S^1$ or on $C\times S^1$ for C a threefold, there is an elliptic, non-constant J-h...

conjecture | open | MathDB 页面 | 原文线索

V2-024839 | MATHDB-334236

Conjecture on the codimension of singular loci of foliations

摘录 / EXCERPT: Let X be a complex projective manifold and let $\mathscr{G}\subset T_X$ be a foliation of codimension q on X with numerically trivial canonical class. Denote by...

open_problem | open | MathDB 页面 | 原文线索

V2-024840 | MATHDB-334247

The rank- cut-locus conjecture for free step-two Carnot groups

摘录 / EXCERPT: Let $\mathbb{G}_k$ be the free step-two Carnot group of rank k, identified with $\mathbb{R}^k\oplus\wedge^2\mathbb{R}^k$, and let $\mathrm{Cut}_0(\mathbb{G}_k)$ denote its cut lo...

conjecture | open | MathDB 页面 | 原文线索

V2-024841 | MATHDB-334254

Hausdorff dimension conjecture for attractors of non-quadratic Pisot numbers

摘录 / EXCERPT: Let β be a Pisot number, meaning an algebraic integer greater than 1 whose other Galois conjugates have modulus less than 1, and let A_β denote the associated attra...

conjecture | open | MathDB 页面 | 原文线索

V2-024842 | MATHDB-334263

Hyperbolisation conjecture for closed 3-manifolds

摘录 / EXCERPT: Hyperbolisation conjecture. If $\pi_1(M)$ is infinite and does not contain $\mathbb{Z}\times\mathbb{Z}$, then M is hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-024843 | MATHDB-334266

The Gromov-Witten invariant energy-bound conjecture

摘录 / EXCERPT: Let (X,ω) be a compact symplectic manifold and let $g\in\mathbb Z$. For fixed classes $H_1,\ldots,H_k\in H^*(X)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-024844 | MATHDB-334323

The torsion conjecture for skein modules of 3-manifolds

摘录 / EXCERPT: Let M be a compact oriented 3-manifold with nonempty boundary. Suppose that every closed incompressible surface in M is parallel to the boundary of M. The torsion conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-024845 | MATHDB-334329

Markov type monotonicity under finite-fiber submetries

摘录 / EXCERPT: Finite-fiber submetry conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-024846 | MATHDB-334331

Kawamata's LG-model conjecture on FCY admissible subcategories

摘录 / EXCERPT: Let (Y_1,w_1) and (Y_2,w_2) be gauged Landau-Ginzburg models with Y_i smooth, a group G acting on each Y_i, and w_i a section of $\operatorname{\mathcal O}(\chi)$ for a...

conjecture | open | MathDB 页面 | 原文线索

V2-024847 | MATHDB-334343

The monodromy-compatibility conjecture for phase tropical hypersurfaces

摘录 / EXCERPT: Monodromy-compatibility conjecture. The diagram with horizontal maps ψ, vertical maps ξ_η and $\mathcal T\xi_\eta$, and the corresponding map ψ on the bottom commu...

conjecture | open | MathDB 页面 | 原文线索

V2-024848 | MATHDB-334359

Demailly-Peternell-Schneider conjecture on the Albanese morphism of compact Kähler manifolds

摘录 / EXCERPT: Demailly-Peternell-Schneider conjecture. The Albanese morphism of X is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-024849 | MATHDB-334380

The Kaufman exceptional set conjecture for planar projections

摘录 / EXCERPT: Let $K\subset\mathbb{R}^{2}$ be a Borel set, and let $\pi_e\colon\mathbb{R}^{2}\to\mathbb{R}$ be the orthogonal projection $\pi_e(x)=x\cdot e$ for $e\in S^{1}$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-024850 | MATHDB-334390

The log-convex density conjecture on hyperbolic space

摘录 / EXCERPT: Let $\mathbb{H}^n$ be hyperbolic space with a smooth, radial, log-convex density, and let spheres about the origin mean geodesic spheres centered at the origin. The log-convex dens...

conjecture | open | MathDB 页面 | 原文线索

V2-024851 | MATHDB-334399

Conjecture relating strongly Gauduchon spaces to class SG

摘录 / EXCERPT: Strongly Gauduchon space conjecture. Any strongly Gauduchon space belongs to class $\mathscr{SG}$.

conjecture | open | MathDB 页面 | 原文线索

V2-024852 | MATHDB-334400

Ampleness conjectures for compact complex manifolds with hyperbolicity or negative holomorphic sectional curvature

摘录 / EXCERPT: Ampleness conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-024853 | MATHDB-334401

Vanishing real bisectional curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Vanishing real bisectional curvature conjecture. If the real bisectional curvature is constantly equal to c, then $c=0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-024854 | MATHDB-334402

Geometric consequences of positive or nonpositive real bisectional curvature

摘录 / EXCERPT: Real bisectional curvature conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-024855 | MATHDB-334412

Calabi-Yau fibration conjecture for the stable-sheaf moduli space

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, let $\beta>0$, and let $\operatorname{Sh}_{\beta}(X)$ be the moduli space used to define the Gopakumar-Vafa invariants, equipped with its d-cri...

conjecture | open | MathDB 页面 | 原文线索

V2-024856 | MATHDB-334434

Bucher-Connell-Lafont simplicial-volume conjecture for affine manifolds

摘录 / EXCERPT: Let M be a closed manifold supporting an affine structure, meaning a closed manifold admitting an atlas whose transition maps are restrictions of affine isomorphisms. Let $\|M\|$...

conjecture | open | MathDB 页面 | 原文线索

V2-024857 | MATHDB-334455

Generalized Sudakov and dual Sudakov minoration conjecture

摘录 / EXCERPT: Generalized Sudakov minoration conjecture. There is a universal constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024858 | MATHDB-334462

Weak positivity conjecture for the inverse σ_k-flow

摘录 / EXCERPT: Weak positivity conjecture. Under these assumptions, this class contains a smooth strictly positive $(n-1,n-1)$-form. The conjecture is a weaker consequence of the full Lejmi-Széke...

conjecture | open | MathDB 页面 | 原文线索

V2-024859 | MATHDB-334486

Han-Li conjecture for the constant scalar curvature and mean curvature problem

摘录 / EXCERPT: Let (M,g_0) be a smooth compact Riemannian manifold of dimension $n\geq 3$ with boundary ∂M. For constants $c_1,c_2\in\mathbb{R}$, consider the problem ... where u is r...

conjecture | open | MathDB 页面 | 原文线索

V2-024860 | MATHDB-334488

Uniqueness conjecture for the v_m-Yamabe problem

摘录 / EXCERPT: Uniqueness conjecture. Under these hypotheses, the normalized solution of the v_m-Yamabe equation is unique. The conjecture is suggested by the variational and geodesic-convexity...

conjecture | open | MathDB 页面 | 原文线索

V2-024861 | MATHDB-334494

The SYZ conjecture for nef parabolic line bundles on hyperkähler manifolds

摘录 / EXCERPT: Let X be a compact hyperkähler manifold, and let L be a nef line bundle. Let q denote the Beauville-Bogomolov-Fujiki form; L is parabolic when $q(L,L)=0$. SYZ conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-024862 | MATHDB-334495

The algebraic-dimension conjecture for compact hyperkähler manifolds

摘录 / EXCERPT: Let X be a non-projective compact hyperkähler manifold and let L be a non-trivial nef line bundle. Let $a(X)$ be the transcendence degree of the field of meromorphic functions...

conjecture | open | MathDB 页面 | 原文线索

V2-024863 | MATHDB-334496

The algebraic-dimension values conjecture for compact hyperkähler manifolds

摘录 / EXCERPT: Let X be a compact hyperkähler manifold of dimension $2n$, and let $a(X)$ denote its algebraic dimension. Algebraic-dimension values conjecture. Then ... The source reports that th...

conjecture | open | MathDB 页面 | 原文线索

V2-024864 | MATHDB-334497

The adjoint strict-nefness conjecture

摘录 / EXCERPT: Let X be a projective manifold of dimension n, and let L be a strictly nef divisor, meaning that $L\cdot C>0$ for every irreducible curve C on X. Adjoint strict-nefness c...

conjecture | open | MathDB 页面 | 原文线索

V2-024865 | MATHDB-334544

Immersed version of the uniform curvature estimate for stable free boundary minimal surfaces

摘录 / EXCERPT: Immersed curvature estimate conjecture. The uniform curvature estimate

conjecture | open | MathDB 页面 | 原文线索

V2-024866 | MATHDB-334573

The Fano endomorphism conjecture for Picard number one

摘录 / EXCERPT: Fano endomorphism conjecture. X should be a projective space.

conjecture | open | MathDB 页面 | 原文线索

V2-024867 | MATHDB-334591

The conjecture that all Takagi function graphs have Assouad dimension 2

摘录 / EXCERPT: Assouad-dimension conjecture. For all such Takagi functions,

conjecture | open | MathDB 页面 | 原文线索

V2-024868 | MATHDB-334598

Supersingular reduction conjecture for non-isotrivial K3 surfaces

摘录 / EXCERPT: Let k be an algebraically closed field of positive characteristic $p>0$, and let C be a proper smooth curve over k with function field $K=k(C)$. Let X be a non-isotrivial,...

conjecture | open | MathDB 页面 | 原文线索

V2-024869 | MATHDB-334601

The converse Markov inequality conjecture for compact convex domains

摘录 / EXCERPT: Let $K\subset \mathbb{C}$ be a compact convex domain, let $q\geq 1$, and let $n\in\mathbb{N}$. Write $\mathcal{P}_n(K)$ for the polynomials of degree at most n on K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024870 | MATHDB-334616

Finiteness conjecture for constant mean curvature surfaces in flat 3-tori

摘录 / EXCERPT: Let $H>0$ and $g\in\mathbb{N}\cup\{0\}$. Consider the moduli space of non-congruent, connected, closed H-surfaces of genus at most g in a fixed flat 3-torus. Finiteness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-024871 | MATHDB-334617

Properness and cubic area growth conjecture for finite-genus constant mean curvature surfaces

摘录 / EXCERPT: Let N be a complete flat 3-manifold, and let M be a complete connected H-surface embedded in N with finite genus. Let W denote its closed mean-convex complement when...

conjecture | open | MathDB 页面 | 原文线索

V2-024872 | MATHDB-334623

Semistability conjecture for nonsplit tangent sheaves of codimension-one foliations on projective three-space

摘录 / EXCERPT: Let ${\mathscr{F}}$ be a codimension one foliation on ${\mathbb{P}}^{3}$, with tangent sheaf $T_{\mathscr{F}}$. A coherent sheaf is split if it is a direct sum of line bundles. Sem...

conjecture | refuted | MathDB 页面 | 原文线索

V2-024873 | MATHDB-334631

The hexagonal-grid realization conjecture for ball-packing contact numbers

摘录 / EXCERPT: Hexagonal-grid realization conjecture. The exact (maximal) value for each $n\in\mathbb{N}$ can be achieved in some hexagonal grid.

conjecture | open | MathDB 页面 | 原文线索

V2-024874 | MATHDB-334646

Conjecture on extensions of nonflat sets in CN2 manifolds

摘录 / EXCERPT: Let M^n be a complete CN2 manifold with finite volume, let V be its set of nonflat points, and let an extension of V be an open set carrying a complete flat totally geodesic...

conjecture | open | MathDB 页面 | 原文线索

V2-024875 | MATHDB-334647

Conjecture on compact CN2 manifolds with signed scalar curvature

摘录 / EXCERPT: A CN2 manifold is a Riemannian manifold whose curvature operator has nullity of codimension at most two. Let M be a compact CN2 manifold with finite volume and scalar curvature t...

conjecture | open | MathDB 页面 | 原文线索

V2-024876 | MATHDB-334649

Quadrisection bound for convex odd-gons

摘录 / EXCERPT: Let R be a convex polygon with $2n+1$ sides. A quadrisection is a partition of R into four regions of equal area by three concurrent cevians. Quadrisection bound conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-024877 | MATHDB-334663

Persistence conjecture for symplectic mapping groups of the five-point blow-up of the projective plane

摘录 / EXCERPT: Let $X=\mathbb{C}P^2\#5\overline{\mathbb{C}P^2}$ be the five-point blow-up of the complex projective plane, equipped with any symplectic form. Let $\Gamma(X,\omega)$ denote the sym...

conjecture | open | MathDB 页面 | 原文线索

V2-024878 | MATHDB-334695

Strict positivity of one-form weights for conical resolutions

摘录 / EXCERPT: Let $\pi:X\to Y$ be a conical resolution over a field of any characteristic, with ... and ... The multiplicative group is denoted by $\mathbb{G}_m$, and it acts on $H^0(X,\Omega_X^1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-024879 | MATHDB-334699

Local uniformization conjecture for codimension-one foliations

摘录 / EXCERPT: Local uniformization conjecture. For every valuation ν of K/k, there exists a projective model M of K/k on which the foliation $\mathcal{F}$ is log-final at the center of...

conjecture | open | MathDB 页面 | 原文线索

V2-024880 | MATHDB-334703

Iitaka-Viehweg conjecture for algebraic fiber spaces

摘录 / EXCERPT: Iitaka-Viehweg conjecture. The Kodaira dimension of X satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024881 | MATHDB-334715

Holomorphic convexity conjecture for universal covers of higher-dimensional arrangement orbifolds

摘录 / EXCERPT: Holomorphic convexity question. Is the universal covering space of $\mathcal{M}_N(\mathcal{A})$ holomorphically convex? This is posed as the higher-dimensional analogue of the Shaf...

open_problem | open | MathDB 页面 | 原文线索

V2-024882 | MATHDB-334723

The boundedness conjecture for intrinsic volumes

摘录 / EXCERPT: Let $n\in\mathbb{N}$, $D>0$, and $\kappa\in\mathbb{R}$. For a closed smooth connected Riemannian manifold (M^n,g), write $V_i(M)$ for its intrinsic volumes. The boundedness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-024883 | MATHDB-334724

The convergence conjecture for intrinsic volumes under noncollapsed limits

摘录 / EXCERPT: Let $\{M_l^n\}_{l=1}^{\infty}$ be smooth n-dimensional closed connected Riemannian manifolds with a uniform lower bound on sectional curvature, converging in the Gromov-Hausdorff...

conjecture | open | MathDB 页面 | 原文线索

V2-024884 | MATHDB-334734

Wolff's conjecture on the dimension of Furstenberg sets

摘录 / EXCERPT: Let $0 \leq s \leq 1$. A Furstenberg s-set is a compact set $K \subset \mathbb{R}^{2}$ such that there is a set of unit vectors $S_K \subset S^1$ with positive length and, for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-024885 | MATHDB-334735

Exceptional-set conjecture for orthogonal projections

摘录 / EXCERPT: Fix $0\leq t/2\leq s\leq t\leq 1$, and let $K\subset\mathbb{R}^2$ be a Borel set with $\dim_{\mathrm{H}}K\geq t$. For $e\in S^1$, let $\pi_e:\mathbb{R}^2\to\mathbb{R}$ be the ortho...

conjecture | open | MathDB 页面 | 原文线索

V2-024886 | MATHDB-334742

Numerical abundance conjecture for klt pairs

摘录 / EXCERPT: Let (X,Δ) be a Kawamata log terminal (klt) pair. The numerical abundance conjecture asserts that the numerical and Iitaka dimensions of the log canonical divisor coincide: .....

conjecture | open | MathDB 页面 | 原文线索

V2-024887 | MATHDB-334762

Exponential concentration implies a Poincaré inequality under non-negative Ricci curvature

摘录 / EXCERPT: Exponential-concentration Poincaré conjecture. There exists a constant $C(M)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024888 | MATHDB-334780

The strong (B) property conjecture for volume intersections with the Euclidean ball

摘录 / EXCERPT: Strong (B) property conjecture. The function $\phi(t)$ is log-concave in t; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-024889 | MATHDB-334791

Local symmetry conjecture for finite-volume manifolds satisfying (P)

摘录 / EXCERPT: Let (N,g) be a finite-volume Riemannian manifold satisfying property (P), meaning that every geodesic in its universal covering has rank at least two. Local symmetry conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-024890 | MATHDB-334798

Densely elliptic rational approximability for Euclidean spaces

摘录 / EXCERPT: Densely elliptic rational approximability for Euclidean spaces. There exists an elliptic rational set $X\subset\mathbb{R}^k$ that is dense in $\mathbb{R}^k$.

conjecture | open | MathDB 页面 | 原文线索

V2-024891 | MATHDB-334799

Densely 2-hyperbolic rational approximability for unit spheres

摘录 / EXCERPT: Densely 2-hyperbolic rational approximability for unit spheres. There exists a 2-hyperbolic rational set $X\subset S^k$ that is dense in S^k; equivalently, there exists a $2r$-...

conjecture | open | MathDB 页面 | 原文线索

V2-024892 | MATHDB-334836

Higher-dimensional Shokurov-type conjecture for standard conic bundles

摘录 / EXCERPT: Let $\pi \colon X \to S$ be an n-dimensional standard conic bundle with $n \ge 3$, and let Δ denote the discriminant divisor of π. Higher-dimensional discriminant conj...

conjecture | open | MathDB 页面 | 原文线索

V2-024893 | MATHDB-334838

The projectivity conjecture for fundamental groups of compact Kähler manifolds

摘录 / EXCERPT: Projectivity conjecture. The fundamental group $cpi_1(X)$ is projective, meaning that there exists a projective manifold M such that

conjecture | open | MathDB 页面 | 原文线索

V2-024894 | MATHDB-334839

The bimeromorphic algebraic approximation conjecture for non-uniruled compact Kähler manifolds

摘录 / EXCERPT: Bimeromorphic algebraic approximation conjecture. There exists a bimeromorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-024895 | MATHDB-334852

Mirror Landau-Ginzburg model conjecture for quasi-Fano manifolds

摘录 / EXCERPT: Let X be a quasi-Fano n-fold and let $Z\in\lvert-K_X\rvert$ be a smooth anti-canonical divisor. A Landau-Ginzburg model is a pair (Y,W) consisting of a Kähler manifold Y an...

conjecture | open | MathDB 页面 | 原文线索

V2-024896 | MATHDB-334866

Neumann's arc-count conjecture for non-Desarguesian projective planes

摘录 / EXCERPT: Neumann's arc-count conjecture. The following equivalent inequalities should hold: for every finite non-Desarguesian plane,

conjecture | open | MathDB 页面 | 原文线索

V2-024897 | MATHDB-334878

Demailly-Peternell-Schneider conjecture on Albanese maps with nef anticanonical bundle

摘录 / EXCERPT: Let X be a compact Kähler manifold such that $-K_X$ is nef, and let $p:X\rightarrow Y$ be its Albanese map. Demailly-Peternell-Schneider conjecture. The map p is locally trivia...

conjecture | open | MathDB 页面 | 原文线索

V2-024898 | MATHDB-334879

Folk conjecture on Einstein four-manifolds with non-negative sectional curvature

摘录 / EXCERPT: Let (M,g) be a smooth four-dimensional Einstein manifold satisfying ... for a constant $\lambda>0$, and suppose that its sectional curvature is non-negative. The metrics g_0,...

conjecture | open | MathDB 页面 | 原文线索

V2-024899 | MATHDB-334884

Aschenbrenner-Friedl-Wilton's virtual-retract conjecture for fully relatively quasiconvex subgroups

摘录 / EXCERPT: Aschenbrenner-Friedl-Wilton's conjecture. If H is a fully relatively quasiconvex subgroup of π_1M, then H is a virtual retract. In particular, H is separable.

conjecture | open | MathDB 页面 | 原文线索

V2-024900 | MATHDB-334894

Finite asymptotic dimension conjecture for finite-rank coarse median spaces

摘录 / EXCERPT: Finite asymptotic dimension conjecture. Every geodesic coarse median space with finite rank has finite asymptotic dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-024901 | MATHDB-334904

Andersen-Kashaev's conjecture on Teichmüller TQFT knot invariants

摘录 / EXCERPT: Let M be a closed oriented 3-manifold. For any hyperbolic knot $K\thinspace\subset M$, consider fully balanced shaped ideal triangulations X of the complement of K in M and...

conjecture | open | MathDB 页面 | 原文线索

V2-024902 | MATHDB-334906

Akutsu-Kashaev invariant conjecture for hyperbolic knots

摘录 / EXCERPT: Akutsu-Kashaev conjecture. There exists such a family with the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-024903 | MATHDB-334931

The shifted cotangent polyvector-field conjecture

摘录 / EXCERPT: Polyvector-field conjecture. The resulting $\mathbb{P}_{n+1}^{gr}$-algebra structure on $\Gamma\big(X,S(\mathbb{T}_X[-n])\big)$ is equivalent to the $\mathbb{P}_{n+1}^{gr}$-algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-024904 | MATHDB-334932

The filtered shifted symplectic and Lagrangian structure conjecture

摘录 / EXCERPT: Filtered shifted-structure conjecture. There exists a relative n-shifted symplectic structure $\widetilde{\omega}$ on $\widetilde{Y}_f$ over $[\mathbb{A}^1/\mathbb{G}_m]$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-024905 | MATHDB-334933

The Lagrangian-Poisson correspondence conjecture

摘录 / EXCERPT: Lagrangian-Poisson correspondence conjecture. The map $f\mapsto\pi_f$ provides an equivalence between the space of n-shifted Lagrangian morphisms $f:X\to Y$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-024906 | MATHDB-334941

Galkin's birationality conjecture for symplectic rational maps

摘录 / EXCERPT: Let $\mathbf{k}$ be a field of characteristic 0, and let $\varphi:\mathbb{A}^2\dashrightarrow\mathbb{A}^2$ be a dominant rational map. The logarithmic volume form is ... A rational...

conjecture | open | MathDB 页面 | 原文线索

V2-024907 | MATHDB-334942

The logarithmic Milnor K-theory birationality conjecture

摘录 / EXCERPT: Let $\mathbf{k}$ be a field of characteristic 0. For a field extension $F/\mathbf{k}$, write ... A dominant rational map induces pullback on Milnor K-groups. The logarithmic Miln...

conjecture | open | MathDB 页面 | 原文线索

V2-024908 | MATHDB-334954

Zaks-Perles-Wilkes volume conjecture for lattice polytopes

摘录 / EXCERPT: Zaks-Perles-Wilkes volume conjecture. Every k-point d-dimensional lattice polytope P satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024909 | MATHDB-334955

Three-dimensional delta-vector bounds for non-hollow polytopes

摘录 / EXCERPT: Let P be a non-hollow three-dimensional lattice polytope, and let $(1,\delta_1,\delta_2,\delta_3)$ be its δ-vector. Three-dimensional delta-vector conjecture. The inequali...

conjecture | open | MathDB 页面 | 原文线索

V2-024910 | MATHDB-335000

Surgery quasi-isomorphism for partially wrapped Fukaya categories

摘录 / EXCERPT: Let $\Lambda=\bigsqcup_{v\in\Gamma}\Lambda_v$ be a Legendrian submanifold, with $\Gamma=\Gamma^+\cup\Gamma^-$, and let C_v denote the associated co-core disks. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-024911 | MATHDB-335002

The Kähler cone theorem for non-nef canonical classes

摘录 / EXCERPT: Kähler cone conjecture. If K_X is not nef, then there exists a rational curve C such that

conjecture | open | MathDB 页面 | 原文线索

V2-024912 | MATHDB-335003

The Kähler Kawamata-Viehweg vanishing conjecture

摘录 / EXCERPT: Kähler vanishing conjecture. If $q>n- u(L)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-024913 | MATHDB-335004

The modified Kodaira conjecture for compact Kähler manifolds

摘录 / EXCERPT: Modified Kodaira conjecture. There exists a bimeromorphic model

conjecture | open | MathDB 页面 | 原文线索

V2-024914 | MATHDB-335005

The pseudoeffectivity criterion for uniruled compact Kähler manifolds

摘录 / EXCERPT: Pseudoeffectivity-uniruledness conjecture. The canonical class K_X is pseudoeffective if and only if X is not uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-024915 | MATHDB-335015

A positive lower bound for the scalar curvature of non-flat Ricci shrinkers

摘录 / EXCERPT: Let $n\geq 4$ be a positive integer, and let (M,p,g,f) be a Ricci shrinker. Denote its scalar curvature by R, and write $\sup_{x\in M}R(x)$ for its supremum. Uniform scalar-cur...

conjecture | open | MathDB 页面 | 原文线索

V2-024916 | MATHDB-335021

The conjecture on tritangent planes to tropical sextic curves

摘录 / EXCERPT: Let Γ be a tropical sextic curve of genus 4 in $\mathbb{R}^3$. Two tritangent planes to Γ are equivalent when they determine equivalent tangency divisors on...

conjecture | open | MathDB 页面 | 原文线索

V2-024917 | MATHDB-335052

Tangent-tube counting conjecture

摘录 / EXCERPT: Let $B_R\subset{\mathbb R}^n$ be a ball of radius R, and let Ω be a collection of $(R^{1/2},R)$-tubes in B_R with $R^{-1/2}$-separated directions. Suppose the tubes are...

conjecture | open | MathDB 页面 | 原文线索

V2-024918 | MATHDB-335066

Tosatti-Weinkove's canonical surgical contraction conjecture for the Chern-Ricci flow

摘录 / EXCERPT: Tosatti-Weinkove's canonical surgical contraction conjecture. There exists a smooth maximal solution ω_t of the Chern-Ricci flow on Y for $t\in(T,T_Y)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-024919 | MATHDB-335068

Uniqueness conjecture for finite-radius Bradlow vortices on a flat disc

摘录 / EXCERPT: Bradlow vortex uniqueness conjecture. The only solution satisfying the Bradlow equation on $\mathbb{D}^2$ with $u(R)=0$ is the axially symmetric solution above, with all vortex pos...

conjecture | open | MathDB 页面 | 原文线索

V2-024920 | MATHDB-335071

Equal-separation converse conjecture for planar point sets

摘录 / EXCERPT: Let X be a set of N points, let K be the number of parts in an equal separation, and let $d(Y)$ denote the minimum degree of a curve through a subset $Y\subseteq X$. Equal-se...

conjecture | open | MathDB 页面 | 原文线索

V2-024921 | MATHDB-335089

The transition-value conjecture for odd vertical-strip partitions

摘录 / EXCERPT: Transition-value conjecture. For any odd integer $k\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-024922 | MATHDB-335090

The Klee-Wagon conjecture on intersections of congruent balls

摘录 / EXCERPT: Let ${\bf p}=(p_1,\ldots,p_N)$ and ${\bf q}=(q_1,\ldots,q_N)$ be configurations of N points in ${\mathbb E}^d$. The configuration ${\bf q}$ is a contraction of ${\bf p}$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-024923 | MATHDB-335133

Gukov-Putrov-Vafa conjecture on the superconformal-index factorization

摘录 / EXCERPT: Index-factorization conjecture. The index satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-024924 | MATHDB-335136

The hemisphere volume conjecture for metric balls

摘录 / EXCERPT: Let M be a complete Riemannian manifold of dimension n, let $B(p,R)$ be a metric ball, and let $inj(M)$ denote the injectivity radius. A hemisphere of intrinsic radius R is a...

conjecture | open | MathDB 页面 | 原文线索

V2-024925 | MATHDB-335156

Pseudo-effectivity of the second Chern class for strictly nef terminal threefolds

摘录 / EXCERPT: Pseudo-effectivity conjecture. The class $c_2(X)$ is pseudo-effective.

conjecture | open | MathDB 页面 | 原文线索

V2-024926 | MATHDB-335157

Preservation of the F-bound under flips

摘录 / EXCERPT: Flip F-bound conjecture. If $F(Y)\leq 24$, then

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V2-024927 | MATHDB-335173

The spherical-cap simplex conjecture

摘录 / EXCERPT: Let $\mathbb S^{n-1}$ be the unit sphere, and consider $N=n+1$ spherical caps of radius R on it. The spherical-cap simplex conjecture. For every $R>0$, the maximal area of the un...

conjecture | open | MathDB 页面 | 原文线索

V2-024928 | MATHDB-335174

The Gaussian simplex measure conjecture

摘录 / EXCERPT: Let $B_0(r)$ be the ball of radius r centered at the origin, and let S range over simplices containing this ball. Let the Gaussian measure have density proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-024929 | MATHDB-335256

Thin monodromy conjecture for invariant isotropic planes

摘录 / EXCERPT: Thin monodromy conjecture. There exists a closed invariant subset $\mathcal{M'}$ in $\mathcal{M}$ such that every 2-plane P in $\mathcal{M'}$ intersects the 2-plane...

conjecture | open | MathDB 页面 | 原文线索

V2-024930 | MATHDB-335265

Two-annulus conjecture for linked horizontal geodesics in Berger spheres

摘录 / EXCERPT: Two-annulus conjecture. There are exactly two embedded minimal annuli bounded by h_1 and h_2; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-024931 | MATHDB-335266

Tilted-unduloid conjecture for the spherical-helicoid deformation

摘录 / EXCERPT: Tilted-unduloid conjecture. For every $\alpha\in I\setminus\{\alpha_0\}$, there exists $\Sigma^\alpha\in\mathcal{M}(h_1,h_2^\alpha)$ such that $\widetilde{\Sigma^\alpha}$ extends t...

conjecture | open | MathDB 页面 | 原文线索

V2-024932 | MATHDB-335274

Same-level sign-change reducibility conjecture for involutions

摘录 / EXCERPT: Let p be a positive integer, let $m\geq 2$ be even, and let $R\in SO(p)$ be an involution of level m. A sign-change matrix is an element of the relevant sign-change subgroup, a...

conjecture | open | MathDB 页面 | 原文线索

V2-024933 | MATHDB-335275

The wrapped microlocal sheaves conjecture for Weinstein manifolds

摘录 / EXCERPT: Wrapped microlocal sheaves conjecture. There is a cosheaf of ... , on $$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-024934 | MATHDB-335286

Conjecture on the uniqueness of the type of Bach-flat Kähler metrics

摘录 / EXCERPT: Let (M,J) be a compact complex surface. A Bach-flat Kähler metric is a Kähler metric compatible with J whose Bach tensor vanishes. The type of such a metric is one of the types...

conjecture | open | MathDB 页面 | 原文线索

V2-024935 | MATHDB-335287

Conjecture on Calabi-energy minimization by Bach-flat Kähler metrics

摘录 / EXCERPT: Let (M,J) be a compact complex surface, and let g be a Bach-flat Kähler metric, meaning a Kähler metric compatible with J whose Bach tensor vanishes. Let $\mathcal{C}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-024936 | MATHDB-335289

Akbulut-Kirby conjecture on concordance from homeomorphic 0-surgeries

摘录 / EXCERPT: Let K and K' be knots, and let $S^3_0(K)$ and $S^3_0(K')$ denote their 0-surgeries. Akbulut-Kirby conjecture. If K and K' have homeomorphic 0-surgeries, then K and K' a...

conjecture | open | MathDB 页面 | 原文线索

V2-024937 | MATHDB-335290

Revised Akbulut-Kirby conjecture on concordance up to knot orientation

摘录 / EXCERPT: Let K and K' be knots, and write $S^3_0(K)$ and $S^3_0(K')$ for their 0-surgeries. Reversing the orientation of either knot means replacing it by the same knot with the opposit...

conjecture | open | MathDB 页面 | 原文线索

V2-024938 | MATHDB-335291

Abe's conjecture on concordance from diffeomorphic 0-traces

摘录 / EXCERPT: Let K and K' be knots. For a knot K, its 0-trace $X_0(K)$ is the 4-manifold obtained by attaching a 0-framed 2-handle to B^4 along K. Abe's conjecture. If K and K' ha...

conjecture | open | MathDB 页面 | 原文线索

V2-024939 | MATHDB-335293

Realization conjecture for the discrete Dirac nullity of triangulated spheres

摘录 / EXCERPT: Let an abstract triangulated sphere be realized in $\mathbb{R}^3$, and let D denote its discrete Dirac operator. A triangulated-sphere realization conjecture. Ever...

conjecture | open | MathDB 页面 | 原文线索

V2-024940 | MATHDB-335334

Precise Gevrey regularity of invariant manifolds at parabolic infinity in restricted three-body problems

摘录 / EXCERPT: In the Sitnikov problem, let $e\in(0,1)$ denote the eccentricity, and in the restricted planar three-body problem (RPTBP), let $\mu\in(0,1/2]$ denote the mass parameter. An invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-024941 | MATHDB-335379

Total geodesy conjecture for compact splitting submanifolds of rank-2 type-I quotients

摘录 / EXCERPT: Rank-2 type-I total geodesy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-024942 | MATHDB-335380

Total geodesy conjecture for compact splitting submanifolds of type-IV quotients

摘录 / EXCERPT: Type-IV total geodesy conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-024943 | MATHDB-335396

Chern's affine Bernstein conjecture

摘录 / EXCERPT: Let $u\colon\mathbb{R}^2\to\mathbb{R}$ be smooth and locally uniformly convex, and suppose its graph is an affine maximal graph, meaning that u satisfies the affine maximal surfa...

conjecture | open | MathDB 页面 | 原文线索

V2-024944 | MATHDB-335397

Trudinger-Wang affine Bernstein conjecture

摘录 / EXCERPT: Trudinger-Wang's conjecture. If $n\leq 9$, then u is an elliptic quadratic polynomial. If $n\geq 10$, then there exists a smooth, locally uniformly convex solution of the equatio...

conjecture | open | MathDB 页面 | 原文线索

V2-024945 | MATHDB-335398

Trudinger-Wang conjecture on the affine mean-convex first boundary value problem

摘录 / EXCERPT: Trudinger-Wang's conjecture. The first boundary value problem has a unique smooth solution u if ϕ is affine mean convex. This conjecture concerns existence and uniqueness fo...

conjecture | open | MathDB 页面 | 原文线索

V2-024946 | MATHDB-335440

Positivity conjecture for tropical slab functions

摘录 / EXCERPT: Let X satisfy the generic condition assumed in the construction. For a primitive relative class γ, let l_{γ} be an affine line labeled by γ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-024947 | MATHDB-335452

Sormani's scalar curvature conjecture for intrinsic flat limits

摘录 / EXCERPT: Let M_j^3 be a sequence of three-dimensional Riemannian manifolds. Write ${\rm Scal}_j$ for the scalar curvature, $\operatorname{Vol}(M_j)$ for their volume, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024948 | MATHDB-335475

Stable-pairs descendent correspondence for relative 3-folds

摘录 / EXCERPT: Let X/D be a relative 3-fold, let β be a curve class, let μ be a relative boundary condition, and let $\gamma_i\in H^*(X)$. The barred relative descendent insertion i...

conjecture | open | MathDB 页面 | 原文线索

V2-024949 | MATHDB-335476

Virasoro-constraint conjecture for stable pairs on projective 3-space

摘录 / EXCERPT: Let $\mathbb D^+$ be the descendent operator space, let $\mathcal L_k$ be the operators defined in the source, let $\mathsf D\in\mathbb D^+$, and let $d\mathsf L$ range over curve...

conjecture | open | MathDB 页面 | 原文线索

V2-024950 | MATHDB-335478

Maximal systolic ratio conjecture for genus two abelian differentials

摘录 / EXCERPT: Maximal systolic ratio conjecture. The supremum of the systolic ratio over genus two surfaces (X,d_{ω}) equals

conjecture | open | MathDB 页面 | 原文线索

V2-024951 | MATHDB-335517

Arthur's parametrization conjecture for the discrete spectrum of metaplectic symplectic covers

摘录 / EXCERPT: Fix a distinguished character $\overline{\chi}^0$. Let $\mathcal{A}_\epsilon(\overline{\operatorname{Sp}}_{2r}(\mathbb A))$ denote the relevant discrete-spectrum representation spa...

conjecture | open | MathDB 页面 | 原文线索

V2-024952 | MATHDB-335524

The manifold inheritance conjecture for isospectral lens orbifolds

摘录 / EXCERPT: The manifold inheritance conjecture. If L and L' are p-isospectral for some $0<p\leq n-1$ and L is a lens space, then L' is also a lens space.

conjecture | open | MathDB 页面 | 原文线索

V2-024953 | MATHDB-335525

The classification conjecture for five-dimensional 2-isospectral lens orbifolds

摘录 / EXCERPT: The five-dimensional $\{2\}$-isospectrality conjecture. The families of non-isometric $\{2\}$-isospectral 5-dimensional lens orbifolds are precisely given by pairs

conjecture | open | MathDB 页面 | 原文线索

V2-024954 | MATHDB-335526

Divisibility conjecture for seven-dimensional isospectral lens orbifolds

摘录 / EXCERPT: The divisibility conjecture. If $L(q;s)$ and $L(q;s')$ form a pair of non-isometric $\{2\}$-isospectral 7-dimensional lens orbifolds, then 3 divides q; respectively, if they...

conjecture | open | MathDB 页面 | 原文线索

V2-024955 | MATHDB-335534

The geodesic noncontainment conjecture for the compactified Jacobian locus

摘录 / EXCERPT: Let J_g be the Jacobian locus in $\mathcal A_g$ and let $\overline J_g$ be its closure. Geodesic noncontainment conjecture. For a sufficiently large integer g, there does not e...

conjecture | open | MathDB 页面 | 原文线索

V2-024956 | MATHDB-335536

The geodesic noncontainment conjecture for universal Jacobian loci

摘录 / EXCERPT: Let $J_{g,h}$ be the universal Jacobian locus in $\mathcal A_{g,h}$, equipped with the Riemannian metric $ds^2_{g,h;A,B}$. Universal geodesic noncontainment conjecture. For a suffi...

conjecture | open | MathDB 页面 | 原文线索

V2-024957 | MATHDB-335567

Generalized tautological conjecture for moduli of hyperkähler manifolds

摘录 / EXCERPT: Let $\mathcal{F}_h$ be the moduli space under consideration of hyperkähler manifolds, and let ${\rm NL}^\ast(\mathcal{F}_h)$ be the ring generated by Noether-Lefschetz cycles. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024958 | MATHDB-335606

Shapiro's conjecture on rational coordinates for log-canonical Poisson brackets

摘录 / EXCERPT: Shapiro's conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-024959 | MATHDB-335643

Mukai's generalized conjecture for Fano manifolds

摘录 / EXCERPT: Bonavero-Casagrande-Debarre-Druel conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-024960 | MATHDB-335674

Existence of smooth epsilon-flat triangulations of compact Riemannian manifolds

摘录 / EXCERPT: Let N be a compact smooth Riemannian manifold. A triangulation $T:K\to N$ is smooth ϵ-flat when its simplices satisfy the paper's ϵ-flatness condition. Existenc...

conjecture | open | MathDB 页面 | 原文线索

V2-024961 | MATHDB-335675

Density conjecture for boundary generic and traversally generic metrics

摘录 / EXCERPT: Let M be a smooth manifold, let $\mathcal G(M)$ be the space of gradient-type metrics, and let $\mathcal G^\dagger(M)$ and $\mathcal G^\ddagger(M)$ denote the two metric classes...

conjecture | open | MathDB 页面 | 原文线索

V2-024962 | MATHDB-335676

Isotopic boundary genericity conjecture for Riemannian metrics

摘录 / EXCERPT: Let M be an open smooth n-manifold with a Riemannian metric g, and let $\alpha:N\subset M$ be a smooth compact codimension-one submanifold. An isotopy $A:N\times[0,1]\to M$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-024963 | MATHDB-335710

Conjecture on strong quasipositivity and Morse-Smale characteristic foliations

摘录 / EXCERPT: Let (Y,ξ) be a contact 3-manifold and let a transverse link in (Y,ξ) be given. A Seifert surface for the link has a Morse-Smale characteristic foliation with positive sin...

conjecture | open | MathDB 页面 | 原文线索

V2-024964 | MATHDB-335722

The spherical-segment covering conjecture for the unit sphere

摘录 / EXCERPT: Spherical-segment covering conjecture. The total width of the spherical segments is at least π.

conjecture | open | MathDB 页面 | 原文线索

V2-024965 | MATHDB-335723

The spherical Goodman covering conjecture for caps

摘录 / EXCERPT: Spherical Goodman conjecture. The given caps can be covered by a cap of spherical radius

conjecture | open | MathDB 页面 | 原文线索

V2-024966 | MATHDB-335724

Bezdek's annular plank conjecture

摘录 / EXCERPT: Bezdek's conjecture. The total width of the planks is at least 1.

conjecture | open | MathDB 页面 | 原文线索

V2-024967 | MATHDB-335725

Bezdek-Schneider's spherical inradius covering conjecture

摘录 / EXCERPT: Bezdek-Schneider's conjecture. The total inradius of the domains is at least r.

conjecture | open | MathDB 页面 | 原文线索

V2-024968 | MATHDB-335728

Bernardara-Marcolli-Tabuada's noncommutative nilpotence conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a smooth and proper dg category, and let $K_0(\mathcal{A})$ be the Grothendieck group of its category of compact objects. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024969 | MATHDB-335729

Kozlov and Fomenko's nonexistence conjecture for higher-degree integrals on the two-sphere

摘录 / EXCERPT: On the two-dimensional sphere, consider Riemannian metrics whose geodesic flows are integrable by means of an integral of degree $n>4$ in the momenta. Kozlov and Fomenko's conjectu...

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V2-024970 | MATHDB-335761

Conjecture on the epsilon-SYZ and epsilon-mirror maps

摘录 / EXCERPT: ϵ-SYZ map conjecture. The ϵ-SYZ map coincides with the inverse of the ϵ-mirror map.

conjecture | open | MathDB 页面 | 原文线索

V2-024971 | MATHDB-335783

Gardner-Giannopoulos cross-section body conjecture for \(s\)-concave densities

摘录 / EXCERPT: Cross-section body conjecture. If $s\in[-1/n,\infty]$, then $C_p(f)$ is a symmetric convex body. This is the geometric reformulation of the reverse Rényi entropy power norm conject...

conjecture | open | MathDB 页面 | 原文线索

V2-024972 | MATHDB-335824

Kobayashi's conjecture on fundamental groups of positively curved semi-Riemannian manifolds

摘录 / EXCERPT: Let n and q be positive integers with $n \geq 2q$. Let (M,g) be an n-dimensional geodesically complete semi-Riemannian manifold of index q, and suppose its sectional curv...

conjecture | open | MathDB 页面 | 原文线索

V2-024973 | MATHDB-335854

Jakobson-Levitin-Nadirashvili-Nigam-Polterovich conjecture for the Bolza surface

摘录 / EXCERPT: Let B be the Bolza surface, the closed Riemann surface of genus two ... and let $g_B\colon B\to \overline{{\mathbb{C}}}$ be the degree-two meromorphic function $g_B(z,w)=z$. Pull b...

conjecture | open | MathDB 页面 | 原文线索

V2-024974 | MATHDB-335858

The Levi-Hadwiger covering conjecture for convex bodies

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^{n}$ be a convex body with nonempty interior, and let $N(K,L)$ denote the least number of translates of L needed to cover K. A parallelotope is an aff...

conjecture | open | MathDB 页面 | 原文线索

V2-024975 | MATHDB-335871

The Gaussian Brunn-Minkowski conjecture for convex sets containing the origin

摘录 / EXCERPT: Let γ_n be the standard Gaussian measure on $\mathbb{R}^n$, defined by ... For $\lambda\in(0,1)$ and sets $A,B\subset\mathbb{R}^n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-024976 | MATHDB-335885

Poisson commutativity of the conserved quantities for the B_r Q-system

摘录 / EXCERPT: Let $A_1,\dots,A_{2r}$ be coordinates on the phase space of the B_r Q-system, and let B be the signed adjacency matrix of its quiver. Define the Poisson bracket by ... where ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-024977 | MATHDB-335892

Exponential-facet threshold conjecture for volume approximation of the Euclidean ball

摘录 / EXCERPT: Let D_n be the Euclidean unit ball in $\mathbb{R}^n$, let N be a positive integer, and let $P^b_{n,N}$ be a polytope with at most N facets that is best-approximating for...

conjecture | open | MathDB 页面 | 原文线索

V2-024978 | MATHDB-335896

The singularity exact-sequence conjecture for Poisson D-modules

摘录 / EXCERPT: Let $X=\{f=0\}\subseteq\mathbf C^3$ be a surface with finite singular locus $Z=\{s_1,\ldots,s_k\}$, let $j:X\setminus Z\to X$ be the open embedding, and let $M(X)_{\mathrm{ind}}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-024979 | MATHDB-335898

The projective Lichnerowicz-Obata conjecture

摘录 / EXCERPT: Projective Lichnerowicz-Obata conjecture. Either G acts on M by affine transformations, or (M^n,g) is Riemannian with positive constant sectional curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-024980 | MATHDB-335901

The optimal rank-form extension of Yau's conjecture

摘录 / EXCERPT: The optimal rank-form extension of Yau's conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-024981 | MATHDB-335914

Chen's regularity conjecture for K-energy minimizers

摘录 / EXCERPT: Let $(M,[\omega],J)$ be a compact Kähler manifold, let ... and let $\mathcal K$ be the K-energy on Kähler potentials. A $C^{1,\bar 1}$ potential is a weak Kähler potential whose comp...

conjecture | open | MathDB 页面 | 原文线索

V2-024982 | MATHDB-335922

ABPS conjecture for split reductive groups

摘录 / EXCERPT: Let F be a non-archimedean local field, let G be the F-points of a connected reductive split algebraic group over F, and let $\mathfrak{s}$ be a Bernstein component. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-024983 | MATHDB-335923

La Nave-Tian's Gromov-Hausdorff convergence conjecture for the continuity method

摘录 / EXCERPT: La Nave-Tian's conjecture. As $t\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-024984 | MATHDB-335941

Orthogonality conjecture for low-multiplicity spherical zone coverings

摘录 / EXCERPT: Let $d\geq 3$ and $n\geq 1$ be integers, and let S^{d-1} be covered by the union of n congruent spherical zones. Suppose each point of S^2 belongs to the interior of at most...

conjecture | open | MathDB 页面 | 原文线索

V2-024985 | MATHDB-335954

Polterovich's Poisson bracket invariant conjecture

摘录 / EXCERPT: Polterovich's conjecture. There exists a constant $C=C(M,\omega)>0$, depending only on the symplectic manifold, such that

conjecture | open | MathDB 页面 | 原文线索

V2-024986 | MATHDB-335962

The abundance conjecture for compact complex manifolds

摘录 / EXCERPT: Let X be a compact complex manifold in class $\mathcal C$, meaning that X is bimeromorphic to a compact Kähler manifold. Let K_X denote its canonical bundle. A line bundle is...

conjecture | open | MathDB 页面 | 原文线索

V2-024987 | MATHDB-335963

The positive total scalar curvature conjecture

摘录 / EXCERPT: Let X be a compact complex manifold in class $\mathcal C$, meaning that X is bimeromorphic to a compact Kähler manifold. A Gauduchon metric is a Hermitian metric whose associat...

conjecture | open | MathDB 页面 | 原文线索

V2-024988 | MATHDB-335964

The Kähler uniruledness and positive scalar curvature conjecture

摘录 / EXCERPT: Let X be a compact Kähler manifold. Say that X is uniruled if it is covered by rational curves, and let a smooth Kähler metric mean a smooth positive Kähler form on X. Kähler...

conjecture | open | MathDB 页面 | 原文线索

V2-024989 | MATHDB-335989

Collapsing-limit conjecture for hyperkähler fibrations

摘录 / EXCERPT: Let $(M^m,\tilde\omega_t)$ be a collapsing family as above, with $[\alpha_0]=f^*[\omega_N]$ for a holomorphic surjective map with connected fibers $f:M\to N$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-024990 | MATHDB-335998

The Ricci-flat projective-induction conjecture

摘录 / EXCERPT: A Ricci-flat metric is a Kähler metric whose Ricci curvature vanishes, and a metric is projectively induced if it admits a Kähler immersion into some projective space...

conjecture | open | MathDB 页面 | 原文线索

V2-024991 | MATHDB-335999

The Ricci-flat TYZ-coefficient conjecture

摘录 / EXCERPT: Let (M,g) be a Kähler metric on an n-dimensional complex manifold, and let a_j denote the coefficients associated to g in the Tian-Yau-Zelditch expansion. The Ricci-flat TY...

conjecture | open | MathDB 页面 | 原文线索

V2-024992 | MATHDB-336003

Conjecture on Springer fiber graph duality for special orbits in odd orthogonal and symplectic Lie algebras

摘录 / EXCERPT: Let $\mathbf{k}$ be the ground field, and let $N\in\operatorname{Lie}(SO_{2n+1})(\mathbf{k})$ and $N'\in\operatorname{Lie}(Sp_{2n})(\mathbf{k})$ lie in special nilpotent orbits rel...

open_problem | open | MathDB 页面 | 原文线索

V2-024993 | MATHDB-336041

Minimal skeleton conjecture for stacky dlt compactifications

摘录 / EXCERPT: Minimal skeleton conjecture. The homeomorphism type of the dual complex of $(\mathcal{X},\sum_{1\le i\le s}\mathcal{D}_{i})$ does not depend on the choice of such compactification.

conjecture | open | MathDB 页面 | 原文线索

V2-024994 | MATHDB-336043

Conjectural Mori chamber decomposition of the one-point blow-up of a Grassmannian

摘录 / EXCERPT: Let $\mathbb G(r,n)_1$ be the blow-up of the Grassmannian $\mathbb G(r,n)$ at one point, let H denote the pullback of the hyperplane class, and let E denote the exceptional div...

conjecture | open | MathDB 页面 | 原文线索

V2-024995 | MATHDB-336058

The one-point deletion conjecture for planar point configurations

摘录 / EXCERPT: One-point deletion conjecture. There exist $p_i\in P$ and $q_i\in Q$, together with some $A\in O(2)$ and $\vec{t}\in\mathbb{R}^2$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-024996 | MATHDB-336077

Maulik-Pandharipande determination conjecture for quintic threefold invariants

摘录 / EXCERPT: Let Q be the quintic threefold in $\boldsymbol{^4}$. Let $N_{g,d}$ denote its Gromov-Witten invariants, and let the degeneration formula for $(V,W)=(\mathbb{P}^4,Q)$ relate the r...

conjecture | open | MathDB 页面 | 原文线索

V2-024997 | MATHDB-336078

Degree-bound conjecture for recursive determination of quintic Gromov-Witten invariants

摘录 / EXCERPT: Let $S_{g,d}$ and $S'_{g,d}$ be the indexing sets used in the paper, let $B_{g,d}(\rho)$ and C_ρ be the corresponding terms and coefficients in the degeneration equations, and...

conjecture | open | MathDB 页面 | 原文线索

V2-024998 | MATHDB-336087

Higher-dimensional geometric quantization conjecture for almost toric manifolds

摘录 / EXCERPT: Let M be a $2m$-dimensional closed almost toric manifold with singularities of any corank and of Williamson type $(k_e,0,k_f)$, where $k_f<2$. Let BS_r and BS_{f-r} be the im...

conjecture | open | MathDB 页面 | 原文线索

V2-024999 | MATHDB-336088

Geometric quantization conjecture for non-degenerate integrable systems with hyperbolic components

摘录 / EXCERPT: Let M be a closed symplectic manifold equipped with a non-degenerate integrable system whose singularities have any corank and Williamson type (k_e,k_h,h_f), with $k_h<2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-025000 | MATHDB-336092

Barthe's symmetric Gaussian surface-area minimizer conjecture

摘录 / EXCERPT: Barthe's conjecture. If Ω minimizes this quantity, then, after a rotation, there exist $r>0$ and $0\leq k\leq n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025001 | MATHDB-336118

Intrinsic flat almost rigidity of the hyperbolic positive mass theorem

摘录 / EXCERPT: Let M_j^m be a sequence of asymptotically hyperbolic Riemannian manifolds with scalar curvature $R\geq -m(m-1)$ and mass tending to zero. Assume that M_j^m contains no closed i...

conjecture | open | MathDB 页面 | 原文线索

V2-025002 | MATHDB-336149

Conjecture on the sharp comparison of the u_{d53d^n}-metric and hyperbolic metric

摘录 / EXCERPT: Let $\mathbb{B}^n$ be the unit ball in $\mathbb{R}^n$, and let $u_{\mathbb{B}^n}$ and $\rho_{\mathbb{B}^n}$ denote the u_D-metric and hyperbolic metric on $\mathbb{B}^n$, respect...

conjecture | open | MathDB 页面 | 原文线索

V2-025003 | MATHDB-336160

Rigidity conjecture for equality in the higher-form Hersch-Payne-Schiffer inequality

摘录 / EXCERPT: Let M be a compact oriented $(2p+2)$-dimensional Riemannian manifold with nonempty boundary. For $m>0$, write ... for the higher-form Hersch-Payne-Schiffer inequality, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-025004 | MATHDB-336190

Generic conformal field theoretic Hodge-elliptic genus conjecture

摘录 / EXCERPT: Generic Hodge-elliptic genus conjecture. The two genera agree:

conjecture | open | MathDB 页面 | 原文线索

V2-025005 | MATHDB-336191

Mirror symmetry conjecture for chiral Hodge-elliptic genera

摘录 / EXCERPT: Mirror symmetry conjecture. The corresponding chiral Hodge-elliptic genera obey

conjecture | open | MathDB 页面 | 原文线索

V2-025006 | MATHDB-336192

SHGH conjecture for equal multiplicities at general points

摘录 / EXCERPT: Let $Z=p_1+\cdots+p_s \subset \mathbb P^2$ be a union of general points, where either s is a square or $s>8$, and let $I(mZ)$ denote the degree-m fat-point ideal. SHGH conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-025007 | MATHDB-336219

Spectral rigidity conjecture for spherically symmetric manifolds

摘录 / EXCERPT: Spectral rigidity conjecture. Under certain geometric hypotheses, a spherically symmetric manifold is uniquely determined by its length spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-025008 | MATHDB-336221

Glover-Homer conjecture for real and complex generalized flag manifolds

摘录 / EXCERPT: Let $n_1,\ldots,n_k$ be distinct positive integers, let $n=n_1+\cdots+n_k$, and for $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$ define ... where $U_{\mathbb{R}}(n)=O(n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025009 | MATHDB-336228

The q conjecture for second largest complete arcs

摘录 / EXCERPT: The $\sqrt{q}$ conjecture. The second largest complete arc in ${\mathrm{PG}}(2,q)$ has size

conjecture | open | MathDB 页面 | 原文线索

V2-025010 | MATHDB-336238

Existence of purely axial torsion-wave solutions in quadratic metric-affine gravity

摘录 / EXCERPT: In quadratic metric-affine gravity, consider metric-compatible spacetimes with purely axial torsion waves and the field equations obtained by varying the action with respect to the...

conjecture | open | MathDB 页面 | 原文线索

V2-025011 | MATHDB-336239

Purely axial torsion pp-wave conjecture in quadratic metric-affine gravity

摘录 / EXCERPT: Consider quadratic metric-affine gravity with a pp-metric and explicitly given purely axial torsion whose Ricci curvature is parallel. Purely axial torsion pp-wave conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-025012 | MATHDB-336271

Rubinstein's conjecture on strongly irreducible Heegaard splittings

摘录 / EXCERPT: Rubinstein's conjecture. H is either isotopic to a minimal surface or isotopic to the oriented double cover of a non-orientable minimal surface with a vertical handle attached.

conjecture | open | MathDB 页面 | 原文线索

V2-025013 | MATHDB-336303

Symplectic singularities conjecture for modified complex symplectic quotients

摘录 / EXCERPT: Symplectic singularities conjecture. For all (V,G), the modified complex symplectic quotient has symplectic singularities and is graded Gorenstein.

conjecture | open | MathDB 页面 | 原文线索

V2-025014 | MATHDB-336305

Completion conjecture for filtered A_{∞} categories

摘录 / EXCERPT: Let $\mathscr C$ be a strict filtered A_{∞} category, and let $(\frak c_i)$ be a Cauchy sequence of objects of $\mathscr C$ with respect to the Hofer distance. Suppose the i...

conjecture | open | MathDB 页面 | 原文线索

V2-025015 | MATHDB-336325

The converse to the cylinder-stopped lemma

摘录 / EXCERPT: Let Y be a fully stopped Liouville sector, and let F be a Liouville manifold. Converse conjecture. If Y is fully stopped, then Y is deformation equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-025016 | MATHDB-336349

The Lagarias--Wang periodic tiling conjecture

摘录 / EXCERPT: Let V be a finite tile in $\mathbb{Z}^n$. A tiling by translations is fully periodic if its translation set has a finite-index period subgroup. Lagarias--Wang conjecture. If V...

conjecture | open | MathDB 页面 | 原文线索

V2-025017 | MATHDB-336350

The mixed-radius Lee sphere tiling conjecture

摘录 / EXCERPT: A Lee sphere in $\mathbb{R}^n$ is a cross-polytope-type region defined by a Lee radius, and a tiling means a covering of $\mathbb{R}^n$ by translates of such spheres with disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-025018 | MATHDB-336368

Bondal-Orlov-Van den Bergh conjecture on derived equivalence of crepant resolutions and NCCRs

摘录 / EXCERPT: Let R be a Gorenstein $\operatorname{\mathbb{C}}$-algebra. A crepant resolution of R is a crepant resolution of the corresponding singularity, and an NCCR of R is a non-commu...

conjecture | open | MathDB 页面 | 原文线索

V2-025019 | MATHDB-336378

Metric Vietoris-Rips thickening conjecture for spheres

摘录 / EXCERPT: Let S^n be the n-sphere, let r_n be the scale under consideration, and let $\mathrm{VR}^m_{\leq}$, $\mathrm{VR}^m_{<}$, $\mathrm{VR}_{<}$, and $\mathrm{VR}_{\leq}$ denote the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-025020 | MATHDB-336379

Metric Latschev theorem for infinite metric spaces

摘录 / EXCERPT: Let M be a closed Riemannian manifold, and let d_{GH} denote the Gromov-Hausdorff distance. For sufficiently small r, the metric Vietoris-Rips thickening...

conjecture | open | MathDB 页面 | 原文线索

V2-025021 | MATHDB-336391

Finite similar shapes conjecture for MICP representable sets

摘录 / EXCERPT: Let $S\subseteq\mathbb{R}^n$ be MICP representable, so that ... where $C\subseteq\mathbb{R}^d$ and each $A_{\bm{z}}\subseteq\mathbb{R}^d$ is a projection of a closed convex set. A fi...

conjecture | open | MathDB 页面 | 原文线索

V2-025022 | MATHDB-336406

Conjecture on higher normalized eigenvalues of the projective plane

摘录 / EXCERPT: Let $\Lambda_k(\mathbb{RP}^2)$ denote the supremum of the normalized k-th Laplace eigenvalue over metrics on the real projective plane. A sequence of metrics is said to degenerat...

conjecture | open | MathDB 页面 | 原文线索

V2-025023 | MATHDB-336412

Four-hyperplane partition conjecture in four-dimensional space

摘录 / EXCERPT: Let P be a finite set of points in $\mathbb{R}^4$. Four-hyperplane partition conjecture. There exist four hyperplanes such that every one of the resulting 16 open regions conta...

conjecture | open | MathDB 页面 | 原文线索

V2-025024 | MATHDB-336414

Higher-dimensional ray-shooting depth conjecture

摘录 / EXCERPT: For a finite set P of n points in $\mathbb{R}^d$, define ray-shooting depth as the maximum, over points q, of the minimum number of $(d-1)$-simplices spanned by P met by a...

conjecture | open | MathDB 页面 | 原文线索

V2-025025 | MATHDB-336419

Monotonicity conjecture for Schrödinger operators under geometric flows

摘录 / EXCERPT: Let S be a Schrödinger operator evolving with a geometric flow $\mathcal{F}$. Let Δ denote the Laplace-Beltrami operator, and suppose that the eigenvalues of $-\Delta$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-025026 | MATHDB-336420

Unitary-equivalence conjecture for spectral monotonicity under geometric flows

摘录 / EXCERPT: Let S be a Schrödinger operator evolving with a geometric flow $\mathcal{F}$, and suppose that its eigenvalues are non-decreasing under $\mathcal{F}$. Let T be an operator unit...

conjecture | open | MathDB 页面 | 原文线索

V2-025027 | MATHDB-336464

Ribbon equivalence conjecture for the even induction functor

摘录 / EXCERPT: Let $\mathrm N$ be the parameter specifying the number of symplectic fermion pairs, let β be the scalar used in the ribbon structure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025028 | MATHDB-336473

Global convergence conjecture for the normalized Kähler-Ricci flow on minimal models of general type

摘录 / EXCERPT: Let X be a smooth minimal model of general type, and let the normalized Kähler-Ricci flow be given by ... Let X_{can} denote the canonical model of X, and let ω_{KE} be...

conjecture | open | MathDB 页面 | 原文线索

V2-025029 | MATHDB-336483

The polynomial growth conjecture for simple closed geodesics on nonorientable surfaces

摘录 / EXCERPT: Simple-closed-geodesic growth conjecture. There exists a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025030 | MATHDB-336485

The closure-equality conjecture for mapping-class-group orbit closures

摘录 / EXCERPT: Closure-equality conjecture. The inclusions in Theorems and are equalities, except when λ is the bounding geodesic of the genus-three closed surface.

conjecture | open | MathDB 页面 | 原文线索

V2-025031 | MATHDB-336486

Beauville-Bogomolov decomposition conjecture for manifolds with nef anticanonical class

摘录 / EXCERPT: Let X be a compact Kähler manifold with nef anticanonical class. Denote its universal cover by $\widetilde{X}$. Decomposition conjecture. The universal cover decomposes as a prod...

conjecture | open | MathDB 页面 | 原文线索

V2-025032 | MATHDB-336493

Wilhelm's conjecture on the fiber dimension of positively curved submersions

摘录 / EXCERPT: Wilhelm's conjecture. Every such submersion satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-025033 | MATHDB-336514

Uniform diameter bound for normalized metrics on Lefschetz fibres

摘录 / EXCERPT: Uniform diameter bound. There is a constant C such that, independently of t and the fibre X_y,

conjecture | open | MathDB 页面 | 原文线索

V2-025034 | MATHDB-336537

Maximal valuation conjecture for sufficiently general valuations

摘录 / EXCERPT: Let ξ be a sufficiently general valuation parameter and let $t\ge 8+1/36$. Let $\operatorname{ord}_{\xi,t}$ denote the valuation written in the source, and let maximality mean...

conjecture | open | MathDB 页面 | 原文线索

V2-025035 | MATHDB-336539

Maximal Nazarov-Sodin constant conjecture for random plane waves and arithmetic random waves

摘录 / EXCERPT: Maximal Nazarov-Sodin constant conjecture. For every $\mu\in\mathcal{P}_{\text{symm}}$ that is a weak-$*$ limit of $\{\mu_n\}$, the maximal value $d_{\max}$ is uniquely attained by...

conjecture | open | MathDB 页面 | 原文线索

V2-025036 | MATHDB-336562

Mixon's divisibility conjecture for complex equiangular tight frames

摘录 / EXCERPT: Mixon's divisibility conjecture. There exists an n-vector equiangular tight frame in $\mathbb{C}^d$ only if one of these quantities divides the product of the other two.

conjecture | open | MathDB 页面 | 原文线索

V2-025037 | MATHDB-336589

Rational factorization for tropical polynomials

摘录 / EXCERPT: A tropical polynomial is a tropical polynomial in any number of variables, and it is rationally factorizable when it admits the rational factorization considered in the paper. Rati...

conjecture | open | MathDB 页面 | 原文线索

V2-025038 | MATHDB-336604

Minkowski's lattice product conjecture

摘录 / EXCERPT: Minkowski's conjecture. The stated inequality holds for every such orthonormal basis, lattice, and vector t. This is a famous equivalent formulation of Minkowski's conjecture. It...

conjecture | open | MathDB 页面 | 原文线索

V2-025039 | MATHDB-336659

Donaldson-Sun canonicity conjecture for metric tangent cones

摘录 / EXCERPT: Let M_∞ be a Gromov-Hausdorff limit of Kähler-Einstein Fano manifolds, let $o\in M_\infty$, and let $C=C_oM_\infty$ be a metric tangent cone at o. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025040 | MATHDB-336687

Quasi-Möbius homeomorphisms of Morse boundaries are induced by quasi-isometries

摘录 / EXCERPT: Let X and Y be proper, cocompact CAT(0) spaces, and let $f:\partial_*X\to\partial_*Y$ be a homeomorphism, where ∂_*X and ∂_*Y denote their Morse boundaries. A...

conjecture | open | MathDB 页面 | 原文线索

V2-025041 | MATHDB-336724

Conjecture on norm classes of maximum scattered subspaces in the new family

摘录 / EXCERPT: Norm-class conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-025042 | MATHDB-336727

Foscolo-Haskins conjecture on cohomogeneity-one nearly Kähler six-manifolds

摘录 / EXCERPT: A compact nearly Kähler manifold is cohomogeneity one if it is equipped with an isometric action of a compact Lie group whose generic orbits have codimension one. Foscolo-Haskins c...

conjecture | open | MathDB 页面 | 原文线索

V2-025043 | MATHDB-336729

Stable Property R conjecture for links

摘录 / EXCERPT: Let U_n be the n-component zero-framed unlink in S^3. A link L has Stable Property R if, for some integers r and m, the split union $L\sqcup U_r$ is handleslide equival...

conjecture | open | MathDB 页面 | 原文线索

V2-025044 | MATHDB-336734

The mutation-deformation correspondence for Fano polygons and del Pezzo surfaces

摘录 / EXCERPT: Mutation-deformation correspondence. There is a canonical bijection between the set of mutation equivalence classes of Fano polygons and deformation families of $\mathbb{Q}$-Gorens...

conjecture | open | MathDB 页面 | 原文线索

V2-025045 | MATHDB-336743

Uniqueness and shooting classification of non-collapsed steady solitons on line bundles

摘录 / EXCERPT: Let L_k be a line bundle over $(\hat{M},\omega)$, with $k>p(\hat{M},\omega)$, and consider the class of metrics specified in the paper, normalized by $b(0)=1$. A steady soliton i...

conjecture | open | MathDB 页面 | 原文线索

V2-025046 | MATHDB-336769

Kawamata-Namikawa extension to noncommutative d-semistable spaces

摘录 / EXCERPT: Let Γ be a cell complex and let $\mathscr{S}_\Gamma$ be a constructible sheaf of categories whose fibers are smooth and proper dg categories. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025047 | MATHDB-336799

Refined Weil-Petersson conjecture for the stringy Kähler moduli space

摘录 / EXCERPT: Let X be a projective Calabi-Yau n-fold, let $\mathcal{M}_{\mathrm{Kah}}(X)$ be its stringy Kähler moduli space, and let $\mathcal{D}_X=\mathrm{D^bCoh}(X)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025048 | MATHDB-336814

Kobayashi-Lang conjecture on ampleness of the canonical bundle

摘录 / EXCERPT: Kobayashi-Lang conjecture. The canonical line bundle K_X is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-025049 | MATHDB-336821

Uniform volume doubling for left-invariant metrics on compact Lie groups

摘录 / EXCERPT: Uniform volume-doubling conjecture. There is a constant $D(K)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025050 | MATHDB-336827

Classification conjecture for double bubbles in radial log-convex density

摘录 / EXCERPT: Higher-dimensional double-bubble classification conjecture. A perimeter-minimizing double bubble is either

conjecture | open | MathDB 页面 | 原文线索

V2-025051 | MATHDB-336836

The abelianity conjecture for fields generated by periodic points of the Pappus-Steiner map

摘录 / EXCERPT: Let $n>2$, and let $\{[\alpha_i,\beta_i]:i=1,\dots,r\}$ be the set of n-periodic points of the Pappus-Steiner map π over $\mathbb{Q}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025052 | MATHDB-336842

Density of orthogonal shapes of integral points on a quadratic surface

摘录 / EXCERPT: Let $Q(v_1,v_2,v_3):=2v_1v_3-v_2^2$ be a quadratic form of signature $(2,1)$, and let ... Write $V_Q^1(\mathbb{Z}):=V_Q^1\cap\mathbb{Z}^3$. For a rank-2 discrete subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-025053 | MATHDB-336844

Density of directional lattices over isotropic planes

摘录 / EXCERPT: Let $Q(v_1,v_2,v_3):=2v_1v_3-v_2^2$ and ... Let $\mathcal{C}\subset\operatorname{Gr}_2(\mathbb{R}^3)$ be the circle of isotropic subspaces, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025054 | MATHDB-336885

Infinitely many triangle-minimizing digon-free pseudocircle arrangements

摘录 / EXCERPT: Let n be a positive integer, and let a digon-free intersecting arrangement be an arrangement of n pseudocircles in which every pair intersects and no cell is a digon. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025055 | MATHDB-336894

RC-positivity conjecture for exterior powers under positive holomorphic sectional curvature

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension $n>2$. RC-positivity conjecture. If X has a Hermitian metric with positive holomorphic sectional curvature, then...

conjecture | open | MathDB 页面 | 原文线索

V2-025056 | MATHDB-336896

Mumford's conjecture with equivalent uniruledness and curvature conditions

摘录 / EXCERPT: Let X be a projective manifold. Mumford's equivalent-conditions conjecture. If ... for every $m\geq 1$, then each of the following holds: X is uniruled; K_X is not pseudo-effec...

conjecture | open | MathDB 页面 | 原文线索

V2-025057 | MATHDB-336897

RC-positivity conjecture for ample vector bundles

摘录 / EXCERPT: Let E be an ample vector bundle over a projective manifold X, and let $\operatorname{rank}(E)$ denote its rank. RC-positivity conjecture. There exists a smooth Hermitian metric...

conjecture | open | MathDB 页面 | 原文线索

V2-025058 | MATHDB-336934

The conjecture relating the two filtrations on the Tate module

摘录 / EXCERPT: Let X have a strictly semistable formal model over $\mathcal{O}_K$, and let $V_p(T)$ and $V_p(\hat{P})$ be the subrepresentations arising from the canonical two-step filtration a...

conjecture | open | MathDB 页面 | 原文线索

V2-025059 | MATHDB-336950

The Gu-Xu normalized scalar-curvature sphere conjecture

摘录 / EXCERPT: Gu-Xu's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025060 | MATHDB-336951

The Xu-Gu Ricci-pinching sphere conjecture for submanifolds

摘录 / EXCERPT: Xu-Gu's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025061 | MATHDB-337002

Fractal dimension conjecture for optimal branched-transport shapes

摘录 / EXCERPT: Let A be an optimal unit-volume set for irrigation from a single source at the origin in branched transport, and let $\beta=d\left(\alpha-\left(1-1/d\right)\right)\in(0,1)$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-025062 | MATHDB-337021

The flat Gauduchon connection conjecture

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold, and let ∇^s denote its s-Gauduchon connection. A connection is flat when its curvature tensor vanishes identically. Flat Gau...

conjecture | open | MathDB 页面 | 原文线索

V2-025063 | MATHDB-337043

Feng-Huang-Rao's algebraic dependence conjecture for affine embeddings of Cantor sets

摘录 / EXCERPT: Let F and E be totally disconnected self-similar sets in $\mathbb{R}^d$, generated by the iterated function systems $\Phi=\lbrace\phi_i\rbrace_{i=1}^l$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025064 | MATHDB-337055

Maubach-Poloni's GLIN conjecture for polynomial automorphism groups

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, and let ${\rm GA}_n(\mathbb{K})$ denote the group of polynomial automorphisms of affine n-space over $\mathbb{K}$. Let ${\rm GLIN}_n(\mathbb{K})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-025065 | MATHDB-337074

Transcendental base-point-free conjecture for nef classes

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[\alpha]$ be a nef $(1,1)$ class on X. Suppose that $\lambda[\alpha]-c_1(K_X)$ is nef and big for some $\lambda>0$. Transcendental b...

conjecture | open | MathDB 页面 | 原文线索

V2-025066 | MATHDB-337133

Goodrick's conjecture on collapsibility of star-shaped complexes

摘录 / EXCERPT: Let C be a simplicial complex, and let $|C|$ denote its underlying space. The space $|C|$ is star-shaped if it is star-shaped in the relevant Euclidean space. Goodrick's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-025067 | MATHDB-337137

Integer-distance conjecture in dimensions two and three

摘录 / EXCERPT: Let $P \subset {\mathbb R}^d$ be a finite point set of size n, where $d=2$ or $d=3$. Write $|p-p'|$ for the Euclidean distance between points $p,p'\in P$. Integer-distance conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-025068 | MATHDB-337138

Lyapunov-spectrum characterization of negatively curved locally symmetric spaces

摘录 / EXCERPT: Lyapunov-spectrum characterization conjecture. Let Y be a closed negatively curved Riemannian manifold, $\dim Y \geq 3$. Suppose that for each periodic point p of g^{t}_{Y} t...

conjecture | open | MathDB 页面 | 原文线索

V2-025069 | MATHDB-337147

The spherical representation conjecture for smooth valuations

摘录 / EXCERPT: Let i be a nonnegative integer, let $\mathbb{R}^{i+1}$ be Euclidean space, and let $\operatorname{Val}_i^{\infty}(\mathbb{R}^{i+1})$ denote the smooth valuations of degree i on...

conjecture | open | MathDB 页面 | 原文线索

V2-025070 | MATHDB-337181

Equality of the big and finitistic test ideals

摘录 / EXCERPT: Let R be an excellent Noetherian domain of characteristic $p>0$. Denote by $\tau_{\rm fg}(R)$ the finitistic test ideal and by $\tau_{\rm b}(R)$ the big test ideal of R. Test i...

conjecture | open | MathDB 页面 | 原文线索

V2-025071 | MATHDB-337214

Guth's macroscopic Schoen conjecture for hyperbolic manifolds

摘录 / EXCERPT: Let M be a closed hyperbolic manifold, let g be an auxiliary Riemannian metric on M, and let $V_{hyp}(R)$ denote the volume of a geodesic ball of radius R in hyperbolic spa...

conjecture | open | MathDB 页面 | 原文线索

V2-025072 | MATHDB-337215

Conjecture on powers of the Jensen-Shannon divergence

摘录 / EXCERPT: Conjecture on powers of the Jensen-Shannon divergence. The quantity $[D_{JS}(P,Q)]^\alpha$ is not a metric for $1/2<\alpha<1$.

conjecture | open | MathDB 页面 | 原文线索

V2-025073 | MATHDB-337227

The Clifford torus energy minimization conjecture

摘录 / EXCERPT: Clifford torus energy minimization conjecture. The minimum of the energy functional is attained on the Clifford torus.

conjecture | open | MathDB 页面 | 原文线索

V2-025074 | MATHDB-337230

The double smoothness conjecture for Alexandrov spaces with maximal obtuse constant

摘录 / EXCERPT: Double smoothness conjecture. If M is compact and ${\rm ob}(M)=\pi/2$, then $D(M)$ has no singular points. Similarly, if M is noncompact and ${\rm ob}_{\infty}(M)=\pi/2$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-025075 | MATHDB-337231

The comparison obtuse constant conjecture for Alexandrov spaces

摘录 / EXCERPT: Comparison obtuse constant conjecture. The conclusions of the relevant compact results would remain valid with $\widetilde{\rm ob}(M)$ in place of ${\rm ob}(M)$ when M is compact...

conjecture | open | MathDB 页面 | 原文线索

V2-025076 | MATHDB-337251

Uniqueness conjecture for generalized bridge trisections inducing isotopic trisections

摘录 / EXCERPT: Let $(X,\mathcal K)$ be a pair consisting of a 4-manifold and a knotted surface. A generalized bridge trisection is a generalized bridge decomposition of this pair, and two such tr...

conjecture | open | MathDB 页面 | 原文线索

V2-025077 | MATHDB-337252

Uniqueness conjecture for generalized bridge trisections with fixed underlying trisection

摘录 / EXCERPT: Let $(X,\mathcal K)$ be a pair consisting of a 4-manifold and a knotted surface, and fix a trisection of X. A generalized bridge trisection of $(X,\mathcal K)$ is compared with a...

conjecture | open | MathDB 页面 | 原文线索

V2-025078 | MATHDB-337266

The Kleiman-Piene conjecture for nodal curves

摘录 / EXCERPT: Let B be a family of curves satisfying the hypotheses ${\bf DIM}_{\mathrm{KP}}$: the locus of non-reduced curves has codimension greater than δ, and for every equisingular...

conjecture | open | MathDB 页面 | 原文线索

V2-025079 | MATHDB-337286

The pure-EKR conjecture for flag simplicial manifolds

摘录 / EXCERPT: Flag-manifold conjecture. Every flag simplicial manifold is pure-EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-025080 | MATHDB-337287

The pure-EKR conjecture for flag pseudo-manifolds

摘录 / EXCERPT: Flag pseudo-manifold conjecture. Every flag pseudo-manifold is pure-EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-025081 | MATHDB-337303

The timelike incompleteness criterion for CMC slices

摘录 / EXCERPT: Let $\mathcal{V}$ be a cosmological spacetime. A timelike path is incomplete when it has finite length. Timelike incompleteness criterion. If every future or every past timelike pa...

conjecture | open | MathDB 页面 | 原文线索

V2-025082 | MATHDB-337304

The constant signed mean-curvature conjecture

摘录 / EXCERPT: Let $\mathcal{V}$ be a cosmological spacetime. A slice has constant signed mean curvature if its mean curvature is constant with a fixed sign convention, while a CMC slice is a sli...

conjecture | open | MathDB 页面 | 原文线索

V2-025083 | MATHDB-337324

Vanishing discriminant conjecture for curve semistable Higgs bundles

摘录 / EXCERPT: Vanishing discriminant conjecture. Every curve semistable Higgs bundle (E,θ) on (X,H) has vanishing discriminant:

conjecture | open | MathDB 页面 | 原文线索

V2-025084 | MATHDB-337327

The maximal-resolution conjecture for G-constellation moduli spaces

摘录 / EXCERPT: Let G be a finite subgroup of $\operatorname{GL}(2, {\mathbb C})$ and let $X={\mathbb C}^2/G$ be the quotient with boundary divisor B. Let $Y\to X$ be any resolution of singula...

conjecture | open | MathDB 页面 | 原文线索

V2-025085 | MATHDB-337343

Characterization conjecture for L_p-surface area measures

摘录 / EXCERPT: Characterization conjecture. The measure μ is the L_p-surface area measure of a convex body $K\in\mathcal K_0^n$ if and only if $\operatorname{supp}\,\mu$ is not a pair of an...

conjecture | open | MathDB 页面 | 原文线索

V2-025086 | MATHDB-337353

Klartag-Kolesnikov's Ricci curvature conjecture for Kähler-Einstein Hessian metrics

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body with barycenter at the origin, and let Φ solve the Kähler-Einstein equation ... For the simplex ... the corresponding Hessian metric...

conjecture | open | MathDB 页面 | 原文线索

V2-025087 | MATHDB-337362

Perrin-Riou's conjecture on Euler systems for geometric Galois representations

摘录 / EXCERPT: Let T be a global Galois representation arising in geometry, with $d_-(T)$ denoting the relevant minus-dimension, and let $\mathcal{K}/K$ be the extension used to define Euler sy...

conjecture | open | MathDB 页面 | 原文线索

V2-025088 | MATHDB-337370

Holmsen-Kynčl-Valculescu colored partition conjecture

摘录 / EXCERPT: Holmsen-Kynčl-Valculescu conjecture. If X has a partition into n subsets of size ℓ such that each subset contains points of at least d colors, then X has such a partit...

conjecture | open | MathDB 页面 | 原文线索

V2-025089 | MATHDB-337383

Poisson and symplectic-leaf conjecture for the relative compactified Jacobian

摘录 / EXCERPT: Let $\mathcal{C}/\mathcal{B}'$ be the family of curves and let $\mathcal{J}(\mathcal{C}/\mathcal{B}')$ be its relative compactified Jacobian. Let h denote the projection to...

conjecture | open | MathDB 页面 | 原文线索

V2-025090 | MATHDB-337408

The zero-Euler-characteristic bounding conjecture for Haken 3-manifolds

摘录 / EXCERPT: Let M^{3} be a closed, oriented Haken 3-manifold. A compact, oriented Haken 4-manifold W^{4} is required to have boundary $\partial W^{4}=M^{3}$, with boundary understood t...

conjecture | open | MathDB 页面 | 原文线索

V2-025091 | MATHDB-337409

The strengthened Euler characteristic-signature inequality conjecture for aspherical 4-manifolds

摘录 / EXCERPT: Let X^{4} be a closed, oriented, aspherical 4-manifold. Write $\chi(X^{4})$ for its Euler characteristic and $\sigma(X^{4})$ for its signature. Strengthened Euler characteristi...

conjecture | open | MathDB 页面 | 原文线索

V2-025092 | MATHDB-337429

Non-constructibility conjecture for common points of investigated conics

摘录 / EXCERPT: Non-constructibility conjecture. If $x',y'\ne 0$, then one can choose the parameters a and b above so that the common point of the investigated conics does not admit a planar c...

conjecture | open | MathDB 页面 | 原文线索

V2-025093 | MATHDB-337431

Flop factorization conjecture for K-equivalent terminal pairs

摘录 / EXCERPT: Flop factorization conjecture. There should be a sequence of projective $\mathbf{Q}$-factorial terminal pairs $\{(X_i,B_i)\}_{0\leq i\leq n}$ with $(X_0,B_0)=(X,B)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025094 | MATHDB-337432

Elementary factorization conjecture for K-inequivalent terminal pairs

摘录 / EXCERPT: Elementary factorization conjecture. If $(X,B)\geq_K(Y,C)$ by a birational map $f:X\dashrightarrow Y$, then there is a sequence of pairs $\{(X_i,B_i)\}_{0\leq i\leq n}$ from...

conjecture | open | MathDB 页面 | 原文线索

V2-025095 | MATHDB-337459

Hausel's conjecture on L^2 harmonic forms of the Hitchin moduli space

摘录 / EXCERPT: Let ${\mathcal M}_{{\rm SL}(2,\mathbb C)}$ be the Hitchin moduli space of ${\rm SL}(2,\mathbb C)$-Higgs bundles. Hausel's conjecture. There are no non-trivial L^2 harmonic forms...

conjecture | open | MathDB 页面 | 原文线索

V2-025096 | MATHDB-337460

Extension of the Hitchin-Calabi-Yau correspondence to general singular elliptic Calabi-Yau threefolds

摘录 / EXCERPT: Let a compact singular elliptically fibered Calabi-Yau 3-fold have singular fibers associated to ${\mathcal G}$-symmetry in codimension 1 over the base, admit a crepant resolut...

conjecture | open | MathDB 页面 | 原文线索

V2-025097 | MATHDB-337461

Flat-degeneration conjecture from parabolic to wild Hitchin moduli

摘录 / EXCERPT: Let ${\mathcal M}_{\rm par}$ be the moduli space of singular parabolic Higgs bundles with n simple poles in their connection, and let ${\mathcal M}_{\rm wild}$ be the moduli spac...

conjecture | open | MathDB 页面 | 原文线索

V2-025098 | MATHDB-337494

The tropicalization count conjecture for bitangents of tropical quartics

摘录 / EXCERPT: Let C be an algebraic curve over the field of Puiseux series, and let $\Gamma = \operatorname{Trop}(C)$. Let $[\Lambda]$ be an equivalence class of bitangents of Γ of tota...

conjecture | open | MathDB 页面 | 原文线索

V2-025099 | MATHDB-337531

Dolnikov-Eckhoff conjecture on line transversals to unit disks

摘录 / EXCERPT: Let $B = \{(x,y)\in\mathbb R^2:x^2+y^2\leq 1\}$ be the unit disk, and let $\lambda(B,3)$ denote the least dilation factor such that every finite point set $X\subset\mathbb R^2$ who...

conjecture | open | MathDB 页面 | 原文线索

V2-025100 | MATHDB-337532

Finite Dolnikov-Eckhoff conjecture for elliptical disks

摘录 / EXCERPT: Finite Dolnikov-Eckhoff conjecture. The set $R\cap E$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-025101 | MATHDB-337536

Connelly's Strong Bellows Conjecture for flexible polyhedra

摘录 / EXCERPT: Let P be a flexible polyhedron in one of the constant-curvature spaces $\mathbb{E}^n$, $\mathbb{S}^n$, or $\mathbb{\Lambda}^n$. Two polyhedra are scissors congruent if they can...

conjecture | open | MathDB 页面 | 原文线索

V2-025102 | MATHDB-337540

Schinzel's multivariate pullback factorization conjecture

摘录 / EXCERPT: Multivariate pullback factorization conjecture. There is a finite set of matrices $\Omega\subset\mathbb{Z}^{n\times n}$ such that, for each $A\in\mathbb{Z}^{n\times k}$, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-025103 | MATHDB-337560

Differentiability and origin behavior of the first Steklov invariant surface for cuboids

摘录 / EXCERPT: Let the cuboid be $[-a,a]\times[-b,b]\times[-1,1]$, with $a,b\leq 1$, and consider the surface plotting its first Steklov invariant in the first octant. Differentiability and origi...

conjecture | open | MathDB 页面 | 原文线索

V2-025104 | MATHDB-337561

First Steklov eigenvalue and eigenspace conjecture for cuboids

摘录 / EXCERPT: Let $0\leq a\leq b\leq c=1$, and let σ_1 denote the first Steklov eigenvalue of the cuboid. Define λ_1 and λ_2 as the parameters occurring in the separated e...

conjecture | open | MathDB 页面 | 原文线索

V2-025105 | MATHDB-337564

The Euclidean-ball maximizer conjecture for the even spectral gap

摘录 / EXCERPT: Let $K\in\mathcal{K}^{2}_{+,e}$ be an origin-symmetric smooth strictly convex body, and let $\lambda_{1,e}(-L_K)$ denote the first nonzero eigenvalue of the Hilbert-Brunn-Minkowski...

conjecture | open | MathDB 页面 | 原文线索

V2-025106 | MATHDB-337579

Untwisted algebraic 1-torsion in the Corollary 8.4 examples

摘录 / EXCERPT: Let Y be the unit cotangent bundle of a hyperbolic surface, and let $(M=\Sigma\times Y,\xi_k)$ be the corresponding 5-dimensional contact manifold of Theorem 5d. These examples...

conjecture | open | MathDB 页面 | 原文线索

V2-025107 | MATHDB-337581

Pansu's isoperimetric conjecture for the sub-Riemannian Heisenberg group

摘录 / EXCERPT: Let $H(\mathbb R)$ be the Heisenberg group with its sub-Riemannian Carnot-Carathéodory metric. Let S be a surface enclosing a region E, let λ denote Haar measure, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-025108 | MATHDB-337582

Generalized Pansu conjecture for sub-Finsler Heisenberg groups

摘录 / EXCERPT: Let $(H(\mathbb R),d_Q,\lambda)$ be the Heisenberg group equipped with Haar measure λ and an arbitrary Carnot-Carathéodory metric d_Q arising from a norm on the horizonta...

conjecture | open | MathDB 页面 | 原文线索

V2-025109 | MATHDB-337583

Pansu's bubble-set conjecture for Heisenberg isoperimetrices

摘录 / EXCERPT: Let $H(\mathbb R)$ be the Heisenberg group with a Carnot-Carathéodory metric arising from a norm on the horizontal plane. In the sub-Riemannian case, Pansu's bubble set is the uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-025110 | MATHDB-337602

Demailly's Hodge conjecture for strongly positive currents

摘录 / EXCERPT: Demailly's Hodge conjecture for strongly positive currents. The current $\mathscr{T}$ is a weak limit

conjecture | open | MathDB 页面 | 原文线索

V2-025111 | MATHDB-337620

Affine-space Abhyankar valuation conjecture for log canonical thresholds

摘录 / EXCERPT: Affine-space Abhyankar valuation conjecture. The following versions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-025112 | MATHDB-337621

Steklov spectral determination conjecture for cuboids

摘录 / EXCERPT: Let a cuboid be a Euclidean rectangular box, and let two cuboids be Steklov isospectral when their Steklov spectra, counted with multiplicities, coincide. Steklov spectral determin...

conjecture | open | MathDB 页面 | 原文线索

V2-025113 | MATHDB-337624

Conjecture on equality of minimal convex and open convex embedding dimensions in dimension one

摘录 / EXCERPT: Dimension-one embedding conjecture. A code $\mathcal{C}$ has minimal convex embedding dimension 1 if and only if it has minimal open convex embedding dimension 1.

conjecture | open | MathDB 页面 | 原文线索

V2-025114 | MATHDB-337625

Conjecture on dimension two for open convex codes

摘录 / EXCERPT: Dimension-two embedding conjecture. If $\mathcal{C}$ has minimal open convex embedding dimension 2, then its minimal convex embedding dimension is 2.

conjecture | open | MathDB 页面 | 原文线索

V2-025115 | MATHDB-337636

Semmes' positive-measure bi-Lipschitz piece conjecture for Carnot groups

摘录 / EXCERPT: Semmes' conjecture. There is a subset $E\subset A$ such that $\mathcal{H}^Q(E)>0$ and f is bi-Lipschitz on E. This conjecture concerns when a Lipschitz map from a PI space into...

conjecture | refuted | MathDB 页面 | 原文线索

V2-025116 | MATHDB-337637

Semmes' weak-tangent conjecture for Lipschitz maps into Carnot groups

摘录 / EXCERPT: Semmes' weak-tangent conjecture. There is a weak tangent of f that is bi-Lipschitz. This is a formulation of the positive-rigidity questions for Lipschitz maps from PI spaces int...

conjecture | open | MathDB 页面 | 原文线索

V2-025117 | MATHDB-337640

The thick-thin conjecture for nonpositively curved manifolds

摘录 / EXCERPT: Let M be a tame, complete, finite-volume n-manifold of bounded nonpositive curvature. The thick-thin conjecture. There is a compact subset $C\twoheadrightarrow M$ that cannot b...

conjecture | open | MathDB 页面 | 原文线索

V2-025118 | MATHDB-337645

Uniqueness conjecture for closed zero-energy minimizers in the Heisenberg group

摘录 / EXCERPT: Uniqueness conjecture. The shifted Heisenberg spheres above are the only closed minimizers for E_1 with zero energy in $\mathcal{H}_1$. The source establishes that these spheres...

conjecture | open | MathDB 页面 | 原文线索

V2-025119 | MATHDB-337646

Uniqueness conjecture for the Clifford torus among CR torus minimizers

摘录 / EXCERPT: Let S^3 be the three-sphere, let E_1 be the energy functional considered in the source, and let $\Sigma_{\frac{\sqrt{2}}{2}}\subset S^3$ denote the Clifford torus. A surface of...

conjecture | open | MathDB 页面 | 原文线索

V2-025120 | MATHDB-337657

Converse of Kra's conjecture for holomorphic retracts of Teichmüller space

摘录 / EXCERPT: Let $\mathcal{T}:=\mathcal{T}_{g,n}$ be the Teichmüller space of a finite-type orientable surface, and let $\tau:\mathbb{H}\to\mathcal{T}$ be a Teichmüller disk generated by a holo...

conjecture | open | MathDB 页面 | 原文线索

V2-025121 | MATHDB-337686

Representation-independent Toda Frobenius manifold mirror

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let $\bar i$ be a marked simple root, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025122 | MATHDB-337687

Toda mirror conjecture for polynomial orbifold projective lines

摘录 / EXCERPT: Orbifold mirror conjecture. With the notation of the preceding Toda mirror construction,

conjecture | open | MathDB 页面 | 原文线索

V2-025123 | MATHDB-337712

Boucksom's Fujiki class conjecture for positive currents

摘录 / EXCERPT: Let X be a compact complex manifold and let T be a closed positive $(1,1)$-current on X. Write T_{ac} for the absolutely continuous part of T in its Lebesgue decompositio...

conjecture | open | MathDB 页面 | 原文线索

V2-025124 | MATHDB-337745

Morita's conjecture on the graded Lie algebras of the Torelli group

摘录 / EXCERPT: Let g be the genus, let $\mathfrak{t}_g$ be the relevant graded Lie algebra associated to the Torelli group, and let $\mathfrak{m}_g$ be the Lie algebra of symplectic derivations...

conjecture | open | MathDB 页面 | 原文线索

V2-025125 | MATHDB-337755

Székelyhidi's Poincaré-type metric conjecture for polarized triples

摘录 / EXCERPT: Let (X,L,Z) be a polarized triple, and let $F(c)$ and $F_{\chi}(c)$ denote the modified and relative modified Donaldson-Futaki invariants associated with the degeneration to the...

conjecture | open | MathDB 页面 | 原文线索

V2-025126 | MATHDB-337768

Local-system conjecture for derived categories of GLSM phases

摘录 / EXCERPT: Let X_{β} be the phase associated with a VGIT parameter $\beta \in L^{\vee}_{\mathbb{R}}$, and let FIPS denote the Fayet-Iliopoulos parameter space of the corresponding gauge...

conjecture | open | MathDB 页面 | 原文线索

V2-025127 | MATHDB-337770

The generalized lower bound conjecture for centrally symmetric simplicial spheres

摘录 / EXCERPT: The centrally symmetric lower bound conjecture. If Δ is a simplicial sphere, a connected simplicial manifold, or a normal pseudomanifold, then

conjecture | open | MathDB 页面 | 原文线索

V2-025128 | MATHDB-337771

Symmetric infinitesimal rigidity conjecture for centrally symmetric complexes

摘录 / EXCERPT: Symmetric infinitesimal rigidity conjecture. If Δ is a simplicial sphere, a connected simplicial manifold, or a normal pseudomanifold and $d-1\geq 3$, then there exists a sy...

conjecture | open | MathDB 页面 | 原文线索

V2-025129 | MATHDB-337772

Zahariuc's multiplicity conjecture for elliptic surfaces

摘录 / EXCERPT: Let ${\mathbb K}$ be an algebraically closed field, let E be a smooth elliptic curve over ${\mathbb K}$, let S be the elliptic surface and F_q,E_∞ the divisors defined i...

conjecture | open | MathDB 页面 | 原文线索

V2-025130 | MATHDB-337793

Strict convexity conjecture for the Mabuchi functional

摘录 / EXCERPT: Let ${\mathcal M}$ be the Mabuchi functional and let ${\mathcal G}$ be a $C^{1,1}$-geodesic segment. Strict convexity conjecture. If ${\mathcal M}$ is linear along ${\mathcal G}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-025131 | MATHDB-337795

Reznikov's Lindelöf-type conjecture for geodesic period integrals

摘录 / EXCERPT: Reznikov's Lindelöf-type conjecture. The geodesic period integrals on compact hyperbolic surfaces satisfy the estimate above. This would improve the known uniform bound and is pres...

conjecture | open | MathDB 页面 | 原文线索

V2-025132 | MATHDB-337796

Hultgren-Witt Nyström's coupled Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Let X be a compact Kähler manifold with $c_1(X)>0$. A decomposition of $c_1(X)$ is a k-tuple of Kähler classes $(\alpha_1,\ldots,\alpha_k)$ satisfying ... Such a decomposition ad...

conjecture | open | MathDB 页面 | 原文线索

V2-025133 | MATHDB-337828

Canonical lift volume-minimizing conjecture for filling closed geodesics

摘录 / EXCERPT: Let Σ be a hyperbolic surface, let γ be a filling closed geodesic on Σ, let $\hat{\gamma}$ denote its canonical lift, and let $M_{\tilde{\gamma}}$ be the compl...

conjecture | open | MathDB 页面 | 原文线索

V2-025134 | MATHDB-337829

Batyrev's conjecture on the nef cone of curves

摘录 / EXCERPT: Let (X,Δ) be a projective KLT pair. Write $\overline{\operatorname{NE}}(X)_{K_X+\Delta>0}$ for the part of the closed cone of effective curves on which $K_X+\Delta$ is posit...

conjecture | open | MathDB 页面 | 原文线索

V2-025135 | MATHDB-337846

Global isoperimetric conjecture for centered circles in the Holmes-Thompson plane

摘录 / EXCERPT: Global Holmes-Thompson isoperimetric conjecture. The circle c_0 encloses the maximal Holmes-Thompson area among all such curves.

conjecture | open | MathDB 页面 | 原文线索

V2-025136 | MATHDB-337850

Microlocal mirror symmetry conjecture for logarithmic coherent sheaves

摘录 / EXCERPT: Let $\mathcal{S}$ be a Lagrangian torus fibration over a real base B which is part of an SYZ model. Assume that the dual torus fibration $\mathcal{S}^\vee$ over B has integrabl...

conjecture | open | MathDB 页面 | 原文线索

V2-025137 | MATHDB-337864

V. Arnold's distinctness conjecture for torus decompositions of Klein sails

摘录 / EXCERPT: V. Arnold's conjecture. Torus decompositions of integer noncongruent Klein sails are distinct.

conjecture | open | MathDB 页面 | 原文线索

V2-025138 | MATHDB-337902

Subadditivity conjecture for the EHZ capacity

摘录 / EXCERPT: Let $K\subset \mathbb{R}^{2n}$ be a convex body covered by a finite set of convex bodies $\{K_i\}$. Subadditivity conjecture. One has ... This is a special case of the subadditivity...

conjecture | open | MathDB 页面 | 原文线索

V2-025139 | MATHDB-337927

Homogeneity conjecture for projectively induced Kähler-Einstein manifolds

摘录 / EXCERPT: Let (M,g) be a complete Kähler-Einstein manifold, and let $\mathbb{C}\mathrm{P}^N$ be complex projective space, where N may be finite or infinite. A Kähler immersion is an isom...

conjecture | refuted | MathDB 页面 | 原文线索

V2-025140 | MATHDB-337928

Compactness conjecture for locally projectively induced Kähler-Einstein manifolds

摘录 / EXCERPT: Let (M,g) be a complete Kähler-Einstein manifold. A local Kähler immersion is a Kähler immersion defined on neighborhoods, and $\mathbb{C}\mathrm{P}^N$ denotes complex projective...

conjecture | open | MathDB 页面 | 原文线索

V2-025141 | MATHDB-337929

Loi-Zedda-Zuddas conjecture on the Taub-NUT metric

摘录 / EXCERPT: Let g_m be the Taub-NUT metric on $\mathds{C}^2$, with $m>0$, and let α be a positive real number. A metric is projectively induced if it admits a Kähler immersion into co...

conjecture | open | MathDB 页面 | 原文线索

V2-025142 | MATHDB-337941

Birational functoriality conjecture for destabilising test configurations

摘录 / EXCERPT: Birational functoriality conjecture. There is a K-unstable pair $(X”,L”)$ destabilized by a test configuration $(\mathcal X”,\mathcal L”)$, a birational map...

conjecture | open | MathDB 页面 | 原文线索

V2-025143 | MATHDB-337958

Shokurov's rationality criterion for standard conic bundles

摘录 / EXCERPT: Let S be a rational surface, let $\Delta\subset S$ be a discriminant curve, and let $\pi:X\to S$ be a standard conic bundle. Write K_S for the canonical divisor of S,...

conjecture | open | MathDB 页面 | 原文线索

V2-025144 | MATHDB-337959

Kantor's conjecture on index-one congruences of rational curves

摘录 / EXCERPT: A congruence of curves in $\mathbb{P}^3$ is an irreducible two-dimensional family of curves covering $\mathbb{P}^3$; its index is the number of curves through a general point. Kant...

conjecture | open | MathDB 页面 | 原文线索

V2-025145 | MATHDB-337960

Kantor's fiberwise formulation for rational standard conic bundles

摘录 / EXCERPT: Let $\pi:X\to S$ be a standard conic bundle, and suppose that there is a birational map ... A Cremona transformation is a birational self-map of $\mathbb{P}^3$. Kantor's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-025146 | MATHDB-337964

Constant-term spectral-action conjecture for noncommutative tori

摘录 / EXCERPT: Constant-term spectral-action conjecture. The constant term of the spectral action of $\mathcal{D}_A$ on the noncommutative n-torus is proportional to the constant term of the sp...

conjecture | open | MathDB 页面 | 原文线索

V2-025147 | MATHDB-337988

Boucksom's transcendental bigness conjecture

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension n, and let T be a closed positive $(1,1)$-current on X. Write T_{ac} for its absolutely continuous part and...

conjecture | open | MathDB 页面 | 原文线索

V2-025148 | MATHDB-337989

Boucksom-Demailly-Păun-Peternell holomorphic Morse inequality conjecture

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension n. Let α be a closed real $(1,1)$-form, and let $X(\alpha,\leq 1)$ be the set where α is nondegenerate an...

conjecture | open | MathDB 页面 | 原文线索

V2-025149 | MATHDB-338003

The polynomial specialization conjecture for solid-partition weights

摘录 / EXCERPT: Polynomial specialization conjecture. The following properties hold: (a) $L_\pi(0,0,0,-d)\mathsf{w}_\pi$ has no pole at $\lambda_4=-(\lambda_1+\lambda_2+\lambda_3)$; (b) its specia...

conjecture | open | MathDB 页面 | 原文线索

V2-025150 | MATHDB-338004

Nekrasov's conjecture for zero-dimensional DT4 invariants of $\mathbb{C}^4$

摘录 / EXCERPT: Nekrasov's conjecture. There exist choices of orientation such that

conjecture | open | MathDB 页面 | 原文线索

V2-025151 | MATHDB-338005

Rodriguez Villegas's refined counting conjecture for global nilpotent-cone polytopes

摘录 / EXCERPT: Let r be a positive integer, let d be an integer, and let $\mathbf r\vdash r$ be a partition. Write $\mathcal P^\mathbb Z_{\mathbf r,d}$ for the integral points of the polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-025152 | MATHDB-338015

Strong properness conjecture for K-energy on Sasaki manifolds

摘录 / EXCERPT: Let (M,g) be a $(2n+1)$-dimensional Sasaki manifold with transverse Kähler form ... Suppose that M admits a transverse Sasaki-Einstein metric. Let K be a maximal compact subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-025153 | MATHDB-338022

Compactness conjecture for ADHM$_{1,k}$ monopoles

摘录 / EXCERPT: Let M be the underlying manifold, and let $(\varepsilon_i,\Psi_i,{\bm\xi}_i,A_i)$ be a sequence of solutions of the blown-up ADHM$_{1,k}$ Seiberg--Witten equation satisfying ... wi...

conjecture | open | MathDB 页面 | 原文线索

V2-025154 | MATHDB-338023

Three-way desingularization conjecture for associative T^2-cone singularities

摘录 / EXCERPT: Let $(\psi_t)_{t\in(-T,T)}$ be a 1-parameter family of closed, tamed, definite 4-forms on Y. Let $\hat P$ be an unobstructed singular associative submanifold in (Y,ψ_0)...

conjecture | open | MathDB 页面 | 原文线索

V2-025155 | MATHDB-338044

Folklore conjecture on automorphisms of affine cubic hypersurfaces

摘录 / EXCERPT: Let X be a smooth cubic hypersurface in the complex projective space$\mathbb{P}^4$, and let S_X be its hyperplane section. The complement $X\setminus S_X$ is an affine cubic hy...

conjecture | open | MathDB 页面 | 原文线索

V2-025156 | MATHDB-338066

The conformal embedding conjecture for causally simple spacetimes

摘录 / EXCERPT: Let (X,g) be a causally simple spacetime. A conformal embedding is an embedding preserving the conformal class of the metric. The conformal embedding conjecture. The spacetime...

conjecture | open | MathDB 页面 | 原文线索

V2-025157 | MATHDB-338067

The open contact submanifold conjecture for spaces of light rays

摘录 / EXCERPT: Let (X,g) be a causally simple spacetime that can be conformally embedded as an open subset into some globally hyperbolic spacetime $(\widetilde X,\widetilde g)$. Let the space o...

conjecture | open | MathDB 页面 | 原文线索

V2-025158 | MATHDB-338082

Shokurov's conjecture on singularities of bases of Fano type fibrations

摘录 / EXCERPT: Shokurov's conjecture. There is $\delta\in\mathbb R^{>0}$ such that one can write

conjecture | open | MathDB 页面 | 原文线索

V2-025159 | MATHDB-338087

One-sided concentration conjecture for random spherical coverings

摘录 / EXCERPT: Let $\epsilon=\epsilon(n)$ be the radius of the geodesic balls, let $N_{c,\epsilon(n)}$ be the number of independently and uniformly selected centers used for the covering, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025160 | MATHDB-338088

The sharp norm conjecture for the (j,k)-plane transform

摘录 / EXCERPT: The sharp norm conjecture for the (j,k)-plane transform. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025161 | MATHDB-338089

The generalized Busemann intersection inequality

摘录 / EXCERPT: Generalized Busemann intersection inequality. More generally, for an arbitrary real number m such that the integral on the right-hand side exists in the Lebesgue sense,

conjecture | open | MathDB 页面 | 原文线索

V2-025162 | MATHDB-338096

Finite-volume quotient conjecture for bounded convex domains

摘录 / EXCERPT: Let D be a bounded convex domain with a finite-volume manifold quotient. Finite-volume quotient conjecture. The domain D is biholomorphic to a bounded symmetric domain. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-025163 | MATHDB-338097

Yau's finite-volume quotient conjecture for bounded pseudoconvex domains

摘录 / EXCERPT: Let $D earrow\mathbb{C}^n$ $(n\geq 2)$ be a bounded pseudoconvex domain whose boundary is C^2-smooth. Assume that D has an open quotient of finite volume in the sense of Kähler...

conjecture | open | MathDB 页面 | 原文线索

V2-025164 | MATHDB-338102

Donaldson's destabilizing geodesic ray conjecture for cscK metrics

摘录 / EXCERPT: Donaldson's conjecture. The following statements are equivalent: (1) there is no constant scalar curvature Kähler metric in $\mathcal H$; (2) there is a potential...

conjecture | open | MathDB 页面 | 原文线索

V2-025165 | MATHDB-338103

Donaldson's metric-space conjecture for Kähler potentials

摘录 / EXCERPT: Donaldson's conjecture. The space $\mathcal H$ is a genuine metric space with the pathwise distance induced by this L^2 inner product.

conjecture | open | MathDB 页面 | 原文线索

V2-025166 | MATHDB-338104

Chen's geodesic properness conjecture for the K-energy

摘录 / EXCERPT: Chen's properness conjecture. A constant scalar curvature Kähler metric exists if and only if the K-energy is proper with respect to geodesic distance.

conjecture | open | MathDB 页面 | 原文线索

V2-025167 | MATHDB-338106

Chen's convexity conjecture for the K-energy

摘录 / EXCERPT: Chen's convexity conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-025168 | MATHDB-338147

The volume-condition extension of the strong maximum principle

摘录 / EXCERPT: Let M, β, θ, ρ, $ u$, b, and the conditions defining the range ... be as in Theorem ... is replaced by ... , but the supplied text gives no resolution or further co...

conjecture | open | MathDB 页面 | 原文线索

V2-025169 | MATHDB-338161

Higher-degree local-curve equivariant GW/DT4 conjecture

摘录 / EXCERPT: Let C be a smooth projective curve, let L_1,L_2,L_3 be line bundles on C satisfying ... and let $d\geq1$. Higher-degree local-curve conjecture. After substituting...

conjecture | open | MathDB 页面 | 原文线索

V2-025170 | MATHDB-338170

Hartshorne-Hirschowitz conjecture for the maximal genus of space curves

摘录 / EXCERPT: For integers $kb5$ and $f\a0\in [k-1,2k-6]$, define ... ... where $\delta(c)=3$ if $c=1$ or 3, $\delta(c)=1$ if $c\equiv2\pmod 3$, and $\delta(c)=0$ otherwise. Let $G(d,k)$ be the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-025171 | MATHDB-338186

Cerveau-Lins Neto's transverse-projectivity conjecture for compact complex foliations

摘录 / EXCERPT: Let $d4d5$ be a codimension one foliation on a compact complex manifold. A foliation is transversely projective when its transverse structure is projective, and $d4d5$ is everywher...

conjecture | open | MathDB 页面 | 原文线索

V2-025172 | MATHDB-338207

Maximal equiangular line sizes in dimensions 14, 16, 17, 18, 19, and 20

摘录 / EXCERPT: Maximal equiangular line sizes conjecture. The maximal numbers of equiangular lines in the specified dimensions are given by the table above.

conjecture | open | MathDB 页面 | 原文线索

V2-025173 | MATHDB-338208

Moreira's question on intervals in the Lagrange and Markov spectra

摘录 / EXCERPT: Moreira's question. For every $\delta>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-025174 | MATHDB-338220

Maximal volume conjecture for truncated tetrahedra with equal edge lower bounds

摘录 / EXCERPT: Let ${\mathcal{T}}_\ell$ denote the space of truncated hyperbolic tetrahedra whose edge lengths satisfy the constraint defining ${\mathcal{T}}_\ell$, and let Δ_ℓ be the r...

conjecture | open | MathDB 页面 | 原文线索

V2-025175 | MATHDB-338221

Dihedral-angle averaging conjecture for truncated tetrahedra

摘录 / EXCERPT: Let $\ell\in\mathbb{R}$, let $\Delta\in{\mathcal{T}}_{\ell}$, and let θ_{ij}, $1\leq i<j\leq4$, be the dihedral angles of Δ. Set ... and let $\Delta'\in{\mathcal{T}}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-025176 | MATHDB-338222

Symmetric-orbit convex-hull conjecture for truncated tetrahedra

摘录 / EXCERPT: Symmetric-orbit convex-hull conjecture. The intersection

conjecture | open | MathDB 页面 | 原文线索

V2-025177 | MATHDB-338251

The infinite staircase conjecture for ellipsoid embeddings into polydisks

摘录 / EXCERPT: Let $n\geq 2$ and $k\geq 0$. For each $i\geq 0$, let $A_{i,n}^{(k)}$ and $\hat{A}_{i,n}^{(k)}$ be the associated classes, and let $C_{L_{n,k}}(\alpha)$, $L_{n,k}$, $c_{i,n,k}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-025178 | MATHDB-338275

The K-energy properness criterion for constant scalar curvature Kähler metrics

摘录 / EXCERPT: Let (M,ω,J) be a compact Kähler manifold, and let K be a maximal compact subgroup of $\operatorname{Aut}_0(M)$. Let ${\mathcal H}$ be the space of Kähler potentials and le...

conjecture | open | MathDB 页面 | 原文线索

V2-025179 | MATHDB-338283

The reduced-cuspidal correspondence conjecture for Gromov-Witten invariants

摘录 / EXCERPT: Reduced-cuspidal correspondence conjecture. Reduced invariants with insertions γ_i are equal to cuspidal invariants with insertions γ_i.

conjecture | open | MathDB 页面 | 原文线索

V2-025180 | MATHDB-338286

Gordon's Cosmetic Surgery Conjecture in the hyperbolic case

摘录 / EXCERPT: Let $\mathcal{M}$ be a 1-cusped hyperbolic 3-manifold. Let $\mathcal{M}(p/q)$ and $\mathcal{M}(p'/q')$ be its p/q- and p'/q'-Dehn-filled manifolds, respectively, and suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-025181 | MATHDB-338294

Nondegeneracy conjecture for mixed Monge-Ampère measures

摘录 / EXCERPT: Assume that k is nontrivially valued, X is geometrically integral, and L is ample. Let $f\colon X_{\mathrm{div}}\to\mathbf{R}$ be a function such that ... for any FS metrics...

conjecture | open | MathDB 页面 | 原文线索

V2-025182 | MATHDB-338296

The coincidence conjecture for discriminants of irreducible tuples

摘录 / EXCERPT: Let A be an irreducible tuple, and let the three discriminant sets be those of Definition 2.33, consisting of the discriminant notions defined earlier in the paper. Coincidence c...

conjecture | open | MathDB 页面 | 原文线索

V2-025183 | MATHDB-338305

Strengthened Gromov conjecture on simplicial volume

摘录 / EXCERPT: Strengthened Gromov conjecture. Then M has positive simplicial volume:

conjecture | open | MathDB 页面 | 原文线索

V2-025184 | MATHDB-338317

Uniqueness conjecture for solutions of the Fu-Yau equation

摘录 / EXCERPT: Let φ be a solution of the Fu-Yau equation in Theorem. Uniqueness conjecture. The solution φ is unique. The conjecture asserts that the condition in the uniqueness...

conjecture | open | MathDB 页面 | 原文线索

V2-025185 | MATHDB-338446

Concavity conjecture for the complex Gursky-Streets operators

摘录 / EXCERPT: Let $u:{\mathbb R}\times{\mathbb C}^n\to{\mathbb R}$ be real-valued, and set ... where $\partial\bar\partial u$ is an $n\times n$ Hermitian matrix. For $1\leq k\leq n$, define ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-025186 | MATHDB-338452

The Volume Density Conjecture for biperiodic alternating links

摘录 / EXCERPT: Let $\mathcal L$ be a biperiodic alternating link with alternating quotient link L, and let K_n be alternating hyperbolic links satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-025187 | MATHDB-338484

Edel-Landjev-Xiang conjecture on the highest intersections with Hermitian surfaces

摘录 / EXCERPT: Edel-Landjev-Xiang conjecture. For each i, there exist linear homogeneous polynomials $H_1,\ldots,H_d$ such that the hyperplanes $V(H_1),\ldots,V(H_d)$ contain a common line and

conjecture | open | MathDB 页面 | 原文线索

V2-025188 | MATHDB-338532

Positive orthogonal Ricci curvature vanishing conjecture

摘录 / EXCERPT: Positive orthogonal Ricci curvature conjecture. If $Ric^{\perp}>0$ everywhere, then for every $1\leq p\leq m$ there is no nontrivial global holomorphic p-form; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-025189 | MATHDB-338547

Lê's asymptotic volume conjecture

摘录 / EXCERPT: Let Γ be the fundamental group of a connected, orientable, irreducible, compact 3-manifold whose boundary is empty or a collection of tori. For finite-index subgroups...

conjecture | open | MathDB 页面 | 原文线索

V2-025190 | MATHDB-338553

Homeomorphism conjecture for the generalized Kähler-Einstein metric space

摘录 / EXCERPT: Let X be an n-dimensional Kähler manifold with semi-ample canonical line bundle K_X. For sufficiently large ℓ, let $\pi:X\rightarrow X_{can}\subset\mathbb{C}P^N$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-025191 | MATHDB-338554

Uniform Ricci curvature bound conjecture along the Kähler-Ricci flow

摘录 / EXCERPT: Let X be an n-dimensional Kähler manifold with semi-ample canonical line bundle, let $\pi:X\rightarrow X_{can}$ be the associated holomorphic Calabi-Yau fibration, and let S...

conjecture | open | MathDB 页面 | 原文线索

V2-025192 | MATHDB-338671

The causal Legendrian-linking conjecture outside the Y^x_{ℓ} case

摘录 / EXCERPT: Causal Legendrian-linking conjecture. Assume that one cannot put a Y^x_{ℓ} Riemann metric on the Cauchy surface M. Then two events $x,y\in X$ are causally related if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-025193 | MATHDB-338676

Balletti-Kasprzyk conjecture on maximal-volume lattice polytopes with prescribed interior points

摘录 / EXCERPT: Let $d \ge 3$ and $k \ge 1$. Define the Sylvester sequence by ... and let ... Let $\mathcal{P}^d(k)$ be the family of all d-dimensional lattice polytopes with k interior lattice po...

conjecture | open | MathDB 页面 | 原文线索

V2-025194 | MATHDB-338688

The SKT metric conjecture on Frölicher spectral sequence degeneration

摘录 / EXCERPT: SKT degeneration conjecture. If X admits an SKT metric, then its Frölicher spectral sequence degenerates at the second page E_2.

conjecture | open | MathDB 页面 | 原文线索

V2-025195 | MATHDB-338694

Schoen's genus conjecture for index-one minimal surfaces

摘录 / EXCERPT: Let M^3 be a closed three-manifold with positive Ricci curvature, and let Σ be an orientable minimal surface in M^3. Schoen's conjecture. If Σ has index one, then...

conjecture | open | MathDB 页面 | 原文线索

V2-025196 | MATHDB-338700

Enumeration conjecture for width-2 positive-integer symplectic 2-friezes

摘录 / EXCERPT: A positive-integer symplectic 2-frieze of width w is a symplectic 2-frieze of width w whose entries are positive integers. Enumeration conjecture. There exist exactly 112 pos...

conjecture | open | MathDB 页面 | 原文线索

V2-025197 | MATHDB-338701

Infinitude conjecture for positive-integer symplectic 2-friezes

摘录 / EXCERPT: For an integer $w\geq 1$, consider symplectic 2-friezes of width w with positive-integer entries. The associated cluster algebra is of type...

conjecture | open | MathDB 页面 | 原文线索

V2-025198 | MATHDB-338704

The transversal-dimension conjecture for colorful Helly families

摘录 / EXCERPT: Transversal-dimension conjecture. For every integer k with $1\leq k\leq d$, there exist numbers $h(k,d)$ such that, for every d-colored family $\mathcal{F}$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-025199 | MATHDB-338714

The security conjecture for closed Riemannian manifolds

摘录 / EXCERPT: Security conjecture. A closed Riemannian manifold is secure if and only if it is flat.

conjecture | open | MathDB 页面 | 原文线索

V2-025200 | MATHDB-338741

The finite-monodromy criterion for equality of Kodaira-Iitaka and numerical dimensions

摘录 / EXCERPT: Finite-monodromy criterion. Finite monodromy provides the necessary and sufficient condition for

conjecture | open | MathDB 页面 | 原文线索

V2-025201 | MATHDB-338767

Marques-Neves-Schoen index-Betti number conjecture

摘录 / EXCERPT: Marques-Neves-Schoen conjecture. There exists $C>0$ such that, for all such hypersurfaces Σ, one has

conjecture | open | MathDB 页面 | 原文线索

V2-025202 | MATHDB-338769

Conjectural product formula for normalized volumes of klt singularities

摘录 / EXCERPT: Let $(x_i i (X_i,D_i))$ be klt singularities for $i=1,2$, and write $n_i:=\dim X_i$. Let $\pi_i:X_1\times X_2\to X_i$ be the projections. Conjectural product formula. ... The formula...

conjecture | open | MathDB 页面 | 原文线索

V2-025203 | MATHDB-338771

Local connectedness conjecture for planar self-similar sets

摘录 / EXCERPT: Let $\mathcal{F}$ be an IFS on $\mathbb{R}^2$ and let E be its attractor. Suppose that E satisfies the open set condition and that every contraction in $\mathcal{F}$ involves n...

conjecture | open | MathDB 页面 | 原文线索

V2-025204 | MATHDB-338784

The Type I pointwise scalar-curvature conjecture for shrinking Ricci flows

摘录 / EXCERPT: Let $\{g(t)\}_{t\in [0,T)}$ be the solution of the Ricci flow on a closed Riemannian manifold M^n with $n\geq 3$, and suppose that ... as $t\to T$. Type I scalar-curvature conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-025205 | MATHDB-338787

Polterovich's Poisson bracket conjecture for displaceable covers

摘录 / EXCERPT: Let $(M,c9)$ be a symplectic manifold. For a positive collection $\mathcal{F}$, let $pb(\mathcal{F})$ denote its Poisson bracket invariant, and for an open cover $\mathcal{U}$ let...

conjecture | open | MathDB 页面 | 原文线索

V2-025206 | MATHDB-338788

Weak Poisson bracket conjecture for displaceable covers

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold of dimension $2n$, and let $\mathcal{U}$ be an open cover constituted of displaceable sets. For $pb(\mathcal{U})$ as above and the symplec...

conjecture | open | MathDB 页面 | 原文线索

V2-025207 | MATHDB-338797

The Cantor graph equality conjecture for complex dimensions

摘录 / EXCERPT: Cantor graph complex-dimension conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-025208 | MATHDB-338846

Uniqueness conjecture for the period condition

摘录 / EXCERPT: Let a and b be the parameters occurring in the period condition, with $a<b$, and let t be the parameter in that condition. Uniqueness conjecture. If $a<b$, then there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-025209 | MATHDB-338850

Polyhedrality conjecture for linkage matching fields

摘录 / EXCERPT: Polyhedrality conjecture. A left linkage matching field is polyhedral if and only if it satisfies the compatible right submatching property.

conjecture | open | MathDB 页面 | 原文线索

V2-025210 | MATHDB-338863

Kodaira classification conjecture for birationally equivalent elliptic threefolds

摘录 / EXCERPT: Let $X\to B$ be a relatively minimal elliptic threefold with section, and let Σ be its stratified discriminant locus. The associated nonabelian gauge algebras and their repr...

conjecture | open | MathDB 页面 | 原文线索

V2-025211 | MATHDB-338879

Noncommutative Bondal-Orlov conjecture for noncommutative crepant resolutions

摘录 / EXCERPT: Let S be a normal noetherian Gorenstein domain, and let a noncommutative crepant resolution of S be an S-algebra of finite global dimension of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-025212 | MATHDB-338882

Local homogeneity conjecture for holomorphic Riemannian metrics

摘录 / EXCERPT: Local homogeneity conjecture. Any holomorphic Riemannian metric on X is locally homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-025213 | MATHDB-338883

Local homogeneity conjecture for affine-type geometric structures

摘录 / EXCERPT: Local homogeneity conjecture. Any such structure ϕ is locally homogeneous. This implies that if ϕ is rigid, then the fundamental group of X is infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-025214 | MATHDB-338896

Maupertuis' conjecture for cubic polynomial integrals on the torus

摘录 / EXCERPT: Let T^2 be the two-dimensional torus with a Riemannian metric, and consider its geodesic flow. A polynomial integral of degree 3 in momenta is a nontrivial function on the cota...

conjecture | open | MathDB 页面 | 原文线索

V2-025215 | MATHDB-338902

Khesin-Soloviev integrability conjecture for diagonal generalized pentagram maps

摘录 / EXCERPT: Let (v_i) be a twisted polygon in $b2b2\text{?}b2b2$ and let I be a multi-index defining a generalized pentagram map $T_{I,J}$. The map $T_{I,I}$ is the diagonal case in which...

conjecture | open | MathDB 页面 | 原文线索

V2-025216 | MATHDB-338903

Generalized argument shift conjecture for finite-dimensional Lie algebras

摘录 / EXCERPT: Let $b2b2\mathfrak g$ be a finite-dimensional Lie algebra, let $b2b2\mathfrak g^*$ be its dual, and let $a\in\mathfrak g^*$ be regular. The dual carries the linear Lie-Poisson brac...

conjecture | open | MathDB 页面 | 原文线索

V2-025217 | MATHDB-338904

Pelayo-Vũ Ngọc inverse spectral conjecture for semitoric systems

摘录 / EXCERPT: A semitoric system is an integrable system $F=(J,H)$, with semiclassical joint spectrum given by the points $(\lambda,\nu)\in\mathbb R^2$ where λ is an eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-025218 | MATHDB-338937

Makai Jr.'s reverse isodiametric conjecture

摘录 / EXCERPT: Makai Jr.'s conjecture. For every $K\in\mathcal{K}^n$ there is a linear transformation $A\in\operatorname{GL}_n(\mathbb{R})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025219 | MATHDB-338938

Dvoretzky-Rogers equality conjecture for the reverse isodiametric problem

摘录 / EXCERPT: Dvoretzky-Rogers equality conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-025220 | MATHDB-338942

The K-theory presentation conjecture for torus manifold bundles

摘录 / EXCERPT: K-theory presentation conjecture. The ring $K^*(E(X))$ is a free $K^*(B)$-module of rank $m=\chi(X)$ and is isomorphic to $\mathfrak{S}/\mathfrak{J}$.

conjecture | open | MathDB 页面 | 原文线索

V2-025221 | MATHDB-338957

Powell's conjecture on Powell equivalence of Heegaard-splitting trivializations

摘录 / EXCERPT: Powell's conjecture. Any h,h' are Powell equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-025222 | MATHDB-338959

The eyeglass-twist conjecture for standard Heegaard surfaces

摘录 / EXCERPT: Eyeglass-twist conjecture. Any eyeglass twist on T_g is a Powell move.

conjecture | open | MathDB 页面 | 原文线索

V2-025223 | MATHDB-338960

The Goeritz-conjugacy eyeglass-twist conjecture

摘录 / EXCERPT: Goeritz-conjugacy conjecture. Any eyeglass twist on T_g is Goeritz conjugate to a Powell move.

conjecture | open | MathDB 页面 | 原文线索

V2-025224 | MATHDB-338997

Szőnyi's minimum-size conjecture for non-trivial blocking sets

摘录 / EXCERPT: Szőnyi's conjecture. The blocking set B has at least

conjecture | open | MathDB 页面 | 原文线索

V2-025225 | MATHDB-339023

The stronger Arnold-Thom conjecture for unit tangents

摘录 / EXCERPT: Let f be an analytic function and let $x(t)$ be a gradient flow line of f. Suppose that $x(t)$ has a limit point x_{∞} as $t\to\infty$. Stronger Arnold-Thom conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-025226 | MATHDB-339028

Finiteness conjecture for touching points of self-touching CMC surfaces

摘录 / EXCERPT: Let M be a three-dimensional Riemannian manifold with positive Ricci curvature, and let S be a self-touching constant mean curvature (CMC) surface in M. Finiteness conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-025227 | MATHDB-339029

Geodesic touching-set conjecture for CMC surfaces

摘录 / EXCERPT: Let M be a three-dimensional Riemannian manifold, and let S be a constant mean curvature (CMC) surface in M whose touching set is one-dimensional. Geodesic touching-set conje...

conjecture | open | MathDB 页面 | 原文线索

V2-025228 | MATHDB-339030

No-neck-pinching conjecture for kissing CMC spheres

摘录 / EXCERPT: Let M be a compact three-dimensional manifold with positive Ricci curvature. Suppose that S_1 and S_2 are embedded constant mean curvature (CMC) spheres in M that kiss at a...

conjecture | open | MathDB 页面 | 原文线索

V2-025229 | MATHDB-339037

Conjecture on eventual coincidence of the Assouad spectra

摘录 / EXCERPT: Let $F\subseteq\mathbb{R}^d$. Denote by $\overline{\dim}_{\mathrm{A}}^\theta F$ and $\dim_{\mathrm{A}}^\theta F$ the upper and ordinary Assouad spectra, respectively. Eventual coin...

conjecture | open | MathDB 页面 | 原文线索

V2-025230 | MATHDB-339041

The negative second scalar curvature conjecture for rationally connected Kähler manifolds

摘录 / EXCERPT: Negative second scalar curvature conjecture. The manifold M cannot be rationally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-025231 | MATHDB-339052

The equal-edge projection conjecture for cube projections

摘录 / EXCERPT: Equal-edge projection conjecture. The maximum volume of a projection of C^n onto a subspace is attained when the projections of all edges of the cube have the same...

conjecture | open | MathDB 页面 | 原文线索

V2-025232 | MATHDB-339085

The rational points conjecture for the 3x+1 set

摘录 / EXCERPT: Let M be the map used to embed $\mathbb Z_2$ into $[0,2]$, and let Q be the 3x+1 conjugacy map. Define the 3x+1 set by ... A point of this set has rational coordinates when both...

conjecture | open | MathDB 页面 | 原文线索

V2-025233 | MATHDB-339103

Meier's conjecture on trisections of the 4-sphere

摘录 / EXCERPT: Meier's conjecture. Every trisection of S^4 is either the $(0,0)$-trisection or a stabilization of the $(0,0)$-trisection.

conjecture | open | MathDB 页面 | 原文线索

V2-025234 | MATHDB-339108

The degree-one characterization conjecture for amplituhedron cells

摘录 / EXCERPT: Let $f\in\tilde{S_n}(-k,n-k)$ be an affine permutation with $\operatorname{inv}(f)=k\ell$. A cell f is (n,k,m)-admissible when it satisfies the source's admissibility condition...

conjecture | open | MathDB 页面 | 原文线索

V2-025235 | MATHDB-339112

Polynomial numerator conjecture for amplituhedron forms

摘录 / EXCERPT: Polynomial numerator conjecture. There exists a polynomial $H(S,A,B)$ in the entries of S,A,B such that

conjecture | open | MathDB 页面 | 原文线索

V2-025236 | MATHDB-339119

Calderón's problem for the Schrödinger equation on Riemannian manifolds

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold with smooth boundary, and assume that $q_1, q_2 \in C^{\infty}(M)$ so that 0 is not a Dirichlet eigenvalue of $\Delta_g + q_j$ in M...

conjecture | open | MathDB 页面 | 原文线索

V2-025237 | MATHDB-339120

Linearized Calderón problem for the Schrödinger equation

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold with smooth boundary. Let $f \in C^{\infty}(M)$ vanish to infinite order at ∂M and satisfy ... for all $u_j \in C^{\infty}(M)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025238 | MATHDB-339127

Continuity conjecture for semipositive envelopes

摘录 / EXCERPT: Let L) be an ample line bundle and let P be the semipositive envelope operator defined as the pointwise limit $P(\mphi)=\sup_m P_m(\phi)$ of the projection operators P_m. A m...

conjecture | open | MathDB 页面 | 原文线索

V2-025239 | MATHDB-339128

Regularization conjecture for asymptotically Fubini-Study metrics

摘录 / EXCERPT: Let (ϕ_α) be a family of asymptotically Fubini-Study metrics, uniformly bounded above. The upper semicontinuous upper envelope is $\left(\sup_\alpha\phi_\alpha\right)^*$,...

conjecture | open | MathDB 页面 | 原文线索

V2-025240 | MATHDB-339186

The coercivity form of the Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Coercivity form of the Yau-Tian-Donaldson conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-025241 | MATHDB-339195

Representations are sheaves conjecture for Legendrian links

摘录 / EXCERPT: Let $n\geq 1$, let $\mathbbm{k}$ be a field, and let $\Lambda\subset\mathbb{R}^3$ be a Legendrian link. The representation-sheaf conjecture. The A_∞ category...

conjecture | open | MathDB 页面 | 原文线索

V2-025242 | MATHDB-339232

Sharp width-dependent Turán conjecture for convex domains

摘录 / EXCERPT: Width-dependent Turán conjecture. There exists an absolute constant $c>0$ such that, for every compact convex domain $K\Subset\mathbb C$ and every $p\in\mathcal P_n(K)$,

conjecture | open | MathDB 页面 | 原文线索

V2-025243 | MATHDB-339237

Conjecture on contact type for positive magnetic curvature

摘录 / EXCERPT: Let (S^2,g,σ) be a symplectic magnetic system. For $m>0$, define the magnetic curvature on the energy level Σ_m by ... Assume that K_m is positive. Contact-type conje...

conjecture | open | MathDB 页面 | 原文线索

V2-025244 | MATHDB-339250

The main volume-Mahler measure conjecture for biperiodic alternating links

摘录 / EXCERPT: Let $\mathcal{L}$ be any biperiodic alternating link, with toroidally alternating Λ-quotient link L. Let $p(z,w)$ be the characteristic polynomial of the toroidal dimer m...

conjecture | open | MathDB 页面 | 原文线索

V2-025245 | MATHDB-339251

The volume bound after augmentation of an alternating link

摘录 / EXCERPT: Let K be a hyperbolic alternating link, and let B be an augmented unknot around any two parallel strands of K. Augmentation volume conjecture. ... This conjecture would give an...

conjecture | open | MathDB 页面 | 原文线索

V2-025246 | MATHDB-339253

The anisotropic partial Calderón conjecture for singular metrics

摘录 / EXCERPT: Anisotropic partial Calderón conjecture. It should follow that $g=\tilde{g}$ up to the diffeomorphism gauge invariance if $\dim M\geq 3$, and up to the diffeomorphism and conformal...

conjecture | open | MathDB 页面 | 原文线索

V2-025247 | MATHDB-339271

Renormalized-volume asymptotics conjecture for polytopes in double hyperbolic space

摘录 / EXCERPT: Renormalized-volume asymptotics conjecture. The quantities $\mu_{u,\epsilon}(P_{+})$, $\mu_{h,\epsilon}(P_{+})$, and $\mu'_{h,\epsilon}(P_{+})$ can all be expressed by these expans...

conjecture | open | MathDB 页面 | 原文线索

V2-025248 | MATHDB-339287

Compatibility of the vertical-collar filtration with the s-dot operad

摘录 / EXCERPT: Compatibility conjecture. The filtration of the vertical-collar theorem defines a map of colored planar ∞-operads to the colored planar ∞-operad obtained from the s...

conjecture | open | MathDB 页面 | 原文线索

V2-025249 | MATHDB-339288

Paracyclic compatibility of the vertical-collar filtration

摘录 / EXCERPT: Paracyclic compatibility conjecture. The map is compatible with the paracyclic action on the s-dot construction.

conjecture | open | MathDB 页面 | 原文线索

V2-025250 | MATHDB-339299

S.-T. Yau's rigidity conjecture for negatively curved Kähler manifolds

摘录 / EXCERPT: Let M be a negatively curved compact Kähler manifold of complex dimension at least 2. Yau's rigidity conjecture. The complex structure of M is unique. This conjecture general...

conjecture | open | MathDB 页面 | 原文线索

V2-025251 | MATHDB-339300

Zheng's hyperbolicity conjecture for nonpositively curved Kähler manifolds

摘录 / EXCERPT: Let M be a nonpositively curved compact Kähler manifold. Zheng's conjecture. If M is of general type, then it must be Kobayashi hyperbolic, meaning that every holomorphic map .....

conjecture | open | MathDB 页面 | 原文线索

V2-025252 | MATHDB-339301

Conjecture on scalar-curvature convergence along the Kähler-Ricci flow

摘录 / EXCERPT: Let X be an n-dimensional Kähler manifold with nef canonical bundle K_X and positive Kodaira dimension. For an initial Kähler metric ω_0, let $\omega(t)$ solve the nor...

conjecture | open | MathDB 页面 | 原文线索

V2-025253 | MATHDB-339328

Kontsevich's smooth DG algebra degeneration conjecture

摘录 / EXCERPT: Let A be a smooth DG algebra. Let $\operatorname{ch}:K_0(A\otimes A^{op})\to (HH_{\bullet}(A)\otimes HH_{\bullet}(A^{op}))_0$ be the Chern character, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025254 | MATHDB-339329

Kaledin-Efimov generalized degeneration conjecture for DG categories

摘录 / EXCERPT: Let $\mathcal B$ and $\mathcal C$ be small DG categories over a field $\mathrm k$ of characteristic zero. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025255 | MATHDB-339331

Classification conjecture for positively curved Alexandrov 4-spaces with circle symmetry

摘录 / EXCERPT: Let T^1 act isometrically and effectively on X^4, where X^4 is a 4-dimensional, closed, positively curved, orientable Alexandrov space. Classification conjecture. Up to equ...

conjecture | open | MathDB 页面 | 原文线索

V2-025256 | MATHDB-339336

Transitivity conjecture for Nil-Killing Lie algebras on CSI Lorentzian spaces

摘录 / EXCERPT: Transitivity conjecture. If M is an n-dimensional Lorentzian space with all curvature invariants constant, then there exists a transitive set $\mathcal{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-025257 | MATHDB-339340

Optimality of k equal balls for Robin eigenvalues

摘录 / EXCERPT: Fix a dimension $d\geq 2$ and an integer $k\geq 3$. Let $\Omega\subset\mathbb{R}^d$ be a sufficiently smooth domain of volume 1, and let $\mathcal{B}_k$ denote the disjoint union...

conjecture | open | MathDB 页面 | 原文线索

V2-025258 | MATHDB-339341

Equal-disks conjecture for sums of Robin eigenvalues

摘录 / EXCERPT: Fix positive numbers A and α. For a Lipschitz domain $\Omega\subset\mathbb{R}^2$ of area A, let $\lambda_j(\Omega,\alpha)$ denote its jth Robin eigenvalue. Let the can...

conjecture | open | MathDB 页面 | 原文线索

V2-025259 | MATHDB-339347

Miao's rigidity conjecture for nonnegative scalar curvature

摘录 / EXCERPT: Let (M,g) be a Riemannian n-manifold with nonnegative scalar curvature, boundary $\partial M=S^{n-1}(1)$, and second fundamental form satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-025260 | MATHDB-339356

Existence of constant-curvature ball packings from equality of extended and ordinary Yamabe invariants

摘录 / EXCERPT: Yamabe-invariant equality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025261 | MATHDB-339362

Equality of rooftop and model envelopes for regularizable psh metrics

摘录 / EXCERPT: Let L be an ample line bundle and let $\operatorname{PSH}^{\uparrow}(L)$ be the space of psh metrics regularizable from below. For two metrics...

conjecture | open | MathDB 页面 | 原文线索

V2-025262 | MATHDB-339372

The dimension-seven conjecture for tritangents of space sextics

摘录 / EXCERPT: Dimension-seven conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025263 | MATHDB-339413

Stationary GW/PT descendent correspondence beyond toric threefolds

摘录 / EXCERPT: Stationary GW/PT correspondence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025264 | MATHDB-339415

Equivariant DT/GW descendent correspondence modulo the cubic relation

摘录 / EXCERPT: Equivariant DT/GW correspondence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025265 | MATHDB-339416

Equivariant DT descendent series Frankenstein algebra conjecture

摘录 / EXCERPT: Equivariant DT descendent algebra conjecture. For $\gamma_i\in H^*_{\mathsf{T}}(X,\mathbb{C})$ and $\beta\in H_2(X,\mathbb{Z})$,

conjecture | open | MathDB 页面 | 原文线索

V2-025266 | MATHDB-339429

Miller's exact geometric conjecture for the heat observability constant

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be a Riemannian manifold and let $\omega\subset\mathcal{M}$ satisfy $\overline{\omega}\neq\mathcal{M}$. Define ... Let ${\mathfrak K}_{heat}(\omega)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-025267 | MATHDB-339434

Birational weak Zariski decomposition conjecture for generalized polarized pairs

摘录 / EXCERPT: Birational weak Zariski decomposition conjecture. Every such pair admits a birational NQC weak Zariski decomposition over Z.

conjecture | open | MathDB 页面 | 原文线索

V2-025268 | MATHDB-339437

Realizability of internal divisors on smooth tropicalizations

摘录 / EXCERPT: Let Γ be a smooth tropicalization of an algebraic curve C. An internal divisor is a divisor lying in a cell σ of $|\mathcal{D}_{\Gamma}|$ parametrizing divisors wit...

conjecture | open | MathDB 页面 | 原文线索

V2-025269 | MATHDB-339491

Meeks-Pérez-Ros conjecture on stable two-sided minimal surfaces

摘录 / EXCERPT: Let (M,g) be a closed Riemannian 3-manifold. A closed embedded minimal surface is one whose two-sided covering is stable when its lifted surface has non-negative second variati...

conjecture | open | MathDB 页面 | 原文线索

V2-025270 | MATHDB-339504

Tropical Griffiths-type defect relation for tropical hypersurfaces

摘录 / EXCERPT: Tropical Griffiths-type defect conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-025271 | MATHDB-339505

Truncated tropical second main theorem for meromorphic functions

摘录 / EXCERPT: Truncated tropical second main theorem conjecture. These truncated inequalities and equality are conjectured to hold as stated. They are tropical analogues of the truncated classic...

conjecture | open | MathDB 页面 | 原文线索

V2-025272 | MATHDB-339506

Truncated tropical Cartan second main theorem for hypersurfaces

摘录 / EXCERPT: Truncated tropical Cartan conjecture. The displayed truncated second main theorem is proposed as the tropical counterpart of the truncated Cartan second main theorem. The source gi...

conjecture | open | MathDB 页面 | 原文线索

V2-025273 | MATHDB-339511

Compactification conjecture for wrapped Fukaya categories

摘录 / EXCERPT: Let $\check{Y}$ be a Calabi--Yau manifold, let $\check{X}\subset\check{Y}$ be a smooth ample divisor, and set $\check{V}=\check{Y}\setminus\check{X}$. Let ${\mathcal W}(\check V)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025274 | MATHDB-339521

Grinenko's K-condition conjecture for degree-1 del Pezzo fibrations

摘录 / EXCERPT: Let $X\to\mathbb{P}^1$ be a del Pezzo fibration of degree 1 with at most terminal singularities. It satisfies the K-condition when $-K_X$ is not in the interior of the movable...

conjecture | open | MathDB 页面 | 原文线索

V2-025275 | MATHDB-339531

The genus-two classification conjecture for 3-manifolds almost in a trisection spine

摘录 / EXCERPT: Let M be a closed orientable 3-manifold that is not of the form $\#^k S^1 \times S^2$, and let M embed so as to almost lie in the spine of a genus-2 trisection of a 4-man...

conjecture | open | MathDB 页面 | 原文线索

V2-025276 | MATHDB-339532

The universal orientable 3-manifold embedding conjecture for stabilized S^2-bundles

摘录 / EXCERPT: Let M be an orientable 3-manifold. Universal embedding conjecture. There is a positive integer n such that M embeds almost in the spine of a minimal-genus trisection of ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-025277 | MATHDB-339587

Generalized finite-dimensionality conjecture for (v,p)-polynomial-growth harmonic functions

摘录 / EXCERPT: Let M be a Riemannian manifold with non-negative Ricci curvature, let $x_0\in M$ be a fixed point, and write $r(x)=d(x,x_0)$. Let v be a real function. A harmonic function has...

conjecture | open | MathDB 页面 | 原文线索

V2-025278 | MATHDB-339602

Ohnita's conjecture on minimal two-spheres in quaternionic projective space

摘录 / EXCERPT: Let $f:S^2\to\textbf{HP}^n$ be a proper minimal immersion, meaning that $f(S^2)$ is not contained in any totally geodesic submanifold of $\textbf{HP}^n$, and suppose that f has c...

conjecture | open | MathDB 页面 | 原文线索

V2-025279 | MATHDB-339624

Polystable reduction conjecture over valuation rings

摘录 / EXCERPT: Polystable reduction conjecture. There exists a modification $Y\to X$ such that $Y_\eta=X_\eta$ and Y is polystable over ${\mathcal O}$.

conjecture | open | MathDB 页面 | 原文线索

V2-025280 | MATHDB-339625

The hyperplane asymptotic conjecture for non-geodesic horizontal projections

摘录 / EXCERPT: Let G be a Carnot group with first layer V_1, let $\pi:G\to G/[G,G]$ be the quotient map, and let d denote the relevant distance. Suppose that $\gamma:\mathbb{R}\to G$ is a g...

conjecture | open | MathDB 页面 | 原文线索

V2-025281 | MATHDB-339647

Characterization conjecture for MTP2-tidy configurations

摘录 / EXCERPT: Let X be a finite min-max closed configuration of points in $\mathbb{R}^d$ such that ... Here, X is MTP2-tidy when the set of feasible tent-function heights yielding supermodular...

conjecture | open | MathDB 页面 | 原文线索

V2-025282 | MATHDB-339659

Classification conjecture for compact Kähler four-manifolds with positive orthogonal Ricci curvature

摘录 / EXCERPT: Classification conjecture. A compact complex manifold M^4 admits a Kähler metric with $Ric^{\perp}>0$ if and only if M^4 is biholomorphic to ${\mathbb P}^4$, ${\mathbb Q}^4$, o...

conjecture | open | MathDB 页面 | 原文线索

V2-025283 | MATHDB-339693

Average error estimate for the spectral counting function on singular surfaces

摘录 / EXCERPT: Average error estimate. We have

conjecture | open | MathDB 页面 | 原文线索

V2-025284 | MATHDB-339701

Reznikov's period integral conjecture for closed geodesics on hyperbolic surfaces

摘录 / EXCERPT: Let (M,g) be a compact hyperbolic surface, let e_λ be an eigenfunction of $\sqrt{-\Delta_g}$ with eigenvalue λ, and let γ be a periodic geodesic or a geode...

conjecture | open | MathDB 页面 | 原文线索

V2-025285 | MATHDB-339702

Reznikov's Fourier coefficient conjecture for restrictions to geodesics

摘录 / EXCERPT: Let (M,g) be a compact hyperbolic surface, let e_λ be an eigenfunction of $\sqrt{-\Delta_g}$ with eigenvalue λ, and let γ be a closed geodesic or a geodesi...

conjecture | open | MathDB 页面 | 原文线索

V2-025286 | MATHDB-339705

The complex-manifold conjecture for q-stability conditions

摘录 / EXCERPT: Let $\operatorname{QStab}\operatorname{\mathcal{D}}_\mathbb{X}$ be the union of the spaces of q-stability conditions over all parameters $s\in\mathbb{C}$, and let n be the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-025287 | MATHDB-339719

The fixed-component closure conjecture for framed instanton sheaves on projective three-space

摘录 / EXCERPT: Let ${\mathbb{P}}^{3}$ carry a lifted toric action, and consider the moduli space of framed instantons, including its fixed components and its locally free component. Fixed-compone...

conjecture | open | MathDB 页面 | 原文线索

V2-025288 | MATHDB-339769

Infinite-dimensionality conjecture for unimodular trees with infinitely many ends

摘录 / EXCERPT: Let $[\boldsymbol{T},\boldsymbol{o}]$ be a unimodular tree with infinitely many ends, endowed with its graph-distance metric. Infinite-dimensionality conjecture. One has ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-025289 | MATHDB-339770

Stable-domain conjecture for eternal Galton-Watson tree dimensions

摘录 / EXCERPT: Let $[\boldsymbol{T},\boldsymbol{o}]$ be a unimodular eternal Galton-Watson tree. Suppose its offspring distribution is in the domain of attraction of an α-stable distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-025290 | MATHDB-339777

Regular CW complex conjecture for the positive orthogonal Grassmannian

摘录 / EXCERPT: Let $\operatorname{OG}_{\ge0}(n,2n)$ be the positive orthogonal Grassmannian, with cell decomposition ... where the cells are indexed by matchings τ on $[2n]$, and the cell inde...

conjecture | open | MathDB 页面 | 原文线索

V2-025291 | MATHDB-339782

Quasi-positive holomorphic sectional curvature and rational connectedness

摘录 / EXCERPT: Quasi-positive curvature conjecture. Then X is projective and rationally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-025292 | MATHDB-339783

RC-positivity, ampleness, and positivity of tautological bundles

摘录 / EXCERPT: RC-positivity equivalence conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-025293 | MATHDB-339784

RC-positive tangent bundle conjecture for rationally connected manifolds

摘录 / EXCERPT: RC-positive tangent bundle conjecture. If X is rationally connected, then T_X is RC-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-025294 | MATHDB-339789

Finite Tanaka prolongation conjecture for locally contactomorphic polarised Lie groups

摘录 / EXCERPT: Let G be a stratified Lie group and let S be a polarised Lie group that is locally contactomorphic to G. Finite Tanaka prolongation conjecture. There is a finite Tanaka prolo...

conjecture | open | MathDB 页面 | 原文线索

V2-025295 | MATHDB-339800

Marques-Neves multiplicity one conjecture for min-max minimal hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be a closed manifold with a generic Riemannian metric, where $3\le n+1\le 7$. Consider closed minimal hypersurfaces obtained by min-max methods, and call a component...

conjecture | open | MathDB 页面 | 原文线索

V2-025296 | MATHDB-339807

Alexandrov-Fenchel inequality for odd curvature integrals

摘录 / EXCERPT: Alexandrov-Fenchel inequality. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-025297 | MATHDB-339818

Generalized Prokhorov-Shokurov conjecture for generalized pairs

摘录 / EXCERPT: Let $(X',B'+M')$ be a generalized sub-pair with data $X\to X'$ and M, where B', M' and M are $\mathbb{Q}$-divisors. Let $f\colon X'\to Z'$ be a contraction such that ... Assu...

conjecture | open | MathDB 页面 | 原文线索

V2-025298 | MATHDB-339834

Stable quotient holomorphic anomaly and polynomiality conjecture for local b2c0^2

摘录 / EXCERPT: Let $K\mathbb{P}^2$ be the local projective plane, let q be the degree variable, and let $T=I^{K\mathbb{P}^2}_1(q)$ be the mirror coordinate with $Q(q)=\exp(T)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025299 | MATHDB-339835

Stable quotient holomorphic anomaly equations for local \u03b2c0^2

摘录 / EXCERPT: Let $K\mathbb{P}^2$ be the local projective plane, let $T=I^{K\mathbb{P}^2}_1(q)$ be the mirror coordinate, and let $\mathcal{F}^{\mathsf{SQ}}_g$ denote its genus-g stable quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-025300 | MATHDB-339836

Stable quotient polynomiality conjecture for local \u03b2c0^3

摘录 / EXCERPT: Let $K\mathbb{P}^3$ be the local projective three-space, let $T=I^{K\mathbb{P}^3}_1(q)$ be the mirror coordinate, and let $\mathcal{F}^{\mathsf{SQ}}_{g,a+b}a,b$ denote the stable q...

conjecture | open | MathDB 页面 | 原文线索

V2-025301 | MATHDB-339838

Equivariant Birkhoff-factorization coefficient conjecture for local \u03b2c0^2

摘录 / EXCERPT: Let $R_{k,i}$ be the coefficients in the asymptotic expansion of the restriction of the equivariant I-function to $H=\lambda_i$ for local $\mathbb{P}^2$, and let $\mathds{G}_2$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-025302 | MATHDB-339841

Conjecture that the dilatation condition can be removed from the spherical-area theorem

摘录 / EXCERPT: Let $f=h+\overline g\in\mathcal S_H^0$ be a normalized univalent, sense-preserving harmonic mapping, and let $\Omega=f(\mathbb D)$. Theorem 2 in the paper proves the stated spheric...

conjecture | open | MathDB 页面 | 原文线索

V2-025303 | MATHDB-339859

The strengthened lower bound for non-hexagonal disk packings

摘录 / EXCERPT: Strengthened packing conjecture. In the conclusion of the hexagonal packing uniqueness conjecture, the assertion $\delta(S)>0$ can be replaced by the stronger assertion...

conjecture | open | MathDB 页面 | 原文线索

V2-025304 | MATHDB-339862

The higher-order curve-cutting conjecture

摘录 / EXCERPT: Let $k\geq 1$ be an integer, let $\varepsilon>0$, and let $\mathcal{C}$ be a set of n irreducible algebraic curves in $\mathbb{R}^2$. A cutting of the curves means decomposing th...

conjecture | open | MathDB 页面 | 原文线索

V2-025305 | MATHDB-339863

The point-curve incidence conjecture

摘录 / EXCERPT: Let $\mathcal{P}\subset\mathbb{R}^2$ be a set of n points and let $\mathcal{C}$ be a set of n irreducible algebraic plane curves. An incidence is a pair...

conjecture | open | MathDB 页面 | 原文线索

V2-025306 | MATHDB-339864

The finite-field higher-order tangency bound conjecture

摘录 / EXCERPT: Let F be a field, let $k\geq 1$ be an integer, and let $\mathcal{C}$ be a set of n irreducible curves in F^2. For a point $p\in F^2$, let $m_{k,\mathcal{C}}(p)$ be the number...

conjecture | open | MathDB 页面 | 原文线索

V2-025307 | MATHDB-339899

Existence of Milnor vector fields for real analytic map germs

摘录 / EXCERPT: Milnor vector field conjecture. If both fibrations exist under these conditions, then they are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-025308 | MATHDB-339904

Existence conjecture for special Kähler structures with prescribed cubic differentials

摘录 / EXCERPT: Let $frak b=(\beta_1,dots,\beta_k)$ be a tuple as in Definition ref{defn:moduli}, and let $gamma>1$ be an integer. Assume that the numerical condition eqref{EqNecessCondMixedSing}...

conjecture | open | MathDB 页面 | 原文线索

V2-025309 | MATHDB-339926

Toric mirror equivalence for punctured-disk categories

摘录 / EXCERPT: Let $\calY=\mathbb C^n/T$ and $\calY^*=\mathbb C^n/T_F^{\vee}$ be the dual toric pair considered in the source. Toric mirror equivalence conjecture. There are equivalences of facto...

conjecture | open | MathDB 页面 | 原文线索

V2-025310 | MATHDB-339939

The Mori-surface conjecture for Hirzebruch surfaces

摘录 / EXCERPT: Mori-surface conjecture. The surface $\mathbb{F}_r$ is a Mori surface.

conjecture | open | MathDB 页面 | 原文线索

V2-025311 | MATHDB-339941

Blowup-blowdown classification conjecture for elliptic algebras

摘录 / EXCERPT: Let T and T' be elliptic algebras with equal graded quotient rings ... A blowup or blowdown is the corresponding birational operation on elliptic algebras described in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-025312 | MATHDB-339946

Uniqueness conjecture for the period condition in the orthorhombic deformation family

摘录 / EXCERPT: Let a and b be the parameters appearing in the period condition, and let t denote the deformation parameter. Uniqueness conjecture. If $a<b$, then there exists a unique t t...

conjecture | open | MathDB 页面 | 原文线索

V2-025313 | MATHDB-339955

Gromov-Hausdorff distance conjecture for a regular polygon and the circle

摘录 / EXCERPT: Let S^1 be the circle of unit radius in $\mathbb{R}^2$ centered at the origin, and let P_n be an inscribed regular polygon with n sides; equip both with the Euclidean metric....

conjecture | open | MathDB 页面 | 原文线索

V2-025314 | MATHDB-340033

ACC conjecture for log canonical thresholds

摘录 / EXCERPT: Let $I\subseteq[0,1]$ and $J\subseteq\mathbb{R}_{>0}$ be sets satisfying the descending chain condition. For a log pair (X,Δ) of dimension n over an arbitrary field, with...

conjecture | open | MathDB 页面 | 原文线索

V2-025315 | MATHDB-340041

Scaling nonvanishing conjecture for isometries of noncollapsed Ricci-limit spaces

摘录 / EXCERPT: Scaling nonvanishing conjecture. Given n and $v>0$, there is a positive function $\Phi(\delta,n,v)$ such that every isometry of M is scaling $\Phi(\delta,n,v)$-nonvanishing at...

conjecture | open | MathDB 页面 | 原文线索

V2-025316 | MATHDB-340042

Bounded-index abelian subgroup conjecture for nonnegatively Ricci-curved manifolds

摘录 / EXCERPT: Nonnegative-Ricci abelian subgroup conjecture. There exists a constant $C(n)$ such that $\pi_1(M)$ contains an abelian subgroup of index at most $C(n)$. Moreover, this subgroup can...

conjecture | open | MathDB 页面 | 原文线索

V2-025317 | MATHDB-340051

Unimodular Frostman measure equality conjecture

摘录 / EXCERPT: Let $\boldsymbol{D}$ be the random metric object in the unimodular Frostman setting, let $\mathcal H^{\alpha}_{M}(\boldsymbol{D})$ denote its unimodular Hausdorff measure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025318 | MATHDB-340058

The irregular Fukaya category realization conjecture

摘录 / EXCERPT: Irregular Fukaya realization conjecture. There exists a version of Fukaya category $\Fuk_m(T^*Z)$ that modifies Nadler--Zaslow's category and realizes

conjecture | open | MathDB 页面 | 原文线索

V2-025319 | MATHDB-340059

The irregular local-system Lagrangian realization conjecture

摘录 / EXCERPT: Irregular local-system realization conjecture. The Lagrangian $L_{\cM}$ defines an object of $\Fuk_m(T^*X)$ independent of the choice of f. The full subcategory of $\Fuk_m(T^*X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025320 | MATHDB-340062

Bernig-Fu-Solanes angularity conjecture for curvature measures

摘录 / EXCERPT: Let M be a Riemannian manifold. The space of angular curvature measures is denoted by $\mathcal{A}(M)$, and the Lipschitz-Killing algebra by $\mathcal{L}\mathcal{K}(M)$. Here a v...

conjecture | open | MathDB 页面 | 原文线索

V2-025321 | MATHDB-340063

Bernig-Fu-Solanes characterization conjecture for angular valuations

摘录 / EXCERPT: Let M be a Riemannian manifold. A valuation μ is called angular when ... where $\mathcal{A}(M)$ is the space of angular curvature measures and $\cdot$ denotes the Alesker produ...

conjecture | open | MathDB 页面 | 原文线索

V2-025322 | MATHDB-340070

Universal identity between refined Mochizuki and Vafa-Witten formulas

摘录 / EXCERPT: For integers $(\beta_1,\beta_2,\beta_3,\beta_4)$ and $v\geq0$, let $F_v(\beta_1,\beta_2,\beta_3,\beta_4,y)$ and $G_v(\beta_1,\beta_2,\beta_3,\beta_4,y)$ be the universal functions...

conjecture | open | MathDB 页面 | 原文线索

V2-025323 | MATHDB-340098

Lutz-Kühnel's minimality conjecture for tight triangulations

摘录 / EXCERPT: Lutz-Kühnel minimality conjecture. Tight triangulations of M minimize every entry of the f-vector among all triangulations of M.

conjecture | open | MathDB 页面 | 原文线索

V2-025324 | MATHDB-340119

Eigenvalue pinching conjecture near the sphere under a lower Ricci bound

摘录 / EXCERPT: Let $n\geq 2$, and let $\epsilon>0$ and $K>0$. For an n-dimensional closed Riemannian manifold (M,g), write $\lambda_{n+1}(\bar{\Delta}^{E},M)$ for the $(n+1)$-st eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-025325 | MATHDB-340141

Beck et al.'s triangulation conjecture for the lecture hall cone

摘录 / EXCERPT: Let $\mathscr{R}_n$ be the lecture hall cone in the special case associated with $(1,2,\ldots,n)$, let $\mathfrak{S}_{n-1}$ be the symmetric group, and let $\mathrm{des}(\pi)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-025326 | MATHDB-340145

Gromov's approximation and interpolation conjectures for elliptic manifolds

摘录 / EXCERPT: Gromov's conjectures. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-025327 | MATHDB-340147

Higher-dimensional ordinary-hyperplane structure conjecture

摘录 / EXCERPT: Higher-dimensional ordinary-hyperplane structure conjecture. One of the following holds: (i) all but at most $O(K)$ points of $\mathcal{S}$ are contained in a single hyperplane; or...

conjecture | open | MathDB 页面 | 原文线索

V2-025328 | MATHDB-340159

Polya's conjecture for Dirichlet and Neumann Laplace spectra

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be compact with Lipschitz boundary, and let w be the single-term Weyl function of Ω. The Dirichlet Laplace operator is superspectral to w...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-025329 | MATHDB-340160

Neumann subspectrality for finite covers by normal manifolds

摘录 / EXCERPT: Let M be a normal manifold, and let $\{\Gamma_i\}$ be a finite open cover of M by normal manifolds. Give the interior boundary components Neumann conditions. The finite-cover s...

conjecture | open | MathDB 页面 | 原文线索

V2-025330 | MATHDB-340161

Equal-area subspectrality conjecture for rectangles

摘录 / EXCERPT: Equal-area subspectral rectangles conjecture. There exists $s>0$ such that, for every $0<\epsilon<s$, R_ϵ is Dirichlet-superspectral to R and Neumann-subspectral to R....

conjecture | open | MathDB 页面 | 原文线索

V2-025331 | MATHDB-340206

Existence and invariance of the curved filtered A-infinity algebra for a Lagrangian complement

摘录 / EXCERPT: Let $(X,[?]$

conjecture | open | MathDB 页面 | 原文线索

V2-025332 | MATHDB-340209

The maximal projection volume conjecture for cross-polytopes

摘录 / EXCERPT: Let $\Diamond^n$ be the standard cross-polytope in $\mathbb{R}^n$, let H_k be a k-dimensional subspace of $\mathbb{R}^n$, and let $\Diamond^n|H_k$ denote the orthogonal project...

conjecture | open | MathDB 页面 | 原文线索

V2-025333 | MATHDB-340210

The normalized frame inequalities for cube sections and cross-polytope projections

摘录 / EXCERPT: Let $S=\{v_1,\ldots,v_n\}$ be an (n,k)-frame spanning $\mathbb{R}^k$, let ... and define $P=\operatorname{co}\{\pm v_1,\ldots,\pm v_n\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025334 | MATHDB-340228

The inscribed-face conjecture for edge-preserving symmetries of convex polytopes

摘录 / EXCERPT: Inscribed-face conjecture. P realizes all its own edge-preserving symmetries.

conjecture | open | MathDB 页面 | 原文线索

V2-025335 | MATHDB-340230

Sabitov's metric automorphism conjecture for convex 3-polytopes

摘录 / EXCERPT: Sabitov's metric automorphism conjecture. Every metric automorphism of P extends to an isometry of ambient space. In particular, every metric automorphism is an edge-preserving s...

conjecture | open | MathDB 页面 | 原文线索

V2-025336 | MATHDB-340233

Affine rational ray conjecture for complements of finite polyhedral unions

摘录 / EXCERPT: Affine rational ray conjecture. Then $cmathbb{R}^n\setminus K$ contains an affine rational ray.

conjecture | open | MathDB 页面 | 原文线索

V2-025337 | MATHDB-340254

ACC conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $A={\mathbb{A}}_k^N$ be affine space over an algebraically closed field k, with origin $0\in A$, and let ${\frak{a}}^e$ be a multiideal whose exponent set is e. ACC conject...

conjecture | open | MathDB 页面 | 原文线索

V2-025338 | MATHDB-340255

Uniform boundedness conjecture for multiindices computing minimal log discrepancies

摘录 / EXCERPT: Let $A={\mathbb{A}}_k^N$ be affine space over an algebraically closed field k of arbitrary characteristic, with origin $0\in A$, and let $e\subset {\mathbb{R}}_{>0}$ be finite. A...

conjecture | open | MathDB 页面 | 原文线索

V2-025339 | MATHDB-340264

Commutative quantisation conjecture for the algebra ${\mathcal Z}$

摘录 / EXCERPT: Let σ be an involution of ${\mathfrak g}$, let $H_1,\ldots,H_l$ be the chosen basic invariants with degrees $d_1,\ldots,d_l$, and write $(H_j)_{(i,d_j-i)}$ for their bihomog...

conjecture | open | MathDB 页面 | 原文线索

V2-025340 | MATHDB-340308

Hosono's Gamma Conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a Calabi-Yau manifold equipped with a symplectic form ω, and let $\{Z_t\}_{t\in \Delta^*}$ be a family of Calabi-Yau manifolds parametrized by t in a small punc...

conjecture | open | MathDB 页面 | 原文线索

V2-025341 | MATHDB-340331

The Strominger-Bismut Kähler-like conjecture

摘录 / EXCERPT: Let M be a compact complex manifold endowed with a Hermitian metric, and let its Strominger-Bismut connection be the Hermitian connection associated with that metric. The Stromin...

conjecture | open | MathDB 页面 | 原文线索

V2-025342 | MATHDB-340332

The Gauduchon canonical-connection Kähler-like conjecture

摘录 / EXCERPT: Let M be a compact complex manifold endowed with a Hermitian metric, and let $ abla^{\varepsilon}$ be a canonical connection in the Gauduchon family, distinct from the Strominger...

conjecture | open | MathDB 页面 | 原文线索

V2-025343 | MATHDB-340365

The order conjecture for Morita-equivalent mapping torus categories

摘录 / EXCERPT: Let $\mathcal{A}$ be as in Theorem. Let ϕ and ϕ' be auto-equivalences satisfying the stated conditions, and let M_ϕ and $M_{\phi'}$ be their mapping torus categories...

conjecture | open | MathDB 页面 | 原文线索

V2-025344 | MATHDB-340366

The abelian subgroup conjecture for doubly twisted mapping torus categories

摘录 / EXCERPT: Let $\mathcal{A}$ be a dg category with strictly commuting strict dg auto-equivalences ϕ and ψ, and let $M_{\phi,\psi}$ be the associated doubly twisted mapping torus cat...

conjecture | open | MathDB 页面 | 原文线索

V2-025345 | MATHDB-340368

Borisov-type criticality conjecture for positively curved surfaces

摘录 / EXCERPT: Let Σ be a 2-dimensional compact Riemannian manifold, possibly with boundary, with positive Gauss curvature. A $C^{1,\alpha}$ isometric embedding is an isometric embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-025346 | MATHDB-340403

Folklore conjecture on Einstein four-manifolds with non-negative sectional curvature

摘录 / EXCERPT: Let M be a simply connected Einstein four-manifold with positive scalar curvature and non-negative sectional curvature. Folklore conjecture. The manifold must be one of the follo...

conjecture | open | MathDB 页面 | 原文线索

V2-025347 | MATHDB-340404

Kollár's divisor criterion for genus one fibrations of Calabi-Yau manifolds

摘录 / EXCERPT: Kollár's conjecture. X is genus one (or elliptically) fibered if and only if there exists a divisor $D\in H^{2}(X,\mathbb{Q})$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-025348 | MATHDB-340425

The negative curvature conjecture for Hitchin and maximal representations

摘录 / EXCERPT: Negative curvature conjecture. The sectional curvature k_σ^N is negative at every immersed point. Consequently, the induced curvature κ is negative at every immersed...

conjecture | open | MathDB 页面 | 原文线索

V2-025349 | MATHDB-340432

Equivalence of generalized and ordinary Haantjes integrability for commuting operators

摘录 / EXCERPT: Let M be a differentiable manifold and let $\boldsymbol{A},\boldsymbol{B}:TM\rightarrow TM$ be two $(1,1)$ tensor fields. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-025350 | MATHDB-340433

The critical beta-value conjecture for four-manifolds

摘录 / EXCERPT: Let M^4 be a smooth oriented four-manifold. For a conformal class $[g]$, let $\beta(M^4,[g])$ denote the conformal curvature invariant, and let ... Here $\mathcal{Y}_2^{+}(M^4)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-025351 | MATHDB-340434

The subcritical beta classification conjecture for four-manifolds

摘录 / EXCERPT: Let M^4 be a smooth oriented four-manifold. Let $\mathcal{Y}_2^{+}(M^4)$ denote the class of metrics with positive second Yamabe constant, and let $\beta(M^4,[g])$ be the conform...

conjecture | open | MathDB 页面 | 原文线索

V2-025352 | MATHDB-340475

Legendrian representative conjecture for knots in lens spaces with 3-sphere surgeries

摘录 / EXCERPT: Let $L(p,q)$ be a lens space with universally tight contact structure ξ_{UT}, and let $K\subset L(p,q)$ be a knot admitting a surgery to the 3-sphere. A Legendrian representati...

conjecture | open | MathDB 页面 | 原文线索

V2-025353 | MATHDB-340476

Grid realization conjecture for Legendrian knots in universally tight lens spaces

摘录 / EXCERPT: Let $K\subset L(p,q)$ be a knot and let $m(K)\subset L(p,p-q)$ denote its mirror. Consider Legendrian representatives of these knots in the universally tight contact structures on...

conjecture | open | MathDB 页面 | 原文线索

V2-025354 | MATHDB-340480

The asymptotic maximal first eigenvalue conjecture for closed Riemann surfaces

摘录 / EXCERPT: Let $\mathcal{M}_g$ denote the moduli space of closed Riemann surfaces of genus g, and let $\lambda_1(X)$ be the first positive eigenvalue of the Laplace operator on X. Asympto...

conjecture | open | MathDB 页面 | 原文线索

V2-025355 | MATHDB-340510

Strichartz's almost-periodic spectral asymptotics conjecture for nonpositive curvature

摘录 / EXCERPT: Strichartz's conjecture. There exists a uniformly almost periodic function g such that

conjecture | open | MathDB 页面 | 原文线索

V2-025356 | MATHDB-340511

The structure conjecture for compact Kähler manifolds with semi-positive holomorphic sectional curvature

摘录 / EXCERPT: Let X be a compact Kähler manifold with semi-positive holomorphic sectional curvature. Structure conjecture. There should exist a smooth locally trivial morphism $X\to Y$ whose f...

conjecture | open | MathDB 页面 | 原文线索

V2-025357 | MATHDB-340562

Finite-generation and virtually abelian fundamental groups under Euclidean volume growth

摘录 / EXCERPT: Fundamental-group conjecture. There exists a constant $C(n,L)$ such that, whenever M is an open n-manifold with $\mathrm{Ric}\ge 0$ and its Riemannian universal cover has Eucli...

conjecture | open | MathDB 页面 | 原文线索

V2-025358 | MATHDB-340575

Lipschitz nodal-set conjecture for Neumann eigenfunctions on symmetric convex bodies

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be bounded, open, convex, and symmetric, and let u be a Neumann eigenfunction with the least non-zero eigenvalue. Lipschitz nodal-set conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-025359 | MATHDB-340578

Convex sandwich conjecture for isoperimetric sets of strictly log-concave measures

摘录 / EXCERPT: Let μ be a finite, strictly log-concave measure on $\mathbb{R}^n$, and let E be an isoperimetric set with volume fraction $0<\alpha<1/2$. Convex sandwich conjecture. There ar...

conjecture | open | MathDB 页面 | 原文线索

V2-025360 | MATHDB-340579

Two hyperplane conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be convex, bounded, and centrally symmetric, and let E be an isoperimetric subset of Ω with $|E|/|\Omega|=1/2$. Two hyperplane conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-025361 | MATHDB-340607

Erdős's density conjecture for distance-1-avoiding sets in the plane

摘录 / EXCERPT: In the normed vector space $(\mathbb{R}^n,\|\cdot\|)$, a measurable set $A\subseteq\mathbb{R}^n$ avoids distance 1 if $\|x-y\|\neq 1$ for all $x,y\in A$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025362 | MATHDB-340656

Melo-Winter's large-intersection conjecture for hypercubes

摘录 / EXCERPT: Melo-Winter's large-intersection conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025363 | MATHDB-340657

Melo-Winter's small-intersection conjecture for hypercubes

摘录 / EXCERPT: Let $\mathbb{H}^n=\{0,1\}^n$ be the vertices of the n-dimensional hypercube, and let $H(n,k)$ be the set of intersection sizes $|\mathbb{H}^n\cap S|$ as S ranges over the k-d...

conjecture | open | MathDB 页面 | 原文线索

V2-025364 | MATHDB-340660

The Abhyankar-Sathaye conjecture for affine hypersurfaces

摘录 / EXCERPT: Abhyankar-Sathaye conjecture. The isomorphism Φ extends to an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-025365 | MATHDB-340664

Green-function divergence conjecture under logarithmic volume growth

摘录 / EXCERPT: Let M be an arbitrary complete connected Riemannian manifold, let $B(o,r)$ denote the geodesic ball centered at o, let μ be the Riemannian measure, let $G(x,o)$ be the Gree...

conjecture | open | MathDB 页面 | 原文线索

V2-025366 | MATHDB-340665

Li-Ni's virtual Betti number conjecture for reducible 3-manifolds

摘录 / EXCERPT: Li-Ni's conjecture. If M is a closed oriented reducible 3-manifold, then

conjecture | open | MathDB 页面 | 原文线索

V2-025367 | MATHDB-340669

Abramovich-Karu's polyhedral semistable reduction conjecture

摘录 / EXCERPT: Let $f\colon X \to B$ be a map of conical complexes. A map of conical complexes is semistable if B is regular, every cone maps into a cone of B, the induced map on the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-025368 | MATHDB-340677

RC-positivity conjecture for maps to Kobayashi hyperbolic manifolds

摘录 / EXCERPT: RC-positivity conjecture. There is no non-constant holomorphic map from M to N.

conjecture | open | MathDB 页面 | 原文线索

V2-025369 | MATHDB-340685

The strict linear precision classification conjecture for lattice polytopes

摘录 / EXCERPT: Let P be a lattice polytope. A polytope has strict linear precision when its toric blending parametrization satisfies the strict linear precision condition described in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-025370 | MATHDB-340694

$C^{1,1}$ uniform geodesic stability conjecture for constant scalar curvature Kähler metrics

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold, let $G:=\operatorname{Aut}_0(X)$ be the identity component of its holomorphic automorphism group, let $\mathcal E^1_\omega$ be the fi...

conjecture | open | MathDB 页面 | 原文线索

V2-025371 | MATHDB-340748

Harmonic diffeomorphism conjecture for ideal two-dimensional simplicial complexes

摘录 / EXCERPT: Let X be the ideal two-dimensional simplicial complex under consideration, let T be any face of X, and let $u\in W^{1,2}$ and $u_D\in\overline{\mathcal{D}}$ be the maps defin...

conjecture | open | MathDB 页面 | 原文线索

V2-025372 | MATHDB-340756

The p-adic Gross-Kohnen-Zagier conjecture for Big Heegner points

摘录 / EXCERPT: Let $\mathcal{R}$ be a primitive branch of a Hida family. Its Big Heegner points determine proportionality coefficients in the fraction field of $\mathcal{R}$, and let $\mathbb{S}$...

conjecture | open | MathDB 页面 | 原文线索

V2-025373 | MATHDB-340757

Cristea-Steinsky conjecture on arc lengths in mixed labyrinth fractals

摘录 / EXCERPT: Let $\{{\cal A}_k\}_{k=1}^{\infty}$ be a sequence of both horizontally and vertically blocked labyrinth patterns, with $m_k\ge 4$. Let L_{∞} be the resulting limit set, and...

conjecture | open | MathDB 页面 | 原文线索

V2-025374 | MATHDB-340767

Convexity question for the bounding function of normalized asymptotic translation length

摘录 / EXCERPT: Convexity question. Is the function g convex?

open_problem | open | MathDB 页面 | 原文线索

V2-025375 | MATHDB-340787

Affine Weyl group conjecture for Harish-Chandra bimodules between quantizations

摘录 / EXCERPT: Let Y be a conical symplectic singularity, let $\lambda,\lambda'\in\mathfrak{h}_Y^*$, and let $\overline{\operatorname{HC}}(\mathcal{A}_{\lambda'},\mathcal{A}_{\lambda})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-025376 | MATHDB-340788

Continuity of the geometric realization map for polarized K3 surfaces

摘录 / EXCERPT: Let $\mathcal{F}_{2d}$ be the moduli space of polarized K3 surfaces of degree $2d$, and let $\overline{\mathcal{F}_{2d}}^{{\rm Sat},\tau_{\rm ad}}$ be its Satake compactification f...

conjecture | open | MathDB 页面 | 原文线索

V2-025377 | MATHDB-340789

Continuity of the geometric realization map for Ricci-flat K3 metrics

摘录 / EXCERPT: Let $\mathcal{M}_{\rm K3}$ be the moduli space of Ricci-flat Kähler K3 surfaces modulo hyperKähler rotation, and let $\overline{\mathcal{M}_{\rm K3}}^{\rm Sat,\tau_{\rm ad}}$ be it...

conjecture | open | MathDB 页面 | 原文线索

V2-025378 | MATHDB-340803

Mean convex neighborhood conjecture

摘录 / EXCERPT: Let $X=(x,t)$ be a space-time point of the mean curvature flow of closed embedded surfaces. A tangent flow at X is a backwardly selfsimilar limit obtained by translating X to t...

conjecture | open | MathDB 页面 | 原文线索

V2-025379 | MATHDB-340804

Nonfattening conjecture for cylindrical and spherical singularities

摘录 / EXCERPT: Let $F_t(M)$ denote the level set flow starting from a closed embedded surface M. Its singularities are called cylindrical or spherical when their tangent flows have the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-025380 | MATHDB-340805

Uniqueness conjecture for mean curvature flow of embedded two-spheres

摘录 / EXCERPT: Consider mean curvature flows starting from embedded two-spheres and allowing singularities. Uniqueness conjecture for mean curvature flow of embedded two-spheres. The mean curvatu...

conjecture | open | MathDB 页面 | 原文线索

V2-025381 | MATHDB-340824

Arnold's conjecture on constant-curvature curves on Riemannian two-spheres

摘录 / EXCERPT: Let a Riemannian two-sphere be a two-dimensional sphere equipped with a Riemannian metric, and let c satisfy $0<c<\infty$. A curve is smoothly immersed if it is smooth and its pa...

conjecture | open | MathDB 页面 | 原文线索

V2-025382 | MATHDB-340825

Novikov's conjecture on embedded constant-curvature curves

摘录 / EXCERPT: Let a Riemannian two-sphere be a two-dimensional sphere equipped with a Riemannian metric, and let c satisfy $0<c<\infty$. A curve is smoothly embedded if it is a smooth embeddin...

conjecture | open | MathDB 页面 | 原文线索

V2-025383 | MATHDB-340830

The totally geodesic orbit conjecture for positively curved manifolds

摘录 / EXCERPT: Let G be a compact Lie group acting freely by isometries on a compact manifold M. Suppose that M has positive sectional curvature. Totally geodesic orbit conjecture. Then M...

conjecture | open | MathDB 页面 | 原文线索

V2-025384 | MATHDB-340839

The ascending chain condition conjecture for minimal log discrepancies

摘录 / EXCERPT: Ascending chain condition conjecture for minimal log discrepancies. The set

conjecture | open | MathDB 页面 | 原文线索

V2-025385 | MATHDB-340842

Efficient trisection conjecture for simply-connected complex four-manifolds

摘录 / EXCERPT: Efficient trisection conjecture. Every simply-connected, complex four-manifold admits an efficient trisection.

conjecture | open | MathDB 页面 | 原文线索

V2-025386 | MATHDB-340843

Trisected 11/8-conjecture

摘录 / EXCERPT: Trisected 11/8-conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025387 | MATHDB-340856

Fractal-set Fourier-dimension conjecture for scalar products

摘录 / EXCERPT: Let d be a positive integer, let $\beta\beta\in[0,d]$, and let $RY=\{ry:r\in R,\ y\in Y\}$ for $R\subseteq(0,\infty)$ and $Y\subseteq\mathbb{R}^d$. Fractal-set conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-025388 | MATHDB-340857

Dual fractal-set Fourier-dimension conjecture for scalar products

摘录 / EXCERPT: Let d be a positive integer, let $RY=\{ry:r\in R,\ y\in Y\}$ for $R\subseteq(0,\infty)$ and $Y\subseteq\mathbb{R}^d$. Dual fractal-set conjecture. For every $\alpha\in[0,1]$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-025389 | MATHDB-340864

Strong convexity of the energy functional for negatively curved targets

摘录 / EXCERPT: Let M be a Riemannian manifold and let N be a Riemannian manifold of negative sectional curvature. Consider the energy functional on a connected component of...

conjecture | open | MathDB 页面 | 原文线索

V2-025390 | MATHDB-340865

The compatible-triple conjecture for real forms of holomorphic inner product Lie algebras

摘录 / EXCERPT: Let $(\mathfrak{g}^{\mathbb{C}}, g^{\mathbb{C}})$ be a holomorphic inner product space, with $\mathfrak{g}^{\mathbb{C}}$ a complex Lie algebra. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025391 | MATHDB-340902

Generalized Chern conjecture for constant-mean-curvature hypersurfaces

摘录 / EXCERPT: Let M be a closed hypersurface with constant mean curvature and constant scalar curvature in the unit sphere $\mathbb{S}^{n + 1}$. Generalized Chern conjecture. For each n and...

conjecture | open | MathDB 页面 | 原文线索

V2-025392 | MATHDB-340932

Upper-semicontinuity conjecture for orthogonal complex structures

摘录 / EXCERPT: Let M be a smooth manifold equipped with a Riemannian metric g. Consider the space of complex structures on M that are orthogonal to g, together with its dimension and numb...

conjecture | open | MathDB 页面 | 原文线索

V2-025393 | MATHDB-340962

Conjecture on initial degrees of symbolic powers of general point sets

摘录 / EXCERPT: Let $Z\subset\mathbb{P}^n$ be a set of $n+3$ points in linearly general position, with n odd, and let I_Z be its homogeneous ideal. For a homogeneous ideal J, let $\alpha(J)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025394 | MATHDB-341006

Conjecture on linear-strand Betti numbers of balanced normal pseudomanifolds

摘录 / EXCERPT: Let Δ be a $(d-1)$-dimensional balanced normal pseudomanifold, with $d\geq 4$, and suppose that $f_0(\Delta)=kd$ for an integer $k\geq 2$. Let $\mathbb{F}[\Delta]$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-025395 | MATHDB-341019

The semisimple monodromy conjecture for vanishing p-curvature

摘录 / EXCERPT: Semisimple monodromy conjecture. If almost all p-curvatures of (V,∇) vanish, then (V,∇) has semisimple monodromy.

conjecture | open | MathDB 页面 | 原文线索

V2-025396 | MATHDB-341024

Folklore stability conjecture for supersymmetric branes

摘录 / EXCERPT: Let X be a Calabi-Yau manifold with mirror $\check{X}$, let L be a Lagrangian in $\check{X}$, and let E be a holomorphic vector bundle on X. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-025397 | MATHDB-341042

Moore's extrinsic product conjecture for Riemannian products

摘录 / EXCERPT: Let $M^{n_1}_1$ and $M^{n_2}_2$ be Riemannian manifolds. Suppose that $M^{n_1}_1$ can be locally isometrically immersed in $\mathbb{R}^{n_1+p_1}$ but in no lower-dimensional Euclid...

conjecture | open | MathDB 页面 | 原文线索

V2-025398 | MATHDB-341043

Perimeter-scaled Robin eigenvalue lower bound for the disk

摘录 / EXCERPT: Let Ω be a convex bounded planar domain, let $L(\Omega)$ denote its perimeter, and let $\lambda_1(\Omega;\alpha/L(\Omega))$ be its first Robin eigenvalue with perimeter-scal...

conjecture | open | MathDB 页面 | 原文线索

V2-025399 | MATHDB-341044

Conformal-radius maximality conjecture for the second Robin eigenvalue

摘录 / EXCERPT: Suppose $F:\mathbb D\to\Omega$ is a conformal map of the unit disk onto a Jordan-Lipschitz domain Ω, and let $L(\Omega)$ denote the perimeter of Ω. For $\alpha>0$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-025400 | MATHDB-341076

The volume conjecture for the Teichmüller TQFT

摘录 / EXCERPT: The volume conjecture for the Teichmüller TQFT. There exists a smooth function $J_{M,K}(\hbar,x)$ on $\mathbb{R}_{>0}\times\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025401 | MATHDB-341094

Irreducibility and divisibility conjecture for the polynomials $C_n(x)$

摘录 / EXCERPT: Polynomial conjecture. The polynomials satisfy both of the following conditions:

conjecture | open | MathDB 页面 | 原文线索

V2-025402 | MATHDB-341095

Incommensurability conjecture for fully augmented pretzel and rotated links

摘录 / EXCERPT: Incommensurability conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-025403 | MATHDB-341097

Kebekus-Kovács pair version of the Viehweg hyperbolicity conjecture

摘录 / EXCERPT: Let $f\colon (U,D)\to V$ be a log smooth family with V quasi-projective, with all coefficients of D in $(0,1)$, and suppose that every fiber (U_y,D_y) is of log general type....

conjecture | open | MathDB 页面 | 原文线索

V2-025404 | MATHDB-341164

The horofunction Morse characterization via ideal triangle points

摘录 / EXCERPT: Morse characterization. The horofunction $h=b_c$ is Morse if and only if there exists a δ such that, for every $x\in X$, $(i_{\infty})_x$ lies in the δ-neighborhood o...

conjecture | open | MathDB 页面 | 原文线索

V2-025405 | MATHDB-341165

The claim that the CAT(0) assumption can be dropped

摘录 / EXCERPT: CAT(0)-assumption claim. The CAT(0) assumption in the previous theorem can be dropped.

conjecture | open | MathDB 页面 | 原文线索

V2-025406 | MATHDB-341170

Generalised Shokurov conjecture under bounded fibre volume

摘录 / EXCERPT: Generalised Shokurov conjecture under bounded fibre volume. There is a positive real number δ, depending only on d,v,ϵ, such that the coefficients of the discrimina...

conjecture | open | MathDB 页面 | 原文线索

V2-025407 | MATHDB-341172

Novikov's common-level quasiperiodicity conjecture

摘录 / EXCERPT: Let M be a compact manifold and let several closed 1-forms on M be given. Their common level is the intersection of the corresponding level sets at a generic choice of levels...

conjecture | open | MathDB 页面 | 原文线索

V2-025408 | MATHDB-341175

Novikov's exceptional-set dimension conjecture

摘录 / EXCERPT: Let $f:{\mathbb T}^3\to\mathbb R$ be a function, and let ${\cal E}(f)$ be its exceptional set of directions, namely the directions not belonging to the stability-zone set. Novikov'...

conjecture | open | MathDB 页面 | 原文线索

V2-025409 | MATHDB-341185

The Hausdorff-measure conjecture for intersections of invariant sets

摘录 / EXCERPT: Let $A_2,A_3\subset[0,1]$ be closed sets invariant under multiplication by 2 and 3 modulo 1, respectively, and suppose ... For $g(x)=\exp(-(-\log x)^s)$, Hausdorff-measure conj...

conjecture | open | MathDB 页面 | 原文线索

V2-025410 | MATHDB-341193

The curvature bound conjecture for adaptive anisotropic total variation eigenfunctions

摘录 / EXCERPT: Let $C\subset\mathbb{R}^2$ be a bounded ellipse whose indicator function is an eigenfunction in the sense of the adaptive anisotropic total variation eigenfunction equation, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-025411 | MATHDB-341223

Busemann-Petty centroid inequality formulation of the slicing conjecture

摘录 / EXCERPT: Let $G\subset\mathbb{R}^d$ be a convex symmetric body in isotropic position, and let $L(G)$ denote its isotropic constant. Slicing conjecture. There is a constant $C>0$ independent...

conjecture | open | MathDB 页面 | 原文线索

V2-025412 | MATHDB-341241

The maximum-width conjecture for hollow three-dimensional convex bodies

摘录 / EXCERPT: A hollow convex body is a three-dimensional convex body whose interior contains no lattice point. The width of a convex body is the minimum, over nonzero integer linear functionals...

conjecture | open | MathDB 页面 | 原文线索

V2-025413 | MATHDB-341286

Nonexistence conjecture for injective linear series on very general curves

摘录 / EXCERPT: Let X be a smooth curve of genus g, and let an injective linear series g^2_d on X mean a base-point-free, not necessarily complete, two-dimensional linear series of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-025414 | MATHDB-341309

Quasi-Möbius characterization of snowflakes of Heisenberg groups

摘录 / EXCERPT: Let $\mathcal{A}=\{\mathsf{R},\mathsf{C},\mathsf{H},\mathsf{O}\}$ be the collection of normed division algebras. For ${\mathsf K}\in\mathcal{A}$ and $n\in\mathsf N$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-025415 | MATHDB-341316

Measurability criterion for CM-semistability

摘录 / EXCERPT: Let $X\subset\mathbb{P}^N$ be a projective embedding of a complex manifold with $\operatorname{Pic} X\simeq\mathbb{Z}$, and let $\omega_{FS}\vert_X$ denote the restriction of the F...

conjecture | open | MathDB 页面 | 原文线索

V2-025416 | MATHDB-341324

Matsumoto's 11/8-conjecture for spin 4-manifold intersection forms

摘录 / EXCERPT: Let p be a nonnegative integer and q a natural number. The form ... can be realized as the intersection form of a closed smooth spin 4-manifold. Matsumoto's 11/8-conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-025417 | MATHDB-341333

Artin's conjecture on birational classes of noncommutative surfaces

摘录 / EXCERPT: Let A be a connected graded domain with $\operatorname{GKdim}(A)=3$, and let ... be its graded quotient division ring. Artin's conjecture. One of the following holds: 1. D is fin...

conjecture | open | MathDB 页面 | 原文线索

V2-025418 | MATHDB-341335

Fujita's log spectrum conjecture for pseudo-effective thresholds

摘录 / EXCERPT: Fujita's log spectrum conjecture. Under this notation,

conjecture | open | MathDB 页面 | 原文线索

V2-025419 | MATHDB-341352

Finite linear parametrization conjecture for equal-volume Dehn fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be a 1-cusped hyperbolic 3-manifold, and write $\mathcal{M}_{p/q}$ for its hyperbolic Dehn filling along slope p/q. Finite linear parametrization conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-025420 | MATHDB-341353

Finite coordinatewise parametrization conjecture for equal-volume multi-cusped fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be an n-cusped hyperbolic 3-manifold, and let $\mathcal{M}_{(p_1/q_1,\dots,p_n/q_n)}$ denote the corresponding hyperbolic Dehn filling. Finite coordinatewise...

conjecture | open | MathDB 页面 | 原文线索

V2-025421 | MATHDB-341354

Uniform boundedness conjecture for equal-volume Dehn fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be an n-cusped hyperbolic 3-manifold. Uniform boundedness conjecture. There exists a constant $c=c(\mathcal{M})$ such that, for every volume v, the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-025422 | MATHDB-341366

Tate shifted symplecticity of multiplicative connection moduli

摘录 / EXCERPT: Suppose C is either an elliptic curve or $\mathbb{CP}^1$ with a fixed framing at ∞. Let G be the group used to define the moduli stacks, and let Λ_i be the stabi...

conjecture | open | MathDB 页面 | 原文线索

V2-025423 | MATHDB-341400

Generation of metric scalable groups by the first layer

摘录 / EXCERPT: First-layer generation conjecture. The first layer $V_1(G)$ generates G.

conjecture | open | MathDB 页面 | 原文线索

V2-025424 | MATHDB-341463

Leung's spherical minimal-submanifold homeomorphism conjecture

摘录 / EXCERPT: Leung's conjecture. If n is odd and $\sigma(W)\leq\frac{n}{n-1}$, then W is homeomorphic to S^n.

conjecture | open | MathDB 页面 | 原文线索

V2-025425 | MATHDB-341464

Leung's stronger spherical minimal-submanifold homeomorphism conjecture

摘录 / EXCERPT: Leung's stronger conjecture. If n is odd and $\sigma(W)<\frac{n+1}{n-1}$, then W is homeomorphic to S^n.

conjecture | open | MathDB 页面 | 原文线索

V2-025426 | MATHDB-341483

Gross-Hacking-Keel mirror construction conjecture for maximal intersection log Calabi-Yau pairs

摘录 / EXCERPT: Gross-Hacking-Keel mirror construction conjecture. The free R-module V with basis $U^{\mathrm{trop}}(\mathbb Z)$ has a natural finitely generated R-algebra structure whose st...

conjecture | open | MathDB 页面 | 原文线索

V2-025427 | MATHDB-341509

Free associative cancellation conjecture

摘录 / EXCERPT: Let R be a $\mathbb{K}$-algebra, and let $\mathbb{K}\langle x_1,\ldots,x_n\rangle$ and $\mathbb{K}\langle y\rangle$ denote free associative algebras. Free associative cancellatio...

conjecture | open | MathDB 页面 | 原文线索

V2-025428 | MATHDB-341539

Anderson's non-existence conjecture for minimal foliations in closed hyperbolic three-manifolds

摘录 / EXCERPT: Anderson's conjecture. There does not exist a local 1-parameter family of closed minimal surfaces in M.

conjecture | open | MathDB 页面 | 原文线索

V2-025429 | MATHDB-341554

The Poisson moment assumption for Haar-distributed lattices

摘录 / EXCERPT: Let $\mathcal B$ be a symmetric set in $\bR^n$ of volume $V=2^{\mathcal O(n)}$, let $k=cn$ for a constant $0<c<1$, let μ be the Haar distribution on lattices, and let Y be Po...

conjecture | open | MathDB 页面 | 原文线索

V2-025430 | MATHDB-341560

Propeller conjecture

摘录 / EXCERPT: Let $m>3$, let $a_1,\dots,a_m>0$ satisfy $\sum_{i=1}^m a_i=1$, and let $w\in\mathbb{R}^{n+1}$ be the vector defined in the paper. Set $\overline{w}^{(i)}=w/a_i$. Consider measurabl...

conjecture | open | MathDB 页面 | 原文线索

V2-025431 | MATHDB-341562

The quadrics conjecture for locally polynomially integrable strictly convex hypersurfaces

摘录 / EXCERPT: Let M be a strictly convex smooth connected hypersurface in $\mathbb R^n$, and let $\widehat M$ be its convex hull. For each $a\in M$, let $A_M^a(t)$ denote the $(n-1)$-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-025432 | MATHDB-341566

The Complete Overlap Conjecture for self-similar attractors

摘录 / EXCERPT: Let $\mathcal{S}=\left\{S_i\right\}_{i=1}^{m}$ be a self-similar iterated function system on $\mathbb{R}^d$, with attractor Λ, similarity dimension s, and symbolic space...

conjecture | refuted | MathDB 页面 | 原文线索

V2-025433 | MATHDB-341576

Guzman's group-theoretic conjecture for subgroups of free groups

摘录 / EXCERPT: Let F be a free group, and let $H,K\le F$ be subgroups with equal rank $m\ge 2$. If ... then Guzman's group-theoretic conjecture. ... Here $H\vee K$ is the subgroup generated by H...

conjecture | open | MathDB 页面 | 原文线索

V2-025434 | MATHDB-341593

Conjecture on transverse actions preserving a foliation and line bundle

摘录 / EXCERPT: Let X be a projective manifold with $\kappa(X)\geq 0$, let $\mathcal F$ be a foliation on X, and let L be a line bundle on X. Suppose that L admits a singular Hermitian m...

conjecture | open | MathDB 页面 | 原文线索

V2-025435 | MATHDB-341601

A positive eigenvalue lower bound under a Ricci lower bound

摘录 / EXCERPT: Let (M^n,g) be a compact Riemannian manifold satisfying $\operatorname{Ric}(M)>K$, and let v be a one-form with $\|v\|_\infty<C$. Suppose that there exists $u\in W^{2,p}(M)$ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-025436 | MATHDB-341609

t-Fourier uniformity conjecture for the Liouville function

摘录 / EXCERPT: Let $t\in\mathbb{N}$, let G be a nilpotent Lie group of step t, let Γ be a cocompact lattice in G, and let $F\colon G/\Gamma\rightarrow\mathbb{C}$ be continuous. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025437 | MATHDB-341644

Oscillating overlap-statistics conjecture for parameterised iterated function systems

摘录 / EXCERPT: Let $U\subset\mathbb R^k$ be an open parameter space, let Φ_u be the parameterised iterated function systems described above, and let X_u be their attractors. For a slowly d...

conjecture | open | MathDB 页面 | 原文线索

V2-025438 | MATHDB-341651

Gompf's embedding conjecture for closed oriented 3-manifolds

摘录 / EXCERPT: Let Y be a closed oriented 3--manifold. Gompf's embedding conjecture. Y admits a smooth embedding into a symplectic 4--manifold. Such an embedding would provide a route tow...

conjecture | open | MathDB 页面 | 原文线索

V2-025439 | MATHDB-341661

Conjecture on density of metrics with countably many singular minimal hypersurfaces

摘录 / EXCERPT: Let M be the closed manifold under consideration, let $\Gamma_\infty(M)$ be its space of smooth metrics, and let $\mathcal{M}_c$ be the set of metrics for which there are only co...

conjecture | open | MathDB 页面 | 原文线索

V2-025440 | MATHDB-341671

The Erdős-Mordell inequality for interior points of arbitrary triangles in taxicab geometry

摘录 / EXCERPT: Erdős-Mordell open problem. In taxicab geometry, the inequality

open_problem | open | MathDB 页面 | 原文线索

V2-025441 | MATHDB-341696

Piecewise smooth isometric immersion conjecture for Riemannian surfaces

摘录 / EXCERPT: Piecewise smooth immersion conjecture. There exists a piecewise smooth isometric immersion

conjecture | open | MathDB 页面 | 原文线索

V2-025442 | MATHDB-341697

Conjecture on the Friedlander-Nadirashvili invariant of even-genus non-orientable surfaces

摘录 / EXCERPT: Let $I_k(\gamma)$ denote the k-th Friedlander-Nadirashvili invariant for a non-orientable surface of genus γ, and suppose that γ is even. Even-genus conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-025443 | MATHDB-341698

Friedlander-Nadirashvili cobordism invariance conjecture

摘录 / EXCERPT: Let M and M' be closed manifolds of the same dimension. They are cobordant if there exists a manifold with boundary W such that $\partial W=M\sqcup M'$, and M is null-cobor...

conjecture | open | MathDB 页面 | 原文线索

V2-025444 | MATHDB-341714

LTP-strong conjecture for crepant Weierstrass fibrations

摘录 / EXCERPT: Let B be a smooth compact manifold, let W be a Weierstrass fibration in ... and let $\widetilde{W}\to W$ be a crepant resolution obtained by a sequence of blowups of Z in smoot...

conjecture | open | MathDB 页面 | 原文线索

V2-025445 | MATHDB-341722

The projective deformation-limit conjecture for Moishezon manifolds

摘录 / EXCERPT: Let $\pi:\mathcal{X}\rightarrow\Delta$ be a holomorphic family of compact complex manifolds of dimension n over an open disk $\Delta\subset\mathbb{C}$, with fibers...

conjecture | open | MathDB 页面 | 原文线索

V2-025446 | MATHDB-341759

Wang-Xia sharp lower-bound conjecture for the first fourth-order Steklov eigenvalue

摘录 / EXCERPT: Let (M^n,g), $n\geq 2$, be a connected compact smooth Riemannian manifold with boundary. Assume that $\operatorname{Ric}_g\geq 0$ and that the principal curvatures of the boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-025447 | MATHDB-341779

Finiteness conjecture for minimal symplectic fillings of planar contact manifolds

摘录 / EXCERPT: Planar filling finiteness conjecture. Every planar contact 3-manifold has at most finitely many distinct deformation classes of minimal symplectic fillings.

conjecture | open | MathDB 页面 | 原文线索

V2-025448 | MATHDB-341787

The sectional-curvature characterization of Alexandrov spaces among RCD spaces

摘录 / EXCERPT: Sectional-curvature characterization conjecture. The following conditions are equivalent: $({\rm M},{\sf d})$ is an n-dimensional Alexandrov space of curvature bounded from below...

conjecture | open | MathDB 页面 | 原文线索

V2-025449 | MATHDB-341794

The compact-vector conjecture for constant-length Killing fields

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold with Lie algebra $\mathfrak{g}$ of Killing fields, and let $X e 0$ be a Killing field of constant length. A vector X is compact in...

conjecture | open | MathDB 页面 | 原文线索

V2-025450 | MATHDB-341795

The nilradical containment conjecture for constant-length Killing fields

摘录 / EXCERPT: Let $(G/H,g)$ be a geodesic orbit Riemannian space, let $\mathfrak{g}=\operatorname{Lie}(G)$, and let $X\in\mathfrak{g}$ be a Killing field of constant length. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-025451 | MATHDB-341818

Sormani's intrinsic-flat stability conjecture for almost nonnegative scalar curvature

摘录 / EXCERPT: Let M_j be a sequence of Riemannian manifolds diffeomorphic to a 3-torus such that ... where ... If the scalar curvature $R(M_j)$ satisfies $R(M_j)\geq -\frac{1}{j}$, then Sormani'...

conjecture | open | MathDB 页面 | 原文线索

V2-025452 | MATHDB-341838

Global bi-Lipschitz regularity conjecture for minimisers between doubly connected domains

摘录 / EXCERPT: Let D and Ω be doubly connected domains with smooth boundaries, and let ρ be a smooth non-vanishing metric defined on the closure of Ω. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-025453 | MATHDB-341859

Loi-Zedda optimal L^2 rigidity conjecture for projective manifolds

摘录 / EXCERPT: Let M be an n-dimensional projective manifold in $\mathbb{P}^{n+r}$ with the induced metric. Let σ be its second fundamental form, let $\ast1$ be the volume form, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-025454 | MATHDB-341873

Finite-time blowup conjecture for incompressible Euler flow on compact manifolds

摘录 / EXCERPT: Finite-time blowup conjecture. There exists such a compact Riemannian manifold (M,g), of some dimension $d>2$, and such a smooth solution that cannot be smoothly continued to the...

conjecture | open | MathDB 页面 | 原文线索

V2-025455 | MATHDB-341898

Birational rowmotion order conjecture for minuscule posets

摘录 / EXCERPT: Minuscule birational rowmotion order conjecture. The order of $\mathrm{row}^{\mathrm B}$ is h.

conjecture | open | MathDB 页面 | 原文线索

V2-025456 | MATHDB-341899

Birational rowmotion order conjecture for coincidental root posets

摘录 / EXCERPT: Coincidental root-poset birational rowmotion order conjecture. The order of $\mathrm{row}^{\mathrm B}$ divides $2h$.

conjecture | open | MathDB 页面 | 原文线索

V2-025457 | MATHDB-341915

Existence of embedded minimal n-noids with arbitrarily many ends

摘录 / EXCERPT: Embedded n-noid conjecture. There exist embedded minimal n-noids in H^3 for every $n\in\mathbb{N}^{\geq 2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-025458 | MATHDB-341919

The index-one Clifford hypersurface classification conjecture for projective spaces

摘录 / EXCERPT: Index-one Clifford hypersurface classification conjecture. The only compact minimal hypersurfaces with index one in $\mathbb{K}P^{r-1}$ are the minimal Clifford hypersurfaces. In p...

conjecture | open | MathDB 页面 | 原文线索

V2-025459 | MATHDB-341930

Willmore-energy and continuity-equation conjecture for skew-mean-curvature flow

摘录 / EXCERPT: Let $F_t:\Sigma^n\rightarrow\mathbb R^{n+2}$ be a codimension 2 submanifold moving by the skew-mean-curvature flow ... Write $\rho=|H|^2$ for the square of its mean curvature. Will...

conjecture | open | MathDB 页面 | 原文线索

V2-025460 | MATHDB-341940

Lowest-entropy conjecture for generalized cylindrical shrinkers

摘录 / EXCERPT: Lowest-entropy conjecture for generalized cylinders. For any codimension and $n\leq 4$, the round generalized cylinders $\mathbb S^k_{\sqrt{2k}}\times\mathbb R^{n-k}$ are the shrin...

conjecture | open | MathDB 页面 | 原文线索

V2-025461 | MATHDB-341946

Conjecture on finite-index geometric automorphism groups of symplectic 4-manifolds

摘录 / EXCERPT: Finite-index geometric automorphism conjecture. Suppose M has symplectic structures and $A(\Gamma_M)$ is infinite. Then $D(M)$ is of finite index if and only if M is either a K...

conjecture | open | MathDB 页面 | 原文线索

V2-025462 | MATHDB-341957

Mirror construction conjecture for Calabi-Yau threefolds from projective joins

摘录 / EXCERPT: Mirror construction conjecture. Mirror Calabi-Yau threefolds of linear-section Calabi-Yau threefolds in projective joins of del Pezzo manifolds are given by Schoen Calabi-Yau three...

conjecture | open | MathDB 页面 | 原文线索

V2-025463 | MATHDB-341978

Log Iitaka conjecture for log canonical fibrations

摘录 / EXCERPT: Log Iitaka conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025464 | MATHDB-341992

Symplectic linear independence conjecture

摘录 / EXCERPT: Symplectic LIC. The polynomials p_α, for $\alpha\in I$, are linearly independent over $\mathbb{C}$. This is the symplectic analogue of the Atiyah-Sutcliffe LIC for the root...

conjecture | open | MathDB 页面 | 原文线索

V2-025465 | MATHDB-341993

Symplectic Atiyah-Sutcliffe determinant lower-bound conjecture

摘录 / EXCERPT: Let $\mathcal{C}_m(\mathbb{R}^3)$ and the symplectic determinant $D_S:\mathcal{C}_m(\mathbb{R}^3)\to\mathbb{C}$ be defined from the Hopf-lift polynomials p_α using the order...

conjecture | open | MathDB 页面 | 原文线索

V2-025466 | MATHDB-342043

Ni-Zheng's CQB conjecture for Kähler C-spaces

摘录 / EXCERPT: Ni-Zheng's CQB conjecture. Every Kähler C-space has nonnegative CQB and positive $^d\!$CQB.

conjecture | open | MathDB 页面 | 原文线索

V2-025467 | MATHDB-342044

The holomorphic sectional curvature conjecture for Kähler C-spaces

摘录 / EXCERPT: Holomorphic sectional curvature conjecture. Every Kähler C-space has positive holomorphic sectional curvature H.

conjecture | open | MathDB 页面 | 原文线索

V2-025468 | MATHDB-342045

The CQB characterization conjecture for Kähler C-spaces

摘录 / EXCERPT: CQB characterization conjecture. Then M is biholomorphic to a Kähler C-space.

conjecture | open | MathDB 页面 | 原文线索

V2-025469 | MATHDB-342048

Covering-radius conjecture for lattice polytopes with prescribed interior points

摘录 / EXCERPT: Let $k,d\in\mathbb{N}$ be nonnegative integers with $k\geq2$. For every lattice d-polytope P with k interior lattice points, the prescribed-interior-points conjecture. ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-025470 | MATHDB-342055

Conjecture on monotonicity of symplectic Kodaira dimension under unchaining

摘录 / EXCERPT: Unchaining monotonicity conjecture. If (X',ω') is obtained from (X,ω) by unchaining, then

conjecture | open | MathDB 页面 | 原文线索

V2-025471 | MATHDB-342061

Conjecture on the Martinez-Maure hyperbolic hedgehog as a singular minimizer

摘录 / EXCERPT: Hyperbolic hedgehog minimizer conjecture. The support function h is a one-homogeneous singular minimizer of a functional of the type described in the source, and D_F consists e...

conjecture | open | MathDB 页面 | 原文线索

V2-025472 | MATHDB-342087

Local flatness conjecture for the vacuum Bartnik mass

摘录 / EXCERPT: Let Ω be a compact domain with boundary data induced by a nonnegative-scalar-curvature metric, and let ... where $\mathcal C_{(\gamma,H)}$ is the set of asymptotically flat ex...

conjecture | open | MathDB 页面 | 原文线索

V2-025473 | MATHDB-342101

ACC conjecture for a-lc thresholds of exceptional singularities

摘录 / EXCERPT: Let d be a positive integer, a a non-negative real number, and Γ a DCC set. For an lc germ $(X\ni x,B)$ and an $\mathbb{R}$-Cartier divisor D, let...

conjecture | open | MathDB 页面 | 原文线索

V2-025474 | MATHDB-342102

ACC conjecture for canonical and regularity thresholds

摘录 / EXCERPT: Let $d>0$ and $-1\leq c\leq d-1$ be integers. Let $\Gamma\subset[0,1]$ and $\Gamma'\subset[0,+\infty)$ be DCC sets. For an lc pair over a contraction $X\to Z$, with $z\in Z$, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-025475 | MATHDB-342132

The angular-factor conjecture for the Atiyah-Sutcliffe determinant

摘录 / EXCERPT: Let $C_n(\mathbb{R}^3)$ be the configuration space of n distinct points, and for $a\ne b$ let $v_{ab}=(\mathbf{x}_b-\mathbf{x}_a)/\|\mathbf{x}_b-\mathbf{x}_a\|\in S^2$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-025476 | MATHDB-342133

The four-point angular formula conjecture for the Atiyah-Sutcliffe determinant

摘录 / EXCERPT: Let $v_{ab}\in S^2$ be the unit vectors associated with a configuration of four distinct points in $\mathbb{R}^3$, let D be the normalized Atiyah-Sutcliffe determinant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025477 | MATHDB-342142

Extreme Kaleidocycle conjecture

摘录 / EXCERPT: Let $N\ge 6$. Recall that $\mathcal{M}_N$ is the configuration space of N-Kaleidocycles, $\pi_c:\mathcal{M}_N\to\mathbb{R}$ is the projection onto the c-axis, where $c=\cos\nu$...

conjecture | open | MathDB 页面 | 原文线索

V2-025478 | MATHDB-342143

Optimal reverse isodiametric inequality for planar bisections

摘录 / EXCERPT: Optimal reverse isodiametric inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-025479 | MATHDB-342144

Optimal reverse isodiametric inequality for centrally symmetric planar bisections

摘录 / EXCERPT: Centrally symmetric reverse isodiametric inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-025480 | MATHDB-342156

The filtered Cat-HA version of the non-abelian Hodge correspondence

摘录 / EXCERPT: Let X be a smooth projective complex curve. The pro-completed category...

conjecture | open | MathDB 页面 | 原文线索

V2-025481 | MATHDB-342170

Finiteness conjecture for manifolds with boundary

摘录 / EXCERPT: Let M be a compact oriented 3-manifold, possibly with boundary. Let $K(M,\mathbb{Q}(A))$ denote its Kauffman bracket skein module, and let $K(\partial M,\mathbb{Q}(A))$ act on...

conjecture | open | MathDB 页面 | 原文线索

V2-025482 | MATHDB-342188

Kontsevich's cosheaf conjecture for Fukaya categories of Liouville sectors

摘录 / EXCERPT: Kontsevich's cosheaf conjecture. Fukaya categories of Liouville sectors form a cosheaf of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-025483 | MATHDB-342265

The toric transversality conjecture for Frobenius structure counts

摘录 / EXCERPT: Toric transversality conjecture. In the definition of $N_{\beta}(q_1,\ldots,q_s)$, all curves in $\mathcal{M}^{\log}_{0,\Delta_{\mathbf q}}(\widetilde{Y}^{\dagger},\beta)$ satisfyi...

conjecture | open | MathDB 页面 | 原文线索

V2-025484 | MATHDB-342275

Higher-dimensional extension of the Football theorem

摘录 / EXCERPT: Higher-dimensional Football conjecture. For every such manifold,

conjecture | open | MathDB 页面 | 原文线索

V2-025485 | MATHDB-342278

Barba's Ham-Sandwich line conjecture

摘录 / EXCERPT: Let R, B, and G be sets of lines in $\mathbb{R}^3$ in general position, each containing an even number of lines. For two lines in general position, say that one lies above th...

conjecture | open | MathDB 页面 | 原文线索

V2-025486 | MATHDB-342279

Langerman's conjecture on bisection by hyperplanes

摘录 / EXCERPT: Let a mass distribution in $\mathbb{R}^d$ be a measure for which the bisection notion is defined. A set of n hyperplanes bisects a mass distribution when, after orienting the hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-025487 | MATHDB-342280

The simpler-proof conjecture for Proposition 2

摘录 / EXCERPT: The paper considers ideal branched triangulations of the relevant drilled 3-manifold and composite ideal branched transits between them, with admissible boundary configurations as...

conjecture | open | MathDB 页面 | 原文线索

V2-025488 | MATHDB-342296

Generic unboundedness of balanced vertices for planar geodesic nets

摘录 / EXCERPT: Let N_0 be a positive integer and let S be an N_0-tuple of points in $\bigl(\mathbb{R}^2\bigr)^{N_0}$. Generic unboundedness conjecture. There exist N_0 and an N_0-tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-025489 | MATHDB-342297

Gromov's boundedness conjecture for geodesic nets

摘录 / EXCERPT: Let a geodesic net in the Euclidean plane have a finite number of unbalanced vertices, each with an imbalance, and let the total imbalance be the sum of those imbalances. Gromov's...

conjecture | open | MathDB 页面 | 原文线索

V2-025490 | MATHDB-342298

Length-diameter boundedness conjecture for geodesic multinets

摘录 / EXCERPT: Let a geodesic multinet in the Euclidean plane have n unbalanced vertices, diameter D, total length L, and some number of balanced vertices. Length-diameter boundedness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-025491 | MATHDB-342299

Boundedness conjecture for geodesic nets on Riemannian manifolds

摘录 / EXCERPT: Let M be a complete Riemannian manifold. For a geodesic net G on M, let L be its total length, $\widetilde L$ its adjusted length, D its diameter, and n the number of u...

conjecture | open | MathDB 页面 | 原文线索

V2-025492 | MATHDB-342300

Nonrepetition conjecture for the suspension-angle derivatives

摘录 / EXCERPT: For the finite layered construction $G_n(\varphi)$, let $\varphi_i(\varphi)$ denote the suspension angle at layer i, and let $\varphi_i'(0)$ be its derivative at $\varphi=0$. Sus...

conjecture | open | MathDB 页面 | 原文线索

V2-025493 | MATHDB-342301

Distinct-angle perturbation conjecture for the layered geodesic-net construction

摘录 / EXCERPT: For the finite sequence of layer angles $\varphi_i(\varphi)$ in the construction $G_n(\varphi)$, take a sufficiently small nonzero parameter $\varphi\in(-\epsilon,\epsilon)$. Disti...

conjecture | open | MathDB 页面 | 原文线索

V2-025494 | MATHDB-342315

The H-trivial line-bundle and piecewise-linear-function equivalence for toric DM stacks

摘录 / EXCERPT: The H-trivial line-bundle and piecewise-linear-function conjecture. There are infinitely many H-trivial line bundles on $\mathbb{P}_{\mathbf{\Sigma}}$ if and only if there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-025495 | MATHDB-342316

The H-trivial line-bundle and Picard-region equivalence for toric DM stacks

摘录 / EXCERPT: The H-trivial line-bundle and Picard-region conjecture. There are infinitely many H-trivial line bundles on $\mathbb{P}_{\mathbf{\Sigma}}$ if and only if there exists a nonzero ele...

conjecture | open | MathDB 页面 | 原文线索

V2-025496 | MATHDB-342320

Boldt-Lauret-Schüth conjecture on Dirac-isospectral spin lens spaces

摘录 / EXCERPT: Boldt-Lauret-Schüth conjecture. If the two spin lens spaces are Dirac isospectral, then they are isometric. The claim is supported by computational evidence for spin lens spaces, w...

conjecture | open | MathDB 页面 | 原文线索

V2-025497 | MATHDB-342321

Covering conjecture for 0-isospectral lens orbifolds

摘录 / EXCERPT: Let q be a positive integer, let $s_1,\dots,s_n,s'_1,\dots,s'_n$ be the parameters defining the lens orbifolds $L(q;s_1,\dots,s_n)$ and $L(q;s'_1,\dots,s'_n)$, and let d be a d...

conjecture | open | MathDB 页面 | 原文线索

V2-025498 | MATHDB-342338

Dimension conjecture for Thurston's equal-systole locus

摘录 / EXCERPT: Dimension conjecture. For every $\varepsilon>0$, there exists an integer $g\geq 2$ such that $\mathcal{X}_g$ has dimension at least $(6-\varepsilon)g$.

conjecture | open | MathDB 页面 | 原文线索

V2-025499 | MATHDB-342340

The linear-general-position projection conjecture

摘录 / EXCERPT: Let $Z\subset\mathbb P^3$ be a set of at least 5 points in linear general position, meaning that no 3 points of Z are collinear and no 4 points are coplanar. Projection con...

conjecture | open | MathDB 页面 | 原文线索

V2-025500 | MATHDB-342371

Finite generation conjecture for canonical algebras of projective log canonical pairs

摘录 / EXCERPT: Soit (X,Δ) une paire projective à singularités log canoniques, et soit son algèbre canonique ... où $\left\lfloor l\Delta\right\rfloor$ est le diviseur de Weil obtenu en arron...

conjecture | open | MathDB 页面 | 原文线索

V2-025501 | MATHDB-342376

The Chern-Yamabe flow curvature-bound conjecture

摘录 / EXCERPT: Let X be the manifold underlying a Chern-Yamabe flow, and suppose the flow exists on ... for some $T>0$. Denote its Chern scalar curvature by $S(x;t)$. Chern-Yamabe flow curvature-...

conjecture | open | MathDB 页面 | 原文线索

V2-025502 | MATHDB-342410

Finite-set distance conjecture for Frostman-distributed balls

摘录 / EXCERPT: Let $\alpha>1$, let E be a set of $\delta^{-\alpha}$ δ-balls in $[0,1]^2$, and suppose that every ball of radius $S\delta$ contains at most...

conjecture | open | MathDB 页面 | 原文线索

V2-025503 | MATHDB-342433

Minimal volume conjecture for non-classical virtual links

摘录 / EXCERPT: Minimal volume conjecture. The minimal volume non-classical virtual link is the virtual trefoil, with a volume of approximately 5.33349.

conjecture | open | MathDB 页面 | 原文线索

V2-025504 | MATHDB-342441

Weak Hermitian Campana-Peternell conjecture

摘录 / EXCERPT: Let (M,g) be a Hermitian manifold. Denote its Chern curvature by Ω and its first Chern-Ricci form by ρ. Assume that Ω is Griffiths semipositive and that ρ...

conjecture | open | MathDB 页面 | 原文线索

V2-025505 | MATHDB-342475

The Kähler-Ricci flow convergence conjecture for smooth minimal models

摘录 / EXCERPT: Let X be a smooth Kähler manifold with nef canonical bundle, and let g_0 be any initial Kähler metric on X. Let $g(t)$ be the solution of the unnormalized Kähler-Ricci flow....

conjecture | open | MathDB 页面 | 原文线索

V2-025506 | MATHDB-342483

Nikolayevsky's compact-vector conjecture for constant-length Killing fields

摘录 / EXCERPT: Let G be a semi-simple Lie group acting effectively and isometrically on a connected Riemannian manifold M, and let $X\in\mathfrak{g}=\operatorname{Lie}(G)$. Suppose that X d...

conjecture | open | MathDB 页面 | 原文线索

V2-025507 | MATHDB-342519

Sottile-Williams conjecture on simple Bruhat interval polytopes

摘录 / EXCERPT: Let $v,w\in\mathfrak{S}_n$ with $v\leq w$, and let $Q_{v,w}$ be the Bruhat interval polytope whose vertices are the permutations $u\in[v,w]$. For each $u\in[v,w]$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-025508 | MATHDB-342553

Compatible correspondence conjecture for nondegenerate shifted coisotropic correspondences

摘录 / EXCERPT: Let $\mathrm{CompCorr}_{n}^{s,\operatorname{nd}}$ be the (∞,n)-category of nondegenerate s-shifted compatible coisotropic correspondences, let...

conjecture | open | MathDB 页面 | 原文线索

V2-025509 | MATHDB-342562

Optimal Gromov capacity bound for Vianna's exotic Lagrangian tori

摘录 / EXCERPT: Let $T_{a,b,c}\subset \mathbb{C}P^2$ be Vianna's monotone Lagrangian torus associated with a Markov triple $(a,b,c)\in\mathfrak M$. A capacity conjecture. There is no monotone symp...

conjecture | open | MathDB 页面 | 原文线索

V2-025510 | MATHDB-342564

Vanishing Weinstein width conjecture for Vianna's exotic Lagrangian tori

摘录 / EXCERPT: For each Markov triple $(a,b,c)\in\mathfrak M$, let $T_{a,b,c}\subset\mathbb{C}P^2$ be Vianna's monotone Lagrangian torus, and let ${\frak w}_{\text{DW}}(T_{a,b,c};\mathbb{C}P^2)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025511 | MATHDB-342574

Genericity conjecture for avoiding gimbal lock in hyperbolic triangulations

摘录 / EXCERPT: Let $\mathcal{T}$ be a finite, orientable triangulation with o vertices and m edges admitting a hyperbolic structure. A hyperbolic structure is generic when the stated property...

conjecture | open | MathDB 页面 | 原文线索

V2-025512 | MATHDB-342580

Projection conjecture for large arcs and common zeros of quadrics

摘录 / EXCERPT: Let X be an arc of size $2k+3$ in $\mathrm{PG}(k-1,\mathbb F)$, and let U be the subspace of quadratic forms that vanish on X. If ... then projection conjecture. The projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-025513 | MATHDB-342589

De Philippis-Gigli conjecture on weakly non-collapsed RCD spaces

摘录 / EXCERPT: De Philippis-Gigli conjecture. If $(X,\mathsf{d},\mathfrak{m})$ is a weakly non-collapsed $\operatorname{RCD}(K,N)$ space, then

conjecture | open | MathDB 页面 | 原文线索

V2-025514 | MATHDB-342593

The index conjecture for log Calabi-Yau pairs

摘录 / EXCERPT: Let d be a natural number and let $\mathfrak{R}\subset [0,1]$ be a finite set of rational numbers. For a finite set $\mathfrak{R}$, write $\Phi(\mathfrak{R})$ for the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-025515 | MATHDB-342595

The index conjecture for klt log Calabi-Yau pairs with positive boundary

摘录 / EXCERPT: The klt index conjecture. There exists a positive integer n, depending only on d and $\mathfrak{R}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-025516 | MATHDB-342596

The bounded birational automorphism representation conjecture for dlt pairs

摘录 / EXCERPT: Let n and d be natural numbers, and let (X,B) be a d-dimensional projective dlt pair such that ... Let $\operatorname{Bir}(X,B)$ be the group of B-birational automorphisms...

conjecture | open | MathDB 页面 | 原文线索

V2-025517 | MATHDB-342605

The harmonic-degree range conjecture for normal face-to-face mosaics

摘录 / EXCERPT: Let $\mathcal{M}$ be a normal, face-to-face mosaic in $\mathbb{R}^d$, and let $\bar h(\mathcal{M})$ denote its harmonic degree. Harmonic-degree range conjecture. ... This formulation...

conjecture | open | MathDB 页面 | 原文线索

V2-025518 | MATHDB-342606

The harmonic-degree range conjecture for nonconvex mosaics

摘录 / EXCERPT: Let $d\geq 2$ and let $\mathcal{M}$ be a normal, face-to-face, possibly nonconvex mosaic in $\mathbb{R}^d$. Write $\bar h(\mathcal{M})$ for its harmonic degree. Nonconvex harmonic-...

conjecture | open | MathDB 页面 | 原文线索

V2-025519 | MATHDB-342614

Finiteness conjecture for digit-special integers

摘录 / EXCERPT: A real number $x\geq 0$ is digit-special if, for every integer base $b\geq 3$, its base-b expansion omits at least one digit from $0,1,\dots,b-1$. Finiteness conjecture. There ar...

conjecture | open | MathDB 页面 | 原文线索

V2-025520 | MATHDB-342632

Gusakova-Zaporozhets uniqueness conjecture for ellipsoids

摘录 / EXCERPT: Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be ellipsoids in $\mathbb{R}^n$. For a bounded convex set $K\subset\mathbb{R}^n$, let $V_0(K),\ldots,V_n(K)$ denote its intrinsic volumes, d...

conjecture | open | MathDB 页面 | 原文线索

V2-025521 | MATHDB-342635

Lee's global pseudo-Einstein structure conjecture

摘录 / EXCERPT: Lee's conjecture. Any closed strictly pseudoconvex CR $(2n+1)$-manifold with vanishing first Chern class

conjecture | open | MathDB 页面 | 原文线索

V2-025522 | MATHDB-342636

Fefferman-Hirachi vanishing CR Q-curvature conjecture

摘录 / EXCERPT: Fefferman-Hirachi conjecture. There exists a contact form θ in the conformal contact class

conjecture | open | MathDB 页面 | 原文线索

V2-025523 | MATHDB-342674

Conjecture on normals of complete minimal surfaces with planar ends

摘录 / EXCERPT: Let $X:\Sigma\setminus\{p_1,\ldots,p_m\}\to\mathbb{R}^3$ be a complete minimal surface with m embedded planar ends $\{p_1,\ldots,p_m\}$. Let $n_X(p_i)$ denote the normal at the e...

conjecture | open | MathDB 页面 | 原文线索

V2-025524 | MATHDB-342684

Marcolli-Tabuada's noncommutative Standard Conjecture D

摘录 / EXCERPT: Marcolli-Tabuada's noncommutative Standard Conjecture D. If $\mathcal{C}$ is a smooth and proper differential $\mathbb{Z}$-graded category over a field k of characteristic 0, h...

conjecture | open | MathDB 页面 | 原文线索

V2-025525 | MATHDB-342699

Yang's Hermitian generalization for exterior powers of the tangent bundle

摘录 / EXCERPT: Yang's conjecture. For every integer p with $1<p<n$, the bundle $\bigwedge^p T_X$ admits a smooth RC-positive metric. In particular, if X is Kähler, then X is a projective an...

conjecture | open | MathDB 页面 | 原文线索

V2-025526 | MATHDB-342700

Conjecture on quasi-positive holomorphic sectional curvature

摘录 / EXCERPT: Conjecture on quasi-positive curvature. If X has a Kähler metric with quasi-positive holomorphic sectional curvature, then X is a projective and rationally connected manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-025527 | MATHDB-342702

Evenness conjecture for planar-ended minimal surfaces on the sphere

摘录 / EXCERPT: Let S^2 be the Riemann sphere, let $p_1,\ldots,p_d$ be distinct points, and let ... be a non-planar minimal surface with embedded planar ends. Evenness conjecture. Then $d\geq 4$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-025528 | MATHDB-342754

Uniform boundedness conjecture for the reduced fracture flux

摘录 / EXCERPT: Uniform boundedness conjecture. For $\bar q(u)$ given by Eq.,

conjecture | open | MathDB 页面 | 原文线索

V2-025529 | MATHDB-342757

Monotonicity of the Robin spectral ratio

摘录 / EXCERPT: Spectral-ratio monotonicity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-025530 | MATHDB-342758

Concavity conjecture for the second Robin eigenvalue

摘录 / EXCERPT: Second-eigenvalue concavity conjecture. The function $\lambda_2(\Omega;\alpha)$ is concave in $\alpha>0$. Concavity is known for domains with a suitable symmetry, including balls a...

conjecture | open | MathDB 页面 | 原文线索

V2-025531 | MATHDB-342759

Rayleigh-Bossel conjecture for the first Robin eigenvalue

摘录 / EXCERPT: Rayleigh-Bossel conjecture. For each $\alpha\in\mathbb R$, the scale-invariant quantity

conjecture | open | MathDB 页面 | 原文线索

V2-025532 | MATHDB-342760

Freitas-Laugesen conjecture for the second Robin eigenvalue on convex domains

摘录 / EXCERPT: Freitas-Laugesen conjecture. The ball maximizes

conjecture | open | MathDB 页面 | 原文线索

V2-025533 | MATHDB-342761

Andrews-Clutterbuck-Hauer Robin gap conjecture

摘录 / EXCERPT: Andrews-Clutterbuck-Hauer conjecture. The Robin spectral gap is minimized by the degenerate box, identified with the line segment $(0,D)$:

conjecture | open | MathDB 页面 | 原文线索

V2-025534 | MATHDB-342781

CR Frankel conjecture for spherical CR manifolds

摘录 / EXCERPT: Let (M,J,θ) be a closed, spherical, strictly pseudoconvex CR $(2n+1)$-manifold with vanishing first Chern class ... where $n\geq 2$. A contact form has CR-pluriharmonic Q-cu...

conjecture | open | MathDB 页面 | 原文线索

V2-025535 | MATHDB-342786

The infinite \(\mathcal{KO}\)-width index conjecture

摘录 / EXCERPT: Let M be a closed spin manifold. Its Rosenberg index is the index class associated with the spin Dirac operator, and M is said to have infinite $\mathcal{KO}$-width when its...

conjecture | open | MathDB 页面 | 原文线索

V2-025536 | MATHDB-342820

Soltan's covering conjecture for convex bodies

摘录 / EXCERPT: Soltan's conjecture. For every natural number d,

conjecture | open | MathDB 页面 | 原文线索

V2-025537 | MATHDB-342821

Generalized Soltan conjecture for translative homothetic coverings

摘录 / EXCERPT: Generalized Soltan conjecture. For all natural d and all α such that $0\leq\alpha\leq d+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-025538 | MATHDB-342823

Bezdek's illumination conjecture for polytopes

摘录 / EXCERPT: Bezdek's conjecture. For every non-negative integer α and every d-dimensional convex polytope B,

conjecture | open | MathDB 页面 | 原文线索

V2-025539 | MATHDB-342873

Rational analytic Novikov conjecture

摘录 / EXCERPT: Analytic Novikov conjecture. The map e induces an injection

conjecture | open | MathDB 页面 | 原文线索

V2-025540 | MATHDB-342888

Jacobian ideal containment conjecture for points in general linear position

摘录 / EXCERPT: Let $\mathbb X\subseteq \mathbb P^2$ be a set of $s=\binom{d+1}{2}$ points in general linear position, and let I be the Jacobian ideal of $I(\mathbb X)$. Jacobian ideal containme...

conjecture | open | MathDB 页面 | 原文线索

V2-025541 | MATHDB-342898

Genus-two reduced splitting conjecture for quintic stable quasimaps

摘录 / EXCERPT: Let $Q\subset \mathbb{P}^4$ be the quintic Calabi-Yau manifold, and let $N_{2,0}^{\operatorname{red}}$ denote the reduced genus-two degree-d quasimap invariant defined from a res...

conjecture | open | MathDB 页面 | 原文线索

V2-025542 | MATHDB-342901

Generality conjecture for Fano 10-tuples of planes

摘录 / EXCERPT: Let $\underline{\Lambda}$ be a Fano 10-tuple of planes in $\mathbb{P}^5$, and let the objects associated with it be as in Proposition cited in the source, including the EPW sexti...

conjecture | open | MathDB 页面 | 原文线索

V2-025543 | MATHDB-342910

Uniqueness and characterization of optimal measures for negatively curved shells

摘录 / EXCERPT: Uniqueness conjecture. The optimal measures μ are unique and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-025544 | MATHDB-342915

The weight-gcd conjecture for tightly constrained convex toric domains

摘录 / EXCERPT: Weight-gcd conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025545 | MATHDB-342917

The cross-curvature flow conjecture for negatively curved three-manifolds

摘录 / EXCERPT: Let M be a three-manifold equipped with a negatively curved metric, and let the Cross Curvature Flow (XCF) evolve this metric. A hyperbolic metric is a metric of constant negativ...

conjecture | open | MathDB 页面 | 原文线索

V2-025546 | MATHDB-342918

The model class image-containment conjecture for secondary obstruction complexes

摘录 / EXCERPT: Let $(Y,T^*_{Y,-})$ be a gt-model. The secondary obstruction complex has differentials ... Let $\Theta(T^*_{Y,-})$ denote the model class and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025547 | MATHDB-342941

The Extended Courant Property

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with piecewise smooth boundary, or a compact Riemannian surface, with or without boundary. Let Δ be the Laplace-Beltram...

conjecture | open | MathDB 页面 | 原文线索

V2-025548 | MATHDB-342953

Alesker-Verbitsky's quaternionic Calabi conjecture

摘录 / EXCERPT: Let (M,I,J,K) be an HKT manifold of quaternionic dimension n, with HKT form Ω and operators ∂ and $\partial_J=J^{-1}\circ\overline{\partial}\circ J$. A q-positi...

conjecture | open | MathDB 页面 | 原文线索

V2-025549 | MATHDB-342967

Strong dimension conjecture for dual cells

摘录 / EXCERPT: Strong dimension conjecture. For every dual k-cell D there is a k-dimensional lattice Λ such that there is a k-dimensional cell in the Delone tessellation of...

conjecture | open | MathDB 页面 | 原文线索

V2-025550 | MATHDB-342992

Existence of a universal strict formal group over a ring

摘录 / EXCERPT: Universal strict formal-group conjecture. There exists a universal strict ${\mathcal O}_K^\triangleright$-formal group over some ring S.

conjecture | open | MathDB 页面 | 原文线索

V2-025551 | MATHDB-343013

Powell's conjecture on generators of the genus-g Goeritz group

摘录 / EXCERPT: Let S^3 have its standard genus g Heegaard surface $a3_g$, and let $\mathcal{G}_g$ be the genus g Goeritz group, consisting of orientation-preserving homeomorphisms of S^3...

conjecture | open | MathDB 页面 | 原文线索

V2-025552 | MATHDB-343032

Mabuchi energy expansion conjecture for polarized Riemann surfaces

摘录 / EXCERPT: Mabuchi energy expansion conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-025553 | MATHDB-343033

Strong Mabuchi energy expansion conjecture for beta-ensembles on the Riemann sphere

摘录 / EXCERPT: Strong beta-ensemble expansion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025554 | MATHDB-343044

Demailly's volume formula conjecture for nef classes on compact Kähler manifolds

摘录 / EXCERPT: Let (X^n,ω) be a compact Kähler manifold and let α be a closed real $(1,1)$ form. For a smooth real-valued function u, write...

conjecture | open | MathDB 页面 | 原文线索

V2-025555 | MATHDB-343045

Volume formula conjecture for nef classes on compact Hermitian manifolds

摘录 / EXCERPT: Let (X^n,ω) be a compact Hermitian manifold and let α be a closed real $(1,1)$ form whose Bott-Chern class $[\alpha]$ is nef, meaning that for every $\varepsilon>0$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-025556 | MATHDB-343054

The sharp first Steklov eigenvalue conjecture for planar domains

摘录 / EXCERPT: Let $\widehat \sigma_1^*$ denote the supremum of the first nonzero Steklov eigenvalue multiplied by the boundary length, over planar domains of area π, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025557 | MATHDB-343062

Isoperimetric comparability conjecture for Neumann eigenvalue counts

摘录 / EXCERPT: Isoperimetric comparability conjecture. There exist constants $c_1,c_2>0$, depending only on n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-025558 | MATHDB-343076

One-parameter family conjecture for Calabi-Yau metrics on \(\mathbf{C}^3\) with tangent cone \(\mathbf{C}\times A_2\)

摘录 / EXCERPT: Let ω be a Calabi-Yau metric on $\mathbf{C}^3$, and suppose its tangent cone at infinity is $\mathbf{C}\times A_2$. Metrics are considered up to scaling and isometry. One-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-025559 | MATHDB-343077

Classification and uniqueness conjecture for exact Calabi-Yau manifolds with tangent cone \(\mathbf{C}\times A_1\)

摘录 / EXCERPT: Let $n>4$. A $\partial\bar\partial$-exact Calabi-Yau manifold is a Calabi-Yau manifold of complex dimension n whose Calabi-Yau metric is $\partial\bar\partial$-exact. Suppose its...

conjecture | open | MathDB 页面 | 原文线索

V2-025560 | MATHDB-343085

Finiteness conjecture for polarised hyperkähler manifolds with non-pseudoeffective twisted cotangent bundles

摘录 / EXCERPT: Let $2n$ be an even natural number. Let (X,H) be a polarised hyperkähler manifold, where X has dimension $2n$ and H is an ample Cartier divisor. Write ... for the projectivised...

conjecture | open | MathDB 页面 | 原文线索

V2-025561 | MATHDB-343090

No optimal planar integral point set with a unit edge for at least six points

摘录 / EXCERPT: Let $\mathfrak{M}(2,n)$ denote planar integral point sets with n points, and call M optimal when it attains the minimum possible diameter among sets in $\mathfrak{M}(2,n)$. No-...

conjecture | open | MathDB 页面 | 原文线索

V2-025562 | MATHDB-343091

Bound on the number of unit distances in higher-dimensional integral point sets

摘录 / EXCERPT: Let $M\in\mathfrak{M}(m,n)$ and let $M_1,M_2\in M$ satisfy $|M_1M_2|=1$. A blowup of a facher set is the construction referred to by the source, and a unit distance is said to occu...

conjecture | open | MathDB 页面 | 原文线索

V2-025563 | MATHDB-343118

The fan structure conjecture for diagrams and strong Khovanskii bases

摘录 / EXCERPT: Let Σ be a fan, let I be the ideal defining the fiber data, and let $\Delta(\Sigma, \operatorname{Trop}_\star(I))$ denote the set of diagrams for bundles over the toric va...

conjecture | open | MathDB 页面 | 原文线索

V2-025564 | MATHDB-343149

Perelman's uniqueness conjecture for compact non-collapsed ancient solutions on S^3

摘录 / EXCERPT: Let $(S^3,g(t))$ be a compact, ancient κ-noncollapsed solution to the Ricci flow on S^3, where κ-noncollapsed means that there is some $\kappa>0$ such that every me...

conjecture | open | MathDB 页面 | 原文线索

V2-025565 | MATHDB-343157

Boundary preservation under pointed limits of non-collapsed RCD spaces

摘录 / EXCERPT: Let $K\in\mathbb{R}$, $N\in\mathbb{N}$, and $v>0$. Let $\{(X_i,\mathsf d_i,\mathcal{H}^{N}_{\mathsf d_i})\}_{i\in\mathbb{N}}$ be a sequence of non-collapsed $\mathsf{RCD}(K,N)$ spa...

conjecture | open | MathDB 页面 | 原文线索

V2-025566 | MATHDB-343169

Reznikov's bounded-width conjecture for Heegaard splittings

摘录 / EXCERPT: Reznikov's bounded-width conjecture. For any $g>1$, there are finitely many Dehn twists $\delta_1,\delta_2,\ldots,\delta_N$ on S_g such that every double coset in

conjecture | refuted | MathDB 页面 | 原文线索

V2-025567 | MATHDB-343191

Non-smash-product conjecture for the generalized symmetry bialgebroid

摘录 / EXCERPT: Let X be the finite set underlying the function algebra $C(X)$, and let H be the smallest subalgebra of $C(X \times X \times X \times X)$ containing $C(X)$ and the set H_0 of...

conjecture | open | MathDB 页面 | 原文线索

V2-025568 | MATHDB-343203

Ellipsoid-hyperboloid intersection conjecture

摘录 / EXCERPT: Ellipsoid-hyperboloid conjecture. If the diagonal of M is $\lambda\mathbf{1}$ and the ellipsoid $\mathcal{E}$ meets every branch of $\mathcal{H}$, then $\lambda\leqslant1$.

conjecture | open | MathDB 页面 | 原文线索

V2-025569 | MATHDB-343303

Berger's conjecture on the isoperimetric profile of projective spaces

摘录 / EXCERPT: Let $\mathbb{K}\in\{\mathbb{R},\mathbb{C},\mathbb{H}\}$, and let $I_{\mathbb{P}^{n+1}(\mathbb{K})}$ denote the isoperimetric profile of $\mathbb{P}^{n+1}(\mathbb{K})$, defined by .....

conjecture | open | MathDB 页面 | 原文线索

V2-025570 | MATHDB-343309

F. H. Lin's Lipschitz regularity conjecture for energy-minimizing maps on Alexandrov spaces

摘录 / EXCERPT: Let Ω be a bounded open domain in an n-dimensional Alexandrov space with curvature bounded from below by k, and let N be a compact smooth Riemannian manifold. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-025571 | MATHDB-343313

Akopyan-Karasev-Petrov conjecture on subadditivity of the Ekeland-Hofer-Zehnder capacity

摘录 / EXCERPT: Let $K,K_1,\ldots,K_m$ be convex bodies in $T^*\mathbb R^n$. Akopyan-Karasev-Petrov conjecture. If ... then ... The conjecture is a subadditivity principle for the Ekeland-Hofer-Zehnde...

conjecture | open | MathDB 页面 | 原文线索

V2-025572 | MATHDB-343314

The conjecture on Hofer-Zehnder subadditivity for convex coverings

摘录 / EXCERPT: Let $K_1,\ldots,K_m$ be convex bodies in $T^*\mathbb R^n$. Hofer-Zehnder subadditivity conjecture. The Hofer-Zehnder capacity of their union should satisfy ... This is stronger than...

conjecture | open | MathDB 页面 | 原文线索

V2-025573 | MATHDB-343316

Flat degeneration between parabolic and wild Higgs-bundle moduli spaces

摘录 / EXCERPT: Flat degeneration conjecture. There exists a flat morphism between the parabolic moduli space and the wild moduli space.

conjecture | open | MathDB 页面 | 原文线索

V2-025574 | MATHDB-343338

Hitchin-Kobayashi conjecture for optimal symplectic connections

摘录 / EXCERPT: Let $\pi:(X,H)\to(B,L)$ be a fibration, where H and L denote the relevant polarizations, and let an optimal symplectic connection mean a connection satisfying the optimal sympl...

conjecture | open | MathDB 页面 | 原文线索

V2-025575 | MATHDB-343339

Uniqueness conjecture for optimal symplectic connections

摘录 / EXCERPT: Let $\pi:X\to B$ be a fibration, and let ω_X and ω_X' be optimal symplectic connections. Let $\operatorname{Aut}(X,H)$ denote the automorphism group preserving H. U...

conjecture | open | MathDB 页面 | 原文线索

V2-025576 | MATHDB-343347

The Haefliger-type h-principle conjecture for totally nonparallel immersions

摘录 / EXCERPT: Let M be a smooth closed manifold of dimension n, and let $F_2(M)$ denote its ordered configuration space of two distinct points. The unordered configuration space is...

conjecture | open | MathDB 页面 | 原文线索

V2-025577 | MATHDB-343368

Horizontal diameter rigidity conjecture for singular Riemannian foliations on spheres

摘录 / EXCERPT: Horizontal diameter rigidity conjecture. For any singular Riemannian foliation on a unit sphere $\mathbb{S}^{n}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-025578 | MATHDB-343420

Bøgvad's conjecture on quadratic generation for smooth polytopes

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ be a lattice polytope, and let ${\mathcal A}_P=\{(\alpha,1)\in\mathbb{Z}^{d+1}:\alpha\in P\cap\mathbb{Z}^d\}$. The toric ideal of P is the kernel of th...

conjecture | open | MathDB 页面 | 原文线索

V2-025579 | MATHDB-343421

Lutwak's affine quermassintegral inequality

摘录 / EXCERPT: For $0 < i < n$, let $A_{n-i}(K)$ denote the affine quermassintegral of a convex body K, with $A_0(K):=V_n(K)$ and $A_n(K):=\kappa_n$. Lutwak's conjecture. For $0 \leq i < j < n$...

conjecture | open | MathDB 页面 | 原文线索

V2-025580 | MATHDB-343425

The classification conjecture for surfaces on the Severi line

摘录 / EXCERPT: Classification conjecture. The surface X is on the Severi line if and only if its canonical model X_{can} is a flat double cover of an Abelian surface.

conjecture | open | MathDB 页面 | 原文线索

V2-025581 | MATHDB-343433

Ganatra's relative smooth Calabi-Yau conjecture for Orlov functors

摘录 / EXCERPT: Let (M,σ) be a Weinstein pair such that M and σ are individually Calabi-Yau. The relative smooth Calabi-Yau conjecture says that the Orlov functor $\imath_\sigma$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-025582 | MATHDB-343440

Existence of full finite-entropy ancient mean curvature flows

摘录 / EXCERPT: Existence conjecture. There exists an n-dimensional, connected, full ancient solution to the mean curvature flow in $\mathbb{R}^N$ with finite entropy.

conjecture | open | MathDB 页面 | 原文线索

V2-025583 | MATHDB-343449

Boundary area conjecture under nonnegative Ricci curvature and convexity

摘录 / EXCERPT: Let M^n be a compact Riemannian manifold with $Ric\geq 0$ and second fundamental form $\Pi\geq 1$ on ∂M. Boundary area conjecture. ... The text presents this as a consequ...

conjecture | open | MathDB 页面 | 原文线索

V2-025584 | MATHDB-343450

Boundary area rigidity conjecture under positive Ricci curvature

摘录 / EXCERPT: Let M^n be a compact Riemannian manifold with boundary $\Sigma=\partial M$, $Ric\geq n-1$, and second fundamental form $\Pi\geq 0$. Boundary area rigidity conjecture. ... Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-025585 | MATHDB-343454

Heil's minimum-volume conjecture for bodies of given thickness

摘录 / EXCERPT: Let d be a prescribed thickness, and consider convex bodies in three-dimensional space with thickness d. The Heil body is the convex hull of six circular arcs of radius d, ce...

conjecture | open | MathDB 页面 | 原文线索

V2-025586 | MATHDB-343455

Campi-Colesanti-Gronchi conjecture for bodies of revolution with constant brightness

摘录 / EXCERPT: Let b be a prescribed brightness, and consider convex bodies of revolution in three-dimensional space with constant brightness b.
Campi-Colesanti-Gronchi conjecture. The body c...

conjecture | open | MathDB 页面 | 原文线索

V2-025587 | MATHDB-343456

The conjecture that non-disc analytic domains have tunneling Steklov problems

摘录 / EXCERPT: Let $\Omega\subset \mathbb{R}^2$ be a bounded, simply-connected domain with real analytic boundary that is not equal to $B(x,r)$ for any $x\in \mathbb{R}^2$, $r>0$. A tunneling Ste...

conjecture | open | MathDB 页面 | 原文线索

V2-025588 | MATHDB-343479

Cut locus smeariness conjecture for real projective spaces

摘录 / EXCERPT: Let $m\geq 2$, and let F and G denote the Fréchet functions associated with the intrinsic and comparison distance settings, respectively, near the origin 0 in the real projec...

conjecture | open | MathDB 页面 | 原文线索

V2-025589 | MATHDB-343480

Global Fréchet mean conjecture for the smeary means in the curse of dimensionality theorem

摘录 / EXCERPT: Let X be a random variable on $\mathbb{S}^m$ whose local Fréchet mean is smeary, as in the curse of dimensionality theorem, and let the corresponding local Fréchet means be those...

conjecture | open | MathDB 页面 | 原文线索

V2-025590 | MATHDB-343487

The exact quasi-hyperbolicity constant conjecture for snowflaked Euclidean space

摘录 / EXCERPT: Let $0<\alpha<1$ and let $n\geq 2$. Equip $\mathbb{R}^n$ with the metric d_2^α, where d_2 is the metric induced by the standard Euclidean norm, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025591 | MATHDB-343488

Lott's pinched Ricci curvature conjecture for complete 3-manifolds

摘录 / EXCERPT: Lott's conjecture. If (M,g) is c-Ricci pinched, then (M,g) is flat or M is compact.

conjecture | open | MathDB 页面 | 原文线索

V2-025592 | MATHDB-343504

The compact hyperkähler P=W conjecture

摘录 / EXCERPT: Let M be a compact hyperkähler manifold admitting a holomorphic Lagrangian torus fibration $\ell:M\to B$, and let β be the pullback of an ample divisor on B. Let M_1 be...

conjecture | open | MathDB 页面 | 原文线索

V2-025593 | MATHDB-343517

Nonexistence of locally conformal Kähler metrics on compact non-Kähler SKL manifolds

摘录 / EXCERPT: Nonexistence conjecture. If $ne5 3$, then M^n does not admit any locally conformal Kähler metric.

conjecture | open | MathDB 页面 | 原文线索

V2-025594 | MATHDB-343518

Strominger-flatness conjecture for compact SKL CYT manifolds

摘录 / EXCERPT: Strominger-flatness conjecture. Then g is Strominger flat.

conjecture | open | MathDB 页面 | 原文线索

V2-025595 | MATHDB-343519

Nilmanifold deformation conjecture for compact non-Kähler SKL manifolds

摘录 / EXCERPT: Nilmanifold deformation conjecture. Then (M^n,g) can be deformed to a complex nilmanifold (N^n,h) of this form. Moreover, the step of G is at most two, the left invariant com...

conjecture | open | MathDB 页面 | 原文线索

V2-025596 | MATHDB-343535

The maximality conjecture for Laplacian algebras

摘录 / EXCERPT: Let V be a finite-dimensional real inner-product space, and let $A\subseteq\mathbb{R}[V]$ be a Laplacian algebra, meaning a graded subalgebra preserved by the Laplacian. Maximali...

conjecture | open | MathDB 页面 | 原文线索

V2-025597 | MATHDB-343578

The tropical Ehrhart coefficient bound by tropical minors

摘录 / EXCERPT: Let $\mathbb{T}\mathbb{N}$ denote the tropical natural numbers, let $M \in \mathbb{T}\mathbb{N}^{d \times m}$, and let $P=\operatorname{tconv}(M)$ be the corresponding tropical lat...

conjecture | open | MathDB 页面 | 原文线索

V2-025598 | MATHDB-343582

The critical-eigenvector conjecture for tropical Perron-Frobenius asymptotics

摘录 / EXCERPT: Critical-eigenvector conjecture. If, for every other $w=(0,w_1,w_2)\in\sigma_{max+}(A)$, there exists an $\alpha>0$ with $\alpha\in\mathbb{R}$ such that $v_1=w_1+\alpha$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025599 | MATHDB-343583

The max-plus eigenspace invariance conjecture for Perron-Frobenius asymptotics

摘录 / EXCERPT: Let A and B be matrices, let $\sigma_{(max+)}(A)$ and $\sigma_{(max+)}(B)$ denote their max-plus eigenspaces, and let $\mathscr{P}_{\infty}(A)$ and $\mathscr{P}_{\infty}(B)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-025600 | MATHDB-343635

A singleton-corona conjecture for asymptotic dimension structures

摘录 / EXCERPT: Singleton-corona conjecture. If the asymptotic dimension of $LS(\omega)$ does not equal 0, then there is an unbounded subset C of X whose corona consists of exactly one point...

conjecture | open | MathDB 页面 | 原文线索

V2-025601 | MATHDB-343658

Regular simplex partition number formula

摘录 / EXCERPT: For $r\geq 3$ and $d\geq r-1$, let $N_{(\Delta_{r-1};d)}$ denote the least number of vertices needed so that every map $f:\Delta_{N-1}\rightarrow\mathbb{R}^d$ admits a partition in...

conjecture | open | MathDB 页面 | 原文线索

V2-025602 | MATHDB-343662

Positivity conjecture for the free boundary Brown-York mass

摘录 / EXCERPT: Let (M,g) be a compact Riemannian three-manifold with boundary $\partial M=\Sigma\cup S$, where S and Σ meet orthogonally, Σ has positive Gauss curvature, and its...

conjecture | open | MathDB 页面 | 原文线索

V2-025603 | MATHDB-343679

Atiyah-Sutcliffe's determinant bound conjecture for hyperbolic configurations

摘录 / EXCERPT: Atiyah-Sutcliffe's determinant bound conjecture. For every $\mathbf{x}\in C_n(H^3)$,

conjecture | open | MathDB 页面 | 原文线索

V2-025604 | MATHDB-343684

Normal generation conjecture for pseudo-Anosov monodromies in a fibered cone

摘录 / EXCERPT: Normal generation conjecture. For all but finitely many primitive classes $(S_\alpha,\psi_\alpha)\in\mathscr{C}_F$, ψ_α normally generates $\operatorname{Mod}(S_\alpha)$.

conjecture | open | MathDB 页面 | 原文线索

V2-025605 | MATHDB-343701

Sharp entropy constant conjecture for ancient curve shortening flows

摘录 / EXCERPT: Sharp constant conjecture. The sharp value of the universal constant C_1 in the Colding-Minicozzi codimension bound is

conjecture | open | MathDB 页面 | 原文线索

V2-025606 | MATHDB-343718

Zariski-geometry conjecture for simple groups of finite Morley rank

摘录 / EXCERPT: Let G be a simple group of finite Morley rank definable in a Zariski geometry. Zariski-geometry algebraicity conjecture. The group G has a Lie algebra and is a simple algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-025607 | MATHDB-343725

Refined Joyce conjecture for one-dimensional sheaves on $\mathbb P^2$

摘录 / EXCERPT: For $d\in\mathbb Z_{>0}$ and $\chi\in\mathbb Z$, let $M_{d,\chi}$ be the moduli space of one-dimensional Gieseker semistable sheaves, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025608 | MATHDB-343727

Blanc invariant conjecture for wrapped microlocal sheaf categories

摘录 / EXCERPT: Let Q be a d-dimensional Spin^c manifold, let $\Lambda\subset T^*Q$ be a conic Lagrangian, and let $U\subset Q$ be open. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-025609 | MATHDB-343737

The 3-dimensional manifold Charney-Davis conjecture

摘录 / EXCERPT: Let Δ be a flag 3-manifold, and let $\beta_1(\Delta)$ denote its first Betti number. 3-dimensional manifold Charney-Davis conjecture. ... The conjecture extends the Davis-Okun...

conjecture | open | MathDB 页面 | 原文线索

V2-025610 | MATHDB-343749

The L^p-Brunn-Minkowski conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: The L^p-Brunn-Minkowski conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025611 | MATHDB-343750

The log-Brunn-Minkowski conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: The log-Brunn-Minkowski conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025612 | MATHDB-343751

The local L^p-Brunn-Minkowski inequality for C^2_+ bodies

摘录 / EXCERPT: The local L^p-Brunn-Minkowski inequality. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025613 | MATHDB-343755

Genus-two relations conjecture for the graded Johnson homomorphism

摘录 / EXCERPT: Let H be the standard symplectic representation in genus 2, let $V_\boxplus$ be the weight -2 ${\mathrm{Sp}}(H)$-module in the minimal presentation ... and, after decomposing a...

conjecture | open | MathDB 页面 | 原文线索

V2-025614 | MATHDB-343763

Equidistribution conjecture for intersections of closed geodesics

摘录 / EXCERPT: Intersection equidistribution conjecture. As $D\rightarrow\infty$, the multiset $I_q(D)$ is equidistributed on $\tilde{\ell}_q$ with respect to the hyperbolic metric.

conjecture | open | MathDB 页面 | 原文线索

V2-025615 | MATHDB-343821

CR Yau finite-generation conjecture for Sasakian manifolds

摘录 / EXCERPT: Let M be a complete noncompact Sasakian $(2n+1)$-manifold with nonnegative pseudohermitian bisectional curvature, and let $\mathcal{O}_p^{CR}(M)$ denote the ring of all CR-holomo...

conjecture | open | MathDB 页面 | 原文线索

V2-025616 | MATHDB-343822

CR Yau uniformization conjecture for Sasakian manifolds

摘录 / EXCERPT: Let M be a complete noncompact Sasakian $(2n+1)$-manifold with positive pseudohermitian bisectional curvature. Let $\mathbf{H}_n=\mathbb{C}^n\times\mathbb{R}$ be the standard...

conjecture | open | MathDB 页面 | 原文线索

V2-025617 | MATHDB-343833

Dimension conjecture for discrete holomorphic quadratic differentials

摘录 / EXCERPT: Let $Q^{\mathbb{C}}(\operatorname{cr})$ and $Q^{\mathbb{R}}(\operatorname{cr})$ denote the complex and real spaces of discrete holomorphic quadratic differentials associated with a...

conjecture | open | MathDB 页面 | 原文线索

V2-025618 | MATHDB-343854

Equality of the Thurston distance and asymmetric Finsler distance for negatively curved metrics

摘录 / EXCERPT: Distance equality conjecture. The distances d_T and d_F coincide on these isometry classes. The inequality $d_T\leq d_F$ is established, and equality is known on Teichmüller sp...

conjecture | open | MathDB 页面 | 原文线索

V2-025619 | MATHDB-343856

Shokurov's ascending chain condition conjecture for minimal log discrepancies

摘录 / EXCERPT: Shokurov's conjecture. For every positive integer d, the set

conjecture | open | MathDB 页面 | 原文线索

V2-025620 | MATHDB-343857

The conjecture on the nonexistence of large strong crescent configurations

摘录 / EXCERPT: Strong crescent configuration conjecture. For sufficiently large N, there do not exist strong crescent configurations of size n in $||\cdot||$.

conjecture | open | MathDB 页面 | 原文线索

V2-025621 | MATHDB-343858

The conjecture on the nonexistence of large weak crescent configurations

摘录 / EXCERPT: Weak crescent configuration conjecture. For sufficiently large N, there do not exist weak crescent configurations of size n in $||\cdot||$.

conjecture | open | MathDB 页面 | 原文线索

V2-025622 | MATHDB-343859

The conjecture on large line-like configurations in generic norms

摘录 / EXCERPT: Large line-like configuration conjecture. There exists N such that, for every $n\geq N$, the only line-like configurations of size n in $||\cdot||$ are n equally spaced point...

conjecture | open | MathDB 页面 | 原文线索

V2-025623 | MATHDB-343860

The conjecture on line-like crescent configurations in non-strictly convex norms

摘录 / EXCERPT: Perpendicular perturbation conjecture. There exists N such that every line-like crescent configuration of size $n\geq N$ in $||\cdot||$ is a perpendicular perturbation in...

conjecture | open | MathDB 页面 | 原文线索

V2-025624 | MATHDB-343881

Huisman's conjecture for unramified real curves in even-dimensional projective space

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n}$ be a smooth, geometrically integral, nondegenerate real curve with $X(\mathbb{R})\neq\emptyset$. A real curve is unramified if every real hyperplane...

conjecture | open | MathDB 页面 | 原文线索

V2-025625 | MATHDB-343884

Degeneration conjecture for the eh-proét spectral sequence of rigid spaces

摘录 / EXCERPT: Let K be the base field and let X be a rigid space over K. Write π_X for the structural morphism and $\Omega^j_{\mathrm{\acute{e}h}}$ for the sheaf of eh-differential j...

conjecture | open | MathDB 页面 | 原文线索

V2-025626 | MATHDB-343920

Weyl law improvement conjecture for products of spheres

摘录 / EXCERPT: Weyl law improvement conjecture. There exists $\delta>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025627 | MATHDB-343921

Hadwiger-Boltyanski illumination conjecture for convex bodies

摘录 / EXCERPT: Let $\mathbf{K}$ be a d-dimensional convex body in $\mathbb E^d$, where $d\geq 3$, and let $I(\mathbf{K})$ denote the smallest number of directions in $\mathbb S^{d-1}$ that illu...

conjecture | open | MathDB 页面 | 原文线索

V2-025628 | MATHDB-343938

Closedness conjecture for logarithmic projected-volume vectors

摘录 / EXCERPT: Closedness conjecture. The set ψ_n is closed.

conjecture | open | MathDB 页面 | 原文线索

V2-025629 | MATHDB-343939

Scaling conjecture for projected-volume vectors

摘录 / EXCERPT: Scaling conjecture. If $v\in\psi_n$ and $\lambda>1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-025630 | MATHDB-343940

Conjecture on singular spherical and hyperbolic orbifold limits of Einstein manifolds

摘录 / EXCERPT: Let (M_o,g_o) be a singular spherical or hyperbolic orbifold, and let (M_i,g_i) be smooth Einstein manifolds. Spherical and hyperbolic orbifold limit conjecture. Singular spher...

conjecture | open | MathDB 页面 | 原文线索

V2-025631 | MATHDB-343969

Conjecture on the maximal number of bi-infinite branches with a common past direction

摘录 / EXCERPT: Let d be the dimension, let BI denote the set of bi-infinite branches, and consider their asymptotic directions toward the past, represented by limits...

conjecture | open | MathDB 页面 | 原文线索

V2-025632 | MATHDB-343990

Geodesic-coordinate conjecture for the pressure metric on the Hitchin component

摘录 / EXCERPT: Let S be a closed oriented surface with genus $g\geq 2$ and $n\geq 4$. For any point $\sigma\in \mathcal{T}(S) \subset \mathcal{H}_{n}(S)$, let X be the Riemann surface corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-025633 | MATHDB-344019

Connectedness and maximal dimension of the tropical double ramification locus

摘录 / EXCERPT: Let $DR_{g,a}$ denote the double ramification locus associated with a genus g and a weight vector a. A tropical cone is maximal if it is maximal under inclusion among the cones...

conjecture | open | MathDB 页面 | 原文线索

V2-025634 | MATHDB-344020

Hang-Wang-Yan conjecture on the isoperimetric quotient over scalar-flat conformal classes

摘录 / EXCERPT: Let (M,g) be a smooth compact Riemannian manifold of dimension $n\ge 3$ with nonempty smooth boundary ∂M. Define the conformal Laplacian by ... and let $\lambda_1(L_g)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-025635 | MATHDB-344025

The coarse geometric ℓ^p-Novikov conjecture

摘录 / EXCERPT: The coarse geometric ℓ^p-Novikov conjecture. If Γ is a discrete metric space with bounded geometry, then the index map e_* is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-025636 | MATHDB-344040

The finiteness conjecture for spherical singularities in mean curvature flow

摘录 / EXCERPT: Let $M\subset\mathbb{R}^3$ be a smooth closed embedded mean-convex surface, and let $\mathcal{M}=\{M_t\}_{t\geq 0}$ be its mean curvature flow. A spherical singularity is a singula...

conjecture | open | MathDB 页面 | 原文线索

V2-025637 | MATHDB-344051

Uniqueness conjecture for hd-compactifications of real-reductive Lie groups

摘录 / EXCERPT: Let G be a real reductive Lie group with compact center, and let an hd-compactification of G be a compact manifold with corners $\overline{G}$ containing G as its interior an...

conjecture | open | MathDB 页面 | 原文线索

V2-025638 | MATHDB-344065

Its-Lisovyy-Prokhorov conjecture on the monodromy symplectic form

摘录 / EXCERPT: Let A_j and G_j be the residue and diagonalizing data of the Fuchsian system, let t_j be its pole positions, and let $\mathcal M$ be the monodromy manifold. Define the 1-form...

conjecture | open | MathDB 页面 | 原文线索

V2-025639 | MATHDB-344074

Haddley and Worsley's conjecture on monohedral tilings of the circular disc

摘录 / EXCERPT: Haddley and Worsley's conjecture. Every monohedral tiling is a subtiling of a radially generated tiling.

conjecture | open | MathDB 页面 | 原文线索

V2-025640 | MATHDB-344093

Symplectic Schur-times-Schur conjecture for skew semistandard oscillating tableaux

摘录 / EXCERPT: For partitions λ,μ,ν of length at most m, let χ_λ be the irreducible character of $Sp(2m)$ indexed by λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025641 | MATHDB-344096

Finiteness conjecture for low-volume hyperbolic knot complements with hidden symmetries

摘录 / EXCERPT: Finiteness conjecture. For any $R>0$, at most finitely many hyperbolic knot complements have hidden symmetries and volume less than R.

conjecture | open | MathDB 页面 | 原文线索

V2-025642 | MATHDB-344101

The maximal symmetry rank conjecture

摘录 / EXCERPT: Maximal symmetry rank conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025643 | MATHDB-344103

van den Berg's localization conjecture for first eigenfunctions of convex domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded convex domain, and let u be a first Dirichlet eigenfunction of Ω. Write $\operatorname{diam}(\Omega)$ for its diameter and...

conjecture | open | MathDB 页面 | 原文线索

V2-025644 | MATHDB-344106

Classification conjecture for Lagrangian disks with conformal Maslov form

摘录 / EXCERPT: Let D be a disk immersed in $4\mathbb{R}^4$. Assume that the immersion is Lagrangian, has conformal Maslov form, and has Legendrian capillary boundary on $4\mathbb{S}^3$. Classif...

conjecture | open | MathDB 页面 | 原文线索

V2-025645 | MATHDB-344107

Embedded annulus classification conjecture for minimal Lagrangian surfaces

摘录 / EXCERPT: Let an annulus-type surface be an embedded surface whose underlying domain is an annulus. Assume it is minimal and Lagrangian in the unit ball, with Legendrian capillary boundary o...

conjecture | open | MathDB 页面 | 原文线索

V2-025646 | MATHDB-344129

The Bobkov-Madiman-Wang volume monotonicity conjecture for Minkowski sums

摘录 / EXCERPT: Let A be a compact set in d^d for some $d\in\mathbb{N}$. For $k\in\mathbb{N}$, write ... The volume is the Lebesgue measure, denoted by vol. Bobkov-Madiman-Wang...

conjecture | open | MathDB 页面 | 原文线索

V2-025647 | MATHDB-344130

The Bobkov-Madiman-Wang mixed Minkowski-sum inequality

摘录 / EXCERPT: Let $k\geq 2$, and let $A_1,A_2,\ldots,A_{k+1}$ be compact sets in $\mathbb{R}^d$. For a finite family of sets, $\sum_i A_i$ denotes their Minkowski sum, and vol d...

conjecture | open | MathDB 页面 | 原文线索

V2-025648 | MATHDB-344140

Quiver formula for the extended Goldman Poisson bracket

摘录 / EXCERPT: Following Fock and Goncharov, consider a triangulation Σ_0 and its subtriangulation into sub-triangles, with the associated extended quiver $\mathscr Q$. The nodes of...

conjecture | open | MathDB 页面 | 原文线索

V2-025649 | MATHDB-344141

Griffiths' conjecture on simultaneous normalization of periods

摘录 / EXCERPT: Let $f:\mathfrak{X}\to S$ be an algebraic family of polarized algebraic manifolds over a quasi-projective manifold S, with period map $\Phi:S\to\Gamma\backslash D$ and lifted per...

conjecture | open | MathDB 页面 | 原文线索

V2-025650 | MATHDB-344150

Toric regulator graph construction conjecture

摘录 / EXCERPT: Toric regulator graph construction conjecture. The projected toric regulator equals the graph-theoretic regulator:

conjecture | open | MathDB 页面 | 原文线索

V2-025651 | MATHDB-344171

Boundedness and effective birationality conjecture for canonical models of foliated surfaces

摘录 / EXCERPT: Let $P:\mathbb Z_{\geq 0}\to\mathbb Z$ be an integer-valued function. A canonical model is a complete foliated surface $(X,\mathcal F)$ with ${\rm kod}(\mathcal F)=2$ and ... for all...

open_problem | open | MathDB 页面 | 原文线索

V2-025652 | MATHDB-344176

Rigidity conjecture for spacelike entire self-shrinking graphs

摘录 / EXCERPT: Rigidity conjecture. Then $u^\alpha(x)$ is a linear function for each $\alpha=1,\ldots,n$.

conjecture | open | MathDB 页面 | 原文线索

V2-025653 | MATHDB-344181

The Ruelle invariant conjecture for ECH capacities of generic star-shaped domains

摘录 / EXCERPT: Let X be a nice star-shaped domain in ${\mathbb R}^4$. Write $e_k(X)$ for the error term in the asymptotic formula recovering the volume of X from its ECH capacities, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025654 | MATHDB-344183

Poissonization functor conjecture for Dirac structures

摘录 / EXCERPT: Let E) be a Courant algebroid and let $L\subset E$ be a Dirac structure admitting a Lagrangian complement $L'\subset E$. A Poissonization functor for Dirac structures assigns a s...

conjecture | open | MathDB 页面 | 原文线索

V2-025655 | MATHDB-344184

Dirac reduction conjecture for spanning Poisson structures

摘录 / EXCERPT: Let E be a Courant algebroid with a pair of transversal Dirac structures $L,L'\subset E$. Assume that E reduces to E_r and that L and L' reduce to Dirac structures...

conjecture | open | MathDB 页面 | 原文线索

V2-025656 | MATHDB-344205

The short-word discreteness conjecture for complex hyperbolic triangle groups

摘录 / EXCERPT: The short-word discreteness conjecture. A (n_1,n_2,n_3)-complex hyperbolic triangle group is a discrete embedding in $PU(2,1)$ if neither W_A nor W_B is non-elliptic.

conjecture | open | MathDB 页面 | 原文线索

V2-025657 | MATHDB-344232

Mirror-spectrum interval conjecture for CIFSes

摘录 / EXCERPT: Let $K\subset[0,\infty)$, and write its mirror image as $\sup(K)-K=\{\sup(K)-x:x\in K\}$. A dimension spectrum is a set arising as the set of Hausdorff dimensions of the limit sets...

conjecture | open | MathDB 页面 | 原文线索

V2-025658 | MATHDB-344233

Continuity and monotonicity conjecture for local dimensions of CIFS spectra

摘录 / EXCERPT: Local-dimension regularity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-025659 | MATHDB-344234

Three-type classification conjecture for CIFS dimension spectra

摘录 / EXCERPT: Let $F\subset[0,\infty)$ be the dimension spectrum of a CIFS, and let $\dim_x(F)$ denote its local Hausdorff dimension at $x\in F$. Three-type classification conjecture. Exactly on...

conjecture | open | MathDB 页面 | 原文线索

V2-025660 | MATHDB-344257

Rational linear independence conjecture for volumes of large Dehn fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be a 1-cusped hyperbolic 3-manifold with non-quadratic cusp shapes. For $n>0$, choose n distinct Dehn filling coefficients, all sufficiently large, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025661 | MATHDB-344280

Ricci curvature rigidity conjecture for isoparametric tubes

摘录 / EXCERPT: Let $(N^{n+1},\mathbf{g})$ be a compact Riemannian manifold with boundary ∂N, and let $(M_c,\bar{\mathbf{g}}|_{M_c})$ be an isoparametric hypersurface in the unit sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-025662 | MATHDB-344281

Grove's conjecture on DDBD structures

摘录 / EXCERPT: Let M be a closed simply connected manifold admitting a Riemannian metric of nonnegative sectional curvature. Grove's conjecture. Then M has a DDBD structure, that is, an isopa...

conjecture | open | MathDB 页面 | 原文线索

V2-025663 | MATHDB-344282

Leafwise curvature conjecture for isoparametric foliations

摘录 / EXCERPT: Let $(N^n,\mathfrak{F})$ be an isoparametric foliation. Suppose its leaves have nonnegative or positive leafwise sectional curvature, $K_{\mathrm{sec}}^\mathcal{F}\geqslant 0$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-025664 | MATHDB-344283

Tang's conjecture on nonnegative curvature of isoparametric leaves

摘录 / EXCERPT: Let M be an isoparametric hypersurface or a focal submanifold of the unit sphere $\mathbb{S}^n$. Tang's conjecture. Every such isoparametric hypersurface and focal submanifold ad...

conjecture | open | MathDB 页面 | 原文线索

V2-025665 | MATHDB-344344

Lebedev-Milin-type inequality for the critical conformal curvature functional

摘录 / EXCERPT: Let (B^{n+1},dx^2) be the unit ball in Euclidean $(n+1)$-space, and let $k=\frac{n+1}{2}$. Define ... For $g=e^{2u}dx^2$, let $\mathcal{F}_k(g)$ denote the critical conformal primi...

conjecture | open | MathDB 页面 | 原文线索

V2-025666 | MATHDB-344361

The faithfulness conjecture for dominant embeddings

摘录 / EXCERPT: Let Γ be the object studied in the paper, and let $e:\Gamma\to\mathrm{PG}(V)$ be an embedding. An embedding is dominant when it is dominant in the sense defined in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-025667 | MATHDB-344362

The Kashaev-trisection invariant conjecture for closed 4-manifolds

摘录 / EXCERPT: Let X be an oriented closed 4-manifold, let $\chi(X)$ be its Euler characteristic, and let $N\geq 2$. Denote by $\mathcal{K}_N(X)$ the Kashaev invariant and by...

conjecture | open | MathDB 页面 | 原文线索

V2-025668 | MATHDB-344365

The Alexandrov gluing conjecture

摘录 / EXCERPT: Alexandrov gluing conjecture. The gluing produces an Alexandrov space Y if and only if the gluing is by isometry and, for every $p\in Y$, $T_p(Y)$ is a metric cone over...

conjecture | open | MathDB 页面 | 原文线索

V2-025669 | MATHDB-344366

The Alexandrov globalization conjecture

摘录 / EXCERPT: Alexandrov globalization conjecture. Then $(X,d_X)\in\operatorname{Alex}^n(\kappa)$.

conjecture | open | MathDB 页面 | 原文线索

V2-025670 | MATHDB-344377

The BEV bound for linear codes in projective spaces

摘录 / EXCERPT: Let $\mathbb{P}_q(n)$ be the projective space of subspaces of $\mathbb{F}_q^n$, equipped with the notion of linear code introduced by Braun, Etzion, and Vardy. The BEV bound states...

conjecture | open | MathDB 页面 | 原文线索

V2-025671 | MATHDB-344378

Unique indecomposable basis conjecture for linear subspace codes

摘录 / EXCERPT: Let $\mathcal{U}$ be a linear code in the projective space $\mathbb{P}_q(n)$. An indecomposable basis is a basis of the vector space over $\mathbb{F}_2$ formed by the code whose el...

conjecture | open | MathDB 页面 | 原文线索

V2-025672 | MATHDB-344379

The projective-space metric-thickening stability conjecture

摘录 / EXCERPT: Projective-space stability conjecture. There exists a sufficiently small $\varepsilon>0$ such that, for every $0<\delta<\varepsilon$, both...

conjecture | open | MathDB 页面 | 原文线索

V2-025673 | MATHDB-344450

Optimality conjecture for the regular heptagon tile of area $\pi/3$

摘录 / EXCERPT: Let S be a closed hyperbolic surface, and let ${R}_7$ denote the regular heptagon with angles of 120°, which has area $\pi/3$. A tile is non-equivalent to ${R}_7$ when it...

conjecture | open | MathDB 页面 | 原文线索

V2-025674 | MATHDB-344451

Regular polygon optimality conjecture for area $(n-6)\pi/3$

摘录 / EXCERPT: Let n be an integer for which a regular hyperbolic n-gon with angles of 120° has area $(n-6)\pi/3$, and compare it with tiles of the same area. Regular polygon optimalit...

conjecture | open | MathDB 页面 | 原文线索

V2-025675 | MATHDB-344452

Angle-sum criterion for tiling hyperbolic quadrilaterals

摘录 / EXCERPT: Let a quadrilateral have distinct angles θ_i, and let k_i be positive integers that are congruent modulo 2; that is, they are either all odd or all even. Quadrilateral t...

conjecture | open | MathDB 页面 | 原文线索

V2-025676 | MATHDB-344489

Existence of nonsoliton ancient flows from nonround self-shrinkers

摘录 / EXCERPT: Let Σ be a compact, embedded, nonround self-shrinker in $\mathbb{R}^{n+1}$. An associated-flow conjecture. There exists an associated nonsoliton ancient flow ... This proposes...

conjecture | open | MathDB 页面 | 原文线索

V2-025677 | MATHDB-344491

Baker's conjecture on concordance of tight fibered knots

摘录 / EXCERPT: Let K_{0} and K_{1} be fibered knots in S^{3} supporting the tight contact structure. Baker's conjecture. If K_{0} and K_{1} are concordant, then $K_{0}=K_{1}$. The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-025678 | MATHDB-344502

Catalan correction-factor formula for the enumerative Euler class

摘录 / EXCERPT: Let $n\geq 2$, let V be the vector bundle and let $\sigma=\alpha\wedge\beta$ be the section considered in the paper. Write ... where ... is the $(n-1)$st Catalan number and $i(n)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-025679 | MATHDB-344512

McDuff's restricted stabilized ellipsoid embedding conjecture

摘录 / EXCERPT: Let $E(a,b)$ denote the four-dimensional symplectic ellipsoid, let $B^4(c)=E(c,c)$ be the round four-ball, and for $N\in\mathbb{Z}_{\geq 1}$ define ... Also let $f_0(x)$ be the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-025680 | MATHDB-344521

The BCOV genus-one mirror symmetry conjecture

摘录 / EXCERPT: Let X be a Calabi--Yau manifold and let $\varphi\colon\mathcal X^\vee\to\mathbf D^\times$ be a mirror family, where $\mathbf D^\times=(\mathbb D^\times)^d$. Let τ_{BCOV} den...

conjecture | open | MathDB 页面 | 原文线索

V2-025681 | MATHDB-344522

The functorial Grothendieck--Riemann--Roch conjecture for the BCOV line

摘录 / EXCERPT: Let $f\colon X\to S$ be a projective family of connected Calabi--Yau manifolds. Let $\lambda_{BCOV}(X/S)$ be the BCOV line bundle, formed from the determinants of the Hodge bundles...

conjecture | open | MathDB 页面 | 原文线索

V2-025682 | MATHDB-344523

The refined BCOV mirror-symmetry conjecture for degenerating Calabi--Yau families

摘录 / EXCERPT: Let $f\colon X\to\mathbf D^\times=(\mathbb D^\times)^d$ be a projective family of Calabi--Yau n-folds with $d=h^{1,n-1}$, effectively parametrized deformation space, and strongly...

conjecture | open | MathDB 页面 | 原文线索

V2-025683 | MATHDB-344531

The no-exotic-obstructions conjecture for deformations of super Riemann surfaces

摘录 / EXCERPT: Let $(\mathcal{U}, \vartheta)$ be an algebraic deformation of a super Riemann surface $S(C, T^*_{C, -})$ over $\mathbb{A}^{0|n}_\mathbb{C}$. Its primary obstruction...

conjecture | open | MathDB 页面 | 原文线索

V2-025684 | MATHDB-344536

Tosatti's conjecture on the Gromov-Hausdorff limit of collapsing Calabi-Yau metrics

摘录 / EXCERPT: Let $N_0=N\backslash D$ and let ω be the twisted Kähler-Einstein metric on N_0. Let (X,d_X) be the metric completion of (N_0,ω), and set $S_X=X\backslash N_0$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-025685 | MATHDB-344551

Uniform Lipschitz extension conjecture for Hadamard manifolds

摘录 / EXCERPT: Let $C<1$. For Hadamard manifolds X and Y satisfying ... write $\mathcal{L}_{X,Y}(C)$ for the optimal Lipschitz constant needed to extend every C-Lipschitz map from a subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-025686 | MATHDB-344552

Power-law Lipschitz extension conjecture for Hadamard manifolds

摘录 / EXCERPT: Let X and Y be Hadamard manifolds, let $C<1$, and let $\mathcal{L}_{X,Y}(C)$ denote the optimal Lipschitz constant for extending every C-Lipschitz map from a subset of X to...

conjecture | open | MathDB 页面 | 原文线索

V2-025687 | MATHDB-344581

The disjoint curves conjecture for Kähler manifolds and orbifolds

摘录 / EXCERPT: Let X be a Kähler manifold or a Kähler orbifold with first Betti number $b_1=0$ and second Betti number $b_2\geq 2$. Suppose that X contains disjoint complex curves spanning...

conjecture | open | MathDB 页面 | 原文线索

V2-025688 | MATHDB-344602

Geodesic-minimal ruled surface conjecture in spaces of oriented geodesics

摘录 / EXCERPT: Let $c\in\{1,-1\}$, let $\mathbb S^{n+1}(c)$ be the real space form of constant sectional curvature c, and let $\mathbb L(\mathbb S^{n+1}(c))$ denote its space of oriented geodes...

conjecture | open | MathDB 页面 | 原文线索

V2-025689 | MATHDB-344612

The volume-product conjecture for finite metric spaces

摘录 / EXCERPT: Volume-product conjecture for finite metric spaces. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025690 | MATHDB-344615

The supersolvable arrangement double-point conjecture

摘录 / EXCERPT: Supersolvable arrangement double-point conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025691 | MATHDB-344616

The supersolvable Sylvester-Gallai conjecture over the complex numbers

摘录 / EXCERPT: Supersolvable Sylvester-Gallai conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025692 | MATHDB-344617

The nonfree line-arrangement double-point conjecture

摘录 / EXCERPT: Nonfree line-arrangement double-point conjecture. If ${\mathcal A}$ is not free, then

conjecture | open | MathDB 页面 | 原文线索

V2-025693 | MATHDB-344618

The deletion double-point conjecture for free line arrangements

摘录 / EXCERPT: Deletion double-point conjecture. If ${\mathcal A}\setminus\{H\}$ is not free, then

conjecture | open | MathDB 页面 | 原文线索

V2-025694 | MATHDB-344624

Positive Ricci curvature conjecture for compatible metrics on contact 3-manifolds

摘录 / EXCERPT: Let (M,ξ) be a contact 3-manifold equipped with a compatible metric g. Write $Ricci(g)>0$ for positivity of its Ricci curvature, and let $(\mathbb{S}^3,\xi_{std})$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-025695 | MATHDB-344625

Tightness conjecture for nowhere Reeb-invariant compatible metrics

摘录 / EXCERPT: Let (M,ξ) be a contact 3-manifold. Let X_α be a Reeb vector field and J a complex structure on the contact distribution, and let θ' denote the instantaneous rot...

conjecture | open | MathDB 页面 | 原文线索

V2-025696 | MATHDB-344628

The Lagrangian Delzant conjecture for monotone Lagrangians

摘录 / EXCERPT: Let P_1 and P_2 be Delzant Fano polytopes in $\mathbb{R}^k$ with $n+k$ facets, and let L_1 and L_2 be the corresponding embedded monotone Lagrangian submanifolds of...

conjecture | open | MathDB 页面 | 原文线索

V2-025697 | MATHDB-344669

Rigidity conjecture for ancient spherical mean curvature flows with isoparametric curvature ratio

摘录 / EXCERPT: Let $f(t,\theta)$ be an ancient solution to the spherical mean curvature flow for smooth compact hypersurfaces in S^{n+1}, and let $A(t,\theta)$ and $H(t,\theta)$ denote its shap...

conjecture | open | MathDB 页面 | 原文线索

V2-025698 | MATHDB-344670

Rigidity conjecture for ancient spherical mean curvature flows satisfying isoparametric pinching estimates

摘录 / EXCERPT: Let $f(t,\theta)$ be an ancient solution to the spherical mean curvature flow for smooth compact hypersurfaces in S^{n+1}. Let $A(t,\theta)$ and $H(t,\theta)$ be its shape operat...

conjecture | open | MathDB 页面 | 原文线索

V2-025699 | MATHDB-344673

The optimal symplectic connection conjecture for fibrations

摘录 / EXCERPT: Let $(X,H)\to(B,L)$ be a polarised fibration, with each fibre equipped with its fibrewise cscK geometry. An optimal symplectic connection is a fibrewise constant scalar curvature K...

conjecture | open | MathDB 页面 | 原文线索

V2-025700 | MATHDB-344678

Chengbo Yue's gap conjecture for convex-cocompact subgroups

摘录 / EXCERPT: Let $G=\operatorname{Aut}(\mathbf B^n)$, and let $\Gamma<G$ be a convex-cocompact torsion-free subgroup. Let δ_Γ denote its critical exponent. Chengbo Yue's gap conject...

conjecture | open | MathDB 页面 | 原文线索

V2-025701 | MATHDB-344679

The nonarithmetic lattice existence conjecture for complex hyperbolic groups

摘录 / EXCERPT: Let n be a positive integer. Nonarithmetic lattice existence conjecture. For each n, the group $\operatorname{PU}(n,1)$ contains a nonarithmetic lattice. Nonarithmetic lattices...

conjecture | open | MathDB 页面 | 原文线索

V2-025702 | MATHDB-344680

Bruce Hunt's hybrid construction conjecture for nonarithmetic lattices

摘录 / EXCERPT: For every $n\ge 2$, consider arithmetic lattices $\Gamma_1,\Gamma_2<\operatorname{SU}(n-1,1)$ and $\Gamma_3<\operatorname{SU}(n-2,1)$. Suppose Γ_3 is isomorphic to subgroups...

conjecture | open | MathDB 页面 | 原文线索

V2-025703 | MATHDB-344685

The ascending chain conjecture for canonical thresholds

摘录 / EXCERPT: Canonical-threshold ACC conjecture. The set $\mathcal{T}_{n}^{\textup{can}}$ satisfies the ascending chain condition.

conjecture | open | MathDB 页面 | 原文线索

V2-025704 | MATHDB-344701

The characterization of 1-connected compact metric spaces by snowflake kernels

摘录 / EXCERPT: Let $4\mathcal{M}^14$ denote the collection of all 1-connected compact metric spaces, and let $4\mathfrak{S}_p4$ be the operator whose kernel consists of the spaces mapped to the...

conjecture | open | MathDB 页面 | 原文线索

V2-025705 | MATHDB-344725

Continuity and asymptotic limit of the Weyl-law constant for the p-Laplacian

摘录 / EXCERPT: Let M be a closed n-dimensional Riemannian manifold, let p be the exponent in the p-Laplacian, and let $c_n(p)$ be the universal constant in the asymptotic law ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-025706 | MATHDB-344750

The intersection-theoretic formulas for affine invariant submanifolds

摘录 / EXCERPT: The affine-invariant-submanifold intersection formulas. The area Siegel-Veech constant and the sum of Lyapunov exponents are

conjecture | open | MathDB 页面 | 原文线索

V2-025707 | MATHDB-344761

Colding-Minicozzi conjectures on the frequency of harmonic functions

摘录 / EXCERPT: Let M^n be a complete Riemannian manifold with nonnegative Ricci curvature and maximal volume growth, and let $\mathscr{F}_u(r)$ denote the frequency function of a harmonic funct...

conjecture | open | MathDB 页面 | 原文线索

V2-025708 | MATHDB-344780

Existence and parametrization of structures over generically semisimple F-manifolds

摘录 / EXCERPT: Let (M,t^0) be an irreducible germ of a generically semisimple F-manifold. A generically semisimple F-manifold structure conjecture. The existence and parametrization assertions...

conjecture | open | MathDB 页面 | 原文线索

V2-025709 | MATHDB-344853

The intrinsic-volume upper bound for intersections of congruent balls

摘录 / EXCERPT: Let ${\mathbb E}^d$ be Euclidean d-space, let $P=\{\mathbf{p}_1,\mathbf{p}_2,\dots,\mathbf{p}_N\}\subseteq{\mathbb E}^d$, and let P^r denote the intersection of the closed ball...

conjecture | open | MathDB 页面 | 原文线索

V2-025710 | MATHDB-344865

Stable irrationality conjecture for very general cubic hypersurfaces

摘录 / EXCERPT: Let $n\geq 3$ and let $X\subset \mathbb P^{n+1}$ be a very general cubic hypersurface. Stable irrationality conjecture. X is not stably rational. The conjecture asks whether the...

conjecture | open | MathDB 页面 | 原文线索

V2-025711 | MATHDB-344868

Wang's uniqueness conjecture for nonlinear boundary problems on convex manifolds

摘录 / EXCERPT: Let (M^{n},g) be a smooth compact Riemannian manifold with boundary Σ, with $\operatorname{Ric}\geq 0$ and principal curvature tensor $\Pi\geq 1$ on Σ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025712 | MATHDB-344869

Wang's type II Yamabe rigidity conjecture

摘录 / EXCERPT: Let (M^n,g) be a smooth compact Riemannian manifold with $Ric\geq 0$ and second fundamental form $\Pi\geq 1$ on its boundary Σ. Let $u\in C^{\infty}(M)$ be a positive solu...

conjecture | open | MathDB 页面 | 原文线索

V2-025713 | MATHDB-344870

Escobar's Steklov eigenvalue conjecture under nonnegative Ricci curvature

摘录 / EXCERPT: Let (M^n,g) be a compact Riemannian manifold with boundary, with $Ric\geq 0$ and second fundamental form $\Pi\geq 1$ on its boundary Σ. Escobar's Steklov eigenvalue conjec...

conjecture | solved | MathDB 页面 | 原文线索

V2-025714 | MATHDB-344877

Failure of Generic Green's conjecture in characteristic 3

摘录 / EXCERPT: Let g denote the genus of a curve over a field of characteristic 3. Generic Green failure conjecture. Generic Green's conjecture fails in characteristic 3 for genera 9 and...

conjecture | open | MathDB 页面 | 原文线索

V2-025715 | MATHDB-344882

Nikolayevsky-Nikonorov's characterization conjecture for Ricci-negative solvable Lie algebras

摘录 / EXCERPT: Let $\mathfrak{s}$ be a solvable Lie algebra with nilradical $\mathfrak{n}$. For a nilpotent Lie algebra $\mathfrak{n}$, let $\mathfrak{t}(\mathfrak{n})$ be a maximal torus of diag...

conjecture | open | MathDB 页面 | 原文线索

V2-025716 | MATHDB-344883

Nikolayevsky-Nikonorov's rank-one reduction conjecture for Ricci-negative solvable Lie algebras

摘录 / EXCERPT: Let $\mathfrak{s}$ be a solvable Lie algebra with nilradical $\mathfrak{n}$, and let $Y\in\mathfrak{s}$. The subspace $\mathbb{R}Y\oplus\mathfrak{n}$ is the corresponding rank-one...

conjecture | open | MathDB 页面 | 原文线索

V2-025717 | MATHDB-344884

Real-part conjecture for Ricci-negative derivations

摘录 / EXCERPT: Let $\mathfrak{n}$ be a nilpotent Lie algebra and let $\operatorname{Der}(\mathfrak{n})_{\mathrm{rn}}$ denote the derivations whose associated solvable extension admits a Ricci-neg...

conjecture | open | MathDB 页面 | 原文线索

V2-025718 | MATHDB-344885

Convexity conjecture for strongly Ricci-negative derivations

摘录 / EXCERPT: Let $\mathfrak{n}$ be a nilpotent Lie algebra, let $\mathfrak{t}(\mathfrak{n})$ be a maximal torus of diagonalizable derivations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025719 | MATHDB-344886

Equality conjecture for the strongly Ricci-negative cone

摘录 / EXCERPT: Let $\mathfrak{n}$ be a nilpotent Lie algebra, let $\mathfrak{t}(\mathfrak{n})$ be a maximal torus of diagonalizable derivations, let $\mathfrak{t}(\mathfrak{n})_{\mathrm{srn}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-025720 | MATHDB-344922

Assouad dimension conjecture for continuum trees

摘录 / EXCERPT: Assouad dimension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025721 | MATHDB-344928

Density of Chern slopes for Brody hyperbolic surfaces with prescribed fundamental group

摘录 / EXCERPT: Let Y be a Brody hyperbolic nonsingular projective surface, and let S be a hyperbolic nonsingular projective surface. Write $c_1^2(S)/c_2(S)$ for the Chern slope of S. Hyperb...

conjecture | open | MathDB 页面 | 原文线索

V2-025722 | MATHDB-344956

Characterization of complex space forms by the Delta-property

摘录 / EXCERPT: Let (M,g) be a complete and simply-connected Kähler manifold. Say that M satisfies the Delta-property if, for every point $x\in M$, there is a coordinate system centered at x...

conjecture | open | MathDB 页面 | 原文线索

V2-025723 | MATHDB-344957

The nontriviality conjecture for barbell implantations in the 4-sphere

摘录 / EXCERPT: Let S^4 be the 4-sphere and let $\operatorname{Diff}_0(S^4)$ denote its diffeomorphism group fixing the relevant boundary data. For a reduced word w in the generators x,y of...

conjecture | open | MathDB 页面 | 原文线索

V2-025724 | MATHDB-344958

The kernel conjecture for isotopy extension and barbell implantations

摘录 / EXCERPT: Let $\operatorname{Emb}(S^1,S^1\times S^3;S^1_0)$ be the based embedding space, and let $\operatorname{Diff}(S^1\times B^3\operatorname{fix}\partial)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025725 | MATHDB-344959

The genus-g handlebody detection conjecture in the 4-sphere

摘录 / EXCERPT: For each $g\geq 0$, let $V_0,V_1\subset S^4$ be 3-dimensional genus-g handlebodies with the same boundary, $\partial V_0=\partial V_1$. A simple closed curve in ∂V_0 is...

conjecture | open | MathDB 页面 | 原文线索

V2-025726 | MATHDB-344961

Fu's Weyl principle for Holmes-Thompson intrinsic volumes

摘录 / EXCERPT: Let V_j be finite-dimensional real vector spaces equipped with smooth norms F_j, and let $e_j\colon M\to V_j$, $j=1,2$, be smooth immersions of a smooth manifold M. For a val...

conjecture | open | MathDB 页面 | 原文线索

V2-025727 | MATHDB-344970

The generating-level bound for the Hodge filtration of a free divisor

摘录 / EXCERPT: Let X be a complex manifold, let D be a divisor satisfying the assumptions of Theorem and Corollary, and let $b_D(s)$ be the relevant Bernstein-Sato polynomial. For a well-filt...

conjecture | open | MathDB 页面 | 原文线索

V2-025728 | MATHDB-344987

Chen-Yang Turaev-Viro volume conjecture

摘录 / EXCERPT: Chen-Yang Turaev-Viro volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025729 | MATHDB-344989

Detcherry-Kalfagianni-Yang colored-Jones volume question for hyperbolic links

摘录 / EXCERPT: Detcherry-Kalfagianni-Yang colored-Jones volume question.

open_problem | open | MathDB 页面 | 原文线索

V2-025730 | MATHDB-345023

The hyperbolic knot volume conjecture

摘录 / EXCERPT: Let K be a hyperbolic knot, and let $J'_N(K;t)$ denote its N-th normalized colored Jones polynomial evaluated at t. The normalization is chosen so that $J'_N(U,t)=1$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-025731 | MATHDB-345039

Cooper-Tillmann's generalized cusp conjecture for properly convex structures

摘录 / EXCERPT: Let M be an orientable non-compact 3-manifold that is the interior of a compact manifold whose boundary is a finite union of disjoint 2-tori. Let $\mathcal{CP}(M)$ be the spa...

conjecture | open | MathDB 页面 | 原文线索

V2-025732 | MATHDB-345045

Iskovskikh's stability conjecture for tangent bundles of Fano manifolds

摘录 / EXCERPT: Iskovskikh's stability conjecture. The tangent bundle Θ_X is (semi)stable.

conjecture | open | MathDB 页面 | 原文线索

V2-025733 | MATHDB-345046

Diagonal Ricci flow conjecture for cohomogeneity one manifolds

摘录 / EXCERPT: Let (M,G) be a cohomogeneity one manifold, let $\mathcal{B}$ be a stably Ricci-diagonal basis for it, and let g_0 be a metric on M that is diagonal with res...

conjecture | open | MathDB 页面 | 原文线索

V2-025734 | MATHDB-345100

The Calabi conjecture for compact Kähler manifolds

摘录 / EXCERPT: Let (X,g,ω) be a compact Kähler manifold, and fix $R\in\Omega^{1,1}(X)$ such that $[R]=c_1(T_X)\in H^{1,1}(X)$. Then there exists a unique Kähler metric $\widetilde{g}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-025735 | MATHDB-345104

Entropy slope formula conjecture for maximal geodesic rays

摘录 / EXCERPT: Let Φ be a maximal geodesic ray, with non-Archimedean limit $\Phi_{\rm NA}$, and let ${\bf H}'^\infty(\Phi)$ and ${\bf H}^{\rm NA}(\Phi_{\rm NA})$ denote respectively the asym...

conjecture | open | MathDB 页面 | 原文线索

V2-025736 | MATHDB-345105

Boucksom-Jonsson's regularization conjecture

摘录 / EXCERPT: Let (X,L) be a polarized manifold, let $\mathcal{E}^{1,\rm NA}(L)$ be the finite-energy non-Archimedean metrics, let ${\mathcal H}^{\rm NA}(L)$ be the smooth positive non-Archime...

conjecture | open | MathDB 页面 | 原文线索

V2-025737 | MATHDB-345124

Orponen's direction-set conjecture for planar Borel sets

摘录 / EXCERPT: Orponen's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025738 | MATHDB-345125

The strong pinned radial-projection conjecture for planar sets

摘录 / EXCERPT: Let E and F be Borel sets in the plane with $\dim_{\mathcal{H}}E>0$ and $\dim_{\mathcal{H}}F>0$, and suppose that F is not contained in a line. For y in the plane, let...

conjecture | open | MathDB 页面 | 原文线索

V2-025739 | MATHDB-345132

The Willmore functional lower-bound conjecture for genus-zero hypersurfaces

摘录 / EXCERPT: Let M^n be a compact hypersurface of genus 0 in $\mathbb{R}^{n+1}$ whose volume equals that of the unit sphere, and let $\mathcal{W}(M)$ denote its Willmore functional. Willmor...

conjecture | open | MathDB 页面 | 原文线索

V2-025740 | MATHDB-345141

Fu's power series conjecture for invariant valuations on complex space forms

摘录 / EXCERPT: Let $b\mathbb{CP}^n_\lambdad$ be the complex space form of constant holomorphic curvature 4λ, with invariant valuation algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-025741 | MATHDB-345145

Gaiotto-Moore-Neitzke's exponential decay conjecture for parabolic $SL(2,\mathbb{C})$-Higgs bundles

摘录 / EXCERPT: Fix a Higgs bundle $({\overline{\partial}}_E,\varphi)$ in the regular locus ${\mathcal M}'_{\mathrm{Higgs}}$, together with a deformation $(\dot{\eta},\dot{\varphi})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-025742 | MATHDB-345153

The non-Stein characterization for critical exponent two

摘录 / EXCERPT: Let $\Gamma<\operatorname{Aut}(\mathbf B^n)$ be a convex-cocompact, torsion-free subgroup, and let $M_\Gamma=\mathbf B^n/\Gamma$. Suppose that $\delta(\Gamma)=2$. The non-Stein cha...

conjecture | open | MathDB 页面 | 原文线索

V2-025743 | MATHDB-345178

Boileau-Boyer-Cebanu-Walsh rigid cusp conjecture

摘录 / EXCERPT: Rigid Cusp Conjecture. If a hyperbolic knot complement $\mathbb{S}^3\setminus K$ covers an orbifold with a rigid cusp, then $\mathbb{S}^3\setminus K$ covers an orbifold with an...

conjecture | open | MathDB 页面 | 原文线索

V2-025744 | MATHDB-345192

The conifold gap conjecture for local \(\mathbb{P}^2\) relative to an elliptic curve

摘录 / EXCERPT: Let t_{con} be the flat coordinate near the conifold point, and let $F_{g,\mathrm{con}}^{\mathbb{P}^2\!/\!E}$ denote the genus-g free energy expressed as a function of...

conjecture | open | MathDB 页面 | 原文线索

V2-025745 | MATHDB-345193

Marché's integral structure conjecture for skein modules

摘录 / EXCERPT: Let M be a closed compact oriented 3-manifold. For $n\in\mathbb{Z}$, define $\lbrace n\rbrace=A^n-A^{-n}$. Marché's integral structure conjecture. There exist an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-025746 | MATHDB-345201

The sharp Morrey embedding conjecture for nested fractals

摘录 / EXCERPT: Morrey embedding conjecture. On nested fractals, $\delta_{\mathcal{E}}=1$ and, for every $p>1$, there exists $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-025747 | MATHDB-345202

The sharp Morrey embedding conjecture for Sierpinski carpet

摘录 / EXCERPT: Morrey embedding conjecture. For the Sierpinski carpet,

conjecture | open | MathDB 页面 | 原文线索

V2-025748 | MATHDB-345219

LI conjecture on exact symplectic central vanishing order

摘录 / EXCERPT: Let $L/\mathbb Q$ be a number-field extension and let $L_0/\mathbb Q$ be its Galois closure. Let $W(\chi)$ be the root number of the Artin L-function attached to χ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025749 | MATHDB-345234

The sharp Gromov-Hausdorff distance conjecture for the circle

摘录 / EXCERPT: Circle distance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025750 | MATHDB-345236

The four minimal 2-spheres conjecture for Riemannian 3-spheres

摘录 / EXCERPT: Let (M^3,g) be a Riemannian manifold diffeomorphic to $\bbS^3$. Four minimal 2-spheres conjecture. M^3 contains at least 4 embedded minimal 2-spheres. This is a higher-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-025751 | MATHDB-345237

Torsion characterization conjecture for the Kauffman bracket skein module

摘录 / EXCERPT: Let M be a compact oriented 3-manifold. Write $\mathcal{S}_{2,\infty}(M)$ for its Kauffman bracket skein module. An incompressible surface is understood in the usual sense, and...

conjecture | open | MathDB 页面 | 原文线索

V2-025752 | MATHDB-345238

Witten's finite-dimensionality conjecture for Kauffman bracket skein modules

摘录 / EXCERPT: Let M be a closed oriented 3-manifold, and let $\mathcal{S}(M;\mathbb{C}(A))$ denote its Kauffman bracket skein module over $\mathbb{C}(A)$, the field of rational functions in...

conjecture | open | MathDB 页面 | 原文线索

V2-025753 | MATHDB-345239

Finite-rank conjecture for atoroidal 3-manifold skein modules

摘录 / EXCERPT: Let M be an oriented atoroidal 3-manifold, and let $\mathcal{S}_{2,\infty}(M)$ denote its Kauffman bracket skein module. Finite-rank conjecture. The Kauffman bracket skein modu...

conjecture | open | MathDB 页面 | 原文线索

V2-025754 | MATHDB-345241

Marklof's gap conjecture for arithmetic hyperbolic 3-orbifolds

摘录 / EXCERPT: Let $\Gamma_\mathfrak{D}$ be an arithmetic quaternion group. The complex length spectrum of $\mathbb{H}^3/\Gamma_\mathfrak{D}$ has gaps, and let $\mathcal{G}(x)$ denote the number...

conjecture | open | MathDB 页面 | 原文线索

V2-025755 | MATHDB-345281

Blickle-Mustaţă-Smith equality conjecture for F-pure thresholds over an algebraic closure

摘录 / EXCERPT: Blickle-Mustaţă-Smith's equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025756 | MATHDB-345309

The conjecture on fillings of terminal quotient singularity links

摘录 / EXCERPT: Let G be a finite nontrivial subgroup of $U(n)$ such that the isolated quotient singularity ${\mathbb C}^n/G$ is terminal, and let its link be the corresponding contact manifold....

conjecture | open | MathDB 页面 | 原文线索

V2-025757 | MATHDB-345322

Projectivity conjecture for compact Kähler manifolds with a rational pseudoeffective dual class

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n. Let $\mathrm{Psef}(X)$ be the pseudoeffective cone in $H^{1,1}(X,\mathbf{R})$, and let $\mathrm{Int}(\mathrm{Psef}(X)^\vee)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025758 | MATHDB-345330

Positivity and convex-hull conjecture for converted matrix weighted rational Bézier curves

摘录 / EXCERPT: Positivity and convex-hull conjecture. The real weights are positive, and the convex hull of the original matrix weighted rational Bézier curve can be computed as the convex hull o...

conjecture | open | MathDB 页面 | 原文线索

V2-025759 | MATHDB-345342

Minimum-weight codeword conjecture for degree-4 maximal arcs

摘录 / EXCERPT: Let D be a design associated with a maximal arc of degree 4 in $PG(2,2^m)$, where $m\ge 2$, and let A be its incidence matrix. Let C be the binary code spanned by the rows...

conjecture | open | MathDB 页面 | 原文线索

V2-025760 | MATHDB-345368

Pitts-Rubinstein conjecture on strongly irreducible Heegaard surfaces

摘录 / EXCERPT: Let a strongly irreducible Heegaard surface lie in a Riemannian three-manifold. Pitts-Rubinstein conjecture. It is either isotopic to a minimal surface of index at most 1, or iso...

conjecture | open | MathDB 页面 | 原文线索

V2-025761 | MATHDB-345397

Coadjoint-orbit conjecture for moduli of weighted prequantizable Lagrangian embeddings

摘录 / EXCERPT: Coadjoint-orbit conjecture. The moduli space of weighted prequantizable Lagrangian embeddings is a symplectic manifold whose connected components are related to the coadjoint orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-025762 | MATHDB-345425

Poisson Noether's problem for finite symplectic reflection groups

摘录 / EXCERPT: Let Γ be a finite symplectic reflection group, and consider the Poisson-Weyl algebra associated with its natural symplectic representation. Poisson Noether conjecture. Poiss...

conjecture | open | MathDB 页面 | 原文线索

V2-025763 | MATHDB-345435

The mixed-volume Minkowski-sum volume conjecture

摘录 / EXCERPT: Let d be a positive integer, let m be a real number, and let $K_1,\ldots,K_d$ be convex bodies in $\mathds{R}^d$ satisfying ... For an integer ℓ with $1\leq\ell\leq d$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-025764 | MATHDB-345437

Abernethy et al.'s gauge-set characterization of strongly convex bodies

摘录 / EXCERPT: Abernethy et al.'s conjecture. K is a gauge set with respect to its gauge $\|\cdot\|_K$ if and only if it is strongly convex with respect to K itself.

conjecture | open | MathDB 页面 | 原文线索

V2-025765 | MATHDB-345458

Genus-zero bridge trisection conjecture for properly embedded surfaces

摘录 / EXCERPT: Let S be a properly embedded smooth surface in B^4. Genus-zero bridge trisection conjecture. Every properly embedded smooth surface S in B^4 has a genus 0 bridge trisecti...

conjecture | open | MathDB 页面 | 原文线索

V2-025766 | MATHDB-345475

The noncommutative Hodge conjecture for saturated dg-categories

摘录 / EXCERPT: Let $\operatorname{\mathcal{A}}$ be a saturated dg-category over $\operatorname{\mathbb{C}}$, and let $\operatorname{Hdg}(\operatorname{\mathcal{A}})_{\operatorname{\mathbb{Q}}}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-025767 | MATHDB-345531

Positive tropical Grassmannian fan conjecture for amplituhedron subdivisions

摘录 / EXCERPT: Let $\operatorname{Trop}^+Gr_{k+1,n}$ be the positive tropical Grassmannian, whose cones carry their natural partial order. Positive tropical Grassmannian fan conjecture. The regul...

conjecture | open | MathDB 页面 | 原文线索

V2-025768 | MATHDB-345567

Collapsed Ricci limit spaces are non-collapsed RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf{d},x,\mathfrak{m})$ be a collapsed Ricci limit space of Hausdorff dimension k, with its limit measure satisfying ... for some $b,k\in(0,\infty)$. Collapsed Ricci li...

conjecture | open | MathDB 页面 | 原文线索

V2-025769 | MATHDB-345568

Cheeger--Colding volume-ratio conjecture for collapsed Ricci limit spaces

摘录 / EXCERPT: Let (X,d) be the metric limit under consideration, let $k=\dim_{\mathcal{H}}(X,\mathsf{d})$, and let $\mathcal{H}^k$ denote its k-dimensional Hausdorff measure. For...

conjecture | open | MathDB 页面 | 原文线索

V2-025770 | MATHDB-345569

De Philippis--Gigli characterization of non-collapsed RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf{d},\mathfrak{m})$ be an $\operatorname{RCD}(K,N)$ space, and let $\dim(X,\mathsf{d},\mathfrak{m})$ be its essential dimension. De Philippis--Gigli conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-025771 | MATHDB-345570

RCD normalization conjecture for Hausdorff measure

摘录 / EXCERPT: RCD normalization conjecture. Then $(X,\mathsf{d},\mathcal{H}^k)$ is a non-collapsed $\operatorname{RCD}(K,k)$ space. This is the RCD formulation of the paper's first conjecture an...

conjecture | open | MathDB 页面 | 原文线索

V2-025772 | MATHDB-345571

Gigli's Laplacian-Hessian trace characterization conjecture

摘录 / EXCERPT: Let $(X,\mathsf{d},\mathfrak{m})$ be an $\operatorname{RCD}(K,N)$ space. For $f\in D(\Delta)$, let Hess_f be its Hessian, viewed as a $(0,2)$-type L^2 tensor, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-025773 | MATHDB-345576

The quadratic-form conjecture for the $n-2$ Hessian equation

摘录 / EXCERPT: The quadratic-form conjecture. For every $n4$-dimensional vector $\xi=(\xi_1,\ldots,\xi_n)\in\mathbb{R}^n4$, the quadratic form

conjecture | open | MathDB 页面 | 原文线索

V2-025774 | MATHDB-345594

Tomanov's crystallographic actions conjecture for real algebraic homogeneous spaces

摘录 / EXCERPT: Let G be a real algebraic group, let $H\subgroup G$ contain a maximal reductive subgroup of G, and let $\Gamma\subgroup G$. Say that Γ acts crystallographically on G/H...

conjecture | open | MathDB 页面 | 原文线索

V2-025775 | MATHDB-345604

The converse conjecture for the weak Erdős-Ginzburg-Ziv constant and hollow polytopes

摘录 / EXCERPT: A polytope $P\subset\mathbb Q^d$ is the convex hull of a finite set of points in $\mathbb Q^d$. A lattice $\Lambda\subset\mathbb Q^d$ is a discrete subset that is an affine image o...

conjecture | open | MathDB 页面 | 原文线索

V2-025776 | MATHDB-345618

Thomas's existence and stability conjecture for special Lagrangians

摘录 / EXCERPT: Let (X,ω,Ω) be a Calabi--Yau manifold and let L be a compact zero Maslov Lagrangian in it. A Lagrangian is stable when it satisfies the stability condition under disc...

conjecture | open | MathDB 页面 | 原文线索

V2-025777 | MATHDB-345655

Log-Minkowski conjecture for origin-symmetric convex bodies

摘录 / EXCERPT: Let K and L be origin-symmetric convex bodies in $\mathbb{R}^n$, with $n\geq2$, and let h_K,h_L be their support functions, $V(K)$ their volumes, and dV_K the cone-volume m...

conjecture | open | MathDB 页面 | 原文线索

V2-025778 | MATHDB-345657

Perelman's compact ancient solution conjecture on the three-sphere

摘录 / EXCERPT: Perelman's compact ancient solution conjecture. Then $g(t)$ is either a family of shrinking spheres or Perelman's ancient solution.

conjecture | open | MathDB 页面 | 原文线索

V2-025779 | MATHDB-345694

The analytic contraction conjecture for nef and big classes on Calabi-Yau manifolds

摘录 / EXCERPT: Analytic contraction conjecture. There is a surjective holomorphic and bimeromorphic map $\pi:X\to Y$ with connected fibers onto a normal compact Kähler analytic space Y such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-025780 | MATHDB-345695

The bounded-potential conjecture for nef classes on Calabi-Yau manifolds

摘录 / EXCERPT: Let X^n be a Calabi-Yau manifold and let α be a closed real $(1,1)$-form such that $[\alpha]\in\partial\mathcal{C}_X$ is nef. Bounded-potential conjecture. There is a boun...

conjecture | open | MathDB 页面 | 原文线索

V2-025781 | MATHDB-345696

The Gromov-Hausdorff limit conjecture for collapsing Ricci-flat metrics in Setup I.A

摘录 / EXCERPT: In Setup I.A, let (X,ω_t) be the corresponding family of Ricci-flat manifolds and let Z denote its Gromov-Hausdorff limit. Gromov-Hausdorff limit conjecture. The limit Z...

conjecture | open | MathDB 页面 | 原文线索

V2-025782 | MATHDB-345697

The smooth convergence conjecture for collapsing Ricci-flat metrics

摘录 / EXCERPT: In Setup I.B.1, let $f:X\to Y$ be the fiber space and let S be the singular-fiber locus. Smooth convergence conjecture. The Ricci-flat Kähler metrics satisfy ... in...

open_problem | open | MathDB 页面 | 原文线索

V2-025783 | MATHDB-345698

The compactification conjecture for collapsing Ricci-flat metrics

摘录 / EXCERPT: Compactification conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-025784 | MATHDB-345699

The point-limit conjecture for collapsing Ricci-flat metrics on K3 surfaces

摘录 / EXCERPT: Let X be a $K3$ surface and let $[\alpha]\in\partial\mathcal{C}_X$ be as in Setup I.B.2. Point-limit conjecture. The Ricci-flat manifolds (X,ω_t) converge to a point in th...

conjecture | open | MathDB 页面 | 原文线索

V2-025785 | MATHDB-345702

The logarithmic diameter growth conjecture for Calabi-Yau degenerations

摘录 / EXCERPT: Let $\mathfrak{X}$ be a Calabi-Yau family as in Setup II that does not satisfy the equivalent conditions in Theorem. Logarithmic diameter growth conjecture. There exist constants...

open_problem | open | MathDB 页面 | 原文线索

V2-025786 | MATHDB-345707

Singer's L^2-Betti-number conjecture

摘录 / EXCERPT: Let M^n be a closed aspherical manifold. Singer's conjecture. The rational L^2-Betti numbers of its fundamental group vanish outside the middle dimension: ... This is a basic van...

conjecture | open | MathDB 页面 | 原文线索

V2-025787 | MATHDB-345712

Higher-dimensional Clifford hypersurface focal-radius conjecture

摘录 / EXCERPT: Let (M^n,g) be a n-sphere with $\operatorname{sec}(g)\geq 1$, where $n=p+q+1$ for $p,q\in\mathbf N_+$. Suppose that $i:\Sigma\approx\mathbf S^p\times\mathbf S^q\to M$ is an imme...

conjecture | open | MathDB 页面 | 原文线索

V2-025788 | MATHDB-345713

Clifford-band width conjecture

摘录 / EXCERPT: Let $M_{p,q}=\mathbf S^p\times\mathbf S^q\times[-1,1]$ be a Clifford band, and let g be a smooth metric on $M_{p,q}$. Clifford-band width conjecture. If ... then ... This is a propos...

conjecture | open | MathDB 页面 | 原文线索

V2-025789 | MATHDB-345763

Steinerberger's Vaserstein distance conjecture for Laplace eigenfunctions

摘录 / EXCERPT: Let (M,g) be a smooth, compact Riemannian manifold without boundary. Let ϕ be an L^2-normalised eigenfunction of the Laplacian with eigenvalue L, satisfying ... on (M,g)...

conjecture | open | MathDB 页面 | 原文线索

V2-025790 | MATHDB-345768

Stolz conjecture for the localized spin bordism group

摘录 / EXCERPT: Let (L,f,h) represent an element of $R^{\textup{spin}}_n(B\Gamma)$, where L is an n-dimensional spin manifold with boundary carrying a positive-scalar-curvature metric h, t...

conjecture | open | MathDB 页面 | 原文线索

V2-025791 | MATHDB-345769

Generalized Stolz conjecture for proper equivariant spin bordism

摘录 / EXCERPT: Generalized Stolz conjecture. The index map Θ is an isomorphism. This is the proper-action analogue of the Stolz conjecture, replacing $B\Gamma$ by the universal proper...

conjecture | open | MathDB 页面 | 原文线索

V2-025792 | MATHDB-345779

Eriksson-Freixas i Montplet-Mourougane's birational invariance conjecture for the BCOV invariant

摘录 / EXCERPT: Eriksson-Freixas i Montplet-Mourougane's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025793 | MATHDB-345783

Faber-Krahn conjecture for the auxiliary operator H_E

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^2$ be a C^∞ simply connected domain, and let $\mu^\Omega(E)$ be the first eigenvalue of the auxiliary operator H_E^Ω. Let $\mathbb{D}$...

conjecture | open | MathDB 页面 | 原文线索

V2-025794 | MATHDB-345821

The (p,q)-conjecture for log-concave measures

摘录 / EXCERPT: Let $p\in[0,1]$, let μ be an even log-concave measure on $\mathbb R^n$, let K,L be nonempty symmetric convex sets, and let $\lambda\in[0,1]$. For $p>0$, define ... with $p=0$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-025795 | MATHDB-345845

The conjecture that the ud-category is an infinity category

摘录 / EXCERPT: Let L and X be the objects for which the ud-category $\mathscr{UD}(L,X)$ is defined. Ud-category infinity-categorification conjecture. The category $\mathscr{UD}(L,X)$ can be r...

conjecture | open | MathDB 页面 | 原文线索

V2-025796 | MATHDB-345847

RCD gluing conjecture for noncollapsed spaces with boundary

摘录 / EXCERPT: RCD gluing conjecture. The glued metric measure space

conjecture | open | MathDB 页面 | 原文线索

V2-025797 | MATHDB-345853

Multi-plank subadditivity conjecture for relative inradii

摘录 / EXCERPT: Let B be an open bounded convex set containing the origin, let K be a convex set, and let $C_1,\ldots,C_N$ be convex sets whose union covers K. For $1\leq k\leq n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-025798 | MATHDB-345865

Uniform potential convergence conjecture for boundary classes on K3 surfaces

摘录 / EXCERPT: Let X be a $K3$ surface, let α be a closed real $(1,1)$-form with $0\neq[\alpha]\in\partial\mathcal{C}_X$ and $\int_X\alpha^2=0$, and let ω be a Kähler metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-025799 | MATHDB-345872

Boros-Füredi-Kelly conjecture on skew lines in four-dimensional space

摘录 / EXCERPT: Let Σ be a finite family of pairwise skew lines in $\mathbb{R}^4$. Suppose that the three-dimensional affine span (hull) containing any two of the lines contains at least tw...

conjecture | open | MathDB 页面 | 原文线索

V2-025800 | MATHDB-345887

Uniform normalization conjecture for fiberwise plurisubharmonic functions

摘录 / EXCERPT: Let $f:X\to Z$ be as in the setting, with fibers X_t, forms θ_t and ω_t, and let V denote the common volume appearing in the fiberwise averages. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-025801 | MATHDB-345890

Hirachi's decomposition conjecture for pseudohermitian scalar invariants

摘录 / EXCERPT: Let I be a natural pseudohermitian scalar invariant whose total integral is a secondary CR invariant. Here, a secondary CR invariant is a total integral invariant under changes o...

conjecture | open | MathDB 页面 | 原文线索

V2-025802 | MATHDB-345910

The inertia-group conjecture for 4-dimensional complete intersections

摘录 / EXCERPT: Let $X_4(\underline d)$ be a 4-dimensional complete intersection, let $v_2(X_4(\underline d))$ and $v_4(X_4(\underline d))$ denote its Wu classes, regarded as elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-025803 | MATHDB-345921

The Gluing Conjecture for multiple Alexandrov spaces

摘录 / EXCERPT: Let $X\in\operatorname{Alex}_{\amalg}^n(\kappa)$ satisfy the paper's condition, let Y be a connected length metric space, and let $f\colon X\to Y$ be a 1-Lipschitz onto map. Glui...

conjecture | open | MathDB 页面 | 原文线索

V2-025804 | MATHDB-345922

The globalization conjecture for Alexandrov spaces

摘录 / EXCERPT: Let Y be an n-dimensional length metric space. For a subset $\mathcal{U}\subseteq Y$, set $\mathcal{A}=Y\setminus\mathcal{U}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-025805 | MATHDB-345934

The zero-defect criterion for geodesically harmonic metrics

摘录 / EXCERPT: Let (M,g) be a connected compact Riemannian manifold, and define $\mathcal D(g)$ as the infimum of the defect $\delta(g^\bullet)$ over Riemannian metrics $g^\bullet$ on SM of v...

conjecture | open | MathDB 页面 | 原文线索

V2-025806 | MATHDB-345935

The Lyapunov volume conjecture

摘录 / EXCERPT: Lyapunov volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025807 | MATHDB-345936

The geodesic Swiss cheese existence conjecture

摘录 / EXCERPT: Let (N,g) be a compact Riemannian manifold. A geodesic Swiss cheese model is the complement $M_{\mathcal B}=N\setminus\bigl(\bigcup_\alpha\operatorname{int}(B_\alpha)\bigr)$ of a...

conjecture | open | MathDB 页面 | 原文线索

V2-025808 | MATHDB-345952

Derived equivalence conjecture for the Lehn-Lehn-Sorger-van Straten eightfold

摘录 / EXCERPT: Let Y be a cubic fourfold, let Z be its associated symplectic eightfold, and suppose that $Y\in\mathcal C_d$ for some d satisfying the relevant condition. Then Z is isomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-025809 | MATHDB-345974

Nonattainment conjecture for the planar fixed-perimeter truncated-Laplacian problem

摘录 / EXCERPT: Fixed-perimeter nonattainment conjecture. The minimization problem

conjecture | open | MathDB 页面 | 原文线索

V2-025810 | MATHDB-345975

Existence conjecture for the planar fixed-diameter truncated-Laplacian problem

摘录 / EXCERPT: Fixed-diameter existence conjecture. The minimization problem

conjecture | open | MathDB 页面 | 原文线索

V2-025811 | MATHDB-345976

Reuleaux-triangle minimizer conjecture for the planar fixed-diameter problem

摘录 / EXCERPT: Reuleaux-triangle minimizer conjecture. The Reuleaux triangle is a minimizer for the fixed-diameter problem.

conjecture | open | MathDB 页面 | 原文线索

V2-025812 | MATHDB-346003

Uniform boundedness conjecture for log-pluricanonical maps

摘录 / EXCERPT: Uniform boundedness conjecture. There is a positive integer $N=N(n,I)$ depending only on n and I such that the linear system

conjecture | open | MathDB 页面 | 原文线索

V2-025813 | MATHDB-346012

Coarsest Dressian structure conjecture for tropical Grassmannians

摘录 / EXCERPT: Let $\operatorname{TGr}_0(k,n)$ denote the characteristic-zero tropical Grassmannian, let $\operatorname{Dr}(k,n)$ denote the Dressian, and let $\mathcal I_{n,k}$ be the Plücker id...

conjecture | open | MathDB 页面 | 原文线索

V2-025814 | MATHDB-346013

Speyer-Williams cluster-complex conjecture for positive tropicalizations

摘录 / EXCERPT: Let $\mathcal S_{k,n}$ denote the possibly infinite cluster complex of the Grassmannian $\operatorname{Gr}(k,n)$. Let $\operatorname{TGr}_0^+(k,n)$ denote its positive tropicalizat...

conjecture | open | MathDB 页面 | 原文线索

V2-025815 | MATHDB-346077

Topping's conjecture for closed immersed surfaces

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^3$ be an immersed connected closed surface. Topping's conjecture. ... This conjecture asks for the optimal lower bound for the total absolute mean curvat...

conjecture | open | MathDB 页面 | 原文线索

V2-025816 | MATHDB-346091

The GM series surgery conjecture for links

摘录 / EXCERPT: The link surgery conjecture. Whenever the right-hand side makes sense, for some $\epsilon\in\{\pm1\}$ and $d\in\mathbb{Q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-025817 | MATHDB-346126

Sticky-geodesic conjecture for Teichmüller geodesics with curve endpoints

摘录 / EXCERPT: Let γ be a Teichmüller geodesic with endpoints γ^- and γ^+ in the Thurston boundary, and call it sticky when it satisfies the sticky-geodesic condition that ge...

conjecture | open | MathDB 页面 | 原文线索

V2-025818 | MATHDB-346151

Global Nakano positivity conjecture for direct images with multiplier ideals

摘录 / EXCERPT: Let $f\colon X\to Y$ be a projective surjective morphism between complex manifolds. Let (L,h) be a holomorphic line bundle over X equipped with a singular Hermitian metric of s...

conjecture | open | MathDB 页面 | 原文线索

V2-025819 | MATHDB-346184

Existence and uniqueness of associative submanifolds with three K3 ends

摘录 / EXCERPT: Let X be a K3 manifold, let $\alpha_1,\alpha_2,\alpha_3$ be -2 classes on X satisfying ... and let $H\subset H^2(X)$ be a maximal positive subspace corresponding to a hyperkähl...

conjecture | open | MathDB 页面 | 原文线索

V2-025820 | MATHDB-346195

Finite generation and bounded exponent conjecture for local class groups of dlt singularities

摘录 / EXCERPT: Finite generation and bounded exponent conjecture. The quotient of the class group by its p-primary torsion is finitely generated, and the p-primary torsion has bounded exponen...

conjecture | open | MathDB 页面 | 原文线索

V2-025821 | MATHDB-346205

Additivity conjecture for the regular genus under connected sum

摘录 / EXCERPT: Let M^n_1 and M^n_2 be two closed orientable n-manifolds. The regular genus additivity conjecture. ... Regular genus is known to be subadditive under connected sum, and the equ...

conjecture | open | MathDB 页面 | 原文线索

V2-025822 | MATHDB-346229

Complex Hopf conjecture for compact Kähler manifolds

摘录 / EXCERPT: Let (M,ω) be an n-dimensional compact Kähler manifold with nonpositive holomorphic bisectional curvature, and suppose its Ricci curvature is quasi-negative, meaning nonpos...

conjecture | open | MathDB 页面 | 原文线索

V2-025823 | MATHDB-346249

Dolgachev-Weisfeiler conjecture over affine space

摘录 / EXCERPT: Let $X\to \mathbb{A}^m$ be an $\mathbb{A}^n$-fibration, meaning that every fibre is isomorphic to $\mathbb{A}^n$. Dolgachev-Weisfeiler conjecture. The fibration is isomorphic to th...

conjecture | open | MathDB 页面 | 原文线索

V2-025824 | MATHDB-346260

Stable Harbourne-Huneke containment conjecture

摘录 / EXCERPT: Let $\mathbbm k$ be a field, let $\mathbb P^N_{\mathbbm k}$ be projective space, and let $I\subseteq\mathbbm k[\mathbb P^N_{\mathbbm k}]$ be a homogeneous radical ideal of big heig...

conjecture | open | MathDB 页面 | 原文线索

V2-025825 | MATHDB-346264

Positivity conjecture for the Hull-Strominger moduli-space metric

摘录 / EXCERPT: Let (X,P) be a Calabi-Yau threefold with bundle P admitting a solution of the Hull-Strominger system. Let $\dot{\mathfrak{a}}\in H^{1,1}_A(X)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025826 | MATHDB-346265

Kähler-potential conjecture for the Hull-Strominger moduli space

摘录 / EXCERPT: Kähler-potential conjecture. The function K defines the Kähler potential for a Kähler metric on this moduli space.

conjecture | open | MathDB 页面 | 原文线索

V2-025827 | MATHDB-346277

The projective compactification conjecture for canonical moduli-part period maps

摘录 / EXCERPT: Let $W_0\subset W$ be the base locus of the relevant smooth family, let $p^c\colon W_0\to D^c/\Gamma$ be the period map of the canonical moduli part, and set ... Let $\mathcal M^c_W$...

conjecture | open | MathDB 页面 | 原文线索

V2-025828 | MATHDB-346286

Weighted uniform K-stability conjecture for the weighted Abreu equation

摘录 / EXCERPT: Let $\triangle$ be a polytope, let A be a smooth function on $\triangle$ satisfying the moment condition, and let f be a positive affine function. The triple $(\triangle,A,f)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025829 | MATHDB-346287

Mori-Reid-Tian conjecture on disjoint positive-genus curves on complex surfaces

摘录 / EXCERPT: Let X be a smooth complex surface with $b_1(X)=0$ and $b_2(X)>1$. Suppose that D_i are smooth, pairwise disjoint complex curves on X, each of positive genus $g_i>0$, and that...

conjecture | open | MathDB 页面 | 原文线索

V2-025830 | MATHDB-346289

Reflexive polygon conjecture for infinite staircases in rational convex toric domains

摘录 / EXCERPT: Reflexive polygon conjecture. If the ellipsoid embedding function of a rational convex toric domain has an infinite staircase, then its moment polygon is a scaling of a reflexive p...

conjecture | open | MathDB 页面 | 原文线索

V2-025831 | MATHDB-346291

The maximum-scattered-subspace count conjecture for the family U_b

摘录 / EXCERPT: The maximum-scattered-subspace count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025832 | MATHDB-346294

Euclidean and geodesic distance propagation conjecture

摘录 / EXCERPT: Euclidean and geodesic distance propagation conjecture. There exists such an obstacle K and datum (J,f) for which the limiting solution u_∞ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-025833 | MATHDB-346299

The reflexive polygon conjecture for rational convex toric domains

摘录 / EXCERPT: Reflexive polygon conjecture. The embedding function $c_\Omega(a)$ has infinitely many singular points only if some scaling of Ω is reflexive.

conjecture | open | MathDB 页面 | 原文线索

V2-025834 | MATHDB-346300

Attainment conjecture for the conformal Steklov invariant

摘录 / EXCERPT: Let Σ be a compact surface, and let $I^\sigma_k(\Sigma)$ denote the infimum of the normalized k-th Steklov eigenvalue over conformal classes on Σ. Attainment conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-025835 | MATHDB-346301

Strict conformal Steklov rigidity conjecture for non-discs

摘录 / EXCERPT: Let Σ be a compact surface, let c be a conformal class on Σ, and let $\sigma^*_1(\Sigma,c)$ denote the normalized first Steklov eigenvalue in that conformal class....

conjecture | open | MathDB 页面 | 原文线索

V2-025836 | MATHDB-346306

Exact overlap conjecture for self-similar sets on the real line

摘录 / EXCERPT: Exact overlap conjecture. If there is a dimension drop for Φ, then Φ has exact overlaps.

conjecture | open | MathDB 页面 | 原文线索

V2-025837 | MATHDB-346337

The unranked graphic order polytope conjecture

摘录 / EXCERPT: Unranked graphic order polytope conjecture. The order polytope $\operatorname{\mathcal{O}}(\operatorname{\mathscr{P}})$ is graphic.

conjecture | open | MathDB 页面 | 原文线索

V2-025838 | MATHDB-346360

Stable inversion configurations for odd numbers of weighted rays

摘录 / EXCERPT: Let $m=2k+1$, let $c_1,\ldots,c_m$ be specified weights, let $\gamma_1,\ldots,\gamma_m$ denote ray directions, and let $\mathcal{S}$ be the corresponding star transform. A configur...

conjecture | open | MathDB 页面 | 原文线索

V2-025839 | MATHDB-346372

Packing-minima transference conjecture

摘录 / EXCERPT: Packing-minima transference conjecture. For $1\leq j\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-025840 | MATHDB-346373

Covering product conjecture

摘录 / EXCERPT: Covering product conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025841 | MATHDB-346375

Hu's surjectivity conjecture for the symplectic Schur-Weyl map

摘录 / EXCERPT: Hu's surjectivity conjecture. The map $\psi_{f,K}$ is surjective. This conjecture asks for the symplectic double-centralizer property on the quotient by the f-th Brauer ideal; it...

conjecture | open | MathDB 页面 | 原文线索

V2-025842 | MATHDB-346376

Hu's generalized surjectivity conjecture for symplectic Schur-Weyl duality

摘录 / EXCERPT: Hu's generalized surjectivity conjecture. The natural map $\psi'_{f,K}$ is surjective. The paper states that this conjecture implies the preceding algebraically closed-field conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-025843 | MATHDB-346380

The non-domability conjecture for the isosceles triangle with sides $(2,2,1)$

摘录 / EXCERPT: Let Δ be an isosceles triangle with side lengths $(2,2,1)$. The $(2,2,1)$ triangle conjecture. The triangle Δ cannot be domed. This conjecture is a central obstructio...

conjecture | open | MathDB 页面 | 原文线索

V2-025844 | MATHDB-346381

The rigidity conjecture for closed domes

摘录 / EXCERPT: Let $S\subset\mathbb R^3$ be a PL-surface homeomorphic to a sphere whose faces are unit triangles. A closed dome S is rigid if it has no continuous family of intrinsically isomet...

conjecture | open | MathDB 页面 | 原文线索

V2-025845 | MATHDB-346382

The countability conjecture for domable planar unit rhombi

摘录 / EXCERPT: Let $\mathcal A$ be the set of all $a\geq 0$ such that the planar rhombus $\rho(a,\sqrt{4-a^2})$ can be domed. Countability conjecture for $\mathcal A$. The set $\mathcal A$ is cou...

conjecture | open | MathDB 页面 | 原文线索

V2-025846 | MATHDB-346383

The non-domability conjecture for the planar rhombus $\rho(1/2,\sqrt{15}/2)$

摘录 / EXCERPT: Let $\mathcal A$ be the set of $a\geq 0$ for which the planar rhombus $\rho(a,\sqrt{4-a^2})$ can be domed, and write $\rho_\lozenge:=\rho(1/2,\sqrt{15}/2)$. The $\rho_\lozenge$ con...

conjecture | open | MathDB 页面 | 原文线索

V2-025847 | MATHDB-346385

The non-domability conjecture for two unit triangles

摘录 / EXCERPT: Let $\Delta_1,\Delta_2\subset\mathbb R^3$ be unit triangles, and let $\Upsilon=\Delta_1\cup\Delta_2$. Two-triangle non-domability conjecture. There exist unit triangles Δ_1...

conjecture | open | MathDB 页面 | 原文线索

V2-025848 | MATHDB-346386

The rhombus reduction conjecture for integral curves

摘录 / EXCERPT: Let $\mathcal M_n$ denote the space of integral curves with n edges, and let $\mathcal M_4$ denote the corresponding space of unit rhombi. Rhombus reduction conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-025849 | MATHDB-346387

The finite-rhombi conjecture for integral curves

摘录 / EXCERPT: Let $\mathcal M_n$ denote the space of integral curves with n edges, and let $\mathcal M_4$ denote the corresponding space of unit rhombi. Finite-rhombi conjecture. For every int...

conjecture | open | MathDB 页面 | 原文线索

V2-025850 | MATHDB-346412

Index-difference conjecture for psc-metrics on spin manifolds with fb-singularities

摘录 / EXCERPT: Let X_{Σ} be a spin manifold with a $(\eta\text{-}\mathrm{fb})$-singularity of dimension $n\geq 9$. Assume that $\beta X\neq\varnothing$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-025851 | MATHDB-346433

Petty's conjecture on equilateral sets in normed spaces

摘录 / EXCERPT: Petty's conjecture. For all normed spaces X of dimension n,

conjecture | open | MathDB 页面 | 原文线索

V2-025852 | MATHDB-346441

Lutwak's Alexandrov-Fenchel conjecture for affine quermassintegrals

摘录 / EXCERPT: For a convex body $K\subset\mathbb{R}^n$, define ... where B_K is the Euclidean ball with the same volume as K. Lutwak's Alexandrov-Fenchel conjecture. For all $p\in[-n,0]$, ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-025853 | MATHDB-346444

The SYZ-Gross-Siebert affine structure conjecture for Calabi-Yau toric degenerations

摘录 / EXCERPT: SYZ-Gross-Siebert affine structure conjecture. The limit of the complex affine structures of the special Lagrangian fibration coincides with the affine structures on the dual inter...

conjecture | open | MathDB 页面 | 原文线索

V2-025854 | MATHDB-346461

The $\widehat Z$-invariant as a Rozansky-Witten invariant

摘录 / EXCERPT: Let M_3 be a 3-manifold, let G be the relevant group, and let ... be the moduli space associated with the disk D^2, viewed conceptually as the cotangent bundle of the affine Gr...

conjecture | open | MathDB 页面 | 原文线索

V2-025855 | MATHDB-346466

The universal-invariant conjecture for Gieseker semistable sheaves on surfaces

摘录 / EXCERPT: Let X be a projective complex surface with geometric genus $p_g=0$, and let $\mathcal A=\operatorname{Coh}(X)$. For an ample line bundle on X, let τ denote the resulting G...

conjecture | open | MathDB 页面 | 原文线索

V2-025856 | MATHDB-346468

The universal-invariant conjecture for Donaldson theory with b-plus equal to one

摘录 / EXCERPT: Let X be a compact oriented 4-manifold with $b_+^2(X)=1$, without assuming that X is simply connected, and consider its Donaldson-theoretic moduli spaces and geometric data a...

conjecture | open | MathDB 页面 | 原文线索

V2-025857 | MATHDB-346478

The period-function monotonicity and endpoint-limit conjecture for G_α

摘录 / EXCERPT: Let $P(\beta)$ be the period function in G_α, with β the parameter used to specify the initial tangent vector. The period-function conjecture. ... and ... The source estab...

conjecture | open | MathDB 页面 | 原文线索

V2-025858 | MATHDB-346479

The midpoint-coordinate limit conjecture for G_α

摘录 / EXCERPT: For $\alpha\in(0,1]$, define ... Here $b_{x_0,\alpha}$ is the coordinate function and $P(x_0,\alpha)$ is the associated period. The midpoint-coordinate limit conjecture. $L(\alpha)$...

conjecture | open | MathDB 页面 | 原文线索

V2-025859 | MATHDB-346483

Mass-gap estimate for finite-dimensional sigma-model approximations

摘录 / EXCERPT: Let S^k be the target sphere of radius R, and let $\tilde{\mathcal{L}}_{N,U}(S^k)$ be the manifold with boundary obtained by removing a tubular neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-025860 | MATHDB-346501

The colorful interior characterization conjecture for compact convex sets

摘录 / EXCERPT: Let $S_1,\dots,S_n$ be a collection of n compact convex sets of $\mathbb{R}^{n-1}$. Write $\mathcal{S}=(S_i)_{i\in[n]}$, and let a family of sets be separated if no hyperplane in...

conjecture | open | MathDB 页面 | 原文线索

V2-025861 | MATHDB-346514

Modular lifting conjecture for Lagrangian 3-tori and 3-spheres

摘录 / EXCERPT: Let $\tilde X$ be the mirror quintic threefold, and let δ be a primitive class in $H_3(\tilde X,\mathbb{Z})$. Write $\operatorname{mod}_2(\delta)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-025862 | MATHDB-346515

The sliding-relations conjecture for the Kauffman bracket skein module of two handlebodies

摘录 / EXCERPT: Let H_n H_m denote the connected sum of handlebodies of genera n and m, let $\mathcal{S}_{2,\infty}(H_2)$ be the Kauffman bracket skein module of the genus-two handlebody,...

conjecture | open | MathDB 页面 | 原文线索

V2-025863 | MATHDB-346527

The projective Lichnerowicz conjecture for holomorphic pseudo-groups

摘录 / EXCERPT: Let M be a compact complex manifold bearing a holomorphic projective connection ϕ. Assume that M does not admit any global holomorphic torsionfree affine connection projec...

conjecture | open | MathDB 页面 | 原文线索

V2-025864 | MATHDB-346539

Uniqueness conjecture for monotone special Lagrangian torus fibres

摘录 / EXCERPT: Let ω_i be the sequence of symplectic forms considered in the paper, let B_0 be the base of the special Lagrangian fibration, and let $\mathcal{U}_i\subseteq B_0$ be a cha...

conjecture | open | MathDB 页面 | 原文线索

V2-025865 | MATHDB-346559

Strong Factorization Conjecture for compact complex manifolds

摘录 / EXCERPT: Strong Factorization Conjecture. Any bimeromorphism ψ between compact complex manifolds can be factorized as

conjecture | open | MathDB 页面 | 原文线索

V2-025866 | MATHDB-346561

Nakamura's boundedness conjecture for divisors computing minimal log discrepancies

摘录 / EXCERPT: Let $\Gamma\subseteq[0,1]$ be a finite set and $X\ni x$ a klt germ. An lc germ $(X\ni x,B)$ has boundary coefficients in Γ. Nakamura's conjecture. There exists an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-025867 | MATHDB-346563

Local stability conjecture for R-Cartier boundaries

摘录 / EXCERPT: Let $\Gamma\subseteq[0,1]$ be a finite set and let $X\ni x$ be a germ. Local stability conjecture. There exists a positive real number τ depending only on Γ and...

open_problem | open | MathDB 页面 | 原文线索

V2-025868 | MATHDB-346618

Trace asymptotics conjecture for geometric quantizations

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold, and let $\{T_k:C^\infty(M)\to\mathcal{L}(H_k)\}_{k\in\mathbb{N}}$ be a geometric quantization satisfying the trace axiom and star...

conjecture | open | MathDB 页面 | 原文线索

V2-025869 | MATHDB-346619

Classification conjecture for Ricci shrinkers with weakly PIC

摘录 / EXCERPT: Let (M^n,g,f) be a Ricci shrinker with weakly PIC. A Ricci shrinker with weakly PIC is expected to satisfy the following classification: Classification conjecture. (M,g) is iso...

conjecture | open | MathDB 页面 | 原文线索

V2-025870 | MATHDB-346620

Weakly PIC2 conjecture for complete ancient Ricci flows

摘录 / EXCERPT: Let $(M^n,g(t))_{t \in (-\infty,0]}$ be a complete ancient solution to the Ricci flow with weakly PIC, meaning the stated curvature condition from the source. Weakly PIC2 conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-025871 | MATHDB-346626

Conjecture on normalized symplectic and real symplectic capacities

摘录 / EXCERPT: Let D be a convex domain in $\mathbb{R}^{2n}$ invariant under a fixed linear anti-symplectic involution. Capacity coincidence conjecture. All normalized symplectic capacities and...

conjecture | open | MathDB 页面 | 原文线索

V2-025872 | MATHDB-346677

Equality of cut time and Maxwell time on the Cartan group

摘录 / EXCERPT: Cut-time equality conjecture. For any $\lambda \in C$

conjecture | open | MathDB 页面 | 原文线索

V2-025873 | MATHDB-346706

Smits's conjecture on Robin spectral-gap monotonicity for convex domains

摘录 / EXCERPT: Smits's conjecture. If Ω is a convex bounded domain, then $\Lambda^{\Omega}(\alpha)$ is an increasing function for $\alpha\geq0$. The conjecture concerns monotonicity of the...

conjecture | open | MathDB 页面 | 原文线索

V2-025874 | MATHDB-346715

Uniform index conjecture under Euclidean volume growth

摘录 / EXCERPT: Given n and $L\in(0,1]$, let M be an open n-manifold with $\operatorname{Ric}\ge 0$, and suppose that the Riemannian universal cover of M has Euclidean volume growth of con...

conjecture | open | MathDB 页面 | 原文线索

V2-025875 | MATHDB-346752

Cap covering inradius conjecture

摘录 / EXCERPT: Cap covering inradius conjecture. The sum of the inradii of all domains in the collection is at least α.

conjecture | open | MathDB 页面 | 原文线索

V2-025876 | MATHDB-346803

Conjecture on compact locally homogeneous simple pseudo-Riemannian Einstein manifolds

摘录 / EXCERPT: The conjecture. The only compact locally homogeneous simple pseudo-Riemannian Einstein manifolds are of the form $\Gamma\backslash G/H$, determined by either some standard triple...

conjecture | open | MathDB 页面 | 原文线索

V2-025877 | MATHDB-346839

Transversality conjecture for biharmonic almost complex structures

摘录 / EXCERPT: Let (g,J) be a compatible almost Hermitian structure, and define ... A biharmonic almost complex structure is a zero of B. Transversality conjecture. The map B is transverse to...

conjecture | open | MathDB 页面 | 原文线索

V2-025878 | MATHDB-346851

Existence of smooth Sasaki-Einstein manifolds from iterated S^3 joins

摘录 / EXCERPT: An iterated S^3 join in the Gorenstein case is a smooth manifold of odd dimension. Existence conjecture. In the Gorenstein case, the iterated S^3 joins admit smooth Sasaki-Eins...

conjecture | open | MathDB 页面 | 原文线索

V2-025879 | MATHDB-346852

The logarithmic stable-pairs rationality and DT/PT correspondence conjecture

摘录 / EXCERPT: Let (X,D) be a logarithmic pair, let β be a curve class, let μ be a relative insertion, and let $\tau_{k_i}(\gamma_i)$ be descendent insertions. Denote the logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-025880 | MATHDB-346897

Kähler-Einstein existence and relative Kähler-Einstein potentials for Fano manifolds

摘录 / EXCERPT: Kähler-Einstein relative singularity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-025881 | MATHDB-346901

Perverse-sheaf Singer-Hopf conjecture for aspherical projective manifolds

摘录 / EXCERPT: Let X be an aspherical projective manifold and let $\mathcal{P}$ be a perverse sheaf on X. Perverse-sheaf Singer-Hopf conjecture. The Euler characteristic of $\mathcal{P}$ is s...

conjecture | open | MathDB 页面 | 原文线索

V2-025882 | MATHDB-346935

The characterization conjecture for the image of the Fujita space

摘录 / EXCERPT: Let $\mathcal{X}$ be the space of smooth birational models, let $\operatorname{\mathcal{F}_{\Sigma}}(\mathcal{X})$ be the Fréchet space defined in the source, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-025883 | MATHDB-347012

Wilking's completeness conjecture for dual foliations

摘录 / EXCERPT: Let M be a complete Riemannian manifold with nonnegative sectional curvature, and let $\mathcal{F}$ be a singular Riemannian foliation on M. The dual foliation is the foliation...

conjecture | open | MathDB 页面 | 原文线索

V2-025884 | MATHDB-347018

The regularity criterion for holonomy groupoids of Haefliger structures

摘录 / EXCERPT: Let M be a manifold, let q be a nonnegative integer, and let H be a codimension-q Haefliger structure defining a foliation $\mathcal{F}$ of M. The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-025885 | MATHDB-347035

Milnor's conjecture on complete umbilic-free surfaces

摘录 / EXCERPT: Milnor's conjecture. Either the Gauss curvature K changes sign on S, or $K=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-025886 | MATHDB-347050

Demailly-Păun-type conjecture for null loci of asymptotically conical Kähler classes

摘录 / EXCERPT: Suppose X is an asymptotically conical Kähler manifold and $[b1] b5 H^{1,1}_{bd}(X,\mathbb{R})$ is a limit of bd-almost compactly supported Kähler classes. Then there is a func...

conjecture | open | MathDB 页面 | 原文线索

V2-025887 | MATHDB-347051

Tangent-cone conjecture for partial resolutions of $Y_{p,p}$

摘录 / EXCERPT: Let $1\leqslant k<p-1$, and let $\overline{Y}$ be the partial resolution obtained by blowing up k of the lines $x=z-\zeta^jw=0$. Its isolated singularity is modeled by ... Let d...

conjecture | open | MathDB 页面 | 原文线索

V2-025888 | MATHDB-347052

Tangent-cone conjecture for the blowdown of a projective bundle

摘录 / EXCERPT: Tangent-cone conjecture. At the singular point $z\in Z$, the tangent cone to (Z,d) is isometric to the blowdown of the zero section in $\mathcal{O}_{\mathbb{P}(V)}(-1)$, equipped...

conjecture | open | MathDB 页面 | 原文线索

V2-025889 | MATHDB-347065

A genus-one FJRW invariant generating-function conjecture

摘录 / EXCERPT: Generating-function conjecture. According to the established cases $i=1,2$, one conjectures that

conjecture | open | MathDB 页面 | 原文线索

V2-025890 | MATHDB-347067

Conjecture on the geodesic-covering number of standard regular surfaces

摘录 / EXCERPT: Let Y_g be a standard regular hyperbolic surface of genus $g\geq 2$, and let K_{Y_g} denote its geodesic-covering number. Geodesic-covering conjecture for standard regular surf...

conjecture | open | MathDB 页面 | 原文线索

V2-025891 | MATHDB-347068

Conjecture on geodesic-covering numbers of finite-index modular subgroups

摘录 / EXCERPT: Let Γ be a subgroup of $\mathrm{PSL}_2(\mathbb{Z})$ of finite index, and let its geodesic-covering number be defined using a fundamental domain. Finite-index modular subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-025892 | MATHDB-347069

Conjecture on geodesic-covering numbers of geometrically finite Fuchsian groups

摘录 / EXCERPT: Let Γ be a discrete subgroup of $\mathrm{PSL}_2(\mathbb{R})$ whose fundamental domain has finitely many sides; such a group is geometrically finite. Geometrically finite geo...

conjecture | open | MathDB 页面 | 原文线索

V2-025893 | MATHDB-347072

Generic multiplicity-two conjecture for Laplace eigenvalues of ellipses

摘录 / EXCERPT: Consider the family of ellipses ... For each such ellipse, consider the multiplicities of the eigenvalues of its Laplacian. Generic multiplicity-two conjecture. For a generic class o...

conjecture | open | MathDB 页面 | 原文线索

V2-025894 | MATHDB-347076

The Laurent-polynomial classification conjecture for smoothing components

摘录 / EXCERPT: Conjecture A. There is a canonical bijective function

conjecture | open | MathDB 页面 | 原文线索

V2-025895 | MATHDB-347092

Greenfield-Wallach-Katok conjecture on globally hypoelliptic flows

摘录 / EXCERPT: Let M be a smooth connected compact manifold, and let a smooth flow be a smooth $\mathbb R$ action on M. The flow is globally hypoelliptic if its associated leafwise Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-025896 | MATHDB-347093

Homogeneous higher-rank nilmanifold conjecture

摘录 / EXCERPT: Let ρ be a homogeneous $\mathbb R^k$ action on a finite-volume space G/D, where G is a connected Lie group and D is a closed subgroup of G. An r-step nilmanifold is...

conjecture | open | MathDB 页面 | 原文线索

V2-025897 | MATHDB-347094

Akopyan-Vysotsky perimeter-diameter inequality for planar curves

摘录 / EXCERPT: Akopyan-Vysotsky conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025898 | MATHDB-347107

The metric Strominger-Yau-Zaslow conjecture for maximally degenerate Calabi-Yau families

摘录 / EXCERPT: Let $(X_t,g_t,J_t,\omega_t,\Omega_t)$ be a 1-parameter maximally degenerate family of polarized n-dimensional Calabi-Yau manifolds of holonomy $SU(n)$ over the punctured disc...

conjecture | open | MathDB 页面 | 原文线索

V2-025899 | MATHDB-347156

Li-Wang-Weng conjecture on annular minimal Lagrangian surfaces

摘录 / EXCERPT: Let Σ be an embedded annulus-type minimal Lagrangian surface in $\mathbb{B}^4$ with Legendrian capillary boundary on $\mathbb{S}^3$. Such a boundary has constant contact ang...

conjecture | open | MathDB 页面 | 原文线索

V2-025900 | MATHDB-347168

The ordinary-line conjecture for configurations on many lines

摘录 / EXCERPT: Ordinary-line conjecture. If each of those lines contains more than $C(m)$ points, then the set admits an ordinary line, that is, a line containing exactly two points of the set.

conjecture | open | MathDB 页面 | 原文线索

V2-025901 | MATHDB-347169

The Hesse configuration conjecture for complex Steiner triple systems

摘录 / EXCERPT: Hesse configuration conjecture. Aside from the Hesse configuration, there are no complex Sylvester-Gallai configurations where each line passes through exactly three points.

conjecture | open | MathDB 页面 | 原文线索

V2-025902 | MATHDB-347175

Effective adjunction conjecture for lc-trivial fibrations

摘录 / EXCERPT: Let (X,B) be an lc pair such that K_X+B is a Q-Cartier divisor, and let $f:(X,B)\to Z$ be an lc-trivial fibration. Let ${\mathbf D}_{\mo}$ denote its moduli b-divisor. Effect...

conjecture | open | MathDB 页面 | 原文线索

V2-025903 | MATHDB-347176

The strong and weak Gamma-effective adjunction conjectures

摘录 / EXCERPT: Let (X,D) be an klt (respectively, lc) pair with dimension d, and let $f:(X,D)\to Z$ be an lc-trivial fibration. An $\mathbb R$-b-divisor ${\bf M}$ is Γ-base-point free...

conjecture | open | MathDB 页面 | 原文线索

V2-025904 | MATHDB-347177

Boundedness conjecture for log canonical models with fixed Iitaka volume

摘录 / EXCERPT: Fix $d\in\mathbb N$, $v\in\mathbb R_{>0}$, and a DCC set $I\subset(0,1]$. For a projective klt pair (X,D) with $\dim X=d$ and $D\in I$, let $\operatorname{Ivol}(K_X+D)$ be its Ii...

conjecture | open | MathDB 页面 | 原文线索

V2-025905 | MATHDB-347178

DCC conjecture for Iitaka volumes of klt pairs

摘录 / EXCERPT: Fix $d\in\mathbb N$ and a DCC set $I\subset(0,1]$. For every projective klt pair (X,D) with $\dim X=d$ and coefficients of D in I, let $\operatorname{Ivol}(K_X+D)$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-025906 | MATHDB-347182

The Einstein metric conjecture for compact nilmanifolds

摘录 / EXCERPT: Let M be a compact nilmanifold and let g be an Einstein metric on M. The Einstein metric conjecture. The metric g is flat, and M is a torus. This concerns the possible Ei...

conjecture | open | MathDB 页面 | 原文线索

V2-025907 | MATHDB-347183

The hemisphere scalar-curvature rigidity conjecture

摘录 / EXCERPT: Let M^n be a compact spin manifold with simply connected boundary ∂M, and let g be a Riemannian metric on M such that ∂M is totally geodesic in M and ha...

conjecture | open | MathDB 页面 | 原文线索

V2-025908 | MATHDB-347185

The symmetric-cone characterization of ellipsoids

摘录 / EXCERPT: Let $B_1(n)$ denote the unit ball in $\mathbb{R}^n$, and let $\mathbb{S}^{n-1}$ denote the unit sphere. For a convex body $K ot i \mathbb{S}^{n-1}$ contained in $B_1(n)$, and each...

conjecture | open | MathDB 页面 | 原文线索

V2-025909 | MATHDB-347186

The line-axis characterization of rotational ellipsoids

摘录 / EXCERPT: Let $B_1(n)$ denote the unit ball in $\mathbb{R}^n$, let $\mathbb{S}^{n-1}$ denote the unit sphere, and let $L oindent oise$ be a line. For each $x otin K$, let C_x be the cone c...

conjecture | open | MathDB 页面 | 原文线索

V2-025910 | MATHDB-347188

Strictness conjecture for inclusions of BPF and nef b-divisor spaces

摘录 / EXCERPT: Strictness conjecture. All the inclusions in this sequence are strict.

conjecture | open | MathDB 页面 | 原文线索

V2-025911 | MATHDB-347196

Hypergeometric formulas for ordered multi-degree contributions

摘录 / EXCERPT: Hypergeometric contribution conjecture. The ordered multi-degree contributions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-025912 | MATHDB-347197

Integrality conjecture for ordered multi-degree contributions

摘录 / EXCERPT: Integrality conjecture. The ordered multi-degree contributions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-025913 | MATHDB-347211

The Hemisystem Conjecture for point-primitive generalised quadrangles

摘录 / EXCERPT: Let G be a group acting point-primitively on the points of a generalised quadrangle $\mathcal{Q}$. The lines of $\mathcal{Q}$ are the objects on which G acts, and a line-orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-025914 | MATHDB-347242

Gusakova-Zaporozhets conjecture on ellipsoids and intrinsic volumes

摘录 / EXCERPT: Let $\mathcal E_1$ and $\mathcal E_2$ be two ellipsoids in $\mathbb R^n$. For an ellipsoid $\mathcal E$, let $V_i(\mathcal E)$ denote its i-th intrinsic volume. Gusakova-Zaporozh...

conjecture | open | MathDB 页面 | 原文线索

V2-025915 | MATHDB-347282

Asymptotic conjecture for smoothed connected essential-surface counts

摘录 / EXCERPT: Let M be a manifold satisfying the hypotheses of Theorem, and define the smoothed count ... where $\atil_M(n)=a_M(n+1)$. Asymptotic conjecture. Either $\atil_M(n)=0$ for all suffic...

conjecture | open | MathDB 页面 | 原文线索

V2-025916 | MATHDB-347311

The bi-Lipschitz rigidity conjecture for 2-attractors

摘录 / EXCERPT: Bi-Lipschitz rigidity conjecture. If 2-attractors are not affinely similar, then they are not bi-Lipschitz equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-025917 | MATHDB-347312

The smooth regularity conjecture for \f-isoperimetric sets

摘录 / EXCERPT: Let ϕ be a norm of class C^∞_+, and let ϕ-isoperimetric sets be isoperimetric sets for the norm ϕ in $\mathbb{H}^1$. Smooth regularity conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-025918 | MATHDB-347357

Ridge-unfolding conjecture for regular convex polytopes

摘录 / EXCERPT: Let P be a regular convex polytope, and consider a ridge unfolding obtained by cutting ridges of P so that its facets can be unfolded into Euclidean space without overlapping....

conjecture | open | MathDB 页面 | 原文线索

V2-025919 | MATHDB-347375

Tropicalization surjectivity conjecture for splitting type loci

摘录 / EXCERPT: Tropicalization surjectivity conjecture. The map Trop is surjective. Moreover, if $g=\lvert\bm{\mu}\rvert$, then it is a bijection.

conjecture | open | MathDB 页面 | 原文线索

V2-025920 | MATHDB-347387

The power-of-2-or-3 conjecture for rigid-type partial geometries

摘录 / EXCERPT: Power-of-2-or-3 conjecture. If $\mathcal{S}$ is a $pg(s,t,\alpha)$ of rigid type, then

conjecture | open | MathDB 页面 | 原文线索

V2-025921 | MATHDB-347399

The automorphism conjecture for complements of smooth hypersurfaces

摘录 / EXCERPT: Let X be a smooth hypersurface in $\mathbb P^n$ of degree $d\geqslant 3$. The smooth hypersurface complement conjecture. Every automorphism of $\mathbb P^n\setminus X$ is induced...

conjecture | open | MathDB 页面 | 原文线索

V2-025922 | MATHDB-347410

Rigidity conjecture for compact hyperbolic 3-manifolds with convex boundary

摘录 / EXCERPT: Let M^1 and M^2 be compact hyperbolic 3-manifolds with convex boundaries, and let $f:M^1\rightarrow M^2$ be a homeomorphism whose restriction to the boundary is isotopic to an...

conjecture | open | MathDB 页面 | 原文线索

V2-025923 | MATHDB-347411

Convex-core metric rigidity conjecture for geometrically finite hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a geometrically finite hyperbolic 3-manifold, meaning that its convex core has finite volume. Its convex core is the intersection of all closed totally convex subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-025924 | MATHDB-347444

Laugwitz's conjecture on flat Hessian metrics of Minkowski norms

摘录 / EXCERPT: Let F be a Minkowski norm on $\mathbb{R}^n$, with $n\geq 3$, and let ... be its Hessian metric on $\mathbb{R}^n\setminus\{0\}$. Laugwitz's conjecture. If g is flat on...

conjecture | open | MathDB 页面 | 原文线索

V2-025925 | MATHDB-347445

Landsberg's unicorn conjecture

摘录 / EXCERPT: Let F be a Finsler metric, and let its Landsberg curvature be the corresponding non-Riemannian curvature tensor. A Finsler metric is Berwald when its Berwald parallel transport i...

conjecture | open | MathDB 页面 | 原文线索

V2-025926 | MATHDB-347453

CAT(0) and acylindrical-hyperbolicity conjectures for Artin groups

摘录 / EXCERPT: Artin groups' geometric conjectures. For every Artin group A_{Γ}: (1) A_{Γ} is CAT(0); and (2) the central quotient $A_{\Gamma}/Z(A_{\Gamma})$ is acylindrically hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-025927 | MATHDB-347458

The submanifold conjecture for maximal-phase stability conditions

摘录 / EXCERPT: The submanifold conjecture. $\operatorname{Stab}_{\mathfrak{P}(\sigma)}\operatorname{\mathcal{D}}$ is a real submanifold of $\operatorname{Stab}\operatorname{\mathcal{D}}$ on whose...

conjecture | open | MathDB 页面 | 原文线索

V2-025928 | MATHDB-347461

Finiteness conjecture for isospectral normal homogeneous spaces

摘录 / EXCERPT: A normal homogeneous space is a homogeneous Riemannian space $(G/H,m)$ whose metric is induced by a bi-invariant metric on the compact Lie group G, and its Laplace spectrum is th...

conjecture | open | MathDB 页面 | 原文线索

V2-025929 | MATHDB-347482

The exact Berezin-kernel norm conjecture in the hyperbolic plane

摘录 / EXCERPT: Let $s\geq 1$, and let $\mathcal{N}_{s,2}$ denote the quantity associated with the Berezin kernel on the hyperbolic plane defined in the paper. Berezin-kernel norm conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-025930 | MATHDB-347491

Robin-mass difference conjecture for random assignment costs

摘录 / EXCERPT: Let Ω and Ω' be regular two-dimensional manifolds. Define the Robin masses by ... Under the assumption that regularization by the parameter δ acts in the same wa...

conjecture | open | MathDB 页面 | 原文线索

V2-025931 | MATHDB-347492

Universal-function and Robin-mass conjecture for assignment-cost asymptotics

摘录 / EXCERPT: Let Ω be a regular two-dimensional manifold, and let $E_{\Omega}(N)$ be its expected optimal assignment cost. Let R_{Ω} denote its Robin mass. Universal-function and...

conjecture | open | MathDB 页面 | 原文线索

V2-025932 | MATHDB-347517

Domokos-Várkonyi approximation conjecture for monostatic polyhedra

摘录 / EXCERPT: Domokos-Várkonyi approximation conjecture. For any centered convex body $K\subset\mathbb R^3$, K can be uniformly approximated by monostable convex polyhedra if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-025933 | MATHDB-347521

Poisson BF theory as the twist of 5d supergravity

摘录 / EXCERPT: Consider 5-dimensional $\mathcal N=1$ supergravity on $\mathbb R^5$. Let $\mathbb C^2$ carry its canonical holomorphic symplectic structure, and let Poisson BF theory be defined on...

conjecture | open | MathDB 页面 | 原文线索

V2-025934 | MATHDB-347522

Poisson BF theory as the twist of pure 4d supergravity

摘录 / EXCERPT: Consider pure 4-dimensional $\mathcal N=1$ supergravity on $\mathbb R^4$. Let $\mathbb C^2$ carry its standard holomorphic symplectic structure, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-025935 | MATHDB-347527

Gompf's conjecture on Brieskorn spheres bounding Stein domains

摘录 / EXCERPT: Gompf's conjecture. No nontrivial Brieskorn sphere, with either orientation, arises as the boundary of a Stein domain in $\mathbb{C}^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-025936 | MATHDB-347530

The model complex projective-plane twist conjecture

摘录 / EXCERPT: Let $\varphi\in\operatorname{Symp}_{ct}(T^*\mathbb{C}\mathbb{P}^2)$ be the compactly supported symplectomorphism constructed from the Lefschetz fibration in the paper. The complex...

conjecture | open | MathDB 页面 | 原文线索

V2-025937 | MATHDB-347587

Sziklai-type bound for absolutely irreducible non-degenerate space curves

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field with q elements. Let $C \subseteq \mathbb{P}^n$, with $n \geq 3$, be an absolutely irreducible, non-degenerate algebraic curve of degree d...

conjecture | open | MathDB 页面 | 原文线索

V2-025938 | MATHDB-347598

Existence conjecture for complete free boundary minimal planes

摘录 / EXCERPT: Let $M=(\mathbb R_{+}^{3},g)$ be an asymptotically flat three-dimensional Riemannian manifold with boundary, and let $p\in\partial M$. A free boundary minimal plane is a complete,...

conjecture | open | MathDB 页面 | 原文线索

V2-025939 | MATHDB-347617

Failure of the fully marked surface theorem for C2 taut foliations

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold, let $\mathcal{F}$ be a C^2 taut foliation on M, and let S be a fully marked surface. C^2 failure conjecture. There exists such a...

conjecture | open | MathDB 页面 | 原文线索

V2-025940 | MATHDB-347622

Arc-length and dimension conjecture for monotone graphs

摘录 / EXCERPT: Monotone-graph dimension conjecture. The graph has Hausdorff dimension 1 and arc length between $\sqrt{2}$ and 2.

conjecture | open | MathDB 页面 | 原文线索

V2-025941 | MATHDB-347623

Hausdorff-dimension conjecture for the V-question-mark graph

摘录 / EXCERPT: Assume that the binary question-mark function $?_V\colon[0,1]\to[0,1]$ is continuous and monotonically increasing, with $?_V(0)=0$ and $?_V(1)=1$. V-question-mark graph dimension c...

conjecture | open | MathDB 页面 | 原文线索

V2-025942 | MATHDB-347638

Yang's conjecture on weakly RC-positive vector bundles

摘录 / EXCERPT: Let X be the underlying complex manifold and let E be a holomorphic vector bundle on X. Assume that E is weakly RC-positive, in the sense used in the paper, and that E is...

conjecture | open | MathDB 页面 | 原文线索

V2-025943 | MATHDB-347658

Polygonal attractors are parallelepipeds

摘录 / EXCERPT: Polygonal-attractor conjecture. Every, not necessarily convex, attractor in $\mathbb{R}^d$ which is a polygon is a parallelepiped.

conjecture | open | MathDB 页面 | 原文线索

V2-025944 | MATHDB-347660

Decreasing intersection quotients characterize quasi-geodesic origami edge-paths

摘录 / EXCERPT: Intersection-quotient conjecture. Under this condition, the edge-path is a quasi-geodesic.

conjecture | open | MathDB 页面 | 原文线索

V2-025945 | MATHDB-347686

Dual minimum-distance conjecture for the code of the design ${\mathbb{D}}$

摘录 / EXCERPT: Let ${\mathbb{D}}$ be the design defined earlier in the paper, and let ${\mathsf{C}}_p({\mathbb{D}})^\perp$ denote the dual of its code over the field of characteristic p. Dual m...

conjecture | open | MathDB 页面 | 原文线索

V2-025946 | MATHDB-347694

Godberson's conjecture on mixed volumes of a convex body and its reflection

摘录 / EXCERPT: Godberson's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-025947 | MATHDB-347703

The Frobenius-manifold and stability-condition isomorphism conjecture for the -Kronecker quiver

摘录 / EXCERPT: Frobenius-manifold and stability-condition isomorphism conjecture. There should exist an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-025948 | MATHDB-347714

Ferri-Gagliardi-Grasselli's additivity conjecture for regular genus

摘录 / EXCERPT: Let M and N be closed simply-connected PL 4-manifolds, and let $\mathcal{G}(M)$ denote the graph-defined PL invariant regular genus of M. Ferri-Gagliardi-Grasselli's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-025949 | MATHDB-347715

Contracted-triangulation conjecture for closed simply-connected PL 4-manifolds

摘录 / EXCERPT: Let M be a closed simply-connected PL 4-manifold. A contracted triangulation is a triangulation satisfying the hypothesis of Theorem (b). Contracted-triangulation conjecture. E...

conjecture | open | MathDB 页面 | 原文线索

V2-025950 | MATHDB-347741

EQ-type equals VEL-type for plane triangulation graphs

摘录 / EXCERPT: Let $4\mathcal{T}4$ be a plane triangulation graph, with EQ-type, VEL-type, and CP-type denoting the discrete types associated respectively with equilateral triangulation geometry,...

conjecture | open | MathDB 页面 | 原文线索

V2-025951 | MATHDB-347742

Singular nilpotent fibration conjecture

摘录 / EXCERPT: Suppose a sequence $\{(M_i,g_i)\}\subset \widetilde{\mathcal{M}}_{\rm Rc}(m,\rho,v)$ collapses to a lower-dimensional compact metric space X, with ... where k is the dimension in...

conjecture | open | MathDB 页面 | 原文线索

V2-025952 | MATHDB-347743

Noncollapsed-cover quantitative space-form rigidity conjecture

摘录 / EXCERPT: Let $(M,g)\in\mathcal{M}_{\rm Rc}(m)$, and let $(\widetilde M,\widetilde g)$ be its Riemannian universal covering. The quantitative space-form rigidity theorem concerns lower bound...

conjecture | open | MathDB 页面 | 原文线索

V2-025953 | MATHDB-347744

Collapsed pseudo-locality conjecture with scalar-curvature control

摘录 / EXCERPT: Collapsed pseudo-locality conjecture. The asserted curvature estimate should hold even when the initial unit ball is Gromov--Hausdorff close to a lower-dimensional Euclidean ball....

conjecture | open | MathDB 页面 | 原文线索

V2-025954 | MATHDB-347745

Torus-bundle conjecture over controlled Riemannian orbifolds

摘录 / EXCERPT: Torus-bundle conjecture. There exists $\varepsilon=\varepsilon(m,I,\iota)>0$ such that these assumptions imply that M is a torus bundle over X. This is presented as an open pro...

conjecture | open | MathDB 页面 | 原文线索

V2-025955 | MATHDB-347747

Stable pairs descendent rationality conjecture

摘录 / EXCERPT: Let X be a nonsingular projective 3-fold, let $\beta\in H_2(X,\mathbb{Z})$, and let ... be the stable pairs descendent series, where $\gamma_i\in H^*(X)$ and $k_i\geq 0$. Stable pa...

conjecture | open | MathDB 页面 | 原文线索

V2-025956 | MATHDB-347751

Unicorn-space symmetry conjecture for uniquely optimal codes

摘录 / EXCERPT: Let (M,d) be a unicorn space, with unicorn sequence $\{C_i\}_{i=1}^\infty$, and let G be a nontrivial isometry group of (M,d). Unicorn-space symmetry conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-025957 | MATHDB-347752

Partial-symmetry conjecture for codes obtained by deleting one point

摘录 / EXCERPT: Let (M,d) be a unicorn space with unicorn sequence $\{C_i\}$, nontrivial isometry group G, and symmetry strength $t\in[0,\infty)$. Partial-symmetry conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-025958 | MATHDB-347753

Four-point optimal-code conjecture for the Hilbert cube

摘录 / EXCERPT: Hilbert-cube four-point conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-025959 | MATHDB-347772

Uniqueness of the projective plane underlying a partial geometry from a maximal arc

摘录 / EXCERPT: Let G be a partial geometry with parameters (s,t,α) as in the paper, for integers $d\ge 2$ and $d'\ge 2$, and let $q=dd'$. Suppose that G arises from a maximal arc of de...

conjecture | open | MathDB 页面 | 原文线索

V2-025960 | MATHDB-347773

2-rank preservation for partial geometries from maximal arcs

摘录 / EXCERPT: Let G be a partial geometry with parameters (s,t,α) as in the paper, where $d=2^i\ge 2$ and $d'=2^j\ge 2$, and let $\mathcal{P}$ be a projective plane of order $q=2^{i+j}$...

conjecture | open | MathDB 页面 | 原文线索

V2-025961 | MATHDB-347778

Uniform foliation conjecture for fibers of the pure N-soliton manifold

摘录 / EXCERPT: Uniform foliation conjecture. The fibers $\mathbf M_{\mathbf z}$ provide a uniformly smooth foliation of $\mathbf M_U^N$.

conjecture | open | MathDB 页面 | 原文线索

V2-025962 | MATHDB-347791

Colorful Bárány-Katchalski-Pach diameter conjecture

摘录 / EXCERPT: Colorful Bárány-Katchalski-Pach conjecture. Under these hypotheses, there is an index $k\in[2d]$ whose entire family has intersection of diameter at least $c d^{-1/2}$, for some ab...

conjecture | open | MathDB 页面 | 原文线索

V2-025963 | MATHDB-347794

The stable Gromov-Lawson-Rosenberg conjecture

摘录 / EXCERPT: Stable Gromov-Lawson-Rosenberg conjecture. A spin manifold stably admits a positive scalar curvature metric if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-025964 | MATHDB-347796

The KO-width-Rosenberg index conjecture

摘录 / EXCERPT: KO-width-Rosenberg index conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-025965 | MATHDB-347797

Teragaito-Motegi conjecture on bi-orderability and generalized torsion in 3-manifold groups

摘录 / EXCERPT: Let G be the fundamental group of a 3--manifold. A generalized torsion element is a non-trivial element whose non-empty finite product of conjugates is the identity, and G is...

conjecture | open | MathDB 页面 | 原文线索

V2-025966 | MATHDB-347799

Kobayashi's reductive constructor conjecture for compact Clifford-Klein forms

摘录 / EXCERPT: Reductive constructor conjecture. If the homogeneous space G/H of reductive type admits compact Clifford-Klein forms, that is, G/H admits a reductive constructor.

conjecture | open | MathDB 页面 | 原文线索

V2-025967 | MATHDB-347833

Unique Virasoro-compatible deformation conjecture for super tau-covers

摘录 / EXCERPT: Let M be a semisimple Frobenius manifold, let its principal hierarchy have tau function Z_M, and let L_m be the operators defining the Virasoro symmetries by ... A super tau-co...

conjecture | open | MathDB 页面 | 原文线索

V2-025968 | MATHDB-347856

Continuous filling area conjecture for self-reverse Finsler surfaces

摘录 / EXCERPT: Let M be a surface with a self-reverse Finsler metric, filling isometrically a circle of length $2L$. The Holmes--Thompson area of M is the area of the surface used here, norma...

conjecture | open | MathDB 页面 | 原文线索

V2-025969 | MATHDB-347857

Discrete filling area conjecture for walled surfaces

摘录 / EXCERPT: Let (M,W) be a walled surface whose boundary is a single closed curve of length $2n$, with area $[?]$ defined as the number of self-crossings of the wallsystem. The boundary metr...

conjecture | open | MathDB 页面 | 原文线索

V2-025970 | MATHDB-347858

Discrete filling area conjecture for square-celled surfaces

摘录 / EXCERPT: Let $C=C_{2n}$ be a cycle graph of length $2n$, and let M be a square-celled surface with boundary C. Distances between vertices are measured in the skeleton graph. Discrete FA...

conjecture | open | MathDB 页面 | 原文线索

V2-025971 | MATHDB-347859

Directed Finsler filling area conjecture

摘录 / EXCERPT: Let M be a surface with a directed Finsler semimetric F, filling without shortcuts a Finsler closed curve (C,G). Let C^+ and C^- denote the two orientations of C, with...

conjecture | open | MathDB 页面 | 原文线索

V2-025972 | MATHDB-347860

Directed discrete filling area conjecture for fine surfaces

摘录 / EXCERPT: Let $C=C_{a,b}$ be a fine cycle whose two oriented lengths are a and b, and let M be a regular fine surface with boundary C. Its area is the number of fine triangles it con...

conjecture | open | MathDB 页面 | 原文线索

V2-025973 | MATHDB-347861

Fine torus conjecture for discretizing integral seminorms

摘录 / EXCERPT: Let $K\subseteq\mathbb R^d$ be a nondegenerate integral convex polytope containing the origin. A fine structure on $\mathbb R^d$ is required to be $\mathbb Z^d$-periodic, and...

conjecture | open | MathDB 页面 | 原文线索

V2-025974 | MATHDB-347894

The wrapped Fukaya interpretation of hypertoric universal deformations

摘录 / EXCERPT: Let $\mathcal{V}$ be a polarized arrangement, not necessarily rational, with associated affine complexification $(V+\eta)_{\mathbb{C}}$ and complexified hyperplane arrangement...

conjecture | open | MathDB 页面 | 原文线索

V2-025975 | MATHDB-347919

Volume conjecture for relative Turaev-Viro invariants

摘录 / EXCERPT: Volume conjecture. As r ranges over all odd integers and $q=e^{\frac{2\pi i}{r}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-025976 | MATHDB-347927

Symplectic Thom conjecture

摘录 / EXCERPT: Let (X,ω) be a symplectic 4-manifold, and let $\Sigma\subset (X,\omega)$ be a smoothly embedded symplectic surface. Genus is measured among smoothly embedded surfaces repr...

conjecture | open | MathDB 页面 | 原文线索

V2-025977 | MATHDB-347950

BCFG classification of symmetric Lagrangian fillings

摘录 / EXCERPT: BCFG classification of Lagrangian fillings. The following symmetric cases have the stated numbers of fillings:

conjecture | open | MathDB 页面 | 原文线索

V2-025978 | MATHDB-347952

The orientation-vanishing conjecture for Seiberg-Witten invariants

摘录 / EXCERPT: Let X be a simply connected, closed, oriented, smooth 4-manifold with $b_2^+(X)>1$ and $b_2^-(X)>1$. Orientation-vanishing conjecture. The Seiberg-Witten invariant vanishes for...

conjecture | open | MathDB 页面 | 原文线索

V2-025979 | MATHDB-347981

Axis-symmetry conjecture for largest small even polygons

摘录 / EXCERPT: For even $n\geq 6$, let $\mathtt{U}_n$ denote a largest small n-gon, and let its diameter graph be the graph whose vertices are the polygon's vertices and whose edges join pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-025980 | MATHDB-348002

Invariance of the Hard Lefschetz condition for compact completely solvable manifolds

摘录 / EXCERPT: Let M be a compact completely solvable manifold, and let ω be a symplectic structure on M. The Hard Lefschetz invariance conjecture. If ω satisfies the Hard Lefsc...

conjecture | open | MathDB 页面 | 原文线索

V2-025981 | MATHDB-348004

The strictly upper triangular holonomy conjecture for a model manifold with density

摘录 / EXCERPT: Let M^n carry the metric ... and density potential $\phi=x_1+x_2+\cdots+x_n$. Compute the holonomy Lie algebra at the origin $(0,0,\ldots,0)$. Strictly upper triangular holonomy co...

conjecture | open | MathDB 页面 | 原文线索

V2-025982 | MATHDB-348012

Lunts's equality conjecture for hyperkähler manifolds

摘录 / EXCERPT: Let X be a projective hyperkähler manifold. The Lunts equality conjecture. There is an equality of Lie subalgebras of $\operatorname{End}(H^\bullet(X,\mathbb{Q}))$: ... This conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-025983 | MATHDB-348023

Joshi's lifting conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a smooth proper Calabi-Yau threefold over k. Joshi's conjecture. Then X lifts to characteristic zero if and only if 1. $H^0(X, \Omega^1)=0$, and 2. X is classical....

conjecture | refuted | MathDB 页面 | 原文线索

V2-025984 | MATHDB-348047

LeBrun-Salamon conjecture for Fano contact manifolds

摘录 / EXCERPT: LeBrun-Salamon conjecture. The manifold Z must be homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-025985 | MATHDB-348082

The LCF Liouville conjecture for nonnegative scalar curvature

摘录 / EXCERPT: Let (M^n,g), $n\geq 3$, be a complete, locally conformally flat manifold with scalar curvature $R\geq 0$, and let $\Phi:M^n\to S^n$ be a conformal map. LCF Liouville conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-025986 | MATHDB-348083

The positive mass conjecture with arbitrary ends

摘录 / EXCERPT: Let N^n be a complete manifold diffeomorphic to $\mathbb{R}^n$, let X^n be a complete connected non-compact manifold, and let g be a smooth complete metric on ... with nonnegat...

conjecture | open | MathDB 页面 | 原文线索

V2-025987 | MATHDB-348087

Circle-type convex-hull realization conjecture for genus-zero hyperbolic surfaces

摘录 / EXCERPT: Circle-type realization conjecture. There exists a circle type closed set $X\subset\mathbb{C}\cup\{\infty\}$ such that Ω is isometric to the boundary of the convex hull of...

conjecture | open | MathDB 页面 | 原文线索

V2-025988 | MATHDB-348088

Uniqueness conjecture for circle-type convex-hull realizations

摘录 / EXCERPT: Circle-type convex-hull uniqueness conjecture. If these two boundaries are isometric, then X and Y differ by a Mf6bius transformation.

conjecture | open | MathDB 页面 | 原文线索

V2-025989 | MATHDB-348089

Rigidity conjecture for locally finite hexagonal square tilings

摘录 / EXCERPT: Hexagonal square-tiling rigidity conjecture. All squares S_i have the same size.

conjecture | open | MathDB 页面 | 原文线索

V2-025990 | MATHDB-348095

The layer-number conjecture for integer grids

摘录 / EXCERPT: Grid layer-number conjecture. For every $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-025991 | MATHDB-348115

Gromov-D’Ambra classification conjecture for rigid geometric structures

摘录 / EXCERPT: Let M be a compact manifold with a rigid geometric structure ω, and let H be a sufficiently large transformation group. Gromov-D’Ambra conjecture. All triples...

conjecture | open | MathDB 页面 | 原文线索

V2-025992 | MATHDB-348125

Microregular germs conjecture for horizontal curves

摘录 / EXCERPT: Microregular germs conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-025993 | MATHDB-348149

Ramos's Grünbaum-Hadwiger-Ramos mass partition conjecture

摘录 / EXCERPT: Ramos's conjecture. The triple (d,k,m) is a solution if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-025994 | MATHDB-348152

Kano's orthogonal-curve mass bisection conjecture

摘录 / EXCERPT: Kano's conjecture. There exists a curve in $\mathds{R}^2$ formed by horizontal and vertical segments, with at most $m-1$ turns, that bisects every measure.

conjecture | open | MathDB 页面 | 原文线索

V2-025995 | MATHDB-348153

Erdős-Lovász-Simmons-Strausz halving-line conjecture

摘录 / EXCERPT: Erdős-Lovász-Simmons-Strausz conjecture. For every $\varepsilon>0$, the number of halving lines determined by a set of n planar points is $O(n^{1+\varepsilon})$.

conjecture | open | MathDB 页面 | 原文线索

V2-025996 | MATHDB-348156

Kollár's non-uniruledness conjecture for general very ample divisors

摘录 / EXCERPT: Kollár's conjecture. A general member $H\in |H|$ is not uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-025997 | MATHDB-348165

The D-equivalence conjecture for birational Calabi-Yau manifolds

摘录 / EXCERPT: Let X and Y be birationally equivalent projective Calabi-Yau (CY) manifolds. Write $\operatorname{DCoh}(X)$ and $\operatorname{DCoh}(Y)$ for their bounded derived categories of...

conjecture | open | MathDB 页面 | 原文线索

V2-025998 | MATHDB-348166

Uniqueness conjecture for the SY class of a curve

摘录 / EXCERPT: Let X be a smooth projective curve in the setting above, and let the SY torsor of X parametrize the choices used to construct its SY bundles. Each element of this torsor determ...

conjecture | open | MathDB 页面 | 原文线索

V2-025999 | MATHDB-348185

The solvable Homeomorphism Conjecture for moduli spaces of fundamental groups

摘录 / EXCERPT: Let $p>0$ be the characteristic. Retain the Frobenius-quotiented moduli space $\overline{\mathfrak{M}}_{g,n}$ of pointed stable curves of type (g,n) and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026000 | MATHDB-348186

Kotrbatý's Hodge-Riemann conjecture for valuations

摘录 / EXCERPT: Kotrbatý's Hodge-Riemann conjecture. Let $i\leq n/2$. If $\phi\in Val_i^\infty(W)$ is a non-zero primitive even valuation, then $Q(\phi)>0$; if it is a non-zero primitive odd valua...

conjecture | open | MathDB 页面 | 原文线索

V2-026001 | MATHDB-348187

Conjectured Lipschitz bound for Möbius transformations of the unit disk in the t-metric

摘录 / EXCERPT: Let $\mathbb{B}^2=\{z\in\mathbb{C}:|z|<1\}$ be the unit disk, and for $a\in\mathbb{B}^2$ define the Möbius transformation ... Here $t_{\mathbb{B}^2}$ denotes the t-metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-026002 | MATHDB-348194

The coherent Springer sheaf heart conjecture

摘录 / EXCERPT: Let G be the reductive group under consideration, let $\widetilde{G}$ denote its Grothendieck resolution, and let ${\mathcal N}$ be the nilpotent cone in its Lie algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-026003 | MATHDB-348224

Etnyre-Min-Mukherjee embedding conjecture for 3-manifolds in symplectic 4-manifolds

摘录 / EXCERPT: Let Y be a closed, oriented, smooth 3-manifold. Etnyre-Min-Mukherjee embedding conjecture. Every such Y smoothly embeds in a symplectic 4-manifold. This conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-026004 | MATHDB-348255

Kertész's density conjecture for TS-packings

摘录 / EXCERPT: Kertész's density conjecture. If ${\bf Q}$ contains a TS-packing of N unit balls, then

conjecture | open | MathDB 页面 | 原文线索

V2-026005 | MATHDB-348260

Intrinsic sign-stability conjecture for mutation loops

摘录 / EXCERPT: Let $\gamma_i:v_i \to v'_i$ for $i=1,2$ be two paths representing the same mutation loop $\phi=[\gamma_1]_{\boldsymbol{s}}=[\gamma_2]_{\boldsymbol{s}}$. Intrinsic sign-stability co...

conjecture | open | MathDB 页面 | 原文线索

V2-026006 | MATHDB-348267

Conjecture on the regularity and lightlikeness of Hitchin-Li-Mochizuki quasicircles

摘录 / EXCERPT: Let $h_{\mathbf{H}^2}$ be the Hitchin map and let $\sigma_{\mathbf{H}^2}$ be the Hitchin-Li-Mochizuki section, whose image parametrizes quasisymmetric maps associated with the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-026007 | MATHDB-348271

Marques-Neves linear Betti-number bound for minimal hypersurfaces

摘录 / EXCERPT: Let (M,g) be an ambient manifold with positive Ricci curvature, and let Σ be an embedded orientable compact minimal hypersurface of index I. Marques-Neves' Betti-number...

conjecture | open | MathDB 页面 | 原文线索

V2-026008 | MATHDB-348274

Existence of the theory of infinity-harmonic functions for general Riemannian metrics

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold, and consider ∞-harmonic functions and the associated notions of viscosity solutions, comparison with cones, gradient estimates, and reg...

conjecture | open | MathDB 页面 | 原文线索

V2-026009 | MATHDB-348276

Uniqueness and equal distribution conjecture for least-gradient sections

摘录 / EXCERPT: Let L be the flat affine bundle and let $v:M\to L$ be the BV section from Theorem, obtained as a limit of the q-harmonic maps. Least-gradient section conjecture. The BV section...

conjecture | open | MathDB 页面 | 原文线索

V2-026010 | MATHDB-348277

Least-gradient primitive conjecture for measured laminations

摘录 / EXCERPT: Let M be the surface under consideration and let λ be an arbitrary oriented lamination on M with a transverse measure. Let $\tilde v:\tilde M\to\mathbb R$ be a primitiv...

conjecture | open | MathDB 页面 | 原文线索

V2-026011 | MATHDB-348283

Foliation Conjecture for the oper stratification

摘录 / EXCERPT: Let $\mathbb{S}_\alpha\rightarrow V_\alpha$ be the projections associated with the oper stratification of the de Rham moduli space. Foliation Conjecture. The Lagrangian fibers of t...

conjecture | open | MathDB 页面 | 原文线索

V2-026012 | MATHDB-348306

The positive-geometry conjecture for amplituhedra

摘录 / EXCERPT: For integers satisfying $k e 0$ and $k+m e n$, let Z be a totally positive matrix and let $\mathcal{A}_{n,k,m}(Z)$ be the image of $\mathop{\rm Gr}_+(k,n)$ under the induced map....

conjecture | open | MathDB 页面 | 原文线索

V2-026013 | MATHDB-348344

Contractible open 3-manifold conjecture for nonnegative scalar curvature

摘录 / EXCERPT: Nonnegative scalar curvature conjecture. If M admits a complete Riemannian metric of nonnegative scalar curvature, then it is diffeomorphic to $\mathbb{R}^{3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-026014 | MATHDB-348346

The two-component conjecture for non-klt loci of nef pairs

摘录 / EXCERPT: Let (X,B) be a pair, and let $f \colon X \rightarrow S$ be a contraction, meaning a projective morphism with $f_*\mathcal{O}_X=\mathcal{O}_S$. Write $\operatorname{Nklt}(X,B)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-026015 | MATHDB-348358

Conjecture on spherical convex floating bodies

摘录 / EXCERPT: Spherical floating-body conjecture. If K_δ is a Euclidean ball, then K is a Euclidean ball.

conjecture | open | MathDB 页面 | 原文线索

V2-026016 | MATHDB-348361

Ellipse-locus conjecture for triangle centers on focus-mounted triangles

摘录 / EXCERPT: Let $\mathcal{E}$ be an ellipse with foci f_1 and f_2, and let $P(t)$ sweep $\mathcal{E}$. For the triangle $\mathcal{T}_f(t)=f_1f_2P(t)$, denote by X_k the kth triangle ce...

conjecture | open | MathDB 页面 | 原文线索

V2-026017 | MATHDB-348372

Effective semi-ampleness conjecture of the moduli part

摘录 / EXCERPT: Let $f\colon X\to Z$ be a klt-trivial fibration of relative dimension k, with (X,B) a klt projective pair whose boundary B has rational coefficients. Let r be the positive...

conjecture | open | MathDB 页面 | 原文线索

V2-026018 | MATHDB-348378

The Fano-manifold maximal local entropy conjecture

摘录 / EXCERPT: Fano-manifold maximal local entropy conjecture. Among all such Fano manifolds,

conjecture | open | MathDB 页面 | 原文线索

V2-026019 | MATHDB-348379

The maximal local entropy conjecture for closed manifolds

摘录 / EXCERPT: Maximal local entropy conjecture for closed manifolds. Among all such manifolds,

conjecture | open | MathDB 页面 | 原文线索

V2-026020 | MATHDB-348390

Parameter-independence conjecture for umbilic indices on superellipsoids

摘录 / EXCERPT: Let $a,b,c>0$ and let $k\in\mathds{Z}_{>1}$. Consider the surface ... An umbilic point is a point where the principal curvatures coincide, and its index is the index of the principal...

conjecture | open | MathDB 页面 | 原文线索

V2-026021 | MATHDB-348391

Umbilic-index conjecture for a quartic surface with a<b

摘录 / EXCERPT: Let $a<b$ and let ϵ satisfy ... Consider the surface ... An umbilic point is a point where the principal curvatures coincide, and its index is the index of the principal direc...

conjecture | open | MathDB 页面 | 原文线索

V2-026022 | MATHDB-348398

North Pole maximal directed discrepancy conjecture for Polar Coordinates

摘录 / EXCERPT: Let $n\in\mathds{Z}^+$, and consider the point configuration called Polar Coordinates with parameter n. For a direction on the sphere, let the directed Discrepancy be the discrep...

conjecture | open | MathDB 页面 | 原文线索

V2-026023 | MATHDB-348407

Uniform projective-basis conjecture for the leading term of minimal codewords

摘录 / EXCERPT: Uniform projective-basis conjecture. Subject to

conjecture | open | MathDB 页面 | 原文线索

V2-026024 | MATHDB-348421

Classification of totally irreducible $(2,0;1)$-bridge trisections

摘录 / EXCERPT: Classification conjecture. There are exactly three, up to mirroring, totally irreducible $(2,0;1)$-bridge trisections.

conjecture | open | MathDB 页面 | 原文线索

V2-026025 | MATHDB-348425

Conjecture on curvature critical points under curve-shortening flow

摘录 / EXCERPT: Let Γ be a generic plane curve evolving under the curve-shortening flow, and let κ denote its signed curvature. Write $n_\kappa(t)$ for the number of critical points...

conjecture | open | MathDB 页面 | 原文线索

V2-026026 | MATHDB-348513

The L^1 scalar-curvature bound under scalar-curvature and entropy lower bounds

摘录 / EXCERPT: Scalar-curvature bound conjecture. There exists $B(n,A)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-026027 | MATHDB-348522

Curvature-dimension conjecture for the α-Grushin plane

摘录 / EXCERPT: Curvature-dimension conjecture. For $b1 geqslant 1$, the $b1$-Grushin plane satisfies $\mathrm{MCP}(K,N)$ if and only if $K\leqslant 0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-026028 | MATHDB-348533

The crystalline kernel conjecture for Delone sets

摘录 / EXCERPT: Let $X\subset\mathbb R^3$ be a Delone set with parameter R. For each point $x\in X$, let $S_x(2R)$ be its local group, and let K be the subset of points whose local groups cont...

conjecture | open | MathDB 页面 | 原文线索

V2-026029 | MATHDB-348534

The 5-gonal symmetry conjecture for Delone sets

摘录 / EXCERPT: Let $X\subset\mathbb R^3$ be a Delone set with parameter R. For each point $x\in X$, let $S_x(2R)$ be its local group, and let Y be the subset of points whose local groups cont...

conjecture | open | MathDB 页面 | 原文线索

V2-026030 | MATHDB-348535

The weak 5-gonal symmetry conjecture for Delone sets

摘录 / EXCERPT: Let $X\subset\mathbb R^3$ be a Delone set with mutually equivalent $2R$-clusters. For each point $x\in X$, let $S_x(2R)$ denote its local group. Weak 5-gonal symmetry conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-026031 | MATHDB-348562

The metric conjecture for the function $w_{\mathbb{B}^n}$

摘录 / EXCERPT: Let $\mathbb{B}^n$ be the unit ball in $\mathbb{R}^n$. For $x,y\in\mathbb{B}^n\setminus\{0\}$, define $\widetilde{x}=x(2-|x|)/|x|$ and $\widetilde{y}=y(2-|y|)/|y|$, and set ... Also...

conjecture | open | MathDB 页面 | 原文线索

V2-026032 | MATHDB-348563

The sharp comparison conjecture for the triangular ratio and $w_{\mathbb{B}^2}$ metrics

摘录 / EXCERPT: Let $\mathbb{B}^2$ be the unit disk, and let $s_{\mathbb{B}^2}$ and $w_{\mathbb{B}^2}$ denote the triangular ratio metric and the function defined in the paper, respectively. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-026033 | MATHDB-348566

Limiting metric conjecture for mathfrak{sl}(2)-type symplectic Higgs bundles

摘录 / EXCERPT: Limiting metric conjecture. The solution $h_{dc}(E,\Phi,\omega)$ to the decoupled Hitchin equation is a limiting metric: h_t converges to h_∞ in C^∞ on every compac...

conjecture | open | MathDB 页面 | 原文线索

V2-026034 | MATHDB-348567

Limiting metric conjecture for odd orthogonal Higgs bundles

摘录 / EXCERPT: Odd orthogonal limiting metric conjecture. The solutions to the decoupled Hitchin equation for $\SO(2n+1,\C)$ constructed in Theorem are limiting metrics.

conjecture | open | MathDB 页面 | 原文线索

V2-026035 | MATHDB-348577

Minimal-volume conjecture for fullerenes with 10k atoms

摘录 / EXCERPT: Let C_N be the class of fullerene isomers with N atoms, and let $N=10k$ with $k\geq 2$. A nanotubical fullerene has a tubular part and two caps; a cap of type (a) is one of the...

conjecture | open | MathDB 页面 | 原文线索

V2-026036 | MATHDB-348578

Minimal-volume conjecture for fullerenes with 6k-4 atoms

摘录 / EXCERPT: Let C_N be the class of fullerene isomers with N atoms. Suppose $N=6k-4\neq 10s$, where $k\geq 5$ and $s\geq 2$. A nanotubical fullerene has a tubular part and two caps, and ca...

conjecture | open | MathDB 页面 | 原文线索

V2-026037 | MATHDB-348579

Minimal-volume conjecture for exceptional-order fullerenes

摘录 / EXCERPT: Let C_N be the class of fullerene isomers with N atoms. Assume $N\neq 20,24,26,28,34$, $N\neq 10s$, and $N\neq 6k-4$, where $k\geq 5$ and $s\geq 2$. A nanotubical fullerene gra...

conjecture | open | MathDB 页面 | 原文线索

V2-026038 | MATHDB-348589

The non-unimodular thick/thin dichotomy for connected Lie groups

摘录 / EXCERPT: Let G be a connected Lie group. The notions of algebraic and analytic thickness and thinness are those used in the paper, and Theorem is the thick/thin dichotomy for unimodular c...

conjecture | open | MathDB 页面 | 原文线索

V2-026039 | MATHDB-348613

The independent packing conjecture for regular symmetric bodies

摘录 / EXCERPT: Independent packing conjecture. There exists an independent C-packing with contact graph G.

conjecture | open | MathDB 页面 | 原文线索

V2-026040 | MATHDB-348623

Nakajima's monopole bubbling conjecture for Coulomb branches

摘录 / EXCERPT: Let G_c be the compact form of a gauge group, and let mu and $lambda$ be coweights with $lambda$ dominant. For dominant nu satisfying $mu\leqnu\leqlambda$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-026041 | MATHDB-348628

The projectively induced Kähler-Einstein flag-manifold conjecture

摘录 / EXCERPT: Flag-manifold conjecture. If (M,g) is a Kähler-Einstein manifold endowed with a projectively induced metric, then it is an open subset of a complex flag manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-026042 | MATHDB-348639

Abate's full omega-limit conjecture for self-intersecting geodesics

摘录 / EXCERPT: Let $S^o=\mathbb{P}^1(\mathbb{C})\setminus\{p_0,p_1,\dots,p_r\}$, where $p_0,p_1,\dots,p_r$ are the poles of a meromorphic connection ∇ on $\mathbb{P}^1(\mathbb{C})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026043 | MATHDB-348640

Abate and Tovena's resonant Fuchsian-pole conjecture

摘录 / EXCERPT: Abate and Tovena's conjecture. There is a neighborhood U of p_0 such that every geodesic ray which enters into U tends to p_0. The two rays of any geodesic which stays in...

conjecture | open | MathDB 页面 | 原文线索

V2-026044 | MATHDB-348655

Bilinear standard simplex conjecture for positive correlation

摘录 / EXCERPT: Let $m\geq3$, let $0<\rho<1$, and let $(\Omega_i)_{i=1}^m$ and $(\Omega_i')_{i=1}^m$ minimize the bilinear Gaussian noise-stability problem with equal prescribed measures, under th...

conjecture | open | MathDB 页面 | 原文线索

V2-026045 | MATHDB-348702

Noncommutative perfectness conjecture for equivariant period complexes

摘录 / EXCERPT: Noncommutative perfectness conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-026046 | MATHDB-348703

Noncommutative perfectness conjecture for Fréchet-Stein coefficient algebras

摘录 / EXCERPT: Fréchet-Stein perfectness conjecture. The complexes satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-026047 | MATHDB-348709

The Klemm-Pandharipande integrality conjecture for Calabi-Yau fourfolds

摘录 / EXCERPT: Klemm-Pandharipande integrality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026048 | MATHDB-348763

Non-hyperbolicity of projective manifolds with trivial real first Chern class

摘录 / EXCERPT: Non-hyperbolicity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026049 | MATHDB-348765

Cecil-Ryan conjecture on compact proper Dupin hypersurfaces

摘录 / EXCERPT: Let M be a compact, connected proper Dupin hypersurface in S^n or ${\bf R}^n$. Cecil-Ryan conjecture. M is Lie equivalent to an isoparametric hypersurface in S^n. The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-026050 | MATHDB-348781

Extension property for maps from projective manifolds to measure-hyperbolic manifolds

摘录 / EXCERPT: Extension property. Every non-degenerate holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-026051 | MATHDB-348814

Fixed-locus Euler characteristic conjecture for invariant compactified Jacobians

摘录 / EXCERPT: Let A be an Abelian surface of Picard rank one, let τ be the involution defining the quotient, and let $C\subset A$ be an integral τ-invariant curve of geometric genus...

conjecture | open | MathDB 页面 | 原文线索

V2-026052 | MATHDB-348830

LG/CY correspondence conjecture for tt* geometries

摘录 / EXCERPT: Let M_{mar} be the marginal-deformation subspace of the deformation space M. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026053 | MATHDB-348831

Conjecture on real structures in the LG/CY correspondence

摘录 / EXCERPT: Let R_f be the Milnor ring, let Δ_f be the Landau-Ginzburg Laplacian, and let $r:\bigoplus_{a=0}^{\infty}R_f^{(n+2)a}\to H^n_{\operatorname{prim}}(X_f)$ be given by ... Supp...

conjecture | open | MathDB 页面 | 原文线索

V2-026054 | MATHDB-348833

Conjecture on tt* structures for quasi-homogeneous Calabi-Yau orbifolds

摘录 / EXCERPT: Let f be a quasi-homogeneous polynomial determining a Calabi-Yau orbifold, and let the monodromy-invariant part of the corresponding Landau-Ginzburg model carry its Landau-Ginzbu...

conjecture | open | MathDB 页面 | 原文线索

V2-026055 | MATHDB-348901

Nonexistence of Whitney towers with the framed Arf tree invariant

摘录 / EXCERPT: Let $A:S^2\cup S^2\looparrowright B^4$ support a Whitney tower $\mathcal W$, and let $t(\mathcal W)$ denote its associated tree invariant. Let T be the sum of trees in Figure 8 o...

conjecture | open | MathDB 页面 | 原文线索

V2-026056 | MATHDB-348902

Pulling-apart criterion for four components in a simply connected 4-manifold

摘录 / EXCERPT: Let X be a simply connected 4-manifold, and let $A=A_1,A_2,A_3,A_4$ support an order-2 non-repeating Whitney tower $\mathcal W$. Let $\lambda_2(A)$ be the order-2 intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-026057 | MATHDB-348942

Generic Schwarzschild Arnold-diffusion conjecture near the photon sphere

摘录 / EXCERPT: Let N_ε denote the perturbed normally hyperbolic invariant manifold associated with the photon sphere, and let L and L_z be the angular-momentum quantities used in...

conjecture | open | MathDB 页面 | 原文线索

V2-026058 | MATHDB-348944

Generic Kerr Arnold-diffusion conjecture near the photon shell

摘录 / EXCERPT: Let N_ε be the perturbed normally hyperbolic invariant manifold associated with the Kerr photon shell, and let (a,M,E,Q,L_z) satisfy the conditions in assumption (2)...

conjecture | open | MathDB 页面 | 原文线索

V2-026059 | MATHDB-348971

The planarity and semi-dilation bounds for open books

摘录 / EXCERPT: The open-book conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026060 | MATHDB-348973

The hierarchy invariant conjecture for links of terminal quotient singularities

摘录 / EXCERPT: The terminal-singularity conjecture. If the singularity $\mathbb{C}^n/G$ is terminal, then

conjecture | open | MathDB 页面 | 原文线索

V2-026061 | MATHDB-348999

Maximal-minor truncation conjecture for Jacobian dual matrices

摘录 / EXCERPT: Let $S=k[x_1,\dots,x_n]$ be a polynomial ring, let $I=(f_1,\dots,f_r)$ be a linearly presented $\operatorname{\mathfrak{m}}$-primary ideal of S, and let ... be the rational map def...

conjecture | open | MathDB 页面 | 原文线索

V2-026062 | MATHDB-349005

Pólya-Szegő conjecture for the fundamental tone of regular polygons

摘录 / EXCERPT: Let $n\geq 3$. An n-gon is a polygon with n sides, and let λ_1 denote its first Dirichlet eigenvalue. Among all n-gons with fixed area, the regular n-gon is the pol...

conjecture | open | MathDB 页面 | 原文线索

V2-026063 | MATHDB-349006

Strong Pólya-Szegő conjecture for finite spectral determination of regular polygons

摘录 / EXCERPT: Let $n\geq 3$, let Ω be a convex n-gon, and let P be a regular n-gon. Write $\lambda_i(\Omega)$ for the ith eigenvalue of the Dirichlet Laplacian on Ω, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-026064 | MATHDB-349007

Antunes-Freitas conjecture on three eigenvalues determining a triangle

摘录 / EXCERPT: Let T_1 and T_2 be triangles, and write $\lambda_i(T_j)$ for their ith eigenvalues of the Dirichlet Laplacian. Antunes-Freitas conjecture. If ... then T_1 and T_2 are ident...

conjecture | open | MathDB 页面 | 原文线索

V2-026065 | MATHDB-349008

Bat conjecture for finite length spectra of convex polygons

摘录 / EXCERPT: Let $n\geq 3$, and let Ω_1 and Ω_2 be convex n-gons. A primitive closed geodesic is a closed geodesic that is not a nontrivial iterate of a shorter closed geodesic;...

conjecture | open | MathDB 页面 | 原文线索

V2-026066 | MATHDB-349012

The conjecture that every hyperkähler manifold has motive of Abelian type

摘录 / EXCERPT: Let X be a hyperkähler manifold, and let $\mathcal{M}_{rat}(\mathbf{C})$ denote the category of rational motives over $\mathbf{C}$. A motive is of Abelian type if it belongs to t...

conjecture | open | MathDB 页面 | 原文线索

V2-026067 | MATHDB-349022

Existence and boundedness of n-complements

摘录 / EXCERPT: Existence and boundedness of n-complements conjecture. The cited theorems should hold without assuming that $X/Z\ni o$ has wFt. The connectedness assumption on X_o may instead...

conjecture | open | MathDB 页面 | 原文线索

V2-026068 | MATHDB-349024

Effective b-semiampleness conjecture

摘录 / EXCERPT: Effective b-semiampleness conjecture. There exists a positive integer $I=I(d,\Gamma)$ such that, for every such 0-contraction with $\dim X=d$ and $D^{\mathrm h}\in\Gamma$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026069 | MATHDB-349025

Gap conjecture for minimal log discrepancies

摘录 / EXCERPT: Gap conjecture. There exists a positive integer $a<1$ such that, whenever $X\ni P$ has dimension d and is a-klt, it is canonical:

conjecture | open | MathDB 页面 | 原文线索

V2-026070 | MATHDB-349026

a-n-complements conjecture

摘录 / EXCERPT: a-n-complements conjecture. Under the additional assumption that $(X/Z\ni o,B)$ has an a-lc over o $\mathbb R$-complement, the cited complement theorems should yield an a...

conjecture | open | MathDB 页面 | 原文线索

V2-026071 | MATHDB-349027

Exponential wave trace remainder conjecture for asymptotically hyperbolic manifolds

摘录 / EXCERPT: Exponential wave trace remainder conjecture. The remainder A is smooth on $(0,\infty)$, and there exist $C,k>0$, depending only on n, k_1, and k_2, such that

conjecture | open | MathDB 页面 | 原文线索

V2-026072 | MATHDB-349046

Seminormal extension of the finite-projectivity result for Hodge-Iwasawa modules

摘录 / EXCERPT: Seminormal extension conjecture. If the space X is seminormal, then the same coadmissibility and finite-projectivity result holds.

conjecture | open | MathDB 页面 | 原文线索

V2-026073 | MATHDB-349074

Toroidal volume conjecture for hyperbolic links

摘录 / EXCERPT: Toroidal volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026074 | MATHDB-349075

Toroidal simplicial-volume conjecture for irreducible link complements

摘录 / EXCERPT: Toroidal simplicial-volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026075 | MATHDB-349092

Fujita-transform conjecture on birational invariance of higher direct images

摘录 / EXCERPT: Fujita-transform conjecture. The derived pushforward

conjecture | open | MathDB 页面 | 原文线索

V2-026076 | MATHDB-349093

Grauert-Riemenschneider vanishing conjecture over simple normal crossing pairs

摘录 / EXCERPT: Grauert-Riemenschneider vanishing conjecture. For every $i>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-026077 | MATHDB-349097

Minimality conjecture for semisimple subgroup foliations on semi-Riemannian Lie groups

摘录 / EXCERPT: Let (G,g) be a semi-Riemannian Lie group with a subgroup K generating a left-invariant conformal foliation $\mathcal F$ on G of codimension two. Semisimple subgroup minimalit...

conjecture | open | MathDB 页面 | 原文线索

V2-026078 | MATHDB-349103

Petty's polar projection problem

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be an o-symmetric convex body, and let ΠK be its projection body, defined by ... Its polar projection body is $\Pi^\circ K=(\Pi K)^\circ$. Petty's c...

conjecture | open | MathDB 页面 | 原文线索

V2-026079 | MATHDB-349117

Jian-Shi-Song's convergence conjecture for cscK metrics on manifolds with semi-ample canonical bundle

摘录 / EXCERPT: Let (M,ω_0) be a compact Kähler manifold with semi-ample canonical bundle K_M. Let $c_1(M)=[\operatorname{Ric}(\omega_0)]$ and consider any sequence of cscK metrics in the...

conjecture | open | MathDB 页面 | 原文线索

V2-026080 | MATHDB-349138

Conjecture that the generic fiber is deformation equivalent to a Barlow surface

摘录 / EXCERPT: Let X^{ε} be the generic fiber of the degeneration constructed in the paper, and let a deformation class mean an equivalence class of surfaces connected by a deformatio...

conjecture | open | MathDB 页面 | 原文线索

V2-026081 | MATHDB-349139

Conjecture on geometrically maximal degenerations of simply-connected surfaces

摘录 / EXCERPT: Let S be a simply-connected surface with Hodge numbers $h^{1,0}=h^{2,0}=0$. A geometrically maximal degeneration is a degeneration having the geometric maximality property used i...

conjecture | open | MathDB 页面 | 原文线索

V2-026082 | MATHDB-349164

The minimizer conjecture for the pluripotential stability threshold

摘录 / EXCERPT: Let (X,L) be a polarized manifold, and let δ and δ_{pp} denote the stability threshold and pluripotential-theoretic stability threshold, respectively. A plur...

conjecture | open | MathDB 页面 | 原文线索

V2-026083 | MATHDB-349165

The pluricomplex Green function minimizer conjecture

摘录 / EXCERPT: Let X be a Fano manifold and let $L=-K_X$. Assume that $\delta<1$, and let G be a pluricomplex Green function obtained as an L^1 limit of normalized potentials along Aubin's c...

conjecture | open | MathDB 页面 | 原文线索

V2-026084 | MATHDB-349166

Finite entropy and non-Archimedean energy conjecture

摘录 / EXCERPT: Let (X,L) be a polarized manifold, let $\mathcal{R}^1$ be the space of test curves, and let $\mathcal{E}^{1,\mathrm{NA}}(L)$ be the subspace consisting of those...

conjecture | open | MathDB 页面 | 原文线索

V2-026085 | MATHDB-349167

The entropy identity for non-Archimedean test curves

摘录 / EXCERPT: Let X be a compact Kähler manifold with Kähler form ω, let $\mathcal{E}^{1,\mathrm{NA}}$ be the relevant non-Archimedean energy space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026086 | MATHDB-349168

Funk-Mahler volume conjecture for convex bodies with centroid at the origin

摘录 / EXCERPT: Let $\mathcal K(\mathbb{R}^n)$ denote the class of convex bodies, let $K\in\mathcal K(\mathbb{R}^n)$ have centroid 0, and for $0\leq\rho<1$ let $\widetilde M_\rho(K,0)$ be its mo...

conjecture | open | MathDB 页面 | 原文线索

V2-026087 | MATHDB-349169

Functional Funk-Mahler volume conjecture for centered integrable functions

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}_+$ be an integrable Borel function with zero first moment, $\int x f(x)\,dx=0$, and let $\widetilde M_\rho(f,0)$ denote its functional modified Fun...

conjecture | open | MathDB 页面 | 原文线索

V2-026088 | MATHDB-349175

Balmus-Montaldo-Oniciuc constant-mean-curvature conjecture for proper biharmonic submanifolds

摘录 / EXCERPT: Balmus-Montaldo-Oniciuc constant-mean-curvature conjecture. Any proper biharmonic submanifold in $\mathbb{S}^{m+1}$ is CMC.

conjecture | open | MathDB 页面 | 原文线索

V2-026089 | MATHDB-349183

Viterbo's weak systolic conjecture for convex contact forms

摘录 / EXCERPT: Viterbo's weak conjecture. The systolic ratio satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026090 | MATHDB-349184

Zauner's conjecture for complex equiangular lines

摘录 / EXCERPT: Let $d\geq 2$. An equiangular line system in $\mathbb{C}^d$ is a set of lines through the origin whose unit-norm representatives have constant squared pairwise inner-product magnit...

conjecture | open | MathDB 页面 | 原文线索

V2-026091 | MATHDB-349198

Anderson's codimension-two conjecture for the Einstein orbifold boundary

摘录 / EXCERPT: Let $\mathbf{E}(M^4)$ be the moduli space of Einstein metrics on a fixed smooth four-manifold, let $\partial_o\mathbf{E}(M^4)$ denote the subspace of singular Einstein orbifold met...

conjecture | open | MathDB 页面 | 原文线索

V2-026092 | MATHDB-349199

The self-duality conjecture for Ricci-flat orbifold limits bubbling Eguchi-Hanson metrics

摘录 / EXCERPT: Let $(M_n,\mathbf{g}_n)_n$ be a sequence of compact smooth Ricci-flat metrics converging in the Gromov-Hausdorff sense to a Ricci-flat orbifold $(M_o,\mathbf{g}_o)$, while Eguchi-H...

conjecture | open | MathDB 页面 | 原文线索

V2-026093 | MATHDB-349204

Picard number conjecture for non-symmetric Kähler-Einstein toric log del Pezzo surfaces

摘录 / EXCERPT: Picard number conjecture. Then S has Picard number one.

conjecture | open | MathDB 页面 | 原文线索

V2-026094 | MATHDB-349206

Barycentric transformation conjecture for smooth Fano polytopes

摘录 / EXCERPT: Let P be a smooth Fano polytope, and let $B(P)$ denote its barycentric transformation. A Fano polytope is of type B_∞ when the barycentric transformation can be applied in...

conjecture | open | MathDB 页面 | 原文线索

V2-026095 | MATHDB-349207

Hacking-Kim conjecture on nonsymmetric Kähler-Einstein Fano polygons

摘录 / EXCERPT: Let P be a Fano polygon, that is, a two-dimensional Fano polytope. Call P symmetric when its symmetry group is nontrivial. Hacking-Kim conjecture. If P is a Kähler-Einstein F...

conjecture | open | MathDB 页面 | 原文线索

V2-026096 | MATHDB-349235

Uniruledness conjecture for parameter spaces of stable Higgs bundles

摘录 / EXCERPT: Uniruledness conjecture. The Kodaira dimension of S is $-\infty$. In particular, if S is a projective manifold, then S is uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-026097 | MATHDB-349278

Metric characterization of ultrametric spaces generated by labeled trees

摘录 / EXCERPT: Metric characterization conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026098 | MATHDB-349297

Generalized cabling conjecture for connected sums of lens spaces

摘录 / EXCERPT: Generalized cabling conjecture. If Dehn surgery on K yields a connected sum of more than $n-1$ lens spaces, then there is a Seifert fibering of Y where K is a regular fiber.

conjecture | open | MathDB 页面 | 原文线索

V2-026099 | MATHDB-349300

Wiseman-Wilson conjecture on ordinary degree-d curves

摘录 / EXCERPT: Let $d\in\mathbb{Z}^+$ and let $A\in\mathcal{P}(\mathbb{R}^2)$ be finite. For a finite point set A, write $\mathcal{O}_d(A)$ for the set of degree-d curves containing exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-026100 | MATHDB-349304

Continuity and strict monotonicity of the top boundary for a self-affine attractor

摘录 / EXCERPT: Top-boundary conjecture. For all $0 < \lambda < \mu < 1$ with $\lambda + \mu > 1$, the set $\partial_{top}(A_{\lambda,\mu})$ is the graph of a continuous, strictly increasing funct...

conjecture | open | MathDB 页面 | 原文线索

V2-026101 | MATHDB-349306

Equivariant Scott-Wojciechowski determinant formula

摘录 / EXCERPT: Let H be a group, let $h\in H$, and let D_P and $D_{\mathcal P_M}$ be the operators associated with the boundary conditions P and ${\mathcal P}_M$. Let A be the relevant bo...

conjecture | open | MathDB 页面 | 原文线索

V2-026102 | MATHDB-349322

Uniform bounded generation conjecture for principal congruence subgroups

摘录 / EXCERPT: Uniform bounded generation conjecture. There is a minimal constant $K:=K(\Phi)$ proportional to ${\rm rank}(\Phi)$ and independent of R and I such that

conjecture | open | MathDB 页面 | 原文线索

V2-026103 | MATHDB-349326

Uniqueness of power-law hyperbolic centers in isodynamic tetrahedra

摘录 / EXCERPT: Let a tetrahedron be isodynamic, and let a center function assign a point on each face. A center function is hyperbolic when the cevians to the corresponding face centers form a hy...

conjecture | open | MathDB 页面 | 原文线索

V2-026104 | MATHDB-349327

Isoscelesness of the reference tetrahedron from its central tetrahedron

摘录 / EXCERPT: Let a reference tetrahedron and its associated central tetrahedron be given. A tetrahedron is isosceles when it has the corresponding symmetry/equality of edge lengths intended in...

conjecture | open | MathDB 页面 | 原文线索

V2-026105 | MATHDB-349328

Regularity of the reference tetrahedron from its central tetrahedron

摘录 / EXCERPT: Let a reference tetrahedron and its associated central tetrahedron be given. A tetrahedron is regular when all its edges are equal. Regular central-tetrahedron conjecture. If the c...

conjecture | open | MathDB 页面 | 原文线索

V2-026106 | MATHDB-349329

Centroid characterization when central and reference tetrahedra are similar

摘录 / EXCERPT: Let a reference tetrahedron and its associated central tetrahedron be given, and suppose the two tetrahedra are similar, meaning that corresponding edge lengths are proportional. L...

conjecture | open | MathDB 页面 | 原文线索

V2-026107 | MATHDB-349330

Isoscelesness from equal lengths of cevians to corresponding face centers

摘录 / EXCERPT: Let a tetrahedron have corresponding face centers, and consider the cevians joining the vertices to those face centers. Equal-cevian-length conjecture. If the cevians to the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-026108 | MATHDB-349353

Facet-defining conjecture for rigid tropical rays

摘录 / EXCERPT: Facet-defining conjecture. For every rigid ray $\tau\in \operatorname{Trop}(X)\cap\mathbb{R}_{\geq 0}^p$,

conjecture | open | MathDB 页面 | 原文线索

V2-026109 | MATHDB-349367

Density conjecture for non-uniqueness times in the KPZ fixed point

摘录 / EXCERPT: Let $\mathfrak{h}_0$ satisfy Assumption, and let $\mathcal{T}_\infty(\mathfrak{h})$ be the set of times at which the KPZ fixed point has non-unique maximizers. Density conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-026110 | MATHDB-349371

Jian-Song metric identification conjecture for Kähler-Ricci flow limits

摘录 / EXCERPT: Let X be an n-dimensional Kähler manifold with semi-ample canonical line bundle K_X, and let $g(t)$ be the long-time solution of the normalized Kähler-Ricci flow. For a seque...

conjecture | open | MathDB 页面 | 原文线索

V2-026111 | MATHDB-349404

Cecil-Chi-Jensen conjecture on proper Dupin hypersurfaces with constant Lie curvatures

摘录 / EXCERPT: Let M be a compact, connected proper Dupin hypersurface in S^n with either four or six principal curvatures. Assume that its Lie curvatures are constant; these are the Lie-sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-026112 | MATHDB-349444

Demailly's conjecture on Waldschmidt constants of point ideals

摘录 / EXCERPT: Suppose that k is an algebraically closed field of characteristic 0, and let I be the defining ideal of a set of points $X\subseteq \mathbb{P}^n_k$. For an ideal J, let...

conjecture | open | MathDB 页面 | 原文线索

V2-026113 | MATHDB-349452

Sign conjecture for the skew algebroids of smooth loopoids

摘录 / EXCERPT: Sign conjecture. The left and right skew algebroids of G differ by sign:

conjecture | open | MathDB 页面 | 原文线索

V2-026114 | MATHDB-349461

Ashbaugh's eigenvalue ratio conjecture for Euclidean domains

摘录 / EXCERPT: Let Ω be a bounded domain in the n-dimensional Euclidean space $\mathbb{R}^{n}$. Let Λ_i denote the i-th Dirichlet eigenvalue of the Laplace operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026115 | MATHDB-349462

Universal eigenvalue-ratio conjecture for Euclidean domains

摘录 / EXCERPT: Let Ω be a bounded domain with piecewise smooth boundary in the n-dimensional Euclidean space $\mathbb{R}^{n}$. Let Λ_j be the j-th Dirichlet eigenvalue of the L...

conjecture | open | MathDB 页面 | 原文线索

V2-026116 | MATHDB-349463

Closed-spectrum eigenvalue inequality conjecture for minimal submanifolds of spheres

摘录 / EXCERPT: Let M^n be an n-dimensional compact minimal submanifold of the unit sphere $S^{n+p}(1)$, and let $\overline{\Lambda}_j$ denote the eigenvalues of the closed Laplace-Beltrami pr...

conjecture | open | MathDB 页面 | 原文线索

V2-026117 | MATHDB-349485

Vanishing of the residue Q-curvature at zero

摘录 / EXCERPT: Vanishing conjecture for residue Q-curvatures. For every $N\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-026118 | MATHDB-349488

Nonexistence of bounded geodesics on odd-dimensional special linear groups

摘录 / EXCERPT: Nonexistence conjecture. If n is odd, then $\mathsf{SL}(n)$ has no bounded geodesics.

conjecture | open | MathDB 页面 | 原文线索

V2-026119 | MATHDB-349489

Constant-norm geodesics are shifted exponential solutions

摘录 / EXCERPT: Constant-norm geodesic conjecture. Shifted exponential solutions of the form Be^{tC} are the only geodesics with constant norm when viewed as curves in the vector space...

conjecture | open | MathDB 页面 | 原文线索

V2-026120 | MATHDB-349490

The covering conjecture for well-rounded unimodular lattices

摘录 / EXCERPT: Covering conjecture. The covering radius satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026121 | MATHDB-349499

Parallel k-partition conjecture for incidence configurations

摘录 / EXCERPT: Parallel k-partition conjecture. For every fixed integer $k>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-026122 | MATHDB-349509

The decomposition theorem for projective maps of rigid spaces

摘录 / EXCERPT: Let $K/\mathbf{Q}_p$ be a finite extension, and let $f:X\to Y$ be a projective map of rigid spaces over K with Y qcqs. Let $\mathrm{IC}_{X,\mathbf{Q}_\ell}$ be the intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-026123 | MATHDB-349572

Normal rational curve completeness conjecture

摘录 / EXCERPT: Let $2\le N\le q-2$. In $\mathrm{PG}(N,q)$, a normal rational curve is a $(q+1)$-arc projectively equivalent to ... A $(q+1)$-arc is complete if no further point can be added while p...

conjecture | open | MathDB 页面 | 原文线索

V2-026124 | MATHDB-349618

Triangular-number conjecture for the smallest triangle containing unit disks

摘录 / EXCERPT: Let n unit disks be given, and let m be a positive integer such that ... The smallest triangle containing these disks is an equilateral triangle of side length $2(m-1)+2\sqrt{3}$...

conjecture | open | MathDB 页面 | 原文线索

V2-026125 | MATHDB-349619

Erdős-Oler-type conjecture for triangular numbers

摘录 / EXCERPT: Let $n>3$ be a triangular number, and consider the smallest triangle containing n disks and the smallest triangle containing $n-1$ disks. Erdős-Oler-type conjecture. The smallest...

conjecture | open | MathDB 页面 | 原文线索

V2-026126 | MATHDB-349620

Centred-hexagonal-number conjecture for the smallest hexagon containing disks

摘录 / EXCERPT: Let $n\in\mathbb{N}$ be a centred hexagonal number, and consider the smallest hexagon containing n disks and the smallest hexagon containing $n-1$ disks. Centred-hexagonal-number...

conjecture | open | MathDB 页面 | 原文线索

V2-026127 | MATHDB-349622

The simple normal crossing degeneration conjecture for Moishezon morphisms

摘录 / EXCERPT: The simple normal crossing degeneration conjecture. Under these assumptions, g is Moishezon.

conjecture | open | MathDB 页面 | 原文线索

V2-026128 | MATHDB-349623

The rational-singularity limit conjecture for Moishezon fibers

摘录 / EXCERPT: The rational-singularity limit conjecture. Then X_0 is Moishezon.

conjecture | open | MathDB 页面 | 原文线索

V2-026129 | MATHDB-349624

The general-type component conjecture for Moishezon families

摘录 / EXCERPT: The general-type component conjecture. Then g is Moishezon, and all other fibers are of general type over a possibly smaller disc.

conjecture | open | MathDB 页面 | 原文线索

V2-026130 | MATHDB-349625

The fiberwise birational model conjecture for Moishezon morphisms

摘录 / EXCERPT: The fiberwise birational model conjecture. Then g is fiberwise birational to a flat, projective morphism $g^{\rm p}:X^{\rm p}\to\mathbb D$ such that, respectively, $X_0^{\rm p}$...

conjecture | open | MathDB 页面 | 原文线索

V2-026131 | MATHDB-349658

Extension of the finite-disk area bound for generalized Minkowski arrangements

摘录 / EXCERPT: Finite-disk area-bound conjecture. The displayed statement in Remark 3 holds for every μ-arrangement of finitely many disks with $\sqrt{3}-1<\mu<1$.

conjecture | open | MathDB 页面 | 原文线索

V2-026132 | MATHDB-349659

Fejes Tóth's hexagonal density conjecture for generalized Minkowski arrangements

摘录 / EXCERPT: Fejes Tóth's hexagonal density conjecture. For any $\mu\in(\sqrt{3}-1,1)$ and any μ-arrangement $\mathcal{F}$ in $\mathbb{R}^2$,

conjecture | open | MathDB 页面 | 原文线索

V2-026133 | MATHDB-349663

Area comparison conjecture for minimal surfaces in hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold with hyperbolic metric g_{hyp}, let g be a metric on M whose scalar curvature is at least -6, and let Σ be a fixed closed...

conjecture | open | MathDB 页面 | 原文线索

V2-026134 | MATHDB-349707

The higher-dimensional contact-star union conjecture

摘录 / EXCERPT: Let $\mathbb P^n$ be projective n-space, and let $\mathbb{S}(\ell_1,\ldots,\ell_r)$ and $\mathbb{S}(m_1,\ldots,m_s)$ denote contact star configurations formed from distinct hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-026135 | MATHDB-349719

The conformal symplectic Weinstein conjecture for rational exact lcs structures

摘录 / EXCERPT: Let M be a closed manifold of dimension at least 4, and let (λ,α) be an exact locally conformally symplectic structure, with α a rational Lee form. Define t...

conjecture | open | MathDB 页面 | 原文线索

V2-026136 | MATHDB-349720

Existence of Reeb 2-curves in rational exact lcs manifolds

摘录 / EXCERPT: Let M be a closed manifold of dimension at least 4, and let $\omega=d^{\alpha}\lambda$ be an exact locally conformally symplectic form on M whose Lee form α is rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-026137 | MATHDB-349721

The Q-curvature volume comparison conjecture for closed hyperbolic manifolds

摘录 / EXCERPT: Q-curvature volume comparison conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026138 | MATHDB-349723

Feng-Hochman-Rogers contraction-ratio conjecture for affine embeddings

摘录 / EXCERPT: Let $E,F\subset\mathbb{R}^d$ be two totally disconnected non-trivial self-similar sets, generated by iterated function systems $\Phi=\{\varphi_i\}_{i\in I}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026139 | MATHDB-349736

Extension of geometrically constructed wrapped sheaves to the cotangent bundle

摘录 / EXCERPT: Extension conjecture. The construction $\operatorname{\mathfrak{w} sh}_\Lambda$ should extend from the zero section to the cotangent bundle $T^\ast M$.

conjecture | open | MathDB 页面 | 原文线索

V2-026140 | MATHDB-349759

Monotonicity conjecture for the lower Paneitz eigenvalues on the annulus model

摘录 / EXCERPT: Let $\tau>0$, and for each $\ell\geq 1$ let λ^-_ℓ denote the smaller of the two eigenvalues associated with the ℓ-th projection in the annulus model...

conjecture | open | MathDB 页面 | 原文线索

V2-026141 | MATHDB-349769

The spectral characterization conjecture for convex subsets of the plane

摘录 / EXCERPT: Let $\mathcal C$ be a proper closed convex subset of $\mathbb R^2$ with positive Lebesgue measure. Assume that $\operatorname{Op}^w(\mathbf 1_{\mathcal C})$ is a bounded self-adjoi...

conjecture | open | MathDB 页面 | 原文线索

V2-026142 | MATHDB-349774

Conjecture on smooth limits when all higher coefficients vanish

摘录 / EXCERPT: Let d_m be the coefficients appearing in the rescaled mean curvature flow expansion, and suppose that the successive approximations are parametrized by Q_m and v_m. Smooth-li...

conjecture | open | MathDB 页面 | 原文线索

V2-026143 | MATHDB-349784

The plane configuration conjecture for Cayley-Bacharach point sets

摘录 / EXCERPT: Plane configuration conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026144 | MATHDB-349802

Invariant powers of fixed points for circumcircles and Euler circles

摘录 / EXCERPT: Consider a non-concentric, non-axis-aligned pair of ellipses admitting Poncelet 3-periodics. For each 3-periodic, let its circumcircle and Euler circle be the circles associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-026145 | MATHDB-349803

Characterization of triangle centers with elliptic loci in concentric Poncelet pairs

摘录 / EXCERPT: Let $\mathcal{X}$ be a triangle center, and consider its locus as the triangle ranges over Poncelet 3-periodics in a concentric pair of non-axis-aligned ellipses. Elliptic-locus ch...

conjecture | solved | MathDB 页面 | 原文线索

V2-026146 | MATHDB-349805

Chi Li's Fujita approximation conjecture for big models

摘录 / EXCERPT: Let (X,L) be a polarized projective manifold, and let $({\mathcal X},{\mathcal L})$ be a big model of (X,L). Thus $({\mathcal X},{\mathcal L})$ is a test configuration-type big...

conjecture | open | MathDB 页面 | 原文线索

V2-026147 | MATHDB-349842

Smoothness conjecture for spaces of codimension-one foliations

摘录 / EXCERPT: Smoothness conjecture. The foliation $\mathcal F$ corresponds to a smooth point of $\textsf{Fol}_{d}(\mathbb P^n)$; equivalently, there exists a unique irreducible component of...

conjecture | open | MathDB 页面 | 原文线索

V2-026148 | MATHDB-349849

Conway-Sloane's tropical theta invariant conjecture

摘录 / EXCERPT: Conway-Sloane's conjecture. The map Θ is injective on any fundamental domain of the $\operatorname{GL}_n(\mathbb{Z})$-action on $S^n_{>0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-026149 | MATHDB-349869

Bezdek's Rhombic Dodecahedral Conjecture for parallelohedra

摘录 / EXCERPT: Let P be a parallelohedron in $\mathbb R^3$ with unit inradius. Bezdek's Rhombic Dodecahedral Conjecture. The surface area of P is at least ... which is the surface area of the r...

conjecture | open | MathDB 页面 | 原文线索

V2-026150 | MATHDB-349893

Radical-polynomial sectional volume conjecture for convex bodies

摘录 / EXCERPT: Let $K \subset \mathbb R^n$ be a compact body with C^{∞} boundary ∂K. For $\xi \in \mathbb R^n$ with $|\xi|=1$ and $t \in \mathbb R$, let ... be the sectional volume...

conjecture | open | MathDB 页面 | 原文线索

V2-026151 | MATHDB-349897

The conjecture on infinitely many geometric ideal triangulations of cusped hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a finite-volume cusped hyperbolic 3--manifold, meaning a noncompact hyperbolic 3--manifold of finite volume. A geometric ideal triangulation of M is a decompositio...

conjecture | open | MathDB 页面 | 原文线索

V2-026152 | MATHDB-349899

Folklore conjecture on Ricci-flow singularity models

摘录 / EXCERPT: A gradient shrinking soliton is a Riemannian manifold (M,g) equipped with a function $f\in C^\infty(M)$ satisfying ... Such a soliton generates a self-similar Ricci flow and is inv...

conjecture | open | MathDB 页面 | 原文线索

V2-026153 | MATHDB-349900

Refined folklore conjecture on Ricci-flow singularity models

摘录 / EXCERPT: A gradient shrinking soliton is a Riemannian manifold (M,g) with a potential function f satisfying ... A soliton may be degenerate, and its singular set may have codimension at l...

conjecture | open | MathDB 页面 | 原文线索

V2-026154 | MATHDB-349903

Beauville's splitting conjecture for compact Kähler manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold equipped with a holomorphic decomposition ... of the tangent bundle such that each subbundle $\bigoplus_{j\in J} \mathcal{F}_j$, for...

conjecture | open | MathDB 页面 | 原文线索

V2-026155 | MATHDB-349905

Scaling-limit conjecture for the canonical evaporated UST component

摘录 / EXCERPT: Let $({\bf T},r)$ be a rooted uniform spanning tree of a graph, and obtain ${\bf t}^{<n}(r)$ by successively removing outgoing edges at independent uniformly chosen non-root vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-026156 | MATHDB-349915

Rigidity conjecture for conjugation-invariant Lorentzian distance functions

摘录 / EXCERPT: Let (M,xi) be a closed contact manifold and let $\mathrm{Cont}_0(M)$ denote the identity component of its contactomorphism group. Write $\llcurly$ for the chronological relation...

conjecture | open | MathDB 页面 | 原文线索

V2-026157 | MATHDB-349927

The admissible collapsing-level conjecture for classical W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be either a symplectic or orthogonal simple Lie algebra, let f be a nilpotent element, and let k be an admissible level. Exhaustiveness conjecture. The cases...

conjecture | open | MathDB 页面 | 原文线索

V2-026158 | MATHDB-349941

The conjecture that all compact spaces feature smeariness

摘录 / EXCERPT: Let X be a compact metric space, and consider random variables taking values in X whose Fréchet means and sample Fréchet means are defined. A distribution is smeary when its Fr...

conjecture | open | MathDB 页面 | 原文线索

V2-026159 | MATHDB-349960

Single-valued multiple zeta value conjecture for the coefficients C_n

摘录 / EXCERPT: Let $C_n(\lambda)$ be the polynomial coefficient arising in the asymptotic expansion studied in the paper, and let ${\mathfrak{Z}}_n^{\mathrm{sv}}$ denote the space of single-value...

conjecture | open | MathDB 页面 | 原文线索

V2-026160 | MATHDB-349987

Non-rationality conjecture for Fano threefolds of type X_(1,1,1,1)

摘录 / EXCERPT: Non-rationality conjecture. Then X is never ${\mathsf{k}}$-rational.

conjecture | open | MathDB 页面 | 原文线索

V2-026161 | MATHDB-349989

Symplectic automorphism conjecture for the Albanese kernel of zero-cycles

摘录 / EXCERPT: Let S be a smooth projective surface. A symplectic automorphism is an automorphism acting trivially on the space of holomorphic two-forms $H^{2,0}(S)$. It acts on the Albanese ke...

conjecture | open | MathDB 页面 | 原文线索

V2-026162 | MATHDB-350005

Bárány-Katchalski-Pach conjecture on intersections of halfspaces

摘录 / EXCERPT: Bárány-Katchalski-Pach conjecture. This intersection contains a point of norm $\sqrt d$.

conjecture | open | MathDB 页面 | 原文线索

V2-026163 | MATHDB-350013

Girao's weighted Alexandrov inequality conjecture for strictly convex hypersurfaces

摘录 / EXCERPT: Let $\Sigma\subset\mathbb S^n$ be a closed, orientable, connected embedded hypersurface, and let H denote its mean curvature. For $x\in\mathbb S^n$, define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-026164 | MATHDB-350036

Gutt-Hutchings consistency conjecture for capacities of star-shaped domains

摘录 / EXCERPT: Let U be a bounded star-shaped domain, and let $c_k^{\text{EH}}(U)$, $c_k^{\text{GH}}(U)$, and $c_k(U)$ denote the Ekeland-Hofer, Gutt-Hutchings, and capacities considered in the...

conjecture | open | MathDB 页面 | 原文线索

V2-026165 | MATHDB-350042

Euclidean discrete uniformization conjecture for weighted marked surfaces

摘录 / EXCERPT: Let (M,V,ε) be a closed connected weighted marked surface, where $\varepsilon:V\rightarrow\{0,1\}$, $\chi(M)=0$, and d is a piecewise linear metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-026166 | MATHDB-350046

Helly-type intersection conjecture for projection-direction sets

摘录 / EXCERPT: Let $\mathcal{F}$ be an intersecting family of convex sets in $\mathbb{R}^3$. For each triple $T=\{F_1,F_2,F_3\}\in {\mathcal{F}\choose 3}$, let $D_T\subset S^2$ consist of the dir...

conjecture | open | MathDB 页面 | 原文线索

V2-026167 | MATHDB-350060

Variance conjecture for closed geodesics in annuli

摘录 / EXCERPT: Let D be a squarefree fundamental discriminant. For $0\leq r<R$, let $A_{r,R}(w)$ be the hyperbolic annulus centered at w with inner radius r and outer radius R, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026168 | MATHDB-350070

Generalized deformation invariance conjecture for 2-step nilmanifolds

摘录 / EXCERPT: Let M be a 2-step nilmanifold with abelian complex structure. Suppose that there exists an ascending basis ... such that $d\omega^{n+1}$ is non-degenerate. Let $\Gamma_1\in H^2(M)$...

conjecture | open | MathDB 页面 | 原文线索

V2-026169 | MATHDB-350076

Dimension conjecture for arcs in mixed labyrinth fractals

摘录 / EXCERPT: Let $f0 f0$?

conjecture | open | MathDB 页面 | 原文线索

V2-026170 | MATHDB-350080

Lei-Xu's optimal pinching conjecture for mean curvature flow in spheres

摘录 / EXCERPT: Lei-Xu's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026171 | MATHDB-350081

Refined Chern conjecture on isoparametric minimal hypersurfaces

摘录 / EXCERPT: Let M^n be a closed immersed minimal hypersurface of the sphere $\mathbb{S}^{n+1}$ with constant scalar curvature. Refined Chern conjecture. Then M^n is isoparametric. The text...

conjecture | open | MathDB 页面 | 原文线索

V2-026172 | MATHDB-350084

Polynomial-time recognition conjecture for ordered curve pseudo-visibility graphs

摘录 / EXCERPT: An ordered curve pseudo-visibility graph is the ordered graph arising from the pseudo-visibility representation by curves discussed in the paper. Recognition conjecture. Ordered cu...

conjecture | open | MathDB 页面 | 原文线索

V2-026173 | MATHDB-350087

Weyl-line intersection conjecture for effective divisors on X^4_s

摘录 / EXCERPT: Weyl-line intersection conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026174 | MATHDB-350090

The Gopakumar-Vafa finiteness conjecture

摘录 / EXCERPT: Let (X,ω) be a closed symplectic 6-manifold. For every $A\in H_2(X,\mathbf Z)$ with $A\neq 0$ and $c_1(A)=0$, let $\mathrm{BPS}_{A,g}(X,\omega)$ be the numbers defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-026175 | MATHDB-350102

Campana-Cao-Matsumura's slope rationally connected quotient conjecture

摘录 / EXCERPT: Campana-Cao-Matsumura's conjecture. There exists an orbifold morphism $\rho:(X,D)\to(R,D_R)$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-026176 | MATHDB-350103

Conjecture on smooth slope rationally connected quotients for nef logarithmic tangent bundles

摘录 / EXCERPT: Let (X,D) be a log smooth pair, meaning that X is smooth and D has simple normal-crossing support. Its logarithmic tangent bundle is denoted by $T_X(-\log D)$. An sRC-quotien...

conjecture | open | MathDB 页面 | 原文线索

V2-026177 | MATHDB-350106

The disk minimisation conjecture for the lowest positive Dirac eigenvalue

摘录 / EXCERPT: Let $m\geq 0$, let $\Omega\subset\mathbb{R}^2$ be an open connected set with locally Lipschitz boundary, and let $\lambda_1(\Omega)$ denote the lowest positive eigenvalue of the Di...

conjecture | open | MathDB 页面 | 原文线索

V2-026178 | MATHDB-350107

The square minimisation conjecture for Dirac rectangles

摘录 / EXCERPT: Let $\Omega_{a,b}$ be a rectangle with side lengths a,b, and write ... for the lowest positive eigenvalue of the Dirac operator with infinite-mass boundary conditions. Square minim...

conjecture | open | MathDB 页面 | 原文线索

V2-026179 | MATHDB-350108

The symmetry-minimising-eigenfunction conjecture for the area-constrained rectangle

摘录 / EXCERPT: Let $\Omega_{a,b}$ be a rectangle, and consider the variational problem defining the lowest positive Dirac eigenvalue under the area constraint, with right-hand side denoted by the...

conjecture | open | MathDB 页面 | 原文线索

V2-026180 | MATHDB-350114

Capacity uniqueness conjecture for convex bodies

摘录 / EXCERPT: Let A be a convex body in $\mathbb R^{2m}$, and let c be a symplectic capacity. Capacity uniqueness conjecture. All symplectic capacities coincide on convex bodies. The claim i...

conjecture | open | MathDB 页面 | 原文线索

V2-026181 | MATHDB-350115

Convex non-squeezing conjecture in intermediate dimensions

摘录 / EXCERPT: Let $1<k<m$, let $\varphi\in\operatorname{Symp}(\mathbb R^{2m})$, and let $P_{\mathbb R^{2k}}:\mathbb R^{2m}\to\mathbb R^{2k}$ be projection onto the first $2k$ coordinates. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-026182 | MATHDB-350117

Green-Griffiths conjecture on algebraic degeneracy of holomorphic curves

摘录 / EXCERPT: Green-Griffiths conjecture. X contains no algebraically non-degenerate holomorphic curves.

conjecture | open | MathDB 页面 | 原文线索

V2-026183 | MATHDB-350134

Unit-ball sunflower conjecture in three dimensions

摘录 / EXCERPT: For each integer $r\geq 3$, let $V\subset\mathbb R^3$ and let $\mathcal F$ be a family of subsets of V, each of size k, such that every member of $\mathcal F$ is the intersecti...

conjecture | open | MathDB 页面 | 原文线索

V2-026184 | MATHDB-350150

O-singular locus conjecture for quantized symplectic resolutions

摘录 / EXCERPT: $\mathcal{O}$-singular locus conjecture. The locus $\mathfrak{c}^{\mathcal{O}-\mathrm{sing}}$ coincides with the union of singular hyperplanes.

conjecture | open | MathDB 页面 | 原文线索

V2-026185 | MATHDB-350152

Divisibility conjecture for twisted characteristic numbers of the Hopkins-Mahowald manifold

摘录 / EXCERPT: Let M_1 be the 24-dimensional String manifold of Hopkins-Mahowald. For non-negative integers i, j, and k, let $T^i\otimes\wedge^j\otimes S^k$ denote the corresponding ten...

conjecture | open | MathDB 页面 | 原文线索

V2-026186 | MATHDB-350210

Gromov's square inequality for spin manifolds with cube-like boundaries

摘录 / EXCERPT: Let (X,g) be an n-dimensional compact connected orientable manifold with boundary, and let $\underline{X}_\bullet$ be a closed orientable manifold of dimension $n-m$. Suppose .....

conjecture | open | MathDB 页面 | 原文线索

V2-026187 | MATHDB-350233

Conformal maximality conjecture for free boundary harmonic annuli

摘录 / EXCERPT: For any $T\geqslant T_1$, let $\mathbb{A}_T=[0,T]\times\mathbb{S}^1$ and let $\mathcal{C}_T$ be the conformal class containing the flat metric on $[0,T]\times\mathbb{S}^1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026188 | MATHDB-350237

Perron's conjecture on the mod-p extension of the Casson invariant

摘录 / EXCERPT: Perron's conjecture. The map γ_p is a well-defined invariant on $\mathcal{S}^3[p]$. Perron proposed this as an extension of the Casson invariant to the mod-p Torelli group...

conjecture | open | MathDB 页面 | 原文线索

V2-026189 | MATHDB-350258

Birman-Solomyak Weyl-law conjecture for pseudodifferential operators on noncommutative tori

摘录 / EXCERPT: Let $P=P^*\in \Psi^{-m}(\mathbb{T}^n_\theta)$ with $m>0$ have principal symbol $\rho_{-m}(\xi)$, and set $p=nm^{-1}$. For a selfadjoint compact operator, write...

conjecture | open | MathDB 页面 | 原文线索

V2-026190 | MATHDB-350282

The constant holomorphic sectional curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Constant holomorphic sectional curvature conjecture. A compact Hermitian manifold with constant holomorphic sectional curvature is Kähler when the constant is non-zero and Chern fl...

conjecture | open | MathDB 页面 | 原文线索

V2-026191 | MATHDB-350296

Xu-Deng's homogeneous Landsberg conjecture

摘录 / EXCERPT: A Finsler manifold (M,F) is homogeneous if its group of isometries acts transitively on M. A Finsler metric is Landsberg if its Landsberg curvature vanishes, and it is Berwald...

conjecture | open | MathDB 页面 | 原文线索

V2-026192 | MATHDB-350390

Karasev's smallest-solid-angle conjecture for simplices

摘录 / EXCERPT: Let ${\mathcal P}\subset\mathbb{R}^d$ be a d-dimensional simplex, and let its solid angle be measured in the usual way. Karasev's smallest-solid-angle conjecture. Every d-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-026193 | MATHDB-350391

Betke-Henk-Wills discretized second Minkowski conjecture

摘录 / EXCERPT: Let $K\subset\mathbb{R}^d$ be a centrally symmetric convex body, let ${\mathcal L}\subset\mathbb{R}^d$ be a lattice, and let $\lambda_k(K,\mathcal L)$ be its successive minima. Bet...

conjecture | open | MathDB 页面 | 原文线索

V2-026194 | MATHDB-350433

Voronoi's conjecture on parallelohedra

摘录 / EXCERPT: Let P be a d-dimensional parallelohedron, meaning a convex polytope that tiles $\mathbb R^d$ by translations. A Voronoi polytope of a d-dimensional lattice Λ is the s...

conjecture | open | MathDB 页面 | 原文线索

V2-026195 | MATHDB-350450

Hadwiger's covering conjecture for convex bodies

摘录 / EXCERPT: Hadwiger's covering conjecture. For each $K\in\mathcal{K}^n$, $c(K)$ is bounded from above by 2^n, and this upper bound is attained only by parallelotopes.

conjecture | open | MathDB 页面 | 原文线索

V2-026196 | MATHDB-350459

The transcendental-lattice conjecture for Nikulin orbifolds and fixed K3 surfaces

摘录 / EXCERPT: Let X be a fourfold of $K3^{[2]}$-type admitting a symplectic involution σ, let Y be the partial resolution of X/σ, and let S be the K3 surface contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-026197 | MATHDB-350473

The right-angle 2-chain conjecture in three dimensions

摘录 / EXCERPT: Let $E\subset\mathbb R^3$ be a large finite point set of n points. An angle 2-chain consists of three consecutive angles determined by four points, and $\Lambda_2(E)$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-026198 | MATHDB-350475

Cyclic dilation conjecture for Milnor fibers of log Kodaira dimension minus infinity

摘录 / EXCERPT: Let M be a Milnor fiber of an isolated hypersurface singularity whose log Kodaira dimension is $-\infty$. Cyclic dilation conjecture. Then M admits a cyclic dilation with $h=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-026199 | MATHDB-350514

Linear blow-up conjecture for the ultrametric skeleton theorem

摘录 / EXCERPT: Let (X,d) be a metric space satisfying the hypotheses of Theorem, and let C_ε denote the balls' blow-up parameter in its measure-growth conclusion. Linear blow-up con...

conjecture | open | MathDB 页面 | 原文线索

V2-026200 | MATHDB-350515

Higher hyperbolic rank rigidity conjecture

摘录 / EXCERPT: A closed Riemannian manifold M has higher hyperbolic rank if every geodesic $c(t)$ in M has a Jacobi field $J(t)$ such that $J(t)\neq 0$ and the sectional curvature of the plan...

conjecture | open | MathDB 页面 | 原文线索

V2-026201 | MATHDB-350537

Uniqueness conjecture for invariant Ricci solitons on the four-sphere

摘录 / EXCERPT: Uniqueness conjecture. Any such solution is the round sphere, up to diffeomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-026202 | MATHDB-350538

Projective-geometric correspondence conjecture for WDVV systems

摘录 / EXCERPT: A WDVV system is a system of first-order PDEs considered together with its projective-geometric structures. A quadratic line complex is the associated family of lines in projective...

conjecture | open | MathDB 页面 | 原文线索

V2-026203 | MATHDB-350575

The asymptotic scaling conjecture for the volume loss function

摘录 / EXCERPT: Volume-loss scaling conjecture. For any $\ell,r>0$, the limit

conjecture | open | MathDB 页面 | 原文线索

V2-026204 | MATHDB-350576

The generalized volume-loss asymptotic conjecture for bounded domains

摘录 / EXCERPT: Let $X\subset\mathbb{R}^4$ be a bounded domain, and define ... where the infimum is over symplectic embeddings $\Phi\colon RX\to\mathbb{R}^4$ with Lipschitz constant at most L, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026205 | MATHDB-350577

Conjecture on the Gromov-Hausdorff distance between consecutive spheres

摘录 / EXCERPT: Let $D54A^m$ denote the unit m-sphere with its intrinsic spherical metric, let d_{GH} be the Gromov-Hausdorff distance, and let $D701_m$ be the quantity defined...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-026206 | MATHDB-350615

The Fusco-Lachand-Robert conjecture for the Dirichlet eigenvalue and perimeter

摘录 / EXCERPT: Let $B\subset\mathbb{R}^2$ be the unit ball, with volume $\omega_n=|B|$, and let $P(\Omega)$ denote the De Giorgi perimeter of a set Ω. For an open, bounded, convex set...

conjecture | open | MathDB 页面 | 原文线索

V2-026207 | MATHDB-350619

Embedding conjecture for null pseudoconvex strongly causal spacetimes

摘录 / EXCERPT: Embedding conjecture. Every null pseudoconvex strongly causal spacetime can be conformally embedded as an open subset into some globally hyperbolic spacetime.

conjecture | open | MathDB 页面 | 原文线索

V2-026208 | MATHDB-350626

Weighted quermassintegral conjecture for static convex domains

摘录 / EXCERPT: Weighted quermassintegral conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-026209 | MATHDB-350627

Weighted Alexandrov-Fenchel conjecture for star-shaped k-convex domains

摘录 / EXCERPT: Let $0\leq k\leq n$ and $0\leq m\leq k$. Let Ω be a star-shaped and k-convex domain in $\mathbb H^{n+1}$ with smooth boundary M. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026210 | MATHDB-350673

Hausdorff-boundedness and Wasserstein-realizability of Gromov-Wasserstein geodesics

摘录 / EXCERPT: Let $p\in[1,\infty)$, and let an ℓ^p-Gromov-Wasserstein geodesic be a geodesic for the ℓ^p-Gromov-Wasserstein distance. A geodesic is Hausdorff-bounded if it satisfies th...

conjecture | open | MathDB 页面 | 原文线索

V2-026211 | MATHDB-350674

Three-space common ambient realization conjecture for Gromov-Hausdorff distance

摘录 / EXCERPT: Let $\mathcal{M}$ denote the collection of compact metric spaces. Given three compact metric spaces $X_1,X_2,X_3\in\mathcal{M}$, an isometric embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-026212 | MATHDB-350675

Three-space compatible optimal correspondences conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ denote the collection of compact metric spaces, and for $i,j\in\{1,2,3\}$ let $e_{ij}:X_1\times X_2\times X_3\to X_i\times X_j$ be the coordinate projection. A co...

conjecture | open | MathDB 页面 | 原文线索

V2-026213 | MATHDB-350690

Conjecture on stability of near-class Type-II mean curvature flow

摘录 / EXCERPT: Near-class stability conjecture. Mean curvature flow solutions starting from initial data sets in the near class have the same singular behavior as the unperturbed solution, their...

conjecture | open | MathDB 页面 | 原文线索

V2-026214 | MATHDB-350691

Conjecture on critical-class Type-II degenerate neckpinches

摘录 / EXCERPT: Critical-class neckpinch conjecture. A solution in the critical class develops a local Type-II degenerate neckpinch whose precise asymptotics are obtained in Angenent and Velázquez...

conjecture | open | MathDB 页面 | 原文线索

V2-026215 | MATHDB-350699

Bourgain--Demeter's additive-energy conjecture for the 2-sphere

摘录 / EXCERPT: Let $\mathcal{S}_{3,1}=\{\bm{x}\in\mathbb{R}^3:x_1^2+x_2^2+x_3^2=1\}$, and let $E_{2,2}(A)$ denote the number of quadruples $(\bm{x}_1,\bm{x}_2,\bm{x}_3,\bm{x}_4)\in A^4$ satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-026216 | MATHDB-350704

Non-vanishing conjecture for hyperkähler manifolds

摘录 / EXCERPT: Non-vanishing conjecture. One has $\kappa(L)\geq 0$.

conjecture | solved | MathDB 页面 | 原文线索

V2-026217 | MATHDB-350706

Finiteness conjecture for Kauffman bracket skein modules of manifolds with boundary

摘录 / EXCERPT: Finiteness conjecture. Let M be a compact oriented 3-manifold. Then $\mathcal{S}(M)$ is a finitely generated $\mathcal{S}(partial M, \mathbb{Q}(A))$-module.

conjecture | open | MathDB 页面 | 原文线索

V2-026218 | MATHDB-350738

Alesker-Verbitsky conjecture for the quaternionic Monge-Ampère equation

摘录 / EXCERPT: Alesker-Verbitsky conjecture. For every such f, the quaternionic Monge-Ampère equation admits a unique smooth solution ϕ, up to addition of a constant.

conjecture | open | MathDB 页面 | 原文线索

V2-026219 | MATHDB-350745

Leaf-wise intersection conjecture for special hypersurfaces in weakly monotone toric line bundles

摘录 / EXCERPT: Let E be a weakly monotone toric negative line bundle with negativity constant k. Let $\mathfrak{p}:E\to M$ be the bundle projection, and suppose the action of...

conjecture | open | MathDB 页面 | 原文线索

V2-026220 | MATHDB-350750

Brakke's multiplicity-one conjecture for mean curvature flow

摘录 / EXCERPT: Let the initial surface be closed and embedded, and consider the associated flow through singularities. Brakke's multiplicity-one conjecture. All blowup limits have multiplicity on...

conjecture | open | MathDB 页面 | 原文线索

V2-026221 | MATHDB-350751

The no-cylinder conjecture for embedded self-shrinkers

摘录 / EXCERPT: A self-shrinker is a complete embedded shrinker for mean curvature flow. The no-cylinder conjecture. The only complete embedded shrinker with a cylindrical end is the round cylinde...

open_problem | open | MathDB 页面 | 原文线索

V2-026222 | MATHDB-350752

Uniqueness conjecture for tangent flows

摘录 / EXCERPT: Let tangent flows be obtained by rescaling a mean curvature flow at a singularity. The uniqueness conjecture for tangent flows. Tangent flows are independent of the choice of seque...

conjecture | open | MathDB 页面 | 原文线索

V2-026223 | MATHDB-350753

Perelman's bounded diameter conjecture for mean curvature flow

摘录 / EXCERPT: Consider a mean curvature flow approaching its first singular time. Perelman's bounded diameter conjecture. The intrinsic diameter stays uniformly bounded as one approaches the fir...

conjecture | open | MathDB 页面 | 原文线索

V2-026224 | MATHDB-350754

Genericity conjecture for neck and spherical singularities

摘录 / EXCERPT: For an initial surface, call the data generic when they form an open and dense set. The genericity conjecture. For generic initial data all singularities are of neck-type or spheri...

conjecture | open | MathDB 页面 | 原文线索

V2-026225 | MATHDB-350755

Partial regularity conjecture for mean curvature flow

摘录 / EXCERPT: Let the initial surface be closed and embedded, and let the singular set be considered with its parabolic metric. The partial regularity conjecture. The parabolic Hausdorff dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-026226 | MATHDB-350756

Finiteness conjecture for singular times in mean curvature flow

摘录 / EXCERPT: Consider a mean curvature flow with its singular times recorded in the time interval of existence. The finiteness conjecture for singular times. There are only finitely many singul...

conjecture | open | MathDB 页面 | 原文线索

V2-026227 | MATHDB-350757

Two-sphere conjecture for mean curvature flow

摘录 / EXCERPT: Let the initial surface be an embedded two-sphere in $\mathbb{R}^3$. The two-sphere conjecture. Mean curvature flow of embedded two-spheres is well-posed. The source states that, u...

conjecture | open | MathDB 页面 | 原文线索

V2-026228 | MATHDB-350758

Lusternik-Schnirelman conjecture for minimal two-spheres

摘录 / EXCERPT: Let the three-sphere be equipped with an arbitrary Riemannian metric. The Lusternik-Schnirelman type conjecture. The three-sphere equipped with any Riemannian metric contains at le...

conjecture | open | MathDB 页面 | 原文线索

V2-026229 | MATHDB-350779

Kronheimer-Ricci-flat metric coincidence conjecture for the resolved threefold

摘录 / EXCERPT: Kronheimer-Ricci-flat coincidence conjecture. The two metrics are different on $Y^\Gamma_{[3]}$, but they coincide on the exceptional divisor $\mathcal{ED}$.

conjecture | open | MathDB 页面 | 原文线索

V2-026230 | MATHDB-350783

Mustăță-Nakamura bounded blow-up conjecture for minimal log discrepancies

摘录 / EXCERPT: Mustăță-Nakamura conjecture. Fix N and the exponent e of real ideals. Then, there exists a number $\ell_{N,e}\in{\mathbb N}$ depending only on N and e such that for any rea...

conjecture | open | MathDB 页面 | 原文线索

V2-026231 | MATHDB-350788

The local constancy conjecture for automorphic categories in a family of curves

摘录 / EXCERPT: Let M be a real manifold, let $\pi_M:\mathfrak{X}_M\to M$ be the induced family of curves, and let $s\in M$. Write ... where $\widetilde{\mathcal{N}}_{\Pi_M}$ is the universal nilp...

conjecture | open | MathDB 页面 | 原文线索

V2-026232 | MATHDB-350822

Finiteness criterion for affinoid Liu algebras

摘录 / EXCERPT: Finiteness criterion for affinoid Liu algebras. The following assertions should hold: (i) $\calA$ is affinoid if and only if $\tilde{\calA}_{\mathrm{gr}}$ is a finitely generated...

conjecture | open | MathDB 页面 | 原文线索

V2-026233 | MATHDB-350831

Wang's polynomial-growth conjecture for fundamental groups of klt pairs

摘录 / EXCERPT: Let (X,Δ) be a projective klt pair with nef anti-log canonical divisor $-(K_X+\Delta)$. Here X_{reg} denotes the regular locus of X, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026234 | MATHDB-350850

Completeness conjecture for the sequence pseudometric

摘录 / EXCERPT: Let (X,d) be an unbounded metric space and let $\tilde{r}=(r_n)_{n\in\mathbb{N}}$ be a scaling sequence. Define ... for $\tilde{x},\tilde{y}\in Seq(X,\tilde{r})$; this is a pseudom...

conjecture | open | MathDB 页面 | 原文线索

V2-026235 | MATHDB-350851

Conjecture on asymptotic isometry and asymptotic equivalence

摘录 / EXCERPT: Let Y and Z be unbounded metric spaces, and let $\tilde{r}=(r_n)_{n\in\mathbb{N}}$ be a scaling sequence. For a metric space W, write $Seq(W,\tilde{r})$ for its sequence spac...

conjecture | open | MathDB 页面 | 原文线索

V2-026236 | MATHDB-350864

Rational barycentric coordinates imply elliptic triangle-center loci

摘录 / EXCERPT: Let a triangle center be defined by barycentric coordinates that are rational functions of the squares of the triangle's sidelengths, and consider its locus over the relevant famil...

conjecture | open | MathDB 页面 | 原文线索

V2-026237 | MATHDB-350865

Convexity and symmetry conjecture for the incenter locus

摘录 / EXCERPT: Let a pair of ellipses admit Poncelet 3-periodics, and let X_1 denote the incenter of a member of this family. If the pair is concentric at O, call O the common center; other...

conjecture | open | MathDB 页面 | 原文线索

V2-026238 | MATHDB-350883

Effective maximum-principle conjecture for eigenfunctions

摘录 / EXCERPT: Effective maximum-principle conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-026239 | MATHDB-350951

Completeness conjecture for Kähler-Einstein metrics near isolated log canonical singularities

摘录 / EXCERPT: Completeness conjecture. The metric g_{KE} is complete near p; more precisely, there is an open neighborhood $V_{q,R}$ of p such that for every $x\in V_{q,R}\setminus\{p\}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-026240 | MATHDB-350971

Franke-Muthiah conjecture on two-dimensional convex embeddings

摘录 / EXCERPT: Let $\mathcal{C}$ be a code. Its minimal convex embedding dimension is the least dimension in which it admits a convex realization, and its minimal open convex embedding dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-026241 | MATHDB-351001

Global Zariski main theorem for quasi-finite morphisms of non-Archimedean analytic spaces

摘录 / EXCERPT: Let k be a complete non-Archimedean valued field, and let $f:X\rightarrow S$ be a quasi-finite morphism of quasi-compacted, separated k-analytic spaces. We seek a factorization...

open_problem | open | MathDB 页面 | 原文线索

V2-026242 | MATHDB-351106

Betke-Weil reverse Alexandrov-Fenchel conjecture for mixed volumes

摘录 / EXCERPT: Betke-Weil conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026243 | MATHDB-351119

Roots-of-unity conjecture for the three-variable power-sum system

摘录 / EXCERPT: Assume $n=3$, $\gcd(a_1,a_2,a_3)=1$, and let $\mathcal{A}_p=\{a_1\bmod p,a_2\bmod p,a_3\bmod p\}$ for $p=2,3$. Let ζ be a primitive cube root of unity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026244 | MATHDB-351130

L-space conjecture for detected slopes

摘录 / EXCERPT: Let M_1 and M_2 be knot manifolds, and let $W=M_1\cup_f M_2$ be their union along their torus boundaries. For $*\in\{LO,CTF\}$, a rational slope is $*$-detected when it has the...

conjecture | open | MathDB 页面 | 原文线索

V2-026245 | MATHDB-351131

The generalized torsion characterization of bi-orderable 3-manifold groups

摘录 / EXCERPT: Let G be the fundamental group of a 3-manifold. A non-trivial element $g\in G$ is a generalized torsion element if some non-empty finite product of conjugates of g equals the...

conjecture | open | MathDB 页面 | 原文线索

V2-026246 | MATHDB-351144

Stability conjecture for higher Laplace eigenvalues on the sphere

摘录 / EXCERPT: Let g_0 be a round metric of curvature one on $\mathbb{S}^2$. Given $k\geq 2$, let μ be an admissible measure such that $\lambda_k(\mathbb{S}^2,\mu)=2$. Then there exists a c...

conjecture | open | MathDB 页面 | 原文线索

V2-026247 | MATHDB-351170

Maulik-Nekrasov-Okounkov-Pandharipande relative GW/DT correspondence

摘录 / EXCERPT: Let X be a smooth projective threefold with a smooth divisor D, let $\beta\in H_2(X,\mathbb{Z})$, and let η be a relative partition. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-026248 | MATHDB-351183

Homeomorphism and codimension conjecture for Kähler-Ricci flow limits

摘录 / EXCERPT: Let M^m be a compact Kähler manifold with semiample canonical bundle and Kodaira dimension $n=\kappa(M)$ satisfying $0<n<m$. Let $f:M\to N$ be its Iitaka fibration onto a normal...

conjecture | open | MathDB 页面 | 原文线索

V2-026249 | MATHDB-351191

The mapping-content factorization conjecture through trees

摘录 / EXCERPT: Mapping-content factorization conjecture. If $\mathcal{H}^{2,m}_\infty(f,Q_0)=0$, then f factors through a tree. Moreover, for every $\epsilon>0$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-026250 | MATHDB-351198

The semistability conjecture for tangent bundles of Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold with Picard number one. The semistability conjecture. The tangent bundle TX is semistable. This conjecture was disproved by Kanemitsu, who found smooth...

conjecture | refuted | MathDB 页面 | 原文线索

V2-026251 | MATHDB-351199

The simplicity conjecture for tangent bundles of Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold with Picard number one. A vector bundle is simple if its only endomorphisms are scalar multiplications. The simplicity conjecture. The tangent bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-026252 | MATHDB-351200

The unbendable-rational-curves conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold with Picard number one. A rational curve $f:\mathbb P^1\to X$ is unbendable when f^*TX decomposes as ... for some nonnegative integers p,q. The unbenda...

open_problem | open | MathDB 页面 | 原文线索

V2-026253 | MATHDB-351213

The positive mass conjecture for asymptotically flat gravitational instantons

摘录 / EXCERPT: Positive mass conjecture. To any AF gravitational instanton $(M,\mathbf{g})$, one can assign a geometric invariant called the mass m, coinciding with the invariant m for toric...

conjecture | open | MathDB 页面 | 原文线索

V2-026254 | MATHDB-351245

Clopen predicate conjecture for d-minimal expansions of the real field

摘录 / EXCERPT: Let $\mathcal R$ be a d-minimal expansion of $\bar{\mathbb{R}}$. Let $X \subseteq \mathbb{R}^n$ be a manifold definable in $\mathcal R$, and let $Y \subseteq X$ be clopen in X. C...

conjecture | open | MathDB 页面 | 原文线索

V2-026255 | MATHDB-351255

Burns-Katok marked length spectrum rigidity conjecture

摘录 / EXCERPT: Burns-Katok conjecture. The marked length spectrum map L is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-026256 | MATHDB-351265

Non-recovery conjecture for geometric planted matchings

摘录 / EXCERPT: Let n be the number of points, d the dimension, σ^2 the noise parameter, and $\mathcal{E}$ the set of errors made by the maximum-likelihood estimator. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-026257 | MATHDB-351269

Non-vanishing conjecture for Mellin transformations

摘录 / EXCERPT: Let $U\subset X$ be complex manifolds satisfying the conditions in Theorem. Let $\mathcal F\in D^b_c(X,A)$ be a constructible complex such that $\mathcal F|_U\neq 0$. Non-vanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-026258 | MATHDB-351279

Irreducible non-realizability conjecture for spherical occlusion diagrams

摘录 / EXCERPT: Irreducible non-realizability conjecture. There is an irreducible SOD satisfying the axioms that is not the visibility map $S_\mathcal P$ of any polyhedron $\mathcal P$ with respec...

conjecture | open | MathDB 页面 | 原文线索

V2-026259 | MATHDB-351280

Two-clockwise-and-two-counterclockwise swirls conjecture for spherical occlusion diagrams

摘录 / EXCERPT: Two-clockwise-and-two-counterclockwise swirls conjecture. Every SOD has at least two clockwise and two counterclockwise swirls.

conjecture | open | MathDB 页面 | 原文线索

V2-026260 | MATHDB-351298

Perverse-sheaf and BV-quantization conjecture for shifted symplectic stacks

摘录 / EXCERPT: Perverse-sheaf and BV-quantization conjecture. There is a canonical BV quantization $\Delta_\hbar$ of $\det(\mathbb{L}_\mathsf{X})^{\frac12}$ and a quasi-isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-026261 | MATHDB-351337

The taut-foliation extension conjecture for punctured 3-manifolds

摘录 / EXCERPT: Let M be a closed orientable irreducible 3-manifold such that $\pi_1(M)$ is left orderable, and let $M_0=M-\operatorname{Int}(B^3)$. Let $\mathcal{F}$ be a foliation of M_0 c...

conjecture | open | MathDB 页面 | 原文线索

V2-026262 | MATHDB-351343

A dimension bound for circle actions with an odd number of fixed points

摘录 / EXCERPT: Dimension-bound conjecture. The dimension of M is bounded above by a function of k; more precisely,

conjecture | open | MathDB 页面 | 原文线索

V2-026263 | MATHDB-351344

Möbius uniqueness conjecture for edge-scribable polytopes

摘录 / EXCERPT: Let $d\ge 2$ and let P be an edge-scribable $(d+1)$-polytope. A polytope is Möbius unique if its edge-scribed realizations are equivalent under Möbius transformations. Möbius uni...

conjecture | open | MathDB 页面 | 原文线索

V2-026264 | MATHDB-351364

Haefliger-type conjecture for coupled embeddings

摘录 / EXCERPT: Let M and N be smooth closed manifolds of dimensions p and q, respectively. Write ... A $(\mathbb{Z}/2)^2$-equivariant map is understood with respect to the involutions excha...

conjecture | open | MathDB 页面 | 原文线索

V2-026265 | MATHDB-351375

The Blaschke-Santaló bounds for the Steklov-Neumann functional

摘录 / EXCERPT: Blaschke-Santaló bounds. For every such Ω,

conjecture | open | MathDB 页面 | 原文线索

V2-026266 | MATHDB-351376

Nonattainment and collapsing rectangles for the infimum of the Steklov-Neumann functional

摘录 / EXCERPT: Nonattainment conjecture. The displayed minimization problem has no solution. In particular, every minimizing sequence Ω_ϵ must be of the form of collapsing rectangle...

conjecture | open | MathDB 页面 | 原文线索

V2-026267 | MATHDB-351377

Rectangular minimizers among convex quadrilaterals

摘录 / EXCERPT: Rectangular minimizer conjecture. For every $0<C\leq 4\sigma_1(\mathbb{S})$, the solution of this minimization problem is given by a rectangle.

conjecture | open | MathDB 页面 | 原文线索

V2-026268 | MATHDB-351390

Constant linear-combination conjecture for the symplectic forms on the moduli space of connections

摘录 / EXCERPT: Let ${\mathcal H}_g(r)$ be the moduli space of the vector-bundle and connection data considered in the paper, equipped with the holomorphic symplectic form from Corollary (1). Th...

conjecture | open | MathDB 页面 | 原文线索

V2-026269 | MATHDB-351396

Braun-Etzion-Vardy conjecture on the maximum size of binary linear projective-space codes

摘录 / EXCERPT: Let $\mathbb{P}_2(n)$ be the projective space consisting of the subspaces of $\mathbb{F}_2^n$, and let a linear code be a subset of $\mathbb{P}_2(n)$ endowed with the linear struct...

conjecture | open | MathDB 页面 | 原文线索

V2-026270 | MATHDB-351436

Equidistribution conjecture for stationary geodesic nets

摘录 / EXCERPT: Let M be a closed manifold, let g be a Riemannian metric on M, and let $\gamma_i:\Gamma_i\rightarrow M$ be stationary geodesic nets. A sequence of such nets is equidistribute...

conjecture | open | MathDB 页面 | 原文线索

V2-026271 | MATHDB-351440

Conjecture on Kähler-Ricci flow near deformed complex structures

摘录 / EXCERPT: Let (M,J_0) be a Fano manifold admitting a Kähler-Ricci soliton $\omega_{KS}\in 2\pi c_1(M,J_0)$. Let H_1 be the operator whose negative eigenspaces define the linear subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-026272 | MATHDB-351442

Chern-Ricci flow convergence conjecture for manifolds of general type

摘录 / EXCERPT: Let M^n be a compact complex manifold with nef canonical bundle K_M and Kodaira dimension $\mathrm{Kod}(M)=n$. Let $\omega(t)$ solve the Chern-Ricci flow starting at an arbitra...

conjecture | open | MathDB 页面 | 原文线索

V2-026273 | MATHDB-351443

Smooth convergence conjecture for the Chern-Ricci flow on elliptic bundles

摘录 / EXCERPT: Let $f:M\to\Sigma$ be an elliptic bundle over a Riemann surface Σ of genus at least 2, and let $\omega(t)$ solve the Chern-Ricci flow on M from an arbitrary Gauduchon me...

conjecture | open | MathDB 页面 | 原文线索

V2-026274 | MATHDB-351453

The surface-construction conjecture for filtered deformations of elliptic algebras

摘录 / EXCERPT: Let $B_{V,\Psi}$ be an elliptic algebra, and let a filtered deformation mean a nontrivial filtered deformation of $B_{V,\Psi}$. Let A^+_M denote the Rees algebra associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-026275 | MATHDB-351454

The trivial-center-deformation conjecture for untwisted elliptic algebras

摘录 / EXCERPT: Let $B_{V,\Psi}$ be an untwisted elliptic algebra, and let its center be denoted by $Z(B_{V,\Psi})$. A filtered deformation induces a trivial deformation of the center when the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-026276 | MATHDB-351473

Popov's nonconjugate Borel subgroups conjecture for the plane Cremona group

摘录 / EXCERPT: Let $n\geq 5$. The Cremona group ${\rm Bir}({\mathbb P}^n)$ is the group of birational transformations of complex projective n-space, and a Borel subgroup is a maximal closed con...

conjecture | open | MathDB 页面 | 原文线索

V2-026277 | MATHDB-351474

Nonconjugate Borel subgroups conjecture for Cremona groups in dimension at least two

摘录 / EXCERPT: Let $n\geq 2$. The Cremona group ${\rm Bir}({\mathbb P}^n)$ consists of birational transformations of complex projective n-space, and a Borel subgroup is a maximal closed connect...

conjecture | open | MathDB 页面 | 原文线索

V2-026278 | MATHDB-351479

The support-cone graze characterization of ellipsoids

摘录 / EXCERPT: Let $K,L\subset\mathbb R^n$ be convex bodies with $L\subset\operatorname{int}K$. For each $x\in\partial K$, let $\Sigma(L,x)$ denote the graze of L from x, namely the intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-026279 | MATHDB-351480

The centroid-projection symmetry conjecture

摘录 / EXCERPT: Let $K\subset\mathbb R^n$ be a convex body, with $n\geq 3$, and let $q\in\operatorname{int}K$. For each $u\in\mathbb S^{n-1}$, let π_u denote orthogonal projection in direction...

conjecture | open | MathDB 页面 | 原文线索

V2-026280 | MATHDB-351495

Cornea's convergence conjecture for Lagrangian submanifolds

摘录 / EXCERPT: Fix a connected symplectic manifold (M,ω) that is closed or convex at infinity, an ω-compatible almost complex structure J such that g_J has uniformly bounded sec...

conjecture | open | MathDB 页面 | 原文线索

V2-026281 | MATHDB-351505

Fukui-Kurdyka-Paunescu conjecture on real analytic multiplicity modulo 2

摘录 / EXCERPT: Let $X,Y\subset\mathbb{R}^n$ be two germs at the origin of irreducible real analytic subsets. Let $h\colon(\mathbb{R}^n,0)\to(\mathbb{R}^n,0)$ be a germ of a subanalytic, arc-analy...

conjecture | open | MathDB 页面 | 原文线索

V2-026282 | MATHDB-351509

Anisotropic curvature flow convergence conjecture

摘录 / EXCERPT: Let ϕ be a smooth, strictly convex norm on $\mathbf{R}^{n+1}$, with Wulff shape $\mathcal{W}^\phi=\{x\in\mathbf{R}^{n+1}:\phi^*(x)=1\}$. Let $X_0:M\to\mathbf{R}^{n+1}$ be a cl...

conjecture | open | MathDB 页面 | 原文线索

V2-026283 | MATHDB-351510

Anisotropic curvature-measure characterization of Wulff shapes

摘录 / EXCERPT: Let ϕ be the norm defining the anisotropic curvature measures $\mathcal{C}_m^\phi(C,\cdot)$ of a convex body $C\subseteq\mathbf{R}^{n+1}$. Anisotropic curvature-measure charac...

conjecture | open | MathDB 页面 | 原文线索

V2-026284 | MATHDB-351511

Doubly distributive hyperfields satisfy the multiplicity bound

摘录 / EXCERPT: Multiplicity-bound conjecture. All hyperfields with the doubly distributive property satisfy the multiplicity bound.

conjecture | open | MathDB 页面 | 原文线索

V2-026285 | MATHDB-351512

Doubly distributive hyperfields satisfy the inheritance property

摘录 / EXCERPT: Inheritance-property conjecture. All hyperfields with the doubly distributive property satisfy the inheritance property.

conjecture | open | MathDB 页面 | 原文线索

V2-026286 | MATHDB-351557

Simplicity conjecture for canonical quantizations

摘录 / EXCERPT: Simplicity conjecture. The algebra $\cA_0$ is simple.

conjecture | open | MathDB 页面 | 原文线索

V2-026287 | MATHDB-351571

The critical-dimension crossover conjecture for parabolic Anderson valleys

摘录 / EXCERPT: Let u be the solution of the parabolic Anderson equation with multiplicative noise, started from bounded initial data u_0. For β as in the definition of the macroscopic H...

conjecture | open | MathDB 页面 | 原文线索

V2-026288 | MATHDB-351597

Conjectural volume bounds for toric Gorenstein singularities

摘录 / EXCERPT: Let an n-dimensional toric Gorenstein singularity be associated with a polytope Δ_{n-1}, either reflexive or non-reflexive. Let Y be the associated Sasaki-Einstein base...

conjecture | open | MathDB 页面 | 原文线索

V2-026289 | MATHDB-351603

The bic-II stationary-locus criterion for conic loci of triangle centers

摘录 / EXCERPT: Let X be a triangle center that does not always lie on a triangle's circumcircle. Consider the bic-II family and its bic-I (poristic) subfamily. A locus is stationary when the co...

conjecture | open | MathDB 页面 | 原文线索

V2-026290 | MATHDB-351604

The bic-III convexity conjecture for the incenter locus

摘录 / EXCERPT: Let $P_1P_2P_3$ be a bic-III family of triangles inscribed in an outer circle $\mathcal{C}$, with P_1P_2 and P_1P_3 tangent to distinct circles $\mathcal{C}'$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026291 | MATHDB-351605

The bic-III non-conic conjecture for triangle-center loci

摘录 / EXCERPT: Let $P_1P_2P_3$ be a bic-III family of triangles inscribed in an outer circle $\mathcal{C}$, with each side tangent to a distinct in-pencil circle, and assume all four in-pencil ci...

conjecture | open | MathDB 页面 | 原文线索

V2-026292 | MATHDB-351607

Munteanu-Sung-Wang curvature-decay dichotomy for steady gradient Ricci solitons

摘录 / EXCERPT: Let (M,g,f) be a complete non Ricci flat steady gradient Ricci soliton with $|\operatorname{Rm}|\to 0$ as $x\to\infty$, where r denotes the distance function on M. Munteanu-S...

conjecture | open | MathDB 页面 | 原文线索

V2-026293 | MATHDB-351612

Sparse approximation conjecture for John decompositions

摘录 / EXCERPT: Let $u_1,\ldots,u_n\in S^{d-1}$ be unit vectors for which there exist positive numbers $\alpha_1,\ldots,\alpha_n$ satisfying ... where I_d is the identity operator on...

conjecture | open | MathDB 页面 | 原文线索

V2-026294 | MATHDB-351619

Sullivan's minimal Betti number conjecture for closed complex manifolds

摘录 / EXCERPT: Let M be a closed complex manifold of real dimension $2n$, and let the total Betti number of M mean the sum of all its Betti numbers. Sullivan's conjecture. The minimal total B...

conjecture | open | MathDB 页面 | 原文线索

V2-026295 | MATHDB-351625

Simplex fair-cut index conjecture

摘录 / EXCERPT: Let Δ_n be a simplex with nonempty interior in Euclidean space $\mathbb R^m$, and let $\Phi(\Delta_n)$ denote its fair-cut index. Simplex fair-cut index conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-026296 | MATHDB-351628

The homogeneous harmonic-curvature conjecture

摘录 / EXCERPT: Homogeneous harmonic-curvature conjecture. Any homogeneous Riemannian manifold M with harmonic curvature is Ricci-parallel.

conjecture | open | MathDB 页面 | 原文线索

V2-026297 | MATHDB-351631

The Suzuki-group construction has edge-girth parameter q-1

摘录 / EXCERPT: The edge-girth conjecture. The family of graphs $\mathcal G_q(V,E)$ is a family of

conjecture | open | MathDB 页面 | 原文线索

V2-026298 | MATHDB-351665

AOUV's two Gauduchon connections conjecture

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold, and let D^r and $D^{r'}$ be Gauduchon connections for real parameters r and r'. A connection is Kähler-like when its curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-026299 | MATHDB-351666

Zhao-Zheng's two canonical connections conjecture for different Hermitian metrics

摘录 / EXCERPT: Let (M^n,g,g') be a compact Hermitian manifold with possibly different Hermitian metrics g and g'. Let D^r_s be the canonical metric connection associated with g, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026300 | MATHDB-351667

The pedal-polygon equivalence conjecture for n-gons

摘录 / EXCERPT: Pedal-polygon equivalence conjecture. For integer $n\ge 5$, if $G(n)$ is the set of n-gons, then

conjecture | open | MathDB 页面 | 原文线索

V2-026301 | MATHDB-351692

The Fock-Goncharov basis conjecture for upper cluster algebras

摘录 / EXCERPT: Fock-Goncharov conjecture. The upper cluster algebra $\mathcal{U}$ possesses a basis parametrized by the tropical points.

conjecture | open | MathDB 页面 | 原文线索

V2-026302 | MATHDB-351706

Geometric P=W conjecture for Lagrangian tori

摘录 / EXCERPT: Let $\mathcal{X}\xymatrix@1@=15pt{\ar[r]&}\Delta$ be a projective minimal divisorial log terminal type III degeneration of hyperkähler manifolds, let $\mathcal{X}_t$ be a smooth fi...

conjecture | open | MathDB 页面 | 原文线索

V2-026303 | MATHDB-351720

Detcherry-Kalfagianni simplicial-volume conjecture for Turaev-Viro invariants

摘录 / EXCERPT: Let M be a compact and orientable 3-manifold with empty or toroidal boundary. For odd integers r, set ... Let $v_3\approx 1.0149$ be the volume of a regular ideal tetrahedron,...

conjecture | open | MathDB 页面 | 原文线索

V2-026304 | MATHDB-351743

Weak Dirac conjecture on ordinary lines

摘录 / EXCERPT: Weak Dirac conjecture. Every set $\mathcal{P}$ of n non-collinear points in the plane contains a point incident to at least

conjecture | open | MathDB 页面 | 原文线索

V2-026305 | MATHDB-351744

The exact linear bound conjecture for few-angle point sets

摘录 / EXCERPT: Exact linear bound conjecture. The lower bound on $P(k)$ in Theorem is tight. Namely,

conjecture | open | MathDB 页面 | 原文线索

V2-026306 | MATHDB-351760

Aubin's conjecture on isoperimetric inequalities in nonpositive curvature

摘录 / EXCERPT: Let (M^m,g) be a smooth complete simply-connected Riemannian manifold with sectional curvature at most κ, where $\kappa\leq 0$. For a bounded domain with smooth boundary i...

conjecture | open | MathDB 页面 | 原文线索

V2-026307 | MATHDB-351766

Integral monodromy conjecture for singular special Kähler metrics

摘录 / EXCERPT: Let d denote the dimension parameter in the collapsing hyperkähler construction, and suppose the associated singular special Kähler metric arises under the uniform energy bound c...

conjecture | open | MathDB 页面 | 原文线索

V2-026308 | MATHDB-351781

A universal linear lower bound for sphere self-intersections

摘录 / EXCERPT: Universal linear bound conjecture. There is a universal constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-026309 | MATHDB-351787

Payne's nodal line conjecture for planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a bounded domain, and consider the second eigenfunction of the Laplacian on Ω with Dirichlet boundary condition. Payne's nodal line conject...

conjecture | open | MathDB 页面 | 原文线索

V2-026310 | MATHDB-351794

Liquid and gas contributions determined by amoeba geometry

摘录 / EXCERPT: Let $P(z,w)=k-p(z,w)$ be a Newton polynomial with fixed $p(z,w)$. Let $m_l(P)$ and $m_g(P)$ denote the liquid and gas contributions to the Mahler measure, and let $\mathcal{A}(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-026311 | MATHDB-351801

Non-Archimedean regularization conjecture for entropy and energy

摘录 / EXCERPT: Let X_{NA} be the non-Archimedean analytification, let $\mathcal{E}^{1}(X_{NA})$ be the finite-energy space, let $\mathcal{H}(X_{NA})$ be the space of non-Archimedean model poten...

conjecture | open | MathDB 页面 | 原文线索

V2-026312 | MATHDB-351803

Palindromicity conjecture for twisted Neumann-Zagier determinant polynomials

摘录 / EXCERPT: Palindromicity conjecture. The Laurent polynomials $\det(\mathbf{A}(t))$ and $\det(\mathbf{B}(t))$ are palindromic.

conjecture | open | MathDB 页面 | 原文线索

V2-026313 | MATHDB-351860

The torus-fibration conjecture for closed manifolds with flat degenerate metrics

摘录 / EXCERPT: Torus-fibration conjecture. M is finitely covered by a manifold diffeomorphic to a fiber bundle over an m-dimensional torus.

conjecture | open | MathDB 页面 | 原文线索

V2-026314 | MATHDB-351903

Doran-Thompson mirror Clemens-Schmid conjecture

摘录 / EXCERPT: Let X be an n-dimensional Calabi-Yau manifold undergoing a Type N degeneration $\mathcal{X}\to\Delta$, where Type N means that the logarithm of monodromy has nilpotency ind...

conjecture | open | MathDB 页面 | 原文线索

V2-026315 | MATHDB-351911

Mellin--Barnes vanishing conjecture for non-representational taut angle structures

摘录 / EXCERPT: Let M be a connected, non-compact, orientable 3-manifold with one toroidal end, let $\mathcal{T}$ be an ideal triangulation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026316 | MATHDB-351917

Zariski-closedness conjecture for diminished base loci of klt pairs

摘录 / EXCERPT: Diminished-base-locus conjecture. The diminished base locus

conjecture | open | MathDB 页面 | 原文线索

V2-026317 | MATHDB-351918

Zariski-closedness conjecture for diminished base loci of generalized klt pairs

摘录 / EXCERPT: Generalized diminished-base-locus conjecture. The diminished base locus

conjecture | open | MathDB 页面 | 原文线索

V2-026318 | MATHDB-351939

Bianchi-Böröczky-Colesanti-Yang existence conjecture for the L_p-surface area measure

摘录 / EXCERPT: Bianchi-Böröczky-Colesanti-Yang conjecture. There exists a convex body K in $\mathbb{R}^n$ such that

conjecture | solved | MathDB 页面 | 原文线索

V2-026319 | MATHDB-351950

Greb-Kebekus conjecture on orbifold Chern classes and torus quotients

摘录 / EXCERPT: Greb-Kebekus conjecture. 1. $\mathrm{c}_{1}(X)=0\in \HH2.X.{\mathbb{R}}.$, and there exists a Kähler class $\alpha\in \HH2.X.{\mathbb{R}}.$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-026320 | MATHDB-351962

Arapura-Wang's non-negative Euler characteristic conjecture for perverse sheaves

摘录 / EXCERPT: Arapura-Wang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026321 | MATHDB-351969

The existence of an aspherical 4-manifold with Euler characteristic one

摘录 / EXCERPT: Aspherical 4-manifold Euler-characteristic problem. There exists an oriented closed aspherical 4-manifold M with

open_problem | open | MathDB 页面 | 原文线索

V2-026322 | MATHDB-352010

The variant of the Leinster-Willerton conjecture for magnitude and intrinsic volumes

摘录 / EXCERPT: Variant of the Leinster-Willerton conjecture. There are universal constants $\gamma_{0,n},\gamma_{1,n},\ldots,\gamma_{n,n}$ such that, for every such X,

conjecture | open | MathDB 页面 | 原文线索

V2-026323 | MATHDB-352021

Coalgebra-symmetry conjecture for Liouville integrability of Poisson maps

摘录 / EXCERPT: Let $T\colon n\mapsto n+1$ be a Poisson map admitting a coalgebra symmetry $(\mathfrak{A},\Delta)$. Let the generators of the coalgebra symmetry satisfy the closure relation referr...

conjecture | open | MathDB 页面 | 原文线索

V2-026324 | MATHDB-352044

Invariance conjecture for bordered manifolds with torus boundary

摘录 / EXCERPT: Let M be a bordered 3-manifold with n toroidal boundary components, equipped with a system of arcs $\mathscr{A}\subseteq M$. Let $\mathcal{X}(M,\mathscr{A})^{\mathcal{L}}$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-026325 | MATHDB-352053

Equatorial conjecture for equivariant minimal hyperspheres

摘录 / EXCERPT: Let $n\geq4$, let $G=\operatorname{O}(n)\times\operatorname{O}(n)$, and consider G-equivariant minimal embeddings of the sphere S^{2n-1} into the round sphere $\mathbb{S}^{2n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-026326 | MATHDB-352075

Optimality of the fairy bread sandwich theorem

摘录 / EXCERPT: Optimality conjecture. The result referred to as the fairy bread sandwich should be optimal for every such k, d, and permutation π.

conjecture | open | MathDB 页面 | 原文线索

V2-026327 | MATHDB-352105

Discreteness conjecture for immersed submanifolds with bounded mean curvature

摘录 / EXCERPT: Discreteness conjecture. Under either of these alternatives, the spectrum of the Laplace-Beltrami operator on M is discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-026328 | MATHDB-352137

Saint-Venant's conjecture for pentagons

摘录 / EXCERPT: Let Ω range over planar pentagons with a fixed area, and let $T(\Omega)$ denote their Torsional rigidity. Saint-Venant's conjecture. Among all pentagons of given area, the r...

conjecture | open | MathDB 页面 | 原文线索

V2-026329 | MATHDB-352145

Characterization of projective spaces by positivity of Chern characters

摘录 / EXCERPT: The projective-space characterization conjecture. Then $X\cong \mathbb{P}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-026330 | MATHDB-352146

Uniform log Yau-Tian-Donaldson conjecture for cscK cone metrics

摘录 / EXCERPT: Uniform log Yau-Tian-Donaldson conjecture. The polarised pair $((X,L);D)$ admits a cscK cone metric if and only if it is uniformly log K-stable.

conjecture | open | MathDB 页面 | 原文线索

V2-026331 | MATHDB-352152

The ellipse conjecture for interior points associated with circumcenters

摘录 / EXCERPT: Ellipse conjecture. The set of all interior points D of the triangle ABC satisfying this condition is an ellipse.

conjecture | open | MathDB 页面 | 原文线索

V2-026332 | MATHDB-352177

Support theorem conjecture for complementary-rank hyperbolic Radon transforms

摘录 / EXCERPT: Let $0\leq j<k\leq n-1$ satisfy $j+k=n-1$, and let $r>0$. Let $S(\Gamma_H(n,d))$ denote the space of rapidly decreasing smooth functions on the hyperbolic Grassmannian, with all de...

conjecture | open | MathDB 页面 | 原文线索

V2-026333 | MATHDB-352201

Two-ended odd-genus self-shrinker conjecture

摘录 / EXCERPT: For sufficiently large $n\in\mathbb{N}$, consider a complete, embedded, noncompact self-shrinker $\Phi_g\subset\mathbb{R}^3$ for mean curvature flow, where $g=2n-1$ denotes its gen...

conjecture | open | MathDB 页面 | 原文线索

V2-026334 | MATHDB-352209

The Poncelet directrix-envelope parabola conjecture

摘录 / EXCERPT: Let a Poncelet triangle family be inscribed in a circle, and consider the directrices of its isogonal circumparabolas (CPs). Directrix-envelope parabola conjecture. Over any Poncel...

conjecture | open | MathDB 页面 | 原文线索

V2-026335 | MATHDB-352210

The Poncelet Simson-line envelope point conjecture

摘录 / EXCERPT: Let a Poncelet triangle family be inscribed in a circle. Fix a point F on the circumcircle, let V be the perpendicular projection of F onto the Simson line of a member of the...

conjecture | open | MathDB 页面 | 原文线索

V2-026336 | MATHDB-352211

The Poncelet inellipse concentricity conjecture for Simson-line envelopes

摘录 / EXCERPT: Let a Poncelet triangle family be inscribed in a circle with inconic or caustic, and let F vary over the circumcircle. For each F, let W be the envelope point of the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-026337 | MATHDB-352255

Neumann-Pichon's conjecture on metrically conical complex analytic sets

摘录 / EXCERPT: Neumann-Pichon's conjecture. X is metrically conical at 0 if and only if X is locally metrically conical at 0.

conjecture | open | MathDB 页面 | 原文线索

V2-026338 | MATHDB-352278

Non-Archimedean SYZ fibration conjecture

摘录 / EXCERPT: Let $\mathcal{R}$ be a simple eigenray diagram and let $\mathcal{Y}_{\mathcal{R}}\to B_{\mathcal{R}}$ be the Stein continuous map asserted in the Stein mirror fibration conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-026339 | MATHDB-352299

Isotopy of orientation-preserving diffeomorphisms of the 4-sphere

摘录 / EXCERPT: Let $f\colon S^4\to S^4$ be an orientation-preserving self-diffeomorphism. Smale conjecture, special case. Every orientation-preserving self-diffeomorphism of S^4 is smoothly iso...

open_problem | open | MathDB 页面 | 原文线索

V2-026340 | MATHDB-352301

Infinite-volume composition conjecture for tg-hyperbolic virtual knots and links

摘录 / EXCERPT: Infinite-volume composition conjecture. There exists a sequence of compositions W_i such that

conjecture | open | MathDB 页面 | 原文线索

V2-026341 | MATHDB-352321

Izumiya's Lagrangian stability conjecture for generic Blaschke affine normal congruences

摘录 / EXCERPT: Izumiya's Lagrangian stability conjecture. Germs of generic Blaschke affine normal congruences at any point are Lagrangian stable.

conjecture | open | MathDB 页面 | 原文线索

V2-026342 | MATHDB-352339

The upper-bound conjecture for diameters of small resolutions of line arrangements

摘录 / EXCERPT: Diameter upper-bound conjecture. The number of diameters of $\mathcal R$ does not exceed

conjecture | open | MathDB 页面 | 原文线索

V2-026343 | MATHDB-352340

Finiteness conjecture for degree-one maps to four-manifolds with finite fundamental group

摘录 / EXCERPT: Finiteness conjecture. Every closed orientable 4-manifold X 1-dominates at most finitely many closed orientable 4-manifolds Y with finite $\pi_1(Y)$, up to homeomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-026344 | MATHDB-352342

Extension of Theorem 3 to non-normal monohedral tilings

摘录 / EXCERPT: Let K be a convex disc with a smooth boundary, and consider monohedral tilings of K. A monohedral tiling is non-normal when it does not satisfy the paper's normality condition....

conjecture | open | MathDB 页面 | 原文线索

V2-026345 | MATHDB-352381

Polar duality conjecture for position and momentum ellipsoids

摘录 / EXCERPT: Polar duality conjecture. The ellipsoids Ω_X and Ω_P form a dual pair:

conjecture | open | MathDB 页面 | 原文线索

V2-026346 | MATHDB-352403

Dumas-Sanders fiber-bundle conjecture for Kapovich-Leeb-Porti manifolds

摘录 / EXCERPT: Let S be the closed surface underlying the quasi-Hitchin representations under consideration, let $M_{\mathbb C}$ be the manifold associated with a quasi-Hitchin representation i...

conjecture | open | MathDB 页面 | 原文线索

V2-026347 | MATHDB-352464

Ciliberto-Ottaviani conjecture on the birationality of Hessian maps

摘录 / EXCERPT: Ciliberto-Ottaviani conjecture. The map $h_{d,n}$ induces a birational morphism onto its image when $d\geq 3$, $n\geq 2$, and $(d,n)\neq(3,2)$. This conjecture concerns the biratio...

conjecture | open | MathDB 页面 | 原文线索

V2-026348 | MATHDB-352526

Greb-Wong conjecture on Stein and affine canonical extensions

摘录 / EXCERPT: Let M be a compact Kähler manifold, and let $\alpha\in H^1(M,\Omega_M)$ be a Kähler class. Consider the extension ... and the associated canonical extension ... Here, T_M is the ta...

conjecture | open | MathDB 页面 | 原文线索

V2-026349 | MATHDB-352528

The torus quotient conjecture for Stein canonical extensions

摘录 / EXCERPT: Let M be a compact Kähler manifold that is not uniruled, and let Z_M be a canonical extension associated with a Kähler class on M. Torus quotient conjecture. If Z_M is Stei...

conjecture | open | MathDB 页面 | 原文线索

V2-026350 | MATHDB-352544

Finiteness conjecture for crepant-birational fibres of maximal-variation fibrations

摘录 / EXCERPT: Finiteness conjecture. There is an open set $U\subseteq Y$ such that for every $y\in U$ the set

conjecture | open | MathDB 页面 | 原文线索

V2-026351 | MATHDB-352558

The weak Whitney theorems for harmonic surfaces

摘录 / EXCERPT: Weak Whitney theorems. The space of harmonic immersions in $\operatorname{\mathfrak{M}}$ is open and dense, and connected if $\dim M\geq 5$; moreover, the space of harmonic embeddi...

conjecture | open | MathDB 页面 | 原文线索

V2-026352 | MATHDB-352569

Duality invariance conjecture for enumerative invariants on projective surfaces

摘录 / EXCERPT: Let X be a connected projective complex surface with $p_g>0$. For a class $\alpha\in C(\operatorname{Coh}(X))$ with $\operatorname{rank}\alpha>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-026353 | MATHDB-352578

Coherence conjecture for multiplier ideal sheaves of Griffiths semi-positive metrics

摘录 / EXCERPT: Coherence conjecture. If h is Griffiths semi-positive, then $\mathcal{E}(h)$ is coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-026354 | MATHDB-352583

Numerical nonvanishing conjecture for generalized polarized pairs

摘录 / EXCERPT: Numerical nonvanishing conjecture. There exists an effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D such that

conjecture | open | MathDB 页面 | 原文线索

V2-026355 | MATHDB-352602

The Realisation Conjecture for Jacobi algebras of noncommutative functions

摘录 / EXCERPT: The Realisation Conjecture. Every $f\in\mathbb C\langle\kern -2.5pt\langle \mathsf{x}\rangle\kern -2.5pt\rangle$ whose Jacobi algebra satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026356 | MATHDB-352646

Pentagram rigidity conjecture for convex Poncelet polygons

摘录 / EXCERPT: A convex Poncelet polygon is a polygon in the projective plane whose vertices lie on one conic and whose edges are tangent to another conic; convexity means convexity in an affine...

conjecture | open | MathDB 页面 | 原文线索

V2-026357 | MATHDB-352648

Integrability conjecture for invariant generalized complex structures on maximal real flag manifolds

摘录 / EXCERPT: Integrability conjecture. No maximal real flag manifold of type C_4 or D_4 admits integrable K-invariant generalized almost complex structures.

conjecture | open | MathDB 页面 | 原文线索

V2-026358 | MATHDB-352649

Classification conjecture for invariant generalized complex structures on real flag manifolds

摘录 / EXCERPT: Classification conjecture. A real flag manifold $\mathbb{F}_Theta=K/K_Theta$ admits a K-invariant generalized complex structure if and only if it is of type C_l and

conjecture | open | MathDB 页面 | 原文线索

V2-026359 | MATHDB-352656

The conjectural description of framed diffeomorphisms of the 3-torus

摘录 / EXCERPT: The 3-torus diffeomorphism conjecture. There are identifications among continuous groups:

conjecture | open | MathDB 页面 | 原文线索

V2-026360 | MATHDB-352661

The p coarse BaumConnes conjecture

摘录 / EXCERPT: Let X be a proper metric space, and let $\{C_k\}_{k=0}^\infty$ be an anti-Cech system of X. For each k, let N_{C_k} be the nerve space of C_k, and let $B_L^p(N_{C_k})$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-026361 | MATHDB-352673

Conjecture on higher Ekeland-Hofer capacities of symplectic p-products

摘录 / EXCERPT: The higher-capacity p-product conjecture. The displayed formula should hold for every such pair of star-shaped domains and every admissible p. The conjecture generalizes the know...

conjecture | open | MathDB 页面 | 原文线索

V2-026362 | MATHDB-352684

Alekseevskii's conjecture for homogeneous Einstein spaces

摘录 / EXCERPT: Let G/H be a connected homogeneous space of dimension n. An Einstein metric has Ricci tensor proportional to the metric, and negative scalar curvature means that this proportio...

conjecture | open | MathDB 页面 | 原文线索

V2-026363 | MATHDB-352699

The superadditivity conjecture for smooth loci of algebraic fiber spaces

摘录 / EXCERPT: The superadditivity conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-026364 | MATHDB-352733

Spectral classification conjecture for hyperbolic orbifolds

摘录 / EXCERPT: Spectral classification conjecture. The set E is bounded above by 44.8883537. The only orbifolds with $\ell_1>\ell_1^{(0)}$ are isometric to $[0;2,3,7]$, $[0;2,4,5]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026365 | MATHDB-352742

Budney-Gabai conjecture on non-isotopic compressing curve equivalent handlebodies

摘录 / EXCERPT: Let H_1 and H_2 be 3-dimensional genus-g handlebodies embedded in S^4, for an integer $g\geq 0$, with common boundary ... They are compressing curve equivalent if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-026366 | MATHDB-352744

Characterisation of extreme points of the Thurston-norm unit sphere

摘录 / EXCERPT: Extreme-point conjecture. The set of stretch vectors in $\mathbf{S}_x$ of stretch paths with respect to maximal chain-recurrent laminations characterise the set of extreme points o...

conjecture | open | MathDB 页面 | 原文线索

V2-026367 | MATHDB-352766

Generalized quasi-affine action conjecture under rigidity or dominated splitting

摘录 / EXCERPT: Let G be a higher-rank simple Lie group, let $\Gamma<G$ be a lattice, let M be a compact manifold, and let $\rho:\Gamma\to\operatorname{Diff}(M)$ be an action. Assume that the...

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V2-026368 | MATHDB-352780

Asymptotic square-root deviation conjecture for knot slope and signature

摘录 / EXCERPT: Slope-signature deviation conjecture. There are constants b and c such that, for any hyperbolic knot K in S^3,

conjecture | open | MathDB 页面 | 原文线索

V2-026369 | MATHDB-352781

The monotone Lagrangian cuplength conjecture

摘录 / EXCERPT: Let L be a compact nondisplaceable monotone Lagrangian submanifold of a tame symplectic manifold (M,ω), let $\varphi\in\mathcal{H}am_c(M,\omega)$, and let $\mathbb{F}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-026370 | MATHDB-352784

Gromov's finite-dimensionality conjecture for curvature homogeneous metrics

摘录 / EXCERPT: Gromov's conjecture. This space is finite dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-026371 | MATHDB-352785

Equality of the two pseudo-isotopy obstruction subgroups

摘录 / EXCERPT: Let X be a 4-manifold with $k_1 X=0$. The subgroups $\Theta(\mathcal{J}(X))$ and $\Theta(\mathcal{J}(X)\cap\ker\Sigma)$ are defined in the target of the pseudo-isotopy invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-026372 | MATHDB-352786

Invariance of the image of the K3 characteristic map under the involution

摘录 / EXCERPT: Let X be a 4-manifold. The involution on $\operatorname{Wh}_1(\pi_1 X;\mathbb{Z}_2\times\pi_2 X)$ is induced by the involution on $(\mathbb{Z}_2\times\pi_2 X)[\pi_1 X]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026373 | MATHDB-352791

Projective equivalence conjecture for generalized pentagram maps

摘录 / EXCERPT: Let P be a $(k+3)$-gon in $\mathbb{RP}^k$, where $k\geq 2$, and let $J=(1,\ldots,1)$. For even k, let $I=(1,\ldots,1,2,1,\ldots,1)$ with the same number of 1's before and aft...

conjecture | open | MathDB 页面 | 原文线索

V2-026374 | MATHDB-352814

Positivity of signed Chern character numbers for irreducible hyper-Kähler manifolds

摘录 / EXCERPT: Chern-character positivity conjecture. All signed Chern character numbers $(-1)^n\operatorname{Ch}_{2\lambda}$, where $|\lambda|=n$, of irreducible hyper-Kähler manifolds are posit...

conjecture | open | MathDB 页面 | 原文线索

V2-026375 | MATHDB-352815

Nishikawa's curvature operator of the second kind conjecture

摘录 / EXCERPT: Let (M,g) be a closed, simply connected Riemannian manifold. Let $d40$ denote the curvature operator of the second kind, restricted as a bilinear form to the trace-free symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-026376 | MATHDB-352832

The non-ellipsoid positive John family conjecture

摘录 / EXCERPT: Non-ellipsoid positive John family conjecture. The determinant of the matrix component of this family is not identically constant: there is no constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-026377 | MATHDB-352861

Spin Euler-characteristic ratio conjecture for minimal strata

摘录 / EXCERPT: Let $\mathcal{M}_g(2g-1)$ be the minimal stratum of differentials, let $\chi(\mathcal{M}_g(2g-1))^{\rm spin}$ denote its spin Euler characteristic, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026378 | MATHDB-352862

Spin double ramification cycle conjecture for strata of differentials

摘录 / EXCERPT: Let $g,n\geq 0$, let $k>0$, and let $a\in\mathbb{Z}^n$ satisfy $|a|=k(2g-2+n)$, $a\notin k\mathbb{Z}_{>0}^n$, and the condition that all entries of a and k are odd. For odd sta...

conjecture | open | MathDB 页面 | 原文线索

V2-026379 | MATHDB-352876

Tor-non-enlargeability of foliations with positive scalar curvature

摘录 / EXCERPT: Let V be a closed n-manifold and let $\mathscr F$ be a smooth foliation equipped with a leafwise Riemannian metric $g_{\mathscr F}$, meaning a smooth bundle metric on the integ...

conjecture | open | MathDB 页面 | 原文线索

V2-026380 | MATHDB-352879

Weak geometric compatibility of T-junction extensions

摘录 / EXCERPT: Weak geometric T-junction conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-026381 | MATHDB-352880

Gromov's conjecture on convergence of almost nonnegatively scalar-flat tori

摘录 / EXCERPT: Gromov's conjecture. If no bubbling occurs along the sequence, or if the bubbles forming along the sequence can be cut out, then a subsequence of M_j converges in some sense to t...

conjecture | open | MathDB 页面 | 原文线索

V2-026382 | MATHDB-352886

The polynomial zero-avoidance conjecture on the unit ball

摘录 / EXCERPT: Polynomial zero-avoidance conjecture. There exists a point $p\in B^d\subset\mathbb R^d$ such that, for every $k=1,\ldots,N$, the point p is at distance at least δ_k from t...

conjecture | open | MathDB 页面 | 原文线索

V2-026383 | MATHDB-352887

The non-rigidity conjecture for Fano threefolds of codimension and index at least 2

摘录 / EXCERPT: Non-rigidity and non-solidity conjecture. Then X is not birationally rigid. Moreover, if the codimension is at least 3, then X is non-solid.

conjecture | open | MathDB 页面 | 原文线索

V2-026384 | MATHDB-352893

Finiteness conjecture for non-uniform points of hypersurfaces

摘录 / EXCERPT: Non-uniform point conjecture. The locus $\mathcal{W}(X)$ is finite unless X is a cone.

conjecture | open | MathDB 页面 | 原文线索

V2-026385 | MATHDB-352908

Conjecture on spherical objects of the derived category of a log Calabi-Yau surface

摘录 / EXCERPT: Let Y be the surface under consideration, and let $i:C\hookrightarrow Y$ denote the inclusion of a (-2) curve. Define the collection of spherical objects ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026386 | MATHDB-352909

Conjectural description of the smoothing-component moduli space

摘录 / EXCERPT: Assume that D is negative definite. Let $D_1,\ldots,D_k$ be the irreducible components of D, and define ... Let $\mathcal{C}$ be the interior of...

conjecture | open | MathDB 页面 | 原文线索

V2-026387 | MATHDB-352910

Generation conjecture for compactly supported autoequivalences

摘录 / EXCERPT: Let Y be the surface, let $D\subset Y$ be the boundary divisor, let $U=Y\setminus D$, and let Z be the union of all (-2) curves in Y. Write $Z'=Z\cap U$, let $\bar{Q}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-026388 | MATHDB-352942

Nishikawa's classification conjecture for curvature operator of the second kind

摘录 / EXCERPT: Nishikawa's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026389 | MATHDB-352950

Donaldson-Sun conjecture on asymptotic Bergman kernel bounds

摘录 / EXCERPT: Let $n\in\mathbb{N}$, $c,d>0$, and $\eta\in(0,1)$. Let $(M,\omega,\mathcal{L},h)$ be a polarized Kähler manifold with $|\operatorname{Ric}|\leq 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026390 | MATHDB-352958

Ustinovskiy's weak Campana-Peternell conjecture

摘录 / EXCERPT: Let (M,g) be a compact Hermitian manifold with Griffiths non-negative Chern curvature and positive first Chern--Ricci curvature. A rational homogeneous manifold is a quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-026391 | MATHDB-352959

Ustinovskiy's convergence conjecture for left-invariant HCF+ solutions

摘录 / EXCERPT: Let G be a simple complex Lie group. A left-invariant solution is a solution of the Hermitian curvature flow equation that is invariant under left translations, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026392 | MATHDB-352975

The QALE conjecture for the hyperpolygon space $X_5(\theta)$

摘录 / EXCERPT: Let $X_5(\theta)$ be the hyperpolygon space associated with a generic parameter θ. A smooth noncompact hyperkähler manifold is quasi-asymptotically locally Euclidean when it...

conjecture | open | MathDB 页面 | 原文线索

V2-026393 | MATHDB-352988

Conjecture on unit simplices in diameter graphs

摘录 / EXCERPT: Let $\mathrm{DIAM}(d,k,n)$ be the maximum number of unit k-simplices spanned by an n-point set in $\mathbb{R}^d$ of diameter 1. Conjecture on unit simplices in diameter graph...

conjecture | open | MathDB 页面 | 原文线索

V2-026394 | MATHDB-352998

Reflection-involution characterization conjecture for length-equivalent orthogeodesics

摘录 / EXCERPT: Let $alpha$ and $beta$ be orthogeodesics. They are length equivalent when they have equal lengths. A reflection involution is an involution reflection acting on the relevant hyperb...

conjecture | open | MathDB 页面 | 原文线索

V2-026395 | MATHDB-353006

Existence of a section after a degenerate twistor deformation of a Lagrangian fibration

摘录 / EXCERPT: Degenerate twistor deformation conjecture. There exists a degenerate twistor deformation of π admitting a holomorphic section. This conjecture would remove the torsion-freeness...

conjecture | open | MathDB 页面 | 原文线索

V2-026396 | MATHDB-353017

The divisible-set union conjecture

摘录 / EXCERPT: Divisible-set union conjecture. There exists a function $f\colon \mathbb{N}\to\mathbb{N}$ such that every 2^r-divisible set of $f(r)\cdot [r+1]_2$ points over $\mathbb{F}_2$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-026397 | MATHDB-353022

Classification conjecture for Hermitian non-Kähler ALF gravitational instantons

摘录 / EXCERPT: Let $(\mathcal{M}, g_{ab})$ be a Hermitian, non-Kähler, ALF gravitational instanton, meaning a complete Ricci-flat Riemannian four-manifold with the stated Hermitian, non-Kähler, a...

conjecture | open | MathDB 页面 | 原文线索

V2-026398 | MATHDB-353024

Monotonicity of Gaussian tropical distance to the diagonal hyperplane

摘录 / EXCERPT: Monotonicity conjecture. The expectation above monotonically decreases with k. Consequently, as $\sigma\to0$, the hyperplane that best fits X converges to H_0, meaning that t...

conjecture | open | MathDB 页面 | 原文线索

V2-026399 | MATHDB-353025

Best-fit Stiefel tropical linear space at the Gaussian center

摘录 / EXCERPT: Best-fit Stiefel tropical linear space conjecture. As $\sigma\to0$, the quantity

conjecture | open | MathDB 页面 | 原文线索

V2-026400 | MATHDB-353039

Weak birational Serrano conjecture for almost strictly nef divisors

摘录 / EXCERPT: Weak birational Serrano conjecture. The divisor

conjecture | open | MathDB 页面 | 原文线索

V2-026401 | MATHDB-353043

The abundance conjecture for exterior powers of the cotangent bundle

摘录 / EXCERPT: Generalized abundance conjecture. For every X and every integer $p>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-026402 | MATHDB-353085

Calogero-Moser fixed-point and unipotent cuspidality conjecture

摘录 / EXCERPT: Cuspidality conjecture. Then p is τ_d-cuspidal.

conjecture | open | MathDB 页面 | 原文线索

V2-026403 | MATHDB-353086

Calogero-Moser and Harish-Chandra symplectic-leaf correspondence conjecture

摘录 / EXCERPT: Harish-Chandra compatibility conjecture. The maps from irreducible characters of the normalizer to Calogero-Moser fixed points, from those characters to unipotent representations v...

conjecture | open | MathDB 页面 | 原文线索

V2-026404 | MATHDB-353087

The fixed-component Calogero-Moser isomorphism conjecture

摘录 / EXCERPT: Fixed-component Calogero-Moser isomorphism conjecture. Assume that τ is regular. There exists a linear map

conjecture | open | MathDB 页面 | 原文线索

V2-026405 | MATHDB-353093

The Rhombicosidodecahedron non-Rupert conjecture

摘录 / EXCERPT: The Rhombicosidodecahedron is the point-symmetric Archimedean solid considered here, and Rupert's property means that a congruent copy can pass through a hole cut into the original...

conjecture | open | MathDB 页面 | 原文线索

V2-026406 | MATHDB-353096

Gromov's compactness conjecture for smooth spin manifolds

摘录 / EXCERPT: Gromov's compactness conjecture. If the $\sideset{}{^1}\varprojlim$ higher index of the Dirac operator of X vanishes, then X admits a complete Riemannian metric with scalar cur...

conjecture | open | MathDB 页面 | 原文线索

V2-026407 | MATHDB-353097

Unstable relative Gromov-Lawson-Rosenberg conjecture

摘录 / EXCERPT: Unstable relative Gromov-Lawson-Rosenberg conjecture. If the relative higher index of D is zero in

conjecture | open | MathDB 页面 | 原文线索

V2-026408 | MATHDB-353106

The intersection conjecture for logarithmic Voronoi polytopes

摘录 / EXCERPT: Let $\M$ be the model under consideration, let F be a face of $\M$ of dimension $d-k$ for some $k\geq 2$, and let $G_1,\ldots,G_m$ be the faces of dimension $d-k+1$ containing...

conjecture | open | MathDB 页面 | 原文线索

V2-026409 | MATHDB-353131

The X55 reflection-triangle concentric-circle conjecture

摘录 / EXCERPT: Let T be a reference triangle, and let T' be its X_{55}-reflection triangle. The six vertices of the P-hyperbolas of T' passing through X_{55} of T are the six points a...

conjecture | open | MathDB 页面 | 原文线索

V2-026410 | MATHDB-353136

The rigidity criterion for maximally mutable Laurent polynomials

摘录 / EXCERPT: Let $\mathcal N$ be a lattice and let $f i\tslashin i \mathbb C[\mathcal N]$ be a Laurent polynomial such that $\operatorname{Newt}(f)$ is a Fano polytope and the constant term of...

conjecture | open | MathDB 页面 | 原文线索

V2-026411 | MATHDB-353144

Bonnafé's conjecture on symplectic-leaf closures of fixed-point Calogero-Moser spaces

摘录 / EXCERPT: Let W be a complex reflection group, let τ be a finite-order element of the normalizer of W in ${\mathrm{GL}}_{\mathbb{C}}(V)$, and let $c\in{\mathcal{C}}$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-026412 | MATHDB-353189

The Doubling Conjecture for positive scalar curvature

摘录 / EXCERPT: Let X be a compact manifold with boundary, let $M=\operatorname{Dbl}(X,\partial X)$ be its double, and let a metric of positive scalar curvature mean a Riemannian metric whose sc...

conjecture | open | MathDB 页面 | 原文线索

V2-026413 | MATHDB-353197

The higher genus log/open correspondence for quasi-tame Looijenga pairs

摘录 / EXCERPT: The higher genus log/open correspondence. One has

conjecture | open | MathDB 页面 | 原文线索

V2-026414 | MATHDB-353200

Maximum-curvature conjecture for optimal control of moving sets

摘录 / EXCERPT: Let $\Omega(t)$ be the evolving set with boundary parametrized by $x(t,\xi)$, and let the optimal control be the normal velocity control acting along this boundary. Denote the curv...

conjecture | open | MathDB 页面 | 原文线索

V2-026415 | MATHDB-353205

Scattering rigidity conjecture for compact Riemannian manifolds of Anosov type

摘录 / EXCERPT: Let M_1 and M_2 be compact Riemannian manifolds of Anosov type, each with scattering data. Scattering rigidity conjecture. If M_1 and M_2 have the same scattering data, the...

conjecture | open | MathDB 页面 | 原文线索

V2-026416 | MATHDB-353222

Equivariant smoothing conjecture for antisymplectic cusp singularities

摘录 / EXCERPT: Let $(p\in X)$ be a cusp singularity equipped with an antisymplectic involution ι. Let D denote the dual cycle of the cusp. An equivariant smoothing conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-026417 | MATHDB-353223

Equivariant smoothing criterion for antisymplectic cusp singularities

摘录 / EXCERPT: Let $(p\in X)$ be a cusp singularity equipped with an antisymplectic involution ι fixed point free on $X\setminus\{p\}$, and let D be its dual cycle. Let Y be a smooth ra...

conjecture | open | MathDB 页面 | 原文线索

V2-026418 | MATHDB-353243

Lipschitz nice-dimensions conjecture

摘录 / EXCERPT: Lipschitz nice-dimensions conjecture. The Lipschitz nice dimensions contain Mather's nice dimensions and their boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-026419 | MATHDB-353244

Sharpness conjecture for Lipschitz nice dimensions

摘录 / EXCERPT: Sharpness conjecture for Lipschitz nice dimensions. If (n,p) lies in the complement of the nice dimensions or in their boundary, then $\mathrm{S}^{Lip}(N,P)$ is not dense in...

conjecture | open | MathDB 页面 | 原文线索

V2-026420 | MATHDB-353245

Bi-Lipschitz equivalence conjecture at the boundary of extra-nice dimensions

摘录 / EXCERPT: Boundary extra-nice-dimensions conjecture. Any two generic immersions g and g' are bi-Lipschitz $\mathcal{K}_{V}$-equivalent, and the germs f and f' they define are bi-Lips...

conjecture | open | MathDB 页面 | 原文线索

V2-026421 | MATHDB-353264

Kapovich's unknotted Menger-curve and hyperbolic-quotient conjecture

摘录 / EXCERPT: Let $G_{6,3}=\langle a_0,\cdots,a_5\mid a_i^3=id,\ a_ia_{i+1}=a_{i+1}a_i,\ i\in\mathbb{Z}/6\mathbb{Z}\rangle$ be the hyperbolic group with boundary the Menger curve, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026422 | MATHDB-353322

The degree-four generalized Fujiki constant formula for hyperkähler manifolds

摘录 / EXCERPT: Degree-four generalized Fujiki constant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026423 | MATHDB-353323

The Rozansky--Witten relation for generalized Fujiki constants in dimension at least eight

摘录 / EXCERPT: Rozansky--Witten generalized Fujiki relation.

conjecture | open | MathDB 页面 | 原文线索

V2-026424 | MATHDB-353324

The positivity conjecture for products of generalized Fujiki constants

摘录 / EXCERPT: Generalized Fujiki positivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026425 | MATHDB-353325

The asymptotic-cone orbit conjecture for nonnegative Ricci curvature

摘录 / EXCERPT: Let (M,p) be an open n-manifold with $\mathrm{Ric}\ge 0$ and $E(M,p)\not=1/2$. Suppose that $\pi_1(M)$ contains a torsion-free nilpotent subgroup of nilpotency length l. An a...

conjecture | open | MathDB 页面 | 原文线索

V2-026426 | MATHDB-353333

The hyper-Kähler SYZ deformation conjecture

摘录 / EXCERPT: Hyper-Kähler SYZ deformation conjecture. Any hyper-Kähler manifold can be deformed into a hyper-Kähler manifold with a Lagrangian fibration.

conjecture | open | MathDB 页面 | 原文线索

V2-026427 | MATHDB-353334

The numerical Riemann-Roch polynomial conjecture for hyper-Kähler manifolds

摘录 / EXCERPT: Numerical Riemann-Roch conjecture. The Huybrechts-Riemann-Roch polynomial of X satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026428 | MATHDB-353348

Generalized affine plank conjecture for translative coverings

摘录 / EXCERPT: Generalized affine plank conjecture. Assume that the convex sets $C_1, \ldots, C_n \subset \mathbb{R}^d$ permit a translative covering of the convex body $B \in \mathcal{K}^d$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-026429 | MATHDB-353349

Soltan's homothetic covering conjecture

摘录 / EXCERPT: Soltan's homothetic covering conjecture. Assume that $K \in \mathcal{K}^d$ and that $\lambda_1 K, \ldots, \lambda_n K$ permit a translative covering of K with...

conjecture | open | MathDB 页面 | 原文线索

V2-026430 | MATHDB-353358

Michel's conjecture on boundary rigidity of simple manifolds

摘录 / EXCERPT: Michel's conjecture. All simple manifolds are boundary rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-026431 | MATHDB-353359

Burago-Ivanov conjecture on strict minimal filling of simple manifolds

摘录 / EXCERPT: Burago-Ivanov conjecture. Every simple manifold is a strict minimal filling.

conjecture | open | MathDB 页面 | 原文线索

V2-026432 | MATHDB-353366

Spherical distance-map conjecture for spheres

摘录 / EXCERPT: Let $n\geq 1$ be an integer, and let $\theta,\theta'\in\mathbb{S}^n$. The map $\Phi_{\mathbb{S}^n}$ is defined through the classical multidimensional scaling construction, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026433 | MATHDB-353395

Gordon's conjecture on connected sums of Heegaard splittings

摘录 / EXCERPT: Gordon's conjecture. The connected sum of two unstabilized Heegaard splittings is unstabilized.

conjecture | solved | MathDB 页面 | 原文线索

V2-026434 | MATHDB-353410

Asymptotic length-saturation for subarithmetic hyperbolic groups

摘录 / EXCERPT: Let $\G$ be a subarithmetic hyperbolic group with growth exponent $\ga$, and let $\cO$ be the ring of integers of its trace field. The group $\G$ asymptotically length-saturates if...

conjecture | open | MathDB 页面 | 原文线索

V2-026435 | MATHDB-353416

Leinster-Willerton magnitude conjecture for convex bodies

摘录 / EXCERPT: Let $X\subset\mathbb{R}^n$ be compact and convex, and let $\mathcal{M}_X(R):=\operatorname{mag}(X,R\cdot\mathrm{d})$ for $R>0$. For $0\leq k\leq n$, let $V_k(X)$ denote the intrins...

conjecture | open | MathDB 页面 | 原文线索

V2-026436 | MATHDB-353419

Virtual pushforward formula for stable pairs on holomorphic symplectic 4-folds

摘录 / EXCERPT: Let X be a holomorphic symplectic 4-fold and let $\beta\in H_2(X,\mathbb{Z})$ be an irreducible curve class. Let $P_n(X,\beta)$ be the moduli space of stable pairs and...

conjecture | open | MathDB 页面 | 原文线索

V2-026437 | MATHDB-353454

Moduli-space conjecture for optimal symplectic connections

摘录 / EXCERPT: A holomorphic fibration is a holomorphic submersion whose fibres are polarised compact Kähler manifolds. An optimal symplectic connection is a symplectic connection satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-026438 | MATHDB-353475

Regularity conjecture for complex Monge-Ampère equations with bounded or Hölder right-hand side

摘录 / EXCERPT: Let φ solve the complex Monge-Ampère equation considered in the paper, with right-hand side F. The question concerns estimates for the regularity of φ under assum...

open_problem | open | MathDB 页面 | 原文线索

V2-026439 | MATHDB-353506

Johansson's uniqueness conjecture for the parabolically shifted Airy process

摘录 / EXCERPT: Johansson's conjecture. Almost surely, $x\mapsto\mathcal{A}(x)-x^2$ attains its maximum at a unique point $Y\in\mathbb{R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-026440 | MATHDB-353507

Conjecture on the Hausdorff dimensions of KPZ fixed-point non-uniqueness times

摘录 / EXCERPT: Non-uniqueness-times conjecture. For nice enough initial conditions h_0, almost surely $\mathcal{T}_{\geq2}(h_0)$ is dense in $[0,\infty)$, $\mathcal{T}_{\geq5}(h_0)$ is empty, a...

conjecture | open | MathDB 页面 | 原文线索

V2-026441 | MATHDB-353520

Infinite transitivity criterion for affine toric surfaces

摘录 / EXCERPT: Infinite transitivity conjecture for affine toric surfaces. The group G acts on the open orbit of X infinitely transitively if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-026442 | MATHDB-353547

Characterization of the universal isometry group of q-map spaces

摘录 / EXCERPT: Let $\overline{N}\cong \mathcal{H}\times L$ be the q-map space, with $g_{L}(p)$ the left-invariant metric on the fiber $L_p\cong L$ parametrized by $p\in\mathcal{H}$. Let G be th...

conjecture | open | MathDB 页面 | 原文线索

V2-026443 | MATHDB-353567

Intermediate-Jacobian non-isomorphism conjecture for quartic double solids and GM threefolds

摘录 / EXCERPT: Let Y be a quartic double solid and let X be a GM threefold. Let $J(Y)$ and $J(X)$ denote their intermediate Jacobians, each with its principal polarization. Intermediate-Jacob...

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V2-026444 | MATHDB-353576

The 2^n multi-tangent-planes conjecture

摘录 / EXCERPT: Let $A_1,\dots,A_n$ be the subsets considered under the conditions of the paper's main theorem, and call them not thin when each A_i contains an open subset. Multi-tangent-planes...

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V2-026445 | MATHDB-353580

Extension of the hyperbolic closed-geodesic characterization to reversible metrics

摘录 / EXCERPT: Hyperbolic closed-geodesic characterization conjecture. The characterization theorem for Finsler metrics, and for Riemannian metrics on closed surfaces, should also hold for revers...

conjecture | open | MathDB 页面 | 原文线索

V2-026446 | MATHDB-353584

Bounded discrepancy conjecture for divisors computing minimal log discrepancies

摘录 / EXCERPT: Bounded discrepancy conjecture. The divisor computing the minimal log discrepancy can be chosen with discrepancy for $(X,0)$ bounded uniformly in terms of d and Γ. The cla...

conjecture | open | MathDB 页面 | 原文线索

V2-026447 | MATHDB-353602

Factorization conjecture for quasi-isometries through collapsing maps

摘录 / EXCERPT: Factorization conjecture. Any quasi-isometry can be expressed as the composition of a collapsing map and Lipschitz map.

conjecture | open | MathDB 页面 | 原文线索

V2-026448 | MATHDB-353603

Generalized collapsing-map conjecture for function-bounded quasi-isometries

摘录 / EXCERPT: Generalized collapsing-map conjecture. Then the generalized form of Theorem 2 and Corollary 2.1 holds.

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V2-026449 | MATHDB-353623

The multiplicity-two conjecture for the second Dirichlet eigenvalue on simply connected planar domains

摘录 / EXCERPT: Multiplicity-two conjecture.

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V2-026450 | MATHDB-353683

Hikita-Iyama isomorphism conjecture for the symmetric group center

摘录 / EXCERPT: Let S_n be the symmetric group, let $Z(\mathbb{C}S_n)$ be its center, and let $\Theta\colon Z(\mathbb{C}S_n)\xrightarrow{\sim}\mathbb{C}[\mathcal{C}(n)]^{\mathbb{T}}$ be the isom...

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V2-026451 | MATHDB-353687

Fibration stability implies adiabatic K-semistability

摘录 / EXCERPT: Let $f:(X,H)\to(B,L)$ be a smooth polarized fibration. The fibration is fibration stable, and (B,L) has a twisted constant scalar curvature Kähler metric with respect to the Weil...

conjecture | open | MathDB 页面 | 原文线索

V2-026452 | MATHDB-353718

Dervan-Sektnan's polystability conjecture for holomorphic fibrations

摘录 / EXCERPT: Let $\pi: (X,H)\to (B,L)$ be a holomorphic fibration, where H and L are the relevant polarisation classes. An optimal symplectic connection is a connection associated with the...

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V2-026453 | MATHDB-353734

Minimax conjecture for Calabi energy

摘录 / EXCERPT: Suppose (X,L) has only klt singularities. The non-archimedean Calabi energy should satisfy ... with an appropriate definition of the right-hand side. This would identify the non-ar...

conjecture | open | MathDB 页面 | 原文线索

V2-026454 | MATHDB-353792

The Sasaki-Einstein existence conjecture for Smale 5-manifolds

摘录 / EXCERPT: Sasaki-Einstein existence conjecture. For each integer $k\leq 8$ and $n\geq 2$, the Smale 5-manifold

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V2-026455 | MATHDB-353802

Thomas-Yau's categorical semistability correspondence conjecture

摘录 / EXCERPT: Let X be a Calabi-Yau manifold, let $D^bFuk(X)$ denote its derived Fukaya category, and let Joyce's conjectural Bridgeland stability be a stability condition on this category. An...

conjecture | open | MathDB 页面 | 原文线索

V2-026456 | MATHDB-353803

Thomas-Yau's Harder-Narasimhan obstruction conjecture

摘录 / EXCERPT: Let X be a Calabi-Yau manifold with Kähler metric. Let L be an almost calibrated exact Lagrangian brane in $D^bFuk(X)$ with a Harder-Narasimhan decomposition ... whose distinguis...

conjecture | open | MathDB 页面 | 原文线索

V2-026457 | MATHDB-353804

Solomon's functional well-definedness conjecture for unobstructed Lagrangians

摘录 / EXCERPT: Let X be a compact almost Calabi-Yau manifold. Let L_0 be a reference Lagrangian, and consider graded immersed unobstructed Lagrangians in the same $D^bFuk(X)$ class as L_0....

conjecture | open | MathDB 页面 | 原文线索

V2-026458 | MATHDB-353805

Thomas-Yau's existence conjecture for special Lagrangians

摘录 / EXCERPT: Let (X,ω,Ω) be an exact Calabi-Yau manifold, and let L_0 be a nontrivial unobstructed exact Lagrangian brane with quantitatively almost calibrated phase function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026459 | MATHDB-353809

Thomas-Yau's uniqueness conjecture for weak special Lagrangians

摘录 / EXCERPT: Let X be a Calabi-Yau manifold, and let L and L' be special Lagrangian integral currents with the same phase angle $\hat{\theta}$. Equip them with suitable unobstructed brane...

conjecture | open | MathDB 页面 | 原文线索

V2-026460 | MATHDB-353839

Campana's numerical-Kodaira dimension conjecture for canonical bundles

摘录 / EXCERPT: Let X be a compact Kähler manifold. The numerical-Kodaira dimension conjecture. One has ... This is the classical equality conjectured for the canonical bundle and is presented as...

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V2-026461 | MATHDB-353840

Campana's cotangent-dimension equality conjecture

摘录 / EXCERPT: Let X be a compact connected Kähler manifold and let $p>0$. Write ... The cotangent-dimension equality conjecture. One has ... This extends the canonical-bundle equality to all exter...

conjecture | open | MathDB 页面 | 原文线索

V2-026462 | MATHDB-353866

The conjecture on densest congruent geodesic ball packings in Thurston geometries

摘录 / EXCERPT: Densest-arrangement conjecture. The ball arrangement $\mathcal{B}_{\mathrm{opt}}(R_4,K_4)$ provides the densest congruent geodesic ball packing for the Thurston geometries.

conjecture | open | MathDB 页面 | 原文线索

V2-026463 | MATHDB-353874

Bang's affine plank conjecture

摘录 / EXCERPT: Let C be a convex body, and let $P_1,\ldots,P_n$ be planks covering it. If H_i is parallel to the boundary hyperplanes of P_i, define the relative width of P_i by...

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V2-026464 | MATHDB-353875

Fejes Tóth's zone covering conjecture

摘录 / EXCERPT: A zone is the spherical neighbourhood of a great circle, with width equal to twice its maximal spherical distance from that great circle. Let zones of widths $w_1,\ldots,w_n$ cover...

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V2-026465 | MATHDB-353894

The continued-fraction length conjecture for staircase classes

摘录 / EXCERPT: Continued-fraction length conjecture. (i) The continued-fraction expansion of the center p/q of ${\bf E}_{\delta}$ has CF-length...

conjecture | open | MathDB 页面 | 原文线索

V2-026466 | MATHDB-353895

The higher special-point obstruction conjecture

摘录 / EXCERPT: Higher special-point obstruction conjecture. For $i>1$, c_{H_b} has no descending staircase. Furthermore, there exists $\varepsilon>0$ such that, for

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V2-026467 | MATHDB-353898

Enumeration conjecture for maximal isotropic subbundles of general rank-five orthogonal bundles

摘录 / EXCERPT: The rank-five maximal-isotropic enumeration conjecture. If $g\equiv1\pmod 4$ and $w_2(V)=0$, so that $V\cong S_W$ for a general $W\in\mathcal M\mathcal S_C(4,\mathcal O_C)$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-026468 | MATHDB-353899

Rank comparison conjecture for maximal isotropic subbundles

摘录 / EXCERPT: Rank comparison conjecture. For all $r\ge3$, one has

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V2-026469 | MATHDB-353900

Equal-component conjecture for maximal isotropic subbundles

摘录 / EXCERPT: Equal-component conjecture. Exactly half of the $N^0(g,2n)$ maximal isotropic subbundles define sections of $\mathrm{OG}(V)_0$, and exactly half define sections of...

conjecture | open | MathDB 页面 | 原文线索

V2-026470 | MATHDB-353902

Layered-scattering determination of intrinsic invariants of a submanifold

摘录 / EXCERPT: Layered-scattering determination conjecture. The averages $\mathcal L^{\mathsf{av}}_{0\leq j\leq\lceil k/2\rceil}$ determine the intrinsic global invariants...

conjecture | open | MathDB 页面 | 原文线索

V2-026471 | MATHDB-353913

Flop invariance of the K-theoretic PT four-point function

摘录 / EXCERPT: Flop invariance conjecture. The normalized four-point function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026472 | MATHDB-353925

Projective 7-hole existence conjecture

摘录 / EXCERPT: A finite set of points in $\operatorname{\mathbb{R}\mathcal{P}^2}$ is in general position, and a projective k-hole is a projective convex hull of k points that contains no othe...

conjecture | open | MathDB 页面 | 原文线索

V2-026473 | MATHDB-353949

Realization conjecture for almost toric fibrations and symplectic log Calabi-Yau divisors

摘录 / EXCERPT: Let $(X,c\omega)$ be a symplectic 4-manifold. An almost toric fibration is a Lagrangian fibration with only nodal and elliptic singularities. Let $\mathcal{ATF}(X,\omega)$ denote...

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V2-026474 | MATHDB-353959

Kahn-Marković asymptotic counting conjecture for surface subgroups

摘录 / EXCERPT: For a hyperbolic 3-manifold M, let $S(M,g)$ denote the number of conjugacy classes of surface subgroups in $\pi_1(M)$ of genus g. Kahn-Marković's conjecture. There exists a c...

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V2-026475 | MATHDB-353982

Uniform non-negativity conjecture for constructible functions

摘录 / EXCERPT: Let X be an aspherical complex projective manifold. A constructible function on X has an effective characteristic cycle when its characteristic cycle is effective. Uniform non-...

conjecture | open | MathDB 页面 | 原文线索

V2-026476 | MATHDB-353996

Strong tangency conjecture for separatrices of non-dicritical foliations

摘录 / EXCERPT: Let X be a quasi-compact complex manifold, let $\mathcal{F}$ be a codimension-1 foliation on X given by an algebraic 1-form, and let Z be the union of its separatrices. S...

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V2-026477 | MATHDB-353997

Jet-density conjecture for separatrices of non-dicritical foliations

摘录 / EXCERPT: Let X be a quasi-compact complex manifold, let $\mathcal{F}$ be a codimension-1 foliation on X, let $\mathfrak{V}(\mathcal{F})$ denote the set of hypersurfaces that are smoot...

conjecture | open | MathDB 页面 | 原文线索

V2-026478 | MATHDB-353998

Converse order bound for sub-exceptional separatrices

摘录 / EXCERPT: Let $\mathcal{F}$ be a codimension-1 foliation with local defining 1-form whose coordinate functions are b_i, and let g define the union of its separatrices as in the prece...

conjecture | open | MathDB 页面 | 原文线索

V2-026479 | MATHDB-354039

Lagrangian embedding conjecture for cubic fourfold hyperplane sections

摘录 / EXCERPT: Lagrangian embedding conjecture. induces a rational map

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V2-026480 | MATHDB-354046

Polyhedrality and extreme divisors of the blow-up of the Losev-Manin space

摘录 / EXCERPT: Let $\phi: \overline{M}_{0,7}\rightarrow \operatorname{Bl}_e \overline{LM}_7$ be the natural reduction map. Let the divisors in Tables 1, 2, and 3 denote the specified S_7-equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-026481 | MATHDB-354047

Viterbo's isoperimetric conjecture for symplectic capacities

摘录 / EXCERPT: Let $\Sigma \in \mathcal{K}^{2n}$ be a convex domain in $(\mathbb{R}^{2n},\omega_0)$, let c be a symplectic capacity, and let $\operatorname{vol}(\Sigma)$ denote its symplectic v...

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V2-026482 | MATHDB-354061

Ball's conjecture on the $SL(n)$-invariant quantity $B(K)$

摘录 / EXCERPT: For a convex body $K\subseteq\mathbb{R}^n$, define ... where K° is the polar body. Ball proved that $B(K)\leq B(B_2^n)$ for unconditional convex bodies. Ball's conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-026483 | MATHDB-354064

Existence of exotic diffeomorphisms relative to any closed oriented 3-manifold boundary

摘录 / EXCERPT: Let Y be a closed, connected, oriented 3-manifold. A compact, oriented smooth 4-manifold W with boundary $\partial W=Y$ has mapping class groups of diffeomorphisms and home...

conjecture | open | MathDB 页面 | 原文线索

V2-026484 | MATHDB-354168

The Mori cone generation conjecture for the mirror quintic

摘录 / EXCERPT: Mori cone generation conjecture. The Mori cone M of X_ψ is generated by the classes of the curves $\ell_{\mathbf{t}}$, $\sigma_{\mathbf{s},\mathbf{t}}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026485 | MATHDB-354177

Classification conjecture for Lee-class sets of class VII surfaces

摘录 / EXCERPT: Let S be a compact complex surface in the Kodaira class ${\rm VII}$. Write ${\mathcal T}(S)$ for its set of Lee classes, viewed in $H^1_{dR}(S,\mathbb R)$ and ordered along...

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V2-026486 | MATHDB-354179

Marked lens rigidity conjecture for manifolds of Anosov type

摘录 / EXCERPT: Let M be a smooth manifold with boundary, and let g_1 and g_2 be smooth metrics of Anosov type on M such that ... Assume that g_1 and g_2 have the same marked lens data,...

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V2-026487 | MATHDB-354180

Lens rigidity conjecture for manifolds of Anosov type

摘录 / EXCERPT: Let (M_1,g_1) and (M_2,g_2) be smooth Riemannian manifolds of Anosov type whose boundary metrics agree: ... Let (ℓ_g,S_g) denote the lens data. Lens rigidity conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-026488 | MATHDB-354181

Solenoidal injectivity conjecture for the X-ray transform on manifolds of Anosov type

摘录 / EXCERPT: Let (M,g) be a smooth Riemannian manifold of Anosov type. For symmetric m-tensors, let I_m^g be the X-ray transform, and call it solenoidal injective when ... Here D_g is the...

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V2-026489 | MATHDB-354221

The geometrically bounded implies spectrally bounded conjecture

摘录 / EXCERPT: Let N be a closed Riemannian manifold, and let ... An exact Lagrangian L is geometrically bounded when it is contained in DT^*N. Geometrically bounded implies spectrally bounde...

conjecture | open | MathDB 页面 | 原文线索

V2-026490 | MATHDB-354222

The spectral boundedness conjecture for bounded gamma-support

摘录 / EXCERPT: Let N be a closed Riemannian manifold and let $DT^*N=\{(q,p)\in T^*N\mid |p|_g\leq 1\}$. For an element $L\in\widehat{\mathfrak L}(T^*N)$, its γ-support is denoted by...

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V2-026491 | MATHDB-354230

Illumination conjecture for 1-unconditionally symmetric cap bodies

摘录 / EXCERPT: Let $\mathbf{K}\subset\mathbb{E}^d$ be a 1-unconditionally symmetric cap body, meaning that $\mathbf{K}$ is symmetric about each coordinate hyperplane of $\mathbb{E}^d$, and let...

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V2-026492 | MATHDB-354233

Burau's non-linearizability conjecture for even-dimensional webs

摘录 / EXCERPT: Let $n\geq 2$ be even, and let $\boldsymbol{\mathcal W}_{0,n+3}$ be the web by rational curves induced by the forgetful maps on $\mathcal M_{0,n+3}$. Burau's conjecture. The web...

conjecture | open | MathDB 页面 | 原文线索

V2-026493 | MATHDB-354234

Conjecture on abelian relations of even-dimensional webs

摘录 / EXCERPT: Let $n\geq 2$ be even, let Π_n be the index set used in the source, and let $AR_\nu(\boldsymbol{\mathcal E}_n)$ be the abelian relation associated with $\nu\in\Pi_n$. Abelian-r...

conjecture | open | MathDB 页面 | 原文线索

V2-026494 | MATHDB-354237

Relative McKernan-Prokhorov boundedness conjecture for log Calabi-Yau fibrations

摘录 / EXCERPT: Relative McKernan-Prokhorov conjecture. The set of base-polarized fibrations $X\to(Z,A)$ such that there exists a (d,r,ϵ)-log Calabi-Yau rationally connected fibration...

conjecture | open | MathDB 页面 | 原文线索

V2-026495 | MATHDB-354269

The Dirac conjecture for point-line configurations

摘录 / EXCERPT: Dirac's conjecture. There is a constant $c>0$ such that some point of S is incident to at least $c n$ lines determined by S.

conjecture | open | MathDB 页面 | 原文线索

V2-026496 | MATHDB-354270

The Dirac--Goodman--Pollack conjecture for pseudolines

摘录 / EXCERPT: Dirac--Goodman--Pollack conjecture. There is a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-026497 | MATHDB-354279

Unobstructedness and B-realizability conjecture for tropical curves

摘录 / EXCERPT: Unobstructedness and B-realizability conjecture. The geometric Lagrangian lift L_V is unobstructed if and only if V is B-realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-026498 | MATHDB-354280

Fiberwise isotopy conjecture for tropical Lagrangian lifts

摘录 / EXCERPT: Fiberwise isotopy conjecture. The subspace of $H^1(L_V,\mathbb R)$ arising from flux classes of fiberwise Lagrangian isotopies is identified with...

conjecture | open | MathDB 页面 | 原文线索

V2-026499 | MATHDB-354281

Superabundance and non-wide Lagrangian lift conjecture

摘录 / EXCERPT: Superabundance and non-wide conjecture. The tropical curve V is superabundant if and only if its Lagrangian lift L_V is not wide.

conjecture | open | MathDB 页面 | 原文线索

V2-026500 | MATHDB-354307

Canonical bijection between degenerate tetrahedra and planar quadruples

摘录 / EXCERPT: A quadruple of points in the special affine plane determines triangle and Varignon-parallelogram areas. Consider the corresponding quadruple of Euclidean-plane points for which the...

conjecture | open | MathDB 页面 | 原文线索

V2-026501 | MATHDB-354321

Matheron's covariogram conjecture for convex bodies

摘录 / EXCERPT: Let K be a convex body, and let g_K denote its covariogram. Matheron's covariogram conjecture. The covariogram g_K determines a convex body K, among all convex bodies, up t...

conjecture | open | MathDB 页面 | 原文线索

V2-026502 | MATHDB-354357

Curvature flow convergence for Markovian weighted graphs

摘录 / EXCERPT: Curvature flow convergence conjecture. The curvature flow converges for any initial condition (G,P_0), that is, $(G,P(t))$ has a well-defined limit

conjecture | open | MathDB 页面 | 原文线索

V2-026503 | MATHDB-354368

The conjectured value of the line-piercing number $L(k)$

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of compact, convex sets in the plane, and let $L(k)$ denote the minimum number of lines guaranteed to pierce $\mathcal{F}$ when $\mathcal{F}$ contains...

conjecture | open | MathDB 页面 | 原文线索

V2-026504 | MATHDB-354383

Farey-word conjecture for discrete two-generator orbifold groups

摘录 / EXCERPT: Let Γ be a discrete representation of $\langle a,b\mid a^p=b^q=1\rangle \cong \mathbb{Z}_p*\mathbb{Z}_q$ in $\operatorname{PSL}(2,\mathbb{C})$, and suppose Γ is not i...

conjecture | open | MathDB 页面 | 原文线索

V2-026505 | MATHDB-354387

Bölcskei's uniqueness conjecture for unilateral equitransitive cube tilings

摘录 / EXCERPT: A tiling of $\mathbb{R}^n$ is a partition into closed sets without holes, called tiles, whose pairwise intersections have Lebesgue measure zero. A tiling of $\mathbb{R}^n$ into hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-026506 | MATHDB-354398

The intrinsic mirror construction conjecture

摘录 / EXCERPT: Let Y be a Fano manifold with smooth anticanonical divisor D. For a suitable submonoid $P\subset \operatorname{Pic}(Y)$, let (X,W) be the intrinsic mirror dual, where ... is a...

conjecture | open | MathDB 页面 | 原文线索

V2-026507 | MATHDB-354399

The FOOO obstruction-term conjecture for the open mirror map

摘录 / EXCERPT: Let Y be a symplectic Fano manifold with smooth anticanonical divisor D, such that $Y\setminus D$ supports a Lagrangian torus fibration collapsing an S^1 along D. Let L b...

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V2-026508 | MATHDB-354451

Parallelogram characterization for central quadrilaterals of general quadrilaterals

摘录 / EXCERPT: Let $ABCD$ be a general quadrilateral, and let a triangle center be specified by a center function in the angles of each component triangle. Two center functions are equivalent if...

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V2-026509 | MATHDB-354452

Orthodiagonal characterization for central quadrilaterals of general quadrilaterals

摘录 / EXCERPT: Let $ABCD$ be a general quadrilateral, and let a triangle center be specified by a center function. The central quadrilateral is formed by placing the chosen center in the four com...

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V2-026510 | MATHDB-354453

Cyclic characterization for centers of tangential quadrilaterals

摘录 / EXCERPT: Let $ABCD$ be a tangential quadrilateral, and form the central quadrilateral by placing the chosen triangle center in the four component triangles determined by the diagonals. Cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-026511 | MATHDB-354454

Nonexistence of a universal rectangular center for general quadrilaterals

摘录 / EXCERPT: Let $ABCD$ be a general quadrilateral, and form the central quadrilateral by selecting one triangle center, specified by a single center function, in each of the four half triangle...

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V2-026512 | MATHDB-354455

Uniqueness of the orthocenter for area-preserving central quadrilaterals

摘录 / EXCERPT: Let $ABCD$ be a general quadrilateral, and form the central quadrilateral from the four half triangles determined by its diagonals, using the same triangle center in each. Area-pre...

conjecture | open | MathDB 页面 | 原文线索

V2-026513 | MATHDB-354456

Aspinwall-Plesser-Wang semiorthogonal-decomposition conjecture for secondary-polytope edges

摘录 / EXCERPT: Let $A\subset N=\mathbb Z^d$ and $Q=\operatorname{conv}(A)$, and let F be an edge of the secondary polytope. Let Z_F be the associated space, let $n_{\Gamma,F}$ be the multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-026514 | MATHDB-354468

Full generic regularity II for min-max minimal hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be a closed smooth manifold, and let g be a Baire C^∞-generic Riemannian metric on M. A min-max g-minimal hypersurface is a minimal hypersurface generate...

conjecture | open | MathDB 页面 | 原文线索

V2-026515 | MATHDB-354470

Metric convex-hull equality for transition attractors

摘录 / EXCERPT: Let $({\mathbb X},d)$ be a metric space, and call $[x,y]:=\{z\in{\mathbb X}:d(x,z)+d(z,y)=d(x,y)\}$ the segment with ends x,y. A set is metrically convex if it contains $[x,y]$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-026516 | MATHDB-354472

Fukaya's gradient-flow-tree conjecture for Fourier modes of Maurer-Cartan solutions

摘录 / EXCERPT: Let X admit a Lagrangian torus fibration ... and let the semi-flat SYZ mirror carry Maurer-Cartan solutions whose Fourier modes are taken along its Lagrangian torus fibers. Near a...

conjecture | open | MathDB 页面 | 原文线索

V2-026517 | MATHDB-354479

Detcherry-Kalfagianni Turaev-Viro simplicial-volume conjecture

摘录 / EXCERPT: Detcherry-Kalfagianni simplicial-volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026518 | MATHDB-354480

Geometric Sen perfectoidness conjecture

摘录 / EXCERPT: Geometric Sen perfectoidness conjecture. The morphism $\theta_{\widetilde{X}}$ is surjective if and only if $\widetilde{X}$ is a perfectoid space.

conjecture | open | MathDB 页面 | 原文线索

V2-026519 | MATHDB-354507

Lattice polytope face-count conjecture for dense faces

摘录 / EXCERPT: Lattice polytope face-count conjecture. The number of t-dense faces of $\operatorname{conv}(S)$, counting faces of every dimension, is

conjecture | open | MathDB 页面 | 原文线索

V2-026520 | MATHDB-354549

Li-Nirenberg symmetry conjecture under the Main Assumption and Condition S

摘录 / EXCERPT: Let M be a smooth compact connected embedded hypersurface in $\mathbb R^{n+1}$. The Main Assumption is that whenever $(X',A),(X',B)\in M$ satisfy $A\le B$ and the segment...

conjecture | open | MathDB 页面 | 原文线索

V2-026521 | MATHDB-354550

Symmetry conjecture for hypersurfaces with ordered m-th curvature

摘录 / EXCERPT: The curvature-symmetry conjecture. Then M must be symmetric about a hyperplane $X_{n+1}=c$. This generalizes the mean-curvature setting to the m-th elementary symmetric curvatu...

conjecture | open | MathDB 页面 | 原文线索

V2-026522 | MATHDB-354551

General curvature-function symmetry conjecture under Condition S

摘录 / EXCERPT: Let $n\ge2$ and let (M,Γ,g) satisfy the paper's condition ... . The general curvature-function symmetry conjecture. Then M must be symmetric about a hyperplane $X_{n+1}=c$....

conjecture | open | MathDB 页面 | 原文线索

V2-026523 | MATHDB-354555

Waldhausen-type conjecture for trisections of S^4

摘录 / EXCERPT: Waldhausen-type conjecture. Every trisection of S^4 is isotopic to either the genus 0 trisection or its stabilization.

conjecture | open | MathDB 页面 | 原文线索

V2-026524 | MATHDB-354560

Li-Zheng's Hopf-type conjecture for Kähler manifolds

摘录 / EXCERPT: Let M be an n-dimensional Kähler manifold with non-positive holomorphic bisectional curvature, and suppose that its Ricci curvature is quasi-negative. Li-Zheng's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-026525 | MATHDB-354583

Yue's critical exponent gap conjecture for complex hyperbolic Kleinian groups

摘录 / EXCERPT: Let m be a positive integer, let $\mathbf{H}_{\mathbb{C}}^{m}$ be complex hyperbolic m-space, and let Γ be a convex cocompact isometry group of...

conjecture | open | MathDB 页面 | 原文线索

V2-026526 | MATHDB-354606

Uniqueness conjecture for minimal representations of irreducible tropical rational functions

摘录 / EXCERPT: Let $X\subset \mathbb{R}^d$ be the locus of non-linearity of a sign-balanced polyhedral fan of dimension $d-1$, with positive and negative parts X_+ and X_-. Construct minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-026527 | MATHDB-354610

Le Donne's conjecture that Carnot groups have no periodic geodesics

摘录 / EXCERPT: Le Donne's conjecture. Carnot groups do not have periodic geodesics.

conjecture | open | MathDB 页面 | 原文线索

V2-026528 | MATHDB-354620

Metric-line classification conjecture for jet spaces

摘录 / EXCERPT: Let $J^k(\mathbb{R},\mathbb{R})$ be the space of k-jets of real functions of one real variable, equipped with its sub-Riemannian distance. A geodesic...

conjecture | open | MathDB 页面 | 原文线索

V2-026529 | MATHDB-354633

The symplectic hook-content product conjecture

摘录 / EXCERPT: Let λ range over partitions, let $h^{\lambda}(u)$ denote the hook length of a cell $u\in\lambda$, and let $c_{sp}^{\lambda}(u)$ be its symplectic content. For an indetermin...

conjecture | open | MathDB 页面 | 原文线索

V2-026530 | MATHDB-354634

The symplectic and orthogonal squared-content identity conjecture

摘录 / EXCERPT: Let λ range over partitions, let $h^{\lambda}(u)$ denote the hook length of a cell $u\in\lambda$, and let $c_{sp}^{\lambda}(u)$ and $c_{O}^{\lambda}(u)$ denote its symplect...

conjecture | open | MathDB 页面 | 原文线索

V2-026531 | MATHDB-354639

Finite-field radial projection exceptional-set conjecture

摘录 / EXCERPT: Finite-field radial projection exceptional-set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026532 | MATHDB-354649

Generating-series correspondence conjecture for lambda-class invariants with point insertions

摘录 / EXCERPT: Let C be a curve class, let g_0 be fixed, and define the generating series ... Let $R_{g_0,C}(q)$ denote the corresponding refined tropical generating series, and set $q=e^{iu}$....

conjecture | open | MathDB 页面 | 原文线索

V2-026533 | MATHDB-354650

FLS generating-series correspondence conjecture for lambda-class invariants

摘录 / EXCERPT: Let C be a curve class, let g_0 be fixed, and define the FLS generating series ... Let $R^{FLS}_{g_0,C}(q)$ denote the corresponding refined tropical FLS generating series, and s...

conjecture | open | MathDB 页面 | 原文线索

V2-026534 | MATHDB-354651

Tropical disk counting reconstruction conjecture for wall structures

摘录 / EXCERPT: Tropical disk counting reconstruction conjecture. For every $k\in\mathbb N$, the sets $\mathfrak{p}^k[x]$ are either empty or polyhedral subsets of codimension at least one, and, u...

conjecture | open | MathDB 页面 | 原文线索

V2-026535 | MATHDB-354663

Conjecture that recovered q-series are invariants for arbitrary plumbings

摘录 / EXCERPT: Let a plumbed 3-manifold be represented by an arbitrary, possibly non-reduced, plumbing, and let the associated (q,t)-series be specialized by taking $t\rightarrow 1$ to obtain a...

conjecture | open | MathDB 页面 | 原文线索

V2-026536 | MATHDB-354667

Non-homeomorphism conjecture for infinity harmonic maps

摘录 / EXCERPT: Let u be a limit infinity harmonic map between hyperbolic surfaces, and let λ be the canonical geodesic lamination with image μ. Non-homeomorphism conjecture. The lim...

conjecture | open | MathDB 页面 | 原文线索

V2-026537 | MATHDB-354670

Metric-monotone form of Min-Oo's conjecture

摘录 / EXCERPT: Let $(\boldsymbol{S}^n_+,\hat g)$ be the standard n-dimensional hemisphere, and let g be the metric appearing in Min-Oo's conjecture. The notation $g\geq\hat g$ means that g...

conjecture | open | MathDB 页面 | 原文线索

V2-026538 | MATHDB-354689

Periodic geodesic conjecture for manifolds with locally convex ends

摘录 / EXCERPT: Let M be a complete non-compact Riemannian manifold with locally convex ends, meaning that each end can be cut off by a bounded hypersurface that is locally convex toward infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-026539 | MATHDB-354699

Stein-fillability conjecture for the edge contact structures on Brieskorn spheres

摘录 / EXCERPT: For $n>2$, consider the tight contact structures $\eta^n_{i,j}$ on $-\Sigma(2,3,6n-1)$ with $0\leq i\leq n-2$, $|j|\leq n-i-2$, and $j\equiv n-i\pmod{2}$, and the tight contact str...

conjecture | open | MathDB 页面 | 原文线索

V2-026540 | MATHDB-354707

Miyaoka-type Chern class inequalities for generically nef sheaves

摘录 / EXCERPT: Let X be a compact Kähler manifold, let $\mathcal{E}$ be a torsion-free coherent sheaf of rank r, and let $\alpha_1,\ldots,\alpha_{n-2}$ be Kähler classes on X. A sheaf is ge...

conjecture | open | MathDB 页面 | 原文线索

V2-026541 | MATHDB-354708

Generic nefness conjecture for the cotangent and tangent bundles

摘录 / EXCERPT: Let X be a compact Kähler manifold, and let $\alpha_1,\ldots,\alpha_{n-1}$ be Kähler classes. For a coherent sheaf, generically nef means generically nef with respect to the indi...

conjecture | open | MathDB 页面 | 原文线索

V2-026542 | MATHDB-354714

Dehn-surgery conjecture for the manifolds at infinity of $\Delta_{3,4,\infty;n}$

摘录 / EXCERPT: Dehn-surgery conjecture. The manifold at infinity of the even subgroup of $\Delta_{3,4,\infty;n}$ is the hyperbolic 3-manifold obtained by Dehn surgery on the first cusp of...

conjecture | open | MathDB 页面 | 原文线索

V2-026543 | MATHDB-354741

Rosenberg-Stolz conjecture for products with the plane

摘录 / EXCERPT: Rosenberg-Stolz conjecture. $Y\times\mathbb{R}^2$ does not admit a complete metric with uniformly positive scalar curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-026544 | MATHDB-354748

Ben-Zvi's singular-support conjecture for chains

摘录 / EXCERPT: Let X be a smooth ambient space and let $Z(f)\subset X$ be the zero locus of f. Model chains on $Z(f)$ using currents on X, and call a current smooth when it is represented b...

conjecture | open | MathDB 页面 | 原文线索

V2-026545 | MATHDB-354750

A hyperbolic Michael-Simon inequality for higher mean curvatures

摘录 / EXCERPT: Higher-mean-curvature hyperbolic Michael-Simon conjecture. For $1\le k\le n$, the inequality

conjecture | open | MathDB 页面 | 原文线索

V2-026546 | MATHDB-354754

Factoriality conjecture for sextic double solids with few mild singularities

摘录 / EXCERPT: Factoriality conjecture for sextic double solids. If X has at most four singular points, then X is factorial.

conjecture | open | MathDB 页面 | 原文线索

V2-026547 | MATHDB-354763

The Laurent deformation conjecture for Tyurin-degenerate Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let the central Hirzebruch scroll be the limiting member of the deformation family of Hirzebruch scrolls described above, and let its Calabi-Yau hypersurface be Tyurin-degenerate,...

conjecture | open | MathDB 页面 | 原文线索

V2-026548 | MATHDB-354773

Gromov-Witten reconstruction conjecture from Fukaya-category splittings

摘录 / EXCERPT: Let X be a symplectic manifold whose Fukaya category splits as ... where the Y_i are not necessarily monotone symplectic manifolds. Gromov-Witten reconstruction conjecture. The G...

conjecture | open | MathDB 页面 | 原文线索

V2-026549 | MATHDB-354795

Symmetry conjecture for positive and negative volumes of Laplace eigenfunctions

摘录 / EXCERPT: Let (M,g) be a smooth Riemannian manifold, and let ψ_{λ} be a real Laplace eigenfunction satisfying ... Write vol for Riemannian volume. Symmetry conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-026550 | MATHDB-354796

Asymptotic L^p symmetry conjecture for positive and negative parts of eigenfunctions

摘录 / EXCERPT: Let (M,g) be a given Riemannian manifold, let ψ_{λ} be a real Laplace eigenfunction, and let $1<p<\infty$. For a set $A\subseteq M$, write χ_A for its characteris...

open_problem | open | MathDB 页面 | 原文线索

V2-026551 | MATHDB-354797

The symmetry conjecture for spherical harmonics

摘录 / EXCERPT: Symmetry conjecture. The limit

conjecture | refuted | MathDB 页面 | 原文线索

V2-026552 | MATHDB-354815

RLF conjecture for isotropic divisors on projective irreducible holomorphic symplectic manifolds

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic (IHS) manifold and let D be an integral divisor on X that is isotropic with respect to the Beauville-Bogomolov-Fujiki...

conjecture | open | MathDB 页面 | 原文线索

V2-026553 | MATHDB-354816

Conjecture on the pseudo-effective cone of projective irreducible holomorphic symplectic manifolds

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic (IHS) manifold with Picard number greater than 3. Let $\operatorname{Neg}(X)$ denote the set of negative divisors used...

conjecture | open | MathDB 页面 | 原文线索

V2-026554 | MATHDB-354832

The nearly G_2 conifold desingularization obstruction conjecture

摘录 / EXCERPT: Let $(\overline{X},\varphi_{CS})$ be a nearly G_2 conifold with a singularity at x_0 modelled on the G_2 cone $(C=C(L),\varphi_C)$. An obstruction conjecture for nearly G_2...

conjecture | open | MathDB 页面 | 原文线索

V2-026555 | MATHDB-354833

The nearly Kähler conifold desingularization obstruction conjecture

摘录 / EXCERPT: Let $(\overline{X},\omega_{CS},\rho_{CS})$ be a nearly Kähler conifold with a singularity at x_0 modelled on the Calabi--Yau cone $(C=C(L),\omega_C,\rho_C)$. An obstruction conje...

conjecture | open | MathDB 页面 | 原文线索

V2-026556 | MATHDB-354855

Equality of regular and weak order-detected slopes

摘录 / EXCERPT: Let M be a knot manifold. The sets of regular order-detected and weakly order-detected slopes on its boundary are denoted by $4\mathcal{D}_{ord}(M)4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026557 | MATHDB-354856

Order-detection converse for left-orderable gluings

摘录 / EXCERPT: Let M_1 and M_2 be knot manifolds, and let ... where $f:\partial M_1\xrightarrow{\;\cong\;}\partial M_2$ is a homeomorphism. A slope $[\alpha_i]$ belongs to the set...

conjecture | open | MathDB 页面 | 原文线索

V2-026558 | MATHDB-354857

Floer simplicity detected by weak order-detected slopes

摘录 / EXCERPT: Let M be a knot manifold. Write $\mathcal{S}(M)$ for the set of slopes on ∂M, $\mathcal{D}_{ord}^{wk}(M)$ for the set of weakly order-detected slopes, and $LO(M)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-026559 | MATHDB-354860

Elekes' six-points-on-a-conic conjecture

摘录 / EXCERPT: Elekes' conjecture. There exists $n_0=n_0(C)$ such that, if $|A|\geq n_0$ and the number of directions determined by A is at most $C|A|$, then A contains six points on a possib...

conjecture | open | MathDB 页面 | 原文线索

V2-026560 | MATHDB-354864

Properness conjecture for moment maps of complete toric Kähler-Ricci solitons

摘录 / EXCERPT: Properness conjecture. The moment map $\bm\mu$ is proper.

conjecture | open | MathDB 页面 | 原文线索

V2-026561 | MATHDB-354870

Supermodularity conjecture for a convex body and compact summands

摘录 / EXCERPT: Let $n\geq 2$. For a convex body A and compact sets $B,C\subset\mathbb{R}^n$, consider their Minkowski sums. Supermodularity conjecture. ... This extends the known one-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-026562 | MATHDB-354891

Vanishing Einstein spectral functional for suitably regular spectral triples of dimension 2

摘录 / EXCERPT: Vanishing Einstein functional conjecture. The Einstein spectral functional vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-026563 | MATHDB-354893

First-order magnitude conjecture for small convex bodies in hypermetric normed spaces

摘录 / EXCERPT: Let $E=(\mathbb{R}^n,\left\Vert\cdot\right\Vert)$ be a hypermetric normed space, and let $K\in\mathcal{K}^n$, the family of compact convex bodies in $\mathbb{R}^n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026564 | MATHDB-354902

Brazitikos-McIntyre conjecture on vector Maclaurin inequalities

摘录 / EXCERPT: Let $x_1,x_2,\ldots,x_n\in\mathbb{R}^d$ be given with $1\leq d\leq n$. For $p\in[0,\infty]$ and $2\leq k\leq d$, consider the k-fold exterior products...

conjecture | open | MathDB 页面 | 原文线索

V2-026565 | MATHDB-354924

The normalized tangent bundle classification conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold of Picard number 1 with dimension at least 3. The normalized tangent bundle of X is pseudoeffective if and only if X is one of the following vari...

conjecture | open | MathDB 页面 | 原文线索

V2-026566 | MATHDB-354944

The modified SYZ conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Modified SYZ conjecture. Given any Calabi-Yau manifold X, there exist a Lagrangian fibration $\pi:X\to B$ and a tropically continuous surjection $f:\mathscr Y\to B$ from an analy...

conjecture | open | MathDB 页面 | 原文线索

V2-026567 | MATHDB-354960

Generic codimension additivity for subdivisions of graphical-model supports

摘录 / EXCERPT: Generic codimension additivity conjecture. If $\mathcal{S}_{G,\mathcal{X}}$ and all tent functions $h_{\mathcal{X}_C,y_C}$ are sufficiently generic, then

conjecture | open | MathDB 页面 | 原文线索

V2-026568 | MATHDB-355000

Stable-cone conjecture for universal hypersurfaces in the plane

摘录 / EXCERPT: Let $X_{2,d}$ be the universal degree-d hypersurface over $\mathbb{P}^2$, and let $\mathbb{P}^2_{\mathrm{stab}}$ denote the stable universal-hypersurface space. Write classes in...

conjecture | open | MathDB 页面 | 原文线索

V2-026569 | MATHDB-355024

Connelly's densest triangulated circle-packing conjecture

摘录 / EXCERPT: Connelly's conjecture. There exists no circle packing of the triangulated plane with r greater than the ratio attained by the reference packing shown in the figure.

conjecture | open | MathDB 页面 | 原文线索

V2-026570 | MATHDB-355032

Milnor's Hausdorff-dimension conjecture for exceptional hyperbolic component boundaries

摘录 / EXCERPT: Milnor's Hausdorff-dimension conjecture. If $\mathcal H$ is not exceptional, then $\partial\mathcal H$ is a fractal set in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-026571 | MATHDB-355036

The flag-manifold conjecture for projectively immersed Kähler-Einstein manifolds

摘录 / EXCERPT: A Kähler-Einstein manifold is a Kähler manifold whose Ricci tensor is proportional to its metric, and a Kähler immersion is a holomorphic isometric immersion into a complex space f...

conjecture | open | MathDB 页面 | 原文线索

V2-026572 | MATHDB-355037

The classification conjecture for projectively induced S^1-invariant Kähler-Einstein manifolds

摘录 / EXCERPT: An S^1-invariant Kähler manifold is a Kähler manifold admitting, in suitable local holomorphic coordinates, a Kähler potential depending only on $(|z_1|^2,\dots,|z_n|^2)$. A Kähl...

conjecture | open | MathDB 页面 | 原文线索

V2-026573 | MATHDB-355050

Székelyhidi's parametrization conjecture for Calabi-Yau metrics with tangent cone \(\mathbf{C} \times A_2\)

摘录 / EXCERPT: Let $\mathbf{C}^3$ carry a Calabi-Yau metric whose tangent cone at infinity is $\mathbf{C} \times A_2$. Identify metrics that differ by biholomorphism and scaling. Székelyhidi's pa...

conjecture | open | MathDB 页面 | 原文线索

V2-026574 | MATHDB-355074

Convergence of the Mabuchi functional on canonical approximants

摘录 / EXCERPT: Let $(X,B;L)$ be a polarized pair, let $R(X,L)$ be its section ring, and let $ $ be a rational divisorial norm. For sufficiently divisible d, let $ _d$ be the norm on $R(X,dL)$ g...

conjecture | open | MathDB 页面 | 原文线索

V2-026575 | MATHDB-355107

Dual-complex sphere conjecture for log Calabi-Yau pairs

摘录 / EXCERPT: Let (X,B) be an n-dimensional log Calabi-Yau pair, and let $\mathcal{D}(X,B)$ denote its dual complex. A finite cover $\pi\colon Y\rightarrow X$ is equipped with the log pull-b...

conjecture | open | MathDB 页面 | 原文线索

V2-026576 | MATHDB-355145

Yamanouti-set conjecture for the Cheeger constant, perimeter, and minimal width

摘录 / EXCERPT: Let $\mathcal{K}^2$ denote the class of planar convex sets, and let $\omega(\Omega)$, $h(\Omega)$, and $P(\Omega)$ denote respectively the minimal width, Cheeger constant, and peri...

conjecture | open | MathDB 页面 | 原文线索

V2-026577 | MATHDB-355146

Nonagon conjecture for the Cheeger constant at fixed diameter and circumradius

摘录 / EXCERPT: Let $\mathcal{K}^2$ denote the class of planar convex sets, and let $h(\Omega)$, $d(\Omega)$, and $R(\Omega)$ denote respectively the Cheeger constant, diameter, and circumradius....

conjecture | open | MathDB 页面 | 原文线索

V2-026578 | MATHDB-355157

Integrality and symmetry conjecture for higher-genus BPS invariants

摘录 / EXCERPT: For a symplectic log Calabi-Yau fourfold (X,D), let $A\in H_2(X,\mathbb Z)$ and $g\geq 1$. Define ... where μ is the Möbius function, $q=\operatorname{e}^{\mathrm{i}h/2}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026579 | MATHDB-355162

The relative Morrison-Kawamata cone conjecture for Calabi-Yau fiber spaces

摘录 / EXCERPT: Let $(X,\Delta) \to S$ be a klt Calabi-Yau fiber space. Let Γ_B and Γ_A be the images of the pseudo-automorphism group $\operatorname{PsAut}(X/S,\Delta)$ and the auto...

conjecture | open | MathDB 页面 | 原文线索

V2-026580 | MATHDB-355164

Finiteness of the relative kernel group for Calabi-Yau fibrations

摘录 / EXCERPT: Finiteness conjecture. The group Γ_W is always finite. The preceding lemma proves finiteness under the stronger assumption that X has terminal singularities; the conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-026581 | MATHDB-355190

Villani's conjecture on convexity of injectivity domains under the MTW condition

摘录 / EXCERPT: Villani's conjecture. If the squared-distance cost satisfies the MTW condition, then all injectivity domains of M are convex.

conjecture | open | MathDB 页面 | 原文线索

V2-026582 | MATHDB-355195

Li's conjecture on the threshold for positive curvature operator of the second kind

摘录 / EXCERPT: Let M^n be a closed n-dimensional Riemannian manifold, and let $n+\frac{n-2}{n}$ be the positivity threshold for its curvature operator of the second kind. Li's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-026583 | MATHDB-355199

The earthquake homeomorphism conjecture for cluster algebras of finite type

摘录 / EXCERPT: Earthquake homeomorphism conjecture. There is a $\Gamma_{\boldsymbol{s}}$-equivariant homeomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-026584 | MATHDB-355225

Holomorphic Lefschetz formula for transversal correspondences

摘录 / EXCERPT: Holomorphic Lefschetz conjecture for correspondences. The displayed formula holds. This is intended as the correspondence analogue of the holomorphic Lefschetz fixed point formula,...

conjecture | open | MathDB 页面 | 原文线索

V2-026585 | MATHDB-355239

Larson-Vogt conjecture on signed real bitangent counts of plane quartics

摘录 / EXCERPT: Let $V(Q)$ be a real quartic such that for each bitangent L, $L\cap V(Q)\cap L_\infty=\emptyset$, where L_∞ denotes the line at infinity. For a real bitangent L, let...

conjecture | open | MathDB 页面 | 原文线索

V2-026586 | MATHDB-355240

Sharp inclusion conjecture for triangular ratio and hyperbolic metric balls

摘录 / EXCERPT: Sharp inclusion conjecture. For $x\in G$ and $R>0$, $B_s(x,t_0)\subseteq B_\rho(x,R)\subseteq B_s(x,t_1)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-026587 | MATHDB-355267

Generalised abundance conjecture for klt pairs with nef adjoint divisors

摘录 / EXCERPT: Generalised abundance conjecture. If K_X+B+M is nef, then K_X+B+M is num-semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-026588 | MATHDB-355298

Sufficiency of the higher-Lie-bracket condition for mirror-flow simulation

摘录 / EXCERPT: Let M be a smooth submanifold of $\mathbb{R}^D$, let $G:M\to\mathbb{R}^d$ be a smooth parametrization, and let $\mathrm{Lie}^{\geq 2}(\partial G)$ denote the span of Lie brackets...

conjecture | open | MathDB 页面 | 原文线索

V2-026589 | MATHDB-355299

Perfectoidness conjecture for the spaces W_{n,d}

摘录 / EXCERPT: Let n be a positive integer and, for each integer d with $0<d<n$, let $W_{n,d}$ be the space defined in the paper from the corresponding local geometric construction. Perfectoi...

conjecture | open | MathDB 页面 | 原文线索

V2-026590 | MATHDB-355302

Quermassintegral inequalities conjecture for spacelike domains in de Sitter space

摘录 / EXCERPT: Quermassintegral inequalities conjecture. There hold

conjecture | open | MathDB 页面 | 原文线索

V2-026591 | MATHDB-355330

Polygonal Ashbaugh-Benguria-Payne-Pólya-Weinberger conjecture for regular polygons

摘录 / EXCERPT: Let $P_k\subset\mathbb{R}^2$ be an open k-gon, and let $P_k^*\subset\mathbb{R}^d$ be an open regular k-gon. For $k\geq 3$, define the fundamental ratio by...

conjecture | open | MathDB 页面 | 原文线索

V2-026592 | MATHDB-355331

Concavity conjecture for the fundamental ratio on triangle moduli space

摘录 / EXCERPT: Let $\xi(D)=\lambda_2(D)/\lambda_1(D)$ be the fundamental ratio of the first two Dirichlet-Laplacian eigenvalues. Consider the moduli space of triangles after quotienting by the sy...

conjecture | open | MathDB 页面 | 原文线索

V2-026593 | MATHDB-355359

Lipschitz regularity at infinity and affine linearity conjecture

摘录 / EXCERPT: Let $X\subset\mathbb{C}^n$ be a pure d-dimensional entire complex analytic subset. Suppose that there exists a sequence $T=\{t_j\}_{j\in\mathbb{N}}$ of real positive numbers such...

conjecture | open | MathDB 页面 | 原文线索

V2-026594 | MATHDB-355381

Ekedahl-Shepherd-Barron-Taylor conjecture on algebraic integrability of foliations

摘录 / EXCERPT: Ekedahl-Shepherd-Barron-Taylor conjecture. The foliation $\mathcal F$ is algebraically integrable if and only if $\mathcal F_{\mathfrak p}$ is p-closed for every closed point...

conjecture | open | MathDB 页面 | 原文线索

V2-026595 | MATHDB-355382

Logarithmic-component conjecture for irreducible foliation components

摘录 / EXCERPT: Logarithmic-component conjecture. If $\delta_2(\Sigma)=d$, then there exist integers d_1 and d_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-026596 | MATHDB-355383

Algebraic-integrability or logarithmic-component conjecture at second degree defect one

摘录 / EXCERPT: Algebraic-integrability or logarithmic-component conjecture. If

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V2-026597 | MATHDB-355395

Volume conjecture for indecomposable summands of toric NCCRs

摘录 / EXCERPT: Toric summand-count conjecture. The number of indecomposable summands in the reflexive module defining an NCCR of R is equal to

conjecture | open | MathDB 页面 | 原文线索

V2-026598 | MATHDB-355403

Quantified asymptotic Euclideanity conjecture for RCD spaces

摘录 / EXCERPT: Quantified asymptotic Euclideanity conjecture. There exists $\delta(N,\epsilon)>0$ such that, for every $\delta\in(0,\delta(N,\epsilon)]$, there exists $R_0\geq 1$ for which...

conjecture | open | MathDB 页面 | 原文线索

V2-026599 | MATHDB-355416

Qin's dual-torus Fukaya-category equivalence conjecture

摘录 / EXCERPT: Qin's dual-torus Fukaya-category conjecture. The Fukaya category of $(T,B+i\omega)$ should be equivalent to the Fukaya category of its dual $(T^\vee,(B+i\omega)^{-1})$.

conjecture | open | MathDB 页面 | 原文线索

V2-026600 | MATHDB-355418

Petersen-Wilhelm fiber dimension conjecture

摘录 / EXCERPT: Let $F\hookrightarrow M\rightarrow B$ be a Riemannian submersion from a positively curved closed manifold M. Petersen-Wilhelm's fiber dimension conjecture. Then ... This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-026601 | MATHDB-355419

Petersen-Wilhelm fiber dimension conjecture for Riemannian submersions

摘录 / EXCERPT: Let $\pi\colon (M^{n+k},\textsl g)\to (B^n,b)$ be a Riemannian submersion. Suppose that $(M,\textsl g)$ is compact and has positive sectional curvature. Petersen-Wilhelm's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-026602 | MATHDB-355441

Buser's conjecture on Bers constants for hyperbolic surfaces

摘录 / EXCERPT: Buser's conjecture. The surface M has a pants decomposition in which each curve has length at most

conjecture | open | MathDB 页面 | 原文线索

V2-026603 | MATHDB-355483

Conjecture that planar grazes characterize ellipsoids

摘录 / EXCERPT: Let $K,L\subset\mathbb R^n$ be convex bodies with $L\subset\operatorname{int}K$. For a point $x\in\mathbb R^n\setminus L$, let $\Sigma(L,x)$ denote the graze of L from x, namel...

conjecture | open | MathDB 页面 | 原文线索

V2-026604 | MATHDB-355518

Pentagram recutting singularity type conjecture

摘录 / EXCERPT: Let $m\geq 1$, let $\tilde{f}:\mathbb{Z}^2\rightarrow {\mathbb{R}\mathrm{P}}^2$ be a $2m$-closed pentagram lattice map, equivalently let...

conjecture | open | MathDB 页面 | 原文线索

V2-026605 | MATHDB-355570

Finite global Khovanskii basis conjecture for cluster algebras

摘录 / EXCERPT: Let $A(\Gamma)$ be a cluster algebra satisfying the assumptions of Theorem, and suppose it has a finite global Khovanskii basis, meaning a finite set of algebra generators that is...

conjecture | open | MathDB 页面 | 原文线索

V2-026606 | MATHDB-355602

Dihedral rigidity conjecture for hyperbolic polyhedra

摘录 / EXCERPT: For a unit vector $cvec{a}=(a^1,a^2,a^3)$ and $s oindent\text{in}cmathbb{R}$, let ... be the totally umbilic planes in the upper-half-space model $(cmathbb{R}^3_+,b)$ of hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-026607 | MATHDB-355605

The canonical-bundle Orlov equivalence for Fukaya categories

摘录 / EXCERPT: Let Z be a general type symplectic manifold. Let $P(K_Z)$ be the principal $\mathbb{C}^{\times}$-bundle associated to the canonical bundle K_Z, and let $\mathrm{Fuk}(Z)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-026608 | MATHDB-355607

The Orlov equivalence for a principal canonical bundle

摘录 / EXCERPT: Let Z be a symplectic manifold of general type, with principal canonical bundle $P_Z=K_Z\setminus Z$ and grading form Ω_Z. Let $\mathrm{RFuk}^{\mathrm{gr}}(P_Z,\Omega_Z)$...

conjecture | open | MathDB 页面 | 原文线索

V2-026609 | MATHDB-355608

Lekili-Ueda's conjecture for complements of smooth ample divisors

摘录 / EXCERPT: Let V be a compact Calabi-Yau manifold, and let $Z\subset V$ be a smooth ample divisor. Set $U=V\setminus Z$, which is a Weinstein manifold. Let $\mathrm{Fuk}^{w}(U)$ be the grad...

conjecture | open | MathDB 页面 | 原文线索

V2-026610 | MATHDB-355610

The Rabinowitz Fukaya equivalence for a nodal divisor complement

摘录 / EXCERPT: Let Z be a nodal divisor in a compact symplectic surface, let U be its complement, and let P_Z be the twisted principal canonical bundle obtained from a smoothing by cutting...

conjecture | open | MathDB 页面 | 原文线索

V2-026611 | MATHDB-355613

Equality of spectral-radius limits for generalized affine fractal interpolation functions

摘录 / EXCERPT: Let $I=[0,1]$ be the domain, let S be the vertical scaling function, and let M_k and M'_k be the matrices associated with the k-th subdivision, with spectral radii denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-026612 | MATHDB-355616

The tautologicality conjecture for effective classes of a logarithmic target

摘录 / EXCERPT: Let $\infty_\mathcal{X}$ be a logarithmic target with Gromov--Witten classes, and let ∞ be the associated target whose effective classes are under consideration. A class is...

conjecture | open | MathDB 页面 | 原文线索

V2-026613 | MATHDB-355619

The near-three-halves threshold conjecture for the scale L2-norm

摘录 / EXCERPT: Let E be a subset of $\mathbb{F}_q^2$, and let S_E denote the set of scales determined by pairs of points of E, with $L^2(S_E)$ its second moment. Near-three-halves threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-026614 | MATHDB-355632

No semistability at infinity for Calabi-Yau metrics without curvature decay

摘录 / EXCERPT: Let X be a complete Calabi-Yau manifold in the setting of Theorem ... should hold without the quadratic curvature decay assumption. This is the expected extension of the paper's ma...

conjecture | open | MathDB 页面 | 原文线索

V2-026615 | MATHDB-355645

The positive-geometry conjecture for Grassmann polytopes and amplituhedra

摘录 / EXCERPT: Let $Z:{\mathbb R}^n\to{\mathbb R}^{k+m}$ be a linear map with $n\geq k+m$, and let $P=Z({\mathrm{Gr}}(k,n)_{\geq 0})$ be the resulting Grassmann polytope when Z is well-defined...

conjecture | open | MathDB 页面 | 原文线索

V2-026616 | MATHDB-355646

The birational simplex-like subdivision conjecture for positive geometries

摘录 / EXCERPT: Let $(X,X_{\geq 0})$ be a positive geometry. A subdivision is a collection of positive geometries $(X_1,X_{1,\geq 0}),\ldots,(X_r,X_{r,\geq 0})$ with pairwise disjoint interiors an...

conjecture | open | MathDB 页面 | 原文线索

V2-026617 | MATHDB-355652

Iwaniec-Sarnak conjecture for negatively curved surfaces

摘录 / EXCERPT: Let M be a negatively curved surface, let $K\subseteq M$ be compact, and let g be the eigenfunction and λ_g its eigenvalue parameter appearing in the Sogge L^p-norm b...

conjecture | open | MathDB 页面 | 原文线索

V2-026618 | MATHDB-355662

Eliashberg's non-exact-fillability conjecture for real projective contact spaces

摘录 / EXCERPT: The standard contact structure on real projective space is denoted by $(\mathbb{R}\mathbb{P}^{2n-1},\xi_{\rm std})$. Eliashberg's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-026619 | MATHDB-355663

Terminal quotient-link conjecture on non-exact fillability

摘录 / EXCERPT: Let $\mathbb{C}^n/G$ be an isolated terminal singularity. Its contact link is $(S^{2n-1}/G,\xi_{\rm std})$. Terminal quotient-link conjecture. The contact link...

conjecture | open | MathDB 页面 | 原文线索

V2-026620 | MATHDB-355664

Terminal quotient-link conjecture on uniqueness of orbifold fillings

摘录 / EXCERPT: Let $\mathbb{C}^n/G$ be an isolated terminal singularity, and let $(S^{2n-1}/G,\xi_{\rm std})$ be its contact link. Terminal quotient-link orbifold-filling conjecture. Exact orbifo...

conjecture | open | MathDB 页面 | 原文线索

V2-026621 | MATHDB-355667

Binomial-transform vanishing conjecture

摘录 / EXCERPT: Let $\{b_n\}_{n\geq0}$ be a complex number sequence such that, for some integer $l\geq1$, $b_nb_{n+l}=0$ for every $n\in\mathbb{N}$, and define ... If $\lim_{n\to\infty}a_n=0$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-026622 | MATHDB-355668

Higher-dimensional projector conjecture for smooth noncommutative Euclidean spaces

摘录 / EXCERPT: Let $P_k(A^\infty(\mathbb{R}^{2n}_\Theta))$ denote the class of projectors of level k in the smooth noncommutative Euclidean algebra, where $k\geq0$, $n\geq1$, and Θ is th...

conjecture | open | MathDB 页面 | 原文线索

V2-026623 | MATHDB-355669

K-theory conjecture for smooth noncommutative Euclidean spaces

摘录 / EXCERPT: Let $A^\infty(\mathbb{R}^{2n}_\Theta)$ be the smooth noncommutative Euclidean algebra, with $n\in\mathbb{N}^*$ and deformation parameter Θ. The smooth noncommutative Euclide...

conjecture | open | MathDB 页面 | 原文线索

V2-026624 | MATHDB-355670

Triviality conjecture for diagonal smooth projectors

摘录 / EXCERPT: Let $n\in\mathbb{N}^*$, and let $\{\mathfrak{b}_{m,n}\}_{m\geq0}$ be a sequence satisfying ... These sequences encode the diagonal elements of $P_0(A^\infty(\mathbb{R}^{2n}_\Theta))$...

conjecture | open | MathDB 页面 | 原文线索

V2-026625 | MATHDB-355689

Kalai's affine-stress reconstruction conjecture up to affine type

摘录 / EXCERPT: Kalai's affine-type reconstruction conjecture. If P has no missing $(d-1)$-faces, then its graph and the space of affine 2-stresses determine P up to affine equivalence. More...

conjecture | open | MathDB 页面 | 原文线索

V2-026626 | MATHDB-355704

Cao's rigidity conjecture for gradient shrinking Ricci solitons

摘录 / EXCERPT: Let (M,g,f) be a gradient shrinking Ricci soliton with constant scalar curvature. In the terminology recalled in the source, such a soliton is rigid when it is a flat bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-026627 | MATHDB-355707

The metric-completion conjecture for LCK surfaces with proper potential

摘录 / EXCERPT: Let M be a compact LCK complex surface with proper potential, and let $\widetilde M$ be its Kähler $\mathbb Z$-covering. The metric-completion conjecture. The metric completion o...

conjecture | open | MathDB 页面 | 原文线索

V2-026628 | MATHDB-355740

The two embedded geodesic lines conjecture for complete Riemannian planes

摘录 / EXCERPT: Let M be a complete Riemannian plane, meaning a complete Riemannian surface homeomorphic to the plane. An embedded geodesic line is the image $c(\mathbb R)$ of an injective, prop...

conjecture | open | MathDB 页面 | 原文线索

V2-026629 | MATHDB-355741

The two disjoint geodesic lines conjecture for complete Riemannian cylinders

摘录 / EXCERPT: Let S be a complete Riemannian surface homeomorphic to the cylinder $S^1\times\mathbb R$. An embedded geodesic line is the image $c(\mathbb R)$ of an injective, proper geodesic...

open_problem | open | MathDB 页面 | 原文线索

V2-026630 | MATHDB-355753

Generalized Bernstein-Schwarzman conjecture for weak weighted projective spaces

摘录 / EXCERPT: Let Γ and Γ_1 be commensurable complex crystallographic groups acting on $\mathbb{C}^n$, with equal linear parts $\mathrm{d}\Gamma_1=\mathrm{d}\Gamma$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-026631 | MATHDB-355767

The disconnected contact-poset conjecture

摘录 / EXCERPT: Disconnected contact-poset conjecture. $\mathcal{P}$ is contact if and only if it is a disjoint sum of two Frobenius, toral posets.

conjecture | open | MathDB 页面 | 原文线索

V2-026632 | MATHDB-355768

The connected contact-poset torality conjecture

摘录 / EXCERPT: Connected contact-poset torality conjecture. If $\mathfrak{g}_A(\mathcal{P})$ is contact, then $\mathcal{P}$ is toral.

conjecture | open | MathDB 页面 | 原文线索

V2-026633 | MATHDB-355773

Cheltsov's nonrationality conjecture for Fano-Enriques threefolds

摘录 / EXCERPT: Cheltsov's conjecture. In each of these three cases, X is not rational.

conjecture | open | MathDB 页面 | 原文线索

V2-026634 | MATHDB-355784

The pseudo complex volume-complex volume equivalence conjecture for Dehn fillings

摘录 / EXCERPT: Let $\mathcal{M}$ be an n-cusped hyperbolic 3-manifold. Let $\mathcal{M}_{p_1/q_1, \dots, p_n/q_n}$ and $\mathcal{M}_{p'_1/q'_2,\dots, p'_n/q'_n}$ be two Dehn fillings of...

conjecture | open | MathDB 页面 | 原文线索

V2-026635 | MATHDB-355785

Ramakrishnan's conjecture relating complex volume and the Bloch invariant

摘录 / EXCERPT: Let $\mathcal{M}$ and $\mathcal{N}$ be hyperbolic 3-manifolds. Write $\operatorname{vol}_{\mathbb{C}}$ for complex volume and β for the Bloch invariant. Ramakrishnan's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-026636 | MATHDB-355786

Cylindrical ball neighborhood conjecture for isolated critical points

摘录 / EXCERPT: Let $f\colon M\to\mathbb{R}$ be a smooth function on a manifold M without boundary, and let $x_0\in M$ be an isolated critical point. A cylindrical ball neighborhood is a cylindr...

conjecture | open | MathDB 页面 | 原文线索

V2-026637 | MATHDB-355800

Bounded fiber multiplicities for Fano fibrations over curves

摘录 / EXCERPT: Let d be a positive integer and ϵ a positive real number. Let (X,B) be a log pair of dimension d and let $\pi:X\to C$ be a contraction, where X is of Fano type ove...

conjecture | open | MathDB 页面 | 原文线索

V2-026638 | MATHDB-355823

Iarrobino's conjecture on Waldschmidt constants of very general points

摘录 / EXCERPT: Iarrobino's conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-026639 | MATHDB-355831

The geometric gluing conjecture for BV-BFV field spaces

摘录 / EXCERPT: Let M be cut by a closed $(n-1)$-submanifold Σ into two n-manifolds with boundary, M_1 and M_2. Let $\mathcal{F}_M$ be the space of fields on M, let Φ_Σ...

conjecture | open | MathDB 页面 | 原文线索

V2-026640 | MATHDB-355832

The half-monodromy rotation-angle conjecture for nonplanar bicycling geodesics

摘录 / EXCERPT: Let $({\bf x}(t),{\bf y}(t))$ be a nonplanar bicycling geodesic with monodromy having rotation angle $\Delta\theta\in[0,2\pi)$ and translation $\Delta z>0$ about its axis. Let I...

conjecture | open | MathDB 页面 | 原文线索

V2-026641 | MATHDB-355853

Geodesic--argmax rigidity conjecture for the directed landscape

摘录 / EXCERPT: Geodesic--argmax rigidity conjecture. For every $\tau\in(0,1)$ and $T>0$, conditionally given $\mathcal L(0,0;0,T)=L$, as $L\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-026642 | MATHDB-355878

The $N/6$ guard conjecture for integral orthogonal polygons

摘录 / EXCERPT: Let P be an integral orthogonal polygon, meaning that all its edges have integer length, and let N be its perimeter. The $N/6$ guard conjecture. The polygon P can always be g...

conjecture | open | MathDB 页面 | 原文线索

V2-026643 | MATHDB-355884

Strong optimality conjecture for the Desarguesian ovoid code

摘录 / EXCERPT: Let C_O be the code associated with the Desarguesian ovoid in $Q^+(7,q)$, with $q\geq 3$. A code is strongly optimal when it is both P-optimal and S-optimal. Strong optimalit...

conjecture | open | MathDB 页面 | 原文线索

V2-026644 | MATHDB-355886

Positive entropy conjecture for composite symplectic Dehn twists

摘录 / EXCERPT: Let S_1 and S_2 be Lagrangian n-spheres in an A_2^n-configuration, intersecting transversely at a single point, and let τ_1 and τ_2 be symplectic Dehn twists alon...

conjecture | open | MathDB 页面 | 原文线索

V2-026645 | MATHDB-355889

The quadratic distinct-angle conjecture in general position

摘录 / EXCERPT: For $d\in\{2,3\}$, let $A_{\mathrm{gen}}^{(d)}(n)$ denote the minimum number of distinct angles formed by a set of n points in general position in $\mathbb R^d$, where general po...

conjecture | open | MathDB 页面 | 原文线索

V2-026646 | MATHDB-355890

The quadratic pinned-endpoint angle conjecture

摘录 / EXCERPT: Let A be a pinned point in a configuration of n points in general position in $\mathbb R^2$ or $\mathbb R^3$, where general position means no three points are collinear and no...

conjecture | open | MathDB 页面 | 原文线索

V2-026647 | MATHDB-355903

Cube-normalized capacity uniqueness conjecture for convex domains

摘录 / EXCERPT: Cube-normalized capacity uniqueness conjecture. All cube-normalized symplectic capacities coincide on X.

conjecture | open | MathDB 页面 | 原文线索

V2-026648 | MATHDB-355907

The compact framed conical approximation conjecture

摘录 / EXCERPT: Tame diagrams approximate all diagrams. Every compact framed conical stratification $(\bl^n,f)$ has an arbitrarily close approximation by a compact manifold n-diagram.

conjecture | open | MathDB 页面 | 原文线索

V2-026649 | MATHDB-355920

The generic polygon image conjecture for polynomial almost-complex curves

摘录 / EXCERPT: Generic polygon image conjecture. The map $\alpha: \mathsf{TS}_{k}\rightarrow\mathsf{TP}_{k+6}$ has image in $\mathsf{TP}^{\mathsf{gen}}_{k+6}$ and is a homeomorphism onto its imag...

conjecture | open | MathDB 页面 | 原文线索

V2-026650 | MATHDB-355924

Shadow-flux lower-bound conjecture for Lagrangian cobordisms

摘录 / EXCERPT: Let L_0 and L_1 be Lagrangian isotopic submanifolds, and let $K:L_1\rightsquigarrow L_0$ be a Lagrangian cobordism. Let $d(L_0,L_1)$ denote the minimal flux distance, defined b...

conjecture | open | MathDB 页面 | 原文线索

V2-026651 | MATHDB-355925

The indecomposable-real-element conjecture for superunitary regions

摘录 / EXCERPT: Let $\mathcal{A}$ be a cluster algebra with a good basis $\mathfrak{B}$, and let $\mathfrak{B}_{ir}$ be the subset of indecomposable real elements, where an element b is real if...

conjecture | open | MathDB 页面 | 原文线索

V2-026652 | MATHDB-355933

The Frobenius-manifold Lyashko-Looijenga degree conjecture

摘录 / EXCERPT: Frobenius-manifold Lyashko-Looijenga degree conjecture. The degree of the map $LL(\mathcal{F}_g)$ equals

conjecture | open | MathDB 页面 | 原文线索

V2-026653 | MATHDB-355972

Conjecture on convex functions and Busemann functions

摘录 / EXCERPT: Let M be a symmetric space of noncompact type. A function $f:M\to\mathbb{R}$ is convex if it is convex in the sense used for functions on M, and a Busemann function is a functi...

conjecture | open | MathDB 页面 | 原文线索

V2-026654 | MATHDB-355990

The Fukaya-category comparison conjecture with descendant Gromov-Witten invariants

摘录 / EXCERPT: Let X be a compact symplectic manifold of real dimension $2d$, let $\mathcal{C}=\operatorname{\mathsf{Fuk}}(X)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026655 | MATHDB-356013

Persson-type spectral minimal partition principle for domains

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^d$, $d \geq 2$, be a domain, possibly equal to $\mathbb{R}^d$, and let $V:\Omega \to \mathbb{R}$ be a sufficiently smooth potential bounded from belo...

conjecture | open | MathDB 页面 | 原文线索

V2-026656 | MATHDB-356035

Minimum-volume conjecture for spherical knotoids

摘录 / EXCERPT: Minimum-volume conjecture. The unique spherical knotoid of smallest volume is the knotoid 2_1, and its volume is approximately $5.33349\dots$.

conjecture | open | MathDB 页面 | 原文线索

V2-026657 | MATHDB-356046

The geodesic growth conjecture for compact Ricci-flat Calabi-Yau manifolds

摘录 / EXCERPT: Let $(X,\tilde{g})$ be a compact, Ricci-flat Calabi-Yau manifold, where a Calabi-Yau manifold means a Kähler manifold (X,g) with vanishing first Chern class $c_1(X)=0$ and trivia...

conjecture | open | MathDB 页面 | 原文线索

V2-026658 | MATHDB-356074

Characterization of the centroid by square preservation

摘录 / EXCERPT: Centroid characterization conjecture. If $FGHI$ is a square independently of the choice of E, then the common center function must be the centroid.

conjecture | open | MathDB 页面 | 原文线索

V2-026659 | MATHDB-356080

Equal-action characteristics symplectomorphic-ball conjecture

摘录 / EXCERPT: Equal-action characteristics conjecture. If all characteristics of K are closed with the same action, then K is a possibly nonlinear symplectomorphic image of a ball.

conjecture | open | MathDB 页面 | 原文线索

V2-026660 | MATHDB-356082

Finite generation conjecture for log canonical pairs over complex analytic spaces

摘录 / EXCERPT: Finite generation conjecture. The graded $\mathcal O_Y$-algebra $R(X/Y,K_X+\Delta)$ is locally finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-026661 | MATHDB-356119

The classification conjecture for complete self-shrinkers with constant squared second fundamental form

摘录 / EXCERPT: Let $X:M^n\rightarrow\mathbb{R}^{n+1}$ be an n-dimensional complete self-shrinker, meaning that its mean curvature vector satisfies ... Suppose that the squared norm of its second...

conjecture | open | MathDB 页面 | 原文线索

V2-026662 | MATHDB-356131

Generic toral eigensheaf conjecture

摘录 / EXCERPT: Let G be a reductive group, let $\Lambda\in\{\overline{\mathbb{F}}_{\ell},\overline{\mathbb{Z}}_{\ell},\overline{\mathbb{Q}}_{\ell}\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026663 | MATHDB-356133

Hertling's conjecture on TERP structures and eigenvalues of \mathcal{Q}

摘录 / EXCERPT: Hertling's conjecture. For any $p\in M$, the set R does not contain the $\mathcal{E}+\overline{\mathcal{E}}$ orbit of p. Far enough along the flow of...

conjecture | open | MathDB 页面 | 原文线索

V2-026664 | MATHDB-356185

Lipnowski-Wright's conjecture on simple and non-simple geodesics

摘录 / EXCERPT: Lipnowski-Wright's conjecture. As $g\to\infty$, on most closed hyperbolic surfaces in $\mathcal{M}_g$, most geodesics of length significantly less than $\sqrt g$ are simple and non...

conjecture | open | MathDB 页面 | 原文线索

V2-026665 | MATHDB-356191

Injectivity conjecture for the geodesic Wilson loop operator

摘录 / EXCERPT: Wilson-loop injectivity conjecture. If (M,g) is a closed odd-dimensional Riemannian manifold with Anosov geodesic flow, then the Wilson loop operator

conjecture | open | MathDB 页面 | 原文线索

V2-026666 | MATHDB-356203

Vojta's general abc conjecture for holomorphic curves

摘录 / EXCERPT: Vojta's general abc conjecture. (a) If $f:\mathbb C\to X$ is an algebraically nondegenerate analytic map, then

conjecture | open | MathDB 页面 | 原文线索

V2-026667 | MATHDB-356226

Bounded Novikov conjecture for foliations

摘录 / EXCERPT: Let $(M,\mathcal{F})$ be a foliation on a compact manifold, let BG be the classifying space of its holonomy groupoid, and let $\mathcal{L}^{lf}$ be the cosheaf assigning to each...

conjecture | open | MathDB 页面 | 原文线索

V2-026668 | MATHDB-356227

Bounded geometry Borel conjecture for foliations

摘录 / EXCERPT: Let $(M,\mathcal{F})$ be a foliation on a compact manifold, let BG be the classifying space of its holonomy groupoid, and let S_x be the leaf through $x\in BG$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026669 | MATHDB-356228

Vanishing of higher Pontryagin classes for rationally contractible foliations

摘录 / EXCERPT: Let M be a smooth manifold and let $E\subset TM$ be an integrable subbundle defining a foliation $\mathcal{F}$ whose leaves are rationally contractible, meaning that for every le...

conjecture | open | MathDB 页面 | 原文线索

V2-026670 | MATHDB-356240

ACC conjecture for minimal log discrepancies of exceptional non-canonical pairs

摘录 / EXCERPT: ACC conjecture for mlds of enc pairs. If Γ satisfies the DCC, then $\operatorname{eMLD}_{d}(\Gamma)$ satisfies the ACC. If Γ is finite, then...

conjecture | open | MathDB 页面 | 原文线索

V2-026671 | MATHDB-356247

The weak Stoker conjecture for polyhedra

摘录 / EXCERPT: Let P_1 and P_2 be polyhedra with the same 1-skeleton Γ. Suppose that, for every edge e of Γ, the dihedral angles of P_1 and P_2 at the corresponding edge...

conjecture | open | MathDB 页面 | 原文线索

V2-026672 | MATHDB-356263

Moduli-space conjecture for Kähler manifolds with complexified polarisation

摘录 / EXCERPT: Moduli-space conjecture. There exists a Hausdorff complex space $\mathfrak{M}_{\gamma}(X,[\omega^{\mathbb{C}}])$ serving as a moduli space of Kähler manifolds with a complexified p...

conjecture | open | MathDB 页面 | 原文线索

V2-026673 | MATHDB-356272

The smoothness and symplecticity conjecture for Betti moduli spaces

摘录 / EXCERPT: Smoothness and symplecticity conjecture. Let $\kappa\in H$. For $\nu>0$, the derived structure on ${}_{\kappa}\mathcal{M}^{0}_{\operatorname{Bet},\psi}$ is trivial, and...

conjecture | open | MathDB 页面 | 原文线索

V2-026674 | MATHDB-356288

Lutwak's smooth even logarithmic Minkowski uniqueness conjecture

摘录 / EXCERPT: Let f be a positive even C^∞ function on S^{n-1}, and consider the logarithmic Minkowski problem, written in the source as equation ... has a unique even solution. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-026675 | MATHDB-356289

The centered logarithmic Brunn-Minkowski conjecture

摘录 / EXCERPT: Logarithmic Brunn-Minkowski conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-026676 | MATHDB-356290

The centered L_p-Minkowski uniqueness conjecture

摘录 / EXCERPT: Let $p\in(0,1)$, and let K and C be centered convex bodies in $\mathbb{R}^n$. Let $S_{K,p}$ denote the L_p surface area measure of K. Centered L_p-Minkowski uniqueness co...

conjecture | open | MathDB 页面 | 原文线索

V2-026677 | MATHDB-356291

The centered L_p-Minkowski inequality conjecture

摘录 / EXCERPT: Let $p\in(0,1)$ and let $K,C\in\mathcal{K}^n$ be centered convex bodies. Let $V_p(K,C)$ be the L_p mixed volume and $S_{K,p}$ the L_p surface area measure. Centered L_p-Minko...

conjecture | open | MathDB 页面 | 原文线索

V2-026678 | MATHDB-356306

Universal minimal-entropy conjecture for 2-fold branched covers of links

摘录 / EXCERPT: For a link L in S^3, let $M_L\to S^3$ be the 2-fold branched cover branched over L. The quantity $\ell_g(M_L)$ denotes the minimal entropy associated with M_L. Universal...

conjecture | open | MathDB 页面 | 原文线索

V2-026679 | MATHDB-356312

Generic nonexistence of analytic geodesics in analytic function spaces

摘录 / EXCERPT: Nonexistence conjecture for analytic geodesics. There exists an open dense subset of $(E\cap\mathcal{H})^2$ such that no pair of elements in this subset is linked by a geodesic in...

conjecture | open | MathDB 页面 | 原文线索

V2-026680 | MATHDB-356367

Conjectural decomposition of critical extrinsic Q-curvature

摘录 / EXCERPT: Conjectural decomposition. If n is even, then ${\bf Q}_n(g)$ is a linear combination of $Q_n(h)$, local conformal invariants of the embedding $M\hookrightarrow X$, and a total di...

conjecture | open | MathDB 页面 | 原文线索

V2-026681 | MATHDB-356372

Classification conjecture for positive singular solutions of critical poly-harmonic equations

摘录 / EXCERPT: Let $m\in\mathbb N^+$, set $N=2m$, and let $n>N$. Consider positive singular solutions $u\in C^N(\mathbb R^n\setminus\{0\})$ of ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-026682 | MATHDB-356380

The L-space conjecture for closed orientable irreducible 3-manifolds

摘录 / EXCERPT: L-space conjecture. The following statements are equivalent for M: (1) M is a non-L-space. (2) $\pi_1(M)$ is left orderable. (3) M admits a co-orientable taut foliation...

conjecture | open | MathDB 页面 | 原文线索

V2-026683 | MATHDB-356384

Duchin-Fisher compact-set conjecture for Thurston-boundary geodesics

摘录 / EXCERPT: Let S be a surface, and let α and β be simple closed curves that fill S. Let $\mathbf{G}$ be a Teichmüller geodesic ending at α and β. For sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-026684 | MATHDB-356420

Paracyclic lift conjecture for microlocal sheaf categories

摘录 / EXCERPT: Paracyclic lift conjecture. This simplicial ∞-category can be lifted to a paracyclic ∞-category.

conjecture | open | MathDB 页面 | 原文线索

V2-026685 | MATHDB-356425

Milnor's finite generation conjecture for fundamental groups of RCD spaces

摘录 / EXCERPT: Let $(X,d,\mathfrak{m})$ be an $RCD^{\ast}(0,N)$ space, where $RCD^{\ast}(0,N)$ denotes the relevant synthetic non-negative Ricci-curvature condition. Milnor's conjecture. The fund...

conjecture | open | MathDB 页面 | 原文线索

V2-026686 | MATHDB-356426

Fukaya-Yamaguchi conjecture on bounded virtual abelianness for finite fundamental groups

摘录 / EXCERPT: Let $(X,d,\mathfrak{m})$ be an $RCD^{\ast}(K,N;D)$ space. If its fundamental group $\pi_1(X)$ is finite, then Fukaya-Yamaguchi conjecture. It contains an abelian subgroup A of in...

conjecture | open | MathDB 页面 | 原文线索

V2-026687 | MATHDB-356452

The hyperbolicity criterion via exact positive currents

摘录 / EXCERPT: Hyperbolicity criterion. If all exact positive $(1,1)$-currents T on X are in $L^2_{-1}(X)$ and satisfy $I(T)=0$, then X is hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-026688 | MATHDB-356453

The Lelong-number conjecture for currents outside the L^2_{-1} space

摘录 / EXCERPT: Lelong-number conjecture. If X admits an exact positive $(1,1)$-current not in $L^2_{-1}(X)$, then there exists on X an exact positive current with a non-vanishing Lelong numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-026689 | MATHDB-356454

The pointwise Lelong-number refinement for the L^2_{-1}-locus

摘录 / EXCERPT: Pointwise Lelong-number refinement. If T is not in $L^2_{-1}(X)$ and $x otin L^2_{-1}(T|X)$, then there exists on X an exact positive $(1,1)$-current with non-vanishing Lelong...

conjecture | open | MathDB 页面 | 原文线索

V2-026690 | MATHDB-356458

Curvature continuity conjecture for PN-modified Catmull-Clark subdivision

摘录 / EXCERPT: Curvature continuity conjecture. The PN-modified Catmull-Clark subdivision can generate curvature-continuous subdivision surfaces, and the extraordinary points of the resulting sur...

conjecture | open | MathDB 页面 | 原文线索

V2-026691 | MATHDB-356466

M-theory-F-theory duality for a class of TCS G_2-compactifications

摘录 / EXCERPT: M-theory-F-theory duality conjecture. The following physical theories are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026692 | MATHDB-356468

The compact TCS G_2 duality conjecture with one D3-brane

摘录 / EXCERPT: Compact TCS G_2 duality conjecture. The following theories are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026693 | MATHDB-356469

The rank-n E_8 compact duality conjecture

摘录 / EXCERPT: Rank-n E_8 compact duality conjecture. The following theories are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026694 | MATHDB-356496

Volume-preserving classification conjecture for lattice actions

摘录 / EXCERPT: Let $n\geq 3$, let $\Gamma<\operatorname{SL}(n,{\bf R})$ be a lattice, and let M be a compact manifold of dimension n. A volume-preserving blow-up or two-sided blow-up is one i...

conjecture | open | MathDB 页面 | 原文线索

V2-026695 | MATHDB-356503

Thurston's geometrization conjecture for closed orientable irreducible 3-manifolds

摘录 / EXCERPT: Thurston's geometrization conjecture. Each component resulting from the geometric decomposition of $\mathcal{M}$ is geometric.

conjecture | open | MathDB 页面 | 原文线索

V2-026696 | MATHDB-356508

Irreducible curvatures of mapping tori of free group endomorphisms

摘录 / EXCERPT: Irreducible curvatures of mapping tori of free group endomorphisms. Then $\rho_+(X)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-026697 | MATHDB-356509

Surface curvatures of mapping tori of free group endomorphisms

摘录 / EXCERPT: Surface curvatures of mapping tori of free group endomorphisms. Then $\sigma_+(X)<0$ unless $\pi_1(X)$ has a Baumslag--Solitar subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-026698 | MATHDB-356510

Graphs of free groups with cyclic edge groups

摘录 / EXCERPT: Graphs of free groups with cyclic edge groups. Then $\rho_+(X)\leq 0$. Furthermore, $\sigma_+(X)<0$ unless $\pi_1(X)$ has a Baumslag--Solitar subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-026699 | MATHDB-356511

Non-positive irreducible curvature implies coherence

摘录 / EXCERPT: Non-positive irreducible curvature implies coherence. If $\rho_+(X)\leq 0$ then $\pi_1(X)$ is coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-026700 | MATHDB-356512

Negative irreducible curvature implies locally quasiconvex

摘录 / EXCERPT: Negative irreducible curvature implies locally quasiconvex. If $\rho_+(X)<0$ then $\pi_1(X)$ is a locally quasiconvex hyperbolic group.

conjecture | open | MathDB 页面 | 原文线索

V2-026701 | MATHDB-356513

Non-positive surface curvature implies asphericity

摘录 / EXCERPT: Non-positive surface curvature implies asphericity. If $\sigma_+(X)\leq 0$ then X is aspherical.

conjecture | open | MathDB 页面 | 原文线索

V2-026702 | MATHDB-356514

Non-positive surface curvature implies solvable word problem

摘录 / EXCERPT: Non-positive surface curvature implies solvable word problem. If $\sigma_+(X)\leq 0$ then the word problem is solvable in $\pi_1(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-026703 | MATHDB-356515

Negative surface curvature implies hyperbolicity

摘录 / EXCERPT: Negative surface curvature implies hyperbolicity. If $\sigma_+(X)<0$ then $\pi_1(X)$ is word-hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-026704 | MATHDB-356516

Positive surface curvature implies no nonpositive-Euler-characteristic surface subgroups

摘录 / EXCERPT: Positive surface curvature implies no nonpositive-Euler-characteristic surface subgroups. If $\sigma_-(X)>0$ then no subgroup of $\pi_1(X)$ is isomorphic to $\pi_1(S)$.

conjecture | open | MathDB 页面 | 原文线索

V2-026705 | MATHDB-356533

Cecil-Chi-Jensen revised conjecture for proper Dupin hypersurfaces

摘录 / EXCERPT: Let M be a compact, connected proper Dupin hypersurface in S^n, with g distinct principal curvatures. The hypersurface has constant Lie curvatures when its Lie curvatures are...

conjecture | open | MathDB 页面 | 原文线索

V2-026706 | MATHDB-356545

Griffiths' conjecture on positivity of Chern-Weil forms

摘录 / EXCERPT: Let $(E,h)\to X$ be a Griffiths semipositive Hermitian holomorphic vector bundle. Let P be a non-negative combination of Schur polynomials, and let $P\big(c_{\bullet}(E,h)\big)$...

conjecture | open | MathDB 页面 | 原文线索

V2-026707 | MATHDB-356550

Finite determination conjecture for cubical coarse median structures on right-angled Coxeter groups

摘录 / EXCERPT: Let W_{Γ} be a right-angled Coxeter group. A cubical coarse median rigidity conjecture. 1. There are only finitely many cubical coarse median structures on W_{Γ}. 2....

conjecture | open | MathDB 页面 | 原文线索

V2-026708 | MATHDB-356570

Ambro-Shokurov ACC conjecture for minimal log discrepancies

摘录 / EXCERPT: Ambro-Shokurov's ACC conjecture. The set Ω_k is an ACC-set: every increasing sequence $\alpha_1\leq\alpha_2\leq\cdots$ in Ω_k eventually becomes stationary.

conjecture | open | MathDB 页面 | 原文线索

V2-026709 | MATHDB-356571

Ambro-Shokurov lower semicontinuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Ambro-Shokurov's lower semicontinuity conjecture. For each d, the function

conjecture | open | MathDB 页面 | 原文线索

V2-026710 | MATHDB-356572

Unbounded degree of irrationality for very general polarized K3 surfaces

摘录 / EXCERPT: Let S_d be a very general polarized $K3$ surface with polarization L_d such that $L_d^2=2d-2$, for each $d\in\mathbb N$. Unbounded irrationality conjecture. ... The degree of irr...

conjecture | open | MathDB 页面 | 原文线索

V2-026711 | MATHDB-356597

Reboulet's conjecture on Chebyshev potentials and Mabuchi geodesics

摘录 / EXCERPT: Let X be a projective manifold and let $L\to X$ be a line bundle with $c_1(L)$ a Kähler class. Let $\mathcal{H}_L$ be the space of Kähler potentials whose associated Kähler forms...

conjecture | open | MathDB 页面 | 原文线索

V2-026712 | MATHDB-356625

Degree-three irrationality conjecture for eight Fano threefold deformation families

摘录 / EXCERPT: Let X_d be a quasismooth member of one of the eight deformation families numbered textnumero 1, textnumero 19, textnumero 28, textnumero 39, textnumero 49, textnumero 59, textnum...

conjecture | open | MathDB 页面 | 原文线索

V2-026713 | MATHDB-356633

Polydisk-Hirzebruch staircase correspondence conjecture

摘录 / EXCERPT: Staircase correspondence conjecture. For every $\beta\geq\sqrt{3}$, the function c_β has an infinite staircase if and only if the function c_{H_b} has an infinite staircase...

conjecture | open | MathDB 页面 | 原文线索

V2-026714 | MATHDB-356634

Inner-corner obstruction conjecture for four-dimensional polydisks

摘录 / EXCERPT: Inner-corner obstruction conjecture. At these intersections,

conjecture | open | MathDB 页面 | 原文线索

V2-026715 | MATHDB-356635

Usher's obstruction-supremum conjecture for ellipsoid embedding functions

摘录 / EXCERPT: Usher's conjecture. Between the center of $A_{0,n}$ and $\mathrm{accum}(L_{n,0})$, the function $c_{L_{n,0}}$ equals

conjecture | open | MathDB 页面 | 原文线索

V2-026716 | MATHDB-356636

Brahmagupta-move staircase conjecture for the polydisk

摘录 / EXCERPT: Brahmagupta-move staircase conjecture. The functions c_{β_i} have infinite staircases.

conjecture | open | MathDB 页面 | 原文线索

V2-026717 | MATHDB-356640

Extension of vanishing and positivity of Chern classes beyond small unbounded loci

摘录 / EXCERPT: Extension conjecture. The vanishing assertion $c_s(\hat{E})\cap T=0$ for $s>\operatorname{rank}E$ and the positivity assertion that $c_s(\hat{E})$ is a closed positive (s,s)-curr...

conjecture | open | MathDB 页面 | 原文线索

V2-026718 | MATHDB-356657

The sparse zonotope approximation conjecture

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^d$ be a zonotope and let $0<\varepsilon\leq\frac12$. A zonotope is a Minkowski sum of finitely many line segments; its number of segments is the number of...

open_problem | open | MathDB 页面 | 原文线索

V2-026719 | MATHDB-356658

The sparse $\ell1$ representation conjecture for zonotopes

摘录 / EXCERPT: Let $A\in\mathbb{R}^{m\times d}$ with $m\geq d$ and $0<\varepsilon\leq\frac12$. Sparse ℓ_1 representation conjecture. There exists a universal constant $C>0$ and a matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-026720 | MATHDB-356662

Cassels-Swinnerton-Dyer's conjecture on periodic multidimensional continued fractions

摘录 / EXCERPT: Cassels-Swinnerton-Dyer's conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026721 | MATHDB-356674

Fueter 2-category conjecture for holomorphic symplectic manifolds

摘录 / EXCERPT: Let M be a holomorphic symplectic manifold with complex structure I and holomorphic symplectic form Ω_I. For I-holomorphic Lagrangian submanifolds $L_1,L_2\subset M$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026722 | MATHDB-356675

Cotangent-bundle Fueter-Fukaya-Seidel conjecture

摘录 / EXCERPT: Let L be an exact Kähler manifold, let W be a holomorphic Morse function on L, and let L' be the graph of dW in T^*L. Cotangent-bundle Fueter-Fukaya-Seidel conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-026723 | MATHDB-356676

Non-commutative Hodge-theoretic interpretation of holomorphic Floer theory

摘录 / EXCERPT: Let (M,Ω_I) be holomorphic symplectic, set $\omega_u=\operatorname{Re}(u^{-1}\Omega_I)$ and $B_u=\operatorname{Im}(u^{-1}\Omega_I)$, and let $F(M,\omega_u,B_u)$ be the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-026724 | MATHDB-356682

Laugesen-Polterovich conjecture for the planar convex Neumann eigenvalue

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a planar convex domain, let $P(\Omega)$ denote its perimeter, and let $\mu_1(\Omega)$ denote the second Neumann eigenvalue of the Laplacian. Laug...

conjecture | open | MathDB 页面 | 原文线索

V2-026725 | MATHDB-356721

Cui-Zhao's hyperbolic Michael-Simon inequality for higher mean curvatures

摘录 / EXCERPT: Let M be a compact hypersurface in hyperbolic space $\mathbb{H}^{n+1}$, possibly with boundary ∂M, and let f be a positive smooth function on M. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-026726 | MATHDB-356755

The normal-crossing-divisor generalization conjecture for Lagrangian Floer theory

摘录 / EXCERPT: Let (X,ω) be a compact symplectic manifold, let $\mathcal D=\bigcup_{i=1}^m\mathcal D_i$ be a normal crossing divisor, and let $L\subset X\setminus\mathcal D$ be a compact r...

conjecture | open | MathDB 页面 | 原文线索

V2-026727 | MATHDB-356758

Semi-parallelism and parallelism conjecture for locally strongly convex affine hyperspheres

摘录 / EXCERPT: Semi-parallelism and parallelism conjecture. The two conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026728 | MATHDB-356760

Connelly's conjecture on optimality of triangulated disc packings

摘录 / EXCERPT: Let a finite set of discs be given. A packing is triangulated saturated if it is both triangulated and saturated, meaning that its contact structure is a triangulation and no furth...

conjecture | open | MathDB 页面 | 原文线索

V2-026729 | MATHDB-356772

Greb-Wong-Höring-Peternell conjecture on canonical extensions and tangent-bundle positivity

摘录 / EXCERPT: Greb-Wong-Höring-Peternell conjecture. The tangent bundle $d4af_X$ is nef, respectively big and nef, if and only if some canonical extension Z_X of X is Stein, respectively aff...

conjecture | open | MathDB 页面 | 原文线索

V2-026730 | MATHDB-356780

Bachman's index-two minimal surface conjecture

摘录 / EXCERPT: Let M be a non-Haken 3-manifold, and let $a3_1$ and $a3_2$ be strongly irreducible Heegaard surfaces in M that are not isotopic. A lowest genus stabilization is a common stab...

conjecture | open | MathDB 页面 | 原文线索

V2-026731 | MATHDB-356835

Metric-line criterion for normal trajectories in metabelian Carnot groups

摘录 / EXCERPT: Let G be a metabelian Carnot group, let $A\triangleleft G$ be an abelian subgroup with $[G,G]\subseteq A$, and let V_1 denote the first layer of $\operatorname{Lie}(G)$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-026732 | MATHDB-356837

The vanishing cosine-sum conjecture for modular geodesics

摘录 / EXCERPT: Let $A,B\in\operatorname{PSL}_2(\mathbb{Z})$ be primitive hyperbolic elements, and let $[\alpha_q]$ and $[\beta_q]$ be the associated modular geodesics for the Fuchsian representat...

conjecture | open | MathDB 页面 | 原文线索

V2-026733 | MATHDB-356843

Fundamental gap conjecture

摘录 / EXCERPT: Fundamental gap conjecture. The fundamental gap is at least $3\pi^2/D^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-026734 | MATHDB-356846

Huisken-Ilmanen Gromov-Hausdorff stability conjecture for the positive mass theorem

摘录 / EXCERPT: Let M_i be a sequence of asymptotically flat 3-manifolds with nonnegative scalar curvature and ADM mass tending to zero. Huisken-Ilmanen's conjecture. There is a set...

conjecture | open | MathDB 页面 | 原文线索

V2-026735 | MATHDB-356854

Horospherical p-Brunn-Minkowski inequalities for modified quermassintegrals

摘录 / EXCERPT: Let $n\geq 1$ and $0\leq k\leq n$ be integers, let $\frac12\leq p\leq 2$, and let $a,b\geq 0$ satisfy $a+b\geq 1$. For smooth uniformly h-convex bounded domains...

conjecture | open | MathDB 页面 | 原文线索

V2-026736 | MATHDB-356855

Horospherical p-Minkowski inequalities of types I and II

摘录 / EXCERPT: Let $n\geq1$, let $0\leq k\leq n$, and let $K,L\subset\mathbb{H}^{n+1}$ be smooth uniformly h-convex bounded domains. Write φ_K,φ_L for their horospherical support fu...

conjecture | open | MathDB 页面 | 原文线索

V2-026737 | MATHDB-356856

Weighted horospherical p-Minkowski inequalities

摘录 / EXCERPT: Let $K,L\subset\mathbb H^{n+1}$ be smooth origin-symmetric, uniformly h-convex bounded domains, with $n\geq1$. Let φ_K,φ_L be their horospherical support functions, l...

conjecture | open | MathDB 页面 | 原文线索

V2-026738 | MATHDB-356857

Weighted horospherical p-Minkowski inequality

摘录 / EXCERPT: Let $n\geq1$, let $p>0$, and let $\Omega\subset\mathbb H^{n+1}$ be a smooth uniformly h-convex bounded domain. Let r be the hyperbolic distance from the origin, $\tilde u$ the re...

conjecture | open | MathDB 页面 | 原文线索

V2-026739 | MATHDB-356869

Family Floer mirror identification conjecture

摘录 / EXCERPT: Let $\pi_\mathcal{R}:M_\mathcal{R}\to B_\mathcal{R}$ be the symplectic fibration, let $p_\mathcal{R}:\mathcal{Y}_\mathcal{R}\to B_\mathcal{R}$ be the corresponding rigid analytic m...

conjecture | open | MathDB 页面 | 原文线索

V2-026740 | MATHDB-356899

Generalized Kähler Calabi-Yau conjecture

摘录 / EXCERPT: Generalized Kähler Calabi-Yau conjecture. There exists a unique generalized Calabi-Yau geometry $(g_{CY},b_{CY},I,J_{CY})\in[(g,b,I,J)]$, and furthermore (g_{CY},I) and...

conjecture | open | MathDB 页面 | 原文线索

V2-026741 | MATHDB-356900

Generalized Kähler-Ricci-flow convergence conjecture

摘录 / EXCERPT: Generalized Kähler-Ricci-flow convergence conjecture. The generalized Kähler-Ricci flow preserves the generalized Kähler class, exists for all time, and converges to the unique gen...

conjecture | open | MathDB 页面 | 原文线索

V2-026742 | MATHDB-356903

The boundedness of B-representations conjecture

摘录 / EXCERPT: Let d and I be positive integers, and let (X,B) be a projective d-dimensional klt log Calabi-Yau pair such that $I(K_X+B)\sim 0$. The birational automorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-026743 | MATHDB-356905

NP-hardness conjecture for limited-capacity covering and packing

摘录 / EXCERPT: NP-hardness conjecture. C-k-COVER is NP-hard for every convex body C and every integer $k\geq 3$. Similarly, C-k-PACK is NP-hard for every convex body C and every integ...

conjecture | open | MathDB 页面 | 原文线索

V2-026744 | MATHDB-356913

CodeParade's conjecture on centers of tangent conics to a quadrilateral

摘录 / EXCERPT: Let Q be a quadrilateral, and consider all conics tangent to the four extended sides of Q. CodeParade's conjecture. The set of their centers is a straight line. The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-026745 | MATHDB-356914

Minthorn's conjecture on centers of conics through a quadrilateral's vertices

摘录 / EXCERPT: Let Q be a quadrilateral, and consider all conics passing through its four vertices. Minthorn's conjecture. If Q is convex, the set of centers of these conics is a hyperbola; i...

conjecture | open | MathDB 页面 | 原文线索

V2-026746 | MATHDB-356919

The quarter-size upper-bound conjecture for crossing families of elbows

摘录 / EXCERPT: Let S be a set of n points in the plane in general position. An elbow is an orthogonal geometric-graph edge consisting of one horizontal and one vertical line segment, and a cr...

conjecture | open | MathDB 页面 | 原文线索

V2-026747 | MATHDB-356932

Glazyrin-Karasev-Polyanskii polynomial plank conjecture

摘录 / EXCERPT: Polynomial plank conjecture. There exists a point $p\in B^d$ such that, for every $k=1,\ldots,N$, the distance from p to the zero set of P_k is at least δ_k.

conjecture | open | MathDB 页面 | 原文线索

V2-026748 | MATHDB-356935

The conjecture on non-contractible closed geodesics on the Riemannian real projective plane

摘录 / EXCERPT: Let $coperatorname{\mathbb{R}P}^2$ be a real projective plane endowed with a Riemannian metric g. A closed geodesic is non-contractible if its underlying loop represents a nontri...

conjecture | open | MathDB 页面 | 原文线索

V2-026749 | MATHDB-356936

The log-local correspondence for nef log Calabi-Yau surface pairs

摘录 / EXCERPT: Let (X,D) be a nef log Calabi-Yau surface pair, with $D=D_1+\cdots+D_l$, and let d be a curve class. Denote by $N_{0,d}^{\rm log}(X,D)$ and $N_{0,d}^{\rm loc}(X,D)$ the genus-z...

conjecture | open | MathDB 页面 | 原文线索

V2-026750 | MATHDB-356937

The genus-zero log-open correspondence for nef Looijenga pairs

摘录 / EXCERPT: Let (X,D) be a nef log Calabi-Yau surface pair with $D=D_1+\cdots+D_l$, and let (Y,L) be the associated toric Calabi-Yau threefold with framed Aganagic-Vafa branes...

conjecture | open | MathDB 页面 | 原文线索

V2-026751 | MATHDB-356939

Forni's equidistribution conjecture for Teichmüller geodesic translates

摘录 / EXCERPT: Let $g\geq 1$, let $\alpha=(\alpha_1,\dots,\alpha_n)$ be a partition of $2g-2$, let $\mathcal H_1(\alpha)$ be the space of unit-area surfaces in the corresponding stratum of abelia...

conjecture | open | MathDB 页面 | 原文线索

V2-026752 | MATHDB-356943

The Stein-degree boundedness conjecture for horizontal boundary components

摘录 / EXCERPT: Stein-degree boundedness conjecture. The quantity $\operatorname{sdeg}(S/Z)$ is bounded from above depending only on d and t.

conjecture | open | MathDB 页面 | 原文线索

V2-026753 | MATHDB-356954

Dinh-Nguyễn's Hodge-Riemann conjecture for Griffiths positive matrices

摘录 / EXCERPT: Let $V^{p,q}$ denote the space of (p,q)-forms on $\mathbb{C}^n$, and let $M=(\alpha_{i,j})$ be a Hermitian $k\times k$ matrix with entries in $V^{1,1}$. Assume that M is Griffi...

conjecture | open | MathDB 页面 | 原文线索

V2-026754 | MATHDB-356965

Existence and directional uniqueness of ODD integral curves

摘录 / EXCERPT: Existence and directional uniqueness conjecture. There exists $\epsilon>0$ and at least one ODD regular integral curve $\gamma\colon(-\epsilon,\epsilon)\to M$ with $\gamma(0)=p$. I...

conjecture | open | MathDB 页面 | 原文线索

V2-026755 | MATHDB-356966

Existence of ODD geodesics

摘录 / EXCERPT: Existence of ODD geodesics conjecture. There exists a unique ODD geodesic γ through p in direction v.

conjecture | open | MathDB 页面 | 原文线索

V2-026756 | MATHDB-356973

The Kähler wild automorphism conjecture for compact complex spaces

摘录 / EXCERPT: Let X be a compact Kähler space. An automorphism $\sigma\in\operatorname{Aut}(X)$ is wild if every non-empty analytic subset $Z\subseteq X$ satisfying $\sigma(Z)=Z$ is equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-026757 | MATHDB-357005

Symplectic reduction conjecture for self-polar convex bodies

摘录 / EXCERPT: Symplectic reduction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026758 | MATHDB-357019

Equality characterization for the planar boundary-data eigenvalue inequality

摘录 / EXCERPT: Let Ω be a convex domain in $\mathbb{R}^2$ such that $\kappa_1=\mu_2$. Here κ_1 denotes the first eigenvalue of the variational problem defined in the paper, μ_2...

conjecture | open | MathDB 页面 | 原文线索

V2-026759 | MATHDB-357020

Even-mode spectral gap conjecture for symmetric planar domains

摘录 / EXCERPT: Let Ω be a symmetric domain in $\mathbb{R}^2$ with a line of symmetry. Let μ_2 be the second Neumann Laplacian eigenvalue, and suppose that all modes associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-026760 | MATHDB-357027

Incidence conjecture for Klein's arrangements of lines and conics

摘录 / EXCERPT: Let $\mathrm{GR}(21_4)$ be the Grünbaum-Rigby configuration, decomposed into subconfigurations $\mathrm S_{12}$, $\mathrm S_{13}$, and $\mathrm S_{23}$ with point orbits...

conjecture | open | MathDB 页面 | 原文线索

V2-026761 | MATHDB-357123

The Bismut Alekseevski-Kimelfeld conjecture

摘录 / EXCERPT: Let (M^n,g) be a compact (or complete) Hermitian manifold. Its Bismut connection is denoted by ∇^b, and a connection is Ambrose-Singer when its torsion and curvature are p...

conjecture | open | MathDB 页面 | 原文线索

V2-026762 | MATHDB-357124

The canonical-connection Ambrose-Singer conjecture

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold. For real numbers t and s, let $\nabla^{(t)}=(1-\frac{t}{2})\nabla+\frac{t}{2}\nabla^b$ be the t-Gauduchon connection, and defin...

conjecture | open | MathDB 页面 | 原文线索

V2-026763 | MATHDB-357178

The conjectured sharp perimeter bound for convex decagons

摘录 / EXCERPT: Decagon perimeter conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-026764 | MATHDB-357179

Lipschitz conjecture for Möbius transformations and the generalized point pair function

摘录 / EXCERPT: Let $T_a:\mathbb{B}^2\to\mathbb{B}^2$ be the Möbius transformation ... For $x,y,a\in\mathbb{B}^2$ and $\alpha>0$, the generalized point pair function conjecture. ... Computer tests sug...

conjecture | open | MathDB 页面 | 原文线索

V2-026765 | MATHDB-357214

Effective log Iitaka fibration conjecture

摘录 / EXCERPT: Effective log Iitaka fibration conjecture. There exists a positive integer m depending only on d and τ such that, whenever (X,B) is an lc pair of dimension d with B...

conjecture | open | MathDB 页面 | 原文线索

V2-026766 | MATHDB-357215

Decomposable Iitaka fibration conjecture

摘录 / EXCERPT: Decomposable Iitaka fibration conjecture. There exist a positive integer m, a finite set $\Gamma_0\subset(0,1]$, and a DCC set $\Gamma'\subset[0,1]$, depending only on d and...

conjecture | open | MathDB 页面 | 原文线索

V2-026767 | MATHDB-357218

Weak Hom-version conjecture for ordinary moduli spaces of curves

摘录 / EXCERPT: Let $M_{g,n}^{\rm ord}$ be the ordinary moduli space of smooth pointed stable curves, and let $M_{g,n,\mathbb{F}_p}$ be its characteristic-p counterpart. For arbitrary points...

conjecture | open | MathDB 页面 | 原文线索

V2-026768 | MATHDB-357219

Tamagawa's weak Isom-version conjecture for ordinary moduli spaces

摘录 / EXCERPT: Let $M_{g,n}$ be the ordinary moduli space of smooth pointed stable curves and let $M_{g,n,\mathbb{F}_p}$ be the corresponding characteristic-p moduli space. For arbitrary points...

conjecture | open | MathDB 页面 | 原文线索

V2-026769 | MATHDB-357220

Pointed-collection conjecture for finite quotients of tame fundamental groups

摘录 / EXCERPT: Let $M_{g,n}^{\rm ord}$ be the ordinary moduli space, let $q_{\rm gen}$ be its generic point, and let $\pi_A^{\rm t}(q)$ denote the set of finite quotients associated with the tame...

conjecture | open | MathDB 页面 | 原文线索

V2-026770 | MATHDB-357231

Refined macroscopic dimension conjecture for the Riemann invariant

摘录 / EXCERPT: Refined macroscopic dimension conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-026771 | MATHDB-357235

Grove's Double Soul Conjecture

摘录 / EXCERPT: Let M be a closed simply connected smooth manifold admitting a Riemannian metric of non-negative sectional curvature. A double disk bundle is a smooth closed manifold obtained by...

conjecture | open | MathDB 页面 | 原文线索

V2-026772 | MATHDB-357251

Oscillation-energy conjecture for translated points on compact contact manifolds

摘录 / EXCERPT: Oscillation-energy conjecture. If φ is a contactomorphism of an arbitrary compact contact manifold Y, then the implication above holds.

conjecture | open | MathDB 页面 | 原文线索

V2-026773 | MATHDB-357257

The rational-simplex conjecture for nonempty rational quadratic-form graphs

摘录 / EXCERPT: Let $q\in\mathcal{Q}_n$, and let $G_0(\operatorname{\mathbb{Q}}^n,q)$ be the connected component containing $\operatorname{\textbf{0}}$ in the distance graph associated with q. A...

conjecture | open | MathDB 页面 | 原文线索

V2-026774 | MATHDB-357266

The vanishing-metric conjecture for Legendrian lifts of exact Lagrangians

摘录 / EXCERPT: Let $(M,d\lambda)$ be an exact symplectic manifold, and let L be a closed exact Lagrangian submanifold with vanishing Chekanov-Hofer metric. Let Λ be the Legendrian lift...

conjecture | open | MathDB 页面 | 原文线索

V2-026775 | MATHDB-357267

Nonexistence of metriplectic formulations with optical rotation

摘录 / EXCERPT: A polarized radiative transfer equation is considered in a basis of the polarization plane, with the optical rotation term denoted by ${\mathcal{N}}$. A metriplectic formulation is...

conjecture | open | MathDB 页面 | 原文线索

V2-026776 | MATHDB-357285

Type-II blow-up conjecture for complete mean curvature flow translators

摘录 / EXCERPT: Let the complete translators constructed in Theorem be the finite-entropy translating hypersurfaces obtained there. Type-II blow-up conjecture. The complete translators constructed...

conjecture | open | MathDB 页面 | 原文线索

V2-026777 | MATHDB-357286

Fattening dichotomy conjecture for mean curvature flow translators

摘录 / EXCERPT: Let $T[\Sigma_0]$ denote the level-set flow associated with the translator Σ_0, and suppose that $T[\Sigma_0]$ fattens. A complete translator is called unstable or mathcal{I...

conjecture | open | MathDB 页面 | 原文线索

V2-026778 | MATHDB-357302

The analytic local uniformization conjecture for rig-smooth spaces

摘录 / EXCERPT: Analytic local uniformization conjecture. Every point $x\in X^{\operatorname{ad}}$ is uniformizable.

conjecture | open | MathDB 页面 | 原文线索

V2-026779 | MATHDB-357303

The compact analytic semistable covering conjecture

摘录 / EXCERPT: Compact covering conjecture. There exists a finite extension l/k and a finite covering

conjecture | open | MathDB 页面 | 原文线索

V2-026780 | MATHDB-357306

Characterization of cyclic curves by the signature volume formula

摘录 / EXCERPT: Cyclic-curve characterization. The path $\mathbf{x}$ is cyclic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-026781 | MATHDB-357307

The stability-manifold and autoequivalence conjecture for Kodaira curves

摘录 / EXCERPT: The stability-manifold conjecture. The action of $\mathrm{Auteq}(C)$ on $\mathrm{Stab}(C)$ preserves $\mathrm{Stab}^{dagger}(C)$, and $\mathrm{Stab}^{dagger}(C)$ is simply connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-026782 | MATHDB-357340

The sigma-deformed 1-loop polynomial conjecture

摘录 / EXCERPT: The sigma-deformed 1-loop polynomial conjecture. The polynomial agrees with the torsion polynomial up to multiplication by a nonzero complex scalar and a monomial in t:

conjecture | open | MathDB 页面 | 原文线索

V2-026783 | MATHDB-357341

The sharp threshold conjecture for a linear-sized reconstructible set

摘录 / EXCERPT: Sharp threshold conjecture. The threshold for the existence of a reconstructible set of size linear in n is sharp and occurs at

conjecture | open | MathDB 页面 | 原文线索

V2-026784 | MATHDB-357347

Nilpotency conjecture for special quasi-Kähler manifolds

摘录 / EXCERPT: Let X be a special quasi-Kähler manifold, and let $\pi_1(X)$ denote its fundamental group. A quasi-Kähler manifold is slope-rationally connected when it has the property intended...

conjecture | open | MathDB 页面 | 原文线索

V2-026785 | MATHDB-357357

Zonotope reformulation of the Lonely Runner Conjecture for $T_{\mathbf n}$

摘录 / EXCERPT: Let $\mathbf{n}\in\mathbb{Z}^d_{>0}$ be a velocity vector, and let ... Here an interior lattice point means a point $\mathbf w\in\operatorname{int}(T_{\mathbf n})\cap\mathbb{Z}^d$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-026786 | MATHDB-357376

Nonbirationality conjecture for Type i KSS pairs

摘录 / EXCERPT: Let (A_i,B_i), for $i=1,\ldots,12$, be one of the twelve Type i KSS pairs of smooth elliptic Calabi--Yau 3-folds over $\mathbb{P}^2$. The pair is called nonbirational when no...

conjecture | open | MathDB 页面 | 原文线索

V2-026787 | MATHDB-357408

Log version of Serrano's conjecture

摘录 / EXCERPT: Let (X,Δ) be a projective log canonical pair of dimension n, and let L be a strictly nef Cartier divisor on X. Log version of Serrano's conjecture. The $\mathbb Q$-Car...

conjecture | open | MathDB 页面 | 原文线索

V2-026788 | MATHDB-357412

The uniqueness conjecture for cubic-surface lines and quartic-curve bitangents

摘录 / EXCERPT: Uniqueness conjecture. For $r,s\geq 1$, let $F:{\mathcal C}_{3,3}\to\operatorname{UConf}_r(\mathbb{G}(1,3))$ and $G:{\mathcal C}_{4,2}\to\operatorname{UConf}_s(\mathbb{G}(1,2))$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-026789 | MATHDB-357423

Huisken-Ilmanen stability conjecture for the positive mass theorem

摘录 / EXCERPT: Let (M_i,g_i) be a sequence of asymptotically flat 3-manifolds with nonnegative scalar curvature, and let $m(g_i)$ denote their ADM masses, with $m(g_i)\to 0$. Huisken-Ilmanen...

conjecture | open | MathDB 页面 | 原文线索

V2-026790 | MATHDB-357424

Ilmanen's quantitative stability conjecture for the positive mass theorem

摘录 / EXCERPT: Let (M_i,g_i) be a sequence of asymptotically flat 3-manifolds with nonnegative scalar curvature and ADM mass $m(g_i)$ tending to 0. Suppose $Z_i\subset M_i$ are the subsets...

conjecture | open | MathDB 页面 | 原文线索

V2-026791 | MATHDB-357429

Tverberg-Vrećica conjecture for regression depth

摘录 / EXCERPT: Let $1\leq k\leq d$ be integers, and let $A_1,\dots,A_{d-k+1}$ be finite arrangements of hyperplanes in $\mathbb{R}^d$. Assume that ... for positive integers r_i, for each...

conjecture | open | MathDB 页面 | 原文线索

V2-026792 | MATHDB-357447

Enumerative mirror symmetry for zero-degree quasimap invariants

摘录 / EXCERPT: Let ${\mathsf r}\geq 2$, and let $\check{\mathsf{QM}}^{\mathsf a}_{{\mathsf d},0}$ and $\hat{\mathsf{QM}}^{\mathsf a}_{{\mathsf d},0}$ denote the zero-degree quasimap invariants in...

conjecture | open | MathDB 页面 | 原文线索

V2-026793 | MATHDB-357458

Generalized permutahedron property of the second Symanzik polynomial for exceptional Euclidean kinematics

摘录 / EXCERPT: Generalized permutahedron conjecture. The Newton polytope $\mathbf{N}[\mathcal{F}]$ is a generalized permutahedron in the Euclidean regime for all kinematics, including exceptional...

conjecture | open | MathDB 页面 | 原文线索

V2-026794 | MATHDB-357470

Characteristic-vector conjecture for Hurwitz polytopes of smooth toric surfaces

摘录 / EXCERPT: Let A be a point configuration $A\subset\mathbb{Z}^{2}$, and let Q be its convex hull. For the associated toric surface X_A, the Hurwitz vectors are the vectors associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-026795 | MATHDB-357478

Semisimplicity conjecture for FJRW theory of two-variable invertible polynomials

摘录 / EXCERPT: Let (W,G) be an admissible Landau-Ginzburg pair, where W is an invertible polynomial with at most two variables and G is an admissible group. Semisimplicity conjecture. The F...

conjecture | open | MathDB 页面 | 原文线索

V2-026796 | MATHDB-357518

Peternell-Höring nonvanishing conjecture for cotangent bundles

摘录 / EXCERPT: Let X be a projective manifold, let q satisfy $1 e q$ and $q e \dim X$ only if excluded?

conjecture | open | MathDB 页面 | 原文线索

V2-026797 | MATHDB-357519

Numerically flat or positively twisted subsheaf conjecture for abelian fibrations

摘录 / EXCERPT: Let X be a projective manifold such that the cotangent bundle Ω_X is pseudoeffective. Assume that X admits an abelian fibration $f:X\to C$ onto a curve C, and let F...

conjecture | open | MathDB 页面 | 原文线索

V2-026798 | MATHDB-357530

Isosceles right-triangle conjecture for the lowest positive Dirac eigenvalue

摘录 / EXCERPT: Let $\Omega_{a,b}\subset\mathbb{R}^2$ be the right triangle with vertices $O=(0,0)$, $A=(a,0)$ and $B=(0,b)$, where $a,b>0$, and write $\lambda_1(a,b)=\lambda_1(\Omega_{a,b})$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-026799 | MATHDB-357532

The exceptional-set projection conjecture in the plane

摘录 / EXCERPT: The exceptional-set projection conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-026800 | MATHDB-357534

The sharp bounds conjecture for the exceptional-set exponent

摘录 / EXCERPT: Let $T(a,s)$ denote the exponent governing the exceptional-set estimate, and let the lower bounds for $T(a,s)$ be those established in Propositions 1 and 2. Sharp bounds conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-026801 | MATHDB-357537

Kollár-Takagi-Tanaka-Witaszek-Yasuda-Yoshikawa log-resolution lifting conjecture for log del Pezzo pairs

摘录 / EXCERPT: Let (X,Δ) be a log del Pezzo pair over a perfect field of characteristic $p>5$, with standard coefficients. A log resolution is a resolution $f\colon Y\to X$ such that the r...

conjecture | open | MathDB 页面 | 原文线索

V2-026802 | MATHDB-357544

The Stein-manifold conjecture for positive constant Bergman curvature

摘录 / EXCERPT: Let M be a Stein manifold. Let $A^2(M)$ be its Bergman space, assumed to be base-point free, meaning that its sections do not vanish simultaneously at any point, and to separate...

conjecture | open | MathDB 页面 | 原文线索

V2-026803 | MATHDB-357545

The lower-positive-curvature conjecture for Stein manifolds

摘录 / EXCERPT: Let M be a Stein manifold of complex dimension at least two, and let $A^2(M)$ be its Bergman space. Assume that $A^2(M)$ is base-point free and separates holomorphic directions....

conjecture | open | MathDB 页面 | 原文线索

V2-026804 | MATHDB-357562

Compatible Poisson pencil for the complex trigonometric spin Ruijsenaars-Schneider system

摘录 / EXCERPT: Poisson-pencil conjecture. There exists a pencil of compatible Poisson brackets on ${\mathcal M}_{n,d,q}^{\times}/\!/\operatorname{GL}_n(\mathbb C)$ containing the Arutyunov-Olivuc...

conjecture | open | MathDB 页面 | 原文线索

V2-026805 | MATHDB-357565

Unduloid conjecture for the three-dimensional cylinder-to-ball transition state

摘录 / EXCERPT: Let $\omega\in((4\pi)^{-1},\pi/6)$ be the fixed volume fraction, and let c and K be the phase-field parameters. An unduloid is a constant-mean-curvature surface of revolution i...

conjecture | open | MathDB 页面 | 原文线索

V2-026806 | MATHDB-357567

Saddle-node conjecture for the two-dimensional Ren-Wei branch

摘录 / EXCERPT: Let χ be the fissility parameter, let χ_* be the bifurcation value, and let $\chi_4\approx0.904\pm0.001$. The Ren-Wei branch is the two-dimensional equilibrium branch eme...

conjecture | open | MathDB 页面 | 原文线索

V2-026807 | MATHDB-357587

Assumption-free discrepancy and vanishing conjecture for derived blow-ups

摘录 / EXCERPT: Assumption-free discrepancy and vanishing conjecture. The assumptions that $\mathbb{B}l_f$ is smooth and that $\pi_0(Y)-\pi_0(X)$ is nonempty can be removed from the discrepancy fo...

conjecture | open | MathDB 页面 | 原文线索

V2-026808 | MATHDB-357608

Subpolynomial relative asymptotic dimension of real hyperbolic space

摘录 / EXCERPT: Subpolynomial asymptotic-dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026809 | MATHDB-357609

Relative asymptotic dimension of rank-one symmetric spaces

摘录 / EXCERPT: Rank-one symmetric-space conjecture. For every such X and every d,

conjecture | open | MathDB 页面 | 原文线索

V2-026810 | MATHDB-357622

The convex-hull conjecture for Thurston unit balls of chained-link complements

摘录 / EXCERPT: Let $C(n,p)$ be the n-chained link with p half-twists on its first component, and let $M(n,p)$ be its complement. For $p<0$, let $V(n,p)$ be a set of points in the Thurston uni...

conjecture | open | MathDB 页面 | 原文线索

V2-026811 | MATHDB-357626

Maximal-time conjecture for the adapted Chern-Ricci flow

摘录 / EXCERPT: Maximal-time conjecture. There exists a unique maximal solution to the adapted Chern-Ricci flow on $[0,T)$.

conjecture | open | MathDB 页面 | 原文线索

V2-026812 | MATHDB-357634

Minkowski's uniqueness conjecture for the multiplicative covering radius

摘录 / EXCERPT: Let $d\geq 2$, let $G=\operatorname{SL}_d(\mathbb{R})$, let $\Gamma=\operatorname{SL}_d(\mathbb{Z})$, and let $X_d=G/\Gamma$, identified with the space of unimodular lattices in...

conjecture | open | MathDB 页面 | 原文线索

V2-026813 | MATHDB-357646

The sharp Lipschitz bound for Möbius transformations of the Hilbert metric in the unit ball

摘录 / EXCERPT: The conjectured Lipschitz bound. For all $x,y,a\in\mathbb{B}^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-026814 | MATHDB-357659

Isomorphism between the normal-space quotient and the tangent bundle of an orbit space

摘录 / EXCERPT: Let M be a smooth manifold with a proper action of a group G. For each $x\in M$, let $N_x(M,G\cdot x)$ be the normal space to the orbit through x, and define ... Let G_x be t...

conjecture | open | MathDB 页面 | 原文线索

V2-026815 | MATHDB-357675

Finiteness conjecture for cohomogeneity-one Einstein metrics on even-dimensional spheres

摘录 / EXCERPT: Finiteness conjecture. Only finitely many even-dimensional spheres admit non-round Einstein metrics of cohomogeneity one.

conjecture | open | MathDB 页面 | 原文线索

V2-026816 | MATHDB-357691

Yau's conjecture for the Hull-Strominger system

摘录 / EXCERPT: Let X be a compact Calabi-Yau threefold with holomorphic volume form Ω, and let $\mathfrak{b}_0$ be a balanced class on X. Let V be a holomorphic vector bundle over...

conjecture | open | MathDB 页面 | 原文线索

V2-026817 | MATHDB-357696

Weinstein's conjecture on tangential star products for Lie algebra duals

摘录 / EXCERPT: Weinstein's conjecture. The dual $\mathfrak{g}^*$ admits a tangential star product if and only if there is a flat torsion-free affine connection on a neighborhood of the identity i...

conjecture | open | MathDB 页面 | 原文线索

V2-026818 | MATHDB-357742

Betke-Henk-Wills-Malikiosis discrete analogue of Minkowski's second theorem

摘录 / EXCERPT: Betke-Henk-Wills-Malikiosis conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-026819 | MATHDB-357745

The 3/2n flip-distance conjecture for non-crossing spanning trees

摘录 / EXCERPT: Let T_1 and T_2 be non-crossing spanning trees on the same set of n points. A flip replaces one edge of a non-crossing spanning tree by another edge so that the result is aga...

conjecture | open | MathDB 页面 | 原文线索

V2-026820 | MATHDB-357748

Chen ranks conjecture for Kähler groups

摘录 / EXCERPT: Let X be a compact Kähler manifold. For each $g\geq 2$, let $\mathcal{E}(X)$ be the set of relevant fibrations $f\colon X\to B_f$ and define ... Here Π_g denotes the fundamenta...

conjecture | open | MathDB 页面 | 原文线索

V2-026821 | MATHDB-357754

Scalar-flatness conjecture for complete critical metrics of the scalar-curvature functional

摘录 / EXCERPT: Scalar-flatness conjecture. If (M^{n},g) is a complete critical metric of $\mathfrak{S}^{2}$ with finite energy, then (M^{n},g) is scalar flat, and hence is a global minimum of...

conjecture | open | MathDB 页面 | 原文线索

V2-026822 | MATHDB-357765

Crystal conjecture for toric non-archimedean μ-entropy maximizers

摘录 / EXCERPT: Let $\mathcal{P}$ be a system, let $\lambda\in\mathbb{R}$, and let q be a maximizer of the toric non-archimedean μ-entropy functional $\bm{\check{\mu}}_{\mathrm{NA}}^\lambda$...

conjecture | open | MathDB 页面 | 原文线索

V2-026823 | MATHDB-357775

Entropy-volume conjecture for triangle-group continued fraction transformations

摘录 / EXCERPT: Entropy-volume conjecture. For all $n\geq 3$ and all $\alpha\in(0,1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-026824 | MATHDB-357785

Maximal symmetry rank conjecture for non-negatively curved manifolds

摘录 / EXCERPT: Let T^k act isometrically and effectively on M^n, where M^n is a closed, simply connected, non-negatively curved Riemannian manifold. Define ... where $n_i\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026825 | MATHDB-357798

Complex analyticity of Gromov-Hausdorff limits

摘录 / EXCERPT: Let (X_j,ω_j,p_j) be a sequence of n-dimensional Kähler-Einstein manifolds with $\operatorname{Ric}(\omega_j)=c_j\omega_j$, $|c_j|\leq 1$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-026826 | MATHDB-357801

Finiteness of the Calabi volume of the Bergman space

摘录 / EXCERPT: The Calabi metric on the space of Kähler metrics in a fixed Kähler class is the restriction of the deWitt-Ebin metric, and its finite-dimensional approximations are domains in fini...

conjecture | open | MathDB 页面 | 原文线索

V2-026827 | MATHDB-357817

Schoen's conjecture for non-compact area-minimizing boundaries

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold of dimension $3\leq n\leq 7$ that is asymptotically flat of rate $\tau=n-2$ and has non-negative scalar curvature. Suppose that there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-026828 | MATHDB-357820

The converse déjà vu conjecture for skies of causally related points

摘录 / EXCERPT: Let $\mathcal{X}$ be a globally hyperbolic spacetime whose Cauchy surface is not homeomorphic to the sphere. Let $x,y\in\mathcal{X}$ be causally related points, and consider future...

conjecture | open | MathDB 页面 | 原文线索

V2-026829 | MATHDB-357821

The angle-partition field and doubling the cube

摘录 / EXCERPT: Let $\mathcal{T}_1$ be the field associated with the geometric construction system under consideration, and let $\mathcal{P}$ denote the field obtained by allowing arbitrary angle...

conjecture | open | MathDB 页面 | 原文线索

V2-026830 | MATHDB-357822

The angle-partition field conjecture for algebraic numbers in $\mathcal{T}_1$

摘录 / EXCERPT: Let $\mathcal{P}$ be the field created by allowing arbitrary partition of angles, and let $\mathcal{T}_1$ be the relevant construction field. The algebraic numbers in...

open_problem | open | MathDB 页面 | 原文线索

V2-026831 | MATHDB-357823

The field-equivalence problem for transcendental-curve intersections

摘录 / EXCERPT: Consider the fields obtained by allowing general intersections with the transcendental curves mentioned in the source, namely trigonometric functions, the Quadratrix of Hippias, an...

open_problem | open | MathDB 页面 | 原文线索

V2-026832 | MATHDB-357824

The algebraic-power conjecture for transcendental-curve intersections

摘录 / EXCERPT: Let $\mathcal{T}_1$ be the construction field from the preceding discussion, and consider allowing general intersections with any of the transcendental curves under consideration....

open_problem | open | MathDB 页面 | 原文线索

V2-026833 | MATHDB-357833

Uniform rational polytopes conjecture for functional divisors on foliations

摘录 / EXCERPT: Let $v_1^0,\dots,v_m^0$ be positive integers and set $\boldsymbol{v}_0=(v_1^0,\dots,v_m^0)$. The rational envelope of $\boldsymbol{v}_0$ is the ambient rational-affine object in wh...

conjecture | open | MathDB 页面 | 原文线索

V2-026834 | MATHDB-357867

Adams's conjecture on global theta lifts and A-parameters

摘录 / EXCERPT: Let F be the global field implicit in the automorphic setting, let W be a symplectic space, let V be a quadratic space, and let $sigma\subset \mathcal{A}_{cusp}({\rm Sp}(W))$...

conjecture | open | MathDB 页面 | 原文线索

V2-026835 | MATHDB-357872

Gromov's long-neck conjecture for area non-increasing maps

摘录 / EXCERPT: Gromov's conjecture. Then ${\rm deg}(\Phi)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-026836 | MATHDB-357905

Markus conjecture for closed flat affine manifolds

摘录 / EXCERPT: Markus conjecture. Any connected closed flat affine manifold with parallel volume is complete.

conjecture | open | MathDB 页面 | 原文线索

V2-026837 | MATHDB-357909

Souam's conjecture on Schiffer domains in the 2-sphere

摘录 / EXCERPT: Souam's conjecture. If $\Omega\subset S^2$ is a sufficiently regular Schiffer domain, then Ω is either a geodesic disk or a round symmetric annulus.

conjecture | open | MathDB 页面 | 原文线索

V2-026838 | MATHDB-357914

Uniqueness conjecture for constant Gaussian-Minkowski data in higher dimensions

摘录 / EXCERPT: Higher-dimensional constant-data uniqueness conjecture. Then h must be a constant solution.

conjecture | open | MathDB 页面 | 原文线索

V2-026839 | MATHDB-357922

Ehrhart's lattice-point conjecture for centered convex bodies

摘录 / EXCERPT: Let $K\theta?$.

conjecture | open | MathDB 页面 | 原文线索

V2-026840 | MATHDB-357923

Mahler's successive-minima inequality for polars

摘录 / EXCERPT: Let $K\in\mathcal K_{os}^n$ be an origin-symmetric convex body and let $\Lambda\in\mathcal L^n$ be a lattice. Write $K^\star$ and $\Lambda^\star$ for the polar body and dual lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-026841 | MATHDB-357924

Makai's lattice-width volume conjecture

摘录 / EXCERPT: Let $K\in\mathcal K^n$ be a full-dimensional convex body and let $\Lambda\in\mathcal L^n$. Write K_s for the symmetrization of K, and $K^\star$ and $\Lambda^\star$ for the pola...

conjecture | open | MathDB 页面 | 原文线索

V2-026842 | MATHDB-357925

Álvarez Paiva et al.'s equality conjecture for lattice width

摘录 / EXCERPT: Let $K\in\mathcal K_o^n$ be an origin-centered convex body and let $\Lambda\in\mathcal L^n$. Write $K^\star$ and $\Lambda^\star$ for the polar body and dual lattice. Álvarez Paiva...

conjecture | open | MathDB 页面 | 原文线索

V2-026843 | MATHDB-357926

Betke-Henk-Wills-Malikiosis discrete Minkowski conjecture

摘录 / EXCERPT: Let $K\in\mathcal K^n$ be a full-dimensional convex body and let $\Lambda\in\mathcal L^n$. Let K_s be the symmetrization of K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026844 | MATHDB-357930

Soprunov-Zvavitch extremizer conjecture for the Bezout ratio

摘录 / EXCERPT: Let $K\in\mathcal{K}_0^n$ be a convex body, where $\mathcal{K}_0^n$ denotes the convex bodies in $\mathbb{R}^n$ containing the origin in their interiors. Define ... Here V_n is the...

conjecture | open | MathDB 页面 | 原文线索

V2-026845 | MATHDB-357939

Derived spectrum invariance for 3-manifolds

摘录 / EXCERPT: A derived spectrum is the spectrum associated with the derived critical-locus construction of a potential for the moduli space of $\mathop{SL}(2,\mathbb{C})$ representations of a 3...

conjecture | open | MathDB 页面 | 原文线索

V2-026846 | MATHDB-357940

Derived spectrum invariance under potentials with the same derived critical locus

摘录 / EXCERPT: Let X be a derived critical locus, and let f range over potentials whose derived critical locus is X. Derived spectrum equivalence conjecture. The derived spectrum is the sam...

conjecture | open | MathDB 页面 | 原文线索

V2-026847 | MATHDB-357961

Mixed Hodge compatibility of the relative-completion map

摘录 / EXCERPT: Let M be a simply connected compact Kähler manifold, let ${\mathscr M}$ be the moduli space of algebraic structures on the manifold underlying M, and let ${\mathcal G}_M$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-026848 | MATHDB-357969

Non-existence conjecture for Hermitian non-Kähler ALE gravitational instantons

摘录 / EXCERPT: Non-existence conjecture. There exist no Hermitian non-Kähler ALE gravitational instantons, without any restrictions on the structure group, except for the Eguchi-Hanson metric wit...

conjecture | open | MathDB 页面 | 原文线索

V2-026849 | MATHDB-357997

Matsumura-Wu structure conjecture for compact Kähler manifolds with nef anti-canonical bundle

摘录 / EXCERPT: Let X be a compact Kähler manifold with nef anti-canonical bundle $-K_X$. A fibration is a surjective holomorphic map with connected fibers, and it is locally constant when its f...

conjecture | open | MathDB 页面 | 原文线索

V2-026850 | MATHDB-358014

Hodge-Riemann and equality conjecture for mixed-volume valuations

摘录 / EXCERPT: Hodge-Riemann and equality conjecture. If all convex bodies in the tuples E_i are smooth and strictly convex, then the valuation

conjecture | open | MathDB 页面 | 原文线索

V2-026851 | MATHDB-358021

Cap-counting conjecture for Galois affine planes

摘录 / EXCERPT: Let I'_q be the maximum number of caps in a Galois affine plane $A(2,q)$. A cap is a set of points containing no three collinear points. Cap-counting conjecture. As $q\to\infty$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026852 | MATHDB-358041

The genus-three cluster-algebra description of the _2-quotient of Teichmller space

摘录 / EXCERPT: Let ${\mathcal T}_{3,0}$ be the Teichmuller space of flat $SL_2(\mathbb C)$-connections on a smooth Riemann surface of genus three. Consider the cluster algebra constructed in Sect...

conjecture | open | MathDB 页面 | 原文线索

V2-026853 | MATHDB-358058

The distinct-distances conjecture for pairs of algebraic curves

摘录 / EXCERPT: Distinct-distances conjecture. For every such pair of sets, $P_1\times P_2$ spans $\Omega(n^{3/2})$ distinct distances, unless each of γ_1 and γ_2 is an algebraic hel...

conjecture | open | MathDB 页面 | 原文线索

V2-026854 | MATHDB-358059

The bipartite rigidity conjecture for complete graphs

摘录 / EXCERPT: Bipartite rigidity conjecture. The framework $K(A,B)$ is rigid in $\mathbb{R}^d$, unless either the points in A or the points in B are contained in a hyperplane in...

conjecture | open | MathDB 页面 | 原文线索

V2-026855 | MATHDB-358062

Arnold's chord conjecture for Legendrian submanifolds of the contact sphere

摘录 / EXCERPT: Let α be a contact form on the standard contact sphere $(S^{2n-1},\xi_{\mathrm{std}})$. A Reeb chord of a closed Legendrian submanifold Λ is a Reeb trajectory of...

conjecture | open | MathDB 页面 | 原文线索

V2-026856 | MATHDB-358075

The G2-congruence conjecture for closed dancing polygons

摘录 / EXCERPT: Let a closed dancing pair be a pair of closed dancing n-gons as defined in the paper, and let G_2 act by the stated symmetries. G2-congruence conjecture. For $n=6,7$,...

conjecture | open | MathDB 页面 | 原文线索

V2-026857 | MATHDB-358097

The curve-quiver potential conjecture for crepant resolutions of abelian quotient singularities

摘录 / EXCERPT: Let G be a finite abelian subgroup of $\operatorname{SL}_3(\mathbb{C})$, and let Q be the curve quiver for a crepant resolution of $\mathbb{C}^3/G$. A minimal potential on Q...

conjecture | open | MathDB 页面 | 原文线索

V2-026858 | MATHDB-358114

Codimension-one threshold manifold conjecture for the cubic wave equation

摘录 / EXCERPT: Let $\mathcal{H}^s=\dot H^s(\mathbb{R}^3)\times\dot H^{s-1}(\mathbb{R}^3)$ denote the phase space, and consider initial data for the cubic wave equation. The self-similar profile i...

conjecture | open | MathDB 页面 | 原文线索

V2-026859 | MATHDB-358124

The minimal-volume conjecture for orientable hyperbolic 3-manifolds with 3 cusps

摘录 / EXCERPT: Let M be an orientable hyperbolic 3-manifold with three cusps, and let C_3 denote the 3-chain link complement. Minimal-volume conjecture. The minimal volume of a 3-cusped orien...

conjecture | open | MathDB 页面 | 原文线索

V2-026860 | MATHDB-358125

The minimal-volume conjecture for the acylindrical taut sutured manifold (N_3,γ)

摘录 / EXCERPT: Let (N_3,γ) be the acylindrical taut sutured manifold obtained by cutting the 3-chain link complement along a 3-punctured sphere. Its volume is $5.33\ldots$. Minimal-volume...

conjecture | open | MathDB 页面 | 原文线索

V2-026861 | MATHDB-358127

The zero-condition conjecture for degenerate pyramids

摘录 / EXCERPT: Zero-condition conjecture. The conditions from the Clean Condition are satisfied only when $F(i)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-026862 | MATHDB-358148

The algebraic torus fibration conjecture

摘录 / EXCERPT: Let $X\to Y$ be an algebraic torus fibration, with discriminant divisor $D\subset Y$, and let $\mathcal{W}_{T}(X)_{-\mathbf{1}}$ denote the spectral component of its equivariant wr...

conjecture | open | MathDB 页面 | 原文线索

V2-026863 | MATHDB-358160

Convergence of annuloids to translating catenoids

摘录 / EXCERPT: Let $s>0$ be fixed, and let M_i be an annuloid in the family $\mathcal A$ from Corollary, with necksize s and inner width $b_i\to\infty$. After suitable vertical translations, c...

conjecture | open | MathDB 页面 | 原文线索

V2-026864 | MATHDB-358178

The Archimedean solid perfection conjecture

摘录 / EXCERPT: An Archimedean solid is a three-dimensional convex polytope whose faces are regular polygons and whose symmetry group acts transitively on its vertices. Archimedean solid perfectio...

conjecture | open | MathDB 页面 | 原文线索

V2-026865 | MATHDB-358218

Bourgain's slicing conjecture

摘录 / EXCERPT: Bourgain's slicing conjecture. There exists a constant $c>0$ independent of dimension such that

conjecture | open | MathDB 页面 | 原文线索

V2-026866 | MATHDB-358244

Exact formula for the minimal length of a geodesic with at least k self-intersections

摘录 / EXCERPT: Let M_k be the infimum of lengths of geodesics with self-intersection number at least k among all finite-type hyperbolic surfaces. A minimal-length geodesic conjecture asserts...

conjecture | open | MathDB 页面 | 原文线索

V2-026867 | MATHDB-358267

The diminished-base-locus conjecture for log canonical centres in MMPs

摘录 / EXCERPT: Let (X_1,B_1+M_1) be a projective $\mathbb{Q}$-factorial NQC dlt generalised pair such that $K_{X_1}+B_1+M_1$ is pseudoeffective. Consider a $(K_{X_1}+B_1+M_1)$-MMP: ... Let S be...

conjecture | open | MathDB 页面 | 原文线索

V2-026868 | MATHDB-358298

Existence conjecture for gradient expanding solitons on connected sums

摘录 / EXCERPT: Let $\gamma = dr^2 + r^2 h$ be an orientable Riemannian cone metric with non-negative scalar curvature. A gradient expanding soliton is a gradient expanding Ricci soliton; the clai...

conjecture | open | MathDB 页面 | 原文线索

V2-026869 | MATHDB-358339

Full-spectrum conjecture for affine spaces over $\mathbb{F}_2$

摘录 / EXCERPT: Let $[n,m]\rightarrow[k,t]$ mean that every m-element subset of $\mathrm{AG}(n,2)$ has a k-dimensional affine subspace containing exactly t points, and let $\rho(n;k,t)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-026870 | MATHDB-358360

Long's elliptic closed-geodesic conjecture for Finsler two-spheres

摘录 / EXCERPT: Let F be a Finsler metric on the two-sphere S^2, and let a closed geodesic be a periodic geodesic of F. Long's elliptic-geodesic conjecture. There exists at least one ellipti...

conjecture | open | MathDB 页面 | 原文线索

V2-026871 | MATHDB-358397

Smoothability conjecture for superfat points

摘录 / EXCERPT: Smoothability conjecture. Z is smoothable for all $m,n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-026872 | MATHDB-358398

Maximal Hilbert function conjecture for superfat points

摘录 / EXCERPT: Maximal Hilbert function conjecture. For every $m,n\geq 2$, there exists an m-superfat point in $\mathbb{P}^2$ having maximal Hilbert function.

conjecture | open | MathDB 页面 | 原文线索

V2-026873 | MATHDB-358405

Virtual Clemens conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Virtual Clemens conjecture. For the purpose of computing genus zero enumerative invariants on Calabi-Yau threefolds X, one may assume that there are only finitely many isolated r...

conjecture | open | MathDB 页面 | 原文线索

V2-026874 | MATHDB-358431

Bloch's conjecture for symplectic automorphisms of K3 surfaces

摘录 / EXCERPT: Let X be a smooth projective K3 surface, and let ${\rm Aut}^s(X)$ denote its group of symplectic automorphisms. For $\phi\in{\rm Aut}^s(X)$, write ... for the induced morphism. Blo...

conjecture | open | MathDB 页面 | 原文线索

V2-026875 | MATHDB-358442

Nandakumar and Ramana-Rao's equal-area and equal-perimeter conjecture

摘录 / EXCERPT: Let P be a convex polygon in the plane and let n be a prescribed natural number. Nandakumar and Ramana-Rao's conjecture. The polygon P can be partitioned into n convex piec...

conjecture | open | MathDB 页面 | 原文线索

V2-026876 | MATHDB-358445

Generic lens spaces are spectrally unique

摘录 / EXCERPT: Generic lens-space spectral uniqueness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026877 | MATHDB-358457

Sun-Yang-Zuo's twisted Higgs-de Rham flow conjecture

摘录 / EXCERPT: Let $D=0+1+\lambda+\infty$ on $\mathbb{P}^1$, and consider logarithmic graded semistable Higgs bundles on $(\mathbb{P}^1,D)$ whose underlying graded vector bundle is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-026878 | MATHDB-358471

Laugesen's perimeter-normalized Neumann eigenvalue conjecture for convex planar domains

摘录 / EXCERPT: Let Ω be a convex domain in $\mathbb{R}^2$, let $L(\Omega)$ denote its perimeter, and let $\mu_2(\Omega)$ be its first nonzero Neumann eigenvalue. Laugesen's conjecture. ... E...

conjecture | open | MathDB 页面 | 原文线索

V2-026879 | MATHDB-358503

Zero-set and extension conjecture for globally subanalytic arc-analytic functions

摘录 / EXCERPT: Let $E\subset\mathbb{R}^n$ be an $\mathscr{A{\\!}R}$-closed set. A function on E is globally subanalytic arc-analytic when it is globally subanalytic and its composition with eve...

conjecture | open | MathDB 页面 | 原文线索

V2-026880 | MATHDB-358504

The Strominger-Yau-Zaslow conjecture for mirror Calabi-Yau manifolds

摘录 / EXCERPT: Let X and Y be “mirror” Calabi-Yau manifolds. There should exist a base manifold B and surjective continuous maps ... with fibers $L_q=\pi^{-1}(q)$ and $F_q=f^{-1}(q)$, and a c...

conjecture | open | MathDB 页面 | 原文线索

V2-026881 | MATHDB-358509

Cao and Zheng's tropical Griffiths conjecture

摘录 / EXCERPT: Cao and Zheng's tropical Griffiths conjecture. The defect sum satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026882 | MATHDB-358529

Cao's classification conjecture for positively curved steady Kähler-Ricci solitons

摘录 / EXCERPT: Let (M,J,g,X) be a simply connected complete steady gradient Kähler-Ricci soliton, meaning that g is a complete Kähler metric and X is a vector field generating the soliton....

conjecture | open | MathDB 页面 | 原文线索

V2-026883 | MATHDB-358537

Non-triviality conjecture for Kauffman bracket skein modules

摘录 / EXCERPT: Let M be an orientable 3-manifold, and let $S(M)$ be its Kauffman bracket skein module over $\mathbb{Q}(A)$. Non-triviality conjecture. For every orientable 3-manifold M, ...

conjecture | open | MathDB 页面 | 原文线索

V2-026884 | MATHDB-358538

Non-Haken finite-generation conjecture for skein modules

摘录 / EXCERPT: Let M be a closed 3-manifold that is irreducible and contains no incompressible surface. Let $S(M)$ denote its Kauffman bracket skein module over $\mathbb{Q}[A^{\pm1}]$. Non-Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-026885 | MATHDB-358549

Dol’nikov's colorful piercing conjecture for arbitrarily many families

摘录 / EXCERPT: Dol’nikov's generalized conjecture. There exists $j \in \{1,2,\dots,n\}$ such that $\bigcup_{i \neq j} \mathcal{F}_{i}$ can be pierced by 3 points.

conjecture | open | MathDB 页面 | 原文线索

V2-026886 | MATHDB-358550

Colorful four-point piercing conjecture for two families of circles

摘录 / EXCERPT: Colorful four-point piercing conjecture. Either $\mathcal{F}_{1}$ or $\mathcal{F}_{2}$ can be pierced by 4 points.

conjecture | open | MathDB 页面 | 原文线索

V2-026887 | MATHDB-358554

Gehring-Martin-Palka conjecture on the conformal capacity of linked curves

摘录 / EXCERPT: Gehring-Martin-Palka conjecture. Every pair of linked curves satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-026888 | MATHDB-358555

Freedman-He conjecture on the conformal capacity of solid tori

摘录 / EXCERPT: Freedman-He conjecture. For every solid torus $T\subset\mathbb S^3$,

conjecture | open | MathDB 页面 | 原文线索

V2-026889 | MATHDB-358573

The Tosatti-Weinkove conjecture on logarithmic poles in nef classes

摘录 / EXCERPT: Let X be a compact n-dimensional complex manifold, and let $\{\beta\}\in H^{1,1}(X,\mathbb{R})$ be nef with $\int_X\beta^n>0$. Fix points $x_1,\ldots,x_N\in X$ and positive rea...

conjecture | open | MathDB 页面 | 原文线索

V2-026890 | MATHDB-358574

The Demailly-Păun conjecture on nef classes with positive volume

摘录 / EXCERPT: Let (X,ω) be a compact Hermitian manifold of complex dimension n. Let $\{\beta\}\in H^{1,1}(X,\mathbb{R})$ be a real $(1,1)$-class with smooth representative β. The...

conjecture | open | MathDB 页面 | 原文线索

V2-026891 | MATHDB-358591

Higher-dimensional ℓp global rigidity and identifiability conjecture

摘录 / EXCERPT: Let p be an even positive integer with $p\neq 2$, let $n\geq 3$, and let G be a connected graph with n vertices. Write ${\cal CM}_n^p$ for the relevant ℓ_p-configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-026892 | MATHDB-358595

The quadratic difference conjecture for symplectic pairs

摘录 / EXCERPT: Let p and q be monic polynomials of degree 2, let V be a vector space, and let (b,u) be a symplectic pair on V. A symplectic (p,q)-difference is a symplectic pair sat...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-026893 | MATHDB-358622

Demazure crystal decomposition conjecture for symplectic bounded reduced factorizations

摘录 / EXCERPT: Symplectic Demazure crystal conjecture. The crystal $\mathrm{BRF}^{\mathsf{Sp}}_n(z,\phi)$ is a direct sum of Demazure $\mathfrak q_n$-crystals.

conjecture | open | MathDB 页面 | 原文线索

V2-026894 | MATHDB-358623

Symplectic reduced-tableau Demazure crystal conjecture

摘录 / EXCERPT: Symplectic reduced-tableau conjecture. There is a $\mathfrak q_n$-crystal isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-026895 | MATHDB-358636

Montiel-Urbano conjecture for the Clifford torus in the complex projective plane

摘录 / EXCERPT: Montiel-Urbano conjecture. The Clifford torus achieves the minimum of the functional $\mathcal{W}^-$, and hence of $\mathcal{W}$, either among all tori in $\mathbb{C}P^2$ or among...

conjecture | open | MathDB 页面 | 原文线索

V2-026896 | MATHDB-358642

The non-geodesicity conjecture for metric spaces of Reeb graphs

摘录 / EXCERPT: Non-geodesicity conjecture. The space of isomorphism classes of Reeb graphs with metric d is not a geodesic space.

conjecture | open | MathDB 页面 | 原文线索

V2-026897 | MATHDB-358643

Large totally geodesic hypersurface rigidity conjecture

摘录 / EXCERPT: Large totally geodesic hypersurface rigidity conjecture. For every $D>0$ and $k\in\mathbb{N}$, there exists $A=A(D,k)$ such that if N contains a totally geodesic hyperbolic hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-026898 | MATHDB-358644

Large-area rigidity conjecture for minimal surfaces in hyperbolic 3-manifolds

摘录 / EXCERPT: Large-area rigidity conjecture. There exists $A=A(M)$ such that every minimal surface Σ with area larger than A is rigid.

open_problem | open | MathDB 页面 | 原文线索

V2-026899 | MATHDB-358664

Non-degeneracy conjecture for the pseudo-metric on the deformation space

摘录 / EXCERPT: Let $V_{(J,A)}$ be the tangent space described by the relevant system of PDEs, and let $\mathcal{L}_{L^2}$ be the space of L^2-regularity tensors representing the infinitesimal d...

conjecture | open | MathDB 页面 | 原文线索

V2-026900 | MATHDB-358678

Li-Naber quantitative stratification conjecture for Alexandrov spaces

摘录 / EXCERPT: Let $\text{Alex}^n(-1)$ denote the n-dimensional Alexandrov spaces with curvature at least -1. For $0\le k\le n-1$, let $\mathcal{S}^k_\epsilon(X)$ be the set of points whose t...

conjecture | open | MathDB 页面 | 原文线索

V2-026901 | MATHDB-358680

Naber's curvature-measure convergence conjecture

摘录 / EXCERPT: Let (M_i,g_i) be a sequence of n-dimensional manifolds with lower Ricci or sectional curvature bounds, and suppose that M_i Gromov-Hausdorff converge without collapsing to...

conjecture | open | MathDB 页面 | 原文线索

V2-026902 | MATHDB-358682

Continuity conjecture for synthetic curvature measures

摘录 / EXCERPT: Let $X_i\in\text{Alex}^n(-1)$ be a sequence of Alexandrov spaces converging to X in the Gromov-Hausdorff sense without collapsing. Let μ_{X_i} and μ_X denote their synthe...

conjecture | open | MathDB 页面 | 原文线索

V2-026903 | MATHDB-358745

Mochizuki's absolute anabelian conjecture without a fixed Galois-group isomorphism

摘录 / EXCERPT: Let K and L be finite extensions of $\mathbb{Q}_p$, and let X/L and Y/K be hyperbolic curves. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-026904 | MATHDB-358747

Cheltsov-Rubinstein's Kähler-Einstein edge metric conjecture for strongly asymptotically log del Pezzo surfaces

摘录 / EXCERPT: Let (S,C) be a strongly asymptotically log del Pezzo surface with $r=1$, where S is a smooth surface and C is a smooth irreducible curve. The notation $(K_S+C)^2$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-026905 | MATHDB-358763

Thurston's convex-core bending-lamination conjecture

摘录 / EXCERPT: Thurston's conjecture. Given any pair of measured laminations $L^{+},L^{-}\in\mathcal{MGL}_{\pi}(S)$ that fill S, there is a unique quasi-Fuchsian manifold Q, up to isotopy, su...

conjecture | open | MathDB 页面 | 原文线索

V2-026906 | MATHDB-358777

Hanner polytope conjecture for relative Funk volume growth

摘录 / EXCERPT: Let Ω be a centrally symmetric convex body, and let H be a Hanner polytope of the same dimension. For $s>0$, let $B_\Omega(s)$ and $B_H(s)$ be the corresponding forward Fu...

conjecture | open | MathDB 页面 | 原文线索

V2-026907 | MATHDB-358782

González-Acuña's trivial Artin presentation conjecture

摘录 / EXCERPT: González-Acuña's conjecture. If A is an Artin n-presentation such that $|A|=1$, then $A\sim T$. This is one of the conjectures relating Artin presentations of the trivial group...

conjecture | open | MathDB 页面 | 原文线索

V2-026908 | MATHDB-358783

The rat-group knot-group conjecture

摘录 / EXCERPT: A rat-group is obtained by deleting the final relator from an Artin n-presentation of the trivial group, so it has a deficiency-one presentation. A knot group is characterized by...

conjecture | open | MathDB 页面 | 原文线索

V2-026909 | MATHDB-358784

Discrepancy formula for derived blow-ups

摘录 / EXCERPT: Assume the setting of the derived blow-up $\mathbb{B}l_{Z/X}$ over S, with morphism $pr_{Z/X}:\mathbb{B}l_{Z/X}\to X$ and integer r as in the main theorem. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026910 | MATHDB-358785

Determinant formula for virtual Cartier divisors

摘录 / EXCERPT: Let $f:Z\to X$ be a virtual Cartier divisor of derived Artin stacks, and let $\mathcal{F}\in\operatorname{Perf}(Z)$ be a perfect complex of rank l. Determinant conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-026911 | MATHDB-358786

Halpern-Leistner's determinant-dualizing-complex conjecture

摘录 / EXCERPT: Let $f:X\to Y$ be a morphism between quasi-smooth, locally almost of finite presentation derived Artin stacks. Let $L_{X/Y}$ be the relative cotangent complex and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026912 | MATHDB-358787

Pushforward formula for the structure sheaf of a derived blow-up

摘录 / EXCERPT: Assume the setting of the derived blow-up $\mathbb{B}l_{Z/X}$ and let $pr_f:\mathbb{B}l_{Z/X}\to X$ be the relevant morphism. Let $\mathfrak{W}_{1-r,1-r}^{f}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026913 | MATHDB-358788

Calaque-Căldăraru-Tu's derived blow-up L-infinity algebroid conjecture

摘录 / EXCERPT: L-infinity algebroid conjecture. There exists a minimal L_{∞}-algebroid structure on $C_{Z/X}^{\vee}$ whose Chevalley-Eilenberg algebra is equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-026914 | MATHDB-358797

Critical-index global existence conjecture for EPDiff equations

摘录 / EXCERPT: Critical-index conjecture. All solutions exist for all time if $k\geq \frac{n}{2}+1$, whereas for $k<\frac{n}{2}+1$ there are always some solutions that break down in finite time....

conjecture | open | MathDB 页面 | 原文线索

V2-026915 | MATHDB-358814

The minimum-weight-sector conjecture for veering triangulations

摘录 / EXCERPT: Let M be a mapping torus equipped with a veering triangulation. A minimum weight sector is a sector of minimum weight in the associated veering data, and the hypothesis of ... is s...

conjecture | open | MathDB 页面 | 原文线索

V2-026916 | MATHDB-358832

The strong regularity-radius conjecture for Delone sets

摘录 / EXCERPT: Let X be a Delone set in $\mathbb{R}^d$ with parameters (r,R), where R is the radius of a largest empty ball, and let $\hat{\rho}_d=\hat{\rho}_d(r,R)$ be the smallest radius...

conjecture | open | MathDB 页面 | 原文线索

V2-026917 | MATHDB-358845

Weak convergence of curvature tensors under Gromov-Hausdorff convergence

摘录 / EXCERPT: Let $\mathcal M(n,r,V)$ be the class of n-dimensional complete Riemannian manifolds with sectional curvature at most 1, injectivity radius at least $r>0$, and volume at most...

conjecture | open | MathDB 页面 | 原文线索

V2-026918 | MATHDB-358874

The plane star-forest covering conjecture for complete geometric graphs

摘录 / EXCERPT: Plane star-forest covering conjecture. There is no complete geometric graph with n vertices that can be decomposed into fewer than

conjecture | open | MathDB 页面 | 原文线索

V2-026919 | MATHDB-358881

Polyhedrality conjecture for uniformly robust norms

摘录 / EXCERPT: A norm on a finite-dimensional real vector space is uniformly robust when its Fermat-Weber problems have bounded robustness under the contamination model considered in the paper. P...

conjecture | open | MathDB 页面 | 原文线索

V2-026920 | MATHDB-358896

Hermitian classification conjecture for non-Kähler ALF instantons

摘录 / EXCERPT: Hermitian classification conjecture. $(\mathcal{M},g_{ab})$ is one of Kerr, Chen--Teo, Taub-bolt, or Taub--NUT with the orientation opposite to the hyperkähler orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-026921 | MATHDB-358898

Hermitianity conjecture for ALF gravitational instantons

摘录 / EXCERPT: ALF Hermitianity conjecture. Every ALF gravitational instanton is Hermitian.

conjecture | open | MathDB 页面 | 原文线索

V2-026922 | MATHDB-358904

Enumerative mirror symmetry formulas for rank-two quasimap invariants

摘录 / EXCERPT: Let $g\geq 2$ be the genus, let ${\mathsf{r}}=2$ be the rank, and let ${\mathsf{d}}$ be the degree. Define the generating series $\check{\mathsf{QM}}^i_{\mathsf{d}}(q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-026923 | MATHDB-358927

The boundedness conjecture for the conformal maximizer

摘录 / EXCERPT: Let $g'=u^4g$ be the maximizing conformal metric appearing in the proof of the conformal conjecture, where u is harmonic. Boundedness conjecture. The maximizer is bounded above:...

conjecture | open | MathDB 页面 | 原文线索

V2-026924 | MATHDB-358951

Decomposition conjecture for compact Fujiki manifolds with vanishing first Chern class

摘录 / EXCERPT: Decomposition Conjecture. Every compact Fujiki manifold with vanishing first Chern class admits such a finite étale cover and decomposition, together with the stated bimeromorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-026925 | MATHDB-358954

Absolute universality conjecture for the natural embedding of the Lie incidence geometry

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, let $A_{n,\{1,n\}}(\mathbb{F})$ be the Lie incidence geometry, and let $\varepsilon_{\mathrm{nat}}$ denote its natural embedding. An embedding is absol...

conjecture | open | MathDB 页面 | 原文线索

V2-026926 | MATHDB-358993

The symplectic embedding-configuration space duality conjecture for up to eight balls

摘录 / EXCERPT: Let $n\leq 8$ and let ${\mathbf c}=c_1\geq\cdots\geq c_n>0$ be admissible capacities. Write $\Im\!\operatorname{Emb}_n({\mathbf c},{\mathbf C}{\mathbf P}^2)$ for the space of sympl...

conjecture | open | MathDB 页面 | 原文线索

V2-026927 | MATHDB-359015

Geometricity of low-energy sheaf quantizations

摘录 / EXCERPT: Let X be a symplectic manifold, let $\bm L$ be a Lagrangian brane, and let $c_{\bm L}>0$ be as in the classification theorem. A sheaf quantization of $\bm L$ in $(X,u<c_{\bm L})$...

conjecture | open | MathDB 页面 | 原文线索

V2-026928 | MATHDB-359018

Smooth regularization conjecture for plurisubharmonic functions on compact Kähler spaces

摘录 / EXCERPT: Let (X,ω) be a compact Kähler space, and let φ be an ω-psh function on X. Smooth regularization conjecture. The function φ can be written as the poi...

conjecture | open | MathDB 页面 | 原文线索

V2-026929 | MATHDB-359023

Failure of the curvature-dimension condition in sub-Finsler Carnot groups

摘录 / EXCERPT: Failure of the curvature-dimension condition conjecture. The metric measure space $(G,\mathsf d_{SF},\mathfrak m)$ does not satisfy the $\mathsf{CD}(K,N)$ condition for any...

conjecture | open | MathDB 页面 | 原文线索

V2-026930 | MATHDB-359024

The four minimal spheres conjecture

摘录 / EXCERPT: Let S^3 be the three-dimensional sphere equipped with an arbitrary Riemannian metric. The four minimal spheres conjecture. S^3 contains at least 4 embedded minimal 2-spheres. T...

conjecture | open | MathDB 页面 | 原文线索

V2-026931 | MATHDB-359025

The three free-boundary minimal disks conjecture

摘录 / EXCERPT: Let B be a strictly convex ball in $\mathbb{R}^3$ or, more generally, a strictly convex ball in a 3-manifold with nonnegative Ricci curvature. A free-boundary minimal 2-disk is a...

conjecture | open | MathDB 页面 | 原文线索

V2-026932 | MATHDB-359040

Saturation Gap Conjecture for chopped ideals of general points

摘录 / EXCERPT: Let S be the homogeneous coordinate ring of $\mathbb{P}^n$, let Z be a general set of r points, and let $I_{\langle d\rangle}=\langle I(Z)_d\rangle_S$. For positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-026933 | MATHDB-359047

The sporadic-set conjecture for Apollonian circle packings

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a primitive Apollonian circle packing. Let $S_{\mathcal{A}}$ be the set of missing curvatures that do not lie in any of the quadratic or quartic obstruction...

conjecture | open | MathDB 页面 | 原文线索

V2-026934 | MATHDB-359060

Cascini-Spicer's basepoint free conjecture for algebraically integrable foliations

摘录 / EXCERPT: Cascini-Spicer's basepoint free conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-026935 | MATHDB-359068

Ranganathan-Wise conjecture on Gorenstein curves and smoothable differentials

摘录 / EXCERPT: Ranganathan-Wise conjecture. The differential η is smoothable if and only if, for every level i:

conjecture | open | MathDB 页面 | 原文线索

V2-026936 | MATHDB-359072

The equivalent genus-function conjecture for the exotic four-manifolds Z_n

摘录 / EXCERPT: Let AP be the four-manifold constructed in the paper, and let Z_n be the infinite family of exotic four-manifolds obtained by replacing the final Luttinger surgery with torus s...

conjecture | open | MathDB 页面 | 原文线索

V2-026937 | MATHDB-359073

The equivalent genus-function conjecture for the exotic four-manifolds V_n

摘录 / EXCERPT: Let BK be the symplectic four-manifold constructed in the paper, and let V_n be the sequence of exotic four-manifolds obtained from BK, with $V_1=BK$. For a smooth four-manif...

conjecture | open | MathDB 页面 | 原文线索

V2-026938 | MATHDB-359080

Log-Minkowski inequality for centrally symmetric convex bodies

摘录 / EXCERPT: Let $C,D\in\mathrm{Conv}_s(\mathbb{R}^{n+1})$ be centrally symmetric convex bodies of unit volume, and let h_C,h_D denote their support functions. Let ν_C be the cone measure...

conjecture | open | MathDB 页面 | 原文线索

V2-026939 | MATHDB-359111

Pasteczka's conjecture on Jensen candidates and Jensen domains

摘录 / EXCERPT: Pasteczka's conjecture. Every Jensen candidate is also a Jensen domain.

conjecture | open | MathDB 页面 | 原文线索

V2-026940 | MATHDB-359118

Chudnovsky's conjecture for finite sets of points

摘录 / EXCERPT: Chudnovsky's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-026941 | MATHDB-359168

The central-interior conjecture for differences of central Cantor sets

摘录 / EXCERPT: Let $a=(a_n)_{n\in\mathbb N}\in(0,1)^{\mathbb N}$, and let $C(a)$ denote the central Cantor set determined by the sequence a. Write int for interior in...

conjecture | open | MathDB 页面 | 原文线索

V2-026942 | MATHDB-359169

Multi-bubble isoperimetric conjecture on the sphere

摘录 / EXCERPT: Let q satisfy $2\leq q\leq n+2$, and let Ω be a q-cluster on $\mathbb{S}^n$ with prescribed volume $V(\Omega)=v\in\operatorname{int}\Delta^{(q-1)}_{1}$. A standard bubbl...

conjecture | open | MathDB 页面 | 原文线索

V2-026943 | MATHDB-359205

Larman-point conjecture for convex bodies

摘录 / EXCERPT: Let $K\subset \mathbb{R}^{n}$, $n\geq 3$, be a convex body. A point $p\in\mathbb{R}^{n}$ is a Larman point of K if, for every hyperplane Π passing through p, the section...

conjecture | open | MathDB 页面 | 原文线索

V2-026944 | MATHDB-359242

The bounded-geometry characterization by the five gradients inequality

摘录 / EXCERPT: Bounded-geometry conjecture. The manifold $({\mathcal{M}},g)$ is of bounded geometry if and only if the five gradients inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-026945 | MATHDB-359243

The RCD characterization by the five gradients inequality

摘录 / EXCERPT: RCD characterization conjecture. The metric-measure space (X,d,μ) is $RCD(K,\infty)$ if and only if the five gradients inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-026946 | MATHDB-359249

Fujino's injectivity conjecture for log-canonical pairs

摘录 / EXCERPT: Let X be a compact Kähler manifold and $D=\sum_{i=1}^{N}D_i$ be a simple-normal-crossing (snc) divisor on X. Let F be a semi-positive line bundle on X, meaning that it admi...

conjecture | open | MathDB 页面 | 原文线索

V2-026947 | MATHDB-359275

Henrot-Michetti conjecture on the Neumann-Steklov eigenvalue ratio for convex planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded, open, convex set, and let ... where $\mu_1(\Omega)$ and $\sigma_1(\Omega)$ are respectively the first nonzero Neumann and Steklov eigenv...

conjecture | open | MathDB 页面 | 原文线索

V2-026948 | MATHDB-359281

The splitting conjecture for non-compact Einstein manifolds with cocompact isometry groups

摘录 / EXCERPT: Splitting conjecture. M^n splits isometrically as a product of a compact Einstein manifold and a non-compact homogeneous Einstein space.

conjecture | open | MathDB 页面 | 原文线索

V2-026949 | MATHDB-359297

The spun-lens-space conjecture for the trisection pants invariant

摘录 / EXCERPT: Spun-lens-space conjecture. The $\mathcal{L}^*_3$-invariant of spun lens spaces is equal to 18.

conjecture | open | MathDB 页面 | 原文线索

V2-026950 | MATHDB-359298

The pants invariant conjecture for connected sums of $S^2\times S^2$

摘录 / EXCERPT: Connected-sum pants-invariant conjecture. The $\mathcal{L}^*_3$-invariant of Y_m is equal to $9m$.

conjecture | open | MathDB 页面 | 原文线索

V2-026951 | MATHDB-359301

Second Robin eigenvalue conjecture for planar exterior domains

摘录 / EXCERPT: Second Robin eigenvalue conjecture. If $\mathcal B$ is a disk, then

conjecture | open | MathDB 页面 | 原文线索

V2-026952 | MATHDB-359309

Stoker's conjecture on rigidity of hyperbolic strictly-convex polyhedra

摘录 / EXCERPT: A hyperbolic strictly-convex polyhedron is a strictly-convex polyhedron in hyperbolic space, with dihedral angles measured along its edges. Stoker's conjecture. Every hyperbolic st...

conjecture | open | MathDB 页面 | 原文线索

V2-026953 | MATHDB-359356

Trisection genus invariance for exotic smooth 4-manifolds

摘录 / EXCERPT: Trisection genus invariance conjecture. If two smooth 4-manifolds X and Y are exotic, then

conjecture | open | MathDB 页面 | 原文线索

V2-026954 | MATHDB-359385

Götsche-Kool-Laarakker horizontal universality conjecture for virtual Euler characteristics

摘录 / EXCERPT: Let $r>1$. The notation $\overline{\Delta}(q)$ and ϵ_r is as defined in the introduction. For a smooth polarized surface (X,H) with $b_1(X)=0$ and $h^{2,0}(X)>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-026955 | MATHDB-359390

Conjectured unbounded imbalance for stretchable point configurations

摘录 / EXCERPT: For a finite set of points in the plane, consider every line determined by at least two points of the set, and compare the numbers of points on its two sides. Stretchable-configura...

conjecture | open | MathDB 页面 | 原文线索

V2-026956 | MATHDB-359397

The local p-refined Friedberg-Jacquet conjecture

摘录 / EXCERPT: Let π be symplectic, let P be a spin parabolic, and let $\tilde\pi^P$ be a P-refinement. Let χ be finite order of conductor $p^\beta>1$, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-026957 | MATHDB-359404

Comparison conjecture for complex and almost complex threefold moduli

摘录 / EXCERPT: Complex-almost complex comparison conjecture. This map is an isomorphism. The conjecture is intended to permit almost complex methods to study invariants of complex threefolds, but...

conjecture | open | MathDB 页面 | 原文线索

V2-026958 | MATHDB-359407

Planar central-symmetry conjecture for equichordal supporting chords

摘录 / EXCERPT: Let $K,L\subset\mathbb R^2$ be convex bodies with $L\subset\operatorname{int}K$, and suppose that L is centrally symmetric. Planar central-symmetry conjecture. If every pair of p...

conjecture | open | MathDB 页面 | 原文线索

V2-026959 | MATHDB-359408

Equichordal projection conjecture for convex bodies

摘录 / EXCERPT: Let $K,L\subset\mathbb R^3$ be convex bodies with $L\subset\operatorname{int}K$, and suppose that L is strictly convex. For $u,w\in\mathbb S^2$ with $u\bot w$, let π_u denote...

conjecture | open | MathDB 页面 | 原文线索

V2-026960 | MATHDB-359409

Equichordal projection conjecture for a point

摘录 / EXCERPT: Let $K\subset\mathbb R^3$ be a convex body and let p be a point in its interior. For $u\in\mathbb S^2$, let π_u denote orthogonal projection in direction u. Point equichord...

conjecture | open | MathDB 页面 | 原文线索

V2-026961 | MATHDB-359410

Minkowski's covering-radius conjecture for unimodular lattices

摘录 / EXCERPT: Let X_d be the space of lattices under consideration, let A be the group acting on X_d, and let $\mu=\operatorname{CovRad}_N$ be the covering radius associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-026962 | MATHDB-359411

The singleton Minkowski spectrum conjecture for function-field lattices

摘录 / EXCERPT: Let $\mathcal{L}_d$ be the space of function-field lattices and define the Minkowski spectrum by ... Here q is the parameter appearing in the function-field absolute value. Singlet...

conjecture | open | MathDB 页面 | 原文线索

V2-026963 | MATHDB-359418

Montejano-Roldán-Pensado-Swanepoel realization conjecture for involutive polyhedral graphs

摘录 / EXCERPT: Let $G=(V,E)$ be an involutive polyhedral graph: a self-dual polyhedral graph equipped with an involution τ satisfying $v\notin\tau(v)$ and $u\in\tau(v)\iff v\in\tau(u)$. A Re...

conjecture | open | MathDB 页面 | 原文线索

V2-026964 | MATHDB-359422

The kissing-number equality for finite unit-distance configurations

摘录 / EXCERPT: For a bounded set $K\subset\mathbb{R}$, let ... Let kiss_d be the kissing number in $\mathbb{R}^d$. Kissing-number conjecture. For every dimension d, ... The evident lower...

conjecture | open | MathDB 页面 | 原文线索

V2-026965 | MATHDB-359473

The UGVL characterization conjecture for totally bounded ultrametric spaces

摘录 / EXCERPT: UGVL characterization conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-026966 | MATHDB-359486

Smooth conjecture for enumerative integrals on spaces of maps

摘录 / EXCERPT: Fix cycles c_i^X in the source $X=\mathbb{P}^k$ and cycles c_i^Y in the target $Y=\mathbb{P}^n$, for $i=1,\ldots,l$. Let $\mathrm{Maps}_d(\mathbb{P}^k,\mathbb{P}^n)$ be the spa...

conjecture | open | MathDB 页面 | 原文线索

V2-026967 | MATHDB-359516

The torus first Dirac eigenvalue optimisation conjecture for the trivial spin structure

摘录 / EXCERPT: Let (a,b) satisfy $0\leq a\leq 1/2$ and $a^2+b^2\geq 1$, let $\mathbb{T}^2=\mathbb{R}^2/\Gamma_{a,b}$ be the corresponding flat torus with flat metric $g_{a,b}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-026968 | MATHDB-359529

Modified maximal-stress conjecture for axially symmetric convex planar domains

摘录 / EXCERPT: Let Ω be an axially symmetric planar convex domain centered at the origin, and let the fail points be the points where the stress $|\nabla u|$ is maximal, with u the torsi...

conjecture | open | MathDB 页面 | 原文线索

V2-026969 | MATHDB-359531

Joyce's natural morphism conjecture for oriented Lagrangian morphisms

摘录 / EXCERPT: Joyce's conjecture. There is a natural morphism

conjecture | open | MathDB 页面 | 原文线索

V2-026970 | MATHDB-359559

Maximal integral packing conjecture for product tori on the blown-up projective plane

摘录 / EXCERPT: For each integer $m\geq 2$, let (Y_m,ω_m) be the toric symplectic manifold obtained from $\mathbb{CP}^{2}$ with lines of area $m+2$ by symplectically blowing up along the st...

conjecture | open | MathDB 页面 | 原文线索

V2-026971 | MATHDB-359580

Hasselblatt's stable-unstable slice dimension conjecture for hyperbolic sets

摘录 / EXCERPT: Let f be a smooth diffeomorphism with a hyperbolic set Λ. For each $x\in\Lambda$, let the stable and unstable slices be the intersections of the local stable and unstable...

conjecture | open | MathDB 页面 | 原文线索

V2-026972 | MATHDB-359597

The base projective-space conjecture for holomorphic Lagrangian fibrations

摘录 / EXCERPT: Let X^{2n} be a hyperkähler manifold, and let $f:X\to B$ be a holomorphic Lagrangian fibration, where B is its base. Base projective-space conjecture. If X is a hyperkähler m...

conjecture | open | MathDB 页面 | 原文线索

V2-026973 | MATHDB-359608

Quantitative Hadwiger-type conjecture for disjoint cylinders in three-dimensional space

摘录 / EXCERPT: Let $\mathcal{B} = \{B_1,\ldots, B_n\}$ be a finite family of disjoint cylinders in $\mathbb{R}^3$. A set of points $P = \{p_1,\ldots, p_n\}$ spans $\mathbb{R}^k$ if its affine spa...

conjecture | open | MathDB 页面 | 原文线索

V2-026974 | MATHDB-359609

Three-set quantitative transversal conjecture for overlapping convex sets

摘录 / EXCERPT: For each $1\leq i\leq n$, let f_i be a monotone function and let α_i be a specified value. A family of sets has a (f,α)-transversal when one transversal satisfies t...

conjecture | open | MathDB 页面 | 原文线索

V2-026975 | MATHDB-359615

Higher-order Bol inequality for normal solutions with bounded Q-curvature

摘录 / EXCERPT: Higher-order Bol inequality conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-026976 | MATHDB-359616

Dual higher-order Bol inequality for normal solutions with lower-bounded Q-curvature

摘录 / EXCERPT: Dual higher-order Bol inequality conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-026977 | MATHDB-359621

Cusp-shape norm conjecture for equal-volume Dehn fillings

摘录 / EXCERPT: Let M be a 1-cusped hyperbolic 3-manifold with cusp shape c_{1}. Cusp-shape norm conjecture. If ... for sufficiently large $|p|+|q|$, then ... The conjecture proposes that, asy...

conjecture | open | MathDB 页面 | 原文线索

V2-026978 | MATHDB-359622

Rational-real-part conjecture for cusp shapes of equal-volume fillings

摘录 / EXCERPT: Let M be a 1-cusped hyperbolic 3-manifold. A rational-real-part cusp shape means that the cusp shape has rational real part for some choice of basis $\mathbf{m},\mathbf{l}$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-026979 | MATHDB-359646

Tosatti's conjecture on independence of Kähler-Ricci flow singularity type

摘录 / EXCERPT: Tosatti's conjecture. The singularity type at infinity does not depend on the choice of the initial metric ω_0.

conjecture | open | MathDB 页面 | 原文线索

V2-026980 | MATHDB-359654

Lattice-minimality conjecture for asymptotic distinct-triangle counts

摘录 / EXCERPT: Lattice-minimality conjecture. This lattice spans at least

conjecture | open | MathDB 页面 | 原文线索

V2-026981 | MATHDB-359662

The cuplength lower bound for Rouquier dimension of wrapped Fukaya categories

摘录 / EXCERPT: Let M be a smooth manifold, and let $\operatorname{Cuplength}(M)$ denote the cuplength of $H^*(M;k)$ for any field k. Let $\mathcal C(M)$ denote the wrapped Fukaya category ass...

conjecture | open | MathDB 页面 | 原文线索

V2-026982 | MATHDB-359677

The Monge-Ampère determination conjecture for Ricci-flat metrics on resolved threefolds

摘录 / EXCERPT: Let $Y_{[3]}^\Gamma$ be a resolved threefold with exceptional divisor $\mathcal{ED}$, and let the Ricci-flat metric and the Kähler metric on $\mathcal{ED}$ be those considered in t...

conjecture | open | MathDB 页面 | 原文线索

V2-026983 | MATHDB-359685

Eisenhart-type product conjecture for affinely nonrigid sub-Riemannian metrics

摘录 / EXCERPT: Eisenhart-type conjecture. If a sub-Riemannian metric g is not affinely rigid near a point q_0, meaning that it admits a locally affinely equivalent non-constantly proportional...

conjecture | open | MathDB 页面 | 原文线索

V2-026984 | MATHDB-359754

The missing-digits product-set interval conjecture

摘录 / EXCERPT: Let K be a missing digits set, meaning a compact set determined by restricting the digits in a fixed integer base. Folklore conjecture. If ... then $K\cdot K$ contains intervals. T...

conjecture | open | MathDB 页面 | 原文线索

V2-026985 | MATHDB-359755

The strongly non-degenerate manifold covering conjecture for missing digits sets

摘录 / EXCERPT: Let $n\geq 1$ be an integer, let $M\subset\mathbb{R}^n$ be a strongly non-degenerate analytic manifold, let $p>2$ be an integer, and let $D\subsetneq\{0,\dots,p-1\}^n$ be a choice...

conjecture | open | MathDB 页面 | 原文线索

V2-026986 | MATHDB-359757

The Hausdorff-dimension neighbourhood estimate for missing digits measures

摘录 / EXCERPT: Let $n\geq2$ be an integer and let $M\subset\mathbb{R}^n$ be a manifold of finite type. Neighbourhood estimate conjecture. There exists a number $\sigma=\sigma(M)>0$ such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-026987 | MATHDB-359766

Tropical Todd genus conjecture for compact tropical manifolds

摘录 / EXCERPT: Let X be a compact tropical manifold of dimension n. The Todd class of X is obtained from the formal power series in its Chern classes by substituting...

conjecture | open | MathDB 页面 | 原文线索

V2-026988 | MATHDB-359768

The transcendental base-point-free conjecture for nef (1,1)-classes

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $[\alpha]$ be a nef $(1,1)$-class such that ... is nef and big for some $\lambda\in\mathbb{R}_{>0}$. Transcendental base-point-free conje...

conjecture | open | MathDB 页面 | 原文线索

V2-026989 | MATHDB-359770

The numerical abundance conjecture for nef line bundles on projective Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a projective Calabi-Yau manifold, meaning a compact Kähler manifold with $c_1(K_X)=0$ in $H^{1,1}(X,\mathbb{R})$, and let $L\to X$ be a nef line bundle. Numerical abunda...

conjecture | open | MathDB 页面 | 原文线索

V2-026990 | MATHDB-359772

The generalized bounded-potential conjecture for nef (1,1)-classes

摘录 / EXCERPT: Let X be a compact Kähler manifold with K_X pseudoeffective, and let $[\alpha]$ be a nef $(1,1)$-class such that ... is nef for some $\lambda\in\mathbb{R}_{>0}$. Generalized boun...

conjecture | open | MathDB 页面 | 原文线索

V2-026991 | MATHDB-359774

The WPIC1 Ricci-flow existence conjecture

摘录 / EXCERPT: Let (M^n,g_0) be a complete WPIC1 manifold with $n\geq 3$. WPIC1 Ricci-flow existence conjecture. There exists a complete WPIC1 Ricci flow $g(t)$ on M for $t\in[0,T)$, for some...

conjecture | open | MathDB 页面 | 原文线索

V2-026992 | MATHDB-359794

Palis' sumset conjecture for missing digits sets

摘录 / EXCERPT: Let $A,B\subset[0,1]$ be missing digits sets. Palis' conjecture. Suppose that ... Then, generically, $A+B$ contains a non-trivial interval. In the paper's missing-digits setting, the...

conjecture | open | MathDB 页面 | 原文线索

V2-026993 | MATHDB-359795

Universal interior conjecture for sums of incommensurable missing digits sets

摘录 / EXCERPT: Let $A,B\subset[0,1]$ be missing digits sets with bases p_A and p_B, and define $T_x(y)=xy$. Universal interior conjecture. Suppose that ... and ... Then $A+T_x(B)$ contains a non-...

conjecture | open | MathDB 页面 | 原文线索

V2-026994 | MATHDB-359797

Cherowitzo's translation hyperoval existence conjecture

摘录 / EXCERPT: A translation plane is a finite projective plane containing a group of elations acting transitively on its affine points; a translation hyperoval is a hyperoval invariant under thi...

conjecture | open | MathDB 页面 | 原文线索

V2-026995 | MATHDB-359811

Asymptotically harmonic analogue of Lichnerowicz's conjecture

摘录 / EXCERPT: Asymptotically harmonic classification conjecture. If M is an asymptotically harmonic manifold, then M is either flat or a rank-one symmetric space of noncompact type.

conjecture | open | MathDB 页面 | 原文线索

V2-026996 | MATHDB-359812

Canonical-bundle conjecture for the divisibility-one fixed locus

摘录 / EXCERPT: Let (X,λ) be a polarized irreducible holomorphic symplectic manifold of $\mathrm{K3}^{[n]}$-type with $q_X(\lambda)=2$ and $\operatorname{div}(\lambda)=1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-026997 | MATHDB-359819

The anticanonical bigness conjecture for projective splinters

摘录 / EXCERPT: Anticanonical bigness conjecture. If X is a splinter, then $-K_X$ is big.

conjecture | open | MathDB 页面 | 原文线索

V2-026998 | MATHDB-359848

Sormani's minA-IF stability conjecture for Riemannian 3-tori

摘录 / EXCERPT: Let $\{(M_i,g_i)\}_{i=1}^\infty$ be a sequence of Riemannian 3-tori. Assume that $\operatorname{diam}(M_i,g_i)$ is uniformly bounded above, while $\operatorname{vol}(M_i,g_i)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-026999 | MATHDB-359849

Finite cop number for game spaces with finite doubling constant

摘录 / EXCERPT: A game space is a compact geodesic metric space equipped with the Cops and Robber game. Its doubling constant is the least $k\in\mathbb{N}\cup\{\infty\}$ such that every ball...

conjecture | open | MathDB 页面 | 原文线索

V2-027000 | MATHDB-359850

Finite cop number for homogeneous game spaces

摘录 / EXCERPT: A metric space (X,d) is homogeneous if for every $x,y\in X$ there is an isometry $i:X\to X$ with $i(x)=y$. Let X be a game space, namely a compact geodesic metric space equippe...

conjecture | open | MathDB 页面 | 原文线索

V2-027001 | MATHDB-359870

Li-Zhan polyhedral-covering conjecture for Calabi-Yau fiber spaces

摘录 / EXCERPT: Polyhedral-covering conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027002 | MATHDB-359871

Brunella's mixed Nevanlinna-current conjecture

摘录 / EXCERPT: Brunella's mixed Nevanlinna-current conjecture. Some entire curve shall produce a Nevanlinna current $\mathsf{T}=\mathsf{T}_{\text{alg}}+\mathsf{T}_{\text{diff}}$ with both nontriv...

conjecture | open | MathDB 页面 | 原文线索

V2-027003 | MATHDB-359872

Universal entire-curve conjecture for Nevanlinna and Ahlfors currents

摘录 / EXCERPT: Universal entire-curve conjecture. There should be some entire curve $g:\mathbb{C}\rightarrow X$ such that the concentric discs

conjecture | open | MathDB 页面 | 原文线索

V2-027004 | MATHDB-359917

Flat point conjecture for complete minimal surfaces in Euclidean three-space

摘录 / EXCERPT: Flat point conjecture. If X has at least one flat point, then its Gauss map omits at most 3 values.

conjecture | open | MathDB 页面 | 原文线索

V2-027005 | MATHDB-359922

Bi-Lipschitz multiplicity conjecture for complex analytic germs

摘录 / EXCERPT: Let $X\subset \mathbb{C}^n$ and $Y\subset \mathbb{C}^m$ be complex analytic sets with $\dim X=\dim Y=d$. Their germs at zero are bi-Lipschitz homeomorphic if there is a bi-Lipschit...

conjecture | open | MathDB 页面 | 原文线索

V2-027006 | MATHDB-359933

The density-property characterization of RR-spaces

摘录 / EXCERPT: Density-property characterization. X satisfies the density property if and only if it is an RR-space.

conjecture | open | MathDB 页面 | 原文线索

V2-027007 | MATHDB-359934

Nontriviality of the identity component of transitive automorphism groups

摘录 / EXCERPT: Automorphism-group component conjecture. The connected component $\operatorname{Aut}^0(X)$ is non-trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-027008 | MATHDB-359957

Greene-Wu conjecture on uniform equivalence of the Kobayashi metric

摘录 / EXCERPT: Let (M,h) be a simply-connected complete Kähler manifold satisfying ... for two positive constants A and B. Let $\mathfrak{K}$ denote the Kobayashi metric on M. Greene-Wu's c...

conjecture | open | MathDB 页面 | 原文线索

V2-027009 | MATHDB-359979

Reeb 1-curve conjecture for closed exact lcs four-manifolds

摘录 / EXCERPT: Reeb 1-curve conjecture. The manifold M has a Reeb 1-curve.

conjecture | open | MathDB 页面 | 原文线索

V2-027010 | MATHDB-359981

Motegi's boundary-slope bound conjecture for exceptional surgeries

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3. A boundary slope is a slope represented by the boundary curves of an essential embedded surface in the exterior of K, and an exceptional su...

conjecture | open | MathDB 页面 | 原文线索

V2-027011 | MATHDB-359982

Boundary-slope interval conjecture for exceptional surgeries

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3 that admits non-trivial exceptional surgeries. A NIT boundary slope is a boundary slope that is non-integral or toroidal. For $b e{}$ an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-027012 | MATHDB-359995

Symplectic-ball conjecture for Lagrangian products of a simplex and parallelohedra

摘录 / EXCERPT: Let S be a three-dimensional simplex and let P be any one of the five three-dimensional parallelohedra, namely the cube, hexagonal prism, elongated dodecahedron, rhombic dodeca...

conjecture | open | MathDB 页面 | 原文线索

V2-027013 | MATHDB-360003

Siu's invariance of plurigenera conjecture for Kähler families

摘录 / EXCERPT: Let $\pi:\mathcal{X}\longrightarrow\Delta$ be a holomorphic family of compact Kähler manifolds $X_t:=\pi^{-1}(t)$ over the open unit disc $\Delta\subset\mathbb{C}$. For a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-027014 | MATHDB-360024

Dominance conjecture for the Blaschke-Santaló diagram of Minkowski-centered triangles

摘录 / EXCERPT: Dominance conjecture. The diagram of any triangle is dominating: for every Minkowski-centered C,

conjecture | open | MathDB 页面 | 原文线索

V2-027015 | MATHDB-360027

Generic transversality conjecture for opers with real monodromy

摘录 / EXCERPT: Generic transversality conjecture. The two real submanifolds $\mathrm{Op}_P(\mathscr{X},G)$ and $\mathrm{Rep}_P^0(\Gamma,G(\mathbb{R}))$ intersect transversally at every point. Equ...

conjecture | open | MathDB 页面 | 原文线索

V2-027016 | MATHDB-360077

Hebey-Vaugon conjecture on the strict equivariant Yamabe bound

摘录 / EXCERPT: Hebey-Vaugon conjecture. If (M^n,g) is not conformal to the standard metric on S^n, or if the action of G has no fixed point, then

conjecture | open | MathDB 页面 | 原文线索

V2-027017 | MATHDB-360078

Schoen-Yau's $K(\pi,1)$ conjecture for positive scalar curvature

摘录 / EXCERPT: Let M^n be a closed n-dimensional $K(\pi,1)$ manifold, meaning an aspherical manifold whose universal cover is contractible. A Riemannian metric has positive scalar curvature i...

conjecture | open | MathDB 页面 | 原文线索

V2-027018 | MATHDB-360079

Weak $K(\pi,1)$ conjecture for manifolds with boundary

摘录 / EXCERPT: Let M^n be a compact n-manifold with non-empty boundary, and let its double be the closed manifold obtained by gluing two copies of M^n along their common boundary. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027019 | MATHDB-360080

Rosenberg's S^1-stability conjecture for the Yamabe invariant

摘录 / EXCERPT: Let M^n be a closed, connected n-manifold, let S^1 be the circle, and let σ denote the Yamabe invariant. Rosenberg's S^1-stability conjecture. The Yamabe invariants...

conjecture | open | MathDB 页面 | 原文线索

V2-027020 | MATHDB-360090

Sign and vanishing conjecture for boundary strata of Reeb flows

摘录 / EXCERPT: Boundary-strata conjecture. If j is even, then $\beta|_{\mathsf{int}(\partial_j^+X(v_\beta))}$ is a contact form away from a set of measure zero, and

conjecture | open | MathDB 页面 | 原文线索

V2-027021 | MATHDB-360127

The zero simplicial volume characterization for nonpositively curved four-manifolds

摘录 / EXCERPT: Let M be a closed nonpositively curved 4-manifold, with simplicial volume $||M||$ and Euler characteristic $\chi(M)$. Zero simplicial volume characterization conjecture. Then .....

conjecture | open | MathDB 页面 | 原文线索

V2-027022 | MATHDB-360147

The K-global F-regularity conjecture for semiample anticanonical pairs

摘录 / EXCERPT: Let (X,B) be a normal projective complex pair with klt singularities such that $-K_X-B$ is semiample. K-global F-regularity conjecture. For every model...

conjecture | open | MathDB 页面 | 原文线索

V2-027023 | MATHDB-360155

Generalized Fefferman-space reconstruction conjecture

摘录 / EXCERPT: Let $(\widetilde{\mathcal{M}},\widetilde{\mathbf{c}},\widetilde{K})$ be an optical geometry of dimension $2m+2>4$ with twisting non-shearing congruence of null geodesics...

conjecture | open | MathDB 页面 | 原文线索

V2-027024 | MATHDB-360170

Belov-Kanel-Kontsevich conjecture on Weyl algebra automorphisms

摘录 / EXCERPT: Let n be a positive integer, let $A_n(\mathbb{C})$ denote the Weyl algebra of index n over $\mathbb{C}$, and equip $\mathbb{C}[x_1,\dots,x_n,y_1,\dots,y_{2n}]$ with its standar...

conjecture | open | MathDB 页面 | 原文线索

V2-027025 | MATHDB-360171

The groupoid isomorphism conjecture for Kleinian singularities

摘录 / EXCERPT: Let V be a two-dimensional complex vector space and let Γ be a finite subgroup of $SL(2,\mathbb{C})$, so that V/Γ is a Kleinian singularity. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027026 | MATHDB-360229

The Type (1) face orbit conjecture for K3-fibered threefolds

摘录 / EXCERPT: Type (1) face orbit conjecture. The pseudoautomorphism group $\operatorname{PsAut}(Y)$ acts with finitely many orbits on the Type (1) faces of $\operatorname{Mov}^e(Y)$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-027027 | MATHDB-360230

Weak R-equivalence conjecture for weighted homogeneous polynomials

摘录 / EXCERPT: Let f,g be weighted homogeneous polynomials on $\mathbb{C}^n$ of the same degree d, with weight vectors ${\bf w}_f$ and ${\bf w}_g$. Two such polynomials are weakly R-equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-027028 | MATHDB-360231

Sasakian Hodge number bound for Calabi-Yau links

摘录 / EXCERPT: Let f be a ${\bf w}$-homogeneous polynomial defining a Calabi-Yau 3-fold, and let the associated link have Sasakian Hodge number $h_S^{2,1}$, while the Calabi-Yau 3-fold has...

conjecture | open | MathDB 页面 | 原文线索

V2-027029 | MATHDB-360264

Q-torus characterization conjecture for zero entropy groups

摘录 / EXCERPT: Let X be a compact Kähler manifold such that $\kappa(X)=0$ and $b_1(X)\ne 0$, and let $G\leq \operatorname{Aut}(X)$ be a zero entropy subgroup. The Q-torus characterization conje...

conjecture | open | MathDB 页面 | 原文线索

V2-027030 | MATHDB-360266

Convergence conjecture for deformations of several smoothable edges

摘录 / EXCERPT: Let $\mathsf{X}=\mathbb{P}^{n-1}\times\mathbb{D}^m$ and let $\pi(\varepsilon)=\sum_{k=0}^{\infty}\varepsilon^k\pi_k$ be the formal Poisson deformation whose existence is asserted f...

conjecture | open | MathDB 页面 | 原文线索

V2-027031 | MATHDB-360276

The generalized Smale conjecture for spherical three-manifolds

摘录 / EXCERPT: Let (M,g) be a closed orientable Riemannian three-manifold, with g a metric of constant sectional curvature ±1. Let $\operatorname{Isom}(M)$ be the isometry group of M an...

conjecture | open | MathDB 页面 | 原文线索

V2-027032 | MATHDB-360282

The birational-sheet degree conjecture for Sommers-Achar stabilizers

摘录 / EXCERPT: Birational-sheet degree conjecture. The degree of μ equals the order of the stabilizer:

conjecture | open | MathDB 页面 | 原文线索

V2-027033 | MATHDB-360283

The multiple-cover contribution formula for a nodal cubic

摘录 / EXCERPT: Let L_0 be the line in the central fibre considered in the paper, and let $\operatorname{C}_{\operatorname{ord}}(d,0,0)$ denote the contribution of degree-d covers of L_0. Mu...

conjecture | open | MathDB 页面 | 原文线索

V2-027034 | MATHDB-360286

The microlocal schober model for 2-categorical category O

摘录 / EXCERPT: Let $(X\supset\mathbb L)$ be a sectorial conical symplectic resolution, and let $\mathbb L$ denote its distinguished Lagrangian. Write $2\mathcal O_{\mathsf A}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027035 | MATHDB-360287

Fully extended 3d mirror symmetry for sectorial symplectic resolutions

摘录 / EXCERPT: Let $(X\supset\mathbb L)$ and $(X^\vee\supset\mathbb L^\vee)$ be dual pairs of sectorial conical symplectic resolutions. Let $2\mathcal O_{\mathsf A}$ and $2\mathcal O_{\mathsf B}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027036 | MATHDB-360300

The Albanese fibration conjecture for compact Kähler manifolds with nef anticanonical bundle

摘录 / EXCERPT: Albanese fibration conjecture. The Albanese morphism of X is a locally trivial fibration with connected fibers.

conjecture | open | MathDB 页面 | 原文线索

V2-027037 | MATHDB-360301

The locally constant rationally connected fibration conjecture for Kähler manifolds with nef anticanonical bundle

摘录 / EXCERPT: Structural conjecture. There exists a fibration $\varphi:X\to Y$ such that φ is locally constant, Y is a compact Kähler manifold with $c_1(Y)=0$, and F is rationally co...

conjecture | open | MathDB 页面 | 原文线索

V2-027038 | MATHDB-360331

Finite-time scalar curvature and fibre diameter conjecture for Kähler-Ricci flow

摘录 / EXCERPT: Let $g(t)$ be the smooth solution of the Kähler-Ricci flow on $X\times[0,T)$, where $T<\infty$, and let $\Phi:X\to Y$ be the map induced by the limiting semi-ample class. For...

conjecture | open | MathDB 页面 | 原文线索

V2-027039 | MATHDB-360336

Uniqueness conjecture for compact convex ancient G-flows

摘录 / EXCERPT: Let G be the curvature function defining the flow, and consider convex ancient G-flows that are noncollapsing, uniformly two-convex, and compact. Uniqueness conjecture. Up to p...

conjecture | open | MathDB 页面 | 原文线索

V2-027040 | MATHDB-360337

Unique geodesics and metric convexity for the space of compact spacelike Cauchy hypersurfaces

摘录 / EXCERPT: Geodesic uniqueness conjecture. Given two spacelike Cauchy hypersurfaces $S_0,S_1\in\mathfrak{C}(M,g)$, there is a unique geodesic path $c:[0,1]\to\mathfrak{C}(M,g)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-027041 | MATHDB-360359

Fractional weighted isoperimetric ratio strict inequality conjecture

摘录 / EXCERPT: Let n be the boundary dimension, let $\gamma\in(0,\frac{1}{2}]$ with $n>2\gamma$, and let $(\overline{X},\bar{g})$ be a smooth compactification that is not conformally diffeomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-027042 | MATHDB-360372

Lawson-Simons conjecture on stable minimal submanifolds in strictly quarter-pinched manifolds

摘录 / EXCERPT: Let M be a compact, simply connected Riemannian manifold whose sectional curvature lies in $(1/4,1]$. A closed minimal submanifold of M is stable when its second variation sati...

conjecture | open | MathDB 页面 | 原文线索

V2-027043 | MATHDB-360392

Derived equivalence of crepant resolutions and noncommutative crepant resolutions

摘录 / EXCERPT: Let R be an equidimensional normal Gorenstein ring. A noncommutative crepant resolution of R is an endomorphism ring of a modifying R-module with finite global dimension, whi...

conjecture | open | MathDB 页面 | 原文线索

V2-027044 | MATHDB-360436

BTP metrics are BAS under homogeneous universal-cover hypotheses

摘录 / EXCERPT: BTP-to-BAS conjecture. Under these assumptions, g must be BAS.

conjecture | open | MathDB 页面 | 原文线索

V2-027045 | MATHDB-360437

Haslhofer's classification conjecture for four-dimensional noncollapsed steady solitons

摘录 / EXCERPT: Haslhofer's classification conjecture. Such a soliton can only be the 4-dimensional Bryant soliton, the product of the 3-dimensional Bryant soliton with a line, or one of the...

conjecture | open | MathDB 页面 | 原文线索

V2-027046 | MATHDB-360441

Hodge-theoretic Singer-Hopf conjecture for compact Kähler manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold of complex dimension n, and write ... where Ω_X^p is the sheaf of holomorphic p-forms. Assume that X is aspherical or has a nef cota...

conjecture | open | MathDB 页面 | 原文线索

V2-027047 | MATHDB-360442

Mixed-Hodge-module extension of the Hodge-theoretic Singer-Hopf conjecture

摘录 / EXCERPT: Let X be a compact Kähler or complex projective manifold that is aspherical or has a nef cotangent bundle, and let $\mathcal{M}$ be a mixed Hodge module on X. For an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-027048 | MATHDB-360443

Hodge-theoretic Hopf conjecture for compact Kähler manifolds with nef tangent bundle

摘录 / EXCERPT: Hodge-theoretic Hopf conjecture. For every p with $0\leq p\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-027049 | MATHDB-360470

Splitting conjecture for tangent cones at infinity of manifolds with non-negative Ricci curvature

摘录 / EXCERPT: Splitting conjecture. Every tangent cone at infinity M_∞ splits isometrically as

conjecture | open | MathDB 页面 | 原文线索

V2-027050 | MATHDB-360499

Bonnesen-Fenchel conjecture on volume minimization by Meissner tetrahedra

摘录 / EXCERPT: Let a three-dimensional body have fixed constant width. Meissner tetrahedra are obtained by applying a smoothing procedure to a Reuleaux tetrahedron, itself formed by intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-027051 | MATHDB-360512

Multicolour Leader-Long antipodal path conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube. A geodesic path is a shortest path between its endpoints, and a colour change occurs when consecutive edges receive different colours. M...

conjecture | open | MathDB 页面 | 原文线索

V2-027052 | MATHDB-360515

Pandharipande-Thomas rationality conjecture for descendent partition functions

摘录 / EXCERPT: Pandharipande-Thomas rationality conjecture. The partition function $\operatorname{Z_{PT}}\left(M;q\mid\prod_{i=1}^r\tau_{k_i}(\gamma_i)\right)_\beta$ is the Laurent expansion of a...

conjecture | open | MathDB 页面 | 原文线索

V2-027053 | MATHDB-360518

Order-k Delaunay triangulations maximize triangles among local-angle hypertriangulations

摘录 / EXCERPT: Order-k Delaunay optimality conjecture. The hypertriangulation P_k has the maximum number of triangles if and only if P_k is the order-k Delaunay triangulation of A.

conjecture | open | MathDB 页面 | 原文线索

V2-027054 | MATHDB-360519

Maximal hypertriangulations are maximum

摘录 / EXCERPT: Maximality conjecture. Any two maximal level-k hypertriangulations of A have the same, and therefore maximum, number of triangles. Equivalently, every maximal level-k hypertr...

conjecture | open | MathDB 页面 | 原文线索

V2-027055 | MATHDB-360528

The centralizer conjecture for Poisson brackets associated with elliptic curves in $G(2,5)$

摘录 / EXCERPT: Let V be a 5-dimensional vector space, and let $W\subset \bigwedge^2V$ be a 5-dimensional subspace such that E_W is an elliptic curve. Define ... The quotient of T_W by...

conjecture | open | MathDB 页面 | 原文线索

V2-027056 | MATHDB-360533

Bounded integer spectral norm conjecture for pseudo-rotations

摘录 / EXCERPT: Let (M,ω) be a monotone symplectic manifold admitting a pseudo-rotation. Here $\gamma_\mathbb{Z}$ denotes the spectral norm over the integers, and ϕ^k denotes the k-t...

conjecture | refuted | MathDB 页面 | 原文线索

V2-027057 | MATHDB-360556

The injectivity conjecture for positive Whitehead doubles of slice disks

摘录 / EXCERPT: Let K be a knot, and let D_1 and D_2 be slice disks for K in B^4. Their positive Whitehead doubles are the slice disks obtained by applying the positive Whitehead satelli...

conjecture | open | MathDB 页面 | 原文线索

V2-027058 | MATHDB-360581

The square-root stability conjecture for Brunn-Minkowski

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geq 2$, let $t\in(0,1/2)$, and let $A,B\subset\mathbb{R}^n$ be measurable sets of equal measure satisfying ... where $\delta<d_{n,t}$. Square-root stabil...

conjecture | open | MathDB 页面 | 原文线索

V2-027059 | MATHDB-360582

Optimal parameter-dependence conjecture for Brunn-Minkowski convex-hull defects

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and $t\in(0,1/2]$, and let $A,B\subset\mathbb{R}^n$ be measurable sets of equal volume satisfying ... where $\delta\in[0,d_{n,t}]$. Optimal parameter-dependence...

conjecture | open | MathDB 页面 | 原文线索

V2-027060 | MATHDB-360594

Moduli-space conjecture for noncommutative deformations of elliptic root systems

摘录 / EXCERPT: Let R be as in the exceptional series, let $d4e_R/\mathfrak h^{ell}$ be a family of dg-categories, and let $U\subset \operatorname{Stab}(d4e_R)$ be a space of relative stability...

conjecture | open | MathDB 页面 | 原文线索

V2-027061 | MATHDB-360604

McKernan's singularity-boundedness conjecture for Fano contractions

摘录 / EXCERPT: McKernan's conjecture. Fix a positive integer r and a real number $0<\epsilon\leq 1$. There exists $\delta>0$ depending only on r,ϵ such that, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-027062 | MATHDB-360609

The integral scalar-curvature stability conjecture for Riemannian tori

摘录 / EXCERPT: Let $K\in\mathbb{R}$, $n\in\mathbb{N}$, and $v,D>0$. For every $\epsilon>0$, consider Riemannian n-tori M satisfying $\operatorname{ric}_M\geq K$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027063 | MATHDB-360615

The coarse assembly injectivity conjecture for bounded-geometry spaces

摘录 / EXCERPT: The coarse assembly injectivity conjecture. The map μ is injective. Injectivity of the coarse assembly map is the coarse Novikov property; the paper establishes this property f...

conjecture | open | MathDB 页面 | 原文线索

V2-027064 | MATHDB-360654

The geometric pre-Calabi-Yau conjecture for Chekanov-Eliashberg dg-algebras

摘录 / EXCERPT: Let W be a stopped Weinstein manifold and let Λ be a singular Legendrian submanifold in its boundary. Its Chekanov-Eliashberg dg-algebra is the dg-algebra associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-027065 | MATHDB-360655

Positive Casimir energy conjecture for the (2,3,7)-triangle group orbifold

摘录 / EXCERPT: Let $\Delta(2,3,7)$ be the $(2,3,7)$-triangle group acting on the hyperbolic plane $\mathbb{H}$, and let ℓ_n denote the lengths of the relevant closed geodesics, subject to th...

conjecture | open | MathDB 页面 | 原文线索

V2-027066 | MATHDB-360669

Thurston's CMC foliation conjecture for almost Fuchsian manifolds

摘录 / EXCERPT: An almost Fuchsian manifold is a quasi-Fuchsian hyperbolic three-manifold containing a closed minimal surface homeomorphic to a fixed closed oriented surface of genus at least two,...

conjecture | open | MathDB 页面 | 原文线索

V2-027067 | MATHDB-360686

Even log-concave Brunn-Minkowski conjecture for origin-symmetric convex sets

摘录 / EXCERPT: Let ν be an even log-concave measure on $\mathbb{R}^{n}$. Let $K_{0},K_{1}\subseteq\mathbb{R}^{n}$ be convex and origin-symmetric, meaning $K_{0}=-K_{0}$ and $K_{1}=-K_{1}$. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-027068 | MATHDB-360727

Flatness conjecture for Type IIb scalar-flat Kähler 4-manifolds

摘录 / EXCERPT: Flatness conjecture. Then M^4 is flat.

conjecture | open | MathDB 页面 | 原文线索

V2-027069 | MATHDB-360759

Strong Sarkisov program

摘录 / EXCERPT: Strong Sarkisov program. The sequence of Sarkisov links constructed in the strong Sarkisov link construction terminates. Equivalently, the birational map...

conjecture | open | MathDB 页面 | 原文线索

V2-027070 | MATHDB-360789

Nontriviality conjecture for the constructed diffeomorphism of the 4-sphere

摘录 / EXCERPT: Nontriviality conjecture. The diffeomorphism f is not smoothly isotopic to the identity. This would provide a potentially nontrivial pseudo-isotopy of S^4 arising from the disc...

conjecture | open | MathDB 页面 | 原文线索

V2-027071 | MATHDB-360825

Hot spots conjecture for simply-connected planar and convex domains

摘录 / EXCERPT: Hot spots conjecture. The function u attains its maximum only on the boundary $\partial\Omega$. The source presents this as the restricted formulation motivated by counterexample...

conjecture | refuted | MathDB 页面 | 原文线索

V2-027072 | MATHDB-360837

Spectral-gap conjecture for compact hyperbolic spin orbifolds

摘录 / EXCERPT: Let X be a hyperbolic orbifold, and let $\lambda^{(0)}_1(X)$ denote the first non-zero eigenvalue of its Laplacian. Let $E_{\mathtt{spin}}\subset\mathbb{R}_{>0}$ be the set of th...

conjecture | open | MathDB 页面 | 原文线索

V2-027073 | MATHDB-360848

The conjecture that reflexive simplices have no ML degree drops

摘录 / EXCERPT: Let Δ be a reflexive simplex, and let $\mathrm{mldeg}(\Delta)$ and $\mathrm{deg}(\Delta)$ denote its maximum likelihood degree and degree, respectively, under the standard s...

conjecture | open | MathDB 页面 | 原文线索

V2-027074 | MATHDB-360852

Galloway-Ling conjecture on CMC Cauchy surfaces

摘录 / EXCERPT: Galloway-Ling conjecture. Then M contains a CMC Cauchy surface.

conjecture | open | MathDB 页面 | 原文线索

V2-027075 | MATHDB-360854

The generalized filling radius conjecture

摘录 / EXCERPT: Generalized filling radius conjecture. The filling radius of (M,g) is no greater than $C(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-027076 | MATHDB-360860

The pentagon symmedian-hyperbolic barycenter coincidence conjecture

摘录 / EXCERPT: Let $\mathbf{P}=(p_1,\dots,p_5)$ be a pentagon, and let $S_\mathbf{P}$ and $L_\mathbf{P}$ denote the symmedian and hyperbolic barycenters, respectively. For cyclically indexed vert...

conjecture | open | MathDB 页面 | 原文线索

V2-027077 | MATHDB-360870

The soul conjecture for open Alexandrov spaces

摘录 / EXCERPT: Soul conjecture. If X contains an open subset where the curvature is bounded below by a positive constant, then S is a point.

conjecture | open | MathDB 页面 | 原文线索

V2-027078 | MATHDB-360871

The fiber-bundle conjecture for Sharafutdinov projections

摘录 / EXCERPT: Fiber-bundle conjecture. If all points in S are weakly m-strained, then $\phi:X\to S$ is a fiber bundle map.

conjecture | open | MathDB 页面 | 原文线索

V2-027079 | MATHDB-360872

The weak-integrability conjectures for Alexandrov submetries

摘录 / EXCERPT: Weak-integrability conjectures. (1) If $\partial Y=\emptyset$ and f is weakly integrable, then f is integrable. (2) If $\Sigma\in \operatorname{Alex}^n(1)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027080 | MATHDB-360873

The splitting conjecture for integrable Sharafutdinov projections

摘录 / EXCERPT: Splitting conjecture. If ϕ is integrable, then

conjecture | open | MathDB 页面 | 原文线索

V2-027081 | MATHDB-360880

The injectivity radius--spherical radius comparison conjecture

摘录 / EXCERPT: Let (M^n,g) be a complete Riemannian manifold. Write $\operatorname{Inj}(M)$ for its injectivity radius and $\operatorname{Rad}_{\mathbb S^n}(M)$ for its spherical radius. The in...

conjecture | open | MathDB 页面 | 原文线索

V2-027082 | MATHDB-360881

Gromov's noncompact injectivity radius conjecture

摘录 / EXCERPT: Gromov's noncompact injectivity radius conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-027083 | MATHDB-360886

Web parametrization conjecture for positive tropical PGL$_3$ points

摘录 / EXCERPT: Let $\hat{S}$ be a compact oriented surface with marked boundary points and punctures. Let $\mathcal{P}_{{\rm PGL}_3,\hat{S}}^+(\mathbb{Z}^t)$ denote the integer tropical points sa...

conjecture | open | MathDB 页面 | 原文线索

V2-027084 | MATHDB-360889

Rubinstein's Ricci iteration convergence conjecture

摘录 / EXCERPT: Rubinstein's Ricci iteration convergence conjecture. The iteration exists for all $i\in\mathbb{N}$ and converges in an appropriate sense to a constant scalar curvature metric. The...

conjecture | open | MathDB 页面 | 原文线索

V2-027085 | MATHDB-360891

Fractional-to-integral transversal bound for fat convex sets

摘录 / EXCERPT: Fractional transversal conjecture. There is a function f such that

conjecture | open | MathDB 页面 | 原文线索

V2-027086 | MATHDB-360966

Yau's projectivity and rational connectedness conjecture for positively curved Kähler manifolds

摘录 / EXCERPT: Let M be a compact complex manifold admitting a Kähler metric ω with positive holomorphic sectional curvature. The holomorphic sectional curvature along a nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-027087 | MATHDB-360968

Agol-Storm-Thurston scalar curvature and volume entropy conjecture

摘录 / EXCERPT: Let (M,g) be a closed Riemannian 3-manifold, let R_g denote its scalar curvature, and let $h(g)$ denote its volume entropy. Agol-Storm-Thurston conjecture. If ... then ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-027088 | MATHDB-360973

The complexified volume conjecture for hyperbolic links

摘录 / EXCERPT: Let L be a hyperbolic link, let $J_N(L;q)$ be its colored Jones polynomial, let $\operatorname{Vol}(L)$ be the hyperbolic volume of $S^3\setminus L$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027089 | MATHDB-360992

Erdős-Szekeres convex polygon conjecture

摘录 / EXCERPT: Let n be an integer, and let a set of points in the plane be in general position, meaning that no three points are collinear. Erdős-Szekeres convex polygon conjecture. Every set...

conjecture | open | MathDB 页面 | 原文线索

V2-027090 | MATHDB-360993

Simplex-is-the-best conjecture for fixed terminal count

摘录 / EXCERPT: Let $d\geq 1$, and consider finite point configurations of $d+1$ points in Euclidean space. Their Steiner ratio is the ratio of the cost of the optimal Steiner tree to the cost of...

conjecture | open | MathDB 页面 | 原文线索

V2-027091 | MATHDB-361004

Conjecture on the curve of maxima of covering constants for Minkowski balls

摘录 / EXCERPT: Let p range over $1<p<\infty$, and consider the curve whose value at p is the maximum covering constant for the Minkowski ball D_p. The curve of maxima conjecture. The curve...

conjecture | open | MathDB 页面 | 原文线索

V2-027092 | MATHDB-361011

Dolgachev's conjecture on p-power orders in Cremona groups

摘录 / EXCERPT: Let k be a field of characteristic $p>0$, and let $\mathrm{Cr}_n(k)=\mathrm{Bir}_k(\mathbb{P}^n_k)$ be the Cremona group of birational automorphisms of projective n-space. An e...

conjecture | open | MathDB 页面 | 原文线索

V2-027093 | MATHDB-361035

Semistability characterization by the minimal J-slope

摘录 / EXCERPT: Let (X,α,β) be a compact Kähler manifold with two Kähler classes. Let ζ_{min} be its minimal J-slope and let μ be the J-slope of the triple. Minimal J-slope se...

conjecture | open | MathDB 页面 | 原文线索

V2-027094 | MATHDB-361077

Ambro's lower semi-continuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Lower semi-continuity conjecture. The function $x\mapsto \operatorname{mld}_x(X,\mathfrak{a})$ is lower semi-continuous.

conjecture | open | MathDB 页面 | 原文线索

V2-027095 | MATHDB-361083

Fournais-Helffer conjecture on the magnetic Neumann eigenvalue of planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded, simply-connected C^∞-smooth domain, let $b>0$ be the intensity of a homogeneous magnetic field, and let $\mu_1^b(\Omega)>0$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-027096 | MATHDB-361084

Stable surface hyperbolicity conjecture for rank-four and rank-two forms

摘录 / EXCERPT: Let (P,λ) be a hermitian, sesquilinear form on a free $\mathbb{Z}[\mathbb{Z}]$-module of rank $2g$, where $g\in\{1,2\}$. A form is surface hyperbolic if it is isometric to...

conjecture | open | MathDB 页面 | 原文线索

V2-027097 | MATHDB-361087

Banaji-Fraser conjecture on characterizing Hausdorff dimension by Hölder stability

摘录 / EXCERPT: Banaji-Fraser conjecture. Under these assumptions, $\dim$ is the Hausdorff dimension. This proposes that adding Hölder stability to the standard properties of a dimension character...

conjecture | open | MathDB 页面 | 原文线索

V2-027098 | MATHDB-361091

The ordered-source conjecture for positive orbits of GV-invariants

摘录 / EXCERPT: Ordered-source conjecture. Positive orbits are in one-to-one correspondence with ordered source vectors, and each positive orbit contains an ordered source vector as a vector of mi...

conjecture | open | MathDB 页面 | 原文线索

V2-027099 | MATHDB-361106

Flenner-Zaidenberg rigidity conjecture for affine Pham-Brieskorn hypersurfaces

摘录 / EXCERPT: Flenner-Zaidenberg rigidity conjecture. The affine Pham-Brieskorn hypersurface $X_{a_0,\dots,a_n}$ is rigid if and only if $\min\{a_0,\dots,a_n\}\geq 2$ and at most one element i...

conjecture | open | MathDB 页面 | 原文线索

V2-027100 | MATHDB-361108

The skein module-DT sheaf conjecture

摘录 / EXCERPT: Let M be a connected closed oriented 3-manifold with a basepoint $x\in M$, and let G be a reductive algebraic group equipped with the extra data described in the source. For...

conjecture | open | MathDB 页面 | 原文线索

V2-027101 | MATHDB-361118

Lower edge-density conjecture for normal convex mosaics in three-dimensional space

摘录 / EXCERPT: Let $\mathcal{M}$ be a normal, convex mosaic in $\mathbb R^3$ whose cells have unit volume, and let $\underline{\rho}_1(\mathcal{M})$ denote its lower edge density. Lower edge-dens...

conjecture | open | MathDB 页面 | 原文线索

V2-027102 | MATHDB-361146

Quasi-transitive bounded-crossing conjecture for coarse planarity

摘录 / EXCERPT: Bounded-crossing conjecture. Every quasi-transitive graph of bounded local crossing number and degree is quasi-isometric to a planar graph.

conjecture | open | MathDB 页面 | 原文线索

V2-027103 | MATHDB-361147

Quasi-transitive 1-planar coarse-planarity conjecture

摘录 / EXCERPT: 1-planar coarse-planarity conjecture. Every quasi-transitive 1-planar graph of bounded degree is quasi-isometric to a planar graph.

conjecture | open | MathDB 页面 | 原文线索

V2-027104 | MATHDB-361148

Bounded-distance crossings conjecture for quasi-transitive 1-planar graphs

摘录 / EXCERPT: Bounded-distance crossings conjecture. There is an integer k such that for every pair of crossing edges uv,xy in G, we have

conjecture | open | MathDB 页面 | 原文线索

V2-027105 | MATHDB-361199

Nill's facet bound conjecture for reflexive polytopes

摘录 / EXCERPT: Let ${\mathcal P}$ be a d-dimensional reflexive polytope, meaning a lattice polytope whose polar dual is also a lattice polytope after placing the origin in its interior. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027106 | MATHDB-361256

DCC conjecture for volumes of orbifold pairs on rational double points

摘录 / EXCERPT: DCC conjecture. The set of volumes of such orbifold pairs satisfies the descending chain condition, and its minimum nonzero value is

conjecture | open | MathDB 页面 | 原文线索

V2-027107 | MATHDB-361257

DCC conjecture for volumes of RDP orbifold pairs

摘录 / EXCERPT: DCC conjecture. The set of volumes of these orbifold pairs satisfies the descending chain condition, and its minimum nonzero value is

conjecture | open | MathDB 页面 | 原文线索

V2-027108 | MATHDB-361330

Detcherry's finiteness conjecture for skein modules of 3-manifolds with boundary

摘录 / EXCERPT: Let M be an oriented compact 3-manifold with possibly empty boundary $\mathrm{partial}M$. Let $\mathscr{S}_q(M)$ be its skein module, with the boundary skein algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-027109 | MATHDB-361383

Fried-Ghys almost-equivalence conjecture for transitive Anosov flows

摘录 / EXCERPT: Let ϕ_1^t and ϕ_2^t be transitive Anosov flows whose stable and unstable line bundles are orientable. Fried-Ghys conjecture. The flows ϕ_1^t and ϕ_2^t are almos...

conjecture | open | MathDB 页面 | 原文线索

V2-027110 | MATHDB-361384

Necessity of transitivity for almost equivalence after horizontal Goodman surgery

摘录 / EXCERPT: Let ϕ^t be a pseudo-Anosov flow on a closed oriented 3-manifold, let c be a positive/negative horizontal surgery curve, and let n be a positive/negative integer, respect...

conjecture | open | MathDB 页面 | 原文线索

V2-027111 | MATHDB-361385

Correspondence conjecture between horizontal Goodman surgery and veering triangulations

摘录 / EXCERPT: A veering triangulation Δ on an oriented 3-manifold N determines a transitive pseudo-Anosov flow $\phi^t(\Delta,M)$ on a closed oriented 3-manifold M, and a transiti...

conjecture | open | MathDB 页面 | 原文线索

V2-027112 | MATHDB-361392

Green function rigidity conjecture for closed hypersurfaces

摘录 / EXCERPT: Green function rigidity conjecture. Then (M,g) is a round sphere.

conjecture | open | MathDB 页面 | 原文线索

V2-027113 | MATHDB-361395

Strict cone conjecture for the L^p-spectrum of the Laplacian on forms

摘录 / EXCERPT: Let M be a complete noncompact Riemannian manifold with finite exponential rate of volume growth $\gamma<\infty$. Let Δ denote the Laplacian on k-forms, and assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-027114 | MATHDB-361418

Simon’s exact coincidence conjecture for self-similar iterated function systems

摘录 / EXCERPT: Let $\Phi = \{x\mapsto r_i x + a_i\}_{i\in {\mathcal I}}$ be a self-similar iterated function system on $\mathbb R$, with $|r_i|\in (0,1)$, and let Λ_Φ be its attractor....

conjecture | open | MathDB 页面 | 原文线索

V2-027115 | MATHDB-361425

Conjecture on intersections of support cones characterizing homothetic ellipsoids

摘录 / EXCERPT: Let $L,K\subset \mathbb{R}^{n}$ be convex bodies, with $n\geq 3$, and suppose that $L\subset \operatorname{int}K$. For $x\in\mathbb{R}^{n}\setminus K$, let $S(L,x)$ denote the boun...

open_problem | open | MathDB 页面 | 原文线索

V2-027116 | MATHDB-361435

Ghys's holomorphic rigidity conjecture for holomorphic Anosov diffeomorphisms

摘录 / EXCERPT: Let M be a compact complex manifold and let $f:M\to M$ be a holomorphic Anosov diffeomorphism. A complex infra-nilmanifold is a quotient of a complex nilmanifold by a finite grou...

conjecture | open | MathDB 页面 | 原文线索

V2-027117 | MATHDB-361436

Classification conjecture for three-dimensional holomorphic partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let f be a holomorphic partially hyperbolic diffeomorphism on a complex 3-manifold. Three-dimensional classification conjecture. Up to passing to a finite cover, f is holomor...

conjecture | open | MathDB 页面 | 原文线索

V2-027118 | MATHDB-361438

Holomorphicity conjecture for center distributions on Kähler manifolds

摘录 / EXCERPT: Let M be a Kähler manifold and let f be a holomorphic partially hyperbolic diffeomorphism of M, with center distribution E^c. Kähler center-holomorphicity conjecture. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-027119 | MATHDB-361452

Regularity conjecture for horospherical tangent cones of symmetric spaces

摘录 / EXCERPT: Consider the horospherical tangent cones at infinity of symmetric spaces, with restricted root systems whose factors determine the relevant types. Regularity conjecture. The horosp...

conjecture | open | MathDB 页面 | 原文线索

V2-027120 | MATHDB-361465

Three-dimensional Gamma-limit conjecture for membranes, tubes, and micelles

摘录 / EXCERPT: Three-dimensional Gamma-limit conjecture. For $n=3$ and $\rho\to0$, if $\zeta\in(\zeta_0,\zeta_1)$, the Gamma-limit of F'_ρ is a quadratic form in the principal curvatures for...

conjecture | open | MathDB 页面 | 原文线索

V2-027121 | MATHDB-361467

Finiteness conjecture for leading terms over a polyhedron

摘录 / EXCERPT: Finiteness conjecture. If $I\subset K\{\mathbf{X};P\}$ is an ideal, then the set $\mathrm{Terms}_P(I)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-027122 | MATHDB-361475

The four-facet conjecture for three-dimensional staircase Kunz semigroups

摘录 / EXCERPT: Let N be a numerical semigroup with associated face F_N, and suppose that N is staircase and Kunz. When $k=3$, the four-facet conjecture. F_N has at most 4 facets. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-027123 | MATHDB-361480

Heine's conjecture for volume polynomials of three lattice polygons

摘录 / EXCERPT: Let M be a 3×3 symmetric matrix with non-negative integer entries and let ... be the corresponding ternary form. Assume that $\det(M)\geq 0$ and $\det(M_I)\leq 0$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-027124 | MATHDB-361481

The discrete Shephard conjecture for volume polynomials of two lattice polytopes

摘录 / EXCERPT: Let $F(x,y)=\sum_{i=0}^d{d\choose i}a_ix^iy^{d-i}$ be a degree d form, where $a_0,\dots,a_d$ are non-negative integers satisfying ... for all $0\leq i\leq j\leq d$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027125 | MATHDB-361525

The maximal-contact cubic count for general plane curves

摘录 / EXCERPT: Maximal-contact cubic count.

conjecture | open | MathDB 页面 | 原文线索

V2-027126 | MATHDB-361531

Cohen-Macaulayness of the symplectic decomposition poset

摘录 / EXCERPT: Cohen-Macaulayness conjecture. The poset $\operatorname{\mathcal{S}}((V,\Psi),\mathbin{\perp})$ is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-027127 | MATHDB-361537

Affine linearity of Yager combination

摘录 / EXCERPT: Let Bel and Bel_i be belief functions on a binary frame, and let $(\alpha_i)_i$ be affine coefficients, so that $\sum_i\alpha_i=1$. Write $\textcircled{\scriptstyle{Y}}$ for Ya...

conjecture | open | MathDB 页面 | 原文线索

V2-027128 | MATHDB-361561

Symmetric L^p-Mahler conjecture

摘录 / EXCERPT: Let $p\in(0,\infty)$, and let $\mathcal{M}_p$ denote the L^p-Mahler volume. For a symmetric convex body $K\subseteq\mathbb{R}^n$, the symmetric L^p-Mahler conjecture. ... The con...

conjecture | open | MathDB 页面 | 原文线索

V2-027129 | MATHDB-361562

Bourgain's hyperplane conjecture

摘录 / EXCERPT: For $u\in\partial B_2^n$, define $u^\perp=\{x\in\mathbb{R}^n:\langle x,u\rangle=0\}$. For a convex body $K\subseteq\mathbb{R}^n$, let $b(K)=\int_K x\,dx/\lvert K\rvert$ be its bary...

conjecture | open | MathDB 页面 | 原文线索

V2-027130 | MATHDB-361563

The isotropic constant conjecture

摘录 / EXCERPT: For a convex body $K\subseteq\mathbb{R}^n$, let L_K denote its isotropic constant. The isotropic constant conjecture. There is a universal constant $c>0$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-027131 | MATHDB-361564

Strong isotropic constant conjecture for general convex bodies

摘录 / EXCERPT: Let $\mathcal{C}(K)=\lvert K\rvert^2/\det\operatorname{Cov}(K)=1/L_K^{2n}$ for a convex body $K\subseteq\mathbb{R}^n$, and let $\Delta_{n,0}$ be the standard simplex. The strong is...

conjecture | open | MathDB 页面 | 原文线索

V2-027132 | MATHDB-361565

Strong isotropic constant conjecture for symmetric convex bodies

摘录 / EXCERPT: Let $\mathcal{C}(K)=\lvert K\rvert^2/\det\operatorname{Cov}(K)=1/L_K^{2n}$ for a convex body $K\subseteq\mathbb{R}^n$. The strong isotropic constant conjecture for symmetric convex...

conjecture | open | MathDB 页面 | 原文线索

V2-027133 | MATHDB-361566

Conormal-bundle conjecture for holomorphic Lagrangians

摘录 / EXCERPT: Let M_{st} be the moduli space of stable bundles, let M_H be the Hitchin moduli space, and let ... be the embedding identifying $T^{\star}M_{st}$ with the cotangent-bundle locus....

conjecture | open | MathDB 页面 | 原文线索

V2-027134 | MATHDB-361569

Legendrian Willmore conjecture for positive-genus surfaces

摘录 / EXCERPT: Let $\mathbb{S}^5\subset\mathbb{C}^3$ be the unit sphere, and let $\Sigma_H\subset\mathbb{S}^5$ be the hexagonal Legendrian minimal torus. For a Legendrian surface...

conjecture | open | MathDB 页面 | 原文线索

V2-027135 | MATHDB-361642

Ranganathan-Wise differential descent conjecture for multiscale differentials

摘录 / EXCERPT: Let (C,η) be a generalised multiscale differential with tropicalisation λ. For every level i, let λ_i denote the truncation of λ to that level, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-027136 | MATHDB-361644

Kählerian almost rigidity conjecture for the first eigenvalues

摘录 / EXCERPT: Let (M,ω) be a compact Kähler manifold of complex dimension n, with Ricci curvature satisfying ... Write λ_j for the eigenvalues of the relevant Laplacian on...

conjecture | open | MathDB 页面 | 原文线索

V2-027137 | MATHDB-361663

Removal of semipositivity in the taming criterion for almost complex 4-manifolds

摘录 / EXCERPT: Let (X,J) be a closed almost complex 4-manifold. Suppose there exists a closed 2-form ϕ such that, for every $x\in X$, there is a closed connected pseudoholomorphic curv...

conjecture | open | MathDB 页面 | 原文线索

V2-027138 | MATHDB-361670

The hot spots conjecture for mixed eigenfunctions in triangles

摘录 / EXCERPT: Let T be a triangle with mixed boundary conditions: Neumann conditions on two sides and a Dirichlet condition on the remaining side. Consider the first positive non-trivial eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-027139 | MATHDB-361672

The gap and non-gap conjecture for minimal Terracini loci in projective space

摘录 / EXCERPT: For $n\ge 3$, let $\mathbb{T}(n,d;x)'$ denote the set of minimally Terracini sets of x points in $\mathbb{P}^n$ with respect to $\mathcal{O}_{\mathbb{P}^n}(d)$: these are sets...

conjecture | open | MathDB 页面 | 原文线索

V2-027140 | MATHDB-361676

Quaternionic Haiman conjecture for zero-fiber rings

摘录 / EXCERPT: Quaternionic Haiman conjecture. There is a $(g+1)^n$-dimensional quotient ring of the zero-fiber ring of W.

conjecture | open | MathDB 页面 | 原文线索

V2-027141 | MATHDB-361679

The doubled-prefix conjecture for reduced banned words

摘录 / EXCERPT: Let $1/2<\rho<1$ be a scaling factor, and let $\{\alpha^i\}_{i\in I}$ be the reduced banned words, enumerated in nondecreasing order of length, with $I\subseteq\mathbb{N}$. For eac...

conjecture | open | MathDB 页面 | 原文线索

V2-027142 | MATHDB-361690

De Smet-Dillen-Verstraelen-Vrancken normal scalar curvature conjecture

摘录 / EXCERPT: Let M^n be an immersed submanifold of a real space form with constant sectional curvature κ. The normalized scalar curvature is denoted by ρ, the normalized normal sc...

conjecture | open | MathDB 页面 | 原文线索

V2-027143 | MATHDB-361748

The existence of TGS links in closed orientable 3-manifolds

摘录 / EXCERPT: Let M be a closed orientable 3-manifold. A TGS link is a link K embedded in a 3-manifold such that $M\setminus K$ is hyperbolic and K has at least one totally geodesic sp...

conjecture | open | MathDB 页面 | 原文线索

V2-027144 | MATHDB-361756

The normalized-volume decay conjecture for matrix-preserving polynomial cones

摘录 / EXCERPT: Normalized-volume decay conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027145 | MATHDB-361761

Montgomery-Yang conjecture for smooth 4-manifolds with lens-space boundary

摘录 / EXCERPT: Let M be a simply-connected, compact, smooth 4-manifold with $H_2(M;\mathbb{Z})=\mathbb{Z}$, and suppose that its boundary is a connected sum ... of lens spaces, where...

conjecture | open | MathDB 页面 | 原文线索

V2-027146 | MATHDB-361763

Exactness of the second semidefinite lifting for the complex cut polytope

摘录 / EXCERPT: Exactness conjecture. The second semidefinite lifting is exact for CUT^4_∞, i.e.,

conjecture | open | MathDB 页面 | 原文线索

V2-027147 | MATHDB-361776

Rigidity of periodic unstable Lyapunov exponents for Anosov geodesic flows

摘录 / EXCERPT: Let M be a complete Riemannian manifold with finite volume, whose geodesic flow is Anosov. The periodic-exponent rigidity conjecture. If the unstable Lyapunov exponents are const...

conjecture | open | MathDB 页面 | 原文线索

V2-027148 | MATHDB-361803

The equivariant DT-PT$_0$ correspondence

摘录 / EXCERPT: Let X be a Calabi-Yau 4-fold with an algebraic torus T acting while preserving the Calabi-Yau volume form, and let L be a T-equivariant line bundle on X. For...

conjecture | open | MathDB 页面 | 原文线索

V2-027149 | MATHDB-361804

The toric fixed-locus reducedness conjecture for PTq moduli on Calabi-Yau 4-folds

摘录 / EXCERPT: Let X be a toric Calabi-Yau 4-fold and let $\curly P=\curly P_v^{(q)}(X)$ for $q\in\{-1,0,1\}$. Toric fixed-locus conjecture. If $\curly P^{T_X}$ is zero-dimensional, then...

conjecture | open | MathDB 页面 | 原文线索

V2-027150 | MATHDB-361814

Local smoothing conjecture for holomorphic sections and c9-trace positivity

摘录 / EXCERPT: Let E be a holomorphic vector bundle with a singular Hermitian metric h, and let $c9$ be the Kähler metric from Proposition cref{smoothing of tr-c9 semi-posi for constant secti...

open_problem | open | MathDB 页面 | 原文线索

V2-027151 | MATHDB-361815

Peternell's generic positivity conjecture for tangent bundles

摘录 / EXCERPT: Let X be a projective manifold with nef anticanonical bundle $-K_X$. For ample divisors $H_1,\ldots,H_{n-1}$, define generic $(H_1,\ldots,H_{n-1})$-semipositivity of T_X by req...

conjecture | open | MathDB 页面 | 原文线索

V2-027152 | MATHDB-361834

Gromov's optimal systolic constant conjecture for essential manifolds

摘录 / EXCERPT: Gromov's systolic conjecture. There exists an absolute constant C such that, for every essential closed Riemannian manifold M^n,

conjecture | open | MathDB 页面 | 原文线索

V2-027153 | MATHDB-361865

Hsiang's uniqueness conjecture for Type E constant mean curvature hypersurfaces

摘录 / EXCERPT: Let $p,q\geq2$, and let a global solution curve $\gamma(t)=(x(t),y(t))$ of ... where $W(t)=\sqrt{(x'(t))^2+(y'(t))^2}$, generate an $O(p)\times O(q)$-invariant constant mean curvatur...

conjecture | open | MathDB 页面 | 原文线索

V2-027154 | MATHDB-361901

Existence of equivariant indecomposable minimal surfaces for negatively curved manifolds

摘录 / EXCERPT: Existence conjecture. There exist an orthogonal representation $\rho:\Gamma\to \operatorname{End}(H)$ weakly equivalent to the regular representation λ_Γ, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-027155 | MATHDB-361924

Bounded relative-global klt complements for Fano fibrations

摘录 / EXCERPT: Bounded relative-global klt complement conjecture. For every positive integer d and positive real number ϵ, there exists a positive integer N, depending only on d an...

conjecture | open | MathDB 页面 | 原文线索

V2-027156 | MATHDB-361927

Freeness conjecture for self-projective sets

摘录 / EXCERPT: Freeness conjecture. If $b[?mG_{\mathcal{A}}b[0m$ is a free semigroup and $b[?m\Lambda_{\mathcal{A}}b[0m$ is non-empty and not a singleton, then

conjecture | open | MathDB 页面 | 原文线索

V2-027157 | MATHDB-361944

Lawson surfaces minimize area among minimal surfaces of fixed genus

摘录 / EXCERPT: Let $\mathcal{A}_\gamma$ be the minimum area among all minimal surfaces in $\mathbb{S}^3$ of genus γ, and let $\xi_{\gamma,1}$ be the Lawson surface of genus γ. Lawso...

conjecture | open | MathDB 页面 | 原文线索

V2-027158 | MATHDB-361949

Point-separation conjecture for first Steklov eigenfunction maps

摘录 / EXCERPT: Let $N^2\subset\mathbb{B}^n$ be a free boundary branched minimal immersion by first Steklov eigenfunctions, with ∂N having at least two components. Steklov point-separati...

conjecture | open | MathDB 页面 | 原文线索

V2-027159 | MATHDB-361976

Ashbaugh-Benguria eigenvalue-sum conjecture for Euclidean domains

摘录 / EXCERPT: Let Ω be a bounded domain in Euclidean space $\mathbb{R}^{n}$, and let λ_i be the i-th Dirichlet eigenvalue of the Laplacian on Ω. Let $\mathbb{B}^{n}$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-027160 | MATHDB-361977

Dirac eigenvalue-sum conjecture for genus-one surfaces in the three-sphere

摘录 / EXCERPT: Let $M^{2}\subset\mathbb{S}^{3}(1)$ be an embedded closed surface of genus one, and suppose that the curvature endomorphism satisfies $\Re|_{\slashed{E}}=0$. Let $\bar{\Gamma}_{1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027161 | MATHDB-361978

Weighted tropical Fermat-Weber polytrope conjecture

摘录 / EXCERPT: Weighted tropical Fermat-Weber polytrope conjecture. The set $\mathrm{FW}_W(\mathcal{S})$ forms a polytrope in $\mathbb{R}^{d-1}\cong\mathbb{R}^{d}/\mathbb{R}\mathbf{1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027162 | MATHDB-361984

Curvature exponent five characterization for sub-Finsler Heisenberg groups

摘录 / EXCERPT: Curvature exponent five conjecture. The metric measure space $(\mathbb{H},\mathsf d,\mathscr{L}^3)$ satisfies $\mathsf{MCP}(0,5)$ if and only if the reference norm is the ℓ^2-...

conjecture | open | MathDB 页面 | 原文线索

V2-027163 | MATHDB-361985

Curvature-dimension failure conjecture for sub-Finsler Carnot groups

摘录 / EXCERPT: Sub-Finsler Carnot group curvature-dimension conjecture. The metric measure space $(G,\mathsf d_{SF},\mathfrak m)$ does not satisfy the $\mathsf{CD}(K,N)$ condition for any...

conjecture | open | MathDB 页面 | 原文线索

V2-027164 | MATHDB-361993

The pentachoron lower-bound conjecture for simply-connected 4-manifolds

摘录 / EXCERPT: Let M be a closed, simply-connected, smooth 4-manifold, and let $\mathcal{T}(M)$ be a triangulation of M with p pentachora. Write χ for the Euler characteristic of M....

conjecture | open | MathDB 页面 | 原文线索

V2-027165 | MATHDB-362003

The amplituhedron positive geometry conjecture

摘录 / EXCERPT: Let $\mathcal{A}_{n,k,m}(Z)$ be the amplituhedron in $\operatorname{Gr}_{k,k+m}(\mathbb{C})$. The amplituhedron positive geometry conjecture. The pair ... should be a positive geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-027166 | MATHDB-362004

The positive geometry conjecture for amplituhedron tiles

摘录 / EXCERPT: Let Z_S be a tile of the amplituhedron $\mathcal{A}_{n,k,m}(Z)$, and let $\tilde{\Omega}(Z_S)$ denote its candidate canonical form, defined by pulling back the canonical form of...

conjecture | open | MathDB 页面 | 原文线索

V2-027167 | MATHDB-362005

The tiling-independence conjecture for amplituhedron canonical forms

摘录 / EXCERPT: Let $\mathcal{A}_{n,k,m}(Z)$ be an amplituhedron and let $\{Z_S\}_{S\in\mathcal{C}}$ be a tiling of it. Canonical form from tilings. The canonical form of the amplituhedron is obta...

conjecture | open | MathDB 页面 | 原文线索

V2-027168 | MATHDB-362010

Liouville integrability conjecture for odd-gon maps and vector fields

摘录 / EXCERPT: Let n be odd, let ${\mathcal T}_n$ be the symplectic space of planar n-gons, let f_t be the maps on this space, and let ξ be the vector field. Let ${\mathcal A}$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-027169 | MATHDB-362011

Invariant-center integrals conjecture for polygon maps and vector fields

摘录 / EXCERPT: Let f_t be the maps on planar polygon spaces and let ξ be the vector field. The polygons evolving under these maps and under the flow of ξ appear to move around invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-027170 | MATHDB-362012

Second-integral conjecture for the triangle map

摘录 / EXCERPT: Let ${\mathcal S}_3$ be the space of unit-area triangles, let $f_t:{\mathcal S}_3\to{\mathcal S}_3$ be the triangle map, and let ${\mathcal A}$ be its area integral. Second-integra...

conjecture | open | MathDB 页面 | 原文线索

V2-027171 | MATHDB-362018

The toroidal link conjecture for cyclic branched covers

摘录 / EXCERPT: Toroidal link conjecture. All cyclic branched covers $cSigma_{\psi}(L)$ of L are NLS, LO, and CTF.

conjecture | open | MathDB 页面 | 原文线索

V2-027172 | MATHDB-362033

Positivity conjecture for lower-order Q-curvatures of conformal metrics

摘录 / EXCERPT: Positivity conjecture for lower-order Q-curvatures. If $Q^{(2m)}_g$ is positive and has a slow decay barrier with rate $-2m<s\leq 0$ at infinity, then, for every $1\leq k\leq m-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-027173 | MATHDB-362036

Borde-Sorkin conjecture on causal continuity of Morse spacetimes

摘录 / EXCERPT: Let a Morse spacetime be a singular Lorentzian spacetime built from a Morse function, whose critical points have an index and a coindex. Borde-Sorkin conjecture. A Morse spacetime...

conjecture | open | MathDB 页面 | 原文线索

V2-027174 | MATHDB-362067

Arithmetic-dynamic simplicity dichotomy for geodesics

摘录 / EXCERPT: Consider a lattice $F\subset \operatorname{PSL}_2(\mathbb{R})$ with invariant trace field $\K=\mathbb{Q}(\{\operatorname{disc}(\gamma)\mid \gamma\in F\})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027175 | MATHDB-362069

The balanced Gauduchon conjecture for Ricci-flat metrics

摘录 / EXCERPT: Let X be a compact complex manifold of dimension n with holomorphic volume form Ω. Let ω be a Hermitian metric satisfying ... and denote its balanced class by...

conjecture | open | MathDB 页面 | 原文线索

V2-027176 | MATHDB-362096

Demailly's conjecture on initial degrees of symbolic powers

摘录 / EXCERPT: Demailly's conjecture. For every integer $s\geqslant 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-027177 | MATHDB-362103

Non-degeneracy conjecture for adjoints of generic polygonal Wachspress models

摘录 / EXCERPT: Adjoint non-degeneracy conjecture. For a generic n-gon $Q=B^T\cdot P_b$, the adjoint α_Q is non-degenerate, with Newton polytope equal to the Newton polytope of

conjecture | open | MathDB 页面 | 原文线索

V2-027178 | MATHDB-362120

Conicality of Minkowski-like space-time manifolds

摘录 / EXCERPT: Minkowski-like conicality conjecture. The manifold M is conical.

conjecture | open | MathDB 页面 | 原文线索

V2-027179 | MATHDB-362122

Negative curvature from nonpositive factors in Cartesian graph products

摘录 / EXCERPT: Let $G^{(1)}$ and $G^{(2)}$ be graphs. For vertices indexed by i and j, let ${\boldsymbol p}_i^{(1)}$ and ${\boldsymbol p}_j^{(2)}$ denote their node resistance curvatures, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027180 | MATHDB-362124

Birkar-Shokurov's conjecture on lc divisors over Fano fibrations

摘录 / EXCERPT: Let d be a natural number and let ϵ be a positive real number. Assume (X,B) is an ϵ-lc pair of dimension d, $f:X\to Z$ is a contraction with $\dim Z>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027181 | MATHDB-362127

Chern's gap conjecture for minimal submanifolds of spheres

摘录 / EXCERPT: Chern's conjecture. The value of $|B|^2$ must lie in a discrete subset of $\mathbb{R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027182 | MATHDB-362133

Perdomo's conjecture on the total curvature of minimal hypersurfaces

摘录 / EXCERPT: Perdomo's conjecture. One has $\sigma\geq n$. Moreover, $\sigma=n$ if and only if $\left\lvert A\right\rvert^2\equiv n$ and M is a Clifford torus.

conjecture | open | MathDB 页面 | 原文线索

V2-027183 | MATHDB-362173

Cao's uniqueness conjecture for positively curved steady Kähler-Ricci solitons

摘录 / EXCERPT: A steady gradient Kähler-Ricci soliton is a complete Kähler manifold with a metric satisfying the steady gradient Ricci soliton equation. Cao's conjecture. Every complete steady gr...

conjecture | open | MathDB 页面 | 原文线索

V2-027184 | MATHDB-362178

Ihringer's conjecture on largest s-EM-families of k-spaces

摘录 / EXCERPT: Let an s-EM-family of k-spaces be a family of k-dimensional subspaces of $\mathbb{F}_q^n$ with matching number at most s, and let an intersecting family be one whose member...

conjecture | open | MathDB 页面 | 原文线索

V2-027185 | MATHDB-362180

q-analogue of the Erdős-Chvátal simplex conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q^n$ be the n-dimensional vector space over the finite field with q elements, and let a d-simplex of k-dimensional subspaces mean the corresponding configur...

open_problem | open | MathDB 页面 | 原文线索

V2-027186 | MATHDB-362181

q-analogue of the Erdős Matching Conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q^n$ be the n-dimensional vector space over the finite field with q elements, and let $\nu_q(\mathcal{V})$ denote the largest size of a matching of pairwise dis...

open_problem | open | MathDB 页面 | 原文线索

V2-027187 | MATHDB-362194

The hyperkähler SYZ conjecture for Lagrangian fibrations

摘录 / EXCERPT: Hyperkähler SYZ conjecture. 1. X has a Lagrangian fibration if and only if it admits a non-trivial isotropic nef line bundle L. 2. X admits a rational Lagrangian fibration if...

conjecture | open | MathDB 页面 | 原文线索

V2-027188 | MATHDB-362201

Conjecture on Fano planes in non-classical unitals of even order

摘录 / EXCERPT: Fano-plane conjecture. Every non-classical unital ${\mathcal U}$ in ${\operatorname{PG}}(2,q^2)$ contains a Fano plane if $q\geq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-027189 | MATHDB-362218

The symplectic Bogomolov-Miyaoka-Yau conjecture

摘录 / EXCERPT: Symplectic Bogomolov-Miyaoka-Yau conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027190 | MATHDB-362232

Multiplicity-one generation criterion for blow-ups of projective toric surfaces

摘录 / EXCERPT: Let $\mathbb P=\mathbb P_\Sigma$ be a projective toric surface with Cox ring $\mathbb C[x_1,\dots,x_n]$ and lattice ideal $I_{\mathbb P}$. Let e denote the identity of the torus...

conjecture | open | MathDB 页面 | 原文线索

V2-027191 | MATHDB-362249

Dense-domain smooth functions form an algebraifold

摘录 / EXCERPT: For a smooth manifold M, let $C^\infty_d(M)$ be the $\mathbb{R}$-algebra of densely defined smooth functions $f:(D_f\subseteq M)\to\mathbb{R}$, where each domain D_f is an embe...

conjecture | open | MathDB 页面 | 原文线索

V2-027192 | MATHDB-362250

Almost-invertible localization of smooth functions is an algebraifold

摘录 / EXCERPT: For a smooth manifold M, let ... be the set of almost invertible smooth functions. Since S contains 1, is closed under multiplication, and contains no zero divisors, define the...

conjecture | open | MathDB 页面 | 原文线索

V2-027193 | MATHDB-362252

The boundary complex conjecture for tropical unramified p-covers

摘录 / EXCERPT: Let $\mathcal{M}_{g,\mathbb{Z}/p}$ be the moduli space of smooth genus-g curves equipped with an unramified $\mathbb{Z}/p$-cover, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027194 | MATHDB-362256

Existence conjecture for asymptotically Euclidean isometric immersions

摘录 / EXCERPT: Asymptotically Euclidean immersion conjecture. There exists an asymptotically Euclidean isometric immersion

conjecture | open | MathDB 页面 | 原文线索

V2-027195 | MATHDB-362257

Geometric inequality conjecture for asymptotically Euclidean immersions

摘录 / EXCERPT: Geometric inequality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027196 | MATHDB-362261

Ball and positive-geometry conjecture for the $m=2$ amplituhedron

摘录 / EXCERPT: For ${\mathcal{N}} \in Q_{n,2}$, let $A_{\mathcal{N}}:=A_{n,k,2}\cap {\mathring{\Pi}}_{\mathcal{N}}$, where $A_{n,k,2}$ is the $m=2$ amplituhedron and...

conjecture | open | MathDB 页面 | 原文线索

V2-027197 | MATHDB-362262

The triple conjecture for multiple mirror models of transpolar VEX multitopes

摘录 / EXCERPT: Let L be a lattice and let $(\pD^{\star}_{X},\pD_{X})$ be a transpolar pair of VEX multitopes as in Claim. The origin-enclosing sub-simplex reductions encode transpose mirror Cal...

conjecture | open | MathDB 页面 | 原文线索

V2-027198 | MATHDB-362264

Precomplex torus-manifold conjecture for flip-folded multitopes

摘录 / EXCERPT: Let $\pD^{\star}_X$ be a flip-folded multitope and let $\pD_X\subseteq\pD^{\star}_{\widetilde X}$ be its transpolar counterpart. Precomplex torus-manifold conjecture. Flip-folded m...

conjecture | open | MathDB 页面 | 原文线索

V2-027199 | MATHDB-362270

Levinson-Ullery conjecture on Cayley-Bacharach sets

摘录 / EXCERPT: Levinson-Ullery conjecture. For $d\geq 1$, if

conjecture | open | MathDB 页面 | 原文线索

V2-027200 | MATHDB-362271

A weighted quermassintegral inequality involving weighted quermassintegrals for strictly convex domains in the sphere

摘录 / EXCERPT: Let Ω be a strictly convex domain with smooth boundary M in $\mathbb{S}^{n+1}$. For integers k and l satisfying $0\leqslant k\leqslant n$ and $0\leqslant l\leqslant k$...

conjecture | open | MathDB 页面 | 原文线索

V2-027201 | MATHDB-362323

The inverse-distance Lipschitz conjecture for sub-Finsler Carnot groups

摘录 / EXCERPT: Let $\mathbb G$ be a sub-Finsler Carnot group of step 2, with Carnot-Carathéodory distance d_{cc}, abnormal set Abn, and a fixed Riemannian metric...

conjecture | open | MathDB 页面 | 原文线索

V2-027202 | MATHDB-362324

Euclidean rectifiability of metric spheres in sub-Finsler Carnot groups

摘录 / EXCERPT: Let $\mathbb G$ be a sub-Finsler Carnot group, and let its metric spheres be the level sets of the Carnot-Carathéodory distance associated with a sub-Finsler structure. Euclidean r...

conjecture | open | MathDB 页面 | 原文线索

V2-027203 | MATHDB-362333

The reverse Faber-Krahn conjecture for the magnetic Neumann Laplacian

摘录 / EXCERPT: Let $\beta>0$, and let $\Omega\subset\mathbb R^2$ be an open, bounded and simply connected set with a C^∞ smooth boundary. Let Ω^* be the disk in $\mathbb R^2$ center...

conjecture | open | MathDB 页面 | 原文线索

V2-027204 | MATHDB-362346

Full Relative Aspherical Conjecture for positive scalar curvature

摘录 / EXCERPT: Full Relative Aspherical Conjecture. If $n\ne 4$, Y is aspherical relative to X, and X admits no metric of positive scalar curvature, then Y admits no metric of positive sc...

conjecture | open | MathDB 页面 | 原文线索

V2-027205 | MATHDB-362356

Shapiro's weighted tropical bound for real zeros

摘录 / EXCERPT: Let $P(x)=\sum_{k=0}^n a_kx^k$ be a polynomial with positive coefficients. Define its weighted tropical polynomial by ... A corner is a point where this piecewise-linear function is...

conjecture | open | MathDB 页面 | 原文线索

V2-027206 | MATHDB-362415

Macbeath point inclusion conjecture

摘录 / EXCERPT: Macbeath point conjecture. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-027207 | MATHDB-362416

Conjectured volume formula for parabolic-taxicab balls

摘录 / EXCERPT: Parabolic-taxicab ball volume conjecture. The measure of this ball is

conjecture | open | MathDB 页面 | 原文线索

V2-027208 | MATHDB-362423

Idempotent completeness conjecture for the wrapped Fukaya category of a toric boundary divisor

摘录 / EXCERPT: Let $\alpha\notin V(A)$ and let $D_\alpha=f_\alpha^{-1}(0)$ be the corresponding Liouville manifold, with wrapped Fukaya category ${\cal W}{\cal F}(D_\alpha)$ and derived category...

conjecture | open | MathDB 页面 | 原文线索

V2-027209 | MATHDB-362424

Lazarev map conjecture for toric boundary divisors

摘录 / EXCERPT: Let $D_\alpha=f_\alpha^{-1}(0)$ be the toric boundary divisor under consideration. Let L be the Lazarev map, and let $\operatorname{L}_{\mathbb C}$ denote its comp...

conjecture | open | MathDB 页面 | 原文线索

V2-027210 | MATHDB-362456

Bruno's non-crossing chord diagram conjecture for phi^4 regions

摘录 / EXCERPT: Let n be even, let $x=(x_0,cldots,x_{n-3})cincmathbb{R}^{n-2}$, and define ... A non-crossing chord diagram is a perfect matching of $n-2$ ordered points, with chords $ctheta_{ab}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027211 | MATHDB-362460

The tropical generation conjecture for Lagrangian cobordism of bielliptic surfaces

摘录 / EXCERPT: Let $\mathcal K$ denote the symplectic bielliptic surface, and let $\operatorname{Cob}^{trop}(\mathcal K)$ be the subgroup of the Lagrangian cobordism group...

conjecture | open | MathDB 页面 | 原文线索

V2-027212 | MATHDB-362469

Symmetry of the counting function in a unitary polar-space construction

摘录 / EXCERPT: Counting-function symmetry conjecture. For every $\kappa\in\mathbb{F}_q^*$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-027213 | MATHDB-362502

Krahn-Szegő conjecture for the second Dirac eigenvalue

摘录 / EXCERPT: Krahn-Szegő conjecture. The minimiser of $\lambda_2(\Omega)$ is the union of two disjoint disks, each having area $\frac12$.

conjecture | open | MathDB 页面 | 原文线索

V2-027214 | MATHDB-362503

Non-disk minimiser conjecture for the third Dirac eigenvalue

摘录 / EXCERPT: Non-disk minimiser conjecture. There exist masses $m\geq0$ such that the minimiser of $\lambda_3(\Omega)$ is not the disk.

conjecture | open | MathDB 页面 | 原文线索

V2-027215 | MATHDB-362504

Ashbaugh-Benguria-type ratio conjecture for Dirac eigenvalues

摘录 / EXCERPT: Ashbaugh-Benguria-type conjecture. The disk is a solution of the maximisation problem

conjecture | open | MathDB 页面 | 原文线索

V2-027216 | MATHDB-362510

Almaraz-Barbosa-de Lima positive mass conjecture with non-compact boundary

摘录 / EXCERPT: Let (M,g) be an asymptotically flat Riemannian manifold with non-compact boundary, scalar curvature R and boundary mean curvature H. Almaraz-Barbosa-de Lima conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-027217 | MATHDB-362511

Min-Oo's conjecture for manifolds with boundary

摘录 / EXCERPT: Let (M,g) be an n-dimensional compact Riemannian manifold with boundary, scalar curvature R, and boundary ∂M. Assume that the boundary is isometric to the standard...

conjecture | open | MathDB 页面 | 原文线索

V2-027218 | MATHDB-362515

Boltyanski's illumination conjecture

摘录 / EXCERPT: Let C be a convex body in $\mathbb R^n$, let ∂C denote its boundary, and let $\operatorname{int}(C)$ denote its interior. Boltyanski's illumination conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-027219 | MATHDB-362517

Cone decomposition conjecture with Euclidean factors

摘录 / EXCERPT: Let Z be the conical manifold and let $\targetcat$ be the target category appearing in the cone decomposition theorem. For each integer $d\geq 0$, consider the restriction functo...

conjecture | open | MathDB 页面 | 原文线索

V2-027220 | MATHDB-362521

Tropical Riemann-Roch conjecture for divisors in moderate position

摘录 / EXCERPT: Moderate-position tropical Riemann-Roch conjecture. If D is in moderate position on X, then

conjecture | open | MathDB 页面 | 原文线索

V2-027221 | MATHDB-362523

Adjunction conjecture for Chern classes of tropical manifolds

摘录 / EXCERPT: Tropical Chern-class adjunction conjecture. For every $k\in\mathbb{Z}_{\geq 0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-027222 | MATHDB-362540

The boundary-value conjecture for LCS structures and Green functions

摘录 / EXCERPT: Let S be a compact complex surface in class VII_0, and let $\mathcal T(S)=]a,b[$, with $-\infty\leq a$ and $b<0$, be the moduli space of locally conformally symplectic structur...

conjecture | open | MathDB 页面 | 原文线索

V2-027223 | MATHDB-362544

The uniform Dirichlet-to-Neumann bound for Riemannian manifolds with boundary

摘录 / EXCERPT: Let M be a Riemannian manifold with a boundary that need not be compact. The Dirichlet-to-Neumann operator and the associated mass-dependent quantities are as in Assumption ... hol...

conjecture | open | MathDB 页面 | 原文线索

V2-027224 | MATHDB-362550

Nontriviality of the spin^c index difference homomorphism

摘录 / EXCERPT: Let (M,L) be a closed spin^c non-spin manifold with $\dim M\geq 5$ and with $\mathcal R^{{\mathrm{gen}+}}(M,L)\neq\emptyset$. Fix a basepoint...

conjecture | open | MathDB 页面 | 原文线索

V2-027225 | MATHDB-362555

Metricity conjecture for the geometric mean distance on a slit Euclidean space

摘录 / EXCERPT: Let n be the ambient dimension, let $u,v\in\mathbb{R}^n$ be distinct points, and let $[u,v]$ denote the line segment joining them. For a parameter c, write...

conjecture | open | MathDB 页面 | 原文线索

V2-027226 | MATHDB-362566

Futer-Wise cubulation approximation conjecture for hyperbolic manifolds

摘录 / EXCERPT: Let $M= obreak\backslash \mathbb{H}^n$ be a closed hyperbolic n-manifold with fundamental group Γ, where either $n\leq 3$ or M is arithmetic of simplest type. A...

conjecture | open | MathDB 页面 | 原文线索

V2-027227 | MATHDB-362595

Song-Zhu conjecture on integral closure of lattice polytopes

摘录 / EXCERPT: Let M be a lattice and let $P\subset M_{\mathbb{R}}$ be a lattice polytope of dimension n. Define the lattice length $l(P)$ to be the minimum, over all edges of P, of the num...

conjecture | open | MathDB 页面 | 原文线索

V2-027228 | MATHDB-362674

Dimension-dependent boundedness conjecture for torus exceptional degree

摘录 / EXCERPT: Let $\mathcal{T}(X,B)$ denote the torus exceptional degree of a log Calabi--Yau pair (X,B). Torus exceptional degree boundedness conjecture. For every positive integer n, there...

conjecture | open | MathDB 页面 | 原文线索

V2-027229 | MATHDB-362676

Formal principle with convergence for general bracket-generating families of rational curves

摘录 / EXCERPT: Let X be a complex manifold and let ${\mathcal K}$ be a connected family of smooth rational curves on X with semipositive normal bundles. The family is bracket-generating when...

conjecture | open | MathDB 页面 | 原文线索

V2-027230 | MATHDB-362721

The cotangent-bundle Hofer-Zehnder capacity conjecture for non-aspherical manifolds

摘录 / EXCERPT: Let Q be a closed oriented manifold that is not a $K(\pi,1)$. The cotangent-bundle conjecture. Then ... and the almost existence property for closed contractible characteristics ho...

conjecture | open | MathDB 页面 | 原文线索

V2-027231 | MATHDB-362726

Ricci-flat tangent cone rigidity for complete Kähler manifolds

摘录 / EXCERPT: Ricci-flat tangent cone conjecture. If one tangent cone at infinity is Ricci flat, then M is Ricci flat. Consequently, the tangent cone at infinity should be unique; analogous qu...

conjecture | open | MathDB 页面 | 原文线索

V2-027232 | MATHDB-362727

Vanishing averaged scalar-curvature conjecture for complete Kähler manifolds

摘录 / EXCERPT: Averaged scalar-curvature conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027233 | MATHDB-362747

Conjecture on convex level sets in Alexandrov spaces

摘录 / EXCERPT: Conjecture on convex level sets in Alexandrov spaces. The level set $\{f=c\}$ is an Alexandrov space of curvature at least κ.

conjecture | open | MathDB 页面 | 原文线索

V2-027234 | MATHDB-362793

The BDPP conjecture for compact Kähler manifolds

摘录 / EXCERPT: BDPP conjecture. The canonical class K_X is pseudoeffective if and only if X is not uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-027235 | MATHDB-362794

The generalized Kähler canonical bundle formula conjecture

摘录 / EXCERPT: Generalized Kähler canonical bundle formula conjecture. If $f:(X,B+\boldsymbol\beta)\to Z$ is a generalized klt (respectively, lc) pair as above, then $(Z,B_Z+\boldsymbol\beta^Z)$...

conjecture | open | MathDB 页面 | 原文线索

V2-027236 | MATHDB-362808

Shiffman-Zelditch's finiteness conjecture for Calabi volume of Bergman spaces

摘录 / EXCERPT: Let M be a projective manifold, let L be a very ample line bundle on M, and for each $k\in\mathbb{N}$ let $\mathcal{B}_{M,L^k}$ be the corresponding Bergman space with Calabi...

conjecture | open | MathDB 页面 | 原文线索

V2-027237 | MATHDB-362809

Unweighted normal distribution conjecture for geodesic periods

摘录 / EXCERPT: Let f be a weight k Maaß form for a general Fuchsian group Γ, let $\operatorname{Conj}(\Gamma)$ denote the conjugacy classes of Γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027238 | MATHDB-362818

Meeks-Pérez-Ros mean-curvature bound conjecture for CMC foliations

摘录 / EXCERPT: Meeks-Pérez-Ros conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-027239 | MATHDB-362819

Ricci lower-bound conjecture for mean curvature of CMC foliations

摘录 / EXCERPT: Ricci lower-bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-027240 | MATHDB-362824

The rigid del Pezzo surface unicuspidal curve-almost toric fibration conjecture

摘录 / EXCERPT: Let M be a rigid del Pezzo surface. An index zero (p,q)-unicuspidal rational symplectic curve in M is as in the source, and an almost toric fibration is a fibration...

conjecture | open | MathDB 页面 | 原文线索

V2-027241 | MATHDB-362825

The optimality conjecture for the folding symplectic embeddings

摘录 / EXCERPT: Let a be the ellipsoid parameter, and let a_{acc} be the accumulation point of the corresponding staircase. Consider the symplectic embeddings constructed in Pro...

conjecture | open | MathDB 页面 | 原文线索

V2-027242 | MATHDB-362826

Nonvanishing conjecture for sesquicuspidal curves in the complex projective plane

摘录 / EXCERPT: Let d be an integer and let $p/q>1$ be a reduced fraction. The count $\mathbf{T}_{d}^{p/q}$ is the index-zero finite-energy holomorphic-plane count associated with degree d cur...

conjecture | open | MathDB 页面 | 原文线索

V2-027243 | MATHDB-362827

Monotonicity conjecture for the ellipsoidal superpotential count

摘录 / EXCERPT: For each degree d, let $\mathbf{T}_{d}^{a}$ denote the count associated with the ellipsoidal superpotential for $\CP^2$ with parameter $a>1$. Monotonicity conjecture. The count...

conjecture | open | MathDB 页面 | 原文线索

V2-027244 | MATHDB-362843

Nguyen-Stancu-Wei fundamental gap conjecture for horoconvex hyperbolic domains

摘录 / EXCERPT: Let N be the dimension, let $ \mathbb H^N$ be hyperbolic N-space, and let $\Omega\subset\mathbb H^N$ be a horoconvex domain. Write $\Gamma(\Omega)$ for its fundamental gap, nam...

conjecture | open | MathDB 页面 | 原文线索

V2-027245 | MATHDB-362855

The generalized Kakeya conjecture

摘录 / EXCERPT: Generalized Kakeya conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027246 | MATHDB-362856

The theta-function mutation criterion for Q-deformation equivalence

摘录 / EXCERPT: Theta-function mutation criterion. In any bounded chamber of the scattering diagram, $\overline{\Delta}^\star$ is a Fano polytope, and X admits a $\mathbb{Q}$-Gorenstein degenera...

conjecture | open | MathDB 页面 | 原文线索

V2-027247 | MATHDB-362883

Non-degeneracy conjecture for the pulled-back Goldman symplectic form

摘录 / EXCERPT: Let $P(\Theta)$ be the space of circle patterns with a fixed Delaunay angle structure Θ, let $P(S_g)$ be the space of marked complex projective structures, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027248 | MATHDB-362885

Plane-wave conjecture for reductive Lorentzian homogeneous spaces

摘录 / EXCERPT: Let G/H be a reductive Lorentzian homogeneous space of dimension $m\ge 4$, with isotropy group H acting indecomposably but non-irreducibly. A plane wave is a Lorentzian manifol...

conjecture | open | MathDB 页面 | 原文线索

V2-027249 | MATHDB-362888

Compactification conjecture for polynomially asymptotically Calabi-Yau manifolds

摘录 / EXCERPT: Compactification conjecture. For any $\kappa_1>\lambda$ and $\kappa_2>\mu$, (X,I,ω,Ω) can be compactified complex analytically to a weak Fano manifold. Furthermore, the...

conjecture | open | MathDB 页面 | 原文线索

V2-027250 | MATHDB-362914

Nonexistence of a mapping class group-invariant earthquake metric on Teichmüller space

摘录 / EXCERPT: Nonexistence conjecture. There is no asymmetric metric on Teichmüller space which is mapping class group invariant and for which left earthquakes are geodesics.

conjecture | open | MathDB 页面 | 原文线索

V2-027251 | MATHDB-362915

Existence conjecture for connected contactifications of compact symplectic manifolds

摘录 / EXCERPT: Let (N,ω) be a compact symplectic manifold, and let $\hbar>0$. The Dirac quantization condition is ... A contactification of (N,ω) is a contact-geometric lift as define...

conjecture | open | MathDB 页面 | 原文线索

V2-027252 | MATHDB-362956

Maximal Poisson-commutative subalgebra conjecture for affine homogeneous spaces

摘录 / EXCERPT: Let G/H be an affine homogeneous space, let ${\mathfrak m}$ be the corresponding isotropy representation, and let c and r be the invariants for which...

conjecture | open | MathDB 页面 | 原文线索

V2-027253 | MATHDB-362973

Its-Prokhorov symplectic-form conjecture for isomonodromic systems

摘录 / EXCERPT: Its-Prokhorov symplectic-form conjecture. There exists a constant $\gamma\in\operatorname{\mathbb{C}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027254 | MATHDB-362974

Its-Prokhorov rational primitive conjecture for the extended isomonodromic form

摘录 / EXCERPT: Its-Prokhorov rational primitive conjecture. There exists a function $G(P_a,Q_a,t_k)$, rational in the variables $\{P_a\}$, $\{Q_a\}$ and $\{t_k\}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-027255 | MATHDB-362982

Reduction of Entov-Polterovich quasimorphisms to Donaldson divisors

摘录 / EXCERPT: Reduction conjecture. The reduction of μ_X^{EP} to Σ is also an Entov-Polterovich quasimorphism: there exists an Entov-Polterovich quasimorphism

conjecture | open | MathDB 页面 | 原文线索

V2-027256 | MATHDB-362986

The A-polynomial realization conjecture for once-cusped hyperbolic three-manifolds

摘录 / EXCERPT: Let M be a complete finite-volume once-cusped hyperbolic three-manifold. An ideal triangulation of M is a triangulation whose vertices lie at the cusp, and an ideal triangulati...

conjecture | open | MathDB 页面 | 原文线索

V2-027257 | MATHDB-362994

Discrete singularity conjecture for four-dimensional noncollapsed Ricci limit spaces

摘录 / EXCERPT: Let (X^4,d) be a noncollapsed Ricci limit space, and consider the set of points whose tangent-cone cross-sections are not homeomorphic to S^3. Discrete singularity con...

conjecture | open | MathDB 页面 | 原文线索

V2-027258 | MATHDB-362995

Yang-Zheng's conjecture on real bisectional curvature

摘录 / EXCERPT: Let M be a Hermitian manifold with a Hermitian metric g. Assume that g has positive or quasi-positive real bisectional curvature. Yang-Zheng's conjecture. Then M is project...

conjecture | open | MathDB 页面 | 原文线索

V2-027259 | MATHDB-362996

The asymptotic volume growth conjecture under positive scalar curvature

摘录 / EXCERPT: Asymptotic volume growth conjecture. There is a universal constant $C(n)>0$ such that, if $R\geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-027260 | MATHDB-363003

Song-Tian's Ricci curvature bound conjecture away from singular fibers

摘录 / EXCERPT: Let X be a compact Kähler manifold with semiample canonical bundle K_X and intermediate Kodaira dimension $0<m<\dim X$, let ω_0 be a Kähler metric on X, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027261 | MATHDB-363010

Marché's decomposition conjecture for Kauffman bracket skein modules

摘录 / EXCERPT: Let M be a closed oriented 3-manifold, and let $\mathcal S_{2,\infty}(M)$ denote its Kauffman bracket skein module over $\mathbb Z[A^{\pm 1}]$. Marché's decomposition conjectur...

conjecture | refuted | MathDB 页面 | 原文线索

V2-027262 | MATHDB-363011

Nondecomposition conjecture for skein modules of connected sums

摘录 / EXCERPT: Let M and N be nontrivial closed oriented 3-manifolds, and let $\mathcal S_{2,\infty}(M\#N)$ denote the Kauffman bracket skein module of their connected sum over...

conjecture | open | MathDB 页面 | 原文线索

V2-027263 | MATHDB-363012

Geodesic connectivity criterion for Cegrell class E

摘录 / EXCERPT: Let Ω be the domain under consideration, let $\mathcal{E}$ be the Cegrell class of negative plurisubharmonic functions on Ω, and let g_u denote the Green-Poisson re...

conjecture | open | MathDB 页面 | 原文线索

V2-027264 | MATHDB-363016

Canonical limit current conjecture for the Kähler-Ricci flow

摘录 / EXCERPT: Canonical limit current conjecture. There is a quasi-plurisubharmonic function φ_∞ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027265 | MATHDB-363031

Arithmetic Fukaya equivalence for the intersection of two quadrics

摘录 / EXCERPT: Let X be the intersection of two quadrics in $\mathbb{C}P^5$, and let Σ_2 be a genus-two surface. For a field K of characteristic not 2, write...

open_problem | open | MathDB 页面 | 原文线索

V2-027266 | MATHDB-363033

Interior angle sum conjecture for horizontal-like geodesic triangles in Sol geometry

摘录 / EXCERPT: Interior angle sum conjecture. The sum of the interior angles of any horizontal-like geodesic triangle is greater than π:

conjecture | open | MathDB 页面 | 原文线索

V2-027267 | MATHDB-363054

The exact overlap conjecture for one-dimensional iterated function systems

摘录 / EXCERPT: Exact overlap conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027268 | MATHDB-363073

Maldacena's negative-volume Schottky filling conjecture

摘录 / EXCERPT: Maldacena's conjecture. Every connected Riemann surface of genus at least 2 is the asymptotic boundary of a Schottky manifold with negative renormalized volume.

conjecture | open | MathDB 页面 | 原文线索

V2-027269 | MATHDB-363076

The conjecture on asymptotically special nef divisors on blow-ups of projective space

摘录 / EXCERPT: Conjecture on asymptotically special nef divisors. The blow-up X^n_r does not contain nef divisors which are asymptotically special.

conjecture | open | MathDB 页面 | 原文线索

V2-027270 | MATHDB-363084

Metric-line classification conjecture for Carnot groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a Carnot group with left-invariant sub-Riemannian structure. A curve $\gamma(t):\mathbb{R}\to\mathbb{G}$ is a metric line if it is a globally minimizing sub-Rie...

conjecture | open | MathDB 页面 | 原文线索

V2-027271 | MATHDB-363094

Critical-orbit bound for unstable manifolds of renormalization

摘录 / EXCERPT: Let $\mathcal{R}$ be a compact analytic renormalization operator with a hyperbolic fixed point, and let $\mathcal{W}^u_{\mathrm{loc}}$ be its local unstable manifold. Assume that e...

conjecture | open | MathDB 页面 | 原文线索

V2-027272 | MATHDB-363095

The Riemannian constant holomorphic sectional curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold of complex dimension $n\geq 2$, and let H^r denote the Riemannian holomorphic sectional curvature, with R^r the curvature of the R...

conjecture | open | MathDB 页面 | 原文线索

V2-027273 | MATHDB-363096

The constant holomorphic sectional curvature conjecture for t-Gauduchon connections

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold of complex dimension $n\geq 2$. For a real number t, let $\nabla^{(t)}=(1-t)\nabla+t\nabla^b$ be the t-Gauduchon connection, let...

conjecture | open | MathDB 页面 | 原文线索

V2-027274 | MATHDB-363103

Systolic lower-bound conjecture for Pachner-graph growth

摘录 / EXCERPT: Let M be a cusped hyperbolic 3-manifold, and let its systole be the length of its shortest geodesic. Consider the growth rate of the Pachner graph of M, whose vertices are tr...

conjecture | open | MathDB 页面 | 原文线索

V2-027275 | MATHDB-363110

Five-part orbit conjecture for symplectic finite models

摘录 / EXCERPT: Let $e\in\mathfrak{sp}_{2n}$ be a nilpotent element whose partition has five parts, and let S_e be the set of centrally extended orbits generated by the good multiplication rules...

conjecture | open | MathDB 页面 | 原文线索

V2-027276 | MATHDB-363123

Perdomo's index conjecture for minimal hypersurfaces in spheres

摘录 / EXCERPT: Let $x:M^n\rightarrow\mathbb{S}^{n+1}$ be a full, closed, minimal hypersurface in $\mathbb{S}^{n+1}$, with $n\geq 3$. Here $\operatorname{Ind}(M)$ denotes the Morse index of M. P...

conjecture | open | MathDB 页面 | 原文线索

V2-027277 | MATHDB-363141

Nonpositive-curvature metric conjecture for relative Artin complexes

摘录 / EXCERPT: Let A_S be an Artin group and let $S'\subset S$ be an irreducible almost spherical subset. Equip the relative Artin complex $\Delta_{S,S'}$ with the metric described in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-027278 | MATHDB-363146

Mirzakhani's twist torus conjecture

摘录 / EXCERPT: Mirzakhani's twist torus conjecture. For any pants decomposition γ,

conjecture | open | MathDB 页面 | 原文线索

V2-027279 | MATHDB-363147

The Kähler positive scalar curvature-uniruledness conjecture

摘录 / EXCERPT: Let (M,ω) be a compact Kähler manifold of arbitrary dimension. A compact complex manifold M is uniruled if every point of M lies on a rational curve, meaning a smooth co...

conjecture | open | MathDB 页面 | 原文线索

V2-027280 | MATHDB-363156

Euclidean dimension bounded by VC-dimension for ideals of convex geometries

摘录 / EXCERPT: Let $\mathcal{I}=(U,\mathcal{C'})$ be an ideal of a convex geometry, with Euclidean dimension $\operatorname{dim_{\mathbb{E}}}(\mathcal{I})$ and VC-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-027281 | MATHDB-363168

The exact value of the normalized nine-point quadrilateral area

摘录 / EXCERPT: Let $\Delta_4'(n)$ denote the minimum, over configurations of n points in the normalized setting considered in the paper, of the area of a convex quadrilateral containing four po...

conjecture | open | MathDB 页面 | 原文线索

V2-027282 | MATHDB-363202

Log Abundance conjecture for klt pairs

摘录 / EXCERPT: Let (X,Δ) be a projective klt pair of dimension n. Let $\operatorname{Nef}(X)$ be the nef cone, let $\operatorname{Sem}(X)$ be the cone of semiample divisors, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027283 | MATHDB-363203

Generalized log nonvanishing and abundance conjecture for movable divisor cones

摘录 / EXCERPT: Let (X,Δ) be a projective klt pair of dimension n, and fix an integer k with $0\leq k\leq n-1$. Let $\operatorname{Mov}^k(X)$ be the cone generated by numerical classes...

conjecture | open | MathDB 页面 | 原文线索

V2-027284 | MATHDB-363212

The directed-landscape geodesic upper-tail rate conjecture

摘录 / EXCERPT: The directed-landscape geodesic upper-tail rate conjecture. For $t\in (0,1/2]$, as $a\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-027285 | MATHDB-363228

The contact L-space conjecture for Reebless foliations

摘录 / EXCERPT: Let M be a 3-manifold distinct from $S^1\times S^2$. The contact L-space conjecture. The manifold M carries a Reebless $\mathcal{C}^0$-foliation if and only if it carries a...

conjecture | open | MathDB 页面 | 原文线索

V2-027286 | MATHDB-363231

The tight positive contact-pair approximation conjecture

摘录 / EXCERPT: Let (ξ_-,ξ_+) be a tight positive contact pair on a 3-manifold M, with unstable plane field η_u. The tight positive contact-pair approximation conjecture. The plane...

conjecture | open | MathDB 页面 | 原文线索

V2-027287 | MATHDB-363249

Khovanskii's format-bound conjecture for Noetherian functions

摘录 / EXCERPT: Let ${\mathcal P}$ be the domain of a Noetherian chain and let $P_1,\ldots,P_n:{\mathcal P}\to\mathbb C$ be Noetherian functions of format ${\mathscr F}$. Khovanskii's format-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-027288 | MATHDB-363260

Binary geometry existence conjecture for polygon face complexes

摘录 / EXCERPT: Let Δ be the face complex of an n-gon. A binary geometry for Δ is a binary geometry whose underlying flag complex is Δ. Binary geometry existence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027289 | MATHDB-363273

Anderson's generic regularity conjecture for the boundary map of Einstein 3-manifolds

摘录 / EXCERPT: Anderson's conjecture. For each fixed Λ, the boundary map Π is regular at generic metrics, meaning that its linearization is surjective with split kernel at those metri...

conjecture | open | MathDB 页面 | 原文线索

V2-027290 | MATHDB-363280

Arthamonov's conjecture on modified double Poisson brackets

摘录 / EXCERPT: Let $\mathbb{K}$ be the ground field and let $A=\mathbb{K}\langle x_1,x_2,x_3\rangle$ be the free associative algebra on generators x_1,x_2,x_3. Define two operations on pairs of...

conjecture | open | MathDB 页面 | 原文线索

V2-027291 | MATHDB-363281

Geometricity conjecture for morphisms of 1-dimensional tropical fans

摘录 / EXCERPT: Geometricity conjecture. Every morphism $\nu:\varphi_Y\to\varphi_X$ is geometric.

conjecture | open | MathDB 页面 | 原文线索

V2-027292 | MATHDB-363282

Geometricity conjecture for homomorphisms from Laurent-generated semirings

摘录 / EXCERPT: Geometricity conjecture. Any semiring homomorphism $\nu:R\to\mathbb{Z}^B_{\mathrm{pos}}\cup\{-\infty\}$ is geometric.

conjecture | open | MathDB 页面 | 原文线索

V2-027293 | MATHDB-363301

Arnold--Thom gradient conjecture

摘录 / EXCERPT: Let $gamma(t)$ be a gradient flow line for an analytic function, and suppose that it has a limit point. Arnold--Thom gradient conjecture. The limit ... exists. This is a stronger, fi...

conjecture | open | MathDB 页面 | 原文线索

V2-027294 | MATHDB-363305

The uniform spectral gap conjecture for the Hodge Laplacian on one forms

摘录 / EXCERPT: Let M be a closed Riemannian n-manifold with $ u^{H,1}$ denoting the first positive eigenvalue of the Hodge Laplacian on one forms. Assume that $\operatorname{Ric}\geq -1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027295 | MATHDB-363315

No-interior-extrema conjecture for first mixed eigenfunctions

摘录 / EXCERPT: Let Ω be a simply connected Lipschitz domain, and let D be a line segment in the boundary. Consider the first mixed Dirichlet-Neumann eigenvalue problem, with Dirichlet co...

conjecture | open | MathDB 页面 | 原文线索

V2-027296 | MATHDB-363354

Rigidity of compactly supported perturbations of rank-one symmetric spaces

摘录 / EXCERPT: Rigidity conjecture. Any Riemannian metric g which coincides with g_0 outside of D is isometric to g_0.

conjecture | open | MathDB 页面 | 原文线索

V2-027297 | MATHDB-363355

The stable-manifold conjecture for blowup in the quadratic nonlinear Schrödinger equation

摘录 / EXCERPT: Stable-manifold conjecture. Such initial data are globally well-posed under the nonlinear Schrödinger dynamics if and only if $u_0\notin\mathcal{W}^s(0)$.

conjecture | open | MathDB 页面 | 原文线索

V2-027298 | MATHDB-363372

Pólya functional bounds for bounded convex planar domains

摘录 / EXCERPT: Pólya functional bounds conjecture. For every bounded convex planar domain D,

conjecture | open | MathDB 页面 | 原文线索

V2-027299 | MATHDB-363423

Esnault's finite-fundamental-group conjecture for Kähler manifolds

摘录 / EXCERPT: Esnault's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027300 | MATHDB-363424

Infinite fundamental group under weak semiampleness and intermediate Kodaira dimension

摘录 / EXCERPT: Infinite fundamental-group conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027301 | MATHDB-363425

Wu-Zheng's structure conjecture for Kähler manifolds with nef cotangent bundle

摘录 / EXCERPT: Wu-Zheng's conjecture. There exists a finite étale cover X' of X admitting a smooth fibration in complex tori

conjecture | open | MathDB 页面 | 原文线索

V2-027302 | MATHDB-363431

Saint-Venant's conjecture on fail points

摘录 / EXCERPT: Saint-Venant's conjecture. Fail points occur at the contact points of the largest inscribed circle.

conjecture | refuted | MathDB 页面 | 原文线索

V2-027303 | MATHDB-363432

Ramaswamy's conjecture on fail points

摘录 / EXCERPT: Ramaswamy's conjecture. The fail points occur either at the contact points of the largest inscribed circle or at points of minimal curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-027304 | MATHDB-363487

The hyperbolic-lattice Schwarzian-limit conjecture

摘录 / EXCERPT: Let L be the curvature length scale and ℓ_0 the lattice edge length for regular or semi-regular hyperbolic tessellations, and let n be the polygon length. Hyperbolic-latti...

conjecture | open | MathDB 页面 | 原文线索

V2-027305 | MATHDB-363488

The analytically continued hyperbolic matrix-model conjecture

摘录 / EXCERPT: Hyperbolic matrix-model conjecture. The disk partition function admits an analytic continuation in q near $q=4$ such that, with

conjecture | open | MathDB 页面 | 原文线索

V2-027306 | MATHDB-363502

Levi negativity conjecture for the L^1-norm on quadratic differentials

摘录 / EXCERPT: Let $q_0\in\mathcal{Q}_g$ be generic, meaning that q_0 lies in the principal stratum. Let $T^H_{q_0}\mathcal{Q}_g$ denote a horizontal lift determined by a guiding frame, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027307 | MATHDB-363503

Plurisubharmonicity conjecture for the Teichmüller metric

摘录 / EXCERPT: Let $\mathcal{T}_g$ be Teichmüller space and let $T\mathcal{T}_g$ be its tangent bundle, equipped with the complex structure induced from the holomorphic tangent bundle. The Teichm...

conjecture | open | MathDB 页面 | 原文线索

V2-027308 | MATHDB-363548

Steinhaus's conjecture on the isoperimetric ratio of Euclidean tilings

摘录 / EXCERPT: Steinhaus's conjecture. The maximum isoperimetric ratio among the cells of $\mathcal{T}$ is minimal when $\mathcal{T}$ is a regular hexagonal tiling. The source recalls this as Pro...

conjecture | open | MathDB 页面 | 原文线索

V2-027309 | MATHDB-363559

Gudmundsson's local and global harmonic morphism conjecture for irreducible symmetric spaces

摘录 / EXCERPT: Gudmundsson's conjecture. For each point $p\in M$, there exists a complex-valued harmonic morphism

conjecture | open | MathDB 页面 | 原文线索

V2-027310 | MATHDB-363576

Singer-Hopf conjecture for aspherical closed manifolds

摘录 / EXCERPT: Singer-Hopf conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027311 | MATHDB-363578

The balanced Chern-Ricci-flat metric conjecture

摘录 / EXCERPT: Let M be a compact complex manifold with vanishing Bott-Chern first Chern class $c_1^{BC}=0$ and a balanced metric ω_0. A Hermitian metric is balanced if...

conjecture | open | MathDB 页面 | 原文线索

V2-027312 | MATHDB-363579

The form-type Calabi-Yau equation solvability conjecture

摘录 / EXCERPT: Form-type Calabi-Yau equation solvability conjecture. There exist a constant b and a function φ satisfying these equations. The equation is introduced as the form-type Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-027313 | MATHDB-363581

Di Scala's conjecture on minimal immersions into nonpositively curved space forms

摘录 / EXCERPT: Di Scala's conjecture. The immersion f must be totally geodesic.

conjecture | open | MathDB 页面 | 原文线索

V2-027314 | MATHDB-363629

Berkovich correspondence conjecture for deeper bubbles

摘录 / EXCERPT: Let $\pi\colon\mathcal{X}\to\Delta^{\rm conv}(\mathbb{C})$ be an affine finite-type log-terminal faithfully flat morphism, where...

conjecture | open | MathDB 页面 | 原文线索

V2-027315 | MATHDB-363651

Barja-González-Naranjo modified Xiao conjecture for the relative irregularity

摘录 / EXCERPT: Barja-González-Naranjo's modified conjecture. For every non-isotrivial fibred surface $f\colon S\to B$ of genus $g\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-027316 | MATHDB-363658

Local L^{∞} Bernstein conjecture for Laplace eigenfunctions

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold of dimension d, and let $\varphi_{\lambda}\in C^{\infty}(M)$ satisfy ... For sufficiently small $r>0$, write $B_g(x,r)$ for the geodesi...

conjecture | open | MathDB 页面 | 原文线索

V2-027317 | MATHDB-363668

Positive-relatedness conjecture for Riemannian g.o. metrics

摘录 / EXCERPT: Let $g_1,\dots,g_k$ be a family of positively related Riemannian metrics on a homogeneous space. A family is positively related when its metrics admit a common decomposition ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-027318 | MATHDB-363685

Przytycki's torsion conjecture for Kauffman bracket skein modules

摘录 / EXCERPT: Przytycki's torsion conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-027319 | MATHDB-363695

Extension of norms from subsets of integer vectors

摘录 / EXCERPT: Norm-extension conjecture. Then $\lVert\cdot\rVert$ is the restriction to G of a norm on $\mathbb Z^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-027320 | MATHDB-363697

The volume growth conjecture for Ricci-nonnegative manifolds

摘录 / EXCERPT: Let (M^n,g) be a complete non-compact Riemannian manifold, let $B_R(p)$ denote the geodesic ball of radius R centered at $p\in M$, and define ... Write Ric and...

conjecture | open | MathDB 页面 | 原文线索

V2-027321 | MATHDB-363714

Convergence of random graph Laplacian eigenmaps on an interval

摘录 / EXCERPT: Let ϕ_k denote the k^th eigenfunction of the Laplacian on $[-1,1]$, and let $\phi_{\epsilon,n,k}$ denote the k^th eigenfunction of a graph Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-027322 | MATHDB-363715

Stability of eigenmaps for random averaging Laplacians

摘录 / EXCERPT: Stability conjecture. Eigenmaps of the random averaging Laplacians give numerically stable locally uniform approximations to the corresponding eigenmap of $\mathcal{L}_{(g)}$. This...

conjecture | open | MathDB 页面 | 原文线索

V2-027323 | MATHDB-363716

Resistance-form discretization conjecture for eigenmaps on the real line

摘录 / EXCERPT: Resistance-form discretization conjecture. Numerically accurate locally uniform approximations to the Laplacian eigenmap can be obtained by discretizing the left- and right-hand si...

conjecture | open | MathDB 页面 | 原文线索

V2-027324 | MATHDB-363717

Cellular-semimetric averaging conjecture on the Sierpiński gasket

摘录 / EXCERPT: Let SG be the Sierpiński gasket with its standard Laplacian $\mathcal{L}$, and define the cellular semimetric by ... Let $\mathcal{L}_\epsilon$ be the averaging operator o...

conjecture | open | MathDB 页面 | 原文线索

V2-027325 | MATHDB-363719

Symmetry-invariant random rotations of Sierpiński-gasket eigenmaps

摘录 / EXCERPT: Let $\mathcal{L}$ be the Laplacian on the Sierpiński gasket, whose relevant Neumann eigenspace has multiplicity two, and consider eigenmaps arising from random graph approximations...

conjecture | open | MathDB 页面 | 原文线索

V2-027326 | MATHDB-363720

Uniform spectral convergence of Euclidean graph approximations on the Sierpiński gasket

摘录 / EXCERPT: Let SG be the Sierpiński gasket with Laplacian $\mathcal{L}$. Consider the rescaled averaging Laplacian and the rescaled graph Laplacian at random sample points, both de...

conjecture | open | MathDB 页面 | 原文线索

V2-027327 | MATHDB-363721

Weak but not strong Euclidean averaging convergence on the Sierpiński gasket

摘录 / EXCERPT: Let SG be the Sierpiński gasket with Laplacian $\mathcal{L}$, and suppose $\mathcal{L}f$ is Hölder continuous on SG. Let $\mathcal{L}_\epsilon$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027328 | MATHDB-363752

Closure-order conjecture for symplectic juggling-pattern cells

摘录 / EXCERPT: Closure-order conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027329 | MATHDB-363772

Entropy lower-bound conjecture for positive-genus self-shrinkers

摘录 / EXCERPT: Let $\Sigma\subset \mathbb R^3$ be a closed self-shrinker of positive genus, and let $\Sigma_C\subset \mathbb S^3$ be the Clifford torus. One should have ... This conjecture is motiv...

conjecture | open | MathDB 页面 | 原文线索

V2-027330 | MATHDB-363800

Optimality conjecture for trigonometric moment curve correspondences

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let $\gamma_{2k}:\mathbb{S}^1\to\mathbb{S}^{2k}$ and $\gamma_{2k+1}:\mathbb{S}^1\to\mathbb{S}^{2k+1}$ be the trigonometric moment curve embeddings defined...

conjecture | open | MathDB 页面 | 原文线索

V2-027331 | MATHDB-363801

Optimal distortion conjecture for the TMC projection

摘录 / EXCERPT: Let $\psi_{2k+1}:\mathbb{S}^{2k+1}\to\mathbb{S}^1$ and $\psi_{2k}:\mathbb{S}^{2k}\to\mathbb{S}^1$ be the projections associated with the trigonometric moment curve embeddings, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027332 | MATHDB-363802

Distortion conjecture for the variant correspondence induced by α

摘录 / EXCERPT: Let $\alpha:\mathbb{S}^1\to\mathbb{S}^2$ be the embedding ... and let R_α be its induced correspondence. Variant correspondence conjecture. ... This correspondence arose from co...

conjecture | open | MathDB 页面 | 原文线索

V2-027333 | MATHDB-363803

Distortion conjecture for the tetrahedral spherical-harmonic correspondence

摘录 / EXCERPT: In spherical coordinates on $\mathbb{S}^2$, define ... and ... where $(x(\phi,\theta),y(\phi,\theta),z(\phi,\theta))=(\cos(\phi)\sin(\theta),\sin(\phi)\sin(\theta),\cos(\theta))$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027334 | MATHDB-363804

Covering-radius conjecture for the odd trigonometric moment curve

摘录 / EXCERPT: Let $\gamma_{2k+1}:\mathbb{S}^1\to\mathbb{S}^{2k+1}$ be the odd trigonometric moment curve embedding, let $\Pi_{2k+1}:\mathbb{S}^{2k+1}\to\gamma_{2k+1}(\mathbb{S}^1)$ be the neares...

conjecture | open | MathDB 页面 | 原文线索

V2-027335 | MATHDB-363805

Monotonicity conjecture for distortion of odd trigonometric moment curves

摘录 / EXCERPT: For each positive integer k, let $\gamma_{2k+1}:\mathbb{S}^1\to\mathbb{S}^{2k+1}$ be the odd trigonometric moment curve embedding, and let $\mathrm{dis}(\gamma_{2k+1})$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-027336 | MATHDB-363811

Meeks-Pérez-Ros conjecture on mean curvature of foliations

摘录 / EXCERPT: Let $\mathcal{F}$ be a codimension one foliation of a complete Riemannian manifold $\overline{M}^{n+1}$. Assume that there is $K_0 \geq 0$ such that $\overline{M}$ has Ricci curvat...

conjecture | open | MathDB 页面 | 原文线索

V2-027337 | MATHDB-363826

Iosevich-Wyman polynomial improvement conjecture for products

摘录 / EXCERPT: Let $M:=M_1\times\ldots\times M_n$ be the product of at least two compact Riemannian manifolds without boundary, and let d be its dimension. Denote by $N_M(\lambda)$ the eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-027338 | MATHDB-363869

Gromov's rational essentialness conjecture for positive scalar curvature

摘录 / EXCERPT: Gromov's rational essentialness conjecture. A closed $\mathbb{Q}$-essential n-manifold does not admit any Riemannian metric with positive scalar curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-027339 | MATHDB-363871

Existence of complements conjecture for log canonical pairs

摘录 / EXCERPT: Existence of Complements Conjecture. There exists a positive integer n, depending only on d and Γ, such that every such pair $(X/Z\ni z,B)$ has an n-complement...

conjecture | open | MathDB 页面 | 原文线索

V2-027340 | MATHDB-363929

The metric SYZ conjecture for maximally degenerate Calabi-Yau families

摘录 / EXCERPT: Metric SYZ conjecture. For any $0<\varepsilon\ll1$, there exists $\delta>0$ such that if $|t|<\delta$, then there exists a special Lagrangian $\mathbb{T}^d$-fibration on an open su...

conjecture | solved | MathDB 页面 | 原文线索

V2-027341 | MATHDB-363930

Popovici's conjecture on limits of Kähler manifolds

摘录 / EXCERPT: Popovici's conjecture. A limit of Kähler manifolds is of Fujiki class $\mathcal C$, that is, it is bimeromorphic to a Kähler manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-027342 | MATHDB-363940

Conjecture that all Weil-Petersson geodesics on Calabi-Yau threefold moduli are stable

摘录 / EXCERPT: Let the moduli space of a Calabi-Yau threefold be equipped with its Weil-Petersson geometry, and consider its geodesics. Stability conjecture. All Weil-Petersson geodesics on the m...

conjecture | open | MathDB 页面 | 原文线索

V2-027343 | MATHDB-363943

The weak bi-anabelian m-step solvable Grothendieck conjecture in characteristic zero

摘录 / EXCERPT: Let k be a finitely generated field of characteristic zero with absolute Galois group G_k, let m be a non-negative integer, and let U_i be hyperbolic curves over k of typ...

conjecture | open | MathDB 页面 | 原文线索

V2-027344 | MATHDB-363944

The weak bi-anabelian m-step solvable Grothendieck conjecture in positive characteristic

摘录 / EXCERPT: Let k be a finitely generated field of characteristic $p>0$ with absolute Galois group G_k, let m be a non-negative integer, and let U_i be hyperbolic curves over k for...

conjecture | open | MathDB 页面 | 原文线索

V2-027345 | MATHDB-363949

Hang-Wang's width and first Neumann eigenvalue conjecture

摘录 / EXCERPT: Let $\Omega\subseteq \mathbb{R}^2$ be a bounded convex open subset with smooth boundary. Its width is ... Write $\operatorname{diam}(\Omega)$ for the diameter of Ω, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027346 | MATHDB-363964

Gerritzen-van der Put's position conjecture for split degenerate superelliptic curves

摘录 / EXCERPT: Gerritzen-van der Put's conjecture. With this setup, the $(2g+2)$-element subsets $S \subset \mathbb{P}_K^1$ and $\mathcal{B} \subset \mathbb{P}_K^1$ have the same position.

conjecture | open | MathDB 页面 | 原文线索

V2-027347 | MATHDB-363980

Condensed and bornological geometry model conjecture

摘录 / EXCERPT: Condensed and bornological geometry model conjecture. The models of geometry in condensed mathematics, such as analytic stacks, and the models of geometry based on bornological rin...

conjecture | open | MathDB 页面 | 原文线索

V2-027348 | MATHDB-363983

Conjecture on kissing numbers of three- and four-dimensional Minkowski spheres

摘录 / EXCERPT: For real $c>1$, consider the Minkowski spheres ... The kissing number is the maximum number of non-overlapping congruent spheres tangent to a given sphere. Conjecture on Minkowski-sp...

conjecture | open | MathDB 页面 | 原文线索

V2-027349 | MATHDB-363995

The flag-manifold conjecture for projectively induced Kähler-Einstein manifolds

摘录 / EXCERPT: Flag-manifold conjecture. An n-dimensional complex manifold endowed with a projectively induced Kähler-Einstein metric is an open subset of a complex flag manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-027350 | MATHDB-364000

The positive Chern character conjecture for Fano manifolds

摘录 / EXCERPT: Positive Chern character conjecture. If ${\rm ch}_k(X)$ is positive for every $k\le n$, then

conjecture | open | MathDB 页面 | 原文线索

V2-027351 | MATHDB-364001

The chain-length characterization conjecture for Fano manifolds of Picard number one

摘录 / EXCERPT: Chain-length characterization conjecture. 1. If

conjecture | open | MathDB 页面 | 原文线索

V2-027352 | MATHDB-364040

Critical-scale conjecture for Vietoris-Rips complexes of spheres

摘录 / EXCERPT: Let S^n be the n-sphere, and write $\mathrm{VR}(S^n;r)$ for its Vietoris-Rips complex at scale $r\ge 0$. Critical-scale conjecture. There are countably many critical scales ... s...

conjecture | open | MathDB 页面 | 原文线索

V2-027353 | MATHDB-364054

Conjecture on asymptotically harmonic manifolds

摘录 / EXCERPT: Asymptotically harmonic manifold conjecture. If (M,g) is asymptotically harmonic, then M is either flat or a rank-one symmetric space of noncompact type.

conjecture | open | MathDB 页面 | 原文线索

V2-027354 | MATHDB-364078

Isotriviality conjecture for families over polydisks with minimal fibers

摘录 / EXCERPT: Isotriviality conjecture for minimal manifolds. If S is minimal with $\dim S\geq 4$, then every fiber of π is biholomorphic to S.

conjecture | open | MathDB 页面 | 原文线索

V2-027355 | MATHDB-364092

Covariance domination conjecture for the uniform prior over the $\ell 1$ ball

摘录 / EXCERPT: Let C be the $\mathbf{\ell}_1$ ball, let $p(w\mid\xi)$ be the posterior density under the uniform prior on C, and write $\operatorname{Cov}_{\mathrm{Uni}(C)}(w)$ for the covari...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-027356 | MATHDB-364098

Cuspidal-principality conjecture for the oddball maximal order

摘录 / EXCERPT: Let $\mathcal{O}_{5,!}$ be the oddball maximal order, and let $\operatorname{Vec}(\mathcal{O}_{5,!})$ denote its vector part. A right ideal generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-027357 | MATHDB-364112

Voisin's Chern-class extension of the weak splitting conjecture

摘录 / EXCERPT: Let X be a projective hyper-Kähler manifold, and let C^* be the subalgebra of $CH^*(X)$ generated by divisors and Chern classes. Voisin's conjecture. The cycle class map is inj...

conjecture | open | MathDB 页面 | 原文线索

V2-027358 | MATHDB-364123

Goddyn's conjecture on periods of iterated circumcenter sequences

摘录 / EXCERPT: Let $d\geq 2$ be an integer, and let $p=(p_i)_{i=1}^\infty$ be a sequence of points in $\mathbb{R}^d$ such that, for every $i\geq 1$, the points $p_i,p_{i+1},\dots,p_{i+d}$ are dis...

conjecture | open | MathDB 页面 | 原文线索

V2-027359 | MATHDB-364136

The perimeter bound for NS-families of translates

摘录 / EXCERPT: Let $F4D0\mathcal{F}$ be an NS-family of $n\geq 1$ translates of an origin-symmetric convex domain $F4D0\mathbf{M}$. Then ... This bound generalizes the preceding successive-perimete...

conjecture | open | MathDB 页面 | 原文线索

V2-027360 | MATHDB-364137

Didid's zone-width and inradius conjecture

摘录 / EXCERPT: Let $F4D0\mathbf{Z}_1,\dots,\mathbf{Z}_n\subset \mathbb{S}^{d-1}$, with $d>2$, be zones of widths $2\beta_1,\dots,2\beta_n$, respectively. Suppose that ... consists of $2m$ spherical...

conjecture | open | MathDB 页面 | 原文线索

V2-027361 | MATHDB-364138

Fejes Tóth's sausage conjecture for TS-packings of unit balls

摘录 / EXCERPT: Let $N>1$ and $3\leq d\leq 41$. Consider an arbitrary TS-packing of N unit balls in $\mathbb{E}^d$. Fejes Tóth's sausage conjecture. The volume of their convex hull is at least t...

conjecture | open | MathDB 页面 | 原文线索

V2-027362 | MATHDB-364154

The minimal-degree uniqueness conjecture for quantizations of principal-orbit points

摘录 / EXCERPT: Let $\mathcal{O}_1$ be the principal orbit in $\mathbb{P}^2(\mathbb{Q})$, and let $\mathcal{Q}(p)$ be the set of quantizations of $p\in\mathcal{O}_1$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-027363 | MATHDB-364223

The Keel-Yu mirror detropicalization conjecture

摘录 / EXCERPT: Keel-Yu mirror detropicalization conjecture. (1) In the presence of multiple open tori, the integral points of $\mathrm{sk}(U)$ can be equipped with the structure of a polyptych la...

conjecture | open | MathDB 页面 | 原文线索

V2-027364 | MATHDB-364232

The higher algebroid morphism conjecture

摘录 / EXCERPT: Higher algebroid morphism conjecture. The map $\mathrm{\Theta}^k:A^{[k]}\rightarrow E^k$ is a higher algebroid morphism.

conjecture | open | MathDB 页面 | 原文线索

V2-027365 | MATHDB-364243

Glazyrin-Pak cobordism conjecture for integral curves

摘录 / EXCERPT: Glazyrin-Pak cobordism conjecture. For every integral curve γ, there is a unit rhombus ρ and a dome over $\gamma\cup\rho$.

conjecture | open | MathDB 页面 | 原文线索

V2-027366 | MATHDB-364258

Osipov's uniqueness conjecture for Sharygin triangles in regular polygons

摘录 / EXCERPT: Let a Sharygin triangle be a non-isosceles triangle whose bisectral triangle, formed by the intersections of its internal angle bisectors with the opposite sides, is isosceles. Sup...

conjecture | open | MathDB 页面 | 原文线索

V2-027367 | MATHDB-364266

The finiteness conjecture for maximal totally geodesic submanifolds

摘录 / EXCERPT: Let (M,g) be a closed Riemannian manifold with negative sectional curvature, of dimension $n\geq 3$. An immersed totally geodesic submanifold N in M is maximal if it is not c...

conjecture | open | MathDB 页面 | 原文线索

V2-027368 | MATHDB-364269

Equivariant rigidity conjecture for convex co-compact diagonal actions

摘录 / EXCERPT: Let (X_1,d_1) and (X_2,d_2) be negatively curved Hadamard manifolds, and let Γ be a torsion-free group acting properly discontinuously and co-compactly on X_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-027369 | MATHDB-364276

Wegner's piercing conjecture for axis-parallel rectangles

摘录 / EXCERPT: Wegner's conjecture. Every finite family of axis-parallel rectangles in $\mathbb{R}^2$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027370 | MATHDB-364277

The reduced piercing conjecture for projected rectangle families

摘录 / EXCERPT: Projected-family piercing conjecture. There exists a constant A such that every finite family of axis-parallel rectangles satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027371 | MATHDB-364278

The fractional line-piercing conjecture for intersecting convex sets in three dimensions

摘录 / EXCERPT: Fractional line-piercing conjecture. There exists a constant c such that some line intersects at least $c|\mathcal{F}|$ members of $\mathcal{F}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027372 | MATHDB-364310

The conjecture on arbitrarily large Weinstein \(\mathcal{L}\)-invariants

摘录 / EXCERPT: Arbitrarily large Weinstein $\mathcal{L}$-invariant conjecture. The Weinstein domains in Example 1 and Example 2 have arbitrarily large Weinstein $\mathcal{L}$-invariants under thi...

conjecture | open | MathDB 页面 | 原文线索

V2-027373 | MATHDB-364313

The Fukaya-to-sheaf quantization conjecture for cotangent bundles

摘录 / EXCERPT: Let $\bK$ be a field, let M be a real manifold, and let $\bR_t$ be the real line with coordinate t. Write $\mathrm{Sh}_{\tau>0}^{\bR^\delta}(M\times \bR_t)$ for the equivariant...

conjecture | open | MathDB 页面 | 原文线索

V2-027374 | MATHDB-364314

The generic sheaf quantizability conjecture for rotated holomorphic Lagrangians

摘录 / EXCERPT: Let C be a Riemann surface and let $L\subset T^*C$ be a holomorphic Lagrangian. For a generic angle θ, Solomon-Verbitsky theory places $e^{2\pi i\theta}\cdot L$ in the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-027375 | MATHDB-364345

The closedness conjecture for maximal FL zones

摘录 / EXCERPT: Closedness conjecture. Maximal FL zones are closed.

conjecture | open | MathDB 页面 | 原文线索

V2-027376 | MATHDB-364361

Hyperbolic fiberwise rearrangement conjecture for the first Dirichlet eigenvalue

摘录 / EXCERPT: Hyperbolic fiberwise rearrangement conjecture. The first Dirichlet eigenvalue satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027377 | MATHDB-364394

Wrapped Fukaya category and relative Fukaya category conjecture

摘录 / EXCERPT: Relative Fukaya category conjecture. The full subcategory of $\mathcal{W}(M,D)$ whose objects are compact Lagrangians is filtered quasi-equivalent to $\mathcal{F}(M,D)$.

conjecture | open | MathDB 页面 | 原文线索

V2-027378 | MATHDB-364396

Grünbaum's conjecture on rational realizations of n_3 configurations

摘录 / EXCERPT: Grünbaum's conjecture. Any n_3 configuration with a real realization also has a rational realization.

conjecture | open | MathDB 页面 | 原文线索

V2-027379 | MATHDB-364424

Rudyak's conjecture for distributional category

摘录 / EXCERPT: Let M and N be closed orientable n-manifolds, and let $f:M\to N$ be a map of degree ±1. Write $\operatorname{\mathsf{dcat}}$ for the distributional category of a manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-027380 | MATHDB-364445

Feng-Huang-Rao logarithmic commensurability conjecture for self-similar sets

摘录 / EXCERPT: Logarithmic commensurability conjecture. The contraction ratios are logarithmically commensurable, meaning

conjecture | open | MathDB 页面 | 原文线索

V2-027381 | MATHDB-364474

The RCD gluing conjecture along intrinsic boundaries

摘录 / EXCERPT: Let X_0 and X_1 be noncollapsed $RCD(K,n)$ spaces with nonempty boundaries ∂X_0 and ∂X_1, where the boundaries are understood in the sense of Gigli and Pasq...

conjecture | open | MathDB 页面 | 原文线索

V2-027382 | MATHDB-364475

The doubling conjecture for noncollapsed RCD spaces

摘录 / EXCERPT: Let X be a noncollapsed $RCD(K,n)$ space with nonempty boundary ∂X. Doubling conjecture. The doubled space $\operatorname{Dbl}(X)$ satisfies the condition $RCD(K,n)$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-027383 | MATHDB-364476

The curvature criterion for doubling geodesically convex subsets

摘录 / EXCERPT: Let X be an $RCD(K,N)$ space, and let Ω be a geodesically convex open subset of X with $\operatorname{m}(\partial\Omega)=0$. Let $\hat{\operatorname{m}}$ be the measure...

conjecture | open | MathDB 页面 | 原文线索

V2-027384 | MATHDB-364493

McKernan's stronger ACC conjecture for interpolated lc thresholds

摘录 / EXCERPT: Let (X,B) be a pair of dimension d, let $\mathcal{F}$ be a foliation on X, and write $B=B^{\operatorname{ninv}}+B^{\operatorname{inv}}$ so that...

conjecture | open | MathDB 页面 | 原文线索

V2-027385 | MATHDB-364494

The ACC conjecture for lc thresholds of adjoint foliated structures

摘录 / EXCERPT: Let $(X,\mathcal{F},B,t)$ be an adjoint foliated structure, with $B=B^{\operatorname{ninv}}+B^{\operatorname{inv}}$, and let D be an $\mathbb{R}$-divisor. The ACC conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-027386 | MATHDB-364495

The uniform interpolated lc rational polytope conjecture

摘录 / EXCERPT: Let d be a positive integer, let $\Gamma_0\subset[0,1]\cap\mathbb{Q}$ be finite, and let $t_0\in[0,1]$ be irrational. Uniform interpolated lc rational polytope conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-027387 | MATHDB-364504

Concurrent normals conjecture for convex bodies

摘录 / EXCERPT: Let C be a convex body in $\mathbb{R}^n$ with smooth boundary. An inward normal to ∂C is a line segment or ray normal to the boundary and directed inward. The concurren...

conjecture | open | MathDB 页面 | 原文线索

V2-027388 | MATHDB-364518

Thurston's conjecture on horospherical Penrose tilings

摘录 / EXCERPT: Let the $3,5,3 honeycomb be a tessellation of $operatorname{mathbb{H}}^3...

conjecture | open | MathDB 页面 | 原文线索

V2-027389 | MATHDB-364568

Small sets are nonbarriers for the Gromov width of a ball

摘录 / EXCERPT: Small-sets nonbarrier conjecture. The Gromov width of $B^{2n}_1\setminus A$ equals π.

conjecture | open | MathDB 页面 | 原文线索

V2-027390 | MATHDB-364582

Periodic tiling conjecture for translational monotile tilings

摘录 / EXCERPT: Let T be a tile in $\mathbb{Z}^n$. A tiling by translated copies of T is periodic if it is invariant under a nonzero translation of $\mathbb{Z}^n$. Periodic tiling conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027391 | MATHDB-364594

Lutwak's affine quermassintegral inequality for real, complex, and quaternionic convex bodies

摘录 / EXCERPT: Let $\mathbb{F}$ be one of $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, let $p=\dim_{\mathbb{R}}\mathbb{F}$, and let $K\subset \mathbb{F}^n$ be a convex body with non-empty interio...

conjecture | open | MathDB 页面 | 原文线索

V2-027392 | MATHDB-364611

The systolic-volume bound for KVol on translation surfaces

摘录 / EXCERPT: Let X be a translation surface. Write $\mathop{\mathrm{KVol}}(X)$ for its KVol and $\mathop{\mathrm{SysVol}}(X)$ for its systolic volume, defined as the volume divided by the squ...

conjecture | open | MathDB 页面 | 原文线索

V2-027393 | MATHDB-364617

The octahedron conjecture for the perverse-Hodge octahedron

摘录 / EXCERPT: Let X be a compact hyperkähler manifold, and define ... The perverse-Hodge octahedron conjecture. The convex hull of the non-zero $h^{i,k,d}$ is an octahedron. This is stronger tha...

conjecture | open | MathDB 页面 | 原文线索

V2-027394 | MATHDB-364635

Hyperbolic Weinstock inequality for convex domains

摘录 / EXCERPT: Hyperbolic Weinstock conjecture. Is it true that

open_problem | open | MathDB 页面 | 原文线索

V2-027395 | MATHDB-364648

Higher-dimensional Neumann monotonicity conjecture

摘录 / EXCERPT: Higher-dimensional Neumann monotonicity conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-027396 | MATHDB-364649

Dirichlet monotonicity conjecture with volume scaling

摘录 / EXCERPT: Dirichlet monotonicity with volume scaling. The function

conjecture | open | MathDB 页面 | 原文线索

V2-027397 | MATHDB-364650

Neumann monotonicity conjecture with volume scaling

摘录 / EXCERPT: Neumann monotonicity with volume scaling. The function

conjecture | open | MathDB 页面 | 原文线索

V2-027398 | MATHDB-364677

Conjecture on tangent-map morphisms for rational curves in projective space

摘录 / EXCERPT: Let $d\geq n+1$ be integers. Set ... For a morphism ... where $C\xrightarrow{\phi}\mathbb{P}^n$ denotes a rational curve of degree d, the following assertions hold. Tangent-map conje...

conjecture | open | MathDB 页面 | 原文线索

V2-027399 | MATHDB-364704

Exact characterization conjecture for boundary hyperplane covers of quadratic sets

摘录 / EXCERPT: Let $B=\{\mathbf{x}:\|\mathbf{x}\|_2\leq 1\}$ and $C=\{\mathbf{x}:g(\mathbf{x})\leq 0\}$, where g is a quadratic polynomial. Suppose that g has one of the forms ... or ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-027400 | MATHDB-364740

Small-ball spectral convergence for sub-Riemannian diffusions

摘录 / EXCERPT: A sub-Riemannian diffusion killed when it exits a sufficiently small metric ball is expected to exhibit an analogous small-ball spectral convergence to the corresponding nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-027401 | MATHDB-364744

Fano data conjecture on positivity and integrality of the naive mirror map

摘录 / EXCERPT: Let I index vectors $\boldsymbol{v}_{ij}$, let K_0 be the monoid used to define the formal power series $\boldsymbol{z}^{\boldsymbol{k}}$, and let $\boldsymbol{v}_{ij}$ be Fano...

conjecture | open | MathDB 页面 | 原文线索

V2-027402 | MATHDB-364745

Integrality conjecture for the true mirror map of Fano data

摘录 / EXCERPT: Let K_{ij} be the monoid defined by allowing the (i,j)-coordinate to be negative subject to the stated sum condition, let τ_{ij} be the associated formal series, and defin...

conjecture | open | MathDB 页面 | 原文线索

V2-027403 | MATHDB-364758

Lutz-Nevo collapse conjecture for flag triangulations of the 3-sphere

摘录 / EXCERPT: The three-dimensional case of Lutz and Nevo's conjecture. For every flag triangulation T of S^3 with $\gamma_2(T)=0$, there is a sequence of such collapses that reduces T to...

conjecture | open | MathDB 页面 | 原文线索

V2-027404 | MATHDB-364760

Ball characterization conjecture for Stein spaces with standard contact boundary

摘录 / EXCERPT: Let W be a Stein space of dimension $n\geq 3$ with at most finitely many isolated normal singularities. Assume that W has a compact strongly pseudoconvex smooth boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-027405 | MATHDB-364780

Streets-Tian-Fino-Vezzoni intersection conjecture

摘录 / EXCERPT: Streets-Tian-Fino-Vezzoni intersection conjecture. If M admits a Hermitian-symplectic metric and a balanced metric, then it must admit a Kähler metric.

conjecture | open | MathDB 页面 | 原文线索

V2-027406 | MATHDB-364792

The Generalised Argument Shift Conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional Lie algebra, let $a\in\mathfrak{g}^*$ be a regular element, and let $\left\{\,, \,\right\}$ and $\left\{\,, \,\right\}_a$ denote respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-027407 | MATHDB-364795

Concentration of lattice polytopes near the boundary volume

摘录 / EXCERPT: Let B_r^d be the convex body used in the paper, and let H_r^d be the set of lattice polytopes contained in B_r^d, with $H_r^d(V/d!\geqslant\mathrm{vol}(B_r^d)-c(r))$ denoting...

conjecture | open | MathDB 页面 | 原文线索

V2-027408 | MATHDB-364802

Folklore rigidity conjecture for AC Ricci-flat Kähler metrics on affine space

摘录 / EXCERPT: Let g be a complete asymptotically conical Ricci-flat Kähler metric on $\mathbb{C}^n$, asymptotic to a Kähler cone $\mathcal{C}$ whose link is smooth. The AC metric rigidity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-027409 | MATHDB-364803

Allais' spectral selector conjecture for orderable contact manifolds

摘录 / EXCERPT: Let (M,ξ) be a closed orderable contact manifold, meaning that it admits no positive contractible loop of contactomorphisms. Let α be any contact form supporting ξ,...

conjecture | open | MathDB 页面 | 原文线索

V2-027410 | MATHDB-364804

Allais' spectrality conjecture for the contact spectral invariant

摘录 / EXCERPT: Let $(M,\ker\alpha)$ be a closed orderable contact manifold, and let $\widetilde{\operatorname{Cont}_0}(M,\ker\alpha)$ denote the universal cover of the identity component of its c...

conjecture | open | MathDB 页面 | 原文线索

V2-027411 | MATHDB-364819

Connell-Nguyen-Spatzier higher hyperbolic rank conjecture

摘录 / EXCERPT: Let (M^n,g) be a closed negatively curved manifold with higher hyperbolic rank, meaning that for every geodesic there is a nonvanishing Jacobi field whose span with the geodesic...

conjecture | open | MathDB 页面 | 原文线索

V2-027412 | MATHDB-364829

The planar finite-field Furstenberg set lower-bound conjecture

摘录 / EXCERPT: Fix $s\in(0,1]$ and $t\in(0,2]$. Let E be an $(s,t;2,1)$-set over $\mathbb{F}_p$, meaning that E is a union of subsets $Y(V)$ over a family of affine 1-planes V containing...

conjecture | open | MathDB 页面 | 原文线索

V2-027413 | MATHDB-364845

Spectral invariance conjecture for periods of Reeb orbits

摘录 / EXCERPT: Let (M,D,g) be a three-dimensional contact manifold with sub-Riemannian Laplacian Δ_{sR} and Reeb flow associated with the canonical contact form determined by (D,g). A...

conjecture | open | MathDB 页面 | 原文线索

V2-027414 | MATHDB-364865

Marques-Neves width stability conjecture for rotationally symmetric three-spheres

摘录 / EXCERPT: Marques-Neves' width stability conjecture. Then $(\mathbb S^3,g_k)$ converges in the Sormani-Wenger intrinsic flat sense to the round unit sphere $(\mathbb S^3,g_{rd})$. The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-027415 | MATHDB-364866

The weak flatness conjecture for induced nilpotent orbits in orthogonal Lie algebras

摘录 / EXCERPT: Let $\fg$ be an orthogonal Lie algebra of type B or D, of rank n, and write a Levi subalgebra as ... with $b_1\geqslant\cdots\geqslant b_k\geqslant1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027416 | MATHDB-364897

Colorful quantitative volume Helly conjecture

摘录 / EXCERPT: Let $\mathcal{F}_1, \dots, \mathcal{F}_{2d}$ be finite families of convex sets in $\mathbb{R}^d$. Assume that for any choice...

open_problem | open | MathDB 页面 | 原文线索

V2-027417 | MATHDB-364922

The bipartiteness and maximum-degree conjecture for mathcal{C}_3-free BonnetMyers sharp graphs

摘录 / EXCERPT: Bipartiteness and maximum-degree conjecture. 1. G is bipartite. 2. For every edge uv,

conjecture | open | MathDB 页面 | 原文线索

V2-027418 | MATHDB-364936

Jordan's Langlands duality conjecture for generic skein modules

摘录 / EXCERPT: Let G be a reductive group, G^{L} its Langlands dual, and M a closed, connected, oriented 3-manifold. Let $\operatorname{Sk}^{\operatorname{gen}}_{G}(M)$ denote the skein m...

conjecture | open | MathDB 页面 | 原文线索

V2-027419 | MATHDB-364940

Fejes Tóth's conjecture on the sum of acute angles

摘录 / EXCERPT: Fejes Tóth's conjecture. This energy is maximal when $x_1,\dots,x_{d+1}$ are mutually orthogonal and $x_k=x_{k-(d+1)}$ for every k with $d+1<k\le N$. Equivalently, periodically r...

conjecture | open | MathDB 页面 | 原文线索

V2-027420 | MATHDB-365003

Conjecture on conic loci of the Gergonne and Nagel points

摘录 / EXCERPT: Let $\mathcal{L}_7$ and $\mathcal{L}_8$ denote the loci of the Gergonne point X_7 and Nagel point X_8, respectively, in a Poncelet triangle family. The pair of Poncelet conics...

conjecture | open | MathDB 页面 | 原文线索

V2-027421 | MATHDB-365004

Conjecture on stationary orthocenters in Poncelet triangle families

摘录 / EXCERPT: Let X_4 be the orthocenter of a Poncelet triangle family interscribed between two conics. The configurations called focal-X_4, MacBeath, and Dual are the three configurations i...

conjecture | open | MathDB 页面 | 原文线索

V2-027422 | MATHDB-365005

Conjecture on conserved half-angle tangents for polar Harmonic families

摘录 / EXCERPT: Let $N>3$, and consider the polar image of the Harmonic family, a Poncelet family of N-gons. For each polygon, let θ_i denote its relevant angles, so that the half-angle t...

conjecture | open | MathDB 页面 | 原文线索

V2-027423 | MATHDB-365035

The corrected local-to-global conjecture for Apollonian curvatures

摘录 / EXCERPT: Let ${\mathcal{P}^{\operatorname{full}}}$ be a primitive integral packing, and let Σ be the union of the residue classes modulo 24 that have representatives in...

conjecture | open | MathDB 页面 | 原文线索

V2-027424 | MATHDB-365048

Conjecture on Legendre duality for subharmonicity and superharmonicity

摘录 / EXCERPT: Legendre-duality conjecture. The Legendre transform should also turn ω-subharmonicity into $\tilde{\omega}$-superharmonicity, and this fact together with the preceding duali...

conjecture | open | MathDB 页面 | 原文线索

V2-027425 | MATHDB-365105

CAT(0) conjecture for the stratified Euclidean metric on polynomial space

摘录 / EXCERPT: Let ${{\normalfont\textsc{Poly}}}_d^{mt}$ be the space of all monic degree-d complex polynomials up to translation, equipped with its stratified Euclidean metric. CAT(0) conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-027426 | MATHDB-365106

CAT(0) equivalence conjecture for polynomial space and the dual braid complex

摘录 / EXCERPT: Let ${{\normalfont\textsc{Poly}}}_d^{mt}$ be the space of all monic degree-d complex polynomials up to translation with its stratified Euclidean metric, and let K_d be the dual...

conjecture | open | MathDB 页面 | 原文线索

V2-027427 | MATHDB-365120

Minimal-perimeter conjecture for ordinary reduced hyperbolic polygons

摘录 / EXCERPT: Let $P\subset H^2$ be an ordinary reduced n-gon of minimal width w. Let $\widetilde{P}$ denote the regular ordinary reduced n-gon of width w, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-027428 | MATHDB-365163

Honda-Mondino spectral gap conjecture for the first Poincaré constant

摘录 / EXCERPT: Let M^n be a closed Riemannian manifold with ... Here $C_{P,1}$ denotes the Poincaré constant for 1-forms. Honda-Mondino's spectral gap conjecture. Under these assumptions, ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-027429 | MATHDB-365164

Hodge-Betti convergence conjecture for non-collapsed Ricci limit spaces

摘录 / EXCERPT: Let ... be a non-collapsed sequence of closed Riemannian manifolds with $\mathrm{Ric}\ge -(n-1)$, converging in the Gromov-Hausdorff sense to a compact Ricci limit space X^n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027430 | MATHDB-365165

Sectional-curvature spectral gap conjecture for Hodge Laplacians

摘录 / EXCERPT: Let M^n be a closed Riemannian manifold satisfying ... Here $C_{P,k}$ denotes the Poincaré constant for k-forms. Sectional-curvature spectral gap conjecture. For every $k\ge 1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027431 | MATHDB-365166

Perelman's bi-Lipschitz stability conjecture for Alexandrov spaces

摘录 / EXCERPT: Let ... be a non-collapsed Gromov-Hausdorff-convergent sequence of compact Alexandrov spaces of dimension n with curvature bounded below by -1. Perelman's bi-Lipschitz stability...

conjecture | open | MathDB 页面 | 原文线索

V2-027432 | MATHDB-365168

Ducat's conjecture on toric models of rational log Calabi-Yau threefolds

摘录 / EXCERPT: Let (X,B) be a log canonical log Calabi-Yau pair, meaning that K_X+B is numerically trivial, such that $\dim X=3$, $\operatorname{coreg}(X,D)=0$, and X is rational. Ducat's c...

conjecture | open | MathDB 页面 | 原文线索

V2-027433 | MATHDB-365191

Minimal model program conjecture for three-dimensional generalized foliated quadruples

摘录 / EXCERPT: Generalized foliated quadruple MMP conjecture. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-027434 | MATHDB-365194

Morrison's nef cone conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let Y be a Calabi-Yau manifold, meaning that Y has trivial canonical bundle and $h^1(\mathcal{O}_Y)=0$. Let $\mathrm{Nef}^+(Y)$ denote the convex hull of the rational points of...

conjecture | open | MathDB 页面 | 原文线索

V2-027435 | MATHDB-365195

Kawamata's movable cone conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let Y be a Calabi-Yau manifold, and let $\overline{\mathrm{Mov}}^+(Y)$ denote the rational hull of its closed movable cone. A pseudo-automorphism of Y is a small modification...

conjecture | open | MathDB 页面 | 原文线索

V2-027436 | MATHDB-365197

The linear lower-bound conjecture for small eigenvalues of cusped hyperbolic surfaces

摘录 / EXCERPT: Linear small-eigenvalue conjecture. There are constants $a,b>0$ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-027437 | MATHDB-365200

Existence of embedded immortal solutions for higher-order area-preserving curvature flows

摘录 / EXCERPT: Let $m\geq1$. Consider the area-preserving curvature flow of order m for embedded initial curves with length L_0 and enclosed area 1. Immortal-solution conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-027438 | MATHDB-365201

Generic finite-time blowup for higher-order area-preserving curvature flows

摘录 / EXCERPT: Let $m\geq0$ and $N_0\geq2$, where N_0 is the rotation number of the initial immersed curve. Consider the area-preserving curvature flow of order m. Finite-time blowup conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-027439 | MATHDB-365202

Stabilized Lagrangian isotopy conjecture for smoothly ribbon-isotopic cobordisms

摘录 / EXCERPT: Stabilized Lagrangian isotopy conjecture. If L_1 and L_2 are smoothly ribbon isotopic, then after stabilizing both cobordisms, they become Lagrangian isotopic.

conjecture | open | MathDB 页面 | 原文线索

V2-027440 | MATHDB-365203

The abundance conjecture for projective manifolds

摘录 / EXCERPT: Abundance conjecture. The manifold X admits a good minimal model X' such that

conjecture | open | MathDB 页面 | 原文线索

V2-027441 | MATHDB-365215

The super-Jacobian conjecture for affine superspaces

摘录 / EXCERPT: Let $\mathbb{F}$ be either $\mathbb{R}$ or an algebraically closed field of characteristic 0, and consider the morphism phi of the affine superspace $\mathbb{A}^{m|n}$ over...

conjecture | open | MathDB 页面 | 原文线索

V2-027442 | MATHDB-365222

Akl et al.'s connectivity conjecture for flip graphs of plane spanning paths

摘录 / EXCERPT: Let S be a point set in general position. A plane spanning path on S is a noncrossing straight-line spanning path, and the flip graph has these paths as vertices, with adjacenc...

conjecture | open | MathDB 页面 | 原文线索

V2-027443 | MATHDB-365227

Magnetic isoperimetric non-vanishing conjecture

摘录 / EXCERPT: Let (M^m,g) be a compact manifold with smooth boundary ∂M, and let $\eta\in\Omega^1(M)$. Let $\mathfrak{B}_M$ denote the relevant gauge-trivial subspace, and for an adm...

conjecture | open | MathDB 页面 | 原文线索

V2-027444 | MATHDB-365240

Product decomposition conjecture for higher-rank group manifolds

摘录 / EXCERPT: Let G_0 be a real linear simple Lie group of real rank at least 2, let K be a maximal compact subgroup of G_0, and let Γ be a discrete subgroup of $G_0\times G_0$ ac...

conjecture | open | MathDB 页面 | 原文线索

V2-027445 | MATHDB-365268

Kähler-Ricci shrinker asymptotic conicity conjecture

摘录 / EXCERPT: Asymptotic conicity conjecture. A Kähler-Ricci shrinker is weakly asymptotically conical in the sense that it has a unique asymptotic cone at infinity, which is a Kähler cone metri...

conjecture | open | MathDB 页面 | 原文线索

V2-027446 | MATHDB-365296

Streets-Tian conjecture on Hermitian-symplectic compact complex manifolds

摘录 / EXCERPT: Let M be a compact complex manifold. A Hermitian-symplectic metric on M is a Hermitian metric whose Kähler form is the $(1,1)$-part of a closed 2-form. Equivalently, there is...

conjecture | open | MathDB 页面 | 原文线索

V2-027447 | MATHDB-365304

Casson's volume conjecture for ideal triangulations

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with toroidal boundary and let X be an ideal triangulation of M. Suppose the space of angle structures $\mathscr A_X$ is non-empty. Casson...

conjecture | open | MathDB 页面 | 原文线索

V2-027448 | MATHDB-365305

Casson's cone-volume conjecture for ideal triangulations

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with toroidal boundary, let X be an ideal triangulation of M, and let $\gamma\in\pi_1(\partial M)$ be a simple closed curve. For...

conjecture | open | MathDB 页面 | 原文线索

V2-027449 | MATHDB-365308

Petrunin-Tuschmann conjecture on simply connected ends of conical asymptotically flat 4-manifolds

摘录 / EXCERPT: Let X be a conical $\mathcal{AF}$ 4-manifold with a simply connected end, and let its asymptotic cone be the metric cone at infinity. Petrunin-Tuschmann conjecture. There is no c...

conjecture | open | MathDB 页面 | 原文线索

V2-027450 | MATHDB-365310

Type II classification conjecture for gravitational instantons

摘录 / EXCERPT: Type II classification conjecture. All Type II gravitational instantons are toric and must be ALF or AF, except the Eguchi-Hanson metric with the reversed orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-027451 | MATHDB-365318

The weighted Laurent-polynomial spectrum conjecture

摘录 / EXCERPT: Weighted Laurent-polynomial spectrum conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-027452 | MATHDB-365319

The consistent coarse Baum-Connes conjecture with filtered coefficients

摘录 / EXCERPT: Let A be a filtered $C^{\ast}$-algebra, let X be a proper metric space, and let N_X be a locally finite net in X. Let $\bigsqcup_{\mathbb{N}}X_i$ denote the separated coars...

conjecture | open | MathDB 页面 | 原文线索

V2-027453 | MATHDB-365320

The coarse Baum-Connes conjecture along a factor with filtered coefficients

摘录 / EXCERPT: Let A be a filtered $C^{\ast}$-algebra, let X and Y be proper metric spaces, and let N_X be a locally finite net in X. For each $k\geq 0$, let $P_k(N_X)$ be the k-Rips...

conjecture | open | MathDB 页面 | 原文线索

V2-027454 | MATHDB-365332

Levi-Hadwiger illumination conjecture

摘录 / EXCERPT: Levi-Hadwiger illumination conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-027455 | MATHDB-365333

Complex illumination conjecture for the polydisc

摘录 / EXCERPT: Complex illumination conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-027456 | MATHDB-365363

The conjecture on separated boundary components for Steklov maximizers

摘录 / EXCERPT: Let Σ be a surface with disconnected boundary $\partial\Sigma$, and let g be a maximizer of the normalized first Steklov eigenvalue $\bar{\sigma}_1(\Sigma,\cdot)$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-027457 | MATHDB-365372

Tropicalization conjecture for the type C cluster configuration space

摘录 / EXCERPT: Tropicalization conjecture. The tropicalization $\operatorname{Trop}\mathcal{M}_{\mathrm{C}_{n-1}}$ is the space of ASPTs: the fan formed by the cones $C_{T,\varphi}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-027458 | MATHDB-365392

Pólya-Szegő conjecture on the logarithmic-to-Newtonian capacity ratio

摘录 / EXCERPT: Let $K\subset\mathbb R^2$ be compact. The logarithmic capacity $\operatorname{Cap}_0(K)$ and Newtonian capacity $\operatorname{Cap}_1(K)$ are the corresponding Riesz capacities. Pó...

conjecture | open | MathDB 页面 | 原文线索

V2-027459 | MATHDB-365394

Regular simplex conjecture for Riesz capacity ratios

摘录 / EXCERPT: Let $n\geq 3$, let $p<q\leq -2$, and let $K\subset\mathbb R^n$ be compact with more than one point. The quantities $\operatorname{Cap}_p(K)$ and $\operatorname{Cap}_q(K)$ are the c...

conjecture | open | MathDB 页面 | 原文线索

V2-027460 | MATHDB-365396

Novaga-Ruffini conjecture on the Brunn-Minkowski inequality for Riesz capacity

摘录 / EXCERPT: Let $n\geq1$ and let $p\in(n-2,n)$. For compact sets $A,B\subset\mathbb R^n$ and $\lambda\in[0,1]$, write ... Let Cap_p denote the Riesz p-capacity. Novaga-Ruffini...

conjecture | open | MathDB 页面 | 原文线索

V2-027461 | MATHDB-365400

The finite-frequency zero mass conjecture for plurisubharmonic functions

摘录 / EXCERPT: Let $u\in {\mathcal P}(B_1)$ be a plurisubharmonic function whose alternating part has finite frequencies, meaning that near the origin, in a proper complex Hopf-coordinate, it can...

conjecture | open | MathDB 页面 | 原文线索

V2-027462 | MATHDB-365410

Realizability conjecture for volume growth, escape time and martingale dimensions

摘录 / EXCERPT: Let $b1 b2$ and d_m satisfy ... and ... Here $b1$ is the volume growth exponent, $b2$ is the escape time exponent, and d_m is the martingale dimension, also called the martingale i...

conjecture | open | MathDB 页面 | 原文线索

V2-027463 | MATHDB-365418

The stable ALE desingularization conjecture for spherical and hyperbolic orbifolds

摘录 / EXCERPT: Stable ALE desingularization conjecture. There exists an ancient, respectively immortal, renormalized Ricci flow on $M_o\#_{\mathbb{S}^3/\Gamma}N$ with limit (M_o,g_o) at...

conjecture | open | MathDB 页面 | 原文线索

V2-027464 | MATHDB-365420

Lie monoid actions in Morse theory

摘录 / EXCERPT: Let G,G',H,H' be compact Lie monoids, let X,X' be closed smooth manifolds acted on by $G\times H$ and $G'\times H'$ respectively, let $\varphi\colon G\to G'$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027465 | MATHDB-365427

Generalized barycentric projection as a Monge mapping for Gromov-Wasserstein distance

摘录 / EXCERPT: Generalized barycentric projection conjecture. The map $cmathcal{T}_{cgamma^*}$ is a Monge mapping for the $coperatorname{GW}(cmathbb{S}, ctilde{cmathbb{X}}_{cgamma^*})$ problem. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-027466 | MATHDB-365443

Uniqueness of the complex 4×8 equiangular tight frame

摘录 / EXCERPT: 4×8 uniqueness conjecture. There exists a unique complex 4×8 ETF up to switching equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-027467 | MATHDB-365444

Bogomolov-Fu-Tschinkel conjecture on common torsion points of elliptic curves

摘录 / EXCERPT: Let E_1 and E_2 be elliptic curves over $\mathbb{C}$, and for each $j=1,2$ let ι be the involution of E_j and choose a double cover $\pi_j:E_j\to\mathbb{P}^{1}$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-027468 | MATHDB-365450

Donaldson-Scaduto conjecture on collapsing special Lagrangians

摘录 / EXCERPT: Let X_t be a 1-parameter family of compact Calabi-Yau manifolds with metrics g_t admitting special Lagrangian fibration structures and with bounded diameters, and let C_t b...

conjecture | open | MathDB 页面 | 原文线索

V2-027469 | MATHDB-365451

Donaldson-Scaduto conjecture on collapsing special Lagrangian submanifolds

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold with a K3-fibration $\pi:X\rightarrow Y$, let ω_X and ω_Y be Kähler forms on X and Y, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027470 | MATHDB-365452

Donaldson-Scaduto realizability conjecture for gradient cycles

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold with a Lefschetz K3-fibration $\pi:X\rightarrow Y$, and let a gradient cycle be a graph in Y whose edges are decorated by 2-cycles in the K3 fi...

conjecture | open | MathDB 页面 | 原文线索

V2-027471 | MATHDB-365461

Milman-Szarek geometric lemma for easily covered polytopes

摘录 / EXCERPT: Let $r\in(0,1]$. A polytope $P=\operatorname{conv}(V)\subset\mathbb{R}^n$ contains rB_2^n when $rB_2^n\subset P$, and $N(P,B_2^n)$ denotes the covering number of P by translate...

conjecture | open | MathDB 页面 | 原文线索

V2-027472 | MATHDB-365465

Weyl chamber and stable-base-locus decomposition conjecture for X^3_8 and X^5_9

摘录 / EXCERPT: Let $X=X^5_9$ or $X=X^3_8$. Write $\overline{\operatorname{Eff}}(X)$ for the pseudoeffective cone of divisors, $\operatorname{WCD}(\overline{\operatorname{Eff}}(X))$ for its Weyl c...

conjecture | open | MathDB 页面 | 原文线索

V2-027473 | MATHDB-365483

The Futaki upper-bound conjecture for toric Calabi-Yau threefolds

摘录 / EXCERPT: Let $\mathcal{X}$ be a toric Calabi-Yau 3-fold with no factors of $\mathbb{C}$, let ζ_R be the Reeb symmetry, let ζ_p be the grading symmetry from GLSM-field degrees,...

conjecture | open | MathDB 页面 | 原文线索

V2-027474 | MATHDB-365484

The divisor-volume lower bound for Futaki invariants

摘录 / EXCERPT: Let $\mathcal{X}$ be a toric Calabi-Yau 3-fold with no factors of $\mathbb{C}$, let ζ_R be the Reeb symmetry, and let η_h be the test symmetry associated with a genera...

conjecture | open | MathDB 页面 | 原文线索

V2-027475 | MATHDB-365504

Verbitsky's conjecture on exotic hypercomplex structures

摘录 / EXCERPT: A hypercomplex structure (X,I,J,K) is exotic if (X,I) is holomorphically symplectic, but for no choice of $\alpha \in \operatorname{Kahl}(X)$ does the corresponding hyperkähler...

conjecture | open | MathDB 页面 | 原文线索

V2-027476 | MATHDB-365509

Connell-Wang's pointwise negative Ricci curvature conjecture for positive simplicial volume

摘录 / EXCERPT: Let M be a manifold. Suppose M admits a Riemannian metric with nonpositive sectional curvature everywhere and negative definite Ricci curvature at some point. Connell-Wang's co...

conjecture | open | MathDB 页面 | 原文线索

V2-027477 | MATHDB-365540

Exterior scalar curvature comparison rigidity conjecture for convex domains

摘录 / EXCERPT: Let Ω be a convex smooth domain in $\mathbb R^n$, with $n\geq 3$, and let $(\mathbb R^n\setminus\Omega,g)$ be an asymptotically flat Riemannian manifold. Let H_g and H_0...

conjecture | open | MathDB 页面 | 原文线索

V2-027478 | MATHDB-365574

Huybrechts's Fourier-Mukai partners conjecture for cubic fourfolds

摘录 / EXCERPT: Let X and X' be smooth cubic fourfolds, and let $\mathcal{A}_X$ and $\mathcal{A}_{X'}$ denote their Kuznetsov components. Huybrechts's conjecture. If ... then X and X' are bi...

conjecture | open | MathDB 页面 | 原文线索

V2-027479 | MATHDB-365591

Berge's conjecture on lens space knots

摘录 / EXCERPT: Let $K\subset S^3$ be a lens space knot, meaning that some positive integer surgery on K produces a lens space. Berge's construction gives a family of such knots. Berge conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-027480 | MATHDB-365603

Conjecture for the dimension function $\mathfrak{f}(s,t)$

摘录 / EXCERPT: Let $\mathfrak{f}(s,t)$ denote the dimension quantity defined in the paper, with s and t in the parameter ranges below. The conjecture is the dimension-function conjecture: for...

conjecture | open | MathDB 页面 | 原文线索

V2-027481 | MATHDB-365618

The weak Atiyah-Sutcliffe conjecture for configurations in three-dimensional space

摘录 / EXCERPT: Let $\operatorname{Conf}_n(\mathbb{R}^3)$ be the configuration space of n distinct points in $\mathbb{R}^3$. For a configuration $(p_1,\dots,p_n)$, let $\ell_i(p_1,\dots,p_n)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-027482 | MATHDB-365620

The strong real Atiyah-Sutcliffe conjecture

摘录 / EXCERPT: Let D be the normalized determinant associated with a configuration in $\operatorname{Conf}_n(\mathbb{R}^2)$, and let $F_n^{\mathbb{R}}$ denote the space of projectively independ...

conjecture | open | MathDB 页面 | 原文线索

V2-027483 | MATHDB-365621

The real Atiyah-Sutcliffe conjecture for Fulton-MacPherson compositions

摘录 / EXCERPT: Let $FM_2(n)$ be the Fulton-MacPherson operad space, let $F_{k_i}^{\mathbb{R}}\subseteq D_{k_i}(BO(1))$ consist of projectively independent real labels, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027484 | MATHDB-365630

Commutation criterion for compatible vector bundle structures on homogeneity manifolds

摘录 / EXCERPT: Let (E,∇) be a homogeneity manifold, and let a vector bundle structure on E have Euler vector field ∇_E. Compatibility means that this vector bundle structure is co...

conjecture | open | MathDB 页面 | 原文线索

V2-027485 | MATHDB-365655

Long-time diffusion mean conjecture for compact symmetric spaces

摘录 / EXCERPT: Let M be a connected compact symmetric space with a smooth isometric embedding into a Euclidean space that is unique up to Euclidean isometries. Let the diffusion mean set be the...

conjecture | open | MathDB 页面 | 原文线索

V2-027486 | MATHDB-365661

Lattice-preserving invariance conjecture for higher-genus Göttsche-Schroeter invariants

摘录 / EXCERPT: Let Δ and Δ' be two h-transverse polygons. A lattice-preserving transformation is an element of the affine group of $\mathbb{R}^2$ preserving the lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-027487 | MATHDB-365664

Index lower-bound conjecture for branched CMC immersions

摘录 / EXCERPT: Let $u:\Sigma^2\to N^3$ be a branched CMC h-immersion, where N^3 has strictly positive sectional curvature, and let g denote the genus of Σ. Write $i_{\mathcal{A}_h}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027488 | MATHDB-365731

Donald's a priori estimate conjecture for the symplectic Calabi-Yau equation

摘录 / EXCERPT: Donald's conjecture. For any $k\geqslant 0$, one can bound $\|\tilde{\omega}\|_{C^k(M,g)}$ by a constant depending only on k and on bounds for σ, ω, and (M,J), wh...

conjecture | open | MathDB 页面 | 原文线索

V2-027489 | MATHDB-365741

The zonotopal Lonely Runner Conjecture

摘录 / EXCERPT: Let $Z=\sum_{i=1}^n[\mathbf{0},\mathbf{u}_i]\subseteq\mathbb{R}^{n-1}$ be a lattice zonotope of dimension $n-1$ with n generators, whose every $n-1$ generators are linearly indep...

conjecture | open | MathDB 页面 | 原文线索

V2-027490 | MATHDB-365743

The cosimple zonotope covering-radius conjecture

摘录 / EXCERPT: Let a finite collection of lattice vectors spanning $\mathbb{R}^d$ be cosimple if it has a linear dependence whose coefficients are all non-zero and have pairwise different absolut...

conjecture | open | MathDB 页面 | 原文线索

V2-027491 | MATHDB-365757

Pólya's conjecture for Dirichlet and Neumann Laplacian counting functions

摘录 / EXCERPT: Pólya's conjecture. The eigenvalue counting function of the Dirichlet Laplacian on Ω can be estimated from above by the leading term of Weyl's law, while the eigenvalue coun...

conjecture | open | MathDB 页面 | 原文线索

V2-027492 | MATHDB-365796

The global Weinstein splitting conjecture for compact Kähler Poisson manifolds

摘录 / EXCERPT: Global Weinstein splitting conjecture. There exist a holomorphic symplectic manifold Y of dimension d, a holomorphic Poisson manifold Z, and a holomorphic P...

conjecture | open | MathDB 页面 | 原文线索

V2-027493 | MATHDB-365797

The Poisson vanishing conjecture for compact Kähler manifolds

摘录 / EXCERPT: Poisson vanishing conjecture. The Hodge numbers $h^{2j,0}(\mathsf{X})$ are nonzero for every $j\leq\tfrac{d}{2}$. In particular, if $h^{2,0}(\mathsf{X})=0$, then every holomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-027494 | MATHDB-365798

The product-cover conjecture for foliations with numerically trivial tangent bundle

摘录 / EXCERPT: Product-cover conjecture. There exist possibly noncompact Kähler manifolds Y and Z, with ω_Y trivial, and a covering map

conjecture | open | MathDB 页面 | 原文线索

V2-027495 | MATHDB-365799

Two-ends Furstenberg conjecture

摘录 / EXCERPT: Let $\delta\in(0,1)$, and let $(L,Y)_\delta$ be a set of directionally δ-separated lines in $\mathbb{R}^n$ with an $(\varepsilon_1,\varepsilon_2)$ two-ends, λ-dense...

conjecture | open | MathDB 页面 | 原文线索

V2-027496 | MATHDB-365824

Bezdek's intrinsic-volume conjecture for ball bodies with fixed inradius

摘录 / EXCERPT: Let $d>1$, $r>0$, and $0<k\leq d-2$. An r-ball body is an intersection of finitely many balls of radius r in $\mathbb{E}^d$; its inradius is the radius of the largest ball cont...

conjecture | open | MathDB 页面 | 原文线索

V2-027497 | MATHDB-365825

Bezdek's intrinsic-volume conjecture for ball bodies with fixed circumradius

摘录 / EXCERPT: Let $r>r_0>0$, $N>1$, $d>k>0$, and let $P=\{\mathbf p_1,\dots,\mathbf p_N\}\subset\mathbb{E}^d$ satisfy $r_{cr}(P)=r_0$, where r_{cr} is the circumradius. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027498 | MATHDB-365826

Strongly self-dual graph realization conjecture for Reuleaux polyhedra

摘录 / EXCERPT: A planar, 3-connected graph G is strongly self-dual if it has a self-duality isomorphism $\tau:G\to G^*$ such that, for every pair of vertices u,v, $u\in\tau(v)$ if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-027499 | MATHDB-365828

Functional symmetric L^p-Mahler conjecture

摘录 / EXCERPT: Functional symmetric L^p-Mahler conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027500 | MATHDB-365829

Functional L^p-Mahler lower-bound conjecture

摘录 / EXCERPT: Functional L^p-Mahler lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027501 | MATHDB-365831

Euclidean-ball asymptotic L^p-Mahler conjecture

摘录 / EXCERPT: Euclidean-ball asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027502 | MATHDB-365833

Banyaga's properness conjecture for strong symplectic homeomorphisms

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold. Write $G_{\omega}(M)$ for the identity component of the group of symplectic diffeomorphisms, and let $SG_{\omega}(M)$ be the group...

conjecture | open | MathDB 页面 | 原文线索

V2-027503 | MATHDB-365845

Good postulation conjecture for general codimension 2 linear subspaces

摘录 / EXCERPT: Good postulation conjecture. A finite union of general codimension 2 linear subspaces in $\mathbb P^N$ has good postulation.

conjecture | open | MathDB 页面 | 原文线索

V2-027504 | MATHDB-365847

Yazdi's virtual Euler class one conjecture

摘录 / EXCERPT: Yazdi's virtual Euler class one conjecture. For any even integral lattice point $w\in H^2(M;{\mathbb R})$ of dual Thurston norm 1, there exists some finite cover $\tilde{M}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-027505 | MATHDB-365861

Oertel's conjecture on the mixed-integer volume

摘录 / EXCERPT: Let $C\subset\mathbb{R}^{n+d}$ be a convex body and let $S=C\cap(\mathbb{Z}^n\times\mathbb{R}^d)$. For a mixed-integer set S, write … for its Oertel radius. Oertel's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027506 | MATHDB-365868

Non-abelian nilpotent Lie groups are not weakly Einstein

摘录 / EXCERPT: Nilpotent weakly Einstein conjecture. The manifold G with this metric is not weakly Einstein.

conjecture | open | MathDB 页面 | 原文线索

V2-027507 | MATHDB-365870

Classification conjecture for 5-regular Ricci-flat graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a 5-regular Ricci-flat graph. A graph is of Cartesian product type if it is the Cartesian product of a Ricci-flat 3-regular graph and a Ricci-flat 2-regular...

conjecture | open | MathDB 页面 | 原文线索

V2-027508 | MATHDB-365883

Lovett's incidence-graph conjecture for points and hyperplanes

摘录 / EXCERPT: Lovett's incidence-graph conjecture. If the graph has at least $(1-\varepsilon)n^2$ edges, then it contains a complete balanced bipartite subgraph of size at least

conjecture | open | MathDB 页面 | 原文线索

V2-027509 | MATHDB-365904

The ℓ^p-localization-algebra formulation of the coarse Baum-Connes conjecture

摘录 / EXCERPT: The ℓ^p-coarse Baum-Connes conjecture. The induced map ev_* is a K-theoretic isomorphism. This is the ℓ^p localization-algebra formulation of the conject...

conjecture | open | MathDB 页面 | 原文线索

V2-027510 | MATHDB-365916

Curvature-corrected asymptotic conjecture for magnetic Dirichlet-to-Neumann ground states

摘录 / EXCERPT: Let Ω be a regular domain in $\mathbb{R}^2$, let A_0 be a magnetic potential whose magnetic field is constant and has norm 1, and let $\lambda^{\rm DN}(bA_0,\Omega)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-027511 | MATHDB-365919

The Peano-circle conjecture for zippers

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold, let $Z^\pm$ be a zipper or P-zipper for M, and let S^1_{univ} be its universal circle. The group $\pi_1(M)$ acts on...

conjecture | open | MathDB 页面 | 原文线索

V2-027512 | MATHDB-365920

The converse flow conjecture for minimal P-zippers

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold. A minimal P-zipper is a minimal P-zipper $Z^\pm$ for M, with endpoint maps denoted $e^\pm$ and corresponding spaces $P^\pm$. The converse flow...

conjecture | open | MathDB 页面 | 原文线索

V2-027513 | MATHDB-365921

Uniformity conjecture for Galois-conjugate hyperbolic representations

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold and let $\rho:G\to \operatorname{SL}(2,\mathbb C)$ be a lift of the discrete faithful representation associated to its hyperbolic structure. Supp...

conjecture | open | MathDB 页面 | 原文线索

V2-027514 | MATHDB-365927

Leading-term conjecture for stable orbital integrals in unitary and symplectic Lie algebras

摘录 / EXCERPT: Let F be a non-Archimedean local field, let $\boldsymbol{o}$ be its ring of integers, and use the notation $B(\gamma)^{\mathrm{irred}}$, $B(\gamma)^{\mathrm{split}}$, d_i,...

conjecture | open | MathDB 页面 | 原文线索

V2-027515 | MATHDB-365947

Conjecture distinguishing connected Julia sets from Kleinian limit sets

摘录 / EXCERPT: Let $\mathcal J$ be the Julia set of a rational map and let Λ be the limit set of a Kleinian group. Suppose that $\mathcal J$ and Λ are connected and neither is hom...

conjecture | open | MathDB 页面 | 原文线索

V2-027516 | MATHDB-365972

Equivariant coarse Novikov conjecture

摘录 / EXCERPT: Equivariant coarse Novikov conjecture. The map μ_X^{Γ} is rationally injective. This is the injectivity part of the equivariant coarse Baum-Connes conjecture and implies t...

conjecture | open | MathDB 页面 | 原文线索

V2-027517 | MATHDB-365973

Equivariant coarse Baum-Connes conjecture

摘录 / EXCERPT: Let Γ be a countable discrete group acting properly on a proper metric space X. Let $KX_*^{\Gamma}(X)$ denote the universal K^{Γ}-group assembling the equivariant i...

conjecture | open | MathDB 页面 | 原文线索

V2-027518 | MATHDB-365974

Analytic equivariant coarse Novikov conjecture

摘录 / EXCERPT: Analytic equivariant coarse Novikov conjecture. The composition of π_* and the assembly map, namely the Miščenko-Kasparov assembly map ν_X^{Γ}, is injective. This is a...

conjecture | open | MathDB 页面 | 原文线索

V2-027519 | MATHDB-365991

The product-body universality conjecture for random polytopes

摘录 / EXCERPT: Let $m\in\mathbb{N}$. For $i\in\{1,\dotsc,m\}$, let $K_i\subset \mathbb{R}^{d_i}$ be a convex body whose boundary is twice differentiable with positive Gaussian curvature everywh...

conjecture | open | MathDB 页面 | 原文线索

V2-027520 | MATHDB-365992

The all-face-dimensions conjecture for block-beta random polytopes

摘录 / EXCERPT: Let $d=d_1+\cdots+d_m$, let $\bbbeta$ be the block-parameter vector, and let $\mathcal{P}_{d,d}^{\bbbeta}$ be the associated block-beta random polytope. All-face-dimensions conject...

conjecture | open | MathDB 页面 | 原文线索

V2-027521 | MATHDB-366004

Differential-transcendence classification for symmetrically self-similar graphs

摘录 / EXCERPT: Let X be a symmetrically self-similar graph with bounded geometry, and let its Green's function be the generating function for walks on X. A star is a graph consisting of finit...

conjecture | open | MathDB 页面 | 原文线索

V2-027522 | MATHDB-366020

Birational invariance conjecture for the BCOV-type invariant of involutions on irreducible holomorphic symplectic fourfolds

摘录 / EXCERPT: Let L_2 be the Beauville-Bogomolov lattice of a manifold of $K3^{[2]}$-type. Let M be an admissible sublattice of L_2, let...

conjecture | open | MathDB 页面 | 原文线索

V2-027523 | MATHDB-366021

Canonical lamination realization conjecture for the stretch unit sphere

摘录 / EXCERPT: Let Σ_g be a closed surface of genus g, let $\sigma\in\mathscr T_g$ be a hyperbolic structure, and let v be a vector in the stretch unit sphere of $T_\sigma\mathscr T_g$...

conjecture | open | MathDB 页面 | 原文线索

V2-027524 | MATHDB-366025

Non-adjacent-node merging conjecture for BMT-derived graphs

摘录 / EXCERPT: A BMT-derived graph is a graph obtained from the tree model by merging color classes of edges. Two internal nodes are non-adjacent when they are not joined by an edge in the underl...

conjecture | open | MathDB 页面 | 原文线索

V2-027525 | MATHDB-366046

The symmetry conjecture for tropical theta functions

摘录 / EXCERPT: Let (X,E) be a smooth Looijenga pair with affine interior $U=X\setminus E$, let $u:\mathcal T\to X$ be the universal torsor, and set $G=\mathcal T|_U$. Let V be the mirror to...

conjecture | open | MathDB 页面 | 原文线索

V2-027526 | MATHDB-366047

The tropical minimum conjecture for theta-function combinations

摘录 / EXCERPT: Let Θ_v be the theta functions indexed by the relevant tropical points, and let a_v be non-zero coefficients. For a linear combination $\sum a_v\Theta_v$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-027527 | MATHDB-366053

Integral Kähler currents and generalized Moishezon-ness for weakly pseudoconvex subsets

摘录 / EXCERPT: Integral Kähler-current conjecture. For relatively compact weakly pseudoconvex subsets, the existence of integral Kähler currents, equivalently singular positive line bundles, shou...

open_problem | open | MathDB 页面 | 原文线索

V2-027528 | MATHDB-366054

Besicovitch-type product inequality for closed geodesics on higher-dimensional spheres

摘录 / EXCERPT: Let $n>2$, and let (S^n,g) be an n-dimensional Riemannian sphere. A Besicovitch-type product inequality asserts that there is a constant $C(n)>0$, depending only on the dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-027529 | MATHDB-366055

Besicovitch-type inequality for a geodesic and minimal hypersurface on higher-dimensional spheres

摘录 / EXCERPT: Let $n>2$, and let (S^n,g) be an n-dimensional Riemannian sphere. A Besicovitch-type geodesic-hypersurface inequality asserts that there is a constant $C(n)>0$, depending only...

conjecture | open | MathDB 页面 | 原文线索

V2-027530 | MATHDB-366057

The Haag-Kertzer-Rickards-Stange conjecture on sporadic curvatures

摘录 / EXCERPT: Haag-Kertzer-Rickards-Stange conjecture. The sporadic set $S({\mathcal P})$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-027531 | MATHDB-366063

Projectivization by toric flips, flops, and anti-flips

摘录 / EXCERPT: Projectivization conjecture. There exists a finite sequence consisting of flips, flops, or anti-flips

conjecture | open | MathDB 页面 | 原文线索

V2-027532 | MATHDB-366066

Directed-hyperarea vector identity in $\mathbb{R}^d$

摘录 / EXCERPT: Directed-hyperarea vector identity. The directed-hyperarea vector satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027533 | MATHDB-366080

The no-(d+2)-on-a-sphere conjecture for lattice cubes

摘录 / EXCERPT: No-$(d+2)$-on-a-sphere conjecture. There is an admissible subset of $[n]^d$ with

conjecture | open | MathDB 页面 | 原文线索

V2-027534 | MATHDB-366084

Biautomaticity conjecture for the fundamental group of X_2

摘录 / EXCERPT: Let X_2 be the specific compact locally CAT(0) triangle-square complex constructed in the paper. Biautomaticity conjecture for X_2. The fundamental group $\pi_1(X_2)$ is biauto...

conjecture | open | MathDB 页面 | 原文线索

V2-027535 | MATHDB-366107

Multiplicity-one conjecture for unstable minimal-surface limits of mean curvature flow

摘录 / EXCERPT: Multiplicity-one conjecture. The convergence to the unstable minimal surface must have multiplicity 1.

conjecture | open | MathDB 页面 | 原文线索

V2-027536 | MATHDB-366125

Soberón-Takahashi conjecture on bisection by parallel hyperplanes

摘录 / EXCERPT: Soberón-Takahashi conjecture. Any $d+k-1$ finite absolutely continuous measures on $\mathbb{R}^d$ can be simultaneously bisected by the chessboard coloring induced by k or fewer...

conjecture | open | MathDB 页面 | 原文线索

V2-027537 | MATHDB-366134

The normed Calabi-Yau conjecture for toric Landau-Ginzburg models

摘录 / EXCERPT: Normed Calabi-Yau conjecture. For each Delzant polytope $\triangle$, the ∞-trace Θ on $MF(W_\triangle)$ is a normed Calabi-Yau structure.

conjecture | open | MathDB 页面 | 原文线索

V2-027538 | MATHDB-366136

The open Gromov-Witten comparison conjecture for Dirac matrix factorizations

摘录 / EXCERPT: Open Gromov-Witten comparison conjecture. There is a homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-027539 | MATHDB-366149

Unbalanced difference-body conjecture for simplices

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body, let $\lambda\in(0,1)$, and define the unbalanced difference body ... Let Δ be an n-dimensional simplex. Unbalanced difference-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-027540 | MATHDB-366174

Magnetic second-eigenvalue minimizer phase-transition conjecture

摘录 / EXCERPT: Let $B\geq0$ be a constant magnetic field, let $\Omega\subset\mathbb{R}^2$ be a bounded open domain with $|\Omega|=1$, and let $\lambda_2(\Omega,B)$ be the second eigenvalue of the...

conjecture | open | MathDB 页面 | 原文线索

V2-027541 | MATHDB-366175

General magnetic eigenvalue disk-minimizer conjecture

摘录 / EXCERPT: Let $n\geq2$, let B be a constant magnetic field, and let $\lambda_n(\Omega,B)$ denote the nth eigenvalue of the magnetic Dirichlet Laplacian on a bounded open domain...

conjecture | open | MathDB 页面 | 原文线索

V2-027542 | MATHDB-366186

Uniqueness conjecture for almost transverse pseudo-Anosov flows

摘录 / EXCERPT: Let $\mathcal{F}$ be a taut foliation of an atoroidal 3-manifold M which is not $\mathbb{R}$-covered. A foliation is almost transverse to a flow when it satisfies the almost-tran...

conjecture | open | MathDB 页面 | 原文线索

V2-027543 | MATHDB-366187

Bijection conjecture between almost transverse pseudo-Anosov flows and universal circles

摘录 / EXCERPT: Let $\mathcal{F}$ be a taut foliation of an atoroidal 3-manifold M which is not $\mathbb{R}$-covered. Let $\mathrm{PA}(\mathcal{F})$ be the set of pseudo-Anosov flows almost tran...

conjecture | open | MathDB 页面 | 原文线索

V2-027544 | MATHDB-366234

Universally tight toric-annulus conjecture for adding Giroux torsion

摘录 / EXCERPT: Let T be a convex incompressible torus in a tight contact 3-manifold (Y,ξ) such that every toric annulus neighborhood of T with convex boundary is universally tight. Conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-027545 | MATHDB-366247

The super-elliptic equality conjecture for linear subalgebras

摘录 / EXCERPT: Super-elliptic equality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027546 | MATHDB-366251

Backward-geodesic discrepancy limiting-distribution conjecture

摘录 / EXCERPT: Let $D_N(t)$ be the discrepancy between the backward-geodesic endpoint index $N(t\downarrow0)$ and the corresponding minimization index in the exclusion-process setting considered...

conjecture | open | MathDB 页面 | 原文线索

V2-027547 | MATHDB-366253

Universal backward-geodesic endpoint argmax conjecture

摘录 / EXCERPT: Universal backward-geodesic endpoint conjecture. There exists a universal constant c_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-027548 | MATHDB-366269

Mahler's conjecture for the restricted class with s-concave power in \mathcal{F}

摘录 / EXCERPT: Let $s\in(-\frac{1}{n},0)$ and let $g:\mathbb{R}^{n}\to\mathbb{R}_{+}$ be even with $g^{s}\in\mathcal{F}$, where $\mathcal{F}$ is the set of convex lower semicontinuous functions...

conjecture | open | MathDB 页面 | 原文线索

V2-027549 | MATHDB-366281

Böhm-Wilking conjecture on invariant curvature cones

摘录 / EXCERPT: Böhm-Wilking's conjecture. For every $a\in[0,n/4]$, the curvature cone $\Omega(a)$ is invariant under the Hamilton ODE and preserved by Ricci flow.

conjecture | open | MathDB 页面 | 原文线索

V2-027550 | MATHDB-366289

Realizability conjecture for orthopositroids

摘录 / EXCERPT: Realizability conjecture for orthopositroids. Every orthopositroid $\mathscr{M}$ of type (k,n) is realizable: there exists $X\in\operatorname{OGr}_{+}(k,n)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-027551 | MATHDB-366296

Marked length spectrum rigidity conjecture

摘录 / EXCERPT: Let X be a closed manifold, and let h_1 and h_2 be Riemannian metrics on X with negative sectional curvature. For each such metric, let MLS_h assign to every con...

conjecture | open | MathDB 页面 | 原文线索

V2-027552 | MATHDB-366331

Nonexistence conjecture for phantom categories on noncommutative projective planes

摘录 / EXCERPT: Let A be a noncommutative projective-plane algebra, so that its noncommutative projective plane is represented by the category $\operatorname{qgr}(A)$. A phantom category is a no...

conjecture | open | MathDB 页面 | 原文线索

V2-027553 | MATHDB-366358

Conjectured eigenvalue monotonicity for a ball with an off-center hole

摘录 / EXCERPT: Ball-hole eigenvalue monotonicity conjecture. The first and second nonzero Steklov eigenvalues $\sigma_1(\Omega_{(t_1,t_2)})$ and $\sigma_2(\Omega_{(t_1,t_2)})$ decrease as the dis...

conjecture | open | MathDB 页面 | 原文线索

V2-027554 | MATHDB-366359

Conjectured eigenvalue monotonicity for an ellipse with an off-center hole

摘录 / EXCERPT: Ellipse-hole eigenvalue monotonicity conjecture. The eigenvalues $\sigma_1(\Omega_{(t_1,t_2)})$ and $\mu_2(\Omega_{(t_1,t_2)})$ decrease with respect to the distance between the ce...

conjecture | open | MathDB 页面 | 原文线索

V2-027555 | MATHDB-366373

The Reuleaux-frame conjecture for closed equilateral polygonal chains

摘录 / EXCERPT: Let C be a closed equilateral polygonal chain in the Euclidean plane, and let its diameter be the supremum of distances between points of C. Reuleaux-frame conjecture. If the d...

conjecture | open | MathDB 页面 | 原文线索

V2-027556 | MATHDB-366374

The union-of-Reuleaux-frames diameter conjecture

摘录 / EXCERPT: Union-of-Reuleaux-frames diameter conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-027557 | MATHDB-366400

Positive rotation-number conjecture for convex bicycle tracks

摘录 / EXCERPT: Let Γ be a simple strictly convex curve with hyperbolic or parabolic monodromy, and let γ be the corresponding rear track with rotation number $\rho(\gamma)$. Positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-027558 | MATHDB-366408

Alexander's sharp isoperimetric conjecture for minimal submanifolds

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^{n+m}$ be an n-dimensional smooth minimal submanifold with smooth boundary $\partial\Sigma$. Write $|\Sigma|$ for its n-dimensional volume,...

conjecture | open | MathDB 页面 | 原文线索

V2-027559 | MATHDB-366411

Donaldson-Scaduto conjecture on associative pairs of pants

摘录 / EXCERPT: Let $(X^4,\omega_1,\omega_2,\omega_3)$ be a hyperkähler $K3$ surface, and let $\alpha_1,\alpha_2,\alpha_3$ be irreducible (-2)-classes in $H^2(X^4;\mathbb{Z})$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-027560 | MATHDB-366429

Wall-chamber decomposition conjecture for the J-equation

摘录 / EXCERPT: Wall-chamber decomposition conjecture. There exists a locally finite collection of closed real submanifolds W_σ of codimension one in the Kähler cone $\mathcal K_X$-meaning...

conjecture | open | MathDB 页面 | 原文线索

V2-027561 | MATHDB-366436

The normalized BGW-number nesting conjecture

摘录 / EXCERPT: Let $g\ge2$ and let $\mathbf d$ be a partition of $g-1$. Write $C(\mathbf d)$ for the normalized BGW number associated with $\mathbf d$. Nesting conjecture. ... The inequalities asse...

conjecture | open | MathDB 页面 | 原文线索

V2-027562 | MATHDB-366448

Monotonicity conjecture for the Cheeger constant of parallel bodies

摘录 / EXCERPT: Let $\Omega\in\mathcal{K}^n$ be a convex body, let $r(\Omega)$ denote its inradius, and write $\Omega_t=\Omega\oplus tB_1$ for its parallel body. The Cheeger constant of a set A...

conjecture | open | MathDB 页面 | 原文线索

V2-027563 | MATHDB-366449

Monotonicity conjecture for the first Dirichlet eigenvalue of parallel bodies

摘录 / EXCERPT: Let $\Omega\in\mathcal{K}^n$ be a convex body, let $r(\Omega)$ denote its inradius, and write $\Omega_t=\Omega\oplus tB_1$ for its parallel body. Let $\lambda_1(\Omega)$ be the fir...

conjecture | open | MathDB 页面 | 原文线索

V2-027564 | MATHDB-366452

Quadratic norm conjecture for invariant twistor 2-forms

摘录 / EXCERPT: Quadratic norm conjecture. The norm squared

conjecture | open | MathDB 页面 | 原文线索

V2-027565 | MATHDB-366459

Uniform higher Theta vanishing for Bondal-Thomsen elements

摘录 / EXCERPT: Uniform higher Θ vanishing. If $-d\in\Theta$ and $i,j\in\{1,\dots,r\}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-027566 | MATHDB-366465

The degree-uniform bounded negativity conjecture for smooth surfaces in projective three-space

摘录 / EXCERPT: Degree-uniform bounded negativity conjecture. There exists a constant c_d, depending only on d, such that every smooth surface of degree d in $\mathbb{P}^3$ has negativity bo...

conjecture | open | MathDB 页面 | 原文线索

V2-027567 | MATHDB-366476

The manifold-with-corners conjecture for augmented stability conditions

摘录 / EXCERPT: Manifold-with-corners conjecture. The map $\ell_{(-)}(E_1,\ldots,E_n)$ induces a homeomorphism from an open neighborhood of σ onto an open subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-027568 | MATHDB-366478

The manifold-with-corners conjecture at admissible points

摘录 / EXCERPT: Manifold-with-corners conjecture. There is an open neighborhood U of σ such that this map is a homeomorphism from U onto an open subset of $\mathcal{A}^{\mathbb{R}}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-027569 | MATHDB-366481

The Hermitian space-form conjecture

摘录 / EXCERPT: Hermitian space-form conjecture. If the Chern connection of M has constant holomorphic sectional curvature, then M must be either Kähler, hence a complex space form, or Chern f...

conjecture | open | MathDB 页面 | 原文线索

V2-027570 | MATHDB-366485

Minimal separation dimension for Ahlfors-regular fractals

摘录 / EXCERPT: Minimal separation dimension. Define the minimal separation dimension of F by

conjecture | open | MathDB 页面 | 原文线索

V2-027571 | MATHDB-366486

Maximum-perimeter convex subset conjecture for triangles

摘录 / EXCERPT: Let Ω be a triangle with vertices A, B, and C. Assume that its diameter is the side $[AB]$ and that $\widehat{CAB}\geq\widehat{CBA}$. For $c\in[0,|\Omega|]$, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-027572 | MATHDB-366494

The elliptic cluster structure conjecture for q-isomonodromic systems

摘录 / EXCERPT: Let q-isomonodromic systems have monodromy manifolds, and let elliptic dilogarithms be the generating functions associated with the proposed transformations. Elliptic cluster str...

conjecture | open | MathDB 页面 | 原文线索

V2-027573 | MATHDB-366510

Volume monotonicity conjecture for alternating hyperbolic n-bongles

摘录 / EXCERPT: Let an alternating hyperbolic n-bongle have k innies and $n-k$ outies, where $n\geq 3$ and $k\leq n/2$. Volume monotonicity conjecture. For fixed n, the volume increases as...

conjecture | solved | MathDB 页面 | 原文线索

V2-027574 | MATHDB-366511

Asymptotic volume conjecture for hyperbolic n-bongles

摘录 / EXCERPT: Let $(v_3,v_4,v_5,\dots)$ be a sequence such that v_n is the volume of some hyperbolic n-bongle. Asymptotic volume conjecture. The normalized volumes satisfy ... The stated limit...

conjecture | solved | MathDB 页面 | 原文线索

V2-027575 | MATHDB-366525

Chen-Zheng conjecture for Bismut connections

摘录 / EXCERPT: Chen-Zheng conjecture. If the holomorphic sectional curvature of ∇^b is a non-zero constant, then g must be Kähler, hence a complex space form.

conjecture | open | MathDB 页面 | 原文线索

V2-027576 | MATHDB-366538

Analytic termination conjecture for the minimal model program for log canonical pairs

摘录 / EXCERPT: Analytic MMP termination conjecture. After shrinking Y around W, there exists a finite sequence of steps of a $(K_X+\Delta)$-minimal model program over Y around...

conjecture | open | MathDB 页面 | 原文线索

V2-027577 | MATHDB-366542

The constant mixed curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Let (M^{n},g) be a compact Hermitian manifold with $n\geq2$. Let α and β be fixed real numbers with $\beta\neq0$, and let $\mathcal{C}_{\alpha,\beta}$ denote the mix...

conjecture | open | MathDB 页面 | 原文线索

V2-027578 | MATHDB-366576

Differentiability and orthogonality conjectures for positive intersection products

摘录 / EXCERPT: Differentiability and orthogonality conjectures. The following two assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-027579 | MATHDB-366587

Linear bound conjecture for weighted Euclidean-distance equilibria in the plane

摘录 / EXCERPT: Linear weighted-distance conjecture. The number of equilibria of the weighted Euclidean distance function defined by n points with positive real weights in $\mathbb{R}^2$ is at m...

conjecture | open | MathDB 页面 | 原文线索

V2-027580 | MATHDB-366588

Linear slice conjecture for electrostatic potentials

摘录 / EXCERPT: Linear slice conjecture. The restriction of V to any straight line or flat plane in $\mathbb{R}^3$ has at most $O(n)$ equilibria.

conjecture | open | MathDB 页面 | 原文线索

V2-027581 | MATHDB-366596

Mean-curvature blow-up conjecture for embedded flows in dimensions at most six

摘录 / EXCERPT: Let M_t be a closed smooth embedded mean curvature flow in $\mathbb{R}^{n+1}$, with $n dleq 6$, and suppose the flow develops a first finite singular time. Mean-curvature blow-up...

conjecture | open | MathDB 页面 | 原文线索

V2-027582 | MATHDB-366612

The centroid-section conjecture for pairs of convex bodies

摘录 / EXCERPT: Let $K_1,K_2\subset\mathbb R^n$, $n\geq 2$, be different convex bodies. For every affine hyperplane $H\in\operatorname{Graff}_{n-1}(\mathbb R^n)$ that intersects both bodies, consi...

conjecture | open | MathDB 页面 | 原文线索

V2-027583 | MATHDB-366631

RCD no-\texorpdfstring{$\mathbb{RP}^2$}{RP2} cone conjecture

摘录 / EXCERPT: RCD no-$\mathbb{RP}^2$ cone conjecture. Then $Y\approx S^2$ or $Y\approx \overline{D}^2$. If the X_i have empty boundaries, only the first possibility can occur.

conjecture | open | MathDB 页面 | 原文线索

V2-027584 | MATHDB-366632

RCD versus smooth positive Ricci curvature conjecture

摘录 / EXCERPT: RCD versus smooth curvature conjecture. Such an n and M^n exist.

conjecture | open | MathDB 页面 | 原文线索

V2-027585 | MATHDB-366665

Conjecture on generic continuity of magnitude for finite subsets of positive definite normed spaces

摘录 / EXCERPT: Let U be a finite-dimensional, positive definite normed space. For a finite subset $X\subseteq U$, write $|X|$ for its magnitude, and equip the set of finite subsets of U with...

conjecture | open | MathDB 页面 | 原文线索

V2-027586 | MATHDB-366666

Signed Euler characteristic conjecture for Kähler manifolds with nonpositive bisectional curvature

摘录 / EXCERPT: Signed Euler characteristic conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-027587 | MATHDB-366678

The Higgs-branch conjecture for Sicilian theories

摘录 / EXCERPT: Let G be a simple group, let g be the genus of a Riemann surface, and let $e^{(1)},\dots,e^{(p)}$ be nilpotent labels for its punctures. Write $\mathcal{M}(G^{\vee},V)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-027588 | MATHDB-366697

The AB-graph conjecture for maximal k-covers

摘录 / EXCERPT: Let $S\subseteq \mathbb{F}_2^n$ be a k-cover, and suppose that ... An AB function is a function $F\colon\mathbb{F}_2^{n/2}\to\mathbb{F}_2^{n/2}$ whose graph is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-027589 | MATHDB-366728

The canonical bundle formula conjecture

摘录 / EXCERPT: Canonical bundle formula conjecture. There is

conjecture | open | MathDB 页面 | 原文线索

V2-027590 | MATHDB-366732

The non-solvsoliton conjecture for deformed symmetric supergravity c-map spaces

摘录 / EXCERPT: Non-solvsoliton conjecture. If the one-loop deformation has $c>0$ and dimension $4n>4$, then every hypersurface orbit of its cohomogeneity-one action, with the induced metric, is n...

conjecture | open | MathDB 页面 | 原文线索

V2-027591 | MATHDB-366736

Conjecture on the Euclidean relative sigma-constant of the solid torus

摘录 / EXCERPT: Let T be a solid torus, i.e., a handlebody with one handle. Define ... where the supremum is taken over all smooth, bounded domains $\Omega\subset\mathbb{R}^3$ whose closure is dif...

conjecture | open | MathDB 页面 | 原文线索

V2-027592 | MATHDB-366738

The CFKRS recipe prediction for symplectic shifted moments

摘录 / EXCERPT: CFKRS conjecture. If $|\operatorname{Re}(s)-\frac12|\ll 1/\log X$ for all $s\in S$, then as $X\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-027593 | MATHDB-366748

Fractal-time local smoothing conjecture for the half-wave propagator

摘录 / EXCERPT: Fractal-time local smoothing conjecture. For every $s>s_E(p,q)$, there exists a constant $C_{s,p,q}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027594 | MATHDB-366749

Functoriality of algebraic Lagrangian cobordism under correspondences

摘录 / EXCERPT: Let M and M' be symplectic manifolds, and let $\overline{M}$ denote M with its symplectic form reversed. Let $Y\subset\overline{M}\times M'$ be an oriented Lagrangian corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-027595 | MATHDB-366765

Reverse Faber-Krahn conjecture for the negative logarithmic eigenvalue

摘录 / EXCERPT: Let Ω be a bounded domain and let H be a polarizer. The reverse Faber-Krahn conjecture. ... The inequality is immediate in the cases discussed in the source when the transfi...

conjecture | open | MathDB 页面 | 原文线索

V2-027596 | MATHDB-366766

Reverse Faber-Krahn conjecture for the logarithmic potential operator

摘录 / EXCERPT: Let Ω be a bounded domain, let Ω^* denote the comparison domain of the same area, and let B_R be the comparison ball for a domain whose transfinite diameter is R....

conjecture | open | MathDB 页面 | 原文线索

V2-027597 | MATHDB-366790

Generic density conjecture for wave fronts on Riemannian manifolds

摘录 / EXCERPT: Let M be a d-dimensional Riemannian manifold with or without boundary, where $d\geq 2$, and let $W_t(P)$ denote the wave front at time t from a point $P\in M$. The wave front...

conjecture | open | MathDB 页面 | 原文线索

V2-027598 | MATHDB-366791

Wave-front density conjecture for smooth metrics on tori

摘录 / EXCERPT: Let $\mathbb{T}^d$ be the d-dimensional torus with $d>1$, equipped with a smooth Riemannian metric, and let $W_t(P)$ denote the wave front at time t from a point...

conjecture | open | MathDB 页面 | 原文线索

V2-027599 | MATHDB-366807

Markov Sail Duality conjecture for Markov-polynomial coefficients

摘录 / EXCERPT: Markov Sail Duality conjecture. The coefficients of Markov polynomials satisfy these two Markov-sail duality identities. This would determine almost all interior sail coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-027600 | MATHDB-366808

Location of 4 conjecture for Markov sails

摘录 / EXCERPT: Let $b/a=[a_1,a_2,\dots,a_n]$ be the continued-fraction expansion of the rational number b/a, and let A_m,B_m be the corresponding penultimate-continuant vertices of the Markov...

conjecture | open | MathDB 页面 | 原文线索

V2-027601 | MATHDB-366813

Druel's conjecture on the dimension of singular loci of foliations

摘录 / EXCERPT: Druel's conjecture. The stated dimension bound holds for the singular locus of every such foliation.

conjecture | open | MathDB 页面 | 原文线索

V2-027602 | MATHDB-366815

Beauville-Bondal conjecture on low-rank loci of Poisson manifolds

摘录 / EXCERPT: Beauville-Bondal conjecture. The stated dimension bound holds for every such low-rank locus of a projective Poisson manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-027603 | MATHDB-366824

Characterization conjecture for axially symmetric Helfrich genus-zero immersions

摘录 / EXCERPT: Let Σ be a closed genus-zero surface and let $X:\Sigma\longrightarrow\mathbb{R}^3$ be an axially symmetric immersion with non-constant mean curvature. Let $\mathcal{H}$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-027604 | MATHDB-366832

The singular-locus conjecture for the ramification hypersurface of a genus-sixteen K3 surface

摘录 / EXCERPT: Let S be a generic K3 surface of genus 16, with associated discriminant hypersurface D_γ and dual ramification hypersurface D^*_γ. Let X be the associated varie...

conjecture | open | MathDB 页面 | 原文线索

V2-027605 | MATHDB-366838

Bandyopadhyay-Dacorogna-Matveev-Troyanov conjecture on flat degenerate metrics

摘录 / EXCERPT: Let M be a closed manifold equipped with a flat, possibly degenerate, non-negative definite metric g of constant rank m. Bandyopadhyay-Dacorogna-Matveev-Troyanov conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027606 | MATHDB-366873

Real-analyticity conjecture for coefficients of two-dimensional superintegrable metrics

摘录 / EXCERPT: Real-analyticity conjecture. On the complement of a discrete set of points, the functions λ, a_i, and b_j are real-analytic in the variables x_1,x_2.

conjecture | open | MathDB 页面 | 原文线索

V2-027607 | MATHDB-366882

Nonsmooth splitting conjecture for infinitesimally Minkowskian TCD spaces

摘录 / EXCERPT: Let $(\mathsf{M},l,\mathfrak{m})$ be a metric measure spacetime. It is infinitesimally Minkowskian if, for all l-causal functions u and v, its maximal weak subslope satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-027608 | MATHDB-366893

Yau's conjecture on the Strominger system through conifold transitions

摘录 / EXCERPT: Let $\hat{X}$ be a Calabi-Yau threefold, and let ... be a conifold transition. Let (Ω_t,ω_t,h_t) denote the holomorphic volume form, Hermitian form, and bundle metric inv...

conjecture | open | MathDB 页面 | 原文线索

V2-027609 | MATHDB-366912

Pointwise Weyl-law improvement for selected surfaces of revolution

摘录 / EXCERPT: Pointwise Weyl-law improvement conjecture. There exists a quantifiable constant $\kappa>0$ such that, for every $x\in K_\varepsilon$,

conjecture | open | MathDB 页面 | 原文线索

V2-027610 | MATHDB-366927

Fujita's conjecture A_n on Kähler compactifications of contractible manifolds

摘录 / EXCERPT: Let U) be an n-dimensional contractible complex manifold and let (M,D) be a Kähler smooth compactification. Fujita's conjecture A_n. Then ... and (M,D) is the standard exam...

conjecture | open | MathDB 页面 | 原文线索

V2-027611 | MATHDB-366951

Godbersen's conjecture for mixed volumes of a convex body and its reflection

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body, and let V denote mixed volume, with $K[k]$ denoting k copies of K. Godbersen's conjecture. For each convex body...

conjecture | open | MathDB 页面 | 原文线索

V2-027612 | MATHDB-366952

The higher-rank Godbersen conjecture for Cartesian powers

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body. Let $\Delta_p:\mathbb{R}^n\to(\mathbb{R}^n)^p$ be the diagonal embedding, and let $\iota_i:\mathbb{R}^n\to(\mathbb{R}^n)^p$ be inclusio...

conjecture | open | MathDB 页面 | 原文线索

V2-027613 | MATHDB-366958

Cao-Höring's conjecture on twisted relative canonical bundles

摘录 / EXCERPT: Let $p:X\to Y$ be a holomorphic surjective map between two compact Kähler manifolds. Let (L,h_L) be a line bundle over X endowed with a possibly singular metric h_L such that...

conjecture | open | MathDB 页面 | 原文线索

V2-027614 | MATHDB-366996

Complex-constructible wrapping conjecture

摘录 / EXCERPT: Complex-constructible wrapping conjecture. For every object $\mathcal E$ of $\Sh_{\mathbb C\text{-}c}(M)$, there exists an inductive system $\{\mathcal E_i\}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-027615 | MATHDB-366997

Microsheaf wrapping conjecture for complex Lagrangians

摘录 / EXCERPT: Microsheaf wrapping conjecture. There exists an inductive system $\{\mathcal E_i\}$ in $\mu\operatorname{sh}_{\mathbb C\text{-}c}(X)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027616 | MATHDB-366998

Hodge lift conjecture for GPS objects

摘录 / EXCERPT: Hodge lift conjecture. For any complex Lagrangian L, the object

conjecture | open | MathDB 页面 | 原文线索

V2-027617 | MATHDB-366999

Core--cocore Koszul duality conjecture

摘录 / EXCERPT: Core--cocore Koszul duality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027618 | MATHDB-367042

The simplicial-volume conjecture for closed affine manifolds

摘录 / EXCERPT: Let M be a closed affine manifold, and let $\|M\|$ denote its simplicial volume. Simplicial-volume conjecture. The simplicial volume of M vanishes: ... This is a stronger form of...

conjecture | open | MathDB 页面 | 原文线索

V2-027619 | MATHDB-367059

The prescribed mean curvature infinitude conjecture

摘录 / EXCERPT: Let (M^{n+1},g) be the ambient Riemannian manifold under consideration, let $h\in C^{\infty}(M)$, and let H_{Y_p} denote the mean curvature of a hypersurface $Y_p\subset M$. Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-027620 | MATHDB-367067

Conjectural trichotomy for bicontact structures supporting Anosov flows

摘录 / EXCERPT: Let (xi^+,xi^-) be a bicontact structure supporting an Anosov flow X. The bicontact trichotomy conjecture. Exactly one of the following holds: 1. xi^+ is Anosov contact, it a...

conjecture | open | MathDB 页面 | 原文线索

V2-027621 | MATHDB-367068

Conjecture on skew Anosov flows and regulating Reeb flows

摘录 / EXCERPT: Let (ξ^+,ξ^-) be a bicontact structure supporting an Anosov flow X. A Reeb flow is bitransverse when it has the bitransversality property required in the paper, and a flow...

conjecture | open | MathDB 页面 | 原文线索

V2-027622 | MATHDB-367070

Existence conjecture for negative scalar curvature metrics on manifolds of positive type

摘录 / EXCERPT: Existence conjecture. Given $K<0$ and $c>\sqrt{-(n-2)K/n}$, there exists a smooth solution $u>0$ to equations.

conjecture | open | MathDB 页面 | 原文线索

V2-027623 | MATHDB-367071

Uniform a priori estimate conjecture for the boundary conformal equation

摘录 / EXCERPT: Uniform a priori estimate conjecture. There exists $C(M,g,\delta)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027624 | MATHDB-367080

Berglund's transpolar VEX multitope conjecture

摘录 / EXCERPT: Berglund's VEX multitope conjecture. For an L-lattice VEX multitope $\mathcal D$: (1) $\mathcal D^\widetilde{\ }\subset L^\vee_\mathbb R$ is a VEX multitope; (2)...

conjecture | open | MathDB 页面 | 原文线索

V2-027625 | MATHDB-367100

The rational curves conjecture for projective IHS manifolds

摘录 / EXCERPT: Let (X,H) be a projective irreducible holomorphic symplectic manifold over the complex numbers, with H a polarization. The rational curves conjecture for projective IHS manifol...

conjecture | open | MathDB 页面 | 原文线索

V2-027626 | MATHDB-367143

Oberlin's restricted projection conjecture in three dimensions

摘录 / EXCERPT: Let $t\in[0,3]$ and let $s\in\left[0,\min\{t,1\}\right]$. For a Borel set $A\subseteq\mathbb{R}^3$ with $\dim A=t$, write $\rho_\theta(A)$ for its restricted projection at angle...

conjecture | open | MathDB 页面 | 原文线索

V2-027627 | MATHDB-367144

Furstenberg dimension conjecture for sine wave and circular sets

摘录 / EXCERPT: Let $0<u\leq1$ and $0\leq v\leq3$. A sine wave (u,v)-Furstenberg set or circular (u,v)-Furstenberg set is a set $F\subseteq\mathbb{R}^2$ of the type specified in the paper, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-027628 | MATHDB-367145

Spectral convergence conjecture for random covers of negatively curved surfaces

摘录 / EXCERPT: Let X be a closed negatively curved surface with universal cover $\widetilde{X}$, and let $\lambda_0=\lambda_0(\widetilde{X})$ be the bottom of the L^2-spectrum of the Laplacia...

conjecture | open | MathDB 页面 | 原文线索

V2-027629 | MATHDB-367152

Gross-Pavanelli mirror conjecture for the Heisenberg quotient Calabi-Yau threefold

摘录 / EXCERPT: Gross-Pavanelli's mirror conjecture. The mirror of X is $\check{X}$, and the mirror of $\check{X}$ is Y.

conjecture | open | MathDB 页面 | 原文线索

V2-027630 | MATHDB-367161

Conjecture on a split limit mixed Hodge structure at an ideal point

摘录 / EXCERPT: Let M be the relevant hyperbolic three-manifold, let $X=\tilde{X}(M)_0^{\mathrm{sm}}$, and let P be an ideal point of X. Choose a local uniformizer z at P with...

conjecture | open | MathDB 页面 | 原文线索

V2-027631 | MATHDB-367233

Detcherry's finiteness conjecture for link exteriors

摘录 / EXCERPT: Let $R=Q(q^{\frac{1}{2}})$, let $L\subset S^3$ be a link with n components, and let E_L denote its exterior. For each component, let $t_i\in\mathcal{S}(E_L)$ be the meridian of...

conjecture | open | MathDB 页面 | 原文线索

V2-027632 | MATHDB-367249

Generation conjecture for the centers of compatible Poisson brackets on $\mathfrak{gl}(n)$

摘录 / EXCERPT: Generation conjecture. The families of functions listed in the cited theorems and remarks generate the corresponding centers. Equivalently, it suffices to verify that, in each case...

conjecture | open | MathDB 页面 | 原文线索

V2-027633 | MATHDB-367257

Perfect-pairing conjecture for logarithmic K-theory

摘录 / EXCERPT: Let X be as above, let $d=\dim(X)$, and take integers m,m',r,r' with $m,m'\geq 0$, $m+m'=2d$, and $m-2r=m'-2r'\geq 0$. The construction in the source gives a pairing ... Perfect-...

conjecture | open | MathDB 页面 | 原文线索

V2-027634 | MATHDB-367259

Semisimple decomposition conjecture for saturated log motives

摘录 / EXCERPT: Let $\mathfrak{M}_{\mathrm{sat}}$ be the category of saturated log motives, let E be a log Tate curve, and let M be an object of $\mathfrak{M}_{\mathrm{sat}}$. Saturated log-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-027635 | MATHDB-367302

The complex-curve criterion for Endo-Pajitnov manifolds

摘录 / EXCERPT: Complex-curve criterion. T_M admits complex curves if and only if the components of a are not linearly independent over $\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027636 | MATHDB-367316

Nash-Semple conjecture on resolution by Nash transformations

摘录 / EXCERPT: Let X be a pure d-dimensional $\mathbb{K}$-analytic set, where $\mathbb{K}$ is $\mathbb{C}$ or $\mathbb{R}$, and let $\mathcal{N}(X)$ denote its Nash transformation. A Nash-Sem...

conjecture | refuted | MathDB 页面 | 原文线索

V2-027637 | MATHDB-367341

The AL-structure characterization conjecture via normally hyperbolic skeleta

摘录 / EXCERPT: Let $(V,\d7omega,\d7lambda)$ be a 4-dimensional finite type Liouville manifold. Its skeleton $\d7\mathfrak{skel}$ is the set used in the source, and the Liouville flow is the flo...

conjecture | open | MathDB 页面 | 原文线索

V2-027638 | MATHDB-367344

The flat-graze conjecture for convex bodies

摘录 / EXCERPT: Let M,K be convex bodies in $\mathbb{R}^n$ with $M\subset\operatorname{int}K$ and $n\geq 3$. For $x\in\operatorname{bd}K$, let the graze $\Sigma(M,x)$ be the set of contact point...

conjecture | open | MathDB 页面 | 原文线索

V2-027639 | MATHDB-367345

Choi-Greene-Lee-Marquis local surjectivity conjecture for weakly orderable orbifolds

摘录 / EXCERPT: Let P be a polyhedron with a set of vertices $\mathcal{V}$, and suppose that P is weakly $\mathcal{V}$-orderable. Let $\mathcal{O}$ be a Coxeter orbifold structure on P with...

conjecture | open | MathDB 页面 | 原文线索

V2-027640 | MATHDB-367366

Separating-line conjecture for two point sets in the Cartesian plane

摘录 / EXCERPT: Let X_{0} and X_{1} be two sets of points in the Cartesian plane. Assume that no two line segments determined by four distinct points, with exactly two points from one set, int...

conjecture | open | MathDB 页面 | 原文线索

V2-027641 | MATHDB-367422

Semi-Riemannian Bakry-Émery warped-product formula

摘录 / EXCERPT: Let $B\times_f^{N}F$ be an N-warped product of a Riemannian manifold B, a positive warping function f, and a weighted Riemannian manifold F. The weighted semi-Riemannian an...

conjecture | open | MathDB 页面 | 原文线索

V2-027642 | MATHDB-367423

Ketterer’s synthetic curvature conjecture for general N-warped products

摘录 / EXCERPT: Let B be a metric-measure base of dimension d, let F be an essentially non-branching metric-measure space, and let f be a Lipschitz nonnegative warping function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027643 | MATHDB-367424

Euclidean cone curvature-dimension conjecture

摘录 / EXCERPT: Let X be an essentially non-branching metric-measure space, and let $C(X)$ denote its Euclidean N-cone. Euclidean cone curvature-dimension conjecture. The Euclidean N-cone...

conjecture | open | MathDB 页面 | 原文线索

V2-027644 | MATHDB-367425

Lorentzian warped-product timelike curvature equivalence

摘录 / EXCERPT: Lorentzian warped-product curvature conjecture. The Lorentzian warped product $-I\times_fF$ has timelike CBB by K if and only if f is $-Kf$-concave and F has CBB by K_F. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-027645 | MATHDB-367426

Lorentzian warped-product timelike curvature-dimension conjecture

摘录 / EXCERPT: Lorentzian warped-product timelike curvature-dimension conjecture. If $f''-Kf\leq0$ and $(F,d_F,\operatorname{m}_F)$ is non-branching and satisfies $\mathsf{CD}(K_F(N-1),N)$, then t...

conjecture | open | MathDB 页面 | 原文线索

V2-027646 | MATHDB-367444

Laugesen-Siudeja's simplicity conjecture for the second Dirichlet eigenvalue

摘录 / EXCERPT: Let T be a bounded triangle in the Euclidean plane, and let the Dirichlet eigenvalues of its Laplacian, listed with multiplicity, be ... An equilateral triangle is a triangle whose...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-027647 | MATHDB-367497

Gromov-Lück inequality for closed aspherical 4-manifolds

摘录 / EXCERPT: Let X be a closed, oriented, aspherical 4-manifold, and let $\tau(X)$ denote its signature. Gromov-Lück inequality. One has ... This inequality is a refined form of the Hopf prob...

conjecture | open | MathDB 页面 | 原文线索

V2-027648 | MATHDB-367499

Counterexamples to the Modified Bellows Conjecture from exotic flexible cross-polytopes

摘录 / EXCERPT: Let $n\ge 4$, and consider exotic flexible cross-polytopes in the sphere $\mathbb{S}^n$. These are flexible cross-polytopes whose dihedral-angle tangents have the exotic parametriz...

conjecture | refuted | MathDB 页面 | 原文线索

V2-027649 | MATHDB-367501

The asymptotic two-smallest-distances conjecture for planar point sets

摘录 / EXCERPT: Let S be a set of n points in the plane. Let $m_1(S)$ and $m_2(S)$ be the numbers of occurrences of the smallest and second smallest distances in S, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027650 | MATHDB-367523

The Nikodym dimension conjecture for constant-curvature manifolds

摘录 / EXCERPT: Let (M^d,g) be a Riemannian manifold of dimension d and constant sectional curvature, and let a Nikodym set be a Borel set whose associated Nikodym enlargement has positive mea...

conjecture | open | MathDB 页面 | 原文线索

V2-027651 | MATHDB-367581

Classification conjecture for compact simply-connected 4-dimensional Ricci solitons

摘录 / EXCERPT: Classification conjecture. If a compact simply-connected 4-dimensional Ricci soliton admits an invariant cohomogeneity one action, then it is isometric to a rescaling of one of t...

conjecture | open | MathDB 页面 | 原文线索

V2-027652 | MATHDB-367619

Quantized volume comparison conjecture for K-semistable Fano manifolds

摘录 / EXCERPT: Quantized volume comparison conjecture. For every $m\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-027653 | MATHDB-367635

The converse Alexandrov-to-distributional sectional curvature conjecture for C^1 metrics

摘录 / EXCERPT: Converse curvature conjecture. The distributional sectional curvature of (M,g) is bounded from below (respectively, above) by k.

conjecture | open | MathDB 页面 | 原文线索

V2-027654 | MATHDB-367646

The full-symmetric monodromy conjecture for optimal triangulation

摘录 / EXCERPT: Full-symmetric monodromy conjecture. The cover π has full-symmetric Galois/monodromy group, and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-027655 | MATHDB-367658

Corti-Filip-Petracci smoothing-component conjecture

摘录 / EXCERPT: Corti-Filip-Petracci smoothing-component conjecture. Smoothing components of $\operatorname{Def}(X_0)$ are in bijection with zero mutable Laurent polynomials f with Newton polyto...

conjecture | open | MathDB 页面 | 原文线索

V2-027656 | MATHDB-367659

Compatible-extraction conjecture for zero mutable Laurent polynomials

摘录 / EXCERPT: Compatible-extraction conjecture. If the partitions $(\mathbf p_e)_e$ are given by $\mathbf{divstep}_e(f)^\vee$, then there exists a compatible collection $Y_i\to X_i$ of divisoria...

conjecture | open | MathDB 页面 | 原文线索

V2-027657 | MATHDB-367660

Spike case of the compatible-extraction conjecture

摘录 / EXCERPT: Spike-case conjecture. The compatible-extraction conjecture holds for all cases in the classification theorem called Spike.

conjecture | open | MathDB 页面 | 原文线索

V2-027658 | MATHDB-367670

The orthosymplectic tensor-power conjecture for the quotient enveloping algebra

摘录 / EXCERPT: Let U be the superspace equipped with its nondegenerate quadratic element ω, let E be a vector space of dimension n, and set ... The elements of S^2E are central, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027659 | MATHDB-367678

Generalized Gompf conjecture for 4-dimensional 2-handlebodies

摘录 / EXCERPT: Let W and W' be 4-dimensional 2-handlebodies. Two such handlebodies are 2-equivalent when they are related by a finite sequence of isotopies, handle slides, and creations...

conjecture | open | MathDB 页面 | 原文线索

V2-027660 | MATHDB-367701

Existence and uniqueness conjecture for prescribed curvature equations with a subsolution

摘录 / EXCERPT: Existence and uniqueness conjecture. There exists a unique k-admissible solution u satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-027661 | MATHDB-367702

The slice Akbulut-Kirby conjecture

摘录 / EXCERPT: Let K and K' be knots in S^3, and let $S^3_0(K)$ denote the 3-manifold obtained by zero surgery on K. A knot is slice if it bounds a smoothly embedded disk in B^4. Slic...

conjecture | open | MathDB 页面 | 原文线索

V2-027662 | MATHDB-367703

The obstruction conjecture for strongly exotic H-slice trace surgeries

摘录 / EXCERPT: Let K and K' be knots in S^3, and let $S^3_0(K)$ denote the 3-manifold obtained by zero surgery on K. A knot is H-slice in a 4-manifold X if it bounds a smoothly embe...

conjecture | open | MathDB 页面 | 原文线索

V2-027663 | MATHDB-367704

The contact annulus twist-Luttinger surgery conjecture for Stein traces

摘录 / EXCERPT: Let a Stein trace be the Stein 4-manifold obtained from a Legendrian knot by attaching a Stein 2-handle. Contact annulus twists modify the knot and its contact data, while Lutt...

conjecture | open | MathDB 页面 | 原文线索

V2-027664 | MATHDB-367705

The local systems of categories embedding conjecture

摘录 / EXCERPT: Let M be a manifold. Let $\mathrm{LocSysCat}(M)$ denote the $(\infty,2)$-category of local systems of categories on M, and let $^{2}\mathrm{Fuk}(\mathrm{T}^{*}[1]M)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-027665 | MATHDB-367706

The Nadler-Zaslow categorification conjecture for schobers

摘录 / EXCERPT: Let M be a manifold and let $\mathcal{S}$ be a stratification of M. Let $\mathrm{Schobers}(M,\mathcal{S})$ denote the $(\infty,2)$-category of schobers on M constructible wit...

conjecture | open | MathDB 页面 | 原文线索

V2-027666 | MATHDB-367717

L-equivalence implies derived equivalence for projective hyperkähler manifolds

摘录 / EXCERPT: L-equivalence conjecture for hyperkähler manifolds. If X and Y are L-equivalent, then their bounded derived categories of coherent sheaves are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-027667 | MATHDB-367725

Countable-exceptional-set conjecture for projections of planar self-similar sets

摘录 / EXCERPT: Let K be a self-similar set in the plane, and call a direction $\pi\in G(2,1)$ exceptional for orthogonal projections of K if the projection dimension equality does not hold. C...

conjecture | open | MathDB 页面 | 原文线索

V2-027668 | MATHDB-367735

Drézet's homogeneity conjecture for intermediate-rank uniform vector bundles

摘录 / EXCERPT: Drézet's conjecture. Every such uniform vector bundle E is homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-027669 | MATHDB-367747

The existence conjecture for codimension-one singular Riemannian foliations

摘录 / EXCERPT: The existence conjecture. Every simply connected Riemannian manifold $(M,\mathtt{g})$ with positive (respectively, non-negative) sectional curvature admits a codimension-one singul...

conjecture | open | MathDB 页面 | 原文线索

V2-027670 | MATHDB-367748

Weak contact-geometric CROSS conjecture for Reeb flows

摘录 / EXCERPT: Let M be a manifold of dimension at least 2. Let $(S^{\ast}M,\xi)$ be its spherical cotangent bundle with the standard contact structure, let α be a contact form with...

conjecture | open | MathDB 页面 | 原文线索

V2-027671 | MATHDB-367749

Weak positive-Legendrian-isotopy CROSS conjecture

摘录 / EXCERPT: Let M be a manifold of dimension at least 2. Let $(S^{\ast}M,\xi)$ be its spherical cotangent bundle with the standard contact structure, and let ϕ_t be a positive Legendr...

conjecture | open | MathDB 页面 | 原文线索

V2-027672 | MATHDB-367754

The p-curvature and Frobenius-constant quantization correspondence

摘录 / EXCERPT: Let $M_{eq,reg}$ be the D-module of twisted traces, let $\mathcal{A}_0^2$ be the degree-2 part of the relevant quantization algebra, and let s be the Artin--Schreier map defi...

conjecture | open | MathDB 页面 | 原文线索

V2-027673 | MATHDB-367832

Finite four-point obstruction conjecture for labeled star graph ultrametric spaces

摘录 / EXCERPT: Let (X^*,d^*) be a finite ultrametric space. The space X^* is a US-space when it belongs to the class ${\bf US}$ of ultrametric spaces generated by labeled star graphs. A four-...

conjecture | open | MathDB 页面 | 原文线索

V2-027674 | MATHDB-367856

Gardner-Zvavitch conjecture for even log-concave measures

摘录 / EXCERPT: Let ν be an even log-concave measure on $\mathbb{R}^{n}$. For origin-symmetric convex bodies $K_{0},K_{1}\subseteq\mathbb{R}^{n}$ and $t\in[0,1]$, define ... The measure ν is...

conjecture | open | MathDB 页面 | 原文线索

V2-027675 | MATHDB-367871

The bidisk conclusion for domains without totally real boundary points

摘录 / EXCERPT: Let M be a hyperbolic complex surface, and let $\Omega\subset\subset M$ be a subdomain with $C^{1,1}$ boundary. A compact quotient of Ω means that Ω admits a quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-027676 | MATHDB-367873

The internal stabilization conjecture for exotically knotted surfaces

摘录 / EXCERPT: Let B^{4} be the smooth 4-ball. For an exotically knotted pair of closed surfaces (Σ_{1},Σ_{2}) in B^{4}, let $d(\Sigma_{1},\Sigma_{2})$ denote the least nonnegat...

conjecture | open | MathDB 页面 | 原文线索

V2-027677 | MATHDB-367874

The external stabilization conjecture for exotic 4-manifolds

摘录 / EXCERPT: Let (X_{0},X_{1}) be an exotic pair of closed, simply-connected smooth 4-manifolds, meaning that they are homeomorphic but not diffeomorphic. Let $X\#Y$ denote the connected su...

conjecture | open | MathDB 页面 | 原文线索

V2-027678 | MATHDB-367908

Small tree-foliation conjecture for positively curved Riemannian 3-spheres

摘录 / EXCERPT: Let (S^3,g) be a Riemannian 3-sphere with scalar curvature $R_g\geq \Lambda_0>0$. A tree-foliation $\{\Sigma_t\}_{t\in T}$ is a foliation of (S^3,g) by surfaces whose parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-027679 | MATHDB-367929

Global Lipschitz continuity of the spherical cap discrepancy

摘录 / EXCERPT: Let $\mathbb{S}^{(d-1)}$ be the sphere and let $\Delta:\mathbb{S}^{(d-1)}\to [0,1]$ denote the spherical cap discrepancy. Global Lipschitz continuity conjecture. The spherical cap...

conjecture | open | MathDB 页面 | 原文线索

V2-027680 | MATHDB-367950

Existence and uniqueness conjecture for submanifolds of nearly Kähler \(\mathbb{C}P^3\)

摘录 / EXCERPT: Existence and uniqueness conjecture. There exists an isometric immersion $f:M\to(\mathbb{C}P^3,g,J)$ and a vector bundle isometry $\varphi:\mathcal{E}\to T^\perp M$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-027681 | MATHDB-368006

Ball-support conjecture for simple fractional-Laplacian eigenvalue minimizers

摘录 / EXCERPT: Let $k\geq 2$, let A be a λ_k-minimizer, and let u_k be the corresponding eigenfunction. Assume that λ_k is simple. Ball-support conjecture. For every copy...

conjecture | open | MathDB 页面 | 原文线索

V2-027682 | MATHDB-368021

Turaev-Viro invariants volume conjecture for compact orientable 3-manifolds

摘录 / EXCERPT: Turaev-Viro invariants volume conjecture. For every such M,

conjecture | open | MathDB 页面 | 原文线索

V2-027683 | MATHDB-368026

Existence of good minimal models for klt adjoint foliated structures

摘录 / EXCERPT: Let $(X,\mathcal{F},t)$ be a projective klt adjoint foliated structure, and set ... Assume that K is pseudo-effective and $t<1$. Existence of good minimal models. The structure...

conjecture | open | MathDB 页面 | 原文线索

V2-027684 | MATHDB-368040

Absolute profinite rigidity conjecture for lattices in ${\rm PSL}(2,\mathbb R)$ and ${\rm PSL}(2,\mathbb C)$

摘录 / EXCERPT: Absolute profinite rigidity conjecture for lattices. All lattices in

conjecture | open | MathDB 页面 | 原文线索

V2-027685 | MATHDB-368066

Yang's BC-positivity characterization conjecture

摘录 / EXCERPT: Let X be a compact Kähler manifold, let T_X denote its holomorphic tangent bundle, and let BC-1 positivity refer to the paper's curvature-positivity notion for T_X. Yang's co...

conjecture | open | MathDB 页面 | 原文线索

V2-027686 | MATHDB-368071

Sufficiency of the FLM inequality's necessary exponent condition

摘录 / EXCERPT: Let $a,b,c>0$ and suppose ... For a sequence of polytopes $P_n\subset\mathbb{R}^n$, write V_n and $\mathcal{F}_n$ for its vertices and facets, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027687 | MATHDB-368115

The sufficient condition for finite generation of simple irrational polyhedral cones

摘录 / EXCERPT: Finite-generation conjecture. The cone C is (R,G)-finitely generated if and only if there exists $A\in\mathbb{Q}^{n\times n}$ and $\lambda_1,\dots,\lambda_n\in\mathbb{R}_{>0}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027688 | MATHDB-368170

Aubry et al.'s point-count conjecture for weighted hypersurfaces

摘录 / EXCERPT: Aubry et al.'s point-count conjecture. If $1=w_0\leq w_1\leq w_2\leq\cdots\leq w_n$ and $\operatorname{lcm}(w_1,w_2,\ldots,w_n)\mid d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-027689 | MATHDB-368171

Rupert Li's congruence conjecture for weighted hypersurfaces

摘录 / EXCERPT: Let $S=\mathbb{F}_q[x_0,\ldots,x_n]$ be graded by positive weights w_i, and let $X=V(F)\subseteq\mathbb{P}(w_0,\ldots,w_n)$ be defined by a nonzero weighted homogeneous polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-027690 | MATHDB-368185

Rifford's curvature exponent-geodesic dimension conjecture for Carnot groups

摘录 / EXCERPT: Let N_{CE} be the curvature exponent of a Carnot group, namely the least N for which the measure contraction property $\mathrm{MCP}(0,N)$ holds. Let N_{GEO}...

conjecture | open | MathDB 页面 | 原文线索

V2-027691 | MATHDB-368186

The curvature exponent-lower-bound conjecture for Carnot groups

摘录 / EXCERPT: Let N_{CE} be the curvature exponent of a Carnot group, namely the least N for which the measure contraction property $\mathrm{MCP}(0,N)$ holds. Let N_0 be the lower...

conjecture | open | MathDB 页面 | 原文线索

V2-027692 | MATHDB-368187

The nonintegral curvature exponent conjecture for step-two Carnot groups

摘录 / EXCERPT: Let N_{CE} denote the curvature exponent of a step-two Carnot group, namely the least N for which the measure contraction property $\mathrm{MCP}(0,N)$ holds. Nonintegr...

conjecture | open | MathDB 页面 | 原文线索

V2-027693 | MATHDB-368253

Drutu-Nowak conjecture on warped cones and coarse Baum-Connes surjectivity

摘录 / EXCERPT: Drutu-Nowak conjecture. The map μ_c is not surjective for $\mathcal O_\Gamma M$. The conjecture makes precise Roe’s proposed use of warped cones to produce counterexamples to t...

conjecture | open | MathDB 页面 | 原文线索

V2-027694 | MATHDB-368255

Calderón Moreno-Mond-Narváez Macarro-Castro Jiménez conjecture for free divisors

摘录 / EXCERPT: Let X be a complex analytic manifold of dimension n, and let $D\subset X$ be a free divisor. Say that D satisfies LCT when the inclusion of logarithmic into meromorphic de Rh...

conjecture | open | MathDB 页面 | 原文线索

V2-027695 | MATHDB-368257

Conjecture on the spark and bender of Paley equiangular tight frames

摘录 / EXCERPT: Paley ETF spark and bender conjecture. The spark of Φ_q is $((p+1)/2)^s$ and the bender is a $(p^s, ((p+1)/2)^s,\lambda)$-BIBD.

conjecture | open | MathDB 页面 | 原文线索

V2-027696 | MATHDB-368278

Generic projective rigidity conjecture for convex polytopes

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ be a convex polytope of dimension $d\geq 3$. A projective transformation is generic when it avoids the exceptional algebraic conditions relevant to the r...

conjecture | open | MathDB 页面 | 原文线索

V2-027697 | MATHDB-368281

The CD condition conjecture for sub-Finsler Carnot groups

摘录 / EXCERPT: A sub-Finsler Carnot group is a stratified, nilpotent Lie group equipped with a left-invariant distance. Its homogeneous dimension is denoted by $\mathcal{Q}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027698 | MATHDB-368285

Generalized Chen-Yang volume conjecture for 3-manifolds

摘录 / EXCERPT: Let M be a compact orientable 3-manifold with empty or toroidal boundary. Define ... where r runs over all odd integers, and let $\operatorname{Vol}(M)$ be the simplicial volume,...

conjecture | open | MathDB 页面 | 原文线索

V2-027699 | MATHDB-368334

Special termination near the divisorial image of the boundary floor

摘录 / EXCERPT: Let (X,B) be a projective $\mathbb{Q}$-factorial effective dlt pair. The divisor $\lfloor B\rfloor$ is the integral part of the boundary, and its image under the relevant biratio...

conjecture | open | MathDB 页面 | 原文线索

V2-027700 | MATHDB-368335

The log abundance conjecture for projective lc pairs

摘录 / EXCERPT: Let (X,B) be a projective lc pair. The divisor K_X+B is assumed to be pseudo-effective, meaning that its numerical class lies in the closure of the cone generated by effective...

conjecture | open | MathDB 页面 | 原文线索

V2-027701 | MATHDB-368340

Geometric bounds for spanning tree entropy of planar lattice graphs

摘录 / EXCERPT: Let $\mathcal G$ be a planar lattice graph with spanning tree entropy $z_{\mathcal G}$. Let $G=\mathcal G/\Lambda$ be the quotient graph on the torus, let L be the associated lin...

conjecture | open | MathDB 页面 | 原文线索

V2-027702 | MATHDB-368341

Geometric bounds conjecture for spanning trees of finite planar graphs

摘录 / EXCERPT: Let Γ be a finite connected planar graph, let $\tau(\Gamma)$ be its number of spanning trees, and let ${\rm vol}(\Gamma)={\rm vol}(S^3-K)$ for the alternating link K assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-027703 | MATHDB-368349

Uniqueness conjecture for maximally circlable sets and circles

摘录 / EXCERPT: Call a nonnegative integer n maximally circlable if there exists a largest circle enclosing exactly n lattice points in its interior. Fix the key triangle with vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-027704 | MATHDB-368350

Density conjectures for the two special classes of maximally circlable numbers

摘录 / EXCERPT: Let $f(k)$ and $g(k)$ be the integer-valued sequences defined in the paper's first and second special classes, respectively. Call a maximally circlable number strong if it is immed...

conjecture | open | MathDB 页面 | 原文线索

V2-027705 | MATHDB-368351

Conjectures on radii, densities, and symmetries of maximally circlable circles

摘录 / EXCERPT: For a nonnegative integer n, an MC-circle is a largest circle enclosing exactly n lattice points in its interior, and R_n is its radius; an integer is MC if it is maximally c...

conjecture | open | MathDB 页面 | 原文线索

V2-027706 | MATHDB-368356

Weak pinned distance conjecture

摘录 / EXCERPT: Weak pinned distance conjecture. For every $\varepsilon>0$, there exists a constant $C_\varepsilon>0$ such that every finite point set $P\subseteq\mathbb R^2$ contains a point...

conjecture | open | MathDB 页面 | 原文线索

V2-027707 | MATHDB-368357

Weak unit distance conjecture

摘录 / EXCERPT: Weak unit distance conjecture. For every $\varepsilon>0$, there exists a constant $C_\varepsilon>0$ such that every finite point set $P\subseteq\mathbb R^2$ determines at most

conjecture | open | MathDB 页面 | 原文线索

V2-027708 | MATHDB-368362

FMP stability conjecture for big movable classes

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n, and let $\alpha,\beta\in\operatorname{Mov}^1(X)$ be big movable classes. Let $F(\alpha,\beta)$ denote their relative asymmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-027709 | MATHDB-368363

Submodularity conjecture for numerical dimensions of nef and movable classes

摘录 / EXCERPT: Let X be a compact Kähler manifold. For a nef class L, write $\operatorname{nd}(L)$ for its numerical dimension, and let M be another nef class and α a movable class....

conjecture | open | MathDB 页面 | 原文线索

V2-027710 | MATHDB-368364

Pullback description conjecture for movable classes annihilated by a semiample-type class

摘录 / EXCERPT: Let $\varphi:X\rightarrow Y$ be a surjective holomorphic map between compact Kähler manifolds of dimensions n and k, respectively, and let A be a Kähler class on Y. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-027711 | MATHDB-368365

Hall-Rado criterion for positive products of movable classes

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n, and let $L_1,\ldots,L_k$ be movable classes, where $1\leq k\leq n$. For $I\subseteq [k]$, write $L_I=\sum_{i\in I}L_i$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-027712 | MATHDB-368366

FMP stability conjecture on a Lorentzian class

摘录 / EXCERPT: Let X be a compact Kähler manifold, let Ω be a Lorentzian class of dimension d on X, and let α,β be big nef classes on Ω. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-027713 | MATHDB-368371

Boundary determines odd-dimensional middle-index handlebodies

摘录 / EXCERPT: Boundary-determination conjecture. If ∂X and $\partial X'$ are diffeomorphic, then X and X' are diffeomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-027714 | MATHDB-368387

Viterbo's volume-capacity conjecture for convex domains

摘录 / EXCERPT: Let c be a symplectic capacity and let $X\subset\mathbb{R}^{2n}$ be a convex domain. Write $\operatorname{Vol}(X)$ for the Euclidean volume of X. Viterbo's conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-027715 | MATHDB-368388

The strong Viterbo conjecture for convex domains

摘录 / EXCERPT: Let $X\subset\mathbb{R}^{2n}$ be a convex domain. A symplectic capacity is normalized if it takes the same value on the unit ball and the unit symplectic cylinder. The strong Viter...

conjecture | open | MathDB 页面 | 原文线索

V2-027716 | MATHDB-368400

The star-surface support-cone characterization of ellipsoids

摘录 / EXCERPT: Let $S\subset \mathbb{R}^{n}$ be an embedding of $\mathbb{S}^{n-1}$ in $\mathbb{R}^{n}$ that is O-star, and let $\phi:S\to S$ map each point to the unique point on the ray from...

open_problem | open | MathDB 页面 | 原文线索

V2-027717 | MATHDB-368404

Convex-body characterization of Lorentzian projection volumes

摘录 / EXCERPT: Let $d\geq 3$, and let (p_{ij}) be a real symmetric $d\times d$ matrix with nonnegative off-diagonal entries and zero diagonal entries. For $i\ne j$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-027718 | MATHDB-368410

Wang-Wu's two-ends Furstenberg conjecture

摘录 / EXCERPT: Let $\delta\in(0,1)$. Let $(L,Y)_\delta$ be a set of directional δ-separated lines in $\mathbb{R}^n$ with an $(\varepsilon_1,\varepsilon_2,C_Y)$-two-ends, λ-dense sh...

conjecture | open | MathDB 页面 | 原文线索

V2-027719 | MATHDB-368419

Borisenko's isoperimetric conjecture for ball bodies

摘录 / EXCERPT: Borisenko's conjecture. Among all such sets, the sets maximizing surface area are precisely the lenses of volume V.

conjecture | open | MathDB 页面 | 原文线索

V2-027720 | MATHDB-368421

The contraction conjecture for dual volumes of ball bodies

摘录 / EXCERPT: Contraction dual-volume conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-027721 | MATHDB-368422

Bergström's conjecture on the Euclidean Steinitz constant

摘录 / EXCERPT: Let B_2^d denote the Euclidean unit ball in $\mathbb{R}^d$, and let $S(B_2^d)$ be its Steinitz constant: the smallest C such that every finite set of vectors $V\subset B_2^d$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-027722 | MATHDB-368427

Failure of the measure contraction property with strong Goh-Legendre geodesics

摘录 / EXCERPT: Failure of the measure contraction property with strong Goh-Legendre geodesics. If there is a strong-Goh-Legendre geodesic γ, then $\mathsf{d}^2:M\times M\to\mathbb{R}_+$ fa...

conjecture | open | MathDB 页面 | 原文线索

V2-027723 | MATHDB-368436

Yau's scalar-curvature growth conjecture for three-manifolds

摘录 / EXCERPT: Let (M^3,g) be a three-dimensional complete manifold with Ricci curvature $\operatorname{Ric}\geq 0$, and let $B_p(R)$ denote the geodesic ball of radius R centered at $p\in M$...

conjecture | open | MathDB 页面 | 原文线索

V2-027724 | MATHDB-368439

The non-archimedean Calabi-Yau metric convergence conjecture

摘录 / EXCERPT: Let $X\to \mathbb{D}^*_t$ be a polarised meromorphic degeneration of n-dimensional Calabi-Yau manifolds, with relatively ample line bundle L, and let X_K^{an} be the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-027725 | MATHDB-368440

The metric SYZ conjecture

摘录 / EXCERPT: Let $(X_t,\omega_{CY,t})$ be a polarised degeneration family of n-dimensional Calabi-Yau manifolds near the large complex structure limit, and let μ_t denote the normalised C...

conjecture | open | MathDB 页面 | 原文线索

V2-027726 | MATHDB-368442

The generalized Lichnerowicz conjecture

摘录 / EXCERPT: Let (M,g) be a pseudo-Riemannian manifold. Its conformal group is denoted by $\mathrm{Conf}^\uparrow(M,g)$, and this group is essential if there is no metric conformal to g for...

conjecture | open | MathDB 页面 | 原文线索

V2-027727 | MATHDB-368443

The NOH conformal flatness conjecture

摘录 / EXCERPT: Let (M,g) be a causal spacetime satisfying the No Observer Horizons condition (NOH). Its conformal group is denoted by $\mathrm{Conf}^\uparrow(M,g)$, and it is essential if there...

conjecture | open | MathDB 页面 | 原文线索

V2-027728 | MATHDB-368460

Equivariant Gordan's lemma for symmetric cones

摘录 / EXCERPT: Equivariant Gordan's lemma. The monoid M is a Sym-equivariantly finitely generated normal monoid.

conjecture | open | MathDB 页面 | 原文线索

V2-027729 | MATHDB-368480

The spherical-distance formula for the rational-map energy functional

摘录 / EXCERPT: Spherical-distance formula. The functional $\mathcal{A}(f)$ agrees, up to scale or something...?, with

conjecture | open | MathDB 页面 | 原文线索

V2-027730 | MATHDB-368498

Face-avoidance conjecture for FM-tetrahedra

摘录 / EXCERPT: An FM-tetrahedron is a tetrahedron formed by four pairwise disjoint spheres, with a support sphere tangent to all four and satisfying the FM-tetrahedron conditions described in the...

conjecture | open | MathDB 页面 | 原文线索

V2-027731 | MATHDB-368511

Expected transversality conjecture for incidence loci of complete quasimaps

摘录 / EXCERPT: Expected transversality conjecture. The intersection I is generically smooth and pure of codimension

conjecture | open | MathDB 页面 | 原文线索

V2-027732 | MATHDB-368516

Tosatti's homeomorphism conjecture for Gromov-Hausdorff limits of Calabi-Yau fibrations

摘录 / EXCERPT: Let M be a projective manifold with a semiample line bundle L inducing a holomorphic map $f:M\to X$, and let $X^\circ\subset X$ be a Zariski open subset over which f is a hol...

conjecture | open | MathDB 页面 | 原文线索

V2-027733 | MATHDB-368517

The RCD conjecture for singular Kähler metric completions

摘录 / EXCERPT: Let (X,ω_X) be an n-dimensional compact normal Kähler space with smooth Kähler metric ω_X, and let $\omega=\omega_X+\sqrt{-1}\partial\bar\partial u$ be a singular K...

conjecture | open | MathDB 页面 | 原文线索

V2-027734 | MATHDB-368546

Filling-count conjecture for twist-spun Legendrians

摘录 / EXCERPT: Let ρ be the twisting monodromy, let ρ^k denote its k-fold iterate, and let $\Sigma_{\rho^k}(\lambda(2,n))$ be the corresponding twist-spun Legendrian. Let $f(k)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-027735 | MATHDB-368547

Decomposability threshold conjecture for Legendrian doubles

摘录 / EXCERPT: Let λ be a Legendrian link such that $\mathcal{M}_1(\lambda)$ has a cluster structure. Let L_1 and L_2 be exact Lagrangian fillings of λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027736 | MATHDB-368560

Classification conjecture for compact reducible Riemannian manifolds with conformal product structures

摘录 / EXCERPT: Let (M,g) be a compact connected Riemannian manifold with reducible holonomy, and let D be a conformal product structure on (M,g) different from the Levi-Civita connection of...

conjecture | open | MathDB 页面 | 原文线索

V2-027737 | MATHDB-368561

Vanishing of the top log Bando-Futaki invariant for a momentum-constructed conical higher cscK metric

摘录 / EXCERPT: Let X) be the minimal ruled surface considered in the paper, with distinguished sections S_0 and S_∞, and let ω be a conical higher cscK metric on X with cone a...

conjecture | open | MathDB 页面 | 原文线索

V2-027738 | MATHDB-368563

Jonsson-Mustaţă's weak and strong conjectures for asymptotic log canonical thresholds

摘录 / EXCERPT: Algebraic version of Jonsson-Mustaţă's conjecture. There exists a quasi-monomial valuation $v\in\mathrm{Val}_X^*$ that computes...

conjecture | open | MathDB 页面 | 原文线索

V2-027739 | MATHDB-368564

Jonsson-Mustaţă's weighted weak and strong conjectures for log canonical thresholds

摘录 / EXCERPT: Jonsson-Mustaţă's weighted conjecture. The weak version asserts that there exists a quasi-monomial valuation $v\in\mathrm{Val}_X^*$ computing...

conjecture | open | MathDB 页面 | 原文线索

V2-027740 | MATHDB-368610

Characterisation of normal crossing divisors by weak log transversality

摘录 / EXCERPT: Let M be a complex manifold and let D be a reduced free divisor of M. A constructible function γ is weakly log transverse to D when it satisfies the weak log transve...

conjecture | open | MathDB 页面 | 原文线索

V2-027741 | MATHDB-368611

Weak log transversality for free divisors satisfying the logarithmic comparison theorem

摘录 / EXCERPT: Let M be a complex manifold, let $D\subset M$ be a free divisor, and suppose that D satisfies the logarithmic comparison theorem. Let M be regarded as the ambient constructib...

conjecture | open | MathDB 页面 | 原文线索

V2-027742 | MATHDB-368629

Shokurov's boundedness of complements conjecture

摘录 / EXCERPT: Shokurov's boundedness of complements conjecture. There exists a constant n depending only on d, p, and R such that $K_{X'}+B'+M'$ has an n-complement...

conjecture | open | MathDB 页面 | 原文线索

V2-027743 | MATHDB-368667

Large-genus Brownian-surface geodesic length spectrum conjecture

摘录 / EXCERPT: Large-genus Brownian-surface geodesic length spectrum conjecture. As $g\to\infty$, the random point process $c\,\Xi_g$ converges in distribution to $\mathcal{P}$:

conjecture | open | MathDB 页面 | 原文线索

V2-027744 | MATHDB-368668

Non-Minkowski measurability of the family C^r for all $r>2$

摘录 / EXCERPT: Let C^r denote the self-similar fractal family defined in the paper, and let Minkowski measurability mean existence of a finite, positive Minkowski content. The conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-027745 | MATHDB-368672

Generalized-cone characterization of the timelike curvature-dimension condition

摘录 / EXCERPT: Generalized-cone characterization conjecture. The space Y satisfies $\textnormal{\textsf{TCD}}(-\kappa N,N+1)$ if and only if $f”-\kappa f\leq0$ in the distributional sense in...

conjecture | open | MathDB 页面 | 原文线索

V2-027746 | MATHDB-368673

The BPS polynomial and K-theoretic refined BPS correspondence

摘录 / EXCERPT: Let S be a smooth projective surface, let $\beta\in H_2(S,\mathbb{Z})$ satisfy ... and let K_S be the total space of the canonical line bundle over S, viewed as a non-compact C...

conjecture | open | MathDB 页面 | 原文线索

V2-027747 | MATHDB-368693

Debarre-Huybrechts-Macrì-Voisin conjecture on numerical invariants of hyper-Kähler manifolds

摘录 / EXCERPT: Debarre-Huybrechts-Macrì-Voisin conjecture. Then $c_X=(2n-1)!!$ and the Huybrechts-Riemann-Roch polynomial of X is

conjecture | open | MathDB 页面 | 原文线索

V2-027748 | MATHDB-368696

Pólya's small-area conjecture for domains in the infinite cylinder

摘录 / EXCERPT: Let Ω be a bounded domain in $\mathbb{S}^{1}\times\mathbb{R}$, and write $|\Omega|$ for its area. Pólya's small-area conjecture. There exists a positive number A_{0} such...

conjecture | open | MathDB 页面 | 原文线索

V2-027749 | MATHDB-368712

Finite-subcomplex relaxation conjecture for ideal polyhedra

摘录 / EXCERPT: Finite-subcomplex relaxation conjecture. Condition (3) can be relaxed so that it is required to hold only outside a finite sub-complex of $\mathcal{D}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027750 | MATHDB-368713

Equivalence conjecture for infinite ideal polyhedra and circle patterns

摘录 / EXCERPT: Equivalence conjecture for infinite ideal polyhedra. The following statements are equivalent, with the Euclidean alternatives applying to IIP and the hyperbolic alternatives applyi...

conjecture | open | MathDB 页面 | 原文线索

V2-027751 | MATHDB-368746

Equilateral triangle conjecture for torsional rigidity under minimal width

摘录 / EXCERPT: Equilateral triangle torsional-rigidity conjecture. The equilateral triangle of unit width minimizes the torsional rigidity among shapes having unit minimal width.

conjecture | open | MathDB 页面 | 原文线索

V2-027752 | MATHDB-368783

Gromov's positivity conjecture for negatively Ricci-curved manifolds

摘录 / EXCERPT: Let M be a closed, connected, oriented n-dimensional manifold. Its simplicial volume is ... A Riemannian metric on M has negative definite Ricci curvature when its Ricci curvat...

conjecture | open | MathDB 页面 | 原文线索

V2-027753 | MATHDB-368784

Connell-Ruan-Wang's equivalence conjecture for nonpositively curved four-manifolds

摘录 / EXCERPT: Let M be a closed nonpositively curved 4-manifold, and let $\|M\|$ denote its simplicial volume and $\chi(M)$ its Euler characteristic. Connell-Ruan-Wang's equivalence conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-027754 | MATHDB-368791

Rigidity conjecture for noncompact Einstein almost Kähler manifolds

摘录 / EXCERPT: Let M be a noncompact Einstein almost Kähler manifold, and suppose that every conformal vector field on M is Killing. Rigidity Conjecture. Then M must be isometric to complex...

conjecture | open | MathDB 页面 | 原文线索

V2-027755 | MATHDB-368794

Uniqueness conjecture for capillary Gauss curvature solitons

摘录 / EXCERPT: Let $\mathcal{C}_{\theta}$ be the spherical cap and consider the capillary soliton equation referred to as Eq. ... . The preceding theorem shows convergence of the normalized capilla...

conjecture | open | MathDB 页面 | 原文线索

V2-027756 | MATHDB-368795

Uniqueness conjecture for power capillary Gauss curvature solitons

摘录 / EXCERPT: Let $\mathcal{C}_{\theta}$ be the spherical cap, let $\alpha>\frac{1}{n+2}$ and let $\theta\in(0,\pi)$. Suppose that h is a positive, strictly convex function on...

conjecture | open | MathDB 页面 | 原文线索

V2-027757 | MATHDB-368824

Lott's lower-bound conjecture for Hodge-Laplacian eigenvalues

摘录 / EXCERPT: For given $m \in \mathop{\mathrm{{\Bbb N}}}\nolimits$, $\kappa \in \mathop{\mathrm{{\Bbb R}}}\nolimits$ and $v,D>0$, consider connected oriented closed Riemannian manifolds...

conjecture | open | MathDB 页面 | 原文线索

V2-027758 | MATHDB-368841

Kollár's fiberwise bimeromorphic model conjecture for canonical families

摘录 / EXCERPT: Let $g:\mathcal X\to\Delta$ be a flat, proper, Moishezon morphism over the unit disc Δa in $\mathbb C$. Assume that X_0 has only canonical singularities. A morphism is fi...

conjecture | open | MathDB 页面 | 原文线索

V2-027759 | MATHDB-368842

Kollár's canonical model conjecture after a base change

摘录 / EXCERPT: Let $g:\mathcal X\to\Delta$ be a flat, proper, Moishezon morphism, where Δ is the unit disc in $\mathbb C$. Assume that X_0 has only canonical singularities. A morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-027760 | MATHDB-368844

Generalised Rayleigh-Faber-Krahn inequality for polyharmonic eigenvalues

摘录 / EXCERPT: Let Ω be a domain of finite measure, let $\mathbb B$ denote the unit ball in $\mathbb R^d$, and let $\omega_d=|\mathbb B|$. For $m,t\in\mathbb N$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-027761 | MATHDB-368868

Generic invariance of Fano type in families

摘录 / EXCERPT: Generic invariance conjecture. The geometric generic fiber $X_{\overline{\eta}}$ is of Fano type.

conjecture | open | MathDB 页面 | 原文线索

V2-027762 | MATHDB-368911

The standard quotient conjecture for reductive homogeneous spaces

摘录 / EXCERPT: Standard quotient conjecture. The homogeneous space G/H admits a cocompact properly discontinuous group if and only if G/H admits a compact standard quotient.

conjecture | open | MathDB 页面 | 原文线索

V2-027763 | MATHDB-368912

The Space Form Conjecture

摘录 / EXCERPT: Space Form Conjecture. There exists a compact, complete pseudo-Riemannian manifold of signature (p,q) with constant sectional curvature 1 if and only if (p,q) lies in the fol...

conjecture | open | MathDB 页面 | 原文线索

V2-027764 | MATHDB-368940

Uniqueness of the centroid-preserving triangle center for arbitrary quadrilaterals

摘录 / EXCERPT: Centroid-preservation uniqueness conjecture. The centroid of $ABCD$ coincides with the centroid of $EFGH$ if and only if $X=X_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-027765 | MATHDB-368948

Large-dimensional hyperbolic spectral exponential sum conjecture

摘录 / EXCERPT: Hyperbolic spectral exponential sum conjecture. As $X,T\to\infty$, for every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-027766 | MATHDB-368955

Light profiniteness conjecture for p-adic analytic smooth Artin stacks

摘录 / EXCERPT: Light profiniteness conjecture. Étale locally, the associated light condensed anima is a light profinite anima.

conjecture | open | MathDB 页面 | 原文线索

V2-027767 | MATHDB-368960

The localization-algebra formulation of the Baum-Connes conjecture with coefficients

摘录 / EXCERPT: Let Γ be a discrete countable group with a length function $|\cdot|$, let $P_d(\Gamma)$ be its Rips complex at scale d, and let $C^{\ast}(P_d(\Gamma),A)^{\Gamma}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027768 | MATHDB-368984

Finite-diameter conjecture for the entropy metric on outer space

摘录 / EXCERPT: Let $r\geq 4$ and consider the outer space of rank-r free groups equipped with the entropy metric. Finite-diameter conjecture. The entropy metric has finite diameter on the outer...

conjecture | open | MathDB 页面 | 原文线索

V2-027769 | MATHDB-369062

Kotschick's conjecture on nonvanishing one-forms on compact Kähler manifolds

摘录 / EXCERPT: Kotschick's conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-027770 | MATHDB-369066

Donaldson's almost-Kähler a priori bounds conjecture

摘录 / EXCERPT: Let $(M,J, omega)$ be a compact almost-Kähler manifold, where J is an almost-complex structure and $ omega$ is its fundamental 2-form. Let $ sigma$ be a smooth volume form on...

conjecture | open | MathDB 页面 | 原文线索

V2-027771 | MATHDB-369067

Classification conjecture for Griffiths semi-positive quasi-Kähler flag manifolds

摘录 / EXCERPT: Let M be a flag manifold with an almost-complex structure J and a quasi-Kähler metric λ whose curvature is Griffiths semi-positive. Let (M_i,g_i) denote Hermitian sym...

conjecture | open | MathDB 页面 | 原文线索

V2-027772 | MATHDB-369081

The structure conjecture for compact Kähler manifolds with nef anti-canonical bundle

摘录 / EXCERPT: Let X be a compact Kähler manifold with nef anti-canonical bundle $-K_X$. A locally constant fibration is a fibration that is locally constant in the sense of Definition 2.3 of M...

conjecture | open | MathDB 页面 | 原文线索

V2-027773 | MATHDB-369083

The conjecture on coisotropic structures and shifted Lagrangian reduction

摘录 / EXCERPT: Coisotropic reduction conjecture. The space of n-shifted coisotropic structures on f is equivalent to the space of such diagrams, where π_Y is an $(n+1)$-shifted Lagrangian...

conjecture | open | MathDB 页面 | 原文线索

V2-027774 | MATHDB-369123

Finiteness conjecture for low-volume knot complements with hidden symmetries

摘录 / EXCERPT: For $R>0$, consider hyperbolic knot complements in $\mathbb{S}^3$ with hidden symmetries and volume less than R. Finiteness conjecture. For any $R>0$, at most finitely many hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-027775 | MATHDB-369124

Reid-Walsh conjecture on knot complements in a commensurability class

摘录 / EXCERPT: Let M be a hyperbolic knot complement in $\mathbb{S}^3$. Two finite-volume hyperbolic 3-manifolds are commensurable when they share a common finite-degree cover; the commensurabi...

conjecture | open | MathDB 页面 | 原文线索

V2-027776 | MATHDB-369126

Sierksma's conjecture on the minimum number of Tverberg partitions

摘录 / EXCERPT: Let $m(d,r)$ denote the minimum number of unordered Tverberg partitions of a set of $(d+1)(r-1)+1$ points in $\mathbb{R}^d$, where a Tverberg partition of order r is a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-027777 | MATHDB-369129

The holomorphic-disk area conjecture for Lagrangians in balls

摘录 / EXCERPT: Let $B^{2n}(a)$ be the symplectic ball of capacity a, let L be a Lagrangian submanifold in $B^{2n}(a)$, and let J be an admissible almost-complex structure. Holomorphic-disk...

conjecture | open | MathDB 页面 | 原文线索

V2-027778 | MATHDB-369140

The Gromov-Lawson-Rosenberg conjecture for totally non-spin spin^c manifolds

摘录 / EXCERPT: Let (M,σ) be a closed connected totally non-spin spin^c manifold with $\dim M=n$, fundamental group π, and classifying map ... Let L be the associated spin^c line bu...

conjecture | open | MathDB 页面 | 原文线索

V2-027779 | MATHDB-369151

The symplectic-structure conjecture for globally hyperbolic four-manifolds

摘录 / EXCERPT: Let (M,g) be an orientable 4-dimensional globally hyperbolic Lorentzian manifold with an orthogonal decomposition of its tangent bundle $TM=N_\star\oplus N$, where N...

conjecture | open | MathDB 页面 | 原文线索

V2-027780 | MATHDB-369192

Bogulavsky's conjecture on degrees of common zero sets

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $V(W)$ be the common zero set of a subspace W of $S_d(m,\mathbb{F}_q)$, and let $\omega_r(d,m)=(\beta_1,\dots,\beta_{m+1})$ be the rth...

conjecture | open | MathDB 页面 | 原文线索

V2-027781 | MATHDB-369197

Austin's common fixed-point conjecture for commuting contractions

摘录 / EXCERPT: Austin's conjecture. If $\{f_1,\ldots,f_r\}$ is a pairwise commuting family of λ-contractions on X, then the maps have a common fixed point; that is, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-027782 | MATHDB-369225

Rectangular inscription conjecture for spherical Jordan curves of diameter π

摘录 / EXCERPT: Let S^2 be the unit round sphere, and let γ be a smooth Jordan curve in S^2. Its diameter is the supremum of the round distances between points of γ; assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-027783 | MATHDB-369236

Hadwiger's illumination conjecture for convex bodies

摘录 / EXCERPT: Let $n\geq 3$ and let K be an n-dimensional convex body. A positive homothetic copy of K is a set of the form $x+\lambda K$ with $x\in\mathbb{R}^n$ and $0<\lambda<1$. Hadwige...

conjecture | open | MathDB 页面 | 原文线索

V2-027784 | MATHDB-369244

Bershadsky-Cecotti-Ooguri-Vafa mirror symmetry conjecture

摘录 / EXCERPT: Let M be a Kähler manifold, let $F_g^{\mathrm{BCOV}}(M)$ be the genus-g BCOV quantity defined by summing the Cauchy principal value graph integrals over connected trivalent gra...

conjecture | open | MathDB 页面 | 原文线索

V2-027785 | MATHDB-369246

The sharp first-eigenvalue conjecture for compact Kähler manifolds with HSC at least 2

摘录 / EXCERPT: First-eigenvalue conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027786 | MATHDB-369312

Gao-Vitale conjecture on intrinsic-volume decay

摘录 / EXCERPT: Gao-Vitale conjecture. Either

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V2-027787 | MATHDB-369313

Conjecture on ultra-log-concave intrinsic-volume sequences

摘录 / EXCERPT: Conjecture on ultra-log-concave intrinsic-volume sequences. The sequence $(a_k)_{k\geq 0}$ is the intrinsic-volume sequence of some infinite-dimensional GB-set if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-027788 | MATHDB-369408

The conjecture on Fujiki manifolds with holomorphic Cartan geometries

摘录 / EXCERPT: Fujiki-manifold conjecture. If a Fujiki manifold admits a holomorphic Cartan geometry, then it admits a flat holomorphic Cartan geometry with the same model, and the Fujiki manifol...

conjecture | open | MathDB 页面 | 原文线索

V2-027789 | MATHDB-369420

Collapse conjecture for integrable polygonal dynamics

摘录 / EXCERPT: Let k be an algebraically closed complete field whose metric comes from a non-trivial valuation. Let T be a polygonal transformation in $\mathbb{P}^d(k)$ such that the induced...

conjecture | open | MathDB 页面 | 原文线索

V2-027790 | MATHDB-369449

Conjecture on homogeneous Kenmotsu manifolds

摘录 / EXCERPT: Homogeneous Kenmotsu manifold conjecture. Every homogeneous Kenmotsu manifold is isometric to the real hyperbolic space.

conjecture | open | MathDB 页面 | 原文线索

V2-027791 | MATHDB-369472

Microlocal endomorphism conjecture for singular support

摘录 / EXCERPT: Microlocal endomorphism conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027792 | MATHDB-369476

The \overline\partial-Robin Laplacian eigenvalue minimization conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded domain with C^2 boundary, and for $a>0$ let $\mathcal{R}_a$ be the $\overline\partial$-Robin Laplacian on $L^2(\Omega)$, with first e...

conjecture | open | MathDB 页面 | 原文线索

V2-027793 | MATHDB-369504

Schneider's support conjecture for mixed area measures

摘录 / EXCERPT: Schneider's conjecture. The support of the mixed area measure satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027794 | MATHDB-369528

Tian-Jun Li's symplectic cone conjecture for minimal Kähler surfaces

摘录 / EXCERPT: Tian-Jun Li's symplectic cone conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027795 | MATHDB-369529

Hamilton's closure symplectic cone conjecture for minimal Kähler surfaces

摘录 / EXCERPT: Hamilton's closure symplectic cone conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027796 | MATHDB-369533

Lê's conjecture on injective holomorphic map germs

摘录 / EXCERPT: Let $f\colon (\mathbb{C}^n,0)\to (\mathbb{C}^{n+1},0)$ be an injective holomorphic map germ. A map germ is corank 1 when the differential df_0 has corank 1. Lê's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027797 | MATHDB-369539

Family band width conjecture

摘录 / EXCERPT: Let $N\to B$ be a fiber bundle with base B a connected closed manifold, and with all fibers diffeomorphic to a fixed connected closed manifold Z. Suppose there is no fiberwise...

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V2-027798 | MATHDB-369540

Incidence bound for circles of different radii

摘录 / EXCERPT: Let $I_C(m,n)$ denote the maximum number of incidences between m points and n circles. The circle-incidence conjecture. For some positive constant c, ... This is a well-known i...

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V2-027799 | MATHDB-369541

Covering an integer grid by circles of distinct radii

摘录 / EXCERPT: Let n be a positive integer, and consider the $n\times n$ grid. Suppose it is covered by m circles, with no two circles having equal radius. Distinct-radii circle-covering conj...

conjecture | open | MathDB 页面 | 原文线索

V2-027800 | MATHDB-369542

Elliptic identity lifting conjecture for higher-genus multiple zeta values

摘录 / EXCERPT: Elliptic identity lifting conjecture. For any $j\in\{1,\ldots,h\}$ and any genus $h\geq1$, the corresponding higher-genus multiple zeta values satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-027801 | MATHDB-369565

The biharmonicity conjecture for non-CMC W-surfaces with flat normal bundle

摘录 / EXCERPT: Let $\varphi:M^2\to N^4(\epsilon)$ be a non-CMC W-surface with flat normal bundle. Assume that $3f^2+K-\epsilon\neq 0$ and $\nabla^\perp E_3\neq 0$ at any point. Biharmonicity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-027802 | MATHDB-369566

The generalized Chen conjecture for proper biharmonic surfaces in four-dimensional space forms

摘录 / EXCERPT: Let $\varphi:M^2\to N^4(\epsilon)$ be a proper biharmonic immersion. Generalized Chen conjecture. Then $\epsilon>0$, so $N^4(\epsilon)$ is the four-dimensional sphere...

open_problem | open | MathDB 页面 | 原文线索

V2-027803 | MATHDB-369630

Conjecture on the circumcircle envelope of Poncelet triangles

摘录 / EXCERPT: Let $\mathcal{T}$ be the family of Poncelet triangles, and let $\mathcal{E}_c$ be the inner Poncelet caustic. Consider the envelope of the circumcircle as the triangles range over...

conjecture | open | MathDB 页面 | 原文线索

V2-027804 | MATHDB-369631

Conjecture on the envelope of the circumcircle-incenter line for Poncelet triangles

摘录 / EXCERPT: Let $\mathcal{T}_{\circ}$ be the Poncelet family whose circumcircle is constrained by the circular caustic, and let Γ denote the line associated with the circumcircle-incent...

conjecture | open | MathDB 页面 | 原文线索

V2-027805 | MATHDB-369649

Conjecture on self-annihilating invariance vector fields for Lie-Poisson systems

摘录 / EXCERPT: Self-annihilating invariance-vector-field conjecture. Among the invariance vector fields there exists a special vector field Z which, acting in the system of natural Lie-algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-027806 | MATHDB-369662

Reverse Faber-Krahn conjecture for origin-symmetric convex bodies

摘录 / EXCERPT: Let $K ot i 0$ be an origin-symmetric convex body in $\mathbb R^m$, meaning that $x\in K$ if and only if $-x\in K$. Write $\lambda(K)$ for the first Dirichlet eigenvalue of K, an...

conjecture | open | MathDB 页面 | 原文线索

V2-027807 | MATHDB-369663

Weak isomorphic reverse isoperimetry conjecture

摘录 / EXCERPT: Let $K\subseteq\mathbb R^m$ be an origin-symmetric convex body, meaning that $x\in K$ if and only if $-x\in K$. Let $\mathsf{SL}_m(\mathbb R)$ denote the group of volume-preserving...

conjecture | open | MathDB 页面 | 原文线索

V2-027808 | MATHDB-369720

Kobayashi's ampleness conjecture for compact Kobayashi-hyperbolic manifolds

摘录 / EXCERPT: Kobayashi's conjecture. The canonical bundle K_X is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-027809 | MATHDB-369722

The equivalence conjecture for Kähler and sG hyperbolicity on compact complex surfaces

摘录 / EXCERPT: Kähler-sG hyperbolicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-027810 | MATHDB-369723

The SKT-to-Kähler hyperbolicity conjecture

摘录 / EXCERPT: SKT-to-Kähler hyperbolicity conjecture. If X admits such a metric ω, then X is Kähler hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-027811 | MATHDB-369740

The solvability conjecture for nilpotent hypercomplex Lie algebras

摘录 / EXCERPT: Let $(\mathfrak{g}, \{J_{\alpha}\})$ be a nilpotent hypercomplex Lie algebra. Here, $\mathbb{H}$-solvability means that the descending $\mathbb{H}$-central series defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-027812 | MATHDB-369741

Short Geodesic Conjecture for arithmetic hyperbolic orbifolds

摘录 / EXCERPT: Let an arithmetic hyperbolic orbifold of the first type be a quotient of hyperbolic space by an arithmetic lattice of the first type. For a hyperbolic element γ, write...

conjecture | open | MathDB 页面 | 原文线索

V2-027813 | MATHDB-369757

The distinct-direction conjecture for point sets in Euclidean space

摘录 / EXCERPT: Let a d-dimensional set of n points in $ obreak\mathbb{R}^d$ be a set whose affine span has dimension d. Two lines have pairwise distinct directions when no two are parallel....

conjecture | open | MathDB 页面 | 原文线索

V2-027814 | MATHDB-369758

The typical-norm distinct-distances conjecture for full-dimensional point sets

摘录 / EXCERPT: A d-norm is a norm $\|\cdot\|$ on $\mathbb{R}^d$, identified with its unit ball. A set of points is full-dimensional when it is not contained in an affine hyperplane, and a set o...

conjecture | open | MathDB 页面 | 原文线索

V2-027815 | MATHDB-369790

The strict Steklov inequality for non-disk surfaces

摘录 / EXCERPT: Let (Σ,g) be a compact Riemannian surface with boundary that is not diffeomorphic to the disk. Let $\sigma_1(\Sigma,[g])$ denote its first normalized Steklov eigenvalue, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027816 | MATHDB-369808

D8-brane partition-function DT/PT identity

摘录 / EXCERPT: Let n,m be nonnegative integers indexing D8/D8$'$ and opposite-brane pairs, and let $v_\alpha=e^{\mathfrak{a}_\alpha}$, $w_\alpha=e^{\mathfrak{b}_\alpha}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027817 | MATHDB-369809

DT/PT correspondence for mixed four-fold vertices with leg boundary conditions

摘录 / EXCERPT: Let $\vec{\pi}_{a,+}$ be n-tuples of plane partitions and $\vec{\pi}_{a,-}$ be m-tuples of plane partitions for each of the four legs. Let $v_\alpha=e^{\mathfrak{a}_\alpha}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027818 | MATHDB-369810

DT/PT correspondence for mixed four-fold vertices with surface boundary conditions

摘录 / EXCERPT: Let $\vec{\lambda}_{A,+}$ be n-tuples of plane partitions and $\vec{\lambda}_{A,-}$ be m-tuples of plane partitions for the six surfaces. Let $v_\alpha=e^{\mathfrak{a}_\alpha}$...

conjecture | open | MathDB 页面 | 原文线索

V2-027819 | MATHDB-369812

Nontriviality of the Nori fundamental group of fake projective planes in positive characteristic

摘录 / EXCERPT: Let X be a fake projective plane over an algebraically closed field of characteristic $p>0$, meaning a smooth projective surface with the same Betti numbers as the projective pla...

conjecture | open | MathDB 页面 | 原文线索

V2-027820 | MATHDB-369826

Birational rigidity conjectures for codimension-two Fano threefold WCIs

摘录 / EXCERPT: Let X be a quasismooth Fano threefold weighted complete intersection of codimension 2 and index 1, belonging to a deformation family indexed by I. Let I_{br}, I_F, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027821 | MATHDB-369827

Birational rigidity conjecture for quartic threefolds with cA1 singularities

摘录 / EXCERPT: Let X be a $mathbb Q$-factorial quartic threefold, equivalently a factorial quartic threefold, whose singular points are all of type cA_1. Quartic cA1 rigidity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-027822 | MATHDB-369828

Rigidity conjectures for sextic double solids with cA singularities

摘录 / EXCERPT: Let X be a $mathbb Q$-factorial sextic double solid, meaning a double cover of $mathbb P^3$ branched over a sextic surface, with singularities of the indicated types. Sextic doub...

conjecture | open | MathDB 页面 | 原文线索

V2-027823 | MATHDB-369872

Lelong-number vanishing conjecture for non-pluripolar powers

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n, and let $alpha in H^{1,1}(X,\mathbb{R})$ be a big class. For a current of minimal singularities in $alpha$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-027824 | MATHDB-369873

Generalized Lelong-number vanishing conjecture in nef codimension

摘录 / EXCERPT: Let X be a compact Kähler manifold of dimension n, and let $k\in\mathbb{N}$ satisfy $1\leq k\leq n-1$. Let $\alpha\in H^{1,1}(X,\mathbb{R})$ be a big class. Say that α i...

conjecture | open | MathDB 页面 | 原文线索

V2-027825 | MATHDB-369916

Quantitative colorful Tverberg conjecture with $2d$ color classes

摘录 / EXCERPT: For integers $d,r\geq 1$, let $cmathcal F_1,\dots,\mathcal F_{2d}$ be families in $cmathbb R^d$, each containing $f(d,r)$ convex sets of volume 1. A colorful Tverberg conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027826 | MATHDB-369917

Optimal colorful homogeneous selection with volume

摘录 / EXCERPT: Let $cmathcal F_1,\dots,\mathcal F_{d+1}$ be collections of volume-1 convex sets in $cmathbb R^d$, with $|\mathcal F_i|=n$ for every $i\in[d+1]$. A colorful homogeneous selection...

conjecture | open | MathDB 页面 | 原文线索

V2-027827 | MATHDB-369927

Joyce's Bridgeland stability conjecture for special Lagrangians

摘录 / EXCERPT: Let (M^n,ω,Ω) be a possibly non-compact Calabi-Yau manifold, with Kähler form ω and holomorphic volume form Ω, suitably convex at infinity. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027828 | MATHDB-369929

The p-parity conjecture for geometric symplectic self-dual representations

摘录 / EXCERPT: Let F be a number field, let L be a p-adic local field, and let V be a geometric p-adic representation of G_F over L, meaning that it is unramified outside finitely m...

conjecture | open | MathDB 页面 | 原文线索

V2-027829 | MATHDB-369930

The relative p-parity conjecture

摘录 / EXCERPT: Let F be a number field, let L be a p-adic local field, and let V and V' be geometric symplectic self-dual p-adic representations of G_F over L. For each representa...

conjecture | open | MathDB 页面 | 原文线索

V2-027830 | MATHDB-369965

Gaitsgory-Lysenko metaplectic spectral action conjecture

摘录 / EXCERPT: Let G be a reductive group, let κ be a Kac-Moody level, and let $(H,\mathcal G_Z,\epsilon)$ be the associated metaplectic Langlands dual datum, where H is reductive,...

conjecture | open | MathDB 页面 | 原文线索

V2-027831 | MATHDB-369999

Quasi-polynomial behavior for counts of maximal lattice slices

摘录 / EXCERPT: Let P be a lattice d-polytope, and let $\ell \in [d]$. Define $\operatorname{LD}_P^{\ell}(k)$ to be the number of ℓ-dimensional slices of P that contain the most lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-027832 | MATHDB-370048

The lemon bipartite-space conjecture for almost Finsler manifolds

摘录 / EXCERPT: Let $(M,S,F^{\pm})$ be a bipartite space, where S is the slit and F^+ is its positive partial Finsler function. The lemon bipartite-space conjecture. A bipartite space of the f...

conjecture | open | MathDB 页面 | 原文线索

V2-027833 | MATHDB-370049

The apple bipartite-space conjecture for almost Finsler manifolds

摘录 / EXCERPT: Let $(M,S,F^{\pm})$ be a bipartite space, where S is the slit, $S_x=\ker(s_x)$ is its fibre at $x\in M$, and F^- is the negative partial Finsler function. The apple bipartite-s...

conjecture | open | MathDB 页面 | 原文线索

V2-027834 | MATHDB-370061

Conjecture on k-positive symmetrized curvature operators and projective space

摘录 / EXCERPT: Symmetrized curvature conjecture. If $1\leq k\leq \left[\frac{n}{2}\right]$ and M has k-positive symmetrized curvature operator $\mathcal R$, then M is biholomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-027835 | MATHDB-370086

Rigidity conjecture for super-exponentially decaying cylindrical flows

摘录 / EXCERPT: Let $\mathcal{M}$ be a rescaled mean curvature flow in $\mathbb{R}^{n+1}$ without boundary. The source distinguishes asymptotic cases for such flows, including a case of super-expo...

conjecture | open | MathDB 页面 | 原文线索

V2-027836 | MATHDB-370112

The restricted interleaving metric conjecture for merge trees

摘录 / EXCERPT: Restricted interleaving metric conjecture. When $n=1,2$,

conjecture | open | MathDB 页面 | 原文线索

V2-027837 | MATHDB-370140

The conjecture that the cut time equals the first coordinate cut time on Grushin spaces

摘录 / EXCERPT: Cut-time conjecture. The cut time t_{cut} is identically

conjecture | open | MathDB 页面 | 原文线索

V2-027838 | MATHDB-370207

A volume inequality for convex polytopes with d+2 vertices on the sphere

摘录 / EXCERPT: The volume inequality. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027839 | MATHDB-370212

Chen's conjecture on biharmonic submanifolds in Euclidean spaces

摘录 / EXCERPT: Let M be a biharmonic submanifold of a Euclidean space. Chen's conjecture. Every such submanifold is minimal. Chen, Ishikawa, and Jiang proved the claim for biharmonic surfaces i...

conjecture | open | MathDB 页面 | 原文线索

V2-027840 | MATHDB-370213

Generalized Chen's conjecture on biharmonic submanifolds in non-positively curved manifolds

摘录 / EXCERPT: Let M be a biharmonic submanifold of a non-positively curved Riemannian manifold. Generalized Chen's conjecture. The submanifold M is minimal. Ou and Tang constructed counterex...

conjecture | open | MathDB 页面 | 原文线索

V2-027841 | MATHDB-370214

BMO conjecture on biharmonic submanifolds in spheres

摘录 / EXCERPT: Let M be a biharmonic submanifold of a sphere. BMO conjecture. The submanifold M has constant mean curvature. The source presents this as one of the well-known open conjectures...

conjecture | open | MathDB 页面 | 原文线索

V2-027842 | MATHDB-370262

Conjecture on super-exponential condensation in analytic iterated function systems

摘录 / EXCERPT: Super-exponential condensation conjecture. Any analytic IFS which has super-exponential condensation but no exact overlaps must be sub-conjugated to a self-similar IFS.

conjecture | open | MathDB 页面 | 原文线索

V2-027843 | MATHDB-370265

Jerrad et al.'s conjecture that every convex polyhedron is Rupert

摘录 / EXCERPT: A Rupert compact convex set $S\subseteq\mathbb{R}^3$ admits a passage in which a second copy of S passes straight through a hole in the interior of the first, with rescaling fact...

conjecture | open | MathDB 页面 | 原文线索

V2-027844 | MATHDB-370266

The tetrahedron's exact Nieuwland constant conjecture

摘录 / EXCERPT: For a compact convex set $S\subseteq\mathbb{R}^3$, its Nieuwland constant μ_S is the largest rescaling factor for which a second copy of S can pass straight through a hole in...

conjecture | open | MathDB 页面 | 原文线索

V2-027845 | MATHDB-370267

Steininger and Yurkevich's negative Rupert conjecture for three Archimedean solids

摘录 / EXCERPT: A compact convex set $S\subseteq\mathbb{R}^3$ is Rupert if a second identical copy can pass straight through a hole in the interior of the first with rescaling factor $\mu_S>1$. St...

conjecture | open | MathDB 页面 | 原文线索

V2-027846 | MATHDB-370275

Existence and uniqueness conjecture for semi-flat cscK currents on minimal elliptic surfaces

摘录 / EXCERPT: Existence and uniqueness conjecture. There is a semi-flat cscK current on X whose regular fibers have volume 1 and whose scalar curvature is -3; moreover, if ω_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-027847 | MATHDB-370276

Uniqueness conjecture for semi-flat cscK currents on minimal elliptic surfaces

摘录 / EXCERPT: Let $\Phi:X\rightarrow\Sigma_g$ be a minimal elliptic surface, and let $H(X,\Phi)$ be the group of pairs of automorphisms preserving Φ: ... A semi-flat cscK current has fiber vo...

conjecture | open | MathDB 页面 | 原文线索

V2-027848 | MATHDB-370286

Yau-Tian-Donaldson conjecture for cscK metrics

摘录 / EXCERPT: Let (X,L) be a smooth polarized manifold, where $c_1(L)$ denotes the first Chern class of L, and let K-polystability mean that every test configuration has nonnegative Donaldso...

conjecture | open | MathDB 页面 | 原文线索

V2-027849 | MATHDB-370300

Conjectural Mirzakhani volume formula for ideal n-gons

摘录 / EXCERPT: Let $\mathbb{D}_n$ denote the moduli space of ideal n-gons, and let $V_{\mathbb{D}_n}$ be its Mirzakhani volume. For an integer $n\geqslant 3$, the ideal-polygon volume conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-027850 | MATHDB-370308

Kanai's smooth-rigidity conjecture for frame flows

摘录 / EXCERPT: Kanai's conjecture. If the distribution $\mathbb{H}$ is $\mathcal{C}^1$, then (M,g) is locally symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-027851 | MATHDB-370311

The Strichartz conjecture on the real line times the three-sphere

摘录 / EXCERPT: The Strichartz conjecture on $\mathbb{R}\times\mathbb{S}^3$. One has

conjecture | open | MathDB 页面 | 原文线索

V2-027852 | MATHDB-370312

The finiteness conjecture for homogeneous Einstein metrics

摘录 / EXCERPT: Let $M=\mathsf G/\mathsf H$ be a compact homogeneous space whose isotropy representation consists of pairwise inequivalent irreducible summands. In particular, this includes the ca...

conjecture | open | MathDB 页面 | 原文线索

V2-027853 | MATHDB-370314

The refined closed-manifold 3D index conjecture

摘录 / EXCERPT: Let M be a closed 3-manifold obtained by Dehn filling a 1-cusped 3-manifold N with a 1-efficient ideal triangulation $\mathcal{T}$. Applying the Gang-Yonekura formula d...

conjecture | open | MathDB 页面 | 原文线索

V2-027854 | MATHDB-370361

The folklore conjecture generalizing Kontsevich-Soibelman's conjecture on collapsing boundaries

摘录 / EXCERPT: Folklore conjecture generalizing Kontsevich-Soibelman's conjecture. The collapsing boundary densely parametrizes singular tropical metric spaces homeomorphic to essential skeleta...

conjecture | open | MathDB 页面 | 原文线索

V2-027855 | MATHDB-370365

Partial-tilting pullback conjecture for quotient stacks

摘录 / EXCERPT: Let X, G, and $\mathcal{E}$ be as in the surrounding construction, with G finite, and let $\pi:[\mathbb{A}(\mathcal{E})/G]\rightarrow[X/G]$ be the induced projection. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-027856 | MATHDB-370366

Tilting-object conjecture for canonical bundles over reflexive polytopes

摘录 / EXCERPT: Let P be a reflexive polytope. Let Σ be a simplicial fan whose ray primitive generators are the vertices of P, and let $\mathcal{V}$ be the fan of the canonical bundle o...

conjecture | open | MathDB 页面 | 原文线索

V2-027857 | MATHDB-370367

Face-extension conjecture for non-commutative crepant resolutions

摘录 / EXCERPT: Let $Q\subset\mathbb{R}^k$ be a lattice polytope with associated Gorenstein cone ... Suppose Q is lattice equivalent to a face F of a lattice polytope $P\subset\mathbb{R}^n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027858 | MATHDB-370386

Lorentzian Lichnerowicz conjecture

摘录 / EXCERPT: Let M be a closed Lorentzian conformal manifold. It is essential if its conformal transformation group does not preserve any metric in the conformal class. Lorentzian Lichnerowic...

conjecture | open | MathDB 页面 | 原文线索

V2-027859 | MATHDB-370390

Conjectural square-root error bound for hyperbolic lattice counting along a geodesic

摘录 / EXCERPT: Square-root error conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-027860 | MATHDB-370400

Huybrechts's symplectic-action conjecture for zero-cycles on K3 surfaces

摘录 / EXCERPT: Let X be a $K3$ surface, and let ${\operatorname{Aut}}_s(X)$ be the group of symplectic automorphisms of X, meaning automorphisms that act trivially on a nonzero holomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-027861 | MATHDB-370402

Mendson-Pereira conjecture for codimension-one foliations on projective space

摘录 / EXCERPT: Let $\mathcal F$ be a codimension one foliation on $\mathbb P^n$, with $n\ge 3$. Let $\Omega^1_{\mathcal F}$ denote its sheaf of foliated 1-forms. A foliation is rational if it i...

conjecture | open | MathDB 页面 | 原文线索

V2-027862 | MATHDB-370403

Mendson-Pereira jet-splitting conjecture for foliations on projective space

摘录 / EXCERPT: Let $\mathcal F$ be a codimension one foliation on $\mathbb P^n$, with $n\ge 3$, and let $N^*_{\mathcal F}$ be its conormal sheaf. Let ∇_B be the Bott connection on...

conjecture | open | MathDB 页面 | 原文线索

V2-027863 | MATHDB-370424

Symplectic Zauner's conjecture for real symplectic equiangular tight frames

摘录 / EXCERPT: Let d be the dimension of a real symplectic vector space, and let a $d\times n$ ETF denote an equiangular tight frame in that space. Symplectic Zauner's conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-027864 | MATHDB-370447

The Calabi-Yau ALE metric conjecture for M_ζ

摘录 / EXCERPT: Let M_ζ be the ALE resolution constructed from the quotient V/Γ, and let ω_ζ denote its Kähler form. The Calabi-Yau ALE metric conjecture. The ALE space...

conjecture | open | MathDB 页面 | 原文线索

V2-027865 | MATHDB-370449

Infirri's Ricci-flatness conjecture for the quotient Kähler metric

摘录 / EXCERPT: Let M_ζ be the Calabi-Yau ALE resolution in complex dimension 3, let g_ζ be its quotient Kähler metric, and let $(\mathcal{H}^{0,1}_\alpha)^\Gamma$ denote the invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-027866 | MATHDB-370450

The universal Calabi-Yau ALE structure conjecture

摘录 / EXCERPT: Let $\kappa:\mathbb{M}\to\Theta_{\operatorname{Im}(\mathbb{C})}$ be the universal Kähler ALE bundle, with vertical forms $\omega_{\mathbb{M};RF}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-027867 | MATHDB-370452

The Spin(7) SYZ fibration conjecture

摘录 / EXCERPT: Let (X,Φ) be a $\operatorname{Spin}(7)$-manifold, and let a toric Cayley mean a Cayley four-fold of the toric type intended in the source. The Spin(7) SYZ fibration conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-027868 | MATHDB-370453

The SYZ mirror-resolution conjecture for Spin(7) orbifolds

摘录 / EXCERPT: Let $(\widehat{X},\widehat{\Phi})$ be an SYZ-fibred $\operatorname{Spin}(7)$-manifold, let τ be fibrewise multiplication by -1, and let S be the compact four-manifold base...

conjecture | open | MathDB 页面 | 原文线索

V2-027869 | MATHDB-370462

Nilpotent geodesic-flow asymptotic expansion conjecture

摘录 / EXCERPT: Let M be a compact Riemann manifold with negative curvature. Let Γ be a finitely generated torsion-free nilpotent group, let $\Phi:\pi_1(M)\to\Gamma$ be surjective, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-027870 | MATHDB-370474

Bezdek's axial-symmetry conjecture for plane sections of convex bodies

摘录 / EXCERPT: Let $K\subset \mathbb{R}^3$ be a convex body. A plane section means the intersection of K with an affine plane, and an axis of symmetry is a line in that plane whose reflection p...

conjecture | open | MathDB 页面 | 原文线索

V2-027871 | MATHDB-370475

Refinement of Bezdek's conjecture for Larman points

摘录 / EXCERPT: Let $K\subset \mathbb{R}^{n}$ be a convex body with $n\geq 3$, and let $p\in\operatorname*{int}K$. For every hyperplane Π passing through p, suppose the section $\Pi\cap K$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-027872 | MATHDB-370491

Kravchuk-Mazáč-Pal conjecture on the bass note spectrum

摘录 / EXCERPT: For a cocompact lattice Γ in $G=\operatorname{PSL}_2(\mathbf{R})$, let $\lambda_1(\Gamma\backslash\mathbf{H})$ be the first Laplace eigenvalue of the compact hyperbolic 2-...

conjecture | open | MathDB 页面 | 原文线索

V2-027873 | MATHDB-370525

The orientable exact Lagrangian fillability conjecture for twist-spun torus links

摘录 / EXCERPT: Let k,n,ℓ and d be as in the construction of the twist-spun torus link, and let $\Sigma_{\psi^\ell}(\Lambda(k,n-k))$ denote that twist-spun link. An orientably exact Lagrang...

conjecture | open | MathDB 页面 | 原文线索

V2-027874 | MATHDB-370548

The even capillary Blaschke-Santaló conjecture

摘录 / EXCERPT: Let $\thetain (0,\frac{\pi}{2})$ and let Σ be an even, smooth, strictly convex capillary hypersurface. Write $\widehat{\Sigma}$ for the region enclosed by Σ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-027875 | MATHDB-370552

Aluffi's covolume conjecture for adjoint polynomials of polyhedral cones

摘录 / EXCERPT: Let $C\subseteq\mathbb{R}^{d+1}$ be a polyhedral cone whose cross section with the hyperplane $\{x_0=1\}$ is a polytope $P\subseteq\mathbb{R}^d$. The adjoint polynomial of C is t...

conjecture | open | MathDB 页面 | 原文线索

V2-027876 | MATHDB-370639

Abundance conjecture for rational nef classes on projective Calabi-Yau manifolds

摘录 / EXCERPT: Abundance conjecture. The class $[\alpha]$ is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-027877 | MATHDB-370640

Tosatti's bounded-potential conjecture for nef classes

摘录 / EXCERPT: Tosatti's bounded-potential conjecture. There is $\varphi\in L^\infty(X)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-027878 | MATHDB-370641

Tosatti's minimal-singularities conjecture for degenerate Ricci-flat limits

摘录 / EXCERPT: Tosatti's minimal-singularities conjecture. If $\eta\in\mathcal{T}_{[\alpha]}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-027879 | MATHDB-370642

Tosatti's uniqueness conjecture for irrational nef isotropic classes on K3 surfaces

摘录 / EXCERPT: Tosatti's uniqueness conjecture. The class $[\alpha]$ contains a unique closed positive $(1,1)$-current.

conjecture | open | MathDB 页面 | 原文线索

V2-027880 | MATHDB-370673

Calabi's conjecture on prescribed Ricci forms

摘录 / EXCERPT: Let M be a compact Kähler manifold with Kähler metric $ \omega$, and let α be a real $(1,1)$-form representing $c_{1}(M)$. Calabi's conjecture. There exists a unique Kähle...

conjecture | open | MathDB 页面 | 原文线索

V2-027881 | MATHDB-370675

Thurston's hyperbolization conjecture for closed 3-manifolds

摘录 / EXCERPT: Let M be a closed, orientable, atoroidal 3-manifold, meaning that it contains no essential torus. Thurston's hyperbolization conjecture. If M has infinite fundamental group, th...

conjecture | open | MathDB 页面 | 原文线索

V2-027882 | MATHDB-370693

Bounded intrinsic diameter conjecture for mean curvature flow

摘录 / EXCERPT: Consider a mean curvature flow of surfaces approaching its first singular time, and let intrinsic diameter denote the diameter measured in the evolving surface's intrinsic metric....

conjecture | open | MathDB 页面 | 原文线索

V2-027883 | MATHDB-370694

Isolation conjecture for mean curvature flow singularities

摘录 / EXCERPT: Consider the outer or inner flow of an embedded surface through singularities. A singularity is an isolated singularity if it is isolated in the relevant spacetime singular set. Is...

conjecture | open | MathDB 页面 | 原文线索

V2-027884 | MATHDB-370695

Ilmanen's multiplicity-one conjecture for mean curvature flow

摘录 / EXCERPT: Consider the mean curvature flow of embedded hypersurfaces, and let a blowup limit mean a limit obtained by parabolic rescaling near a singularity, counted with its varifold multip...

conjecture | open | MathDB 页面 | 原文线索

V2-027885 | MATHDB-370696

Huisken's genericity conjecture for mean curvature flow

摘录 / EXCERPT: Consider mean curvature flow with generic initial data, meaning initial data outside an appropriate exceptional subset. A spherical singularity has spherical tangent model, and a c...

conjecture | open | MathDB 页面 | 原文线索

V2-027886 | MATHDB-370697

Ilmanen's mean-convex neighborhood conjecture

摘录 / EXCERPT: Consider a cylindrical singularity of mean curvature flow. A neighborhood is mean-convex when the flow in that neighborhood has the corresponding one-sided mean-convex orientation....

conjecture | open | MathDB 页面 | 原文线索

V2-027887 | MATHDB-370713

Legendrian Gauss-map rigidity under contact homeomorphisms

摘录 / EXCERPT: Legendrian Gauss-map rigidity conjecture. The diagram relating the Lagrangian Gauss maps of Λ and $\varphi_\infty(\Lambda)$ commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-027888 | MATHDB-370714

Loose Legendrian rigidity under contact homeomorphisms

摘录 / EXCERPT: Loose Legendrian rigidity conjecture. The Legendrian $\varphi_\infty(\Lambda)$ is still loose.

conjecture | open | MathDB 页面 | 原文线索

V2-027889 | MATHDB-370722

The aspherical-manifold virtual fibring conjecture

摘录 / EXCERPT: Let M be a closed aspherical manifold with RFRS fundamental group $G=\pi_1(M)$, and let $\beta_i^{(2)}(G;\mathbb K)$ denote the ith L^2-Betti number with coefficients in a fi...

conjecture | open | MathDB 页面 | 原文线索

V2-027890 | MATHDB-370723

The odd-dimensional hyperbolic virtual fibring conjecture

摘录 / EXCERPT: A finite-volume hyperbolic manifold is a quotient of real hyperbolic space by a group of isometries acting freely and properly discontinuously. Odd-dimensional hyperbolic virtual f...

conjecture | open | MathDB 页面 | 原文线索

V2-027891 | MATHDB-370724

The Dodziuk conjecture over arbitrary fields

摘录 / EXCERPT: Let G be the fundamental group of a finite-volume hyperbolic manifold of odd dimension, assume that G is RFRS, and let $\beta_i^{(2)}(G;\mathbb K)$ denote its ith L^2-Betti...

conjecture | open | MathDB 页面 | 原文线索

V2-027892 | MATHDB-370760

Subgroup tameness conjecture for hyperbolic mapping tori of free groups

摘录 / EXCERPT: Let $\mathbb{F}$ be a finitely generated free group and let $\psi\to \mathbb{F}$ be a fully irreducible monomorphism, meaning that no proper nontrivial finitely generated free fact...

conjecture | open | MathDB 页面 | 原文线索

V2-027893 | MATHDB-370790

The abundance conjecture for klt pairs

摘录 / EXCERPT: Let (X,B) be a klt pair. Write $ u(K_X+B)$ for the numerical dimension and $\kappa_{\iota}(K_X+B)$ for the invariant Iitaka dimension of K_X+B. Abundance conjecture. One has ……

conjecture | open | MathDB 页面 | 原文线索

V2-027894 | MATHDB-370791

The Schoen-Yau-Gromov-Lawson conjecture on positive scalar curvature and aspherical manifolds

摘录 / EXCERPT: An aspherical manifold is a manifold with contractible universal cover. The Schoen-Yau-Gromov-Lawson conjecture. There is no Riemannian metric of positive scalar curvature on a clo…

conjecture | solved | MathDB 页面 | 原文线索

V2-027895 | MATHDB-370797

Classification conjecture for three-dimensional manifolds with invariant horospherical mean curvature

摘录 / EXCERPT: Three-dimensional classification conjecture. If (X,g) is a 3-dimensional simply connected manifold with invariant horospherical mean curvature function, then X is, up to scal…

conjecture | open | MathDB 页面 | 原文线索

V2-027896 | MATHDB-370808

Generic single-flex conjecture for twinned polyhedra

摘录 / EXCERPT: Let P be a triangulated polyhedron with distinguished vertices A,B,A',B' satisfying … Form the twinned polyhedron T of P around the equator $ABA'B'$, as in the construction…

conjecture | open | MathDB 页面 | 原文线索

V2-027897 | MATHDB-370809

Classification conjecture for compact non-Kähler almost Hermitian 4-manifolds of class $\mathcal{AH}_1$

摘录 / EXCERPT: Classification conjecture. If (M^4,g,J,ω) is a compact 4-dimensional almost Hermitian manifold of class $\mathcal{AH}_1$, then either the manifold is Kähler, or it is one of…

conjecture | open | MathDB 页面 | 原文线索

V2-027898 | MATHDB-370814

Tang's constant mixed curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Let (M^{n},g) be a compact Hermitian manifold with $n\geq2$. For a $(1,0)$-vector X, let $\mathcal{C}_{\alpha,\beta}(X)$ denote the mixed curvature, formed from the relevant Ch…

conjecture | open | MathDB 页面 | 原文线索

V2-027899 | MATHDB-370815

The constant kth-mixed curvature conjecture for compact Hermitian manifolds

摘录 / EXCERPT: Let (M^{n},g) be a compact Hermitian manifold with $n\geq2$. For $k\in\{1,2,3,4\}$ and $\beta\neq0$, define the kth-mixed curvature by … where $Ric^{(k)}$ is the kth Chern Ricci…

conjecture | open | MathDB 页面 | 原文线索

V2-027900 | MATHDB-370832

The higher-dimensional star-shaped Reeb-orbit two-or-infinity conjecture

摘录 / EXCERPT: Let Σ^{2n-1} bound a star-shaped domain in $\mathbb{R}^{2n}$. Higher-dimensional two-or-infinity conjecture. For each such Σ, … The dimension-four case is solved by t…

conjecture | open | MathDB 页面 | 原文线索

V2-027901 | MATHDB-370834

The elliptic closed Reeb-orbit conjecture for strictly convex hypersurfaces

摘录 / EXCERPT: Let Σ^{2n-1} bound a strictly convex domain in $\mathbb{R}^{2n}$. Elliptic orbit conjecture. There exists at least one elliptic closed Reeb orbit on Σ. The source des…

conjecture | open | MathDB 页面 | 原文线索

V2-027902 | MATHDB-370836

Completeness conjecture for the Bollobás-Riordan polynomial of lens-space Heegaard graphs

摘录 / EXCERPT: Bollobás-Riordan completeness conjecture. The Bollobás-Riordan polynomial of the Heegaard graph of lens spaces is a complete invariant for lens spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-027903 | MATHDB-370837

Volume growth and asymptotic cone dimension conjecture

摘录 / EXCERPT: Volume growth and asymptotic cone dimension conjecture. (1) If

conjecture | open | MathDB 页面 | 原文线索

V2-027904 | MATHDB-370848

Multi-Bubble Isoperimetric Conjecture in Gaussian space

摘录 / EXCERPT: Let $\mathbb{G}^n$ denote Gaussian space, and let $\Omega^{k+1}[\mathbb{G}^n]$ be a standard simplicial k-bubble, meaning a $(k+1)$-partition obtained by translating a centered s…

conjecture | open | MathDB 页面 | 原文线索

V2-027905 | MATHDB-370892

The Gibbons-Bando-Kasue-Nakajima conjecture on simply-connected Ricci-flat ALE instantons

摘录 / EXCERPT: A simply-connected Ricci-flat ALE instanton is a complete Riemannian manifold with vanishing Ricci curvature and asymptotically locally Euclidean geometry. Gibbons-Bando-Kasue-Naka…

conjecture | open | MathDB 页面 | 原文线索

V2-027906 | MATHDB-370923

Orbit-closure dichotomy for Apollonian circle packings

摘录 / EXCERPT: Let $\mathcal P$ be an Apollonian circle packing with symmetry group $\Gamma<\operatorname{PSL}_2(\mathbb C)$. For any circle $C\in \mathcal C_\Lambda^*$, consider its orbit…

conjecture | open | MathDB 页面 | 原文线索

V2-027907 | MATHDB-370950

Finiteness conjecture for Forman-Ricci-positive 5-polytopes

摘录 / EXCERPT: Let P be a polytope, and call it Forman-Ricci-positive when its Forman-Ricci curvature satisfies $\kappa_F(e)>0$ for every edge e. Finiteness conjecture for Forman-Ricci-positi…

conjecture | open | MathDB 页面 | 原文线索

V2-027908 | MATHDB-370958

Giroux-Pardon conjecture on regular Lagrangians in Lefschetz fibrations

摘录 / EXCERPT: Giroux-Pardon conjecture. There is a Weinstein Lefschetz fibration

conjecture | open | MathDB 页面 | 原文线索

V2-027909 | MATHDB-370970

Positive scalar-curvature rigidity conjecture for small-curvature noncompact manifolds

摘录 / EXCERPT: Let $C_s,v_0>0$ and $n\geq 3$. Let (M^n,g_0) be a complete non-compact Riemannian manifold. Assume that ... ... and $\mathcal{R}(g_0)\geq 0$. Positive scalar-curvature rigidity con...

conjecture | open | MathDB 页面 | 原文线索

V2-027910 | MATHDB-370979

Arzela-Ascoli conjecture with addresses for Gromov-Hausdorff converging spaces

摘录 / EXCERPT: Let (X_j,d_{X_j}) and (Y_j,d_{Y_j}) be sequences of compact metric spaces such that ... Suppose that $F_j:X_j\to Y_j$ are uniformly bounded K-Lipschitz functions for some $K>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-027911 | MATHDB-370980

Equality of the three-dimensional abelian Cremona group classes

摘录 / EXCERPT: Let $\mathcal{A}_3$ and $\mathcal{B}_3$ be the classes of finite abelian groups arising in the paper, and let ... while ... where $G_1=(\mathbb{Z}/4)^4$,...

conjecture | open | MathDB 页面 | 原文线索

V2-027912 | MATHDB-370981

Yau's conjecture on Kähler-Einstein Bergman metrics

摘录 / EXCERPT: Let Ω be a bounded domain that is not a product domain. A domain is homogeneous if its automorphism group acts transitively, meaning that for any $x,y e \Omega$ there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-027913 | MATHDB-370982

Cheng's conjecture on Kähler-Einstein Bergman metrics

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}^n$ be a bounded strictly pseudoconvex domain with C^∞ boundary, and let ω_B denote its Bergman metric. Cheng's conjecture. If ω_B...

conjecture | open | MathDB 页面 | 原文线索

V2-027914 | MATHDB-370994

Frankel's conjecture for compact Kähler manifolds

摘录 / EXCERPT: Let (M,ω) be a compact n-dimensional Kähler manifold with positive holomorphic bisectional curvature. Frankel's conjecture. Then M is biholomorphic to $\mathbb{CP}^{n}$....

conjecture | open | MathDB 页面 | 原文线索

V2-027915 | MATHDB-370995

Linear lower-bound conjecture for distinct dot products of planar point sets

摘录 / EXCERPT: Linear distinct-dot-products conjecture. For every such sequence,

conjecture | open | MathDB 页面 | 原文线索

V2-027916 | MATHDB-371000

Finiteness conjecture for projective weakly log canonical klt models

摘录 / EXCERPT: Finiteness conjecture for wlc klt models. The number of projective wlc klt models in a fixed 0-class is finite up to log isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-027917 | MATHDB-371003

Maulik-Ranganathan's logarithmic Donaldson-Thomas conjecture

摘录 / EXCERPT: Maulik-Ranganathan's conjecture. The generating functions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-027918 | MATHDB-371032

The Milnor-Wood inequality for Euler numbers of hyperbolic 4-manifold bundles

摘录 / EXCERPT: Let $M=M(e,g)$ be homeomorphic to an $\mathbb R^2$-bundle over a closed orientable surface of genus $g>1$, with Euler number $e=\langle[F],[F]\rangle$ given by the self-intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-027919 | MATHDB-371033

Dropping incompressibility in the intersection estimate

摘录 / EXCERPT: Let M be the Riemannian 4-manifold and let the cycles in Theorem 1 be the cycles whose intersection pairing is being estimated. Incompressibility conjecture. The condition that...

conjecture | open | MathDB 页面 | 原文线索

V2-027920 | MATHDB-371036

Uniform volume-doubling failure for domains with vanishing angles

摘录 / EXCERPT: Let U be a bounded domain in Euclidean space or in a Riemannian manifold, and let φ_U denote its first Dirichlet eigenfunction. A family of domains satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-027921 | MATHDB-371063

Root-count conjecture for mutations in admissible Dynkin biagrams

摘录 / EXCERPT: Let (Γ, Δ) be an admissible Dynkin biagram with r vertices. Let h_{Γ} and h_{Δ} denote the Coxeter numbers associated to the respective components, and...

conjecture | open | MathDB 页面 | 原文线索

V2-027922 | MATHDB-371064

Mutation bound conjecture for affine-by-finite admissible biagrams

摘录 / EXCERPT: Let (Γ, Δ) be an admissible biagram with r vertices, with every component of Γ an affine Dynkin diagram and every component of Δ a Dynkin diagram. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027923 | MATHDB-371103

Desarguesian conjecture for cyclic projective planes

摘录 / EXCERPT: A cyclic projective plane is a projective plane admitting a regular action by a cyclic group. A projective plane is Desarguesian when it satisfies Desargues's theorem, and a perfec...

conjecture | open | MathDB 页面 | 原文线索

V2-027924 | MATHDB-371122

Böröczky-Lutwak-Yang-Zhang even L_p Brunn-Minkowski conjecture

摘录 / EXCERPT: Böröczky-Lutwak-Yang-Zhang conjecture. The L_p Brunn-Minkowski inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-027925 | MATHDB-371123

Even L_p-Minkowski uniqueness conjecture

摘录 / EXCERPT: Even L_p-Minkowski uniqueness conjecture. For $p\in[0,1)$, the Monge-Ampère equation

conjecture | open | MathDB 页面 | 原文线索

V2-027926 | MATHDB-371124

Kolesnikov-Milman local L_p-Minkowski conjecture

摘录 / EXCERPT: Kolesnikov-Milman local L_p-Minkowski conjecture. For every $z\in C^2_e(S^n)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-027927 | MATHDB-371135

Local homogeneity implies Bergman logarithmic flatness

摘录 / EXCERPT: Let $(M, g; L, h)$ be a polarized manifold, and let S be the circle bundle of (L^*, h^*). The manifold (M,g) is locally homogeneous if its local biholomorphic isometries act...

conjecture | open | MathDB 页面 | 原文线索

V2-027928 | MATHDB-371176

Deformation invariance of semiampleness and generalized plurigenera

摘录 / EXCERPT: Deformation invariance conjecture. For every $t\in\Delta$, K_{X_t} is semiample and $K_{\mathcal{X}}$ is π-semiample over $\pi^{-1}(U_t)$, where U_t is a Zariski neighborho...

conjecture | open | MathDB 页面 | 原文线索

V2-027929 | MATHDB-371196

Khan-Marković conjecture on parabolic twists for flute surfaces

摘录 / EXCERPT: Khan-Marković conjecture. For every such sequence $(\ell_n)_{n=1}^{\infty}$, there exists a choice of twists $(t_n)_{n=1}^{\infty}$ such that the flute surface...

conjecture | open | MathDB 页面 | 原文线索

V2-027930 | MATHDB-371240

Rosenberg's product conjecture for positive scalar curvature

摘录 / EXCERPT: Let Σ be a closed manifold of dimension at least 3, with $\dim\Sigma\ne4$. A positive scalar curvature metric is a Riemannian metric whose scalar curvature is everywhere p...

conjecture | open | MathDB 页面 | 原文线索

V2-027931 | MATHDB-371275

Yang's refined minimal-degree conjecture for Kähler manifolds

摘录 / EXCERPT: Let (M^n,g) be a complete noncompact Kähler manifold with nonnegative bisectional curvature, and assume that the universal cover of M does not split. For a point p, let...

conjecture | open | MathDB 页面 | 原文线索

V2-027932 | MATHDB-371276

Yang's Kähler-Ricci-flow Lyapunov-exponent conjecture

摘录 / EXCERPT: Let (M^n,g) be a complete noncompact Kähler manifold with nonnegative curvature and maximal volume growth. Let $g(t)$, $t\in[0,+\infty)$, be the unique complete solution of the K...

conjecture | open | MathDB 页面 | 原文线索

V2-027933 | MATHDB-371277

Rigidity conjecture for complete balanced Hermitian manifolds

摘录 / EXCERPT: Let (M,ω) be a complete balanced Hermitian manifold of complex dimension n, and suppose that the equality case in the first-eigenvalue estimate of Theorem 1.1 holds, so th...

conjecture | open | MathDB 页面 | 原文线索

V2-027934 | MATHDB-371296

Fefferman-Graham flatness conjecture for Poincaré-Einstein representatives

摘录 / EXCERPT: Fefferman-Graham flatness conjecture. There exists a metric representative in $(M_+,\boldsymbol{c})$ that is formally to all orders Poincaré-Einstein.

conjecture | open | MathDB 页面 | 原文线索

V2-027935 | MATHDB-371311

Ganatra-Kuo-Li-Wu conjecture on proper sheaf kernels

摘录 / EXCERPT: Let X and Y be manifolds, let $[?]$ be conic subsets specifying the microsupported sheaf categories, and let $K \in \Sh_{-\Lambda \times \Sigma}(X \times Y)$ be a sheaf kernel....

conjecture | open | MathDB 页面 | 原文线索

V2-027936 | MATHDB-371312

Ganatra-Kuo-Li-Wu's positive-wrapping conjecture for sheaf kernels

摘录 / EXCERPT: Let X and Y be manifolds, let Λ and Σ be conic conditions defining the relevant microsupported sheaf categories, and let $L \in \Sh(X \times Y)$ be a constructib...

conjecture | open | MathDB 页面 | 原文线索

V2-027937 | MATHDB-371315

Injectivity of inherent scalar variance for projected normal variables on Stiefel manifolds

摘录 / EXCERPT: Injectivity conjecture. The projected normal random variable has inherent average $\mathbb{E}[X]$. Moreover, there exists an injective function

conjecture | open | MathDB 页面 | 原文线索

V2-027938 | MATHDB-371341

Finiteness conjecture for extendable mapping classes of embedded surfaces

摘录 / EXCERPT: Let F_g be a closed orientable surface of genus g embedded in S^4. An orientation-preserving self-homeomorphism of F_g is extendable if it extends to a homeomorphism of the...

conjecture | open | MathDB 页面 | 原文线索

V2-027939 | MATHDB-371358

Gardner-Zvavitch dimensional Brunn-Minkowski conjecture

摘录 / EXCERPT: Let cmu be an even log-concave measure on $cmathbb{R}^d$, let $clambdacin(0,1)$, and let $K,Lcsubsetcmathbb{R}^d$ be convex and centrally symmetric. Gardner-Zvavitch's dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-027940 | MATHDB-371386

Constancy of sections in Calabi-Yau Kähler families

摘录 / EXCERPT: Let $p:\mathcal X\to\mathbb D$ be a nonsingular, proper family of Calabi-Yau Kähler manifolds with $K_X\equiv 0$, where X is the central fiber. Let $L\to\mathcal X$ be a holomorp...

open_problem | open | MathDB 页面 | 原文线索

V2-027941 | MATHDB-371436

Yau's finite-dimensionality conjecture for polynomial-growth harmonic functions

摘录 / EXCERPT: Let M be a complete noncompact (open) Riemannian manifold with nonnegative Ricci curvature. For a fixed growth rate, consider the vector space of harmonic functions on M with p...

conjecture | open | MathDB 页面 | 原文线索

V2-027942 | MATHDB-371437

Finite-dimensionality conjecture for bounded biharmonic functions on manifolds with nonnegative Ricci curvature

摘录 / EXCERPT: Let M be an open manifold with nonnegative Ricci curvature, and let u be a bounded biharmonic function on M, meaning that $ abla^4 u=0$ in the source's terminology via the bi...

conjecture | open | MathDB 页面 | 原文线索

V2-027943 | MATHDB-371438

Finite-dimensionality conjecture for polynomial-growth biharmonic functions

摘录 / EXCERPT: Let M be an open manifold with nonnegative Ricci curvature, and consider the vector space of biharmonic functions on M, where biharmonic means satisfying ... For a fixed growth r...

conjecture | open | MathDB 页面 | 原文线索

V2-027944 | MATHDB-371444

Wang's Liouville conjecture for positive harmonic functions with nonlinear boundary conditions

摘录 / EXCERPT: Let (M^n,g), with $n\geq 3$, be a compact Riemannian manifold with nonnegative Ricci curvature, and let the second fundamental form satisfy $\Pi\geq 1$ on ∂M. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027945 | MATHDB-371451

Local spectral rigidity of Liouville metrics on the torus

摘录 / EXCERPT: Let g_0 be a Liouville metric on $\mathbb{T}^d$, let r be the regularity exponent appearing in the C^r norm, and let g be another Riemannian metric. Say that g is Laplace...

conjecture | open | MathDB 页面 | 原文线索

V2-027946 | MATHDB-371473

Cubic skein determination of wrapping number

摘录 / EXCERPT: Let L be a link in the solid torus. Its wrapping number is the minimum geometric intersection number of L with a meridional disk of the solid torus. Cubic skein determination o...

conjecture | open | MathDB 页面 | 原文线索

V2-027947 | MATHDB-371487

Generalization of the main gradient-flow theorems to Hadamard manifolds

摘录 / EXCERPT: The paper considers generalized gradient flows and convex optimization on entanglement polytopes in Hadamard manifolds. In particular, the preceding discussion uses unitary matrice...

conjecture | open | MathDB 页面 | 原文线索

V2-027948 | MATHDB-371503

Mukai's conjecture on syzygies of adjoint line bundles on surfaces

摘录 / EXCERPT: Let X be a surface and let A be an ample line bundle on X. For an integer $p\geq 0$ and an integer ℓ, set $L=K_X+\ell A$. Mukai's conjecture. The line bundle L satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-027949 | MATHDB-371544

Bounded-deviation strengthening for closed tracking geodesics

摘录 / EXCERPT: Bounded-deviation strengthening. There exists $C>0$ such that if an orbit $\widetilde y,\dots,\widetilde f^{n_0}(\widetilde y)$ of $\widetilde f$ crosses $V_C(\widetilde\gamma)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-027950 | MATHDB-371558

Mahler's symmetric volume-product conjecture

摘录 / EXCERPT: Let K be a centrally symmetric convex body in $cmathbb{R}^{n}$, and let $\mathcal{P}(K)$ denote its volume product. Let B_{∞}^{n} be the unit ball of the ℓ_{∞} n...

conjecture | open | MathDB 页面 | 原文线索

V2-027951 | MATHDB-371559

Makai Jr.'s non-symmetric difference-body volume-product conjecture

摘录 / EXCERPT: Let K be a convex body in $\mathbb{R}^{n}$, and let $(K-K)^{\circ}$ be the polar body of its difference body. Makai Jr.'s conjecture. ... Equality should hold if and only if K is...

conjecture | open | MathDB 页面 | 原文线索

V2-027952 | MATHDB-371572

The geography conjecture for closed aspherical 4-manifolds

摘录 / EXCERPT: Let X be a closed oriented aspherical 4-manifold. Write $\chi(X)$ for its Euler characteristic and $\sigma(X)$ for its signature. Geography conjecture. One should have ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-027953 | MATHDB-371576

Existence of smooth ancient Kähler-Ricci flows desingularizing orbifold shrinking solitons

摘录 / EXCERPT: Let (M^4,g,f) be a Kähler Ricci shrinking soliton orbifold singular at p_o, with singularity group in $U(2)$. Desingularization conjecture. There exists a smooth ancient Kähler...

conjecture | open | MathDB 页面 | 原文线索

V2-027954 | MATHDB-371580

Gibbs polystability criterion for log Fano manifolds

摘录 / EXCERPT: Let (X,Δ) be a log Fano manifold. Gibbs polystability conjecture. (X,Δ) is Gibbs polystable if and only if it is uniformly Gibbs polystable if and only if it is K-pol...

conjecture | open | MathDB 页面 | 原文线索

V2-027955 | MATHDB-371581

Analytic and microscopic stability threshold conjecture

摘录 / EXCERPT: Let (X,Δ) be a log Fano manifold with vanishing Futaki character. Let $\gamma(X,\Delta)^{\mathcal{G}}$ denote the reduced Gibbs stability threshold and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027956 | MATHDB-371582

K-reduced partition function and empirical measure conjecture

摘录 / EXCERPT: Let (X,Δ) be a log Fano manifold admitting a Kähler-Einstein metric. Let $Z_{\boldsymbol{m}_{N}}$ be the K-reduced partition function, let $\mathcal{M}$ be the Mabuchi funct...

conjecture | open | MathDB 页面 | 原文线索

V2-027957 | MATHDB-371583

Divisorial polystability threshold conjecture

摘录 / EXCERPT: Let (X,Δ) be a log Fano manifold with vanishing Futaki character. Define the divisorial polystability threshold $\delta(X,\Delta)_{\mathrm{ps}}$ as the infimum of the ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-027958 | MATHDB-371585

Distance-exponent monotonicity conjecture for CLE carpet metrics

摘录 / EXCERPT: For $\kappa \in (8/3,4)$, let $\theta(\kappa)$ denote the distance exponent associated with the canonical geodesic metric on the $\operatorname{CLE}_\kappa$ carpet. Distance-expone...

conjecture | open | MathDB 页面 | 原文线索

V2-027959 | MATHDB-371586

Almost-sure uniqueness conjecture for typical CLE carpet geodesics

摘录 / EXCERPT: Let X be the carpet of a non-nested $\operatorname{CLE}_\kappa$. Given X, let x and y be independent samples from the canonical conformally covariant measure μ_X, norma...

conjecture | open | MathDB 页面 | 原文线索

V2-027960 | MATHDB-371590

The Bergman density rigidity conjecture for polarized hyperbolic Riemann surfaces

摘录 / EXCERPT: Bergman density rigidity conjecture. Then Φ is either holomorphic and

conjecture | open | MathDB 页面 | 原文线索

V2-027961 | MATHDB-371601

Eells' conjecture on harmonic maps from the 3-sphere to the 2-sphere

摘录 / EXCERPT: Let $f:\mathbb{S}^3\to\mathbb{S}^2$ be a harmonic map, where both spheres have their round metrics. Let $\pi:\mathbb{S}^3\to\mathbb{S}^2$ be the Hopf fibration. Eells' conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027962 | MATHDB-371604

The homogeneous reduction of the Jacobian Conjecture

摘录 / EXCERPT: Let n be a positive integer, and let $f:{\mathbb{C}}^n\rightarrow {\mathbb{C}}^n$ be a polynomial map with Jacobian $J(f)=1$. Suppose that, for each i, $f_i=x_i-h_i$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-027963 | MATHDB-371626

The normalized Nash blowup conjecture

摘录 / EXCERPT: Let M be a lattice, let p be the characteristic of the base field, and let $\mathcal{K}_{p}(\mathsf{M})$ be the directed graph whose vertices are pointed full-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-027964 | MATHDB-371627

The normalized Nash blowup conjecture on the N-side

摘录 / EXCERPT: Let N be a lattice and let $\sigma\subset\mathsf{N}_{\mathbf{R}}$ be a cone. The Nash subdivision of σ is obtained from the dual cone...

conjecture | open | MathDB 页面 | 原文线索

V2-027965 | MATHDB-371648

The positivity conjecture for skein partition functions

摘录 / EXCERPT: Let M be a closed, oriented 3-manifold, and write its skein partition function as an Euler expansion with integer exponents c_k. Positivity conjecture for skein partition fun...

conjecture | open | MathDB 页面 | 原文线索

V2-027966 | MATHDB-371653

Conjecture relating trisection invariants to Kashaev's invariant

摘录 / EXCERPT: Kashaev's invariant conjecture. The invariants satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-027967 | MATHDB-371662

Bismut constant holomorphic sectional curvature conjecture

摘录 / EXCERPT: Bismut space-form conjecture. If a compact Hermitian manifold (M^n,g) has constant Bismut holomorphic sectional curvature c and $c\neq 0$, then g must be Kähler.

conjecture | open | MathDB 页面 | 原文线索

V2-027968 | MATHDB-371663

Chen-Nie conjecture for canonical Hermitian connections

摘录 / EXCERPT: Chen-Nie conjecture. Let (M^n,g) be a compact Hermitian manifold. Assume that the holomorphic section curvature of D^r_s is a constant c. If $c\neq 0$, then g must be Kähle...

conjecture | open | MathDB 页面 | 原文线索

V2-027969 | MATHDB-371689

Expected-gonality conjecture for smooth tropical plane curves

摘录 / EXCERPT: Let P be a lattice polygon and let Γ be a smooth tropical plane curve with Newton polygon P. The expected-gonality conjecture asserts that ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-027970 | MATHDB-371690

Dimension-equality conjecture for tropical plane-curve gonality loci

摘录 / EXCERPT: Dimension-equality conjecture. For every $g\geq 0$ and $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-027971 | MATHDB-371696

The asymptotic-velocity conjecture for metric lines in Carnot groups

摘录 / EXCERPT: Asymptotic-velocity conjecture. The metric lines in $\mathbb{G}$ are precisely the sub-Riemannian geodesics $\gamma(t)$ parameterized by arc length for which there exists a unit...

conjecture | open | MathDB 页面 | 原文线索

V2-027972 | MATHDB-371716

Continuity of the Seshadri constant at regular points of arbitrary codimension

摘录 / EXCERPT: Fix a model X_π of X, a point $p\in X_\pi$ whose local ring $\mathcal{O}_{X_\pi,p}$ is regular of dimension $c\le \dim X$, and a regular system of parameters at p. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027973 | MATHDB-371753

The universal L^2-torsion characterization of free-group isomorphisms

摘录 / EXCERPT: Let F_1 and F_2 be finitely generated free groups, let $\varphi\colon F_1\to F_2$ be a homomorphism, and let $\tau^{(2)}_u(\varphi)$ denote its universal L^2-torsion. Univers...

conjecture | open | MathDB 页面 | 原文线索

V2-027974 | MATHDB-371780

Uniqueness conjecture for metric spaces with discrete distance sets

摘录 / EXCERPT: Uniqueness conjecture. Every metric space X with a closed and discrete set $\operatorname{dist} X$ has the property of uniqueness in the class $\operatorname{\mathcal{G\!H}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-027975 | MATHDB-371787

A conjectured local Liakopoulos-Meyer inequality for convex bodies

摘录 / EXCERPT: Let $K\text{ be a convex body in }\mathbb{R}^n$ and let $(\sigma_1,\dots,\sigma_m)$ be an s-cover of $\sigma\subset[n]$. For each $\tau\subset[n]$, let H_τ denote the coordi...

conjecture | open | MathDB 页面 | 原文线索

V2-027976 | MATHDB-371789

Kinetic model for the many-particle PCC model in the normalized case

摘录 / EXCERPT: Let M be the state manifold, let $R>0$, let v denote the velocity variable, and let ${\mathcal K}$ be the interaction kernel. Let μ_t^N be the empirical measure of the norm...

conjecture | open | MathDB 页面 | 原文线索

V2-027977 | MATHDB-371794

Tightness is symplectically determined

摘录 / EXCERPT: Let (Y,ξ) be a contact 3-manifold, and consider the half-symplectization $Y\times(-\infty,0]$ with symplectic form $d(e^t\alpha)$, where $\xi=\ker\alpha$. A compact properly em...

conjecture | open | MathDB 页面 | 原文线索

V2-027978 | MATHDB-371804

Adjoint positivity and embedding conjecture for weakly pseudoconvex complex spaces

摘录 / EXCERPT: Let X be a non-compact weakly pseudoconvex complex space of pure dimension n, let $\pi:\widetilde{X}\longrightarrow X$ be a canonical desingularization, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-027979 | MATHDB-371806

The folklore conjecture on projectively induced Kähler-Einstein metrics

摘录 / EXCERPT: Folklore conjecture. Then (X,g) is a homogeneous space.

conjecture | open | MathDB 页面 | 原文线索

V2-027980 | MATHDB-371822

Chen's conjecture on rationally convex hypersurfaces in rr^4

摘录 / EXCERPT: Let $Y\subset \mathbb{R}^4$ be a hypersurface with first Betti number $b_1(Y)=0$, and let W be the domain bounded by Y. The domain W is rationally convex if it satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-027981 | MATHDB-371838

Weight-λ Newton-stratum classicality conjecture

摘录 / EXCERPT: Let G be the group in the source, let μ be the specified minuscule cocharacter, let $GDR^{\Psi-\mathrm{la}}_{\lambda}$ be the geometric locally analytic de Rham complex of we...

conjecture | open | MathDB 页面 | 原文线索

V2-027982 | MATHDB-371843

The relative Hitchin metric gluing conjecture

摘录 / EXCERPT: Let S be a smooth complex curve, let $X\to S$ be a polarized smooth family of smooth projective curves of genus at least 2, and let $g:M^{s}_{Hig}(X/S,r)\to S$ be the relative...

conjecture | open | MathDB 页面 | 原文线索

V2-027983 | MATHDB-371848

Stable linearisability conjecture for G-surfaces with permutation atoms

摘录 / EXCERPT: Let G act on a surface X. An atom is a component in the atomic semi-orthogonal decomposition associated with the G-surface, and an atom is non-twisted permutation if it has t...

conjecture | open | MathDB 页面 | 原文线索

V2-027984 | MATHDB-371849

Birationality conjecture for geometrically rational surfaces with the same atoms

摘录 / EXCERPT: Let X and Y be geometrically rational surfaces, and let nontrivial atoms mean the atoms other than the trivial pieces in their atomic semi-orthogonal decompositions. Birational...

conjecture | open | MathDB 页面 | 原文线索

V2-027985 | MATHDB-371850

Stable rationality conjecture for smooth projective surfaces

摘录 / EXCERPT: Let X be a smooth projective surface. The surface is stably rational if $X\times\mathbb{P}^1$ is rational. Stable rationality conjecture for surfaces. If ... is rational, then X...

conjecture | open | MathDB 页面 | 原文线索

V2-027986 | MATHDB-371860

Conjecture on the trivial central-character skein module at roots of unity

摘录 / EXCERPT: Trivial-character skein-module conjecture. If $X(M)$ is finite, then

conjecture | open | MathDB 页面 | 原文线索

V2-027987 | MATHDB-371892

The Gauss-Bonnet Penrose inequality

摘录 / EXCERPT: Let M be the mass, A the horizon area, α the Einstein-Gauss-Bonnet coupling, and $\chi(\Sigma)$ the Euler characteristic of the horizon. Gauss-Bonnet Penrose conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-027988 | MATHDB-371941

Kivinen's hyper-Kähler equivalence conjecture for Calogero-Moser spaces

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra and let $\mathcal{CM}_c=\operatorname{Spec}Z(H^{\mathrm{rat}}_{c,0})$ be its Calogero-Moser space, with c as in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-027989 | MATHDB-371969

The Polynomial Wolff axioms conjecture

摘录 / EXCERPT: Polynomial Wolff axioms conjecture. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-027990 | MATHDB-371994

Arithmetic holonomy conjecture for maximal volume ideal polyhedra

摘录 / EXCERPT: Let S be the surface associated with a maximal volume ideal polyhedron, and let ... be its holonomy representation. An arithmetic Fuchsian group is a subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-027991 | MATHDB-371997

Barker's tensor-product characterization of Choquet simplices

摘录 / EXCERPT: Barker's conjecture. $K_1 \otimes_* K_2 = K_1 \otimes^* K_2$ if and only if one of K_1 and K_2 is a Choquet simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-027992 | MATHDB-372004

The Kannan-Lovász-Simonovits conjecture for logconcave measures

摘录 / EXCERPT: KLS conjecture. The KLS constant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-027993 | MATHDB-372005

The thin-shell conjecture for isotropic logconcave measures

摘录 / EXCERPT: Thin-shell conjecture. The thin-shell constant is universally bounded:

conjecture | open | MathDB 页面 | 原文线索

V2-027994 | MATHDB-372006

The slicing conjecture for convex bodies

摘录 / EXCERPT: Slicing conjecture. The slicing constant is universally bounded:

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-027995 | MATHDB-372008

Conjectured coefficients in the transfer-matrix expansion for Camassa-Holm peakons

摘录 / EXCERPT: Conjectured coefficient formula. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-027996 | MATHDB-372015

Nonexistence conjecture for nonnegative weighted scalar curvature on Oeljeklaus-Toma manifolds

摘录 / EXCERPT: Let $X(K,U)$ be an Oeljeklaus-Toma manifold. For a pluriclosed metric, let $R^{H,psi}$ denote the weighted scalar curvature determined by the associated generalized Ricci-flow data...

conjecture | open | MathDB 页面 | 原文线索

V2-027997 | MATHDB-372019

The averaged Zaremba counting conjecture

摘录 / EXCERPT: The averaged Zaremba conjecture. For $N\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-027998 | MATHDB-372044

Nitsche-Fraser-Li conjecture for free boundary minimal annuli

摘录 / EXCERPT: Let $\mathbb{B}^3$ be the Euclidean unit ball, and let an embedded free boundary minimal annulus be an embedded minimal annulus whose boundary meets $\partial\mathbb{B}^3$ orthogon...

conjecture | open | MathDB 页面 | 原文线索

V2-027999 | MATHDB-372045

Rotational symmetry conjecture for free boundary minimal annuli in rescaled spherical caps

摘录 / EXCERPT: Let $R\in[0,\frac{\pi}{2}]$ and let $\mathbb{M}^3_{\kappa_R}$ be the space form of curvature $\kappa_R=\sin^2 R$. An embedded free boundary minimal annulus in...

conjecture | open | MathDB 页面 | 原文线索

V2-028000 | MATHDB-372051

Jensen-Len tropical Martens conjecture for metric graphs

摘录 / EXCERPT: Let Γ be a metric graph of genus g, and let d,r be integers satisfying $0<2r\leq d<g$. The Brill-Noether rank $w_d^r(\Gamma)$ is the tropical Brill-Noether rank. Jensen-...

conjecture | open | MathDB 页面 | 原文线索

V2-028001 | MATHDB-372055

Hard Lefschetz and Hodge-Riemann conjecture for products of Chern classes of Kähler vector bundles

摘录 / EXCERPT: Hard Lefschetz and Hodge-Riemann conjecture. The class

conjecture | open | MathDB 页面 | 原文线索

V2-028002 | MATHDB-372058

Volume rigidity conjecture for closed hyperbolic manifolds

摘录 / EXCERPT: Volume rigidity conjecture. The metric g is isometric to the hyperbolic metric g_H.

conjecture | open | MathDB 页面 | 原文线索

V2-028003 | MATHDB-372059

Generalized Einstein rigidity conjecture for pointwise smaller metrics

摘录 / EXCERPT: Let (M^n,g_E) be a closed Einstein n-manifold with $n\geq 3$ and negative scalar curvature $R^*<0$. Let g be another metric on M satisfying ... Here $g\leq g_E$ means that...

conjecture | open | MathDB 页面 | 原文线索

V2-028004 | MATHDB-372070

Totally geodesic rigidity conjecture for negatively curved manifolds

摘录 / EXCERPT: Let M be a compact Riemannian manifold with negative sectional curvature and dimension at least 3. Suppose that, for some $k\geq 2$, there exist infinitely many distinct immers...

conjecture | open | MathDB 页面 | 原文线索

V2-028005 | MATHDB-372075

Flapan-Macrì-O'Grady-Saccà conjecture on the semiorthogonal decomposition of the Fano eightfold

摘录 / EXCERPT: Flapan-Macrì-O'Grady-Saccà conjecture. The Fano eightfold F admits the semiorthogonal decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-028006 | MATHDB-372076

Eisenbud-Harris conjecture on the genus of curves of nearly maximal genus

摘录 / EXCERPT: Let $C\subseteq\mathbb{P}^r$ be a nondegenerate projective curve of degree d and genus g. For nonnegative integers k,m, say that C satisfies the property $\propa[k,m]$ if i...

conjecture | open | MathDB 页面 | 原文线索

V2-028007 | MATHDB-372077

Petrakiev's conjecture on quadratic relations of points in symmetric position

摘录 / EXCERPT: Let $\Gamma\subseteq\mathbb{P}^{e}$ be a set of d points in symmetric position. For nonnegative integers k,m, say that Γ satisfies the property $\propa[k,m]$ if it is no...

conjecture | open | MathDB 页面 | 原文线索

V2-028008 | MATHDB-372078

The boundary-point conjecture for the Heilbronn triangle problem

摘录 / EXCERPT: Boundary-point conjecture. For any optimal solution with at least nine points, each edge of the unit square $[0,1]^2$ contains exactly two points.

conjecture | open | MathDB 页面 | 原文线索

V2-028009 | MATHDB-372079

Complement determination conjecture for Brunnian theta-curves

摘录 / EXCERPT: Let Γ_1 and Γ_2 be Brunnian θ-curves, meaning spatial θ-curves in S^3 whose proper subgraphs are unknotted. Suppose their complements...

conjecture | open | MathDB 页面 | 原文线索

V2-028010 | MATHDB-372088

Banuelos and Burdzy's width conjecture for multiple Neumann eigenvalues

摘录 / EXCERPT: Ba{n}uelos and Burdzy's conjecture. If $\mu_1(\Omega)$ has multiplicity 2, then

conjecture | open | MathDB 页面 | 原文线索

V2-028011 | MATHDB-372102

Existence of diffeomorphic but not 1-isotopic ribbon surfaces

摘录 / EXCERPT: A ribbon surface is a ribbon surface carried as a 2-dimensional sub-2-handlebody of a 4-dimensional handlebody. A 1-isotopy is a $(1,2)$-deformation between ribbon surfaces,...

conjecture | open | MathDB 页面 | 原文线索

V2-028012 | MATHDB-372106

Birationality conjecture for minimal bipartite graphs

摘录 / EXCERPT: Let G be a minimal bipartite graph on a torus whose Newton polygon has $d+2$ vertices, and let $\mathcal F \colon \mathcal B \to \mathcal S$ be the map from the space of white da...

conjecture | open | MathDB 页面 | 原文线索

V2-028013 | MATHDB-372138

Existence of logarithmic-ratio attaining sequences for compact metric spaces

摘录 / EXCERPT: Conjecture 2.8. If

conjecture | open | MathDB 页面 | 原文线索

V2-028014 | MATHDB-372155

Classification of Poncelet triangle families with stationary orthocenter

摘录 / EXCERPT: Stationary-orthocenter classification conjecture. A Poncelet triangle family maintains X_4 stationary only if the conics are configured as focal-X_4, MacBeath, or Dual.

conjecture | open | MathDB 页面 | 原文线索

V2-028015 | MATHDB-372156

Conservation of half-angle tangents for polar Harmonic families

摘录 / EXCERPT: Polar Harmonic-family conjecture. The sum of half-angle tangents is conserved for the polar image of the Harmonic family, for all $N>3$.

conjecture | open | MathDB 页面 | 原文线索

V2-028016 | MATHDB-372206

Conjecture that Gauduchon hyperbolicity implies balanced hyperbolicity

摘录 / EXCERPT: Gauduchon-to-balanced hyperbolicity conjecture. If X admits a balanced metric ω which is Gauduchon hyperbolic, then X is balanced hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-028017 | MATHDB-372207

Conjecture that p-SKT and p-Kähler manifolds coincide under the ∂∂̄-lemma

摘录 / EXCERPT: p-SKT/p-Kähler conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-028018 | MATHDB-372222

Global hyperbolicity conjecture for cone structures of positive contactomorphism paths

摘录 / EXCERPT: Let Σ be a smooth manifold, let $ST^{\ast}\Sigma$ denote its unit cosphere bundle, and let $(f_t)_{t\in\mathbb R}$ be a positive path of contactomorphisms of...

conjecture | open | MathDB 页面 | 原文线索

V2-028019 | MATHDB-372240

Uniqueness conjecture for trisections of the 4-sphere

摘录 / EXCERPT: A trisection of a closed 4-manifold is a decomposition into three 4-dimensional 1-handlebodies whose pairwise intersections are 3-dimensional handlebodies and whose triple intersec...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028020 | MATHDB-372252

Codimension-5 conjecture for noncollapsed Einstein 5-manifold limits

摘录 / EXCERPT: Codimension-5 conjecture. The set $\mathcal{S}_{\mathrm{orb}}$ is countable. In particular, away from a countable set of points, (X,d) has the structure of a smooth Einstein 5-...

conjecture | open | MathDB 页面 | 原文线索

V2-028021 | MATHDB-372254

Ambrosetti-Malchiodi-Ni concentration-layer conjecture

摘录 / EXCERPT: Let Γ be a k-dimensional submanifold in $\mathbb R^\mathcal{N}$ and a nondegenerate critical point of the functional ... Assume $1<p<\frac{n+2-k}{n-2+k}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-028022 | MATHDB-372270

The commensurable-sides conjecture for $2\pi/3$ tilings

摘录 / EXCERPT: Commensurable-sides conjecture. The tiling must have commensurable sides.

conjecture | open | MathDB 页面 | 原文线索

V2-028023 | MATHDB-372282

Wang-Xia's conjecture on the first Steklov eigenvalue

摘录 / EXCERPT: Wang-Xia's conjecture. The eigenvalue ξ_1 satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028024 | MATHDB-372289

Generalized FAMED conjecture for ideal triangulations

摘录 / EXCERPT: Let K be a hyperbolic knot in ${\mathbb S}^3$, let X be an ordered ideal triangulation of ${\mathbb S}^3{\smallsetminus} K$, and suppose that X admits an angle structure. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028025 | MATHDB-372290

Generalized FAMED semi-geometric triangulation conjecture

摘录 / EXCERPT: Let K be a hyperbolic knot in ${\mathbb S}^3$, and let l and m denote the preferred longitude and meridian, respectively. A semi-geometric triangulation is an ideal triangula...

conjecture | open | MathDB 页面 | 原文线索

V2-028026 | MATHDB-372297

Filip-Tosatti's non-semipositivity conjecture for the line bundle on Grauert's example

摘录 / EXCERPT: Let L be the line bundle in Grauert's example, a projective surface example discussed by Filip and Tosatti. Filip-Tosatti's conjecture. The line bundle L is nef and big, althou...

conjecture | open | MathDB 页面 | 原文线索

V2-028027 | MATHDB-372322

Uniform bounded geometry conjecture for hyperbolic components

摘录 / EXCERPT: Let $\mathcal{M}$ be the Mandelbrot set, let $\mathcal{H}\subset\mathcal{M}$ be a hyperbolic component, and let $\mathcal{M}_{\mathcal{H}}$ be the associated small copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-028028 | MATHDB-372346

Folklore openness conjecture for pseudo-Brody hyperbolicity

摘录 / EXCERPT: Let $f:\mathscr{X}\to\mathbb{D}$ be a holomorphic proper submersion from a complex manifold to the unit disk with connected fibers, and write $X_t=f^{-1}(t)$. A fiber is pseudo-Bro...

conjecture | open | MathDB 页面 | 原文线索

V2-028029 | MATHDB-372427

Strong Green function rigidity conjecture for embedded hypersurfaces

摘录 / EXCERPT: Strong Green function rigidity conjecture. If there exists such a point $p\in M$, then M must be a round sphere.

conjecture | open | MathDB 页面 | 原文线索

V2-028030 | MATHDB-372472

Filtration monotonicity under refinement for canonical relation lattices

摘录 / EXCERPT: Filtration monotonicity under refinement. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028031 | MATHDB-372474

The geometric fibration conjecture for reductive homogeneous spaces

摘录 / EXCERPT: Geometric fibration conjecture. The group Γ is the fundamental group of a closed aspherical manifold M, and there exists a smooth Γ-equivariant fiber bundle

conjecture | open | MathDB 页面 | 原文线索

V2-028032 | MATHDB-372476

The precise geometric fibration conjecture for compact quotients

摘录 / EXCERPT: Precise geometric fibration conjecture. There exists a Γ-invariant contractible smooth submanifold $\widetilde{M}$ of G/K, of dimension $\dim_+(G/H)$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-028033 | MATHDB-372488

Gromov's upper bound conjecture for total mean curvature of spin fill-ins

摘录 / EXCERPT: Gromov's upper bound conjecture. There exists a constant $C(M,\sigma)$ such that for every such X,

conjecture | open | MathDB 页面 | 原文线索

V2-028034 | MATHDB-372523

Splitting conjecture for Floer spectra of del Pezzo complements

摘录 / EXCERPT: Let M be a del Pezzo surface of degree 1, 5, or 7, and let D be a generic smooth representative of M's anticanonical divisor. A stable $\mathbb{R}$-polarization of X...

conjecture | open | MathDB 页面 | 原文线索

V2-028035 | MATHDB-372565

Directional dichotomy conjecture for sets of size p in finite affine 3-space

摘录 / EXCERPT: Let $A\subseteq\mathbb{F}_p^3$ be a set with $|A|=p$. A determined direction is a direction of a line passing through at least two points of A. Directional dichotomy for sets of...

conjecture | open | MathDB 页面 | 原文线索

V2-028036 | MATHDB-372567

Yau's uniformization conjecture for complete noncompact Kähler manifolds

摘录 / EXCERPT: Yau's uniformization conjecture. If M has these properties, then M is biholomorphic to the standard m-dimensional complex Euclidean space

conjecture | open | MathDB 页面 | 原文线索

V2-028037 | MATHDB-372568

Sasakian uniformization conjecture for positive transverse holomorphic bisectional curvature

摘录 / EXCERPT: Sasakian uniformization conjecture. Then M is CR biholomorphic to the standard Heisenberg group $\mathbb{H}_{n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-028038 | MATHDB-372574

Kummer-faithfulness conjecture for finite extensions of random fixed fields

摘录 / EXCERPT: Kummer-faithfulness conjecture. Any finite extension of $\overline{K}(\sigma)$ is Kummer-faithful for almost all $\sigma\in G_K^e$.

conjecture | open | MathDB 页面 | 原文线索

V2-028039 | MATHDB-372624

Lott's conjecture on lower bounds for Hodge-Laplacian eigenvalues

摘录 / EXCERPT: Lott's conjecture. There exists a positive constant $C(n,\kappa,\nu,D)>0$, depending only on n, κ, ν, and D, such that

conjecture | open | MathDB 页面 | 原文线索

V2-028040 | MATHDB-372644

Closed-ball characterization via distance sets in ultrametric spaces

摘录 / EXCERPT: Let (X,d) be an ultrametric space. For each $p\to X$, define ... Let $\overline{\bf B}_X$ denote the set of all closed balls of (X,d), and let $\bf Cs_X$ denote the family of ce...

conjecture | open | MathDB 页面 | 原文线索

V2-028041 | MATHDB-372645

Three-point characterization of ultrametric spaces whose subsets are centered spheres

摘录 / EXCERPT: Let (Y,ρ) be an ultrametric space with $|Y|\geq3$. A centered sphere is a sphere with a center in the underlying space. The three-point centered-sphere conjecture. The followi...

conjecture | open | MathDB 页面 | 原文线索

V2-028042 | MATHDB-372652

The Calabi-Yau criterion for the Maurer-Cartan algebra of an immersed Lagrangian

摘录 / EXCERPT: Let $\mathbb{L}$ be a compact, relatively spin, graded, oriented immersed Lagrangian of dimension n in a symplectic manifold M, and let $\mathcal{A}_{\mathbb{L}}$ be its Maurer...

conjecture | open | MathDB 页面 | 原文线索

V2-028043 | MATHDB-372655

Perturbation conjecture for distinct distances in strictly normed planes

摘录 / EXCERPT: Perturbation conjecture. For every $\epsilon>0$, there is a set $S'=\{p'_1,p'_2,\ldots,p'_n\}$ in bijective correspondence with S such that each corresponding pair satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-028044 | MATHDB-372658

Near-optimal spectral-gap meta-conjecture for random hyperbolic surfaces

摘录 / EXCERPT: Let $(X_k)_k$ be a sequence of random finite-area hyperbolic surfaces such that ... Write $\lambda_1(X_k)$ for the first positive Laplace eigenvalue. Random hyperbolic-surface spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-028045 | MATHDB-372659

Higher-dimensional cNTV spacetimes refute the two-light-line axiom

摘录 / EXCERPT: Let $n \geq 3$ and let M be a cNTV n-dimensional spacetime. Write $\langle M, \mathrel{\scalebox{0.85}{$oldalpha$}} \rangle$ for its frame with the causal relation...

conjecture | open | MathDB 页面 | 原文线索

V2-028046 | MATHDB-372674

Macroscopic scalar curvature Urysohn width conjecture

摘录 / EXCERPT: Macroscopic scalar curvature conjecture. There exist dimensional constants $C_d,\sigma_d>0$ such that, for every closed Riemannian manifold M^d satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-028047 | MATHDB-372677

Hauswirth-Pérez-Romon-Ros isoperimetric conjecture for flat torus products

摘录 / EXCERPT: Hauswirth-Pérez-Romon-Ros conjecture. The only solutions of the isoperimetric problem in $\mathbb{T}^2\times\mathbb{R}$ are spheres, cylinders, and pairs of horizontal planes.

conjecture | open | MathDB 页面 | 原文线索

V2-028048 | MATHDB-372678

Uniformization conjecture for bounded domains with parallel Bergman curvature

摘录 / EXCERPT: Let $\Omega \subset \mathbb{C}^n$, $n \geq 1$, be a bounded domain. Write R^Ω for the curvature tensor of its Bergman metric, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028049 | MATHDB-372681

Codenotti-Santos-Schymura conjecture on covering radii of non-hollow lattice polytopes

摘录 / EXCERPT: Let $P\subseteq \mathbb{R}^d$ be a non-hollow lattice polytope. A full-dimensional lattice polytope is called terminal if it is a direct sum of translates of terminal simplices, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-028050 | MATHDB-372682

Merino-Schymura conjecture on covering minima of terminal simplices

摘录 / EXCERPT: Let $T_d=\operatorname{conv}(-\mathbb{1}_d,e_1,\dots,e_d)$ be the terminal simplex in dimension d, and let $i\in[d]$. Merino-Schymura conjecture. For every $d\in\mathbb{N}$ and e...

conjecture | open | MathDB 页面 | 原文线索

V2-028051 | MATHDB-372683

Codenotti-Santos-Schymura conjecture on covering minima of weighted terminal simplices

摘录 / EXCERPT: Let $\omega=(\omega_0,\dots,\omega_d)\in \mathbb{R}^{d+1}_{>0}$ satisfy $\omega_0\leqslant\dots\leqslant\omega_d$, and define the weighted terminal simplex ... For $i\in[d]$, Codenott...

conjecture | open | MathDB 页面 | 原文线索

V2-028052 | MATHDB-372687

The converse to the surplus condition criterion for point configurations

摘录 / EXCERPT: The converse to the surplus condition criterion. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028053 | MATHDB-372696

Bilinear smoothing conjecture for Fourier integral operators

摘录 / EXCERPT: Let $d\geq2$. Let $T_a^{\phi_1,\phi_2}$ be a bilinear Fourier integral operator whose non-degenerate phase functions satisfy the cinematic curvature condition, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028054 | MATHDB-372750

Improved eigenvalue asymptotics conjecture for critical-order operators on Carnot manifolds

摘录 / EXCERPT: Let (M,H) be a compact Carnot manifold, and let $T\in \Psi^{-d_H}_H(M)$ be a pseudodifferential operator on (M,H) in the van Erp-Yuncken sense. Let ν be a density on M, i...

conjecture | open | MathDB 页面 | 原文线索

V2-028055 | MATHDB-372753

Martínez-Roldán-Pensado-Rubin's 2-colored line-transversal conjecture

摘录 / EXCERPT: Let $\mathcal{K}=\mathcal{K}_1\uplus\mathcal{K}_2$ be a finite 2-colored family of convex sets in $\mathbb{R}^3$. Suppose that every $K_1\in\mathcal{K}_1$ and $K_2\in\mathcal{K}_2$...

conjecture | open | MathDB 页面 | 原文线索

V2-028056 | MATHDB-372754

Fractional line-transversal conjecture for pairwise intersecting convex sets in three dimensions

摘录 / EXCERPT: Let $\mathcal{K}$ be a finite family of n convex sets in $\mathbb{R}^3$, and suppose that every two members $K_1,K_2\in\mathcal{K}$ satisfy $K_1\cap K_2\neq\emptyset$. Fractional...

conjecture | open | MathDB 页面 | 原文线索

V2-028057 | MATHDB-372755

Constant-line transversal conjecture for pairwise intersecting convex sets in three dimensions

摘录 / EXCERPT: Let $\mathcal{K}$ be a finite family of pairwise intersecting convex sets in $\mathbb{R}^3$. Constant-line transversal conjecture. The family $\mathcal{K}$ admits a transversal by...

open_problem | open | MathDB 页面 | 原文线索

V2-028058 | MATHDB-372804

Tropical quasi-automorphism relations extend to the dual canonical basis

摘录 / EXCERPT: Dual-canonical-basis extension conjecture. The relations

conjecture | open | MathDB 页面 | 原文线索

V2-028059 | MATHDB-372805

Tropical cyclic rotation agrees with tableau promotion

摘录 / EXCERPT: Promotion conjecture. For any tableau $T\in\operatorname{SSYT}(k,[n])$ without frozen factors,

conjecture | open | MathDB 页面 | 原文线索

V2-028060 | MATHDB-372806

Tropical reflection identities for Grassmannian cluster variables

摘录 / EXCERPT: Tropical reflection conjecture. For any cluster variable b in $\mathbb{C}[\operatorname{Gr}(k,n)]$, all of the identities listed in the source hold, namely

conjecture | open | MathDB 页面 | 原文线索

V2-028061 | MATHDB-372807

Tropical reflection agrees with evacuation of tableaux

摘录 / EXCERPT: Evacuation conjecture. For every $T\in\operatorname{SSYT}(k,[n])$,

conjecture | open | MathDB 页面 | 原文线索

V2-028062 | MATHDB-372808

Tropical tableau reflection equals evacuation

摘录 / EXCERPT: Tilde-theta evacuation conjecture. For every $T\in\operatorname{SSYT}(k,[n])$,

conjecture | open | MathDB 页面 | 原文线索

V2-028063 | MATHDB-372815

Linear lower-bound conjecture for the bicolored MST crossing number

摘录 / EXCERPT: Let P be a generic set of n points in the plane. The linear lower-bound conjecture. ... The authors identify this as the most important problem for improving the lower bound for...

conjecture | open | MathDB 页面 | 原文线索

V2-028064 | MATHDB-372826

Non-splitting conjecture for symplectic monodromy of intersections of two quadrics

摘录 / EXCERPT: Let $X:=Q_0\cap Q_1\subset\mathbb{P}^{2g+1}$ be the intersection of two smooth quadric hypersurfaces with its canonical monotone symplectic form, and let $\mathcal{M}_X$ be the mod...

conjecture | open | MathDB 页面 | 原文线索

V2-028065 | MATHDB-372843

Nikodym set size conjecture for finite fields

摘录 / EXCERPT: Let $d\geq 3$, let $\mathbb{F}_q^d$ be the d-dimensional vector space over the finite field with q elements, and let a weak Nikodym set be a subset of $\mathbb{F}_q^d$ containi...

conjecture | open | MathDB 页面 | 原文线索

V2-028066 | MATHDB-372866

The proximal limit set characterization for Coxeter polytopes of negative type

摘录 / EXCERPT: Proximal limit set characterization.

conjecture | open | MathDB 页面 | 原文线索

V2-028067 | MATHDB-372867

The density conjecture for extreme points of quasiperfect Vinberg domains

摘录 / EXCERPT: Extreme-point density conjecture. The extreme points of $\overline{\Omega_P}$ are dense in $\partial\Omega_P$ unless W_P is affine of type $\widetilde{A}$.

conjecture | open | MathDB 页面 | 原文线索

V2-028068 | MATHDB-372901

Existence of mono-monostatic convex bodies in smooth strictly convex normed spaces

摘录 / EXCERPT: Let $\mathbb{M}^3$ be a 3-dimensional normed space whose unit ball has a C^2-class differentiable boundary and strictly positive Gaussian curvature. Existence conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-028069 | MATHDB-372903

Karabegov’s conjecture on finite planar point configurations with three closest neighbors

摘录 / EXCERPT: Let a finite planar point configuration be a finite set of points in the plane, where the closest neighbors of a point are the points minimizing Euclidean distance from it. Karabeg...

conjecture | open | MathDB 页面 | 原文线索

V2-028070 | MATHDB-372930

Complete Kähler metric question for regular loci and complements

摘录 / EXCERPT: Let X be a weakly pseudoconvex Kähler complex space. Its regular locus is denoted by X_{reg}, and an analytic subset of a complex space is denoted by A. Comple...

open_problem | open | MathDB 页面 | 原文线索

V2-028071 | MATHDB-372931

Nadel vanishing for exhaustion singular-positive line bundles

摘录 / EXCERPT: Let X be a weakly pseudoconvex manifold, let $L\longrightarrow X$ be a holomorphic line bundle, and let $\mathscr{J}$ be an ideal sheaf on X. Assume that L is exhaustion sing...

conjecture | open | MathDB 页面 | 原文线索

V2-028072 | MATHDB-372952

Siu's conjecture on deformation invariance of plurigenera for compact Kähler manifolds

摘录 / EXCERPT: Siu's conjecture. The m-genus $P_m(X_t)$ is independent of $t\in\Delta$. The problem is widely open for smooth Kähler families, although deformation invariance is known in severa...

conjecture | open | MathDB 页面 | 原文线索

V2-028073 | MATHDB-372954

Deformation invariance of volumes of adjoint classes in smooth Kähler families

摘录 / EXCERPT: Let $f:X\to S$ be a smooth family from a Kähler manifold X onto a smooth, connected, relatively compact curve S. Assume that $(X,B+\boldsymbol{\beta})$ is a generalized klt pai...

conjecture | open | MathDB 页面 | 原文线索

V2-028074 | MATHDB-372979

The higher-dimensional obstruction-tensor conjecture at null infinity

摘录 / EXCERPT: Let $\{\mathscr{I},\bm{\gamma},\bm\ell,\ell^{(2)},\sigma,\mathfrak q\}$ be $\mathscr{I}$-structure data of odd dimension $\mathfrak n\ge 7$ admitting a cross-section...

conjecture | open | MathDB 页面 | 原文线索

V2-028075 | MATHDB-372998

Kähler rigidity for manifolds with continuous symmetries

摘录 / EXCERPT: Let $(M_1,g_1,J_1,omega_1)$ and $(M_2,g_2,J_2,omega_2)$ be compact Kähler manifolds. Let $\operatorname{Isom}_{\mathrm{Kah}}}(M_i)$ denote the Lie group of holomorphic isometries of...

conjecture | open | MathDB 页面 | 原文线索

V2-028076 | MATHDB-373000

Clean lex tuple conjecture for monic Weierstrass hypersurfaces

摘录 / EXCERPT: Let g and h be the base polynomials in the monic Weierstrass family of hypersurfaces ... and let the clean lex tuple be ... A bounded-delay descent condition means that improvement...

conjecture | open | MathDB 页面 | 原文线索

V2-028077 | MATHDB-373001

Discretized ranking-function conjecture for purely inseparable hypersurface singularities

摘录 / EXCERPT: Let Π be the discretization map from the paper, and let $f_0,\ldots,f_{25}$ be the intrinsic invariants used to construct a five-component raw rank function...

conjecture | open | MathDB 页面 | 原文线索

V2-028078 | MATHDB-373010

Toric characterization of mean-width minimizers under symplectomorphisms

摘录 / EXCERPT: Toric mean-width conjecture. A convex body $K\subset\mathbb{R}^{2n}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028079 | MATHDB-373048

Gamma conjecture for asymptotics of mirror Lagrangian periods

摘录 / EXCERPT: Let X be a Calabi--Yau manifold equipped with a symplectic form ω, and let $\{Z_t\}_{t\in\Delta^\ast}$ be a family of Calabi--Yau manifolds parametrized by t in a small...

conjecture | open | MathDB 页面 | 原文线索

V2-028080 | MATHDB-373064

Extension of the projected closed-loop ISS result to the full tangent bundle

摘录 / EXCERPT: Extension of the projected closed-loop ISS result. Under standard model-order-reduction assumptions, the closed-loop dynamics on the submanifold...

conjecture | open | MathDB 页面 | 原文线索

V2-028081 | MATHDB-373089

Integral scalar-curvature formula for three-manifolds with nonnegative Ricci curvature

摘录 / EXCERPT: Let (M^3,g) be a complete non-compact Riemannian manifold, let $q\in M$, and suppose that $Rc\geq 0$ and M has maximum volume growth. Writing R for the scalar curvature and...

conjecture | open | MathDB 页面 | 原文线索

V2-028082 | MATHDB-373090

Hamilton's flatness conjecture for Ricci-pinched noncompact three-manifolds

摘录 / EXCERPT: Let (M^3,g) be a complete, non-compact Riemannian manifold. Say that (M^3,g) is Ricci pinched when it satisfies the Ricci-pinching condition used in Hamilton's conjecture. Hami...

conjecture | open | MathDB 页面 | 原文线索

V2-028083 | MATHDB-373125

The higher-genus open r-spin Witten conjecture

摘录 / EXCERPT: Let r be the spin parameter, let $g\geq 2$, let $F_g^{x^r,o}$ denote the open r-spin potential, and let ϕ_g denote the corresponding closed r-spin potential. The notatio...

conjecture | open | MathDB 页面 | 原文线索

V2-028084 | MATHDB-373163

Equivariant multiplicity inequality for wobbly components

摘录 / EXCERPT: Let $\theta:\mathcal{M}\to\mathbb{A}$ be an integrable system, with virtual and genuine equivariant multiplicities $\mu_F(t)$ and $m_F(t)$ for fixed components F. Wobbly-componen...

conjecture | open | MathDB 页面 | 原文线索

V2-028085 | MATHDB-373185

Bi-Ricci curvature conjecture for immersions into the unit ball

摘录 / EXCERPT: Let X^4 be a closed manifold and let $f:X^4\to\mathbb{B}^N$ be an immersion. Write $g=f^*g_{\mathbb{R}^N}$ for the induced metric, and let $\mathfrak{c}(f)$ denote its minimal no...

conjecture | open | MathDB 页面 | 原文线索

V2-028086 | MATHDB-373186

Ricci eigenvalue conjecture for immersions into the unit ball

摘录 / EXCERPT: Let X^n be a closed manifold and let $f:X^n\to\mathbb{B}^N$ be an immersion. Write $g=f^*g_{\mathbb{R}^N}$ for the induced metric, and let $\mathfrak{c}(f)$ denote its minimal no...

conjecture | open | MathDB 页面 | 原文线索

V2-028087 | MATHDB-373242

The discrete filling-area density conjecture

摘录 / EXCERPT: The discrete filling-area density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-028088 | MATHDB-373243

The logarithmic Brunn-Minkowski conjecture for origin-symmetric convex bodies

摘录 / EXCERPT: Let $K,L\in\mathcal{K}_e(\mathbb{R}^n)$ be origin-symmetric convex bodies containing the origin in their interiors, and let K^tL^{1-t} denote their logarithmic combination, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-028089 | MATHDB-373244

The local L^p-Brunn-Minkowski conjecture

摘录 / EXCERPT: Let $K,L\in\mathcal{K}_e(\mathbb{R}^n)$ be origin-symmetric convex bodies containing the origin in their interiors. Let V denote the mixed volume functional, let S_K be the sur...

conjecture | open | MathDB 页面 | 原文线索

V2-028090 | MATHDB-373252

Vanishing Haantjes torsion for mutual symmetries with a gl-regular combination

摘录 / EXCERPT: Let $K_1,\dots,K_n$ be n-dimensional operator fields that are mutual symmetries, meaning that they satisfy the mutual-symmetry condition, and are linearly independent at every po...

conjecture | open | MathDB 页面 | 原文线索

V2-028091 | MATHDB-373257

Positivity conjecture for the fractional Bakry-Émery curvature constant

摘录 / EXCERPT: Let $\kappa(\gamma)$ denote the limiting curvature constant associated with the fractional Laplacian parameter γ, as defined in the surrounding results. Positivity conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-028092 | MATHDB-373270

The stacky Batyrev-Manin conjecture

摘录 / EXCERPT: Stacky Batyrev-Manin conjecture. There exists a thin set $T\subset\mathcal{X}\langle F\rangle$ and $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-028093 | MATHDB-373274

Song-Tian's Gromov-Hausdorff convergence conjecture for Kähler-Ricci flow

摘录 / EXCERPT: Let (X^n,ω_0) be a compact Kähler manifold with semiample canonical bundle K_X. Let $\omega(t)$ solve the normalized Kähler-Ricci flow ... Let $f:X\to Y$ be the canonical-mo...

conjecture | open | MathDB 页面 | 原文线索

V2-028094 | MATHDB-373283

Ball's Santaló type inequality for symmetric convex bodies

摘录 / EXCERPT: Let Ksubseteq$\mathbb R^n$ be a symmetric convex body, and define its polar body by ... Let $B_2^n=\{x\in\mathbb R^n:\|x\|\leq 1\}$ be the Euclidean unit ball. Ball's conjecture. .....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-028095 | MATHDB-373288

Hausdorff measure conjecture for regular complex infinite IFS limit sets

摘录 / EXCERPT: Hausdorff measure conjecture. 1. If $h_{\mathcal{F}}<2\theta_{\mathcal{F}}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-028096 | MATHDB-373289

Finite Hausdorff measure conjecture for large-parameter complex infinite IFSs

摘录 / EXCERPT: Finite Hausdorff measure conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-028097 | MATHDB-373309

Existence of non-commutative crepant resolutions for Gorenstein toric algebras

摘录 / EXCERPT: Let R be a Gorenstein toric algebra, meaning the coordinate ring associated to a Gorenstein cone. An NCCR of R is an algebra of the form $\Lambda=\operatorname{End}_R(M)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-028098 | MATHDB-373339

The face-map conjecture for local-model toric strata

摘录 / EXCERPT: Let ${\rm Adm}^{\mathcal{G}}(\{\mu\})$ be the admissible set and let $|\Delta|$ be the divisor map from this poset to the face poset ${\mathscr{F}}(\sigma_{{\mathcal{G}},\mu})$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-028099 | MATHDB-373361

The area-invariance conjecture for concentric circular 3-Poncelet pairs

摘录 / EXCERPT: Area-invariance conjecture. The area $\mathrm{Area}(\triangle ABC)$ is independent of the choice of $\triangle ABC\in\mathcal{P}$ if and only if $\mathcal{C}$ and $\mathcal{D}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-028100 | MATHDB-373362

Betke-Henk-Wills conjecture on lattice point enumerators

摘录 / EXCERPT: Betke-Henk-Wills conjecture. For any $K\in\mathcal{K}^d_0$ and $\Lambda\in\mathcal{L}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-028101 | MATHDB-373363

Corrected Guy-Kelly asymptotic conjecture for the no-three-in-line problem

摘录 / EXCERPT: Let S_n be the set of n^2 points in $\mathbb{R}^2$ with integer coordinates (x,y) satisfying $1\leq x,y<n$, and let f_n be the maximum size of a subset of S_n containing...

conjecture | open | MathDB 页面 | 原文线索

V2-028102 | MATHDB-373388

Verstraelen's isoparametricity conjecture for minimal hypersurfaces

摘录 / EXCERPT: Let M^{n} be a closed immersed minimal hypersurface of the unit sphere $\mathbb{S}^{n+1}$ with constant scalar curvature. A hypersurface is isoparametric if it has constant princ...

conjecture | open | MathDB 页面 | 原文线索

V2-028103 | MATHDB-373401

The finite-field Erdős-Falconer distance conjecture in even dimensions

摘录 / EXCERPT: Erdős-Falconer distance conjecture. If $|E|\geq Cq^{d/2}$ for a sufficiently large constant C independent of q, then

conjecture | open | MathDB 页面 | 原文线索

V2-028104 | MATHDB-373436

Determinant asymptotic conjecture for the Corner System

摘录 / EXCERPT: Let F be a skew finite subset of $\boldsymbol{\backslashell}_1^N$, and let $r\in(0,\operatorname{skew}(F)/2)$. Define ... so that k is the number of 1-dimensional faces in...

conjecture | open | MathDB 页面 | 原文线索

V2-028105 | MATHDB-373455

Nonintegrability conjecture for twists of the standard almost complex structure on the 6-sphere

摘录 / EXCERPT: Let S^6 carry its standard almost complex structure J, and let $\operatorname{Aut}(TS^6)$ denote the automorphisms of its tangent bundle. For $\psi\in\operatorname{Aut}(TS^6)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028106 | MATHDB-373460

The p-adic non-abelian Hodge correspondence for smooth projective rigid analytic spaces

摘录 / EXCERPT: Let X be a smooth projective rigid analytic space over $\mathbb{C}_p$, and let G be a reductive group defined over $\mathbb{C}_p$. A $G(\mathbb{C}_p)$-representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-028107 | MATHDB-373494

Logarithmic GW/PT correspondence for primary insertions

摘录 / EXCERPT: Let $(Y\mid\partial Y)$ be a toric threefold pair, let β be a curve class, let $\bm\mu$ be a vector of partitions, and let $\upalpha$ be an insertion. Write d_β for the...

conjecture | open | MathDB 页面 | 原文线索

V2-028108 | MATHDB-373495

Stratum logarithmic GW/PT correspondence

摘录 / EXCERPT: Let $(Y\mid\partial Y)$ be a toric threefold pair, let β be a curve class, let $\bm\mu$ be a vector of partitions, and let $[[![\sigma,\vartheta,\upalpha][!]]$ be a stratum c...

conjecture | open | MathDB 页面 | 原文线索

V2-028109 | MATHDB-373498

Borisenko's reverse isoperimetric conjecture for lambda-convex bodies

摘录 / EXCERPT: Let $n\geqslant 2$, and let $M^n(c)$ be an n-dimensional model space of constant sectional curvature $c\in\mathbb{R}$. Let $K\subset M^n(c)$ be a λ-convex body, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028110 | MATHDB-373507

The Geometric Riemann Hypothesis for the fractal zero set

摘录 / EXCERPT: Let Z_F be the fractal zero set constructed from the positive imaginary parts of the non-trivial zeros of $\zeta(s)$ by the recursive four-subinterval construction described in t...

conjecture | open | MathDB 页面 | 原文线索

V2-028111 | MATHDB-373508

The canonical ordering conjecture for the fractal zero set

摘录 / EXCERPT: Let Z_F be constructed from the non-trivial zeta zeros, and consider orderings of those zeros, including the standard ordering by increasing imaginary part. Canonical Ordering co...

conjecture | open | MathDB 页面 | 原文线索

V2-028112 | MATHDB-373527

Darmon-Vonk antisymmetry conjecture for Hecke-translated real quadratic singular moduli

摘录 / EXCERPT: Let $\tau,\omega\in\mathcal{H}_p$ be RM points with ω not in the G-orbit of τ, let $\mathrm{T}\in\mathcal{I}_p$ be a Hecke operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028113 | MATHDB-373542

Besse's spherical CPE conjecture

摘录 / EXCERPT: Let (M^n,g,f) be a CPE metric, meaning that (M^n,g) is a closed, oriented Riemannian manifold of dimension $n\geq 3$ with constant scalar curvature and $f:M^n\to\mathbb{R}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-028114 | MATHDB-373565

The non-big movable divisor conjecture for smooth Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a smooth Calabi-Yau threefold, and let its effective cone be the cone generated by effective divisor classes and its movable cone be the cone generated by divisor classe...

conjecture | open | MathDB 页面 | 原文线索

V2-028115 | MATHDB-373566

The Hilbert-basis-hole conjecture for Kreuzer-Skarke Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a smooth Calabi-Yau threefold arising from the Kreuzer-Skarke database, and let a non-trivial Hilbert basis element be a divisor class in the Hilbert basis under conside...

conjecture | open | MathDB 页面 | 原文线索

V2-028116 | MATHDB-373583

The equivariant CR Yamabe conjecture

摘录 / EXCERPT: Let (M,J,θ) be a compact strictly pseudoconvex CR manifold of real dimension $2n+1$, and let G be a compact subgroup of $I(M,\theta)$, the group of pseudo-Hermitian transf...

conjecture | open | MathDB 页面 | 原文线索

V2-028117 | MATHDB-373591

Viterbo's isocapacitary conjecture for convex bodies

摘录 / EXCERPT: Let $K,Q e othing$ be convex bodies in $\mathbb{R}^n$, and let $c(K\times Q)$ denote the symplectic capacity of their Lagrangian product. Viterbo's isocapacitary conjecture. The La...

conjecture | open | MathDB 页面 | 原文线索

V2-028118 | MATHDB-373593

Meylender-Thalmaier optimal-cover conjecture for truncated squares

摘录 / EXCERPT: Meylender-Thalmaier conjecture. The systolic ratio is strictly less than 1 for every convex shape K, and it is maximized with respect to K if and only if K is a translate o...

conjecture | open | MathDB 页面 | 原文线索

V2-028119 | MATHDB-373600

Lawson's Clifford torus uniqueness conjecture

摘录 / EXCERPT: Let $T\subset S^3$ be an embedded minimal torus, and let T_{Cl} denote the Clifford torus in S^3. Lawson's conjecture. There exists an isometry F of S^3 such that...

conjecture | open | MathDB 页面 | 原文线索

V2-028120 | MATHDB-373616

The higher-order GJMS geometric Aubin set conjecture

摘录 / EXCERPT: The higher-order GJMS geometric Aubin set conjecture. The set $\mathscr{A}_{n,2k}$ is a geometric Aubin set for L_{2k}.

conjecture | open | MathDB 页面 | 原文线索

V2-028121 | MATHDB-373617

The optimal dimension bound for higher-order GJMS Aubin sets

摘录 / EXCERPT: Let $k,m\in\mathbb{N}$ and let $N(k,m)$ be the dimension threshold appearing in the higher-order GJMS geometric Aubin set conjecture for conformal manifolds virtually conformal to...

conjecture | open | MathDB 页面 | 原文线索

V2-028122 | MATHDB-373661

The local Langlands parametrization conjecture for disconnected even orthogonal groups

摘录 / EXCERPT: Let H be a p-adic symplectic group or split special orthogonal group, and in the special even orthogonal case let H^+ be the corresponding full even orthogonal group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028123 | MATHDB-373663

The peak heat flux conjecture for convex planar domains

摘录 / EXCERPT: Let Ω be a bounded convex domain in the plane, let λ_1 be its first Dirichlet eigenvalue, and let u_1 be the corresponding L^2-normalized eigenfunction. Define .....

conjecture | open | MathDB 页面 | 原文线索

V2-028124 | MATHDB-373665

Birational invariance conjecture for the K3 atom's BPS spectrum

摘录 / EXCERPT: Let $X \subset \mathbb{P}^5$ be a smooth cubic fourfold with Kuznetsov component $\mathcal{A}_X$, and let $\widetilde{X} \to X$ be a birational modification obtained by a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-028125 | MATHDB-373792

Adic and differential structure conjecture for potentially unramified towers

摘录 / EXCERPT: Potentially unramified tower conjecture. The following assertions hold: (1) $\hat{\tilde{X}}$ is an adic space and $(\hat{\tilde{X}})^\diamond=\tilde{X}^\diamond$; (2) the natural...

conjecture | open | MathDB 页面 | 原文线索

V2-028126 | MATHDB-373794

Minuscule bi-analytic Ax-Schanuel conjecture

摘录 / EXCERPT: Let $G/\mathbb{Q}_p$ be a reductive group, let $[\mu]$ be a minuscule conjugacy class of cocharacters of $G_{\overline{\mathbb{Q}}_p}$, and let $b\in G(\breve{\mathbb{Q}}_p)$ repre...

conjecture | open | MathDB 页面 | 原文线索

V2-028127 | MATHDB-373816

Uniform affine Dimension Growth Conjecture for hypersurfaces

摘录 / EXCERPT: Let $H\subset\mathbb{A}^n$ be the hypersurface defined by a polynomial $F\in\mathbb{Z}[X_1,\ldots,X_n]$ of degree $d\geq 2$, with absolutely irreducible leading form. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028128 | MATHDB-373838

Higher-degree denominator conjecture for membrane indices on $\mathcal{O}(-1)\oplus\mathcal{O}(3)$ over $\mathbb{P}^2$

摘录 / EXCERPT: Higher-degree denominator conjecture. For every integer $d>1$,

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V2-028129 | MATHDB-373839

Laurent-polynomial conjecture for membrane indices on $\mathcal{O}(-2)\oplus\mathcal{O}(2)$ over $\mathbb{P}^3$

摘录 / EXCERPT: Laurent-polynomial conjecture. For every degree d, $\Omega_{d[L]}$ is a Laurent polynomial in q_0, q_1, q_2, and q_3.

conjecture | open | MathDB 页面 | 原文线索

V2-028130 | MATHDB-373841

Conjectural unboundedness of third covariant Ricci derivatives along Kähler-Ricci flow

摘录 / EXCERPT: Unboundedness conjecture. Under the assumptions above, the quantity

conjecture | open | MathDB 页面 | 原文线索

V2-028131 | MATHDB-373859

Existence and non-degeneracy conjecture for the Steklov maximizer among convex polygons

摘录 / EXCERPT: Existence and non-degeneracy conjecture. The supremum is attained by a convex polygon Ω^* with exactly N sides. In particular, the maximizer cannot be a degenerate polygon...

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V2-028132 | MATHDB-373860

E. Oudet's regular-polygon conjecture for the Steklov maximizer

摘录 / EXCERPT: Let $\mathcal{C}_{N}$ be the class of convex polygons in $\mathbb{R}^2$ with at most N sides and perimeter 2π, and let Ω^* be a maximizer of the first nonzero Steklov...

conjecture | open | MathDB 页面 | 原文线索

V2-028133 | MATHDB-373877

The weak-similarity characterization of maximal centers of distances

摘录 / EXCERPT: The weak-similarity characterization. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-028134 | MATHDB-373878

The isometry characterization by centers of distances

摘录 / EXCERPT: The isometry characterization. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-028135 | MATHDB-373879

The realization conjecture for finite distance centers of ultrametric spaces

摘录 / EXCERPT: The realization conjecture. There exists a finite ultrametric space (X,d) such that

conjecture | open | MathDB 页面 | 原文线索

V2-028136 | MATHDB-373893

Gorman's peel restriction conjecture for finite metric spaces

摘录 / EXCERPT: Let d be a metric of strict negative type on a finite space, let $\mathcal{I}$ be a subset of $\operatorname{peel}(d)$, and let $d|_{\mathcal{I}}$ denote the restriction of d t...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028137 | MATHDB-373894

Euclidean peel restriction conjecture

摘录 / EXCERPT: Let d be the metric induced on a finite subset of $\mathbb{R}^m$, let $\mathcal{I}$ be a subset of $\operatorname{peel}(d)$, and let $d|_{\mathcal{I}}$ denote the restriction of...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028138 | MATHDB-373895

General-position peel restriction conjecture

摘录 / EXCERPT: Let d be a metric of strict negative type on a finite space in general position, let $\mathcal{I}$ be a subset of $\operatorname{peel}(d)$, and let $d|_{\mathcal{I}}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-028139 | MATHDB-373896

The smooth Cayley fibration non-existence conjecture for full-holonomy Spin(7)-manifolds

摘录 / EXCERPT: A torsion-free ${\mathrm{Spin}}(7)$-manifold is a Riemannian manifold with torsion-free ${\mathrm{Spin}}(7)$-structure, and a smooth Cayley fibration is a smooth fibration whose fi...

conjecture | open | MathDB 页面 | 原文线索

V2-028140 | MATHDB-373897

The boundary Dehn twist conjecture for the Case 2 Cayley-fibration fiber

摘录 / EXCERPT: Let F be a smooth manifold homeomorphic to ... Let $B\subset F$ be a small 4-ball. A boundary Dehn twist is the boundary-supported diffeomorphism of $F\setminus B$ associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-028141 | MATHDB-373898

Property (S) conjecture for manifolds homeomorphic to K3#K3#(S2×S2)

摘录 / EXCERPT: Let F be a smooth manifold homeomorphic to ... Such a manifold satisfies Property (S) when it has the spinnability property for fiber bundles defined earlier in the paper. Property...

conjecture | open | MathDB 页面 | 原文线索

V2-028142 | MATHDB-373915

Hellinger-kernel conjecture for trained diffusion-model NTKs

摘录 / EXCERPT: Let Θ be the neural tangent kernel of a diffusion model and let k_H denote the Hellinger kernel of its data distribution. A Hellinger kernel is understood here as the uniq...

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V2-028143 | MATHDB-373923

Staticity conjecture for noncollapsed ancient Ricci flows with Ricci-flat tangent cones

摘录 / EXCERPT: Let $(M,g_t)_{t\in(-\infty,0]}$ be a noncollapsed ancient Ricci flow with bounded curvature within each compact time interval. Suppose ... on $M\times(-\infty,0]$, and suppose every...

conjecture | open | MathDB 页面 | 原文线索

V2-028144 | MATHDB-373947

Relative quasi-convergence and compactification of the stability manifold

摘录 / EXCERPT: Let R be a ring and let $\mathcal{D}$ be an R-linear, smooth and proper category. A continuous path $\sigma_\bullet:[0,+\infty)\rightarrow\operatorname{Stab}(\mathcal{D}/R)$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-028145 | MATHDB-373952

Flat shadow boundary characterization of ellipsoids

摘录 / EXCERPT: Let $K\subset\mathbb R^n$, $n\geq 3$, be a convex body and let S be a hypersurface that is the image of an embedding of the sphere $\mathbb S^{n-1}$, with K contained in the in...

conjecture | open | MathDB 页面 | 原文线索

V2-028146 | MATHDB-373953

Bianchi-Gruber conjecture on centrally symmetric sections

摘录 / EXCERPT: Let $K\subset\mathbb R^n$, $n\geq 3$, be a strictly convex body, and let $\mathcal S$ be a family of hyperplanes that is the image of an embedding of the tangent bundle of the sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-028147 | MATHDB-373955

Central symmetry from inversely homothetic sections

摘录 / EXCERPT: Let $K,S\subset\mathbb R^n$, $n\geq 3$, be convex bodies with $S\subset\operatorname{int}K$. Inversely homothetic sections conjecture. If, for every pair of parallel hyperplanes...

conjecture | open | MathDB 页面 | 原文线索

V2-028148 | MATHDB-373957

Local abundance of flat shadow boundaries implies an ellipsoid

摘录 / EXCERPT: Let $K\subset\mathbb R^n$, $n\geq 3$, be a convex body with boundary of class C^3. Local light-source conjecture. If for every $p\in\partial K$ there are at least $n+1$ point lig...

conjecture | open | MathDB 页面 | 原文线索

V2-028149 | MATHDB-373958

Two-body flat-shadow characterization of concentric homothetic ellipsoids

摘录 / EXCERPT: Let $K,L\subset\mathbb R^n$, $n\geq 3$, be convex bodies with $L\subseteq K$. Two-body flat-shadow conjecture. If for every $p\in\partial K$ there are at least $L(n)$ point light s...

conjecture | open | MathDB 页面 | 原文线索

V2-028150 | MATHDB-373959

Orthogonal tangent intersections characterize quadrics

摘录 / EXCERPT: Let $M\subset\mathbb R^n$, $n\geq 3$, be an open patch of a hypersurface of class C^2, and let Γ_1,Γ_2 be space curves. Orthogonal-intersection conjecture. If for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-028151 | MATHDB-373960

The analogue of the strong Viterbo conjecture for systolically convex domains

摘录 / EXCERPT: Let $(T^*\mathbb T^2,\omega_{\rm can})$ be the cotangent bundle of the two-torus, and let $\mathcal S$ denote the class of systolically convex domains defined in the source. A symp...

conjecture | open | MathDB 页面 | 原文线索

V2-028152 | MATHDB-373991

Conjecture on infinitely many half-volume constant-mean-curvature hypersurfaces

摘录 / EXCERPT: Let (M^{n+1},g) be a closed Riemannian manifold with $3\le n+1\le 7$. A closed hypersurface encloses half the volume of M when one of the regions it bounds has volume equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-028153 | MATHDB-374031

Birational invariance conjecture for the image of the Hitchin morphism

摘录 / EXCERPT: Birational invariance conjecture. The set-theoretic image $\operatorname{im}(h^r_X)$ of the Hitchin morphism is invariant under smooth birational modifications.

conjecture | open | MathDB 页面 | 原文线索

V2-028154 | MATHDB-374032

The local relative p-adic monodromy conjecture around rank-1 points

摘录 / EXCERPT: Let X be a smooth rigid-analytic space over a p-adic field K, and let $\mathbb{L}$ be a de Rham $\mathbb{Q}_p$-local system on X. A point $x\in X$ has rank 1 if it is a r...

conjecture | open | MathDB 页面 | 原文线索

V2-028155 | MATHDB-374040

The abundance conjecture for Kähler log canonical pairs

摘录 / EXCERPT: Let (X,Δ) be a Kähler log canonical pair. A divisor is semiample if some positive multiple is base-point free. Abundance conjecture. If $K_X+\Delta$ is nef, then...

conjecture | open | MathDB 页面 | 原文线索

V2-028156 | MATHDB-374048

Optimality of the constant-deviation density bound for continuous triangulations

摘录 / EXCERPT: Let $\mathcal{T}$ be any locally finite triangulation of $\mathbb{R}^2$ and let f be a continuous piecewise linear ε-approximation of $F(x,y)=xy$ with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-028157 | MATHDB-374080

Yang's conjecture on quasi-positive holomorphic sectional curvature and rational connectedness

摘录 / EXCERPT: Let X be a compact Kähler manifold, and let TX denote its holomorphic tangent bundle. A Hermitian metric $c9$ on TX is assumed to be uniformly RC-quasi-positive, or equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-028158 | MATHDB-374081

Rationality and symmetry conjecture for stable-pair generating series

摘录 / EXCERPT: Stable-pair rationality and symmetry conjecture. The series $Z_\beta^\mathsf{PT}(q\mid D)$ is the Laurent expansion of a rational function, more precisely of $q^{d_\beta/2}f(q)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-028159 | MATHDB-374096

Purity implies formality for quantised orientation modules

摘录 / EXCERPT: Let (X,σ) be a holomorphic symplectic manifold and let $\mathfrak{L}$ be a countable collection of orientable compact Lagrangian submanifolds such that the weight f...

conjecture | open | MathDB 页面 | 原文线索

V2-028160 | MATHDB-374097

The de Rham-perverse-sheaf spectral sequence conjecture

摘录 / EXCERPT: Let X be a holomorphic symplectic manifold, and let $\mathscr{D}_\mathrm{L}$ and $\mathscr{D}_\mathrm{M}$ be quantised orientation modules associated to Lagrangian subma...

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V2-028161 | MATHDB-374098

Kapustin's hyperkähler Fukaya-deformation-quantisation correspondence

摘录 / EXCERPT: Let $(\mathrm{X},\mathrm{I},\mathrm{J},\mathrm{K})$ be a hyperkähler manifold with symplectic form ω_J, and set ... Let $\mathcal{DF}(\mathrm{X},\omega_\mathrm{J})$ b...

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V2-028162 | MATHDB-374099

Ivan Smith's formality conjecture for hyperkähler Fukaya categories

摘录 / EXCERPT: Let $(\mathrm{X},\mathrm{I},\mathrm{J},\mathrm{K},g)$ be a hyperkähler manifold, and let $\mathfrak{L}$ be a collection of compact spin I-holomorphic Lagrangian submanif...

conjecture | open | MathDB 页面 | 原文线索

V2-028163 | MATHDB-374100

The Solomon-Verbitsky Fukaya/deformation-quantisation quasi-isomorphism conjecture

摘录 / EXCERPT: Let $\mathfrak{L}$ be a Solomon-Verbitsky collection of Lagrangian submanifolds in a holomorphic symplectic manifold (X,σ). Let $\widehat{\mathcal{A}}_\mathfrak{L}$...

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V2-028164 | MATHDB-374101

Hexality conjecture for bicontact forms on the surgered 3-torus

摘录 / EXCERPT: Let $T^3=\boldsymbol{R}^3/\boldsymbol{Z}^3$ and suppose $n_x>0$, $n_y<0$, and $n_z>0$. Let M_{n_x,n_y,n_z} be the result of the specified surgeries on the coordinate curves, with...

conjecture | open | MathDB 页面 | 原文线索

V2-028165 | MATHDB-374102

Hexality conjecture for swapped contact structures and Anosov flows

摘录 / EXCERPT: Suppose $n_x>0$, $n_y<0$, and $n_z>0$, and let M_{n_x,n_y,n_z} carry the Anosov flows ϕ_{xy} and ϕ_{yz} from the two surgery descriptions. A contact-structure swapping...

conjecture | open | MathDB 页面 | 原文线索

V2-028166 | MATHDB-374106

Bondal's conjecture on degeneracy loci of Fano Poisson manifolds

摘录 / EXCERPT: Bondal's conjecture. For every $k\in\{0,1,\dots,r-1\}$, $D_{2k}(\pi)$ has an irreducible component of dimension at least $2k+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-028167 | MATHDB-374115

Thurston's Euler class realization conjecture for taut foliations

摘录 / EXCERPT: Let M be a closed orientable irreducible atoroidal 3-manifold with positive first Betti number. An integral class is a class $a\in H^2(M;\mathbb{R})$ arising from integral cohomo...

conjecture | open | MathDB 页面 | 原文线索

V2-028168 | MATHDB-374121

Mesa limit and uniform projection conjecture

摘录 / EXCERPT: Mesa limit and uniform projection conjecture. The flows should converge to a Hele-Shaw or mesa-type free-boundary evolution whose terminal state is the Wasserstein projection of...

conjecture | open | MathDB 页面 | 原文线索

V2-028169 | MATHDB-374138

Candidate anharmonic reactive width from the classical normal form

摘录 / EXCERPT: Let $K_{\mathrm{CNF}}(I,J_2,\dots,J_n)$ be a truncated classical normal form near an index-1 saddle, where I is the reaction integral and J_k are the bath actions. On the norma...

conjecture | open | MathDB 页面 | 原文线索

V2-028170 | MATHDB-374169

Existence conjecture for the moduli spaces of L-manifolds

摘录 / EXCERPT: A family of spaces (or, more precisely, Artin stacks) $\mathcal{L}_{0,n}$ should exist, equipped with the following structure. The spaces form a cyclic operad and have “forgetful”...

open_problem | open | MathDB 页面 | 原文线索

V2-028171 | MATHDB-374185

Classification conjecture for totally magnetic submanifolds of ellipsoids

摘录 / EXCERPT: Classification conjecture. Every closed totally magnetic submanifold of $E(\mathbf a)$ is of the form

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V2-028172 | MATHDB-374278

Geometric stability of the Schoen-Yau zero mass rigidity theorem

摘录 / EXCERPT: Let M_j^3 be a sequence of three-dimensional manifolds in the class $\mathcal{M}$, and let $m_{ADM}(M_j^3)$ denote their ADM masses. Let ${\mathbb E}^3$ denote Euclidean three-sp...

conjecture | open | MathDB 页面 | 原文线索

V2-028173 | MATHDB-374279

Volume convergence consequence of the Schoen-Yau zero mass stability conjecture

摘录 / EXCERPT: Let $\Omega_j(R)$ be the isoperimetric regions associated with a sequence satisfying the hypotheses of the zero mass stability conjecture, and let $B_0(R)\subset{\mathbb E}^3$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-028174 | MATHDB-374281

Capacity convergence consequence of the Schoen-Yau zero mass stability conjecture

摘录 / EXCERPT: Let Ω_j be the isoperimetric regions associated with a sequence satisfying the hypotheses of the zero mass stability conjecture, and let $B_0(R)\subset{\mathbb E}^3$ be the...

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V2-028175 | MATHDB-374286

The Generalized Property R Conjecture for links

摘录 / EXCERPT: Let $L=K_1\cup\dots\cup K_n$ be an n-component link in S^3. A 0-framed surgery on L means surgery with framing zero on every component, and let $\#_n(S^2\times S^1)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-028176 | MATHDB-374287

The Weak Generalized Property R Conjecture

摘录 / EXCERPT: An R-link is a link L such that 0-surgery on L yields a connected sum of copies of $S^1\times S^2$. A 0-framed unlink is an unlink whose components all have framing zero, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-028177 | MATHDB-374298

Intertwining symplectomorphism for the compactified Ruijsenaars-Schneider system

摘录 / EXCERPT: Let U and $\widehat U$ be open neighbourhoods of the zero section of T^*S^3 and of the fiber $\hat \beta^{-1}(I(2))$ in $P(\mu_0(y))$, respectively. Let X and W be the inte...

conjecture | open | MathDB 页面 | 原文线索

V2-028178 | MATHDB-374300

Projected-stationarity conjecture for complete Das-Dennis grids

摘录 / EXCERPT: Projected-stationarity conjecture. For every $H\geq 2$, the complete Das-Dennis grid G_H is projected-stationary for anchored-box magnitude on Δ_2. More precisely:

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V2-028179 | MATHDB-374305

Donaldson's conjecture on hypersymplectic structures and hyperkähler structures

摘录 / EXCERPT: Donaldson's conjecture. For every such M and $\underline{\omega}$, there exists a family with these properties.

conjecture | open | MathDB 页面 | 原文线索

V2-028180 | MATHDB-374308

Conjecture on Steklov spectral determination of convex polygonal domains

摘录 / EXCERPT: Let P be a convex polygonal domain, and let Ω be a simply-connected smoothly bounded domain. Two domains are Steklov isospectral if their Steklov spectra, counted with mul...

conjecture | open | MathDB 页面 | 原文线索

V2-028181 | MATHDB-374309

Optimality of the lower d_1-J comparison constant

摘录 / EXCERPT: Optimality conjecture. The constant $m=\frac{2}{n+1}\left(\frac{n}{n+1}\right)^n$ is optimal for all Kähler manifolds (X,ω).

conjecture | open | MathDB 页面 | 原文线索

V2-028182 | MATHDB-374310

Stability of approximable Kähler geodesic rays under interpolation

摘录 / EXCERPT: Interpolation conjecture. For each $s\in[0,1]$, the ray $\{u_t^s\}$ is also approximable.

conjecture | open | MathDB 页面 | 原文线索

V2-028183 | MATHDB-374311

Rauch's hot spot conjecture for convex and simply connected planar domains

摘录 / EXCERPT: Let Ω be a domain in $\mathbb{R}^n$, and let u be a second Neumann eigenfunction on Ω. Rauch's hot spot conjecture. If Ω is convex, or if Ω is a simpl...

conjecture | open | MathDB 页面 | 原文线索

V2-028184 | MATHDB-374325

Sharp symmetric-body bound for Hausdorff distance from convexity

摘录 / EXCERPT: Let $K\subset\mathbb{R}^2$ be a symmetric convex body with strictly convex C^1 boundary. For nonempty compact sets $A,B\subset\mathbb{R}^2$, let $\mathcal{R}_K(a,b)$ be the famil...

conjecture | open | MathDB 页面 | 原文线索

V2-028185 | MATHDB-374328

Nontrivial-domain existence conjecture for finite-time overdetermination

摘录 / EXCERPT: Finite-time nontriviality conjecture. For every $m\in\mathbb{N}$, this problem admits nontrivial solutions, meaning solutions whose domains are not Euclidean balls.

conjecture | open | MathDB 页面 | 原文线索

V2-028186 | MATHDB-374343

Nonexistence conjecture for isolated cylindrical singularities with stable connected-link cones

摘录 / EXCERPT: Let Y be an exact special Lagrangian submanifold in $\mathbf C^{n+1}$ with tangent cone $C\times\mathbf R$ at 0. Assume that C is stable, meaning that the only linear and qua...

conjecture | open | MathDB 页面 | 原文线索

V2-028187 | MATHDB-374352

Admissibility criterion for regular spherical polygons

摘录 / EXCERPT: Admissibility criterion. A regular spherical polygon is admissible if and only if the inner angle of its polar is at least

conjecture | open | MathDB 页面 | 原文线索

V2-028188 | MATHDB-374364

Kourouniotis's bending conjecture for compact hyperbolic manifolds

摘录 / EXCERPT: Let M be a compact hyperbolic n-manifold with $n>2$, and consider deformations of its complete hyperbolic structure. Kourouniotis's conjecture. The only nontrivial deformations...

conjecture | open | MathDB 页面 | 原文线索

V2-028189 | MATHDB-374393

McKernan's ACC conjecture for interpolated lc thresholds

摘录 / EXCERPT: McKernan's ACC conjecture. There exists an ACC set $\Gamma\subset [0,1]$ depending only on d such that, for every such $\mathcal{F}$ on X, the threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-028190 | MATHDB-374413

Uniform quasicircle conjecture for CaTherine wheels

摘录 / EXCERPT: Let $R>r>0$, let G be a Kleinian group, and let f be a CaTherine wheel invariant under G. Assume that the injectivity radius of $\bbH^3/G$ lies everywhere in the interval...

conjecture | open | MathDB 页面 | 原文线索

V2-028191 | MATHDB-374414

No-wandering-continua conjecture for cocompact Kleinian groups

摘录 / EXCERPT: Let G be a cocompact Kleinian group acting on the sphere at infinity S^2_∞, and let $K\subset S^2_\infty$ be a compact connected set disjoint from every G-translate of i...

conjecture | open | MathDB 页面 | 原文线索

V2-028192 | MATHDB-374423

Rational linearity conjecture for resonance components of formal CDGAs

摘录 / EXCERPT: Let (A,d) be a connected, q-finite, and q-formal $\mathbb{k}$-CDGA. For $i\leq q$ and $s\geq 1$, let $\mathcal{R}^{i,s}(A)$ and $\mathcal{R}_{i,s}(A)$ denote the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-028193 | MATHDB-374428

Weight-Two Classification for positive tropical Grassmannians

摘录 / EXCERPT: Weight-Two Classification. The point

conjecture | open | MathDB 页面 | 原文线索

V2-028194 | MATHDB-374430

Maximal-rank conjecture for focal maps of equiclassical plane curves

摘录 / EXCERPT: Let W be an irreducible component of the real equiclassical family of curves in $(\mathbb{P}^2)^\vee$ of degree c, genus g, and class d, and let G be a general smooth poi...

conjecture | open | MathDB 页面 | 原文线索

V2-028195 | MATHDB-374445

Uniqueness of the conic-sweeping triangle center for inversive Poncelet triangles

摘录 / EXCERPT: Let $\mathcal{E}$ and $\mathcal{E}_c$ be nested ellipses admitting a family $\mathcal{T}$ of Poncelet triangles, and let $T=ABC$ be a triangle in $\mathcal{T}$. Let $T'=A'B'C'$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-028196 | MATHDB-374453

Jerrard-Wetzel-Yuan conjecture on convex polyhedra

摘录 / EXCERPT: Let $\mathcal{P}$ be a convex polyhedron, meaning the convex hull of a finite set of points in $\mathbb{R}^3$ in convex position. A polyhedron is Rupert if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-028197 | MATHDB-374454

Steininger-Yurkevich conjecture on the rhombicosidodecahedron

摘录 / EXCERPT: Let $\mathcal{P}$ be a polyhedron. It is Rupert if there exist $R_1,R_2\in\operatorname{SO}(3)$ and $t\in\mathbb{R}^2$ such that ... where $P:\mathbb{R}^3\to\mathbb{R}^2$ drops the...

conjecture | open | MathDB 页面 | 原文线索

V2-028198 | MATHDB-374455

Conjecture that the stellated tetrahedron $\mathcal{P}_{11/20}$ is not Rupert

摘录 / EXCERPT: Let $\mathcal{P}_a$ be the stellated tetrahedron with vertices ... ... where edges join all p_i,p_j and p_i,q_j for $i\ne j$, and $a\in(0.5,0.57)$. A polyhedron is Rupert if it...

conjecture | open | MathDB 页面 | 原文线索

V2-028199 | MATHDB-374461

Width-one conjecture for rational toric hypersurfaces

摘录 / EXCERPT: Width-one conjecture. Under the stated hypothesis, P has width one.

conjecture | open | MathDB 页面 | 原文线索

V2-028200 | MATHDB-374469

Kawamata-Matsuki finiteness conjecture for wlc klt models

摘录 / EXCERPT: Finiteness of minimal models. The number of (log) isomorphism classes of projective wlc klt models in a fixed 0-class is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-028201 | MATHDB-374470

Kawamata-Morrison-Totaro arithmetic cone conjecture

摘录 / EXCERPT: Arithmetic cone conjecture. Suppose (X,B) is klt. Then there is a rational polyhedral cone $\Pi\subseteq\textnormal{Nef}^e(X/Y)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-028202 | MATHDB-374471

Kawamata-Morrison-Totaro geometric cone conjecture

摘录 / EXCERPT: Geometric cone conjecture. Suppose (X,B) is klt and X is $\mathbb{Q}$-factorial. Then the number of $\textnormal{Aut}(X/Y,B)$-equivalence classes of faces of...

conjecture | open | MathDB 页面 | 原文线索

V2-028203 | MATHDB-374482

Isosceles Trapezoid Peg Problem

摘录 / EXCERPT: Let a Jordan curve be a simple closed curve $\gamma\subset\mathbb{R}^2$. An isosceles trapezoid is a quadrilateral whose vertices are the intersection set of two parallel lines and...

conjecture | open | MathDB 页面 | 原文线索

V2-028204 | MATHDB-374506

Kára-Pór-Wood Big-Line-Big-Clique Conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}^2$ be a finite planar point set. Two points of A are visible if the open segment between them contains no other point of A; a mutually visible subset is...

conjecture | open | MathDB 页面 | 原文线索

V2-028205 | MATHDB-374507

Ambient cubic-container problem for visibility cliques

摘录 / EXCERPT: Let $A\subset\mathbb{R}^2$ be a finite point set. A visibility patch is a subset $W\subset\Gamma(\mathbb{R})$ of a real cubic on which the relevant pairs of points are considered,...

open_problem | open | MathDB 页面 | 原文线索

V2-028206 | MATHDB-374508

The uniqueness and nonexistence conjecture for Einstein metrics on branched hyperbolic coverings

摘录 / EXCERPT: Let M be a closed hyperbolic manifold of dimension $n\geq 4$, and let X be a nontrivial finite covering of M branched along a closed totally geodesic submanifold of codimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-028207 | MATHDB-374546

Gromov's (α,β,λ)-convergence conjecture for scalar curvature

摘录 / EXCERPT: Let X and X_i be closed oriented Riemannian manifolds, and let $W_{\alpha_i,\beta_i,\lambda_i}\colon X_i\to X$ denote Riemannian $(\alpha_i,\beta_i,\lambda_i)$-cobordisms, mean...

conjecture | open | MathDB 页面 | 原文线索

V2-028208 | MATHDB-374551

Kontsevich's Fukaya-category reconstruction conjecture for Gromov-Witten invariants

摘录 / EXCERPT: Let X be a symplectic manifold, let $\mathcal{F}uk(X)$ denote its Fukaya category, and let the Gromov-Witten invariants of X be its symplectic enumerative invariants. Kontsevic...

conjecture | open | MathDB 页面 | 原文线索

V2-028209 | MATHDB-374569

Laugesen-Siudeja's sharp area-perimeter eigenvalue conjecture for triangles

摘录 / EXCERPT: Laugesen-Siudeja's conjecture. The functional $\mathcal{F}(\triangle)$ is minimized uniquely by the equilateral triangle, and its minimum value is $7\sqrt{3}\pi^2/12$. This is a sh...

conjecture | open | MathDB 页面 | 原文线索

V2-028210 | MATHDB-374570

Parini's sharp Cheeger inequality conjecture for triangles

摘录 / EXCERPT: Let $\triangle$ be a triangle, let $\lambda_1(\triangle)$ be its first Dirichlet eigenvalue, and let ... be its Cheeger constant. Parini's triangle conjecture. For any triangle...

conjecture | open | MathDB 页面 | 原文线索

V2-028211 | MATHDB-374574

Wehler's weakened local-isomorphism conjecture for smooth families

摘录 / EXCERPT: Let S be a complex manifold, and let $\pi:{\mathcal X}\to S$ and $\pi':{\mathcal Y}\to S$ be smooth families of compact complex manifolds. Write ${\mathcal X}_U=\pi^{-1}(U)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-028212 | MATHDB-374598

The smallest-eigenvalue conjecture for Hodge Laplacians on homogeneous 3-spheres

摘录 / EXCERPT: Let $g_{(a,b,c)}$ be the homogeneous metric on the Lie group $\operatorname{SU}(2)$ or $\operatorname{SO}(3)$ determined by positive parameters a,b,c. For the Hodge Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-028213 | MATHDB-374655

Georgakopoulos-Papasoglu's coarse Kuratowski conjecture

摘录 / EXCERPT: A length space is a metric space in which the distance between two points is the infimum of the lengths of paths joining them. A length space is quasi-isometric to a graph when the...

conjecture | open | MathDB 页面 | 原文线索

V2-028214 | MATHDB-374685

Pre-compactness of globally hyperbolic spacetimes with Ricci lower bounds

摘录 / EXCERPT: Let (M,h) be an n-dimensional Riemannian manifold, and let g be a Lorentzian metric such that (M,g) is a globally hyperbolic spacetime. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-028215 | MATHDB-374687

Almost nonnegative curvature operator Betti-number conjecture

摘录 / EXCERPT: Let m be a positive integer, and let (M^m,g) be a compact Riemannian manifold with diameter $\operatorname{diam}(M)=1$. Write $\mathcal{R}m$ for its curvature operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028216 | MATHDB-374710

Kinoshita conjecture on reducibility of knotted projective planes

摘录 / EXCERPT: A smoothly embedded projective plane in S^4 is knotted if it is not the standard unknotted embedding, and it is reducible if it admits an unknotted summand that is not a 2-sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-028217 | MATHDB-374746

The toric quotient conjecture for hybrid fans

摘录 / EXCERPT: Let $W \subset \mathrm{GL}(V)$ be a finite reflection group preserving a lattice $M \subset V$ and an M-rational complete simplicial fan $\Sigma \subset V$. Let X_Σ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-028218 | MATHDB-374754

Growth-rate conjecture for primitive Eisenstein circle packings

摘录 / EXCERPT: Let E) be a moiety of a bounded primitive Eisenstein circle packing. Let N be a positive real number, and let c_E denote a constant depending on E. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028219 | MATHDB-374773

Soldatenkov-Verbitsky conjecture for smooth families of holomorphically symplectic manifolds

摘录 / EXCERPT: Let X_t be a smooth family of compact holomorphically symplectic manifolds over a disk Δ. Assume that X_t is hyperkähler for all $t\neq 0$. Soldatenkov-Verbitsky conject...

conjecture | open | MathDB 页面 | 原文线索

V2-028220 | MATHDB-374786

Conjecture on the normalized volume spectrum of compact right-angled hyperbolic polyhedra

摘录 / EXCERPT: Let $\Omega(\mathcal R_{comp})$ denote the spectrum of normalized volumes of compact right-angled hyperbolic polyhedra, let $\operatorname{vol}(R(5))$ be the volume of the right-an...

conjecture | open | MathDB 页面 | 原文线索

V2-028221 | MATHDB-374791

The generic Hilbert function conjecture for general unions of flat fat points

摘录 / EXCERPT: Let $Z_{s,m}$ be a general union of s flat fat points of multiplicity m in $\mathbb P^2$. Assume $m\geq 4$ and $s\geq 2m-3$. Its Hilbert function should satisfy ... A flat fat po...

conjecture | open | MathDB 页面 | 原文线索

V2-028222 | MATHDB-374797

The simple-manifold decomposition conjecture

摘录 / EXCERPT: Simple-manifold decomposition conjecture. X_n is either bimeromorphically irreducible symplectic or Kummer. In particular, X_n is Kummer if n is odd.

conjecture | open | MathDB 页面 | 原文线索

V2-028223 | MATHDB-374798

The isotriviality conjecture for fibrations with simple fibres

摘录 / EXCERPT: Isotriviality conjecture for simple fibres. If X_y is simple and $a(Y)=0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-028224 | MATHDB-374834

Hyperkähler rotation conjecture for affine Toda moduli spaces

摘录 / EXCERPT: Let G be a semisimple group over $\a9\a9$ and let G^{sc} be its simply-connected cover. Let $\mathcal{M}^{\circ}_G$ denote the neutral component of the affine Toda mod...

conjecture | open | MathDB 页面 | 原文线索

V2-028225 | MATHDB-374844

S.-T. Yau's four minimal two-spheres conjecture

摘录 / EXCERPT: Let M be a manifold diffeomorphic to S^3, equipped with a Riemannian metric. An embedded minimal two-sphere in M is an embedded minimal submanifold diffeomorphic to S^2. S....

conjecture | open | MathDB 页面 | 原文线索

V2-028226 | MATHDB-374876

Tosatti-Weinkove's Chern-Ricci flow convergence conjecture for minimal surfaces

摘录 / EXCERPT: Let X^n be a compact complex manifold with K_X nef and big, and let $\tilde\omega(s)$ be the solution of the Chern-Ricci flow ... starting at an arbitrary Hermitian metric...

conjecture | open | MathDB 页面 | 原文线索

V2-028227 | MATHDB-374901

Schechtman's zonotope sparsification conjecture

摘录 / EXCERPT: Schechtman's sparsification conjecture. Is there a universal constant C such that there is a zonotope $\widetilde Z$ generated by at most

open_problem | open | MathDB 页面 | 原文线索

V2-028228 | MATHDB-374906

Hirschowitz's formal principle conjecture for smooth unbendable rational curves

摘录 / EXCERPT: Let A be a smooth unbendable rational curve in a complex manifold X. The curve A satisfies the formal principle: whenever a compact complex submanifold $\widetilde{A}$ in a c...

conjecture | open | MathDB 页面 | 原文线索

V2-028229 | MATHDB-374912

Milman's spectral conjecture for positively Ricci-curved spheres

摘录 / EXCERPT: Let d be a positive integer, let $(\mathbb{S}^d,g)$ be a Riemannian sphere, and let vol_g denote its volume measure. Suppose that ... for some $\rho>0$. Let...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028230 | MATHDB-374913

Milman's contracting transport conjecture for positively Ricci-curved spheres

摘录 / EXCERPT: Milman's contracting transport conjecture. There exists a map

conjecture | open | MathDB 页面 | 原文线索

V2-028231 | MATHDB-374914

Milman's Gaussian spectral conjecture under the CD condition

摘录 / EXCERPT: Let d be a positive integer, let $(\mathbb{R}^d,g,\mu)$ satisfy the curvature-dimension condition $\operatorname{CD}(\rho,\infty)$ for some $\rho>0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028232 | MATHDB-374921

Mixed-volume asymptotic conjecture for amoeba contours

摘录 / EXCERPT: Let $V_d=\{f_{1,d}=\cdots=f_{r,d}=0\}\subset(\mathbb C^\ast)^n$ be the complete intersection under consideration, and let $\Delta_{i,d}$ be the Newton polytope of $f_{i,d}$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-028233 | MATHDB-374923

Gromov-Sormani MinA scalar compactness conjecture

摘录 / EXCERPT: Gromov-Sormani MinA scalar compactness conjecture. A subsequence converges in the volume-preserving intrinsic flat sense to a three-dimensional rectifiable limit space M_∞....

conjecture | open | MathDB 页面 | 原文线索

V2-028234 | MATHDB-374924

Convexity conjecture for illumination bodies in nonpositive-curvature space forms

摘录 / EXCERPT: Let $n\geq 3$, let $\lambda\leq 0$, and let $K\in\mathcal{K}_0(\mathrm{Sp}^n(\lambda))$ be a convex body. Illumination-body convexity conjecture. For every $\delta>0$, the illumina...

conjecture | open | MathDB 页面 | 原文线索

V2-028235 | MATHDB-374925

Yamabe product conjecture for $M^{n-1}\times S^1$ and $S^2\times T^2$

摘录 / EXCERPT: Yamabe product conjecture. The Yamabe constant $Y(g(T))$ is minimized by $u_{n,T}$, and

conjecture | open | MathDB 页面 | 原文线索

V2-028236 | MATHDB-374932

Separate polynomiality of symmetric-power Chern coefficients

摘录 / EXCERPT: Separate polynomiality conjecture. For each fixed k and each partition $\lambda\vdash k$, the coefficient $f_\lambda(n,d)$ is polynomial in d for fixed n, and polynomial in...

conjecture | open | MathDB 页面 | 原文线索

V2-028237 | MATHDB-374933

Strong binomiality conjecture for symmetric-power Chern coefficients

摘录 / EXCERPT: Strong binomiality conjecture. For each fixed k and each partition $\lambda\vdash k$, the coefficient $f_\lambda(n,d)$ is a polynomial in finitely many binomial coefficients of t...

conjecture | open | MathDB 页面 | 原文线索

V2-028238 | MATHDB-374935

Shifted-binomial log-concavity conjecture for Plücker coefficients

摘录 / EXCERPT: Shifted-binomial log-concavity conjecture. For every partition λ without parts equal to 1, and every admissible index i, the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-028239 | MATHDB-374936

Shifted Plücker value log-concavity conjecture

摘录 / EXCERPT: Let $\lambda=(2^{e_2},\ldots,m^{e_m})$ be a partition without parts equal to 1, set $N:=|\lambda|$ and $M:=|\widetilde\lambda|$, and write ... For each admissible pair...

conjecture | open | MathDB 页面 | 原文线索

V2-028240 | MATHDB-374942

Engliš-Zhang conjecture for the Fefferman-Szegő kernel

摘录 / EXCERPT: Let $\Omega\subset\mathbb C^n$ be a C^∞-smooth bounded strongly pseudoconvex domain, and let $\partial\Omega$ denote its boundary. Consider the logarithmic term in the asymp...

conjecture | open | MathDB 页面 | 原文线索

V2-028241 | MATHDB-374994

Schwartz's discreteness conjecture for complex hyperbolic triangle groups

摘录 / EXCERPT: Schwartz's conjecture. The $(p,q,r;\alpha)$ representation of $\Delta_{p,q,r}$ is discrete and faithful if and only if W_A and W_B are non-elliptic. Furthermore, if $p<10$, it...

conjecture | open | MathDB 页面 | 原文线索

V2-028242 | MATHDB-375001

Limbeek-Zimmer conjecture on divisible domains in flag manifolds

摘录 / EXCERPT: Let G/P be a flag manifold, where G/P is neither real projective space nor the conformal sphere. A proper divisible domain is a proper domain in G/P admitting a properly disc...

conjecture | open | MathDB 页面 | 原文线索

V2-028243 | MATHDB-375002

Limbeek-Zimmer's reformulated conjecture for almost-homogeneous domains

摘录 / EXCERPT: Let G be a simple Lie group and let P be a parabolic subgroup of G. A domain in G/P is proper almost-homogeneous if it is proper and admits an automorphism group with a com...

conjecture | open | MathDB 页面 | 原文线索

V2-028244 | MATHDB-375010

The multiple cover formula for reduced Gromov-Witten invariants of symplectic surfaces

摘录 / EXCERPT: Let S be a symplectic surface, and let $\beta\in H_2(S,\mathbb{Z})$ be an effective curve class. For every positive integer k dividing β, let ... be a degree-preserving...

conjecture | open | MathDB 页面 | 原文线索

V2-028245 | MATHDB-375021

Li's compactness conjecture for manifolds with uniformly convex boundary

摘录 / EXCERPT: Let (M^n,g) be a complete Riemannian manifold with smooth nonempty boundary ∂M. Suppose M has nonnegative Ricci curvature and ∂M is uniformly convex, meanin...

conjecture | open | MathDB 页面 | 原文线索

V2-028246 | MATHDB-375024

The generalized one-layer conjecture for framed graph-layer presentations

摘录 / EXCERPT: Let ... be a graph-layer word in the twists T_{ij}, and let $\mathbf m=(m_1,\ldots,m_g)\in\mathbb Z^g$. Put ... The associated framed graph-layer presentation is ... A framed graph-lay...

conjecture | open | MathDB 页面 | 原文线索

V2-028247 | MATHDB-375041

The generalized LeBrun equality for compact complex manifolds

摘录 / EXCERPT: Generalized LeBrun equality. The equality above should hold for any compact complex n-manifold, at least when K_M is big.

conjecture | open | MathDB 页面 | 原文线索

V2-028248 | MATHDB-375064

Hyperkähler cone conjecture for conical symplectic singularities

摘录 / EXCERPT: Hyperkähler cone conjecture. Any conical symplectic singularity is a hyperkähler cone.

conjecture | open | MathDB 页面 | 原文线索

V2-028249 | MATHDB-375070

Fino-Vezzoni conjecture for the pluriclosed flow when $c_1=0$

摘录 / EXCERPT: Fino-Vezzoni flow conjecture. The pluriclosed flow exists for all time and converges smoothly to a Kähler metric. In particular, M is Kähler.

conjecture | open | MathDB 页面 | 原文线索

V2-028250 | MATHDB-375124

Maximal restricted sections of the cross-polytope

摘录 / EXCERPT: Let $B_1^n=\operatorname{conv}\{\pm e_1,\ldots,\pm e_n\}$ be the cross-polytope, and let ... For $a\in\mathcal{M}$, consider the central hyperplane section $B_1^n\cap a^\perp$. Maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-028251 | MATHDB-375128

General-k upper-bound conjecture for temporal Wasserstein forecasting

摘录 / EXCERPT: Let $k\ge1$. On a regular problem with an equispaced dense design, consider a degree-k temporal local-polynomial forecaster on the tangent bundle, obtained by geodesic or barycen...

conjecture | open | MathDB 页面 | 原文线索

V2-028252 | MATHDB-375146

Conjecture on smooth Kähler structures for formally Kähler manifolds

摘录 / EXCERPT: Let $p\gg 0$, and let X be a $W^{2,p}\times W^{1,p}$ formally Kähler manifold with almost-complex structure J and nearby form ω_J. Smooth Kähler structure conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-028253 | MATHDB-375147

Conjectural low-regularity Newlander-Nirenberg theorem

摘录 / EXCERPT: Let X be a $W^{2,p}\times W^{1,p}$ formally Kähler manifold with complex structure J and metric g. Low-regularity Newlander-Nirenberg conjecture. The Newlander-Nirenberg theo...

conjecture | open | MathDB 页面 | 原文线索

V2-028254 | MATHDB-375157

Transverse lattice-point counting conjecture for spherical bands

摘录 / EXCERPT: Let $A_{k,d,\lambda}$ denote the maximal number of lattice points in the intersection of k unit-width, ν-transverse bands on the sphere λS^{d-1}, with...

conjecture | open | MathDB 页面 | 原文线索

V2-028255 | MATHDB-375162

Boundary curvature-integral conjecture for Alexandrov spaces

摘录 / EXCERPT: Boundary curvature-integral conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028256 | MATHDB-375168

Minimal model conjecture for generalized pairs of relative log numerical dimension zero

摘录 / EXCERPT: Minimal model conjecture. Then $(X/Z,B,\mathbf{M})$ has a minimal model.

conjecture | open | MathDB 页面 | 原文线索

V2-028257 | MATHDB-375170

Haiden-Katzarkov-Kontsevich-Pandit conjecture for Lagrangian mean curvature flow

摘录 / EXCERPT: Let (X,ω,Ω) be a compact Calabi-Yau manifold, and let $L_1,\ldots,L_r$ be special Lagrangian submanifolds in X with the same phase that intersect transversely. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028258 | MATHDB-375171

Conjecture on iterated balancing filtrations and Kähler-Ricci flow

摘录 / EXCERPT: Let X be a K-semistable Fano manifold, and let ω denote the initial Kähler form for the Kähler-Ricci flow. The iterated balancing filtration is the filtration referred to...

conjecture | open | MathDB 页面 | 原文线索

V2-028259 | MATHDB-375189

Scalar-curvature extension conjecture for the Kähler-Ricci flow coupled with a (1,1)-form

摘录 / EXCERPT: Let $(\mathcal{X},\omega)$ be a closed Kähler manifold and let α be a closed nonnegative $(1,1)$-form satisfying ... Let $(\omega(t),\alpha(t))$ solve the Kähler-Ricci flow co...

conjecture | open | MathDB 页面 | 原文线索

V2-028260 | MATHDB-375190

Maximal-time conjecture for the Kähler-Ricci flow coupled with a (1,1)-form

摘录 / EXCERPT: Let $(\mathcal{X},\omega)$ be a closed Kähler manifold, let α be a closed nonnegative $(1,1)$-form, and let $(\omega(t),\alpha(t))$ solve the coupled flow on its maximal int...

conjecture | open | MathDB 页面 | 原文线索

V2-028261 | MATHDB-375191

Bryant's small-torsion convergence conjecture for the Laplacian flow

摘录 / EXCERPT: Let φ be a closed G_2-structure, meaning a positive closed three-form, on a closed 7-manifold. Consider the Laplacian flow ... A closed G_2-structure has sufficiently s...

conjecture | open | MathDB 页面 | 原文线索

V2-028262 | MATHDB-375192

Yau's compactification conjecture for complete Ricci-flat Kähler manifolds

摘录 / EXCERPT: Let M be a complete Ricci-flat Kähler manifold. Yau's compactification conjecture. M can be compactified into a compact complex manifold by adding a divisor at infinity. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-028263 | MATHDB-375193

Metric-cone conjecture for asymptotic cones of gravitational instantons

摘录 / EXCERPT: Let (M,g) be a gravitational instanton, and let an asymptotic cone mean a pointed-Gromov-Hausdorff limit obtained by rescaling (M,g) along a sequence tending to infinity. Metri...

conjecture | open | MathDB 页面 | 原文线索

V2-028264 | MATHDB-375194

Uniqueness conjecture for asymptotic cones of gravitational instantons

摘录 / EXCERPT: Let (M,g) be a gravitational instanton. Asymptotic-cone uniqueness conjecture. The gravitational instanton (M,g) has a unique asymptotic cone. Uniqueness is known for complete...

conjecture | open | MathDB 页面 | 原文线索

V2-028265 | MATHDB-375196

Bando-Kasue-Nakajima conjecture for ALE gravitational instantons

摘录 / EXCERPT: Let (M,g) be an ALE gravitational instanton. Bando-Kasue-Nakajima conjecture. The gravitational instanton (M,g) is of Type I, up to orientation reversal. Partial results are kn...

conjecture | open | MathDB 页面 | 原文线索

V2-028266 | MATHDB-375197

Schwarzschild uniqueness conjecture for non-flat AF0 gravitational instantons

摘录 / EXCERPT: Let (M,g) be a non-flat AF_0 gravitational instanton. Schwarzschild uniqueness conjecture. The metric g is, up to isometry, the Schwarzschild metric. Evidence...

conjecture | open | MathDB 页面 | 原文线索

V2-028267 | MATHDB-375200

Dyn-Farkhi conjecture on Hausdorff convexification under Minkowski addition

摘录 / EXCERPT: Dyn-Farkhi conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028268 | MATHDB-375242

Sharp timelike asymptotic volume ratio inequality for singularity formation

摘录 / EXCERPT: Let $(X,\mathsf d,\mathfrak m, \ll, \leq, \tau)$ be a measured Lorentzian space, let $V\subset X$ be the achronal set occurring in the hypotheses of Theorem, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028269 | MATHDB-375250

The Calabi-Yau critical-value conjecture

摘录 / EXCERPT: Calabi-Yau critical-value conjecture. If $c^1(X)=0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-028270 | MATHDB-375259

Unobstructedness conjecture for ellipsoid embeddings into H_b

摘录 / EXCERPT: Unobstructedness conjecture. The function $c_b(z)$ is unobstructed for $z<z_[\infty^b$ if and only if $b=m/d$ for some $(d;m;p,q)\in (H)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028271 | MATHDB-375294

The inherited-arc conjecture for tangent conics in André planes of order q^3

摘录 / EXCERPT: Let ${\textup{PG}}(2,q^3)$ be a projective plane with q even and $q>2$. Let $\mathcal{C}$ be a conic with nucleus N tangent to the line ℓ_∞ at T, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028272 | MATHDB-375305

Symmetric minimal covering body classification conjecture

摘录 / EXCERPT: Symmetric minimal covering body classification conjecture. There exists a unimodular transformation T such that

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V2-028273 | MATHDB-375307

Minkowski-type criterion for three-dimensional lattice coverings

摘录 / EXCERPT: Minkowski-type criterion for three-dimensional lattice coverings. The arrangement $K+\mathbb{Z}^3$ is a lattice covering of $\mathbb{R}^3$ if and only if there exist a unimodular t...

conjecture | open | MathDB 页面 | 原文线索

V2-028274 | MATHDB-375308

Algebraic intersection criterion for symmetric lattice coverings

摘录 / EXCERPT: Algebraic intersection criterion. The arrangement $\lambda K+\mathbb{Z}^3$ is a lattice covering of $\mathbb{R}^3$ if and only if there exists a unimodular transformation...

conjecture | open | MathDB 页面 | 原文线索

V2-028275 | MATHDB-375328

Abrishami-Czyżewska-Kluk-Pilipczuk-Pilipczuk-Rzążewski coarse separator conjecture

摘录 / EXCERPT: Abrishami-Czyżewska-Kluk-Pilipczuk-Pilipczuk-Rzążewski's conjecture. For every $k,r\in\mathbb{N}$, there exist $k',r'\in\mathbb{N}$ such that if G admits (k,r)-balanced separat...

conjecture | open | MathDB 页面 | 原文线索

V2-028276 | MATHDB-375329

Fixed-radius coarse separator strengthening

摘录 / EXCERPT: Fixed-radius coarse separator strengthening. For every $k,r\in\mathbb{N}$, there exists $k'\in\mathbb{N}$ such that if G admits (k,r)-balanced separators, then G admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-028277 | MATHDB-375336

Luo-Wu's discrete uniformization conjecture for hyperbolic convex hulls

摘录 / EXCERPT: Discrete uniformization conjecture. Suppose that S is a genus zero complete hyperbolic surface with countably many ends, all but at most one of which are cusp ends. Then S is i...

conjecture | open | MathDB 页面 | 原文线索

V2-028278 | MATHDB-375338

Non-1-convexity and non-n-convexity conjecture for ball bundles

摘录 / EXCERPT: Let M be a compact Kähler manifold of dimension k, and let E be a holomorphic $\mathbb{B}^n$-bundle over M embedded in the associated $\mathbb{CP}^n$-bundle $\widehat{E}$....

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V2-028279 | MATHDB-375348

Equivariant tilting-generator conjecture for affine Grassmannian convolution resolutions

摘录 / EXCERPT: Let G be a connected reductive algebraic group, let $\operatorname{Gr}=G(\mathcal K)/G(\mathcal O)$ be its affine Grassmannian, and let $\lambda_1^\vee,\ldots,\lambda_k^\vee$ be...

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V2-028280 | MATHDB-375355

Song-Zhu's integral closedness conjecture for lattice polytopes

摘录 / EXCERPT: Let M be a lattice and let $M_{\mathbb{R}}=M\otimes_{\mathbb{Z}}\mathbb{R}$. For a lattice polytope $P\subseteq M_{\mathbb{R}}$, let $L(P)$ denote its lattice length, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-028281 | MATHDB-375364

The tropical chromatic number conjecture for Euclidean space

摘录 / EXCERPT: Let $d4a3^n$ be equipped with the tropical metric, and let $d4a4^n$ denote its integer lattice with the induced tropical metric. The tropical chromatic number conjecture. The tropi...

conjecture | open | MathDB 页面 | 原文线索

V2-028282 | MATHDB-375427

Gromov's critical decay conjecture for positive scalar curvature metrics

摘录 / EXCERPT: Let M be an orientable n-manifold equipped with a complete Riemannian metric of positive scalar curvature. For a complete Riemannian metric on M, say that it has at most C-...

conjecture | open | MathDB 页面 | 原文线索

V2-028283 | MATHDB-375431

Cieliebak-Mohnke's ellipsoid capacity conjecture

摘录 / EXCERPT: Let $0<a_1\le a_2\le\cdots\le a_n<\infty$, and let $E^{2n}(a_1,\dots,a_n)\subset\mathbb{C}^n$ be the ellipsoid defined by ... For a closed Lagrangian torus $L\subset(X,\omega)$, set...

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V2-028284 | MATHDB-375434

The Lagrangian embedding capacity identity for ellipsoids

摘录 / EXCERPT: For $a\ge1$, define the embedding capacity ... Here $E^4(1,a)$ and $B^k(r)$ are the standard symplectic ellipsoid and ball, respectively. The Lagrangian embedding capacity identity....

conjecture | open | MathDB 页面 | 原文线索

V2-028285 | MATHDB-375453

Podestà-Zuddas conjecture on invariant Bismut-parallel metrics on complex flag manifolds

摘录 / EXCERPT: Podestà-Zuddas conjecture. Every G-invariant BTP metric on $(G/H,J)$ is either a Kähler metric or a multiple of the standard metric g_o.

conjecture | open | MathDB 页面 | 原文线索

V2-028286 | MATHDB-375462

Schoen's conjecture on singular metrics with nonnegative scalar curvature

摘录 / EXCERPT: Let M be a closed smooth manifold, and let g be an L^∞-metric on M, meaning that for some smooth reference Riemannian metric g_0 and constant $\Lambda>0$,...

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V2-028287 | MATHDB-375473

Integral Stokes coefficient conjecture for complex geodesics and heat kernels

摘录 / EXCERPT: Integral Stokes coefficient conjecture. Each γ_i determines such a normalized formal heat solution, and there exists a coefficient $S_{ij}^{+}\in\mathbb Z$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-028288 | MATHDB-375475

Bartnik's cosmological splitting conjecture in terms of Lorentz-distance families

摘录 / EXCERPT: Let (M,g) be a globally hyperbolic, timelike geodesically complete spacetime with compact Cauchy surfaces satisfying the strong energy condition. For a Cauchy temporal function...

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V2-028289 | MATHDB-375512

The strong Arnol'd chord conjecture for uniformly convex domains

摘录 / EXCERPT: Let Ω be a uniformly convex domain in $\mathbb{R}^{2n}$ with $n>1$, and let $c_k(\Omega)$ denote its Gutt--Hutchings capacities. The first capacity $c_1(\Omega)$ is the syst...

conjecture | open | MathDB 页面 | 原文线索

V2-028290 | MATHDB-375548

Linking and hull containment conjecture for elliptic and hyperbolic components

摘录 / EXCERPT: Let K_1 be an elliptic component and K_2 a hyperbolic component of a link $L=K_1\cup K_2$, and let M be the embedded manifold whose holomorphic hull is denoted by...

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V2-028291 | MATHDB-375549

Degeneracy and perturbation conjecture for isolated complex tangents

摘录 / EXCERPT: Let $M^3\hookrightarrow\mathbb{C}^3$ be an embedding, and suppose it has an isolated complex tangent. Isolated complex tangent conjecture. Every isolated complex tangent of the emb...

conjecture | open | MathDB 页面 | 原文线索

V2-028292 | MATHDB-375551

Upper-bound conjecture for lines on Weddle surfaces

摘录 / EXCERPT: Let Z be a set of six points in linearly general position in $\mathbb{P}^3$, and let $\mathcal W(Z)$ denote its Weddle surface. Upper-bound conjecture for lines on Weddle surface...

conjecture | open | MathDB 页面 | 原文线索

V2-028293 | MATHDB-375565

Mixed hard Lefschetz and Hodge-Riemann conjecture for non-Archimedean valuations

摘录 / EXCERPT: Mixed hard Lefschetz and Hodge-Riemann conjecture. $Val^\infty(V)$ satisfies the mixed hard Lefschetz theorem for the subset ${\cal K}$, and it satisfies the mixed Hodge-Riemann re...

conjecture | solved | MathDB 页面 | 原文线索

V2-028294 | MATHDB-375566

Analytic Hodge-Riemann positivity conjecture for non-Archimedean valuations

摘录 / EXCERPT: Let V have dimension n, let $Val_i^\infty(V)$ denote the degree-i smooth translation-invariant valuations, and identify valuations represented by functions on the relevant Gr...

conjecture | open | MathDB 页面 | 原文线索

V2-028295 | MATHDB-375612

Xia-Wang reciprocal Neumann eigenvalue conjecture for non-Euclidean space forms

摘录 / EXCERPT: Let M^n_κ be the simply connected n-dimensional Riemannian manifold of constant sectional curvature $\kappa\in\{-1,1\}$, and let $\Omega\subset M^n_\kappa$ be a bounded co...

conjecture | open | MathDB 页面 | 原文线索

V2-028296 | MATHDB-375656

Tosatti's transcendental basepoint-free conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold. A class $\alpha \in H^{1,1}(X,\mathbb{R})$ is semi-ample if there exists a holomorphic contraction $\rho:X\to X'$ to a normal analytic space...

conjecture | open | MathDB 页面 | 原文线索

V2-028297 | MATHDB-375667

Kolesnikov-Werner's many-body functional Blaschke-Santaló conjecture

摘录 / EXCERPT: Let $n\ge1$ and $k\ge2$. Let $\rho:\mathbb R\to\mathbb R_+$ be non-increasing, and let $f_1,\ldots,f_k:\mathbb R^n\to\mathbb R_+$ be even integrable functions. For ... assume ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-028298 | MATHDB-375668

Kalantzopoulos-Saroglou's geometric many-body Blaschke-Santaló conjecture

摘录 / EXCERPT: Let $n\ge1$ and $k\ge2$, and let $K_1,\ldots,K_k\subset\mathbb R^n$ be origin-symmetric convex bodies. Kalantzopoulos-Saroglou's conjecture. If ... for all $x_i\in K_i$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028299 | MATHDB-375681

Hasegawa's biholomorphism conjecture for nilpotent Lie groups

摘录 / EXCERPT: Let G be a simply connected, nilpotent Lie group of dimension $2n$ and let J be a left-invariant complex structure on G. Hasegawa's biholomorphism conjecture. The complex man...

conjecture | open | MathDB 页面 | 原文线索

V2-028300 | MATHDB-375705

Le Donne's BLD embedding conjecture for compact length metric spaces

摘录 / EXCERPT: Let $(\mathsf{X},\mathsf{d}_{\mathsf{X}})$ be a compact length metric space with finite Hausdorff dimension. A bounded-length-distortion map, or BLD map, is a continuous map into a...

conjecture | open | MathDB 页面 | 原文线索

V2-028301 | MATHDB-375717

Ashbaugh-Benguria reciprocal-gap conjecture for Dirichlet eigenvalues

摘录 / EXCERPT: Let $N\ge2$, let $\Omega\subset\mathbb R^N$ be a bounded domain, and let $\nu=(N-2)/2$. Denote the Dirichlet eigenvalues of Ω by...

conjecture | open | MathDB 页面 | 原文线索

V2-028302 | MATHDB-375728

Kosniowski's conjecture on fixed points of unitary circle manifolds

摘录 / EXCERPT: Kosniowski's conjecture. If M^{2n} is not a boundary, then the number of fixed points is greater than $f(n)$. Kosniowski conjectured that the candidate function is

conjecture | open | MathDB 页面 | 原文线索

V2-028303 | MATHDB-375785

Rogers--Shephard conjecture for asymmetric L_p-zonoids

摘录 / EXCERPT: Let $p\geq1$ and let $Z\subset\mathbb{R}^n$ be an asymmetric L_p-zonoid. Let q satisfy $1/p+1/q=1$. Asymmetric L_p-zonoid Rogers--Shephard conjecture. One has ... Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-028304 | MATHDB-375786

Planar equality conjecture for symmetric convex bodies

摘录 / EXCERPT: Let $K\subset\mathbb{R}^2$ be a convex body with a center of symmetry and containing the origin, and let $p>1$. Planar symmetric-body equality conjecture. Equality in the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-028305 | MATHDB-375807

Campana-Demailly-Verbitsky conjecture on simple compact Kähler manifolds

摘录 / EXCERPT: Campana-Demailly-Verbitsky conjecture. Either X has a finite étale cover which is bimeromorphic to a complex torus, or $H^0(X,\Omega_X^2)$ is generated by a holomorphic 2-form...

conjecture | open | MathDB 页面 | 原文线索

V2-028306 | MATHDB-375824

Lift conjecture for universal torsors of projective Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a projective Calabi-Yau manifold, meaning that its canonical bundle is torsion and ... and let $\mathcal U$ be a universal torsor for X. Lift conjecture. The action of...

conjecture | open | MathDB 页面 | 原文线索

V2-028307 | MATHDB-375848

Selberg's eigenvalue conjecture for principal congruence surfaces

摘录 / EXCERPT: Selberg's eigenvalue conjecture. For every N,

conjecture | open | MathDB 页面 | 原文线索

V2-028308 | MATHDB-375855

Second pinching conjecture for closed minimal hypersurfaces

摘录 / EXCERPT: Second pinching conjecture. If $S>n$, then $S\geq 2n$.

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V2-028309 | MATHDB-375856

Bryant's conjecture for complete minimal hypersurfaces in the 4-sphere

摘录 / EXCERPT: Bryant's conjecture. M is isoparametric.

conjecture | open | MathDB 页面 | 原文线索

V2-028310 | MATHDB-375858

Poisson K-polystability and the semiclassical Yau-Tian-Donaldson correspondence

摘录 / EXCERPT: Semiclassical Yau-Tian-Donaldson conjecture. The triple (X,σ,α) is Poisson K-polystable if and only if there exists $\epsilon>0$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-028311 | MATHDB-375859

The polarized semiclassical Yau-Tian-Donaldson conjecture for holomorphic Poisson tensors

摘录 / EXCERPT: Polarized semiclassical Yau-Tian-Donaldson conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-028312 | MATHDB-375870

Brakke's radial log-convex density conjecture

摘录 / EXCERPT: Let $\mathbb{R}^{n}$ be equipped with a smooth, radial, log-convex density, and consider regions of prescribed volume. Brakke's conjecture. Balls centered at the origin are isoperi...

conjecture | open | MathDB 页面 | 原文线索

V2-028313 | MATHDB-375910

The positive spun triangulation conjecture for the hyperbolic manifold Vol3

摘录 / EXCERPT: The positive spun triangulation conjecture for Vol3. For any embedded closed geodesic γ, the pair (N,γ) does not admit a positive spun triangulation.

conjecture | open | MathDB 页面 | 原文线索

V2-028314 | MATHDB-375911

The generalised Casson volume inequality conjecture

摘录 / EXCERPT: The generalised Casson conjecture. For every $\alpha\in\mathcal{A}(\mathcal{T},\mathbf{s})$,

conjecture | open | MathDB 页面 | 原文线索

V2-028315 | MATHDB-375920

Boltyanskii's illumination conjecture for convex bodies

摘录 / EXCERPT: Let $K\in\mathcal{K}^d$ be a convex body, and say that a set of directions illuminates K if every point of $\operatorname{bd}K$ is illuminated by at least one direction in the se...

conjecture | open | MathDB 页面 | 原文线索

V2-028316 | MATHDB-375943

Yau's nodal volume conjecture

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold without boundary, and let φ be an eigenfunction satisfying ... Write $\mathcal{L}_{\varphi}(M)$ for its nodal volume. Yau's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-028317 | MATHDB-375948

Jollivet-Sharafutdinov inverse Steklov spectral conjecture

摘录 / EXCERPT: Jollivet-Sharafutdinov inverse Steklov spectral conjecture. The metrics g_1 and g_2 have the same Steklov spectrum if and only if they are σ-isometric. This conjecture i...

conjecture | open | MathDB 页面 | 原文线索

V2-028318 | MATHDB-375949

Flat inverse Steklov spectral conjecture

摘录 / EXCERPT: Let two flat surfaces be equipped with their Steklov spectra. Flat inverse Steklov spectral conjecture. Two flat surfaces are Steklov isospectral if and only if they are isometric....

conjecture | open | MathDB 页面 | 原文线索

V2-028319 | MATHDB-375955

The BAS characterization conjecture for pseudo-Anosov flows

摘录 / EXCERPT: Let phi be a transitive pseudo-Anosov flow on an oriented closed 3-manifold. Let $\operatorname{sing}(phi)$ denote its singular orbits, and let positive and negative lozenges b...

conjecture | open | MathDB 页面 | 原文线索

V2-028320 | MATHDB-375956

The rigidity conjecture for branching orbits of pseudo-Anosov flows

摘录 / EXCERPT: Let $Lambda^s$ and $Lambda^u$ denote the stable and unstable foliations of a pseudo-Anosov flow $varphi$. A branching orbit is a closed orbit that lies on a leaf of $Lambda^s$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-028321 | MATHDB-375957

Guth's normalized volume conjecture for uniform hyperbolic ball comparison

摘录 / EXCERPT: For a closed oriented connected n-dimensional manifold M admitting a hyperbolic metric, let g be a Riemannian metric on M, let $\|M\|_{\Delta}$ denote its Gromov simplicial...

conjecture | open | MathDB 页面 | 原文线索

V2-028322 | MATHDB-375964

Chern's discreteness conjecture for minimal submanifolds with constant scalar curvature

摘录 / EXCERPT: Let M^n be a closed, minimally immersed submanifold in the unit sphere $\mathbb{S}^{n+m}$ with constant scalar curvature, equivalently with constant length of the second fundamen...

conjecture | open | MathDB 页面 | 原文线索

V2-028323 | MATHDB-375965

Stronger Chern conjecture for minimal hypersurfaces

摘录 / EXCERPT: Let M^n be a closed, minimally immersed hypersurface of the unit sphere $\mathbb{S}^{n+1}$ with constant scalar curvature. Stronger version of Chern's conjecture. Then M^n is i...

conjecture | open | MathDB 页面 | 原文线索

V2-028324 | MATHDB-375970

Conjectural characterization of tropical characteristic polynomial coefficients for symmetric matrices

摘录 / EXCERPT: Conjectural characterization. For every m with $3\leqslant m\leqslant n$, at least one of these two inequalities holds. The inequalities express a concavity-type regularity in th...

conjecture | open | MathDB 页面 | 原文线索

V2-028325 | MATHDB-375979

Normalization of tropicalizations of singular curves

摘录 / EXCERPT: Normalization conjecture for tropical curves. There is an embedding of C with an ideal I having the property that the integral closure of B in its field of fractions is obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-028326 | MATHDB-375985

Low-curvature skill fixed-point conjecture for LASKO systems

摘录 / EXCERPT: Let $\mathcal{A}\to\mathsf{WfMd}$ be a Lie algebroid of controlled intervention modes over a typed Markdown workflow space, with anchor $\rho:\mathcal{A}\to T\mathsf{WfMd}$. For co...

conjecture | open | MathDB 页面 | 原文线索

V2-028327 | MATHDB-375988

The noncompact extension of the orbifold Hecke algebra isomorphism

摘录 / EXCERPT: Noncompact extension conjecture. The isomorphism of Theorem also holds for such noncompact X and G.

conjecture | open | MathDB 页面 | 原文线索

V2-028328 | MATHDB-375992

Sarnak-Chiu conjecture on the height of flat tori

摘录 / EXCERPT: Let $L=C\mathbb{Z}^n\subset\mathbb{R}^n$ be a lattice with Gram matrix $G=C^{\mathsf T}C$ and associated flat torus $T_L=\mathbb{R}^n/L$. Normalize to unit volume, so $\det G=1$. I...

conjecture | open | MathDB 页面 | 原文线索

V2-028329 | MATHDB-376015

Darvas-Di Nezza-Lu ceiling-envelope conjecture

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold, and let θ be a smooth closed real $(1,1)$-form representing a big class. For $u\in\operatorname{PSH}(X,\theta)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-028330 | MATHDB-376016

Lu's idempotence conjecture for the singularity envelope

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold, let θ be a smooth closed real $(1,1)$-form representing a big class, and let $P_\theta[u]$ denote the singularity envelope of...

conjecture | open | MathDB 页面 | 原文线索

V2-028331 | MATHDB-376031

Chang's sharp relative comparison conjecture for Poincaré-Einstein manifolds

摘录 / EXCERPT: Let (X^N,g_+), with $N=n+1\ge 3$, be a smooth Poincaré-Einstein manifold with conformal infinity $(M^n,[h])$. Assume that $Y(M,[h])>0$, and let $(\overline X,[\bar g])$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-028332 | MATHDB-376036

Hopf conjecture on the Euler characteristic of nonpositively curved manifolds

摘录 / EXCERPT: Let (X,g) be a closed $2n$-dimensional Riemannian manifold with sectional curvature sec_g. Write $\chi(X)$ for its Euler characteristic. Hopf conjecture. ... and ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-028333 | MATHDB-376043

The absence of a universal upper bound for the first eigenvalue in terms of mean distance

摘录 / EXCERPT: Mean-distance upper-bound conjecture. There is no universal constant $C>0$ such that

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V2-028334 | MATHDB-376044

The absence of a universal upper bound for the first eigenvalue in terms of inradius

摘录 / EXCERPT: Inradius upper-bound conjecture. There is no universal constant $C>0$ such that

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V2-028335 | MATHDB-376055

Stoker-type conjecture for infinitesimal dihedral-angle rigidity of polytopes

摘录 / EXCERPT: Let $S(t)$ be a differentiable one-parameter family of oriented piecewise-linear surfaces. For each facet F, let n_F be its unit normal and V_F its volume, and define the dih...

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V2-028336 | MATHDB-376056

Strong stress-flex conjecture for coned frameworks over closed PL surfaces

摘录 / EXCERPT: Let S be a closed piecewise-linear surface, and let $(G_S^\star,\boldsymbol p)$ be its coned framework: the framework has the vertices and edges of S, together with a cone vert...

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V2-028337 | MATHDB-376064

Special-leaf correspondence for Coulomb and Higgs branches

摘录 / EXCERPT: Let G be a reductive group, let $\mathbf{N}$ be a representation, and let $\mathcal{M}(G,\mathbf{N})$ and $\mathcal{M}_{\mathsf{Higgs}}(G,\mathbf{N})$ denote the associated Coulo...

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V2-028338 | MATHDB-376081

Bounded-degree polynomial growth conjecture for nonnegative Bakry-Émery graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with degree bound d_*, and let $B(x,r)$ denote the open ball of radius r about x. Assume the degree and curvature hypotheses referred to as Assumptio...

conjecture | open | MathDB 页面 | 原文线索

V2-028339 | MATHDB-376122

Gromov's Euclidean C^0-rigidity conjecture

摘录 / EXCERPT: Let g be a smooth complete metric on $\mathbb R^n$, where $n\geq 3$, and let g_{euc} denote the Euclidean metric. Assume that ... and ... as $x\to\infty$. Gromov's Euclide...

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V2-028340 | MATHDB-376129

Thomas-Yau-Joyce stability conjecture for special Lagrangians and dHYM solutions

摘录 / EXCERPT: Let $\check{\mathcal{X}}$ be the mirror geometry, let L be a Lagrangian in $\check{\mathcal{X}}$, and let $\mathcal{E}$ be a holomorphic vector bundle on $\mathcal{X}$. A Bridgel...

conjecture | open | MathDB 页面 | 原文线索

V2-028341 | MATHDB-376146

Yau's pinching problem for sectional and normalized scalar curvature

摘录 / EXCERPT: Let (M^{n},g) be a closed, simply connected Riemannian manifold of dimension $n\ge 4$. Define the minimum sectional curvature and normalized scalar curvature at $x\in M$ by ... Her...

conjecture | open | MathDB 页面 | 原文线索

V2-028342 | MATHDB-376148

Strong nilpotency criterion for special multi-flag germs

摘录 / EXCERPT: Let a special m-flag germ have singularity class ... Let c^i_j denote the constants appearing in its EKR normal form. Strong nilpotency conjecture. The germ is strongly nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-028343 | MATHDB-376149

Strong nilpotency conjecture for Goursat germs

摘录 / EXCERPT: A Goursat germ is a germ of a Goursat distribution, and a tangential germ is one belonging to the tangential class. Mormul's Goursat conjecture. Goursat germs are strongly nilpoten...

conjecture | open | MathDB 页面 | 原文线索

V2-028344 | MATHDB-376153

Streets-Tian conjecture on closed Hermitian-symplectic complex manifolds

摘录 / EXCERPT: Streets-Tian conjecture. Any closed Hermitian-symplectic complex manifold (X,J) admits a Kähler metric.

conjecture | open | MathDB 页面 | 原文线索

V2-028345 | MATHDB-376158

Singular apparent-space conjecture for half-period configurations

摘录 / EXCERPT: Let $\tau\in\mathbb{H}$, let $\Sigma=E_\tau[2]$, and let $(\alpha_1,\ldots,\alpha_4)\in(2\mathbb{Z})^4$. The apparent space $AP_\Sigma(\tau)$ parametrizes the relevant apparent dat...

conjecture | open | MathDB 页面 | 原文线索

V2-028346 | MATHDB-376159

Singular apparent-space conjecture for even configurations on elliptic curves

摘录 / EXCERPT: Let E_τ be the elliptic curve with parameter $\tau\in\mathbb{H}$, and let $\Sigma=\{p_1,\ldots,p_n\}\subseteq E_\tau$, where n is even. Let $AP_\Sigma(\tau)$ be the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-028347 | MATHDB-376161

Existence classification of collinear relative equilibria in positive curvature

摘录 / EXCERPT: Existence classification conjecture. There exist $\kappa_i=\kappa_i(\mu)\in(0,\pi^2/4)$ with $\kappa_1<\kappa_2<\kappa_3$ such that the curved restricted three-body problem has exa...

conjecture | open | MathDB 页面 | 原文线索

V2-028348 | MATHDB-376162

Existence classification of triangular relative equilibria in positive curvature

摘录 / EXCERPT: Triangular existence classification conjecture. There exist $\kappa_4(\mu)<\kappa_5(\mu)$ in $(0,\pi^2/4)$ such that there are two triangular relative equilibria,...

conjecture | open | MathDB 页面 | 原文线索

V2-028349 | MATHDB-376163

Stability classification of collinear relative equilibria in positive curvature

摘录 / EXCERPT: Collinear stability classification conjecture. $\operatorname{L}_1,\operatorname{L}_2,\operatorname{E}_3$, and A_1 are center-saddles whenever they exist;...

conjecture | open | MathDB 页面 | 原文线索

V2-028350 | MATHDB-376203

Lu's second-gap conjecture for minimal submanifolds

摘录 / EXCERPT: Let M^n be a closed minimal submanifold of the unit sphere $\mathbb S^{n+m}$. With respect to a local orthonormal normal frame, let $\mathcal A$ be the fundamental matrix of the...

conjecture | open | MathDB 页面 | 原文线索

V2-028351 | MATHDB-376211

Higher-dimensional Cheeger log-convexity conjecture

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body, let $P>0$, and let $h(K)=\inf_{E\subset K}\operatorname{Per}(E)/|E|$ be its Cheeger constant. Higher-dimensional Cheeger log-convexity...

conjecture | open | MathDB 页面 | 原文线索

V2-028352 | MATHDB-376212

Schmuckenschläger's reverse Faber-Krahn bound

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body, and let $\Lambda_{\mathrm{FK}}(K)$ be its normalized minimal first Dirichlet eigenvalue. Schmuckenschläger's reverse Faber-Krahn bound....

conjecture | open | MathDB 页面 | 原文线索

V2-028353 | MATHDB-376239

Yang-Zheng's ampleness conjecture for Hermitian manifolds

摘录 / EXCERPT: Let X be a compact complex manifold, let K_X denote its canonical bundle, and let holomorphic sectional curvature refer to the curvature of a Hermitian metric on X. A Hermiti...

conjecture | open | MathDB 页面 | 原文线索

V2-028354 | MATHDB-376266

Conjecture on the Moussong metric for virtual Deligne complexes

摘录 / EXCERPT: Let Γ be a Coxeter graph, and let vD_{Γ} be the virtual Deligne complex associated with the virtual Artin group $\mathrm{VA}[\Gamma]$. Equip...

conjecture | open | MathDB 页面 | 原文线索

V2-028355 | MATHDB-376281

Existence of log terminal models for compact Kähler gklt pairs

摘录 / EXCERPT: Existence of log terminal models. The pair $(X,B+\boldsymbol{\beta})$ has a log terminal model.

conjecture | open | MathDB 页面 | 原文线索

V2-028356 | MATHDB-376282

Termination of flips for compact Kähler gklt pairs

摘录 / EXCERPT: Termination of flips. Any sequence of $K_X+B+\boldsymbol{\beta}_X$ flips is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-028357 | MATHDB-376289

The equivariant Hodge index conjecture for arbitrary weights

摘录 / EXCERPT: Let Setup be fixed. For a weight $v \colon P \to \mathbb{R}_{>0}$, let $C_v>0$ be a constant, and let $\mathcal{E}$ be a $\mathbb{G}$-equivariant torsion-free sheaf. The equivarian...

conjecture | open | MathDB 页面 | 原文线索

V2-028358 | MATHDB-376290

The Bogomolov-Gieseker conjecture for stable sheaves and arbitrary weights

摘录 / EXCERPT: Let Setup be fixed, let $v \colon P \to \mathbb{R}_{>0}$ be a weight, and let $\mathcal{E}$ be a reflexive sheaf of rank r that is v-stable. Bogomolov-Gieseker conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-028359 | MATHDB-376292

Donaldson's conjecture on the Calabi functional and non-Archimedean K-energy

摘录 / EXCERPT: Let (X,L) be a polarized manifold. Let $\mathcal{H}$ be the space of Kähler metrics in the class $c_1(L)$, let $\mathcal{H}^{\operatorname{NA}}$ be the space of smooth non-Archim...

conjecture | open | MathDB 页面 | 原文线索

V2-028360 | MATHDB-376295

Noncompact Bonnet-Myers conjecture for Q-curvature and scalar curvature

摘录 / EXCERPT: Let (M^4,g) be a complete four-dimensional Riemannian manifold. Noncompact Bonnet-Myers conjecture. If ... then M^4 is compact. This extends the corresponding proposition from co...

conjecture | open | MathDB 页面 | 原文线索

V2-028361 | MATHDB-376296

Nonnegative Q-curvature and scalar curvature imply nonnegative Ricci curvature

摘录 / EXCERPT: Let (M^4,g) be a complete four-dimensional Riemannian manifold. Nonnegative curvature implication conjecture. If ... then ... This is motivated by results for conformally flat four-m...

conjecture | open | MathDB 页面 | 原文线索

V2-028362 | MATHDB-376312

Metric cone conjecture for the Ricci-flow limit of Kähler-Ricci shrinkers

摘录 / EXCERPT: Metric cone conjecture. If the Kähler-Ricci shrinker has maximal volume growth, then (Z_0,d^Z_0) is a metric cone and coincides with the metric completion of...

conjecture | open | MathDB 页面 | 原文线索

V2-028363 | MATHDB-376315

The Euler-line concurrence classification for circumcircle centers

摘录 / EXCERPT: Euler-line concurrence conjecture. A center on the circumcircle is concurrent for the Euler line if and only if it lies in the orbit of X_{99}.

conjecture | open | MathDB 页面 | 原文线索

V2-028364 | MATHDB-376316

Product Hermitian Jacobi eigenfunction conjecture on the simplex

摘录 / EXCERPT: Let m be a positive integer, let k be the number of matrix variables, let $(\kappa_1,\dots,\kappa_{k+1})$ be the parameters of the Hermitian Jacobi operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028365 | MATHDB-376317

Strong triangularity conjecture for the Jacobi operator on the Hermitian matrix simplex

摘录 / EXCERPT: Let $\mathcal{G}_{\kappa_1,\dots,\kappa_{k+1}}^m$ be the Hermitian Jacobi operator on the simplex, and let the multivariate Schur polynomials be the basis of unitary-conjugation-in...

conjecture | open | MathDB 页面 | 原文线索

V2-028366 | MATHDB-376349

The Neoplatonic conjecture for Euclidean realizations

摘录 / EXCERPT: Neoplatonic conjecture. Any 6-net σ has a realization, unique up to isometry, as an undented Euclidean polyhedron $\Delta(\sigma)$ built from equilateral triangles of side...

conjecture | open | MathDB 页面 | 原文线索

V2-028367 | MATHDB-376350

The ideal neoplatonic conjecture

摘录 / EXCERPT: Ideal neoplatonic conjecture. Every 6-net σ has a realization $\Delta(\sigma,\infty)$, unique up to isometry, as an ideal neoplatonic.

conjecture | open | MathDB 页面 | 原文线索

V2-028368 | MATHDB-376361

The convex-domain local-minimality conjecture for symmetric Maxwell eigenvalue functions

摘录 / EXCERPT: Let B denote the ball in $\mathbb{R}^3$, and let the first three Maxwell eigenvalues of a domain $Omega$ be $lambda_1(Omega),lambda_2(Omega),lambda_3(Omega)$. For $j=1,2,3$, let...

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V2-028369 | MATHDB-376362

The convex-domain local-maximality conjecture for the first Maxwell eigenvalue

摘录 / EXCERPT: Let B denote the ball in $\mathbb{R}^3$, and let $lambda_1(Omega)$ be the first Maxwell eigenvalue of a domain $Omega$. Consider domains subject to either a fixed-volume constrai...

conjecture | open | MathDB 页面 | 原文线索

V2-028370 | MATHDB-376378

The optimal partial plank covering conjecture for convex bodies

摘录 / EXCERPT: Optimal partial plank covering conjecture. There exists a plank P of width W such that, for every such finite family,

conjecture | open | MathDB 页面 | 原文线索

V2-028371 | MATHDB-376390

Milman's measure-preserving contraction conjecture for positively Ricci-curved spheres

摘录 / EXCERPT: Milman's contraction conjecture. There exists a map

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V2-028372 | MATHDB-376395

Classification conjecture for codimension-four \mathfrak{bms}_3 Poisson pairs

摘录 / EXCERPT: Let $\mathfrak{bms}_3^*$ be the dual of the centrally extended $\mathfrak{bms}_3$ algebra, let c_1 be its first central charge, and let $\{,\}_\mathrm{LP}^{\mathfrak{bms}_3^*}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-028373 | MATHDB-376405

Conjecture on invariant sums of squared side areas for cyclic polygons

摘录 / EXCERPT: Invariant side-square area conjecture. The sum of the areas of the constructed squares remains invariant throughout $\mathcal P$ if and only if the circumcenter of the triangle coi...

conjecture | open | MathDB 页面 | 原文线索

V2-028374 | MATHDB-376419

Bolsinov-Konyaev-Matveev conjecture on Haantjes integrability

摘录 / EXCERPT: Let M be an n-dimensional manifold and let $K_1,\dots,K_n$ be mutual symmetries on M that are linearly independent at every point. For operator fields L,M, define the...

conjecture | open | MathDB 页面 | 原文线索

V2-028375 | MATHDB-376421

Alessandrini-Bassanelli conjecture on the hierarchy of p-Kähler structures

摘录 / EXCERPT: Let (M,J) be a complex manifold. A p-Kähler structure on (M,J) is a closed, pointwise transverse (p,p)-form. Alessandrini-Bassanelli conjecture. If (M,J) admits a p-Käh...

conjecture | open | MathDB 页面 | 原文线索

V2-028376 | MATHDB-376432

The normal-crossings mirror symmetry conjecture

摘录 / EXCERPT: Let Y be a Calabi-Yau manifold with $h^{2,0}(Y)=0$, and let $\pi:\mathcal{Y}\to S$ be a family of complex structures on Y whose Kodaira-Spencer map is an isomorphism everywhere...

conjecture | open | MathDB 页面 | 原文线索

V2-028377 | MATHDB-376436

The integral mirror-map conjecture for toric Calabi-Yau complete intersections

摘录 / EXCERPT: Integral mirror-map conjecture. These are the ∇_{AV}-flat coordinates on $\tilde H^2$, and every coefficient in their expansions as power series in the ∇_{AP}-flat co...

conjecture | open | MathDB 页面 | 原文线索

V2-028378 | MATHDB-376440

The monomial-divisor mirror map conjecture for toric Kähler and complex moduli

摘录 / EXCERPT: Monomial-divisor mirror map conjecture. There is an open set $U\subset(\mathcal{D}_{\mathop{\rm toric}\nolimits}(\widehat{X})/\Gamma_{\mathop{\rm toric}\nolimits})^-$ containing...

conjecture | open | MathDB 页面 | 原文线索

V2-028379 | MATHDB-376443

Mirror symmetry conjecture for dual reflexive Gorenstein cones

摘录 / EXCERPT: Let $\frac{}{}$ be a pair of $\bar d$-dimensional dual reflexive Gorenstein cones $\frac{}{}$ and $\frac{}{}$ of index r. The associated generalized Calabi-Yau manifolds arise as...

conjecture | open | MathDB 页面 | 原文线索

V2-028380 | MATHDB-376447

Ramachandran's Bochner-Hartogs or surface-group dichotomy

摘录 / EXCERPT: Ramachandran's conjecture. One of the following holds: the universal cover $\widetilde{X}$ has the Bochner-Hartogs property, or $\pi_{1}(X)$ is commensurable with the fundamental g...

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V2-028381 | MATHDB-376453

The symplectic connected-sum conjecture for simply connected four-manifolds

摘录 / EXCERPT: A smooth, closed, oriented and simply connected 4-manifold is a smooth manifold of dimension four with the stated properties. A symplectic manifold is a smooth manifold equipped...

conjecture | open | MathDB 页面 | 原文线索

V2-028382 | MATHDB-376454

The minimal conjecture for four-manifolds

摘录 / EXCERPT: A smooth, closed and oriented 4-manifold is a smooth manifold of dimension four with the stated properties. A symplectic manifold is a smooth manifold equipped with a symplectic...

conjecture | open | MathDB 页面 | 原文线索

V2-028383 | MATHDB-376456

Noguchi's finiteness conjecture for hyperbolic complex spaces

摘录 / EXCERPT: Let X be a Zariski open subset of an irreducible compact complex space. A compact irreducible complex space Y is hyperbolic in the sense used by Noguchi, and $f:X\rightarrow Y$...

conjecture | open | MathDB 页面 | 原文线索

V2-028384 | MATHDB-376460

Kotschick-Morgan wall-crossing polynomiality conjecture

摘录 / EXCERPT: Let X be a smooth simply connected 4-manifold. For a class $x\in H^2(X,\mathbb Z)$ defining a wall, an integer N, and the associated wall-crossing term $\delta^X_{\xi,N}$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-028385 | MATHDB-376478

Twisted cotangent description of the Mukai-dual section space

摘录 / EXCERPT: Twisted cotangent conjecture. Under these assumptions, there exists a natural isomorphism of complex manifolds

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V2-028386 | MATHDB-376481

The SYZ compactification conjecture for dual torus fibrations

摘录 / EXCERPT: Let Y be a Calabi--Yau manifold with a special Lagrangian T^n-fibration, and let ${\cal M}_D(T^n,Y)$ be the family of dual tori over ${\cal M}_{sL}(T^n,Y)$. The SYZ compactific...

conjecture | open | MathDB 页面 | 原文线索

V2-028387 | MATHDB-376482

The complex and Ricci-flat structure conjecture for the dual torus family

摘录 / EXCERPT: Let Y be a Calabi--Yau manifold with a special Lagrangian T^n-fibration, and let ${\cal M}_D(T^n,Y)$ be the family of dual tori over ${\cal M}_{sL}(T^n,Y)$. The dual torus fami...

conjecture | open | MathDB 页面 | 原文线索

V2-028388 | MATHDB-376483

The geometric mirror pair conjecture

摘录 / EXCERPT: Let (X,Y) be a geometric mirror pair, meaning that there is a parameter space U and, for each $t\in U$, a correspondence $Z_t\subset X\times Y$, sections of the projections, Ri...

conjecture | open | MathDB 页面 | 原文线索

V2-028389 | MATHDB-376492

The generic smoothness and Euler-number conjecture for the McKay polyhedra

摘录 / EXCERPT: Let C_ζ be the polyhedra associated with the parameter ζ, let n denote the ambient dimension, and let $\mathcal{S}^k$ be the configurations of codimension k whose a...

conjecture | open | MathDB 页面 | 原文线索

V2-028390 | MATHDB-376494

Bounded-facet conjecture for cones of Gale-transform fans

摘录 / EXCERPT: Let $A\in{\bf Z}^{d\times n}$ be a unimodular matrix of rank d, and let B be a Gale transform of A. The fan $\Gamma(B)$ is the fan associated with the Gale transform B. Spe...

conjecture | open | MathDB 页面 | 原文线索

V2-028391 | MATHDB-376511

Toroidal lifting after modifications

摘录 / EXCERPT: Toroidal lifting conjecture. There exist modifications $X'\to X$ and $B'\to B$, each a composition of blowings up with smooth centers lying over Σ, and a lifting $X'\to B'$...

conjecture | open | MathDB 页面 | 原文线索

V2-028392 | MATHDB-376522

Enumerative interpretation of Gromov-Witten invariants for the cubic degeneration and [?]2

摘录 / EXCERPT: Let X be either the Hirzebruch surface $\mathbb F_2$ or the surface B' obtained by blowing up six points on a conic, and let E be its (-2)-curve. Let D be an effective di...

conjecture | open | MathDB 页面 | 原文线索

V2-028393 | MATHDB-376527

Invariance conjecture for points at infinity under changes of cuts

摘录 / EXCERPT: Let M be a complex curve equipped with two families of cuts satisfying the conditions in the preceding conjecture. For each integer $n\geq0$, let the associated set of C^n-poin...

conjecture | open | MathDB 页面 | 原文线索

V2-028394 | MATHDB-376537

Stringy Euler number decreases under three-dimensional flips

摘录 / EXCERPT: Stringy flip conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028395 | MATHDB-376538

The polyhedral semistable reduction conjecture

摘录 / EXCERPT: Let $f:\Delta_X\rightarrow\Delta_B$ be a surjective morphism of rational conical polyhedral complexes such that $f^{-1}(0)=\{0\}$. A projective subdivision replaces the complexes b...

conjecture | open | MathDB 页面 | 原文线索

V2-028396 | MATHDB-376539

Shokurov's exceptional-index conjecture for complements

摘录 / EXCERPT: For each dimension d, let N_d denote the bounded set of complement indices in the complements conjecture, and define the exceptional indices by $EN_d=N_2\setminus N_{d-1}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028397 | MATHDB-376616

Structural conjecture for minimal pseudo-codewords of projective-plane Tanner graphs

摘录 / EXCERPT: Structural conjecture. Every minimal pseudo-codeword is a sum of a few minimal pseudo-codewords, with one or two low-value components changed so that they become the large componen...

conjecture | open | MathDB 页面 | 原文线索

V2-028398 | MATHDB-376617

Conjecture on the type of minimum-weight non-codeword minimal pseudo-codewords

摘录 / EXCERPT: Type conjecture. The type $\boldsymbol{t}$ of these pseudo-codewords has t_0 non-negative, t_1 positive, t_2 positive, and $t_\ell=0$ otherwise. If this is false, the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-028399 | MATHDB-376620

Small-type minimal pseudo-codeword conjecture for projective-plane Tanner graphs

摘录 / EXCERPT: Let $\mathbf{H}=\mathbf{H}_{\operatorname{PG}(2,q)}$. Consider minimal pseudo-codewords having minimum AWGNC pseudo-weight among all minimal pseudo-codewords that are not multiples...

conjecture | open | MathDB 页面 | 原文线索

V2-028400 | MATHDB-376634

Fekete's aperture-angle approximation conjecture

摘录 / EXCERPT: Let C be a compact convex figure in the plane, let $\alpha(C,k)$ denote the best worst-case aperture-angle approximation of C by an inscribed convex k-gon, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028401 | MATHDB-376635

Brass's subpolygon Hausdorff-approximation conjecture

摘录 / EXCERPT: Let $\mathcal{P}$ be a family of convex polygons in $\mathbb{R}^{2}$ that is closed under taking subpolygons, where a subpolygon is the convex hull of a subset of the polygon's ver...

conjecture | open | MathDB 页面 | 原文线索

V2-028402 | MATHDB-376637

Rousseeuw-Hubert partition conjecture for regression depth

摘录 / EXCERPT: Let P be a set of n points, and let regression depth be the minimum number of points intersected by a hyperplane during any continuous motion taking it to a vertical hyperplane...

conjecture | open | MathDB 页面 | 原文线索

V2-028403 | MATHDB-376642

Conjectural formula for the multivariate regression-depth constant

摘录 / EXCERPT: Let d and $0\leq k<d$ be constants. For a set of points with k independent and $d-k$ dependent degrees of freedom, let $R(d,k)$ denote a constant such that some k-flat has re...

conjecture | open | MathDB 页面 | 原文线索

V2-028404 | MATHDB-376644

Taubes' decomposition conjecture for smooth oriented four-manifolds

摘录 / EXCERPT: A smooth, compact, oriented 4-manifold without boundary is understood as a connected sum of symplectic manifolds, allowing either the symplectic or the opposite orientation, and...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028405 | MATHDB-376645

Taubes' simply connected symplectic decomposition conjecture

摘录 / EXCERPT: Let X be a smooth, compact, oriented, simply connected 4-manifold without boundary. A connected sum here means a decomposition into smooth four-manifolds, and both orientations...

conjecture | open | MathDB 页面 | 原文线索

V2-028406 | MATHDB-376646

Simple-type conjecture for Seiberg-Witten invariants

摘录 / EXCERPT: Let X be a manifold of simple type. For a choice of $Spin_c$-structure L, let M_L be the Seiberg-Witten moduli space and let $d=\dim(M_L)$. Simple-type conjecture. The only n...

conjecture | open | MathDB 页面 | 原文线索

V2-028407 | MATHDB-376650

The finite-dimensional Groenewold-Van Hove conjecture

摘录 / EXCERPT: Groenewold-Van Hove conjecture. There is no nontrivial strong quantization of

conjecture | open | MathDB 页面 | 原文线索

V2-028408 | MATHDB-376651

The dense basic-set existence conjecture for quantization

摘录 / EXCERPT: Dense basic-set existence conjecture. There exists a nontrivial quantization of

conjecture | open | MathDB 页面 | 原文线索

V2-028409 | MATHDB-376652

The Poisson-normalizer extension conjecture

摘录 / EXCERPT: Poisson-normalizer extension conjecture. Every integrable irreducible representation of $\wp({\cal B})$ can be extended to a quantization of

conjecture | open | MathDB 页面 | 原文线索

V2-028410 | MATHDB-376654

Metric-independence conjecture for delocalized L^2-invariants of hyperbolic manifolds

摘录 / EXCERPT: Metric-independence conjecture. For every such metric g, the delocalized L^2-invariants are well-defined and independent of the choice of g.

conjecture | open | MathDB 页面 | 原文线索

V2-028411 | MATHDB-376658

Bounded-diameter Yamabe sequence conjecture for Kodaira surfaces

摘录 / EXCERPT: Let M be the underlying 4-manifold of a complex algebraic surface of Kodaira dimension ≥ 0. A sequence of unit-volume constant-scalar-curvature metrics on M has uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-028412 | MATHDB-376659

Yamabe invariant conjecture for compact complex surfaces of Kodaira dimension 0 or 1

摘录 / EXCERPT: Let M be the underlying 4-manifold of a compact complex manifold of Kodaira dimension 0 or 1, and let $b_1(M)$ denote its first Betti number. Yamabe invariant conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-028413 | MATHDB-376660

Negative Yamabe invariant conjecture for compact complex surfaces

摘录 / EXCERPT: Let (M,J) be a compact complex surface, where M is its underlying 4-manifold and $Y(M)$ is its Yamabe invariant. Negative Yamabe invariant conjecture. If ... then M is of gener...

conjecture | open | MathDB 页面 | 原文线索

V2-028414 | MATHDB-376668

Pidstrigach-Tyurin reduction formula for PU(2) monopoles

摘录 / EXCERPT: Let X be a four-manifold, let E be a rank-two bundle, and let $z\in\mathbb A(X)$. Let n_a, d_a, $\overline{\operatorname{Red}}(W,E)$, $\overline V(z)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028415 | MATHDB-376673

Uniqueness conjecture for metric-affine torsion waves

摘录 / EXCERPT: Let c_j be the weights in the metric-affine action, and let Minkowski space be equipped with connections of the form ... Here $R(x)$ denotes the curvature of the connection, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-028416 | MATHDB-376674

Uniqueness conjecture for Riemannian metric-affine solutions

摘录 / EXCERPT: Consider the metric-affine Euler-Lagrange problem for the action with weights c_j, restricted to Riemannian spacetimes, meaning spacetimes with metric-compatible connections. Tho...

conjecture | open | MathDB 页面 | 原文线索

V2-028417 | MATHDB-376675

Uniqueness conjecture for Riemannian solutions of the metric-affine Bach problem

摘录 / EXCERPT: In the Bach case, the weights satisfy ... The Bach action is the squared norm of the Weyl curvature, and the problem is varied independently with respect to the metric and connection...

conjecture | open | MathDB 页面 | 原文线索

V2-028418 | MATHDB-376677

Legendrian-linking conjecture for causality in (3+1)-dimensional spacetimes

摘录 / EXCERPT: Let (M,g) be a globally hyperbolic $(3+1)$-dimensional spacetime with a Cauchy surface diffeomorphic to a subset of $\mathbb{R}^3$, and let N be its manifold of light rays. For...

conjecture | open | MathDB 页面 | 原文线索

V2-028419 | MATHDB-376683

The CSI conjecture on locally homogeneous representatives

摘录 / EXCERPT: Let a Lorentzian spacetime $\mathcal{M}$ be a constant scalar invariant (CSI) spacetime, with curvature invariants $\mathcal{I}$. A spacetime is CSI when all scalar polynomial curv...

conjecture | open | MathDB 页面 | 原文线索

V2-028420 | MATHDB-376684

The CSI_F conjecture on canonical forms of CSI spacetimes

摘录 / EXCERPT: Let a spacetime have a Riemann tensor R and all of its covariant derivatives. A null frame is a frame adapted to the Lorentzian structure, and the boost order and boost weight re...

conjecture | open | MathDB 页面 | 原文线索

V2-028421 | MATHDB-376685

The CSI_K conjecture for higher-dimensional Kundt spacetimes

摘录 / EXCERPT: Let a spacetime be a constant scalar invariant (CSI) spacetime, meaning that all scalar polynomial curvature invariants are constant. The higher-dimensional Kundt CSI class consist...

conjecture | open | MathDB 页面 | 原文线索

V2-028422 | MATHDB-376686

The CSI construction conjecture from locally homogeneous and VSI spacetimes

摘录 / EXCERPT: Let a spacetime be a constant scalar invariant (CSI) spacetime, meaning that all scalar polynomial curvature invariants are constant. A VSI spacetime is one for which all scalar po...

conjecture | open | MathDB 页面 | 原文线索

V2-028423 | MATHDB-376687

Strong cosmic censorship for spherical cosmological spacetimes with positive cosmological constant

摘录 / EXCERPT: Let k be the curvature of the symmetric surfaces and let Λ be the cosmological constant. Consider surface-symmetric cosmological spacetimes with collisionless matter and...

open_problem | open | MathDB 页面 | 原文线索

V2-028424 | MATHDB-376690

Future causal geodesic completeness near expanding surface-symmetric cosmologies

摘录 / EXCERPT: Consider the setting of Theorem ... , the spacetime is future causally geodesically complete. The surrounding discussion says that the answer is expected to be affirmative for...

conjecture | open | MathDB 页面 | 原文线索

V2-028425 | MATHDB-376698

The generating-function conjecture for $G(t,-k)$

摘录 / EXCERPT: Let $G(t,-k)$ denote the generating function considered in the paper, with t a formal variable and k an integer. Generating-function conjecture. For every integer $k\geq 1$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-028426 | MATHDB-376700

Exceptional-divisor monodromy conjecture

摘录 / EXCERPT: Let X be a Calabi-Yau space, let E be an irreducible divisor in X that collapses to a point, and let $i:E\hookrightarrow X$ be the inclusion map. Let ${\Scr K}^K_{{\Scr E}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-028427 | MATHDB-376705

The holonomic-module model conjecture for the Fukaya category

摘录 / EXCERPT: Let X be a symplectic manifold, let $F^{naive}(X,\omega)$ denote its naive Fukaya category, and let $D^b_{\infty}(hol(X))$ be the derived A_{∞}-category of holonomic modul...

conjecture | open | MathDB 页面 | 原文线索

V2-028428 | MATHDB-376711

The EZ-transformation massless D-brane conjecture

摘录 / EXCERPT: Let X be a Calabi-Yau threefold containing a subspace E, let $i:E\hookrightarrow X$ be the inclusion, and let $q:E\to Z$ be the map contracting E to a lower-dimensional subsp...

conjecture | open | MathDB 页面 | 原文线索

V2-028429 | MATHDB-376712

The EZ-transformation monodromy autoequivalence conjecture

摘录 / EXCERPT: EZ-transformation monodromy autoequivalence conjecture. The monodromy around the discriminant in the wall associated with a phase transition given by an EZ-transformation should ac...

conjecture | open | MathDB 页面 | 原文线索

V2-028430 | MATHDB-376729

The prepotential reconstruction conjecture for noncompact toric Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a noncompact toric Calabi-Yau threefold with $\dim H_4(X,\mathbb{Z})=0$, and let $\{l^1,\dots,l^m\}$ define X via symplectic quotient. Define ... where ... and ... If...

conjecture | open | MathDB 页面 | 原文线索

V2-028431 | MATHDB-376763

The toric integrability conjecture for metrics on the torus

摘录 / EXCERPT: Let g be a Riemannian metric on the torus $\mathbb{R}^n/\mathbb{Z}^n$, and suppose that its geodesic flow is toric integrable. Toric integrability conjecture. Then g is flat. T...

conjecture | open | MathDB 页面 | 原文线索

V2-028432 | MATHDB-376765

The flatness conjecture for completely integrable geodesic flows

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold whose geodesic flow is completely integrable. Let Δ_g+V be a Schrödinger operator on (M,g), and suppose that all of its orthon...

conjecture | open | MathDB 页面 | 原文线索

V2-028433 | MATHDB-376766

The generic-potential bounded-eigenfunction problem on flat manifolds

摘录 / EXCERPT: Let (M,g) be a flat manifold and let Δ_g+V be a Schrödinger operator on (M,g). Generic-potential bounded-eigenfunction problem. For generic V, are the eigenfunctions u...

open_problem | open | MathDB 页面 | 原文线索

V2-028434 | MATHDB-376771

The realisability conjecture for equivalence of spinor structures

摘录 / EXCERPT: Realisability conjecture. The bundles Q and Q' are equivalent if and only if φ is realisable.

conjecture | open | MathDB 页面 | 原文线索

V2-028435 | MATHDB-376779

Conjectural presentation of the first-order calculus for the Kronecker foliation spectral triple

摘录 / EXCERPT: Let $\cA=C^\infty(\mathbb{T}^2)\rtimes\mathbb{R}$, with generators u_1,u_2 for the torus part and v_t for the crossed-product action, and let D be the Dirac-type operator def...

conjecture | open | MathDB 页面 | 原文线索

V2-028436 | MATHDB-376784

General-position conjecture for all simple mechanical systems

摘录 / EXCERPT: For each manifold M and each simple mechanical system with kinetic energy T and potential energy V, let $\psi(M,T,V)$ be the homomorphism from the universal Lie algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-028437 | MATHDB-376785

Exact symplectic characterization of Lie algebras with no coadjoint invariants

摘录 / EXCERPT: Exact symplectic characterization. A Lie algebra $\mathfrak{g}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028438 | MATHDB-376788

Lee-Weinberg-Yi specialization conjecture for monopole metrics

摘录 / EXCERPT: Let $\mu_0\in\mathfrak g$ be nonzero, let $\mu:[0,\delta]\to\mathfrak g$ be a regular deformation of μ_0, meaning that $\mu(t)$ is regular for all $t>0$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-028439 | MATHDB-376799

Mishchenko-Fomenko conjecture for homogeneous geodesic flows

摘录 / EXCERPT: Let G be a compact Lie group, let H be a subgroup, and let $\mathfrak v$ denote the corresponding tangent-space representation used to form the invariant polynomial algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-028440 | MATHDB-376812

Petz's conjecture on scalar-curvature monotonicity for BKM metrics

摘录 / EXCERPT: Petz's conjecture. The scalar curvature of the BKM metric is a Schur-increasing function; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-028441 | MATHDB-376813

The commutative alpha-geometry conjecture on scalar-curvature Schur monotonicity

摘录 / EXCERPT: Commutative alpha-geometry conjecture. Suppose $n>2$. If $p\in(1,2)$, then Scal_p is strictly Schur-decreasing; if $p\in(2,+\infty]$, then Scal_p...

conjecture | open | MathDB 页面 | 原文线索

V2-028442 | MATHDB-376830

Existence and K-theory conjecture for projective quantization dg-modules

摘录 / EXCERPT: Let M be a symplectic manifold, let ${\cal W}$ be its Weyl algebra bundle, and let $(\Omega(M, {\cal W}), \nabla, R)$ be the associated Q-algebra. A finitely generated projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-028443 | MATHDB-376875

The Groenewold-Van Hove conjecture for finite-dimensional basic algebras

摘录 / EXCERPT: Let M be a symplectic manifold with a finite-dimensional basic algebra ${\mathfrak b}$. Let $P({\mathfrak b})$ denote its polynomial algebra, and suppose that condition (D1),...

conjecture | open | MathDB 页面 | 原文线索

V2-028444 | MATHDB-376876

The torus quantization conjecture for infinite-dimensional basic algebras

摘录 / EXCERPT: Let M be a symplectic manifold and ${\mathfrak b}$ a basic algebra such that $P^1({\mathfrak b})$ is dense in $C^{\infty}(M)$, where $P^1({\mathfrak b})$ denotes the polynomial a...

conjecture | open | MathDB 页面 | 原文线索

V2-028445 | MATHDB-376878

Local embeddability conjecture for CR 3-spaces

摘录 / EXCERPT: Local embeddability conjecture. A CR 3-space admits locally a closed, non-zero section of its canonical bundle if and only if it is locally embeddable.

conjecture | open | MathDB 页面 | 原文线索

V2-028446 | MATHDB-376883

Localization of symplectic quotient invariants to subtori of fixed dimension

摘录 / EXCERPT: Let T be a torus and let $\leq i\leq \dim T$. For each subtorus $\tau\subset T$, let $\mathcal{A}_\tau$ denote the corresponding summand of the module $\mathcal{A}$, and set ... Le...

conjecture | open | MathDB 页面 | 原文线索

V2-028447 | MATHDB-376889

Characterisation of Cayley hypersurfaces by affine symmetry and parallel normals

摘录 / EXCERPT: Cayley hypersurface characterisation conjecture. Then, in a suitable affine coordinate system, Σ is given by the Cayley hypersurface equation.

conjecture | open | MathDB 页面 | 原文线索

V2-028448 | MATHDB-376892

The C_∞-equivalence conjecture for links in 3-manifolds

摘录 / EXCERPT: Let M be a 3-manifold, and let two links in M be given. They are C_∞-equivalent when they are C_k-equivalent for every $k\geq 1$. C_∞-equivalence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-028449 | MATHDB-376908

The q=3 sharpened Severi inequality conjecture

摘录 / EXCERPT: Let S be a minimal surface of general type with Albanese dimension 2 and irregularity $q(S)=3$. The q=3 conjecture. Then ... and equality holds if and only if S is the symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-028450 | MATHDB-376909

The q≥4 sharpened Severi inequality conjecture

摘录 / EXCERPT: Let S be a minimal surface of general type with Albanese dimension 2 and irregularity $q(S)\ge 4$. The q≥4 conjecture. Then ... Moreover, equality holds if and only if S is a p...

conjecture | open | MathDB 页面 | 原文线索

V2-028451 | MATHDB-376912

Minimal-volume conjecture for complex surfaces of general type

摘录 / EXCERPT: Let M be a compact complex surface of general type, and let $\operatorname{Vol}_{K}(M)$ and $\operatorname{Vol}_{s}(M)$ denote its minimal volume defined using sectional curvatur...

conjecture | open | MathDB 页面 | 原文线索

V2-028452 | MATHDB-376932

Mirror expansion formula for the logarithmic period

摘录 / EXCERPT: Let $z:=\phi^3$, let $\theta:=z\frac{d}{dz}$, and let I_3 be the logarithmic solution of the Picard-Fuchs equation described in the source. Define ... where $I^{(0)}_2$ is the holo...

conjecture | open | MathDB 页面 | 原文线索

V2-028453 | MATHDB-376973

Donaldson's energy lower-bound conjecture for the Kähler metric space

摘录 / EXCERPT: Let V be a Kähler manifold, let $\mathcal{H}$ be the space of Kähler potentials, and let $\mathcal{H}_0$ denote its normalized subspace. For a normalized $\phi\in\mathcal{H}_0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028454 | MATHDB-377009

Congruence conjecture for Hurwitz numbers in local Calabi-Yau geometry

摘录 / EXCERPT: Hurwitz-number congruence conjecture. If d is not divisible by 4, 6, or 9, then

conjecture | open | MathDB 页面 | 原文线索

V2-028455 | MATHDB-377021

Conjecture on homogeneous and real analytic Poisson vector bundles

摘录 / EXCERPT: Poisson vector bundle conjecture. (1) Every homogeneous Hermitian Poisson vector bundle over ${\mathfrak g}^*$ is Poisson-trivial. (2) There is a one-to-one correspondence between...

conjecture | open | MathDB 页面 | 原文线索

V2-028456 | MATHDB-377022

The characterization of self-bumping representations by shuffle immersions

摘录 / EXCERPT: Let M be the hyperbolic 3-manifold under consideration, let B be its relevant deformation space, and let $\mathcal C$ be a non-empty collection of curves in M. Set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028457 | MATHDB-377032

Donaldson's long exact sequence conjecture for global monodromy

摘录 / EXCERPT: Let (E,π) be an exact Morse fibration over D with relative Maslov map $\delta_{E/D}$, let M be the fibre over the base point, and let $\mu\in \operatorname{Symp}^e(M)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-028458 | MATHDB-377037

Joyce's asymptotic exactness conjecture for the operator D

摘录 / EXCERPT: Let M be a hyperkähler manifold that is AC of order $l\geq 1$, and let $k\geq -1$ be an integer. Let $x:M\rightarrow\mathbb{H}$ be a smooth function such that $D(x)$ is a 1-for...

conjecture | open | MathDB 页面 | 原文线索

V2-028459 | MATHDB-377038

The SGH-algebra conjecture for hyperkähler cones

摘录 / EXCERPT: Let C be a hyperkähler cone, and let B be the graded H-algebra defined in Definition 3.3. SGH-algebra conjecture. The graded H-algebra B is an SGH-algebra. This is the algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-028460 | MATHDB-377039

Joyce's coadjoint-orbit conjecture for hyperkähler structures

摘录 / EXCERPT: Let G be a compact Lie group with Lie algebra $\mathfrak g$, let G^c be its complexification, and let Ω be the data defining the examples associated to G. Denote the r...

conjecture | open | MathDB 页面 | 原文线索

V2-028461 | MATHDB-377049

Classification conjecture for (1,2)-symplectic Borel metrics on full flag manifolds

摘录 / EXCERPT: Let $F(n)$ be a full flag manifold with $n\geq 5$, equipped with a complex structure J and its associated tournament ${\cal T}(J)$. Let ds^2_{Λ} denote the Borel metric....

conjecture | open | MathDB 页面 | 原文线索

V2-028462 | MATHDB-377050

Dimension conjecture for non-Kähler families of (1,2)-symplectic metrics on full flag manifolds

摘录 / EXCERPT: Let $F(n)$ be a full flag manifold, and consider families of (1,2)-symplectic metrics on $F(n)$ that are different from the family of Kähler metrics. Dimension conjecture. All such...

conjecture | open | MathDB 页面 | 原文线索

V2-028463 | MATHDB-377066

Maximal-degeneration Gromov-Hausdorff limit conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let ${{\cal X}}_{mer}$ be a maximally degenerate Calabi-Yau family at $q=0$, let X_q^{new} be the rescaled family, and let n be the complex dimension. Maximal-degeneration limi...

conjecture | open | MathDB 页面 | 原文线索

V2-028464 | MATHDB-377067

Asymptotic torus-bundle model conjecture for maximal Calabi-Yau degenerations

摘录 / EXCERPT: Let $({\cal X}_{mer},vol)=(X_q,vol_q)$ be a one-parameter family of maximally degenerate Calabi-Yau manifolds, let X_q^{new} be the rescaled family, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028465 | MATHDB-377068

Dual Monge-Ampère structure conjecture for maximal Calabi-Yau degenerations

摘录 / EXCERPT: Let Y be the smooth part of the limiting base of a maximal degeneration of a Calabi-Yau family, equipped with its Monge-Ampère metric and integral affine structure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028466 | MATHDB-377073

The Gamma-analogue of the spherical Harish-Chandra isomorphism

摘录 / EXCERPT: Gamma spherical Harish-Chandra conjecture. There is a filtration-preserving algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-028467 | MATHDB-377079

Joyce's generic local-model conjecture for codimension-one singularities

摘录 / EXCERPT: An SL fibration is a special Lagrangian fibration, and a singularity has codimension one when its image in the base has codimension one. The fibrations F and F' are the local m...

conjecture | open | MathDB 页面 | 原文线索

V2-028468 | MATHDB-377080

Joyce's generic local-model conjecture for codimension-two singularities

摘录 / EXCERPT: An SL fibration is a special Lagrangian fibration, and a singularity has codimension two when its image in the base has codimension two. The fibration $\hat F$ is the local model c...

conjecture | open | MathDB 页面 | 原文线索

V2-028469 | MATHDB-377081

Joyce's piecewise-smoothness conjecture for generic special Lagrangian fibrations

摘录 / EXCERPT: Let $f:M\rightarrow B$ be a generic special Lagrangian fibration of an almost Calabi-Yau 3-fold, including fibres with singularities, where generic' has the meaning specified in...

conjecture | open | MathDB 页面 | 原文线索

V2-028470 | MATHDB-377082

Joyce's nonidentification conjecture for generic mirror discriminants

摘录 / EXCERPT: Let M and $\hat M$ be generic mirror Calabi-Yau 3-folds. Suppose there exist special Lagrangian fibrations ... Let $\Delta_f\subset B$ and $\Delta_{\hat f}\subset\hat B$ be their...

conjecture | open | MathDB 页面 | 原文线索

V2-028471 | MATHDB-377102

Donaldson's conjecture on stabilization of symplectic four-manifolds

摘录 / EXCERPT: Let (X,ω_X) and (Y,ω_Y) be symplectic four-manifolds that are homeomorphic. Write ω_{st} for the standard Kähler form on projective space, and identify their s...

conjecture | open | MathDB 页面 | 原文线索

V2-028472 | MATHDB-377119

The distance-set dimension improvement conjecture

摘录 / EXCERPT: For a compact set $K\subset\mathbb{R}^2$, let $\operatorname{dist}(K)=\{|x-y|:x,y\in K\}$ be its distance set, and let ${\text{\roman dim}}$ denote Hausdorff dimension. Distance-se...

conjecture | open | MathDB 页面 | 原文线索

V2-028473 | MATHDB-377121

The discretized Furstenburg conjecture

摘录 / EXCERPT: Let $0<\delta\ll1$, let Ω be a δ-separated set of directions, and for each $\omega\in\Omega$ let R_ω be a $(\delta,1/2)_2$ set contained in a rectangle of dime...

conjecture | open | MathDB 页面 | 原文线索

V2-028474 | MATHDB-377123

Thurston's conjecture on convex hyperbolic metrics with prescribed boundary metric

摘录 / EXCERPT: Thurston's conjecture. There is a unique hyperbolic metric g on M which induces h on the boundary, and for which the boundary is convex.

conjecture | open | MathDB 页面 | 原文线索

V2-028475 | MATHDB-377125

Virtual Fibration Conjecture for finite-volume hyperbolic 3-manifolds

摘录 / EXCERPT: Virtual Fibration Conjecture. M has a finite cover which is a surface bundle over S^1.

conjecture | open | MathDB 页面 | 原文线索

V2-028476 | MATHDB-377126

Hopf-Chern Euler characteristic conjecture for closed aspherical manifolds

摘录 / EXCERPT: Euler characteristic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-028477 | MATHDB-377204

The symplectic invariance and Taubes correspondence conjecture for surface counts

摘录 / EXCERPT: Symplectic invariance and Taubes correspondence conjecture. Once f is of sufficiently high degree, $\mathcal{I}_{(X,f)}$ should be independent of f and define a symplectic inva...

conjecture | open | MathDB 页面 | 原文线索

V2-028478 | MATHDB-377219

The ample-normal-bundle conjecture for compact Kähler submanifolds

摘录 / EXCERPT: Ample-normal-bundle conjecture. The algebraic dimension of X should be at least one greater than the dimension of any positive-dimensional compact submanifold with ample normal b...

conjecture | open | MathDB 页面 | 原文线索

V2-028479 | MATHDB-377226

The moduli conjecture for strongly asymptotically conical special Lagrangian folds

摘录 / EXCERPT: Let L be a strongly asymptotically conical special Lagrangian m-fold in $\mathbb C^m$ with cone C, and let k be the number of ends of C at infinity. Let $\mathcal M^s_L$...

conjecture | open | MathDB 页面 | 原文线索

V2-028480 | MATHDB-377227

The moduli conjecture for weakly asymptotically conical special Lagrangian folds

摘录 / EXCERPT: Let L be a weakly asymptotically conical special Lagrangian m-fold in $\mathbb C^m$ with cone C, and let k be the number of ends of C at infinity. Let $\mathcal M^w_L$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-028481 | MATHDB-377249

Classification of invariant Poisson brackets on the dual of a simple Lie algebra

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra. The polynomial algebra on $\mathfrak{g}^*$ is $S\mathfrak{g}$. An invariant Poisson bracket is a Poisson bracket on $S\mathfrak{g}$ inva...

open_problem | open | MathDB 页面 | 原文线索

V2-028482 | MATHDB-377259

Orbifold additivity conjecture C^{orb}_{n,m}

摘录 / EXCERPT: Let $g:(Y/H)\to Z$ be a prepared and high holomorphic fibration between manifolds, with $Y\in\mathcal C$. Let $\Delta(g,H)$ be its base orbifold, let $z\in Z$ be general, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-028483 | MATHDB-377263

Conjecture IV_H on the Kobayashi metric of the core

摘录 / EXCERPT: Let $X\in\mathcal C$ be a compact complex manifold with core $c_X:X\to C(X)$, and let $d_{C(X)}^*$ be the induced pseudometric on the core. Conjecture IVH. The pseudometric...

conjecture | open | MathDB 页面 | 原文线索

V2-028484 | MATHDB-377309

The uniform systole conjecture for arithmetic locally symmetric manifolds

摘录 / EXCERPT: Fix a symmetric space S of non-compact type without Euclidean de Rham factors, and let an arithmetic S-manifold be a quotient $M=\Gamma\backslash S$ by a torsion-free arithmeti...

conjecture | open | MathDB 页面 | 原文线索

V2-028485 | MATHDB-377317

Vanishing conjecture for reverse-oriented complex-hyperbolic 4-manifolds

摘录 / EXCERPT: Let $M^{4}=\overline{{\mathbb C}{\mathcal H}_{2}/\Gamma}$ be a reverse-oriented compact complex-hyperbolic 4-manifold. Reverse-oriented complex-hyperbolic vanishing conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-028486 | MATHDB-377318

Vanishing conjecture for compact anti-self-dual Einstein 4-manifolds

摘录 / EXCERPT: Let (M^{4},g) be a compact anti-self-dual Einstein manifold with negative scalar curvature. Anti-self-dual Einstein vanishing conjecture. All the Seiberg-Witten invariants vanish...

conjecture | open | MathDB 页面 | 原文线索

V2-028487 | MATHDB-377320

Bounded-discrepancy hypersurface singularity conjecture

摘录 / EXCERPT: Bounded-discrepancy hypersurface singularity conjecture. There exists a hyperplane section H such that (X,H) is plt (lc). Hence $(X\ni P)$ is not weakly exceptional (not except...

conjecture | open | MathDB 页面 | 原文线索

V2-028488 | MATHDB-377343

Symplectic circle-action conjecture for 4-manifolds

摘录 / EXCERPT: Symplectic circle-action conjecture. Every symplectic 4-manifold which admits a circle action also admits (possibly a different) symplectic form and circle action which are symplec...

conjecture | open | MathDB 页面 | 原文线索

V2-028489 | MATHDB-377344

Sheats's order-independence conjecture for symplectic jeu de taquin

摘录 / EXCERPT: Let T be an admissible skew tableau. At each step, choose an inside corner and apply the symplectic jeu de taquin sliding procedure, iterating until a symplectic tableau is obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-028490 | MATHDB-377357

The 2-blow-up conjecture for totally nondegenerate embeddings

摘录 / EXCERPT: Let X and $\widetilde{X}$ be nonsingular projective toric 4-folds, and let $\psi:\widetilde{X}\rightarrow X$ be a 2-blow-up, meaning an equivariant blow-up along a T_N-invari...

conjecture | open | MathDB 页面 | 原文线索

V2-028491 | MATHDB-377358

Generic finite-graph conjecture for singular sets of two-dimensional Hamilton-Jacobi solutions

摘录 / EXCERPT: Generic finite-graph conjecture. For a generic such function $A(x)$, the closure of the singular set of u is a finite graph.

conjecture | open | MathDB 页面 | 原文线索

V2-028492 | MATHDB-377406

Spin and characteristic-class conjecture for the commutator subgroup

摘录 / EXCERPT: Let X be a simply connected symplectic manifold, let $G_k^0(X,\omega)$ be the reduced stabilized fundamental group, and let L_k and K_X be the classes appearing in the constr...

conjecture | open | MathDB 页面 | 原文线索

V2-028493 | MATHDB-377407

Arnold's signature conjecture for domains in real projective space

摘录 / EXCERPT: Consider a domain $U\subset{\mathbb R}P^n$ bounded by a connected smooth hypersurface B. Suppose that the second fundamental form of B is non-degenerate at every point and has...

conjecture | open | MathDB 页面 | 原文线索

V2-028494 | MATHDB-377408

Arnold's hypothesis on hyperbolic hypersurfaces

摘录 / EXCERPT: A smooth hypersurface in $\mathbb{RP}^{n+1}$ is (k,l)-hyperbolic if its second quadratic form has signature (k,l). Let $U\subset\mathbb{R}P^n$ be a domain bounded by a connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-028495 | MATHDB-377414

Nodal-line conjecture for a simple third Dirichlet eigenfunction

摘录 / EXCERPT: Let $\Omega\subseteq\mathbb{R}^2$ be a connected planar domain, and let u_3 be an eigenfunction corresponding to a simple third eigenvalue λ_3 of the Dirichlet Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-028496 | MATHDB-377419

Critical surfaces can be isotoped to index 2 minimal surfaces

摘录 / EXCERPT: A critical surface is a Heegaard surface with the criticality property defined in the paper, and an index 2 minimal surface is a minimal surface whose second variation has index 2....

conjecture | open | MathDB 页面 | 原文线索

V2-028497 | MATHDB-377429

The entropy rigidity conjecture

摘录 / EXCERPT: Let M be a closed manifold which admits a locally symmetric Riemannian metric g_{loc} with nonpositive sectional curvature. Assume that (M,g_{loc}) has no local factors isome...

conjecture | open | MathDB 页面 | 原文线索

V2-028498 | MATHDB-377430

The degree conjecture in nonpositive curvature

摘录 / EXCERPT: Let M be a closed n-manifold with nonpositive sectional curvature, negative Ricci curvature, and no local $\mathbb{R}$ factors. Let N be any closed Riemannian manifold and le...

conjecture | open | MathDB 页面 | 原文线索

V2-028499 | MATHDB-377431

Gromov's positivity conjecture for the Gromov norm

摘录 / EXCERPT: Let M be a closed, nonpositively curved, locally symmetric manifold with no local $\mathbb{R}$ factors. The Gromov norm of M, namely the simplicial volume, is positive. Gromov'...

conjecture | open | MathDB 页面 | 原文线索

V2-028500 | MATHDB-377453

The Bismut holonomy existence conjecture for compact complex manifolds

摘录 / EXCERPT: Let M be a compact complex manifold of real dimension $2n$ with vanishing first Chern class. A hermitian structure on M is a Riemannian metric compatible with its complex struc...

conjecture | open | MathDB 页面 | 原文线索

V2-028501 | MATHDB-377474

The two-skew-lines conjecture for CI-liaison classes in projective four-space

摘录 / EXCERPT: Let C be a set of two skew lines in $\mathbb{P}^{4}$, spanning a hyperplane H, and let C' be another set of two skew lines in $\mathbb{P}^{4}$, spanning a hyperplane H'. Tw...

conjecture | open | MathDB 页面 | 原文线索

V2-028502 | MATHDB-377490

The DK hypothesis: D-equivalence and K-equivalence coincide

摘录 / EXCERPT: The DK hypothesis. The following conditions are equivalent: $D(X)$ and $D(Y)$ are equivalent as triangulated categories, and X and Y are K-equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-028503 | MATHDB-377493

Mixed toric residue mirror conjecture for Calabi-Yau complete intersections

摘录 / EXCERPT: Let $\Delta=\Delta_1+\cdots+\Delta_r$ and $\nabla=\nabla_1+\cdots+\nabla_r$ be dual nef-partitions, let $\mathcal T$ be a suitable coherent triangulation, and let $\mathbb P$ and t...

conjecture | open | MathDB 页面 | 原文线索

V2-028504 | MATHDB-377501

The derived equivalence conjecture for the compactified Fukaya category

摘录 / EXCERPT: Let M be an exact symplectic manifold embedded in a closed symplectic manifold X as in Assumption. Let $\mathcal F(M \subset X)_{gen}$ be the generic fibre of the A_∞-de...

conjecture | open | MathDB 页面 | 原文线索

V2-028505 | MATHDB-377505

Lipeomorphic invariance of metric-coordinate dimension

摘录 / EXCERPT: Lipeomorphic invariance conjecture. Metric-coordinate dimension is invariant under lipeomorphisms; that is, if M and N are lipeomorphic, then they have the same metric-coordina...

conjecture | open | MathDB 页面 | 原文线索

V2-028506 | MATHDB-377506

Kawamata's categorical resolution conjecture for K-equivalent stacks

摘录 / EXCERPT: Let $\mathcal{X}$ and $\mathcal{Y}$ be smooth Deligne--Mumford stacks with Gorenstein moduli spaces X and Y. Suppose there are birational maps ... such that $f^*K_X=g^*K_Y$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-028507 | MATHDB-377513

Generalized flip and flop derived-functor conjecture

摘录 / EXCERPT: Generalized flip and flop derived-functor conjecture. For any generalized flip $X\stackrel{\mathrm{fl}}{\dashrightarrow}X^+$ there is an exact fully faithful functor

conjecture | open | MathDB 页面 | 原文线索

V2-028508 | MATHDB-377514

Minimal noncommutative desingularization conjecture for canonical singularities

摘录 / EXCERPT: Minimal noncommutative desingularization conjecture. The pair $(\operatorname{id}_X,\mathcal{E}nd(q_*\mathcal{O}_Y))$ is a minimal desingularization of X.

conjecture | open | MathDB 页面 | 原文线索

V2-028509 | MATHDB-377519

Tverberg's conjecture on common flats and convex hull partitions

摘录 / EXCERPT: Let $0\leq k\leq d-1$ and let $S_0,S_1,\ldots,S_k$ be finite sets of points in $\mathbb R^d$, with ... for $i=0,1,\ldots,k$. A common convex-hull transversal is a k-flat meeting ea...

conjecture | open | MathDB 页面 | 原文线索

V2-028510 | MATHDB-377573

The generalized E6-subfactor conjecture on distinguishing lens spaces

摘录 / EXCERPT: Let $L(7,1)$ and $L(7,2)$ be the indicated lens spaces, and suppose there exists a generalized E_6-subfactor with group symmetry $G=\mathbb{Z}/7\mathbb{Z}$. Let the Turaev-Viro-O...

conjecture | open | MathDB 页面 | 原文线索

V2-028511 | MATHDB-377575

The polyhedral realization conjecture for the bialgebra PROP differential

摘录 / EXCERPT: Let m,n be positive integers, let ξ^m_n denote the generator indexed by m inputs and n outputs, and let ∂ be the perturbed differential in the minimal model of t...

conjecture | open | MathDB 页面 | 原文线索

V2-028512 | MATHDB-377605

Khashin's Mori-structure conjecture for the double space U

摘录 / EXCERPT: Let U be the double space considered in the source, and let ${\mathcal S}$ denote the family of curves used there to construct the fibrations $V_l/\mathbb{P}^1$. Khashin's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-028513 | MATHDB-377607

Conformal-flat connected-sum conjecture for manifolds with vanishing Stiefel-Whitney classes

摘录 / EXCERPT: Let M^n be a closed connected smooth n-manifold whose Stiefel-Whitney classes all vanish. A Riemannian metric is conformally flat when its Weyl tensor vanishes. Conformal-flat...

conjecture | open | MathDB 页面 | 原文线索

V2-028514 | MATHDB-377614

The rank two bundle conjecture

摘录 / EXCERPT: Let $\mathbb{P}^N$ be a projective space of dimension at least six, and let $\mathcal{E}$ be a vector bundle of rank two on it. The rank two bundle conjecture. The bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-028515 | MATHDB-377615

The higher infinitesimal-neighbourhood conjecture for codimension-two linear spaces

摘录 / EXCERPT: Let $\mathbb{P}^N$ with $N\geq 6$ be a linear subspace of codimension two in $\mathbb{P}^{N+2}$. Let M denote the integer occurring in the infinitesimal-neighbourhood notation, a...

conjecture | open | MathDB 页面 | 原文线索

V2-028516 | MATHDB-377620

Normal holonomy conjecture for nonnegatively curved connection metrics

摘录 / EXCERPT: Normal holonomy conjecture. The normal holonomy group is isomorphic to a subgroup of $\operatorname{SO}(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028517 | MATHDB-377621

The Weak Hopf Conjecture for nonnegatively curved metrics on S^n times R^k

摘录 / EXCERPT: Let $S^n\times\mathbb{R}^k$ carry a complete metric of nonnegative sectional curvature, with soul and a small metric tube about the soul. Weak Hopf Conjecture. There does not exist...

conjecture | open | MathDB 页面 | 原文线索

V2-028518 | MATHDB-377646

Conjecture on the moduli of Zoll metrics and totally real Lagrangian embeddings

摘录 / EXCERPT: A Zoll metric on S^2 is a Riemannian metric whose geodesics are all simple closed curves of equal length. A marked Zoll structure is a Zoll structure together with a choice of an...

conjecture | open | MathDB 页面 | 原文线索

V2-028519 | MATHDB-377647

Bers-Sullivan-Thurston Density Conjecture

摘录 / EXCERPT: Let N be a complete hyperbolic 3-manifold with finitely generated fundamental group. An algebraic limit of geometrically finite hyperbolic 3-manifolds is a manifold obtained as t...

conjecture | open | MathDB 页面 | 原文线索

V2-028520 | MATHDB-377649

Jørgensen's strong convergence conjecture

摘录 / EXCERPT: Let M be the fixed manifold parametrizing the deformation space ${\mathcal D}(M)$. A sequence $\rho_i\to\rho$ is type-preserving if parabolic elements are preserved throughout th...

conjecture | open | MathDB 页面 | 原文线索

V2-028521 | MATHDB-377650

Mori's orbifold ansatz for orbifold Mori fibrations

摘录 / EXCERPT: Let X^∂ be a smooth projective effective orbifold with $\kappa(X^\partial)=-\infty$. An orbifold Mori fibration is a fibration whose general fibers are orbifold rationally...

conjecture | open | MathDB 页面 | 原文线索

V2-028522 | MATHDB-377659

Waldhausen's virtual positive first Betti number conjecture for 3-manifolds

摘录 / EXCERPT: Waldhausen's conjecture. There is a finite cover $\tilde M\to M$ with $b_1(\tilde M)>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-028523 | MATHDB-377662

Orthogonal Plücker-type conjecture for simple forms

摘录 / EXCERPT: Let $\mathbb{V}$ be a finite-dimensional real vector space with a euclidean or lorentzian inner product, let $p\geq 2$, and let $F\in\Lambda^p\mathbb{V}^*$. For...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028524 | MATHDB-377668

Hamilton's nonexistence conjecture for Type I-like ancient Ricci flows

摘录 / EXCERPT: A complete noncompact ancient solution to the Ricci flow is a Ricci flow $(\mathcal{M}^{n},g(t))$ defined for $-\infty<t<\omega$ on a complete noncompact manifold. It is Type-I lik...

conjecture | open | MathDB 页面 | 原文线索

V2-028525 | MATHDB-377677

Weak adjunction conjecture for log canonical pairs

摘录 / EXCERPT: Let $f\colon X\to Z$ be a fiber space and let D be a $\mathbb{Q}$-divisor on X such that (X,D) is log canonical and K_X+D is $\mathbb{Q}$-linearly trivial over Z. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028526 | MATHDB-377685

Bers density conjecture for singly degenerate Kleinian groups

摘录 / EXCERPT: Bers density conjecture. If M is singly degenerate, then $\Gamma\in\overline{B}_X$, where X is the conformal boundary of M.

conjecture | open | MathDB 页面 | 原文线索

V2-028527 | MATHDB-377688

The complete type-edge-pair description of the solution set for Sp_4

摘录 / EXCERPT: Complete type-edge-pair conjecture. The union over all type-edge pairs equals the solution set:

conjecture | open | MathDB 页面 | 原文线索

V2-028528 | MATHDB-377708

The Bers-Sullivan-Thurston density conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a complete hyperbolic 3-manifold with finitely generated fundamental group. A hyperbolic 3-manifold is geometrically finite if its associated Kleinian group is geome...

conjecture | open | MathDB 页面 | 原文线索

V2-028529 | MATHDB-377720

Density conjecture for rational curves on K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, and let $\Sigma=\bigcup D_\alpha$ be the union of all rational curves on S. Let $\overline{\Sigma}$ denote the closure of Σ in the analytic topolo...

conjecture | open | MathDB 页面 | 原文线索

V2-028530 | MATHDB-377728

Curvature-graph conjecture for tight graphs of type 3_n

摘录 / EXCERPT: Curvature-graph conjecture. The graph of curvatures of any tight graph of type 3_n is the graph in the first of those three cases.

conjecture | open | MathDB 页面 | 原文线索

V2-028531 | MATHDB-377743

Meeks-Sullivan parabolicity conjecture for finite-genus proper minimal surfaces

摘录 / EXCERPT: Meeks-Sullivan conjecture. The surface M is parabolic.

conjecture | open | MathDB 页面 | 原文线索

V2-028532 | MATHDB-377755

The Calabi conjecture for bi-polyhedral Kähler affine structures

摘录 / EXCERPT: The Calabi conjecture. There is a unique bi-polyhedral Kähler affine structure on $\Sigma\backslash D$ of type (λ,ν) such that the metric is Monge-Ampère:

conjecture | open | MathDB 页面 | 原文线索

V2-028533 | MATHDB-377772

Free-action conjecture for quantization of quasi-Poisson manifolds

摘录 / EXCERPT: Free-action quantization conjecture. A quasi-Poisson manifold X for $({\mathfrak{g}},Z)$ can be quantized if the ${\mathfrak{g}}$-action on X is free, i.e. if X is a principa...

conjecture | open | MathDB 页面 | 原文线索

V2-028534 | MATHDB-377781

Pu's relative 1-systole conjecture

摘录 / EXCERPT: Pu's relative 1-systole conjecture. For any such S and $\hat\phi$,

conjecture | open | MathDB 页面 | 原文线索

V2-028535 | MATHDB-377782

Croke-Dairbekov-Sharafutdinov filling-volume rigidity conjecture

摘录 / EXCERPT: Croke-Dairbekov-Sharafutdinov filling-volume conjecture. The filling volume of ∂X with its chordal metric equals $\operatorname{vol}(X)$, and X is the unique volume-min...

conjecture | open | MathDB 页面 | 原文线索

V2-028536 | MATHDB-377783

Croke-Dairbekov-Sharafutdinov marked-length-spectrum volume conjecture

摘录 / EXCERPT: Croke-Dairbekov-Sharafutdinov marked-length-spectrum volume conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-028537 | MATHDB-377784

Universal stable 2-systolic inequality in dimension four

摘录 / EXCERPT: Universal stable 2-systolic inequality conjecture. There is a numerical constant C, independent of X, such that for every metric $\mathbf{g}$ on X,

open_problem | open | MathDB 页面 | 原文线索

V2-028538 | MATHDB-377785

Pu-Loewner inequality conjecture for the four-torus-projective-plane product

摘录 / EXCERPT: Pu-Loewner inequality conjecture. Every such metric satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028539 | MATHDB-377789

Kronheimer's adjunction inequality conjecture for taut foliations on 4-manifolds

摘录 / EXCERPT: Let M^4 be a 4-manifold, let ${\mathcal F}$ be a taut foliation on M, and let $J_{\mathcal F}$ be an almost-complex structure induced by ${\mathcal F}$. For an embedded surfa...

conjecture | open | MathDB 页面 | 原文线索

V2-028540 | MATHDB-377792

Foliation adjunction inequality for taut foliations on four-manifolds

摘录 / EXCERPT: Let M^4 be a four-manifold with a taut foliation ${\mathcal F}$, and let $J_{\mathcal F}$ be an almost-complex structure induced by ${\mathcal F}$. For an embedded surface S in...

conjecture | open | MathDB 页面 | 原文线索

V2-028541 | MATHDB-377806

Index-one conjecture for the biharmonic map from the generalised Clifford torus

摘录 / EXCERPT: Let $\mathbb S^{l}(\frac{1}{2})\times\mathbb S^{l}(\frac{1}{2})$ be the generalised Clifford torus, viewed as a submanifold of $\mathbb S^{2l+2}$, and consider the associated bihar...

conjecture | open | MathDB 页面 | 原文线索

V2-028542 | MATHDB-377814

Shokurov's finite generation conjecture for weak log Fano contractions

摘录 / EXCERPT: Let $(Y/T,B)$ be a weak log Fano contraction, in particular Klt, and let $\{\mathcal{D}_i\}_{i=1}^{\infty}$ be a system of b-divisors. Assume that $\{\mathcal{D}_i\}$ satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-028543 | MATHDB-377815

Shokurov's canonical confinement conjecture for log-canonically saturated b-divisors

摘录 / EXCERPT: Let $(Y',B_{Y'})$ be a pair, and let $\mathfrak{M}(Y',B_{Y'})$ denote the set of b-free b-divisors that are log canonically saturated, meaning ... A model has canonically confined si...

conjecture | open | MathDB 页面 | 原文线索

V2-028544 | MATHDB-377821

Joyce's generic Kähler perturbation conjecture for special Lagrangian moduli spaces

摘录 / EXCERPT: Let (M,J,ω,Ω) be an almost Calabi-Yau m-fold, let X be a compact special Lagrangian m-fold in M with conical singularities, and let $\mathcal I_{X'}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-028545 | MATHDB-377832

Meylan-Mir-Zaitsev approximation conjecture for real analytic targets

摘录 / EXCERPT: Let $M\subset\mathbb C^N$ be a real analytic CR submanifold, let $p\in M$ be a point of finite type, and let K be a positive integer. Let $M'\subset\mathbb C^{N'}$ be a real anal...

conjecture | open | MathDB 页面 | 原文线索

V2-028546 | MATHDB-377833

CR-submanifold formal-equivalence conjecture of Baouendi, Rothschild, and Zaitsev

摘录 / EXCERPT: Let $M\subset\mathbb C^N$ be a real analytic CR submanifold. Let $p\in M$, let $M'\subset\mathbb C^N$ be a real analytic submanifold with $\dim_{\mathbb R}M=\dim_{\mathbb R}M'$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-028547 | MATHDB-377834

Formal-map convergence conjecture for real analytic hypersurfaces in complex dimension two

摘录 / EXCERPT: Let M and M' be connected real analytic hypersurfaces through 0 in $\mathbb C^2$, and suppose that M is of finite type at some point. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028548 | MATHDB-377835

Meylan-Mir-Zaitsev convergence conjecture for real analytic targets

摘录 / EXCERPT: Let $M\subset\mathbb C^N$ be a real analytic generic submanifold and let $p\in M$ be a point of finite type. Let $M'\subset\mathbb C^{N'}$ be a real analytic set containing no nont...

conjecture | open | MathDB 页面 | 原文线索

V2-028549 | MATHDB-377836

Strictly pseudoconvex formal-embedding convergence conjecture

摘录 / EXCERPT: Let N and N' be integers with $2\le N\le N'$. Let $M\subset\mathbb C^N$ and $M'\subset\mathbb C^{N'}$ be real-analytic strictly pseudoconvex hypersurfaces, and let a formal emb...

conjecture | open | MathDB 页面 | 原文线索

V2-028550 | MATHDB-377867

Discreteness interval conjecture for complex reflection triangle groups

摘录 / EXCERPT: Let $\rho_t(p,q,r)$ be the one-parameter family of representations of the (p,q,r)-reflection triangle group by complex reflections, with $p\leq q\leq r$ and ρ_0 stabilizing...

conjecture | open | MathDB 页面 | 原文线索

V2-028551 | MATHDB-377868

Type classification conjecture for complex reflection triangle groups

摘录 / EXCERPT: Let (p,q,r) be a triple with $p\leq q\leq r$, and let $\rho_t(p,q,r)$ be the associated one-parameter family. Call (p,q,r) type A when the endpoints of the critical interval oc...

conjecture | open | MathDB 页面 | 原文线索

V2-028552 | MATHDB-377869

Extra discrete representations conjecture for complex reflection triangle groups

摘录 / EXCERPT: Let (p,q,r) be a triple with $p\leq q\leq r$, let $\rho_t(p,q,r)$ be the associated family, and classify the triple as type A or type B according to the critical-interval definit...

conjecture | open | MathDB 页面 | 原文线索

V2-028553 | MATHDB-377883

The two-dimensional wave maps global well-posedness conjecture

摘录 / EXCERPT: Let (M,g) be a complete Riemannian manifold, and consider wave maps from $\mathbb R^2\times\mathbb R$ into (M,g). The critical initial-data space is $H^1\times L^2$. Wave maps...

conjecture | open | MathDB 页面 | 原文线索

V2-028554 | MATHDB-377896

The conjecture that the maximum score of a decomposition-star space is $8\,\mathrm{pt}$

摘录 / EXCERPT: Maximum-score conjecture. The maximum of σ on X is the constant

conjecture | open | MathDB 页面 | 原文线索

V2-028555 | MATHDB-377909

The Blaschke manifold classification conjecture

摘录 / EXCERPT: Blaschke manifold classification conjecture. If M is a Riemannian manifold such that

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V2-028556 | MATHDB-377935

The symplectic-resolution deformation-equivalence conjecture

摘录 / EXCERPT: Symplectic-resolution deformation-equivalence conjecture. Then Z_1 is deformation equivalent to Z_2.

conjecture | open | MathDB 页面 | 原文线索

V2-028557 | MATHDB-377942

The SYZ integrable-system and B-field mirror-symmetry conjecture

摘录 / EXCERPT: SYZ and integrable-system conjecture. Near the large-complex-structure/large-volume limit: the SYZ picture holds on $\mathcal M_{\mathbb R}$; for a B-field B' on X', $X_{B'}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-028558 | MATHDB-377950

Arnold's Four Cusp Conjecture for wave-front eversions

摘录 / EXCERPT: Let γ_t, $t\in[0,1]$, be an eversion of a smooth circular wave front with no dangerous self tangencies: a family of generic oriented wave fronts in $\mathbb{R}^2$ whose endp...

conjecture | open | MathDB 页面 | 原文线索

V2-028559 | MATHDB-377951

Smooth-sliceness conjecture for zero-twisted Whitehead doubles

摘录 / EXCERPT: Let K be a knot, and let $W_0(K)$ denote its zero-twisted Whitehead double. Smooth-sliceness conjecture for zero-twisted Whitehead doubles. The knot K is smoothly slice if and...

conjecture | open | MathDB 页面 | 原文线索

V2-028560 | MATHDB-377969

The unit-coefficient conjecture for the resultant complex

摘录 / EXCERPT: Unit-coefficient conjecture. The constant $c=\pm1$.

conjecture | open | MathDB 页面 | 原文线索

V2-028561 | MATHDB-377970

The irreducibility conjecture for toric subresultants

摘录 / EXCERPT: Let ${\mathbb A}$ be the coefficient ring, let $P_h\in{\mathbb A}$ denote the factor associated with a monomial h of critical degree, and let ℓ be the index of the lattice s...

conjecture | open | MathDB 页面 | 原文线索

V2-028562 | MATHDB-377986

Birational invariance of condition star for dP3 fibrations

摘录 / EXCERPT: Birational invariance conjecture for condition star. Then $X'/\mathbb{P}^1$ also satisfies condition $(\ast)$. This predicts that condition $(\ast)$ is preserved under square-birat...

conjecture | open | MathDB 页面 | 原文线索

V2-028563 | MATHDB-378009

Finiteness conjecture for Euler characteristics of Stein fillings

摘录 / EXCERPT: Finiteness conjecture. The set $\mathcal{C}_{(M,\xi)}$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-028564 | MATHDB-378014

The meromorphic-degree conjecture for hyper-Kähler manifolds

摘录 / EXCERPT: Let N be a hyper-Kähler manifold of complex dimension $2n$, and suppose its Néron-Severi group has rank one and is generated by an isotropic element of $H^{2}(N,\mathbb{Z})$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-028565 | MATHDB-378026

The logarithmic Picard group conjecture for toric degenerations

摘录 / EXCERPT: Let $f:\mathcal{X}\rightarrow S$ be a toric degeneration, and write $X=\mathcal{X}_0$. Let Z be the singular set of the degeneration, and let $\mathcal{M}_X$ denote the induced l...

conjecture | open | MathDB 页面 | 原文线索

V2-028566 | MATHDB-378076

Taubes' conjecture on symplectic products and fibred 3-manifolds

摘录 / EXCERPT: Let M be a closed oriented 3--manifold. Taubes' conjecture. The product $S^1 \times M$ admits a symplectic structure if and only if M fibers over S^1. The conjecture relates...

conjecture | open | MathDB 页面 | 原文线索

V2-028567 | MATHDB-378083

Lecouvey's charge formula conjecture for type C_n Kostka-Foulkes polynomials

摘录 / EXCERPT: Let β be a dominant weight for the root system C_n, let $B(\beta)$ be the corresponding crystal, and let ... be its subset of tableaux of weight μ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028568 | MATHDB-378089

The relative-invariant hierarchy conjecture for toric Fano surfaces

摘录 / EXCERPT: Let X be a toric Fano surface. Let D be a toric invariant divisor, and let D^+ and D^- be disjoint toric invariant divisors. Consider the suitably defined generating series...

conjecture | open | MathDB 页面 | 原文线索

V2-028569 | MATHDB-378100

Polynomial bound for maps between fixed curves

摘录 / EXCERPT: Let X and Y be fixed smooth compact complex curves of genus at least 2, and let the genus of X be g. A polynomial bound conjecture. There is a polynomial function $B(g)$...

conjecture | open | MathDB 页面 | 原文线索

V2-028570 | MATHDB-378102

Thurston's Elliptization Conjecture for closed 3-manifolds

摘录 / EXCERPT: Thurston's Elliptization Conjecture. Every closed 3-manifold with finite fundamental group can be modeled on the spherical geometry $(S^3,SO(4))$.

conjecture | open | MathDB 页面 | 原文线索

V2-028571 | MATHDB-378104

Log crepant derived equivalence conjecture for associated Deligne-Mumford stacks

摘录 / EXCERPT: Log crepant derived equivalence conjecture. Then there exists an equivalence of triangulated categories

conjecture | open | MathDB 页面 | 原文线索

V2-028572 | MATHDB-378106

The generic-metric conjecture for harmonic spinors

摘录 / EXCERPT: Generic-metric conjecture. For a generic Riemannian metric on any compact spin manifold, the space of harmonic spinors is no larger than forced by the index theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-028573 | MATHDB-378107

The arbitrary finite Dirac-spectrum prescription conjecture

摘录 / EXCERPT: Arbitrary finite Dirac-spectrum prescription conjecture. Apart from these algebraic constraints and the index-theoretic constraints on the zero eigenvalue, there is a Riemannian me...

conjecture | open | MathDB 页面 | 原文线索

V2-028574 | MATHDB-378121

The noncovering conjecture for the exceptional manifolds N_i

摘录 / EXCERPT: Noncovering conjecture. The manifolds N_i for $i=1,2,3,4,5,6$ do not nontrivially cover any manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-028575 | MATHDB-378122

Finiteness conjecture for covering spectra of Laplace-isospectral manifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be a set of Laplace-isospectral manifolds with a uniform upper bound on diameter. Finiteness conjecture. There are only finitely many distinct covering spectra fo...

conjecture | open | MathDB 页面 | 原文线索

V2-028576 | MATHDB-378132

The orbit lattice conjecture for platycosms

摘录 / EXCERPT: Orbit lattice conjecture for platycosms. All 10 platycosms have the orbit lattice property.

conjecture | open | MathDB 页面 | 原文线索

V2-028577 | MATHDB-378133

Monotonicity conjecture for the intersection angle

摘录 / EXCERPT: Monotonicity conjecture. The angle $\alpha(p)$ is a monotone function for

conjecture | open | MathDB 页面 | 原文线索

V2-028578 | MATHDB-378141

Almost toric neighborhood conjecture for symplectic four-manifolds

摘录 / EXCERPT: Let (M,ω) be a symplectic four-manifold containing a top-dimensional submanifold with boundary that is diffeomorphic to an almost toric manifold whose base has a one-stratum...

conjecture | open | MathDB 页面 | 原文线索

V2-028579 | MATHDB-378149

Equivistal subdivision conjecture for convex polyhedra

摘录 / EXCERPT: Equivistal subdivision conjecture. The equivalence relation induced by equivistality constitutes a convex polyhedral subdivision of S. Moreover, the number of open regions in thi...

conjecture | open | MathDB 页面 | 原文线索

V2-028580 | MATHDB-378161

The cluster-algebra tropical fan conjecture

摘录 / EXCERPT: Tropical fan conjecture. If the lineality space of $\operatorname{Trop}^+ \operatorname{Spec} \mathcal{A}$ has dimension $|C|$, then the quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-028581 | MATHDB-378188

Abelianization conjecture for braid-group monodromy of symplectic branched covers

摘录 / EXCERPT: Let (X,ω) be a simply connected symplectic 4-manifold. For sufficiently large k, let G_k^0 be the relevant subgroup associated with the branch curve of the degree-...

conjecture | open | MathDB 页面 | 原文线索

V2-028582 | MATHDB-378197

Gray-Wolf conjecture on homogeneous strictly nearly Kähler spaces

摘录 / EXCERPT: Gray-Wolf conjecture. Every homogeneous strictly nearly Kähler space is a naturally reductive 3-symmetric space equipped with its canonical almost-complex structure.

conjecture | open | MathDB 页面 | 原文线索

V2-028583 | MATHDB-378200

Density conjecture for the real parts of complex dimensions of nonlattice strings

摘录 / EXCERPT: Density conjecture. The real parts of the complex dimensions of $\mathcal{L}$ form a set that is dense in the connected interval $[\sigma_l,D]$.

conjecture | open | MathDB 页面 | 原文线索

V2-028584 | MATHDB-378205

Gillet-Soulé's determinant bound conjecture for line bundles on curves

摘录 / EXCERPT: Let C be a compact complex curve with Hermitian metric g^{C}, and let l be a line bundle over C with associated Hermitian metric g^{l}. Write $A(g^{C},g^{l})$ for the log...

conjecture | open | MathDB 页面 | 原文线索

V2-028585 | MATHDB-378206

Sharp determinant conjecture at the canonical metric

摘录 / EXCERPT: Let C be a smooth closed curve with canonical constant-curvature metric g_{0}^{C}, and let l be a holomorphic line bundle with induced canonical metric g_{0}^{l}. For a Her...

conjecture | open | MathDB 页面 | 原文线索

V2-028586 | MATHDB-378207

Sharp two-dimensional Szegő limit conjecture on the sphere

摘录 / EXCERPT: Let S^{2} be the two-sphere, let μ be the measure used in the definitions of the functionals A_{n} and B_{n}, and let $A_{n}(\varphi)$ and $B_{n}(\varphi)$ be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-028587 | MATHDB-378211

Manifold conjecture for Barvinok rank-two matrix complexes

摘录 / EXCERPT: For positive integers m and n, let $B_{m,n}$ denote the complex of $m\times n$ matrices of Barvinok rank two. Manifold conjecture. For all m and n, the complex $B_{m,n}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-028588 | MATHDB-378229

Independence of the intersection tree from the Whitney tower

摘录 / EXCERPT: Let $A_1,\dots,A_m$ be the original maps, let $\mathcal{W}$ be a Whitney tower on them, and let $\tau_n(\mathcal{W})$ be its intersection tree, regarded as an element of the quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-028589 | MATHDB-378247

Improved symmetry-rank conjecture for positive quaternionic Kähler manifolds of odd quaternionic dimension

摘录 / EXCERPT: Let M be a positive quaternionic Kähler manifold of dimension $8m+4$, and let $\operatorname{rank}(M)$ denote the rank of its isometry group. Improved symmetry-rank conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-028590 | MATHDB-378249

Schwartz's conjectural picture for complex hyperbolic triangle groups

摘录 / EXCERPT: Schwartz's conjectural picture. A (p_1,p_2,p_3)-representation is a discrete embedding if and only if neither w_A nor w_B is elliptic. The corresponding parameter values form...

conjecture | open | MathDB 页面 | 原文线索

V2-028591 | MATHDB-378266

The geometric realization conjecture for ideal triangulations of 3-manifolds

摘录 / EXCERPT: Geometric realization conjecture. The triangulation T is realized by hyperbolic partially truncated tetrahedra.

conjecture | open | MathDB 页面 | 原文线索

V2-028592 | MATHDB-378283

The Hurwitz-quadrangulation correspondence conjecture for Painlevé I

摘录 / EXCERPT: Hurwitz-quadrangulation correspondence conjecture. The constants satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-028593 | MATHDB-378287

CM placement conjecture for ramified quadratic extensions at 7

摘录 / EXCERPT: CM placement conjecture at 7. If

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V2-028594 | MATHDB-378288

CM placement conjecture for ramified quadratic extensions at 13

摘录 / EXCERPT: CM placement conjecture at 13. If

conjecture | open | MathDB 页面 | 原文线索

V2-028595 | MATHDB-378289

CM residue-disk placement conjecture for supersingular reductions

摘录 / EXCERPT: CM residue-disk placement conjecture. The j-invariants of all such curves E lie in $2(p+1)/i$ residue disks inside the corresponding Atkin-Lehner circle, with $(p+1)/i$ disks f...

conjecture | open | MathDB 页面 | 原文线索

V2-028596 | MATHDB-378290

Stable supersingular-component conjecture for X_0(p^3)

摘录 / EXCERPT: Stable supersingular-component conjecture. There is a semistable covering of $X_0(p^3)$ defined over R_p such that, for each such A, one connected component of the supersingula...

conjecture | open | MathDB 页面 | 原文线索

V2-028597 | MATHDB-378310

Reflection-function extension conjecture for CR mappings

摘录 / EXCERPT: Let $M,M'\subset\mathbb{C}^n$, with $n\geq 2$, be globally minimal real analytic hypersurfaces. Let $h:M\to M'$ be a continuous CR mapping whose holomorphic extension to a global o...

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V2-028598 | MATHDB-378321

Gordon's stabilization conjecture for connected sums of Heegaard splittings

摘录 / EXCERPT: Let M_1 and M_2 be closed, orientable 3-manifolds, and let H_i be a Heegaard surface in M_i for $i=1,2$. The connected sum $H_1\# H_2$ is a Heegaard surface in...

conjecture | open | MathDB 页面 | 原文线索

V2-028599 | MATHDB-378322

F.-N.'s deformation conjecture for symplectic resolutions

摘录 / EXCERPT: F.-N.'s conjecture. There are flat deformations

conjecture | open | MathDB 页面 | 原文线索

V2-028600 | MATHDB-378355

Uniform lattice conjecture for the Lipschitz extension constant

摘录 / EXCERPT: Uniform lattice conjecture. Every lattice $\Gamma\subset M$ with the $\mathcal{LE}$ is uniform. The source places this definition immediately before the conjectural discussion, but...

conjecture | open | MathDB 页面 | 原文线索

V2-028601 | MATHDB-378364

The nef square-zero divisor criterion for abelian fibrations

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold, and let q_X denote its Beauville-Bogomolov quadratic form. A divisor is nef if it has nonnegative intersection with eve...

conjecture | open | MathDB 页面 | 原文线索

V2-028602 | MATHDB-378380

Sparla's existence conjecture for neighborly triangulations of sphere products

摘录 / EXCERPT: Sparla's existence conjecture. There are centrally $(r+1)$-neighborly triangulations of $S^r\times S^r$ on $4r+4$ vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-028603 | MATHDB-378400

Campana's torus-submersion conjecture for Stein universal covers

摘录 / EXCERPT: Campana's torus-submersion conjecture. Up to a finite étale cover of X, the manifold X admits a torus submersion over a projective manifold Y such that K_Y is ample and the...

conjecture | open | MathDB 页面 | 原文线索

V2-028604 | MATHDB-378402

Stretching cusps conjecture for arithmetic locally symmetric manifolds

摘录 / EXCERPT: Stretching cusps conjecture. The sequence $\{(M,g_i)\}$ Gromov-Hausdorff converges, in the appropriate sense, to a disjoint union of two isometric copies of N.

conjecture | open | MathDB 页面 | 原文线索

V2-028605 | MATHDB-378405

The pointed Brody principle for the Kobayashi-Royden pseudometric

摘录 / EXCERPT: Pointed Brody conjecture. If the infinitesimal Kobayashi-Royden pseudometric is degenerate on T_xX, then there exists a non-constant holomorphic map

conjecture | open | MathDB 页面 | 原文线索

V2-028606 | MATHDB-378410

The atoroidality conjecture for bricks

摘录 / EXCERPT: Let a brick be one of the building blocks in the brick decomposition of a 3-manifold used in the source, and call a 3-manifold atoroidal when it contains no essential torus. Atoroi...

conjecture | open | MathDB 页面 | 原文线索

V2-028607 | MATHDB-378411

The conjecture on Sasakian-Einstein metrics for odd-dimensional spheres

摘录 / EXCERPT: An odd-dimensional sphere is the standard sphere S^{2n+1}, and a parallelizable manifold is a manifold with trivial tangent bundle. A Sasakian-Einstein metric is a Sasakian metri...

conjecture | open | MathDB 页面 | 原文线索

V2-028608 | MATHDB-378418

Liouville conjecture for compact generalized Calabi-Yau manifolds

摘录 / EXCERPT: Liouville conjecture. The function f is constant. Equivalently, a compact generalized Calabi-Yau manifold has no nonconstant nowhere-zero smooth rescaling of a global closed gene...

conjecture | open | MathDB 页面 | 原文线索

V2-028609 | MATHDB-378429

Deligne's freeness conjecture for the graded Galois Lie algebra of the thrice-punctured projective line

摘录 / EXCERPT: Let $\mathbb P^1-\{0,1,\infty\}$ be the thrice-punctured projective line over $\mathbb Q$, let $Gal(\overline{\mathbb Q}/\mathbb Q)$ be its absolute Galois group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028610 | MATHDB-378454

Relative Pu conjecture for orientable surfaces with an involution

摘录 / EXCERPT: Relative Pu conjecture. There is a point $p\in X$ such that

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V2-028611 | MATHDB-378476

Quantization conjecture for the conformal energy of simplicial Willmore spheres

摘录 / EXCERPT: Let W denote the conformal energy of a simplicial Willmore sphere. Quantization conjecture. The conformal energy is quantized: ... This conjecture is motivated by numerical experim...

conjecture | open | MathDB 页面 | 原文线索

V2-028612 | MATHDB-378490

The k-distance set bound and equality conjecture

摘录 / EXCERPT: Let S be a k-distance set in an n-dimensional normed space, meaning that the set of distances between distinct points of S contains at most k numbers. k-distance set co...

conjecture | open | MathDB 页面 | 原文线索

V2-028613 | MATHDB-378503

Kontsevich-Soibelman conjecture on maximal degeneration limits

摘录 / EXCERPT: Let ${X}_{mer}$ be an algebraic n-dimensional Calabi-Yau manifold over the field of meromorphic germs, with a non-vanishing volume form, and let (X_t,Ω_t) be the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-028614 | MATHDB-378504

Kontsevich-Soibelman conjecture on dual integral Monge-Ampère limits

摘录 / EXCERPT: Let a dual pair of Calabi-Yau families have Gromov-Hausdorff limits, and let the smooth parts of these limits carry the induced integral Monge-Ampère structures. A Monge-Ampère man...

conjecture | open | MathDB 页面 | 原文线索

V2-028615 | MATHDB-378506

Steinness conjecture for the non-archimedean collapse map

摘录 / EXCERPT: Let $\pi:{X}^{an}\to B$ be the continuous map from the analytic Calabi-Yau space to its limiting base constructed in the non-archimedean collapse picture. Steinness conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-028616 | MATHDB-378507

Conjecture on codimension-one affine singularities

摘录 / EXCERPT: Assume that the singular set of the limiting space has a stratification $B^{sing}=B^{sing}_{n-1}\cup B^{sing}_{\le n-2}$, where B^{sing}_{n-1} consists of the $(n-1)$-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-028617 | MATHDB-378508

Conjecture on deformation to PL affine compactifications

摘录 / EXCERPT: Let B be a compact space with a ${\bf Z}$-affine structure with singularities, with actual and potential singular sets equal, $B^{sing}=B^{pre-sing}$, and assume this set has cod...

conjecture | open | MathDB 页面 | 原文线索

V2-028618 | MATHDB-378509

Monodromy factorization conjecture for K3 affine structures

摘录 / EXCERPT: Let ${\cal F}_v$ be a leaf indexed by a positive vector v in the K3 period description, let Γ_v be the corresponding stabilizer group, and let $Aut_{{\bf Z}PL,v}(S^2)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-028619 | MATHDB-378510

Intrinsic period-lattice conjecture for analytic K3 surfaces

摘录 / EXCERPT: Let $K={{\bf C}}((t))$ and let X^{an}/K be an analytic K3 surface admitting an analytic torus fibration $X^{an}\to S^2$ with standard singularities. Intrinsic period-lattice conj...

conjecture | open | MathDB 页面 | 原文线索

V2-028620 | MATHDB-378512

Independence conjecture for the constructed analytic K3 surface

摘录 / EXCERPT: Let (X^{an},Ω) be the K-analytic K3 surface with volume form constructed from a chosen set ${\cal L}$ of lines. Independence conjecture. The isomorphism class of the pair...

conjecture | open | MathDB 页面 | 原文线索

V2-028621 | MATHDB-378518

The kernel-generation conjecture for symplectic Lefschetz pencils

摘录 / EXCERPT: Kernel-generation conjecture. For pencils of sufficiently large degree $(k\gg 0)$, the kernel of ρ is generated by the three types of elements described in Propositions 3.2--3...

conjecture | open | MathDB 页面 | 原文线索

V2-028622 | MATHDB-378520

Iarrobino's generalised Nagata conjecture for equal fat points

摘录 / EXCERPT: Let (r,d) be integers satisfying ... and $(r,d)\notin\{(7,2),(8,2),(9,3)\}$. Let $I(d,\delta,\mu^r)$ denote the dimension of the space of homogeneous degree-δ polynomials on...

conjecture | open | MathDB 页面 | 原文线索

V2-028623 | MATHDB-378521

Iarrobino's strengthened generalised Nagata conjecture for equal fat points

摘录 / EXCERPT: Let (r,d) be integers satisfying ... and $(r,d)\notin\{(7,2),(8,2),(9,2),(8,3),(9,3)\}$. Let $I(d,\delta,\mu^r)$ denote the dimension of the space of homogeneous degree-δ po...

conjecture | open | MathDB 页面 | 原文线索

V2-028624 | MATHDB-378522

The corner-tetrahedra lower-bound conjecture for triangulations of prisms

摘录 / EXCERPT: Let an m-prism be triangulated using t corner tetrahedra. Corner-tetrahedra conjecture. Such a triangulation has at least ... tetrahedra, and therefore at least ... interior triang...

conjecture | open | MathDB 页面 | 原文线索

V2-028625 | MATHDB-378525

The geometric lemma for convex hulls and mean width

摘录 / EXCERPT: Let D denote the Euclidean unit ball in $\mathbb{R}^n$, and let $M^*(A)$ denote the mean width functional used in the paper. For a set $S=\{x_i\}_{i=1}^{2^k}\subset\mathbb{R}^n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028626 | MATHDB-378546

Family Donaldson-Smith invariants are independent of the Lefschetz fibration

摘录 / EXCERPT: Let (X,ω) be a symplectic 4-manifold and let $\alpha\in H^2(X;\mathbb{Z})$. Let $f\colon X'\to S^2$ be a Lefschetz fibration obtained from a sufficiently high-degree Lefsche...

conjecture | open | MathDB 页面 | 原文线索

V2-028627 | MATHDB-378549

Sedgwick's conjecture on infinitely many irreducible Heegaard splittings

摘录 / EXCERPT: Sedgwick's conjecture. If M contains infinitely many irreducible Heegaard splittings that are pairwise non-isotopic, then M is Haken.

conjecture | open | MathDB 页面 | 原文线索

V2-028628 | MATHDB-378550

Generalized Waldhausen conjecture for fixed-genus Heegaard splittings

摘录 / EXCERPT: Generalized Waldhausen conjecture. If M contains infinitely many Heegaard splittings, pairwise non-isotopic, all of the same genus, then M is toroidal.

conjecture | open | MathDB 页面 | 原文线索

V2-028629 | MATHDB-378551

Conjecture on irreducible Heegaard splittings of arbitrarily large genus

摘录 / EXCERPT: Large-genus Heegaard splitting conjecture. If M contains irreducible Heegaard splittings of arbitrarily large genus, then M is Haken.

conjecture | open | MathDB 页面 | 原文线索

V2-028630 | MATHDB-378561

The unique-centre conjecture for simplices

摘录 / EXCERPT: Unique-centre conjecture. If a simplex has a unique simplex centre, then it is equifacetal.

conjecture | open | MathDB 页面 | 原文线索

V2-028631 | MATHDB-378589

Schäffer's girth duality conjecture for normed spaces

摘录 / EXCERPT: Schäffer's girth duality conjecture. The girth of X equals the girth of X^*.

conjecture | open | MathDB 页面 | 原文线索

V2-028632 | MATHDB-378590

Finsler uniformization conjecture for reversible metrics on the projective plane

摘录 / EXCERPT: Let L be a reversible Finsler metric on ${\mathbb R}P^2$. A smooth positive function ρ on ${\mathbb R}P^2$ is a function whose product with L defines the conformally resca...

conjecture | open | MathDB 页面 | 原文线索

V2-028633 | MATHDB-378609

The disk's nonexistence conjecture for Dirichlet-Neumann isospectrality

摘录 / EXCERPT: A bounded planar domain is equipped with a decomposition of its boundary into Dirichlet and Neumann parts, and Dirichlet-Neumann isospectrality means that swapping these boundary c...

conjecture | open | MathDB 页面 | 原文线索

V2-028634 | MATHDB-378611

The quarter-sphere first-eigenvalue inequality conjecture

摘录 / EXCERPT: Let Q be either of the two isospectral mixed Dirichlet-Neumann problems on a quarter-sphere shown in the source, and let $Q_{a,2}$ be the axisymmetric problem with the Dirichlet...

conjecture | open | MathDB 页面 | 原文线索

V2-028635 | MATHDB-378623

Farrell-Zdravkovska geometric conjecture on bounding flat and almost flat manifolds

摘录 / EXCERPT: Let M^n be a closed manifold. In part (a), assume that M^n is a flat Riemannian manifold. In part (b), assume that M^n admits an almost flat structure. Let W^{n+1} be a com...

conjecture | open | MathDB 页面 | 原文线索

V2-028636 | MATHDB-378662

Joyce's special Lagrangian invariant conjecture for Calabi-Yau 3-folds

摘录 / EXCERPT: Let (M,J,ω,Ω) be an (almost) Calabi-Yau 3-fold with compact underlying 6-manifold M, complex structure J, Kähler form ω, and holomorphic volume form...

conjecture | open | MathDB 页面 | 原文线索

V2-028637 | MATHDB-378665

Matsuki-Gindikin conjecture on domains associated with flag-manifold orbits

摘录 / EXCERPT: Matsuki-Gindikin conjecture. For every $K_\mathbb C$-orbit S of nonholomorphic type on X, one has

conjecture | open | MathDB 页面 | 原文线索

V2-028638 | MATHDB-378695

The universal-cover criterion for smooth triviality of elliptic 3-manifold s-cobordisms

摘录 / EXCERPT: Universal-cover criterion. A smooth s-cobordism of elliptic 3-manifolds is smoothly trivial if and only if its universal cover is smoothly trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-028639 | MATHDB-378700

Conditions 1-6 imply condition 7

摘录 / EXCERPT: Let U be the constructed polygonal surface and let conditions 1-7 be the preceding geometric and arithmetic conditions imposed in the construction of a sail for a periodic contin...

conjecture | open | MathDB 页面 | 原文线索

V2-028640 | MATHDB-378705

Arnold's characteristic reconstruction conjecture for multidimensional continued fractions

摘录 / EXCERPT: A two-dimensional (or, more generally, n-dimensional) continued fraction is represented by a sail, whose facets have homeomorphic types, adjacency relations, integer volumes and...

conjecture | open | MathDB 页面 | 原文线索

V2-028641 | MATHDB-378717

Conjecture on stabilized fundamental groups of branch curve complements

摘录 / EXCERPT: Let X be a simply connected algebraic surface, let $D\subset\mathbb{CP}^2$ be the branch curve of the relevant projection, and let G^0, K, $\bar{G}$, ϕ, Λ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-028642 | MATHDB-378720

Codimension conjecture for irredundant affine hyperplane coverings

摘录 / EXCERPT: Codimension conjecture. The codimension of

conjecture | open | MathDB 页面 | 原文线索

V2-028643 | MATHDB-378739

Density-property characterization of affine space

摘录 / EXCERPT: Let X be a Stein manifold diffeomorphic to ${\mathbb C}^n$ and having the density property, meaning that the closure of the Lie algebra generated by the complete holomorphic vect...

conjecture | open | MathDB 页面 | 原文线索

V2-028644 | MATHDB-378741

Minimal Hilbert function for the first infinitesimal neighborhood of a k-configuration

摘录 / EXCERPT: Linear-configuration conjecture. The Hilbert function of $C_{\underline h}$ is $\underline h^{\min}$.

conjecture | open | MathDB 页面 | 原文线索

V2-028645 | MATHDB-378760

The hypertight-contact-structure conjecture for manifolds with taut foliations

摘录 / EXCERPT: Hypertight-contact-structure conjecture. Every closed, orientable manifold that carries a taut foliation also carries a hypertight contact structure.

conjecture | open | MathDB 页面 | 原文线索

V2-028646 | MATHDB-378764

Calderón-Moreno et al.'s LCT implies strong Euler homogeneity conjecture for free divisors

摘录 / EXCERPT: Let D be a free divisor in a complex manifold X. The logarithmic comparison conjecture. If the logarithmic comparison theorem holds for D, then D is strongly Euler homogene...

conjecture | open | MathDB 页面 | 原文线索

V2-028647 | MATHDB-378788

The derived-equivalence conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let X and Y be Calabi-Yau threefolds, and write $D(X)$ and $D(Y)$ for their bounded derived categories. Derived-equivalence conjecture for Calabi-Yau threefolds. If ... then X...

conjecture | open | MathDB 页面 | 原文线索

V2-028648 | MATHDB-378801

Parity-vanishing surjectivity conjecture for Hilbert-manifold cobordism

摘录 / EXCERPT: Parity-vanishing surjectivity conjecture. If $\mathcal U^{\operatorname{ev}}(X)=0$ or $\mathcal U^{\mathrm{odd}}(X)=0$, then ρ is a surjection.

conjecture | open | MathDB 页面 | 原文线索

V2-028649 | MATHDB-378803

The lattice-point maximality conjecture for the Sylvester reflexive simplex

摘录 / EXCERPT: Let $d\geq 4$, and let $S_{Q'_d}$ be the reflexive simplex defined from the Sylvester-sequence weight system in the source. A d-dimensional lattice polytope is assumed to have ex...

conjecture | open | MathDB 页面 | 原文线索

V2-028650 | MATHDB-378820

Isolated singularities conjecture for properly embedded minimal surfaces

摘录 / EXCERPT: Let B be a closed geodesic ball in a Riemannian three-manifold, and let M be a minimal surface with boundary $\partial M \subset \partial B$. Suppose that M is properly embedde...

conjecture | open | MathDB 页面 | 原文线索

V2-028651 | MATHDB-378828

The branching-diagram conjecture for Richardson elements of type (b) symplectic parabolics

摘录 / EXCERPT: Let $\mathfrak{sp}_{2n}$ be the symplectic Lie algebra, and consider a parabolic subalgebra of type (b) with block parameters $d=(d_1,\ldots,d_r,d_{r+1},d_r,\ldots,d_1)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028652 | MATHDB-378835

The conformal Ricci-flatness conjecture for one-ended manifolds

摘录 / EXCERPT: Let (M,g) be a complete, noncompact manifold with only one end: outside a compact set it is diffeomorphic to a spherical shell $S^{n-1}\times\mathbb{R}$. Let g_0 denote the Euc...

conjecture | open | MathDB 页面 | 原文线索

V2-028653 | MATHDB-378836

The paraboloid-end quasi-isometry conjecture

摘录 / EXCERPT: Let (M,g) be a complete, noncompact manifold with finitely many ends, each end outside a compact set diffeomorphic to a spherical shell. Paraboloid-end conjecture. Each of the sp...

conjecture | open | MathDB 页面 | 原文线索

V2-028654 | MATHDB-378837

The one-endedness conjecture for positively Ricci-curved manifolds

摘录 / EXCERPT: Let (M,g) be a complete, noncompact manifold with finitely many ends, each end outside a compact set diffeomorphic to a spherical shell. Assume that M has everywhere positive R...

conjecture | open | MathDB 页面 | 原文线索

V2-028655 | MATHDB-378838

The paraboloid rigidity conjecture

摘录 / EXCERPT: Let (M,g) satisfy the hypotheses of the preceding one-endedness conjecture, and suppose that its Ricci curvature agrees with that of a paraboloid for $r\leq r_0$. Paraboloid rigi...

conjecture | open | MathDB 页面 | 原文线索

V2-028656 | MATHDB-378839

The higher-dimensional Ricci compactness conjecture

摘录 / EXCERPT: Let M be complete, let O be a fixed origin, and suppose that for some $r_0>0$ the Ricci curvature is bounded below by $(n-1)/(4r^2)$ whenever $r\geq r_0$. Higher-dimensional Ri...

conjecture | open | MathDB 页面 | 原文线索

V2-028657 | MATHDB-378854

Cubications versus immersions' conjecture

摘录 / EXCERPT: Let M^n be a closed n-manifold. A marked cubical decomposition of M^n is a cubical decomposition together with the marking data used to distinguish its cubical cells, and two...

conjecture | open | MathDB 页面 | 原文线索

V2-028658 | MATHDB-378860

Schoen's Yamabe invariant conjecture for closed 3-manifolds

摘录 / EXCERPT: Let M be a closed 3-manifold that admits a constant-curvature metric h. Here $Y(M)$ denotes the Yamabe invariant of M, and $E(h)$ denotes the normalized Einstein-Hilbert en...

conjecture | open | MathDB 页面 | 原文线索

V2-028659 | MATHDB-378891

Non-trivial realization conjecture for virtual link diagrams with zero intersection pairs

摘录 / EXCERPT: Non-trivial realization conjecture. Then there exists a minimal genus representation resulting in a non-trivial realization.

conjecture | open | MathDB 页面 | 原文线索

V2-028660 | MATHDB-378903

Kawamata's fully faithful derived embedding conjecture for log K-related pairs

摘录 / EXCERPT: Kawamata's fully faithful embedding conjecture. Under these assumptions, there exists a fully faithful exact functor

conjecture | open | MathDB 页面 | 原文线索

V2-028661 | MATHDB-378907

Compact integration criterion for Poisson linearization around symplectic leaves

摘录 / EXCERPT: Poisson linearization conjecture. If the restricted Lie algebroid $T^*_S M$ integrates to a compact, source 1-connected Lie algebroid, then pi is linearizable around S.

conjecture | open | MathDB 页面 | 原文线索

V2-028662 | MATHDB-378908

Non-bounding Milnor manifolds are not bordant to Dold manifolds

摘录 / EXCERPT: Let $H(m,n)$ denote a Milnor manifold and $P(r,s)$ a Dold manifold. The Milnor manifold $H(m,n)$ is non-bounding if it does not bound, and it is among those not considered in Propo...

conjecture | open | MathDB 页面 | 原文线索

V2-028663 | MATHDB-378913

Eastwood-Ezhov's LSA characterization conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a complete abelian left-symmetric algebra (LSA) with a nondegenerate Hessian-type inner product. Let $\operatorname{sDer}(\mathcal A)$ denote its space of infin...

conjecture | open | MathDB 页面 | 原文线索

V2-028664 | MATHDB-378958

Existence of good geometric triangulations for hyperbolic 3-manifolds with geodesic boundary

摘录 / EXCERPT: Good geometric triangulation conjecture. Every hyperbolic N with non-empty geodesic boundary admits a good geometric triangulation.

conjecture | open | MathDB 页面 | 原文线索

V2-028665 | MATHDB-378974

The symplectic Poincaré-type uniqueness conjecture for minimizers

摘录 / EXCERPT: Let G be a group, and let M be a symplectic 4-manifold with $\pi_1(M)\cong G$ that minimizes $\chi:\mathfrak{M}(G)\rightarrow {\mathbb Z}$. Let Q_M denote the intersection fo...

conjecture | open | MathDB 页面 | 原文线索

V2-028666 | MATHDB-378983

Cylinder conjecture for skew branes over non-centrally symmetric hypersurfaces

摘录 / EXCERPT: Let $M^{2n-1}\be2 {\bf R}^{2n}$ be a non-centrally symmetric hypersurface. A skew brane is an $(2n-1)$-dimensional submanifold of ${\bf R}^{2n+1}={\bf R}^{2n}\be2 {\bf R}$; the ver...

conjecture | open | MathDB 页面 | 原文线索

V2-028667 | MATHDB-379002

Birationality criterion for residual linear type monomial ideals

摘录 / EXCERPT: Let $k[\mathbf{x}]=k[x_1,\ldots,x_n]$, and let F be a finite set of $q\geq n$ monomials of the same degree $d\geq 2$. Suppose that the ideal $(F)\subset k[\mathbf{x}]$ is of resi...

conjecture | open | MathDB 页面 | 原文线索

V2-028668 | MATHDB-379008

The nilpotent quotient conjecture for nef Kähler manifolds

摘录 / EXCERPT: Let M be a nef Kähler manifold of dimension n. Suppose there is an epimorphism ... where Γ is a torsion-free nilpotent group of rank at least $2n$. Nilpotent quotient conj...

conjecture | open | MathDB 页面 | 原文线索

V2-028669 | MATHDB-379014

Birational factorization conjecture for elliptic fibrations on weighted Fano threefold hypersurfaces

摘录 / EXCERPT: Let X be the weighted Fano threefold hypersurface considered in the paper, with numerical parameter N, and let a_{2} and a_{3} be the weights defining the weighted projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-028670 | MATHDB-379024

Uniqueness of variational limits under measured and pointed Gromov-Hausdorff convergence

摘录 / EXCERPT: Let $\mathcal{M}$ be the family of source spaces and $\mathcal{Y}$ the family of pointed target spaces. Suppose that a net $M_i\in\mathcal{M}$ converges in the measured Gromov-Haus...

conjecture | open | MathDB 页面 | 原文线索

V2-028671 | MATHDB-379026

The hyperbolic Ehrenpreis conjecture

摘录 / EXCERPT: Let $\epsilon>0$ and let S_1,S_2 be two closed hyperbolic surfaces. A finite-sheeted locally isometric cover of a surface is a finite-sheeted covering surface equipped with the p...

conjecture | open | MathDB 页面 | 原文线索

V2-028672 | MATHDB-379038

Weil-Petersson completeness conjecture for Calabi-Yau threefold moduli

摘录 / EXCERPT: A polarized Calabi-Yau manifold is a pair (X,ω) consisting of a compact algebraic manifold X with zero first Chern class and a Kähler form $\omega\in H^2(X,\mathbb Z)$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-028673 | MATHDB-379043

Maximum colourful simplicial depth conjecture

摘录 / EXCERPT: Maximum colourful simplicial depth conjecture. The maximum colourful simplicial depth of any point in the interior of the core is

conjecture | open | MathDB 页面 | 原文线索

V2-028674 | MATHDB-379061

Cieliebak-Mohnke's presentation conjecture for the normalized Lagrangian capacity in all dimensions

摘录 / EXCERPT: Let $\bar c_L$ be the normalized Lagrangian capacity on $2n$-dimensional ellipsoids, let $P(1/n,\dots,1/n)$ denote the cube with all n radius parameters equal to $1/n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028675 | MATHDB-379062

Resolution conjecture for reduced analytic vector fields in dimension three

摘录 / EXCERPT: Let χ be a reduced analytic vector field on a real analytic manifold M without boundary, and let $U\subset M$ be relatively compact. Write $L_\chi|_U$ for the line field ind...

conjecture | open | MathDB 页面 | 原文线索

V2-028676 | MATHDB-379078

Storm's characterization of geodesic boundary by local volume minimality

摘录 / EXCERPT: Storm's local-minimality conjecture. If M is locally minimal among such manifolds, then its boundary is totally geodesic.

conjecture | open | MathDB 页面 | 原文线索

V2-028677 | MATHDB-379080

Finite-volume scalar-curvature volume inequality

摘录 / EXCERPT: Finite-volume volume conjecture. The volume satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028678 | MATHDB-379089

Walkup-type characterization conjecture for the 3-torus

摘录 / EXCERPT: 3-torus f-vector conjecture. The f-vectors of triangulations of the 3-torus T^3 are characterized by

conjecture | open | MathDB 页面 | 原文线索

V2-028679 | MATHDB-379090

Walkup-type characterization conjecture for the lens space L(3,1)

摘录 / EXCERPT: $L(3,1)$ f-vector conjecture. The f-vectors of triangulations of the lens space $L(3,1)$ are characterized by

conjecture | open | MathDB 页面 | 原文线索

V2-028680 | MATHDB-379093

Realization conjecture for complexes as direct images on projective space

摘录 / EXCERPT: Let A be a ring and let ... be a bounded complex of finitely generated A-modules. Realization conjecture. Every such complex arises as the direct image complex of a family of she...

conjecture | open | MathDB 页面 | 原文线索

V2-028681 | MATHDB-379099

Ehrenpreis conjecture for finite hyperbolic covers

摘录 / EXCERPT: Let Σ and Σ' be closed Riemann surfaces of genus greater than 1, regarded as Riemannian manifolds with metrics of curvature -1. The Ehrenpreis conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-028682 | MATHDB-379105

Quantitative symmetry bound for universal covers

摘录 / EXCERPT: Let M be a closed Riemannian manifold, let $\widetilde{M}$ be its universal cover, and let $\operatorname{Isom}(\widetilde{M})$ be its isometry group containing $\pi_1(M)$. Quant...

conjecture | open | MathDB 页面 | 原文线索

V2-028683 | MATHDB-379111

Minimal isometric immersion conjecture for locally homogeneous or Einstein manifolds

摘录 / EXCERPT: Minimal immersion conjecture. The immersion f must be totally geodesic.

conjecture | open | MathDB 页面 | 原文线索

V2-028684 | MATHDB-379140

The non-transverse monotone conjecture for Grassmannian Schubert problems

摘录 / EXCERPT: Let $\alpha=\{\alpha_1<\dotsb<\alpha_k\}$ with $\alpha_k<n$, and let $(w_1,\dotsc,w_m)$ be a Grassmannian Schubert problem for $\mathbb{F}\ell(\alpha;n)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028685 | MATHDB-379143

The monotone conjecture for arbitrary Schubert problems on flag manifolds

摘录 / EXCERPT: Let $\alpha=\{\alpha_1<\dotsb<\alpha_k\}$ with $\alpha_k<n$, and let $(w_1,\dotsc,w_m)$ be a Schubert problem for $\mathbb{F}\ell(\alpha;n)$. For each permutation w_i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-028686 | MATHDB-379156

The rational density conjecture for iterated line intersections

摘录 / EXCERPT: Let S be a finite set of points in the plane with rational coordinates. Write $\mathcal{T}(S)$ for the set of points that are intersections of two lines, each determined by a pai...

conjecture | open | MathDB 页面 | 原文线索

V2-028687 | MATHDB-379173

Weak BMY Conjecture for symplectic 4-manifolds

摘录 / EXCERPT: Weak BMY Conjecture. For each finitely presented group G,

conjecture | open | MathDB 页面 | 原文线索

V2-028688 | MATHDB-379175

Existence of flips for kawamata log terminal pairs

摘录 / EXCERPT: Existence of flips. The flip of f exists.

conjecture | open | MathDB 页面 | 原文线索

V2-028689 | MATHDB-379192

Qin's punctual-partition generating-function conjecture

摘录 / EXCERPT: Let $r\ge2$, and let $P_r(n)$ and $\widetilde P_r(n)$ denote the two partition-counting functions defined earlier in the paper. Qin's punctual-partition generating-function conject...

conjecture | open | MathDB 页面 | 原文线索

V2-028690 | MATHDB-379221

Torquato-Stillinger conjecture on high-dimensional disordered packing terminal density

摘录 / EXCERPT: Torquato-Stillinger conjecture. The function $g_2({\bf r})$ is a pair correlation function of a translationally invariant disordered sphere packing in $\mathbb{R}^d$ at number dens...

conjecture | open | MathDB 页面 | 原文线索

V2-028691 | MATHDB-379225

Distance-realization property for geodesics in the Heisenberg group

摘录 / EXCERPT: Distance-realization property. Then the distance $\operatorname{dist}(P,S_R)$ is realized by the point

open_problem | open | MathDB 页面 | 原文线索

V2-028692 | MATHDB-379226

Perelman's codimension-one manifold conjecture for non-generic NLS blow-up

摘录 / EXCERPT: Consider the critical focusing nonlinear Schrödinger equation in one spatial dimension and its standing-wave blow-up solutions. The generalized root space of the linearized operato...

conjecture | open | MathDB 页面 | 原文线索

V2-028693 | MATHDB-379241

Viehweg-Zuo's factorization and stability conjecture for Arakelov-equality variations

摘录 / EXCERPT: Factorization and stability conjecture. There exists a unique $i\in\{1,\ldots,s\}$ such that the Higgs field factors through

conjecture | open | MathDB 页面 | 原文线索

V2-028694 | MATHDB-379255

The isotropic line bundle conjecture for Lagrangian fibrations on holomorphic symplectic manifolds

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold of dimension $2g$, and let q_X denote its Beauville-Bogomolov form. A line bundle L on X is called non-trivial if it...

conjecture | open | MathDB 页面 | 原文线索

V2-028695 | MATHDB-379277

The quarter-sphere mixed-boundary spectral conjecture

摘录 / EXCERPT: Quarter-sphere spectral conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-028696 | MATHDB-379278

The CAT(0) conjecture for Coxeter simplex splittings

摘录 / EXCERPT: CAT(0) conjecture. The universal cover of the v-splitting of W determined by S' is $\operatorname{CAT}(0)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028697 | MATHDB-379296

The characterization of maximum complete and empty subgraphs in the quadrance graph

摘录 / EXCERPT: Maximum subgraph conjecture. If U spans a complete subgraph or an empty subgraph of order q in V_q, then U is a line in $F_q\times F_q$.

conjecture | open | MathDB 页面 | 原文线索

V2-028698 | MATHDB-379314

The tropical discriminant is a union of cones in the secondary fan

摘录 / EXCERPT: Tropical discriminant conjecture. For any point configuration A, the tropical discriminant $\tau(X_A^*)$ is a union of cones in the secondary fan $\Sigma(A)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028699 | MATHDB-379322

Friedlander-Nadirashvili invariant conjecture for closed surfaces

摘录 / EXCERPT: Let Σ be a closed surface, and let $I(\Sigma)$ denote its Friedlander-Nadirashvili invariant, defined as the infimum over conformal classes C on Σ of the supremum...

conjecture | open | MathDB 页面 | 原文线索

V2-028700 | MATHDB-379331

Deformation-equivalence conjecture for symplectic resolutions

摘录 / EXCERPT: Deformation-equivalence conjecture. The resolutions satisfy that π_1 is deformation equivalent to π_2, and Z_1 is diffeomorphic to Z_2.

conjecture | solved | MathDB 页面 | 原文线索

V2-028701 | MATHDB-379334

Conjecture on deformation equivalence of symplectic resolutions

摘录 / EXCERPT: Let $X\to Y$ and $X^+\to Y$ be proper morphisms. They are deformation equivalent if there are deformations $\mathcal X\to\mathcal Y$ and $\mathcal X^+\to\mathcal Y$ over a pointed...

conjecture | open | MathDB 页面 | 原文线索

V2-028702 | MATHDB-379336

Sharp lower-bound conjecture for odd-order planar Besicovitch sets

摘录 / EXCERPT: Let q be odd, and write $\ell(m,b)$ and $\ell(\infty,a)$ for the lines $y=mx+b$ and $x=a$ in $\mathbb{F}_q^2$. A Besicovitch set is a set $B\subset\mathbb{F}_q^2$ containing one...

conjecture | open | MathDB 页面 | 原文线索

V2-028703 | MATHDB-379337

Multiplicity-line conjecture for small planar Besicovitch sets

摘录 / EXCERPT: Multiplicity-line conjecture. If B is small, then there is some $j\in\mathbb{F}_q\cup\{\infty\}$ such that every point $P\in B$ with $m_P\geq3$ lies on the line $\ell(j,b_j)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-028704 | MATHDB-379350

The bounded isometry conjecture for symplectomorphism groups

摘录 / EXCERPT: Bounded isometry conjecture. For all (M,ω),

conjecture | open | MathDB 页面 | 原文线索

V2-028705 | MATHDB-379351

The extension conjecture for the Hofer metric on symplectomorphism groups

摘录 / EXCERPT: Extension conjecture. For all symplectic manifolds (M,ω) such that

conjecture | open | MathDB 页面 | 原文线索

V2-028706 | MATHDB-379356

The Tameness Conjecture for train-track generic automorphisms

摘录 / EXCERPT: Tameness Conjecture. The automorphism f can be isotoped so that it preserves the splitting and the induced automorphism g on H_1 is train-track generic and has a tame one-dim...

conjecture | open | MathDB 页面 | 原文线索

V2-028707 | MATHDB-379361

The (C_d) conjecture for almost strictly nef line bundles

摘录 / EXCERPT: Let F be a projective manifold of dimension d with Kodaira dimension $\kappa(F)=0$, and let L be an almost strictly nef line bundle, meaning that L is the pullback of a str...

conjecture | open | MathDB 页面 | 原文线索

V2-028708 | MATHDB-379389

J. Simon's finiteness conjecture for epimorphisms of knot groups

摘录 / EXCERPT: J. Simon's finiteness conjecture. There are only finitely many knots $k_i \in {\mathbb S}^3$ for which such an epimorphism exists.

conjecture | open | MathDB 页面 | 原文线索

V2-028709 | MATHDB-379411

Generalized spiral-point conjecture for Donaldson sections

摘录 / EXCERPT: Generalized spiral-point conjecture. After this normalization, the sections $(p_k\mathbf{w}^k,q_k\mathbf{w}^k)$ satisfy the conditions in Theorem. The construction is supported by...

conjecture | open | MathDB 页面 | 原文线索

V2-028710 | MATHDB-379418

Milnor's continuity and vanishing conjecture for simplex volumes

摘录 / EXCERPT: Milnor's continuity and vanishing conjecture. The volume function V admits a continuous extension to $\overline{\Omega}$. Furthermore, the points on $\partial\Omega$ where V va...

conjecture | open | MathDB 页面 | 原文线索

V2-028711 | MATHDB-379420

The period-map surjectivity conjecture for four-manifolds

摘录 / EXCERPT: Let X be a closed oriented 4-manifold with $b_2^+(X)=1$. The period map sends a Riemannian metric to the self-dual line H^+ in the cone $\mathcal{P}$ of elements of positive...

conjecture | open | MathDB 页面 | 原文线索

V2-028712 | MATHDB-379425

Morita autoequivalence conjecture for Weyl algebras

摘录 / EXCERPT: Let $A_{n,\mathbb C}$ be the Weyl algebra of index n over $\mathbb C$, and let $P_{n,\mathbb C}$ be the Poisson algebra of polynomial functions on the $2n$-dimensional affine sym...

conjecture | open | MathDB 页面 | 原文线索

V2-028713 | MATHDB-379427

Canonical Planck-parameter lift conjecture for polynomial symplectomorphisms

摘录 / EXCERPT: Let R be a finitely generated smooth commutative algebra over $\mathbb Z$, let $P_{n,R}$ be the polynomial Poisson algebra, and let $A^{\hbar}_{n,R[\hbar]}$ be the algebra over...

conjecture | open | MathDB 页面 | 原文线索

V2-028714 | MATHDB-379455

The quadratic Dirac-kernel estimate for two-spheres

摘录 / EXCERPT: Quadratic Dirac-kernel estimate. For every Dirac operator on a two-sphere, the estimate

conjecture | open | MathDB 页面 | 原文线索

V2-028715 | MATHDB-379459

Uniform conformal-volume conjecture for elliptic orbifolds

摘录 / EXCERPT: Let $n\leq 16$, and let $\mathcal{O}$ be an elliptic n-orbifold. Write $V_{c}(\mathcal{O})$ for its conformal volume. Uniform conformal-volume conjecture. There is a function...

conjecture | open | MathDB 页面 | 原文线索

V2-028716 | MATHDB-379481

Adiabatic asymptotic formula for arbitrary foliations

摘录 / EXCERPT: Let ${\mathcal F}$ be an arbitrary foliation on a compact Riemannian manifold. In the above notation, let Δ_F and Δ_H denote the leafwise and transverse Laplacians, r...

conjecture | open | MathDB 页面 | 原文线索

V2-028717 | MATHDB-379518

Global versal-family conjecture for proper rigid analytic spaces

摘录 / EXCERPT: Let X be a proper rigid analytic space over K. A formal versal deformation of X is the formal deformation supplied by the local deformation theory of X. Global versal-famil...

conjecture | open | MathDB 页面 | 原文线索

V2-028718 | MATHDB-379552

Bondal-Orlov conjecture on fully faithful functors for birational maps

摘录 / EXCERPT: Bondal-Orlov conjecture. If K_{X_2/X_1} is effective, then there is a fully faithful functor

conjecture | open | MathDB 页面 | 原文线索

V2-028719 | MATHDB-379639

Rigidity conjecture for holomorphic parabolic geometries

摘录 / EXCERPT: Let G be a complex simple Lie group and let $P\subset G$ be a maximal parabolic subgroup. Assume that G/P is not a compact Hermitian symmetric space, or, if G/P is a compact...

conjecture | open | MathDB 页面 | 原文线索

V2-028720 | MATHDB-379640

Gudmundsson-Svensson conjecture on local and global harmonic morphisms from irreducible symmetric spaces

摘录 / EXCERPT: Let (M^m,g) be an irreducible Riemannian symmetric space of dimension $m\ge 2$. For each point $p\in M$, a complex valued harmonic morphism is a harmonic morphism ... defined on an...

conjecture | open | MathDB 页面 | 原文线索

V2-028721 | MATHDB-379641

Existence conjecture for harmonic morphisms from irreducible Riemannian symmetric spaces

摘录 / EXCERPT: Existence conjecture. There exists a complex-valued harmonic morphism

conjecture | open | MathDB 页面 | 原文线索

V2-028722 | MATHDB-379643

Generalized Calabi-Markus conjecture for positive sectional curvature

摘录 / EXCERPT: Generalized Calabi-Markus conjecture. The fundamental group $\pi_1(M)$ is finite, and M is noncompact.

conjecture | open | MathDB 页面 | 原文线索

V2-028723 | MATHDB-379644

Auslander conjecture for compact flat pseudo-Riemannian manifolds

摘录 / EXCERPT: Auslander conjecture, special case. The fundamental group $\pi_1(M)$ contains a solvable subgroup of finite index.

conjecture | open | MathDB 页面 | 原文线索

V2-028724 | MATHDB-379661

Weak generalized Property R conjecture for 4-dimensional 2-handlebodies

摘录 / EXCERPT: Weak generalized Property R conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-028725 | MATHDB-379670

Minimal layered-triangulation conjecture for the solid torus

摘录 / EXCERPT: Let p/q be a reduced rational number with $0\leq p\leq q$, including the exceptional forms $0/1$ and $1/1$. A p/q-layered-triangulation is a layered-triangulation of the solid...

conjecture | open | MathDB 页面 | 原文线索

V2-028726 | MATHDB-379671

Minimal layered-triangulation conjecture for lens spaces

摘录 / EXCERPT: Let M be a lens space, and call a triangulation minimal if it has the smallest number of tetrahedra among triangulations of M. A minimal layered-triangulation of M is a layer...

conjecture | open | MathDB 页面 | 原文线索

V2-028727 | MATHDB-379673

Conger's Steiner-point degree conjecture for elliptic Minkowski spaces

摘录 / EXCERPT: Let $\mathbb{M}^n$ be a piece-wise differentiable, elliptic Minkowski space, and let $s(\mathbb{M}^n)$ denote the maximum degree of a Steiner point in a Steiner minimal tree in...

conjecture | open | MathDB 页面 | 原文线索

V2-028728 | MATHDB-379674

A sharp two-sided eigenvalue bound for triangles

摘录 / EXCERPT: Triangle eigenvalue bound. The eigenvalue satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028729 | MATHDB-379685

The hierarchy conjecture for k-symmetric Lorentzian manifolds

摘录 / EXCERPT: Let $D\subset \mathcal V$ be a simply connected domain of an n-dimensional k-symmetric Lorentzian manifold $(\mathcal V,g)$. A vector field is null covariantly constant if it i...

conjecture | open | MathDB 页面 | 原文线索

V2-028730 | MATHDB-379686

The hierarchy conjecture for k-recurrent Lorentzian manifolds

摘录 / EXCERPT: Let $D\subset \mathcal V$ be a simply connected domain of an n-dimensional k-recurrent Lorentzian manifold $(\mathcal V,g)$. A vector field is null recurrent if it is null and...

conjecture | open | MathDB 页面 | 原文线索

V2-028731 | MATHDB-379714

The generic injectivity conjecture for the moduli map of self-associated sets in \({\bf P}^7\)

摘录 / EXCERPT: Generic injectivity conjecture. The map σ is generically injective.

conjecture | open | MathDB 页面 | 原文线索

V2-028732 | MATHDB-379721

Conjecture on infinitely many primes preserving the log canonical threshold

摘录 / EXCERPT: Let A be the coefficient ring in the reduction-modulo-p setup, let $\frak{a}\subset A[x_1,\ldots,x_n]$ be an ideal, let c be its log canonical threshold, and let c_p denote...

conjecture | open | MathDB 页面 | 原文线索

V2-028733 | MATHDB-379727

The Liouvillianity conjecture for abelian relations of webs

摘录 / EXCERPT: Let $\mathcal W$ be a web, and let its abelian relations mean the relations among the defining foliations obtained by summing closed one-forms along the leaves. A Liouvillian exten...

conjecture | open | MathDB 页面 | 原文线索

V2-028734 | MATHDB-379736

Weak monodromy nilpotency conjecture for degenerations of irreducible symplectic manifolds

摘录 / EXCERPT: Let $\pi:\mathcal X\to\Delta$ be a degeneration of an irreducible symplectic $2n$-fold. For each m, let T_m be the associated monodromy operator on...

conjecture | open | MathDB 页面 | 原文线索

V2-028735 | MATHDB-379739

Hartl's arcwise connectedness conjecture for the intrinsic period domain

摘录 / EXCERPT: Hartl's connectedness conjecture. The space $\breve{\cal{F}}^0_b$ is arcwise connected.

conjecture | open | MathDB 页面 | 原文线索

V2-028736 | MATHDB-379753

Uniform stable commutator length conjecture for arithmetic hyperbolic manifolds

摘录 / EXCERPT: Uniform stable commutator length conjecture. For each n there is a constant $C(n)>0$ such that, for every arithmetic hyperbolic n-manifold M and every non-parabolic...

conjecture | open | MathDB 页面 | 原文线索

V2-028737 | MATHDB-379779

Unique constant-curvature simultaneity distribution for conformally flat polar ends

摘录 / EXCERPT: Let $(M,\mathcal{C})$ be a Riemann-Lorentz conformal space with polar end D^{∞}. Suppose that D^{∞} is conformal-flat. A simultaneity distribution is a distribution w...

conjecture | open | MathDB 页面 | 原文线索

V2-028738 | MATHDB-379783

The Weeks manifold smallest-volume conjecture

摘录 / EXCERPT: Let M_W be the manifold obtained by $(5,1)$ and $(5,2)$ surgery on the two components of the Whitehead link. The Weeks manifold smallest-volume conjecture. M_W is the unique or...

conjecture | open | MathDB 页面 | 原文线索

V2-028739 | MATHDB-379784

The complete Mom-4 classification conjecture

摘录 / EXCERPT: Let M be a hyperbolic manifold and let $(M,T,Delta)$ be a Mom-2, Mom-3, or Mom-4 for some T and $Delta$. A Mom-4 is strict when it is not a Mom-2 or Mom-3. The comp...

conjecture | open | MathDB 页面 | 原文线索

V2-028740 | MATHDB-379785

The eight multi-cusped Mom-manifolds conjecture

摘录 / EXCERPT: Let M be a hyperbolic manifold with boundary $partial M$, and let $(M,T,Delta)$ be a Mom-n for some integer n with $2\le n\le4$. The eight multi-cusped Mom-manifolds conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-028741 | MATHDB-379791

The volume lower-bound conjecture for 0-symmetric lattice polytopes

摘录 / EXCERPT: Volume lower-bound conjecture. Every $P\in\mathcal{P}_o^n$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-028742 | MATHDB-379800

The projective-space conjecture for compact Kähler manifolds with positive orthogonal bisectional curvature

摘录 / EXCERPT: Let M be an irreducible compact Kähler manifold with positive orthogonal bisectional curvature. Projective-space conjecture. Is M necessarily biholomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-028743 | MATHDB-379804

The log canonical adjunction, effective log abundance, and effective adjunction conjectures

摘录 / EXCERPT: Let $f\colon X\to Z$ be the fibre space and let D, Θ, D_{div}, and D_{mod} be as in the paper's adjunction construction. Assume that D...

conjecture | open | MathDB 页面 | 原文线索

V2-028744 | MATHDB-379810

The proper Dupin hypersurface conjecture in the four-curvature case

摘录 / EXCERPT: Proper Dupin hypersurface conjecture. Every compact embedded proper Dupin hypersurface in S^5 with four distinct principal curvatures is diffeomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-028745 | MATHDB-379812

The high-distance handlebody-gluing conjecture for Heegaard genus

摘录 / EXCERPT: Let X have a single boundary component F with $g(F)\geq 2$, and let $V\cup_H W$ be a minimal-genus Heegaard splitting of X with F contained in the compression body V. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028746 | MATHDB-379821

Nonpositive-curvature obstruction conjecture for the moduli orbifold

摘录 / EXCERPT: For $g\geq 2$, let $\M_g$ be the moduli orbifold of genus-g Riemann surfaces. Nonpositive-curvature conjecture. The orbifold $\M_g$ admits no complete, finite-volume Riemannian m...

conjecture | open | MathDB 页面 | 原文线索

V2-028747 | MATHDB-379822

Tangent-cone conjecture for moduli space

摘录 / EXCERPT: Let $\M_g$ be moduli space with the Teichmüller metric, and let ${\cal C}_g$ be the curve complex of a closed genus-g surface: its vertices are nontrivial, nonperipheral isotopy...

conjecture | open | MathDB 页面 | 原文线索

V2-028748 | MATHDB-379828

Formality conjecture for Hochschild complexes of orbifolds

摘录 / EXCERPT: Formality conjecture. The Hochschild complex of the algebra $C^\infty(M)\rtimes G$ is a formal differential graded Lie algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-028749 | MATHDB-379840

Douglas-Reinbacher-Yau strengthened Bogomolov inequality

摘录 / EXCERPT: Douglas-Reinbacher-Yau strengthened Bogomolov conjecture. The Chern classes of E satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-028750 | MATHDB-379871

Twistor correspondence conjecture for singular self-dual Zollfrei conformal structures

摘录 / EXCERPT: Twistor correspondence conjecture. There is a natural one-to-one correspondence between equivalence classes of these singular self-dual Zollfrei conformal structures and equivalenc...

conjecture | open | MathDB 页面 | 原文线索

V2-028751 | MATHDB-379894

Non-log-exp conjecture for non-integrable SR Martinet spheres

摘录 / EXCERPT: Let β be the parameter appearing in the non-integrable SR Martinet problem, and let an SR Martinet sphere denote the corresponding sub-Riemannian sphere. A function is pfaffi...

conjecture | open | MathDB 页面 | 原文线索

V2-028752 | MATHDB-379896

Cut-locus stratification and branching conjecture in Martinet geometry

摘录 / EXCERPT: In the Martinet case, the cut-locus is the set of endpoints where minimizing geodesics cease to be minimizing; in the generic case, the northern hemisphere and the end-point of the...

conjecture | open | MathDB 页面 | 原文线索

V2-028753 | MATHDB-379897

Universal optimality of the hexagonal, E8, and Leech lattices among periodic configurations

摘录 / EXCERPT: Let Λ_2, Λ_8, and Λ_{24} be the hexagonal lattice in $\mathbb{R}^2$, the E_8 root lattice in $\mathbb{R}^8$, and the Leech lattice in $\mathbb{R}^{24}$, r...

conjecture | open | MathDB 页面 | 原文线索

V2-028754 | MATHDB-379920

The converse to Karpenko's theorem on ruled quadrics

摘录 / EXCERPT: Let q be an anisotropic quadratic form over a field k. Its first Witt index is the Witt index of q over the function field of its associated projective quadric. Converse to K...

conjecture | open | MathDB 页面 | 原文线索

V2-028755 | MATHDB-379942

Conjecture on dilating positive-weight actions on transversal symplectic slices

摘录 / EXCERPT: Let Y be a symplectic singularity, let $y\in Y$ be a closed point, and let $Y_{y,0}$ be the transversal symplectic slice in the formal product decomposition ... A $\mathbb{G}_m$-ac...

conjecture | open | MathDB 页面 | 原文线索

V2-028756 | MATHDB-379943

Kashiwara's D-affinity conjecture for formal quantizations

摘录 / EXCERPT: Let $\pi:{\mathfrak X}\to S[[h]]$ and the quantization sheaf $\operatorname{\cal O}_h$ satisfy the assumptions of Theorem. Consider the global sections functor ... from sheaves of fi...

conjecture | open | MathDB 页面 | 原文线索

V2-028757 | MATHDB-379947

The genus-zero classification conjecture for N=2 super-Riemann surfaces

摘录 / EXCERPT: An $N=2$ super-Riemann surface with genus-zero compact body is an $N=2$ super-Riemann surface whose underlying compact body has genus zero. The $N=2$ super-Riemann sphere is denote...

conjecture | open | MathDB 页面 | 原文线索

V2-028758 | MATHDB-379958

Tapp's conjecture on inverse-linear paths of nonnegatively curved metrics

摘录 / EXCERPT: Let G be a compact Lie group with a bi-invariant metric h_0 and Lie algebra $\mathfrak g$. For a left-invariant metric h on G, let Φ be the unique h_0-self-adjoint p...

conjecture | open | MathDB 页面 | 原文线索

V2-028759 | MATHDB-379959

The inverse-linear conjecture for nonnegatively curved left-invariant metrics

摘录 / EXCERPT: Inverse-linear conjecture. If h has nonnegative curvature, then the inverse-linear path from h_0 to $h_1=h$ consists entirely of metrics with nonnegative curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-028760 | MATHDB-379968

Eventual equality of the Thurston and higher-order Alexander norms

摘录 / EXCERPT: Let M be a 3-manifold, let $\pi_1(M)_{r}^{(n)}$ denote the nth term of its rational derived series, and define ... For each $n\in\mathbb{N}$, let $||\,\cdot\,||_n$ be the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-028761 | MATHDB-379972

The infinite-value conjecture for Perelman's invariant on a homeomorphism type

摘录 / EXCERPT: Infinite-value conjecture. There is a suitable homeomorphism type on which Perelman's invariant takes on infinitely many distinct values.

open_problem | open | MathDB 页面 | 原文线索

V2-028762 | MATHDB-379992

Weil's global Torelli conjectures for K3 surfaces

摘录 / EXCERPT: Let M be a fixed differentiable manifold underlying a K3 surface, and let ... and $'$. Write $[]$ and $[']$ for their periods in ${\mathbb P}(H^2(M,{\mathbb C}))$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028763 | MATHDB-380036

Sabbah's conjecture on good formal structures for surface connections

摘录 / EXCERPT: Sabbah's conjecture. There is a finite sequence of point blow-ups

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V2-028764 | MATHDB-380042

ACC conjecture for singularity-type sets

摘录 / EXCERPT: Let $a\geq0$ and d be a nonnegative integer. The sets $\mathcal S_d\text{(global)}$, $\mathcal S_d^0\text{(global)}$, $\mathcal S_{a,d}\text{(local)}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028765 | MATHDB-380045

Realization conjecture for colored regular graphs

摘录 / EXCERPT: Let (Γ,α) be a finite connected regular $(n+1)$-valent graph whose edges are colored by $(\mathbb Z_2)^{n+1}$, with α a “good” coloring. A moment graph is the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-028766 | MATHDB-380071

Necessary local normal form conjecture for conformally flat transverse Riemann-Lorentz manifolds

摘录 / EXCERPT: Let $(M,\mathcal{C})$ be a transverse Riemann-Lorentz conformal manifold of dimension $m\geq 4$, with singular set Σ. Around each singular point $p\in\Sigma$, consider a coo...

conjecture | open | MathDB 页面 | 原文线索

V2-028767 | MATHDB-380086

Filling-radius comparison with the round sphere

摘录 / EXCERPT: Let (M^n,g) be a complete Riemannian manifold, and let (S^n,g_0) be a round sphere. Choose the radius of the round sphere so that the filling radius of (M^n,g) equals that of...

conjecture | open | MathDB 页面 | 原文线索

V2-028768 | MATHDB-380087

The all-radii hyperbolic volume comparison conjecture

摘录 / EXCERPT: Let (M^n,hyp) be a closed hyperbolic manifold, and let g be another metric on M with $Vol(M,g)<Vol(M,hyp)$. Hyperbolic volume comparison conjecture. ... for all R. This stren...

conjecture | open | MathDB 页面 | 原文线索

V2-028769 | MATHDB-380135

Rivin-type realization conjecture for non-contractible angled blocks

摘录 / EXCERPT: Let $\pp$ be a compact, oriented, irreducible, atoroidal 3-manifold with boundary equipped with an angled block structure, and let $\tilde{\pp}$ denote its universal cover. The a...

conjecture | open | MathDB 页面 | 原文线索

V2-028770 | MATHDB-380146

Mahler's lower-bound conjecture for centrally symmetric convex bodies

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^n$ be a centrally symmetric convex body, and let K° be its polar body. Define the Mahler volume by ... Let $C_n=[-1,1]^n$ be the cube. Mahler's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-028771 | MATHDB-380147

The ellipsoidal maximization conjecture for the polar second moment

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^n$ be centrally symmetric, let K° be its polar body, and let $(\vec{x},\vec{y})$ be uniformly distributed on $K\times K^\circ$. Polar second-moment...

conjecture | open | MathDB 页面 | 原文线索

V2-028772 | MATHDB-380153

The extension conjecture for intersection forms of smooth four-manifolds with infinite cyclic fundamental group

摘录 / EXCERPT: Extension conjecture. The intersection form on π_2M is extended from the integers.

conjecture | open | MathDB 页面 | 原文线索

V2-028773 | MATHDB-380176

Existence of overtwisted contact structures on odd-dimensional spheres

摘录 / EXCERPT: Let S^{2n+1} denote the $(2n+1)$-dimensional sphere, and let a contact manifold mean a manifold equipped with a contact structure. Sphere overtwistedness conjecture. The sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-028774 | MATHDB-380178

Extrinsic Berger conjecture for complex submanifolds of projective space

摘录 / EXCERPT: Let $M\hookrightarrow \mathbb{C} P^N$ be a complete, connected, and full complex submanifold, meaning that it is not contained in a proper hyperplane. Suppose its normal holonomy g...

conjecture | open | MathDB 页面 | 原文线索

V2-028775 | MATHDB-380182

The base-subset collineation conjecture for polar Grassmannians

摘录 / EXCERPT: Let ${\mathcal G}_k$ be the Grassmannian of k-dimensional singular subspaces of a building of type $\textsf{X}_{n}=\textsf{C}_{n},\textsf{D}_{n}$, and let ${\mathfrak G}_k$ be it...

conjecture | open | MathDB 页面 | 原文线索

V2-028776 | MATHDB-380209

Handlebody basis conjecture for the three-variable skein module

摘录 / EXCERPT: Handlebody basis conjecture. For every handlebody H_n, ${\mathcal S}_3(H_n)$ is the free $Z[x_0^{\mp 1},x_1^{\mp 1},x_2^{\mp 1}]$-module with basis

conjecture | open | MathDB 页面 | 原文线索

V2-028777 | MATHDB-380219

Meeks-Pérez-Ros genus-zero conjecture for properly embedded minimal surfaces

摘录 / EXCERPT: Genus-zero conjecture. M is a Riemann minimal example.

conjecture | open | MathDB 页面 | 原文线索

V2-028778 | MATHDB-380220

Lee-Naor nonembedding conjecture for the Heisenberg group

摘录 / EXCERPT: Let $(\mathbb H,d^{\mathbb H})$ be the Heisenberg group equipped with its Carnot-Carathéodory metric. Lee-Naor conjecture. The metric space $(\mathbb H,d^{\mathbb H})$ does not adm...

conjecture | open | MathDB 页面 | 原文线索

V2-028779 | MATHDB-380229

Symplectic replacement conjecture for friendly Calabi-Yau algebras

摘录 / EXCERPT: Let (D,ω,ξ) be the symplectic data used in the universal construction, and let A be a friendly algebra satisfying the stated Calabi-Yau condition. Theorem main(ii) repre...

conjecture | open | MathDB 页面 | 原文线索

V2-028780 | MATHDB-380230

Presentation conjecture for friendly Calabi-Yau algebras of dimension three

摘录 / EXCERPT: Let R be the base algebra, let A be a friendly Calabi-Yau R-algebra of dimension 3, let F be a smooth R-algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028781 | MATHDB-380231

Surface-group algebra presentation conjecture via the noncommutative Weil representation

摘录 / EXCERPT: Let C be a genus-g Heegaard surface with fundamental group $\pi_1(C)$, let X be the associated three-manifold, and let L_+ be one of the two lattices in the Heegaard splitt...

conjecture | open | MathDB 页面 | 原文线索

V2-028782 | MATHDB-380233

The B-model partition-function conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: The B-model of a Calabi-Yau threefold has a partition function associated with its monodromy action. B-model partition-function conjecture. The analogue of Theorem holds for the B-...

conjecture | open | MathDB 页面 | 原文线索

V2-028783 | MATHDB-380237

The ample-cone conjecture for moduli spaces of stable curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable n-pointed curves of genus g, let $NE_1(\overline{\mathcal{M}}_{g,n})$ denote its Mori cone of curves, and let a...

conjecture | open | MathDB 页面 | 原文线索

V2-028784 | MATHDB-380242

Arithmetic GFERF conjecture for quadratic-form hyperbolic manifolds

摘录 / EXCERPT: A hyperbolic manifold is arithmetic when its fundamental group is an arithmetic lattice. A subgroup is geometrically finite if its convex core has finite volume, and a lattice is G...

conjecture | open | MathDB 页面 | 原文线索

V2-028785 | MATHDB-380265

The conjecture that exotic tori admit no symplectic structures

摘录 / EXCERPT: Exotic tori carry no symplectic structures. There are no symplectic structures on exotic tori.

conjecture | open | MathDB 页面 | 原文线索

V2-028786 | MATHDB-380266

Turbiner's non-separability conjecture for flat Schrödinger operators

摘录 / EXCERPT: A quasi-exactly-solvable or exactly-solvable problem in $\mathbb{R}^2$ is one containing the Laplace-Beltrami operator with a flat-space metric tensor; its variables are non-separa...

conjecture | open | MathDB 页面 | 原文线索

V2-028787 | MATHDB-380267

Turbiner's Euclidean-geometry separation conjecture

摘录 / EXCERPT: Turbiner's second conjecture. If the symbol of H engenders a Euclidean geometry, then the spectral equation $H\psi=E\psi$ can be solved by separation of variables. The paper stat...

conjecture | open | MathDB 页面 | 原文线索

V2-028788 | MATHDB-380269

Gray and Wolf's conjecture on homogeneous nearly Kähler manifolds

摘录 / EXCERPT: Gray and Wolf's conjecture. Every nearly Kähler homogeneous manifold is a 3-symmetric space equipped with its canonical almost complex structure.

conjecture | open | MathDB 页面 | 原文线索

V2-028789 | MATHDB-380270

The compact nearly Kähler 3-symmetric-space conjecture

摘录 / EXCERPT: Compact nearly Kähler conjecture. Every compact nearly Kähler manifold is a 3-symmetric space.

conjecture | open | MathDB 页面 | 原文线索

V2-028790 | MATHDB-380298

The obstruction to achiral Lefschetz fibrations on closed four-manifolds

摘录 / EXCERPT: BLF achirality conjecture. Not all closed, smooth, oriented 4-manifolds admit BLFs; in particular, it is possible that $\mathbb{C}P^2$ is necessarily achiral as a fibration, even...

conjecture | open | MathDB 页面 | 原文线索

V2-028791 | MATHDB-380340

GMSY's Fano intersection-number conjecture

摘录 / EXCERPT: Let V be a smooth Fano manifold of complex dimension $n-1$, and let $I(V)\in\mathbb{N}$ be its Fano index. GMSY's Fano inequality conjecture. ... with equality if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-028792 | MATHDB-380341

Yau's Kähler-Einstein stability conjecture for Fano manifolds

摘录 / EXCERPT: Let V be a Fano manifold, and let stable mean stable in the sense of Geometric Invariant Theory. Yau's conjecture. The manifold V admits a Kähler-Einstein metric if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-028793 | MATHDB-380368

Schmutz-Schaller's multiplicity conjecture for the simple length spectrum

摘录 / EXCERPT: A once-punctured torus is a hyperbolic surface whose simple closed geodesics have a length spectrum, with multiplicity counting how many distinct simple closed geodesics have a giv...

conjecture | open | MathDB 页面 | 原文线索

V2-028794 | MATHDB-380369

Markoff's isometry conjecture for the modular torus

摘录 / EXCERPT: Let $\mathbf{M}$ be the modular torus, the hyperbolic once-punctured torus described in the paper. A pair of simple closed geodesics is a pair of simple geodesics on $\mathbf{M}$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-028795 | MATHDB-380437

The Tango-curve characterization of Kawamata-Viehweg counterexamples on surfaces

摘录 / EXCERPT: Tango-curve characterization. If there is a counterexample to the Kawamata-Viehweg vanishing theorem on X, then there exists a dominant rational map

conjecture | open | MathDB 页面 | 原文线索

V2-028796 | MATHDB-380438

The ruled-surface characterization of Kawamata-Viehweg counterexamples

摘录 / EXCERPT: Ruled-surface characterization. There exists a counterexample to the Kawamata-Viehweg vanishing theorem on some geometrically ruled surface over C if and only if C is a Tango c...

conjecture | open | MathDB 页面 | 原文线索

V2-028797 | MATHDB-380439

The integral-Tango characterization of Kodaira counterexamples

摘录 / EXCERPT: Integral-Tango characterization. If there is a counterexample to the Kodaira vanishing theorem on Y, then there exists a fibration

conjecture | open | MathDB 页面 | 原文线索

V2-028798 | MATHDB-380440

Real-root and sign-distribution conjecture for characteristic polynomials in hyperbolic space

摘录 / EXCERPT: Characteristic-polynomial conjecture. All roots of $f(x)$ are real numbers. If $m=n+2$, the points are in general position in H^n, and exactly s of the α_i are positive,...

conjecture | open | MathDB 页面 | 原文线索

V2-028799 | MATHDB-380445

Hexagon polyomino convex-hull area bound

摘录 / EXCERPT: Let n congruent regular unit hexagons form an edge-to-edge connected system. Hexagon polyomino area bound. The area of its convex hull is at most ... The statement is presented in...

open_problem | open | MathDB 页面 | 原文线索

V2-028800 | MATHDB-380477

Steiner polynomial root-location conjecture

摘录 / EXCERPT: Let $K,E\in\mathcal{K}^n$ be convex bodies, and let ... be their relative Steiner polynomial. Let $a_1\leq\dots\leq a_n$ be the real parts of the roots of $f(K,E,s)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028801 | MATHDB-380495

The isomorphism-isometry conjecture for symplectic quandles over \mathbb{Z}_n

摘录 / EXCERPT: Let Q and Q' be symplectic quandles of the same dimension over the ring $\mathbb{Z}_n$. They are isomorphism-isometry conjecture. Isomorphic as quandles if and only if they are...

conjecture | open | MathDB 页面 | 原文线索

V2-028802 | MATHDB-380497

Conjecture on Stein universal covers and deformation invariance for solvmanifolds

摘录 / EXCERPT: Solvmanifold deformation conjecture. (i) Every left-invariant complex structure on G is Stein, respectively biholomorphic to ${\bf C}^n$ in the nilpotent case. (ii) Every small d...

conjecture | open | MathDB 页面 | 原文线索

V2-028803 | MATHDB-380500

Extension conjecture for neutral fibre functors on hyperbolic curves

摘录 / EXCERPT: Extension conjecture. Every $\rho\in{\sf Fib}_k(U)$ extends to a functor in ${\sf Fib}_k(V)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028804 | MATHDB-380506

Fukaya's gradient-line and pseudoholomorphic-disc correspondence near a caustic

摘录 / EXCERPT: Let a point of the caustic be fixed, and consider the moduli space of gradient lines and the moduli space of pseudoholomorphic discs associated with the Lagrangian fibration near t...

conjecture | open | MathDB 页面 | 原文线索

V2-028805 | MATHDB-380516

Properness conjecture for automorphism groups of strictly pseudoconvex CR manifolds

摘录 / EXCERPT: Properness conjecture. The action of $\scr A(M)$ on M should be proper unless M is CR equivalent to the sphere or to the Heisenberg group with its standard CR structure.

conjecture | open | MathDB 页面 | 原文线索

V2-028806 | MATHDB-380530

Complete saturation and reduction existence conjecture for bodies

摘录 / EXCERPT: Complete saturation and reduction conjecture. Every body K in $\mathbb{E}^d$ (respectively, in $\mathbb{H}^d$) admits a completely saturated packing and a completely reduced cove...

conjecture | open | MathDB 页面 | 原文线索

V2-028807 | MATHDB-380539

Hyperbolization Conjecture for closed irreducible 3-manifolds

摘录 / EXCERPT: Let M be a closed, connected, ${\mathbf{P}^2}$-irreducible 3-manifold with infinite fundamental group. Hyperbolization Conjecture. If $\pi_1(M)$ contains no subgroup isomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-028808 | MATHDB-380540

Closed Virtual Bundle Conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a closed, connected, ${\mathbf{P}^2}$-irreducible 3-manifold with infinite fundamental group. Closed Virtual Bundle Conjecture. If M is hyperbolic, then some finite-...

conjecture | open | MathDB 页面 | 原文线索

V2-028809 | MATHDB-380543

Bounded Virtual Bundle Conjecture for finite-volume hyperbolic 3-manifolds

摘录 / EXCERPT: Let N be a compact, connected, ${\mathbf{P}^2}$-irreducible 3-manifold whose boundary is a non-empty collection of incompressible tori and Klein bottles. Assume that every subg...

conjecture | open | MathDB 页面 | 原文线索

V2-028810 | MATHDB-380547

The irrationality conjecture for Chern-Simons invariants

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with invariant trace field $k=k(M)$, and let $\overline{k}$ denote its complex conjugate. The Chern-Simons invariant $CS(M)$ lies in...

conjecture | open | MathDB 页面 | 原文线索

V2-028811 | MATHDB-380550

Dehn invariant sufficiency conjecture for hyperbolic and spherical 3-space

摘录 / EXCERPT: Let $\operatorname{\mathcal P}(\mathbb X)$ be the scissors congruence group of polytopes in a 3-dimensional geometry $\mathbb X$, and let ... be the kernel of the Dehn invariant. Deh...

conjecture | open | MathDB 页面 | 原文线索

V2-028812 | MATHDB-380559

General Dirac-potential integral conjecture for spheres

摘录 / EXCERPT: Let $\mathcal E$ be the spinor bundle over a sphere, let $\mathcal D$ be a Dirac operator on $\mathcal E$, and write $U(z,\bar z)$ for its potential in a Weierstrass representation...

conjecture | open | MathDB 页面 | 原文线索

V2-028813 | MATHDB-380576

Uniqueness of the scalar-curvature sign for Einstein metrics on closed smooth 4-manifolds

摘录 / EXCERPT: Let M be a closed smooth 4-manifold. Uniqueness conjecture. M admits Einstein metrics for at most one sign of the scalar curvature. The conjecture proposes that, for a fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-028814 | MATHDB-380580

Barannikov-Kontsevich Frobenius submanifold conjecture for Landau-Ginzburg orbifolds

摘录 / EXCERPT: Barannikov-Kontsevich conjecture. The Frobenius structure $(S',\eta_F|_{S'},\Phi_F|_{S'})$ should be a Frobenius submanifold of the Frobenius manifold constructed by Barannikov-Kon...

conjecture | open | MathDB 页面 | 原文线索

V2-028815 | MATHDB-380581

Toric Fano Landau-Ginzburg mirror conjecture

摘录 / EXCERPT: Toric Fano mirror conjecture. There should exist a primitive form for f given by

conjecture | open | MathDB 页面 | 原文线索

V2-028816 | MATHDB-380602

GPSS conjecture for geometric progress singularity-series

摘录 / EXCERPT: GPSS conjecture. All such cyclic Gorenstein quotient singularities admit torus-equivariant projective, crepant, full resolutions. The theorem immediately preceding the conjecture p...

conjecture | open | MathDB 页面 | 原文线索

V2-028817 | MATHDB-380609

Uniqueness conjecture for the parameter-dependent canonical trace

摘录 / EXCERPT: Let M be a compact manifold and let $\Gamma\subset\mathbb{R}^p$ be a connected cone with nonempty interior and $H^1_{\rm dR}(\Gamma)=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028818 | MATHDB-380629

Existence criterion for codimension-two monadic complexes on projective space

摘录 / EXCERPT: Let $k\geq 2$, and consider the complex ... Assume that β is surjective and that α degenerates in codimension 2; equivalently, the cokernel in degree 0 is...

conjecture | open | MathDB 页面 | 原文线索

V2-028819 | MATHDB-380630

The generic-metric bound on intersecting mediatrices

摘录 / EXCERPT: Let $(X,\tilde g)$ be the universal cover of a d-dimensional smooth Riemannian manifold (M,g). A mediatrix is the locus equidistant from the reference point and one of its nont...

conjecture | open | MathDB 页面 | 原文线索

V2-028820 | MATHDB-380643

Conformal invariance of higher DSII integrals for surfaces

摘录 / EXCERPT: Let the DSII hierarchy be the integrable hierarchy governing deformations of surfaces immersed in $\mathbb{R}^4$, $\mathbb{R}^{3,1}$, and $\mathbb{R}^{2,2}$ via generalized Weierst...

conjecture | open | MathDB 页面 | 原文线索

V2-028821 | MATHDB-380644

Intrinsic characterization conjecture for codimension-one embeddings of nonnegatively curved spheres

摘录 / EXCERPT: Intrinsic characterization conjecture. $(\mathbb S^n,g)$ admits a smooth global isometric embedding

conjecture | open | MathDB 页面 | 原文线索

V2-028822 | MATHDB-380645

Moishezon's conjecture on Moishezon 4-manifolds and symplectic 4-manifolds

摘录 / EXCERPT: A Moishezon 4-manifold is the smooth 4-manifold constructed from a semi-algebraic cuspidal curve in $\mathbb P^2$ and an epimorphism to a symmetric group, with a smooth map to...

conjecture | open | MathDB 页面 | 原文线索

V2-028823 | MATHDB-380650

Conjecture on intersection lattices of symplectic fillings

摘录 / EXCERPT: Intersection-lattice conjecture. The conclusion of Corollary still holds, under the same assumptions, if X is allowed to range over the set of all symplectic fillings of Y.

conjecture | open | MathDB 页面 | 原文线索

V2-028824 | MATHDB-380651

Kotschick-Morgan conjecture on the structure of Donaldson wallcrossing terms

摘录 / EXCERPT: Let S be a rational algebraic surface, let C be the first Chern class of a differentiable complex vector bundle on S, and let e be the corresponding Donaldson-invariant gra...

conjecture | open | MathDB 页面 | 原文线索

V2-028825 | MATHDB-380660

The noncommutative residue formula for differential operators on line bundles

摘录 / EXCERPT: Let M be a compact complex manifold, let $ obreak\lambda$ be a line bundle on M, and let $\boldsymbol1_\lambda$ denote its identity differential operator. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-028826 | MATHDB-380681

The density conjecture for Kleinian groups

摘录 / EXCERPT: Let G be a finitely generated group. Let $\operatorname{AH}(G)$ denote the space of algebraic limits of discrete faithful representations and let $\operatorname{MP}(G)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-028827 | MATHDB-380682

Jørgensen's algebraic-geometric limits conjecture

摘录 / EXCERPT: Let Γ be a finitely generated, torsion-free, non-elementary Kleinian group, and let $\{\rho_n\}$ be a sequence in $\mathcal D(\Gamma)$ converging to a representation ρ....

conjecture | open | MathDB 页面 | 原文线索

V2-028828 | MATHDB-380683

McMullen's injectivity-radius conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let G be a finitely generated group with g generators. For a hyperbolic 3-manifold N, write $\operatorname{inj}_N(x)$ for the injectivity radius at x, and let the convex...

conjecture | open | MathDB 页面 | 原文线索

V2-028829 | MATHDB-380687

The handle-number bound for components of maximal surface decompositions

摘录 / EXCERPT: Let M be a closed, irreducible 3-manifold and let $\mathcal F$ be a maximal embedded collection of orientable, incompressible surfaces. If $M-N(\mathcal F)$ has n components,...

conjecture | open | MathDB 页面 | 原文线索

V2-028830 | MATHDB-380691

Kuperberg's bottleneck conjecture

摘录 / EXCERPT: Bottleneck conjecture. For convex bodies K in n dimensions with n fixed, the volume $D(K)$ is uniquely minimized when K is an ellipsoid.

conjecture | open | MathDB 页面 | 原文线索

V2-028831 | MATHDB-380693

Indefinite energy minimization conjecture

摘录 / EXCERPT: Indefinite energy conjecture. The quantity $Q(\operatorname{\stackrel{\longrightarrow}{Vol}}N)$ is uniquely minimized when N is the intersection of Q with an a-plane in V c...

conjecture | open | MathDB 页面 | 原文线索

V2-028832 | MATHDB-380694

Hamilton's Harnack-estimate conjecture for arbitrary initial metrics

摘录 / EXCERPT: Let $(M,g(t))$ be a solution to the Ricci flow on a compact 3-manifold with arbitrary initial metric. Hamilton's Harnack-estimate conjecture. A Harnack-type estimate holds for th...

conjecture | open | MathDB 页面 | 原文线索

V2-028833 | MATHDB-380710

Symplecticity conjecture for transverse hyperplane sections of nearly Kähler embeddings

摘录 / EXCERPT: Symplecticity conjecture. For sufficiently large k, the transverse hyperplane sections of F_k are symplectic submanifolds of X.

conjecture | open | MathDB 页面 | 原文线索

V2-028834 | MATHDB-380713

Vanishing conjecture for PU(2) monopole link intersections

摘录 / EXCERPT: Let X be a closed, oriented, smooth four-manifold with $b_2^+(X)>0$. Let ${\mathfrak{s}}={\mathfrak{s}}_0\otimes L_1$ be a Spin^c structure defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-028835 | MATHDB-380747

The birational base change conjecture for discriminant divisors

摘录 / EXCERPT: The Base Change Conjecture. Then K_Y+B_Y is $\mathbb{Q}$-Cartier and, if $f':(X',B^{X'})\to Y'$ is induced by a birational base change $\sigma:Y'\to Y$, then

conjecture | open | MathDB 页面 | 原文线索

V2-028836 | MATHDB-380768

Włodarczyk's weak factorization conjecture

摘录 / EXCERPT: Weak factorization conjecture. Every such birational map admits this factorization, with the stated preservation of the open set and normal-crossings boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-028837 | MATHDB-380773

Strong factorization conjecture for birational maps

摘录 / EXCERPT: Strong factorization conjecture. There exists a diagram

conjecture | open | MathDB 页面 | 原文线索

V2-028838 | MATHDB-380794

Shokurov's codimension bound conjecture for minimal log discrepancies

摘录 / EXCERPT: Shokurov's codimension bound conjecture. The following inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-028839 | MATHDB-380799

Volume bound for guts of hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a hyperbolic manifold containing an incompressible surface S. For a manifold N with acylindrical boundary, let $Vol(N)$ denote the volume of the components of N th...

conjecture | open | MathDB 页面 | 原文线索

V2-028840 | MATHDB-380810

McMullen's convex-core embedded-ball conjecture

摘录 / EXCERPT: McMullen's conjecture. The convex core $C(N)$ does not contain an embedded ball of radius L, where L depends on the number of generators of $\pi_1(N)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028841 | MATHDB-380825

The knot-surgery diffeomorphism conjecture for the K3 surface

摘录 / EXCERPT: Let $E(2)$ be the elliptic K3 surface, and let X_{K_1} and X_{K_2} denote the manifolds obtained from $E(2)$ by knot surgery using knots K_1 and K_2, respectively. Knot-sur...

conjecture | open | MathDB 页面 | 原文线索

V2-028842 | MATHDB-380828

Conway-Sloane fiber conjecture for dense sphere packings

摘录 / EXCERPT: Let n satisfy $2\leq n\leq 8$, and let 2^k be the largest power of 2 strictly less than n. A sphere packing is weakly recurrent and dense when it has the corresponding recu...

conjecture | open | MathDB 页面 | 原文线索

V2-028843 | MATHDB-380832

The analytic approximation conjecture for Kronecker webs

摘录 / EXCERPT: Analytic approximation conjecture. There is an open subset $U\subset U$ containing 0 and a real-analytic Kronecker web $\cal W'$ on U' whose k-jet at 0 agrees with the k...

conjecture | open | MathDB 页面 | 原文线索

V2-028844 | MATHDB-380833

The extension conjecture for Kronecker jet-webs

摘录 / EXCERPT: Extension conjecture. There are an open neighborhood $U\subset\Bbb C^n$ of 0 and a Kronecker web $\cal W'$ on U whose k-jet at 0 agrees with $\cal W$.

conjecture | open | MathDB 页面 | 原文线索

V2-028845 | MATHDB-380840

Boundedness of B-canonical representations

摘录 / EXCERPT: Boundedness of B-canonical representations. There exist positive integers B_l and B'_l such that

conjecture | open | MathDB 页面 | 原文线索

V2-028846 | MATHDB-380867

Weak Besse conjecture on rigidly scalar-flat manifolds

摘录 / EXCERPT: Let M be a compact, locally irreducible Riemannian manifold that is rigidly scalar-flat, meaning that it admits a scalar-flat metric but no metric of positive scalar curvature. I...

conjecture | open | MathDB 页面 | 原文线索

V2-028847 | MATHDB-380868

The Cayley characterization conjecture for GKZ-rational configurations

摘录 / EXCERPT: Let $A_0,A_1,\dots,A_r$ be vector configurations in $\mathbb{Z}^r$, and let A be their Cayley configuration ... where $e_0,\dots,e_r$ is the standard basis of $\mathbb{Z}^{r+1}$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-028848 | MATHDB-380869

The boundary-format hyperdeterminant conjecture

摘录 / EXCERPT: Let Δ_k denote the k-simplex, and let ... where $k_0\geq\cdots\geq k_r$. The configuration A is GKZ-rational when its associated GKZ hypergeometric system has the rational...

conjecture | open | MathDB 页面 | 原文线索

V2-028849 | MATHDB-380870

The iterated-derivative toric residue conjecture

摘录 / EXCERPT: Let A be a configuration, let f be a rational A-hypergeometric function in variables $x_1,\dots,x_s$, and let a facial subset be a subset of A arising as a face of the conf...

conjecture | open | MathDB 页面 | 原文线索

V2-028850 | MATHDB-380871

Aspinwall-Morrison conjecture on non-birational fibres of a Calabi-Yau family

摘录 / EXCERPT: Let $\mathcal{Y}\to B$ be the one-parameter family of smooth, multiply connected Calabi-Yau threefolds over ... where ξ is a primitive fifth root of unity, and write Y_t for th...

conjecture | open | MathDB 页面 | 原文线索

V2-028851 | MATHDB-380878

The generalized Ten Martini problem for hyperbolic magnetic operators

摘录 / EXCERPT: Let Γ be a Fuchsian group with fundamental class $[\Gamma]$, let σ be a multiplier on Γ satisfying $\delta(\sigma)=0$, and define ... Here a...

open_problem | open | MathDB 页面 | 原文线索

V2-028852 | MATHDB-380879

Stringy category structure-constants conjecture

摘录 / EXCERPT: Let $R^k(\alpha)=\sum_{g=0}^{\infty}R^k_g\alpha^g$ be the formal power series whose coefficients are defined by counts of holomorphic maps from a genus-g curve with k boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-028853 | MATHDB-380880

Almost-complex-structure invariance conjecture for stringy category elements

摘录 / EXCERPT: Let J_1 and J_2 be choices of almost complex structure, and let ${\Bbb R}_1$ and ${\Bbb R}_2$ be the corresponding elements of the completed exterior-algebra construction. Let…

conjecture | open | MathDB 页面 | 原文线索

V2-028854 | MATHDB-380881

Compactification and uniform orientability conjecture for open-curve moduli spaces

摘录 / EXCERPT: Let M be a manifold with symplectic form ω and compatible almost complex structure J, let $L_1(j),\dots,L_{p_j}(j)$ be transversely meeting Lagrangian submanifolds, and…

conjecture | open | MathDB 页面 | 原文线索

V2-028855 | MATHDB-380884

Atiyah's faithfulness conjecture for the hyperbolic-space action on the sphere-at-infinity measure

摘录 / EXCERPT: Let ω be the symmetric measure on the sphere at infinity S^2 of hyperbolic space ${\mathbb H}^3$. For $f\in C^0_c({\mathbb H}^3)$, define the action … where x is regarde…

conjecture | open | MathDB 页面 | 原文线索

V2-028856 | MATHDB-380889

The hyperbolic cosmetic filling conjecture

摘录 / EXCERPT: Let N be a cusped hyperbolic 3-manifold, and consider fillings of N along distinct slopes that produce hyperbolic manifolds. A filling pair is cosmetic if the resulting manifol…

conjecture | open | MathDB 页面 | 原文线索

V2-028857 | MATHDB-380902

The complement conjecture for terminal threefold contractions

摘录 / EXCERPT: Let $f\colon X\to Z\ni o$ be a contraction from a threefold with only terminal singularities such that $-K_X$ is f-nef and f-big. An n-complement of K_X is a complement wit…

conjecture | open | MathDB 页面 | 原文线索

V2-028858 | MATHDB-380920

Boundedness conjecture for complements

摘录 / EXCERPT: Let ${\EuScript{N}}_d({\Phi})$ denote the set of positive integers occurring as complement indices in the setting of the assumptions referred to in the source. Boundedness conjectu…

conjecture | open | MathDB 页面 | 原文线索

V2-028859 | MATHDB-380922

The hyperbolic complex line arrangement realization conjecture for quasipositive Seifert surfaces

摘录 / EXCERPT: Let $F\subset S^3$ be a quasipositive Seifert surface. A proper arc in F is an arc with endpoints on ∂F and interior in the interior of F. Let $A(O,-1)$ be the surfac…

conjecture | open | MathDB 页面 | 原文线索

V2-028860 | MATHDB-380949

Traczyk's infinite-generation conjecture for skein modules of simply connected 3-manifolds

摘录 / EXCERPT: Let M be a simply connected 3-manifold, possibly with holes, and let $S_{2,\infty}(M)$ denote its Kauffman bracket skein module at parameter 2 and infinity. Traczyk's conject…

conjecture | open | MathDB 页面 | 原文线索

V2-028861 | MATHDB-380957

Semi-transition relations for semi-supermanifolds

摘录 / EXCERPT: Semi-supermanifold semi-transition conjecture. On $U_\alpha\cap U_\beta$,

conjecture | open | MathDB 页面 | 原文线索

V2-028862 | MATHDB-380958

Ordinary obstruction for obstructed semi-manifolds

摘录 / EXCERPT: Obstructed semi-manifold obstruction conjecture. Obstructed semi-manifolds may have nonvanishing ordinary obstruction, and this obstruction can be calculated by extending standard…

conjecture | open | MathDB 页面 | 原文线索

V2-028863 | MATHDB-380959

Kontsevich's Formality Conjecture for the Hochschild complex

摘录 / EXCERPT: Let $A=C^\infty(X)$ be the algebra of smooth functions on a smooth real manifold X, and let $C^\bullet=C^\bullet(A,A)[1]$ be the suspended local Hochschild complex, whose cohomol…

conjecture | open | MathDB 页面 | 原文线索

V2-028864 | MATHDB-381010

The eigenvector-map conjecture for generic ultra-discrete Toda fibers

摘录 / EXCERPT: Let $\mathcal{T}$ be the phase space of the ultra-discrete $(g+1)$-periodic Toda lattice, let $\mathcal{C}$ be the moduli space of compact tropical curves Γ_C, and let … be…

conjecture | open | MathDB 页面 | 原文线索

V2-028865 | MATHDB-381016

The semipositive representative conjecture for nef and big classes on Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a compact Kähler Calabi-Yau manifold, and let $\alpha\in H^{1,1}_{\mathbb{R}}(X)$ be a class that is nef and big but not Kähler. A smooth $(1,1)$-form representing…

conjecture | open | MathDB 页面 | 原文线索

V2-028866 | MATHDB-381045

Calabi-Yau domain conjecture for the properly embedded one-manifold [?]

摘录 / EXCERPT: Let $\mathbb{B}$ be the unit ball and let $\Delta\subset\mathbb{B}$ be the properly embedded one-manifold given in the proof of Theorem. Let $\overline{B}$ be a smooth compact Riem…

conjecture | open | MathDB 页面 | 原文线索

V2-028867 | MATHDB-381047

The Amerik-Kollár-Peternell conjecture on endomorphisms of Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold of Picard number 1, and let $f:X\to X$ be a surjective endomorphism. Here, $\deg f$ denotes the degree of f, and $\mathbb{P}^n$ denotes projective...

conjecture | open | MathDB 页面 | 原文线索

V2-028868 | MATHDB-381058

The volume lower-bound conjecture for closed hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold, and let $b_1(M)$ denote its first Betti number. Volume lower-bound conjecture. If ... then ... This conjectural lower bound is used to esta...

conjecture | open | MathDB 页面 | 原文线索

V2-028869 | MATHDB-381100

Mosco convergence of the diffusion large-deviation functional

摘录 / EXCERPT: Mosco-convergence conjecture. For every fixed $\bar\rho\in\mathcal P_2^{\mathcal S}(\mathbb R^d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-028870 | MATHDB-381111

Neumann's volume realization conjecture for boundary-unipotent representations

摘录 / EXCERPT: Let M be a 3-manifold, let n be a positive integer, and let ρ be a boundary-unipotent representation of $\pi_1(M)$ in $\mathrm{SL}(n,\mathbb C)$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-028871 | MATHDB-381113

The rooted-tree and oriented-tree correspondence in Joyce-Song wall-crossing

摘录 / EXCERPT: Let $\mathcal{T}'$ be a Γ-labelled tree. In the Gaiotto-Moore-Neitzke asymptotic expansion, consider the total contribution to wall crossing obtained by choosing every possi...

conjecture | open | MathDB 页面 | 原文线索

V2-028872 | MATHDB-381116

The small-characteristic bound for Poisson traces of quasihomogeneous surface singularities

摘录 / EXCERPT: Let A be the algebra of regular functions on a quasihomogeneous isolated surface singularity X defined by $Q(x,y,z)=0$ over a field of characteristic p, with the notation...

conjecture | open | MathDB 页面 | 原文线索

V2-028873 | MATHDB-381123

The asymptotic hexagonal limit conjecture for spectral equipartitions

摘录 / EXCERPT: Let Ω be a bounded domain in $\mathbb{R}^2$, and let $\mathfrak L_k(\Omega)$ denote the infimum of the maximal first Dirichlet eigenvalue over all k-partitions of Ω...

conjecture | open | MathDB 页面 | 原文线索

V2-028874 | MATHDB-381134

CR Obata conjecture for closed pseudohermitian manifolds

摘录 / EXCERPT: Let (M,θ) be a closed pseudohermitian $(2n+1)$-manifold with $n\geq 2$. Assume, in addition, that the CR-Paneitz operator is nonnegative if $n=1$. Suppose there is a positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-028875 | MATHDB-381136

The local-global conjecture for curvatures in an integral Apollonian gasket

摘录 / EXCERPT: Let $\mathcal{G}$ be the fixed primitive integral Apollonian gasket, let $\mathcal{B}=\mathcal{B}(\mathcal{G})$ be the set of its circle curvatures, ... and call an integer represented if it lies in $\mathcal{B}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028876 | MATHDB-381152

Extension of the platonic robustness theorem to downward robustness

摘录 / EXCERPT: In the setting of Theorem, concerning robustness classes of convex solids and platonic solids, let downward external robustness and downward full robustness be the corresponding do...

open_problem | open | MathDB 页面 | 原文线索

V2-028877 | MATHDB-381165

The anisotropic Calderón conjecture

摘录 / EXCERPT: Let (M,g_1) and (M,g_2) be compact Riemannian manifolds with smooth boundary, with $\dim(M)\geq 3$. For each metric, let C_{g_i} denote the Cauchy data set of harmonic functi...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-028878 | MATHDB-381194

Properness of analytic stacks without unramified diagonal

摘录 / EXCERPT: The theorem concerns a morphism of k-analytic stacks whose diagonal is assumed to be unramified in the stated result. Properness conjecture. The theorem that this morphism is pro...

conjecture | open | MathDB 页面 | 原文线索

V2-028879 | MATHDB-381231

Finite minimal-arc representation conjecture for ball hulls

摘录 / EXCERPT: Let $K=\{p_1,p_2,\dots,p_n\}$ be a finite set in a normed plane $\mathbb{M}^2$, and let $\lambda\geq\lambda_K$. A minimal-arc representation conjecture asserts that there exist bal...

conjecture | open | MathDB 页面 | 原文线索

V2-028880 | MATHDB-381236

The moduli-space conjecture for enriched curves

摘录 / EXCERPT: Let Y be a nodal curve with $b4$ nodes, let Γ be its dual graph, and let $\mathcal B$ be the set of bonds of Γ. Define the vector-weight collection...

conjecture | open | MathDB 页面 | 原文线索

V2-028881 | MATHDB-381249

Benedetti-Guadagnini conjecture on CMC marked spectra

摘录 / EXCERPT: Benedetti-Guadagnini conjecture. For every $\gamma\in\pi_1(M)$,

conjecture | open | MathDB 页面 | 原文线索

V2-028882 | MATHDB-381276

Stability conjecture for contact numbers of nearly congruent disks

摘录 / EXCERPT: Let $n\ge 2$ and consider a packing of n circular disks whose radii all lie in an interval $[1-\epsilon,1]$. Stability conjecture. There exists an $\epsilon>0$ such that, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-028883 | MATHDB-381282

Distinctness conjecture for the tori Θ^n_s

摘录 / EXCERPT: Let $s,s_i\in(1/2,1)$, let k_i be positive even integers for $i=1,\dots,l$, and assume $n=\sum_i k_i$. Consider the Lagrangian tori Θ^n_s and...

conjecture | open | MathDB 页面 | 原文线索

V2-028884 | MATHDB-381306

The higher BV quantization conjecture for shifted quadratic modules

摘录 / EXCERPT: Higher BV quantization conjecture. There is a natural equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-028885 | MATHDB-381340

Convex geodesic Swiss cheese conjecture

摘录 / EXCERPT: Convex geodesic Swiss cheese conjecture. There exists such a finite disjoint union whose removal from N delivers a metric g_M of gradient type on M.

conjecture | open | MathDB 页面 | 原文线索

V2-028886 | MATHDB-381355

Local-global conjecture for curvatures in integral circle packings

摘录 / EXCERPT: Let $\mathcal{P}$ be an integral circle packing, let $\mathcal{K}\subset\mathbb{Z}$ be its set of curvatures, and let $\mathcal{K}_a$ be the set of integers passing all local obstr...

conjecture | open | MathDB 页面 | 原文线索

V2-028887 | MATHDB-381388

Cherry's hyperbolicity equivalence conjecture for non-Archimedean analytic spaces

摘录 / EXCERPT: Let k be a non-Archimedean complete valued field that is nontrivially valued and algebraically closed, and let X be a smooth compact boundaryless k-analytic space. An entire...

conjecture | open | MathDB 页面 | 原文线索

V2-028888 | MATHDB-381397

The generalized vector space partition conjecture for $A_q(2k,2k-2;k)$

摘录 / EXCERPT: Let $A_q(n,d;k)$ denote the maximum size of a constant-dimension k-subspace code in $\mathbb{F}_q^n$ with minimum subspace distance at least d. Generalized vector space partiti...

conjecture | open | MathDB 页面 | 原文线索

V2-028889 | MATHDB-381407

A simple generalized permutahedron conjecture for Speer fans

摘录 / EXCERPT: Simple generalized permutahedron conjecture. There is a natural map

conjecture | open | MathDB 页面 | 原文线索

V2-028890 | MATHDB-381427

Expansion conjecture for Seifert fibered spaces over the 2-sphere

摘录 / EXCERPT: Expansion conjecture. The space Y smoothly embeds in S^4 if and only if it is obtained by a possibly empty sequence of expansions from some Y' of the displayed form, with

conjecture | open | MathDB 页面 | 原文线索

V2-028891 | MATHDB-381443

Density conjecture for metrics with empty-interior critical-width spectrum

摘录 / EXCERPT: Let M be the closed manifold and let $\Gamma_\infty(M)$ denote the space of smooth Riemannian metrics on M. Let $\mathcal{M}_{ei}$ be the set of metrics for which the set of cr...

conjecture | open | MathDB 页面 | 原文线索

V2-028892 | MATHDB-381473

Sublinear multiplicity bound for the one-half eigenvalue of the drifted Jacobi operator

摘录 / EXCERPT: Sublinear multiplicity conjecture. There exist $\alpha<1$ and a constant $C_{\alpha}=C_{\alpha}(\alpha,\gamma)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-028893 | MATHDB-381478

The three-dimensional flatness conjecture for hollow convex bodies

摘录 / EXCERPT: Three-dimensional flatness conjecture. No hollow convex 3-body has width larger than $$. That is,

conjecture | open | MathDB 页面 | 原文线索

V2-028894 | MATHDB-381486

The order-two bordism conjecture for M-theory on unoriented manifolds

摘录 / EXCERPT: Order-two bordism conjecture. The group $\pi_{11}M\mathfrak m_c$ is cyclic of order 2, the bordism class of $(N,0)$ represents its generator, and the mod-2 index of the...

conjecture | open | MathDB 页面 | 原文线索

V2-028895 | MATHDB-381501

Separation of scissors congruence classes by volume and generalized Dehn invariants

摘录 / EXCERPT: Volume-Dehn separation conjecture. In Euclidean, spherical, and hyperbolic geometries, do the volume and generalized Dehn invariant separate the scissors congruence classes of poly...

open_problem | open | MathDB 页面 | 原文线索

V2-028896 | MATHDB-381522

The log-Brunn-Minkowski conjecture

摘录 / EXCERPT: Let K and L be convex bodies in $\mathbb{R}^n$. For $\lambda\in(0,1)$, their geometric mean is defined, for origin-symmetric bodies, by ... Here V denotes volume,...

conjecture | open | MathDB 页面 | 原文线索

V2-028897 | MATHDB-381528

The Burns-Epstein invariant formula conjecture for strictly pseudoconvex CR manifolds

摘录 / EXCERPT: Fix a positive integer n. Let $(M,T^{1,0}M)$ be a closed strictly pseudoconvex CR manifold of dimension $2n+1$ admitting a pseudo-Einstein contact form. For each partition...

conjecture | open | MathDB 页面 | 原文线索

V2-028898 | MATHDB-381548

Polynomiality of coefficients in the covariant derivatives of eigenfunctions

摘录 / EXCERPT: Polynomiality conjecture. Each coefficient $a^{i_1i_3\cdots i_{2l-1}}_{i_2i_4\cdots i_{2l}}$ is a polynomial in θ of degree l. Conjecturally, the constant in the displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-028899 | MATHDB-381587

Burns-Knieper flatness conjecture for bounded-distance geodesic foliations

摘录 / EXCERPT: Let M be a simply connected surface with a complete Riemannian metric with no conjugate points. Suppose that $\mathcal F$ is a foliation on M whose leaves are all geodesic line...

conjecture | open | MathDB 页面 | 原文线索

V2-028900 | MATHDB-381589

The full-mass characterization of the Monge-Ampère mass for the associated plurisubharmonic singularity

摘录 / EXCERPT: Full-mass conjecture. $e_n(\varphi)=1$ if and only if $u\in\mathcal{E}(\omega_0)$.

conjecture | open | MathDB 页面 | 原文线索

V2-028901 | MATHDB-381593

Uniqueness conjecture for high-temperature μ-cscK metrics

摘录 / EXCERPT: High-temperature uniqueness conjecture. For $\lambda\leq 0$, μ^λ-cscK metrics are unique modulo the automorphism group.

conjecture | open | MathDB 页面 | 原文线索

V2-028902 | MATHDB-381602

Euler continuant quasi-bisymplecticity conjecture for the localized doubled-quiver algebra

摘录 / EXCERPT: Quasi-bisymplecticity conjecture. The triple $({\mathcal{B}}(\Gamma_n),\Phi,\omega_n)$ is a quasi-bisymplectic algebra. Furthermore, the double quasi-Poisson bracket...

conjecture | open | MathDB 页面 | 原文线索

V2-028903 | MATHDB-381608

Extension of Teissier's results to non-isolated singularities

摘录 / EXCERPT: Let $f:({\mathbb C}^n,0)\rightarrow({\mathbb C},0)$ be a holomorphic function, with $n\ge 2$, and allow f to have non-isolated singularities. The quantities $Q(f)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-028904 | MATHDB-381610

Regularized formula for positive integer surgery

摘录 / EXCERPT: Regularized $+p$-surgery formula. When the $+p$-surgery formula of Gukov and Manolescu converges, it should be used; when it does not converge, the displayed identity should regula...

conjecture | open | MathDB 页面 | 原文线索

V2-028905 | MATHDB-381612

Rigidity conjecture for essentially disjoint plurisupported currents

摘录 / EXCERPT: Rigidity conjecture. Either $T=0$ or $S=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-028906 | MATHDB-381641

Conjecture on the dimension of partial Okounkov bodies

摘录 / EXCERPT: Let X be the compact complex manifold, let θ be the closed $(1,1)$-form, and let θ-plurisubharmonic function $\varphi\in\operatorname{PSH}(X,\theta)$ be as above, w...

conjecture | open | MathDB 页面 | 原文线索

V2-028907 | MATHDB-381651

Cameron-Liebler conjecture for Boolean degree 1 functions on Grassmann graphs

摘录 / EXCERPT: Let $V=\mathbb{F}_q^n$ and let $J_q(n,k)$ be the Grassmann graph whose vertices are the k-dimensional subspaces of V. A function f on the vertices is Boolean degree 1 if it...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028908 | MATHDB-381667

Space form conjecture for compact pseudo-Riemannian manifolds

摘录 / EXCERPT: A pseudo-Riemannian manifold of signature (p,q) is understood here to be compact and complete, with constant sectional curvature 1. Space form conjecture. There exists such a m...

conjecture | open | MathDB 页面 | 原文线索

V2-028909 | MATHDB-381695

Gromov's average area ratio conjecture for hyperbolic manifolds

摘录 / EXCERPT: Let (M,h_0) be a closed hyperbolic manifold of dimension $n\geq 3$, and let N be a closed n-manifold with Riemannian metric g whose scalar curvature satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-028910 | MATHDB-381699

Four-periodic infinite staircase conjecture for four-dimensional polydisks

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{\geq 2}$ and define ... Let c_{β_n} denote the capacity function associated with the four-dimensional polydisk, and let $\operatorname{accum}(\beta_n)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-028911 | MATHDB-381714

Polterovich's Poisson bracket invariant lower-bound conjecture

摘录 / EXCERPT: Polterovich's conjecture. There exists a constant C_M, depending only on the symplectic manifold (M,ω), such that for every finite open cover $\mathcal{U}=\{U_i\}$ of M,

conjecture | open | MathDB 页面 | 原文线索

V2-028912 | MATHDB-381715

Persistence-rank conjecture for symplectic rational surfaces

摘录 / EXCERPT: Let $X=\mathbb{C}P^2\# n\overline{\mathbb{C}P^2}$ carry a symplectic form ω, and let $\Gamma(X,\omega)$ be its Lagrangian root system. Write $PR[\Gamma(X,\omega)]$ for the p...

conjecture | open | MathDB 页面 | 原文线索

V2-028913 | MATHDB-381722

The negative-equals-reverse conjecture for Schoenflies balls

摘录 / EXCERPT: Let Δ be a Schoenflies ball, and let $-\Delta$ denote its complementary Schoenflies ball in S^4. Write $\bar\Delta$ for Δ with the opposite orientation. Negative-eq...

conjecture | open | MathDB 页面 | 原文线索

V2-028914 | MATHDB-381723

The Poincare ball 5-ball conjecture

摘录 / EXCERPT: A Poincare 4-ball is a contractible 4-manifold whose boundary is S^3. A Schoenflies 4-ball is a Poincare 4-ball that embeds in S^4. Poincare ball 5-ball conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-028915 | MATHDB-381724

The Gluck ball 5-ball conjecture

摘录 / EXCERPT: A Gluck ball is the Poincare 4-ball associated with a Gluck-twisted S^4. Gluck ball 5-ball conjecture. If Δ^4 is a Gluck ball, then ... This is described as a special case o...

conjecture | open | MathDB 页面 | 原文线索

V2-028916 | MATHDB-381725

The 0-, 1-, 2-handle conjectures for contractible manifolds

摘录 / EXCERPT: Let Δ^4 be a Poincare ball, and let Δ^5 be a contractible 5-manifold. A handle decomposition has only 0-, 1- and 2-handles when all its handles have these indices. 0-...

conjecture | open | MathDB 页面 | 原文线索

V2-028917 | MATHDB-381726

The periodic Higgs conjecture for parabolic Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve over $\mathbb{C}$, and let $D\subset C$ be a reduced effective divisor. A parabolic Higgs bundle over (C,D) is called periodic when it is per...

conjecture | open | MathDB 页面 | 原文线索

V2-028918 | MATHDB-381744

Fradelizi--Meyer's affine-invariant Mahler conjecture

摘录 / EXCERPT: Fradelizi--Meyer's conjecture. For every convex body $K\subset\mathbb{R}^n$,

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V2-028919 | MATHDB-381745

Viterbo's symplectic isoperimetric conjecture

摘录 / EXCERPT: Viterbo's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-028920 | MATHDB-381746

Rigidity conjecture for positively Ricci-curved Bryant-asymptotic steady solitons

摘录 / EXCERPT: Let (M^n,g) be a steady gradient Ricci soliton with positive Ricci curvature. Suppose there are a ball $B_R(0)\subset\mathbb R^n$ and a diffeomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-028921 | MATHDB-381772

Semi-ampleness and Kodaira-dimension conjecture for the moduli part of maximal-moduli fibrations

摘录 / EXCERPT: Variant of Shokurov's b-semi-ampleness conjecture. For all $m\gg 0$ and any $\epsilon\in\mathbb Q_{>0}$, the divisor

conjecture | open | MathDB 页面 | 原文线索

V2-028922 | MATHDB-381782

Böröczky's polyhedral norm conjecture

摘录 / EXCERPT: Böröczky's conjecture. There is a point $x\in P$ with norm

conjecture | open | MathDB 页面 | 原文线索

V2-028923 | MATHDB-381794

The simplicity conjecture for convex hulls of Wachs permutations

摘录 / EXCERPT: Let $m\in\mathbb{P}$, and let $c(\operatorname{\mathcal{W}}(S_{2m}))$ be the convex hull of the Wachs permutations in $\mathbb{R}^{2m}$. Simplicity conjecture. The polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-028924 | MATHDB-381795

The simplicity conjecture for convex hulls of signed Wachs permutations

摘录 / EXCERPT: Let $m\in\mathbb{P}$, and let $c(\operatorname{\mathcal{W}}(B_{2m}))$ be the convex hull of the signed Wachs permutations in $\mathbb{R}^{2m}$. Simplicity conjecture. The polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-028925 | MATHDB-381829

Functorial correspondence between class field towers and resolved curves

摘录 / EXCERPT: Let $(0,0)$ be an isolated singular point of an algebraic curve over a number field k, with local ring $\mathcal{O}=k[[x,y]]/(f)$ and multiplicity m. Let C be the nonsingular...

conjecture | open | MathDB 页面 | 原文线索

V2-028926 | MATHDB-381846

The conjecture that algebraically integrable hypersurfaces are quadrics

摘录 / EXCERPT: Quadraticity conjecture. Every smooth polynomially integrable hypersurface is a strictly convex quadric in $\mathbb R^{2k+1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-028927 | MATHDB-381847

The conjecture on powers of sectional-volume functions

摘录 / EXCERPT: Power-polynomial conjecture. Then ∂K is an ellipsoid; consequently, one can take $m=1$ when n is odd and $m=2$ when n is even.

conjecture | open | MathDB 页面 | 原文线索

V2-028928 | MATHDB-381848

The interior problem conjecture for the Radon transform

摘录 / EXCERPT: Interior problem conjecture. There exists a function $f\in C_c^{\infty}(\overline D)$, not identically zero in K, such that $\mathcal R f(L)$ vanishes for every line L that int...

open_problem | open | MathDB 页面 | 原文线索

V2-028929 | MATHDB-381855

The octonionic Calabi-Yau conjecture for compact sixteen-dimensional manifolds

摘录 / EXCERPT: Octonionic Calabi-Yau conjecture. There exist a C^∞-smooth function $u\colon M\longrightarrow\mathbb{R}$ and a constant $A>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-028930 | MATHDB-381857

Giordano's spectral-sequence collapse conjecture for twisted polynomial superfunctors

摘录 / EXCERPT: Spectral-sequence collapse conjecture. For all $F,G\in\mathcal{P}$, this spectral sequence collapses at the second page. In particular, there is a graded isomorphism, not necessari...

conjecture | open | MathDB 页面 | 原文线索

V2-028931 | MATHDB-381868

A hyperbolic distance identity for intersections of lines in the Poincaré disk

摘录 / EXCERPT: Let a,b,c,d be four complex points on the unit circle S^1 in this order, with the Euclidean lines $L(a,b)$ and $L(c,d)$ nonparallel. Let h be an arbitrary point on the Euclid...

conjecture | open | MathDB 页面 | 原文线索

V2-028932 | MATHDB-381918

Perry's noncommutative integral Hodge conjecture

摘录 / EXCERPT: Perry's noncommutative integral Hodge conjecture. The homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-028933 | MATHDB-381963

Curvature flow of the octahedron

摘录 / EXCERPT: Octahedral curvature-flow conjecture. The normalized curvature flow on the octahedron converges to a limit described by the stated classification of totally degenerate non-lazy cur...

conjecture | open | MathDB 页面 | 原文线索

V2-028934 | MATHDB-382013

Lie point symmetry conjecture for the Ricci flow

摘录 / EXCERPT: Let (M^n,g) be a Riemannian manifold with $n\geq 2$, and write $g=(g_{ij})$ in local coordinates $(x^1,\ldots,x^n)$. Let t denote time, and let $\xi^1,\ldots,\xi^n$ be arbitrar...

conjecture | open | MathDB 页面 | 原文线索

V2-028935 | MATHDB-382088

Carnot tangent characterization of biLipschitz rectifiability

摘录 / EXCERPT: Carnot tangent characterization. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-028936 | MATHDB-382105

Boundary Polyakov-Alvarez asymptotic formula for Brownian loop measure

摘录 / EXCERPT: Let (M,g) be a fixed compact smooth two-dimensional Riemannian manifold with smooth boundary, and let μ^{loop} denote the Brownian loop measure on (M,g). Let K be...

open_problem | open | MathDB 页面 | 原文线索

V2-028937 | MATHDB-382109

Asymptotic separation between systolic bounds for Ramanujan surfaces

摘录 / EXCERPT: Let g denote the genus of a hyperbolic surface, let $\sys(M)$ denote its systole, and let $\Nsmall$ and $\lambda_1(M)$ be the quantities used in the paper. The first quantity bel...

conjecture | open | MathDB 页面 | 原文线索

V2-028938 | MATHDB-382140

GGLR's Moishezon conjecture for the period-map Stein factor

摘录 / EXCERPT: Let D be a Mumford--Tate domain parameterizing pure, effective, weight w, Q-polarized Hodge structures on a finite-dimensional rational vector space V. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-028939 | MATHDB-382145

Dickenstein-Nill Cayley decomposition conjecture for lattice polytopes

摘录 / EXCERPT: Dickenstein-Nill's Cayley decomposition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-028940 | MATHDB-382154

Lascoux expansion conjecture for symplectic involution Grothendieck polynomials

摘录 / EXCERPT: Let I^{fpf}_∞ denote the fixed-point-free involutions, let $\mathfrak G^{\mathsf{Sp}}_z$ be the corresponding symplectic involution Grothendieck polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-028941 | MATHDB-382167

Polystability conjecture for polarised fibrations and optimal symplectic connections

摘录 / EXCERPT: Let $(X,H)\to(B,L)$ be a polarised fibration, meaning a fibration equipped with relatively ample polarisation H and a polarisation L on the base. An optimal symplectic connecti...

conjecture | open | MathDB 页面 | 原文线索

V2-028942 | MATHDB-382190

Asymptotic product conjecture for the Calabi ansatz metric on \(\mathbb C^k\)

摘录 / EXCERPT: Let ω be the metric on $\mathbb C^k$ determined by Lemma, and let $\omega_p=iD_0\partial\bar{\partial}s^p$, where $s=|z|^2$ on $\mathbb C^k$. Say that a metric is asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-028943 | MATHDB-382224

Sheaf quantization and Fukaya category conjectures for Liouville manifolds

摘录 / EXCERPT: Sheaf quantization and Fukaya category conjectures. For each Lagrangian brane $\bm L$ in X, there is an almost quasi-isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-028944 | MATHDB-382225

Smooth-to-holonomic equivalence conjecture over the Novikov field

摘录 / EXCERPT: Smooth-to-holonomic equivalence conjecture. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-028945 | MATHDB-382233

Non-Kähler nef intersection conjecture

摘录 / EXCERPT: Let X be a compact complex manifold, let $\mathcal{N}\subset H^{1,1}(X,\mathbb{R})$ be its cone of nef $(1,1)$-classes, let $[\alpha]\in\mathcal{N}$, and let $V\subset X$ be a cl...

open_problem | open | MathDB 页面 | 原文线索

V2-028946 | MATHDB-382255

Gilbert-Pollak Steiner ratio conjecture

摘录 / EXCERPT: For a finite point configuration $P\subset\mathbb{R}^2$, let its Steiner ratio be the ratio of the cost of an optimal Steiner tree for P to the cost of a minimum spanning tree fo...

conjecture | refuted | MathDB 页面 | 原文线索

V2-028947 | MATHDB-382261

Universal analytic Gröbner basis conjecture

摘录 / EXCERPT: Let $K\{\mathbf{X};P\}$ be the analytic power-series ring over the valued field K, with P the parameter set of valuation vectors, and let $I\subset K\{\mathbf{X};P\}$ be an ide...

conjecture | open | MathDB 页面 | 原文线索

V2-028948 | MATHDB-382279

Improved parameter range for the eta-expanded identifiability theorem

摘录 / EXCERPT: Improved eta-expanded theorem. The thesis of Theorem ...

conjecture | open | MathDB 页面 | 原文线索

V2-028949 | MATHDB-382281

The uniform Lagrangian spectral-width conjecture

摘录 / EXCERPT: Let N be a closed manifold, let g be a Riemannian metric on N, and let $D_1T^*N=\{(q,p)\mid |p|_g\leq 1\}$. Let $\mathfrak L(T^*N)$ denote the relevant class of closed exact...

conjecture | open | MathDB 页面 | 原文线索

V2-028950 | MATHDB-382303

Kuznetsov-Polishchuk fullness conjecture for isotropic Grassmannians

摘录 / EXCERPT: Let V be a $2n$-dimensional symplectic vector space, let ${\mathsf{IGr}}(k,V)$ be the isotropic Grassmannian, and let ${\mathcal{A}}_t$ be the admissible subcategories defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-028951 | MATHDB-382305

Angella-Otal-Ugarte-Villacampa conjecture for compact Bismut Kähler-like manifolds

摘录 / EXCERPT: Let (M^n,g) be a compact Hermitian manifold. A Hermitian manifold is Bismut Kähler-like (BKL) if its Bismut curvature has the Kähler symmetries ... for all type $(1,0)$ tangent vec...

conjecture | open | MathDB 页面 | 原文线索

V2-028952 | MATHDB-382306

Strongly Gauduchon conjecture for compact non-balanced BTP manifolds

摘录 / EXCERPT: Strongly Gauduchon conjecture. The manifold M does not admit any strongly Gauduchon metric. The claim extends the known obstruction for compact non-Kähler BKL manifolds; the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-028953 | MATHDB-382316

Consecutive eigenvalue gap conjecture for elliptic divergence-form operators

摘录 / EXCERPT: Let M^n be a complete Riemannian manifold, let $\Omega\subset M^n$ be a bounded domain, and let λ_k denote the Dirichlet eigenvalues of the operator $\mathcal{L}$ from Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-028954 | MATHDB-382321

Thurston's modified conjecture on horospherical sections of the {3,5,3} honeycomb

摘录 / EXCERPT: Take the $\{3,5,3\}$ honeycomb in $\operatorname{\mathbb{H}}^3$ and cut it along a horosphere. Consider the union of all icosahedra interior to, or intersecting, the horosphere; th...

conjecture | open | MathDB 页面 | 原文线索

V2-028955 | MATHDB-382325

The Kawamata-Morrison-Totaro cone conjecture for klt Calabi-Yau pairs

摘录 / EXCERPT: Kawamata-Morrison-Totaro cone conjecture. The group $\operatorname{Aut}(X/S,\Delta)$ acts on $\mathcal{A}^e(X/S)$ with finitely many orbits of Mori faces, while...

conjecture | open | MathDB 页面 | 原文线索

V2-028956 | MATHDB-382343

Nonexistence of higher-degree smooth absolute minima in the standard contact 3-sphere model

摘录 / EXCERPT: Nonexistence conjecture. There are no smooth absolute minima of E of degree greater than 1, irrespective of the coupling constant.

conjecture | open | MathDB 页面 | 原文线索

V2-028957 | MATHDB-382372

Conjecture on the 2-rank of the matrix M_2

摘录 / EXCERPT: Conjecture on the 2-rank of M_2.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-028958 | MATHDB-382374

The divisorial valuation and rational-rank-one conjecture for symplectic singularities

摘录 / EXCERPT: Let $x\thicksim X$ be a symplectic singularity. Let v_X be the valuation associated with x in Theorem AGlc, and let $r(\xi)$ denote the rational rank associated with ξ in T...

conjecture | open | MathDB 页面 | 原文线索

V2-028959 | MATHDB-382380

The homeomorphism extension for thin points and points in the double locus

摘录 / EXCERPT: Homeomorphism extension. The homeomorphism version of Theorem would also hold for any thin point x with $f_*={\rm id}$ and for any $x\in X_0^2$ if the ball...

conjecture | open | MathDB 页面 | 原文线索

V2-028960 | MATHDB-382428

The Weinstein-domain identification conjecture for Lagrangian skeleta

摘录 / EXCERPT: Let Z be the open Batyrev-Borisov complete intersection with skeleton ... where $S=\partial\widetilde{C}_{0,\beta}\subset M_\R$ is the codimension-r sphere introduced above, and...

conjecture | open | MathDB 页面 | 原文线索

V2-028961 | MATHDB-382437

Non-APN conjecture for the functions $f_{b,c,r}$

摘录 / EXCERPT: Let $m\ge 8$ be an even integer, let $b,c\in\mathbb{F}_{2^m}$, and let r satisfy $r\le m/2$ and $\gcd(r,m)=1$. Let $f_{b,c,r}$ denote the function defined in Theorem. Non-APN con...

conjecture | open | MathDB 页面 | 原文线索

V2-028962 | MATHDB-382470

Extension of the face theorem to non-toric NCCRs

摘录 / EXCERPT: Non-toric NCCR extension conjecture. The statement of the theorem holds also if Λ' is a non-toric NCCR.

conjecture | open | MathDB 页面 | 原文线索

V2-028963 | MATHDB-382490

Equality of the transition heights for symmetric and antisymmetric second Neumann eigenfunctions

摘录 / EXCERPT: Let h_0 and h_1 be the positive heights from part (2) of Theorem, where the ordering of the symmetric and antisymmetric second Neumann eigenvalues changes as the height varies....

conjecture | open | MathDB 页面 | 原文线索

V2-028964 | MATHDB-382509

Catino-Mastrolia-Monticelli scalar-flatness conjecture for critical metrics

摘录 / EXCERPT: Let (M^n,g) be a complete noncompact Riemannian manifold with $n\ge5$. Write R_g for its scalar curvature, and let $\mathcal S^2$ denote the L^2-scalar curvature functional....

conjecture | open | MathDB 页面 | 原文线索

V2-028965 | MATHDB-382520

Generalized Liu-Terng conjecture for mean curvature flows in compact symmetric spaces

摘录 / EXCERPT: Generalized Liu-Terng conjecture. (i) If $A(t)$ and $H(t)$ satisfy the inequality referred to as for some $\epsilon\in(0,1/(2\kappa))$ instead of ϵ_F, then M is an equif...

conjecture | open | MathDB 页面 | 原文线索

V2-028966 | MATHDB-382523

Rousseeuw and Hubert's conjecture on maximum regression depth

摘录 / EXCERPT: Let S be a set of n points in d-dimensional space. The regression depth of a hyperplane is the minimum number of points intersected by the hyperplane as it undergoes a contin...

conjecture | open | MathDB 页面 | 原文线索

V2-028967 | MATHDB-382532

Donald's uniqueness and variational conjecture for the homogeneous complex Monge-Ampère equation

摘录 / EXCERPT: Let V be the compact complex manifold underlying the forms in the statement, let R be a compact Riemann surface with boundary, and let $\rho:V\times\partial R\to\mathbf{R}$ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-028968 | MATHDB-382540

Mori structures conjecture for double spaces of index 2

摘录 / EXCERPT: Mori structures conjecture. The Mori structures of U are precisely U itself and

conjecture | open | MathDB 页面 | 原文线索

V2-028969 | MATHDB-382546

Iitaka-Severi conjecture on dominant rational maps to manifolds of general type

摘录 / EXCERPT: Iitaka-Severi conjecture. The number of birational equivalence classes of such manifolds Y is finite. This conjecture generalizes the de Franchis-Severi finiteness theorem for ma...

conjecture | open | MathDB 页面 | 原文线索

V2-028970 | MATHDB-382547

Specialness criterion for standard linear systems in projective three-space

摘录 / EXCERPT: Specialness criterion. The system $\mathcal L$ is special if and only if at least one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-028971 | MATHDB-382556

The dimension conjecture for simple null symplectic triple systems in characteristic 3

摘录 / EXCERPT: A symplectic triple system over a field of characteristic 3 is called null when its alternating bilinear form is zero. Dimension conjecture. The dimension of any simple null symp...

conjecture | open | MathDB 页面 | 原文线索

V2-028972 | MATHDB-382561

Classical representation conjecture for null Lagrangians on diffeomorphism groups

摘录 / EXCERPT: Let M be a manifold, let $nl^{0}(M)$ denote the space of null Lagrangians of order zero on M, and let $J_{c}(M)$ denote the relevant compactly supported jet space. A classical...

conjecture | open | MathDB 页面 | 原文线索

V2-028973 | MATHDB-382565

The classification of numerically elementary (-1)-curves with $b4<2bc_1$

摘录 / EXCERPT: Classification conjecture. The only numerically elementary (-1)-curves satisfying $\delta<2\mu_1$ are the lines through two points.

conjecture | open | MathDB 页面 | 原文线索

V2-028974 | MATHDB-382969

Fino-Vezzoni conjecture on balanced Bismut torsion-parallel manifolds

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension n. If X admits a balanced Hermitian metric with fundamental form ω_{B} satisfying $d(\omega_{B}^{n-1})=0$ and...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-028975 | MATHDB-383029

Erdős distinct-distances problem in R3

摘录 / EXCERPT: Does there exist an absolute constant $c>0$ such that, for every finite set $P\subset\mathbb{R}^3$ with $|P|=N$, the number of distinct pairwise distances satisfies...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-028976 | MATHDB-383290

Chern-do Carmo-Kobayashi conjecture

摘录 / EXCERPT: For every $n,m\ge 1$, the set of possible constant values of $S=\lvert h\rvert^2$ for closed, minimally immersed n-dimensional submanifolds $M^n\hookrightarrow \mathbb{S}^{n+m}$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-028977 | MATHDB-383489

Weak Simplex Conjecture

摘录 / EXCERPT: For every integer $n\ge 1$, energy $E>0$, and noise standard deviation $\sigma>0$, let $x_0,\ldots,x_n\in\mathbb{R}^n$ satisfy $\lVert x_i\rVert^2=E$ for all i, and let the signa...

conjecture | solved | MathDB 页面 | 原文线索

V2-028978 | MATHDB-383506

Morimoto-Nagano realization problem

摘录 / EXCERPT: For every real parameter $t>1$, does there exist a real-analytic CR-embedding $F_t:M_t^3\hookrightarrow\mathbb{C}^3$ such that $F_t(M_t^3)$ is a compact real-analytic strongly pseu...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-028979 | MATHDB-383538

Gil Kalai’s Helly-Type Question for Two-Component Convex Sets

摘录 / EXCERPT: For integers $d\ge 2$ and $n\ge d+2$, let $F_1,\ldots,F_n\subseteq\mathbb{R}^d$ be sets, each of which is the union of exactly two disjoint, nonempty, closed convex sets. If, for e...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-028980 | MATHDB-383556

Hopf product conjecture

摘录 / EXCERPT: ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-028981 | MATHDB-383680

Bray’s conjecture

摘录 / EXCERPT: For every integer $n\ge 3$, there exists a constant $\varepsilon_n\in(0,1)$ such that, for every connected, closed, smooth Riemannian manifold (M^n,g), if...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-028982 | MATHDB-383791

Totally Anosov TNS action

摘录 / EXCERPT: Let α be a smooth action on a closed manifold X by $\mathbb{Z}^{k}$, $k\ge 2$. Assume that every elements in $\mathbb{Z}^{k}$ is an Anosov diffeomorphism and the action is...

open_problem | open | MathDB 页面 | 原文线索

V2-028983 | MATHDB-383793

Perturbation of rank one factor

摘录 / EXCERPT: Let α be a C^{∞} $\mathbb{R}^{k}$ action on a manifold X with C^{∞} rank one factor $\mathbb{R}$ action β on Y. Does any perturbation of β lift...

open_problem | open | MathDB 页面 | 原文线索

V2-028984 | MATHDB-383794

Perturbation of product action

摘录 / EXCERPT: Let $G=\mathbb{Z}^{k}$ or $\mathbb{R}^{k}$ with $k\ge 2$. Assume that G action α on a closed manifold X is Anosov. Fix a closed manifold Y. When is every perturbation...

open_problem | open | MathDB 页面 | 原文线索

V2-028985 | MATHDB-383796

Actions on n manifolds by lattices in $\textrm{SL}_{n}(\mathbb{R})$

摘录 / EXCERPT: Let Γ be a lattice in $\textrm{SL}_{n}(\mathbb{R})$. Let $\alpha:\Gamma\to \textrm{Diff}(M)$ be a smooth action on n manifold M by Γ. Assume that there is an isol...

open_problem | open | MathDB 页面 | 原文线索

V2-028986 | MATHDB-383797

Higher rank lattice actions (problem 2)

摘录 / EXCERPT: Let Γ be a lattice in $G=\textrm{SL}_{n}(\mathbb{R})$. Let $\alpha:\Gamma\to \textrm{Diff}(M)$ be a smooth action on n manifold M by Γ. Assume that α prese...

open_problem | open | MathDB 页面 | 原文线索

V2-028987 | MATHDB-383798

Higher rank lattice actions (problem 3)

摘录 / EXCERPT: Let Γ be a lattice in $\textrm{SL}_{n}(\mathbb{R})$. Let $\alpha:\Gamma\to \textrm{Diff}(M)$ be a smooth action on n manifold M by Γ. Assume that there is an...

open_problem | open | MathDB 页面 | 原文线索

V2-028988 | MATHDB-383800

Actions on manifolds by lattices in p-adic group

摘录 / EXCERPT: Let Γ be a lattice in $\textrm{SL}_{n}(\mathbb{Q}_{p})$ with $n\ge 3$. Does Γ act smoothly non-isometrically on a closed manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-028989 | MATHDB-383801

Lattices in infinite center Lie group

摘录 / EXCERPT: Let Γ be a lattice in $G=\widetilde{\textrm{Sp}_{2g}(\mathbb{R})}$. begin{enumerate} item Does G act faithfully on a compact manifold? item If not, does Γ act faith...

open_problem | open | MathDB 页面 | 原文线索

V2-028990 | MATHDB-383803

Complex manifold

摘录 / EXCERPT: Let G be a higher rank semisimple Lie group and Γ be a lattice in it. Does same question hold when we consider biholomorphic actions on an almost complex manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-028991 | MATHDB-383804

Bi-Lipschitz action

摘录 / EXCERPT: Let Γ be a lattice in $\textrm{SL}_{n}(\mathbb{R})$ with $n\ge 3$. Let X be a manifold with $\textrm{dim}(X)\le n-2$. Let α be a bi-lipschitz action on X by...

open_problem | open | MathDB 页面 | 原文线索

V2-028992 | MATHDB-383806

Higher rank lattice actions (problem 11)

摘录 / EXCERPT: Is there a locally free $\textrm{SL}_{3}(\mathbb{R})$ action on a closed 9 dimensional manifold X with at least one non-closed orbit?

open_problem | open | MathDB 页面 | 原文线索

V2-028993 | MATHDB-385078

Are the fundamental groups of hyperbolic ... -manifolds almost treeable?

摘录 / EXCERPT: Are the fundamental groups of hyperbolic 3-manifolds almost treeable?

open_problem | open | MathDB 页面 | 原文线索

V2-028994 | MATHDB-385143

Analytic questions (problem 8)

摘录 / EXCERPT: Fix $n\ge4$. Does there exist a closed, negatively curved n-manifolds M_i such that the conformal dimension of $\partial\widetilde M_i$ goes to infinity?

open_problem | open | MathDB 页面 | 原文线索

V2-028995 | MATHDB-385144

Other

摘录 / EXCERPT: Let M_0,M_1 be 3-manifolds with curvature $K\le -1$ and some point/plane with curvature -1. Suppose the induced metrics on S^2 have the same Hausdorff dimension. Must they be...

open_problem | open | MathDB 页面 | 原文线索

V2-028996 | MATHDB-385148

Quasi-isometric rigidity

摘录 / EXCERPT: Let M compact $K<0$ and $\partial M\neq\varnothing$ totally geodesic. Is every quasi-isometry of $\widetilde M$ bounded distance from an isometry? What about when M is one of t...

open_problem | open | MathDB 页面 | 原文线索

V2-028997 | MATHDB-385167

Group analogues to manifold constructions

摘录 / EXCERPT: Is there a group theoretic notion of drilling? (If ∂G is connected, then the new Bowditch boundary should have no cut points.) If G is Kleinian, what happens to...

open_problem | open | MathDB 页面 | 原文线索

V2-028998 | MATHDB-385176

Manifold questions

摘录 / EXCERPT: Let M be a locally CAT(-1) closed manifold with $\partial\widetilde M=S^{n-1}$. Does M admit a Riemannian metric with $K<0$?

open_problem | open | MathDB 页面 | 原文线索

V2-028999 | MATHDB-385233

Non-positive curvature (problem 5)

摘录 / EXCERPT: For $a,b \in A$, say $a \leq b$ if there is a positive word c with $a = bc$. Is $(A,\leq)$ a join-semilattice?

open_problem | open | MathDB 页面 | 原文线索

V2-029000 | MATHDB-385237

What kind of non-positive curvature might these complexes have?

摘录 / EXCERPT: What kind of non-positive curvature might these complexes have? (systolic, Helly, CUB, $\mathrm{CAT}(0)$?)

open_problem | open | MathDB 页面 | 原文线索

V2-029001 | MATHDB-385240

If one of the complexes has non-positive curvature and all cell stabilizers have solvable word problem, does this mean the whole group has solvable word problem?

摘录 / EXCERPT: If one of the complexes has non-positive curvature and all cell stabilizers have solvable word problem, does this mean the whole group has solvable word problem? Can this be genera...

open_problem | open | MathDB 页面 | 原文线索

V2-029002 | MATHDB-385266

Relatively hyperbolic case

摘录 / EXCERPT: If G is a LERF relatively hyperbolic free-by-cyclic group which is not hyperbolic then G is the fundamental group of a hyperbolic 3-manifold.

open_problem | open | MathDB 页面 | 原文线索

V2-029003 | MATHDB-385283

Geometry of free-by-cyclic groups (problem 2)

摘录 / EXCERPT: Let $F = F(a,b,c)$ be a free group of rank 3. Define an automorphism $\phi \in \mathrm{Aut}(F)$ such that ... Does $G = F \rtimes_{\phi} \mathbb{Z}$ act geometrically on a CAT(0) cub...

open_problem | open | MathDB 页面 | 原文线索

V2-029004 | MATHDB-385296

Ricci Flow (problem 3)

摘录 / EXCERPT: Given a hypersurface $M\subset \mathbb{R}^{n+1}$, define ... ... Question: Is there a useful notion of μ-stability for Ricci Flow self-shrinkers analogous to Colding-Minicozzi for...

open_problem | open | MathDB 页面 | 原文线索

V2-029005 | MATHDB-385299

Ricci Flow (problem 6)

摘录 / EXCERPT: Does $\max_M R(\cdot,t)$ blow up as $t\to T$ the singular time?

open_problem | open | MathDB 页面 | 原文线索

V2-029006 | MATHDB-385307

Mean Curvature Flow (problem 3)

摘录 / EXCERPT: Pick an integer N, and onsider a smooth embedded curve $\gamma\subset{\\mathbb{R}}^2$ constructed as follows. Let $\alpha_\pm:[0,2\pi N]\rightarrow{\\mathbb{R}}^2$ be a pair of d...

open_problem | open | MathDB 页面 | 原文线索

V2-029007 | MATHDB-385310

Other Geometric Flows (problem 2)

摘录 / EXCERPT: Let S^2 be the 2 dimension unit sphere and g be a metric on S^2 with positive Gauss curvature. Take $g=e^{2u} ds^2$. Consider the flow ... where $\overline{K_u}$ is the average...

open_problem | open | MathDB 页面 | 原文线索

V2-029008 | MATHDB-385311

What are closed manifolds with almost nonnegative curvature operator?

摘录 / EXCERPT: What are closed manifolds with almost nonnegative curvature operator? ...

open_problem | open | MathDB 页面 | 原文线索

V2-029009 | MATHDB-385314

Riemannian Geometry (problem 4)

摘录 / EXCERPT: What are ANCO(almost nonnegative curvature operator, i. e., there exists metric with $\text{Riem} \cdot \text{diam}^2 \geq -\epsilon$) manifolds? Constructions: 1. compact symmetri...

open_problem | open | MathDB 页面 | 原文线索

V2-029010 | MATHDB-385315

Riemannian Geometry (problem 5)

摘录 / EXCERPT: Can we smooth metrics with $\text{Riem} \geq -\text{Id}$ on volume scale, i.e., want nearby metrics with $|\nabla^k Rm| \leq C_k C_p^{-k-2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-029011 | MATHDB-385317

Existence of high genus free boundary minimal surface in ... .

摘录 / EXCERPT: Existence of high genus free boundary minimal surface in B^3.

open_problem | open | MathDB 页面 | 原文线索

V2-029012 | MATHDB-385319

Do ... -dimension shrinkers have bounded scalar curvature?

摘录 / EXCERPT: Do 4-dimension shrinkers have bounded scalar curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029013 | MATHDB-385320

Ricci Solitons (problem 2)

摘录 / EXCERPT: Are there examples of shrinkers other than the one on $\hat{\mathbb{C}}$ in 4 dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-029014 | MATHDB-385324

Are compact shrinkers with positive sectional curvature Einstein?

摘录 / EXCERPT: Are compact shrinkers with positive sectional curvature Einstein?

open_problem | open | MathDB 页面 | 原文线索

V2-029015 | MATHDB-385351

Open problems (problem 26)

摘录 / EXCERPT: Torelli theorems on $\mathcal{M}_{dR}$ with parabolic structure.

open_problem | open | MathDB 页面 | 原文线索

V2-029016 | MATHDB-385353

The Lax project

摘录 / EXCERPT: For each classical integrable system there is often a "Lax pair", which is basically a meromorphic Higgs bundle on the trivial bundle on the Riemann sphere. Thus we can consider th...

open_problem | open | MathDB 页面 | 原文线索

V2-029017 | MATHDB-385357

The soft dodecahedral conjecture

摘录 / EXCERPT: Prove the soft dodecahedral conjecture: The density of a Voronoi cell in a soft packing is maximized when the Voronoi cell is a regular circumscribing dodecahedron.

open_problem | open | MathDB 页面 | 原文线索

V2-029018 | MATHDB-385358

Phase transitions

摘录 / EXCERPT: Describe the behavior of optimal arrangements of soft balls as λ varies.

open_problem | open | MathDB 页面 | 原文线索

V2-029019 | MATHDB-385359

Translative Packings

摘录 / EXCERPT: What is the lowers dimension for which the densest translative packing of a convex body is denser than the densest lattice packing?

open_problem | open | MathDB 页面 | 原文线索

V2-029020 | MATHDB-385365

Potentials

摘录 / EXCERPT: There are other potentials that could reasonably describe a "soft packing." Can we fit them into this framework?

open_problem | open | MathDB 页面 | 原文线索

V2-029021 | MATHDB-385367

Crystal Growth From Clusters

摘录 / EXCERPT: Can connections be drawn between cluster structures that appear during complex crystal growth, and finite clusters that are optimal under other conditions? Are these finite cluster...

open_problem | open | MathDB 页面 | 原文线索

V2-029022 | MATHDB-385372

Delone Sets (problem 3)

摘录 / EXCERPT: Can this result be improved for $\rho < 10R$, e.g. $\rho = 4R$? Is it possible to show similar results for full dimensional r-sets with conditions weaker than the R condition?

open_problem | open | MathDB 页面 | 原文线索

V2-029023 | MATHDB-385373

Cycles in skeletal complexes

摘录 / EXCERPT: What are the applications and physical significance of the cycles in skeletal polygonal complexes?

open_problem | open | MathDB 页面 | 原文线索

V2-029024 | MATHDB-385376

What is the maximal number of symmetries for a radial arboreal lagrangian in ... ?

摘录 / EXCERPT: What is the maximal number of symmetries for a radial arboreal lagrangian in $\mathbb{C}^n$? When does this maximal symmetry occur? And does the ribbon of the arboreal link have th...

open_problem | open | MathDB 页面 | 原文线索

V2-029025 | MATHDB-385379

Can one produce a finite list of all arboreal skeleta for a given Weinstein manifold?

摘录 / EXCERPT: Can one produce a finite list of all arboreal skeleta for a given Weinstein manifold? And is there a good notion of a wall-and-chamber structure separating these in the moduli spac...

open_problem | open | MathDB 页面 | 原文线索

V2-029026 | MATHDB-385380

Can we find bordered generating function models for arboreal singularities?

摘录 / EXCERPT: Can we find bordered generating function models for arboreal singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-029027 | MATHDB-385381

Can any arboreal singularity can be obtained as a symplectic reduction of a product of trivalent vertices?

摘录 / EXCERPT: Can any arboreal singularity can be obtained as a symplectic reduction of a product of trivalent vertices?

open_problem | open | MathDB 页面 | 原文线索

V2-029028 | MATHDB-385382

For a Weinstein manifold ... , what are the obstructions to finding a skeleton with a restricted class of singularities?

摘录 / EXCERPT: For a Weinstein manifold W, what are the obstructions to finding a skeleton with a restricted class of singularities? What if we restrict to certain signed arboreal singularities...

open_problem | open | MathDB 页面 | 原文线索

V2-029029 | MATHDB-385383

For a Weinstein manifold ... , what are the obstructions to finding a skeleton without strictly generalized arboreal singularities?

摘录 / EXCERPT: For a Weinstein manifold W, what are the obstructions to finding a skeleton without strictly generalized arboreal singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-029030 | MATHDB-385385

What restrictions are imposed on an arboreal singularity by the requirement that it be the skeleton of the complement of a Donaldon divisor?

摘录 / EXCERPT: What restrictions are imposed on an arboreal singularity by the requirement that it be the skeleton of the complement of a Donaldon divisor? (For instance, can you say something ab...

open_problem | open | MathDB 页面 | 原文线索

V2-029031 | MATHDB-385386

Examples

摘录 / EXCERPT: What is an instructive, manageable playground for dealing with arboreal singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-029032 | MATHDB-385387

What are naturally occurring examples of the ... singularity in skeleta of Weinstein manifolds?

摘录 / EXCERPT: What are naturally occurring examples of the D_4 singularity in skeleta of Weinstein manifolds?

open_problem | open | MathDB 页面 | 原文线索

V2-029033 | MATHDB-385388

Give some notion of metric on the space of skeleta so that arboreal things are generic.

摘录 / EXCERPT: Give some notion of metric on the space of skeleta so that arboreal things are generic.

open_problem | open | MathDB 页面 | 原文线索

V2-029034 | MATHDB-385389

Is there a notion of a "Weinstein join" of Weinstein manifolds which on the skeleton is really the join?

摘录 / EXCERPT: Is there a notion of a "Weinstein join" of Weinstein manifolds which on the skeleton is really the join?

open_problem | open | MathDB 页面 | 原文线索

V2-029035 | MATHDB-385390

How can we understand the resulting very singular skeleton?

摘录 / EXCERPT: How can we understand the resulting very singular skeleton? For instance, how do we calculate its symplectic invariants?

open_problem | open | MathDB 页面 | 原文线索

V2-029036 | MATHDB-385391

Describe the category of microlocal sheaves along the skeleton of the Weinstein manifold obtained from ... by Weinstein handle attachment, as well as the effects of surgery on this category.

摘录 / EXCERPT: Describe the category of microlocal sheaves along the skeleton of the Weinstein manifold obtained from $T^*\Sigma_g$ by Weinstein handle attachment, as well as the effects of surge...

open_problem | open | MathDB 页面 | 原文线索

V2-029037 | MATHDB-385392

Provide general constructions and explanations describing these sort of relations between skeleta of 4- and 6-dimensional Weinstein manifolds.

摘录 / EXCERPT: Provide general constructions and explanations describing these sort of relations between skeleta of 4- and 6-dimensional Weinstein manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029038 | MATHDB-385393

Can any smooth stable ... -category be realized as a finite colimit of categories associated to arboreal singularities?

摘录 / EXCERPT: Can any smooth stable ∞-category be realized as a finite colimit of categories associated to arboreal singularities? (This turns out to be true for tautological reasons, so...

open_problem | open | MathDB 页面 | 原文线索

V2-029039 | MATHDB-385394

What are the Reeb dynamics on the boundary of a tubular neighborhood of a conical arboreal Lagrangian in ... ?

摘录 / EXCERPT: What are the Reeb dynamics on the boundary of a tubular neighborhood of a conical arboreal Lagrangian in $\mathbb{C}^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-029040 | MATHDB-385395

Geometric applications (problem 2)

摘录 / EXCERPT: Generalizing Dehn twists and t^ete-a-t^ete symplectomorphisms, can we associate symplectomorphisms to more general classes of arboreal Lagrangians?

open_problem | open | MathDB 页面 | 原文线索

V2-029041 | MATHDB-385396

Is there a notion of arboreal skeleton for a contact manifold?

摘录 / EXCERPT: Is there a notion of arboreal skeleton for a contact manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-029042 | MATHDB-385398

Geometric applications (problem 5)

摘录 / EXCERPT: Can the t^ete-a-t^ete automorphism be expressed as a product of right-handed Dehn twists?

open_problem | open | MathDB 页面 | 原文线索

V2-029043 | MATHDB-385399

Is there a theory of cobordism for closed arboreal spaces?

摘录 / EXCERPT: Is there a theory of cobordism for closed arboreal spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-029044 | MATHDB-385400

Give the correct definition of "microlocal vector bundles" on an arboreal space.

摘录 / EXCERPT: Give the correct definition of "microlocal vector bundles" on an arboreal space.

open_problem | open | MathDB 页面 | 原文线索

V2-029045 | MATHDB-385401

Is it possible to set up Floer theory with boundary on an arboreal skeleton?

摘录 / EXCERPT: Is it possible to set up Floer theory with boundary on an arboreal skeleton? What are arboreal Lagrangian boundary conditions?

open_problem | open | MathDB 页面 | 原文线索

V2-029046 | MATHDB-385402

Structures in arboreal geometry (problem 4)

摘录 / EXCERPT: Suppose a Weinstein manifold W, with arboreal skeleton L, has a holomorphic symplectic structure. How can we use this structure to enhance the arboreal space L with extra dat...

open_problem | open | MathDB 页面 | 原文线索

V2-029047 | MATHDB-385404

Are there any fully stopped compact contact manifolds-with-convex-boundary other than ... ?

摘录 / EXCERPT: Are there any fully stopped compact contact manifolds-with-convex-boundary other than $F\times[0,1]$?

open_problem | open | MathDB 页面 | 原文线索

V2-029048 | MATHDB-385405

Liouville sectors (problem 2)

摘录 / EXCERPT: Can we recover F from Λ? Can we do so uniquely?

open_problem | open | MathDB 页面 | 原文线索

V2-029049 | MATHDB-385406

Liouville sectors (problem 3)

摘录 / EXCERPT: Find a geometric description of the adjoint of the functor $\mathcal{W}(X)\to\mathcal{W}(X')$ for an inclusion $X\hookrightarrow X'$ of Liouville sectors.

open_problem | open | MathDB 页面 | 原文线索

V2-029050 | MATHDB-385407

Liouville sectors (problem 4)

摘录 / EXCERPT: Construct the embedding $\mathcal{W}(X)\otimes\mathcal{W}(X')\to\mathcal{W}(X\times X')$ for X,X' Liouville sectors.

open_problem | open | MathDB 页面 | 原文线索

V2-029051 | MATHDB-385408

Is there a "local-to-global" principle for verifying "low-action generation"?

摘录 / EXCERPT: Is there a "local-to-global" principle for verifying "low-action generation"?

open_problem | open | MathDB 页面 | 原文线索

V2-029052 | MATHDB-385409

Liouville sectors (problem 6)

摘录 / EXCERPT: What is the generation criterion for $\mathcal{W}(X)$, X a Liouville sector?

open_problem | open | MathDB 页面 | 原文线索

V2-029053 | MATHDB-385410

Liouville sectors (problem 7)

摘录 / EXCERPT: Given a Liouville hypersurface $F\subset\partial_\infty \bar{X}$, is there a Lefschetz fibration $\pi:\bar{X}\to\mathbb{C}$ for which F is a Liouville subdomain of the fiber of...

open_problem | open | MathDB 页面 | 原文线索

V2-029054 | MATHDB-385411

Liouville sectors (problem 8)

摘录 / EXCERPT: Is there any precise sense in which $\text{colim}_n\mathcal{W}(X\times T^*B^n)$ is better behaved than $\mathcal{W}(X)$?

open_problem | open | MathDB 页面 | 原文线索

V2-029055 | MATHDB-385412

Liouville sectors (problem 9)

摘录 / EXCERPT: Suppose that X is a Liouville manifold with smooth Lagrangian core L. Is X equivalent to T^*L as exact symplectic manifolds? Can we define the "fibers" as objects of...

open_problem | open | MathDB 页面 | 原文线索

V2-029056 | MATHDB-385413

Eigenvalues and Curves

摘录 / EXCERPT: Do the eigenvalues of $a^w(x,hD)$ lie on or cluster around curves in the complex plane $\mathbb{C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-029057 | MATHDB-385416

Hyper-K\"{a}hler Structure on $T^* M$

摘录 / EXCERPT: Let M be a K"{a}hler manifold. Must T^*M admit a hyper-K"{a}hler structure?

open_problem | open | MathDB 页面 | 原文线索

V2-029058 | MATHDB-385418

Monge-Ampère equations (problem 1)

摘录 / EXCERPT: Let u plurisubharmonic satisfy ... on $B_1(0)$ with Dirichlet boundary conditions, then u is rotationally symmetric

open_problem | open | MathDB 页面 | 原文线索

V2-029059 | MATHDB-385419

Monge-Ampère equations (problem 2)

摘录 / EXCERPT: Let u strictly convex with Dirichlet boundary conditions solve ... then $u=\varphi+w$ for φ smooth uniformly convex cone, and w smooth.

open_problem | open | MathDB 页面 | 原文线索

V2-029060 | MATHDB-385420

Monge-Ampère equations (problem 3)

摘录 / EXCERPT: For u strictly convex, determine the existence and regularity of solutions to ...

open_problem | open | MathDB 页面 | 原文线索

V2-029061 | MATHDB-385421

Monge-Ampère equations (problem 4)

摘录 / EXCERPT: (textit{Conjecture of Nadirashvili-Vlu{a}duc{t}}) Any viscosity solution of the special Lagrangian equation on the unit ball is $C^1(B)$ for sufficiently smooth boundary data.

open_problem | open | MathDB 页面 | 原文线索

V2-029062 | MATHDB-385423

Monge-Ampère equations (problem 6)

摘录 / EXCERPT: Let u be a solution to ... on $T^n\times[0,1]$ with convex boundary data. Prove a C^2 estimate for $u(\cdot,t)$ for t fixed.

open_problem | open | MathDB 页面 | 原文线索

V2-029063 | MATHDB-385425

Monge-Ampère equations (problem 8)

摘录 / EXCERPT: Let u be plurisubharmonic solution to $\mathrm{det}(u_{\overline k j}) =1$ on $\mathbb C^n$ satisfying ... then u is quadratic.

open_problem | open | MathDB 页面 | 原文线索

V2-029064 | MATHDB-385426

Monge-Ampère equations (problem 9)

摘录 / EXCERPT: Let $u:\mathbb R^2 \to \mathbb R$ be a solution of ... If $u(x) \leq C(1 + |x|^2)$, then u must be quadratic. Can the growth condition be removed? If one considers instead solution...

open_problem | open | MathDB 页面 | 原文线索

V2-029065 | MATHDB-385427

Monge-Ampère equations (problem 10)

摘录 / EXCERPT: Let $\mathrm{det}(u_{\overline k j}) = \frac {f(z)}{|z|^\alpha}$ for $f(z)$ smooth. What is the optimal regularity of u?

open_problem | open | MathDB 页面 | 原文线索

V2-029066 | MATHDB-385429

Complex geometry (problem 2)

摘录 / EXCERPT: (textit{Goldberg Conjecture}) Is a compact, almost K"ahler, Einstein manifold necessarily K"ahler?

open_problem | open | MathDB 页面 | 原文线索

V2-029067 | MATHDB-385430

Complex geometry (problem 3)

摘录 / EXCERPT: (Question of Donaldson) Find smooth solutions to the equation ... for $\lambda>0$ on $M \times [0,1]$ with $\varphi|_{M\times\{0\}}=0$, $\varphi|_{M\times\{0\}}=\varphi_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-029068 | MATHDB-385432

Find a counterexample to the maximal rank conjecture.

摘录 / EXCERPT: Find a counterexample to the maximal rank conjecture. textbf{emph{Conjecture}} (Maximal Rank Conjecture) Let $A\subseteq\mathbb C$ be the annulus...

open_problem | open | MathDB 页面 | 原文线索

V2-029069 | MATHDB-385433

Complex geometry (problem 6)

摘录 / EXCERPT: Let u be a plurisubharmonic function satisfying ... with $\psi>0$ and smooth. Assume u is $C^{1,\alpha}$ for $\alpha > 1 - \frac 2 n$, then $u\in C^\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-029070 | MATHDB-385439

Geometric flows (problem 2)

摘录 / EXCERPT: What is the behavior of $g_{j \overline k}(t)$ solving the Calabi flow, ... Can long time existence be established for toric surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-029071 | MATHDB-385440

Geometric flows (problem 3)

摘录 / EXCERPT: Consider the flow of inverse Hermitian metrics on a compact complex manifold: ... One has short time existence, and it is known that at the maximal time of existence ... Can this be im...

open_problem | open | MathDB 页面 | 原文线索

V2-029072 | MATHDB-385444

Discrete geometry and automorphic forms (problem 3)

摘录 / EXCERPT: Which lattices Λ in $\mathbb R^d$ of determinant 1 maximize ... For $s \to \infty$ this is the covering problem. On S^{n-1} this is called polarization.

open_problem | open | MathDB 页面 | 原文线索

V2-029073 | MATHDB-385449

To obtain LP bound in ... numerically, use linear combinations of different Gaussian (with complex parameters).

摘录 / EXCERPT: To obtain LP bound in $\mathbb R^n$ numerically, use linear combinations of different Gaussian (with complex parameters).

open_problem | open | MathDB 页面 | 原文线索

V2-029074 | MATHDB-385450

Discrete geometry and automorphic forms (problem 9)

摘录 / EXCERPT: Let $A_-(d)$ be the minimum or r for which there exists a Schwartz function $f \colon \mathbb R^d \to \mathbb R$ such that $f(0) = \hat f(0) = 1$, $f(x) \leq 0$ for $|x| \geq r$,...

open_problem | open | MathDB 页面 | 原文线索

V2-029075 | MATHDB-385451

Discrete geometry and automorphic forms (problem 10)

摘录 / EXCERPT: It is know that $A_+(12) = \sqrt{2}$ from the paper by Kumar and Gonc{c}alves. Numerically it seems $A_+'(12) = \sqrt{8}/63 = (\sqrt{2})^9/504$. Can we prove this?

open_problem | open | MathDB 页面 | 原文线索

V2-029076 | MATHDB-385452

Discrete geometry and automorphic forms (problem 11)

摘录 / EXCERPT: Uniqueness of 4-dimensional kissing configuration.

open_problem | open | MathDB 页面 | 原文线索

V2-029077 | MATHDB-385454

Discrete geometry and automorphic forms (problem 13)

摘录 / EXCERPT: Consider the following 3-point bound generalization of the Cohn-Elkies bound: ... where ... This gives an upper bound on the optimal Lattice sphere packing center density. Computatio...

open_problem | open | MathDB 页面 | 原文线索

V2-029078 | MATHDB-385455

Discrete geometry and automorphic forms (problem 14)

摘录 / EXCERPT: Minimize energy for $f(r) = e^{-\alpha r^2}$ in $\mathbb{R}^n$ as $\alpha \rightarrow 0$

open_problem | open | MathDB 页面 | 原文线索

V2-029079 | MATHDB-385458

Discrete geometry and automorphic forms (problem 17)

摘录 / EXCERPT: Numerical computation of LP bound in $\mathbb{H}^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-029080 | MATHDB-385460

Prove there is no lattice whose shells are spherical ... -designs

摘录 / EXCERPT: Prove there is no lattice whose shells are spherical 12-designs

open_problem | open | MathDB 页面 | 原文线索

V2-029081 | MATHDB-385463

Discrete geometry and automorphic forms (problem 22)

摘录 / EXCERPT: Suppose we have a radial function f on $\mathbb{R}^n$ ($0 < n < 10$) such that f has a single root at $\sqrt{2}$, double roots at $\sqrt{2k}$ for $k > 1$, and $\hat f$ has doub...

open_problem | open | MathDB 页面 | 原文线索

V2-029082 | MATHDB-385471

Noncompact Phase Spaces (problem 5)

摘录 / EXCERPT: Extend the results of to noncompact spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-029083 | MATHDB-385479

Generalize the results of Bufetov and Gurevich in to the space of quadratic differentials.

摘录 / EXCERPT: Generalize the results of Bufetov and Gurevich in to the space of quadratic differentials.

open_problem | open | MathDB 页面 | 原文线索

V2-029084 | MATHDB-385486

Define a decomposition and pressure gap for … geodesic flows, and extend for these flows.

摘录 / EXCERPT: Define a decomposition and pressure gap for $\text{CAT}(0)$ geodesic flows, and extend for these flows.

open_problem | open | MathDB 页面 | 原文线索

V2-029085 | MATHDB-385487

Can one simplify the assumptions and/or arguments of to the setting of ?

摘录 / EXCERPT: Can one simplify the assumptions and/or arguments of to the setting of ?

open_problem | open | MathDB 页面 | 原文线索

V2-029086 | MATHDB-385488

Extending the Burns-Climenhaga-Fisher-Thompson Technology (problem 4)

摘录 / EXCERPT: Extend to the case of no focal points. In particular, can one apply this to the case of the geometric potential? In particular, the Donnay-type sphere examples of surfaces with “sp…

open_problem | open | MathDB 页面 | 原文线索

V2-029087 | MATHDB-385489

Can one find a simplified proof of Burns-Gelfert in using, possibly with stronger conclusions?

摘录 / EXCERPT: Can one find a simplified proof of Burns-Gelfert in using, possibly with stronger conclusions? Similarly for the results of Paulin-Policott-Schapira in.

open_problem | open | MathDB 页面 | 原文线索

V2-029088 | MATHDB-385492

Other Directions (problem 3)

摘录 / EXCERPT: Let $f: \mathbb{T}^2 \to \mathbb{T}^2$ be a nonuniformly hyperbolic diffeomorphism. Let μ be the unstable SRB measure, ν be the measure of maximal entropy and m be the me…

open_problem | open | MathDB 页面 | 原文线索

V2-029089 | MATHDB-385498

Develop the theory for weighting orbits in a subadditive (rather than additive) way.

摘录 / EXCERPT: Develop the theory for weighting orbits in a subadditive (rather than additive) way. {rm Ask Kiho for references}

open_problem | open | MathDB 页面 | 原文线索

V2-029090 | MATHDB-385499

Inversive Distance Packings

摘录 / EXCERPT: Let T be a triangulation of the unit sphere with real weights w_{ij} on the edges. Decide if it is possible to produce a a mapping (realization) of vertices to circles such tha…

open_problem | open | MathDB 页面 | 原文线索

V2-029091 | MATHDB-385500

Disk Packing for Laman Graphs

摘录 / EXCERPT: Let G be a planar Laman graph. Does there exist a packing of disks with contact graph G that has generic radii?

open_problem | open | MathDB 页面 | 原文线索

V2-029092 | MATHDB-385501

Disk Packing on the Unit Sphere

摘录 / EXCERPT: Given a packing of disks on the unit sphere, each with radius $\leq \epsilon$, such that the packing has a nontrivial infinitesimal flex, is the packing unjammed for some…

open_problem | open | MathDB 页面 | 原文线索

V2-029093 | MATHDB-385502

Maximum Contacts in Disk Packing

摘录 / EXCERPT: Find a collection of n objects (e.g. disks of given radii) such that all packings have k contacts where k is $\leq$ the number required for isostaticity for that object but t…

open_problem | open | MathDB 页面 | 原文线索

V2-029094 | MATHDB-385503

Sticky Sphere Packing in 3D

摘录 / EXCERPT: begin{enumerate} item Can there be an edge-length equilibrium stress of a sticky sphere packing in $\mathbb{R}^3$? item Under what conditions does the neighborhood of a point in…

open_problem | open | MathDB 页面 | 原文线索

V2-029095 | MATHDB-385504

Periodic Graphs (problem 1)

摘录 / EXCERPT: Which d-periodic graphs $(\tilde{G}, \Gamma)$ can be obtained by this construction from some finite simple graph G? If so, can G be constructed from the d-periodic graph?

open_problem | open | MathDB 页面 | 原文线索

V2-029096 | MATHDB-385505

Periodic Graphs (problem 2)

摘录 / EXCERPT: Suppose that $(\tilde{G},\tilde{p})$ is periodic in $\mathbb{R}^2$ (with a fixed lattice L), such that there exists an equilibrium stress which is strictly positive on all edges…

open_problem | open | MathDB 页面 | 原文线索

V2-029097 | MATHDB-385506

Periodic Packings

摘录 / EXCERPT: Let $G=(V,E)$ be a multigraph with $|V|>1$ and let (G,ψ) be the Γ-labelled quotient gain graph of the contact graph $\tilde G$ of a periodic packing in $\mathbb{R}^2$ w…

open_problem | open | MathDB 页面 | 原文线索

V2-029098 | MATHDB-385507

Rigidity (problem 1)

摘录 / EXCERPT: Suppose G is redundantly rigid in $\mathbb{R}^d$. If H is another graph with n vertices and m edges with $M_G = M_H$ then is $G \simeq H$?

open_problem | open | MathDB 页面 | 原文线索

V2-029099 | MATHDB-385508

Rigidity (problem 2)

摘录 / EXCERPT: If G is a 4-connected triangulation of a non-spherical closed 2-surface then is G generically globally rigid in dimension 3?

open_problem | open | MathDB 页面 | 原文线索

V2-029100 | MATHDB-385509

Rigidity (problem 3)

摘录 / EXCERPT: Extend Fogelsanger's proof that every triangulation of a closed, non-spherical, surface is infinitesimally rigid in 3-dimensions to apply in the case when two vertices u,v, such...

open_problem | open | MathDB 页面 | 原文线索

V2-029101 | MATHDB-385510

Rigidity (problem 4)

摘录 / EXCERPT: Let $f(n)$ be a function such that for any unit length graph G with n vertices, if G has at least $f(n)$ edges then G contains a non-trivial rigid subgraph (trivial means a...

open_problem | open | MathDB 页面 | 原文线索

V2-029102 | MATHDB-385514

Does the Grove-Petersen-Wu Finiteness Theorem hold in dimension ... , i.e.

摘录 / EXCERPT: Does the Grove-Petersen-Wu Finiteness Theorem hold in dimension 4, i.e. are there only finitely many diffeomorphism types in the class of 4-dimensional Riemannian manifolds sat...

open_problem | open | MathDB 页面 | 原文线索

V2-029103 | MATHDB-385515

Diameter Pinching (problem 2)

摘录 / EXCERPT: If $\sec_M \geq 1$ and $\operatorname{diam}(M) > \frac{\pi}{2}$, must M be diffeomorphic to S^n?

open_problem | open | MathDB 页面 | 原文线索

V2-029104 | MATHDB-385516

Diameter Pinching (problem 3)

摘录 / EXCERPT: Let M^n be a manifold for which $\sec_M \geq 1$ and $\operatorname{diam} (M) \geq \frac{\pi}{2} - \varepsilon(n)$. Is M homeomorphic to a manifold M', where...

open_problem | open | MathDB 页面 | 原文线索

V2-029105 | MATHDB-385518

Collapse and Alexandrov Geometry (problem 2)

摘录 / EXCERPT: Understand DC-structures on manifolds. In particular, does Perelman's Stability Theorem hold in the DC-category. Is PL = DC always?

open_problem | open | MathDB 页面 | 原文线索

V2-029106 | MATHDB-385520

Collapse and Alexandrov Geometry (problem 4)

摘录 / EXCERPT: Suppose X is the non-collapsed Gromov-Hausdorff limit of (M_i^n, g_i), where $|\sec_{M_i}| \leq 1$, and that along every geodesic on M_i one hits a conjugate point before...

open_problem | open | MathDB 页面 | 原文线索

V2-029107 | MATHDB-385521

Collapse and Alexandrov Geometry (problem 5)

摘录 / EXCERPT: Is there a sequence of simply-connected, pointwise strictly $\frac{1}{4}$-pinched manifolds M_i^n, $n > 2$, that collapse?

open_problem | open | MathDB 页面 | 原文线索

V2-029108 | MATHDB-385523

Collapse and Alexandrov Geometry (problem 7)

摘录 / EXCERPT: Study the collapse of Alexandrov spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-029109 | MATHDB-385524

Collapse and Alexandrov Geometry (problem 8)

摘录 / EXCERPT: Loosen this notion to get a “brotherhood” on the manifolds M_0 and a property of such M_0 {it not} known to be possessed by all manifolds of $\sec \geq K$,...

open_problem | open | MathDB 页面 | 原文线索

V2-029110 | MATHDB-385525

Collapse and Alexandrov Geometry (problem 9)

摘录 / EXCERPT: Give an Alexandrov analogue of rational ellipticity. In particular, are manifolds with almost non-negative sectional curvature rationally elliptic?

open_problem | open | MathDB 页面 | 原文线索

V2-029111 | MATHDB-385526

Collapse and Alexandrov Geometry (problem 10)

摘录 / EXCERPT: Given a non-collapsing Gromov-Hausdorff convergence $M_i \longrightarrow X$, can one find a “tangent bundle” structure on X that is sensitive to the diffeomorphism class of the...

open_problem | open | MathDB 页面 | 原文线索

V2-029112 | MATHDB-385528

Is every finite dimensional Alexandrov space a limit of Riemannian manifolds with ... ?

摘录 / EXCERPT: Is every finite dimensional Alexandrov space a limit of Riemannian manifolds with $\sec \geq K$?

open_problem | open | MathDB 页面 | 原文线索

V2-029113 | MATHDB-385530

Collapse and Alexandrov Geometry (problem 14)

摘录 / EXCERPT: Can an n-dimensional torus collapse to an interval?

open_problem | open | MathDB 页面 | 原文线索

V2-029114 | MATHDB-385531

Study the collapse of Riemannian manifolds with boundary which have ... on the interior and controlled boundary concavity.

摘录 / EXCERPT: Study the collapse of Riemannian manifolds with boundary which have $\sec \geq K$ on the interior and controlled boundary concavity.

open_problem | open | MathDB 页面 | 原文线索

V2-029115 | MATHDB-385536

Can one reduce the structure group of a fat principal ... -bundle?

摘录 / EXCERPT: Can one reduce the structure group of a fat principal G-bundle?

open_problem | open | MathDB 页面 | 原文线索

V2-029116 | MATHDB-385540

Classify Riemannian submersions from a Lie group with a bi-invariant metric.

摘录 / EXCERPT: Classify Riemannian submersions from a Lie group with a bi-invariant metric.

open_problem | open | MathDB 页面 | 原文线索

V2-029117 | MATHDB-385544

Study the existence and non-existence of metrics with non-negative curvature on manifolds of cohomogeneity-one.

摘录 / EXCERPT: Study the existence and non-existence of metrics with non-negative curvature on manifolds of cohomogeneity-one.

open_problem | open | MathDB 页面 | 原文线索

V2-029118 | MATHDB-385546

Classify cohomogeneity-one manifolds with ... and at least one totally geodesic principal orbit.

摘录 / EXCERPT: Classify cohomogeneity-one manifolds with $\sec \geq 0$ and at least one totally geodesic principal orbit.

open_problem | open | MathDB 页面 | 原文线索

V2-029119 | MATHDB-385552

Vector Bundles (problem 2)

摘录 / EXCERPT: Which vector bundles over $S^2 \times S^2$ or ${\\mathbb{C}} P^2 \# \pm {\\mathbb{C}} P^2$ where the structure group does not reduce to a torus admit $\sec \geq 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-029120 | MATHDB-385553

Vector Bundles (problem 3)

摘录 / EXCERPT: Classify metrics with $\sec \geq 0$ on $S^2 \times {\\mathbb{R}}^4$ (or, more generally, on $S^n \times {\\mathbb{R}}^k$).

open_problem | open | MathDB 页面 | 原文线索

V2-029121 | MATHDB-385554

Vector Bundles (problem 4)

摘录 / EXCERPT: Suppose $E \longrightarrow M$ is a vector bundle over a compact, simply-connected manifold M for which $\sec_M \geq 0$. Does $E \oplus {\\mathbb{R}}^k \longrightarrow M$ have...

open_problem | open | MathDB 页面 | 原文线索

V2-029122 | MATHDB-385555

Which theorems from ... carry over to positive curvature on an open dense set?

摘录 / EXCERPT: Which theorems from $\sec > 0$ carry over to positive curvature on an open dense set?

open_problem | open | MathDB 页面 | 原文线索

V2-029123 | MATHDB-385556

Quasi-positive Curvature and Positive Curvature on an Open Dense Set (problem 2)

摘录 / EXCERPT: Suppose G is a compact Lie group with a left-invariant metric. Are there any new examples $H \backslash G$ with quasi-positive curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029124 | MATHDB-385558

Quasi-positive Curvature and Positive Curvature on an Open Dense Set (problem 4)

摘录 / EXCERPT: Fix $k \in {\\mathbb{N}}$. Is there a $n_0 = n_0 (k)$ such that for any quasi-positively curved manifold (M^n, g) with $n \geq n_0$ and $\operatorname{cohom}(M^n,g) \leq k$, ther...

open_problem | open | MathDB 页面 | 原文线索

V2-029125 | MATHDB-385562

Is a “generic” manifold a ... -space (where “generic” is to be determined)?

摘录 / EXCERPT: Is a “generic” manifold a $K(\pi, 1)$-space (where “generic” is to be determined)?

open_problem | open | MathDB 页面 | 原文线索

V2-029126 | MATHDB-385563

Does positive sectional curvature imply that the manifold is formal (in the sense of Sullivan's minimal model)?

摘录 / EXCERPT: Does positive sectional curvature imply that the manifold is formal (in the sense of Sullivan's minimal model)?

open_problem | open | MathDB 页面 | 原文线索

V2-029127 | MATHDB-385566

Miscellaneous Problems (problem 5)

摘录 / EXCERPT: Is there a $\delta(n) > 0$ such that any n-dimensional, positively curved manifold carries a $\delta(n)$-pinched metric?

open_problem | open | MathDB 页面 | 原文线索

V2-029128 | MATHDB-385569

Dualize the theorem. If we replace generalized zonoids by "generalized intersection bodies", Minkowski sums by radial sums, etc, does a similar theorem hold?

摘录 / EXCERPT: Dualize the theorem. If we replace generalized zonoids by "generalized intersection bodies", Minkowski sums by radial sums, etc, does a similar theorem hold?

open_problem | open | MathDB 页面 | 原文线索

V2-029129 | MATHDB-385572

In ... , how many segments do we need to approximate a zonoid by zonotopes?

摘录 / EXCERPT: In $\mathbb{R}^n$, how many segments do we need to approximate a zonoid by zonotopes? Let K be a zonoid and Z a zonotope that is the sum of M segments. If...

open_problem | open | MathDB 页面 | 原文线索

V2-029130 | MATHDB-385582

Minimum of slabs of the cube B^n_∞

摘录 / EXCERPT: There are numbers t_1 and t_2 such that for $2\sqrt{2}-2 < t < t_1$, the minimum is in the direction of $(1,1,0,\ldots,0)$; for $t_1 < t < t_2$, the minimum is in the direction...

open_problem | open | MathDB 页面 | 原文线索

V2-029131 | MATHDB-385591

Open Problems (problem 1)

摘录 / EXCERPT: Suppose we are given a closed 4-manifold $(M^4, [g])$ which is not conformally equivalent to the standard 4-sphere, and for which P_g is nonnegative, has trivial kernel, and...

open_problem | open | MathDB 页面 | 原文线索

V2-029132 | MATHDB-385592

Open Problems (problem 2)

摘录 / EXCERPT: Let (X^{n+1}, g_+) be a Poincar'e Einstein manifold and assume $Y^{k+1}\subseteq X^{n+1}$, k odd, is minimal and regular at infinity. Does there exist a local conformal invaria...

open_problem | open | MathDB 页面 | 原文线索

V2-029133 | MATHDB-385593

Open Problems (problem 3)

摘录 / EXCERPT: (A) Generalize formula of G-M-T (see below) to higher dimensions. ... (B) Generalize (A) to higher co-dimension. Dimension of X_+ is even, dimension of Y is odd.

open_problem | open | MathDB 页面 | 原文线索

V2-029134 | MATHDB-385594

Open Problems (problem 4)

摘录 / EXCERPT: Consider a closed 6-manifold M^6. The space $\mathcal{S}$ of global conformal invariants is 4-dimensional: $\mathcal{S}= \text{span}\{I_1,\cdots, I_4\}$. Theorem (Case, 2022):...

open_problem | open | MathDB 页面 | 原文线索

V2-029135 | MATHDB-385596

Are there any “curvature quantity" which transforms under conformal changes via the p-laplace operator?

摘录 / EXCERPT: Are there any “curvature quantity" which transforms under conformal changes via the p-laplace operator? Note: $\int |\nabla u|^n\;dv_g$ is conformally invariant. In general, is the...

open_problem | open | MathDB 页面 | 原文线索

V2-029136 | MATHDB-385597

Open Problems (problem 7)

摘录 / EXCERPT: X^{n+1} compact manifold with $\partial X^{n+1}\not = \emptyset$. Define ... ... Consider (X^{n+1},g_+) CCE, $[g_o] = $ conformal infinity, and assume $Y(\partial X,[g_o])>0$. (Che...

open_problem | open | MathDB 页面 | 原文线索

V2-029137 | MATHDB-385600

Open Problems (problem 10)

摘录 / EXCERPT: Suppose we are given a 4-manifold $(M^4,[g])$ with $\partial M^4\not = \emptyset$ and a 2-dimensional corner Σ. Does there exist a unique $\tilde g = e^{2w}g$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-029138 | MATHDB-385601

Open Problems (problem 11)

摘录 / EXCERPT: Let g_o be a conformally compact metric defined on $\mathbb{B}^4$. Assume $Y(\mathbb{S}^3,[\rho^2 g_o])>0$, where $\rho>0$ is a defining function. Consider the normalized Ricci f...

open_problem | open | MathDB 页面 | 原文线索

V2-029139 | MATHDB-385611

The peripheral locus (problem 2)

摘录 / EXCERPT: When P is nonempty, when do we expect the complement of P to be codimension 2?

open_problem | open | MathDB 页面 | 原文线索

V2-029140 | MATHDB-385616

How does this relate to compactifications coming from tropical geometry?

摘录 / EXCERPT: How does this relate to compactifications coming from tropical geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-029141 | MATHDB-385618

Does the MV-basis equal the ... -basis?

摘录 / EXCERPT: Does the MV-basis equal the Θ-basis?

open_problem | open | MathDB 页面 | 原文线索

V2-029142 | MATHDB-385621

Why does there exist an affine springer fiber with the same betti numbers and a cell decomposition?

摘录 / EXCERPT: Why does there exist an affine springer fiber with the same betti numbers and a cell decomposition? Do the rings match?

open_problem | open | MathDB 页面 | 原文线索

V2-029143 | MATHDB-385623

What do the cluster structures correspond to on the springer fiber?

摘录 / EXCERPT: What do the cluster structures correspond to on the springer fiber?

open_problem | open | MathDB 页面 | 原文线索

V2-029144 | MATHDB-385639

L^1 norm of first eigenfunction.

摘录 / EXCERPT: Let $\Omega\subset \mathbb{R}^n$ be an open set with finite measure, $\lambda_1(\Omega)$ be the first eigenvalue of the Laplacian with Dirichlet boundary conditions, and let ... wher...

open_problem | open | MathDB 页面 | 原文线索

V2-029145 | MATHDB-385640

Maximal gradient of the torsion

摘录 / EXCERPT: This is a report of a question that was raised in (Hoskins, Steinerberger - 2021), and the following statement is based on this paper. Let $\Omega \subset \mathbb{R}^2$ be a convex...

open_problem | open | MathDB 页面 | 原文线索

V2-029146 | MATHDB-385641

Shape of the ground state

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded convex set, and let u be the lowest eigenfunction (ground state) of the Laplacian with Dirichet boundary conditions, with unit L^2...

open_problem | open | MathDB 页面 | 原文线索

V2-029147 | MATHDB-385642

Curl-curl eigenvalue problem

摘录 / EXCERPT: Let Ω be a smooth compact domain of $\mathbb{R}^3$, we define $(\vec{E_k},\lambda_k)_{k\geq 1}$ to be the eigenpair of the Maxwell equation in Ω, as given by the equa...

open_problem | open | MathDB 页面 | 原文线索

V2-029148 | MATHDB-385643

Curl eigenvalue

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz set with volume 1, consider the curl eigenvalue problem % ... More details on the definition may be found in [Enciso, Gerne...

open_problem | open | MathDB 页面 | 原文线索

V2-029149 | MATHDB-385644

Exterior Robin problem

摘录 / EXCERPT: Let $n \geq 3$ and let $\Omega\subset\mathbb{R}^3$ be a smooth compact domain. For a given $\alpha \in \mathbb{R}$, we consider the Robin eigenvalue problem on the complement of...

open_problem | open | MathDB 页面 | 原文线索

V2-029150 | MATHDB-385645

Maximizing Steklov σ_1 on N-gons

摘录 / EXCERPT: One of the “well-known” conjecture of shape optimization is the optimization of λ_1 (first eigenvalue of the Laplacian with Dirichlet boundary condition) among polygonal se...

open_problem | open | MathDB 页面 | 原文线索

V2-029151 | MATHDB-385646

Minimization of p-capacity

摘录 / EXCERPT: Let $p\in (1,2)$, and $\Omega\subset \mathbb{R}^2$ a bounded convex set, we define its p-capacity ... Note that this notion of capacity is not well-defined for $p=2$ (by the scale-...

open_problem | open | MathDB 页面 | 原文线索

V2-029152 | MATHDB-385647

Cheeger's inequality on convex sets

摘录 / EXCERPT: For an open set of finite measure $\Omega\subset\mathbb{R}^n$, denote $\lambda_1(\Omega)$ the first eigenvalue of the Laplacian with Dirichlet boundary conditions, and $h(\Omega)$...

open_problem | open | MathDB 页面 | 原文线索

V2-029153 | MATHDB-385648

Slicing problem in 3 dimensions

摘录 / EXCERPT: Suppose $\Omega \subset \mathbb{R}^3$ is convex, normalized to have volume 1, and with its center of mass at the origin. Let ... The ball minimizes $\det A$. The problem is to find...

open_problem | open | MathDB 页面 | 原文线索

V2-029154 | MATHDB-385649

Rearrangement problem

摘录 / EXCERPT: Let $F:\mathbb{R}\to\mathbb{R}$ be an even convex function, then it is known (from Xi, Zhao (2022)) that for any function $f,g\in L^1(\mathbb{R}^n)$ (for some $n\geq 1$), denoting...

open_problem | open | MathDB 页面 | 原文线索

V2-029155 | MATHDB-385651

Is the map from Goncahrov's projectivised scissors congruence groups to the Hopf algebra of polylogarithms surjective?

摘录 / EXCERPT: Is the map from Goncahrov's projectivised scissors congruence groups to the Hopf algebra of polylogarithms surjective?

open_problem | open | MathDB 页面 | 原文线索

V2-029156 | MATHDB-385652

Scissors Congruence (problem 3)

摘录 / EXCERPT: There are two different spectral sequences arising from Sah & Goncharov; what is their relationship? Relation to the Minkowski Filtration?

open_problem | open | MathDB 页面 | 原文线索

V2-029157 | MATHDB-385653

Is there a direct scanning map proof that scissors congruence ... -theory of polytopes is a Thom spectrum?

摘录 / EXCERPT: Is there a direct scanning map proof that scissors congruence K-theory of polytopes is a Thom spectrum?

open_problem | open | MathDB 页面 | 原文线索

V2-029158 | MATHDB-385661

Kahler Currents

摘录 / EXCERPT: Show that singular Kahler-Einstein metrics (or more generally singular cscK metrics) are in fact Kahler currents. This was shown in [GGZ] (2022) under suitable assumptions. This is...

open_problem | open | MathDB 页面 | 原文线索

V2-029159 | MATHDB-385662

Regularity of Kahler-Einstein Metrics

摘录 / EXCERPT: Are the local potentials of a singular Kahler-Einstein metric Holder continuous (with respect to the Fubini-Study metric), given that the singularity is smoothable? This is known i...

open_problem | open | MathDB 页面 | 原文线索

V2-029160 | MATHDB-385663

Convexity of the Regular Set

摘录 / EXCERPT: Is the regular set of a singular Kahler-Einstein manifold (X,ω_{KE}) convex? That is, is it true that for any two points $x,y\in X_{\text{reg}}$, there is a minimal geodesic...

open_problem | open | MathDB 页面 | 原文线索

V2-029161 | MATHDB-385664

Uniform Bounds in Families

摘录 / EXCERPT: Problem 5. Let X be a connected component of the moduli space of Kahler-Einstein metrics on a compact complex manifold Y of general type, and let $\pi:\mathcal{X}\to X$ be the co...

open_problem | open | MathDB 页面 | 原文线索

V2-029162 | MATHDB-385665

Volume Bounds for Sublevel Sets

摘录 / EXCERPT: It is known that given a bounded pseudoconvex domain $\Omega\subseteq\mathbb{C}^{n}$, there exist $C=C(\Omega)$ and $\alpha=\alpha(\Omega)$ such that for any $u\in PSH(\Omega)$ sat...

open_problem | open | MathDB 页面 | 原文线索

V2-029163 | MATHDB-385666

C^0 Bounds for Complex Hessian Equations

摘录 / EXCERPT: Suppose (X,ω_{X}) is a Kahler manifold and ω_{0} is another Kahler metric on X. Suppose $C_{1}<\infty$ satisfies ... Let $u\in C^{\infty}(X)$, and let $\lambda[u]$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-029164 | MATHDB-385667

Gradient Estimate

摘录 / EXCERPT: Consider the equation ... where ω_{0},ω_{X} are nonnegative Hermitian $(1,1)$-forms and $\lambda[u]$ are the eigenvalues of...

open_problem | open | MathDB 页面 | 原文线索

V2-029165 | MATHDB-385668

Liouville Property

摘录 / EXCERPT: Suppose $\det(\partial_{i}\partial_{\overline{j}}u)=1$ on $\mathbb{C}^{n}$ and ... for all $z\in\mathbb{C}^{n}$. Is it true that u It known if, in addition,...

open_problem | open | MathDB 页面 | 原文线索

V2-029166 | MATHDB-385669

Flow Convergence from Perturbation of Elliptic Solution

摘录 / EXCERPT: Suppose (M,ω) is a complete Kaher manifold with bounded geometry, and $u\in C^{\infty}(M)$ solves ... and satisfies ... ... as $d_{\omega}(z_{0},z)\to\infty$, for some fixed point...

open_problem | open | MathDB 页面 | 原文线索

V2-029167 | MATHDB-385670

Kahler-Ricci Tangent Flows

摘录 / EXCERPT: Given a closed Kahler-Ricci flow $(M,(\omega_{t})_{t\in[0,T)})$, under what conditions are tangent flows complex analytic spaces? It is known if M is Fano or if $n=2$. It is conj...

open_problem | open | MathDB 页面 | 原文线索

V2-029168 | MATHDB-385671

Maximal Existence Time for Pluriclosed Flow

摘录 / EXCERPT: A pluriclosed metric is a positive $(1,1)$-form η such that $\sqrt{-1}\partial\overline{\partial}\eta=0$. The pluriclosed flow is ... where Rc_{B} is the Ricci curvature of th...

open_problem | open | MathDB 页面 | 原文线索

V2-029169 | MATHDB-385672

Expanding Solitons from Divisorial Contractions

摘录 / EXCERPT: Consider a Kahler-Ricci Flow $(M^{2},(g_{t})_{t\in[0,T)})$ developing a finite-time singularity with $\lim_{t\to T}\text{Vol}_{\omega_{t}}(M)>0$. Doing a Type-I blowup backwards in...

open_problem | open | MathDB 页面 | 原文线索

V2-029170 | MATHDB-385673

Long-Time Existence of Complete Noncompact Kahler-Ricci Flow

摘录 / EXCERPT: If $\sqrt{-1}\partial\overline{\partial}u$ is complete and $U(n)$-invariant Kahler metric on $\mathbb{C}^{n}$, does there exist a solution to the Kahler-Ricci flow? This is known i...

open_problem | open | MathDB 页面 | 原文线索

V2-029171 | MATHDB-385674

Diameter of Moduli of Kahler-Einstein Metrics of General Type

摘录 / EXCERPT: Does the moduli space of Kahler-Einstein manifolds of general type equipped with the Weil-Petersen metric have bounded diameter? It is known that local potentials are continuous up...

open_problem | open | MathDB 页面 | 原文线索

V2-029172 | MATHDB-385675

Properness of Harmonic Map Energy

摘录 / EXCERPT: Given a Teichmuller space of Kahler-Einstein manifolds (this is the universal cover of the moduli space) of general type, the energy functional for maps into a Riemannian manifold...

open_problem | open | MathDB 页面 | 原文线索

V2-029173 | MATHDB-385681

Can we understand geometrically the Serre functor of ... ?

摘录 / EXCERPT: Can we understand geometrically the Serre functor of D_{Cox}?

open_problem | open | MathDB 页面 | 原文线索

V2-029174 | MATHDB-385682

Questions about Cox Category (problem 3)

摘录 / EXCERPT: Is there a Landau-Ginzburg mirror to $D_{Cox}?$

open_problem | open | MathDB 页面 | 原文线索

V2-029175 | MATHDB-385683

Questions about Cox Category (problem 4)

摘录 / EXCERPT: Is there a notion of Koszul duality for $D_{Cox}?$.

open_problem | open | MathDB 页面 | 原文线索

V2-029176 | MATHDB-385684

Questions about Cox Category (problem 5)

摘录 / EXCERPT: Given a wall in the secondary fan, one can consider a sub-GIT problem with the corresponding D_{Cox} category. Is there a functorial relationship between these D_{Cox} categori...

open_problem | open | MathDB 页面 | 原文线索

V2-029177 | MATHDB-385685

Do there exist analogs of ... for other spaces?

摘录 / EXCERPT: Do there exist analogs of D_{Cox} for other spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-029178 | MATHDB-385686

Can we implement computations in M2 for toric stacks?

摘录 / EXCERPT: Can we implement computations in M2 for toric stacks?

open_problem | open | MathDB 页面 | 原文线索

V2-029179 | MATHDB-385688

Can M2 compute the exceptional collection constructed in the work of Favero, Ballard, Katzarkov?

摘录 / EXCERPT: Can M2 compute the exceptional collection constructed in the work of Favero, Ballard, Katzarkov?

open_problem | open | MathDB 页面 | 原文线索

V2-029180 | MATHDB-385689

Can we build functoriality in M2 to work with secondary fans?

摘录 / EXCERPT: Can we build functoriality in M2 to work with secondary fans?

open_problem | open | MathDB 页面 | 原文线索

V2-029181 | MATHDB-385690

Can we implement symplectic/sheaf theoretic computations of linking disks/microstalks in M2?

摘录 / EXCERPT: Can we implement symplectic/sheaf theoretic computations of linking disks/microstalks in M2?

open_problem | open | MathDB 页面 | 原文线索

V2-029182 | MATHDB-385691

How do the choices of stop data in the cotangent bundle relate to the choices of the irrelevant ideal in the Cox ring?

摘录 / EXCERPT: How do the choices of stop data in the cotangent bundle relate to the choices of the irrelevant ideal in the Cox ring?

open_problem | open | MathDB 页面 | 原文线索

V2-029183 | MATHDB-385692

Clarifying and Expanding Toric Mirror Symmetry (problem 2)

摘录 / EXCERPT: Given a choice of a mirror skeleton, one can consider its canonical probe generators. What is the geometric meaning of the mutated collection of these generators? Can we find anoth...

open_problem | open | MathDB 页面 | 原文线索

V2-029184 | MATHDB-385693

Does Morelli's algebro-geometric cobordism in the context of weak factorizations have a mirror interpretation?

摘录 / EXCERPT: Does Morelli's algebro-geometric cobordism in the context of weak factorizations have a mirror interpretation? What about elementary cobordism?

open_problem | open | MathDB 页面 | 原文线索

V2-029185 | MATHDB-385694

What is the algebro-geometric "mirror" to Pontryagin classes of Lagrangians in ... ?

摘录 / EXCERPT: What is the algebro-geometric "mirror" to Pontryagin classes of Lagrangians in $(T^{*}\mathbb{T}^n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-029186 | MATHDB-385696

Is Bondal-Thompsen collection related to Bridgeland stability conditions?

摘录 / EXCERPT: Is Bondal-Thompsen collection related to Bridgeland stability conditions?

open_problem | open | MathDB 页面 | 原文线索

V2-029187 | MATHDB-386503

Mappings between balls (problem 2)

摘录 / EXCERPT: Assume $M \subseteq \mathbb{C}^{n}$ is a real-analytic generic submanifold, $f = \frac{p}{q}:\mathbb{C}^{n} \rightarrow \mathbb{C}^{N}$ is the germ of a meromorphic map at...

open_problem | open | MathDB 页面 | 原文线索

V2-029188 | MATHDB-386854

Is there an explicit description of the ... classes only in terms of the geometry of the source curve, which is coherent?

摘录 / EXCERPT: Is there an explicit description of the ψ classes only in terms of the geometry of the source curve, which is coherent?

open_problem | open | MathDB 页面 | 原文线索

V2-029189 | MATHDB-388278

Antipodes of symmetric convex bodies

摘录 / EXCERPT: On a centrally symmetric convex body, must every pair of points at maximum intrinsic surface distance be antipodal? Resolve this even for rectangular boxes.

open_problem | open | MathDB 页面 | 原文线索

V2-029190 | MATHDB-388775

Boundaries of Groups and Kleinian Groups - Problem 35

摘录 / EXCERPT: Are diffeomorphisms Rn →Rn dense in the space of all quasiconformal maps?

open_problem | open | MathDB 页面 | 原文线索

V2-029191 | MATHDB-388981

Closed geodesics

摘录 / EXCERPT: Is this true without the bumpy assumption?

open_problem | open | MathDB 页面 | 原文线索

V2-029192 | MATHDB-390595

Feldman-Ilmanen-Knopf conjecture

摘录 / EXCERPT: Let Y be a compact Kähler manifold, let $p\in Y$, and let $\pi:M=\operatorname{Bl}_pY\to Y$ be the blow-up at p. Suppose that a Kähler-Ricci flow on M with initial Kähler cla...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029193 | MATHDB-392670

Realization of connected or countable discrete groups as automorphism groups

摘录 / EXCERPT: Let G be a finite-dimensional real connected Lie group or a countable discrete group. Then there exists a Stein, hyperbolic, connected complex manifold X such that G is isomo...

open_problem | open | MathDB 页面 | 原文线索

V2-029194 | MATHDB-392676

Solvability criterion for nonlinear boundary interpolation

摘录 / EXCERPT: Given a smooth family of manifolds M_t for $t\in\mathbb{T}$ and prescribed data $t_j\in\mathbb{T}$, $w_j\in M_{t_j}$, $j=0,\ldots,k$, determine a necessary-and-sufficient criteri...

open_problem | open | MathDB 页面 | 原文线索

V2-029195 | MATHDB-392726

McKay equivalence for symplectic quotient singularities

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic 0, let V be a finite-dimensional K-vector space equipped with a non-degenerate skew-symmetric form...

open_problem | open | MathDB 页面 | 原文线索

V2-029196 | MATHDB-392758

Convergence of the Goodwillie-Weiss Taylor tower for 3-dimensional long knots

摘录 / EXCERPT: Let M be a smooth manifold with nonempty boundary, and let $\operatorname{Emb}(I,M;\partial)$ be the space of long knots defined in the report. The report proves that if...

open_problem | open | MathDB 页面 | 原文线索

V2-029197 | MATHDB-392776

Standardness of the complex flag manifold $SU(5)/S(U(3)\times U(2))$

摘录 / EXCERPT: Every locally smooth action of $SU(5)$ on the complex flag manifold $SU(5)/S(U(3)\times U(2))$ is conjugate to the standard action by left translation.

open_problem | open | MathDB 页面 | 原文线索

V2-029198 | MATHDB-392777

Albanese-type fibration conjecture in the modification case of the Shafarevich conjecture

摘录 / EXCERPT: Let X be a complex projective manifold whose universal cover is in the modification case of the Shafarevich map, meaning that through the general point of the universal cover the...

conjecture | open | MathDB 页面 | 原文线索

V2-029199 | MATHDB-392778

Campana’s modified Kodaira-dimension conjecture

摘录 / EXCERPT: Conjecture: For every projective manifold X that is not uniruled, $\ba^+(X)=\ba(X)$, where $\ba^+(X)$ denotes the modified Kodaira dimension introduced in the report.

conjecture | open | MathDB 页面 | 原文线索

V2-029200 | MATHDB-392782

Removing smoothness from the submanifold hypothesis in the L^2 extension theorem

摘录 / EXCERPT: Can the smoothness assumption on the submanifold in the L^2 extension theorem be removed without strengthening the curvature condition?

conjecture | open | MathDB 页面 | 原文线索

V2-029201 | MATHDB-392786

The CR Yamabe problem

摘录 / EXCERPT: Given a compact strictly pseudoconvex CR manifold of dimension $2n+1$ with a contact form, find a contact form in its conformal class whose Webster pseudo-scalar curvature is const...

open_problem | open | MathDB 页面 | 原文线索

V2-029202 | MATHDB-392790

Volumes of complex hyperbolic manifolds with cusps

摘录 / EXCERPT: Let $P(...

open_problem | open | MathDB 页面 | 原文线索

V2-029203 | MATHDB-392792

Shafarevich conjecture of holomorphic convexity

摘录 / EXCERPT: The universal covering space of a complex projective manifold should be holomorphically convex; equivalently, it should admit a proper holomorphic map to a Stein space.

conjecture | open | MathDB 页面 | 原文线索

V2-029204 | MATHDB-392812

Strominger-Yau-Zaslow conjecture on mirror Calabi-Yau fibrations

摘录 / EXCERPT: If a Calabi-Yau manifold has a mirror, then it must be a fibration with general fibers being three-tori that are special Lagrangian submanifolds. Furthermore, the mirror manifold h...

conjecture | open | MathDB 页面 | 原文线索

V2-029205 | MATHDB-392976

Positivity of the TERP-structure associated with a tame function

摘录 / EXCERPT: Let Y be an affine algebraic manifold and let $f:Y\to\mathbb{C}$ be an algebraic function with only isolated singularities and tame behavior at infinity. Is the associated...

open_problem | open | MathDB 页面 | 原文线索

V2-029206 | MATHDB-393007

Algebraic dimension of a hyperkähler manifold

摘录 / EXCERPT: Is the algebraic dimension of a hyperkähler manifold M necessarily ...

open_problem | open | MathDB 页面 | 原文线索

V2-029207 | MATHDB-393026

Farrell’s criterion for fibering over the circle

摘录 / EXCERPT: Let M be a closed, connected smooth manifold with $\dim M\ge 6$, and let $\xi\in H^1(M;\mathbb Z)$ (equivalently, a class represented by a map $f\colon M\to S^1$). Then ξ is...

open_problem | open | MathDB 页面 | 原文线索

V2-029208 | MATHDB-393029

Marden’s conjecture on tameness of hyperbolic 3-manifolds

摘录 / EXCERPT: Every hyperbolic 3-manifold with finitely generated fundamental group is tame; that is, it is homeomorphic to the interior of a compact manifold with boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-029209 | MATHDB-393030

Multiplicativity of the signature modulo 4 for fibre bundles

摘录 / EXCERPT: For a fibre bundle of orientable manifolds $F\to E\to B$, the signatures are multiplicative modulo 4: ... where $\sigma=0$ when the manifold dimension is not congruent to 0 modul...

open_problem | open | MathDB 页面 | 原文线索

V2-029210 | MATHDB-393032

Criterion for infinitely many components of the deformation space

摘录 / EXCERPT: Let M be a compact hyperbolizable 3-manifold with nonempty incompressible boundary. Then $AH(M)$ has infinitely many components if and only if the characteristic submanifold of...

open_problem | open | MathDB 页面 | 原文线索

V2-029211 | MATHDB-393035

Minimum number of coincidence components for maps between manifolds

摘录 / EXCERPT: Let M and N be smooth connected manifolds without boundary, of strictly positive dimensions m and n, respectively, with M compact. For continuous maps $f_1,f_2:M\to N$, d...

open_problem | open | MathDB 页面 | 原文线索

V2-029212 | MATHDB-393045

Natural direct-limit C*-algebra for a general curved Hilbert manifold

摘录 / EXCERPT: For a general curved Hilbert manifold M with Riemannian metric g, construct a natural direct-limit C^*-algebra ... where the direct limit is taken over finite-dimensional subma...

open_problem | open | MathDB 页面 | 原文线索

V2-029213 | MATHDB-393071

Projective Lichnerowicz Conjecture

摘录 / EXCERPT: Let a connected Lie group G act on a complete connected Riemannian manifold (M^n,g) of dimension $n \ge 2$ by projective transformations. Then G acts by affine transformation...

conjecture | open | MathDB 页面 | 原文线索

V2-029214 | MATHDB-393072

Nonvanishing perturbation of a 1-form on a zero-Euler-characteristic manifold

摘录 / EXCERPT: Conjecture: Let M be a closed manifold with zero Euler characteristic, and let α be a differential 1-form on M such that $d\alpha\ne 0$. Then there exists a function...

open_problem | open | MathDB 页面 | 原文线索

V2-029215 | MATHDB-393073

Existence of a 3-dimensional totally skew disc in 7-dimensional space

摘录 / EXCERPT: Does there exist a 3-dimensional totally skew disc in 7-dimensional space, i.e. in $\cmathbb{R}^7$? Here, a totally skew submanifold $M^n\subset\mathbb{R}^N$ is one for which, fo...

open_problem | open | MathDB 页面 | 原文线索

V2-029216 | MATHDB-393080

Integrability of tempered Jacobi fields on constant-mean-curvature surfaces

摘录 / EXCERPT: Are tempered (sub-exponential-growth) Jacobi fields on constant mean curvature surfaces always integrable, in the sense that each is the velocity vector field of a one-parameter fa...

open_problem | open | MathDB 页面 | 原文线索

V2-029217 | MATHDB-393083

Polylogarithmic expression for volumes of odd-dimensional hyperbolic manifolds

摘录 / EXCERPT: Let M^{2n-1} be a $(2n-1)$-dimensional hyperbolic manifold of finite volume. Then ... for some $z_i\in\overline{\mathbb{Q}}$ satisfying ...

open_problem | open | MathDB 页面 | 原文线索

V2-029218 | MATHDB-393116

Isoperimetric minimization of the lowest eigenvalue for closed plane curves

摘录 / EXCERPT: For a smooth closed curve $C\subset\mathbb{R}^2$ of length 2π and positive curvature, determine the curve C that minimizes the lowest eigenvalue $\lambda_1(g,C)$ of the opera...

open_problem | open | MathDB 页面 | 原文线索

V2-029219 | MATHDB-393121

Ovals isoperimetric conjecture for closed plane curves

摘录 / EXCERPT: Let C be a smooth closed curve in $\d53d^2$ of length 2π with positive curvature $\d705(s)$. Define ... on $L^2(C)$ with periodic boundary conditions, and let $\u0001d706_1(C)$...

conjecture | open | MathDB 页面 | 原文线索

V2-029220 | MATHDB-393122

Isoperimetric inequality for the lowest eigenvalue of a curvature-dependent Schrödinger operator

摘录 / EXCERPT: Let C be a smooth closed curve in $\mathbb{R}^2$ of length 2π with positive curvature $\kappa(s)$. Define, with periodic boundary conditions on $L^2(C)$, ... where s is arc l...

open_problem | open | MathDB 页面 | 原文线索

V2-029221 | MATHDB-393123

Isoperimetric lower bound for the lowest curvature Schrödinger eigenvalue

摘录 / EXCERPT: Let C be a smooth closed curve in $\a1^2$ of length $2\c0$ with positive curvature $\ba(s)$, where s is arclength. Let ... act on $L^2(C)$ with periodic boundary conditions, and...

open_problem | open | MathDB 页面 | 原文线索

V2-029222 | MATHDB-393124

Eigenvalue sequences associated with finitely many nondegenerate curvature maxima

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a smooth, bounded, simply connected domain, and let H_h be the magnetic Neumann operator defined by ... with magnetic Neumann boundary condition....

open_problem | open | MathDB 页面 | 原文线索

V2-029223 | MATHDB-393125

Asymptotic splitting of the two lowest magnetic Neumann eigenvalues

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a smooth, bounded, simply connected domain. Assume that its boundary curvature $\kappa(s)$ has a unique maximum at s_0, with...

open_problem | open | MathDB 页面 | 原文线索

V2-029224 | MATHDB-393173

Larsen’s conjecture on moments of reductive subgroups

摘录 / EXCERPT: Let $V=\mathbb{C}^{d}$ with $d>4$. Fix a nondegenerate quadratic form on V and, if d is even, a nondegenerate symplectic form, and let $\mathcal{G}$ be one of $GL(V)$, $O(V)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-029225 | MATHDB-393189

Upper bound for the total curvature of the central trajectory

摘录 / EXCERPT: Conjecture: $F(\infty,0)=O\!\left(n^{3.5}\log\left(\bar{\chi}^{*}_{A}+n\right)\right)$, where F is the central-trajectory curvature integral defined in the preceding pages.

open_problem | open | MathDB 页面 | 原文线索

V2-029226 | MATHDB-393199

Milnor's finite-generation conjecture

摘录 / EXCERPT: Every complete Riemannian manifold with non-negative Ricci curvature has a finitely generated fundamental group.

conjecture | open | MathDB 页面 | 原文线索

V2-029227 | MATHDB-393236

Birational Kodaira problem for compact Kähler manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold. Does there exist a bimeromorphic model X' of X which deforms to a projective complex manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-029228 | MATHDB-393251

Dimension of loci of chains of minimal rational curves

摘录 / EXCERPT: Compute the dimension d_k of the locus of the family of length-k chains of minimal rational curves passing through a fixed, general point of a Fano manifold.

open_problem | open | MathDB 页面 | 原文线索

V2-029229 | MATHDB-393260

Interpolation on a smooth manifold with prescribed points and endpoint velocities

摘录 / EXCERPT: Let M be a smooth n-dimensional manifold embedded into $\mathbb{R}^N$ so that, for every $p\in M$, the tangent space T_pM can be considered as an affine subspace of...

open_problem | open | MathDB 页面 | 原文线索

V2-029230 | MATHDB-393380

Global convergence of the Yamabe flow on compact three-manifolds

摘录 / EXCERPT: For a compact three-dimensional manifold of positive conformal Yamabe invariant, the Yamabe flow ... started from an arbitrary initial metric converges smoothly to a unique limit met...

open_problem | open | MathDB 页面 | 原文线索

V2-029231 | MATHDB-393382

Existence theorem for prescribed Q-curvature on the round 4-sphere

摘录 / EXCERPT: Let $f:S^4\to\mathbb{R}$ be positive, have only nondegenerate critical points, and satisfy $\Delta_{g_{S^4}}f(p)\ne 0$ at every critical point p. Define ... If there is no solution...

open_problem | open | MathDB 页面 | 原文线索

V2-029232 | MATHDB-393425

Existence of conformal metrics with constant Q-curvature

摘录 / EXCERPT: Let (M,g) be a four-dimensional manifold. Suppose $\ker P_g=\{\text{constants}\}$ and assume that $k_P\ne 8k\pi^2$ for $k=1,2,\ldots$. Then M admits a conformal metric with con...

open_problem | open | MathDB 页面 | 原文线索

V2-029233 | MATHDB-393453

Maximum eigenfunction blowup at recurrent points

摘录 / EXCERPT: Does one always have the maximum blowup rate $e_j(x)=\Omega\!\left(\lambda_j^{(n-1)/2}\right)$ at recurrent points of the geodesic flow?

open_problem | open | MathDB 页面 | 原文线索

V2-029234 | MATHDB-393458

The general Weinstein conjecture in dimension three

摘录 / EXCERPT: For every closed three-dimensional manifold M equipped with a contact form $\[200~\lambda$, the associated Reeb vector field X_λ has a periodic trajectory.

conjecture | open | MathDB 页面 | 原文线索

V2-029235 | MATHDB-393459

General three-dimensional Weinstein problem via a modified pseudoholomorphic-curve equation

摘录 / EXCERPT: In the general, non-planar case, does the problem $(*)$ admit a solution for a closed three-dimensional contact manifold (M,λ)? Namely, with (S,j) a closed Riemann surfac...

open_problem | open | MathDB 页面 | 原文线索

V2-029236 | MATHDB-393468

Schwartzman’s criterion for Lyapunov 1-forms with empty zero set

摘录 / EXCERPT: Let (varphi^t) be a smooth flow on a smooth closed manifold (M), generated by the vector field (V). For a finite positive invariant measure (mu), define the asymptotic cycle (Amu:H...

open_problem | open | MathDB 页面 | 原文线索

V2-029237 | MATHDB-393469

Existence of smooth Lyapunov 1-forms relative to an isolated invariant set

摘录 / EXCERPT: Let φ^t be a smooth flow on a smooth closed manifold M, and let $Z\subset M$ be an isolated invariant set. Let ... A smooth Lyapunov 1-form for (φ^t,Z) representi...

open_problem | open | MathDB 页面 | 原文线索

V2-029238 | MATHDB-393470

Anosov’s multiplicity question for closed geodesics on Finsler spheres

摘录 / EXCERPT: Does every irreversible Finsler metric on the sphere S^n have at least $2\lfloor n/2\rfloor$ closed geodesics?

open_problem | open | MathDB 页面 | 原文线索

V2-029239 | MATHDB-393475

Closed characteristics in contact-type hypersurfaces of cotangent bundles

摘录 / EXCERPT: Let M be a closed, connected smooth manifold of finite dimension, and let $M\subset W\subset T^*M$, where W is open with compact closure and smooth boundary $Q=\partial W$ of c...

open_problem | open | MathDB 页面 | 原文线索

V2-029240 | MATHDB-393483

Virtual Betti Number Conjecture

摘录 / EXCERPT: Let M be a compact, connected, orientable 3-manifold with infinite $\\pi_1(M)$. Then there exists a finite cover $\widetilde{M}\to M$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-029241 | MATHDB-393509

Local structure near multiplicity-two branch points of stable minimal hypersurfaces

摘录 / EXCERPT: Determine the local structure, including the singularities, of an orientable immersed stable minimal hypersurface in Euclidean space (or a smooth Riemannian manifold) of arbitrary...

open_problem | open | MathDB 页面 | 原文线索

V2-029242 | MATHDB-393511

A conformal gap conjecture for Bach-flat four-manifolds

摘录 / EXCERPT: Conjecture: There exists a positive constant ε such that the only Bach-flat conformal structure in dimension four with positive operators L and P satisfying ... is th...

conjecture | open | MathDB 页面 | 原文线索

V2-029243 | MATHDB-393517

Stein manifolds diffeomorphic to a ball without a one-critical-point plurisubharmonic Morse function

摘录 / EXCERPT: There exist Stein manifolds diffeomorphic to the ball but admitting no plurisubharmonic Morse function having a single critical point.

open_problem | open | MathDB 页面 | 原文线索

V2-029244 | MATHDB-393518

Global rigidity of hyperbolic cone-3-manifolds

摘录 / EXCERPT: Let X and X' be hyperbolic cone-3-manifolds with cone-angles at most $\c0$. If there exists a homeomorphism of pairs $(X,\Sigma)\cong(X',\Sigma')$ such that the cone-angles aro...

open_problem | open | MathDB 页面 | 原文线索

V2-029245 | MATHDB-393520

Invariant positive scalar curvature on fixed-point-free S^1-manifolds

摘录 / EXCERPT: Theorem B. Let M be a connected closed oriented fixed-point-free S^1-manifold such that all normal bundles around H-fixed components, where $H\subset S^1$ is a closed subgrou...

open_problem | open | MathDB 页面 | 原文线索

V2-029246 | MATHDB-393521

Tian’s properness conjecture for the Mabuchi energy

摘录 / EXCERPT: Tian conjectured that the existence of a constant scalar curvature Kähler metric is equivalent to properness of the Mabuchi energy. The equivalence holds when the first Chern class...

conjecture | open | MathDB 页面 | 原文线索

V2-029247 | MATHDB-393522

Tian’s K-stability conjecture for Fano manifolds

摘录 / EXCERPT: When $c_1(M)>0$, so that M is Fano, the existence of a Kähler-Einstein metric on M is equivalent to Tian’s K-stability of M.

conjecture | open | MathDB 页面 | 原文线索

V2-029248 | MATHDB-393528

Milnor conjecture for open manifolds with nonnegative Ricci curvature

摘录 / EXCERPT: Milnor conjecture. The fundamental group of every open Riemannian manifold with nonnegative Ricci curvature is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-029249 | MATHDB-393556

Universal strict inequality for the first positive Dirac eigenvalue

摘录 / EXCERPT: Is it true that ... holds for all manifolds not conformal to the standard sphere S^n?

open_problem | open | MathDB 页面 | 原文线索

V2-029250 | MATHDB-393569

Constant-scalar-curvature Kähler metrics on blow-ups at points

摘录 / EXCERPT: Let $(M,\d714_0)$ be a compact constant-scalar-curvature Kähler manifold of complex dimension $m\ge 2$. Given distinct points $p_1,\ldots,p_k\in M$, is the blow-up...

open_problem | open | MathDB 页面 | 原文线索

V2-029251 | MATHDB-393570

Lower bound for the off-diagonal spectral function

摘录 / EXCERPT: Let X be a compact Riemannian manifold of dimension $n\ge 2$, let $\Delta\phi_i=\lambda_i\phi_i$ be an orthonormal eigenbasis, and define ... If $x,y\in X$ are two points that are...

open_problem | open | MathDB 页面 | 原文线索

V2-029252 | MATHDB-393618

Finiteness of 3-manifolds dominated by a fixed closed orientable 3-manifold

摘录 / EXCERPT: Given a closed orientable 3-manifold M, are there at most finitely many 3-manifolds N in G, up to homeomorphism, dominated by M? Here G is the set of closed orientabl...

open_problem | open | MathDB 页面 | 原文线索

V2-029253 | MATHDB-393620

Effective local rigidity for hyperbolic 3-manifolds with smooth convex boundary

摘录 / EXCERPT: Give an effective proof that hyperbolic 3-manifolds with smooth convex boundary are locally rigid relative to the metric on their boundaries.

open_problem | open | MathDB 页面 | 原文线索

V2-029254 | MATHDB-393621

Model harmonic deformations and local rigidity of convex hyperbolic 3-manifold boundaries

摘录 / EXCERPT: Find model harmonic deformations for the boundary of a convex hyperbolic 3-manifold and prove local rigidity results using them.

open_problem | open | MathDB 页面 | 原文线索

V2-029255 | MATHDB-393623

Generalized Waldhausen conjecture

摘录 / EXCERPT: Every closed, orientable, irreducible, and atoroidal 3-manifold has only finitely many Heegaard splittings in each genus, up to isotopy.

conjecture | open | MathDB 页面 | 原文线索

V2-029256 | MATHDB-393624

Finite branched-surface collection for strongly irreducible Heegaard surfaces

摘录 / EXCERPT: Let M be a closed, orientable, irreducible and atoroidal 3-manifold, and suppose M is not a small Seifert fiber space. Then M has a finite collection of branched surfaces s...

open_problem | open | MathDB 页面 | 原文线索

V2-029257 | MATHDB-393625

Existence of spherical decompositions for open 3-manifolds

摘录 / EXCERPT: Does every open, orientable 3-manifold admit a spherical decomposition, meaning a locally finite collection of pairwise disjoint embedded 2-spheres such that, after splitting alo...

open_problem | open | MathDB 页面 | 原文线索

V2-029258 | MATHDB-393628

Surfaces in finite covers and the group determinant: Theorem 1

摘录 / EXCERPT: Let Y be a hyperbolic 3-manifold with one torus boundary component. Then either: 1. for all $n\in\mathbb{N}$, there is a regular cover $\widetilde{Y}\to Y$ and a slope α...

open_problem | open | MathDB 页面 | 原文线索

V2-029259 | MATHDB-393630

Deforming Euclidean cone-3-manifolds

摘录 / EXCERPT: Let X be a closed, orientable Euclidean cone-3-manifold with cone angles at most $\\pi$. If X is not almost product, then for every multiangle $\bar{\alpha}\in(0,\pi)^N$ ther...

open_problem | open | MathDB 页面 | 原文线索

V2-029260 | MATHDB-393644

Consistent deformation of Riemannian geometry

摘录 / EXCERPT: Can one consistently deform Riemannian geometry into a noncommutative Riemannian geometry?

open_problem | open | MathDB 页面 | 原文线索

V2-029261 | MATHDB-393648

Differential calculus from spectral data in q-deformed geometry

摘录 / EXCERPT: What is the differential calculus obtained from the spectral data?

open_problem | open | MathDB 页面 | 原文线索

V2-029262 | MATHDB-393693

(0,2) mirror symmetry

摘录 / EXCERPT: The conjectured “(0,2) mirror symmetry” exchanges pairs consisting of complex Kähler manifolds with holomorphic vector bundles.

open_problem | open | MathDB 页面 | 原文线索

V2-029263 | MATHDB-393704

Relation between the Frobenius manifolds \(\Gamma_G\) and the Frobenius manifold associated to \(G\)

摘录 / EXCERPT: It seems an interesting problem to see the relations between the Frobenius manifold Γ_G and the one associated to G, arising from Dubrovin’s almost duality ([15]).

open_problem | open | MathDB 页面 | 原文线索

V2-029264 | MATHDB-393706

Range of L^p-boundedness for the Riesz transform

摘录 / EXCERPT: Let M be a complete Riemannian manifold with infinite measure, and let $T=d\Delta^{-1/2}$ be its Riesz transform, where Δ is the positive Laplace operator. Determine the r...

open_problem | open | MathDB 页面 | 原文线索

V2-029265 | MATHDB-393761

Decay and spectral representation for scalar waves in the Kerr geometry

摘录 / EXCERPT: For the scalar wave equation in the Kerr geometry with smooth, compactly supported Cauchy data outside the event horizon, the solution can be represented as a superposition of solu...

open_problem | open | MathDB 页面 | 原文线索

V2-029266 | MATHDB-393766

Existence classification for an unresolved Einstein-scalar-field Lichnerowicz class

摘录 / EXCERPT: For compact a0n-dimensional manifolds, determine whether the Einstein-scalar-field Lichnerowicz equation … has a positive solution for conformal data in the class … where W sat...

open_problem | open | MathDB 页面 | 原文线索

V2-029267 | MATHDB-393770

Classification of homogeneous complex supermanifolds with flag-manifold reduction

摘录 / EXCERPT: For a given complex flag manifold M, describe all homogeneous compact complex supermanifolds $(M,\mathcal O)$ whose reduction is M.

open_problem | open | MathDB 页面 | 原文线索

V2-029268 | MATHDB-393771

Classification of complex supermanifolds with a given retract

摘录 / EXCERPT: Classify, up to isomorphism, all complex supermanifolds with a given retract.

open_problem | open | MathDB 页面 | 原文线索

V2-029269 | MATHDB-393806

Conjecture 2: locally bounded solutions under vanishing p-curvature

摘录 / EXCERPT: Let (M,∂_M) be a global D-module over a global D-ring (O'_F,∂_F) with vanishing p-curvature for almost all primes $p\in\mathbb{Z}$. Then the solutions of...

conjecture | open | MathDB 页面 | 原文线索

V2-029270 | MATHDB-393863

More precise Novikov-Shubin estimates for finite-volume hyperbolic interiors

摘录 / EXCERPT: Find more precise estimates for the Novikov-Shubin invariants of manifolds whose interior admits a complete hyperbolic metric of finite volume. Led by the general result about...

open_problem | open | MathDB 页面 | 原文线索

V2-029271 | MATHDB-393868

Multiplicity conjecture for the second Schrödinger eigenvalue

摘录 / EXCERPT: For any manifold, the maximal multiplicity of the second eigenvalue of any Schrödinger operator is $C(X)-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-029272 | MATHDB-393882

Unified definition of discrete CMC nets in space forms

摘录 / EXCERPT: The problem posed is to find a unified integrable discretization of surfaces of constant mean curvature H in every space form $Q^3_\ba$, valid for all values of H, with analogu...

open_problem | open | MathDB 页面 | 原文线索

V2-029273 | MATHDB-393885

Partial unfoldings realizing closed oriented manifolds in dimensions at most three

摘录 / EXCERPT: For each closed oriented manifold M with $d := \dim M \le 3$, there is a triangulation Δ of the d-sphere such that M is homeomorphic with the partial unfolding of...

open_problem | open | MathDB 页面 | 原文线索

V2-029274 | MATHDB-393898

Link between integrator symplecticity and optimization behavior

摘录 / EXCERPT: Find a direct link between the symplecticity of the integrator and the behaviour of the optimization problem.

open_problem | open | MathDB 页面 | 原文线索

V2-029275 | MATHDB-393926

H-principle for the space of framed functions

摘录 / EXCERPT: For a closed smooth manifold M, determine whether the space $L(M)$ of framed functions on M satisfies an h-principle.

open_problem | open | MathDB 页面 | 原文线索

V2-029276 | MATHDB-394001

Asymptotic counting of simple closed geodesics in a mapping-class-group orbit

摘录 / EXCERPT: For a compact hyperbolic surface $X\in\mathcal T_g$ and $\gamma\in\mathcal{ML}_g(\mathbb Z)$, define ... Then ... where $n_\gamma(X)$ is a smooth proper function of $X\in\mathcal M_g$.

open_problem | open | MathDB 页面 | 原文线索

V2-029277 | MATHDB-394002

Negative convergence of stretch lines to Thurston’s boundary

摘录 / EXCERPT: For every complete geodesic lamination μ on Σ with non-empty stump γ, and every $g\in\mathcal T(\Sigma)$, determine whether the associated stretch line...

open_problem | open | MathDB 页面 | 原文线索

V2-029278 | MATHDB-394086

Equidistribution of spherical domains evolved by the geodesic flow

摘录 / EXCERPT: Under the standing assumptions that M is compact with strictly negative sectional curvature, let $p\in\widetilde M$ and let $A\subset S_p\widetilde M$ have, say, piecewise smooth...

open_problem | open | MathDB 页面 | 原文线索

V2-029279 | MATHDB-394095

Uniqueness of the Hofer norm among invariant norms

摘录 / EXCERPT: Let $(M,\fc\t\omega)$ be a closed symplectic manifold, let $G:=\operatorname{Ham}(M,\omega)$, and identify the Lie algebra of G with the space $\mathcal F$ of all smooth function...

open_problem | open | MathDB 页面 | 原文线索

V2-029280 | MATHDB-394097

Boundedness of Hofer norms of commutators

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold and let $G=\operatorname{Ham}(M,\omega)$. For $f,g\in G$, write $[f,g]=fgf^{-1}g^{-1}$, and define ... where ρ_H is the Hofer di...

open_problem | open | MathDB 页面 | 原文线索

V2-029281 | MATHDB-394111

Expected obstruction from the contact invariant \(\tau_\xi(K)\)

摘录 / EXCERPT: In the case where (Y,ξ) admits a symplectic filling, the report expects $\tau_\xi(K)$ to provide an obstruction for (Y,K) to arise as the intersection of Y with a J-holom...

open_problem | open | MathDB 页面 | 原文线索

V2-029282 | MATHDB-394112

Symplectic structures on $S^1\times N$ and fibrations of N

摘录 / EXCERPT: Conjecture 1. Let N be a closed 3-manifold. If $S^1\times N$ is symplectic, then there exists $\phi\in H^1(N;\mathbb Z)$ such that (N,ϕ) fibers over S^1. Here,...

open_problem | open | MathDB 页面 | 原文线索

V2-029283 | MATHDB-394113

Thurston’s surface-subgroup separability conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: All (surface) subgroups of hyperbolic 3-manifolds are separable.

conjecture | open | MathDB 页面 | 原文线索

V2-029284 | MATHDB-394119

Minimal right-veering open books and tightness

摘录 / EXCERPT: We might optimistically conjecture that a minimal (i.e. not destabilizable) right-veering open book defines a tight contact structure.

open_problem | open | MathDB 页面 | 原文线索

V2-029285 | MATHDB-394138

Stability of steady marine ice-sheet solutions under bed geometry

摘录 / EXCERPT: Understand how bed geometry affects the stability of steady solutions to the marine ice sheet problem.

open_problem | open | MathDB 页面 | 原文线索

V2-029286 | MATHDB-394143

Existence of polarized toric degenerations from polarized tropical manifolds

摘录 / EXCERPT: Every polarized tropical manifold $(B,\mathcal P,\varphi)$ arises as the intersection complex of a polarized toric degeneration as defined in [3, Section 4.2].

open_problem | open | MathDB 页面 | 原文线索

V2-029287 | MATHDB-394144

Existence and uniqueness of continuous solutions to a degenerate complex Monge-Ampère equation

摘录 / EXCERPT: Theorem A. Let X be a compact Kähler manifold, and let $\c9$ be a semipositive $(1,1)$-form such that $\int_X \omega^n>0$. Let $0\leq f\in L^p(X,\omega^n)$, with $p>1$, be a dens...

open_problem | open | MathDB 页面 | 原文线索

V2-029288 | MATHDB-394145

Ricci-curvature lower bound for the Kähler cone of a Calabi-Yau manifold

摘录 / EXCERPT: Let X be a Calabi-Yau n-fold, let K be its Kähler cone, and let W be the index cone determined by the degree-n cup-product form. Set $W_1:=\{D\in W:D^n=1\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029289 | MATHDB-394147

Rigid targets for Fano manifolds of Picard number one

摘录 / EXCERPT: Let X be a Fano manifold with second Betti number $b_2=1$, which is different from the projective space. Then X is a rigid target.

open_problem | open | MathDB 页面 | 原文线索

V2-029290 | MATHDB-394148

Immersed projective spaces in Fano manifolds with Picard number one

摘录 / EXCERPT: Let X be a Fano manifold with $b_2=1$, different from the projective space. Then X contains no immersed $\[200~\mathbb{P}^k$ with trivial normal bundle for any $1<k<\dim X$.

open_problem | open | MathDB 页面 | 原文线索

V2-029291 | MATHDB-394176

Existence of almost complex 4-manifolds with vanishing first Chern class

摘录 / EXCERPT: Given a number σ divisible by 16 and an integer $b_1\ge 0$, does there exist a 4-manifold realizing these characteristic numbers and supporting an almost complex structure...

open_problem | open | MathDB 页面 | 原文线索

V2-029292 | MATHDB-394179

Commensurability classification of non-geometric graph-manifold groups

摘录 / EXCERPT: Classify the fundamental groups of non-geometric graph manifolds up to commensurability.

open_problem | open | MathDB 页面 | 原文线索

V2-029293 | MATHDB-394183

Information contained in the reduced index obstruction

摘录 / EXCERPT: Let M be a closed smooth spin manifold of dimension m. If $\c0_1(M)$ is torsion-free and satisfies the strong Novikov conjecture, meaning that the Baum-Connes map...

open_problem | open | MathDB 页面 | 原文线索

V2-029294 | MATHDB-394192

Asymptotic behavior of systolic volume under connected sums

摘录 / EXCERPT: For a closed orientable aspherical n-manifold M, determine the asymptotic behavior, as $k\to\infty$, of the systolic volume $\sigma(\#_k M)$, where $\#_k M$ is the connected su...

open_problem | open | MathDB 页面 | 原文线索

V2-029295 | MATHDB-394195

Isoperimetricity of CMC spheres in the Heisenberg group Nil

摘录 / EXCERPT: Are the constant-mean-curvature (CMC) spheres in Nil isoperimetric surfaces for all volumes?

open_problem | open | MathDB 页面 | 原文线索

V2-029296 | MATHDB-394249

Existence of harmonic spinors on closed spin manifolds

摘录 / EXCERPT: On any closed spin manifold of dimension greater than 2, there exists a Riemannian metric with harmonic spinors.

open_problem | open | MathDB 页面 | 原文线索

V2-029297 | MATHDB-394252

Metrics with arbitrarily many harmonic spinors

摘录 / EXCERPT: For every compact spin manifold M of dimension $n\ge 3$ and every $N\in\mathbb{N}$, there is a Riemannian metric g on M such that $k_g\ge N$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-029298 | MATHDB-394253

Weak unitary equivalence of Fredholm modules detects conformal metrics

摘录 / EXCERPT: Let M be a compact manifold with a fixed spin structure, and let g and g' be two Riemannian metrics on M. The associated Fredholm modules ... are weakly unitarily equivalent...

open_problem | open | MathDB 页面 | 原文线索

V2-029299 | MATHDB-394265

Alexandrov’s uniqueness hypothesis for smooth convex surfaces

摘录 / EXCERPT: Let $K\subset\mathbb{R}^3$ be a smooth convex body. If there is a constant C such that, at every point of ∂K, the principal curvature radii satisfy $R_1\le C\le R_2$, m…

conjecture | open | MathDB 页面 | 原文线索

V2-029300 | MATHDB-394301

Varolin-Toth conjecture on detecting complex Euclidean space

摘录 / EXCERPT: If X is a Stein manifold with the density property and X is diffeomorphic to $\^{2n}$, then X is biholomorphic to $\^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-029301 | MATHDB-394405

Optical tomography on simple Riemannian manifolds

摘录 / EXCERPT: Let $M\subset\mathbb{R}^n$ be a bounded open domain with smooth boundary and let g be a Riemannian metric on M. For particles of unit speed, let f satisfy … where Ω_xM…

open_problem | open | MathDB 页面 | 原文线索

V2-029302 | MATHDB-394406

Uniqueness in optical tomography on a two-dimensional Riemannian manifold

摘录 / EXCERPT: Let (M,g) be a two-dimensional Riemannian manifold with smooth, strictly convex boundary satisfying M1-M3: its curvature is bounded above by $\[...]\[0m\kappa_0$, it has no focal…

open_problem | open | MathDB 页面 | 原文线索

V2-029303 | MATHDB-394407

Unique determination in optical tomography on simple Riemannian surfaces

摘录 / EXCERPT: Let (M,g) be a two-dimensional Riemannian manifold with smooth, strictly convex boundary satisfying M1-M3: the sectional curvature of (M,g) is bounded above by $\κ_0$; if…

open_problem | open | MathDB 页面 | 原文线索

V2-029304 | MATHDB-394511

Existence of non-regular Poisson manifolds of compact type

摘录 / EXCERPT: Does there exist a non-regular Poisson manifold of compact type (PMCT), where a PMCT is a Poisson manifold whose cotangent Lie algebroid is integrable and whose canonical integrati…

open_problem | open | MathDB 页面 | 原文线索

V2-029305 | MATHDB-394516

Problem A: Isometric immersions of scaled reflective submanifolds

摘录 / EXCERPT: Suppose given a simply connected, immersed, reflective submanifold $N:=\exp(p)$ of a semisimple Riemannian symmetric space $\bar N$. Thus, N_R is a reflective submanifold of…

open_problem | open | MathDB 页面 | 原文线索

V2-029306 | MATHDB-394521

Willmore energy infimum for immersed tori

摘录 / EXCERPT: Does the infimum of the Willmore energy, with $W=\int H^2$ the integral of the square of the mean curvature, over a torus T^2 immersed in 3-space equal $2\pi^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-029307 | MATHDB-394593

Thurston’s geometrization conjecture for closed irreducible 3-manifolds

摘录 / EXCERPT: If M is a closed, orientable, irreducible smooth 3-manifold, then M is hyperbolic, Seifert-fibered, or contains an incompressible torus.

conjecture | open | MathDB 页面 | 原文线索

V2-029308 | MATHDB-394594

Conditional geometrization theorem for aspherical 3-manifolds

摘录 / EXCERPT: Let M be a closed, orientable, irreducible, aspherical 3-manifold. Assume that M carries a sequence of Riemannian metrics g_n such that: 1.…

open_problem | open | MathDB 页面 | 原文线索

V2-029309 | MATHDB-394595

Sharp quadratic isoperimetric constant implies Gromov hyperbolicity

摘录 / EXCERPT: Let X be a geodesic metric space. Suppose there exists $ \varepsilon>0$ such that every sufficiently long Lipschitz loop γ in X bounds a singular Lipschitz disc Σ…

open_problem | open | MathDB 页面 | 原文线索

V2-029310 | MATHDB-394598

Relation between the Ptolemy inequality and non-positive curvature

摘录 / EXCERPT: How does the Ptolemy inequality relate to concepts of non-positive curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029311 | MATHDB-394599

Ptolemy inequality and CAT(0) geometry

摘录 / EXCERPT: A metric space is CAT(0) if and only if it is Ptolemy and Busemann convex.

open_problem | open | MathDB 页面 | 原文线索

V2-029312 | MATHDB-394601

Asymptotic metric theorem for periodic distances on solvable Lie groups

摘录 / EXCERPT: Let S be a simply connected solvable Lie group of polynomial growth, and let ρ be a coarsely geodesic periodic distance on S. Then there exists a left-invariant Carnot-Car…

open_problem | open | MathDB 页面 | 原文线索

V2-029313 | MATHDB-394602

Perelman’s Stability Theorem

摘录 / EXCERPT: Let X^n be a compact n-dimensional Alexandrov space of curvature $\geq\kappa$. Then there exists an $\epsilon=\u0000epsilon(X)>0$ such that, for any n-dimensional Alexandrov…

open_problem | open | MathDB 页面 | 原文线索

V2-029314 | MATHDB-394605

Geometric Lehmer conjecture for closed arithmetic hyperbolic 3-manifolds

摘录 / EXCERPT: There exists some ε such that every closed arithmetic hyperbolic 3-manifold has injectivity radius at least ε.

conjecture | open | MathDB 页面 | 原文线索

V2-029315 | MATHDB-394608

Systematic formalization of end obstructions from suitable coverings

摘录 / EXCERPT: In some sense, the proof of the theorem uses the ends of suitable coverings of the manifold in question and pins down the obstruction to positive scalar curvature near those. One s...

open_problem | open | MathDB 页面 | 原文线索

V2-029316 | MATHDB-394611

Wild Dynamics: existence of flows with 2-wild trajectories

摘录 / EXCERPT: Every boundaryless 3-manifold admits a flow with a discrete set of fixed points such that every non-trivial trajectory is 2-wild.

open_problem | open | MathDB 页面 | 原文线索

V2-029317 | MATHDB-394639

Uniqueness of spheres among compact hypersurfaces with constant Levi curvatures

摘录 / EXCERPT: Are spheres the unique compact hypersurfaces with constant Levi curvatures?

open_problem | open | MathDB 页面 | 原文线索

V2-029318 | MATHDB-394649

Reproducing subgroups of the metaplectic-Heisenberg group

摘录 / EXCERPT: Determine which Lie subgroups H of the semidirect-product group G formed from the (double cover of the) symplectic group $Sp(d,\mathbb R)$ and the Heisenberg group H_d, modul...

open_problem | open | MathDB 页面 | 原文线索

V2-029319 | MATHDB-394695

Existence of exact Lagrangian submanifolds beyond the standard examples

摘录 / EXCERPT: For a closed manifold M, is every closed exact Lagrangian submanifold $L\hookrightarrow T^*M$ the graph $L_f=\{(q,df_q)\mid q\in M\}$ of a function $f:M\to\mathbb{R}$, or the ima...

open_problem | open | MathDB 页面 | 原文线索

V2-029320 | MATHDB-394705

Spectral gaps for periodic magnetic Schrödinger operators with a one-dimensional magnetic zero set

摘录 / EXCERPT: Let M be a noncompact oriented two-dimensional Riemannian manifold with a properly discontinuous cocompact action of a finitely generated discrete group $\93$, and suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-029321 | MATHDB-394712

Invertible double for a Dirac-type operator in the product-near-boundary case

摘录 / EXCERPT: Let M be a smooth compact connected Riemannian manifold with boundary, and let D be a Dirac-type operator on M acting between sections of the Hermitian vector bundle E. Ass...

open_problem | open | MathDB 页面 | 原文线索

V2-029322 | MATHDB-394713

Existence of first-order elliptic operators with nontrivial ghost solutions

摘录 / EXCERPT: Does there exist a smooth compact connected Riemannian manifold M with boundary, Hermitian vector bundles E,F over M, and a first-order elliptic differential operator...

open_problem | open | MathDB 页面 | 原文线索

V2-029323 | MATHDB-394775

Generic nefness and pseudo-effectivity of the canonical bundle

摘录 / EXCERPT: Let X be a projective manifold of dimension n. A line bundle L on X is generically nef if $L\cdot H_1\cdots H_{n-1}\ge 0$ for all ample line bundles $H_1,\ldots,H_{n-1}$. S...

open_problem | open | MathDB 页面 | 原文线索

V2-029324 | MATHDB-394776

Generic nefness of the cotangent bundle on maximal covering families

摘录 / EXCERPT: Let $(C_t)_{t\in T}$ be a covering family of curves of a projective manifold X, and suppose that the family is maximal, i.e. the parameter space T is an irreducible family of t...

open_problem | open | MathDB 页面 | 原文线索

V2-029325 | MATHDB-394777

Generic nefness of the tangent bundle under nef anticanonical class

摘录 / EXCERPT: Let X be a projective manifold with $-K_X$ nef. Is T_X generically nef for some/all polarizations?

open_problem | open | MathDB 页面 | 原文线索

V2-029326 | MATHDB-394778

Sufficient nefness of the tangent bundle for nef anticanonical class

摘录 / EXCERPT: Let X be a projective manifold with $-K_X$ nef. Then T_X is sufficiently nef, meaning that through every point $x\in X$ there is a covering family (C_t) of curves passing thr...

open_problem | open | MathDB 页面 | 原文线索

V2-029327 | MATHDB-394779

Albanese submersion for manifolds with nef anticanonical bundle

摘录 / EXCERPT: Let X be a projective manifold with $-K_X$ nef. Is the Albanese map of X a (surjective) submersion?

open_problem | open | MathDB 页面 | 原文线索

V2-029328 | MATHDB-394784

Small deformations of nilmanifolds with left-invariant complex structure

摘录 / EXCERPT: For every nilmanifold $M=\Gamma\backslash G$ with a left-invariant complex structure J, are all sufficiently small deformations of (M,J) again nilmanifolds with left-invariant...

open_problem | open | MathDB 页面 | 原文线索

V2-029329 | MATHDB-394868

SYZ mirror construction and superpotential

摘录 / EXCERPT: The SYZ conjecture: given a manifold with a singular special Lagrangian torus fibration over a base B, the mirror manifold is expected to be the “dual” torus fibration over the s...

open_problem | open | MathDB 页面 | 原文线索

V2-029330 | MATHDB-395080

Virtual pushforwards for morphisms of 1-dg-manifolds

摘录 / EXCERPT: Conjecture: the virtual pushforward can be naturally defined for morphisms of 1-dg-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029331 | MATHDB-395083

Ample-cone criterion for holomorphic symplectic manifolds of K3${}^{[n]}$-type

摘录 / EXCERPT: Let (F,g) be a polarized holomorphic symplectic manifold deformation equivalent to $S^{[n]}$, where S is a K3 surface. Conjecture: a divisor class h is ample if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-029332 | MATHDB-395089

Existence of weak solutions for the biharmonic map heat flow

摘录 / EXCERPT: Let $4\le n\le 8$, let $\Omega\subset\mathbb{R}^n$ be an open, bounded domain with smooth boundary, and let N be a compact, smooth Riemannian manifold isometrically embedded in...

open_problem | open | MathDB 页面 | 原文线索

V2-029333 | MATHDB-395090

Zero mean curvature of eternal blow-up limits at isolated singularities

摘录 / EXCERPT: In the rescaling setup described in the report, if the singularities of the Lagrangian mean curvature flow are isolated, is the resulting eternal solution...

open_problem | open | MathDB 页面 | 原文线索

V2-029334 | MATHDB-395091

Future stability for power-law expansion in the Einstein-nonlinear scalar field system

摘录 / EXCERPT: Let $V(\varphi)=V_0e^{-\lambda\varphi}$, where $V_0>0$ and $\lambda\in(0,\sqrt{2})$. Let M be a connected and simply connected 3-dimensional manifold, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029335 | MATHDB-395092

Future stability of Einstein-nonlinear scalar field solutions

摘录 / EXCERPT: Given a globally hyperbolic and future causally geodesically complete solution to the Einstein-nonlinear scalar field equations (1)-(4), do sufficiently small perturbations of the...

open_problem | open | MathDB 页面 | 原文线索

V2-029336 | MATHDB-395095

Improved uniqueness criterion for the harmonic map heat flow in two dimensions

摘录 / EXCERPT: Let M be a closed Riemannian surface and N a compact Riemannian manifold. For weak solutions $u\in H^{1}([0,T]\times M)$ of the harmonic map heat flow (1), assume -...

open_problem | open | MathDB 页面 | 原文线索

V2-029337 | MATHDB-395096

Realization of a prescribed-geometry minimal surface

摘录 / EXCERPT: If $\kappa(\gamma)>0$, the surface (S^2,γ) can be realized as a stably embedded minimal surface in a null scalar curvature manifold.

open_problem | open | MathDB 页面 | 原文线索

V2-029338 | MATHDB-395121

Local minimization of absolute Gaussian curvature for a polyhedral cone

摘录 / EXCERPT: Given a piecewise-linear embedding C of S^1 into $\[3$ whose vertices have rational coordinates, find a point $q\in\mathbb{R}^3\setminus C$ such that the piecewise-linear surfa...

open_problem | open | MathDB 页面 | 原文线索

V2-029339 | MATHDB-395136

Foliations of asymptotically flat manifolds by surfaces of Willmore type

摘录 / EXCERPT: For fixed constants C' and $m>0$, there is $\ta_0=\ta_0(m)>0$ such that, for every $\sigma>0$, suitable constants λ_0 and C exist with the following property. If...

open_problem | open | MathDB 页面 | 原文线索

V2-029340 | MATHDB-395148

Hildebrandt continuity conjecture for conformally invariant variational problems

摘录 / EXCERPT: Let ω be a two-dimensional domain and let N^k be a closed submanifold of Euclidean space. The report states that critical points $u\in W^{1,2}(\omega,N^k)$ of continuously...

conjecture | open | MathDB 页面 | 原文线索

V2-029341 | MATHDB-395171

Optimal Bochner-Riesz exponent for asymptotically conic manifolds

摘录 / EXCERPT: For a given $p\ne 2$, determine the optimal range of $\b1$ for which ...

open_problem | open | MathDB 页面 | 原文线索

V2-029342 | MATHDB-395173

Yau’s conjecture relating constant scalar curvature Kähler metrics to stability

摘录 / EXCERPT: For a polarized compact Kähler manifold X with polarization Lto X, the existence of a metric of constant scalar curvature in the Kähler class [omega0] should be equivalent to the s...

conjecture | open | MathDB 页面 | 原文线索

V2-029343 | MATHDB-395174

Finite-time accessibility of motion camouflage

摘录 / EXCERPT: For the two-particle planar system with pursuer speed 1, evader speed $0<\nu<1$, continuous evader curvature u_e satisfying bounded $|u_e|$, initial conditions $\Gamma(0)<1$ an...

open_problem | open | MathDB 页面 | 原文线索

V2-029344 | MATHDB-395175

Integrable discretization of completely integrable flows

摘录 / EXCERPT: Given a completely integrable flow ... with Hamilton function H on a Poisson manifold P with Poisson bracket $\{\cdot,\cdot\}$, find a family of diffeomorphisms $P\to P$, ... depen...

open_problem | open | MathDB 页面 | 原文线索

V2-029345 | MATHDB-395177

Relative Expander Monotonicity and MCF with Singular Initial Data

摘录 / EXCERPT: Let M_0 be a hypersurface in $\mathbb{R}^{n+1}$ with isolated point singularities modeled on regular hypercones, and let $(M_t)_{t\ge 0}$ be a mean curvature flow with initial da...

open_problem | open | MathDB 页面 | 原文线索

V2-029346 | MATHDB-395179

Sharp threshold for Ricci-flow-invariant curvature cones

摘录 / EXCERPT: Conjecture: Theorem 1 remains valid with the threshold dimension set to $n_0=12$. Namely, for every $n\ge 12$, if n is even then $C_{d(n)}$ is invariant under the Ricci flow, whi...

open_problem | open | MathDB 页面 | 原文线索

V2-029347 | MATHDB-395183

Entropy bound for n-dimensional Hilbert geometries

摘录 / EXCERPT: Conjecture: For any n-dimensional Hilbert geometry, $\operatorname{Ent}_K \le n-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-029348 | MATHDB-395235

Algebraic dimensions and algebraic reductions of hyperkähler manifolds

摘录 / EXCERPT: Let X be a hyperkähler manifold of dimension $2n$. Then its algebraic dimension takes only the values 0, n, and $2n$. Moreover, if $a(X)=n$, then the algebraic reduction has...

open_problem | open | MathDB 页面 | 原文线索

V2-029349 | MATHDB-395236

Păun's non-vanishing theorem

摘录 / EXCERPT: Let X be a smooth projective manifold, and let B be an $\mathbb{R}$-divisor such that: (i) the pair (X,B) is klt and B is big; (ii) the adjoint bundle K_X+B is pseudo-eff...

open_problem | open | MathDB 页面 | 原文线索

V2-029350 | MATHDB-395254

Lower semicontinuity of the spontaneous-curvature energy

摘录 / EXCERPT: For a fixed constant H_0, consider the functional ... where $H=\vec H\cdot\nu$ is the scalar mean curvature of a smooth boundary Γ, with ν a continuous unit normal field...

open_problem | open | MathDB 页面 | 原文线索

V2-029351 | MATHDB-395256

Non-domination of irreducible locally symmetric spaces by products

摘录 / EXCERPT: If P is any non-trivial product of closed manifolds and N is a closed irreducible locally symmetric space of non-compact type, then $P \ngeq N$, where $P \geq N$ means that the...

open_problem | open | MathDB 页面 | 原文线索

V2-029352 | MATHDB-395260

Can the homogeneity assumption be removed from the area-space rigidity theorem?

摘录 / EXCERPT: Let M and M' be compact Riemannian manifolds, possibly with boundary, of dimension at least 3. The report proves that every homogeneous isometry...

open_problem | open | MathDB 页面 | 原文线索

V2-029353 | MATHDB-395262

Continuity of the Poisson bracket under C^0 convergence

摘录 / EXCERPT: Let (F_n) and (G_n) be sequences of smooth functions on a symplectic manifold $(M,\c9)$ such that $F_n\to F$, $G_n\to G$, and $\{F_n,G_n\}\to H$ in the C^0 sense. Must...

open_problem | open | MathDB 页面 | 原文线索

V2-029354 | MATHDB-395266

C^0-invariance of characteristic foliations under symplectic homeomorphisms

摘录 / EXCERPT: Let S and S' be smooth hypersurfaces of symplectic manifolds (M,ω) and (M',ω'). If a symplectic homeomorphism between M and M' sends S to S', does it tran...

open_problem | open | MathDB 页面 | 原文线索

V2-029355 | MATHDB-395267

Determine the embedding capacity on the interval \([7,8]\)

摘录 / EXCERPT: Determine the function ... for $a\in[7,8]$, where $E(1,a)\subset\mathbb C^2$ is the four-dimensional symplectic ellipsoid and $B^4(A)=E(A,A)$.

open_problem | open | MathDB 页面 | 原文线索

V2-029356 | MATHDB-395283

Geodesicity of the compression-body tunnel

摘录 / EXCERPT: Let C be the compression body with genus-2 positive boundary and torus negative boundary described in the report, and let τ be the arc running through its compressible han...

open_problem | open | MathDB 页面 | 原文线索

V2-029357 | MATHDB-395290

Gordon's exceptional Dehn filling conjecture

摘录 / EXCERPT: For any hyperbolic knot manifold M, we have $\#E(M)\le 10$ and $\Delta(E(M))\le 8$. Moreover, if $M\neq M_1,M_2,M_3,M_4$, then $\#E(M)\le 7$ and $\Delta(E(M))\le 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-029358 | MATHDB-395292

Infinitely many smooth structures from nontrivial Seiberg-Witten invariants

摘录 / EXCERPT: If the Seiberg-Witten invariants of a simply-connected smooth 4-manifold X are non-trivial, then X has infinitely many distinct smooth structures.

open_problem | open | MathDB 页面 | 原文线索

V2-029359 | MATHDB-395315

Schmidt conjecture for lattice points on smooth convex surfaces

摘录 / EXCERPT: For every integer dimension $d\ge 3$ and every compact smooth convex surface $\partial K\subset\mathbb{R}^d$ with everywhere non-vanishing Gaussian curvature, one has ... for dilatio...

conjecture | open | MathDB 页面 | 原文线索

V2-029360 | MATHDB-395464

Cox ring of a generic K3 surface with Picard number two

摘录 / EXCERPT: Let X be a generic K3 surface with a non-symplectic involution and associated double cover $X\to Y$, with $\rho(X)=2$. Then ... where the degree of T_i is the i-th column of th...

open_problem | open | MathDB 页面 | 原文线索

V2-029361 | MATHDB-395471

Koszul duality for categories associated to symplectic dual pairs

摘录 / EXCERPT: For a symplectic dual pair (M,T,η,ξ) and $(M^\vee,T^\vee,\eta^\vee,\xi^\vee)$, the certain abelian categories associated to these data-categories of modules over quantizatio...

open_problem | open | MathDB 页面 | 原文线索

V2-029362 | MATHDB-395473

A local Ricci-curvature notion for general metric spaces

摘录 / EXCERPT: Find a definition of Ricci curvature valid for spaces more general than Riemannian manifolds, with the definition being local rather than global.

open_problem | open | MathDB 页面 | 原文线索

V2-029363 | MATHDB-395489

Non-self-intersecting developments of piecewise-geodesic curves on S^2

摘录 / EXCERPT: Given a simple piecewise-geodesic curve on S^2, develop it onto the plane (same lengths and angles). Find conditions which guarantee that the developed image does not intersect i...

open_problem | open | MathDB 页面 | 原文线索

V2-029364 | MATHDB-395573

The Ovals Problem for the curvature Schrödinger operator

摘录 / EXCERPT: Consider a smooth closed curve γ of length 2π in $\mathbb{R}^3$, let $\kappa(s)$ be its curvature as a function of arclength, and let $H_C=-\frac{d^2}{ds^2}+\kappa^2$ ac...

open_problem | open | MathDB 页面 | 原文线索

V2-029365 | MATHDB-395623

When is a control system mechanical?

摘录 / EXCERPT: Given a control-affine system ... where $F,G_1,\ldots,G_m$ are C^∞ vector fields on a $2n$-dimensional manifold M, determine when Σ is state-equivalent to a mechanica...

open_problem | open | MathDB 页面 | 原文线索

V2-029366 | MATHDB-395709

Nikolski’s lemma connecting the Operator Corona Problem with analytic projections

摘录 / EXCERPT: Let Ω be a domain in $\mathbb{C}^n$ (in fact, one may let Ω be a manifold). Let $F\in H^\infty_{E^*\to E}(\Omega)$ satisfy ... Then F is left invertible in...

open_problem | open | MathDB 页面 | 原文线索

V2-029367 | MATHDB-395716

Conjecture 2: Localized L^2 extension from an analytic set

摘录 / EXCERPT: Let M be a complex manifold of dimension n, let E be a holomorphic vector bundle on M, let $K=K_M$ be the canonical line bundle, let dV be a C^{∞} volume form, let...

conjecture | open | MathDB 页面 | 原文线索

V2-029368 | MATHDB-395717

Demailly's conjecture on transcendental Morse inequalities

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension n. Suppose there exists a d-closed $(1,1)$-current T on X such that $T \geq 0$ and $\int_X T_{\mathrm{ac}}^n>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-029369 | MATHDB-395720

Nonvanishing from a decomposition of a multiple of the canonical bundle

摘录 / EXCERPT: Suppose X is a projective manifold, and suppose a decomposition $N K_X=A+B$ with some positive integer N, an effective divisor A (one may assume A spanned) and a pseudo-eff...

open_problem | open | MathDB 页面 | 原文线索

V2-029370 | MATHDB-395723

Non-algebraicity criterion for foliations of general type

摘录 / EXCERPT: Let X be a complex projective manifold and let $\F\subset T_X$ be a rank-one foliation, meaning a line subbundle of T_X. Let $K=\mathcal F^*$ be its canonical bundle, and suppo...

open_problem | open | MathDB 页面 | 原文线索

V2-029371 | MATHDB-395794

Hebey-Vaugon conjecture for the equivariant Yamabe problem

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold of dimension $n\ge 3$, and let G be a subgroup of the isometry group $I(M,g)$. Denote by $\mathcal O_G(Q)$ the G-orbit of $Q\in M$,...

conjecture | open | MathDB 页面 | 原文线索

V2-029372 | MATHDB-395822

Closed aspherical manifold realization for hyperbolic groups with spherical boundary

摘录 / EXCERPT: Theorem. Let G be a torsion-free hyperbolic group. If $\partial G \cong S^{n-1}$ for some integer $n \ge 6$, then there is an n-dimensional closed aspherical manifold M such...

open_problem | open | MathDB 页面 | 原文线索

V2-029373 | MATHDB-395827

Naive local index formula for Dirac operators coupled to superconnections

摘录 / EXCERPT: For a compact spin Riemannian manifold X and a finite-dimensional complex Hermitian $\\mathbb{Z}/2\mathbb{Z}$-graded vector bundle $V\to X$ with unitary superconnection ∇,...

open_problem | open | MathDB 页面 | 原文线索

V2-029374 | MATHDB-395832

Stolz conjecture on positive Ricci curvature and the Witten genus

摘录 / EXCERPT: Let M^n be a spin manifold with $\frac{1}{2}p_1(M)=0\in H^4(M;\mathbb Z)$. If M admits a metric of positive Ricci curvature, then the Witten genus of M is zero.

conjecture | open | MathDB 页面 | 原文线索

V2-029375 | MATHDB-395833

Vanishing of the tmf-valued invariant under positive Ricci curvature and vanishing canonical 3-form

摘录 / EXCERPT: Let M^n be a spin manifold with $\frac{p_1}{2}(M)=0\u2208 H^4(M;\mathbb{Z})$. If M admits a string class $\mathcal{S}$ and a metric g such that g has positive Ricci curvatu...

open_problem | open | MathDB 页面 | 原文线索

V2-029376 | MATHDB-395870

Smoothly integrable geodesic flows on compact surfaces of genus greater than one

摘录 / EXCERPT: Do there exist compact surfaces of genus greater than one carrying geodesic flows that admit a C^∞-smooth integral, despite admitting no analytically integrable geodesic flo...

open_problem | open | MathDB 页面 | 原文线索

V2-029377 | MATHDB-395875

McDuff’s ellipsoid-ball packing equivalence

摘录 / EXCERPT: For each $k\in\mathbb N$, $E(1,k)$ symplectically embeds into $B^4(A)=E(A,A)$ if and only if the disjoint union of k copies of $B^4(1)$ symplectically embeds into $B^4(A)$.

open_problem | open | MathDB 页面 | 原文线索

V2-029378 | MATHDB-395924

Fully nonlinear Yamabe problem on locally conformally flat manifolds with umbilic boundary

摘录 / EXCERPT: Let 4(M^n,g)4 be a locally conformally flat Riemannian manifold with umbilic boundary and dimension 4nge 34. Assume that, for some integer 2le kle n, the Schouten tensor ... satisfie...

open_problem | open | MathDB 页面 | 原文线索

V2-029379 | MATHDB-395965

Signature operator theorem for Witt spaces

摘录 / EXCERPT: Let $\widehat X$ be any smoothly stratified pseudomanifold satisfying the Witt hypothesis, and let g be any iterated conic metric on the regular part X of $\widehat X$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-029380 | MATHDB-395976

Actions of non-Hausdorff groupoids on C*-algebras

摘录 / EXCERPT: Let G be a non-Hausdorff, locally Hausdorff groupoid, and let EG be its universal proper space, expected to be a non-Hausdorff, locally Hausdorff manifold bundle over G^0. Co...

open_problem | open | MathDB 页面 | 原文线索

V2-029381 | MATHDB-396001

Lehn’s conjecture on symplectic reductions of irreducible stable polar representations

摘录 / EXCERPT: Let (G,V) be an irreducible, stable, polar representation of a simple algebraic group, and let $\mathfrak c\subseteq V$ be a Cartan subspace. Then there is a subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-029382 | MATHDB-396002

Characterizing when symplectic reduction yields a symplectic singularity

摘录 / EXCERPT: Symplectic reduction does not always lead to a symplectic singularity. Can we make precise when it does?

open_problem | open | MathDB 页面 | 原文线索

V2-029383 | MATHDB-396027

4π-Conjecture (Meeks-Yau-Nitsche)

摘录 / EXCERPT: If Γ is a simple closed curve in $\mathbb{R}^3$ with total curvature at most 4π, then Γ bounds a unique compact, orientable, branched minimal surface, and this un...

conjecture | open | MathDB 页面 | 原文线索

V2-029384 | MATHDB-396045

Lamination Closure Theorem

摘录 / EXCERPT: If M is a minimal surface in a 3-manifold N with injectivity radius bounded away from zero, i.e. $\\operatorname{inj} M \ge C>0$, then M is a minimal lamination.

open_problem | open | MathDB 页面 | 原文线索

V2-029385 | MATHDB-396046

Asymptotic uniqueness of the helicoid for one-ended embedded minimal surfaces

摘录 / EXCERPT: Every nonplanar, properly embedded, one-ended minimal surface $M\subset\mathbb{R}^3$ with finite genus and infinite total curvature is conformally a compact Riemann surface punctur...

open_problem | open | MathDB 页面 | 原文线索

V2-029386 | MATHDB-396047

Asymptotic classification of complete embedded minimal annuli with infinite total curvature

摘录 / EXCERPT: Every complete, embedded minimal annulus $E\subset\mathbb{R}^3$ with compact boundary and infinite total curvature satisfies: (a) E is properly embedded in $\mathbb{R}^3$ and con...

open_problem | open | MathDB 页面 | 原文线索

V2-029387 | MATHDB-396048

Duality between CMC graphs in Bianchi-Cartan-Vranceanu spaces

摘录 / EXCERPT: There exists a duality between the moduli space of graphs with constant mean curvature H in $E^3(\\kappa,\\tau)$ and the moduli space of spacelike graphs with constant mean curva...

open_problem | open | MathDB 页面 | 原文线索

V2-029388 | MATHDB-396049

Nonexistence of entire spacelike maximal graphs in Lorentzian Heisenberg space

摘录 / EXCERPT: There exists no entire spacelike graph with zero mean curvature in the Lorentzian Heisenberg space $\operatorname{Nil}^{1}_{3}(\tau)$ for $\tau\ne 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-029389 | MATHDB-396184

Payne upper bound for the first buckling/Krein eigenvalue

摘录 / EXCERPT: For a domain Ω whose boundary $\partial\Omega$ is everywhere of nonnegative mean curvature, the report states the solved estimate ... Here $\lambda_{K,1}(\Omega)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029390 | MATHDB-396265

Natural product-\SL_2 action on symplectic branching multiplicity spaces

摘录 / EXCERPT: For $(\mu,\lambda)\in\Lambda_{n-1,n}=\Lambda_{n-1}\times\Lambda_n\$, let ... with $\lambda_{n+1}=\mu_0=0$ and F_r the $(r+1)$-dimensional irreducible SL_2-module. Is the...

open_problem | open | MathDB 页面 | 原文线索

V2-029391 | MATHDB-396266

Canonical product action on symplectic branching multiplicity spaces

摘录 / EXCERPT: Let Λ_n be the set of partitions of length n, let $M=\mathcal O(U_n\backslash G_n/U_{n-1})$ be the branching algebra for $G_n=\operatorname{Sp}_{2n}$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029392 | MATHDB-396309

Giambelli formula for symplectic Grassmannians

摘录 / EXCERPT: Let $0\le k\le n$, and let $\operatorname{IG}(n-k,2n)$ be the symplectic Grassmannian of $(n-k)$-dimensional isotropic subspaces of $\mathbb{C}^{2n}$. Its Schubert classes are inde...

open_problem | open | MathDB 页面 | 原文线索

V2-029393 | MATHDB-396325

Contractibility of the generalized Kakimizu complex

摘录 / EXCERPT: Let E be a compact, connected, orientable, irreducible, and $\\partial$-irreducible 3-manifold. Let γ be a union of oriented, disjoint simple closed curves on...

open_problem | open | MathDB 页面 | 原文线索

V2-029394 | MATHDB-396327

Ballmann-Buyalo rank-rigidity conjecture

摘录 / EXCERPT: Let X be a locally compact geodesically complete CAT(0) space and let Γ be a discrete group acting properly and cocompactly by isometries on X. Then one of the followi...

conjecture | open | MathDB 页面 | 原文线索

V2-029395 | MATHDB-396330

Quasi-isometric classification of graph manifolds

摘录 / EXCERPT: If M and M' are non-geometric graph manifolds, possibly with boundary, then the following are equivalent: 1. The universal covers $[U+007F]M$ and $[U+007F]M'$ are bilipschitz homeomorphic. 2...

open_problem | open | MathDB 页面 | 原文线索

V2-029396 | MATHDB-396334

Embeddedness and uniqueness of index-one CMC spheres in Solv(3)

摘录 / EXCERPT: Prove that a (possibly immersed) CMC sphere Σ_H in $\operatorname{Solv}(3)$ with mean curvature H and index 1 is actually embedded and unique up to congruence.

open_problem | open | MathDB 页面 | 原文线索

V2-029397 | MATHDB-396336

Noether-theoretic origin of the generalized Hopf differential

摘录 / EXCERPT: Does the existence of the generalized Hopf differential for CMC surfaces in homogeneous 3-manifolds $M^3_{\d4f,\d49}$ also follow from Noether's theorem?

open_problem | open | MathDB 页面 | 原文线索

V2-029398 | MATHDB-396338

Are all constant-mean-curvature spheres in \(\mathrm{Sol}_3\) isoperimetric?

摘录 / EXCERPT: Is it true that all CMC spheres are isoperimetric in Sol_3?

open_problem | open | MathDB 页面 | 原文线索

V2-029399 | MATHDB-396348

Homogeneity of isoparametric hypersurfaces with six principal curvatures

摘录 / EXCERPT: Isoparametric hypersurfaces in S^{13} with $(g,m)=(6,2)$ are homogeneous and are given by the adjoint G_2 orbits in its Lie algebra.

open_problem | open | MathDB 页面 | 原文线索

V2-029400 | MATHDB-396350

Theorem 2: minimization of the Sadowsky functional for a fixed tangent image

摘录 / EXCERPT: Let $T:[0,\tilde L]\to S^2$ be an arclength-parametrized curve with curvature λ. Among all strips $\gamma:[0,\tilde L]\to\mathbb{R}^3$ having tangent image T, the strip .....

open_problem | open | MathDB 页面 | 原文线索

V2-029401 | MATHDB-396352

Classification of properly embedded mod-2 area-minimizing surfaces in complete flat 3-manifolds

摘录 / EXCERPT: Classify properly embedded area-minimizing surfaces (mod 2) in complete flat 3-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029402 | MATHDB-396354

Constructing periodic mean-curvature functions

摘录 / EXCERPT: The report asks how to obtain a periodic function satisfying ... and ... It answers that, for the curvature $k(s)$ of any planar smooth closed curve Γ, the function...

open_problem | open | MathDB 页面 | 原文线索

V2-029403 | MATHDB-396384

Uniqueness of instantaneously complete Ricci flows

摘录 / EXCERPT: Let M^2 be a surface and let g_0 be an arbitrary smooth metric on M, possibly incomplete and possibly of unbounded curvature. The report conjectures that the smooth Ricci flo...

open_problem | open | MathDB 页面 | 原文线索

V2-029404 | MATHDB-396385

Generalized Chern-Milnor conjecture for closed aspherical manifolds

摘录 / EXCERPT: The conjecture is that closed aspherical manifolds with nonzero Euler characteristic admit no flat structure.

conjecture | open | MathDB 页面 | 原文线索

V2-029405 | MATHDB-396386

Min-Oo's conjecture for the spherical hemisphere

摘录 / EXCERPT: If g is a metric on the standard hemisphere S_+^n whose scalar curvature is greater than or equal to $n(n-1)$, and g coincides with the standard metric in a neighborhood of t...

conjecture | open | MathDB 页面 | 原文线索

V2-029406 | MATHDB-396387

Local collapsing implies graph-manifold structure

摘录 / EXCERPT: The report states the following as its main result. An orientable 3-manifold M is a graph manifold if there is a finite disjoint collection of embedded tori...

open_problem | open | MathDB 页面 | 原文线索

V2-029407 | MATHDB-396388

Donaldson's Calabi-Yau estimates on symplectic four-manifolds

摘录 / EXCERPT: Let 4(M,omega)u00024 be a compact symplectic four-manifold equipped with an almost-complex structure 4Ju00024 tamed by omega. Let sigma be a smooth volume form on M. If widetildeom...

open_problem | open | MathDB 页面 | 原文线索

V2-029408 | MATHDB-396389

Donaldson's Conjecture 4: tamed-to-compatible symplectic forms

摘录 / EXCERPT: Let (M^4,ω) be a compact symplectic 4-manifold with $b^+(M)=1$, and let J be an almost-complex structure on M tamed by ω. Then there exists another symplectic f...

conjecture | open | MathDB 页面 | 原文线索

V2-029409 | MATHDB-396390

Epsilon-regularity conjecture for the Calabi-Yau equation on symplectic four-manifolds

摘录 / EXCERPT: Let $(M,\c9)$ be a compact symplectic four-manifold equipped with an almost-complex structure J tamed by $\c9$. Let $\c3$ be a smooth volume form on M. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-029410 | MATHDB-396391

ε-regularity theorem for the Calabi-Yau equation on symplectic four-manifolds

摘录 / EXCERPT: In the setting of Conjecture 3-that is, for a compact symplectic four-manifold $(M,\a0\omega)$ with an almost-complex structure J tamed by ω, a smooth volume form σ...

open_problem | open | MathDB 页面 | 原文线索

V2-029411 | MATHDB-396393

Canonical surgical contraction of exceptional divisors under the Kähler-Ricci flow

摘录 / EXCERPT: Let (X,g_0) be a compact Kähler manifold. Suppose there exists a holomorphic map $\[X \to Y$ blowing down disjoint exceptional divisors $E_1,\ldots,E_k$ on X, where Y is a c...

open_problem | open | MathDB 页面 | 原文线索

V2-029412 | MATHDB-396410

Identification of the residual term with the Brüning-Ma anomaly

摘录 / EXCERPT: Under the notation of Theorem 1, for the bounded generalized cone $M^m=(0,1]\times N$ with $g_M=dx^2\oplus x^2g_N$ over a closed oriented Riemannian manifold (N^n,g_N), the resid...

open_problem | open | MathDB 页面 | 原文线索

V2-029413 | MATHDB-396411

Metric invariance of analytic torsion for simple edge singularities

摘录 / EXCERPT: Let (M,g_M) be an odd-dimensional compact Riemannian manifold with a simple edge singularity B. Let $U=(0,1)\times Y$ be an open neighborhood of B, where Y is the total spa...

open_problem | open | MathDB 页面 | 原文线索

V2-029414 | MATHDB-396417

Positive density of curvatures in integer Apollonian circle packings

摘录 / EXCERPT: Let $\kappa(P,X)$ denote the number of distinct curvatures less than X of circles in an integer Apollonian packing P. Then, for X sufficiently large, $\kappa(P,X)\gg X$, wher...

open_problem | open | MathDB 页面 | 原文线索

V2-029415 | MATHDB-396431

Flaring for models constructed from Weil-Petersson geodesics

摘录 / EXCERPT: It would be interesting to know if models constructed from Weil-Petersson geodesics satisfy such a flaring condition, and thereby give rise to models of hyperbolic 3-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029416 | MATHDB-396451

Symplectic ball-packing criterion for ellipsoid embeddings

摘录 / EXCERPT: For the weight expansion $w(a)=(w_1(a),\ldots,w_k(a))$ associated with a rational $a\ge 1$, the embedding problem $E(1,a)\mathrel{\hookrightarrow_s}B^4(\lambda)$ is equivalent to t...

open_problem | open | MathDB 页面 | 原文线索

V2-029417 | MATHDB-396452

Symplectic embedding of $E(1,2)$ into a four-dimensional ball

摘录 / EXCERPT: Determine the smallest $\[1m\lambda>0\[0m$ such that the closed ellipsoid $E(1,2)$ symplectically embeds into the closed ball $B^4(\lambda)=E(\lambda,\lambda)$. The report solves t...

open_problem | open | MathDB 页面 | 原文线索

V2-029418 | MATHDB-396453

Symplectic embedding capacity into the ellipsoid $E(\lambda,2\lambda)$ (equivalently, the cube $C^4(\lambda)$)

摘录 / EXCERPT: For a rational $a\ge 1$ with weight expansion $w(a)=(w_1(a),\ldots,w_k(a))$, the embedding-capacity problem for $E(1,a)\overset{s}{\hookrightarrow}E(\lambda,2\lambda)$ is equivalen...

open_problem | open | MathDB 页面 | 原文线索

V2-029419 | MATHDB-396455

Smooth rigidity at infinity of negatively curved manifolds

摘录 / EXCERPT: Let M be a closed Riemannian manifold of negative curvature, and let $\widetilde M$ be its universal cover, compactified by its ideal sphere at infinity $\partial\widetilde M$. T...

open_problem | open | MathDB 页面 | 原文线索

V2-029420 | MATHDB-396459

Are compact partially hyperbolic geodesic flows Anosov?

摘录 / EXCERPT: Let (M,g) be a compact Riemannian manifold and let $\phi^t:SM\to SM$ be a partially hyperbolic geodesic flow. Is it true that ϕ^t is Anosov?

open_problem | open | MathDB 页面 | 原文线索

V2-029421 | MATHDB-396461

Commutator width one for identity-component diffeomorphism groups

摘录 / EXCERPT: Is it true that, for any smooth (C^∞) compact manifold M, every element of $\mathrm{Diff}_0(M)$-the group of all smooth diffeomorphisms of M isotopic to the identity-can...

open_problem | open | MathDB 页面 | 原文线索

V2-029422 | MATHDB-396464

Counterexample to Min-Oo’s conjecture

摘录 / EXCERPT: Let M be a compact Riemannian manifold of dimension n whose boundary ∂M is totally geodesic and isometric to the standard sphere S^{n-1}. Min-Oo’s conjecture asser...

conjecture | open | MathDB 页面 | 原文线索

V2-029423 | MATHDB-396466

Min-Oo’s conjecture for the hemisphere

摘录 / EXCERPT: Min-Oo’s conjecture. Let M be a compact Riemannian manifold of dimension n whose boundary $\[?]$ is totally geodesic and is isometric to the standard sphere S^{n-1}. If the s...

conjecture | open | MathDB 页面 | 原文线索

V2-029424 | MATHDB-396504

Virtual Fibration Conjecture (Thurston)

摘录 / EXCERPT: If M is a finite-volume hyperbolic 3-manifold, then M has a finite cover which fibers over the circle, i.e. is a surface bundle over the circle.

conjecture | open | MathDB 页面 | 原文线索

V2-029425 | MATHDB-396509

Volume rigidity conjecture for representations into a quotient of \(\mathrm{SL}(n,\mathbb C)\)

摘录 / EXCERPT: Let M be a complete, oriented, hyperbolic 3-manifold of finite volume, and let $\rho:\pi_1(M)\to \mathrm{SL}(n,\mathbb C)/\{\pm I\}$ be a representation. We have … with equality…

conjecture | open | MathDB 页面 | 原文线索

V2-029426 | MATHDB-396510

Integral combinations of Galois-conjugate hyperbolic volumes

摘录 / EXCERPT: Every exotic volume is an integral linear combination of (Galois-conjugate) volumes of hyperbolic manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029427 | MATHDB-396514

Short Geodesic Conjecture

摘录 / EXCERPT: There is a universal positive lower bound for the length of geodesics of arithmetic hyperbolic 2- and 3-orbifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-029428 | MATHDB-396515

Subpolynomial volume growth for short systoles in dimensions n ≥ 4

摘录 / EXCERPT: It is unknown whether, for $n\ge 4$, there exist hyperbolic n-manifolds M with $\operatorname{Syst}_1(M)\to 0$ and $\operatorname{Vol}(M)$ growing slower than a polynomial in...

open_problem | open | MathDB 页面 | 原文线索

V2-029429 | MATHDB-396524

Secretly Seifert-fibered hyperbolic 3-manifolds

摘录 / EXCERPT: Which closed hyperbolic 3-manifolds are secretly Seifert-fibered? Here a closed oriented 3-manifold M is secretly Seifert-fibered if M is not Seifert-fibered, but there exists…

open_problem | open | MathDB 页面 | 原文线索

V2-029430 | MATHDB-396577

Genus bound for families of canonically polarized manifolds

摘录 / EXCERPT: If a compact curve C parameterizes a non-isotrivial family of canonically polarized manifolds, then its genus satisfies $g(C)>1$.

open_problem | open | MathDB 页面 | 原文线索

V2-029431 | MATHDB-396619

Nonexistence of Kervaire invariant one elements for $j\ge 7$

摘录 / EXCERPT: The element $\theta_j\in \pi_{2^{j+1}-2}(S^0)$, representing h_{2^j} in the Adams spectral sequence and corresponding to a framed manifold of the same dimension with nontrivial K...

open_problem | open | MathDB 页面 | 原文线索

V2-029432 | MATHDB-396620

Existence of smooth framed manifolds with nontrivial Kervaire invariant

摘录 / EXCERPT: For which m is there a smooth framed manifold of dimension $4m+2$ with nontrivial Kervaire invariant?

open_problem | open | MathDB 页面 | 原文线索

V2-029433 | MATHDB-396645

Rationality of the Fano threshold

摘录 / EXCERPT: Let (X,E) satisfy the report's Setup: X is a complex projective Fano manifold of index i_X, E is a rank-two vector bundle on X, $\rho(X)=1$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-029434 | MATHDB-396749

Asymptotic Nielsen realization under bounded geometry

摘录 / EXCERPT: Let R be a Riemann surface satisfying the bounded geometry condition, meaning that the injectivity radii at all points of R are uniformly bounded from above and below. Then eve...

open_problem | open | MathDB 页面 | 原文线索

V2-029435 | MATHDB-396917

Multiple Haefliger-Wu Invariant Conjecture

摘录 / EXCERPT: Conjecture 7 (refuted in the report). In the PL category, the multiple Haefliger-Wu invariant is bijective for any connected PL n-manifold N and any $m \ge n+3$.

conjecture | open | MathDB 页面 | 原文线索

V2-029436 | MATHDB-396927

Additivity of \(\Gamma\) under boundary connected sum

摘录 / EXCERPT: For a 3-manifold M with boundary, define ... where Δ ranges over simplicial triangulations of M. If $M\sqcup N$ denotes the boundary connected sum obtained by gluing two s...

open_problem | open | MathDB 页面 | 原文线索

V2-029437 | MATHDB-396939

Chazarain's wave-trace singular-support theorem

摘录 / EXCERPT: Let (X,g) be a compact Riemannian manifold, let $\Delta=d^*d$ be its Laplace operator, and define the length spectrum by ... Let ... where the trace is understood in the distribution...

open_problem | open | MathDB 页面 | 原文线索

V2-029438 | MATHDB-396942

Rationality of L^2-Betti numbers

摘录 / EXCERPT: Are the L^2-Betti numbers of a compact manifold M, defined from harmonic L^2-forms on its universal covering, always rational?

open_problem | open | MathDB 页面 | 原文线索

V2-029439 | MATHDB-396944

Asymptotic behavior of analytic torsion in a degenerating Kähler family

摘录 / EXCERPT: Let X be a connected Kähler manifold of dimension $n+1$, let $S\subset\mathbb{C}$ be the unit disc, and let $\pi:X\to S$ be a proper surjective holomorphic map with connected fib...

open_problem | open | MathDB 页面 | 原文线索

V2-029440 | MATHDB-396982

Bartnik's static Einstein metric extension conjecture

摘录 / EXCERPT: The Bartnik conjecture concerns asymptotically flat manifolds with nonnegative scalar curvature, no closed minimal hypersurfaces, and an inner boundary B. It is conjectured that...

conjecture | open | MathDB 页面 | 原文线索

V2-029441 | MATHDB-397001

Bott-Grove conjecture for almost nonnegatively curved manifolds

摘录 / EXCERPT: Are almost nonnegatively curved manifolds rationally elliptic?

conjecture | open | MathDB 页面 | 原文线索

V2-029442 | MATHDB-397003

Gromoll-Meyer theorem for isometry-invariant geodesics

摘录 / EXCERPT: Let M be a compact simply connected Riemannian manifold and let A be an isometry. If the Betti numbers of the A-invariant free loop space LM^A are unbounded, then M admit...

open_problem | open | MathDB 页面 | 原文线索

V2-029443 | MATHDB-397005

Vector bundles over products with tori and nonnegative curvature

摘录 / EXCERPT: Given a manifold $B=C\times T$ of sectional curvature ≥ 0, which vector bundles over B admit metrics of nonnegative sectional curvature? Here C denotes a simply-connected c...

open_problem | open | MathDB 页面 | 原文线索

V2-029444 | MATHDB-397009

Curvature conditions yielding finite-cover splittings with a cuplength bound

摘录 / EXCERPT: Are there other curvature conditions on closed manifolds that give product splittings up to a finite cover with the cuplength condition stated above?

open_problem | open | MathDB 页面 | 原文线索

V2-029445 | MATHDB-397011

Nonexistence, in general, of geodesics between points in the space of Kähler metrics

摘录 / EXCERPT: Lempert presented a result showing that, in general, two given points in the space of Kähler metrics cannot be connected by a geodesic. The report states that this solved a long-st...

open_problem | open | MathDB 页面 | 原文线索

V2-029446 | MATHDB-397017

Closedness of the strongly Gauduchon property under holomorphic deformations

摘录 / EXCERPT: Let $\pi:X\to\Delta$ be a complex analytic family of compact complex manifolds, meaning that π is a proper holomorphic submersion from a complex manifold X to an open disc...

open_problem | open | MathDB 页面 | 原文线索

V2-029447 | MATHDB-397021

Plurisubharmonic exhaustion of locally pseudoconvex domains over Kähler manifolds

摘录 / EXCERPT: Let D be a locally pseudoconvex domain with twice continuously differentiable boundary in a complex manifold X. If X admits a Kähler metric, does D admit a plurisubharmonic...

open_problem | open | MathDB 页面 | 原文线索

V2-029448 | MATHDB-397025

Radon transform for smooth algebraic submanifolds of a linearly concave domain

摘录 / EXCERPT: The report states the following theorem. Let $D\subset\mathbb{CP}^{n}$ be a linearly concave domain with $n\ge 2$, and let $V\subset\mathbb{CP}^{n}$ be an $(n-m)$-dimensional algeb...

open_problem | open | MathDB 页面 | 原文线索

V2-029449 | MATHDB-397032

Necessity of the sign choice in real enumerative geometry

摘录 / EXCERPT: Give a mathematical justification that the sign choice leading to the observed cancellation between even-degree disks and cross-caps is necessary, rather than merely supported by p...

open_problem | open | MathDB 页面 | 原文线索

V2-029450 | MATHDB-397094

Sharp closed-geodesic counting formula for compact hyperbolic surfaces

摘录 / EXCERPT: Let M be a compact hyperbolic surface, and let $\\pi_T$ denote the number of closed geodesics of length at most T. The report presents the conjecture ... where the a_j form a s...

open_problem | open | MathDB 页面 | 原文线索

V2-029451 | MATHDB-397114

Asymptotic dimension versus ordinary dimension of closed aspherical manifolds

摘录 / EXCERPT: Can the asymptotic dimension of a closed aspherical manifold exceed its (ordinary) dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-029452 | MATHDB-397116

Dranishnikov’s asymptotic-dimension dichotomy problem

摘录 / EXCERPT: Is the asymptotic dimension of an aspherical n-manifold always either n or $\\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-029453 | MATHDB-397142

BKMP/remodeling conjecture for open Gromov-Witten invariants

摘录 / EXCERPT: Let X be a 3-dimensional Calabi-Yau manifold with a toric symmetry, and let $L\subset X$ be a toric sublagrangian manifold. For the open Gromov-Witten invariants...

conjecture | open | MathDB 页面 | 原文线索

V2-029454 | MATHDB-397159

Euler characteristic bounded by simplicial volume for aspherical manifolds

摘录 / EXCERPT: Does there exist a constant $C(n)$ depending only on the dimension n such that $\lvert\chi(M)\rvert\le C(n)\,\lVert M\rVert$ for any aspherical manifold M?

open_problem | open | MathDB 页面 | 原文线索

V2-029455 | MATHDB-397161

Surface bundles over closed hyperbolic manifolds with hyperbolic fundamental group

摘录 / EXCERPT: Does there exist a surface bundle over a closed hyperbolic n-manifold whose fundamental group is Gromov hyperbolic, for some $n\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-029456 | MATHDB-397188

Critical-exponent rigidity for cocompact actions on ACu(-1) spaces

摘录 / EXCERPT: Let X be a proper geodesic Gromov-hyperbolic space that is an $AC_u(-1)$-space. Suppose that a group $\b3$ acts on X isometrically, properly discontinuously, and cocompactly. L...

open_problem | open | MathDB 页面 | 原文线索

V2-029457 | MATHDB-397191

Polar actions of cohomogeneity two on positively curved manifolds

摘录 / EXCERPT: Every polar action on a simply connected, compact, positively curved manifold of cohomogeneity two is equivariantly diffeomorphic to a polar action on a compact rank-one symmetric...

open_problem | open | MathDB 页面 | 原文线索

V2-029458 | MATHDB-397196

Obstructions to smoothing PL negatively curved metrics

摘录 / EXCERPT: Are there any obstructions to smoothing a PL metric of negative curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029459 | MATHDB-397197

Tightness under contact surgery

摘录 / EXCERPT: If (M,ξ) is a tight contact manifold and (M',ξ') is obtained from (M,ξ) by contact surgery, then (M',ξ') is also tight.

open_problem | open | MathDB 页面 | 原文线索

V2-029460 | MATHDB-397198

Examples with all possible orders of algebraic torsion in all dimensions

摘录 / EXCERPT: Are there examples in all dimensions of contact manifolds with all possible orders of algebraic torsion?

open_problem | open | MathDB 页面 | 原文线索

V2-029461 | MATHDB-397200

Bounded geodesics and closed geodesics in asymptotically Euclidean metrics

摘录 / EXCERPT: Suppose g is a Riemannian metric on $\bb R^n$ that is asymptotic to the Euclidean metric at infinity. Assume there exists a non-constant geodesic $c:\mathbb R\to(\bb R^n,g)$ that...

open_problem | open | MathDB 页面 | 原文线索

V2-029462 | MATHDB-397201

White’s question on stationary integral varifolds

摘录 / EXCERPT: Let N be a compact Riemannian manifold with k-convex boundary that contains a non-zero stationary k-varifold, where $1 \le k \le \dim N-1$. Does this imply that N contains...

open_problem | open | MathDB 页面 | 原文线索

V2-029463 | MATHDB-397203

Bounded-or-infinite multiplicity of closed geodesics on compact Finsler manifolds

摘录 / EXCERPT: For each positive integer n, there exist positive integers $1\le p_n\le q_n$, with $p_n\to+\infty$ as $n\to+\infty$, such that, for every compact Finsler manifold (M,F) with...

open_problem | open | MathDB 页面 | 原文线索

V2-029464 | MATHDB-397246

Spectral multiplier bounds for first-order self-adjoint operators

摘录 / EXCERPT: Let M be a compact manifold with smooth measure, let G be a Hermitian vector bundle over M, and let D be a first-order self-adjoint differential operator acting on sections...

open_problem | open | MathDB 页面 | 原文线索

V2-029465 | MATHDB-397269

No cancellation for three-dimensional positive-scalar-curvature manifolds

摘录 / EXCERPT: The report conjectures that if M_j are three-dimensional manifolds with positive scalar curvature and no interior closed minimal surfaces, then there is no cancellation; conseque...

open_problem | open | MathDB 页面 | 原文线索

V2-029466 | MATHDB-397271

An optimal embeddedness criterion for stable codimension 1 integral varifolds

摘录 / EXCERPT: Let V be a stationary stable integral n-varifold on a smooth $(n+1)$-dimensional Riemannian manifold B. Stability on the regular part is understood as ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-029467 | MATHDB-397282

Okounkov bodies in tropical geometry

摘录 / EXCERPT: Can we construct Okounkov bodies in tropical setting?

open_problem | open | MathDB 页面 | 原文线索

V2-029468 | MATHDB-397303

The joints problem in three-dimensional incidence geometry

摘录 / EXCERPT: Given a set L of n lines in $\r\mathbb{R}^3$, determine a sharp upper bound for the number of points incident to at least three non-coplanar lines (the joints of L). The repo...

open_problem | open | MathDB 页面 | 原文线索

V2-029469 | MATHDB-397351

Index problem for the contact hypoelliptic operator P_γ

摘录 / EXCERPT: Let X be a compact contact manifold without boundary of dimension $2n+1$, with contact form θ. Let $\gamma:X\to\mathbb{C}\setminus\{\ldots,-n-4,-n-2,-n,n,n+2,n+4,\ldots\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-029470 | MATHDB-397364

Identification of the Radul cocycle class

摘录 / EXCERPT: Let M be a closed, not necessarily orientable, smooth manifold. Let $\mathrm{CS}(M)=\mathrm{CL}(M)/\mathrm{CL}^{-\infty}(M)$ be the algebra of formal classical pseudodifferential...

open_problem | open | MathDB 页面 | 原文线索

V2-029471 | MATHDB-397384

Logarithmic version of the Tian-Yau-Donaldson conjecture

摘录 / EXCERPT: There is a constant scalar curvature conic Kähler metric on (X,Y) if and only if (X,Y) is log-K-stable.

conjecture | open | MathDB 页面 | 原文线索

V2-029472 | MATHDB-397387

Beauville’s question on Lagrangian fibrations

摘录 / EXCERPT: Let X be a hyperkähler manifold and $L\subset X$ a Lagrangian submanifold biholomorphic to a complex torus. Is L a fibre of a (meromorphic) Lagrangian fibration...

open_problem | open | MathDB 页面 | 原文线索

V2-029473 | MATHDB-397440

Local well-posedness for mean-curvature flow with higher-dimensional triple junctions

摘录 / EXCERPT: Let $(\Gamma_0^1,\Gamma_0^2,\Gamma_0^3)$ be a $C^{3+\alpha}$ surface cluster with a $C^{3+\alpha}$ triple-junction curve γ. Assume that the initial cluster satisfies the ang...

open_problem | open | MathDB 页面 | 原文线索

V2-029474 | MATHDB-397441

Global existence for graphical mean curvature flow with locally Lipschitz initial data

摘录 / EXCERPT: Consider the Cauchy problem ... where $T\in(0,\infty]$. The reported result is that, whenever u_0 is locally Lipschitz (in particular, $u_0\in C^1(\mathbb{R}^n)$ suffices), this pr...

open_problem | open | MathDB 页面 | 原文线索

V2-029475 | MATHDB-397448

Elliptic regularization minimizers converge to curves of maximal slope

摘录 / EXCERPT: Let (X,d) be a complete metric space and let $\phi:X\to(-\infty,+\infty]$ be proper, lower semicontinuous, λ-geodesically convex, and have compact sublevels. For...

open_problem | open | MathDB 页面 | 原文线索

V2-029476 | MATHDB-397456

Adapting Witten’s positive-mass proof to nonspin manifolds

摘录 / EXCERPT: Can one adapt Witten’s proof of the positive mass theorem to nonspin manifolds?

open_problem | open | MathDB 页面 | 原文线索

V2-029477 | MATHDB-397470

Invariant positive-Ricci metrics beyond cohomogeneities 3 and 5

摘录 / EXCERPT: Can one find compact manifolds with more than two non-principal orbits and a group-invariant metric of positive Ricci curvature in cohomogeneities other than 3 and 5?

open_problem | open | MathDB 页面 | 原文线索

V2-029478 | MATHDB-397471

Maximal symmetry-rank conjecture for simply connected nonnegatively curved manifolds

摘录 / EXCERPT: If M is a compact, simply connected, nonnegatively curved Riemannian n-manifold, then its symmetry rank satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-029479 | MATHDB-397472

Classification of closed simply connected nonnegatively curved fixed-point homogeneous 5-manifolds

摘录 / EXCERPT: Let M^5 be a closed, simply connected, 5-dimensional nonnegatively curved fixed-point homogeneous G-manifold. Then G is $\\mathrm{SO}(5)$, $\mathrm{SO}(4)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-029480 | MATHDB-397473

Polyhedral analogue of Frankel conjecture (Conjecture 1)

摘录 / EXCERPT: Conjecture 1. Consider a non-negatively curved non-flat polyhedral manifold M. Suppose that the holonomy of the metric on M is irreducible and $b_2(M)>0$. Then M has a natura...

conjecture | open | MathDB 页面 | 原文线索

V2-029481 | MATHDB-397477

Extension of the Hopf conjecture with symmetry to all even dimensions

摘录 / EXCERPT: Does the following extension of Theorem 1 hold in all even dimensions? Let M^n be a closed Riemannian manifold with positive sectional curvature and an effective, isometric T^r...

conjecture | open | MathDB 页面 | 原文线索

V2-029482 | MATHDB-397481

Constancy of the elliptic genus under positive sectional curvature

摘录 / EXCERPT: Let (M,g) be a closed Riemannian Spin manifold with $\operatorname{sec}(M)>0$. Is the elliptic genus constant as a modular function, i.e. does $\varphi(M)=\operatorname{sign}(M)$...

open_problem | open | MathDB 页面 | 原文线索

V2-029483 | MATHDB-397505

Continuous-time limit of the thermalized rate-independent process

摘录 / EXCERPT: For a finite-dimensional smooth Riemannian manifold (Q,g), an energetic potential $E:[0,T]\times Q\to\mathbb{R}$, and a continuous, convex, non-degenerate dissipation potential...

open_problem | open | MathDB 页面 | 原文线索

V2-029484 | MATHDB-397604

Infinite-dimensional geometry for general Kac-Moody algebras and groups

摘录 / EXCERPT: There is a rich infinite-dimensional geometry lurking behind all Kac-Moody algebras and groups, which parallels the finite-dimensional and affine theory.

open_problem | open | MathDB 页面 | 原文线索

V2-029485 | MATHDB-397605

E10 conjecture for eleven-dimensional supergravity

摘录 / EXCERPT: The E10 conjecture asserts that the values of all fields of 11-dimensional supergravity and their spatial gradients can be mapped to a null geodesic motion in the infinite-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-029486 | MATHDB-397644

Equality of the Picard-Fuchs and Lagrangian-Floer superpotentials

摘录 / EXCERPT: Let X be a toric manifold with numerically effective anticanonical line bundle $-K_X$, and let W^{PF} and W^{LF} be the superpotentials in the mirror introd...

open_problem | open | MathDB 页面 | 原文线索

V2-029487 | MATHDB-397646

Finiteness of Hodge numbers of irreducible Calabi-Yau manifolds

摘录 / EXCERPT: Are there only finitely many Hodge numbers of n-dimensional irreducible Calabi-Yau manifolds over $\mathbb{C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-029488 | MATHDB-397668

Mirror symmetry between orbifold projective lines and cusp singularities

摘录 / EXCERPT: Let $A=(a_1,a_2,a_3)$ be a triple of positive integers. Set ... and let ... There should exist isomorphisms of Frobenius manifolds among $M_{\mathbb P^1_A}$, the Frobenius manifold con...

open_problem | open | MathDB 页面 | 原文线索

V2-029489 | MATHDB-397671

Taut angle structure on an orientable 3-manifold triangulation

摘录 / EXCERPT: Given an orientable 3-manifold triangulation T, determine whether there exists a taut angle structure on T. Here a taut angle structure assigns interior angles...

open_problem | open | MathDB 页面 | 原文线索

V2-029490 | MATHDB-397673

Lens space recognition

摘录 / EXCERPT: Given a triangulated 3-manifold (M,Δ), determine whether M is homeomorphic to a lens space, i.e. to one of the manifolds ... where $p,q\in\mathbb N$, $1\le q\le p$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-029491 | MATHDB-397676

Additivity of \[1m\[0m\(\Gamma\) under connected sum

摘录 / EXCERPT: For closed manifolds M_1 and M_2, with Γ defined as in the report, is ...

open_problem | open | MathDB 页面 | 原文线索

V2-029492 | MATHDB-397680

Geometric triangulation conjecture for hyperbolic 3-manifolds with torus boundary

摘录 / EXCERPT: It is conjectured that every hyperbolic 3-manifold with torus boundary components has a decomposition into positive-volume ideal hyperbolic tetrahedra.

conjecture | open | MathDB 页面 | 原文线索

V2-029493 | MATHDB-397688

Finite-ring \a0PSL(2,R) separation for compact 3-manifold groups

摘录 / EXCERPT: If M is a compact 3-manifold and $\gamma \in \pi_1(M) \setminus \{1}$, there exists a finite commutative ring R with identity and a homomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-029494 | MATHDB-397693

Explicit example of a collapsible manifold that is not a ball

摘录 / EXCERPT: Construct explicitly an example of a collapsible manifold with boundary that is not homeomorphic to a ball.

open_problem | open | MathDB 页面 | 原文线索

V2-029495 | MATHDB-397701

Geometric triangulations of cusped hyperbolic 3-manifolds

摘录 / EXCERPT: Every hyperbolic 3-manifold with torus boundary components has a decomposition into positive-volume ideal hyperbolic tetrahedra (a geometric triangulation).

open_problem | open | MathDB 页面 | 原文线索

V2-029496 | MATHDB-397706

Geometric ideal triangulations of cusped hyperbolic 3-manifolds

摘录 / EXCERPT: If M^3 is a cusped hyperbolic manifold, does it admit a triangulation into geometric ideal tetrahedra with positive volume?

open_problem | open | MathDB 页面 | 原文线索

V2-029497 | MATHDB-397708

Collapsible simplicial manifolds and CAT(0) polyhedral metrics

摘录 / EXCERPT: Are there collapsible simplicial manifolds whose underlying space does not admit any CAT(0) polyhedral metric?

open_problem | open | MathDB 页面 | 原文线索

V2-029498 | MATHDB-397710

Cauchy problem for parallel spinors

摘录 / EXCERPT: Given a spin manifold (M,g) with a spinor Φ and a symmetric endomorphism field W satisfying the generalized Killing-spinor equation $∇^M_X\Phi = \frac{1}{2}W(X)·\Phi$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-029499 | MATHDB-397711

Removal of the product-structure assumption in the singular Cheeger-Müller reduction

摘录 / EXCERPT: Can the product-structure assumption in Theorem 2 be dispensed with? Equivalently, for a singular manifold decomposed as $M=U\cup_{\partial M_1}M_1$, does the stated reduction of t...

open_problem | open | MathDB 页面 | 原文线索

V2-029500 | MATHDB-397712

Smooth Cauchy problem for Riemannian Einstein metrics

摘录 / EXCERPT: Let (M^n,g) be a smooth Riemannian manifold and let $W\in\operatorname{End}(TM)$ be a smooth symmetric endomorphism field satisfying ... Does there exist, locally near M, an...

open_problem | open | MathDB 页面 | 原文线索

V2-029501 | MATHDB-397713

Existence of hypersurface tensors near zero Gaussian curvature

摘录 / EXCERPT: For a Riemannian surface (M,g), determine whether, locally near points where the Gaussian curvature vanishes, there exists a symmetric endomorphism field...

open_problem | open | MathDB 页面 | 原文线索

V2-029502 | MATHDB-397714

Analytic Cauchy problem for parallel spinors

摘录 / EXCERPT: Let (M,g) be a real-analytic spin manifold carrying a real-analytic generalized Killing spinor Φ with respect to a symmetric endomorphism field W, meaning...

open_problem | open | MathDB 页面 | 原文线索

V2-029503 | MATHDB-397722

Delocalized coarse APS-index theorem in the remaining dimensions

摘录 / EXCERPT: Does the delocalized coarse APS-index theorem hold in general, beyond the model calculation established only when m is odd? Namely, if X is a spin manifold of dimension $m+1$ w...

open_problem | open | MathDB 页面 | 原文线索

V2-029504 | MATHDB-397723

Instantaneous curvature boundedness for nonnegatively curved Ricci flows

摘录 / EXCERPT: The question whether the Ricci flow instantaneously has bounded curvature has a negative answer: there exists an immortal 3-dimensional nonnegatively curved complete Ricci flow...

open_problem | open | MathDB 页面 | 原文线索

V2-029505 | MATHDB-397732

Unique limiting cylinders for mean-curvature-flow neckpinch singularities

摘录 / EXCERPT: Mean-curvature-flow neckpinch singularities have unique limiting cylinders.

open_problem | open | MathDB 页面 | 原文线索

V2-029506 | MATHDB-397742

The Willmore inequality and equality case for embedded surfaces in S^3

摘录 / EXCERPT: Let $\Sigma \subset S^3$ be an embedded closed surface of genus $g\ge 1$. With Willmore energy ... where H is the mean curvature of Σ in S^3, one has ... Equality holds if a...

open_problem | open | MathDB 页面 | 原文线索

V2-029507 | MATHDB-397743

Generalized Hitchin-Thorpe inequality for Einstein edge-cone metrics

摘录 / EXCERPT: Let (M,Σ) be a pair consisting of a smooth compact oriented 4-manifold and a fixed smoothly embedded compact oriented surface. If (M,Σ) admits an Einstein edge-cone...

open_problem | open | MathDB 页面 | 原文线索

V2-029508 | MATHDB-397744

Eventual cessation of surgeries in long-time Ricci flow with surgery

摘录 / EXCERPT: In the generic long-time case, namely when Perelman’s Ricci flow with surgery on a closed 3-manifold exists on $[0,\infty)$, is it always true that only finitely many surgeries occ...

open_problem | open | MathDB 页面 | 原文线索

V2-029509 | MATHDB-397747

Compactness for the Yamabe problem on manifolds with umbilic boundary

摘录 / EXCERPT: Let (M^n,g) be a smooth compact Riemannian manifold of dimension $3\le n\le 24$ with umbilic boundary, and suppose that (M,g) is not conformally equivalent to the standard hemi...

open_problem | open | MathDB 页面 | 原文线索

V2-029510 | MATHDB-397748

Failure of the boundary Yamabe compactness theorem in dimensions n ≥ 25

摘录 / EXCERPT: Does the compactness conclusion of Theorem 1 remain valid in dimensions $n\ge 25$? More precisely, for every smooth compact Riemannian manifold (M^n,g) with umbilic boundary that...

open_problem | open | MathDB 页面 | 原文线索

V2-029511 | MATHDB-397754

Embeddedness in the CMC doubling construction

摘录 / EXCERPT: Assume that $n\ge 2$, (M,g) is a compact $(n+1)$-dimensional Riemannian manifold, and Λ is a smooth, compact, orientable, nondegenerate minimal hypersurface immersed in...

open_problem | open | MathDB 页面 | 原文线索

V2-029512 | MATHDB-397755

Surgery stability of curvature conditions

摘录 / EXCERPT: Let M^n be a Riemannian manifold. A p-surgery replaces an embedded region $S^{n-p-1}\times B^{p+1}$ by $B^{n-p}\times S^p$, identifying their common boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-029513 | MATHDB-397762

Parity theory for knots in arbitrary 3-manifolds

摘录 / EXCERPT: Is there any parity theory for knots in arbitrary 3-manifolds?

open_problem | open | MathDB 页面 | 原文线索

V2-029514 | MATHDB-397766

Legendrian surgery preserves tightness

摘录 / EXCERPT: If (M,ξ) is any tight contact manifold and (M',ξ') is obtained from (M,ξ) by Dehn surgery along a Legendrian knot with framing -1 relative to the canonical contact fr...

open_problem | open | MathDB 页面 | 原文线索

V2-029515 | MATHDB-397795

Symplectic finite-order automorphisms act trivially on the Chow group of a K3 surface

摘录 / EXCERPT: Let X be a complex projective K3 surface and let $f:X\to X$ be a symplectic automorphism of finite order, meaning that $f^*\sigma=\sigma$ for a nonzero regular two-form σ...

open_problem | open | MathDB 页面 | 原文线索

V2-029516 | MATHDB-397827

Necessity of the genericity assumption for equivariant spectral determination

摘录 / EXCERPT: Are there examples of non-generic, non-isometric symplectic toric orbifolds with the same equivariant spectrum?

open_problem | open | MathDB 页面 | 原文线索

V2-029517 | MATHDB-397828

Equivariant inverse spectral determination of labelled moment polytopes

摘录 / EXCERPT: Does the equivariant spectrum of a symplectic toric orbifold equipped with a toric metric determine the labelled moment polytope of the orbifold?

open_problem | open | MathDB 页面 | 原文线索

V2-029518 | MATHDB-397831

Maximization of Steklov eigenvalues over Riemannian metrics

摘录 / EXCERPT: Let Ω be a smooth compact connected Riemannian manifold with boundary $\Sigma=\partial\Omega$, and let $n=\dim\Sigma$. For a fixed $k\in\mathbb N$, maximize $\sigma_k(g)$ ov...

open_problem | open | MathDB 页面 | 原文线索

V2-029519 | MATHDB-397860

Geodesic rays escaping to infinity on Euclidean cone-surfaces

摘录 / EXCERPT: Let M be a metrically complete non-compact Euclidean surface with finitely many cone singularities of negative curvature. Is it true that for every point $x\in M$ there exists a...

open_problem | open | MathDB 页面 | 原文线索

V2-029520 | MATHDB-397864

The Lichnerowicz conjecture for general harmonic manifolds

摘录 / EXCERPT: Does the Lichnerowicz conjecture, modified to account for the Damek-Ricci counterexamples, hold in full generality for harmonic manifolds without any homogeneity or cocompactness a...

conjecture | open | MathDB 页面 | 原文线索

V2-029521 | MATHDB-397865

Modified Lichnerowicz conjecture for asymptotically harmonic manifolds

摘录 / EXCERPT: Does the Lichnerowicz conjecture, as modified in view of the Damek-Ricci counterexamples, hold in full generality for asymptotically harmonic Cartan-Hadamard manifolds, without any...

conjecture | open | MathDB 页面 | 原文线索

V2-029522 | MATHDB-397866

Classification of asymptotically harmonic manifolds

摘录 / EXCERPT: Does the modified Lichnerowicz conjecture hold for asymptotically harmonic manifolds in full generality, without any homogeneity or cocompactness assumption-namely, must every such...

open_problem | open | MathDB 页面 | 原文线索

V2-029523 | MATHDB-397867

Local middle-dimensional non-squeezing for symplectic images of balls

摘录 / EXCERPT: Let $\varphi:\mathbb{R}^{2n}\to\mathbb{R}^{2n}$ be a symplectomorphism and let $x\in\mathbb{R}^{2n}$. For $1<k<n$, is it true that ... for every sufficiently small $r>0$? Here...

open_problem | open | MathDB 页面 | 原文线索

V2-029524 | MATHDB-397869

Aspherical groups and manifolds with extreme properties

摘录 / EXCERPT: Every finitely generated group with a recursive aspherical presentation (i.e. a recursive aspherical 2-dimensional $K(\cdot,1)$) embeds into a finitely presented group with a finit...

open_problem | open | MathDB 页面 | 原文线索

V2-029525 | MATHDB-397876

Explicit full packing of the four-torus by one ball

摘录 / EXCERPT: Explicitly construct a full symplectic packing of the four-torus $T^4=\mathbb{R}^4/\mathbb{Z}^4$ by one ball; equivalently, construct a symplectic embedding of $B^4(\sqrt{2})$ into...

open_problem | open | MathDB 页面 | 原文线索

V2-029526 | MATHDB-397877

Symplectic embeddings of four-dimensional ellipsoids into balls

摘录 / EXCERPT: For the open ellipsoid ... determine the smallest capacity of a four-dimensional ball into which it symplectically embeds. After scaling to $a_1=1$, this is the problem of determinin...

open_problem | open | MathDB 页面 | 原文线索

V2-029527 | MATHDB-397878

Uniform Poincaré inequalities on horospheres

摘录 / EXCERPT: Let M be a closed, negatively curved $(n+1)$-dimensional Riemannian manifold, let $\widetilde M$ be its universal cover, and let X be a horosphere of $\widetilde M$ with its in...

open_problem | open | MathDB 页面 | 原文线索

V2-029528 | MATHDB-397879

Poincaré inequalities on horospheres in negatively curved manifolds

摘录 / EXCERPT: Let M be a closed negatively curved manifold of dimension $n+1$, let $\widetilde M$ be its universal cover, and let X be a horosphere of $\widetilde M$ endowed with its induced...

open_problem | open | MathDB 页面 | 原文线索

V2-029529 | MATHDB-397880

Growth of the free rank in congruence covers of arithmetic hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be an arithmetic hyperbolic 3-manifold defined by a quadratic form, and let $M_i\to M$ be a sequence of its congruence covers. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-029530 | MATHDB-397881

Gromov’s PL realization question for hyperbolic orbifold quotients

摘录 / EXCERPT: Let M^n be an arbitrary compact PL manifold without boundary. Is there a cocompact lattice Γ in $O(n,1)$ such that the quotient $\mathbb{H}^n/\Gamma$ is PL-isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-029531 | MATHDB-397882

Symplectic Calabi-Yau six-manifolds with prescribed fundamental group

摘录 / EXCERPT: For any finitely presented group G there exists a compact symplectic six-manifold with $c_1=0$ and with fundamental group equal to G.

open_problem | open | MathDB 页面 | 原文线索

V2-029532 | MATHDB-397887

Boundedness and word-length equivalence of the ball Hofer norm

摘录 / EXCERPT: Let $B\subset(M,\omega)$ be a ball in a symplectic manifold, let $x_0\in B$, and let $\gamma\in\pi_1(M,x_0)$. Define ... Assume that $D_B(\gamma)\neq\varnothing$ for every...

open_problem | open | MathDB 页面 | 原文线索

V2-029533 | MATHDB-397890

Quasi-isometry of horospheres in negatively curved manifolds

摘录 / EXCERPT: Let (X^{n+1},g) be a closed manifold with negative sectional curvature, and let $(\widetilde X^{n+1},\widetilde g)$ be its universal cover. Are all horospheres...

open_problem | open | MathDB 页面 | 原文线索

V2-029534 | MATHDB-397891

Quasi-isometry of horospheres in negatively curved universal covers

摘录 / EXCERPT: Let (X^{n+1},g) be a closed manifold with negative sectional curvature, and let $(\widetilde X^{n+1},\widetilde g)$ be its universal cover. Are all the horospheres...

open_problem | open | MathDB 页面 | 原文线索

V2-029535 | MATHDB-397901

Nonembedded stable minimal surfaces in strictly mean convex 3-manifolds

摘录 / EXCERPT: Let M be a compact, orientable, strictly mean convex 3-manifold. Then there exists a simple closed curve $\Gamma\subset\partial M$ that bounds a stable minimal surface in M w...

open_problem | open | MathDB 页面 | 原文线索

V2-029536 | MATHDB-397902

Embeddedness of stable minimal disks in mean-convex 3-manifolds

摘录 / EXCERPT: For every mean-convex 3-manifold M and every simple closed curve $\Gamma\subset\partial M$, is every stable minimal disk in M with boundary Γ embedded?

open_problem | open | MathDB 页面 | 原文线索

V2-029537 | MATHDB-397904

Nontriviality of Type I Ricci-flow blow-up limits

摘录 / EXCERPT: Let $(M^n,g(t),p)$, $t\in[0,T)$, $p\in\Sigma$, be a complete Type I Ricci flow, where Σ is the set of points having no neighborhood on which the curvature remains bounded as...

open_problem | open | MathDB 页面 | 原文线索

V2-029538 | MATHDB-397907

Existence of immersed spheres minimizing curvature functionals in compact 3-manifolds

摘录 / EXCERPT: Let M be a compact 3-dimensional Riemannian manifold. On the class $[S^2,M]$ of immersions $f:S^2\to M$, define ... Assume that: 1. there exists an $f\in[S^2,M]$ with $E(f)<4\pi$...

open_problem | open | MathDB 页面 | 原文线索

V2-029539 | MATHDB-397908

Prescribed Gauss curvature by conformal metrics on closed surfaces of genus greater than one

摘录 / EXCERPT: For a closed Riemann surface (M,g_0) with $g(M)>1$, where g_0 has constant Gauss curvature k_0 and normalized volume, characterize exactly those functions $f\in C^\infty(M)$...

open_problem | open | MathDB 页面 | 原文线索

V2-029540 | MATHDB-397913

Isoperimetric character of the Huisken-Yau foliation leaves

摘录 / EXCERPT: Whether the leaves of the Huisken-Yau foliation through stable constant mean curvature spheres, [U+007F]$\{\Sigma_H\}_{0<H<H_0}$, are in fact isoperimetric.

open_problem | open | MathDB 页面 | 原文线索

V2-029541 | MATHDB-397990

Virtual positivity of hyperbolic volume and Gromov simplicial volume

摘录 / EXCERPT: A closed 3-manifold has a virtually positive hyperbolic volume if and only if its Gromov simplicial volume is positive.

open_problem | open | MathDB 页面 | 原文线索

V2-029542 | MATHDB-397992

Optimal volume bounds by cusp type

摘录 / EXCERPT: What are the optimal values of positive constants v_1 and v_2 such that every hyperbolic 3-manifold having r_1 Klein-bottle cusps and r_2 torus cusps satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-029543 | MATHDB-397993

Linearity of the normalized-volume function for hyperbolic 3-manifolds

摘录 / EXCERPT: For a rational number $t\in[0,1]$, let $A(t)$ be the infimum of $\nu(M):=\operatorname{vol}(M)/r$ over all hyperbolic 3-manifolds M with r cusps, of which r_1 are Klein-bot...

open_problem | open | MathDB 页面 | 原文线索

V2-029544 | MATHDB-398002

Virtual positive Betti number in a specific arithmetic case

摘录 / EXCERPT: Let $M=\mathbb{H}^3/\Gamma$ be a compact arithmetically defined hyperbolic 3-manifold. Suppose that the defining field k contains a subfield k' such that the field extension...

open_problem | open | MathDB 页面 | 原文线索

V2-029545 | MATHDB-398007

Kobayashi’s conjecture: hyperbolicity implies general type

摘录 / EXCERPT: Every compact projective complex manifold X that is Kobayashi hyperbolic is of general type.

conjecture | open | MathDB 页面 | 原文线索

V2-029546 | MATHDB-398010

Symmetric differentials and finite images of fundamental-group representations

摘录 / EXCERPT: Let X be a compact connected Kähler manifold with no global holomorphic symmetric differentials. Then every finite-dimensional representation of $\\pi_1(X)$ over any field has fi...

open_problem | open | MathDB 页面 | 原文线索

V2-029547 | MATHDB-398011

Characterization of compact projective manifolds by their bounded-domain universal covers

摘录 / EXCERPT: Given a bounded domain $\Omega \subset \mathbb{C}^n$, determine how to tell when a projective manifold X with ample canonical divisor K_X has universal covering biholomorphic t...

open_problem | open | MathDB 页面 | 原文线索

V2-029548 | MATHDB-398014

Existence of compact non-Kähler Hermitian-symplectic manifolds

摘录 / EXCERPT: Does there exist a compact complex Hermitian-symplectic manifold that does not admit a Kähler structure?

open_problem | open | MathDB 页面 | 原文线索

V2-029549 | MATHDB-398018

Real ample polarizations versus rational multipolarizations for slope stability

摘录 / EXCERPT: For the projective manifold considered in the report, slope stability with respect to any real ample polarization $H\in\operatorname{Amp}_{\mathbb R}(X)$ is equivalent to slope sta...

open_problem | open | MathDB 页面 | 原文线索

V2-029550 | MATHDB-398019

Surjectivity of the Albanese morphism for compact Kähler manifolds with nef anticanonical bundle

摘录 / EXCERPT: Let X be a compact Kähler manifold such that $-K_X$ is nef. Then its Albanese morphism $\alpha_X:X\to\operatorname{Alb}(X)$ is surjective.

open_problem | open | MathDB 页面 | 原文线索

V2-029551 | MATHDB-398037

The virtually Haken conjecture

摘录 / EXCERPT: Every closed aspherical 3-manifold has a finite-sheeted cover that is Haken.

conjecture | open | MathDB 页面 | 原文线索

V2-029552 | MATHDB-398063

Local properties of symplectic mixed polynomial singularities

摘录 / EXCERPT: Do the following properties hold locally near an isolated singularity at 0 for every mixed polynomial mapping $f:\mathbb{C}^{2}\to\mathbb{C}$ that is locally symplectic near the...

open_problem | open | MathDB 页面 | 原文线索

V2-029553 | MATHDB-398065

Contact structure on the link of a mixed homogeneous covering lift

摘录 / EXCERPT: Assume that $f(z)$ is a convenient non-degenerate holomorphic function and let $g=\cphi_{a,b}^{*}f$ be its mixed homogeneous covering lift, where ... Assume $a>b>0$ and define the li...

open_problem | open | MathDB 页面 | 原文线索

V2-029554 | MATHDB-398215

Geodesic of length at least the average degree in a subgraph of the hypercube

摘录 / EXCERPT: Every subgraph G of Q_n of average degree d contains a geodesic of length at least d.

open_problem | open | MathDB 页面 | 原文线索

V2-029555 | MATHDB-398236

Hard Lefschetz theorem for translation-invariant curvature measures

摘录 / EXCERPT: Let V be a vector space with $\dim V=n$. Let L be the derivation operator on curvature measures, defined in the report by ... For every integer k satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-029556 | MATHDB-398308

Finite-time full faceting in planar crystalline curvature flow

摘录 / EXCERPT: For the planar anisotropic curvature flow $V=M(\vec n)(\kappa_\gamma+\sigma)$ with crystalline interfacial energy density γ_0, positive constant driving force σ, and...

open_problem | open | MathDB 页面 | 原文线索

V2-029557 | MATHDB-398390

Pinkall-Sterling conjecture for embedded CMC tori

摘录 / EXCERPT: Every embedded constant mean curvature torus in the three-dimensional sphere is axially symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-029558 | MATHDB-398396

Willmore conjecture for immersed tori in \(\mathbb{R}^3\)

摘录 / EXCERPT: The integral of the square of the mean curvature of a torus immersed in $\mathbb{R}^3$ is at least $2\pi^2$. Equality is achieved by the torus of revolution whose generating circle...

conjecture | open | MathDB 页面 | 原文线索

V2-029559 | MATHDB-398409

Symplectic measure of hyperbolic links

摘录 / EXCERPT: What is $A([L])$? Is it positive when L is a hyperbolic link?

open_problem | open | MathDB 页面 | 原文线索

V2-029560 | MATHDB-398436

Viehweg-Zuo hyperbolicity statement for maximal-variation families

摘录 / EXCERPT: Let $f:Z\to B$ be an algebraic proper submersion between connected quasi-projective manifolds. Assume that all fibres of f have semi-ample canonical bundle, and that the variatio...

open_problem | open | MathDB 页面 | 原文线索

V2-029561 | MATHDB-398443

Uniqueness of equivariant compactification structures on Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold of dimension n with Picard number 1, different from $\mathbb{P}^n$. Assume that X has a family of minimal rational curves whose VMRT...

open_problem | open | MathDB 页面 | 原文线索

V2-029562 | MATHDB-398454

Nonexistence of a compact moduli space from asymptotic Chow semistability

摘录 / EXCERPT: Given a family of canonically polarized manifolds, does asymptotically Chow semistability yield a compact moduli space?

open_problem | open | MathDB 页面 | 原文线索

V2-029563 | MATHDB-398509

Lott’s secondary index theorem conjecture

摘录 / EXCERPT: There should be a natural transformation $KR^{0}\to KR_{\\mathbb{R}/\mathbb{Z}}^{-1}$ such that, for a proper submersion of manifolds $\pi:W\to B$ equipped with the stated metric a...

conjecture | open | MathDB 页面 | 原文线索

V2-029564 | MATHDB-398510

First part of Lott’s secondary index conjecture

摘录 / EXCERPT: Let R be the ring of integers in a number field. For every smooth manifold M, Lott’s secondary K-group admits a natural transformation ... under which the generator...

conjecture | open | MathDB 页面 | 原文线索

V2-029565 | MATHDB-398524

Diameter growth for long-time 3-dimensional Ricci flows on a single geometric piece

摘录 / EXCERPT: Is the diameter of the manifold bounded by $C\sqrt{t}$ for all sufficiently large t whenever the underlying manifold consists of a single geometric piece?

open_problem | open | MathDB 页面 | 原文线索

V2-029566 | MATHDB-398526

Chern’s conjecture for closed affine manifolds

摘录 / EXCERPT: The Euler characteristic of every closed affine manifold vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-029567 | MATHDB-398528

Gluing Einstein metrics to obtain critical metrics of quadratic curvature functionals

摘录 / EXCERPT: Is it possible to glue together Einstein metrics and produce a critical point of a certain Riemannian functional generalizing the Einstein-Hilbert functional?

open_problem | open | MathDB 页面 | 原文线索

V2-029568 | MATHDB-398530

Is every CMC sphere in Sol$_3$ isoperimetric or stable?

摘录 / EXCERPT: In $\\mathrm{Sol}_3$ with its standard left-invariant metric, is every compact constant-mean-curvature sphere a solution to the isoperimetric problem, or even stable?

open_problem | open | MathDB 页面 | 原文线索

V2-029569 | MATHDB-398531

Classification of compact stable CMC surfaces in Nil_3

摘录 / EXCERPT: Classify the compact stable constant-mean-curvature surfaces in the Heisenberg group Nil3; the report conjectures that every such surface is a rotational sphere.

open_problem | open | MathDB 页面 | 原文线索

V2-029570 | MATHDB-398532

Positive sectional curvature on $S^2\times S^2$

摘录 / EXCERPT: Does $S^2\times S^2$ carry a metric with $K>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-029571 | MATHDB-398579

Infinitely many minimal embedded hypersurfaces in positive Ricci curvature

摘录 / EXCERPT: Every manifold of dimension no bigger than eight having a metric of positive Ricci curvature admits an infinite number of minimal embedded hypersurfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-029572 | MATHDB-398583

Classification of planar critical curves for the Möbius energy

摘录 / EXCERPT: Theorem 1. Let $\Gamma\subset\mathbb{R}^2$ be a properly embedded open or closed smooth curve of bounded curvature satisfying

open_problem | open | MathDB 页面 | 原文线索

V2-029573 | MATHDB-398585

Willmore conjecture for immersed tori in Euclidean three-space

摘录 / EXCERPT: For every torus immersed in Euclidean three-space, the integral of the square of its mean curvature is at least $2\pi^2$. In the report’s convention, this is...

conjecture | open | MathDB 页面 | 原文线索

V2-029574 | MATHDB-398592

Monotonicity of the entropy under mean curvature flow

摘录 / EXCERPT: For a smooth family of bounded domains $(\Omega_t)_{t\in(0,T)}$ whose boundary $M_t=\partial\Omega_t$ evolves with normal speed $\beta=H_{M_t}$ (mean curvature), and with...

open_problem | open | MathDB 页面 | 原文线索

V2-029575 | MATHDB-398593

Li-Yau-Hamilton Harnack inequalities for entropy minimizers on mean-curvature-flow domains

摘录 / EXCERPT: Establish Li-Yau-Hamilton-type Harnack inequalities for solutions f of the Neumann problem associated with the Euler-Lagrange equation for $[U+007F]mu_\beta(\Omega_t,\tau(t))$, in parti...

open_problem | open | MathDB 页面 | 原文线索

V2-029576 | MATHDB-398645

Lower bound for the rank of the moduli group of positive-scalar-curvature metrics

摘录 / EXCERPT: Let M be a closed spin manifold with $\pi_1(M)=\Gamma$ and $\dim M=2k+1\geq 5$, which carries a positive scalar curvature metric. Then the rank of the abelian group...

open_problem | open | MathDB 页面 | 原文线索

V2-029577 | MATHDB-398647

Rank lower bound for the moduli group when the fundamental group has torsion

摘录 / EXCERPT: Let M be a closed spin manifold which carries a positive scalar curvature metric, with $\dim M=2k+1\ge 5$. If $\Gamma=\pi_1(M)$ is not torsion free, then the rank of the abelian...

open_problem | open | MathDB 页面 | 原文线索

V2-029578 | MATHDB-398719

Salamon-LeBrun conjecture

摘录 / EXCERPT: Is every compact quaternion-Kähler manifold with positive scalar curvature (equivalently, with Einstein constant $ \lambda>0$) symmetric?

conjecture | open | MathDB 页面 | 原文线索

V2-029579 | MATHDB-398723

Are all contact lines on a contact Fano manifold standard?

摘录 / EXCERPT: Let X be a contact Fano manifold of dimension $2n+1$, with $X\not\simeq\mathbb P^{2n+1}$. A contact line is a rational curve $C\subset X$ with normalization...

open_problem | open | MathDB 页面 | 原文线索

V2-029580 | MATHDB-398725

VMRT Conjecture 1′ for Fano contact manifolds

摘录 / EXCERPT: Let X be a Fano manifold with $b_2(X)=1$ equipped with a contact structure $F\subset TX$. If C_x at a general point $x\in X$ is a connected nonlinear Legendrean submanifold of...

conjecture | open | MathDB 页面 | 原文线索

V2-029581 | MATHDB-398726

Homogeneity conjecture for Fano manifolds via VMRT hypersurfaces

摘录 / EXCERPT: Let X be a Fano manifold with $b_2(X)=1$. If C_x at a general point $x\in X$ is a nonlinear smooth hypersurface of PT_xX, then X is homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-029582 | MATHDB-398727

LeBrun-Salamon conjecture in real dimension 12

摘录 / EXCERPT: Every quaternionic Kähler manifold of real dimension 12 with positive scalar curvature is a symmetric space.

conjecture | open | MathDB 页面 | 原文线索

V2-029583 | MATHDB-398728

LeBrun-Salamon conjecture for positive quaternionic Kähler manifolds

摘录 / EXCERPT: Every quaternionic Kähler manifold with positive scalar curvature is a symmetric space.

conjecture | open | MathDB 页面 | 原文线索

V2-029584 | MATHDB-398729

Characteristic-number relations and the Salamon conjecture

摘录 / EXCERPT: For a positive quaternionic Kähler manifold M of quaternionic dimension $n\ge 2$ and positive scalar curvature $\kappa>0$, can one prove sufficiently many relations among the fou...

conjecture | open | MathDB 页面 | 原文线索

V2-029585 | MATHDB-398730

Geometric formality and rational ellipticity of positive quaternion-Kähler manifolds

摘录 / EXCERPT: A positive quaternion-Kähler manifold is geometrically formal and rationally elliptic.

open_problem | open | MathDB 页面 | 原文线索

V2-029586 | MATHDB-398822

Symplectic periods for the class \(\mathcal G'\)

摘录 / EXCERPT: Let $\mathcal L'$ be the class of ladder representations. Define $\mathcal G'$ to consist of representations of the form $\theta=Q(\Delta_1,\ldots,\Delta_{2r})\in\mathcal L'$ such...

open_problem | open | MathDB 页面 | 原文线索

V2-029587 | MATHDB-398831

Explicit determination of reducibility data for supercuspidal representations of symplectic groups

摘录 / EXCERPT: Describe explicitly the set $\operatorname{Red}(\pi)=\{(\rho,s_\rho)\mid s_\rho\ge 1\}$ in terms of the local data used to construct the irreducible supercuspidal representation...

open_problem | open | MathDB 页面 | 原文线索

V2-029588 | MATHDB-398836

Theory of finite-time invariant manifolds and flow barriers

摘录 / EXCERPT: Develop a theory of finite-time invariant manifolds that specifically locates finite-time flow barriers and establishes connections with any diagnostic method.

open_problem | open | MathDB 页面 | 原文线索

V2-029589 | MATHDB-398844

Determination of the minimal field of definition of a projective residual representation

摘录 / EXCERPT: Let $\rho_\bullet=(\rho_\lambda)_\lambda$ be the compatible symplectic system described in the report, with $\rho_\lambda:G_{\mathbb Q}\to\operatorname{GSp}_{2n}(L_\lambda)$ and wi...

open_problem | open | MathDB 页面 | 原文线索

V2-029590 | MATHDB-398879

Finiteness of Euler characteristics of Stein fillings

摘录 / EXCERPT: For a contact 3-manifold (M,ξ), the set $C(M,\xi)=\{\chi(X)\mid X\text{ is a Stein filling of }(M,\xi)\}$ is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-029591 | MATHDB-398904

Whether every Kähler group is projective

摘录 / EXCERPT: Let a Kähler group mean a group occurring as the fundamental group of a closed Kähler manifold. Is the class of Kähler groups strictly larger than the class of fundamental groups o...

open_problem | open | MathDB 页面 | 原文线索

V2-029592 | MATHDB-398906

Finiteness of the fundamental group from vanishing symmetric differentials

摘录 / EXCERPT: Let X be a compact connected Kähler manifold such that $H^{0}(X,S^{>0}\Omega^{1}_{X})=0$. The expectation is that X has a finite étale fundamental group.

open_problem | open | MathDB 页面 | 原文线索

V2-029593 | MATHDB-398912

A factorization theorem for harmonic maps of low rank

摘录 / EXCERPT: Let X be a compact Kähler manifold with universal cover $\widetilde{X}$. Let G be a connected Lie group with simple Lie algebra and trivial center, with associated symmetric sp...

open_problem | open | MathDB 页面 | 原文线索

V2-029594 | MATHDB-398972

Unconditional uniqueness for equivariant wave maps on admissible curved backgrounds

摘录 / EXCERPT: Under the hypotheses of Theorem 1-namely, the equivariant wave-map Cauchy problem (3)-(4) on rotationally invariant manifolds with admissible base manifold, with the parameters and...

open_problem | open | MathDB 页面 | 原文线索

V2-029595 | MATHDB-398978

Continuity of Lyapunov exponents under convergence of affine measures

摘录 / EXCERPT: Theorem 5 asserts: Let (N_n) be a sequence of affine manifolds, and suppose that the affine measures ν_{N_n} converge to the affine measure ν as in Theorem 4. Then the Ly...

open_problem | open | MathDB 页面 | 原文线索

V2-029596 | MATHDB-398984

Classification of higher-rank orbit closures in the odd component of \(\mathcal H(4)\)

摘录 / EXCERPT: The only proper higher-rank affine invariant submanifold of $\mathcal H^{\mathrm{odd}}(4)$ is the Prym locus $\widetilde Q(3,-1^3)$, the canonical double cover of the stratum of qu...

open_problem | open | MathDB 页面 | 原文线索

V2-029597 | MATHDB-398994

Cutting sequences on Bouw-Möller surfaces

摘录 / EXCERPT: Characterize all possible cutting sequences of geodesic trajectories on the 24(m,n)4 Bouw-Möller surfaces with labeled edges, using the symmetries of these surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-029598 | MATHDB-399096

Natural moment maps from polarized manifolds to Okounkov bodies

摘录 / EXCERPT: Given a polarized manifold (X,L), a positive Hermitian metric on L, and an admissible flag used to define the Okounkov body $\Delta(L)$, does there exist, in general, a natural...

open_problem | open | MathDB 页面 | 原文线索

V2-029599 | MATHDB-399097

Comparison of gradient-flow constructions with symplectic contraction maps

摘录 / EXCERPT: Determine whether the gradient-flow construction corresponding to some valuation v_h coincides with the contraction map Φ_X in known cases.

open_problem | open | MathDB 页面 | 原文线索

V2-029600 | MATHDB-399111

Analyticity of the singularity formation set for finite-time Kähler-Ricci flow

摘录 / EXCERPT: Let (X) be a compact Kähler manifold, and let (omega(t)), (tin[0,T)), be a finite-time Kähler-Ricci flow singularity with (T<infty). Set (alpha=[omega0]-2pi T c1(X)), and define [...

open_problem | open | MathDB 页面 | 原文线索

V2-029601 | MATHDB-399113

Surface subgroups in closed rank-one locally symmetric manifolds

摘录 / EXCERPT: Let M be a closed rank one locally symmetric manifold. Then $\pi_1(M)$ contains surface subgroups.

open_problem | open | MathDB 页面 | 原文线索

V2-029602 | MATHDB-399115

Removing the Kähler assumption in the uniqueness theorem for the four-dimensional Stenzel/Eguchi-Hanson space

摘录 / EXCERPT: For $n=2$, is every complete Ricci-flat Riemannian manifold asymptotic at infinity to the cone (M_0,g_0) necessarily, up to scaling, the Stenzel manifold (M_1,g_1), without ass...

open_problem | open | MathDB 页面 | 原文线索

V2-029603 | MATHDB-399116

Smoothing polyhedral spaces by nonnegatively cosectionally curved Riemannian orbifolds

摘录 / EXCERPT: Any polyhedral space admits smoothing by Riemannian orbifolds with nonnegative cosectional curvature.

open_problem | open | MathDB 页面 | 原文线索

V2-029604 | MATHDB-399118

Evolution of complete noncompact warped product manifolds under Ricci flow

摘录 / EXCERPT: Let N be a flat, complete, connected Riemannian manifold of dimension $n\ge 2$. Let $h_0=f_0^2dx^2+g_0^2g_N$ be a complete warped product metric on $\mathbb{R}\times N$ with boun...

open_problem | open | MathDB 页面 | 原文线索

V2-029605 | MATHDB-399119

Global existence for the Teichmüller harmonic map flow into nonpositively curved targets

摘录 / EXCERPT: Let $M=M_\\gamma$ be a smooth closed orientable surface of genus $\\gamma\\ge 2$, let (N,G) be a smooth compact Riemannian manifold of any dimension with nonpositive sectional cu...

open_problem | open | MathDB 页面 | 原文线索

V2-029606 | MATHDB-399120

A generalization of Gromov’s almost flat manifold theorem

摘录 / EXCERPT: For every $n\in\mathbb{N}$ and every $D<\infty$, there exists $\varepsilon=\varepsilon(n,D)>0$ such that if (M,g) is a Riemannian n-dimensional manifold satisfying ... then M i...

open_problem | open | MathDB 页面 | 原文线索

V2-029607 | MATHDB-399121

Obstructions to anti-self-dual metrics

摘录 / EXCERPT: Find new obstructions to the existence of anti-self-dual metrics on smooth compact oriented 4-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029608 | MATHDB-399122

Anti-self-dual metrics and Weyl-functional minimizers on blow-ups of the projective plane

摘录 / EXCERPT: Conjecture. For any $k\le 9$, the 4-manifold $M=\mathbb{CP}^{2}\#k\,\overline{\mathbb{CP}}^{2}$ does not admit any anti-self-dual metrics. Moreover, for $k=0,\ldots,9$,...

open_problem | open | MathDB 页面 | 原文线索

V2-029609 | MATHDB-399123

Anti-self-dual metrics on the nine-point blow-up of the complex projective plane

摘录 / EXCERPT: Conjecture: The smooth oriented 4-manifold $M=\mathbb{CP}^{2}\#9\,\overline{\mathbb{CP}}^{2}$ does not admit any anti-self-dual metrics.

open_problem | open | MathDB 页面 | 原文线索

V2-029610 | MATHDB-399160

Complete stable minimal hypersurfaces in asymptotically flat manifolds

摘录 / EXCERPT: Problem (A). Are there complete, stable minimal hypersurfaces in asymptotically flat manifolds? Here, “complete” is understood in the report to implicitly refer to unbounded minima...

open_problem | open | MathDB 页面 | 原文线索

V2-029611 | MATHDB-399161

Corollary on Plateau solutions with diverging boundaries

摘录 / EXCERPT: Let (M,g) be an asymptotically Schwarzschildean 3-manifold of non-negative scalar curvature and positive ADM mass, let $\a0\Pi$ be a hyperplane, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029612 | MATHDB-399167

Existence of large conformal-metric solutions with the second blow-up profile

摘录 / EXCERPT: Let (M,g_0) be a closed Riemann surface of genus at least 2, with g_0 smooth, of constant curvature, and normalized volume 1. Let $f_0\\le 0$ be smooth and nonconstant, wit...

open_problem | open | MathDB 页面 | 原文线索

V2-029613 | MATHDB-399197

Optimal localization of asymptotically flat metrics with nonnegative scalar curvature

摘录 / EXCERPT: What is the optimal localization of an asymptotically flat metric of non-negative scalar curvature? For instance, can one construct an asymptotically flat metric of non-negative sc...

open_problem | open | MathDB 页面 | 原文线索

V2-029614 | MATHDB-399198

Are the large stable constant mean curvature spheres isoperimetric?

摘录 / EXCERPT: Are the large stable constant mean curvature spheres in the Euclidean regions in fact isoperimetric?

open_problem | open | MathDB 页面 | 原文线索

V2-029615 | MATHDB-399199

Positive mass for closed simply-connected non-spin manifolds in dimensions at least 8

摘录 / EXCERPT: For every integer $n\ge 8$, does there exist a closed, simply-connected, non-spin manifold M of dimension n such that, for every Riemannian metric g on M whose conformal La...

open_problem | open | MathDB 页面 | 原文线索

V2-029616 | MATHDB-399200

Compactness for embedded minimal hypersurfaces with bounded area and index

摘录 / EXCERPT: Let (N^{n+1},g) be an orientable, closed Riemannian manifold, and let $\{M_k^n\}\subset N$ be a sequence of stable, smooth, embedded minimal hypersurfaces with...

open_problem | open | MathDB 页面 | 原文线索

V2-029617 | MATHDB-399201

Existence of manifolds with infinite \ba-invariant

摘录 / EXCERPT: Does the case $\kappa(M)=\infty$ occur?

open_problem | open | MathDB 页面 | 原文线索

V2-029618 | MATHDB-399202

Improving the constant in the \widehat{A}-genus bound

摘录 / EXCERPT: Can one replace the factor 2^{2m-1} in Theorem 4 by 2^m? Here Theorem 4 states that if M is a compact spin manifold of dimension $n=4m$, then...

open_problem | open | MathDB 页面 | 原文线索

V2-029619 | MATHDB-399203

The positive mass theorem for distributional curvature

摘录 / EXCERPT: Let M be a smooth n-manifold, $n\ge 3$, endowed with a spin structure and a $C^0\cap W^{1,n}_{-q}$-regular, asymptotically flat Riemannian metric g, with $q\ge (n-2)/2$. If t...

open_problem | open | MathDB 页面 | 原文线索

V2-029620 | MATHDB-399212

Constructing and classifying solutions of the Einstein constraint equations

摘录 / EXCERPT: Construct and classify solutions of the constraint system ... for initial data consisting of a manifold M, a metric $\hat g$ on M, and a symmetric 2-tensor $\hat K$ on M.

open_problem | open | MathDB 页面 | 原文线索

V2-029621 | MATHDB-399214

Existence of metrics of positive scalar curvature

摘录 / EXCERPT: Given a smooth closed manifold M, when is the space of Riemannian metrics of positive scalar curvature on M non-empty; equivalently, when does M admit a metric of positive sc...

open_problem | open | MathDB 页面 | 原文线索

V2-029622 | MATHDB-399215

Michel’s boundary rigidity conjecture

摘录 / EXCERPT: Every simple compact Riemannian manifold with boundary is boundary rigid: the Riemannian metric is determined, up to isometry, by the boundary distance function $d_g(x,y)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-029623 | MATHDB-399218

Foliation condition for simple manifolds

摘录 / EXCERPT: Does every compact simple Riemannian manifold with boundary satisfy the foliation condition by strictly convex hypersurfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-029624 | MATHDB-399234

Finiteness of hyperbolic 3-manifolds with bounded volume and trace-field degree

摘录 / EXCERPT: For a given number $D>0$, are there only finitely many hyperbolic 3-manifolds whose volume and degree of its trace field are bounded by D?

open_problem | open | MathDB 页面 | 原文线索

V2-029625 | MATHDB-399235

Finiteness of Dehn fillings with bounded trace-field degree

摘录 / EXCERPT: For an n-cusped manifold M and a constant $D>0$, are there finitely many Dehn fillings of M whose trace fields have degree $[U+007F]\le D$?

open_problem | open | MathDB 页面 | 原文线索

V2-029626 | MATHDB-399239

Which symplectic-group representations arise from natural-characteristic Integral SO(3)-TQFT?

摘录 / EXCERPT: Which irreducible representations of $\operatorname{Sp}(2g,\mathbb{F}_p)$ arise as the representations F^{odd} and F/F^{odd} obtained from the first approxima…

open_problem | open | MathDB 页面 | 原文线索

V2-029627 | MATHDB-399254

Deformational invariance of plurigenera for compact Kähler manifolds

摘录 / EXCERPT: For a compact Kähler manifold X, is $\dim_{\mathbb C} H^0(X,mK_X)$ unchanged under holomorphic deformations of X for every integer $m a0 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-029628 | MATHDB-399255

Calabi-Yau theorem for the complex Monge-Ampère equation

摘录 / EXCERPT: Let X be a compact Kähler manifold of complex dimension n, let $ … Then the equation … admits a unique solution $[1mvarphiu001b[0min C^infty(X,mathbb{R})…

open_problem | open | MathDB 页面 | 原文线索

V2-029629 | MATHDB-399257

Finite Hodge-monodromy orbits of primitive MBM classes

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic (hyperkähler) manifold as defined in the report. Does the Hodge monodromy group act with finitely many orbits on the set of primiti…

open_problem | open | MathDB 页面 | 原文线索

V2-029630 | MATHDB-399258

Cone conjecture for projective irreducible holomorphic symplectic manifolds

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic manifold with Picard number at least 4. Then the automorphism group $\[?]$ acts with finitely many orbits on the set of…

conjecture | open | MathDB 页面 | 原文线索

V2-029631 | MATHDB-399259

Kawamata-Morrison cone conjecture for irreducible holomorphic symplectic manifolds

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold, and let $K(X)$ denote its Kähler cone. Then the group $\operatorname{Aut}(X)$ has only finitely many orbits on the set of…

conjecture | open | MathDB 页面 | 原文线索

V2-029632 | MATHDB-399262

Extension of Bezdek’s approach beyond Euclidean geometry

摘录 / EXCERPT: Is it possible to extend Bezdek’s approach to hyperbolic or spherical geometry, or to Riemannian and Finsler manifolds?

open_problem | open | MathDB 页面 | 原文线索

V2-029633 | MATHDB-399302

Gromov’s simplicial-volume/Euler-characteristic conjecture

摘录 / EXCERPT: Let M be an oriented, closed, connected aspherical manifold. Does vanishing simplicial volume imply vanishing Euler characteristic, i.e. does $\lVert M\rVert=0$ imply $\chi(M)=0$…

conjecture | open | MathDB 页面 | 原文线索

V2-029634 | MATHDB-399326

Induced Einstein geometry on the solvable factor

摘录 / EXCERPT: Equipping G_2 with the geometry it picks up as a subspace of G/K, is this an Einstein space? This question is closely related to the question of how A and G_1 fit together.…

open_problem | open | MathDB 页面 | 原文线索

V2-029635 | MATHDB-399336

Einstein solvmanifolds are maximally symmetric

摘录 / EXCERPT: If a solvable Lie group admits a left-invariant Einstein metric g of negative Ricci curvature, then g is maximally symmetric: for every other left-invariant Riemannian metric…

open_problem | open | MathDB 页面 | 原文线索

V2-029636 | MATHDB-399337

The Blaschke conjecture

摘录 / EXCERPT: Any Blaschke manifold is isometric to a compact rank-one-symmetric space ("CROSS").

conjecture | open | MathDB 页面 | 原文线索

V2-029637 | MATHDB-399374

Mostow rigidity theorem for finite-volume hyperbolic manifolds

摘录 / EXCERPT: The report states that the canonical map … from isometries between finite-volume hyperbolic manifolds M and N to isomorphisms between their fundamental groups modulo compositio…

open_problem | open | MathDB 页面 | 原文线索

V2-029638 | MATHDB-399418

Extension of the mean-curvature bound to domains with connected boundary

摘录 / EXCERPT: The report asks whether the inequality ... with equality if and only if Ω is a ball, can be extended from star-shaped domains with C^2 boundary to regular domains with conne...

open_problem | open | MathDB 页面 | 原文线索

V2-029639 | MATHDB-399512

Uniqueness of K-polystable degenerations of Fano Kähler-Einstein manifolds

摘录 / EXCERPT: Let $X\to C$ be a flat family over a pointed smooth curve $(C,0)$ with $0\in C$. Suppose: (1) $-K_{X/C}$ is relatively ample; (2) for every $t\in C^\circ:=C\setminus\{0\}$, X_t i...

open_problem | open | MathDB 页面 | 原文线索

V2-029640 | MATHDB-399518

Rationality of the Ueno-type manifold \(X_{4,6}\)

摘录 / EXCERPT: Let k be a field of characteristic different from $2,3$ containing a primitive third root of unity, let $E_6=\{(x,y)\mid y^2=x^3-1\}$, and let $X_{4,6}$ be the minimal resolution...

open_problem | open | MathDB 页面 | 原文线索

V2-029641 | MATHDB-399633

Ghys’s conjecture on Jordan diffeomorphism groups

摘录 / EXCERPT: For every smooth compact manifold M, the diffeomorphism group $\operatorname{Diff}(M)$ is Jordan; that is, there exists a constant C such that every finite subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-029642 | MATHDB-399667

Mathematical foundation of the bivariational principle

摘录 / EXCERPT: Since the bivariational functional E is not below bounded, can one be sure that a critical value of E, restricted to the manifold N, is in fact an approximation to an eigenva...

open_problem | open | MathDB 页面 | 原文线索

V2-029643 | MATHDB-399669

Liberation of general compact real algebraic manifolds

摘录 / EXCERPT: For a general compact real algebraic manifold $X\subset\mathbb{C}^N$, solve the “liberation” question: understand how the correspondence $X_{\mathrm{class}}\to X$ can arise, where...

open_problem | open | MathDB 页面 | 原文线索

V2-029644 | MATHDB-399702

Embedding Lemma

摘录 / EXCERPT: Let M be a connected n-manifold, $1\le p<n-1$, and $g\in\pi_1(M)$. Suppose that $\alpha\in\pi_p(M)$ is represented by an embedded sphere with a nowhere-zero normal vector field...

open_problem | open | MathDB 页面 | 原文线索

V2-029645 | MATHDB-399704

Universal L^2-torsion and the Thurston polytope

摘录 / EXCERPT: Let M be a compact connected orientable irreducible 3-manifold with infinite fundamental group $\pi=\pi_1(M)$ and empty or incompressible torus boundary, and suppose that M i...

open_problem | open | MathDB 页面 | 原文线索

V2-029646 | MATHDB-399716

Rigidity of measurable maps preserving the chain geometry

摘录 / EXCERPT: Let $p>1$ and $1<m<n$. If $\varphi:\partial\mathbb H^p_{\mathbb C}\to S_{m,n}$ is a measurable map whose essential image is Zariski dense, and for almost every triple (x,y,z) wit...

open_problem | open | MathDB 页面 | 原文线索

V2-029647 | MATHDB-399717

Finiteness of surface-bundle structures on a 4-manifold

摘录 / EXCERPT: Every 4-manifold E fibers as a surface bundle over a surface in finitely many ways. The report takes fiberings to be equivalent up to fiberwise diffeomorphism for these purposes.

open_problem | open | MathDB 页面 | 原文线索

V2-029648 | MATHDB-399722

Gauduchon’s prescribed Chern-Ricci form conjecture

摘录 / EXCERPT: Conjecture (Gauduchon). Let (M,ω) be a compact Gauduchon manifold, meaning that $\partial\bar\partial(\omega^{n-1})=0$. Given $\Psi\in c^{1}_{\mathrm{BC}}(M)$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-029649 | MATHDB-399724

Vanishing of the \(\widehat A\)-genus for almost non-negatively curved spin manifolds

摘录 / EXCERPT: Does every compact spin manifold M admitting a sequence of Riemannian metrics g_i with $\operatorname{diam}(M,g_i)=1$ and sectional curvature $K_{g_i}\ge -1/i$ necessarily have...

open_problem | open | MathDB 页面 | 原文线索

V2-029650 | MATHDB-399732

Universal upper bound for the total curvature of minimizing geodesics on convex surfaces

摘录 / EXCERPT: Does there exist a universal upper bound for the total curvature of minimizing geodesics on convex surfaces in 3-dimensional Euclidean space?

open_problem | open | MathDB 页面 | 原文线索

V2-029651 | MATHDB-399789

Rate of convergence of the volume-normalized Yamabe flow

摘录 / EXCERPT: Let (M^n,g_0) be a closed Riemannian manifold with $n\geq 3$, and let $g(t)$ solve the volume-normalized Yamabe flow ... Assume that $g(t)$ converges in $C^{2,\alpha}(M,g_\infty)$...

open_problem | open | MathDB 页面 | 原文线索

V2-029652 | MATHDB-399790

Geodesics in the space of conformal metrics with positive or negative curvature

摘录 / EXCERPT: Theorem 1. Let (M^2,g) be a compact Riemann surface. Then $(\Gamma_1^{\pm},d)$ is a length space, with any two points connected by a unique regularizable $C^{1,1}$ geodesic. More...

open_problem | open | MathDB 页面 | 原文线索

V2-029653 | MATHDB-399794

Bounded index should imply bounded volume in positively curved ambient manifolds

摘录 / EXCERPT: Let 0N^{n+1} be a smooth, closed Riemannian manifold in the reports ambient setting, with strictly positive sectional curvatures $ 0\sigma_N>0$, and let $M\in\mathcal{M}_n(N)$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-029654 | MATHDB-399795

Interior regularity of two-dimensional semi-calibrated currents

摘录 / EXCERPT: Let Σ be a Riemannian manifold of class C^{3,ε_0}, and let T be a two-dimensional integer rectifiable current in Σ that is semi-calibrated by a two-dime...

open_problem | open | MathDB 页面 | 原文线索

V2-029655 | MATHDB-399796

Weighted \c3_2-curvature and the Hitchin\-Thorpe inequality

摘录 / EXCERPT: Conjecture 1. Let (M^n,g) be a compact Riemannian manifold. If $\sup_{\Gamma_2^\infty} W_2<\infty$, then there is a metric $\widehat g\in[g]$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-029656 | MATHDB-399798

Non-degeneracy of the constructed constant-mean-curvature hypersurfaces

摘录 / EXCERPT: Are the hypersurfaces that we construct non-degenerate?

open_problem | open | MathDB 页面 | 原文线索

V2-029657 | MATHDB-399799

Extending the CMC hypersurface gluing construction to more general manifolds

摘录 / EXCERPT: Can the gluing construction for constant-mean-curvature hypersurfaces be extended to more general manifolds? What requirements will be needed for the background structure Γ,...

open_problem | open | MathDB 页面 | 原文线索

V2-029658 | MATHDB-399800

Isoperimetric inequality under total Q-curvature bounds

摘录 / EXCERPT: Suppose $(M^n,g)=(\mathbb{R}^n,e^{2u}|dx|^2)$ is a noncompact complete Riemannian manifold with normal metric. If its Q-curvature satisfies ... then the manifold satisfies the isoper...

open_problem | open | MathDB 页面 | 原文线索

V2-029659 | MATHDB-399856

Demailly’s strong openness conjecture for multiplier ideal sheaves

摘录 / EXCERPT: Let X be a complex manifold and let $\\varphi$ be a plurisubharmonic function on X. Define the multiplier ideal sheaf $\mathcal{I}(\\varphi)$ by taking, at each $x\in X$, the g...

conjecture | open | MathDB 页面 | 原文线索

V2-029660 | MATHDB-399857

Minimal ambient dimension for complete bounded embedded complex submanifolds

摘录 / EXCERPT: For any $k\in\mathbb{N}$, there are complete bounded embedded complex k-dimensional submanifolds of $\mathbb{C}^{2k}$. Is $N=2k$ the minimal possible ambient dimension in this ca...

open_problem | open | MathDB 页面 | 原文线索

V2-029661 | MATHDB-399858

Yang’s question: complete closed complex submanifolds of the unit ball

摘录 / EXCERPT: For each pair of integers k,N with $1\le k<N$, there exists a complete, closed, k-dimensional complex submanifold of the unit ball $B_N\subset\mathbb{C}^N$.

open_problem | open | MathDB 页面 | 原文线索

V2-029662 | MATHDB-399884

Recognition of the 4-sphere

摘录 / EXCERPT: Given a (PL) compact connected 4-manifold N, decide whether N is PL homeomorphic to S^4.

open_problem | open | MathDB 页面 | 原文线索

V2-029663 | MATHDB-399885

NP certification of Thurston geometries

摘录 / EXCERPT: Determine how to certify, in NP, that a three-manifold has a particular Thurston geometry. The geometries S^3, $S^2 \times \mathbb{R}$, $\mathbb{E}^3$, Nil, and Sol are stated to...

open_problem | open | MathDB 页面 | 原文线索

V2-029664 | MATHDB-399918

Cluster supermanifold structure from superfriezes

摘录 / EXCERPT: Use superfriezes to define a structure of cluster supermanifold on the moduli spaces $M0.n$ of n-gons in the contact projective line $P^{1|1}$.

open_problem | open | MathDB 页面 | 原文线索

V2-029665 | MATHDB-399923

Continuous analogue of the Hirsch conjecture for central-path curvature

摘录 / EXCERPT: Deza, Terlaky, and Zinchenko conjectured that the total curvature of the central path of a linear program in dimension d with $n>d$ constraints should be bounded by $O(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-029666 | MATHDB-399924

Counterexample to the continuous Hirsch-type curvature conjecture

摘录 / EXCERPT: For every integer $r\ge 1$ and every sufficiently large real parameter t, consider the linear program ... It has $3r+4$ inequalities in the $2r+2$ variables...

conjecture | open | MathDB 页面 | 原文线索

V2-029667 | MATHDB-399944

Triangulations realizing vanishing corrected h-numbers from low-index handle decompositions

摘录 / EXCERPT: If X is a PL-manifold with boundary and handle decomposition using handles of index i or less, does there exist a triangulation $\[1m\Delta\u001b[0m$ of X with...

open_problem | open | MathDB 页面 | 原文线索

V2-029668 | MATHDB-399951

On a triangulation of a manifold, is there always a unique optimal discrete Morse vector?

摘录 / EXCERPT: Is there a finite simplicial complex C and a positive integer k such that: (1) C is homeomorphic to some manifold; (2) C admits no discrete Morse function with fewer than...

open_problem | open | MathDB 页面 | 原文线索

V2-029669 | MATHDB-399955

Camere’s conjecture on symplectic involutions of manifolds of $K3^{[2]}$-type

摘录 / EXCERPT: Every symplectic involution on a hyperkähler manifold of $K3^{[2]}$-type can be deformed, as part of a pair, to an involution on $S^{[2]}$ induced by a symplectic involution of a K...

conjecture | open | MathDB 页面 | 原文线索

V2-029670 | MATHDB-399970

Classification of degenerations of higher-dimensional hyperkähler manifolds

摘录 / EXCERPT: Is it possible to classify degenerations of hyperkähler manifolds of higher dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-029671 | MATHDB-399971

Degenerations and compactifications of higher-dimensional hyper-Kähler manifolds

摘录 / EXCERPT: Question 26. What happens for hyper-Kähler manifolds of higher dimension? Is it possible to classify their degenerations? How many examples of degenerate hyper-Kähler manifolds are...

open_problem | open | MathDB 页面 | 原文线索

V2-029672 | MATHDB-400001

Error estimates for nonlinear systems of hyperbolic conservation laws on manifolds

摘录 / EXCERPT: How can error estimates for nonlinear systems of hyperbolic conservation laws on manifolds be obtained?

open_problem | open | MathDB 页面 | 原文线索

V2-029673 | MATHDB-400015

Dual Minkowski problem for dual curvature measures

摘录 / EXCERPT: Suppose k is a fixed integer such that $1\le k\le n$. If μ is a finite Borel measure on S^{n-1}, find necessary and sufficient conditions on μ so that μ is the k-...

open_problem | open | MathDB 页面 | 原文线索

V2-029674 | MATHDB-400050

Fueter-section model for bubbling of G_2-instantons

摘录 / EXCERPT: Let (A_n) be a sequence of G_2-instantons on a closed G_2-manifold Y whose bubbling set is a smooth three-dimensional associative submanifold M. For each $m\\in M$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-029675 | MATHDB-400053

Existence of asymptotically conical Ricci-flat Kähler resolutions

摘录 / EXCERPT: Given a Sasaki-Einstein manifold (L,g_L) with metric cone $C(L)$, does there exist a complete Ricci-flat Kähler metric on a non-compact manifold X that is asymptotically conica...

open_problem | open | MathDB 页面 | 原文线索

V2-029676 | MATHDB-400300

Non-compactness of the space of embedded minimal hypersurfaces

摘录 / EXCERPT: Is the space of embedded minimal hypersurfaces non-compact for a manifold with an analytic metric of positive Ricci curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029677 | MATHDB-400301

Existence of a closed embedded minimal hypersurface in a thick noncompact manifold

摘录 / EXCERPT: Let (N^n,g) be a complete non-compact Riemannian manifold of dimension $n\le 7$. Suppose that: - N contains a bounded open subset Ω such that Ω is a manifold with...

open_problem | open | MathDB 页面 | 原文线索

V2-029678 | MATHDB-400302

Mean curvature flow with surgery in closed Riemannian 3-manifolds

摘录 / EXCERPT: Let X be a closed smooth Riemannian 3-manifold, and let the initial surface be a closed, embedded 2-dimensional hypersurface with positive mean curvature. The report states t...

open_problem | open | MathDB 页面 | 原文线索

V2-029679 | MATHDB-400305

Are the min-max minimal hypersurfaces isometrically distinct in the equality case?

摘录 / EXCERPT: Let (M,g) be a closed manifold with positive Ricci curvature. In the case that $ \omega_p=\omega_{p+1}$ for some p, are the minimal hypersurfaces obtained by the construction o...

open_problem | open | MathDB 页面 | 原文线索

V2-029680 | MATHDB-400306

Betti-number convergence under bounded Ricci curvature

摘录 / EXCERPT: Let n be a positive integer and $K>0$. Let (X_i) be a sequence of n-dimensional compact Riemannian manifolds satisfying $|\operatorname{Ric}_{X_i}|\le K$, and suppose that...

open_problem | open | MathDB 页面 | 原文线索

V2-029681 | MATHDB-400307

Higher-codimension Weyl law for widths

摘录 / EXCERPT: Is it true that, for a compact Riemannian manifold M and $k<n$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-029682 | MATHDB-400309

Equality of Colding-Naber dimension and Hausdorff dimension

摘录 / EXCERPT: Let $(M_i,p_i)_{i\in\mathbb{N}}$ be a collapsing sequence of pointed complete connected n-dimensional Riemannian manifolds with $\operatorname{Ric}_{M_i}\geq -(n-1)$, converging...

open_problem | open | MathDB 页面 | 原文线索

V2-029683 | MATHDB-400310

Classification of stable CMC hypersurfaces with free boundary in a ball

摘录 / EXCERPT: Let $B\subset\mathbb{R}^{n+1}$ be a closed ball. Are the totally umbilical ones the only immersed stable constant mean curvature hypersurfaces with free boundary in B?

open_problem | open | MathDB 页面 | 原文线索

V2-029684 | MATHDB-400312

Donaldson’s question for the Calabi-Yau equation on almost-Kähler four-manifolds

摘录 / EXCERPT: Let 4(M,J,comega)4 be a compact symplectic 4-manifold with a compatible almost complex structure, and assume that $b_2^+(M)=1$. Given $F dir C^{\inf}(M)$ satisfying ... does there ex...

open_problem | open | MathDB 页面 | 原文线索

V2-029685 | MATHDB-400396

Non-round asymptotic metrics for inverse mean curvature flow in hyperbolic space

摘录 / EXCERPT: There exists a star-shaped, mean-convex, closed surface Σ_0 in hyperbolic 3-space $\mathbb{H}^3$ such that, if Σ_t is the inverse mean curvature flow of Σ_0,...

open_problem | open | MathDB 页面 | 原文线索

V2-029686 | MATHDB-400398

Generalization of the Bochner inequality from a manifold to path space

摘录 / EXCERPT: Is there any way to generalize the Bochner inequality (1) from M to PM?

open_problem | open | MathDB 页面 | 原文线索

V2-029687 | MATHDB-400404

Large-volume isoperimetric regions in asymptotically flat 3-manifolds

摘录 / EXCERPT: Let (M,g) be a 3-manifold that is asymptotically flat at rate $\tau>\frac12$, with integrable scalar curvature, non-negative scalar curvature, no closed minimal surfaces other...

open_problem | open | MathDB 页面 | 原文线索

V2-029688 | MATHDB-400437

Strong involutions in centralisers for symplectic and orthogonal groups

摘录 / EXCERPT: Let G be an n-dimensional finite symplectic or orthogonal group of odd characteristic, let $t\in G$ be a strong involution (that is, both eigenspaces $E_+(t)$ and $E_-(t)$ have...

open_problem | open | MathDB 页面 | 原文线索

V2-029689 | MATHDB-400502

Optimal lower bounds for densities of intermediate-dimensional orthonormal subspaces

摘录 / EXCERPT: Let (M,g) be a smooth, compact, boundaryless Riemannian manifold of dimension $N\ge 2$, let Δ_g be its non-negative Laplace-Beltrami operator, and define ... For...

open_problem | open | MathDB 页面 | 原文线索

V2-029690 | MATHDB-400520

Symplectic Chevalley restriction for visible polar representations

摘录 / EXCERPT: Let f4c9(G,V) be a polar representation with Cartan subspace f4c9c and Weyl group f4c9W, and let f4c9V^* be the dual representation, with Cartan subspace f4c9c^*. The nat...

open_problem | open | MathDB 页面 | 原文线索

V2-029691 | MATHDB-400542

Beresnevich’s theorem on badly approximable points on analytic non-degenerate manifolds

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$ with $1\le m\le n$, let B be an open ball in $\mathbb{R}^m$, and let $\mathcal{F}_n(B)$ be a finite family of analytic non-degenerate maps...

open_problem | open | MathDB 页面 | 原文线索

V2-029692 | MATHDB-400543

Badly approximable points on nondegenerate submanifolds

摘录 / EXCERPT: Prove that for any nondegenerate submanifold $M\subseteq\mathbb{R}^n$ and any $\mathbf r=(r_1,\ldots,r_n)\in\mathbb{R}_{\ge 0}^n$ with $\sum_{i=1}^n r_i=1$, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-029693 | MATHDB-400544

Inhomogeneous weighted badly approximable points on analytic manifolds

摘录 / EXCERPT: Let $\theta=(\theta_1,\ldots,\theta_n)\in\mathbb R^n$ and let $r=(r_1,\ldots,r_n)\in\mathbb R_{\b7 0}^n$ satisfy $\sum_{i=1}^n r_i=1$. Define $\operatorname{Bad}(r;\theta)$ to be t...

open_problem | open | MathDB 页面 | 原文线索

V2-029694 | MATHDB-400567

Explicit description of the pregeometry jcl

摘录 / EXCERPT: Is there a more explicit description of the pregeometry jcl?

open_problem | open | MathDB 页面 | 原文线索

V2-029695 | MATHDB-400615

A cellular Sharbly-type resolution for the symplectic group

摘录 / EXCERPT: Construct, for the symplectic group, a resolution of its Steinberg module that reflects the cellular structure in [16], rather than merely an arbitrary resolution. The report state...

open_problem | open | MathDB 页面 | 原文线索

V2-029696 | MATHDB-400653

Monotonicity of the W-entropy on RCD$*$(0,N) spaces

摘录 / EXCERPT: Let (X,d,m) be a Polish geodesic metric measure space such that $m(B_r(x))\in(0,\infty)$ for every $x\in X$ and $r>0$, and suppose that (X,d,m) is an $\mathrm{RCD}^*(0,N)$ spac...

open_problem | open | MathDB 页面 | 原文线索

V2-029697 | MATHDB-400715

Integration theory for semialgebraic differential forms on surreal submanifolds

摘录 / EXCERPT: Can one obtain an integration theory for semialgebraic differential forms on semialgebraic $\No$-submanifolds, including Stokes theorem?

open_problem | open | MathDB 页面 | 原文线索

V2-029698 | MATHDB-400867

Four-point bounds for the spherical code problem $A(4,\arccos(1/3))$

摘录 / EXCERPT: Compute four-point bounds for packing problems in discrete geometry, with particular interest in the spherical code problem $A(4,\arccos(1/3))$. A construction of 14 points exist...

open_problem | open | MathDB 页面 | 原文线索

V2-029699 | MATHDB-400937

Kühnel's conjecture on embedding simplex skeleta

摘录 / EXCERPT: If the k-skeleton $\Delta_n^{(k)}$ of an n-dimensional simplex embeds into a $(k-1)$-connected compact $2k$-dimensional manifold M, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-029700 | MATHDB-401036

Conformally covariant symmetry breaking operators on hypersurfaces

摘录 / EXCERPT: Let X be a Riemannian manifold and Y a hypersurface. Construct and classify conformally covariant symmetry breaking operators from $E^i(X)$ to $E^j(Y)$, i.e. linear maps...

open_problem | open | MathDB 页面 | 原文线索

V2-029701 | MATHDB-401050

Density-one sup-norm bound for eigenfunctions on negatively curved manifolds

摘录 / EXCERPT: For a closed Riemannian manifold Y of strictly negative curvature, it is reasonable to expect that the strong bound ... holds with density one.

open_problem | open | MathDB 页面 | 原文线索

V2-029702 | MATHDB-401082

Infinitely many components of the moduli space of positive Ricci curvature metrics

摘录 / EXCERPT: Does $\mathcal M_{\mathrm{Ric}>0}(M)$ have infinitely many components for a closed spin manifold M of dimension $4n-1\ge 7$ and of positive Ricci curvature?

open_problem | open | MathDB 页面 | 原文线索

V2-029703 | MATHDB-401086

Recognition of fiber bundles over S^1 with fiber $\mathbb{CP}^3$ via positive curvature

摘录 / EXCERPT: Let M be a smooth fiber bundle over S^1 with fiber $\mathbb{CP}^3$. If M admits positively curved metrics on each fiber, varying smoothly, then is M diffeomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-029704 | MATHDB-401087

Spaces of positive-scalar-curvature metrics on non-spin manifolds

摘录 / EXCERPT: What can be said about the spaces $R(M)$, in particular the space $Riem^+(M)$, if M is a simply-connected non-spin manifold, or more generally a manifold whose universal covering...

open_problem | open | MathDB 页面 | 原文线索

V2-029705 | MATHDB-401089

Liouville measure of flat geodesics on a nonpositively curved surface

摘录 / EXCERPT: Let d4a2S,gd4a3 be a non-positively curved surface of genus d4a5d4a5d4a5 2. Equip d4a5Td4a5d4a5d4a5S with the Liouville measure, and let d4e6d4e6d4e6d4e6 d4a5Td4a5d4a5d4a5S be the...

open_problem | open | MathDB 页面 | 原文线索

V2-029706 | MATHDB-401129

Sharp Mihlin-Hörmander thresholds for sub-elliptic operators on manifolds

摘录 / EXCERPT: For more general sub-elliptic operators L on manifolds, determine the sharp threshold $\varsigma(L)$.

open_problem | open | MathDB 页面 | 原文线索

V2-029707 | MATHDB-401142

Existence of closed CMC hypersurfaces with prescribed mean curvature

摘录 / EXCERPT: Let M^{n+1} be a smooth, closed Riemannian manifold with $3\le n+1\le 7$. Given any $c\in\mathbb{R}$, there exists a nontrivial, smooth, closed, almost embedded hypersurface...

open_problem | open | MathDB 页面 | 原文线索

V2-029708 | MATHDB-401150

Rigidity of Schwarzschild manifold

摘录 / EXCERPT: Assume $n\ge 3$. Let M^n be a smooth, connected, n-dimensional manifold with nonempty, possibly disconnected, smooth, compact inner boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-029709 | MATHDB-401151

Stability of ALE Ricci-flat manifolds under Ricci flow

摘录 / EXCERPT: Let (M^n,g_0) be an ALE Ricci-flat manifold. Assume that it is linearly stable and integrable. Then, for every $\b5>0$, there exists $\b4>0$ such that, for every metric...

open_problem | open | MathDB 页面 | 原文线索

V2-029710 | MATHDB-401152

Goldberg conjecture for closed almost Kähler four-manifolds

摘录 / EXCERPT: Let M be a closed almost Kähler four-manifold. If M is an Einstein manifold, then the almost complex structure must be integrable.

conjecture | open | MathDB 页面 | 原文线索

V2-029711 | MATHDB-401154

Asymptotically planar minimal surfaces of Morse index one

摘录 / EXCERPT: Whether asymptotically flat manifolds may or should in fact contain asymptotically planar minimal surfaces with Morse index equal to one.

open_problem | open | MathDB 页面 | 原文线索

V2-029712 | MATHDB-401155

Existence of non-totally-geodesic minimal surfaces in asymptotically Schwarzschildean manifolds

摘录 / EXCERPT: Does there exist a single-ended asymptotically Schwarzschildean Riemannian manifold containing a non-compact minimal surface that is not totally geodesic?

open_problem | open | MathDB 页面 | 原文线索

V2-029713 | MATHDB-401156

Existence of non-totally-geodesic minimal surfaces in single-ended asymptotically Schwarzschildean manifolds

摘录 / EXCERPT: Does there exist a single-ended asymptotically Schwarzschildean Riemannian manifold (M^3,g) of non-negative scalar curvature containing a non-compact minimal surface that is not...

open_problem | open | MathDB 页面 | 原文线索

V2-029714 | MATHDB-401157

Infinitely many properly embedded geodesics on every complete Riemannian plane

摘录 / EXCERPT: Does every complete Riemannian metric g on $\b2^2$ contain infinitely many properly embedded geodesics $\b3:\mathbb{R}\to(\b2^2,g)$?

open_problem | open | MathDB 页面 | 原文线索

V2-029715 | MATHDB-401158

Embedded mountain-pass geodesic proposition

摘录 / EXCERPT: Let (N,g) be a complete boundaryless Riemannian 2-manifold. For distinct points $p,q\in N$, suppose there are two embedded geodesics $\gamma_1,\gamma_2:[0,1]\to N$ bounding a d...

open_problem | open | MathDB 页面 | 原文线索

V2-029716 | MATHDB-401161

Cauchy problem for extending generalised imaginary Killing spinors to parallel null spinors

摘录 / EXCERPT: Let (M,g,φ) be a smooth Riemannian spin manifold with a smooth spinor field satisfying ... where W is the Weingarten tensor appearing in the initial-data conditions. Then t...

open_problem | open | MathDB 页面 | 原文线索

V2-029717 | MATHDB-401163

Differentiable stability near maximal injectivity radius under scalar-curvature lower bounds

摘录 / EXCERPT: Does there exist an $\epsilon=\epsilon(n)>0$ such that every closed n-dimensional Riemannian manifold M with $\mathrm{scal}_M\thirdot\ge n(n-1)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029718 | MATHDB-401178

Cosmetic Surgery Conjecture (Hyperbolic Case)

摘录 / EXCERPT: Let M be a 1-cusped hyperbolic 3-manifold. Let $M(p/q)$ and $M(p'/q')$ be the p/q- and p'/q'-Dehn filled manifolds of M, respectively, which are also hyperbolic. If...

conjecture | open | MathDB 页面 | 原文线索

V2-029719 | MATHDB-401189

Betti numbers of closed manifolds

摘录 / EXCERPT: Which choices of Betti numbers with $b_0=1$, $b_{4k}=1$, and b_{2k} odd are realized by closed $4k$-manifolds? The report concentrates on the case $b_1=\cdots=b_{2k-1}=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-029720 | MATHDB-401192

Explicit upper bounds for normalized volumes of cusped hyperbolic 3-manifolds

摘录 / EXCERPT: Let $\mathcal{C}_{\mathrm{cm}}$ be the context whose objects are finite-volume cusped hyperbolic 3-manifolds. For an n-cusped manifold M, with r_1 Klein-bottle cusps and...

open_problem | open | MathDB 页面 | 原文线索

V2-029721 | MATHDB-401200

Cerveau-Lins Neto conjecture on codimension-one foliations

摘录 / EXCERPT: Let $\mathcal{F}$ be a codimension-one holomorphic foliation on a projective manifold X. Then one of the following assertions holds true: (1) The foliation $\mathcal{F}$ is trans...

conjecture | open | MathDB 页面 | 原文线索

V2-029722 | MATHDB-401202

Griffiths conjecture

摘录 / EXCERPT: The Griffiths conjecture asserts that any ample vector bundle over a compact complex manifold is Griffiths positive.

conjecture | open | MathDB 页面 | 原文线索

V2-029723 | MATHDB-401204

Chern’s conjecture for special affine manifolds

摘录 / EXCERPT: If X is a connected closed special affine manifold, then $\chi(X)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-029724 | MATHDB-401208

Viehweg’s general-type conjecture for bases of families of polarized manifolds

摘录 / EXCERPT: Let $\pi:X\to S$ be a holomorphic family of polarized manifolds of type (a), (b), or (c) over a quasi-projective manifold S. If π has maximal variation, then S is of log ge...

conjecture | open | MathDB 页面 | 原文线索

V2-029725 | MATHDB-401209

Isotriviality over special bases for canonically polarized families

摘录 / EXCERPT: Let $\pi:X\to S$ be a holomorphic family of canonically polarized manifolds over a quasi-projective manifold S. If S is special, then π is isotrivial.

open_problem | open | MathDB 页面 | 原文线索

V2-029726 | MATHDB-401244

Uniqueness of Cartan subalgebras in uniform Roe algebras

摘录 / EXCERPT: For a bounded-geometry discrete metric space X, to what extent is the canonical Cartan subalgebra $\ell^\infty(X)\subseteq C_u^*(X)$ unique?

open_problem | open | MathDB 页面 | 原文线索

V2-029727 | MATHDB-401344

Complete classification of Kähler groups (Serre’s problem)

摘录 / EXCERPT: A Kähler group is a group that can be realized as the fundamental group of a connected, compact Kähler manifold. It is an open problem, attributed to Serre, to completely classify...

open_problem | open | MathDB 页面 | 原文线索

V2-029728 | MATHDB-401346

Boyer-Gordon-Watson equivalence for graph manifolds

摘录 / EXCERPT: For every closed, connected, orientable graph manifold M, M is not an L-space if and only if $\\pi_1(M)$ is left-orderable.

open_problem | open | MathDB 页面 | 原文线索

V2-029729 | MATHDB-401351

Kähler groups that are fundamental groups of closed 3-manifolds

摘录 / EXCERPT: Let G be the fundamental group of a closed 3-manifold. Then G is Kähler if and only if G is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-029730 | MATHDB-401506

Rigidity theorem for warped-cone actions (Theorem 3)

摘录 / EXCERPT: Let Γ and Λ be finitely presented groups acting isometrically, freely, and minimally on closed manifolds M and N, respectively. Assume that neither Γ nor...

open_problem | open | MathDB 页面 | 原文线索

V2-029731 | MATHDB-401518

Warped cones with wall structures from manifold actions

摘录 / EXCERPT: Is there a free and measure-preserving action of a free group on a manifold such that the associated warped cone admits a wall structure?

open_problem | open | MathDB 页面 | 原文线索

V2-029732 | MATHDB-401549

Existence conditions for cubic spline interpolation on Riemannian manifolds

摘录 / EXCERPT: Given interpolation times $t_1,\,\,\,t_m i[0,1]$ and data points $x_1,\,\,\,x_m i M$, determine conditions on the Riemannian manifold (M,g) and the interpolation data under which...

open_problem | open | MathDB 页面 | 原文线索

V2-029733 | MATHDB-401551

Closed-form rank-one subspace update on the Grassmann manifold

摘录 / EXCERPT: Given a rank-one update $X_{\mathrm{new}}=X+ab^{T}=UW+ab^{T}$ with $U\in\operatorname{St}(n,p)$, define ... ... Then $\Delta=qw^{T}$ is a unit tangent vector at $[U]$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-029734 | MATHDB-401670

Log-concavity inequality for non-pluripolar products of positive currents

摘录 / EXCERPT: Let X be a compact connected Kähler manifold of complex dimension n. For any closed positive $(1,1)$-currents $T_1,\ldots,T_n$ on X, one has ... In particular, the function ... i...

open_problem | open | MathDB 页面 | 原文线索

V2-029735 | MATHDB-401671

Equidistribution of minimal hypersurfaces for generic metrics

摘录 / EXCERPT: Let M^{n+1} be a closed manifold with $2\le n\le 6$. For a generic C^∞ Riemannian metric g on M in the sense of Baire, there exists a sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-029736 | MATHDB-401675

Boundary-connected approximation for the first Steklov eigenvalue

摘录 / EXCERPT: Given any compact Riemannian manifold Ω^n with non-empty boundary and $n\ge 3$, and given any $\varepsilon>0$, there exists a smooth subdomain Ω_ε of...

open_problem | open | MathDB 页面 | 原文线索

V2-029737 | MATHDB-401676

Bounded diameter conjecture for two-convex mean curvature flow

摘录 / EXCERPT: If $\{M_t\subset\mathbb{R}^{n+1}\}_{t\in[0,T)}$ is a mean curvature flow of two-convex closed embedded hypersurfaces, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-029738 | MATHDB-401677

Hopf conjecture under isometry-group rank at least five

摘录 / EXCERPT: Every even-dimensional compact Riemannian manifold admitting a metric with positive sectional curvature and whose isometry group has rank at least five has positive Euler character...

conjecture | open | MathDB 页面 | 原文线索

V2-029739 | MATHDB-401678

Generalized Hopf conjecture for compact symmetric spaces

摘录 / EXCERPT: No compact, simply connected Riemannian symmetric space of rank greater than one admits a Riemannian metric with positive sectional curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-029740 | MATHDB-401706

Stickiness or smeariness of the maximum-likelihood mean

摘录 / EXCERPT: For the maximum-likelihood (ML) mean on a manifold, does it exhibit properties analogous to the stickiness or smeariness effects known for Fréchet means in the non-smooth category?

open_problem | open | MathDB 页面 | 原文线索

V2-029741 | MATHDB-401738

Certified bound for the first coexact 1-form eigenvalue of the Meyerhoff manifold

摘录 / EXCERPT: Theorem 2 gives the numerically certified bound for manifold 1 in the Hodgson-Weeks census, also known as the Meyerhoff manifold, ...

open_problem | open | MathDB 页面 | 原文线索

V2-029742 | MATHDB-401776

Hausdorff measure of the singular set of an area-stationary submanifold

摘录 / EXCERPT: For general n-dimensional area-stationary submanifolds M, is $\mathcal{H}^n(\operatorname{sing} M)=0$?

open_problem | open | MathDB 页面 | 原文线索

V2-029743 | MATHDB-401815

Equality of the two bounded-plurisubharmonic hulls

摘录 / EXCERPT: Let M be a complex manifold and let $w_0\in M$. Define ... and ... where $\operatorname{PSH}_{cb}(M)$ is the space of bounded continuous plurisubharmonic functions on M. Is ...

open_problem | open | MathDB 页面 | 原文线索

V2-029744 | MATHDB-401816

Polynomially convex smooth embedding dimension of compact real manifolds

摘录 / EXCERPT: For a given dimension m, what is the least integer n such that every m-dimensional compact real manifold admits a smooth polynomially convex embedding into $\[1m\mathbb{C}^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-029745 | MATHDB-401848

Geodesic embedding under doubling for the arc distance

摘录 / EXCERPT: Let S be a surface with non-empty boundary, and let S^d be its double. The natural embedding...

open_problem | open | MathDB 页面 | 原文线索

V2-029746 | MATHDB-401849

Geodesic embedding for the doubled surface with respect to the Lipschitz distance

摘录 / EXCERPT: Let S be a surface with boundary, let S^d be its double, and let the natural embedding send a hyperbolic structure X on S to its double X^d on S^d. Is ... a geodesic embe...

open_problem | open | MathDB 页面 | 原文线索

V2-029747 | MATHDB-401859

Equality of three asymmetric metrics on Teichmüller space with geodesic boundary

摘录 / EXCERPT: Let S be a compact finite-type surface of genus g with $r>0$ geodesic boundary components, and let $\mathcal{T}(S)$ denote its Teichmüller space. Define ... where the supremum is...

open_problem | open | MathDB 页面 | 原文线索

V2-029748 | MATHDB-401862

Branched local-homeomorphism property of the holonomy map

摘录 / EXCERPT: Suppose that M is a closed manifold. Consider the holonomy map $\operatorname{Def}_{(G,X)}(M) \longrightarrow \operatorname{Rep}(\pi_1(M),G)$ from the deformation space of...

open_problem | open | MathDB 页面 | 原文线索

V2-029749 | MATHDB-401863

Limit points of universal Weil-Petersson geodesics in the Thurston boundary

摘录 / EXCERPT: Let $T=T(\mathbb H^2)$ be the universal Teichmüller space with Thurston boundary $\mathbb{PML}_{\mathrm{bdd}}=\mathbb{PML}_{\mathrm{bdd}}(\mathbb H^2)$, the projectivized bounded m...

open_problem | open | MathDB 页面 | 原文线索

V2-029750 | MATHDB-401991

Characterization of angular valuations

摘录 / EXCERPT: For a Riemannian manifold M, the algebra of angular valuations on M equals the Lipschitz-Killing algebra $LK(M)$; equivalently, the algebra of valuations μ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-029751 | MATHDB-402094

Fried conjecture for flows

摘录 / EXCERPT: Let X be a closed manifold and let $(\varphi_t)_{t\in\mathbb R}$ satisfy A.1-A.3: it has no trivial closed orbits; all nontrivial closed orbits are nondegenerate; and there is...

conjecture | open | MathDB 页面 | 原文线索

V2-029752 | MATHDB-402156

Duality between tropical cocycles and tropical cycles on tropical manifolds

摘录 / EXCERPT: The notion of tropical cocycles is dual to the notion of tropical cycles on tropical manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029753 | MATHDB-402176

Taubes compactness theorem for sequences of Kapustin-Witten solutions with unbounded Higgs fields

摘录 / EXCERPT: Let X be a closed, oriented, smooth 4-manifold and let P be a principal $SU(2)$- or $SO(3)$-bundle. Consider a sequence of solutions (A_i,a_i) to the Kapustin-Witten system...

open_problem | open | MathDB 页面 | 原文线索

V2-029754 | MATHDB-402208

Full large-time convergence and asymptotic profile for a degenerate level-set eikonal-curvature equation

摘录 / EXCERPT: Let $a\ge 0$, let $r:\mathbb{R}^N\to[0,\infty)$ be continuous with compact support, and let u be the global-in-time viscosity solution with zero initial data of ... Find the asympt...

open_problem | open | MathDB 页面 | 原文线索

V2-029755 | MATHDB-402218

Higher-dimensional classification of constant Gaussian curvature hypersurfaces

摘录 / EXCERPT: Can one classify constant Gaussian curvature hypersurfaces in Minkowski space of dimension at least $3+1$?

open_problem | open | MathDB 页面 | 原文线索

V2-029756 | MATHDB-402229

Possible ranks of the action spectrum of a contact form

摘录 / EXCERPT: Let (Y,λ) be a closed three-manifold with a contact form. What ranks are possible?

open_problem | open | MathDB 页面 | 原文线索

V2-029757 | MATHDB-402230

Reeb orbits for rank-two contact forms

摘录 / EXCERPT: Let (Y,λ) be a closed three-manifold with a contact form. If the action spectrum $\mathcal{A}(Y,\lambda)$ has rank 2, must there be exactly two geometrically distinct Ree...

open_problem | open | MathDB 页面 | 原文线索

V2-029758 | MATHDB-402232

Continuity and extension of the barcode map

摘录 / EXCERPT: Let (M,ω) be closed, connected, and symplectically aspherical. The mapping $B:(\operatorname{Ham}(M,\omega),d_{C^0})\to(\widehat{\mathrm{Barcodes}},d_{\mathrm{bottle}})$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-029759 | MATHDB-402243

Whether Δ-wings arise as mean-curvature-flow blowups

摘录 / EXCERPT: Do the Δ-wings-namely, the complete strictly convex translating graphs $u_b:\mathbb{R}\times(-b,b)\to\mathbb{R}$ for $b>\pi/2$-occur as blowups of mean curvature flows?

open_problem | open | MathDB 页面 | 原文线索

V2-029760 | MATHDB-402245

The dihedral rigidity conjecture

摘录 / EXCERPT: Let M^n be a convex polyhedron in Euclidean space, and let g be a metric on M. Suppose (M,g) has nonnegative scalar curvature and weakly mean-convex faces, and that along t...

conjecture | open | MathDB 页面 | 原文线索

V2-029761 | MATHDB-402246

Dihedral rigidity conjecture for n-dimensional cubes

摘录 / EXCERPT: For every integer n with $3\le n\le 7$, suppose (M,g) is an n-dimensional cube with nonnegative scalar curvature, weakly mean convex faces, and with the dihedral angle along...

conjecture | open | MathDB 页面 | 原文线索

V2-029762 | MATHDB-402257

Multiplicity One Conjecture in Min-max theory

摘录 / EXCERPT: Let (M^{n+1},g) be a closed orientable Riemannian manifold with $3\le n+1\le 7$, and suppose that g is bumpy, meaning that every closed immersed minimal hypersurface is non-deg...

conjecture | open | MathDB 页面 | 原文线索

V2-029763 | MATHDB-402273

Fragmentation property for compactly supported sections

摘录 / EXCERPT: Let $\pi:E\to M$ be a Serre fibration over a smooth n-manifold, with base section s_0. Let $\mathrm{Sect}_c(\pi)$ be the space of compactly supported sections, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029764 | MATHDB-402274

C-principle from fragmentation

摘录 / EXCERPT: Theorem (N). Let F be a good presheaf on n-dimensional manifolds. Assume that $F(\mathbb{R}^n)$ is at least $(n-1)$-connected and that F has the fragmentation property. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-029765 | MATHDB-402332

Half-partial-data X-ray transform on a boundary

摘录 / EXCERPT: Given the integrals of a function over all geodesics with one endpoint in a small open subset of the boundary and the other endpoint free to vary across the entire boundary, determ...

open_problem | open | MathDB 页面 | 原文线索

V2-029766 | MATHDB-402364

Conjecture 2: geometric criterion for unicity of types

摘录 / EXCERPT: There exists a bounded convex neighbourhood Γ of x, containing $B(G)^H$, such that, for every $z\in B(G)\setminus\Gamma$, the geodesic $[x,z]$ has non-empty intersection w...

conjecture | open | MathDB 页面 | 原文线索

V2-029767 | MATHDB-402375

Explicit Plancherel formula for the symplectic quotient

摘录 / EXCERPT: Let F be a local field, $G=\mathrm{GL}_{2n}(F)$, $H=\mathrm{Sp}_n(F)$, and $G'=\mathrm{GL}_n(F)$. For $\pi\in\operatorname{Irr}_{\mathrm{temp}}(G')$, let $S(\pi)$ be the Langland...

open_problem | open | MathDB 页面 | 原文线索

V2-029768 | MATHDB-402402

Essential self-adjointness of the Laplace-Beltrami operator on the regular region

摘录 / EXCERPT: Let M be a smooth n-dimensional manifold, and let $X_1, ... The vector fields induce a sub-Riemannian distance d on M; assume that (M,d) is complete. On R, let $g...

open_problem | open | MathDB 页面 | 原文线索

V2-029769 | MATHDB-402404

Essential self-adjointness at tangency points in almost-Riemannian geometry

摘录 / EXCERPT: For the almost-Riemannian structure on $\mathbb{R}^2$ generated by ... with singular region $S=\{z-x^2=0\}$, regular region $R=\mathbb{R}^2\setminus S$, $\phi(x,z)=z-x^2$, and natura...

open_problem | open | MathDB 页面 | 原文线索

V2-029770 | MATHDB-402409

Zimmer’s conjecture for higher-rank lattices

摘录 / EXCERPT: A higher-rank lattice Γ has only finite-image homomorphisms $\rho:\Gamma\to\mathrm{Diff}(M)$, where M is a compact manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-029771 | MATHDB-402425

Detection of non-isotopic surface embeddings by Z_C

摘录 / EXCERPT: Determine whether the extended TQFT Z_C can detect non-isotopic embeddings of surfaces into closed three-manifolds. The report answers this negatively for embedded spheres: a sph...

open_problem | open | MathDB 页面 | 原文线索

V2-029772 | MATHDB-402428

Canonical comparison of Kähler and real geometric quantizations

摘录 / EXCERPT: Given a prequantized closed symplectic manifold (X,ω,L,∇,h), an ω-compatible complex structure J, and a Lagrangian fibration $\mu:X\to B$, can one provide a ca...

open_problem | open | MathDB 页面 | 原文线索

V2-029773 | MATHDB-402433

Existence of squeezing symplectomorphisms in infinite dimensions

摘录 / EXCERPT: Does a symplectic diffeomorphism in infinite dimensions which is squeezing exist?

open_problem | open | MathDB 页面 | 原文线索

V2-029774 | MATHDB-402434

Fourier restriction estimates in the optimal range

摘录 / EXCERPT: For a Schwartz function $f:\mathbb{R}^{1+n}\to\mathbb{C}$, with $R f=\widehat f\vert_C$ the restriction of its Fourier transform to a submanifold $C\subset\mathbb{R}^{1+n}$, prove...

open_problem | open | MathDB 页面 | 原文线索

V2-029775 | MATHDB-402438

Convergence of Dziuk’s evolving finite element method for mean curvature flow

摘录 / EXCERPT: Prove convergence of Dziuk’s method or related evolving finite element methods for mean curvature flow of closed two-dimensional surfaces (or higher-dimensional hypersurfaces).

open_problem | open | MathDB 页面 | 原文线索

V2-029776 | MATHDB-402466

Commensurability of non-arithmetic hyperbolic manifolds with identical geodesic length spectra

摘录 / EXCERPT: For non-arithmetic hyperbolic 2- and 3-manifolds, are manifolds with the same geodesic length spectra necessarily commensurable?

open_problem | open | MathDB 页面 | 原文线索

V2-029777 | MATHDB-402494

Seshadri positivity of the tangent sheaf and characterization of projective space

摘录 / EXCERPT: For an n-dimensional projective manifold X, if there exists a point $x_0\in X$ such that $\varepsilon(T_X;x_0)>0$, then $X\simeq \mathbb{P}^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-029778 | MATHDB-402518

Gaussian uniformization: attainment of the conformal walk dimension

摘录 / EXCERPT: Let a complete, locally compact, geodesic metric space (X,d) carry a Radon measure m with full support and a strongly local regular Dirichlet form $(\mathcal E,\mathcal F)$. As...

open_problem | open | MathDB 页面 | 原文线索

V2-029779 | MATHDB-402596

Strominger-Yau-Zaslow conjecture for maximally degenerating Calabi-Yau families

摘录 / EXCERPT: Let $X\to\Delta^*$ be a maximally degenerating projective family of Calabi-Yau manifolds of dimension n. The SYZ conjecture states that a general fibre X_t admits a fibration...

conjecture | open | MathDB 页面 | 原文线索

V2-029780 | MATHDB-402600

Dual complex of a maximally unipotent good minimal dlt degeneration

摘录 / EXCERPT: Let X be a maximally unipotent good minimal dlt degeneration of compact hyper-Kähler manifolds over $\\mathbb{C}((t))$. Then the dual complex of the special fibre of X is homeo...

open_problem | open | MathDB 页面 | 原文线索

V2-029781 | MATHDB-402703

Rational strong Novikov conjecture for countable subgroups of diffeomorphism groups

摘录 / EXCERPT: For a closed smooth manifold N and a density ω on N, prove the rational strong Novikov conjecture for every countable subgroup $\Gamma\leq\operatorname{Diff}(N,\omega)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-029782 | MATHDB-402712

Automorphism and birational groups of an IHS manifold with Picard rank one

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic manifold. Assume that $\operatorname{rank}\operatorname{Pic}(X)=1$, i.e. $\operatorname{Pic}(X)=\mathbb{Z}H$ with $H^2>0$. Can one...

open_problem | open | MathDB 页面 | 原文线索

V2-029783 | MATHDB-402732

Geometry of Pappas-Zhu local models

摘录 / EXCERPT: Let $(G,\{\mu\},G_f)$ be a triple as in the report: G is a reductive $\mathbb{Q}_p$-group, $\{\mu\}$ is a minuscule conjugacy class of geometric cocharacters defined over a finit...

open_problem | open | MathDB 页面 | 原文线索

V2-029784 | MATHDB-402746

Multiplicity one for Allen-Cahn min-max hypersurfaces under positive Ricci curvature

摘录 / EXCERPT: Let N^{n+1}, $n\ge 2$, be a compact Riemannian manifold whose metric has positive Ricci curvature. Then the Allen-Cahn one-parameter min-max construction developed in [4] produce...

open_problem | open | MathDB 页面 | 原文线索

V2-029785 | MATHDB-402759

Borel hyperbolicity for quasi-projective manifolds with quasi-finite period map

摘录 / EXCERPT: Let Y be a quasi-projective manifold, which admits a polarized variation of Hodge structures, whose period map is quasi-finite. Then Y is Borel hyperbolic.

open_problem | open | MathDB 页面 | 原文线索

V2-029786 | MATHDB-402761

Campana’s specialness-dense entire curve conjecture

摘录 / EXCERPT: Let X be a connected compact complex manifold of dimension n. Conjecture (F. Campana): X is special if and only if there exists an entire curve $h:\mathbb{C}\to X$ with dense...

conjecture | open | MathDB 页面 | 原文线索

V2-029787 | MATHDB-402762

Lang’s conjecture relating maximal Kodaira dimension and entire curves

摘录 / EXCERPT: Let X be a connected compact complex manifold of dimension n. Then $\kappa(X)=n$ if and only if there exists a proper Zariski-closed subset $W e X$ such that the image of every...

conjecture | open | MathDB 页面 | 原文线索

V2-029788 | MATHDB-402767

Extension of the precise Yau-Tian-Donaldson result to spherical manifolds

摘录 / EXCERPT: Can the more precise Yau-Tian-Donaldson result of Donaldson for toric surfaces be upgraded to spherical manifolds of dimension three and rank two?

open_problem | open | MathDB 页面 | 原文线索

V2-029789 | MATHDB-402782

Higher-order Fourier uniformity of the Möbius function on nilmanifolds

摘录 / EXCERPT: Let G/Γ be a nilmanifold of degree $k\ge 1$ and let $H=H(X)\to\infty$ as $X\to\infty$. Then, for any fixed Lipschitz map $F:G/\Gamma\to\mathbb C$, ... for almost all...

open_problem | open | MathDB 页面 | 原文线索

V2-029790 | MATHDB-402868

Universal shortest-path discovery for the ant reinforcement-learning model

摘录 / EXCERPT: For the ant process on an arbitrary finite graph $G=(V,E)$ with distinguished nest and food vertices N and F, using the loop-erased or uniform-geodesic reinforcement rule descr...

open_problem | open | MathDB 页面 | 原文线索

V2-029791 | MATHDB-402872

Complex structure on the six-sphere

摘录 / EXCERPT: Does the six-sphere S^6 carry the structure of a complex manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-029792 | MATHDB-402874

Criteria for rational pseudoholomorphic curves on symplectic manifolds

摘录 / EXCERPT: Let (M,ω) be a compact symplectic smooth differentiable manifold of dimension $2n$, and let J be a compatible almost complex structure on M. The first Chern class...

open_problem | open | MathDB 页面 | 原文线索

V2-029793 | MATHDB-402876

Existence of non-Kähler Hermitian-symplectic complex manifolds

摘录 / EXCERPT: Do there exist non-Kähler Hermitian-symplectic complex manifolds X with $\\dim_{\\mathbb C} X \\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-029794 | MATHDB-402878

Sullivan’s conjecture on Betti numbers of compact complex manifolds

摘录 / EXCERPT: Let M be a compact complex manifold with $\\dim_{\\mathbb C} M \ge 3$. Then $b(M) \ge 4$, where $b(M)$ denotes the sum of the Betti numbers of M.

conjecture | open | MathDB 页面 | 原文线索

V2-029795 | MATHDB-402880

Almost Kähler Kodaira-Spencer problem on the Kodaira-Thurston manifold

摘录 / EXCERPT: There exist almost complex structures on the Kodaira-Thurston manifold such that $h^{0,1}$ varies with different choices of almost Kähler metrics.

open_problem | open | MathDB 页面 | 原文线索

V2-029796 | MATHDB-402903

Comparison of global optima under a smooth lift

摘录 / EXCERPT: Let E be a Euclidean space, let $X\subseteq E$ be a possibly nonsmooth set, let $f:E\to\mathbb{R}$ be smooth, and suppose a smooth manifold M and a smooth map $\phi:M\to E$ sat...

open_problem | open | MathDB 页面 | 原文线索

V2-029797 | MATHDB-402904

Nonexistence of a global manifold thin-plate-spline minimizer

摘录 / EXCERPT: For the manifold thin-plate-spline minimization problem ... there exist a Riemannian manifold M and data $d_1,\ldots,d_K\in M$ such that this problem has no global solution.

open_problem | open | MathDB 页面 | 原文线索

V2-029798 | MATHDB-402909

Consistency rate for polyhedral discretizations of tangent-point energy

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^m$ be a submanifold of positive reach. For the tangent-point energy E_q^{2q}, does the polyhedral discretization have consistency rate $O(h)$ as the...

open_problem | open | MathDB 页面 | 原文线索

V2-029799 | MATHDB-402965

High-dimensional Rosenberg conjecture for odd-order finite fundamental groups

摘录 / EXCERPT: Let M be a closed connected smooth manifold of dimension at least 5 with finite fundamental group of odd order. If the universal cover of M admits a Riemannian metric of posi...

conjecture | open | MathDB 页面 | 原文线索

V2-029800 | MATHDB-402966

Positive scalar curvature for spin manifolds with vanishing \b1-invariants

摘录 / EXCERPT: Conjecture: Theorem 4 should also hold for closed, connected, smooth manifolds of dimension at least 5 with odd-order abelian fundamental group that are spin, p-atoral for every pr...

open_problem | open | MathDB 页面 | 原文线索

V2-029801 | MATHDB-402967

Hyperbolicity criterion for fibered 3-manifolds

摘录 / EXCERPT: Assume throughout that Σ is a surface of genus at least 2. Let $f\in\operatorname{Homeo}^{+}(\Sigma)$, and let $M\cong\Sigma*_f$ be the mapping torus obtained by gluing th...

open_problem | open | MathDB 页面 | 原文线索

V2-029802 | MATHDB-402973

Benjamini-Schramm approximation of first Betti numbers

摘录 / EXCERPT: Let $d\in\mathbb{N}$ with $d\ge 3$, and let M_n be a sequence of triangulated compact d-dimensional manifolds such that the 1-skeleta of the triangulations have uniformly bou...

open_problem | open | MathDB 页面 | 原文线索

V2-029803 | MATHDB-402974

Integral approximation for simplicial volume of closed 3-manifolds with zero simplicial volume

摘录 / EXCERPT: Let M be an oriented closed connected 3-manifold with $[U+007F]M[U+007F]=0$. Then the following are equivalent: 1. The simplicial volume of M satisfies integral approximation. 2. The manif...

open_problem | open | MathDB 页面 | 原文线索

V2-029804 | MATHDB-402980

Density-one obstruction to surgery descriptions of Motegi graph manifolds

摘录 / EXCERPT: There exists a set $S\subset\mathbb N$ of density zero such that, if $T_{a,b}$ is a nontrivial torus knot with $a,b>2$ and $ab\notin S$, then ... is not obtainable by Dehn surgery on...

open_problem | open | MathDB 页面 | 原文线索

V2-029805 | MATHDB-402987

The L-space conjecture: left orders and co-oriented taut foliations

摘录 / EXCERPT: For any closed, oriented, irreducible 3-manifold M, M admits a co-oriented taut foliation if and only if its fundamental group $\\pi_1(M)$ admits a left multiplication-invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-029806 | MATHDB-402995

Definite 4-manifolds with multiple smooth structures

摘录 / EXCERPT: Does there exist any closed orientable compact definite 4-manifold with multiple smooth structures?

open_problem | open | MathDB 页面 | 原文线索

V2-029807 | MATHDB-402996

Distinct knots with diffeomorphic knot-trace boundaries

摘录 / EXCERPT: For which knots $K\subset S^3$ does there exist a distinct knot $J\subset S^3$ such that ... Here $X(K)$ is the 4-manifold obtained by attaching a 0-framed 2-handle to B^4 along...

open_problem | open | MathDB 页面 | 原文线索

V2-029808 | MATHDB-403141

Non-almost-Kähler Hermitian metrics and the value of $h^{1,1}_{\partial}$

摘录 / EXCERPT: Let (M,J) be a compact almost-complex 4-dimensional manifold which admits an almost-Kähler structure. Does it have a non-almost-Kähler Hermitian metric such that...

open_problem | open | MathDB 页面 | 原文线索

V2-029809 | MATHDB-403142

An intermediate value for $h^{1,1}_{\bar\partial}$ on almost-complex 4-manifolds

摘录 / EXCERPT: Is there an example of a compact almost-complex 4-dimensional manifold admitting a Hermitian metric ω such that $h^{1,1}_{\bar\partial} \ne b^-$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029810 | MATHDB-403144

Compactness of the \[?25l\bar{\partial}\u001b[?25h-Neumann operator on domains with bounded intrinsic geometry

摘录 / EXCERPT: Suppose $\Omega \subset \mathbb{C}^d$ is a bounded domain with bounded intrinsic geometry, and let g_Ω be the Bergman metric on Ω. Then the following are equivalent:...

open_problem | open | MathDB 页面 | 原文线索

V2-029811 | MATHDB-403151

Tropical analogues of natural classes on moduli spaces of curves

摘录 / EXCERPT: The construction of tropical analogues of the other natural classes on the moduli space of curves, and their relevance to enumerative geometry questions, is an interesting question...

open_problem | open | MathDB 页面 | 原文线索

V2-029812 | MATHDB-403157

Asymptotic growth of the maximal systole of closed hyperbolic surfaces

摘录 / EXCERPT: Determine an asymptotic equivalent for ... as $v\to\infty$, where $\mathcal H(n,v)$ is the set of connected, oriented, closed hyperbolic n-manifolds of volume at most v. In parti...

open_problem | open | MathDB 页面 | 原文线索

V2-029813 | MATHDB-403159

Asymptotic maximum of the first Laplace eigenvalue on hyperbolic surfaces

摘录 / EXCERPT: For $\Lambda(2,v)=\max_{M\in\mathcal H(2,v)}\{\lambda_1(M)\}$, where $\mathcal H(n,v)$ is the set of connected, oriented, closed hyperbolic n-manifolds of volume at most v, det...

open_problem | open | MathDB 页面 | 原文线索

V2-029814 | MATHDB-403161

Power-saving error terms for counting non-simple closed geodesics

摘录 / EXCERPT: For a closed, orientable hyperbolic surface X of genus $g\ge 2$ and a non-simple closed curve β on X, prove a quantitative asymptotic estimate with a power-saving error t...

open_problem | open | MathDB 页面 | 原文线索

V2-029815 | MATHDB-403171

Existence of a hyper-Kähler model for a Lagrangian fibration with integral fibers

摘录 / EXCERPT: Let $\pi\colon M\to\mathbb{P}^{n}$ be a Lagrangian fibration with integral fibers. Then there exists a projective hyper-Kähler manifold A, with a Lagrangian fibration...

open_problem | open | MathDB 页面 | 原文线索

V2-029816 | MATHDB-403190

Gromov’s long neck problem

摘录 / EXCERPT: Let (M,g) be a compact connected n-dimensional Riemannian manifold with boundary such that $\\operatorname{scal}_g\ge n(n-1)$. Suppose that $\Phi:M\to S^n$ is a smooth area non...

open_problem | open | MathDB 页面 | 原文线索

V2-029817 | MATHDB-403191

Long neck problem for spin manifolds with boundary

摘录 / EXCERPT: Theorem 2 (cf. [1]). Let (M,g) be a compact connected n-dimensional Riemannian spin manifold with boundary such that $\operatorname{scal}_g\ge n(n-1)$ on M, where $n\ge 2$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-029818 | MATHDB-403194

Rigidity for positive scalar curvature metrics on $S^n\setminus\Sigma$

摘录 / EXCERPT: Let Σ be a subset of the standard unit sphere S^n with the wrapping property: for all sufficiently small $\varepsilon>0$, the complement of the ε-neighborhood...

open_problem | open | MathDB 页面 | 原文线索

V2-029819 | MATHDB-403195

Converse extension problem for initial data satisfying the dominant energy condition

摘录 / EXCERPT: Given an initial data set (M,g,k) subject to $\rho \ge \|j\|$, does M embed as a spacelike hypersurface into some time-oriented Lorentzian manifold $(\overline M,\overline g)$...

open_problem | open | MathDB 页面 | 原文线索

V2-029820 | MATHDB-403196

Rigidity for the band-width equality case

摘录 / EXCERPT: Let Y be a closed manifold of dimension $n-1\ne 4$ which does not admit a metric with positive scalar curvature. Let g be a Riemannian metric on $X=Y\times[-1,1]$ and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029821 | MATHDB-403197

Bartnik’s static extension conjecture

摘录 / EXCERPT: Let $n\ge 3$ and let (Ω,g_0) be a compact n-dimensional Riemannian manifold with scalar curvature $R_{g_0}\ge 0$. Suppose the mean curvature H_{g_0} is not everywhere...

conjecture | open | MathDB 页面 | 原文线索

V2-029822 | MATHDB-403198

Gromov’s scalar-curvature fill-in problem (Problem 7)

摘录 / EXCERPT: Let (Y,h) be a closed $(n-1)$-dimensional Riemannian manifold and let φ be a smooth function on Y. Find conditions such that, for a given number σ, there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-029823 | MATHDB-403199

Heat kernel estimates for the Lichnerowicz Laplacian on Ricci-flat ALE manifolds

摘录 / EXCERPT: Theorem 1 (Heat kernel estimates). Let (M^n,g) be a Ricci-flat ALE manifold equipped with a parallel spinor. Let $1<p\le q<\infty$, $k\in\mathbb N_0$, and $\varepsilon>0$. Then:

open_problem | open | MathDB 页面 | 原文线索

V2-029824 | MATHDB-403200

Explicit characterization of nonnegative scalar curvature at the origin for a prescribed second-order metric form

摘录 / EXCERPT: Suppose $g=g_{ij}\,dx^i\otimes dx^j$ is a metric on a neighborhood of the origin in $\mathbb{R}^n$, and that we may write ... where the G_{ij} are functions on...

open_problem | open | MathDB 页面 | 原文线索

V2-029825 | MATHDB-403202

Spacetime positive energy theorem for asymptotically Euclidean initial data

摘录 / EXCERPT: Let (M,g) be an asymptotically Euclidean Riemannian manifold, where M is the union of a compact set and of an asymptotically flat region, of dimension $n\ge 3$. Suppose that th...

open_problem | open | MathDB 页面 | 原文线索

V2-029826 | MATHDB-403203

Positive energy for an asymptotically hyperbolic end

摘录 / EXCERPT: Let (M,g) denote an n-dimensional Riemannian manifold which is the union of a compact set and an AH end. If the scalar curvature $R(g)$ satisfies $R(g)\ge -n(n-1)$, then the en...

open_problem | open | MathDB 页面 | 原文线索

V2-029827 | MATHDB-403206

Negative ADM mass on an asymptotically flat end

摘录 / EXCERPT: Let (M,g) be a complete Riemannian manifold with $ obreak\operatorname{scal}_g\ge 0$. Suppose E is an asymptotically flat end of M. Can E have negative ADM mass?

open_problem | open | MathDB 页面 | 原文线索

V2-029828 | MATHDB-403213

Asymptotic dimension of geodesic planes

摘录 / EXCERPT: Is the asymptotic dimension of a plane at most two for any geodesic metric?

open_problem | open | MathDB 页面 | 原文线索

V2-029829 | MATHDB-403215

Unbounded torsion in discrete isometry groups of negatively curved Hadamard manifolds

摘录 / EXCERPT: For each $n \ge 4$ and $\epsilon>0$, there exists an n-dimensional Hadamard manifold X with sectional curvature $-1-\epsilon \le K_X \le -1$ and a discrete, finitely generated...

open_problem | open | MathDB 页面 | 原文线索

V2-029830 | MATHDB-403216

Kapovich’s conjecture on cusps and torsion for negatively curved Hadamard manifolds

摘录 / EXCERPT: 1. Let Γ be a finitely generated discrete isometry group of a negatively curved Hadamard manifold X with negatively pinched sectional curvature, $-b^2 \le K_X \le -1$. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-029831 | MATHDB-403217

Finiteness of cusps and bounded torsion for small critical exponent

摘录 / EXCERPT: Let Γ be a finitely generated discrete isometry group of a negatively curved Hadamard manifold X with negatively pinched sectional curvature $-b^2 \le K_X \le -1$. Assume...

open_problem | open | MathDB 页面 | 原文线索

V2-029832 | MATHDB-403218

Metric SYZ conjecture: special Lagrangian torus fibrations in the generic region

摘录 / EXCERPT: The Strominger-Yau-Zaslow conjecture, subject to some interpretation, asks for a special Lagrangian T^n-fibration at least in the generic region of a Calabi-Yau manifold, where “...

conjecture | open | MathDB 页面 | 原文线索

V2-029833 | MATHDB-403219

Smooth asymptotics for collapsing Calabi-Yau metrics

摘录 / EXCERPT: Let $f:X^{m+n}\to B^m$ be a surjective holomorphic map of compact Kähler manifolds, with $c_1(X)=0$, and let the generic fiber be a smooth compact Calabi-Yau manifold. For Kähler f...

open_problem | open | MathDB 页面 | 原文线索

V2-029834 | MATHDB-403223

Hamilton-Tian conjecture for the Kähler-Ricci flow

摘录 / EXCERPT: Let M be a Fano manifold of complex dimension n, and let $\omega(t)$ solve ... where ω_0 is a Kähler metric with Kähler class $2\pi c_1(M)$. Conjecture 1. Any sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-029835 | MATHDB-403240

The $r=1$ local-conjugacy codimension conjecture for interval exchange transformations

摘录 / EXCERPT: Determine whether, for $r=1$, the C^1 local conjugacy class of a Roth-type interval exchange transformation is a submanifold of codimension $(d-1)+(g-1)$, where d is the number...

conjecture | open | MathDB 页面 | 原文线索

V2-029836 | MATHDB-403245

Polterovich’s Poisson-bracket lower-bound conjecture

摘录 / EXCERPT: Conjecture (Polterovich). Let (M,ω) be a closed symplectic manifold, and let $\mathcal U=\{U_i\}_{i\in I}$ be a finite open cover of M by displaceable sets. Then there exi...

conjecture | open | MathDB 页面 | 原文线索

V2-029837 | MATHDB-403248

Does a Lagrangian in the Humilière completion equal its support?

摘录 / EXCERPT: Let $L\in\widehat L(T^*N)$ be such that $\operatorname{supp}(L)$ is a smooth submanifold. Is $L=\operatorname{supp}(L)$?

open_problem | open | MathDB 页面 | 原文线索

V2-029838 | MATHDB-403282

Smooth convergence of collapsing Ricci-flat Kähler metrics

摘录 / EXCERPT: Let X^{m+n} be a compact Calabi-Yau manifold with $c_1(X)=0$ in $H^2(X,\mathbb R)$, admitting a surjective holomorphic map $f:X\to B$ with connected fibers onto a compact Kähler...

open_problem | open | MathDB 页面 | 原文线索

V2-029839 | MATHDB-403286

Schoen-Yau conjecture on positive scalar curvature of closed aspherical manifolds

摘录 / EXCERPT: Let $n \ge 4$ and let M^n be a closed smooth aspherical manifold. Then M does not admit any Riemannian metric of positive scalar curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-029840 | MATHDB-403295

Unbounded trajectories in the unstable manifolds of the nontrivial equilibria

摘录 / EXCERPT: For the nonlinear Schrödinger equation (1), does the unstable manifold of either of the two nontrivial equilibria from Theorem 3 contain an isolated trajectory that grows without b...

open_problem | open | MathDB 页面 | 原文线索

V2-029841 | MATHDB-403322

Formal degree and Plancherel-density conjectures for symplectic and even special orthogonal groups

摘录 / EXCERPT: Let F be a local field of characteristic zero, let $\psi:F\to\mathbb{C}^{\times}$ be a continuous non-trivial unitary character, and let H be a symplectic group or an even spec...

conjecture | open | MathDB 页面 | 原文线索

V2-029842 | MATHDB-403386

Divisibility of signatures of twisted intersection forms

摘录 / EXCERPT: Let 4M4 be a spin 4-manifold, let 4lambda:Ptimes Ptomathbb Z4 be a nonsingular symmetric form on a finitely generated free abelian group 4P4, and let 4alpha:pi1(M)tooperatorname{Au...

open_problem | open | MathDB 页面 | 原文线索

V2-029843 | MATHDB-403387

Evenness of twisted intersection forms on spin 4-manifolds

摘录 / EXCERPT: Let $\lambda:P\times P\to\mathbb{Z}$ be a non-singular symmetric form on a finitely generated free abelian group P, let M be a 4-manifold, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-029844 | MATHDB-403390

Classification of irreducible 4-manifolds of trisection genus three

摘录 / EXCERPT: For every closed, smooth, connected, oriented, irreducible 4-manifold X with trisection genus $g_t(X)=3$, X is either the spin of a lens space, or a Gluck twist on a specific...

open_problem | open | MathDB 页面 | 原文线索

V2-029845 | MATHDB-403425

Sharp L^p estimates for oscillatory integral operators of arbitrary signature

摘录 / EXCERPT: For a hypersurface $\Sigma=\{(\omega,\Sigma(\omega)):\omega\in B^{n-1}(0,1)\}\subset\mathbb{R}^n$ with non-vanishing Gaussian curvature, define ... for $f:B^{n-1}(0,1)\to\mathbb{C}$...

open_problem | open | MathDB 页面 | 原文线索

V2-029846 | MATHDB-403437

Wall-crossing computation of symplectic monodromy for isolated hypersurface singularities

摘录 / EXCERPT: Compute the symplectic monodromy representation of the universal unfolding of an isolated complex hypersurface singularity by a wall-crossing method.

open_problem | open | MathDB 页面 | 原文线索

V2-029847 | MATHDB-403445

Finiteness of resolution data for normal surface singularities with fixed link

摘录 / EXCERPT: Let M be a real 3-manifold. There exist finitely many triplets $(\[1m\Gamma,L,P\u001b[0m)$, where $\[1m\Gamma\[0m$ is a weighted graph and...

open_problem | open | MathDB 页面 | 原文线索

V2-029848 | MATHDB-403524

Katok-Spatzier conjecture on higher-rank Anosov actions

摘录 / EXCERPT: Conjecture (Katok-Spatzier). All C^{∞} Anosov actions of $\mathbb{R}^{k}$ or $\mathbb{Z}^{k}$, $k\ge 2$ (higher rank), on any compact manifold without C^{∞} rank one...

conjecture | open | MathDB 页面 | 原文线索

V2-029849 | MATHDB-403530

Calderón-Zygmund inequalities for $p>2$ without injectivity-radius control

摘录 / EXCERPT: Theorem 2 [8]. Let M be a complete Riemannian manifold satisfying condition (H): (i) $\\operatorname{Ric}\geq K$ for some constant K; and (ii) there exist functions K_1,K_2 i...

open_problem | open | MathDB 页面 | 原文线索

V2-029850 | MATHDB-403582

The rational-essential-manifold positive-scalar-curvature conjecture

摘录 / EXCERPT: Conjecture: Rationally essential manifolds do not admit a metric of positive scalar curvature. Here a closed oriented manifold is rationally essential if its classifying map sends...

conjecture | open | MathDB 页面 | 原文线索

V2-029851 | MATHDB-403662

Khintchine and Jarník-Besicovitch theorems on non-degenerate submanifolds

摘录 / EXCERPT: Generalise Khintchine’s theorem and the Jarník-Besicovitch theorem from $S_n(\psi)$ and $S_n(v)$ to $S_f(\psi)$ and $S_f(v)$, respectively, for suitably chosen smooth maps...

open_problem | open | MathDB 页面 | 原文线索

V2-029852 | MATHDB-403733

Hamilton conjecture

摘录 / EXCERPT: Let (M,g) be a smooth, complete manifold with bounded and uniformly pinched non-negative Ricci curvature. Thus $R\ge 0$, and the eigenvalues $\lambda_1\le \lambda_2\le \lambda_3$...

conjecture | open | MathDB 页面 | 原文线索

V2-029853 | MATHDB-403738

Zhou-Zhu existence theorem for prescribed mean curvature hypersurfaces

摘录 / EXCERPT: Theorem 1 (Zhou and Zhu). Assume M is a closed Riemannian manifold of dimension between 3 and 7. Then for an admissible function $h\colon M\to\mathbb{R}$, there exists a smoo...

open_problem | open | MathDB 页面 | 原文线索

V2-029854 | MATHDB-403739

Min-max bound for constant-mean-curvature surfaces in asymptotically flat 3-manifolds

摘录 / EXCERPT: Conjecture: Let (M^3,g) be a complete, asymptotically flat Riemannian 3-manifold without boundary. For a constant $c>0$, define ... where the infimum is over smoothly varying fam...

open_problem | open | MathDB 页面 | 原文线索

V2-029855 | MATHDB-403741

Short-time existence of Ricci flow from arbitrary complete three-manifolds with nonnegative Ricci curvature

摘录 / EXCERPT: It is a long-standing open problem as to whether one can start the Ricci flow with an arbitrary complete three-dimensional Riemannian manifold with nonnegative Ricci curvature.

open_problem | open | MathDB 页面 | 原文线索

V2-029856 | MATHDB-403745

Classification of simply connected, positively curved, closed Riemannian manifolds

摘录 / EXCERPT: Classify the simply connected, positively curved, closed Riemannian manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-029857 | MATHDB-403779

Endpoint $L^{p_b}\to L^{q_b}$ boundedness for smooth Radon-like transforms

摘录 / EXCERPT: For the Radon-like transform T in (1), with $\c6$ smooth, $D_x\c6$ of rank k at the relevant point, and Q the trilinear curvature functional defined in (3), is nondegeneracy...

open_problem | open | MathDB 页面 | 原文线索

V2-029858 | MATHDB-403809

Gromov’s width conjecture under a scalar-curvature lower bound

摘录 / EXCERPT: Let M be a closed connected manifold of dimension $n-1\ne 4$ such that M does not admit a metric of positive scalar curvature. Let g be a Riemannian metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-029859 | MATHDB-403810

Interior Kasparov product formula for \-classes on general foliations

摘录 / EXCERPT: In the setup of the reporta smooth closed manifold with spin foliations and an inclusion :F0to F1, with associated monodromy groupoids and longitudinal positive-scalar-curvature me...

open_problem | open | MathDB 页面 | 原文线索

V2-029860 | MATHDB-403917

Coarse nodal-domain bound for spectral subspaces

摘录 / EXCERPT: Let M be a compact connected n-dimensional Riemannian manifold; if $ ... where C depends only on M and k.

open_problem | open | MathDB 页面 | 原文线索

V2-029861 | MATHDB-404047

Linear growth of reinforced edge weights identifies geodesic edges

摘录 / EXCERPT: On any finite graph with two marked nodes, the nest N and the food F, the only edges whose weight will grow linearly are those lying on a geodesic between N and F.

open_problem | open | MathDB 页面 | 原文线索

V2-029862 | MATHDB-404088

Kotschick’s conjecture on holomorphic one-forms without zeros

摘录 / EXCERPT: Let X be a compact Kähler manifold. Then the following are equivalent: 1. There is a holomorphic one-form without zeros. 2. There is a closed real 1-form...

conjecture | open | MathDB 页面 | 原文线索

V2-029863 | MATHDB-404092

Type-I behavior of Yamabe-flow singularities on asymptotically Euclidean manifolds

摘录 / EXCERPT: It would be interesting to determine whether, unlike in the general noncompact case, all (necessarily infinite-time) singularities of the Yamabe flow on asymptotically Euclidean ma...

open_problem | open | MathDB 页面 | 原文线索

V2-029864 | MATHDB-404096

Beauville-Voisin conjecture for Lagrangian-fibered projective hyper-Kähler manifolds

摘录 / EXCERPT: Let $f\colon X\to B$ be a Lagrangian fibration of a projective hyper-Kähler manifold, and let $L\in\operatorname{Pic}(X)\cong CH^1(X)$ be the pullback of a generator of...

conjecture | open | MathDB 页面 | 原文线索

V2-029865 | MATHDB-404130

Scalar curvature rigidity for spheres with subsets removed

摘录 / EXCERPT: Which subsets $\Sigma \subset S^n$, $n \ge 3$, are “scalar curvature rigid” in the sense that each Riemannian metric g on $S^n \setminus \Sigma$ with $g \ge g_{S^n}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029866 | MATHDB-404133

Improved generic regularity for minimizing hypersurfaces

摘录 / EXCERPT: Let $\Gamma^{n-1}\subset \mathbb{R}^{n+1}$ be a smooth, closed, oriented submanifold. There exist arbitrarily small perturbations Γ' of Γ such that every minimizing i...

open_problem | open | MathDB 页面 | 原文线索

V2-029867 | MATHDB-404134

Existence of smooth least-area Plateau minimizers

摘录 / EXCERPT: Let Γ be a smooth, closed, oriented, $(n-1)$-dimensional submanifold of $\mathbb{R}^{n+1}$. Among all smooth, compact, oriented hypersurfaces $M\subset\mathbb{R}^{n+1}$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-029868 | MATHDB-404138

Nonnegativity of the limiting C^0 local mass

摘录 / EXCERPT: If g satisfies the hypotheses of item (2) of Theorem 2, and moreover g has nonnegative scalar curvature in the sense of Ricci flow everywhere on M, do we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-029869 | MATHDB-404140

Hamilton-Lott conjecture in dimension three

摘录 / EXCERPT: Let (M^3,g) be a smooth, complete Riemannian manifold with bounded curvature and uniformly Ricci-pinched curvature, meaning that $\\operatorname{Ric}(g)\ge 0$ and there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-029870 | MATHDB-404141

Macroscopic volume lower bound for rationally essential manifolds

摘录 / EXCERPT: There exists a dimensional constant $\varepsilon(d)>0$ with the following property: for every rationally essential Riemannian manifold (M,g) of dimension d and every $R>0$, ... w...

open_problem | open | MathDB 页面 | 原文线索

V2-029871 | MATHDB-404146

Trisections containing a non-separating 3-sphere

摘录 / EXCERPT: If X is a smooth 4-manifold containing a non-separating 3-sphere, then every trisection of X is reducible.

open_problem | open | MathDB 页面 | 原文线索

V2-029872 | MATHDB-404149

Coincidence of LO-, CTF-, and NLS-detected slopes

摘录 / EXCERPT: The sets of LO-detected slopes, CTF-detected slopes, and NLS-detected slopes on the boundary of a knot manifold coincide.

open_problem | open | MathDB 页面 | 原文线索

V2-029873 | MATHDB-404150

Double branched covers of hyperbolic L-space knots

摘录 / EXCERPT: For a hyperbolic L-space knot K, is the 2-fold cyclic branched cover $\Sigma_2(K)$ an NLS manifold, i.e. not a Heegaard Floer L-space?

open_problem | open | MathDB 页面 | 原文线索

V2-029874 | MATHDB-404155

Connected sums of non-loose knots

摘录 / EXCERPT: Suppose K_1 and K_2 are two non-trivial non-loose knots in an overtwisted contact 3-manifold (M,ξ). Is $K_1\#K_2$ non-loose?

open_problem | open | MathDB 页面 | 原文线索

V2-029875 | MATHDB-404157

Smooth triviality of the Dehn twist for closed spin simply connected smooth 4-manifolds

摘录 / EXCERPT: For which closed, spin, simply connected, smooth 4-manifolds X is the Dehn twist t_X smoothly isotopic to the identity?

open_problem | open | MathDB 页面 | 原文线索

V2-029876 | MATHDB-404165

Marked boundary rigidity for surfaces of Anosov type

摘录 / EXCERPT: Let Σ be a smooth compact connected oriented manifold with boundary, and let g_1,g_2 be two metrics of Anosov type on Σ: the boundary is strictly convex, the metric...

open_problem | open | MathDB 页面 | 原文线索

V2-029877 | MATHDB-404168

Existence of Birkhoff sections for all Reeb vector fields

摘录 / EXCERPT: Does every Reeb vector field of a given contact structure on a closed 3-manifold admit a Birkhoff section?

open_problem | open | MathDB 页面 | 原文线索

V2-029878 | MATHDB-404191

Higher-dimensional critical-regularity actions of elementary subgroups

摘录 / EXCERPT: Let M be a compact, connected, smooth manifold, and let $r\in[1,\infty)$. Let $\Gamma_{r,M}\le \mathrm{Diff}^r(M)$ be an elementary subgroup. Then, for $s>r$, is every action of...

open_problem | open | MathDB 页面 | 原文线索

V2-029879 | MATHDB-404194

Existence and uniqueness of maximal submanifolds in pseudo-hyperbolic space

摘录 / EXCERPT: Let Λ be any non-negative $(p-1)$-sphere in $\partial_\infty \mathbb H^{p,q}$. Then there exists a unique complete maximal submanifold Σ in $\mathbb H^{p,q}$ such th...

open_problem | open | MathDB 页面 | 原文线索

V2-029880 | MATHDB-404199

Equivariant parametrized stable h-cobordism splitting

摘录 / EXCERPT: For a compact smooth G-manifold M, is there a splitting ... analogous to the nonequivariant splitting, where ...

open_problem | open | MathDB 页面 | 原文线索

V2-029881 | MATHDB-404220

Inner radius of nodal domains in high dimensions

摘录 / EXCERPT: Let (M,g) be a closed smooth Riemannian manifold of dimension $d\ge 3$. Let u_λ be an eigenfunction of the positive Laplace-Beltrami operator $-\Delta_g$ with eigenvalue...

open_problem | open | MathDB 页面 | 原文线索

V2-029882 | MATHDB-404225

Maximality of the geodesic disk for the second Robin eigenvalue in hyperbolic space

摘录 / EXCERPT: For hyperbolic domains with positive Robin parameter, in the second quadrant of Figure 2, does the geodesic disk still maximize the second Robin eigenvalue...

open_problem | open | MathDB 页面 | 原文线索

V2-029883 | MATHDB-404227

Construction of singular Riemannian structures with prescribed Weyl asymptotics

摘录 / EXCERPT: Let M be an n-dimensional compact manifold, let $S\subset M$ be a closed submanifold, and let $\upsilon:\mathbb{R}_+\to\mathbb{R}_+$ be a non-decreasing slowly varying function...

open_problem | open | MathDB 页面 | 原文线索

V2-029884 | MATHDB-404268

Diagrammatic description of diffeomorphisms of 4-manifolds

摘录 / EXCERPT: Show that any diffeomorphism between smooth 4-manifolds can be described, up to isotopy, by a sequence of moves on the framed, marked, generalized trisection diagrams introduced in...

open_problem | open | MathDB 页面 | 原文线索

V2-029885 | MATHDB-404271

Existence of quadrisections for closed smooth 5-manifolds

摘录 / EXCERPT: Theorem 3 (Ben Aribi, Courte, Golla, M.). Any closed smooth 5-manifold admits a quadrisection.

open_problem | open | MathDB 页面 | 原文线索

V2-029886 | MATHDB-404325

Hard Lefschetz and Hodge-Riemann properties for Kähler Schur classes

摘录 / EXCERPT: Let X be a compact Kähler manifold of complex dimension n, let $\omega_1,\dots,\omega_e$ be Kähler classes on X, and let $\lambda\in\Lambda(n-2,e)$. Then the Schur class...

open_problem | open | MathDB 页面 | 原文线索

V2-029887 | MATHDB-404423

Rigidity of high-order Fano manifolds

摘录 / EXCERPT: Let X be a k-Fano manifold of dimension n. If $k \ge \lceil \log_2(n+1)\rceil$, then $X \simeq \mathbb{P}^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-029888 | MATHDB-404424

K-polystability of the general member of a smooth Fano family

摘录 / EXCERPT: For each deformation family of smooth Fano manifolds, is the general member of the family K-polystable? Equivalently, is the associated component of $M^{Kps}_{n,V}$ non-empty?

open_problem | open | MathDB 页面 | 原文线索

V2-029889 | MATHDB-404501

Vanishing-mass Kähler metrics developing gravitational wells

摘录 / EXCERPT: Find a family of asymptotically Euclidean Kähler metrics with nonnegative, integrable scalar curvature and vanishing ADM mass that develops a gravitational well.

open_problem | open | MathDB 页面 | 原文线索

V2-029890 | MATHDB-404502

Gromov’s Weyl-law conjecture for the volume spectrum

摘录 / EXCERPT: For each pair of integers $n\ge 1$ and $0\le k<n+1$, does there exist a universal constant $a(n,k)>0$ such that every compact $(n+1)$-dimensional Riemannian manifold (M,g) satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-029891 | MATHDB-404607

Vogan’s conjecture on Arthur packets for split odd orthogonal and symplectic groups

摘录 / EXCERPT: If G is a split odd orthogonal or symplectic group and $ ... Moreover, ... and the function ... coincides with the function ... defined by Arthur’s theory.

conjecture | open | MathDB 页面 | 原文线索

V2-029892 | MATHDB-404672

VADB stability for sequences not diffeomorphic to the limit manifold

摘录 / EXCERPT: Can one prove a version of Theorem 8 for sequences which are not diffeomorphic to the limit manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-029893 | MATHDB-404675

Removing the spin assumption from Llarull’s rigidity theorem

摘录 / EXCERPT: Can the spin assumption be dispensed with in Theorem 1, namely the rigidity statement for a closed connected Riemannian manifold admitting a smooth distance-nonincreasing map of no...

open_problem | open | MathDB 页面 | 原文线索

V2-029894 | MATHDB-404680

DEC extension of initial data sets

摘录 / EXCERPT: Given a manifold M equipped with a Riemannian metric g and a symmetric 2-tensor k satisfying the initial-data dominant energy condition ... does there exist a time-oriented L...

open_problem | open | MathDB 页面 | 原文线索

V2-029895 | MATHDB-404681

Schoen’s conjecture on noncompact area-minimizing boundaries

摘录 / EXCERPT: Let (M,g) be an asymptotically flat Riemannian manifold of dimension $3\le n\le 7$ with nonnegative scalar curvature. Suppose that there exists a noncompact area-minimizing bound...

conjecture | open | MathDB 页面 | 原文线索

V2-029896 | MATHDB-404686

Preservation of nonnegative scalar curvature under SWIF convergence

摘录 / EXCERPT: Suppose a sequence of three-dimensional Riemannian manifolds (M_j^3,g_j) with $\operatorname{Scal}\ge 0$ converges in the Sormani-Wenger intrinsic flat (SWIF) sense to a smooth R...

open_problem | open | MathDB 页面 | 原文线索

V2-029897 | MATHDB-404687

Preservation of nonnegative scalar curvature under SWIF limits with a MinA bound

摘录 / EXCERPT: Open Question 2. If (M_j^3,g_j) have $\operatorname{Scal}\ge 0$, satisfy the MinA hypothesis, and converge in the SWIF sense to a smooth (M_0,g_0), does (M_0,g_0) have...

open_problem | open | MathDB 页面 | 原文线索

V2-029898 | MATHDB-404690

A convergence notion capturing volume-preserving SWIF and $W^{1,p}$ limits

摘录 / EXCERPT: Is there a notion of convergence for sequences of distinct Riemannian manifolds, (M_j,g_j), which implies volume-preserving SWIF convergence and also captures the information enc...

open_problem | open | MathDB 页面 | 原文线索

V2-029899 | MATHDB-404693

Finite-time extinction and classification for null mean curvature flow on the Minkowski lightcone

摘录 / EXCERPT: Let (Σ_0,γ_0) be a spacelike cross section of the future-pointing standard lightcone N in $3+1$-dimensional Minkowski spacetime. Then the solution of null mean curvat...

open_problem | open | MathDB 页面 | 原文线索

V2-029900 | MATHDB-404715

Sensitivity of isoperimetric mass to scalar curvature and boundary minimality

摘录 / EXCERPT: How can the definition of isoperimetric mass be fine-tuned to make it more sensitive to nonnegative scalar curvature and minimality of the boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-029901 | MATHDB-404716

Multiplicity one for unstable components in long-time mean curvature-flow limits

摘录 / EXCERPT: In the setting of Theorem 2, can one avoid the finite piecewise perturbations and perturb only the initial data? Equivalently, is it true that the multiplicity of every unstable mi...

open_problem | open | MathDB 页面 | 原文线索

V2-029902 | MATHDB-404718

σ_k-Yamabe problem in the larger cone

摘录 / EXCERPT: Let (M,g_0) be a compact Riemannian manifold and let $[g_0]$ be its conformal class. For $k\ge 2$, define $\mathcal C_{k-1}([g_0])=\Gamma_{k-1}^+\cap[g_0]$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-029903 | MATHDB-404719

Isoperimetric inequality between total Q-curvatures

摘录 / EXCERPT: Conjecture. Let $0<l<k<n/2$. The inequality

open_problem | open | MathDB 页面 | 原文线索

V2-029904 | MATHDB-404729

Stolz-Teichner conjecture on the spinor bundle over loop space

摘录 / EXCERPT: It was conjectured that a compact spin manifold M of dimension $n \ge 5$ is string if and only if its loop space LM is spin, with the loop-space spinor bundle admitting a certa...

conjecture | open | MathDB 页面 | 原文线索

V2-029905 | MATHDB-404734

Existence of fundamental solutions for the fractional porous medium equation on noncompact manifolds

摘录 / EXCERPT: Let M be a complete, connected, noncompact N-dimensional Riemannian manifold satisfying $\\operatorname{Ric}(M)\ge -(N-1)k$ for some $k>0$ and the Faber-Krahn inequality stated...

open_problem | open | MathDB 页面 | 原文线索

V2-029906 | MATHDB-404830

Multiplicity One Conjecture for mean curvature flow in \(\mathbb{R}^3\)

摘录 / EXCERPT: If M is an almost regular mean curvature flow in $\mathbb{R}^3$, then every tangent flow is a multiplicity-one shrinking soliton.

conjecture | open | MathDB 页面 | 原文线索

V2-029907 | MATHDB-404833

Contractibility of spaces of positive scalar curvature metrics with symmetry

摘录 / EXCERPT: Let Γ be a compact Lie group and let M be a compact connected smooth Γ-manifold without boundary. Suppose that Γ contains a normal S^1-subgroup whose fixed...

open_problem | open | MathDB 页面 | 原文线索

V2-029908 | MATHDB-404836

Contractible 3-manifolds with complete nonnegative scalar curvature

摘录 / EXCERPT: If M is a contractible 3-manifold and admits a complete metric of nonnegative scalar curvature, is M diffeomorphic to $\mathbb{R}^3$?

open_problem | open | MathDB 页面 | 原文线索

V2-029909 | MATHDB-404838

The isospectral problem for p-widths

摘录 / EXCERPT: For which manifolds do the values of $\{\omega_p\}$ determine (M,g)?

open_problem | open | MathDB 页面 | 原文线索

V2-029910 | MATHDB-404841

Critical Rate of Decay Conjecture

摘录 / EXCERPT: There exists a dimensional constant $C_n>0$ such that the following holds. Let M be an orientable n-manifold that admits a complete Riemannian metric of positive scalar curvatu...

conjecture | open | MathDB 页面 | 原文线索

V2-029911 | MATHDB-404842

Contractible 4-manifolds with nonnegative Ricci curvature and Euclidean volume growth

摘录 / EXCERPT: Conjecture 1. Let (M^4,g) be a smooth, complete, contractible 4-manifold with $\\operatorname{Ric} \ge 0$ and Euclidean volume growth, meaning that there exist $c>0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-029912 | MATHDB-404844

Structure near flat branch points for two-dimensional mod(q) area-minimizing currents

摘录 / EXCERPT: Let T be a 2-dimensional area-minimizing current mod (q) in a smooth Riemannian 3-manifold. Then, for any density-$q/2$ flat singular point p of T, there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-029913 | MATHDB-404848

Monotonicity of Liouville entropy along the normalized Ricci flow on negatively curved surfaces

摘录 / EXCERPT: Let M be a smooth closed orientable surface of negative Euler characteristic, and let g_0 be a smooth Riemannian metric on M of non-constant negative Gaussian curvature. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-029914 | MATHDB-404856

Geometric counting and rigidity of quasi-Fuchsian k-surfaces

摘录 / EXCERPT: Let (M,h_0) be a closed hyperbolic 3-manifold, let h be a Riemannian metric on M with $\operatorname{sect}_h\le -1$, and fix $0<k<1$. For $[\Gamma]$ in the set QF of conj...

open_problem | open | MathDB 页面 | 原文线索

V2-029915 | MATHDB-404860

Counting rational points near general nondegenerate submanifolds

摘录 / EXCERPT: Let $M\subset\mathbb{R}^{M}$ be a bounded smooth immersed submanifold with boundary, of dimension n and codimension R, satisfying the mild nondegeneracy condition referred to i...

open_problem | open | MathDB 页面 | 原文线索

V2-029916 | MATHDB-404869

Conjectured index bound for min-max geodesics

摘录 / EXCERPT: It is expected that the collection of geodesics $\{\gamma_i\}_{i=1}^{N_p}$ achieving the p-width can be chosen to satisfy ... where $\operatorname{ind}(\gamma_i)$ is the Morse inde...

conjecture | open | MathDB 页面 | 原文线索

V2-029917 | MATHDB-404873

Genericity of nondegenerate neckpinch singularities

摘录 / EXCERPT: The report expects that if a mean curvature flow starting from an initial hypersurface M_0 develops a degenerate neckpinch singularity, then there is a sequence of perturbations...

open_problem | open | MathDB 页面 | 原文线索

V2-029918 | MATHDB-404875

Schulze’s uniqueness conjecture for Ricci flows coming out of metric cones

摘录 / EXCERPT: Let $C(X)$ be a metric cone. A Ricci flow $g(t)$ on a smooth manifold M is said to be coming out of $C(X)$ if, for some $x_0\in M$, $(M,d_{g(t)},x_0)$ converges to...

conjecture | open | MathDB 页面 | 原文线索

V2-029919 | MATHDB-404876

Generic evanescence of strongly isolated singularities

摘录 / EXCERPT: Given a closed manifold M of dimension $N\ge 3$, there exists a generic subset $\mathcal G_0$ of the space of smooth metrics on M such that, for every $g\in\mathcal G_0$, any...

open_problem | open | MathDB 页面 | 原文线索

V2-029920 | MATHDB-404878

Classification of minimal links with Morse index equal to the ambient dimension

摘录 / EXCERPT: Determine all $(n-1)$-dimensional minimal submanifolds $\Sigma\subset S^{N-1}$ whose Morse index is equal to N.

open_problem | open | MathDB 页面 | 原文线索

V2-029921 | MATHDB-404880

Path-connectedness of the moduli space of mean-convex embedded two-spheres

摘录 / EXCERPT: Let (M,g) be a complete, orientable Riemannian three-manifold with nonnegative Ricci curvature. Then the moduli space $\mathcal{M}_{H>0}(S^2,M)$ of mean-convex embedded two-sided...

open_problem | open | MathDB 页面 | 原文线索

V2-029922 | MATHDB-404979

QHD minimal symplectic fillings versus QHD smoothings

摘录 / EXCERPT: It was conjectured that QHD minimal symplectic fillings can exist only when smoothings do.

open_problem | open | MathDB 页面 | 原文线索

V2-029923 | MATHDB-404995

Asymptotic expansion conjecture for WRT invariants of closed oriented 3-manifolds

摘录 / EXCERPT: For a closed, oriented 3-manifold M, the surgery formula $Z_k(M)$ for the Witten-Reshetikhin-Turaev invariant should have, as $k\to\infty$, the asymptotic expansion ... where the s...

conjecture | open | MathDB 页面 | 原文线索

V2-029924 | MATHDB-405034

Cannon Conjecture

摘录 / EXCERPT: Every word-hyperbolic group whose boundary at infinity is a two-sphere has a finite-index subgroup which is the fundamental group of a closed hyperbolic 3-manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-029925 | MATHDB-405072

Threshold nature of the DSS profile U_2

摘录 / EXCERPT: Conjecture 2 (Threshold nature of U_2). The stable manifold of the DSS profile U_2 is locally a threshold for generic blowup. More precisely, the stable manifold of U_2 separ...

open_problem | open | MathDB 页面 | 原文线索

V2-029926 | MATHDB-405112

Nonlinear least-squares finite element method for the prescribed Gaussian curvature problem

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a bounded convex polygonal domain. Assume $K\in H^2(\Omega)$ is strictly positive on $\overline\Omega$, $\phi\in H^4(\Omega)$ is strictly convex,...

open_problem | open | MathDB 页面 | 原文线索

V2-029927 | MATHDB-405170

Truncated Octahedral Conjecture

摘录 / EXCERPT: Let $\mathcal{P}_3$ be the class of convex three-dimensional parallelohedra, namely convex polyhedra that tile $\mathbb{R}^3$ by translations. For every $V>0$ and every...

conjecture | claimed_solved | MathDB 页面 | 原文线索

B06 拓扑 / Topology | 5,704 条

V2-029928 | MATHDB-315464

Cohn-Elkies high-dimensional sphere-packing rate

摘录 / EXCERPT: What is the exponential rate of the Cohn--Elkies linear-programming bound for sphere packing as $d\to\infty$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029929 | MATHDB-315476

Positivity for arbitrary coherent quotients on Deligne-Mumford stacks

摘录 / EXCERPT: Can the torsion-free hypothesis be removed from the positivity theorem for coherent quotients of tensor powers of $\Omega^1_{\mathcal X}(\log\Delta)$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029930 | MATHDB-315477

Twelve common flex lines in a general pencil of cubics

摘录 / EXCERPT: Does a general pencil of plane cubics over $\mathbb C$ have exactly 12 common flex lines?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029931 | MATHDB-315515

Thin-triangle Kakeya optimization at 128 slopes

摘录 / EXCERPT: For the fixed slopes $a_i=i/128$, choose rational intercepts b_i minimizing the area of the union of the prescribed thin triangular neighborhoods of the lines $y=a_ix+b_i$.

open_problem | open | MathDB 页面 | 原文线索

V2-029932 | MATHDB-315520

Kirby Problem 5.16 for noncommutative semifree DGAs

摘录 / EXCERPT: For semifree noncommutative differential graded algebras, are stable tame isomorphism, quasi-isomorphism, or derived Morita equivalence algorithmically decidable?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-029933 | MATHDB-315531

Spin parity for k-differentials

摘录 / EXCERPT: Let $k\geq 3$ be an odd integer. For any integer n, let $N_k(n)$ be the number of pairs of positive integers (b_1,b_2) such that $b_1,b_2\leq (k-1)/2$, $b_1+b_2\geq (k+1)/2$, a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029934 | MATHDB-315592

Piecewise-affine Möbius-band width

摘录 / EXCERPT: For the benchmark's G_β-invariant clean triangulations and squeeze maps of the strip, prove that $\sqrt 3$ is the infimum of the realized values of β.

open_problem | open | MathDB 页面 | 原文线索

V2-029935 | MATHDB-315596

Contractibility of a crossing-matching complex

摘录 / EXCERPT: For a reducible quasi-reduced free-group word w, let F_w be the CW complex whose cells encode crossing matchings and their resolutions. Must F_w be contractible?

open_problem | open | MathDB 页面 | 原文线索

V2-029936 | MATHDB-315608

IRIS Conjecture 6.1 on simple 3-polytopes

摘录 / EXCERPT: Conjecture 6.1 (NuevaMirada). If P is a simple 3-polytope with face vector $(p_3,p_4,\ldots,p_m)$, such that $\sum_{k\geq 7}p_k\geq 3$, then

open_problem | open | MathDB 页面 | 原文线索

V2-029937 | MATHDB-315676

Boucksom’s local analytic Bertini conjecture

摘录 / EXCERPT: In Theorem 1.1 of the cited paper, is $\Lambda\setminus\mathcal{G}$ a pluripolar subset of Λ, where $\Lambda\simeq\mathbb{P}^{N}$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029938 | MATHDB-315680

Kuperberg’s six-cylinder conjecture

摘录 / EXCERPT: How many pairwise non-overlapping infinite unit cylinders can simultaneously touch a unit ball?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029939 | MATHDB-315681

Bellman’s lost-in-a-forest problem for the golden gnomon

摘录 / EXCERPT: Let $G=\operatorname{conv}\{(-c,0),(c,0),(0,s)\}$ be the isosceles triangle whose equal sides have length 1, whose base angles are 36°, and whose apex angle is...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029940 | MATHDB-315727

Nearly linear lattice coverings of arbitrary convex bodies

摘录 / EXCERPT: Does there exist an absolute constant $C>0$ such that, for every positive integer n and every n-dimensional convex body K, the lattice covering density satisfies ...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-029941 | MATHDB-315732

Complete model spaces do not force Lie-group regularity

摘录 / EXCERPT: Is every infinite-dimensional Lie group modelled on a complete locally convex space regular?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029942 | MATHDB-315736

Shokurov global index conjecture for threefold foliations

摘录 / EXCERPT: Is the index of a numerically trivial log-canonical foliated log Calabi--Yau triple uniformly bounded in dimension three?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029943 | MATHDB-315739

Weighted homogeneity via logarithmic vector fields

摘录 / EXCERPT: Machado-Seade conjecture. The following statements are equivalent: $(D,0)$ is weighted homogeneous; there exists a holomorphic vector field tangent to D with an isolated singular...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029944 | MATHDB-315765

Conant’s mod-4 Kawauchi conjecture

摘录 / EXCERPT: For every amphicheiral knot K, does there exist $f(z)\in\mathbb{Z}[z]$ such that ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029945 | MATHDB-315785

Erdős unit-distance conjecture

摘录 / EXCERPT: Let N points be given in the Euclidean plane, and let a unit distance mean a pair of points whose Euclidean distance is 1. The unit distance conjecture. The number of unit dist...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029946 | MATHDB-315874

Homological conjectures in commutative algebra

摘录 / EXCERPT: In mathematics, homological conjectures have been a focus of research activity in commutative algebra since the early 1960s. They concern a number of interrelated conjectures relat...

conjecture | open | MathDB 页面 | 原文线索

V2-029947 | MATHDB-316141

Berge conjecture

摘录 / EXCERPT: Let $K \subset S^3$ be a knot, let γ be a slope on $\partial N(K)$, where $N(K)$ is a tubular neighbourhood of K, and suppose that the Dehn surgery $S^3_\gamma(K)$ obtaine...

conjecture | open | MathDB 页面 | 原文线索

V2-029948 | MATHDB-316143

Borel conjecture

摘录 / EXCERPT: Let M and N be closed topological manifolds that are aspherical, i.e. connected with $\pi_n(M)=\pi_n(N)=0$ for all $n\ge 2$ (equivalently, with contractible universal covers)....

conjecture | open | MathDB 页面 | 原文线索

V2-029949 | MATHDB-316146

Novikov conjecture

摘录 / EXCERPT: Let G be a discrete group and let BG be its classifying space, an Eilenberg-MacLane space of type $K(G,1)$ (unique up to homotopy equivalence as a CW complex). For a closed ori...

conjecture | open | MathDB 页面 | 原文线索

V2-029950 | MATHDB-316147

Telescope conjecture

摘录 / EXCERPT: In mathematics, the Ravenel conjectures are a set of mathematical conjectures in the field of stable homotopy theory posed by Douglas Ravenel at the end of a paper published in 198...

conjecture | open | MathDB 页面 | 原文线索

V2-029951 | MATHDB-316149

Volume conjecture

摘录 / EXCERPT: Let O be the unknot in the 3-sphere S^{3}, and for a knot $K\subset S^{3}$ and an integer $N\ge 2$ let $J_{K,N}(q)$ denote the N-colored Jones polynomial of K. Define the...

conjecture | open | MathDB 页面 | 原文线索

V2-029952 | MATHDB-316151

Zeeman conjecture

摘录 / EXCERPT: Let L be a finite simplicial complex. A simplex $\sigma\in L$ is called a free face of L if σ is a proper face of exactly one simplex $\tau\in L$; in that case...

conjecture | open | MathDB 页面 | 原文线索

V2-029953 | MATHDB-316176

Tameness conjecture

摘录 / EXCERPT: In mathematics, the tameness theorem states that every complete hyperbolic 3-manifold with finitely generated fundamental group is topologically tame, in other words homeomorphic t...

conjecture | open | MathDB 页面 | 原文线索

V2-029954 | MATHDB-316218

Virtual Haken conjecture

摘录 / EXCERPT: In topology, an area of mathematics, the virtually Haken conjecture states that every compact, orientable, irreducible three-dimensional manifold with infinite fundamental group is...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029955 | MATHDB-316219

Hsiang-Lawson's conjecture

摘录 / EXCERPT: In differential geometry, Lawson's conjecture states that the Clifford torus is the only minimally embedded torus in the 3-sphere S3. The conjecture was featured by the Australian...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029956 | MATHDB-316220

Ehrenpreis conjecture

摘录 / EXCERPT: In mathematics, the Ehrenpreis conjecture of Leon Ehrenpreis states that for any K greater than 1, any two closed Riemann surfaces of genus at least 2 have finite-degree covers whi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029957 | MATHDB-316221

Atiyah conjecture

摘录 / EXCERPT: In mathematics, the Atiyah conjecture is a collective term for a number of statements about restrictions on possible values of -Betti numbers.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-029958 | MATHDB-316222

Cobordism hypothesis

摘录 / EXCERPT: Fix an integer $n\geq 1$. Let $\mathrm{Bord}_n^{\mathrm{fr}}$ denote the symmetric monoidal (∞,n)-category of framed bordisms: its objects are 0-dimensional framed manifol...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-029959 | MATHDB-316224

Poincaré conjecture

摘录 / EXCERPT: Let M be a topological manifold of dimension 3 (a second-countable Hausdorff space each of whose points has a neighborhood homeomorphic to an open subset of $\mathbb{R}^3$). Su...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029960 | MATHDB-316226

Nikiel's conjecture

摘录 / EXCERPT: Let X be a compact Hausdorff space. Call a linearly ordered set $(L,<)$ with its open-interval topology a linearly ordered topological space, and say that X is a continuous ima...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029961 | MATHDB-316227

Ganea conjecture

摘录 / EXCERPT: Ganea's conjecture is a now disproved claim in algebraic topology. It states that

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-029962 | MATHDB-316251

Fibrewise homotopy invariance conjecture for the signature L^2-rho form

摘录 / EXCERPT: Let $\pi\colon M\to B$ and $\theta\colon N\to B$ be smooth fibre bundles of compact manifolds over the same base, with normal Γ-coverings $p\colon \widetilde M\to M$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-029963 | MATHDB-316262

Cohomological collapse conjecture for finite cyclic semidirect products

摘录 / EXCERPT: Let G be a finite cyclic group and let L be a finitely generated $\mathbb ZG$-lattice. For the semidirect product extension ... consider its Lyndon-Hochschild-Serre spectral sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-029964 | MATHDB-316263

Tait's writhe conjecture for minimal alternating diagrams

摘录 / EXCERPT: Tait's conjecture II. Minimal crossing number diagrams of the same alternating knot have the same writhe.

conjecture | open | MathDB 页面 | 原文线索

V2-029965 | MATHDB-316264

Tait's flyping conjecture for alternating knot diagrams

摘录 / EXCERPT: Tait's conjecture III. Any two alternating diagrams of the same knot are related by a finite sequence of flypes.

conjecture | open | MathDB 页面 | 原文线索

V2-029966 | MATHDB-316300

The strong weight complex functor conjecture

摘录 / EXCERPT: Let $(\underline{C},w)$ be a triangulated category with a weight structure w, let $\underline{Hw}=\underline{C}^{w=0}$ be its heart, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-029967 | MATHDB-316369

Incidence Property criterion for singularities

摘录 / EXCERPT: Let $\xi\subset J^k(n,p)$ be a singularity, and let $e(\xi)=[\xi]|_\xi$ denote its normal Euler class. Say that ξ has the Incidence Property when its incidence class is nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-029968 | MATHDB-316380

Hausel's nonequivariant abelianization conjecture

摘录 / EXCERPT: Let G be a group with maximal torus T, Weyl group $W=N(T)/T$, and associated symplectic quotients $\mathfrak{M}_{\alpha,0}(G)$ and $\mathfrak{M}_{\alpha,0}(T)$. Let $\Phi_0(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-029969 | MATHDB-316381

The intersection-cohomology ring conjecture for nonabelian hypertoric quotients

摘录 / EXCERPT: Let G be a group with maximal torus T and Weyl group W, and let $\mathfrak{M}_{\alpha,0}(G)$ and $\mathfrak{M}_{\alpha,0}(T)$ be the corresponding symplectic quotients. Assum...

conjecture | open | MathDB 页面 | 原文线索

V2-029970 | MATHDB-316394

Ozsváth-Szabó simple-type conjecture for symplectic 4-manifolds

摘录 / EXCERPT: Let X be a symplectic 4-manifold with $b^+(X)\geq 2$. The manifold X has Ozsváth-Szabó simple type if every spin^c structure $\mathfrak{s}$ for which...

conjecture | open | MathDB 页面 | 原文线索

V2-029971 | MATHDB-316437

Ando's equivariant Witten genus conjecture

摘录 / EXCERPT: Let X be a compact, differentiable, G-equivariant spin manifold of dimension $24d$. Assume that $\frac{p_1}{2}(X)=0$ and that ... where π is the map from X to a point and...

conjecture | open | MathDB 页面 | 原文线索

V2-029972 | MATHDB-316450

The relative homological mirror symmetry conjecture for the pair (X,D)

摘录 / EXCERPT: Let X be a complex manifold with divisor D, let M be the mirror moduli space, let M_D be the mirror of D, let $\mathcal{F}(M,M_D)$ be the relevant Fukaya category of M...

conjecture | open | MathDB 页面 | 原文线索

V2-029973 | MATHDB-316462

The orientation asymmetry conjecture for proper L-surgery structures on Brieskorn homology spheres

摘录 / EXCERPT: Let $\Sigma(p,q,r)$ be any Brieskorn homology sphere. Its orientation reversal is denoted by $\overline{\Sigma(p,q,r)}$. A proper L-surgery structure conjecture. $\Sigma(p,q,r)$ ca...

conjecture | open | MathDB 页面 | 原文线索

V2-029974 | MATHDB-316471

Generalized Jones' conjecture for braid representatives

摘录 / EXCERPT: Let $\mathcal{K}$ be a knot, let $\mathcal{B}_{\mathcal{K}}$ denote the set of braid diagrams representing $\mathcal{K}$, and let b_D and w_D be the braid index and writhe of...

conjecture | open | MathDB 页面 | 原文线索

V2-029975 | MATHDB-316472

Maximal Bennequin number conjecture

摘录 / EXCERPT: Let $\mathcal{K}$ be a knot, let $\mathcal{B}_{\mathcal{K}}$ be the set of braid diagrams representing it, and for $D\in\mathcal{B}_{\mathcal{K}}$ define the Bennequin number by...

conjecture | open | MathDB 页面 | 原文线索

V2-029976 | MATHDB-316486

Berthelot's coherence and overconvergence conjecture for relative rigid cohomology

摘录 / EXCERPT: Let $(S,S,\mathcal{S})$ be a triple endowed with a lift of Frobenius on $\mathcal{S}$, and let ... be a diagram of pairs such that $f:\overline{X}\to\overline{Y}$ is proper,...

conjecture | open | MathDB 页面 | 原文线索

V2-029977 | MATHDB-316517

Adem's integral collapse conjecture for cyclic group extensions

摘录 / EXCERPT: Let C_n be a cyclic group of order n and let L be a finite-rank integral lattice. For a split group extension ... let $\{E_*,d_*\}$ denote its Lyndon-Hochschild-Serre spectral...

conjecture | open | MathDB 页面 | 原文线索

V2-029978 | MATHDB-316521

Ext comparison conjecture for smooth and injective representations

摘录 / EXCERPT: Ext comparison conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-029979 | MATHDB-316522

Every 2-bridge knot has a Fourier representation with two z-frequencies

摘录 / EXCERPT: 2-bridge Fourier representation conjecture. Every 2-bridge knot can be expressed as a Fourier-$(1,1,k)$ knot with $n_x=2$ and $k\leq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-029980 | MATHDB-316523

The explicit Fourier representation conjecture for torus knots T_{2,q}

摘录 / EXCERPT: Torus-knot representation conjecture. The torus knot $T_{2,q}$ can be represented as a Fourier-$(1,1,2)$ knot with frequencies $n_x=2$, $n_y=q$, $n_{z,1}=2$, and $n_{z,2}=q-2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-029981 | MATHDB-316524

The Fourier representation conjecture for torus knots

摘录 / EXCERPT: Torus-knot Fourier representation conjecture. The torus knot $T_{p,q}$, with $0<p<q$, can be represented as a Fourier-$(1,1,2)$ knot with frequencies $n_x=p$, $n_y=q$, $n_{z,1}=p$,...

conjecture | open | MathDB 页面 | 原文线索

V2-029982 | MATHDB-316528

Conjugacy conjecture for minimal-trip Lorenz link representations

摘录 / EXCERPT: Let $\mathcal{L}$ be a Lorenz link with representations $\mathbb{M}_1,\mathbb{M}_2\in B_t$, where t is the trip number of $\mathcal{L}$. Conjugacy conjecture. The braids...

conjecture | open | MathDB 页面 | 原文线索

V2-029983 | MATHDB-316539

Emerton's cohomological dimension conjecture for reductive groups

摘录 / EXCERPT: Let $\mathbf{G}/\mathbb{Q}$ be a reductive group whose centre has rational rank zero. Let $X_{\mathbf{G}}$ be the associated symmetric space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-029984 | MATHDB-316543

Contactomorphism conjecture for negative definite plumbing graphs

摘录 / EXCERPT: Let Γ be a negative definite plumbing graph. Let ξ_C be the contact structure induced on the boundary of an ω-convex neighborhood of the plumbing configuration, a...

conjecture | open | MathDB 页面 | 原文线索

V2-029985 | MATHDB-316545

Conjecture on the symmetric homology of polynomial rings

摘录 / EXCERPT: Let $k[t_1,t_2,\dots,t_n]$ be the polynomial ring in n indeterminates over k, and let C_∞ denote the monad associated to an E_∞ operad. Write S^i for the i-sp...

conjecture | open | MathDB 页面 | 原文线索

V2-029986 | MATHDB-316553

Conjecture on integrable compatible and tamed cones of four-dimensional manifolds

摘录 / EXCERPT: Let M be a smooth manifold, and let ${\mathcal K}_{Int}^t(M)$ and ${\mathcal K}_{Int}^c(M)$ denote its integrable tamed and integrable compatible cones, respectively. In dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-029987 | MATHDB-316558

Kusner-Sullivan's composite-knot energy conjecture

摘录 / EXCERPT: Let $[K_1]$ and $[K_2]$ be isotopy classes of knots, and let $[K_1\sharp K_2]$ be their composite isotopy class. For an isotopy class $[K]$, write $E^{(2)}_{\circ}([K])$ for the in...

conjecture | open | MathDB 页面 | 原文线索

V2-029988 | MATHDB-316594

Non-containment of profinite closures of products of conjugacy classes

摘录 / EXCERPT: Let F be a finitely generated free group. For $g_1,g_2,\ldots,g_k\in F$, write $[g_i]^F$ for the conjugacy class of g_i in F, let $N(g_1,g_2,\ldots,g_k)$ be the normal subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-029989 | MATHDB-316599

Six-pattern conjecture for lens surgeries on the Poincare homology sphere

摘录 / EXCERPT: Let $\Sigma(2,3,5)$ be the Poincare homology sphere, let $K\subset\Sigma(2,3,5)$ be a knot, and suppose ... For $h\in\mathcal{H}(p,K)$, where $\mathcal{H}(p,K)$ is the set defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-029990 | MATHDB-316600

The virtual cohomological dimension conjecture for homological subgroups of the mapping class group

摘录 / EXCERPT: Let S_g be a closed oriented surface of genus g, let $\Mod(S_g)$ be its mapping class group, and let $\I^k(S_g)$ be the subgroup consisting of elements that act trivially on a...

conjecture | open | MathDB 页面 | 原文线索

V2-029991 | MATHDB-316603

Sen's monopole L^2 cohomology dimension conjecture

摘录 / EXCERPT: Let M_k^0 be the hyperkähler moduli space of certain ${\rm SU}(2)$ magnetic monopoles on $\mathbb R^3$, and let $\widetilde{M}_k^0$ be its universal cover. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-029992 | MATHDB-316604

Vafa-Witten's L^2 harmonic form dimension conjecture for instanton moduli spaces

摘录 / EXCERPT: Let M^{k,c_1}_ϕ be the smooth completion of the moduli space of ${\rm U}(n)$ Yang--Mills instantons of first Chern class c_1, energy k, and framing ϕ on a four-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-029993 | MATHDB-316618

The perfect-knot conjecture for knot Floer homology in the double branched cover

摘录 / EXCERPT: Perfect-knot conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-029994 | MATHDB-316632

The geometric characterization of knots trivial to all finite orders

摘录 / EXCERPT: Geometric characterization conjecture. A knot K is n-trivial for all $n\in\mathbb{N}$ if and only if it is n-hyperbolic for all $n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-029995 | MATHDB-316640

Positive area conjecture for hyperbolic two-component links

摘录 / EXCERPT: Let $L=C_1\cup C_2$ be a two-component link, and let $[L]$ denote its isotopy class. Define ... Suppose that L is a hyperbolic link and that each component is a non-trivial knot. P...

conjecture | open | MathDB 页面 | 原文线索

V2-029996 | MATHDB-316659

Product selection-principle theorem for spaces of countable selection rank

摘录 / EXCERPT: Product selection-principle theorem. If Y has ${\sf S}_c(\mathcal{O},\mathcal{O})$ and ${\sf S}_{fin}(\Omega,\mathcal{O}^{gp})$, then $X\times Y$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-029997 | MATHDB-316669

Cyclotomic cup-product and modular-symbol conjecture

摘录 / EXCERPT: Let K be the cyclotomic field containing the relevant Np-power roots of unity, let $X_K^{\circ}(1)$ be the primitive part of the relevant second cohomology group, and let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-029998 | MATHDB-316670

Finite-level cyclotomic cup-product and modular-symbol conjecture

摘录 / EXCERPT: Let $F_r=\mathbf{Q}(\mu_{Np^r})$, let $\nu_r:H^2_{\mathrm{cts}}(G_{F_r,S},\mathbf{Z}_p(2))^{\circ}\to(Y_r^-/I_rY_r^-)(1)$ be the canonical map, and let $\bar\xi_r(u:v)$ be the imag...

conjecture | open | MathDB 页面 | 原文线索

V2-029999 | MATHDB-316671

Cyclotomic cup-product spanning conjecture

摘录 / EXCERPT: Let $F_r=\mathbf{Q}(\mu_{Np^r})$ and let ... be the projected cup-product pairing on the relevant S-unit cohomology, with values in...

conjecture | open | MathDB 页面 | 原文线索

V2-030000 | MATHDB-316672

Eisenstein Hecke-action conjecture for the cyclotomic reciprocity map

摘录 / EXCERPT: Let $\varpi_r:H_1(X_1^r(N),C_1^r(N);\mathbf{Z}_p)^+\to H^2_{\mathrm{cts}}(G_{F_r,S},\mathbf{Z}_p(2))^{\circ}$ be the homomorphism sending Manin symbols to cup products of cyclotomi...

conjecture | open | MathDB 页面 | 原文线索

V2-030001 | MATHDB-316673

Universal reciprocity-map and p-adic L-function conjecture

摘录 / EXCERPT: Let M be a positive divisor of N. Let $\Xi_N=\phi_1\otimes\phi_2$ be the map from $X_K^{\circ}\otimes_{\mathbf{Z}_p}\mathfrak{X}_K^-$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-030002 | MATHDB-316674

Twisted cyclotomic cup-product and p-adic L-value conjecture

摘录 / EXCERPT: Let $M\mid N$, let ψ and γ be finite even arithmetic characters on $\mathbf{Z}_{p,N}$, set $\theta=\omega\psi\gamma$, and suppose that $p\mid B_{1,\theta}$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-030003 | MATHDB-316675

Standard p-adic L-value cup-product conjecture

摘录 / EXCERPT: Let θ and ψ be finite arithmetic characters on $\mathbf{Z}_{p,N}$, with θ odd and ψ even, suppose that $p\mid B_{1,\theta}$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-030004 | MATHDB-316677

The sigma-model example conjecture for symmetric open-closed topological conformal field theories

摘录 / EXCERPT: Sigma-model example conjecture. Sigma model produces an example of a symmetric OC TCFT.

conjecture | open | MathDB 页面 | 原文线索

V2-030005 | MATHDB-316679

The algebraic g-conjecture for homology spheres

摘录 / EXCERPT: Let Δ be a homology sphere. Let $k[\Delta]$ be its Stanley-Reisner ring, let Θ be a linear system of parameters, and define $L(\Delta)$ to be the set of pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-030006 | MATHDB-316680

The manifold algebraic g-conjecture

摘录 / EXCERPT: Let Δ be a homology manifold, and let $L^i_s(\Delta)$ denote the set of pairs consisting of a linear system of parameters and a one-form satisfying the partial Lefschetz con...

conjecture | open | MathDB 页面 | 原文线索

V2-030007 | MATHDB-316684

Scharlemann's conjecture on reducible refills of a genus-two handlebody

摘录 / EXCERPT: Let W be a genus-two handlebody embedded in a 3-manifold M, and let $[?]$ and $[?]$ be essential discs in W whose refills produce exteriors $M[\alpha]$ and $M[\beta]$. Suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-030008 | MATHDB-316696

Baker-Grigsby-Hedden conjecture on simple Floer homology knots

摘录 / EXCERPT: Let $L(p,q)$ be a lens space and let $K'\subset L(p,q)$ be a knot. A knot has simple Floer homology when its knot Floer homology has the same total rank as the Heegaard Floer homol...

conjecture | open | MathDB 页面 | 原文线索

V2-030009 | MATHDB-316697

Baker-Grigsby-Hedden conjecture on the two exceptional knots in a lens space

摘录 / EXCERPT: Let $L(p,q)$ be a lens space, and let $T\subset L(p,q)$ be a knot. Baker-Grigsby-Hedden's conjecture. There are exactly two knots $T_R,T_L\subset L(p,q)$ which satisfy ... These conj...

conjecture | open | MathDB 页面 | 原文线索

V2-030010 | MATHDB-316698

Floer-homological characterization conjecture for simple knots

摘录 / EXCERPT: Let $L(p,q)$ be a lens space, and let $(L(p,q),G)$ be any simple knot. For a knot $K'\subset L(p,q)$, write $\widehat{HFK}_*(L(p,q),K',\underline{\mathfrak s}_i)$ for the knot Floe...

conjecture | open | MathDB 页面 | 原文线索

V2-030011 | MATHDB-316699

The Berge conjecture in grid-number-one form

摘录 / EXCERPT: Berge conjecture. If surgery on $K\subset L(p,q)$ yields the three-sphere, then K has grid number one.

conjecture | open | MathDB 页面 | 原文线索

V2-030012 | MATHDB-316700

Simple knot Floer homology implies grid number one

摘录 / EXCERPT: Simple knot Floer homology conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-030013 | MATHDB-316701

The rank-p+2 conjecture for knots in lens spaces

摘录 / EXCERPT: Rank-$p+2$ conjecture. There are exactly two knots in $L(p,q)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-030014 | MATHDB-316706

The connective K-theory spectrum conjecture for surface groups

摘录 / EXCERPT: Let M^g be a Riemann surface of genus g, let $\pi_1 M$ be its fundamental group, and let $K^{\mathrm{def}}(\pi_1 M)$ denote the deformation K-theory spectrum of this group. L...

conjecture | open | MathDB 页面 | 原文线索

V2-030015 | MATHDB-316734

Khovanov-Rozansky comparison conjecture for algebraic knot homology

摘录 / EXCERPT: Let L be a link, and let $H^{i,j}_{\text{alg}}(L)$ denote the bigraded algebraic knot homology defined from the functor Ψ, while $H^{i,j}_{\text{KR}}(L)$ denotes Khovanov-Ro...

conjecture | open | MathDB 页面 | 原文线索

V2-030016 | MATHDB-316750

Chas's algebraic characterization conjecture for simple conjugacy classes

摘录 / EXCERPT: Chas's conjecture. The conjugacy class α is simple if and only if either

conjecture | open | MathDB 页面 | 原文线索

V2-030017 | MATHDB-316752

Non-framed-homotopy of the two knots shown in Figure 2

摘录 / EXCERPT: Let $\mathcal {M}$ be the space of long-knot diagrams, let $\Sigma^{(1)}_{\mathrm{iness}}$ be the union of inessential codimension-one strata, and let ... A generic path in...

conjecture | open | MathDB 页面 | 原文线索

V2-030018 | MATHDB-316753

Regular-isotopy invariance of Cross for unknotting-number-one long knots

摘录 / EXCERPT: Let K be a framed long knot of unknotting number one. Let h be a homotopy which unknots K and satisfies ... or, respectively, $ind(h)=-1$. Let $Cross(h)$ denote the crossing-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-030019 | MATHDB-316767

The generalized non-commutative main conjecture for Tate motives

摘录 / EXCERPT: Let $K\in\mathcal{F}^+$ be unramified outside a finite set of places Σ, and let C_K be the associated compactly supported arithmetic complex. Generalized Tate-motive main...

conjecture | open | MathDB 页面 | 原文线索

V2-030020 | MATHDB-316773

Trotman's tubular-neighborhood conjecture for b-regular stratified spaces

摘录 / EXCERPT: Let (A,Σ) be a differentiable stratified space b-regularly embedded by p into $\mathbb R^n$: for strata $Y<X$, whenever sequences in X and Y converge to a point of...

conjecture | open | MathDB 页面 | 原文线索

V2-030021 | MATHDB-316792

The nonvanishing characteristic-class conjecture for skew-framed immersions

摘录 / EXCERPT: Let $b=b(k)$ be the number of ones in the binary expansion of k, and assume $n+2\geq 2^{b(k)}$. Let $s=n-2b(k)+k+3$. Consider a skew-framed immersion ... of a manifold $M^{b(k)-1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-030022 | MATHDB-316793

The rationality conjecture for the crepant-resolution shift

摘录 / EXCERPT: Rationality conjecture for the shift. Then

conjecture | open | MathDB 页面 | 原文线索

V2-030023 | MATHDB-316794

The modified cohomological crepant resolution conjecture with a flat gerbe

摘录 / EXCERPT: Modified cohomological crepant resolution conjecture. There is a flat gerbe ... -twisted small quantum product of Y by analytic continuation in $Q_{s+1},\ldots,Q_r$, if necessary,...

conjecture | open | MathDB 页面 | 原文线索

V2-030024 | MATHDB-316795

The modified Ruan crepant resolution conjecture with a flat gerbe

摘录 / EXCERPT: Modified Ruan crepant resolution conjecture. There are a flat gerbe $$ over Y and elements

conjecture | open | MathDB 页面 | 原文线索

V2-030025 | MATHDB-316797

Nontriviality conjecture for the associahedral class in the curve complex

摘录 / EXCERPT: Nontriviality conjecture. For $g \geq 1$, the class

conjecture | open | MathDB 页面 | 原文线索

V2-030026 | MATHDB-316810

The Gorenstein quotient conjecture for homology manifolds

摘录 / EXCERPT: Gorenstein quotient conjecture. The ring $\overline{\mathbf{k}[\Delta]}$ should be Gorenstein. This is presented as a closely related, potentially weaker conjecture to the precedin...

conjecture | open | MathDB 页面 | 原文线索

V2-030027 | MATHDB-316822

A metrizable Menger bounded group separating neighborhood and open-cover selection

摘录 / EXCERPT: A topological group $(G,*)$ is Menger bounded when it satisfies the relevant neighborhood selection property, and ${\rm S}_c(obreak\text{\mathcal O}_{\rm nbd},\mathcal O)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030028 | MATHDB-316823

A metrizable Hurewicz bounded group separating neighborhood and open-cover selection

摘录 / EXCERPT: A topological group $(G,*)$ is Hurewicz bounded when it satisfies ${\rm S}_1(\Omega_{\rm nbd},\mathcal O^{\rm gp})$. The principle ${\rm S}_c(\mathcal O_{\rm nbd},\mathcal O)$ conc...

conjecture | open | MathDB 页面 | 原文线索

V2-030029 | MATHDB-316824

A metrizable Menger group separating neighborhood and open-cover selection

摘录 / EXCERPT: A topological group is Menger when it satisfies the Menger selection property, equivalently ${\rm S}_{\rm fin}(\mathcal O,\mathcal O)$. The principles...

conjecture | open | MathDB 页面 | 原文线索

V2-030030 | MATHDB-316825

A metrizable Menger bounded group with a Menger bounded square lacking neighborhood selection

摘录 / EXCERPT: Let $(G,*)$ be a metrizable topological group, and let $G^2=G\times G$ denote its square. The conjecture. There is a metrizable Menger bounded group $(G,*)$ with property ... such th...

conjecture | open | MathDB 页面 | 原文线索

V2-030031 | MATHDB-316826

A metrizable group with a Menger square lacking neighborhood selection

摘录 / EXCERPT: Let $(G,*)$ be a metrizable topological group, and write $G^2=G\times G$ for its square. The conjecture. There is a metrizable group $(G,*)$ which has the property ... and G^2 has...

conjecture | open | MathDB 页面 | 原文线索

V2-030032 | MATHDB-316832

Menger-screenability conjecture for the Haver property

摘录 / EXCERPT: Let X be a metrizable space. The Menger-screenability conjecture asserts that there is such an X with the Menger property and ... which does not have the Haver property in some m...

conjecture | open | MathDB 页面 | 原文线索

V2-030033 | MATHDB-316836

The free-loop-space Hochschild cohomology Gerstenhaber algebra isomorphism conjecture

摘录 / EXCERPT: Let M be a compact oriented d-dimensional smooth manifold, let $LM=map(S^1,M)$ be its free loop space, and let $S^*(M)$ denote the differential graded algebra of singular cocha...

conjecture | open | MathDB 页面 | 原文线索

V2-030034 | MATHDB-316845

Polynomial-growth conjecture at minimal Alexander roots

摘录 / EXCERPT: Let K be a knot, and let $\Delta(K;t)$ be its Alexander polynomial. Define ... Choose $c\in\Lambda(K)$ such that $|c|=\min\{|z|\mid z\in\Lambda(K)\}$. Polynomial-growth conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030035 | MATHDB-316881

The conjecture relating the modified multivariable invariant to the ADO invariant

摘录 / EXCERPT: Modified-invariant conjecture. The modified invariant for $\mathfrak{sl}(N-1|1)$ should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-030036 | MATHDB-316885

The shifted-complex polyhedral product wedge decomposition conjecture

摘录 / EXCERPT: Let K be a shifted simplicial complex, let (CX_i,X_i,x_i) be based pairs indexed by the vertices of K, and write $Z(K;(\underline{CX},\underline{X}))$ for the associated poly...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-030037 | MATHDB-316886

Oriented-homotopy invariance conjecture for higher string topology operations

摘录 / EXCERPT: Oriented-homotopy invariance conjecture. The operations of the main theorem and the conjectured relative string topology operations are oriented-homotopy invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-030038 | MATHDB-316889

Stable equality conjecture for the Torelli lower central series and Y-filtration

摘录 / EXCERPT: Stable equality conjecture. For any $g\geq 0$ and all $j\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-030039 | MATHDB-316907

Conjecture on reducing links using (2,2)-moves and sixth-power braid moves

摘录 / EXCERPT: A link is a disjoint union of circles in S^3. A $7(2,2)$-move is a $(2,2)$-move or its mirror image, and a $(\sigma_1\sigma_2)^6$ move is the local move associated with the sixth...

open_problem | open | MathDB 页面 | 原文线索

V2-030040 | MATHDB-316923

Gordon's conjecture on knots yielding lens spaces

摘录 / EXCERPT: Let k be a knot, and let a lens space surgery mean a Dehn surgery on k whose result is a lens space. A Berge knot is a knot arising from Berge's primitive/primitive constructio...

conjecture | open | MathDB 页面 | 原文线索

V2-030041 | MATHDB-316928

Toric mirror symmetry conjecture for the $\frac{\infty}{2}$ variation of Hodge structure

摘录 / EXCERPT: Let the toric data satisfy $\hat{\rho}\in \operatorname{cl}(\tilde{C}_\mathcal{X})$. Let $\mathcal{F}_{\rm B}$ be the B-model $\frac{\infty}{2}$ variation of Hodge structure associ...

conjecture | open | MathDB 页面 | 原文线索

V2-030042 | MATHDB-316933

Generalised Arnold conjecture for reducible mapping classes

摘录 / EXCERPT: Let M be a compact connected surface, let g be a reducible mapping class with finite-type component g_{fin} and pseudo-Anosov components g^i_{pA}, $i=1,\ldots,s$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030043 | MATHDB-316963

Finite topological type conjecture for shrinking Ricci solitons

摘录 / EXCERPT: Let (M,g) be a shrinking Ricci soliton satisfying ... where $f\in C^{\infty}(M)$. Finite topological type conjecture. Any shrinking Ricci soliton has finite topological type, meani...

conjecture | open | MathDB 页面 | 原文线索

V2-030044 | MATHDB-316965

Menichi's string topology compatibility conjecture for classifying spaces

摘录 / EXCERPT: Let G) be a connected compact Lie group of dimension d, let LBG denote the free loop space of its classifying space, and let $S_*(G)$ and $S^*(G)$ denote the singular chains...

conjecture | open | MathDB 页面 | 原文线索

V2-030045 | MATHDB-316966

Orlik's torsion conjecture for links of weighted homogeneous polynomials

摘录 / EXCERPT: Let $L_f({\bf u},{\bf v})$ be a link with fractional weights $({\bf u},{\bf v})$, and let $d_1,\ldots,d_r$ be the positive integers produced by Orlik's algorithm from these fractio...

conjecture | open | MathDB 页面 | 原文线索

V2-030046 | MATHDB-316982

Conjecture on universal covers of three-manifolds with homotopic tight contact structures

摘录 / EXCERPT: Universal-cover conjecture. If V is a closed manifold carrying a pair of homotopic tight contact structures, then the universal cover of V is diffeomorphic to $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-030047 | MATHDB-317004

Zariski-type connectedness conjecture for totally geodesic submanifolds

摘录 / EXCERPT: Let M be an m-dimensional Riemannian manifold of positive sectional curvature. Let N and H be totally geodesic closed submanifolds of dimensions n and h, respectively,...

conjecture | open | MathDB 页面 | 原文线索

V2-030048 | MATHDB-317021

Multiple disjunction conjecture for spaces of Poincare embeddings

摘录 / EXCERPT: Let N be a smooth compact n-dimensional manifold. Let $(Q_1,\dots,Q_r)$, with $r\geq 1$, be pairwise disjoint compact submanifolds of N transverse to ∂N, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030049 | MATHDB-317022

Multiple disjunction conjecture for spaces of embeddings

摘录 / EXCERPT: Let N be a smooth compact n-dimensional manifold and assume $s\geq 2$. For $1\leq i\leq s$, let $(Q_i,\partial_0Q_i,\partial_1Q_i)$ be a smooth compact manifold triad, with the...

conjecture | open | MathDB 页面 | 原文线索

V2-030050 | MATHDB-317040

Ladder correspondence for reduced Khovanov complexes

摘录 / EXCERPT: Let $(D',P)=\tau(D,P)$ be obtained from a connected link diagram with basepoint by component-preserving Conway mutation. Let $\widetilde{C}(D)$ denote the reduced Khovanov complex,...

conjecture | open | MathDB 页面 | 原文线索

V2-030051 | MATHDB-317042

The generalized complexified Volume Conjecture

摘录 / EXCERPT: Generalized complexified Volume Conjecture. There exists an open set $\mathcal{O}$ of $\mathbb{C}$ such that, for every $u\in\mathcal{O}$, the series

conjecture | open | MathDB 页面 | 原文线索

V2-030052 | MATHDB-317066

McMullen's g-conjecture for homology spheres

摘录 / EXCERPT: A homology sphere is a pure simplicial complex L such that every face link has the homology of a sphere of the same dimension. Its g-vector is the sequence derived from its h...

conjecture | open | MathDB 页面 | 原文线索

V2-030053 | MATHDB-317067

The Lefschetz-to-g hierarchy conjecture for homology spheres

摘录 / EXCERPT: Let L be a homology $(d-1)$-sphere. Write $\binom{a}{b}$ only through the interval notation in the claim, where $[a,b]$ denotes the corresponding set of integers, and let ... Delta...

conjecture | open | MathDB 页面 | 原文线索

V2-030054 | MATHDB-317080

Nonholonomic Moser conjecture for symplectic structures

摘录 / EXCERPT: Let M be a manifold, let ω_0 and ω_1 be symplectic structures in the same cohomology class, and suppose they can be connected by a path of symplectic structures in...

conjecture | open | MathDB 页面 | 原文线索

V2-030055 | MATHDB-317089

Tsygan's cyclic formality conjecture

摘录 / EXCERPT: Tsygan's cyclic formality conjecture. There exists a natural $\mathbb C[[u]]$-linear quasi-isomorphism of L_∞-modules

conjecture | open | MathDB 页面 | 原文线索

V2-030056 | MATHDB-317090

Kontsevich's formality conjecture for cyclic cochains

摘录 / EXCERPT: Kontsevich's cyclic formality conjecture. For each volume form $\Omega\in\Omega^d(M)$ there exists an L_∞-quasi-isomorphism of L_∞-algebras

conjecture | open | MathDB 页面 | 原文线索

V2-030057 | MATHDB-317097

Modified Adolphson-Sperber conjecture for the exponent of a polytope

摘录 / EXCERPT: Modified Adolphson-Sperber conjecture. If $p\equiv 1\pmod{I(\Delta)}$, then $U_p(\Delta)$ is Zariski dense in $N(\Delta,\mathbb{F}_p)$; in particular, $U_p(\Delta,T)$ is Zariski de...

conjecture | open | MathDB 页面 | 原文线索

V2-030058 | MATHDB-317098

Equality of generic Newton polygons for finite-order and T-adic characters

摘录 / EXCERPT: Let ${\rm GNP}_p(\Delta,\psi)$ denote the generic Newton polygon associated with a non-trivial finite-order character ψ, and let ${\rm GNP}_p(\Delta,T)$ denote the generic abs...

conjecture | solved | MathDB 页面 | 原文线索

V2-030059 | MATHDB-317099

Limiting generic Newton polygon conjecture

摘录 / EXCERPT: Let $M(\Delta)$ be the relevant monoid associated with Δ, let $E(\Delta)$ be the monoid generated by the lattice points of Δ, and let ${\rm HP}(\Delta)$ be the absolu...

conjecture | open | MathDB 页面 | 原文线索

V2-030060 | MATHDB-317114

Directional isoperimetric conjecture for cycles in Euclidean space

摘录 / EXCERPT: Directional isoperimetric conjecture. There is such a $(k+1)$-chain y for which, for every $(k+1)$-tuple I,

conjecture | open | MathDB 页面 | 原文线索

V2-030061 | MATHDB-317122

The Dwork-congruence criterion for divisible Hodge F-crystals

摘录 / EXCERPT: Let ${\cal P}$ be a CY(4)-operator with normalized holomorphic solution ... and let ${\cal Q}$ be the associated CY(5)-operator with normalized holomorphic solution ... The Dwork congr...

conjecture | open | MathDB 页面 | 原文线索

V2-030062 | MATHDB-317133

The equivariant bordism formula for Nielsen numbers of iterates

摘录 / EXCERPT: Let M be a connected closed manifold, let $f\colon M\to M$ be a self-map, and let $n\geq 2$. For the homotopy n-periodic point space, let $\ell_n^k(f)$ denote the relevant comp...

conjecture | open | MathDB 页面 | 原文线索

V2-030063 | MATHDB-317140

Strong fillability implies nonvanishing contact homology

摘录 / EXCERPT: Nonvanishing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030064 | MATHDB-317141

Connected sums preserve algebraic overtwistedness

摘录 / EXCERPT: Connected-sum conjecture. If (N,η) is algebraically overtwisted, then

conjecture | open | MathDB 页面 | 原文线索

V2-030065 | MATHDB-317143

Left-handed stabilization has vanishing contact homology

摘录 / EXCERPT: Left-handed stabilization conjecture. The left-handed stabilization of (M,α) along L has vanishing contact homology:

conjecture | open | MathDB 页面 | 原文线索

V2-030066 | MATHDB-317166

FPRAS hardness conjecture for Jones polynomial evaluation

摘录 / EXCERPT: Jones-polynomial FPRAS conjecture. There is no FPRAS for evaluating $J_L(q^2)$ except at the special points $\ell\in\{1,2,3,4,6\}$, provided $RP\neq NP$.

conjecture | open | MathDB 页面 | 原文线索

V2-030067 | MATHDB-317177

The monodromy invariant-subspace conjecture for polynomial fibers

摘录 / EXCERPT: Monodromy invariant-subspace conjecture. If there exists a non-trivial subspace of $H_1(F^{-1}(t),\mathbb C)$ which is invariant under the monodromy, then either the polynomial f...

conjecture | open | MathDB 页面 | 原文线索

V2-030068 | MATHDB-317218

Repovš's near-embedding conjecture for codimension-one manifolds

摘录 / EXCERPT: Let $f\colon M^{n-1} \to M^n$ be any continuous, possibly surjective map from a closed $(n-1)$-manifold M^{n-1} into a closed n-manifold M^n, where $n\ge 3$. Define the singu...

conjecture | open | MathDB 页面 | 原文线索

V2-030069 | MATHDB-317227

Trace-field symmetry conjecture for $(-2,3,n)$ pretzel knots

摘录 / EXCERPT: Let n be an odd, negative integer, and let the trace field of a hyperbolic knot complement mean its invariant trace field. Trace-field symmetry conjecture. The $(-2,3,n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030070 | MATHDB-317230

Haagerup property conjecture for groups of local similarities of compact ultrametric spaces

摘录 / EXCERPT: Let X be a compact ultrametric space, and let $LS(X)$ denote the locally compact group of local similarities on X. Haagerup property conjecture. The group $LS(X)$ has the Haage...

conjecture | open | MathDB 页面 | 原文线索

V2-030071 | MATHDB-317241

Mather-Thurston connectivity conjecture for the homotopy fiber of the universal normal bundle

摘录 / EXCERPT: Let $F\Gamma_q^r$ be the homotopy fiber of the classifying map of the universal normal bundle, $\nu\colon B\Gamma_q^r\to BO(q)$. The space $F\Gamma_q^r$ classifies codimension-q,...

conjecture | open | MathDB 页面 | 原文线索

V2-030072 | MATHDB-317267

Exact-sequence conjecture for Spaces, the D-particle field category, and Hochschild cohomology

摘录 / EXCERPT: Let ${\bf Spaces}$ be the category of spaces, let ${\rm ker}\,\mathbf{Q}$ be the proposed L_∞ category, and let $HH({\bf Spaces})$ denote the Hochschild cohomology of...

conjecture | open | MathDB 页面 | 原文线索

V2-030073 | MATHDB-317277

Nonvanishing of the top Ext homomorphism for local cohomology

摘录 / EXCERPT: Nonvanishing conjecture. The homomorphism ϕ is in general non-zero, equivalently, a surjection.

open_problem | open | MathDB 页面 | 原文线索

V2-030074 | MATHDB-317286

Ruan's conjecture on orbifold symplectic flops

摘录 / EXCERPT: Let X and Y be orbifolds related by an orbifold symplectic flop. The orbifold quantum cohomology rings are denoted by $QH^\ast_{CR}(X)$ and $QH^\ast_{CR}(Y)$. Ruan's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030075 | MATHDB-317298

Xu's non-vanishing conjecture for connected sums with a four-torus

摘录 / EXCERPT: Let X_i for $i=1,2,3$ be either a 4-torus T^4, or a closed oriented almost complex 4-manifold with ... ... and $SW_{X_i}(\Gamma_{X_i})\equiv 1\pmod 2$, where Γ_{X_i} is a sp...

conjecture | open | MathDB 页面 | 原文线索

V2-030076 | MATHDB-317303

Smooth conjugacy conjecture for Anosov diffeomorphisms of the three-torus

摘录 / EXCERPT: Let f and g be topologically conjugate C^r, $r>1$, Anosov diffeomorphisms of $\mathbb T^3$ with coinciding periodic data, meaning that the differentials of corresponding retu...

conjecture | open | MathDB 页面 | 原文线索

V2-030077 | MATHDB-317305

Rate-separation conjecture for partially hyperbolic toral diffeomorphisms

摘录 / EXCERPT: Let L be the hyperbolic automorphism under consideration, and let f be homotopic to L and partially hyperbolic in the strongest sense described by the source. Denote the rate...

conjecture | open | MathDB 页面 | 原文线索

V2-030078 | MATHDB-317315

Complexity conjecture for torus knots in lens spaces

摘录 / EXCERPT: Let $L(p,q)$ be a lens space and let $K(\ell,m)$ be the torus knot represented by the slope ℓ/m. Assume that $K(\ell,m)$ is $(0,1,2)$-irreducible, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030079 | MATHDB-317322

Morse nondegeneracy conjecture for heteroclinic-orbit flows

摘录 / EXCERPT: Let $\mathcal{H}$ be the space of heteroclinic orbits of the flow associated with ... is restricted to $\mathcal{H}$, all equilibria become nondegenerate critical points in the sense...

conjecture | open | MathDB 页面 | 原文线索

V2-030080 | MATHDB-317324

Floer boundary-operator conjecture for connecting manifolds

摘录 / EXCERPT: Let $C_k(R)$ be the free R-module generated by the k-dimensional connecting manifolds of ... , there are maps ... such that each ∂_k is an R-module homomorphism,...

conjecture | open | MathDB 页面 | 原文线索

V2-030081 | MATHDB-317325

Floer homology conjecture for heteroclinic-orbit spaces

摘录 / EXCERPT: Let E be the space of heteroclinic orbits of ... , ... Thus the Floer complex should compute the homology of the space of heteroclinic orbits. The source presents this as a reasonabl...

conjecture | open | MathDB 页面 | 原文线索

V2-030082 | MATHDB-317340

Mosaic representatives as a complete invariant of tame knots

摘录 / EXCERPT: Let k_1 and k_2 be tame knots or links, and let K_1 and K_2 be arbitrary chosen mosaic representatives of k_1 and k_2, respectively. Two mosaics have the same knot mosa...

conjecture | open | MathDB 页面 | 原文线索

V2-030083 | MATHDB-317345

Borel's higher order cohomology conjecture

摘录 / EXCERPT: Let G be a reductive Lie group with maximal compact subgroup K, Lie algebra ${\mathfrak g}$, arithmetic subgroup Γ, and let (σ,E) be a finite-dimensional represen...

conjecture | open | MathDB 页面 | 原文线索

V2-030084 | MATHDB-317356

The delta-grading conjecture for spanning tree complexes

摘录 / EXCERPT: Let D be a diagram of a knot or link K, let ${ \mathcal H}_2(D)$ denote the associated Heegaard diagram, and let $\widehat{CF}({\mathcal H}_2(D))$ be its Heegaard Floer chain c...

conjecture | open | MathDB 页面 | 原文线索

V2-030085 | MATHDB-317410

Kontsevich-Soibelman noncommutative Hodge structure conjecture

摘录 / EXCERPT: Let X be a smooth and compact noncommutative space over $\mathbb{C}$. The periodic cyclic homology $HP_{\bullet}(C_X)$ is defined from the category C_X associated with X, and...

conjecture | open | MathDB 页面 | 原文线索

V2-030086 | MATHDB-317445

Light-tail Cannings convergence to tree-valued Fleming-Viot dynamics

摘录 / EXCERPT: Light-tail Cannings conjecture. Under a suitable time rescaling, $(\mathcal V^N)_{N\in\mathbb N}$ converges weakly to the tree-valued Fleming-Viot dynamics in the Skorohod topology...

conjecture | open | MathDB 页面 | 原文线索

V2-030087 | MATHDB-317477

The Local Realization Conjecture for finite-degree Steenrod modules

摘录 / EXCERPT: Let M be an $\mathcal A$-module concentrated in finitely many degrees, where $\mathcal A$ is the mod 2 Steenrod algebra. Local Realization Conjecture. For all sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-030088 | MATHDB-317513

Hyperbolic manifolds as maximal domination classes

摘录 / EXCERPT: Let $n\geq 2$, and consider the domination relation $\geq$ on closed oriented manifolds, where $N\geq M$ means that there is a map $N\longrightarrow M$ of non-zero degree. Maximal...

conjecture | open | MathDB 页面 | 原文线索

V2-030089 | MATHDB-317515

Equivalences for the remaining MOY relations in graded matrix factorizations

摘录 / EXCERPT: Let ${\mathrm{HMF}}^{gr}$ be the category of graded matrix factorizations used to construct intertwiners of the fundamental representations of the quantum group...

conjecture | open | MathDB 页面 | 原文线索

V2-030090 | MATHDB-317536

Finite determination conjecture for quantum knot invariants

摘录 / EXCERPT: Let $\mathfrak g$ be a Lie algebra, let K be a knot, and let $J_{\mathfrak g,K}$ be its quantum invariant. Let W be the Weyl group, and let $I_{\mathfrak g,K}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030091 | MATHDB-317542

Maslov grading compatibility conjecture for sutured Floer homology

摘录 / EXCERPT: Let $(\Sigma,{\text{\boldmath α}},{\text{\boldmath β}})$ be a sutured Heegaard diagram for ${\text{\boldmath Σ}}(D\times I,T)$. Write $\mathbb{T}_{\alpha}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030092 | MATHDB-317547

Fukaya-module refinement of the symplectic Gysin sequence

摘录 / EXCERPT: Assume that the monotone Fukaya A_∞-category $\EuScript{F}_N$ is rigorously formulated, and fix a monotone Lagrangian $L\in\operatorname{Ob}(\EuScript{F}_N)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030093 | MATHDB-317551

The nonisolating contact invariant conjecture over the integers

摘录 / EXCERPT: Nonisolating contact invariant conjecture. Over $\mathbb{Z}$-coefficients, the following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-030094 | MATHDB-317557

Integrality conjecture for the previously defined $\mathfrak{sl}(N)$-link homology

摘录 / EXCERPT: Let $N\geq 4$ and consider the $\mathfrak{sl}(N)$-link homology defined in the preceding sections over the rational theory, with its relations and homomorphisms suggested to admit...

conjecture | open | MathDB 页面 | 原文线索

V2-030095 | MATHDB-317560

The nowhere-dense conjecture for subthreshold permutation-class growth rates

摘录 / EXCERPT: Let λ be the unique real root of $x^5-2x^4-2x^2-2x-1$, and let S be the set of growth rates of permutation classes that are below λ. Nowhere-dense growth-rate con...

conjecture | open | MathDB 页面 | 原文线索

V2-030096 | MATHDB-317583

Finite presentability conjecture for symplectic mapping class groups of right-angled Artin groups

摘录 / EXCERPT: Let Γ be a graph, let A_Γ be its right-angled Artin group, and let (w,Q) be a symplectic structure on A_Γ. Write $\mathrm{Mod}(\Gamma,w,Q)$ for the subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-030097 | MATHDB-317586

The structure of the second differential cohomology of \(\mathcal{K}(1)\)-modules of symbols

摘录 / EXCERPT: Second-cohomology structure statement. The second differential cohomology has the structure

conjecture | open | MathDB 页面 | 原文线索

V2-030098 | MATHDB-317588

Sutured monopole-Heegaard-instanton homology conjecture

摘录 / EXCERPT: Let $(M,b3)$ be a balanced sutured manifold. Denote its monopole, Heegaard, and instanton sutured homology groups by $aSHM(M,b3)$, $aSFH(M,b3)$, and $aSHI(M,b3)$, respectively. Sut...

conjecture | open | MathDB 页面 | 原文线索

V2-030099 | MATHDB-317589

Instanton knot homology comparison conjecture

摘录 / EXCERPT: Let $K\subset S^{3}$ be a knot. Denote the knot homologies defined by Ozsváth-Szabó and Rasmussen by their respective knot homology groups, and denote Floer's instanton homology fo...

conjecture | open | MathDB 页面 | 原文线索

V2-030100 | MATHDB-317590

Instanton foliation-depth conjecture for knots

摘录 / EXCERPT: Let $K\subset S^{3}$ be a knot. Consider the irreducible homomorphisms $c1:c0_{1}(S^{3}\setminus K)\to\operatorname{SU}(2)$ that map a chosen meridian m to the element...

conjecture | open | MathDB 页面 | 原文线索

V2-030101 | MATHDB-317598

Nollet-Xavier hyperplane-preimage injectivity conjecture

摘录 / EXCERPT: Nollet-Xavier conjecture. If the pre-image $f^{-1}(H)$ of every affine hyperplane $H\subseteq\mathbb{R}^n$ is connected, possibly empty, then f is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-030102 | MATHDB-317636

The narrow-wide dichotomy for monotone Lagrangian submanifolds

摘录 / EXCERPT: A monotone Lagrangian submanifold is one with monotonicity constant and minimal Maslov number $N_L\geq 2$; it is called narrow or wide according to the corresponding alternatives d...

conjecture | open | MathDB 页面 | 原文线索

V2-030103 | MATHDB-317637

The disjoint Lagrangians conjecture for point-invertible manifolds

摘录 / EXCERPT: Let M be a point-invertible manifold, and let $L_0,L_1\subset M$ be monotone Lagrangian submanifolds with $L_0\cap L_1=\varnothing$. Disjoint Lagrangians conjecture. At least one...

conjecture | open | MathDB 页面 | 原文线索

V2-030104 | MATHDB-317638

Baez-Dolan-Lurie cobordism hypothesis for framed manifolds

摘录 / EXCERPT: Let $\mathscr{C}$ be a symmetric monoidal infinity n-category. A fully dualizable object of $\mathscr{C}$ is an object satisfying the dualizability conditions at every categorica...

conjecture | open | MathDB 页面 | 原文线索

V2-030105 | MATHDB-317639

Integrality conjecture for the dimensional reduction of a 1-2-3 theory

摘录 / EXCERPT: Let F' be the dimensional reduction of a 1-2-3 theory F, obtained by evaluating F on products with a circle; in particular, for a closed 2-manifold Y,...

conjecture | open | MathDB 页面 | 原文线索

V2-030106 | MATHDB-317640

K-theoretic refinement conjecture for the dimensional reduction of a 1-2-3 theory

摘录 / EXCERPT: Let F' be the dimensional reduction of a 1-2-3 theory F, and let K denote K-theory, viewed as an E^{∞} ring spectrum. K-theoretic refinement conjecture. The dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-030107 | MATHDB-317645

Linear cusp-area and slope-length bounds for alternating knots

摘录 / EXCERPT: Alternating-knot cusp bounds. The area of a maximal cusp metric on K is bounded above and below by linear functions of the twist number of the diagram. Likewise, the length of th...

conjecture | open | MathDB 页面 | 原文线索

V2-030108 | MATHDB-317649

The mod 4 characterization of continued fractions with $\mu(r)=-1$

摘录 / EXCERPT: Mod 4 characterization conjecture. $\mu(r)=-1$ if and only if r is (mod 4)-equivalent to either $[p]$ or $[3p]$.

conjecture | open | MathDB 页面 | 原文线索

V2-030109 | MATHDB-317650

Silver-Williams conjecture for total twisted Alexander polynomials

摘录 / EXCERPT: Silver-Williams conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030110 | MATHDB-317651

The generalized evaluation conjecture for twisted Alexander polynomial quotients

摘录 / EXCERPT: Generalized evaluation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030111 | MATHDB-317652

The representation-polynomial divisibility conjecture for epimorphisms of 2-bridge knot groups

摘录 / EXCERPT: Representation-polynomial divisibility conjecture. If $f_2(z)$ divides $f_1(z)$, then there exists an epimorphism

conjecture | open | MathDB 页面 | 原文线索

V2-030112 | MATHDB-317673

The differential-square-zero conjecture for the coloured HOMFLY-PT complexes

摘录 / EXCERPT: For a crossing with colours i and j, let $d_k^{\pm}$ be the successive differentials in the associated complex, with $0\leq k\leq i-2$. Differential-square-zero conjecture. ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-030113 | MATHDB-317696

Smooth-in-time unfolding conjecture for smooth simple closed curves

摘录 / EXCERPT: Let $\mathbf f:\mathbb R/2\pi\longrightarrow\mathbb C$ be a smooth unit-speed simple closed curve, and let $\mathcal D$ denote the space of deformations satisfying conditions (1)--...

conjecture | open | MathDB 页面 | 原文线索

V2-030114 | MATHDB-317698

Polygon-free proof conjecture for the unfolding theorem

摘录 / EXCERPT: Let Theorem (main) denote the unfolding theorem established in the paper. Polygon-free proof conjecture. There exists a proof of Theorem (main) that does not rely on approximation...

open_problem | open | MathDB 页面 | 原文线索

V2-030115 | MATHDB-317727

The Ditters conjecture on polynomial generators for the algebra of secondary operations

摘录 / EXCERPT: Let ${\mathscr M}$ be the algebra with basis elements $M_{d_1,\ldots,d_n}$, and let ${\mathrm{ESL}}({\mathbb Z})$ denote the set of elementary Lyndon basis elements, namely element...

conjecture | open | MathDB 页面 | 原文线索

V2-030116 | MATHDB-317730

Local crepant resolution conjecture for quotient singularities

摘录 / EXCERPT: Let $G\subset SL(n,\mathbb{C})$ be finite with $n=2$ or 3, let $\mathcal{X}=[\mathbb{C}^n/G]$, and let ... be the standard crepant resolution. For an exceptional rational curve...

conjecture | open | MathDB 页面 | 原文线索

V2-030117 | MATHDB-317742

The quantum and delta grading conjecture for the Khovanov-Heegaard Floer spectral sequence

摘录 / EXCERPT: Let L be a link, let $\Sigma(L)$ denote the double cover of S^3 branched along L, and let $\{E^k(L),D^k\}$ be the spectral sequence whose E^2 term is the reduced Khovanov h...

conjecture | open | MathDB 页面 | 原文线索

V2-030118 | MATHDB-317763

Integral homology conjecture for quotients of tori by the involution

摘录 / EXCERPT: The integral homology conjecture. For $n\geq 1$ and $n\geq 2i\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-030119 | MATHDB-317764

Strong deformation retraction of metric graph moduli spaces in genus 1

摘录 / EXCERPT: Let X_n and Y_n be the spaces appearing in the construction, with X_n homotopy equivalent to $MG_{1,n}^{\mathfrak{v}}$ and Y_n homotopy equivalent to $TM_{1,n}$. Strong def...

conjecture | open | MathDB 页面 | 原文线索

V2-030120 | MATHDB-317772

The isomorphism conjecture for homology of level mapping class groups

摘录 / EXCERPT: Isomorphism conjecture. For fixed $L \geq 2$ and $k \geq 1$, there exists some N such that if $g \geq N$, then this map is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-030121 | MATHDB-317773

The weak stability conjecture for level mapping class groups

摘录 / EXCERPT: Stability conjecture. For fixed $L \geq 2$ and $k \geq 1$, there exists some N such that if $g \geq N$ and γ is a simple closed nonseparating curve on Σ_g, then thi...

conjecture | open | MathDB 页面 | 原文线索

V2-030122 | MATHDB-317785

The topological fundamental-group conjecture for higher-rank actions

摘录 / EXCERPT: Let G be a simple Lie group of real rank at least two acting faithfully and preserving volume on a compact manifold M. The topological fundamental-group conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-030123 | MATHDB-317789

Zimmer's volume-preserving low-dimensional action conjecture

摘录 / EXCERPT: Let G be a semisimple Lie group with property (T) and let Γ be a lattice in G. Let d be the least dimension of an infinite-image linear representation and n the le...

conjecture | open | MathDB 页面 | 原文线索

V2-030124 | MATHDB-317806

Alev's Poisson-Hochschild homology dimension conjecture for types B and D

摘录 / EXCERPT: Alev's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030125 | MATHDB-317822

Canonicity conjecture for the Lee functor on framed link cobordisms

摘录 / EXCERPT: Let ${\mathcal Cob}_{f/i}^4$ be the quotient of ${\mathcal Cob}_f^4$ by framed movie moves. Assume the chain transformations in the Lee functor construction are made canonical, ind...

conjecture | open | MathDB 页面 | 原文线索

V2-030126 | MATHDB-317865

Grothendieck's cofiniteness conjecture for local cohomology

摘录 / EXCERPT: Let R be a commutative noetherian ring, let $\mathfrak{a}$ be an ideal of R, and let M be a finite R-module. For each integer n, consider the local cohomology module...

conjecture | open | MathDB 页面 | 原文线索

V2-030127 | MATHDB-317872

Jones-Conway polynomial detection conjecture for cables of non-isotopic simple knots

摘录 / EXCERPT: Let K_1 and K_2 be non-isotopic simple knots. For integers p and q, the cable-detection conjecture asserts that there exist numbers p and q such that the (p,q)-cables...

conjecture | open | MathDB 页面 | 原文线索

V2-030128 | MATHDB-317873

Supersignature unknotting-number bound

摘录 / EXCERPT: Let u,v be parameters and let L be a link for which the supersignature $\sigma_{u,v}(L)$ exists and is finite. Let $u(L)$ denote the unknotting number, and let T_n be the tri...

conjecture | open | MathDB 页面 | 原文线索

V2-030129 | MATHDB-317874

JCK polynomial detection conjecture for closed 3-braids

摘录 / EXCERPT: Let γ be a braid on three strands, let $\hat{\gamma}$ be its closure, and let $\overline{\gamma}$ denote the braid obtained by taking the mirror operation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030130 | MATHDB-317877

Conjecture on stability from the canonical Poisson cohomology map

摘录 / EXCERPT: Stability conjecture. If Φ vanishes, then any Poisson structure close enough to π admits at least one nearby symplectic leaf diffeomorphic to S.

conjecture | open | MathDB 页面 | 原文线索

V2-030131 | MATHDB-317878

Divisibility by a binomial power of the first Gram determinant

摘录 / EXCERPT: Let G_n be the Gram matrix in the paper, and let divisibility be taken in its polynomial ring. The first-determinant divisibility conjecture. For every integer $n\geq 1$, ... The c...

conjecture | open | MathDB 页面 | 原文线索

V2-030132 | MATHDB-317879

Divisibility by powers of the new factors of Gram determinants

摘录 / EXCERPT: New-factor divisibility conjecture. For the resulting sequence of factors,

conjecture | open | MathDB 页面 | 原文线索

V2-030133 | MATHDB-317880

Symmetric factorization conjecture for Gram determinants of planar curves

摘录 / EXCERPT: Symmetric-factorization conjecture. The determinant has the following factorization properties: (1) $\det G_n=u^2-v^2$ for some $u\in R_1$ and $v\in R_2$; (2) it factors as

conjecture | open | MathDB 页面 | 原文线索

V2-030134 | MATHDB-317894

Generalized lower bound conjecture for complexes with bounded missing-face dimension

摘录 / EXCERPT: Let $d ot obreak\geq 0$ and $0<i\leq d+1$ be integers. Let $\mathcal{C}(i,d)$ be the family of d-dimensional simplicial complexes with nonzero reduced d-homology and no missing...

conjecture | open | MathDB 页面 | 原文线索

V2-030135 | MATHDB-317901

Harary-Kauffman conjecture for alternating knot diagrams

摘录 / EXCERPT: Let ${\mathcal D}$ be an alternating knot diagram with no nugatory crossings. The determinant of ${\mathcal D}$ is the order of its first homology group of the double branched cove...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-030136 | MATHDB-317906

Thurston's virtual first Betti number conjecture for hyperbolic lattices

摘录 / EXCERPT: Let Γ be any lattice in $\mathop{\mathrm{SO}}\nolimits(1,n)$. Thurston's conjecture. There are a finite-index subgroup Γ' of Γ and a surjective homomorphism .....

conjecture | open | MathDB 页面 | 原文线索

V2-030137 | MATHDB-317914

Bordism invariance conjecture for the instanton line bundles

摘录 / EXCERPT: Bordism invariance conjecture. The bordism $\hat{W}$ between $\hat{\Sigma}_l$ and $\hat{\Sigma}_l'$ gives a natural isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-030138 | MATHDB-317915

Variant of the Atiyah-Floer conjecture for symplectic instanton homology

摘录 / EXCERPT: Let Y be an integral homology sphere. Let $HSI(Y)$ denote symplectic instanton homology, and let $\widetilde{HF}(Y)$ denote Donaldson-Floer homology, both viewed with their relat...

conjecture | open | MathDB 页面 | 原文线索

V2-030139 | MATHDB-317917

Generalized manifolds resolution conjecture

摘录 / EXCERPT: Let X be a generalized n-manifold with $n\geq 3$. Generalized manifolds resolution conjecture. The space X has a resolution. The conjecture is false in higher dimensions: non...

conjecture | open | MathDB 页面 | 原文线索

V2-030140 | MATHDB-317918

Alternate Bing-Borsuk conjecture via approximate fibrations

摘录 / EXCERPT: Let X be nicely embedded in $\mathbb R^{m+n}$ for some $m\geq 3$, and suppose that X has a mapping cylinder neighborhood $N=C_\phi$ of a map $\phi:\partial N\to X$, with mappin...

conjecture | open | MathDB 页面 | 原文线索

V2-030141 | MATHDB-317919

Bryant-Ferry-Mio-Weinberger homogeneity conjecture

摘录 / EXCERPT: Let X be a generalized n-manifold with $n\geq 7$ satisfying the disjoint disks property. Bryant-Ferry-Mio-Weinberger conjecture. The space X is homogeneous. If true, this wou...

conjecture | open | MathDB 页面 | 原文线索

V2-030142 | MATHDB-317920

Busemann G-spaces resolution conjecture

摘录 / EXCERPT: Let X be an $(n\geq 5)$-dimensional Busemann G-space. Busemann G-spaces resolution conjecture. The space X has a resolution. The conjecture is related to the unresolved hig...

conjecture | open | MathDB 页面 | 原文线索

V2-030143 | MATHDB-317921

de Groot's absolute cone conjecture

摘录 / EXCERPT: Let X be an n-dimensional absolute cone, meaning a compact finite-dimensional metric space that is a cone with respect to every point. de Groot's absolute cone conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-030144 | MATHDB-317922

Moore's codimension-one manifold factor conjecture

摘录 / EXCERPT: Let X be a resolvable generalized manifold. A codimension-one manifold factor is a space such that $X\times\mathbb R$ is a topological manifold. Moore conjecture. Every resolvabl...

conjecture | open | MathDB 页面 | 原文线索

V2-030145 | MATHDB-317939

The P2' property conjecture for the constructed Chern-Simons propagator

摘录 / EXCERPT: Let M be the manifold and let $\textsf{P1},\textsf{P2},\textsf{P2'},\textsf{P3},\textsf{P4},\textsf{P5}$ denote the propagator properties defined in the paper. Suppose that an od...

conjecture | open | MathDB 页面 | 原文线索

V2-030146 | MATHDB-317941

Non-reduced SNC divisor spectral sequence degeneration conjecture

摘录 / EXCERPT: Let X be a Kähler manifold and let $D=sum D_i$ be a reduced simple normal crossings divisor such that every intersection $D_{i_0}\cap\cdots\cap D_{i_p}$ is irreducible. Let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-030147 | MATHDB-317942

Cohomological vanishing conjecture for dual complexes of rational singularities

摘录 / EXCERPT: Let $f:X\to Y$ be a good resolution of an isolated rational singularity $o\in Y$, and let the exceptional divisor be ... with effective simple normal crossings support. Let Δ...

conjecture | open | MathDB 页面 | 原文线索

V2-030148 | MATHDB-317943

Rational contractibility conjecture for dual complexes of rational hypersurface singularities

摘录 / EXCERPT: Let D be the exceptional divisor of a resolution of an isolated rational hypersurface singularity of dimension at least 3, and let $\Delta=\Delta(D)$ be its dual complex. Ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-030149 | MATHDB-317950

Green-Lazarsfeld's refinement for general Prym-canonical bundles

摘录 / EXCERPT: Let C be a general curve of genus $g=2d-2\geq6$, let $\eta\in\operatorname{Pic}^0(C)$ be a general line bundle, and let $K_{p,q}(C,L)$ denote Koszul cohomology. Green-Lazarsfeld'...

conjecture | open | MathDB 页面 | 原文线索

V2-030150 | MATHDB-317952

The Minimal Syzygy Conjecture for general curves

摘录 / EXCERPT: Minimal Syzygy Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030151 | MATHDB-317954

The Hurewicz-set translate conjecture in ${\mathbb R}^{\mathbb N}$

摘录 / EXCERPT: Hurewicz-set translate conjecture. If $H\subset {\mathbb R}^{\mathbb N}$ is a Hurewicz set, then for each countable dimensional set $C\subset {\mathbb R}^{\mathbb N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-030152 | MATHDB-317958

Vafa-Witten conjecture for the L^2 cohomology of Nakajima's metric

摘录 / EXCERPT: Vafa-Witten conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-030153 | MATHDB-317988

Coordinate-product conjecture for transversely embedded linked-twist cylinders

摘录 / EXCERPT: Let M be a smooth two-dimensional manifold, let C_i be cylinders, and let $E_i:C_i\to A\subset M$ be embeddings for $i=1,2$. Assume that the embedded cones $E_1(C_1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030154 | MATHDB-318062

The homotopy-equivalence conjecture for normalized algebraic-map spaces

摘录 / EXCERPT: Let $A_d^{\mathbb{C}}(m,n)$ be the space of normalized $(n+1)$-tuples of homogeneous complex polynomials of degree d defining algebraic maps from $\mathbb{R}\operatorname{P}^{m}$...

conjecture | open | MathDB 页面 | 原文线索

V2-030155 | MATHDB-318082

Jp-speciality conjecture for alternating knots and links

摘录 / EXCERPT: Jp-speciality conjecture. All alternating knots or links, except those belonging to the family given by Conway symbol n-the torus links $(2,n)$, $n\ge2$-are not Jp-special.

conjecture | open | MathDB 页面 | 原文线索

V2-030156 | MATHDB-318083

Jp-speciality conjecture for adequate non-alternating links

摘录 / EXCERPT: Jp-speciality conjecture. Every adequate non-alternating link is Jp-special.

conjecture | open | MathDB 页面 | 原文线索

V2-030157 | MATHDB-318084

Jp-speciality conjecture for minimal positive braids

摘录 / EXCERPT: Jp-speciality conjecture. All minimal positive braids are Jp-special.

conjecture | open | MathDB 页面 | 原文线索

V2-030158 | MATHDB-318085

Odd-homology thickness conjecture for reduced Montesinos links

摘录 / EXCERPT: Montesinos thickness conjecture. A Montesinos link of the form $p_1,\ldots,p_m,-q$ is $\overline{Kh}'$-thick if it has a minimal-crossing Conway symbol satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-030159 | MATHDB-318086

Odd-homology thickness conjecture for mixed rational-tangle algebraic links

摘录 / EXCERPT: Mixed-tangle thickness conjecture. The link is $\overline{Kh}'$-thick if $m\ge3$, $q>1$, each p_i of length at least 2 satisfies $\overline p_i=1$, and

conjecture | open | MathDB 页面 | 原文线索

V2-030160 | MATHDB-318087

Odd-homology thickness conjecture for algebraic links with a positive tail

摘录 / EXCERPT: Positive-tail thickness conjecture. The link is $\overline{Kh}'$-thick if $q>1$ and either

conjecture | open | MathDB 页面 | 原文线索

V2-030161 | MATHDB-318089

Topological minimality exclusion conjecture for the 3-sphere and 3-ball

摘录 / EXCERPT: Let M be either the 3-sphere S^3 or the 3-ball B^3, and let a topologically minimal surface mean a surface whose disk complex is empty or non-contractible. Topological minima...

conjecture | open | MathDB 页面 | 原文线索

V2-030162 | MATHDB-318090

Topological index three conjecture for common Heegaard stabilizations

摘录 / EXCERPT: Let M be a 3-manifold containing unstabilized Heegaard surfaces F and G, neither of which has topological index 1. Suppose that the minimal-genus common stabilization of...

conjecture | open | MathDB 页面 | 原文线索

V2-030163 | MATHDB-318091

Product-region classification conjecture for topologically minimal surfaces

摘录 / EXCERPT: Let F be a surface of positive genus, and let I denote the closed interval. A surface is topologically minimal when its disk complex is empty or non-contractible. Product-regio...

conjecture | open | MathDB 页面 | 原文线索

V2-030164 | MATHDB-318092

Geometric minimality conjecture for topological-index surfaces

摘录 / EXCERPT: Let H be a surface with topological index n. A surface is geometrically minimal if it is minimal with respect to the geometric notion used in the paper. Geometric minimality co...

conjecture | open | MathDB 页面 | 原文线索

V2-030165 | MATHDB-318095

Generic Newton polygon conjecture for T-adic C-functions

摘录 / EXCERPT: Let $C_f(s,T)$ be the C-function of the T-adic exponential sums associated with a polynomial f, and let its absolute T-adic Newton polygon mean the T^a-adic Newton polygo...

conjecture | open | MathDB 页面 | 原文线索

V2-030166 | MATHDB-318096

Independence-of-m conjecture for generic Newton polygons of C-functions

摘录 / EXCERPT: Let $C_f(s,\pi_m;\mathbb{F}_{p^a})$ be the specialization of the C-function at a root π_m, and call its absolute π_m^a-adic Newton polygon the absolute Newton polygon of...

conjecture | open | MathDB 页面 | 原文线索

V2-030167 | MATHDB-318108

The compactness conjecture for type spaces of finitely forcible graphons

摘录 / EXCERPT: Let ${\cal W}_0$ be the space of graphons under consideration, and let $T(W)$ denote the type space associated with a graphon W. Compactness conjecture. If W is finitely forcib...

conjecture | open | MathDB 页面 | 原文线索

V2-030168 | MATHDB-318129

Geometric realization conjecture for the symplectic Verlinde algebra completion

摘录 / EXCERPT: Geometric realization conjecture. There is an increasing function $N:\mathbb{N}\to\mathbb{N}$ and a collection of isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-030169 | MATHDB-318131

Equality of Matveev and modified Heegaard complexity

摘录 / EXCERPT: Let M be a compact connected orientable 3-manifold. The Matveev complexity $c(M)$ and the modified Heegaard complexity $\widetilde{c}(M)$ are defined by minimizing the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-030170 | MATHDB-318151

Schmid's cohomological conjecture for non-regular p-groups

摘录 / EXCERPT: Schmid's cohomological conjecture. For every integer n, one has

conjecture | open | MathDB 页面 | 原文线索

V2-030171 | MATHDB-318152

Determinantal formula for the Baker-Akhiezer kernel

摘录 / EXCERPT: Let x be the spectral-curve coordinate and let $\widetilde{K}(z_1,z_2)$ be the formal Baker-Akhiezer kernel defined by ... where the primed sum excludes all terms with...

conjecture | open | MathDB 页面 | 原文线索

V2-030172 | MATHDB-318162

The genus-one open-book fillability conjecture

摘录 / EXCERPT: Let (M,ξ) be a closed contact three-manifold supported by a genus one open book $(S_{1,r},\phi)$, where ϕ is pseudo-Anosov and the associated foliations have two singulari...

conjecture | open | MathDB 页面 | 原文线索

V2-030173 | MATHDB-318173

The Seifert-state-cycle nontriviality conjecture

摘录 / EXCERPT: Let α be a state cycle associated to the Seifert state of a knot, and suppose its quantum grading is $s-1$, where s denotes Rasmussen's invariant. The Seifert-state-cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-030174 | MATHDB-318179

Higher-genus deformation-space non-local-connectivity conjecture

摘录 / EXCERPT: Let S be a compact surface with boundary. Write $AH(S\times[0,1],\partial S\times[0,1])$ for the deformation space of marked hyperbolic structures on the pared 3-manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-030175 | MATHDB-318184

Strong Novikov conjecture for the analytic assembly map

摘录 / EXCERPT: Strong Novikov conjecture. The analytic assembly map α is a monomorphism. This is the injectivity statement connecting KO-homology of the classifying space to the K-theo...

conjecture | open | MathDB 页面 | 原文线索

V2-030176 | MATHDB-318217

Degree-one symmetric homology conjecture for truncated polynomial algebras

摘录 / EXCERPT: Let k be a commutative ground ring and let n be a nonnegative integer. Consider the truncated polynomial algebra $k[t]/(t^n)$ and its degree-one symmetric homology...

conjecture | open | MathDB 页面 | 原文线索

V2-030177 | MATHDB-318218

Layered degree-one symmetric homology conjecture for the polynomial algebra

摘录 / EXCERPT: Let k be a commutative ring, let $k[t]$ be the polynomial algebra, and for $m\geq 0$ let $HS_1(k[t])_{t^m}$ denote its t^m-layered symmetric homology, obtained from the decompo...

conjecture | open | MathDB 页面 | 原文线索

V2-030178 | MATHDB-318224

Alternating-knot unknotting-crossing conjecture

摘录 / EXCERPT: Let $K\subset S^3$ be an alternating knot, and let an alternating diagram be a diagram in which over- and under-crossings alternate along each component. An unknotting crossing is...

conjecture | open | MathDB 页面 | 原文线索

V2-030179 | MATHDB-318232

Thurston's virtual fibration conjecture for finite-volume hyperbolic 3-manifolds

摘录 / EXCERPT: Thurston's virtual fibration conjecture. Every finite-volume hyperbolic 3-manifold has a finite cover that is fibered.

conjecture | open | MathDB 页面 | 原文线索

V2-030180 | MATHDB-318237

The conjecture that finite Poincaré complexes are manifolds up to homotopy

摘录 / EXCERPT: A finite Poincaré complex is a finite complex satisfying Poincaré duality. Aspherical Poincaré complexes conjecture. Every finite Poincaré complex is homotopy equivalent to a close...

conjecture | open | MathDB 页面 | 原文线索

V2-030181 | MATHDB-318238

The characterization of Poincaré duality groups by manifold groups

摘录 / EXCERPT: A finitely presented group G is an n-dimensional Poincaré duality group when its group cohomology satisfies Poincaré duality in dimension n. Poincaré duality groups conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-030182 | MATHDB-318254

Ščepin's local triviality conjecture for Serre fibrations with low-dimensional manifold fibers

摘录 / EXCERPT: Let $p\colon E\to B$ be a Serre fibration of metric spaces, where B is locally arcwise connected, and suppose that every fiber of p is homeomorphic to a given manifold M^n of...

conjecture | open | MathDB 页面 | 原文线索

V2-030183 | MATHDB-318255

Shchepin's local triviality conjecture for Serre fibrations

摘录 / EXCERPT: Let $p\colon E\to B$ be a Serre fibration, where B is a locally arcwise connected metric space, and suppose that every fiber of p is homeomorphic to a fixed manifold M^n of d...

conjecture | open | MathDB 页面 | 原文线索

V2-030184 | MATHDB-318256

Homotopic Hopf-Galois extension Quillen equivalence conjecture

摘录 / EXCERPT: Let H be a bimonoid in a monoidal model category $\mathbf M$. Suppose that the category $\mathbf{Alg}_{H}$ of H-comodule algebras admits a model structure for which the coinvar...

conjecture | open | MathDB 页面 | 原文线索

V2-030185 | MATHDB-318259

Primality conjecture for Chebyshev knots

摘录 / EXCERPT: Let T_n denote the degree-n Chebyshev polynomial of the first kind, and let (i,j,k) be a reduced triple as in the source. Primality conjecture. Every knot of the form ... for t...

conjecture | open | MathDB 页面 | 原文线索

V2-030186 | MATHDB-318261

Crossing-number conjecture for consecutive Chebyshev degrees

摘录 / EXCERPT: Let T_n denote the degree-n Chebyshev polynomial of the first kind. For an odd integer $i\ge 3$, the consecutive-degree crossing-number conjecture. The parametrization ... parame...

conjecture | open | MathDB 页面 | 原文线索

V2-030187 | MATHDB-318262

Crossing-number conjecture for Chebyshev degrees of common difference two

摘录 / EXCERPT: Let T_n denote the degree-n Chebyshev polynomial of the first kind. For an odd integer $i\ge 3$, the difference-two crossing-number conjecture. The parametrization ... parametriz...

conjecture | open | MathDB 页面 | 原文线索

V2-030188 | MATHDB-318280

Bott ellipticity conjecture for positively curved manifolds

摘录 / EXCERPT: Let M^n be a simply connected positively curved manifold, and let ΛM denote its loop space. The Betti numbers of ΛM are the homology ranks with coefficients in...

conjecture | open | MathDB 页面 | 原文线索

V2-030189 | MATHDB-318288

Nonvanishing Rabinowitz Floer homology above Mañé's critical value

摘录 / EXCERPT: Let M be a closed manifold with Tonelli Hamiltonian H, magnetic form, Mañé critical value c, and energy level $\Sigma_k=H^{-1}(k)$. For $k>c$, consider the Rabinowitz Floer h...

conjecture | open | MathDB 页面 | 原文线索

V2-030190 | MATHDB-318301

Hirasawa-Murasugi metabelian twisted Alexander polynomial conjecture

摘录 / EXCERPT: Let K be a knot with knot group $G(K)$, and suppose there is a homomorphism $f:G(K)\twoheadrightarrow M(n\mid p,k)$. Let $N=\langle s\rangle=\{1,s,s^2,\ldots,s^{n-1}\}$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030191 | MATHDB-318303

Approximation of completed homology classes by sums of geodetic circles

摘录 / EXCERPT: Geodetic approximation conjecture. For every $c\in\widehat{H'_1}$, there is a Cauchy sequence $(\chi_i)_{i\in\mathbb N}\in c$ such that every χ_i is the homology class of a fi...

conjecture | open | MathDB 页面 | 原文线索

V2-030192 | MATHDB-318304

Geodetic circles generate the homology of locally finite graphs

摘录 / EXCERPT: Geodetic-generation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030193 | MATHDB-318327

Existence of the algebra of higher cohomology operations

摘录 / EXCERPT: Let $R=\mathbb{Z}/p^2$, and let $\mathcal{M}^N$ be the category of finitely generated free graded R-modules concentrated in degree less than N. For a bigraded differential alge...

conjecture | open | MathDB 页面 | 原文线索

V2-030194 | MATHDB-318328

Simon-type conjecture for link-group quotients

摘录 / EXCERPT: Let $L\subset S^3$ be a non-trivial link of $n\geq 2$ components. A homology boundary link is a link whose fundamental group surjects onto a non-abelian free group of rank equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-030195 | MATHDB-318329

Simon-type conjecture for knot exteriors in rational homology spheres

摘录 / EXCERPT: Let X be a knot exterior in a closed orientable 3-manifold such that $H_1(X;\mathbf{Q})\cong\mathbf{Q}$. Simon-type knot-exterior conjecture. Then $\pi_1(X)$ surjects onto only...

conjecture | open | MathDB 页面 | 原文线索

V2-030196 | MATHDB-318360

Abdenur-Bonatti-Crovisier transitivity conjecture for homoclinic classes

摘录 / EXCERPT: Let M be a compact connected boundaryless manifold of dimension d, and let $\operatorname{Diff}^{1}(M)$ be the space of diffeomorphisms of M with the C^1 topology. A homocl...

conjecture | open | MathDB 页面 | 原文线索

V2-030197 | MATHDB-318361

Asymptotic expansion conjecture for TQFT curve operators

摘录 / EXCERPT: Let M be the moduli space, let ${\mathcal T}$ be the parameter space, and let P be a formal trivialization of the formal Hitchin connection, with expansion...

conjecture | open | MathDB 页面 | 原文线索

V2-030198 | MATHDB-318365

Talelli's conjecture on torsion-free groups with periodic cohomology

摘录 / EXCERPT: Let Γ be a group. Its cohomological dimension is ... The group Γ has periodic cohomology after k steps if there exists an integer $q>0$ such that $H^i(\Gamma,-)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030199 | MATHDB-318386

Equivalent locally finite form of Kuhn's unbounded strong realization conjecture

摘录 / EXCERPT: Locally finite form of the unbounded strong realization conjecture. If R_sM is of finite weight for each s and M is topologically realizable, then M is locally fin...

conjecture | open | MathDB 页面 | 原文线索

V2-030200 | MATHDB-318389

The two-generator conjecture for the mod 2 cohomology of BE_8

摘录 / EXCERPT: Let E_8 be the exceptional Lie group, and let BE_8 be its classifying space. Write ... for the adjoint representation, and let $w_{128}(\rho_8)$ denote its 128-th Stiefel-Whitney...

conjecture | open | MathDB 页面 | 原文线索

V2-030201 | MATHDB-318391

Nguyễn Hưng's eventual isomorphism conjecture for the squaring operation on Steenrod-algebra Ext

摘录 / EXCERPT: Let $\mathcal{A}$ be the Steenrod algebra over $\mathbb{F}_2$, and let ... be the operation induced by the Frobenius map on the dual Steenrod algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030202 | MATHDB-318417

The universal extension first-limit isomorphism for abelian groups

摘录 / EXCERPT: Universal extension limit conjecture. The above map is a natural isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-030203 | MATHDB-318430

The two-orientation conjecture for strongly quasipositive links

摘录 / EXCERPT: Let L be a non-splittable unoriented link in S^3. An orientation of L is strongly quasipositive if the oriented link bounds a strongly quasipositive surface. Two-orientation...

conjecture | open | MathDB 页面 | 原文线索

V2-030204 | MATHDB-318431

The category conjecture for unordered Euclidean configuration spaces

摘录 / EXCERPT: Category conjecture. For all n and k,

conjecture | open | MathDB 页面 | 原文线索

V2-030205 | MATHDB-318444

Goussarov-Habiro conjecture for finite type invariants of string links

摘录 / EXCERPT: Let string links be proper tangles without closed components, and let two string links have the same number of components. They are Goussarov-Habiro conjecture. Two string links of...

conjecture | open | MathDB 页面 | 原文线索

V2-030206 | MATHDB-318455

Satellite Lorenz knot conjecture

摘录 / EXCERPT: Let K be a Lorenz knot that is a satellite knot. Then the satellite Lorenz knot conjecture. K is a satellite of a torus knot lying on a Lorenz knot, according to the cabling pa...

conjecture | open | MathDB 页面 | 原文线索

V2-030207 | MATHDB-318469

Dense-copy conjecture for nonabelian limit groups

摘录 / EXCERPT: Let G be a locally compact group, and let F_2 denote the free group of rank two. A group has a dense copy in G if it contains a subgroup isomorphic to that group whose image...

conjecture | open | MathDB 页面 | 原文线索

V2-030208 | MATHDB-318496

Equality of left and right weak Gorenstein global dimensions

摘录 / EXCERPT: Equality conjecture. For any ring R,

conjecture | open | MathDB 页面 | 原文线索

V2-030209 | MATHDB-318500

Flip-and-slip uniqueness conjecture for simplified broken Lefschetz fibrations

摘录 / EXCERPT: Let M be a smooth 4-manifold and let $f_1,f_2:M\rightarrow S^2$ be homotopic simplified broken Lefschetz fibrations. A flip-and-slip modification is a modification of a simplif...

conjecture | open | MathDB 页面 | 原文线索

V2-030210 | MATHDB-318503

The one-negative-crossing conjecture for link signatures

摘录 / EXCERPT: Let L be a nontrivial link admitting a diagram with at most one negative crossing.
One-negative-crossing conjecture. The link L has negative signature, that is, $\sigma(L)<0$....

conjecture | open | MathDB 页面 | 原文线索

V2-030211 | MATHDB-318507

The cubical complex conjecture for homology of asynchronous transition systems

摘录 / EXCERPT: Cubical homology conjecture. If $M(E,I)$ is locally finite-dimensional, then for all integers $n\geqslant 0$, the groups $\underrightarrow{\lim}_n^{K(\Sigma)}F$ are isomorphic to t...

conjecture | open | MathDB 页面 | 原文线索

V2-030212 | MATHDB-318512

Moore's conjecture on projectivity over finite-index subgroups

摘录 / EXCERPT: Moore's conjecture. If M is projective over $\mathbb{Z}H$, then it is projective over $\mathbb{Z}\Gamma$. In particular, for torsion-free Γ, projectivity over...

conjecture | open | MathDB 页面 | 原文线索

V2-030213 | MATHDB-318513

Homotopy commutativity of the homotopy shuffle product

摘录 / EXCERPT: Homotopy shuffle commutativity conjecture. The homotopy shuffle product is homotopy commutative for every such $A^\bullet$. Consequently, the induced product

conjecture | open | MathDB 页面 | 原文线索

V2-030214 | MATHDB-318514

Habiro's integrality conjecture for boundary bottom tangles

摘录 / EXCERPT: Set $v=\exp\frac{h}{2}$ and $q=v^2$, and let J_T denote the universal sl_2 invariant of a bottom tangle. For an n-component bottom tangle, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030215 | MATHDB-318515

Conjecture on universal invariants of bottom tangles concordant to boundary tangles

摘录 / EXCERPT: Set $v=\exp\frac{h}{2}$ and $q=v^2$, and let J_T denote the universal sl_2 invariant of a bottom tangle. Let $(\bar U_q^{ev})\;\hat {}^{\;\hat \otimes n}$ be the subalgebra int...

conjecture | open | MathDB 页面 | 原文线索

V2-030216 | MATHDB-318526

Ilmanen's outermost-horizon topology conjecture

摘录 / EXCERPT: Let (M^n,g,k) and $\bar\Sigma$ be as in Ilmanen's conjecture: the data are complete and asymptotically flat, $3\le n\le 7$, and $\bar\Sigma$ is the unique outermost generalized t...

conjecture | open | MathDB 页面 | 原文线索

V2-030217 | MATHDB-318539

The higher-degree comparison-map conjecture for semisimple Lie groups

摘录 / EXCERPT: Higher-degree comparison-map conjecture. The comparison map

conjecture | open | MathDB 页面 | 原文线索

V2-030218 | MATHDB-318564

The integral second cohomology conjecture for $(n-2,1^2)$ Specht modules

摘录 / EXCERPT: Let $n\geq 4$, let Σ_n be the symmetric group, and let $S^{(n-2,1^2)}_{\mathbb Z}$ be the integral Specht module indexed by the partition $(n-2,1^2)$. The integral cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-030219 | MATHDB-318566

The integral second cohomology conjecture for $(n-3,2,1)$ Specht modules

摘录 / EXCERPT: Let $n\geq 5$, let Σ_n be the symmetric group, and let $S^{(n-3,2,1)}_{\mathbb Z}$ be the integral Specht module indexed by $(n-3,2,1)$. The integral cohomology conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030220 | MATHDB-318571

Seiberg-Witten-Furuta-Ohta invariant conjecture for homology $S^1\times S^3$ four-manifolds

摘录 / EXCERPT: Let X be a smooth 4-manifold with the $\mathbb Z[\mathbb Z]$ homology of $S^1\times S^3$. Denote by $\lambda_{\,\operatorname{SW}\,}(X)$ the Seiberg-Witten invariant and by...

conjecture | open | MathDB 页面 | 原文线索

V2-030221 | MATHDB-318572

The waist bound for almost alternating knots

摘录 / EXCERPT: Let K be an m-almost alternating knot, meaning that it has a diagram which becomes alternating after changing some m crossings but no such diagram after changing fewer crossi...

conjecture | open | MathDB 页面 | 原文线索

V2-030222 | MATHDB-318573

The waist bound by braid index

摘录 / EXCERPT: Let K be a knot, and let $braid(K)$ denote its braid index. The braid-index waist conjecture. ... The inequality is presented as an expected bound based on work cited by the source...

conjecture | open | MathDB 页面 | 原文线索

V2-030223 | MATHDB-318593

The Montgomery-Yang problem for pseudofree circle actions on the 5-sphere

摘录 / EXCERPT: Montgomery-Yang problem. Every pseudofree $\mathbb{S}^1$-action on $\mathbb{S}^5$ has at most 3 non-free orbits.

open_problem | open | MathDB 页面 | 原文线索

V2-030224 | MATHDB-318641

The equivariant generating hypothesis for finite G-spectra

摘录 / EXCERPT: Equivariant generating hypothesis. The restriction of $\underline{\pi}_*^G(-)$ to the subcategory of finite G-spectra is faithful. Equivalently, if a map $f:X\to Y$ between finit...

conjecture | open | MathDB 页面 | 原文线索

V2-030225 | MATHDB-318679

The Kirby-calculus reformulation of the smooth four-dimensional Poincaré conjecture

摘录 / EXCERPT: Let $L=L_1\cup L_2$ be a link in S^3, with L_1 a dotted p-component unlink and L_2 a framed link of $p+q$ components. Suppose that L_2 normally generates $\pi_1(S^3-L_1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-030226 | MATHDB-318680

Stable diffeomorphism implies diffeomorphism for homology 4-spheres

摘录 / EXCERPT: Let $L=L_1\cup L_2$ satisfy the hypotheses of the preceding Kirby-calculus formulation, except that L_2 is required only to generate $H_1(S^3-L_1;\mathbb Z)$ rather than...

conjecture | open | MathDB 页面 | 原文线索

V2-030227 | MATHDB-318681

The no odd-index handles conjecture for simply connected 4-manifolds

摘录 / EXCERPT: Let $L=L_1\cup L_2$ satisfy the hypotheses of the preceding Kirby-calculus formulation, but impose no restriction on the number of components of L_2. If $p+q\ne0$, then L can b...

conjecture | open | MathDB 页面 | 原文线索

V2-030228 | MATHDB-318682

Fukaya-string topology equivalence conjecture for cotangent bundles

摘录 / EXCERPT: Let M be the manifold under consideration, and let $Fuk_{conor}(T^*M)$ be the full subcategory of the Fukaya category generated by conormal bundles of closed, connected submanifo...

conjecture | open | MathDB 页面 | 原文线索

V2-030229 | MATHDB-318710

The open solid handlebody conjecture for complete affine 3-manifolds

摘录 / EXCERPT: Let M^3 be a 3-dimensional complete affine manifold with free fundamental group. The open solid handlebody conjecture. Then M is homeomorphic to an open solid handlebody. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-030230 | MATHDB-318711

The crooked fundamental domain conjecture for nonsolvable affine actions

摘录 / EXCERPT: Let ${\boldsymbol E}^3_1$ be a 3-dimensional Lorentzian affine space, and let $amma\subset\operatorname{Aff}({\boldsymbol E}^3_1)$ be a discrete group acting properly on...

conjecture | open | MathDB 页面 | 原文线索

V2-030231 | MATHDB-318716

Conjecture on limiting deficits of cables

摘录 / EXCERPT: Let ${\cal K}$ be a knot or link, and let ${\cal K}_{p,q}$ denote its (p,q)-cable. Let $D_{{\cal K}_{p,q}}$ be the deficit of this cable from the relevant braid-theoretic bound.\...

conjecture | open | MathDB 页面 | 原文线索

V2-030232 | MATHDB-318717

The Signature Growth conjecture for positive links

摘录 / EXCERPT: For a link L, let $n'(L)$ denote its number of components and let $\chi(L)$ be its Euler characteristic. Define ... Let $\sg(L)$ denote the signature of L. Signature Growth conje...

conjecture | open | MathDB 页面 | 原文线索

V2-030233 | MATHDB-318744

The Gerstenhaber algebra conjecture for singular chains and free loop homology

摘录 / EXCERPT: Let G be a topological group such that M is a classifying space of G, and let $S_*(G)$ denote the normalized singular-chain algebra of G. Write $LM=map(S^1,M)$ for the free...

conjecture | open | MathDB 页面 | 原文线索

V2-030234 | MATHDB-318745

The Gerstenhaber algebra conjecture for singular cochains and free loop homology

摘录 / EXCERPT: Let M be simply connected and let ${\Bbbk}$ be a field. Let $LM=map(S^1,M)$ be the free loop space, let d be the dimension parameter of M, and let $S^*(M)$ denote the algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-030235 | MATHDB-318752

Conjectured bounds for orders of fundamental groups in hyperbolic lens space surgeries

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3 that yields the lens space $L(p,q)$ by Dehn surgery, and let g be the Seifert genus of K. Conjectured bounds for hyperbolic lens space sur...

conjecture | open | MathDB 页面 | 原文线索

V2-030236 | MATHDB-318804

The commutative 2-cocycle conjecture for classical Lie algebras in characteristic 3

摘录 / EXCERPT: Commutative 2-cocycle conjecture. The space of commutative 2-cocycles on L coincides with the space of symmetric bilinear invariant forms and is therefore 1-dimensional, assu...

conjecture | open | MathDB 页面 | 原文线索

V2-030237 | MATHDB-318829

The duality conjecture for Lagrangian-fibration mirror symmetry

摘录 / EXCERPT: Let $f:X\to B$ be a Lagrangian fibration of class $\mathcal C$, let σ be a Lagrangian section, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030238 | MATHDB-318830

The twist conjecture for Lagrangian-fibration mirror symmetry

摘录 / EXCERPT: Let $f:X\to B$ be a Lagrangian fibration with section σ, let $m_{f,\sigma}:\mathcal D^b\operatorname{Fuk}(X)\simeq\mathcal D^b\operatorname{Coh}(Y_f)$ be the mirror functor,...

conjecture | open | MathDB 页面 | 原文线索

V2-030239 | MATHDB-318831

Invariance of delta-groupoid homology under the 2-3 Pachner move

摘录 / EXCERPT: Pachner-move invariance conjecture. The corresponding homology groups are isomorphic:

conjecture | open | MathDB 页面 | 原文线索

V2-030240 | MATHDB-318843

Hatcher's stable rational homology conjecture for automorphism groups of free groups

摘录 / EXCERPT: Let F_n be the free group of rank n, let $\operatorname{Aut}(F_\infty)=\varinjlim_{n\to\infty}\operatorname{Aut}(F_n)$, and let $B\operatorname{Aut}(F_\infty)$ be its classifyi...

conjecture | open | MathDB 页面 | 原文线索

V2-030241 | MATHDB-318845

Evenness conjecture for the trivializing number of virtual shadows

摘录 / EXCERPT: Evenness conjecture. For every virtual shadow S, $\text{tr}(S)$ is even.

conjecture | open | MathDB 页面 | 原文线索

V2-030242 | MATHDB-318917

Finiteness conjecture for irreducible 4-manifolds of fixed complexity

摘录 / EXCERPT: Let c be a natural number, and let a complexity-c irreducible 4-manifold mean an irreducible 4-manifold having complexity c. Finiteness conjecture. For every natural number...

conjecture | open | MathDB 页面 | 原文线索

V2-030243 | MATHDB-318936

Conjectural spectral sequence behavior for torus knots T(3,6n ± 1)

摘录 / EXCERPT: Let $T(3,6n+1)$ and $T(3,6n-1)$ be the indicated families of torus knots, and let E^k denote the pages of the link surgery spectral sequence associated to a choice of analytic da...

conjecture | open | MathDB 页面 | 原文线索

V2-030244 | MATHDB-318937

Conjectural higher Khovanov, Jones, and δ-polynomials for T(3,6n ± 1)

摘录 / EXCERPT: Let $\mathcal{S}^n=T(3,6n+1)$ and $\mathcal{T}^n=T(3,6n-1)$ for $n greater than or equal to 1$. Define the auxiliary polynomials $f_n(t,q)=sum_{k=0}^{n-1}t^8q^{12}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030245 | MATHDB-318948

The toral rank conjecture

摘录 / EXCERPT: Toral rank conjecture. If T^r acts almost freely on X, then

conjecture | open | MathDB 页面 | 原文线索

V2-030246 | MATHDB-318952

Alternating knot representativity conjecture

摘录 / EXCERPT: Let K be an alternating knot, and let $r(K)$ denote its representativity. Alternating knot representativity conjecture. It holds that ... The source gives no further discussion of...

conjecture | open | MathDB 页面 | 原文线索

V2-030247 | MATHDB-318956

The UTP conjecture for fibered knots

摘录 / EXCERPT: Let K be a fibered knot in S^3. The UTP conjecture. K fails the uniform thickness property if and only if $xi_K \cong xi_{std}$, and hence if and only if K is fibered stron...

conjecture | open | MathDB 页面 | 原文线索

V2-030248 | MATHDB-318965

Kalai's strong Lefschetz conjecture for homology manifolds

摘录 / EXCERPT: Let Δ be a $(d-1)$-dimensional orientable homology manifold over k, let $k[\Delta]$ be its face ring, and let $\Theta\subseteq k[\Delta]$ be a linear system of parameters....

conjecture | open | MathDB 页面 | 原文线索

V2-030249 | MATHDB-318968

Boundary determination conjecture for mixed cup products

摘录 / EXCERPT: Let M be a compact Riemannian manifold with boundary, let $(\partial M,\Lambda)$ denote its boundary data, and let $[\alpha]$ and $[\beta]$ be absolute and relative cohomology cl...

conjecture | open | MathDB 页面 | 原文线索

V2-030250 | MATHDB-319007

Mañé's weak conjecture on generic Mather sets

摘录 / EXCERPT: Let M be a manifold and let L be a Tonelli Lagrangian on TM. For a cohomology class $c\in H^1(M,\mathbb{R})$, write $\mathcal{M}(L,c)$ for the Mather set associated with L...

conjecture | open | MathDB 页面 | 原文线索

V2-030251 | MATHDB-319009

Mañé's weak Aubry-set conjecture

摘录 / EXCERPT: Let M be a manifold and let L be a Tonelli Lagrangian on TM. For a cohomology class $c\in H^1(M,\mathbb{R})$, let the Aubry set of $(L+f,c)$ denote the Aubry set associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-030252 | MATHDB-319013

Conjecture on the suspension-kernel classes generated by $[\eta_i]_6$

摘录 / EXCERPT: Suspension-kernel conjecture. The class $[\eta_i]_6$ dies under σ_*. Consequently, the subalgebra of $H_*Q_0S^{-2^i+6}$ generated by the classes $Q^I[\eta_i]_6$ belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-030253 | MATHDB-319019

Bonatti's robust-cycle denseness conjecture

摘录 / EXCERPT: Let M be a smooth manifold, and let C^1-diffeomorphisms of M be considered with the C^1 topology. A diffeomorphism has a C^1-robust cycle if it exhibits either a homoclin...

conjecture | open | MathDB 页面 | 原文线索

V2-030254 | MATHDB-319021

Low-dimensional relative K-theory computation for the coordinate axes

摘录 / EXCERPT: Let $A=\mathbb{Z}[x,y]/(xy)$ and let $I=(x,y)$. The low-dimensional relative K-theory conjecture. ... and ... These computations reduce to calculations of equivariant topological cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-030255 | MATHDB-319022

The conjectural computation of $\operatorname{TR}^3_{9-\lambda_4}(\mathbb{Z};2)$

摘录 / EXCERPT: Let $\operatorname{TR}^3_{9-\lambda_4}(\mathbb{Z};2)$ be the indicated equivariant topological Hochschild homology group, where λ_4 is the relevant $RO(S^1)$-representation...

conjecture | open | MathDB 页面 | 原文线索

V2-030256 | MATHDB-319023

The conjectural computation of $\operatorname{TR}^3_{13-\lambda_4}(\mathbb{Z};2)$

摘录 / EXCERPT: Let $\operatorname{TR}^3_{13-\lambda_4}(\mathbb{Z};2)$ be the indicated equivariant topological Hochschild homology group, where λ_4 is the relevant $RO(S^1)$-representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-030257 | MATHDB-319027

Homotopy equivalence for connected branched polymers under arrangement inequalities

摘录 / EXCERPT: Let ${\mathcal A}=\{H_1,\ldots,H_N\}$ be the hyperplane arrangement with $H_i=\{\mathbf{x}\in\mathbb C^r\mid h_i(\mathbf{x})=0\}$. Define $T_{\mathcal A}\subset\mathbb R_{>0}^N$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-030258 | MATHDB-319058

Realization of the cofibrations as cofibrations of spaces

摘录 / EXCERPT: The preceding constructions involve spectra $\mathbf L(i)\wedge \mathbf T(2^{n-i}-1)$, together with the spectra appearing in the complex of Theorem; the final cofibration has sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-030259 | MATHDB-319077

The braid Floer-Morse-Conley homology correspondence conjecture

摘录 / EXCERPT: Braid Floer-Morse-Conley homology correspondence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030260 | MATHDB-319087

Infinite generation conjecture for homology of power subgroups of mapping class groups

摘录 / EXCERPT: Let Σ_g be a closed surface of genus g, let M_g be its mapping class group, and let $M_g[D]$ be the subgroup generated by the D-th powers of Dehn twists about all simp...

conjecture | open | MathDB 页面 | 原文线索

V2-030261 | MATHDB-319106

Equivariant Chern class formula for arc spaces of the general linear group

摘录 / EXCERPT: Let $G=GL_n$ and let $V_\bullet$ be the universal flag representation used to define the contact loci $\operatorname{Cont}^{\geq\lambda}(V_\bullet)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030262 | MATHDB-319119

Non-Stein-fillability conjecture for the contact structures on $-\Sigma(2,3,6n-1)$

摘录 / EXCERPT: Non-Stein-fillability conjecture. The contact structures $\eta_{i,j}^n$ are not Stein fillable if $i>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-030263 | MATHDB-319122

Solenoidal Hodge conjecture

摘录 / EXCERPT: Let M be a compact Kähler manifold of complex dimension n. A complex immersed solenoid in M is a solenoid whose leaves are mapped to complex immersed submanifolds; such a sol...

conjecture | open | MathDB 页面 | 原文线索

V2-030264 | MATHDB-319141

The BKMP topological recursion conjecture for the framed mirror curve of $\mathbb{C}^3$

摘录 / EXCERPT: Fix a framing parameter f and let the framed mirror curve C be given by ... Let $y(x)$ be the formal power-series inverse specified in the source, let W_g be the diffe...

conjecture | open | MathDB 页面 | 原文线索

V2-030265 | MATHDB-319144

Existence of nondestabilizable Legendrian knots with vanishing contact homology

摘录 / EXCERPT: A nondestabilizable Legendrian knot is a Legendrian knot that cannot be obtained from another Legendrian knot by stabilization. Its Legendrian contact homology is the homology asso...

conjecture | open | MathDB 页面 | 原文线索

V2-030266 | MATHDB-319154

Chatterji-Mislin's centralizer formula for complete Euler characteristics

摘录 / EXCERPT: Let G be a group of type FP over $\mathbb C$, and suppose that the centralizer of every finite-order element of G has finite L^2-Betti numbers. Let $E(G)(s)$ denote the s-c...

conjecture | open | MathDB 页面 | 原文线索

V2-030267 | MATHDB-319166

Mom-4 conjecture

摘录 / EXCERPT: A hyperbolic Mom-4 manifold is a hyperbolic 3-manifold equipped with a Mom-4 structure. Mom-4 conjecture. The collection of hyperbolic Mom-4 manifolds is exactly the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-030268 | MATHDB-319179

Self-dual Bott-index obstruction conjecture for almost commuting triples

摘录 / EXCERPT: Let H_1,H_2,H_3 be self-dual Hermitian matrices, and let $[H_i,H_j]=H_iH_j-H_jH_i$ denote their commutators. Let I be the identity matrix, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030269 | MATHDB-319191

Colin-Ghiggini-Honda-Hutchings conjecture on sutured Heegaard Floer and embedded contact homology

摘录 / EXCERPT: Let (M,Γ,ξ) be a contact 3-manifold with convex or sutured boundary and dividing set Γ. Let $\mathfrak s_{\xi}$ denote the relative Spin^c-structure determined...

conjecture | open | MathDB 页面 | 原文线索

V2-030270 | MATHDB-319192

Invariance conjecture for sutured embedded contact homology

摘录 / EXCERPT: Let $(M,\Gamma,U(\Gamma),\xi)$ be a sutured contact 3-manifold with adapted contact form α, let $(M^{\ast},\alpha^{\ast})$ be its completion, and let J be an almost comp...

conjecture | open | MathDB 页面 | 原文线索

V2-030271 | MATHDB-319194

Kontsevich-Soibelman conjecture on A_{∞}-pre-categories

摘录 / EXCERPT: Kontsevich-Soibelman conjecture. Quasi-equivalence classes of A_{∞}-pre-categories over k are in bijection with quasi-equivalence classes of A_{∞}-categories over...

conjecture | open | MathDB 页面 | 原文线索

V2-030272 | MATHDB-319201

Generic arithmetic polygon conjecture for T-adic exponential sums

摘录 / EXCERPT: Let $k<d$ be fixed positive integers, and let ... Let $p_{d,[0,k]}$ denote the arithmetic polygon of $\{d\}\cup[0,k]$, and let $C_f(s,T)$ be the associated T-adic characteristic fu...

conjecture | open | MathDB 页面 | 原文线索

V2-030273 | MATHDB-319202

Uniform limiting polygon conjecture for T^{p-1}-adic Newton polygons

摘录 / EXCERPT: Let $k<d$ be fixed positive integers, and let ... Let $C_f(s,T)$ be the associated characteristic function and $H_{[0,d]}^{\infty}$ the limiting Hodge polygon appearing in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-030274 | MATHDB-319203

Conjecture on jump cohomology and finite cohomological dimension for groups without R-torsion

摘录 / EXCERPT: Let G) be a group without R-torsion, where R is a commutative ring with a unit, and let k be a nonnegative integer. Jump cohomology conjecture. G has jump cohomology of h...

conjecture | open | MathDB 页面 | 原文线索

V2-030275 | MATHDB-319220

The crosscut lower-bound conjecture for diagramless link homology

摘录 / EXCERPT: Let k be the number of crosscuts, let N be the number of copies of Khovanov homology in the diagramless homology of a link with k crosscuts, and let the minimum crossing numb...

conjecture | open | MathDB 页面 | 原文线索

V2-030276 | MATHDB-319221

The diagram-count bounds conjecture for diagramless link homology

摘录 / EXCERPT: Let L be a non-split link and let k be a fixed positive integer. Let $n_k(L)$ be the number of distinct k-crossing link diagrams for L up to 2-space isotopy. Let N be the...

conjecture | open | MathDB 页面 | 原文线索

V2-030277 | MATHDB-319239

Mazur's conjecture on the topology of positive-rank fibers

摘录 / EXCERPT: Mazur's conjecture. One of the following two conditions holds:

conjecture | open | MathDB 页面 | 原文线索

V2-030278 | MATHDB-319268

Bloch-Kato conjecture on Galois symbols

摘录 / EXCERPT: Bloch-Kato conjecture. The map $h_{k,n}^i$ is bijective. This conjecture identifies Milnor K-theory modulo n with the corresponding Galois cohomology group; its resolution is a...

conjecture | open | MathDB 页面 | 原文线索

V2-030279 | MATHDB-319271

The Cosmetic Surgery Conjecture for nontrivial knots

摘录 / EXCERPT: Let K be a nontrivial knot in a closed oriented three-manifold Y. Two surgeries along K with slopes r and r' are nonequivalent if no orientation-preserving homeomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-030280 | MATHDB-319289

Vanishing higher homology conjecture for symmetric-group Hurwitz spaces

摘录 / EXCERPT: Let G be a symmetric group on more than two letters, and let c be the conjugacy class of transpositions. For each integer $i\geq 2$, consider the rational homology of the conne...

conjecture | open | MathDB 页面 | 原文线索

V2-030281 | MATHDB-319296

Lee-Pandharipande-Xiao-Xu universal vanishing equations for Gromov-Witten invariants

摘录 / EXCERPT: Let X be a smooth projective target, let $x_i,y_i\in H^*(X)$, and let p_i,q_i be descendant indices; use dual cohomology bases $\{\gamma_a\}$ and $\{\gamma^a\}$. Let g,r,s be...

conjecture | open | MathDB 页面 | 原文线索

V2-030282 | MATHDB-319297

Lee-Pandharipande-Xiao-Xu two-point vanishing and Hodge identity

摘录 / EXCERPT: Let X be the target, let T_a and T^a be dual cohomology bases, let g be a genus, and let B_{2g} be the Bernoulli number. Two-point universal-equations conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-030283 | MATHDB-319325

Rivin's bounded multiplicity conjecture for simple length spectra

摘录 / EXCERPT: Rivin's conjecture. The multiplicity in the simple length spectrum is bounded above by a constant depending only on the topology of the surface.

conjecture | open | MathDB 页面 | 原文线索

V2-030284 | MATHDB-319340

Five-ball construction conjecture for h-vectors

摘录 / EXCERPT: A five-ball is a triangulated ball of dimension five, with h-vector written as $h=(1,h_1,h_2,h_3,h_4,h_5,0)$. Theorem's construction gives a collection of h-vectors described b...

conjecture | open | MathDB 页面 | 原文线索

V2-030285 | MATHDB-319368

Reverse Harnack conjecture for non-null-homotopic harmonic mappings

摘录 / EXCERPT: Let $\Omega\subset\mathbb C$ be a doubly connected domain, and let $\mathbb C_\circ=\mathbb C\setminus\{0\}$. Suppose $h\colon\Omega\to\mathbb C_\circ$ is harmonic and is not homot...

conjecture | open | MathDB 页面 | 原文线索

V2-030286 | MATHDB-319400

The decomposition conjecture for the twisted K-theory group of a regular automorphism

摘录 / EXCERPT: Let A be a smooth $*$-algebra, let σ be a regular automorphism of A, and let F be its fixed point algebra. Let ${\mathcal S}$ be the semigroup of modular homotopy clas...

conjecture | open | MathDB 页面 | 原文线索

V2-030287 | MATHDB-319431

The non-manifold conjecture for hyperbolizable 3-manifold deformation spaces

摘录 / EXCERPT: Non-manifold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030288 | MATHDB-319432

The finite-components conjecture for deformation spaces

摘录 / EXCERPT: Finite-components conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030289 | MATHDB-319440

Absolute continuity and exponent rigidity for large measures

摘录 / EXCERPT: Let M be an infranilmanifold, let α_0 be a totally non-symplectic $\mathbb{Z}^k$ action by automorphisms of M, and let α be an action with homotopy data...

conjecture | open | MathDB 页面 | 原文线索

V2-030290 | MATHDB-319446

Benson's strong regularity conjecture for group cohomology

摘录 / EXCERPT: Benson's strong regularity conjecture. The cohomology ring $H^*(G,k)$ always contains a very strongly quasi-regular system of parameters.

conjecture | open | MathDB 页面 | 原文线索

V2-030291 | MATHDB-319452

The all-positive-degrees conjecture for local cohomology after generic linear reduction

摘录 / EXCERPT: All-positive-degrees conjecture. The same formula holds for every integer $i>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-030292 | MATHDB-319462

The primitive-character L-function formula for the Artin-Schreier hypersurface

摘录 / EXCERPT: Let F be a nonarchimedean local field with residue field of cardinality q, let h be a positive integer, let X be the Artin-Schreier hypersurface over ${\bf F}_{q^h}$ descri...

conjecture | open | MathDB 页面 | 原文线索

V2-030293 | MATHDB-319477

Weakly unital strict groupoids model truncated spaces

摘录 / EXCERPT: Let n be a nonnegative integer. Consider the category of weakly unital strict n-groupoids, its equivalences, and the category of n-truncated spaces with weak equivalences. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-030294 | MATHDB-319478

Weakly unital strict categories model weak higher categories

摘录 / EXCERPT: Let n be a nonnegative integer. Consider weakly unital strict n-categories, the equivalences between them, and the models of weak n-categories due to Tamsamani, Baez-Dolan, a...

conjecture | open | MathDB 页面 | 原文线索

V2-030295 | MATHDB-319491

The twisted Morava theory differential conjecture for $W_7+[H_7]$

摘录 / EXCERPT: Twisted Morava theory differential conjecture. The cohomology class $W_7+[H_7]$ corresponds to a differential in twisted Morava K-theory and twisted Morava E-theory, where...

conjecture | open | MathDB 页面 | 原文线索

V2-030296 | MATHDB-319499

Höhn-Stolz conjecture for the Witten genus

摘录 / EXCERPT: Let M be a closed string manifold. The Höhn-Stolz conjecture. If M admits a Riemannian metric of positive Ricci curvature, then the Witten genus $W(M)$ vanishes. The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030297 | MATHDB-319524

Minimality conjecture for 2-bridge knots with four boundary slopes

摘录 / EXCERPT: Let K be a 2-bridge knot, and suppose that its boundary-slope set has exactly four distinct elements. A 2-bridge knot K_1 is minimal with respect to the Silver-Whitten partial...

conjecture | open | MathDB 页面 | 原文线索

V2-030298 | MATHDB-319530

Stability of identities following from associativity under ultrafilter algebras

摘录 / EXCERPT: Let an identity be a formula in the language of a groupoid, and consider the algebra of ultrafilters associated with a compact left topological groupoid. Ultrafilter-stability conj...

conjecture | open | MathDB 页面 | 原文线索

V2-030299 | MATHDB-319536

Trivial Lie bracket conjecture for Ext of braided bialgebras

摘录 / EXCERPT: Trivial Lie bracket conjecture. The Lie algebra of this Gerstenhaber algebra is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-030300 | MATHDB-319537

Kontsevich's 3-algebra conjecture for Gerstenhaber-Schack cohomology

摘录 / EXCERPT: Kontsevich's conjecture. The graded algebra $H_{GS}(H,H)$ is a 3-algebra, that is, an algebra over the homology of the little 3-cubes operad $\mathcal{C}_3$.

conjecture | open | MathDB 页面 | 原文线索

V2-030301 | MATHDB-319538

3-algebra conjecture for Ext of braided bialgebras

摘录 / EXCERPT: 3-algebra conjecture for braided bialgebras. The graded algebra $\operatorname{Ext}_A^*(\mathbb{F},\mathbb{F})$ is a 3-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-030302 | MATHDB-319539

Cohen's Gerstenhaber structure conjecture for double loop spaces

摘录 / EXCERPT: Cohen's Gerstenhaber structure conjecture. There is an isomorphism of Gerstenhaber algebras

conjecture | open | MathDB 页面 | 原文线索

V2-030303 | MATHDB-319548

Quasi-isomorphism conjecture for link and braid diagram complexes

摘录 / EXCERPT: Quasi-isomorphism conjecture. The cochain maps from Theorem 1 are quasi-isomorphisms for $n>3$. This conjecture asserts that the diagram complexes calculate the cohomology of space...

conjecture | open | MathDB 页面 | 原文线索

V2-030304 | MATHDB-319549

Cokernel polynomial conjecture for quasi-alternating links

摘录 / EXCERPT: Let $\kappa=\kappa_\beta$ be the closure of a braid $\beta\in Br_m$ with writhe w, and let $P_{\mathrm{coker}}(t)$ denote the associated cokernel polynomial. Cokernel polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-030305 | MATHDB-319560

The fractional linear generation conjecture for higher Chow groups

摘录 / EXCERPT: Let K be a field satisfying the rank conjecture of Suslin, and let $CH^p(K,p+q)$ denote its cubical higher Chow groups, where $q\geq 0$. A fractional linear generation conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030306 | MATHDB-319568

The summability converse for spectral triples on the Sierpiński gasket

摘录 / EXCERPT: Let SG denote the Sierpiński gasket and let (A,H,D) be a spectral triple for $C(SG)$ inducing the geodesic distance on the gasket. Suppose that it is summable for every...

conjecture | open | MathDB 页面 | 原文线索

V2-030307 | MATHDB-319572

Galois representations for Hecke eigenclasses in arithmetic-group homology

摘录 / EXCERPT: The paper considers the usual group cohomology, Tate-Farrell cohomology, Steinberg homology, and sharbly homology of arithmetic subgroups, all equipped with Hecke actions. Galois-r...

conjecture | open | MathDB 页面 | 原文线索

V2-030308 | MATHDB-319585

Makeev's four-point conjecture on the sphere

摘录 / EXCERPT: Let S^2 be a unit sphere in $\mathbb R^3$. Suppose we are given 4 points $x_1, \ldots, x_4\in S^2$ lying on a single circle, and a continuous function $f:S^2\to\mathbb R$. Make...

conjecture | refuted | MathDB 页面 | 原文线索

V2-030309 | MATHDB-319604

Ozsváth-Szabó's triple-cup-product conjecture for the Heegaard Floer spectral sequence

摘录 / EXCERPT: Ozsváth-Szabó's conjecture. The differential d_3 is given by

conjecture | open | MathDB 页面 | 原文线索

V2-030310 | MATHDB-319627

Dunfield-Gukov-Rasmussen conjecture on triply graded knot homology

摘录 / EXCERPT: Dunfield-Gukov-Rasmussen conjecture. There exists a homology theory with the stated Euler-characteristic, differential, and symmetry properties such that for all $N>1$ the homology...

conjecture | open | MathDB 页面 | 原文线索

V2-030311 | MATHDB-319628

Marked-Dyck-path formula at $q=1$ for torus-knot homology

摘录 / EXCERPT: Let $\mathcal{P}_{s}^{2k}(T_{n,m})(q,t)$ denote the relevant coefficient of the triply graded Poincaré polynomial, and let the sum range over all marked Dyck paths in the...

conjecture | open | MathDB 页面 | 原文线索

V2-030312 | MATHDB-319629

Algebra-action generation conjecture for stable torus-knot homology

摘录 / EXCERPT: Let $\mathcal{A}_{n,m}$ be algebras represented on the stable homology space $\mathcal{H}(T_{n,m})$, and let $e_1,\ldots,e_n$ denote the even polynomial generators and...

conjecture | open | MathDB 页面 | 原文线索

V2-030313 | MATHDB-319633

The global-to-local injectivity conjecture for H²

摘录 / EXCERPT: Let K be a number field, let X/K be smooth and projective, let S be a finite set of places as in the source, and let m,s be integers. Global-to-local injectivity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030314 | MATHDB-319653

Existence of exact Lagrangian embeddings underlying cotangent contact inclusions

摘录 / EXCERPT: Let L and N be simply connected manifolds, and let ST^*L denote the unit cotangent sphere bundle of L, viewed as a contact manifold embedded in T^*N. An exact Lagrangian...

conjecture | open | MathDB 页面 | 原文线索

V2-030315 | MATHDB-319671

Shioya-Yamaguchi's Alexandrov sphere conjecture

摘录 / EXCERPT: Let Y^3_∞ be a 3-dimensional compact, simply connected, non-negatively curved Alexandrov space without boundary, and suppose that Y^3_∞ is a topological manifold. S...

conjecture | open | MathDB 页面 | 原文线索

V2-030316 | MATHDB-319675

The stable lower central series conjecture for the Torelli group

摘录 / EXCERPT: Stable lower central series conjecture. The lower central series of the Torelli group stably coincides with the restriction of the Y-filtration:

conjecture | open | MathDB 页面 | 原文线索

V2-030317 | MATHDB-319678

Gluing conjecture for Fukaya-category bimodules of sutured manifolds

摘录 / EXCERPT: Let F,F',F'' be connected Riemann surfaces and let Z be a finite subset of the identified boundaries. Let Y_1,Y_2 be sutured manifolds with ... and let $Y=Y_1\cup_{F'}Y_2$ be ob...

conjecture | open | MathDB 页面 | 原文线索

V2-030318 | MATHDB-319681

Noncommutative Hodge-to-de Rham comparison conjecture

摘录 / EXCERPT: Let k be a ring containing ${\mathbb Q}$, and let $A^{\:\raisebox{3pt}{\text{\circle*{1.5}}}}$ be a smooth and proper DG algebra over k. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030319 | MATHDB-319697

The generalized Freyd conjecture for a triangulated category

摘录 / EXCERPT: Let $\mathscr{T}$ be a triangulated category, let $\mathscr{B}$ be a small full subcategory closed under the shift functor Σ, and let $\mathscr{C}$ be the thick full subcate...

conjecture | open | MathDB 页面 | 原文线索

V2-030320 | MATHDB-319717

Meta-isomorphism conjecture for assembly maps

摘录 / EXCERPT: Meta-isomorphism conjecture. The assembly map $A_{\mathcal{F}}$ is an isomorphism for every $n\in\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-030321 | MATHDB-319718

McL conjecture on algebraic characterisation of planar Peano continua

摘录 / EXCERPT: Let X be a compact, locally connected, metrizable space that is locally embeddable in S^2. A simple set of circles in X is a set S of circles with the simplicity property u...

conjecture | open | MathDB 页面 | 原文线索

V2-030322 | MATHDB-319719

Metric realisation conjecture for first Čech homology

摘录 / EXCERPT: Let X be a Peano continuum, that is, a compact, connected, locally connected metrizable space. For a compatible metric d on X, let $\hat{H}_d$ denote the associated metric-de...

conjecture | open | MathDB 页面 | 原文线索

V2-030323 | MATHDB-319740

Lax monoidal product conjecture for derived endomorphism ring spectra

摘录 / EXCERPT: Let a category have a lax or partial lax monoidal product whose structure maps are weak equivalences, and let its unit have a derived endomorphism ring spectrum. Lax monoidal produ...

conjecture | open | MathDB 页面 | 原文线索

V2-030324 | MATHDB-319745

Berglund-Hübsch-Kontsevich homological mirror symmetry conjecture for invertible polynomials

摘录 / EXCERPT: Let W be an invertible polynomial in n variables, let $R=\mathbb{C}[x_1,\dots,x_n]/(W)$, and let $W^\ast$ be its transpose polynomial. Write $D^b_{\mathrm{sing}}(R)$ for the bo...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-030325 | MATHDB-319746

Kapustin-Katzarkov-Orlov-Yotov Fano-general-type Fukaya embedding conjecture

摘录 / EXCERPT: Let F and G be projective manifolds. Suppose F is a Fano manifold and G is a manifold of general type or a Calabi-Yau manifold. Suppose there is a fully faithful functor ……

conjecture | open | MathDB 页面 | 原文线索

V2-030326 | MATHDB-319747

Kapustin-Katzarkov-Orlov-Yotov A-to-B Landau-Ginzburg mirror conjecture

摘录 / EXCERPT: Let $(\underline{X},\omega)$ be the symplectic manifold underlying a projective manifold X of general type, with $c_1(X)=[-\omega]$, and let ${\sf w}:Y\to\mathbb{C}$ be a holomor...

conjecture | open | MathDB 页面 | 原文线索

V2-030327 | MATHDB-319753

MacPherson's conjecture on the birational invariance of L^2 arithmetic genus

摘录 / EXCERPT: MacPherson's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030328 | MATHDB-319766

Modularity conjecture for odd Galois representations valued in dual groups

摘录 / EXCERPT: Modularity conjecture. There is a Hecke eigenclass in $H^*(\Gamma_0(N),M_{\mathbf{x}})$ that matches ρ.

conjecture | open | MathDB 页面 | 原文线索

V2-030329 | MATHDB-319784

Conjectured upper bound for the mosaic number of a knot

摘录 / EXCERPT: Mosaic-number upper-bound conjecture. The mosaic number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-030330 | MATHDB-319786

Rozansky-Witten conjecture on monoidal deformations of coherent-sheaf categories

摘录 / EXCERPT: Let Y be the complex manifold under consideration. Write T^*_fY for its formal holomorphic cotangent neighborhood, and let β be an inhomogeneous $(0,q)$-form with odd q...

conjecture | open | MathDB 页面 | 原文线索

V2-030331 | MATHDB-319790

Non-finite generation conjecture for the cohomology of Johnson filtration subgroups

摘录 / EXCERPT: Let F_n be the free group of rank n, let Γ^{s+1} be the $(s+1)$st term of its lower central series, and let J_n^s denote the kernel of the homomorphism ... For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-030332 | MATHDB-319793

M. Reid's conjecture on components of the moduli space of numerical Godeaux surfaces

摘录 / EXCERPT: Let $\mathfrak{M}_{(1,1)}^{can}$ denote the moduli space of canonical surfaces with invariants $K^2=1$ and $\chi=1$, and let $\pi_1(S)$ be the topological fundamental group of a nu...

conjecture | open | MathDB 页面 | 原文线索

V2-030333 | MATHDB-319794

Finiteness conjecture for fundamental groups of numerical Campedelli surfaces

摘录 / EXCERPT: Let S be a numerical Campedelli surface, meaning a minimal surface of general type with $p_g(S)=q(S)=0$ and $K_S^2=2$, and let $\pi_1(S)$ denote its topological fundamental group...

conjecture | open | MathDB 页面 | 原文线索

V2-030334 | MATHDB-319801

Topological invariance conjecture for sutured embedded contact homology

摘录 / EXCERPT: Sutured ECH invariance conjecture. The sutured embedded contact homologies satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-030335 | MATHDB-319802

Sutured ECH-sutured Floer homology conjecture

摘录 / EXCERPT: Sutured ECH-SFH conjecture. There should be an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-030336 | MATHDB-319828

Cohomological conjecture for Higgs invariants of Hitchin-pair moduli spaces

摘录 / EXCERPT: Cohomological conjecture. The refined Higgs invariants satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-030337 | MATHDB-319837

Conjecture on the abelianized characteristic algebra as a knot invariant

摘录 / EXCERPT: Let K be a knot and let $\mathcal{K}$ be any Legendrian representative of K with maximal Thurston-Bennequin number. Let $\mathcal{C}'(Ch(\mathcal{K}))$ denote the ungraded abel...

conjecture | open | MathDB 页面 | 原文线索

V2-030338 | MATHDB-319840

Gabrielov's regular triangulation conjecture for definable functions

摘录 / EXCERPT: Gabrielov's regular triangulation conjecture. There exists a definable triangulation of K such that, for each $n\leq\dim K$ and every open n-simplex Δ of the triangulati...

conjecture | open | MathDB 页面 | 原文线索

V2-030339 | MATHDB-319856

Generic strange duality for moduli spaces on elliptic surfaces

摘录 / EXCERPT: Let X be a simply connected elliptic surface with a section and at worst nodal fibers, and let v and w be orthogonal topological types of ranks $r,s\geq 2$ satisfying ... and .....

conjecture | open | MathDB 页面 | 原文线索

V2-030340 | MATHDB-319857

The fp-injective module conjecture

摘录 / EXCERPT: Let A be a ring, and let $\operatorname{Fpinj} A$ be the category of fp-injective A-modules. Write $\mathbf K(\operatorname{Inj} A)$ for the homotopy category of complexes of i...

conjecture | open | MathDB 页面 | 原文线索

V2-030341 | MATHDB-319867

Getzler's surjectivity conjecture for the tautological ring

摘录 / EXCERPT: Getzler's surjectivity conjecture. The cycle map is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-030342 | MATHDB-319891

The weighted-homogeneity conjecture for rational surface singularities with rational homology disk smoothings

摘录 / EXCERPT: Weighted-homogeneity conjecture. Suppose a rational surface singularity has a $\mathbb Q$HD smoothing. Then the singularity is weighted homogeneous, hence is one of the known examp...

conjecture | open | MathDB 页面 | 原文线索

V2-030343 | MATHDB-319894

The volume conjecture for four-dimensional Liouville domains

摘录 / EXCERPT: Volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030344 | MATHDB-319895

The strict upper bound for the distinguished ECH spectrum

摘录 / EXCERPT: Strict ECH volume bound.

conjecture | open | MathDB 页面 | 原文线索

V2-030345 | MATHDB-319896

The general ECH volume conjecture for graded classes

摘录 / EXCERPT: General ECH volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030346 | MATHDB-319914

Carlson-Toledo conjecture in terms of the Hurewicz morphism

摘录 / EXCERPT: Let Γ be an infinite Kähler group. Carlson-Toledo conjecture. There exists a compact Kähler manifold M with $\pi_1(M)$ a finite-index subgroup of Γ such that the Hu...

conjecture | open | MathDB 页面 | 原文线索

V2-030347 | MATHDB-319922

The collapsing conjecture for twisting spectral sequences

摘录 / EXCERPT: Let F and G be strict polynomial functors, and for $r\geq 0$ let $E(F,G,r)$ denote the twisting spectral sequence associated with them, whose abutment computes...

conjecture | open | MathDB 页面 | 原文线索

V2-030348 | MATHDB-319950

Homotopy CW-complex conjecture for ascending and descending links

摘录 / EXCERPT: In the situation of Theorem ... k^-:{cal K}^-V(p)to{cal L}^-V(p),qquad k^+:{cal K}^+V(p)to{cal L}^+V(p), ... h^-V(p):K^-V(p)to L^-V(p),qquad h^+V(p):K^+V(p)to L^+V(p). ... , analogous to...

conjecture | open | MathDB 页面 | 原文线索

V2-030349 | MATHDB-319951

Goresky-MacPherson cellular deformation-retraction conjecture

摘录 / EXCERPT: Goresky-MacPherson cellular deformation-retraction conjecture. There exists a ∇f-like vector field V on X such that, with

conjecture | open | MathDB 页面 | 原文线索

V2-030350 | MATHDB-319968

Multiple Yang-Mills solutions in every topological component

摘录 / EXCERPT: Multiple-solutions conjecture. For a rather general family of boundary values, multiple solutions to $(\mathcal{D}_{\epsilon})$ should exist in each component $\mathcal{A}_k(A_0)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-030351 | MATHDB-320006

The inverse Galois conjecture for semi-topological coverings

摘录 / EXCERPT: Semi-topological inverse Galois conjecture. There exists a Weierstrass polynomial f on X such that the splitting covering $E_f \stackrel{q}{\rightarrow}X$ is equivalent to the...

conjecture | open | MathDB 页面 | 原文线索

V2-030352 | MATHDB-320072

The periodic-representation detection conjecture for nonfibered knots

摘录 / EXCERPT: Periodic-representation detection conjecture. The knot k is nonfibered if and only if there exist an integer $N>0$ and a periodic representation $\rho\in\Phi_{S_N}({\cal G})$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-030353 | MATHDB-320083

Grid-move conjecture for minimal grid diagrams of $(p,2)$ torus knots

摘录 / EXCERPT: Let $T_{p,2}$ be a torus knot, let K_L be a Legendrian knot of topological type $T_{p,2}$, and let $\alpha(T_{p,2})$ denote the grid number of $T_{p,2}$. Consider a grid diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-030354 | MATHDB-320084

Cube-number conjecture distinguishing extreme Legendrian representatives of left-handed torus knots

摘录 / EXCERPT: Let K_{max} and K_{min} be left-handed Legendrian torus knots with $p\geq 5$ and maximal Thurston-Bennequin number, such that the rotation number of K_{max} is maximal and th...

conjecture | open | MathDB 页面 | 原文线索

V2-030355 | MATHDB-320121

Periodicity conjecture for torsion-free crystallographic groups

摘录 / EXCERPT: Let $\Gamma<\operatorname{Isom}(\mathbb{R}^k)$ be a torsion-free crystallographic group. Its quotient $\mathbb{R}^k/\Gamma$ is a closed aspherical manifold, and...

conjecture | open | MathDB 页面 | 原文线索

V2-030356 | MATHDB-320124

The null-by-finite conjecture for small nm-stable compact G-rings

摘录 / EXCERPT: Let (R,G) be a small, nm-stable compact G-ring. Null-by-finite conjecture. R is null-by-finite. The conjecture is presented as an intermediate conjecture concerning the str...

conjecture | open | MathDB 页面 | 原文线索

V2-030357 | MATHDB-320142

Noncommuting one-parameter subgroups conjecture for continuous interval exchange actions

摘录 / EXCERPT: Let $\mathcal{E}$ be the topological group of continuous interval exchange actions, and let $F=\{f_t\}$ and $G=\{g_s\}$ be one-parameter subgroups of $\mathcal{E}$. Noncommuting on...

conjecture | open | MathDB 页面 | 原文线索

V2-030358 | MATHDB-320151

The Orthogonal LMOV conjecture for orthogonal Chern-Simons invariants

摘录 / EXCERPT: Let $\mathcal{L}$ be a link with L components, and let $Z_{CS}^{SO}(\mathcal{L},q,t)$ be its orthogonal Chern-Simons partition function. Write ... and define ... where $\mu(k)$ is th...

conjecture | open | MathDB 页面 | 原文线索

V2-030359 | MATHDB-320152

The valuation bound for orthogonal reformulated invariants

摘录 / EXCERPT: Let $\mathcal{L}$ be a link with L components, let $F_{\overrightarrow{\mu}}^{SO}$ be the coefficient of the free energy associated with a nonzero tuple of partitions...

conjecture | open | MathDB 页面 | 原文线索

V2-030360 | MATHDB-320212

Simon's conjecture on covers of compact irreducible 3-manifolds

摘录 / EXCERPT: Let M be a compact irreducible 3-manifold, and let N be a cover of M with finitely generated fundamental group. Simon's conjecture. The interior of N is topologically tam...

conjecture | open | MathDB 页面 | 原文线索

V2-030361 | MATHDB-320224

Existence conjecture for energy-minimal diffeomorphisms of multiply connected domains

摘录 / EXCERPT: Let Ω and Ω^* be bounded k-connected domains in $\mathbb{C}$, where $k\geqslant 2$. Suppose that $\Omega\subset\Omega^*$ and that the inclusion is a homotopy equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-030362 | MATHDB-320230

Conjecture on rigid classes and cyclic commensurability of hyperbolic knots

摘录 / EXCERPT: A commensurability class of hyperbolic 3-manifolds is called rigid when its unique minimal element has a single cusp whose horospherical cross-section is a Euclidean turnover. Tw...

conjecture | open | MathDB 页面 | 原文线索

V2-030363 | MATHDB-320233

Positivity conjecture for completed 2-dimensional manifold pairings

摘录 / EXCERPT: Positivity conjecture. For every compact 2-dimensional surface S, this quadratic function has no kernel: if

conjecture | open | MathDB 页面 | 原文线索

V2-030364 | MATHDB-320245

Uniform representation stability for cohomology of pure string motion groups

摘录 / EXCERPT: Let $P\Sigma_n$ be the pure string motion group and let W_n be the hyperoctahedral group acting on it by permuting labels and reversing orientations. For each fixed $i\geq 0$, co...

conjecture | open | MathDB 页面 | 原文线索

V2-030365 | MATHDB-320247

Stable finite-dimensionality of Torelli and IA_n homology

摘录 / EXCERPT: Let $\mathcal{I}_{n,1}$ be the Torelli group of a genus-n surface with one boundary component and let IA_n be the IA-automorphism group of the free group F_n....

conjecture | open | MathDB 页面 | 原文线索

V2-030366 | MATHDB-320248

Finite generation of Torelli and IA_n homology modules

摘录 / EXCERPT: Let $\mathcal{I}_{n,1}$ be the Torelli group of a genus-n surface with one boundary component and let IA_n be the IA-automorphism group of the free group of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-030367 | MATHDB-320249

Periodic stability for homology of congruence subgroups

摘录 / EXCERPT: Let $\Gamma_n(p)$ and $\Gamma^{\operatorname{Sp}}_{2n}(p)$ denote the congruence subgroups considered in the source, with homology over $\mathbb{F}_p$ carrying the natural actions...

conjecture | open | MathDB 页面 | 原文线索

V2-030368 | MATHDB-320250

Periodic stability for homology of mapping-class-group congruence subgroups

摘录 / EXCERPT: Let $\operatorname{Mod}_{g,1}(p)$ be the level-p congruence subgroup of the mapping class group of a genus-g surface with one boundary component. Its integral homology carries...

conjecture | open | MathDB 页面 | 原文线索

V2-030369 | MATHDB-320286

Twisted ECH functoriality for capping and decoupling cobordisms

摘录 / EXCERPT: Twisted ECH cobordism conjecture. If $c(M_+,xi_+;Omega_+)$ vanishes, then $c(M_-,xi_-;Omega_-)$ also vanishes. Moreover, if $c(M_+,xi_+;Omega_+)$ lies in the image of U^k on...

conjecture | open | MathDB 页面 | 原文线索

V2-030370 | MATHDB-320369

Goodwillie-Whitehead conjecture: strong form

摘录 / EXCERPT: For each k, let ... be the map in Kuhn's Kahn-Priddy sequence, and let ... be the k-invariant map of the Goodwillie tower for S^1. Goodwillie-Whitehead conjecture, strong form. .....

conjecture | open | MathDB 页面 | 原文线索

V2-030371 | MATHDB-320381

Conjecture on complex Gaussian distribution of quantum invariants of random 3-manifolds

摘录 / EXCERPT: Complex Gaussian distribution conjecture. The invariant $Z(M)$ is distributed by a standard complex Gaussian.

conjecture | open | MathDB 页面 | 原文线索

V2-030372 | MATHDB-320394

Vanishing conjecture for special injective modules on the infinitesimal p-adic topos

摘录 / EXCERPT: Let $\calldaginf$ be a special injective left $\protect\calfragile D\sp{\dagger}_{X\sp{\dagger}\sb{\protect\mathop{\rm inf}\nolimits}/K}$-module. The Zariski sheaf...

conjecture | open | MathDB 页面 | 原文线索

V2-030373 | MATHDB-320396

Converse overtwistedness conjecture for the ECH invariant f

摘录 / EXCERPT: Let (Y,ξ) be a closed contact 3-manifold. Let (λ,J) range over ECH data for (Y,ξ), and let $f(Y,\lambda,J)$ be the invariant defined in the paper. Converse overtw...

conjecture | open | MathDB 页面 | 原文线索

V2-030374 | MATHDB-320397

Torsion-freeness conjecture for the cohomology of affine Weyl toric arrangement complements

摘录 / EXCERPT: Let $\widetilde{W}$ be an affine Weyl group, and let $\mathcal{T}_{\widetilde{W}}$ be the corresponding toric arrangement. Its complement is the space obtained by removing the arra...

conjecture | open | MathDB 页面 | 原文线索

V2-030375 | MATHDB-320411

The ribbon-category conjecture for the full category of non-restricted quantum-group modules

摘录 / EXCERPT: Let $\mathcal{D}^{H}$ be the category of modules under consideration, and let $\varpi\in\mathcal{U}^{H}$ be the central element defined by $\varpi=uS(u)^{-1}K_{2\rho}^{2(r-1)}$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-030376 | MATHDB-320429

Chas-Sullivan conjecture for the center of the Goldman Lie algebra

摘录 / EXCERPT: Let S be a connected oriented surface, let $\hat{\pi}=[S^1,S]$ be the set of free homotopy classes of oriented loops on S, and let $\mathbb{Q}\hat{\pi}$ be the Goldman Lie alge...

conjecture | open | MathDB 页面 | 原文线索

V2-030377 | MATHDB-320430

Dual Goodwillie calculus conjecture for simplicial model categories

摘录 / EXCERPT: Dual Goodwillie calculus conjecture. There should be a cooperad $\partial_*(\Sigma^\infty_\mathcal{C}\Omega^\infty_\mathcal{C})$ and an equivalence of operads

conjecture | open | MathDB 页面 | 原文线索

V2-030378 | MATHDB-320449

Nonexistence of PL isometric embeddings for some PL manifolds

摘录 / EXCERPT: Let $n\geq 3$ and let M^n be an n-dimensional PL-manifold. A PL isometric embedding is an embedding of M^n into Euclidean space that preserves its piecewise-linear metric....

conjecture | open | MathDB 页面 | 原文线索

V2-030379 | MATHDB-320484

Finite-time constant-topology conjecture for heterogeneous Hegselmann-Krause dynamics

摘录 / EXCERPT: Let $r=\{r_1,\dots,r_n\}\in\mathbb{R}^n_{>0}$ be the agents' confidence bounds and let $x(t)\in\mathbb{R}^n$ evolve according to the heterogeneous Hegselmann-Krause system ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-030380 | MATHDB-320506

Zupan's width conjecture for satellite knots

摘录 / EXCERPT: Let K be a satellite knot with companion J and pattern $\hat{K}$ having wrapping number n. The width of a knot, denoted by $w(\cdot)$, is the knot width in the sense of Gabai...

conjecture | open | MathDB 页面 | 原文线索

V2-030381 | MATHDB-320519

The superpotential conjecture for fundamental group algebras of acyclic manifolds

摘录 / EXCERPT: Let M be an acyclic oriented compact manifold of dimension d, and let $k[\pi_1(M)]$ denote its fundamental group algebra. Superpotential conjecture. If M is an acyclic orient...

conjecture | open | MathDB 页面 | 原文线索

V2-030382 | MATHDB-320520

The exact Calabi-Yau structure conjecture for acyclic manifold group algebras

摘录 / EXCERPT: Let M be a d-dimensional compact orientable acyclic manifold, and let $k[\pi_1(M)]$ be its fundamental group algebra. Write $\operatorname{HH}_d(k[\pi_1(M)])$ for its degree-...

conjecture | open | MathDB 页面 | 原文线索

V2-030383 | MATHDB-320521

The circle-bundle superpotential conjecture for acyclic manifold group algebras

摘录 / EXCERPT: Let M be an acyclic compact orientable manifold, let $\pi(M)$ denote its fundamental group, and let $k[\pi(M)]$ be the corresponding group algebra. Circle-bundle superpotential c...

conjecture | open | MathDB 页面 | 原文线索

V2-030384 | MATHDB-320532

The specialization conjecture for rigid and crystalline cohomology

摘录 / EXCERPT: Let $X\to S$ be proper, flat and generically smooth, with special fibre X_s over k and geometric generic fibre $X_{\bar\eta}$. Let $H^i_{\operatorname{rig}}(X_s/k)$ be rigid co...

conjecture | open | MathDB 页面 | 原文线索

V2-030385 | MATHDB-320533

The regular-model specialization conjecture for rigid and crystalline cohomology

摘录 / EXCERPT: Let $X\to S$ be proper, flat, generically smooth and regular. Let sp' be the specialization map from rigid cohomology to crystalline cohomology, and let W_{i-1}...

conjecture | open | MathDB 页面 | 原文线索

V2-030386 | MATHDB-320534

The weight-filtered p-adic invariant-cycles conjecture

摘录 / EXCERPT: Let X be regular, and let W_{i-1} denote the indicated weight filtration on the p-adic étale cohomology groups of the special and geometric generic fibres. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030387 | MATHDB-320535

The weight-zero p-adic invariant-cycles conjecture

摘录 / EXCERPT: Let X be regular, let W_0 be the sum of generalized ϕ-eigenspaces whose eigenvalues are roots of unity, and let sp be the étale specialization map. Weight...

conjecture | open | MathDB 页面 | 原文线索

V2-030388 | MATHDB-320538

Minimal grid representative conjecture for Legendrian knots

摘录 / EXCERPT: A grid diagram represents a Legendrian knot, with its grid size measuring the complexity of that representative. For a topological knot K, let $\overline{tb}(K)$ denote its maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-030389 | MATHDB-320541

The Minimal Resolution Conjecture for syzygies of general curves

摘录 / EXCERPT: Minimal Resolution Conjecture. Fix integers $g,r,d,p\geq1$ such that $g-d+r\geq0$, and assume

conjecture | open | MathDB 页面 | 原文线索

V2-030390 | MATHDB-320554

K(pi,1) conjecture for the mixed elliptic motive algebra

摘录 / EXCERPT: Let E be the elliptic curve and let ${\mathcal A}_{{\frak M}(E)}$ be the algebra of mixed elliptic motives equipped with its grading and differential structure. K(pi,1) conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-030391 | MATHDB-320595

The cohomological spectrum-to-spectrum conjecture for arithmetic locally symmetric spaces

摘录 / EXCERPT: The cohomological spectrum conjecture. The lattices Γ_{K_1} and Γ_{K_2} are L-equivalent in $G_{1,\infty}\simeq G_{2,\infty}$.

conjecture | open | MathDB 页面 | 原文线索

V2-030392 | MATHDB-320616

The Poincaré-sphere surgery realization conjecture

摘录 / EXCERPT: Let $L(p,q)$ be a lens space, let $K\subset L(p,q)$ be a knot, let $g(K)$ denote its genus, and let P^3 denote the Poincaré sphere. An integer surgery is Dehn surgery with an int...

conjecture | open | MathDB 页面 | 原文线索

V2-030393 | MATHDB-320632

The quasi-stabilization mapping-cone conjecture for link Floer complexes

摘录 / EXCERPT: Let $\bar{\mathcal H}=(\Sigma,\bar{\boldsymbol\alpha},\bar{\boldsymbol\beta},\bar{\mathbf w},\mathbf z,\bar\tau)$ be a Heegaard diagram, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030394 | MATHDB-320645

Avramov-Buchweitz-Iyengar rank conjecture for differential modules

摘录 / EXCERPT: Let R be a regular local ring of dimension d, and let F be a differential R-module admitting a finite free flag. If the homology $H(F)=\ker\delta/\operatorname{im}\delta$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-030395 | MATHDB-320668

Deitmar-Horozov conjecture on restriction of closed iterated integrals

摘录 / EXCERPT: Deitmar-Horozov conjecture. The restriction map

conjecture | open | MathDB 页面 | 原文线索

V2-030396 | MATHDB-320685

Essentialness conjecture for positive scalar curvature

摘录 / EXCERPT: Essentialness conjecture. An essential manifold does not admit a metric of positive scalar curvature. The paper explains that essential spin^c manifolds are K-theoretically essen...

conjecture | open | MathDB 页面 | 原文线索

V2-030397 | MATHDB-320719

Functoriality conjecture for the map from virtual knots to classical knots

摘录 / EXCERPT: Let g be the map from virtual knots to classical knots obtained by reducing to a minimal-genus representative and applying the parity projection. For virtual knots K_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-030398 | MATHDB-320729

Cohen-Heap-Pettet cohomological finiteness conjecture for the Johnson filtration

摘录 / EXCERPT: Let F_n be the free group of rank n, and let J^s_n be the s-th term of the Andreadakis-Johnson filtration on $\operatorname{Aut}(F_n)$. Cohen-Heap-Pettet conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-030399 | MATHDB-320755

Trapezoidal conjecture for Conway polynomial coefficients of two-bridge links

摘录 / EXCERPT: Trapezoidal conjecture. There exists an integer $n\leq\lfloor m/2\rfloor$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-030400 | MATHDB-320756

The injectivity conjecture for the homology-cylinder maps a_k

摘录 / EXCERPT: For every $k\geq 1$, consider the homomorphism ... from the exact sequence associated with the A_n-filtration of homology cylinders, where ${\sf K}^{\sf Y}_{4k-3}$ is the kernel ap...

conjecture | open | MathDB 页面 | 原文线索

V2-030401 | MATHDB-320798

Stable topological invariants for soft sphere representations in the tenfold way classes

摘录 / EXCERPT: Let H_r, for $r=1,\ldots,d+1$, be matrices in one of the ten random-matrix-theory universality classes that give a soft representation of the sphere S^d. In the stable limit, c...

conjecture | open | MathDB 页面 | 原文线索

V2-030402 | MATHDB-320799

Independence of the Bott index from the polynomial or logarithmic map

摘录 / EXCERPT: Let U_1 and U_2 be approximately commuting unitary matrices, with $ obreak\bigl\Vert[U_1,U_2]\bigr\Vert\leq ecdelta$, and let H_1,H_2,H_3 be the associated Hermitian matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-030403 | MATHDB-320808

Rational collapse conjecture for graph-space spectral sequences

摘录 / EXCERPT: Rational collapse conjecture. The spectral sequences for the pointed spaces rationally collapse in total degrees $2* \leq n-3$, while the corresponding spectral sequences for the u...

conjecture | open | MathDB 页面 | 原文线索

V2-030404 | MATHDB-320836

Labastida-Mariño-Ooguri-Vafa integrality conjecture for colored link invariants

摘录 / EXCERPT: Labastida-Mariño-Ooguri-Vafa conjecture. For any $\vec{A}\in\mathcal{P}^{L}$, there exist $P_{\vec{B}}(q,t)$ for $\vec{B}\in\mathcal{P}^{L}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-030405 | MATHDB-320839

Lobb's proportionality conjecture for Gornik invariants

摘录 / EXCERPT: Let K be a knot, and let $s_n(K)\in 2\mathbb{Z}$ denote the even integer determined by the Gornik homology of K for each $n\geq 2$. Lobb's proportionality conjecture. For any k...

conjecture | open | MathDB 页面 | 原文线索

V2-030406 | MATHDB-320840

Divisibility conjecture for the Gornik invariants

摘录 / EXCERPT: Let K be a knot, and let $s_n(K)\in 2\mathbb{Z}$ denote the even integer determined by the Gornik homology of K for each $n\geq 2$. Divisibility conjecture for the Gornik invar...

conjecture | open | MathDB 页面 | 原文线索

V2-030407 | MATHDB-320848

Polynomial-time solvability of Knot Genus for orientable 3-manifolds with vanishing second Betti number

摘录 / EXCERPT: Let Y be an orientable 3-manifold with $b_2(Y)=0$, and let Knot Genus ask, given a knot K embedded in the 1-skeleton of a triangulation of Y and an integer g_0, whether...

conjecture | open | MathDB 页面 | 原文线索

V2-030408 | MATHDB-320862

Planar graph detection conjecture via ribbon graph homology

摘录 / EXCERPT: Let (G,S) be a ribbon graph embedded in S^3, and let H_i denote its homology groups. Planar graph detection conjecture. The ribbon graph (G,S) is isotopic to a planar graph...

conjecture | open | MathDB 页面 | 原文线索

V2-030409 | MATHDB-320869

Classification conjecture for genus-1 simplified broken Lefschetz fibrations

摘录 / EXCERPT: Let M be a 4-manifold admitting a genus-1 simplified broken Lefschetz fibration (SBLF) structure with non-empty round singular locus. Write $\sharp$ for connected sum, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-030410 | MATHDB-320873

Unknotting number conjecture for connected sums of opposite torus knots

摘录 / EXCERPT: Let $T(2,k)$ be the $(2,k)$-torus link, let $T(2,-k)$ be its mirror image, and let $u(L)$ denote the unknotting number of a link L. For an odd integer k, consider the connected...

conjecture | open | MathDB 页面 | 原文线索

V2-030411 | MATHDB-320890

The metrizable locally quasi-convex Mackey group conjecture

摘录 / EXCERPT: Metrizable LQC Mackey conjecture. Every metrizable $(G,\mu)\in \operatorname{LCS}$ is a Mackey group in LQC.

conjecture | open | MathDB 页面 | 原文线索

V2-030412 | MATHDB-320891

The discreteness conjecture for Mackey precompact groups

摘录 / EXCERPT: Discreteness conjecture. If $G\in \operatorname{MAP}$ is a Mackey group in MAP, then G is a discrete group.

conjecture | open | MathDB 页面 | 原文线索

V2-030413 | MATHDB-320892

The finest precompact topology conjecture

摘录 / EXCERPT: Finest precompact topology conjecture. If for (G,ν) there exists the MAP-Mackey topology associated with ν, then ν is the finest precompact group topol...

conjecture | open | MathDB 页面 | 原文线索

V2-030414 | MATHDB-320932

The super matrix factorization version of the Hodge conjecture

摘录 / EXCERPT: Let a supermanifold be described by a matrix-factorization category, and let its Jacobian ideal be viewed inside Hochschild cohomology. A (p,p) class is a Hodge-type class of bid...

conjecture | open | MathDB 页面 | 原文线索

V2-030415 | MATHDB-320934

The supercohomology determination conjecture

摘录 / EXCERPT: Let supercohomology be the cohomological invariant under consideration, let the Jacobian ring encode the relevant algebraic data, let gaps refer to gaps in the associated categoric...

conjecture | open | MathDB 页面 | 原文线索

V2-030416 | MATHDB-320947

The algebraic-to-topological hermitian K-theory conjecture for Banach algebras

摘录 / EXCERPT: Let A be a real or complex commutative Banach algebra with involution, let ε denote the hermitian sign, and let m be the coefficient modulus. The algebraic-to-topol...

conjecture | open | MathDB 页面 | 原文线索

V2-030417 | MATHDB-320975

Layered-torus-bundle conjecture for non-flat non-orientable torus bundles

摘录 / EXCERPT: A non-flat non-orientable torus bundle over the circle is a non-orientable torus bundle over S^1 that is not flat, and a minimal triangulation is a triangulation using the fewest...

conjecture | open | MathDB 页面 | 原文线索

V2-030418 | MATHDB-320976

Plugged-I-bundle conjecture for non-orientable Seifert fibred spaces

摘录 / EXCERPT: A non-flat non-orientable Seifert fibred space over $\mathbb{R}P^2$ or $\bar{D}$ with two exceptional fibres is a Seifert fibred 3-manifold in the stated class, where $\bar{D}$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-030419 | MATHDB-320981

The Murasugi-Przytycki braid-index conjecture for alternating diagrams

摘录 / EXCERPT: Murasugi-Przytycki conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030420 | MATHDB-320982

The wave-move conjecture for unknot diagrams

摘录 / EXCERPT: A wave move is a local move on a knot diagram that may preserve or reduce its crossing number. Wave-move conjecture. Crossing-number-preserving and crossing-number-reducing wave mo...

conjecture | open | MathDB 页面 | 原文线索

V2-030421 | MATHDB-320983

The crossing-number and braid-index inequality for special alternating links

摘录 / EXCERPT: Let L be a special alternating link. Write $c(L)$ for its crossing number, $g(L)$ for its genus, and $b(L)$ for its braid index. Special alternating link inequality. One has ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-030422 | MATHDB-320984

The log-concavity conjecture for Alexander and Jones polynomials

摘录 / EXCERPT: For a Laurent polynomial $X=\sum_k X_{[k]}t^k\in\mathbb Z[t^{\pm1}]$, call X log-concave if ... for every $k\in\mathbb Z$. Let $n(K)$ be the number of components of a link K, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030423 | MATHDB-321002

Characterization of D-separable compact spaces and equality of cardinal invariants

摘录 / EXCERPT: Characterization and cardinal-invariant conjecture. begin{enumerate} item A compact space X is D-separable if and only if X has a σ-disjoint π-base. item For every c...

open_problem | open | MathDB 页面 | 原文线索

V2-030424 | MATHDB-321017

Brunnian conjecture for the family $L_{i,j,n}$

摘录 / EXCERPT: Brunnian conjecture. All links in the family $L_{i,j,n}$ are Brunnian.

conjecture | open | MathDB 页面 | 原文线索

V2-030425 | MATHDB-321018

Existence of convex Brunnian links beyond the family $L_{i,j,n}$

摘录 / EXCERPT: Existence conjecture. There exist convex Brunnian links that are not isotopic to a link of the form $L_{i,j,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-030426 | MATHDB-321019

Round approximation conjecture for convex Brunnian links

摘录 / EXCERPT: Round approximation conjecture. All convex Brunnian links can be made arbitrarily close to perfectly round in this manner.

conjecture | open | MathDB 页面 | 原文线索

V2-030427 | MATHDB-321033

Extension conjecture for strictly colored sphere triangulations

摘录 / EXCERPT: Let T be a triangulation of the n-sphere $4S^n$ and let $4{\Greekmath 0120}$ be a strict $n+2$-coloring of T. A triangulation T' of the disk $4\mathbb{D}^{n+1}$ extends T...

conjecture | open | MathDB 页面 | 原文线索

V2-030428 | MATHDB-321058

The infinite-order polynomial bounded K-theory class conjecture

摘录 / EXCERPT: Let G be the solvable group described in the source, with BG homotopy equivalent to a closed oriented 3-manifold, and let $E G$ be the homogeneous bar resolution. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030429 | MATHDB-321061

Colored HOMFLY-PT homology limit conjecture

摘录 / EXCERPT: Let $B^{(m)}$ be the m-colored version of a braid diagram B, and let $H_N(B^{(m)})$ denote its colored $athfrak{sl}(N)$ link homology. Denote by H the colored HOMFLY-PT link...

conjecture | open | MathDB 页面 | 原文线索

V2-030430 | MATHDB-321062

Jones' conjecture on the writhe of minimal braid representations

摘录 / EXCERPT: Let L be a link of braid index b, and let B_1 and B_2 be braid diagrams of L, each with b strands. The writhe of a braid diagram is the signed crossing number. Jones' c...

conjecture | open | MathDB 页面 | 原文线索

V2-030431 | MATHDB-321063

Folklore conjecture on maximum measures for surface diffeomorphisms

摘录 / EXCERPT: Let $f:M\to M$ be a C^∞ surface diffeomorphism, and let $h_{\operatorname{top}}(f)$ denote its topological entropy. A maximum measure is an invariant probability measure who...

conjecture | open | MathDB 页面 | 原文线索

V2-030432 | MATHDB-321064

Collapse versus free deformation retraction for compact polyhedra

摘录 / EXCERPT: Let P be a compact n-polyhedron. A free deformation retraction of P onto a subspace Q is a homotopy $h_t:P\to P$ with $h_0=\operatorname{id}$, h_1 a retraction onto Q,...

conjecture | open | MathDB 页面 | 原文线索

V2-030433 | MATHDB-321144

Rosenberg-Weinberger positive scalar curvature conjecture for non-spin universal covers

摘录 / EXCERPT: Let M be a compact oriented manifold whose universal covering does not admit a spin structure. Write $\Gamma=\pi_1(M)$, let $n=\dim M\ge 5$, and let ... be the composition of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-030434 | MATHDB-321151

Generalized Property R conjecture

摘录 / EXCERPT: Generalized Property R conjecture. Every knot has Property nR for any $n\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-030435 | MATHDB-321152

Nice-balance conjecture for balanced links

摘录 / EXCERPT: Let L be a link that is balanced in the sense of the paper's balance criterion, and let a link be nicely balanced when its kink measure is represented by a kink tension function...

conjecture | open | MathDB 页面 | 原文线索

V2-030436 | MATHDB-321158

Kontsevich-Soibelman conjecture on the structure of the cohomological Hall algebra of the m-loop quiver

摘录 / EXCERPT: Kontsevich-Soibelman conjecture. The bigraded algebra $\mathcal{H}$ is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-030437 | MATHDB-321172

Upper semicontinuity conjecture for invariants of singularities

摘录 / EXCERPT: Let (Y,y) be a singularity, and let $w_{j}^i(y)$, $\bar{w}_{j}^i(y)$, and $\ell_{j}^i(y)$ denote the invariants defined from the mixed Hodge structures associated with the singul...

conjecture | open | MathDB 页面 | 原文线索

V2-030438 | MATHDB-321190

Topological invariance of the quantum gluing ideal for knot complements

摘录 / EXCERPT: Topological invariance conjecture. The construction of $\hat{\mathcal I}_M$ produces a topological invariant of M, independent of the triangulation and of any other choices made.

conjecture | open | MathDB 页面 | 原文线索

V2-030439 | MATHDB-321217

The cytoplasmic decomposition conjecture for separable surfaces

摘录 / EXCERPT: Cytoplasmic decomposition conjecture. Any separable 2-manifold with countable fundamental group has a cytoplasmic decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-030440 | MATHDB-321218

The non-metric three-dimensional hairy ball conjecture

摘录 / EXCERPT: Non-metric three-dimensional hairy ball conjecture. Any simply-connected $c9$-bounded non-metric 3-manifold has the fpp for flows.

conjecture | open | MathDB 页面 | 原文线索

V2-030441 | MATHDB-321219

The omega-bounded Lefschetz-space conjecture

摘录 / EXCERPT: Omega-bounded Lefschetz-space conjecture. $c9$-bounded n-manifolds have finite-dimensional (singular) homology over the rationals, and are Lefschetz spaces, i.e., any self-map wi...

conjecture | open | MathDB 页面 | 原文线索

V2-030442 | MATHDB-321220

The bagpipe-surface fixed-point conjecture

摘录 / EXCERPT: Bagpipe-surface fixed-point conjecture. Let $M=M_{g,n}$ be a bagpipe surface of genus g with n long pipes (without further specification, so that M is equivalently an $c9$-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-030443 | MATHDB-321221

The Densification-Inflation conjecture for separable manifolds

摘录 / EXCERPT: Densification-Inflation conjecture. Assume that M is a connected separable manifold. Then for each non-empty open set U in M, there is an open set V dense in M and homeom...

conjecture | open | MathDB 页面 | 原文线索

V2-030444 | MATHDB-321222

The transitivity conjecture for separable manifolds

摘录 / EXCERPT: Transitivity conjecture for separable manifolds. Any separable connected 3-manifold (or of higher dimensions) is transitive.

conjecture | open | MathDB 页面 | 原文线索

V2-030445 | MATHDB-321223

The Lefschetz homology conjecture for omega-bounded manifolds

摘录 / EXCERPT: Lefschetz homology conjecture. Let M be an $c9$-bounded manifold. Then $H_p(M,\mathbb Q)$ is finite dimensional for each p (and of course vanish above the dimension of M (Mil...

conjecture | open | MathDB 页面 | 原文线索

V2-030446 | MATHDB-321224

The omega-bounded Lefschetz fixed-point conjecture

摘录 / EXCERPT: Omega-bounded Lefschetz fixed-point conjecture. Let $f\colon M \to M$ be a (continuous) map of an $c9$-bounded manifold. If the Lefschetz number $b7_f$ is non-zero, then f has a...

conjecture | open | MathDB 页面 | 原文线索

V2-030447 | MATHDB-321225

The topological Hopf brushing conjecture

摘录 / EXCERPT: Topological Hopf brushing conjecture. A closed topological manifold with vanishing Euler characteristic admits a brushing.

conjecture | open | MathDB 页面 | 原文线索

V2-030448 | MATHDB-321226

The bordered-manifold Hopf brushing conjecture

摘录 / EXCERPT: Bordered-manifold Hopf brushing conjecture. Let W be a compact bordered manifold (first C^{∞} and then C^0). Assume that $c7(W)=0$, and when W is odd-dimensional that...

conjecture | open | MathDB 页面 | 原文线索

V2-030449 | MATHDB-321305

The Euler-to-homological convolution equivalence conjecture

摘录 / EXCERPT: Let $\mathrm{\mathbf{econv}}(\mathrm{Gr})_0$ be the subcategory of constructible-function convolutions generated by the relevant Satake-fibre functions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030450 | MATHDB-321308

Symmetry conjecture for the ideal trefoil

摘录 / EXCERPT: Let $\gamma_{3_1}:\mathbb{S}\longrightarrow\mathbb{R}^3$ denote the ideal trefoil, and let its contact functions be defined by the contact-chord construction described above. Symme...

conjecture | open | MathDB 页面 | 原文线索

V2-030451 | MATHDB-321309

Nine-cycle conjecture for the ideal trefoil contact function

摘录 / EXCERPT: Let $\gamma_{3_1}:\mathbb{S}\longrightarrow\mathbb{R}^3$ be the ideal trefoil, parameterized with constant speed so that $\gamma_{3_1}(0)$ is the outer point on a symmetry axis, an...

conjecture | open | MathDB 页面 | 原文线索

V2-030452 | MATHDB-321310

Attractor conjecture for the ideal trefoil's nine-cycle

摘录 / EXCERPT: Let $b_n\in(\mathbb{S})^n$ be a cycle for the contact function $\sigma:\mathbb{S}\longrightarrow\mathbb{S}$. Call b_n an attractor if, for every $t\in\mathbb{S}$ and fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-030453 | MATHDB-321336

The independence of real conjugation from virtual L-moves

摘录 / EXCERPT: A virtual braid is a braid diagram whose crossings may be classical or virtual, considered up to virtual braid isotopy. Real conjugation independence conjecture. Real conjugation i...

conjecture | open | MathDB 页面 | 原文线索

V2-030454 | MATHDB-321353

Finiteness conjecture for highly connected minimal non-embeddable cell complexes

摘录 / EXCERPT: For an integer n, consider n-dimensional cell complexes K whose underlying spaces are $(n-1)$-connected and do not embed in S^{2n}. An h-minor is a cell complex obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-030455 | MATHDB-321358

Geisser's finiteness conjecture for Kato homology with $\mathbb Q/\mathbb Z$ coefficients

摘录 / EXCERPT: Let X be smooth and projective over a finite field, and let $H_i^K(X,\mathbb Q/\mathbb Z(1))$ denote Kato homology in weight 1. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030456 | MATHDB-321364

Vogel's antisymmetry conjecture for three-legged diagrams

摘录 / EXCERPT: Let $\mathcal{X}^3$ be the space of fixed Jacobi diagrams with three legs, and let $\Lambda\subseteq\mathcal{X}^3$ be the subspace of elements antisymmetric under permutations of t...

conjecture | open | MathDB 页面 | 原文线索

V2-030457 | MATHDB-321425

Random stable commutator length conjecture for closed hyperbolic manifolds

摘录 / EXCERPT: Let M be a closed hyperbolic d-manifold. Fix some $\delta>0$. Let γ be a random geodesic of length in $[n-\delta,n+\delta]$ conditioned to be homologically trivial, and...

conjecture | open | MathDB 页面 | 原文线索

V2-030458 | MATHDB-321438

Generic vanishing conjecture for the J-anti-invariant cohomology

摘录 / EXCERPT: Let M be a compact 4-manifold and let J be an almost complex structure on M. Denote by H_J^- the J-anti-invariant part of the second cohomology and set...

conjecture | open | MathDB 页面 | 原文线索

V2-030459 | MATHDB-321439

Integrability conjecture from J-anti-invariant cohomology

摘录 / EXCERPT: Let M be a compact 4-manifold, let J be an almost complex structure on M, and let H_J^- denote the J-anti-invariant part of the second cohomology, with...

conjecture | open | MathDB 页面 | 原文线索

V2-030460 | MATHDB-321447

C. McA. Gordon's exceptional-slope bound conjecture

摘录 / EXCERPT: C. McA. Gordon's conjecture. For any hyperbolic knot manifold M, we have $\#\mathcal{E}(M) \leq 10$ and $\Delta(\mathcal{E}(M)) \leq 8$. Moreover, if $M \ne M_1,M_2,M_3,M_4$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-030461 | MATHDB-321448

C. McA. Gordon's small Seifert-toroidal distance conjecture

摘录 / EXCERPT: C. McA. Gordon's conjecture. Suppose that M is a hyperbolic knot manifold M and α, β are slopes on ∂M such that $M(\alpha)$ is small Seifert and $M(\beta)$...

conjecture | open | MathDB 页面 | 原文线索

V2-030462 | MATHDB-321458

Finiteness conjecture for the Tate-Shafarevich group of the K-theoretic obstruction

摘录 / EXCERPT: Conjecture de finitude de l'obstruction K-théorique. Le groupe $\cyr{X}\mathcal N(V)$ est fini. Cette conjecture est un principe local-global en K-théorie algébrique; son statu...

conjecture | open | MathDB 页面 | 原文线索

V2-030463 | MATHDB-321462

The PFA(S)[S] conjecture on first-countable perfect pre-images of ω_1

摘录 / EXCERPT: PFA(S)[S] conjecture. Under $\mathrm{PFA}(S)[S]$, every first-countable perfect pre-image of ω_1 includes a copy of ω_1.

conjecture | open | MathDB 页面 | 原文线索

V2-030464 | MATHDB-321492

Converse to the H(2)-unknotting characterization for composite links

摘录 / EXCERPT: Converse conjecture. Every composite link with H(2)-unknotting number one has the form described in Proposition.

conjecture | open | MathDB 页面 | 原文线索

V2-030465 | MATHDB-321493

The simple loop conjecture for 3-manifold groups

摘录 / EXCERPT: Let M be an orientable 3-manifold, and let Σ be a closed orientable surface. The simple loop conjecture. The kernel of every non-injective homomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-030466 | MATHDB-321503

Nonvanishing of the cusp differential in cohomology of Bianchi groups

摘录 / EXCERPT: Nonvanishing conjecture. This restricted differential is nonzero.

conjecture | open | MathDB 页面 | 原文线索

V2-030467 | MATHDB-321504

Atiyah's conjecture on Yang-Mills and loop-space Morse homology

摘录 / EXCERPT: Let G be a compact Lie group and let P be a principal G-bundle over a Riemann surface Σ. Write $\mathcal A(P)$ for the space of connections on P, $\mathcal G_0(P)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-030468 | MATHDB-321505

Bounded genus conjecture for polyhedra of fixed dimension

摘录 / EXCERPT: Bounded genus conjecture. For every positive integer n there is an integer c_n such that

conjecture | open | MathDB 页面 | 原文线索

V2-030469 | MATHDB-321506

Uniform sphere-retraction conjecture for polyhedra of fixed dimension

摘录 / EXCERPT: Uniform sphere-retraction conjecture. For every integer d there is an integer $m>0$ such that, for every polyhedron X of dimension d, there are maps

conjecture | open | MathDB 页面 | 原文线索

V2-030470 | MATHDB-321509

Asymptotic Expansion Conjecture for Witten-Reshetikhin-Turaev invariants

摘录 / EXCERPT: Let X be a general oriented compact three-manifold and let ${\mathcal Z}^{(k)}_G(X)$ denote its Reshetikhin-Turaev invariant at level k for a semisimple, simply connected Lie g...

conjecture | open | MathDB 页面 | 原文线索

V2-030471 | MATHDB-321521

Extension of the planar Peano continuum homomorphism theorem

摘录 / EXCERPT: Let X and Y be Peano continua, with X planar and Y a planar Peano continuum, and suppose the homomorphism conclusion of Theorem ... should remain true when X is planar inst...

conjecture | open | MathDB 页面 | 原文线索

V2-030472 | MATHDB-321526

The Grassmann-algebraic identity for the 2-to-4 Pachner move

摘录 / EXCERPT: Consider the $2\to4$ Pachner move replacing the 4-simplices (12345) and (12346) by (12356), (12456), (13456), and (23456). Let $\tilde{\mathcal W}_{ijklm}$ denote the c...

conjecture | open | MathDB 页面 | 原文线索

V2-030473 | MATHDB-321527

The Grassmann-algebraic invariant of 4-dimensional Pachner moves

摘录 / EXCERPT: The Grassmann-algebraic Pachner-move invariant. The expression

conjecture | open | MathDB 页面 | 原文线索

V2-030474 | MATHDB-321556

The constant-map free-energy formula for the framed mirror curve of $\mathbb{C}^3$

摘录 / EXCERPT: Constant-map formula. For $g\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-030475 | MATHDB-321572

Virtual Haken conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let M be a finite-volume hyperbolic 3-manifold. A 3-manifold is Haken if it contains a properly embedded incompressible surface. Virtual Haken conjecture. The manifold M ha...

conjecture | open | MathDB 页面 | 原文线索

V2-030476 | MATHDB-321586

Noncommutative Lefschetz standard conjecture for saturated dg-categories

摘录 / EXCERPT: Let T be a saturated dg-category. Consider its algebraic classes in Hochschild homology, namely classes of the form $\alpha_E(\operatorname{Id}_E)$ for objects $E\in T$, and call...

conjecture | open | MathDB 页面 | 原文线索

V2-030477 | MATHDB-321593

Unique ergodicity of minimal systems without nontrivial finite-independent pairs

摘录 / EXCERPT: Let (X,T) be a minimal topological dynamical system, and let ${\rm Ind}_{fip}(X,T)$ denote its set of finite-independent pairs. Write $\Delta_X=\{(x,x):x\in X\}$. Unique ergodici...

conjecture | open | MathDB 页面 | 原文线索

V2-030478 | MATHDB-321595

Nöbeling-space model conjecture for ending lamination spaces

摘录 / EXCERPT: Let S be a p-punctured surface of genus g, and set $n=3g+p-4$. Define k by $k=0$ if $g=0$, $k=g-1$ if both $p\neq 0$ and $g\neq 0$, and $k=g-2$ if $p=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030479 | MATHDB-321596

Duality conjecture for ending lamination spaces and curve complexes

摘录 / EXCERPT: Let S be a p-punctured surface of genus g. Write $\mathcal{C}(S)$ for its curve complex, $\mathcal{EL}(S)$ for its ending lamination space, and...

conjecture | open | MathDB 页面 | 原文线索

V2-030480 | MATHDB-321597

Homological and cohomological inclusion conjecture for curve-complex models

摘录 / EXCERPT: Let S be a finite-type surface. Let $\mathcal{C}(S)_{\operatorname{sub}}$ be the subspace model of the curve complex, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030481 | MATHDB-321607

Exceptional surgery conjecture for hyperbolic knots of wrapping number 2

摘录 / EXCERPT: Let V be a solid torus and let K be a hyperbolic knot in V. Exclude the Whitehead knot and the wrapped Montesinos knot $K^1(-1/2, 1/3)$. A hyperbolic wrapping-number-2 surger...

conjecture | open | MathDB 页面 | 原文线索

V2-030482 | MATHDB-321612

Reis-Weiss's conjecture on rational Pontryagin classes of TOP bundles

摘录 / EXCERPT: Let ξ be a $\mathrm{TOP}(n)$-bundle over a space B, and let $p_i(\xi)\in H^{4i}(B;\mathbb{Q})$ denote its i-th rational Pontryagin class. Reis-Weiss's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-030483 | MATHDB-321613

Adem-Ge-Pan-Petrosyan collapse conjecture for semidirect products

摘录 / EXCERPT: Let $G\cong \mathbb{Z}/m$ be a finite cyclic group, let $L\cong\mathbb{Z}^n$ be a finitely generated free abelian group, let $\rho\colon G\to\operatorname{aut}_{\mathbb{Z}}(L)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-030484 | MATHDB-321628

Semilocal simple connectivity of orbit spaces of smooth vector fields

摘录 / EXCERPT: Let M be a Hausdorff manifold with a countable basis and empty boundary, and let v be an arbitrary smooth vector field on M. Let $M/{\sim}$ denote the orbit space of v. Orb...

conjecture | open | MathDB 页面 | 原文线索

V2-030485 | MATHDB-321650

Bux-Wortman coarse filling conjecture for arithmetic groups

摘录 / EXCERPT: Bux-Wortman's coarse filling conjecture. Every such coarse manifold in G can be replaced by one in the arithmetic group with the same boundary and topological type, with volume b...

conjecture | open | MathDB 页面 | 原文线索

V2-030486 | MATHDB-321681

The Direct Summand Conjecture

摘录 / EXCERPT: Let $R \rightarrow S$ be an injective ring homomorphism, where R is a regular local ring and S is module-finite over R. Direct Summand Conjecture. Then R is a direct summan...

conjecture | open | MathDB 页面 | 原文线索

V2-030487 | MATHDB-321682

The Cohen-Macaulayness of Direct Summands Conjecture

摘录 / EXCERPT: Let $A \rightarrow R$ be a ring homomorphism such that A is a direct summand of R, and suppose that R is regular. Cohen-Macaulayness of Direct Summands conjecture. Then A i...

conjecture | open | MathDB 页面 | 原文线索

V2-030488 | MATHDB-321683

Tripp's conjecture for alternating knots

摘录 / EXCERPT: A knot is an ambient isotopy class of an oriented smoothly embedded circle in S^3. For a knot K, let $c(K)$ denote its minimal crossing number, let $g_c(K)$ denote its canonica...

conjecture | open | MathDB 页面 | 原文线索

V2-030489 | MATHDB-321701

Unimodality conjecture for Betti tables of powers

摘录 / EXCERPT: Let $I\subseteq R$ be an equigenerated homogeneous ideal generated in degree r. For each pair of indices (i,j), consider the graded Betti numbers $\beta_{i,j+dr}(I^d)$ as d v...

conjecture | open | MathDB 页面 | 原文线索

V2-030490 | MATHDB-321706

Grothendieck's finiteness conjecture for local cohomology

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a noetherian local ring, let $I\subseteq R$ be an ideal, and let M be a finitely generated R-module. Grothendieck's finiteness conjecture. The module...

conjecture | open | MathDB 页面 | 原文线索

V2-030491 | MATHDB-321745

The weak Palis conjecture for vector fields

摘录 / EXCERPT: Let X be a vector field, and let C^r denote the differentiability topology. A horseshoe is the standard hyperbolic chaotic invariant set, and a Morse-Smale vector field has hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-030492 | MATHDB-321753

Conjecture on minimally knotted cycles in canonical book representations

摘录 / EXCERPT: Minimal knotted-cycle conjecture. The canonical book representation $\widetilde{K}_n$ contains the fewest total number of knotted cycles possible in any embedding of K_n.

conjecture | open | MathDB 页面 | 原文线索

V2-030493 | MATHDB-321757

The cohomological decomposition conjecture for crystallographic groups with cyclic holonomy

摘录 / EXCERPT: Let G be a finite cyclic group and L a finitely generated $\mathbb{Z}G$-lattice. For the split crystallographic group $L\rtimes G$, consider its integral cohomology and the coh...

conjecture | open | MathDB 页面 | 原文线索

V2-030494 | MATHDB-321758

Heawood's empire conjecture for orientable surfaces

摘录 / EXCERPT: Heawood's empire conjecture. The inequality is sharp:

conjecture | open | MathDB 页面 | 原文线索

V2-030495 | MATHDB-321779

Postnikov-Reiner-Williams monotonicity conjecture for subdivisions of flag homology spheres

摘录 / EXCERPT: Let Δ and Δ' be flag homology spheres, and suppose that Δ' geometrically subdivides Δ. Let $\gamma(\Delta)$ and $\gamma(\Delta')$ denote their γ-...

conjecture | open | MathDB 页面 | 原文线索

V2-030496 | MATHDB-321786

Fukaya's Floer-theoretic homological mirror symmetry conjecture

摘录 / EXCERPT: Let X be a symplectic manifold with associated filtered A_{∞} category $\mathcal{LAG}(X)$, let $\hat X$ be its mirror, and let (L,b) be a pair consisting of a Lagrangian...

conjecture | open | MathDB 页面 | 原文线索

V2-030497 | MATHDB-321787

Fukaya's Hochschild cohomology-quantum cohomology conjecture

摘录 / EXCERPT: Let X be a symplectic manifold and let $\mathcal C=\mathcal{LAG}(X)$ be its associated filtered A_{∞} category. Write $\operatorname{Hom}(1_{\mathcal C},1_{\mathcal C})$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-030498 | MATHDB-321788

Fukaya's coherent-sheaf construction from Floer cohomology

摘录 / EXCERPT: Assume the Strominger-Yau-Zaslow fibration conjecture holds, let (X,W) be the relevant mirror-symmetry data, and let (L,b) be a Lagrangian brane. For each point $p\in\hat X$, i...

conjecture | open | MathDB 页面 | 原文线索

V2-030499 | MATHDB-321789

The Fukaya category decomposition by critical values

摘录 / EXCERPT: Let X be a toric symplectic manifold, let Λ_0 be the Novikov ring, let $\mathcal{LAG}(X)$ be the filtered A_{∞} category, and let $W=\mathfrak{PO}$ be the potentia...

conjecture | open | MathDB 页面 | 原文线索

V2-030500 | MATHDB-321790

The matrix-factorization mirror conjecture for Fukaya category summands

摘录 / EXCERPT: Let X be a toric symplectic manifold with potential W on its mirror $\hat X$. Suppose that c is a critical value of W and write $\hat X_c=W^{-1}(c)$. Let the direct-sum fac...

conjecture | open | MathDB 页面 | 原文线索

V2-030501 | MATHDB-321791

Seidel's homological mirror symmetry conjecture for directed Fukaya categories

摘录 / EXCERPT: Let (X,W) be a symplectic manifold with a function satisfying the exactness conditions required to define the directed A_{∞} category $\mathcal{LAG}(X,W)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030502 | MATHDB-321792

The Gross-Siebert algebra and homological mirror symmetry for log Calabi-Yau surfaces

摘录 / EXCERPT: Let Y be a smooth projective surface, let D be an anticanonical divisor supporting an ample divisor, let $R=\Bbbk[\operatorname{NE}(Y)]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030503 | MATHDB-321793

The symplectic-cohomology description of the mirror

摘录 / EXCERPT: Let $U=Y\setminus D$ and let X be the mirror fiber of the constructed family over $s=\exp(2\pi i(\mathbf B+i\omega))$. Symplectic-cohomology mirror conjecture. There is an isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-030504 | MATHDB-321794

Wrapped homological mirror symmetry in the negative semi-definite case

摘录 / EXCERPT: Let $D\subset Y$ be a cycle of (-2)-curves, let $w\colon U=Y\setminus D\to\mathbb C$ be the elliptic fibration, let $X=Z\setminus E$ be the corresponding open mirror fiber, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-030505 | MATHDB-321795

The Jacobian-ring and quantum-cohomology mirror conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let Y be a del Pezzo surface with $-K_Y$ ample, let $W\colon X\to\mathbb C$ be the mirror Landau-Ginzburg potential, let $J(X,W)=\Gamma(\mathcal O_{\operatorname{Sing}W})$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-030506 | MATHDB-321801

The conjecture that interleaving distance is optimal over any field

摘录 / EXCERPT: Interleaving optimality conjecture. For any field k and $i\in\mathbb Z_{\geq 0}$, d_I is i-optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-030507 | MATHDB-321829

Covering and stabilisation conjecture for Fukaya categories

摘录 / EXCERPT: Let M be a Liouville manifold, let $\widetilde M\to M$ be a cover, and let $W:M\times D^2\to D^2$ be given by $W(m,z)=z^2$. Let $\mathcal F$ denote the relevant Fukaya categories...

conjecture | open | MathDB 页面 | 原文线索

V2-030508 | MATHDB-321830

Truncated-polynomial cohomology conjecture for cotangent-bundle symmetries

摘录 / EXCERPT: Truncated-polynomial cohomology conjecture. If ρ is non-trivial, then $H^*(Q;\mathbb Z)$ is a truncated polynomial ring. This is a more precise formulation of the preceding Bo...

conjecture | open | MathDB 页面 | 原文线索

V2-030509 | MATHDB-321839

The Alexander polynomial vanishing conjecture for even-component amphicheiral links

摘录 / EXCERPT: Let $L=K_1\cup\cdots\cup K_r$ be an r-component link, and let ... where t_i corresponds to a meridian of K_i. Suppose that L is algebraically split and component-preservingly...

conjecture | open | MathDB 页面 | 原文线索

V2-030510 | MATHDB-321840

Kawauchi's Alexander polynomial conjecture for even-component amphicheiral links

摘录 / EXCERPT: Let L be an r-component link whose components have pairwise linking number zero, and suppose that L is component-preservingly amphicheiral, meaning that an orientation-revers...

conjecture | open | MathDB 页面 | 原文线索

V2-030511 | MATHDB-321863

The knot-state Lagrangian conjecture near abelian representations

摘录 / EXCERPT: Knot-state Lagrangian conjecture. At every regular abelian point, the knot state is a Lagrangian state; more precisely, for a suitable neighborhood V and the local data specified...

conjecture | open | MathDB 页面 | 原文线索

V2-030512 | MATHDB-321866

Irreducible knot-state asymptotics

摘录 / EXCERPT: Let K be a knot with exterior E_K and peripheral torus Σ. Let ${\mathcal M}^s(\Sigma)$ and ${\mathcal M}^s(E_K)$ denote the smooth loci of the corresponding character mo...

conjecture | open | MathDB 页面 | 原文线索

V2-030513 | MATHDB-321867

Abelian knot-state asymptotics

摘录 / EXCERPT: Let K be a knot with exterior E_K, let $E=H_1(\Sigma,\mathbb R)$ for the peripheral torus Σ, and let λ be the longitude direction. Let a regular point of...

conjecture | open | MathDB 页面 | 原文线索

V2-030514 | MATHDB-321872

Polynomial invariants distinguish alternating links within a family

摘录 / EXCERPT: Let F be a family of knots or links, and let L_1 and L_2 be two alternating, nonisotopic KLs belonging to F. Let P be a polynomial invariant. Family-distinguishing conj...

conjecture | open | MathDB 页面 | 原文线索

V2-030515 | MATHDB-321874

Amenable minimal matchbox manifolds are affable

摘录 / EXCERPT: Affability conjecture. Then ${\mathfrak{M}}$ is affable.

conjecture | open | MathDB 页面 | 原文线索

V2-030516 | MATHDB-321885

The contact-form existence problem for odd-dimensional manifolds with non-degenerate two-forms

摘录 / EXCERPT: Contact-form existence problem. Does M admit a contact form?

open_problem | open | MathDB 页面 | 原文线索

V2-030517 | MATHDB-321909

Zero-in-the-spectrum conjecture for finite aspherical complexes

摘录 / EXCERPT: Let M be a finite aspherical CW-complex (or weakly a closed, connected, oriented and aspherical Riemannian manifold) with fundamental group π. Write $C_{r}^{\ast }(\pi)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-030518 | MATHDB-321920

The pullback correspondence for equivariant slice towers

摘录 / EXCERPT: Pullback slice-tower conjecture. The slice tower for a G/N-spectrum pulls back to the slice tower for the pullback G-spectrum, where the k-slice becomes the...

conjecture | open | MathDB 页面 | 原文线索

V2-030519 | MATHDB-321924

The first author's conjecture on the stable abelianization of \(\mathfrak{h}_{g,1}\)

摘录 / EXCERPT: Stable abelianization conjecture. The stable abelianization of the Lie algebra $\mathfrak{h}_{g,1}$ vanishes; namely

conjecture | open | MathDB 页面 | 原文线索

V2-030520 | MATHDB-321959

The genus-independence conjecture for the monopole Floer Legendrian invariant

摘录 / EXCERPT: Let $ \mathcal{K}\subset(Y,\xi)$ be a Legendrian knot, and let $ \ell_g(\mathcal{K})\in KHM(-Y,K)\otimes\mathcal{R}$ denote the invariant constructed using a closure of genus g,...

conjecture | open | MathDB 页面 | 原文线索

V2-030521 | MATHDB-321960

The stabilization conjecture for the monopole Floer Legendrian invariant

摘录 / EXCERPT: Let $ \mathcal{K}\subset Y$ be a Legendrian knot. Let $S_{+}(\mathcal{K})$ and $S_{-}(\mathcal{K})$ denote its positive and negative stabilizations, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030522 | MATHDB-321961

The arbitrary-genus Lagrangian cobordism functoriality conjecture

摘录 / EXCERPT: Let $ \mathcal{K}_0$ and $ \mathcal{K}_1$ be Legendrian knots in (Y,ξ), and suppose that $ \mathcal{K}_0\prec_{\Sigma}\mathcal{K}_1$ is a Lagrangian cobordism of arbitrary genu...

conjecture | open | MathDB 页面 | 原文线索

V2-030523 | MATHDB-321965

Cheeger-Simons injectivity conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Let $H_3(\operatorname{PSL}(2,\mathbb C);\mathbb Z)$ be the third homology group of $\operatorname{PSL}(2,\mathbb C)$ with integer coefficients. The Cheeger-Simons class is the hom...

conjecture | open | MathDB 页面 | 原文线索

V2-030524 | MATHDB-321982

Conjecture on the trigonometric form of curve operators

摘录 / EXCERPT: Let $t=(t_1,\ldots,t_n)$ be a family in $([0,\pi]\cap\mathbb{Q}\pi)^n$, let D be the common denominator of the rational numbers t_j/π, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030525 | MATHDB-321983

The subprincipal-symbol conjecture for curve operators

摘录 / EXCERPT: Let γ be a multicurve, let f_γ be its trace function, and let (τ,θ) be action-angle coordinates on the relevant open subset of $\mathcal M(\Sigma,t)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-030526 | MATHDB-321991

Asymptotic growth conjecture for graph-complex homotopy ranks

摘录 / EXCERPT: Let $\overline{\operatorname{Emb}}_c(\mathbb{R}^m,\mathbb{R}^n)$ denote the homotopy fibre of the inclusion of embeddings into immersions, and let $\pi_*^{(t-\ell,t)}$ denote its c...

conjecture | open | MathDB 页面 | 原文线索

V2-030527 | MATHDB-322016

The condensed three-braid Jones polynomial coefficient conjecture

摘录 / EXCERPT: Condensed three-braid conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030528 | MATHDB-322036

The generalized Strong Bass conjecture for rapid decay group algebras

摘录 / EXCERPT: Generalized Strong Bass conjecture. For each non-elliptic conjugacy class $\langle x\rangle$, the image of

conjecture | open | MathDB 页面 | 原文线索

V2-030529 | MATHDB-322067

Tunnel Classification Conjecture for tunnel number 1 knots

摘录 / EXCERPT: Let K be a knot of tunnel number 1. A regular tunnel is a tunnel that is not a $(1,1)$-tunnel, and a $(1,1)$-position is a presentation of K as a $(1,1)$-knot with its associ...

conjecture | open | MathDB 页面 | 原文线索

V2-030530 | MATHDB-322112

Leopoldt conjecture for p-adic Lie extensions

摘录 / EXCERPT: Let K_∞/K be a p-adic Lie extension with Galois group G, let V be a Galois representation, and let T be a $\mathbb{Z}_p$-lattice in V. The equivalent conditions of...

conjecture | open | MathDB 页面 | 原文线索

V2-030531 | MATHDB-322129

Conjecture on the limiting density of the sets V_n

摘录 / EXCERPT: Limiting V_n-density conjecture. The limiting density is

conjecture | open | MathDB 页面 | 原文线索

V2-030532 | MATHDB-322130

Conjecture on the limiting lower bound for Wecken-map density

摘录 / EXCERPT: Limiting lower-bound conjecture. The sequence of lower bounds satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-030533 | MATHDB-322131

Projection-map conjecture for spaces of real projective maps

摘录 / EXCERPT: Projection-map conjecture. These projection maps are homotopy equivalences. This conjecture concerns the topology of spaces of equivariant algebraic maps between real projective sp...

conjecture | open | MathDB 页面 | 原文线索

V2-030534 | MATHDB-322158

Nonvanishing conjecture for the determinant \(\mathbb{D}(d)\)

摘录 / EXCERPT: Nonvanishing conjecture. For all odd numbers $d>1$, $\mathbb{D}(d)\ne 0.$ This determinant arises from the study of surgery curves for genus one slice knots. The source provides no...

conjecture | open | MathDB 页面 | 原文线索

V2-030535 | MATHDB-322202

Codimension-one manifold conjecture for near-soliton wave maps

摘录 / EXCERPT: Consider equivariant wave-map solutions that are close to a soliton in a stronger topology, with additional regularity and decay at infinity. Type 1 solutions are finite-time blow-...

conjecture | open | MathDB 页面 | 原文线索

V2-030536 | MATHDB-322233

$(n+1)$-oidal Deligne's conjecture for homotopy centers

摘录 / EXCERPT: Let an n-oidal category be a category equipped with n monoidal structures related by interchange morphisms and coherence morphisms between unit objects. Let ${\mathtt N}$ be an...

conjecture | open | MathDB 页面 | 原文线索

V2-030537 | MATHDB-322244

Seidel's homological mirror symmetry conjecture at the large complex structure and volume limits

摘录 / EXCERPT: Let Z^∞ be the large complex structure limit of Z, and let $Z^\vee_\infty$ be the large volume limit of $Z^\vee$, an affine hypersurface in an algebraic torus. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030538 | MATHDB-322248

Farkas's pure-resolution conjecture for special curves

摘录 / EXCERPT: Farkas's conjecture. The curve C has a pure resolution, equivalently it satisfies property N_p.

conjecture | solved | MathDB 页面 | 原文线索

V2-030539 | MATHDB-322256

Nadler-Tanaka's conjecture on the endomorphism spectrum of the point

摘录 / EXCERPT: Nadler-Tanaka's endomorphism-spectrum conjecture. The spectrum $\cE$ is connective and

conjecture | open | MathDB 页面 | 原文线索

V2-030540 | MATHDB-322311

The L-space conjecture for 3-manifolds

摘录 / EXCERPT: Let Y be a closed, connected 3-manifold. A group is left-orderable if it is nontrivial and admits a total order invariant under left multiplication, and Y is an L-space if it...

conjecture | open | MathDB 页面 | 原文线索

V2-030541 | MATHDB-322318

Szabó's spectral sequence and the Heegaard Floer spectral sequence

摘录 / EXCERPT: Let K be a knot in S^3. Denote by $\widetilde{E}^k(K)$ the reduced Szabó spectral sequence and by $E_{HF}^k(K)$ the Heegaard Floer spectral sequence associated to K. Szabó-He...

conjecture | open | MathDB 页面 | 原文线索

V2-030542 | MATHDB-322319

Distinguished-component independence conjecture for Szabó's reduced spectral sequence

摘录 / EXCERPT: Let L be a link, and let c and c' be components of L. Denote by $\widetilde{E}^k(L,c)$ the reduced Szabó spectral sequence associated to L with distinguished component...

conjecture | open | MathDB 页面 | 原文线索

V2-030543 | MATHDB-322320

Twin Arrows conjecture for Szabó's spectral sequence

摘录 / EXCERPT: Let L be a link. Denote by $E^k(L)$ the unreduced Szabó spectral sequence and by $\widetilde{E}^k(L)$ its reduced version; braces indicate quantum grading shifts. Twin Arrows con...

conjecture | open | MathDB 页面 | 原文线索

V2-030544 | MATHDB-322321

Agreement of Szabó spectral sequences from the two differentials

摘录 / EXCERPT: Let L be a knot or link. Let $E^k(L)$ be the Szabó spectral sequence and let $E'^k(L)$ be the spectral sequence arising from the alternate differential defined using mirror confi...

conjecture | open | MathDB 页面 | 原文线索

V2-030545 | MATHDB-322343

The self-replication conjecture for the determinantal kernel

摘录 / EXCERPT: Self-replication conjecture. The spinor ψ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-030546 | MATHDB-322344

The BKMP remodeling conjecture for toric Calabi-Yau threefolds

摘录 / EXCERPT: BKMP remodeling conjecture. The generating series of Gromov-Witten invariants of $\mathfrak{X}$ are given by topological recursion applied to $\mathcal{S}_{\mathfrak{X}}$; in parti...

conjecture | open | MathDB 页面 | 原文线索

V2-030547 | MATHDB-322377

Conjecture on special-cycle generation of cuspidal cohomology

摘录 / EXCERPT: Let Y be the arithmetic manifold, let $E(\lambda)$ be the coefficient system, and let $H^{n\times q}_{\mathrm{cusp}}(Y,E(\lambda))$ denote the indicated cuspidal cohomology subsp...

conjecture | open | MathDB 页面 | 原文线索

V2-030548 | MATHDB-322378

Folklore conjecture on Arthur packets and Adams-Johnson packets

摘录 / EXCERPT: Let G be the relevant real reductive group, let L be a Levi subgroup, and let Ψ be an Adams-Johnson parameter, meaning a local Arthur parameter ... that factors through...

conjecture | open | MathDB 页面 | 原文线索

V2-030549 | MATHDB-322385

Hard Lefschetz conjecture for Lawson homology

摘录 / EXCERPT: Hard Lefschetz conjecture for Lawson homology. The map $h^k\bullet$ is injective for $k\geq1$. This conjecture is presented as a Lawson-homology analogue of hard Lefschetz; the sup...

conjecture | open | MathDB 页面 | 原文线索

V2-030550 | MATHDB-322389

Spectral-radius conjecture for pseudo-Anosov subsurface mapping classes

摘录 / EXCERPT: Let Σ be a surface and let ψ be a mapping class which is pseudo-Anosov when restricted to some connected subsurface of Σ. A finite cover is a covering map...

conjecture | open | MathDB 页面 | 原文线索

V2-030551 | MATHDB-322391

Toric quantum sheaf cohomology ring conjecture

摘录 / EXCERPT: Toric quantum sheaf cohomology ring conjecture. There is a toric quantum sheaf cohomology ring $QH^*_{\sheaf E}(X)^{\mathrm{toric}}$ of the form

conjecture | open | MathDB 页面 | 原文线索

V2-030552 | MATHDB-322419

The dimension conjecture for spaces of real places of rational function fields

摘录 / EXCERPT: Dimension conjecture. For every such K and n,

conjecture | open | MathDB 页面 | 原文线索

V2-030553 | MATHDB-322449

The global-dimension extension for trace monoid diagram categories

摘录 / EXCERPT: Global-dimension extension. This is true for all Abelian categories with coproducts.

conjecture | open | MathDB 页面 | 原文线索

V2-030554 | MATHDB-322452

The ABP conjecture for periodic cyclic homology

摘录 / EXCERPT: Let ${\mathcal G}$ be a reductive p-adic group and let ${\mathfrak s}$ be a point in the Bernstein spectrum of ${\mathcal G}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030555 | MATHDB-322453

The recurrence-set characterization of d-step almost automorphy

摘录 / EXCERPT: Let (X,T) be a minimal topological dynamical system, let $x\in X$, and let $d\in\mathbb{N}$. Let $\mathcal{F}_{P_d}$ and $\mathcal{F}_{B_d}$ denote, respectively, the families of...

conjecture | open | MathDB 页面 | 原文线索

V2-030556 | MATHDB-322490

Optimality conjecture for knots obtained by braid substitution

摘录 / EXCERPT: Let L be a link with a free decomposing surface S, and replace the trivial braids in $N(S)\cap L$ by an arbitrary collection of braids. The resulting link is obtained by braid...

conjecture | open | MathDB 页面 | 原文线索

V2-030557 | MATHDB-322544

The type D loop-space quotient conjecture

摘录 / EXCERPT: Let $\rho_{n-1}^{(1)}$ and $\rho_{n-1}^{(2)}$ be the two specified affine Grassmannian elements, let ... and let $\Gamma_{(n)}^D=\mathbb{Z}[Q_1,\ldots,Q_{n-1},2Q_n,\ldots,2Q_{2n-2}]$...

conjecture | open | MathDB 页面 | 原文线索

V2-030558 | MATHDB-322555

Completeness of the cohomology obstruction for strong contextuality

摘录 / EXCERPT: A measurement scenario consists of a set of measurements together with a family of contexts, and a probabilistic empirical model assigns compatible probability distributions to the...

conjecture | open | MathDB 页面 | 原文线索

V2-030559 | MATHDB-322573

González-Zarati lower-bound conjecture for biequivariant maps

摘录 / EXCERPT: Let $\alpha(m)$ denote the number of ones in the binary expansion of m, and let $b(n,e)$ be the smallest positive integer r for which there is a $\mathbb{Z}_{2^e}$-biequivarian...

conjecture | open | MathDB 页面 | 原文线索

V2-030560 | MATHDB-322574

Conjecture on the ku-filtration of the toral class

摘录 / EXCERPT: Let ku be the connective cover of complex K-theory, and let $v_1\in ku_2$ correspond to Bott periodicity. For $k\geq e$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030561 | MATHDB-322575

Non-annihilation conjecture for the toral class

摘录 / EXCERPT: Let ku be the connective cover of complex K-theory, let $v_1\in ku_2$ correspond to Bott periodicity, and let $\tau_{k,e}\in ku_2B(\mathbb{Z}_{2^k}\times\mathbb{Z}_{2^e})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-030562 | MATHDB-322582

Vanishing homology conjecture for semi-lattices with a projector

摘录 / EXCERPT: Let L be a semi-lattice with a projector, and let $H_n(L,t)$ denote its degree-n distributive homology with coefficients determined by scalars (a,b). Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-030563 | MATHDB-322583

Orbit invariance conjecture for commutative spindles

摘录 / EXCERPT: Let $(X,\star)$ be a commutative spindle, meaning a set equipped with a self-distributive binary operation satisfying the spindle axioms. For $x,t\in X$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-030564 | MATHDB-322584

Homology decomposition conjecture for skew lattices with one extremum

摘录 / EXCERPT: Let L be a finite skew lattice, and let $L/\!\!\sim$ denote its quotient by the relevant equivalence relation. Assume that L has either a unique minimum or a unique maximum. Ho...

conjecture | open | MathDB 页面 | 原文线索

V2-030565 | MATHDB-322594

The punctured-torus skeleton conjecture for the Fukaya category

摘录 / EXCERPT: Let (Σ,Γ_Σ) be a punctured Riemann surface together with its skeleton, and let $\mathrm{CPM}(-)$ be the sheaf of dg categories on Γ_Σ whose local restr...

conjecture | open | MathDB 页面 | 原文线索

V2-030566 | MATHDB-322613

The mod-2 Betti-number characterization of splitting manifolds

摘录 / EXCERPT: Let M be a manifold. Say that M splits when it is divided by every divisor, and let β_1 be the first Betti number of M with coefficients in $\mathbb{Z}/2\mathbb{Z}$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-030567 | MATHDB-322614

Hausdorffness conjecture for leaf-spaces of omega-bounded surfaces

摘录 / EXCERPT: Let S be a foliated simply-connected surface, and write $S/\mathcal F$ for its leaf-space. A surface is ω-bounded when every countable subset has compact closure. The leaf...

conjecture | open | MathDB 页面 | 原文线索

V2-030568 | MATHDB-322615

Parallel-leaves conjecture for omega-bounded foliated absolutes

摘录 / EXCERPT: An absolute is a simply-connected surface, and a pseudo-line is a connected divisor in an absolute that is not a circle. Three pseudo-lines are parallel when one of them divides th...

conjecture | open | MathDB 页面 | 原文线索

V2-030569 | MATHDB-322616

The Euler obstruction conjecture for foliations of omega-bounded surfaces

摘录 / EXCERPT: Let S be an ω-bounded surface, meaning that every countable subset has compact closure, and let $\chi(S)$ denote its Euler characteristic. The Euler obstruction conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030570 | MATHDB-322617

Foliation conjecture for pseudo-Moore surfaces

摘录 / EXCERPT: A pseudo-Moore surface is a separable, simply-connected, non-metric surface. The pseudo-Moore foliation conjecture. Every pseudo-Moore surface admits a foliation. The source presen...

conjecture | open | MathDB 页面 | 原文线索

V2-030571 | MATHDB-322627

DAHA-Khovanov-Rozansky correspondence conjecture

摘录 / EXCERPT: Let $H\!D_{r,s}(b;q,t,a)$ have positive coefficients, let $\mathcal{H}=\bigoplus\mathcal{H}_{i,j,k}$ be the triply graded space for the torus knot $\{r,s\}$ and weight b, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030572 | MATHDB-322637

The homological hyperbolicity conjecture for geometric realization

摘录 / EXCERPT: Let Φ be a substitution with tiling space Ω_{Φ}, and let H^1_{hyp} be the largest subgroup of $H^1(\Omega_{\Phi};\mathbb Z)$ containing the Φ^*-invariant subg...

conjecture | open | MathDB 页面 | 原文线索

V2-030573 | MATHDB-322639

The essential cohomology conjecture for geometric realization

摘录 / EXCERPT: Let Φ be a substitution with tiling space Ω_{Φ}. Let $H^1_{hyp}(\Omega_{\Phi};\mathbb Z)$ be the hyperbolic part of first cohomology, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030574 | MATHDB-322640

The essential homological Pisot conjecture

摘录 / EXCERPT: Let Φ be a unimodular Pisot family substitution with expansion Λ, tiling space Ω_{Φ}, and essential cohomology $H^1_{ess}(\Omega_{\Phi};\mathbb Z)$. The esse...

conjecture | open | MathDB 页面 | 原文线索

V2-030575 | MATHDB-322647

The smash-operation conjecture for path-connected topological operads

摘录 / EXCERPT: Let $\mathscr{C}$ be a path-connected topological operad with a basepoint, let Y be a $\mathscr{C}$-space, and let $\alpha\in [S^l,\mathscr{C}(k)]$ be such that all $d_i\alpha$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-030576 | MATHDB-322648

The central-element conjecture for smash operations

摘录 / EXCERPT: Let $\mathscr{C}$ be an operad with a basepoint, let $k\geq 2$, let $l\geq 1$, and let Y be a $\mathscr{C}$-space. Write $\mathcal{Z}_k[S^l,\mathscr{C}]$ for the relevant k-th...

conjecture | open | MathDB 页面 | 原文线索

V2-030577 | MATHDB-322649

The conjecture that the maps ϕ_n and ψ_n are homotopic

摘录 / EXCERPT: Let X be the space in the preceding construction, let $n\geq 2$, and let ϕ_n and ψ_n be the two maps defined there from the relevant smash-product construction into...

conjecture | open | MathDB 页面 | 原文线索

V2-030578 | MATHDB-322650

The conjecture that the induced smash operation equals the Whitehead product

摘录 / EXCERPT: Let X be a pointed space, let $n\geq 1$, and let $\widetilde{\theta}_{\iota}$ be the operation on homotopy groups induced by the distinguished class ι. For $i,j\geq 1$, ide...

conjecture | open | MathDB 页面 | 原文线索

V2-030579 | MATHDB-322695

Conjecture on tame local monodromy at zero

摘录 / EXCERPT: Local monodromy conjecture at zero. Under ρ_k, a topological generator of the tame inertia group $I_0^{\operatorname{tame}}$ gets mapped to an element conjugate to...

conjecture | open | MathDB 页面 | 原文线索

V2-030580 | MATHDB-322696

Conjecture on cohomological rigidity of the exceptional local system

摘录 / EXCERPT: Cohomological rigidity conjecture. The local system $\mathcal{E}$ is cohomologically rigid, i.e.

conjecture | open | MathDB 页面 | 原文线索

V2-030581 | MATHDB-322726

The cobracket criterion for generalized Dehn twists

摘录 / EXCERPT: Cobracket criterion conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-030582 | MATHDB-322727

The Zariski-closure conjecture for the mapping class group image

摘录 / EXCERPT: Zariski-closure conjecture. The Zariski closure of $\tau(\mathcal{I}(S,E))$ equals

conjecture | open | MathDB 页面 | 原文线索

V2-030583 | MATHDB-322729

The mirror duality conjecture for the Khovanov stable homotopy type

摘录 / EXCERPT: Let L be a link, let $m(L)$ denote its mirror, and let $X^\vee$ denote the Spanier-Whitehead dual of a spectrum X. The Khovanov stable homotopy type in quantum grading j is d...

conjecture | open | MathDB 页面 | 原文线索

V2-030584 | MATHDB-322730

The disjoint-union formula for the Khovanov stable homotopy type

摘录 / EXCERPT: Let L_1 and L_2 be links, and let $L_1\amalg L_2$ denote their disjoint union. Write $\mathcal{X}_\mathit{Kh}^j(L)$ for the Khovanov stable homotopy type in quantum grading j...

conjecture | open | MathDB 页面 | 原文线索

V2-030585 | MATHDB-322731

The connected-sum formula for reduced Khovanov stable homotopy types

摘录 / EXCERPT: Let L_1 and L_2 be based links, and let $L_1\#L_2$ denote their connected sum taken near the basepoints. Write $\widetilde{\mathcal{X}}_\mathit{Kh}^j(L)$ for the reduced Khovan...

conjecture | open | MathDB 页面 | 原文线索

V2-030586 | MATHDB-322732

The module tensor-product conjecture for Khovanov stable homotopy types

摘录 / EXCERPT: Let L_1 and L_2 be based links. Let $\mathcal{X}_\mathit{Kh}(U)$ denote the Khovanov stable homotopy type of the unknot, and regard each $\mathcal{X}_\mathit{Kh}(L_i)$ as a mod...

conjecture | open | MathDB 页面 | 原文线索

V2-030587 | MATHDB-322743

Injectivity and differentiability conjecture for the renormalization operator

摘录 / EXCERPT: Let $\bb$ be the space of maps, let $\td$ be the domain of the renormalization operator $\tt$, let $\bb_0$ be the topology space, and let $\rr$ be the renormalization operator with...

conjecture | open | MathDB 页面 | 原文线索

V2-030588 | MATHDB-322774

The entropy conjecture for exactly Devaney chaotic maps on totally regular continua

摘录 / EXCERPT: Let X be a totally regular continuum, meaning a continuum in which every two points can be joined by an arc of finite length. Write $I^{\operatorname{ED}}(X)$ for the infimum of...

conjecture | open | MathDB 页面 | 原文线索

V2-030589 | MATHDB-322789

Dikranjan's conjecture on the coarsest Hausdorff topology of permutation groups

摘录 / EXCERPT: Dikranjan's conjecture. The topology of pointwise convergence is the coarsest Hausdorff group topology on G.

conjecture | open | MathDB 页面 | 原文线索

V2-030590 | MATHDB-322825

The strongly star-Hurewicz game nonimplication

摘录 / EXCERPT: Let X be a topological space. The strongly star-Hurewicz property means that for every sequence $(\mathcal U_n:n\in\mathbb N)$ of open covers of X, there are finite sets...

conjecture | open | MathDB 页面 | 原文线索

V2-030591 | MATHDB-322847

Smoothness conjecture for homotopy fiber bundles with smooth total space

摘录 / EXCERPT: Let $f:X\to Y$ be a proper morphism of complex spaces. It is a homotopy fiber bundle if locally over Y each fiber inclusion is a homotopy equivalence; it is a ${\mathbb Z}$-homol...

conjecture | open | MathDB 页面 | 原文线索

V2-030592 | MATHDB-322873

Heegaard Floer correction-term conjecture for finite knot surgeries

摘录 / EXCERPT: A finite manifold Y is a rational homology sphere with finite fundamental group, and $d(Y,\mathfrak{t})$ denotes its Heegaard Floer correction term for a $b\mathfrak{t}d$ spin...

conjecture | open | MathDB 页面 | 原文线索

V2-030593 | MATHDB-322894

Extension of the deformed Pachner relation to the 2-to-4 move

摘录 / EXCERPT: Extension conjecture. The analogues of these two theorems hold also for a $2\to 4$ Pachner move.

conjecture | open | MathDB 页面 | 原文线索

V2-030594 | MATHDB-322905

Axiomatization of dynamic topological logic over minimal systems

摘录 / EXCERPT: Let $\mathcal{DTL}^\ast$ be the dynamic topological logic with infinitary temporal modalities, and let $\mathsf L^\ast_{\Diamond f [f]\forall}$ be its monadic language with a unive...

conjecture | open | MathDB 页面 | 原文线索

V2-030595 | MATHDB-322933

Cohomological finiteness conjecture for wrapped categories of negative line bundles

摘录 / EXCERPT: Let $E\to B$ be a monotone negative line bundle. Suppose that the open-closed map ... hits an invertible element for some full subcategory $\mathcal{S}\subset\mathcal{F}(B)$. Cohomol...

conjecture | open | MathDB 页面 | 原文线索

V2-030596 | MATHDB-322934

Non-nilpotence of the first Chern class for closed monotone toric manifolds

摘录 / EXCERPT: Let B be a closed monotone toric manifold, and let $c_1(TB)\in QH^*(B)$ denote the first Chern class of its tangent bundle. Non-nilpotence conjecture. The element...

conjecture | open | MathDB 页面 | 原文线索

V2-030597 | MATHDB-322971

The universal tunnel-number bound conjecture for Dehn surgeries

摘录 / EXCERPT: Universal tunnel-number bound conjecture. There is a function

conjecture | open | MathDB 页面 | 原文线索

V2-030598 | MATHDB-322990

Norbury-Scott conjecture on stationary Gromov-Witten invariants of the projective line

摘录 / EXCERPT: Let $W_{g,n}^{\mathbb{P}^1}(t_1,\dots,t_n)$ denote the Eynard-Orantin differential forms associated with the stationary Gromov-Witten theory of $\mathbb{P}^1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030599 | MATHDB-322991

Infinite-rank conjecture for elliptic curves over fields with finitely generated Galois group

摘录 / EXCERPT: Let K) be a field that is not locally finite, and let ... be topologically finitely generated. Let E/K be a non-trivial elliptic curve. Infinite-rank conjecture. The elliptic cur...

conjecture | open | MathDB 页面 | 原文线索

V2-030600 | MATHDB-323025

Bridge-index conjecture for discrete Morse vectors of knots

摘录 / EXCERPT: Let K be a knot with bridge index b. Say that a d-complex has discrete Morse vector $(c_0,\ldots,c_d)$ if it admits a discrete Morse function with exactly c_i critical face...

conjecture | open | MathDB 页面 | 原文线索

V2-030601 | MATHDB-323037

Common-limit conjecture for renormalization fixed points of one-dimensional maps

摘录 / EXCERPT: Suppose a topological class of one-dimensional maps consists of maps with one critical value, and suppose that, for each critical exponent in a sequence tending to ∞, limits...

conjecture | open | MathDB 页面 | 原文线索

V2-030602 | MATHDB-323087

Serre goodness conjecture for braid groups of finite complex reflection groups

摘录 / EXCERPT: Let W be an irreducible finite complex reflection group, let B be its braid group, and let P be its pure braid group. A group is good in the sense of Serre when its cohomolog...

conjecture | open | MathDB 页面 | 原文线索

V2-030603 | MATHDB-323088

Nontriviality and Ohtsuki-invariant recovery of the higher-order Fukaya invariant

摘录 / EXCERPT: Let M be a homology 3-sphere and, for each $k\geq 1$, let $\widehat{Z}_{2k,3k}(M)$ denote the invariant constructed from the higher-order configuration-space integral. Nontrivi...

conjecture | open | MathDB 页面 | 原文线索

V2-030604 | MATHDB-323089

Axelrod-Singer-Kontsevich identification conjecture for the higher-order Fukaya invariant

摘录 / EXCERPT: Let M be a homology 3-sphere and let $\widehat{Z}_{2k,3k}(M)$ be the invariant defined from the higher-order configuration-space construction. The configuration-space integral...

conjecture | open | MathDB 页面 | 原文线索

V2-030605 | MATHDB-323110

The 1-loop torsion conjecture for hyperbolic knot complements

摘录 / EXCERPT: Let M be a cusped hyperbolic 3-manifold, let τ_M be the topological invariant obtained from the 1-loop invariant on the Epstein-Penner component, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030606 | MATHDB-323116

The horizontal nonvanishing conjecture for big Heegner points

摘录 / EXCERPT: Horizontal nonvanishing conjecture. The cohomology class $\mathfrak{Z}_0$ is not R-torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-030607 | MATHDB-323120

Generalized Tait conjecture IV on amphicheiral links

摘录 / EXCERPT: Let L be an amphicheiral link in S^3, meaning that an orientation-reversing homeomorphism of S^3 maps each component of L to a component of L, possibly reversing orientat...

conjecture | open | MathDB 页面 | 原文线索

V2-030608 | MATHDB-323145

Serre's intersection multiplicity conjecture

摘录 / EXCERPT: Serre's intersection multiplicity conjecture. The following assertions hold: 1. $\operatorname{Kdim}M+\operatorname{Kdim}N\leq\operatorname{Kdim}R$. 2. If...

conjecture | open | MathDB 页面 | 原文线索

V2-030609 | MATHDB-323161

Equality of totally positive and quantum Schubert positive subsets

摘录 / EXCERPT: Positivity equality conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-030610 | MATHDB-323189

The straightbrush model conjecture for the residual compact set

摘录 / EXCERPT: Let K_{addr} be the addresses odometer, identified with the Cantor set of connected components of K_∞. A subset $B\subset K_{\mathrm{addr}}\times[0,1]$ is require...

conjecture | open | MathDB 页面 | 原文线索

V2-030611 | MATHDB-323200

Pi-formality conjecture for bordered Floer algebras

摘录 / EXCERPT: Let $\mathcal{Z}$ be an arc diagram and let i be an integer. Pi-formality conjecture. The algebra $\mathcal{A}(\mathcal{Z},i)$ is π-formal, in the sense of Definition. This a...

conjecture | open | MathDB 页面 | 原文线索

V2-030612 | MATHDB-323207

Higher dimensional de Rham Theorem

摘录 / EXCERPT: Let X be a smooth manifold, let $x_0\in X$, and let $B_s^n(X)_{x_0}^{hom}$ denote the space of homotopy-invariant higher-dimensional iterated integrals of length at most s on b...

conjecture | open | MathDB 页面 | 原文线索

V2-030613 | MATHDB-323214

The uniform homeomorphism group conjecture for Euclidean spaces

摘录 / EXCERPT: Let $\mathcal H^u(\mathbb R^n)_b$ denote the group of bounded uniform homeomorphisms of Euclidean space $\mathbb R^n$, equipped with the sup-metric, and let ℓ_∞ be the Ba...

conjecture | open | MathDB 页面 | 原文线索

V2-030614 | MATHDB-323217

Bazzoni-Glaz conjecture on the weak global dimension of Gaussian rings

摘录 / EXCERPT: Let R be a commutative ring that is Gaussian, meaning that every polynomial in $R[x]$ satisfies $c(f)c(g)=c(fg)$ for all $g\in R[x]$, where $c(f)$ is the ideal generated by the c...

conjecture | open | MathDB 页面 | 原文线索

V2-030615 | MATHDB-323248

Colin-Honda growth-rate conjecture for contact homology

摘录 / EXCERPT: Let M be a closed 3-manifold with a geometric structure, and let ξ be a universally tight contact structure on M. Consider the growth rate of contact homology associated...

conjecture | open | MathDB 页面 | 原文线索

V2-030616 | MATHDB-323250

The 5-torsion conjecture for matching complexes

摘录 / EXCERPT: Let n,d,k,r be related by ... and let M_n denote the matching complex of the complete graph on n vertices. 5-torsion conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-030617 | MATHDB-323251

The split exactness conjecture for matching-complex deletion sequences

摘录 / EXCERPT: Split exactness conjecture. This sequence, obtained by cutting the long exact 0-e-2 sequence, is split exact for every $n\geq3$ and every d.

conjecture | open | MathDB 页面 | 原文线索

V2-030618 | MATHDB-323252

Shareshian-Wachs conjecture on torsion in chessboard-complex homology

摘录 / EXCERPT: Let $\mathsf{M}_{m,n}$ denote the chessboard complex, and let $\tilde{H}_{d}(\mathsf{M}_{m,n};\mathbb{Z})$ be its reduced homology. For the parametrization used in the source, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030619 | MATHDB-323253

Jonsson's 3-torsion conjecture for matching complexes

摘录 / EXCERPT: Let $n\geq 1$ and write ... Thus the relevant reduced homology group is $\tilde{H}_{k-1+r}(\mathsf{M}_{2k+1+3r};\mathbb{Z})$. Jonsson's 3-torsion conjecture. The group contains eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-030620 | MATHDB-323254

The 3-rank conjecture for matching complexes

摘录 / EXCERPT: For $k\geq 0$, consider the reduced homology group $\tilde{H}_{k+1}(\mathsf{M}_{2k+7};\mathbb{Z})$, where M_n denotes the matching complex on n vertices. The 3-rank co...

conjecture | open | MathDB 页面 | 原文线索

V2-030621 | MATHDB-323256

The 7-torsion conjecture for matching complexes

摘录 / EXCERPT: Let $n=2k+1+3r$ and $d=k-1+r$, so that the relevant reduced homology group is $\tilde{H}_{k-1+r}(\mathsf{M}_{2k+1+3r};\mathbb{Z})$. The 7-torsion conjecture. The group contains ele...

conjecture | open | MathDB 页面 | 原文线索

V2-030622 | MATHDB-323263

Vanishing of the tail-homology direct limit for non-B-adequate diagrams

摘录 / EXCERPT: Let D be a link diagram. For each N, let $\widetilde{\mathrm{H}}^{\mathrm{Kh}}(\xDclv{N})$ be its shifted Khovanov homology, and let ... be the direct limit of the degree-preserv...

conjecture | open | MathDB 页面 | 原文线索

V2-030623 | MATHDB-323264

Tail stabilization of the colored Jones polynomial

摘录 / EXCERPT: Let L be a link and let $\nu\in\{0,1\}$. The colored Jones polynomial is denoted by $J_N(L;q)$. Tail-stabilization conjecture. There exists a series...

conjecture | open | MathDB 页面 | 原文线索

V2-030624 | MATHDB-323265

Vanishing of the two-variable series for non-B-adequate links

摘录 / EXCERPT: Let L be a framed link, and let $J_{\thicksim}(L)$ denote the two-variable Laurent series associated with L in the source. A framed link is B-adequate if it has the relevant B-...

conjecture | open | MathDB 页面 | 原文线索

V2-030625 | MATHDB-323298

Kotschick-Löh's domination conjecture by hyperbolic manifolds

摘录 / EXCERPT: Let M^n be an oriented closed manifold, where $n\ge 2$. A manifold is hyperbolic if it admits a Riemannian metric of constant negative curvature. Kotschick-Löh's conjecture. Ever...

conjecture | open | MathDB 页面 | 原文线索

V2-030626 | MATHDB-323300

Existence of hyperbolic URC-manifolds in every dimension

摘录 / EXCERPT: For each integer $n\ge 2$, let a hyperbolic URC-manifold mean a URC-manifold admitting a Riemannian metric of constant negative curvature. Hyperbolic URC-manifold conjecture. In ev...

conjecture | open | MathDB 页面 | 原文线索

V2-030627 | MATHDB-323336

Regularity of subFinsler spheres in Carnot groups

摘录 / EXCERPT: Regularity conjecture. The unit sphere of d_∞ is rectifiable with respect to every Riemannian distance on G_∞. In particular, if G_∞ has topological dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-030628 | MATHDB-323340

Projection weak homotopy equivalence conjecture for twisted master operators

摘录 / EXCERPT: Let Γ be the finite graph under consideration, let $\mathcal M_{\Gamma}$ be its parameter space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030629 | MATHDB-323341

Equality of the Chern-class slant-product homomorphisms

摘录 / EXCERPT: Equality of the Chern-class slant-product homomorphisms. The homomorphisms $c_1(\eta)/$ and $c_1(\xi)/$ coincide. This asserts that the analytically constructed line bundle and the...

conjecture | open | MathDB 页面 | 原文线索

V2-030630 | MATHDB-323345

The homotopy-lifting conjecture for secondary constructions

摘录 / EXCERPT: Homotopy-lifting conjecture. All secondary constructions come from suitable homotopy structures at the level of characteristic cochains.

conjecture | open | MathDB 页面 | 原文线索

V2-030631 | MATHDB-323350

Heegaard Floer rank invariance under genus two mutation of three-manifolds

摘录 / EXCERPT: Let M be a closed, oriented three-manifold with an embedded genus two surface F. Let τ be the hyperelliptic involution of F, and let M^τ denote the genus two mutant...

open_problem | open | MathDB 页面 | 原文线索

V2-030632 | MATHDB-323403

Separability conjecture for projective Banach lattices with a topological order unit

摘录 / EXCERPT: A projective Banach lattice is a Banach lattice with the projective lifting property, and a topological order unit is an element whose principal order ideal is dense in the lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-030633 | MATHDB-323404

Freeness conjecture for projective Banach lattices without a topological order unit

摘录 / EXCERPT: Let X be a projective Banach lattice, and let a topological order unit mean an element whose principal order ideal is dense in X. A Banach lattice is free when it is a free Ban...

conjecture | open | MathDB 页面 | 原文线索

V2-030634 | MATHDB-323424

Even-degree homology conjecture for graph moduli spaces

摘录 / EXCERPT: Even-degree homology conjecture. For any graph Γ and every odd integer i,

conjecture | open | MathDB 页面 | 原文线索

V2-030635 | MATHDB-323449

The Dyck's surface conjecture for non-integral surgery on hyperbolic knots

摘录 / EXCERPT: Let K' be a hyperbolic knot in S^3, let γ be a surgery slope, and let μ be a meridian of K'. Suppose that $K'(\gamma)$ contains an embedded Dyck's surface, meaning...

conjecture | open | MathDB 页面 | 原文线索

V2-030636 | MATHDB-323455

Surface homological eigenvalue conjectures for pseudo-Anosov mapping classes

摘录 / EXCERPT: Surface conjectures. There exists a lift $\widetilde{\psi}$ whose action on $H_1(\widetilde{S},\mathbb{C})$ has infinite order. In fact, there exists a lift $\widetilde{\psi}$ whos...

conjecture | open | MathDB 页面 | 原文线索

V2-030637 | MATHDB-323456

Torsion homology growth conjectures for hyperbolic mapping tori

摘录 / EXCERPT: Torsion conjectures. The manifold T has exponential torsion homology growth with respect to some tower of finite covers. In fact, T has exponential torsion homology growth with...

conjecture | open | MathDB 页面 | 原文线索

V2-030638 | MATHDB-323457

Universal abelian-cover torsion growth conjecture for pseudo-Anosov mapping tori

摘录 / EXCERPT: Torsion conjecture (T1.5). There exists a finite cover X of T_ψ and a universal tower of finite abelian covers $\{X_i\}$ of X which have exponential torsion homology growt...

conjecture | open | MathDB 页面 | 原文线索

V2-030639 | MATHDB-323461

Soul bundle conjecture for topologically nice Alexandrov spaces

摘录 / EXCERPT: Soul bundle conjecture. There exists a positive number ϵ such that

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V2-030640 | MATHDB-323499

The non-perturbative topological-recursion solution conjecture

摘录 / EXCERPT: Let $\mathfrak{M}$ be a hyperbolic 3-manifold with one cusp, let $(\mathcal{C}_0,u,v)$ be an A-spectral curve from an irreducible component of its A-polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030641 | MATHDB-323523

Gal's classification conjecture for gamma-vectors of flag 3-GHSs

摘录 / EXCERPT: Let K be a flag generalized homology 3-sphere, or flag 3-GHS, with γ-vector $(1,\gamma_1,\gamma_2)$. A join of two polygons is the simplicial join of the boundary comp...

conjecture | open | MathDB 页面 | 原文线索

V2-030642 | MATHDB-323549

Qazaqzeh-Chbili-Qublan's characterization conjecture for quasi-alternating Montesinos links

摘录 / EXCERPT: A Montesinos link is a link of the form $M(e;t_1,dots,t_p)$, with the parameters and notation as in Theorem 5.3 and Theorem 5.1 of the source. Qazaqzeh-Chbili-Qublan's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030643 | MATHDB-323563

Equivalence of unitality and homotopy unitality for A_∞-morphisms

摘录 / EXCERPT: Unitality-homotopy unitality equivalence. An A_∞-morphism is unital if and only if it admits a homotopy unital structure; equivalently, any unital A_∞-morphism admits...

conjecture | open | MathDB 页面 | 原文线索

V2-030644 | MATHDB-323576

Typical-measure multifractal spectrum conjecture for compact subsets of Euclidean space

摘录 / EXCERPT: Let $K\subset\mathbb{R}^{d}$ be a compact set, and let $\mathcal{M}(K)$ be the space of probability measures on K with its weak topology. For a measure $\mu\in\mathcal{M}(K)$, wr...

conjecture | open | MathDB 页面 | 原文线索

V2-030645 | MATHDB-323601

Topology-refinement reformulation of the strong Jayne-Rogers conjecture

摘录 / EXCERPT: Let X be an analytic space, Y a separable metrizable space, $1<n<m<\omega$, and $f\in\Sigma_{m,n}(X,Y)$. Topology-refinement reformulation of the strong Jayne-Rogers conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-030646 | MATHDB-323607

Higher-dimensional black-hole topology conjecture with maximal rotational symmetry

摘录 / EXCERPT: Let $(\mathbb{T}{M},g)$ be a spin spacetime that is asymptotically flat or Kaluza-Klein, whose Ricci tensor satisfies the null-convergence condition, and that admits an action of...

conjecture | open | MathDB 页面 | 原文线索

V2-030647 | MATHDB-323612

Geometric lifts conjecture for zero-dimensional homology interleavings

摘录 / EXCERPT: Geometric lifts conjecture. There exists a CW-complex X and continuous functions $ $gammaM, $ $ $\gamma_N:X\to \mathbb R^n$ $ such that

conjecture | open | MathDB 页面 | 原文线索

V2-030648 | MATHDB-323615

Fukaya's cohomology F-manifold mirror symmetry conjecture

摘录 / EXCERPT: Let M and $M^{\vee}$ be open Calabi-Yau manifolds that are mirror to each other in the sense of homological mirror symmetry. Write $SH^*(M)$ for the symplectic cohomology of M,...

conjecture | open | MathDB 页面 | 原文线索

V2-030649 | MATHDB-323633

Gukov-Walcher potentials for colored homology of the unknot

摘录 / EXCERPT: Let $W_{\mathfrak{sl}(N),S^k}(x_1,\ldots,x_k)$ denote the potential associated with the totally symmetric representation S^k of $\mathfrak{sl}(N)$. Gukov-Walcher conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-030650 | MATHDB-323635

Topological invariance conjecture for the partially abelianized characteristic algebra

摘录 / EXCERPT: Topological invariance conjecture. Up to equivalence, the ungraded partially abelianized characteristic algebra of a Legendrian knot K depends only on the smooth knot type and...

conjecture | open | MathDB 页面 | 原文线索

V2-030651 | MATHDB-323641

Ohtsuki's asymptotic expansion conjecture for quantum invariants

摘录 / EXCERPT: Let M be a closed oriented 3-manifold and let $G=\operatorname{SU}(N)$. Set $r=k+N$, and let $\{c_0=0,\dots,c_m\}$ be the finitely many values of the Chern-Simons action on the...

conjecture | open | MathDB 页面 | 原文线索

V2-030652 | MATHDB-323643

Stretch factor conjecture for quantum invariants of mapping tori

摘录 / EXCERPT: Let $G=\operatorname{SU}(N)$, let Σ_g be a closed surface of genus g, and let $\varphi\in\Gamma_g$ be a mapping class with stretch factor λ. Write T_{φ^n}...

conjecture | open | MathDB 页面 | 原文线索

V2-030653 | MATHDB-323644

A formula for correction terms of Brieskorn spheres

摘录 / EXCERPT: Let $p>1$ be an odd integer, let k be an integer satisfying $\gcd(k,2p)=1$ and $k\not\equiv\pm1\pmod{2p}$, and let n be an integer. For a rational homology sphere Y, write...

conjecture | open | MathDB 页面 | 原文线索

V2-030654 | MATHDB-323649

Nucinkis's conjecture on finite-dimensional classifying spaces for proper actions

摘录 / EXCERPT: Nucinkis's conjecture. Every group of finite $\operatorname{\mathfrak{F}}$-cohomological dimension admits a finite-dimensional model for...

conjecture | open | MathDB 页面 | 原文线索

V2-030655 | MATHDB-323658

Grothendieck's homotopy hypothesis for infinity-groupoids

摘录 / EXCERPT: Grothendieck's conjecture. The functor $\overline{\Pi_\infty}$ is an equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-030656 | MATHDB-323672

Kontsevich's mapping-spectrum conjecture for smooth noncommutative spaces

摘录 / EXCERPT: Kontsevich's conjecture. There is a natural equivalence of spectra

conjecture | open | MathDB 页面 | 原文线索

V2-030657 | MATHDB-323705

Special polynomial conjecture for colored HOMFLY polynomials

摘录 / EXCERPT: Special polynomial conjecture. The following identities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-030658 | MATHDB-323712

The higher-jet transversality codimension conjecture

摘录 / EXCERPT: Higher-jet transversality codimension conjecture. The set $\tilde A$ is a countable union of smooth submanifolds of codimension more than $n=\dim X$ in $J^{p+1}(X,Y)$.

conjecture | open | MathDB 页面 | 原文线索

V2-030659 | MATHDB-323746

Gournay dimension-zero characterization by invariant exhaustions

摘录 / EXCERPT: Gournay's dimension-zero conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030660 | MATHDB-323770

Convolution convergence conjecture for affine-Grassmannian clasps

摘录 / EXCERPT: Let $\mathcal{K}_{\mathrm{Gr},m}$ be the geometric categorical $\mathfrak{sl}_\infty$ 2-representation, let $\mathcal{K}^{\pm}_{\mathrm{Gr},m}$ denote its bounded-sided homotopy ca...

conjecture | open | MathDB 页面 | 原文线索

V2-030661 | MATHDB-323777

Nandakumar-Ramana Rao conjecture for convex bodies

摘录 / EXCERPT: Let K be a planar convex body and let $n>1$ be a natural number. A partition of the plane into n convex pieces is a collection of convex pieces $P_1,\ldots,P_n$ whose union is...

conjecture | open | MathDB 页面 | 原文线索

V2-030662 | MATHDB-323778

Generalized Nandakumar-Ramana Rao conjecture for convex bodies and measures

摘录 / EXCERPT: Let K be a convex body in $\mathbb{R}^d$, let μ be an absolutely continuous probability measure on $\mathbb{R}^d$, let $n>1$ be a natural number, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-030663 | MATHDB-323785

Infinite loop space machine for E-infinity ring G-categories

摘录 / EXCERPT: An E_{∞} ring G-category is a G-category with an action by an E_{∞} operad pair $(\mathcal{P}_G,\mathcal{R}_G)$ whose multiplicative action extends from...

conjecture | open | MathDB 页面 | 原文线索

V2-030664 | MATHDB-323788

Equivalence conjecture for the hammock localization of decorated correspondences

摘录 / EXCERPT: Let ${\mathcal F}_n$ be the category of decorated objects, let ${\mathcal E}_n$ be the category in which the underlying objects live, and let L^H denote hammock localization at t...

conjecture | open | MathDB 页面 | 原文线索

V2-030665 | MATHDB-323804

The stable reduced sl_N homology conjecture for torus knots

摘录 / EXCERPT: Stable $\mathfrak{sl}_N$ homology conjecture. The stable reduced $\mathfrak{sl}_N$ homology of $T(n,\infty)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-030666 | MATHDB-323806

Bryan-Gholampour's type D quantum McKay formula

摘录 / EXCERPT: Let $G\subset SO(3)$ be a polyhedral subgroup, let $\mathcal{X}=[\mathbb{C}^{3}/G]$, and let $F_{0}^{\mathcal{X}}(x_1,\ldots,x_n)$ denote its $\mathbb{C}^{*}$-equivariant genus-zer...

conjecture | open | MathDB 页面 | 原文线索

V2-030667 | MATHDB-323841

Jorgensen's direct-sum conjecture for modules detecting finite projective dimension

摘录 / EXCERPT: Jorgensen's conjecture. If $\operatorname{NT}(R)$ is closed under direct sums, then R is a complete intersection of codimension at least two. This conjecture concerns when closur...

conjecture | open | MathDB 页面 | 原文线索

V2-030668 | MATHDB-323931

Schneider's conjecture on divisible Galois cohomology

摘录 / EXCERPT: Schneider's conjecture. $i_n(F)=0$ for all $n\ne 1$. This predicts the vanishing of the divisible part of $H^2(G_S,W^n)$ outside the exceptional twist $n=1$; the supplied text give...

conjecture | open | MathDB 页面 | 原文线索

V2-030669 | MATHDB-323952

Non-closedness of transitive cellular automata in the uniform Bernoulli space

摘录 / EXCERPT: Let $\mathtt{CA}$ denote the space of cellular automata equipped with the uniform Bernoulli topology, and call a cellular automaton transitive if it has the transitivity property u...

conjecture | open | MathDB 页面 | 原文线索

V2-030670 | MATHDB-323962

Vanishing conjecture for the top cohomology of $\operatorname{SL}_n\mathbb{Z}$

摘录 / EXCERPT: Let $\operatorname{SL}_n\mathbb{Z}$ be the special linear group over the integers, and let $H^*(-;\mathbb{Q})$ denote rational group cohomology. Vanishing conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030671 | MATHDB-323963

Steinberg-module stability conjecture for $\operatorname{SL}_n\mathbb{Z}$

摘录 / EXCERPT: Let $\operatorname{SL}_n\mathbb{Z}$ be the special linear group over the integers, and let $\operatorname{St}(\operatorname{SL}_n\mathbb{Z})$ be its Steinberg module. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-030672 | MATHDB-323964

Stable instability conjecture for mapping class groups

摘录 / EXCERPT: Let Mod_g be the mapping class group of a closed, oriented surface of genus $g\geq 2$, and let $H^*(-;\mathbb{Q})$ denote rational group cohomology. Stable instabi...

conjecture | open | MathDB 页面 | 原文线索

V2-030673 | MATHDB-323965

Vanishing conjecture for the top cohomology of mapping class groups

摘录 / EXCERPT: Let Mod_g be the mapping class group of a closed, oriented surface of genus $g\geq 2$, and let $H^*(-;\mathbb{Q})$ denote rational group cohomology. Vanishing conj...

conjecture | open | MathDB 页面 | 原文线索

V2-030674 | MATHDB-323966

Boundary-forgetting conjecture for the Steinberg homology of mapping class groups

摘录 / EXCERPT: Let Mod_g^1 be the mapping class group of a genus-g surface with one boundary component, let Mod_g be the corresponding closed-surface mapping c...

conjecture | open | MathDB 页面 | 原文线索

V2-030675 | MATHDB-323967

Parity stability conjecture for the top cohomology of $\operatorname{Aut}(F_n)$

摘录 / EXCERPT: Let F_n be the free group of rank $n\geq 2$, let $\operatorname{Aut}(F_n)$ be its automorphism group, and let $H^*(-;\mathbb{Q})$ denote rational group cohomology. Parity stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-030676 | MATHDB-324005

Finiteness and Calabi-Yau structure conjecture for the wrapped Fukaya category

摘录 / EXCERPT: Finiteness and Calabi-Yau structure conjecture. For any object L in the wrapped Fukaya category, the natural morphism

conjecture | open | MathDB 页面 | 原文线索

V2-030677 | MATHDB-324016

The orbifold quantum cohomology-Jacobian ring correspondence

摘录 / EXCERPT: Let $\mathcal{X}$ be a Gorenstein toric orbifold, and let $W^{LF}_{\mathcal{X}}$ be its Lagrangian Floer superpotential. Denote by $QH^*_{\mathrm{orb}}(\mathcal{X})$ its orbifold q...

conjecture | open | MathDB 页面 | 原文线索

V2-030678 | MATHDB-324056

The planar-or-bounded-high-degree structure conjecture for graphs without a topological K_5

摘录 / EXCERPT: Planar-or-bounded-high-degree conjecture. There exist constants D and a such that every graph G that does not contain K_5 as a topological minor can be expressed as a cliqu...

conjecture | open | MathDB 页面 | 原文线索

V2-030679 | MATHDB-324062

Annulus-twist invariance conjecture for the 4-ball genus

摘录 / EXCERPT: Annulus-twist invariance conjecture. The 4-ball genera satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-030680 | MATHDB-324082

The pathwise C^0-Flux conjecture for Hamiltonian isotopies

摘录 / EXCERPT: Pathwise C^0-Flux conjecture. $PHam(M)$ is C^0-closed in $PSymp(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-030681 | MATHDB-324092

Kauffman's conjecture on detecting knot chirality by the Kauffman polynomial

摘录 / EXCERPT: Let K be a knot, let $\overline{K}$ denote its mirror image, and let $-\overline{K}$ denote the mirror image with reversed orientation. The Kauffman polynomial of K is denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-030682 | MATHDB-324104

Orthogonality conjecture for the bimodule summands of a tilted algebra

摘录 / EXCERPT: Let C be a tilted algebra, and let $E=\operatorname{Ext}_{C}^{2}(DC,C)$. Regard E as a C-C-bimodule and decompose it into indecomposable summands. Orthogonality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030683 | MATHDB-324105

The Lawvere-theoretic conjecture for homotopy conilpotent groups

摘录 / EXCERPT: Let n be a nonnegative integer, let $\mathcal{P}^n$ be a theory in the sense of Lawvere, and let its objects be the free homotopy conilpotent groups of class n, namely product...

conjecture | open | MathDB 页面 | 原文线索

V2-030684 | MATHDB-324106

The characterization conjecture for local LS-category and cocategory

摘录 / EXCERPT: Let X be a space and let $n>2$. Define $LS_2(X)\leq n-2$ when X has Lusternik-Schnirelmann category at most n and admits an open cover $U_1,\ldots,U_n$ such that each map...

conjecture | open | MathDB 页面 | 原文线索

V2-030685 | MATHDB-324107

Klein's cup-one length conjecture

摘录 / EXCERPT: Let X be a space and let $n>2$. Assume that $LS_2(X)\leq n-2$, where LS_2 is the local LS-category invariant defined using an n-set cover whose pairwise unions map null homot...

conjecture | open | MathDB 页面 | 原文线索

V2-030686 | MATHDB-324116

Homological stability for stabilized topological chiral homology

摘录 / EXCERPT: Homological stability conjecture. If A has homological stability, then there is a range of dimensions tending to infinity below which t_b induces an isomorphism on homology. Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-030687 | MATHDB-324117

Topological invariance of complete integrability for foliations by curves

摘录 / EXCERPT: Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be foliations by curves on $(\mathbb{C}^3,0)$. Suppose they are topologically equivalent and do not admit invariant surfaces over which the...

conjecture | open | MathDB 页面 | 原文线索

V2-030688 | MATHDB-324184

Quantum Kirwan map conjecture in the semipositive case

摘录 / EXCERPT: Let (M,ω,μ) satisfy (H), be convex at infinity, and be semipositive. Let φ be a map ... satisfying the equivariant Novikov condition, and let φ_* be the indu...

conjecture | open | MathDB 页面 | 原文线索

V2-030689 | MATHDB-324191

The Seidel-element and Jacobian-generator conjecture for compact toric manifolds

摘录 / EXCERPT: Seidel-element and Jacobian-generator conjecture. The isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-030690 | MATHDB-324204

Homological mirror symmetry conjecture for the Barlow surface

摘录 / EXCERPT: Let S be a Barlow surface equipped with a generic Kähler form. An orthogonal decomposition of its Fukaya category is a decomposition into mutually orthogonal full subcategories,...

conjecture | open | MathDB 页面 | 原文线索

V2-030691 | MATHDB-324211

Katok-Katok conjecture on cohomology of standard partially hyperbolic actions

摘录 / EXCERPT: Let α be a standard partially hyperbolic action of $\mathbb{Z}_+^d$, $\mathbb{Z}^d$, or $\mathbb{R}^d$, where $d\geq2$. For n-cohomology, the smooth n-cohomology is the...

conjecture | open | MathDB 页面 | 原文线索

V2-030692 | MATHDB-324214

Kontsevich's homological mirror symmetry conjecture for toric Fano stacks

摘录 / EXCERPT: Let S be the relevant symplectic surface, let $\boldsymbol{\Delta}=(\Delta_1,\ldots,\Delta_m)$ be the distinguished collection of Lefschetz thimbles, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030693 | MATHDB-324215

Kontsevich-Seidel conjecture on the algebraic and symplectic mirror rings

摘录 / EXCERPT: Let $\mathcal{A}$ and $\mathcal{B}$ be the rings defined from the symplectic and algebraic mirror constructions, and let $\mathfrak{T}$ and $\mathfrak{S}$ be their respective ring...

conjecture | open | MathDB 页面 | 原文线索

V2-030694 | MATHDB-324238

Symplectic duality conjecture for Poisson homology and intersection cohomology

摘录 / EXCERPT: Let $\mathfrak{N}$ and $\mathfrak{N}^!$ be a symplectic dual pair of cones admitting projective symplectic resolutions $\mathfrak{M}$ and $\mathfrak{M}^!$. Let $\mathscr{N}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-030695 | MATHDB-324239

Quantized symplectic duality conjecture for Hochschild homology

摘录 / EXCERPT: Let $\mathfrak{N}$ and $\mathfrak{N}^!$ be as in the symplectic duality conjecture, let A be a quantization of $\mathscr{N}$, and let A_0 be its central quotient quantizing...

conjecture | open | MathDB 页面 | 原文线索

V2-030696 | MATHDB-324322

Aganagic-Vafa conjecture relating the augmentation and Q-deformed A-polynomials

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $\mathbf{A}_K(e^x,e^p,Q)$ denote the three-variable Q-deformed A-polynomial associated with the mirror construction for the resolved conif...

conjecture | open | MathDB 页面 | 原文线索

V2-030697 | MATHDB-324323

HOMFLY-PT specialization conjecture from the augmentation polynomial

摘录 / EXCERPT: Let K be a knot in S^3, and let $f(U)$ be the polynomial determined by the branch of the zero locus of Aug_K near $(\lambda,\mu,U)=(0,U,U)$ through ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-030698 | MATHDB-324329

The product-space conjecture for D-separability

摘录 / EXCERPT: Product-space conjecture. The space

conjecture | open | MathDB 页面 | 原文线索

V2-030699 | MATHDB-324378

The Manifold Semistability Conjecture for universal covers

摘录 / EXCERPT: Let M be a closed aspherical manifold of dimension greater than 1, and let $\widetilde{M}$ denote its universal cover. A space is strongly connected at infinity when its end ha...

conjecture | open | MathDB 页面 | 原文线索

V2-030700 | MATHDB-324388

The L-space homology-sphere classification conjecture

摘录 / EXCERPT: A rational homology 3-sphere Y is an L-space if the rank of its Heegaard Floer homology group $\widehat{\operatorname{HF}}(Y)$ equals the order of $H_1(Y;\mathbb{Z})$. An irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-030701 | MATHDB-324389

The incompressible-torus obstruction to L-spaces

摘录 / EXCERPT: Let Y be an irreducible homology sphere, and suppose that it contains an incompressible torus, meaning an embedded torus whose inclusion induces an injective map on fundamental g...

conjecture | open | MathDB 页面 | 原文线索

V2-030702 | MATHDB-324394

Hopf-Thurston Euler characteristic sign conjecture

摘录 / EXCERPT: Hopf-Thurston sign conjecture. The Euler characteristic of M satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-030703 | MATHDB-324396

The integrality conjecture for Seifert fibered surgery on hyperbolic knots

摘录 / EXCERPT: A Seifert fibered surgery is a Dehn surgery on a knot whose resulting 3-manifold is Seifert fibered. The integrality conjecture. Any Seifert fibered surgery on a hyperbolic knot...

conjecture | open | MathDB 页面 | 原文线索

V2-030704 | MATHDB-324407

Topological simplicity conjecture for acylindrically hyperbolic groups

摘录 / EXCERPT: Let G be an acylindrically hyperbolic group with no non-trivial finite normal subgroups. A group topology on G is topologically simple if its only closed normal subgroups are t...

conjecture | open | MathDB 页面 | 原文线索

V2-030705 | MATHDB-324427

The PL surface projection moves conjecture

摘录 / EXCERPT: PL surface projection moves conjecture. The moves for singularities of projection of a PL surface should be the analogues of the moves for the singularities of projection of a smoo...

conjecture | open | MathDB 页面 | 原文线索

V2-030706 | MATHDB-324437

The thick tube conjecture for k-expanding embeddings

摘录 / EXCERPT: Let p, m, k, δ, and R be such that I is a k-expanding embedding of the product $S^p(\delta)\times B^{m-p}(R)$ into the unit m-ball. Thick tube conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-030707 | MATHDB-324438

The null-homotopy conjecture for small k-dilation maps between spheres

摘录 / EXCERPT: Let m, n, c, and k be integers satisfying the hypotheses below, and let F be a map from the unit m-sphere to the unit n-sphere. Write $\operatorname{Dil}_k(F)$ for it...

conjecture | open | MathDB 页面 | 原文线索

V2-030708 | MATHDB-324457

Stability of global hyperbolicity for cone-bundle sections

摘录 / EXCERPT: Let F be the bundle whose sections define the relevant causal structures, and let $\mathrm{\Gamma}(F)$ be its space of sections with the Whitney fine C^0 topology. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030709 | MATHDB-324478

Universal interval conjecture for taut foliations in knot complements

摘录 / EXCERPT: Let k be a nontrivial knot in S^3, and let $M=S^3\setminus\operatorname{int}(N(k))$ be its exterior. For slopes $s\in(-a,b)$, require that M have a taut foliation whose restr...

conjecture | open | MathDB 页面 | 原文线索

V2-030710 | MATHDB-324493

Atiyah-Hitchin approximation conjecture for foliated topological gravity on exotic R4

摘录 / EXCERPT: Let e be an exotic $\mathbb{R}^{4}$ belonging to the DeMichelis-Freedman radial family, and let the foliated topological limit of general relativity (FTL of GR) on e be evaluat...

conjecture | open | MathDB 页面 | 原文线索

V2-030711 | MATHDB-324506

Heegaard genus lower bound for gluing 3-manifolds

摘录 / EXCERPT: Let M_1 and M_2 be compact orientable 3-manifolds with incompressible boundary components F_1 and F_2, respectively, and let $h:F_1\rightarrow F_2$ be any homeomorphism....

conjecture | open | MathDB 页面 | 原文线索

V2-030712 | MATHDB-324512

Fully faithful functor conjecture for Hochschild cohomology of fully dualizable objects

摘录 / EXCERPT: Let $\mathcal{C}$ be a fully dualizable object, and let $Z(\mathcal{C})=HH^\bullet(\mathcal{C})$ denote its Hochschild cohomology. Consider a category of fully dualizable objects a...

conjecture | open | MathDB 页面 | 原文线索

V2-030713 | MATHDB-324513

Williamson-Janos conjecture on complete admissible metrics

摘录 / EXCERPT: A complete admissible metric d for a σ-compact, locally compact space X is a metric inducing the topology of X and complete as a metric space. A Heine-Borel metric is...

conjecture | open | MathDB 页面 | 原文线索

V2-030714 | MATHDB-324515

The vanishing conjecture for negative powers of the contact line bundle

摘录 / EXCERPT: Vanishing conjecture. For every $k\leq -1$ and every $i\leq n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-030715 | MATHDB-324595

Vanishing of negative semi-topological K-groups for smooth proper dg-categories

摘录 / EXCERPT: Let T be a smooth proper dg-category over $\mathbb{C}$, and let $\mathbf{K}^{\mathrm{st}}_i(T)$ denote its semi-topological K-groups. Vanishing conjecture. ... This is presented as...

conjecture | open | MathDB 页面 | 原文线索

V2-030716 | MATHDB-324596

Comparison of algebraic and semi-topological K-theory with finite coefficients

摘录 / EXCERPT: Let T be a dg-category over $\mathbb{C}$ and let $n>0$ be an integer. Write $\mathbf{K}(T)/n$ and $\mathbf{K}^{\mathrm{st}}(T)/n$ for the reductions modulo n of the algebraic a...

conjecture | open | MathDB 页面 | 原文线索

V2-030717 | MATHDB-324599

Uncountably many inequivalent unknotted solenoid embeddings

摘录 / EXCERPT: Unknotted-embedding conjecture. For every solenoid, there are uncountably many inequivalent unknotted embeddings in S^3.

conjecture | open | MathDB 页面 | 原文线索

V2-030718 | MATHDB-324600

Distinct knotted solenoid embeddings have distinct complement fundamental groups

摘录 / EXCERPT: Knotted-embedding conjecture. If a solenoid is embedded in two “different” knotted ways, the fundamental groups of the complements are different:

conjecture | open | MathDB 页面 | 原文线索

V2-030719 | MATHDB-324612

Živaljević's conjecture on LC triangulations of simply connected smooth manifolds

摘录 / EXCERPT: Živaljević's conjecture. Every simply connected smooth d-manifold admits some LC triangulations.

conjecture | open | MathDB 页面 | 原文线索

V2-030720 | MATHDB-324620

Canonical-transversal preservation conjecture for homeomorphisms of tiling spaces

摘录 / EXCERPT: Canonical-transversal preservation conjecture. There is a homeomorphism h' isotopic to h such that, for any canonical transversal Ξ in Ω, $h'(\Xi)$ is a canonical tr...

conjecture | open | MathDB 页面 | 原文线索

V2-030721 | MATHDB-324635

Conjecture on semiperfect quasi-π_∞-regularity of finitely presented algebra extensions

摘录 / EXCERPT: Semiperfect quasi-π_∞-regularity conjecture. Then R is semiperfect and quasi-π_∞-regular with respect to some topology. The condition on J ensures that the to...

conjecture | open | MathDB 页面 | 原文线索

V2-030722 | MATHDB-324648

Botvinnik's Topological Conjecture on concordances and positive scalar curvature metrics

摘录 / EXCERPT: Let M be a closed manifold equipped with a metric g of positive scalar curvature. Let H be a non-trivial concordance on M, meaning a concordance that is not isotopic to the...

conjecture | refuted | MathDB 页面 | 原文线索

V2-030723 | MATHDB-324662

Abdenur-Andersson's wicked homeomorphism conjecture

摘录 / EXCERPT: Abdenur-Andersson's conjecture. A homeomorphism f which is generic in the set of homeomorphisms of X without any open trapping set is wicked: it is not uniquely ergodic, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-030724 | MATHDB-324667

Polynomial intersection conjecture for curve graph distance

摘录 / EXCERPT: Let S be a surface with complexity $\xi(S)$, let k be an integer, and let α and β be simple closed curves on S. Write $d_{\mathcal{C}}$ for curve graph distance...

conjecture | open | MathDB 页面 | 原文线索

V2-030725 | MATHDB-324682

Randomized N-path density conjecture for hyperbolic polynomials

摘录 / EXCERPT: Randomized N-path density conjecture. (1) For a given $p\in\mathcal{H}$, the set of L such that

conjecture | open | MathDB 页面 | 原文线索

V2-030726 | MATHDB-324689

Realization conjecture for homology triplets by complexes of coherent sheaves

摘录 / EXCERPT: Let (B,H,C) be a homology triplet of type n, with degrees of B given by ... and strand starts of H and C given respectively by ... A complex of coherent sheaves...

conjecture | open | MathDB 页面 | 原文线索

V2-030727 | MATHDB-324690

Uniqueness conjecture for Hilbert polynomials of homology-triplet complexes

摘录 / EXCERPT: Let (B,H,C) be a homology triplet, and let $\mathcal{E}^{\bullet}$ be a complex of coherent sheaves corresponding to it as in the realization conjecture. Hilbert-polynomial uniqu...

conjecture | open | MathDB 页面 | 原文线索

V2-030728 | MATHDB-324694

Jacquet-Langlands equality of localized new homology

摘录 / EXCERPT: Let $Y(K_{\Sigma})$ and $Y'(K_{\Sigma})$ be a Jacquet-Langlands pair, let $\mathbf{T}$ be the relevant Hecke algebra, and let $\mathfrak{m}$ be a non-Eisenstein maximal ideal. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-030729 | MATHDB-324695

Multiplicity-one conjecture for localized new homology

摘录 / EXCERPT: Let $\overline{\rho}:G_F\rightarrow\operatorname{GL}_2(k)$ be an absolutely irreducible Galois representation, let $\mathfrak{m}$ be the associated maximal ideal, and let d_F be...

conjecture | open | MathDB 页面 | 原文线索

V2-030730 | MATHDB-324696

Eisenstein homology conjecture for S-arithmetic groups

摘录 / EXCERPT: Let T be a finite set of primes with $|T|=d$, and let $Y(K[1/T])$ be the associated S-arithmetic space. Eisenstein homology conjecture. Then $H_i(Y(K[1/T]),\mathbf{Z})$ is Eise...

conjecture | open | MathDB 页面 | 原文线索

V2-030731 | MATHDB-324697

Cyclotomic Eisenstein conjecture for S-arithmetic homology

摘录 / EXCERPT: Let T be a finite set of primes with $|T|=d$, let $K[1/T]$ denote the corresponding S-arithmetic level, and let $\mathcal{F}$ be the coefficient system used in the paper. Cyclo...

conjecture | open | MathDB 页面 | 原文线索

V2-030732 | MATHDB-324698

Eisenstein deformation criterion via essential homology

摘录 / EXCERPT: Let R be the universal deformation ring of the cyclotomic residual representation described in the source, let $\mathfrak{m}$ be the corresponding Eisenstein maximal ideal, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-030733 | MATHDB-324714

Console-Fino conjecture on invariant Dolbeault cohomology of nilmanifolds

摘录 / EXCERPT: Console-Fino conjecture. These maps of complexes are quasi-isomorphisms, equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-030734 | MATHDB-324716

Weiss tower obstruction conjecture for embedding immersions

摘录 / EXCERPT: Let P and N be manifolds of dimensions p and n, respectively, and let $f\colon P\to N$ be an immersion equipped with a lift $f_{j-1}\in E_{j-1}(P,N)$. For $j\geq 2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-030735 | MATHDB-324730

Noetherian type conjecture for Pixley-Roy hyperspaces

摘录 / EXCERPT: Let (X,τ) be a topological space, and let $PR(X)=[X]^{<\omega}$ be the set of finite subsets of X equipped with the Pixley-Roy topology, whose basic open sets are ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-030736 | MATHDB-324740

Productive Menger and Hurewicz properties imply their weak counterparts

摘录 / EXCERPT: Productive weak-property conjectures. If a space is productively Menger then it is productively weakly Menger. If a space is productively Hurewicz then it is productively weakly Hu...

conjecture | open | MathDB 页面 | 原文线索

V2-030737 | MATHDB-324770

The bar-construction homotopy-equivalence conjecture for holomorphic bundles on the blown-up plane

摘录 / EXCERPT: Let $I\subset\mathbb C^2$ be finite. Write $\mathfrak M_I$ for the stable moduli space of holomorphic bundles associated with I, and let ... be the bar construction, with its natur...

conjecture | open | MathDB 页面 | 原文线索

V2-030738 | MATHDB-324857

Montesinos's four-dimensional branched covering conjecture

摘录 / EXCERPT: Let M^4 be a closed, oriented, connected 4-manifold. A simple branched covering is a branched covering whose local monodromies around the branch set are transpositions; let...

conjecture | open | MathDB 页面 | 原文线索

V2-030739 | MATHDB-324872

The convex-concave conjecture for Alexander polynomials of two-bridge knots

摘录 / EXCERPT: Let ... be the Alexander polynomial of a two-bridge knot. Convex-concave conjecture. There exists an integer $k\leq n$ such that $(a_0,\ldots,a_k)$ is convex and $(a_k,\ldots,a_n)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-030740 | MATHDB-324887

The interval-shape conjecture for left-orderable surgery slopes

摘录 / EXCERPT: Interval-shape conjecture. The set $\mathcal{S}_{LO}(K)$ coincides with one of

conjecture | open | MathDB 页面 | 原文线索

V2-030741 | MATHDB-324889

The inverse-limit conjecture for singular quotient 3-manifolds

摘录 / EXCERPT: A singular quotient 3-manifold is a quotient space of a topological 3-manifold by a continuum that generates a singular point. An inverse limit of closed 3-manifolds is a spa...

conjecture | open | MathDB 页面 | 原文线索

V2-030742 | MATHDB-324903

Arnold's exact Lagrangian conjecture for cotangent bundles

摘录 / EXCERPT: Let N be a compact closed manifold, and let $L\subset T^*N$ be a compact closed exact Lagrangian submanifold, where T^*N has its standard Liouville form. Arnold's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030743 | MATHDB-324905

Bouquet-of-spheres conjecture for random simplicial complexes

摘录 / EXCERPT: Let $X\sim X(n,p)$ be a random simplicial complex, let $d\geq 3$, and fix ... Set $p=n^{-\alpha}$. Bouquet-of-spheres conjecture. With high probability, X is homotopy equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-030744 | MATHDB-324919

The tropical eigenwave hard Lefschetz conjecture

摘录 / EXCERPT: Tropical eigenwave conjecture. This map is an isomorphism. This is a tropical analogue of hard Lefschetz, relating tropical homology groups with complementary coefficient degrees....

conjecture | open | MathDB 页面 | 原文线索

V2-030745 | MATHDB-324920

The tropical intersection-form non-degeneracy conjecture

摘录 / EXCERPT: Tropical intersection-form conjecture. This form is non-degenerate. The preceding theorem establishes that the product descends to a symmetric bilinear form on homology. The suppli...

conjecture | open | MathDB 页面 | 原文线索

V2-030746 | MATHDB-325021

Ghys's almost commensurability conjecture for transitive Anosov flows

摘录 / EXCERPT: Let ϕ and ϕ' be transitive Anosov flows on manifolds M and M', respectively. Two flows are topologically almost commensurable if, after removing finite collections of...

conjecture | open | MathDB 页面 | 原文线索

V2-030747 | MATHDB-325041

Simple closed geodesics realizing the intersection-form constant on hyperbolic surfaces

摘录 / EXCERPT: Simple-geodesics realization conjecture. For any $s>1$, for almost every (M,g) in $\mathcal{M}_s$, $K(M,g)$ is realized by the homology classes of simple closed geodesics.

conjecture | open | MathDB 页面 | 原文线索

V2-030748 | MATHDB-325068

Characterization of Y-equivalence for bordered spin 3-manifolds

摘录 / EXCERPT: Main conjecture. The following conditions are equivalent: (M,ϕ,s) and (M',ϕ',s') are Y-equivalent; and there is a homology isomorphism from (M,ϕ) to (M',ϕ') s...

conjecture | open | MathDB 页面 | 原文线索

V2-030749 | MATHDB-325088

Proper uniformly bounded representations with nontrivial cohomology for hyperbolic groups

摘录 / EXCERPT: Shalom's conjecture. Every non-elementary hyperbolic group admits a uniformly bounded representation on a Hilbert space for which $H^1(G,c0)\neq 0$ and for which there exists a pro...

open_problem | open | MathDB 页面 | 原文线索

V2-030750 | MATHDB-325089

Gromov's conjecture on reduced \(\ell_p\)-cohomology of amenable groups

摘录 / EXCERPT: Gromov's conjecture. The reduced ℓ_p-cohomology of G vanishes for

conjecture | open | MathDB 页面 | 原文线索

V2-030751 | MATHDB-325093

Greene's doubly primitive conjecture for knots in $S^1 \times S^2$

摘录 / EXCERPT: Let K be a knot in $S^1 \times S^2$ and suppose that a longitudinal surgery on K produces a lens space. Greene's conjecture. The knot K is doubly primitive. This is the broad...

conjecture | open | MathDB 页面 | 原文线索

V2-030752 | MATHDB-325094

Homology-class classification for longitudinal $S^1 \times S^2$ surgeries

摘录 / EXCERPT: Let L be a lens space, let $[?]$ be a knot in L representing $[?]$ and let $[?]$ denote the homology class of a Heegaard-solid-torus core. Homology-class conjecture. If the kno...

conjecture | open | MathDB 页面 | 原文线索

V2-030753 | MATHDB-325127

The equivariant cohomological realization conjecture for local Dahmen-Micchelli modules

摘录 / EXCERPT: Equivariant cohomological realization conjecture. For every semilocal $\mathcal T$, there is an isomorphism of graded $S[\mathfrak g^*]$-modules

conjecture | open | MathDB 页面 | 原文线索

V2-030754 | MATHDB-325138

Functoriality of knot Floer homology under marked smooth surface isotopy

摘录 / EXCERPT: Grid-cobordism functoriality conjecture. Grid diagram maps induce maps on HFK^- that are invariant with respect to marked smooth isotopy classes of surfaces.

conjecture | open | MathDB 页面 | 原文线索

V2-030755 | MATHDB-325146

Stroppel's generalized sutured Khovanov-Hochschild homology conjecture

摘录 / EXCERPT: Let $\sigma \in {\mathfrak{B}}_{n}$, and let $m(\widehat{\sigma}) \subset A \times I$ be the mirror of its annular closure. The notation $A^{k,n-k}$ denotes the Chen-Khovanov-Strop...

conjecture | open | MathDB 页面 | 原文线索

V2-030756 | MATHDB-325149

Furuta-Ohta and Seiberg-Witten invariant correspondence conjecture

摘录 / EXCERPT: Furuta-Ohta-Seiberg-Witten correspondence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030757 | MATHDB-325186

Belavin-Drinfeld cohomology conjecture for the Drinfeld-Jimbo r-matrix

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let $G=\operatorname{Ad}(\mathfrak{g})$ be its adjoint group, and let r_{DJ} be the Drinfeld-Jimbo r-matrix. For a Belavin-Drinfeld...

conjecture | open | MathDB 页面 | 原文线索

V2-030758 | MATHDB-325196

Kauffman's Strong Conjecture for genus-one Seifert surfaces

摘录 / EXCERPT: Let K be a slice knot, and let F be a genus-one Seifert surface for K. For a simple closed curve d on F, write d^+ for its positive push-off and...

conjecture | open | MathDB 页面 | 原文线索

V2-030759 | MATHDB-325197

Kauffman's Weak Conjecture for genus-one Seifert surfaces

摘录 / EXCERPT: Let K be a slice knot, and let F be a genus-one Seifert surface for K. For a simple closed curve d on F, write d^+ for its positive push-off and define its self-linking...

conjecture | open | MathDB 页面 | 原文线索

V2-030760 | MATHDB-325198

Existence of a surgery curve with vanishing Levine-Tristram signature

摘录 / EXCERPT: Let K be a slice knot and let F be a genus-one Seifert surface for K. A Levine-Tristram signature function is the signature function associated to a knot, denoted in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-030761 | MATHDB-325199

Stable Kauffman Conjecture for Levine surfaces

摘录 / EXCERPT: Let K be a slice knot. A Seifert surface F stabilizes to a Seifert surface F' if F' is obtained from F by finitely many trivial one-handle attachments. A Levine surface i...

conjecture | open | MathDB 页面 | 原文线索

V2-030762 | MATHDB-325211

The Continuum Hypothesis conjecture on nontrivial corona automorphisms

摘录 / EXCERPT: Let $\mathcal{A}$ be a separable, nonunital C-algebra, and let its corona be the quotient $\mathcal{Q}(\mathcal{A})=\mathcal{M}(\mathcal{A})/\mathcal{A}$ of its multiplier algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-030763 | MATHDB-325218

Ottaviani-Paoletti conjecture for syzygies of Veronese embeddings

摘录 / EXCERPT: Ottaviani-Paoletti conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030764 | MATHDB-325219

Homology concentration conjecture for matching complexes

摘录 / EXCERPT: Matching-complex homology conjecture. The only non-zero homology group of C_{nd}^d for $n=1,\ldots,3d-1$ is

conjecture | open | MathDB 页面 | 原文线索

V2-030765 | MATHDB-325222

Conjecture on differentiability of the equilibrium measure

摘录 / EXCERPT: Let $\tilde{\gamma}_j$ be neighborhoods of the connected components γ_j of the support of the equilibrium measure, and set $\tilde{\Gamma}=\coprod_{j=1}^r\tilde{\gamma}_j$....

conjecture | open | MathDB 页面 | 原文线索

V2-030766 | MATHDB-325225

Integral locality conjecture for perfectoid affinoid algebras

摘录 / EXCERPT: Let K be a perfectoid field, let (A,A^+) be a complete affinoid K-algebra for which A^+ has the p-adic topology, and set $X=\operatorname{Spa}(A,A^+)$. Assume that X ha...

conjecture | open | MathDB 页面 | 原文线索

V2-030767 | MATHDB-325286

HOMFLYPT quantum one-cocycle evaluation conjecture

摘录 / EXCERPT: HOMFLYPT quantum one-cocycle conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030768 | MATHDB-325287

Quantum one-cocycle characterization of torus knots

摘录 / EXCERPT: Quantum one-cocycle torus-knot conjecture. $\bar R^{(1)}(\operatorname{hat}(K))$ is a nonzero integer multiple of $\bar R^{(1)}(\operatorname{rot}(K))$ if and only if K is a toru...

conjecture | open | MathDB 页面 | 原文线索

V2-030769 | MATHDB-325302

Classification conjecture for absolutely cartesian cubes of spaces

摘录 / EXCERPT: Let an n-cube of spaces be a diagram indexed by the powerset of $[n]=\{0,1,\ldots,n\}$, and call it absolutely cartesian when every homotopy functor sends it to a homotopy cartes...

conjecture | open | MathDB 页面 | 原文线索

V2-030770 | MATHDB-325303

Absolute cartesian-cocartesian equivalence conjecture for cubes of spaces

摘录 / EXCERPT: Let an n-cube be a diagram of spaces indexed by the powerset of $[n]=\{0,1,\ldots,n\}$. A diagram is absolutely cartesian if every homotopy functor sends it to a homotopy cartesi...

conjecture | open | MathDB 页面 | 原文线索

V2-030771 | MATHDB-325316

Characterization of loops with realizable generalized Dehn twists

摘录 / EXCERPT: Let S be the surface under discussion, let γ be an unoriented loop on S, and let t_{γ} denote the generalized Dehn twist along γ. A loop is homotopic to a...

conjecture | open | MathDB 页面 | 原文线索

V2-030772 | MATHDB-325335

Cherednik's stabilization conjecture for DAHA superpolynomials

摘录 / EXCERPT: Let $\mathcal{P}_{n,m}^{\lambda,N}(q,t)$ be the DAHA superpolynomial associated with a partition λ in rank N. Cherednik's stabilization conjecture. There exists a polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-030773 | MATHDB-325336

Cherednik's duality conjecture for reduced superpolynomials

摘录 / EXCERPT: Let $\mathcal{P}_{n,m}^{\lambda}(u,q,t)$ be the stabilized superpolynomial and define its reduced version by ... Cherednik's duality conjecture. For the transposed partition...

conjecture | open | MathDB 页面 | 原文线索

V2-030774 | MATHDB-325380

Quantum cohomology relations for Lagrangian Grassmannians

摘录 / EXCERPT: Quantum cohomology relation. The following relation holds in the quantum cohomology of $LG(m)$ for every $1\leq l\leq m-1$:

conjecture | open | MathDB 页面 | 原文线索

V2-030775 | MATHDB-325381

Quantum-connection identification for the Lagrangian Grassmannian Landau-Ginzburg model

摘录 / EXCERPT: Quantum-connection identification conjecture. The differential operators $\hbar\,{}^B\nabla_{\partial_t}$ and $\hbar\,{}^B\nabla_{\hbar\partial_\hbar}$ preserve $\bar G_0^{W_t}$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-030776 | MATHDB-325412

The cord-algebra isomorphism conjecture for knot contact homology

摘录 / EXCERPT: Let Φ be the chain map from the degree-zero knot contact-homology complex at $Q=1$ to the cord algebra of a knot. Cord-algebra isomorphism conjecture. The map Φ induces a...

conjecture | open | MathDB 页面 | 原文线索

V2-030777 | MATHDB-325441

Non-invariance of order for finite-type entire functions

摘录 / EXCERPT: Non-invariance of order conjecture. There exist entire functions $f,g\in\mathcal{S}$ such that f and g are topologically equivalent, but

conjecture | open | MathDB 页面 | 原文线索

V2-030778 | MATHDB-325456

Triangulated telescope conjecture

摘录 / EXCERPT: Let $\mathscr{T}$ be a compactly generated triangulated category with all coproducts, and let ... be a smashing localization, meaning that its right adjoint preserves coproducts. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-030779 | MATHDB-325480

The two-site mutation conjecture for rational subtangle replacements

摘录 / EXCERPT: Let τ be a tangle, and let α_0 and α_1 be sites for a rational subtangle replacement (RSR) on τ. Suppose their lifts to the branched double cover of τ...

conjecture | open | MathDB 页面 | 原文线索

V2-030780 | MATHDB-325504

Periodicity conjecture for the quantum invariants $d_l(\alpha_{m,n})$

摘录 / EXCERPT: Let $l\geq 1$ and let m,n satisfy ... For each such pair, let $d_l(\alpha_{m,n})$ be the quantity defined in the paper. Periodicity conjecture. In this range, $d_l(\alpha_{m,n})$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-030781 | MATHDB-325513

The Calabi-Yau conjecture for the wrapped Fukaya category of the conifold

摘录 / EXCERPT: Let Y^0 be the conifold complement, let $SH^0(Y^0)$ denote its degree-zero symplectic cohomology, and let ... be the open-closed string map. Regard the derived category of the wrap...

conjecture | open | MathDB 页面 | 原文线索

V2-030782 | MATHDB-325529

Serre fibration conjecture for flexible Weinstein structures

摘录 / EXCERPT: Let η be a nonempty homotopy class of nondegenerate 2-forms on a domain or manifold of dimension $2n>4$. Let $\mathfrak{Morse}_n$ be the space of Morse functions whose criti...

conjecture | open | MathDB 页面 | 原文线索

V2-030783 | MATHDB-325575

Generic shadowing and weak specification for non-Anosov area-preserving maps

摘录 / EXCERPT: Let M be a compact surface, and consider the space of C^1 area-preserving maps of M with its C^1 topology. A map is Anosov if it has a uniformly hyperbolic invariant splitt...

conjecture | open | MathDB 页面 | 原文线索

V2-030784 | MATHDB-325577

Pseudocenter conjecture for entropy on Mandelbrot wakes

摘录 / EXCERPT: Let $\theta_1<\theta_2$ be external angles whose parameter rays $R_M(\theta_1)$ and $R_M(\theta_2)$ land at the same parameter on the boundary of the Mandelbrot set. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030785 | MATHDB-325632

Local cyclic index theorem conjecture

摘录 / EXCERPT: Local cyclic index theorem conjecture. For any local elliptic operator A,

conjecture | open | MathDB 页面 | 原文线索

V2-030786 | MATHDB-325634

Catenarity conjecture for rings with Cohen-Macaulay blowings up

摘录 / EXCERPT: Let R be a Noetherian ring. A Cohen-Macaulay blowing up of $\operatorname{Spec}(R)$ is a proper birational morphism whose source is Cohen-Macaulay. Catenarity conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-030787 | MATHDB-325674

Homotopy equivalence conjecture for rational associahedra

摘录 / EXCERPT: Let ${\mathsf{Ass}}(a,b)$ be the rational associahedron simplicial complex, and let ${\widehat{\mathsf{Ass}}}(a,b)$ be the simplicial complex whose faces are collections of mutuall...

conjecture | open | MathDB 页面 | 原文线索

V2-030788 | MATHDB-325677

Uniqueness conjecture for bridge surfaces of iterated torus knots

摘录 / EXCERPT: Iterated torus knot bridge-surface conjecture. For each $g=0,\dots,n+1$, the knot K_n has a unique irreducible genus g bridge surface, and every (g',b)-surface with...

conjecture | open | MathDB 页面 | 原文线索

V2-030789 | MATHDB-325691

Kapustin's closed-string states and Hochschild cohomology conjecture

摘录 / EXCERPT: Let $\mathcal A$ be an associative algebra equipped with a differential ... For a theory with a single D-brane, the deformation data are encoded by $(\mathcal A,Q)$, and their equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-030790 | MATHDB-325692

Bulk-boundary OPE and deformation matching conjecture for branes

摘录 / EXCERPT: Bulk-boundary deformation conjecture. Such a mathematical pairing should realize a bulk-boundary operator product expansion under deformation of the BRST operator, and mathematical...

conjecture | open | MathDB 页面 | 原文线索

V2-030791 | MATHDB-325697

Mauricio's conjecture on lens-space connected-sum surgeries

摘录 / EXCERPT: Mauricio's conjecture. Surgery on K never produces

conjecture | open | MathDB 页面 | 原文线索

V2-030792 | MATHDB-325709

The Rank Growth Conjecture for distributive homology

摘录 / EXCERPT: Let $(X,\operatorname{\star})$ be a shelf, with distributive homology groups $H_n(X)$ and rank denoted by rk. Rank Growth Conjecture. For $n\geqslant |X|-2$, one h...

conjecture | open | MathDB 页面 | 原文线索

V2-030793 | MATHDB-325710

The Normalized Rank Growth Conjecture for spindle homology

摘录 / EXCERPT: Let $(X,\operatorname{\star})$ be a spindle, and let $\widetilde H^{N}_n(X)$ denote its normalized homology, with rank denoted by rk. Normalized Rank Growth Conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-030794 | MATHDB-325711

The Growth Conjecture for distributive homology including torsion

摘录 / EXCERPT: Let $(X,\operatorname{\star})$ be a shelf, with distributive homology groups $H_n(X)$. A spindle is a shelf satisfying the additional spindle condition, and let...

conjecture | refuted | MathDB 页面 | 原文线索

V2-030795 | MATHDB-325730

The Borel conjecture for closed Haken manifolds

摘录 / EXCERPT: Borel conjecture for Haken manifolds. The map f is homotopic to a homeomorphism.

open_problem | open | MathDB 页面 | 原文线索

V2-030796 | MATHDB-325805

The r-Bouchard-Mariño conjecture for r-Hurwitz numbers

摘录 / EXCERPT: For each g,n, let $H_{g,r}(x_1,\dots,x_n)$ be the generating function formed from the invariants $h_{g,r;k_1,\dots,k_n}$, and let D denote the differential operator ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030797 | MATHDB-325806

The cohomological criterion for exceptional syzygy bundles to be Steiner

摘录 / EXCERPT: Cohomological criterion for syzygy bundles. The two cohomology groups are isomorphic if and only if they are both zero; consequently, the syzygy bundles F_i are Steiner if and on...

conjecture | open | MathDB 页面 | 原文线索

V2-030798 | MATHDB-325838

Cohomological Landau-Ginzburg/Calabi-Yau correspondence

摘录 / EXCERPT: Let (W,G) be of Calabi-Yau type: W is nondegenerate, not necessarily invertible, with weights q_j satisfying ... and G contains $\langle j_W\rangle$ and lies in SL_W. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-030799 | MATHDB-325856

Fixed-degree bound for zeros of alternating-knot Alexander polynomials

摘录 / EXCERPT: Let Γ_n be the set of Alexander polynomials of degree n arising from alternating knots. Fixed-degree boundedness conjecture. There exists a real number $\delta_n>0$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-030800 | MATHDB-325857

Signature-controlled unimodality for alternating-knot Alexander polynomials

摘录 / EXCERPT: Let ... be the Alexander polynomial of an alternating knot, and let $\sigma(K)$ be its signature. Signature-controlled coefficient conjecture. If $\sigma(K)=0$, then ... More generally...

conjecture | open | MathDB 页面 | 原文线索

V2-030801 | MATHDB-325859

Large-parameter c-stability for a family of 2-bridge knots

摘录 / EXCERPT: Large-parameter stability conjecture. There exists a positive integer N_m such that: if m is even, then $K(r_m)$ is c-stable for $k\geq N_m$; and if m is odd, then $K(r_m)$...

conjecture | open | MathDB 页面 | 原文线索

V2-030802 | MATHDB-325861

Lunar-domain conjecture for zeros of the Alexander polynomials of $K[n]$

摘录 / EXCERPT: Let ... be a real zero of $\Delta_{K[1]}(t)=t^2-3t+1$, and let C be the circle with centre $(\frac{\alpha-1}{2},0)$ and radius $\frac{\alpha+1}{2}$. Lunar-domain conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-030803 | MATHDB-325873

Emerton's derived ordinary parts conjecture

摘录 / EXCERPT: Emerton's conjecture. Le morphisme

conjecture | open | MathDB 页面 | 原文线索

V2-030804 | MATHDB-325902

Polynomial-time fixed-topology minimum-bead tree conjecture

摘录 / EXCERPT: Let N be a set of terminals embedded in the Euclidean plane, and let G be any tree topology on N whose Steiner points have degree at least three. Fixed-topology bead-minimiza...

conjecture | open | MathDB 页面 | 原文线索

V2-030805 | MATHDB-325903

Curtis-Madsen conjecture on the mod 2 Hurewicz homomorphism

摘录 / EXCERPT: Let QS^0 denote the basepoint component of the stable homotopy space of spheres, and let $f\in\pi_{>0}QS^0$ be a stable map of positive degree. The Curtis-Madsen conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-030806 | MATHDB-325919

Compatibility conjecture for quantum characteristic classes and categorical characteristic classes

摘录 / EXCERPT: Let (M,ω) be a monotone symplectic manifold, let $P\to X$ be a Hamiltonian fibration with fiber M, and let $Fuk(M)$ be its rational, pre-triangulated A_∞ Fukaya cat...

conjecture | open | MathDB 页面 | 原文线索

V2-030807 | MATHDB-325926

Dense spectral finiteness for constrained matrix tuples

摘录 / EXCERPT: Let $\mathbb{A}\in\{0,1\}^{K\times K}$ be any matrix, and let ... with its natural subspace topology. Here spectral finiteness restricted to $\mathbb{A}$ means that the joint spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-030808 | MATHDB-325927

Equivariant homological mirror symmetry for torus fibrations

摘录 / EXCERPT: Equivariant torus-fibration mirror-symmetry conjecture. There should be equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-030809 | MATHDB-325928

Mirror symmetry for quotient torus fibrations

摘录 / EXCERPT: Quotient-cover mirror-symmetry conjecture. In this setting, there should be equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-030810 | MATHDB-325945

Strengthened Dehn lemma conjecture for the zero-framed surgery invariant

摘录 / EXCERPT: Strengthened Dehn lemma conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030811 | MATHDB-325951

Sign conjecture for Bloch-Kato Selmer groups

摘录 / EXCERPT: Let F be an imaginary quadratic field, let π be a regular algebraic cuspidal automorphic representation of GL_n over F satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-030812 | MATHDB-325992

Negative semi-topological K-theory vanishing for smooth proper dg-categories

摘录 / EXCERPT: Let T be a smooth proper dg-category over $\mathbb{C}$. Write $\mathbf{K}^{\mathrm{st}}_i(T)$ for the degree-i semi-topological K-theory group of T. The vanishing conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-030813 | MATHDB-325993

Modular comparison conjecture for algebraic and semi-topological K-theory

摘录 / EXCERPT: Let T be a dg-category over $\mathbb{C}$ and let $n>0$ be an integer. Write $\mathbf{K}(T)$ for algebraic K-theory and $\mathbf{K}^{\mathrm{st}}(T)$ for semi-topological K-theory...

conjecture | open | MathDB 页面 | 原文线索

V2-030814 | MATHDB-325994

Main conjecture on cohomological invariants of normal surface singularities

摘录 / EXCERPT: Let $(X,E)\rightarrow (V,0)$ be the minimal good resolution of a normal surface singularity, not an RDP. Write S_X for the sheaf of logarithmic derivations on X. Main conjectur...

conjecture | solved | MathDB 页面 | 原文线索

V2-030815 | MATHDB-326000

The discrete localisation conjecture for two-dimensional cobordism categories

摘录 / EXCERPT: Let $\mathcal{N}$, $\mathcal{O}$, $\overline{\mathcal{N}}$, and $\mathcal{K}$ denote the cobordism categories considered in the paper, and let their orientable subcategories and re...

conjecture | open | MathDB 页面 | 原文线索

V2-030816 | MATHDB-326011

The quasi-positivity and sharp HOMFLY bound conjecture for fillable knots

摘录 / EXCERPT: Let K be a smooth knot type. The HOMFLY bound on the maximum Thurston-Bennequin number of K is the upper bound obtained from the HOMFLY polynomial, and K is fillable if it ad...

conjecture | open | MathDB 页面 | 原文线索

V2-030817 | MATHDB-326038

Homotopy finiteness conjecture for coherent derived categories in positive characteristic

摘录 / EXCERPT: Homotopy finiteness conjecture. The DG category $D^b_{\operatorname{coh}}(X)$ is hfp.

conjecture | open | MathDB 页面 | 原文线索

V2-030818 | MATHDB-326054

Nonvanishing conjecture for the generating polynomial H_r

摘录 / EXCERPT: Fix integers d, s, and r, and define $H_{r,n}(X)$ and $H_r=\sum_{n=1}^{d-1}H_{r,n}(X_{r,n})$ as in the source. Nonvanishing conjecture for H_r. For every integer r with...

conjecture | open | MathDB 页面 | 原文线索

V2-030819 | MATHDB-326079

Nonexistence of mirror-image isotopies for sufficiently large polygon spaces

摘录 / EXCERPT: Nonexistence conjecture. For an arbitrary n with $n\geq n_0$, where n_0 is sufficiently large, the isotopy of Theorem 1 is not valid, even without the additional requirements t...

conjecture | open | MathDB 页面 | 原文线索

V2-030820 | MATHDB-326094

The positivity conjecture for K-orbit multidegrees

摘录 / EXCERPT: Let γ be a clan and let $\Upsilon_{\gamma}(X;Y)$ denote the polynomial associated to the K-orbit closure Y_{γ}. K-orbit multidegree conjecture. Theorem(III) holds f...

conjecture | solved | MathDB 页面 | 原文线索

V2-030821 | MATHDB-326148

The weak polynomial degree conjecture for homology of elementary matrix groups

摘录 / EXCERPT: Weak polynomial degree conjecture. For every nonunital ring I and every integer $n\in\mathbb{N}$, the functor $H_n(\Gamma_I)$ is weakly polynomial of degree at most $2n$.

conjecture | open | MathDB 页面 | 原文线索

V2-030822 | MATHDB-326156

Uniqueness of higher deloopings in the Weiss tower

摘录 / EXCERPT: Let V be a real inner-product space, let $QBE_k^{(\mathbb R\oplus V)\otimes_{\mathbb R}\rho_k}$ denote the source spaces in the Weiss tower, and let hoNat denote...

conjecture | open | MathDB 页面 | 原文线索

V2-030823 | MATHDB-326163

Homological mirror symmetry via the higher-dimensional wedge-contraction mirror functor

摘录 / EXCERPT: Let X be the relevant Fermat hypersurface, let $\bar{L}$ be Sheridan's immersed Lagrangian, and let W be the Landau-Ginzburg superpotential. Assume that the elements of V are...

conjecture | open | MathDB 页面 | 原文线索

V2-030824 | MATHDB-326170

The reduced structure-group rank conjecture for finitely embeddable groups

摘录 / EXCERPT: Let M be a compact oriented manifold, let $S(M)$ be its structure group, and let $S_0(M)$ be the subgroup generated by elements $[(f,M')]-[(\psi\circ f,M')]$, where $f:M'\to M$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-030825 | MATHDB-326213

Conjecture on the Euler characteristic of finite CW-complexes

摘录 / EXCERPT: Let X be a simply-connected finite CW-complex, let ${\Bbbk}$ be a field, and let $i:\Omega X\rightarrow LX$ be the inclusion of based loops into free loops. Suppose that ... is ont...

conjecture | open | MathDB 页面 | 原文线索

V2-030826 | MATHDB-326235

Non-sliceness conjecture for the exceptional pretzel family

摘录 / EXCERPT: For integers $m\geq 0$, odd integers a and q_i with $a,|q_i|\geq3$ and $a\equiv1,11,37,47,49,59\pmod{60}$, define ... Let $P(p_1,\dots,p_n)$ be a pretzel knot with one even param...

conjecture | open | MathDB 页面 | 原文线索

V2-030827 | MATHDB-326236

Ribbon-algorithm characterization for pretzel knots

摘录 / EXCERPT: Let $P(p_1,\dots,p_n)$ be a pretzel knot with $|p_i|>1$ for every i. The paper's algorithm, given by Proposition, detects certain orderings of the parameters by exhibiting a ribb...

conjecture | open | MathDB 页面 | 原文线索

V2-030828 | MATHDB-326244

Greenlees's universality conjecture for equivariant complex cobordism

摘录 / EXCERPT: Greenlees's conjecture. There is a unique homomorphism of rings

conjecture | open | MathDB 页面 | 原文线索

V2-030829 | MATHDB-326255

Contractibility conjecture for geometric joins

摘录 / EXCERPT: Let $X_1,\dots,X_m$ be subsets of $\mathbb R^d$, and let $X_{[m]}$ denote their geometric join. Contractibility conjecture. The geometric join $X_{[m]}$ is contractible whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-030830 | MATHDB-326278

Barratt-Eccles sub-operad conjecture for indexing systems

摘录 / EXCERPT: Barratt-Eccles sub-operad conjecture. If ${\mathcal C}$ is an indexing system and ${\mathcal F}$ is the associated sequence of families, then $\mathbb O^{{\mathcal F}}$ is a sub-op...

conjecture | open | MathDB 页面 | 原文线索

V2-030831 | MATHDB-326279

Tensor product conjecture for N_∞ operads

摘录 / EXCERPT: Tensor product conjecture. Subject to suitably cofibrancy conditions, ${\mathcal O}\otimes{\mathcal O}'$ is an N_∞ operad and

conjecture | open | MathDB 页面 | 原文线索

V2-030832 | MATHDB-326299

Fresse's bar-cobar equivalence conjecture for E_n-algebras

摘录 / EXCERPT: Let A be an $4\mathcal E_n$ algebra, and let $B A$ denote its bar construction, regarded as an $4\mathcal E_{n-1}$ Hopf algebra. Let $\Omega B A$ be the cobar construction applie...

conjecture | open | MathDB 页面 | 原文线索

V2-030833 | MATHDB-326335

Functoriality conjecture for the knot homology theory

摘录 / EXCERPT: Functoriality conjecture. This assignment of a map to a cobordism is independent of the choice of Morse function, i.e. this makes the knot homology theory $\mathcal{K}(-)$ functori...

conjecture | open | MathDB 页面 | 原文线索

V2-030834 | MATHDB-326341

Ruan's crepant resolution conjecture for quantum D-modules

摘录 / EXCERPT: Let $\mathcal{X}$ and Y be the orbifold and its crepant resolution, with $\mathrm{QDM}(\mathcal{X})$ and $\mathrm{QDM}(Y)$ their quantum D-modules. A global quantum D-module...

conjecture | open | MathDB 页面 | 原文线索

V2-030835 | MATHDB-326349

The higher-limit map for abelian-group presentations

摘录 / EXCERPT: Higher-limit conjecture. The above map is a natural isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-030836 | MATHDB-326387

Cohomological dimension conjecture for the diagrammatic Clifford category

摘录 / EXCERPT: Cohomological dimension conjecture. For $d535a,d535b\in\mathcal{E}(\mathcal{CL})$,

conjecture | open | MathDB 页面 | 原文线索

V2-030837 | MATHDB-326426

High-entropy winning-shift conjecture

摘录 / EXCERPT: Fix a finite alphabet S. For a subshift $X\subset S^\mathbb{X}$, let $h(X)$ denote its topological entropy and let $W(X)$ denote its winning shift. High-entropy winning-shift con...

conjecture | open | MathDB 页面 | 原文线索

V2-030838 | MATHDB-326430

The proposed relations for the coefficient ring of the 2-typization of the universal Abel formal group law

摘录 / EXCERPT: Let $\mathcal{F}_{Ab}$ be the universal Abel formal group law, and let its 2-typization have coefficient ring presented as a quotient of the localized Brown-Peterson coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-030839 | MATHDB-326452

Boyer-Zhang's integer finite surgery conjecture for hyperbolic knots

摘录 / EXCERPT: Let $K\subset S^3$ be a hyperbolic knot, and let $S^3_K(\frac{p}{q})$ denote the result of $\frac{p}{q}$-Dehn surgery on K. A finite surgery is a surgery whose resulting manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-030840 | MATHDB-326453

Unit-root Alexander polynomial conjecture for knots with L-space surgeries

摘录 / EXCERPT: Let $K\subset S^3$ be a knot. An L-space surgery is a Dehn surgery on K whose resulting rational homology sphere is an L-space, and $\Delta_K(t)$ denotes the Alexander polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-030841 | MATHDB-326477

Levin's stable intersection conjecture for compacta

摘录 / EXCERPT: Levin's stable intersection conjecture. Compacta X and Y do not admit a stable intersection in $\mathbb{R}^n$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-030842 | MATHDB-326501

Chen-Chen orthogonal LMOV conjecture

摘录 / EXCERPT: Let $\mathfrak{g}=so_{2N+1}$, let $\mathcal{L}$ be a link with L components, and let $g_{overrightarrow{mu}}(\mathcal{L};q,t)$ be the reformulated invariants defined from the ort...

conjecture | open | MathDB 页面 | 原文线索

V2-030843 | MATHDB-326504

Ozsváth-Szabó's conjecture relating absolute and Seiberg-Witten invariants

摘录 / EXCERPT: Let X be a closed oriented smooth 4-manifold with $b_2^+(X)\geq 2$, let $\mathfrak{s}\in\operatorname{Spin}^c(X)$, and let $d(\mathfrak{s})$ be the expected dimension of the Seib...

conjecture | open | MathDB 页面 | 原文线索

V2-030844 | MATHDB-326505

Szabó's Conway mutation conjecture for knot Floer homology

摘录 / EXCERPT: Let K and K' be knots related by Conway mutation, and let the δ-grading on $\widehat{HFK}$ be the difference between the Alexander and homological gradings. Szabó's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-030845 | MATHDB-326507

Némethi's lattice homology conjecture

摘录 / EXCERPT: Let G be a negative definite plumbing tree, let Y_G be its corresponding 3-manifold, and let $\mathbb{HF}^-(G)$ denote the lattice homology associated to G. Némethi's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-030846 | MATHDB-326570

Postnikov-tower mixed Hodge complex conjecture for perfect complexes

摘录 / EXCERPT: Postnikov mixed Hodge complex conjecture. For each i there exists a unipotent proalgebraic group $\mathcal{U}_i$ with a mixed Hodge structure, and the groups $\mathcal{U}_i$ form...

conjecture | open | MathDB 页面 | 原文线索

V2-030847 | MATHDB-326571

Mixed Hodge complex conjecture for the homotopy groups of very presentable stacks

摘录 / EXCERPT: Mixed Hodge complex conjecture. There exist proalgebraic groups

conjecture | open | MathDB 页面 | 原文线索

V2-030848 | MATHDB-326580

Cohomology F-manifold structure on free-loop-space homology

摘录 / EXCERPT: Free-loop-space F-manifold conjecture. These ring isomorphisms can be lifted to isomorphisms of cohomology F-manifolds. This conjecture extends the known algebraic structures relat...

conjecture | open | MathDB 页面 | 原文线索

V2-030849 | MATHDB-326605

The quasi-isomorphism conjecture for configuration space integrals on long knot spaces

摘录 / EXCERPT: Quasi-isomorphism conjecture for configuration space integrals. The map $I_{{\mathcal{K}}^n}$ is a quasi-isomorphism. The diagram complex is known to have the same homology as the...

conjecture | open | MathDB 页面 | 原文线索

V2-030850 | MATHDB-326613

The conjecture on Kronheimer-Mrowka differentials and the -grading

摘录 / EXCERPT: Kronheimer-Mrowka differential conjecture. For any knot, all non-trivial differentials in the Kronheimer-Mrowka spectral sequence strictly lower the $$ -grading.

conjecture | open | MathDB 页面 | 原文线索

V2-030851 | MATHDB-326626

Topological T-duality for Real and equivariant circle bundles

摘录 / EXCERPT: Let X be a space with a $\mathbb{Z}/2$-action. A T-dual pair consists of a pair (E,h), where $\mathbb{Z}/2$ acts equivariantly on the circle bundle $pi:E\to X$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030852 | MATHDB-326627

T-duality isomorphism for twisted Real K-theory

摘录 / EXCERPT: Let (E,h) and $(\hat E,\hat h)$ be T-dual pairs of a $\mathbb{Z}/2$-equivariant circle bundle and a Real circle bundle, with twists $h\in H^3_\pm(E)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-030853 | MATHDB-326637

The Borel conjecture for shape-equivalent matchbox manifolds

摘录 / EXCERPT: Borel conjecture for matchbox manifolds. If ${\mathfrak{M}}_1$ and ${\mathfrak{M}}_2$ are equicontinuous, ${\mathcal A}_B$-like matchbox manifolds that are shape equivalent, then...

conjecture | open | MathDB 页面 | 原文线索

V2-030854 | MATHDB-326638

The pro-fundamental-group Borel conjecture for matchbox manifolds

摘录 / EXCERPT: Pro-fundamental-group Borel conjecture. If ${\mathfrak{M}}_1$ and ${\mathfrak{M}}_2$ are equicontinuous, ${\mathcal A}_B$-like matchbox manifolds that have isomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-030855 | MATHDB-326655

The structural decomposition conjecture for equivariant Khovanov homology

摘录 / EXCERPT: Let (K,h) be a strongly invertible knot. For some $\delta\in\mathbb{Z}$, let ... Here the second grading is regarded as a relative $\mathbb{Z}$-grading, and $V[m,n]$ denotes the co...

conjecture | open | MathDB 页面 | 原文线索

V2-030856 | MATHDB-326656

The rank invariance conjecture for strong inversions

摘录 / EXCERPT: Let K be a knot, and let h_1 and h_2 be strong inversions on K. Rank invariance conjecture. The dimensions of the associated equivariant Khovanov homology groups are equal:...

conjecture | open | MathDB 页面 | 原文线索

V2-030857 | MATHDB-326657

The Khovanov-theoretic characterization conjecture for L-space knots

摘录 / EXCERPT: Recall that an L-space is a rational homology sphere Y satisfying ... and a knot in S^3 admitting an L-space surgery is called an L-space knot. A strong inversion on a knot is an...

conjecture | open | MathDB 页面 | 原文线索

V2-030858 | MATHDB-326660

Linear signature bound for non-trivial positive braids

摘录 / EXCERPT: Positive-braid signature conjecture. For every non-trivial positive braid β,

conjecture | open | MathDB 页面 | 原文线索

V2-030859 | MATHDB-326661

Homogenized signature bound for positive braids

摘录 / EXCERPT: Homogenized signature conjecture. For every positive braid β,

conjecture | open | MathDB 页面 | 原文线索

V2-030860 | MATHDB-326678

Thickness bounds imply recursively bounded refinement complexity

摘录 / EXCERPT: Let K^n be a simplicial complex, let ϵ be positive, and let $B^d_1\subset\mathbb{R}^d$ denote the unit ball. A piecewise smooth embedding of K^n into $\mathbb{R}^d$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-030861 | MATHDB-326682

Visible satellite torus conjecture for semi-adequate diagrams

摘录 / EXCERPT: Let $D(K)$ be a semi-adequate diagram of a satellite link K. Visible satellite torus conjecture. Either $D(K)$ or its mirror image can be obtained using the construction describe...

conjecture | open | MathDB 页面 | 原文线索

V2-030862 | MATHDB-326687

Conjectural twisted Stong calculation for connective twisted KO-theory

摘录 / EXCERPT: Let $k_2o\langle r\rangle$ denote the indicated connective twisted KO-theory spectrum, let $\underline{\mathcal A(1)}$ be the twisted Steenrod subalgebra, let φ be the em...

conjecture | open | MathDB 页面 | 原文线索

V2-030863 | MATHDB-326715

The dimension bound conjecture for infinite-dimensional completed-cohomology modules

摘录 / EXCERPT: Let $K=\mathrm{SL}_n(\mathbf{Z}_p)$ and let $K(p)$ be its principal congruence subgroup. Write $\Lambda=\mathbf{Z}_p[[K(p)]]$. For a finitely generated Λ-module M, let it...

conjecture | open | MathDB 页面 | 原文线索

V2-030864 | MATHDB-326716

The dimension bound conjecture for smooth admissible representations of general linear groups

摘录 / EXCERPT: Let π be an irreducible infinite-dimensional smooth admissible representation of $\mathrm{GL}_n(\mathbf{Q}_p)$, and let $\pi^{\vee}=\operatorname{Hom}(\pi,\mathbf{F}_p)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030865 | MATHDB-326777

The PL-embeddability form of the balanced Kalai-Sarkaria conjecture

摘录 / EXCERPT: Let K be a d-dimensional balanced complex, and let $<$ be a $(2,\ldots,2)$-admissible order. Let $K^{b,<}$ denote the corresponding balanced shift. Balanced PL-embeddability co...

conjecture | open | MathDB 页面 | 原文线索

V2-030866 | MATHDB-326778

The balanced shifting conjecture for the van Kampen obstruction

摘录 / EXCERPT: Let K be a d-dimensional balanced complex, and let $\mathfrak{o}(K)$ denote its van Kampen obstruction to PL embeddability in $\mathbb{S}^{2d}$. Balanced van Kampen-obstruction...

conjecture | open | MathDB 页面 | 原文线索

V2-030867 | MATHDB-326785

The central-extension relabeling conjecture for Mathieu moonshine

摘录 / EXCERPT: Let $g\in M_{23}\subset M_{24}$, let α satisfy the cocycle conditions referred to in the source, and let $C^\alpha_{M_{24}}(g)$ be the corresponding central extension, with...

conjecture | open | MathDB 页面 | 原文线索

V2-030868 | MATHDB-326786

Conjecture on codimension-one singular Riemannian foliations for non-negatively curved manifolds

摘录 / EXCERPT: Let M be a closed simply connected Riemannian manifold with non-negative sectional curvature. A singular Riemannian foliation of codimension 1 is a singular Riemannian foliation...

conjecture | open | MathDB 页面 | 原文线索

V2-030869 | MATHDB-326788

Almgren-Pitts existence conjecture for epsilon-min-max varifolds

摘录 / EXCERPT: Let X be a compact Riemannian manifold of dimension n isometrically embedded in $\mathbb{R}^N$, and let $1\leq k\leq n$. For $\varepsilon>0$, let $[\Pi]$ be a homotopy class of...

conjecture | open | MathDB 页面 | 原文线索

V2-030870 | MATHDB-326790

The filling pair systole lower-bound conjecture

摘录 / EXCERPT: Let S_g be a closed orientable surface of genus g, let $\mathcal{M}_g$ denote its moduli space of hyperbolic metrics, and let m_g be the minimum perimeter of a regular, right...

conjecture | open | MathDB 页面 | 原文线索

V2-030871 | MATHDB-326793

Differential systolic inequality for embedded one-holed tori

摘录 / EXCERPT: Let $K\subset {\mathbb{R}}^N$ be an embedded torus with one boundary component, and suppose that ∂K is a unit circle. A closed curve in K is non-null-homotopic if it is...

conjecture | open | MathDB 页面 | 原文线索

V2-030872 | MATHDB-326794

Daverman's shrinkability conjecture for upper semicontinuous decompositions of Euclidean space

摘录 / EXCERPT: Let G be a u.s.c. decomposition of E^n, let H_G be the set of nondegenerate elements of G, and let N_G denote the set used in the decomposition-space formulation. For eac...

open_problem | open | MathDB 页面 | 原文线索

V2-030873 | MATHDB-326798

Kohn's unknotting-crossing conjecture for minimal diagrams

摘录 / EXCERPT: Kohn's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-030874 | MATHDB-326799

Uniqueness conjecture for unknotting crossings of alternating knots

摘录 / EXCERPT: Uniqueness conjecture. If D contains more than one unknotting crossing of the same sign, then K is a clasp knot.

conjecture | open | MathDB 页面 | 原文线索

V2-030875 | MATHDB-326800

Seifert fibering conjecture for non-integer surgeries

摘录 / EXCERPT: Seifert fibering conjecture. If non-integer surgery on κ is Seifert fibred, then κ is a torus knot or a cable of a torus knot.

conjecture | open | MathDB 页面 | 原文线索

V2-030876 | MATHDB-326808

The bounded-intersection hub curve conjecture

摘录 / EXCERPT: Bounded-intersection hub curve conjecture. For every t, the intersection graph of $\mathcal{F}$ is chi-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-030877 | MATHDB-326820

Hecke equivariance conjecture for base change on cohomology

摘录 / EXCERPT: Hecke equivariance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030878 | MATHDB-326835

Torsion-freeness of Stone-Čech cohomology for simply connected open manifolds

摘录 / EXCERPT: Let N be a simply connected open n-manifold, let $\beta(N)$ be its Stone-Čech compactification, and let $\mathcal S$ be a locally constant sheaf on $\beta(N)$ whose stalk is a...

conjecture | open | MathDB 页面 | 原文线索

V2-030879 | MATHDB-326837

The convex-tile inverse Grünbaum-Shephard conjecture for spherical monohedral tilings

摘录 / EXCERPT: Convex-tile inverse Grünbaum-Shephard conjecture. For every normal spherical monohedral tiling topologically a Platonic solid, an Archimedean dual, an n-gonal bipyramid, or an...

conjecture | open | MathDB 页面 | 原文线索

V2-030880 | MATHDB-326842

The finite-quandle separation conjecture for knot symmetry classes

摘录 / EXCERPT: Finite-quandle separation conjecture. There exists such a sequence $\mathcal S$ satisfying, for all $K,K'\in\mathcal K$,

conjecture | open | MathDB 页面 | 原文线索

V2-030881 | MATHDB-326843

The universal quandle-coloring separation conjecture

摘录 / EXCERPT: Universal quandle-coloring separation conjecture. There exists such a sequence satisfying, for all knots $K,K' $,

conjecture | open | MathDB 页面 | 原文线索

V2-030882 | MATHDB-326845

The chain-level framed little disc operad structure on symplectic cohomology

摘录 / EXCERPT: Let M be a manifold, let $SC^{*}(M)$ denote a chain complex computing its symplectic cohomology, and let E_{2}^{f} be the framed little disc operad. Chain-level operad structur...

conjecture | open | MathDB 页面 | 原文线索

V2-030883 | MATHDB-326855

Naturality conjecture for filtered Seiberg-Witten cobordism maps and ECH

摘录 / EXCERPT: Naturality conjecture. The diagram of filtered Seiberg-Witten Floer cohomology groups, with horizontal maps given by the chain maps of and vertical maps given by the canonical iden...

conjecture | open | MathDB 页面 | 原文线索

V2-030884 | MATHDB-326866

Blanc-Toën Chern character conjecture for smooth proper DG categories

摘录 / EXCERPT: Blanc-Toën conjecture. If $A^{\bullet}$ is smooth and proper, the Chern character map is a quasi-isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-030885 | MATHDB-326879

Jablan-Sazdanović conjecture on unknotting numbers of positive odd pretzel knots

摘录 / EXCERPT: Let $P(a_1,\dots,a_n)$ be the pretzel knot with n odd and positive odd integers $a_1,\dots,a_n$, ordered so that $a_1\leq a_2\leq\dots\leq a_n$. Jablan-Sazdanović conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-030886 | MATHDB-326928

The hypermap topological-recursion conjecture

摘录 / EXCERPT: For a positive integer a, let $M_{g,n}^{[a]}(b_1,\ldots,b_n)$ denote the enumeration coefficients of a-hypermaps, and let $\omega_{g,n}$ be the correlation differentials produc...

conjecture | open | MathDB 页面 | 原文线索

V2-030887 | MATHDB-326942

Uniqueness conjecture for highly twisted plat projections

摘录 / EXCERPT: Uniqueness conjecture. K has a unique such highly twisted plat projection. This asks whether sufficiently high twisting determines a unique highly twisted plat projection, extend...

conjecture | open | MathDB 页面 | 原文线索

V2-030888 | MATHDB-326944

Pivotal fusion categories admit combed field theory structures

摘录 / EXCERPT: Let $\mathcal C$ be a pivotal fusion category over a field of characteristic zero. An $SO(2)$ homotopy fixed point structure on $\mathcal C$ is the additional structure correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-030889 | MATHDB-326945

Fusion categories admit spin field theory structures

摘录 / EXCERPT: Let $\mathcal C$ be a fusion category over an algebraically closed field of characteristic zero. A $Spin(3)$ homotopy fixed point structure on $\mathcal C$ is the additional struct...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-030890 | MATHDB-326950

Munson-Volić collapse conjecture for the long-link spectral sequence

摘录 / EXCERPT: Let $\mathcal{K}^{m\bullet}_d$ be the cosimplicial space of long links of m strands in $\mathbb{R}^d$ constructed by Munson and Volić, and let its homology Bousfield-Kan spectral...

conjecture | open | MathDB 页面 | 原文线索

V2-030891 | MATHDB-326951

Positive radius conjecture for the long-knot Poincaré series

摘录 / EXCERPT: Let R denote the radius of convergence of the Poincaré series of the space of long knots modulo immersions. Positive-radius conjecture. The radius of convergence is positive; tha...

conjecture | open | MathDB 页面 | 原文线索

V2-030892 | MATHDB-326963

Homotopy equivalence conjecture for irreducible euclidean Artin groups

摘录 / EXCERPT: Let A be an irreducible Artin group of euclidean type, and let $B A$ be the classifying space constructed in the paper. Let M be the standard topological space with fundamental...

conjecture | open | MathDB 页面 | 原文线索

V2-030893 | MATHDB-326981

Low-dimensional sandpile toppling-probability exponent

摘录 / EXCERPT: Let ν be the infinite-volume sandpile measure on $\{0,1,\ldots,2d-1\}^{\mathbb{Z}^d}$, let $\mathrm{Av}_o=\{x\in\mathbb{Z}^d:n(o,x;\cdot)>0\}$ be the avalanche set generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-030894 | MATHDB-327016

Kauffman-Przytycki conjecture on detection of minimal-genus surfaces by the surface bracket polynomial

摘录 / EXCERPT: Kauffman-Przytycki conjecture. For a virtual knot L, if a diagram of L is realized in a surface of the minimal genus, then this fact is detected by the surface bracket polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-030895 | MATHDB-327027

Purity conjecture for the fundamental domain of an affine Springer fiber

摘录 / EXCERPT: Fundamental-domain purity conjecture. The cohomology of F_{γ} is pure in the sense of Deligne.

conjecture | open | MathDB 页面 | 原文线索

V2-030896 | MATHDB-327029

Formal Poincare polynomial conjecture for torus actions

摘录 / EXCERPT: Formal Poincare polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-030897 | MATHDB-327038

The minimum b_2 conjecture for right-angled Artin groups

摘录 / EXCERPT: Let G be a right-angled Artin group, and let $h(G)$ be the minimum of $b_2(M)$ over all closed, oriented topological 4-manifolds M with fundamental group isomorphic to G. L...

conjecture | open | MathDB 页面 | 原文线索

V2-030898 | MATHDB-327108

Random mapping-torus volume and torsion growth limits

摘录 / EXCERPT: Consider random mapping tori whose monodromies are random words, and let the genus of the fiber tend to infinity. Let the expected growth rates refer to growth as a function of the...

conjecture | open | MathDB 页面 | 原文线索

V2-030899 | MATHDB-327116

The Baire power conjecture for Black's winning strategy

摘录 / EXCERPT: Let X be a topological space, let $\pi(X)$ be the least cardinality of a π-base for X, and let $\operatorname{BM}(X)$ denote the Banach-Mazur game on X. For positive card...

conjecture | open | MathDB 页面 | 原文线索

V2-030900 | MATHDB-327134

The PFA triviality conjecture for Stone-Čech remainders

摘录 / EXCERPT: Let X and Y be locally compact Polish spaces. For a locally compact space X, write $X^* = \beta X\setminus X$ for its Stone-Čech remainder, and call a homeomorphism between s...

conjecture | open | MathDB 页面 | 原文线索

V2-030901 | MATHDB-327135

The CH abundance result for nontrivial autohomeomorphisms

摘录 / EXCERPT: Let X be a locally compact, non-compact, separable metrizable space, and let X^* denote its Stone-Čech remainder. An autohomeomorphism of X^* is a homeomorphism from X^* to...

conjecture | open | MathDB 页面 | 原文线索

V2-030902 | MATHDB-327137

Hyperbolic spherical space form surgery conjecture

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3, and let $p>0$ be an integer such that the p-surgery on K is a spherical space form of $\mathbf{T}$-, $\mathbf{O}$-, or $\mathbf{I}$-type....

conjecture | open | MathDB 页面 | 原文线索

V2-030903 | MATHDB-327168

The twisted big J-function conjecture for GIT quotients

摘录 / EXCERPT: Twisted big J-function conjecture. The function $\tilde{\mathds{J}}^{\theta,\varepsilon,E}$ lies on the Lagrangian cone encoding the genus-zero, ${\bf T}$-equivariant,...

conjecture | open | MathDB 页面 | 原文线索

V2-030904 | MATHDB-327178

The NLS-detection gluing conjecture for graph manifolds

摘录 / EXCERPT: Let W be a rational homology 3-sphere graph manifold with JSJ tori, and let m be the number of these tori. For each i, let $[alpha_i]$ be a rational slope associated to the...

conjecture | open | MathDB 页面 | 原文线索

V2-030905 | MATHDB-327208

Gabrielov-Vorobjov approximation conjecture without separability

摘录 / EXCERPT: Let $S \subset K \subset \mathbb R^n$ be a bounded definable set with a monotone family of compact sets $\{S_\delta\}_{\delta>0}$ satisfying $S=\bigcup_\delta S_\delta$. Suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-030906 | MATHDB-327219

Linear homotopy complexity conjecture for locally symmetric spaces

摘录 / EXCERPT: Linear homotopy complexity conjecture. There are constants α and d, depending only on X, such that every such M is homotopically equivalent to a simplicial complex...

conjecture | open | MathDB 页面 | 原文线索

V2-030907 | MATHDB-327231

The second cohomology classification conjecture for generalized groupoids

摘录 / EXCERPT: Second cohomology classification conjecture. The bijective correspondence of the previous proposition descends to a bijective correspondence between the set of homotopy equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-030908 | MATHDB-327240

Loday's conjectural Leibniz homology theory for groups

摘录 / EXCERPT: Let $\mathbf{Grp}$ be the category of groups, let $\mathbf{Zinb}^c$ denote coZinbiel coalgebras, and let $\mathbf{Com}^c$ denote cocommutative coalgebras. For a group G, write...

conjecture | open | MathDB 页面 | 原文线索

V2-030909 | MATHDB-327241

Loday's conjecture on Leibniz homology of abelian groups

摘录 / EXCERPT: Let G be an abelian group, let $\Bbbk$ be a field, let $\mathrm{HL}_n(G,\Bbbk)$ denote Leibniz homology, and let $\mathrm{H}_n(G,\Bbbk)$ denote Eilenberg-MacLane homology. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030910 | MATHDB-327242

Loday's conjecture on Leibniz homology of the linear group

摘录 / EXCERPT: Let R be a unital associative ring and let $\mathrm{GL}(R)$ denote its infinite linear group. Write $\mathrm{HL}_{\bullet}(\mathrm{GL}(R),\Bbbk)$ for the Leibniz homology of this...

conjecture | open | MathDB 页面 | 原文线索

V2-030911 | MATHDB-327251

Hom-module generator bound for torsion-free modules

摘录 / EXCERPT: Hom-module generator conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-030912 | MATHDB-327255

Polynomial bound relating homological degree and regularity

摘录 / EXCERPT: Homological-degree regularity conjecture. There exists a polynomial $f(t)$ of degree d such that

conjecture | open | MathDB 页面 | 原文线索

V2-030913 | MATHDB-327277

Bounded profile conjecture for nilpotent cohomology

摘录 / EXCERPT: Let X be a space such that the reduced cohomology $\tilde H^*X$ is nilpotent. Let $q_X=w_{QH^*X}$ be the profile function of the unstable module of indecomposable elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-030914 | MATHDB-327279

The volume conjecture for the colored Jones polynomial

摘录 / EXCERPT: Let K be a knot in S^3, and let $J_N(K;q)\in\mathbb{Z}[q,q^{-1}]$ be its colored Jones polynomial associated with the irreducible N-dimensional representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-030915 | MATHDB-327288

The handlebody conjecture for self-affine 3-manifolds

摘录 / EXCERPT: The handlebody conjecture. Every self-affine 3-manifold is homeomorphic to a handlebody.

conjecture | open | MathDB 页面 | 原文线索

V2-030916 | MATHDB-327318

The S^1-stability conjecture for positive scalar curvature

摘录 / EXCERPT: S^1-stability conjecture. The manifold M admits a metric of positive scalar curvature if and only if its product with the circle does:

conjecture | open | MathDB 页面 | 原文线索

V2-030917 | MATHDB-327343

The conjecture on de Rham representations in analytic cohomology of the Lubin-Tate tower

摘录 / EXCERPT: Let $\rho_p:G_{\mathbb{Q}_p}\rightarrow\operatorname{GL}_2(E)$ be a continuous de Rham Galois representation, and let $B(\rho_p)$ denote the associated p-adic local Langlands rep...

conjecture | open | MathDB 页面 | 原文线索

V2-030918 | MATHDB-327358

The module-model conjecture for 2-bridge knot skein modules

摘录 / EXCERPT: Let K be a 2-bridge knot, and let M be the module defined in the source by the referenced construction. Let the nonsymmetric skein module of K be regarded as a module over th...

conjecture | open | MathDB 页面 | 原文线索

V2-030919 | MATHDB-327410

Ellipticity and geometric formality conjecture for positive quaternion Kähler manifolds

摘录 / EXCERPT: Ellipticity and formality conjecture. Every positive quaternion Kähler manifold is rationally elliptic and geometrically formal.

conjecture | open | MathDB 页面 | 原文线索

V2-030920 | MATHDB-327430

Handle-decomposition independence of the sutured monopole Floer contact map

摘录 / EXCERPT: Let (M,Γ,ξ_M) be a sutured contact manifold contained in (M',Γ'), and let ξ be a contact structure on $M'\smallsetminus \operatorname{int}(M)$ with dividing set...

conjecture | open | MathDB 页面 | 原文线索

V2-030921 | MATHDB-327431

Isomorphism between sutured monopole and sutured Floer contact invariants

摘录 / EXCERPT: Let (M,Γ,ξ) be a sutured contact manifold, and let $\psi(M,\Gamma,\xi)\in\underline{SHM}(-M,-\Gamma)$ and $EH(M,\Gamma,\xi)\in SFH(-M,-\Gamma)$ denote the monopole Floer a...

conjecture | open | MathDB 页面 | 原文线索

V2-030922 | MATHDB-327453

Golden-ratio conjecture for normalized dilatations

摘录 / EXCERPT: Let S be a closed surface and let ϕ be a pseudo-Anosov mapping class with dilatation $\lambda(\phi)$. For a fibered cohomology class, let $L(S,\phi)$ denote the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-030923 | MATHDB-327457

Finite-type conjecture for Polyak-Viro 1-cocycles of knots

摘录 / EXCERPT: Finite-type conjecture. Every 1-cocycle presented by an arrow germ formula of degree n is of finite type, of degree at most n.

conjecture | open | MathDB 页面 | 原文线索

V2-030924 | MATHDB-327484

Farrell-Ontaneda conjecture on the topological triviality of negatively curved bundles

摘录 / EXCERPT: Let X be a compact simply connected manifold or a simply connected finite simplicial complex, and let $M\to E\stackrel{p}{\to}X$ be a negatively curved bundle. A bundle is topolo...

conjecture | open | MathDB 页面 | 原文线索

V2-030925 | MATHDB-327486

Conjecture on topological triviality from low-dimensional center dynamics

摘录 / EXCERPT: Let X be a closed manifold or finite simplicial complex, and let $p\colon E\to X$ be a fiber homotopically trivial bundle with closed-manifold fiber M. Assume that E admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-030926 | MATHDB-327487

Conjecture on topological triviality from irreducible partially hyperbolic dynamics

摘录 / EXCERPT: Let X be a simply connected closed manifold or a simply connected finite simplicial complex, and let $p\colon E\to X$ be a fiber bundle with closed-manifold fiber M. Assume tha...

conjecture | open | MathDB 页面 | 原文线索

V2-030927 | MATHDB-327500

Chromatic redshift conjecture for the K-theory of algebraic integers

摘录 / EXCERPT: Let $n\ge1$, let E_n be the $K(n)$-local Lubin-Tate ring spectrum with connective cover e_n, and let Ω_n be the p-completed homotopy colimit of the connected commutati...

conjecture | open | MathDB 页面 | 原文线索

V2-030928 | MATHDB-327503

Kono's conjecture on Samelson products in p-regular exceptional Lie groups

摘录 / EXCERPT: Kono's conjecture. For $i,j\in\mathtt{t}(G)$, there exists $k\in\mathtt{t}(G)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-030929 | MATHDB-327525

Conjecture on cyclic homology of the crossed product by \(\mathbb Z_2\)

摘录 / EXCERPT: Let $\theta\notin\mathbb Q$, let $\mathcal A_\theta$ be the algebraic irrational rotational algebra, and let $\mathcal A_\theta\rtimes\mathbb Z_2$ denote its crossed product by the...

conjecture | open | MathDB 页面 | 原文线索

V2-030930 | MATHDB-327534

The recurring fantasy on cup products for ordinary surfaces

摘录 / EXCERPT: The recurring-fantasy conjecture. For every nonzero $v\in H^1(X,W\mathcal O_X)$, the map $\langle v,-\rangle$ is injective, and for every nonzero $v'\in H^0(X,W\Omega^1_X)$, the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-030931 | MATHDB-327535

The Calabi-Yau lifting conjecture in terms of Hodge-Witt numbers

摘录 / EXCERPT: Let X be a smooth, projective Calabi-Yau threefold over a field of positive characteristic. Let $H^0(X,\Omega^1_X)$ denote its global one-forms and let $h^{1,2}_W$ be its $(1,2)$...

conjecture | open | MathDB 页面 | 原文线索

V2-030932 | MATHDB-327552

Athanasia's gamma-vector monotonicity conjecture for flag homology subdivisions

摘录 / EXCERPT: Athanasia's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-030933 | MATHDB-327566

Doeraene-El Haouari conjecture on sectional and relative category

摘录 / EXCERPT: Let $f:Y\rightarrow X$ be a map. A homotopy retraction of f is a map $r:X\rightarrow Y$ such that $rf\simeq \operatorname{id}_Y$. The sectional category of f, denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-030934 | MATHDB-327567

Iwase-Sakai conjecture on monoidal topological complexity

摘录 / EXCERPT: Let X be a locally finite simplicial complex, and let $\operatorname{TC}(X)$ denote its topological complexity and $\operatorname{TC}^M(X)$ its monoidal topological complexity. T...

conjecture | open | MathDB 页面 | 原文线索

V2-030935 | MATHDB-327572

Hess-Hirsch's homotopy-sphere conjecture for the reduced weak separation complex

摘录 / EXCERPT: For $2\leq k\leq n-2$, let $\widetilde{\Delta}^{Sep}_{k,n}$ be the reduced complex obtained from the weak separation complex of weakly separated k-subsets of $[n]$ by removing it...

conjecture | open | MathDB 页面 | 原文线索

V2-030936 | MATHDB-327576

Conjecture on limits of area-minimizing surfaces with non-tall asymptotic boundary

摘录 / EXCERPT: Limit-boundary conjecture. The curve $\widehat{\Gamma}=\partial_\infty\Sigma$ is tall, and $\Gamma\mathbin{\triangle}\widehat{\Gamma}$ is a collection of nullhomotopic short curves...

conjecture | open | MathDB 页面 | 原文线索

V2-030937 | MATHDB-327579

The Kähler-Ricci flow Gromov-Hausdorff convergence conjecture

摘录 / EXCERPT: Kähler-Ricci flow convergence conjecture. The metric spaces $(X,g(t))$ converge to the metric completion (X_{can},g_∞) in Gromov-Hausdorff topology as $t\to\infty$. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-030938 | MATHDB-327581

The compactness conjecture for negatively curved Kähler-Einstein manifolds

摘录 / EXCERPT: Compactness conjecture. For any such sequence, after passing to a subsequence, the sequence converges in pointed Gromov-Hausdorff topology to a singular Kähler-Einstein metric leng...

conjecture | open | MathDB 页面 | 原文线索

V2-030939 | MATHDB-327584

Stable sl_N homology is generated by the x_k and μ_k

摘录 / EXCERPT: The generation conjecture. The homology of d_N is generated as an algebra by x_k and mu_k. This gives a proposed algebraic description of stable sl_N-homolog...

conjecture | open | MathDB 页面 | 原文线索

V2-030940 | MATHDB-327599

Lagrangian-torus C^0-convergence conjecture

摘录 / EXCERPT: Let L and L' be smooth manifolds, let $i_k:L\hookrightarrow T^*L'$ be smooth Lagrangian embeddings for $k=1,2,\ldots$, and let $\iota:L\hookrightarrow T^*L'$ be a topological e...

conjecture | open | MathDB 页面 | 原文线索

V2-030941 | MATHDB-327628

Purity conjecture for fundamental groups of L_n-local E_infinity-rings

摘录 / EXCERPT: Purity conjecture. For every L_n-local $\mathbf{E}_{\infty}$-ring R, the map $R\to L_1R$ induces an isomorphism on fundamental groups.

conjecture | open | MathDB 页面 | 原文线索

V2-030942 | MATHDB-327636

The Jacobian conjecture reformulation for two-variable polynomial maps

摘录 / EXCERPT: Let k be a field with $\operatorname{char}(k)=0$. Take $A=k[X,Y]$, $B=k[f,g]$, and let $\varepsilon:B\to A$ be determined by $\varepsilon(f)=f(X,Y)$ and $\varepsilon(g)=g(X,Y)$....

conjecture | open | MathDB 页面 | 原文线索

V2-030943 | MATHDB-327637

Stošić's color-shift conjecture for rational two-component links

摘录 / EXCERPT: Changing color on unknot components only shifts q-grading. There exist rational functions $F_j(L)\in \mathbb{Z}a^{\pm 1},s^{\pm 1},t^{\pm 1}$ such that, for all $i\geq j$,

conjecture | open | MathDB 页面 | 原文线索

V2-030944 | MATHDB-327638

The power conjecture for colored HOMFLY homology of rational knots and links

摘录 / EXCERPT: Power conjecture. There exists a corrected Q-grading such that

conjecture | open | MathDB 页面 | 原文线索

V2-030945 | MATHDB-327711

Propagation of the main hypothesis through categorical Bockstein reduction

摘录 / EXCERPT: Let k be a complete discrete valuation ring with uniformizing element $l\in k$. For objects $X,Y\in\mathcal E_{k,0}^{G/H}$, suppose that the natural maps ... are surjective for all...

conjecture | open | MathDB 页面 | 原文线索

V2-030946 | MATHDB-327715

Palis's density conjecture for vector fields

摘录 / EXCERPT: Let M be a manifold and let X be a vector field on M. Say that X is accumulated by a class of vector fields if it lies in the closure of that class in the relevant topology...

conjecture | open | MathDB 页面 | 原文线索

V2-030947 | MATHDB-327721

Bracketed-foliation conjecture for taut oriented C^0-foliations

摘录 / EXCERPT: Let M be a closed oriented 3-manifold with $M\ne S^1\times S^2$, and let $\mathcal F$ be a taut oriented C^0-foliation of M. A foliation is bracketed if, for some volume-pres...

conjecture | open | MathDB 页面 | 原文线索

V2-030948 | MATHDB-327756

Finiteness conjecture for knots sharing the Floer homology of an L-space knot

摘录 / EXCERPT: Let K be an L-space knot, meaning a knot for which $S^3_n(K)$ is an L-space for some $n>0$. Let $\widehat{HFK}(K)$ denote its hat knot Floer homology. L-space-knot finiteness con...

conjecture | open | MathDB 页面 | 原文线索

V2-030949 | MATHDB-327757

Strong inversion conjecture for L-space knots

摘录 / EXCERPT: Let K be an L-space knot, meaning a knot for which $S^3_n(K)$ is an L-space for some $n>0$. A strong inversion is an involution of the ambient 3-manifold preserving the knot wh...

conjecture | open | MathDB 页面 | 原文线索

V2-030950 | MATHDB-327759

Li-Ni conjecture on L-space-knot Floer homology and iterated torus knots

摘录 / EXCERPT: Let K be a knot whose Floer homology agrees with that of an L-space knot, and suppose the Alexander polynomial of that L-space knot has all its roots on the unit circle. Li-Ni co...

conjecture | open | MathDB 页面 | 原文线索

V2-030951 | MATHDB-327776

Conjectural graded dimension formula for the torus state space

摘录 / EXCERPT: Let $\Sigma=S^1\times S^1$ and let $\omega\in H^1(S^1\times S^1;{\mathbb Z}/2{\mathbb Z})$. For a graded vector space, write $\dim_s$ for the sum over $k\in{\mathbb Z}$ of s^k ti...

conjecture | open | MathDB 页面 | 原文线索

V2-030952 | MATHDB-327783

Conjecture on Ricci-positive metrics after connected sum with a homotopy sphere

摘录 / EXCERPT: Let M be a highly connected manifold admitting a positive scalar curvature metric and with vanishing Witten genus. A homotopy-sphere Ricci-positivity conjecture. There exists a h...

conjecture | open | MathDB 页面 | 原文线索

V2-030953 | MATHDB-327789

Lidman-Moore conjecture on essential Conway spheres in L-space knots

摘录 / EXCERPT: Let K be a knot in S^3. An essential Conway sphere is an embedded sphere intersecting K transversally in four points such that the resulting four-punctured sphere is incompre...

conjecture | open | MathDB 页面 | 原文线索

V2-030954 | MATHDB-327790

Lidman-Moore conjecture on tangle primeness of L-space knots

摘录 / EXCERPT: Let K be a knot in a 3--manifold Y. An essential n--string tangle decomposition is an embedded sphere Q intersecting K transversally in $2n$ points such that...

conjecture | open | MathDB 页面 | 原文线索

V2-030955 | MATHDB-327809

The lattice-cohomological index conjecture for rational cuspidal plane curves

摘录 / EXCERPT: Let $L=S^3_{-d}(K)$ be the link of a superisolated surface singularity corresponding to a rational cuspidal projective plane curve of degree d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030956 | MATHDB-327815

Kanenobu knots crossing-number upper-bound conjecture

摘录 / EXCERPT: Kanenobu knots crossing-number upper-bound conjecture. The crossing number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-030957 | MATHDB-327836

Stable cohomology conjecture for finite p-groups

摘录 / EXCERPT: Stable-cohomology top-degree conjecture. There exists a G-module A with trivial action such that

conjecture | open | MathDB 页面 | 原文线索

V2-030958 | MATHDB-327849

The equivariant Eilenberg spectral sequence algebra conjecture

摘录 / EXCERPT: Equivariant Eilenberg spectral sequence algebra conjecture. In some cases of interest, this trigraded equivariant Eilenberg spectral sequence is a spectral sequence of algebras. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-030959 | MATHDB-327864

Nilpotence and boundedness conjecture for the Cooper-Krushkal projectors

摘录 / EXCERPT: For each $1\leq k\leq n$, let P_n be the Cooper-Krushkal projector and let $U_k^{(n)}\in\operatorname{END}(P_n)$ denote the map from the paper's polynomial action. Nilpotence and...

conjecture | open | MathDB 页面 | 原文线索

V2-030960 | MATHDB-327880

Breuil-Herzig global ordinary-part conjecture for indecomposable points

摘录 / EXCERPT: Let X be the parameter space appearing in the global setting, let L be the coefficient field, and let Σ_p be the set of places above p. For a generic ordinary point...

conjecture | open | MathDB 页面 | 原文线索

V2-030961 | MATHDB-327882

Langlands correspondence conjecture for rigid automorphic data

摘录 / EXCERPT: Langlands correspondence conjecture for rigid objects. The local system $\mathcal{F}_{\pi}$ is cohomologically rigid. Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-030962 | MATHDB-327886

Canard-cycle conjecture for simple unconnected slow curves

摘录 / EXCERPT: Canard-cycle conjecture. For every desired number $l\in\mathbb{N}$ there exists a local topologically generic set of systems such that the slow curve is nondegenerate and simple un...

conjecture | open | MathDB 页面 | 原文线索

V2-030963 | MATHDB-327888

Effectiveness conjecture for Legendrian and transverse knot invariants

摘录 / EXCERPT: Let (K,p) be a based, oriented Legendrian or transverse knot in a contact 3-manifold (Y,ξ), and let $\mathcal{L}$ and $\mathcal{T}$ denote the associated Legendrian and tra...

conjecture | open | MathDB 页面 | 原文线索

V2-030964 | MATHDB-327889

Handle-decomposition independence conjecture for sutured instanton cobordism maps

摘录 / EXCERPT: Let (M,Γ) be a sutured submanifold of (M',Γ'), and let ξ be a contact structure on $M'\setminus\operatorname{int}(M)$ with dividing set $\Gamma\cup\Gamma'$. Given...

conjecture | open | MathDB 页面 | 原文线索

V2-030965 | MATHDB-327910

Odd-primary characteristic conjecture for $S/\!/p^r,\alpha_1$

摘录 / EXCERPT: Odd-primary characteristic conjecture. $S/\!/p^r,\alpha_1$ is a characteristic for $H\mathbb{Z}/p^r$.

conjecture | open | MathDB 页面 | 原文线索

V2-030966 | MATHDB-327911

Characteristic conjecture for $S/\!/2^r,\eta$

摘录 / EXCERPT: Characteristic conjecture. $S/\!/2^r,\eta$ is a characteristic for $H\mathbb{Z}/2^r$.

conjecture | open | MathDB 页面 | 原文线索

V2-030967 | MATHDB-327912

Characteristic morphisms for $S/\!/\eta,\sigma$, $S/\!/\nu,\sigma$, and $S/\!/\sigma$

摘录 / EXCERPT: Characteristic morphism conjecture. Each of the three displayed maps is a characteristic morphism.

conjecture | open | MathDB 页面 | 原文线索

V2-030968 | MATHDB-327913

Epimorphism conjecture for a characteristic morphism to tmf

摘录 / EXCERPT: Epimorphism conjecture. A 2-local characteristic morphism

conjecture | open | MathDB 页面 | 原文线索

V2-030969 | MATHDB-327920

AJ conjecture for the recurrence polynomial of a knot

摘录 / EXCERPT: Let K) be a knot in the 3-sphere, and let $calpha_K(t,M,L)$ be its recurrence polynomial. Let the A-polynomial of K be the classical polynomial in the meridian and longitud...

conjecture | open | MathDB 页面 | 原文线索

V2-030970 | MATHDB-327921

Kauffman's conjecture on pseudo-alternating links and alternative links

摘录 / EXCERPT: Kauffman's conjecture. Every pseudo-alternating link is alternative.

conjecture | open | MathDB 页面 | 原文线索

V2-030971 | MATHDB-327922

Finite residuality of involutory quandles of knots

摘录 / EXCERPT: Involutory quandles are finitely residual. If $IQ(D)$ is non-trivial, then there is a finite non-trivial involutory quandle Q and a surjective homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-030972 | MATHDB-327923

Signature-index invariance under oriented homotopy equivalence at infinity

摘录 / EXCERPT: Signature-index invariance conjecture. The large-scale indices should coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-030973 | MATHDB-327951

The no-other-contributions conjecture for orbifold Gromov-Witten invariants of \(\mathbb{P}^1_{3,6,6}\)

摘录 / EXCERPT: Let $h_8(q)$ and $h_9(q)$ denote the generating functions for the orbifold insertions described above, and let $F_{0,3}$ and $F_{1,3}$ be the corresponding decompositions by the no...

conjecture | open | MathDB 页面 | 原文线索

V2-030974 | MATHDB-327970

Non-degeneracy conjecture for the knot-dragging bilinear form

摘录 / EXCERPT: Non-degeneracy conjecture. The form $B_{\mathbb{D}^1}$ is non-degenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-030975 | MATHDB-327971

Entropy conjecture for positive pseudo-Anosov 4-braids

摘录 / EXCERPT: Let $\beta=\beta^+\Delta$ be a positive pseudo-Anosov 4-braid containing a half-twist and whose closure is a knot. Let $\lambda_{R(\beta)}$ be the 1-cocycle spectral radius, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-030976 | MATHDB-327986

Finite-value conjecture for the normalized 4d-TQFT invariant

摘录 / EXCERPT: Finite-value conjecture. For a given oriented closed 4-manifold X, the normalized quantum invariant, considered as a function on the set of all complex roots of unity, takes on...

conjecture | open | MathDB 页面 | 原文线索

V2-030977 | MATHDB-327993

The algorithmic orientation-reversing symmetry problem for irreducible 3-manifolds

摘录 / EXCERPT: Let M be a closed, orientable, irreducible 3-manifold. Algorithmic orientation-reversing symmetry conjecture. There is an algorithm that determines whether M admits an orient...

conjecture | open | MathDB 页面 | 原文线索

V2-030978 | MATHDB-328029

Mosaic number conjecture for the knots 6_{1} and 6_{3}

摘录 / EXCERPT: For a positive integer n, an n-mosaic is an $n\times n$ mosaic-tile board representing a link or knot, and the mosaic number of a knot is the smallest natural number m for wh...

conjecture | open | MathDB 页面 | 原文线索

V2-030979 | MATHDB-328030

Chirality conjecture for the E_{2k+1} knot family

摘录 / EXCERPT: For each integer $k\geq 1$, let E_{2k+1} denote the family of alternating knots that embed on an M_{2k+1} mosaic with $|D|=2$, where D is the relevant dual structure. A knot...

conjecture | open | MathDB 页面 | 原文线索

V2-030980 | MATHDB-328066

Satellite L-space knot converse conjecture

摘录 / EXCERPT: Let P be a pattern in the solid torus and let K be a knot, with satellite knot $P(K)$. An L-space knot is a knot admitting a positive Dehn surgery yielding an L-space. Satellit...

conjecture | open | MathDB 页面 | 原文线索

V2-030981 | MATHDB-328077

Tait's minimal alternating diagram conjecture

摘录 / EXCERPT: Tait's conjecture. Every alternating link has an alternating diagram with minimal crossing number among all diagrams of that link.

conjecture | open | MathDB 页面 | 原文线索

V2-030982 | MATHDB-328078

Adequate-knot homology and mutation conjecture

摘录 / EXCERPT: Adequate-knot homology and mutation conjecture. For any adequate diagram of K, the ranks of both Khovanov homology and knot Floer homology in each δ-grading are given by...

conjecture | open | MathDB 页面 | 原文线索

V2-030983 | MATHDB-328098

The Whittaker conditions for fundamental classes of Uhlenbeck spaces

摘录 / EXCERPT: Let $|1^d\rangle=[\mathcal U_G^d]$ be the fundamental class in equivariant intersection cohomology, and let $\widetilde W^{(\kappa)}_n$ denote the positive modes of the generators...

conjecture | open | MathDB 页面 | 原文线索

V2-030984 | MATHDB-328112

The small perturbation Seiberg-Witten conjecture for property $(\ast)$

摘录 / EXCERPT: Small perturbation Seiberg-Witten conjecture. Under these assumptions, M satisfies property $(\ast)$.

conjecture | open | MathDB 页面 | 原文线索

V2-030985 | MATHDB-328126

Conjecture on a noncircular unknotted curve that cannot form Borromean rings

摘录 / EXCERPT: Let three unknotted curves in $\R^3$ be given, and suppose that at least one of them is not a circle. Noncircular obstruction conjecture. It is possible to choose such three curves...

conjecture | open | MathDB 页面 | 原文线索

V2-030986 | MATHDB-328127

Planar-curve conjecture for forming Borromean rings

摘录 / EXCERPT: Let C_1, C_2, and C_3 be any three planar curves. Planar-curve conjecture. They can be used to form the Borromean rings provided that at least one of them is not a circle. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-030987 | MATHDB-328201

The compatibility conjecture for Beilinson-Tate and TLF residues

摘录 / EXCERPT: Let K be an n-dimensional topological local field over $\boldsymbol{k}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-030988 | MATHDB-328203

Continuity conjecture for the Beilinson-Tate residue

摘录 / EXCERPT: Let K be a topological local field over $\boldsymbol{k}$, and let $\operatorname{Res}^{\mathrm{BT}}_{K / \boldsymbol{k}}:\Omega^n_{K / \boldsymbol{k}}\to\boldsymbol{k}$ be its Be...

conjecture | open | MathDB 页面 | 原文线索

V2-030989 | MATHDB-328207

Realization conjecture for topological conformal dimension

摘录 / EXCERPT: For every $d \in[2,\infty]$, let X be a metric space and denote its topological conformal dimension by $\dim_{tC}X$. Realization conjecture. For every $d\in[2,\infty]$ there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-030990 | MATHDB-328234

The diffeomorphism conjecture for the moment-angle manifold associated to K

摘录 / EXCERPT: Diffeomorphism conjecture. $\mathcal{Z}_{K}$ is diffeomorphic to the connected sum of sphere products in Proposition 1.

conjecture | open | MathDB 页面 | 原文线索

V2-030991 | MATHDB-328238

The restriction-map conjecture for localized Steenrod-algebra Ext

摘录 / EXCERPT: Let $A(2)\subset A$ be the indicated subalgebra of the mod-2 Steenrod algebra, and consider the localization map ... Let a_1 and v_n denote the classes used in the source. Rest...

conjecture | open | MathDB 页面 | 原文线索

V2-030992 | MATHDB-328261

Conjecture on hyperbolic knots admitting Nil Seifert fibred surgery

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3. A surgery on K is a Dehn surgery whose resulting manifold is a Nil Seifert fibred space. Nil surgery conjecture. If K admits a surgery yi...

conjecture | open | MathDB 页面 | 原文线索

V2-030993 | MATHDB-328262

Finiteness conjecture for Seifert surgeries with fixed hyperbolic base orbifold

摘录 / EXCERPT: Fix a 2-orbifold $S^2(k,l,m)$ with $k,l,m>1$. Consider Dehn surgeries on hyperbolic knots in S^3 that produce Seifert fibred spaces having $S^2(k,l,m)$ as their base orbifold....

conjecture | open | MathDB 页面 | 原文线索

V2-030994 | MATHDB-328273

Equivariant Yoneda-center conjecture for symplectic duality

摘录 / EXCERPT: Let $\mathfrak{M}$ and $\mathfrak{M}^!$ be a symplectic dual pair, let E be the Yoneda algebra of the category $\mathcal{O}$ for $\mathfrak{M}$, and let $\widetilde E$ be its uni...

conjecture | open | MathDB 页面 | 原文线索

V2-030995 | MATHDB-328284

The tidy-pair vanishing conjecture

摘录 / EXCERPT: Let (M^{2k+1},F^{2k}) be a tidy pair, with $i:F\hookrightarrow M$ the inclusion and $i_*:L^2H_k(F,\partial F)\to L^2H_k(M,\partial M)$ the induced map. Tidy-pair vanishing conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-030996 | MATHDB-328285

Connes-Karoubi regulator conjecture for generalized Dirac operators

摘录 / EXCERPT: Let X be a closed Riemannian manifold of odd dimension d equipped with a generalized Dirac operator $\slashed{D}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-030997 | MATHDB-328289

Unmarked-boundary-loop diameter growth conjecture for modular flip-graphs

摘录 / EXCERPT: Unmarked-loop diameter growth conjecture. For every $\varepsilon>0$, there exists k_ε such that, if Σ is a surface with k_ε unmarked boundary non-pri...

conjecture | open | MathDB 页面 | 原文线索

V2-030998 | MATHDB-328290

Filling-surface growth-rate finiteness conjecture

摘录 / EXCERPT: Filling-surface growth-rate finiteness conjecture. The number of topological types of filling surfaces with the same growth rate is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-030999 | MATHDB-328295

Displaceability conjecture for pre-Lagrangian toric fibers

摘录 / EXCERPT: Let V be a compact toric contact manifold. A contact manifold is orderable if its identity component of the contactomorphism group admits no positive contractible loop, and a pre...

conjecture | open | MathDB 页面 | 原文线索

V2-031000 | MATHDB-328303

The schematized opposite-groupoid mirror conjecture

摘录 / EXCERPT: Let Y be a compact Kähler manifold, let Y_{top} denote its underlying topological space, and let $\mathcal{F}$ be a Kan-complex fibrant replacement of the singular simplicial s...

conjecture | open | MathDB 页面 | 原文线索

V2-031001 | MATHDB-328304

The symmetry actions and dualizability conjectures for higher categories

摘录 / EXCERPT: Let Cat be the category of categories, let $(\infty,n)\operatorname{Cat}$ denote an (∞,n)-category of higher categories, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031002 | MATHDB-328316

The equality case for the improved positive-writhe inequality

摘录 / EXCERPT: Let the assumptions of Lemma(c) hold, and consider the inequality in Remark. Equality conjecture. Under these assumptions, the inequality is an equality. This proposed equality wou...

conjecture | open | MathDB 页面 | 原文线索

V2-031003 | MATHDB-328329

The rank-genus conjecture for knot complements

摘录 / EXCERPT: For a knot $K\subset S^3$, let $r(K)$ be the minimal number of generators of $\pi_1(S^3-K)$, and let $E(K,S^3)$ denote its exterior. The rank-genus conjecture. For all knots...

conjecture | open | MathDB 页面 | 原文线索

V2-031004 | MATHDB-328345

Algebraicity conjecture for perturbative invariants of fiber knots

摘录 / EXCERPT: Algebraicity conjecture. The extension L can be chosen depending only on the ambient manifold and the Lie-algebra series $X\in\{A,BCD\}$, and, at least for fiber knots, the pertu...

conjecture | open | MathDB 页面 | 原文线索

V2-031005 | MATHDB-328347

Algebraicity conjecture for perturbative link invariants in Seifert spaces

摘录 / EXCERPT: Link-invariant algebraicity conjecture. All perturbative colored HOMFLY invariants of links in M belong to L, and all perturbative colored Kauffman invariants of links in M b...

conjecture | open | MathDB 页面 | 原文线索

V2-031006 | MATHDB-328350

Functoriality conjecture for marked link Floer homology

摘录 / EXCERPT: Functoriality conjecture. Link Floer homology is functorial with respect to marked smooth isotopy classes of surfaces embedded in four-space: a marked surface with boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-031007 | MATHDB-328356

Katok's conjecture on DUE cohomologically stable diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold and let $f\in Diff^{\infty}(M)$ be a diffeomorphism. The diffeomorphism f is distributionally uniquely ergodic (DUE) when its invariant distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-031008 | MATHDB-328357

Givental's conjecture on the R-matrix action for semisimple CohFTs

摘录 / EXCERPT: Let V be the state space of a cohomological field theory with metric η, and let $R=1+O(z)$ be an endomorphism-valued power series satisfying $\eta(R(z)v,R(-z)w)=\eta(v,w)$....

conjecture | open | MathDB 页面 | 原文线索

V2-031009 | MATHDB-328384

Quandle-coloring distinguishability conjecture for knots modulo reversed mirrors

摘录 / EXCERPT: Let K and K' be knots, and let $rm(K)$ denote the reversed mirror of K. A finite quandle is a set with a self-distributive binary operation, and ${\rm Col}_{X}(K)$ denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-031010 | MATHDB-328385

End-monochromaticity conjecture for selected Rig quandles

摘录 / EXCERPT: End-monochromaticity conjecture. Every knot K is end monochromatic with X.

conjecture | open | MathDB 页面 | 原文线索

V2-031011 | MATHDB-328407

Poincaré embedding multiple disjunction conjecture

摘录 / EXCERPT: Poincaré embedding multiple disjunction conjecture. If the handle codimension restrictions are omitted and $r\geq1$, then the r-cube $E(P,N-Q_\bullet)$ is $(1-p+\Sigma)$-cartesia...

conjecture | open | MathDB 页面 | 原文线索

V2-031012 | MATHDB-328408

Poincaré embedding-to-mapping conjecture

摘录 / EXCERPT: Let N and P be Poincaré spaces, and let $Q_1,\dots,Q_r$ be Poincaré spaces embedded in N. Let $\mathit{EF}(P,N-Q_\bullet)$ denote the $(r+1)$-cube obtained from the cube of P...

conjecture | open | MathDB 页面 | 原文线索

V2-031013 | MATHDB-328409

Multiple disjunction conjecture for embeddings of several Poincaré spaces

摘录 / EXCERPT: Multiple disjunction conjecture for Poincaré embeddings. If $r\geq2$, then the r-cube $E(Q_\bullet,N)$ is $(3-n+\Sigma)$-cartesian. This is one of the paper's conjectural extensi...

conjecture | open | MathDB 页面 | 原文线索

V2-031014 | MATHDB-328429

Sойбельman's conjecture on the full COHA of the quiver A_1

摘录 / EXCERPT: Sойбельman's conjecture. The full COHA for the quiver A_1 is isomorphic to Cl_c, with mixed generators satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-031015 | MATHDB-328432

Berge-type conjecture on longitudinal surgery duals

摘录 / EXCERPT: Berge-type conjecture. If knots $K_1\subset L(p_1,q_1)$ and $K_2\subset L(p_2,q_2)$ are longitudinal surgery duals, then, up to reindexing, K_2 is a $(1,1)$-knot and...

conjecture | open | MathDB 页面 | 原文线索

V2-031016 | MATHDB-328463

The cabling conjecture for reducible surgeries on knots

摘录 / EXCERPT: Let K be a knot in S^3, and let $M_K(\gamma)$ denote the manifold obtained by Dehn surgery on K with integer slope γ. A knot is a (p,q)-cable if it is a cable knot w...

conjecture | open | MathDB 页面 | 原文线索

V2-031017 | MATHDB-328464

The conjectural p-adic Adams operations on algebraic K-theory

摘录 / EXCERPT: Let R be an E_{∞} ring spectrum. Write $K(R)^{\scriptscriptstyle\wedge}_{p}$ for its p-completed algebraic K-theory spectrum, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031018 | MATHDB-328465

The compatibility of the cyclotomic trace with p-adic Adams operations

摘录 / EXCERPT: Let R be a connective E_{∞} ring spectrum, and let ... be the cyclotomic trace, where both spectra are p-completed. The cyclotomic-trace compatibility conjecture. The cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-031019 | MATHDB-328473

Dranishnikov's rational inessentiality conjecture for macroscopic smallness

摘录 / EXCERPT: Let M be a closed oriented n-dimensional manifold, let $\pi=\pi_1(M)$, and let $B\pi$ be a classifying space. If $f\colon M\to B\pi$ classifies the universal bundle, call M r...

conjecture | open | MathDB 页面 | 原文线索

V2-031020 | MATHDB-328485

The homological determinant conjecture for noetherian connected graded AS Gorenstein algebras

摘录 / EXCERPT: Let A be a noetherian connected graded Artin-Schelter Gorenstein algebra, and let μ_A denote its Nakayama automorphism. Let hdet denote the homological deter...

conjecture | open | MathDB 页面 | 原文线索

V2-031021 | MATHDB-328516

Cantor attractor conjecture for random piecewise isometries

摘录 / EXCERPT: Cantor attractor conjecture. If A is an interval, the topological attractor $X(w)$ is almost surely a Cantor set.

conjecture | open | MathDB 页面 | 原文线索

V2-031022 | MATHDB-328521

The topological extension conjecture for total ideals of C^*-algebras

摘录 / EXCERPT: Topological extension conjecture. The functors $\widetilde{\mathscr{T}}$ and $\mathcal{I}$ are naturally isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-031023 | MATHDB-328529

The thickness conjecture for unstable modules with finite injective hulls

摘录 / EXCERPT: Let $\mathcal U$ be the category of unstable modules, and consider the full subcategory whose objects are unstable modules whose injective hull is a finite direct sum of indecompos...

conjecture | open | MathDB 页面 | 原文线索

V2-031024 | MATHDB-328544

The non-delooping conjecture for mapping spaces of classifying-space skeleta

摘录 / EXCERPT: Let G be a non-commutative compact connected Lie group, let B_nG denote the n-th stage of the classifying-space construction, and let $\iota_n\colon B_nG\to BG$ be the canoni...

conjecture | open | MathDB 页面 | 原文线索

V2-031025 | MATHDB-328546

Guillemin-Ginzburg-Karshon's conjecture on equivariant cohomology Chern numbers

摘录 / EXCERPT: Guillemin-Ginzburg-Karshon's conjecture. An element $\beta\in\Omega_*^{U,G}$ is zero if and only if all integral equivariant cohomology Chern numbers of β vanish.

conjecture | open | MathDB 页面 | 原文线索

V2-031026 | MATHDB-328556

The growth rate conjecture for Witten-Reshetikhin-Turaev invariants

摘录 / EXCERPT: Let X be an oriented closed 3-manifold and L a framed link in X. For conjugacy classes $\bar c=(c_1,\ldots,c_n)$, let $\mathcal M_{X,L,\bar c}^{q_j}$ be the union of compon...

conjecture | open | MathDB 页面 | 原文线索

V2-031027 | MATHDB-328563

Recurrence transfer conjecture for nilsystems

摘录 / EXCERPT: Let $s\geq 1$, and let $R\subset\mathbb N$ be a set of s-recurrence for s-step nilsystems. Recurrence transfer conjecture. Then R is a set of s-recurrence for all t-step...

conjecture | open | MathDB 页面 | 原文线索

V2-031028 | MATHDB-328602

Homological critical-probability conjecture for hyperbolic percolation

摘录 / EXCERPT: Let $G(m)$ be the hyperbolic graph considered in the paper, let p_c denote its critical probability, and let ... where $D(p)$ is the rank difference function defined from the limit...

conjecture | open | MathDB 页面 | 原文线索

V2-031029 | MATHDB-328615

Conjecture on stable ergodicity and dominated splittings

摘录 / EXCERPT: Let M be the connected manifold in the source, let $r>1$, and consider the volume-preserving diffeomorphism space $\operatorname{Diff}^r_m(M)$ with its C^1 topology. A diffeomo...

conjecture | open | MathDB 页面 | 原文线索

V2-031030 | MATHDB-328624

Equivariant quantum K-theory presentation for Grassmannians

摘录 / EXCERPT: Let $qh_n^{\ast}$ be the generalized quantum cohomology ring for the Grassmannian $\limfunc{Gr}_{n,N}=\limfunc{Gr}_n(\mathbb{C}^N)$, and let $\mathbb{T}$ denote the torus used for...

conjecture | open | MathDB 页面 | 原文线索

V2-031031 | MATHDB-328639

Y. Matsumoto's 11/8-conjecture for smooth 4-manifolds

摘录 / EXCERPT: Matsumoto's $11/8$-conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-031032 | MATHDB-328640

Equivalent 11/8-conjecture for the bounding genus and ds-invariant

摘录 / EXCERPT: Let Y be a homology 3-sphere with $(Y)=0$, and let $\frak{ds}(Y)$ be finite. The bounding genus $|Y|$ is the invariant defined using homologically 1-connected bounding 4-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-031033 | MATHDB-328670

The ^1-connectedness conjecture for reductive algebraic groups

摘录 / EXCERPT: The conjecture. For every field extension F of k,

conjecture | open | MathDB 页面 | 原文线索

V2-031034 | MATHDB-328680

Conjecture on the classification of L-space two-bridge links

摘录 / EXCERPT: Classification conjecture. The set of all L-space two-bridge links is

conjecture | open | MathDB 页面 | 原文线索

V2-031035 | MATHDB-328697

The new intersection conjecture

摘录 / EXCERPT: Intersection conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-031036 | MATHDB-328699

Trace characterization for traversally generic fields

摘录 / EXCERPT: Trace characterization. The restriction of $\ker(\mathcal L_v)$ to the boundary $\partial_1X\subset X$ coincides with

conjecture | open | MathDB 页面 | 原文线索

V2-031037 | MATHDB-328730

The torus-generator chain-map conjecture for convex toric domains

摘录 / EXCERPT: Let X_Ω be a smooth convex toric domain with boundary Y. Let $C_*^{T^3}$ be the chain complex freely generated over $\mathbb{Z}/2$ by torus generators, with differential g...

conjecture | open | MathDB 页面 | 原文线索

V2-031038 | MATHDB-328734

Gerko's conjecture on semidualizing dimensions

摘录 / EXCERPT: Let R be the ring under consideration, let C be a semidualizing module, and let X be any R-module. Write $\mathcal{G}_C\mathop{\rm \text{-}dim}\nolimits X$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-031039 | MATHDB-328745

The splicing conjecture for knot complements and L-spaces

摘录 / EXCERPT: Splicing conjecture. The manifold Y is an L-space if and only if all of the following conditions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-031040 | MATHDB-328751

The bounded height conjecture in geometric topology

摘录 / EXCERPT: Let M be a k-cusped hyperbolic 3-manifold. A Dehn filling point is a point corresponding to a Dehn filling of M, and its height is the height used in the associated Dehn fi...

conjecture | open | MathDB 页面 | 原文线索

V2-031041 | MATHDB-328790

Connectivity criterion for homological stability in homogeneous categories

摘录 / EXCERPT: Let $(\mathcal{C},\oplus,0)$ be a homogeneous category and let A and X be objects of $\mathcal{C}$. Set $G_n=\operatorname{Aut}(A\oplus X^{\oplus n})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031042 | MATHDB-328805

Euler-Poincaré-Dirac index conjecture for Harish-Chandra modules

摘录 / EXCERPT: Let $\widetilde K$ be the spin cover of a maximal compact subgroup K, and let $\mathsf{I}_{\mathrm{Dir}}(X)$ denote the Dirac index of a finite-length Harish-Chandra module X....

conjecture | open | MathDB 页面 | 原文线索

V2-031043 | MATHDB-328835

Balanced first Betti number lower bound conjecture

摘录 / EXCERPT: Let Δ be a balanced connected simplicial complex of dimension $d-1 \geq 3$. If Δ is a ${\bf k}$-homology manifold that is orientable over ${\bf k}$, then the balanced...

conjecture | open | MathDB 页面 | 原文线索

V2-031044 | MATHDB-328836

Balanced first Betti number equality conjecture

摘录 / EXCERPT: Let Δ be a balanced connected ${\bf k}$-homology manifold that is orientable over ${\bf k}$ of dimension $d-1 \geq 3$. Let $\mathcal{BH}^d$ denote the class of balanced conn...

conjecture | open | MathDB 页面 | 原文线索

V2-031045 | MATHDB-328849

Cobordism hypothesis for topological quantum field theories with moduli level m

摘录 / EXCERPT: Let $m\geq 0$, and let ${\rm Bord}^{G}(n)$ denote the symmetric monoidal bordism (∞,n)-category with G-structure. Let $(m+n)\text{-}\mathrm{Vect}$ be the symmetric monoida...

conjecture | open | MathDB 页面 | 原文线索

V2-031046 | MATHDB-328850

Reshetikhin-Turaev comparison conjecture for modular tensor categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a modular tensor category. Let $(\mathcal{A},q)$ be an isomorphism between Vect and $\mathcal{C}$ in ${\rm Braid}^{\otimes}$, equivalently an equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-031047 | MATHDB-328887

The overtwistedness conjecture for positive contact surgery on Legendrian knots

摘录 / EXCERPT: Let L be a null-homologous Legendrian knot with Thurston-Bennequin invariant $tb(L)\leq -2$, and let n be a positive integer. Overtwistedness conjecture. If $n<|tb(L)|$, then c...

conjecture | open | MathDB 页面 | 原文线索

V2-031048 | MATHDB-328901

Freedman-He-Wang conjecture for Möbius cross energy of links

摘录 / EXCERPT: Freedman-He-Wang conjecture. The Möbius energy is minimized, among the class of all nontrivial links in $\mathbb R^3$, by the stereographic projection of the standard Hopf link in...

conjecture | open | MathDB 页面 | 原文线索

V2-031049 | MATHDB-328979

Marino's LMOV integrality conjecture for composite invariants

摘录 / EXCERPT: Marino's conjecture. The reformulated quantities

conjecture | open | MathDB 页面 | 原文线索

V2-031050 | MATHDB-328980

The congruent skein relation for reformulated colored HOMFLYPT invariants

摘录 / EXCERPT: Congruent skein relation. If the crossing is a self-crossing of a knot, then

conjecture | open | MathDB 页面 | 原文线索

V2-031051 | MATHDB-328999

Coarse metrizability of proper bounded coarse structures

摘录 / EXCERPT: Let X be a set and let ${\operatorname{D}}$ be a family of pseudometrics on X. The topology induced by ${\operatorname{D}}$ is the topology generated by the pseudometric balls,...

conjecture | open | MathDB 页面 | 原文线索

V2-031052 | MATHDB-329008

Metric convergence at finite-time singularities of the continuity method

摘录 / EXCERPT: Finite-time metric convergence conjecture. As $t\to T$, (M,ω_t) converges in the GromovHausdorff topology to a compact metric space (M_T,d_T) satisfying the following:...

conjecture | open | MathDB 页面 | 原文线索

V2-031053 | MATHDB-329010

Kähler-Einstein limit conjecture in the general-type case

摘录 / EXCERPT: General-type Kähler-Einstein limit conjecture. If $\kappa(M)=\dim M=n$, then $t^{-1}\omega_t$ should converge in the smooth topology on $M\setminus D$ to a Kähler-Einstein metric...

conjecture | open | MathDB 页面 | 原文线索

V2-031054 | MATHDB-329011

Generalized Kähler-Einstein limit conjecture for intermediate Kodaira dimension

摘录 / EXCERPT: Generalized Kähler-Einstein limit conjecture. The rescaled metrics $(M,t^{-1}\omega_t)$ should converge in the Gromov-Hausdorff topology to a compact metric space...

conjecture | open | MathDB 页面 | 原文线索

V2-031055 | MATHDB-329019

The conjecture that the stated monodromy condition characterizes infinite distortion

摘录 / EXCERPT: Let $X\to B$ be an almost Postnikov pair with $\pi_1X=\pi_1B=\Gamma$, monodromy representation ρ, and Euler class eu. Suppose that for some $\gamma\in\Gamma$,...

conjecture | refuted | MathDB 页面 | 原文线索

V2-031056 | MATHDB-329020

Two Summands Conjecture for reducible Dehn surgery

摘录 / EXCERPT: Let K be a nontrivial knot in S^3, and let $S^3_r(K)$ denote the result of r-Dehn surgery on K, for $r\in\mathbb{Q}$. A 3-manifold is reducible if it contains an essential...

conjecture | open | MathDB 页面 | 原文线索

V2-031057 | MATHDB-329051

Full embedded contact knot homology conjecture for knots

摘录 / EXCERPT: Let K be a knot in a contact three-manifold Y, and let α be any contact form on Y adapted to K. Denote by $ECK(K,Y,\alpha)$ the full embedded contact knot homology a...

conjecture | open | MathDB 页面 | 原文线索

V2-031058 | MATHDB-329052

Embedded contact knot homology conjecture for links

摘录 / EXCERPT: Let L be a link in a contact three-manifold Y, and let α be any contact form on Y adapted to L. Denote by $ECK(L,Y,\alpha)$ and $\widehat{ECK}(L,Y,\alpha)$ the full...

conjecture | open | MathDB 页面 | 原文线索

V2-031059 | MATHDB-329071

Path-connectedness of spaces of polynomial knots

摘录 / EXCERPT: Polynomial-knot path-connectedness conjecture. For every $d\geq3$, the space $\boldsymbol{P}_d$ is path connected.

conjecture | open | MathDB 页面 | 原文线索

V2-031060 | MATHDB-329081

Unramifiedness and nilpotent ramification conjecture for Hilbert modular cohomology

摘录 / EXCERPT: Unramifiedness and nilpotent ramification conjecture. The representation $\bar\rho$ is unramified at $\mathfrak{p}$, and there exists a positive integer n depending on g such t...

conjecture | open | MathDB 页面 | 原文线索

V2-031061 | MATHDB-329090

Postnikov-Reiner-Williams subdivision conjecture for gamma-vectors

摘录 / EXCERPT: Let Δ and Δ' be flag homology spheres, and suppose that Δ' geometrically subdivides Δ. Let $\gamma(\Delta)$ and $\gamma(\Delta')$ denote their gamma-vec...

conjecture | open | MathDB 页面 | 原文线索

V2-031062 | MATHDB-329101

The full L^2-Alexander torsion conjecture for genus, fiberedness and monodromy entropy

摘录 / EXCERPT: Let K be a knot. The full L^2-Alexander torsion conjecture. (1) The full L^2-Alexander torsion $d_0f^{(2)}(K)(t)$ determines the genus of K. (2) The full L^2-Alexander to...

conjecture | open | MathDB 页面 | 原文线索

V2-031063 | MATHDB-329102

Dunfield--Friedl--Jackson conjecture for the discrete faithful representation

摘录 / EXCERPT: Let $K\subset S^3$ be a hyperbolic oriented knot, and let π_K denote its knot group. Let $\alpha\colon \pi_K\to \operatorname{SL}(2,\mathbb{C})$ be a lift of the discrete and f...

conjecture | open | MathDB 页面 | 原文线索

V2-031064 | MATHDB-329103

Existence conjecture for genus-detecting higher-order Alexander torsion

摘录 / EXCERPT: Let K be a knot. Genus-detecting higher-order torsion conjecture. There exists an epimorphism $\gamma\colon\pi_K\to\Gamma$ onto a torsion-free elementary-amenable group such that...

conjecture | open | MathDB 页面 | 原文线索

V2-031065 | MATHDB-329104

Regularity conjecture for L^2-Alexander torsion

摘录 / EXCERPT: Let K be an oriented knot, let π_K denote its knot group, and let $\gamma\colon\pi_K\to\Gamma$ be an admissible epimorphism. L^2-Alexander torsion regularity conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-031066 | MATHDB-329105

L^2-Alexander torsion and higher-order torsion degree conjecture

摘录 / EXCERPT: Let K be an oriented knot, let π_K denote its knot group, and let $\gamma\colon\pi_K\to\Gamma$ be an epimorphism onto a nontrivial torsion-free elementary-amenable group....

conjecture | open | MathDB 页面 | 原文线索

V2-031067 | MATHDB-329106

Full L^2-Alexander torsion conjecture for knot genus

摘录 / EXCERPT: Let K be an oriented knot. Full L^2-Alexander genus conjecture. The full L^2-Alexander torsion satisfies ... This would make the full L^2-Alexander torsion a sharp genus inva...

conjecture | open | MathDB 页面 | 原文线索

V2-031068 | MATHDB-329139

The cohomological-weight and Grothendieck-Teichmüller isomorphism conjecture

摘录 / EXCERPT: Let $A(M^{\delta}_0)$ be the cooperadic algebra associated with Brown's compactified moduli spaces, let coGer be the coGerstenhaber cooperad, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031069 | MATHDB-329193

The indecomposability conjecture for a Lagrangian concordance from the trefoil

摘录 / EXCERPT: Let $W\subset J^1(S^1)$ be the Legendrian Whitehead double pattern, let U be the standard Legendrian unknot, and let Λ be the Legendrian $\overline{9_{46}}$ knot with a L...

conjecture | open | MathDB 页面 | 原文线索

V2-031070 | MATHDB-329199

Topological Whitehead double conjecture

摘录 / EXCERPT: Topological Whitehead double conjecture. The Whitehead double $\Wh(L)$ of a link L is freely slice if and only if L is homotopically trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-031071 | MATHDB-329200

Homotopy ribbon-slice conjecture

摘录 / EXCERPT: Homotopy ribbon-slice conjecture. A knot is slice if and only if it is homotopy ribbon.

conjecture | open | MathDB 页面 | 原文线索

V2-031072 | MATHDB-329201

Topological classification conjecture for the underlying space of the orbifold $\calO$

摘录 / EXCERPT: Topological classification conjecture for $|\calO|$. The underlying space $|\calO|$ is homeomorphic to either

conjecture | open | MathDB 页面 | 原文线索

V2-031073 | MATHDB-329227

Trivalent partner conjecture for loop graphs in odd graph cohomology

摘录 / EXCERPT: Let GC_3 be the graph complex, and let L_k denote its loop graphs. The partner classes are those classes killed by the loop graphs in the spectral sequence described i...

conjecture | open | MathDB 页面 | 原文线索

V2-031074 | MATHDB-329228

Effective quasi-hyperbolic splitting conjecture for branched surfaces

摘录 / EXCERPT: Let N be a compact atoroidal 3-manifold, and let B be a branched surface in N. A branched surface is quasi-hyperbolic if it carries no sphere or torus but fully carries a sur...

conjecture | open | MathDB 页面 | 原文线索

V2-031075 | MATHDB-329233

Rigidity conjecture for baby tricorn-like sets

摘录 / EXCERPT: Rigidity conjecture for baby tricorn-like sets. Any two baby tricorn-like sets are not homeomorphic unless they are symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-031076 | MATHDB-329235

The rational cohomology characterization of complex tori

摘录 / EXCERPT: Let X be a compact Kähler manifold of complex dimension n, and write $H^*(X,\mathbb{Q})$ for its rational cohomology algebra. The rational cohomology conjecture for complex tor...

conjecture | open | MathDB 页面 | 原文线索

V2-031077 | MATHDB-329293

Twisted-cohomological criterion for isotopy of Legendrian foliations

摘录 / EXCERPT: Let F_t be a C^0-small homotopy of foliations and let L be a fixed transverse line field. Let F_t have defining one-forms $l_t=l+b_t$, with $b_0=0$, where $b_t(L)=0$ for al...

conjecture | open | MathDB 页面 | 原文线索

V2-031078 | MATHDB-329321

The Tor generator conjecture for Cohen-Macaulay quotients

摘录 / EXCERPT: Let (R,m) be a regular local ring with Krull dimension n. Suppose $I+J$ is an m-primary ideal in R and R/I and R/J are Cohen-Macaulay. The Tor generator conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-031079 | MATHDB-329322

The Artinian first Koszul homology generator conjecture

摘录 / EXCERPT: Let (A,m) be an Artinian local ring, and let $x_1,\ldots,x_n\in m$. Write $H_1(x_1,\ldots,x_n;A)$ for the first Koszul homology module of the sequence on A. Artinian Koszul hom...

conjecture | open | MathDB 页面 | 原文线索

V2-031080 | MATHDB-329330

Homological modification of mutation for flips

摘录 / EXCERPT: Work in the setting of Cohen-Macaulay rational singularities. Let $N=H^0(\mathcal{V}_X)$, where $\mathcal{V}_X$ is the object from the paper’s contraction and tilting setup, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-031081 | MATHDB-329345

Uniform-transcendental master-integral conjecture

摘录 / EXCERPT: A master integral is one of the finitely many basis integrals to which the integrals of a topology reduce through integration-by-parts identities. A master integral is uniformly tr...

conjecture | open | MathDB 页面 | 原文线索

V2-031082 | MATHDB-329354

Castle-Evans-Hyde conjecture on nontrivial planar graph embeddings on the torus

摘录 / EXCERPT: Let G be a planar graph embedded nontrivially on the torus. A nontrivial knot is a knotted embedded circle, and a nonsplit link is a nonsplit link with at least two components. C...

conjecture | open | MathDB 页面 | 原文线索

V2-031083 | MATHDB-329358

The determinant-volume inequality for alternating hyperbolic links

摘录 / EXCERPT: Let K be an alternating hyperbolic link, and write ${\rm vol}(K)$ for its hyperbolic volume and $\det(K)$ for its determinant. The determinant-volume conjecture. ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-031084 | MATHDB-329359

The alternating-diagram maximal volume conjecture

摘录 / EXCERPT: Let K be an alternating hyperbolic knot, and let K' be obtained from K by changing crossings in a diagram. The alternating-diagram maximal volume conjecture. (a) If any cross...

conjecture | open | MathDB 页面 | 原文线索

V2-031085 | MATHDB-329385

Fukaya's Whitehead torsion conjecture for Floer complexes

摘录 / EXCERPT: Let L_1 and L_2 be exact Lagrangians with Floer homology $HF(L_1,L_2)$ and Floer complex $CF(L_1,L_2)$, whose Whitehead torsion is denoted by $\tau(CF(L_1,L_2))$. Fukaya's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-031086 | MATHDB-329397

Extension of the monoidal algebraic model to rational torus-power spectra

摘录 / EXCERPT: Let $r>1$, and consider the product $\torus^r$ of r copies of $\torus$. The algebraic model for rational $\torus^r$-equivariant spectra has an injective model structure in which...

conjecture | open | MathDB 页面 | 原文线索

V2-031087 | MATHDB-329398

Kameko's dimension conjecture for the hit problem

摘录 / EXCERPT: Let E^k be an elementary abelian 2-group of rank k, let BE^k be its classifying space, and set ... Write $\mathbb F_2\underset{\mathcal A}\otimes P_k$ for the quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-031088 | MATHDB-329399

Characterization of weakly aleph-spaces for spaces of continuous functions

摘录 / EXCERPT: Weak-space characterization conjecture. $C_c(X)$ is a weakly $\aleph$-space if and only if $C_c(X)$ is a weakly $\aleph_0$-space if and only if X is a countable $\aleph_0$-space.

conjecture | open | MathDB 页面 | 原文线索

V2-031089 | MATHDB-329421

Open Calabi-Yau mirror symmetry for cohomology F-manifolds

摘录 / EXCERPT: Let M and $M^{\vee}$ be open Calabi-Yau manifolds that are mirror to each other in the sense of homological mirror symmetry. A ring isomorphism is an isomorphism preserving the p...

conjecture | open | MathDB 页面 | 原文线索

V2-031090 | MATHDB-329422

Seidel's quintic divisor-complement conjecture

摘录 / EXCERPT: Let $\overline{M}$ be the quintic three-fold, let $D\subset\overline{M}$ be the smooth hyperplane section used in the paper, and set $M=\overline{M}\backslash D$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031091 | MATHDB-329424

The slope conjecture on the linear degree terms of colored Jones polynomials

摘录 / EXCERPT: For a knot $K\subset S^3$, write the maximal and minimal degrees of its colored Jones function as quadratic quasi-polynomials ... A knot is called mono-sloped when each...

conjecture | open | MathDB 页面 | 原文线索

V2-031092 | MATHDB-329439

Self-replication conjecture for the formal Baker-Akhiezer kernel

摘录 / EXCERPT: Let ${\cal S}$ be a spectral curve, let $\psi({\cal S},\hbar;p_1,p_2)$ be its formal Baker-Akhiezer kernel, and define $\Omega_p(q)=\frac{B(p,q)}{dx(p)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031093 | MATHDB-329440

Colored Jones Baker-Akhiezer kernel conjecture

摘录 / EXCERPT: Let $\mathfrak{K}$ be a knot with spectral curve ${\cal S}$ defined by its A-polynomial, and let $J_N(q)$ be the colored-Jones polynomial associated with a representation...

conjecture | open | MathDB 页面 | 原文线索

V2-031094 | MATHDB-329441

Kameko's dimension conjecture for the Peterson hit problem

摘录 / EXCERPT: Let P_k be the polynomial algebra on k generators over $\mathbb{F}_2$, let QP_k denote its indecomposable quotient under the Steenrod algebra, and write $(QP_k)_n$ for its de...

conjecture | open | MathDB 页面 | 原文线索

V2-031095 | MATHDB-329452

Homotopy-Oni composition identity

摘录 / EXCERPT: Let H_0 be a C^2-small Hamiltonian, let H be a Hamiltonian, and let $CF_*(H_0)$ and $CF_*(H)$ denote the corresponding Floer complexes. Let $\mathrm{id}|_{CF_*(H_0)}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-031096 | MATHDB-329453

Factorization conjecture for the Oni operator

摘录 / EXCERPT: Let H be a Hamiltonian satisfying ... so that its time-one map admits a square root. Let n and $\kappa(\omega)$ be the parameters appearing in the paper, and let ν be even an...

conjecture | open | MathDB 页面 | 原文线索

V2-031097 | MATHDB-329467

The suspension detection conjecture for maps inducing isomorphisms in homology and fundamental group

摘录 / EXCERPT: Suspension detection conjecture. Then $g\simeq*$.

conjecture | open | MathDB 页面 | 原文线索

V2-031098 | MATHDB-329475

The topological two-contact-point conjecture for cheese domains

摘录 / EXCERPT: Let Ω_0 be a compact convex subset of $\mathbb{R}^2$ with non-empty interior. Let Ω_1 be a convex open set such that ... Set $\Omega=\Omega_0\setminus\Omega_1$. For a b...

conjecture | open | MathDB 页面 | 原文线索

V2-031099 | MATHDB-329488

Linear linking-number bound for polygonal unknots

摘录 / EXCERPT: Let $b3$ be a piecewise linear unknot with vertices $p_i,dots,p_n$. Let $varepsilon>0$ be small enough that $N_varepsilon(b3)$ is a tubular neighborhood. For each choice of points...

conjecture | open | MathDB 页面 | 原文线索

V2-031100 | MATHDB-329508

Stable non-nullity conjecture for quasitoric manifolds

摘录 / EXCERPT: Stable non-nullity conjecture. For any quasitoric manifold M, $\Sigma^\infty\pi$ is not null homotopic.

conjecture | open | MathDB 页面 | 原文线索

V2-031101 | MATHDB-329522

Finite Whitehead conjecture for finite aspherical 2-complexes

摘录 / EXCERPT: Let L be a finite, aspherical 2-complex and let K be a subcomplex of L. Finite Whitehead conjecture. K is aspherical. This is the finite case singled out because Howie's re...

conjecture | open | MathDB 页面 | 原文线索

V2-031102 | MATHDB-329559

The symplectic-invariant formula for the irregular spectral curve

摘录 / EXCERPT: Symplectic-invariant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031103 | MATHDB-329567

The Atiyah-Floer conjecture for pillowcase homology

摘录 / EXCERPT: Let (X,K) be a knot in a homology 3-sphere, and choose a 2-tangle decomposition $(X,K)=(Y,T)\cup(D,U)$ as in the paper. Let P denote the pillowcase, let $R_\pi(Y,T)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-031104 | MATHDB-329570

The HOMFLY slope conjecture

摘录 / EXCERPT: HOMFLY slope conjecture. For any knot K,

conjecture | open | MathDB 页面 | 原文线索

V2-031105 | MATHDB-329575

The classification conjecture for square-zero sphere classes

摘录 / EXCERPT: Let X be the smooth, closed, oriented 4-manifold under consideration, with signature σ and modified Euler characteristic $\tilde \chi$, and let Γ denote its inter...

conjecture | open | MathDB 页面 | 原文线索

V2-031106 | MATHDB-329605

Strong Slope Conjecture for knots

摘录 / EXCERPT: Let K be a knot, let M_K be its exterior, and let js_K be the set of Jones slopes obtained from the cluster points of $4n^{-2}d_+[J_K(n)]$. Let jx_K be the cluster-point se...

conjecture | open | MathDB 页面 | 原文线索

V2-031107 | MATHDB-329606

Conjecture on the linear terms of colored Jones degrees

摘录 / EXCERPT: For every non-trivial knot $K\subset S^3$, let $b_K(n)$ denote the periodic linear coefficient in the eventual quadratic quasi-polynomial expression for $d_+[J_K(n)]$. Linear-term...

conjecture | open | MathDB 页面 | 原文线索

V2-031108 | MATHDB-329611

SL(2, C)-twisted homology product conjecture for taut sutured manifolds

摘录 / EXCERPT: Let $M=(M,R_-,R_+,\gamma)$ be a taut sutured manifold. A representation $\alpha\colon\pi_1(M)\to\operatorname{SL}_2(\mathbb C)$ makes M a homology product when the inclusion-indu...

conjecture | open | MathDB 页面 | 原文线索

V2-031109 | MATHDB-329614

Neumann-Reid conjecture on hyperbolic knots with hidden symmetries

摘录 / EXCERPT: Neumann-Reid conjecture. The figure-eight knot and the two dodecahedral knots are the only hyperbolic knots in S^3 admitting hidden symmetries.

conjecture | open | MathDB 页面 | 原文线索

V2-031110 | MATHDB-329615

Rigid cusp conjecture for non-arithmetic knot complements with hidden symmetries

摘录 / EXCERPT: Rigid cusp conjecture. The minimal orientable orbifold covered by a non-arithmetic knot complement with hidden symmetries has a rigid cusp of type $S^2(2,3,6)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031111 | MATHDB-329617

The assembly-generation conjecture below virtual cohomological dimension

摘录 / EXCERPT: Let $\Gamma_{n,s}$ denote the relevant graph automorphism group, and let A_ϕ be the assembly map associated to a gluing ϕ producing the graph $X_{n,s}$. Write $2n+s-3$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-031112 | MATHDB-329620

Vanishing conjecture for assembled classes under rank stabilization

摘录 / EXCERPT: Let ϕ be a gluing and let A_ϕ be its assembly map for the homology groups of the graph automorphism groups $\Gamma_{n,s}$. A class is positive-dimensional when it has hom...

conjecture | open | MathDB 页面 | 原文线索

V2-031113 | MATHDB-329650

Mirzakhani's arithmetic affine-manifold conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ be an affine manifold in a stratum of translation surfaces. Its rank is the dimension of its tangent space projected to absolute cohomology divided by 2, and...

conjecture | open | MathDB 页面 | 原文线索

V2-031114 | MATHDB-329653

The derived change-of-groups conjecture for tori

摘录 / EXCERPT: Let $j:T\longrightarrow S$ be an inclusion of tori, and write $\lambda=j^*:H^*(BS)\longrightarrow H^*(BT)$. The change-of-groups functors between S-spectra and T-spectra are gi...

conjecture | open | MathDB 页面 | 原文线索

V2-031115 | MATHDB-329668

Vanishing of the first Tate cohomology of the idèle class group

摘录 / EXCERPT: Idèle class cohomology vanishing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031116 | MATHDB-329681

Calabi extension conjecture for Hamiltonian homeomorphisms of the disc

摘录 / EXCERPT: Let D^2 be the two-disc, with $[?]$ its area form, and let $Hameo(D^2,\partial D^2)$ be the subgroup of Hamiltonian homeomorphisms inside $Homeo^\Omega(D^2,\partial D^2)$. The Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-031117 | MATHDB-329682

Vanishing conjecture for the extended Calabi invariant on contractible loops

摘录 / EXCERPT: Let λ be a contractible topological Hamiltonian loop based at the identity, and let $\overline{\operatorname{Cal}}^{path}(\lambda)$ denote the extended Calabi invariant of...

conjecture | open | MathDB 页面 | 原文线索

V2-031118 | MATHDB-329683

Vanishing conjecture for the limit basic phase function on the two-disc

摘录 / EXCERPT: Let $M=S^2$ and let $B=D^2_-$ be the lower hemisphere. Let ... be a hameotopy contracting a topological Hamiltonian loop ϕ_F, where $F=H(1)$, such that ... for all $t,s\in[0,1]$....

conjecture | open | MathDB 页面 | 原文线索

V2-031119 | MATHDB-329684

Polynomial Sarnak's conjecture for minimal systems II

摘录 / EXCERPT: A minimal topological dynamical system (X,T) has polynomial entropy zero when its sequential topological entropy is zero along all polynomial sequences. Let (X,T) be a minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-031120 | MATHDB-329685

Polynomial Sarnak's conjecture for minimal systems III

摘录 / EXCERPT: Let $p:\mathbb{N}\to\mathbb{N}_0$ be a polynomial and let (X,T) be a minimal topological dynamical system. For $f\in C(X)$ and $x\in X$, consider the sequence $(f(T^{p(n)}x))$. P...

conjecture | open | MathDB 页面 | 原文线索

V2-031121 | MATHDB-329704

The linear-term conjecture for colored Jones polynomials

摘录 / EXCERPT: Linear-term conjecture. For every non-trivial knot $K\subset S^3$,

conjecture | open | MathDB 页面 | 原文线索

V2-031122 | MATHDB-329719

The Laurent-polynomial mirror theorem for orbifold del Pezzo surfaces

摘录 / EXCERPT: Conjecture B. There is an affine-linear isomorphism $\varphi\colon L_P^T\to H_X^{\mathrm{ts}}$, the mirror map, such that the regularized quantum period...

conjecture | open | MathDB 页面 | 原文线索

V2-031123 | MATHDB-329721

The quantum-period conjecture for class TG del Pezzo surfaces

摘录 / EXCERPT: Quantum-period conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-031124 | MATHDB-329736

Conjecture on positivity and integrality of the multi-backbone ancestor coefficients

摘录 / EXCERPT: For genus g and the general multi-backbone case, let $\widehat b^{(g)}_{\vec k,\vec\beta}$ denote the coefficients identified with combinations of ancestor invariants, where...

conjecture | open | MathDB 页面 | 原文线索

V2-031125 | MATHDB-329737

Craw-Quintero-Vélez's toric cell complex conjecture for consistent toric algebras

摘录 / EXCERPT: Craw-Quintero-Vélez's conjecture. If the global dimension of A equals the dimension of Y, then the toric cell complex of A exists and is constructed as in their work, recover...

conjecture | open | MathDB 页面 | 原文线索

V2-031126 | MATHDB-329743

Characterization of trivial twisted Alexander polynomials for transverse curve arrangements

摘录 / EXCERPT: Let C be a curve whose components all intersect transversely, let ρ range over the representations under consideration, and let Δ_1 and Δ_0 denote the twisted...

conjecture | open | MathDB 页面 | 原文线索

V2-031127 | MATHDB-329785

The Singer-transfer Lannes-Zarati conjecture

摘录 / EXCERPT: The Singer-transfer Lannes-Zarati conjecture. The composition j_s vanishes in every positive degree t for $s\geq 3$. This is presented as a weak version of the graded-associate...

conjecture | open | MathDB 页面 | 原文线索

V2-031128 | MATHDB-329786

The augmentation-ideal conjecture for the Singer-transfer complex

摘录 / EXCERPT: Let A be the mod p Steenrod algebra, let E_s be the elementary abelian group of rank s, and let $\mathscr{B}[s]$ be the subobject defined in the paper inside H^*BE_s. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-031129 | MATHDB-329787

Hopkins's equivariant presentation conjecture for the Morava E-theory homotopy groups

摘录 / EXCERPT: Let $K\subset \mathbb{S}_n$ be a finite subgroup. Let $\rho\in E_n$ be placed in degree -2, let $S(\rho)$ denote the graded symmetric algebra, let N be a trivial representation...

conjecture | open | MathDB 页面 | 原文线索

V2-031130 | MATHDB-329808

Isomorphism conjecture for twisted Lie bialgebra structures corresponding to Belavin-Drinfeld matrices

摘录 / EXCERPT: Let r_{BD} be an r-matrix from the list in Lemma, and let $\mathfrak{g}$ be the corresponding Lie algebra. Twisted Lie bialgebra structures on $\mathfrak{g}$ corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-031131 | MATHDB-329813

Connectivity conjecture for simplicial complexes of projective-cube walk powers

摘录 / EXCERPT: Let r and k be integers with $r\geq k$. Consider the simplicial complex associated to the graph $\mathcal{PC}_{2r}^{(2k-1)}$, where $\mathcal{PC}_{2r}^{(2k-1)}$ is the $(2k-1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-031132 | MATHDB-329872

Conjecture that minimizing hypercones are strictly area-minimizing

摘录 / EXCERPT: A minimizing hypercone is a codimension-one cone that is area-minimizing, equivalently mass-minimizing. The paper excludes trivial examples in low dimensions. Strict area-minimalit...

conjecture | open | MathDB 页面 | 原文线索

V2-031133 | MATHDB-329882

Lin's symmetry-preserving almost-commuting matrices conjecture

摘录 / EXCERPT: Let H and X be self-adjoint matrices in a matrix algebra, and suppose that (X,H) belongs to an Altland-Zirnbauer symmetry class. Lin's theorem states that sufficiently almost...

conjecture | open | MathDB 页面 | 原文线索

V2-031134 | MATHDB-329884

Compactness characterization by nonsurjective locally contractive maps

摘录 / EXCERPT: Let (X,ρ) be an infinite Polish ultrametric space. A map $\varphi:X\rightarrow X$ is locally contractive if, for every $x\in X$, there is a neighborhood on which distances fro...

conjecture | open | MathDB 页面 | 原文线索

V2-031135 | MATHDB-329899

The algebraic H-conjecture for elliptic Sullivan minimal algebras

摘录 / EXCERPT: Let $(\Lambda V,d)$ be an elliptic Sullivan minimal algebra, so its cohomology and generating graded vector space are finite-dimensional. The algebraic H-conjecture. One has ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-031136 | MATHDB-329902

The quantum volume conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $A_K(m,\ell)=0$ be its A-polynomial curve. Put $(m,\ell)=(e^u,e^v)$, and let $S_0(u)$ be defined by the wave-function equations in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-031137 | MATHDB-329903

The topological-recursion quantisation conjecture for the A-polynomial

摘录 / EXCERPT: Let $K\subset S^3$ be a knot with A-polynomial curve $A_K(m,\ell)=0$. Let $\hat A_K$ be the quantisation obtained from topological recursion: calculate the functions $S_k(u)$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-031138 | MATHDB-329908

Relative Donaldson-Thomas 4 virtual class conjecture for degenerations

摘录 / EXCERPT: Relative virtual class conjecture. There exists a D-orbifold associated with $\mathfrak{I}^{P}_{\mathfrak{X}/\mathfrak{C}}$ and, if it is orientable, a Borel-Moore homology class

conjecture | open | MathDB 页面 | 原文线索

V2-031139 | MATHDB-329926

Blagojević-Frick-Ziegler's balanced Tverberg partition conjecture

摘录 / EXCERPT: Blagojević-Frick-Ziegler's conjecture. There are r pairwise disjoint faces $\sigma_1,\ldots,\sigma_r$ of Δ_N such that

conjecture | open | MathDB 页面 | 原文线索

V2-031140 | MATHDB-329944

Reyes-Rogalski-Zhang's homological determinant conjecture for Nakayama automorphisms

摘录 / EXCERPT: Let S be an AS-regular algebra, and let u be its Nakayama automorphism. The homological determinant conjecture. If ν is the Nakayama automorphism of an AS-regular algebra,...

conjecture | open | MathDB 页面 | 原文线索

V2-031141 | MATHDB-329949

The 4-adjacency cycle conjecture for digital images

摘录 / EXCERPT: 4-adjacency cycle conjecture. The image X is non-rigid and irreducible if and only if X is isomorphic to C_n for some $n>4$.

conjecture | open | MathDB 页面 | 原文线索

V2-031142 | MATHDB-330001

The noncommutative cohomology conjecture for dg-categories

摘录 / EXCERPT: Let $\mathbb{TR}$ be a candidate (co)homology theory on dg-categories over $\mathbf{F}_{p}$, with $\mathsf{dg}\mathsf{Cat}_{\mathbf{F}_{p}}$ the category of dg-categories over...

conjecture | open | MathDB 页面 | 原文线索

V2-031143 | MATHDB-330015

Baldwin-Lobb's collapse conjecture for the Khovanov spectral sequence

摘录 / EXCERPT: For a link diagram, let $\operatorname{Kh}_{p,q}^r$ denote the bigraded vector spaces on the pages of the Khovanov spectral sequence, with...

conjecture | open | MathDB 页面 | 原文线索

V2-031144 | MATHDB-330023

The topological structure conjecture for constructions

摘录 / EXCERPT: Topological structure conjecture. The topological space $F(p)$ is Hausdorff and is a CW-complex.

conjecture | open | MathDB 页面 | 原文线索

V2-031145 | MATHDB-330024

The building realization conjecture

摘录 / EXCERPT: Building realization conjecture. The topological space $F(p)$ has a natural structure of an $\mathbb{R}$-building modeled on the affine space A.

conjecture | open | MathDB 页面 | 原文线索

V2-031146 | MATHDB-330031

Ambrosio-De Lellis-Mantegazza relaxation conjecture for line energies

摘录 / EXCERPT: Ambrosio-De Lellis-Mantegazza conjecture. If $f(t)=t^p$ with $1\leq p\leq3$, then $\overline{\mathcal{I}_f}$ is lower semicontinuous for the strong topology in L^1.

conjecture | open | MathDB 页面 | 原文线索

V2-031147 | MATHDB-330058

Wall's D(2) problem

摘录 / EXCERPT: Let X be a finite 3-dimensional CW complex whose cohomological dimension is at most 2. Wall's D(2) problem. X is homotopy equivalent to a 2-dimensional CW complex. The pr...

open_problem | open | MathDB 页面 | 原文线索

V2-031148 | MATHDB-330059

The stable D(2,n) problem

摘录 / EXCERPT: Let G be a finitely presentable group and let $n\geq 0$. For a finite 3-dimensional CW complex X with cohomological dimension at most 2 and fundamental group...

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V2-031149 | MATHDB-330068

Lafont's bounded-cohomology surjectivity conjecture for higher-rank symmetric spaces

摘录 / EXCERPT: Let G be a connected noncompact simple Lie group and X an irreducible symmetric space. Given a geodesic γ in X, let $F(\gamma)$ be the union of all geodesics in X pa...

conjecture | open | MathDB 页面 | 原文线索

V2-031150 | MATHDB-330071

Chen-Yang unknot detection conjecture for Turaev-Viro invariants

摘录 / EXCERPT: Let K be a knot in S^3, and let $TV_r(K)$ denote the corresponding Turaev-Viro invariant, evaluated at the allowed roots of unity q. Chen-Yang unknot detection conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-031151 | MATHDB-330076

The positive Euler characteristic conjecture for two-dimensional Coxeter groups

摘录 / EXCERPT: Let G be a finitely generated infinite Coxeter group of dimension at most 2. Its Euler characteristic is denoted by $4\chi(G)4$. Positive Euler characteristic conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-031152 | MATHDB-330085

Gross-Matessi conjecture on Lagrangian sections and line bundles

摘录 / EXCERPT: Let $\check X$ be a toric Calabi-Yau threefold and let X be its mirror with Lagrangian fibration $f:X\to\mathbb R^3$. For a collection of twisting numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-031153 | MATHDB-330092

Continuity of topological entropy for partially hyperbolic diffeomorphisms with one-dimensional center

摘录 / EXCERPT: Let M be a compact manifold, and let f be a C^1 partially hyperbolic diffeomorphism of M with a decomposition $TM=E^s\oplus E^c\oplus E^u$ into invariant continuous sub-bun...

conjecture | open | MathDB 页面 | 原文线索

V2-031154 | MATHDB-330093

Continuity of topological entropy for three-dimensional partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let M be a three-dimensional manifold and let f be a partially hyperbolic diffeomorphism of M, with stable, center, and unstable bundles. Three-dimensional continuity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-031155 | MATHDB-330104

The $K(\pi,1)$ conjecturette for diagram homotopy

摘录 / EXCERPT: $K(\pi,1)$ conjecturette. All diagrams for the chosen Coxeter group or braid group are homotopic to the empty diagram through a series of the aforementioned allowed transformations...

conjecture | open | MathDB 页面 | 原文线索

V2-031156 | MATHDB-330139

Conjectural spectrum-level splittings for the unoriented Thom spectra

摘录 / EXCERPT: Spectrum-level splitting conjecture. As a spectrum, Mj_2 is a wedge of $k\mathrm{O}$-module spectra, Mj_3 is a wedge of tmf-module spectra and Mj^{c} is a...

conjecture | open | MathDB 页面 | 原文线索

V2-031157 | MATHDB-330140

The epimorphism conjecture for the $\E_\infty$ orientation from Mj_3 to tmf

摘录 / EXCERPT: Epimorphism conjecture. The $\mathcal{E}_\infty$ orientation $Mj_3\to\mathrm{tmf}$ induces a ring epimorphism

conjecture | open | MathDB 页面 | 原文线索

V2-031158 | MATHDB-330141

The $\E_\infty$ Thom spectrum conjecture for $\mathrm{tmf}_1(3)$

摘录 / EXCERPT: $\mathcal{E}_\infty$ Thom spectrum conjecture. There is an $\mathcal{E}_\infty$ morphism $Mj\to\mathrm{tmf}_1(3)$ which factors through $Mj/\!/w_4,w_8,w_{12}$, and the induced morp...

conjecture | open | MathDB 页面 | 原文线索

V2-031159 | MATHDB-330179

Equality of module sectional category and rational sectional category

摘录 / EXCERPT: Let $f\colon X\rightarrow Y$ be a map whose rationalisation f_0 admits a homotopy retraction. Let ${\rm secat}(f_0)$ denote the sectional category of the rationalisation and...

conjecture | open | MathDB 页面 | 原文线索

V2-031160 | MATHDB-330194

Conjectures on thick ideals generated by the cofiber of the Hopf map

摘录 / EXCERPT: Thick-ideal conjectures for $C\eta$. The following assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-031161 | MATHDB-330195

Winning-strategy conjecture for selective strong screenability games on cubes

摘录 / EXCERPT: Winning-strategy conjecture. For each positive integer n, ONE has a winning strategy in

conjecture | open | MathDB 页面 | 原文线索

V2-031162 | MATHDB-330203

The Hodge-theoretic F-nilpotence conjecture

摘录 / EXCERPT: Let $(x \in X)$ be an n-dimensional normal isolated singularity over $\mathbb{C}$. The local cohomology groups $H^i_{\{x\}}(X_{\rm an},\mathbb{C})$ carry canonical mixed Hodge st...

conjecture | open | MathDB 页面 | 原文线索

V2-031163 | MATHDB-330248

Existence of Tychonoff spaces separating successive diagonal ranks

摘录 / EXCERPT: A rank n-diagonal is a diagonal satisfying the rank-n condition described in the surrounding theory. Diagonal-rank separation conjecture. For every natural number n there is...

conjecture | open | MathDB 页面 | 原文线索

V2-031164 | MATHDB-330258

Extension and self-linking bound for the Ozsváth-Szabó-Rasmussen invariant

摘录 / EXCERPT: Let K be a knot in S^3, let $\tau(K)$ denote the concordance invariant from knot Floer homology, and define ... For an oriented link L, let $sl(L)$ denote its transverse self-l...

conjecture | open | MathDB 页面 | 原文线索

V2-031165 | MATHDB-330279

The noncommutative Brouwer fixed-point conjecture for compact spaces

摘录 / EXCERPT: Noncommutative Brouwer fixed-point conjecture. Every continuous map

conjecture | open | MathDB 页面 | 原文线索

V2-031166 | MATHDB-330289

The Hsiang conjecture on rationalized classical assembly

摘录 / EXCERPT: Hsiang's conjecture. For every torsion-free group G and every $n\in\mathbb{Z}$, this rationalized classical assembly map is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-031167 | MATHDB-330352

Large-slope doubly primitive conjecture for knots in the Poincare homology sphere

摘录 / EXCERPT: Let K be a knot in the Poincaré homology sphere, and let $g(K)$ denote its Seifert genus. Suppose that p-surgery on K produces a lens space, where p is an integer satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-031168 | MATHDB-330353

Tange-Hedden classification conjecture for doubly primitive knots in the Poincare homology sphere

摘录 / EXCERPT: Let K be a knot in the Poincaré homology sphere. A knot is doubly primitive if it lies on a genus 2 Heegaard surface and represents a generator of the fundamental group of each...

conjecture | open | MathDB 页面 | 原文线索

V2-031169 | MATHDB-330392

Canonical local identifications of transgressed flat bundles

摘录 / EXCERPT: Let G be a compact Lie group acting smoothly on a compact manifold M, and let $b1\in H^3_G(M,\mathbb Z)$. For each $g\in G$, let $P^g\to M^g$ be the associated G-equivariant...

conjecture | open | MathDB 页面 | 原文线索

V2-031170 | MATHDB-330393

The twisted equivariant cyclic-to-de Rham quasi-isomorphism

摘录 / EXCERPT: Let G be a compact Lie group acting smoothly on a compact manifold M, let $\alpha\in H^3_G(M,\mathbb Z)$, let H be the relevant transformation groupoid, and let L be the as...

conjecture | open | MathDB 页面 | 原文线索

V2-031171 | MATHDB-330394

The delocalized twisted equivariant Chern character isomorphism

摘录 / EXCERPT: Let G be a compact Lie group acting smoothly on a compact manifold M, let $\alpha\in H^3_G(M,\mathbb Z)$, let $R(G)$ be the representation ring, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031172 | MATHDB-330409

High-distance neighbors conjecture for bridge splittings

摘录 / EXCERPT: Let K be a knot in a 3-manifold M with a (g,b)-splitting, where $g\geq 1$. Let (g',b',n) be a triple of non-negative integers with $g'\geq g$ and $b'\geq b$. High-distance...

conjecture | open | MathDB 页面 | 原文线索

V2-031173 | MATHDB-330438

Positivity conjecture for rectangular DAHA-superpolynomials

摘录 / EXCERPT: Rectangular DAHA-superpolynomial positivity conjecture. The coefficients of $H\!D_{r,s}(j\omega_i;q,t,a)$ are positive integers. After the displayed transformation, the resulting p...

conjecture | open | MathDB 页面 | 原文线索

V2-031174 | MATHDB-330445

Tate's homological isomorphism conjecture for motives

摘录 / EXCERPT: Let $M_{\mathbb Q}$ and $M'_{\mathbb Q}$ be pure $\mathbb Q$-motives over F, let M_p and M'_p be their p-adic étale realizations, and let G_F be the absolute Galois group...

conjecture | open | MathDB 页面 | 原文线索

V2-031175 | MATHDB-330476

Global extension conjecture for semisimple Frobenius manifolds

摘录 / EXCERPT: Let M be an N-dimensional semisimple Frobenius manifold such that it possesses a holomorphic genus one potential $\mathrm dG$. Global extension conjecture. There exists a conve...

conjecture | open | MathDB 页面 | 原文线索

V2-031176 | MATHDB-330499

The Cabling Conjecture for hyperbolic knots in lens spaces

摘录 / EXCERPT: Baker's conjecture. If K is hyperbolic, then

conjecture | open | MathDB 页面 | 原文线索

V2-031177 | MATHDB-330500

The stabilized-unknot formula for the annular Khovanov invariant

摘录 / EXCERPT: Let $\sigma_n^{\epsilon_n}\sigma_{n-1}^{\epsilon_{n-1}}\cdots\sigma_1^{\epsilon_1}$ be a stabilized unknot, where each $\epsilon_i=\pm1$, and let W_0 denote the corresponding inv...

conjecture | open | MathDB 页面 | 原文线索

V2-031178 | MATHDB-330501

The collapse conjecture for the sutured-annular-to-Khovanov spectral sequence

摘录 / EXCERPT: Let W_k denote the invariants in the decomposition associated with the spectral sequence from sutured annular Khovanov homology to ordinary Khovanov homology. Collapse conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-031179 | MATHDB-330543

Castle, Evans and Hyde's conjecture on polyhedral toroidal molecules

摘录 / EXCERPT: A polyhedral molecule has an underlying graph that is abstractly planar, 3-connected, and simple; a molecule is toroidal when its graph embedding on the standard torus does not emb...

conjecture | open | MathDB 页面 | 原文线索

V2-031180 | MATHDB-330558

The untwistedness conjecture for simply connected closed manifolds

摘录 / EXCERPT: Let M be a simply connected closed manifold. A closed manifold is untwisted when the CDGA $C(0)$ is a model of its configuration space of two points $F(M,2)$. Untwistedness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-031181 | MATHDB-330559

The manifold g''-conjecture for balanced homology manifolds

摘录 / EXCERPT: Let Δ be a balanced, orientable $\mathbf{k}$-homology manifold without boundary. Define $g”_j(\Delta)=h”_j(\Delta)-h”_{j-1}(\Delta)$. Balanced manifold $g”$-conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-031182 | MATHDB-330561

The balanced generalized lower bound conjecture

摘录 / EXCERPT: Let Δ be a balanced d-polytope, or more generally a balanced $\mathbf{k}$-homology $(d-1)$-sphere. Balanced GLBC. ... The inequalities were established for all balanced poly...

conjecture | open | MathDB 页面 | 原文线索

V2-031183 | MATHDB-330562

Nevo's cone containment conjecture for flag and balanced spheres

摘录 / EXCERPT: Let H_{bal} be the cone generated by the h-vectors of balanced d-polytopes or balanced homology $(d-1)$-spheres, and let H_{flag} be the cone generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-031184 | MATHDB-330563

Lutz-Nevo upper bound conjecture for flag spheres

摘录 / EXCERPT: Let $d=2k\geq 4$, and let Δ be a flag homology $(d-1)$-sphere on n vertices. Let $\mathcal{J}_k(n)$ be the flag sphere obtained by joining k graph cycles, each having ei...

conjecture | open | MathDB 页面 | 原文线索

V2-031185 | MATHDB-330583

The conjecture that BCOV theory on \(\mathbb{C}^5\) twists type IIB supergravity

摘录 / EXCERPT: BCOV theory is a closed-string field theory for the topological B-model, formulated here on complex affine space. Type IIB supergravity is the ten-dimensional supergravity theory...

conjecture | open | MathDB 页面 | 原文线索

V2-031186 | MATHDB-330611

Vanishing probability conjecture for fixed knots in the Chebyshev billiard model

摘录 / EXCERPT: Let $T(3,n+1)$ be the random knot model obtained from Chebyshev billiard table diagrams, and let $P(T^{(n)}_K)$ denote the probability that the resulting diagram represents a fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-031187 | MATHDB-330628

Generalized non-commutative degeneration conjecture

摘录 / EXCERPT: Generalized non-commutative degeneration conjecture. For every pair of small DG categories ${\mathcal B}$ and ${\mathcal C}$ over k, one has $\varphi_n=0$ for $n\leq 0$. This con...

conjecture | open | MathDB 页面 | 原文线索

V2-031188 | MATHDB-330629

Smooth categorical compactification conjecture for homotopically finite DG categories

摘录 / EXCERPT: Let k be a field, let $\operatorname{Ho}_M(\operatorname{dgcat}_k)$ be the homotopy category of small DG categories over k with respect to the Morita model structure, and let a...

conjecture | open | MathDB 页面 | 原文线索

V2-031189 | MATHDB-330630

Boundary-map reformulation of the Kontsevich-Soibelman conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be a smooth and compact DG category, and let $HH_{\bullet}({\mathcal A})$ and $HC^-_{\bullet}({\mathcal A})$ denote its Hochschild and negative cyclic homology....

conjecture | open | MathDB 页面 | 原文线索

V2-031190 | MATHDB-330667

Unoriented string-diagram model conjecture

摘录 / EXCERPT: Let $\mathcal{SD}^{\mathrm{u}}$ be the space of unoriented string diagrams, and let $\mathcal{M}$ be the moduli space of Riemann surfaces with parameterized boundary. Unoriented st...

conjecture | open | MathDB 页面 | 原文线索

V2-031191 | MATHDB-330668

Bödigheimer compactification conjecture for unoriented string diagrams

摘录 / EXCERPT: Let $\mathcal{SD}^{\mathrm{u}}$ be the space of unoriented string diagrams, let $\mathcal{SD}^{\mathrm{u}}/\sim$ be its quotient by the stated equivalence relation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031192 | MATHDB-330669

Triviality conjecture for the orientation double cover of string diagrams

摘录 / EXCERPT: Orientation-cover triviality conjecture. This double cover is trivial. The source does not provide a proof or status evidence for this assertion.

conjecture | open | MathDB 页面 | 原文线索

V2-031193 | MATHDB-330685

Formality conjecture for the Turaev cobracket and Kashiwara-Vergne solutions

摘录 / EXCERPT: Let a genus 0 compact surface be equipped with a group-like expansion arising from a solution to the Kashiwara-Vergne problem, and consider the tensorial description of its Turae...

conjecture | open | MathDB 页面 | 原文线索

V2-031194 | MATHDB-330706

Volume monotonicity under crossing changes for alternating hyperbolic links

摘录 / EXCERPT: Volume monotonicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031195 | MATHDB-330709

Hedden-Watson finiteness conjecture for knot Floer homology

摘录 / EXCERPT: Let K be an L-space knot with knot Floer homology $\widehat{\operatorname{HFK}}(K)$. Hedden-Watson's finiteness conjecture. There are only finitely many other knots whose Floer h...

conjecture | open | MathDB 页面 | 原文线索

V2-031196 | MATHDB-330710

Finiteness conjecture for L-space knots of bounded genus

摘录 / EXCERPT: Let $N\geq 0$ be an integer, and let $g(K)$ denote the genus of a knot K. Finiteness conjecture for bounded genus. There are only finitely many L-space knots K with...

conjecture | open | MathDB 页面 | 原文线索

V2-031197 | MATHDB-330735

The torsion-free Whitehead group conjecture

摘录 / EXCERPT: Vanishing of Whitehead Groups. If π is torsion-free, then

conjecture | open | MathDB 页面 | 原文线索

V2-031198 | MATHDB-330750

Kenyon's determinant density upper-bound conjecture

摘录 / EXCERPT: Let K be a knot or link, let $c(K)$ denote its crossing number, let $\det(K)$ denote its determinant, and let $v_{\rm oct}$ denote the volume of the regular ideal octahedron. Ken...

conjecture | open | MathDB 页面 | 原文线索

V2-031199 | MATHDB-330751

Volume is bounded by determinant for alternating hyperbolic links

摘录 / EXCERPT: Let K be an alternating hyperbolic link, let $\operatorname{vol}(K)$ denote its hyperbolic volume, and let $\det(K)$ denote its determinant. Alternating volume-determinant conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-031200 | MATHDB-330753

Equality of volume, determinant and quantum density spectra

摘录 / EXCERPT: Let $\operatorname{Spec}_{\rm vol}$, $\operatorname{Spec}_{\rm det}$ and Spec_q denote respectively the derived sets of volume densities, determinant densities, an...

conjecture | open | MathDB 页面 | 原文线索

V2-031201 | MATHDB-330768

Chen's conjecture on the cohomological invariance of the twisted path threshold

摘录 / EXCERPT: Let $\chi>0$ be a closed positive $(1,1)$-form, and define $R(\chi)$ to be the supremum of times $t\in[0,1]$ for which the twisted continuity-path equation is solvable. Chen's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-031202 | MATHDB-330787

Continuum of bifurcation-diagram germs in three-parameter families

摘录 / EXCERPT: Continuum conjecture for bifurcation diagrams. There exists an open set in the space of three-parameter families such that the set of pairwise topologically different germs of bifu...

conjecture | open | MathDB 页面 | 原文线索

V2-031203 | MATHDB-330789

Wilson loop observable and shadow invariant conjecture

摘录 / EXCERPT: Wilson loop-shadow invariant conjecture. For every L fulfilling Assumption,

conjecture | open | MathDB 页面 | 原文线索

V2-031204 | MATHDB-330790

Toda recursion and Gromov-Witten correspondence conjecture for ADE geometries

摘录 / EXCERPT: Let $\Gamma\subset \mathrm{SL}(2,\mathbb{C})$ be a cyclic or binary polyhedral group, with McKay Dynkin diagram $\mathcal{D}_\Gamma$ and associated simply connected, simply-laced L...

conjecture | open | MathDB 页面 | 原文线索

V2-031205 | MATHDB-330805

The GHM conjecture on the Fredholm determinant and modified grand potential

摘录 / EXCERPT: Let ρ_S be the inverse of the quantized mirror-curve operator acting on $L^2(\mathbb R)$, let $\Xi_S(\kappa,\boldsymbol{\xi},\hbar)$ denote its Fredholm determinant, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-031206 | MATHDB-330824

Convexity conjecture for the infinitesimal strip map of small surfaces

摘录 / EXCERPT: Let S be the hyperbolic surface, let $\mathcal{T}$ be its Teichmüller space, let $[g]inmathcal{T}$, and let $\overline{X}$ be the relevant compactified weighted arc complex with...

conjecture | open | MathDB 页面 | 原文线索

V2-031207 | MATHDB-330828

The annular web evaluation equivalence conjecture

摘录 / EXCERPT: Annular web evaluation conjecture. There is an equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-031208 | MATHDB-330851

Sarkar's formula conjecture for the knot basepoint-moving map

摘录 / EXCERPT: Let K be an oriented, null-homologous knot in a three-manifold Y, and let $\mathcal{H}$ be a choice of Heegaard data for K. Write ... for the differential on...

conjecture | open | MathDB 页面 | 原文线索

V2-031209 | MATHDB-330864

L-space branched-cover conjecture

摘录 / EXCERPT: Let Y be a rational homology 3-sphere, let K be a knot in Y, and let $\Sigma(Y,K)$ denote the double cover of Y branched over K. L-space branched-cover conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-031210 | MATHDB-330865

The general local base-change criterion via Tate cohomology

摘录 / EXCERPT: Let $F\supset E$ be a degree l extension of characteristic-zero non-archimedean local fields with residue characteristic p, where p and l are distinct primes. Let π be...

conjecture | open | MathDB 页面 | 原文线索

V2-031211 | MATHDB-330875

Casson-Gordon signature characterization of ribbon 2-bridge knots

摘录 / EXCERPT: Let $K_{m^2,q}$ be a 2-bridge knot, and let $\mathcal{R}$ denote the family of ribbon knots. The Casson-Gordon signature invariants associated to the double branched cover are the...

conjecture | open | MathDB 页面 | 原文线索

V2-031212 | MATHDB-330885

Mackey functor characterization of n-excisive functors on spectra

摘录 / EXCERPT: Let $\mathcal{M}=\mathcal{F}_s^{\leq n}$ be the category of finite sets and surjections of cardinality at most n, as in Example, and let $\mathbf{C}$ be a semiadditive presentabl...

conjecture | open | MathDB 页面 | 原文线索

V2-031213 | MATHDB-330890

Su's h-positivity conjecture for truncated even-rank partition-poset homology

摘录 / EXCERPT: Su's h-positivity conjecture. The action of S_{2n} on the homology of $\Pi_{2n}^e(\bar{k})$ is a permutation module that can be written as a sum of induced modules whose stabilis...

conjecture | open | MathDB 页面 | 原文线索

V2-031214 | MATHDB-330891

Coefficientwise domination conjecture for even-rank partition-poset homology

摘录 / EXCERPT: Coefficientwise domination conjecture. The symmetric function $\alpha_{2n}-R_{2n}$ is a nonnegative integer combination of homogeneous symmetric functions h_λ indexed by pa...

conjecture | open | MathDB 页面 | 原文线索

V2-031215 | MATHDB-330921

Gonzales Acuña-Short conjecture on reducible Dehn surgeries

摘录 / EXCERPT: Let K be a nontrivial knot in S^3, and let $S^3_r(K)$ denote surgery on K with slope r. A knot is cabled if it has the form $J_{p,q}$, the (p,q)-cable of a knot J, with...

conjecture | open | MathDB 页面 | 原文线索

V2-031216 | MATHDB-330922

Two-summands conjecture for reducible surgeries

摘录 / EXCERPT: Let K be a nontrivial knot in S^3, and let $S^3_r(K)$ denote surgery on K with slope r. A 3-manifold is reducible if it has an essential two-sphere, and an irreducible 3-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-031217 | MATHDB-330923

Slice cable conjecture

摘录 / EXCERPT: Let K be the (p,q)-cable of a knot J. A knot is slice if it bounds a smooth, properly embedded disk in B^4. Slice cable conjecture. The knot K is slice if and only if J...

conjecture | open | MathDB 页面 | 原文线索

V2-031218 | MATHDB-330926

The local lattice conjecture for $U(3)$ arithmetic manifolds

摘录 / EXCERPT: Let ρ be the Galois representation attached to an automorphic representation of a unitary group of semisimple rank 2, and let $\tau_v\subset\pi_v$ be the K-type correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-031219 | MATHDB-330950

Riley's real-root conjecture for parabolic representations of two-bridge knots

摘录 / EXCERPT: Let K be a two-bridge knot, and let $\Phi_K(2,y)$ be its Riley polynomial specialized at meridian trace 2. The knot signature is denoted by $\sigma(K)$. Riley's conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-031220 | MATHDB-330974

The infinitely categorified absolute Galois group conjecture for the real numbers

摘录 / EXCERPT: The field $\mathbb R$ has an absolute Galois group whose categorification is related to the extension of real vector spaces by complex super vector spaces. The paper proposes an infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-031221 | MATHDB-330985

The torsion-translate conjecture for local-system jump loci of hypersurface singularity complements

摘录 / EXCERPT: Let $X=\mathbb{C}^n$, let $f:X\to\mathbb{C}$ be holomorphic, let $x\in f^{-1}(0)$, and for a sufficiently small ball B_x centered at x set ... Thus $U_{f,x}$ is the complement of...

conjecture | open | MathDB 页面 | 原文线索

V2-031222 | MATHDB-330992

The Thurston-Bennequin number conjecture for unknotted minimal embeddings of K_5

摘录 / EXCERPT: Let K_5 be the complete graph on five vertices, and consider an unknotted minimal embedding of K_5 as a Legendrian graph. For each 5-cycle in the embedding, let tb denote i...

conjecture | open | MathDB 页面 | 原文线索

V2-031223 | MATHDB-331005

Permanent-cycle conjecture for $a_{\sigma_n}u_{\rho_n-\sigma_n}$

摘录 / EXCERPT: Let C_{2^n} be the cyclic group of order 2^n, and let a_{σ_n} and $u_{\rho_n-\sigma_n}$ denote the Euler and orientation classes associated to the representations...

conjecture | open | MathDB 页面 | 原文线索

V2-031224 | MATHDB-331006

Stable-filtration order conjecture

摘录 / EXCERPT: Let an element in the relevant equivariant homotopy group have an order, and suppose that its filtration eventually reaches a lowest value, called its stable filtration. Stable-fil...

conjecture | open | MathDB 页面 | 原文线索

V2-031225 | MATHDB-331031

The homeomorphism conjecture for connected sums of moment-angle complexes

摘录 / EXCERPT: Let K_1 and K_2 be generalized homology spheres of dimension $n-1$, with m_1 and m_2 vertices, respectively. Let ... The operation $\mathcal{G}$ on manifolds is the gyration,...

conjecture | open | MathDB 页面 | 原文线索

V2-031226 | MATHDB-331066

The sliceness conjecture for 2-bridge knots

摘录 / EXCERPT: Sliceness conjecture for 2-bridge knots. For all 2-bridge knots K, $g_{4}(K)=0$ if and only if $g_{4}^{{\rm top}}(K)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-031227 | MATHDB-331069

Nonexistence of Stein structures on the contractible 4-manifolds $X_{n,k}$

摘录 / EXCERPT: Let $X_{n,k}$ be the compact contractible oriented smooth 4-manifold whose boundary is Stein fillable. Nonexistence conjecture. The manifold $X_{n,k}$ does not admit any Stein st...

conjecture | open | MathDB 页面 | 原文线索

V2-031228 | MATHDB-331070

Equivariant positivity conjecture for Chern-Schwartz-MacPherson classes of Schubert cells

摘录 / EXCERPT: Let G/B be a generalized flag manifold with Weyl group W, and write the equivariant Chern-Schwartz-MacPherson class of the Schubert cell $X(w)^\circ$ as ... where the coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-031229 | MATHDB-331108

The no-isolated-points conjecture for circular orders on free products

摘录 / EXCERPT: Free-product no-isolated-points conjecture. The space $\operatorname{CO}(G*H)$ either has no isolated points or is finite. In particular, if it is infinite, then...

conjecture | open | MathDB 页面 | 原文线索

V2-031230 | MATHDB-331145

The log_p-conjecture for chromatic inclusions in equivariant stable homotopy theory

摘录 / EXCERPT: The $\log_p$-conjecture. For all $n\geq r$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-031231 | MATHDB-331199

Hanselman's generalized splicing conjecture for L-spaces

摘录 / EXCERPT: Let K_1 and K_2 be knots in S^3, and let $M_i=S^3\smallsetminus\nu(K_i)$ for $i=1,2$. Let h identify the torus boundaries of M_1 and M_2, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031232 | MATHDB-331215

Equality of the Tits-class obstruction for absolutely simple groups

摘录 / EXCERPT: Equality conjecture. Equality holds in the obstruction

conjecture | open | MathDB 页面 | 原文线索

V2-031233 | MATHDB-331241

Algorithmic simulation of Ricci flow with surgery in ER

摘录 / EXCERPT: Let M be a closed Riemannian 3-manifold. Ricci-flow simulation conjecture. Ricci flow with surgery on M can be accurately simulated in ER. In other words, Perelman's...

conjecture | open | MathDB 页面 | 原文线索

V2-031234 | MATHDB-331244

Berthelot's original conjecture on rigid cohomology

摘录 / EXCERPT: Berthelot's conjecture. There is a unique object

conjecture | open | MathDB 页面 | 原文线索

V2-031235 | MATHDB-331245

Tsuzuki's conjecture on relative overconvergent isocrystals

摘录 / EXCERPT: Tsuzuki's conjecture. There is a unique object

conjecture | open | MathDB 页面 | 原文线索

V2-031236 | MATHDB-331246

Shiho's conjecture on relative overconvergent isocrystals

摘录 / EXCERPT: Shiho's conjecture. There exists a unique overconvergent (F)-isocrystal $\widetilde E$ on $(S,\overline{S})$ whose restriction to...

conjecture | open | MathDB 页面 | 原文线索

V2-031237 | MATHDB-331247

Shiho's stronger conjecture on framewise relative cohomology

摘录 / EXCERPT: Stronger Shiho conjecture. There exists a unique overconvergent (F)-isocrystal $\widetilde E$ on $(S,\overline{S})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-031238 | MATHDB-331259

The planar web dimension conjecture for instanton homology

摘录 / EXCERPT: Let K be a web in $\R^{3}$, meaning an embedded trivalent graph, and let $\Jsharp(K)$ be the associated finite-dimensional vector space over the field $\F$ of two elements. A Tai...

conjecture | open | MathDB 页面 | 原文线索

V2-031239 | MATHDB-331260

The well-definedness conjecture for the pre-foam evaluation

摘录 / EXCERPT: Let S be a closed pre-foam, namely an abstract finite 2-dimensional cell complex with foam local structure, possibly with tetrahedral points and dots. Define $\Jflat(S)\in\F$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-031240 | MATHDB-331261

The closed-foam evaluation conjecture for instanton homology

摘录 / EXCERPT: Let Σ be a closed foam in $\R^{3}$, and let S be its underlying closed pre-foam. Let $\Jsharp(\Sigma)$ denote the instanton-homological evaluation of Σ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031241 | MATHDB-331308

Existence of virtually equivalent, non-isotopic knotted spheres

摘录 / EXCERPT: Let $n\geq 2$. A knotted n-sphere is an embedded S^n in $D^{n+1}\times [0,1]$, and two such knots are virtually equivalent when they become equivalent after the allowed virtual...

conjecture | open | MathDB 页面 | 原文线索

V2-031242 | MATHDB-331309

Faithfulness of the spinning map for virtual links

摘录 / EXCERPT: Let N be a fixed n-manifold, and let $\Phi_N:V_m\rightarrow V_{m+n}$ be the construction sending $(F\times [0,1],L)$ to $(F\times N\times [0,1],L\times N)$. Here V_m denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-031243 | MATHDB-331336

The subtop cohomology conjecture for metaplectic Whittaker models

摘录 / EXCERPT: Let ρ be the quadratic form on the coweight lattice associated with the metaplectic data. For $\bar c$ the residue-field point and $\bar\theta$ the image of θ in the ad...

conjecture | open | MathDB 页面 | 原文线索

V2-031244 | MATHDB-331350

Minkowski-relation presentation of the face-part cohomology

摘录 / EXCERPT: Minkowski-relation presentation conjecture. As a vector space, $\mathcal{F}^{2k}(X)$ is generated by the elements $\{x_I\}_{I\in K}$ subject to the Minkowski relations

conjecture | open | MathDB 页面 | 原文线索

V2-031245 | MATHDB-331378

Szabó's conjecture on the homology of the geometric spectral sequence

摘录 / EXCERPT: Szabó's conjecture. The homology of this complex is isomorphic to two copies of the Heegaard Floer homology of the branched double cover of L with reversed orientation:

conjecture | open | MathDB 页面 | 原文线索

V2-031246 | MATHDB-331432

Quasi-invertibility conjecture for bordered Koszul-duality bimodules

摘录 / EXCERPT: Quasi-invertibility conjecture. Either or both of the DD bimodules

conjecture | open | MathDB 页面 | 原文线索

V2-031247 | MATHDB-331436

A-theoretic isomorphism conjecture for families of groups

摘录 / EXCERPT: Let $\cF$ be a family of groups, let G be a countable discrete group, and let W be a free G-CW-complex. Let $E_\cF G$ denote the classifying space of G for the family $\cF$...

conjecture | open | MathDB 页面 | 原文线索

V2-031248 | MATHDB-331460

Homological sphere conjecture for differences of Minkowski sums

摘录 / EXCERPT: Let A and B be convex bodies in $\mathbb R^n$. The set $(A+B)\setminus A$ is the difference between a Minkowski sum and one summand. Homological sphere conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-031249 | MATHDB-331461

Minkowski summand conjecture from acyclic translates

摘录 / EXCERPT: Let A and B be convex bodies in $\mathbb{R}^n$. For every vector $t\in\mathbb R^n$, consider the translate $A+t$ and its difference from B. Minkowski summand conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-031250 | MATHDB-331468

Universal product formula and sign conjecture for the coefficients of higher antibrackets

摘录 / EXCERPT: Let A be the (super)commutative algebra and let $f\colon A\to A$ be a linear operator. For every integer $n>0$, let $c_1^n,\ldots,c_n^n$ be the unique rational coefficients in th...

conjecture | open | MathDB 页面 | 原文线索

V2-031251 | MATHDB-331475

Conjecture on the range of topological conformal dimension

摘录 / EXCERPT: Range conjecture. For every $d\in[2,\infty]$ there is a metric space X with

conjecture | open | MathDB 页面 | 原文线索

V2-031252 | MATHDB-331476

Conjecture on topological conformal dimensions of fractal rugs

摘录 / EXCERPT: Fractal-rug conjecture. For any $c\geq 1$ there is a Jordan arc V_{c-1} such that

conjecture | open | MathDB 页面 | 原文线索

V2-031253 | MATHDB-331486

Sub-Vietoris embedding conjecture for non-type-definable equivalence relations

摘录 / EXCERPT: Suppose E,p,X,Y are as in Theorem: E is an equivalence relation with the stated hypotheses of that theorem, p is a complete type, and X,Y are the associated sets. Let Y_M...

conjecture | open | MathDB 页面 | 原文线索

V2-031254 | MATHDB-331541

The symplectic Bar-Natan homology conjecture

摘录 / EXCERPT: Let L be a link, let $C_{\mathit{BN}}(L)$ be Bar-Natan's chain complex over $\mathbb F_2[H]$, and let $\mathit{HF}_{\mathbb Z/2}(\mathcal K'_A,\mathcal K_B)$ be the $\mathbb Z/2$...

conjecture | open | MathDB 页面 | 原文线索

V2-031255 | MATHDB-331542

Seidel-Smith-Szabó equivariant symplectic Khovanov conjecture

摘录 / EXCERPT: Let L be a link, let $\tau(u,v)=(u,-v)$ generate the $\mathbb Z/2=\{1,\tau\}$ action, and let $\mathit{HF}_{\{1,\tau\}}(\mathcal K'_A,\mathcal K_B)$ be the resulting equivariant...

conjecture | open | MathDB 页面 | 原文线索

V2-031256 | MATHDB-331563

Fernández de Bobadilla's conjecture on non-splitting of Milnor-fiber cohomology

摘录 / EXCERPT: Let $\mathcal{U}$ be an open neighborhood of the origin in $\mathbb{C}^{n+1}$, let $f:(\mathcal{U},\mathbf{0})\to(\mathbb{C},0)$ be a complex analytic function with...

conjecture | open | MathDB 页面 | 原文线索

V2-031257 | MATHDB-331573

Vanishing of higher-order degrees for complements of two transversally intersecting curves

摘录 / EXCERPT: Let C be a curve formed by two curves that intersect transversally, and let δ_n denote the n-th higher-order degree of the complement of C. Vanishing conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-031258 | MATHDB-331575

Bivariant polynomial coefficient homology conjecture

摘录 / EXCERPT: Let ${\mathcal{C}}$ be a subcategory of $\mathbf{S}(\mathbf{gr})$ containing $\mathbf{S}_c(\mathbf{gr})$. Let β and λ denote the restrictions to ${\mathcal{C}}$ of th...

conjecture | open | MathDB 页面 | 原文线索

V2-031259 | MATHDB-331582

Dąbrowski's suspension conjecture for free involutions

摘录 / EXCERPT: Let A be a unital C^*-algebra with a free action of $\mathbb{Z}/2\mathbb{Z}$. Its unreduced suspension is ... The action extends to ΣA by...

conjecture | open | MathDB 页面 | 原文线索

V2-031260 | MATHDB-331586

Thurston's triple crossing diagram conjecture for oriented surfaces with boundary

摘录 / EXCERPT: Let S be a disk, and let a triple crossing diagram on S be a collection of curves with triple crossing points. A diagram is minimal when it has no monogons or bad digons, equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-031261 | MATHDB-331633

Champanerkar-Kofman-Purcell determinant density upper-bound conjecture

摘录 / EXCERPT: Determinant density upper-bound conjecture. For any link L,

conjecture | open | MathDB 页面 | 原文线索

V2-031262 | MATHDB-331634

Bridge-number density bounds conjecture for hyperbolic links

摘录 / EXCERPT: Bridge-number density bounds conjecture. The sequences $\{\beta_{vol}^k\}_{k\in\mathbb{N}}$ and $\{\beta_{det}^k\}_{k\in\mathbb{N}}$ are strictly increasing, and both converge to...

conjecture | open | MathDB 页面 | 原文线索

V2-031263 | MATHDB-331636

Circular elastic knot conjecture

摘录 / EXCERPT: Let ${\mathcal K}$ be a tame knot class, let $\operatorname{bra}{\mathcal K}$ denote its braid index, and let $\operatorname{bri}{\mathcal K}$ denote its bridge index. An a-times...

conjecture | open | MathDB 页面 | 原文线索

V2-031264 | MATHDB-331653

Fintushel--Stern knot surgery symplecticity conjecture for torus Lefschetz fibrations

摘录 / EXCERPT: Suppose that X^4 is a closed 4--manifold admitting a Lefschetz fibration whose regular fibers are tori. Let $T\subset X$ be a regular fiber of the fibration, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-031265 | MATHDB-331688

Duality conjecture for critical-handle Legendrian homology

摘录 / EXCERPT: Let W be a Liouville cobordism corresponding to attaching critical handles along a collection Λ of Legendrian spheres, and let $L{\mathbb H}^{\text{Ho}}(\Lambda)_*$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031266 | MATHDB-331689

Bourgeois-Ekholm-Eliashberg cyclic surgery conjecture

摘录 / EXCERPT: Bourgeois-Ekholm-Eliashberg cyclic surgery conjecture. Under this isomorphism, the surgery exact triangle for cyclic Legendrian contact homology is identified with the exact triang...

conjecture | open | MathDB 页面 | 原文线索

V2-031267 | MATHDB-331696

Nawata-Oblomkov cyclotomic expansion conjecture for $SU(n)$ invariants

摘录 / EXCERPT: Nawata-Oblomkov cyclotomic expansion conjecture. There exist $\widetilde{H}_k(\mathcal{K})\in\mathbb{Z}[q,q^{-1}]$, independent of N for $N\geq 0$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-031268 | MATHDB-331700

Hamiltonian non-displaceability conjecture for the tori T_a in $\mathbb{C} P^2$

摘录 / EXCERPT: Let $T_a\subset \mathbb{C} P^2$ be the family of Lagrangian tori obtained as the images of the tori $\hat{T}_a\subset \mathbb{C} P^1\times \mathbb{C} P^1$ under the 2:1 cover branc...

conjecture | open | MathDB 页面 | 原文线索

V2-031269 | MATHDB-331716

Sharpness of torus detection for invertibility of 2-dimensional topological field theories

摘录 / EXCERPT: Sharpness conjecture. There exists a symmetric monoidal bicategory $\mathcal C$ and a 2-framed 2-dimensional topological field theory

conjecture | open | MathDB 页面 | 原文线索

V2-031270 | MATHDB-331737

The generalized Milnor invariant generation conjecture for link-map cohomology

摘录 / EXCERPT: Let m be a positive integer and let $d>3$. Consider the space of link maps ... A product of Milnor invariants of type n gives, via configuration-space integrals, a generalized Mi...

conjecture | open | MathDB 页面 | 原文线索

V2-031271 | MATHDB-331760

Factorization conjecture for lazy cohomology of the Drinfeld double

摘录 / EXCERPT: Let G be a group, let k be the underlying field, and let ${DG}^*$ denote the dual of the Drinfeld double. Write $\mathrm{H}^2_L({DG}^*)$ for its lazy second cohomology group,...

conjecture | open | MathDB 页面 | 原文线索

V2-031272 | MATHDB-331784

Floer simple knot complement conjecture for L-spaces

摘录 / EXCERPT: Let X be an L-space, and let a knot complement in X mean the complement $X\setminus\nu(K)$ of a knot $K\subset X$. Such a knot complement is Floer simple when its set of L-spac...

conjecture | open | MathDB 页面 | 原文线索

V2-031273 | MATHDB-331796

Batalin-Vilkovisky semi-infinite pairing conjecture

摘录 / EXCERPT: Let n be the dimension parameter specified in the source, and let $BV^i(N,N')$ denote the cohomology of the total Koszul bicomplex associated with the finite-dimensional approxim...

conjecture | open | MathDB 页面 | 原文线索

V2-031274 | MATHDB-331799

Compatibility and limiting local cohomology conjectures for arc algebras

摘录 / EXCERPT: Let $J[0^N,(1)^{N'}]$ and $A[0^N,(1)^{N'}]$ be the finite arc algebras, let $\mathfrak{a}$ be the local-cohomology support ideal, and let th denote the homomorphisms ind...

conjecture | open | MathDB 页面 | 原文线索

V2-031275 | MATHDB-331801

Perfect local-cohomology pairing for finite arc algebras

摘录 / EXCERPT: Let J^N be the finite arc algebra and let $\mathfrak{i}_k$ and $\mathfrak{j}_k$ be the support ideals generated respectively by the initial and terminal arc coefficients as defin...

conjecture | open | MathDB 页面 | 原文线索

V2-031276 | MATHDB-331809

The simple loop conjecture for 3-manifolds

摘录 / EXCERPT: Simple loop conjecture. If F_* is not injective, then there is an essential simple loop in Σ that represents an element of the kernel of F_*. This generalizes the Loop T...

conjecture | open | MathDB 页面 | 原文线索

V2-031277 | MATHDB-331885

The unitary induction conjecture for Dirac cohomology

摘录 / EXCERPT: Let π be a unitary representation. A representation is induced by parabolic induction when it is obtained from a representation of a parabolic subgroup in this way. Unitary Dir...

conjecture | open | MathDB 页面 | 原文线索

V2-031278 | MATHDB-331901

Bernstein-Sato support conjecture for Sabbah's specialization complex

摘录 / EXCERPT: Let $F=(f_1,\ldots,f_r):X\rightarrow\mathbb{C}^r$ be a collection of non-invertible analytic functions on a complex manifold such that $f^{-1}(0)$ admits a finite Whitney stratific...

conjecture | open | MathDB 页面 | 原文线索

V2-031279 | MATHDB-331904

Beilinson-Bloch-Kato conjecture for polarized Chow motives

摘录 / EXCERPT: Let K be a number field and let M be a polarized Chow motive in $\mathsf{Mot}_K^\mathrm{rat}$. Let $\operatorname{CH}(\mathsf{M})^0$ be the homologically trivial degre...

conjecture | open | MathDB 页面 | 原文线索

V2-031280 | MATHDB-331914

Rigidity conjecture for maximal-rank torus actions on rationally elliptic manifolds

摘录 / EXCERPT: Rigidity conjecture. M^n is equivariantly diffeomorphic to an effective, linear action of T^k on a manifold of one of the following forms:

conjecture | open | MathDB 页面 | 原文线索

V2-031281 | MATHDB-331917

Connectedness conjecture for the marked μ-constant moduli space

摘录 / EXCERPT: Let f_0 be a fixed isolated hypersurface singularity. For the bc-homotopy class of f_0, let M^{mar}_bc be the moduli space of right equivalence classes of marked singularit...

conjecture | open | MathDB 页面 | 原文线索

V2-031282 | MATHDB-331925

Manolescu's conjecture on singular-resolution knot Floer and HOMFLY-PT homology

摘录 / EXCERPT: Manolescu's conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-031283 | MATHDB-331952

Kauffman's genus-one slice derivative conjecture

摘录 / EXCERPT: Kauffman's conjecture. There exists an essential simple closed curve d on F such that

conjecture | open | MathDB 页面 | 原文线索

V2-031284 | MATHDB-331974

Hom boundedness implies cone finiteness

摘录 / EXCERPT: A triangulated category D is Hom bounded if for every pair of objects $A,B\in\mathsf{D}$, the dimensions of $\operatorname{Hom}(A,B)$ are bounded independently of the ch...

conjecture | open | MathDB 页面 | 原文线索

V2-031285 | MATHDB-331985

Hu'ng-Peterson algebraic conjecture on spherical classes

摘录 / EXCERPT: Algebraic conjecture on spherical classes. The homomorphism φ_k is zero for all $k>2$.

conjecture | open | MathDB 页面 | 原文线索

V2-031286 | MATHDB-331986

Hu'ng's weak algebraic conjecture on spherical classes

摘录 / EXCERPT: Let A be the mod-2 Steenrod algebra, let D_k be the Dickson algebra, and let ... be the Lannes-Zarati homomorphism. Let ... be Singer's algebraic transfer, where...

conjecture | open | MathDB 页面 | 原文线索

V2-031287 | MATHDB-331998

The Valdivia compacta ccc dichotomy conjecture

摘录 / EXCERPT: Let K be a nonempty Valdivia compact space satisfying the countable chain condition (ccc). Valdivia compacta ccc dichotomy conjecture. Either K has a G_δ point or K ad...

conjecture | open | MathDB 页面 | 原文线索

V2-031288 | MATHDB-332074

Baldwin-Levine-Sarkar's graded pointed-link Khovanov-knot Floer conjecture

摘录 / EXCERPT: Let $(L,\mathbf{p})$ be a pointed link, let (X,D^0) be the associated Alexander-graded complex, and let $\operatorname{CKh}(L,\mathbf{p})$ denote the pointed Khovanov chain compl...

conjecture | open | MathDB 页面 | 原文线索

V2-031289 | MATHDB-332075

Rasmussen's Khovanov-knot Floer rank conjecture for pointed links

摘录 / EXCERPT: Let $L\subset S^3$ be an l-component link, let $p\in L$ be a basepoint, and let $\mathbb{F}$ be any field. Write $\widetilde{\operatorname{Kh}}(L,p;\mathbb{F})$ for reduced Khova...

conjecture | open | MathDB 页面 | 原文线索

V2-031290 | MATHDB-332079

Eynard-Orantin construction of dispersive universal Whitham tau-functions

摘录 / EXCERPT: Let a spectral curve be the spectral-curve datum to which Eynard-Orantin topological recursion applies, and let the associated dispersive universal Whitham hierarchy be the hierarc...

conjecture | open | MathDB 页面 | 原文线索

V2-031291 | MATHDB-332095

Torsion lower-bound conjecture for homology of finite spaces

摘录 / EXCERPT: Torsion lower-bound conjecture. If $H_k(X)$ has torsion, then

conjecture | open | MathDB 页面 | 原文线索

V2-031292 | MATHDB-332106

The critical-rank vanishing conjecture for Borel regulator classes

摘录 / EXCERPT: Let k be an integer with $k\geq 1$. The Borel regulator class β_{2k+1} is a cohomology class in $H^{4k+1}(\mathrm{GL}(2k+2,\mathbb{Z});\mathbb{R})$. Critical-rank vanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-031293 | MATHDB-332107

The Morita-pairing conjecture for secondary Borel classes

摘录 / EXCERPT: Let $\mu_k\in H_{4k}(\mathrm{Out}\,F_{2k+2};\mathbb{Q})$ be the Morita class, and let $\overset{\circ}{\beta}_{2k+1}$ be the secondary class defined at the critical rank $2k+2$, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-031294 | MATHDB-332108

The dual Borel-pairing conjecture for Morita classes

摘录 / EXCERPT: Assume $(p_0)_*(\mu_k)=0$, and construct a cycle $u\in Z_{4k+1}(\mathrm{GL}(2k+3,\mathbb{Z});\mathbb{Q})$ by bounding the relevant abelian cycle in $\mathrm{Out}\,F_{2k+3}$ and its...

conjecture | open | MathDB 页面 | 原文线索

V2-031295 | MATHDB-332125

The free Picard n-groupoid conjecture for the sphere spectrum

摘录 / EXCERPT: Let $\mathbb{S}=\Sigma^\infty S^0$ be the sphere spectrum, and let $\tau_{[0,n]}\mathbb{S}$ denote its truncation to degrees from 0 through n. Let $\mathcal{F}_L^{(n)}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-031296 | MATHDB-332137

The cyclotomic expansion conjecture for colored Kauffman superpolynomials

摘录 / EXCERPT: The Kauffman cyclotomic expansion conjecture. There exists an integer-valued invariant $\beta(\mathcal{K})\in\mathbb{Z}$ and coefficient functions...

conjecture | open | MathDB 页面 | 原文线索

V2-031297 | MATHDB-332138

The volume conjecture for $SU(n)$-specialized HOMFLY-PT superpolynomials

摘录 / EXCERPT: Let $\mathcal{K}$ be a hyperbolic knot, let n and b satisfy $b\geq1$ and $(n-1-b)/2$ not be a positive integer, and let $\mathcal{P}_N(\mathcal{K};a,q,t)$ be the reduced colore...

conjecture | open | MathDB 页面 | 原文线索

V2-031298 | MATHDB-332150

Katthän's weakened Stanley-depth conjecture

摘录 / EXCERPT: Let S be a polynomial ring over a field and let $I\subset S$ be a monomial ideal. Write depth for homological depth and sdepth for Stanley depth...

conjecture | open | MathDB 页面 | 原文线索

V2-031299 | MATHDB-332178

Kawauchi's 4-move conjecture for link-homotopic links

摘录 / EXCERPT: Let two links be link-homotopic if one can be obtained from the other by finitely many self-crossing changes. Two links are 4-equivalent if one can be reached from the other by fin...

conjecture | open | MathDB 页面 | 原文线索

V2-031300 | MATHDB-332189

Kühnel-Lutz conjecture on tight triangulations and strong minimality

摘录 / EXCERPT: Let $\mathbb{F}$ be a field. A simplicial complex X is $\mathbb{F}$-tight if it is connected and, for every induced subcomplex Y of X, the inclusion-induced map ... is injectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-031301 | MATHDB-332190

Topological classification conjecture for Poincaré-type foliation germs

摘录 / EXCERPT: Topological classification conjecture. The two foliation germs are topologically equivalent if and only if: (1) the ray configurations of their eigenvalue tuples are equivalent; an...

conjecture | open | MathDB 页面 | 原文线索

V2-031302 | MATHDB-332196

The restricted cohomological-dimension-one structure conjecture for Lie p-algebras

摘录 / EXCERPT: Restricted cohomological-dimension-one structure conjecture. Any Lie p-algebra L with $\operatorname{cd}_*(L)=1$ is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-031303 | MATHDB-332278

Q-graded correlator conjecture for open intersection numbers

摘录 / EXCERPT: Q-graded correlator conjecture. These correlators are the Q-graded correlators for open intersection numbers. The proposed grading is intended to separate contributions by bounda...

conjecture | open | MathDB 页面 | 原文线索

V2-031304 | MATHDB-332306

Baldwin-Levine conjecture on mutation invariance of delta-graded link Floer homology

摘录 / EXCERPT: Let L be a link and let L' be obtained from L by Conway mutation. Write $\operatorname{\widehat{HFL}}(L)$ and $\operatorname{\widehat{HFL}}(L')$ for their link Floer homologi...

conjecture | open | MathDB 页面 | 原文线索

V2-031305 | MATHDB-332313

The quantum-dimension criterion for braiding universality of anyons

摘录 / EXCERPT: Quantum-dimension criterion. The anyon a is braiding universal if and only if $\dim(a)^2$ is not an integer.

conjecture | open | MathDB 页面 | 原文线索

V2-031306 | MATHDB-332353

The converse characterization of the right-veering arc relation

摘录 / EXCERPT: Let S be a surface with a set of marked points P, let C be a relevant boundary component, and let $\alpha,\beta\in\mathcal{A}_{C}(S,P)$ be arcs. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031307 | MATHDB-332358

Linial-Meshulam torsion-freeness conjecture for first homology

摘录 / EXCERPT: Let $Y_d(n,p)$ denote the Linial-Meshulam random d-dimensional simplicial complex, and let c_2 be the constant appearing in the homological phase transition. A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-031308 | MATHDB-332415

Farrell-Ontaneda conjecture on topological triviality of negatively curved fiber bundles

摘录 / EXCERPT: Let $M\to E\to B$ be a smooth fiber bundle whose fiber admits a negatively curved Riemannian metric. Assume that B is a paracompact, Hausdorff space with the homotopy type of a s...

conjecture | open | MathDB 页面 | 原文线索

V2-031309 | MATHDB-332417

The colored stable homotopy type conjecture for the 3-colored unknot

摘录 / EXCERPT: Let U_3 be the 0-framed 3-colored unknot, and let $\mathcal{X}_{\rm col}^q(U_3)$ denote its colored Khovanov stable homotopy type in quantum degree q. Write $X(\eta2,n)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-031310 | MATHDB-332442

Tian's modified properness conjecture for cscK metrics

摘录 / EXCERPT: Let (X,ω) be a Kähler manifold, let $\mathcal H$ be the normalized space of Kähler potentials cohomologous to ω, let $G=\operatorname{Aut}_0(X,J)$ act on $\mathcal H$...

conjecture | open | MathDB 页面 | 原文线索

V2-031311 | MATHDB-332455

Strongly expansive models for higher-dimensional Furstenberg systems

摘录 / EXCERPT: Let $p_1,\dots,p_d$ be the integers defining the joint action of $\times p_1,\dots,\times p_d$ on $[0,1)$. Let μ be a nonatomic Borel probability measure on $[0,1)$ that is erg...

conjecture | open | MathDB 页面 | 原文线索

V2-031312 | MATHDB-332456

The irrational-ray conjecture for the mutation fan of the four-punctured sphere

摘录 / EXCERPT: Irrational-ray conjecture. The mutation fan ${\mathcal F}_B$ is the union of ${\mathcal F}_{\mathbb Q}(T_0)$ with the set of all irrational rays in P.

conjecture | open | MathDB 页面 | 原文线索

V2-031313 | MATHDB-332457

The support conjecture for the rational quasi-lamination fan of the four-punctured sphere

摘录 / EXCERPT: Support conjecture. The complement ${\mathbb R}^6\setminus|{\mathcal F}_{\mathbb Q}(T_0)|$ is the union of the relative interiors of all irrational rays in P.

conjecture | open | MathDB 页面 | 原文线索

V2-031314 | MATHDB-332458

The real universal-coefficients conjecture for the four-punctured sphere

摘录 / EXCERPT: Let T_0 be the triangulation shown in Figure, let $B=B(T_0)$, and let P be the plane defined by ... Let the vectors described in Items 1--3 of Theorem be the previously deter...

conjecture | open | MathDB 页面 | 原文线索

V2-031315 | MATHDB-332462

Classical knotoid embedding conjecture in virtual knotoids

摘录 / EXCERPT: Classical knotoid embedding conjecture. If two knotoid diagrams in S^2 are virtually equivalent, then they are equivalent to each other by finitely many Ω-moves and isotop...

conjecture | open | MathDB 页面 | 原文线索

V2-031316 | MATHDB-332471

The all-dimensional fundamental-group injectivity conjecture for Rips-to-shadow projections

摘录 / EXCERPT: Let X be a finite subset of Euclidean space, let $\mathcal{R}(X)$ be its Rips complex, and let $\mathcal{S}(X)$ be its shadow complex. Let ... be the projection. The all-dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-031317 | MATHDB-332494

Weighted Singer Conjecture

摘录 / EXCERPT: Weighted Singer Conjecture. The weighted L^2-homology vanishes above the middle dimension:

conjecture | open | MathDB 页面 | 原文线索

V2-031318 | MATHDB-332516

Characterization of local knots by satellite operators

摘录 / EXCERPT: Let Y be an integral homology sphere and let $K\in\mathcal{K}(Y)$ be a knot. A knot is local if it lies in a ball in Y. Local-knot characterization. The almost-concordance clas...

conjecture | open | MathDB 页面 | 原文线索

V2-031319 | MATHDB-332521

Classification of invertible bosonic GQLs by modular extensions

摘录 / EXCERPT: Let G be a symmetry group and let $\cE=\operatorname{Rep}(G)$. A modular extension $\cM$ of $\cE$ is a UMTC containing $\cE$ fully faithfully and satisfying the defining centrali...

conjecture | open | MathDB 页面 | 原文线索

V2-031320 | MATHDB-332522

Classification of invertible fermionic GQLs by modular extensions

摘录 / EXCERPT: Let G^f be a fermionic symmetry group and let $\cE=\operatorname{sRep}(G^f)$ be the corresponding symmetric fusion category. Let $\cM$ be a modular extension of $\cE$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031321 | MATHDB-332523

Classification of 2+1D GQLs by bulk category, modular extension and central charge

摘录 / EXCERPT: Let $\cE$ be the symmetry category, let $\cC$ be a non-degenerate unitary braided fusion category over $\cE$, and let $\cM$ be a modular extension of $\cC$. Let c be the central...

conjecture | open | MathDB 页面 | 原文线索

V2-031322 | MATHDB-332524

Classification of 2+1D GQLs by fusion, spin and modular-extension data

摘录 / EXCERPT: Let $(\widetilde N^{ab}_c,\widetilde s_a;N^{ij}_k,s_i;\mathcal N^{IJ}_K,\mathcal S_I;c)$ denote the specified fusion, spin, modular-extension and central-charge data, subject to th...

conjecture | open | MathDB 页面 | 原文线索

V2-031323 | MATHDB-332553

Stable and topological ranks of real-symmetric continuous function algebras

摘录 / EXCERPT: Let X be a compact Hausdorff space, and let τ be a topological involution of X. Denote the set of fixed points of τ by E. The algebra $C(X,\tau)$ is the algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-031324 | MATHDB-332567

Finite-dimensional Frobenius algebra conjecture for multiplication-comultiplication axioms

摘录 / EXCERPT: Finite-dimensional Frobenius algebra conjecture. Then A admits a unit for μ and a counit for δ, and is, in fact, a Frobenius algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-031325 | MATHDB-332575

Acyclicity conjecture for the extra differential on hairy graph complexes with m = -1

摘录 / EXCERPT: Let $\mathrm{HGC}_{-1,n}$ be the hairy graph complex with hair degree parameter $m=-1$ and arity parameter n, equipped with its standard differential δ and an extra differ...

conjecture | open | MathDB 页面 | 原文线索

V2-031326 | MATHDB-332605

The cohomology-ring criterion for degree-d irreducible polynomials

摘录 / EXCERPT: Cohomology-ring criterion. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-031327 | MATHDB-332657

The homological-smallness conjecture for Hopf actions

摘录 / EXCERPT: Homological-smallness conjecture. If the H-action on R has trivial homological determinant, then

conjecture | open | MathDB 页面 | 原文线索

V2-031328 | MATHDB-332679

The faithful representation version of Thurston's Euler class one conjecture

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold with infinite first homology, and let $a\in H^2(M;\mathbb{R})$ be an integral class satisfying the parity condition and having dual norm e...

conjecture | open | MathDB 页面 | 原文线索

V2-031329 | MATHDB-332733

The orientifold CoHA freeness conjecture

摘录 / EXCERPT: Orientifold CoHA freeness conjecture. If $\mathcal{H}_Q$ is supercommutative, then the CoHA action map

conjecture | open | MathDB 页面 | 原文线索

V2-031330 | MATHDB-332735

The orientifold CoHA structure conjecture

摘录 / EXCERPT: Orientifold CoHA structure conjecture. The CoHA action map

conjecture | open | MathDB 页面 | 原文线索

V2-031331 | MATHDB-332736

The orientifold primitive-cohomology conjecture

摘录 / EXCERPT: Orientifold primitive-cohomology conjecture. The surjection

conjecture | open | MathDB 页面 | 原文线索

V2-031332 | MATHDB-332782

Dwyer-Hess conjecture for little-cubes bimodules

摘录 / EXCERPT: Dwyer-Hess conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-031333 | MATHDB-332792

Tkachenko's conjecture on weakly pseudocompact precompact groups

摘录 / EXCERPT: Let G be a subgroup of a compact group. A space is weakly pseudocompact if it is G_δ-dense in some compact space, meaning that every non-empty G_δ-subset of that co...

conjecture | open | MathDB 页面 | 原文线索

V2-031334 | MATHDB-332797

Artin-Hasse explicit-subspace conjecture

摘录 / EXCERPT: Artin-Hasse subspace conjecture. If $(V,J)\in W'(\chi_1,\chi_2)$ for some J, then

conjecture | open | MathDB 页面 | 原文线索

V2-031335 | MATHDB-332801

The Naive Conjecture on approximate fibering modulo tight torsion

摘录 / EXCERPT: Let $p\colon M\to B$ be a continuous map satisfying the standing assumptions: M is a connected closed topological manifold; B is an aspherical closed topological manifold admit...

conjecture | open | MathDB 页面 | 原文线索

V2-031336 | MATHDB-332802

The MAF Rigidity Conjecture for approximate fibrations

摘录 / EXCERPT: Let B be a closed aspherical manifold, and let $p\colon M\to B$ and $q\colon N\to B$ be MAFs. MAF Rigidity Conjecture. The maps p and q are controlled homeomorphic if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-031337 | MATHDB-332821

The torsion-freeness conjecture for two-term rack homology of associative shelves

摘录 / EXCERPT: An associative shelf is a set equipped with an associative, self-distributive binary operation. Its two-term (rack) homology groups are the homology groups obtained from the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-031338 | MATHDB-332824

The height additivity conjecture for knots

摘录 / EXCERPT: Height additivity conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-031339 | MATHDB-332825

The trunk additivity conjecture for knots

摘录 / EXCERPT: Trunk additivity conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-031340 | MATHDB-332826

The rank versus genus conjecture for knot exteriors

摘录 / EXCERPT: Rank versus genus conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-031341 | MATHDB-332827

The meridional rank versus bridge number conjecture

摘录 / EXCERPT: Meridional rank versus bridge number conjecture. The minimum number of meridional generators of $\pi_1(E(K))$ is equal to

conjecture | open | MathDB 页面 | 原文线索

V2-031342 | MATHDB-332828

Morse-Novikov number additivity conjecture for knots

摘录 / EXCERPT: Morse-Novikov additivity conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-031343 | MATHDB-332829

The two slopes conjecture for knots

摘录 / EXCERPT: Two slopes conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-031344 | MATHDB-332830

The cabling conjecture for Dehn surgery on knots

摘录 / EXCERPT: Cabling conjecture. Then K is a cabled knot, and the surgery slope is the boundary slope of the cabling annulus.

conjecture | open | MathDB 页面 | 原文线索

V2-031345 | MATHDB-332831

The planar surface conjecture for knot exteriors

摘录 / EXCERPT: Planar surface conjecture. A non-meridional essential planar surface in a knot exterior is either a disk or an annulus.

conjecture | open | MathDB 页面 | 原文线索

V2-031346 | MATHDB-332832

Lopez's conjecture on small knots in small 3-manifolds

摘录 / EXCERPT: Lopez conjecture. There exists a small knot in every small closed 3-manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-031347 | MATHDB-332833

The Neuwirth conjecture for knots

摘录 / EXCERPT: Neuwirth conjecture. There exists a closed surface F such that F contains K non-separatingly and is essential in the exterior of K.

conjecture | open | MathDB 页面 | 原文线索

V2-031348 | MATHDB-332835

The coil number conjecture for alternating knots

摘录 / EXCERPT: Alternating-knot coil number conjecture. For an alternating knot K, it holds that

conjecture | open | MathDB 页面 | 原文线索

V2-031349 | MATHDB-332857

Duality conjecture for all pages of the local Hodge-de Rham spectral sequences

摘录 / EXCERPT: Let A be a complete local ring with coefficient field k of characteristic zero, and let R be as in the construction of the local Hodge-de Rham homology and cohomology spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-031350 | MATHDB-332872

The minimal-crossing conjecture for highly twisted plat diagrams

摘录 / EXCERPT: Consider a knot or link diagram satisfying the conditions of Theorem; in particular, the diagram is a sufficiently long plat with at least three crossings in each twist region. Min...

conjecture | open | MathDB 页面 | 原文线索

V2-031351 | MATHDB-332886

The integral homology conjecture for the graph complexes \(\mathcal{G}^1_n(X)\) and \(\mathcal{G}_n(X)\)

摘录 / EXCERPT: Integral homology conjecture. The map $\mathcal{G}^1_n(X)\to\mathcal{G}_n(X)$ induces an isomorphism on homology in degrees $2*\leq n-3$.

conjecture | open | MathDB 页面 | 原文线索

V2-031352 | MATHDB-332911

Cohomological classification conjecture for 2-representations of crossed modules

摘录 / EXCERPT: Let $\mathcal{K}$ be a crossed module with finite fundamental group $\pi_1(\mathcal{K})$, let $\widetilde{\mathcal{K}}$ denote its associated 2-group, and let $\mathcal{X}$ be a cl...

conjecture | open | MathDB 页面 | 原文线索

V2-031353 | MATHDB-332929

Hom-Lidman-Vafaee conjecture on satellite L-space knots

摘录 / EXCERPT: Given a pattern knot P and a companion knot C, both in the three-sphere, let $P(C)$ denote the resulting satellite knot. Hom-Lidman-Vafaee conjecture. If $P(C)$ is an L-space k...

conjecture | open | MathDB 页面 | 原文线索

V2-031354 | MATHDB-332957

Asymptotic coordinate-ratio conjecture for Apéry solutions

摘录 / EXCERPT: Let A_0 and A_γ be the quantum D-module solutions associated with 1 and a coprimitive cohomology class γ, and let $A_0^{(k)}$ and $A_\gamma^{(k)}$ denote their...

conjecture | open | MathDB 页面 | 原文线索

V2-031355 | MATHDB-332968

Oriented Knot Complement Conjecture

摘录 / EXCERPT: Let M be a closed oriented 3-manifold, and let K_1 and K_2 be knots in M whose complements are homeomorphic by an orientation preserving homeomorphism and are not homeomo...

conjecture | open | MathDB 页面 | 原文线索

V2-031356 | MATHDB-332978

Generalized vanishing conjecture for the square of a curve's ideal sheaf

摘录 / EXCERPT: Let $C\subset\mathbf{P}^r$ be a curve of genus g, embedded by a complete linear system $|L|$ of degree d, and suppose that $h^1(L)\leq 1$. Generalized vanishing conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-031357 | MATHDB-333001

Reflection-positive invertible field theory classification conjecture

摘录 / EXCERPT: Let H_n be the symmetry group in dimension n, let MTH be the Thom spectrum associated to this symmetry, and let $I\mathbb Z$ denote the Anderson dual of the sphere spectrum....

conjecture | open | MathDB 页面 | 原文线索

V2-031358 | MATHDB-333060

Rubinstein-Weng-Wormald linear-error conjecture for minimum-topology approximate Steiner trees

摘录 / EXCERPT: Let an ε-approximate Steiner tree be a tree whose angles at Steiner points lie in $[2\pi/3-\varepsilon,2\pi/3+\varepsilon]$. For $d\geqslant 2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-031359 | MATHDB-333069

The satellite conjecture for pro-homotopy

摘录 / EXCERPT: Let X be a space, let $NORM(X)$ be the site of normal coverings of X, and let $OPEN(X)$ be the standard site. Let S be a set, write $S_{\#}$ for the associated constant coshe...

conjecture | open | MathDB 页面 | 原文线索

V2-031360 | MATHDB-333084

The stable homotopy type determines the move-equivalence class of a framed flow category

摘录 / EXCERPT: Let $(\mathscr{C}_1,\iota_1,\varphi_1)$ and $(\mathscr{C}_2,\iota_2,\varphi_2)$ be framed flow categories. Their associated stable homotopy types are the stable homotopy types dete...

conjecture | open | MathDB 页面 | 原文线索

V2-031361 | MATHDB-333092

Meeks-Rosenberg finite topology conjecture II for embedded minimal surfaces

摘录 / EXCERPT: Meeks-Rosenberg finite topology conjecture II. For every positive integer g, there exists a $\Sigma_g\in{\cal M}$ with one end and genus g, unique up to congruences and homoth...

conjecture | open | MathDB 页面 | 原文线索

V2-031362 | MATHDB-333152

Regularity characterization conjecture for correctable patterns in tensor-product topologies

摘录 / EXCERPT: Regularity characterization conjecture. An erasure pattern E is correctable for $T_{m\times n}(a,b,0)$ if and only if it is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-031363 | MATHDB-333170

Unit-root crystal conjecture for higher Hasse-Witt matrices

摘录 / EXCERPT: Let f define a nonsingular hypersurface $X_f=\{f=0\}$, and let $H_f=H^{\cdot}_{dR}(X_f)$ be its de Rham cohomology, equipped with the Gauss-Manin connection and Frobenius maps...

conjecture | open | MathDB 页面 | 原文线索

V2-031364 | MATHDB-333186

Positselski's conjecture on Koszul algebras of finite global dimension

摘录 / EXCERPT: Positselski's conjecture. Every such algebra A satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-031365 | MATHDB-333195

Asymptotic vanishing conjecture for syzygies of large-degree embeddings

摘录 / EXCERPT: Asymptotic vanishing conjecture. Fix $q\in[2,n]$. In the situation of the nonvanishing theorem referenced in the source, there is a constant C_3, depending on X, A, B and...

conjecture | open | MathDB 页面 | 原文线索

V2-031366 | MATHDB-333204

The B-model R-matrix conjecture for primitive-form phases

摘录 / EXCERPT: Let $\tilde W:\mathbb{C}^N\to\mathbb{C}$ be a hypersurface singularity, and let ζ_∞ and ζ_0 be the two specified choices of primitive form. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031367 | MATHDB-333205

The conjecture on the stable gap for sequential topological complexity of real projective spaces

摘录 / EXCERPT: Assume $m\equiv2^e-1\pmod{2^{e+1}}$ with $e\geq0$, so e is the length of the block of consecutive ones ending the binary expansion of m. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031368 | MATHDB-333272

Macdonald-operator identities for normalized link symmetric functions

摘录 / EXCERPT: Let $v\in\{0,1\}^n$, let $L_v(x;q,t)$ be the link symmetric function, and set ... For a partition μ, let $\widetilde{H}_\mu$ be the modified Macdonald polynomial, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-031369 | MATHDB-333273

Bergeron's conjectures for link symmetric functions

摘录 / EXCERPT: Let u,v be binary words of compatible lengths, let $L_v(x;q,t)$ be the link symmetric function, and let $\widetilde{L}_v(x;q,t)=(1-q)^{n-|v|}L_v(x;q,t)$. Let $\widetilde{H}_\mu$...

conjecture | open | MathDB 页面 | 原文线索

V2-031370 | MATHDB-333274

Non-Artinianness conjecture for top local cohomology modules

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a Noetherian local ring, let M be a finitely generated R-module of dimension n, and let I be an ideal of R such that ... Here,...

conjecture | open | MathDB 页面 | 原文线索

V2-031371 | MATHDB-333290

The difference-of-roots conjecture for the Enriques lattice

摘录 / EXCERPT: Let $\Lambda=U\oplus \mathbb{E}_8(-1)$ be the even unimodular lattice of signature $(1,9)$, identified with the cohomology lattice of an Enriques surface up to torsion. Difference-...

conjecture | open | MathDB 页面 | 原文线索

V2-031372 | MATHDB-333299

Higher-smoothness entropy conjecture for cycles of basic sets

摘录 / EXCERPT: Let f be a C^r-diffeomorphism of a manifold M, and let p be a hyperbolic periodic point belonging to a cycle of basic sets having no dominated splitting of any index. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031373 | MATHDB-333308

The capacity conjecture for stably non-displaceable subsets

摘录 / EXCERPT: The capacity conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-031374 | MATHDB-333338

Conjecture that all Lagrangian homology classes of simply connected Calabi-Yau manifolds lie in ${\mathcal S}$

摘录 / EXCERPT: Let N be a simply connected closed Calabi-Yau manifold of any complex dimension, and let ${\mathcal S}$ denote the subgroup generated by special Lagrangian cycles. Lagrangian hom...

conjecture | open | MathDB 页面 | 原文线索

V2-031375 | MATHDB-333351

Salvetti-Serventi connected homology graph conjecture for real arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ be a real affine line arrangement, and let $\Gamma({\mathcal A})$ be the graph whose vertices are the lines $H\in{\mathcal A}$, with an edge (H,H') exactly whe...

conjecture | open | MathDB 页面 | 原文线索

V2-031376 | MATHDB-333363

Homotopy equivalence of bounded worldsheet cylinders and worldsheet tori

摘录 / EXCERPT: Let a worldsheet cylinder be the rolled-up lateral surface of a bounded worldsheet, and let a worldsheet torus be obtained by bending a worldsheet cylinder until its ends meet. Wor...

conjecture | open | MathDB 页面 | 原文线索

V2-031377 | MATHDB-333372

The local-cohomology regularity equality conjecture for degree-wise coherent FI_G-modules

摘录 / EXCERPT: Let V be a degree-wise coherent ${\mathscr{FI}}_G$-module, let $H^i_\mathfrak{m}(V)$ denote its local cohomology modules, and let $\operatorname{reg}(V)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031378 | MATHDB-333438

Conjecture $\mathbb{Z}$ for knot exteriors

摘录 / EXCERPT: Conjecture $\mathbb{Z}$. If F is a compact orientable and non-separating surface properly embedded in a knot exterior E, then

conjecture | open | MathDB 页面 | 原文线索

V2-031379 | MATHDB-333452

The reduced-genus conjecture for the length of the $\mathcal{D}$-module generated by $1/f$

摘录 / EXCERPT: Reduced-genus conjecture. The length of the left $\mathcal{D}$-submodule of $H^1_f(R)$ generated by $\frac{1}{f}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-031380 | MATHDB-333466

Topological invariance of the Betti automorphic category

摘录 / EXCERPT: Let G be a complex reductive group and let X be a smooth complex projective curve. Let $\mathcal Shv_{\mathcal N}(Bun_G(X))$ be the Betti category of nilpotent sheaves. Topolog...

conjecture | open | MathDB 页面 | 原文线索

V2-031381 | MATHDB-333468

Elliptic character-sheaf Hochschild homology conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let E be an elliptic curve, and let $\mathcal H_G$ be the affine Hecke category. Let $\mathcal Shv_{\mathcal N}(Bun_G(E))$ denote the catego...

conjecture | open | MathDB 页面 | 原文线索

V2-031382 | MATHDB-333479

Vanishing homology and intersection homology under constant Euler characteristic

摘录 / EXCERPT: Let $G=(G_1,\ldots,G_{n-1}):\mathbb{C}^n\to\mathbb{C}^{n-1}$ be a polynomial mapping with $n\geq 2$ and $K_0(G)=\emptyset$. Assume that there exists a very good projection with res...

conjecture | open | MathDB 页面 | 原文线索

V2-031383 | MATHDB-333523

AMU conjecture for surface groups

摘录 / EXCERPT: Let Σ be a closed surface of genus at least 2, let $p=2r$ be an even integer, and consider the quantum representation ... Here $V_p(\Sigma,n)$ is the vector space assigned b...

conjecture | open | MathDB 页面 | 原文线索

V2-031384 | MATHDB-333574

Mahowald's conjecture on higher associativity of cofiber spectra

摘录 / EXCERPT: Let τ be a nonzero element of $\pi_{k-1}(\mathbb{S}^0)$, and let $C\tau$ denote its cofiber. Mahowald's conjecture. The spectrum $C\tau$ does not admit an A_{∞}-structu...

conjecture | open | MathDB 页面 | 原文线索

V2-031385 | MATHDB-333582

The derived-groups conjecture for lattice cohomology of plumbed 3-manifolds

摘录 / EXCERPT: Derived-groups conjecture. The groups A' and $A”$ are isomorphic, up to a grading shift, to the groups B' and $B”$, respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-031386 | MATHDB-333601

Completeness conjecture for quandle 2-cocycle invariants of oriented knots

摘录 / EXCERPT: Let K_1 and K_2 be oriented knots. For a finite quandle Q and a 2-cocycle ϕ, write $\Phi_\phi(K)$ for the associated quandle 2-cocycle invariant. Completeness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-031387 | MATHDB-333602

Completeness conjecture for quandle coloring enhancements of oriented knots

摘录 / EXCERPT: Let K be an oriented knot and let Q be a finite connected quandle. Denote by $\Psi^e_Q(K)$ the invariant associated with Q. Completeness conjecture. The invariant Ψ^e_Q...

conjecture | open | MathDB 页面 | 原文线索

V2-031388 | MATHDB-333620

Updated conjectural picture for finite surgeries on hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3. A finite-surgery conjectural picture consists of the following assertions: 1. (The Berge conjecture) If K admits a nontrivial cyclic surger...

conjecture | open | MathDB 页面 | 原文线索

V2-031389 | MATHDB-333634

The equivalence conjecture for arrangement complements and boundary manifolds

摘录 / EXCERPT: Let ${\mathcal{A}}$ and ${\mathcal{A}}'$ be central arrangements in $\mathbb{C}^3$. Write $U({\mathcal{A}})$ for the complement, ${\partial{{\overline{U}}}}({\mathcal{A}})$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-031390 | MATHDB-333635

The Milnor-fiber equivalence conjecture for central arrangements

摘录 / EXCERPT: Let ${\mathcal{A}}$ and ${\mathcal{A}}'$ be central arrangements in $\mathbb{C}^3$, let $F({\mathcal{A}})$ denote the Milnor fiber, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031391 | MATHDB-333637

The Meta-Isomorphism Conjecture for functors from spaces to spectra

摘录 / EXCERPT: Let ${\mathbf S} \colon \operatorname{SPACES}\to \operatorname{SPECTRA}$ be a covariant functor respecting weak equivalences and disjoint unions. For a group G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-031392 | MATHDB-333638

The Meta-Isomorphism Conjecture for spaces-to-spectra functors with coefficients

摘录 / EXCERPT: Let ${\mathbf S}\colon\operatorname{SPACES}\to\operatorname{SPECTRA}$ be a covariant functor respecting weak equivalences and disjoint unions. Let G be a group, ${\mathcal F}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-031393 | MATHDB-333639

The Meta-Isomorphism Conjecture with coefficients and finite wreath products

摘录 / EXCERPT: Let ${\mathbf S}\colon\operatorname{SPACES}\to\operatorname{SPECTRA}$ be a covariant functor respecting weak equivalences and disjoint unions. Let ${\mathcal C}$ be a class of grou...

conjecture | open | MathDB 页面 | 原文线索

V2-031394 | MATHDB-333640

The Fibered Meta-Isomorphism Conjecture for equivariant homology theories

摘录 / EXCERPT: Let Γ be a group, let (G,ψ) be a group over Γ with homomorphism $\psi\colon G\to\Gamma$, let ${\mathcal H}^?_*$ be an equivariant homology theory over Γ,...

conjecture | open | MathDB 页面 | 原文线索

V2-031395 | MATHDB-333641

The Fibered Meta-Isomorphism Conjecture for spaces-to-spectra functors with coefficients

摘录 / EXCERPT: Let ${\mathbf S}\colon\operatorname{SPACES}\to\operatorname{SPECTRA}$ respect weak equivalences and disjoint unions. Let G be a group and ${\mathcal F}$ a family of subgroups of...

conjecture | open | MathDB 页面 | 原文线索

V2-031396 | MATHDB-333646

Torsion-freeness conjecture for homology of random simplicial complexes

摘录 / EXCERPT: Let $Y_d(n,p)$ be the random d-dimensional simplicial complex, and let c_d^* be the real number such that $p=c_d^*/n$ is a one-sided sharp threshold for the non-triviality of...

conjecture | open | MathDB 页面 | 原文线索

V2-031397 | MATHDB-333679

String transformation conjecture for quasimap and Gromov-Witten S-operators

摘录 / EXCERPT: Let $E\to E\sslash \textbf{G}$ be the GIT quotient in the paper, let $H^*(E\sslash \textbf{G},\Lambda)$ be its cohomology with coefficient ring Λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031398 | MATHDB-333693

Width-minimizing positions that fail to minimize trunk or bridge number

摘录 / EXCERPT: Let K and K' be knots, with embeddings k and k', respectively. An embedding is width-minimizing when it realizes the knot's minimum width; similarly, trunk-minimizing and b...

conjecture | open | MathDB 页面 | 原文线索

V2-031399 | MATHDB-333694

Thin-position conjecture for the knots $K_{2,1,3,7}$ and $K_{4,1,3,3}$

摘录 / EXCERPT: Let B_i be the braids occurring in the paper's construction, and let $K_{2,1,3,7}$ and $K_{4,1,3,3}$ denote the corresponding knots. A thin position is an embedding realizing min...

conjecture | open | MathDB 页面 | 原文线索

V2-031400 | MATHDB-333723

Snaith's conjecture on the Kervaire problem in dimension 126

摘录 / EXCERPT: Let $n=126$. The Kervaire problem asks whether there exists a framed manifold of dimension $2^{n+1}-2$ with Kervaire invariant one. Snaith's conjecture. The Kervaire problem is neg...

conjecture | open | MathDB 页面 | 原文线索

V2-031401 | MATHDB-333733

Pole structure conjecture for generalized BGW correlation differentials

摘录 / EXCERPT: Pole structure conjecture. All $\omega_{g,n}$ are meromorphic differentials on the spectral curve and have poles of finite degree only at the branch point $z_j=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-031402 | MATHDB-333758

Jin's quadrisecant approximation conjecture

摘录 / EXCERPT: Jin's conjecture. The quadrisecant approximation $\hat{K}$ has the knot type of K, and K and $\hat{K}$ have the same set of quadrisecants.

conjecture | open | MathDB 页面 | 原文线索

V2-031403 | MATHDB-333759

Essential quadrisecants conjecture for the figure-8 and 5_2 knots

摘录 / EXCERPT: Essential quadrisecants conjecture. The figure-8 knot has at least one essential flipped quadrisecant, and the 5_2 knot has at least one essential simple quadrisecant.

conjecture | open | MathDB 页面 | 原文线索

V2-031404 | MATHDB-333782

Linear growth conjecture for minimal contact triangulations

摘录 / EXCERPT: Let M be a 3-manifold with contact structure xi, and let d^2 and d^3 denote the relative contact-structure invariants used to compare xi' with xi. Let xi' be a contac...

conjecture | open | MathDB 页面 | 原文线索

V2-031405 | MATHDB-333788

One-dimensionality conjecture for the cohomology of conformally invariant loop cocycles

摘录 / EXCERPT: A configuration is either a pair (ℓ,Σ) consisting of a Riemann surface Σ and a loop $\ell\subset\Sigma$, or a triple (ℓ,Σ_1,Σ_2) consisting of Riema...

conjecture | open | MathDB 页面 | 原文线索

V2-031406 | MATHDB-333801

Conant's integral Conway polynomial splitting conjecture for amphicheiral knots

摘录 / EXCERPT: Let K be an amphicheiral knot, and let $C_K(z)$ denote its Conway polynomial. Conant's integral splitting conjecture. For some $\phi\in\mathbb Z[z]$, one has ... The conjecture str...

conjecture | open | MathDB 页面 | 原文线索

V2-031407 | MATHDB-333802

Stoimenow's square leading-coefficient conjecture for amphicheiral knots

摘录 / EXCERPT: Let K be an amphicheiral knot, and let Δ_K denote its Alexander polynomial; equivalently, consider the leading coefficient of its Conway polynomial. Stoimenow's square lea...

conjecture | open | MathDB 页面 | 原文线索

V2-031408 | MATHDB-333803

Stoimenow's sign conjecture for amphicheiral Alexander polynomials

摘录 / EXCERPT: Let K be an amphicheiral knot, and let Δ_K be its Alexander polynomial. Stoimenow's sign conjecture. The sign of the leading coefficient of Δ_K is ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-031409 | MATHDB-333806

Existence of loop-filling rays that are not ray-filling

摘录 / EXCERPT: A ray in the surface is loop-filling if it intersects every nontrivial loop, and ray-filling if it intersects every short ray. Existence conjecture. There are loop-filling rays whi...

conjecture | open | MathDB 页面 | 原文线索

V2-031410 | MATHDB-333811

Purity conjecture for Borel-Moore homology of algebraic-symplectic stacks

摘录 / EXCERPT: Purity conjecture. The Hodge structure on $H_i^{BM}(\mathcal{C}oh(X)_P^{H-\operatorname{ss}})$ is pure for any H, and, in general, $H_i^{BM}(\mathcal{X})$ is pure for any algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-031411 | MATHDB-333854

The naive conjecture on cohomology restriction thresholds

摘录 / EXCERPT: Let Conf_n be the configuration space of n disks, let $\operatorname{Conf}_{n,r}$ be the subspace in which the disks have radius at least r, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031412 | MATHDB-333855

The modified conjecture for cohomology classes of fixed forest shape

摘录 / EXCERPT: Let Conf_n be the configuration space of n disks, and let $H^{\underline{m}}(\operatorname{Conf}_n)$ be the subspace of cohomology generated by ordered forests w...

conjecture | open | MathDB 页面 | 原文线索

V2-031413 | MATHDB-333856

The strong conjecture on disk configuration cohomology thresholds

摘录 / EXCERPT: Let $\underline{m}$ be a finite multiset of component sizes, and let $r_{\min}(\underline{m})$ and $r_{\max}(\underline{m})$ denote the corresponding injectivity and nonvanishing t...

conjecture | open | MathDB 页面 | 原文线索

V2-031414 | MATHDB-333857

The conjecture on the critical segment length

摘录 / EXCERPT: Let Seg_n be the configuration space of n segments, let $\operatorname{Seg}_{n,r}$ be the subspace of segments of length at least r, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031415 | MATHDB-333858

The k-diagonal kernel conjecture for disk configuration spaces

摘录 / EXCERPT: Let Conf_n be the configuration space of n disks, and for $2\leq k\leq n$ let $\Delta_k\subset\operatorname{Conf}_n$ be the locus where at least k points are c...

conjecture | open | MathDB 页面 | 原文线索

V2-031416 | MATHDB-333863

Conjecture on compositions of left and right interleavings

摘录 / EXCERPT: Composition conjecture. Any composition of suitable left and right interleavings yields an equivalence with interleavings.

conjecture | open | MathDB 页面 | 原文线索

V2-031417 | MATHDB-333868

The fiberedness conjecture for Floer simple knots in L-spaces

摘录 / EXCERPT: Let Y be an L-space, and let $K\subset Y$ be a Floer simple knot with irreducible complement. Write $[K]^{\perp}$ for the orthogonal complement of the homology class $[K]$ under...

conjecture | open | MathDB 页面 | 原文线索

V2-031418 | MATHDB-333874

Dao-Kurano's dimension-bound vanishing conjecture for the Hochster theta pairing

摘录 / EXCERPT: Dao-Kurano's conjecture. If $\dim(M)\leqslant \frac{d}{2}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-031419 | MATHDB-333876

Rationality conjecture for the derived Hecke action on automorphic cohomology

摘录 / EXCERPT: Rationality conjecture. The action of $\mathsf{V}_{\mathbf{Q}}$ preserves $H^*(Y(1),\mathbf{Q})_\Pi$.

conjecture | open | MathDB 页面 | 原文线索

V2-031420 | MATHDB-333898

Buzzard-Diamond-Jarvis conjecture for the cohomology of Shimura curves

摘录 / EXCERPT: Let F be a totally real field, let D be an indefinite quaternion algebra over F, and let $S^D(\mathbb F)$ be the direct limit of the mod p étale cohomology of the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-031421 | MATHDB-333910

The generalized-torsion conjecture for 3-manifold groups

摘录 / EXCERPT: Generalized-torsion conjecture. G is bi-orderable if and only if G has no generalized torsion element.

conjecture | open | MathDB 页面 | 原文线索

V2-031422 | MATHDB-333919

The final unstable homology conjecture for general linear groups over \(\mathbb{F}_2\)

摘录 / EXCERPT: Final unstable homology conjecture. For every $m\geqslant 1$, the group $H_m(\operatorname{GL}_{2m}(\mathbb{F}_2);\mathbb{F}_2)$ is a single copy of $\mathbb{F}_2$ generated by the...

conjecture | open | MathDB 页面 | 原文线索

V2-031423 | MATHDB-333920

Decomposability conjecture beyond the stable range

摘录 / EXCERPT: Let $(G_p)_{p\geqslant 0}$ be a family of groups with multiplication, and let SP_n denote the splitting complex associated with this family. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-031424 | MATHDB-333948

Conjecture D on the degree-one Poitou-Tate sequence for knot groups

摘录 / EXCERPT: Let G be a knot group, let $H\leq G$ be a peripheral subgroup, and let $U\trianglelefteq G$ be a normal subgroup of finite index. Write $E_\mathcal{G}G/U$ for the associated clas...

conjecture | open | MathDB 页面 | 原文线索

V2-031425 | MATHDB-333962

Thin-monodromy equality conjecture for the 14 Calabi-Yau hypergeometric systems

摘录 / EXCERPT: Consider the 14 rank-4 hypergeometric local systems that could arise as the middle cohomology of families of Calabi-Yau threefolds with $h^{2,1}=1$. For these systems, let ... be t...

conjecture | open | MathDB 页面 | 原文线索

V2-031426 | MATHDB-334029

The pressure threshold conjecture for resonance density

摘录 / EXCERPT: Let $N(\Omega):=N_{P(h)}(\Omega)$ be the resonance counting function, and let $\mathcal P_E(s)$ denote the topological pressure from the source. Pressure threshold conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-031427 | MATHDB-334030

The resonance concentration conjecture near the pressure line

摘录 / EXCERPT: Let $N(\Omega):=N_{P(h)}(\Omega)$ be the resonance counting function, and let $\mathcal P_E(s)$ be the topological pressure. Resonance concentration conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-031428 | MATHDB-334054

Gompf's standardness conjecture for twisted doubles of infinite order corks

摘录 / EXCERPT: Let $C=C(r,s;m)$ be Gompf's cork, and define the twisted double ... Here r,s,m are integers with $r,s>0>m$, and k is a nonzero integer. Gompf's conjecture. The homotopy 4-sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-031429 | MATHDB-334057

Renormalization conjecture for infinitely renormalizable Lorenz-map classes

摘录 / EXCERPT: Fix a topological class $\mathcal{T}$ of infinitely (a,b)-renormalizable Lorenz maps, with a and b sufficiently large. Let $\mathcal{U}\subset\mathcal{T}$ be the nonempty ope...

conjecture | open | MathDB 页面 | 原文线索

V2-031430 | MATHDB-334072

General geometric conjecture for character-valued parabolic-bundle polynomials

摘录 / EXCERPT: General geometric conjecture. There exists an action of $\mathfrak{H}$ on $H_c^i(\mathcal{X}_{\bf S}^{\mathcal{E}},\mathbb{C})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-031431 | MATHDB-334073

Trace formula conjecture for logarithmic generating functions and cohomology

摘录 / EXCERPT: Trace formula conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-031432 | MATHDB-334088

Finite historic-measure mass implies zero entropy

摘录 / EXCERPT: Zero-entropy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-031433 | MATHDB-334089

Full-size wild historic set conjecture

摘录 / EXCERPT: Full-size wild historic set conjecture. In this class of examples, the set of wild historic points has full Hausdorff dimension, full topological entropy and full topological press...

conjecture | open | MathDB 页面 | 原文线索

V2-031434 | MATHDB-334123

Langlands' conjecture on motives attached to cohomological automorphic representations

摘录 / EXCERPT: Langlands' motive conjecture. Given a cohomology class as above, there exists a $\widehat{G}$-motive X over $\mathbf{Q}$ such that, for every nonarchimedean λ, the Galois...

conjecture | open | MathDB 页面 | 原文线索

V2-031435 | MATHDB-334142

Kahn's low-dimensional unramified Witt-ring conjecture

摘录 / EXCERPT: Let F be a field of characteristic different from 2, let φ be an anisotropic quadratic form over F, and let $F(\varphi)$ be the function field of its projective quadr...

conjecture | open | MathDB 页面 | 原文线索

V2-031436 | MATHDB-334157

The inequality of the Thurston norm

摘录 / EXCERPT: Let M and N be admissible 3-manifolds, and let $f\colon M\to N$ be a map that is surjective on $\pi_1(N)$ and induces isomorphisms ... for every $n\geq 0$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-031437 | MATHDB-334158

Residual local indicability conjecture for admissible 3-manifold groups

摘录 / EXCERPT: Let M be an admissible 3-manifold with first Betti number $b_1(M)\geq 1$. Residual local indicability conjecture. The fundamental group $\pi_1(M)$ is residually locally indicab...

conjecture | open | MathDB 页面 | 原文线索

V2-031438 | MATHDB-334224

Weierstrass quantum-curve conjecture

摘录 / EXCERPT: Consider the Weierstrass spectral curve ... Let $\hat{P}(\hat{x},\hat{y};\hbar)$ be a quantum curve of the form ... where the $A_i(x)$ and $B_j(x)$ are polynomials in x. A Weierstras...

conjecture | open | MathDB 页面 | 原文线索

V2-031439 | MATHDB-334232

Exponential formula for reconstructing the wave function by topological recursion

摘录 / EXCERPT: Exponential formula. The solution should have the WKB expansion

conjecture | open | MathDB 页面 | 原文线索

V2-031440 | MATHDB-334237

Vanishing simplicial volume conjecture for affine manifolds

摘录 / EXCERPT: Vanishing simplicial volume conjecture. If M is a closed manifold supporting an affine structure, then

conjecture | open | MathDB 页面 | 原文线索

V2-031441 | MATHDB-334244

Associative Cayley-Dickson imaginaroid extension conjecture

摘录 / EXCERPT: Cayley-Dickson extension conjecture. The space A' can be given the structure of a Cayley-Dickson imaginaroid; moreover, this structure is associative if A is commutative.

conjecture | open | MathDB 页面 | 原文线索

V2-031442 | MATHDB-334245

Auslander-type conjecture for semisimple Hopf actions on AS regular algebras

摘录 / EXCERPT: Let A be a noetherian AS regular algebra admitting an inner faithful action of a semisimple Hopf algebra H with trivial homological determinant. The Auslander-type conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-031443 | MATHDB-334261

Thurston's geometrisation conjecture

摘录 / EXCERPT: Let M be an irreducible orientable compact 3-manifold with possibly empty boundary consisting of tori. A geometric 3-manifold is one whose interior has a finite-volume complete g...

conjecture | open | MathDB 页面 | 原文线索

V2-031444 | MATHDB-334262

Elliptisation conjecture for closed 3-manifolds

摘录 / EXCERPT: Elliptisation conjecture. Every closed 3-manifold M with finite $\pi_1(M)$ is elliptic.

conjecture | open | MathDB 页面 | 原文线索

V2-031445 | MATHDB-334275

The Seifert surgery conjecture for knots in the three-sphere

摘录 / EXCERPT: Let K be a knot in S^3, let $p/q\in\mathbb{Q}$ be a surgery slope with $q\geq 2$, and write $S_{p/q}^3(K)$ for the manifold obtained by p/q-surgery on K. A Seifert fibred s...

conjecture | open | MathDB 页面 | 原文线索

V2-031446 | MATHDB-334293

Chataur-Menichi conjecture on the Gerstenhaber algebra of free-loop homology

摘录 / EXCERPT: Let G be a connected compact Lie group of dimension d, let BG be its classifying space, and let LBG denote the free loop space of BG. Write $S_*(G)$ for the singular chai...

conjecture | open | MathDB 页面 | 原文线索

V2-031447 | MATHDB-334311

Equality of topological and analytical local cyclic indices

摘录 / EXCERPT: Let M be a quasi-conformal closed manifold. Let $\mathcal{A}_{\mu}$ be the algebra of pseudo-differential operators on M, localized by the support of operators about the diagon...

conjecture | open | MathDB 页面 | 原文线索

V2-031448 | MATHDB-334340

Juhász-Zemke conjecture on agreement of link Floer TQFT maps

摘录 / EXCERPT: Juhász-Zemke conjecture. The maps $\widehat{F}_{W,\mathcal{F}}$ coincide with the maps $F_{W,\mathcal{F}}$ defined by Juhász using sutured Floer homology.

conjecture | open | MathDB 页面 | 原文线索

V2-031449 | MATHDB-334394

Topological equivalence of the Lorenz flow and the modular geodesic flow

摘录 / EXCERPT: Let the Lorenz flow be the flow defined by the Lorenz equations at the primary T-point parameters $\rho_0\approx 30.8680$, $\sigma_0\approx 10.1673$, and $\beta_0=\frac{8}{3}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031450 | MATHDB-334428

Integral-current realization conjecture for Lipschitz chains

摘录 / EXCERPT: Integral-current realization conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-031451 | MATHDB-334429

Yamaguchi's mass comparison conjecture for Lipschitz simplices

摘录 / EXCERPT: Yamaguchi's mass comparison conjecture. The mass $\widetilde{\mathrm{mass}}(f)$ in Yamaguchi's sense coincides with the mass defined in the paper.

conjecture | open | MathDB 页面 | 原文线索

V2-031452 | MATHDB-334430

Positive synthetic Ricci curvature conjecture for orientable Alexandrov spaces

摘录 / EXCERPT: Positive synthetic Ricci curvature conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-031453 | MATHDB-334441

The Pin(2)-monopole Floer spectral sequence conjecture for branched double covers

摘录 / EXCERPT: Let $L\subset S^3$ be a link, let $\Sigma(L)$ denote its branched double cover, and let $\bar{L}$ be the mirror of L. Write $\widetilde{\mathrm{BN}}^3_{*,*}(\bar{L})$ for the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-031454 | MATHDB-334442

Berest-Samuelson's DAHA action conjecture for knot skein modules

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $\mathrm{sk}_q(S^3\setminus K)$ be the Kauffman bracket skein module of its complement, and let $\mathrm{SH}_{q,t_1,t_2,1,1}$ denote the spherical double...

conjecture | open | MathDB 页面 | 原文线索

V2-031455 | MATHDB-334446

Open-closed Landau-Ginzburg TFT axioms for compact critical locus

摘录 / EXCERPT: Let (X,W) be a B-type open-closed topological Landau-Ginzburg theory, with X a non-compact Calabi-Yau manifold and W a holomorphic function whose critical set Z_W is compac...

conjecture | open | MathDB 页面 | 原文线索

V2-031456 | MATHDB-334490

Symmetric hyperbolic 4-cobordism conjecture

摘录 / EXCERPT: Let M be a hyperbolic 4-cobordism with boundary components N_0 and N_1. Suppose that N_0 and N_1 are highly symmetric to each other, both topologically and geometricall...

conjecture | open | MathDB 页面 | 原文线索

V2-031457 | MATHDB-334503

Rozansky's trivial tail homology conjecture for non-A-adequate diagrams

摘录 / EXCERPT: Rozansky's conjecture. If a diagram D is not A-adequate, then $H^{\infty}(D)$ is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-031458 | MATHDB-334510

Homology decomposition conjecture for the two-term Yang-Baxter complex

摘录 / EXCERPT: Homology decomposition conjecture. For every n,

conjecture | open | MathDB 页面 | 原文线索

V2-031459 | MATHDB-334532

Surjectivity conjecture for the hyperkähler Kirwan map

摘录 / EXCERPT: Hyperkähler Kirwan surjectivity conjecture. The hyperkähler Kirwan map is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-031460 | MATHDB-334533

VGIT-ring conjecture for hyperkähler Kirwan images

摘录 / EXCERPT: VGIT-ring conjecture. Under these assumptions, the image of the hyperkähler Kirwan map is equal to the VGIT ring:

conjecture | open | MathDB 页面 | 原文线索

V2-031461 | MATHDB-334543

Quillen equivalence for simplicial cyclic operads

摘录 / EXCERPT: Let simplicial cyclic operads carry the model structure in which weak equivalences and fibrations are defined levelwise, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031462 | MATHDB-334572

Quantitative nullhomotopy conjecture via Sullivan-model depth

摘录 / EXCERPT: Let X have dimension at most n, let Y be simply connected, and let $f:X\to Y$ be nullhomotopic with Lipschitz constant at most L. Let ... be a filtration of the indecomposabl...

conjecture | open | MathDB 页面 | 原文线索

V2-031463 | MATHDB-334575

Franks-Misiurewicz conjecture for rotation sets of torus homeomorphisms

摘录 / EXCERPT: Let $f\colon\mathbb{T}^2\to\mathbb{T}^2$ be a homeomorphism homotopic to the identity and let $\tilde f\colon\mathbb{R}^2\to\mathbb{R}^2$ be a lift of f. Suppose that the rotatio...

conjecture | open | MathDB 页面 | 原文线索

V2-031464 | MATHDB-334602

Conjecture on the limiting expected volume per crossing of random alternating link diagrams

摘录 / EXCERPT: Let Vol_n denote the hyperbolic volume of a random alternating link diagram with n crossings, and let $\mathbb{E}[\mathit{Vol}_n]$ be its expectation. Limiting expecte...

conjecture | open | MathDB 页面 | 原文线索

V2-031465 | MATHDB-334609

The torsion-freeness conjecture for lbo homology

摘录 / EXCERPT: Let X be a shelf, and let $H_*^{\mathrm{lbo}}(X)$ denote its lbo homology groups. Torsion-freeness conjecture. There is no torsion in $H_*^{\mathrm{lbo}}(X)$. The conjecture is s...

conjecture | open | MathDB 页面 | 原文线索

V2-031466 | MATHDB-334613

SU(2) representation conjecture for closed 3-manifolds

摘录 / EXCERPT: SU(2) representation conjecture. If $Y\not\cong S^3$, then there is a nontrivial homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-031467 | MATHDB-334635

The reduced LMOV conjecture for the framed unknot

摘录 / EXCERPT: Let $\tau\in\mathbb{Z}$ be a framing and let ... be the reduced open string partition function of $(\mathbb{C}^3,D_{\tau})$. The reduced LMOV conjecture. There exist nonnegative inte...

conjecture | open | MathDB 页面 | 原文线索

V2-031468 | MATHDB-334636

The LMOV conjecture for a framed knot

摘录 / EXCERPT: Let $\mathcal{K}_{\tau}$ be a framed knot with $\tau\in\mathbb{Z}$, and let $f_m(\mathcal{K}_{\tau};q,a)$ be the reduced free-energy coefficients defined from the reduced Chern-Sim...

conjecture | open | MathDB 页面 | 原文线索

V2-031469 | MATHDB-334664

The periodic square peg problem

摘录 / EXCERPT: Periodic square peg conjecture. The union $\sigma_1(\mathbb{R}/L\mathbb{Z})\cup\sigma_2(\mathbb{R}/L\mathbb{Z})$ inscribes a square.

conjecture | open | MathDB 页面 | 原文线索

V2-031470 | MATHDB-334666

The area inequality for four periodic curves

摘录 / EXCERPT: Area inequality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-031471 | MATHDB-334691

The second cohomology generation conjecture for differential operator modules of \(\mathfrak{osp}(2|2)\)

摘录 / EXCERPT: Let $\mathfrak{osp}(2|2)$ be the orthosymplectic Lie superalgebra and let $\mathfrak{D}_{\lambda,\mu}$ denote the corresponding module of differential operators, with...

conjecture | open | MathDB 页面 | 原文线索

V2-031472 | MATHDB-334714

Celoria's almost-concordance conjecture for knots in 3-manifolds

摘录 / EXCERPT: Celoria's almost-concordance conjecture. Unless $Y=S^3$, or $(Y,x)=(S^1\times S^2,[S^1\times\{\operatorname{pt}\}])$, there are infinitely many distinct almost-concordance classes...

conjecture | open | MathDB 页面 | 原文线索

V2-031473 | MATHDB-334743

The connectedness conjecture for twelve unit spheres

摘录 / EXCERPT: Let $\operatorname{Conf}(12)[1]$ be the configuration space of 12 distinct labeled points on the 2-sphere whose pairwise spherical angles are at least $\frac{\pi}{3}$. Connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-031474 | MATHDB-334744

The strong connectedness conjecture for twelve-sphere configurations

摘录 / EXCERPT: Let $\operatorname{Conf}(12)[r]$ denote the configuration space of 12 labeled equal spheres of radius r touching a central unit sphere. Strong connectedness conjecture. The rad...

conjecture | open | MathDB 页面 | 原文线索

V2-031475 | MATHDB-334745

The FCC and HCP bottleneck conjecture

摘录 / EXCERPT: Let $\operatorname{BConf}(12)[1]$ be the reduced configuration space of twelve labeled spheres of radius 1 touching a central unit sphere, and let DOD,...

conjecture | open | MathDB 页面 | 原文线索

V2-031476 | MATHDB-334746

The two connected components conjecture for radii above one

摘录 / EXCERPT: Let R_1 be the upper critical radius defined in the paper, and let $\operatorname{Conf}(12)[r]$ be the configuration space of twelve labeled equal spheres of radius r. Two conn...

conjecture | open | MathDB 页面 | 原文线索

V2-031477 | MATHDB-334747

The non-dodecahedral components conjecture

摘录 / EXCERPT: Let $\operatorname{BConf}(12)[r]$ be the reduced configuration space of twelve equal spheres of radius r touching a central unit sphere, and let DOD denote a dod...

conjecture | open | MathDB 页面 | 原文线索

V2-031478 | MATHDB-334756

Uniqueness conjecture for the four-region sandwich cluster

摘录 / EXCERPT: Let $\mathcal C(1,1,1,1)$ denote the stationary planar clusters with four regions of area 1, and let a sandwich-topology cluster be one with the topology described in the source....

conjecture | open | MathDB 页面 | 原文线索

V2-031479 | MATHDB-334764

Telescopic red-shift conjecture

摘录 / EXCERPT: Let R be a commutative ring spectrum with telescopic complexity n, meaning that the thick subcategory determining its telescopic height is $\mathcal{T}_R=\mathfrak{C}_{n-1}$. H...

conjecture | open | MathDB 页面 | 原文线索

V2-031480 | MATHDB-334770

Nori's Homotopy Conjecture for lifting surjections

摘录 / EXCERPT: Homotopy Conjecture. There exists a surjective map $F:P[T]\twoheadrightarrow J$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-031481 | MATHDB-334785

Nevo--Petersen's flag-complex interpretation conjecture for gamma-vectors

摘录 / EXCERPT: Let Δ be a flag homology sphere. Write its γ-vector as $\gamma^\Delta=(1,\gamma_1,\gamma_2,\ldots)$. A simplicial complex is flag when its faces are exactly the cliqu...

conjecture | open | MathDB 页面 | 原文线索

V2-031482 | MATHDB-334789

Spectral sequence conjecture for symplectic instanton homology

摘录 / EXCERPT: Soit (Y,c) une 3-variété munie d'une classe $c\in H_1(Y,\mathbb{Z}/ 2 \mathbb{Z})$. Pour un scindement de Heegaard, notons $L(H_0,c_0)$ et $L(H_1,c_1)$ les deux Lagrangiennes déf...

conjecture | open | MathDB 页面 | 原文线索

V2-031483 | MATHDB-334822

The crossing-number additivity conjecture for knots

摘录 / EXCERPT: Let τ_1 and τ_2 be knots, let $\tau_1#\tau_2$ denote their connected sum, and let $\operatorname{cr}(K)$ denote the crossing number of a knot K. Crossing-number additiv...

conjecture | open | MathDB 页面 | 原文线索

V2-031484 | MATHDB-334823

The satellite-knot crossing-number conjecture

摘录 / EXCERPT: Let K be a satellite knot and let P be its companion. Let $\operatorname{cr}(X)$ denote the crossing number of a knot X. Satellite-knot crossing-number conjecture. The crossi...

conjecture | open | MathDB 页面 | 原文线索

V2-031485 | MATHDB-334824

The composite-knot factor crossing-number conjecture

摘录 / EXCERPT: Let K be a composite knot with factors $P_1,\ldots,P_r$, and let $\operatorname{cr}(X)$ denote the crossing number of a knot X. Composite-knot factor crossing-number conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-031486 | MATHDB-334827

The positive-density weak property PT conjecture for hyperbolic knots

摘录 / EXCERPT: Let a knot have weak property PT when it satisfies the property defined in the paper. Positive-density weak property PT conjecture. There exist $\varepsilon>0$ and $N>0$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-031487 | MATHDB-334837

Buch-Mihalcea's uniqueness conjecture for minimal degrees in quantum products

摘录 / EXCERPT: Let $QH^*(X)=H^*(X)\otimes_{\mathbb{Z}}\mathbb{Z}[q]$ be the quantum cohomology ring, with Schubert classes σ_u indexed by $u\in W$ and minimal-degree set $\delta_P(u,v)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-031488 | MATHDB-334840

May's rational acyclicity conjecture for H-infinity-ring spectra

摘录 / EXCERPT: May's conjecture. Then R is $K(n)$-acyclic for all $n>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-031489 | MATHDB-334857

Homotopy complexity conjecture for arithmetic manifolds

摘录 / EXCERPT: Homotopy complexity conjecture. There are $\alpha>0$ and $\delta>0$ such that every arithmetic X-manifold M is homotopy equivalent to a $(\delta,\alpha\operatorname{vol}(M))$-s...

conjecture | open | MathDB 页面 | 原文线索

V2-031490 | MATHDB-334875

The plactic monoid cohomology exterior-algebra conjecture

摘录 / EXCERPT: Let A be a finite ordered set, let $\Bbbk$ be a commutative unital ring, and let $\mathbf{Pl}_A$ denote the plactic monoid on A. Theorem gives an injective algebra map from the...

conjecture | open | MathDB 页面 | 原文线索

V2-031491 | MATHDB-334886

Non-vanishing stable instability conjecture for complex projective spaces

摘录 / EXCERPT: Let k be a positive integer, and let $\operatorname{Conf}^n\mathbb{C}P^k$ denote the configuration space of n ordered distinct points in complex projective k-space. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031492 | MATHDB-334896

The eight-vector regular octagon conjecture for the knot 5_1

摘录 / EXCERPT: Let a collection of 8 vectors in $\mathbb{R}^2$ project to the sides of a regular 8-gon. Reordering the vectors means arranging them tip-to-tail to obtain a polygonal knot projecti...

conjecture | open | MathDB 页面 | 原文线索

V2-031493 | MATHDB-334905

Andersen-Kashaev's equivalence conjecture for the new Teichmüller TQFT model

摘录 / EXCERPT: Let X be a levelled shaped triangulated oriented pseudo 3-manifold, and let $Z_{\hbar}^{\mathrm{new}}(X)$ be the partition function defined from the Boltzmann weights in the prop...

conjecture | open | MathDB 页面 | 原文线索

V2-031494 | MATHDB-334910

Hajós' 4-coloring conjecture for graphs without a topological K_5

摘录 / EXCERPT: Hajós' conjecture. Every graph containing no TK_5 is 4-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-031495 | MATHDB-334911

Fox-Lee-Sudakov lower bound for topological cliques

摘录 / EXCERPT: Fox-Lee-Sudakov conjecture. There is a constant $c>0$ such that every graph G with $\chi(G)=k$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-031496 | MATHDB-334916

Baum-Dąbrowski-Hajac conjecture on equivariant maps from joins

摘录 / EXCERPT: Let X be a compact Hausdorff space with a continuous free action of a nontrivial compact Hausdorff group G. Let $X*G$ denote the join equipped with the diagonal action of G....

conjecture | open | MathDB 页面 | 原文线索

V2-031497 | MATHDB-334921

Topological amenability conjecture on connected components

摘录 / EXCERPT: Topological amenability conjecture. If G is amenable, then

conjecture | open | MathDB 页面 | 原文线索

V2-031498 | MATHDB-334922

Definable topological amenability conjecture with a definable neighborhood basis

摘录 / EXCERPT: Definable topological amenability conjecture. If G is definably topologically amenable, then

conjecture | open | MathDB 页面 | 原文线索

V2-031499 | MATHDB-334956

The Rigidity Conjecture for topological classes of dynamical systems

摘录 / EXCERPT: Rigidity Conjecture. The topological class is a finite-codimension manifold which is stratified by probabilistic rigidity classes.

conjecture | open | MathDB 页面 | 原文线索

V2-031500 | MATHDB-334977

The non-trivial-knot conjecture for instanton knot homology

摘录 / EXCERPT: Let K be a knot in S^3, and write $\operatorname{KHI}(K)$ for its instanton knot homology. The instanton knot homology conjecture. If K is a non-trivial knot in S^3, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-031501 | MATHDB-334980

The quantum-cohomology conjecture for the invariant Floer algebra

摘录 / EXCERPT: Let $V_{g,d}^{N}=\ker(\epsilon-1)$ be the invariant subalgebra of the relevant Floer algebra, and let $QH^*({\mathcal N}_{N,d})$ be the quantum cohomology ring of the moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-031502 | MATHDB-334984

Baldwin-Levine's mutation-invariance conjecture for delta-graded knot Floer homology

摘录 / EXCERPT: Let L and L' be a mutant pair of links. Write ${\rm {\widehat{HFK}}}_{\delta}(L)$ for the knot Floer homology obtained by collapsing the bigrading along diagonals $m-a=\delta$....

conjecture | open | MathDB 页面 | 原文线索

V2-031503 | MATHDB-334985

The positive-mutation conjecture for bigraded knot Floer homology

摘录 / EXCERPT: Let L and L' be mutant links, and let T be a tangle satisfying the hypotheses of the paper's mutation-invariance theorem for delta-graded knot Floer homology. Positive-mutati...

conjecture | open | MathDB 页面 | 原文线索

V2-031504 | MATHDB-335001

Quasi-isomorphism conjecture for the surgery chain map

摘录 / EXCERPT: Surgery chain-map conjecture. The chain-map Ψ_1 induces an isomorphism on homology. The map is intended to be analyzed using the action/energy filtration, where its triangular...

conjecture | open | MathDB 页面 | 原文线索

V2-031505 | MATHDB-335019

Arnol'd's fixed-point conjecture for Hamiltonian diffeomorphisms

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold of dimension $2n$, and let ϕ be a Hamiltonian diffeomorphism whose fixed points are all non-degenerate. Arnol'd's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-031506 | MATHDB-335063

The homotopy-orbit conjecture for the framed bordism bicategory

摘录 / EXCERPT: Let $SO(2)$ act on the symmetric monoidal bicategory $\mathbb{F}_{cfd}$ of framed two-dimensional bordisms with fully dualizable objects. Write $(\mathbb{F}_{cfd})_{SO(2)}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-031507 | MATHDB-335083

Prasad-Saeki braid conjecture for torsion in Khovanov homology

摘录 / EXCERPT: Prasad-Saeki braid conjecture. The Khovanov homology of a closed 3-braid can have only $\mathbb{Z}_2$-torsion. The Khovanov homology of a closed 4-braid cannot have odd torsion...

conjecture | open | MathDB 页面 | 原文线索

V2-031508 | MATHDB-335084

Conjecture on 2-cabling torus knots and powers of two torsion

摘录 / EXCERPT: 2-cabling torsion conjecture. The flat 2-cabling of $T(2,2s+1)$ has $\mathbb{Z}_{2^{s'}}$-torsion for every $0<s'\leq s$ in both stated families of bidegrees.

conjecture | open | MathDB 页面 | 原文线索

V2-031509 | MATHDB-335095

Generic transitivity for pairs of volume-preserving diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold and let $r\in\mathbb N\cup\{\infty\}$. An iterated function system consists of a finite family of homeomorphisms of M; it is transitive if the semig...

conjecture | open | MathDB 页面 | 原文线索

V2-031510 | MATHDB-335107

The Poisson-uncertainty conjecture for the Poisson-bracket function

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold. For each ball capacity c, let $\operatorname{pb}(c)$ denote the Poisson-bracket invariant associated with the corresponding con...

conjecture | open | MathDB 页面 | 原文线索

V2-031511 | MATHDB-335108

The topology conjecture for the small-capacity Poisson-bracket limit

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold, and let $\operatorname{pb}(c)$ be the Poisson-bracket function associated with ball capacity c. The topology conjecture. Assumi...

conjecture | open | MathDB 页面 | 原文线索

V2-031512 | MATHDB-335115

The branching conjecture for partition complexes

摘录 / EXCERPT: Let $\Lcal_n$ be the partition complex, let $\Pcal_n^\diamond$ denote the unreduced suspension of the partition complex associated to the partition lattice, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031513 | MATHDB-335187

Whitney smoothing conjecture for derived manifolds

摘录 / EXCERPT: Let $l\geq k>0$, fix an o-minimal structure, and let a derived C^k-manifold be a manifold in the corresponding derived differentiable setting. Whitney smoothing conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-031514 | MATHDB-335210

Grothendieck's conjecture on models of weak infinity-groupoids

摘录 / EXCERPT: A globular weak ∞-groupoid model is any of the unspecified models of globular weak ∞-groupoids considered in the conjecture. Let the category of such models be equipp...

conjecture | open | MathDB 页面 | 原文线索

V2-031515 | MATHDB-335227

The adiabatic-limit correspondence for gauged Witten invariants

摘录 / EXCERPT: Let $\bar{X}_W$ be the symplectic reduction in the geometric phase, let $H_{\rm CR}^*(\bar{X}_W;\Lambda)$ denote its Chen-Ruan cohomology, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031516 | MATHDB-335243

Torsion-point conjecture for cohomology jump loci of quasi-compact Kähler manifolds

摘录 / EXCERPT: Torsion-point conjecture. Each irreducible component of $\Sigma^i_k(X)$ contains a torsion point.

conjecture | open | MathDB 页面 | 原文线索

V2-031517 | MATHDB-335248

The trivial h-cobordism conjecture for torsion-free fundamental groups

摘录 / EXCERPT: Let M be a closed, connected, smooth manifold of dimension at least 5 with torsion-free fundamental group. An h-cobordism over M is a smooth cobordism over M for which bo...

conjecture | open | MathDB 页面 | 原文线索

V2-031518 | MATHDB-335249

The Whitehead vanishing conjecture for torsion-free groups

摘录 / EXCERPT: For a group G, the Whitehead group $\operatorname{Wh}(G)$ is the quotient of $K_1(\mathbb{Z}[G])$ by the subgroup represented by the units $\pm g$ for $g\in G$. The Whitehead van...

conjecture | open | MathDB 页面 | 原文线索

V2-031519 | MATHDB-335288

Kohno's generalized equivalence for framed braid group representations

摘录 / EXCERPT: Generalized Kohno equivalence. There is an open dense subset $U\subset\mathbb{C}\times\mathbb{C}^*$ such that, whenever $(\lambda,\kappa)\in U$, the monodromy representation...

conjecture | open | MathDB 页面 | 原文线索

V2-031520 | MATHDB-335292

Topological Akbulut-Kirby conjecture on concordance from 0-surgeries

摘录 / EXCERPT: Let K and K' be knots, and let $S^3_0(K)$ and $S^3_0(K')$ denote their 0-surgeries. Reversing the orientation of either knot means replacing it by the same knot with the opposi...

conjecture | open | MathDB 页面 | 原文线索

V2-031521 | MATHDB-335294

The Hochschild-Floer correspondence conjecture for mapping classes

摘录 / EXCERPT: Let Σ_g be a genus-g surface with one boundary component $\partial\Sigma_g=S^1=U_1$, and let $MCG_0(\Sigma_g,\partial\Sigma_g=S^1=U_1)$ denote the mapping classes fixing t...

conjecture | open | MathDB 页面 | 原文线索

V2-031522 | MATHDB-335314

Smoothing conjecture for closed cubical manifolds in codimension two

摘录 / EXCERPT: Smoothing conjecture. Every such N is smoothable. More precisely, N admits a transverse field of 2-planes and therefore there is an arbitrarily small topological isotopy carryi...

conjecture | open | MathDB 页面 | 原文线索

V2-031523 | MATHDB-335432

Buryak's DR/DZ equivalence conjecture

摘录 / EXCERPT: Let a semisimple CohFT be given, and let its associated Dubrovin-Zhang (DZ) and double ramification (DR) hierarchies be the corresponding tau-symmetric integrable hierarchies. Bury...

conjecture | open | MathDB 页面 | 原文线索

V2-031524 | MATHDB-335439

The integrality conjecture for tropical generalized Donaldson-Thomas invariants

摘录 / EXCERPT: Let X satisfy the generic condition assumed in the construction, let Γ be the relevant relative homology lattice, let W'^{trop}_{γ} be the wall locus for a class...

conjecture | open | MathDB 页面 | 原文线索

V2-031525 | MATHDB-335451

Nonexistence of an E-infinity structure on p-local Brown-Peterson spectra at odd primes

摘录 / EXCERPT: Nonexistence conjecture. The spectrum $BP_{(p)}$ does not admit the structure of an E_∞-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-031526 | MATHDB-335461

Dunfield-Gukov-Rasmussen spectral sequence conjecture

摘录 / EXCERPT: Dunfield-Gukov-Rasmussen conjecture. There is a spectral sequence from HOMFLY-PT homology to knot Floer homology.

conjecture | open | MathDB 页面 | 原文线索

V2-031527 | MATHDB-335462

Cube-compatible identification conjecture for HOMFLY-PT and knot Floer homologies

摘录 / EXCERPT: Cube-compatible identification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031528 | MATHDB-335472

Kawasaki's capacity equality conjecture for stably non-displaceable subsets

摘录 / EXCERPT: Let X be a stably non-displaceable compact subset of a closed symplectic manifold (M,ω). Kawasaki's conjecture. For any $R>0$ and $\ell\in\mathbb{Z}$, one has ... This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-031529 | MATHDB-335481

The LRS metric conjecture for minimal Cantor systems

摘录 / EXCERPT: Let (C,T) be a minimal dynamical system on a Cantor set C, and let ρ be a metric on C equivalent to its original metric. The (LRS) property means that T is locally rad...

conjecture | open | MathDB 页面 | 原文线索

V2-031530 | MATHDB-335542

Zero-mode criterion for fusion of boundary defects

摘录 / EXCERPT: Let $\mathcal{A}_i$ and $\mathcal{A}_j$ be Lagrangian algebras, and let τ_{ij} denote a boundary defect between them. Let t_i be a simple object of the boundary fusion categ...

conjecture | open | MathDB 页面 | 原文线索

V2-031531 | MATHDB-335568

Cohomological tautological conjecture for moduli of hyperkähler manifolds

摘录 / EXCERPT: Let ${\rm NL}^\ast_{\rm hom}(\mathcal{F}_h)$ be the image of the Noether-Lefschetz ring in $H^\ast(\mathcal{F}_h,\mathbb{Q})$, and let $\mathrm{R}^\ast_{\rm hom}(\mathcal{F}_h)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-031532 | MATHDB-335600

Buchsbaumness conjecture for Stanley-Reisner rings with non-isolated singularities

摘录 / EXCERPT: Buchsbaumness conjecture. If the singularities of Δ are not homologically isolated, then

conjecture | open | MathDB 页面 | 原文线索

V2-031533 | MATHDB-335627

Projective-dimension characterization of split links

摘录 / EXCERPT: Split-link characterization. B represents an n-split link if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-031534 | MATHDB-335645

Generic absence of sky catastrophes for homotopies of Reeb vector fields

摘录 / EXCERPT: Let M be a compact contact manifold, and let $\{X_t\}$, $t\in[0,1]$, be a smooth homotopy through Reeb vector fields. A sky catastrophe is the phenomenon excluded by the stated g...

conjecture | open | MathDB 页面 | 原文线索

V2-031535 | MATHDB-335695

Minimum-genus Bennequin surface conjecture for transverse links

摘录 / EXCERPT: Minimum-genus Bennequin surface conjecture. Every transverse link $\mathcal{T}$ in (S^3,ξ_{std}) is represented by a braid word w whose Bennequin surface F_w contains...

conjecture | open | MathDB 页面 | 原文线索

V2-031536 | MATHDB-335696

Minimum-genus alpha-Bennequin surface conjecture

摘录 / EXCERPT: Minimum-genus alpha-Bennequin surface conjecture. If $\delta(\mathcal{T},\alpha)\geq 0$, then $\mathcal{T}$ bounds a minimum-genus α-Bennequin surface with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-031537 | MATHDB-335771

Champanerkar-Kofman-Purcell volume-determinant conjecture for alternating knots

摘录 / EXCERPT: Let K be a hyperbolic alternating knot. Here, $\operatorname{vol}(K)$ denotes the hyperbolic volume of the complement $S^3\backslash K$, and $\det(K)$ denotes the determinant of...

conjecture | open | MathDB 页面 | 原文线索

V2-031538 | MATHDB-335829

Nonexistence conjecture for surjections between braid groups of higher-genus surfaces

摘录 / EXCERPT: Let $S=S_{g,p}$ be a surface with $p\ge 0$ and $g>1$. For positive integers m and n with $m\neq n$, consider homomorphisms between the braid groups $B_n(S)$ and $B_m(S)$. Nonex...

conjecture | open | MathDB 页面 | 原文线索

V2-031539 | MATHDB-335834

Higher-cohomology vanishing and positivity for quiver Kostka-Shoji polynomials

摘录 / EXCERPT: Let Q be the quiver under consideration, let $(\mathbf{i},\mathbf{a})$ be the associated data, and let $\mu(\bullet)$ be a weight whose concatenated weight $\mu^{(i)}$ is weakly...

conjecture | open | MathDB 页面 | 原文线索

V2-031540 | MATHDB-335857

Sharpness conjecture for the complexity of non-orientable Seifert fibre spaces

摘录 / EXCERPT: Let $M=\left\{b;\left(\epsilon,g,(t,k)\right);\left(\ \mid \ \right);\left((p_1,q_1),\ldots,(p_r,q_r)\right)\right\}$ be a non-orientable closed irreducible and $\mathbb P^2$-irre...

conjecture | open | MathDB 页面 | 原文线索

V2-031541 | MATHDB-335870

Donaldson's hyperkähler deformation conjecture for compact hypersymplectic 4-manifolds

摘录 / EXCERPT: Donaldson's conjecture. There exists a deformation of $\underline{\omega}$ through cohomologous hypersymplectic structures to a hyperkähler triple. This conjecture asserts that, up...

conjecture | open | MathDB 页面 | 原文线索

V2-031542 | MATHDB-335895

The Poisson-de Rham homology conjecture for affine symplectic resolutions

摘录 / EXCERPT: Let $\rho:\widetilde X\to X$ be a symplectic resolution, with X affine, and let $\operatorname{\mathsf{HP}}_i^{DR}(X)$ denote the Poisson-de Rham homology of X. The affine symp...

conjecture | open | MathDB 页面 | 原文线索

V2-031543 | MATHDB-335943

Factorization homology with adjoints for solidly framed stratified manifolds

摘录 / EXCERPT: Let $n\geq 0$. Write $\operatorname{{\sf Cat}}_n^{\operatorname{{\sf adj}}}$ for the full ∞-subcategory of (∞,n)-categories in which every k-morphism, for $0<k<n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-031544 | MATHDB-335947

The twisted holography Koszul duality conjecture

摘录 / EXCERPT: Consider a D-brane in type IIA or IIB string theory, a brane in the topological string, or a brane in M-theory, viewed as a defect in the string theory or supergravity theory....

conjecture | open | MathDB 页面 | 原文线索

V2-031545 | MATHDB-335952

The quantum-field-theoretic action on categorified Donaldson-Thomas invariants

摘录 / EXCERPT: Let $\mathcal M_{k,d,N,\alpha}$ be the relevant moduli space of quasi-maps, let $\mathcal P$ be the perverse sheaf lifting the Behrend-function data, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031546 | MATHDB-335953

The ADE instanton quantum field theory action conjecture

摘录 / EXCERPT: Let X_M be the resolution of the A_{M-1} singularity, and let $\mathscr O_{c,\lambda_1,\dots,\lambda_{M-1}}(X_M)$ be its family of deformation-quantizations. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031547 | MATHDB-335988

Generalized strong DR/DZ equivalence conjecture

摘录 / EXCERPT: Let F^{DR} denote the DR potential and let F^{red} denote the reduced potential associated to a cohomological field theory. Generalized strong DR/DZ equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-031548 | MATHDB-336053

Bernhard-Jablan unknotting conjecture

摘录 / EXCERPT: Bernhard-Jablan conjecture. Every knot K possesses a minimal-crossing-number diagram D and a crossing in D such that changing the crossing results in a diagram D' for a kno...

conjecture | open | MathDB 页面 | 原文线索

V2-031549 | MATHDB-336054

Dubrovin's Stokes matrix conjecture for quantum connections

摘录 / EXCERPT: Let X be a Fano manifold, let $\mathcal{H}_X=H^\bullet(X)\otimes\mathcal{O}_{\mathbb{C}_z}$, and let ... be its quantum connection at quantum parameter $\tau=0$. Assume that the qu...

conjecture | open | MathDB 页面 | 原文线索

V2-031550 | MATHDB-336059

The TPM elementary-excitation and connectivity conjecture

摘录 / EXCERPT: Let a $2D$ topological phase of matter (TPM) be an equivalence class of connected stable topological Hamiltonian schemas, and let its topological order be the unitary modular categ...

conjecture | open | MathDB 页面 | 原文线索

V2-031551 | MATHDB-336060

The UTMF realization conjecture for Drinfeld centers

摘录 / EXCERPT: Let V be a unitary topological modular functor (UTMF), and let $\mathcal{B}$ be its associated unitary modular category. Let $\mathcal{C}$ be a unitary fusion category. UTMF real...

conjecture | open | MathDB 页面 | 原文线索

V2-031552 | MATHDB-336061

The Levin-Wen ground-state error-correction conjecture

摘录 / EXCERPT: Let a Levin-Wen (LW) model be the Hamiltonian model determined by a unitary fusion category, and consider its ground-state subspace. Levin-Wen error-correction conjecture. The grou...

conjecture | open | MathDB 页面 | 原文线索

V2-031553 | MATHDB-336062

The absence of CLP realizations for chiral topological phases

摘录 / EXCERPT: A chiral topological phase of matter (TPM) is a TPM whose topological order is a chiral unitary modular category. A CLP Hamiltonian is a commuting local-projector Hamiltonian. Chir...

conjecture | open | MathDB 页面 | 原文线索

V2-031554 | MATHDB-336065

The localization conjecture for simple objects

摘录 / EXCERPT: Let $\mathcal{C}$ be a unitary braided fusion category, and let $X\in\mathcal{C}$ be simple. Localization conjecture. The object X can be localized if and only if ... The source sa...

conjecture | open | MathDB 页面 | 原文线索

V2-031555 | MATHDB-336067

The symmetry-defect braid-representation conjecture

摘录 / EXCERPT: Let G act on a braided fusion category, let X_g be a defect fixed by the G-action, and let $\rho_{X_g,n}$ be the associated projective representation of the braid group...

conjecture | open | MathDB 页面 | 原文线索

V2-031556 | MATHDB-336069

The homotopy-invariance conjecture for unitary (3+1)-TQFTs

摘录 / EXCERPT: Let Z be the partition function of a unitary $(3+1)$-dimensional topological quantum field theory. Homotopy-invariance conjecture. The partition function of any unitary $(3+1)$-T...

conjecture | open | MathDB 页面 | 原文线索

V2-031557 | MATHDB-336093

Mirror interpretation of acceleration functors

摘录 / EXCERPT: Let $f:(\mathbb{C}^*)^n\to\mathbb{C}$ define $H=f^{-1}(0)$, let $\mathcal{F}^\circ((\mathbb{C}^*)^n,f)$ be the Fukaya-Seidel category, and let Z_0 be the distinguished toric divi...

conjecture | open | MathDB 页面 | 原文线索

V2-031558 | MATHDB-336095

Generation of the wrapped Fukaya category of the pair of pants

摘录 / EXCERPT: Let $\Pi_n\subset(\mathbb{C}^*)^{n+1}$ be the n-dimensional pair of pants, and let $L_{\{i\}}$, for $i=0,1,\dots,n+1$, be the specified objects of its wrapped Fukaya category. Pa...

conjecture | open | MathDB 页面 | 原文线索

V2-031559 | MATHDB-336127

Codimension-1 Grassmannian embedding conjecture

摘录 / EXCERPT: Codimension-1 Grassmannian embedding conjecture. The only codimension-1 smooth embedding

conjecture | open | MathDB 页面 | 原文线索

V2-031560 | MATHDB-336128

Large-9 limit for Wilson loops around simple closed curves on compact surfaces

摘录 / EXCERPT: Let Σ be a surface of area $\operatorname{area}(\Sigma)$, let $U\subset\Sigma$ be a topological disk, and let C be a simple closed curve in U. Assign area a to the int...

conjecture | open | MathDB 页面 | 原文线索

V2-031561 | MATHDB-336129

Large-9 limit for winding Wilson loops on compact surfaces

摘录 / EXCERPT: Let Σ, U, C, a, and $c=\operatorname{area}(\Sigma)-a$ be as in the preceding conjecture, with C a simple closed curve in the topological disk U. For...

conjecture | open | MathDB 页面 | 原文线索

V2-031562 | MATHDB-336134

The coloring conjecture for knot singularities in complexes on the 3-sphere

摘录 / EXCERPT: Coloring conjecture. If G is an unknot, then X is 5-colorable; if G is a knot, then there exists $X'\approx X$ such that X' is 6-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-031563 | MATHDB-336199

Non-simple connectivity of the space of light bulbs

摘录 / EXCERPT: Let the space of light bulbs be the space of properly embedded discs in $S^2\times D^2$ that agree with $x_0\times D^2$ near their boundary, with the appropriate topology. Non-simp...

conjecture | open | MathDB 页面 | 原文线索

V2-031564 | MATHDB-336215

Baker-Moore conjecture on tangle decompositions of L-space knots

摘录 / EXCERPT: Let K be a knot in S^3. An essential tangle decomposition is an essential n-string tangle decomposition for some integer n, meaning that there is a 2-sphere intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-031565 | MATHDB-336226

Zvonkine's topological-recursion conjecture for orbifold r-spin Hurwitz numbers

摘录 / EXCERPT: Let $g\geq 0$ and $n\geq 1$. For positive integers $\mu_1,\ldots,\mu_n$, let $h_{g;\mu}^{\circ,q,r}$ be the connected q-orbifold r-spin Hurwitz number, and define the formal sy...

conjecture | open | MathDB 页面 | 原文线索

V2-031566 | MATHDB-336256

The Azumaya-positive knot conjecture for L-space knots

摘录 / EXCERPT: Azumaya-positive knot conjecture. K is not an L-space knot.

conjecture | open | MathDB 页面 | 原文线索

V2-031567 | MATHDB-336320

Clark-Elhamdadi-Saito-Yeatman conjecture on finite quandle coloring invariants

摘录 / EXCERPT: For a knot or link L, let $\mathcal{K}(L)$ denote its fundamental quandle, and let $\chi_L(X):=|\operatorname{Hom}_{\operatorname{quandle}}(\mathcal{K}(L),X)|$ be the quandle cou...

conjecture | open | MathDB 页面 | 原文线索

V2-031568 | MATHDB-336333

Kalai's manifold g-conjecture

摘录 / EXCERPT: Let Δ be an orientable simplicial $\Bbbk$-homology d-manifold without boundary, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031569 | MATHDB-336334

Stress-algebra weak Lefschetz conjecture for homology manifolds

摘录 / EXCERPT: Let Δ be an orientable simplicial $\mathbb{R}$-homology d-manifold without boundary, with $H_1(\Delta;\mathbb{Z}/2\mathbb{Z})=0$ whenever $d\geq2$. Let Υ be a sub...

conjecture | open | MathDB 页面 | 原文线索

V2-031570 | MATHDB-336335

Stress-algebra dimension formula for homology manifolds

摘录 / EXCERPT: Let (Δ,ν) be a PL realization of an orientable simplicial $\mathbb{R}$-homology d-manifold without boundary in $\mathbb{R}^d$, and let $\Psi(\Delta,\nu)$ be its stress a...

conjecture | open | MathDB 页面 | 原文线索

V2-031571 | MATHDB-336340

The infinitesimal rigidity conjecture for centrally symmetric homology spheres

摘录 / EXCERPT: Let Δ be a centrally symmetric simplicial complex of dimension $d-1\geq 3$. Assume that Δ is a homology sphere, a connected homology manifold, or a normal pseudomanif...

conjecture | open | MathDB 页面 | 原文线索

V2-031572 | MATHDB-336354

Cabling conjecture for the primitive bridge spectrum of pretzel knots

摘录 / EXCERPT: Let $K_j=K_j(p_1,p_2,\ldots,p_j)$ be a pretzel knot with stair-step primitive bridge spectrum. Let $T_{m,n}$ be an (m,n)-torus knot, and set ... If...

conjecture | open | MathDB 页面 | 原文线索

V2-031573 | MATHDB-336355

Montesinos-versus-pretzel characterization of m-stair-step bridge spectra

摘录 / EXCERPT: Let M_0 be a Montesinos knot, and let $K={\mathsf {cable}}(T_{m,n},M_0)$ be an (m,n)-cable of M_0. The bridge spectrum of K is the Montesinos-pretzel bridge-spectrum conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-031574 | MATHDB-336356

Cabling conjecture for bridge spectra of knots

摘录 / EXCERPT: Let J be a knot with bridge spectrum ... and let $K={\mathsf {cable}}(T_{m,n},J)$ be a cable of J. The bridge-spectrum cabling conjecture. ... The conjecture expresses the observed...

conjecture | open | MathDB 页面 | 原文线索

V2-031575 | MATHDB-336380

Stability of metric Vietoris-Rips persistence for infinite metric spaces

摘录 / EXCERPT: Let X and Y be metric spaces, let $i\geq 0$, and let d_b denote the bottleneck distance between persistent homology modules. Write $\mathrm{PH}_i(\mathrm{VR}^m(X;-))$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031576 | MATHDB-336382

C. Woodward's spherical quantum $6j$ asymptotic conjecture

摘录 / EXCERPT: Let T be a spherical tetrahedron whose edge lengths are $l_{ij}=\pi\alpha_{ij}$ for $1\leq i<j\leq 4$, with associated dihedral angles θ_{ij}, volume V, and spherical Gra...

conjecture | open | MathDB 页面 | 原文线索

V2-031577 | MATHDB-336383

Symmetry of asymptotics for quantum $6j$ symbols

摘录 / EXCERPT: Let T be a hyperbolic tetrahedron with dihedral angles $\theta_a,\theta_b,\theta_c,\theta_d,\theta_e,\theta_f$ at edges $a,\ldots,f$. Let $a_r,b_r,\ldots,f_r$ be non-negative hal...

conjecture | open | MathDB 页面 | 原文线索

V2-031578 | MATHDB-336384

Jordan triple-system conjecture for the second Veronese power

摘录 / EXCERPT: Jordan triple-system conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-031579 | MATHDB-336422

Conjecture on the $K(2)$-local homotopy groups of spectra in $\mathbb{Z}$

摘录 / EXCERPT: Let $\mathbb{Z}$ be the class of 2-local finite spectra considered in the paper, and let $Z\in\mathbb{Z}$. Write $L_{K(2)}Z$ for the $K(2)$-localization of Z, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031580 | MATHDB-336431

Invariance of involutive bordered Floer modules

摘录 / EXCERPT: Let Y be a bordered 3-manifold with bordered Heegaard diagram $\mathcal{H}$. The involutive type D structure $\widehat{\mathit{CFDI}}(\mathcal{H})$ and involutive A_∞-...

conjecture | open | MathDB 页面 | 原文线索

V2-031581 | MATHDB-336444

The author's stable inequivalence conjecture for torus-knot complexes

摘录 / EXCERPT: Let $T(a,b)$ denote the (a,b)-torus knot, let $T(k,p)\mathbin{\#}T(p,p+1)$ denote their connected sum, and let $\operatorname{CFK}^\infty(K)$ denote the knot Floer complex of a...

conjecture | open | MathDB 页面 | 原文线索

V2-031582 | MATHDB-336475

Generation conjecture for the scaling-limit observable algebra

摘录 / EXCERPT: Let $\mathcal{A}$ be the scaling-limit observable algebra, let $\overline{\mathcal{A}}$ denote its closure under the relevant topology, and let $Y(a,f)$ be the smeared field associ...

conjecture | open | MathDB 页面 | 原文线索

V2-031583 | MATHDB-336496

The topological 5-dimensional s-cobordism conjecture

摘录 / EXCERPT: s-cobordism conjecture. There is a homeomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-031584 | MATHDB-336497

The disc embedding conjecture

摘录 / EXCERPT: Disc embedding conjecture. The circles $f_i(S^1)$ bound disjointly embedded discs in M with transverse spheres, inducing the same framing on $f_i(S^1)$ as the f_i.

conjecture | open | MathDB 页面 | 原文线索

V2-031585 | MATHDB-336541

Ooguri-Vafa integrality conjecture for the open topological string

摘录 / EXCERPT: Ooguri-Vafa integrality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-031586 | MATHDB-336542

Ooguri-Vafa integrality conjecture for the framed unknot

摘录 / EXCERPT: Let $\tau\in\mathbb{Z}$ be a framing, let $Z^{(\mathbb{C}^3,\mathcal{D}_\tau)}(\mathbf{x};q)$ be the framed open-string partition function, and let $f_n^\tau(q)$ be the coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-031587 | MATHDB-336543

Refined Rogers-Ramanujan product conjecture

摘录 / EXCERPT: Let q and x be formal variables, and for each fixed integer $m\geq1$ let $n_{m,k}$ be positive integers indexed by finitely many $k\in I_m$. Refined Rogers-Ramanujan conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-031588 | MATHDB-336550

Factorization conjecture for geometric Langlands Hochschild cochains

摘录 / EXCERPT: Let $\mathbb C$ be the configuration-space curve, let $\operatorname{Ran}(\mathbb C)$ be its Ran space, and for $x\in\operatorname{Ran}(\mathbb C)$ let...

conjecture | open | MathDB 页面 | 原文线索

V2-031589 | MATHDB-336587

The Khovanov-to-annular Khovanov spectral sequence conjecture for 2-periodic links

摘录 / EXCERPT: Let $\tilde L$ be a 2-periodic link in S^3 with quotient link L. Write Kh for Khovanov homology and AKh for annular Khovanov homology, both over $\mathbb{F}_2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031590 | MATHDB-336590

The New Doomsday Conjecture for Sq^0-families

摘录 / EXCERPT: New Doomsday Conjecture. For any Sq^0-family in $\operatorname{Ext}_{A_*}(\mathbb{F}_2,\mathbb{F}_2)$, only finitely many elements survive to the E_∞-page of the classical...

conjecture | open | MathDB 页面 | 原文线索

V2-031591 | MATHDB-336595

The stable-homotopy lifting conjecture for the e-invariant

摘录 / EXCERPT: Let e denote the map constructed in the paper from relative complex K-theory to the relevant stable-homotopy target. A map is weakly homotopic to another map when it represents...

conjecture | open | MathDB 页面 | 原文线索

V2-031592 | MATHDB-336596

The finer index theorem for the APS e-invariant

摘录 / EXCERPT: Let ... , let ... be the APS e-invariant map. The notation $[E,F_{t,}]$ and $[B,F_{t,}]$ denotes homotopy classes of maps into the indicated target. Finer APS index conjecture. The f...

conjecture | open | MathDB 页面 | 原文线索

V2-031593 | MATHDB-336599

Factorization conjecture for colored Jones polynomials of links

摘录 / EXCERPT: Let $\mathcal{L}$ be an m-component link, and suppose there are sub-links $\mathcal{L}_{1}$ and $\mathcal{L}_{2}$ with i and $(m-i)$ components, respectively. For colorings...

conjecture | open | MathDB 页面 | 原文线索

V2-031594 | MATHDB-336608

Generalized symplectic instanton Floer invariant conjecture for 3-manifolds and G-bundles

摘录 / EXCERPT: Let G be a group as in the paper's discussion of extended moduli spaces, let M be a 3-manifold, and let $\mathcal E$ be a G-bundle over M. Generalized symplectic instanto...

conjecture | open | MathDB 页面 | 原文线索

V2-031595 | MATHDB-336609

Daemi-Fukaya extension conjecture for cobordisms with empty boundary

摘录 / EXCERPT: Let $(M,\mathcal E)$ be a cobordism $(\Sigma_1,\mathcal F_1)\to(\Sigma_2,\mathcal F_2)$, where either boundary component may be empty, and let $\mathfrak I_{(M,\mathcal E)}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-031596 | MATHDB-336610

Daemi-Fukaya composition conjecture for cobordisms with empty boundary

摘录 / EXCERPT: Let $(M_1,\mathcal E_1)$ and $(M_2,\mathcal E_2)$ be composable 3-dimensional cobordisms as in part (A-3) of the preceding conjecture, and let $(M,\mathcal E)$ be their compositi...

conjecture | open | MathDB 页面 | 原文线索

V2-031597 | MATHDB-336627

Homotopy-colimit endomorphism conjecture for full-twist eigenmaps

摘录 / EXCERPT: Let FT_n be the full-twist complex and let $\alpha:\1\rightarrow\operatorname{FT}_n$ be an eigenmap. Form the directed system ... with transition maps induced by ten...

conjecture | open | MathDB 页面 | 原文线索

V2-031598 | MATHDB-336666

The homological independence conjecture for toric cycles

摘录 / EXCERPT: Homological independence conjecture. The toric cycles Γ_ν constitute a homologically independent family in the homology group $H_n((\mathbb{C}^{\times})^n\setminus V)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031599 | MATHDB-336706

Polynomiality conjecture for the stable homology of IA-groups

摘录 / EXCERPT: Let F_n be a free group of rank n, and let IA_n be the kernel of the canonical epimorphism ${\rm Aut}(F_n)\twoheadrightarrow GL_n(\mathbb{Z})$. For each $d\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-031600 | MATHDB-336711

Mislin's finiteness-obstruction conjecture for finitely dominated H-spaces

摘录 / EXCERPT: Let X be a finitely dominated H-space, meaning a pointed space with a map $X\times X\to X$ whose restriction to $X\vee X$ is homotopic to the folding map. Its finiteness obstruct...

conjecture | solved | MathDB 页面 | 原文线索

V2-031601 | MATHDB-336721

The moonshine anomaly cohomology conjecture

摘录 / EXCERPT: Let $\mathbb M$ denote the Monster group, and let $H^3(\mathbb M,\mathrm U(1))$ be its third cohomology group with coefficients in $\mathrm U(1)$. Moonshine anomaly cohomology conj...

conjecture | open | MathDB 页面 | 原文线索

V2-031602 | MATHDB-336770

Classification conjecture for irreducible 4-manifolds of trisection genus three

摘录 / EXCERPT: Trisection-genus-three classification conjecture. Every irreducible 4-manifold with trisection genus three is either the spin of a lens space, or a Gluck twist on a specific 2-knot...

conjecture | open | MathDB 页面 | 原文线索

V2-031603 | MATHDB-336812

Rank-one freeness conjecture for ordinary cohomology at non-Eisenstein maximal ideals

摘录 / EXCERPT: Let $\mathfrak{m}$ be a non-Eisenstein maximal ideal in ${\bf h}_{\rm ord}(K_{\infty,\infty};\mathcal{O})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031604 | MATHDB-336820

Murakami and Yasuhara's sharpness conjecture for the non-orientable slice genus

摘录 / EXCERPT: Let K be a knot, let $g_4(K)$ denote its smooth orientable slice genus, and let $b3_4(K)$ denote its smooth non-orientable slice genus. A knot is non-slice if $g_4(K)>0$. Murakam...

conjecture | open | MathDB 页面 | 原文线索

V2-031605 | MATHDB-336878

Embedding conjecture for -actions

摘录 / EXCERPT: Let $(X,\mathbb{Z}^k)$ be a dynamical system, and for each subgroup A of $\mathbb{Z}^k$ define ... The quotient group $\mathbb{Z}^k/A$ acts naturally on X_A; when the quotient is...

conjecture | open | MathDB 页面 | 原文线索

V2-031606 | MATHDB-336942

The algebraic chromatic splitting conjecture

摘录 / EXCERPT: Assume p is sufficiently large with respect to n. Let E be the height-n theory in the source, let I_{n-1} be the corresponding invariant ideal, and let L_{n-1} and...

conjecture | open | MathDB 页面 | 原文线索

V2-031607 | MATHDB-336946

The remodeling conjecture for open Gromov-Witten generating functions

摘录 / EXCERPT: Consider the topological A-model on a local toric Calabi-Yau 3-fold $Y_{\bm{l}}$ with a special Lagrangian submanifold L. Let $F_n^{(g)}(\mathsf{x}_1,\ldots,\mathsf{x}_n)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-031608 | MATHDB-336947

Relative topological Hochschild homology conjecture for the Hopf-Galois extension X(n) to X(n+1)

摘录 / EXCERPT: Relative THH conjecture. The relative topological Hochschild homology spectrum should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-031609 | MATHDB-336962

Cohomology conjecture for Calogero-Moser spaces

摘录 / EXCERPT: Let W be a complex reflection group, let $c\in{\boldsymbol{\mathcal C}}$, and let ${\boldsymbol{\mathcal Z}}_c$ be the corresponding Calogero-Moser space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031610 | MATHDB-336983

The TFT datum conjecture for holomorphic factorizations

摘录 / EXCERPT: Let (X,W) be a Landau-Ginzburg model with compact critical set Z_W. Let $\operatorname{HPV}(X,W)$ be the cohomology algebra of the bulk complex...

conjecture | open | MathDB 页面 | 原文线索

V2-031611 | MATHDB-336985

The Antieau-Williams topological period-index conjecture

摘录 / EXCERPT: Antieau-Williams' conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031612 | MATHDB-336986

The Antieau-Williams sharp topological period-index conjecture

摘录 / EXCERPT: Antieau-Williams' conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031613 | MATHDB-337015

The expansion conjecture for finite topological rank

摘录 / EXCERPT: An $c9$-categorical structure is a structure whose theory has, up to isomorphism, a unique countable model. A structure admits a CIR when it has the property called CIR in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-031614 | MATHDB-337027

Strong Broadhurst-Kreimer-Zagier conjecture for linearised double shuffle Lie algebras

摘录 / EXCERPT: Strong Broadhurst-Kreimer-Zagier conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031615 | MATHDB-337036

Dunfield, Friedl and Jackson's twisted Alexander polynomial conjecture for hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot, let $g(K)$ denote its genus, and let $\rho_{\rm holo}:\pi(K)\to\operatorname{SL}(2,\mathbb{C})$ be a lift of the discrete and faithful representation...

conjecture | open | MathDB 页面 | 原文线索

V2-031616 | MATHDB-337106

Baryshnikov-Ghrist monotonicity conjecture for unimodal p-category

摘录 / EXCERPT: Let X be a topological space and let $f:X\to[0,\infty)$ be a continuous function. For $p\in(0,\infty]$, let $\operatorname{{\bf ucat}}^p(f)$ denote the minimum number of unimodal...

conjecture | open | MathDB 页面 | 原文线索

V2-031617 | MATHDB-337118

Height and min-height bounds for connected sums of knots

摘录 / EXCERPT: Height and min-height bounds conjecture. The connected sum should satisfy both inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-031618 | MATHDB-337119

Strict inequality between min-height and height of a knot

摘录 / EXCERPT: Let $ht(K)$ and $ht_{\min}(K)$ denote the height and min-height of a knot K, respectively. Min-height gap conjecture. There exists a knot K such that ... The authors note that th...

conjecture | open | MathDB 页面 | 原文线索

V2-031619 | MATHDB-337140

The indicator-dimension conjecture for real and discrete polyhedral groups

摘录 / EXCERPT: Indicator-dimension conjecture. The indicator-dimension of Q equals the rank of Q as a free module over the field $\mathbb{R}$ in the real case or the group $\mathbb{Z}$ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-031620 | MATHDB-337154

The almost positive link exact Lagrangian fillability conjecture

摘录 / EXCERPT: Exact Lagrangian fillability conjecture. All almost positive links are exact Lagrangian fillable.

conjecture | open | MathDB 页面 | 原文线索

V2-031621 | MATHDB-337155

Hayden-Sabloff's quasipositivity criterion for exact Lagrangian fillability

摘录 / EXCERPT: Hayden-Sabloff's conjecture. A knot is exact Lagrangian fillable if and only if it is quasipositive and the HOMFLYPT bound on the maximal Thurston-Bennequin number of K is sharp.

conjecture | open | MathDB 页面 | 原文线索

V2-031622 | MATHDB-337173

The higher Sharifi isomorphism conjecture for \(\varpi^{(p^r)}\)

摘录 / EXCERPT: Let Λ_r be the relevant Hecke algebra, let $\left(\widetilde{H}^{(p^r)}\right)_+$ be the plus part of the indicated homology module, let $\widetilde{I}_{\infty}$ be the Eis...

conjecture | open | MathDB 页面 | 原文线索

V2-031623 | MATHDB-337176

The finite generation conjecture for finite-dimensional Hopf algebras

摘录 / EXCERPT: Let A be a finite dimensional Hopf algebra over a field k, and let V be a finite dimensional A-module. Write $H^*(A,k)=\operatorname{Ext}^*_A(k,k)$ for the cohomology algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-031624 | MATHDB-337187

Dolgachev's mirror lattice conjecture for symplectic K3 surfaces

摘录 / EXCERPT: Dolgachev's mirror lattice conjecture. If a symplectic $K3$ (X,ω) is mirror to a complex $K3$ X°, then the above lattice isometry and induced moduli-space isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-031625 | MATHDB-337202

The topological rigidity conjecture

摘录 / EXCERPT: Let ${\cal F}$ be a family of maps, and let $f\in{\cal F}$ be non-hyperbolic. The map f is topologically rigid when it is not topologically conjugate to any other map in a suffic...

conjecture | open | MathDB 页面 | 原文线索

V2-031626 | MATHDB-337247

McCoy's alternating-surgery conjecture

摘录 / EXCERPT: Let K be a knot in S^3. An alternating surgery on K is a non-trivial Dehn surgery whose result is the double branched cover of a non-split alternating link. McCoy's alternati...

conjecture | open | MathDB 页面 | 原文线索

V2-031627 | MATHDB-337248

The banding refinement of McCoy's alternating-surgery conjecture

摘录 / EXCERPT: A banding of the unknot is a band surgery on the unknot that produces an alternating link. An almost alternating diagram of the unknot is an alternating diagram with one dealternat...

conjecture | open | MathDB 页面 | 原文线索

V2-031628 | MATHDB-337253

The flow category moves conjecture

摘录 / EXCERPT: Flow category moves conjecture. If two framed flow categories determine the same stable homotopy type, then they are move equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-031629 | MATHDB-337256

The Chang-space 2-torsion conjecture for Khovanov stable homotopy types

摘录 / EXCERPT: Chang-space 2-torsion conjecture. If a Chang space occurs as a wedge summand of the Khovanov stable homotopy type of a link, then that Chang space has a $\mathbb{Z}/2$ summand in i...

conjecture | open | MathDB 页面 | 原文线索

V2-031630 | MATHDB-337355

Eisenbud and Schreyer's natural vector bundle conjecture on the quadric surface

摘录 / EXCERPT: Eisenbud and Schreyer's conjecture. For any $p(x,y)\in\mathbb{Q}[x,y]$ with $\gamma>0$, there exists a vector bundle E with natural cohomology and Hilbert polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-031631 | MATHDB-337380

Positive lower ample canonical height for Zariski-dense orbits

摘录 / EXCERPT: Positive lower ample canonical height conjecture. If $O_f(x)$ is dense in the Zariski topology, then $\underline h_f(x)>0$. The source describes this as a weaker conjecture and as...

conjecture | open | MathDB 页面 | 原文线索

V2-031632 | MATHDB-337382

The perverse filtration comparison conjecture for compactified Jacobians

摘录 / EXCERPT: Let $C_{q/p}$ be the plane curve under consideration, let $J_{q/p}$ be its compactified Jacobian, and let P be the perverse filtration on $\textup{H}^{*}(J_{q/p})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031633 | MATHDB-337384

Relations conjecture for the affine-flag Springer fiber

摘录 / EXCERPT: Let $\textup{Sp}_{q/p}$ be the affine-flag Springer fiber, and let $\epsilon,\xi_1,\ldots,\xi_p,e_2,\ldots,e_p$ be the equivariant classes and parameters defined in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031634 | MATHDB-337389

Hitting-time conjecture for the giant shadow and homological giant

摘录 / EXCERPT: Let $\mathcal{Y}_d(n)$ be the stochastic Linial-Meshulam process in dimension $d\geq2$. Let $LT(n)$ denote its largest torsion group, and for a d-complex Y with complete...

conjecture | open | MathDB 页面 | 原文线索

V2-031635 | MATHDB-337401

Metric invariance of the homology of traversally generic differential complexes

摘录 / EXCERPT: Metric-invariance conjecture. The homology groups of these differential complexes depend only on the connected component of the space of traversally generic metrics on M containi...

conjecture | open | MathDB 页面 | 原文线索

V2-031636 | MATHDB-337418

Non-symmetric ribbon conjecture for the family \mathcal{A}(S,T)

摘录 / EXCERPT: Let $\mathcal{A}(S,T)$ be the ribbon-knot family considered in the paper, with parameters S,T, and let 12n268 denote the specified knot in the determinant-9 table. A symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-031637 | MATHDB-337419

Non-symmetric ribbon conjecture for the family \mathcal{B}(R,S,T)

摘录 / EXCERPT: Let $\mathcal{B}(R,S,T)$ be the ribbon-knot family considered in the paper, with parameters R,S,T. A ribbon is respected when a transformation to a symmetric union preserves that...

conjecture | open | MathDB 页面 | 原文线索

V2-031638 | MATHDB-337438

The branched-cover L-space conjecture for strongly quasipositive links

摘录 / EXCERPT: The branched-cover L-space conjecture. If some $\Sigma_n(L)$ is an L-space, then L is simply laced arborescent. If $L=K$ is a knot, then K is a $(2,k)$, $(3,4)$, or $(3,5)$ tor...

conjecture | open | MathDB 页面 | 原文线索

V2-031639 | MATHDB-337439

The odd-parameter pretzel knot strong quasipositivity conjecture

摘录 / EXCERPT: Odd-parameter pretzel conjecture. The knot K is strongly quasipositive if and only if either $|d|=n$, or $|d|=n-2$ and, after possibly reordering so that...

conjecture | open | MathDB 页面 | 原文线索

V2-031640 | MATHDB-337443

Eynard-Orantin symplectic invariance conjecture for topological-recursion free energies

摘录 / EXCERPT: Symplectic invariance conjecture. For every $g\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-031641 | MATHDB-337444

Borot-Garcia-Failde's exchange conjecture for fully simple map amplitudes

摘录 / EXCERPT: Fully simple amplitude conjecture. For ordinary maps or maps with loops,

conjecture | open | MathDB 页面 | 原文线索

V2-031642 | MATHDB-337445

Borot-Garcia-Failde's fully simple maps conjecture for loop models

摘录 / EXCERPT: Fully simple loop-model conjecture. After the symplectic transformation

conjecture | open | MathDB 页面 | 原文线索

V2-031643 | MATHDB-337499

The property F conjecture for braided fusion categories

摘录 / EXCERPT: Property F conjecture. The category $\mcC$ has property F if, and only if,

conjecture | open | MathDB 页面 | 原文线索

V2-031644 | MATHDB-337533

Newelski's Ellis group conjecture for definable flows

摘录 / EXCERPT: Let M be a first-order structure and let G be an M-definable group. The space $S_{ext,G}(M)$ is the universal definable flow of G over M, and G^{00} is the smallest typ...

conjecture | open | MathDB 页面 | 原文线索

V2-031645 | MATHDB-337534

Newelski's ideal-subgroup restriction conjecture

摘录 / EXCERPT: Let $M\prec^*N$ be a $*$-elementary extension, meaning that N is obtained from an elementary extension of M_{ext} in the language $\mathcal{L}_{ext,M}$ by reduct to the origina...

conjecture | open | MathDB 页面 | 原文线索

V2-031646 | MATHDB-337537

Minimal-diagram bridge-number realization conjecture for knots

摘录 / EXCERPT: Minimal-diagram bridge-number realization conjecture. For any knot K,

conjecture | open | MathDB 页面 | 原文线索

V2-031647 | MATHDB-337585

Novikov's stability-zone area and topology conjecture

摘录 / EXCERPT: Let $M_g^2\subset\mathbb{T}^3$ be a generic embedded surface, let ℓ be a stability-zone label, and let $\mathcal{D}_{\ell}(M_g^2)$ denote its stability region. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031648 | MATHDB-337586

Novikov and Dynnikov's Hausdorff-dimension conjecture for the exceptional set

摘录 / EXCERPT: Let $f:\mathbb{T}^3\to\mathbb{R}$ be a smooth function and let $\mathcal{E}(f)$ be its exceptional set of directions. Let $\dim_H$ denote Hausdorff dimension. Novikov and Dynnikov'...

conjecture | open | MathDB 页面 | 原文线索

V2-031649 | MATHDB-337587

Novikov and Dynnikov's generic TCI-stability conjecture in the four-torus

摘录 / EXCERPT: Let $\varepsilon:\mathbb{T}^4\to\mathbb{R}$ be a smooth function, let $\varepsilon_c=\varepsilon^{-1}(c)$ be a level hypersurface, and let $\Pi\in G_{4,2}(\mathbb{R})$ be a 2-pla...

conjecture | open | MathDB 页面 | 原文线索

V2-031650 | MATHDB-337635

The Prem Conjecture for smoothly k-realizable maps

摘录 / EXCERPT: The Prem Conjecture. Smoothly k-realizable general-position smooth maps $N^n\to\mathbb{R}^m$ are smooth k-prems, at least in the metastable range

conjecture | open | MathDB 页面 | 原文线索

V2-031651 | MATHDB-337650

The classical volume conjecture for colored Jones polynomials

摘录 / EXCERPT: Let K be a knot, and let $J_N^{(2)}(K;q)$ denote its colored Jones polynomial evaluated at q. The knot complement is $**\simplicial volume**$ measured by...

conjecture | open | MathDB 页面 | 原文线索

V2-031652 | MATHDB-337651

The volume conjecture for colored HOMFLY polynomials

摘录 / EXCERPT: Let K be a hyperbolic knot, and let $J_N^{(n)}(K;q)$ denote the colored HOMFLY polynomial, or $SU(n)$ invariant, associated with the symmetric representation and evaluated at q...

conjecture | open | MathDB 页面 | 原文线索

V2-031653 | MATHDB-337654

The L-space conjecture for irreducible 3-manifolds

摘录 / EXCERPT: L-space conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-031654 | MATHDB-337655

The universal abelian cover criterion for hyperbolic knot surgeries

摘录 / EXCERPT: Universal abelian cover conjecture. If the universal abelian cover of $X(K)(n\mu+q\lambda)$ is not excellent, then K is fibred and

conjecture | open | MathDB 页面 | 原文线索

V2-031655 | MATHDB-337658

Local exactness conjecture for singular BGG complexes

摘录 / EXCERPT: The singular BGG complex is constructed on the twistor space Z' associated with a suitably chosen, but arbitrarily small, open subset X'. Local exactness conjecture. The conclu...

conjecture | open | MathDB 页面 | 原文线索

V2-031656 | MATHDB-337659

Higher-singularity vanishing conjecture for twistor spaces

摘录 / EXCERPT: Let $X \subseteq \operatorname{iGr}(n,2n)$ be the big affine cell, or a ball or a polydisc inside it, and let Z be the corresponding twistor space in the stated double fibration....

conjecture | open | MathDB 页面 | 原文线索

V2-031657 | MATHDB-337663

Lakzian's Euclidean diffeomorphism conjecture for Berwald spaces

摘录 / EXCERPT: Lakzian's conjecture. The space M is diffeomorphic to Euclidean space $\mathbb{R}^{n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-031658 | MATHDB-337713

The Liouville quantum gravity scaling-limit conjecture for quadrangulations of the annulus

摘录 / EXCERPT: Liouville quantum gravity scaling-limit conjecture. The limit in law as $a\to0$ of (u_a,ν_a^{∂}) exists in the space of Radon measures equipped with the topology of wea...

conjecture | open | MathDB 页面 | 原文线索

V2-031659 | MATHDB-337724

Dafermos-Holzegel static uniqueness conjecture for the Eguchi-Hanson mass gap

摘录 / EXCERPT: Dafermos-Holzegel conjecture. There are no static, globally regular asymptotically locally AdS solutions to the five-dimensional Einstein vacuum equations with topology...

conjecture | open | MathDB 页面 | 原文线索

V2-031660 | MATHDB-337741

Descriptive homology nerve boundary-intersection conjecture for planar shapes

摘录 / EXCERPT: Descriptive homology nerve boundary-intersection conjecture. Every such planar shape contains a descriptive homology nerve that intersects the boundary of a hole.

conjecture | open | MathDB 页面 | 原文线索

V2-031661 | MATHDB-337743

Higher-dimensional barcode bound for Laplace eigenfunction combinations

摘录 / EXCERPT: Higher-dimensional barcode-functional conjecture. There exists a constant $\kappa_g>0$ such that, for any $\lambda>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-031662 | MATHDB-337744

The homotopy-theoretic Nearby Lagrangian conjecture

摘录 / EXCERPT: Let N and L be closed, simply connected manifolds of the same dimension. The cotangent bundle T^*N carries its natural symplectic structure, and consider the space of Lagrang...

conjecture | open | MathDB 页面 | 原文线索

V2-031663 | MATHDB-337789

Exponential slow recurrence for piecewise expanding maps with countably many discontinuities

摘录 / EXCERPT: Let $J=[-1/2,1/2]$ and consider the topologically exact Lorenz-type piecewise expanding map described in the paper, with a countable monotonicity partition, singular set consisting...

conjecture | open | MathDB 页面 | 原文线索

V2-031664 | MATHDB-337792

The one-point connectification conjecture for locally connected T_3-spaces

摘录 / EXCERPT: One-point connectification conjecture. Let $i=3$. A locally connected T_i-space has a T_i one-point connectification if and only if it has no compact (or almost compact) compon...

conjecture | open | MathDB 页面 | 原文线索

V2-031665 | MATHDB-337815

Mardešić's conjecture on metrizable factors of products of compact spaces

摘录 / EXCERPT: Let $d,s\geq 1$. Suppose that $L_1,\ldots,L_d$ are linearly ordered compact spaces and there is a continuous surjection ... where $K_1,\ldots,K_{d+s}$ are infinite compact spaces. Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-031666 | MATHDB-337821

The malleability conjecture for algebraic divides

摘录 / EXCERPT: An algebraic divide is a divide arising from a real morsification of a plane curve singularity. A divide is malleable when it is YB-equivalent to a scannable divide, where YB-equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-031667 | MATHDB-337866

Horizontal non-vanishing conjecture for Hecke twists of automorphic Rankin-Selberg L-functions

摘录 / EXCERPT: Let F be a totally real number field, $\mathbb{A}$ its ring of adeles, and π a cuspidal cohomological automorphic representation of $\operatorname{GL}_{2}(\mathbb{A})$. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-031668 | MATHDB-337897

Bernard's homotopy-transfer conjecture for the higher-genus KZB equation

摘录 / EXCERPT: Homotopy-transfer conjecture. The connection can be constructed via the homotopy transfer theorem, using constructions (1) and (2).

conjecture | open | MathDB 页面 | 原文线索

V2-031669 | MATHDB-337904

Monotone invariant conjecture for non-virtually-trivial circle bundles

摘录 / EXCERPT: Monotone invariant conjecture. In every dimension n, there is such a homotopy n-manifold numerical invariant I_n that is positive and finite on every circle bundle over a clo...

conjecture | open | MathDB 页面 | 原文线索

V2-031670 | MATHDB-337914

Fargues-Fontaine cohomology conjecture for smooth rigid spaces

摘录 / EXCERPT: Let C be an algebraically closed p-adic field, let FF_C be the Fargues-Fontaine curve with distinguished point ∞, and let X be a quasicompact separate...

conjecture | open | MathDB 页面 | 原文线索

V2-031671 | MATHDB-337945

Hirasawa-Murasugi conjecture on stable coefficients and signature of alternating knots

摘录 / EXCERPT: Let K be an alternating knot with signature satisfying $|\sigma(K)|=2k$, and let its Alexander polynomial be ... where $a_j>0$. Hirasawa-Murasugi conjecture. The coefficients satis...

conjecture | open | MathDB 页面 | 原文线索

V2-031672 | MATHDB-337976

Arens-Michael envelope conjecture for invertible bimodules

摘录 / EXCERPT: Let A be an algebra and let M be an invertible A-bimodule, with inverse M^{-1}. Write $4\widehat{A}4$, $4\widehat{M}4$, and $4\widehat{M^{-1}}4$ for their Arens-Michael env...

conjecture | open | MathDB 页面 | 原文线索

V2-031673 | MATHDB-337983

The blow-up formula conjecture for Bott-Chern cohomology

摘录 / EXCERPT: Let X be a compact complex manifold with complex dimension n, and let $Z\subseteq X$ be a closed complex submanifold of complex codimension $r\geq 2$. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-031674 | MATHDB-337991

The cohomological-dimension conjecture for commutator subgroups of pure braid groups

摘录 / EXCERPT: Let P_n be the pure braid group of the disc, and let $[P_n,P_n]$ denote its commutator subgroup. The cohomological-dimension conjecture asserts that ... This value would make the l...

conjecture | open | MathDB 页面 | 原文线索

V2-031675 | MATHDB-337992

The disc braid-group conjecture for topological complexity

摘录 / EXCERPT: Let D be the disc, let $C(D,n)$ be its unordered configuration space, and let B_n be the braid group on n strands. The disc braid-group conjecture asserts that ... This would f...

conjecture | open | MathDB 页面 | 原文线索

V2-031676 | MATHDB-338025

Semisimplicity or virtual unipotence conjecture for automorphisms of compact Kähler threefolds

摘录 / EXCERPT: Semisimplicity or virtual unipotence conjecture. The action f^*_2 is either semisimple or virtually unipotent. The mixed case is not possible for automorphisms of compact Kähler...

conjecture | open | MathDB 页面 | 原文线索

V2-031677 | MATHDB-338057

The unrestricted surgery polygon conjecture for ${\rm SU}(N)$-instanton Floer homology

摘录 / EXCERPT: The unrestricted surgery polygon conjecture. Theorem holds without the assumption $N\leq4$. Thus, under the same admissibility hypotheses, ${\rm SU}(N)$-instanton Floer homology as...

conjecture | open | MathDB 页面 | 原文线索

V2-031678 | MATHDB-338064

The Legendrian Low conjecture for causally simple spacetimes

摘录 / EXCERPT: Let (X^{m+1},g), $m>1$, be a causally simple spacetime that does not admit a metric g' such that (X,g') is refocussing and causally simple. A Legendrian link is a link formed...

conjecture | open | MathDB 页面 | 原文线索

V2-031679 | MATHDB-338065

The Legendrian Low conjecture for nonrefocussing globally hyperbolic spacetimes

摘录 / EXCERPT: Let (X^{m+1},g) be a globally hyperbolic spacetime with $m\geq 2$. Two points are causally related when one lies in the causal future or causal past of the other, and their skies...

conjecture | open | MathDB 页面 | 原文线索

V2-031680 | MATHDB-338093

Gromov's sharp growth conjecture for homotopy groups

摘录 / EXCERPT: Let Y be a simply connected finite complex, and let $\rho_Y(L)$ denote the number of homotopy classes in $\bigcup_n\bigl[S^n,Y\bigr]$ that admit representatives with Lipschitz co...

conjecture | open | MathDB 页面 | 原文线索

V2-031681 | MATHDB-338107

Eccles conjecture on unstable Hurewicz images

摘录 / EXCERPT: Let X be a path connected CW-complex with finitely generated homology. For $n>0$, let $f\in\pi_n^sX\simeq\pi_nQX$, and let ... be the unstable Hurewicz homomorphism. Eccles conje...

conjecture | open | MathDB 页面 | 原文线索

V2-031682 | MATHDB-338108

Weak geometric version of the Curtis and Eccles conjectures

摘录 / EXCERPT: Weak geometric version of the Curtis and Eccles conjectures. (i) The image of the first homomorphism is finite. (ii) For a path connected space X, the image of the second homomor...

conjecture | open | MathDB 页面 | 原文线索

V2-031683 | MATHDB-338111

Equivariant Ching conjecture for the little 2-disks operad

摘录 / EXCERPT: Let D_2 be the little 2-disks operad equipped with its S^1-action, let Σ^∞_+ denote suspension spectrum after adjoining a disjoint basepoint, and let K be the d...

conjecture | open | MathDB 页面 | 原文线索

V2-031684 | MATHDB-338115

Vlasenko's p-adic Frobenius and connection matrix conjecture

摘录 / EXCERPT: Let f be a homogeneous polynomial defining a smooth hypersurface family, and let α_s be the matrices associated with the Laurent polynomial construction, with σ the...

conjecture | open | MathDB 页面 | 原文线索

V2-031685 | MATHDB-338183

The global-dimension conjecture for sufficiently nice localizations

摘录 / EXCERPT: Let R be a ring spectrum, let $v\in \pi_*R$ be a homogeneous element, and consider the localization map ... Here $R[v^{-1}]$ denotes the localization obtained by inverting v, and...

conjecture | open | MathDB 页面 | 原文线索

V2-031686 | MATHDB-338187

Regularity suffices for recoverability in product-topology codes

摘录 / EXCERPT: Regularity conjecture. For the topology $T_{m,n}(a,b,0)$, if an erasure pattern is regular, then it is recoverable.

conjecture | open | MathDB 页面 | 原文线索

V2-031687 | MATHDB-338205

Gruson's conjecture on Ext of the fraction field of a polynomial ring

摘录 / EXCERPT: Let k be a field. Define s by $|k|=\aleph_s$ if k is infinite, and let $s=0$ otherwise. For a non-negative integer n, set $R=k[X_1,\ldots,X_n]$ and let $Q(R)$ denote its fr...

conjecture | open | MathDB 页面 | 原文线索

V2-031688 | MATHDB-338247

Curtis conjecture on spherical classes in H_*QS^0

摘录 / EXCERPT: Curtis conjecture. Only the Hopf invariant one and Kervaire invariant one elements survive under h.

conjecture | open | MathDB 页面 | 原文线索

V2-031689 | MATHDB-338248

Eccles's conjecture on spherical classes and detection by homology operations

摘录 / EXCERPT: Eccles conjecture. Under these hypotheses, the stable adjoint of f either is detected by homology or is detected by a primary operation in its mapping cone.

conjecture | open | MathDB 页面 | 原文线索

V2-031690 | MATHDB-338272

Chromatic Vanishing Conjecture for the Lubin-Tate ring

摘录 / EXCERPT: Let R_n be the Lubin-Tate ring associated to the Honda formal group law $\mathbb{H}_n$ of height n over $\mathbb{F}_{p^n}$, with $R_n\cong \mathbb{W}[[u_1,\ldots,u_{n-1}]]$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-031691 | MATHDB-338329

Jablan-Radović conjecture on rational knots and OGC knots

摘录 / EXCERPT: A knot diagram is straight if, after labeling its n crossings in traversal order, its knot word has the form ... where $\sigma\in S_n$; a straight number $\mathtt{str}(K)$ is the m...

conjecture | open | MathDB 页面 | 原文线索

V2-031692 | MATHDB-338336

Generalised Kauffman conjecture on slice derivatives

摘录 / EXCERPT: Let K be a knot, and let a slice derivative mean a derivative of K that is slice in the relevant category. Generalised Kauffman conjecture. If K is a topologically (respectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-031693 | MATHDB-338353

Harris's conjecture on Mantovan cohomology and supercuspidal representations

摘录 / EXCERPT: Let (G,μ) be a local Shimura datum as in the source, let $b\in\mathbf{B}(G,\mu)$, and let $M_S\subset M_b$ be a standard Levi subgroup. Let ρ be a supercuspidal representa...

conjecture | open | MathDB 页面 | 原文线索

V2-031694 | MATHDB-338367

Explicit description conjecture for the class of DG-flat modules

摘录 / EXCERPT: Let R be the DG-algebra under consideration, and let ${\mathcal F}$ be the full subcategory of ${\mathsf{D}}(R)$ consisting of the relevant DG-flat objects. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031695 | MATHDB-338380

Pestov's conjecture on the universal ambit and enveloping semigroup

摘录 / EXCERPT: Pestov's conjecture. The canonical map is an isomorphism if and only if G is precompact. This conjecture concerns when the two canonical dynamical objects associated with a topol...

conjecture | open | MathDB 页面 | 原文线索

V2-031696 | MATHDB-338381

Conjecture on non-isomorphism of universal ambits for closed subgroups of S_∞

摘录 / EXCERPT: Non-isomorphism conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031697 | MATHDB-338394

Syntomic sheaf conjecture for quasisyntomic semiperfectoid rings

摘录 / EXCERPT: Syntomic sheaf conjecture. The sheaf of complexes $\mathbb Z_p(i)$ on $\mathrm{QRSPerfd}$ is locally concentrated in degree 0 and is given by a p-torsion-free sheaf. Equivalent...

conjecture | solved | MathDB 页面 | 原文线索

V2-031698 | MATHDB-338454

Infinite-length arcs conjecture for supermixed labyrinth fractals

摘录 / EXCERPT: Infinite-length arcs conjecture. The supermixed labyrinth fractal L_{∞} generated by $\{\tilde{\mathcal A}_k\}_{k\ge 1}$ has the property that, between any two distinct poin...

conjecture | open | MathDB 页面 | 原文线索

V2-031699 | MATHDB-338461

The generalized invariant conjecture for homeomorphism classes

摘录 / EXCERPT: Generalized invariant conjecture. Knowledge of $G(g,0,X)$ for every such g determines the homeomorphism class of X.

conjecture | open | MathDB 页面 | 原文线索

V2-031700 | MATHDB-338506

Higher inductive type constructions from cofibrant W-types with reductions

摘录 / EXCERPT: Let ${\mathbb C}$ be a topos with natural number object satisfying all the axioms considered by Orton and Pitts, and suppose that initial algebras exist for all cofibrant polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-031701 | MATHDB-338514

Furuta-Ohta invariant and Seiberg-Witten invariant relation

摘录 / EXCERPT: Let X be a closed oriented homology-oriented smooth 4-manifold satisfying ... Let $\lambda_{\rm FO}(X)$ and $\lambda_{\rm SW}(X)$ denote the Furuta-Ohta and Mrowka-Ruberman-Savelie...

conjecture | solved | MathDB 页面 | 原文线索

V2-031702 | MATHDB-338524

Finite-basis conjecture for functions beyond every countable Baire class

摘录 / EXCERPT: For a countable ordinal $\alpha<\omega_1$, consider the class of functions that are not of Baire class α, ordered by topological embeddability. A finite basis for an upward-...

conjecture | open | MathDB 页面 | 原文线索

V2-031703 | MATHDB-338551

The powerful p-star-central subgroup conjecture

摘录 / EXCERPT: Let G be a finite p-group of fixed rank r. A subgroup K is powerful p^*-central with the ΩEP if it has those properties. The powerful p^*-central subgroup conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-031704 | MATHDB-338552

The cohomology conjecture for powerful 4-star-central 2-groups

摘录 / EXCERPT: Let G be a powerful 4^*-central 2-group with the ΩEP property, and let $d=\operatorname{rk}(\Omega_1(G))$. Write Λ for the exterior algebra. The cohomology con...

conjecture | open | MathDB 页面 | 原文线索

V2-031705 | MATHDB-338572

The pretzel-knot surgery left-orderability conjecture

摘录 / EXCERPT: Let $s\ge 3$ be an integer, and let $M_{p/q}$ be the manifold obtained by Dehn surgery with slope $\frac{p}{q}$ along the $(-2,3,2s+1)$-pretzel knot. Pretzel-knot surgery conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-031706 | MATHDB-338587

Injectivity and projectivity of adelic blocks

摘录 / EXCERPT: Adelic-block conjecture. Every adelic block $G\in\mathsf{LCA}_{F}$ is both an injective and projective object.

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V2-031707 | MATHDB-338591

Uniqueness of the topological quandle structure on the closed interval

摘录 / EXCERPT: Let $[0,1]$ be the closed unit interval with its usual topology. A topological quandle structure on $[0,1]$ is a quandle operation that is continuous with respect to this topology....

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V2-031708 | MATHDB-338665

Davis-Okun conjecture on action dimension and L^2-Betti numbers

摘录 / EXCERPT: Davis-Okun conjecture. If the ith L^2-Betti number of G is nonzero, then

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V2-031709 | MATHDB-338764

Vanishing conjecture for log-pluricanonical sheaves with standard coefficients

摘录 / EXCERPT: Vanishing conjecture. The higher cohomology of the m-th reflexive log-pluricanonical sheaf vanishes:

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V2-031710 | MATHDB-338772

Relative Giroux correspondence for contact manifolds with convex boundary

摘录 / EXCERPT: Relative Giroux correspondence. Every compact contact manifold with convex boundary is contactomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-031711 | MATHDB-338774

Gauduchon's conjecture on prescribed Chern-Ricci curvature

摘录 / EXCERPT: Gauduchon's conjecture. There exists a Gauduchon metric ω on M such that

conjecture | open | MathDB 页面 | 原文线索

V2-031712 | MATHDB-338781

Compact Chern character and weighted orbital integral conjecture

摘录 / EXCERPT: Let G be a p-adic group, let ${\rm Sm}$ be the category of smooth modules and ${\rm Adm}\subset {\rm Sm}$ the subcategory of admissible modules. Let $HH_0^c({\rm Sm})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031713 | MATHDB-338789

Presentation conjecture for the skein algebra of a surface

摘录 / EXCERPT: Let Σ be the surface under consideration, let $\bar{\Gamma}(\Sigma)$ be the extended mapping class group, and let $\mathcal{A}(\Gamma(\Sigma))$ be the quotient algebra defin...

conjecture | open | MathDB 页面 | 原文线索

V2-031714 | MATHDB-338793

Existence of an univalent path category of infinity-types over the effective topos

摘录 / EXCERPT: A path category with homotopy Π-types is a category equipped with the path-category structure and dependent products needed to interpret homotopy type theory. A fibration is ca...

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V2-031715 | MATHDB-338828

Wall's conjecture on Poincaré duality groups of dimension three

摘录 / EXCERPT: Let G be a Poincaré duality group of dimension 3. Wall's conjecture. G is the fundamental group of a closed aspherical 3-manifold. This conjecture asks whether the algebrai...

conjecture | open | MathDB 页面 | 原文线索

V2-031716 | MATHDB-338829

The ENR homology-manifold realization conjecture for Poincaré duality groups

摘录 / EXCERPT: Let G be a finitely presented Poincaré duality group. ENR homology-manifold realization conjecture. G is the fundamental group of a closed aspherical ENR homol...

conjecture | open | MathDB 页面 | 原文线索

V2-031717 | MATHDB-338834

Agreement of Bass and Blumberg-Gepner-Tabuada non-connective K-theory

摘录 / EXCERPT: Agreement conjecture. The non-connective K-theory functor K^B agrees with the non-connective K-theory functor of Blumberg, Gepner, and Tabuada.

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V2-031718 | MATHDB-338844

Baum-Dąbrowski-Hajac topological Borsuk-Ulam conjecture

摘录 / EXCERPT: Let X be a compact Hausdorff space with a free action of a nontrivial compact group G. Equip the join $X * G$ with the diagonal action. Baum-Dąbrowski-Hajac's topological Borsu...

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V2-031719 | MATHDB-338884

Rasmussen's spectral sequence conjecture from reduced Khovanov homology to knot Floer homology

摘录 / EXCERPT: Rasmussen's spectral sequence conjecture. There is a spectral sequence from $\overline{Kh}(K)$ to $\widehat{\mathit{HFK}}(K)$.

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V2-031720 | MATHDB-338885

Dowlin's graded sl_n to knot Floer spectral sequence conjecture

摘录 / EXCERPT: Dowlin's graded spectral sequence conjecture. For each n, there are spectral sequences

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V2-031721 | MATHDB-338907

Fontaine-Jannsen semistable conjecture

摘录 / EXCERPT: Let K be the underlying p-adic field, let F be the coefficient field of Hyodo-Kato cohomology, and let X be a proper, log-smooth, fine and saturated...

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V2-031722 | MATHDB-338919

The even-order conjecture for self-homotopy equivalences of A_2^2-polyhedra

摘录 / EXCERPT: Let X be an A_2^2-polyhedron, and let ${\mathcal B}^4(X)$ denote the group under consideration of self-homotopy equivalences of X. Even-order conjecture. If...

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V2-031723 | MATHDB-338936

The straight-number gap conjecture for alternating knots

摘录 / EXCERPT: Let K be an alternating knot, and let $c(K)$ denote its crossing number and $\mathtt{str}(K)$ its straight number. Suppose ... where $n\geq 1$, and let K' be the knot obtained by...

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V2-031724 | MATHDB-338943

Kalfagianni-Tran conjecture on nonpositive linear degree coefficients

摘录 / EXCERPT: Kalfagianni-Tran conjecture. For every such congruence class j, the linear coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-031725 | MATHDB-338980

The asymptotic upper-bound conjecture for torsion realization complexity

摘录 / EXCERPT: Let $d\geq 2$ and let G be a finite abelian group. Write $T_d(G)$ for the minimum number of vertices in a simplicial complex whose $(d-1)$st homology group has torsion part isomo...

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V2-031726 | MATHDB-339003

The pro-homology conjecture for cosheaf homology satellites

摘录 / EXCERPT: Pro-homology conjecture. On the standard site $OPEN(X)$, if X is Hausdorff paracompact, then

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V2-031727 | MATHDB-339004

The locally finite satellite conjecture for singular homology and homotopy

摘录 / EXCERPT: Locally finite satellite conjecture. The left satellites are naturally isomorphic to singular homology:

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V2-031728 | MATHDB-339012

Gałęski's local smooth rank-constrained approximation conjecture

摘录 / EXCERPT: Gałęski's local approximation conjecture. Every such mapping admits, around each point, a uniformly convergent smooth approximation sequence preserving the derivative-rank bound.

conjecture | open | MathDB 页面 | 原文线索

V2-031729 | MATHDB-339013

Rank-one smooth approximation conjecture

摘录 / EXCERPT: Rank-one approximation conjecture. The mapping f can be uniformly approximated, at least locally, by mappings $g\in C^\infty(\mathbb R^n,\mathbb R^k)$ satisfying...

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V2-031730 | MATHDB-339043

Alternating quadrisecant approximation conjecture for knots with nontrivial Conway polynomial

摘录 / EXCERPT: Alternating quadrisecant approximation conjecture. Such sequences exist for every $\varepsilon>0$. The conjecture would connect the existence of alternating quadrisecants with insc...

conjecture | open | MathDB 页面 | 原文线索

V2-031731 | MATHDB-339044

Inscribed right-handed trefoil conjecture

摘录 / EXCERPT: Let $\gamma:\mathbb{R}\to\mathbb{R}^3$ be an analytic $\mathbb{Z}$-periodic function with nonvanishing derivative that parameterizes a right-handed trefoil. A polygonal inscribed t...

conjecture | open | MathDB 页面 | 原文线索

V2-031732 | MATHDB-339079

Mazur's dimension conjecture for universal deformation rings

摘录 / EXCERPT: Mazur's dimension conjecture. The ring R is flat over $W(k)$ and its relative dimension over $W(k)$ is $\delta(\bar{\rho})$. The conjecture predicts that the expected dimension f...

conjecture | open | MathDB 页面 | 原文线索

V2-031733 | MATHDB-339080

Jannsen's conjecture on the dimension of global H^2

摘录 / EXCERPT: Let F be a number field, let S be a finite set of places containing all places above p and all infinite places, and let V be a p-adic representation of $G_{F,S}$ that is...

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V2-031734 | MATHDB-339088

Meta-Isomorphism Conjecture for spectra

摘录 / EXCERPT: Meta-Isomorphism Conjecture. The induced map

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V2-031735 | MATHDB-339138

The closed-interval conjecture for attainable complexities of mathcal K-analytic spaces

摘录 / EXCERPT: For a space X, let $\operatorname{Compl}(X)$ denote its set of attainable complexities. A space is $\mathcal K$-analytic if it has the standard $\mathcal K$-analyticity property...

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V2-031736 | MATHDB-339139

Classification of attainable complexities for arbitrary spaces

摘录 / EXCERPT: For a topological space X, let $\operatorname{Compl}(X)$ be its set of attainable complexities. Classification conjecture. Exactly one of the following alternatives holds: ... More...

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V2-031737 | MATHDB-339141

Geometric realization conjecture for higher-dimensional hyperkähler moduli

摘录 / EXCERPT: Fix a connected moduli space M of polarized $2n$-dimensional irreducible holomorphic symplectic manifolds (X,L) with second-cohomology lattice Λ, and write the associat...

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V2-031738 | MATHDB-339187

Genus formula conjecture for lens space knots in \Sigma(2,3,7)

摘录 / EXCERPT: Let $K_{p',k}$ be any knot whose parameter (p',k) occurs in the displayed table for lens space surgeries in $\Sigma(2,3,7)$, and let g' be the table's value in its g' column....

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V2-031739 | MATHDB-339188

Completeness conjecture for lens space knots in \Sigma(2,3,5) and \Sigma(2,3,7)

摘录 / EXCERPT: Let $K_{p,k}$ and $K_{p',k}$ denote the parameterized dual knots in the tables for $\Sigma(2,3,5)$ and $\Sigma(2,3,7)$, respectively, and let $g(K)$ denote the knot genus. Complete...

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V2-031740 | MATHDB-339189

Existence conjecture for lens space knots in Brieskorn homology spheres

摘录 / EXCERPT: Let $\Sigma(p,q,r)$ be a Brieskorn homology sphere, and let $K_{p,k}$ denote a parameterized lens space knot. Existence conjecture. Every Brieskorn homology sphere contains a knot...

conjecture | open | MathDB 页面 | 原文线索

V2-031741 | MATHDB-339190

Nonexistence conjecture for lens space knots in oppositely oriented Brieskorn spheres

摘录 / EXCERPT: Let $\Sigma(p,q,r)$ be a Brieskorn homology sphere with its usual orientation, and let $-\Sigma(p,q,r)$ denote the same manifold with the opposite orientation. Let $K_{p,k}$ denote...

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V2-031742 | MATHDB-339229

Two-number and mod-2 Betti-number conjecture for compact symmetric spaces

摘录 / EXCERPT: Let M be a connected compact symmetric space, let $\#_2M$ denote its two-number, and let $H(M,\mathbb Z_2)$ denote its homology with coefficients in $\mathbb Z_2$. Two-number hom...

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V2-031743 | MATHDB-339230

Universal CW-cell conjecture for the two-number of compact symmetric spaces

摘录 / EXCERPT: Let M be a connected compact symmetric space, let $\#_2M$ denote its two-number, and let a CW complex structure on M be a cellular decomposition of M. Universal CW-cell conje...

conjecture | open | MathDB 页面 | 原文线索

V2-031744 | MATHDB-339298

The integrality and mod 4 reduction conjecture for Muñoz's polynomial

摘录 / EXCERPT: The integrality and reduction conjecture. The expression

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V2-031745 | MATHDB-339325

Gal's stability conjecture for dense flag homology manifolds

摘录 / EXCERPT: Let $m\geq 2$, and let Δ be a flag homology $(2m-1)$-manifold with n vertices. The Gal's stability conjecture. There exists a constant $b=b(m)$ such that if ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-031746 | MATHDB-339327

Kontsevich's proper DG algebra degeneration conjecture

摘录 / EXCERPT: Let B be a proper DG algebra over a field $cmathrm k$. The Hochschild homology pairing is a map $HH_{\bullet}(B)\otimes HH_{\bullet}(B^{op})\to \mathrm k$, and let...

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V2-031747 | MATHDB-339342

Kontsevich's period conjecture for relations among periods

摘录 / EXCERPT: Let $\mathcal{P}^i\subset\mathbb{C}$ be the set of i-periods, let ... be the set of effective cohomological periods, and let $\mathcal{P}=\mathcal{P}^{\mathrm{eff}}[\pi^{-1}]$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-031748 | MATHDB-339346

Halperin-Carlsson rank conjecture for semifree differential graded modules

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic 2, let $S=k[x_1,\ldots,x_r]$ be the polynomial algebra in r variables of degree -1, and let (M,∂) be a dif...

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V2-031749 | MATHDB-339373

Hikami-Inoue conjecture on geometric representations from braid solutions

摘录 / EXCERPT: Hikami-Inoue conjecture. There exists a non-degenerate solution $\mathbf{x}^1$ such that the induced representation $\rho_{\mathbf{x}^1}$ is geometric, namely discrete and faithful...

conjecture | open | MathDB 页面 | 原文线索

V2-031750 | MATHDB-339376

The logarithmic Dolbeault equation conjecture for simple normal crossing divisors

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $D=\sum_{i=1}^r D_i$ be a simple normal crossing divisor in X. Let L be a holomorphic bundle over X such that ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-031751 | MATHDB-339407

Frobenius-surjectivity conjecture for degree-zero local cohomology

摘录 / EXCERPT: Let $(R,\mathfrak{m},\mathbb{C})$ be a reduced standard graded $\mathbb{C}$-algebra with an isolated singularity at $\mathfrak{m}$. Let A be a regular subring of $\mathbb{C}$ suc...

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V2-031752 | MATHDB-339464

The extension conjecture for operator K-theory

摘录 / EXCERPT: Let $K\colon \operatorname{KHaus}^{\operatorname{op}}\to\operatorname{Ab}$ be topological K-theory, and let $\tilde{K}\colon \operatorname{uC^*}\to\operatorname{Ab}$ denote its e...

conjecture | open | MathDB 页面 | 原文线索

V2-031753 | MATHDB-339473

The graded homology classification conjecture for finite graphs

摘录 / EXCERPT: Let E and F be finite graphs, and let R be a field. Write $\mathcal{G}_E$ and $\mathcal{G}_F$ for their graph groupoids, and let H_0^{gr} denote graded group...

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V2-031754 | MATHDB-339474

Monoidal categorification conjecture for surface skein algebras

摘录 / EXCERPT: Let $\mathcal{S}$ be an oriented surface of finite type. Let $\mathcal{S}\mathrm{MKh}=\mathcal{S}\mathrm{M}\circ\mathcal{S}\mathrm{Kh}$ be a functor from surface links to a target...

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V2-031755 | MATHDB-339476

Categorified Frohman--Gelca product-to-sum conjecture

摘录 / EXCERPT: Let $\operatorname{Kar}(\mathcal{T}\mathrm{Foam}^{\mathrm{ess}}_0)_{\mathrm{dg}}$ be the proposed monoidal target category, let $*$ denote its monoidal product, and let $(m,n)^F_T$...

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V2-031756 | MATHDB-339522

Saeki's rigidity conjecture for 4-dimensional manifolds admitting special generic maps

摘录 / EXCERPT: Let M and N be 4-dimensional homeomorphic closed and connected manifolds, and suppose that each admits a special generic map into some Euclidean space of dimension smaller th...

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V2-031757 | MATHDB-339536

The torsion conjecture for non-split links

摘录 / EXCERPT: Let L be a non-split link, excluding the trivial knot, the Hopf link, and their connected sums. Let $\mathcal{H}(L)$ denote its integer Khovanov homology. The non-split-link tors...

conjecture | open | MathDB 页面 | 原文线索

V2-031758 | MATHDB-339537

The higher-page Bockstein-Turner differential relation conjecture

摘录 / EXCERPT: Let L be a link, and consider the Turner and Bockstein spectral sequences on its Khovanov homology $\mathcal{H}_{\mathbb{Z}_2}(L)$ over $\mathbb{Z}_2$. For higher pages, denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-031759 | MATHDB-339538

The matching collapse conjecture for Turner and Bockstein spectral sequences

摘录 / EXCERPT: Let L be a link. Consider the Turner and Bockstein spectral sequences associated with $\mathcal{H}_{\mathbb{Z}_2}(L)$, and let r be a positive integer. Matching collapse conjec...

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V2-031760 | MATHDB-339552

Szabó's conjecture on Szabó homology and Heegaard Floer homology

摘录 / EXCERPT: Let $\operatorname{\mathcal{D}}$ be a link diagram. Szabó homology $\operatorname{Sz}(\operatorname{\mathcal{D}})$ is the homology of the filtered chain complex assigned to...

conjecture | open | MathDB 页面 | 原文线索

V2-031761 | MATHDB-339553

Seed's conjectures on reduced Szabó homology and its spectral sequence

摘录 / EXCERPT: Let $\operatorname{\mathcal{D}}$ be a link diagram, let p and p' be points that are not double points of $\operatorname{\mathcal{D}}$, and let...

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V2-031762 | MATHDB-339597

Cohomological-building-block conjecture for integral-character unitary representations of \mathrm{Sp}(p,q)

摘录 / EXCERPT: Cohomological-building-block conjecture. There exist a θ-stable parabolic $\mathfrak{q}=\mathfrak{l}\oplus\mathfrak{u}$ such that

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V2-031763 | MATHDB-339598

Raghu and Haldane's photonic bulk-edge correspondence

摘录 / EXCERPT: Let a two-dimensional photonic crystal have a boundary and bulk band gaps. In each bulk band gap, consider the left- and right-moving boundary modes, and let the bulk Chern number...

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V2-031764 | MATHDB-339601

The weak Palis density conjecture for dynamical systems

摘录 / EXCERPT: Let $r\geq 1$, and consider a certain space of dynamical systems equipped with the C^r topology. Morse-Smale systems and horseshoe systems are the two classes of systems under co...

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V2-031765 | MATHDB-339607

Existence of primitive multiple lines with good cohomology

摘录 / EXCERPT: Let $d \geq 5$ be an integer congruent to 2 modulo 3. A primitive d-line is a primitive locally Cohen-Macaulay d-fold structure supported on a line $L\subset\mathbb{P}^3$;...

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V2-031766 | MATHDB-339645

The homotopy Barr-Beck conjecture for DG-categories

摘录 / EXCERPT: Let A_1 and A_2 be DG-categories, and let F and G be a pair of homotopy adjoint functors between them. Let T denote the homotopy monad induced by the composite of the adj...

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V2-031767 | MATHDB-339652

Infinite injective dimension for rational sheaves on spaces with non-empty perfect hull

摘录 / EXCERPT: Let X be a topological space with finite Cantor-Bendixson rank and non-empty perfect hull. Consider sheaves of $\mathbb{Q}$-modules on X. The conjecture. The injective dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-031768 | MATHDB-339677

Conjecture on the leading coefficient of the L^2-Alexander torsion

摘录 / EXCERPT: Let N be a 3-manifold with empty or toroidal boundary, let $\phi\in H^1(N;\mathbb{Z})$, and let Σ be a Thurston norm minimizing surface dual to ϕ. Let $C(N,\phi)$ d...

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V2-031769 | MATHDB-339678

Xiao-Song Lin's conjecture relating the colored Jones polynomial to L^2-Alexander torsion

摘录 / EXCERPT: Let $K\subset S^3$ be a knot. For every $N\in\mathbb{N}$, let $J_K(N,x)\in\mathbb{Z}[q^{\pm1}]$ be the normalized N-th colored Jones polynomial. For $t\in\mathbb{C}^*$, let...

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V2-031770 | MATHDB-339683

Stability conjecture for Steinberg-module homology of congruence subgroups

摘录 / EXCERPT: Steinberg-module stability conjecture. For each $i,j\geq 0$, the $\mathrm{VB}_{\mathbb{Z}}/\Gamma(p)$-module

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V2-031771 | MATHDB-339746

Generalized semi-infinite cohomology formula at positive level

摘录 / EXCERPT: Fix a positive level κ, set $q=\operatorname{exp}\left(\frac{\pi\sqrt{-1}}{\kappa'-\kappa_{\textup{crit}}}\right)$, and let M be a $U^{\textup{Lus}}_q(\mathfrak{g})$-modul...

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V2-031772 | MATHDB-339752

The characterization of Fréchet-Urysohn free topological groups of length four

摘录 / EXCERPT: Let X be a metrizable space, and let $F_4(X)$ denote the corresponding free topological group of length at most four. Suppose that the set of all non-isolated points of X is co...

conjecture | open | MathDB 页面 | 原文线索

V2-031773 | MATHDB-339758

The cube decomposition conjecture for topological Hochschild homology of Johnson-Wilson spectra

摘录 / EXCERPT: The cube decomposition conjecture. The spectrum $\mathrm{THH}(E(n))$ decomposes as a sum of 2^n factors: one copy of $E(n)$, together with 2^{n-i-1} suspended copies of...

conjecture | open | MathDB 页面 | 原文线索

V2-031774 | MATHDB-339764

Nevo's balanced-complex strengthening of Gal's conjecture

摘录 / EXCERPT: Let Δ be a flag homology sphere, and let $\gamma(\Delta)$ denote its γ-vector. Nevo's strengthening of Gal's conjecture. The vector $\gamma(\Delta)$ is the f-vector...

conjecture | open | MathDB 页面 | 原文线索

V2-031775 | MATHDB-339849

The hammock-space conjecture for simplicial model categories

摘录 / EXCERPT: The hammock-space conjecture. The space $\widehat{A}$ is weakly equivalent to $\emph{Bhaut} A$:

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V2-031776 | MATHDB-339850

Classification of homogeneous functors by classifying spaces of self-equivalences

摘录 / EXCERPT: Classification conjecture. There are bijections

conjecture | open | MathDB 页面 | 原文线索

V2-031777 | MATHDB-339856

Teissier's conjecture on sectional Milnor numbers and topological type

摘录 / EXCERPT: Teissier's conjecture. If the germs have the same topological type, then

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V2-031778 | MATHDB-339917

Cohomological stability under categorical duality

摘录 / EXCERPT: Let k be a field, let $\mathscr{C}$ be a finite tensor category of finite type, and let $\mathscr{M}$ be an exact $\mathscr{C}$-module category. Write...

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V2-031779 | MATHDB-339998

Persistent homology scaling conjecture

摘录 / EXCERPT: Let X_n be a random sample of n points from a compact set $X\subseteq\mathbb{R}^m$, distributed according to a nonsingular probability measure μ, where $m\geq 2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031780 | MATHDB-339999

Universality of persistent homology distributions for uniform measures

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^m$ be compact with positive Lebesgue measure, and let μ be Lebesgue measure restricted to X and rescaled to have total mass one. Let...

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V2-031781 | MATHDB-340026

Positselski's PBW conjecture for Galois cohomology

摘录 / EXCERPT: Let F be a field containing a primitive pth root of unity, and let G_F be its absolute Galois group. The algebra $H^\bullet(G_F,\mathbb{F}_p)$ is a graded mod-p cohomology...

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V2-031782 | MATHDB-340035

Hausdorff-measure conjecture for persistent homology functionals

摘录 / EXCERPT: Hausdorff-measure conjecture. If $\mathit{PH}_0(\operatorname{supp}\mu)$ is trivial and $0<\alpha<d$, then $$

conjecture | open | MathDB 页面 | 原文线索

V2-031783 | MATHDB-340129

Crosscut criterion for extending isotopies of plane compacta

摘录 / EXCERPT: Crosscut extension conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-031784 | MATHDB-340138

Vaes-Wahl conjecture on weakly mixing Maharam extensions of Bernoulli actions

摘录 / EXCERPT: Let G be a countable group with non-vanishing first ℓ^2-cohomology. Equivalently, suppose that there is a function $f:G\to\mathbb{C}$ with $f(0)=0$, which is not in...

conjecture | open | MathDB 页面 | 原文线索

V2-031785 | MATHDB-340148

The multiplicative comparison conjecture for the mod p derived Hecke algebra

摘录 / EXCERPT: Multiplicative comparison conjecture. The cohomology algebra

conjecture | open | MathDB 页面 | 原文线索

V2-031786 | MATHDB-340163

Sphere coloring conjecture for discrete spheres

摘录 / EXCERPT: Sphere coloring conjecture. Every d-sphere has chromatic number $d+1$ or $d+2$.

conjecture | open | MathDB 页面 | 原文线索

V2-031787 | MATHDB-340164

Eulerian edge-refinement conjecture for sphere embeddings

摘录 / EXCERPT: Eulerian edge-refinement conjecture. It is possible to edge refine H, using only edges different from those of G, so that the modified H becomes Eulerian.

conjecture | open | MathDB 页面 | 原文线索

V2-031788 | MATHDB-340165

Eulerian edge-refinement conjecture for discrete spheres

摘录 / EXCERPT: Eulerian edge-refinement conjecture. d-spheres have edge refinements which are Eulerian.

conjecture | open | MathDB 页面 | 原文线索

V2-031789 | MATHDB-340182

Generalized Borel conjecture for aleph-zero-bounded groups

摘录 / EXCERPT: Generalized Borel conjecture. The cardinality of X is no larger than the weight of G:

conjecture | open | MathDB 页面 | 原文线索

V2-031790 | MATHDB-340187

The rough Baum-Connes conjecture for uniformly discrete metric spaces

摘录 / EXCERPT: Rough Baum-Connes conjecture. The map μ_u is an isomorphism. This conjecture asserts that uniform K-homology of the Rips complexes computes the K-theory of the uniform Roe...

conjecture | open | MathDB 页面 | 原文线索

V2-031791 | MATHDB-340194

Non-degeneracy conjecture for the α-metric on closed connected Legendrian submanifolds

摘录 / EXCERPT: Non-degeneracy conjecture. The α-metric δ_{α} on $\mathcal L(N)$ is non-degenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-031792 | MATHDB-340222

Main existence conjecture for pluriclosed flow

摘录 / EXCERPT: Let (M^{2n},g_0,J) be a compact complex manifold with pluriclosed metric, and let $\tau^*(\omega_0)$ be the formal cohomological existence time defined by the condition...

conjecture | open | MathDB 页面 | 原文线索

V2-031793 | MATHDB-340256

Semipositivity conjecture for divisors with topologically trivial normal bundle

摘录 / EXCERPT: Semipositivity conjecture. The line bundle $[Y]$ admits a C^∞ Hermitian metric with semi-positive curvature if and only if the pair (Y,X) is of type (β).

conjecture | open | MathDB 页面 | 原文线索

V2-031794 | MATHDB-340261

Vanishing of second derived limits for nerves of open covers

摘录 / EXCERPT: Let X be a locally compact separable metrizable space, and let N_α be the nerves of its open covers. Conjecture A. For each n, ... This conjecture asserts the vanishing of...

conjecture | open | MathDB 页面 | 原文线索

V2-031795 | MATHDB-340358

Deformation-retract conjecture for Abel-map fibers of splice quotient singularities

摘录 / EXCERPT: Deformation-retract conjecture. The fiber $c^{-1}({\mathcal O}(l'))$ is a topological deformation retract of ${\rm ECa}^{l'}(Z)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031796 | MATHDB-340369

Satellite homomorphism conjecture for the smooth concordance group

摘录 / EXCERPT: Let $\mathcal{C}$ denote a smooth or topological knot concordance group, and let a satellite operator be the self-map $P:\mathcal{C}\to\mathcal{C}$ induced by a knot pattern...

conjecture | open | MathDB 页面 | 原文线索

V2-031797 | MATHDB-340370

Infinite-rank image conjecture for satellite operators

摘录 / EXCERPT: Let $\mathcal{C}$ be a smooth or topological knot concordance group, and let $P:\mathcal{C}\to\mathcal{C}$ be a satellite operator. Call P non-constant if its image is not a sing...

conjecture | open | MathDB 页面 | 原文线索

V2-031798 | MATHDB-340402

The lifting conjecture for the singular Hochschild cohomology morphism

摘录 / EXCERPT: Let A be an algebra, let $C_{sg}(A,A)$ denote its singular Hochschild cochain algebra, and let $\operatorname{\mathsf{sg}}_{dg}(A)$ be the dg singularity category of A. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031799 | MATHDB-340414

The extended conjecture on spheres attached to constrained immersed-curve spaces

摘录 / EXCERPT: Extended conjecture. If $\bar{L}_{i,n}>\bar{D}_{i,n}$ for $i=1,2$, then a sphere $\mathbb{S}^k$ is attached to $\mathcal{L}^{+\kappa_0}_{-\kappa_0}(\pmb{I},\pmb{Q})$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-031800 | MATHDB-340435

Lovász and Szegedy's finite-dimension conjecture for typical vertices

摘录 / EXCERPT: A graphon is a measurable symmetric function $W:[0,1]^2\to[0,1]$. It is finitely forcible if it is uniquely determined, up to weak isomorphism, by the densities of finitely many gr...

conjecture | open | MathDB 页面 | 原文线索

V2-031801 | MATHDB-340438

Sharifi's reciprocity-map inverse conjecture

摘录 / EXCERPT: Fix an even character $\theta:(\mathbb{Z}/Np\mathbb{Z})^{\times}\rightarrow\overline{\mathbb{Q}}_p^{\times}$ and let ω be the Teichmüller character. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-031802 | MATHDB-340441

Tropical-division conjecture for Fukaya-Seidel categories

摘录 / EXCERPT: Tropical-division conjecture. $\mathcal{F}_{\Delta^t}(W_\Sigma)$ is derived equivalent to the Fukaya-Seidel category of W_Σ.

conjecture | open | MathDB 页面 | 原文线索

V2-031803 | MATHDB-340442

Anticanonical tropical-division conjecture for line bundles

摘录 / EXCERPT: Anticanonical tropical-division conjecture. For an appropriate choice of toric Kähler form and any small δ, Δ^t is adapted to Σ_{ac} and...

conjecture | open | MathDB 页面 | 原文线索

V2-031804 | MATHDB-340443

Line-bundles-are-thimbles conjecture

摘录 / EXCERPT: Line-bundles-are-thimbles conjecture. There is a Lagrangian thimble mirror to each line bundle on X.

conjecture | open | MathDB 页面 | 原文线索

V2-031805 | MATHDB-340454

The pro-nilfactor conjecture for topological characteristic factors along arithmetic progressions

摘录 / EXCERPT: Pro-nilfactor conjecture. If (X,T) is a distal minimal system, then its maximal $(d-1)$-step pro-nilfactor is its topological characteristic factor along arithmetic progressions...

open_problem | open | MathDB 页面 | 原文线索

V2-031806 | MATHDB-340459

The flag lower bound conjecture for g-numbers

摘录 / EXCERPT: Let Δ be a flag homology $(d-1)$-sphere, and let $g_i(\Delta)$ denote its g-numbers. Flag lower bound conjecture. ... Equality holds if and only if Δ is the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-031807 | MATHDB-340500

Existence of a low-dimensional optimal solution for the multimarginal mass transportation problem

摘录 / EXCERPT: Let M denote the support of an optimal solution to the multimarginal mass transportation problem with cost function $c(x,y,z)=xyz$ and the prescribed marginals considered in the...

conjecture | open | MathDB 页面 | 原文线索

V2-031808 | MATHDB-340508

The Ginzburg-Kaledin quantized Hochschild-orbifold cohomology conjecture

摘录 / EXCERPT: Let $\mathscr{X}$ be a symplectic quotient orbifold and let $\mathscr{A}$ be an appropriate quantization of $\mathscr{X}$. Ginzburg-Kaledin conjecture. There is an algebra isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-031809 | MATHDB-340509

The unfiltered Hochschild-orbifold cohomology conjecture for projective symplectic quotient orbifolds

摘录 / EXCERPT: Let $\mathscr{X}$ be a projective symplectic quotient orbifold, and write $H\!H^\bullet(\mathscr{X})$ for its Hochschild cohomology and $H^\bullet_\mathrm{orb}(\mathscr{X})$ for i...

conjecture | open | MathDB 页面 | 原文线索

V2-031810 | MATHDB-340548

The Knight Move Conjecture for Khovanov homology

摘录 / EXCERPT: Let K be a knot. Its Khovanov homology over $\mathbb{Q}$ is a bigraded vector space, and write s for Rasmussen's invariant. A pawn move piece is ... and a knight move piece is .....

conjecture | open | MathDB 页面 | 原文线索

V2-031811 | MATHDB-340549

The Poincare-series reformulation of the Knight Move Conjecture

摘录 / EXCERPT: Let K be a knot, let $\operatorname{Kh}(K)(t,q)$ denote its Khovanov Poincare series, and let s be Rasmussen's invariant. A Laurent polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-031812 | MATHDB-340550

Well-balanced classes are asymptotically negligible for repetitive finite-local-complexity tilings

摘录 / EXCERPT: Well-balanced-class conjecture. Every well-balanced class in

conjecture | open | MathDB 页面 | 原文线索

V2-031813 | MATHDB-340554

Takahashi's homological mirror symmetry conjecture for invertible polynomials

摘录 / EXCERPT: Let $f\in S^n$ be an invertible polynomial, and let $\widetilde{f}$ denote its Berglund-Hübsch transpose. Let L_f be the maximal grading of f, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031814 | MATHDB-340555

Lekili-Ueda's tilting-object conjecture for maximally graded matrix factorizations

摘录 / EXCERPT: Let $f\in S^n$ be an invertible polynomial, and let L_f be its maximal grading. Let $\operatorname{HMF}^{L_f}_{S^n}(f)$ be the homotopy category of maximally graded matrix factor...

conjecture | open | MathDB 页面 | 原文线索

V2-031815 | MATHDB-340564

NP-hardness conjecture for unknotting number

摘录 / EXCERPT: Unknotting-number conjecture. The textsc{unknotting number} problem is NP-hard.

conjecture | open | MathDB 页面 | 原文线索

V2-031816 | MATHDB-340572

Möbius function and homotopy type conjecture for weak order on monotone triangles

摘录 / EXCERPT: Homotopy-type conjecture. The complex $\Delta(T',T)$ is contractible unless

conjecture | open | MathDB 页面 | 原文线索

V2-031817 | MATHDB-340583

The defect-zero characterization of strongly quasipositive braids

摘录 / EXCERPT: Defect-zero characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031818 | MATHDB-340633

L-space conjecture of Boyer, Gordon and Watson

摘录 / EXCERPT: Boyer-Gordon-Watson L-space conjecture. M is not an L-space if and only if $\pi_1(M)$ is left-orderable.

conjecture | open | MathDB 页面 | 原文线索

V2-031819 | MATHDB-340643

Dwyer-Hess-Knudsen additivity conjecture for skew little cubes

摘录 / EXCERPT: Let $G\to GL(m)$ and $H\to GL(n)$ be dilation representations. The operads $\mathcal C_m^G$ and $\mathcal C_n^H$ are the skew little-cube operads associated to these representation...

conjecture | open | MathDB 页面 | 原文线索

V2-031820 | MATHDB-340646

Genericity of shadowing for locally connected continua

摘录 / EXCERPT: Genericity conjecture. $\mathcal{T}(X)$ is generic in $\mathcal{C}(X)$ and $\mathcal{S}(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031821 | MATHDB-340651

Cohomological divisibility conjecture for local BPS invariants

摘录 / EXCERPT: Assume that S is a del Pezzo surface and that $\mathcal{M}_\beta$ is the moduli space of one-dimensional stable sheaves on S with class β and holomorphic Euler characteri...

conjecture | open | MathDB 页面 | 原文线索

V2-031822 | MATHDB-340724

The reconstruction conjecture for Vassiliev 1-cocycles from the Kontsevich integral

摘录 / EXCERPT: Let α be a Vassiliev 1-cocycle of degree m, and let $\tilde{\alpha}$ be the 1-cocycle obtained by composing the weight system associated to the principal part of...

conjecture | open | MathDB 页面 | 原文线索

V2-031823 | MATHDB-340778

Homology of cyclic groups with trivial coefficients

摘录 / EXCERPT: Cyclic-group homology conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-031824 | MATHDB-340790

Bruin-Kolyada-Snoha conjecture on minimal maps of the pseudo-circle

摘录 / EXCERPT: Let the pseudo-circle be R.H. Bing's pseudo-circle, an infinite continuum that admits minimal homeomorphisms. A map is noninvertible when it is not a homeomorphism, and a map is mi...

conjecture | open | MathDB 页面 | 原文线索

V2-031825 | MATHDB-340807

Noncentral component conjecture for nondegenerate spherical curves

摘录 / EXCERPT: Noncentral component conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-031826 | MATHDB-340808

Simple connectivity conjecture for components of nondegenerate spherical-curve spaces

摘录 / EXCERPT: Simple connectivity conjecture. For every $z\in\operatorname{CG}_{n+1}$, every connected component of ${\cal L}_n(z)$ is simply connected.

conjecture | open | MathDB 页面 | 原文线索

V2-031827 | MATHDB-340838

Greek-letter family red-shift conjecture for algebraic K-theory

摘录 / EXCERPT: Let R be a commutative ring spectrum. Say that R detects the n-th Greek letter family if each element $\alpha^{(n)}_k$ is nontrivial in the image of the unit map ... Suppose tha...

conjecture | open | MathDB 页面 | 原文线索

V2-031828 | MATHDB-340844

Internal-symmetry homotopy classification conjecture for topological phases

摘录 / EXCERPT: Internal-symmetry classification conjecture. The classification of SPT or SET topological phases in d dimensions with internal symmetry G is given by homotopy classes of maps

conjecture | open | MathDB 页面 | 原文线索

V2-031829 | MATHDB-340845

Crystalline equivalence principle for topological phases

摘录 / EXCERPT: Crystalline equivalence conjecture. The classification of SPT or SET topological phases in d dimensions with spatial symmetry G acting on X is given by homotopy classes of ma...

conjecture | open | MathDB 页面 | 原文线索

V2-031830 | MATHDB-340846

Generalized-cohomology conjecture for invertible topological phases

摘录 / EXCERPT: Generalized-cohomology conjecture. The spaces Θ_d can be chosen to satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-031831 | MATHDB-340851

Dunfield's orderability conjecture for Dehn fillings

摘录 / EXCERPT: Let M be a hyperbolic $\mathbb{Z}$-homology solid torus, and let $M(0)$ denote its longitudinal filling. Assume that $M(0)$ is hyperbolic and that its holonomy representation has...

conjecture | open | MathDB 页面 | 原文线索

V2-031832 | MATHDB-340871

Cabling conjecture for maximal Thurston-Bennequin number

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $K_{p,q}$ denote its (p,q)-cable. Write $\mathcal{L}(K_{p,q})$ for the Legendrian representatives in the knot type of $K_{p,q}$. Cabling con...

conjecture | open | MathDB 页面 | 原文线索

V2-031833 | MATHDB-340872

Uniform thickness criterion for fibered knots

摘录 / EXCERPT: Let $K\subset S^3$ be a fibered knot, and let ξ_K be the contact structure induced by an open-book decomposition of K. Let ξ_{Std} denote the standard contact structure o...

conjecture | open | MathDB 页面 | 原文线索

V2-031834 | MATHDB-340899

Trace and cotrace conjecture for sutured monopole Floer homology

摘录 / EXCERPT: Let (M,γ) be a balanced sutured manifold, and let $\mathcal{W}=(W=M\times[0,1],Z=\partial M\times[0,1],[\xi])$ be the trace sutured cobordism from...

conjecture | open | MathDB 页面 | 原文线索

V2-031835 | MATHDB-340907

The Alishahi-Dowlin spectral sequence conjecture from Khovanov homology to knot Floer homology

摘录 / EXCERPT: Alishahi-Dowlin spectral sequence conjecture. There is a spectral sequence from the Khovanov homology of K to the knot Floer homology of K.

conjecture | open | MathDB 页面 | 原文线索

V2-031836 | MATHDB-340908

The H_{1-1}-HFK_{2} identification conjecture

摘录 / EXCERPT: H_{1-1}-HFK_{2} conjecture. The total homology $H_{1-1}(K)$ is isomorphic to $\mathit{HFK}_{2}(K)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031837 | MATHDB-340963

The general higher semiadditivity conjecture

摘录 / EXCERPT: A presentable stable ∞-category is an ∞-category that is presentable and stable, and 1-semiadditive and ∞-semiadditive refer to the corresponding higher semi...

conjecture | open | MathDB 页面 | 原文线索

V2-031838 | MATHDB-340964

Conjecture on the Ising symmetry defect data

摘录 / EXCERPT: Let $\textbf{Ising}$ be the Ising fusion category, and consider its Deligne product $\textbf{Ising}\boxtimes\textbf{Ising}$ together with a $\mathbb{Z}_2$-crossed extension contain...

conjecture | open | MathDB 页面 | 原文线索

V2-031839 | MATHDB-340967

Existence of a measure of maximal entropy above the entropy threshold

摘录 / EXCERPT: Let f be a C^r surface diffeomorphism, where $1<r<\infty$, and let $h_{\operatorname{top}}(f)$ be its topological entropy. A measure maximizing entropy is an invariant probabil...

conjecture | open | MathDB 页面 | 原文线索

V2-031840 | MATHDB-340998

Nonvanishing conjecture for special values of new p-adic hypergeometric functions

摘录 / EXCERPT: Assume the condition referred to as, with i and j as in the preceding construction. Nonvanishing conjecture. Under this condition, ... This is motivated by the non-vanishing of t...

conjecture | open | MathDB 页面 | 原文线索

V2-031841 | MATHDB-341028

Double Hurwitz topological-recursion conjecture

摘录 / EXCERPT: Let d be a positive integer, let $q_1,\ldots,q_d$ be parameters, and set ... Consider the rational spectral curve ... Let $DH_{g,n}(\mu_1,\ldots,\mu_n)$ denote the associated double...

conjecture | open | MathDB 页面 | 原文线索

V2-031842 | MATHDB-341029

Double Hurwitz polynomial-structure conjecture

摘录 / EXCERPT: Let $F_{g,n}$ be the free-energy generating functions associated with the double Hurwitz numbers, let $a_1,\ldots,a_d$ be the branch points of the spectral curve, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031843 | MATHDB-341037

Formal isotopy detection by contact push-offs

摘录 / EXCERPT: Let n be a positive integer, and let (S^{2n+1},ξ_st) be the standard contact sphere. Let $\Lambda_0,\Lambda_1\subseteq(S^{2n+1},\xi_\text{st})$ be smooth Legendrian su...

conjecture | open | MathDB 页面 | 原文线索

V2-031844 | MATHDB-341038

Non-simplicity of formal contact isotopy classes of isocontact embeddings

摘录 / EXCERPT: Let n be a positive integer, and let (S^{2n+1},ξ_st) be the standard contact sphere. Consider isocontact embeddings of $(\partial T^*S^n,\xi_\text{st})$ into...

conjecture | open | MathDB 页面 | 原文线索

V2-031845 | MATHDB-341070

The 3-ball conjecture for self-affine tiles

摘录 / EXCERPT: 3-ball conjecture. Every self-affine tile satisfying the conditions of Theorem is homeomorphic to a 3-ball.

conjecture | open | MathDB 页面 | 原文线索

V2-031846 | MATHDB-341081

Crystalline Chebotarëv density conjecture

摘录 / EXCERPT: Let U be the curve, let K be the specified finite totally ramified extension, and let ${\mathcal{F}}$ be a convergent F-isocrystal on U. Fix $u\in U({\mathbb{F}}_{q^e})$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-031847 | MATHDB-341082

Component-density variant of the crystalline Chebotarëv conjecture

摘录 / EXCERPT: Let ${\mathcal{F}}$ be a convergent F-isocrystal on the curve U, fix $u\in U({\mathbb{F}}_{q^e})$, and let $\operatorname{Gr}({\mathcal{F}}/U,u)$ be its monodromy group over...

conjecture | open | MathDB 页面 | 原文线索

V2-031848 | MATHDB-341083

Parabolic restriction conjecture for crystalline monodromy groups

摘录 / EXCERPT: Let $f\colon V\hookrightarrow U$ be an open sub-curve, let $u\in V({\mathbb{F}}_{q^e})$ be a base point, and let ${\mathcal{F}}$ be a convergent F-isocrystal on U. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031849 | MATHDB-341106

Topological protection conjecture for edge modes in perturbed honeycomb structures

摘录 / EXCERPT: Let ${\mathscr{P}}_\delta[\zeta]:H^2[\zeta]\to L^2[\zeta]$ be the family of edge operators, with Dirac points $(\xi_\star^A,E_\star)$ and $(\xi_\star^B,E_\star)$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-031850 | MATHDB-341119

Conjecture on the limiting empirical mass distribution of red leaves

摘录 / EXCERPT: Red-leaf empirical-measure conjecture. Under $\mathbb{P}_{x_t}$, this random measure converges in law for the topology of vague convergence, and its limiting law does not depend on...

conjecture | open | MathDB 页面 | 原文线索

V2-031851 | MATHDB-341132

The sutured ECH-Heegaard Floer correspondence

摘录 / EXCERPT: Let $(M,\Gamma,\xi=\ker\alpha)$ be a sutured contact manifold, let $A\in H_1(M;\mathbb{Z})$, and let $\mathfrak{s}_\xi$ be the relative Spin^c structure determined by ξ. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-031852 | MATHDB-341136

Suzuki's unknotting conjecture for 2-knots

摘录 / EXCERPT: Suzuki's unknotting conjecture. Let G be a 2-knot in S^4 with unknotted equator. Then G is smoothly unknotted.

conjecture | open | MathDB 页面 | 原文线索

V2-031853 | MATHDB-341150

Sandon's translated-points conjecture for contactomorphisms

摘录 / EXCERPT: Let (V,ξ) be a closed contact manifold, where ξ is a cooriented contact structure, and let $\operatorname{Cont}_0(V,\xi)$ denote the identity component of its contactomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-031854 | MATHDB-341160

Klee-Novik's equality characterization conjecture for balanced 3-manifolds

摘录 / EXCERPT: Let Δ be a connected 3-dimensional balanced $\mathbb{F}$-homology manifold. The Klee-Novik conjecture. Equality ... holds if and only if Δ belongs to the balanced Wal...

conjecture | open | MathDB 页面 | 原文线索

V2-031855 | MATHDB-341173

Gusein-Zade's Betti-density conjecture for quasiperiodic functions

摘录 / EXCERPT: Let f be a quasiperiodic function, and let $b_f^k(c)$ denote the density of its k-th Betti number at level c, defined by averaging the corresponding Betti number over expandi...

conjecture | open | MathDB 页面 | 原文线索

V2-031856 | MATHDB-341174

Dynnikov's stability-zone size and topology conjecture

摘录 / EXCERPT: Let $M_g^2\subset{\mathbb T}^3$ be a generic embedded surface, let ℓ be a soul label, and let ${\cal D}_{\ell}(M_g^2)$ denote its stability region; write $\|\ell\|$ for the no...

conjecture | open | MathDB 页面 | 原文线索

V2-031857 | MATHDB-341177

Novikov-Dynnikov generic TCI-stability conjecture

摘录 / EXCERPT: Let ε_c be a level hypersurface of a smooth quasiperiodic function on ${\mathbb T}^4$, and let $\Pi\in G_{4,2}(\mathbb R)$ be a 2-plane. The level is topologically co...

conjecture | open | MathDB 页面 | 原文线索

V2-031858 | MATHDB-341198

Closure-interval conjecture for powers of two and irrational rotations

摘录 / EXCERPT: Let α be an irrational number and d be a real number. Consider the sequence $\{2^n d+n\alpha\mod 1\}_{n\geq 1}$ in $[0,1]$. Closure-interval conjecture. The topological cl...

conjecture | open | MathDB 页面 | 原文线索

V2-031859 | MATHDB-341200

Connected excision conjecture for sutured monopole Floer contact elements

摘录 / EXCERPT: Connected excision conjecture. Theorem 1 continues to hold when the disjoint union $(M_1\sqcup M_2,gamma_1\cupgamma_2)$ is replaced by the connected balanced sutured manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-031860 | MATHDB-341201

Weak symplectic cobordism conjecture for monopole Floer contact elements

摘录 / EXCERPT: Weak symplectic cobordism conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-031861 | MATHDB-341202

Trivial-torsion conjecture for four-longitude sutured monopole Floer homology

摘录 / EXCERPT: Trivial-torsion conjecture. The torsion group G_{tor} in Lemma 4 is trivial:

conjecture | open | MathDB 页面 | 原文线索

V2-031862 | MATHDB-341215

Separation of volume classes for dense representations of the free group

摘录 / EXCERPT: Let F_2 be the free group of rank 2, let $\operatorname{PSL}_2\mathbb{C}$ be the group of orientation-preserving isometries of hyperbolic 3-space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031863 | MATHDB-341218

The fiberwise Fuller trace obstruction conjecture

摘录 / EXCERPT: Let B be a finite-dimensional cell complex, and let $E\to B$ be a smooth closed manifold bundle with fiber M of dimension at least $3+\dim B$. Let $f\colon E\to E$ be a family...

conjecture | open | MathDB 页面 | 原文线索

V2-031864 | MATHDB-341219

The fiberwise Fuller trace counterexample conjecture

摘录 / EXCERPT: Let B be a base space supporting a family of endomorphisms $f\colon E\to E$ over B. Write $R_B(f)$ and $R_B(f^2)$ for the corresponding fiberwise Reidemeister traces, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031865 | MATHDB-341221

The period-map local-homeomorphism conjecture for q-quadratic differentials

摘录 / EXCERPT: Let $\log {\mathbf{S}}_\bigtriangleup$ be a topological log surface with its hat homology group $\widehat{\operatorname{\bf H}}_{}(\log {\mathbf{S}}_\bigtriangleup)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031866 | MATHDB-341262

Enhancement uniqueness conjecture for complicial t-structures

摘录 / EXCERPT: Enhancement uniqueness conjecture. The homotopy category $h\mathcal{C}$ admits a unique ∞-categorical enhancement.

conjecture | open | MathDB 页面 | 原文线索

V2-031867 | MATHDB-341266

The dimension-three criterion for infinitely many H-trivial line bundles on toric DM stacks

摘录 / EXCERPT: The dimension-three H-triviality conjecture. There are infinitely many H-trivial line bundles on $\mathbb{P}_{\mathbf{\Sigma}}$ if and only if there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-031868 | MATHDB-341284

The visibility-end conjecture for finite-volume nonpositive-curvature manifolds

摘录 / EXCERPT: Visibility-end conjecture. The fundamental group of each end of M is almost nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-031869 | MATHDB-341299

Logarithmic-decay conjecture for slope filtrations

摘录 / EXCERPT: Let M be an irreducible overconvergent F-isocrystal over $\operatorname{Spec}(F)$. Suppose its slope filtration is ... with graded pieces of slopes...

conjecture | open | MathDB 页面 | 原文线索

V2-031870 | MATHDB-341325

Jones's conjecture on level-(p,q) Furuta-Mahowald classes

摘录 / EXCERPT: Let a level-(p,q) Furuta--Mahowald class mean the class considered in the paper, with p and q its level parameters. Jones's conjecture. For $p\geq 2$, a level-(p,q) Furuta-...

conjecture | open | MathDB 页面 | 原文线索

V2-031871 | MATHDB-341326

Bredon-Löffler-Mahowald-Ravenel conjecture on the Mahowald invariant

摘录 / EXCERPT: Let α be a non-equivariant stable homotopy class of positive degree, and let $M(\alpha)$ denote its Mahowald invariant.
Bredon-Löffler-Mahowald-Ravenel conjecture. The Mahow...

conjecture | open | MathDB 页面 | 原文线索

V2-031872 | MATHDB-341390

Well-defined 3D TQFT conjecture for the assignments of the partition function

摘录 / EXCERPT: Well-defined 3D TQFT conjecture. The above assignments of the values of Z are part of the data of a well-defined 3D TQFT.

conjecture | open | MathDB 页面 | 原文线索

V2-031873 | MATHDB-341428

Kalai-Sarkaria conjecture on algebraic shifting and embeddability

摘录 / EXCERPT: Kalai-Sarkaria conjecture. If K admits an embedding into S^{r}, then

conjecture | open | MathDB 页面 | 原文线索

V2-031874 | MATHDB-341451

The conjecture that closed aspherical Alexandrov 3-spaces are 3-manifolds

摘录 / EXCERPT: An Alexandrov 3-space is a three-dimensional Alexandrov space; it is aspherical when all its higher homotopy groups vanish, equivalently when its universal cover is contractible....

conjecture | open | MathDB 页面 | 原文线索

V2-031875 | MATHDB-341459

Cohomological realization of the Lusztig-Vogan bijection

摘录 / EXCERPT: Suppose we are in the reductive case. Let $\boldsymbol{ u}\in-\boldsymbol{X}^+$, and let $({\mathscr{O}}_{\boldsymbol{ u}},\boldsymbol{\rho}_{\boldsymbol{ u}})\in\Xi$ be the pair c...

conjecture | open | MathDB 页面 | 原文线索

V2-031876 | MATHDB-341466

Simpson's conjecture on strict composition and exchange laws

摘录 / EXCERPT: Simpson's conjecture. Every weak $infty$-category is equivalent to an $infty$-category in which composition and exchange laws are strict and only the unit laws are allowed to hold...

conjecture | open | MathDB 页面 | 原文线索

V2-031877 | MATHDB-341467

Presentability conjecture for associative higher categories

摘录 / EXCERPT: Presentability conjecture. Every presented associative n-category is naturally an associative n-category, up to some arbitrary (contractible) choices for morphisms witnessing h...

conjecture | open | MathDB 页面 | 原文线索

V2-031878 | MATHDB-341468

Presentability equivalence for associative n-categories

摘录 / EXCERPT: Presentability equivalence conjecture. Every associative n-category is equivalent to a presented associative n-category, for a reasonable notion of equivalence between associat...

conjecture | open | MathDB 页面 | 原文线索

V2-031879 | MATHDB-341469

Associative and weak n-category equivalence conjecture

摘录 / EXCERPT: Associative-weak equivalence conjecture. The theory of associative n-categories is equivalent to a theory weak n-categories.

conjecture | open | MathDB 页面 | 原文线索

V2-031880 | MATHDB-341470

Every associative n-category is presented

摘录 / EXCERPT: All-categories-presented conjecture. For any reasonable notion of equivalence of associative n-categories, every associative n-category is equivalent to a presented associative...

conjecture | open | MathDB 页面 | 原文线索

V2-031881 | MATHDB-341471

Manifold-diagram representation of morphisms

摘录 / EXCERPT: Manifold-diagram representation conjecture. Let C be a presented associative n-category. The mapping from labelled singular cubes to colored cubes descends to an injection of s...

conjecture | open | MathDB 页面 | 原文线索

V2-031882 | MATHDB-341472

CW-complex reconstruction from associative higher groupoids

摘录 / EXCERPT: CW reconstruction conjecture. There is a non-unique $infty$-groupoid X such that $X_k=I^X_k$ and, for each $h e I^X_k$, $M(h)$ is equivalent to $\llbracket h\rrbracket$. Converse...

conjecture | open | MathDB 页面 | 原文线索

V2-031883 | MATHDB-341481

Przytycki's Homflypt skein module conjecture for submanifolds of rational homology spheres

摘录 / EXCERPT: Let M be a submanifold of a rational homology sphere. Assume that M contains no closed, oriented incompressible surface. Let ${\cal S}_3(M)$ denote its Homflypt skein module, I...

conjecture | open | MathDB 页面 | 原文线索

V2-031884 | MATHDB-341520

Orbifold Property O multiplicity conjecture

摘录 / EXCERPT: Orbifold Property O multiplicity conjecture. The eigenvalue ρ belongs to $\operatorname{Spec}(\hat c_1)$ with multiplicity equal to the number of conjugacy classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-031885 | MATHDB-341521

Orbifold spectral rotational symmetry conjecture

摘录 / EXCERPT: Orbifold rotational symmetry conjecture. The rotational symmetry group of $\operatorname{Spec}(\hat c_1)$ is cyclic of order equal to the Fano index of $\mathcal{X}$.

conjecture | open | MathDB 页面 | 原文线索

V2-031886 | MATHDB-341538

Krupiński-Pillay's amenability conjecture for topological groups

摘录 / EXCERPT: Let $G(M)$ be a topological group. It is amenable when every $G(M)$-flow admits a $G(M)$-invariant Borel probability measure. Krupiński-Pillay's conjecture. If $G(M)$ is amenable,...

conjecture | open | MathDB 页面 | 原文线索

V2-031887 | MATHDB-341547

Connectedness conjecture for entropy level sets in the unimodal region

摘录 / EXCERPT: Let $\mathcal{M}_2(\Bbb{R})$ be the moduli space of real quadratic rational maps, let $h_\Bbb{R}$ be its real entropy function, and let the unimodal region denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-031888 | MATHDB-341548

Disconnected entropy level sets in the (+-+) bimodal region

摘录 / EXCERPT: Let $\mathcal{M}_2(\Bbb{R})$ be the moduli space of real quadratic rational maps, let $h_\Bbb{R}$ be its real entropy function, and let the $(+-+)$-bimodal region denote the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-031889 | MATHDB-341549

Connected entropy level sets across adjacent unimodal and (-+-) regions

摘录 / EXCERPT: Let $\mathcal{M}_2(\Bbb{R})$ be the moduli space of real quadratic rational maps, let $h_\Bbb{R}$ be its real entropy function, and consider the adjacent unimodal and $(-+-)$-bimod...

conjecture | open | MathDB 页面 | 原文线索

V2-031890 | MATHDB-341569

The Barto-Opršal-Pinsker tractability dichotomy conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Barto-Opršal-Pinsker dichotomy conjecture. Exactly one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-031891 | MATHDB-341621

Conjecture on hyperbolic density in real-polynomial isentropes

摘录 / EXCERPT: Hyperbolic-density conjecture. In the space of polynomials of degree $d>2$ with all critical points real, there are no isentropes of entropy

conjecture | open | MathDB 页面 | 原文线索

V2-031892 | MATHDB-341629

Torsion-volume growth lower bound for biperiodic alternating links

摘录 / EXCERPT: Let $\mathcal L$ be a hyperbolic biperiodic alternating link, and let $\{X(L_n)\}$ be the associated sequence of finite-volume covers described above. Write $TH_1(X(L_n))$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-031893 | MATHDB-341640

Conjecture on band number for ribbon surfaces of fixed genus

摘录 / EXCERPT: For a knot K, let its ribbon genus be the minimum genus of a ribbon surface bounding it, and let $b(K)$ denote its band number. Fixed-genus band number conjecture. For every genu...

conjecture | open | MathDB 页面 | 原文线索

V2-031894 | MATHDB-341666

Conca-De Negri-Gorla local cohomology conjecture for Cartwright-Sturmfels ideals

摘录 / EXCERPT: Let $A=\mathbb{K}[x_1,\dots,x_n]$ be a polynomial ring over a field $\mathbb{K}$, let $I\subseteq A$ be a $\mathbb{Z}^m$-graded Cartwright-Sturmfels ideal, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-031895 | MATHDB-341678

The chromatic splitting conjecture for the zero-line of the descent spectral sequence

摘录 / EXCERPT: Zero-line chromatic splitting conjecture. If p is odd, or if $p=2$ and n is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-031896 | MATHDB-341679

The strong chromatic splitting conjecture

摘录 / EXCERPT: Let E_{n-1}-local spectra be localized with respect to Morava E-theory, let S_p^0 be the p-complete sphere spectrum, and let ... denote the exterior-algebra-type wedge of R...

conjecture | open | MathDB 页面 | 原文线索

V2-031897 | MATHDB-341680

The finiteness conjecture for the exotic Picard group

摘录 / EXCERPT: Hopkins's exotic Picard-group conjecture. The group κ_n is a finite p-group. The source notes that $\kappa_n=0$ when $2(p-1)\geq n^2$, while it is nontrivial in many other...

conjecture | open | MathDB 页面 | 原文线索

V2-031898 | MATHDB-341700

Rationality and nonnegativity conjecture for Serre dimensions

摘录 / EXCERPT: Rationality and nonnegativity conjecture. The two Serre dimensions are rational numbers, and the upper Serre dimension is nonnegative:

conjecture | open | MathDB 页面 | 原文线索

V2-031899 | MATHDB-341702

Vanishing upper Serre dimension conjecture for finite-dimensional algebras

摘录 / EXCERPT: Vanishing upper Serre dimension conjecture. The upper Serre dimension satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-031900 | MATHDB-341703

Pillay-Yıldırım conjecture on weakly generic and almost periodic types

摘录 / EXCERPT: Let T be an NIP theory, let M be a model of T, and let G be a definably amenable group definable in M. Write $S_G(M)$ for the space of types over M concentrating on...

conjecture | open | MathDB 页面 | 原文线索

V2-031901 | MATHDB-341721

The basin equality conjecture for sectional-hyperbolic attracting sets

摘录 / EXCERPT: Basin equality conjecture. For any sectional-hyperbolic attracting set Λ, the topological basin of attraction coincides with the family of stable manifolds through the poin...

conjecture | open | MathDB 页面 | 原文线索

V2-031902 | MATHDB-341727

The Baire-banalytic characterization conjecture for Hausdorff topological groups

摘录 / EXCERPT: Let X be a Hausdorff topological group. Baire-banalytic characterization conjecture. The group X is Polish if and only if X is a Baire space and a banalytic space. This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-031903 | MATHDB-341729

The presentation-form conjecture for D(2)-complexes with quaternion fundamental group

摘录 / EXCERPT: Presentation-form conjecture. Every such Y is homotopy equivalent to a presentation of Q_{4n} of the form

conjecture | open | MathDB 页面 | 原文线索

V2-031904 | MATHDB-341743

Independence-of-chi conjecture for DT4 invariants of irreducible curve classes

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau 4-fold, let $\beta\in H_2(X,\mathbb{Z})$ be an irreducible curve class, and let $\gamma_1,\ldots,\gamma_l$ be integral cohomology classe...

conjecture | open | MathDB 页面 | 原文线索

V2-031905 | MATHDB-341767

Correspondence homomorphism conjecture for Suslin homology

摘录 / EXCERPT: Correspondence homomorphism conjecture. (i) For any $\alpha\in c_{\mathrm{equi}}(X\times Y/X,0)$, tr_{α} satisfies (M1). (ii) For any...

conjecture | open | MathDB 页面 | 原文线索

V2-031906 | MATHDB-341772

Local systolic-diastolic inequality for odd-symplectic forms

摘录 / EXCERPT: Let Σ be the underlying manifold, let $C\in H^2_{\mathrm{dR}}(\Sigma)$, and let Ω_* be a Zoll odd-symplectic form with cohomology class C. For forms...

conjecture | open | MathDB 页面 | 原文线索

V2-031907 | MATHDB-341822

The projective Lorentzian space is a closed Euclidean ball

摘录 / EXCERPT: Let H^d_n be the space of degree-d homogeneous polynomials in n variables, and let L^d_n be the cone of Lorentzian polynomials in H^d_n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-031908 | MATHDB-341848

The noncommutative Hodge-to-de Rham degeneration conjecture

摘录 / EXCERPT: Hodge-to-de Rham degeneration conjecture. The negative cyclic homology $HC_*^-(A)$ is a locally free $\mathbb{K}[[u]]$-module of finite rank.

conjecture | open | MathDB 页面 | 原文线索

V2-031909 | MATHDB-341853

Freyd's generating hypothesis for finite spectra

摘录 / EXCERPT: Generating hypothesis. The map f is nullhomotopic.

conjecture | open | MathDB 页面 | 原文线索

V2-031910 | MATHDB-341854

Hopkins's chromatic splitting conjecture

摘录 / EXCERPT: Chromatic splitting conjecture. If X is the p-completion of a finite spectrum, then a splitting γ_n exists for all n.

conjecture | open | MathDB 页面 | 原文线索

V2-031911 | MATHDB-341867

Expected graded dimension conjecture for marked vinyl graphs

摘录 / EXCERPT: Let Γ be a marked vinyl graph, and let ${\mathcal{S}}_0(\Gamma)$ be the associated homological object with graded dimension $\dim_q$, while ${\mathrm{egd}}(\Gamma)$ denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-031912 | MATHDB-341877

Schleimer's bounded translation distance conjecture for fibered knots

摘录 / EXCERPT: Schleimer's conjecture. There is a constant $t(M)$ such that the monodromy of every fibered knot $K\subset M$ has translation distance in the arc complex of the fiber at most...

conjecture | open | MathDB 页面 | 原文线索

V2-031913 | MATHDB-341884

The Cosmetic Surgery Conjecture

摘录 / EXCERPT: Let Y be a closed, oriented three-manifold and K a knot whose exterior is boundary-irreducible. A purely cosmetic surgery is a pair of distinct surgery slopes on K producing...

conjecture | open | MathDB 页面 | 原文线索

V2-031914 | MATHDB-341932

Log weak Lefschetz conjecture for log isocrystalline cohomologies

摘录 / EXCERPT: Log weak Lefschetz conjecture. Assume that $\mathcal O_Y(E)$ is ample. Then the morphism above is a filtered isomorphism with respect to P if $q\leq d-2$ and strictly injective f...

conjecture | open | MathDB 页面 | 原文线索

V2-031915 | MATHDB-341953

The cohomological concentration conjecture for the flag complex

摘录 / EXCERPT: Let G be a finite abelian group, let $\mathcal G_\bullet$ range over flags of subgroups of G, and let $\mathcal M_n^-(G)$ denote the antisymmetric symbol module. The flags and...

conjecture | open | MathDB 页面 | 原文线索

V2-031916 | MATHDB-341954

The Laplacian invertibility conjecture for the flag complexes

摘录 / EXCERPT: The Laplacian invertibility conjecture. On each term of the complex, the operator $\bm{\Delta}$ is invertible after tensoring with $\mathbb Q$, except on the first term...

conjecture | open | MathDB 页面 | 原文线索

V2-031917 | MATHDB-341955

The cuspidal, interior, and primitive cohomology conjecture

摘录 / EXCERPT: Let G be a nontrivial finite abelian group and $n\ge 1$. In the Borel-Moore homology $H_n^{BM}(\mathbb X_n,\mathcal L_{n,G})$, let $H_{n,\int}(\mathbb X_n,\mathcal L_{n,G})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-031918 | MATHDB-341962

Galatius-Randal-Williams' conjecture on the class of the homotopy sphere Sigma_Q

摘录 / EXCERPT: Let $n\geq 3$ be odd, and let Σ_Q be the homotopy sphere obtained as the boundary of the manifold Q in the relevant bordism group. Its class $[\Sigma_Q]$ lies in...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-031919 | MATHDB-341966

Yoshino's Hilbert-series formula conjecture for tensor products

摘录 / EXCERPT: Let R be a Cohen-Macaulay local ring, let M be a perfect R-module, and let N be a Buchsbaum R-module of maximal dimension. Write $\beta_j(M)$ for the jth Betti number o...

conjecture | open | MathDB 页面 | 原文线索

V2-031920 | MATHDB-341994

Tuschmann's finite-action conjecture for almost non-negatively curved manifolds

摘录 / EXCERPT: Let M be a manifold with almost non-negative sectional curvature, abbreviated ANSC. The fundamental group $\pi_1(M)$ acts on the homotopy group $\pi_2(M)$ by deck transformations...

conjecture | open | MathDB 页面 | 原文线索

V2-031921 | MATHDB-342008

Strong rationality conjecture for cohomological BPS invariants

摘录 / EXCERPT: Let Q be the quiver associated with a noncommutative crepant resolution $A=\mathbb{C}\{Q,W\}$ of the threefold setup, let C be the flopping curve, and let $\Phi(0,1)$ be its di...

conjecture | open | MathDB 页面 | 原文线索

V2-031922 | MATHDB-342021

Fiber homeomorphism conjecture for maps to totally nonnegative spaces

摘录 / EXCERPT: Let $f_{(i_1,\dots,i_d)}$ be one of the maps considered in the paper, let $p\in Y_w^o$, and let $\Delta((i_1,\dots,i_d),w)$ denote the corresponding subword complex. Consider the c...

conjecture | open | MathDB 页面 | 原文线索

V2-031923 | MATHDB-342029

Polynomial fertility conjecture for reduced knot shadows

摘录 / EXCERPT: Let S be a reduced shadow with n crossings. Say that S resolves into a knot type if some assignment of overcrossings and undercrossings to its crossings produces a diagram of...

conjecture | open | MathDB 页面 | 原文线索

V2-031924 | MATHDB-342108

Finiteness conjecture for Lagrangian ribbon disk fillings

摘录 / EXCERPT: Let L be a Legendrian ribbon knot in (S^3,ξ_{st}). A Lagrangian ribbon disk is a Lagrangian disk in B^4 with ribbon singularities. Finiteness conjecture. Then L bo...

conjecture | open | MathDB 页面 | 原文线索

V2-031925 | MATHDB-342119

Kerz-Esnault-Wittenberg p-adic cycle-class conjecture

摘录 / EXCERPT: Let A be a henselian discrete valuation ring of characteristic zero with residue field of characteristic p and function field K. Let X be smooth and projective of relative...

conjecture | open | MathDB 页面 | 原文线索

V2-031926 | MATHDB-342138

Exponential torsion growth conjecture for arithmetic hyperbolic manifolds

摘录 / EXCERPT: Let $d=2n+1$ and let $\Gamma\subset \operatorname{SO}_0(d,1)$ be an arithmetic subgroup. Let $\{\Gamma_i\}$ be a cusp-uniform sequence of torsion free congruence subgroups of...

conjecture | open | MathDB 页面 | 原文线索

V2-031927 | MATHDB-342165

The amplitude inequality for finite-dimensional injective dimension objects over CDGAs

摘录 / EXCERPT: Amplitude inequality. For every such object M,

conjecture | open | MathDB 页面 | 原文线索

V2-031928 | MATHDB-342171

Finiteness conjecture for links

摘录 / EXCERPT: Let $L=\bigcup_{i=1}^{n}L_i\subset S^3$ be an n-component link with $n\geqslant 1$, and let $E_L=S^3\smallsetminus L$ be its exterior. For each component L_i, let m_i be it...

conjecture | open | MathDB 页面 | 原文线索

V2-031929 | MATHDB-342190

Ganatra-Pardon-Shende's colimit conjecture for Fukaya categories

摘录 / EXCERPT: Ganatra-Pardon-Shende's colimit conjecture. The Fukaya category of M is the colimit

conjecture | open | MathDB 页面 | 原文线索

V2-031930 | MATHDB-342218

Matveev complexity conjecture for twist knots as pairs

摘录 / EXCERPT: Matveev complexity conjecture for twist knots as pairs. For every $n\geqslant3$,

conjecture | open | MathDB 页面 | 原文线索

V2-031931 | MATHDB-342248

Quantization of the degree of maps from the sphere to the two-sphere

摘录 / EXCERPT: Let $X_1,X_2,X_3\in\operatorname{Mat}(N\times N,\mathbb{C})$ be self-adjoint operators depending on N and satisfying ... Assume that there is a constant c such that ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-031932 | MATHDB-342259

Untwisted algebraic 1-torsion for higher-dimensional contact manifolds without Giroux torsion

摘录 / EXCERPT: Let Y be the unit cotangent bundle of a hyperbolic surface, and let $(M=\Sigma \times Y,\xi_k)$ be the corresponding 5-dimensional contact manifold from Corollary ... have untwis...

conjecture | open | MathDB 页面 | 原文线索

V2-031933 | MATHDB-342313

The pro-p smoothness conjecture for weak Bloch--Kato pairs

摘录 / EXCERPT: Let $\mathcal{G}=(G,\theta)$ be a torsion-free 1-smooth pro-p pair. Pro-p smoothness conjecture. The pair $\mathcal{G}$ is weakly Bloch--Kato. This is the pro-p version of th...

conjecture | open | MathDB 页面 | 原文线索

V2-031934 | MATHDB-342329

Σ-Leopoldt conjecture for the CM field L

摘录 / EXCERPT: Let T be the relevant p-adic representation, let e_χ denote the χ-idempotent, and let $H^1_f(G_{k_v},T)$ be the finite local cohomology subgroup at v. Σ-Leop...

conjecture | open | MathDB 页面 | 原文线索

V2-031935 | MATHDB-342390

Framed foam functoriality conjecture for annular Khovanov-Rozansky homology

摘录 / EXCERPT: Let S be the surface supporting the tangled webs, let $\mathrm{S}\boldsymbol{\mathrm{Tan}}$ be the category of tangled webs and framed foams in four-dimensional space, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-031936 | MATHDB-342393

Coxeter braid complex decomposition conjecture

摘录 / EXCERPT: Let W be a finite Coxeter group with simple reflections S, let C_ϵ be the horizontal trace of the Rouquier complex associated with the Coxeter braid determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-031937 | MATHDB-342397

Persistent foliar branched surface conjecture for non-L-space knots

摘录 / EXCERPT: Let K be a knot in S^3 that is not an L-space knot, meaning that no nontrivial Dehn surgery on K yields an L-space. A persistently foliar branched surface in the exterior of...

conjecture | open | MathDB 页面 | 原文线索

V2-031938 | MATHDB-342405

Weinstein-trisection characterization of symplectic surfaces

摘录 / EXCERPT: Weinstein-trisection characterization conjecture. The surface $\mathcal{K}$ is isotopic to a symplectic surface if and only if it is isotopic into transverse bridge position with r...

conjecture | open | MathDB 页面 | 原文线索

V2-031939 | MATHDB-342420

The irreducible tangles are persistent conjecture

摘录 / EXCERPT: Irreducible tangles are persistent conjecture. Every irreducible tangle is a persistent tangle.

conjecture | open | MathDB 页面 | 原文线索

V2-031940 | MATHDB-342423

The twisted tropical Lagrangian mirror correspondence conjecture

摘录 / EXCERPT: Twisted Lagrangian mirror correspondence conjecture. The twisted tropical Lagrangians $L(\phi;\sigma)$ are mirror to the direct image of line bundles on the mirror divisor D.

conjecture | open | MathDB 页面 | 原文线索

V2-031941 | MATHDB-342425

Borel-resummation conjecture for knot series

摘录 / EXCERPT: Borel-resummation conjecture. For any knot $K\subset S^3$, Borel resummation of the double series gives a knot invariant $f_K(x,q)$ with integer coefficients, up to multiplication...

conjecture | open | MathDB 页面 | 原文线索

V2-031942 | MATHDB-342426

Quantum A-polynomial conjecture for knot series

摘录 / EXCERPT: Quantum A-polynomial conjecture. For any knot $K\subset S^3$,

conjecture | open | MathDB 页面 | 原文线索

V2-031943 | MATHDB-342462

The L-space conjecture for irreducible rational homology 3-spheres

摘录 / EXCERPT: Let M be a rational homology 3-sphere, meaning that $H_1(M;\mathbf{Q})=0$. Its Heegaard Floer homology has rank at least $\left|H_1(M;\mathbf{Z})\right|$, and M is an L-space...

conjecture | open | MathDB 页面 | 原文线索

V2-031944 | MATHDB-342469

Homological norm criterion for higher-dimensional connectivity

摘录 / EXCERPT: Homological norm criterion conjecture. For all $i\in\mathbb{N}$ and $\epsilon>0$, there exist $r_i,\ell_i,w_i\geq0$ such that, for every middling $p:\mathbb{N}\rightarrow[0,1]$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-031945 | MATHDB-342477

The stable Generalized Property R conjecture

摘录 / EXCERPT: Let L be an R-link, meaning an n-component link in S^3 whose 0-surgery yields $Y_n=\mathop{\#}^n(S^1\times S^2)$. Let U be a 0-framed unlink. The stable Generalized Propert...

conjecture | open | MathDB 页面 | 原文线索

V2-031946 | MATHDB-342478

The weak stable Generalized Property R conjecture

摘录 / EXCERPT: Let L be an R-link, and let U be a 0-framed unlink. A Hopf pair is a Hopf link with one component 0-framed and the other dotted, representing a 4-dimensional 1-handle. The weak...

conjecture | open | MathDB 页面 | 原文线索

V2-031947 | MATHDB-342495

Finiteness conjectures for finite-dimensional functors in positive characteristic

摘录 / EXCERPT: Soit K un corps de caractéristique $p>0$, soit A un anneau, et soit $\mathcal{F}^{df}(A,K)$ la catégorie des foncteurs de la catégorie considérée à valeurs dans des espaces vec...

conjecture | open | MathDB 页面 | 原文线索

V2-031948 | MATHDB-342518

Conjecture on higher vanishing of A-infinity operations for manifolds

摘录 / EXCERPT: Let M be an orientable N-dimensional k-connected compact manifold, with $\dim H^{k+1}(M)=1$. Let $l\geq 3$ and assume ... An A_∞-minimal model of $\Omega^*(M)$ is a mini...

conjecture | open | MathDB 页面 | 原文线索

V2-031949 | MATHDB-342539

Rational chromatic splitting conjecture for the factorization class

摘录 / EXCERPT: Let $\iota'\in[\mathbb S^{d+1},\mathbb X\mathrm P^1]$ be the factor of the fundamental class through $\mathbb X\mathrm P^1$ in the displayed factorization. The factorization-class...

conjecture | open | MathDB 页面 | 原文线索

V2-031950 | MATHDB-342543

Analytic realization conjecture for singular-crossing bimodules

摘录 / EXCERPT: The paper constructs bimodules $\mathcal{X}^{DA}$, $\mathcal{X}^{DD}$, and $\mathcal{X}_i^{DD}$, together with the Heegaard diagram in Figure and its variant globally extended to t...

conjecture | open | MathDB 页面 | 原文线索

V2-031951 | MATHDB-342549

Stable-envelope classes for the attracting stratification

摘录 / EXCERPT: Stable-envelope existence and uniqueness conjecture. There exist unique classes $(S_p)_{p\in\mathfrak M_1^T}$ in $H_{\dim\mathfrak Z}^T(\mathfrak M_1)$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-031952 | MATHDB-342560

The conjecture that satellite knots are not quasi-alternating

摘录 / EXCERPT: Let a satellite knot be a knot whose exterior contains an incompressible torus. Let $\mathcal{Q}$ denote the class of quasi-alternating links, defined as the smallest set containin...

conjecture | open | MathDB 页面 | 原文线索

V2-031953 | MATHDB-342568

The bipolar-filtration conjecture for knot reversal

摘录 / EXCERPT: Let $\mathcal{T}$ be the group of topologically slice knots, let $\mathcal{T}_n$ be its bipolar filtration, and let ... The map ρ denotes knot reversal, and...

conjecture | open | MathDB 页面 | 原文线索

V2-031954 | MATHDB-342569

The free-subgroup conjecture for reversal on the Alexander-trivial quotient

摘录 / EXCERPT: Let $\mathcal{T}$ be the group of topologically slice knots, let $\mathcal{T}_\Delta$ be the subgroup appearing in the Alexander-trivial quotient, and let ρ denote knot revers...

conjecture | open | MathDB 页面 | 原文线索

V2-031955 | MATHDB-342570

The order-two knot-reversal conjecture for topologically slice knots

摘录 / EXCERPT: Let $\mathcal{T}$ be the group of topologically slice knots, let ρ denote knot reversal, and interpret $2K=0$ in the knot concordance group. Order-two reversal conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-031956 | MATHDB-342577

The Lada-Markl enveloping conjecture for L_∞ algebras

摘录 / EXCERPT: Let L be an L_∞ algebra. Write $\mathcal C(L)$ for its Chevalley-Eilenberg coalgebra, $\mathcal U L$ for Lada and Markl's universal enveloping A_∞ algebra, and...

conjecture | open | MathDB 页面 | 原文线索

V2-031957 | MATHDB-342607

Sharpness conjecture for the stabilisation-singular distance inequality

摘录 / EXCERPT: Sharpness conjecture. There exist X, Σ, and Σ' as above such that

conjecture | open | MathDB 页面 | 原文线索

V2-031958 | MATHDB-342627

Goussarov-Polyak-Viro conjecture for finite type invariants of knots

摘录 / EXCERPT: A finite type invariant of classical knots is an invariant filtered by finite-dimensional vector spaces. A finite type invariant of long virtual knots is defined analogously in the...

conjecture | open | MathDB 页面 | 原文线索

V2-031959 | MATHDB-342628

Conjecture on bounded second fundamental form and spherical topology

摘录 / EXCERPT: Spherical embedding conjecture. If, in addition, M satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-031960 | MATHDB-342637

Conjecture on the stable connectivity of the covariant part of the A-theory transformation

摘录 / EXCERPT: Let $\theta=(B,p,xi)$ be a family of Rd-bundles, and let $T^%$ denote the covariant part of the transformation $tau^%$ associated to the extended A-theory characteristic. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-031961 | MATHDB-342646

Mahowald's conjecture on periodic classes in the cohomology of finite sub-Hopf algebras

摘录 / EXCERPT: Mahowald's conjecture. For any natural numbers n and k, the class $v_n^{2^{n+1+k}}$ is defined and nonzero in

conjecture | open | MathDB 页面 | 原文线索

V2-031962 | MATHDB-342664

Functorial transfer conjecture for cohomological representations from $\operatorname{Sp}(4,\mathbb{R})$

摘录 / EXCERPT: Functorial transfer conjecture. The representation $\iota(\pi)$ is cohomological if π is one of the following: the trivial representation; a discrete series representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-031963 | MATHDB-342667

Kirby’s Whitehead-double concordance conjecture

摘录 / EXCERPT: Let K) be a knot, let $D(K)$ denote its positively-clasped Whitehead double, and let $[K]$ and $[D(K)]$ denote their elements in the knot concordance group $\mathcal{C}$. Kirby’s...

conjecture | open | MathDB 页面 | 原文线索

V2-031964 | MATHDB-342682

Bezrukavnikov-Okounkov conjecture for quantum Coulomb branches

摘录 / EXCERPT: Let $\mathfrak{M}_{\mathbb{Q}}$ be the commutative resolution associated with the Coulomb branch, let π be the relevant fundamental group acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-031965 | MATHDB-342695

The L-space conjecture for persistently foliar knots

摘录 / EXCERPT: A knot κ in S^3 is an L-space knot if some non-trivial Dehn surgery on κ yields an L-space, and it is persistently foliar if, for each boundary slope, its complemen...

conjecture | open | MathDB 页面 | 原文线索

V2-031966 | MATHDB-342696

The conjecture that every composite knot is persistently foliar

摘录 / EXCERPT: A composite knot is a nontrivial connected sum of knots, and a knot is persistently foliar if, for each boundary slope, its complement admits a co-oriented taut foliation meeting t...

conjecture | open | MathDB 页面 | 原文线索

V2-031967 | MATHDB-342734

Conjecture on Möbius invariance of O'Hara knot energies

摘录 / EXCERPT: For $d4aa^{\alpha,p}$ an O'Hara knot energy with $d4aa^{\alpha,p}$ defined for $d4aa p\geq 4$ and $p\geq 2$, the parameters $d4aa$ and p determine its scaling and invariance prop...

conjecture | open | MathDB 页面 | 原文线索

V2-031968 | MATHDB-342741

The independence-complex conjecture for products of three complete graphs

摘录 / EXCERPT: Let K_n denote the complete graph on n vertices, let $\times$ denote the graph product, and let ${\rm{Ind}}(G)$ denote the independence complex of a graph G. For $n\geq 2$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-031969 | MATHDB-342780

Lee's conjecture on global pseudo-Einstein structures

摘录 / EXCERPT: Let M be a closed, strictly pseudoconvex CR $(2n+1)$-manifold with vanishing first Chern class ... where $n\geq 2$. A pseudo-Einstein structure is a contact structure whose pseudoh...

conjecture | open | MathDB 页面 | 原文线索

V2-031970 | MATHDB-342796

Finite-rank conjecture for nilpotent locally compact p-groups

摘录 / EXCERPT: Let G be a nilpotent locally compact p-group, and let $\mathrm{rank}_p(G)$ denote its p-rank. The finite-rank conjecture. If ... then $G\in\mathfrak E_{<\infty}$, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-031971 | MATHDB-342799

Bestiary hypothesis for higher categories of vector spaces

摘录 / EXCERPT: Let $d\geq 0$. Consider the homotopy actions of the relevant orthogonal groups on the indicated higher categories of vector spaces. Framing-action triviality conjecture. The homoto...

conjecture | open | MathDB 页面 | 原文线索

V2-031972 | MATHDB-342850

The balanced-presentation conjecture for groups with 4-periodic cohomology

摘录 / EXCERPT: Let G be a group with 4-periodic cohomology. Balanced-presentation conjecture. Then G has a balanced presentation. This would imply that groups with 4-periodic cohomology a...

conjecture | open | MathDB 页面 | 原文线索

V2-031973 | MATHDB-342875

Structure-set rank conjecture for algebraic and geometric structures

摘录 / EXCERPT: Structure-set rank conjecture. The free ranks satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-031974 | MATHDB-342919

The chain-homotopy conjecture for obstruction complexes of model fibers

摘录 / EXCERPT: Let $(X,T^*_{X,-})$ be a model and let $(Y,T^*_{Y,-})$ be a gt-model over a connected Stein base whose fiber is $(X,T^*_{X,-})$, so that $Y=X\times B$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-031975 | MATHDB-342940

The A·B·C·D version of the interval nonvanishing proposition

摘录 / EXCERPT: A·B·C·D version of the interval nonvanishing proposition. (I) $C_{u,s_{\alpha}v}^{w}\neq 0$ when $v<s_{\alpha}v\leq w$. (II) If $\ell(w)<\ell(u)+\ell(v)$, then there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-031976 | MATHDB-342952

Alternating quotients conjecture for periodic knots

摘录 / EXCERPT: Let $K\subset S^3$ be an alternating periodic knot, meaning that K admits a finite-order symmetry of the pair (S^3,K) with nonempty fixed-point set disjoint from K. The quoti...

conjecture | open | MathDB 页面 | 原文线索

V2-031977 | MATHDB-342974

The rigged-Hilbert-space topology conjecture for generalized eigenstates

摘录 / EXCERPT: Let $[...]$ be the dense domain of finite-particle states generated by Schwartz test functions, contained in the Hilbert space $[...]$. For a distribution $F[...]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-031978 | MATHDB-342988

Connected-sum conjecture for powers of three in Khovanov torsion

摘录 / EXCERPT: Let $T(2,3)$ denote the trefoil, let $(\sigma_1\sigma_2\sigma_3)^4\sigma_1\sigma_2$ denote the braid whose closure is used in the connected sum, and let $\#_m$ denote the m-fold...

conjecture | open | MathDB 页面 | 原文线索

V2-031979 | MATHDB-342994

Non-negativity conjecture for the polynomials underlying the quasi-polynomial of lattice-point counts

摘录 / EXCERPT: Non-negativity conjecture. The polynomials underlying the quasi-polynomial $\overline{N}_{g,n}$ have non-negative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-031980 | MATHDB-343018

The cohomological DT/PT correspondence with insertions on Calabi-Yau fourfolds

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau fourfold and let $\beta\in H_2(X,\mathbb Z)$. For a line bundle L on X, write $L^{[n]}$ for its tautological bundle on the relevant mo...

conjecture | open | MathDB 页面 | 原文线索

V2-031981 | MATHDB-343029

Steinhaus-Čech equivalence conjecture in Euclidean space

摘录 / EXCERPT: Steinhaus-Čech equivalence conjecture. The Čech filtration constructed from X is isomorphic to the cover filtration on X constructed from $\check{C}_R(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-031982 | MATHDB-343084

Vanishing conjecture for the obstruction group of selfdual contact instantons

摘录 / EXCERPT: Let M be a compact Sasakian 7-manifold with positive transverse scalar curvature, and let ${\bf E}\to M$ be a Sasakian G-bundle. Let A be an irreducible selfdual contact in...

conjecture | open | MathDB 页面 | 原文线索

V2-031983 | MATHDB-343119

Vernitski's conjecture on knot semigroups of 2-bridge knots

摘录 / EXCERPT: Vernitski's conjecture. The knot semigroup of every 2-bridge knot is isomorphic to an alternating sum semigroup.

conjecture | open | MathDB 页面 | 原文线索

V2-031984 | MATHDB-343120

Conjecture on knot semigroups of tripletwist knots

摘录 / EXCERPT: Tripletwist knot-semigroup conjecture. If $(ml+1)n+m$ is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-031985 | MATHDB-343131

Independence of signature data for equivariant instanton complexes

摘录 / EXCERPT: Let Y be a rational homology sphere, let E be the bundle used to define the equivariant instanton complex, let R be the coefficient ring, and let σ be a choice of sign...

conjecture | open | MathDB 页面 | 原文线索

V2-031986 | MATHDB-343142

The H^1-H^r equivalence conjecture for low-dimensional permutations

摘录 / EXCERPT: Let S_{n+1} be the symmetric group, let ${\mathbf W}_n$ be the set of words, and let $\operatorname{inv}(\sigma)$ denote the inversion number of σ. For a nonidentity permu...

conjecture | open | MathDB 页面 | 原文线索

V2-031987 | MATHDB-343144

The coNP conjecture for recognition of non-spherical small geometric homology spheres

摘录 / EXCERPT: Let $\mathcal{T}$ be a triangulation of a 3-manifold. Call the triangulated manifold small if it does not contain an incompressible surface of positive genus, and call it an integr...

conjecture | open | MathDB 页面 | 原文线索

V2-031988 | MATHDB-343146

Isomorphism conjecture for modules over the stated skein algebra

摘录 / EXCERPT: Fix an odd-order root of unity $\in\mathbb{C}$ of order $N>1$ and a triangulated punctured surface $(\mathbf{\Sigma},\Delta)$. Let Y be the maximal spectrum of the center of the...

conjecture | open | MathDB 页面 | 原文线索

V2-031989 | MATHDB-343148

The classical L-space knot conjecture

摘录 / EXCERPT: Classical L-space knot conjecture. The knot $kappa$ is persistently foliar if and only if it has no non-trivial L-space or reducible surgeries.

conjecture | open | MathDB 页面 | 原文线索

V2-031990 | MATHDB-343155

The codimension-three conjecture for the topologically singular set of RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf d,\mathcal{H}^{N})$ be a non-collapsed $\mathsf{RCD}(K,N)$ space. Denote by ${\mathcal R}_{N}$ its N-regular set. Codimension-three conjecture. There exists an op...

conjecture | open | MathDB 页面 | 原文线索

V2-031991 | MATHDB-343218

T7rkelli's modification for $(T\normal G)$-towers

摘录 / EXCERPT: Let $G\subset S_n$ be transitive, let $T\normal G$, let $(L/K,\iota_B)$ be a B-extension, and let $\pi:G_K\rightarrow G$ be a homomorphism with $\pi\equiv \iota_B\mod T$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-031992 | MATHDB-343239

The Geometric Syzygy Conjecture in Even Genus

摘录 / EXCERPT: The Geometric Syzygy Conjecture in Even Genus. The last linear syzygy space

conjecture | open | MathDB 页面 | 原文线索

V2-031993 | MATHDB-343276

Nonexistence conjecture for even Vassiliev invariants

摘录 / EXCERPT: An even, degree-one Vassiliev invariant is a function V on virtual knots such that $V(K_+)-V(K_-)=0$ for even crossings and $V(K_+)-V(K_-) e 0$ for odd crossings. Nonexistence co...

conjecture | open | MathDB 页面 | 原文线索

V2-031994 | MATHDB-343286

Topological ribbon concordance partial-order conjecture

摘录 / EXCERPT: Let $\geq_{\operatorname{top}}$ denote the topological ribbon concordance relation on knots. Topological ribbon concordance conjecture. The relation $\geq_{\operatorname{top}}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-031995 | MATHDB-343298

The primitive homology conjecture for closed exact Lagrangians

摘录 / EXCERPT: Let M be a Weinstein manifold of dimension $2n$, and let $L\subset M$ be a closed exact Lagrangian submanifold with vanishing Maslov class. Its homology class is...

conjecture | open | MathDB 页面 | 原文线索

V2-031996 | MATHDB-343310

The Slice-Ribbon conjecture for knots

摘录 / EXCERPT: Let K be a knot in S^3. A concordance is a cobordism of minimal intrinsic complexity, and it has minimal extrinsic complexity when it has no index 2 Morse critical points. Sl...

conjecture | open | MathDB 页面 | 原文线索

V2-031997 | MATHDB-343363

Bestvina-Brady conjecture on PL epsilon-maps of finite 2-complexes

摘录 / EXCERPT: Let L be a finite 2-complex equipped with a metric. A map is an ε-map if the pre-image of every point has diameter less than ε. Bestvina-Brady's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-031998 | MATHDB-343373

Conjecture on a local-cohomological submodule for simplicial complexes

摘录 / EXCERPT: The local-cohomological submodule conjecture. There exists a submodule

conjecture | open | MathDB 页面 | 原文线索

V2-031999 | MATHDB-343395

Hyperbolic cosmetic surgery conjecture

摘录 / EXCERPT: Let N be a finite-volume hyperbolic 3--manifold with one or more cusps, and let $\mathbf{s}_1$ and $\mathbf{s}_2$ be tuples of slopes on the cusps of N. A Dehn filling...

conjecture | open | MathDB 页面 | 原文线索

V2-032000 | MATHDB-343423

The fractional Helly number conjecture for families in Euclidean space

摘录 / EXCERPT: Let $b\geq 1$, $d\geq 2$ be integers, and let $\alpha\in(0,1)$. For a family $\mathcal{A}$ of sets in $\mathbb{R}^d$, write $TC_{\lceil d/2\rceil}(\mathcal{A})$ for its topological...

conjecture | open | MathDB 页面 | 原文线索

V2-032001 | MATHDB-343427

Bridge Theorem conjecture for topological and algebraic entropy

摘录 / EXCERPT: Let S be a cancellative left amenable monoid acting on the compact abelian group K on the left by γ. Let $\widehat\gamma$ denote the induced dual action on the Pontryagi...

conjecture | open | MathDB 页面 | 原文线索

V2-032002 | MATHDB-343431

Generalized Mumford conjecture for a nodal-curve moduli space

摘录 / EXCERPT: Let X_0 be an irreducible nodal curve with exactly one node, and let $U_{X_0}(2,\mathcal{L}_0)$ be the moduli space of rank-two semistable sheaves on X_0 with determinant...

conjecture | open | MathDB 页面 | 原文线索

V2-032003 | MATHDB-343434

The E_n-enhancement conjecture for homological epimorphisms

摘录 / EXCERPT: Let $F\colon\mathcal A\to\mathcal B$ be a homological epimorphism of A_∞-algebras, and let $\widetilde{\mathcal A}$ be an E_n-algebra enhancing A. The E_n-enhancement...

conjecture | open | MathDB 页面 | 原文线索

V2-032004 | MATHDB-343438

Equivalence classification conjecture for non-Abelian families

摘录 / EXCERPT: Let ${\cal C}$ be a topological order, and let ${\cal C}_{\boldsymbol a,K}$ and ${\cal C}_{\boldsymbol a',K'}$ be topological orders constructed from ${\cal C}$ using anyon data...

conjecture | open | MathDB 页面 | 原文线索

V2-032005 | MATHDB-343465

Topological slice genus subadditivity conjecture for satellite knots

摘录 / EXCERPT: Let P be a pattern, meaning a knot in a solid torus, and let K be a knot in S^3. Write $P(K)$ for the satellite knot and $P(U)$ for the satellite of the unknot, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032006 | MATHDB-343474

Asymptotic twist-number conjecture for weaving knots

摘录 / EXCERPT: Asymptotic twist-number conjecture. For m sufficiently large,

conjecture | open | MathDB 页面 | 原文线索

V2-032007 | MATHDB-343486

The nonsymmetric skein-module localization conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot. Let $\widehat{K}_q(S^3\setminus K)$ be its nonsymmetric skein module, let $\widehat{K}^{\mathrm{loc}}_q(S^3\setminus K)$ be its localization, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032008 | MATHDB-343513

Kirby's generalized Property R conjecture

摘录 / EXCERPT: Let Y be an integer homology sphere containing a knot K which admits a surgery to $S^1 \times S^2$. Kirby's generalized Property R conjecture. Then K is unique up to ambient...

conjecture | open | MathDB 页面 | 原文线索

V2-032009 | MATHDB-343540

Pirashvili's conjecture for finite-dimensional complex Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a non-trivial finite-dimensional complex Lie algebra. Say that $\mathfrak{g}$ satisfies the Pirashvili conditions if, equivalently, $HL_p(\mathfrak{g})=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-032010 | MATHDB-343544

Dimension inequality for finite-type invariants of weak (1,3) homotopy

摘录 / EXCERPT: Dimension inequality conjecture. The pattern

conjecture | open | MathDB 页面 | 原文线索

V2-032011 | MATHDB-343549

Minimal genus conjecture for minimal crossing diagrams of virtual links

摘录 / EXCERPT: Minimal genus conjecture for virtual links. Given a virtual link L, any minimal crossing diagram for it has minimal genus.

conjecture | open | MathDB 页面 | 原文线索

V2-032012 | MATHDB-343557

Isometric standard cubes in geometric realizations of precubical sets

摘录 / EXCERPT: Let C be a finite-dimensional geometric precubical set, let $n\in\mathbb{N}$, let $c\in C(n)$, and let I^n be the standard n-cube. Write $\greal{C}$ for the geometric realiza...

conjecture | open | MathDB 页面 | 原文线索

V2-032013 | MATHDB-343609

LMOV integrality conjecture for open topological strings

摘录 / EXCERPT: LMOV conjecture. For every $\mu\in\mathcal{P}_+$, the function $F_{\mu}^{(X,\mathcal{D})}$ has the integral expansion given by the right-hand side of the open multiple-covering for...

conjecture | open | MathDB 页面 | 原文线索

V2-032014 | MATHDB-343613

Equator Conjecture for flag homology spheres

摘录 / EXCERPT: Let Δ be a flag homology sphere. An equator is an induced subcomplex that is a flag homology sphere of codimension 1. Equator Conjecture. For every equator E in Δ...

conjecture | open | MathDB 页面 | 原文线索

V2-032015 | MATHDB-343663

C. Monical's Horn-type conjecture for equivariant Schubert calculus

摘录 / EXCERPT: C. Monical's Horn-type conjecture. $O_{,}^{} eq 0$ if and only if $k egthinspace\footnotesize\geq 0$ and, for every $0<r<n$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-032016 | MATHDB-343665

Euler characteristic conjecture for longitudinal Floer homology with rational holonomy

摘录 / EXCERPT: Let K be a knot in an integral homology sphere, let Y_{0} be the result of 0 surgery on K, let $K_{0}\subset Y_{0}$ be the surgery core, and let $HIL(Y_{0},K_{0},\alpha)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-032017 | MATHDB-343675

Grid homotopic curve shortening approximates affine curve-shortening flow

摘录 / EXCERPT: Let δ be an initial curve, and let γ_0 be the shortest curve homotopic to it among curves avoiding the obstacle set $P_n=(\mathbb Z/n)^2$. For $t>0$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-032018 | MATHDB-343676

Random-point homotopic curve shortening approximates affine curve-shortening flow

摘录 / EXCERPT: Let δ be an initial curve contained in a convex region R of area A. For $t>0$, write $\delta'=\delta(t)$ for its affine curve-shortening flow whenever it is defined. For...

conjecture | open | MathDB 页面 | 原文线索

V2-032019 | MATHDB-343690

Witten's conjecture on integral Floer homology for \aSpin^c Lagrangians

摘录 / EXCERPT: Let L be a Lagrangian submanifold admitting a $Spin^c$ structure with associated complex line bundle λ. Suppose that L admits a $Spin^c$ connection whose induced connec...

conjecture | open | MathDB 页面 | 原文线索

V2-032020 | MATHDB-343695

Chernousov-Gille-Pianzola conjecture on Zariski-locally trivial torsors

摘录 / EXCERPT: Let k be a field of characteristic 0, let $R=k[x_1^{\pm 1},\ldots,x_n^{\pm 1}]$, and let G be a loop reductive group over R, meaning that G contains a maximal R-torus....

conjecture | open | MathDB 页面 | 原文线索

V2-032021 | MATHDB-343722

Conjecture on the intersection Betti numbers of moduli spaces of sheaves on the plane

摘录 / EXCERPT: Let $M_{(0,d,\chi)}$ be the moduli space of Gieseker semistable sheaves on $\mathbb P^2$ with Chern character $(0,d,\chi)$, and write its intersection Betti numbers as...

conjecture | open | MathDB 页面 | 原文线索

V2-032022 | MATHDB-343728

Length-filtration conjecture for complex K-theory of manifolds

摘录 / EXCERPT: Let (Q,q_0) and (Q',q'_0) be compact pointed Riemannian manifolds. For each $t\in\mathbf{R}$, let $F_{<t}C_*(\Omega_{q_0}Q;\mathbf{C})$ denote the subcomplex generated by loops...

conjecture | open | MathDB 页面 | 原文线索

V2-032023 | MATHDB-343731

The modulus-of-continuity conjecture for topologies on Urysohn isometry groups

摘录 / EXCERPT: Modulus-of-continuity conjecture. Every group topology on G strictly coarser than τ_{st} is of the form τ_f for some $\mathcal{R}$-modulus of continuity.

conjecture | open | MathDB 页面 | 原文线索

V2-032024 | MATHDB-343745

Closure of approximate L^2-rigidity under joins with diffuse intersection

摘录 / EXCERPT: Let M be a $\rm II_1$ factor and let $Q_1,Q_2\leq M$ be von Neumann subalgebras such that $Q_i\leq M$ is approximately L^2-rigid for $i=1,2$. Suppose that $Q_1\cap Q_2$ is diff...

conjecture | open | MathDB 页面 | 原文线索

V2-032025 | MATHDB-343754

Conjecture relating Ozsváth-Szabó and holomorphic knot Floer complexes

摘录 / EXCERPT: Let $D=\mathsf{p}(b)$ be a plat diagram, let $\mathsf{OS}(D)$ be the curved box tensor product of the Ozsváth-Szabó Type A, Type DA, and Type D structures associated to D, and ch...

conjecture | open | MathDB 页面 | 原文线索

V2-032026 | MATHDB-343756

Stable cohomology isomorphism conjecture for relative completion

摘录 / EXCERPT: For a partition ${\boldsymbol\mu}$ with at most g parts, let $V_{\boldsymbol\mu}$ be the corresponding irreducible ${\mathrm{Sp}}(H)$-module, and let ... be the homomorphism induce...

conjecture | open | MathDB 页面 | 原文线索

V2-032027 | MATHDB-343757

Purity conjecture for the stable cohomology of relative completion

摘录 / EXCERPT: Let ${\mathfrak u}_{g,n+\vec r}$ be the prounipotent Lie algebra of the relative completion, equipped with its weight filtration, and write Gr^W_m for its weight-...

conjecture | open | MathDB 页面 | 原文线索

V2-032028 | MATHDB-343762

Shklyarov's normalization conjecture for the higher residue pairing

摘录 / EXCERPT: Let Q be the polynomial ring and f a polynomial as in the source, let n be the relevant degree, and let $c_f\in\mathbb{C}$ be the constant such that the canonical pairing on...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-032029 | MATHDB-343801

Topology of the BATS tip-solution classification sets

摘录 / EXCERPT: Let Y_μ be the parameter domain for tip solutions. For a subset $Y\subset Y_\mu$, define $A_\mu(Y)$ as the parameters for which ρ_α reaches zero after having...

conjecture | open | MathDB 页面 | 原文线索

V2-032030 | MATHDB-343835

Coincidence point set conjecture for homotopic fixed point sets

摘录 / EXCERPT: Let X be a contractible digital image and let $f:X\longrightarrow X$ be a continuous mapping. Here, $HFS(f,f)$ denotes the homotopic fixed point set for the pair (f,f), $S(f)$...

conjecture | open | MathDB 页面 | 原文线索

V2-032031 | MATHDB-343845

Associative algebraic RDS resolution

摘录 / EXCERPT: Let an RDS be a representable diagrammatic set, and let an associative algebraic RDS mean an algebraic RDS whose composition operations are associative in the sense defined in the...

open_problem | open | MathDB 页面 | 原文线索

V2-032032 | MATHDB-343861

Solid-algebra persistence conjecture for cohomological dimension

摘录 / EXCERPT: Let R be a Noetherian domain, let S be a solid R-algebra, and let I be an ideal of R. Solid-algebra persistence conjecture. One has ... The conjecture asks whether cohomolo...

conjecture | open | MathDB 页面 | 原文线索

V2-032033 | MATHDB-343898

Interval decomposition conjecture for the nesting-poset zigzag module

摘录 / EXCERPT: Let P be a finite poset, and let $\field P$ denote its incidence algebra. For the critical and regular levels $a_0,t_1,a_1,t_2,\dots,t_n,a_n$ associated with the Morse function...

conjecture | open | MathDB 页面 | 原文线索

V2-032034 | MATHDB-343901

Local volume-growth conjecture for quasi-optimal pseudomanifolds

摘录 / EXCERPT: Let (P_n) be a sequence of quasi-optimal pseudomanifolds representing multiples of a homology class, equipped with the piecewise-flat metric in which every m-simplex is isometr...

conjecture | open | MathDB 页面 | 原文线索

V2-032035 | MATHDB-343902

Logarithmic relative-systole alternative for quasi-optimal pseudomanifolds

摘录 / EXCERPT: Let (P_n) be a sequence of quasi-optimal pseudomanifolds representing $n\,\mathbf a$, and let $\varphi_n:\pi_1(P_n)\to\pi_1(X)$ be the induced homomorphism. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032036 | MATHDB-343927

The contramodule Bass-flat equivalence conjecture

摘录 / EXCERPT: Contramodule Bass-flat equivalence conjecture. The following conditions are equivalent to each other and to the conditions listed in the theorem: (i$^\flat$) all Bass flat left...

conjecture | open | MathDB 页面 | 原文线索

V2-032037 | MATHDB-343934

The global fixed-point conjecture for actions of homeomorphism groups

摘录 / EXCERPT: Let M be a simply connected manifold that is not homeomorphic to S^n, and let N be a manifold of dimension $n+k<2n$. The group $\operatorname{Homeo}_0(M)$ acts nontrivially o...

conjecture | open | MathDB 页面 | 原文线索

V2-032038 | MATHDB-343941

Higher-rank F_K conjecture

摘录 / EXCERPT: Let G be a root system of rank r, with Weyl group W, positive roots Δ^+, weight lattice P, root lattice Q, Weyl vector ρ, and dominant weights P_+. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032039 | MATHDB-343942

Higher-rank surgery formula for $\hat{Z}^G$

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let G be a root system with positive roots Δ^+, and let $\mathcal{L}_{p/r}^{(b)}$ denote the surgery transform appearing in the formula. High...

conjecture | open | MathDB 页面 | 原文线索

V2-032040 | MATHDB-343943

HOMFLY-PT analogue of F_K

摘录 / EXCERPT: Let K be a knot, let a and q be parameters, and let $\hat{A}_K(\hat{x},\hat{y},a,q)$ be the a-deformed quantum A-polynomial. HOMFLY-PT analogue conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-032041 | MATHDB-343955

The intersection-triple bound conjecture for closed curves on the pair of pants

摘录 / EXCERPT: The intersection-triple bound conjecture. For any curve α on the pair of pants,

conjecture | open | MathDB 页面 | 原文线索

V2-032042 | MATHDB-343956

The universal k-equivalence descent conjecture for the pair of pants

摘录 / EXCERPT: Universal k-equivalence descent conjecture. On the pair of pants, k-equivalence implies $(k-1)$-equivalence for all positive integers k.

conjecture | open | MathDB 页面 | 原文线索

V2-032043 | MATHDB-343959

The localized Adams differentials conjecture for Z

摘录 / EXCERPT: Consider the localized Adams spectral sequence ... with E_2-term $\mathbb{F}_2[v_2^{\pm},\widetilde{h}_{2,1},h_{3,0},h_{3,1},h_{4,0},h_{4,1},\ldots]$. Differentials conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-032044 | MATHDB-343960

The parabola conjecture for the telescope of Z

摘录 / EXCERPT: Let $\widehat{Z}$ be the telescope of the v_2-self-map on Z, and let $x_i:=h_{i,0}^2$. Parabola conjecture. The localized Adams spectral sequence for Z collapses at E_5, an...

conjecture | open | MathDB 页面 | 原文线索

V2-032045 | MATHDB-343961

The May-Ravenel collapse and torsion conjecture

摘录 / EXCERPT: Consider the May-Ravenel spectral sequence used to compute the relevant tmf-resolution input for Z. Torsion conjecture. The May-Ravenel spectral sequence collapses at...

conjecture | open | MathDB 页面 | 原文线索

V2-032046 | MATHDB-343962

The tmf-ASS differentials conjecture for Z, part 1

摘录 / EXCERPT: Let $x_j=h_{j,0}^2$. In the tmf-Adams spectral sequence for Z, use indices $i\geq3$, $l\geq i+3$, $m<2^{i+1}-1$, $k_j\geq0$, and $\epsilon_j,\bar\epsilon_j\in\{0,1\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-032047 | MATHDB-343963

The hidden-extension conjecture in the tmf-ASS

摘录 / EXCERPT: Let $x_i=h_{i,0}^2$ in the tmf-Adams spectral sequence for Z. Hidden-extension conjecture. There are hidden extensions ... This is suggested by comparing the known...

conjecture | open | MathDB 页面 | 原文线索

V2-032048 | MATHDB-343964

The tmf-ASS differentials conjecture for Z, part 2

摘录 / EXCERPT: In the tmf-Adams spectral sequence for Z, let $\epsilon_j,\bar\epsilon_j\in\{0,1\}$ and let $m\gg0$. Differentials conjecture, part 2. The v_2-families ... support dif...

conjecture | open | MathDB 页面 | 原文线索

V2-032049 | MATHDB-343965

The anti-telescope parabola conjecture for Z

摘录 / EXCERPT: Let $\widehat Z$ be the telescope of the v_2-self-map on Z. Let $x_i=h_{i,0}^2$. Parabola conjecture, height 2. The differentials of the preceding height-2 differentials conj...

conjecture | open | MathDB 页面 | 原文线索

V2-032050 | MATHDB-343966

Generic nondegeneracy conjecture for short star-products

摘录 / EXCERPT: Let $\mathfrak g$ be the Lie algebra under consideration, let s denote the degree-sign involution, and let ${{\mathbf A}}_\lambda$ be the corresponding filtered quantization para...

conjecture | open | MathDB 页面 | 原文线索

V2-032051 | MATHDB-344021

Seidel's generation conjecture for symplectic mapping classes of rational surfaces

摘录 / EXCERPT: Let $X=\mathbb{CP}^2\#k\overline{\mathbb{CP}}^2$. Write $\pi_0\operatorname{Symp}(X)$ for the group of symplectic mapping classes, and call a symplectomorphism a Lagrangian Dehn t...

conjecture | open | MathDB 页面 | 原文线索

V2-032052 | MATHDB-344022

Generic connectedness conjecture for symplectomorphism groups of rational surfaces

摘录 / EXCERPT: Let $X=\mathbb{CP}^2\#k\overline{\mathbb{CP}}^2$, and let ω be a generic symplectic form on X. Generic connectedness conjecture. The symplectomorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-032053 | MATHDB-344023

Rank formula conjecture for symplectic mapping groups of rational surfaces

摘录 / EXCERPT: Let (X,ω) be a rational symplectic surface with $8\leq\chi(X)\leq 11$, and let $Q(X)=\frac{1}{2}(\chi(X)-2)(\chi(X)-3)$. Let $r^+(X,\omega)$ denote the number of homology cl...

conjecture | open | MathDB 页面 | 原文线索

V2-032054 | MATHDB-344024

Hamiltonian classification conjecture for Lagrangian spheres in ADE Milnor fibers

摘录 / EXCERPT: Let M be a surface Milnor fiber of an ADE-type singularity, with its standard zero sections and standard Lagrangian spheres. ADE Lagrangian-sphere conjecture. Every Lagrangian...

conjecture | open | MathDB 页面 | 原文线索

V2-032055 | MATHDB-344027

The Hodge conjecture in Bott-Chern cohomology

摘录 / EXCERPT: The Hodge conjecture in Bott-Chern cohomology. Every element of $H^{p,p}_{BC}(Y;\Q)$ is representable by holomorphic p-chains with $\Q$-coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-032056 | MATHDB-344054

Positive-path conjecture for Gordian adjacency of positive braid knots

摘录 / EXCERPT: Let $\hat\beta_1$ and $\hat\beta_2$ be positive braid knots. A positive path from $\hat\beta_2$ to $\hat\beta_1$ is a sequence of positive braid knots beginning at $\hat\beta_2$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-032057 | MATHDB-344059

The Eisenstein quotient conjecture for the G'-cyclotomic maps

摘录 / EXCERPT: Eisenstein quotient conjecture. The map $\widetilde{\varpi}_{G'}$ factors through the Eisenstein ideal, and $\varpi_{G'}$ induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032058 | MATHDB-344076

Dual minimal slope conjecture for overconvergent F-isocrystals

摘录 / EXCERPT: Let X, K, $\mathcal M^\dagger$, and $\mathcal N^\dagger$ be as in the minimal slope conjecture, with associated convergent F-isocrystals $\mathcal M$ and $\mathcal N$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032059 | MATHDB-344077

Borisov-Horja duality conjecture for GKZ hypergeometric systems

摘录 / EXCERPT: Borisov-Horja duality conjecture. There exists a collection of polynomials $p_{c,d}(x_1,\ldots,x_n)$, indexed by $c\in C$ and $d\in C^{\circ}$, such that only finitely many are non...

conjecture | open | MathDB 页面 | 原文线索

V2-032060 | MATHDB-344146

Conjecture 2 on homology of the free Tits Lie algebra

摘录 / EXCERPT: Conjecture 2. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032061 | MATHDB-344147

Conjecture 3 on invariant homology of the free Tits Lie algebra

摘录 / EXCERPT: Conjecture 3. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032062 | MATHDB-344151

The Galois-invariant homology conjecture for special orthogonal groups

摘录 / EXCERPT: Let k be a field containing $sqrt{2}$ and not containing $i=sqrt{-1}$, and let $k(i)$ be the quadratic extension obtained by adjoining i. The Galois action of $Gal(k(i)/k)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-032063 | MATHDB-344161

The specialization conjecture for symmetry-breaking link homology

摘录 / EXCERPT: Specialization conjecture. Under the algebraic specialization $\wp(x)=x^n$, the groups

conjecture | open | MathDB 页面 | 原文线索

V2-032064 | MATHDB-344178

The positive-diagram formula for the Rasmussen invariant in connected sums of $S^1\times S^2$

摘录 / EXCERPT: Positive-diagram conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032065 | MATHDB-344179

The adjunction inequality for the Rasmussen invariant in negative-definite four-manifolds

摘录 / EXCERPT: Adjunction conjecture for s.

conjecture | open | MathDB 页面 | 原文线索

V2-032066 | MATHDB-344211

The Fukaya-category elliptic-curve summand conjecture for the blowup examples

摘录 / EXCERPT: Fukaya-category elliptic-curve summand conjecture. The Fukaya category of X should contain a direct summand equivalent to the derived category of sheaves on a genus-one curve who...

conjecture | open | MathDB 页面 | 原文线索

V2-032067 | MATHDB-344212

TWG bracket disjointness conjecture for distinct undirected curves

摘录 / EXCERPT: Let $\widetilde{x}$ and $\widetilde{y}$ be distinct free homotopy classes of undirected closed curves on a surface. The TWG Lie bracket is the Lie bracket on classes of undirected...

conjecture | open | MathDB 页面 | 原文线索

V2-032068 | MATHDB-344215

Bounded-spacer factor-isomorphism conjecture for rank-one subshifts

摘录 / EXCERPT: Let (X,T) be a rank-one subshift with bounded spacer parameter, and suppose that (Y,S) is a factor of (X,T). Bounded-spacer factor-isomorphism conjecture. Either Y is finit...

conjecture | open | MathDB 页面 | 原文线索

V2-032069 | MATHDB-344229

The filtration conjecture for algebraic K-theory of endomorphisms

摘录 / EXCERPT: Let A be the ring under consideration, and let $\widetilde{K}(\operatorname{End}(A))$ denote the reduced algebraic K-theory spectrum of its category of endomorphisms. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032070 | MATHDB-344230

The sphere-spectrum V-completion conjecture

摘录 / EXCERPT: Let $\operatorname{Set}^{\mathrm{fin}}_{\cong}$ be the category of finite sets and bijections, let $\widetilde{K}(\operatorname{Set}^{\mathrm{fin}}_{\cong})$ denote its reduced alg...

conjecture | open | MathDB 页面 | 原文线索

V2-032071 | MATHDB-344231

The definition of THH for Waldhausen categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a Waldhausen category, and let $\operatorname{End}(\mathcal{C})$ be its category of endomorphisms. Let $\widetilde{K}(\operatorname{End}(\mathcal{C}))$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-032072 | MATHDB-344236

The top-degree continuous K-theory conjecture for rigid spaces

摘录 / EXCERPT: Let X be a quasi-compact and quasi-separated rigid k-space of dimension d. Write $\operatorname{K}^\mathrm{cont}_{-d}(X)$ for its degree $-d$ continuous K-theory pro-abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-032073 | MATHDB-344254

The 2-torsion conjecture for real flag manifolds and R-spaces

摘录 / EXCERPT: Let $\operatorname{Fl}_{\mathcal{D}}$ be a real flag manifold, and let $H^*(\operatorname{Fl}_{\mathcal{D}};\mathbb{Z})$ denote its integral cohomology. More generally, let an R-...

conjecture | open | MathDB 页面 | 原文线索

V2-032074 | MATHDB-344255

Sharpness conjecture for real Schubert problem lower bounds

摘录 / EXCERPT: Consider a real Schubert problem for which Proposition ... is sharp. The conjecture asserts that the cohomological lower bound can always be attained; the supplied excerpt gives no r...

conjecture | open | MathDB 页面 | 原文线索

V2-032075 | MATHDB-344270

Fixed-signature connectivity conjecture for pairings on surfaces with arbitrary boundary

摘录 / EXCERPT: Fixed-signature connectivity conjecture. All pairings that have this fixed signature on the surface are connected.

conjecture | open | MathDB 页面 | 原文线索

V2-032076 | MATHDB-344290

Singleton conjecture for proper gradient homotopy classes of planar maps of negative degree

摘录 / EXCERPT: Let $\mathcal{P}_{k}^\nabla[\mathbb{R}^{2}]$ denote the set of proper gradient homotopy classes of proper maps $\mathbb{R}^{2}\to\mathbb{R}^{2}$ of degree k. Singleton conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032077 | MATHDB-344342

The Weinstein homotopy conjecture for the Liouville domains $W_{(Y,\Lambda)}$

摘录 / EXCERPT: Let Y be a compact manifold without boundary, let $M\subset J^1Y$ be a closed tubular neighborhood of the zero-section, and let $\Lambda\subset(M,\ker(dz-pdq))$ be a Legendrian s...

conjecture | open | MathDB 页面 | 原文线索

V2-032078 | MATHDB-344355

Component-preserving Conway mutation invariance for Khovanov homology

摘录 / EXCERPT: Component-preserving mutation conjecture. The groups $\widetilde{\operatorname{BN}}(\mathcal{L};\mathbb{Z})$, $\widetilde{\operatorname{Kh}}(\mathcal{L};\mathbb{Z})$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032079 | MATHDB-344356

Kim-Livingston conjecture for rational homology cobordism

摘录 / EXCERPT: Kim-Livingston conjecture. This homomorphism is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-032080 | MATHDB-344363

The veering sphere conjecture for veering triangulations

摘录 / EXCERPT: Let M be the manifold under consideration, let α denote the relevant veering structure, and let $S^2(\alpha)$ be the veering sphere constructed from it. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032081 | MATHDB-344374

Lekili-Evans comparison conjecture for Hamiltonian action maps

摘录 / EXCERPT: Let G act Hamiltonianly on a monotone symplectic manifold X. The action gives an E_2-algebra map ... Lekili and Evans' construction gives a map from the endomorphisms of the co...

conjecture | open | MathDB 页面 | 原文线索

V2-032082 | MATHDB-344383

Lalonde-? higher homotopical stabilization conjecture for symplectic ball embeddings

摘录 / EXCERPT: Higher homotopical stabilization conjecture. For every $m\in\mathbb{Z}^+$, there exists an $\epsilon(m)>0$ such that, whenever $c_i<\frac{\epsilon(m)}{k}$, the map σ is m-...

conjecture | open | MathDB 页面 | 原文线索

V2-032083 | MATHDB-344420

Splitting conjecture for set-theoretic Yang-Baxter homology of Rump right quasigroups

摘录 / EXCERPT: A Rump right quasigroup is a set equipped with the right-quasigroup structure arising in Rump's framework for set-theoretic solutions of the Yang-Baxter equation. Its set-theoretic...

conjecture | open | MathDB 页面 | 原文线索

V2-032084 | MATHDB-344435

The Galvin-Mycielski-Solovay characterization of locally compact Polish groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a Polish group. A set $S\subseteq\mathbb{G}$ is of strong measure zero, and a set $M\subseteq\mathbb{G}$ is meager. Galvin-Mycielski-Solovay characterization co...

conjecture | open | MathDB 页面 | 原文线索

V2-032085 | MATHDB-344436

The strong conjecture on anti-GMS sets in Polish groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a Polish group. A nonempty subset $C\subseteq\mathbb{G}$ is anti-GMS if it is nowhere dense and, for every sequence $\{U_n:n\in\omega\}$ of open neighborhoods o...

conjecture | open | MathDB 页面 | 原文线索

V2-032086 | MATHDB-344459

Equality of arithmetic-progression and regionally proximal relations for minimal distal systems

摘录 / EXCERPT: Let (X,T) be a minimal distal system. The arithmetic-progression proximality conjecture. ... Here ${\bf AP}^{[d]}$ denotes the regionally proximal relation of order d along arith...

conjecture | open | MathDB 页面 | 原文线索

V2-032087 | MATHDB-344513

Meier's classification conjecture for genus-three trisections

摘录 / EXCERPT: A closed smooth 4-manifold is said to admit a genus-three trisection when it admits a trisection of genus three. A 4-manifold is a spun lens space when it is obtained by spinni...

conjecture | open | MathDB 页面 | 原文线索

V2-032088 | MATHDB-344524

The great-circle conjecture for noncompact base directions

摘录 / EXCERPT: Great-circle conjecture. If there exist multiple $u\in U$ for which S_u is noncompact, then the set of such u is contained in some great circle in S^2.

conjecture | open | MathDB 页面 | 原文线索

V2-032089 | MATHDB-344525

The antipodal-pair conjecture for non-1-parameter line fibrations

摘录 / EXCERPT: Let $V:\mathbb R^3\to S^2$ be a unit vector field defining a fibration of $\mathbb R^3$ by oriented lines. Put $U=V(\mathbb R^3)$, and for $u\in U$ let $u^\perp$ be the plane throu...

conjecture | open | MathDB 页面 | 原文线索

V2-032090 | MATHDB-344530

Coincidence point spectrum equality for disconnected digital images

摘录 / EXCERPT: Coincidence spectrum conjecture. The coincidence point spectra satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-032091 | MATHDB-344556

The tangent-cone Milnor-fiber conjecture for embedded topologically equivalent hypersurface germs

摘录 / EXCERPT: Let $f,g:(\mathbb{C}^d,0)\rightarrow(\mathbb{C},0)$ be two germs of holomorphic functions. Their initial forms are their lowest-degree nonzero homogeneous parts, and their Milnor f...

conjecture | open | MathDB 页面 | 原文线索

V2-032092 | MATHDB-344557

Cha's primary decomposition conjecture for the topologically slice concordance group

摘录 / EXCERPT: Cha's primary decomposition conjecture. The canonical homomorphisms $\mathcal{T}^p_\Delta\to\mathcal{T}_\Delta$ induce an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032093 | MATHDB-344561

The WRT invariant formula in terms of chat Z invariants

摘录 / EXCERPT: Let M_3 be a 3-manifold, let b_1 denote its first Betti number, and let a and b label (almost) abelian flat connections on M_3. Write $\operatorname{CS}(a)$ for the Chern...

conjecture | open | MathDB 页面 | 原文线索

V2-032094 | MATHDB-344562

The Borel-resummed invariant conjecture for colored Jones asymptotics

摘录 / EXCERPT: Let K be a knot, let $J_n(q)$ be its n-colored Jones polynomial, and let $F_K(x,q)$ be the two-variable invariant defined by Borel resummation of the colored Jones expansion. F...

conjecture | open | MathDB 页面 | 原文线索

V2-032095 | MATHDB-344563

The 0-surgery formula for chat Z

摘录 / EXCERPT: Let K be a knot in S^3, and write ... Let $\cong$ denote equality up to sign and an overall power of q. The 0-surgery formula. ... In particular, for the torus-knot notation...

conjecture | open | MathDB 页面 | 原文线索

V2-032096 | MATHDB-344564

The residue formula for almost abelian 0-surgery invariants

摘录 / EXCERPT: The residue formula.

conjecture | open | MathDB 页面 | 原文线索

V2-032097 | MATHDB-344565

The residue formula for all almost abelian invariants of twist-knot 0-surgeries

摘录 / EXCERPT: The all-residues formula. For the labels a appearing in the almost abelian invariants,

conjecture | open | MathDB 页面 | 原文线索

V2-032098 | MATHDB-344566

The torsion formula for chat Z

摘录 / EXCERPT: Let M_3 be a plumbed 3-manifold with plumbing matrix Q, let a and b label the corresponding invariants, and let $\tau_{M_3}(b)(a)$ denote the torsion expression in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-032099 | MATHDB-344567

The surgery exact sequence for chat Z homology

摘录 / EXCERPT: Let $K\subset Y$ be a knot in an integral homology 3-sphere Y, and let $Y_0(K)$ and $Y_{+1}(K)$ denote the corresponding 0- and $+1$-surgeries. Let $\mathcal H(M_3)$ be the homol...

conjecture | open | MathDB 页面 | 原文线索

V2-032100 | MATHDB-344614

Twisted-index integrality conjecture for adjoint Reidemeister torsions

摘录 / EXCERPT: Let M be a compact 3-manifold with a torus boundary, and suppose that every irreducible component of $X^\mathrm{irr}(M)$ has dimension 1. For a slope...

conjecture | open | MathDB 页面 | 原文线索

V2-032101 | MATHDB-344639

The Steenrod-algebra kernel-image conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be the mod-2 Steenrod algebra, let β denote the relevant operation, and consider the quotient left $\mathcal{A}$-module $\mathcal{A}/\mathcal{A}\beta$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-032102 | MATHDB-344646

Concrete uncountable Moore-Schmidt conjecture

摘录 / EXCERPT: Let Γ be a discrete group acting concretely on a measure space $X=(X,\mathcal{X},\mu)$, and let K be a compact Hausdorff abelian group. A concrete $K_{\mathcal{B}a}$-value...

conjecture | open | MathDB 页面 | 原文线索

V2-032103 | MATHDB-344702

The embedding criterion for scalable spaces

摘录 / EXCERPT: Let Y be a space, let $H^*(Y;\mathbb{R})$ denote its real cohomology algebra, and let $\bigwedge^*\mathbb{R}^N$ be the exterior algebra on a finite-dimensional real vector space....

conjecture | open | MathDB 页面 | 原文线索

V2-032104 | MATHDB-344741

The signature-Arf invariant conjecture for the $\mathbb{C}P^2$-genus of knots

摘录 / EXCERPT: Signature-Arf invariant conjecture. If K_1 and K_2 have the same signature and Arf invariant, then they have the same $\mathbb{C}P^2$-genus.

conjecture | open | MathDB 页面 | 原文线索

V2-032105 | MATHDB-344753

Alternation conjecture for non-negatively curved odd GKM$_3$ manifolds

摘录 / EXCERPT: Alternation conjecture. The odd GKM$_3$ graph corresponding to M^{2n+1} is alternating.

conjecture | open | MathDB 页面 | 原文线索

V2-032106 | MATHDB-344795

The easier restricted-tile conjecture for hexagonal mosaics

摘录 / EXCERPT: For a knot, let a hexagonal r-mosaic be a knot diagram assembled on the hexagonal r-mosaic arrangement. The interior tiles are the tiles away from the boundary. The four three-...

conjecture | open | MathDB 页面 | 原文线索

V2-032107 | MATHDB-344796

The harder restricted-tile conjecture for hexagonal mosaics

摘录 / EXCERPT: For a knot, let a hexagonal r-mosaic be a knot diagram assembled on the hexagonal r-mosaic arrangement, and let the interior tiles be the tiles away from the boundary. The four...

conjecture | open | MathDB 页面 | 原文线索

V2-032108 | MATHDB-344858

Degree and sign independence conjecture for automorphic \a3-invariants

摘录 / EXCERPT: Let G be a reductive group, let $\a3$ be a sign character, let d be an integer with $0\leq d\leq\delta$, and let $i\in\Delta$ be a root. The automorphic invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-032109 | MATHDB-344927

Density of Chern slopes with fixed fundamental group and ample canonical class

摘录 / EXCERPT: Let G be the (topological) fundamental group of a nonsingular complex projective surface. For a nonsingular projective surface S, write $c_1^2(S)/c_2(S)$ for its Chern slope. D...

conjecture | open | MathDB 页面 | 原文线索

V2-032110 | MATHDB-344931

Schmutz Schaller's automorphism conjecture for the Schmutz Schaller curve graph

摘录 / EXCERPT: Let S be the surface under consideration, and let $\mathcal{SC}(S)$ denote the Schmutz Schaller curve graph, with vertices and edges as defined in the paper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032111 | MATHDB-344960

Lupton's cohomology-dimension conjecture for elliptic Sullivan algebras

摘录 / EXCERPT: Lupton's conjecture. Either

conjecture | open | MathDB 页面 | 原文线索

V2-032112 | MATHDB-345076

Heuristic conjecture on gestural similarity of musical gestures

摘录 / EXCERPT: Let two musical gestures be based on the same skeleton, meaning an abstract structure of points and arrows that is specified into a collection of points and arrows in a topological...

conjecture | open | MathDB 页面 | 原文线索

V2-032113 | MATHDB-345079

Stable $U_{d4e4}(d4f0d4f1_2)$-action conjecture for quantum annular spectra

摘录 / EXCERPT: Let L be an annular link, let $Kh_{\mathbb{A}_\mathfrak{q}}(L)$ denote its quantum annular homology, and let ${\mathcal X}^r_{\mathbb{A}_\mathfrak{q}}(L)$ be the associated spect...

conjecture | open | MathDB 页面 | 原文线索

V2-032114 | MATHDB-345092

Min-max existence conjecture for embedded constant mean curvature surfaces

摘录 / EXCERPT: Let M be a closed 3-manifold, and let S be an embedded surface in M with nontrivial homology. Let $\widehat{C}_{[S]}$ be the lower bound defined using the width of the homo...

conjecture | open | MathDB 页面 | 原文线索

V2-032115 | MATHDB-345093

Polishchuk-Vaintrob homogeneity conjecture

摘录 / EXCERPT: For an r-tuple $\overline{{\boldsymbol\gamma}}=({\boldsymbol\gamma}_1,\ldots,{\boldsymbol\gamma}_r)$ in $G_{\boldsymbol{w}}^r$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-032116 | MATHDB-345111

Uniform random complex top-homology asymptotic conjecture

摘录 / EXCERPT: Let $\mathbb{F}$ be a field and let $\Delta\tooltip\sim \mathcal{U}(n)$, where n is even. Here $\mathcal{U}(n)$ denotes the uniform random simplicial complex on $[n]$. Top-homolog...

conjecture | open | MathDB 页面 | 原文线索

V2-032117 | MATHDB-345155

Classifying-map conjecture for principal simplicial group bundles

摘录 / EXCERPT: Classifying-map conjecture. Under the relevant zigzag of equivalences, the map $N\alpha_*$ is a classifying map for $p:E\to B$: its homotopy class coincides with...

conjecture | open | MathDB 页面 | 原文线索

V2-032118 | MATHDB-345168

Exponential growth conjecture for mapping-torus Turaev-Viro invariants

摘录 / EXCERPT: Let Σ be a compact orientable surface and let $f\in\mathrm{Mod}(\Sigma)$. Let M_f denote the mapping torus of f, and let $\lVert M_f\rVert$ denote its Gromov norm. A com...

conjecture | open | MathDB 页面 | 原文线索

V2-032119 | MATHDB-345211

The g-conjecture for simplicial and rational homology spheres

摘录 / EXCERPT: Let Δ be a simplicial sphere or a rational homology sphere. Its h-vector is subject to the g-theorem conditions: Dehn-Sommerville symmetry, unimodality up to the middle,...

conjecture | open | MathDB 页面 | 原文线索

V2-032120 | MATHDB-345212

The GrunbaumKalaiSarkaria conjecture

摘录 / EXCERPT: Let Δ be a simplicial complex of dimension d. Assume that Δ admits a PL embedding into $\mathbb{R}^{2d}$. GrunbaumKalaiSarkaria conjecture. Then ... This conjecture s...

conjecture | open | MathDB 页面 | 原文线索

V2-032121 | MATHDB-345228

The integral homology sphere conjecture for k-Lê-Yomdin singularities

摘录 / EXCERPT: Integral homology sphere conjecture. No k-Lê-Yomdin singularity with $k>1$ has an integral homology sphere link.

conjecture | open | MathDB 页面 | 原文线索

V2-032122 | MATHDB-345231

Lower semicontinuity of the homotopy type of Vietoris-Rips complexes

摘录 / EXCERPT: Lower semicontinuity conjecture. The complex $\mathrm{VR}_r(X)$ is homotopy equivalent to $\mathrm{VR}_{r-\varepsilon}(X)$ whenever $\varepsilon>0$ is sufficiently small.

conjecture | open | MathDB 页面 | 原文线索

V2-032123 | MATHDB-345232

AAF17's first post-critical homotopy-type conjecture for spheres

摘录 / EXCERPT: AAF17's sphere conjecture. There exists an $\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032124 | MATHDB-345233

The longest Vietoris-Rips persistence interval conjecture for metric manifolds

摘录 / EXCERPT: Longest-interval conjecture. For every $l\in\mathrm{barc}^{\mathrm{VR}}_k(M;\mathbb{F})$ and every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032125 | MATHDB-345235

The ANR version of Hausmann's theorem for Vietoris-Rips complexes

摘录 / EXCERPT: ANR Vietoris-Rips conjecture. There exists $r(X)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032126 | MATHDB-345260

The g-conjecture for rational homology spheres

摘录 / EXCERPT: The g-conjecture. The g-vector of a rational homology sphere is an M-vector.

conjecture | open | MathDB 页面 | 原文线索

V2-032127 | MATHDB-345261

The algebraic g-conjecture for rational homology spheres

摘录 / EXCERPT: The algebraic g-conjecture. Every rational homology sphere has Lefschetz property.

conjecture | open | MathDB 页面 | 原文线索

V2-032128 | MATHDB-345356

The folklore conjecture on symplectic Calabi-Yau four-manifolds

摘录 / EXCERPT: Folklore conjecture. If $\pi_1(M)=0$ or $b_+(M)=3$, then there exists an integrable almost complex structure J on M compatible with ω, making (M,J,ω) a Kähler sur...

conjecture | open | MathDB 页面 | 原文线索

V2-032129 | MATHDB-345357

Donaldson's hypersymplectic isotopy conjecture

摘录 / EXCERPT: Donald'son conjecture. For every hypersymplectic structure $\underline{\omega}$ on a compact 4-manifold M, there is an isotopy $\underline{\omega}(t)$ of cohomologous hypersymple...

conjecture | open | MathDB 页面 | 原文线索

V2-032130 | MATHDB-345369

Effective splitting conjecture for branched surfaces

摘录 / EXCERPT: Let B be a branched surface in a compact triangulated atoroidal irreducible 3-manifold N. A branched surface is quasi-hyperbolic if it does not carry any sphere or torus but...

conjecture | open | MathDB 页面 | 原文线索

V2-032131 | MATHDB-345370

Linear lower bound for compressing-disc boundary lengths in handlebodies

摘录 / EXCERPT: Let H^i_0 and H^i_1 be the handlebodies referred to in the source's preceding remark, and let g denote the relevant genus. Linear boundary-length conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-032132 | MATHDB-345372

The top-degree dimension conjecture for the exterior graded Swiss-Cheese operad

摘录 / EXCERPT: Let V be a finite-dimensional vector space over k, and write $d=\dim_k(V)$. Let ${\Lambda}^{S^2}_V(m)$ denote the degree-m component of the exterior graded Swiss-Cheese opera...

conjecture | open | MathDB 页面 | 原文线索

V2-032133 | MATHDB-345386

Acyclicity conjecture for the Quilt operad

摘录 / EXCERPT: Let $\operatorname{\textit{Quilt}}$ be the operad of quilts, and let $H_k(\operatorname{Quilt})$ denote its homology in degree k. Acyclicity conjecture for the Quilt operad....

conjecture | open | MathDB 页面 | 原文线索

V2-032134 | MATHDB-345408

Vortex-cluster formula for the baryon number

摘录 / EXCERPT: Let $X=\mathbb{R}^3\cup\{\infty\}$ and let $U:X\to S^3$ be a map of degree B. Interpret U as two vortices in ψ_1 and ψ_2 of $\boldsymbol{\psi}\in S^3$, with winding n...

conjecture | open | MathDB 页面 | 原文线索

V2-032135 | MATHDB-345446

Aperiodicity conjecture for disjointness of additive and multiplicative systems

摘录 / EXCERPT: Let (X,T) be an additive topological dynamical system of zero entropy, and let (Y,S) be a finitely generated multiplicative topological dynamical system. Aperiodicity conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-032136 | MATHDB-345447

Uniquely ergodic correlation conjecture for additive and multiplicative systems

摘录 / EXCERPT: Let (X,cmu,T) be a uniquely ergodic additive topological dynamical system of zero entropy, and let (Y,cnu,S) be a finitely generated and strongly uniquely ergodic multiplicativ...

conjecture | open | MathDB 页面 | 原文线索

V2-032137 | MATHDB-345457

The stronger hemispherical-curve conjecture for locally convex curves on the 3-sphere

摘录 / EXCERPT: Let $\mathcal{L}\mathbb S^3(\mathbf{1},-\mathbf{1})$ be the space of locally convex curves on the 3-sphere with the indicated endpoint data, let L denote the associated map to cu...

conjecture | open | MathDB 页面 | 原文线索

V2-032138 | MATHDB-345523

Formality conjecture for the cyclic bar complex of Soergel bimodules

摘录 / EXCERPT: Let W be a Coxeter group with simple reflections $S\subset W$, a realization V over $\mathbb{C}$, and let $\mathrm{SBim}(W)$ be the associated monoidal category of Soergel bimo...

conjecture | open | MathDB 页面 | 原文线索

V2-032139 | MATHDB-345525

Rotator-Connes differential conjecture for derived annular Khovanov-Rozansky invariants

摘录 / EXCERPT: Let Tr be the universal dg monoidal trace, let $\operatorname{Tr}(\mathbb{1})$ be the trace of the monoidal unit, and let X be a twisted complex built out of su...

conjecture | open | MathDB 页面 | 原文线索

V2-032140 | MATHDB-345540

The nondisplaceable-subsurface characterization of geometric normal subgroups

摘录 / EXCERPT: Let Σ be a surface, let $\operatorname{Map}^{\pm}(\Sigma)$ be its extended mapping class group, and let N be a normal subgroup of this group. A finite-type subsurface...

conjecture | open | MathDB 页面 | 原文线索

V2-032141 | MATHDB-345541

Brendle-Margalit conjectures on complexes of regions and geometric normal subgroups

摘录 / EXCERPT: Let Σ be a surface, let $\mathcal{C}_R(\Sigma)$ be a complex of regions, and let N be a normal subgroup of $\operatorname{Map}^{\pm}(\Sigma)$. Brendle-Margalit conjectures...

conjecture | open | MathDB 页面 | 原文线索

V2-032142 | MATHDB-345548

Dimension reduction conjecture for representations of motion groups of links with generalized axes

摘录 / EXCERPT: A link $L\subset\mathbb{S}^3$ has a generalized axis γ with fiber surface F when the complement of γ fibers over $\mathbb{S}^1$ with fiber F. A 1-extended...

conjecture | open | MathDB 页面 | 原文线索

V2-032143 | MATHDB-345550

Koike's infinite-type conjecture for semi-positive hypersurfaces

摘录 / EXCERPT: Let X be a complex manifold and let Y be a non-singular compact Kähler hypersurface of X. Assume that the normal bundle $N_{Y/X}$ is topologically trivial and that the line b...

conjecture | open | MathDB 页面 | 原文线索

V2-032144 | MATHDB-345573

Composite-order Tate cohomology conjecture for graded Brauer characters

摘录 / EXCERPT: Composite-order graded Brauer character conjecture. Each of these series is a linear combination of Hauptmoduls. This predicts a uniform Hauptmodul description for the graded Braue...

conjecture | open | MathDB 页面 | 原文线索

V2-032145 | MATHDB-345577

Finite-to-one extension conjecture for systems without IT-tuples

摘录 / EXCERPT: Finite-to-one extension conjecture. If (X,T) has no k-IT-tuples, then (X,T) is an almost finite-to-one extension of its maximal equicontinuous factor.

conjecture | open | MathDB 页面 | 原文线索

V2-032146 | MATHDB-345578

IT-tuples in proximal non-almost-one-to-one extensions

摘录 / EXCERPT: IT-tuple conjecture. If π is proximal and not almost one-to-one, then for each $k\ge 2$, there is a k-IT tuple.

conjecture | open | MathDB 页面 | 原文线索

V2-032147 | MATHDB-345579

Enumerative P=W conjecture for the defect-homogeneous extension

摘录 / EXCERPT: Enumerative P=W conjecture. In the redundancy range, there exists a unique defect-homogeneous $F\in H^*(Z)$ with $\operatorname{def}(F)=2k-2$ such that, for every defect-homogeneou...

conjecture | solved | MathDB 页面 | 原文线索

V2-032148 | MATHDB-345601

Topological-recursion quantization conjecture for 2-functions

摘录 / EXCERPT: Let $\mathbf{T}$ be the set of 2-functions and let $W\in\mathbf{T}$. Construct a spectral curve $\mathcal{C}_W$ associated with W, apply topological recursion to obtain symplecti...

conjecture | open | MathDB 页面 | 原文线索

V2-032149 | MATHDB-345617

Tree-factorization conjecture for mappings with small $(2,1)$-mapping content

摘录 / EXCERPT: Let $Q_0=[0,1]^3$ and let $f\colon Q_0\rightarrow X$ be a 1-Lipschitz mapping into a metric space. Assume without loss of generality that $X\subseteq\ell_\infty$. A Lipschitz map...

conjecture | open | MathDB 页面 | 原文线索

V2-032150 | MATHDB-345619

Symplectic/contact analogue of the Looijenga conjecture

摘录 / EXCERPT: Let a cusp singularity be a surface singularity whose link carries the contact structure associated with its cusp resolution, and let a rational dual mean that the dual cusp singul...

conjecture | open | MathDB 页面 | 原文线索

V2-032151 | MATHDB-345621

Homological Berglund-Hübsch mirror symmetry for chain type invertible singularities

摘录 / EXCERPT: Homological Berglund-Hübsch mirror symmetry conjecture. There exists a triangulated equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-032152 | MATHDB-345622

The pro-nilsystem conjecture for distal minimal topological dynamical systems

摘录 / EXCERPT: Let (X,T) be a distal minimal topological dynamical system, and let ${\bf Ind}^{[d]}_{ap}(X,T)$ denote its arithmetic-progression independence relation of order d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032153 | MATHDB-345671

The alternating-knot characterization conjecture via colored Jones degrees

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let p_K denote its Jones period, and let $d_+[J_K(n)]$ and $d_-[J_K(n)]$ denote the maximal and minimal degrees of its colored Jones polynomial in...

conjecture | open | MathDB 页面 | 原文线索

V2-032154 | MATHDB-345700

The metric SYZ conjecture for large complex structure limits

摘录 / EXCERPT: Metric SYZ conjecture. As $t\to0$, $(X_t,\widetilde{\omega}_t)$ converges in the Gromov-Hausdorff topology to a compact metric space (Z,d) with an open dense subset Z_0 such th...

conjecture | open | MathDB 页面 | 原文线索

V2-032155 | MATHDB-345701

The essential-skeleton metric limit conjecture

摘录 / EXCERPT: Essential-skeleton metric limit conjecture. As $t\to0$, $(X_t,\widetilde{\omega}_t)$ converges in the Gromov-Hausdorff topology to a compact metric space (Z,d) homeomorphic to th...

conjecture | open | MathDB 页面 | 原文线索

V2-032156 | MATHDB-345706

Lück's torsion-growth conjecture for residually finite aspherical manifolds

摘录 / EXCERPT: Let M^n be a closed aspherical n-manifold with residually finite fundamental group. Let $\Gamma_k\vartriangleleft\pi_1(M^n)$ be any normal chain with $\bigcap_k\Gamma_k=1$. Lüc...

conjecture | open | MathDB 页面 | 原文线索

V2-032157 | MATHDB-345708

Lekili-Ueda conjecture for invertible polynomials

摘录 / EXCERPT: Let $\mathbf{w}$ and $\check{\mathbf{w}}$ be a pair of invertible polynomials, let $\Gamma_{\mathbf{w}}$ be the maximal symmetry group of $\mathbf{w}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032158 | MATHDB-345711

Arnold's structural stability conjecture for finite-parameter families on the two-sphere

摘录 / EXCERPT: Arnold's conjecture. Generic finite-parameter families of vector fields on the two-sphere are structurally stable: close families are weakly topologically equivalent.

conjecture | refuted | MathDB 页面 | 原文线索

V2-032159 | MATHDB-345724

Hilali's conjecture for mixed Hodge and homotopical Poincaré polynomials

摘录 / EXCERPT: Hilali conjecture. The homotopical Poincaré polynomial is bounded above by the Poincaré polynomial:

conjecture | open | MathDB 页面 | 原文线索

V2-032160 | MATHDB-345725

Asymptotic strict mixed Hodge inequality for powers of a space

摘录 / EXCERPT: Asymptotic strict mixed Hodge inequality conjecture. There exists a positive integer n_0 such that, for every $n\geqq n_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-032161 | MATHDB-345727

The local Tamagawa number conjecture for $h^0(K)(j)$

摘录 / EXCERPT: The local Tamagawa number conjecture. For every positive integer j there exists a unique $\mathbb{Z}_p$-basis

conjecture | open | MathDB 页面 | 原文线索

V2-032162 | MATHDB-345728

The Tamagawa number conjecture for $h^0(K)(j)$

摘录 / EXCERPT: The Tamagawa number conjecture. There exists a unique $\mathbb{Z}_p$-basis

conjecture | open | MathDB 页面 | 原文线索

V2-032163 | MATHDB-345730

The generalized Coleman-Ihara formula

摘录 / EXCERPT: Generalized Coleman-Ihara formula. For every odd integer $j>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032164 | MATHDB-345757

Juhász's depth bound conjecture for taut foliations

摘录 / EXCERPT: Juhász's conjecture. The sutured manifold (M,γ) admits a taut foliation of depth at most $2k$.

conjecture | open | MathDB 页面 | 原文线索

V2-032165 | MATHDB-345758

Kronheimer-Mrowka's depth bound conjecture for knot complements

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $S^3(K)$ denote its knot complement. Consider the irreducible homomorphisms ... that map a chosen meridian m to $\mathbf{i}\in SU(2)$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-032166 | MATHDB-345772

Conjecture on minimal diagrams of decomposable handlebody links

摘录 / EXCERPT: Decomposable-handlebody-link conjecture. Minimal diagrams of a decomposable handlebody link have 2-connectivity.

conjecture | open | MathDB 页面 | 原文线索

V2-032167 | MATHDB-345773

Conjecture on 1-connected minimal diagrams of reducible handlebody links

摘录 / EXCERPT: Reducible-handlebody-link conjecture. Every non-split, reducible handlebody link admits a minimal diagram with 1-connectivity.

conjecture | open | MathDB 页面 | 原文线索

V2-032168 | MATHDB-345774

Crossing-number additivity conjecture for order-1 connected sums of handlebody links

摘录 / EXCERPT: Crossing-number additivity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-032169 | MATHDB-345778

Conjecture on the orientability order of the universal complex line bundle for EO-theory

摘录 / EXCERPT: Let Γ be a formal group of height $n=(p-1)k$, and let $\mathbb{C}$ denote the complex line bundle appearing in the definition of the orientability order Θ. Orientabil...

conjecture | open | MathDB 页面 | 原文线索

V2-032170 | MATHDB-345812

The non-negative curvature and Seifert fibration conjecture for 2-connected 7-manifolds

摘录 / EXCERPT: Let M be a 2-connected 7-manifold whose cohomology ring is that of an $\mathbf{S}^3$-bundle over $\mathbf{S}^4$. A non-negative curvature and Seifert fibration conjecture ass...

conjecture | open | MathDB 页面 | 原文线索

V2-032171 | MATHDB-345844

Continuity conjecture for Hardy constants of weighted means

摘录 / EXCERPT: Continuity conjecture. If $\mathscr{M}$ is a $\textbf1$-Hardy mean, then its weighted Hardy constant $\mathscr{H}_{\mathscr{M}}$ is continuous in the pointwise topology.

open_problem | open | MathDB 页面 | 原文线索

V2-032172 | MATHDB-345855

Resolution conjecture for generalized manifolds

摘录 / EXCERPT: Resolution conjecture. Every generalized n-manifold for $n\geq 3$ is resolvable.

conjecture | open | MathDB 页面 | 原文线索

V2-032173 | MATHDB-345856

Homogeneity conjecture for generalized manifolds

摘录 / EXCERPT: Homogeneity conjecture. Every connected generalized n-manifold with the disjoint disks property is homogeneous, for $n\geq 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-032174 | MATHDB-345864

Cantat-McMullen Kummer rigidity conjecture for K3 automorphisms

摘录 / EXCERPT: Let X be a $K3$ surface, let $T:X\to X$ be an automorphism with positive topological entropy, let $\mathrm{dVol}=\Omega\wedge\overline{\Omega}$ be the Lebesgue measure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032175 | MATHDB-345875

Compact generation of local groups from compact generation of the universal group

摘录 / EXCERPT: Let $(\Delta,\mathcal{U}(\boldsymbol F))$ be a right-angled building with universal group determined by local groups F_i, indexed by $i\in I$. Assume that each local group F_i...

open_problem | open | MathDB 页面 | 原文线索

V2-032176 | MATHDB-345876

Resolution conjecture for tensor hierarchies of stringent Lie-Leibniz triples

摘录 / EXCERPT: Let $(\mathfrak{g},V,\Theta)$ be a stringent Lie-Leibniz triple, and let $\mathcal{I}$ denote its ideal of squares. Its associated tensor hierarchy has terms T_{-k} and different...

conjecture | open | MathDB 页面 | 原文线索

V2-032177 | MATHDB-345877

Conjecture on removing assumptions from the uniqueness of Poincaré duality models

摘录 / EXCERPT: Let V be a differential Poincaré duality algebra with $\\H^0(V)=\operatorname{Span}\{1\}$ and $\H^1(V)=0$, and let $\Model_1$ and $\Model_2$ be Poincaré duality models satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-032178 | MATHDB-345879

Equivalence conjecture for algebraic and geometric Chern-Simons constructions

摘录 / EXCERPT: Let M be a connected, oriented, closed Riemannian n-manifold with $\HDR^1(M)=0$. Let $\DR(M)$ be its de Rham complex, $\HDR(M)$ its de Rham cohomology, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032179 | MATHDB-345880

Formality conjecture for the geometric Chern-Simons construction

摘录 / EXCERPT: Let M be a connected, oriented, closed Riemannian manifold with $\HDR^1(M)=0$, and suppose that M is formal. In the situation of the equivalence conjecture, let $\PMC$ be the C...

conjecture | open | MathDB 页面 | 原文线索

V2-032180 | MATHDB-345881

String topology conjecture for closed oriented manifolds

摘录 / EXCERPT: Let M be a closed oriented manifold of dimension n, and let $\H_{\mathrm{dR}}$ be its de Rham cohomology. Let $\DBCyc\,\H_{\mathrm{dR}}[2-n]$ denote a suitable shifted cyclic c...

conjecture | open | MathDB 页面 | 原文线索

V2-032181 | MATHDB-345882

String topology chain-map conjecture for simply-connected manifolds

摘录 / EXCERPT: Let M be an oriented closed manifold of dimension n. Let $C^{\mathrm{sing}}(\Loop_{\Sph{1}}M;\R)$ be the smooth singular chain complex with boundary $\Bdd$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032182 | MATHDB-345909

The Sullivan Conjecture for smooth complete intersections

摘录 / EXCERPT: A complete intersection $X_n(\underline d)\subseteq\mathbb{C}P^{n+k}$ is the transverse intersection of k complex hypersurfaces of multidegree $\underline d=(d_1,\ldots,d_k)$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-032183 | MATHDB-345912

Berger and Turaev's density conjecture for positive-entropy area-preserving disc diffeomorphisms

摘录 / EXCERPT: Let ${\rm Diff}^\infty_\omega(\mathbb{D})$ be the set of C^∞ diffeomorphisms of the disc that preserve area, with its smooth topology. A diffeomorphism has positive metric e...

conjecture | open | MathDB 页面 | 原文线索

V2-032184 | MATHDB-346007

Independence of the elliptic parameter in higher-rank K-theoretic DT theory

摘录 / EXCERPT: Independence conjecture. The series $\mathsf{DT}_{r,k}^{\rm ell}(\mathbb{A}^3,q,t,w;p)$ does not depend on the elliptic parameter p.

conjecture | open | MathDB 页面 | 原文线索

V2-032185 | MATHDB-346009

Qazaqzeh-Qublan-Jaradat crossing-number conjecture for quasi-alternating links

摘录 / EXCERPT: Let L be a quasi-alternating link, and let $c(L)$ denote its crossing number and $\det(L)$ its determinant. Qazaqzeh-Qublan-Jaradat conjecture. Every quasi-alternating link L s...

conjecture | open | MathDB 页面 | 原文线索

V2-032186 | MATHDB-346017

Lurie's dualizability conjecture for E_n-algebras

摘录 / EXCERPT: Let n be a nonnegative integer, let $\mathcal{S}$ denote the ambient symmetric monoidal category of spaces, and let $\mathcal{A}\in\operatorname{Alg}_n(\mathcal{S})$ be an E_n-...

conjecture | open | MathDB 页面 | 原文线索

V2-032187 | MATHDB-346032

Fixed-point conjecture for L-space knots

摘录 / EXCERPT: Let K be an L-space knot, and let μ and λ represent a meridian and a longitude in the knot group. For a homomorphism ρ from $\pi_1(S^3\setminus K)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-032188 | MATHDB-346059

Normalized Jones R-matrix Yang-Baxter homology conjecture

摘录 / EXCERPT: Yang-Baxter homology conjecture. The homology groups satisfy

conjecture | solved | MathDB 页面 | 原文线索

V2-032189 | MATHDB-346075

Cofibrancy preservation conjecture for equivariant colored categories and operads

摘录 / EXCERPT: Cofibrancy preservation conjecture. These pullback functors preserve cofibrations between cofibrant objects. This generalizes the Interval Cofibrancy Theorem, which is the special...

conjecture | open | MathDB 页面 | 原文线索

V2-032190 | MATHDB-346089

The GM series quantum A-polynomial annihilation conjecture

摘录 / EXCERPT: Let K be a knot, let $F_K(x,q)$ be the GM series, and let $\hat{A}_K(\hat{x},\hat{y},q)$ be the quantum A-polynomial annihilating the unreduced colored Jones polynomials. The q...

conjecture | open | MathDB 页面 | 原文线索

V2-032191 | MATHDB-346104

Self-repulsiveness thresholds for regularized alpha-energy

摘录 / EXCERPT: Self-repulsiveness conjecture. The regularized α-energy E_α for smooth closed surfaces is self-repulsive with respect to C^0-topology if and only if $\alpha\leq -4$...

conjecture | open | MathDB 页面 | 原文线索

V2-032192 | MATHDB-346118

Equivalence of the SCC and NSC for flag Gorenstein* complexes

摘录 / EXCERPT: A flag Gorenstein* n-complex K is assumed to satisfy the SCC, and the NSC is the condition that there is no full subcomplex $K_I\subset K$ such that K_I is a suspension an...

conjecture | open | MathDB 页面 | 原文线索

V2-032193 | MATHDB-346125

Gordon's conjecture on toroidal and Seifert Dehn fillings

摘录 / EXCERPT: Let M be a hyperbolic knot manifold, meaning a compact, connected, orientable 3-manifold with torus boundary whose interior admits a complete finite-volume hyperbolic structure...

conjecture | open | MathDB 页面 | 原文线索

V2-032194 | MATHDB-346130

Equivalence of virtual knot theory and virtual mosaic theory

摘录 / EXCERPT: Let L_1 and L_2 be virtual knots or links, and let virtual mosaics represent them. Two virtual mosaics are equivalent when they represent equivalent virtual knots or links. Vir...

conjecture | open | MathDB 页面 | 原文线索

V2-032195 | MATHDB-346146

Orientation-reversal conjecture for the Seiberg-Witten invariant

摘录 / EXCERPT: Let X be a smooth spin rational homology $S^1\times S^3$ which is oriented and homology oriented, and let $-X$ denote X with reversed orientation but the same homology orientat...

conjecture | open | MathDB 页面 | 原文线索

V2-032196 | MATHDB-346162

Non-abelian Leopoldt dimension conjecture for ordinary cohomology

摘录 / EXCERPT: Let Λ_E be the regular local ordinary weight algebra of dimension $r=\operatorname{rk}\mathrm T$, let $F_{\mathfrak m}^{\bullet}$ be the complex interpolating the localized...

conjecture | open | MathDB 页面 | 原文线索

V2-032197 | MATHDB-346164

Smoothness conjecture for the ordinary deformation ring

摘录 / EXCERPT: Let R_ρ^{ord} be the ordinary deformation ring and let $H^2_{\mathcal P_{\mathrm{ord}}}(\operatorname{Gal}_{\mathbb Q,\Sigma},\operatorname{Lie}\check{\mathrm G})$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-032198 | MATHDB-346165

Galatius--Venkatesh compatibility of derived Hecke actions

摘录 / EXCERPT: Let K be the level, let $\mathfrak m$ be the maximal ideal associated to π, and let ... with $f\in H^*(Y(K),E)_\mathfrak m$. Galatius--Venkatesh compatibility conjecture. The d...

conjecture | open | MathDB 页面 | 原文线索

V2-032199 | MATHDB-346173

The topology conjecture for proper dp-minimal expansions of $(\mathbb{Z},+)$

摘录 / EXCERPT: Topology conjecture. Any proper dp-minimal expansion of $(\mathbb{Z},+)$ which eliminates $\exists^\infty$ defines a non-discrete group topology on $(\mathbb{Z},+)$.

conjecture | open | MathDB 页面 | 原文线索

V2-032200 | MATHDB-346179

Density and openness of conical degeneracies for periodic Schrödinger operators

摘录 / EXCERPT: Let $V\in C^\infty(\mathbb{R}^3/\mathbb{Z}^3)$ be a smooth periodic potential, and consider the periodic Schrödinger operator $-\Delta_{\mathbb{R}^3}+V$. A degeneracy of its Bloch...

conjecture | open | MathDB 页面 | 原文线索

V2-032201 | MATHDB-346185

The uniqueness conjecture for PD algebras with nonzero second cohomology

摘录 / EXCERPT: Let V_1 and V_2 be PD algebras satisfying the assumptions of Proposition, except that their second cohomology is allowed to be nonzero, and suppose ... Here a PDGA quasi-isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-032202 | MATHDB-346186

Homological mirror symmetry conjecture for Milnor fibers of simple singularities

摘录 / EXCERPT: Let $\w\in\bC[x_1,\ldots,x_{n+1}]$ be an invertible polynomial, with transpose polynomial $\wv$, and let $\Gamma_{\w}$ be the group acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-032203 | MATHDB-346188

Conjecture on Brieskorn homology spheres from regular Lagrangian slice knots

摘录 / EXCERPT: Let $\Sigma(p,q,r)$ be a non-trivial Brieskorn homology sphere, let n be a natural number, and let K be a regular Lagrangian slice knot. Lagrangian-surgery conjecture. No such...

conjecture | open | MathDB 页面 | 原文线索

V2-032204 | MATHDB-346189

The Homflypt Yang-Baxter homology conjecture for the two-element rack

摘录 / EXCERPT: The Homflypt Yang-Baxter homology conjecture. For $m=2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-032205 | MATHDB-346190

The third Yang-Baxter homology formula for the Homflypt rack family

摘录 / EXCERPT: Let $R_{(m)}$ be the rack considered in the paper, let $H_3(R_{(m)})$ denote its third Yang-Baxter homology over k, and let m be the parameter indexing this rack family. The th...

conjecture | open | MathDB 页面 | 原文线索

V2-032206 | MATHDB-346196

The conjecture on the distribution of writhe in random grid diagrams

摘录 / EXCERPT: Let w denote the writhe of a knot represented by an $n\times n$ grid diagram. The distribution is considered for knots encoded by two n-permutations recording the order of the...

conjecture | open | MathDB 页面 | 原文线索

V2-032207 | MATHDB-346215

The genus-2 mutation invariance conjecture for Floer homology ranks

摘录 / EXCERPT: Floer-rank mutation conjecture. The ranks of $I^\#$ and $\widehat{\mathit{HF}}$ are preserved by genus-2 mutation.

conjecture | open | MathDB 页面 | 原文线索

V2-032208 | MATHDB-346216

The tau-sharp equality conjecture for knots

摘录 / EXCERPT: Tau-sharp equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032209 | MATHDB-346223

Cohen-Moore-Neisendorfer-Gray-Mahowald conjecture on charming maps and Anick spaces

摘录 / EXCERPT: Charming-map conjecture. The following statements are true:

conjecture | open | MathDB 页面 | 原文线索

V2-032210 | MATHDB-346235

The sutured embedded contact homology-sutured Floer homology isomorphism conjecture

摘录 / EXCERPT: Sutured ECH-SFH isomorphism conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032211 | MATHDB-346236

The knot Floer homology and embedded contact homology correspondence conjecture

摘录 / EXCERPT: ECH-knot Floer correspondence conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032212 | MATHDB-346256

Closed string mirror symmetry for compact toric Fano manifolds

摘录 / EXCERPT: Let X be a compact toric Fano manifold, and let W be its Landau-Ginzburg mirror. Write $QH^*(X)$ for the quantum cohomology ring and $Jac(W)$ for the Jacobian ring of W. Clos...

conjecture | open | MathDB 页面 | 原文线索

V2-032213 | MATHDB-346268

The conjecture that all gapped fracton schemas are lax-topological

摘录 / EXCERPT: A lax-topological D-dimensional Hamiltonian schema satisfies $TQO0$, $TQO1$, and $TQO2$, and there exist constants α and β depending only on D such that ... for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-032214 | MATHDB-346269

The conjecture that all state-sum Hamiltonian schemas are topological

摘录 / EXCERPT: A state-sum Hamiltonian schema is obtained from a topological state-sum construction: for a cellulation Δ of a space manifold Y, the state-sum on $Y\times I$ defines an id...

conjecture | open | MathDB 页面 | 原文线索

V2-032215 | MATHDB-346288

Topological-simplicial comodule-contramodule correspondence conjecture

摘录 / EXCERPT: Topological-simplicial correspondence conjecture. The left adjoint functors are either on top or on the left, and the vertical solid arrows are Quillen equivalences.

conjecture | open | MathDB 页面 | 原文线索

V2-032216 | MATHDB-346322

The decomposition conjecture for W-topological fields in characteristic zero

摘录 / EXCERPT: Let (K,τ) be a W-topological field of characteristic 0. Decomposition conjecture. The topology τ is generated by jointly independent topologies $\tau_1,\ldots,\tau_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-032217 | MATHDB-346323

The positive-characteristic decomposition conjecture for W-topologies

摘录 / EXCERPT: Let K be a perfect field of positive characteristic, and let τ be a W_n-topology on K. Positive-characteristic decomposition conjecture. The topology τ is generated...

conjecture | open | MathDB 页面 | 原文线索

V2-032218 | MATHDB-346324

The NTP2 and burden conjecture for W_n-rings

摘录 / EXCERPT: Let R be a W_n-ring on an algebraically closed field. Wn-ring conjecture. The ring R is NTP$_2$, and the burden of R is at most n. The source gives no further context or...

conjecture | open | MathDB 页面 | 原文线索

V2-032219 | MATHDB-346328

Hikita-Nakajima conjecture for BFN Coulomb branches

摘录 / EXCERPT: Hikita-Nakajima conjecture. There is an algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032220 | MATHDB-346346

The graded Drinfeld-center conjecture for holonomy invariants

摘录 / EXCERPT: Let $\mathcal{C}$ be the $\operatorname{SL}_2(\mathbb{C})^*$-graded category described in the paper, let $\mathcal{D}$ be the doubled category constructed from $\mathcal{C}$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-032221 | MATHDB-346347

A graded Drinfeld-double realization of the doubled category

摘录 / EXCERPT: Let $\mathcal{D}$ be the doubled category constructed in the paper, let $\mathcal{U}_i(\mathfrak{sl}_2)$ be the quantum group at i, and let D_G denote a G-graded version of t...

conjecture | open | MathDB 页面 | 原文线索

V2-032222 | MATHDB-346356

Quadrelli-Weigel's Bloch-Kato conjecture for right-angled Artin pro-p groups

摘录 / EXCERPT: Let Γ be a finite simplicial graph, and let $G=G_{\Gamma}$ be the associated right-angled Artin pro-p group. The graph contains an induced subgraph of type C_4 when it h...

conjecture | open | MathDB 页面 | 原文线索

V2-032223 | MATHDB-346374

Log hard Lefschetz conjecture

摘录 / EXCERPT: Log hard Lefschetz conjecture. If $\mathcal{L}$ is ample, then L^r is an isomorphism for all $0\leq r\leq n$.

conjecture | open | MathDB 页面 | 原文线索

V2-032224 | MATHDB-346446

Comparison conjecture for perverse Eilenberg-MacLane spaces

摘录 / EXCERPT: Let ${\mathtt P}=[n]$, and let $\overline{p}$ and $\overline{q}$ be Goresky-MacPherson perversities. Let ${\mathtt K}(R,n,{\mathtt P},\overline{p})$ denote the perverse Eilenberg-M...

conjecture | open | MathDB 页面 | 原文线索

V2-032225 | MATHDB-346447

Free perverse algebra conjecture for perverse Eilenberg-MacLane spaces

摘录 / EXCERPT: Let ${\mathtt P}=[n]$, let $\overline{p}$ be a Goresky-MacPherson perversity, and let m be the differential degree of a generator x. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032226 | MATHDB-346454

The bounded Mackey topology conjecture for $L^{1}[0,1]$ and nonreflexive Banach spaces

摘录 / EXCERPT: Bounded Mackey topology conjecture. For $P=L^{1}\left[0,1\right]$ at least, and possibly for every nonreflexive Banach space P, the topology $\mathrm{bm}\left(P^{\ast},P\right)$...

conjecture | open | MathDB 页面 | 原文线索

V2-032227 | MATHDB-346467

The universal-invariant conjecture for coherent sheaves on Calabi-Yau fourfolds

摘录 / EXCERPT: Let X be a Calabi-Yau 4-fold and let $\mathcal A=\operatorname{Coh}(X)$, with the moduli, stability, orientation, homology, and virtual-class data specified in the universal en...

conjecture | open | MathDB 页面 | 原文线索

V2-032228 | MATHDB-346472

Sign conjecture for the $SL_2(\mathbb{R})$-Casson invariant of Brieskorn homology spheres

摘录 / EXCERPT: Sign conjecture. For every $f\in R^{\rm zd}(\Sigma(m,p,q))$, the quantity ε_f is negative. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-032229 | MATHDB-346500

Seiberg-Witten-Casson-Furuta-Ohta invariant conjecture for homology $S^1\times S^3$

摘录 / EXCERPT: Let X be a smooth oriented homology-oriented 4-manifold with the $\mathbb{Z}[\mathbb{Z}]$-homology of $S^1\times S^3$. The invariants $\lambda_{SW}(X)$ and $\lambda_{FO}(X)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-032230 | MATHDB-346506

The non-integral boundary slope conjecture for hyperbolic link exteriors

摘录 / EXCERPT: Non-integral boundary slope conjecture. There does not exist an essential n-punctured sphere with non-meridional, non-integral boundary slope in a hyperbolic link exterior in the...

conjecture | open | MathDB 页面 | 原文线索

V2-032231 | MATHDB-346508

Holomorphic factorization conjecture for null-homotopic symplectic mappings

摘录 / EXCERPT: Let X be a finite dimensional reduced Stein space, and let $f\colon X\to \operatorname{Sp}_{2m}(\mathbb{C})$ be a holomorphic mapping that is null-homotopic. Let $M_1,\ldots,M_K$...

conjecture | open | MathDB 页面 | 原文线索

V2-032232 | MATHDB-346552

The weak form of the mod-2 cohomology conjecture for decomposable Postnikov towers

摘录 / EXCERPT: Weak-form conjecture. As graded algebras, one has

conjecture | open | MathDB 页面 | 原文线索

V2-032233 | MATHDB-346553

The strong form of the mod-2 cohomology conjecture for decomposable Postnikov towers

摘录 / EXCERPT: Strong-form conjecture. As unstable algebras, one has

conjecture | open | MathDB 页面 | 原文线索

V2-032234 | MATHDB-346650

The conjecture that all tie knots are alternating 2-bridge knots

摘录 / EXCERPT: Tie-knot conjecture. All tie knots are alternating 2-bridge knots.

conjecture | open | MathDB 页面 | 原文线索

V2-032235 | MATHDB-346680

Sharp and ordered phase transitions for giant homological cycles

摘录 / EXCERPT: Let d be the dimension of the torus and let $\lambda_{0,k}$ and $\lambda_{1,k}$ denote the lower and upper thresholds for the appearance of giant k-cycles, while λ_c an...

conjecture | open | MathDB 页面 | 原文线索

V2-032236 | MATHDB-346687

Tresser's period-spectrum conjecture for area-contracting disk embeddings

摘录 / EXCERPT: Tresser's conjecture. Generically, maps in the boundary of positive topological entropy have a set of periodic orbits which, except for a finite subset, consists of infinitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-032237 | MATHDB-346696

Carlson's depth equality conjecture for finite p-groups

摘录 / EXCERPT: Let G be a finite p-group. Define $\omega_a(G)$ as the minimum of $\dim \operatorname{H}^{*}(G;\mathbb{F}_p)/\mathfrak{p}$ over associated primes $\mathfrak{p}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-032238 | MATHDB-346707

Stabilization-free extension theorem for controlled Mather-Thurston constructions

摘录 / EXCERPT: Let $(V,\partial V)$ be a manifold of dimension at least 3. Let $X\to B\to V$ be a bundle with structure group $\operatorname{Homeo}(X)$, where X is any manifold with a point-s...

conjecture | open | MathDB 页面 | 原文线索

V2-032239 | MATHDB-346721

Oberdieck-Pandharipande multiple-cover conjecture for K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let β be a primitive curve class, let m be a positive integer, and let $\gamma_1,\ldots,\gamma_n$ be cohomology insertions. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032240 | MATHDB-346727

The generalized cosmetic crossing conjecture for band surgeries

摘录 / EXCERPT: Let L be a two-component link, let b be a band, and let K_b be the oriented knot obtained from L by band surgery along b. For each $n\in\mathbf{Z}$, let K_{b+n} be the...

conjecture | open | MathDB 页面 | 原文线索

V2-032241 | MATHDB-346731

Finite generation conjecture for homology of no-k-equal configuration spaces

摘录 / EXCERPT: Let j and w be integers, let $\operatorname{config}(n,w)$ denote the relevant configuration space, and let FI_d be the category of finite sets and injections w...

conjecture | open | MathDB 页面 | 原文线索

V2-032242 | MATHDB-346748

Garoufalidis's AJ conjecture for knots

摘录 / EXCERPT: AJ conjecture. For every knot K in S^3, $varepsilon(alpha_K)$ is equal to $A_K(M,L)$, up to multiplication by a polynomial depending only on M.

conjecture | open | MathDB 页面 | 原文线索

V2-032243 | MATHDB-346750

Topological conjugacy invariance of hyperbolicity for polynomial automorphisms of the plane

摘录 / EXCERPT: Let f_0 and f_1 be polynomial automorphisms of ${\mathbb{C}^2}$ with non-trivial dynamics. Write $J_i=J(f_i)$ and $J_i^\star=J^\star(f_i)$ for their Julia sets and assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-032244 | MATHDB-346769

The topological cusp forms conjecture for Mathieu Moonshine

摘录 / EXCERPT: Let $\mathrm{Tcf}^\bullet$ be the spectrum of topological cusp forms, and let $[\overline{V^{f\natural}}]\nu$ denote the $\mathrm M_{24}$-equivariant class refining the nonequivari...

conjecture | open | MathDB 页面 | 原文线索

V2-032245 | MATHDB-346771

The Co1-equivariant non-nullhomotopy conjecture for Mathieu Moonshine

摘录 / EXCERPT: Let $V^{f\natural}$ be the holomorphic SCFT with automorphism group Co_1, and let $\overline{\operatorname{Fer}(3)}$ be the antiholomorphic three-fermion theory. Set ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-032246 | MATHDB-346772

The SQFT spectrum conjecture

摘录 / EXCERPT: Let $\SQFT^\bullet$ be the spectrum formed from spaces of compact unitary $\mathcal N=(0,1)$ SQFTs, with grading determined by the corresponding fermionic anomaly. SQFT spectrum co...

conjecture | open | MathDB 页面 | 原文线索

V2-032247 | MATHDB-346789

Yamaguchi-Yokura's estimate for the homotopical-to-cohomological quotient in elliptic fibrations

摘录 / EXCERPT: Yamaguchi-Yokura's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032248 | MATHDB-346790

Estimate for positively curved manifolds and simply-connected fibrations

摘录 / EXCERPT: Conjectured estimate. Estimate (2) is well-defined and holds.

conjecture | open | MathDB 页面 | 原文线索

V2-032249 | MATHDB-346796

Waldhausen null-homotopy conjecture for rigid analytic spaces

摘录 / EXCERPT: Waldhausen null-homotopy conjecture. The induced map

conjecture | open | MathDB 页面 | 原文线索

V2-032250 | MATHDB-346840

Uniqueness and invariance conjecture for energy-minimizing biharmonic almost complex structures

摘录 / EXCERPT: Let M be a compact, simply-connected, nonspin four-dimensional almost complex manifold. An energy-minimizing biharmonic almost complex structure on M determines a homotopy clas...

conjecture | open | MathDB 页面 | 原文线索

V2-032251 | MATHDB-346953

Global sections conjecture for twisted differential operators on quantized flag manifolds

摘录 / EXCERPT: Global sections conjecture. For every $t\in H$, this homomorphism is an isomorphism and

conjecture | open | MathDB 页面 | 原文线索

V2-032252 | MATHDB-346954

Untwisted global sections conjecture for quantized differential operators

摘录 / EXCERPT: Untwisted global sections conjecture. This homomorphism is an isomorphism and

conjecture | open | MathDB 页面 | 原文线索

V2-032253 | MATHDB-346955

Finite-part global sections conjecture for quantized differential operators

摘录 / EXCERPT: Finite-part global sections conjecture. This homomorphism is an isomorphism and

conjecture | open | MathDB 页面 | 原文线索

V2-032254 | MATHDB-346956

Twisted finite-part global sections conjecture

摘录 / EXCERPT: Twisted finite-part global sections conjecture. This homomorphism is an isomorphism and

conjecture | open | MathDB 页面 | 原文线索

V2-032255 | MATHDB-346969

Polynomial recurrence for totally minimal systems

摘录 / EXCERPT: Let (X,T) be a totally minimal system, and let $P(n)$ be a non-constant integral polynomial. Polynomial recurrence conjecture. There is a dense G_δ subset Ω of X...

conjecture | open | MathDB 页面 | 原文线索

V2-032256 | MATHDB-346970

Simultaneous polynomial recurrence along prescribed residue classes

摘录 / EXCERPT: Let (X,T^k) be minimal for some $k\geq 2$, let $d\in\mathbb N$, and let $P_m(n)$, $1\leq m\leq d$, be non-constant integral polynomials satisfying $P_m(0)=0$. Simultaneous polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-032257 | MATHDB-346971

A minimal non-distal system with trivial order-two regional proximal relation

摘录 / EXCERPT: For a topological dynamical system (X,T), let $\textbf{AP}^{[2]}(X,T)$ denote its order-two approximate proximal relation and let $\Delta_X=\\{(x,x):x\in X\\}$. Approximate-proxi...

conjecture | open | MathDB 页面 | 原文线索

V2-032258 | MATHDB-346978

Finite representable orbispectra as generators of parameterized spectra

摘录 / EXCERPT: Let $R(*)$ be the terminal representable orbispace, and let $\mathrm{RepOrbSp}^f$ be the category of finite genuine representable orbispectra. Finite-orbispectrum generation conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-032259 | MATHDB-346979

Full faithfulness of the global-space functor for finite orbispaces

摘录 / EXCERPT: Global-space full-faithfulness conjecture. For $X\in\mathrm{OrbSpc}^f$ and $F\in\mathrm{GloSpc}$, the set $\operatorname{Hom}(X,F)$ coincides with the morphisms from the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-032260 | MATHDB-346980

Compatibility of the global-space functor with the presheaf equivalence

摘录 / EXCERPT: Let OrbSpc be the category of orbispaces, let $\overline{\mathrm{OrbSpc}}$ be its enlargement obtained by gluing cells under arbitrary maps, and let GloSpc be...

conjecture | open | MathDB 页面 | 原文线索

V2-032261 | MATHDB-346982

Compatibility with the global classifying space of stable vector bundles

摘录 / EXCERPT: Let $\mathbf{bO}$ denote the global space that is the direct limit of the global classifying spaces $\mathbf B O(n)$, and regard $\mathbf{bO}$ also as an orbispace through the func...

conjecture | open | MathDB 页面 | 原文线索

V2-032262 | MATHDB-346999

Nonnegative tensor-power conjecture for homology of rank-selected subword order

摘录 / EXCERPT: Let A be an alphabet of size $n\ge 2$, and let S_n act on A^* by permuting the alphabet. A rank-selected subposet is obtained by retaining words whose ranks belong to a finit...

conjecture | open | MathDB 页面 | 原文线索

V2-032263 | MATHDB-347005

Conjecture on the homotopical depth of higher matching complexes

摘录 / EXCERPT: Let $M_{n-2}(K_n)$ be the matching complex of $(n-2)$-element matchings in the complete graph K_n, and let $M_{n-1}(K_{n,n})$ be the matching complex of $(n-1)$-element matchings...

conjecture | open | MathDB 页面 | 原文线索

V2-032264 | MATHDB-347025

The uniqueness conjecture for Basic Models of Γ surfaces

摘录 / EXCERPT: A Γ surface is a pair (Z,E) obtained from a curve configuration $D\subset \mathbb P^2$ by successive blow-ups, where E is a curve configuration of type Γ and the compone...

conjecture | open | MathDB 页面 | 原文线索

V2-032265 | MATHDB-347043

Three-component algebraically split L-space link surgery conjecture

摘录 / EXCERPT: Let $\mathcal{L}=L_{1}\cup L_{2}\cup L_{3}$ be an algebraically split L-space link such that ... is the Poincaré homology sphere. Three-component L-space link surgery conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-032266 | MATHDB-347057

Decomposition criterion for equivalence of spatial graphs

摘录 / EXCERPT: Decomposition criterion. The graphs Γ and Γ' are equivalent if and only if they admit decompositions

conjecture | open | MathDB 页面 | 原文线索

V2-032267 | MATHDB-347062

Nonexistence of a filling inducing the nontrivial augmentation of the standard Legendrian unknot

摘录 / EXCERPT: Let U be the standard Legendrian unknot, whose unique Reeb chord is denoted by b. A conical Legendrian filling Σ of U is equipped with a 1-graded augmentation...

conjecture | open | MathDB 页面 | 原文线索

V2-032268 | MATHDB-347077

Simple length spectrum rigidity for hyperbolic surfaces

摘录 / EXCERPT: Let S and S' be hyperbolic surfaces, and let the multi-set of lengths of their simple closed geodesics be called their simple length spectrum. Simple length spectrum rigidity c...

conjecture | open | MathDB 页面 | 原文线索

V2-032269 | MATHDB-347078

Regularity-free simple elevation theorem for covers

摘录 / EXCERPT: Let S be a surface and let S_1 and S_2 be covers of S. A closed curve on S has a simple elevation to a cover if one of its elevations is a simple closed curve. Regularity...

conjecture | open | MathDB 页面 | 原文线索

V2-032270 | MATHDB-347088

Kåhrström's left-cell conjecture for projective dimensions

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple complex Lie algebra with Weyl group W, let L_y be a simple module in the principal block $\mathcal{O}_0$, and let θ_x be the indecompo...

conjecture | solved | MathDB 页面 | 原文线索

V2-032271 | MATHDB-347113

Relative Hilali conjecture for maps

摘录 / EXCERPT: Relative Hilali conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-032272 | MATHDB-347114

Hilali's conjecture as the contractible-target case

摘录 / EXCERPT: Hilali's conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-032273 | MATHDB-347115

Generalized relative Hilali conjecture with finite homology target

摘录 / EXCERPT: Generalized relative Hilali conjecture. If the homology rank $\dim H_*(Y;\mathbb Q)$ of the target is finite, then

conjecture | open | MathDB 页面 | 原文线索

V2-032274 | MATHDB-347116

Injectivity conjecture for homological and homotopical maps

摘录 / EXCERPT: Injectivity conjecture. If $H_i(f;\mathbb Q)$ is injective for every $i>1$, then $\pi_i(f)\otimes\mathbb Q$ is injective for every $i>1$. The paper states that the authors have bee...

conjecture | open | MathDB 页面 | 原文线索

V2-032275 | MATHDB-347117

Continuity conjecture for restriction of extensions

摘录 / EXCERPT: Continuity conjecture. For every such S^G, $\imath$, R, and $\mathbb{G}$, the map $\imath^*$ is continuous in the described topologies.

conjecture | open | MathDB 页面 | 原文线索

V2-032276 | MATHDB-347171

The immersed-genus extension of Property G

摘录 / EXCERPT: Let Y be a closed, oriented, connected 3--manifold, let $K\subset Y$ be a null-homologous knot, and let F be a taut Seifert surface for K. Set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032277 | MATHDB-347172

Property G for null-homologous knots in reducible 3-manifolds

摘录 / EXCERPT: Let Y be a closed, oriented, connected, reducible 3--manifold, and let $K\subset Y$ be a null-homologous knot such that $Y\backslash\!\backslash K$ is irreducible. Property G c...

conjecture | open | MathDB 页面 | 原文线索

V2-032278 | MATHDB-347180

Simple-knot detection conjecture for knots in lens spaces

摘录 / EXCERPT: Simple-knot detection conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-032279 | MATHDB-347181

Hyperbolicity conjecture for constrained knots

摘录 / EXCERPT: Hyperbolicity conjecture for constrained knots. If $l>1$ and $v\neq\pm 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-032280 | MATHDB-347206

Isotopy of equivalent knots in oriented 3-manifolds

摘录 / EXCERPT: Let M be an oriented 3-manifold. Write $\operatorname{Mod}^+(M)$ for the group of orientation-preserving mapping classes of M, and call two knots in M equivalent if an orie...

open_problem | open | MathDB 页面 | 原文线索

V2-032281 | MATHDB-347218

Matui's AH conjecture for effective minimal ample groupoids

摘录 / EXCERPT: Let $\mathcal{G}$ be an effective minimal second countable Hausdorff ample groupoid whose unit space $\mathcal{G}^{(0)}$ is a Cantor space. Write $H_0(\mathcal{G})$ and...

conjecture | solved | MathDB 页面 | 原文线索

V2-032282 | MATHDB-347234

The $\a0mu$-vanishing conjecture for modified Iwasawa cohomology

摘录 / EXCERPT: Let L_∞/L be the cyclotomic $\mathbb{Z}_p$-extension, let T be the given p-adic representation, and let Σ be a finite set of places disjoint from S such that...

conjecture | open | MathDB 页面 | 原文线索

V2-032283 | MATHDB-347239

Chbili-Qazaqzeh's Jones polynomial conjecture for quasi-alternating links

摘录 / EXCERPT: Chbili-Qazaqzeh's conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-032284 | MATHDB-347261

The decomposition conjecture for shoreless continua

摘录 / EXCERPT: Let X be a continuum with no shore points, let $p\in X$, and let $X=X_1\oplus\cdots\oplus X_N$ denote a decomposition into subcontinua. A point is non-coastal in a subcontinuum i...

conjecture | open | MathDB 页面 | 原文线索

V2-032285 | MATHDB-347262

The minimal decomposition conjecture for shoreless continua

摘录 / EXCERPT: The minimal decomposition conjecture. The point p is non-coastal when treated as a point of each X_n. This asserts the desired non-coastal behavior for every minimal decomposit...

conjecture | open | MathDB 页面 | 原文线索

V2-032286 | MATHDB-347278

Injectivity conjecture for the map from reduced virtual OU tangles

摘录 / EXCERPT: Let $v\mathcal{ROU}_n$ be the set of reduced virtual OU tangles with n open strands, let $v\mathcal{T}_n$ be the set of virtual tangles with n open strands, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032287 | MATHDB-347300

Weak twisted Leopoldt conjecture for multiple Z_p-extensions

摘录 / EXCERPT: Let F be a number field, let L be a $\mathbb{Z}_p^d$-extension of F, and let $G_S(L)$ be the Galois group of the maximal extension of L unramified outside the primes above...

conjecture | open | MathDB 页面 | 原文线索

V2-032288 | MATHDB-347322

Orbifold Jacobian algebra conjecture before quotienting

摘录 / EXCERPT: Orbifold Jacobian algebra conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-032289 | MATHDB-347329

Homological distance product formula for arbitrary finite-field chain complexes

摘录 / EXCERPT: Arbitrary finite-field homological distance conjecture. For any pair of bounded chain complexes of vector spaces over a finite field, the homological distances...

conjecture | open | MathDB 页面 | 原文线索

V2-032290 | MATHDB-347352

The bounding cohomology conjecture for smooth projective surfaces

摘录 / EXCERPT: Bounding cohomology conjecture. There exists a constant $c_X>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032291 | MATHDB-347359

The knot-complement series and ADO-invariant conjecture

摘录 / EXCERPT: ADO-invariant conjecture. For any knot K in S^3,

conjecture | open | MathDB 页面 | 原文线索

V2-032292 | MATHDB-347360

The t-deformed ADO-polynomial conjecture for torus-knot series

摘录 / EXCERPT: Let K be a knot in S^3, let $F_K(x,q,a,t)$ be the refined knot-complement series, let $\Delta_K(x,y)$ denote the two-variable Alexander polynomial appearing in the formula, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032293 | MATHDB-347361

The quantum A-polynomial recursion conjecture for the knot-complement series

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $F_K(x,q)$ be its knot-complement invariant, and define the normalized series ... Let $\hat{x}$ and $\hat{y}$ be the quantum operators entering the...

conjecture | open | MathDB 页面 | 原文线索

V2-032294 | MATHDB-347362

The geometric Cartan doubled Yangian conjecture for perverse coherent systems

摘录 / EXCERPT: Let X be the local toric Calabi-Yau 3-fold in the source, let Y_∞ be the geometric Cartan doubled Yangian, and let $V:=H^*_c(\fM_\zeta^\chi,\varphi_W\mathbf{C})^\vee$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-032295 | MATHDB-347377

The large-[$*$] U-map isomorphism conjecture for prequantization bundles

摘录 / EXCERPT: Large-parameter U-map conjecture. For $*$ sufficiently large, the U map is an isomorphism on the chain level of the ECH of prequantization bundles, for the chain complex of Pro...

conjecture | open | MathDB 页面 | 原文线索

V2-032296 | MATHDB-347408

The realizability conjecture for unknown points of $T(2,n)$

摘录 / EXCERPT: For an odd integer n, let $T(2,n)$ denote the corresponding torus knot, and let (e,h) denote the pair consisting of the normal Euler number and nonorientable genus of a surface...

conjecture | open | MathDB 页面 | 原文线索

V2-032297 | MATHDB-347409

The unique minimal point conjecture for torus knots

摘录 / EXCERPT: Let K be a torus knot, let $\gamma_4(K)$ denote its smooth nonorientable 4-genus, and let (e,h) denote the normal Euler number and nonorientable genus of a surface bounded by...

conjecture | open | MathDB 页面 | 原文线索

V2-032298 | MATHDB-347415

The degree-one map conjecture for Heegaard genus

摘录 / EXCERPT: Let M and N be closed orientable 3-manifolds. A degree-one map is a map $f\colon M\to N$ of degree one, and $g(M)$ denotes the Heegaard genus of M. The degree-one map conject...

conjecture | open | MathDB 页面 | 原文线索

V2-032299 | MATHDB-347416

The surgery formulation of the degree-one map conjecture

摘录 / EXCERPT: Let $M=\mathcal{W}\cup_\mathcal{T}\mathcal{V}$, where $\mathcal{T}=\partial \mathcal{W}=\partial \mathcal{V}=\mathcal{W}\cap\mathcal{V}$ is a genus-g surface. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032300 | MATHDB-347418

Fox Trapezoidal Conjecture for alternating knots

摘录 / EXCERPT: Let K be an alternating knot with normalized Alexander polynomial ... where g is the genus of K. Fox Trapezoidal Conjecture. The coefficients satisfy ... Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-032301 | MATHDB-347421

The universal covering conjecture for isotropic semisimple groups

摘录 / EXCERPT: Let G be an ${\mathbb A}^1$-connected, isotropic, semisimple, almost simple, simply connected algebraic group over k, split by a finite separable field extension F/k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032302 | MATHDB-347505

Global inverse function conjecture for curved L_∞ spaces

摘录 / EXCERPT: Let $\mathfrak{f}:\mathbb{M}\to\mathbb{N}$ be a morphism of curved L_∞ spaces, with underlying manifolds M and N, curvature maps $\mu_0^\mathbb{M}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-032303 | MATHDB-347508

Effaceability of derived ordinary parts for locally admissible representations

摘录 / EXCERPT: Effaceability conjecture. The functors $\textnormal{H}^i\textnormal{Ord}^G_{P^-}$ are effaceable on $\mathfrak{Rep}^{\textnormal{ladm}}(G)$ for $i>0$. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-032304 | MATHDB-347528

Kollár's conjecture on Brieskorn spheres and integer homology balls

摘录 / EXCERPT: Kollár's conjecture. No Brieskorn sphere $\Sigma(a_1,\ldots,a_n)$ with $n\geq 4$ is the boundary of a smooth integer homology ball.

conjecture | open | MathDB 页面 | 原文线索

V2-032305 | MATHDB-347529

The planar open-book conjecture for tight positive contact structures on Brieskorn homology spheres

摘录 / EXCERPT: Planarity conjecture. No tight, positive contact structure on a Brieskorn homology sphere is supported by a planar open book decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-032306 | MATHDB-347556

Finite dense resolutions for tame modules over real polyhedral groups

摘录 / EXCERPT: Finite dense resolution conjecture. Every tame, semialgebraic, or PL module M over a real polyhedral group Q has finite dense downset and upset resolutions of the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-032307 | MATHDB-347557

Finite minimal resolutions for tame modules over discrete polyhedral groups

摘录 / EXCERPT: Finite minimal resolution conjecture. Every tame module M over a discrete polyhedral group Q has finite minimal downset and upset resolutions of the corresponding type.

conjecture | open | MathDB 页面 | 原文线索

V2-032308 | MATHDB-347558

Indicator-dimension conjecture for polyhedral groups

摘录 / EXCERPT: Indicator-dimension conjecture. The indicator-dimension of any real or discrete polyhedral group Q equals the rank of Q as a free module, over the field $\mathbb{R}$ or the gro...

conjecture | open | MathDB 页面 | 原文线索

V2-032309 | MATHDB-347569

The Frobenius structure conjecture for relative quantum cohomology

摘录 / EXCERPT: Frobenius structure conjecture. The coefficient of $ vartheta_0$ in the product $ vartheta_{p_1} star\text{?} cdots star vartheta_{p_m}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-032310 | MATHDB-347572

The prismatic-de Rham-Witt comparison conjecture for special perfect prisms

摘录 / EXCERPT: Let (A,d) be a perfect prism such that A/d is p-torsion free, and let S be a smooth A/d-algebra. Here $W_n\Omega_{S/(A/d)}^i$ denotes the degree-i de Rham-Witt group, a...

conjecture | open | MathDB 页面 | 原文线索

V2-032311 | MATHDB-347609

Specialization conjecture for multivariable Lie algebra invariants

摘录 / EXCERPT: Let $\mathfrak{k}$ be the Lie algebra obtained by removing m simple roots α_j from a simply-laced Lie algebra $\mathfrak{g}$ of rank n. Suppose the corresponding entries...

conjecture | open | MathDB 页面 | 原文线索

V2-032312 | MATHDB-347611

Regularity up to the cohomological time for pluriclosed flow

摘录 / EXCERPT: Let (M^{2n},g_0,J) be a compact complex manifold with pluriclosed metric. Define ... Here ω_0 is the fundamental form of g_0, c_1 is the first Chern class in Aeppli coho...

conjecture | open | MathDB 页面 | 原文线索

V2-032313 | MATHDB-347616

Failure of the fully marked surface theorem without homology adjustment

摘录 / EXCERPT: Let M be a closed hyperbolic 3-manifold, let $\mathcal{F}$ be a taut foliation on M, and let S be a fully marked surface. Failure conjecture for the fully marked surface theo...

conjecture | open | MathDB 页面 | 原文线索

V2-032314 | MATHDB-347682

Minimal-diagram connected-sum crossing-number conjecture for links in thickened surfaces

摘录 / EXCERPT: Let $L_1\subset\Sigma_1\times I$ and $L_2\subset\Sigma_2\times I$ be links in thickened surfaces, and let D_1,D_2 be minimal-crossing representatives of L_1,L_2, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-032315 | MATHDB-347732

Linear and quadratic loop equations for the quartic Kontsevich model

摘录 / EXCERPT: Let $R:\widehat{\mathbb{C}}\to\widehat{\mathbb{C}}$ be the ramified covering defined by the model, let $\beta_1,\ldots,\beta_{2d}$ be its ramification points, and let σ_i be...

conjecture | open | MathDB 页面 | 原文线索

V2-032316 | MATHDB-347755

Carlsson's algebraic rank conjecture for differential graded polynomial modules

摘录 / EXCERPT: Let k be an algebraically closed field and let $A=k[y_1,\ldots,y_r]$ be the polynomial ring in variables $y_1,\ldots,y_r$, each of degree -1. Let M be a free, finitely genera...

conjecture | open | MathDB 页面 | 原文线索

V2-032317 | MATHDB-347872

Orbit-equivalence conjecture for Anosov flows supported by homotopic bi-contact structures

摘录 / EXCERPT: Orbit-equivalence conjecture. The two Anosov flows are orbit equivalent. Equivalently, two Anosov flows that are homotopic through projectively Anosov flows are orbit equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-032318 | MATHDB-347873

Ciocan-Fontanine-Kim-Sabbah conjecture for quantum abelianisation

摘录 / EXCERPT: Ciocan-Fontanine-Kim-Sabbah conjecture. For $\alpha,\alpha',\zeta,\zeta'$ as above,

conjecture | open | MathDB 页面 | 原文线索

V2-032319 | MATHDB-347898

Quillen's vanishing conjecture for linear groups

摘录 / EXCERPT: Let $GL_j(R)$ be the general linear group and let $D_j(R)\subset GL_j(R)$ be its diagonal subgroup. For group homology with coefficients in k, the canonical inclusion induces ... H...

conjecture | open | MathDB 页面 | 原文线索

V2-032320 | MATHDB-347900

The t-adic refinement of Kaneko-Zagier's conjecture

摘录 / EXCERPT: The t-adic refinement. There is a topological $\mathbb{Q}$-algebra isomorphism from $\mathcal{Z}_{\widehat{\mathcal{A}}}$ onto $\overline{\mathcal{Z}}\llbracket t\rrbracket$ whic...

conjecture | open | MathDB 页面 | 原文线索

V2-032321 | MATHDB-347917

Champanerkar-Kofman-Purcell conjecture on right-angled knots

摘录 / EXCERPT: A knot in S^3 is called right-angled if its complement admits a decomposition into ideal hyperbolic right-angled polyhedra. Champanerkar-Kofman-Purcell conjecture. There does not...

conjecture | open | MathDB 页面 | 原文线索

V2-032322 | MATHDB-347936

Loop-space conjecture for the space of core metrics on spheres

摘录 / EXCERPT: Loop-space conjecture. The space $\operatorname{\mathcal{R}^\text{Cores}}(\operatorname{S}^n)$ equipped with μ is an associative H-space and is homotopy equivalent to a loop...

conjecture | open | MathDB 页面 | 原文线索

V2-032323 | MATHDB-347956

Standard conjecture on the perfectness of algebraic cohomology pairings

摘录 / EXCERPT: Perfectness conjecture. The intersection pairing on algebraic cohomology classes is perfect. This is presented as one of the paper's analogues of Grothendieck's standard conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032324 | MATHDB-347965

McCallum-Sharifi's surjectivity conjecture for the cyclotomic cohomology map

摘录 / EXCERPT: Let $p\nmid N\phi(N)$, let θ be a primitive even Dirichlet character of conductor Np, and let ϖ_{θ} denote the θ-component of Sharifi's homomorphism fro...

conjecture | open | MathDB 页面 | 原文线索

V2-032325 | MATHDB-348024

Donagi-Gao-Schulz reduction and derived-equivalence conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let $\mathcal{X}_{1,3}$ and $\mathcal{X}_{3,1}$ denote the Calabi-Yau threefolds studied by Donagi, Gao, and Schulz, whose Betti cohomology has torsion at primes p dividing that...

conjecture | open | MathDB 页面 | 原文线索

V2-032326 | MATHDB-348067

Low-dimensional rigidity conjecture for stable fractional one-phase solutions

摘录 / EXCERPT: Let $s\in (0,1)$ and let $u\in C^s(\mathbb{R}^n)$ be a global, stable, s-homogeneous solution to the fractional one-phase problem. A solution is one-dimensional if it depends on...

conjecture | open | MathDB 页面 | 原文线索

V2-032327 | MATHDB-348077

Infinite-length and nondifferentiability conjecture for blocked mixed labyrinth fractals

摘录 / EXCERPT: Let $\{{\cal A}_k\}_{k=1}^{\infty}$ be a sequence of labyrinth patterns that are both horizontally and vertically blocked, with $m_k\ge 4$, and let L_{∞} be the resulting li...

conjecture | open | MathDB 页面 | 原文线索

V2-032328 | MATHDB-348104

Jeon-Jin conjecture on knots with superbridge index three

摘录 / EXCERPT: A knot's superbridge index is the minimum, over all embeddings of the knot in Euclidean three-space, of the maximum number of local maxima of the height function over all direction...

conjecture | open | MathDB 页面 | 原文线索

V2-032329 | MATHDB-348155

The improved ball-number bound for links

摘录 / EXCERPT: Improved ball-number conjecture. For every link L,

conjecture | open | MathDB 页面 | 原文线索

V2-032330 | MATHDB-348158

Greenberg's weak Leopoldt conjecture for $T_\Pi^\dagger$

摘录 / EXCERPT: Let Σ be the finite set of places used to define the global Galois cohomology, let ${\mathbb Q}_\infty$ be the cyclotomic extension of ${\mathbb Q}$, and let $T_\Pi^\dagger$...

conjecture | open | MathDB 页面 | 原文线索

V2-032331 | MATHDB-348159

Perrin-Riou's Euler system conjecture for T_Π

摘录 / EXCERPT: Let T_Π be the Galois lattice attached to Π, let $\mathcal O$ be its coefficient ring, and let $\mathcal N$ be the set of integers mp^k where m is a square-free product...

conjecture | open | MathDB 页面 | 原文线索

V2-032332 | MATHDB-348206

Gorenstein New Intersection Theorem

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a noetherian local ring. A finite complex of totally reflexive modules is a bounded complex ... A module is totally reflexive if it is finitely generated, t...

conjecture | open | MathDB 页面 | 原文线索

V2-032333 | MATHDB-348220

The conjectural (r,s) classes for irregularly admissible spectral curves

摘录 / EXCERPT: Let $r\geq 2$ and $s\in [r-1]$ satisfy $r=\pm 1\,\,\operatorname{mod}\,s$. For $a_i\in [r]$ and $g,n\in\mathbb{N}$ with $2g-2+n>0$, let $F_{g,n}$ be the coefficients associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-032334 | MATHDB-348233

The instanton immersed-curve invariants conjecture for pointed 2-tangles

摘录 / EXCERPT: Let T be a pointed 2-tangle and let $P^*\cong\partial(D^3,T)$ be the pillowcase. Write $I(T)$, $I^\natural(T)$ and $I^\sharp(T)$ for the proposed instanton-theoretic immersed cur...

conjecture | open | MathDB 页面 | 原文线索

V2-032335 | MATHDB-348239

Symplectic isotopy conjecture for ruled surfaces

摘录 / EXCERPT: Let $M_g=\Sigma_g\times S^2$ be a minimal topologically trivial ruled surface equipped with a symplectic form ω. A symplectomorphism of (M_g,ω) is symplectically isot...

conjecture | open | MathDB 页面 | 原文线索

V2-032336 | MATHDB-348245

Additivity conjecture for the bad-domain invariant of non-alternating knots

摘录 / EXCERPT: Let K_1 and K_2 be non-alternating knots. For a knot K, let $\beta(K)$ denote the minimum number of bad domains over diagrams representing K, and let $K_1\# K_2$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-032337 | MATHDB-348256

The embedding-calculus universality conjecture for classical knots

摘录 / EXCERPT: Let $I=[0,1]$, let $\Knots(I^3)=\Emb_\partial(I,I^3)$ be the space of long knots, and write $\KK(I^3):=\pi_0\Knots(I^3)$. For each $n\geq 1$, let $\pT_n(I^3)$ be the nth stage of...

conjecture | open | MathDB 页面 | 原文线索

V2-032338 | MATHDB-348264

Blumberg-Lesnick persistent Whitehead conjecture

摘录 / EXCERPT: Let $X,Y: \mathbf{R} \to \mathbf{Top}$ be connected, cofibrant persistent spaces, with each $X(r)$ and $Y(r)$ a CW-complex of dimension at most d. Let $f: X \to Y^\delta$ induce...

conjecture | open | MathDB 页面 | 原文线索

V2-032339 | MATHDB-348266

Higher-degree spectral-curve BPS free-energy conjecture

摘录 / EXCERPT: Higher-degree BPS free-energy conjecture. Suppose the BPS structure obtained from Σ is uncoupled. Then the topological-recursion free energies satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-032340 | MATHDB-348275

Uniqueness conjecture for infinity-harmonic maps to the circle

摘录 / EXCERPT: Let M be the manifold under consideration, and let $u:M\to S^1$ be an ∞-harmonic map in a fixed homotopy class. Uniqueness conjecture. The map u is unique up to rotation...

conjecture | open | MathDB 页面 | 原文线索

V2-032341 | MATHDB-348290

The characterization of admissible Banach spaces by nontrivial type

摘录 / EXCERPT: Admissibility conjecture. A Banach space is admissible if and only if it has nontrivial type.

conjecture | open | MathDB 页面 | 原文线索

V2-032342 | MATHDB-348326

The instanton-Heegaard Floer correspondence conjecture

摘录 / EXCERPT: Let (M,γ) be a balanced sutured manifold. Write SHI for sutured instanton Floer homology, SFH for sutured Heegaard Floer homology, $I^{\sharp}$ for framed instanton Floe...

conjecture | open | MathDB 页面 | 原文线索

V2-032343 | MATHDB-348327

The knot-independence conjecture for the essential instanton Floer summand

摘录 / EXCERPT: Let $(\widehat{Y},\widehat{K})$ satisfy the hypotheses of the source's torsion spin^c decomposition theorem, and let $\widehat{K}'\subset\widehat{Y}$ be another knot satisfying t...

conjecture | open | MathDB 页面 | 原文线索

V2-032344 | MATHDB-348328

The bypass-map compatibility conjecture for surgery cobordisms

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $S^3_{-n}(K)$ be obtained by $(-n)$-surgery, and let W be the natural cobordism from S^3 to $S^3_{-n}(K)$. Let $t_i:H_2(W_{-n})\to\mathbb{Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-032345 | MATHDB-348347

Dupont's conjecture on the comparison map for continuous bounded cohomology

摘录 / EXCERPT: Let G) be a connected, non-compact, semi-simple Lie group with finite center. Write ... for the comparison map from continuous bounded cohomology to continuous cohomology. Dupont's...

conjecture | open | MathDB 页面 | 原文线索

V2-032346 | MATHDB-348399

Gauge homotopy equivalence of the homotopy inverse composite

摘录 / EXCERPT: Let $h_=h\Psi$ be the element defined in the last remark before the appendix of Dotsenko, Shadrin, and Vallette (2016). Let $i_$ and $p_$ be the $_\alpha$-morphisms whose composite...

conjecture | open | MathDB 页面 | 原文线索

V2-032347 | MATHDB-348411

Östlund's conjecture on simplifying generic circle immersions

摘录 / EXCERPT: A knot projection is the image of a generic immersion of a circle into the 2-sphere, and deformations of types 1, 2, and 3 are the local replacements analogous to Reidemeis...

conjecture | open | MathDB 页面 | 原文线索

V2-032348 | MATHDB-348423

Shadow-diagram move conjecture for surface-links

摘录 / EXCERPT: Shadow-diagram move conjecture. The diagrams $\mathfrak D^1$ and $\mathfrak D^2$ can be related by a finite sequence of moves, each of which is one of the listed types.

conjecture | open | MathDB 页面 | 原文线索

V2-032349 | MATHDB-348450

General slice-Bennequin inequality for tight contact 3-manifolds

摘录 / EXCERPT: General slice-Bennequin inequality. (Y,ξ) is tight if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-032350 | MATHDB-348451

General Milnor conjecture for fibered links in tight contact 3-manifolds

摘录 / EXCERPT: General Milnor conjecture. If ξ_L is tight, then Σ maximizes Euler characteristic among all such surfaces.

conjecture | open | MathDB 页面 | 原文线索

V2-032351 | MATHDB-348452

Open-book subsurface Thurston norm conjecture for tight contact manifolds

摘录 / EXCERPT: Open-book subsurface conjecture. (Y,ξ) is tight if and only if every subsurface of the page of each supporting open book is Thurston norm minimizing and, moreover, has minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-032352 | MATHDB-348523

The weak Leopoldt conjecture for the representation T

摘录 / EXCERPT: Weak Leopoldt conjecture. For every $E\in\Omega(\mathcal{K}'/K)$, $H^2_{\mathrm{Iw}}(\mathcal{O}_{E,S(E)},T^*(1))$ is a torsion Λ-module.

conjecture | open | MathDB 页面 | 原文线索

V2-032353 | MATHDB-348525

Axial symmetry conjecture for minimizing disc configurations with elastic boundaries

摘录 / EXCERPT: Let E be the elastic surface energy for topological discs, with positive parameters α, β, c_o and $b\neq 0$, and define ... Assume that $\underline E\geq 0$. Axial s...

conjecture | open | MathDB 页面 | 原文线索

V2-032354 | MATHDB-348528

Lowest-degree degeneration conjecture for Grassmannian K-theoretic structure constants

摘录 / EXCERPT: Let $\mathcal{I}$ be the indexing set for Schubert cells of $\operatorname{Gr}(d,n)$, and let $C^K_{I,J}$ and $c^K_{I,J}$ denote the K-theoretic and cohomological structure constan...

conjecture | open | MathDB 页面 | 原文线索

V2-032355 | MATHDB-348540

Fiedorowicz-Pirashvili-Schwänzl-Vogt-Waldhausen conjecture on MacLane's Q-construction

摘录 / EXCERPT: Let R be a discrete ring. Write $Q(R)$ for the spectrum associated with MacLane's Q-construction, let $\mathbb{Z}$ denote the Eilenberg-Mac Lane spectrum of the integers, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-032356 | MATHDB-348545

The elliptic Omnibus Conjecture

摘录 / EXCERPT: A simply-connected topological space X of finite type is rationally elliptic when its rational cohomology and rational homotopy are both finite dimensional. Elliptic Omnibus Conj...

conjecture | open | MathDB 页面 | 原文线索

V2-032357 | MATHDB-348546

Anick's approximation conjecture for elliptic spaces

摘录 / EXCERPT: Let S be a simply connected finite CW complex, let n be a natural number, and let S be a simply connected finite Postnikov piece when the second formulation is considered. An...

conjecture | open | MathDB 页面 | 原文线索

V2-032358 | MATHDB-348547

Anick's Sullivan-model conjecture

摘录 / EXCERPT: Let $(\Lambda W,\operatorname{d})$ be a minimal Sullivan algebra with $\dim W<\infty$, $W^1=0$, and fix $n\in\mathbb{N}$. A relative Sullivan algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-032359 | MATHDB-348548

The Félix--Jessup--Murillo nilpotence conjecture

摘录 / EXCERPT: Let $(\Lambda V,\operatorname{d})$ be a minimal Sullivan algebra with $V^1=0$, and suppose that both the even-degree part V^{even} and the even-degree cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-032360 | MATHDB-348556

Quillen's conjectures on André-Quillen homology

摘录 / EXCERPT: Let $\varphi\colon R\to S$ be a ring homomorphism, and let $\operatorname{D}_n(S\!\mid\!R,-)$ denote André-Quillen homology with arbitrary coefficients. Quillen's conjectures. Assu...

conjecture | open | MathDB 页面 | 原文线索

V2-032361 | MATHDB-348585

Quantum-liquid TQFTs as complete invariants of bordisms

摘录 / EXCERPT: Let f and g be n-morphisms in the oriented bordism category $\mathbf{Bord}^{\mathrm{or}}_n$. Completeness conjecture. If f and g are not equivalent, then there exists a s...

conjecture | open | MathDB 页面 | 原文线索

V2-032362 | MATHDB-348596

Complete homogeneous basis conjecture for Grassmannian cohomology subalgebras

摘录 / EXCERPT: Complete homogeneous basis conjecture. For $1\leq m\leq\min(\ell,k)$,

conjecture | open | MathDB 页面 | 原文线索

V2-032363 | MATHDB-348601

The extension conjecture for periodic de Rham bundles

摘录 / EXCERPT: Extension conjecture. The canonical parabolic extension of an irreducible periodic de Rham bundle over U is again periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-032364 | MATHDB-348602

The weak periodic de Rham conjecture for parabolic bundles

摘录 / EXCERPT: Weak periodic de Rham conjecture. The natural inclusion functor

conjecture | open | MathDB 页面 | 原文线索

V2-032365 | MATHDB-348603

The periodic de Rham conjecture for completely reducible objects

摘录 / EXCERPT: Periodic de Rham conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032366 | MATHDB-348618

Vanishing conjecture for the second-page spectral-sequence terms

摘录 / EXCERPT: Let F be an infinite field and let $M_\bullet$ be the exact complex of $\operatorname{GL}_n(F)$-modules used to construct the first-quadrant spectral sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-032367 | MATHDB-348643

Lindenstrauss-Tsukamoto embedding conjecture for topological dynamical systems

摘录 / EXCERPT: Let (X,T) be a topological dynamical system, let d be a positive integer, let $\operatorname{mdim}(X,T)$ denote its mean dimension, let dim denote topological...

conjecture | open | MathDB 页面 | 原文线索

V2-032368 | MATHDB-348644

Gutman-Qiao-Tsukamoto embedding conjecture for band-limited flows

摘录 / EXCERPT: Let (X,T) be a topological dynamical system, let a be a positive real parameter, let $\operatorname{mdim}(X,T)$ denote its mean dimension, let dim denote topol...

conjecture | open | MathDB 页面 | 原文线索

V2-032369 | MATHDB-348652

Douglas's K-homology fundamental-class conjecture for algebraic sets

摘录 / EXCERPT: Douglas's conjecture. Conjecture A holds true, and its induced extension class τ_I is identified with the fundamental class of X_I, namely the extension class induced by the...

conjecture | open | MathDB 页面 | 原文线索

V2-032370 | MATHDB-348683

The LSC conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $(X,\mathfrak{a})$ be a log pair, and let $|X|$ denote the set of closed points of X with the Zariski topology. Write $\operatorname{mld}_x(X,\mathfrak{a})$ for the minimal l...

conjecture | open | MathDB 页面 | 原文线索

V2-032371 | MATHDB-348688

Eliashberg-Hofer conjecture on contact homology invariance

摘录 / EXCERPT: Eliashberg-Hofer conjecture. The differential satisfies $d^2=0$, and the homology $H_*(C_*,d)$ is independent of the contact form α for ξ and of the complex structure...

conjecture | open | MathDB 页面 | 原文线索

V2-032372 | MATHDB-348701

Noncommutative finiteness conjecture for Iwasawa cohomology

摘录 / EXCERPT: Noncommutative coadmissibility conjecture. The cohomology $h^\bullet(C_{\psi_I}(M))$ should be coadmissible over

conjecture | open | MathDB 页面 | 原文线索

V2-032373 | MATHDB-348719

The Jones polynomial conjecture for detecting causality

摘录 / EXCERPT: Jones polynomial conjecture. The Jones polynomial detects causality between any two events in X.

conjecture | open | MathDB 页面 | 原文线索

V2-032374 | MATHDB-348749

The Berstein-Schwarz formula for the LS-category of a group homomorphism

摘录 / EXCERPT: Let $f:G\to H$ be a homomorphism between discrete groups. Let $Bf:BG\to BH$ be a map between Eilenberg-MacLane spaces inducing f on fundamental groups, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032375 | MATHDB-348770

The two distinguished-wave-determined slopes conjecture

摘录 / EXCERPT: Let M be a tunnel-number-one orientable 3-manifold with incompressible torus boundary. A tunnel of M determines a distinguished-wave-determined slope on ∂M. The two...

conjecture | open | MathDB 页面 | 原文线索

V2-032376 | MATHDB-348804

Félix-Oprea-Tanré's generalised toral rank conjecture

摘录 / EXCERPT: Let $F\hookrightarrow E\to B$ be a quasi-nilpotent fibration, with fibre F equal to a torus T^r, and suppose that the rational cohomology of B is finite dimensional. Generali...

conjecture | open | MathDB 页面 | 原文线索

V2-032377 | MATHDB-348805

The algebraic toral rank conjecture

摘录 / EXCERPT: Let $(\Lambda V,{\operatorname{d}})$ be a minimal Sullivan algebra. Let ... be a relative minimal model with $\deg t_i=2$ for $1\leq i\leq r$, and suppose that ... is finite dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-032378 | MATHDB-348806

The nilpotent Lie algebra toral rank conjecture

摘录 / EXCERPT: Let L be a finite-dimensional nilpotent Lie algebra defined over the rational numbers, and let $C(L)$ denote its center. Nilpotent Lie algebra toral rank conjecture. Then ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-032379 | MATHDB-348837

The spanning conjecture for the sets of bounded-genus surfaces

摘录 / EXCERPT: Let $A(n)$ be the space defined by the two-dimensional topological theory, and let $\mathscr{A}^m(n)$ be the set of surface diagrams whose connected components satisfy the genus bo...

conjecture | open | MathDB 页面 | 原文线索

V2-032380 | MATHDB-348858

The explicit reciprocity law for Gross-Prasad Selmer classes

摘录 / EXCERPT: Let $\Delta(\Pi\times\underline{\mathcal G},\heartsuit)$ be the cohomology class supplied by the family Selmer conjecture in a region $\heartsuit$ with global root number -1, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032381 | MATHDB-348865

A-polynomial and defining-curve specialization conjecture

摘录 / EXCERPT: A-polynomial and defining-curve specialization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032382 | MATHDB-348871

Conjecture on maximal cohomology for generalized 12-fold tilings

摘录 / EXCERPT: Let $\Omega_{E_{12}^{\gamma}}$ be the tiling space associated with a generalized 12-fold tiling with parameter γ. For generic parameters, meaning those with the maximal numb...

conjecture | open | MathDB 页面 | 原文线索

V2-032383 | MATHDB-348880

Ausoni-Rognes redshift conjecture for algebraic K-theory

摘录 / EXCERPT: Ausoni-Rognes redshift conjecture. The p-completed algebraic K-theory spectrum

conjecture | open | MathDB 页面 | 原文线索

V2-032384 | MATHDB-348887

Galois representation and infinitesimal character conjecture for Hecke eigenspaces

摘录 / EXCERPT: Galois-infinitesimal character conjecture. The center $Z(\mathfrak{g})$ acts on this representation via ζ^C_ρ.

conjecture | open | MathDB 页面 | 原文线索

V2-032385 | MATHDB-348926

Strong sparse non-liftability conjecture for chain complexes

摘录 / EXCERPT: A chain complex over $\mathbb{F}_2$ is sparse when every row and column of every boundary operator has $O(1)$ nonzero entries; a sparse lift to $\mathbb{Z}$ has uniformly bounded r...

conjecture | open | MathDB 页面 | 原文线索

V2-032386 | MATHDB-348927

Stable sparse non-liftability conjecture for 2-complexes

摘录 / EXCERPT: A sparse 2-complex over $\mathbb{F}_2$ is a chain complex in degrees 0, 1, and 2 whose boundary operators are sparse. A stable lift is obtained by adding finitely many cont...

conjecture | open | MathDB 页面 | 原文线索

V2-032387 | MATHDB-348928

Sparse LU obstruction conjecture

摘录 / EXCERPT: A matrix is sparse when its relevant rows and columns have uniformly boundedly many nonzero entries. A sparse LU decomposition is a factorization ... where P and Q are permutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-032388 | MATHDB-348929

Unavoidable torsion in sparse lifts conjecture

摘录 / EXCERPT: A sparse chain complex over $\mathbb{F}_2$ has a sparse lift to integer coefficients when its boundary operators can be lifted while remaining sparse. Torsion obstruction conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-032389 | MATHDB-348930

2-0-to-monotonic connectivity conjecture for special spines

摘录 / EXCERPT: 2-0-to-monotonic connectivity conjecture. Then P is connected to Q by a monotonic sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-032390 | MATHDB-348931

Local monotonicity conjecture for spine moves

摘录 / EXCERPT: Local monotonicity conjecture. Then P is connected to Q by a monotonic sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-032391 | MATHDB-348937

Elliott's conjecture on geodesic curvature flow of noncontractible curves on a cone

摘录 / EXCERPT: Let $\mathcal M$ be a cone and let a closed curve on $\mathcal M$ be not homotopic to a point on $\mathcal M$. Elliott's conjecture. Under geodesic curvature flow, the curve will c...

conjecture | open | MathDB 页面 | 原文线索

V2-032392 | MATHDB-348976

Ball conjecture for projective multiaffine real stable polynomial spaces

摘录 / EXCERPT: Let $\mathbb{P}\underline{\mathrm{S}}^d_n$ be the projective space of all multiaffine homogeneous real stable polynomials of degree d in n variables. Ball conjecture. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-032393 | MATHDB-349014

The torus-knot slope-jump conjecture for the invariant Φ

摘录 / EXCERPT: Torus-knot slope-jump conjecture. For the torus knot $T_{n,n+1}$, the jump in the slope $\Delta \Phi'_{T_{n,n+1}}(\alpha)$ is equal to 0 except when $\alpha=\frac{2+2k}{2+3k}$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-032394 | MATHDB-349076

Halpern-Weaver optimality conjecture for paper Möbius bands

摘录 / EXCERPT: Let $M_{\lambda}=([0,1]\times[0,\lambda])/\sim$ be the flat Möbius band, where $(x,0)\sim(1-x,\lambda)$, and let a paper Möbius band be a smooth isometric embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-032395 | MATHDB-349077

Rigidity conjecture for optimal paper Möbius bands

摘录 / EXCERPT: Let a paper Möbius band be a smooth isometric embedding of the flat Möbius band M_{λ} in $\mathbb{R}^3$, and consider a sequence of smooth embedded paper Möbius bands whose...

conjecture | open | MathDB 页面 | 原文线索

V2-032396 | MATHDB-349095

Compactifiability conjecture for cylindrically stratified morphisms

摘录 / EXCERPT: Let $f:X\to Y$ be a cylindrically stratified morphism of stratified topological spaces. A morphism is compactifiable if it factors as a stratified open embedding followed by a prop...

conjecture | open | MathDB 页面 | 原文线索

V2-032397 | MATHDB-349111

The $P=W$ conjecture for positroid torus quotients and compactified Jacobians

摘录 / EXCERPT: Let $\Pit_{k,n}$ be the torus quotient and let $J_{k,n-k}$ be the compactified Jacobian of the plane curve singularity $x^k=y^{n-k}$. Their cohomology carries a weight filtration a...

conjecture | open | MathDB 页面 | 原文线索

V2-032398 | MATHDB-349112

Equality of top HOMFLY-Khovanov-Rozansky polynomials in the knot case

摘录 / EXCERPT: Equality conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-032399 | MATHDB-349156

Gorenstein and complete-intersection conjecture for modules of derivations

摘录 / EXCERPT: Let R be a local k-algebra such that the R-module of k-derivations $\operatorname{Der}_k(R)$ is finitely generated. Write Gdim for Gorenstein dimension and...

conjecture | open | MathDB 页面 | 原文线索

V2-032400 | MATHDB-349160

SO-type super-A-polynomial branch conjecture

摘录 / EXCERPT: SO-type super-A-polynomial branch conjecture. At $t=-1$, the zero locus

conjecture | open | MathDB 页面 | 原文线索

V2-032401 | MATHDB-349176

The inclusion of ideal-limit points under $\mathcal{I}^{\mathcal{K}}$-convergence

摘录 / EXCERPT: Expected inclusion.

conjecture | open | MathDB 页面 | 原文线索

V2-032402 | MATHDB-349187

Filtered log convergent hard Lefschetz conjecture

摘录 / EXCERPT: Let $f\colon X\to S$ be in the log convergent setting of the paper, with relative dimension d, and let P denote the weight filtration. Let N be the monodromy parameter and as...

conjecture | open | MathDB 页面 | 原文线索

V2-032403 | MATHDB-349202

The cascade conjecture for rational Q-homology projective planes

摘录 / EXCERPT: Let S be a rational $\mathbb{Q}$-homology projective plane with quotient singularities. We say that S admits a cascade if there is a sequence of blow-downs between surfaces who...

conjecture | open | MathDB 页面 | 原文线索

V2-032404 | MATHDB-349212

Conjectural formula for the derivative of the homomorphism sigma

摘录 / EXCERPT: Let $p':= p-b$ and $q':= q-a$. For $\xi_1\in\pi_pSO_{q'}$ and $\xi_2\in\pi_qSO_{p'}$, let $h\colon\pi_iSO_j\to\pi_i(S^{j-1})$ be induced by $SO_j\to S^{j-1}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-032405 | MATHDB-349236

Arnold's fixed-point conjecture for Hamiltonian symplectomorphisms

摘录 / EXCERPT: Arnold's conjecture. The number of fixed points of f satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-032406 | MATHDB-349237

Deligne's conjecture on the operadic action on Hochschild cochains

摘录 / EXCERPT: Deligne's conjecture. There exists a natural action of the operad $\mathsf{Chains}(C_2)$ on the Hochschild complex $\mathcal{C}^\bullet(\mathcal{A},\mathcal{A})$ for any associativ...

conjecture | open | MathDB 页面 | 原文线索

V2-032407 | MATHDB-349238

Conjecture on even pretzel knots with maximal nonorientable concordance genus

摘录 / EXCERPT: Let $n\ge 3$, and let an even n-stranded pretzel knot be a pretzel knot $K=P(a_1,\dots,a_n)$ with even parameters in the sense used for pretzel knots. Write $\gamma_c(K)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-032408 | MATHDB-349252

The zero mean dimension marker property conjecture

摘录 / EXCERPT: A free dynamical system is a topological dynamical system (X,T) in which $T^n x\neq x$ for every $x\in X$ and every nonzero integer n. Its mean dimension is the dynamical dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-032409 | MATHDB-349256

Cohen-Lenstra distribution conjecture for torsion homology of random simplicial complexes

摘录 / EXCERPT: Let $X_\bullet$ be a random simplicial complex, and let $H_\ast(X_\bullet)_{\operatorname{tors}}$ denote the torsion subgroup of its homology. A Cohen-Lenstra distribution conjectu...

open_problem | open | MathDB 页面 | 原文线索

V2-032410 | MATHDB-349257

Landesman's homological stability conjecture for Selmer spaces

摘录 / EXCERPT: For fixed n, let $\operatorname{Sel}_n^d(\mathbb{C})$ be the Selmer space associated with the universal family of Weierstrass models of degree parameter d. A Landesman homologi...

conjecture | open | MathDB 页面 | 原文线索

V2-032411 | MATHDB-349270

Cornwell-Ng-Sivek conjecture on a non-decomposable satellite Lagrangian concordance

摘录 / EXCERPT: Let Λ_{-} and Λ_{+} be the Legendrian knots described in the satellite construction, and let $P\subset J^1S^1$ be the Legendrian clasp tangle used to form their sat...

conjecture | open | MathDB 页面 | 原文线索

V2-032412 | MATHDB-349281

The covering conjecture for localized skein algebras

摘录 / EXCERPT: Let Σ be an unpunctured surface, let Δ be an ideal triangulation of Σ, and for each internal edge $E\in e_{\mathrm{int}}(\Delta)$ let...

conjecture | open | MathDB 页面 | 原文线索

V2-032413 | MATHDB-349282

The 9_{35} divisibility conjecture for HOMFLY F-factors

摘录 / EXCERPT: The 9_{35} divisibility conjecture. For positive integers $i>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032414 | MATHDB-349283

The 9_{46} odd-index F-factor divisibility conjecture

摘录 / EXCERPT: The 9_{46} odd-index divisibility conjecture. For odd positive integers $i>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032415 | MATHDB-349284

The 9_{46} odd-index transformed F-factor conjecture

摘录 / EXCERPT: The 9_{46} transformed-factor conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032416 | MATHDB-349285

The normalized Q^{1} periodicity conjecture for pretzel knots

摘录 / EXCERPT: The normalized Q^1 periodicity conjecture. For the indicated values of c and r,

conjecture | open | MathDB 页面 | 原文线索

V2-032417 | MATHDB-349286

The Q^{r} shift conjecture for symmetric pretzel-knot differences

摘录 / EXCERPT: The Q^r shift conjecture. For all odd integers a,b,c and symmetric representations $[r]$,

conjecture | open | MathDB 页面 | 原文线索

V2-032418 | MATHDB-349289

Lecuona's topological sliceness conjecture for 3-stranded pretzel knots

摘录 / EXCERPT: Lecuona's conjecture. For every odd integer $a\geq 3$, the 3-stranded pretzel knot

conjecture | open | MathDB 页面 | 原文线索

V2-032419 | MATHDB-349299

Chen-Nagano's conjectures on antipodal sets in compact symmetric spaces

摘录 / EXCERPT: Chen-Nagano's conjectures. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-032420 | MATHDB-349309

Nonexistence conjecture for simply-connected strongly inflexible manifolds

摘录 / EXCERPT: Nonexistence conjecture. Simply-connected strongly inflexible manifolds do not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-032421 | MATHDB-349365

Extended Reidemeister move conjecture for knots with discrete diagrams

摘录 / EXCERPT: Define the extended Reidemeister moves to be the standard Reidemeister move set together with a fourth move, in which A is a compact set whose interior remains fixed relative to...

conjecture | open | MathDB 页面 | 原文线索

V2-032422 | MATHDB-349366

The Lazard-ring computation conjecture for local Noetherian rings

摘录 / EXCERPT: Let A be a local Noetherian ring. Write $d53f^*$ for the Lazard ring with cohomological grading. Lazard-ring computation conjecture. The natural map ... is an isomorphism. This con...

conjecture | open | MathDB 页面 | 原文线索

V2-032423 | MATHDB-349374

Injective free-Lie-algebra map from canonical forms to graph cohomology

摘录 / EXCERPT: Let $\Omega^{\bullet}_{\mathrm{can}}$ be the graded space of canonical differential forms, let $\mathbb{L}(\Omega^{\bullet}_{\mathrm{can}})$ be the free Lie algebra on it, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032424 | MATHDB-349388

The conjectural equivalence of sutured instanton and Heegaard Floer homology

摘录 / EXCERPT: Sutured instanton-Heegaard Floer equivalence conjecture. For every balanced sutured manifold (M,γ), there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032425 | MATHDB-349413

Manolescu's HOMFLY-PT to knot Floer homology spectral sequence conjecture

摘录 / EXCERPT: Let L be a link in S^3. Write $\overline{H}(L)$ and $H^-(L)$ for the reduced, middle, and unreduced HOMFLY-PT homologies of L, and $\overline{\operatorname{HFK}}(L)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-032426 | MATHDB-349414

Manolescu's graded HOMFLY-PT to knot Floer homology spectral sequence conjecture

摘录 / EXCERPT: Let L be a link in S^3. The HOMFLY-PT homologies carry gradings gr_T and gr_M, and the grading-shifted knot Floer homologies...

conjecture | open | MathDB 页面 | 原文线索

V2-032427 | MATHDB-349415

Dowlin's sl(n) to knot Floer homology spectral sequence conjecture

摘录 / EXCERPT: Let L be a link in S^3 and let n be a positive integer. Write $\overline{H}_n(L)$ and $H_n(L)$ for reduced and unreduced $\mathfrak{sl}(n)$ homology, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032428 | MATHDB-349423

Golla-Lisca conjecture on universal tightness of contact torus bundles

摘录 / EXCERPT: Let D be a circular spherical divisor, meaning a topological divisor consisting of a cycle of spheres, and let $b^+(D)$ denote the positive index of its intersection form. The as...

conjecture | open | MathDB 页面 | 原文线索

V2-032429 | MATHDB-349431

Conjectural equivariant Poincaré-series formulas for the Higgs-bundle blow-ups

摘录 / EXCERPT: Let g be the genus of the curve, let $\mathrm{SL}(2)$ act on the spaces $\mathbf{R}$, $\mathbf{R}_{1}^{ss}$, $\mathbf{R}_{2}^{s}$, $\mathbb{P}\Upsilon^{-1}(0)^{ss}$, E_2^{ss},...

conjecture | open | MathDB 页面 | 原文线索

V2-032430 | MATHDB-349454

Deligne-Clozel conjecture on algebraicity of critical standard L-values

摘录 / EXCERPT: Let $\mathbb F$ be a totally real field of degree d, let $n=2r$ or $n=2r+1$, and let $\mathit{\Pi}$ be a cohomological irreducible cuspidal automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-032431 | MATHDB-349508

The p-adic intersection cohomology conjecture for rigid spaces

摘录 / EXCERPT: Let $K/\mathbf{Q}_p$ be a finite extension, let G_K be its absolute Galois group, and let X be a rigid space proper over K. Write $IH^i(X_C,\mathbf{Q}_p)$ for its p-adic in...

conjecture | open | MathDB 页面 | 原文线索

V2-032432 | MATHDB-349515

The regular-pattern conjecture for correctability in grid-like topologies

摘录 / EXCERPT: Regular-pattern conjecture. An erasure pattern $\mathcal{E}$ is correctable for the topology $T_{m\times n}(a,b,0)$ if and only if it is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-032433 | MATHDB-349517

Fredholmness and finite-dimensional periodic cohomology conjecture

摘录 / EXCERPT: Let $E_{b4}(Z)$ be the operator associated with the elliptic complex on the periodic end, and let $\widehat{H}^*_z(V)$ denote the corresponding periodic cohomology group, with para...

conjecture | open | MathDB 页面 | 原文线索

V2-032434 | MATHDB-349518

Mod-16 conjecture for the torus Furuta-Ohta invariant

摘录 / EXCERPT: Let X be an integral homology $S^1 \times S^3$ containing an essentially embedded torus $\mathcal{T}$, let Y_0 be the cross-section obtained by the associated 0-surgery, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032435 | MATHDB-349525

Closure equivalence conjecture for metric thickenings

摘录 / EXCERPT: Let $M=(X,K,\phi)$ be a simplicial metric thickening, with metric thickening $\mathcal{M}\subseteq\mathcal{P}(X)$. Let $\hat{M}$ be the closure of $\mathcal{M}$ in $\mathcal{P}(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-032436 | MATHDB-349526

Vietoris-Rips thickening self-reconstruction conjecture

摘录 / EXCERPT: Let X be a metric space, potentially with additional assumptions, let $r\geq 0$, and let ... Here $\operatorname{VR}^{\mathrm{m}}(X;r)$ denotes the metric Vietoris-Rips thickening...

conjecture | open | MathDB 页面 | 原文线索

V2-032437 | MATHDB-349549

The Melvin-Morton-Rozansky conjecture for the knot invariant F_K

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot. Write $F_K(x,q)$ for the knot invariant, set $q={\rm e}^{\hbar}$, and keep $x=e^{n\hbar}$ fixed, where n is the color of K. Let $\Delta_K(x)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-032438 | MATHDB-349550

The AJ conjecture for the knot invariant F_K

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot, let $F_K(x,q)$ be its invariant, and define the normalized series ... Let $\hat A_K(q,\hat x,\hat y)$ denote the quantum A-polynomial of K. AJ conje...

conjecture | open | MathDB 页面 | 原文线索

V2-032439 | MATHDB-349551

The AJ conjecture for the quantum A-polynomial

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot. Let $\hat A_K(t,M,L)$ be the noncommutative quantum A-polynomial obtained from the recursion ideal of the colored Jones polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032440 | MATHDB-349559

Homological Berglund-Hübsch-Henningson mirror symmetry for invertible polynomials

摘录 / EXCERPT: Let $\mathbf{w}$ be an invertible polynomial with admissible symmetry group $\Gamma\subseteq\Gamma_{\mathbf{w}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032441 | MATHDB-349560

Wrapped Fukaya-matrix factorisation mirror symmetry for equivariant Milnor fibres

摘录 / EXCERPT: Let $\mathbf{w}$ be an invertible polynomial with admissible symmetry group $\Gamma\subseteq\Gamma_{\mathbf{w}}$, and let $\widecheck{\Gamma}$ be the corresponding dual group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032442 | MATHDB-349563

The three-summand conjecture for reducible surgeries on hyperbolic knots

摘录 / EXCERPT: Three-summand conjecture. A closed orientable 3--manifold with at least 3 summands does not contain a hyperbolic knot with a non-trivial reducible surgery.

conjecture | open | MathDB 页面 | 原文线索

V2-032443 | MATHDB-349609

The symplectic cap obstruction conjecture for fillable contact structures on L-spaces

摘录 / EXCERPT: Let (Y,ξ) be a contact 3-manifold, where Y is an L-space and ξ is symplectically fillable. A symplectic cap of (Y,ξ) is a symplectic 4-manifold with (Y,ξ) as it...

conjecture | open | MathDB 页面 | 原文线索

V2-032444 | MATHDB-349653

The noncommutative Hodge conjecture for small dg categories

摘录 / EXCERPT: Let $\mathcal{A}$ be a small dg category. Its Hodge classes are the subspace ... where $\operatorname{Ch}^{\operatorname{top}}$ is the topological Chern character, j is the map fro...

conjecture | open | MathDB 页面 | 原文线索

V2-032445 | MATHDB-349698

Infinite free-extension naive lifting conjecture

摘录 / EXCERPT: Infinite free-extension naive lifting conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-032446 | MATHDB-349751

Xu's quantization conjecture for Lie bialgebroids

摘录 / EXCERPT: Let (E,E^*) be a Lie bialgebroid. A quantization is a topological deformation of the standard associative bialgebroid structure on the universal enveloping algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-032447 | MATHDB-349753

Universal quantization conjecture for Lie-quasi bialgebroids

摘录 / EXCERPT: Let a Lie-quasi bialgebroid be a quasi-bialgebroid generalising Lie bialgebroids and quasi-bialgebras, and let a quasi-quantum groupoid be a topological deformation of the standard...

conjecture | open | MathDB 页面 | 原文线索

V2-032448 | MATHDB-349808

Grothendieck's conjecture for homotopy theory

摘录 / EXCERPT: Grothendieck's conjecture for homotopy theory. For all $m\in\mathbb{N}$, the category $\mathbb{M}\operatorname{od}(\Theta^{\infty}_{\mathbb{M}^m})$ is Quillen equivalent to categor...

conjecture | open | MathDB 页面 | 原文线索

V2-032449 | MATHDB-349821

Hofer-small exact isotopy preserves unobstructedness

摘录 / EXCERPT: Let L be an unobstructed Lagrangian with a bounding cochain of valuation c, and let L' be exactly homotopic to L through an exact homotopy whose Hofer norm is less than c...

conjecture | open | MathDB 页面 | 原文线索

V2-032450 | MATHDB-349844

Effective descent conjecture for finitely presented module spectra

摘录 / EXCERPT: Let A,A_1,A_2,A_{12} be noetherian $\mathbb{E}_1$-Banach rings in the indicated simplicial ind-Banach categories. Suppose that their homotopy rings fit into a short strictly exac...

conjecture | open | MathDB 页面 | 原文线索

V2-032451 | MATHDB-349875

Two-minima one-saddle conjecture for alternating links

摘录 / EXCERPT: Two-minima one-saddle conjecture. If L bounds a ribbon surface with two minima and a single saddle point, then L is algorithmically ribbon.

conjecture | open | MathDB 页面 | 原文线索

V2-032452 | MATHDB-349891

The zero mean dimension conjecture for the marker property

摘录 / EXCERPT: Zero mean dimension conjecture. Every aperiodic dynamical system with zero mean dimension has the marker property.

conjecture | open | MathDB 页面 | 原文线索

V2-032453 | MATHDB-349898

Generalized Smale Conjecture for constant-curvature 3-manifolds

摘录 / EXCERPT: Generalized Smale Conjecture. This inclusion is a homotopy equivalence. The conjecture was verified by Kleiner and Bamler, giving a complete description of the relevant homotopy ty...

conjecture | open | MathDB 页面 | 原文线索

V2-032454 | MATHDB-349901

Conjecture on four-dimensional Ricci flows through singularities

摘录 / EXCERPT: Let (M,g) be a closed Riemannian 4-manifold. A Ricci flow through singularities is a flow in which topological change can occur while time-slices may develop controlled singulari...

conjecture | open | MathDB 页面 | 原文线索

V2-032455 | MATHDB-349955

Derived reciprocity conjecture for the Kato class in the $\alpha_g\beta_h=1$ case

摘录 / EXCERPT: Assume $\alpha_g\beta_h=1$, and let P,Q be the points in $(E(H)\otimes V_{gh}^{\vee})^{G_{\mathbb Q}}$. Let $\kappa'(f,g_{\alpha},h_{1/\beta})$ be the derived cohomology class, l...

conjecture | open | MathDB 页面 | 原文线索

V2-032456 | MATHDB-349962

Dunfield-Gukov-Rasmussen symmetry conjecture for reduced Khovanov-Rozansky homology

摘录 / EXCERPT: Let K be a knot and let $\overline{\mathrm{HHH}}(K)=\bigoplus \overline{\mathrm{HHH}}_{i,j,k}(K)$ be its reduced triply graded homology, where i is the a-grading, j is the...

conjecture | open | MathDB 页面 | 原文线索

V2-032457 | MATHDB-349963

Tensor-product compatibility conjecture for Koszul complexes and twisted modules

摘录 / EXCERPT: Let S_n be the symmetric group, let R be the polynomial ring used to define the complexes K_r, and let $\mathcal{A}_w$ be the dg algebra whose modules are being considered. F...

conjecture | open | MathDB 页面 | 原文线索

V2-032458 | MATHDB-349977

Continuous orbit equivalence for bounded speedups of free \(\mathbb{Z}^2\)-odometers

摘录 / EXCERPT: Let $(X_{\mathfrak{G}},\sigma_{\mathfrak{G}})$ be a free $\mathbb{Z}^2$-odometer, and let $\mathbf{S}:\mathbb{Z}^2\curvearrowright X$ be a minimal speedup of...

conjecture | open | MathDB 页面 | 原文线索

V2-032459 | MATHDB-350010

Relative Reshetikhin-Turaev volume conjecture with adjoint torsion

摘录 / EXCERPT: Let M be a closed oriented 3-manifold and let L be a framed hyperbolic link in M with $|L|$ components. Let $\\{\mathbf a^{(r)}\} $ be a sequence of colorings of the compon...

conjecture | open | MathDB 页面 | 原文线索

V2-032460 | MATHDB-350011

Relative Turaev-Viro volume conjecture with adjoint torsion

摘录 / EXCERPT: Let N be a 3-manifold with non-empty boundary and let $\mathcal T$ be an ideal triangulation with edge set E. Let $\{\mathbf b^{(r)}\}$ be a sequence of colorings of...

conjecture | open | MathDB 页面 | 原文线索

V2-032461 | MATHDB-350056

The persistent-homology lifespan upper-bound conjecture

摘录 / EXCERPT: Let G be the filtered simplicial complex under consideration, let $\mathrm{PD}_k(G)$ denote the set of persistence pairs in degree k, and let Ω be a k-dimensional homo...

conjecture | open | MathDB 页面 | 原文线索

V2-032462 | MATHDB-350057

The width lower-bound conjecture for persistent-homology life spans

摘录 / EXCERPT: Let G be the filtered simplicial complex under consideration, let $\mathrm{PD}_k(G)$ denote the degree-k persistence pairs, and let Ω be the associated homology class wi...

conjecture | open | MathDB 页面 | 原文线索

V2-032463 | MATHDB-350063

The higher-rank wreath-product non-embeddability conjecture

摘录 / EXCERPT: Higher-rank wreath-product non-embeddability conjecture. If $d\geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-032464 | MATHDB-350071

The ribbon-move equivalence conjecture

摘录 / EXCERPT: Fix a ribbon knot and consider ribbons spanning it. A ribbon 0-move, ribbon 1-move, finger move, and ribbon 3-move are the ribbon transformations described in the paper's figures;...

conjecture | open | MathDB 页面 | 原文线索

V2-032465 | MATHDB-350082

Avramov-Conca-Iyengar subadditivity conjecture for syzygies of Koszul algebras

摘录 / EXCERPT: Let $R=Q/I$ be a Koszul algebra, and let $t_i(R)$ denote the maximal degree of a minimal generator in homological degree i of the minimal graded free resolution of R over Q....

conjecture | open | MathDB 页面 | 原文线索

V2-032466 | MATHDB-350088

Vanishing and global-generation conjecture for divisors on the resolved fourfold

摘录 / EXCERPT: Vanishing and global-generation conjecture. For every effective divisor D on $\widehat{X^4_8}$:

conjecture | open | MathDB 页面 | 原文线索

V2-032467 | MATHDB-350098

The universality conjecture for linear relations between cohomological invariants and Chern numbers

摘录 / EXCERPT: Let X be a compact complex manifold. Consider $\bbZ$-linear relations and congruences involving its cohomological invariants and Chern numbers, and quotient these by the relation...

conjecture | open | MathDB 页面 | 原文线索

V2-032468 | MATHDB-350116

Lorenz satellite conjecture

摘录 / EXCERPT: Lorenz satellite conjecture. A Lorenz knot that is a satellite has companion a Lorenz knot. Its pattern, when embedded in an unknotted solid torus in S^3, is equivalent to a Lore...

conjecture | open | MathDB 页面 | 原文线索

V2-032469 | MATHDB-350123

The simplicial-volume equality conjecture for manifolds covered by $(\mathbb{H}^2)^n$

摘录 / EXCERPT: Let M be a closed oriented manifold covered by $(\mathbb{H}^2)^n$, and let Θ_n be the bounded volume form defined in the preceding conjecture. Simplicial-volume equality c...

conjecture | open | MathDB 页面 | 原文线索

V2-032470 | MATHDB-350155

Conjectural endoscopic Mantovan formula

摘录 / EXCERPT: Let (G,b,μ) be the local Shimura data appearing in the paper, let $\mathcal{H}^{\mathfrak{e}}$ be an endoscopic datum, and let $\operatorname{Mant}_{G,b,\mu}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-032471 | MATHDB-350160

Kronheimer-Mrowka's framed instanton and monopole Floer homology conjecture

摘录 / EXCERPT: Let Y be a closed oriented 3-manifold. Kronheimer-Mrowka's conjecture. The framed instanton homology and the tilde version of monopole Floer homology satisfy ... This is a specia...

conjecture | open | MathDB 页面 | 原文线索

V2-032472 | MATHDB-350161

Existence and chamber invariance conjecture for Seiberg-Witten $U(2)$ monopole Floer homology

摘录 / EXCERPT: Suppose that Y is a closed oriented 3-manifold, and E is a $U(2)$ bundle with admissibility condition $w=c_1(E)\in H^2(Y;\mathbb{Z})$ a non-torsion odd class. Choose a spin-...

conjecture | open | MathDB 页面 | 原文线索

V2-032473 | MATHDB-350165

Dye's conjecture on checkerboard framings of virtual link diagrams

摘录 / EXCERPT: A checkerboard framing is an extension of a checkerboard coloring for a virtual link diagram; consider any two checkerboard framings of the same diagram. Dye's conjecture. Any two...

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V2-032474 | MATHDB-350168

Balanced negative extensions for topological extriangulated categories

摘录 / EXCERPT: Let $(\mathscr{C},\mathbb{E},\mathfrak{s})$ be a topological extriangulated category, and let a bivariant δ-functor consist of functors $\mathbb{E}^n$ together with covarian...

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V2-032475 | MATHDB-350169

Vanishing bound for negative extensions in finite positive global dimension

摘录 / EXCERPT: Let $(\mathscr{C},\mathbb{E},\mathfrak{s})$ be an extriangulated category of positive global dimension n, and assume the balanced negative extensions conjecture above holds for i...

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V2-032476 | MATHDB-350226

Cohomological concentration and discrete-series archimedean component

摘录 / EXCERPT: Let G be a group such that $G(\mathbb{R})$ admits discrete series, let π^S be an automorphic representation that is not G-Eisenstein, and write an automorphic cuspidal repr...

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V2-032477 | MATHDB-350254

Conjecture on connected simple systems of parameterized Conley indices

摘录 / EXCERPT: Let Z be a Hausdorff space, let $\omega\colon M\to Z$ be continuous, and let $P=(P_1,P_2)$ range over regular index pairs for an isolated invariant set S. The parameterized Con...

conjecture | open | MathDB 页面 | 原文线索

V2-032478 | MATHDB-350278

Simplicity conjecture for the homotopy eigenvalues

摘录 / EXCERPT: Let ${\boldsymbol{M}}_n$ and ${\boldsymbol{S}}_n$ be the matrices described above, and define ... Here ${\boldsymbol{T}}({\varepsilon})$ is the symmetric homotopy matrix and...

conjecture | open | MathDB 页面 | 原文线索

V2-032479 | MATHDB-350279

Linear upper-bound conjecture for the largest singular value

摘录 / EXCERPT: Let $\sigma_{1,n}$ denote the largest singular value of the matrix associated with index n. Largest-singular-value bound. ... This bound is suggested by the numerical homotopy expe...

conjecture | open | MathDB 页面 | 原文线索

V2-032480 | MATHDB-350280

The zero-volume conjecture for knot exteriors and Mahler measure

摘录 / EXCERPT: Let K be a knot, let $\mathcal{M}_{K}$ denote its exterior, let A_K be its A-polynomial, let $\mathrm{Vol}(\mathcal{M}_K)$ be the sum of the volumes of the hyperbolic pieces...

conjecture | open | MathDB 页面 | 原文线索

V2-032481 | MATHDB-350281

The satellite A-polynomial formula for integer pseudo-graph knots

摘录 / EXCERPT: Let $C\in\mathcal{G}_{\mathbb{Z}}$ have strongly detected boundary slopes $\mathcal{DS}_{C}$, and let $K=\operatorname{Sat}(P,C,f)$ be a satellite knot whose embedded pattern knot...

conjecture | open | MathDB 页面 | 原文线索

V2-032482 | MATHDB-350302

Stein Schoenflies conjecture for embedded homotopy 4-balls

摘录 / EXCERPT: Let X be a compact Stein domain of complex dimension 2, and let $B\subset X$ be an embedded homotopy 4-ball bounded by a smoothly embedded S^3. Stein Schoenflies conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032483 | MATHDB-350333

Halperin's conjecture on negative-degree derivations of positively elliptic algebras

摘录 / EXCERPT: Halperin's conjecture. If H^* is a positively elliptic algebra, then H^* does not admit a non-trivial derivation of negative degree.

conjecture | open | MathDB 页面 | 原文线索

V2-032484 | MATHDB-350371

Gorsky-Hogancamp-Simental-Rasmussen conjecture for positroid Catalan invariants

摘录 / EXCERPT: Gorsky-Hogancamp-Simental-Rasmussen conjecture. The knots $\hat\beta(\mathbf{a}(f))$ and $\hat\beta_f$ are isotopic. Up to a monomial in q and t,

conjecture | open | MathDB 页面 | 原文线索

V2-032485 | MATHDB-350392

Complex-structure independence of large black brane cycle volumes

摘录 / EXCERPT: Let X be a Calabi-Yau threefold. An even-dimensional homology class $[\Sigma]$ is called large when, for example, $[\Sigma]=N[\Sigma_0]$ with $N\gg1$. If the associated black bra...

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V2-032486 | MATHDB-350457

NNI-sequence conjecture for tree topologies along tropical line segments

摘录 / EXCERPT: Let T_1 and T_2 be equidistant rooted phylogenetic trees with n leaves in X, and let Γ_{T_1,T_2} be the tropical line segment between them. A nearest neighbor interc...

conjecture | open | MathDB 页面 | 原文线索

V2-032487 | MATHDB-350478

The minimal length conjecture for finite-projective-dimension modules

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a local ring, let $e(R)$ denote its Hilbert-Samuel multiplicity, and let $\ell_R M$ denote the length of an R-module M. Minimal length conjecture. For...

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V2-032488 | MATHDB-350484

Norm functors for compatible indexing categories

摘录 / EXCERPT: Norm functor conjecture. There is a norm functor

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V2-032489 | MATHDB-350485

External Hoyer-Mazur conjecture for bi-incomplete Tambara functors

摘录 / EXCERPT: External Hoyer-Mazur conjecture. For any compatible pair of indexing categories $(\mathcal O_a,\mathcal O_m)$, there is an equivalence of categories between $\mathcal O_m$-commutat...

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V2-032490 | MATHDB-350486

Coinduction preservation conjecture for compatible indexing categories

摘录 / EXCERPT: Coinduction preservation conjecture. Coinduction restricts to a functor

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V2-032491 | MATHDB-350511

Berglund-Hübsch mirror symmetry conjecture for invertible polynomials

摘录 / EXCERPT: Let $A=(a_{ij})$ be an $(n+1)\times(n+1)$ integer matrix with nonzero determinant, and define ... Let $\check{\bf w}$ be the polynomial associated to A^T, let $\Gamma_{\bf w}$ be...

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V2-032492 | MATHDB-350531

Kahle's bouquet of spheres conjecture for random clique complexes

摘录 / EXCERPT: Let d be an integer with $d\geq 3$, let α satisfy ... and let $\Delta\sim\Delta(n,n^{-\alpha})$ be a random clique complex. Kahle's bouquet of spheres conjecture. With high...

conjecture | open | MathDB 页面 | 原文线索

V2-032493 | MATHDB-350557

Greene's branched-double-cover conjecture for alternating links

摘录 / EXCERPT: Greene's conjecture. If a pair of links have homeomorphic branched double-covers, then either both are alternating or both are non-alternating.

conjecture | open | MathDB 页面 | 原文线索

V2-032494 | MATHDB-350578

Connelly's deformation-retraction conjecture for geodesic triangulations

摘录 / EXCERPT: Let S be a closed orientable surface with a constant-curvature metric. Consider the space of geodesic triangulations of S and the subgroup of isometries of S homotopic to the...

conjecture | open | MathDB 页面 | 原文线索

V2-032495 | MATHDB-350579

Finite non-positively curved complex model conjecture for compact two-dimensional spaces

摘录 / EXCERPT: Let X be a compact, two-dimensional, non-positively curved space. Finite complex model conjecture. The space X is homotopy equivalent to a finite, two-dimensional, non-positive...

conjecture | open | MathDB 页面 | 原文线索

V2-032496 | MATHDB-350581

Gorenstein cohomology conjecture for three-variable cochain DG skew polynomial algebras

摘录 / EXCERPT: Let k be the base field, let $M\in M_3(k)$, and let $\mathcal{A}_{\mathcal{O}_{-1}(k^3)}(M)$ denote the associated cochain DG skew polynomial algebra. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032497 | MATHDB-350585

The improperness conjecture for layered origami knots

摘录 / EXCERPT: A layered origami folding consists of a folding together with a layer ordering, and a layered origami knot is a knot obtained from a loop under such a folding by resolving isolated...

conjecture | open | MathDB 页面 | 原文线索

V2-032498 | MATHDB-350597

The colored Alexander polynomial formula for hook partitions

摘录 / EXCERPT: Let $\mathcal{K}$ be a knot, let $A(\mathcal{K};q)$ denote its classical Alexander polynomial, and let $A_{\lambda}(\mathcal{K};q)$ denote the colored Alexander polynomial associat...

conjecture | open | MathDB 页面 | 原文线索

V2-032499 | MATHDB-350598

The refined LMOV conjecture for framed links

摘录 / EXCERPT: Let $\mathcal{L}_{\vec{\tau}}$ be a framed link, let $z=q-q^{-1}$, and let $g^{(h)}_{\vec{\mu}}(\mathcal{L}_{\vec{\tau}};q,a)$ denote its reformulated LMOV function for...

conjecture | open | MathDB 页面 | 原文线索

V2-032500 | MATHDB-350628

Homological mirror symmetry for symmetric squares of punctured spheres

摘录 / EXCERPT: Let $V_{n,k}$ be the symmetric-square-type Liouville space considered in the paper, equipped with its natural integer grading η_0, and let $\mathcal{Z}_{n,k}$ be the special f...

conjecture | open | MathDB 页面 | 原文线索

V2-032501 | MATHDB-350629

Homological mirror symmetry for compactified symmetric squares of punctured spheres

摘录 / EXCERPT: Let $\overline{\mathcal{Y}}_{n,k}$ and $\overline{\mathcal{Z}}_{n,k}$ be the compactifications of $\mathcal{Y}_{n,k}$ and $\mathcal{Z}_{n,k}$ described in the source, let...

conjecture | open | MathDB 页面 | 原文线索

V2-032502 | MATHDB-350642

The cohomology algebra conjecture for oriented right-angled Artin groups

摘录 / EXCERPT: Let Γ be a special graph, let G_Γ be its associated oriented right-angled Artin group, and let $\mathbb F_2(\Gamma)$ be the quadratic $\mathbb F_2$-algebra generated...

conjecture | open | MathDB 页面 | 原文线索

V2-032503 | MATHDB-350646

The mod-p^r hyperbolicity amplification of Moore's conjecture

摘录 / EXCERPT: The mod-p^r hyperbolicity conjecture. Let X be a finite simply-connected CW-complex. If X is rationally hyperbolic, then it is mod-p^r hyperbolic for all primes p and p...

conjecture | open | MathDB 页面 | 原文线索

V2-032504 | MATHDB-350672

Kubrak-Travkin's conical resolution inequality

摘录 / EXCERPT: Let $\mathcal O_K$ be the ring of integers of a finite extension of $\mathbb Q_p$, let k be its residue field, and let $\pi\colon X\to Y$ be a conical resolution over...

conjecture | open | MathDB 页面 | 原文线索

V2-032505 | MATHDB-350685

Varadhan-type decomposition conjecture for invariant closed forms on topological crystals

摘录 / EXCERPT: Let (X,E) be a locale with a free action of a finitely generated abelian group G of rank d, with finite quotient X/G, and let $ u$ be a probability measure supported on S...

conjecture | open | MathDB 页面 | 原文线索

V2-032506 | MATHDB-350686

The two-point thin-filling conjecture for tangles

摘录 / EXCERPT: Let T be a tangle. Write $\operatorname{ThinHF}(T)$ and $\operatorname{ThinKh}(T)$ for the spaces of thin fillings associated respectively to Heegaard Floer homology and Khovanov...

conjecture | open | MathDB 页面 | 原文线索

V2-032507 | MATHDB-350707

Strong finiteness conjecture for Kauffman bracket skein modules

摘录 / EXCERPT: Strong finiteness conjecture. There exists a finite collection $Sigma_1,\dots,Sigma_k$ of essential subsurfaces $Sigma_i\subsetpartial M$ such that, for each i, the dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-032508 | MATHDB-350725

Yang-Wong volume conjecture for relative Reshetikhin-Turaev invariants

摘录 / EXCERPT: Let M be a closed oriented 3-manifold and let L be a framed hyperbolic link in M with n components. For odd integers $r\geq 3$, let $\mathbf m=(m_1,\dots,m_n)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032509 | MATHDB-350726

Yang-Wong asymptotic expansion conjecture for relative Reshetikhin-Turaev invariants

摘录 / EXCERPT: Let M be a closed oriented 3-manifold and let L be a framed hyperbolic link in M with n components. Let $\mathbf m^{(r)}=(m_1^{(r)},\dots,m_n^{(r)})$ be colorings by elem...

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V2-032510 | MATHDB-350730

Conjecture that Betti numbers are determined by the generating series of one-relator binary operads

摘录 / EXCERPT: Let P be a one-relator binary operad, let $G_P(z)$ be its generating series, and let $b_{n,k}$ denote the number of degree-n generators containing exactly k copies of the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-032511 | MATHDB-350735

The simplicial volume conjecture for open 3-manifolds

摘录 / EXCERPT: Simplicial volume conjecture for open 3-manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-032512 | MATHDB-350739

Completeness of the relations for minimal labeled graphs

摘录 / EXCERPT: A minimal labeled graph is a graph appearing in the higher-lamination calculus, with labels determined by flags on the vertices of a triangle. Consider two minimal labeled graphs c...

conjecture | open | MathDB 页面 | 原文线索

V2-032513 | MATHDB-350768

Strict triple-crossing lower bound for nonmonic Alexander polynomials

摘录 / EXCERPT: Let K be a knot. Write $c_3(K)$ for its triple-crossing number and let $\Delta(K)$ denote its Alexander polynomial; write $\operatorname{breadth}(\Delta(K))$ for the breadth of t...

conjecture | open | MathDB 页面 | 原文线索

V2-032514 | MATHDB-350821

Classification by the Bianchi-Massey and pentagonal Massey tensors

摘录 / EXCERPT: Let X and Y be closed $(n{-}1)$-connected m-manifolds, with $n>1$ and $m\leq 5n-2$. Let ... be an isomorphism of cohomology algebras. Write $\mathcal{T}_b$ and $\mathcal{T}_c$...

conjecture | open | MathDB 页面 | 原文线索

V2-032515 | MATHDB-350828

The completeness conjecture for Jones rational coincidences

摘录 / EXCERPT: A rational knot is represented by an integer sequence presentation, and a Jones rational coincidence is a pair of rational knots with the same Jones polynomial. The pivoting-pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-032516 | MATHDB-350834

Djament's isomorphism conjecture for the wheeled PROP of stable cohomology

摘录 / EXCERPT: Let $\mathcal{E}$ and $\mathcal{H}$ be the wheeled PROPs in the paper, and let ... be the morphism of wheeled PROPs whose image is isomorphic to $\mathcal{K}$. Djament's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-032517 | MATHDB-350838

Supercongruence conjecture for hypercubic, hyper-octahedral and simplicial families

摘录 / EXCERPT: Let $F_N(t)=\sum_{n=0}^{N-1}f_nt^n$ be the truncation of $F(t)$ at t^N, and let $t\mapsto t^\sigma$ be the excellent Frobenius lift. The congruence ... holds for every $m,s\geq 1$....

conjecture | open | MathDB 页面 | 原文线索

V2-032518 | MATHDB-350847

Tannakian obstruction conjecture for braided fusion categories

摘录 / EXCERPT: Tannakian obstruction conjecture. The displayed isomorphism holds. This would complete the obstruction theory for $\mathbf{Rep}(G)$ by identifying every class in...

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V2-032519 | MATHDB-350852

The Decomposition Problem for bi-Lipschitz maps of spheres

摘录 / EXCERPT: Let $LIP(\mathbb{S}^n)$ denote the orientation-preserving homeomorphisms of the n-sphere that are bi-Lipschitz. For a bi-Lipschitz map, its isometric distortion is the least...

open_problem | open | MathDB 页面 | 原文线索

V2-032520 | MATHDB-350871

Chain-map conjecture for cubical homology of digital images

摘录 / EXCERPT: Let $X\subset \mathbb{Z}^n$ and $Y\subset \mathbb{Z}^m$ be digital images with c_1-adjacency, and let $f:X\to Y$ be continuous. Chain-map conjecture. The induced homomorphism ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-032521 | MATHDB-350872

Cubical and singular cubical homology conjecture for digital images

摘录 / EXCERPT: Let $X\subset \mathbb{Z}^n$ be a digital image with c_1-adjacency. The groups $\bar H_q(X)$ and $\bar H_q^{c_1}(X)$ are, respectively, the cubical and elementary cubical homology...

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V2-032522 | MATHDB-350873

Homology conjecture for digital cycles

摘录 / EXCERPT: Let C_m denote the digital cycle with m vertices, and let $H_q(C_m)$ and $\bar H_q(C_m)$ be the homology theories used in the paper. Digital-cycle homology conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-032523 | MATHDB-350876

Existence of an HK-realizing principal groupoid with prescribed K-theory

摘录 / EXCERPT: HK-realization conjecture. There should exist a possibly different groupoid $\widetilde{\mathcal{G}}$ such that

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V2-032524 | MATHDB-350877

The rational HK-conjecture for principal groupoids

摘录 / EXCERPT: Rational HK-conjecture. One should have

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V2-032525 | MATHDB-350881

Extended Langer-Singer lower-bound conjecture for long null-homotopic curves

摘录 / EXCERPT: Let N be a Riemannian manifold with sectional curvature at most $\delta<0$, and let γ be a null-homotopic regular closed curve in N whose length is greater than...

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V2-032526 | MATHDB-350891

Attractor invariants conjecture for toric Calabi-Yau threefolds

摘录 / EXCERPT: Attractor invariants conjecture. The attractor invariants are given by

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V2-032527 | MATHDB-350929

Fibered-knot Laurent polynomial conjecture for F_K

摘录 / EXCERPT: Let K be a fibered knot, and write $F_K(x,q)$ as a series in x. Fibered-knot Laurent polynomial conjecture. For any fibered knot K, the coefficients of F_K are Laurent poly...

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V2-032528 | MATHDB-350934

Boucksom-Demailly-Păun-Peternell transcendental holomorphic Morse inequality conjecture

摘录 / EXCERPT: Let X be a compact complex manifold of dimension n, and let $H^{1,1}_{BC}(X,\mathbb C)$ denote its complex Bott-Chern $(1,1)$-cohomology. A class is nef if it lies in the nef c...

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V2-032529 | MATHDB-350963

The identity conjecture for compactly supported symplectomorphisms of flexible Weinstein structures

摘录 / EXCERPT: A flexible Weinstein structure is a Weinstein structure whose underlying Weinstein domain is flexible, and a compactly supported symplectomorphism is a symplectomorphism equal to t...

conjecture | open | MathDB 页面 | 原文线索

V2-032530 | MATHDB-350978

Conjectural computation of topological Hochschild homology of truncated Brown-Peterson spectra

摘录 / EXCERPT: Conjectural THH computation. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032531 | MATHDB-351037

Varbaro's depth and local cohomological dimension conjecture

摘录 / EXCERPT: Let R be a regular local ring containing a field, and let I be an ideal of R. Varbaro's conjecture. If ... then ... This extends the known characteristic-independent cases of the...

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V2-032532 | MATHDB-351054

The L-space conjecture

摘录 / EXCERPT: Let M be an irreducible, rational homology 3-sphere other than $\mathbb{S}^3$. The L-space conjecture. The following are equivalent: 1. The fundamental group of M is left-ord...

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V2-032533 | MATHDB-351055

The circular orderability version of the L-space conjecture

摘录 / EXCERPT: Let M be an irreducible, rational homology 3-sphere that is not a lens space. Circular orderability version of the L-space conjecture. The following are equivalent: 1. The fund...

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V2-032534 | MATHDB-351056

Circular orderability conjecture for rational homology sphere graph manifolds

摘录 / EXCERPT: Let W be a rational homology sphere graph manifold. Circular orderability conjecture. If $\pi_1(W)$ is infinite, then $\pi_1(W)$ is circularly orderable. The conjecture isolates...

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V2-032535 | MATHDB-351071

Graphing injectivity for rational homotopy and homology of string-link spaces

摘录 / EXCERPT: Fix $m\geq 2$, $k\geq 1$, and $n\geq 3$. Let ... be the graphing map. Graphing injectivity conjecture. The induced maps on rational homotopy and rational homology are injective: ... ar...

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V2-032536 | MATHDB-351103

Topological deformation conjecture for Virasoro actions

摘录 / EXCERPT: Consider a deformation of the bihamiltonian structure of the Principal Hierarchy, and let its central invariants be the functions parametrizing the infinitesimal deformation. The a...

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V2-032537 | MATHDB-351110

Polynomial computability and NP-hardness conjecture for multiparameter persistence distances

摘录 / EXCERPT: Let $n\geq 1$ be fixed, and let $d^p_{\mathcal{M}}$ and $d^p_{\mathcal I}$ denote the corresponding distances on finitely presented n-parameter persistence modules, with...

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V2-032538 | MATHDB-351113

Vanishing of higher cohomology for complexes with extreme local rank 1

摘录 / EXCERPT: Vanishing conjecture. If $\ASC$ has extreme local rank 1, then

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V2-032539 | MATHDB-351152

Equivariant E(n)-local spectrum image conjecture

摘录 / EXCERPT: Equivariant E(n)-local spectrum image conjecture. The map φ_G is a homeomorphism onto its image.

conjecture | open | MathDB 页面 | 原文线索

V2-032540 | MATHDB-351154

Boyer-Gordon-Watson L-space/non-left-orderability conjecture

摘录 / EXCERPT: A nontrivial group G is left-orderable if it admits a total ordering invariant under left multiplication. An L-space is a rational homology 3-sphere with minimal Heegaard Flo...

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V2-032541 | MATHDB-351158

Balmer's homological-to-tensor triangular spectrum bijection conjecture

摘录 / EXCERPT: Let $\cat K$ be an essentially small tensor triangulated category. Write $\Spch(\cat K)$ for its homological spectrum, $\Spc(\cat K)$ for its Balmer spectrum, and ... for the canonic...

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V2-032542 | MATHDB-351167

The composition formula for the unstable homotopy classes \(\zeta_{4n+1}\)

摘录 / EXCERPT: Let Δ denote the relevant connecting homomorphism in the EHP sequence, let η_k and ε_k denote the corresponding unstable homotopy classes, and let...

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V2-032543 | MATHDB-351168

The \(\Delta\sigma\)-\(\mu\) relation in the unstable homotopy groups of spheres

摘录 / EXCERPT: Let Δ denote the relevant connecting homomorphism in the EHP sequence, and let σ_k and μ_k denote the standard unstable homotopy classes; write...

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V2-032544 | MATHDB-351208

The Nielsen-Borsuk-Ulam dichotomy conjecture for free involutions on tori

摘录 / EXCERPT: Let ${\mathbb{T}}^n$ be the n-torus, let $f:{\mathbb{T}}^n\to{\mathbb{T}}^n$ be a map, and let τ be a free involution of ${\mathbb{T}}^n$. Nielsen-Borsuk-Ulam dichotomy conj...

conjecture | open | MathDB 页面 | 原文线索

V2-032545 | MATHDB-351214

Injectivity of the Seifert linking form for even-dimensional manifolds

摘录 / EXCERPT: Let N be a closed connected orientable n-manifold with $H_1(N)$ torsion-free, $n\geq 4$, and n even, and let N_0 denote a punctured version of N. For an embedding...

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V2-032546 | MATHDB-351230

Positive-characteristic p-adic interpolation conjecture for analytic automorphisms

摘录 / EXCERPT: Positive-characteristic p-adic interpolation conjecture. Under these hypotheses, the f-periodic points are not dense in $(K^{\circ})^r$ with respect to the t-adic topology.

conjecture | open | MathDB 页面 | 原文线索

V2-032547 | MATHDB-351251

Profinite Ellis group conjecture for groups over p-adic closed fields

摘录 / EXCERPT: Let M be a p-adic closed field and let G be a group definable in M. The profinite Ellis group conjecture. The Ellis group of the flow $S_{G,\operatorname{ext}}(M)$ is abstr...

conjecture | open | MathDB 页面 | 原文线索

V2-032548 | MATHDB-351283

Dubrovin-Zhang hierarchy conjecture for homogeneous CohFTs

摘录 / EXCERPT: Dubrovin-Zhang hierarchy conjecture. (1) For every $1\leq\alpha,\beta\leq N$ and $a,b\geq0$, there is $\Omega_{\alpha,a;\beta,b}\in\widehat{\mathcal A}_{w;0}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032549 | MATHDB-351284

Gallo-Kapovich-Marden grafting conjecture for complex projective structures

摘录 / EXCERPT: Let Σ be an orientable surface with finitely many punctures and no boundary, and let $\mathcal{P}(\Sigma)$ denote its deformation space of complex projective structures. Eac...

conjecture | open | MathDB 页面 | 原文线索

V2-032550 | MATHDB-351317

Conjecture on density of the typical Hodge locus

摘录 / EXCERPT: Density conjecture for the typical Hodge locus. If $\operatorname{HL}(S, {\mathbb V}^{\otimes})_{\operatorname{typ}}$ is nonempty, then it is dense in S for the analytic topology...

conjecture | open | MathDB 页面 | 原文线索

V2-032551 | MATHDB-351349

The colored cable colimit conjecture

摘录 / EXCERPT: Let $\mathbf{K}$ be a framed, oriented knot, and let $\chi\mathbf{K}$ denote the knot with framing increased by one. For $b\geq0$, consider the directed system whose terms alternat...

conjecture | open | MathDB 页面 | 原文线索

V2-032552 | MATHDB-351351

Colored torus links are parity

摘录 / EXCERPT: Let ${\bm{b}}=(b_1,\ldots,b_m)$ and let $T(m,m;{\bm{b}})$ be the ${\bm{b}}$-colored (m,m)-torus link. Colored torus-link parity conjecture. The link $T(m,m;{\bm{b}})$ is parity;...

conjecture | open | MathDB 页面 | 原文线索

V2-032553 | MATHDB-351373

The heavy-SH-heavy equivalence conjecture

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold, and let $K\subset M$ be compact. The subset K is heavy if it satisfies the heavy-set condition defined using the relevant symplec...

conjecture | open | MathDB 页面 | 原文线索

V2-032554 | MATHDB-351374

The multi-intersection problem for three or more sets

摘录 / EXCERPT: Multi-intersection conjecture. Question has an affirmative answer provided there exist elements φ_i lying in the identity component of the group of volume-preserving diffeo...

conjecture | open | MathDB 页面 | 原文线索

V2-032555 | MATHDB-351384

The affine associahedron sphere conjecture

摘录 / EXCERPT: An affine associahedron is the dual of the quotient cluster complex associated with an affine cluster algebra, and its dimension is determined by the rank of that algebra. The affi...

conjecture | open | MathDB 页面 | 原文线索

V2-032556 | MATHDB-351385

The doubly extended exchange-complex homotopy conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a doubly extended cluster algebra of rank $n+2$. Its exchange complex is the complex associated with the exchange structure, and its doubly extended associahed...

conjecture | open | MathDB 页面 | 原文线索

V2-032557 | MATHDB-351394

Localization dichotomy for non-periodic gapped systems

摘录 / EXCERPT: Localization dichotomy conjecture. The following statements are equivalent: (a) P admits a generalized Wannier basis that is exponentially localized; (b) P admits a generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-032558 | MATHDB-351399

Voronoi-realized cohomology decomposition conjecture

摘录 / EXCERPT: Let E be a real quadratic field and let $\Gamma\subset\operatorname{SL}_3(\mathbb{Z})$ be a finite-index subgroup. Let $H(\Gamma,E)$ be the Steinberg-generated subspace, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032559 | MATHDB-351400

Dimension formula conjecture for Steinberg-generated cohomology

摘录 / EXCERPT: Let E be a real quadratic field and let $\Gamma\subset\operatorname{SL}_3(\mathbb{Z})$ be a finite-index subgroup. Let $H(\Gamma,E)$ be the Steinberg-generated subspace of...

conjecture | open | MathDB 页面 | 原文线索

V2-032560 | MATHDB-351402

Random induced subcomplexes of type A buildings are asymptotically spherical

摘录 / EXCERPT: Let $\Delta_{k,n}$ be the finite spherical building of type A_k, let $\mathcal{P}(\Delta_{k,n})$ denote its induced subcomplexes, and let $\mathcal{S}(\Delta_{k,n})$ denote the s...

conjecture | open | MathDB 页面 | 原文线索

V2-032561 | MATHDB-351403

Random induced subcomplexes of thick finite buildings are asymptotically spherical

摘录 / EXCERPT: Let $d \in \mathbb{N}$ and let $(\Delta_n)_{n \in \mathbb{N}}$ be a sequence of finite d-dimensional buildings whose thickness satisfies ... as $n \rightarrow \infty$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032562 | MATHDB-351406

Commutativity conjecture for instanton knot Floer differentials

摘录 / EXCERPT: Suppose $K\subset S^3$ is a quasi-alternating knot. Let $\underline{\operatorname{KHI}}(-S^3,K)$ be the instanton knot Floer homology group, and let d_+ and d_- be maps on it....

conjecture | open | MathDB 页面 | 原文线索

V2-032563 | MATHDB-351412

The conjecture on arbitrarily large topological four-genus of algebraically slice GA-knots

摘录 / EXCERPT: Arbitrarily large four-genus conjecture. There are algebraically slice GA-knots with arbitrary large topological four-genus.

conjecture | open | MathDB 页面 | 原文线索

V2-032564 | MATHDB-351418

Hilali conjecture on rational homotopy and homology dimensions

摘录 / EXCERPT: Let X be a simply connected rationally elliptic space. Write $\boldsymbol{\pi_*(X)}$ for its rational homotopy groups and $H_*(X;\mathbb Q)$ for its rational homology groups. Hil...

conjecture | open | MathDB 页面 | 原文线索

V2-032565 | MATHDB-351419

Transcendence conjecture for the real stabilization threshold of elliptic spaces

摘录 / EXCERPT: Let X be a simply connected elliptic space, and let $\frak{pp}_{\mathbb R}(X;1)$ denote its real stabilization threshold. Let $P(s)=P_X^{\pi}(s)$ and $Q(s)=P_X(s)$. Transcendence...

conjecture | open | MathDB 页面 | 原文线索

V2-032566 | MATHDB-351446

Edge-contraction conjecture for dosp mutation graphs

摘录 / EXCERPT: Let $\mathbb{S}=S_{g,h}$ be a closed surface notation used in the paper, let k be the rank, and let $\mathbb{E}_{\rm good}$ map to the dosp mutation graph via the map...

conjecture | open | MathDB 页面 | 原文线索

V2-032567 | MATHDB-351461

The effective-line conjecture for hyper-Kähler manifolds

摘录 / EXCERPT: Let X be a projective irreducible holomorphic symplectic manifold, and let $\widetilde{H}(X,\mathbb{Q})$ be its extended rational cohomology. A line ℓ in...

conjecture | open | MathDB 页面 | 原文线索

V2-032568 | MATHDB-351469

The Alexander-polynomial normalization of the link series

摘录 / EXCERPT: Let $L\subset S^3$ be a framed oriented link with multivariable Alexander polynomial ∇_L, and suppose ∇_L is nonzero. Let F_L^t be the regularized formal series ass...

conjecture | open | MathDB 页面 | 原文线索

V2-032569 | MATHDB-351496

Homology-dimension conjecture for images of stable perturbations on ICIS

摘录 / EXCERPT: Let $f\colon (X,S)\to(\mathbb{C}^{p},0)$ be a map germ of finite $\mathscr{A}_e$-codimension, where X is an ICIS of dimension $n<p$. Let $d(f)$ and $s(f)$ denote the quantities u...

conjecture | open | MathDB 页面 | 原文线索

V2-032570 | MATHDB-351539

Thick tensor-ideal exhaustion conjecture for dualizable spectra in the \Sp_{k,n}-local category

摘录 / EXCERPT: Let $0 \le i \le n+1$. For $i \le n$, let $\mathcal D_i$ be the full subcategory of dualizable objects X in $\Sp_{k,n}^{\dual}$ that are retracts of $Y \wedge Z$ for some dualiza...

conjecture | open | MathDB 页面 | 原文线索

V2-032571 | MATHDB-351543

Skopenkov's NP-hardness conjecture for geometric embeddability

摘录 / EXCERPT: For integers k,d, let $\operatorname{GEM}_{k \rightarrow d}$ denote the decision problem of determining whether a given k-complex has a geometric embedding in $\mathbb{R}^d$. S...

conjecture | open | MathDB 页面 | 原文线索

V2-032572 | MATHDB-351553

Przytycki-Vojtěchovský conjecture for cyclic biquandle Yang-Baxter homology

摘录 / EXCERPT: Przytycki-Vojtěchovský conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032573 | MATHDB-351580

Conjecture on the two tight parameterizations of a $(1,1)$-knot

摘录 / EXCERPT: Let K be a $(1,1)$-knot in a tight presentation ... A tight parameterization is a parameterization arising from this tight presentation. Two-parameterization conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-032574 | MATHDB-351668

Erlandsson-Parlier conjecture on self-intersection excess of systoles on compact hyperbolic surfaces

摘录 / EXCERPT: Let Σ be a compact hyperbolic surface, and let $I_k(\Sigma)$ denote the maximal self-intersection number of a k-systole on Σ. Erlandsson-Parlier's conjecture. One h...

conjecture | open | MathDB 页面 | 原文线索

V2-032575 | MATHDB-351669

Quantum cohomology as a deformation of symplectic cohomology

摘录 / EXCERPT: Let M be the compactification and X its complement of the divisor, let $QH^*(M;\Lambda):=H^*(M;\Bbbk)\otimes_\Bbbk\Lambda$, and let $SH^*(X;\Bbbk)$ denote symplectic cohomology...

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V2-032576 | MATHDB-351670

Conjecture that the skeleton is relative symplectic-cohomology full

摘录 / EXCERPT: Let $a\in QH^0(M;\Lambda)$ be the idempotent generating the ideal $QH^*(M;\Lambda)_{\mathrm{crit}}$, and let $\mathbb{L}$ be the skeleton. A subset is a-SH-full when its a-re...

conjecture | open | MathDB 页面 | 原文线索

V2-032577 | MATHDB-351671

Maurer-Cartan deformation conjecture for symplectic cochains

摘录 / EXCERPT: Assume $\Bbbk$ has characteristic zero and Hypothesis ... can be taken to be $\widetilde{SC}_{\Lambda}=SC_{\Lambda}$ with differential $\partial=\ell^1_\beta$. This would eliminate t...

conjecture | open | MathDB 页面 | 原文线索

V2-032578 | MATHDB-351672

Tonkonog's cap class as the first Maurer-Cartan component

摘录 / EXCERPT: In Tonkonog's setup, let $\mathcal{BS}\in SH^0(X;\Bbbk)$ be the class obtained by counting holomorphic caps, and write $\beta=\beta_1+\beta_2+\cdots$ for the filtration components...

conjecture | open | MathDB 页面 | 原文线索

V2-032579 | MATHDB-351673

Explicit Maurer-Cartan element in the log Calabi-Yau case

摘录 / EXCERPT: Assume the log Calabi-Yau condition, so that $\lambda_i=2$ for every divisor component, and assume the minimal Chern number of M is at least 2. For each i, let B_i be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-032580 | MATHDB-351675

Dubois-Friedl-Lück conjecture on higher-order torsion and knot genus

摘录 / EXCERPT: Let K be a knot in S^3, let X_K be its exterior, and let $\phi\in H^1(X_K;\mathbb{Z})$ be a generator. A pair (ρ,ϕ) is assumed to be admissible, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032581 | MATHDB-351685

The topological-rank conjecture for L^1 full groups of flows

摘录 / EXCERPT: Let $\mathcal F:\mathbb R\curvearrowright(X,\mu)$ be a measure-preserving flow, and let n be the number of its ergodic components. Write $[\mathcal F]_1$ for its L^{1}...

conjecture | open | MathDB 页面 | 原文线索

V2-032582 | MATHDB-351704

Arnold's filtered invariants conjecture for spherical curves

摘录 / EXCERPT: Arnold's conjecture. There exists a sequence of invariants $\{E_i\}_{i\geq 0}$ with $E_0=J^+ + 2 St$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032583 | MATHDB-351709

Super-exponential growth of minimal-origami orbits

摘录 / EXCERPT: Let N_g denote the number of distinct $SL(2,\mathbb{Z})$-orbits of origamis in $\mathcal{H}(2g-2)$ with $2g-1$ squares, such that each of their horizontal and vertical cylinder d...

conjecture | open | MathDB 页面 | 原文线索

V2-032584 | MATHDB-351714

Partial-order conjecture for ribbon rational homology cobordism

摘录 / EXCERPT: Ribbon rational homology cobordism conjecture. If there exists a ribbon $\mathbb{Q}$-homology cobordism from Y_0 to Y_1 and one from Y_1 to Y_0, then Y_0 and Y_1 are di...

conjecture | open | MathDB 页面 | 原文线索

V2-032585 | MATHDB-351721

Wang-Zhou conjecture on realizations of abstract topological recursion

摘录 / EXCERPT: Let $4\mathcal W_{g,n}4$ be the abstract n-point functions of the abstract quantum field theory of fat graphs. Under a suitable Feynman-rule realization, let $4\mathcal C4$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-032586 | MATHDB-351722

Wang-Zhou conjecture on residue reformulation of abstract topological recursion

摘录 / EXCERPT: Let $4\mathcal W_{g,n}4$ be the abstract n-point functions and let the abstract quantum field theory have spectral curve $4\mathcal C4$. Wang-Zhou's residue-reformulation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-032587 | MATHDB-351741

Characterization of non-alternating Montesinos BB knots

摘录 / EXCERPT: A Montesinos knot is a knot in the Montesinos class, and a BB knot is a knot whose bridge index equals its braid index. The source refers to the non-alternating Montesinos knots de...

conjecture | open | MathDB 页面 | 原文线索

V2-032588 | MATHDB-351749

Analytic lattice cohomology as a submodule of topological lattice cohomology

摘录 / EXCERPT: Analytic lattice cohomology conjecture. The morphism

conjecture | open | MathDB 页面 | 原文线索

V2-032589 | MATHDB-351750

Constancy of analytic lattice cohomology in p_g-constant deformations

摘录 / EXCERPT: Constancy conjecture. The graded $\mathbb Z[U]$-module $\mathbb H^*_{an,0}(X_t,o_t)$ is constant along the deformation.

conjecture | open | MathDB 页面 | 原文线索

V2-032590 | MATHDB-351756

Adolphson-Sperber conjecture on generic ordinarity

摘录 / EXCERPT: Let Δ be a lattice polytope of degree denominator D, and let ${\rm GNP}(\Delta,p)$ and ${\rm HP}(\Delta)$ denote its generic Newton polygon and Hodge polygon, respectively...

conjecture | open | MathDB 页面 | 原文线索

V2-032591 | MATHDB-351758

The generalized-quiver partition-function conjecture for HOMFLY-PT homology

摘录 / EXCERPT: Generalized-quiver partition-function conjecture. After this substitution, the generalized-quiver partition function equals the generating function of Gromov-Witten invariants coun...

conjecture | open | MathDB 页面 | 原文线索

V2-032592 | MATHDB-351784

Jones's conjecture on nontrivial knots with trivial Jones polynomial

摘录 / EXCERPT: Let K be a knot, and let $V(K)$ denote its Jones polynomial. A knot is non-trivial if it is not isotopic to the unknot. Jones's conjecture. There exists a non-trivial knot K su...

conjecture | open | MathDB 页面 | 原文线索

V2-032593 | MATHDB-351785

Colouring construction of the Jones polynomial

摘录 / EXCERPT: The Alexander polynomial can be defined using a method known as colouring. The Jones-polynomial colouring problem. Can the Jones polynomial be defined via colouring? The source pre...

open_problem | open | MathDB 页面 | 原文线索

V2-032594 | MATHDB-351820

The d-invariant characterization of amphichiral 2-bridge knots

摘录 / EXCERPT: Let $p,q\in\mathbb{N}$ with p odd and $(p,q)=1$. Write $L(p,q)$ for the lens space associated to the 2-bridge knot $K(p/q)$, and let its Heegaard Floer correction terms be the va...

conjecture | open | MathDB 页面 | 原文线索

V2-032595 | MATHDB-351821

Strong negative amphichirality criterion for 2-bridge knots

摘录 / EXCERPT: Let K be a 2-bridge knot, and let $Y=\Sigma(S^3,K)$ be its double branched cover. Let $\widetilde{\sigma}$ be a lift of the strong negative amphichiral involution when one exists...

conjecture | open | MathDB 页面 | 原文线索

V2-032596 | MATHDB-351831

The cohomological P=W conjecture

摘录 / EXCERPT: Let C be a punctured Riemann surface, let G be a linear reductive algebraic group over $\mathbb{C}$, and let $\mathcal{M}_B$ and $\mathcal{M}_{Dol}$ be corresponding Betti and...

conjecture | open | MathDB 页面 | 原文线索

V2-032597 | MATHDB-351832

The geometric P=W conjecture for Betti and Dolbeault moduli spaces

摘录 / EXCERPT: Let N_{Dol}^* be the complement of a large Hitchin-map sublevel set in $\mathcal{M}_{Dol}$, let N_B^* be a punctured tubular neighborhood of the boundary divisor in a log compa...

conjecture | open | MathDB 页面 | 原文线索

V2-032598 | MATHDB-351836

The K(π,1) conjecture for mixed Tate motives

摘录 / EXCERPT: Pour un corps F, soit $\mathrm{DM}(F)$ la catégorie triangulée des motifs à coefficients dans $\mathbb{Q}$, et soit $\mathrm{MT}(F)$ sa catégorie des motifs de Tate mixtes. Pour...

conjecture | open | MathDB 页面 | 原文线索

V2-032599 | MATHDB-351846

Higher-order conjectures for the perturbed Gaussian knot invariant

摘录 / EXCERPT: Higher-order conjectures. For $j\geq 1$, $\rho_{k,j}(K)$ is determined by

conjecture | open | MathDB 页面 | 原文线索

V2-032600 | MATHDB-351850

Hochschild functor conjecture for enriched categories and bimodules

摘录 / EXCERPT: Let $(\mathscr{V},\otimes,I)$ be a presentably symmetric monoidal $(\infty,1)$-category. Let $\mathscr{M}\mathrm{od}_{\mathscr{V}}$ be the $(\infty,2)$-category whose objects are...

conjecture | open | MathDB 页面 | 原文线索

V2-032601 | MATHDB-351880

Robustness conjecture for positive topological entropy on closed surfaces

摘录 / EXCERPT: Let Q be a closed surface, let g be a Riemannian metric on Q, and write $h_{\rm top}(g)$ for the topological entropy of its geodesic flow. Say that $h_{\rm top}(g)$ is robust...

conjecture | open | MathDB 页面 | 原文线索

V2-032602 | MATHDB-351892

Merkurjev-Suslin conjecture on the reduced norm in K_{2}

摘录 / EXCERPT: Merkurjev-Suslin conjecture. There is an exact sequence

conjecture | open | MathDB 页面 | 原文线索

V2-032603 | MATHDB-351900

Complexification of Kashaev's conjecture

摘录 / EXCERPT: Let H be a hyperbolic knot in S^3, and let $J_N(H;q)$ be its colored Jones polynomial. Let $\operatorname{CS}(S^3\setminus H)$ denote the $SO(3)$ Chern--Simons invariant associ...

conjecture | open | MathDB 页面 | 原文线索

V2-032604 | MATHDB-351910

Topological invariance of the Mellin--Barnes coefficient for real representations

摘录 / EXCERPT: Let M be a connected, non-compact, orientable 3-manifold with one toroidal end and a complete hyperbolic structure, and let $\rho:\pi_1(M)\to PSL(2,\mathbb{R})$ be an irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-032605 | MATHDB-352041

The special-correlator freedom conjecture for intersections of two quadrics

摘录 / EXCERPT: Let X be an even-dimensional complete intersection of two quadrics in $\mathbb{P}^{n+2}$, and let z denote its special correlator. Let $F(t_0,\dots,t_{2n+3};z)$ be the generati...

conjecture | open | MathDB 页面 | 原文线索

V2-032606 | MATHDB-352042

The quadratic special-correlator conjecture for intersections of two quadrics

摘录 / EXCERPT: Let X be an even-dimensional complete intersection of two quadrics in $\mathbb{P}^{n+2}$. Write $\varepsilon_1,\dots,\varepsilon_{n+3}$ for the specified basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-032607 | MATHDB-352058

Hopkins's characteristic-class conjecture for block upper-triangular groups

摘录 / EXCERPT: Hopkins's block upper-triangular characteristic-class conjecture. This induced map is an isomorphism. The statement appears among results attributed to ideas of Bott, but the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-032608 | MATHDB-352070

Gromov's Uryson width conjecture for smooth domains

摘录 / EXCERPT: Gromov's conjecture. There should exist a continuous self-map $R:X\to X$ such that $R(X)$ has topological dimension $n-2$ and

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V2-032609 | MATHDB-352086

The height n Telescope Conjecture for spectra

摘录 / EXCERPT: The height n Telescope Conjecture. The Bousfield class of $T(n)$ agrees with the Bousfield class of $K(n)$:

conjecture | open | MathDB 页面 | 原文线索

V2-032610 | MATHDB-352089

The BPS invariant connect-sum formula for knots in integral homology spheres

摘录 / EXCERPT: Let Y be an integral homology 3-sphere and let K_1,K_2 be knots in Y. For each knot, let $F_K(x,q)$ be its two-variable series invariant, with $c\in\mathbb{Z}_{+}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-032611 | MATHDB-352090

The Cosmetic Crossing Conjecture

摘录 / EXCERPT: Let K be a knot, and let a crossing circle be the circle encircling the two strands at a crossing. A crossing is nugatory if its crossing circle bounds an embedded disk in the co...

conjecture | open | MathDB 页面 | 原文线索

V2-032612 | MATHDB-352091

The Cosmetic Crossing Conjecture for Conway tangles

摘录 / EXCERPT: Let T be a Conway tangle, and let K_0 and K_1 be the knots obtained from the associated family of tangle fillings. Say that T is horizontally split when it admits the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-032613 | MATHDB-352092

The Generalized Cosmetic Crossing Conjecture for Conway tangles

摘录 / EXCERPT: Let T be a Conway tangle with the specified connectivity and no closed components, and let $\{K_n\}_{n\in\mathbb{Z}}$ be the associated family of knots. Say that T is horizonta...

conjecture | open | MathDB 页面 | 原文线索

V2-032614 | MATHDB-352101

The unknotted-core conjecture for minimal necklaces of length less than 18

摘录 / EXCERPT: A minimal necklace is a necklace satisfying the minimality condition defined in the paper, and its length is the number of components in the necklace. Its core is the core curve as...

conjecture | open | MathDB 页面 | 原文线索

V2-032615 | MATHDB-352132

The integral formula conjecture for the two-dimensional tame symbol

摘录 / EXCERPT: Let f, g, and h be smooth invertible functions on $\mathbb{T}^2$. Define ... and let $w_1(f),w_2(f)\in\mathbb{Z}$ be the winding numbers of $z\mapsto f(z,1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-032616 | MATHDB-352133

The Connes-Karoubi character factorisation conjecture for the two-torus

摘录 / EXCERPT: Let $\mathcal{F}_{\mathbb{T}^2}$ be the 3-summable Fredholm module over $C^\infty(\mathbb{T}^2)$, and let $[c]\in H^3(GL(C^\infty(\mathbb{T}^2)),\mathbb{C}^*)$ be the class of th...

conjecture | open | MathDB 页面 | 原文线索

V2-032617 | MATHDB-352150

Nontrivial knots have no $SU(2)$-abelian surgeries with slope in $[0,5)$

摘录 / EXCERPT: Let $K\subset S^3$ be a nontrivial knot, let $r\in[0,5)$ be a rational number, and denote by $S^3_r(K)$ the result of rational Dehn surgery on K with slope r. A 3-manifold is...

conjecture | open | MathDB 页面 | 原文线索

V2-032618 | MATHDB-352153

Cyclotomic expansion conjecture for colored $SU(n)$ invariants

摘录 / EXCERPT: Cyclotomic expansion conjecture. For any knot $\mathcal{K}$, there exist Laurent polynomials $H_k^{(n)}(\mathcal{K})\in\mathbb{Z}[q,q^{-1}]$, independent of N, such that

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V2-032619 | MATHDB-352156

Boundary Euler characteristic conjecture for the refined Chen-Yang coefficient

摘录 / EXCERPT: Boundary Euler characteristic conjecture. For all $M\in\mathcal{M}$, the coefficient $b(M)$ depends only on the topology of ∂M, as an affine function of...

conjecture | open | MathDB 页面 | 原文线索

V2-032620 | MATHDB-352234

The TWO strategy conjecture for the countable-dimensional game

摘录 / EXCERPT: Let X be a T_1 space, and let $G_c(\mathcal O,\mathcal O)$ denote the compact-open selection game in which ONE and TWO choose open covers and TWO selects finite subfamilies. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-032621 | MATHDB-352252

Minimal-volume conjecture for meridian type vector fields

摘录 / EXCERPT: Let $M^\star$ be the punctured 2-sphere under consideration, and for each positive integer k let $X_{\mathrm{m},k}$ denote the meridian type vector field whose relative homology...

conjecture | open | MathDB 页面 | 原文线索

V2-032622 | MATHDB-352277

Polyvector-field description of Viterbo symplectic cohomology

摘录 / EXCERPT: Let $\mathcal{R}$ be a simple eigenray diagram, let $M_{\mathcal{R}}$ be the associated symplectic manifold, and let $\mathcal{Y}_{\mathcal{R}}$ be its proposed non-Archimedean mir...

conjecture | open | MathDB 页面 | 原文线索

V2-032623 | MATHDB-352305

Slope-bound conjecture for classical coherent cohomology

摘录 / EXCERPT: Let $\be X^\star(T^c)^{M_\mu,+}$ and let $:T(\mathbb{Z}_p)\to\overline{F}^{\times}$ be a finite-order character. For $w\in C^+$, set $=-w^{-1}w_{0,M}(+)-$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032624 | MATHDB-352328

Garoufalidis-Yoon twisted 1-loop conjecture

摘录 / EXCERPT: Let M be a cusped hyperbolic 3-manifold, let $\mathcal{T}$ be an ideal triangulation, and choose an epimorphism $\alpha:\pi_1(M)\twoheadrightarrow\mathbb{Z}$, equivalently an inf...

conjecture | open | MathDB 页面 | 原文线索

V2-032625 | MATHDB-352337

Conjecture on the quantum cohomology of a three-sphere-type Lagrangian

摘录 / EXCERPT: Let L be the total space of the fibration defined by the theorem preceding the conjecture, with $k=1$, $p_1=4$, $p_2=11$, and $n=20$. Thus L is the specific monotone Lagrangian...

conjecture | open | MathDB 页面 | 原文线索

V2-032626 | MATHDB-352387

The chiral-part conjecture for heterotic/type I duality

摘录 / EXCERPT: Let X be a CalabiYau 5-fold. The type I topological string on X is the unoriented topological B-model with its open-string sector, whose gauge theory has structure group...

conjecture | open | MathDB 页面 | 原文线索

V2-032627 | MATHDB-352388

The fiber-base Betti-number conjecture for persistent homology

摘录 / EXCERPT: Fiber-base Betti-number conjecture. The fiber $\mathrm{PH}^{-1}(D_K)$ and K have the same Betti numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-032628 | MATHDB-352389

Zeeman's conjecture on collapsibility after subdivision

摘录 / EXCERPT: Zeeman's conjecture. After taking finitely many barycentric subdivisions, the product

conjecture | open | MathDB 页面 | 原文线索

V2-032629 | MATHDB-352394

Bijection conjecture for mixed and bordered-mixed complexes

摘录 / EXCERPT: A finitely generated mixed complex is a mixed complex of the form used in the paper with finite generation understood in the source, and a bordered-mixed complex is a tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-032630 | MATHDB-352399

Yangian module structure conjecture for superspin-chain sigma models

摘录 / EXCERPT: Let m,n,L be positive integers, let $\mathbf{M}=(M_1,\dotsc,M_{m+n-1})\in\mathbb{Z}_{\geq0}^{m+n-1}$, and let $\CM(\mathbf{M})$ be the Calabi-Yau target of the corresponding effe...

conjecture | open | MathDB 页面 | 原文线索

V2-032631 | MATHDB-352428

The characterizing contact slope conjecture for Legendrian figure-eight and torus knots

摘录 / EXCERPT: A characterizing contact slope for a Legendrian knot is a contact surgery slope that distinguishes the knot from every Legendrian knot with contactomorphic surgery. The claim conce...

conjecture | open | MathDB 页面 | 原文线索

V2-032632 | MATHDB-352494

The J-homomorphism conjecture on cohomology indecomposables

摘录 / EXCERPT: Let J be the J-homomorphism, let β_j denote the indicated cohomology classes associated with SO, and let $ obreak\gamma_{1[1]},\gamma_{1[2]},\gamma_{2[1]}$ denote the c...

conjecture | open | MathDB 页面 | 原文线索

V2-032633 | MATHDB-352495

The sparse-presentation conjecture for the J-induced map on the Nakaoka module

摘录 / EXCERPT: Let the Nakaoka module be presented using generators including $\gamma_{1[1]}^{2^k-1}$, and consider the map induced by the J-homomorphism. Sparse-presentation conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-032634 | MATHDB-352499

Stipsicz's high surgery conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and for a rational number r let $S^3_r(K)$ denote the r-surgery on K. Stipsicz's high surgery conjecture. There is an integer n_K such that fo...

conjecture | open | MathDB 页面 | 原文线索

V2-032635 | MATHDB-352500

Cohomological rigidity conjecture for Airy local systems

摘录 / EXCERPT: Let $j:U\hookrightarrow X=\bP^1_k$ be an open immersion, let $u\in U$ be a geometric point, and let $\mathrm{Ai}_{\hG}$ be the Airy $\hG$-local system with corresponding representa...

conjecture | open | MathDB 页面 | 原文线索

V2-032636 | MATHDB-352513

The shifted edge labeled tableaux rule for specialized equivariant structure coefficients

摘录 / EXCERPT: Let $44\lambda,\mu,\nu\in {\mathcal {SP}}_n$ and let ${\mathfrak d}_{\lambda,\mu}^{\nu}$ be the normalized structure coefficient defined from the coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-032637 | MATHDB-352524

The factorization conjecture for Riley polynomials

摘录 / EXCERPT: Let p/q be a rational slope, and let $Q_{p/q}$ denote the polynomial associated with this slope. If $q=2^n r$ with r odd, then n is the exponent of 2 in the denominator; wh...

conjecture | open | MathDB 页面 | 原文线索

V2-032638 | MATHDB-352559

Equivariant factorization-homology realization of the twisted Dennis trace

摘录 / EXCERPT: Let R be a C_n-ring spectrum. The twisted Dennis trace is a map ... Horev's identification gives ... where $\int_{S^1_{\mathrm{rot}}} R$ is the C_n-equivariant factorization homo...

conjecture | open | MathDB 页面 | 原文线索

V2-032639 | MATHDB-352579

Harer's conjecture on Hopf plumbing characterisation of fibered links

摘录 / EXCERPT: Let L be a link in S^3, and let its fiber surface be the page of a fibration of its complement over S^1. A Hopf plumbing or deplumbing is the operation of adding or removing...

conjecture | open | MathDB 页面 | 原文线索

V2-032640 | MATHDB-352588

Genus-independent topological-recursion formula for bipartite quadrangulations

摘录 / EXCERPT: Bipartite quadrangulation conjecture. For $g>0$, the topological recursion for this spectral curve gives

conjecture | open | MathDB 页面 | 原文线索

V2-032641 | MATHDB-352607

Hopkins' algebraic chromatic splitting conjecture

摘录 / EXCERPT: Let E_* be the coefficient ring of Morava E-theory at height n, let E_*E be its cooperations, and let $v_0=p,v_1,\ldots,v_{n-1}$ denote the usual periodicity elements. For...

conjecture | open | MathDB 页面 | 原文线索

V2-032642 | MATHDB-352613

Analytic continuation conjecture for the quantum modularity cocycle

摘录 / EXCERPT: Let $\gamma=\begin{pmatrix}a&b\\c&d\end{pmatrix}\in\operatorname{SL}_2(\mathbb Z)$ with $c\neq0$, and let W_γ be defined on $\mathbb Q\smallsetminus\{\gamma^{-1}(\infty)\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-032643 | MATHDB-352626

Strong cohomological rigidity conjecture for Bott manifolds

摘录 / EXCERPT: A Bott tower of height n is an iterated $\mathbb C P^1$-bundle whose stages are Bott manifolds; each Bott manifold is a closed manifold of dimension $2n$. Strong cohomological ri...

conjecture | open | MathDB 页面 | 原文线索

V2-032644 | MATHDB-352653

Gordon's purely cosmetic surgery conjecture in the 3-sphere

摘录 / EXCERPT: Let K be a nontrivial knot in S^3. For surgery slopes r and s, the manifolds $S^3_r(K)$ and $S^3_s(K)$ are purely cosmetic when they are homeomorphic as oriented manifolds....

conjecture | open | MathDB 页面 | 原文线索

V2-032645 | MATHDB-352657

The secondary cyclotomic trace conjecture for 2-algebras

摘录 / EXCERPT: Secondary cyclotomic trace conjecture. For each 2-algebra A over $cbkk$, there is a canonical filler in the diagram among spectra:

conjecture | open | MathDB 页面 | 原文线索

V2-032646 | MATHDB-352660

Boundary-map dimension conjecture for real Grassmannians

摘录 / EXCERPT: Let $\delta:H^*(\operatorname{Gr}_d({\mathbb R}^{m-1});Q_n)\to\widetilde H^{*+2^{n+1}-1}(C_d({\mathbb R}^{m});Q_n)$ be the boundary map in Q_n-homology, and let $k_n^\delta(d,m)$...

conjecture | open | MathDB 页面 | 原文线索

V2-032647 | MATHDB-352698

Morita equivalence of weakly and strictly stable weak identity types

摘录 / EXCERPT: Morita equivalence conjecture. The theories of weakly stable weak identity types and strictly stable weak identity types are Morita equivalent: for every I_{ws}-cellular model...

conjecture | open | MathDB 页面 | 原文线索

V2-032648 | MATHDB-352719

Conjecture on the coindex of planar polygon spaces

摘录 / EXCERPT: Let α be a generic n-length vector, and let M_{α} denote its planar polygon space. The smallest gee of α is the smallest gee appearing in its geneti...

conjecture | open | MathDB 页面 | 原文线索

V2-032649 | MATHDB-352764

Del Pezzo-cyclic loop space duality conjecture

摘录 / EXCERPT: Let $\mathbb{B}_k$, for $0 \leq k \leq 8$, and $\mathbb{C}\mathbb{P}^1 \times \mathbb{C}\mathbb{P}^1$ be the indicated series of del Pezzo surfaces. Let $\mathcal{L}^{k}_c S^4$, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-032650 | MATHDB-352767

The exterior-algebra module refinement of Némethi's conjecture

摘录 / EXCERPT: Let G be a plumbing tree, let $Y(G)$ be its associated plumbed 3-manifold, and let $\boldsymbol{\mathbf{\mathit{HF}}}^-(Y(G))$ denote the completed Heegaard Floer homology. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032651 | MATHDB-352820

Categorical quantum-group realization of the A-twisted theory

摘录 / EXCERPT: Fix integers $n\geq 2$ and $k\geq n$, set $q=e^{i\pi/k}$, and let $\mathcal C^{(n,k)}$ be the coherent sheaf of dg categories of line operators of $\mathcal T_{n,k}^A$ over...

conjecture | open | MathDB 页面 | 原文线索

V2-032652 | MATHDB-352848

The algebraic transfer isomorphism conjecture in bidegree $(5,5+n_d)$

摘录 / EXCERPT: Let $\mathscr A_2$ be the mod-2 Steenrod algebra, let n_d be the internal degree sequence used in the paper, and let $Tr_5^{\mathscr A_2}$ denote the fifth algebraic transfer....

conjecture | open | MathDB 页面 | 原文线索

V2-032653 | MATHDB-352857

The classification conjecture for chirally cosmetic surgeries

摘录 / EXCERPT: Let K be a non-trivial knot in S^{3}. A chirally cosmetic surgery is an orientation-reversing homeomorphism between surgeries on inequivalent slopes. The known families are (A)...

conjecture | open | MathDB 页面 | 原文线索

V2-032654 | MATHDB-352858

The type classification conjecture for chirally cosmetic surgeries

摘录 / EXCERPT: Let K be a non-trivial knot in S^{3}. A chirally cosmetic surgery is of 0-type when $S^3_K(r)\cong -S^3_K(-r)$, of $+$-type when $S^3_K(r)\cong -S^3_K(r')$ with $rr'>0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032655 | MATHDB-352859

The alternating-knot nonexistence conjecture for chirally cosmetic surgeries

摘录 / EXCERPT: Let K be an alternating knot in S^{3} that is non-amphicheiral and is not a $(2,p)$-torus knot. Alternating-knot nonexistence conjecture. The knot K does not admit chirally c...

conjecture | open | MathDB 页面 | 原文线索

V2-032656 | MATHDB-352869

Universal-cover-of-balls conjecture for codimension 2 width

摘录 / EXCERPT: Universal-cover-of-balls conjecture. There exist positive dimensional constants δ_n, C_n, and R_n such that, for every $p\in M$,

conjecture | open | MathDB 页面 | 原文线索

V2-032657 | MATHDB-352897

Andersen's asymptotic expansion conjecture for Reshetikhin-Turaev invariants

摘录 / EXCERPT: For an oriented compact three-manifold X, let G be a semisimple, simply connected Lie group, and let $\mathcal{Z}_G^{(K)}(X)$ denote the Reshetikhin-Turaev TQFT invariant at le...

conjecture | open | MathDB 页面 | 原文线索

V2-032658 | MATHDB-352916

Global Selmer class conjecture for twisted Stark-Heegner cycles

摘录 / EXCERPT: Twisted Stark-Heegner cycle conjecture. There exists a global Selmer class

conjecture | open | MathDB 页面 | 原文线索

V2-032659 | MATHDB-352922

The maximally efficient surface braid conjecture

摘录 / EXCERPT: Let $\mathbf{G}$ be the countably infinite set of regular maps on the torus, where each $\mathcal{G}\in\mathbf{G}$ is an embedded graph in the torus that is arc-transitive. For eac...

conjecture | open | MathDB 页面 | 原文线索

V2-032660 | MATHDB-352930

Gonzalez-Acuña-Short Cabling Conjecture

摘录 / EXCERPT: Cabling Conjecture. If K is a knot in S^3 which has a reducible surgery, then K is a cabled knot and the reducing slope is given by the cabling annulus.

conjecture | open | MathDB 页面 | 原文线索

V2-032661 | MATHDB-352965

Topological mirror symmetry conjecture for Langlands dual Hitchin systems

摘录 / EXCERPT: Let C be a smooth projective curve, let ${\mathcal M}$ be the smooth quasi-projective $\SL_n$ Hitchin moduli space, and let ${\mathcal M}^\vee={\mathcal M}/\Gamma$ be the quasi-p...

conjecture | open | MathDB 页面 | 原文线索

V2-032662 | MATHDB-352968

Lower semi-continuity conjecture for generalized minimal log discrepancies

摘录 / EXCERPT: Let $(X,\Delta+M)/Z$ be a generalized pair. Define ... where $|X|_{\mathrm{cl}}$ denotes the set of all closed points of X with the Zariski topology. Lower semi-continuity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-032663 | MATHDB-352979

The invariant-ring sum conjecture for infinite-type Cartan matrices

摘录 / EXCERPT: Let A be an $n\times n$ Cartan matrix of infinite type. Let P be the ring of invariants associated with the Weyl group of A, and let P_i be the corresponding invariant subr...

conjecture | open | MathDB 页面 | 原文线索

V2-032664 | MATHDB-352981

The infinite-generation conjecture for rational cohomology of indefinite Kac-Moody groups

摘录 / EXCERPT: Let $A=(a_{ij})_{n\times n}$ be a Cartan matrix of indefinite type with $n\geq3$, and let $G(A)$ be the associated Kac-Moody group. Infinite-generation conjecture. The rational coh...

conjecture | open | MathDB 页面 | 原文线索

V2-032665 | MATHDB-352992

Indecomposability criterion for Calabi-Yau admissible subcategories

摘录 / EXCERPT: Indecomposability criterion. The category $\mathcal{B}$ is indecomposable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-032666 | MATHDB-352999

Infinitesimal rigidity conjecture for cyclic covers of finite-volume hyperbolic manifolds

摘录 / EXCERPT: Let M be a finite-volume hyperbolic manifold of dimension at least 4. Let $f\colon M\to S^1$ be a non-homotopically trivial smooth map such that $\ker(f_*)$ is finitely generat...

conjecture | open | MathDB 页面 | 原文线索

V2-032667 | MATHDB-353000

Algorithmic ribbonness conjecture for alternating knots

摘录 / EXCERPT: Algorithmic ribbonness conjecture. If an alternating knot admits a ribbon disk with three critical points, then it is algorithmically ribbon.

conjecture | open | MathDB 页面 | 原文线索

V2-032668 | MATHDB-353037

Functional relations under symplectic exchange for topological-recursion differentials

摘录 / EXCERPT: Functional-relations conjecture. For every $2g-2+n\geq 0$, these differentials satisfy the functional relations of Theorem, after those relations are expressed as relations between...

conjecture | open | MathDB 页面 | 原文线索

V2-032669 | MATHDB-353040

Regularity conjecture for big quantum cohomology of homogeneous spaces

摘录 / EXCERPT: Let G be a semisimple algebraic group and let $\mathrm{P}\subset \mathrm{G}$ be a parabolic subgroup. The ring $\operatorname{BQH}(\mathrm{G}/\mathrm{P})$ is the big qua...

conjecture | open | MathDB 页面 | 原文线索

V2-032670 | MATHDB-353060

Conjectural period relations for symmetric-power liftings

摘录 / EXCERPT: Let $\mathit\Pi$ be the cohomological cuspidal automorphic representation under consideration, with central character $\omega_\mathit\Pi$, Whittaker vector $f_\mathit\Pi$, period...

conjecture | open | MathDB 页面 | 原文线索

V2-032671 | MATHDB-353077

Family Floer superpotential critical-value conjecture

摘录 / EXCERPT: Let (X,ω) be a compact symplectic manifold that is Fano, or more generally has $-K_X$ nef, and let $(X^\vee,W^\vee)$ be its Landau-Ginzburg mirror, with $W^\vee$ the superpo...

conjecture | open | MathDB 页面 | 原文线索

V2-032672 | MATHDB-353083

Broué-Malle-Michel conjecture on Deligne-Lusztig cohomology

摘录 / EXCERPT: Broué-Malle-Michel conjecture. There exists such a parabolic subgroup $\mathbf P$ for which the cohomology modules in distinct degrees have no common irreducible constituent, and f...

conjecture | open | MathDB 页面 | 原文线索

V2-032673 | MATHDB-353100

The conjecture that every knot is isotopic to an unknot

摘录 / EXCERPT: Every-knot isotopy conjecture. Every knot is isotopic to an unknot.

conjecture | open | MathDB 页面 | 原文线索

V2-032674 | MATHDB-353107

The non-equivalence conjecture for spaces of locally convex curves

摘录 / EXCERPT: Let $n>3$ and let $q otin Z(\operatorname{Quat}_{n+1})$. Consider the space ${\cal L}_n(1;q)$ of locally convex curves with endpoint data $(1,q)$. If it contains convex curves, let...

conjecture | open | MathDB 页面 | 原文线索

V2-032675 | MATHDB-353116

The trivial-fibration characterization for nice spaces

摘录 / EXCERPT: Let M be the five-point space with two open and three closed points, let Λ be the three-point space with one open and two closed points, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032676 | MATHDB-353118

The finite-test-map factorization conjecture for topological spaces

摘录 / EXCERPT: Let P be a finite set of maps between finite topological spaces. For a string of letters l and r, let $(P)^{sl}$, $(P)^{slr}$, $(P)^{srl}$, and $(P)^{sr}$ denote the iterated...

conjecture | open | MathDB 页面 | 原文线索

V2-032677 | MATHDB-353130

The microbundle Thom-space conjecture for constrained polynomial spaces

摘录 / EXCERPT: Let Θ be a closed constraint, and let $\mathcal B_d^\Theta$ denote the corresponding space of degree-d polynomials satisfying the constraint. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032678 | MATHDB-353142

The paracompactness conjecture for singular cardinals

摘录 / EXCERPT: Let κ be a singular cardinal, and let ${}^{<\kappa}\square\,2^\kappa$ denote the corresponding middle box product space. The paracompactness conjecture for singular cardinal...

conjecture | open | MathDB 页面 | 原文线索

V2-032679 | MATHDB-353159

Petrie's conjecture on Pontryagin classes of homotopy complex projective spaces

摘录 / EXCERPT: Let M be a $2n$-dimensional compact oriented manifold homotopy equivalent to the complex projective space $4\mathbb{CP}^n4$. Suppose that M admits a non-trivial S^1-action. P...

conjecture | open | MathDB 页面 | 原文线索

V2-032680 | MATHDB-353186

Full p-adic coherence conjecture for local-model special fibers

摘录 / EXCERPT: Full p-adic coherence conjecture. This natural morphism is always an isomorphism. The assertion is already known under some hypotheses, while the paper identifies a small number of...

conjecture | open | MathDB 页面 | 原文线索

V2-032681 | MATHDB-353201

Jannsen's injectivity question for geometric p-adic representations

摘录 / EXCERPT: Let F be a global field, let S be the specified finite set of primes, and write $G_S(F)$ for the Galois group unramified outside S. Let V be a p-adic representation of...

open_problem | open | MathDB 页面 | 原文线索

V2-032682 | MATHDB-353221

Weak equivalence conjecture for polynomial and function spaces with prescribed patterns

摘录 / EXCERPT: Pattern-preserving weak equivalence conjecture. The embedding $\mathcal I$ is a weak homotopy equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-032683 | MATHDB-353272

The cohomological product formula for arithmetic coefficients on G-bundle stacks

摘录 / EXCERPT: Let X be a smooth proper curve over $\mathbb{F}_p$, let $G\to X$ satisfy the stated smoothness, affine, semisimplicity, simple-connectedness, and connected-fiber hypotheses, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032684 | MATHDB-353273

Finite-dimensionality of cotangent cohomology for G-bundle stacks

摘录 / EXCERPT: Finite-dimensionality conjecture. Its cohomology

conjecture | open | MathDB 页面 | 原文线索

V2-032685 | MATHDB-353274

The cohomological product formula for rigid Gross G-motives

摘录 / EXCERPT: Let X be a smooth proper curve over $\mathbb{F}_p$, let $G\to X$ satisfy the stated hypotheses, and let $\mathcal{E}\in\mathbb{D}_{\mathrm{hol}}(X)$ be an augmented commutative...

conjecture | open | MathDB 页面 | 原文线索

V2-032686 | MATHDB-353283

Hall's conjecture on Yang-Mills convergence for powers of simple loops

摘录 / EXCERPT: Let (M,a) be a closed, area-weighted topological map, embedded in a closed, connected, orientable surface, and let $G_N=\mathrm{U}(N)$. A loop is included in a disc if its drawin...

conjecture | open | MathDB 页面 | 原文线索

V2-032687 | MATHDB-353332

Isaksen-Wang-Xu's average stable homotopy growth conjecture

摘录 / EXCERPT: Let $\ell_2^{S}(i)$ denote the size function for the stable 2-local homotopy groups of spheres. Isaksen-Wang-Xu's conjecture. There exists a nonzero constant C such that ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-032688 | MATHDB-353385

Lutz-Nevo conjecture on flag PL-spheres with vanishing second gamma coefficient

摘录 / EXCERPT: Lutz-Nevo conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-032689 | MATHDB-353390

Davison's BPS cohomology conjecture for rank 2 degree 0 Higgs bundles

摘录 / EXCERPT: Davison's conjecture. There is an isomorphism of mixed Hodge structures

conjecture | open | MathDB 页面 | 原文线索

V2-032690 | MATHDB-353394

A bridge version of Gordon's conjecture

摘录 / EXCERPT: Bridge version of Gordon's conjecture. The connected sum of two unperturbed bridge positions is unperturbed.

conjecture | open | MathDB 页面 | 原文线索

V2-032691 | MATHDB-353402

The finite homological Lagrangian monodromy classification conjecture

摘录 / EXCERPT: Finite monodromy classification conjecture. Either $\mathcal{H}_L$ is isomorphic to a subgroup of

conjecture | open | MathDB 页面 | 原文线索

V2-032692 | MATHDB-353414

Ivanov's metaconjecture on automorphisms of rich surface objects

摘录 / EXCERPT: Let S be a surface, and let an object be naturally associated to S and have sufficiently rich structure. Ivanov's metaconjecture. Every such object has the extended mapping cla...

conjecture | open | MathDB 页面 | 原文线索

V2-032693 | MATHDB-353418

Contraction conjecture for shortest homology basis loops in planar metric graphs

摘录 / EXCERPT: Let X be a planar metric graph, meaning a geodesic space homeomorphic to a finite 1-dimensional simplicial complex that admits a planar embedding. Let G be a coefficient grou...

conjecture | open | MathDB 页面 | 原文线索

V2-032694 | MATHDB-353425

The forcing-axiom rigidity conjecture for C*-algebra coronas

摘录 / EXCERPT: Corona rigidity conjecture. CH implies that there are automorphisms of $\mathcal Q(A)$ that are not topologically trivial. Forcing Axioms imply that all isomorphisms between...

conjecture | open | MathDB 页面 | 原文线索

V2-032695 | MATHDB-353426

The CH conjecture on nontrivial autohomeomorphisms of Polish remainders

摘录 / EXCERPT: Remainder automorphism conjecture. The remainder $\beta X\setminus X$ has $2^{\mathfrak c}$ algebraically nontrivial autohomeomorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-032696 | MATHDB-353442

The periodic Floer homology conjecture for fibred knots

摘录 / EXCERPT: Let $K\subset Y$ be a fibered knot of genus g, with monodromy φ, and let T_φ denote its mapping torus. Write $\overline{Y}$ and $\overline{K}$ for the orientation...

conjecture | open | MathDB 页面 | 原文线索

V2-032697 | MATHDB-353472

The extension-composition conjecture for additive homogeneous superfunctors

摘录 / EXCERPT: Extension-composition conjecture. There is a graded isomorphism, taking total degree on the left-hand side,

conjecture | open | MathDB 页面 | 原文线索

V2-032698 | MATHDB-353478

Haefliger-Thurston conjecture on the connectivity of diffeomorphism classifying spaces

摘录 / EXCERPT: Haefliger-Thurston conjecture. The map η is a homology isomorphism in degrees at most $\operatorname{dim}(M)$ and is surjective on homology in degree $\operatorname{dim}(M)+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-032699 | MATHDB-353479

Contact Haefliger-Thurston conjecture

摘录 / EXCERPT: Contact Haefliger-Thurston conjecture. The natural map induces a homology isomorphism through degree n and a surjection on homology in degree $n+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-032700 | MATHDB-353491

Surjectivity conjecture for the field-difference homomorphism

摘录 / EXCERPT: Surjectivity conjecture. The homomorphism Σ is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-032701 | MATHDB-353492

Integral Rasmussen invariant conjecture for Whitehead doubles of odd torus knots

摘录 / EXCERPT: Whitehead-double torus-knot conjecture. If n is a positive integer and K is the mirror of

conjecture | open | MathDB 页面 | 原文线索

V2-032702 | MATHDB-353493

Integral Rasmussen invariant conjecture for Whitehead doubles of consecutive torus knots

摘录 / EXCERPT: Consecutive-torus-knot conjecture. For every integer $n\geq 2$, if

conjecture | open | MathDB 页面 | 原文线索

V2-032703 | MATHDB-353502

Terao and Yuzvinsky's algebraic logarithmic comparison conjecture for hyperplane arrangements

摘录 / EXCERPT: Let $f=f_1\cdots f_d\in R=\mathbb{C}[x_1,\dots,x_n]$ define a reduced hyperplane arrangement $\mathscr{A}$ in $X=\mathbb{C}^n$, and let $U=X\setminus\mathscr{A}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032704 | MATHDB-353538

Homology-dimension conjecture for Vietoris-Rips complexes of hypercube graphs

摘录 / EXCERPT: Let $\mathbb{I}_n$ denote the n-dimensional hypercube graph, and let $\mathcal{VR}(\mathbb{I}_n;r)$ be its Vietoris-Rips complex at scale r. For $r\in\{2,3\}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-032705 | MATHDB-353577

The odd-degree Jones coefficient nonvanishing conjecture modulo 8

摘录 / EXCERPT: Odd-degree Jones coefficient nonvanishing conjecture. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-032706 | MATHDB-353588

FDQC equivalence conjecture for commuting projector Hamiltonians

摘录 / EXCERPT: FDQC equivalence conjecture. The Hamiltonians H and H' are FDQC equivalent if and only if their surface topological orders in the presence of a gapped boundary are Witt equival...

conjecture | open | MathDB 页面 | 原文线索

V2-032707 | MATHDB-353591

Integral-coefficient conjecture for higher-dimensional Heegaard Floer homology

摘录 / EXCERPT: Let Σ be the surface appearing in the definition of the higher-dimensional Heegaard Floer algebra, let q_i be the chosen points, and let c denote the parameter in...

conjecture | open | MathDB 页面 | 原文线索

V2-032708 | MATHDB-353601

The optimist's conjecture on positive scalar curvature for non-spin universal covers

摘录 / EXCERPT: Optimist's conjecture. Then M carries a metric of positive scalar curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-032709 | MATHDB-353639

Closed-disk conjecture for the q-deformed Farey tessellation

摘录 / EXCERPT: Let L_q be the proper subset of $\mathbb{H}$ covered by the q-deformed Farey tessellation. Its boundary satisfies $\partial\overline{L}_q=\mathbb{Q}_q\sqcup\mathbb{I}_q$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-032710 | MATHDB-353673

Beilinson-Tate conjecture for the Spin motive of a cohomological representation

摘录 / EXCERPT: Let $\pi = \pi_\infty \otimes \pi_f$ be a cohomological cuspidal automorphic representation of $\operatorname{PGSp}_6(\mathbf{A})$. Let $M(\pi_f)$ be the Spin Chow motive with coef...

conjecture | open | MathDB 页面 | 原文线索

V2-032711 | MATHDB-353679

Anisotropy-refined Borde-Sorkin conjecture for Morse spacetimes

摘录 / EXCERPT: Let $(\mathcal{M},h,f,\zeta)$ be a Morse geometry of dimension n, and let (M,g) be the corresponding Morse spacetime. Assume that f has a single critical point for each criti...

conjecture | open | MathDB 页面 | 原文线索

V2-032712 | MATHDB-353699

Agreement of Čech and infinity-sheaf cohomology for diffeological spaces

摘录 / EXCERPT: Cohomology agreement conjecture. For every integer $k\geq 0$, there are natural isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-032713 | MATHDB-353722

Stable SPT classification group-structure conjecture

摘录 / EXCERPT: Stable SPT classification group-structure conjecture. The operation $*$ is well defined, independently of the representatives, and makes S an abelian group.

conjecture | open | MathDB 页面 | 原文线索

V2-032714 | MATHDB-353736

Properness conjecture for non-archimedean Calabi energy

摘录 / EXCERPT: Assume (X,L) is klt. For $1\le p<2$, equip $\mathcal{E}_{\mathrm{NA}}^2(X,L)$ with the d_p-topology. Then the normalized superlevel set ... should be compact in the d_p-topolog...

conjecture | open | MathDB 页面 | 原文线索

V2-032715 | MATHDB-353762

The converse characterization of squeezed knots by the invariant \ell

摘录 / EXCERPT: Let K be a knot. The invariant $\ell(K)$ is defined for knots, and a knot is called squeezed when it has the squeezing property discussed above. Converse characterization of sque...

conjecture | open | MathDB 页面 | 原文线索

V2-032716 | MATHDB-353781

MRS differentials conjecture

摘录 / EXCERPT: Let $y(h)$ be the spectrum under consideration, and let $h_{2h+i-j,j}$, $b_{h+i,j}$, and q_h denote the classes and coefficient appearing in its localized Adams spectral sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-032717 | MATHDB-353782

MRS collapse conjecture for localized Adams spectral sequences

摘录 / EXCERPT: Let $y(h)$ be the spectrum under consideration, and let $b_{h+i,j}$ denote the indicated classes in its localized Adams spectral sequence. MRS collapse conjecture. The classes...

conjecture | open | MathDB 页面 | 原文线索

V2-032718 | MATHDB-353783

Telescopic growth conjecture for finite type h spectra

摘录 / EXCERPT: Let X be a finite spectrum of type h, and let $\varpi_{\mathrm{T}(h)}(X,n)$ denote the size function measuring its telescopic homotopy groups through degree n. Telescopic gro...

conjecture | open | MathDB 页面 | 原文线索

V2-032719 | MATHDB-353785

Hopkins-Smith vanishing conjecture for the Adams-Novikov vanishing curve

摘录 / EXCERPT: Hopkins-Smith vanishing conjecture. The vanishing curve satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-032720 | MATHDB-353786

Moore's uniform torsion conjecture for finite simply-connected spaces

摘录 / EXCERPT: Let X be a simply-connected finite space. Say that X has a uniform p-torsion bound if there is an integer N such that the p-primary torsion in every homotopy group of X...

conjecture | open | MathDB 页面 | 原文线索

V2-032721 | MATHDB-353819

The tame topology characterization for the usual I^{1} space

摘录 / EXCERPT: Let (I^{1},τ^{1}) be the usual I^{1} space. For each $p>1$, let $i_{p}\colon I^{1}\to I^{p}$ be the inclusion map, and let $i_{w}\colon I^{1}\to I^{1}_{w}$ be the canonical...

conjecture | open | MathDB 页面 | 原文线索

V2-032722 | MATHDB-353847

Equality of the LS-category and cohomological dimension of a homomorphism

摘录 / EXCERPT: Let $\phi:\Gamma\to\pi$ be a homomorphism of discrete groups. The LS-category $\operatorname{cat}\phi$ is the least integer k such that a classifying-space map $B\Gamma\to B\pi$...

conjecture | open | MathDB 页面 | 原文线索

V2-032723 | MATHDB-353856

Cyclic-invariant restriction conjecture for the Hochschild pairing quasi-isomorphism

摘录 / EXCERPT: Let M be a closed oriented $C^$-manifold, let $\Omega(M)$ denote its de Rham algebra, and let $\Omega(M)^\vee[-\dim M]$ be the shifted dual complex. Write...

open_problem | open | MathDB 页面 | 原文线索

V2-032724 | MATHDB-353897

Positive topological entropy conjecture for lemon billiards

摘录 / EXCERPT: Let $\mathcal{Q}(b)$ be the intersection of two unit disks with center distance $b\in(0,2)$, and let ... be the billiard map on the lemon table $\mathcal{Q}(b)$. Lemon billiard entro...

conjecture | open | MathDB 页面 | 原文线索

V2-032725 | MATHDB-353911

The L_2-intersection cohomology concentration conjecture for analytic subsets of Kähler hyperbolic manifolds

摘录 / EXCERPT: Let Z be a singular compact complex space, let $\pi:\widetilde{Z}\to Z$ be its universal covering space, and write $\mathcal{IC}^{\bullet}_Z$ for the intersection cohomology shea...

conjecture | open | MathDB 页面 | 原文线索

V2-032726 | MATHDB-353912

The mixed Hodge structure conjecture for L_2 cohomology of Mixed Hodge Modules

摘录 / EXCERPT: Let X be a complex projective manifold, let $\mathbb{M}$ be a Mixed Hodge Module on X, and let $H^k_2(\widetilde{X},\mathbb{M})$ denote the associated L_2 cohomology, equippe...

conjecture | open | MathDB 页面 | 原文线索

V2-032727 | MATHDB-353944

Hollands-Ishibashi conjecture on the topology of spin domain-of-outer-communications

摘录 / EXCERPT: Hollands-Ishibashi conjecture. The Cauchy surface can be decomposed as

conjecture | open | MathDB 页面 | 原文线索

V2-032728 | MATHDB-353945

Spin torus-action conjecture for simply connected manifolds

摘录 / EXCERPT: Spin torus-action conjecture. The manifold M is homeomorphic to one of S^3, S^4, S^5, or

conjecture | open | MathDB 页面 | 原文线索

V2-032729 | MATHDB-353946

Refined Hollands-Ishibashi conjecture for spin Kaluza-Klein black holes

摘录 / EXCERPT: Refined Hollands-Ishibashi conjecture. There exists a choice of horizon fill-in and a cap for the asymptotic end such that the corresponding compactified domain of outer communicat...

conjecture | open | MathDB 页面 | 原文线索

V2-032730 | MATHDB-353947

Symmetry conjecture for optimal RN topology in cooperative source localization

摘录 / EXCERPT: Let the transmitter node (TN) be uniformly distributed in the environment of interest, and let $\{\mathbf p_1^*,\ldots,\mathbf p_K^*\}$ solve ... where $\hat{\mathbf p}_{\mathrm T}$...

conjecture | open | MathDB 页面 | 原文线索

V2-032731 | MATHDB-353961

The reduced Homological Vanishing Conjecture

摘录 / EXCERPT: Reduced Homological Vanishing Conjecture. For the parameters under consideration, this map is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032732 | MATHDB-353962

Fontaine's crystalline comparison conjecture

摘录 / EXCERPT: Let K be a complete discretely valued p-adic field of characteristic 0 with ring of integers $\mathcal{O}_K$ and perfect residue field k, and let X be a smooth, proper sc...

conjecture | open | MathDB 页面 | 原文线索

V2-032733 | MATHDB-353983

Topological expansion conjecture for BGW and HCIZ integrals

摘录 / EXCERPT: Topological expansion conjecture. There exists a positive constant ε such that, for each nonnegative integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-032734 | MATHDB-354002

Homotopy classification conjecture for spaces of two-dimensional gapped Hamiltonians

摘录 / EXCERPT: Let $\mathcal{S}$ be a connected component of the space of two-dimensional gapped Hamiltonians realizing a fixed topological order. Let invertible states be identified under equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-032735 | MATHDB-354005

Dion-Ray conjecture on vanishing Iwasawa invariants for 2-bridge links

摘录 / EXCERPT: Dion-Ray conjecture. The set $\mathcal{U}^p$ has density

conjecture | open | MathDB 页面 | 原文线索

V2-032736 | MATHDB-354136

Finite-generation conjecture for affine Springer fiber homology

摘录 / EXCERPT: For a regular semisimple $gamma$, define the graded module ... It is a module over the graded algebra $oplus_{d=0}^{cinfty}{}_0\mathcal A_d$ and therefore defines a quasi-coherent sh...

conjecture | open | MathDB 页面 | 原文线索

V2-032737 | MATHDB-354140

Vanishing conjecture for the fundamental cyclic class below the half-Hölder threshold

摘录 / EXCERPT: Let $C^\alpha(S^1)$ denote the Hölder algebra of exponent α, and let z be the coordinate function on S^1. The class $[z\otimes z^{-1}]$ is considered in reduced cyclic h...

conjecture | open | MathDB 页面 | 原文线索

V2-032738 | MATHDB-354141

Non-triviality conjecture for exotic cyclic and Hochschild classes on the two-torus

摘录 / EXCERPT: Non-triviality conjecture. The cyclic cohomology class $[\mathfrak{c}_\omega]$ and the associated Hochschild cohomology class $[\mathfrak{h}_\omega]$ are non-trivial. These classes...

conjecture | open | MathDB 页面 | 原文线索

V2-032739 | MATHDB-354172

Collins-Jiang-Yau conjecture for the deformed Hermitian-Yang-Mills equation

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold of complex dimension n, let $[\alpha]\in H^{1,1}(X,\mathbb R)$ be a real cohomology class, and let $\hat\theta$ be a supercritical p...

conjecture | open | MathDB 页面 | 原文线索

V2-032740 | MATHDB-354251

The Geroch conjecture with arbitrary ends

摘录 / EXCERPT: Let X be an n-manifold, and let T^n denote the n-torus. Write $T^n\sharp X$ for their connected sum. Geroch conjecture with arbitrary ends. There is no smooth complete metr...

conjecture | open | MathDB 页面 | 原文线索

V2-032741 | MATHDB-354256

Garoufalidis's Slope Conjecture for Jones slopes

摘录 / EXCERPT: Garoufalidis's Slope Conjecture. For any knot K in S^3, every Jones slope is a boundary slope:

conjecture | open | MathDB 页面 | 原文线索

V2-032742 | MATHDB-354257

Kalfagianni-Tran's Strong Slope Conjecture

摘录 / EXCERPT: Kalfagianni-Tran's Strong Slope Conjecture. For any knot in S^3, every Jones slope satisfies $SS(n)$ for some $n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-032743 | MATHDB-354258

The nonnegativity conjecture for Jones-slope coefficients

摘录 / EXCERPT: Coefficient nonnegativity conjecture. For any knot in S^3,

conjecture | open | MathDB 页面 | 原文线索

V2-032744 | MATHDB-354272

Tube-polymer link-statistics conjecture

摘录 / EXCERPT: Let $\mathbb T=\mathbb T_{M_1,M_2}$ be a lattice tube, and let L be a non-split link embeddable in $\mathbb T$. Let $p_{\mathbb T,n}(L)$ denote the number of n-edge embeddings...

conjecture | open | MathDB 页面 | 原文线索

V2-032745 | MATHDB-354284

Todorčević's consistency conjecture for regular spaces of countable spread

摘录 / EXCERPT: Todorčević's conjecture. It is relatively consistent with ZFC that if X is a regular space with no uncountable discrete subspace, then $\mathsf{OGA}(X)$ holds.

conjecture | open | MathDB 页面 | 原文线索

V2-032746 | MATHDB-354285

Todorčević's consistency conjecture for regular hereditary Lindelöf spaces

摘录 / EXCERPT: Todorčević's conjecture. It is relatively consistent with ZFC that if X is a regular HL space, then $\mathsf{OGA}(X)$ holds.

conjecture | open | MathDB 页面 | 原文线索

V2-032747 | MATHDB-354306

Harris's topological and analytic finiteness conjectures for special components

摘录 / EXCERPT: Fix an integer $d\geq 5$. Let NL_d be the Noether-Lefschetz locus, and let a component be special when its codimension is strictly less than the maximal value...

conjecture | open | MathDB 页面 | 原文线索

V2-032748 | MATHDB-354378

Homotopy-equivalence conjecture for rational maps in the periodic critical-point locus

摘录 / EXCERPT: Let d and n be positive integers, let $\mathscr M_d$ be the space of degree-d branched self-coverings of the sphere S^2, and let ... After giving S^2 its complex structure,...

conjecture | open | MathDB 页面 | 原文线索

V2-032749 | MATHDB-354400

Topological recursion for higher BGW tau-functions

摘录 / EXCERPT: Higher BGW topological-recursion conjecture. The generating functions of the higher BGW models $\tau^{(m)}$ are given by a version of the Chekhov--Eynard--Orantin topological recur...

conjecture | open | MathDB 页面 | 原文线索

V2-032750 | MATHDB-354408

Asymptotic average genus conjecture for rational knots and links

摘录 / EXCERPT: Asymptotic average genus conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032751 | MATHDB-354417

The harmonic-forms and intersection-cohomology conjecture for F-theory

摘录 / EXCERPT: Harmonic intersection-cohomology conjecture. The vector space of $\mathbb{V}$-valued i-forms that are harmonic and L^2 with respect to the Kähler metric is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-032752 | MATHDB-354443

Graphing conjecture for low-degree homotopy groups of long links

摘录 / EXCERPT: Let n, p, and m be integers with $m\geq 2$ and $1\leq p\leq n-3$. Let $\mathcal{L}_{m\cdot p}^{n}$ denote the space of long links with m components of dimension p in...

conjecture | open | MathDB 页面 | 原文线索

V2-032753 | MATHDB-354457

Homotopic exact controllability conjecture for the controlled wave maps equation

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be a Riemannian manifold. The controlled wave maps equation is a map $\phi:\mathbb{R}\times\mathbb{S}^1\to\mathcal{M}$, with states in...

conjecture | open | MathDB 页面 | 原文线索

V2-032754 | MATHDB-354464

The lifting conjecture for the Huneke-Wiegand property

摘录 / EXCERPT: Huneke-Wiegand lifting conjecture. If $R/(x)$ satisfies (HW), then R satisfies (HW).

conjecture | open | MathDB 页面 | 原文线索

V2-032755 | MATHDB-354465

The Huneke-Wiegand conjecture for Gorenstein quasi-fiber product rings

摘录 / EXCERPT: Huneke-Wiegand quasi-fiber-product conjecture. Gorenstein quasi-fiber product rings satisfy (HW).

conjecture | open | MathDB 页面 | 原文线索

V2-032756 | MATHDB-354466

Split-generation conjecture for the ungraded mirror of the real projective plane

摘录 / EXCERPT: Let $Q_{d53f\mathbb{R}P^2}$ denote the matrix factorization corresponding, under homological mirror symmetry, to the monotone Lagrangian $(\mathbb{R}P^2,b=0)$, and let ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-032757 | MATHDB-354476

Przytycki's freeness conjecture for non-Haken 3-manifolds

摘录 / EXCERPT: Let M be a closed, oriented, irreducible 3-manifold. A manifold is non-Haken when it contains no embedded oriented incompressible surfaces. Przytycki's conjecture. The followin...

conjecture | open | MathDB 页面 | 原文线索

V2-032758 | MATHDB-354484

Invariance of representations up to homotopy under Lie algebroid n-equivalence

摘录 / EXCERPT: Invariance conjecture. There is an equivalence of categories between representations up to homotopy of length at most n of A and of B.

conjecture | open | MathDB 页面 | 原文线索

V2-032759 | MATHDB-354485

Cohomological invariance under Lie algebroid n-equivalence

摘录 / EXCERPT: Cohomological invariance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-032760 | MATHDB-354486

The higher smooth homotopy hypothesis for Lie n-algebroids and Lie n-groupoids

摘录 / EXCERPT: Higher smooth homotopy hypothesis. Lie n-algebroids up to n-equivalence are equivalent to source n-connected Lie n-groupoids up to equivalence. In particular, generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-032761 | MATHDB-354487

The smooth homotopy hypothesis for L-infinity algebroids

摘录 / EXCERPT: Higher integration conjecture. In the category of L_∞-algebroids over Lie infinity-groupoids, $\mathfrak{g}$ is Morita equivalent to $\Pi_{\infty}(\mathfrak{g})$. More gener...

conjecture | open | MathDB 页面 | 原文线索

V2-032762 | MATHDB-354488

The van Est theorem for double Lie groupoids

摘录 / EXCERPT: Double-groupoid van Est conjecture. There is a van Est map from the cohomology of the double Lie groupoid to the cohomology of its LA-groupoid. If differentiation takes place in th...

conjecture | open | MathDB 页面 | 原文线索

V2-032763 | MATHDB-354489

Cohomological invariance under naturally isomorphic Lie groupoid morphisms

摘录 / EXCERPT: Natural-isomorphism invariance conjecture. The natural isomorphism induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032764 | MATHDB-354520

Torus conjecture for toric homotopy path algebras

摘录 / EXCERPT: Torus conjecture. The complex $X_{A_{\mathcal{X}}}$ is homotopic to a real torus $\mathbb{T}$.

conjecture | open | MathDB 页面 | 原文线索

V2-032765 | MATHDB-354521

Torus and rationality conjecture for homotopy path algebras

摘录 / EXCERPT: Torus and rationality conjecture. There exists a subcomplex $Y \subseteq X_A$ homotopic to $\mathbb{T}^n$. Furthermore, $\mathcal{X}$ is rational, with local systems on Y corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-032766 | MATHDB-354532

Farber-Kaufman-Schütz rationality conjecture for higher topological complexity

摘录 / EXCERPT: Farber-Kaufman-Schütz rationality conjecture. The function $\mathcal{F}_X(x)$ is rational with a single pole of order 2 at $x=1$; equivalently, there exists $P\in\mathbb{Z}[X]$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-032767 | MATHDB-354584

The pochette surgery conjecture for homotopy 4-spheres

摘录 / EXCERPT: Let e be an embedding of a pochette into S^4, let S_e denote its associated spun 2-knot, and let $S^4(e,p/q,\varepsilon)$ denote the result of pochette surgery with parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-032768 | MATHDB-354683

Relative Sen comparison conjecture for enhanced log Higgs modules

摘录 / EXCERPT: Let (H,θ_H,ϕ_H) be an arithmetic Higgs module over $R_C[\frac{1}{p}]$ arising from a representation $V\in\operatorname{Rep}_{\Gamma}(\widehat R_{\infty}[\frac{1}{p}])$. R...

conjecture | open | MathDB 页面 | 原文线索

V2-032769 | MATHDB-354694

Diagonal-coinvariant conjecture for the perverse-graded affine Springer-fiber cohomology

摘录 / EXCERPT: Let $G=GL_n$, let γ be the regular element used to define the affine Springer fiber $\Fl^{\gamma}_{\bI}$, let Λ be the lattice acting on it, and let P be the perve...

conjecture | open | MathDB 页面 | 原文线索

V2-032770 | MATHDB-354695

Springer-resolution Hochschild-center conjecture for affine Springer-fiber cohomology

摘录 / EXCERPT: Let $P\subset G$ be a parabolic subgroup, let $\widetilde{N}=T^*(G/P)$ be the corresponding Springer resolution, and let $T\widetilde{N}$ be its tangent bundle. The dilation action...

conjecture | open | MathDB 页面 | 原文线索

V2-032771 | MATHDB-354715

Wavelet determination of the persistence diagram of the phase space

摘录 / EXCERPT: Let C be the phase space for the universal quasiperiodic factorization, with $u=U\circ\Phi$ and $\Phi:S^1\to C$. An appropriate wavelet basis is used to decompose the signal u...

conjecture | open | MathDB 页面 | 原文线索

V2-032772 | MATHDB-354733

Strong Tube Theorem

摘录 / EXCERPT: Strong Tube Theorem. The image of the topological Abel--Jacobi homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032773 | MATHDB-354734

Jacobi-type theorem for the vanishing intermediate Jacobian

摘录 / EXCERPT: Jacobi-type theorem. There is some positive integer k such that the topological Abel--Jacobi mapping

conjecture | open | MathDB 页面 | 原文线索

V2-032774 | MATHDB-354747

Ben-Zvi's resolution conjecture for microlocal homology

摘录 / EXCERPT: Let X be a smooth space, let $\boldsymbol{Z}(f)$ be the derived zero locus associated to f, let $d4aa^f_{\Lambda}$ denote the microlocal homology object associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-032775 | MATHDB-354749

Ben-Zvi's microlocal homology invariance conjecture

摘录 / EXCERPT: Let X be a smooth ambient space, let f define the derived zero locus $\boldsymbol{Z}(f)$, and let $V^{\vee}$ be the ambient dual vector space. The source gives a closed immersi...

conjecture | open | MathDB 页面 | 原文线索

V2-032776 | MATHDB-354774

R-matrix reconstruction conjecture for the intersection of two quadrics

摘录 / EXCERPT: Let X be the complete intersection of two quadric hypersurfaces in $\mathbb{CP}^5$, and let Σ_2 be a genus 2 surface. Suppose an R-matrix relates the two splittings of t...

conjecture | open | MathDB 页面 | 原文线索

V2-032777 | MATHDB-354800

Acyclicity conjecture for components of line transversals

摘录 / EXCERPT: Acyclicity conjecture. Every connected component of the space of line transversals to F is acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-032778 | MATHDB-354808

Reverse stratification conjecture for de Rham prismatic crystals

摘录 / EXCERPT: Reverse stratification conjecture. The expression

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V2-032779 | MATHDB-354809

Formal identity conjecture for the stratification coefficients

摘录 / EXCERPT: Formal identity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-032780 | MATHDB-354825

The conjecture that every non-inane shape is a cube

摘录 / EXCERPT: Let $\Shafinnb$ be the category of shapes, and let ... be the functor sending n to $\cubeinc{n}$. Two objects of $\Shafinnb$ are understood to be equivalent when they are isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-032781 | MATHDB-354847

The integration conjecture for anomaly-free coefficient systems on closed stratified surfaces

摘录 / EXCERPT: Integration conjecture. For every such Σ and A, there is an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-032782 | MATHDB-354867

Degeneration and log mixed Hodge structure conjecture for logarithmic de Rham cohomology

摘录 / EXCERPT: Degeneration and LMH conjecture. (1) This spectral sequence degenerates at E_1, and all $R^qf_*(\omega^p_{X/S})$ are locally free of finite rank as $\cO_S$-modules; hence...

conjecture | open | MathDB 页面 | 原文线索

V2-032783 | MATHDB-354879

Decomposition-independence conjecture for vineyard matching

摘录 / EXCERPT: Let $S=\{\sigma_1,\dots,\sigma_N\}$ be a filtered simplicial complex, and let $\tau\in\mathfrak S_N$ be a permutation of its simplices. Write … For any sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-032784 | MATHDB-354884

Descent spectral sequence surjectivity conjecture for truncated Brown-Peterson spectra

摘录 / EXCERPT: Descent spectral sequence surjectivity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-032785 | MATHDB-354904

The polyhedron-bisecting curve conjecture

摘录 / EXCERPT: Polyhedron-bisecting curve conjecture. For each polyhedron P in $\mathbb{R}^3$ there exists a smooth simple closed curve $\mathcal{C}$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-032786 | MATHDB-354932

The winding-number lower-bound conjecture for satellite knots

摘录 / EXCERPT: Winding-number lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032787 | MATHDB-354945

The Kontsevich-Seidel folklore conjecture on quantum eigenvalues and mirror critical values

摘录 / EXCERPT: Kontsevich-Seidel folklore conjecture. The critical values of the mirror Landau-Ginzburg superpotential on $X^\vee$ are the eigenvalues of quantum multiplication by the first Chern...

conjecture | open | MathDB 页面 | 原文线索

V2-032788 | MATHDB-354946

Irie's strong closing conjecture for ellipsoid boundaries

摘录 / EXCERPT: Let $E \subset \mathbb{C}^n$ be an ellipsoid and let $(\partial E,\lambda|_{\partial E})$ be its contact boundary, where λ is the standard Liouville form. Irie's strong clo...

conjecture | open | MathDB 页面 | 原文线索

V2-032789 | MATHDB-354947

Periodic contact homology conjecture

摘录 / EXCERPT: Let (Y,ξ) be a closed contact manifold with a periodic contact form α. The closed Reeb orbits form components S of the orbit orbifolds, and to each component associate...

conjecture | open | MathDB 页面 | 原文线索

V2-032790 | MATHDB-354948

Periodic spectral-gap conjecture for contact homology

摘录 / EXCERPT: Let (Y,ξ) be a closed contact manifold with a periodic contact form α of period T. Let $CH(Y,\xi)$ denote contact homology, $\mathfrak{s}_\sigma(Y,\alpha)$ the spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-032791 | MATHDB-354949

Periodic Reeb flow and vanishing spectral gap equivalence conjecture

摘录 / EXCERPT: Let (Y,ξ) be a closed contact manifold with contact form α. Write $\mathfrak{gap}_\sigma(Y,\alpha)$ for the spectral gap associated with a class $\sigma\in CH(Y,\xi)$. P...

conjecture | open | MathDB 页面 | 原文线索

V2-032792 | MATHDB-354953

Integral graded Manin-invariant conjecture

摘录 / EXCERPT: Integral Manin-invariant conjecture. If $\operatorname{char}(k)=0$ and R is a $\mathbb{Z}[1/(3d+1)!]$-algebra, or if $\operatorname{char}(k)=p>0$ and R is a...

conjecture | open | MathDB 页面 | 原文线索

V2-032793 | MATHDB-354996

The structure of the poset of rational pointed cones

摘录 / EXCERPT: Let $\operatorname{\textsf{Cones}}(d)$ be the partially ordered set of pointed, rational, finitely generated cones in $\mathbb{R}^d$, ordered by successive addition of nonnegative...

conjecture | open | MathDB 页面 | 原文线索

V2-032794 | MATHDB-354998

Generalized coarse Bézout conjecture for persistent zero sets

摘录 / EXCERPT: Let M be the compact manifold, let D be the relevant geometric parameter, and let $\mathcal{F}_{\lambda_j}$ denote the corresponding eigenspaces. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032795 | MATHDB-355019

Existence of a knot with incompatible stick number and superbridge index realizations

摘录 / EXCERPT: Let K be a knot type. Its stick number is the minimum number of edges in a polygonal realization of K, denoted by $\operatorname{stick}[K]$, and its superbridge index is the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-032796 | MATHDB-355041

The homotopical 2-systole conjecture

摘录 / EXCERPT: Homotopical 2-systole conjecture. Under these hypotheses,

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V2-032797 | MATHDB-355049

Exteriority and collapse conjecture for the $RO(C_2)$-graded homology of K_V

摘录 / EXCERPT: Exteriory and collapse conjecture. The $RO(C_2)$-graded homology of K_V is exterior on generators given by the cycles on the E^2-page of the $RO(C_2)$-graded bar spectral seque...

conjecture | open | MathDB 页面 | 原文线索

V2-032798 | MATHDB-355055

The motivated de Rham-Betti conjecture

摘录 / EXCERPT: Let M be an André motive over K, and let Z_M be the Zariski closure of its comparison isomorphism, Ω_M the smallest K-subtorsor generated by that comparison,...

conjecture | open | MathDB 页面 | 原文线索

V2-032799 | MATHDB-355056

The de Rham-Betti conjecture

摘录 / EXCERPT: Let M be an André motive over K, or a homological motive over K. Write Ω_M^{dRB} for the torsor of de Rham-Betti periods, Ω_M^{And} for the to...

conjecture | open | MathDB 页面 | 原文线索

V2-032800 | MATHDB-355062

Künneth conjecture for Čech homology of product cosheaves

摘录 / EXCERPT: Let M be a manifold, let l be a positive integer, and let $k_1,\ldots,k_l$ be degrees. For a cover $\mathcal{U}$ of M^l, suppose that every intersection of elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-032801 | MATHDB-355088

The conjectural integral surgery triangle for rationally null-homologous knots

摘录 / EXCERPT: Let $\widehat{K}$ be a rationally null-homologous knot in $\widehat{Y}$, and let $\hat{\mu}$ and $\hat{\lambda}$ be the slope data used in the construction. Given $m\in\mathbb{Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-032802 | MATHDB-355089

The general integral surgery formula conjecture

摘录 / EXCERPT: Suppose $\hat{\mu}=q\mu+p\lambda$ with $q>0$, and suppose $\hat{\lambda}=q_0\mu+p_0\lambda$ is chosen as in the setup. For $m\in\mathbb{Z}$ such that $\hat{\lambda}-m\hat{\mu}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-032803 | MATHDB-355125

Bilinearised boundary model conjecture for exterior complexes

摘录 / EXCERPT: Let Λ be a sutured Legendrian with boundary components $\partial_0\Lambda$ and $\partial_1\Lambda$ in (Γ,λ_Γ). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032804 | MATHDB-355152

Log-prismatic extension conjecture after semistable alteration

摘录 / EXCERPT: Log-prismatic extension conjecture. There exists such a morphism

conjecture | open | MathDB 页面 | 原文线索

V2-032805 | MATHDB-355154

Preimage-of-a-cycle conjecture for PL maps in strong general position

摘录 / EXCERPT: Let c,n,d be non-negative integers such that $c<d$. Let C be a c-chain in $d53d^d$, let N be an n-dimensional closed PL manifold, and let $f:N\tod53d^d-\partial C$ be a P...

conjecture | open | MathDB 页面 | 原文线索

V2-032806 | MATHDB-355155

Simplicial-chain realization conjecture with prescribed general position

摘录 / EXCERPT: Let C be a c-chain in $\mathbb{R}^d$ with $c<d$, and let U be a finite set of points in $\mathbb{R}^d$ in strong general position with respect to C. Simplicial-chain realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-032807 | MATHDB-355187

Jacobsson's conjecture on the lower grading of Gottlieb elements

摘录 / EXCERPT: Let X be a formal space with bigraded Sullivan minimal model $(\land V,d)$, where ... and the differential satisfies $d(V_0)=0$ and $d(V_i)\subseteq (\land V)_{i-1}$ for $i\geq 1$....

conjecture | open | MathDB 页面 | 原文线索

V2-032808 | MATHDB-355188

Félix-Halperin-Thomas degree bound problem for rational Gottlieb elements

摘录 / EXCERPT: Let X be a space such that $H^*(X;\mathbb{Q})$ is finite-dimensional, and let ... The rational Gottlieb elements of X are the elements associated to the rational Gottlieb group o...

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V2-032809 | MATHDB-355196

Eventual linearity for regularity and projective dimension of invariant chains

摘录 / EXCERPT: Let $\Pi=\operatorname{Sym}$ or $\Pi=\operatorname{Inc}$, and let ${\mathcal I}=(I_n)_{n\ge 1}$ be a Π-invariant chain of proper homogeneous ideals, with I_n an ideal in...

conjecture | open | MathDB 页面 | 原文线索

V2-032810 | MATHDB-355229

The stable Albanese homology structure conjecture for IO-automorphism groups

摘录 / EXCERPT: Let F_n be a free group of rank n, let IO_n be the subgroup of automorphisms acting trivially on the integral homology modulo two, and let U^O_* be the grade...

conjecture | open | MathDB 页面 | 原文线索

V2-032811 | MATHDB-355230

The Kawazumi-Vespa stable cohomology conjecture for Aut(F_n)

摘录 / EXCERPT: For $p,q\ge0$, let $H^{p,q}=H^{\otimes p}\otimes(H^*)^{\otimes q}$, and let $\mathcal{H}$ and $\mathcal{E}$ be the wheeled PROPs defined from stable twisted cohomology and the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-032812 | MATHDB-355231

The Albanese cohomology comparison conjecture for Aut(F_n)

摘录 / EXCERPT: Let F_n be a free group, let $V_{\boldsymbol\lambda}$ be the algebraic $\operatorname{GL}(n,\mathbb{Q})$-representation associated to a bipartition $\boldsymbol\lambda$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032813 | MATHDB-355232

The stable quadraticity conjecture for Albanese cohomology of IA_n

摘录 / EXCERPT: Let U be the rational abelianization representation associated to IA_n, and let R_2 be the kernel of the map from $H^2(U,\mathbb{Q})$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-032814 | MATHDB-355233

The stable Albanese cohomology-graded Lie algebra conjecture for IA_n

摘录 / EXCERPT: Let $\operatorname{Gr}\operatorname{IA}_n$ be the graded Lie algebra associated to the Andreadakis filtration of IA_n. The stable Lie-algebra comparison conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032815 | MATHDB-355235

The Kawazumi-Morita invariant-cohomology conjecture for Torelli groups

摘录 / EXCERPT: Let $\mathcal{I}_g$ be the Torelli group of a closed genus-g surface, and let e_{2i} denote the Mumford-Morita-Miller class of degree $4i$. Kawazumi-Morita's conjecture. Stably...

conjecture | open | MathDB 页面 | 原文线索

V2-032816 | MATHDB-355241

Crossing-change invariance conjecture for lifts of tree-type webs

摘录 / EXCERPT: Let $\tilde{G}$ and $\tilde{G}'$ be tangled tree diagrams related by crossing changes, and let $G\in\mathsf{Tree}_{\mathfrak{sp}_4,\Sigma}$ be a tree-type elementary web. Crossing-...

conjecture | open | MathDB 页面 | 原文线索

V2-032817 | MATHDB-355245

Rank inequality for satellite knot Floer homology

摘录 / EXCERPT: Let P be a pattern knot in $S^1\times D^2$, let K be a companion knot in S^3, and let U denote the unknot in S^3. Satellite rank inequality. There is an inequality ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-032818 | MATHDB-355269

Nonexistence conjecture for generically universal Cantor sets

摘录 / EXCERPT: Let d be a positive integer. A set is generically universal if every generic set-meaning a dense G_{δ} set whose complement has Lebesgue measure zero-contains a linear and...

conjecture | open | MathDB 页面 | 原文线索

V2-032819 | MATHDB-355270

Nonexistence conjecture for uncountable topologically universal sets

摘录 / EXCERPT: Let d be a positive integer. A set is topologically universal if every dense G_{δ} subset of $\mathbb{R}^d$ contains a linear and translated copy of it. Topological nonexi...

conjecture | open | MathDB 页面 | 原文线索

V2-032820 | MATHDB-355291

Strong universality of persistent homology ell-values

摘录 / EXCERPT: Let $\mathbb{S}$ be a sampling model in the class $\mathcal{U}$. For a complex type $\mathcal{T}$ and homological degree $k\geq 1$, let $\mathcal{L}_n$ be the empirical measure of...

conjecture | open | MathDB 页面 | 原文线索

V2-032821 | MATHDB-355292

The left-skewed Gumbel conjecture for persistent homology

摘录 / EXCERPT: Left-skewed Gumbel conjecture. The universal limiting distribution is

conjecture | open | MathDB 页面 | 原文线索

V2-032822 | MATHDB-355318

Existence of class-0 non-affine knots with genus different from 1

摘录 / EXCERPT: In $\mathbb{R}P^3$, a knot is called class-0 if it has class 0, and a knot is non-affine if it is not affine. Its genus is the minimum genus of a good surface bounding it. Exis...

conjecture | open | MathDB 页面 | 原文线索

V2-032823 | MATHDB-355327

Infinite-dimensionality conjecture for Vafa-Witten homology of closed 3-manifolds

摘录 / EXCERPT: Let M_3 be a closed 3-manifold, and let $\mathcal{H}_{\mathrm{VW}}(M_3)$ denote its Vafa-Witten homology. Infinite-dimensionality conjecture. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-032824 | MATHDB-355328

No-instanton conjecture for Vafa-Witten theory on 3-manifolds

摘录 / EXCERPT: Let M_3 be a 3-manifold and let $\mathcal{H}_{\mathrm{VW}}(M_3)$ denote the Vafa-Witten homology associated with it. No-instanton conjecture. In Vafa-Witten theory on 3-manifolds...

conjecture | open | MathDB 页面 | 原文线索

V2-032825 | MATHDB-355329

Spectral-sequence conjecture for RG flows from 4d N=4 theory

摘录 / EXCERPT: Let a 4d $\mathcal{N}=2$ theory be reachable from 4d $\mathcal{N}=4$ super-Yang-Mills theory by an RG flow, and let M_3 be a 3-manifold. Spectral-sequence conjecture. The RG flow...

conjecture | open | MathDB 页面 | 原文线索

V2-032826 | MATHDB-355376

The strongly star-Scheepers game conjecture for closed-discrete and sigma-compact spaces

摘录 / EXCERPT: The strongly star-Scheepers game conjecture. If X is strongly star-Scheepers, then ONE does not have a winning strategy in the game...

conjecture | open | MathDB 页面 | 原文线索

V2-032827 | MATHDB-355397

Periodic-cyclic-homology formula for the stringy Euler characteristic

摘录 / EXCERPT: Stringy Euler-characteristic conjecture. The stringy Euler characteristic of X can be computed as the Euler characteristic of an NCCR, computed via periodic cyclic homology.

conjecture | open | MathDB 页面 | 原文线索

V2-032828 | MATHDB-355415

Qin's u-part Floer cohomology conjecture for lifted Lagrangian branes

摘录 / EXCERPT: Qin's u-part Floer cohomology conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032829 | MATHDB-355451

The ghost-space equality conjecture for $\Gamma_1(3,p)$

摘录 / EXCERPT: Ghost-space equality conjecture. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-032830 | MATHDB-355488

Topological p-adic Simpson correspondence for moduli spaces

摘录 / EXCERPT: Let K be algebraically closed, let X be a smooth proper rigid space over K, and let n be a positive integer. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032831 | MATHDB-355526

Cubic bound for rank-exact Betti numbers of two-parameter persistence modules

摘录 / EXCERPT: Cubic Betti-number bound. We have

conjecture | open | MathDB 页面 | 原文线索

V2-032832 | MATHDB-355536

Ramos's conjecture for the Grünbaum-Hadwiger-Ramos function

摘录 / EXCERPT: Let $\Delta(j,k)$ be the minimal dimension d of a Euclidean space $\mathbb{R}^d$ such that every collection of j masses in $\mathbb{R}^d$ admits a k-arrangement equiparting a...

conjecture | open | MathDB 页面 | 原文线索

V2-032833 | MATHDB-355541

H-cofibrancy conjecture for regular directed realizations of precubical sets

摘录 / EXCERPT: Let K be a precubical set, and let $|K|^t_{reg}$ denote its regular multipointed d-space realization. A multipointed d-space is called h-cofibrant when it is cofibrant in the...

conjecture | open | MathDB 页面 | 原文线索

V2-032834 | MATHDB-355546

Euler system existence and nonvanishing conjecture for rank-critical Panchishkin representations

摘录 / EXCERPT: Euler system existence and nonvanishing conjecture. If V is r-critical and satisfies the rank r Panchishkin condition for $r\geq 0$, there exists a collection of cohomology c...

conjecture | open | MathDB 页面 | 原文线索

V2-032835 | MATHDB-355548

Quasi-coherence conjecture for diagonal and Hochschild homology sheaves

摘录 / EXCERPT: Let $(X,\mathcal{C}^\infty)$ be a homologically regular reduced differentiable space. Let $\mathscr{E}_\bullet(\mathcal{C}^\infty_X)$ be its diagonal sheaf complex and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032836 | MATHDB-355552

Switch invariance conjecture for the HOMFLY polynomial

摘录 / EXCERPT: Suppose that a planar diagram of an oriented link L intersects the vertical line $x=0$ at $(0,-b)$, $(0,-a)$, $(0,a)$, and $(0,b)$, where $0<a<b$. Suppose that L is directed to...

conjecture | open | MathDB 页面 | 原文线索

V2-032837 | MATHDB-355571

Dimension prediction for the modification process

摘录 / EXCERPT: Let X be a 2-dimensional simplicial complex, let ${\mathcal{F}}$ be a sheaf on X, and use the notation of the modification process, including the subspaces E_r, E'_r,...

conjecture | open | MathDB 页面 | 原文线索

V2-032838 | MATHDB-355572

First-step cup-product dimension conjecture

摘录 / EXCERPT: Let X be a 2-dimensional simplicial complex, let ${\mathcal{F}}$ be a sheaf on X, and use the notation of the modification process. Set ... Assume $M\geq 0$. First-step cup-pro...

conjecture | open | MathDB 页面 | 原文线索

V2-032839 | MATHDB-355589

Minimum folded ribbonlength conjecture for Möbius-band unknots with linking number ±3

摘录 / EXCERPT: Let $\mathcal{U}_{w,F}$ be a folded ribbon unknot that is a Möbius band, with ribbon linking number ${\operatorname{Lk}}(\mathcal{U}_{w,F})=\pm3$. Let ${\operatorname{Rib}}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-032840 | MATHDB-355590

Annular folded ribbonlength conjecture for unknots with prescribed linking number

摘录 / EXCERPT: Let n be a nonnegative integer, and let $\mathcal{U}_{w,F}$ be a folded ribbon unknot that is a topological annulus, with ribbon linking number...

conjecture | open | MathDB 页面 | 原文线索

V2-032841 | MATHDB-355628

The dimension and cohomology conjecture for the maximal ideal-space closure of a Riemann surface

摘录 / EXCERPT: Let X be a connected Riemann surface such that the Banach algebra $H^\infty(X)$ of bounded holomorphic functions on X separates the points of X. Let ${\rm cl}\,X$ be the clos...

conjecture | open | MathDB 页面 | 原文线索

V2-032842 | MATHDB-355672

Diagonal Poincaré embedding conjecture for finitely dominated Poincaré spaces

摘录 / EXCERPT: Let M be a connected Poincaré duality space of dimension d satisfying Hypothesis: either M is homotopy finite, or M is finitely dominated and the Bass trace conjecture hold...

conjecture | open | MathDB 页面 | 原文线索

V2-032843 | MATHDB-355692

The affine partition-of-unity conjecture for simplicial polytopes and homology spheres

摘录 / EXCERPT: Affine partition-of-unity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032844 | MATHDB-355719

The equality of Lusternik-Schnirelmann category and cup-length for special orthogonal groups

摘录 / EXCERPT: For the special orthogonal group $\mathrm{SO}(n)$, let $\operatorname{cat}(\mathrm{SO}(n))$ denote its Lusternik-Schnirelmann category and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032845 | MATHDB-355766

Cappell and Shaneson's meridional rank conjecture

摘录 / EXCERPT: Let $L\subset S^3$ be a link. Its bridge number is the minimum number of local maxima in a bridge presentation, and its meridional rank is the smallest number of meridians needed t...

open_problem | open | MathDB 页面 | 原文线索

V2-032846 | MATHDB-355771

Canonical embedding conjecture for toroidal homology of periodic cell complexes

摘录 / EXCERPT: Let X_n be the periodic cell complex considered in the paper, let I^n_k denote the subgroup of k-cycles associated with the non-toroidal part of the homology filtration, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032847 | MATHDB-355794

Le Stum's conjecture on the p-adic Deligne-Kashiwara equivalence

摘录 / EXCERPT: Le Stum's conjecture. For any pair (X,Y) over k, the functor $\mathrm{sp}_!$ is an equivalence of categories.

conjecture | open | MathDB 页面 | 原文线索

V2-032848 | MATHDB-355815

The Bennequin-sharpness characterization of strongly quasipositive links

摘录 / EXCERPT: Strong quasipositivity conjecture. A link L is strongly quasipositive if and only if it is Bennequin-sharp.

conjecture | open | MathDB 页面 | 原文线索

V2-032849 | MATHDB-355842

The property F conjecture for braiding universality of anyons

摘录 / EXCERPT: Property F conjecture. An anyon X is braiding universal if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-032850 | MATHDB-355866

Geometric realization conjecture for path triangulations in digital images

摘录 / EXCERPT: A path triangulation is a triangulation obtained from paths in a planar cell complex, and a triangulated digital image is a digital image equipped with such a triangulation. Geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-032851 | MATHDB-355867

Free-group presentation conjecture for finitely presented path triangulations

摘录 / EXCERPT: A finitely presented path triangulation free group is a free group associated with a finitely presented path triangulation. A free group presentation is a presentation of a group b...

conjecture | open | MathDB 页面 | 原文线索

V2-032852 | MATHDB-355879

The tau-power conjecture for the rho-Bockstein spectral sequence

摘录 / EXCERPT: Consider the $\operatorname{\mathop{\rm Ext}}$ groups in the rho-Bockstein spectral sequence ... Here, the power of τ is the exponent of τ in the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-032853 | MATHDB-355882

Conjecture on local Rupertness of trapezohedra and reverse Rupertness of antiprisms

摘录 / EXCERPT: A polyhedron is locally Rupert if it admits a Rupert passage after an arbitrarily small alteration of a suitable orientation, and locally reverse Rupert is the analogous reverse-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-032854 | MATHDB-355883

Conjecture that local Rupertness is dual to reverse local Rupertness

摘录 / EXCERPT: Let Q be a polyhedron and let D be its dual. A polyhedron is locally Rupert if it has a local Rupert passage, and reverse locally Rupert if it has the corresponding reverse pas...

conjecture | open | MathDB 页面 | 原文线索

V2-032855 | MATHDB-355909

The PL-manifold structure conjecture for tame tangle manifolds

摘录 / EXCERPT: Tame tangle manifolds are PL manifolds. The canonical PL structures of open or compact tame m-tangles restrict to PL m-manifold structures on tame tangle manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-032856 | MATHDB-355910

The smooth faithfulness conjecture for tame tangles

摘录 / EXCERPT: Framed homeomorphism of tame smooth tangles implies diffeomorphism. If two tame smooth tangles $W\to\mathbb{I}^n$ and $W'\to\mathbb{I}^n$ are framed stratified homeomorphic, then...

conjecture | open | MathDB 页面 | 原文线索

V2-032857 | MATHDB-355911

The smooth framed-isotopy faithfulness conjecture for tame tangles

摘录 / EXCERPT: Smooth framed isotopy of tame smooth tangles implies diffeomorphism. Given smooth m-tame tangles $f:W\to\mathbb{I}^n$ and $f':W'\to\mathbb{I}^n$, if f and f' are smoothly fra...

conjecture | open | MathDB 页面 | 原文线索

V2-032858 | MATHDB-355937

Mirror-duality conjecture for the two second Steenrod squares

摘录 / EXCERPT: Mirror-duality conjecture. The diagram

conjecture | open | MathDB 页面 | 原文线索

V2-032859 | MATHDB-355945

Full support conjecture for the degree-zero Hitchin system

摘录 / EXCERPT: Let $M(n,0)$ be the degree-zero moduli space in the Hitchin fibration, with Hitchin map $\chi(n,0)$ to its basis. The complex $R\chi(n,0)_*\operatorname{IC}(M(n,0),\mathbb{Q})$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-032860 | MATHDB-355967

Cartwright-Strumfels local-cohomology invariance conjecture

摘录 / EXCERPT: Let A be a polynomial ring, let $I\subseteq A$ be a $\mathbb{Z}^m$-graded Cartwright-Strumfels ideal, let $\operatorname{gin}(I)$ be its $\mathbb{Z}^m$-graded generic initial ide...

conjecture | open | MathDB 页面 | 原文线索

V2-032861 | MATHDB-355982

Flag gamma-positivity conjecture for theta polynomials of homology balls

摘录 / EXCERPT: Let Δ be a flag homology ball, and suppose that its boundary $\partial\Delta$ is an induced subcomplex of Δ. Let $\theta(\Delta,x)$ denote the polynomial introduced f...

conjecture | open | MathDB 页面 | 原文线索

V2-032862 | MATHDB-355986

Cohen-Dimca-Orlik type conjecture for integer local systems on complex arrangements

摘录 / EXCERPT: Cohen-Dimca-Orlik type conjecture. The cohomology groups satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-032863 | MATHDB-355996

Deng-Liu-Ngai conjecture on ball-like self-affine tiles

摘录 / EXCERPT: Deng-Liu-Ngai conjecture. Under these hypotheses, the self-affine tile T is homeomorphic to a d-dimensional ball if and only if its interior T° is connected. The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-032864 | MATHDB-355997

Generalized tame-ball conjecture for three-dimensional self-affine tiles

摘录 / EXCERPT: Let p,q,r be integers with absolute value at least 2, let s,t,a_i,b_j be real numbers, and define ... and ... Here a_∞ and b_∞ denote the limiting quantities used i...

conjecture | open | MathDB 页面 | 原文线索

V2-032865 | MATHDB-356060

The coNP conjecture for 3-sphere and lens space recognition

摘录 / EXCERPT: Recognition conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032866 | MATHDB-356066

Conjecture on the lower bound for maximal persistence in random clique complexes

摘录 / EXCERPT: Let $X(n,p)$ be the random clique complex, let $M_k(n)$ denote the maximal persistence over all k-dimensional cycles in $X(n,p)$, let $f_k(n)=n^{1/k(k+1)}(\log n)^{1/(k+1)}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-032867 | MATHDB-356067

Conjecture on the limiting distribution of maximal persistence in random clique complexes

摘录 / EXCERPT: Limiting-distribution conjecture. The random variable

conjecture | open | MathDB 页面 | 原文线索

V2-032868 | MATHDB-356076

The folklore conjecture on stable Torelli-group homology

摘录 / EXCERPT: Folklore conjecture. For each k, there exists some G_k such that for $g \geq G_k$, the homology group $\operatorname{H}_k(\operatorname{\mathcal{I}}_g;\mathbb{Q})$ is finite-di...

conjecture | open | MathDB 页面 | 原文线索

V2-032869 | MATHDB-356113

The universal rank-expansion conjecture for winding-zero satellite operators

摘录 / EXCERPT: Let $\mathcal{C}$ be the smooth knot concordance group. A satellite operator P is rank-expanding along $\{nK\}_{n\in\mathbb{Z}}$ if $\{nK\}_{n\in\mathbb{Z}}$ is a rank-one subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-032870 | MATHDB-356115

The Gap Conjecture for orientation-reversing involutions of the 3-sphere

摘录 / EXCERPT: Gap Conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-032871 | MATHDB-356116

The High-dimensional Bing Conjecture

摘录 / EXCERPT: High-dimensional Bing Conjecture. The theorem and scholium hold after replacing S^3 by S^n, $n\geq4$, and replacing the Bing involution by its spun counterparts.

conjecture | open | MathDB 页面 | 原文线索

V2-032872 | MATHDB-356154

Fock-Goncharov positivity conjecture for surface skein algebras

摘录 / EXCERPT: Let $\mathord{\mathbb{S}}$ be an orientable surface, and let the multicurve basis be the basis of the surface skein algebra given by multicurves. Applying Chebyshev polynomials of...

conjecture | open | MathDB 页面 | 原文线索

V2-032873 | MATHDB-356172

Invariant-cohomology conjecture for non-hyperelliptic curves

摘录 / EXCERPT: For a smooth non-hyperelliptic curve X, write $H^1(X)$ for its first cohomology group and let $\operatorname{Aut}(X)$ act diagonally on $H^1(X)^{\otimes 3}$. Invariant-cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-032874 | MATHDB-356187

Connected sum formula for bent complexes in instanton Floer homology

摘录 / EXCERPT: Let $K_i\subset Y_i$ for $i=1,2$ be rationally null-homologous knots. Write $K_1\# K_2$ for their connected sum, and let $B^\pm(K)$ denote the bent complexes associated to a knot...

conjecture | open | MathDB 页面 | 原文线索

V2-032875 | MATHDB-356241

Morse-Novikov genus conjecture for oriented links

摘录 / EXCERPT: Morse-Novikov genus conjecture. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-032876 | MATHDB-356248

The volume conjecture for hyperbolic polyhedra

摘录 / EXCERPT: Let P be a hyperbolic polyhedron with dihedral angles $\alpha_1,\dots,\alpha_m$ at edges $e_1,\dots,e_m$, and let Γ be its 1-skeleton. Let col_r be a sequence of r-a...

conjecture | open | MathDB 页面 | 原文线索

V2-032877 | MATHDB-356249

The volume conjecture for proper polyhedra

摘录 / EXCERPT: The volume conjecture for polyhedra. Then

conjecture | open | MathDB 页面 | 原文线索

V2-032878 | MATHDB-356271

Ishibashi-Sugawara-Yoshinaga torsion conjecture for arrangement double covers

摘录 / EXCERPT: Let $\mathcal{A}$ be a complex hyperplane arrangement in $\mathbb{C}^n$, and let $M(\mathcal{A})$ be its complement. For a nonzero $\omega\in H^1(M(\mathcal{A}),\mathbb{Z}_2)$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-032879 | MATHDB-356276

Strict proof-theoretic strength of the HIT real axiom

摘录 / EXCERPT: Let $\mathbf{MLTT}^-$ be Martin-Löf type theory without W-types, let $\mathbf{UA}$ denote the univalence axiom, and let $\mathbb{R}_{\mathbf{HIT}}$ denote the axiom that the high...

conjecture | open | MathDB 页面 | 原文线索

V2-032880 | MATHDB-356305

Hurewicz fibration conjecture for randomized simplicial sets

摘录 / EXCERPT: Hurewicz fibration conjecture. The maps

conjecture | open | MathDB 页面 | 原文线索

V2-032881 | MATHDB-356313

The qAGH knot invariance conjecture

摘录 / EXCERPT: qAGH knot invariance conjecture. If $\Bbbk$ is a field of characteristic 0, then ${\mathit{qAGH}}$ is a knot invariant for any q.

conjecture | open | MathDB 页面 | 原文线索

V2-032882 | MATHDB-356325

Whitehead's subcomplex asphericity conjecture

摘录 / EXCERPT: Let X be an aspherical 2-complex, and let Y be a subcomplex of X. Whitehead's conjecture. The subcomplex Y is aspherical. This is a famous conjecture about asphericity inhe...

conjecture | open | MathDB 页面 | 原文线索

V2-032883 | MATHDB-356341

The cord algebra conjecture for tangle contact homology

摘录 / EXCERPT: Let T be a framed oriented r-component tangle in ${\mathbb{R}}^3_{x\geq 0}$, let $N(\varLambda_T)$ be its specified neighborhood, and let H denote the reference subspace used...

conjecture | open | MathDB 页面 | 原文线索

V2-032884 | MATHDB-356344

Auroux's mirror correspondence conjecture for restriction, wrapping, and lifting functors

摘录 / EXCERPT: Auroux's mirror correspondence conjecture. Under homological mirror symmetry, ρ corresponds to the composite of q with the Knörrer periodicity equivalence; α_0 corres...

conjecture | open | MathDB 页面 | 原文线索

V2-032885 | MATHDB-356348

Totaro's cycle-map conjecture for Brown-Peterson cohomology of classifying spaces

摘录 / EXCERPT: Let G be a complex linear algebraic group, let $CH^i(BG)_{(p)}$ be the p-localized Chow group of its classifying space, and let $BP^*(BG)$ denote its Brown-Peterson cohomology....

conjecture | open | MathDB 页面 | 原文线索

V2-032886 | MATHDB-356395

Bobadilla-Kollár's homology-fiber-bundle smoothness conjecture

摘录 / EXCERPT: Bobadilla-Kollár's conjecture. If f is a $\mathbb Z$-homology fiber bundle, then f is smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-032887 | MATHDB-356417

Unbounded handle complexity conjecture for connected sums of the Poincaré sphere

摘录 / EXCERPT: Let $Y_n=\#_n\Sigma(2,3,5)$, where $\Sigma(2,3,5)$ is the Poincaré homology sphere. A bounding four-manifold for $Y_n\#-Y_n$ is a four-manifold whose boundary is $Y_n\#-Y_n$. Unbou...

conjecture | open | MathDB 页面 | 原文线索

V2-032888 | MATHDB-356440

Flexibility conjecture for the entropy of (a,b)-continued fraction transformations

摘录 / EXCERPT: Let ${\mathcal P}$ be the parameter space of pairs (a,b), let $f_{a,b}$ denote the corresponding (a,b)-continued fraction transformation, and let $h_{\rm top}(f_{a,b})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-032889 | MATHDB-356459

Allen's conjecture on sums of torus knots concordant to L-space knots

摘录 / EXCERPT: Allen's conjecture. If a connected sum of possibly several torus knots is concordant to an L-space knot, then it is concordant to a positive torus knot.

conjecture | open | MathDB 页面 | 原文线索

V2-032890 | MATHDB-356460

Brezis's refined topological-degree estimate for circle maps

摘录 / EXCERPT: Brezis's conjecture. There exist constants $C>0$ and $\delta_0>0$ such that the displayed inequality holds for every continuous $f\in C^0({\mathbb S}^1;{\mathbb S}^1)$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-032891 | MATHDB-356461

Equivariant model conjecture for finite regular covers

摘录 / EXCERPT: Equivariant model conjecture. The fixed subalgebra A^{Φ} is a model for X.

conjecture | open | MathDB 页面 | 原文线索

V2-032892 | MATHDB-356506

Non-positive curvature for geometric complexes

摘录 / EXCERPT: Non-positive curvature for geometric complexes. If $\sigma_+(X)>0$ then either $\partial M(X)$ has a spherical component or $\pi_1(X)$ splits freely.

conjecture | open | MathDB 页面 | 原文线索

V2-032893 | MATHDB-356507

Negative curvature for geometric complexes

摘录 / EXCERPT: Negative curvature for geometric complexes. If $\sigma_+(X)=0$ then either $\partial M(X)$ has a toroidal boundary component or $\pi_1(X)$ has a $\mathbb{Z}^2$ subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-032894 | MATHDB-356520

Ault-Fiedorowicz conjecture on the symmetric homology of polynomial algebras

摘录 / EXCERPT: Ault-Fiedorowicz conjecture. For every $n\geq 1$, there is an isomorphism of graded algebras

conjecture | open | MathDB 页面 | 原文线索

V2-032895 | MATHDB-356521

Ault-Fiedorowicz conjecture on the topological formula for symmetric homology

摘录 / EXCERPT: Ault-Fiedorowicz conjecture. For every k-algebra A, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032896 | MATHDB-356577

Removal of the technical condition for orientability of modules over local systems

摘录 / EXCERPT: Let E_n denote the height-n Morava E-theory spectrum, and let $\rho_{ ull}$ denote the ρ-localization functor. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032897 | MATHDB-356581

Two-bridge knot transvergent diagram conjecture

摘录 / EXCERPT: Two-bridge knot transvergent diagram conjecture. A two-bridge knot K has a transvergent diagram with $c(K)$ crossings if and only if one of these two conditions is fulfilled.

conjecture | open | MathDB 页面 | 原文线索

V2-032898 | MATHDB-356638

Uniqueness conjecture for derivations on the Adams E_2 page of the sphere

摘录 / EXCERPT: Let E_2 denote the Adams E_2 page for the sphere, viewed with its algebraic structure, and let its differentials be regarded as derivations. Uniqueness conjecture. The space of...

conjecture | open | MathDB 页面 | 原文线索

V2-032899 | MATHDB-356652

Merkulov-Willwacher's injectivity conjecture for oriented graph cohomology

摘录 / EXCERPT: Let d be an integer, and let $\textsf{OGC}_{2d+1}^{{\:\raisebox{3pt}{\text{\circle*{1.5}}}}}$ be the oriented graph complex and...

conjecture | open | MathDB 页面 | 原文线索

V2-032900 | MATHDB-356656

Isotopy invariance of collapsed generalized reduced Khovanov homology in $S^2 \times S^1$

摘录 / EXCERPT: Let T be an oriented 1-tangle diagram in the annulus, and let L_{T^0} be the link in $S^2 \times S^1$ obtained by closing it with an overpass or underpass arc. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032901 | MATHDB-356703

The integral isomorphism conjecture for bounded cohomology and uniformly finite homology

摘录 / EXCERPT: Let M be a complete Riemannian manifold of dimension n with a properly discontinuous, cocompact, isometric action by a group G, and let ... be the isomorphism obtained from the...

conjecture | open | MathDB 页面 | 原文线索

V2-032902 | MATHDB-356714

The 13-stick classification conjecture for the simple hexagonal lattice

摘录 / EXCERPT: Let $[K]$ be a knot type, let $s_{sh}[K]$ denote its stick number in the simple hexagonal lattice, and let 3_1, 4_1, 5_1, and 5_2 denote the indicated knot types. 13-stick...

conjecture | open | MathDB 页面 | 原文线索

V2-032903 | MATHDB-356715

The crossing-number bound for 14-stick knots in the simple hexagonal lattice

摘录 / EXCERPT: Let $[K]$ be a knot type, let $s_{sh}[K]$ denote its stick number in the simple hexagonal lattice, and let $c[K]$ denote its crossing number. Crossing-number bound conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-032904 | MATHDB-356740

The third Brumer-Stark formula

摘录 / EXCERPT: Let $u_3\in F_{\mathfrak p}^*\otimes\mathbb Z[G]$ be the element defined by the cap product ... Let $u_{\mathfrak p}$ denote the Brumer-Stark element. Third Brumer-Stark formula. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-032905 | MATHDB-356754

The curved filtered A_{∞}-algebra conjecture for Lagrangian cohomology

摘录 / EXCERPT: Suppose (X,ω) and $\mathcal D$ are as in the beginning of the introduction, and let $L\subset X\setminus\mathcal D$ be a relatively spin compact Lagrangian submanifold. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-032906 | MATHDB-356767

Stroppel's intrinsic formality conjecture for extended Khovanov arc algebras

摘录 / EXCERPT: Let ${\mathrm{K}}_m^n$ be the extended Khovanov arc algebra indexed by m and n, and let $\operatorname{HH}^2_{i-2}({\mathrm{K}}_m^n,${\mathrm{K}}m^n)...

conjecture | open | MathDB 页面 | 原文线索

V2-032907 | MATHDB-356781

Multiplicity-one conjecture for smooth fixed-topology min-max surfaces

摘录 / EXCERPT: Let g be a Riemannian metric and let a min-max sequence from a sweepout of genus g surfaces converge as a varifold to ... where the pairwise disjoint, smooth, closed, embedded mi...

conjecture | open | MathDB 页面 | 原文线索

V2-032908 | MATHDB-356801

Polynomial-growth monodromy conjecture for first homology torsion growth

摘录 / EXCERPT: Let F_m be a finitely generated free group, let $varphi$ be an automorphism, and set ... Assume that $varphi$ is polynomially growing of degree d. For a Farber sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-032909 | MATHDB-356861

The $P=C$ conjecture for the free cohomology of local $\mathbb{P}^2$ moduli spaces

摘录 / EXCERPT: Let $M_{d,\chi}$ be the moduli space considered in the paper, with perverse filtration P_k on its cohomology. For $d\geq 3$, let $H^{*\leq 2d-4}(M_{d,\chi},\mathbb{Q})$ be the fr...

conjecture | open | MathDB 页面 | 原文线索

V2-032910 | MATHDB-356867

Lagrangian-cylinder generation conjecture for the wrapped Fukaya category

摘录 / EXCERPT: Let V be the Liouville sector and let $\mathcal L_\gamma$ and $\mathbb{T}_\gamma$ be the Lagrangian cylinder and the torus obtained from it by wrapping and resolving the resultin...

conjecture | open | MathDB 页面 | 原文线索

V2-032911 | MATHDB-356868

Novikov-coefficient open-closed and homological-smoothness conjecture

摘录 / EXCERPT: Assume that $V=[-1,1]\times M$ is a torus bundle domain. Let $\mathcal{W}_0'(V)$ be the wrapped Fukaya category with coefficients in the Novikov ring $\Lambda_\mathbb{Z}$, containi...

conjecture | open | MathDB 页面 | 原文线索

V2-032912 | MATHDB-356870

Closed-string polyvector field conjecture for symplectic cluster manifolds

摘录 / EXCERPT: Let $M_\mathcal{R}$ be the symplectic cluster manifold, let $\mathcal{Y}_\mathcal{R}$ be its mirror, and suppose that all node multiplicities in $\mathcal{R}$ are 1. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032913 | MATHDB-356872

Arnold-type cuplength conjecture for Lagrangian intersections

摘录 / EXCERPT: Let (X,ω) be a symplectic manifold, and let L and L' be Lagrangian submanifolds of X. Assume that X is either closed or a Liouville manifold, and that L is connect...

conjecture | open | MathDB 页面 | 原文线索

V2-032914 | MATHDB-356888

Finite-dimensionality conjecture for finite-codimension subalgebras of algebraic vector fields

摘录 / EXCERPT: Let X be an affine algebraic manifold, let ${\mathcal T}_X$ be its Lie algebra of algebraic vector fields, and let $\mathfrak h\subseteq {\mathcal T}_X$ be a Lie subalgebra of fi...

conjecture | open | MathDB 页面 | 原文线索

V2-032915 | MATHDB-356968

K-theoretic DT/PT correspondence conjecture for local surfaces

摘录 / EXCERPT: K-theoretic computation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032916 | MATHDB-356978

Sharp persistence-exponent conjecture for Hölder functions

摘录 / EXCERPT: Let X) be a compact Riemannian manifold of dimension d, and let $f\in C^{r+\alpha}(X)$, with the persistence quantity $\Pers_p(H_k(X,f))$ defined as the p-persistence of the...

conjecture | open | MathDB 页面 | 原文线索

V2-032917 | MATHDB-356996

The stable Albanese decomposition conjecture for IA-automorphism groups

摘录 / EXCERPT: Stable Albanese decomposition conjecture. The graded algebra map ω_n is a stable isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-032918 | MATHDB-356997

The polynomial invariant-cohomology conjecture for IA-automorphism groups

摘录 / EXCERPT: Polynomial invariant-cohomology conjecture. For every $i\ge0$, this invariant cohomology stabilizes: there is $N\ge0$ such that for all $n\ge N$, inclusion induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-032919 | MATHDB-357008

Conjecture on the equivalence induced by the functor D_X

摘录 / EXCERPT: Equivalence conjecture. The functors D_X and $D_X^{\operatorname{perf}}$ each define an equivalence of categories. This conjecture predicts that the comparison between the overco...

conjecture | open | MathDB 页面 | 原文线索

V2-032920 | MATHDB-357082

The braid-index-bridge-index conjecture for L-space knots

摘录 / EXCERPT: Let K be an L-space knot, meaning a knot admitting a Dehn surgery yielding an L-space. Let $b(K)$ denote its bridge index and $\beta(K)$ its braid index. The L-space knot index c...

conjecture | open | MathDB 页面 | 原文线索

V2-032921 | MATHDB-357083

The Alexander-polynomial conjecture for hyperbolic L-space knots

摘录 / EXCERPT: Let K be a hyperbolic L-space knot, let g be its Seifert genus, and let n be its braid index. Let $Delta_K(t)$ denote its Alexander polynomial. The hyperbolic L-space knot Al...

conjecture | open | MathDB 页面 | 原文线索

V2-032922 | MATHDB-357084

The braid-index-bridge-index conjecture for positive braid knots

摘录 / EXCERPT: Let K be a positive braid knot, meaning a knot represented as the closure of a braid whose crossings are all positive. Let $b(K)$ denote its bridge index and $\beta(K)$ its braid...

conjecture | open | MathDB 页面 | 原文线索

V2-032923 | MATHDB-357097

The universal formula conjecture for the x-y swap in topological recursion

摘录 / EXCERPT: Let $g\geq 0$ and $n\geq 1$. Let $\omega^{(g)}_{0,n}$ be the n-differentials associated with the original variables, let $\omega^{(h)}_{m,0}$ denote the corresponding differentia...

conjecture | open | MathDB 页面 | 原文线索

V2-032924 | MATHDB-357136

The spiral conjecture for relation-avoiding paths

摘录 / EXCERPT: The spiral conjecture. Every relation-avoiding origin path p is weakly homotopic to a relation-avoiding origin path q for which there exist linearly independent vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-032925 | MATHDB-357137

The monotonic spiral conjecture for relation-avoiding paths

摘录 / EXCERPT: Monotonic spiral conjecture. There exists a continuous function $h:[0,1]\times[0,\infty)\to V$ satisfying $\lim_{t\to\infty}h(s,t)=0$ for all s, $h(0,t)=p(t)$, and, for linearly...

conjecture | open | MathDB 页面 | 原文线索

V2-032926 | MATHDB-357159

Conservativity conjecture for realization functors of motives

摘录 / EXCERPT: Let k be a base field and consider the diagram of categories of motives and realization functors described in the paper, with rational coefficients. A morphism of motives is test...

conjecture | open | MathDB 页面 | 原文线索

V2-032927 | MATHDB-357175

The ball number bound for nontrivial nonsplittable links

摘录 / EXCERPT: Ball number conjecture. For any nontrivial and nonsplittable link L,

conjecture | open | MathDB 页面 | 原文线索

V2-032928 | MATHDB-357208

Polynomial-generator conjecture for the cohomology of $K_{1+k(\sigma-1)}$

摘录 / EXCERPT: Polynomial-generator conjecture. $H^{\star}(K_{1+k(\sigma-1)})$ is a polynomial algebra on $k+1$ generators with k relations; the square of each of the first k generators is a...

conjecture | open | MathDB 页面 | 原文线索

V2-032929 | MATHDB-357209

Steenrod-operation polynomiality conjecture for $H^{\star}(K_{r\sigma+s})$

摘录 / EXCERPT: Steenrod-operation polynomiality conjecture. $H^{\star}(K_{r\sigma+s})$ is a polynomial algebra on certain classes

conjecture | open | MathDB 页面 | 原文线索

V2-032930 | MATHDB-357222

Classical blue-shift phenomenon

摘录 / EXCERPT: Classical blue-shift phenomenon. The spectrum $t_{G}({\rm inf}^G_{e}(E))^G$ is $v_{n-s_{G;E}}$-periodic for some positive integer $s_{G;E}$. When $s_{G;E}>n$, the $v_{n-s_{G;E}}$-p...

conjecture | open | MathDB 页面 | 原文线索

V2-032931 | MATHDB-357223

Unstable Adams operation conjecture for finite p-groups

摘录 / EXCERPT: Let G be a finite p-group, let E be a p-complete complex-oriented spectrum with associated formal group of height n, and let $E^2(f)$ denote the induced map in degree-two...

conjecture | open | MathDB 页面 | 原文线索

V2-032932 | MATHDB-357297

Equivariance conjecture for graded roots of delta-constant Gorenstein deformations

摘录 / EXCERPT: Let (C_t,o) be a $delta$-constant flat deformation of Gorenstein curves. The associated graded graph map ... is the map constructed from the deformation, and the graded roots carry...

conjecture | open | MathDB 页面 | 原文线索

V2-032933 | MATHDB-357316

Yumoto-Misumi conjecture on lattice-fermion species and Betti numbers

摘录 / EXCERPT: Yumoto-Misumi conjecture. Under these conditions, the maximal number of fermion species is

conjecture | open | MathDB 页面 | 原文线索

V2-032934 | MATHDB-357368

The heaviness-SH-visibility conjecture

摘录 / EXCERPT: Let $(M,\a9omega)$ be a closed symplectic manifold and let $K\subset M$ be a compact subset. The subset K is SH-visible when its relative symplectic cohomology satisfies ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-032935 | MATHDB-357369

The heaviness-SH-heaviness conjecture

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold and let $K\subset M$ be a compact subset. The subset K is SH-heavy in the sense of the quantum cohomology ideal-valued quasi-meas...

conjecture | open | MathDB 页面 | 原文线索

V2-032936 | MATHDB-357404

Nice-group conjecture for the cross-index of products of free simplicial posets

摘录 / EXCERPT: Let G be a nice group, meaning a cyclic p-group or a generalized quaternion group whose order is a power of 2. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be finite free G-simp...

conjecture | open | MathDB 页面 | 原文线索

V2-032937 | MATHDB-357405

Nice-group conjecture for cross-indices of face posets

摘录 / EXCERPT: Let G be a nice group, meaning a cyclic p-group or a generalized quaternion group whose order is a power of 2. For a finite free G-simplicial complex $\mathcal{K}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-032938 | MATHDB-357413

Tanaka's topological complexity conjecture for Khalimsky circles

摘录 / EXCERPT: Let $\mathbb{S}^1_{2^k}$ be the Khalimsky circle with 2^k points, where $k\geq 1$, and let $\operatorname{tc}(X)$ denote the topological complexity of a space X. Tanaka's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-032939 | MATHDB-357414

Tanaka's strict topological-complexity inequality conjecture for finite spaces

摘录 / EXCERPT: Let X be a finite topological space, let PX be its path space, and let $\operatorname{tc}(X)$ denote its topological complexity. Let $\operatorname{cat}(X)$ denote its Lusterni...

conjecture | open | MathDB 页面 | 原文线索

V2-032940 | MATHDB-357418

Generic smoothness conjecture for isoperimetric regions

摘录 / EXCERPT: Let (M,g) be an 8-dimensional closed Riemannian manifold. For $k=3,4,\ldots$, let $\mathcal{M}^k$ be the class of C^k metrics on M, endowed with the C^k topology. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-032941 | MATHDB-357421

Well-definedness and invariance conjecture for Legendrian embedded contact homology

摘录 / EXCERPT: Let $(Y,Lambda)$ be a pair consisting of a closed contact 3-manifold Y, a closed Legendrian link $Lambda \subset Y$, and a contact form α that is non-degenerate for...

conjecture | open | MathDB 页面 | 原文线索

V2-032942 | MATHDB-357645

The paracyclic stable infinity-category and connective spherical complex correspondence

摘录 / EXCERPT: Paracyclic-spherical correspondence. There is an equivalence of infinity-categories between the infinity-category of paracyclic stable infinity-categories and the infinity-category...

conjecture | open | MathDB 页面 | 原文线索

V2-032943 | MATHDB-357660

The concordance-invariance conjecture for modulo-4 reduced knot-homology ranks

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $\mathcal{A}$ be either $\mathbb{F}_p$ or $\mathbb{Q}$, and let $\mathcal{C}$ denote the knot concordance group. Concordance-invariance conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032944 | MATHDB-357661

The concordance-invariance conjecture for modulo-4 unreduced Khovanov ranks

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $\mathcal{A}$ be either $\mathbb{Q}$ or $\mathbb{F}_p$ with $p\ne 2$. Write $\operatorname{Kh}_{\mathcal{A}}(K)$ for unreduced Khovanov homolo...

conjecture | open | MathDB 页面 | 原文线索

V2-032945 | MATHDB-357695

Shen-Yin's perverse-Hodge symmetry conjecture

摘录 / EXCERPT: Let $\pi\colon M\to B$ be the Lagrangian fibration under consideration, and let ... be the associated complexes of coherent $\mathscr{O}_B$-modules, where i records the cohomologic...

conjecture | open | MathDB 页面 | 原文线索

V2-032946 | MATHDB-357711

The equivalence between torus and reductive Grothendieck-Serre injectivity

摘录 / EXCERPT: Torus-reductive injectivity equivalence. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-032947 | MATHDB-357723

Infinite counterexample conjecture for tunnel edges in tunnel-number-one knots

摘录 / EXCERPT: Infinite tunnel-edge counterexample conjecture. Every knot with tunnel number one admits infinitely many ideal triangulations with no tunnel edges.

conjecture | open | MathDB 页面 | 原文线索

V2-032948 | MATHDB-357725

The dense subgroup density conjecture for compact groups

摘录 / EXCERPT: Let G be a compact topological group. A subgroup $H\leq G$ is dense if its closure in G is G, and $d(H)$ denotes the density character of H, while $w(G)$ denotes the weight...

conjecture | open | MathDB 页面 | 原文线索

V2-032949 | MATHDB-357726

The positive-half TQFT conjecture for \(\mathfrak{psl}(1|1)\)

摘录 / EXCERPT: Positive-half TQFT conjecture. The TQFT assigning $(\operatorname{Rep}(H),\otimes)$ to a point recovers both aspects of the $\mathfrak{psl}(1|1)$ Chern-Simons TQFT and aspects of d...

conjecture | open | MathDB 页面 | 原文线索

V2-032950 | MATHDB-357729

Equivariance conjecture for the instanton skein exact triangle

摘录 / EXCERPT: Let L_+, L_- and L_0 be oriented links, with L_0 the resolution of a distinguished crossing, and suppose the skein exact triangle involves the module ... If L_0 has ℓ...

conjecture | open | MathDB 页面 | 原文线索

V2-032951 | MATHDB-357730

Free-model conjecture for instanton link Floer homology

摘录 / EXCERPT: Let L be an n-component link in a homology 3-sphere, and let $\boldsymbol{\mathbf{n}}$ be a choice of framing. Let $\mathit{CLI}^-(L,\boldsymbol{\mathbf{n}})$ denote the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-032952 | MATHDB-357746

Concordance-implies-isotopy conjecture for positive scalar curvature metrics

摘录 / EXCERPT: Let M be a closed manifold, and let $\mathrm{Riem}^+(M)$ denote the space of positive scalar curvature metrics on M. Two metrics in this space are concordant if there is a posi...

conjecture | open | MathDB 页面 | 原文线索

V2-032953 | MATHDB-357747

Strong concordance-implies-isotopy conjecture for positive scalar curvature metrics

摘录 / EXCERPT: Let $\mathrm{SingMet}$ and $\mathrm{ConcSet}$ be the cubical models of, respectively, the space of positive scalar curvature metrics and the space of positive scalar curvature conc...

conjecture | open | MathDB 页面 | 原文线索

V2-032954 | MATHDB-357812

Plamenevskaya-type invariant survival conjecture for bridgeless plane graphs

摘录 / EXCERPT: Let Γ be a bridgeless plane graph of a connected graph G. Let $\psi(\Gamma)$ be the invariant represented by the all-x^{n-1} tensor in the n-color homology, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032955 | MATHDB-357828

Banach cohomology and comparison conjecture for p-adic period domains

摘录 / EXCERPT: Banach cohomology and comparison conjecture. There is an isomorphism of topological K-vector spaces

conjecture | open | MathDB 页面 | 原文线索

V2-032956 | MATHDB-357839

The geometric volume conjecture for SO(3)-knot states

摘录 / EXCERPT: Let K be a knot in S^3. The geometric quantized space for the $SO(3)$-Witten-Chern-Simons theory of the torus is the alternating subspace ... of holomorphic sections of...

conjecture | open | MathDB 页面 | 原文线索

V2-032957 | MATHDB-357857

The topological concordance conjecture for links in homology spheres

摘录 / EXCERPT: Let $K\subseteq Y$ be a link in a homology sphere. A homology cobordism W from Y to S^3 is a cobordism inducing homology isomorphisms from its boundary components. Two links...

conjecture | open | MathDB 页面 | 原文线索

V2-032958 | MATHDB-357868

Conjecture on the cut locus of lifted Fermat hypersurfaces

摘录 / EXCERPT: Let $\pi:\mathbb{S}^{2n+1}\to\mathbb{CP}^n$ be the quotient map, let ... and set $\tilde{X}(d)=\pi^{-1}(X(d))$. Here $\mathbb{Z}_d^{\star(n+1)}$ denotes the $(n+1)$-fold topological...

conjecture | open | MathDB 页面 | 原文线索

V2-032959 | MATHDB-357874

The simple polygon space conjecture

摘录 / EXCERPT: Let $S(n)$ be the connected component of the space of simple labeled n-gons modulo oriented similarity consisting of positively oriented polygons. As an open subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-032960 | MATHDB-357875

The global cone conjecture for moduli spaces of simple polygons

摘录 / EXCERPT: Global cone conjecture for moduli spaces of simple polygons. ${\mathcal M}(n)$ is homeomorphic to $C(\mathbb{S}^{2n-5}/\langle\Sigma,T\rangle)$. A local version holds in a neighbor...

conjecture | open | MathDB 页面 | 原文线索

V2-032961 | MATHDB-357886

Cohomological distinction conjecture for the groups $G(\bar n)$

摘录 / EXCERPT: Fix, for every integer $n\geq 1$, the fundamental group G_n of an $(n+2)$-dimensional cusped hyperbolic manifold. Let $\bar n$ be a finite collection of integers at least 1, an...

conjecture | open | MathDB 页面 | 原文线索

V2-032962 | MATHDB-357891

Left properness conjecture for q-model structures on multipointed d-spaces

摘录 / EXCERPT: Take ${\mathbf{Top}}$ to be the category of Δ-Hausdorff Δ-generated spaces. Left properness conjecture. The q-model structures of ${\rm{\mathbf{{\mathcal{G}}dTop}}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-032963 | MATHDB-357895

Row-column mirror symmetry for y-ified colored HOMFLYPT homology

摘录 / EXCERPT: Let $P^y_{\boldsymbol{\bigl.\lambda}}$ be y-ifications of the categorified projectors P_{λ}. For an r-component link...

conjecture | open | MathDB 页面 | 原文线索

V2-032964 | MATHDB-357896

Chinburg-Reid-Stover conjecture for once-punctured torus bundles

摘录 / EXCERPT: Let M be a hyperbolic once-punctured torus bundle, let $\widetilde{C}$ denote the relevant character-curve component, and let $\mathcal{A}_{k(\widetilde{C})}$ be its tautological...

conjecture | open | MathDB 页面 | 原文线索

V2-032965 | MATHDB-357913

Real equivariant conjectures for truncated Brown-Peterson spectra

摘录 / EXCERPT: Real equivariant conjectures. The six listed assertions in the source hold: the required $\mathbf{E}_{\rho}\rtimes\mathrm{U}(1)_\mathbf{R}$-algebra and splitting structures exist;...

conjecture | open | MathDB 页面 | 原文线索

V2-032966 | MATHDB-357931

Dunfield-Friedl-Jackson conjecture for hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3, let $E_K=S^3\setminus\nu(K)$ be its exterior, and let $x(K)$ denote its Thurston norm, equivalently ... Let ρ^{geom} be a lift of...

conjecture | open | MathDB 页面 | 原文线索

V2-032967 | MATHDB-357932

Lamm's equivariant crossing number conjecture for two-bridge knots

摘录 / EXCERPT: Lamm's conjecture. $c_t(K)=c(K)$ if and only if K can be represented by a continued fraction expansion $[a_1,\ldots,a_n]$ of type (A) or type (B).

conjecture | open | MathDB 页面 | 原文线索

V2-032968 | MATHDB-357938

Detcherry-Kalfagianni exponential growth conjecture

摘录 / EXCERPT: Let M be a compact, oriented 3-manifold with empty or toroidal boundary, and let ... The manifold M is q-hyperbolic when $\textit{ITV}(M)>0$, and $\lVert M\rVert$ denotes its...

conjecture | open | MathDB 页面 | 原文线索

V2-032969 | MATHDB-357947

Allen's non-orientable surface realizability conjecture for torus knots

摘录 / EXCERPT: For a positive torus knot $T(3,6n+1)$, let (e,h) denote the normal Euler number and non-orientable genus of a smoothly and properly embedded surface in D^4 bounded by the knot....

conjecture | open | MathDB 页面 | 原文线索

V2-032970 | MATHDB-357948

Asymptotic equality conjecture for real monopole Floer invariants

摘录 / EXCERPT: Let K be a knot, let $\#_nK$ denote the connected sum of n copies of K, and let $\underline{\delta}_R$, $\bar{\delta}_R$, and $\delta(K)$ be the invariants used in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-032971 | MATHDB-357949

Manolescu-Lidman-type conjecture for real monopole Floer homology of links

摘录 / EXCERPT: Let L be an oriented link with non-zero determinant. Let $SWF(L)$ be the link Floer stable homotopy object, let $\frak{s}_L$ be the spin structure determined by the orientation o...

conjecture | open | MathDB 页面 | 原文线索

V2-032972 | MATHDB-358015

Reflection-vector conjecture for exceptional objects and collections

摘录 / EXCERPT: Reflection-vector conjecture. (a) If the quantum cohomology of X is convergent and semisimple, then $\Psi_q(\mathcal{E})$ is a reflection vector for every exceptional object...

conjecture | open | MathDB 页面 | 原文线索

V2-032973 | MATHDB-358038

Generalized Sarnak conjecture for non-pretentious multiplicative functions

摘录 / EXCERPT: Let $\mathbb{D}$ be the closed unit disc, and call a sequence $a:\mathbb{N}\to\mathbb{D}$ deterministic if it is generated by a continuous observable on a zero-entropy topological...

conjecture | open | MathDB 页面 | 原文线索

V2-032974 | MATHDB-358096

Freed-Hopkins classification conjecture for reflection-positive invertible QFTs

摘录 / EXCERPT: Let $G=\{G_d,s_d,\rho_d\}_{d\in\mathbb{Z}_{\geq 0}}$ be a tangential structure, and let $I\mathbb{C}^{\times}$ denote the Brown-Comenetz dual of the sphere. Write $I\Omega^G$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-032975 | MATHDB-358104

The conjecture that four twice-punctured tori suffice in hyperbolic knot exteriors

摘录 / EXCERPT: Four-torus bound conjecture. There are at most four pairwise disjoint, non-isotopic, nested, embedded twice-punctured tori with an integral slope in the complement of K.

conjecture | open | MathDB 页面 | 原文线索

V2-032976 | MATHDB-358108

The equality conjecture for iterated local dimension sets

摘录 / EXCERPT: Equality conjecture. If X is almost zero-dimensional, then

conjecture | open | MathDB 页面 | 原文线索

V2-032977 | MATHDB-358122

Homology conjecture for complements of caustics of E7 and E8 singularities

摘录 / EXCERPT: Let f be a real function singularity of type E_7 or E_8, and consider the complement of the caustic in the parameter space of a versal deformation of f. Homology conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-032978 | MATHDB-358123

P8^1 caustic-complement cohomology conjecture

摘录 / EXCERPT: Let $f_0=x^3+y^3+z^3$, and consider the morsification obtained from the perturbation $x^3-3\varepsilon^2x+y^3-3\varepsilon^2y+z^3$ by splitting the critical points with coordinates...

conjecture | open | MathDB 页面 | 原文线索

V2-032979 | MATHDB-358139

External-univalence conjecture for contextual models of SOGATs

摘录 / EXCERPT: Let $a54$ be a SOGAT, and let $a54$ be equipped with weakly stable identity types satisfying function extensionality and saturation, which is the condition called external univalen...

conjecture | open | MathDB 页面 | 原文线索

V2-032980 | MATHDB-358150

The Borman-Sheridan class compatibility conjecture

摘录 / EXCERPT: Borman-Sheridan class compatibility conjecture. Under this equivalence, the action of t on $\mathcal{W}_{S^1}(X)_{-1}$ coincides with the action of τ on...

conjecture | open | MathDB 页面 | 原文线索

V2-032981 | MATHDB-358174

Freed-Teleman-Walker conjecture for semisimple WRT theories

摘录 / EXCERPT: Let $\mathscr{V}$ be a semisimple modular tensor category, set $\mathcal{V}=\operatorname{Ind}(\mathscr{V})$, and let $\mathcal{Z}_{\mathcal{V}}$ be an oriented 4-TQFT with...

conjecture | open | MathDB 页面 | 原文线索

V2-032982 | MATHDB-358175

Freed-Teleman-Walker conjecture for non-semisimple WRT theories

摘录 / EXCERPT: Let $\mathscr{V}$ be a non-semisimple modular tensor category, set $\mathcal{V}=\operatorname{Ind}(\mathscr{V})$, and let $\mathcal{Z}_{\mathcal{V}}$ be an oriented 4-TQFT with...

conjecture | open | MathDB 页面 | 原文线索

V2-032983 | MATHDB-358176

Freed-Teleman-Walker conjecture for non-semisimple WRT theories via non-compact filled cobordisms

摘录 / EXCERPT: Let $\mathscr{V}$ be a non-semisimple modular tensor category. Consider the non-compact filled 321-dimensional cobordism category $Cob^{nc,filled}_{321}$, its quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-032984 | MATHDB-358186

Vigué-Poirrier conjecture on exponential growth of free-loop-space homology

摘录 / EXCERPT: Let X be a simply connected space and let $\mathcal{L}X=\operatorname{Map}(S^1,X)$ be its free loop space. Write ΩX for the based loop space, and call X rationally hype...

conjecture | open | MathDB 页面 | 原文线索

V2-032985 | MATHDB-358195

Twisted symplectic cohomology vanishing for Higgs branch moduli spaces with exact Lagrangians

摘录 / EXCERPT: Twisted symplectic cohomology vanishing conjecture. A deformation argument should imply

conjecture | open | MathDB 页面 | 原文线索

V2-032986 | MATHDB-358235

Blanc's lattice conjecture for smooth proper dg-categories

摘录 / EXCERPT: Let $\mathcal{T}$ be a smooth proper dg-category over $\mathbb{C}$. Write $K_{\operatorname{top}}(\mathcal{T})$ for its topological K-theory spectrum,...

conjecture | open | MathDB 页面 | 原文线索

V2-032987 | MATHDB-358266

Connected sum formula for ECH spectral invariants

摘录 / EXCERPT: Let $(Y_1,lambda_1)$ and $(Y_2,lambda_2)$ be closed connected contact three-manifolds with nonzero contact invariants, and let k be a nonnegative integer. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-032988 | MATHDB-358285

Extension of refined loop-equation existence to hyperelliptic refined spectral curves

摘录 / EXCERPT: Let $\mathcal{S}_{\bm\mu,\bm\kappa}$ be a hyperelliptic refined spectral curve, and let $\omega_{g,n}$ be the multidifferentials constructed by refined topological recursion. Exist...

conjecture | open | MathDB 页面 | 原文线索

V2-032989 | MATHDB-358292

Categorified generalized projector decomposition conjecture

摘录 / EXCERPT: Let κ be the coefficient field, let $\operatorname{Ch}^-(\mathsf{BN}_n)$ be the homotopy category of bounded-above complexes of n-strand Bar-Natan 1-morphisms, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-032990 | MATHDB-358293

Functorial colored knot homology conjecture

摘录 / EXCERPT: Let K be a framed oriented knot, let $\operatorname{Kar}(\mathcal T\mkern-3mu\mathcal L)^\oplus$ be the additive Karoubi envelope of the Temperley-Lieb category, and let V_n de...

conjecture | open | MathDB 页面 | 原文线索

V2-032991 | MATHDB-358294

Categorified Hochschild colored complex conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a framed oriented knot and let $X\in\operatorname{Ch}^-(\mathsf{BN}_n)$ be a complex of n-strand Bar-Natan 1-morphisms. Let $C(K,V_n)$ denote the V_n-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-032992 | MATHDB-358361

Vasconcelos' SCM conjecture for height-three homogeneous ideals

摘录 / EXCERPT: Vasconcelos' SCM conjecture. If I has deviation at least 3, is generically a complete intersection, is not Gorenstein, and the resolution of S/I is pure, then I is not SCM.

conjecture | open | MathDB 页面 | 原文线索

V2-032993 | MATHDB-358379

Finiteness conjecture for top integral cohomology of configuration spaces

摘录 / EXCERPT: Finiteness conjecture. There exist $k_{0}\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032994 | MATHDB-358406

Finite-cohomology generation conjecture for homotopy categories of injectives

摘录 / EXCERPT: Finite-cohomology generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-032995 | MATHDB-358407

Cohomological approximation conjecture for homotopy categories of injectives

摘录 / EXCERPT: Cohomological approximation conjecture. If X belongs to $\mathsf{Klnj}^{b}(\mathsf{k}G)$, then X is cohomologically equivalent to an object of $\mathsf{D}^{b}(\mathsf{k}G)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-032996 | MATHDB-358408

Cohomological realization conjecture for cochains on classifying spaces

摘录 / EXCERPT: Cohomological realization conjecture. If Y belongs to $\mathsf{D}^{b}(C^*BG)$, then there is an object X in $\mathsf{D}^{b}(\mathsf{k}G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-032997 | MATHDB-358426

Avramov's total rank conjecture

摘录 / EXCERPT: Avramov's total rank conjecture. Under the same assumptions as in the Buchsbaum-Eisenbud-Horrocks conjecture,

conjecture | open | MathDB 页面 | 原文线索

V2-032998 | MATHDB-358427

Total rank conjecture for short complexes

摘录 / EXCERPT: For a Noetherian commutative ring R, let P be a short complex, meaning a non-exact complex of finite-rank projective R-modules concentrated in degrees $[0,c]$, where ... For a...

conjecture | open | MathDB 页面 | 原文线索

V2-032999 | MATHDB-358451

Weinberger's contractibility conjecture for preferential attachment clique complexes

摘录 / EXCERPT: Let $X(\infty,\delta,m(\cdot))$ be the clique complex of the graph obtained by adding one node and $m(t)$ preferential-attachment edges at each time t, where...

conjecture | open | MathDB 页面 | 原文线索

V2-033000 | MATHDB-358452

Asymptotic-limit conjecture for expected Betti numbers of preferential attachment clique complexes

摘录 / EXCERPT: For $q\geq 2$ and $m\geq 2q$, let $X(T,\delta,m)$ denote the affine preferential attachment clique complex and let $\beta_q(X(T,\delta,m))$ be its q-th Betti number. If...

conjecture | open | MathDB 页面 | 原文线索

V2-033001 | MATHDB-358453

Diameter-dominance conjecture for Rips complexes of preferential attachment graphs

摘录 / EXCERPT: Let $G(T,\delta,m)$ be the preferential attachment graph and let its Rips complex of radius r be formed using the graph metric. Diameter-dominance conjecture. The Betti numbers o...

conjecture | open | MathDB 页面 | 原文线索

V2-033002 | MATHDB-358463

Conjecture on semicontinuity and local constancy in algebraic representation families

摘录 / EXCERPT: Semicontinuity and local-constancy conjecture. With respect to the Zariski topology on $\operatorname{Spec}(R)$, $m^i(\pi|_x)$ is upper semicontinuous for each $i\in\mathbb{N}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-033003 | MATHDB-358533

Surjectivity conjecture for saddle cobordisms between braid diagrams

摘录 / EXCERPT: Let $D_{n,m}^i$ and $E_{n,m}^i$ be the diagrams defined in the paper, and consider the saddle cobordism ... at the crossing corresponding to the last letter σ_i. Surjectivity...

conjecture | open | MathDB 页面 | 原文线索

V2-033004 | MATHDB-358534

Recursive Poincaré-polynomial conjecture for Khovanov homology of $T(n,n)$

摘录 / EXCERPT: Let $K_n(t,q)$ denote the Poincaré polynomial of the Khovanov homology of $T(n+1,n)$, and let $L_n(t,q)$ denote that of $T(n,n)$. Let $C_n=\frac{1}{n+1}\binom{2n}{n}$ be the nth...

conjecture | open | MathDB 页面 | 原文线索

V2-033005 | MATHDB-358535

Shumakovitch-Turner recurrence conjecture for torus-knot Khovanov polynomials

摘录 / EXCERPT: Let $K_n(t,q)$ denote the Poincaré polynomial of the Khovanov homology of the torus knot $T(n+1,n)$. Shumakovitch-Turner recurrence conjecture. The polynomials satisfy ... and, for...

conjecture | open | MathDB 页面 | 原文线索

V2-033006 | MATHDB-358541

The bordered-curve complement invariance conjecture for knot Floer homology

摘录 / EXCERPT: Let $K\subset Y$ be a knot with complement $M=Y\setminus\nu(K)$. The decorated curves $\widehat{\mathit{HF}}(Y,K)$ are curves in the punctured torus T_M^*, and...

conjecture | open | MathDB 页面 | 原文线索

V2-033007 | MATHDB-358542

The minus bordered-curve complement invariance conjecture

摘录 / EXCERPT: Let $K\subset Y$ be a knot with complement $M=Y\setminus\nu(K)$. Let $\mathit{HF}^{-}(Y,K)$ denote the decorated curves constructed from minus knot Floer homology in the marked tor...

conjecture | open | MathDB 页面 | 原文线索

V2-033008 | MATHDB-358543

The minus immersed-curve pairing conjecture

摘录 / EXCERPT: Let M_1 and M_2 be manifolds with torus boundary, and let $\phi:\partial M_1\to\partial M_2$ be an orientation-reversing gluing map. Let $\mathit{HF}^{-}(M_i)$ denote the decor...

conjecture | open | MathDB 页面 | 原文线索

V2-033009 | MATHDB-358586

Polynomial-time Khovanov homology and spectrum conjecture for fixed-strand closed braids

摘录 / EXCERPT: Let w be a closed braid with a fixed number of strands, and measure complexity with respect to the number of crossings of w. Polynomial-time Khovanov conjecture. Computing the...

conjecture | open | MathDB 页面 | 原文线索

V2-033010 | MATHDB-358596

Universal finite generation for homology of graph configuration spaces

摘录 / EXCERPT: Let $F_n(G)$ be the configuration space of n ordered points in a graph G. For a class of graphs, universal finite generation in degree i and with n particles means that a f...

conjecture | open | MathDB 页面 | 原文线索

V2-033011 | MATHDB-358598

Categorical Robertson conjecture

摘录 / EXCERPT: For $k>0$, let $cmathcal{GTM}_k$ be the full subcategory of topological minor morphisms spanned by graphs not admitting the Robertson chain R_k as a topological minor, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033012 | MATHDB-358599

Finite generation of graph configuration-space homology as a topological graph-injection module

摘录 / EXCERPT: For a graph G, let $F_n(G)$ be the configuration space of n ordered points in G, and let $cmathcal{TGI}$ be the category of topological graph embeddings. The assignment...

conjecture | open | MathDB 页面 | 原文线索

V2-033013 | MATHDB-358695

Local template persistence for the Rössler system near trefoil parameters

摘录 / EXCERPT: Let P be the parameter space, let $p\in P$ be the trefoil parameter, and let F_v denote the vector field corresponding to $v\in P$. Let $L(0,1)$ be the template associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-033014 | MATHDB-358696

Self-linking complexity conjecture for heteroclinic knots in the Rössler system

摘录 / EXCERPT: Let P be the parameter space, let $p\in P$ be a heteroclinic parameter generating a heteroclinic knot K, and let F_p be the corresponding vector field. Assume the conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-033015 | MATHDB-358714

Conjecture on evaluation of prismatic G-F-gauges

摘录 / EXCERPT: Let R be a complete regular local ring over $W(k)$ with residue field k, let $(\mathfrak{S},(\mathcal{E}))=(W(k)[[t_1,\dotsc,t_n]],(\mathcal{E}))$ be a prism of Breuil--Kisin t...

conjecture | open | MathDB 页面 | 原文线索

V2-033016 | MATHDB-358715

Universality conjecture for deformations of prismatic G-F-gauges

摘录 / EXCERPT: Let k be the residue field, let μ be 1-bounded, and let $\mathscr{Q}\in G\mathchar`-F\mathchar`-\mathrm{Gauge}_\mu(k)$. Let $R_{G,\mu}$ represent the deformation functor on c...

conjecture | open | MathDB 页面 | 原文线索

V2-033017 | MATHDB-358726

The characteristic-p analytic cohomology conjecture

摘录 / EXCERPT: Characteristic-p analytic cohomology conjecture. (a) If there is a sequence of subgroups H_j of H forming a neighborhood basis of the identity such that

conjecture | open | MathDB 页面 | 原文线索

V2-033018 | MATHDB-358729

The semistable comparison conjecture for smooth affinoid rigid spaces

摘录 / EXCERPT: Let C be the completed algebraic closure of a p-adic field, and let X be a smooth affinoid rigid space over C. Write $\boldsymbol{\Omega}^j(X)$ for the space of j-forms,...

conjecture | open | MathDB 页面 | 原文线索

V2-033019 | MATHDB-358776

Simultaneous crossing-and-wrapping diagram conjecture for solid torus links

摘录 / EXCERPT: Let L be a solid torus link with crossing number c and wrapping number wr. A diagram of L is said to realize both invariants when it has exactly c crossing...

conjecture | open | MathDB 页面 | 原文线索

V2-033020 | MATHDB-358806

The truncated Anderson dual conjecture for real topological K-theory

摘录 / EXCERPT: The truncated Anderson dual conjecture. There is an equivalence of spectra

conjecture | open | MathDB 页面 | 原文线索

V2-033021 | MATHDB-358807

The Brauer spectrum conjecture for complex K-theory

摘录 / EXCERPT: Brauer spectrum conjecture. There is an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-033022 | MATHDB-358808

The higher units conjecture for complex K-theory

摘录 / EXCERPT: Higher units conjecture. For all $n\geq 0$ there are equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-033023 | MATHDB-358823

The O'Hara conjecture on maximum energies of equilateral convex polygons

摘录 / EXCERPT: Let Γ_n denote the set of convex n-gons with each edge of length 1, and let E_n^α be the energy associated with the increasing continuous function...

conjecture | solved | MathDB 页面 | 原文线索

V2-033024 | MATHDB-358972

Genus bound from the s-invariant for surfaces in the tangent disk bundle of S^2

摘录 / EXCERPT: Let DTS^2 be the unit disk bundle of the tangent bundle of S^2, with boundary $\mathbb{RP}^3$. Let $(\Sigma,K)\subset(DTS^2,\mathbb{RP}^3)$ be a properly embedded orientable co...

conjecture | open | MathDB 页面 | 原文线索

V2-033025 | MATHDB-359006

Distinct topological functionals from inequivalent enriched TQFTs

摘录 / EXCERPT: Let Y be an n-finite space, meaning that π_0Y and all homotopy groups $\pi_q(Y,y)$ for $0\leq q\leq n$ and $y\in Y$ are finite. An n-dimensional bosonic or fermionic Y-...

conjecture | open | MathDB 页面 | 原文线索

V2-033026 | MATHDB-359007

Gaiotto-Johnson-Freyd trivialization conjecture for TQFT target categories

摘录 / EXCERPT: Let $\Sigma^{n-1}\mathsf{Vect}^{fd}$ and $\Sigma^{n-1}\mathsf{sVect}^{fd}$ be the stable suspensions used as codomains for bosonic and fermionic fully extended TQFTs, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-033027 | MATHDB-359008

Anomaly-freedom conjecture for solitonic symmetry

摘录 / EXCERPT: Let a quantum field theory possess solitonic symmetry arising from its target-space topological data. Solitonic anomaly-freedom conjecture. Solitonic symmetry is free of ’t Hooft a...

conjecture | open | MathDB 页面 | 原文线索

V2-033028 | MATHDB-359012

Baez-Dolan cobordism hypothesis for framed extended topological field theories

摘录 / EXCERPT: Let $\mathcal{C}$ be a symmetric monoidal (∞,n)-category with duals. A framed extended n-dimensional TFT valued in $\mathcal{C}$ is a symmetric monoidal functor from the f...

conjecture | open | MathDB 页面 | 原文线索

V2-033029 | MATHDB-359016

Quasi-isomorphism between sheaf-quantization and Floer curved algebras

摘录 / EXCERPT: Let $\bm L$ be a Lagrangian brane, let $CF_{\mathrm{SQ}}(\bm L,\bm L)$ be the colimit curved dga constructed from sheaf quantizations, and let $CF(\bm L,\bm L)$ be the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-033030 | MATHDB-359030

The moderate doodle bordism extension conjecture for the annulus

摘录 / EXCERPT: Let $A=\mathbb R\times[0,1]$. Denote by $\mathbf B^{\mathsf{imm}}_{\mathsf{moderate \leq 2}}(A)$ the abelian group of moderate immersed doodle bordism classes, and by $\mathbb Z_2$...

conjecture | open | MathDB 页面 | 原文线索

V2-033031 | MATHDB-359031

The moderate open-curve doodle extension conjecture for the annulus

摘录 / EXCERPT: Let $A=\mathbb R\times[0,1]$. Denote by $\mathbf{OC}^{\mathsf{imm}}_{\mathsf{moderate \leq 2}}(A)$ the abelian group of moderate immersed open-curve doodle classes, and by...

conjecture | open | MathDB 页面 | 原文线索

V2-033032 | MATHDB-359034

Extension of the \mathscr{Q}-top recursion theorem to arbitrary spectral curves

摘录 / EXCERPT: Extension conjecture. Theorem $$ holds for any Σ.

conjecture | open | MathDB 页面 | 原文线索

V2-033033 | MATHDB-359035

Extension of the refined variational formula to arbitrary spectral curves

摘录 / EXCERPT: Variational extension conjecture. Theorem $$ holds for any Σ.

conjecture | open | MathDB 页面 | 原文线索

V2-033034 | MATHDB-359054

Asymptotic volume-knot Floer homology rank conjecture

摘录 / EXCERPT: Let K be a knot, let $r(K)$ denote the total rank of its knot Floer homology, let $\operatorname{Vol}(K)$ denote its hyperbolic volume, and let c denote the crossing number. As...

conjecture | open | MathDB 页面 | 原文线索

V2-033035 | MATHDB-359056

Sigmoid density conjecture for low-rank knot Floer homology

摘录 / EXCERPT: Fix a small cut-off value d for the total rank of knot Floer homology. For x a volume threshold, let $f(x)$ be the fraction of knots whose total knot Floer homology rank is les...

conjecture | open | MathDB 页面 | 原文线索

V2-033036 | MATHDB-359071

The no essential sphere or torus conjecture for the four-manifold AP

摘录 / EXCERPT: Let AP be the four-manifold constructed in the paper, and let $H_2(AP;\mathbb Z)$ denote its second homology group. An embedded surface in AP is homologically essential if its...

conjecture | open | MathDB 页面 | 原文线索

V2-033037 | MATHDB-359143

The refined volume conjecture for hyperbolic knots

摘录 / EXCERPT: Let $\mathscr{H}\subset S^3$ be a hyperbolic knot. Let $\operatorname{cv}(\mathscr{H})$ be its complex volume, and let $T(\mathscr{H})$ be the adjoint cohomological Reidemeister to...

conjecture | open | MathDB 页面 | 原文线索

V2-033038 | MATHDB-359144

The generalized volume conjecture near the complete hyperbolic structure

摘录 / EXCERPT: Generalized volume conjecture. For a hyperbolic knot $\mathscr{H}$, there exists a neighbourhood $U\in\mathbb{C}$ of 0 such that, if $u\in U\setminus\pi\sqrt{-1}\mathbb{Q}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-033039 | MATHDB-359152

Nontrivial delta move finite type invariants detect unknotting number

摘录 / EXCERPT: Delta move finite type invariant conjecture. For some n, we have

conjecture | open | MathDB 页面 | 原文线索

V2-033040 | MATHDB-359153

Distinct transverse knots remain distinct as virtual transverse knots

摘录 / EXCERPT: Transverse-to-virtual injectivity conjecture. No two transverse knots are the same virtual transverse knot.

conjecture | open | MathDB 页面 | 原文线索

V2-033041 | MATHDB-359154

Virtual transverse finite type invariants detect braided type and self-linking number

摘录 / EXCERPT: Virtual transverse triviality conjecture. Then x and y represent the same element of $U_n(VTK)$ for all n.

conjecture | open | MathDB 页面 | 原文线索

V2-033042 | MATHDB-359155

Nonuniqueness of negative stabilization for virtual transverse knots

摘录 / EXCERPT: Negative stabilization nonuniqueness conjecture. Negative stabilization is not a unique operation on virtual transverse knots. That is, there are two ways to negatively stabilize s...

conjecture | open | MathDB 页面 | 原文线索

V2-033043 | MATHDB-359216

Chi-independence conjecture for BPS sheaves in families of Higgs bundles

摘录 / EXCERPT: Let $f\colon\mathcal{C}\to S$ be a family of smooth projective curves of genus g over a smooth connected base S. Let $\mathfrak{M}_{r,d}^{\operatorname{Dol}}(\mathcal{C})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-033044 | MATHDB-359259

Conjecture on the independence complex of a categorical product of three complete graphs

摘录 / EXCERPT: Let $I(G)$ denote the independence complex of a graph G, let K_n, K_m, and K_l be complete graphs, and let $f(n,m,l)$ denote the unspecified number of spheres occurring in...

conjecture | open | MathDB 页面 | 原文线索

V2-033045 | MATHDB-359264

Bulk-boundary correspondence conjecture for two-dimensional Hamiltonians

摘录 / EXCERPT: Let a two-dimensional Hamiltonian on a spin system satisfy the assumptions denoted by ... , and let $\mathfrak A$ be its net of boundary algebras. Write $\mathsf{DHR}(\mathfrak A)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-033046 | MATHDB-359267

Merkulov's homology isomorphism conjecture for the truncated graph complex

摘录 / EXCERPT: Let G_n be the Kontsevich graph complex, with differential d, and let $\mathrm{G}^{\mathrm{Me}}_n=\{\Gamma\in\mathrm{G}_n\mid \pi_5\Gamma=\pi_5d\Gamma=0\}$ be Merkulov...

conjecture | open | MathDB 页面 | 原文线索

V2-033047 | MATHDB-359268

Morita's nontriviality conjecture for graph homology classes

摘录 / EXCERPT: Morita's conjecture. The Morita homology classes $[\mu_k]\in H\bigl(\mathsf{FG}_0^{2k+2}(0)\bigr)$ are non-trivial for every $k\geq 1$. These classes are part of an infinite family...

conjecture | open | MathDB 页面 | 原文线索

V2-033048 | MATHDB-359269

Odd Morita nontriviality conjecture for the classes W_{2k}

摘录 / EXCERPT: For each $k=2,3,4,\dots$, let $W_{2k}\in\mathsf{FG}_1^{2k\text{-loop}}(0)$ be the cycle described in the preceding lemma, which states that W_{2k} is non-zero and satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-033049 | MATHDB-359273

Andersen-Kashaev's TQFT volume conjecture for hyperbolic knots

摘录 / EXCERPT: Let M be an oriented compact closed 3-manifold, and let K be a hyperbolic knot in M. For a fully balanced shaped ideal triangulation X of the complementary space of K i...

conjecture | open | MathDB 页面 | 原文线索

V2-033050 | MATHDB-359282

The ellipsoidal superpotential nonvanishing conjecture for plane curves

摘录 / EXCERPT: Let $p,q,d in \mathbb{Z}_{\geq 1}$ satisfy ... and let $[L]\in H_2(\mathbb{CP}^2)$ be the line class. The ellipsoidal superpotential nonvanishing conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-033051 | MATHDB-359296

Zupan's distance conjecture for bridge positions of links

摘录 / EXCERPT: Zupan's distance conjecture. For any (g,b)-bridge position of (M,K),

conjecture | open | MathDB 页面 | 原文线索

V2-033052 | MATHDB-359335

The flat-section form of Gamma-conjecture I

摘录 / EXCERPT: Let X be a Fano manifold and let $\mathcal{A}$ be the vector space of flat sections $s(z)$ over the positive real axis such that $e^{T/z}s(z)$ has at most polynomial growth as...

conjecture | open | MathDB 页面 | 原文线索

V2-033053 | MATHDB-359338

Equivalence characterization for the tree σE

摘录 / EXCERPT: Equivalence conjecture for σE.

conjecture | open | MathDB 页面 | 原文线索

V2-033054 | MATHDB-359348

Galashin-Lam conjecture on DAHA/EHA invariants and KR homology of algebraic links

摘录 / EXCERPT: Let C be a convex monotone curve, let L_C be its associated link, and let $\mathcal P^{\mathcal E}_C(a,q,t)$ and $\mathcal P^{\operatorname{KR}}_{L_C}(a,q,t)$ denote respective...

conjecture | open | MathDB 页面 | 原文线索

V2-033055 | MATHDB-359363

Serre's conjecture II for principal homogeneous spaces

摘录 / EXCERPT: Let K be a field of cohomological dimension at most 2, and let Z be a principal homogeneous space under a semisimple simply connected K-group. Serre's conjecture II. The va...

conjecture | open | MathDB 页面 | 原文线索

V2-033056 | MATHDB-359364

Higher Serre's conjecture II for the C^q_{sc} property

摘录 / EXCERPT: Let q be a non-negative integer and let K be a field. Say that K has the C^q_{sc} property if, for every finite extension L/K and every principal homogeneous spa...

conjecture | open | MathDB 页面 | 原文线索

V2-033057 | MATHDB-359366

The cohomology approximation conjecture for random cubical covers

摘录 / EXCERPT: Cohomology approximation conjecture. If $d \geq 4(2k+1)^2$, then, as h tends to infinity, the induced map τ^* on the kth cohomology groups is an isomorphism to...

conjecture | open | MathDB 页面 | 原文线索

V2-033058 | MATHDB-359368

Finiteness conjecture for quasi-alternating links of fixed Jones-polynomial breadth

摘录 / EXCERPT: Let b be an integer, and consider quasi-alternating links whose Jones polynomial has breadth b, where the breadth is the difference between the highest and lowest exponents occ...

conjecture | open | MathDB 页面 | 原文线索

V2-033059 | MATHDB-359373

Roger-Yang skein algebra positivity conjecture for punctured surfaces

摘录 / EXCERPT: Let Σ be a punctured surface with puncture set $\mathcal{M}$, and let B be the $\mathbb{Z}[q^{\pm1/2}]$-basis of the Roger-Yang skein algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-033060 | MATHDB-359381

Conjecture on sensitivity to initial data for minimal solutions of the supercooled Stefan problem

摘录 / EXCERPT: Sensitivity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-033061 | MATHDB-359406

Geometric and equivariant local curve element conjecture

摘录 / EXCERPT: Let a geometric local curve element and an equivariant local curve element be local curve elements of the same topological type in the framework of the paper. Geometric-equivariant...

conjecture | open | MathDB 页面 | 原文线索

V2-033062 | MATHDB-359443

Ganatra-Pomerleano's coordinate-function conjecture for cotangent bundles

摘录 / EXCERPT: Ganatra-Pomerleano's conjecture. If T^*Q admits a coordinate function, then Q is diffeomorphic to $S^1\times\Sigma_g$ for some $g\geq0$. The conjecture is presented as motivate...

conjecture | open | MathDB 页面 | 原文线索

V2-033063 | MATHDB-359444

Ganatra-Pomerleano's cyclic coordinate-function conjecture

摘录 / EXCERPT: Cyclic coordinate-function conjecture. If T^*Q admits a cyclic coordinate function, then Q is diffeomorphic to $S^1\times\Sigma_g$, where $g\geq0$, or a spherical space form. T...

conjecture | open | MathDB 页面 | 原文线索

V2-033064 | MATHDB-359445

Generalized Fukaya conjecture for polarized Kähler manifolds

摘录 / EXCERPT: Let X be a closed Kähler manifold, and let $\Sigma\subset X$ be a complex hypersurface such that $X\setminus\Sigma$ is a Weinstein manifold admitting a cyclic dilation; then...

conjecture | open | MathDB 页面 | 原文线索

V2-033065 | MATHDB-359467

The higher-rank replacement for property T in lattice cohomology

摘录 / EXCERPT: Let G be a connected semisimple Lie group with finite center and without compact factors, and let τ denote the property T assumption in Theorem. Higher-rank replacement conj...

conjecture | open | MathDB 页面 | 原文线索

V2-033066 | MATHDB-359468

Rank-removal conjecture for adelic and arithmetic cohomology

摘录 / EXCERPT: Let k be a number field, let $\mathbf{G}$ be a connected and simply connected almost simple k-algebraic group, and let V be a unitary $\mathbf{G}(k)$-representation. Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-033067 | MATHDB-359470

Higher-rank extension conjecture for semisimple lattice cohomology

摘录 / EXCERPT: Let G be a product of non-compact almost simple groups over local fields, or a connected semisimple Lie group with finite center and without compact factors, and let $\Gamma<G$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-033068 | MATHDB-359471

Property-T removal conjecture for the adelic cohomological dual

摘录 / EXCERPT: Let k be a number field and let $\mathbf{G}$ be a connected, simply connected almost simple k-group. Theorem and Corollary assume that every local group $\mathbf{G}(k_s)$ has p...

conjecture | open | MathDB 页面 | 原文线索

V2-033069 | MATHDB-359485

Atiyah-Floer conjecture for instanton and Lagrangian Floer homology

摘录 / EXCERPT: Let $M=H_g^1\cup_{\Sigma_g}H_g^2$ be a Heegaard decomposition of a homology 3-sphere, and let $I(M)$ denote its instanton Floer homology. For the trivial $SU(2)$ bundles on the h...

conjecture | open | MathDB 页面 | 原文线索

V2-033070 | MATHDB-359517

Sarnak's conjecture on Möbius disjointness along polynomial iterates

摘录 / EXCERPT: Let (X,T) be a minimal topological dynamical system, meaning that for every $x\in X$, the set $\{T^n x\}$ is dense, with zero entropy. Let $x_0\in X$, let $f\in C(X)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033071 | MATHDB-359568

Green-simplicial twisting cochain homotopy equivalence conjecture

摘录 / EXCERPT: Let $i\colon{\mathscr{Green}}\hookrightarrow{\mathscr{sTwist}}$ be the inclusion of the simplicial presheaf of Green complexes into the simplicial presheaf of simplicial twisting c...

conjecture | open | MathDB 页面 | 原文线索

V2-033072 | MATHDB-359578

Chirally cosmetic surgery conjecture for knots

摘录 / EXCERPT: Let K be a knot in S^3. Two surgeries on K with distinct slopes $r,r'\in\mathbb{Q}\cup\{\infty\}$ are chirally cosmetic surgeries when $S^3_r(K)\cong -S^3_{r'}(K)$, where the...

conjecture | open | MathDB 页面 | 原文线索

V2-033073 | MATHDB-359579

Inequality conjecture for positive 2-bridge knot invariants

摘录 / EXCERPT: Let $K=C(p,q)$ be a positive 2-bridge knot with rational number p/q, where $p,q>20$. Let v_3, a_2, a_4, $\det(K)$, and g denote the knot invariants used in the obstructio...

conjecture | open | MathDB 页面 | 原文线索

V2-033074 | MATHDB-359656

The quantum connection's exponential-type singularity and quasi-unipotent formal monodromy at infinity

摘录 / EXCERPT: Quantum-connection conjecture. (i) ∇_{∂_q} has a singularity of unramified exponential type at $q=\infty$. (ii) The regularized formal monodromies at $q=\infty$ are q...

conjecture | open | MathDB 页面 | 原文线索

V2-033075 | MATHDB-359666

Whole-center conjecture for the \(\mathrm{SL}_d\)-skein algebra

摘录 / EXCERPT: Let S be an oriented surface of finite topological type. For each puncture, let $[P_i]$ denote the skein represented by a small loop around that puncture with weight...

conjecture | open | MathDB 页面 | 原文线索

V2-033076 | MATHDB-359669

The volume conjecture for simple generalized hyperbolic polyhedra

摘录 / EXCERPT: The volume conjecture for polyhedra. At $q=e^{\frac{2\pi\sqrt{-1}}{r}}$, as r runs over all the odd integers,

conjecture | open | MathDB 页面 | 原文线索

V2-033077 | MATHDB-359690

The algebraicity conjecture for special values of rigid meromorphic cocycles

摘录 / EXCERPT: Let G be the orthogonal group and let Γ be the arithmetic subgroup under consideration. Write s for the relevant cohomological degree, let $\mathcal{RMC}(\Gamma)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-033078 | MATHDB-359691

The conjectural reciprocity law for rigid meromorphic cocycles

摘录 / EXCERPT: Fix open subgroups $U^p\subseteq G(\mathbb{A}^{p,\infty})$ and $U_p\subseteq G(\mathbb{Q}_p)$ containing $G(\mathbb{Q}_p)^+$, and put $U=U^pU_pG(\mathbb{R})^+$. For a coset...

conjecture | open | MathDB 页面 | 原文线索

V2-033079 | MATHDB-359699

Grant's LS-category conjecture for group homomorphisms

摘录 / EXCERPT: Grant's LS-category conjecture. For any group homomorphism $\phi:\Gamma\to\Lambda$,

conjecture | open | MathDB 页面 | 原文线索

V2-033080 | MATHDB-359717

Changemaker lattice conjecture for integer surgeries

摘录 / EXCERPT: Let L be a definite unimodular lattice, and let $\Lambda(p,q)$ be an L-changemaker lattice. Here $L(p,q)$ denotes the corresponding surgery-related lattice realization. Changem...

conjecture | open | MathDB 页面 | 原文线索

V2-033081 | MATHDB-359724

Weakly-filtered endomorphism algebra conjecture

摘录 / EXCERPT: Let E_0 and E_1 be the objects appearing in the surgery-formula construction, and let $\operatorname{End}_{w\mathrm{Fil}}(\beta_0^{E_0}\oplus\beta_1^{E_1})$ denote the endomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-033082 | MATHDB-359773

The PIC1 open-manifold topology conjecture

摘录 / EXCERPT: Let M be an open, complete, noncompact manifold without boundary whose curvature is PIC1. PIC1 open-manifold topology conjecture. M is diffeomorphic to Euclidean space. This co...

conjecture | open | MathDB 页面 | 原文线索

V2-033083 | MATHDB-359775

Crane-Frenkel categorification conjecture for quantum mathfrak{sl}_2

摘录 / EXCERPT: Crane-Frenkel conjecture. There exists a categorification of the quantum group $U_q(\mathfrak{sl}_2)$ at roots of unity giving rise to a 4D TQFT.

conjecture | open | MathDB 页面 | 原文线索

V2-033084 | MATHDB-359800

Characterizing slopes conjecture for knots

摘录 / EXCERPT: Characterizing slopes conjecture. There exists a constant $C=C(K)$ such that any slope p/q satisfying $|q|\geq 2$ and $|p|+|q|\geq C$ is characterizing for K.

conjecture | open | MathDB 页面 | 原文线索

V2-033085 | MATHDB-359821

Conjecture that geometrical limiting measures are Liouville

摘录 / EXCERPT: Let ${\bf M}$ be a closed hyperbolic 3-manifold. A geometrical limiting measure is a probability measure on ${\mathcal {G}}_2({\bf {M}})$ obtained as a weak limit of geometrical l...

conjecture | open | MathDB 页面 | 原文线索

V2-033086 | MATHDB-359822

Conjecture on the support of topological limiting measures

摘录 / EXCERPT: Let ${\bf M}$ be a closed hyperbolic 3-manifold with a totally geodesic surface. Let $h_0=h_0({\bf M})$ be the genus of a totally geodesic surface of smallest genus in ${\bf M}$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-033087 | MATHDB-359835

Realization conjecture for finite-order obstruction classes of UHF algebra kernels

摘录 / EXCERPT: Let G be a countable discrete group, and let $c\fundin H^3(G,\mathbb{T})$ have non-zero finite order r. A finite-order obstruction realization conjecture asserts that c is re...

conjecture | open | MathDB 页面 | 原文线索

V2-033088 | MATHDB-359862

Chromatic redshift conjecture for complex and quaternionic Mahowald invariants

摘录 / EXCERPT: Chromatic redshift conjecture. If $\alpha \notin M_{\mathbb{K}}(\alpha)$, then $M_{\mathbb{K}}(\alpha)$ consists entirely of v_n-torsion elements. If $w:\Sigma^dY\to Y$ is a powe...

conjecture | open | MathDB 页面 | 原文线索

V2-033089 | MATHDB-359863

Equivariant Mahowald-invariant conjecture for classical compact Lie groups

摘录 / EXCERPT: Equivariant Mahowald-invariant conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-033090 | MATHDB-359875

The generalized Andrews-Curtis conjecture for 2-dimensional CW-pairs

摘录 / EXCERPT: Generalized Andrews-Curtis conjecture. Any two simple-homotopy equivalent 2-dimensional CW-pairs are 2-equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-033091 | MATHDB-359886

Finiteness conjecture for non-destabilizable Legendrian link types

摘录 / EXCERPT: Let $\mathscr L$ be a link type in $\mathbb S^3$. A Legendrian link type is non-destabilizable if it cannot be obtained from another Legendrian link type by a destabilization. Fini...

conjecture | open | MathDB 页面 | 原文线索

V2-033092 | MATHDB-359906

The symmetric-power conjecture for Hochschild homology of smooth proper dg categories

摘录 / EXCERPT: Let $\mathcal{T}$ be a smooth proper dg category, and let $\operatorname{Sym}^n\mathcal{T}$ denote its nth symmetric power in the sense of Ganter-Kapranov. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-033093 | MATHDB-359923

Weak topological rigidity for ${\mathrm{SL}}_2$-quotients

摘录 / EXCERPT: Let G be a simple simply-connected complex algebraic group. For unipotent elements $u,v\in G$, let H_u and H_v be their corresponding subgroups, and write $X_u(G)=G/H_u$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033094 | MATHDB-359924

Strong topological rigidity for ${\mathrm{SL}}_2$-quotients

摘录 / EXCERPT: Let G be a simple simply-connected complex algebraic group. Let $u,v\in G$ be nonidentity unipotent elements, and let $X_u(G)=G/H_u$ and $X_v(G)=G/H_v$. Suppose that ... is a conti...

conjecture | open | MathDB 页面 | 原文线索

V2-033095 | MATHDB-359925

Component-group conjecture for homotopy self-equivalences of ${\mathrm{SL}}_2$-quotients

摘录 / EXCERPT: Let G be a simple simply-connected complex algebraic group, let $u\in G$ be a unipotent element, and set $H=H_u$ and $X_u=G/H$. Every G-equivariant self-map of X_u is induced...

conjecture | open | MathDB 页面 | 原文线索

V2-033096 | MATHDB-359929

The converse of the closed-manifold theorem for regular CW complexes

摘录 / EXCERPT: Let X be a regular CW complex, and let $X^{(n)}$ denote its n-skeleton. Converse of the closed-manifold theorem. If $X=X^{(n)}$, then X is a closed manifold. This would chara...

conjecture | open | MathDB 页面 | 原文线索

V2-033097 | MATHDB-359996

Verjovsky's conjecture on codimension-one Anosov flows

摘录 / EXCERPT: Verjovsky's conjecture. Any codimension-one Anosov flow on a closed manifold of dimension greater than three is topologically equivalent to the suspension flow of a hyperbolic tora...

conjecture | open | MathDB 页面 | 原文线索

V2-033098 | MATHDB-360030

The m-KP integrability conjecture of topological recursion

摘录 / EXCERPT: Let $\mathcal C=(\Sigma,x,y)$ be a spectral curve with boundaries ${\bf b}=\{b_1,\ldots,b_m\}$, and let $\Lambda=(\lambda_1,\ldots,\lambda_m)$ be local coordinates near the boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-033099 | MATHDB-360031

The Laplace-transform conjecture for the J-function

摘录 / EXCERPT: Let $\mathcal C=(\Sigma,x,y)$ be a spectral curve and let $\gamma\subset\Sigma\setminus{\bf b}$ be an admissible path associated with $e^{-x/\mathfrak u}$. Let η be the CohFT...

conjecture | open | MathDB 页面 | 原文线索

V2-033100 | MATHDB-360032

The generalized Witten conjecture for geometric descendent potentials

摘录 / EXCERPT: Let $\mathcal C=(\Sigma,x,y)$ be a spectral curve with m boundaries. Let $\mathcal D({\bf t};\hbar)$ be its geometric total descendent potential, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-033101 | MATHDB-360035

Cohomological Sarnak-Xue density hypothesis

摘录 / EXCERPT: Let G be as above, let $\Pi^{\mathrm{alg}}(G)$ be the finite-dimensional algebraic representations of G, and let $\Pi^{\mathrm{coh}}(G_\infty;E)$ denote the irreducible represe...

conjecture | open | MathDB 页面 | 原文线索

V2-033102 | MATHDB-360037

CSXDH-shape conjecture

摘录 / EXCERPT: Let $E\in\Pi^{\mathrm{alg}}(G)$ and let $\mathcal{M}(G)$ be the set of A-shapes. For $\varsigma\in\mathcal{M}(G)$, let $h(G,\varsigma;q,E)$ be the associated cohomological counti...

conjecture | open | MathDB 页面 | 原文线索

V2-033103 | MATHDB-360064

The higher-dimensional Heegaard Floer-Khovanov correspondence conjecture

摘录 / EXCERPT: Let $\widehat{\sigma}$ be the link associated to a braid σ, and let $Kh^{\sharp,k}(\widehat{\sigma})$ denote the degree-k part of the higher-dimensional Floer homology, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-033104 | MATHDB-360066

Relative equivariant covering type equality

摘录 / EXCERPT: Relative covering type conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033105 | MATHDB-360069

Neighborhood equivariant retract conjecture for twisted products

摘录 / EXCERPT: Let G be a locally compact group, let K be a compact large subgroup of G, and let S be a K-space. The neighborhood equivariant retract conjecture. S is a neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-033106 | MATHDB-360070

Non-commutative crystalline comparison conjecture

摘录 / EXCERPT: Non-commutative crystalline comparison conjecture. There is an isomorphism of B_{crys}-modules

conjecture | open | MathDB 页面 | 原文线索

V2-033107 | MATHDB-360076

Shelukhin's Hofer-Zehnder conjecture for local Floer homology

摘录 / EXCERPT: Shelukhin's Hofer-Zehnder conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-033108 | MATHDB-360106

Untwisted multiplication conjecture for derivative spectra

摘录 / EXCERPT: Untwisted multiplication conjecture. The maps constructed from the monoidal structure make $\widetilde{\Theta^n}F$ into a nonunital E_∞-ring spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-033109 | MATHDB-360110

Davison's stacky PS=WS conjecture

摘录 / EXCERPT: Let $\mathfrak{M}_{Dol}(r,0)$ and $\mathfrak{M}_{B}(r,0)$ be the stacks of semistable Higgs bundles and semisimple representations, respectively. Let $W_\bullet$ be the weight filt...

conjecture | open | MathDB 页面 | 原文线索

V2-033110 | MATHDB-360126

The strengthened Gromov positive simplicial volume conjecture

摘录 / EXCERPT: Let M be a closed, connected, oriented topological n-manifold. Suppose that M admits a Riemannian metric whose sectional curvature is nonpositive everywhere and whose Ricci c...

conjecture | open | MathDB 页面 | 原文线索

V2-033111 | MATHDB-360141

Rank conjecture for symmetric \mathfrak{gl}_1-homology and reduced triply graded homology

摘录 / EXCERPT: Rank conjecture. The total rank of the symmetric $\mathfrak{gl}_1$-homology is equal to the total rank of the reduced triply graded homology.

conjecture | open | MathDB 页面 | 原文线索

V2-033112 | MATHDB-360161

Abelian sector conjecture for the figure-8 region

摘录 / EXCERPT: Let X_8 be the immersed figure-8 annulus in the sphere, let ${\mathcal C}_8$ be the set of superselection sectors detected by X_8, and let d_μ denote the quantum dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-033113 | MATHDB-360162

Strong isomorphism conjecture for immersed regions in two dimensions

摘录 / EXCERPT: Let Ω and Ω' be immersed regions in the sphere $\mathbf{S}^2$, and let $\Sigma(\Omega)$ denote the associated information convex set. Strong isomorphism conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-033114 | MATHDB-360163

Parallel transportation conjecture for immersed annuli

摘录 / EXCERPT: Let ${\mathcal L}$ be an immersed annulus, let ${\mathcal C}_{\mathcal L}$ be the superselection sectors detected by it, and let d_μ be the quantum dimension of μ. Transpor...

conjecture | open | MathDB 页面 | 原文线索

V2-033115 | MATHDB-360164

Strong isomorphism conjecture for immersed regions in arbitrary dimensions

摘录 / EXCERPT: Let Ω and Ω' be immersed regions in the n-sphere $\mathbf{S}^n$, with axioms ${\bf A0}$ and ${\bf A1}$ satisfied everywhere on the reference state. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033116 | MATHDB-360185

Stratified rigidity question for Scott spaces

摘录 / EXCERPT: Let M be a Scott complex or quaternionic manifold with its natural stratification. Let N be a stratified space and let $f:N\to M$ be a stratified homotopy equivalence. Stratifi...

open_problem | open | MathDB 页面 | 原文线索

V2-033117 | MATHDB-360203

Cyclotomic Alexander polynomials for periodic L-space knots

摘录 / EXCERPT: Let K be a periodic or freely periodic L-space knot, and let $\Delta_K(t)$ denote its Alexander polynomial. Cyclotomic Alexander polynomial conjecture. $\Delta_K(t)$ is a product...

conjecture | open | MathDB 页面 | 原文线索

V2-033118 | MATHDB-360204

Quotients of periodic L-space knots are L-space knots

摘录 / EXCERPT: Let K be a periodic L-space knot with quotient knot $\overline{K}$. Quotient L-space knot conjecture. $\overline{K}$ is an L-space knot. The claim concerns how the L-space-knot p...

conjecture | open | MathDB 页面 | 原文线索

V2-033119 | MATHDB-360238

Face invariance of Poincaré series for numerical semigroup algebras

摘录 / EXCERPT: Let C_m be the Kunz cone, let F be a face of C_m, and let S and T be numerical semigroups whose corresponding points lie in F. For a field $\Bbbk$, write $\Bbbk[S]$ and...

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V2-033120 | MATHDB-360246

Fang-Man-Zhang conjecture on finite topological type of shrinking Ricci solitons

摘录 / EXCERPT: Fang-Man-Zhang conjecture. Any shrinking Ricci soliton has finite topological type.

conjecture | open | MathDB 页面 | 原文线索

V2-033121 | MATHDB-360267

Holonomicity conjecture for biresidue matrices

摘录 / EXCERPT: Let $B=(b_{ij})_{i,j=0}^{n-1}$ be a biresidue matrix. It is holonomic if, for every subset $J\subset\{0,1,\dots,n-1\}$ of odd cardinality, the linear span of the rows of the submat...

conjecture | open | MathDB 页面 | 原文线索

V2-033122 | MATHDB-360275

Non-left-orderability conjecture for involutory quandles of quasi-alternating links

摘录 / EXCERPT: Let L be a quasi-alternating link, and let $IQ(L)$ denote its involutory quandle. A quandle is left-orderable if it admits a strict total order invariant under all left translati...

conjecture | open | MathDB 页面 | 原文线索

V2-033123 | MATHDB-360312

Symplectic-quandle separation conjecture for Allen-Swenberg links

摘录 / EXCERPT: Let T be a symplectic quandle, and let the Allen-Swenberg series be the infinite family of links considered in the source, with H the connected sum of two Hopf links. The sympl...

conjecture | open | MathDB 页面 | 原文线索

V2-033124 | MATHDB-360320

The Oblomkov-Rasmussen-Shende conjectural correspondence in affine Springer homology

摘录 / EXCERPT: Let γ be an unramified element with associated link L_{γ}, let Sp_{γ} be its affine Springer fiber, and let gr_P be the specified perverse...

conjecture | open | MathDB 页面 | 原文线索

V2-033125 | MATHDB-360322

Finite-generation conjecture for Coulomb-branch affine Springer homology

摘录 / EXCERPT: Let G be a reductive group, let $\mathcal A_G=\bigoplus_{d=0}^{\infty}{}_0\mathcal A_d$ be the graded Coulomb branch algebra, and for $\gamma\in\mathfrak g$ define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033126 | MATHDB-360323

Equivariant formality conjecture for all diagonal affine Springer fibers

摘录 / EXCERPT: Let ... where $z_i\in\mathbb{C}^*$ are pairwise distinct and $d_i\geq 0$. Equivariant formality conjecture. The affine Springer fiber Sp_{γ} is equivariantly formal fo...

conjecture | open | MathDB 页面 | 原文线索

V2-033127 | MATHDB-360329

Non-loose realizations from non-zero hat Legendrian invariants

摘录 / EXCERPT: Non-loose realization conjecture. If the hat Legendrian invariant of a Legendrian realization is non-zero, then the construction produces a non-loose realization in an overtwisted...

conjecture | open | MathDB 页面 | 原文线索

V2-033128 | MATHDB-360377

The aspect-ratio conjecture for smooth embedded multi-twisted paper Möbius bands

摘录 / EXCERPT: A multi-twisted paper Möbius band is a paper Möbius band with at least two twists, and its aspect ratio is its length divided by its width. The multi-twisted aspect-ratio conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-033129 | MATHDB-360387

Celoria-Friedl-Nagel-Orson-Powell almost-concordance conjecture

摘录 / EXCERPT: Let M be a closed orientable 3-manifold with $M\neq S^3$, and let $x\in[S^1,M]$ be a free homotopy class. A dual 2-sphere for x is a 2-sphere in M representing the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-033130 | MATHDB-360395

The characteristic-rank conjecture for oriented Grassmannians

摘录 / EXCERPT: Characteristic-rank conjecture. The characteristic rank is

conjecture | open | MathDB 页面 | 原文线索

V2-033131 | MATHDB-360414

Integrality and eventual vanishing conjecture for sums of inverse adjoint torsions

摘录 / EXCERPT: Fix a knot K, and let $\tau_{p/r}$ denote the adjoint torsions associated with the p/r surgery on K. Inverse-torsion sum conjecture. The sum of the inverse torsions is always...

conjecture | open | MathDB 页面 | 原文线索

V2-033132 | MATHDB-360438

Generalization of the trefoil-parameter template conjecture for the Rössler system

摘录 / EXCERPT: Generalized trefoil-parameter conjecture. There exists a neighborhood $O\subseteq P$ of p such that, for every $v\in O$, there is a smooth vector field G_v, hyperbolic on its n...

conjecture | open | MathDB 页面 | 原文线索

V2-033133 | MATHDB-360440

Complexity conjecture for templates generated by heteroclinic Rössler knots

摘录 / EXCERPT: Heteroclinic-knot complexity conjecture. The more Λ is knotted with itself, the more knot types occur in the template T_p; in particular, every knot type on T_p, save p...

conjecture | open | MathDB 页面 | 原文线索

V2-033134 | MATHDB-360466

Takahashi's vanishing conjecture for DHKK complexities in singularity categories

摘录 / EXCERPT: Let R be a commutative noetherian ring. Denote by $\operatorname{D_{sg}}(R)$ the singularity category of R, defined as the Verdier quotient of the bounded derived category of f...

conjecture | open | MathDB 页面 | 原文线索

V2-033135 | MATHDB-360493

Čech-de Rham isomorphism for non-Hausdorff manifolds

摘录 / EXCERPT: Čech-de Rham conjecture. For every $q\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-033136 | MATHDB-360498

Finite-type conjecture for symmetric powers of Rep(C_2)-complexes

摘录 / EXCERPT: Let $G=C_2$. A $\operatorname{Rep}(C_2)$-complex is a C_2-space built by attaching cells of the form $D(\mathbf R^{p,q})$, where $p\geq q\geq0$; it is of finite type if, for each...

conjecture | open | MathDB 页面 | 原文线索

V2-033137 | MATHDB-360550

Special alternating knots are ribbon concordance minimal

摘录 / EXCERPT: Let K and K' be knots in S^3, and write $K'\leq K$ when there is a ribbon concordance from K to K'. A knot K is ribbon concordance minimal if $K'\leq K$ implies that...

conjecture | open | MathDB 页面 | 原文线索

V2-033138 | MATHDB-360564

Q-sequence conjecture for systems of parameters

摘录 / EXCERPT: Let R be a Noetherian local ring of dimension d. A sequence $x_1,\ldots,x_d$ is a system of parameters when it generates an ideal whose radical is the maximal ideal; more gener...

conjecture | open | MathDB 页面 | 原文线索

V2-033139 | MATHDB-360585

The NIP Hausdorffness conjecture for Ellis groups

摘录 / EXCERPT: NIP Hausdorffness conjecture. If M has NIP, then $u\mathcal{M}$ is Hausdorff.

conjecture | open | MathDB 页面 | 原文线索

V2-033140 | MATHDB-360677

Schleimer's distance formula conjecture for multiarc and curve graphs

摘录 / EXCERPT: Let Σ be a compact, connected, orientable surface, and let $\mathcal A$ be a multiarc and curve graph on Σ. For a witness W, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033141 | MATHDB-360734

The shuffle-algebra image conjecture for the preprojective CoHA

摘录 / EXCERPT: Shuffle-algebra image conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033142 | MATHDB-360744

The exceptional Thurston-Bennequin conjecture for the knot 9_{42}

摘录 / EXCERPT: For a strongly invertible Legendrian knot, let its Thurston-Bennequin number be denoted by tb, let $\overline{\operatorname{tb}}(K)$ be the maximal Thurston-Benneq...

conjecture | open | MathDB 页面 | 原文线索

V2-033143 | MATHDB-360745

The $m(10_{125})$ strongly invertible Legendrian representative conjecture

摘录 / EXCERPT: Let $\mu(\overrightarrow{\mathscr K})$ denote the Legendrian mirror of an oriented Legendrian knot $\overrightarrow{\mathscr K}$, and let $-\mu(\mathscr K)$ denote its reversed Leg...

conjecture | open | MathDB 页面 | 原文线索

V2-033144 | MATHDB-360746

The 8_5 rotation-number conjecture for strongly invertible Legendrian knots

摘录 / EXCERPT: Let $\overrightarrow{\mathscr K}$ be a strongly invertible oriented Legendrian knot of type 8_5, and let $\operatorname{tb}(\overrightarrow{\mathscr K})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033145 | MATHDB-360751

Rickard's standardness conjecture for derived equivalences of flat algebras

摘录 / EXCERPT: Rickard's conjecture. Any derived equivalence between flat algebras is standard.

conjecture | open | MathDB 页面 | 原文线索

V2-033146 | MATHDB-360764

Rubin's saturation conjecture for linear isometries operads

摘录 / EXCERPT: Rubin's saturation conjecture. For every n there exists some $f(n)$ such that, for any collection of distinct primes $p_1,\ldots,p_n\geqslant f(n)$ and any choice of...

conjecture | open | MathDB 页面 | 原文线索

V2-033147 | MATHDB-360778

Implication of homological mirror symmetry

摘录 / EXCERPT: Let X_b be a fiber of a Calabi-Yau threefold family and let $\hat X_0$ be its mirror. Write $F_B^\bullet$ and $F_A^\bullet$ for the Hodge filtrations on the respective stalks, le...

conjecture | open | MathDB 页面 | 原文线索

V2-033148 | MATHDB-360801

Fausk-Lewis-May conjecture on homotopy representations

摘录 / EXCERPT: Fausk-Lewis-May conjecture. The canonical map

conjecture | open | MathDB 页面 | 原文线索

V2-033149 | MATHDB-360807

C^0 lower semi-continuity conjecture for entropy of Reeb flows

摘录 / EXCERPT: Reeb-flow entropy stability conjecture. The functional $h_{\rm top}$ is lower semi-continuous with respect to d_{C^0}.

conjecture | open | MathDB 页面 | 原文线索

V2-033150 | MATHDB-360808

C^0 lower semi-continuity conjecture for entropy of surface metrics

摘录 / EXCERPT: Surface-metric entropy stability conjecture. The functional $h_{\rm top}$ is lower semi-continuous with respect to $\overline{d}_{C^0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-033151 | MATHDB-360813

The Cochran-Harvey-Leidy fractal conjecture for knot concordance groups

摘录 / EXCERPT: Let $\mathcal{C}^\mathrm{diff}$ and $\mathcal{C}^\mathrm{top}$ denote the smooth and topological knot concordance groups. A set has a fractal structure if it admits self-similariti...

conjecture | open | MathDB 页面 | 原文线索

V2-033152 | MATHDB-360835

Bonahon-Wong-Yang volume conjecture for the four-puncture sphere

摘录 / EXCERPT: Bonahon-Wong-Yang volume conjecture. In the above setup,

conjecture | open | MathDB 页面 | 原文线索

V2-033153 | MATHDB-360843

Action of the deformed W-algebra beyond tautological classes

摘录 / EXCERPT: Let X be a two-sided Hecke pattern of rank r. Write $H_*(X)$ for its Borel-Moore homology, and let $W_{\star,\star}(S)$ denote the algebra with the appropriate choices of indic...

conjecture | open | MathDB 页面 | 原文线索

V2-033154 | MATHDB-360856

The hyperbolic L-space conjecture for universally tight contact structures

摘录 / EXCERPT: Hyperbolic L-space conjecture. M is an L-space if and only if it does not support a universally tight contact structure.

conjecture | open | MathDB 页面 | 原文线索

V2-033155 | MATHDB-360857

Logarithmic primary PT/GW correspondence

摘录 / EXCERPT: Let (X,D) be the logarithmic pair, let β be a curve class, and set ... Let $\zeta_1,\ldots,\zeta_k$ be cohomology insertions and let $\bm{\mu}(\delta)$ be relative conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-033156 | MATHDB-360858

Logarithmic descendent PT/GW correspondence

摘录 / EXCERPT: Let (X,D) be the logarithmic pair, let β be a curve class, and set ... Let $\alpha=(\alpha_1,\ldots,\alpha_k)$ be a partition, let $\zeta_1,\ldots,\zeta_k$ be cohomology clas...

conjecture | open | MathDB 页面 | 原文线索

V2-033157 | MATHDB-360859

Seidel's long exact sequence conjecture for composed Dehn twists

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold of dimension $2d$ with $2c_1(M,\omega)=0$, and let $L_1,\ldots,L_n$ be graded Lagrangian spheres. Let $\mathcal{A}$ be the directed Fukaya...

conjecture | open | MathDB 页面 | 原文线索

V2-033158 | MATHDB-360863

Functoriality of the L-infinity structure under exact Lagrangian cobordisms

摘录 / EXCERPT: Functoriality conjecture. There is an L_∞ morphism from the L_∞ algebra $((\mathcal{A}^+)^{\operatorname{comm}},\{\ell_k^+\})$ of Λ^+ to the L_∞ algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-033159 | MATHDB-360882

Gromov--Lawson conjecture on positive scalar curvature of aspherical manifolds

摘录 / EXCERPT: Let M^n be a closed, aspherical manifold, meaning that $\pi_k(M)=0$ for every $k\geq 2$. Gromov--Lawson conjecture. The manifold M admits no Riemannian metric with positive sca...

conjecture | open | MathDB 页面 | 原文线索

V2-033160 | MATHDB-360897

Global mirror symmetry via descent for symplectic cohomology

摘录 / EXCERPT: Let M be a closed graded manifold with a weakly involutive cover $M=\bigcup_{i=1}^N C_i$, and let $Y=\bigcup_{i=1}^N Y_i$ be a smooth rigid analytic space over Λ with an...

conjecture | open | MathDB 页面 | 原文线索

V2-033161 | MATHDB-360903

Bowden's bounded acyclicity conjecture for the stable mapping class group

摘录 / EXCERPT: Bowden's conjecture. The stable mapping class group Γ_∞ is boundedly acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-033162 | MATHDB-360906

Adelic Zariski dense orbit conjecture

摘录 / EXCERPT: Adelic Zariski dense orbit conjecture. The stated adelic-open-set conclusion holds.

conjecture | open | MathDB 页面 | 原文线索

V2-033163 | MATHDB-360920

Homology isomorphism conjecture for the simplex embedding

摘录 / EXCERPT: Homology isomorphism conjecture. The embedding $\iota:\Delta_n^{d-1}\hookrightarrow F_{n,d}$ defines an isomorphism on homology. The conjecture would identify the homology of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-033164 | MATHDB-360921

Homotopy equivalence conjecture for Liouville stops

摘录 / EXCERPT: Let $\Lambda^{(d)}_n$ be the stop formed from the $n+1$ points of Λ_n by taking their products with $\operatorname{Sym}^{d-1}(\mathbb{D})$, and let $F_{n,d}$ be the symplec...

conjecture | open | MathDB 页面 | 原文线索

V2-033165 | MATHDB-360925

The universal cork conjecture

摘录 / EXCERPT: Universal cork conjecture. There is no universal cork, i.e. no cork (U,f_U) relating every exotic pair of simply-connected, closed 4-manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-033166 | MATHDB-360926

Generation conjecture for the HDHF wrapped Fukaya category of cotangent bundles

摘录 / EXCERPT: Generation conjecture. The category $\mathcal{W}_\kappa(T^*Q)$ is split-generated by any κ-tuple of disjoint cotangent fibers.

conjecture | open | MathDB 页面 | 原文线索

V2-033167 | MATHDB-360928

Refined BPS formula for hypergeometric-type spectral curves

摘录 / EXCERPT: Refined BPS free-energy conjecture. The refined topological recursion free energy satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033168 | MATHDB-360958

The eternal quotient dichotomy for symplectic cohomology

摘录 / EXCERPT: Let W be a convex-at-infinity symplectic manifold. For each contact isotopy φ_t, let $\mathrm{HF}(\varphi_t)$ denote its Floer cohomology, and let $\mathrm{SH}^e(W)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-033169 | MATHDB-360972

The volume conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let $\|K\|$ denote its simplicial volume, known here to be the sum of the hyperbolic volumes of the hyperbolic pieces in the knot complement. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033170 | MATHDB-361021

The Weak Borel conjecture for closed aspherical manifolds

摘录 / EXCERPT: Let M and N be closed aspherical manifolds with isomorphic fundamental groups, and let $\widetilde{M}$ and $\widetilde{N}$ denote their universal covers. Weak Borel conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-033171 | MATHDB-361022

The aspherical manifold stability conjecture

摘录 / EXCERPT: Let M be a closed aspherical manifold and let $\widetilde{M}$ be its universal cover. A manifold is inward tame if it has arbitrarily small neighborhoods of infinity that are fin...

conjecture | open | MathDB 页面 | 原文线索

V2-033172 | MATHDB-361023

Freedman's controlled Mather-Thurston conjecture

摘录 / EXCERPT: Let $(V,\partial V)$ be a manifold with $\dim(V)\geq 3$. Let $X\hookrightarrow B\to V$ be a bundle with structure group $\operatorname{Homeo}(X)$, where X is a manifold with a po...

conjecture | open | MathDB 页面 | 原文线索

V2-033173 | MATHDB-361104

Stoimenow's unknotting-number conjecture for positive fibered knots

摘录 / EXCERPT: Let K be a positive fibered knot. Its unknotting number is denoted by $u(K)$, and its 3-genus by $g(K)$. Stoimenow's conjecture. ... The conjecture extends the equality known for b...

conjecture | open | MathDB 页面 | 原文线索

V2-033174 | MATHDB-361105

The unknotting-number conjecture for the family \(\mathcal{K}_n\)

摘录 / EXCERPT: For each integer $n\geq 1$, let $\mathcal{K}_n$ be the knot represented by the positive diagram D_n in the paper's infinite family. The family unknotting-number conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-033175 | MATHDB-361185

Van Engelen's Borel classification conjecture for zero-dimensional spaces

摘录 / EXCERPT: Van Engelen's conjecture. The same equivalence holds for every zero-dimensional Borel space that is not locally compact.

conjecture | open | MathDB 页面 | 原文线索

V2-033176 | MATHDB-361241

Bosonization conjecture for anomaly theories

摘录 / EXCERPT: Let G be a fermionic symmetry acting on a super modular tensor category, with anomaly $\tilde\alpha$. Let $F_{\tilde\alpha}$ be the Dijkgraaf-Witten-type theory obtained by summi...

conjecture | open | MathDB 页面 | 原文线索

V2-033177 | MATHDB-361243

Invertibility transfer from bosonic convolution to fermionic tensor product

摘录 / EXCERPT: Let $\tilde\beta$ be the bosonic-shadow theory and $\tilde\alpha$ the corresponding fermionic anomaly theory, with a conjectural defect $\tilde z_c$ relating them. Invertibility-tr...

conjecture | open | MathDB 页面 | 原文线索

V2-033178 | MATHDB-361267

Arnol'd's Lagrangian intersection conjecture

摘录 / EXCERPT: Let L be a closed smooth manifold and let T^*L denote its cotangent bundle. For every compactly supported Hamiltonian diffeomorphism ϕ of T^*L, consider the intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-033179 | MATHDB-361295

The invariant-homology vanishing conjecture for free Allison-Benkart-Gao Lie algebras

摘录 / EXCERPT: Invariant-homology conjecture. As $sl_3(k)$-modules,

conjecture | open | MathDB 页面 | 原文线索

V2-033180 | MATHDB-361302

The simplicial freeness conjecture for the Brunnian planar braid subgroup

摘录 / EXCERPT: Let SPT_* be the simplicial group of simplicial pure twin groups, let c_{111} be the 2-simplex with trivial faces, and let K_* be the smallest simplicial subgroup of SPT_*...

conjecture | open | MathDB 页面 | 原文线索

V2-033181 | MATHDB-361314

Hock's x-y swap conjecture for rational spectral curves

摘录 / EXCERPT: Let $\Sigma=\mathbb{C}\mathrm{P}^1$. In Case 1, let $y(z)=z$ be a global affine coordinate and let $x=\log R(z)+Az$, where $R(z)$ is a rational function and $A\in\mathbb{C}$. In Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-033182 | MATHDB-361322

The Lattice Conjecture for smooth proper dg-categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a $\mathbb{C}$-linear dg-category. Write $K_*^{\operatorname{top}}(\mathcal{C})$ for its topological K-theory groups and $HP_*(\mathcal{C})$ for its periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-033183 | MATHDB-361345

Density of 1-unstable periodic points for coupled tent maps

摘录 / EXCERPT: Let $f_a\colon\mathbb R\to\mathbb R$ be the skew tent map and let the coupled tent map have coupling parameter ω. Its topological attractor is the closed invariant set conta...

conjecture | open | MathDB 页面 | 原文线索

V2-033184 | MATHDB-361349

The topological recursion implication for universal equations

摘录 / EXCERPT: Let ... is a consequence of the degree-g topological recursion relations proposed in the cited work. This proposed connection relates universal equations for higher-genus Gromov-Wi...

conjecture | open | MathDB 页面 | 原文线索

V2-033185 | MATHDB-361427

Haefliger-Thurston connectivity conjecture for trivial-normal Haefliger structures

摘录 / EXCERPT: Recall that $\overline{\mathrm{B}\Gamma}^r_k$ classifies codimension k C^r-Haefliger structures with trivial normal bundle. Haefliger-Thurston's conjecture. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-033186 | MATHDB-361428

Haefliger-Thurston homological conjecture for diffeomorphism groups

摘录 / EXCERPT: Let M be an n-manifold, and let $\mathrm{BDiff}_c^{r}(M)^{\delta}$ denote the classifying space of the discrete group of compactly supported C^r diffeomorphisms, while...

conjecture | open | MathDB 页面 | 原文线索

V2-033187 | MATHDB-361464

Atiyah conjecture for torsion-free countable groups

摘录 / EXCERPT: Let G be a torsion-free countable group, and let ${\mathcal{D}}(\mathbb{C}G\subset {\mathcal{U}}G)$ denote the ring appearing in the Atiyah-conjecture construction. Atiyah conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-033188 | MATHDB-361472

Homology-preservation conjecture for link-detected barycentric subcomplexes

摘录 / EXCERPT: Let Δ be a simplicial complex. For an integer $i\geq -1$, let $\mathcal{L}_{i}(\Delta)$ be the induced subcomplex of the barycentric subdivision $\mathcal{B}(\Delta)$ on the...

conjecture | open | MathDB 页面 | 原文线索

V2-033189 | MATHDB-361473

Homology-preservation conjecture for the full link-detected barycentric subcomplex

摘录 / EXCERPT: Let Δ be a simplicial complex, and let $\mathcal{L}(\Delta)$ be the induced subcomplex of the barycentric subdivision $\mathcal{B}(\Delta)$ on the vertices u_σ for wh...

conjecture | open | MathDB 页面 | 原文线索

V2-033190 | MATHDB-361478

Tom Dieck-Petrie conjecture for homology planes

摘录 / EXCERPT: Let X be a homology plane, meaning a smooth quasi-projective surface with the same integral homology as $\mathbb{C}^2$, of log-general type, and let $(\bar{X},D)$ be a minimal SN...

conjecture | open | MathDB 页面 | 原文线索

V2-033191 | MATHDB-361484

Antieau-Gepner-Heller connective K-theory conjecture for bounded t-structures

摘录 / EXCERPT: Let $\sC$ be a small stable ∞-category equipped with a bounded t-structure. A spectrum is connective when its negative homotopy groups vanish. Antieau-Gepner-Heller's connec...

conjecture | open | MathDB 页面 | 原文线索

V2-033192 | MATHDB-361488

Compact-axis conjecture for charged symmetric black holes

摘录 / EXCERPT: Consider ${\rm S}^{1}$-symmetric black holes whose fixed-point set, called the axis, is compact, and suppose the black hole has ends at which its charge is nonzero. Compact-axis bl...

conjecture | open | MathDB 页面 | 原文线索

V2-033193 | MATHDB-361518

The topological-recursion tau-function conjecture for Fay identities

摘录 / EXCERPT: Let ${\cal S}$ be a spectral curve with topological-recursion invariants $F_g({\cal S})$ and $\omega_{g,n}({\cal S})$, and let $\hbar$ be a small parameter. Define the formal tau f...

conjecture | open | MathDB 页面 | 原文线索

V2-033194 | MATHDB-361547

Lê-Sikora's injectivity conjecture for splitting maps

摘录 / EXCERPT: Lê-Sikora's injectivity conjecture. For any pb surface Σ and any ideal arc c, the splitting map Θ_c is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-033195 | MATHDB-361548

Conjecture on finite-to-one maximal equicontinuous factors of minimal actions

摘录 / EXCERPT: Let G be an amenable group, ${G \curvearrowright X}$ a minimal action, and let $k\geq 2$. Write ${\rm IT}_k(X,G)$ for the set of independence tuples of length k, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033196 | MATHDB-361567

Nonnegativity conjecture for double Schubert structure constants

摘录 / EXCERPT: Nonnegativity conjecture. For all u,v,w, $c_{uv}^w(y;z)$ is a polynomial in the differences y_i-z_j with nonnegative integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-033197 | MATHDB-361572

Abouzaid-Blumberg's framed flow-category equivalence conjecture

摘录 / EXCERPT: Let $\Flow^{fr}$ be the homotopy category of framed flow categories, whose objects are framed flow categories and whose morphisms are framed morphisms up to framed bordism. Abouzai...

conjecture | open | MathDB 页面 | 原文线索

V2-033198 | MATHDB-361573

Abouzaid-Blumberg's truncated flow-category equivalence conjecture

摘录 / EXCERPT: Fix an integer n. Let $\Flow_{\tau_{\leq n}}$ be the category of flow categories with morphisms given by homotopy classes of $\tau_{\leq n}$-morphisms, where $\tau_{\leq n}$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-033199 | MATHDB-361576

May's equivariant Landweber exactness conjecture

摘录 / EXCERPT: Let E) be a Landweber exact spectrum, let A be a compact abelian Lie group, and let MU_A denote equivariant complex cobordism for A. The relative smash product...

conjecture | open | MathDB 页面 | 原文线索

V2-033200 | MATHDB-361577

Global May functor existence conjecture

摘录 / EXCERPT: Let E be a Landweber exact spectrum, and let $\mathbf{MU}$ denote the global equivariant analogue of complex cobordism. A global version of May's functor would assign to E the...

conjecture | open | MathDB 页面 | 原文线索

V2-033201 | MATHDB-361578

Fixed-point homotopy-orbit Landweber exactness conjecture

摘录 / EXCERPT: Let A be a compact abelian Lie group, let B be a subgroup of A, let E be a Landweber exact spectrum, and let Φ^B denote geometric fixed points. For an A-space or A...

conjecture | open | MathDB 页面 | 原文线索

V2-033202 | MATHDB-361579

May's equivariant complex K-theory comparison conjecture

摘录 / EXCERPT: Let G be a compact Lie group, let MU_G denote G-equivariant complex cobordism, and let KU_G denote equivariant complex K-theory. The relative smash product...

conjecture | open | MathDB 页面 | 原文线索

V2-033203 | MATHDB-361580

Auslander's super-prolongation conjecture for downstream structures

摘录 / EXCERPT: Auslander's super-prolongation conjecture. The relation P is a downstream structure and is the smallest downstream structure. In particular, every downstream structure P' based...

conjecture | open | MathDB 页面 | 原文线索

V2-033204 | MATHDB-361581

Monotonicity conjecture for stream-structure graphs

摘录 / EXCERPT: Monotonicity conjecture. If $N_1\thicksim_S N_2$ in Down, then $N_1^*\thicksim_S N_2^*$ in Down^*; equivalently, $Gamma$ maps into $Gamma^*$.

conjecture | open | MathDB 页面 | 原文线索

V2-033205 | MATHDB-361590

Topological specialization conjecture for mock Alexander polynomials

摘录 / EXCERPT: Topological specialization conjecture. The two mock Alexander polynomials satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-033206 | MATHDB-361605

Putman's non-isomorphism conjecture for twisted cohomology maps

摘录 / EXCERPT: Let $r\ge 2$, and consider the map ... Here the map is induced by the covering map from the regular cover $S_{\mathcal{D}}\to\Sigma_{g,p}^b$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-033207 | MATHDB-361627

Homeomorphism conjecture for Milnor fibers of ambient-homeomorphic hypersurface singularities

摘录 / EXCERPT: Let two hypersurface singularities be ambient-homeomorphic if their ambient spaces and singularities are related by an ambient homeomorphism, and let F_f denote the Milnor fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-033208 | MATHDB-361640

Coincidence of analog and distributional category and topological complexity

摘录 / EXCERPT: Coincidence conjecture. The analog and distributional notions of category and topological complexity coincide on the class of metrizable spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-033209 | MATHDB-361658

Existence of a knot with distinct minimum and maximum thin-position heights

摘录 / EXCERPT: Height variation conjecture. There is a knot K such that

conjecture | open | MathDB 页面 | 原文线索

V2-033210 | MATHDB-361659

Cohomological lower-bound conjecture for distributional sectional category

摘录 / EXCERPT: Proposed cohomological lower bound. If

conjecture | open | MathDB 页面 | 原文线索

V2-033211 | MATHDB-361665

The infinite quasi-flat conjecture for Hamiltonian groups of Liouville domains

摘录 / EXCERPT: Let (W,ω) be a Liouville domain, and let ${\rm SH}^*(W,\omega)$ denote its symplectic homology. Equip $\operatorname{Ham}(W,\omega)$ with the metric d_{γ}. Infinite q...

conjecture | open | MathDB 页面 | 原文线索

V2-033212 | MATHDB-361722

The cohomology conjecture for blow-up Whitney forms

摘录 / EXCERPT: Let M be a PL manifold with a given triangulation, and let the blow-up Whitney forms on this triangulation be equipped with the continuity conditions described above. Blow-up Whi...

conjecture | open | MathDB 页面 | 原文线索

V2-033213 | MATHDB-361743

Equivariant monoid realization conjecture for G-connected based G-spaces

摘录 / EXCERPT: Equivariant monoid realization conjecture. Any G-connected based G-space X is weakly equivalent to

conjecture | open | MathDB 页面 | 原文线索

V2-033214 | MATHDB-361751

Nontriviality of the BV cycle in the Pachner complex

摘录 / EXCERPT: Let c_Δ^k be the BV cycle defined by the displayed construction referred to as (cDelta^n). BV-cycle nontriviality conjecture. The cycle c_Δ^k is non...

conjecture | open | MathDB 页面 | 原文线索

V2-033215 | MATHDB-361752

Pachner complex model for Zwiebach's moduli space

摘录 / EXCERPT: Let Σ be a surface of genus h with n boundary circles, whose boundary circles are triangulated into $k_1,\ldots,k_n$ intervals. Pachner-complex conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-033216 | MATHDB-361753

Bonahon-Wong-Yang volume conjecture for quantum invariants of surface diffeomorphisms

摘录 / EXCERPT: Let $\varphi:\Sigma\to\Sigma$ be a pseudo-Anosov diffeomorphism, let M_φ be its mapping torus, and let r be a generic φ-invariant character in the smooth part of...

conjecture | open | MathDB 页面 | 原文线索

V2-033217 | MATHDB-361754

Generalized Bonahon-Wong-Yang volume conjecture for hyperbolic cone mapping tori

摘录 / EXCERPT: Let $\varphi:\Sigma\to\Sigma$ be a pseudo-Anosov diffeomorphism with mapping torus M_φ. For each puncture v, let $m_v^{(n)}\in\{-(n-1)/2,-(n-3)/2,\ldots,(n-1)/2\}$ be a s...

conjecture | open | MathDB 页面 | 原文线索

V2-033218 | MATHDB-361760

Reduced Algebraic Montgomery-Yang conjecture for rational homology projective planes

摘录 / EXCERPT: Let S be a rational homology $\mathbb{CP}^2$ with at most 4 cyclic singularities, and let S^0 denote its smooth locus. Reduced Algebraic Montgomery-Yang conjecture. If S^0...

conjecture | open | MathDB 页面 | 原文线索

V2-033219 | MATHDB-361812

The random-curve taut smoothing conjecture

摘录 / EXCERPT: Random-curve taut smoothing conjecture. A random curve has a taut smoothing.

conjecture | open | MathDB 页面 | 原文线索

V2-033220 | MATHDB-361813

The multi-arc smoothing lower-bound conjecture

摘录 / EXCERPT: Multi-arc smoothing lower-bound conjecture. The multi-arc $calpha$ has a smoothing that cannot be homotoped to a multi-arc with fewer than

conjecture | open | MathDB 页面 | 原文线索

V2-033221 | MATHDB-361818

Limiting root-measure conjecture for random positive 3-braids

摘录 / EXCERPT: Limiting root-measure conjecture. There is a compactly supported measure ν_∞ on $\mathbb{C}$ such that, for almost every sequence $(g_i)_{i\in\mathbb{N}}\in\Omega$,...

conjecture | open | MathDB 页面 | 原文线索

V2-033222 | MATHDB-361819

Positive-coefficient conjecture for Alexander polynomials of positive 3-braids

摘录 / EXCERPT: Positive-coefficient conjecture. Every coefficient of h is positive:

conjecture | open | MathDB 页面 | 原文线索

V2-033223 | MATHDB-361820

The revised rational HK-conjecture for étale groupoids

摘录 / EXCERPT: Let $\mathcal{G}$ be a second countable, locally compact, Hausdorff, étale groupoid, and let $\widehat{\mathcal{G}}$ be its blow-up at torsion elements of isotropy groups. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-033224 | MATHDB-361821

The general HK-conjecture for étale groupoids

摘录 / EXCERPT: Let $\mathcal{G}$ be a second countable, étale groupoid, and let $\widehat{\mathcal{G}}$ be the blow-up groupoid with base-related notation $\widehat{X}$. Suppose that c-...

conjecture | open | MathDB 页面 | 原文线索

V2-033225 | MATHDB-361868

Ishii-Morton Gram determinant conjecture for the Möbius-band type $(Mb)_1$

摘录 / EXCERPT: Let $n\geq 1$, let $D_n^{(Mb)_1}$ denote the Gram determinant of type $(Mb)_1$, let T_k be the kth Chebyshev polynomial of the first kind, and set $d=-A^2-A^{-2}$. Ishii-Morton...

conjecture | open | MathDB 页面 | 原文线索

V2-033226 | MATHDB-361869

The determinant conjecture for the remaining Möbius-band block

摘录 / EXCERPT: Let $n\geq 2$, let $\widetilde{G}_{n}^{Mb_{n,1}}$ denote the remaining block in the decomposition of the Gram matrix of type $(Mb)_1$, and let T_{2k} be the Chebyshev polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-033227 | MATHDB-361883

Finiteness conjecture for spectral lifts of powers of Hogancamp's endomorphisms

摘录 / EXCERPT: For each $n\geq 2$, let P_n be the categorified Jones-Wenzl projector and let $U_n\colon \mathrm{q}^{2n}\Sigma^{2n-2}P_n\to P_n$ be Hogancamp's chain map. For each power U_n^k,...

conjecture | open | MathDB 页面 | 原文线索

V2-033228 | MATHDB-361885

Conjectural Poincaré polynomials for the Khovanov skein lasagna module of the negative projective plane

摘录 / EXCERPT: Let $K_n(t,q)$ denote the Poincaré polynomial of $Kh(T(n,n-1);\mathbb{Q})$ for $n>0$. The Khovanov skein lasagna modules in degree 0 and with boundary labels 0 and 1 are deno...

conjecture | open | MathDB 页面 | 原文线索

V2-033229 | MATHDB-361886

Stein 2-handlebody conjecture for Khovanov skein lasagna invariants

摘录 / EXCERPT: Let K be a knot, let $TB(K)$ denote its Thurston-Bennequin number, and let $X=X_n(K)$ be the n-trace on K with $n<TB(K)$. Let $s(X;0)$ and $s(X;1)$ be the corresponding lasag...

conjecture | open | MathDB 页面 | 原文线索

V2-033230 | MATHDB-361899

Asymptotic Gamma-class conjecture for monotone symplectic manifolds

摘录 / EXCERPT: Let X be a monotone symplectic manifold. For Chern roots x_j of TX, let u be the equivariant parameter, let $J_X^d=ev^d_*\left(\frac{1}{1-u\psi}\right)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033231 | MATHDB-361900

Asymptotic J-function conjecture for monotone symplectic manifolds

摘录 / EXCERPT: Asymptotic J-function conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-033232 | MATHDB-361925

Total rank conjecture for finite-length modules

摘录 / EXCERPT: Let $(R,\mathfrak{m},k)$ be a d-dimensional Noetherian local ring, and let M be a finitely generated nonzero R-module. Suppose that M has finite length and finite projectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-033233 | MATHDB-361948

Topological realization in geodesic balls of spherical and hyperbolic space

摘录 / EXCERPT: Let B be any geodesic ball in $\mathbb{S}^3_+$ or $\mathbb{H}^3$. Geodesic-ball realization conjecture. The ball B admits a free boundary minimal surface of every prescribed or...

conjecture | open | MathDB 页面 | 原文线索

V2-033234 | MATHDB-362006

Rank-three Khovanov homology classification for null-homologous knots in \(\mathbb{RP}^3\)

摘录 / EXCERPT: Let K be a null-homologous knot in $\mathbb{RP}^3$. Write $\widetilde{\operatorname{Kh}}(K;\mathbb{Z}/2)$ for its reduced Khovanov homology over $\mathbb{Z}/2$. Rank-three Khovan...

conjecture | open | MathDB 页面 | 原文线索

V2-033235 | MATHDB-362007

Strict rank inequality between Khovanov and instanton Floer homology for non-local knots in \(\mathbb{RP}^3\)

摘录 / EXCERPT: Let K be a non-local null-homologous knot in $\mathbb{RP}^3$. Let $\widetilde{\operatorname{Kh}}(K;\mathbb{Z}/2)$ denote reduced Khovanov homology over $\mathbb{Z}/2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033236 | MATHDB-362017

The ADE Link Conjecture for cyclic branched covers

摘录 / EXCERPT: The ADE Link Conjecture. If L is not an ADE link, then all cyclic branched covers $cSigma_{\psi}(L)$ of L are NLS, LO, and CTF.

conjecture | open | MathDB 页面 | 原文线索

V2-033237 | MATHDB-362026

Emerton's conjecture on the cohomology of the modular tower

摘录 / EXCERPT: Emerton's conjecture. If ρ is unramified outside finitely many places, then

conjecture | open | MathDB 页面 | 原文线索

V2-033238 | MATHDB-362027

Generalized Emerton factorization conjecture for completed cohomology

摘录 / EXCERPT: Generalized Emerton conjecture. For every finite set S, there is a factorization

conjecture | open | MathDB 页面 | 原文线索

V2-033239 | MATHDB-362041

BV double conjectures for Rabinowitz loop homology and symplectic homology

摘录 / EXCERPT: Let Q be a manifold, let W be a Liouville domain of dimension $2n$, and let $\widehat{\mathbb{H}}_*\Lambda$, $\overline{\mathbb{H}}_*\Lambda$, $S\mathbb{H}_*({\partial}W)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-033240 | MATHDB-362042

Factorizable Lie bialgebra conjectures for equivariant Rabinowitz homology

摘录 / EXCERPT: Let Q be a manifold and let W be a Liouville domain. Write $\widehat{\mathbb{H}}_*^{S^1}\Lambda$ for Rabinowitz string homology, $S\mathbb{H}_*^{S^1}({\partial}W)$ for S^1-eq...

conjecture | open | MathDB 页面 | 原文线索

V2-033241 | MATHDB-362062

Finite chromatic defect implies the telescope conjecture

摘录 / EXCERPT: Finite-defect telescope conjecture. Any spectrum with finite chromatic defect satisfies the condition of the telescope conjecture at all heights and primes.

conjecture | open | MathDB 页面 | 原文线索

V2-033242 | MATHDB-362116

Hill-Hopkins-Ravenel transchromatic phenomenon for equivariant slice spectral sequences

摘录 / EXCERPT: Hill-Hopkins-Ravenel transchromatic phenomenon. The slice spectral sequences of higher height theories contain isomorphism regions to the slice spectral sequences of lower height t...

conjecture | open | MathDB 页面 | 原文线索

V2-033243 | MATHDB-362117

Height classification conjecture for differentials in cyclic equivariant slice spectral sequences

摘录 / EXCERPT: Height classification conjecture. The C_{2^n}-slice spectral sequence of $BP^{(C_{2^n})}\langle m\rangle$ contains all differentials in the slice spectral sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-033244 | MATHDB-362118

Surjectivity conjecture for higher pages of cyclic equivariant slice spectral sequences

摘录 / EXCERPT: Surjectivity conjecture. For all $m\geq 1$ and $r\geq 2$, this map induces a surjection on the $\mathcal{E}_r$-page and on the d_r-differentials.

conjecture | open | MathDB 页面 | 原文线索

V2-033245 | MATHDB-362136

The generalised Albert form conjecture

摘录 / EXCERPT: Generalised Albert form conjecture. Every such form φ is a generalised Albert form.

conjecture | open | MathDB 页面 | 原文线索

V2-033246 | MATHDB-362166

The perturbed Conway invariant positivity conjecture

摘录 / EXCERPT: Perturbed Conway invariant positivity conjecture. If $K\subset S^3$ is a positive knot, then all coefficients of $\delta_1(K)$ are non-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-033247 | MATHDB-362168

Generalised Geroch conjecture with boundary

摘录 / EXCERPT: Let $n\geq 2$ be such that the generalised Geroch conjecture without boundary holds for all n-manifolds. Let M be an n-manifold with non-empty boundary, and form $M\# T^n$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-033248 | MATHDB-362170

The homological modularity conjecture for relative fusion categories

摘录 / EXCERPT: Homological modularity conjecture. The category $\mathsf{RFC}(\mathfrak{sl}(N),M_3)$ is modular if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-033249 | MATHDB-362217

Wrappingness and trunkenness additivity under fibered gluing

摘录 / EXCERPT: Let Y_1 and Y_2 be fibered 3-manifolds with boundary, with fibrations $\pi_1:Y_1\to S^1$ and $\pi_2:Y_2\to S^1$, and glue them along a fibered torus-boundary component to obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-033250 | MATHDB-362240

The $P=C$ conjecture for moduli spaces of one-dimensional sheaves

摘录 / EXCERPT: Let $M_{d,c7}$ be the moduli space of one-dimensional sheaves on $\mathbb{P}^2$ with degree d and Euler characteristic χ, and let $P_\bullet H^*(M_{d,\chi},\mathbb{Q})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033251 | MATHDB-362258

The deficiency-one conjecture for oriented Grassmannians

摘录 / EXCERPT: Let $\widetilde{\operatorname{Gr}}_k(n)$ denote the oriented Grassmannian, and let $\delta(\widetilde{\operatorname{Gr}}_k(n))$ denote its deficiency. The previously established ca...

conjecture | open | MathDB 页面 | 原文线索

V2-033252 | MATHDB-362259

The deficiency conjecture for torsion exponents of homogeneous spaces

摘录 / EXCERPT: Let G/K be a homogeneous space, let $\delta_p(G,K)$ denote its deficiency with coefficients in a finite field $\mathbb{F}_p$, and let Tor denote the torsion subg...

conjecture | open | MathDB 页面 | 原文线索

V2-033253 | MATHDB-362260

The 4-torsion conjecture for oriented Grassmannians

摘录 / EXCERPT: Let $\widetilde{\operatorname{Gr}}_k(n)$ be the oriented Grassmannian, and let a_n and d_n denote the anomalous generators used in the source. 4-torsion conjecture. The followi...

conjecture | open | MathDB 页面 | 原文线索

V2-033254 | MATHDB-362263

Optimal torus manifold conjecture for multifans

摘录 / EXCERPT: Let X range over real $2n$-dimensional torus manifolds corresponding to a multifan $\mathcal S_X$. Let $h_k(\mathcal S_X)=\sum_{i=k}^n(-1)^{i-k}\binom{i}{k}e_{n-i}(\mathcal S_X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-033255 | MATHDB-362268

The p-adic formulation of the Hilbert-Smith conjecture

摘录 / EXCERPT: Let ${\mathbb{Z}}_p$ denote the topological group of p-adic integers, and let X be a connected m-manifold. Hilbert-Smith conjecture, p-adic formulation. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-033256 | MATHDB-362269

The almost-periodic homeomorphism conjecture

摘录 / EXCERPT: Let X be a connected manifold, let $\cl G$ be a closed subgroup of $\operatorname{Homeo}(X)$, and let $f\in\cl G$ be almost periodic, meaning that the subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-033257 | MATHDB-362282

Ayala-Francis-Rozenblyum's topological stratified homotopy hypothesis

摘录 / EXCERPT: Let $\mathcal{S}\mathrm{trat}$ denote a homotopy theory of topological stratified spaces, and let $\iCat$ denote the homotopy theory of ∞-categories. The exit path construct...

conjecture | open | MathDB 页面 | 原文线索

V2-033258 | MATHDB-362283

Conjectured monoidal algebraicity equivalence for monochromatic spectra

摘录 / EXCERPT: Conjectured monoidal algebraicity equivalence. Under these assumptions, there is an equivalence of symmetric monoidal stable ∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-033259 | MATHDB-362284

Finite-dimensionality dichotomy for lattice models over background phases

摘录 / EXCERPT: Let B^{n+1} be a background phase and let C^{n+1} be an $(n+1)$D anomaly-free topological order. An MCLP lattice model over B^{n+1} has a local spi...

conjecture | open | MathDB 页面 | 原文线索

V2-033260 | MATHDB-362328

Ichihara-Ito-Saito's chirally cosmetic surgery conjecture for knots in S^3

摘录 / EXCERPT: Let K be a knot in S^3 that is not a $(2,r)$ torus knot, and let $N=S^3-K$. Let (s,s') be a chirally cosmetic pair of slopes on ∂N. Ichihara-Ito-Saito's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-033261 | MATHDB-362347

Strong Relative Aspherical Conjecture for positive scalar curvature

摘录 / EXCERPT: Strong Relative Aspherical Conjecture. If $n\ne 4$, Y is weakly aspherical relative to X, and X admits no metric of positive scalar curvature, then Y admits no metric of po...

conjecture | open | MathDB 页面 | 原文线索

V2-033262 | MATHDB-362348

Schoen-Yau-Schick degree-one dominated S^2-stability conjecture

摘录 / EXCERPT: Schoen-Yau-Schick degree-one dominated S^2-stability conjecture. Every Schoen-Yau-Schick manifold X^n with $n\leq 5$ has the degree-one version of dominated S^2-stability.

conjecture | open | MathDB 页面 | 原文线索

V2-033263 | MATHDB-362373

The injectivity conjecture for the splitting maps

摘录 / EXCERPT: Let σ and $\overline{\sigma}$ be the splitting maps associated with the 3d quantum trace construction. Splitting-map injectivity conjecture. The maps σ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033264 | MATHDB-362374

Injectivity conjecture for the 3d quantum trace map

摘录 / EXCERPT: Let Y be the ideally triangulated 3-manifold considered in the paper, with triangulation $\mathcal{T}$, and let $\operatorname{Tr}_{\mathcal{T}}$ denote its 3d quantum trace ma...

conjecture | open | MathDB 页面 | 原文线索

V2-033265 | MATHDB-362375

The even-link image conjecture for the 3d quantum trace

摘录 / EXCERPT: Assume that $Y=S^3\setminus\mathcal{K}$ is an ideally triangulated knot complement. A link $L\subset S^3\setminus\mathcal{K}$ is even if it represents the zero homology class in .....

conjecture | open | MathDB 页面 | 原文线索

V2-033266 | MATHDB-362377

The q-holonomicity conjecture for quantum invariants of contragredient Lie superalgebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple, finite-dimensional, basic contragredient Lie superalgebra whose even subalgebra is reductive. Let $U_hmathfrak{g}$ be its quantized universal envelo...

conjecture | open | MathDB 页面 | 原文线索

V2-033267 | MATHDB-362420

The inverted Habiro-series formula for knot-complement invariants

摘录 / EXCERPT: Let K be a knot with Alexander polynomial $\Delta_K\neq1$, let $F_K(x,\tau)$ be its knot-complement invariant, and let $a_{-m}(K;q)$ denote inverted Habiro coefficients. Inverted...

conjecture | open | MathDB 页面 | 原文线索

V2-033268 | MATHDB-362425

Relation between the equivariant instanton and Heegaard Floer slice invariants

摘录 / EXCERPT: Let K be a knot in S^3. Let $q_M(K)$ and $q_M^\dagger(K)$ be the invariants defined from equivariant singular instanton Floer theory, let $q_\tau(K)$ be Hendricks-Lipshitz-Sark...

conjecture | open | MathDB 页面 | 原文线索

V2-033269 | MATHDB-362426

Equivariant Seiberg-Witten and Heegaard Floer contact invariants for branched covers

摘录 / EXCERPT: Let K be a transverse knot in S^3, let $\Sigma_2(K)$ be its double branched cover, and let $\widetilde{\xi}$ be the induced contact structure. Let ... and ... be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-033270 | MATHDB-362441

Putman-Wieland conjecture on homology orbits of characteristic surface covers

摘录 / EXCERPT: Let Σ be the surface under consideration, let $\widetilde{\Sigma}\to\Sigma$ be a finite characteristic cover, and let $v\in {\rm H}_1(\widetilde{\Sigma},\mathbb{Q})$ be a no...

conjecture | open | MathDB 页面 | 原文线索

V2-033271 | MATHDB-362443

Sutured Heegaard Floer-embedded contact homology isomorphism conjecture

摘录 / EXCERPT: Let (M,Γ,ξ) be a sutured contact 3-manifold. For $A\in H_1(M;\mathbb{Z})$, let $\mathfrak{s}_\xi$ be the canonical Spin^c-structure determined by ξ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033272 | MATHDB-362455

The half Seiberg-Witten curve topological free-energy conjecture

摘录 / EXCERPT: Let $r>2$ and let $Q_1,\ldots,Q_r$ and Λ be the parameters of the half Seiberg-Witten curve, with B_{2g} denoting the Bernoulli number. Topological free-energy conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-033273 | MATHDB-362496

Primitively generated loop homology conjecture for Davis-Januszkiewicz spaces

摘录 / EXCERPT: Let $\mathcal{K}$ be a simplicial complex and let $\mathbf{k}$ be a ring such that $H_*(\Omega\mathrm{DJ}(\mathcal{K});\mathbf{k})$ is a free $\mathbf{k}$-module. Recall that a Hop...

conjecture | open | MathDB 页面 | 原文线索

V2-033274 | MATHDB-362497

Huang's coformality generalisation for fibrations

摘录 / EXCERPT: Let $F\to E\overset{p}\longrightarrow B$ be a fibration of simply connected spaces of finite type. Suppose that Ωp has a homotopy section and that E is coformal over a ri...

conjecture | open | MathDB 页面 | 原文线索

V2-033275 | MATHDB-362499

Koszul cohomology criterion for formality of flag moment-angle complexes

摘录 / EXCERPT: Let $\mathcal{K}$ be a flag simplicial complex, and consider the grading on $H^*(\mathcal{Z}_\mathcal{K};\mathbb{Q})$ specified in the source's preceding theorem. Koszul formality...

conjecture | open | MathDB 页面 | 原文线索

V2-033276 | MATHDB-362505

Noncontractibility conjecture for configuration spaces of paracompact spaces

摘录 / EXCERPT: Conjetura sobre la no contractibilidad de espacios de configuraciones. Los espacios de configuraciones $\operatorname{Conf}(X,k)$ nunca son contráctiles para $k\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-033277 | MATHDB-362518

Gromov's vanishing conjecture for Lp-cohomology of symmetric spaces

摘录 / EXCERPT: Let G be a semisimple Lie group, let K be a maximal compact subgroup, and let G/K be the associated homogeneous space. Write $\operatorname{rank}_{\mathbb{R}}(G)$ for the rea...

conjecture | open | MathDB 页面 | 原文线索

V2-033278 | MATHDB-362522

Relative tropical Riemann-Roch conjecture for a pair of divisors

摘录 / EXCERPT: Relative tropical Riemann-Roch conjecture.

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V2-033279 | MATHDB-362529

Generalized Trinity conjecture for Witt-equivalent modular tensor categories

摘录 / EXCERPT: Generalized pull-open conjecture. Any stable gapped domain wall ${}_B\CM_B$ is equivalent to

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V2-033280 | MATHDB-362530

Indecomposability conjecture for domain walls obtained by condensation

摘录 / EXCERPT: Domain-wall decomposition conjecture. Any gapped domain wall of the form

conjecture | open | MathDB 页面 | 原文线索

V2-033281 | MATHDB-362537

Algebraic overtwistedness conjecture for positive-braid rainbow closures

摘录 / EXCERPT: Let Λ be a non-trivial Legendrian knot that is the Legendrian rainbow closure of a positive braid. For $k\in {\mathbb N}_+$, let $(S^{3}_{1/k}(\Lambda),\xi_{1/k}(\Lambda))$...

conjecture | open | MathDB 页面 | 原文线索

V2-033282 | MATHDB-362548

The Boyer-Gordon-Watson-Juhász L-space conjecture

摘录 / EXCERPT: The L-space conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-033283 | MATHDB-362549

The Lin geography restriction conjecture for Heegaard Floer homology

摘录 / EXCERPT: Lin geography restriction conjecture. The Heegaard Floer module $\operatorname{HF}^-(Y)$ satisfies the Lin geography restriction for all rational homology spheres Y.

conjecture | open | MathDB 页面 | 原文线索

V2-033284 | MATHDB-362552

Viterbo's boundedness conjecture for the gamma metric

摘录 / EXCERPT: Let N be a manifold, let T^*N carry the cotangent-bundle structure, and let ... $D_rT^*N=\{(q,p)\in T^*N:|p|\leq r\}$. ... is bounded on the space of exact Lagrangians contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-033285 | MATHDB-362567

The conjecture on torsion in first homology of Milnor fiber boundaries

摘录 / EXCERPT: Torsion conjecture for Milnor fiber boundaries. (1) If

conjecture | open | MathDB 页面 | 原文线索

V2-033286 | MATHDB-362573

The conductor test-object conjecture

摘录 / EXCERPT: Let R be a ring, let $\overline{R}$ be its integral closure, and let ... be the conductor ideal. The conductor test-object conjecture. The conductor ideal $\mathfrak{C}_R$ can be u...

conjecture | open | MathDB 页面 | 原文线索

V2-033287 | MATHDB-362586

Uniqueness of the limiting cohomology class of conjugate p-tight forms

摘录 / EXCERPT: Uniqueness conjecture. The family (α_q) has a unique limit as $q\to1$. This asks whether the limiting cohomology class arising from the q-harmonic conjugates is distinguis...

conjecture | open | MathDB 页面 | 原文线索

V2-033288 | MATHDB-362629

Kottwitz conjecture for local shtuka cohomology

摘录 / EXCERPT: Kottwitz conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-033289 | MATHDB-362630

Vanishing conjecture for local shtuka cohomology

摘录 / EXCERPT: Let $b\in B(G)$ be basic, let $\mu_\bullet$ be a finite collection of cocharacters, and let ρ be an irreducible admissible representation of $G_b(\bbQ_p)$ whose parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-033290 | MATHDB-362637

The primitive-cohomology strip conjecture

摘录 / EXCERPT: Primitive-cohomology strip conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033291 | MATHDB-362638

The Prim polynomial conjecture

摘录 / EXCERPT: Prim polynomial conjecture. 1. If $n=3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-033292 | MATHDB-362647

The monodromy conjecture for relatively exact Lagrangians

摘录 / EXCERPT: Let M be a compact or Liouville symplectic manifold, and let $L\subseteq M$ be a compact, relatively exact Lagrangian. Let $\tilde\phi:L\to L$ be a diffeomorphism extending to a...

conjecture | open | MathDB 页面 | 原文线索

V2-033293 | MATHDB-362654

Gompf's conjecture on Cappell-Shaneson matrix equivalence

摘录 / EXCERPT: A Cappell-Shaneson matrix is a matrix $A\in SL(3;\mathbb{Z})$ satisfying $\det(A-I)=1$. Two such matrices are Gompf equivalent when they are related by Gompf's equivalence relation...

conjecture | open | MathDB 页面 | 原文线索

V2-033294 | MATHDB-362655

The parametrized Gompf-equivalence conjecture for Cappell-Shaneson matrices

摘录 / EXCERPT: Let ... The set $\mathcal{C}\mathcal{S}$ parametrizes standard Cappell-Shaneson matrices via $(c,d,n)\mapsto X_{c,d,n}$. Let $\sim$ denote Gompf equivalence on...

conjecture | open | MathDB 页面 | 原文线索

V2-033295 | MATHDB-362661

Stable crossing number conjecture for twist families

摘录 / EXCERPT: Stable crossing number conjecture. The stable crossing number exists and

conjecture | open | MathDB 页面 | 原文线索

V2-033296 | MATHDB-362662

Eventual linearity conjecture for crossing numbers in twist families

摘录 / EXCERPT: Eventual linearity conjecture. For sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-033297 | MATHDB-362663

Positive stable crossing growth conjecture for twist families

摘录 / EXCERPT: Positive stable crossing growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033298 | MATHDB-362664

Positive stable braid index conjecture for non-coherent twist families

摘录 / EXCERPT: Positive stable braid index conjecture. If $\{K_n\}$ is non-coherent, then

conjecture | open | MathDB 页面 | 原文线索

V2-033299 | MATHDB-362685

Katada's stable Albanese cohomology conjecture for IA_n

摘录 / EXCERPT: Katada's conjecture. For $n\gg *$,

conjecture | open | MathDB 页面 | 原文线索

V2-033300 | MATHDB-362716

Ribbon-symmetric union conjecture for ribbon knots

摘录 / EXCERPT: A knot K is a symmetric union if it is obtained from the connected sum $J\#-J$ of a knot J with its mirror $-J$ by replacing trivial 4-tangles intersecting the symmetry axis wi...

conjecture | open | MathDB 页面 | 原文线索

V2-033301 | MATHDB-362741

Conjecture on twisted L-groups and Q-groups from algebraic Wu classes

摘录 / EXCERPT: Let (B_*,β) be a chain bundle with algebraic Wu class $\widehat v_{2k}(\beta)$, and let ev denote evaluation on homology. Let Σ, Σ_8, Λ...

conjecture | open | MathDB 页面 | 原文线索

V2-033302 | MATHDB-362742

Fox's conjecture on crossing number and bridge number

摘录 / EXCERPT: Fox's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033303 | MATHDB-362759

Exact top-dimensional homology rank conjecture for independence complexes of Kneser graphs

摘录 / EXCERPT: Exact homology-rank conjecture. The rank of the p-dimensional homology group of $\operatorname{Ind}(\operatorname{KG}(n,k))$ is

conjecture | open | MathDB 页面 | 原文线索

V2-033304 | MATHDB-362764

Median-resummation conjecture for strong-coupling Stokes constants

摘录 / EXCERPT: Let $f_\infty(y)$ be the generating function in Eq., and let $\tilde{f}_\infty(y)$ be its asymptotic expansion as $y\to0$ with $\Im(y)>0$. Strong-coupling median-resummation conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-033305 | MATHDB-362773

Doyle's conjecture on locally univalent circle packings of the hexagonal triangulation

摘录 / EXCERPT: Doyle's conjecture. There are no locally univalent circle packings of the hexagonal triangulation $\mathcal{T}_H$ other than regular hexagonal packings and Doyle spirals.

conjecture | open | MathDB 页面 | 原文线索

V2-033306 | MATHDB-362775

Gromov-Lees-type homotopy equivalence for polyhedral isotropic immersions

摘录 / EXCERPT: Let Σ be a surface endowed with a triangulation $\mathscr{T}$. Let V be the ambient symplectic vector space in which the immersions take values, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033307 | MATHDB-362799

Huneke-Wiegand conjecture for two-generated ideals of symmetric numerical semigroup rings

摘录 / EXCERPT: Let Γ be a symmetric numerical semigroup, let $\Bbbk$ be a field, and let $R=\Bbbk[\Gamma]$ be its semigroup algebra. Let M be a 2-generated ideal of R, viewed as an R...

conjecture | open | MathDB 页面 | 原文线索

V2-033308 | MATHDB-362889

The generalized Loewy length monotonicity conjecture under flat local extensions

摘录 / EXCERPT: Let $R\to S$ be a flat local extension between Cohen--Macaulay local rings with infinite residue fields. Write $g\ell\ell_R(R)$ and $g\ell\ell_S(S)$ for their generalized Loewy len...

conjecture | open | MathDB 页面 | 原文线索

V2-033309 | MATHDB-362906

A numerical invariant for homotopy Hopf manifolds

摘录 / EXCERPT: Numerical-map conjecture. There is a numerical map N which is constant with value $\frac{1}{12}$ on $\sigma^8_{\textrm{Milnor}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-033310 | MATHDB-362907

A factorization conjecture for the numerical map on homotopy Hopf manifolds

摘录 / EXCERPT: Let σ^8 be the space of homotopy Hopf manifolds, let $\mathcal L$ and p_1 be the maps appearing in the source, and let $MO\langle 8\rangle$ be the Thom spectrum o...

conjecture | open | MathDB 页面 | 原文线索

V2-033311 | MATHDB-362908

A classification conjecture for homotopy Hopf manifolds failing the one-twelfth invariant

摘录 / EXCERPT: Let σ^8 be the space of homotopy Hopf manifolds, let $\widetilde{\sigma}$ be the map appearing in the source, and let θ^7 denote the group of homotopy seven-spheres....

conjecture | open | MathDB 页面 | 原文线索

V2-033312 | MATHDB-362909

The arc-faithful diagram conjecture for hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot, and call a diagram of K arc-faithful when its associated octahedral coloring has the required non-pinched property and holonomy equal to the complet...

conjecture | open | MathDB 页面 | 原文线索

V2-033313 | MATHDB-362910

The monodromy-weight conjecture for projective log smooth morphisms

摘录 / EXCERPT: Monodromy-weight conjecture. For every integer m, $R^mf_*(\overline{\mathbb{Q}}_{\ell})$ is an object of ${\mathcal A}_Y$ of pure W-weight m.

conjecture | open | MathDB 页面 | 原文线索

V2-033314 | MATHDB-362924

Benson's torus-generation conjecture for classifying-space cochains

摘录 / EXCERPT: Benson's torus-generation conjecture. The bounded derived category $\mathcal{D}^b(C^*BG)$ is generated by the module C^*BT.

conjecture | open | MathDB 页面 | 原文线索

V2-033315 | MATHDB-362926

Greene-Owens cubiquity conjecture for double branched covers

摘录 / EXCERPT: Greene-Owens cubiquity conjecture. $\Sigma(L)$ bounds a rational homology 4-ball if and only if $\Lambda(L)$ is cubiquitous.

conjecture | open | MathDB 页面 | 原文线索

V2-033316 | MATHDB-362962

Banach's isometric conjecture for normed spaces

摘录 / EXCERPT: Let V be a normed space over the reals, and let $2\leq k<\dim V$ be an integer. Suppose that all linear k-dimensional subspaces of V are isometric to each other. Banach's iso...

conjecture | solved | MathDB 页面 | 原文线索

V2-033317 | MATHDB-363056

Existence of cts-links distinguished by the parameterized invariant

摘录 / EXCERPT: A cts-link is a singular link considered in the paper, and $\widetilde{\Upsilon}_{x,y,z}$ and $\widetilde{\Upsilon}_{x,y,0}$ are the corresponding link invariants, with the latter...

conjecture | open | MathDB 页面 | 原文线索

V2-033318 | MATHDB-363059

Avramov-Buchweitz total rank conjecture

摘录 / EXCERPT: Let R be a d-dimensional Noetherian local ring, and let M be a finitely generated nonzero R-module. Total rank conjecture. If M has finite length and finite projective di...

conjecture | open | MathDB 页面 | 原文线索

V2-033319 | MATHDB-363060

Herzog-Vasconcelos conjecture on derivation modules

摘录 / EXCERPT: Let R be a Noetherian local ring with residue field k, and let $\operatorname{Der}_k(R)$ denote the R-module of k-derivations of R. Herzog-Vasconcelos conjecture. If ... th...

open_problem | open | MathDB 页面 | 原文线索

V2-033320 | MATHDB-363063

Huneke's conjecture on associated primes of local cohomology modules

摘录 / EXCERPT: Let R be a Noetherian local ring, let M be a finitely generated R-module, let I be an ideal of R, and write ${\rm H}^i_I(M)$ for the i-th local cohomology module of M...

conjecture | open | MathDB 页面 | 原文线索

V2-033321 | MATHDB-363065

Definable Urysohn Metrization Conjecture

摘录 / EXCERPT: Let (X,τ) be a definable topological space in an o-minimal expansion of an ordered field. Call (X,τ) affine if it is definably homeomorphic to a set with the o-minimal Eu...

conjecture | open | MathDB 页面 | 原文线索

V2-033322 | MATHDB-363072

Equality of the tropical Ceresa period and Ceresa-Zharkov class

摘录 / EXCERPT: Let G be a graph of genus g. A hyperelliptic involution is an involution τ of the surface of genus g with the hyperelliptic property. Equality conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-033323 | MATHDB-363079

Boninger-Greene conjecture on positive knots and ribbon concordance

摘录 / EXCERPT: Let $K_0,K_1\subset S^3$ be knots, and write $K_0\leq K_1$ when there is a ribbon concordance from K_1 to K_0. A knot is positive if it admits a diagram in which all crossings...

conjecture | open | MathDB 页面 | 原文线索

V2-033324 | MATHDB-363080

Uniqueness conjecture for positive knots in smooth concordance classes

摘录 / EXCERPT: A smooth concordance class is an equivalence class of knots under smooth concordance. A knot is minimal with respect to ribbon concordance if no distinct knot lies below it in the...

conjecture | open | MathDB 页面 | 原文线索

V2-033325 | MATHDB-363081

Positive knots are rationally anisotropic

摘录 / EXCERPT: Let K be a knot, let $\overline{X}$ be the infinite cyclic cover of its exterior, and consider the associated Seifert-form data over $\mathbb{Q}$. A knot is $\mathbb{Q}$-anisotro...

conjecture | open | MathDB 页面 | 原文线索

V2-033326 | MATHDB-363133

The flat-geodesic finiteness and ergodicity conjecture for nonpositively curved surfaces

摘录 / EXCERPT: Flat-geodesic finiteness conjecture. All flat geodesics are closed, and there are only finitely many homotopy classes of such geodesics. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-033327 | MATHDB-363134

Bounded lensbordant surgeries for knots with nonzero \nu^+

摘录 / EXCERPT: Let K be a knot in S^3. A lensbordant surgery on K is a positive integer surgery that is smoothly rational homology cobordant to a lens space, and let $ u^+(K)$ denote the kn...

conjecture | open | MathDB 页面 | 原文线索

V2-033328 | MATHDB-363139

Variational filtered convergent log hard Lefschetz conjecture

摘录 / EXCERPT: Variational filtered convergent log hard Lefschetz conjecture. For the relevant integers i, the cup product

conjecture | open | MathDB 页面 | 原文线索

V2-033329 | MATHDB-363151

Strong DR/DZ equivalence conjecture

摘录 / EXCERPT: Let w^{α} be the normal coordinates for the Dubrovin-Zhang hierarchy, let $\tilde u^{\alpha}$ be the normal coordinates for Buryak's double ramification hierarchy, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033330 | MATHDB-363165

Quantum Kirwan map conjecture

摘录 / EXCERPT: Let V be a Hamiltonian K-manifold with symplectic quotient $V /\hspace{-0.1cm}/ K$. Write $QH_K^*(V)$ and $QH^*(V /\hspace{-0.1cm}/ K)$ for the equivariant and ordinary quantum...

conjecture | open | MathDB 页面 | 原文线索

V2-033331 | MATHDB-363201

Foam category group completion conjecture

摘录 / EXCERPT: Let $\mathcal{C}_0$ be the decoration data for an E_1-monoidal $(\infty,n-1)$-category $\mathscr{F}_{n-1}(\mathcal{C}_0)$ of singular manifolds in $\mathbb{R}^n$, framed in codim...

conjecture | open | MathDB 页面 | 原文线索

V2-033332 | MATHDB-363227

The slope-set conjecture for taut foliations of knot complements

摘录 / EXCERPT: Let K be a knot in S^3 with Seifert genus $g>0$. Let $\mathcal{S}_K\subset\mathbb{Q}$ be the set of rational slopes s for which there exists a taut $\mathcal{C}^0$-foliation...

conjecture | open | MathDB 页面 | 原文线索

V2-033333 | MATHDB-363229

The algebraically tight contact-pair conjecture

摘录 / EXCERPT: The algebraically tight contact-pair conjecture. The pair (ξ_-,ξ_+) is homotopic through contact pairs to a positive contact pair.

conjecture | open | MathDB 页面 | 原文线索

V2-033334 | MATHDB-363230

The Floer-summand conjecture for Reebless foliations

摘录 / EXCERPT: Let M be an irreducible rational homology sphere, and let $HM_*(M)$ have a direct $\mathbb{F}$-summand, where $\mathbb{F}=\mathbb{Z}/2\mathbb{Z}$. The Floer-summand conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-033335 | MATHDB-363240

The instanton L-space surgery conjecture for knots

摘录 / EXCERPT: Instanton L-space surgery conjecture. Under these assumptions, $S^3_{p/q}(K)$ must be an instanton L-space, meaning

conjecture | open | MathDB 页面 | 原文线索

V2-033336 | MATHDB-363252

A-model conifold eigenvalue conjecture for Fano manifolds

摘录 / EXCERPT: For a Fano manifold X, let $G_X(t)=\sum_{n=0}^{\infty}a_nt^n/n!$ be its quantum period and define the A-model conifold value by ... Let $\hat c_1$ denote quantum multiplication by...

conjecture | open | MathDB 页面 | 原文线索

V2-033337 | MATHDB-363253

Modified Gamma conjecture I, weak form

摘录 / EXCERPT: Let X be a Fano manifold satisfying Property $\mathcal O_A$, where $T_{\mathrm{A,con}}$ is a simple rightmost eigenvalue of $\hat c_1$. Let A_X be the principal asymptotic clas...

conjecture | open | MathDB 页面 | 原文线索

V2-033338 | MATHDB-363257

The rational cohomology conjecture for moduli stacks of principal bundles

摘录 / EXCERPT: Rational cohomology conjecture. The rational cohomology of $\mathscr{B}un_{G,X}^0$ is

conjecture | open | MathDB 页面 | 原文线索

V2-033339 | MATHDB-363266

The normalized cohomology convergence conjecture for semisimple algebraic groups

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{C}$, let Γ be a torsion-free subgroup of type FP_∞, and let $\{\mathsf{W}_k\}$ be a sequence of irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-033340 | MATHDB-363267

The quantitative normalized cohomology conjecture for semisimple algebraic groups

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{C}$, let Γ be a torsion-free subgroup of type FP_∞, and let $\lambda_i=\lambda_i(k)$ be the highest wei...

conjecture | open | MathDB 页面 | 原文线索

V2-033341 | MATHDB-363278

Mixed-parity Hecke and Satake formalism for two-fold Weil coverings

摘录 / EXCERPT: Mixed-parity Hecke conjecture. In the mixed-parity setting this Hecke operator can be defined, its image lies in

conjecture | open | MathDB 页面 | 原文线索

V2-033342 | MATHDB-363302

Twisted quantum Hamiltonian reduction conjecture for Ramond modules

摘录 / EXCERPT: Let $\widehat{\mathfrak g}^{\rm tw}$ be the twisted affine algebra, let $\mathcal O$ be its category of modules, and let H_j denote the homology of the twisted quantum Hamiltonia...

conjecture | open | MathDB 页面 | 原文线索

V2-033343 | MATHDB-363307

Delooping conjecture for polyhedral coproducts

摘录 / EXCERPT: Let K be a shifted complex or a flag complex with chordal 1-skeleton. The decomposition in Theorem is a decomposition of loop spaces that should deloop. However, this does not...

conjecture | refuted | MathDB 页面 | 原文线索

V2-033344 | MATHDB-363358

The revised Newelski conjecture on Hausdorffness of the tau-topology

摘录 / EXCERPT: Let M be a model of an NIP theory, let G be a 0-definable group in M, and let $N\succ M$ be an $|M|^{+}$-saturated elementary extension. Let $S_G^{\operatorname{fs}}(N,M)$...

conjecture | open | MathDB 页面 | 原文线索

V2-033345 | MATHDB-363365

The formality conjecture for hyperkähler Fukaya algebras

摘录 / EXCERPT: Formality conjecture. The Floer--Fukaya A_∞ algebra

conjecture | open | MathDB 页面 | 原文线索

V2-033346 | MATHDB-363403

Cieliebak-Eliashberg's Stein-to-Weinstein weak homotopy equivalence conjecture

摘录 / EXCERPT: Let $\mathfrak{Stein}$ and $\mathfrak{Weinstein}$ denote the spaces of Stein and Weinstein structures, respectively, and let ... be the map sending a Stein structure to its associate...

conjecture | open | MathDB 页面 | 原文线索

V2-033347 | MATHDB-363416

Weak-star separability conjecture for Lipschitz functions

摘录 / EXCERPT: Let M be a metric space, let $\operatorname{dens} M$ denote its density, and let $\operatorname{Lip}_0(M)$ be the Banach space of Lipschitz functions on M vanishing at the dist...

conjecture | open | MathDB 页面 | 原文线索

V2-033348 | MATHDB-363451

The knot-realization conjecture for the figure-eight Lorenz template

摘录 / EXCERPT: Knot-realization conjecture. For any knot appearing as a periodic orbit on the template, there exists a periodic orbit for the original equation at the figure-eight parameter with...

conjecture | open | MathDB 页面 | 原文线索

V2-033349 | MATHDB-363459

Robust non-convexity and heterodimensional cycles

摘录 / EXCERPT: Let (Y,ξ) be a contact manifold and let Σ be a closed hypersurface. Write Σ_ξ for its characteristic foliation. A robust heterodimensional cycle is a cycle of h...

conjecture | open | MathDB 页面 | 原文线索

V2-033350 | MATHDB-363460

Small isotopic perturbations to robustly non-convex hypersurfaces

摘录 / EXCERPT: Let (Y,ξ) be a contact manifold of dimension at least 5, and let Σ be a closed oriented hypersurface in (Y,ξ). Approximation conjecture. The hypersurface Σ...

conjecture | open | MathDB 页面 | 原文线索

V2-033351 | MATHDB-363461

C^1-genericity of convex hypersurfaces

摘录 / EXCERPT: Let (Y,ξ) be a contact manifold. C^1-genericity conjecture. Convex hypersurfaces are C^1-generic in (Y,ξ). This conjecture asserts that convexity is the generic behavio...

conjecture | open | MathDB 页面 | 原文线索

V2-033352 | MATHDB-363471

Demailly's transcendental Morse inequalities conjecture

摘录 / EXCERPT: Let X be an n-dimensional compact complex manifold, and let α be a real d-closed C^∞ $(1,1)$-form on X. Define ... The BottChern cohomology class...

conjecture | open | MathDB 页面 | 原文线索

V2-033353 | MATHDB-363501

Polynomial-size approximation conjecture for Vietoris-Rips subdivisions on bounded-doubling metric spaces

摘录 / EXCERPT: Polynomial-size approximation conjecture. There exists a $(1+\epsilon)$-approximation to $\mathcal{SR}(X)$ whose fixed-dimensional skeleta have size polynomial in $|X|$.

conjecture | open | MathDB 页面 | 原文线索

V2-033354 | MATHDB-363510

Symmetry conjecture for the mock Alexander polynomial of starred knotoids

摘录 / EXCERPT: Let K be a knotoid diagram. Let K_1 and K_2 be the starred knotoid diagrams obtained from K by placing a star in the region incident to the tail and the region incident to...

conjecture | open | MathDB 页面 | 原文线索

V2-033355 | MATHDB-363526

The non-equivalence of the nth Whitehead link under quasi-isotopy

摘录 / EXCERPT: Whitehead-link conjecture. (a) W_n is not n-quasi-isotopic to the unlink. (b) W_n is not strongly 1-quasi-isotopic to the unlink.

conjecture | open | MathDB 页面 | 原文线索

V2-033356 | MATHDB-363532

Minimality conjecture for Sakuma-Weeks triangulations of hyperbolic 2-bridge link complements

摘录 / EXCERPT: Let $K(\Omega)$ be the 2-bridge link generated by the word ... Let $\mathcal{T}=\mathcal{T}(M)$ be the Sakuma-Weeks triangulation of the complement $S^3\backslash K(\Omega)$. Minim...

conjecture | open | MathDB 页面 | 原文线索

V2-033357 | MATHDB-363539

Small collapsed RCD spaces of nearly maximal rank are infranilmanifolds

摘录 / EXCERPT: For $K\rrbracketin\mathbb{R}$ and $N\in\mathbb{N}$, let an $\operatorname{RCD}(K,N+\varepsilon)$ space mean a metric-measure space satisfying the RCD condition with parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-033358 | MATHDB-363546

The low-complexity tautological generation conjecture for the cohomology of moduli spaces of curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable curves of genus g with n markings, and let $\mathcal{H}^{2*}(\overline{\mathcal{M}}_{g,n})$ denote its even coh...

conjecture | open | MathDB 页面 | 原文线索

V2-033359 | MATHDB-363566

Stable intersection-cohomology conjecture for one-dimensional sheaf moduli spaces

摘录 / EXCERPT: Let (S,H) be a polarized surface, let β be an effective divisor, and let $M_{\beta,\chi}$ denote the moduli space of polystable one-dimensional sheaves on S with determin...

conjecture | open | MathDB 页面 | 原文线索

V2-033360 | MATHDB-363574

The comeagre-conjugacy conjecture for countable index in Polish groups

摘录 / EXCERPT: Let G be a Polish group, and let $C\subseteq G$ be a comeagre conjugacy class, meaning a conjugacy class whose complement is meagre. Say that G has the countable index property...

conjecture | open | MathDB 页面 | 原文线索

V2-033361 | MATHDB-363653

Evans-Lekili conjecture on small resolutions of compound Du Val singularities

摘录 / EXCERPT: Let $P \in X$ be a compound Du Val singularity, let $\Uv$ be its Milnor fiber, and let $\SH^i(\Uv)$ denote its symplectic cohomology in degree i. Evans-Lekili conjecture. The sin...

conjecture | open | MathDB 页面 | 原文线索

V2-033362 | MATHDB-363654

Refined symplectic-cohomology formula for small Q-factorializations

摘录 / EXCERPT: Let $Y \to X$ be a small $\bQ$-factorialization of a compound Du Val singularity $P \in X$. Let ℓ be the number of irreducible components of the exceptional locus, let $\Uv$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-033363 | MATHDB-363662

Polynomial relationship conjecture for persistence entropy and shape factor

摘录 / EXCERPT: Polynomial relationship conjecture. For any triply-periodic minimal surface M, there exists a polynomial $P \in \mathbb{R}[t]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-033364 | MATHDB-363663

Density conjecture for 2-paths among piecewise Hölder-differentiable curves

摘录 / EXCERPT: Density conjecture. The subset of piecewise $C^{1,\beta}$ 2-paths is dense in $C^{1,\beta}$.

conjecture | open | MathDB 页面 | 原文线索

V2-033365 | MATHDB-363677

Continuous orthonormal trivialization conjecture for continuous projector families

摘录 / EXCERPT: Continuous orthonormal trivialization conjecture. The bases can be chosen so that the map $T:G\to GL_d(\mathbb{K})$ defined by

conjecture | open | MathDB 页面 | 原文线索

V2-033366 | MATHDB-363678

Topological recursion/quantum curve correspondence

摘录 / EXCERPT: Let $\mathcal{S}$ be an admissible spectral curve with x,y satisfying ... A quantisation of P is an operator $\hat{P}(\hat{x},\hat{y};\hbar)$ of the form ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-033367 | MATHDB-363711

The Mahowald-lift existence conjecture for equivariant self-maps

摘录 / EXCERPT: Suppose that $X\in\mathrm{Sp}^{C_2}_{(2),\omega}$ is a type $(n,n-1)$ complex. Suppose further that the modified inductive procedure produces a v_n-self map as a Mahowald invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-033368 | MATHDB-363712

The $v_{(2,1)}$-periodicity conjecture for $S/(h,v^4_{(1,0)})$

摘录 / EXCERPT: Let $S/(h,v^4_{(1,0)})$ denote the indicated C_2-equivariant complex, let a be the Euler-class element, and let $\widetilde{\bar{\kappa}}$ denote the specified equivariant lift...

conjecture | open | MathDB 页面 | 原文线索

V2-033369 | MATHDB-363744

Quantum trace compatibility for spin generators

摘录 / EXCERPT: Let $\iota:{\rm End}_{U_q(\mathfrak{so}_{2n+1})}(S^{\otimes(m-1)})\rightarrow{\rm End}_{U_q(\mathfrak{so}_{2n+1})}(S^{\otimes m})$ be given by...

conjecture | open | MathDB 页面 | 原文线索

V2-033370 | MATHDB-363745

Stable decategorification formula for spin link homology

摘录 / EXCERPT: Let $\beta\in\mathrm{Br}_m$ and let equation denote the paper's stated trace/decategorification equality between type B spin link homology and the spin-colored...

conjecture | open | MathDB 页面 | 原文线索

V2-033371 | MATHDB-363746

Higher-rank trace relation for spin link homology

摘录 / EXCERPT: Let equation denote the paper's preceding conjectural relation, and let n and k be the rank and generator index appearing there. Higher-rank trace conjecture. Equation also hol...

conjecture | open | MathDB 页面 | 原文线索

V2-033372 | MATHDB-363747

Conjectural braid-index formulas for exceptional Type 3 pretzel links

摘录 / EXCERPT: Conjectural Type 3 braid-index formulas. In the first positive case, $\textbf{b}(\mathcal{L})=2+\sum\alpha_j$; in the second positive case, $\textbf{b}(\mathcal{L})=1+\sum\alpha_j$...

conjecture | open | MathDB 页面 | 原文线索

V2-033373 | MATHDB-363755

The sharpness conjecture for triviality of cochains on spheres

摘录 / EXCERPT: Let $n\geq 1$ and let A be an $\mathbb{E}_{n+2}$-algebra in Sp. For a pointed space X, write $C^*(X;A)$ for the A-valued cochains on X, regarded as an augm...

conjecture | open | MathDB 页面 | 原文线索

V2-033374 | MATHDB-363769

Monod's isomorphism conjecture for bounded cohomology of semisimple Lie groups

摘录 / EXCERPT: Monod's isomorphism conjecture. The continuous bounded cohomology of G with real coefficients is naturally isomorphic to its ordinary continuous cohomology:

open_problem | open | MathDB 页面 | 原文线索

V2-033375 | MATHDB-363792

Complete metrizability of metrics on the uncountable discrete space

摘录 / EXCERPT: Recall that $\aleph_{1}$ denotes the first uncountable cardinal, and let $D_{\aleph_{1}}$ be the discrete space of cardinality $\aleph_{1}$. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-033376 | MATHDB-363798

Weak topological section conjecture for Kodaira fibrations

摘录 / EXCERPT: Weak topological section conjecture. The fibration f admits an algebraic section if and only if its short exact sequence of fundamental groups splits. This is introduced as a con...

conjecture | open | MathDB 页面 | 原文线索

V2-033377 | MATHDB-363799

Existence conjecture for Kodaira fibrations without topological sections

摘录 / EXCERPT: Existence conjecture. There is a Kodaira fibration with no topological section. The source motivates this by the universal family of curves, which has no topological sections, but...

conjecture | open | MathDB 页面 | 原文线索

V2-033378 | MATHDB-363814

Concentration conjecture for localized completed homology

摘录 / EXCERPT: Let U^p be a compact open subgroup away from p, let $\mathcal{O}$ be the coefficient ring, let $\mathbb{T}^S(U^p)$ be the Hecke algebra, and let q_0 be the distinguished homo...

conjecture | open | MathDB 页面 | 原文线索

V2-033379 | MATHDB-363815

Pseudorepresentation compatibility conjecture for patched completed homology

摘录 / EXCERPT: Let R^{ps}_p be the relevant pseudodeformation ring, let $E_{\mathfrak{B}}$ be the corresponding Hecke-module coefficient algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033380 | MATHDB-363852

Hilali conjecture for rationally elliptic spaces

摘录 / EXCERPT: Hilali conjecture. If X is a rationally elliptic space, then

conjecture | open | MathDB 页面 | 原文线索

V2-033381 | MATHDB-363854

Bott conjecture on non-negatively curved manifolds

摘录 / EXCERPT: Bott conjecture. The manifold M is rationally elliptic.

conjecture | open | MathDB 页面 | 原文线索

V2-033382 | MATHDB-363856

Explicit splitting conjecture for truncated de Rham cohomology of the Drinfeld space

摘录 / EXCERPT: Explicit splitting conjecture. For $\ell\in\{2,\dots,n-2\}$, the morphism of complexes

conjecture | open | MathDB 页面 | 原文线索

V2-033383 | MATHDB-363886

Existence of paths detecting every element of the continuous-contingent microbundle space

摘录 / EXCERPT: Let W_{CC} be the space of continuous-contingent microbundles, let $u\in\mathsf{W}_{\mathrm{CC}}$, and let $\mathrm{D}(u)$ denote its domain. A path ω de...

conjecture | open | MathDB 页面 | 原文线索

V2-033384 | MATHDB-363887

Subcanonicity conjecture for the Grothendieck topology of linear microbundles

摘录 / EXCERPT: Let $(\mathbf{Top},T)$ be the underlying site and let $(\mathbb{V},\mathcal{T})$ be the site obtained by augmenting it through the specified data $(\mathbf{Set},C,D,E,H)$. A Grothe...

conjecture | open | MathDB 页面 | 原文线索

V2-033385 | MATHDB-363891

Rigidity for tame cohomology over henselian pairs

摘录 / EXCERPT: Let S be a henselian discrete valuation ring with maximal ideal $\mathfrak m_S=(\pi)$. Let A be a separated essentially of finite type S-algebra such that $(A,I=(\pi)A)$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-033386 | MATHDB-363894

The top-degree cohomology conjecture for p-adic tame Tate twists

摘录 / EXCERPT: Let A be a discrete valuation ring with local parameter π, perfect residue field k of characteristic $p>0$, and fraction field K of characteristic zero. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-033387 | MATHDB-363895

Quantitative separation conjecture for Aarnes and symplectic quasi-states

摘录 / EXCERPT: Let M be a closed connected symplectic manifold with symplectic form ω, and endow M with a metric inducing its topology. Let ${\mathcal{S}}(M,\omega)$ be the collection...

conjecture | open | MathDB 页面 | 原文线索

V2-033388 | MATHDB-363933

Zariski's multiplicity conjecture for holomorphic foliations

摘录 / EXCERPT: Let $\mathcal{F}$ and $\mathcal{G}$ be holomorphic foliations by curves at $(\mathbb{C}^{n},0)$, with algebraic multiplicities denoted by $\nu(\mathcal{F},0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033389 | MATHDB-363958

The nonvanishing conjecture for cuspidal cohomology of $GL(n)$

摘录 / EXCERPT: Let F be a number field, let $G=GL_n$, and let E be a coefficient field. Write $X^+_{00}(\operatorname{Res}_{F/\mathbb{Q}}(T_0)\times E)$ for the set of strongly-pure dominant...

conjecture | open | MathDB 页面 | 原文线索

V2-033390 | MATHDB-363972

Positivity conjecture for cohomology L-classes of projective space

摘录 / EXCERPT: Let n be a positive integer, let $\mathbb{P}^n$ be complex projective space, and let $L^{2j}(\mathbb{P}^n)\in H^{4j}(\mathbb{P}^n,\mathbb{Q})$ denote its degree-$4j$ cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-033391 | MATHDB-363984

Allen's non-realizability conjecture for unknown pairs of T(2,n)

摘录 / EXCERPT: For an odd positive integer n, consider a torus knot $T(2,n)$ and the pairs (e,h) of normal Euler number and first Betti number of smooth surfaces in B^4 bounded by $T(2,n)$....

conjecture | open | MathDB 页面 | 原文线索

V2-033392 | MATHDB-363985

Allen's unique minimal-point conjecture for torus knots

摘录 / EXCERPT: Let K be a torus knot, and let $\gamma_4(K)$ denote its smooth 4-genus. A realizable pair (e,h) records the normal Euler number and first Betti number of a smooth surface in...

conjecture | open | MathDB 页面 | 原文线索

V2-033393 | MATHDB-363996

Lin's cosmetic crossing conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be an oriented knot and let $D\subset S^3$ be an oriented disk intersecting K in two points of opposite sign. A crossing change on K is the operation of perf...

conjecture | open | MathDB 页面 | 原文线索

V2-033394 | MATHDB-364005

Parametrised loop coproduct obstruction conjecture

摘录 / EXCERPT: Let $E\to B$ be a smooth fiber bundle with fiber a smooth closed manifold M, and let $f:E\to M\times B$ be a fiberwise homotopy equivalence over B. Suppose there are spectral o...

conjecture | open | MathDB 页面 | 原文线索

V2-033395 | MATHDB-364006

Higher Reidemeister trace characterization conjecture

摘录 / EXCERPT: Let M be a smooth closed manifold, let $\mathcal{H}(M)$ be its stable h-cobordism space, and suppose the higher Reidemeister trace constructions define a map ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033396 | MATHDB-364007

Higher Whitehead torsion and Reidemeister trace compatibility conjecture

摘录 / EXCERPT: Higher Whitehead torsion and Reidemeister trace compatibility conjecture. The following diagram commutes up to natural homotopy:

conjecture | open | MathDB 页面 | 原文线索

V2-033397 | MATHDB-364013

Kervaire's classification conjecture for 4-connected 10-manifolds

摘录 / EXCERPT: Kervaire's classification conjecture. The 5-th Betti number $B_5(M)$ and Kervaire invariant $\Phi(M)\in \mathbb{Z}_2$ are a complete set of invariants of the homotopy type of M...

conjecture | open | MathDB 页面 | 原文线索

V2-033398 | MATHDB-364025

Formality conjecture for isotropy Lie infinity-algebras of singular foliations

摘录 / EXCERPT: Let $\mathcal F$ be a singular foliation admitting a geometric resolution, and let its isotropy Lie ∞-algebra at a point be the corresponding homotopy-theoretic isotropy obj...

conjecture | open | MathDB 页面 | 原文线索

V2-033399 | MATHDB-364041

Near-r_n homotopy-type conjecture for Vietoris-Rips complexes of spheres

摘录 / EXCERPT: Let S^n be the n-sphere, let r_n be the scale defined in the paper, let $\mathrm{SO}(n+1)$ be the special orthogonal group, let A_{n+2} be the alternating group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033400 | MATHDB-364042

Finite-CW conjecture for Vietoris-Rips complexes of spheres

摘录 / EXCERPT: Let S^n be the n-sphere, let $r\ge 0$, and let $\mathrm{VR}(S^n;r)$ denote the Vietoris-Rips complex formed using the strict-distance convention of the paper. Finite-CW conject...

conjecture | open | MathDB 页面 | 原文线索

V2-033401 | MATHDB-364047

Berarducci-Peterzil-Pillay's interpretability conjecture for finite central extensions

摘录 / EXCERPT: Berarducci-Peterzil-Pillay's conjecture. The natural inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-033402 | MATHDB-364075

BD additivity conjecture

摘录 / EXCERPT: Work over the complete topological $k\llbracket\hbar\rrbracket$-modules, with their symmetric monoidal structure, and let $\mathbb{BD}_0$ and $\mathbb{BD}_1$ denote the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-033403 | MATHDB-364125

Cohomological spectral-sequence conjecture for locally analytic vectors

摘录 / EXCERPT: Let G be a compact p-adic Lie group, let M be a G-module in the setting of the paper, and let $\mathrm{R}_{\mathrm{la}}^j(M)$ denote the jth derived functor of locally an...

conjecture | open | MathDB 页面 | 原文线索

V2-033404 | MATHDB-364141

The gravitational cosmic Galois group conjecture

摘录 / EXCERPT: Let quantum gravity theories be theories whose gravitational path integrals include sums over spacetime topologies, and let gauge redundancies associated with topology change denot...

conjecture | open | MathDB 页面 | 原文线索

V2-033405 | MATHDB-364143

Enochs's conjecture on covering classes of modules

摘录 / EXCERPT: Let $\mathcal{X}$ be a class of modules. A class is covering if every module M has an $\mathcal{X}$-cover, namely an $\mathcal{X}$-precover $f\colon X\to M$ such that every...

conjecture | open | MathDB 页面 | 原文线索

V2-033406 | MATHDB-364238

Invariant-measure variational principle for r-neutralized topological entropy

摘录 / EXCERPT: Let (X,d) be a compact metric space, let $f:X\to X$ be a homeomorphism, and let $\mathcal{M}(f,X)$ be the invariant Borel probability measures for f. For $r>0$, define the r-...

conjecture | open | MathDB 页面 | 原文线索

V2-033407 | MATHDB-364252

The CoHA affine Yangian identification for cyclic quivers

摘录 / EXCERPT: Let $K>2$, and let Y^{K} be the $C[Chbar_1,Chbar_2]$-linear algebra obtained by gluing the two copies of the double deformed current algebra A^K constructed from observables on...

conjecture | open | MathDB 页面 | 原文线索

V2-033408 | MATHDB-364255

Finite-test homotopy equivalence conjecture for finite-dimensional CW complexes

摘录 / EXCERPT: Finite-test homotopy equivalence conjecture. The map f is a homotopy equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-033409 | MATHDB-364259

Gallotti-Pierre-Louis conjecture on circular elastic knots

摘录 / EXCERPT: Let N be a positive integer, and let a knot class have braid index and bridge index both equal to N. An N-fold circle is the corresponding circle traversed N times, and an...

conjecture | open | MathDB 页面 | 原文线索

V2-033410 | MATHDB-364260

Uniqueness conjecture for the elastic teardrop-heart

摘录 / EXCERPT: Let the figure-eight knot class be the knot class represented by the figure-eight knot, and let an elastic knot be understood in the sense of Gerlach-Reiter-von der Mosel. Elastic…

conjecture | open | MathDB 页面 | 原文线索

V2-033411 | MATHDB-364261

Miura's second-energy conjecture for the elastic propeller

摘录 / EXCERPT: For a tame knot class K, let $\mathcal{K}$ be the $W^{2,2}$-closure of the class of unit-length, arclength-parametrized closed curves in K: … A stable elastic knot for K is a…

conjecture | open | MathDB 页面 | 原文线索

V2-033412 | MATHDB-364262

Existence conjecture for stable closed leafed elasticae

摘录 / EXCERPT: For an integer $r\geq4$, a closed r-leafed elastica is a closed elastic curve consisting of r leaves. A stable elastic knot for the unknot class is a curve that locally minimiz…

conjecture | open | MathDB 页面 | 原文线索

V2-033413 | MATHDB-364291

Carlson-Toledo conjecture on virtually positive second Betti number of Kähler groups

摘录 / EXCERPT: Let G be an infinite Kähler group, meaning the fundamental group of a smooth complex Kähler manifold. Carlson-Toledo conjecture. There exists a finite-index subgroup G° of…

conjecture | open | MathDB 页面 | 原文线索

V2-033414 | MATHDB-364328

The topological-generation conjecture for morphisms of generic motives

摘录 / EXCERPT: Let generic motives be the pro-objects considered in the paper, and let morphisms of type $(\mathrm{D}n^*)$ and relations $(\mathrm{R}m^*)$ be the classes defined in the paper. Top…

conjecture | open | MathDB 页面 | 原文线索

V2-033415 | MATHDB-364330

The characteristic-zero comparison conjecture for symplectic annular Khovanov homology

摘录 / EXCERPT: Let L be an annular link, let $k\in\mathbb Z$ be its winding grading, and let $\mathbb F$ be a field of characteristic zero. Denote by…

conjecture | open | MathDB 页面 | 原文线索

V2-033416 | MATHDB-364349

The radial limit conjecture for Costantino-Geer-Patureau-Mirand invariants

摘录 / EXCERPT: The radial limit conjecture. For any sector $S\subset\mathbb{H}$, the limit as $\tau\in S$ tends to 0 satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033417 | MATHDB-364371

The top elementary coefficient conjecture for Segre powers of Boolean lattices

摘录 / EXCERPT: Top elementary coefficient conjecture. The coefficient of e_n is always 1, that is, $a_{(n)}=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-033418 | MATHDB-364372

Irreducible-support stability conjecture for Segre powers of Boolean lattices

摘录 / EXCERPT: Irreducible-support stability conjecture. For $t\geqslant3$, every irreducible appearing in $\tilde{H}_{n-2}(B_n^{(t)})$ also appears in $\tilde{H}_{n-2}(B_n^{(2)})$.

conjecture | open | MathDB 页面 | 原文线索

V2-033419 | MATHDB-364388

The universal upper-bound conjecture for 2-dimensional simplicial complexes

摘录 / EXCERPT: Let X be any 2-dimensional simplicial complex. The topological Turán number is denoted by $\operatorname{ex}_{\hom}(n,X)$. Universal upper-bound conjecture. There is a constant…

conjecture | open | MathDB 页面 | 原文线索

V2-033420 | MATHDB-364389

The lower-bound conjecture for sufficiently complicated 2-complexes

摘录 / EXCERPT: Let X be a 2-dimensional simplicial complex in which every 1-simplex is contained in at least two 2-simplices. The topological Turán number is denoted by…

conjecture | open | MathDB 页面 | 原文线索

V2-033421 | MATHDB-364390

Newman-Pavelka sphere exponent conjecture

摘录 / EXCERPT: Let $d\geq 3$, let $\mathbb S^d$ be the d-dimensional sphere, and let $\operatorname{ex}_{\hom}(n,\mathbb S^d)$ denote its topological Turán number. Newman-Pavelka sphere exponen…

conjecture | open | MathDB 页面 | 原文线索

V2-033422 | MATHDB-364393

Relative symplectic cohomology and Maurer-Cartan deformation conjecture

摘录 / EXCERPT: Relative symplectic cohomology conjecture. There is an isomorphism of graded R-modules

conjecture | open | MathDB 页面 | 原文线索

V2-033423 | MATHDB-364395

The \(\mathcal{QHB}\) conjecture for small Seifert fibered \(L\)-spaces

摘录 / EXCERPT: The $\mathcal{QHB}$ conjecture. If Y admits a symplectic rational homology ball filling, then

conjecture | open | MathDB 页面 | 原文线索

V2-033424 | MATHDB-364412

The Whitehead-product kernel conjecture for shrinking wedges

摘录 / EXCERPT: Let $n\geq 2$ and let … be a shrinking wedge of finite $(n-1)$-connected CW-complexes. Let $\sigma:X\to\prod_{j=1}^{\infty}X_j$ be the inclusion map. Whitehead-product kernel conje…

conjecture | open | MathDB 页面 | 原文线索

V2-033425 | MATHDB-364430

The topological reformulation of Frankl's conjecture

摘录 / EXCERPT: A supratopological space is a pair $(X,\mathcal{F})$ in which X is a finite set and $\mathcal{F}\subseteq\mathcal{P}(X)$ is union-closed with $X\in\mathcal{F}$; the members of...

conjecture | open | MathDB 页面 | 原文线索

V2-033426 | MATHDB-364441

Ribbon number additivity under connected sum

摘录 / EXCERPT: Let K_1 and K_2 be ribbon knots, and let $r(K)$ denote the minimum ribbon number of a ribbon disk for K. Ribbon number additivity conjecture. If K_1 and K_2 are ribbon kn...

conjecture | open | MathDB 页面 | 原文线索

V2-033427 | MATHDB-364442

Ribbon number of a knot and its mirror connected sum

摘录 / EXCERPT: Let K be a knot, let $\overline{K}$ denote its mirror image, let $r(K)$ denote the minimum ribbon number of a ribbon disk for K, and let $c(K)$ denote the crossing number of...

conjecture | open | MathDB 页面 | 原文线索

V2-033428 | MATHDB-364456

Buryak-Rossi-Shadrin bihamiltonian conjecture for the DR hierarchy

摘录 / EXCERPT: Buryak-Rossi-Shadrin's conjecture. The operator K^DR is Poisson and compatible with $\eta^{-1}\partial_x$. It endows the DR hierarchy with a bihamiltonian structure sati...

conjecture | solved | MathDB 页面 | 原文线索

V2-033429 | MATHDB-364477

The Witt-group decomposition conjecture for trivial syllepsis

摘录 / EXCERPT: Let A be a finite abelian group. Write $\mathcal{W}itt(A,\mathsf{triv})$ for the Witt group associated to A with trivial syllepsis, let $\mathcal{W}itt$ denote the ordinary Wit...

conjecture | open | MathDB 页面 | 原文线索

V2-033430 | MATHDB-364482

The dimer-model conjecture for torus knots

摘录 / EXCERPT: Torus-knot dimer-model conjecture. Every (p,q) torus knot admits a dimer model.

conjecture | open | MathDB 页面 | 原文线索

V2-033431 | MATHDB-364489

Equivariant concordance invariance of the Maslov grading invariant

摘录 / EXCERPT: Equivariant concordance conjecture. The number s_τ is an equivariant concordance invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-033432 | MATHDB-364490

Non-collapse of the spectral sequence for strongly invertible knots

摘录 / EXCERPT: Spectral-sequence non-collapse conjecture. There exists a strongly invertible knot $\widetilde{K}$ such that the spectral sequence of Theorem 1 collapses on the E_k page for some...

conjecture | open | MathDB 页面 | 原文线索

V2-033433 | MATHDB-364531

Cohomological identification conjectures for BPS complexes of symmetric quotient stacks

摘录 / EXCERPT: BPS complex conjectures. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-033434 | MATHDB-364532

Classification conjecture for parameterized families of 1+1d gapped systems

摘录 / EXCERPT: Let $\mathcal{C}$ be a fusion category, let $\mathcal{M}$ be a $\mathcal{C}$-module category, and denote the corresponding gapped phase by $\mathcal{T}_{\mathcal{M}}^{\mathcal{C}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-033435 | MATHDB-364533

Non-abelian cohomology conjecture for parameterized 1+1d gapped systems

摘录 / EXCERPT: Let X be a sufficiently nice parameter space, such as a CW complex, let $\mathcal{C}$ be a fusion category, and let $\mathcal{M}$ be a fixed $\mathcal{C}$-module category with co...

conjecture | open | MathDB 页面 | 原文线索

V2-033436 | MATHDB-364534

Fiber 2-functor conjecture for parameterized 2+1d non-chiral SPT phases

摘录 / EXCERPT: Let $\mathcal{C}$ be a fusion 2-category and let $2\operatorname{Vec}$ be the 2-category of finite semisimple 1-categories. A fiber 2-functor is a tensor 2-functor...

conjecture | open | MathDB 页面 | 原文线索

V2-033437 | MATHDB-364550

Zhang's SU(2)-abelian rational homology sphere conjecture

摘录 / EXCERPT: Let Y be a rational homology 3-sphere. Call Y $SU(2)$-abelian if every representation $\pi_1(Y)\to SU(2)$ has abelian image. Zhang's conjecture. If Y is $SU(2)$-abelian, th...

conjecture | open | MathDB 页面 | 原文线索

V2-033438 | MATHDB-364605

Rosenberg's conjecture on topological and algebraic K-theory of C*-algebras

摘录 / EXCERPT: Rosenberg's conjecture. 1. For $n>0$, the map

conjecture | open | MathDB 页面 | 原文线索

V2-033439 | MATHDB-364620

The second-level classification conjecture for low-dimensional forms in I^n

摘录 / EXCERPT: Let F be a field, let $I^n(F)$ denote the nth power of the fundamental ideal in the Witt ring of F, and let $ u$-fold Pfister forms be tensor products of $ u$ binary forms. A...

conjecture | open | MathDB 页面 | 原文线索

V2-033440 | MATHDB-364667

The Sierpinski Tetrahedron universality conjecture for knots

摘录 / EXCERPT: Let K be a knot, and let the Sierpinski Tetrahedron denote the fractal obtained by the iterative construction described in the paper. A finite iteration is one of the finite-stag...

conjecture | open | MathDB 页面 | 原文线索

V2-033441 | MATHDB-364672

The AJ conjecture for the colored Jones polynomial

摘录 / EXCERPT: Let K be a knot. Its colored Jones polynomial is $J_K(n)=J_n(K;q)$ and satisfies a nontrivial recurrence ... where $c_j(q,q^n)\in\mathbb{Z}[q,q^n]$. Define operators E and Q by...

conjecture | open | MathDB 页面 | 原文线索

V2-033442 | MATHDB-364675

The exponential type conjecture for quantum t-connections

摘录 / EXCERPT: Let $(X,comega)$ be a closed monotone symplectic manifold of dimension $2n$, with $[comega]=c_1(X)$. Let t be a formal variable of degree 2, and let ... be the quantum t-connec...

conjecture | open | MathDB 页面 | 原文线索

V2-033443 | MATHDB-364716

Steingart's surjectivity conjecture for Lubin-Tate Iwasawa cohomology

摘录 / EXCERPT: Steingart's surjectivity conjecture. The map comp is surjective after taking H^1, namely

conjecture | open | MathDB 页面 | 原文线索

V2-033444 | MATHDB-364752

Minimal-system nonrecurrence under two transformations

摘录 / EXCERPT: A topological dynamical system is a pair (X,T) consisting of a compact space X and a continuous map $T:X\to X$; it is minimal if every orbit is dense. For two transformations...

conjecture | open | MathDB 页面 | 原文线索

V2-033445 | MATHDB-364753

Polynomial simultaneous nonrecurrence conjecture for minimal systems

摘录 / EXCERPT: Let p and q be integral polynomials vanishing at 0, with $\deg p\ge 2$ and $\deg q\ge 2$. Let (X,T) and (X,S) be systems on the same compact space X. Polynomial simulta...

conjecture | open | MathDB 页面 | 原文线索

V2-033446 | MATHDB-364754

Mixed linear-quadratic nonrecurrence conjecture for minimal systems

摘录 / EXCERPT: Let (X,T) and (X,S) be minimal systems on the same compact space X. Mixed linear-quadratic nonrecurrence conjecture. There are minimal systems (X,T) and (X,S) such that f...

conjecture | open | MathDB 页面 | 原文线索

V2-033447 | MATHDB-364763

The homological contractibility conjecture for generic oper structures

摘录 / EXCERPT: Generic-oper contractibility conjecture. The space of generic oper structures on a given irreducible local system is homologically contractible.

conjecture | open | MathDB 页面 | 原文线索

V2-033448 | MATHDB-364764

The multivariate mean equicontinuity characterization of finite topomorphic extensions

摘录 / EXCERPT: Let (X,φ) be a minimal topological dynamical system, and let (Y,ψ) be its maximal equicontinuous factor (MEF). For $m\in\mathbb{N}$, say that (X,φ) is mean m...

conjecture | open | MathDB 页面 | 原文线索

V2-033449 | MATHDB-364776

Crossing-arc separation conjecture for connected sums

摘录 / EXCERPT: Let $K\#K'$ be a connected sum of knots, let a collection of unknotting crossing arcs for $K\#K'$ be given, and let the 2-sphere specifying the connected sum be fixed. Crossing-arc...

conjecture | open | MathDB 页面 | 原文线索

V2-033450 | MATHDB-364789

Watson's diagonal grading characterization of strongly invertible L-space knots

摘录 / EXCERPT: Watson's conjecture. K is an L-space knot if and only if $[?]$ is supported in a single diagonal grading

conjecture | open | MathDB 页面 | 原文线索

V2-033451 | MATHDB-364790

The at-most-two-slopes conjecture for alternating surgeries

摘录 / EXCERPT: Let Knotin $\mathcal{D}$ be a knot admitting positive alternating surgeries. Write $\mathcal{S}_{\rm alt}(K)$ for the set of alternating surgery slopes, and let N denote the re...

conjecture | open | MathDB 页面 | 原文线索

V2-033452 | MATHDB-364791

The no-alternating-surgery conjecture for knots sharing Alexander polynomials with \(\mathcal{D}\)

摘录 / EXCERPT: Let $K\notin \mathcal{D}$ be a knot, and suppose that there is a knot $K'\in \mathcal{D}$ with ... Here $\Delta_K(x)$ denotes the Alexander polynomial of K, and...

conjecture | open | MathDB 页面 | 原文线索

V2-033453 | MATHDB-364832

Infinite-order monodromy conjecture for positive-geometric-genus surface singularities

摘录 / EXCERPT: Let $V = \{f = 0\}$ be a 2-dimensional weighted-homogeneous isolated hypersurface singularity with rational homology 3-sphere link Y, Milnor fiber M, and monodromy ψ o...

conjecture | open | MathDB 页面 | 原文线索

V2-033454 | MATHDB-364833

Generalized infinite-order monodromy conjecture for surface singularity smoothings

摘录 / EXCERPT: Let (V,p) be a normal isolated surface singularity with rational homology 3-sphere link, and let it have a smoothing with Milnor fiber M and monodromy ψ. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-033455 | MATHDB-364854

The pretzel-knot ribbon number spectrum conjecture

摘录 / EXCERPT: Let $p,q\in\mathbb{Z}$ be odd, let $n\in\mathbb{N}$, and let K be the $(2n+1)$-stranded pretzel knot ... For $0\leq g\leq n$, write $r_g(K)$ for the higher-genus ribbon number and...

conjecture | open | MathDB 页面 | 原文线索

V2-033456 | MATHDB-364869

The Morava K-theory analogue of Yagita's conjecture

摘录 / EXCERPT: Let K be a compact connected Lie group, T its maximal torus, and $\pi\colon K\rightarrow K/T$ the natural projection. Denote by $G=K_{\mathbb C}$ the corresponding split reduct...

conjecture | open | MathDB 页面 | 原文线索

V2-033457 | MATHDB-364896

Homotopy conjecture for the Fukaya category completion functors

摘录 / EXCERPT: Let $\mathcal{N}(\bar{X},F_{0})$ be the Fukaya-type category and let $\widehat{\mathcal{W}}_{F}$ be its categorical formal completion. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-033458 | MATHDB-364911

Divisibility conjecture for hyperbolic alternating knots with the strong roller-coaster property

摘录 / EXCERPT: Divisibility conjecture. If K has the strong roller-coaster property, then the crossing number of K is divisible by four.

conjecture | open | MathDB 页面 | 原文线索

V2-033459 | MATHDB-364912

Minimal ascending number conjecture for reduced alternating diagrams

摘录 / EXCERPT: Minimal ascending number conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-033460 | MATHDB-364934

Kapustin-Witten Langlands duality conjecture for cohomological Donaldson-Thomas invariants

摘录 / EXCERPT: Let G be a reductive group, G^{L} its Langlands dual, and M a closed oriented 3-manifold. Kapustin-Witten Langlands duality conjecture. There is an isomorphism ... This is mo...

conjecture | open | MathDB 页面 | 原文线索

V2-033461 | MATHDB-364957

Completeness conjecture for the S^n functional

摘录 / EXCERPT: Completeness conjecture for the S^n functional. The S^n functional Z is complete positive and complete finite. The conjecture occurs in the paper's discussion of extending th...

conjecture | open | MathDB 页面 | 原文线索

V2-033462 | MATHDB-365015

The super-Kauffman conjecture for Turk’s head knots

摘录 / EXCERPT: Let $Th(p,q)$ denote a Turk’s head knot, with $p\geq 5$ and $q\geq 4$. A knot is super-Kauffman if it admits a minimal alternating diagram whose every crossing has the Kauffman pro...

conjecture | open | MathDB 页面 | 原文线索

V2-033463 | MATHDB-365016

The symmetry-group conjecture for Turk’s head knots

摘录 / EXCERPT: Let p,q be positive integers with $\gcd(p,q)=1$, so that $Th(p,q)$ is a Turk’s head knot, and let D_{2q} denote the dihedral group of order $2q$. Symmetry-group conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-033464 | MATHDB-365017

The slice-genus, unknotting-number, and Seifert-genus conjecture for Turk’s head knots

摘录 / EXCERPT: For a Turk’s head knot $Th(p,q)$, let g_4 denote slice genus, u unknotting number, and g_3 Seifert genus. Turk’s head genus conjecture. ... and equality...

conjecture | open | MathDB 页面 | 原文线索

V2-033465 | MATHDB-365024

The non-T_0 geometric-realisation conjecture for finite spaces

摘录 / EXCERPT: Let Y be a finite topological space, and let $|Y|_\omega$ denote its non-T_0 geometric realisation. Let $\mathsf{W}\to\mathsf{V}$ and $\mathsf{V}\to\{\bullet\}$ be the finite-s...

conjecture | open | MathDB 页面 | 原文线索

V2-033466 | MATHDB-365025

The proper-map characterization by the simplest finite counterexample

摘录 / EXCERPT: Let $\{o\}\to\{o\to c\}$ be the indicated map of finite topological spaces, and let the subscript <5 denote the subclass consisting of maps between spaces with fewer than 5 poi...

conjecture | open | MathDB 页面 | 原文线索

V2-033467 | MATHDB-365026

The finite-map criterion for properness

摘录 / EXCERPT: Let f be a map of finite topological spaces. Say that a fibre has two points bounded above but not below when its two points have a common upper bound but no common lower bound,...

conjecture | open | MathDB 页面 | 原文线索

V2-033468 | MATHDB-365027

The maximality conjecture for proper maps under double orthogonals

摘录 / EXCERPT: Let Q be any class of morphisms of topological spaces, and write Q^{lr} for its double Quillen orthogonal. The proper-map maximality conjecture. If ... then ... The paper proves th...

conjecture | open | MathDB 页面 | 原文线索

V2-033469 | MATHDB-365028

The model-structure conjecture generated by finite topological spaces

摘录 / EXCERPT: Let $\mathsf{W}\to\mathsf{V}$ and $\mathsf{V}\to\{\bullet\}$ be the indicated maps of finite topological spaces, and let Top denote the category of topological spa...

conjecture | open | MathDB 页面 | 原文线索

V2-033470 | MATHDB-365029

The non-Hausdorff Brouwer fixed point conjecture

摘录 / EXCERPT: Let K be a topological space equipped with maps $K\to\{\bullet\}$ lying in both indicated right-orthogonal classes: the class generated by the finite-space maps...

conjecture | refuted | MathDB 页面 | 原文线索

V2-033471 | MATHDB-365030

The finite-space characterization of Lebesgue covering dimension

摘录 / EXCERPT: Let $\partial\Delta^n$ be a finite topological space weakly homotopy equivalent to the sphere $\mathbb S^n$, with $\mathbb S^n\to\partial\Delta^n$ a trivial Serre fibration, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-033472 | MATHDB-365031

The finite-space characterization of fibrations

摘录 / EXCERPT: Let $f:X\to B$ be a cellular map of finite CW complexes. Let P be the finite-space map from the three-point diagram with one distinguished point and two indistinguishable points...

conjecture | open | MathDB 页面 | 原文线索

V2-033473 | MATHDB-365040

The normalized flat Jones-Krushkal polynomial conjecture for almost classical flat knots

摘录 / EXCERPT: Normalized flat Jones-Krushkal polynomial conjecture. For any almost classical flat knot α, the normalized flat Jones-Krushkal polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033474 | MATHDB-365066

Gallagher-Lai-Weber topological Erdős similarity conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}$ be topologically universal if every dense G_δ subset of $\mathbb{R}$ contains an affine copy of X. Gallagher-Lai-Weber's conjecture. No uncounta...

conjecture | open | MathDB 页面 | 原文线索

V2-033475 | MATHDB-365067

Topological universality conjecture for locally compact Polish groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a locally compact Polish group, and call a subset of $\mathbb{G}$ topologically universal if every dense G_δ subset of $\mathbb{G}$ contains its image un...

conjecture | open | MathDB 页面 | 原文线索

V2-033476 | MATHDB-365108

Parity conjecture for Coxeter braids

摘录 / EXCERPT: For integers $0\leq d_1\leq d_2\leq\cdots\leq d_n$, define the Coxeter braid ... where JM_k is a Jucys-Murphy element on the first k strands. A braid is parity whe...

conjecture | open | MathDB 页面 | 原文线索

V2-033477 | MATHDB-365109

Ideal description conjecture for Coxeter braid homology

摘录 / EXCERPT: Let $0\leq d_1\leq\cdots\leq d_1$, let $\beta=\operatorname{JM}_n^{d_1}\ldots\operatorname{JM}_2^{d_{n-1}}$, and set $J=J(d_1,\ldots,d_n)$, where ... The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-033478 | MATHDB-365115

Upper-degree conjecture for generators and relations of 3D space-group cohomology

摘录 / EXCERPT: Let G be a 3D space group and consider its mod-2 cohomology ring $H^*(G,\mathbb{Z}_2)$. Upper-degree conjecture for generators and relations. An independent generator can appear...

conjecture | open | MathDB 页面 | 原文线索

V2-033479 | MATHDB-365116

Module-support conjecture for the q=1,2 rows of the LHS spectral sequence

摘录 / EXCERPT: Let G be a 3D space group, and consider the standard Lyndon-Hochschild-Serre spectral sequence for its translation extension. At the E_2 page, examine the module structures in...

conjecture | open | MathDB 页面 | 原文线索

V2-033480 | MATHDB-365128

The local Hamiltonian nondisplacement conjecture

摘录 / EXCERPT: Let L be a displaceable Lagrangian in a symplectic manifold M, and let d_{C^0} be the C^0 distance on Hamiltonian diffeomorphisms. Local Hamiltonian nondisplacement conject...

conjecture | open | MathDB 页面 | 原文线索

V2-033481 | MATHDB-365184

The strong-edge conjecture for graphs of dynamical systems

摘录 / EXCERPT: Let F be a dynamical system with a compact global attractor. An edge of the graph of F is called strong when it satisfies the source's strong-edge relation. Strong-edge conject...

conjecture | open | MathDB 页面 | 原文线索

V2-033482 | MATHDB-365189

Gukov-Pei-Putrov-Vafa radial limit conjecture for negative definite plumbed manifolds

摘录 / EXCERPT: Gukov-Pei-Putrov-Vafa conjecture. In the above situation,

conjecture | open | MathDB 页面 | 原文线索

V2-033483 | MATHDB-365221

Quantum law of large numbers in the strong operator topology of ℓp

摘录 / EXCERPT: Quantum law of large numbers. Under these assumptions, $e^{A_1t/n}\cdots e^{A_nt/n}$ converges in probability in the strong operator topology of ℓ_p to...

conjecture | open | MathDB 页面 | 原文线索

V2-033484 | MATHDB-365234

The Mahowald-Rezk telescope conjecture for fp-spectra

摘录 / EXCERPT: Let X be a spectrum of fp-type n, and let L_n^f and L_n denote, respectively, finite and ordinary chromatic localization at height n. Mahowald-Rezk conjecture. X satisf...

conjecture | refuted | MathDB 页面 | 原文线索

V2-033485 | MATHDB-365245

The purely noncommutative decorated-tiling fruitfulness conjecture

摘录 / EXCERPT: The decorated-tiling fruitfulness conjecture. Any purely noncommutative decorated tiling of a polygon provides a fruitful seed. More generally, a decorated tiling of an arbitrary s...

conjecture | open | MathDB 页面 | 原文线索

V2-033486 | MATHDB-365252

Euler-characteristic asymptotics for the Fibonacci Lie algebra

摘录 / EXCERPT: Let $\mathcal L$ be the Fibonacci Lie algebra, let $\lambda=(1+\sqrt 5)/2$, and let $\mathbf E(\mathcal L,t)$ denote its Euler characteristic. Euler-characteristic asymptotic conje...

conjecture | open | MathDB 页面 | 原文线索

V2-033487 | MATHDB-365254

Infinite-dimensionality conjecture for higher homology of the Fibonacci Lie algebra

摘录 / EXCERPT: Let $p=2$, let $\mathcal L=\operatorname{Lie}(v_1,v_2)$ be the Fibonacci Lie algebra, and let $H_n(\mathcal L)$ denote its homology groups. Higher-homology conjecture. ... The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-033488 | MATHDB-365264

Birbrair-Gabrielov conjecture on bi-Lipschitz equivalence of LNE surface germs

摘录 / EXCERPT: Birbrair-Gabrielov conjecture. If $(X_0,0)$ and $(X_1,0)$ are ambient topologically equivalent and outer bi-Lipschitz equivalent, then X_0 and X_1 are ambient bi-Lipschitz equi...

conjecture | open | MathDB 页面 | 原文线索

V2-033489 | MATHDB-365267

Finite tensor categories have finitely generated cohomology

摘录 / EXCERPT: Let $\mathbbm{k}$ be an algebraically closed field of characteristic zero, and let $\mathcal{C}$ be a finite tensor category over $\mathbbm{k}$. Write ... for the cohomology of an ob...

conjecture | open | MathDB 页面 | 原文线索

V2-033490 | MATHDB-365306

Andersen-Kashaev Jones function conjecture

摘录 / EXCERPT: Let M be a connected closed oriented 3-manifold and let $K\subset M$ be a hyperbolic knot. Andersen-Kashaev Jones function conjecture. There exist an ideal triangulation X of...

conjecture | open | MathDB 页面 | 原文线索

V2-033491 | MATHDB-365309

Finite-energy versus finite-topology conjecture for complete Ricci-flat 4-manifolds

摘录 / EXCERPT: Finite-energy versus finite-topology conjecture. The manifold (X,g) has finite energy if and only if X has finite topological type.

conjecture | open | MathDB 页面 | 原文线索

V2-033492 | MATHDB-365350

The elliptic stable-basis duality conjecture under symplectic duality

摘录 / EXCERPT: Elliptic stable-basis duality conjecture. For any $p_1,p_2\in X^H$,

conjecture | open | MathDB 页面 | 原文线索

V2-033493 | MATHDB-365351

The existence conjecture for elliptic canonical bases

摘录 / EXCERPT: Let ${\mathfrak{X}}$ be the symplectic-resolution object under consideration, and let $\Xi_{{\mathfrak{X}}}$ be the parameter set for elliptic canonical-basis classes. Let Properti...

conjecture | open | MathDB 页面 | 原文线索

V2-033494 | MATHDB-365388

Berglund-Hübsch homological mirror symmetry for invertible polynomials

摘录 / EXCERPT: Berglund-Hübsch homological mirror symmetry conjecture. There exists a derived equivalence between $\mathcal{F}(W,G)$ and $\mathcal{MF}(W^T,G^T)$.

conjecture | solved | MathDB 页面 | 原文线索

V2-033495 | MATHDB-365397

The Ribbon-SU conjecture for ribbon knots

摘录 / EXCERPT: Ribbon-SU conjecture. Every ribbon knot is an SU knot.

conjecture | open | MathDB 页面 | 原文线索

V2-033496 | MATHDB-365398

The symmetric braid index conjecture for eight small SU knots

摘录 / EXCERPT: Symmetric braid index conjecture. If K is one of these eight SU knots, then

conjecture | open | MathDB 页面 | 原文线索

V2-033497 | MATHDB-365409

Jones polynomial conjecture for neighborhoods of Conway and Kinoshita-Terasaka knots

摘录 / EXCERPT: Let the Conway and Kinoshita-Terasaka knots be considered through their embeddings and their open curve neighborhoods. The Jones polynomial is the knot invariant used to compare th...

conjecture | open | MathDB 页面 | 原文线索

V2-033498 | MATHDB-365448

Embeddability conjecture for skeleta of triangulated complex projective spaces

摘录 / EXCERPT: Embeddability conjecture. If $d=2^k-1$ for some $k\geq 1$, then the d-skeleton of K is embeddable into $\mathbb{R}^{2d}$.

conjecture | open | MathDB 页面 | 原文线索

V2-033499 | MATHDB-365469

The Witt-to-interface conjecture for rational conformal field theories

摘录 / EXCERPT: Let $\mathcal{T}$ and $\mathcal{T}'$ be RCFTs with positive chiral algebras. RCFT Witt-to-interface conjecture. If $\mathcal{T}$ and $\mathcal{T}'$ belong to the same Witt class, t...

conjecture | open | MathDB 页面 | 原文线索

V2-033500 | MATHDB-365490

Boundary characterization for positive matrices with size-2 psd factorizations

摘录 / EXCERPT: Size-2 boundary characterization conjecture. A positive matrix M is on the topological boundary of $\mathcal{M}_{3,2}^{p \times q}$ if and only if any one of the conditions in th...

conjecture | open | MathDB 页面 | 原文线索

V2-033501 | MATHDB-365510

Boyer-Gordon-Hu meridional slope detection conjecture

摘录 / EXCERPT: Let $M\ncong S^1\times D^2$ be an irreducible rational homology solid torus with longitude λ_M integrally null-homologous. A rational slope at distance 1 from λ_M...

conjecture | open | MathDB 页面 | 原文线索

V2-033502 | MATHDB-365511

Baker's Floer-simple knot conjecture

摘录 / EXCERPT: Let L be a lens space and let $K\subset L$ be a knot. A knot is Floer simple when its Floer-theoretic complexity is minimal, and it is simple when it belongs to the standard clas...

conjecture | open | MathDB 页面 | 原文线索

V2-033503 | MATHDB-365512

CTF detection conjecture for non-simple knots in lens spaces

摘录 / EXCERPT: Let L be a lens space and let $K\subset L$ be a non-simple knot. The knot meridian determines a boundary slope of the knot complement, and a slope is CTF-detected if it is realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-033504 | MATHDB-365516

The conjecture on the vanishing of the third James-Hopf obstruction

摘录 / EXCERPT: The conjecture. Under these hypotheses, the map

conjecture | open | MathDB 页面 | 原文线索

V2-033505 | MATHDB-365520

Every augmentation lifts to a curved augmentation

摘录 / EXCERPT: Let $(\mathcal{A},\partial)$ be the Chekanov-Eliashberg algebra of a Legendrian, let $(\mathcal{L},d)$ be its LSFT algebra, and let $\epsilon:\mathcal{A}\to\Bbbk$ be an augmentatio...

conjecture | open | MathDB 页面 | 原文线索

V2-033506 | MATHDB-365526

The extended image conjecture for the Dehn-Nielsen-Baer map

摘录 / EXCERPT: Let S be an infinite-type surface properly homotopy equivalent to a graph $\G$, and let $\widetilde{\Theta}:\operatorname{ExMap}(S)\to\operatorname{Map}(\G)$ be the continuous in...

conjecture | open | MathDB 页面 | 原文线索

V2-033507 | MATHDB-365527

The open-closed string analogue of Deligne's conjecture

摘录 / EXCERPT: Open-closed Deligne conjecture. For any open-closed homotopy algebra $(\mathfrak{l},\mathfrak{q})$, there is a G_∞ algebra structure on the space...

conjecture | open | MathDB 页面 | 原文线索

V2-033508 | MATHDB-365541

The cosmetic surgery conjecture for non-trivial knots

摘录 / EXCERPT: Cosmetic surgery conjecture. If $p/q\neq p'/q'$, then $S^3_{p/q}(K)$ and $S^3_{p'/q'}(K)$ are not orientation-preserving diffeomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-033509 | MATHDB-365568

Stable homology conjecture for Hurwitz spaces

摘录 / EXCERPT: Stable homology conjecture. The only stable homology of these Hurwitz spaces should be the obvious stable homology coming from the configuration space. The conjecture is known for...

conjecture | open | MathDB 页面 | 原文线索

V2-033510 | MATHDB-365588

The algebraic Montgomery-Yang problem for rational homology projective planes

摘录 / EXCERPT: Let S be a rational homology projective plane, meaning a normal projective complex surface whose Betti numbers are the same as those of $\mathbb{CP}^2$. Assume that S has quoti...

conjecture | open | MathDB 页面 | 原文线索

V2-033511 | MATHDB-365608

The 2-knot branched-cover conjecture for odd Khovanov homology

摘录 / EXCERPT: Let $K\subset S^4$ be a smooth 2-knot. Removing a small neighborhood of a point of K gives a smooth link cobordism from the empty link to the unknot U, and hence a map ... For de...

conjecture | open | MathDB 页面 | 原文线索

V2-033512 | MATHDB-365619

The strong Atiyah-Sutcliffe conjecture

摘录 / EXCERPT: Let $D\colon\operatorname{Conf}_n(\mathbb{R}^3)\to\mathbb{C}$ be the normalized determinant associated with the complex lines constructed from a configuration of points. Atiyah-Sut...

conjecture | open | MathDB 页面 | 原文线索

V2-033513 | MATHDB-365640

Universality of the span 2-category for truncated indexing categories

摘录 / EXCERPT: Let $\cX$ be a truncated category, let $\Spant(\cX)$ denote its span 2-category, let $\tA$ be a 2-category, and let $\operatorname{\mbf{Fun}}^{\psa}(\cX^\op,\tA)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-033514 | MATHDB-365667

Flow-category Thom spectrum equivalence

摘录 / EXCERPT: Let F be a space and let $R:=\operatorname{Thom}(\Omega F \to BO \times \mathbb{Z})$ be the associated unital ring spectrum. Let Flow^F denote the stable...

conjecture | open | MathDB 页面 | 原文线索

V2-033515 | MATHDB-365668

Tangential-structure Floer module conjecture

摘录 / EXCERPT: Let X be a Liouville domain with stable tangent bundle classified by $X \to BU$, and let $\Phi \to BU$ be a based space over BU. Suppose the classifying map $X \to BU$ factors...

conjecture | open | MathDB 页面 | 原文线索

V2-033516 | MATHDB-365670

Flow categories as perfect bimodules conjecture

摘录 / EXCERPT: Let Θ be a tangential structure and let R be the ring spectrum defined from it. Let $\operatorname{Flow}^{\Theta}$ denote the category of Θ-oriented flow categories...

conjecture | open | MathDB 页面 | 原文线索

V2-033517 | MATHDB-365671

Flow-category Hochschild homology conjecture

摘录 / EXCERPT: Let Θ be a tangential structure, let R be its associated ring spectrum, and suppose $\operatorname{Flow}^{\Theta} \simeq \operatorname{Perf}({}_R\operatorname{mod}_R)$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-033518 | MATHDB-365708

The wedge-of-spheres conjecture for matching complexes of planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let $\mathcal{M}(G)$ denote its matching complex, whose vertices are the edges of G and whose simplices are sets of pairwise disjoint edges. Planar...

conjecture | open | MathDB 页面 | 原文线索

V2-033519 | MATHDB-365727

Marar-Mond conjecture on good real perturbations

摘录 / EXCERPT: Let $f:(\mathbb{C}^n,0)\to(\mathbb{C}^{n+1},0)$ admit a good real perturbation $f_s^{\mathbb{R}}$, meaning that the changes in homology caused by the perturbation can be observed i...

conjecture | open | MathDB 页面 | 原文线索

V2-033520 | MATHDB-365754

Semistable comparison conjecture for torsion in p-adic étale cohomology

摘录 / EXCERPT: Semistable comparison conjecture. For all $i\in \mathbb{Z}$ and $n\in \mathbb{Z}_{\ge 0}$, there is a natural isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033521 | MATHDB-365755

Entropy-minimal subshift characterization of uniform dimension in arithmetic progressions

摘录 / EXCERPT: Entropy-minimal subshift conjecture. There exists an entropy minimal subshift representing A if and only if, for every arithmetic progression P,

conjecture | open | MathDB 页面 | 原文线索

V2-033522 | MATHDB-365768

Torsion Singer conjecture for homology growth

摘录 / EXCERPT: Let M^n be a closed aspherical n-manifold with residually finite fundamental group, and set $G=\pi_1(M^n)$. Let $G_k\mathrel{\lhd}G$ be any normal chain of finite-index subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-033523 | MATHDB-365802

Purity conjecture for resolved Coulomb branches of connected semisimple groups

摘录 / EXCERPT: Purity conjecture. The rational cohomology of every resolved Coulomb branch associated with connected semisimple groups, in the sense of Braverman-Finkelberg-Nakajima, has pure Hod...

conjecture | open | MathDB 页面 | 原文线索

V2-033524 | MATHDB-365812

Split booklink isotopy conjecture for open book 3-manifolds

摘录 / EXCERPT: Let λ be a booklink in an open book 3-manifold representing a split link. A split booklink is a booklink admitting a splitting sphere intersecting each component of the bin...

conjecture | open | MathDB 页面 | 原文线索

V2-033525 | MATHDB-365813

Concavity conjecture for the bridge-braid spectrum

摘录 / EXCERPT: Let $b(d)$ denote the entries of the bridge-braid spectrum of a knot, indexed by the relevant integer parameter d. Concavity conjecture. The sequence $\{b(d)\}$ is concave upward...

conjecture | open | MathDB 页面 | 原文线索

V2-033526 | MATHDB-365814

Realization conjecture for knot bridge-braid spectra

摘录 / EXCERPT: A bridge-braid spectrum is a sequence of integers associated to a knot, with the entries decreasing and ending in $\{\ldots,1,0\}$. Bridge-braid spectrum realization conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-033527 | MATHDB-365815

Link bridge-braid spectrum realization conjecture

摘录 / EXCERPT: Let n be a positive integer, and let $\{\mathfrak{n},\ldots,1,0\}$ be a sequence. Link spectrum realization conjecture. Any sequence $\{\mathfrak{n},\ldots,1,0\}$ satisfying a se...

conjecture | open | MathDB 页面 | 原文线索

V2-033528 | MATHDB-365846

Unboundedness conjecture for the contactomorphism-group metric d_{cc}

摘录 / EXCERPT: Let W be the completion of a Liouville domain, and let $\overline{\operatorname{Cont}}_{0,c}(W\times S^1)$ be the C^0-completion of the identity component of the group of compa...

conjecture | open | MathDB 页面 | 原文线索

V2-033529 | MATHDB-365848

The vanishing Alexander polynomial conjecture for 3-manifolds

摘录 / EXCERPT: Vanishing Alexander polynomial conjecture. There exists some primitive cohomology class $\psi\in H^1(M;{\mathbb Z})$ such that the Alexander polynomial $\Delta_M^\psi(t)$ vanishes.

conjecture | open | MathDB 页面 | 原文线索

V2-033530 | MATHDB-365852

The Vol-Det conjecture for alternating hyperbolic links

摘录 / EXCERPT: Let K be an alternating hyperbolic link in S^3. Its hyperbolic volume is denoted by $\operatorname{vol}(K)=\operatorname{vol}(S^3\setminus K)$, and its determinant by $\det(K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-033531 | MATHDB-365859

Sheaf property of the structure pre-sheaf on the noncommutative spectrum

摘录 / EXCERPT: Let A be a homotopical ring and let ${\mathrm{Spec}^{\hspace{-0.4mm}\tiny{\mathtt{NC}}}\hspace{-0.5mm}}(A)$ be its noncommutative spectrum equipped with the structure pre-sheaf....

conjecture | open | MathDB 页面 | 原文线索

V2-033532 | MATHDB-365860

Faithfulness conjecture for fine Zariski covers of homotopical rings

摘录 / EXCERPT: Let A and B_i be homotopical rings, and let $\{f_i:A\to B_i\}_{i\in I}$ be a family of homotopical epimorphisms. Suppose that the family is a cover for the fine Zariski topolog...

conjecture | open | MathDB 页面 | 原文线索

V2-033533 | MATHDB-365864

Conjecture that boundary Dehn twists have infinite order for general Seifert fillings

摘录 / EXCERPT: Boundary Dehn twist conjecture. The boundary Dehn twist τ_M has infinite order in the smooth mapping class group

conjecture | open | MathDB 页面 | 原文线索

V2-033534 | MATHDB-365901

Accumulation conjecture for infinite-type maps near stable regions

摘录 / EXCERPT: Let a stable region be a region of stable interval translation maps, and let an infinite-type map be an interval translation map that is not of finite type. Accumulation of infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-033535 | MATHDB-365905

Goncharov's Chern character conjecture for the formal Hopf algebra

摘录 / EXCERPT: Let F be an infinite field, let $1\leq i\leq n$, and let $\mathcal{H}^{\mathrm{f}}(F)$ denote the formal Hopf algebra with weight-n cohomology...

conjecture | open | MathDB 页面 | 原文线索

V2-033536 | MATHDB-365911

Farrell-Zdravkovska conjecture on geometric bounding of flat and almost-flat manifolds

摘录 / EXCERPT: Farrell-Zdravkovska conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033537 | MATHDB-365932

The Galois representation conjecture for Hecke algebras

摘录 / EXCERPT: Let E be the quadratic field under consideration, let h be the Hecke algebra acting on the relevant cohomology, let $\mathfrak{m}\subset h$ be a non-Eisenstein maximal ideal, a...

conjecture | open | MathDB 页面 | 原文线索

V2-033538 | MATHDB-365934

The Borel-Wallach range vanishing conjecture for localized cohomology

摘录 / EXCERPT: Let E be the quadratic field under consideration, let K_f be the chosen level, let $\mathcal{L}_{\bm{n}}(\mathbb{F})$ be the coefficient system, and let $\mathfrak{m}$ be a non...

conjecture | open | MathDB 页面 | 原文线索

V2-033539 | MATHDB-365944

Equivariant chain-level isomorphism conjecture for broken trajectories

摘录 / EXCERPT: Equivariant chain-level isomorphism conjecture. The chain complexes $(C^\epsilon,\partial^\epsilon)$ and $(\boldsymbol{C},\boldsymbol{\partial})$ are isomorphic as $\mathbb{Z}[G]$-...

conjecture | open | MathDB 页面 | 原文线索

V2-033540 | MATHDB-365996

The non-complete-intersection conjecture for formal semigroups of hyperbolic L-space knots

摘录 / EXCERPT: Formal-semigroup conjecture. The formal semigroup S_K is not a complete intersection semigroup.

conjecture | open | MathDB 页面 | 原文线索

V2-033541 | MATHDB-366027

Cohomological lower-bound conjecture for non-Fano hypersurfaces

摘录 / EXCERPT: Let k be an algebraically closed field and let $i\colon X=V(f)\hookrightarrow\mathbb{P}^n_k$ be a closed embedding of a smooth irreducible hypersurface of degree d. For a bound...

conjecture | open | MathDB 页面 | 原文线索

V2-033542 | MATHDB-366028

Przytycki-Silvero conjecture on independence complexes of circle graphs

摘录 / EXCERPT: Przytycki-Silvero conjecture. If G is a circle graph, then $I(G)$ has the homotopy type of a wedge of spheres.

conjecture | open | MathDB 页面 | 原文线索

V2-033543 | MATHDB-366029

Przytycki-Silvero conjecture for bipartite circle graphs

摘录 / EXCERPT: Przytycki-Silvero conjecture for bipartite circle graphs. If G is a bipartite circle graph, then $I(G)$ has the homotopy type of a wedge of spheres.

conjecture | open | MathDB 页面 | 原文线索

V2-033544 | MATHDB-366035

Novikov's conjecture on homotopy invariance of higher signatures

摘录 / EXCERPT: Novikov's conjecture. The higher signatures are oriented homotopy invariant. The conjecture is a central problem in index theory and topology. In the paper's setting, it is related...

conjecture | open | MathDB 页面 | 原文线索

V2-033545 | MATHDB-366040

Generalized Auslander-Reiten conjecture for local rings

摘录 / EXCERPT: Let R be the local ring under consideration and let M be an R-module. Generalized Auslander-Reiten conjecture. If ... for all $n\gg 1$, then M has finite projective dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-033546 | MATHDB-366049

Existence of a nice embedded tree

摘录 / EXCERPT: Let X be a noncompact, metrizable generalized continuum. An infinite rooted tree is an infinite tree with a distinguished root, and a proper embedding is a topological embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-033547 | MATHDB-366051

Generalized statistics are classified by higher-group cohomology

摘录 / EXCERPT: Let p-dimensional excitations have a $(d-p-1)$-form symmetry with finite fusion group G in d spatial dimensions. Let X be a simplicial complex triangulating the d-sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-033548 | MATHDB-366056

Upper-bound conjecture for Δ-invariants of negative-definite plumbed manifolds

摘录 / EXCERPT: Let Y be a negative-definite plumbed manifold, let $\spc{Y}$ denote its set of $spin^c$ structures, and let Δ_b be the corresponding invariant for $b\in \spc{Y}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033549 | MATHDB-366100

Formal Swiss Cheese conjecture for enriched 2-operads

摘录 / EXCERPT: Let V be a symmetric monoidal category and let $\mathcal{P}$ be a V-enriched 2-operad. Let $\mathcal{P}_1$ be the 1-operad obtained by restricting $\mathcal{P}$ to the 1-...

conjecture | open | MathDB 页面 | 原文线索

V2-033550 | MATHDB-366104

Unbounded hardness conjecture for unknot diagrams

摘录 / EXCERPT: Let m be an integer and let $\hat{\mathcal{D}}$ be the 0-crossing diagram of the unknot. A diagram of the unknot $\mathcal{D}$ has n crossings if...

conjecture | open | MathDB 页面 | 原文线索

V2-033551 | MATHDB-366128

Draisma's topological noetherianity conjecture for the affine infinite Grassmannian

摘录 / EXCERPT: Draisma's conjecture. The affine infinite Grassmannian $\widetilde{\operatorname{Gr}}(\infty/2,V^*_\infty)$ is topologically S_∞-noetherian: every descending chain of...

conjecture | open | MathDB 页面 | 原文线索

V2-033552 | MATHDB-366142

The standardness conjecture for the census 4-ball B^4_C

摘录 / EXCERPT: Let C be the specified 2-pentachoron cut-open Cappell--Shaneson complex, and let B^4_C be the topological 4-ball containing C described in the paper. Standardness conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-033553 | MATHDB-366162

The lift of the weak proper Calabi-Yau structure on the Rabinowitz Fukaya category

摘录 / EXCERPT: Let X be the setting of Theorem, let $\mathcal{RW}(X)$ denote its Rabinowitz Fukaya category, and let the weak proper Calabi-Yau structure on $\mathcal{RW}(X)$ be the one supplie...

conjecture | open | MathDB 页面 | 原文线索

V2-033554 | MATHDB-366197

Triangulation invariance of statistics for topological excitations

摘录 / EXCERPT: Triangulation-invariance conjecture. There is a good definition of $T_p(M,G)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-033555 | MATHDB-366198

Eilenberg-MacLane homology conjecture for statistics of topological excitations

摘录 / EXCERPT: Eilenberg-MacLane homology conjecture. The following three groups are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-033556 | MATHDB-366199

Embedding-complex invariance conjecture for statistics of topological excitations

摘录 / EXCERPT: Embedding-complex invariance conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-033557 | MATHDB-366202

Linear-size conjecture for the distilled Vietoris-Rips filtration of manifold samples

摘录 / EXCERPT: Linear-size conjecture. The filtration $\mathcal{D}_{\infty}^{1}(X)$ consists of $O(kn)$ simplices.

conjecture | open | MathDB 页面 | 原文线索

V2-033558 | MATHDB-366211

BK version of the b2-conjecture for Bloch-Kato Lie algebras

摘录 / EXCERPT: BK version of the b_2-conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-033559 | MATHDB-366236

Homological mirror symmetry for Berglund-Hübsch mirror pairs

摘录 / EXCERPT: Let W be an invertible weighted homogeneous polynomial in $n+1$ variables, let $\check{W}$ be its Berglund-Hübsch transpose polynomial, and let Γ_W be the group acting dia...

conjecture | open | MathDB 页面 | 原文线索

V2-033560 | MATHDB-366241

Chen-Yang's Turaev-Viro volume conjecture

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold, either closed, with cusps, or compact with totally geodesic boundary. Chen-Yang's Turaev-Viro volume conjecture. As r varies along the odd n...

conjecture | open | MathDB 页面 | 原文线索

V2-033561 | MATHDB-366273

An upper-bound conjecture for equivariant parametrized topological complexity of configuration spaces

摘录 / EXCERPT: Let $F(\mathbb{R}^2,n)$ denote the ordered configuration space of n distinct points in the plane, and let $\mathrm{TC}^{\Sigma_s}[F(\mathbb{R}^2,s+1)\to F(\mathbb{R}^2,s)]$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-033562 | MATHDB-366318

The unrestricted danceability-bridge index conjecture for virtual knots

摘录 / EXCERPT: Let K be a virtual knot. Denote by $da^{(u)}(K)$ its unrestricted danceability index and by $b_1(K)$ its virtual bridge index. The unrestricted danceability-bridge index conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-033563 | MATHDB-366319

The classical-knot equality of danceability indices

摘录 / EXCERPT: Let K be a classical knot. Write $da(K)$ for the classical danceability index, $da^{(u)}(K)$ for the unrestricted danceability index, and $da^{(c)}(K)$ for the coincident danceab...

conjecture | open | MathDB 页面 | 原文线索

V2-033564 | MATHDB-366327

Scholze's uniqueness conjecture for six-functor formalisms

摘录 / EXCERPT: Scholze's uniqueness conjecture. The six-functor formalisms produced by the Liu-Zheng construction are exactly those for which every morphism in P is cohomologically proper and e...

conjecture | open | MathDB 页面 | 原文线索

V2-033565 | MATHDB-366343

The Z_2 periodicity conjecture

摘录 / EXCERPT: Let X be a connected topological space. A nonzero element $x\in H^k(X;\mathbb{Z}_2)$ induces periodicity if multiplication by x gives isomorphisms ... for all $i\geq 0$; in this...

conjecture | open | MathDB 页面 | 原文线索

V2-033566 | MATHDB-366428

Kinoshita's problem on reducibility of projective-plane knots

摘录 / EXCERPT: Kinoshita's problem. Every P^2-knot is P^2-reducible, that is, every P^2-knot is a connected sum of a standard P^2-knot and a surface-knot homeomorphic to the 2-sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-033567 | MATHDB-366477

The boundedness conjecture for smooth and proper stable categories

摘录 / EXCERPT: Boundedness conjecture. If $\mathcal{C}$ is smooth and proper over k, then every stability condition on $\mathcal{C}$ admits a mass-Hom bound.

conjecture | open | MathDB 页面 | 原文线索

V2-033568 | MATHDB-366498

Homological q-holonomicity of colored HOMFLY homologies

摘录 / EXCERPT: A sequence of chain complexes of filtered graded vector spaces is said to be homologically q-holonomic when it satisfies the proposed notion of homological q-holonomicity. For...

conjecture | open | MathDB 页面 | 原文线索

V2-033569 | MATHDB-366500

Rognes's equivariant common basis complex equivalence conjecture

摘录 / EXCERPT: Rognes's equivariant equivalence conjecture. There should be a $GL_k(R)$-equivariant equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-033570 | MATHDB-366501

The wedge-of-spheres conjecture for minimal spanning posets

摘录 / EXCERPT: Wedge-of-spheres conjecture. The space $\P^\circ$ is a wedge of $2k-3$-spheres.

conjecture | open | MathDB 页面 | 原文线索

V2-033571 | MATHDB-366517

Grigsby's categorified wrapping number conjecture for annular links

摘录 / EXCERPT: Let A= $S^1\times[0,1]$ be the annulus, let $L\subset A\times I$ be an annular link, and let $\mathrm{wrap}(L)$ be the minimal geometric intersection number between L and a mer...

conjecture | open | MathDB 页面 | 原文线索

V2-033572 | MATHDB-366518

Hoste-Przytycki wrapping number conjecture for annular links

摘录 / EXCERPT: Let A be the annulus, and let $L\subset A\times I$ be an annular link. Let $\mathrm{wrap}(L)$ be the minimal geometric intersection number between L and a meridional disk in...

conjecture | open | MathDB 页面 | 原文线索

V2-033573 | MATHDB-366527

The smooth Poincaré conjecture in dimension 4

摘录 / EXCERPT: Smooth Poincaré conjecture in dimension 4. There are no exotic S^4's; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-033574 | MATHDB-366546

Thom's finiteness conjecture for polynomial-map topological types

摘录 / EXCERPT: Let cK be either cC or cR, and let $cPKknp$ denote the space of degree-d polynomial maps from cK^n to cK^p, with topological type defined by composition with homeomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-033575 | MATHDB-366568

Veronese projection conjecture for the constructed Lefschetz bipencil

摘录 / EXCERPT: Veronese projection conjecture. The Lefschetz bipencil constructed in this section is isomorphic to the composition

conjecture | open | MathDB 页面 | 原文线索

V2-033576 | MATHDB-366593

The Evenness Conjecture for Real equivariant homotopy groups

摘录 / EXCERPT: Let G be a compact quasi-abelian Lie group equipped with an augmentation ... Write ρ for the relevant regular representation and $M\mathbb{R}_\eta$ for the associated Real equ...

conjecture | open | MathDB 页面 | 原文线索

V2-033577 | MATHDB-366594

The regularity conjecture for the Real global group law

摘录 / EXCERPT: Let $\mathbf{R}$ denote the Real global group law constructed from the Real globally oriented spectrum $\mathbf{MR}$. For each torus $\mathbb{T}^n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033578 | MATHDB-366630

RCD topological stability conjecture under noncollapsed three-dimensional convergence

摘录 / EXCERPT: RCD stability conjecture. There exists $i_0\in\mathbb{N}$ such that X_i is homeomorphic to X for every $i\ge i_0$.

conjecture | open | MathDB 页面 | 原文线索

V2-033579 | MATHDB-366652

Kono-Yagita nilpotence conjecture for Brown-Peterson cohomology

摘录 / EXCERPT: Let G be a compact Lie group, let BG be its classifying space, and let $BP^*(BG)$ denote its Brown-Peterson cohomology ring. Kono-Yagita's nilpotence conjecture. The ring...

conjecture | open | MathDB 页面 | 原文线索

V2-033580 | MATHDB-366653

Totaro's Chow-Brown-Peterson comparison conjecture

摘录 / EXCERPT: Totaro's conjecture. Under these hypotheses, this map is an isomorphism. The conjecture compares the Chow ring with Brown-Peterson cohomology after localization. No resolution is s...

conjecture | open | MathDB 页面 | 原文线索

V2-033581 | MATHDB-366654

Kono-Yagita even-degree conjecture for Brown-Peterson cohomology

摘录 / EXCERPT: Let G be a compact Lie group and let BG be its classifying space. Kono-Yagita's even-degree conjecture. The Brown-Peterson cohomology $BP^*(BG)$ is concentrated in even degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-033582 | MATHDB-366667

Weak homotopy equivalence for convex disks with Legendrian boundary

摘录 / EXCERPT: Let (M,cxi) be an overtwisted contact 3-manifold, and let the hypotheses of the principal theorem hold. Denote by the inclusion in the main inclusion the map from the space of...

conjecture | open | MathDB 页面 | 原文线索

V2-033583 | MATHDB-366668

Weak homotopy equivalence for Legendrians in a Darboux ball

摘录 / EXCERPT: Let (M,ξ) be an overtwisted contact 3-manifold, and let $\Lambda\subseteq(M,\xi)$ be an embedded Legendrian lying in some Darboux ball. Denote by the inclusion in the Legendr...

conjecture | open | MathDB 页面 | 原文线索

V2-033584 | MATHDB-366669

The Cantor-space conjecture for crowded Baire CDH groups

摘录 / EXCERPT: A topological group is CDH (countable dense homogeneous) if any two countable dense subsets are carried to one another by a homeomorphism of the group. A space is crowded if it has...

conjecture | open | MathDB 页面 | 原文线索

V2-033585 | MATHDB-366675

Chromatic redshift conjecture for algebraic K-theory

摘录 / EXCERPT: Chromatic redshift conjecture. If A has height exactly n, then $\operatorname{K}(A)$ is an $\mathbf{E}_\infty$-ring of height exactly $n+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-033586 | MATHDB-366691

Gabbard's equivariant sliceness conjecture for a strongly invertible 2-knot

摘录 / EXCERPT: Let (J,τ) be the strongly invertible 2-knot referred to in Gabbard's Theorem 5.8. Gabbard's equivariant sliceness conjecture. The knot (J,τ) does not bound an equivaria...

conjecture | open | MathDB 页面 | 原文线索

V2-033587 | MATHDB-366700

Finite-field computation of weight-zero Kato homology

摘录 / EXCERPT: Finite-field computation. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033588 | MATHDB-366702

Global localization conjecture for weight-one Kato homology

摘录 / EXCERPT: Global localization conjecture. For $j>0$, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033589 | MATHDB-366703

Boundedness conjecture for d-invariants on abelian Torelli subgroups

摘录 / EXCERPT: Let S be a genus g surface embedded in a homology sphere Y, and let H be an abelian subgroup of the Torelli group. Boundedness conjecture. The set ... is bounded. This conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-033590 | MATHDB-366715

Przytycki-Sazdanović torsion conjecture for closed 3-braids

摘录 / EXCERPT: Przytycki-Sazdanović conjecture. The Khovanov homology of a closed 3-braid can only have torsion of order 2.

conjecture | open | MathDB 页面 | 原文线索

V2-033591 | MATHDB-366747

Categorification conjecture for colored Hopf-link invariants in real projective three-space

摘录 / EXCERPT: Let L be a Hopf link in $\mathbb{R}P^3$ whose components are colored by an n-dimensional skew-symmetric and/or m-dimensional symmetric representation R of $su(2)$. Speciali...

conjecture | open | MathDB 页面 | 原文线索

V2-033592 | MATHDB-366773

Herzog's conormal module conjecture

摘录 / EXCERPT: Herzog's conormal module conjecture. If I/I^2 has finite projective dimension as an R/I-module, then I is a complete intersection.

conjecture | open | MathDB 页面 | 原文线索

V2-033593 | MATHDB-366814

The singular-locus dimension conjecture for foliations of low dimension

摘录 / EXCERPT: Low-dimensional foliation singular-locus conjecture. The stated dimension bound should hold for every such foliation.

conjecture | open | MathDB 页面 | 原文线索

V2-033594 | MATHDB-366822

Huang-Lian-Shao-Ye conjecture on almost finite-to-one maximal equicontinuous factors

摘录 / EXCERPT: Let (X,G) be a minimal topological dynamical system, where G is an amenable group, and suppose that (X,G) admits no essential K-IT-tuples. Let $\pi_{eq}:X\to X_{eq}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-033595 | MATHDB-366823

Huang-Lian-Shao-Ye conjecture on essential IT-tuples for proximal factors

摘录 / EXCERPT: Let (X,G) be a minimal topological dynamical system, where G is an amenable group, and let $\pi_{eq}:X\to X_{eq}$ be the factor map to the maximal equicontinuous factor. Suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-033596 | MATHDB-366869

Characteristic structures for QFTs with defects

摘录 / EXCERPT: Let M be a manifold with a tangential structure, and let F be an embedded submanifold that is Poincaré dual to a characteristic class of M. A characteristic structure is the...

conjecture | open | MathDB 页面 | 原文线索

V2-033597 | MATHDB-366870

Characteristic long exact sequence conjecture for bordism groups

摘录 / EXCERPT: Let $(\xi, \mathcal P)$ be a pair consisting of a tangential structure and a characteristic class as in the definition of a characteristic pair, and let ξ' be a tangential stru...

conjecture | open | MathDB 页面 | 原文线索

V2-033598 | MATHDB-366874

Topological non-equivalence conjecture for independent dense-phase embeddings

摘录 / EXCERPT: Let $\mathbb{X}\subset\mathbb{S}^2$ be the embedded random surface, and let $\mathbb{X}'$ be an independent copy of $\mathbb{X}$. A neighborhood of $\mathbb{X}$ means a neighborhoo...

conjecture | open | MathDB 页面 | 原文线索

V2-033599 | MATHDB-366875

Alberts' twisted Malle conjecture

摘录 / EXCERPT: Let k be a number field, G a transitive permutation group of degree n, and $T\normal G$ a proper normal subgroup. Let $q:G\to G/T$ be the canonical quotient map, let G_k be...

conjecture | open | MathDB 页面 | 原文线索

V2-033600 | MATHDB-366878

Fourier-transform conjecture for equivariant quantum volumes and reduced quantum volumes

摘录 / EXCERPT: Let X be a Hamiltonian T-manifold with moment map mu, and let $X//_tT=mu^{-1}(t)/T$ be the smooth symplectic reduction. Let $Pi_X^{\rm eq}$ denote the equivariant quantum vol...

conjecture | open | MathDB 页面 | 原文线索

V2-033601 | MATHDB-366904

Monotonicity conjecture for the connectivity of Čech complexes of spheres

摘录 / EXCERPT: For $n\ge 1$, let S^n be the unit n-sphere and let $\mathrm{\check{C}}(S^n;r)$ denote its Čech complex at scale $r\in(0,\pi)$. Write $\operatorname{conn}(K)$ for the connectivi...

conjecture | open | MathDB 页面 | 原文线索

V2-033602 | MATHDB-366905

Finite CW-complex conjecture for Čech complexes of spheres

摘录 / EXCERPT: For each dimension $n\ge 1$ and scale $r>0$, let S^n be the unit n-sphere and let $\mathrm{\check{C}}(S^n;r)$ denote its Čech complex at scale r. Finite CW-complex conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-033603 | MATHDB-366928

Fujita's conjecture B_n on projective manifolds with homology-cell divisors

摘录 / EXCERPT: Let M be an n-dimensional projective manifold and let D be a smooth ample divisor on M. Suppose that the natural homomorphism ... is bijective for $0\leq k\leq 2(n-1)$. Fujit...

conjecture | open | MathDB 页面 | 原文线索

V2-033604 | MATHDB-366929

Fujita's conjecture on Fano manifolds with the cohomology of projective space

摘录 / EXCERPT: Fujita's cohomological characterization conjecture. Then $M\cong\mathbb{P}^n$. This is the strongest of the three related conjectures introduced by Fujita. The source gives no evid...

conjecture | open | MathDB 页面 | 原文线索

V2-033605 | MATHDB-366935

Berry's random wave conjecture for chaotic billiards

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a chaotic billiard, so its billiard flow is ergodic and has positive topological entropy. Let $\{u_n\}$ be an orthonormal sequence of eigenfuncti...

conjecture | open | MathDB 页面 | 原文线索

V2-033606 | MATHDB-366942

The wedge-of-spheres conjecture for noncrossing parking function posets

摘录 / EXCERPT: Wedge-of-spheres conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033607 | MATHDB-367017

Optimal upper slope bound for the cohomology of rigid spaces

摘录 / EXCERPT: Slope-bound conjecture. For all $i>d$, the q-slopes of these eigenvalues are contained in

conjecture | open | MathDB 页面 | 原文线索

V2-033608 | MATHDB-367069

Lozenge characterization of free homotopy classes for bicontact Reeb flows

摘录 / EXCERPT: Let X be an Anosov flow supported by a bicontact structure (ξ^+,ξ^-). For a flow R, let $\mathcal{P}(R)$ denote its set of free homotopy data, and let $\mathbb{R}$-covere...

conjecture | open | MathDB 页面 | 原文线索

V2-033609 | MATHDB-367112

Openness conjecture for nontrivial basic de Rham classes of contact forms

摘录 / EXCERPT: Openness conjecture. The set of contact forms β on X for which

conjecture | open | MathDB 页面 | 原文线索

V2-033610 | MATHDB-367113

Basic harmonic homology conjecture for traversing vector fields

摘录 / EXCERPT: Basic harmonic homology conjecture. For every degree k,

conjecture | open | MathDB 页面 | 原文线索

V2-033611 | MATHDB-367114

Trajectory-space invariance conjecture for basic de Rham cohomology

摘录 / EXCERPT: Trajectory-space invariance conjecture. The v-basic de Rham cohomology

conjecture | open | MathDB 页面 | 原文线索

V2-033612 | MATHDB-367123

Steprāns-Watson conjecture on Baumgartner axioms in finite powers of the reals

摘录 / EXCERPT: Let BA denote Baumgartner's axiom for the real line, and let $\mathsf{BA}(X)$ be its parametrized version for a topological space X. For a positive integer n, write...

conjecture | open | MathDB 页面 | 原文线索

V2-033613 | MATHDB-367130

Kapouleas's boundary conjecture for minimal surfaces in the 3-sphere

摘录 / EXCERPT: Consider minimal surfaces in $\mathbb{S}^3$ with area less than 8π, and take the boundary of this space with respect to the varifold topology. A sequence whose areas tend to...

conjecture | open | MathDB 页面 | 原文线索

V2-033614 | MATHDB-367155

Ringström's topology-obliviousness conjecture for expanding Einstein spacetimes

摘录 / EXCERPT: Let $(\widetilde{M}, \widehat{g})$ be a globally hyperbolic Lorentzian manifold with compact Cauchy hypersurfaces. Say that late-time observers are oblivious to topology if there e...

conjecture | open | MathDB 页面 | 原文线索

V2-033615 | MATHDB-367172

The Orlik-Terao conjecture for graphical configuration spaces

摘录 / EXCERPT: Let Γ be a directed graph, let $X(SU(2),\Gamma)$ be its graphical configuration space, and let $\overline{\operatorname{OT}}(\Gamma)$ be the reduced Orlik-Terao algebra asso...

conjecture | open | MathDB 页面 | 原文线索

V2-033616 | MATHDB-367186

Galkin's lower bound conjecture for the spectral radius of quantum multiplication

摘录 / EXCERPT: For a Fano manifold X, let $QH^\bullet(X):=H^{\rm ev}(X)\otimes_{\mathbb{Q}}\mathbb{Q}[\mathbf{q}]$ be its even quantum cohomology, and let $\hat c_1$ be quantum multiplication b...

conjecture | open | MathDB 页面 | 原文线索

V2-033617 | MATHDB-367187

Galkin-Golyshev-Iritani Conjecture O for Fano manifolds

摘录 / EXCERPT: Galkin-Golyshev-Iritani's Conjecture O. Every Fano manifold X satisfies: ρ is an eigenvalue of $\hat c_1$ with multiplicity one; and for every eigenvalue λ of...

conjecture | open | MathDB 页面 | 原文线索

V2-033618 | MATHDB-367194

Herzog's cotangent homology conjecture

摘录 / EXCERPT: Herzog's cotangent homology conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-033619 | MATHDB-367195

Herzog's generalized eventual cotangent homology conjecture

摘录 / EXCERPT: Herzog's generalized conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-033620 | MATHDB-367211

Conjectured description of the second half of the connective KO-theory summands

摘录 / EXCERPT: Conjectured second-half description. The ν summands in the second half of $\mathcal{A}_k$ are $y_1^{2^{k-2}-2}M_{k+1}$ and all

conjecture | open | MathDB 页面 | 原文线索

V2-033621 | MATHDB-367215

Conjecture on positive Thurston-Bennequin representatives of fibered knots

摘录 / EXCERPT: Let K be a fibered knot, and let ξ_K be the contact structure it supports in the sense of Giroux. Positive Thurston-Bennequin conjecture. In ξ_K, the knot type K has Le...

conjecture | open | MathDB 页面 | 原文线索

V2-033622 | MATHDB-367216

Conjecture on irreducible fibered knots and negative Thurston-Bennequin invariant

摘录 / EXCERPT: Let K be a fibered knot in S^3 with irreducible monodromy, and suppose that K does not support the standard contact structure ξ_{std}. Irreducible-monodromy conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-033623 | MATHDB-367217

Conjecture on non-thickenable tori and supported contact structures

摘录 / EXCERPT: Let K be a fibered knot in a manifold M, and let ξ be a contact structure on M. Non-thickenable-tori conjecture. The knot type K admits non-thickenable tori in ξ if...

conjecture | open | MathDB 页面 | 原文线索

V2-033624 | MATHDB-367227

Characterization of crossing matrices of positive pure braids

摘录 / EXCERPT: Characterization conjecture. An $n\times n$ matrix M is the crossing matrix of some positive pure braid if and only if M is a non-negative integer T0 symmetric matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-033625 | MATHDB-367237

Ooguri-Vafa conjecture for framed links

摘录 / EXCERPT: Let $\mathcal{L}^{\vec{\tau}}$ be a framed L-component link with component framings $\vec{\tau}=(\tau_1,\ldots,\tau_L)$. For nonnegative integers $r_1,\ldots,r_L$, not all zero,...

conjecture | open | MathDB 页面 | 原文线索

V2-033626 | MATHDB-367252

The log Tate conjecture via homotopy K-theory

摘录 / EXCERPT: Let $s=\operatorname{Spec}(k)$ be the standard log point, let X be as above, let ℓ be invertible in k, and let $m\geq 2r$. Define $KH_{i,\lim}(X)=\varinjlim_{X'}KH_i(X')$...

conjecture | open | MathDB 页面 | 原文线索

V2-033627 | MATHDB-367266

The wedge-of-spheres conjecture for independence complexes of circle graphs

摘录 / EXCERPT: Let G be a circle graph, namely the intersection graph of the chords in a chord diagram, and let $I(G)$ be its independence complex, whose simplices are the subsets of pairwise n...

conjecture | open | MathDB 页面 | 原文线索

V2-033628 | MATHDB-367267

The wedge-of-spheres conjecture for independence complexes of bipartite circle graphs

摘录 / EXCERPT: Let G be a bipartite circle graph, namely a circle graph that is bipartite, and let $I(G)$ be its independence complex, whose simplices are the subsets of pairwise non-adjacent v...

conjecture | open | MathDB 页面 | 原文线索

V2-033629 | MATHDB-367271

Nakanishi's 4-move conjecture for knots

摘录 / EXCERPT: A 4-move is the local operation on a link diagram that replaces four consecutive half-twists between two strands by the opposite four half-twists. Nakanishi's 4-move conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-033630 | MATHDB-367287

Zhou's conjecture on CMC hypersurfaces in asymptotically flat manifolds

摘录 / EXCERPT: Let M be an asymptotically flat manifold of low dimension, and let c be a prescribed mean curvature. A closed hypersurface in M has constant mean curvature c when its mean...

conjecture | open | MathDB 页面 | 原文线索

V2-033631 | MATHDB-367300

The dimension-zero-dimension-one commutation conjecture for surface Hall algebras

摘录 / EXCERPT: Let S be the surface in the preceding setup, let C_i be one of the distinguished curves, and let $[\mathfrak Y_{i,(d)}]$ and $[\mathfrak Z_{i,\ell}]$ denote the corresponding d...

conjecture | open | MathDB 页面 | 原文线索

V2-033632 | MATHDB-367317

Free-rank conjecture for Yang-Baxter homology

摘录 / EXCERPT: Let $R_{(m)}$ be the Yang-Baxter operator at level m, let $C_\bullet^{mf}$ and $C_\bullet^m$ be the associated chain complexes, and write $\operatorname{free\,rank}$ for the rank...

conjecture | open | MathDB 页面 | 原文线索

V2-033633 | MATHDB-367318

Yang-Baxter homology conjecture for the level-two operator

摘录 / EXCERPT: Let $R_{(2)}$ be the level-two Yang-Baxter operator, let $H_n(R_{(2)})$ denote its nth homology, and let k be the coefficient ring. Let f_n be the Fibonacci numbers, with...

conjecture | open | MathDB 页面 | 原文线索

V2-033634 | MATHDB-367319

Fifth Yang-Baxter homology conjecture for the HOMFLYPT operator

摘录 / EXCERPT: Let $R_{(m)}$ be the level-m Yang-Baxter operator yielding the HOMFLYPT polynomial, and let $k=\mathbb{Z}[y^2]$. The notation $\binom{m-1}{j}$ denotes a binomial coefficient. Fif...

conjecture | open | MathDB 页面 | 原文线索

V2-033635 | MATHDB-367320

Künneth subcomplex splitting conjecture for Yang-Baxter chain complexes

摘录 / EXCERPT: Fix a decomposition of the alphabet $X_{(m)}=A\cup B$ and let $C_\bullet^{(m,A,B)}$ be the subchain complex generated by words that begin with letters from A and end with letters...

conjecture | open | MathDB 页面 | 原文线索

V2-033636 | MATHDB-367321

Level-six torsion conjecture for the HOMFLYPT Yang-Baxter operator

摘录 / EXCERPT: Let $R_{(m)}$ be the level-m Yang-Baxter operator yielding the HOMFLYPT polynomial, let $k=\mathbb{Z}[y^2]$, and let $a_6(m),b_6(m),c_6(m),d_6(m)$ denote the multiplicities in th...

conjecture | open | MathDB 页面 | 原文线索

V2-033637 | MATHDB-367322

Splitting conjecture for the filtered Yang-Baxter subcomplexes

摘录 / EXCERPT: Filtered-subcomplex splitting conjecture. Over $k=\mathbb{Z}[y^2]$,

conjecture | open | MathDB 页面 | 原文线索

V2-033638 | MATHDB-367340

Hozoori's projective deformation conjecture for Anosov flows

摘录 / EXCERPT: Let M be a manifold, and consider two oriented Anosov flows on M. They are projectively deformation equivalent when they are related through the notion of homotopy through proj...

conjecture | open | MathDB 页面 | 原文线索

V2-033639 | MATHDB-367346

The canonical splitting conjecture for the quantum connection

摘录 / EXCERPT: Canonical splitting conjecture. Given these idempotents, the space $H^*(M)[q^{\pm 1}][[t]]$ carries a canonical splitting into graded $R[q^{\pm 1}][[t]]$-modules, invariant under...

conjecture | open | MathDB 页面 | 原文线索

V2-033640 | MATHDB-367368

Gromov's rational inessentiality conjecture

摘录 / EXCERPT: Gromov's rational inessentiality conjecture. Then M is not rationally essential.

conjecture | open | MathDB 页面 | 原文线索

V2-033641 | MATHDB-367388

Real equivariant almost-commuting sphere matrices conjecture

摘录 / EXCERPT: Real equivariant almost-commuting sphere matrices conjecture. For every $\eta>0$ there is a $\delta>0$, independent of n, such that there is a triple (X_2,Y_2,H_2) of commuting...

conjecture | open | MathDB 页面 | 原文线索

V2-033642 | MATHDB-367406

Farber-Oprea rationality conjecture for the TC-generating function

摘录 / EXCERPT: Farber-Oprea conjecture. The formal power series $f_X(t)$ represents a rational function of the form

conjecture | open | MathDB 页面 | 原文线索

V2-033643 | MATHDB-367413

Donald's deformation conjecture for compact hypersymplectic 4-manifolds

摘录 / EXCERPT: Donald's conjecture. The structure $\underline{\omega}$ can be deformed through cohomologous hypersymplectic structures to the triple of Kähler forms arising from a hyperkähler met...

conjecture | open | MathDB 页面 | 原文线索

V2-033644 | MATHDB-367417

The cup-product formula for cohomology classes of Bier spheres

摘录 / EXCERPT: Cup-product conjecture. For each $I\in\mathcal{I}_i(K)$ and $J\in\mathcal{I}_j(K)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-033645 | MATHDB-367429

Topological invariance conjecture for zigzag multiplicities and Chern numbers

摘录 / EXCERPT: A zigzag multiplicity is the multiplicity of a zigzag summand in the bicomplex of differential forms of a compact complex manifold, and a Chern number is a characteristic number ob...

conjecture | open | MathDB 页面 | 原文线索

V2-033646 | MATHDB-367440

Gamma Conjecture I for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold. Denote by $c_1(X)\star_0$ quantum multiplication by the first Chern class at the origin, and suppose that it has a simple rightmost eigenvalue. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033647 | MATHDB-367555

The equivariant IP assembly-map conjecture

摘录 / EXCERPT: Let Γ be a countable group acting properly and isometrically on the Euclidean space $\mathbb{E}_{\Gamma}$, and let μ_{IP} denote the assembly map for the equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-033648 | MATHDB-367556

The integral-lift conjecture for the top Chern-Dold character of IP-theory

摘录 / EXCERPT: Let IP^d be the degree-d cohomology theory associated with the invertible-phase Ω-spectrum, and let ... be the top Chern-Dold character. Integral-lift conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-033649 | MATHDB-367559

Twisted Connes-Popa rigidity conjecture for ICC property (T) groups

摘录 / EXCERPT: Let G be an ICC property (T) group, let H be an arbitrary countable ICC group, and let $d:G\times G\to\mathbb{T}$ and $c:H\times H\to\mathbb{T}$ be 2-cocycles. For $t>0$, sup...

conjecture | open | MathDB 页面 | 原文线索

V2-033650 | MATHDB-367567

Compactness conjecture for Nyikos inverse topological semigroups

摘录 / EXCERPT: Compactness conjecture. Every Nyikos inverse topological semigroup is compact.

conjecture | open | MathDB 页面 | 原文线索

V2-033651 | MATHDB-367569

Universal-cover homology conjecture for positive intermediate curvature

摘录 / EXCERPT: Universal-cover homology conjecture. Then M does not admit a metric with positive m-intermediate curvature. The conjecture is proposed as a common generalization of the aspheri...

conjecture | open | MathDB 页面 | 原文线索

V2-033652 | MATHDB-367584

Signature and Rasmussen invariant conjecture for thin non-alternating knots

摘录 / EXCERPT: Let K be a thin non-alternating knot, and let $\sigma(K)$ and $s(K)$ denote its signature and Rasmussen invariant, respectively. Signature-Rasmussen conjecture. Their absolute va...

conjecture | open | MathDB 页面 | 原文线索

V2-033653 | MATHDB-367585

Khovanov-homology signature bounds conjecture

摘录 / EXCERPT: Let K be a knot whose integral Khovanov homology $Kh(K,\mathbb{Z})$ is supported on diagonals ... where w is the width of its Khovanov homology, and let $\sigma(K)$ denote its si...

conjecture | open | MathDB 页面 | 原文线索

V2-033654 | MATHDB-367624

Weak admissibility conjecture for exponentially twisted cohomology

摘录 / EXCERPT: Weak admissibility conjecture. The specialization map $\iota_{\widehat{F}}$ is an isomorphism, and

conjecture | open | MathDB 页面 | 原文线索

V2-033655 | MATHDB-367629

Homological initial-form Milnor-fibre conjecture for isolated singularities

摘录 / EXCERPT: Let $f,g\colon(\mathbb{C}^n,0)\to(\mathbb{C},0)$ be holomorphic function-germs with isolated singularities at the origin, and let f_* and g_* be their initial forms. Suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-033656 | MATHDB-367630

Homological Zariski Problem B for initial forms

摘录 / EXCERPT: Let $f,g\colon(\mathbb{C}^n,0)\to(\mathbb{C},0)$ be reduced holomorphic function-germs, and let f_* and g_* be their initial forms. Suppose there is a homeomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-033657 | MATHDB-367631

Parity version of Zariski's multiplicity conjecture

摘录 / EXCERPT: Let $f,g\colon(\mathbb{C}^n,0)\to(\mathbb{C},0)$ be reduced holomorphic function-germs. Suppose there is a homeomorphism $\varphi\colon(\mathbb{C}^n,0)\to(\mathbb{C}^n,0)$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-033658 | MATHDB-367649

Yau's conjecture on the topology of isolated surface singularities

摘录 / EXCERPT: Let $(V_i,0)\subset(\mathbb{C}^3,0)$, for $i=1,2$, be germs of isolated surface singularities. Let K_1,K_2 be their links, and let Δ_1,Δ_2 be the characteristic polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-033659 | MATHDB-367679

Factorization conjecture for semidirect biproduct invariants

摘录 / EXCERPT: Let $U=u_{qc1}\text{sl}_2$ and $H=H(\mathbf{m},t,\mathbf{d},\mathbf{u})$, and let $U\ltimes H$ be their semidirect biproduct. Let W be a 0 to 0 morphism in the category...

conjecture | open | MathDB 页面 | 原文线索

V2-033660 | MATHDB-367697

The chromatic height red-shift conjecture for algebraic K-theory

摘录 / EXCERPT: Let R be a ring spectrum of height n, meaning that $K(n)_*R\neq 0$ and $K(m)_*R=0$ for every $m>n$, where $K(m)$ denotes Morava K-theory. Height red-shift conjecture. The algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-033661 | MATHDB-367698

The THH homotopy-ring conjecture for truncated Brown-Peterson algebras

摘录 / EXCERPT: For $n\geq 0$ and $p>2$, let $A_*=\operatorname{BP}\langle n\rangle_*[x]/x^e$. Let $w_{n+1,i}$, $y_{j,i}$, and $\gamma_{p^m}(\sigma^2x^e)$ denote the classes introduced in the prec...

conjecture | open | MathDB 页面 | 原文线索

V2-033662 | MATHDB-367750

The quantum Hikita conjecture for symplectically dual resolutions

摘录 / EXCERPT: Let X and $X^!$ be symplectically dual conical Hamiltonian symplectic resolutions, with equivariant and Kähler roots, rings of differential operators, and associated D-modules...

conjecture | open | MathDB 页面 | 原文线索

V2-033663 | MATHDB-367752

The Hikita--Nakajima conjecture for symplectically dual resolutions

摘录 / EXCERPT: Let X and $X^!$ be symplectically dual, and let $\mathcal{A}$ be the universal quantization of X. The quantization exact sequence and the equivariant cohomology exact sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-033664 | MATHDB-367753

The p-curvature formula for equivariant quantum Steenrod operators

摘录 / EXCERPT: p-curvature conjecture. The equivariant quantum Steenrod operator satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033665 | MATHDB-367755

The p-curvature and quantum Steenrod correspondence

摘录 / EXCERPT: Let $M_{Kah,reg}$ be the specialized quantum D-module, let $x\in H^2_{T\times\mathbb{G}_m}(X)$, and let s be the Artin--Schreier map defining p-curvature. Quantum-side p-curv...

conjecture | open | MathDB 页面 | 原文线索

V2-033666 | MATHDB-367775

Kato's log l-adic monodromy-weight conjecture

摘录 / EXCERPT: Kato's log l-adic monodromy-weight conjecture. If $\overset{\circ}{X}$ is projective over κ, then, for every $e\in{\mathbb Z}_{\geq 1}$, N induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033667 | MATHDB-367777

The automorphic commutator relation

摘录 / EXCERPT: Automorphic commutator relation. Under $\thref{commsetup}$,

conjecture | open | MathDB 页面 | 原文线索

V2-033668 | MATHDB-367819

Dubrovin-Liu-Yang-Zhang conjecture on the standard form of tau-symmetric deformations

摘录 / EXCERPT: Let a tau-symmetric deformation of the Riemann hierarchy be given. A normal Miura transformation is a Miura transformation of the type defined in the source that preserves the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-033669 | MATHDB-367844

Smith-Thom inequality for topological groupoids with involution

摘录 / EXCERPT: Smith-Thom conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-033670 | MATHDB-367904

Asymptotic empirical-current conjecture for Fekete currents

摘录 / EXCERPT: Asymptotic empirical-current conjecture. The total variation measures $\lVert T^{(r)}\rVert$ converge to μ_E in the weak* topology of measures supported on E. Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-033671 | MATHDB-367957

The global shtuka-Langlands cohomology conjecture

摘录 / EXCERPT: Assume that G is split over ${\mathbb F}_q$, fix an open compact level $K\subset G({\mathbb O}_F)$, and let Q be the set of places where $K_v\neq G({\mathcal O}_v)$. For a fini...

conjecture | open | MathDB 页面 | 原文线索

V2-033672 | MATHDB-367962

The Index-Invariant completeness conjecture for routes to chaos

摘录 / EXCERPT: A route to chaos is a sequence of dynamical systems connected by an isotopy, and its Index-Invariant is the invariant introduced in the paper for such a route. Index-Invariant comp...

conjecture | open | MathDB 页面 | 原文线索

V2-033673 | MATHDB-367963

The Conformal Index conjecture for braid types of routes to chaos

摘录 / EXCERPT: Let C and C' be different routes to chaos, and let their Conformal Index be the endpoint invariant associated with each route. Say that C and C' define the same braid types...

conjecture | open | MathDB 页面 | 原文线索

V2-033674 | MATHDB-367964

The metric conjecture for the Teichmüller-type space of routes to chaos

摘录 / EXCERPT: For each route to chaos C, let $[C]$ be its boundary, namely the collection of limit dilatations of sequences drawn from the dilatation sets of the stages of C. Let T_{chaos}...

conjecture | open | MathDB 页面 | 原文线索

V2-033675 | MATHDB-367994

The topological rigidity conjecture for affine K-systems with hyperbolic centralizers

摘录 / EXCERPT: Let X be a compact homogeneous manifold, let $f_0\in\operatorname{Aff}(X)$ be a K-system, and suppose that $\mathcal{Z}(f_0)$ contains a hyperbolic affine diffeomorphism. If...

conjecture | open | MathDB 页面 | 原文线索

V2-033676 | MATHDB-367999

Characterization of smoothness for topological Anosov flows

摘录 / EXCERPT: Characterization of smoothness. The flow θ is topologically conjugate to a smooth Anosov flow generated by some $Y\in\operatorname{An}(M)$ if and only if X belongs to the...

conjecture | open | MathDB 页面 | 原文线索

V2-033677 | MATHDB-368000

Density of periodic orbits for topological Anosov flows

摘录 / EXCERPT: Density of periodic orbits. Every topological Anosov flow $\theta\in\mathcal{TA}\mathcal{F}(M)$ possesses a dense set of periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-033678 | MATHDB-368001

Positive entropy for topological Anosov flows

摘录 / EXCERPT: Positive topological entropy. Every topological Anosov flow $\theta\in\mathcal{TA}\mathcal{F}(M)$ has

conjecture | open | MathDB 页面 | 原文线索

V2-033679 | MATHDB-368003

Completeness of the space of topological Anosov flows

摘录 / EXCERPT: Completeness. With such a topology, $\mathcal{TA}\mathcal{F}(M)$ forms a complete metric space, or is complete in a relevant sense.

conjecture | open | MathDB 页面 | 原文线索

V2-033680 | MATHDB-368004

Equivalence of the topological Anosov definition with classical hyperbolicity criteria

摘录 / EXCERPT: Equivalence with classical definitions. A continuous flow θ is a topological Anosov flow according to the source's definition if and only if it satisfies such standard crite...

conjecture | open | MathDB 页面 | 原文线索

V2-033681 | MATHDB-368005

Moore's conjecture on rational hyperbolicity and homotopy exponents

摘录 / EXCERPT: Moore's conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-033682 | MATHDB-368008

Minimality conjecture for the double twist knot triangulations

摘录 / EXCERPT: Minimality conjecture. For $p,q2$, the triangulations $\mathcal{T}_{K(p,q)}$ realise the triangulation complexity of the complement of $K(p,q)$; equivalently, they use the minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-033683 | MATHDB-368010

Strong operator law of large numbers for discrete generalized quantum channels

摘录 / EXCERPT: Strong operator law of large numbers. Under these assumptions, $e^{A_1t/n}\ldots e^{A_nt/n}$ converges almost surely in the strong operator topology of $\mathcal{T}_p$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-033684 | MATHDB-368017

Discrete configuration-space models for framed cube complexes

摘录 / EXCERPT: Discrete configuration-space conjecture. For all $m\leq n$, the ordered configuration space of m points in $K(n)$ is homotopy equivalent to the mth discrete ordered configurati...

conjecture | open | MathDB 页面 | 原文线索

V2-033685 | MATHDB-368030

The t-henselian approximation conjecture for algebraic curves

摘录 / EXCERPT: T-henselian approximation conjecture. If τ_1 is t-henselian, then τ_1 and τ_2 induce independent topologies on $C(K)$.

conjecture | open | MathDB 页面 | 原文线索

V2-033686 | MATHDB-368031

The algebraically closed field-topology independence conjecture

摘录 / EXCERPT: Curve-topology independence conjecture. The topologies τ_1 and τ_2 induce independent topologies on $C(K)$.

conjecture | open | MathDB 页面 | 原文线索

V2-033687 | MATHDB-368032

The factor entropy-mean dimension equivalence conjecture

摘录 / EXCERPT: Let (X,T) be a dynamical system. A factor is nontrivial if it is not a one-point dynamical system. Topological entropy measures orbit complexity, while mean dimension measures th...

conjecture | open | MathDB 页面 | 原文线索

V2-033688 | MATHDB-368067

Determination of the graded root by lattice cohomology for curve singularities

摘录 / EXCERPT: Graded-root determination conjecture. The graded root $\big(R(C,o),\chi\big)$ is determined by the lattice cohomology module $\mathbb{H}^0(C,o)$.

conjecture | open | MathDB 页面 | 原文线索

V2-033689 | MATHDB-368068

Equality characterizations for the slice Cromwell inequality of homogeneous links

摘录 / EXCERPT: Let L be a homogeneous link. The slice Cromwell equality conjectures. (i) The slice Cromwell inequality ... is an equality if and only if L is positive. (ii) For a homogeneous di...

conjecture | open | MathDB 页面 | 原文线索

V2-033690 | MATHDB-368089

Periodicity of coexponents of h_0-submodules in the v_1-periodic algebra

摘录 / EXCERPT: Periodicity conjecture. These h_0-submodules have the same h_0-coexponent for all i.

conjecture | open | MathDB 页面 | 原文线索

V2-033691 | MATHDB-368092

The Johnson homomorphism formula for homologous bounding-pair maps

摘录 / EXCERPT: Let Σ_g be a closed oriented surface of genus g, and let γ,γ' be nonseparating simple curves that are homologous and disjoint from pairwise disjoint curves...

conjecture | open | MathDB 页面 | 原文线索

V2-033692 | MATHDB-368139

The mod-8 signature conjecture for 2-adic Poincare complexes

摘录 / EXCERPT: Mod-8 signature conjecture. The evaluation

conjecture | open | MathDB 页面 | 原文线索

V2-033693 | MATHDB-368147

Refined defect conjecture for negative band representatives

摘录 / EXCERPT: Let K be a knot or link type, let $\chi(K)$ be its maximal Euler characteristic, let $\operatorname{SL}(K)$ be its maximal self-linking number, and let $\operatorname{nb}(K)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-033694 | MATHDB-368148

Cassels-Swinnerton-Dyer-Margulis conjecture on bounded diagonal orbits

摘录 / EXCERPT: Let D be the diagonal group of $\operatorname{SL}_n(\mathbb{R})$ where $n\ge 3$. A D-orbit in $\operatorname{SL}_n(\mathbb{R})/\operatorname{SL}_n(\mathbb{Z})$ is relatively co...

conjecture | open | MathDB 页面 | 原文线索

V2-033695 | MATHDB-368152

Non-invertible Conway defects conjecture for K3 sigma models

摘录 / EXCERPT: Let $\Pi^{\natural}\subset \Lambda\otimes\mathbb{R}\subset V^{f\natural}_{tw}$ be a four-dimensional subspace, and let $\mathsf{Top}_{\Pi^{\natural}}$ be the full subcategory of to...

conjecture | open | MathDB 页面 | 原文线索

V2-033696 | MATHDB-368176

Classification of modular-invariant boundary maps

摘录 / EXCERPT: Boundary-map classification conjecture. All maps B satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-033697 | MATHDB-368188

Weak Kahn descent conjecture for quadratic forms

摘录 / EXCERPT: Let q be an anisotropic quadratic form over k, let $K=k(q)$, and let r be a quadratic form defined over K. Let $W_{\mathrm{nr}}(K/k)$ denote the unramified Witt ring. Weak...

conjecture | open | MathDB 页面 | 原文线索

V2-033698 | MATHDB-368258

Fourier-Mukai duality conjecture for cubic curves

摘录 / EXCERPT: Let $C:\operatorname{Spec} R\to \mathcal{M}_{cub}$ be a cubic curve, and let $\widehat{C}$ denote its formal completion, a formal group over R. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-033699 | MATHDB-368292

The multiplicative tom Dieck splitting conjecture for structure Xi

摘录 / EXCERPT: Multiplicative tom Dieck splitting conjecture. There exists a multiplicative structure Ξ such that Ξ-algebras are in particular G-E_{∞} algebras and, in the augmen...

conjecture | open | MathDB 页面 | 原文线索

V2-033700 | MATHDB-368312

The unknot colored Khovanov homology formulas for all nonnegative colors

摘录 / EXCERPT: Let $n\in\mathbb{N}_0$ denote the color of the 1-framed unknot, and let the formulas referred to as the formulas from Observation 1 hold for the colored Khovanov homology in color...

conjecture | open | MathDB 页面 | 原文线索

V2-033701 | MATHDB-368313

The skein lasagna module Poincaré series conjecture for the complex projective plane

摘录 / EXCERPT: Let $P_n(t,q)$ denote the colored Khovanov Poincaré polynomial used in the paper, let $\mathcal{S}_0^2(\overline{\mathbb{CP}^2};h;\mathbb{Q})$ denote the skein lasagna module of...

conjecture | open | MathDB 页面 | 原文线索

V2-033702 | MATHDB-368322

Conjectural classification of rational homology ball fillings of Seifert fibered spaces

摘录 / EXCERPT: Let $Y=Y(e_0;r_1,r_2,\ldots,r_n)$ be a Seifert fibered space that is also a rational homology sphere, and let ξ be a contact structure on Y. Seifert filling classification co...

conjecture | open | MathDB 页面 | 原文线索

V2-033703 | MATHDB-368324

The adjoint-category functor conjecture

摘录 / EXCERPT: Let ${\mathsf{Vect}}_{\operatorname{inj}}$ be the category of finite-dimensional real vector spaces and injective linear maps, and let ${\mathsf{AdjCat}}_n$ denote the $(\infty,1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-033704 | MATHDB-368325

The equivalence conjecture for oriented dagger and disk-like higher categories

摘录 / EXCERPT: Let SO_n-dagger categories be the higher categories equipped with the oriented tangential structure described in the source, and let disk-like n-categories be the structures in...

conjecture | open | MathDB 页面 | 原文线索

V2-033705 | MATHDB-368327

Klein-Williams conjecture on equivariant Nielsen numbers

摘录 / EXCERPT: Let G act on a closed smooth manifold M, let $f:M\to M$ be a G-equivariant map, and let $\ell_G(f)$ be the Klein-Williams invariant. For each conjugacy class of subgroups H...

conjecture | open | MathDB 页面 | 原文线索

V2-033706 | MATHDB-368328

Klein-Williams conjecture on the terms of the equivariant invariant

摘录 / EXCERPT: Let M be a closed smooth manifold, let $f:M\to M$ be a self-map, and let $\ell_n(f)$ lie in the direct sum $\bigoplus_{k\mid n}\mathbb{Z}[\pi_1(M)_{\rho,k}]$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-033707 | MATHDB-368336

Exact formula for the stabilized Kac polynomial

摘录 / EXCERPT: Let Q be the quiver, let d be the dimension vector, and let δ be as in Theorem; write $\operatorname{supp}\delta$ for its support, let d_i denote the components of d...

conjecture | open | MathDB 页面 | 原文线索

V2-033708 | MATHDB-368408

Conjecture on realizing arbitrary-genus symplectic leaves

摘录 / EXCERPT: Arbitrary-genus symplectic-leaf conjecture. For every integer $k\in\mathbb{N}$, an appropriate choice of U produces a sample of symplectic leaves that are topologically Riemann s...

conjecture | open | MathDB 页面 | 原文线索

V2-033709 | MATHDB-368432

Locally analytic de Rham criterion for completed cohomology representations

摘录 / EXCERPT: Let L be the local field appearing in the construction, let E be a finite extension of L containing all embeddings of L into $\overline{\mathbb{Q}}_p$, and let ... be a two-d...

conjecture | open | MathDB 页面 | 原文线索

V2-033710 | MATHDB-368437

Factorised decomposition conjecture for the Harer-Zagier transform

摘录 / EXCERPT: Factorised decomposition conjecture. The transform can be expressed as

conjecture | open | MathDB 页面 | 原文线索

V2-033711 | MATHDB-368476

Root-of-unity conjecture for monodromy eigenvalues of persistent local systems

摘录 / EXCERPT: Root-of-unity conjecture. The eigenvalues of M_i must be roots of unity.

conjecture | open | MathDB 页面 | 原文线索

V2-033712 | MATHDB-368477

Lifted toroidal-cycle conjecture for cycles trivial in the quotient

摘录 / EXCERPT: Lifted toroidal-cycle conjecture. All cycles in K which are trivial in G can be written in terms of lifts of toroidal cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-033713 | MATHDB-368481

Unbounded common surgery slopes among pairs of knots

摘录 / EXCERPT: Unbounded-slope conjecture. For any natural number N there exist pairwise-different integers $n_1,\ldots,n_N$ and non-isotopic knots K and K' such that, for $i=1,\ldots,N$,...

conjecture | open | MathDB 页面 | 原文线索

V2-033714 | MATHDB-368550

Universal Jones invariant lifts over the quantised Habiro ring

摘录 / EXCERPT: Let L be a link, let ${\large \hat{{\Huge {\Gamma}}}^{J}}(L)\in \hat{\mathbb L}^J$ denote its universal Jones link invariant, and let ${\hat{\mathbb L}^J}'$ be the refined univer...

conjecture | open | MathDB 页面 | 原文线索

V2-033715 | MATHDB-368551

The refined universal Jones intersection is a link invariant

摘录 / EXCERPT: Let β be a braid and let L be its closure. Let ${\large \hat{{\Huge {\Gamma}}}^{J,R}}(\beta)\in {\hat{\mathbb L}^J}'$ denote the universal refined Jones intersection, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-033716 | MATHDB-368552

The refined universal ADO intersection is a link invariant

摘录 / EXCERPT: Let β be a braid and let L be its closure. Let ${\large {\Huge {\Gamma^R}}}(L)\in\mathbb L^R$ denote the refined universal ADO intersection, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033717 | MATHDB-368592

The conjecture that the Wada polynomial is the Alexander polynomial at $t=-1$

摘录 / EXCERPT: For a knot, let the Wada polynomial and Alexander polynomial be the knot invariants defined from representations of its braid group by automorphisms of a free group. Specialization...

conjecture | open | MathDB 页面 | 原文线索

V2-033718 | MATHDB-368601

Non-generation conjecture for 1-reduced simplicial sets

摘录 / EXCERPT: Non-generation conjecture. The set $R_{1}(\widetilde{J})$ is not a generating set of acyclic cofibrations for $\mathcal{SS}\mathrm{ets}_{1}$. The question is motivated by the fact...

conjecture | open | MathDB 页面 | 原文线索

V2-033719 | MATHDB-368647

The L_{∞} and d-algebra structure conjecture for CFT deformations

摘录 / EXCERPT: Let ${\cal A}$ be the cooperad governing the OPE data, let H be the state space regarded as an ${\cal A}$-algebra, and let $\mathfrak g$ be the L_{∞}-algebra controlling f...

conjecture | open | MathDB 页面 | 原文线索

V2-033720 | MATHDB-368698

Linear upper bound for the ribbonlength of ribbon knots

摘录 / EXCERPT: Let K be a ribbon knot, let $\operatorname{Rib}(K)$ denote its minimal ribbonlength, and let $c(K)$ denote its crossing number. Linear ribbonlength conjecture. The minimal ribbon...

conjecture | open | MathDB 页面 | 原文线索

V2-033721 | MATHDB-368701

Geometric decomposition conjecture for length-minimising thin physical unknots

摘录 / EXCERPT: Geometric decomposition conjecture. Every length-minimising unknot in $\mathcal{U}_\tau$ for $\tau\in(0,2)$ admits a decomposition into finitely many elementary geometric component...

conjecture | open | MathDB 页面 | 原文线索

V2-033722 | MATHDB-368708

Morsification conjecture for topological exceptional collections of vanishing arcsets

摘录 / EXCERPT: Morsification conjecture for topological exceptional collections. The ordered collection of vanishing cycles obtained from a topological exceptional collection of geometric vanishi...

conjecture | open | MathDB 页面 | 原文线索

V2-033723 | MATHDB-368720

The one-step quantum Frobenius Ext decomposition conjecture

摘录 / EXCERPT: Let F and G be strict polynomial functors, let $F^{(1)_q}$ and $G^{(1)_q}$ denote their quantum Frobenius twists, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033724 | MATHDB-368721

The iterated quantum Frobenius Ext decomposition conjecture

摘录 / EXCERPT: Let F and G be strict polynomial functors, let $F^{(s)_q}$ and $G^{(s)_q}$ denote their s-th quantum Frobenius twists, and define ... For each degree k, the iterated quantum...

conjecture | open | MathDB 页面 | 原文线索

V2-033725 | MATHDB-368722

Kao-Lai-Osting conjecture for the second eigenvalue of the torus

摘录 / EXCERPT: Let T be the torus, and let $\Lambda_2(T)$ denote its second topological eigenvalue. Kao-Lai-Osting conjecture. ... This conjecture concerns the sharp value of the second eigenvalu...

conjecture | open | MathDB 页面 | 原文线索

V2-033726 | MATHDB-368753

Katok's cohomology-free vector field conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ be a compact, connected manifold, and let X be a vector field on $\mathcal{M}$. The vector field X is cohomology free if the associated cohomological equation...

conjecture | open | MathDB 页面 | 原文线索

V2-033727 | MATHDB-368778

Purity and exterior-power basis conjecture for the hypergeometric cohomology of Y

摘录 / EXCERPT: Purity and exterior-power basis conjecture. For every $w\in Z_{r+1}$, the cohomology of $(\Omega^{\bullet}(Y),\nabla_{\tilde{\omega}})$ is pure,

conjecture | open | MathDB 页面 | 原文线索

V2-033728 | MATHDB-368816

The toral rank and Carlsson conjectures for manifolds

摘录 / EXCERPT: Toral rank and Carlsson conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-033729 | MATHDB-368818

Symmetric monoidality conjecture for the Goodwillie derivative functor

摘录 / EXCERPT: Let C be the category appearing above, let $L{\mathsf{Sp}}$ be the corresponding localized category of spectra, and write $\partial^{BB}_\ast$ for the Goodwillie derivative funct...

conjecture | open | MathDB 页面 | 原文线索

V2-033730 | MATHDB-368876

Ghiggini's conjecture on separating half Giroux torsion

摘录 / EXCERPT: Ghiggini's conjecture. The contact invariant vanishes:

conjecture | open | MathDB 页面 | 原文线索

V2-033731 | MATHDB-368906

Non-fiberedness conjecture for non-trivial 2-plat 2-knots

摘录 / EXCERPT: Non-fiberedness conjecture. Every non-trivial 2-plat 2-knot is non-fibered.

conjecture | open | MathDB 页面 | 原文线索

V2-033732 | MATHDB-368907

The exact $mathbb{Z}_2$-Gromov-Hausdorff distance between Euclidean spheres of consecutive dimensions

摘录 / EXCERPT: Let $\mathbb{S}_{\mathrm{E}}^m$ denote the m-dimensional Euclidean sphere equipped with its standard $\mathbb{Z}_2$-action, and let $d_{\mathrm{GH}}^{\mathbb{Z}_2}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-033733 | MATHDB-368952

Spiral-knot parameter uniqueness conjecture

摘录 / EXCERPT: Let $S(p,q,\varepsilon)$ and $S(p',q',\varepsilon')$ be spiral knots, with p,q,p',q' and the signs ε,ε' as in the spiral-knot notation. Parameter uniqueness...

conjecture | open | MathDB 页面 | 原文线索

V2-033734 | MATHDB-368953

Bridge-number conjecture for spiral knots

摘录 / EXCERPT: For a spiral knot $S(p,q,\varepsilon)$, let $b(S(p,q,\varepsilon))$ denote its bridge number, and let $\min(p,q)$ be the minimum of its two integer parameters. Bridge-number conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-033735 | MATHDB-369017

The spherical framed Floer spectrum identification conjecture

摘录 / EXCERPT: Spherical framed Floer spectrum identification conjecture. There is a homotopy equivalence of ring spectra

conjecture | open | MathDB 页面 | 原文线索

V2-033736 | MATHDB-369048

The toroidal small-homology conjecture for SU(2)-abelian 3-manifolds

摘录 / EXCERPT: Let Y be a toroidal rational homology 3-sphere. The toroidal small-homology conjecture. If $|H_1(Y;\mathbb{Z})| \le 4$, then Y is not $SU(2)$-abelian. This conjecture predict...

conjecture | open | MathDB 页面 | 原文线索

V2-033737 | MATHDB-369049

The SU(2)-abelian implies Heegaard Floer L-space conjecture

摘录 / EXCERPT: Let Y be a rational homology 3-sphere. The SU(2)-abelian implies Heegaard Floer L-space conjecture. If Y is $SU(2)$-abelian, then it is a Heegaard Floer L-space. Here, a Heeg...

conjecture | open | MathDB 页面 | 原文线索

V2-033738 | MATHDB-369050

The refined toroidal small-homology conjecture for SU(2)-abelian 3-manifolds

摘录 / EXCERPT: Let Y be a toroidal rational homology 3-sphere. The refined toroidal small-homology conjecture. If $|H_1(Y;\mathbb{Z})| \le 6$, then Y is not $SU(2)$-abelian. This strengthen...

conjecture | open | MathDB 页面 | 原文线索

V2-033739 | MATHDB-369080

Delayed-Fibonacci torsion conjecture for the four-element Alexander quandle

摘录 / EXCERPT: Delayed-Fibonacci torsion conjecture. The torsion subgroup of $\operatorname{H}^{\rm Q}_n(X)$ is isomorphic to $\mathbb{Z}_4^{f_n}\oplus\mathbb{Z}_2^{f'_n}$. The displayed low-degr...

conjecture | open | MathDB 页面 | 原文线索

V2-033740 | MATHDB-369085

Kusner's 16π-conjecture for sphere eversions

摘录 / EXCERPT: A sphere eversion is a regular homotopy $H:\mathbb{S}^2\times[0,1]\to\mathbb{R}^3$ connecting two round spheres of opposite orientation. Kusner's 16π-conjecture. There exists a sph...

conjecture | open | MathDB 页面 | 原文线索

V2-033741 | MATHDB-369101

The algebraic generation conjecture for the homology of d_N

摘录 / EXCERPT: Let u_i and ξ_j be the generators in the polynomial algebra underlying the differential d_N, and for $k=1,\ldots,n-1$ define ... The differential satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-033742 | MATHDB-369102

The algebraic generation conjecture for the homology of d_W

摘录 / EXCERPT: Algebraic generation conjecture. The homology of d_W is generated by u_i and μ_k^W as an algebra. The supplied text establishes that the coefficients μ_i^W are cycles,...

conjecture | open | MathDB 页面 | 原文线索

V2-033743 | MATHDB-369146

The BPS cohomological interpretation of the p-adic integral for SL_n Higgs bundles

摘录 / EXCERPT: Let $S=\operatorname{Spec}\mathcal{O}_F$ where $\mathcal{O}_F$ is a local ring with finite residue field k_F. Let $D=K_\mathcal{C}$ and let $L\in\mathcal{P}ic^d(\mathcal{C})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-033744 | MATHDB-369165

Cellular generation conjecture for flat-pure monomorphisms

摘录 / EXCERPT: Let S be a monoid, and let $\mathcal{F}$-PureMono be the class of pure monomorphisms of right S-acts whose Rees quotient is flat. A class of monomorphisms is cellularly generat...

conjecture | open | MathDB 页面 | 原文线索

V2-033745 | MATHDB-369289

The zero-index characterization of stable short-range entanglement for free-fermion states

摘录 / EXCERPT: Let $\mathcal H=\ell^2(\mathbb Z^2)$ and let P be the Fermi projection of a gapped Hamiltonian H satisfying exponential locality. Let ω_P be the associated quasi-free st...

conjecture | open | MathDB 页面 | 原文线索

V2-033746 | MATHDB-369303

The L-theory Farrell-Jones conjecture for torsionfree groups

摘录 / EXCERPT: Let G be a group, let R be a ring with involution, and let $\mathbf{L}^{\langle-\infty\rangle}(R)$ be the associated L-theory spectrum, whose nth homotopy group is...

conjecture | open | MathDB 页面 | 原文线索

V2-033747 | MATHDB-369305

The manifold-structure conjecture for aspherical Poincaré complexes

摘录 / EXCERPT: Let X be a finitely dominated aspherical Poincaré complex. The manifold-structure conjecture. The complex X is homotopy equivalent to a closed topological manifold. This is one...

conjecture | open | MathDB 页面 | 原文线索

V2-033748 | MATHDB-369306

The L-theoretic Farrell-Jones conjecture with additive-category coefficients

摘录 / EXCERPT: Let G be a group, let $E_{\mathcal{VCY}}(G)$ be the classifying G-space for the family of virtually cyclic subgroups, and let $\mathcal{A}$ be an additive G-category with inv...

conjecture | open | MathDB 页面 | 原文线索

V2-033749 | MATHDB-369321

Theta-invariant conjecture for spherical 3-manifolds

摘录 / EXCERPT: Let N be a spherical 3-manifold with nontrivial fundamental group π. Let $\mathscr{F}(N)$ be the vector space of $\mathbb{Z}\pi$-homology equivalences, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033750 | MATHDB-369346

Perverse-equivalence conjecture for quantum-cohomology eigenlines

摘录 / EXCERPT: Let A be an alcove in the space of real stabilities, and let $\mathcal{E}(A)$ be the set of joint eigenlines for quantum multiplication by equivariant cohomology classes of Y o...

conjecture | open | MathDB 页面 | 原文线索

V2-033751 | MATHDB-369350

Homologically undetectable entropy for Kummer diffeomorphisms

摘录 / EXCERPT: Homologically undetectable entropy. The minimal entropy of any $g\in\operatorname{Diff}(M)$ isotopic to f_T satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033752 | MATHDB-369395

Greedy matching conjecture for torus braids

摘录 / EXCERPT: Greedy matching conjecture. The greedy matching is a Morse matching for all torus braids, that is, the graphs

conjecture | open | MathDB 页面 | 原文线索

V2-033753 | MATHDB-369396

Gorsky-Oblomkov-Rasmussen conjecture for stable Khovanov homology

摘录 / EXCERPT: Gorsky-Oblomkov-Rasmussen conjecture. The stable Khovanov homology $\operatorname{Kh}(T(n,\infty))$ is the dual to the homology of the Koszul complex associated to the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-033754 | MATHDB-369397

Greedy matching conjecture for delooped Khovanov complexes

摘录 / EXCERPT: Greedy matching conjecture. For any tangle diagram T, the greedy matching M_{gr} is a Morse matching on the delooped Khovanov complex...

conjecture | open | MathDB 页面 | 原文线索

V2-033755 | MATHDB-369398

Polynomial-growth conjecture for Khovanov homology of torus links

摘录 / EXCERPT: Polynomial-growth conjecture. The total dimension satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033756 | MATHDB-369402

BNPT geometric-definition conjecture for open Gromov-Witten invariants

摘录 / EXCERPT: Let S be a moduli specification, let $\vec{a}$ be a vector of nonnegative integers, and let $\vec{\gamma}$ be a vector of cohomology classes in $H^*_T(\mathbb{P}^1)$. The combina...

conjecture | open | MathDB 页面 | 原文线索

V2-033757 | MATHDB-369451

The main torsion-length inequality for crystalline and de Rham cohomology

摘录 / EXCERPT: Main conjecture. For all $i\geqslant 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-033758 | MATHDB-369462

Cohomology comparison conjecture for saturable pro-p groups

摘录 / EXCERPT: Let G be a saturable pro-p group with associated $\mathbb{Z}_p$-Lie algebra $\mathfrak{g}$. Here $\operatorname{H}^{*}(G;\mathbb{F}_p)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033759 | MATHDB-369465

Echelonmotion and the Auslander-Reiten permutation for Auslander-regular posets

摘录 / EXCERPT: Let R be a finite poset whose incidence algebra over a field of characteristic 0 is Auslander-regular, and suppose that R is connected. A finite poset is echelon-independent...

conjecture | open | MathDB 页面 | 原文线索

V2-033760 | MATHDB-369479

Adequacy characterization by Jones polynomial span and Turaev genus

摘录 / EXCERPT: Adequacy-span conjecture. A non-split link is adequate if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-033761 | MATHDB-369480

Kauffman-polynomial span bound in terms of Turaev genus

摘录 / EXCERPT: Kauffman-polynomial span conjecture. The spans satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-033762 | MATHDB-369506

Kim-Kim's bounded fundamental class conjecture for pinched negatively curved manifolds

摘录 / EXCERPT: Let M be a pinched negatively curved manifold of infinite volume, of dimension n, and let $[\hat{\omega}]\in H^{n}_b(M,\mathbb{R})$ denote its bounded fundamental class. Kim-Kim'...

conjecture | open | MathDB 页面 | 原文线索

V2-033763 | MATHDB-369552

Topological closure and Kolchin dimension for generic derivations

摘录 / EXCERPT: Let $\mathfrak C$ carry a topology τ satisfying suitable conditions, and let $\tau_{\bar \delta}$ be the topology induced by $x\mapsto\operatorname{Jet}(x)$ into...

conjecture | open | MathDB 页面 | 原文线索

V2-033764 | MATHDB-369654

The instanton Poincaré conjecture

摘录 / EXCERPT: Let Y be a connected, closed, oriented 3-manifold, and let $I^\sharp(Y;\mathbb{F}_2)$ denote its framed instanton homology over the field with two elements. Instanton Poincaré...

conjecture | open | MathDB 页面 | 原文线索

V2-033765 | MATHDB-369721

The Hopf-Chern conjecture on Euler characteristics of negatively curved manifolds

摘录 / EXCERPT: Hopf-Chern conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033766 | MATHDB-369724

Freyd's Generating Hypothesis for the p-completed Spanier-Whitehead category

摘录 / EXCERPT: Freyd's Generating Hypothesis for ${\mathcal{F}}_p$. The functor π_* is faithful. Therefore, ${\mathcal{M}}_p$ is a Freyd category, with ${\mathcal{F}}_p$ as its full subcatego...

conjecture | open | MathDB 页面 | 原文线索

V2-033767 | MATHDB-369728

The width-three, three-highly-twisted plat rigidity conjecture

摘录 / EXCERPT: Let a knot or link have a plat presentation of width greater than or equal to 3, and suppose the presentation is 3-highly twisted. Plat rigidity conjecture. The statement of Th...

conjecture | open | MathDB 页面 | 原文线索

V2-033768 | MATHDB-369729

Rigidity for even and doubly even highly twisted plat diagrams

摘录 / EXCERPT: Let b be a braid word, and consider its even plat closure or doubly even plat closure. For even plats, connect consecutive endpoint pairs at the top and shifted consecutive pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-033769 | MATHDB-369783

The super-$\hat{Z}$ surgery formula conjecture

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot and let $S^{3}_{p/r}(K)$ denote the result of Dehn surgery on K with slope p/r. For any choice of good chamber α_i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033770 | MATHDB-369784

The torus-knot extension conjecture for Proposition 1.3

摘录 / EXCERPT: A torus knot is a knot $T(s,t)\subseteq S^3$ with $\gcd(s,t)=1$. Let Proposition 1.3 denote the proposition preceding this conjecture, which establishes the stated well-defined Lau...

conjecture | open | MathDB 页面 | 原文线索

V2-033771 | MATHDB-369785

The torus-knot extension conjecture for Proposition 5.2

摘录 / EXCERPT: A torus knot is a knot $T(s,t)\subseteq S^3$ with $\gcd(s,t)=1$. Let Proposition 5.2 denote the proposition preceding this claim, concerning the chamber parameters and associated c...

conjecture | open | MathDB 页面 | 原文线索

V2-033772 | MATHDB-369798

Cohomological smoothness conjecture for the non-very-special locus

摘录 / EXCERPT: Cohomological smoothness conjecture. The structure morphism

conjecture | open | MathDB 页面 | 原文线索

V2-033773 | MATHDB-369841

Nontriviality conjecture for the classes X_{4k+2}

摘录 / EXCERPT: Nontriviality conjecture. For every $k=1,2,\dots$, the element X_{4k+2} defines a non-trivial cohomology class in

conjecture | open | MathDB 页面 | 原文线索

V2-033774 | MATHDB-369871

Celoria-Levine-Friedl-Nagel-Orson-Powell concordance conjecture

摘录 / EXCERPT: Let $M\neq S^3$ be a closed orientable 3-manifold, let $[S^1,M]$ denote the set of free homotopy classes of loops in M, and let $\mathcal{C}^M_x$ be the set of concordance clas...

conjecture | open | MathDB 页面 | 原文线索

V2-033775 | MATHDB-369879

Gukov-Manolescu conjecture on the F_K series and quantum A-polynomial

摘录 / EXCERPT: Gukov-Manolescu conjecture. The series satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-033776 | MATHDB-369918

Weak Leopoldt conjecture for a p-adic Galois representation

摘录 / EXCERPT: Let $\mathcal{K}$ be a finite extension of $\mathbb{Q}_p$, let $\mathcal{O}$ be its valuation ring, and let $\mathbf{A}(rho)=\mathbf{V}_rho/\mathbf{T}_rho$ be the p-divisible mod...

conjecture | open | MathDB 页面 | 原文线索

V2-033777 | MATHDB-369939

The universal linear stick-number bound for knots

摘录 / EXCERPT: Let K be a knot type, let $c[K]$ denote its crossing number, and let $\operatorname{stick}[K]$ denote its stick number. Universal linear stick-number conjecture. Every knot with...

conjecture | open | MathDB 页面 | 原文线索

V2-033778 | MATHDB-369941

Malkiewich-Ponto's fiberwise Fuller trace obstruction conjecture

摘录 / EXCERPT: Malkiewich-Ponto's conjecture. The fiberwise Fuller trace

conjecture | open | MathDB 页面 | 原文线索

V2-033779 | MATHDB-369943

Knot-quiver skein relation conjecture

摘录 / EXCERPT: Let K be a knot with corresponding quiver Q_K. For each crossing of type L_+, let $K_{i,\tau}$, with $\tau\binom{}{}$ replaced by τeq? be the knot or link obtained by c...

conjecture | open | MathDB 页面 | 原文线索

V2-033780 | MATHDB-369961

Classical LMO completeness conjecture for integral homology 3-spheres

摘录 / EXCERPT: Let M be an integral homology 3-sphere, and let $\hat{Z}_{\mathrm{LMO}}(M)$ denote its classical LMO invariant with values in the algebra of Jacobi diagrams...

conjecture | open | MathDB 页面 | 原文线索

V2-033781 | MATHDB-369962

Existence conjecture for LMO-trivial manifolds in the Johnson filtration

摘录 / EXCERPT: Let $\mathcal{I}_g$ be the Torelli group, let $\mathcal{I}_g(k)$ be its kth Johnson-filtration subgroup, and let M_{ϕ} denote the integral homology 3-sphere constructed from...

conjecture | open | MathDB 页面 | 原文线索

V2-033782 | MATHDB-369980

Topological mirror symmetry for quasi-BPS categories of Higgs bundles

摘录 / EXCERPT: Let G be a reductive group, let ^{L}G be its Langlands dual, and let $(\chi,w)\in\pi_1(G)\times Z(G)^{\vee}$. Let $\mathbb{T}_{^{L}G}(w)_{-\chi}$ and $\mathbb{T}_G(\chi)_w$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-033783 | MATHDB-369986

Kirby's $SU(2)$ representation conjecture for integer homology 3-spheres

摘录 / EXCERPT: Let Y be a closed oriented 3-manifold satisfying ... If Y is not S^3, then Kirby's $SU(2)$ representation conjecture. there exists a homomorphism ... with non-abelian image. This...

conjecture | open | MathDB 页面 | 原文线索

V2-033784 | MATHDB-370034

Asymptotic saturation conjecture for area-minimizing representatives modulo n

摘录 / EXCERPT: Let M be the setting of the preceding discussion, let d be the chain dimension, and let g be a Riemannian metric. For each n, consider the proportion in the displayed limin...

conjecture | open | MathDB 页面 | 原文线索

V2-033785 | MATHDB-370036

The Hasse principle for size-minimizing flat chains

摘录 / EXCERPT: Let $[\Sigma]$ be an integral homology class, with reductions $[\Sigma]^{\mathbb{R}}$ and $[\Sigma]^{\mathbb{Z}/n\mathbb{Z}}$ in the corresponding coefficient systems. The Hasse pr...

conjecture | open | MathDB 页面 | 原文线索

V2-033786 | MATHDB-370045

Witten's asymptotic expansion conjecture for WRT invariants

摘录 / EXCERPT: Let M be a closed and oriented 3-manifold. For each positive integer k, let $Z_k(M)\in\mathbb{C}$ denote the $\operatorname{SU}(2)$ WRT invariant for M at level k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033787 | MATHDB-370046

Hikami--GPPV--Manolescu radial limit conjecture for WRT invariants

摘录 / EXCERPT: Let M be a rational homology sphere. For each Spin^c structure $b\in\operatorname{Spin}^c(M)$, suppose there are invariants $\Delta_b\in\mathbb{Q}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-033788 | MATHDB-370047

Chern--Simons set identification conjecture for negative definite plumbed manifolds

摘录 / EXCERPT: Let M be the plumbed 3-manifold in the preceding asymptotic expansion, let $\mathcal{S}$ be the finite set of phases appearing there, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033789 | MATHDB-370051

Stabilization conjecture for colored knot Floer homology

摘录 / EXCERPT: Let L_m be the m-fold cable link of a link L, and let $\overline{\mathbf{s}}$ be a normalized Alexander degree. The maps ... are the connecting maps in the directed system defi...

conjecture | open | MathDB 页面 | 原文线索

V2-033790 | MATHDB-370052

Stabilization conjecture for cable links with fixed remainder

摘录 / EXCERPT: Cable stabilization conjecture. For a given normalized Alexander degree $\widetilde{\mathbf{s}}$, the dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-033791 | MATHDB-370053

Colored Rasmussen spectral sequence conjecture for the unknot

摘录 / EXCERPT: Let $\mathrm{HHH}_{S^n}(O_1)$ be the S^n-colored triply graded Khovanov-Rozansky homology of the unknot, with even generators $u_0,\ldots,u_{n-1}$ and odd generators...

conjecture | open | MathDB 页面 | 原文线索

V2-033792 | MATHDB-370056

The obstruction conjecture for Toeplitz speedups

摘录 / EXCERPT: Let (X,T) be a Toeplitz flow with period structure (p_k), and let c be a positive integer. A minimal speedup with orbit number c means a minimal bounded speedup (X,S) of...

conjecture | open | MathDB 页面 | 原文线索

V2-033793 | MATHDB-370057

The characterization conjecture for Toeplitz speedups

摘录 / EXCERPT: Let (X,T) be a Toeplitz flow with period structure (p_k), and let (X,S) be a bounded speedup of (X,T) with orbit number c. For the Kakutani-Rokhlin partition...

conjecture | open | MathDB 页面 | 原文线索

V2-033794 | MATHDB-370073

Hodge-Tate and semisimplicity conjecture for hyperplane sections of rational homogeneous spaces

摘录 / EXCERPT: Hodge-Tate and semisimplicity conjecture. The following should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-033795 | MATHDB-370080

Cellularity conjecture for $K(1)$-local motives over $\overline{\mathbb Q}$

摘录 / EXCERPT: The category of $K(1)$-local motives relative to $\overline{\mathbb Q}$ is the category under consideration, with cellular objects understood as those in the localizing subcategory...

conjecture | open | MathDB 页面 | 原文线索

V2-033796 | MATHDB-370082

Integral resolution conjecture for Steinberg representations

摘录 / EXCERPT: Let W be a finite-rank free $\mathbb{Z}$-module, and let $\operatorname{St}(W)$ be the top reduced homology of its Tits building. For $n\geq 3$, define the integral complex with...

conjecture | open | MathDB 页面 | 原文线索

V2-033797 | MATHDB-370126

Conjecture that Koschorke's and Milnor-type invariants agree for Brunnian embedded links

摘录 / EXCERPT: Invariant agreement conjecture. For $\b7$-Brunnian embedded links, Koschorke's $\b7_\b7(f)$ coincides up to sign with $\b7(\b7(1),\ldots,\b7(d-1),d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-033798 | MATHDB-370127

The conjecture on the -topology

摘录 / EXCERPT: The $\mathscr{E}_K$-topology conjecture. If K is large, then the $\mathscr{E}_K$-topology is the intersection of all gt-henselian topologies on K. Consequently, the...

conjecture | open | MathDB 页面 | 原文线索

V2-033799 | MATHDB-370128

The saturated large-field conjecture for the $\mathscr{E}_K$-topology

摘录 / EXCERPT: Saturated large-field conjecture. The $\mathscr{E}_K$-topology is the intersection of all R-adic topologies as R ranges over henselian local domains $R\subseteq K$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-033800 | MATHDB-370129

The definable-basis criterion for the $\mathscr{E}_K$-topology

摘录 / EXCERPT: Definable-basis conjecture. If K admits a gt-henselian topology with a basis consisting of definable sets, then the $\mathscr{E}_K$-topology is a field topology.

conjecture | open | MathDB 页面 | 原文线索

V2-033801 | MATHDB-370130

The rc-convex hull conjecture for hole-free digital pictures

摘录 / EXCERPT: Let (X,c_2,c_1) be a digital picture with no hole, and let $H_{rc}(X)$ denote its rc-convex hull. Rc-convex hull conjecture. The digital picture $(H_{rc}(X),c_2,c_1)$ is a set of...

conjecture | open | MathDB 页面 | 原文线索

V2-033802 | MATHDB-370139

Internal language conjecture for locally cartesian closed infinity-categories

摘录 / EXCERPT: Let $\mathbf{CompCat}_{\Sigma, \Pi_\text{ext}, \text{Id}}$ be the category of models of intensional Martin-Löf type theory with dependent sums, extensional dependent products, and...

conjecture | open | MathDB 页面 | 原文线索

V2-033803 | MATHDB-370142

Low's conjecture on linked skies

摘录 / EXCERPT: Low Conjecture. The skies of causally related points in X are topologically linked.

conjecture | open | MathDB 页面 | 原文线索

V2-033804 | MATHDB-370146

Stable rational abelianization conjecture for the handlebody hyperelliptic Torelli group

摘录 / EXCERPT: Let $\mathcal{H}_B\mathcal{I}_{g,p}^b$ denote the handlebody hyperelliptic Torelli group, and let $\overline{W}_{\mathbb Q}$ and $W_{\mathbb Q}$ be the rational modules appearing i...

conjecture | open | MathDB 页面 | 原文线索

V2-033805 | MATHDB-370148

HK-good groupoid model conjecture for classifiable C*-algebras

摘录 / EXCERPT: A C^*-algebra is called classifiable if it belongs to the class of algebras classified by its Elliott invariant, and a groupoid model is HK-good when it has the homology-K-theo...

conjecture | open | MathDB 页面 | 原文线索

V2-033806 | MATHDB-370180

Interaction-strength conjecture for Bouw-Möller surfaces with divisible parameters

摘录 / EXCERPT: Interaction-strength conjecture. For every pair of closed curves α,β on $S_{m,n}$,

conjecture | open | MathDB 页面 | 原文线索

V2-033807 | MATHDB-370200

Conjectural description of the isotypic part of cohomology of Shtukas

摘录 / EXCERPT: Let C be the underlying smooth projective curve, let I be a finite index set with $|I|=r$, and let fs_n be a geometrically irreducible Weil local system of rank n on C. W...

conjecture | open | MathDB 页面 | 原文线索

V2-033808 | MATHDB-370204

Stability-independence conjecture for the cohomology rings of compactified Jacobians

摘录 / EXCERPT: Let C_0 be the central curve in the setting of Theorem 0.8, and let ϕ and ϕ' be nondegenerate stability conditions. Write $\overline{J}^{\phi}_{C_0}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033809 | MATHDB-370238

Kahle-Newman Cohen-Lenstra conjecture for random hypertree torsion

摘录 / EXCERPT: Kahle-Newman conjecture. For every finite abelian p-group G,

conjecture | open | MathDB 页面 | 原文线索

V2-033810 | MATHDB-370248

The universal bifurcation diagram conjecture

摘录 / EXCERPT: Universal bifurcation diagram conjecture. There exists a homotopy

conjecture | open | MathDB 页面 | 原文线索

V2-033811 | MATHDB-370256

Improved Beilinson-Flach class conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be the family locus under consideration, let $\kappa_{{\mathbf{g}},{\mathbf h}}\vert_{\mathcal{C}}$ be the restriction of the Beilinson-Flach cohomology class, an...

conjecture | open | MathDB 页面 | 原文线索

V2-033812 | MATHDB-370270

The chain-level \(\mathfrak{gl}_N\) link invariant functoriality conjecture

摘录 / EXCERPT: Chain-level $\mathfrak{gl}_N$ functoriality conjecture. The chain-level $\mathfrak{gl}_N$ link invariant arises as the truncation of an $\mathbb{E}_3$-monoidal functor

conjecture | open | MathDB 页面 | 原文线索

V2-033813 | MATHDB-370295

Neitzke-Yan compatibility conjecture for quantum UV-IR maps

摘录 / EXCERPT: Let $\Gamma=H_1(\widetilde{\Sigma};\mathbb{Z})$ have its standard intersection pairing, and let ... where σ is the $\mathbb{Z}/2$-deck transformation action. Let $Q_{2\Gamma}$...

conjecture | open | MathDB 页面 | 原文线索

V2-033814 | MATHDB-370306

Singer's injectivity conjecture for the algebraic transfer

摘录 / EXCERPT: Singer's conjecture. The homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033815 | MATHDB-370307

Detection of the element t by the sixth algebraic transfer

摘录 / EXCERPT: Detection conjecture. The nonzero element t is detected by the sixth algebraic transfer $Tr_6(\mathbb F_2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-033816 | MATHDB-370323

The projective compactification no-retraction conjecture

摘录 / EXCERPT: Projective compactification no-retraction conjecture. Every projective Γ-compactification with non-trivial boundary X is non-Dirac. In particular, there does not exist a r...

conjecture | open | MathDB 页面 | 原文线索

V2-033817 | MATHDB-370392

Polishchuk-Positselski's generator conjecture for Koszul algebras

摘录 / EXCERPT: Polishchuk-Positselski's conjecture. The algebra A has at least n generators.

conjecture | refuted | MathDB 页面 | 原文线索

V2-033818 | MATHDB-370394

Artin's flat duality conjecture for surfaces

摘录 / EXCERPT: Let $\pi\colon X\rightarrow\operatorname{Spec} k$ be a smooth complete surface over a perfect field k of characteristic $p\neq0$. Let $H_{\mathrm{fl}}^i(X,\mu_p)$ denote flat coh...

conjecture | open | MathDB 页面 | 原文线索

V2-033819 | MATHDB-370395

Artin's precise flat duality conjecture for surfaces

摘录 / EXCERPT: Let $\pi\colon X\rightarrow\operatorname{Spec} k$ be a smooth complete surface over a perfect field k of characteristic $p\neq0$, and let $R\pi_*\mu_{p^n}$ be the derived flat-co...

conjecture | open | MathDB 页面 | 原文线索

V2-033820 | MATHDB-370415

The gluing formula for fhat{Z}-invariants of glued knot complements

摘录 / EXCERPT: Let K_1 and K_2 be knots with well-defined GM series, and let ... be the manifold obtained by gluing with $\gamma=\begin{pmatrix}r&a\\p&b\end{pmatrix}$. Write the GM series as...

conjecture | open | MathDB 页面 | 原文线索

V2-033821 | MATHDB-370416

The general link wavefunction conjecture

摘录 / EXCERPT: Let L be a link with a well-defined GM series ... and let Q be its linking matrix with zero diagonal. General link wavefunction conjecture. The wavefunction in...

conjecture | open | MathDB 页面 | 原文线索

V2-033822 | MATHDB-370417

The general partial surgery formula for GM series

摘录 / EXCERPT: Let L be a link with a well-defined GM series, and let L' be obtained by partial $-1/r$ surgery on an unknot component, or by ∞-surgery in the corresponding case. Partia...

conjecture | open | MathDB 页面 | 原文线索

V2-033823 | MATHDB-370418

The satellite formula for GM series

摘录 / EXCERPT: Let K be a knot and P a pattern whose pattern-defining link P' has a well-defined GM series. Denote the satellite by $C_P(K)$. Satellite Formula. The GM series of $C_P(K)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-033824 | MATHDB-370419

The cabling formula for GM series

摘录 / EXCERPT: Let K be a knot with a well-defined GM series, and let $C_{(tp,sp)}(K)$ denote its (tp,sp)-cabling with $\gcd(s,t)=1$. Cabling Formula. The cabling formula above holds for any...

conjecture | open | MathDB 页面 | 原文线索

V2-033825 | MATHDB-370464

Strong version of Ishii's conjecture for the Links-Gould invariant

摘录 / EXCERPT: Strong version of Ishii's conjecture. The coefficients satisfy $a_{ij}=(-1)^{\mu+i+j+1}|a_{ij}|$ for all i,j, and (a_{ij}) has no interior zeros. The source notes that this str...

conjecture | open | MathDB 页面 | 原文线索

V2-033826 | MATHDB-370496

Goncharov's conjecture on Sah algebra cohomology and algebraic K-theory

摘录 / EXCERPT: Let $\widetilde{\mathcal{P}}(S^{n-1})$ be the reduced spherical scissors congruence group, and let the direct sum ... be equipped with the Sah algebra Hopf structure, whose coproduct...

conjecture | open | MathDB 页面 | 原文线索

V2-033827 | MATHDB-370499

Möbius-band trefoil folded ribbonlength conjecture

摘录 / EXCERPT: Let K be a folded ribbon trefoil knot that is topologically a Möbius band, and let $\operatorname{Rib}([K]_{\mathrm{Möb}})$ denote its infimal folded ribbonlength in the Möbius-b...

conjecture | open | MathDB 页面 | 原文线索

V2-033828 | MATHDB-370500

Five-or-more-half-twist Möbius-band aspect-ratio conjecture

摘录 / EXCERPT: An embedded paper Möbius band with 5 or more half-twists is a paper Möbius band embedded in space with at least five half-twists; its aspect ratio is the ratio of its length to i...

conjecture | open | MathDB 页面 | 原文线索

V2-033829 | MATHDB-370566

Acyclic direction contractad homotopy-equivalence conjecture

摘录 / EXCERPT: Acyclic direction contractad homotopy-equivalence conjecture. The map ψ_d induces a homotopy equivalence of topological graphical collections

conjecture | open | MathDB 页面 | 原文线索

V2-033830 | MATHDB-370569

Safronov's Langlands duality conjecture for critical cohomology of character stacks

摘录 / EXCERPT: Safronov's conjecture. There is a natural isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-033831 | MATHDB-370582

Bergeron-Venkatesh-Lück torsion growth conjecture for 3-manifolds

摘录 / EXCERPT: Let M be a compact connected orientable and irreducible 3-manifold with infinite fundamental group and empty or toroidal boundary. A cofinal tower of regular finite-sheeted cov...

conjecture | open | MathDB 页面 | 原文线索

V2-033832 | MATHDB-370591

Free-monoid description of iterated category constructions on classifying spaces

摘录 / EXCERPT: Let M be an ungroup-like abelian monoid, let X be a topological space, and define $\mathrm{SP}(X,M)$ to be the free strictly commutative topological monoid generated by X-man...

conjecture | open | MathDB 页面 | 原文线索

V2-033833 | MATHDB-370607

Naturality conjecture for the embedding-calculus and Vassiliev isomorphism

摘录 / EXCERPT: Let K_3 be the space of long knots in $\mathbb{R}^3$. Let $U_{\mathcal{T}}$ and Σ be the spaces in the preceding construction, and let ... be the maps defined there. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033834 | MATHDB-370648

Gamma-class overconvergence conjecture for quantum connections

摘录 / EXCERPT: Gamma-class overconvergence conjecture. For the quantum connection on any monotone symplectic manifold, the Frobenius structure with constant term above, where $b=\Gamma_p(TM)$, is...

conjecture | open | MathDB 页面 | 原文线索

V2-033835 | MATHDB-370649

Frobenius valuation conjecture for monotone symplectic manifolds

摘录 / EXCERPT: Frobenius valuation conjecture. The valuations of the eigenvalues of $\Phi(\theta)$ are nonpositive integers; among them, listed with multiplicities, the number of times that each...

conjecture | open | MathDB 页面 | 原文线索

V2-033836 | MATHDB-370657

Modular distribution of invariants for topo-symmetric extensions

摘录 / EXCERPT: Modular distribution of invariants. The invariants of topo-symmetric extensions are distributed uniformly modulo $\gcd(|G|,|H|)$ in the sense above. Numerical experiments for small...

conjecture | open | MathDB 页面 | 原文线索

V2-033837 | MATHDB-370662

The general nilpotent nullhomotopy volume upper-bound conjecture

摘录 / EXCERPT: Let Y be a c-step nilpotent space, and let $\operatorname{V\delta}^{n}_{Y}(L)$ denote its volume of nullhomotopies in degree n at scale L. General nilpotent upper-bound con...

conjecture | open | MathDB 页面 | 原文线索

V2-033838 | MATHDB-370663

The light-factor conjecture for weight-maximizing differential summands

摘录 / EXCERPT: Let M_{n}^{J} be the degree-n piece at filtration level J, let $\mathrm{wt}(\omega)$ denote the weight of ω, and let d_{sim} be the indicated component of the differ...

conjecture | open | MathDB 页面 | 原文线索

V2-033839 | MATHDB-370664

Folklore torsion-freeness conjecture for pure graph braid groups

摘录 / EXCERPT: Let $\mathsf{\Gamma}$ be a graph and let $k\geq 0$. Write $F_k(\mathsf{\Gamma})$ for its k-point configuration space. Folklore torsion-freeness conjecture. The space...

conjecture | open | MathDB 页面 | 原文线索

V2-033840 | MATHDB-370665

Finite universal generators conjecture for graph configuration-space homology

摘录 / EXCERPT: Let $\mathcal{G}$ be a collection of graphs. Say that $\mathcal{G}$ is a set of universal generators in degree i if, for every graph $\mathsf{\Gamma}$, the group...

conjecture | open | MathDB 页面 | 原文线索

V2-033841 | MATHDB-370668

The spectral Sullivan conjecture on symmetric-power layers

摘录 / EXCERPT: The spectral Sullivan conjecture. The connective Bousfield class of $L(k)$ is the same as that of some type k finite spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-033842 | MATHDB-370671

Kohn-type conjecture for 2-adjacent alternating knots

摘录 / EXCERPT: Let K be an alternating non-trivial 2-adjacent knot, and let a minimal diagram mean a diagram of K with the minimum crossing number. A 2-adjacency set is a pair of crossing...

conjecture | open | MathDB 页面 | 原文线索

V2-033843 | MATHDB-370702

The exactness conjecture for local coefficient systems on complete regions

摘录 / EXCERPT: Let Δ be the Bruhat-Tits building of dimension d, let $\mathbb{X}$ be the coefficient system, and let $\mathcal{X}$ be a set of facets. For each $i\leq d$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-033844 | MATHDB-370738

Hoste's conjecture on zeros of Alexander polynomials of alternating knots

摘录 / EXCERPT: Hoste's conjecture. Then $\Re(t) > -1$. This conjecture concerns the location of zeros of Alexander polynomials for alternating knots. The source recalls it while analyzing zeros i...

conjecture | open | MathDB 页面 | 原文线索

V2-033845 | MATHDB-370749

Nontrivial-link conjecture for repelling trajectories in flows on the 3-sphere

摘录 / EXCERPT: Let f^t be a structurally stable, nonsingular flow on $\mathbb{S}^3$ whose non-wandering set consists of an expanding attractor and a finite set of repelling periodic trajectorie...

conjecture | open | MathDB 页面 | 原文线索

V2-033846 | MATHDB-370750

Existence conjecture for a flow on the 3-sphere with a non-figure-eight repeller

摘录 / EXCERPT: A nonsingular flow on $\mathbb{S}^3$ has a non-wandering set consisting of an expanding attractor and a unique repelling periodic trajectory. Non-figure-eight repeller conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-033847 | MATHDB-370774

Real monopole-Heegaard Floer homology isomorphism conjecture

摘录 / EXCERPT: Let (Y,ι) be a real three-manifold with a compatible real $\mathrm{spin^c}$ structure $\mathfrak{s}$. Let $\widetilde{\mathit{HMR}}$, $\widecheck{\mathit{HMR}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-033848 | MATHDB-370783

Rainbow Reidemeister conjecture for isotopic surfaces

摘录 / EXCERPT: Let two rainbow diagrams represent isotopic surfaces. A rainbow moves conjecture. Any two rainbow diagrams representing isotopic surfaces are related by a sequence of rainbow moves...

conjecture | open | MathDB 页面 | 原文线索

V2-033849 | MATHDB-370793

The handlebody positive-scalar-curvature conjecture

摘录 / EXCERPT: Let M^3 be the interior of a genus-γ handlebody, equipped with a complete metric whose scalar curvature satisfies $R>0$. The handlebody positive-scalar-curvature conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-033850 | MATHDB-370794

The solid-torus exhaustion problem for positive scalar curvature

摘录 / EXCERPT: Let M^3 be a three-manifold that has an exhaustion by solid tori and admits a complete metric with scalar curvature $R>0$ and bounded geometry. The solid-torus exhaustion problem...

open_problem | open | MathDB 页面 | 原文线索

V2-033851 | MATHDB-370795

The generic regularity hypothesis for least-area hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be a closed smooth manifold, let $H_n(M;\mathbb{Z})$ denote its codimension-one homology group, and let a representative of a homology class mean a smooth hypersurfac...

conjecture | open | MathDB 页面 | 原文线索

V2-033852 | MATHDB-370799

The Bogomolov inequality for Hodge-Riemann pairs

摘录 / EXCERPT: Let X be a compact complex manifold of dimension d, and let (η_{d-1},η_{d-2}) be a pair of cohomology classes of bidegrees $(d-1,d-1)$ and $(d-2,d-2)$, respectively. A...

conjecture | open | MathDB 页面 | 原文线索

V2-033853 | MATHDB-370800

The Hodge-Riemann-to-Bogomolov pair conjecture

摘录 / EXCERPT: Let X be a compact complex manifold, and let (η_{d-1},η_{d-2}) be a pair of cohomology classes. A pair is Hodge-Riemann if it satisfies the Hodge-Riemann relations, and i...

conjecture | open | MathDB 页面 | 原文线索

V2-033854 | MATHDB-370819

Non-determinacy of the cut-and-choose game on a Bernstein subset of the double arrow space

摘录 / EXCERPT: Non-determinacy problem. The game $G_{\omega}(Y,\mathbb{A})$ is not determined.

open_problem | open | MathDB 页面 | 原文线索

V2-033855 | MATHDB-370822

Order-4 meridian and stable irreducibility conjecture for the Klein bottles \(\mathcal K(l,m,n)\)

摘录 / EXCERPT: Let $\mathcal K(l,m,n)$ denote the Klein bottle constructed from the parameters l,m,n. Its meridians are said to have order 4 when they represent elements of order 4 in the fun...

conjecture | open | MathDB 页面 | 原文线索

V2-033856 | MATHDB-370823

Non-ribbon conjecture for the Klein bottles \(\mathcal K(l,m,n)\)

摘录 / EXCERPT: Let $\mathcal K(l,m,n)$ denote the Klein bottles constructed in the paper. An embedded surface is ribbon when it admits a ribbon presentation, equivalently one with only ribbon sin...

conjecture | open | MathDB 页面 | 原文线索

V2-033857 | MATHDB-370839

The power-structure compatibility conjecture for the -Euler characteristic

摘录 / EXCERPT: Power-structure compatibility conjecture. For every such k, X, and n,

conjecture | open | MathDB 页面 | 原文线索

V2-033858 | MATHDB-370844

SET conjecture for structurally stable glocal families

摘录 / EXCERPT: Let a glocal family of vector fields be structurally stable, meaning that it is equivalent to any nearby family in the relevant C^k topology. SET conjecture. The family has the s...

conjecture | open | MathDB 页面 | 原文线索

V2-033859 | MATHDB-370876

Hyperbolicity conjecture for swatches reducible into knit and purl swatches

摘录 / EXCERPT: Let $L\thinspace\subset T^2\times I$ be an $m\times n$ swatch, and let H be the Hopf link used to form the corresponding link $H\cup f(L)\subset S^3$. A swatch is reducible in te...

conjecture | open | MathDB 页面 | 原文线索

V2-033860 | MATHDB-370877

Additivity conjecture for hyperbolic volume of annulus-summed swatches

摘录 / EXCERPT: Let $L_1,L_2\subset T^2\times I$ be swatches such that the corresponding links $H\cup f(L_1)\subset S^3$ and $H\cup f(L_2)\subset S^3$ are three-component hyperbolic links. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033861 | MATHDB-370878

Multiplicativity conjecture for the multivariable Alexander polynomial of annulus-summed swatches

摘录 / EXCERPT: Let $L_1\subset T^2\times I$ and $L_2\subset T^2\times I$ be an $m_1\times n$ swatch and an $m_2\times n$ swatch, respectively, where $m_1,m_2,n\in\{1,2,\ldots\}$. Let H be the H...

conjecture | open | MathDB 页面 | 原文线索

V2-033862 | MATHDB-370904

The functor-of-points conjecture for $cwidehat{\bbF}_1^\mathrm{Syn}$

摘录 / EXCERPT: Let $\bbS^{\widehat{\mathbbl{\Delta}}}$ be the complement of the zero section in the universal formal group $\widehat{\bbG}_\mathrm{univ}=\bbS^{\widehat{\calN}}$, regarded as an an...

conjecture | open | MathDB 页面 | 原文线索

V2-033863 | MATHDB-370935

Delman-Roberts' persistent foliation conjecture for knots in L-spaces

摘录 / EXCERPT: Let K be a knot in an L-space. The knot K is persistently foliar if, except for one meridional slope, every boundary slope of its complement is strongly realized by a co-orient...

conjecture | open | MathDB 页面 | 原文线索

V2-033864 | MATHDB-370959

Hoffman-Meeks finite topology conjecture for properly minimally embedded surfaces

摘录 / EXCERPT: Let M be a connected surface of finite topology, with genus g and $r>2$ ends. Finite topology conjecture. The surface M can be properly minimally embedded in $\mathbb{R}^3$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-033865 | MATHDB-370965

Generalized cohomology vanishing conjecture for parabolic connections

摘录 / EXCERPT: For $x\in\mathbb{P}^1$, let ξ_x be the bundle on $\overline{\mathcal{M}}$ whose fiber at $(L,\nabla,\varphi;\epsilon\in E)$ is L_x. Let $\mathcal{M}$ denote the moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-033866 | MATHDB-371002

The genus-2 handlebody representation conjecture for the Clifford orbit of W_2

摘录 / EXCERPT: Let W_2 denote the two-anyon-pair state, and let Σ_2 be a genus-2 handlebody. The Clifford group orbit of W_2 is represented by removing two solid tori from the interi...

conjecture | open | MathDB 页面 | 原文线索

V2-033867 | MATHDB-371017

Symmetric monoidality of duality for almost perfect modules over the sphere spectrum

摘录 / EXCERPT: Sphere-spectrum duality claim. The result established for coherent, eventually coconnective commutative dg algebras over $\C$ also holds over $\mathbb{S}$: the restricted duality f...

conjecture | open | MathDB 页面 | 原文线索

V2-033868 | MATHDB-371019

Hecke lifting conjecture for framed links

摘录 / EXCERPT: Let $cmathcal{L}$ be a framed oriented link in S^3 with framing $cvec{ctau}=(ctau^1,cdots,ctau^L)$. For a partition cmu, let $cZ_{cmu}(cmathcal{L};q,a)$ be the reformulated fra...

conjecture | open | MathDB 页面 | 原文线索

V2-033869 | MATHDB-371021

Uniform folded ribbonlength conjecture for (2,q)-torus links

摘录 / EXCERPT: Let $T(2,q)$ denote a $(2,q)$-torus link, let $[T(2,q)]$ denote its link type, and let ${\operatorname{Rib}}([T(2,q)])$ denote its infimal folded ribbonlength. Uniform torus-link c...

conjecture | open | MathDB 页面 | 原文线索

V2-033870 | MATHDB-371022

Uniform folded ribbonlength conjecture for pretzel links

摘录 / EXCERPT: Let L be a three-strand pretzel link $P(p,q,r)$, with folded ribbon realization L_w, and let ${\operatorname{Rib}}(L_w)$ denote its folded ribbonlength. Pretzel-link uniformity...

conjecture | open | MathDB 页面 | 原文线索

V2-033871 | MATHDB-371044

Maurer-Cartan deformation conjecture for ambient symplectic cochains

摘录 / EXCERPT: Let D be a Liouville domain whose boundary $partial D$ has no contractible Reeb orbits. The intrinsic symplectic cochains $SC^*(D)$ and the S^1-equivariant symplectic cochains...

conjecture | open | MathDB 页面 | 原文线索

V2-033872 | MATHDB-371045

Domain-dependent neck-stretching Maurer-Cartan conjecture

摘录 / EXCERPT: Let X be a symplectic manifold, let $V\subset X$ be a symplectic crossing divisor carrying the symplectic form, and let $D\subset X\setminus V$. The intrinsic symplectic cochains...

conjecture | open | MathDB 页面 | 原文线索

V2-033873 | MATHDB-371046

Infinite boundary depth conjecture for Maslov-zero Lagrangian tori

摘录 / EXCERPT: Let $L\subset M$ be a Maslov 0 Lagrangian torus, and let $C^*(L,M)$ be its Floer complex with differential satisfying ... Here $\tau(L)$ denotes the relevant boundary-depth invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-033874 | MATHDB-371052

The Homological Time Complexity Relation

摘录 / EXCERPT: Let L be a computational problem, let $T_L(n)$ denote its time complexity, and let $h(L)$ denote its homological complexity. Homological Time Complexity Relation. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-033875 | MATHDB-371053

The physical realization conjecture for homological complexity

摘录 / EXCERPT: Let L be a computational problem with homological complexity $h(L)$. A physical system is said to solve L efficiently when it computes solutions to L within the relevant effi...

conjecture | open | MathDB 页面 | 原文线索

V2-033876 | MATHDB-371054

The quantum homological obstruction conjecture

摘录 / EXCERPT: Let L be a decision problem in the bounded-error quantum polynomial-time class $\mathcal{BQP}$, and let $h(L)$ denote its homological complexity. Quantum homological obstruction...

conjecture | open | MathDB 页面 | 原文线索

V2-033877 | MATHDB-371055

The ultimate homological characterization of complexity classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a natural complexity class, let $h(L)$ denote the homological complexity of a problem L, and let $S_{\mathcal{C}}$ be a subset of $\mathbb{N}\cup\{\infty\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-033878 | MATHDB-371056

The homological permanent-determinant conjecture

摘录 / EXCERPT: Let $h(\mathrm{perm}_n)$ and $h(\mathrm{det}_n)$ denote the homological complexities associated with the $n\times n$ permanent and determinant, respectively, and let $\mathcal{VP}$...

conjecture | open | MathDB 页面 | 原文线索

V2-033879 | MATHDB-371057

The precise homological complexity characterization of complexity classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a natural complexity class, let $h(L_n)$ denote the homological complexity of the restriction L_n of a problem L to inputs of size n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-033880 | MATHDB-371058

The quantum homological complexity conjecture

摘录 / EXCERPT: Let L be a computational problem, let $h(L)$ be its ordinary homological complexity, and let $h_q(L)$ be a proposed quantum homological complexity measure. Quantum homological co...

conjecture | open | MathDB 页面 | 原文线索

V2-033881 | MATHDB-371059

The homological characterization of one-way functions

摘录 / EXCERPT: Let f be a function, let ... and let $h(L_f)$ denote the homological complexity of the inversion problem associated with f. Homological characterization of one-way functions. The...

conjecture | open | MathDB 页面 | 原文线索

V2-033882 | MATHDB-371060

The homological form of the Church-Turing thesis

摘录 / EXCERPT: Let L be a physically realizable computation and let $h(L)$ denote its homological complexity. Homological Church-Turing thesis. Every physically realizable computation has finit...

conjecture | open | MathDB 页面 | 原文线索

V2-033883 | MATHDB-371061

Peterson's hit problem conjecture

摘录 / EXCERPT: Let P_h be the polynomial algebra on h variables over $\mathbb F_p$, equipped with the action of the Steenrod algebra $\mathscr A_p$. For a positive integer n, write its p-...

conjecture | open | MathDB 页面 | 原文线索

V2-033884 | MATHDB-371109

The homological connectivity threshold conjecture for k-fold coverage

摘录 / EXCERPT: Homological connectivity threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033885 | MATHDB-371117

Lück's homological torsion growth conjecture for aspherical manifolds

摘录 / EXCERPT: Let M be an aspherical closed manifold. Consider a descending chain of subgroups ... such that each G_i is normal in $G=\pi_1(M)$, the index $[G:G_i]$ is finite, and...

conjecture | open | MathDB 页面 | 原文线索

V2-033886 | MATHDB-371125

Continuity conjecture for the first even Hilbert-Brunn-Minkowski eigenvalue

摘录 / EXCERPT: Continuity conjecture for the first even eigenvalue. The map $K\mapsto\lambda_{1,e}(-L_K)$ is continuous in the Hausdorff topology; equivalently, if $h_k\to h$ in L^∞, then

open_problem | open | MathDB 页面 | 原文线索

V2-033887 | MATHDB-371127

Varchenko's path-sum conjecture for stable-envelope integrals

摘录 / EXCERPT: Let $X=T^*Gr(k,n)$, let $\boldsymbol{a}=(a_1,\ldots,a_n)$ be the equivariant parameters, and let $\boldsymbol{a}\to 0$ denote the non-equivariant limit. For a partition $lambda$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-033888 | MATHDB-371171

Relative homological projective duality conjecture for genus-one Calabi-Yau threefolds with a 5-section

摘录 / EXCERPT: Let B be a generalized del Pezzo surface with $b_2(X)=b_2(B)+1$, and let ... be a smooth genus-one fibered Calabi-Yau threefold with a 5-section. Assume that X is the degeneracy...

conjecture | open | MathDB 页面 | 原文线索

V2-033889 | MATHDB-371198

Gauge-theory realization conjecture for Majorana, Dijkgraaf-Witten, and Gu-Wen anomalies

摘录 / EXCERPT: Let k be as in the paper, let x_1 and y specify the twist, and let α_{DW} and α_{GW} be the Dijkgraaf-Witten and Gu-Wen classes in...

conjecture | open | MathDB 页面 | 原文线索

V2-033890 | MATHDB-371203

Connectivity conjecture for random graph neighborhood complexes

摘录 / EXCERPT: Connectivity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-033891 | MATHDB-371204

Vanishing conjecture for random graph neighborhood-complex cohomology

摘录 / EXCERPT: Cohomology-vanishing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-033892 | MATHDB-371241

The positive scalar curvature conjecture for closed aspherical manifolds

摘录 / EXCERPT: Let M be a closed aspherical manifold, meaning that its higher homotopy groups satisfy $\pi_k(M)=0$ for all $k\ge2$. A positive scalar curvature metric is a Riemannian metric who...

conjecture | open | MathDB 页面 | 原文线索

V2-033893 | MATHDB-371250

Segal-Stolz-Teichner conjecture for supersymmetric field theories and TMF

摘录 / EXCERPT: Let X be a space and let n be an integer. Consider fully extended, degree $-n$, 2-dimensional supersymmetric field theories over X, modulo concordance. Segal-Stolz-Teichner c...

conjecture | open | MathDB 页面 | 原文线索

V2-033894 | MATHDB-371297

Glasner-Huang-Shao-Weiss-Ye polynomial odd-recurrence conjecture

摘录 / EXCERPT: Glasner-Huang-Shao-Weiss-Ye conjecture. The set R has nonempty intersection with every infinite arithmetic progression of step size k.

conjecture | open | MathDB 页面 | 原文线索

V2-033895 | MATHDB-371298

Leibman's connected orbit-closure conjecture for polynomial sequences

摘录 / EXCERPT: Leibman's conjecture. The orbit closure

conjecture | open | MathDB 页面 | 原文线索

V2-033896 | MATHDB-371340

The conjecture on strongly topologically nilpotent crystals from rigid analytic geometry

摘录 / EXCERPT: Let R be a topologically finitely generated and p-complete ${\mathcal{O}}_{\mathbf{C}}$-algebra, and let ... be an ${\mathcal{O}}_{\mathbf C}$-homomorphism. A rational Hodge--Tat...

conjecture | open | MathDB 页面 | 原文线索

V2-033897 | MATHDB-371342

Finiteness conjecture for extendable automorphisms of embedded tori

摘录 / EXCERPT: Let T^p be a p-dimensional torus embedded in S^{p+2}. Consider extendable homeomorphisms of the embedding, and distinguish their mapping classes in $\operatorname{MCG}(T^p)$...

conjecture | open | MathDB 页面 | 原文线索

V2-033898 | MATHDB-371343

The decategorification isomorphism conjecture for Hochschild homology

摘录 / EXCERPT: Decategorification isomorphism conjecture. The injective algebra homomorphism above is always an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-033899 | MATHDB-371353

Homological detection conjecture for Vietoris-Rips complexes of integer lattices

摘录 / EXCERPT: Let $\mathbb{Z}^n$ have the Manhattan metric, let $\{0,1\}^n\subseteq\mathbb{Z}^n$ have the induced metric, and let $\widetilde{H}_i(-;\mathbb{Z})$ denote reduced homology with int...

conjecture | open | MathDB 页面 | 原文线索

V2-033900 | MATHDB-371369

Kato-Kuzumaki's cohomological dimension conjecture

摘录 / EXCERPT: Kato-Kuzumaki's conjecture. A field k satisfies C_i^q if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-033901 | MATHDB-371373

The knot complement conjecture for knots in closed oriented 3-manifolds

摘录 / EXCERPT: Knot complement conjecture. If K_1 and K_2 have orientation-preservingly homeomorphic exteriors, then there exists an orientation-preserving homeomorphism of Y mapping K_1...

conjecture | open | MathDB 页面 | 原文线索

V2-033902 | MATHDB-371374

Boileau's nontrivial surgery conjecture for nullhomotopic knots

摘录 / EXCERPT: Boileau's nontrivial surgery conjecture. A non-trivial surgery on K never gives a manifold homeomorphic to Y in an orientation-preserving way.

conjecture | open | MathDB 页面 | 原文线索

V2-033903 | MATHDB-371375

The nullhomotopic knot complement conjecture

摘录 / EXCERPT: Nullhomotopic knot complement conjecture. If K_1 and K_2 have orientation-preservingly homeomorphic exteriors, then there exists an orientation-preserving homeomorphism of Y...

conjecture | open | MathDB 页面 | 原文线索

V2-033904 | MATHDB-371420

Katok's exponential periodic-orbit growth conjecture for surface diffeomorphisms

摘录 / EXCERPT: Let M be a compact surface, let $f\colon M\to M$ be a diffeomorphism, let $P_n(f)=\{x\in M:x=f^n(x)\}$, and let $h_{\operatorname{top}}(f)$ denote the topological entropy of f....

open_problem | open | MathDB 页面 | 原文线索

V2-033905 | MATHDB-371428

The optimistic conjecture on canonical Banach topologies for cohomology

摘录 / EXCERPT: Let F be the coefficient field, let $\mathcal{R}_F$ be its Robba ring, and let $\mathcal{D}$ be a (φ,Γ)-module over $\mathcal{R}_F$. For a representation V of...

conjecture | open | MathDB 页面 | 原文线索

V2-033906 | MATHDB-371429

The pessimistic conjecture on closed images of saturated submodules

摘录 / EXCERPT: Pessimistic conjecture. The image of this map is closed in the Banach-space topology of $H^1(\mathcal{K}_\infty,V)$. This weaker assertion is stated as sufficient for the paper's p...

conjecture | open | MathDB 页面 | 原文线索

V2-033907 | MATHDB-371459

Nonexistence of a genuine splitting of Real spin bordism

摘录 / EXCERPT: Let C_2 be the cyclic group of order two, let $\operatorname{MSpin}^c_{\operatorname{\mathbb{R}}}$ denote Real spin bordism, let $\mathcal{P}$ be the indexing set for the Anderso...

conjecture | open | MathDB 页面 | 原文线索

V2-033908 | MATHDB-371496

Finiteness of high-entropy topological homoclinic classes

摘录 / EXCERPT: Let M be a closed surface and let $f:M\rightarrow M$ be a $\mathcal{C}^{\infty}$ local diffeomorphism. For an f-invariant measure μ, let $\operatorname{HC}(\mu)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-033909 | MATHDB-371513

Homological shift generation conjecture for complementary edge ideals

摘录 / EXCERPT: Let G be a connected graph, and let $I=I_c(G)$ be its complementary edge ideal. For each $i\geq 0$, let $\HS_i(\mathcal{R}(I))$ denote the i-th homological shift ideal of the R...

conjecture | open | MathDB 页面 | 原文线索

V2-033910 | MATHDB-371538

Stability conjecture for higher-arity global dimensions

摘录 / EXCERPT: Let T be a $Gamma$-semiring and, for each integer $n\geq 3$, let $\mathrm{gldim}^{(n)}_\Gamma(T)$ denote its n-ary global dimension. Stability conjecture. For all $n\geq 3$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-033911 | MATHDB-371539

Complexity conjecture for cohomology vanishing over finite Γ-semirings

摘录 / EXCERPT: Complexity conjecture. This problem lies in $\mathbf{coNP}$, and becomes $\mathbf{P}$-complete for cyclic additive semirings. The claim gives a proposed complexity classification f...

conjecture | open | MathDB 页面 | 原文线索

V2-033912 | MATHDB-371556

The converse characterization of R-Poincaré duality spaces by graph cobordisms

摘录 / EXCERPT: Converse characterization conjecture. If $(\mathfrak{I}_{\hat\Psi},\mathfrak{t}_{\hat\Psi})$ is a commutative Frobenius algebra in

conjecture | open | MathDB 页面 | 原文线索

V2-033913 | MATHDB-371592

Transitivity conjecture for beta-transformations in the parameter triangle

摘录 / EXCERPT: Let $\mathcal{T}$ be the parameter triangle for the family of beta-transformations, with parameters (α,β). Transitivity conjecture. For all parameters in $\mathcal{T}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-033914 | MATHDB-371593

Almost-everywhere equivalence of transitivity and LEO for beta-transformations

摘录 / EXCERPT: Let $\mathcal{T}$ be the parameter triangle for the family of beta-transformations, with parameters (α,β). A beta-transformation is topologically LEO when it has the loc...

conjecture | open | MathDB 页面 | 原文线索

V2-033915 | MATHDB-371595

Logarithmic topological mirror symmetry for stringy intersection cohomology

摘录 / EXCERPT: Let C be a smooth projective curve with $g(C)\geq 2$, let G be a connected semisimple group, and let $G^{\vee}$ be its Langlands dual. Let ${\mathcal L}$ be a line bundle on...

conjecture | open | MathDB 页面 | 原文线索

V2-033916 | MATHDB-371596

Orbifold dimensional reduction conjecture for character stacks

摘录 / EXCERPT: Let M be a Seifert-fibred 3-manifold, let G be a semisimple group, and for $n>0$ let $\Sigma_{\mathrm{orb},M}^{[n]}$ be the quotient of M by the S^1-action composed with...

conjecture | open | MathDB 页面 | 原文线索

V2-033917 | MATHDB-371597

Langlands duality for twisted stringy BPS cohomology of character stacks

摘录 / EXCERPT: Let Σ be a compact oriented 2-manifold and let G be a semisimple group, with Langlands dual group $G^{\vee}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033918 | MATHDB-371598

Langlands duality for twisted stringy BPS cohomology of Higgs moduli

摘录 / EXCERPT: Let C be a smooth projective curve and let G be a semisimple group, with Langlands dual group $G^{\vee}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033919 | MATHDB-371656

Ivanov's conjecture on finite-index subgroups of mapping class groups

摘录 / EXCERPT: Let S_g^n be a compact oriented surface of genus g with n boundary components, and let $\mathrm{Mod}(S_g^n)$ be its mapping class group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-033920 | MATHDB-371660

Yau's almost-complex-to-complex conjecture

摘录 / EXCERPT: Yau's conjecture. For $n\geq 3$, any closed $2n$-manifold admitting an almost complex structure will also admit a complex structure.

conjecture | open | MathDB 页面 | 原文线索

V2-033921 | MATHDB-371666

The semisimple below-rank cohomology conjecture for super-reflexive coefficients

摘录 / EXCERPT: Let G be an S-semisimple group of rank r and let $\Gamma<u001b[0mG$ be an irreducible lattice, meaning that its projection to each simple factor of G is dense. Let V be a...

conjecture | open | MathDB 页面 | 原文线索

V2-033922 | MATHDB-371667

The non-uniform lattice extension of Gromov's L^p-cohomology theorem

摘录 / EXCERPT: Let G be a non-compact semisimple group over a local field F with finite center and rank r. The non-uniform lattice extension conjecture. The full statement of Gromov's L^p...

conjecture | open | MathDB 页面 | 原文线索

V2-033923 | MATHDB-371668

The non-vanishing conjecture at twice the rank for product actions

摘录 / EXCERPT: Let G be a non-compact simple group over a local field F with finite center and rank r. The twice-the-rank non-vanishing conjecture. For every sufficiently large p, ... A sim...

conjecture | open | MathDB 页面 | 原文线索

V2-033924 | MATHDB-371671

The finite-cover topological overlap conjecture

摘录 / EXCERPT: Let X be a connected 2-dimensional simplicial complex whose fundamental group has property T. The finite-cover topological overlap conjecture. There exists $c>0$ such that ever...

conjecture | open | MathDB 页面 | 原文线索

V2-033925 | MATHDB-371676

The URS rigidity conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be an irreducible higher-rank lattice, and let a URS mean a minimal subsystem of the space of subgroups of Γ with the Chabauty topology. The URS rigidity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-033926 | MATHDB-371684

Conjecture that semistability is equivalent to cohomological semistability for vector bundles on curves

摘录 / EXCERPT: Semistability-cohomological semistability conjecture. E is semistable if and only if E is cohomologically semistable.

conjecture | open | MathDB 页面 | 原文线索

V2-033927 | MATHDB-371709

Refined correlator-volume correspondence for bordered Klein surfaces

摘录 / EXCERPT: Let $g\in\frac12\mathbb{Z}_{\geq0}$ and $n\in\mathbb{Z}_{\geq1}$ satisfy $2-2g-n<0$. Let $\omega_{g,n}^{\mathfrak b}$ be the refined correlator on the refined spectral curve, let...

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V2-033928 | MATHDB-371715

Sphericality conjecture for upper intervals in partial decomposition posets of buildings

摘录 / EXCERPT: Let Δ be a spherical building, and let $\sigma\in\Delta$ be any non-empty simplex. Let ${\mathcal{OPD}}(\Delta)$ and ${\mathcal{PD}}(\Delta)$ denote the ordered partial deco...

conjecture | open | MathDB 页面 | 原文线索

V2-033929 | MATHDB-371717

Finite-difference conjecture for Ext dimensions of complete intersections

摘录 / EXCERPT: Let R be a complete intersection ring with k relations and n variables, where $k\geq 1$. For any map $f:\mathbb{N}\longrightarrow\mathbb{N}$, define $\Delta^0f(m)=f(m)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033930 | MATHDB-371805

Kawamata-Viehweg-type vanishing for exhaustion singular-positive line bundles

摘录 / EXCERPT: Let X be a weakly pseudoconvex manifold, let $\mathscr{J}$ be an ideal sheaf on X, and let $L\longrightarrow X$ be a holomorphic line bundle. Assume that L is exhaustion sing...

conjecture | open | MathDB 页面 | 原文线索

V2-033931 | MATHDB-371837

Newton-stratum classicality conjecture for geometric de Rham cohomology

摘录 / EXCERPT: Let $G_{\mathbb{Q}_p}:=\operatorname{Res}_{F_w/\mathbb{Q}_p}\mathrm{GL}_{2,F_w}$, let μ be the specified minuscule cocharacter, and let $\mathscr{F}\!\ell^b$ be the Newton stra...

conjecture | open | MathDB 页面 | 原文线索

V2-033932 | MATHDB-371842

The differentiable nonabelian Hodge correspondence for relative moduli spaces

摘录 / EXCERPT: Differentiable nonabelian Hodge conjecture. The map α_S is an isomorphism of differentiable fiber bundles over S. This would upgrade the known fiberwise topological identi...

conjecture | open | MathDB 页面 | 原文线索

V2-033933 | MATHDB-371859

GJS conjecture on skein-module dimension and Abouzaid-Manolescu homology

摘录 / EXCERPT: GJS conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033934 | MATHDB-371901

Criteria for baby Julia sets in generalized McMullen parameter slices

摘录 / EXCERPT: Let W be a region in a one-dimensional subfamily of the generalized McMullen parameter space, parameterized by a. Suppose one critical value $v_+(a)$ is topologically stable, w...

conjecture | open | MathDB 页面 | 原文线索

V2-033935 | MATHDB-371924

Low-dimensional sum-set decomposition conjecture for folded Reed-Solomon list recovery

摘录 / EXCERPT: Low-dimensional sum-set decomposition conjecture. For every choice of $\ell\in\mathbb N$, $\varepsilon>0$, and $R>0$, the decomposition above exists for all sufficiently large fold...

conjecture | open | MathDB 页面 | 原文线索

V2-033936 | MATHDB-371952

Standardness conjecture for Berge-knot homotopy projective planes

摘录 / EXCERPT: Let X be a homotopy $\mathbb{CP}^2$ constructed in Proposition from a Berge knot. Standardness conjecture. The manifold X is diffeomorphic to the standard $\mathbb{CP}^2$. Thes...

conjecture | open | MathDB 页面 | 原文线索

V2-033937 | MATHDB-371989

The universal covering component conjecture for connected types

摘录 / EXCERPT: Universal covering component conjecture. The connected component of $\tilde A$ in $\Coveringn$ is $\trunc{n+1}A$.

conjecture | open | MathDB 页面 | 原文线索

V2-033938 | MATHDB-372000

Universal quantum modularity conjecture for geometric integer homology spheres

摘录 / EXCERPT: Let $M\cong N/\pi_1(M)$ be a geometric integer homology sphere with model geometry N, and let $W_M(\xi)$ be its properly normalized Witten-Reshetikhin-Turaev invariant for a prim...

conjecture | open | MathDB 页面 | 原文线索

V2-033939 | MATHDB-372039

Han's conjecture on Hochschild homology and smoothness

摘录 / EXCERPT: Let B be a finite-dimensional algebra over an algebraically closed field k. Write $\operatorname{HH}_n(B)$ for the Hochschild homology groups of B. The algebra B is (homolo...

conjecture | open | MathDB 页面 | 原文线索

V2-033940 | MATHDB-372083

Dualizability conjecture for unimodular finite braided tensor categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a unimodular finite braided tensor category, and let $\operatorname{Ver}^{\zeta^{1/2}}_{p^{(n)}}$ denote a mixed higher Verlinde category. Regard these categor...

conjecture | open | MathDB 页面 | 原文线索

V2-033941 | MATHDB-372086

Multiplicative splitting conjecture for the perverse filtration on compactified Jacobians

摘录 / EXCERPT: Multiplicative splitting conjecture. There exists such a splitting that is multiplicative under cup product:

conjecture | open | MathDB 页面 | 原文线索

V2-033942 | MATHDB-372100

Abelianness and strict A1-invariance of connected components of reductive groups

摘录 / EXCERPT: Abelianness and strict invariance conjecture. For every reductive group G over k, the sheaf $\pi_0^{\mathbb A}^1(G)$ is a sheaf of abelian groups. Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-033943 | MATHDB-372109

Uniqueness of standard neighborhoods with negative boundary slope

摘录 / EXCERPT: Let T be a transverse knot in a contact manifold. A standard neighborhood of T has a boundary slope, measured with respect to the chosen framing, and standard neighborhoods are...

conjecture | open | MathDB 页面 | 原文线索

V2-033944 | MATHDB-372110

Eventual uniqueness of standard neighborhoods for transverse knot representatives

摘录 / EXCERPT: Fix a smooth knot type K. A transverse representative T of K has a standard neighborhood, whose boundary has a slope, and standard neighborhoods are considered up to contact...

conjecture | open | MathDB 页面 | 原文线索

V2-033945 | MATHDB-372111

Eventual destabilization of low-Thurston-Bennequin Legendrian representatives

摘录 / EXCERPT: Let K be a null-homologous smooth knot type in a tight contact manifold (M,ξ). A Legendrian representative of K has a Thurston-Bennequin invariant $\tb$. Low-Thurston-Benne...

conjecture | open | MathDB 页面 | 原文线索

V2-033946 | MATHDB-372154

Nullity, determinant, and FAMEDness conjectures for ordered geometric triangulations

摘录 / EXCERPT: Nullity, determinant, and FAMEDness conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-033947 | MATHDB-372198

Global asymptotic-shape conjecture for the free boundary free elastic flow

摘录 / EXCERPT: Global asymptotic-shape conjecture. The rescaled flow $\hat\gamma(\cdot,t)$ converges as $t\to\infty$ exponentially fast in the smooth topology to a limit γ_∞. Furtherm...

conjecture | open | MathDB 页面 | 原文线索

V2-033948 | MATHDB-372215

The twining-genus correspondence for Conway defects and K3 sigma models

摘录 / EXCERPT: Let $\Pi^{\natural} \subset \Lambda\otimes\mathbb{R}\subset V^{f\natural}_{tw}$ be a 4-dimensional subspace, and let $\mathsf{Top}_{\Pi^{\natural}}$ be the full subcategory of to...

conjecture | open | MathDB 页面 | 原文线索

V2-033949 | MATHDB-372223

The terminality conjecture for Hochschild Swiss cheese actions

摘录 / EXCERPT: Terminality conjecture. The object $\mathrm{Hoch}(A)$ is terminal in the homotopy category of Swiss cheese actions on A.

conjecture | open | MathDB 页面 | 原文线索

V2-033950 | MATHDB-372232

Purely Cosmetic Surgery Conjecture

摘录 / EXCERPT: Let K be a knot, and let $S^3_r(K)$ denote the manifold obtained by surgery along K with slope r. Purely Cosmetic Surgery Conjecture. If ... via an orientation-preserving homeo...

conjecture | open | MathDB 页面 | 原文线索

V2-033951 | MATHDB-372237

GNSSS color-shifting conjecture for reduced colored HOMFLY homology

摘录 / EXCERPT: Let $\mathcal{L} = \mathcal{L}_1 \sqcup \dots \sqcup \mathcal{L}_d$ be a link with $\mathcal{L}_1$ an unknot and suppose $p \in \mathcal{L}_1$. For colors $k_1,\dots,k_d$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-033952 | MATHDB-372262

Homological-lift conjecture with expected degree

摘录 / EXCERPT: Let μ be a stability condition as in Assumption (stability), let $\alpha\in C(\mathcal A)_\mathrm{pe}$, and let $\varepsilon_\alpha^\mu\in K_\ast(\mathfrak M^\mathrm{rig})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-033953 | MATHDB-372287

Grove-Ziller conjecture on hyperbolic sections and rational hyperbolicity

摘录 / EXCERPT: Let M be a compact, simply connected manifold with a polar action by a compact group G, meaning an isometric action admitting a section S with $\dim S=\dim M/G$ that meets al...

conjecture | open | MathDB 页面 | 原文线索

V2-033954 | MATHDB-372288

Grove-Wilking-Yeager problem on hyperbolic orbit spaces and rational hyperbolicity

摘录 / EXCERPT: Let M be a compact, simply connected manifold with an isometric action by a compact group G. The orbit space M/G is assumed to have dimension two and to be covered by the hyp...

open_problem | open | MathDB 页面 | 原文线索

V2-033955 | MATHDB-372291

Dimofte-Garoufalidis 1-loop conjecture for hyperbolic knot complements

摘录 / EXCERPT: Let $K\subset{\mathbb S}^3$ be a hyperbolic knot, let γ be a simple closed curve on $\partial\nu(K)$, and let ρ_0 be the discrete faithful representation associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-033956 | MATHDB-372300

The universal finite-N N-5 scaling-law conjecture

摘录 / EXCERPT: Let N be the number of rotor directions and let K denote the number of distinct one-dimensional solution branches. The numerical exhaustion method and solver resolution are the...

conjecture | open | MathDB 页面 | 原文线索

V2-033957 | MATHDB-372303

Presentation conjecture for the non-compact 3-dimensional cobordism 2-category

摘录 / EXCERPT: The objects are closed 1-dimensional manifolds equipped with chosen bounding 2-manifolds, the 1-morphisms are 2-dimensional cobordisms equipped with bounding 3-manifolds, and the 2...

conjecture | open | MathDB 页面 | 原文线索

V2-033958 | MATHDB-372307

The Maslov-grading conjecture for strongly quasipositive knots

摘录 / EXCERPT: Let K be a strongly quasipositive knot with genus g, and let $\widehat{\mathit{HFK}}_m(K,g)$ denote the summand of knot Floer homology in Maslov grading m and top Alexander g...

conjecture | open | MathDB 页面 | 原文线索

V2-033959 | MATHDB-372308

The F_K-slope conjecture

摘录 / EXCERPT: Let $F_K(x,q)$ be the F_K invariant, with coefficient expansion $F_K(x,q)=x^{1/2}\sum_{n\geq 0}f_n(q)x^n$. For sufficiently large n, write the minimal q-degree as a quadratic...

conjecture | open | MathDB 页面 | 原文线索

V2-033960 | MATHDB-372326

The instanton-unoriented knot Floer homology correspondence conjecture

摘录 / EXCERPT: Let Y be a closed, oriented 3-manifold and $K\subset Y$ a knot. Let $HFK'_2(Y,K;\mathbb{C})$ be the homology of the mapping-cone complex ... and let $I^\sharp(Y,K;\mathbb{C})$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-033961 | MATHDB-372327

The N-filling formula for unoriented knot Floer homology

摘录 / EXCERPT: Let Y be a 3-manifold and $K\subset Y$ a knot, with meridian μ. Let N be the twisted I-bundle over the Klein bottle, and let Y_N^μ be the N-filling of...

conjecture | open | MathDB 页面 | 原文线索

V2-033962 | MATHDB-372333

Hyodo-Kato splitting conjecture for Drinfeld upper-half space

摘录 / EXCERPT: Let $\mathbb{H}$ be the Drinfeld upper-half space, let $H^{i}_{\operatorname{HK}}(\mathbb{H})$ denote its Hyodo-Kato cohomology, and let $V_{[1,n-1-i]}^{\infty}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-033963 | MATHDB-372336

Hochschild cohomology quotient conjecture for equivariant partially wrapped Fukaya categories

摘录 / EXCERPT: Hochschild cohomology quotient conjecture. There exists a $\mathbb{C}[s]$-module structure on $HH^*(\mathcal{W}(X))$ such that $HH^*(\mathcal{W}(\underline{X}))$ is isomorphic to t...

conjecture | open | MathDB 页面 | 原文线索

V2-033964 | MATHDB-372340

Chern-Hopf-Thurston conjecture on Euler characteristics of aspherical manifolds

摘录 / EXCERPT: Let M be a closed real $2n$-manifold. It is aspherical when its universal covering is contractible. Chern-Hopf-Thurston conjecture. If M is aspherical, then it satisfies ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-033965 | MATHDB-372411

Polynomial-time conjecture for Khovanov homology of fixed-strand closed braids

摘录 / EXCERPT: Let b be a braid with a fixed number of strands, let $\widehat{b}$ denote its closure, and let the number of crossings of b be the input size. Przytycki-Silvero conjecture. Com...

conjecture | open | MathDB 页面 | 原文线索

V2-033966 | MATHDB-372458

Cancellation conjecture for nonorientable 4-manifolds with cyclic fundamental group

摘录 / EXCERPT: Let M_{2p} be a closed topological nonorientable 4-manifold with cyclic fundamental group of order $2p$, where $p>1$ is odd. Theorem Cancellation Homeomorphism Classes asserts...

conjecture | open | MathDB 页面 | 原文线索

V2-033967 | MATHDB-372459

Asymptotic state-integral conjecture for geometrically nondegenerate representations

摘录 / EXCERPT: Let K be a hyperbolic knot, let D be an open diagram of K, and let ρ be a decorated representation of D that is geometrically nondegenerate, meaning that ρ is non...

conjecture | open | MathDB 页面 | 原文线索

V2-033968 | MATHDB-372513

The P=W compatibility conjecture for nonabelian Hodge theory

摘录 / EXCERPT: Let C be a smooth projective curve with underlying Riemann surface Σ_g. Let $\mathcal{H}^{\operatorname{Betti}}$ and $\mathcal{H}^{\operatorname{Dol}}$ be the graded BPS c...

conjecture | open | MathDB 页面 | 原文线索

V2-033969 | MATHDB-372526

Boyd's homological formulation of the $K(\pi,1)$ conjecture

摘录 / EXCERPT: Let A_Γ^+ be the Artin monoid associated with an Artin group A_Γ, and let $\mathbb Z A_\Gamma^+$ and $\mathbb Z A_\Gamma$ be their monoid and group rings. The inclusi...

conjecture | open | MathDB 页面 | 原文线索

V2-033970 | MATHDB-372527

Huang's relative Artin complex conjecture

摘录 / EXCERPT: Let A_Γ be an Artin group with defining graph Γ, and let $\Delta_{S,S'}$ denote the relative Artin complex spanned by vertices of types indexed by S'. Suppose...

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V2-033971 | MATHDB-372529

Qazaqzeh-Chbili-Lowrance conjecture on adequacy and Turaev genus

摘录 / EXCERPT: Let L be a non-split link. Write $c(L)$ for its crossing number, $g_T(L)$ for its Turaev genus, and $v_L(t)$ for its Jones polynomial. Qazaqzeh-Chbili-Lowrance conjecture. The li...

conjecture | open | MathDB 页面 | 原文线索

V2-033972 | MATHDB-372532

The topological exactness conjecture for hard telic problems

摘录 / EXCERPT: Topological exactness conjecture. If T is topologically exact and the system admits a hard telic problem, then there exists an almost-everywhere δ hard-on-average telic pr...

conjecture | open | MathDB 页面 | 原文线索

V2-033973 | MATHDB-372535

The mixing conjecture for transitive systems with intractable telic problems

摘录 / EXCERPT: Mixing conjecture. If (X,T) is transitive and admits an intractable telic problem, then the system is topologically mixing.

conjecture | open | MathDB 页面 | 原文线索

V2-033974 | MATHDB-372558

Empirical rigidity of urban percolation

摘录 / EXCERPT: Let a spatial process be constrained by an urban support, and let the percolation threshold ε^* and fusion curves characterize its spatial connectivity. Empirical rigid...

conjecture | open | MathDB 页面 | 原文线索

V2-033975 | MATHDB-372560

Infinitely many minimal hypersurfaces in a prescribed homology class

摘录 / EXCERPT: Let (M^{n+1},g) be a closed Riemannian manifold and let $[\Sigma_0]\in H_n(M^{n+1};\mathbb Z_2)$ be a homology class represented by a closed embedded hypersurface. Homology-class...

conjecture | open | MathDB 页面 | 原文线索

V2-033976 | MATHDB-372561

Equivariant Yau conjecture in a prescribed homology class

摘录 / EXCERPT: Equivariant Yau conjecture. The manifold M contains infinitely many closed embedded minimal G-hypersurfaces Σ in the given G-homology class. This is the equivariant ge...

conjecture | open | MathDB 页面 | 原文线索

V2-033977 | MATHDB-372601

Index bounds for homeomorphisms of collapsed surfaces

摘录 / EXCERPT: Index-bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-033978 | MATHDB-372619

Non-exact wrapped Fukaya category and deformed multiplicative preprojective algebra

摘录 / EXCERPT: Let X_{res} be the resolution of the A_n surface singularity, let D_r be the indicated divisor, and let s be a formal parameter with coefficient field...

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V2-033979 | MATHDB-372620

Stopped wrapped Fukaya category and a deformed preprojective algebra

摘录 / EXCERPT: Let X_{res} be the resolution of the A_n surface singularity, let D_r be the indicated divisor, and let σ be a stop. Let s be a formal parameter, with coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-033980 | MATHDB-372737

Hogancamp's Hamiltonian-action conjecture for link homology

摘录 / EXCERPT: Let L be an arbitrary link, let $\mathrm{HY}(L)$ denote its homology, and let $\mathcal{E}_k$ and $\mathcal{F}_k$ be the operators acting on it. Let $\mathcal{H}_2$ be the Lie al...

conjecture | open | MathDB 页面 | 原文线索

V2-033981 | MATHDB-372836

Uniqueness of the loop-graph coefficient in topological recursion relations

摘录 / EXCERPT: Let g be a non-negative integer and consider a degree g topological recursion relation of the shape … where $\Gamma^\mathrm{loop}_{0,g,[n]}$ is the stable graph with one genus-...

conjecture | open | MathDB 页面 | 原文线索

V2-033982 | MATHDB-372837

Bouquet-graph identity for topological recursion relations

摘录 / EXCERPT: Bouquet-graph conjecture. For any topological recursion relation

conjecture | open | MathDB 页面 | 原文线索

V2-033983 | MATHDB-372848

Conjecture on asymptotic convergence for immortal elastic and ideal flows

摘录 / EXCERPT: Asymptotic convergence conjecture. The curve $\gamma(\cdot,t)$ converges exponentially fast in the smooth topology to a multiply-covered lemniscate or circle.

conjecture | open | MathDB 页面 | 原文线索

V2-033984 | MATHDB-372891

Full-basin conjecture for a smooth circle covering with a repelling fixed point

摘录 / EXCERPT: Let $\mathcal{D}$ be the set of circle local homeomorphisms $f:\mathbb{S}^1\to\mathbb{S}^1$ that are topologically conjugate to the doubling map. For $f\in\mathcal{D}$, let p be...

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V2-033985 | MATHDB-372949

Cappell-Shaneson conjecture on characteristic classes

摘录 / EXCERPT: Cappell-Shaneson conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-033986 | MATHDB-372977

The codimension Arf invariant vanishing conjecture

摘录 / EXCERPT: For a closed oriented or non-orientable manifold product with an Arf invariant normal map, the resulting surgery obstruction modulo the ordinary Arf-Kervaire invariant is called a...

conjecture | open | MathDB 页面 | 原文线索

V2-033987 | MATHDB-373003

Matui's HK conjecture for ample groupoids

摘录 / EXCERPT: HK conjecture. For $* = 0,1$, there is an isomorphism

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V2-033988 | MATHDB-373014

The four-strand simple pinning hardness conjecture

摘录 / EXCERPT: Hardness-starts-at-four conjecture. For a fixed orientable surface Σ, the problem $\mathsf{SimplePin}(\Sigma,4)$ is $\textsf{NP}$-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-033989 | MATHDB-373015

The non-orientable surface pinning complexity conjecture

摘录 / EXCERPT: Non-orientable surface complexity conjecture. Both $\mathsf{Pin}(\Sigma,s)$ and $\mathsf{SimplePin}(\Sigma,s)$ are in $\textsf{P}$ when $s\leq 3$, and are $\textsf{NP}$-complete wh...

conjecture | open | MathDB 页面 | 原文线索

V2-033990 | MATHDB-373080

The Gorenstein-dimension grade conjecture

摘录 / EXCERPT: Let R be a commutative Noetherian local ring and let M be a finitely generated R-module. Write $\operatorname{Gdim}(M)$ for the Gorenstein dimension,...

conjecture | open | MathDB 页面 | 原文线索

V2-033991 | MATHDB-373082

Burguet's periodic-point growth conjecture for surface diffeomorphisms

摘录 / EXCERPT: Let f be a C^r diffeomorphism, with $r>1$, on a compact surface M. Write $h_{\rm top}(f)$ for its topological entropy and $\lambda^{+}(f)$ for the relevant positive Lyapunov-...

conjecture | open | MathDB 页面 | 原文线索

V2-033992 | MATHDB-373101

The predicted splitting of the hyperreal dual Steenrod algebra

摘录 / EXCERPT: Let G be the group acting in the hyperreal setting, let C_2 be its indicated subgroup, and let $E_{2+\sigma}$ and $BBU_{\mathbb R}$ denote the spaces appearing in the source. W...

conjecture | open | MathDB 页面 | 原文线索

V2-033993 | MATHDB-373119

The evaluation-map conjecture for the HOMFLYPT difference module

摘录 / EXCERPT: Let L be a link, let $\mathsf{H}_q(L)$ be its HOMFLYPT difference module, let $[L]$ be its distinguished link class, and let $\mathsf{ev}\colon \mathsf{H}(L)\to \mathcal{F}_r$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-033994 | MATHDB-373208

Kitaev's conjecture on the spectrum of invertible phases

摘录 / EXCERPT: For each spatial dimension d, let $\mathcal I(\mathbb Z^d)$ be the abelian group of invertible gapped phases, and let $\mathrm{IP}(\mathbb Z^d)$ denote a proposed space-level con...

conjecture | open | MathDB 页面 | 原文线索

V2-033995 | MATHDB-373209

The QCA conjecture on the spectrum of quantum cellular automata

摘录 / EXCERPT: For each spatial dimension d, let $\mathcal Q^*(\mathbb Z^d)$ be the group of quantum cellular automata and let $\mathcal C^*(\mathbb Z^d)$ be its normal subgroup of quantum circ...

conjecture | open | MathDB 页面 | 原文线索

V2-033996 | MATHDB-373220

Strong Cabling Conjecture

摘录 / EXCERPT: Strong Cabling Conjecture. For any even integer $b\ge 6$, the triple $(0,b,1)$ is not realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-033997 | MATHDB-373226

Eventual constancy conjecture for Serre-depth functions of powers

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a Noetherian local ring or a standard graded K-algebra. Let $I\subset R$ be an ideal, homogeneous when R is a K-algebra. The Serre-depth stabilizati...

conjecture | open | MathDB 页面 | 原文线索

V2-033998 | MATHDB-373239

The idempotent classification conjecture for integral Dehn quandles

摘录 / EXCERPT: Let $g\geq 1$, let $\mathcal{D}_g$ be the Dehn quandle of a closed orientable surface of genus g, and let $\mathbb{Z}[\mathcal{D}_g]$ be its integral quandle algebra. An element...

conjecture | open | MathDB 页面 | 原文线索

V2-033999 | MATHDB-373263

Multiplicative decomposition and Chern-class conjecture for Lagrangian fibrations

摘录 / EXCERPT: Lagrangian-fibration conjecture. The morphism f admits a multiplicative decomposition theorem

conjecture | open | MathDB 页面 | 原文线索

V2-034000 | MATHDB-373301

The finite-predecessor conjecture for ribbon concordance

摘录 / EXCERPT: Let K be a knot in S^3, and write $J\leq K$ when the knot J is ribbon concordant to K. Finite-predecessor conjecture. For each knot $K\subset S^3$, there are only finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-034001 | MATHDB-373302

Alpert-Kahle-MacPherson homological stability conjecture for sliding fifteen-puzzle configurations

摘录 / EXCERPT: Alpert-Kahle-MacPherson's conjecture. The inclusion of $SF_n(R_{w,h})$ into $F_n(\mathbb{R}^2)$ induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034002 | MATHDB-373304

Shen et al.'s homotopy conjecture for total cut complexes of powers of cycles

摘录 / EXCERPT: Let C_n^r denote the rth power of the cycle on n vertices, and let $\Delta_d^t(C_n^r)$ denote its total d-cut complex. For $d\geq3$ and $n\geq(2d-1)r+1$, Shen et al.'s conj...

conjecture | open | MathDB 页面 | 原文线索

V2-034003 | MATHDB-373324

The uniform geometric K-homology conjecture

摘录 / EXCERPT: Uniform geometric K-homology conjecture. The natural map

conjecture | open | MathDB 页面 | 原文线索

V2-034004 | MATHDB-373348

The twisted number field counting conjecture

摘录 / EXCERPT: Let k be a number field, let G be a transitive permutation group of degree n, and let $N\normal G$. Write $q:G\to G/N$ for the quotient map and ... for the induced pushforward....

conjecture | open | MathDB 页面 | 原文线索

V2-034005 | MATHDB-373398

The Links-Quivers Correspondence for colored HOMFLY-PT homology

摘录 / EXCERPT: Links-Quivers Correspondence for homology. The colored HOMFLY-PT homologies of L may be represented by chain complexes such that

conjecture | open | MathDB 页面 | 原文线索

V2-034006 | MATHDB-373404

The spectral-gap conjecture for hyperbolic rational homology spheres

摘录 / EXCERPT: Let Y be a hyperbolic rational homology sphere, and let λ_1^* denote the first positive eigenvalue of the Laplacian on coexact 1-forms. Spectral-gap conjecture. There e...

conjecture | open | MathDB 页面 | 原文线索

V2-034007 | MATHDB-373433

Corrected HZ factorisability conjecture for knots of braid index at least four

摘录 / EXCERPT: Corrected HZ factorisability conjecture. If the HOMFLY-PT/Kauffman relation holds, then the HZ transform is factorisable:

conjecture | open | MathDB 页面 | 原文线索

V2-034008 | MATHDB-373465

Latschev-Wendl's existence conjecture for higher-dimensional algebraic torsion

摘录 / EXCERPT: Let $k \ge 1$ and $n \ge 2$ be integers, and consider closed $(2n+1)$-dimensional contact manifolds, as well as algebraic torsion and untwisted algebraic torsion in that dimension....

conjecture | open | MathDB 页面 | 原文线索

V2-034009 | MATHDB-373468

Modular collapse conjecture for locally compact Moufang-type quasigroups

摘录 / EXCERPT: Modular collapse conjecture. The cocycle collapses:

conjecture | open | MathDB 页面 | 原文线索

V2-034010 | MATHDB-373470

The monic Alexander polynomial conjecture for twisted torus knots

摘录 / EXCERPT: Twisted torus knot fiberedness conjecture. A twisted torus knot is fibered if and only if its Alexander polynomial is monic.

conjecture | open | MathDB 页面 | 原文线索

V2-034011 | MATHDB-373509

The generalized cosmetic crossing conjecture for knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, and let c be a crossing of K. A crossing is nugatory if its crossing circle bounds a disk in the complement of K; a generalized crossing change...

conjecture | open | MathDB 页面 | 原文线索

V2-034012 | MATHDB-373522

Conjecture on the sequential topological complexity of quaternionic space forms

摘录 / EXCERPT: Let Q_{8m} denote the generalized quaternion group of order $8m$ acting freely on S^3, and let r-th sequential topological complexity be denoted by TC_r. Sequentia...

conjecture | open | MathDB 页面 | 原文线索

V2-034013 | MATHDB-373525

Schoen-Marques-Neves index-topology conjecture for minimal hypersurfaces

摘录 / EXCERPT: Let (N^{n+1},g) be a closed Riemannian manifold with positive Ricci curvature. A closed embedded minimal hypersurface is a smooth hypersurface $M^n\subset N^{n+1}$ with zero mean...

conjecture | open | MathDB 页面 | 原文线索

V2-034014 | MATHDB-373568

Expected properties of Yang-Mills plus Chern-Simons theory

摘录 / EXCERPT: Expected properties of Yang-Mills plus Chern-Simons theory. The following assertions are expected: (1) the Lagrangian determines a family of Wick-rotated field theories; (2) the si...

conjecture | open | MathDB 页面 | 原文线索

V2-034015 | MATHDB-373612

Classification of conservation-forced primitive substitution tilings

摘录 / EXCERPT: Let $(\mathcal{T},\sigma,M)$ be a primitive substitution tiling system of $\mathbb{R}^d$ with finite local complexity. Call it conservation-forced when it satisfies the conservatio...

conjecture | open | MathDB 页面 | 原文线索

V2-034016 | MATHDB-373652

Lawson homology conjecture for symmetric products of projective spaces

摘录 / EXCERPT: Let $\mathrm{SP}^d(\mathbb{P}^n)$ be the d-th symmetric product of complex projective space, and let cl denote the cycle class map from Lawson homology to singular homology wit...

conjecture | open | MathDB 页面 | 原文线索

V2-034017 | MATHDB-373670

The continuum gluing conjecture for spin foam nets

摘录 / EXCERPT: Gluing conjecture. The continuum amplitudes satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-034018 | MATHDB-373719

Huneke's finiteness conjecture for Bass numbers of local cohomology modules

摘录 / EXCERPT: Huneke's conjecture. The Bass numbers $\mu^i(\mathfrak{p},H^j_{\mathfrak{a}}(R))$ are finite for all $i,j\in\mathbb{N}_0$ and every $\mathfrak{p}\in\operatorname{Spec}(R)$. This co...

conjecture | open | MathDB 页面 | 原文线索

V2-034019 | MATHDB-373729

Gehrmann's conjecture on automorphic $\mathcal{L}$-invariant spaces

摘录 / EXCERPT: Let π be the automorphic representation and let $c\in\{1,\dots,n-1\}$, $0\le r\le\ell$, and $\varepsilon\in\mathcal{E}_\pi$. Let $\mathcal{L}_c^{r,\varepsilon}(\pi)$ be the sub...

conjecture | open | MathDB 页面 | 原文线索

V2-034020 | MATHDB-373751

The generalized Geroch conjecture for connected sums with tori

摘录 / EXCERPT: Let M be a manifold of dimension m, and let T^m denote the m-dimensional torus. Generalized Geroch conjecture. There is no complete metric of positive scalar curvature on .....

conjecture | open | MathDB 页面 | 原文线索

V2-034021 | MATHDB-373786

The fibered strongly quasi-positive knot conjecture on friends

摘录 / EXCERPT: A fibered and strongly quasi-positive knot (or a fibered and strongly quasi-negative knot) is a knot having the stated properties; a friend is a knot associated to it by the trace-...

conjecture | open | MathDB 页面 | 原文线索

V2-034022 | MATHDB-373787

The Piccirillo friend conjecture for twisted Whitehead doubles

摘录 / EXCERPT: Let K be a knot with an unknotting crossing x, and let K_x denote its Piccirillo friend, constructed from K and x. A twisted Whitehead double is a knot obtained by applyi...

conjecture | open | MathDB 页面 | 原文线索

V2-034023 | MATHDB-373788

The Piccirillo friend conjecture for unknotting-number-one knots

摘录 / EXCERPT: Let K be a knot with unknotting number one. For an unknotting crossing x of K, let K_x and K'_x be the associated Piccirillo friends. Piccirillo friend conjecture. The fo...

conjecture | open | MathDB 页面 | 原文线索

V2-034024 | MATHDB-373798

Acyclicity conjecture for Drinfeld-Kohno 2-algebras

摘录 / EXCERPT: For $n\in\mathbb{N}$, let the n^th Drinfeld-Kohno 2-algebra be the associative 2-algebra generated by ... with boundary map determined by ... and subject to the relations (i...

conjecture | open | MathDB 页面 | 原文线索

V2-034025 | MATHDB-373872

Balmer's Nerves of Steel Conjecture

摘录 / EXCERPT: Balmer's Nerves of Steel Conjecture. The map ϕ is bijective. This conjecture seeks to reconcile the homological and triangular spectra of a rigid tensor-triangulated category....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-034026 | MATHDB-373883

The single valuation conjecture for integrally closed ideals in two variables

摘录 / EXCERPT: Let $k[x_1,x_2]$ be the polynomial ring in two variables, and let $\mathcal{I}\triangleleft k[x_1,x_2]$ be a finite-codimensional, integrally closed ideal. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-034027 | MATHDB-373884

The single valuation conjecture for homological dimension zero

摘录 / EXCERPT: Let $\mathcal{O}_2=k[x_1,x_2]$ and let $\mathcal{I}\triangleleft\mathcal{O}_2$ be a finite-codimensional, integrally closed ideal. Let $\mathcal{O}$ denote the ambient module in th...

conjecture | open | MathDB 页面 | 原文线索

V2-034028 | MATHDB-373918

Bergelson's amenability conjecture via recurrence

摘录 / EXCERPT: Let G be a group. A set $R\subset G$ is a set of measurable recurrence if, for every measure-preserving G-system $(X,\mathcal{X},\mu,(T_g))$ and every $A\in\mathcal{X}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-034029 | MATHDB-373921

The nullhomotopy-dependence conjecture for Floer-theoretic nearby Lagrangian torsion

摘录 / EXCERPT: Nullhomotopy-dependence conjecture. The classes $\tau_{2k}(L,\gamma_t^i)$, for $i=1,2$, differ by $p_k(E)$.

conjecture | open | MathDB 页面 | 原文线索

V2-034030 | MATHDB-373922

The Floer-theoretic identification conjecture for nearby Lagrangian torsion

摘录 / EXCERPT: Floer-theoretic identification conjecture. The Floer-theoretic coset is the same as the nearby Lagrangian torsion:

conjecture | open | MathDB 页面 | 原文线索

V2-034031 | MATHDB-373929

Selmer criterion for plectic Heegner classes

摘录 / EXCERPT: Selmer criterion for plectic Heegner classes. If χ is primitive, then

conjecture | open | MathDB 页面 | 原文线索

V2-034032 | MATHDB-373961

Andersen-Masbaum-Ueno conjecture for mapping class groups

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a compact oriented surface, and let ... be its mapping class group, with homeomorphisms and isotopies fixed on the boundary. For an odd integer $r\geq 3$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-034033 | MATHDB-373974

Nekrasov's wheel-condition generation conjecture for tripled quivers

摘录 / EXCERPT: Nekrasov's wheel-condition generation conjecture. The localized wheel-condition algebra equals the localized spherical subalgebra:

conjecture | open | MathDB 页面 | 原文线索

V2-034034 | MATHDB-373980

The disklike characterization conjecture for maximal compatible transfer systems

摘录 / EXCERPT: Disklike characterization conjecture. The maximal compatible transfer system is

conjecture | open | MathDB 页面 | 原文线索

V2-034035 | MATHDB-374017

Futer-Purcell-Schleimer's purely cosmetic surgery conjecture for cusped hyperbolic manifolds

摘录 / EXCERPT: Let M be a cusped hyperbolic manifold with at least one cusp, and let $\mathbf{s}$ and $\mathbf{s'}$ be sets of slopes on the cusps. Futer-Purcell-Schleimer's conjecture. If the...

conjecture | open | MathDB 页面 | 原文线索

V2-034036 | MATHDB-374018

Chirally cosmetic surgery conjecture for chiral knots

摘录 / EXCERPT: A knot K is chiral if it is not equivalent to its mirror image, and $T(2,n)$ denotes the $(2,n)$ torus knot. Chirally cosmetic surgery conjecture. If K is a chiral knot that is...

conjecture | open | MathDB 页面 | 原文线索

V2-034037 | MATHDB-374037

Conjectures on magnitude invariants of real hyperplane arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a real hyperplane arrangement of rank r, let $n=\#\mathcal{A}$, and write its reduced magnitude as ... Let $\mathcal{T}(\mathcal{A})$ denote the tope graph, le...

conjecture | open | MathDB 页面 | 原文线索

V2-034038 | MATHDB-374049

Kähler package conjecture for corrected perverse cohomology

摘录 / EXCERPT: Let $\pi:\mathcal X\to\Delta$ be a projective conifold degeneration, and let $\mathcal P$ be the corrected perverse object constructed from nearby and vanishing cycles. A mixed-Hod...

conjecture | open | MathDB 页面 | 原文线索

V2-034039 | MATHDB-374054

The L-space conjecture for closed irreducible 3-manifolds

摘录 / EXCERPT: L-space conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-034040 | MATHDB-374064

Serre's Conjecture I for smooth connected linear algebraic groups

摘录 / EXCERPT: Serre's Conjecture I. Does

conjecture | open | MathDB 页面 | 原文线索

V2-034041 | MATHDB-374068

General $B(m,n)$ triviality conjecture for inhomogeneous deformations

摘录 / EXCERPT: Let $B(m,n)=\mathfrak{osp}(2m+1|2n)$ with $m+n\geq 2$, and let gb denote the collection of deformation parameters in the γ-deformation. General $B(m,n)$ conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-034042 | MATHDB-374079

Non-vanishing conjecture for Pontryagin classes of orientable flat manifolds

摘录 / EXCERPT: Pontryagin-class non-vanishing conjecture. For the orientable flat manifolds $\widehat{\mathsf{LS}}_n$, the mod 2 reduction of $p_i(\widehat{\mathsf{LS}}_n)$ is non-zero whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-034043 | MATHDB-374092

Hersh-Reiner sharp stability conjecture for rank-selected homology of partition lattices

摘录 / EXCERPT: Let Π_n be the poset of set partitions of $\{1,2,\dots,n\}$ ordered by refinement, and let $S\subseteq\{1,2,\dots,n-2\}$. Write $\beta_S(\Pi_n)$ for the rank-selected homology...

conjecture | open | MathDB 页面 | 原文线索

V2-034044 | MATHDB-374111

Fixed-level bridge finiteness conjecture for links in the 3-sphere

摘录 / EXCERPT: For an integer $n\ge 1$, an n-bridge position of a link in S^3 is a bridge presentation with n bridges, considered up to bridge isotopy. Fixed-level bridge finiteness conject...

conjecture | open | MathDB 页面 | 原文线索

V2-034045 | MATHDB-374135

Conjecture on free and torsion decompositions of Kauffman bracket skein modules

摘录 / EXCERPT: Let M be a closed, oriented 3-manifold, and let $\mathscr S(M)$ denote its Kauffman bracket skein module over $\mathbb Z[A^{\pm1}]$. A module decomposition into free modules an...

conjecture | open | MathDB 页面 | 原文线索

V2-034046 | MATHDB-374159

The Reidemeister-braid-perturbation conjecture for 4-plane diagrams

摘录 / EXCERPT: A 4-plane diagram is a diagram describing a 3-manifold embedded in S^5, and interior Reidemeister moves, mutual braid moves, and 3-manifold perturbations are the diagrammat...

conjecture | open | MathDB 页面 | 原文线索

V2-034047 | MATHDB-374178

Equivariant homology-class conjecture for minimal hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be as in the equivariant setting, with a compact Lie group G acting by isometries and satisfying ... A G-homology class is represented by a fixed G-hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-034048 | MATHDB-374216

Finite generation conjecture for diffeomorphism groups of closed manifolds

摘录 / EXCERPT: Let X be a smooth closed manifold. Write $\mathrm{Diff}(X)$ for its diffeomorphism group, $B\mathrm{Diff}(X)$ for its classifying space, and π_k and $H_k(-;\mathbb{Z})$ for h...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-034049 | MATHDB-374219

Clausen's conjecture on analytic syntomic cohomology of the p-adic complex numbers

摘录 / EXCERPT: For an integer $n\geq 2$, let $R\Gamma(\mathbb{C}_p^\mathrm{Syn},\mathcal{O}\{n\})$ denote analytic syntomic cohomology of $\mathbb{C}_p$ with coefficients in the n-th Tate-twist...

conjecture | open | MathDB 页面 | 原文线索

V2-034050 | MATHDB-374226

Burnside conjecture for exceptional surface homeomorphism groups

摘录 / EXCERPT: Let Σ be one of the four exceptional closed surfaces ... A subgroup is periodic if every one of its elements has finite order. Burnside conjecture for exceptional surfaces. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-034051 | MATHDB-374227

Burnside conjecture for homeomorphism groups of closed manifolds

摘录 / EXCERPT: Let M be a finite-dimensional closed connected manifold, meaning a compact connected manifold without boundary. A subgroup is periodic if every one of its elements has finite ord...

conjecture | open | MathDB 页面 | 原文线索

V2-034052 | MATHDB-374253

Realization conjecture for singularity categories of finite-dimensional graded algebras

摘录 / EXCERPT: Realization conjecture. There exist an exact category $\mathcal{B}$ of finite homological dimension and a group H acting on $\mathsf{D}^{b}_{\mathrm{dg}}(\mathcal{B})$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-034053 | MATHDB-374301

The pure discrete Morse matching complex simplex-boundary homotopy conjecture

摘录 / EXCERPT: Let $cmathcal{M}_P(K)$ denote the pure complex of discrete Morse matchings of a simplicial complex K. Let $cmathrm{cDelta}^n$ be the n-simplex and...

conjecture | open | MathDB 页面 | 原文线索

V2-034054 | MATHDB-374304

Domokos-Horváth-Goriely-Regős conjecture on completely softening polyhedral tilings

摘录 / EXCERPT: Domokos-Horváth-Goriely-Regős conjecture. Every polyhedral tiling can be completely softened.

conjecture | open | MathDB 页面 | 原文线索

V2-034055 | MATHDB-374342

The linear variance conjecture for signatures of 2-bridge knots

摘录 / EXCERPT: Let $T(c)$ be the set of objects with crossing number c considered in the paper, and let the variance of their signatures be computed with respect to the uniform distribution on...

conjecture | open | MathDB 页面 | 原文线索

V2-034056 | MATHDB-374346

Coverage conjecture for real knot Floer homology of a doubled knot

摘录 / EXCERPT: Coverage conjecture. The bigraded group $\widehat{\operatorname{HFK}}(K)$ can be covered from

conjecture | open | MathDB 页面 | 原文线索

V2-034057 | MATHDB-374347

Auxiliary-data invariance conjecture for strongly invertible knots

摘录 / EXCERPT: Auxiliary-data invariance conjecture. The auxiliary data $\fra$ does not affect the real knot Floer groups.

conjecture | open | MathDB 页面 | 原文线索

V2-034058 | MATHDB-374348

Nontrivial real knot Floer homology conjecture for strongly invertible knots

摘录 / EXCERPT: Nontriviality conjecture. If K admits a strong inversion, then there is a strong inversion on K with nontrivial real knot Floer group.

conjecture | open | MathDB 页面 | 原文线索

V2-034059 | MATHDB-374349

Real Floer realization conjecture for connected sums and doubly periodic knots

摘录 / EXCERPT: Real Floer realization conjecture. There is an isomorphism of bigraded groups

conjecture | open | MathDB 页面 | 原文线索

V2-034060 | MATHDB-374365

Menasco-Reid conjecture on totally geodesic surfaces in hyperbolic knot complements

摘录 / EXCERPT: Let $S^{3}\setminus K$ be the complement of a hyperbolic knot K in S^{3}. A closed embedded totally geodesic surface is a closed totally geodesic surface embedded in this hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-034061 | MATHDB-374389

Quantum deformation and Langlands duality for relative Lie algebra cohomology

摘录 / EXCERPT: Quantum deformation and Langlands duality conjecture. There exists a square-zero deformation

conjecture | open | MathDB 页面 | 原文线索

V2-034062 | MATHDB-374412

Local weak contractibility conjecture for pseudo-isotopy spaces

摘录 / EXCERPT: Let $f\taut$ be a map, where $\taut$ denotes the boundary of the space of embeddings, and let $\tP(f)$ denote the corresponding pseudo-isotopy space. Local weak contractibility con...

conjecture | open | MathDB 页面 | 原文线索

V2-034063 | MATHDB-374421

The universal categorical spectrum conjecture for topological stabilizer codes

摘录 / EXCERPT: Let $\mathbf{TS}^*$ be a proposed categorical spectrum of topological stabilizer codes, with $\mathbf{TS}^n$ its degree-n component, and let $\mathbf Q^*(\mathrm{pt})$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-034064 | MATHDB-374466

The sharp writhe conjecture for smoothly hard max-tb Legendrian unknots

摘录 / EXCERPT: Let $\mathrm{minwr}_{\mathbb{R}^2}$ and minwr_{S^2} denote the minimum writhe of a max-tb Legendrian unknot front projection that is smoothly hard on the plane and spher...

conjecture | open | MathDB 页面 | 原文线索

V2-034065 | MATHDB-374475

Finite Quillen-Lichtenbaum dimension for smooth complex-linear stable infinity-categories

摘录 / EXCERPT: Let $\mathcal{A}$ be a smooth $\mathbb{C}$-linear stable ∞-category. The finite Quillen-Lichtenbaum dimension conjecture. Then ... Here $\dim_{\mathrm{QL}}\mathcal{A}$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-034066 | MATHDB-374480

Planar graph moduli and unlabeled configuration space conjecture

摘录 / EXCERPT: Let $\mathcal N_n(\mathbb R^2)$ be the moduli space of planar graphs with n bounded faces, and let $\operatorname{uConf}_n(\mathbb R^2)$ be the configuration space of n unlabel...

conjecture | open | MathDB 页面 | 原文线索

V2-034067 | MATHDB-374481

Embedded and unrestricted Legendrian stop moduli conjecture

摘录 / EXCERPT: Let Y be the ambient space and let Λ_n be the specified Legendrian stop. Write $\mathcal L^{\mathrm{emb}}_{\Lambda_n}(Y)$ for the moduli space of stops with embedded Lagr...

conjecture | open | MathDB 页面 | 原文线索

V2-034068 | MATHDB-374490

Compression lower bound from supertrunk for knot types

摘录 / EXCERPT: Compression lower bound from supertrunk. There is a universal constant $c>0$ such that for every knot type K,

conjecture | open | MathDB 页面 | 原文线索

V2-034069 | MATHDB-374509

The slice decomposition conjecture for the quaternionic norm of real bordism

摘录 / EXCERPT: Let Q_8 be the quaternion group, let C_2 be its indicated subgroup, and let $N_{C_2}^{Q_8} MU\mathbb{R}$ denote the quaternionic norm of real bordism. For a subgroup L of...

conjecture | open | MathDB 页面 | 原文线索

V2-034070 | MATHDB-374535

Algebraic characterization of the smallest closed invariant equivalence relation containing the higher order regionally proximal relation

摘录 / EXCERPT: The conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034071 | MATHDB-374547

Finite witness principle for knot theory

摘录 / EXCERPT: Finite witness principle. For each $\mathcal I\in\mathfrak I$, each knot type K, and each generic projection direction u, there exists a finite witness scale

conjecture | open | MathDB 页面 | 原文线索

V2-034072 | MATHDB-374549

Dospinescu-Paškūnas-Schraen's Hecke-eigenspace infinitesimal character conjecture

摘录 / EXCERPT: Let $x:\mathT^S(K^p)[1/p]\to\overline{\mathbf Q}_p$ be a continuous ring homomorphism, let $\mathfrak m_x=\ker(x)$, and suppose that the global Langlands program supplies a corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-034073 | MATHDB-374560

Local compactness conjecture for convex functionals on surfaces

摘录 / EXCERPT: Let S be a surface, and let $\operatorname{Convex}(S)$ denote the space of convex curve functionals on S, equipped with the product topology inherited from its values on the cu...

conjecture | open | MathDB 页面 | 原文线索

V2-034074 | MATHDB-374592

Loday's proposal for contravariant algebraic K-theory

摘录 / EXCERPT: Loday's proposal. These data should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-034075 | MATHDB-374602

Non-degeneracy of swept-area distance for ropelength-filtered knot spaces

摘录 / EXCERPT: Non-degeneracy conjecture. The swept-area pseudometric is non-degenerate on each admissible component of $Y_{\Lambda}(K)$ modulo $\operatorname{Isom}^+(\mathbb{R}^3)$: zero swept-a...

conjecture | open | MathDB 页面 | 原文线索

V2-034076 | MATHDB-374603

Hölder control of diameter by swept-area distance

摘录 / EXCERPT: Diameter variation conjecture. For each ropelength level Λ, there should exist constants $C(\Lambda)>0$ and $\alpha>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-034077 | MATHDB-374608

Wahl's conjecture on rational homology disk singularities

摘录 / EXCERPT: A rational homology disk singularity is a normal surface singularity whose link bounds a rational homology disk; a Wahl singularity is a cyclic quotient surface singularity of the...

conjecture | open | MathDB 页面 | 原文线索

V2-034078 | MATHDB-374634

Moore's conjecture on rational ellipticity and homotopy exponents

摘录 / EXCERPT: Let X be a finite, simply-connected CW-complex. Call X rationally elliptic if it has finitely many rational homotopy groups, and say that X has a finite homotopy exponent a...

conjecture | open | MathDB 页面 | 原文线索

V2-034079 | MATHDB-374682

Additivity of the log Chern character for distinguished triangles

摘录 / EXCERPT: Let (X,D) be a log smooth pair, and let $E_1\to E_2\to E_3\to E_1[1]$ be a distinguished triangle in $D(X\times^{\log}X)$ whose terms are strong log Fourier-Mukai kernels. Additi...

conjecture | open | MathDB 页面 | 原文线索

V2-034080 | MATHDB-374683

Compatibility of logarithmic and ordinary Hochschild functoriality

摘录 / EXCERPT: Let (X,D) and (X',D') be log smooth pairs, and let E be a strong Fourier-Mukai kernel on $X\times^{\log}X'$. Let $\pi:X\times^{\log}X'\to X\times X'$ be the projection and re...

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V2-034081 | MATHDB-374724

Topological equivalence conjecture for tangles

摘录 / EXCERPT: Let t_1 and t_2 be tame tangles, and let T_1 and T_2 be arbitrarily chosen mosaic representatives of t_1 and t_2, respectively. Topological equivalence conjecture for t...

conjecture | open | MathDB 页面 | 原文线索

V2-034082 | MATHDB-374725

Topological equivalence conjecture for 2,4 tangles

摘录 / EXCERPT: Let l_1 and l_2 be two $\rv{2,4}$ tangles, and let L_1 and L_2 be two arbitrarily chosen $\rv{2,4}$ tangle n-mosaic representatives of l_1 and l_2, respectively, with...

conjecture | open | MathDB 页面 | 原文线索

V2-034083 | MATHDB-374796

Ext-group decomposition conjecture for quantum Frobenius twists

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $q\in\mathbb{K}^{\times}$ be a non-zero scalar, and let F and G be strict polynomial functors. Write $F^{(1)_q}$ and $G^{(1)_q}$ for their quan...

conjecture | open | MathDB 页面 | 原文线索

V2-034084 | MATHDB-374813

The taut 20 conjecture for moduli spaces of stable curves

摘录 / EXCERPT: Taut 20 conjecture. The cohomology groups

conjecture | open | MathDB 页面 | 原文线索

V2-034085 | MATHDB-374814

The taut 10 conjecture for moduli spaces of stable curves

摘录 / EXCERPT: Taut 10 conjecture. All cohomology groups

conjecture | open | MathDB 页面 | 原文线索

V2-034086 | MATHDB-374827

The oriented connected-endomorphism conjecture for the 2-sphere

摘录 / EXCERPT: Let $\mathbb{L}_{SO}^\otimes$ be the proposed oriented ∞-operad, and let $\Omega^2\operatorname{Bord}_3^{SO}$ be the symmetric monoidal ∞-category of oriented 3-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-034087 | MATHDB-374839

The flow-loop count equals the BPS quantum series for fibered links

摘录 / EXCERPT: Flow-loop/BPS-series conjecture. For every fibered link K,

conjecture | open | MathDB 页面 | 原文线索

V2-034088 | MATHDB-374840

Finiteness conjecture for Melvin-Morton-Rozansky expansions of fibered knots

摘录 / EXCERPT: MMR resummation finiteness conjecture. For every fibered knot K, the coefficients of this expansion, which are a priori elements of $\mathbb{Q}[[\hbar]]$, can be resummed into a...

conjecture | open | MathDB 页面 | 原文线索

V2-034089 | MATHDB-374841

Quantum-group compatibility of the abelianization map

摘录 / EXCERPT: Quantum-group abelianization conjecture. The abelianization map Ab is equivalent to the abelianization map constructed using quantum groups.

conjecture | open | MathDB 页面 | 原文线索

V2-034090 | MATHDB-374845

Conjectural weights of incompressible surfaces in odd pretzel knot complements

摘录 / EXCERPT: Pretzel-surface weight conjecture. The nine incompressible surfaces have weights, in that order,

conjecture | open | MathDB 页面 | 原文线索

V2-034091 | MATHDB-374898

The Simple Loop Conjecture for maps from surfaces to 3-manifolds

摘录 / EXCERPT: Let S be an orientable surface, let M be a 3-manifold, and let $f\colon S\to M$ be a map. The induced homomorphism is ... Simple Loop Conjecture. If f_* is not injective, then...

conjecture | open | MathDB 页面 | 原文线索

V2-034092 | MATHDB-374941

Positive lattice amphichirality barrier for cubic lattice knots

摘录 / EXCERPT: Let K be an amphichiral knot type and let ω be a minimal simple cubic lattice representative of K. Consider the minimal BFACF graph and the reflected mirror seed of...

conjecture | open | MathDB 页面 | 原文线索

V2-034093 | MATHDB-374952

Conjecture on intermediate differentials in the genuine C_2-equivariant Adams spectral sequence

摘录 / EXCERPT: Intermediate-differential conjecture. For $j\gg k>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-034094 | MATHDB-375047

Configuration-space integral realization conjecture for long embeddings

摘录 / EXCERPT: Configuration-space integral realization conjecture. There exists a map

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V2-034095 | MATHDB-375063

Equality of ribbon numbers for symmetric ribbon knots of ribbon number two

摘录 / EXCERPT: Ribbon-number-two conjecture. If $r(K)=2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-034096 | MATHDB-375076

The tri-plane crossing number of the spun trefoil

摘录 / EXCERPT: Let the tri-plane crossing number of a 2-knot be the minimum total number of crossings among its tri-plane diagrams. The spun trefoil is the 2-knot obtained by spinning the tre...

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V2-034097 | MATHDB-375077

The low-crossing tri-plane classification conjecture for 2-knots

摘录 / EXCERPT: Let a tri-plane diagram be a diagram describing a bridge trisection of an embedded S^2 in S^4, and count its total crossings across the three constituent plane diagrams. Low-cr...

conjecture | open | MathDB 页面 | 原文线索

V2-034098 | MATHDB-375093

Artemev's singular-part conjecture for dual topological-recursion amplitudes

摘录 / EXCERPT: Artemev's conjecture. Up to a suitable normalization, $A^{(g),\vee}_n(k_1,\dots,k_n)$ coincides with $A^{(g),\mathsf{sing}}_n(k_1,\dots,k_n)$ for $g\geq 0$, $n\geq 1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034099 | MATHDB-375135

Divided-difference conjecture for cohomology classes of balanced two-row Springer fibers

摘录 / EXCERPT: Let $\mathcal{B}_{\boldsymbol{\rho}}$ be the component of the two-row Springer fiber indexed by a standard tableau $\boldsymbol{\rho}$ of shape (ρ,ρ), with cup diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-034100 | MATHDB-375136

Hellus-Lyubeznik-Yildirim's conjecture on local cohomology modules

摘录 / EXCERPT: Let $(R,\mathfrak{m},\mathbb{K})$ be a commutative Noetherian local ring with residue field $\mathbb{K}=R/\mathfrak{m}$, let $I\subset R$ be an ideal, and let $H_I^j(R)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-034101 | MATHDB-375195

Finite-topological-type conjecture for gravitational instantons

摘录 / EXCERPT: Let (M,g) be a gravitational instanton. Finite-topological-type conjecture. The manifold M has finite topological type, meaning that it can be compactified into a smooth manifo...

conjecture | open | MathDB 页面 | 原文线索

V2-034102 | MATHDB-375198

Topological R4 rigidity conjecture for gravitational instantons

摘录 / EXCERPT: Let (M,g) be a gravitational instanton whose underlying manifold M is topologically $\mathbb{R}^4$. Topological $\mathbb{R}^4$ rigidity conjecture. The metric g is either the...

conjecture | open | MathDB 页面 | 原文线索

V2-034103 | MATHDB-375199

Finite-topological-type characterization of gravitational instantons

摘录 / EXCERPT: Let (M,g) be a complete oriented Ricci-flat 4-manifold. Finite-topological-type characterization. The manifold (M,g) is a gravitational instanton if and only if M has finite...

conjecture | open | MathDB 页面 | 原文线索

V2-034104 | MATHDB-375216

Gunningham-Safronov's generic skein-DT sheaf conjecture

摘录 / EXCERPT: Skein-DT sheaf conjecture. There is an isomorphism

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V2-034105 | MATHDB-375218

Gunningham-Safronov's derived skein-DT cohomology conjecture

摘录 / EXCERPT: Derived skein-DT conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034106 | MATHDB-375227

Fox's trapezoidal conjecture for alternating knots

摘录 / EXCERPT: Let K be an alternating knot, and write its Alexander polynomial as $\Delta_K(t)=\sum_i a_i t^i$. A sequence is trapezoidal if the absolute values of its terms increase, possibly...

conjecture | open | MathDB 页面 | 原文线索

V2-034107 | MATHDB-375249

Finiteness conjecture for nontrivial Floer cohomology in a Weinstein family

摘录 / EXCERPT: Finiteness conjecture for Weinstein deformations. Suppose $QH_{\frak b}(X;\Lambda)$ is semi-simple. Then there exist only finitely many u for which there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-034108 | MATHDB-375258

Equivariant Milnor map conjecture

摘录 / EXCERPT: Let T^k be the k-torus and let $\mathbb{Z}_2^k$ be its real subgroup. Write Ω^{U,T^k}_{2*} and $\Omega^{U,\mathbb{Z}_2^k}_{2*}$ for the unitary equivariant bordism group...

conjecture | open | MathDB 页面 | 原文线索

V2-034109 | MATHDB-375272

Injectivity conjecture for the Panitch-Park quantum gluing module map

摘录 / EXCERPT: Let $\mathcal{T}$ be a triangulation, let R be the coefficient ring, and let ... be the surjective R-module homomorphism from the quantum gluing module to the Panitch-Park quantu...

conjecture | open | MathDB 页面 | 原文线索

V2-034110 | MATHDB-375283

Coarse Whitehead conjecture for spaces with finitely many ends

摘录 / EXCERPT: Let X and Y be compact metric spaces with finite shape dimension, let $\mathfrak{a}:cX\to cY$ be a coarse map, and suppose that M_X has $t\in\mathbb{N}$ ends. Choose...

conjecture | open | MathDB 页面 | 原文线索

V2-034111 | MATHDB-375284

Coarse homotopy groups of arbitrary ladders

摘录 / EXCERPT: Let X be a ladder consisting of two rays joined at height 0 and rungs of lengths $l_m\in\mathbb{N}$ at heights $h_m\in\mathbb{N}$, where (h_m) is strictly increasing, and equ...

conjecture | open | MathDB 页面 | 原文线索

V2-034112 | MATHDB-375285

Coarse mapping-space model for coarse homotopy groups

摘录 / EXCERPT: Coarse mapping-space conjecture. The coarse homotopy group $\pi_n^c(X,\omega)$ is isomorphic to the group, or pointed set when appropriate, of base-ray-preserving coarse homotopy c...

conjecture | open | MathDB 页面 | 原文线索

V2-034113 | MATHDB-375286

Coarse Hurewicz map

摘录 / EXCERPT: Let (X,d) be a proper metric space and let $n\in\mathbb{N}_0$. Let $H\mathcal{X}_*(X)$ denote coarse homology. Coarse Hurewicz conjecture. There is a natural morphism ... that make...

conjecture | open | MathDB 页面 | 原文线索

V2-034114 | MATHDB-375287

Milnor exact sequence for coarse homology of nested compacta

摘录 / EXCERPT: Let Y be a compact Hausdorff space and let $(Y_i)_{i\in\mathbb{N}}$ be a decreasing family of closed subspaces, with intersection given the subspace topology. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034115 | MATHDB-375288

Cofibre-colimit model for coarse homology at infinity

摘录 / EXCERPT: Let $E:\mathbf{BornCoarse}\to\mathbf{C}$ be a coarse homology theory, let X be a uniform bornological coarse space, and let $i_\alpha:M_\alpha\to M$ be the canonical inclusions a...

conjecture | open | MathDB 页面 | 原文线索

V2-034116 | MATHDB-375326

Daemi-Lidman-Vela-Vick-Wong rational homology ribbon cobordism partial-order conjecture

摘录 / EXCERPT: Let W_0 and W_1 be closed, connected, oriented 3-manifolds. A rational homology ribbon cobordism from W_0 to W_1 is a ribbon cobordism W from W_0 to W_1 such that t...

conjecture | open | MathDB 页面 | 原文线索

V2-034117 | MATHDB-375381

Continuity conjecture for relative symplectic cohomology

摘录 / EXCERPT: Let $(M,bomega)$ be a closed symplectic manifold and let $K_0\subset M$ be a compact subset. The relative symplectic cohomology $SH_M(K;\Lambda_{\geq 0})$ is defined for compact sub...

conjecture | open | MathDB 页面 | 原文线索

V2-034118 | MATHDB-375449

Ribbon minimality conjecture for the 4-twisted Whitehead link

摘录 / EXCERPT: Let W_4 denote the 4-twisted Whitehead link, a two-component link formed from the fourth twist knot and an unknot. Ribbon minimality conjecture. The link W_4 is ribbon minimal....

conjecture | open | MathDB 页面 | 原文线索

V2-034119 | MATHDB-375450

Conway knot ribbon minimality conjecture

摘录 / EXCERPT: Let $C=11n_{34}$ denote the Conway knot. It has Alexander polynomial 1 and is topologically slice but not smoothly slice. Conway knot ribbon minimality conjecture. The Conway kno...

conjecture | open | MathDB 页面 | 原文线索

V2-034120 | MATHDB-375498

Small square-table conjecture for saddle surfaces

摘录 / EXCERPT: A saddle surface is the graph $G(f)$ of a continuous map $f:\mathbb{R}^{2}\to\mathbb{R}$ satisfying the saddle-surface conditions, with center at the origin. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034121 | MATHDB-375499

Fixed-size rectangular-table conjecture for Fenn graphs

摘录 / EXCERPT: A Fenn graph is the graph $G(f)$ of a continuous map $f:\mathbb{R}^{2}\to\mathbb{R}$ over a compact convex domain $D\subset\mathbb{R}^{2}$ satisfying Fenn's boundary conditions. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-034122 | MATHDB-375510

Church-Farb-Putman conjecture on codimension cohomology of special linear groups

摘录 / EXCERPT: Church-Farb-Putman conjecture. The codimension-i cohomology of $\operatorname{SL}_n(\mathbb Z)$ vanishes for $n\geq i+2$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-034123 | MATHDB-375519

The A-cone conjecture for Morse-Novikov numbers of links

摘录 / EXCERPT: A-cone conjecture. For every natural number k, the set Γ_k is an A-cone.

conjecture | open | MathDB 页面 | 原文线索

V2-034124 | MATHDB-375520

Additivity conjecture for the Morse-Novikov number under connected sum of links

摘录 / EXCERPT: Morse-Novikov additivity conjecture. One should have

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V2-034125 | MATHDB-375524

Exact crystal Frobenius model conjecture

摘录 / EXCERPT: Let the BrainiaK crystal space denote the semantic state space equipped with maps induced by semantic composition and tree decomposition. A finite-dimensional quotient or completio...

conjecture | open | MathDB 页面 | 原文线索

V2-034126 | MATHDB-375535

Separated presentation adjunction for causal and triangular Frobenius presentations

摘录 / EXCERPT: Let $\mathbf{Caus}$ be the category of finite acyclic directed graphs equipped with a chosen topological order, with order-preserving graph homomorphisms or refinements as morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-034127 | MATHDB-375592

The twisted coalgebra conjecture for configuration-space models

摘录 / EXCERPT: Twisted coalgebra conjecture. This zig-zag can be promoted to a zig-zag of equivalences

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V2-034128 | MATHDB-375593

The stronger twisted coalgebra conjecture in right $mathbb{E}_d$-modules

摘录 / EXCERPT: Stronger twisted coalgebra conjecture. For some model of the $\mathbb{E}_d$ operad, this zig-zag can be promoted to a zig-zag of equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-034129 | MATHDB-375609

The contact cosmetic surgery conjecture for Legendrian knots

摘录 / EXCERPT: Contact cosmetic surgery conjecture. Any Legendrian knot in (Y,si) that is not smoothly an unknot admits no cosmetic contact surgeries.

conjecture | open | MathDB 页面 | 原文线索

V2-034130 | MATHDB-375620

Mod-p vanishing conjecture for the relative quantum connection

摘录 / EXCERPT: Mod-p vanishing conjecture. Assume p is chosen so that holds. Then

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V2-034131 | MATHDB-375622

The oriented exact-triangle conjecture for framed real monopole Floer homology

摘录 / EXCERPT: Let (L_+,L_-,L_0) be an oriented skein triple, and let $\mathbf{p}$ be a set of basepoints away from the resolutions. Denote by $\widetilde{\operatorname{HMR}}_*(L,\mathbf{p})$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-034132 | MATHDB-375623

The framed real monopole Floer and real Heegaard Floer isomorphism conjecture

摘录 / EXCERPT: Let $(Y,\tau,\mathbf{p})$ be a multi-based real three-manifold, let $\widetilde{\operatorname{HMR}}_*(Y,\tau,\mathbf{p})$ denote framed real monopole Floer homology, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034133 | MATHDB-375650

Even-degree tautological cohomology conjecture for moduli spaces of curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable curves, let $H^k(\overline{\mathcal{M}}_{g,n})$ denote its cohomology, and let $RH^k(\overline{\mathcal{M}}_{g,n})$...

conjecture | open | MathDB 页面 | 原文线索

V2-034134 | MATHDB-375715

Tier-2 proximity scaling for Yang-Mills-Higgs minimisers

摘录 / EXCERPT: Let M be a closed oriented Riemannian manifold with a non-zero relevant topological invariant, and let $D=\{(x_+,+1),(x_-,-1)\}$ consist of two oppositely labelled points at geod...

conjecture | open | MathDB 页面 | 原文线索

V2-034135 | MATHDB-375744

Pixton's completeness conjecture for the 3-spin relations

摘录 / EXCERPT: Pixton's conjecture. The 3-spin relations are complete in Chow and cohomology:

conjecture | open | MathDB 页面 | 原文线索

V2-034136 | MATHDB-375762

Universal heterodimensional-cycle obstruction conjecture for convexity

摘录 / EXCERPT: Let (Σ,η) be a contact Hamiltonian manifold, where η is a contact Hamiltonian structure on Σ. A heterodimensional cycle is a pair of heteroclinic trajectories...

conjecture | open | MathDB 页面 | 原文线索

V2-034137 | MATHDB-375767

The vexillary equivariant Schubert Lam-Postnikov-Pylyavskyy inequality

摘录 / EXCERPT: Let $u,v\in S_n$ be vexillary permutations, let $\phi(w)$ be the Wachs flag of w, and write $a\preceq b$ when a is a prefix of b. Let $\mathfrak S_w(\mathbf x;\mathbf y)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-034138 | MATHDB-375768

The equivariant Schur Lam-Postnikov-Pylyavskyy inequality

摘录 / EXCERPT: Let λ and μ be partitions, let $\mathbf x_k=(x_1,\ldots,x_k)$, and let $s_\lambda(\mathbf x_k;\mathbf y)$ denote the double Schur polynomial. Write $\lambda\vee\mu$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-034139 | MATHDB-375778

Dominated splitting conjecture for robustly transitive geodesic flows

摘录 / EXCERPT: Let g be a Riemannian metric whose geodesic flow is robustly transitive, meaning that g admits a C^2-neighbourhood such that every metric in that neighbourhood has topologica...

conjecture | open | MathDB 页面 | 原文线索

V2-034140 | MATHDB-375779

Dominated splitting conjecture for robustly transitive contact flows

摘录 / EXCERPT: Let X be a robustly transitive contact flow, meaning a contact flow that remains topologically transitive under the relevant perturbations within the class of contact flows. Cont...

conjecture | open | MathDB 页面 | 原文线索

V2-034141 | MATHDB-375787

Palis's conjecture on generic diffeomorphism dynamics

摘录 / EXCERPT: Let a diffeomorphism or non-singular line field be given, with genericity taken in the C^1 topology. Palis's conjecture. A C^1-generic diffeomorphism (or non-singular line fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-034142 | MATHDB-375793

Miller-Wong conjecture on the odd invariant of smooth 2-knots

摘录 / EXCERPT: Let F be a smooth 2-knot, let $n(F)$ denote its odd Khovanov invariant, and let $\Sigma(F)$ denote the branched double-cover associated to F. Miller-Wong's conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-034143 | MATHDB-375822

The symplectically aspherical positive scalar curvature conjecture

摘录 / EXCERPT: A closed symplectically aspherical manifold is a closed manifold whose symplectic form vanishes on every spherical homology class. The symplectically aspherical positive scalar cur...

conjecture | open | MathDB 页面 | 原文线索

V2-034144 | MATHDB-375826

Strict cyclotomic Deligne conjecture

摘录 / EXCERPT: Strict cyclotomic Deligne conjecture. For every natural number n, the following diagram of genuine S^1-spectra commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-034145 | MATHDB-375838

The homology fiber bundle smoothness conjecture for projective morphisms

摘录 / EXCERPT: Let Y be a complex analytic manifold and let $f:Y\to\Delta$ be a projective morphism, where Δ is a disc. A morphism is a homotopy fiber bundle if it has the homotopy-local...

conjecture | open | MathDB 页面 | 原文线索

V2-034146 | MATHDB-375845

The Fenn graph conjecture for cyclic quadrilaterals

摘录 / EXCERPT: Fenn graph conjecture. For every rectangle, and more generally for every such cyclic quadrilateral Q, there exists a rigid motion T such that

conjecture | open | MathDB 页面 | 原文线索

V2-034147 | MATHDB-375857

Kollár's algebraic Montgomery-Yang conjecture

摘录 / EXCERPT: Let S be a rational homology projective plane with quotient singularities, and write ... The smooth locus S^0 is assumed to be simply connected. Kollár's algebraic Montgomery-Yan...

conjecture | open | MathDB 页面 | 原文线索

V2-034148 | MATHDB-375897

Conjecture on knot Floer homology of Piccirillo friends

摘录 / EXCERPT: Knot Floer homology conjecture. The group $\widehat{HFK}(J)$ is supported nontrivially in exactly g diagonals, shifted up in Maslov grading from K. Hence the corresponding exot...

conjecture | open | MathDB 页面 | 原文线索

V2-034149 | MATHDB-375961

Ba and Clay's cyclic-cover conjecture for circular orderability, foliations, and L-spaces

摘录 / EXCERPT: Let W be an irreducible rational homology 3-sphere that is not a lens space. Ba and Clay's conjecture. The following statements are equivalent: 1. The fundamental group of W...

conjecture | open | MathDB 页面 | 原文线索

V2-034150 | MATHDB-375995

Decorated genus ordering conjecture

摘录 / EXCERPT: Let $\pi=\sigma^{-1}\tau\ne e$, let $L_0(\sigma)$ be the planar loop count, and let $g(\sigma,\pi)$ be the genus determined by the bare ribbon gluing. Let $L_\Gamma(\sigma,\pi)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-034151 | MATHDB-375997

Crepant resolution conjecture for foldings of ADE resolutions

摘录 / EXCERPT: Let $\mathcal{R}$ be an ADE root system, let $\mathcal{R}^{\mathrm{fold}}$ be its folding, and let $\mathcal{X}_{\mathcal{R}^{\mathrm{fold}}}$ be the corresponding orbifold with cr...

conjecture | open | MathDB 页面 | 原文线索

V2-034152 | MATHDB-375998

Iritani's integral crepant resolution conjecture

摘录 / EXCERPT: Let $\mathcal{X}$ be an orbifold with coarse moduli space X, and let $\phi\colon Z\to X$ be a crepant resolution. Iritani's conjecture. There exists an isomorphism ... induced by a...

conjecture | open | MathDB 页面 | 原文线索

V2-034153 | MATHDB-376014

The band-sum unknotting-number conjecture

摘录 / EXCERPT: Band-sum unknotting-number conjecture. Any band sum of two nontrivial knots has unknotting number greater than one.

conjecture | open | MathDB 页面 | 原文线索

V2-034154 | MATHDB-376017

Extended Uniqueness Conjecture for coherent weakly approximable triangulated categories

摘录 / EXCERPT: Extended Uniqueness Conjecture. If there is an exact equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-034155 | MATHDB-376062

Bobadilla-Kollár conjecture on projective morphisms over the disc

摘录 / EXCERPT: Let Y be a complex analytic manifold and let ... be a projective morphism to the complex disc. A homotopy fiber bundle is a morphism whose fibers are homotopy equivalent in the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-034156 | MATHDB-376082

Conjugate-slope property conjecture for L-space knots

摘录 / EXCERPT: Let K be a nontrivial L-space knot in S^3, and let $g(K)$ be its Seifert genus. The conjugate-slope property at n is the property defined in the paper for $n\in\mathbb N$. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-034157 | MATHDB-376083

Characterization of L-space knots by conjugate-slope and cofinal detection properties

摘录 / EXCERPT: Let K be a nontrivial knot in S^3. The conjugate-slope property and the cofinal detection property are the properties defined in the paper. L-space knot characterization conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-034158 | MATHDB-376104

Foxby's conjecture on faithful modules with minimal top Bass number

摘录 / EXCERPT: Foxby's conjecture. If M is faithful and

conjecture | open | MathDB 页面 | 原文线索

V2-034159 | MATHDB-376140

Kochloukova-Lima homological Virtual Surjection Conjecture

摘录 / EXCERPT: Let $2\leqslant k\leqslant m$ be integers. For groups $G_{1},\ldots,G_{m}$, a subgroup $P\leqslant G_{1}\times\cdots\times G_{m}$ virtually surjects on k factors if, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-034160 | MATHDB-376142

Kochloukova-Lima homological n-$(n+1)$-$(n+2)$ conjecture

摘录 / EXCERPT: The homological n-$(n+1)$-$(n+2)$ conjecture. If N_{1} is of type FP_n, G_1 and G_2 are of type FP_{n+1}, and Q is of type...

conjecture | open | MathDB 页面 | 原文线索

V2-034161 | MATHDB-376214

Fidkowski's conjecture on invertible phases entangled by QCAs

摘录 / EXCERPT: Fidkowski's conjecture. Y is a nontrivial QCA if the invertible phase it entangles has a nontrivial partition function on some orientable manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-034162 | MATHDB-376258

Mixed CR minimisers for homological systoles in Berger projective spaces

摘录 / EXCERPT: Let $N\geq 2$, let $1\leq k\leq 2N-2$, set $d=k+1$, and let $\mu>0$. For the Berger metric g_μ on $\mathbb {RP}^{2N-1}$, define the mod-two homological systole by ... For a real...

conjecture | solved | MathDB 页面 | 原文线索

V2-034163 | MATHDB-376278

White's generic regularity conjecture for area-minimizing cycles

摘录 / EXCERPT: Let M be a Riemannian manifold and let a homologically area-minimizing cycle in M have a singular set. White's generic regularity conjecture. Singularities in a homologically a...

open_problem | open | MathDB 页面 | 原文线索

V2-034164 | MATHDB-376293

The first homology conjecture for the handlebody Torelli group

摘录 / EXCERPT: Let g be sufficiently large, let $\mathcal{HI}_g^1$ denote the handlebody Torelli group, and let $B_3^{b_i}$ be the subspace defined from the Birman-Craggs-Johnson homomorphism....

conjecture | open | MathDB 页面 | 原文线索

V2-034165 | MATHDB-376308

Twisted Singer conjecture for closed aspherical odd-dimensional manifolds

摘录 / EXCERPT: Let M be a closed aspherical $(2n+1)$-manifold. For a cohomology class $\phi\in H^1(M;\mathbb{R})$, let $b_i^{(2)}(\widetilde M;\phi)$ denote the twisted L^2-Betti numbers of t...

conjecture | open | MathDB 页面 | 原文线索

V2-034166 | MATHDB-376311

Polarized affine cone conjecture for the Ricci-flow limit of Kähler-Ricci shrinkers

摘录 / EXCERPT: Polarized affine cone conjecture. The space Z_0, with the topology induced by d^Z_0, admits a polarized affine cone structure, and the map Φ realizes the polarized Fano fi...

conjecture | open | MathDB 页面 | 原文线索

V2-034167 | MATHDB-376319

An intrinsic approximation principle characterizing continuity of Scott function spaces

摘录 / EXCERPT: Let X and L be continuous domains, and let T denote the domain used in the Scott function spaces $[X\to T]$ and $[T\to L]$. The paper considers finite-layer truncation maps a...

conjecture | open | MathDB 页面 | 原文线索

V2-034168 | MATHDB-376337

Buchsbaum-Eisenbud-Horrocks conjecture

摘录 / EXCERPT: Let R be a commutative Noetherian ring such that $\operatorname{Spec}(R)$ is connected, and let M be a non-zero, finitely generated R-module of finite projective dimension. F...

conjecture | open | MathDB 页面 | 原文线索

V2-034169 | MATHDB-376353

The Neoplatonic homotopy conjecture

摘录 / EXCERPT: Neoplatonic homotopy conjecture. Every 6-net σ has a family of realizations $\Delta(\sigma,l)$, each unique up to isometry, as undented hyperbolic polyhedra of side length...

conjecture | open | MathDB 页面 | 原文线索

V2-034170 | MATHDB-376402

The Montesinos-Nakanishi 3-move conjecture

摘录 / EXCERPT: Montesinos-Nakanishi 3-move conjecture. Every link can be reduced to a trivial link by 3-moves.

conjecture | open | MathDB 页面 | 原文线索

V2-034171 | MATHDB-376403

The free-rank conjecture for HOMFLYPT Yang-Baxter homology

摘录 / EXCERPT: Free-rank conjecture. The rank of the free part of $H_n(R_{(m)})$ is 2^{m-1} for $n\geq m-1$, while

conjecture | open | MathDB 页面 | 原文线索

V2-034172 | MATHDB-376404

The homology formula conjecture for the level-two HOMFLYPT Yang-Baxter operator

摘录 / EXCERPT: Level-two homology formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034173 | MATHDB-376435

The generalized hypergeometric solution conjecture for toric Calabi-Yau complete intersections

摘录 / EXCERPT: Generalized hypergeometric solution conjecture. The generalized hypergeometric series $\Phi_0(z)$, regarded as a function of ∇_{AP}-flat coordinates z on $\tilde H^2$, is...

conjecture | open | MathDB 页面 | 原文线索

V2-034174 | MATHDB-376437

The rationality conjecture for A-model structure constants of Calabi-Yau threefolds

摘录 / EXCERPT: Rationality conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-034175 | MATHDB-376513

Quillen's vanishing André-Quillen homology conjecture for local rings

摘录 / EXCERPT: Quillen's conjecture. The following are equivalent: $D_s(\Bbb{F}|R)=0$ for $s\gg 0$; $D_s(\Bbb{F}|R)=0$ for $s\geq 3$; and R is a complete intersection. This conjecture character...

conjecture | open | MathDB 页面 | 原文线索

V2-034176 | MATHDB-376514

Turner's simplicial vanishing conjecture for local complete intersections

摘录 / EXCERPT: Let $\Bbb{F}$ be a field and let R be a simplicial commutative ring that is locally finite and locally of non-zero characteristic. Say that R is locally a complete intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-034177 | MATHDB-376526

Homotopy conjecture for families of cuts on a complex curve

摘录 / EXCERPT: Let M be a complex curve. Let $\gamma_{i}^{\left(1\right)}$, $i\in I^{\left(1\right)}$, be a disjoint family of smooth embedded cycles in M such that...

conjecture | open | MathDB 页面 | 原文线索

V2-034178 | MATHDB-376533

The factorial specialization conjecture for Grassmannian quantum series

摘录 / EXCERPT: Let $G(k,n)$ be the Grassmannian, let $P(k,n)$ be its Gorenstein toric degeneration, and let $A_{G(k,n)}(q)$ denote the special solution to the quantum ${\cal D}$-module determined...

conjecture | open | MathDB 页面 | 原文线索

V2-034179 | MATHDB-376534

The Lax-operator conjecture for Grassmannians

摘录 / EXCERPT: Let $G(r,s)$ be the Grassmannian and let $X_{[a,b]}$ be the variables indexed by $1\leq a\leq s-r$ and $1\leq b\leq r$, with $X_{a,b}=0$ when $a>s-r$ or $b>r$. Let q be the quant...

conjecture | open | MathDB 页面 | 原文线索

V2-034180 | MATHDB-376655

Projection-diameter conjecture for space curves

摘录 / EXCERPT: Projection-diameter conjecture. There is some orthogonal projection to $\mathbb R^{n-1}$ in which the image of γ has diameter at most

conjecture | open | MathDB 页面 | 原文线索

V2-034181 | MATHDB-376656

Existence and rigidity conjecture for distortion ropelength minimizers

摘录 / EXCERPT: Distortion ropelength minimizer conjecture. For each nontrivial knot type K, there exists a shortest curve $[?]\subset\mathbb R^3$ of distortion thickness 1 in that knot type....

conjecture | open | MathDB 页面 | 原文线索

V2-034182 | MATHDB-376662

Homotopy invariance of L^2 torsion

摘录 / EXCERPT: Let M and N be compact odd-dimensional manifolds, and let $f:M\to N$ be a homotopy equivalence. Let $\widehat M$ and $\widehat N$ be their normal Γ-covering spaces, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034183 | MATHDB-376663

Existence of L^2 torsion for groups with finite odd-dimensional classifying spaces

摘录 / EXCERPT: Let Γ be a discrete group whose classifying space $B\Gamma$ is a finite CW complex of odd dimension, and let $E\Gamma$ be its universal cover. Let $\phi_{E\Gamma}$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-034184 | MATHDB-376664

Hutchings-Lee conjecture relating Seiberg-Witten and Morse theory invariants

摘录 / EXCERPT: Let X be a closed 3-manifold with $b_1(X)>0$, let o be a homology orientation, and let $\operatorname{Spin}^c(X)$ denote the set of spin-c structures on X. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034185 | MATHDB-376665

Turaev's conjecture relating Seiberg-Witten invariants and torsion

摘录 / EXCERPT: Let X be a closed 3-manifold with $b_1(X)>0$, let o be a homology orientation, and let $\operatorname{Eul}(X)$ be the set of Euler structures. Turaev’s torsion defines a map...

conjecture | open | MathDB 页面 | 原文线索

V2-034186 | MATHDB-376667

Weinstein's conjecture for rotational Beltrami flows

摘录 / EXCERPT: Hydrodynamical Weinstein conjecture. Every smooth rotational Beltrami flow on a Riemannian three-manifold has a closed orbit.

conjecture | open | MathDB 页面 | 原文线索

V2-034187 | MATHDB-376676

Low's causality conjecture for skies in (2+1)-dimensional spacetimes

摘录 / EXCERPT: Let (M,g) be an orientable, time-oriented, globally hyperbolic $(2+1)$-dimensional spacetime with a Cauchy surface $\Sigma\subset M$ diffeomorphic to a subset of $\mathbb{R}^2$....

conjecture | open | MathDB 页面 | 原文线索

V2-034188 | MATHDB-376693

The connectivity-shell hypothesis for the metric/topological picture of space-time

摘录 / EXCERPT: Connectivity-shell hypothesis. When the unfolding process is fully developed, the infinitesimal neighborhood of S_0 is densely connected to it, and connectivity decreases through...

conjecture | open | MathDB 页面 | 原文线索

V2-034189 | MATHDB-376721

Homological mirror symmetry for noncommutative two-tori

摘录 / EXCERPT: Let ${\cal C}^A$ be the cyclic A_∞-category obtained from Fukaya's A_∞-category $(V^F,\eta^F,{\frak m}^F)$ by suspension, and let ${\cal C}^B$ be the cyclic...

conjecture | open | MathDB 页面 | 原文线索

V2-034190 | MATHDB-376737

Hecke R-matrix bidifferential-complex conjecture

摘录 / EXCERPT: Let V be a vector space equipped with a Hecke $\check R$-matrix, and let $K^{\bullet\bullet}(V)$ and the complexes $C_V^\bullet(k)$ be the associated bidifferential and one-varia...

conjecture | open | MathDB 页面 | 原文线索

V2-034191 | MATHDB-376738

Nondegeneracy conjecture for the Berezinian pairing

摘录 / EXCERPT: Berezinian pairing conjecture. The above pairing is always non-degenerate. For q not a root of unity, the text derives nondegeneracy, while it notes that the simpler duality brea...

conjecture | open | MathDB 页面 | 原文线索

V2-034192 | MATHDB-376745

Trialgebra reconstruction conjecture for quantum groups and Donaldson-Floer theory

摘录 / EXCERPT: A trialgebra is an algebraic structure with three compatible algebraic operations, as envisioned here for constructing four-dimensional topological quantum field theories. Trialgeb...

conjecture | open | MathDB 页面 | 原文线索

V2-034193 | MATHDB-376747

Recursive law for cubic curves on Calabi-Yau hypersurfaces

摘录 / EXCERPT: Cubic Calabi-Yau recursion conjecture. The recursive law is

conjecture | open | MathDB 页面 | 原文线索

V2-034194 | MATHDB-376749

Mirror-series and Calabi-Yau quantum-cohomology conjecture from recursive descent

摘录 / EXCERPT: Calabi-Yau mirror-recursion conjecture. Formal iteration of the laws of Conjecture 2 down to $N=k$ should give

conjecture | open | MathDB 页面 | 原文线索

V2-034195 | MATHDB-376755

Generalized mirror-transformation conjecture for projective hypersurfaces

摘录 / EXCERPT: Let P_m denote the set of partitions of m, let $l(\sigma_m)$ be the length of a partition, and let $C_m^d(\sigma_m)$ be the coefficients associated with partitions in the propo...

conjecture | open | MathDB 页面 | 原文线索

V2-034196 | MATHDB-376839

The superfield realization conjecture for engineerable Adinkra topologies

摘录 / EXCERPT: Superfield realization conjecture. For every N and every topology of engineerable Adinkras, there exists a corresponding set of superderivative constraints such that the set of s...

conjecture | open | MathDB 页面 | 原文线索

V2-034197 | MATHDB-376867

The first cohomology conjecture for differential operators on the supercircle

摘录 / EXCERPT: Let $\mathrm{osp}(1|2)$ act on the space ${\cal D}_{\lambda,\mu}$ of differential operators from tensor densities of weight λ to tensor densities of weight μ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034198 | MATHDB-376887

The easier connected-sum superbridge index conjecture

摘录 / EXCERPT: Let K_1 and K_2 be nontrivial knots. Write $s[K]$ for the superbridge index of a knot K, and write $K_1\sharp K_2$ for their connected sum. Easier connected-sum conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-034199 | MATHDB-376888

The strict superbridge-index increase conjecture for connected sums

摘录 / EXCERPT: Let K_1 and K_2 be nontrivial knots. Write $s[K]$ for the superbridge index of a knot K, and write $K_1\sharp K_2$ for their connected sum. Strict increase conjecture. Any co...

conjecture | open | MathDB 页面 | 原文线索

V2-034200 | MATHDB-376890

Kontsevich-Soibelman cell decomposition conjecture for the Fulton-MacPherson operad

摘录 / EXCERPT: For $n\ge 2$, let $FM_2(n)$ be the Fulton-MacPherson space, let $MT(n)$ be the set of isomorphism classes of meta-trees with n tails, and let P be the dg-operad whose standard...

conjecture | open | MathDB 页面 | 原文线索

V2-034201 | MATHDB-376893

Hard Lefschetz conjecture for ample classes on complete fans

摘录 / EXCERPT: Let Φ be a complete fan in a vector space V of dimension n. Let Ω^1_Φ be the sheaf with stalk $\Omega^1_{\Phi,\sigma}=\operatorname{Span}(\sigma)^\perp$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034202 | MATHDB-376900

Functorial invariance of singular-category tangle presentations

摘录 / EXCERPT: Let $\cap_{i,n}$ and $\cup_{i,n}$ be the cap and cup functors between the relevant singular categories, and let $R_{i,n}$ be the cone of the adjointness morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-034203 | MATHDB-376902

Functorial invariance of parabolic-category tangle presentations

摘录 / EXCERPT: Let $D^b(\mathcal O^n)$ be the bounded derived category of the direct sum of the parabolic categories $\mathcal O^{k,n-k}$, and extend the cap, cup, and reflection functors to deri...

conjecture | open | MathDB 页面 | 原文线索

V2-034204 | MATHDB-376927

The Massey-Rolfsen conjecture for links with one nonembedded component

摘录 / EXCERPT: Massey-Rolfsen conjecture. Theorem still holds if one component of the link L is not embedded (but mapped into $\mathbb{R}^4$ disjointly from the other components).

conjecture | open | MathDB 页面 | 原文线索

V2-034205 | MATHDB-376953

Poénaru's conjecture on geometric simple connectivity of contractible 4-manifolds

摘录 / EXCERPT: Let M be a compact contractible 4-manifold whose boundary is a homology sphere. Write $\operatorname{int}(M)$ for its interior, and let g.s.c. denote geometric simple connectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-034206 | MATHDB-376957

The local cohomology conjecture for additive generalized Okamoto-Painlevé pairs

摘录 / EXCERPT: Local cohomology conjecture. Under the same notation and assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-034207 | MATHDB-376998

Finiteness conjecture for HOMFLY polynomials of knots

摘录 / EXCERPT: Let P_K denote the HOMFLY polynomial of a knot K, let $\operatorname{span}_l P_K$ be its span in the variable l, and let $\max\deg_m P_K$ be its maximum degree in the variabl...

conjecture | open | MathDB 页面 | 原文线索

V2-034208 | MATHDB-376999

Conformal concordance implies isotopy of positive conformal classes

摘录 / EXCERPT: Conformal concordance conjecture. If C_0 and C_1 are conformally concordant, then they are isotopic in ${\cal C}^+(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-034209 | MATHDB-377001

Donaldson-Frohman-Nicas TQFT identification conjecture

摘录 / EXCERPT: Let ${\cal V}^{DF}$ be Donaldson's TQFT associated with moduli spaces of connections on a non-trivial $SO(3)$ bundle. For $x_1,x_2,\ldots$, define ... Let ${\cal V}^{(D)}$ be the pol...

conjecture | open | MathDB 页面 | 原文线索

V2-034210 | MATHDB-377023

The hyperbolic rational homology 3-sphere deformation conjecture

摘录 / EXCERPT: Let Y be a hyperbolic rational homology 3-sphere, and let $H^{1,-}_0(Y\times\mathbb R)$ denote the relevant space of bounded solutions on the cylindrical anti-self-dual space...

conjecture | open | MathDB 页面 | 原文线索

V2-034211 | MATHDB-377027

Flat-metric convergence conjecture for degenerating conformal metrics on the 2-sphere

摘录 / EXCERPT: Let M be a topological 2-sphere, and let $g_i=e^{2u_i}g_0$ be a family of conformal metrics with bounded area and energy. Assume ... Suppose there is exactly one bubble point...

conjecture | open | MathDB 页面 | 原文线索

V2-034212 | MATHDB-377028

The cubical knot complex is homotopy equivalent to the Vassiliev simplicial complex

摘录 / EXCERPT: Let CK^* be the space obtained from $CK_b(S^3)$ by attaching an $(n+1)$-cell for every knot $K_{(n-2,1)}$, with $n\geq 2$, whose boundary is given by resolving the double points...

conjecture | open | MathDB 页面 | 原文线索

V2-034213 | MATHDB-377029

Rigidity conjecture for edge-preserving bijections of the cubical knot complex

摘录 / EXCERPT: Let $CK(\mathbf{R}^3)$ be the cubical knot complex, and let $F:CK(\mathbf{R}^3)\to CK(\mathbf{R}^3)$ be the map induced by taking mirror images of knots. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-034214 | MATHDB-377033

Induction conjecture for Floer cohomology of eventually Lagrangian submanifolds

摘录 / EXCERPT: Let c,c' be matching paths, let Λ_c and $\Lambda_{c'}$ be the associated eventually Lagrangian submanifolds, and let $\mathcal A,\mathcal A'$ be directed A_∞-categ...

conjecture | open | MathDB 页面 | 原文线索

V2-034215 | MATHDB-377034

Bouquet conjecture for splitting of isolated singularities

摘录 / EXCERPT: Bouquet conjecture. The Milnor fibre of (X_0,f_0) is homotopy equivalent to the bouquet of the Milnor fibres of the isolated singularities into which it splits.

conjecture | open | MathDB 页面 | 原文线索

V2-034216 | MATHDB-377035

Homotopical constancy conjecture for non-splitting singularities

摘录 / EXCERPT: Homotopical constancy conjecture. If the isolated singularity (X_0,f_0) does not split, then the Milnor fibration of the isolated singularity of (X_t,f_t) is homotopically cons...

conjecture | open | MathDB 页面 | 原文线索

V2-034217 | MATHDB-377042

Nilpotency bounds for groups of self-homotopy equivalences

摘录 / EXCERPT: Nilpotency conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034218 | MATHDB-377047

Goda-Teragaito conjecture on genus-one tunnel-number-one knots

摘录 / EXCERPT: Goda-Teragaito conjecture. K is a 2-bridge knot.

conjecture | open | MathDB 页面 | 原文线索

V2-034219 | MATHDB-377061

Montesinos-Nakanishi conjecture and its basic-tangle generalizations

摘录 / EXCERPT: Montesinos-Nakanishi conjecture. Every link is 3-equivalent to a trivial link. More generally, every 2-tangle is 3-equivalent to a tangle with at most one crossing; every 3...

conjecture | open | MathDB 页面 | 原文线索

V2-034220 | MATHDB-377062

Fourth skein module generation conjecture for tangles

摘录 / EXCERPT: Fourth skein module generation conjecture. The fourth skein module $\mathcal{S}_4(S^3)$ is generated by trivial links; the fourth skein module of 2-tangles in a disk is generated...

conjecture | open | MathDB 页面 | 原文线索

V2-034221 | MATHDB-377063

Conjecture on linear independence of trivial links in the fourth skein module

摘录 / EXCERPT: Linear-independence conjecture. The trivial links T_i are linearly independent in $\mathcal{S}_4(S^3)$ in either of the following cases:

conjecture | open | MathDB 页面 | 原文线索

V2-034222 | MATHDB-377075

Lagrangianity conjecture for the automorphism group of an A-infinity category

摘录 / EXCERPT: Let ${\cal C}$ be an A_{∞}-category such that $HH^1({\cal C})$ is abelian and the connected automorphism group $G_{\cal C}=Aut^0({\cal C})$ is a finite-dimensional commutati...

conjecture | open | MathDB 页面 | 原文线索

V2-034223 | MATHDB-377089

The abelianized characteristic-algebra topological-knot conjecture

摘录 / EXCERPT: Let K be a Legendrian knot with maximal Thurston-Bennequin number. Let its characteristic algebra be abelianized by imposing $vw=wv$ for all elements v,w, and regard the result...

conjecture | open | MathDB 页面 | 原文线索

V2-034224 | MATHDB-377097

Shapovalov's homological Duflo conjecture

摘录 / EXCERPT: Shapovalov's homological Duflo conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-034225 | MATHDB-377099

Temperley-Lieb unfaithfulness conjecture for knots with trivial Jones polynomial

摘录 / EXCERPT: Let B_n be a braid group, and let its Temperley-Lieb representation be the representation into the Temperley-Lieb algebra. A knot is non-trivial if it is not isotopic to the unkn...

conjecture | open | MathDB 页面 | 原文线索

V2-034226 | MATHDB-377100

Powers of a braid conjecture for producing non-trivial knot closures

摘录 / EXCERPT: Let B_n be the braid group on n strands, with standard generators $sigma_1,sigma_2,dots,sigma_{n-1}$. For a braid, its closure is the link obtained by joining corresponding end...

conjecture | open | MathDB 页面 | 原文线索

V2-034227 | MATHDB-377101

Normal-subgroup conjecture for non-trivial braid closures

摘录 / EXCERPT: Let B_n be the braid group on n strands. For a braid, its closure is the link obtained by joining corresponding endpoints. A subgroup H of B_n is non-trivial normal if it i...

conjecture | refuted | MathDB 页面 | 原文线索

V2-034228 | MATHDB-377109

Generic transversality conjecture for nonlinear dynamics

摘录 / EXCERPT: Let M be a compact manifold of dimension m, let $P:M\rightarrow\mathbb{R}^d$ be smooth, and let $\mathfrak{F}$ be the quasi-stratification of M by the level sets...

conjecture | open | MathDB 页面 | 原文线索

V2-034229 | MATHDB-377118

Conjecture on the cyclic cohomology of the smooth dihedral group-algebra subalgebra

摘录 / EXCERPT: Let A^{∞} be the smooth subalgebra of the group C^*-algebra associated with the dihedral group, and let $HC^n(A^{\infty})$ denote its cyclic cohomology in degree n. Cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-034230 | MATHDB-377124

Finiteness conjecture for tight contact structures on atoroidal 3-manifolds

摘录 / EXCERPT: Finiteness conjecture. If M is atoroidal, then M carries only finitely many isotopy classes of tight contact structures.

conjecture | open | MathDB 页面 | 原文线索

V2-034231 | MATHDB-377128

Finite flag subcomplex vanishing conjecture

摘录 / EXCERPT: Finite flag subcomplex vanishing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034232 | MATHDB-377129

Partial-link Singer conjecture for aspherical polyhedra

摘录 / EXCERPT: Partial-link Singer conjecture. If X has spherical links in codimensions at most $2l+1$, where $2l+1\le n$, then

conjecture | open | MathDB 页面 | 原文线索

V2-034233 | MATHDB-377130

Singer-type vanishing conjecture for flag complexes with spherical links

摘录 / EXCERPT: Flag-complex partial-link conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034234 | MATHDB-377136

The extended thin elements conjecture for Kan completions

摘录 / EXCERPT: Let K be a precubical set, let $F(K)$ be the free non-contracting ω-category generated by K, and let $\widehat{F(K)}$ denote its Kan completion. Let C^{gl} and...

conjecture | open | MathDB 页面 | 原文线索

V2-034235 | MATHDB-377137

The thin elements conjecture for formal globular and branching complexes

摘录 / EXCERPT: Let $\mathcal{C}$ be a strict non-contracting globular ω-category. Let $C F^{gl}$ be its formal globular complex and $C F^{\pm}$ its formal branching and merging complexes,...

conjecture | open | MathDB 页面 | 原文线索

V2-034236 | MATHDB-377138

The thin elements conjecture for the branching semi-globular nerve

摘录 / EXCERPT: Let $\mathcal{C}$ be a non-contracting ω-category freely generated by a precubical set. Let $M_n^{gl^-}(\mathcal{C})$ denote the degree-n elements of the branching semi-gl...

conjecture | open | MathDB 页面 | 原文线索

V2-034237 | MATHDB-377144

The mirror-symmetry categorical mapping conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let $({\mathcal Y}\rightarrow T,P)$ be a family of Calabi-Yau manifolds with a complex cusp $P\in\partial T$, let X be the topological mirror of Y with respect to P, and assu...

conjecture | open | MathDB 页面 | 原文线索

V2-034238 | MATHDB-377145

The Fourier-Mukai orientation conjecture for K3 surfaces

摘录 / EXCERPT: Let Z_0 and X_0 be smooth projective K3 surfaces, and let ... be a Fourier-Mukai equivalence. The induced map on integral cohomology is denoted by ... Fix the canonical orientation...

conjecture | open | MathDB 页面 | 原文线索

V2-034239 | MATHDB-377153

Parallel crossing number conjecture for knots and links

摘录 / EXCERPT: Parallel crossing number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034240 | MATHDB-377154

Piecewise analyticity conjecture for ropelength minimizers

摘录 / EXCERPT: Piecewise analyticity conjecture. Ropelength minimizers are piecewise analytic.

conjecture | open | MathDB 页面 | 原文线索

V2-034241 | MATHDB-377155

Gluing conjecture for relative Seiberg-Witten invariants

摘录 / EXCERPT: Let X_1 be a cobordism between three-manifolds Y_1 and Y_2, and let X_2 be a cobordism between Y_2 and Y_3. Denote by $\mathcal{D}_{X_1}$, $\mathcal{D}_{X_2}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034242 | MATHDB-377156

Conjecture on characteristic classes of codimension-p foliations

摘录 / EXCERPT: Characteristic-class conjecture. The characteristic classes of codimension-p foliations on X coincide with the elements of

conjecture | open | MathDB 页面 | 原文线索

V2-034243 | MATHDB-377167

Wan's generic Newton polygon convergence conjecture

摘录 / EXCERPT: Wan's conjecture. There is a Zariski dense subset ${\mathcal{U}}$ in $\mathbb{A}^d$ such that, for every $f(x)\in {\mathcal{U}}$, the limit exists and

conjecture | open | MathDB 页面 | 原文线索

V2-034244 | MATHDB-377174

Conjecture on completion and localization for commutative localized quotients

摘录 / EXCERPT: Let R be a ring spectrum and let E be a commutative localized quotient of R. For a finite cell R-module M, let $\hat{\mathrm{L}}^R_E M$ denote the completed E-localizat...

conjecture | open | MathDB 页面 | 原文线索

V2-034245 | MATHDB-377180

Universality conjecture for the formal group of geometric cobordisms over Hopf algebras

摘录 / EXCERPT: Universality conjecture. The formal group ${\frak F}(x\otimes 1,1\otimes x)$ is the universal object in the category of formal groups over a (topological) Hopf algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-034246 | MATHDB-377181

Vassiliev's first-term stabilization conjecture for the long-knot spectral sequence

摘录 / EXCERPT: Vassiliev's stabilization conjecture. The spectral sequence associated with this filtration and computing the Borel-Moore homology groups of the resolution σ stabilizes over...

conjecture | open | MathDB 页面 | 原文线索

V2-034247 | MATHDB-377182

Vassiliev's homotopy-splitting conjecture for the resolved discriminant

摘录 / EXCERPT: Let σ be the simplicial resolution of the discriminant Σ of the space of long knots, and let ... be its natural filtration. Write $\bar\sigma_i$ and $\bar\sigma$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-034248 | MATHDB-377192

Kronheimer-Mrowka's metric-independence conjecture for equivariant Seiberg-Witten Floer homology

摘录 / EXCERPT: Kronheimer-Mrowka's conjecture. These equivariant Seiberg-Witten Floer homologies are well-defined, meaning that they are independent of the particular choices of metrics and pertu...

conjecture | open | MathDB 页面 | 原文线索

V2-034249 | MATHDB-377208

The complete-relations conjecture for the universal subring of Higgs-bundle cohomology

摘录 / EXCERPT: Let $\mathcal{M}$ be the moduli space above, let $H_I^*(\mathcal{M})\subset H^*(\mathcal{M})^\Gamma$ be the subring generated by the universal classes $\alpha_{\mathcal{M}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-034250 | MATHDB-377210

The Whitehead conjecture for weak S-homotopy equivalences of globular CW-complexes

摘录 / EXCERPT: Let f be a morphism of globular CW-complexes from X to Y. A weak S-homotopy equivalence is a morphism that induces a bijection between the 0-skeleta and, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-034251 | MATHDB-377237

Nakanishi's 2-tangle reduction conjecture

摘录 / EXCERPT: A 2-tangle is a tangle with four boundary endpoints, and 3-move equivalence is generated by 3-moves, inverse moves, and Reidemeister moves. The four basic 2-tangles are the tangles...

conjecture | open | MathDB 页面 | 原文线索

V2-034252 | MATHDB-377238

Harikae-Nakanishi $(2,2)$-move conjecture for links

摘录 / EXCERPT: A $(2,2)$-move is the local link modification shown in Figure 1.10; two links are $(2,2)$-move equivalent if they are related by such moves, their inverses, and Reidemeister moves....

conjecture | open | MathDB 页面 | 原文线索

V2-034253 | MATHDB-377239

The 11-coloring-preserving $(2,5)$ and $(4,-3)$ move conjecture

摘录 / EXCERPT: A rational (m,q)-move is the local move corresponding to the indicated rational tangle, and its inverse and mirror-image moves are also allowed. The Fox 11-coloring space of a...

conjecture | open | MathDB 页面 | 原文线索

V2-034254 | MATHDB-377256

The positive-diagram crossing-number conjecture

摘录 / EXCERPT: Positive-diagram crossing-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034255 | MATHDB-377257

Jones's braid-index uniqueness conjecture

摘录 / EXCERPT: Jones's braid-index uniqueness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-034256 | MATHDB-377272

Neumann-Wahl conjecture for universal abelian covers

摘录 / EXCERPT: Let (X,o) be a ${\mathbb Q}$-Gorenstein normal surface singularity with ${\mathbb Q}$-homology sphere link, and let Δ be the splice diagram associated to its resolution gr...

conjecture | open | MathDB 页面 | 原文线索

V2-034257 | MATHDB-377273

Integral homology sphere splice-diagram conjecture

摘录 / EXCERPT: Let (X,o) be a Gorenstein normal surface singularity whose link M is an integral homology sphere, meaning $H_1(M;{\mathbb Z})=\{0\}$. Let Δ be its splice diagram. Integr...

conjecture | open | MathDB 页面 | 原文线索

V2-034258 | MATHDB-377277

Nilpotence filtration conjecture for the mod-2 cohomology of highly connected spaces

摘录 / EXCERPT: Let X be a $2d$-connected space, with $d\geq 1$, and let $M=\widetilde H^*(X;\mathbf F_2)$ be its reduced mod-2 cohomology. Assume that all cup products in M are trivial, tha...

conjecture | open | MathDB 页面 | 原文线索

V2-034259 | MATHDB-377278

Kuhn's dichotomy conjecture for the functors associated to unstable cohomology

摘录 / EXCERPT: Let X be a space. For each s, let $R_s(H^*X)$ denote the analytic functor associated with the reduced unstable module $R_s(H^*X)$. Kuhn's dichotomy conjecture. Either at least...

conjecture | open | MathDB 页面 | 原文线索

V2-034260 | MATHDB-377279

Degree-growth conjecture for the functors associated to unstable cohomology

摘录 / EXCERPT: Let X be a space, and let d be the least integer such that R_d is a nonconstant functor of finite degree. Assume that $d>1$. Degree-growth conjecture. At least one of the fun...

conjecture | open | MathDB 页面 | 原文线索

V2-034261 | MATHDB-377288

The syntomic regulator formula for discrete valuation subfields

摘录 / EXCERPT: Let $K\subset\mathbb C_p$ be a discrete valuation subfield, meaning that its valuation is induced from the valuation on $\mathbb C_p$, and let R be the valuation ring of K. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-034262 | MATHDB-377289

Cantor-set conjecture for orderings of free groups

摘录 / EXCERPT: Let F_n be the free group on $n>1$ generators. Write $LO(F_n)$ for the space of left-orderings of F_n and $BiO(F_n)$ for the space of bi-orderings, each with the topology inher...

conjecture | open | MathDB 页面 | 原文线索

V2-034263 | MATHDB-377297

The Charney-Davis conjecture for non-positively curved piecewise Euclidean manifolds

摘录 / EXCERPT: Let M^d be a compact, closed, d-dimensional manifold with a piecewise Euclidean structure, and suppose that it is non-positively curved in the sense of Gromov. Let $\chi(M^d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-034264 | MATHDB-377307

The volume-controlled homotopy type conjecture for locally symmetric manifolds

摘录 / EXCERPT: Let S be a symmetric space of non-compact type, and let an S-manifold be a complete Riemannian manifold locally isometric to S. An irreducible S-manifold has no nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-034265 | MATHDB-377332

The interval conjecture for degrees in quantum Schubert products

摘录 / EXCERPT: Let X be the Grassmannian, and let σ_{λ} and σ_{μ} be Schubert classes indexed by Young diagrams. Write $d_{\min}$ and $d_{\max}$ for the smallest and large...

conjecture | open | MathDB 页面 | 原文线索

V2-034266 | MATHDB-377333

The degree-reduction conjecture for quantum Schubert structure constants

摘录 / EXCERPT: Let $d\geq 1$ be an integer, and let λ, μ, and ν be Young diagrams indexing Schubert classes in the Grassmannian. Denote by $\langle\lambda,\mu,\nu\rangle_d$ the co...

conjecture | open | MathDB 页面 | 原文线索

V2-034267 | MATHDB-377346

Conjecture on the cyclic cohomology of the smooth dihedral group algebra

摘录 / EXCERPT: Cyclic cohomology conjecture. The cyclic cohomology of A^{∞} is given by

conjecture | open | MathDB 页面 | 原文线索

V2-034268 | MATHDB-377349

The factorization conjecture for Alexander polynomials of equivariant slice knots

摘录 / EXCERPT: Factorization conjecture. Alexander polynomials of equivariant slice knots satisfy this factorization condition with $b(g,t)=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-034269 | MATHDB-377354

Equality of geometric and representation-theoretic invariants for hyperbolic-like triples

摘录 / EXCERPT: Let (W,L,ρ) be a hyperbolic-like triple, and let ${\mathcal T}_{{\mathcal I}}$ be an ${\mathcal I}$-triangulation of (W,L,ρ). Denote by $c_{{\mathcal I}}(W,L,\rho)$ the c...

conjecture | solved | MathDB 页面 | 原文线索

V2-034270 | MATHDB-377359

The conjecture that all knots are H-restricted

摘录 / EXCERPT: Let K be a knot, and let $\mathcal{C}(K)$ be its complex of graded modules over the algebra $\mathcal{A}$. The knot is H-restricted when the non-acyclic indecomposable summands o...

conjecture | open | MathDB 页面 | 原文线索

V2-034271 | MATHDB-377370

Higher Hopf invariant one conjecture for A_m-spaces

摘录 / EXCERPT: Let G be an A_m-space, and let $P^m(G)$ be the G-projective m-space associated with its given A_m-structure in the sense of Stasheff. Let $\bar{E}^{m+1}(G)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-034272 | MATHDB-377371

Product formula conjecture for Lusternik-Schnirelmann categories of compact Lie groups

摘录 / EXCERPT: Let G be a simply-connected compact Lie group with ... where each H_i is a simple Lie group. Product formula conjecture. The weak category, ordinary category, and strong category...

conjecture | open | MathDB 页面 | 原文线索

V2-034273 | MATHDB-377372

The conjecture that the Lusternik-Schnirelmann category of symplectic groups is $2n-1$

摘录 / EXCERPT: For the compact symplectic group $Sp(n)$, let $\operatorname{cat}(Sp(n))$ denote its Lusternik-Schnirelmann category. Symplectic-group category conjecture. ... This is described in t...

conjecture | open | MathDB 页面 | 原文线索

V2-034274 | MATHDB-377381

Ihara's isomorphism conjecture for the stable derivation algebra

摘录 / EXCERPT: Let $\mathfrak g$ be the graded ${\mathbb Z}_p$-Lie algebra associated with the weight filtration on the pro-p Galois action, let $\mathcal D$ be Ihara's stable derivation algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-034275 | MATHDB-377382

Intersection-cohomology stalk conjecture for Uhlenbeck flag spaces

摘录 / EXCERPT: Intersection-cohomology stalk conjecture. Its stalk at $(\phi,\underline{c},\underline{s})$ is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-034276 | MATHDB-377387

The conjectural cohomology-ring presentation for stable maps of degree three

摘录 / EXCERPT: Cohomology-ring presentation conjecture. The de Rham cohomology ring is

conjecture | open | MathDB 页面 | 原文线索

V2-034277 | MATHDB-377396

Givental's conjecture identifying the Floer and quantum cohomology Heisenberg modules

摘录 / EXCERPT: Givental's conjecture. The Floer $\mathcal{D}$-module and the quantum cohomology $\mathcal{D}$-module are the same; equivalently, $F=G$, with the components of F giving appropria...

conjecture | open | MathDB 页面 | 原文线索

V2-034278 | MATHDB-377397

Weinberger's homotopy invariance conjecture for the rho-invariant

摘录 / EXCERPT: Let M be a closed $(2k-1)$-manifold with torsion-free fundamental group, and let α be a unitary representation used to define the rho-invariant $\rho(M,\alpha)$. Weinberge...

conjecture | refuted | MathDB 页面 | 原文线索

V2-034279 | MATHDB-377398

Weinberger's boundary-relative homotopy invariance conjecture for the rho-invariant

摘录 / EXCERPT: Let $F\colon X\to X'$ be a homotopy equivalence of manifolds with torsion-free fundamental groups that restricts to a diffeomorphism on the boundary. Equip the boundaries with Riem...

conjecture | open | MathDB 页面 | 原文线索

V2-034280 | MATHDB-377400

The simplicial-volume form of Kashaev's conjecture

摘录 / EXCERPT: For any link L, let $J_N(L)$ be the colored Jones polynomial evaluated at $\exp\left(2\pi\sqrt{-1}/N\right)$. Let $\Vert L\Vert$ denote the simplicial volume of the complement of...

conjecture | open | MathDB 页面 | 原文线索

V2-034281 | MATHDB-377409

Charge-and-tightness conjecture for the R^p ropelength energies

摘录 / EXCERPT: Let L be a link, let $R^p(L)$ be the scale-invariant energy defined in the paper, and let $R(L)$ denote ropelength, with thickness measuring the radius of the largest embedded tu...

conjecture | open | MathDB 页面 | 原文线索

V2-034282 | MATHDB-377415

Kronheimer-Manolescu's Seiberg-Witten-Ozsváth-Szabó Floer homology conjecture

摘录 / EXCERPT: Let Y be a three-manifold with $b_1(Y)=1$ and a nontorsion $spin^c$ structure ${\mathfrak{s}}$. Let ${\text{SWF}}$ and ${\text{SWF}}_0$ be the associated periodic Seiberg-Witten...

conjecture | open | MathDB 页面 | 原文线索

V2-034283 | MATHDB-377416

The p-modular TQFT structure conjecture

摘录 / EXCERPT: The p-modular TQFT structure conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034284 | MATHDB-377417

Gilmer's cut-number conjecture

摘录 / EXCERPT: Gilmer's cut-number conjecture. For every closed, connected 3-manifold M,

conjecture | open | MathDB 页面 | 原文线索

V2-034285 | MATHDB-377418

Ozsváth-Szabó's spectral sequence differential conjecture for torsion \a0Spin^c three-manifolds

摘录 / EXCERPT: Spectral sequence differential conjecture. The spectral sequence collapses after the E_3 stage, and its differential

conjecture | open | MathDB 页面 | 原文线索

V2-034286 | MATHDB-377426

Orlik-Randell conjecture on cohomology of generic arrangement Milnor fibers

摘录 / EXCERPT: Orlik-Randell conjecture. For the fiber $X_1=\operatorname{Var}(Q-1)$, there is a finite-dimensional homogeneous vector space $U\subset R_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-034287 | MATHDB-377433

Existence of alternating quadrisecants for nontrivial knots

摘录 / EXCERPT: A quadrisecant of a knot is a straight line that intersects the knot four times. It is alternating when the four intersection points alternate between the north and south poles of...

conjecture | open | MathDB 页面 | 原文线索

V2-034288 | MATHDB-377435

The double midset conjecture for continua

摘录 / EXCERPT: A continuum is a non-degenerate connected compact metric space. For a metric space, the double midset property (DMP) means that every pair of distinct points has a midset consistin...

conjecture | open | MathDB 页面 | 原文线索

V2-034289 | MATHDB-377436

The higher-dimensional double midset conjecture

摘录 / EXCERPT: Let X be a non-degenerate compact metric space, and suppose that every midset of X is homeomorphic to the $(n-1)$-sphere S^{n-1}. Higher-dimensional double midset conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-034290 | MATHDB-377442

Taubes's symplectic-fibration conjecture for 3-manifolds

摘录 / EXCERPT: Let N be a 3-manifold, and let $S^{1}\times N$ admit a symplectic structure ω. Taubes's conjecture. Then N admits a fibration over S^{1}. This conjecture relates sy...

conjecture | open | MathDB 页面 | 原文线索

V2-034291 | MATHDB-377443

Künneth-component fibered-cone conjecture for symplectic products

摘录 / EXCERPT: Let N be a 3-manifold and let $S^{1}\times N$ admit a symplectic structure ω. Write the Künneth component of $[\omega]$ in $H^{1}(N,\mathbb R)$ as the corresponding cla...

conjecture | open | MathDB 页面 | 原文线索

V2-034292 | MATHDB-377471

The cuspidal cohomology decomposition conjecture for GL_r

摘录 / EXCERPT: Fix a cocompact lattice $J\subset Z(\mathbbA)/Z(F)$ and an admissible n-tuple $\overline{\omega}$. Let ... be the cuspidal intersection cohomology representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034293 | MATHDB-377497

The unbounded difference conjecture for unknotting and slicing genus

摘录 / EXCERPT: Let K be a knot, let $u_s(K)$ denote its slicing number, and let $g_s(K)$ denote its slice genus. Unbounded difference conjecture. The difference ... can be arbitrarily large. This...

conjecture | open | MathDB 页面 | 原文线索

V2-034294 | MATHDB-377499

The independence-of-ll conjecture for arithmetic monodromy groups and their representations

摘录 / EXCERPT: Let X be a smooth curve over a finite field of characteristic p, let E be a number field, and let ${\mathbf{L}}=\{{\mathcal{L}}_\lambda\}$ be an E-compatible system of liss...

conjecture | open | MathDB 页面 | 原文线索

V2-034295 | MATHDB-377500

The open-closed isomorphism conjecture for Fukaya categories

摘录 / EXCERPT: Let M be an exact symplectic manifold satisfying Assumption, and let ... be the natural open-closed string map from symplectic cohomology to the Hochschild cohomology of the Fukaya...

conjecture | open | MathDB 页面 | 原文线索

V2-034296 | MATHDB-377509

Existence of knots distinguished by higher-order bordism invariants

摘录 / EXCERPT: Existence conjecture. For each $n\geq 0$ there exist knots K and K_* such that $\mathcal{A}_i(K)\cong\mathcal{A}_i(K_*)$ for $0\leq i<n$, but

conjecture | open | MathDB 页面 | 原文线索

V2-034297 | MATHDB-377543

Exact-sequence conjecture relating homology-cylinder filtrations

摘录 / EXCERPT: Let H be the first homology group used to define the tree and homology-cylinder filtrations, and let $\mathcal G_k^Y(\mathcal H_g)$ and $\mathcal G_k^w(\mathcal H_g)$ denote thei...

conjecture | open | MathDB 页面 | 原文线索

V2-034298 | MATHDB-377544

The Harder-Borel quasi-isomorphism conjecture for generalized automorphic complexes

摘录 / EXCERPT: Harder-Borel conjecture. The inclusions

conjecture | solved | MathDB 页面 | 原文线索

V2-034299 | MATHDB-377546

Consistency of distinct selection principles for tau-covers

摘录 / EXCERPT: Let X be a topological space. Write ${\sf S}_1(\{\mathcal O_n\}_{n\in\mathbb{N}),\mathrm{T})$ for the property that, from every sequence of covers $\mathcal O_n$ of the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-034300 | MATHDB-377574

The conjecture on algebra structures of K(1) at p=2

摘录 / EXCERPT: For $p=2$, let $K(1)$ be the spectrum considered as an algebra over the completed connective complex K-theory spectrum $\widehat{KU}_2$, and let an opposite algebra structure mea...

conjecture | open | MathDB 页面 | 原文线索

V2-034301 | MATHDB-377577

Binary-composition conjecture for strongly regular slices of contractible operads

摘录 / EXCERPT: Binary-composition conjecture. Every weak n-category is weakly equivalent to an A-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-034302 | MATHDB-377580

Koszulness of ideals generated by two-dimensional subspaces

摘录 / EXCERPT: Let F be a field and let $\Theta\subset H^1(G_F,\mathbb Z/2)$ be a two-dimensional subspace. Let I_Θ denote the ideal generated by Θ in the cohomology algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-034303 | MATHDB-377591

Fontaine-Perrin-Riou's local epsilon-conjecture for crystalline representations

摘录 / EXCERPT: La conjecture $C_{EP,F}(V)$. Pour tout $\omega\in\Delta_F(V)$, on a

conjecture | open | MathDB 页面 | 原文线索

V2-034304 | MATHDB-377602

The derivation-space conjecture for components of multiplicative mapping spaces

摘录 / EXCERPT: Let R be a commutative ring spectrum, let A and B be R-algebras, and let $f:A\rightarrow B$ be an R-algebra map. Write $F_{R\text{-alg}}(A,B)$ for the space of R-algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-034305 | MATHDB-377603

The Whitehead-product conjecture for the derivation bracket

摘录 / EXCERPT: Let A be an R-algebra. Write $F_{R\text{-alg}}(A,A)$ for the monoid of R-algebra self-maps, $B F_{R\text{-alg}}(A,A)$ for its classifying space, and ${\bf Der}_R(A,A)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-034306 | MATHDB-377610

The connectivity conjecture for toric fibrations

摘录 / EXCERPT: Let (M,ω) be a symplectic four-manifold equipped with toric fibrations, and let an almost toric fibration mean a fibration of (M,ω) obtained by allowing the nodal sin...

conjecture | open | MathDB 页面 | 原文线索

V2-034307 | MATHDB-377652

General Floer homology and asymptotic growth conjecture for mapping classes

摘录 / EXCERPT: General Floer homology and asymptotic growth conjecture.

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V2-034308 | MATHDB-377670

Equivariant Seiberg-Witten Floer and Heegaard Floer homology correspondence

摘录 / EXCERPT: Equivariant Seiberg-Witten Floer and Heegaard Floer correspondence. For every such Y and $\mathfrak{s}$, there are isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-034309 | MATHDB-377682

Conjecture on geometric deformations and the Hoste-Przytycki module

摘录 / EXCERPT: Geometric deformation conjecture. (a) For each oriented 3-manifold, the epimorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034310 | MATHDB-377683

Conjectures on the comparison of geometric deformation modules

摘录 / EXCERPT: Comparison conjectures. (a) The projection

conjecture | open | MathDB 页面 | 原文线索

V2-034311 | MATHDB-377686

Morimoto's tunnel-number conjecture for connected sums of knots

摘录 / EXCERPT: Morimoto's conjecture. The knots $K_1\subset S^3$, $K_2\subset S^3$, and $K_1\#K_2$ satisfy

conjecture | open | MathDB 页面 | 原文线索

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Peskine-Szpiro dimension inequality conjecture

摘录 / EXCERPT: Peskine-Szpiro conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-034313 | MATHDB-377714

The equivariant multiplicative McKay conjecture

摘录 / EXCERPT: Let V be a symplectic vector space, let G act symplectically on V, and let $X\to V/G$ be a symplectic resolution. The natural dilation action of $\mathbb C^*$ on V lifts to...

conjecture | open | MathDB 页面 | 原文线索

V2-034314 | MATHDB-377740

Finite generation of Ext algebras for finite tensor categories

摘录 / EXCERPT: Let ${\mathcal C}$ be a finite tensor category, let ${\bf 1}$ be its tensor unit, and let $X\in{\mathcal C}$. Write ${\rm Ext}^*({\bf 1},{\bf 1})$ for the graded Ext algebra and...

conjecture | open | MathDB 页面 | 原文线索

V2-034315 | MATHDB-377744

The Gorenstein splice-type conjecture for homology-sphere links

摘录 / EXCERPT: Let (X,o) be a Gorenstein surface singularity with integral homology sphere link, and let its splice diagram encode the topology of the link. Gorenstein splice-type conjecture....

conjecture | solved | MathDB 页面 | 原文线索

V2-034316 | MATHDB-377754

Frobenius test-module conjecture for finite projective dimension

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be a local ring of characteristic $p>0$, let $n>0$, and regard R as the R-module ${}^{\varphi^n}\!R$ via the nth Frobenius map. Let M be an R-modu...

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V2-034317 | MATHDB-377756

The conjecture that comodule equivalence is generated by weak equivalences

摘录 / EXCERPT: Let (A,Γ) and (B,Σ) be flat Hopf algebroids, and let a weak equivalence mean a morphism of flat Hopf algebroids inducing an equivalence between their comodule categor...

conjecture | open | MathDB 页面 | 原文线索

V2-034318 | MATHDB-377766

The Wedderburn-projective formulation of the Idempotent Nakayama Conjecture

摘录 / EXCERPT: Let B be a Noetherian algebra, and let N be a Wedderburn projective B-module. Write $\tau_N(B)$ for the trace of N in B. Wedderburn-projective Idempotent Nakayama Conject...

conjecture | open | MathDB 页面 | 原文线索

V2-034319 | MATHDB-377793

Przytycki's 4-move conjecture

摘录 / EXCERPT: Przytycki's 4-move conjecture. Any knot is 4-equivalent to the unknot.

conjecture | open | MathDB 页面 | 原文线索

V2-034320 | MATHDB-377794

Achiral-generator conjecture for alternating knot series

摘录 / EXCERPT: Achiral-generator conjecture. Then at least one of the following holds: K is achiral; K is an iterated mutant of its obverse; or K has self-conjugate HOMFLY and/or Kauffman p...

conjecture | open | MathDB 页面 | 原文线索

V2-034321 | MATHDB-377795

Cromwell's minimal-genus composite-diagram conjecture

摘录 / EXCERPT: Cromwell's conjecture. Then D is composite.

conjecture | open | MathDB 页面 | 原文线索

V2-034322 | MATHDB-377796

The universal finite-type invariant conjecture for the embedding approximation

摘录 / EXCERPT: Universal finite-type invariant conjecture. The map

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V2-034323 | MATHDB-377799

Cyclic cohomology conjecture for commutative extended Hopf algebras

摘录 / EXCERPT: Cyclic cohomology conjecture. The cyclic cohomology of (H,R) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-034324 | MATHDB-377802

The pull-back conjecture for semi-free equivariant bordism

摘录 / EXCERPT: Pull-back conjecture. The square

conjecture | open | MathDB 页面 | 原文线索

V2-034325 | MATHDB-377805

The conjectural formula for Kontsevich cycles $[W_{n,k}^\ast]$

摘录 / EXCERPT: Let $W_{n,k}^\ast$ be the Kontsevich cycle indexed by nonnegative integers n and k, and let $\widetilde{\kappa}_j$ denote the corresponding modified kappa class. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-034326 | MATHDB-377816

The strata-generation conjecture for Hurwitz-space cohomology

摘录 / EXCERPT: Let ${\mathbb P}{\cal H}_n$ be the projectivized Hurwitz space, and let C_n and M_n be the cohomology classes of the caustic and Maxwell stratum, respectively. Consider the sub...

conjecture | open | MathDB 页面 | 原文线索

V2-034327 | MATHDB-377817

The polynomiality conjecture for strata classes in Hurwitz spaces

摘录 / EXCERPT: Let ${\mathbb P}{\cal H}_n$ be the projectivized Hurwitz space. Let $\Psi_n,\Delta_n\in H^2({\mathbb P}{\cal H}_n)$, where Ψ_n is the distinguished class used for intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-034328 | MATHDB-377822

Murasugi sum conjecture for the braid index of special alternating links

摘录 / EXCERPT: Let L) be the planar Murasugi sum of special alternating links $L_1,\ldots,L_n$. Murasugi sum conjecture. Then ... This conjecture was proposed in connection with the analogous pro...

conjecture | open | MathDB 页面 | 原文线索

V2-034329 | MATHDB-377823

Stoimenow-Kidwell conjecture on the Kauffman polynomial of Whitehead doubles

摘录 / EXCERPT: Let K be a knot, let W_K be its Whitehead double, and let $F(K)$ be the Kauffman polynomial of K in the variables a and z. Stoimenow-Kidwell conjecture. Then ... The questi...

conjecture | open | MathDB 页面 | 原文线索

V2-034330 | MATHDB-377830

Cord-ring conjecture for unit conormal contact homology

摘录 / EXCERPT: Let K be a codimension 2 submanifold of a manifold M, let LK be its unit conormal Legendrian submanifold in the contact manifold ST^*M of unit covectors, and let the cord...

conjecture | open | MathDB 页面 | 原文线索

V2-034331 | MATHDB-377842

Cell-structure conjecture for finite subset spaces of cell complexes

摘录 / EXCERPT: Finite subset-space cell-structure conjecture. The space $\exp_k X$ has a cell structure obtained from $\exp_{k-1} X$ by adding cells of dimensions $k-c\leq i\leq nk$.

conjecture | open | MathDB 页面 | 原文线索

V2-034332 | MATHDB-377845

Conjecture on abelianized monodromy invariants for symplectic branched covers

摘录 / EXCERPT: Let (X,ω) be a simply connected symplectic 4-manifold. For the degree-N_k symplectic branched covering associated with sufficiently large k, let G_k^0 be the relevan...

conjecture | open | MathDB 页面 | 原文线索

V2-034333 | MATHDB-377877

Galois descent conjecture for topological restriction homology

摘录 / EXCERPT: Let K be a field with valuation ring V, residue field k, and let $\bar{K}$ be an algebraic closure of K with Galois group G_K. Let $\bar{V}$ be the integral closure of...

conjecture | open | MathDB 页面 | 原文线索

V2-034334 | MATHDB-377888

Nonexistence of robustly transitive diffeomorphisms on the 3-sphere

摘录 / EXCERPT: Let S^3 denote the three-dimensional sphere. A diffeomorphism is robustly transitive if its transitivity persists under sufficiently small perturbations in the C^1 topology. Sp...

conjecture | open | MathDB 页面 | 原文线索

V2-034335 | MATHDB-377893

Colorability implies bounded-degree disconnectedness of graph homomorphism spaces

摘录 / EXCERPT: Let H be a graph that is k-colorable. For a finite graph G, let ${\rm Hom}(G,H)$ denote the space of graph homomorphisms from G to H, with adjacency given by changing one...

conjecture | open | MathDB 页面 | 原文线索

V2-034336 | MATHDB-377913

The hyperwinding family conjecture for simply connected symplectic four-manifolds

摘录 / EXCERPT: Hyperwinding family conjecture. M is diffeomorphic to the underlying smooth manifold of a $K3$ surface, and the hyperwinding family of symplectic forms is homotopic, through...

conjecture | open | MathDB 页面 | 原文线索

V2-034337 | MATHDB-377918

Feder-Gitler conjecture on degrees of self maps of quaternionic projective spaces

摘录 / EXCERPT: Feder-Gitler conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034338 | MATHDB-377936

Rosenberg's positive scalar curvature conjecture for odd-order fundamental groups

摘录 / EXCERPT: Let M^n be a connected closed manifold whose fundamental group $\pi_1(M)$ is finite of odd order. A metric of positive scalar curvature on the universal cover of M need not des...

conjecture | open | MathDB 页面 | 原文线索

V2-034339 | MATHDB-377946

The AJ conjecture for knots

摘录 / EXCERPT: Let K be a knot in S^3. Let $A(K)(L,M)$ denote its A-polynomial, and let $A_q(K)$ denote the A_q-polynomial of the recursion ideal of its colored Jones function. Under the...

conjecture | open | MathDB 页面 | 原文线索

V2-034340 | MATHDB-377949

Grodal's conjecture on the cohomology of simply connected polyGEMs

摘录 / EXCERPT: Let X be a simply connected polyGEM of finite type at 2, and let $\widetilde{\operatorname{H}}^*X$ denote its reduced cohomology with coefficients in $\mathbb{F}_2$. Grodal's c...

conjecture | open | MathDB 页面 | 原文线索

V2-034341 | MATHDB-377959

The generic Hochschild cohomology conjecture for rank-one double affine Hecke algebras

摘录 / EXCERPT: Let $H(\underline{t};q)$ be the rank-one double affine Hecke algebra with parameters q and $\underline{t}$. Here HH^i denotes its i-th Hochschild cohomology group, and...

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V2-034342 | MATHDB-377968

Torsion characterization by stable torsion length

摘录 / EXCERPT: Stable torsion length conjecture. If

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V2-034343 | MATHDB-377979

Topological invariance of rationality for threefold Mori fibre spaces

摘录 / EXCERPT: Let X/S and Y/T be threefold Mori fibre spaces, with total spaces X and Y diffeomorphic. Topological rationality conjecture. If X is rational, then Y is also rational....

conjecture | open | MathDB 页面 | 原文线索

V2-034344 | MATHDB-377980

Topological invariance of rigidity for threefold Mori fibre spaces

摘录 / EXCERPT: Let X/S and Y/T be threefold Mori fibre spaces, and let rigidity mean birational rigidity of the Mori fibre space. Topological rigidity conjecture. Whether a threefold Mori fib...

conjecture | open | MathDB 页面 | 原文线索

V2-034345 | MATHDB-377981

Topological invariance of pliability for threefold Mori fibre spaces

摘录 / EXCERPT: Let X/S be a threefold Mori fibre space, and let $\mathcal{P}(X/S)$ denote its pliability. Topological pliability conjecture. The pliability $\mathcal{P}(X/S)$ is, in a natural w...

conjecture | open | MathDB 页面 | 原文线索

V2-034346 | MATHDB-377987

Distinctness conjecture for contact structures from Seifert surgery diagrams

摘录 / EXCERPT: Distinctness conjecture. All the tight contact structures defined by Figure and satisfying the assumptions are distinct up to isotopy.

conjecture | open | MathDB 页面 | 原文线索

V2-034347 | MATHDB-377988

Nonfillability conjecture for contact structures from Seifert surgery diagrams

摘录 / EXCERPT: Nonfillability conjecture. No contact structure defined by Figure and satisfying conditions is fillable.

conjecture | open | MathDB 页面 | 原文线索

V2-034348 | MATHDB-377997

Connectedness conjecture for real spherical tight-frame spaces

摘录 / EXCERPT: Let $\mathcal{F}^{\mathbf R}_{k,n}$ denote the space of real spherical tight frames with k frame vectors in dimension n. Connectedness conjecture. For $k,n\in\mathbf{N}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-034349 | MATHDB-378005

The compatibility of the modified HKR maps with Hochschild and harmonic structures

摘录 / EXCERPT: Let X be a compact smooth space. The modified Hochschild-Kostant-Rosenberg maps I^K and I_K are the maps obtained by twisting the Hochschild-Kostant-Rosenberg isomorphism by...

conjecture | open | MathDB 页面 | 原文线索

V2-034350 | MATHDB-378015

The finiteness conjecture for topological types of compact hyper-Kähler manifolds

摘录 / EXCERPT: For each fixed complex dimension, consider compact hyper-Kähler manifolds and identify two such manifolds when they are homeomorphic, equivalently when they have the same topologic...

conjecture | open | MathDB 页面 | 原文线索

V2-034351 | MATHDB-378036

The filtered Hilbert-series conjecture for the Grassmannian cohomology ring

摘录 / EXCERPT: The filtered Hilbert-series conjecture. For $m\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-034352 | MATHDB-378037

The filtration-saturation conjecture for the Grassmannian cohomology ring

摘录 / EXCERPT: The filtration-saturation conjecture. For $1\leq m\leq k$,

conjecture | open | MathDB 页面 | 原文线索

V2-034353 | MATHDB-378038

The exact e_m-killer conjecture for the Grassmannian cohomology ring

摘录 / EXCERPT: The exact e_m-killer conjecture. For $1\leq m\leq k$,

conjecture | open | MathDB 页面 | 原文线索

V2-034354 | MATHDB-378039

The e_m-overkiller conjecture for the Grassmannian cohomology ring

摘录 / EXCERPT: The e_m-overkiller conjecture. For $3\leq m\leq k$,

conjecture | open | MathDB 页面 | 原文线索

V2-034355 | MATHDB-378046

The cohomology-gap conjecture for elliptic minimal algebras

摘录 / EXCERPT: Let $(\Lambda V,d)$ be an elliptic minimal algebra whose differential has homogeneous length. Write ... where e is given by the formula of the theorem on homogeneous-length differe...

conjecture | open | MathDB 页面 | 原文线索

V2-034356 | MATHDB-378047

Sasakian-Einstein metrics on odd-dimensional homotopy spheres bounding parallelizable manifolds

摘录 / EXCERPT: Let M be an odd-dimensional homotopy sphere that bounds a parallelizable manifold. Sasakian-Einstein metric conjecture. Every such M admits a Sasakian-Einstein metric. This wou...

conjecture | open | MathDB 页面 | 原文线索

V2-034357 | MATHDB-378048

Halperin's conjecture on fibrations of F_0-spaces

摘录 / EXCERPT: Halperin's conjecture. The fibration is TNCZ, meaning that j induces a surjection on rational cohomology.

conjecture | open | MathDB 页面 | 原文线索

V2-034358 | MATHDB-378060

Mean curvature flow convergence conjecture for low-oscillation exact Lagrangians

摘录 / EXCERPT: Let (M,ω,Φ) be a Calabi-Yau n-fold, and let η be an n-dimensional homology class in M. For an oriented exact Lagrangian L in M, write ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-034359 | MATHDB-378065

Quantum nonnegative monomial-basis conjecture for crystallographic Coxeter systems

摘录 / EXCERPT: Let (W,S) be a crystallographic Coxeter system, let $qBE(W,S)$ be the associated quantum algebra, and let $\tilde{P}_w\in {\bf S}(V^{\ast})\otimes {\bf R}[q_s]$ be the polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-034360 | MATHDB-378079

Amenability characterized by non-Hausdorff first cohomology

摘录 / EXCERPT: Let G be an infinite, finitely generated group. Let M be a Banach space with norm $||\ ||_M$ that is also a left $\C G$-module, and suppose that G acts on M by continuous...

conjecture | open | MathDB 页面 | 原文线索

V2-034361 | MATHDB-378095

Strominger--Yau--Zaslow conjecture

摘录 / EXCERPT: Suppose M and $\hat M$ are mirror Calabi--Yau 3-folds. The conjecture asserts the existence, under some additional conditions, of a compact topological 3-manifold B and sur...

conjecture | open | MathDB 页面 | 原文线索

V2-034362 | MATHDB-378097

The second cohomology structure conjecture for differential operators on differential forms

摘录 / EXCERPT: Let $\mathrm{Vect}(\mathbb R^n)$ be the Lie algebra of vector fields on $\mathbb R^n$, let $\Omega(\mathbb R^n)$ denote the space of differential forms, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034363 | MATHDB-378099

Hoste-Shanahan commensurability conjecture for 2-bridge knots

摘录 / EXCERPT: Hoste-Shanahan's commensurability conjecture. The complement $\mathbb S^3-J(m,n)$ is commensurable to a fibered knot in a $\mathbb Z/2\mathbb Z$-homology sphere if and only if...

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V2-034364 | MATHDB-378127

Andrews-Curtis conjecture for contractible simple polyhedra

摘录 / EXCERPT: Andrews-Curtis conjecture. Any two contractible simple polyhedra can be related by a sequence of pentagon, hexagon, and 2-0 moves and their inverses, ignoring all gleams.

conjecture | open | MathDB 页面 | 原文线索

V2-034365 | MATHDB-378154

Toën's structure conjecture for Tannakian Segal categories

摘录 / EXCERPT: Let k be a field and let T be a k-Tannakian Segal category. Let $\mathcal{H}(T)$ denote a k-Tannakian category and let $Ind(\mathcal{H}(T))$ be its ind-category with its in...

conjecture | open | MathDB 页面 | 原文线索

V2-034366 | MATHDB-378155

Toën's schematization and Tannakian duality conjecture

摘录 / EXCERPT: Let k be a field and let $(X\otimes k)^{sch}\in LAff_k^{\sim,ffqc}$ be the schematic homotopy type associated with a connected finite CW complex X, and let $LParf(X,k)$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-034367 | MATHDB-378156

Toën's conjecture on schematic homotopy types and Tannakian gerbes

摘录 / EXCERPT: Let k be a field and let $F\in LAff_k^{\sim,ffqc}$ be a schematic homotopy type. Let $i(F)$ be its image among stacks over Hk, and let $LParf(i(F))$ be the associated k-tenso...

conjecture | open | MathDB 页面 | 原文线索

V2-034368 | MATHDB-378157

The Quillen equivalence conjecture for Segal categories and quasi-categories

摘录 / EXCERPT: Let $PrCat$ be the model category of Segal precategories, let $QCat$ be the model category of quasi-categories, and let $\phi:PrCat\to QCat$ and $\psi:QCat\to PrCat$ be the adjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-034369 | MATHDB-378158

Generalized correction-term bounds and the diagonal intersection form conjecture

摘录 / EXCERPT: Let Y be a rational homology sphere with $|H_1(Y;\mathbb Z)|=h$. Suppose that Y bounds a negative-definite four-manifold X whose first homology has no torsion. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-034370 | MATHDB-378163

Novikov-homology criterion for fibred links

摘录 / EXCERPT: Let L be a link, let C_L be its exterior, and let $\xi:\pi_1C_L\to\mathbf{Z}$ be the homomorphism sending each meridian generator to 1. Let Λ be the group ring of...

conjecture | open | MathDB 页面 | 原文线索

V2-034371 | MATHDB-378168

Neveu-Schwarz and Ramond sector correspondence conjecture

摘录 / EXCERPT: Sector correspondence conjecture. The Neveu-Schwarz sectors are in one-to-one correspondence with the one-dimensional twisted sectors, while the Ramond sectors are in one-to-one co...

conjecture | open | MathDB 页面 | 原文线索

V2-034372 | MATHDB-378169

Mod-p cohomological rigidity conjecture for classifying spaces of compact Lie groups

摘录 / EXCERPT: Let G be a compact connected Lie group, let X be a p-complete space, and let $\mathcal{A}_p$ denote the mod-p Steenrod algebra. Suppose that ... as algebras over...

conjecture | open | MathDB 页面 | 原文线索

V2-034373 | MATHDB-378179

The universal $(2,2)$-move reduction conjecture for links

摘录 / EXCERPT: A $(2,2)$-move is the local move shown in the source; a 5-move is described there as a combination of $(2,2)$-moves and their mirror-image $(-2,-2)$-moves. Universal $(2,2)$-move...

conjecture | open | MathDB 页面 | 原文线索

V2-034374 | MATHDB-378221

Euclidean Dehn complex cohomology conjecture

摘录 / EXCERPT: Let F be an algebraically closed field, and let ${\cal E}_{(n)}^{\ast}(F)$ be the degree-n Euclidean Dehn complex of Euclidean scissor congruence groups. For an F-vector spac...

conjecture | open | MathDB 页面 | 原文线索

V2-034375 | MATHDB-378226

The generalized Waldhausen conjecture for strongly irreducible splittings

摘录 / EXCERPT: Let M be a closed, orientable, atoroidal three-manifold. A Heegaard splitting is strongly irreducible if every compressing disk on one side intersects every compressing disk on t...

conjecture | open | MathDB 页面 | 原文线索

V2-034376 | MATHDB-378227

Finiteness conjecture for full Heegaard splittings

摘录 / EXCERPT: Let M be a closed, orientable three-manifold. A Heegaard splitting is full when it does not have the disjoint curve property in the terminology of the source. Finiteness conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-034377 | MATHDB-378236

Morimoto's conjecture on tunnel number and primitive meridians

摘录 / EXCERPT: Let K and K' be knots, let $K\#K'$ denote their connected sum, and let $t(K)$ denote the tunnel number. A knot admits a primitive meridian when its exterior has a minimal genus...

conjecture | open | MathDB 页面 | 原文线索

V2-034378 | MATHDB-378260

Binding conjecture for open books with prescribed negative twists

摘录 / EXCERPT: Let $\mathcal M_k$, for $k\geq 0$, be the set of closed, oriented, smooth 3-manifolds admitting open book decompositions whose monodromies contain at most k negative Dehn twist...

conjecture | open | MathDB 页面 | 原文线索

V2-034379 | MATHDB-378269

The generalized Singer conjecture for weighted L^2-homology

摘录 / EXCERPT: Let (W,S) be a Coxeter system with nerve L, let $ Sigma$ be its associated complex, and suppose that L is a generalized homology sphere of dimension $n-1$, written...

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V2-034380 | MATHDB-378294

Formal equivalence conjecture for extensions of codifferentials

摘录 / EXCERPT: Let $d=d_2+d_k$ be a codifferential, and let $d_e=d_2+d_k+d_l+\cdots$ and $d_e'=d_2+d_k+d_l'+\cdots$ be extensions. Two codifferentials are formally equivalent when a formal automo...

conjecture | open | MathDB 页面 | 原文线索

V2-034381 | MATHDB-378299

Chen's smooth product conjecture for elliptic 3-manifold s-cobordisms

摘录 / EXCERPT: Let M be an elliptic 3-manifold, and let W be a smooth s-cobordism from M to itself. Let $\widetilde{W}$ be its universal cover. Chen's smooth product conjecture. W is...

conjecture | open | MathDB 页面 | 原文线索

V2-034382 | MATHDB-378301

Cohomological rigidity conjecture for Lie algebroids with proper integrations

摘录 / EXCERPT: Let A be a Lie algebroid that admits a proper integrating groupoid $\mathcal{G}$ whose s-fibers are 2-simply connected. Cohomological rigidity conjecture. Then ... This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-034383 | MATHDB-378306

The naive Baum-Connes conjecture in KK-theory

摘录 / EXCERPT: Let G be a topological group with a universal proper space $\mathcal E G$. Let A be a C^*-algebra and let B be a G-algebra. Define the naive topological KK-groups by .....

conjecture | open | MathDB 页面 | 原文线索

V2-034384 | MATHDB-378365

The sheaf Lefschetz injectivity conjecture for irreducible holomorphic symplectic fourfolds

摘录 / EXCERPT: Let X be an irreducible holomorphic symplectic 4-fold, let E be a nef line bundle on X, and let L^* denote the map in the sheaf Lefschetz action. The sheaf Lefschetz inje...

conjecture | open | MathDB 页面 | 原文线索

V2-034385 | MATHDB-378396

The Koszul propérade bar-to-levelled composition quasi-isomorphism

摘录 / EXCERPT: Koszul bar-to-levelled conjecture. The composition

conjecture | open | MathDB 页面 | 原文线索

V2-034386 | MATHDB-378398

The symplectic Khovanov homology conjecture

摘录 / EXCERPT: Let κ be an oriented link, represented as the closure of an m-stranded braid, and let $\mathit{Kh}_{\mathrm{symp}}^*(\kappa)$ denote the graded symplectic invariant define...

conjecture | open | MathDB 页面 | 原文线索

V2-034387 | MATHDB-378404

Moore's conjecture for strongly graded rings

摘录 / EXCERPT: Moore's conjecture. Assume that for every nontrivial element x in Γ, at least one of the following conditions holds: $M1)$ $\langle x\rangle\cap H\neq\{e\}$; or $M2)$...

conjecture | open | MathDB 页面 | 原文线索

V2-034388 | MATHDB-378408

Conjecture on the middle-degree Bockstein and injectivity of the second Bockstein

摘录 / EXCERPT: Bockstein conjecture. For N even,

conjecture | open | MathDB 页面 | 原文线索

V2-034389 | MATHDB-378435

The homology-sphere $(1+t)$-torsion conjecture

摘录 / EXCERPT: Let M be an integral homology sphere, let $K(M)$ denote its Kauffman bracket skein module, and let $\overline{K(M)}$ be its completion. Homology-sphere $(1+t)$-torsion conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-034390 | MATHDB-378436

Generalised Links-Gould polynomial conjecture for the Alexander polynomial

摘录 / EXCERPT: Let L be an oriented link, let m,n be positive integers, let $LG^{m,n}_L(\tau,t)$ denote the Links-Gould invariant, and let Δ_L denote the Alexander polynomial. Generali...

conjecture | open | MathDB 页面 | 原文线索

V2-034391 | MATHDB-378437

GLZ invariant conjecture for the Alexander polynomial

摘录 / EXCERPT: Let L be an oriented link, let n be a positive integer, let $GLZ^n_L(\tau,t)$ denote the invariant associated with the quantum superalgebra $U_q[\operatorname{osp}(2|2n)]$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034392 | MATHDB-378439

Jin-Rolfsen conjecture on Alexander polynomials of rotant links

摘录 / EXCERPT: Let L be an oriented link, and let $r(L)$ denote the link obtained from L by rotation of its rotor. The Jin-Rolfsen conjecture. The Alexander polynomials of L and $r(L)$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-034393 | MATHDB-378442

Torsion existence conjecture for non-split links

摘录 / EXCERPT: Torsion existence conjecture. Except for the trivial knot, the Hopf link, and their connected sums, the Khovanov homology of L has 2-torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-034394 | MATHDB-378443

Powers-of-two torsion conjecture for Khovanov homology

摘录 / EXCERPT: Powers-of-two torsion conjecture. The Khovanov homology of every link has no torsion of order p for any p other than a power of 2.

conjecture | open | MathDB 页面 | 原文线索

V2-034395 | MATHDB-378444

H-thin implies T-thin conjecture for Khovanov homology

摘录 / EXCERPT: H-thin implies T-thin conjecture. Every H-thin link is T-thin. In particular, every non-split alternating link is T-thin.

conjecture | open | MathDB 页面 | 原文线索

V2-034396 | MATHDB-378445

Reduced Khovanov torsion conjecture for T-rich knots

摘录 / EXCERPT: Reduced-torsion conjecture. The knot K is T-rich if and only if its reduced Khovanov homology has torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-034397 | MATHDB-378449

Greenberg's vanishing conjecture for the second cohomology group

摘录 / EXCERPT: Let F be a number field, let F_{∞} be its cyclotomic $\mathbf{Z}_{p}$-extension, let V be a critical nearly ordinary Galois representation, and let $A=V/T$ be a torsion...

conjecture | open | MathDB 页面 | 原文线索

V2-034398 | MATHDB-378470

Stable diffeomorphism conjecture for torsion-free fundamental groups

摘录 / EXCERPT: Let M and N be closed, orientable, smooth 4-manifolds that are homotopy equivalent and have torsion-free fundamental group. Stable diffeomorphism conjecture. Then M and N...

conjecture | open | MathDB 页面 | 原文线索

V2-034399 | MATHDB-378471

Stable homeomorphism conjecture for torsion-free fundamental groups

摘录 / EXCERPT: Let M and N be closed, orientable, topological 4-manifolds that are homotopy equivalent, have torsion-free fundamental group, and have the same Kirby-Siebenmann invariant. St...

conjecture | open | MathDB 页面 | 原文线索

V2-034400 | MATHDB-378474

Kidwell-Stoimenow's degree conjecture for Whitehead doubles

摘录 / EXCERPT: Let K) be a non-trivial knot, and let $W(K)$ be a Whitehead double of K. The Kidwell-Stoimenow conjecture. Is it always the case that ... Kidwell and Stoimenow reported that the...

conjecture | open | MathDB 页面 | 原文线索

V2-034401 | MATHDB-378475

The annular-link degree conjecture for Rudolph polynomials

摘录 / EXCERPT: Let $A(K,f)$ be a framed link, where K is its underlying link and f is its framing. Let $\{K,f\}$ denote the framed-link polynomial used in the source, and let F_K be the Kau...

conjecture | open | MathDB 页面 | 原文线索

V2-034402 | MATHDB-378492

Singh's polynomial-ring conjecture on p-torsion in local cohomology

摘录 / EXCERPT: Singh's conjecture. There exists $k\in\mathbb N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-034403 | MATHDB-378493

Lyubeznik's finite-associated-primes conjecture

摘录 / EXCERPT: Let R be a regular ring, let $\mathfrak{a}$ be an ideal of R, and let $H^i_\mathfrak{a}(R)$ denote its ith local cohomology module. Lyubeznik's conjecture. The module...

conjecture | open | MathDB 页面 | 原文线索

V2-034404 | MATHDB-378514

Non-isomorphism conjecture for generalized groups of the square and granny knots

摘录 / EXCERPT: Let $G_n(SK)$ and $G_n(GK)$ denote the generalized knot groups associated with the square knot SK and granny knot GK, respectively, for integers $n\geq 2$. Non-isomorphism conj...

conjecture | open | MathDB 页面 | 原文线索

V2-034405 | MATHDB-378533

Thurston's virtually fibered conjecture for three-manifolds

摘录 / EXCERPT: A fibered three-manifold is a three-manifold admitting a submersion $p:M\to S^1$; equivalently, it is a mapping torus ... for a surface F and an orientation-preserving isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-034406 | MATHDB-378537

Filling homotopy conjecture for regularly homotopic filling immersions

摘录 / EXCERPT: Filling homotopy conjecture. Regularly homotopic filling immersions of arbitrary surfaces are filling homotopic.

conjecture | open | MathDB 页面 | 原文线索

V2-034407 | MATHDB-378548

The higher-dimensional torus contact-structure fundamental group conjecture

摘录 / EXCERPT: Higher-dimensional torus conjecture. There exists a T^2-invariant contact structure ξ on T^{2n+1} such that

conjecture | open | MathDB 页面 | 原文线索

V2-034408 | MATHDB-378565

Hutchings's twice-integrability conjecture for invariants of singular knots

摘录 / EXCERPT: Let V^r be an invariant of r-singular knots. Say that V^r is once-integrable if there is an invariant V^{r-1} of $(r-1)$-singular knots whose derivative is V^r. Hutchings...

conjecture | open | MathDB 页面 | 原文线索

V2-034409 | MATHDB-378567

Rationality conjecture for bases of simply connected Seifert bundles

摘录 / EXCERPT: Let ... be a 5-dimensional Seifert bundle with L smooth. Assume that L is a simply connected rational homology sphere. Rationality conjecture. Then S is a rational surface. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-034410 | MATHDB-378570

The Gottlieb group conjecture for spheres in dimension difference eight

摘录 / EXCERPT: Proposed conjecture. For $n\equiv 6\pmod 8$ with $n\geq 14$,

conjecture | open | MathDB 页面 | 原文线索

V2-034411 | MATHDB-378594

The freeness conjecture for chessboard-complex homology

摘录 / EXCERPT: Freeness conjecture. The reduced homology group $\widetilde H_i(M_{m,n})$ is free if and only if $i\geq m-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-034412 | MATHDB-378595

The elementary-3 and cyclic-order conjecture for D_n homology

摘录 / EXCERPT: The D_n conjecture. There exists an integer $n_0\geq 8$ such that, if $n\geq n_0$, then $\widetilde H_{\nu_{2n}}(M(D_n))$ is an elementary 3-group. Moreover, if $n\geq n_0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-034413 | MATHDB-378603

Topological invariance of the sign of upper pointwise Lyapunov exponents

摘录 / EXCERPT: Let f and g be topologically conjugate one-dimensional maps, and let x be a point of f that is not attracted to a periodic orbit. The upper pointwise Lyapunov exponent of...

conjecture | open | MathDB 页面 | 原文线索

V2-034414 | MATHDB-378617

The abelian invariant-Alexander polynomial conjecture for links

摘录 / EXCERPT: Let $L\subset S^3$ be a link with N components, let $\Delta_L(t_1,\ldots,t_N)$ be its Alexander polynomial, and let $\rho\colon \pi L\to \operatorname{SO}(3)$ be an abelian repre...

conjecture | open | MathDB 页面 | 原文线索

V2-034415 | MATHDB-378634

The triple-tangency conjecture for the twice tangent sphere map

摘录 / EXCERPT: Let K be a knot and let $\psi_K^{(2,2)}$ be its twice tangent sphere map, which assigns to a pair of distinct points of K the sphere tangent to K at both points, wherever thi...

conjecture | open | MathDB 页面 | 原文线索

V2-034416 | MATHDB-378635

Positive lower-bound conjecture for conformal angle energy of non-trivial knots

摘录 / EXCERPT: Let K be a knot, and let $|csl|(K)$ denote the absolute value of its conformal linking energy. Positive lower-bound conjecture. There is a positive constant C such that, if K...

conjecture | open | MathDB 页面 | 原文线索

V2-034417 | MATHDB-378638

The conjecture on negative transverse curvature for integrable hyperplane distributions

摘录 / EXCERPT: Conjecture on negative transverse curvature. There is a Riemannian metric g on M such that the function $\chi_{g,\bot H}$ is everywhere negative.

conjecture | open | MathDB 页面 | 原文线索

V2-034418 | MATHDB-378669

Shareshian-Wachs conjecture for the matching complex homology representation

摘录 / EXCERPT: Let $n=2k$ be even. The complexes ${\mathsf {NPM}}_n$ and the poset ${\mathcal L}_n^{(2)}$ have reduced homology groups carrying actions of S_n, where ε_n denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-034419 | MATHDB-378670

Bipartite matching complex homology tensor-product conjecture

摘录 / EXCERPT: Let ${\mathsf {BNM}}_k(r,s)$ be the bipartite matching complex on two parts of sizes r and s, with the natural action of $S_r\times S_s$. Let $H(n,l)$ denote the irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-034420 | MATHDB-378671

Resolution conjecture for collections of abelian groups

摘录 / EXCERPT: Resolution conjecture. There exist a compactum Z with $\dim Z\leq n$ and a surjective map $r:Z\longrightarrow X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-034421 | MATHDB-378685

Dimension conjectures for chaotic magnetic-field directions

摘录 / EXCERPT: Let $M_Fcsubset\mathbb T^3$ be a connected smooth two-manifold homologous to zero, and let a magnetic field determine trajectories on it. For a smooth one-parameter family...

conjecture | open | MathDB 页面 | 原文线索

V2-034422 | MATHDB-378686

The Khovanov-basicity conjecture for alternating links

摘录 / EXCERPT: Let L be a link with c components. Call L Khovanov-basic if its Khovanov complex $\operatorname{Kh}(L)$ is homotopy equivalent to a direct sum of basic complexes and exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-034423 | MATHDB-378711

Cancellation conjecture for doubly slice knots

摘录 / EXCERPT: Cancellation conjecture. If knots J and $K\#J$ are doubly slice, then K is doubly slice.

conjecture | open | MathDB 页面 | 原文线索

V2-034424 | MATHDB-378721

Floer-theoretic string topology operations for marked chord diagrams

摘录 / EXCERPT: Floer-theoretic string topology operations conjecture. For every marked chord diagram Γ, there is an umkehr map

conjecture | open | MathDB 页面 | 原文线索

V2-034425 | MATHDB-378722

Legendre-transform equivalence for chord-diagram moduli spaces

摘录 / EXCERPT: Legendre-transform equivalence conjecture. The natural map induced by the Legendre transform,

conjecture | open | MathDB 页面 | 原文线索

V2-034426 | MATHDB-378725

Cohomological special effect conjecture for hypersurface linear systems

摘录 / EXCERPT: Cohomological special effect conjecture. The system ${\mathcal{L}}$ is special if and only if it is cohomologically special.

conjecture | open | MathDB 页面 | 原文线索

V2-034427 | MATHDB-378728

The K-contact energy-minimization conjecture for curl eigenforms

摘录 / EXCERPT: K-contact energy-minimization conjecture. The curl eigenform η defined by a K-contact structure is always energy-minimizing on Ξ_η.

conjecture | open | MathDB 页面 | 原文线索

V2-034428 | MATHDB-378729

Rudolph's characterization conjecture for quasipositive ribbons

摘录 / EXCERPT: Rudolph's characterization conjecture. A ribbon is quasipositive if and only if, for some n, it is ambient isotopic to a subribbon of $\widetilde T_{n,n}$. This would characteriz...

conjecture | open | MathDB 页面 | 原文线索

V2-034429 | MATHDB-378730

Finite self-similarity conjecture for finite simplicial complexes

摘录 / EXCERPT: A finite simplicial complex is the topological space obtained by gluing finitely many simplices along faces. A compact metrizable space is finitely self-similar if it arises from a...

conjecture | open | MathDB 页面 | 原文线索

V2-034430 | MATHDB-378776

Fukaya-category Hochschild-Floer isomorphism conjecture

摘录 / EXCERPT: Fukaya-category Hochschild-Floer isomorphism conjecture. In good circumstances, this map is an isomorphism. This would identify the Hochschild homology of the Fukaya category with...

conjecture | open | MathDB 页面 | 原文线索

V2-034431 | MATHDB-378779

The correspondence conjecture for transverse-link and branched-cover contact invariants

摘录 / EXCERPT: Let L be a transverse representative of an alternating smooth link. Write $\psi(L)\in\widetilde{Kh}(L)$ for the transverse-link invariant in reduced Khovanov homology, let...

conjecture | open | MathDB 页面 | 原文线索

V2-034432 | MATHDB-378789

The eventual monotonicity conjecture for colored Jones growth

摘录 / EXCERPT: For a knot K in S^3, define the sequence ... where $J_K(n)$ is the colored Jones polynomial. The eventual monotonicity conjecture. For every knot K in S^3, the sequence a_n...

conjecture | open | MathDB 页面 | 原文线索

V2-034433 | MATHDB-378795

The global-dimension criterion for Iwasawa algebras

摘录 / EXCERPT: Let G be a compact p-adic analytic group, let H be a closed normal subgroup of G, and write $\Omega_{G,H}$ and $\Lambda_{G,H}$ for the corresponding Iwasawa algebras. For...

conjecture | open | MathDB 页面 | 原文线索

V2-034434 | MATHDB-378796

Lower semicontinuity conjecture for the normalized curve energy

摘录 / EXCERPT: Let $I=[0,1]$, and let $C:I\to\mathbb{R}^n$ and $C_h:I\to\mathbb{R}^n$ be Lipschitz homotopies. Define the length functions ... Assume that $l_h\to l$ uniformly and that...

conjecture | open | MathDB 页面 | 原文线索

V2-034435 | MATHDB-378797

Existence conjecture for minimizing homotopies of the Michor-Mumford energy

摘录 / EXCERPT: Existence conjecture. The energy $E^A(C)$ admits a minimum in the class $\mathbb{C}$ of homotopies connecting c_0 and c_1. The text states that existence of minimal geodesics f...

conjecture | open | MathDB 页面 | 原文线索

V2-034436 | MATHDB-378816

Generalized Novikov additivity for perversity intersection signatures

摘录 / EXCERPT: Let X^n be a pseudomanifold and let $Y\subset X$ be a compact codimension-1 submanifold such that ... where $Y\subset\subset X^{\mathrm{reg}}$. For a perversity function...

conjecture | open | MathDB 页面 | 原文线索

V2-034437 | MATHDB-378817

The L^2 signature and intersection-cohomology conjecture for fibred boundary metrics

摘录 / EXCERPT: Let M be a manifold with boundary fibration ... and suppose that the metric on M is quasi-isometric near the boundary to ... where $R\in[1,\infty)$, $-1\leq c<0$, and k is a symm...

conjecture | open | MathDB 页面 | 原文线索

V2-034438 | MATHDB-378841

Nonnegativity conjecture for Thom polynomials of Thom-Boardman classes

摘录 / EXCERPT: A Thom-Boardman class is a singularity class indexed by a sequence of nonnegative integers, and its Thom polynomial is the universal cohomology class representing the locus of maps...

conjecture | open | MathDB 页面 | 原文线索

V2-034439 | MATHDB-378843

Finiteness conjecture for topologically realizable filtered lambda-ring structures

摘录 / EXCERPT: Let n be an integer with $n\geq 3$. Consider the filtered truncated polynomial ring ... A filtered λ-ring structure is a filtered λ-ring structure on this ring, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034440 | MATHDB-378847

Hajnal-Juhász basis conjecture for regular Hausdorff spaces

摘录 / EXCERPT: Let X be a regular Hausdorff space. A space is hereditarily separable if every subspace is separable, and hereditarily Lindelöf if every subspace is Lindelöf. Hajnal-Juhász basis...

conjecture | open | MathDB 页面 | 原文线索

V2-034441 | MATHDB-378849

Gruenhage's three-element basis conjecture for uncountable regular Hausdorff spaces

摘录 / EXCERPT: A basis of topological spaces means that every space in the class has one of the listed spaces as the relevant basis representative. Gruenhage's three-element basis conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-034442 | MATHDB-378850

Fremlin's two-to-one compact-image conjecture

摘录 / EXCERPT: Let X be a compact topological space, and suppose every closed subset of X is a G_δ set, meaning a countable intersection of open subsets of X. Fremlin's two-to-one co...

conjecture | open | MathDB 页面 | 原文线索

V2-034443 | MATHDB-378853

Harikae-Nakanishi-Uchida conjecture on $(2,2)$-move equivalence

摘录 / EXCERPT: Harikae-Nakanishi-Uchida conjecture. Every link is $(2,2)$-move equivalent to a trivial link.

conjecture | open | MathDB 页面 | 原文线索

V2-034444 | MATHDB-378897

Moore's conjecture on projectivity over finite-index subgroup extensions

摘录 / EXCERPT: Let G be a group, let H be a subgroup of finite index, and let R be an arbitrary ring. Assume that for every nontrivial element x of G, at least one of the following cond...

conjecture | open | MathDB 页面 | 原文线索

V2-034445 | MATHDB-378906

Relative Poisson cohomology criterion for stable symplectic leaves

摘录 / EXCERPT: Relative Poisson cohomology conjecture. Let pi be a Poisson structure with a compact symplectic leaf $S\subset M$. If

conjecture | open | MathDB 页面 | 原文线索

V2-034446 | MATHDB-378910

Morimoto's conjecture on Heegaard genus degeneration of connected sums

摘录 / EXCERPT: A knot exterior X is the complement of an open tubular neighborhood of a knot, and $g(X)$ denotes its Heegaard genus. An exterior admits a primitive meridian if it has a minimal-...

conjecture | open | MathDB 页面 | 原文线索

V2-034447 | MATHDB-378922

The maximal torus normalizer conjecture for connected p-compact groups

摘录 / EXCERPT: Let X be a connected p-compact group with maximal torus T_X. Its maximal torus normalizer is represented by the loop space NT_X. Maximal torus normalizer conjecture. The gr...

conjecture | open | MathDB 页面 | 原文线索

V2-034448 | MATHDB-378928

Infinite-order TQFT representation conjecture for pseudo-Anosov mapping classes

摘录 / EXCERPT: Let ϕ be a pseudo-Anosov mapping class of a surface of arbitrary genus, and let k denote the level of the associated TQFT representation. Infinite-order TQFT representation...

conjecture | open | MathDB 页面 | 原文线索

V2-034449 | MATHDB-378929

The NE-equivalence conjecture for simple homotopy types

摘录 / EXCERPT: Let X and Y be finite simplicial complexes. The NE-equivalence conjecture. There exist finite simplicial complexes X and Y having the same simple homotopy type, such that...

conjecture | open | MathDB 页面 | 原文线索

V2-034450 | MATHDB-378930

The strong NE-equivalence conjecture for simple homotopy types

摘录 / EXCERPT: Let $\{X_i\}_{i=1}^\infty$ be a family of finite simplicial complexes. The strong NE-equivalence conjecture. There exists an infinite family of finite simplicial complexes...

conjecture | open | MathDB 页面 | 原文线索

V2-034451 | MATHDB-378948

Non-triviality conjecture for generalized alternating knots

摘录 / EXCERPT: Non-triviality conjecture. K is not trivial for F.

conjecture | open | MathDB 页面 | 原文线索

V2-034452 | MATHDB-378955

Gromov's vanishing conjecture for functorial seminorms

摘录 / EXCERPT: Gromov's conjecture. Every functorial seminorm on singular homology with real coefficients vanishes for simply connected spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-034453 | MATHDB-378984

Non-vanishing perturbation conjecture for non-closed 1-forms

摘录 / EXCERPT: Let M be a closed manifold with zero Euler characteristic, and let $alpha$ be a non-closed differential 1-form on M. Non-vanishing perturbation conjecture. There exists a smo...

conjecture | open | MathDB 页面 | 原文线索

V2-034454 | MATHDB-378999

Identification of intersection cohomology with L^2-cohomology for unipotent harmonic bundles

摘录 / EXCERPT: Let S be a noncompact smooth curve with smooth compactification $\overline S$, let $j:S\hookrightarrow\overline S$ be the inclusion, and let L_{ρ} be the local system associ...

conjecture | open | MathDB 页面 | 原文线索

V2-034455 | MATHDB-379013

The h-ribbon characterization conjecture for ribbon groups

摘录 / EXCERPT: H-ribbon characterization conjecture. A knot K is h-ribbon with group G if and only if there exists an epimorphism $\varphi\colon \pi_1(M_K)\twoheadrightarrow G$ such that cond...

conjecture | open | MathDB 页面 | 原文线索

V2-034456 | MATHDB-379035

The homotopy-test-graph conjecture for connected bipartite graphs

摘录 / EXCERPT: A graph is a homotopy test graph if its Hom-complexes detect graph colorability through the corresponding homotopy-connectivity bound. Homotopy-test-graph conjecture. Every connect...

conjecture | open | MathDB 页面 | 原文线索

V2-034457 | MATHDB-379040

Gukov-Schwarz-Vafa's superpolynomial conjecture

摘录 / EXCERPT: Let K be a knot, and let $\overline{\mathit{KhR}}_N(q,t)$ denote the graded Poincaré polynomial of the $sl(N)$ Khovanov-Rozansky homology. A finite polynomial has only finitely m...

conjecture | open | MathDB 页面 | 原文线索

V2-034458 | MATHDB-379041

The HOMFLY homology differentials conjecture

摘录 / EXCERPT: HOMFLY homology differentials conjecture. There is a homology theory $\mathcal H_*$ categorifying the HOMFLY polynomial, with differentials $\{d_N\}$ satisfying those three axioms,...

conjecture | open | MathDB 页面 | 原文线索

V2-034459 | MATHDB-379042

The torus-knot $T_{3,m}$ superpolynomial prediction

摘录 / EXCERPT: Let $T_{3,m}$ be a torus knot, and let $\mathcal P(T_{3,m})$ denote its reduced superpolynomial. For $m=3k+1$ or $m=3k+2$, define the three polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-034460 | MATHDB-379047

Thurston's ending lamination conjecture

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with finitely generated fundamental group. Its topological type consists of its underlying topology, and its end invariants encode the geometry...

conjecture | open | MathDB 页面 | 原文线索

V2-034461 | MATHDB-379048

Marden's tameness conjecture

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with finitely generated fundamental group. The interior of a compact manifold with boundary is a topological model for a tame hyperbolic 3-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-034462 | MATHDB-379057

Gerstenhaber conjecture for natural operations on associative cohomology

摘录 / EXCERPT: Let ${\mathcal A{\it ss}}$ be the associative operad, and let ${\mathcal B}_{{\mathcal A{\it ss}}}$ be the operad of natural operations on the cohomology of associative algebras. L...

conjecture | open | MathDB 页面 | 原文线索

V2-034463 | MATHDB-379058

Lie-operad conjecture for natural operations on Lie cohomology

摘录 / EXCERPT: Let ${\mathcal L{\it ie}}$ be the Lie operad, and let ${\mathcal B}_{{\mathcal L{\it ie}}}$ be the operad of natural operations on the cohomology of Lie algebras. Lie-operad conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-034464 | MATHDB-379059

Intrinsic-bracket conjecture for degree-zero natural operations

摘录 / EXCERPT: Intrinsic-bracket conjecture. The homomorphism I induces an isomorphism of operads

conjecture | open | MathDB 页面 | 原文线索

V2-034465 | MATHDB-379066

Regularity conjecture for the equivariant asymptotic solution

摘录 / EXCERPT: Let $R_{\sigma\tau}^{(n)}(t,\lambda)$ be the coefficient of $\hbar^n$ in the asymptotic solution matrix $R_{\sigma\tau}(t,\lambda,\hbar)$ of the equivariant quantum differential eq...

conjecture | open | MathDB 页面 | 原文线索

V2-034466 | MATHDB-379095

Virtually Haken surgery conjecture for hyperbolic knot manifolds

摘录 / EXCERPT: A knot manifold is an irreducible, orientable, compact 3-manifold whose boundary is a single torus. It is hyperbolic if its interior admits a complete hyperbolic metric of finite v...

conjecture | open | MathDB 页面 | 原文线索

V2-034467 | MATHDB-379104

The classification conjecture for surgery in the kernel of a colouring

摘录 / EXCERPT: Classification conjecture. The invariant is a complete invariant of the set of equivalence classes modulo surgery in $\ker\rho$. Equivalently, the theorem asserting that there are...

conjecture | open | MathDB 页面 | 原文线索

V2-034468 | MATHDB-379108

The simple-curve minimum-girth conjecture for surface subgroups

摘录 / EXCERPT: Let H be a handlebody and let S be a compact surface in ∂H. Let $\mathcal{N}(\partial S)$ be the subgroup of $\pi_1(S)$ normally generated by the boundary curves of...

conjecture | open | MathDB 页面 | 原文线索

V2-034469 | MATHDB-379109

Koszul-operad bar-complex resolution conjecture

摘录 / EXCERPT: Bar-complex resolution conjecture. If $\mathcal{P}$ is a Koszul operad, then ${{\mathsf{B}}_{\bullet}^{\mathcal{P}}\!A}$ is a resolution of the left A-module...

conjecture | open | MathDB 页面 | 原文线索

V2-034470 | MATHDB-379121

Torsion criterion for graph cohomology of algebras ${\mathcal A}_m$

摘录 / EXCERPT: Let G be a graph, and let $H^{**}_{{\mathcal A}_m}(G)$ denote its graph cohomology associated with the algebra ${\mathcal A}_m$. A graph is loopless if it has no loops, and a cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-034471 | MATHDB-379125

Fukaya-Kato equivariant integrality conjecture

摘录 / EXCERPT: Let $F/\mathbb Q$ be a p-adic Lie extension with Galois group G, let $\Lambda=\Lambda(G)$ be its Iwasawa algebra, let T_p be a Galois-stable lattice, and put...

conjecture | open | MathDB 页面 | 原文线索

V2-034472 | MATHDB-379135

Morita's non-vanishing conjecture for the Morita classes of free-group automorphisms

摘录 / EXCERPT: For each integer $i\geq1$, let ... be the homology class constructed from Kontsevich's graph-homology construction. Morita's conjecture. The classes μ_i are non-trivial for all...

conjecture | open | MathDB 页面 | 原文线索

V2-034473 | MATHDB-379136

Morita's basis conjecture for the symplectic second cohomology of the derivation Lie algebra

摘录 / EXCERPT: Let $\mathfrak h_{g,1}^{\mathbb Q}$ be the graded Lie algebra under consideration, and let Sp act on it through the symplectic action on $H_{\mathbb Q}$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-034474 | MATHDB-379137

Morita's trace-pullback conjecture for the homology-cobordism group

摘录 / EXCERPT: Let $\mathcal H_{g,1}$ and $\overline{\mathcal H}_{g,1}$ be the homology-cobordism groups in the source, let σ be the homomorphism to the inverse limit of automorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-034475 | MATHDB-379159

Koszulness conjecture for the cohomology algebra of the real moduli space

摘录 / EXCERPT: Let M_n denote the real locus of the moduli space of stable genus-zero curves with n marked points, and let $\Lambda_n=H^*(M_n,\mathbb{Q})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034476 | MATHDB-379160

Freeness conjecture for the Lie and enveloping algebras associated with the cactus group

摘录 / EXCERPT: Let L_n be the quadratic Lie algebra generated by the elements μ_{ijk}, and let $U_n=U(L_n)$ be its universal enveloping algebra. Freeness conjecture. The $\mathbb{Z}$-module...

conjecture | open | MathDB 页面 | 原文线索

V2-034477 | MATHDB-379161

Inductive kernel conjecture for the Lie and enveloping algebras of the cactus group

摘录 / EXCERPT: Let $U_n=U(L_n)$, where L_n is the quadratic Lie algebra generated by the elements μ_{ijk}. Consider the natural morphisms $U_{n+1}\to U_n$ and $L_{n+1}\to L_n$ sending...

conjecture | open | MathDB 页面 | 原文线索

V2-034478 | MATHDB-379162

Rational K(pi,1) conjecture for the real moduli space

摘录 / EXCERPT: Let M_n be the real locus of the moduli space of stable genus-zero curves with n marked points. A connected space is rational $K(\pi,1)$ if its rational completion is a...

conjecture | open | MathDB 页面 | 原文线索

V2-034479 | MATHDB-379182

Owens' H-thin concordance invariant conjecture for delta

摘录 / EXCERPT: Owens' H-thin conjecture. For any H-thin knot K,

conjecture | open | MathDB 页面 | 原文线索

V2-034480 | MATHDB-379183

Restriction equivalence conjecture for semi-infinite, Hilbert, and ordinary spectra

摘录 / EXCERPT: Restriction equivalence conjecture. There is a notion of weak equivalence of semi-infinite spectra and a notion of weak equivalence of Hilbert spectra such that the restriction map...

conjecture | open | MathDB 页面 | 原文线索

V2-034481 | MATHDB-379197

Suppression conjecture for colored Jones polynomials of Whitehead doubles

摘录 / EXCERPT: Let K be a nontrivial knot, let N be a positive integer, and let n satisfy $\left|n-\frac{N}{2}\right|<N^\delta$ for some $\frac{1}{2}<\delta<\frac{2}{3}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-034482 | MATHDB-379204

Minimal-valuation local semistable reduction conjecture for F-isocrystals

摘录 / EXCERPT: Minimal-valuation local semistable reduction conjecture. $\mathcal{E}$ admits local semistable reduction at any minimal valuation v on $k(X)$ centered on $\overline{X}$.

conjecture | open | MathDB 页面 | 原文线索

V2-034483 | MATHDB-379207

Waldhausen's virtual Haken conjecture

摘录 / EXCERPT: Let M be an orientable, irreducible closed 3-manifold with infinite fundamental group. Waldhausen's virtual Haken conjecture. M is finitely covered by a Haken manifold. This...

conjecture | open | MathDB 页面 | 原文线索

V2-034484 | MATHDB-379208

Virtual positive Betti number conjecture for 3-manifolds

摘录 / EXCERPT: Let M be an orientable, irreducible closed 3-manifold with infinite fundamental group. Virtual positive Betti number conjecture. Some finite cover of M has positive first Bet...

conjecture | open | MathDB 页面 | 原文线索

V2-034485 | MATHDB-379209

Virtual infinite Betti number conjecture for 3-manifolds

摘录 / EXCERPT: Let M be an orientable, irreducible closed 3-manifold with infinite fundamental group. Virtual infinite Betti number conjecture. Either $\pi_1(M)$ is virtually solvable or M...

conjecture | open | MathDB 页面 | 原文线索

V2-034486 | MATHDB-379210

Virtual fibering conjecture for 3-manifolds

摘录 / EXCERPT: Let M be an orientable, irreducible closed 3-manifold with infinite fundamental group. Virtual fibering conjecture. M has a finite cover that fibers over the circle. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-034487 | MATHDB-379211

Hesselholt's Galois descent conjecture for topological cyclic homology

摘录 / EXCERPT: Hesselholt's descent conjecture. For all positive integers q, this canonical map is an isomorphism of pro-abelian groups, and the higher continuous cohomology groups of the pro-...

conjecture | open | MathDB 页面 | 原文线索

V2-034488 | MATHDB-379227

The exceptional-arrangement characterization for dual GKZ systems

摘录 / EXCERPT: Let A be an integer matrix, let S_A be its semigroup ring, and let $\mathfrak m$ be its homogeneous maximal ideal. For each $i<\dim(S_A)$, let $H^i_{\mathfrak m}(S_A)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-034489 | MATHDB-379242

The stability-condition and quantum-cohomology correspondence for local \(\mathbb P^2\)

摘录 / EXCERPT: Let $X=\mathcal O_{\mathbb P^2}(-3)$, and let $\operatorname{Stab}^{0}_{n}(X)$ be the normalized connected submanifold of stability conditions defined by $Z(\mathcal O_x)=i$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034490 | MATHDB-379244

End-curve characterization conjecture for splice quotients

摘录 / EXCERPT: Let $(Y,0)$ be a normal surface singularity with rational homology sphere link, and let $\tilde{Y}$ be its minimal good resolution. An end-curve function for an end-curve is a funct...

conjecture | open | MathDB 页面 | 原文线索

V2-034491 | MATHDB-379253

General Lefschetz-operation conjecture for the intersection cohomology of complete fans

摘录 / EXCERPT: Let Δ be a complete fan of dimension n, and let ${\mathcal A}(B\Delta)$ be its associated graded algebra with elements $L_i\in {\mathcal A}(B\Delta)$ of degree e_i, for...

conjecture | open | MathDB 页面 | 原文线索

V2-034492 | MATHDB-379254

Constant-multiplication conjecture in the self-adjoint linear-map lemma

摘录 / EXCERPT: Let $V=\bigoplus_i V_i$ be a finite-dimensional vector space, with each V_i equipped with a non-degenerate symmetric bilinear pairing, and let $K\subset V$ be a subspace such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-034493 | MATHDB-379266

Tamarkin's homotopy-type conjecture for natural Lie operations

摘录 / EXCERPT: Let ${\cal B}_{\it Lie}^*$ be the differential graded operad of natural operations on the Chevalley-Eilenberg complex of a Lie algebra with coefficients in itself, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034494 | MATHDB-379302

The quasitriangular group cohomology isomorphism conjecture

摘录 / EXCERPT: For each n, let QA_n be the quadratic algebra generated by the cohomology classes associated with the generators of the quasitriangular group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034495 | MATHDB-379329

Berestovski\s conjecture on locally compact similarity-homogeneous metric spaces

摘录 / EXCERPT: Berestovskis conjecture. The space X is homeomorphic to the topological product

conjecture | open | MathDB 页面 | 原文线索

V2-034496 | MATHDB-379332

Poisson-cohomology conjecture for symplectic resolutions

摘录 / EXCERPT: Poisson-cohomology conjecture. There is an isomorphism of vector spaces

conjecture | open | MathDB 页面 | 原文线索

V2-034497 | MATHDB-379345

Ichihara's boundary slope diameter conjecture

摘录 / EXCERPT: Ichihara's conjecture. For every knot K in S^3,

conjecture | open | MathDB 页面 | 原文线索

V2-034498 | MATHDB-379346

Ishikawa--Shimokawa conjecture on finite boundary slopes

摘录 / EXCERPT: Ishikawa--Shimokawa conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-034499 | MATHDB-379348

Andersen's topological interpretation conjecture for asymptotic exponents

摘录 / EXCERPT: Let X be a closed oriented 3-manifold and let $G=\operatorname{SU}(2)$. Let $\mathcal M_j$ be the union of the connected components of the moduli space of flat G-connections...

conjecture | open | MathDB 页面 | 原文线索

V2-034500 | MATHDB-379349

Vanishing conjecture for non-contributing Seifert stationary points

摘录 / EXCERPT: Let X be a Seifert manifold with base S^2, and let $(I,\underline{n}')\in\mathcal I_2^b$. Suppose that there do not exist elements $g_1,\ldots,g_n\in\operatorname{SU}(2)$ satis...

conjecture | open | MathDB 页面 | 原文线索

V2-034501 | MATHDB-379352

Eliashberg's torsion obstruction to strong symplectic fillability

摘录 / EXCERPT: Let Y be a 3-manifold and let ξ be a contact structure on Y. The torsion of (Y,ξ), denoted by ... is the supremum of the integers $n\geq 1$ for which there exists a con...

conjecture | open | MathDB 页面 | 原文线索

V2-034502 | MATHDB-379353

Torsion vanishing conjecture for the Ozsváth-Szabó contact invariant

摘录 / EXCERPT: Let Y be a 3-manifold and let ξ be a contact structure on Y. Let ${\rm Tor}(Y,\xi)$ denote the torsion, defined as the supremum of the integers $n\geq 1$ for which there...

conjecture | open | MathDB 页面 | 原文线索

V2-034503 | MATHDB-379358

Excision for periodic dual Hopf cyclic homology of comodule algebras

摘录 / EXCERPT: Let B be a Hopf algebra with an invertible antipode and let X be a stable anti-Yetter-Drinfeld module. Let ... be a sequence of B-comodule algebras such that both J and A a...

conjecture | open | MathDB 页面 | 原文线索

V2-034504 | MATHDB-379370

Asakura's characterization conjecture for the dlog image

摘录 / EXCERPT: Let R be the ring of integers in an unramified extension K of $\mathbb Q_p$, and let $\pi_R\colon X_R\to C_R$ be an elliptic surface over R satisfying condition (Rat). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034505 | MATHDB-379388

Nonvanishing conjecture for the middle homology of torus-knot diagrams

摘录 / EXCERPT: Let $D_{p,p}$ and $D_{p,p+1}$ be the torus-knot diagrams used in the paper, and let $H^{i,j}(D)$ denote their Khovanov homology in homological degree i and quantum degree j. No...

conjecture | open | MathDB 页面 | 原文线索

V2-034506 | MATHDB-379407

Congruence cohomology conjecture for groups with discrete series

摘录 / EXCERPT: Let G be a semisimple group with discrete series. Let $\pi\in\widehat G$ be a cohomological representation, and consider the question of whether there is a lattice Γ in...

conjecture | open | MathDB 页面 | 原文线索

V2-034507 | MATHDB-379408

Vanishing of first Betti numbers for trialitarian groups

摘录 / EXCERPT: Let G be a semisimple algebraic group over ${\mathbb Q}$ obtained by restriction of scalars from a group of type ${}^3D_4$ or ${}^6D_4$ over a totally real number field, and supp...

conjecture | open | MathDB 页面 | 原文线索

V2-034508 | MATHDB-379409

Global occurrence conjecture for cohomological representations

摘录 / EXCERPT: Let G be as above, and let $\pi=A_{\mathfrak q}$ be a cohomological representation of G such that ... where $\mathfrak l'_0$ is a compact Lie algebra and $\mathfrak g_1$ is a non...

conjecture | open | MathDB 页面 | 原文线索

V2-034509 | MATHDB-379415

The cohomology tensor-product conjecture for the Coxeter complex

摘录 / EXCERPT: The cohomology tensor-product conjecture. There is an equality

conjecture | open | MathDB 页面 | 原文线索

V2-034510 | MATHDB-379421

The $\frac{5}{4}$-conjecture for even four-manifolds

摘录 / EXCERPT: Let X be a closed oriented even 4-manifold, meaning that its intersection form is even, and let $\sigma(X)$ denote its signature. The $\frac{5}{4}$-conjecture. One has ... This i...

conjecture | open | MathDB 页面 | 原文线索

V2-034511 | MATHDB-379485

Completeness conjecture for the knot biquandle of virtual links

摘录 / EXCERPT: Knot-biquandle completeness conjecture. The knot biquandle is a complete invariant of virtual link type: if two knotlike biquandle presentations are related by Tietze moves, then t...

conjecture | open | MathDB 页面 | 原文线索

V2-034512 | MATHDB-379487

The Hodge-de Rham conjecture for smooth proper dg categories

摘录 / EXCERPT: Let k be a field of characteristic 0, and let ${\mathcal A}$ be a smooth proper dg category over k. For every integer $n\geq 1$, consider the complex ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-034513 | MATHDB-379489

The proper subgroup conjecture for Hamiltonian homeomorphisms

摘录 / EXCERPT: Let $Hameo(M,\omega)$ be the group of Hamiltonian homeomorphisms and let $Sympeo_0(M,\omega)$ be the identity component of the group of symplectic homeomorphisms. Proper subgroup c...

conjecture | open | MathDB 页面 | 原文线索

V2-034514 | MATHDB-379490

The descent conjecture for the continuous Calabi quasi-morphism on the two-sphere

摘录 / EXCERPT: Let ${\mathcal P}^{ham}(Sympeo(S^2),id)$ denote the space of continuous Hamiltonian paths from the identity, let $\overline\mu^{path}$ be the extension of the homogeneous Calabi qu...

conjecture | open | MathDB 页面 | 原文线索

V2-034515 | MATHDB-379491

The extension conjecture for the Calabi homomorphism on the disk

摘录 / EXCERPT: Calabi extension conjecture. The Calabi homomorphism extends to a homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034516 | MATHDB-379494

The matched-diagram conjecture for links

摘录 / EXCERPT: Matched-diagram conjecture. (i) Not every link has a matched diagram. (ii) Every oriented link is t_3-move equivalent to a link with a matched diagram.

conjecture | open | MathDB 页面 | 原文线索

V2-034517 | MATHDB-379495

Labastida-Mariño-Vafa integrality conjecture for reformulated colored HOMFLY polynomials

摘录 / EXCERPT: Labastida-Mariño-Vafa conjecture. For partitions $\lambda^1,\dots,\lambda^l$,

conjecture | open | MathDB 页面 | 原文线索

V2-034518 | MATHDB-379507

The Tate elliptic curve product formula for the algebraic intermediate Jacobian

摘录 / EXCERPT: Let $X=E_1\times E_2\times E_3$ be a product of three Tate elliptic curves with parameters q_i for $i=1,2,3$. Let r be the rank of the space of integer triples (n_1,n_2,n_3)...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-034519 | MATHDB-379509

The Seifert-surface identification conjecture for knot Floer homology

摘录 / EXCERPT: Let K be a knot in S^3 and let R be a genus g Seifert surface of K. Write $\widehat{HFK}(K,g)$ for the top Alexander-graded knot Floer homology and...

conjecture | open | MathDB 页面 | 原文线索

V2-034520 | MATHDB-379517

Properness conjecture for Hamiltonian homeomorphisms

摘录 / EXCERPT: Let $(M,c9)$ be a closed symplectic manifold. Write $Hameo(M,c9)$ for the group of Hamiltonian homeomorphisms and $Sympeo_0(M,c9)$ for the identity component of the group of symple...

conjecture | open | MathDB 页面 | 原文线索

V2-034521 | MATHDB-379526

Manolescu's symplectic link homology conjecture

摘录 / EXCERPT: Let κ be a link, let $n>0$, and let ${{\mathscr{H}}_{(n)}{\scriptscriptstyle{\mathrm{symp}}}}^k(\kappa)$ denote the link invariant constructed from Lagrangian Floer theor...

conjecture | open | MathDB 页面 | 原文线索

V2-034522 | MATHDB-379530

Neumann-Wahl splice-type equations conjecture

摘录 / EXCERPT: Let $(X,0)$ be a complete intersection surface singularity whose link is an integral homology sphere. Neumann-Wahl's splice-type equations conjecture. The equations defining...

conjecture | open | MathDB 页面 | 原文线索

V2-034523 | MATHDB-379531

Generic non-splice-quotient conjecture for surface singularities

摘录 / EXCERPT: Fix a rational homology sphere link and consider the general' $\mathbb{Q}$-Gorenstein singularity with that link. Suppose that its fundamental cycle has arithmetic genus at least...

conjecture | open | MathDB 页面 | 原文线索

V2-034524 | MATHDB-379542

Nonzero minimal entropy for four-manifolds with hyperbolic-product decompositions

摘录 / EXCERPT: Let M be a smooth orientable four-manifold that admits a proper geometric decomposition into pieces modelled on ${\mathbb H}^2 \times {\mathbb H}^2$, and let ${\rm h}(M)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-034525 | MATHDB-379548

Freedman's conjecture on surgery for free groups

摘录 / EXCERPT: Let M be a topological 4-manifold, let ${\vee}^3 S^1$ denote the wedge of three circles, and let ${\mathcal S}^0(Wh(Bor))$ denote the zero-framed surgery on the Whitehead doubl...

conjecture | open | MathDB 页面 | 原文线索

V2-034526 | MATHDB-379559

Barratt's exponent conjecture for double suspensions

摘录 / EXCERPT: Barratt's exponent conjecture. The image of the induced map

conjecture | open | MathDB 页面 | 原文线索

V2-034527 | MATHDB-379560

Strong Barratt exponent conjecture for double loop maps

摘录 / EXCERPT: Strong Barratt exponent conjecture. The double loop map

conjecture | open | MathDB 页面 | 原文线索

V2-034528 | MATHDB-379561

Casian-Kodama conjecture on Toda blow-up graphs and flag-manifold cohomology

摘录 / EXCERPT: Casian-Kodama conjecture. The graph ${\mathcal G}$ is the graph of incidence numbers for the integral cohomology of the real flag manifold ${\mathcal B}$ ($\hat{\mathcal B}$) in te...

conjecture | open | MathDB 页面 | 原文线索

V2-034529 | MATHDB-379562

Burghelea-Haller conjecture on complex Ray-Singer and Farber-Turaev torsions

摘录 / EXCERPT: Burghelea-Haller conjecture. The Burghelea-Haller quadratic form satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-034530 | MATHDB-379577

Freeness conjecture for relative Kauffman bracket skein modules of closed surfaces

摘录 / EXCERPT: Let F be a closed surface, let $x_i\in F\times\{0\}$ for each i, and let ${\cal S}_{2,\infty}(F\times I,\{x_i\}_1^{2n};R,A)$ denote the relative Kauffman bracket skein module w...

conjecture | open | MathDB 页面 | 原文线索

V2-034531 | MATHDB-379578

Torsion conjecture for Kauffman bracket skein modules of connected sums

摘录 / EXCERPT: Let $M=M_1\# M_2$, where each M_i is a 3-manifold, possibly with holes, and neither M_i is equal to S^3. Let ${\cal S}_{2,\infty}(M)$ denote the Kauffman bracket skein module...

conjecture | open | MathDB 页面 | 原文线索

V2-034532 | MATHDB-379586

The rank conjecture for differential modules with finite free flag

摘录 / EXCERPT: Let R be a local ring and let F be a differential R-module admitting a finite free flag. Suppose that $\operatorname{H}(F)$ has non-zero finite length, and write $d=\dim R$....

conjecture | open | MathDB 页面 | 原文线索

V2-034533 | MATHDB-379595

Casson-Gordon's three-family conjecture for 2-bridge ribbon knots

摘录 / EXCERPT: Casson-Gordon's conjecture. Every 2-bridge ribbon knot is contained in one of the three families of Casson and Gordon.

conjecture | open | MathDB 页面 | 原文线索

V2-034534 | MATHDB-379596

The ribbon-knot symmetric-union conjecture

摘录 / EXCERPT: Ribbon-knot symmetric-union conjecture. Every ribbon knot is a symmetric union.

conjecture | open | MathDB 页面 | 原文线索

V2-034535 | MATHDB-379605

Minor monotonicity of embeddability in spheres

摘录 / EXCERPT: Let H and K be simplicial complexes, and write $H<K$ when H is a minor of K, obtained by admissible contractions and deletions. Let m be a nonnegative integer. Minor mono...

conjecture | open | MathDB 页面 | 原文线索

V2-034536 | MATHDB-379606

Wagner-author non-embeddability conjecture for skeletons with a missing face

摘录 / EXCERPT: Non-embeddability conjecture. The complex L does not embed in $\mathbb{R}^{2d}$. This generalizes the example discussed in the source and is described there as joint work in prog...

conjecture | open | MathDB 页面 | 原文线索

V2-034537 | MATHDB-379607

The smooth Bogomolov-Miyaoka-Yau bound for pseudo-free circle actions

摘录 / EXCERPT: Let L be a 5-dimensional rational homology sphere with $H_1(L,\mathbb Z)=0$ admitting a pseudo-free differentiable circle action. Let $O_1,\dots,O_k$ be the nonfree orbits with s...

conjecture | open | MathDB 页面 | 原文线索

V2-034538 | MATHDB-379608

The smooth Bogomolov-Miyaoka-Yau bound for exceptional circle orbits

摘录 / EXCERPT: Let L be a 5-dimensional rational homology sphere with $H_1(L,\mathbb Z)=0$ admitting a fixed-point-free differentiable circle action. For an orbit O, define its exceptional mu...

conjecture | open | MathDB 页面 | 原文线索

V2-034539 | MATHDB-379609

The H-cobordism bound for Seifert fibered rational homology spheres

摘录 / EXCERPT: Let $M\to S^2$ be a Seifert fibered rational homology sphere with fiber multiplicities $d_1,\dots,d_k$. Seifert H-cobordism conjecture. If M is rationally H-cobordant to zero, th...

conjecture | open | MathDB 页面 | 原文线索

V2-034540 | MATHDB-379610

The simply connected smooth-locus conjecture for rational homology projective planes

摘录 / EXCERPT: Let S be a rational homology $\mathbb C\mathbb P^2$ with quotient singularities, and write ... Assume that S^0 is simply connected. Simply connected smooth-locus conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-034541 | MATHDB-379637

Gukov's A-polynomial conjecture from colored Jones asymptotics

摘录 / EXCERPT: Gukov's conjecture. The pair

conjecture | open | MathDB 页面 | 原文线索

V2-034542 | MATHDB-379646

Strand-number conjecture for braids of maximal entropy

摘录 / EXCERPT: Strand-number conjecture. Braids of maximal entropy belong to $\operatorname{Br}(3)$ or $\operatorname{Br}(4)$. The paper presents this as an empirical conjecture obtained from its...

conjecture | open | MathDB 页面 | 原文线索

V2-034543 | MATHDB-379647

Alternation conjecture for braids of maximal entropy

摘录 / EXCERPT: Alternation conjecture. Braids with maximal entropy are alternated. The conjecture concerns the observed structure of entropy-maximizing braids among those considered by word lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-034544 | MATHDB-379658

Kirby-Zhang surgery conjecture for the Poincaré homology sphere

摘录 / EXCERPT: Let K be a knot in S^3. Suppose there exists a rational number r such that the 3-manifold obtained by r-surgery on K is homeomorphic to the Poincaré homology sphere...

conjecture | open | MathDB 页面 | 原文线索

V2-034545 | MATHDB-379675

Exponential puncture-growth conjecture for Brunnian pseudo-Anosov dilatations

摘录 / EXCERPT: Let $S_{g,p}$ be a genus-g surface with p punctures, let $\brun(S_{g,p})$ denote the set of Brunnian pseudo-Anosov mapping classes on $S_{g,p}$, and let $L(\brun(S_{g,p}))$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-034546 | MATHDB-379677

The aligned meridians conjecture for admissible pairs

摘录 / EXCERPT: Let (M,W) be an admissible pair, with W a genus two handlebody and α,β the specified meridian disks of W; the source defines admissibility and alignment in the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-034547 | MATHDB-379678

The boundary-irreducible aligned meridians conjecture

摘录 / EXCERPT: Let (M,W) be an admissible pair, where W is a genus two handlebody with meridian disks α and β, and suppose that $M-W$ is ∂-irreducible. Boundary-irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-034548 | MATHDB-379681

Parity conjecture for quantum cohomology of Hirzebruch surfaces

摘录 / EXCERPT: Let $F_n=\mathbb P(\mathcal O\oplus\mathcal O(-n))$ be the Hirzebruch surface over $\mathbb P^1$, in the range $n\geq 3$ considered by the source. Parity conjecture. There are two...

conjecture | open | MathDB 页面 | 原文线索

V2-034549 | MATHDB-379688

Contractible definable neighborhood basis for the infinitesimal subgroup

摘录 / EXCERPT: Let G be a definably compact group, let M_0 be the parameter structure over which G is defined, and let G^{00} denote the infinitesimal subgroup. Contractibility conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-034550 | MATHDB-379690

Manin's conjecture for rational points on singular Del Pezzo surfaces

摘录 / EXCERPT: Let S be a Del Pezzo surface over $\mathbb{Q}$ with at most rational double points, and let $S^\circ\subset S$ be a dense open subset in the Zariski topology. Let $-K_S$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-034551 | MATHDB-379694

Unitary representability conjecture for free topological groups

摘录 / EXCERPT: Unitary representability conjecture. For every Tikhonov space X, there exists a metric space M of L^1-type such that $F(X)$ is isomorphic to a subgroup of the group of isomet...

conjecture | open | MathDB 页面 | 原文线索

V2-034552 | MATHDB-379699

Künneth conjecture for directed Fukaya categories of invertible polynomials

摘录 / EXCERPT: Fix a field k and positive integers $p_0,\ldots,p_n>1$. For each i, let $W_{p_i}(X_i)\in\mathbb{C}[X_i]$ be a general polynomial of degree p_i, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034553 | MATHDB-379706

The homological detection conjecture for incompressible surfaces in 3-manifolds

摘录 / EXCERPT: Homological detection conjecture. If

open_problem | open | MathDB 页面 | 原文线索

V2-034554 | MATHDB-379737

The quantum SO_3 spin-Chern-Simons correspondence with BM spin TQFTs

摘录 / EXCERPT: Quantum SO_3 spin-Chern-Simons conjecture. The quantum SO_3 spin-Chern-Simons theory is, in fact, a TQFT and at the 1 (mod 2)-valued SO_3 level $2m-1$, it is isomorphic t...

conjecture | open | MathDB 页面 | 原文线索

V2-034555 | MATHDB-379743

Rozansky-type cyclotomic rationality conjecture for knots

摘录 / EXCERPT: Let K be a knot and let $d\colon \mathbb{N}\to\{0,1,2,\ldots\}$ vanish except at finitely many elements. Define ... Let $\Lambda $ be the cyclotomic completion containing the unifi...

conjecture | open | MathDB 页面 | 原文线索

V2-034556 | MATHDB-379744

Root-of-unity rationality conjecture for the unified knot invariant

摘录 / EXCERPT: Root-of-unity rationality conjecture. For every $k\ge 0$, one has $P_{K,\zeta,k}(t)\in\mathbb{Z}[\zeta][t+t^{-1}]$. The source presents this as implied by the preceding cyclotomic...

conjecture | open | MathDB 页面 | 原文线索

V2-034557 | MATHDB-379745

Non-vanishing conjecture for Witten-Reshetikhin-Turaev invariants

摘录 / EXCERPT: Let M be an integral homology sphere and let $\tau _\zeta(M)$ denote its Witten-Reshetikhin-Turaev invariant at a root of unity $\zeta $. Non-vanishing conjecture. For every root...

conjecture | open | MathDB 页面 | 原文线索

V2-034558 | MATHDB-379746

Non-vanishing conjecture for the eighth WRT invariant

摘录 / EXCERPT: Let M be an integral homology sphere, let $\zeta _8$ be a primitive eighth root of unity, and define $\widetilde{\tau }_8(M)=\sqrt{-1}^{\lambda(M)}\tau _8(M)$. Eighth-invariant n...

conjecture | open | MathDB 页面 | 原文线索

V2-034559 | MATHDB-379747

p-adic non-vanishing conjecture for the WRT function

摘录 / EXCERPT: p-adic non-vanishing conjecture. The function $\tau ^p(M)$ is nowhere zero on $U(\mathcal{O}_p)$. The source presents this as a generalization of the root-of-unity non-vanishing...

conjecture | open | MathDB 页面 | 原文线索

V2-034560 | MATHDB-379748

mod p non-vanishing conjecture for the WRT function

摘录 / EXCERPT: mod p non-vanishing conjecture. The function $\widetilde{\tau }_p(M)$ is nowhere zero on $\mathbb{F}_p^\times$. The source says this is supported by computer calculations and giv...

conjecture | open | MathDB 页面 | 原文线索

V2-034561 | MATHDB-379754

Palis's weak conjecture on generic Morse-Smale or homoclinic dynamics

摘录 / EXCERPT: Let M be a compact manifold and let $\operatorname{Diff}^r(M)$ be the space of C^r-diffeomorphisms, where $r\geq 1$, endowed with the C^r-topology. Let $\mathcal{MS}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-034562 | MATHDB-379756

Genus-recovery conjecture for essential surfaces from the HOMFLY polynomial

摘录 / EXCERPT: Let $L \subset \Sigma \times I$ be a link, where Σ is a surface. An essential surface for L is a surface of smallest genus in which L can be embedded in the thickened su...

conjecture | open | MathDB 页面 | 原文线索

V2-034563 | MATHDB-379768

The Arf-invariant conjecture for twist knots and second-order Lissajous representations

摘录 / EXCERPT: A second-order Lissajous knot is a knot K represented parametrically by ... ... where n_1,n_2,n_3,n_4 are integer frequencies, $\phi_1,\phi_2,\phi_3,\phi_4$ are real phase shifts,...

conjecture | open | MathDB 页面 | 原文线索

V2-034564 | MATHDB-379769

The Lissajous projection conjecture

摘录 / EXCERPT: A knot has a Lissajous projection if its projection to the x-y plane arises from a second-order Lissajous parametrization, with coordinates of the form ... for integer frequencie...

conjecture | open | MathDB 页面 | 原文线索

V2-034565 | MATHDB-379773

Shellability and homotopy conjecture for positive generalized cluster complexes

摘录 / EXCERPT: Let Φ be a crystallographic root system of rank ℓ, let $m\geq 1$, and let $\Delta^m_+(\Phi)$ be the positive part of the generalized cluster complex, with $N^+(\Phi,m-1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-034566 | MATHDB-379776

Exactness conjecture for the homology sequence of general linear groups

摘录 / EXCERPT: Exactness conjecture. This sequence is exact.

conjecture | open | MathDB 页面 | 原文线索

V2-034567 | MATHDB-379780

Commutativity conjecture for iterated vanishing-cycle correspondences

摘录 / EXCERPT: Let Σ be a surface and let $L,L'\subset\Sigma$ be disjoint circles. Write $\widehat{V}_L^n$ and $\widehat{V}_{L'}^n$ for the associated Lagrangian correspondences between th...

conjecture | open | MathDB 页面 | 原文线索

V2-034568 | MATHDB-379781

Floer-theoretic fibred Dehn twist exact sequence conjecture

摘录 / EXCERPT: Fibred Dehn twist exact sequence conjecture. Under conditions which render a and b well-defined, they fit into a long exact sequence

conjecture | open | MathDB 页面 | 原文线索

V2-034569 | MATHDB-379825

Picard-number-one conjecture for moduli spaces with level structure

摘录 / EXCERPT: For $g\geq 3$ and $L\geq 2$, let ${\rm Mod}_g[L]$ be the level-L congruence subgroup acting trivially on $H_1(\Sigma_g,\mathbb Z/L\mathbb Z)$. Picard-number-one conjecture for le...

open_problem | open | MathDB 页面 | 原文线索

V2-034570 | MATHDB-379826

The Jones-polynomial norm conjecture for closed braids

摘录 / EXCERPT: Let $\beta\in B_n$ be an n-braid, let $\widehat{\beta}$ be its closure, and let $V(t)$ denote the Jones polynomial of $\widehat{\beta}$. For a complex number t, choose a square...

conjecture | open | MathDB 页面 | 原文线索

V2-034571 | MATHDB-379827

Hirasawa-Murasugi's fibering conjecture for Dean knots

摘录 / EXCERPT: Let ... be the Dean knot represented by the closed p-braid ... where $p>r>1$ and q,s are nonzero integers with $\gcd(q,p)=1$. Let Δ_K be the Alexander polynomial of K. Hir...

conjecture | open | MathDB 页面 | 原文线索

V2-034572 | MATHDB-379832

Rognes' localization conjecture for Johnson-Wilson spectra

摘录 / EXCERPT: Rognes' localization conjecture. These maps should fit into a cofiber sequence in the stable category

conjecture | open | MathDB 页面 | 原文线索

V2-034573 | MATHDB-379838

Kawauchi-Nakanishi conjecture on t_4-moves and homotopy

摘录 / EXCERPT: Let L_1 and L_2 be links. Two links are t_4, $\bar t_4$ equivalent if one can be obtained from the other by t_4 and $\bar t_4$ moves and isotopy. Kawauchi-Nakanishi conject...

conjecture | open | MathDB 页面 | 原文线索

V2-034574 | MATHDB-379839

Conjecture on t_3 and $\bar t_4$ moves to trivial links

摘录 / EXCERPT: Let a link be t_3, $\bar t_4$ equivalent to another link when it can be obtained from it by t_3 and $\bar t_4$ moves and isotopy. The t_3-$\bar t_4$ trivial-link conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-034575 | MATHDB-379852

Trace-range conjecture for unconditional twisted group-algebra completions

摘录 / EXCERPT: Let Γ be a torsion-free group such that (Γ,σ) satisfies the twisted Bost conjecture, and suppose that the classifying space $B\Gamma$ is a smooth, compact, orien...

conjecture | open | MathDB 页面 | 原文线索

V2-034576 | MATHDB-379856

The quantum cohomology conjecture for crepant resolutions of the anticanonical cone

摘录 / EXCERPT: Let $A=\operatorname{Spec}\bigoplus_{n\geq0}H^0(w\mathbb{P},-nK)$ be the anticanonical affine cone over the weighted projective space $w\mathbb{P}$, and let $\widetilde{A}\to A$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-034577 | MATHDB-379870

Conjecture on closures of beta-surfaces for singular self-dual Zollfrei metrics

摘录 / EXCERPT: Closure conjecture. The set $\overline{S}-S$ is a finite subset of C, and $\overline{S}$ is a topological manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-034578 | MATHDB-379878

Width conjecture for triangle-polygon graphs

摘录 / EXCERPT: Let $P_{t,k}=P_3|P_k$ be the graph with $k+1$ vertices obtained by gluing a triangle and a k-gon along an edge. Let $w^1_{{\mathcal A}_3}(G)$ be the width of the torsion in...

conjecture | open | MathDB 页面 | 原文线索

V2-034579 | MATHDB-379879

Torsion formula for triangle-square chain graphs

摘录 / EXCERPT: Let G_{t,s^k} be the graph obtained by gluing a triangle and k squares in a sequence along edges, with $v=3+2k$ vertices. Torsion formula for G_{t,s^k}. ... This is one of the...

conjecture | open | MathDB 页面 | 原文线索

V2-034580 | MATHDB-379880

Width conjecture for triangle-square chain graphs

摘录 / EXCERPT: Let G_{t,s^k} be the graph obtained by gluing a triangle and k squares in a sequence along edges, and let $w^1_{{\mathcal A}_3}(G)$ denote the width of the torsion in...

conjecture | open | MathDB 页面 | 原文线索

V2-034581 | MATHDB-379881

Width conjecture for wheel graphs

摘录 / EXCERPT: Let W_n be the wheel graph, the cone over an $(n-1)$-gon, and let $w^1_{{\mathcal A}_3}(W_n)$ denote the width of the torsion in $H^{1,*}_{{\mathcal A}_3}(W_n)$. Width conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-034582 | MATHDB-379882

Width conjecture for complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $w^1_{{\mathcal A}_3}(K_n)$ denote the width of the torsion in $H^{1,*}_{{\mathcal A}_3}(K_n)$. Width conjecture for comple...

conjecture | open | MathDB 页面 | 原文线索

V2-034583 | MATHDB-379883

Wheel and broken-wheel cohomology conjecture over \(\mathcal A_5\)

摘录 / EXCERPT: For $n>4$, let W_n be the wheel graph, let W_n^{out} be obtained by deleting an edge from its polygonal part, and let W_n^{in} be obtained by deleting a spike edge. Wheel and...

conjecture | open | MathDB 页面 | 原文线索

V2-034584 | MATHDB-379884

Cohomology conjecture for the square with a diagonal

摘录 / EXCERPT: Let $P_3|P_3$ denote the graph formed by gluing two triangles along an edge, and let $H^{i,j}_{{\mathcal A}_m}(G)$ denote the corresponding graph cohomology. Square-with-diagonal c...

conjecture | open | MathDB 页面 | 原文线索

V2-034585 | MATHDB-379885

Cohomology conjecture for the complete graph \(K_4\)

摘录 / EXCERPT: Let K_4 be the complete graph on four vertices. Complete-graph K_4 conjecture. For every $m\geq4$, ... The paper reports verification for $m\leq14$, but does not establish the st...

conjecture | open | MathDB 页面 | 原文线索

V2-034586 | MATHDB-379886

Cohomology conjecture for the complete graph \(K_5\)

摘录 / EXCERPT: Let K_5 be the complete graph on five vertices. Complete-graph K_5 conjecture. ... The source presents this as a conjectural computational pattern and gives no resolution.

conjecture | open | MathDB 页面 | 原文线索

V2-034587 | MATHDB-379903

Koszulity conjecture for the degree-zero and degree-one subalgebra of the extension algebra

摘录 / EXCERPT: Let A be the algebra governing the principal block of category $\mathcal{O}$, let Δ be the direct sum of the relevant Verma modules, and let $\mathtt{E}$ be the graded ext...

conjecture | open | MathDB 页面 | 原文线索

V2-034588 | MATHDB-379905

The chord-realization conjecture for distortion quotient

摘录 / EXCERPT: Let γ be a curve, let $[\gamma]$ denote its knot type, let $dq_{\gamma}(s,t)$ be the distortion quotient of the chord with distinct endpoints $\gamma(s)$ and $\gamma(t)$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-034589 | MATHDB-379906

Existence of length minimizers in fixed distortion classes

摘录 / EXCERPT: Let $[\gamma]$ be a knot type, let $U_C([\gamma])$ be the set of curves in that knot type whose distortion is at most C, and let $\delta([\gamma])$ be the distortion thickness of...

conjecture | open | MathDB 页面 | 原文线索

V2-034590 | MATHDB-379957

The parameterized Volume Conjecture for colored Jones polynomials

摘录 / EXCERPT: Parameterized Volume Conjecture. There exists an open subset $\mathcal{O}_K\subset\mathbb{C}$ such that for every $u\in\mathcal{O}_K$,

conjecture | open | MathDB 页面 | 原文线索

V2-034591 | MATHDB-379960

Rigidification conjecture for the dual of the adjoint bundle spectrum

摘录 / EXCERPT: Let G be the compact Lie group and EG the universal principal G-bundle, with $Ad(EG)$ its adjoint bundle and T_v its vertical tangent bundle. Let LBG^{-TBG} denote the as...

conjecture | open | MathDB 页面 | 原文线索

V2-034592 | MATHDB-379970

Kálmán's uniqueness conjecture for braid-positive Legendrian links

摘录 / EXCERPT: Let β be a positive braid, let L_β be its braid-positive link closure, and let the corresponding Legendrian closure be represented by the front diagram f_β. A Leg...

conjecture | open | MathDB 页面 | 原文线索

V2-034593 | MATHDB-379974

Vanishing of the zeroth cohomology for symmetric powers

摘录 / EXCERPT: Vanishing conjecture. $H_k^0 = 0$ for all positive integers k. Consequently, $M_k(T)$ is a polynomial for all positive integers k.

conjecture | open | MathDB 页面 | 原文线索

V2-034594 | MATHDB-379979

The geometric presentation conjecture for orientable 3-manifold groups

摘录 / EXCERPT: Geometric presentation conjecture. The group $\langle x_1,\ldots,x_n\mid w\rangle$ is the fundamental group of an orientable 3-manifold if and only if w is a geometric word.

conjecture | open | MathDB 页面 | 原文线索

V2-034595 | MATHDB-379983

The contraction conjecture for the weight-two abelianization of symplectic derivations

摘录 / EXCERPT: Contraction conjecture. The contraction C_{13} induces an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034596 | MATHDB-379988

Quinn's ANR homology 4-manifold conjecture for Poincaré 4-complexes

摘录 / EXCERPT: Quinn's conjecture. If $s(X^4)=0$, then X^4 is (simple) homotopy equivalent to an ANR homology 4--manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-034597 | MATHDB-379989

ANR homology 4-manifold conjecture for Poincaré 4-complexes with free fundamental group

摘录 / EXCERPT: Free-fundamental-group conjecture. Then X^4 is (simple) homotopy equivalent to an ANR homology 4--manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-034598 | MATHDB-380009

Structural stability implies hyperbolicity for rational maps

摘录 / EXCERPT: Let f be a rational map. It is structurally stable if, within a holomorphic family containing it, all sufficiently nearby maps are topologically conjugate to it; it is hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-034599 | MATHDB-380023

Chas-Sullivan's free loop space-Hochschild cohomology conjecture

摘录 / EXCERPT: Let M be a simply connected compact oriented smooth manifold, let $LM=\operatorname{map}(S^1,M)$ be its free loop space, and write $\mathbb{H}_*(LM)=H_{*+d}(LM)$ for shifted free...

conjecture | open | MathDB 页面 | 原文线索

V2-034600 | MATHDB-380038

The Tannakian variation conjecture for graded cohomology of stable G-bundles

摘录 / EXCERPT: Tannakian variation conjecture. The variation of Hodge structure associated to $Gr_W H^i({\rm Bun}^s_{G,C})$ lies in the Tannakian subcategory generated by $R^1\pi_*{\mathbb Q}$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-034601 | MATHDB-380058

The Alexander-polynomial expression conjecture for finite type invariants of long odd-dimensional knots

摘录 / EXCERPT: Let n be an odd integer with $n>1$. For each p with $1\leq p\leq \lfloor n/2\rfloor$, let $\Delta_p(t)$ be the suitably normalized generalized Alexander invariant of a long n...

conjecture | open | MathDB 页面 | 原文线索

V2-034602 | MATHDB-380072

SYZ conjecture on mirrors of Lagrangian sections

摘录 / EXCERPT: Let $M \to B$ and $\check{M} \to B$ be dual torus fibrations arising in the SYZ picture, and let a Lagrangian section of $M \to B$ be a Lagrangian submanifold mapping isomorphicall...

conjecture | open | MathDB 页面 | 原文线索

V2-034603 | MATHDB-380075

Hougardy-Lutz-Zelke realizability conjecture for triangulated orientable surfaces

摘录 / EXCERPT: Hougardy-Lutz-Zelke conjecture. Every triangulation of an orientable surface of genus g with

conjecture | open | MathDB 页面 | 原文线索

V2-034604 | MATHDB-380080

The crepant resolution conjecture for orbifolds

摘录 / EXCERPT: Let $\mathcal{X}$ be an orbifold satisfying the hard Lefschetz condition and admitting a crepant resolution Y. Let F^Y and $F^\mathcal{X}$ denote their potential functions, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-034605 | MATHDB-380093

The degree-two superduflot regular sequence conjecture

摘录 / EXCERPT: Let G be a finite p-group. Its cohomology ring $H^*(G,\mathbb F_p)$ is a graded-commutative, noetherian, local $\mathbb F_p$-algebra. Write $\operatorname{rank}_p Z(G)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-034606 | MATHDB-380094

The field-extension invariance conjecture for degree sequences of regular sequences

摘录 / EXCERPT: Let G be a finite p-group, let k be a positive integer, and let d and $a_1,\dots,a_d$ be integers. A regular sequence has degree sequence $a_1,\dots,a_d$ when its elements...

conjecture | open | MathDB 页面 | 原文线索

V2-034607 | MATHDB-380121

The comparison conjecture for Hopf-cyclic cohomology of algebras with coefficients

摘录 / EXCERPT: Comparison conjecture. There is an isomorphism of homology theories

conjecture | open | MathDB 页面 | 原文线索

V2-034608 | MATHDB-380122

The inverse-limit comparison conjecture for Hopf-cyclic cohomology of coalgebras

摘录 / EXCERPT: Inverse-limit comparison conjecture. The map above is an isomorphism of the cohomology groups.

conjecture | open | MathDB 页面 | 原文线索

V2-034609 | MATHDB-380125

The torus-surgery conjecture for simply connected smooth 4-manifolds

摘录 / EXCERPT: Torus-surgery conjecture. X and Y are related via a sequence of surgeries on tori of square 0.

conjecture | open | MathDB 页面 | 原文线索

V2-034610 | MATHDB-380126

The rational blowup and torus-surgery conjecture for simply connected smooth 4-manifolds

摘录 / EXCERPT: Rational blowup and torus-surgery conjecture. X and Y are related via a sequence of surgeries on tori of square 0, rational blowups, and rational blowdowns.

conjecture | open | MathDB 页面 | 原文线索

V2-034611 | MATHDB-380134

The projective bundle mirror theorem conjecture

摘录 / EXCERPT: Projective bundle mirror theorem conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-034612 | MATHDB-380152

The free abelian link-group conjecture for welded links

摘录 / EXCERPT: Free abelian link-group conjecture. Then L is welded equivalent to the unlink.

conjecture | open | MathDB 页面 | 原文线索

V2-034613 | MATHDB-380185

Finite-projective-dimension decency conjecture

摘录 / EXCERPT: Let R be a local ring and let M be an R-module of finite projective dimension. For every R-module N such that $l(M\otimes_RN)<\infty$, finite-projective-dimension decency...

conjecture | open | MathDB 页面 | 原文线索

V2-034614 | MATHDB-380194

Mumford's conjecture on the stable cohomology of mapping class groups

摘录 / EXCERPT: Mumford's conjecture. This homomorphism should be an isomorphism in the stable range of Harer's stability theorem; in the formulation stated earlier, it is an isomorphism in degree...

conjecture | open | MathDB 页面 | 原文线索

V2-034615 | MATHDB-380199

Automorphism conjecture for Hall algebra Drinfeld doubles

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary finitary category. A derived functor is called friendly when it sends every object into two adjacent cohomological degrees. Automorphism conjectur...

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V2-034616 | MATHDB-380200

Kontsevich-Soibelman non-commutative Hodge-to-de Rham degeneration conjecture

摘录 / EXCERPT: Kontsevich-Soibelman conjecture. For any saturated DG algebra $A^{\:\raisebox{3pt}{\text{\circle*{1.5}}}}$ over a field K of characteristic 0, the Hodge-to-de Rham spectral seq...

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V2-034617 | MATHDB-380210

Niebrzydowski-Przytycki injectivity conjecture for the quandle homology operation h_s

摘录 / EXCERPT: Injectivity conjecture for h_s. For $n>1$, the homomorphism

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V2-034618 | MATHDB-380212

Combined injectivity conjecture for rack homology operations on odd dihedral quandles

摘录 / EXCERPT: Combined injectivity conjecture. For odd k and $n>1$, the direct-sum map

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V2-034619 | MATHDB-380213

Combined injectivity conjecture for quandle homology operations on odd dihedral quandles

摘录 / EXCERPT: Let R_k be an odd dihedral quandle, and let $(h_s)_*$ and $(h'_{(s,0)})_*$ be the induced operations ... ... Here $h'_{(s,0)}=h'_0h_s$. Combined quandle injectivity conjecture. For o...

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V2-034620 | MATHDB-380232

Kazarian's rational homotopy conjecture for the Kazarian space

摘录 / EXCERPT: Let τ be a collection of singularity types, let X_τ be the classifying space for τ-cobordisms, let $\mathcal K_\tau$ be the Kazarian space, let $\mathcal K^+_\tau$ b...

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V2-034621 | MATHDB-380241

Arbitrarily short geodesics in closed hyperbolic manifolds of every dimension

摘录 / EXCERPT: For an integer n, a closed hyperbolic n-manifold is a compact manifold locally isometric to n-dimensional hyperbolic space, and a closed geodesic is a geodesic that is period...

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V2-034622 | MATHDB-380249

The Čech cohomology invariance conjecture for the o-minimal spectrum

摘录 / EXCERPT: Let $M\preccurlyeq N$, let $X=\operatorname{Spec}(A)$ be the o-minimal spectrum under consideration, let $Z\subseteq X$ be definably closed, and let $W:=\theta^{-1}(Z)$, where...

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V2-034623 | MATHDB-380250

The definable T_5 spectrum conjecture

摘录 / EXCERPT: Let M be a first-order topological structure, and let $\operatorname{Spec}(A)$ denote the Stone spectrum of a definable set $A\subseteq M^k$. Say that M is definably T_5 when...

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V2-034624 | MATHDB-380319

Wall's realization conjecture for Poincaré duality groups

摘录 / EXCERPT: Let Γ be a finitely presented n-dimensional Poincaré duality group over $\mathbb Z$. Wall's conjecture. There exists a closed n-dimensional manifold M which is...

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V2-034625 | MATHDB-380324

The virtual first Betti number conjecture for hyperbolic lattices

摘录 / EXCERPT: Let $\Gamma\subset \operatorname{Mob}(\mathbb S^n)$ be a lattice. Virtual first Betti number conjecture. There exists a finite-index subgroup $\Gamma'\subset\Gamma$ such that ... Equ...

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V2-034626 | MATHDB-380327

Generalized amalgamation conjecture for irreducible 3-manifolds

摘录 / EXCERPT: Let M_i ($i=1,2$) be irreducible 3-manifolds with connected boundary, with $\partial M_1\cong\partial M_2\cong F$. Let $\mathcal{D}_i$ be the set of essential curves in F that...

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V2-034627 | MATHDB-380334

Zero-divisors-cup-length conjecture for hyperplane arrangement complements

摘录 / EXCERPT: Let an arrangement of finitely many complex central hyperplanes have complement M. Let the zero-divisors-cup-length of $H^*(M,\mathbf C)$ mean the largest number of zero-divisors...

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V2-034628 | MATHDB-380357

The entropy-or-flatness conjecture for compact manifolds without conjugate points

摘录 / EXCERPT: Entropy-or-flatness conjecture. Either M is flat or its geodesic flow has positive topological entropy.

conjecture | open | MathDB 页面 | 原文线索

V2-034629 | MATHDB-380363

Prasad's comparison conjecture for second cohomology of p-adic analytic groups

摘录 / EXCERPT: Let P be a pro-p group, and write ... Let $\varphi^2:H_{\operatorname{cont}}^{2}(P)\to H_{\operatorname{disc}}^{2}(P)$ be the comparison map. Prasad's conjecture. For every p-a...

conjecture | open | MathDB 页面 | 原文线索

V2-034630 | MATHDB-380364

Morimoto's conjecture on Heegaard genus under connected sum

摘录 / EXCERPT: Morimoto's conjecture. The inequality

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V2-034631 | MATHDB-380370

Determination of Gromov-Witten invariants after blowing up a curve

摘录 / EXCERPT: Let Y be the blow-up of a smooth projective threefold X along a curve C such that $g(C)\geq 1$, or $g(C)=0$ and $-K_X.C\geq 0$. The associativity equations of the quantum pro...

conjecture | open | MathDB 页面 | 原文线索

V2-034632 | MATHDB-380382

Minimax volume conjecture for Steenrod-square cup powers on rectangles

摘录 / EXCERPT: Minimax volume conjecture. The minimax volume satisfies, up to a constant factor depending only on n,

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V2-034633 | MATHDB-380387

Connectivity conjecture for the two-component link-map comparison map

摘录 / EXCERPT: Connectivity conjecture. The map l_2 is $(2n-\max\{2p+q,2q+p\}-3)$-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-034634 | MATHDB-380388

Three-component link-map connectivity conjecture

摘录 / EXCERPT: Three-component connectivity conjecture. The map l_3 is $(3n-\max\{2p_1+p_2+p_3,p_1+2p_2+p_3,p_1+p_2+2p_3\}-5)$-connected.

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V2-034635 | MATHDB-380391

The PL sphere-or-ball conjecture for intervals with the NOF property

摘录 / EXCERPT: PL sphere-or-ball conjecture. If $[u,v]$ is full, then (u,v) is a PL sphere of dimension $\rho(v)-\rho(u)-2$. Otherwise, (u,v) is a PL ball of the same dimension.

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V2-034636 | MATHDB-380447

Gottlieb's centralizer conjecture for nonzero-degree maps

摘录 / EXCERPT: Let (Y,y) be a finite aspherical CW-complex which is not acyclic, and let $f\colon (X,x)\to (Y,y)$ be a continuous map. Define the degree of f by letting $\deg(f)$ be the least...

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V2-034637 | MATHDB-380456

Shereshevsky's finite-entropy conjecture for multidimensional cellular automata

摘录 / EXCERPT: Let S be a finite set, let $d>1$, and let $X=S^{\mathbb{Z}^d}$. A d-dimensional cellular automaton is a continuous map $T:X\to X$ commuting with the shift action of...

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V2-034638 | MATHDB-380458

Fukaya-dimer quasi-equivalence conjecture for convex lattice polygons

摘录 / EXCERPT: Let Δ be a convex lattice polygon containing the origin in its interior. A Laurent polynomial W has Newton polygon Δ, and a pair (G,D) consists of a consistent di...

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V2-034639 | MATHDB-380469

The topological volume-preservation problem for Anosov flows

摘录 / EXCERPT: Let Φ be an Anosov flow. It is topologically equivalent to a volume-preserving flow when it is topologically equivalent to an Anosov flow that preserves a volume. Volume-prese...

open_problem | open | MathDB 页面 | 原文线索

V2-034640 | MATHDB-380471

Friedlander-Milnor conjecture for Lie groups

摘录 / EXCERPT: Let G be a Lie group. Regard BG as the classifying space of the underlying discrete topological group, and let $BG^{\rm top}$ be the classifying space with its usual topology....

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V2-034641 | MATHDB-380488

Homotopy invariance of L2-torsion

摘录 / EXCERPT: Let G be a group, let $\operatorname{Wh}(G)$ be its Whitehead group, and let $A\in GL_n(\mathbb ZG)$. Let $r_A^{(2)}$ be the G-equivariant bounded operator on $\ell^2(G)^n$ giv...

conjecture | open | MathDB 页面 | 原文线索

V2-034642 | MATHDB-380496

Smoothness conjecture for the singular part of the distributional leafwise trace

摘录 / EXCERPT: The smoothness conjecture. If $\operatorname{codim}\mathcal{F}>0$ and the leaves are dense, then

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V2-034643 | MATHDB-380502

Jannsen's local vanishing conjecture

摘录 / EXCERPT: Let F be a number field, let $\mathfrak{p}$ be a finite place, and let $F_{\mathfrak{p}}$ be the completion of F at $\mathfrak{p}$ with algebraic closure...

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V2-034644 | MATHDB-380503

Jannsen's vanishing conjecture for Hecke-character realizations

摘录 / EXCERPT: Let K be the imaginary quadratic field and let $M_{\theta\mathbb{Q}_p}(n)$ be the p-adic realization of the Hecke-character motive defined in the source, with weight w. Let...

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V2-034645 | MATHDB-380508

Residual representability of the Heegaard category

摘录 / EXCERPT: Let ${\cal H}$ be the Heegaard category, and let $T(V)$ denote the relevant target associated with a vector space V. Residual representability conjecture. If a and b are dist...

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V2-034646 | MATHDB-380512

Telgarsky's separation conjecture for Banach-Mazur k-tactics

摘录 / EXCERPT: Telgarsky's conjecture. For each positive integer k there is a topological space (X_k,τ_k) such that TWO does not have a winning k-tactic, but does have a winning $k+1$-ta...

conjecture | open | MathDB 页面 | 原文线索

V2-034647 | MATHDB-380513

The 2-tactic conjecture for the strongly monotonic game

摘录 / EXCERPT: Let κ be an infinite cardinal. In the game $SMG([\kappa]^{<\aleph_{0}})$, player ONE plays an increasing sequence of members of the generated ideal and TWO responds with mem...

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V2-034648 | MATHDB-380515

Baum-Douglas elliptic boundary condition conjecture

摘录 / EXCERPT: Let M be a compact oriented smooth manifold with smooth boundary ∂M, let E_0 and E_1 be smooth complex Hermitian vector bundles over M, and let...

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V2-034649 | MATHDB-380523

Formal smallness and torsoriality conjecture for Gen

摘录 / EXCERPT: Let $\text{\bf Gen}$ be the category of topological spaces under consideration, equipped with its pseudoétale and torsorial topologies. The formal smallness conjecture. The pseudoé...

conjecture | open | MathDB 页面 | 原文线索

V2-034650 | MATHDB-380524

Invariance of torsorial and pseudoétale morphisms under cardinal restriction

摘录 / EXCERPT: Let S be an infinite set, and let f be a $\text{\bf Gen}[S]$-morphism, where $\text{\bf Gen}[S]$ is the subcategory of objects admitting a cover by open sets of cardinality at...

conjecture | open | MathDB 页面 | 原文线索

V2-034651 | MATHDB-380525

Agreement of pseudoétale and elementary finite-to-one topologies on diffuse manifolds

摘录 / EXCERPT: Let $\mathcal{M}$ be the subcategory of C^n-manifolds, where $n\in\mathbb{N}$ or $n=\infty$, consisting of all manifolds but only diffuse C^n-functions. Let the elementary fini...

conjecture | open | MathDB 页面 | 原文线索

V2-034652 | MATHDB-380535

The Universal Covering Conjecture for closed irreducible orientable 3-manifolds

摘录 / EXCERPT: Let X be a closed, connected, irreducible, orientable 3-manifold with infinite fundamental group. Its universal covering space is denoted by $\widetilde{X}$. Universal Covering C...

conjecture | open | MathDB 页面 | 原文线索

V2-034653 | MATHDB-380536

Special Equivariant Plane Conjecture

摘录 / EXCERPT: Let G be a finitely generated group of covering translations acting on a Whitehead manifold W, with quotient a 3-manifold M. A proper plane Π in W is equivariant if, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-034654 | MATHDB-380565

The Bousfield lattice quotient by strange classes

摘录 / EXCERPT: The quotient-isomorphism conjecture. The epimorphism r' is an isomorphism.

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V2-034655 | MATHDB-380566

The minimality conjecture for the Bousfield classes of A(n)

摘录 / EXCERPT: The minimality conjecture for A(n). If $n\geq 2$, then $\langle A(n)\rangle$ is a minimal nonzero Bousfield class. Moreover, every such E is Bousfield equivalent to a finite wedg...

conjecture | open | MathDB 页面 | 原文线索

V2-034656 | MATHDB-380567

The Bousfield class decomposition conjecture for D

摘录 / EXCERPT: The decomposition conjecture for D. One has

conjecture | open | MathDB 页面 | 原文线索

V2-034657 | MATHDB-380568

The finite detection conjecture for the Brown-Comenetz dual of the sphere

摘录 / EXCERPT: Finite detection conjecture for I. If $E\wedge I\neq 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-034658 | MATHDB-380569

The Dichotomy Conjecture for finite locals and finite acyclics

摘录 / EXCERPT: The Dichotomy Conjecture. Every spectrum has either a finite local or a finite acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-034659 | MATHDB-380570

The Bousfield-class formulation of the Dichotomy Conjecture

摘录 / EXCERPT: Bousfield-class dichotomy conjecture. Then

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V2-034660 | MATHDB-380571

The adjacency conjecture for X(n) and X(n+1)

摘录 / EXCERPT: X(n)-adjacency conjecture. The spectra $X(n)$ and $X(n+1)$ are adjacent in the Bousfield lattice: if

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V2-034661 | MATHDB-380572

The X(n)-telescope meet conjecture

摘录 / EXCERPT: X(n)-telescope meet conjecture. For all n and k,

conjecture | refuted | MathDB 页面 | 原文线索

V2-034662 | MATHDB-380573

The sphere and complex projective space Bousfield-equivalence conjecture

摘录 / EXCERPT: Sphere-projective-space conjecture. Their Bousfield classes are equal:

conjecture | refuted | MathDB 页面 | 原文线索

V2-034663 | MATHDB-380574

The principal-localizing-subcategory conjecture

摘录 / EXCERPT: Principal-localizing-subcategory conjecture. Every localizing subcategory is the collection of E-acyclics for some spectrum E, and is therefore principal.

conjecture | open | MathDB 页面 | 原文线索

V2-034664 | MATHDB-380575

Milnor's homotopy-sphere conjecture for Brieskorn manifolds

摘录 / EXCERPT: Let $a_0, a_1,\dots,a_n\geq 2$, and let … be the $(2n-1)$-dimensional Brieskorn link. Let $\Gamma(a_0,a_1,\dots,a_n)$ be the graph with vertices $0,1,\dots,n$, joining i and j…

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V2-034665 | MATHDB-380597

The linked-pair lower bound for the second Vassiliev invariant

摘录 / EXCERPT: Let D be a positive bireduced knot diagram, and let $lk(D)$ denote the number of linked pairs in D. Let v_2 be the second Vassiliev invariant. The linked-pair lower-bound con…

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V2-034666 | MATHDB-380598

The degree equality conjecture for almost positive knots

摘录 / EXCERPT: Let K be an almost positive knot. Write Δ_K for its Alexander polynomial and V_K for its Jones polynomial, with $\max\deg \Delta_K$ and $\min\deg V_K$ denoting the corre…

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V2-034667 | MATHDB-380599

The pretzel-minimizer conjecture for 2-almost positive diagrams

摘录 / EXCERPT: A knot is 2-almost positive if the minimal number of negative crossings among all its diagrams is 2. Let L be a 2-almost positive diagram with even crossing number that minimiz…

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V2-034668 | MATHDB-380600

Semi-splitting conjecture for symplectic fibre bundles over spheres

摘录 / EXCERPT: Semi-splitting conjecture. The deformation of $H^*(M \times N)$ determined by $H^*(E)$ is semi-split with respect to $H^*(M)$. This conjecture proposes a general cohomological spli…

conjecture | open | MathDB 页面 | 原文线索

V2-034669 | MATHDB-380608

General equivariant higher signature homotopy invariance conjecture

摘录 / EXCERPT: Let M be a closed oriented manifold with an effective S^1-action. For each connected component F of M^{S^1}, assume the hypotheses on Γ_F, its smooth subalgebra, and…

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V2-034670 | MATHDB-380613

Mimura's spherical resolution conjecture for E_7/F_4

摘录 / EXCERPT: Localize all spaces at 3. Let $B_2(27,35)$ denote the space occurring in the proposed spherical resolution, and let α_2 be the element generating…

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V2-034671 | MATHDB-380614

Divisibility conjecture for Conway polynomials of links in rational homology spheres

摘录 / EXCERPT: Let K be a null-homologous oriented k-component link with zero pairwise linking numbers in a rational homology sphere $Sigma$, and let L be an admissible link of ell compon…

conjecture | open | MathDB 页面 | 原文线索

V2-034672 | MATHDB-380618

Positive twist knot diagrams minimize the Casson-Vassiliev invariant

摘录 / EXCERPT: Positive twist-knot minimization conjecture. Positive twist knot diagrams minimize v_3 among all connected irreducible positive diagrams with odd crossing number.

conjecture | open | MathDB 页面 | 原文线索

V2-034673 | MATHDB-380619

Murakami-Przytycki's unknotting-number conjecture for positive fibered knots

摘录 / EXCERPT: Murakami-Przytycki's conjecture. For every positive fibered knot K,

conjecture | open | MathDB 页面 | 原文线索

V2-034674 | MATHDB-380620

The linked-pair lower bound for the Casson invariant of positive bireduced diagrams

摘录 / EXCERPT: Linked-pair lower-bound conjecture. One has

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V2-034675 | MATHDB-380624

Eguchi-Hori-Xiong Virasoro constraints for the quantum cohomology partition function

摘录 / EXCERPT: Eguchi-Hori-Xiong conjecture. The partition function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-034676 | MATHDB-380628

The generic local-system cohomology conjecture for Orlik-Solomon algebras

摘录 / EXCERPT: Generic local-system cohomology conjecture. For every p and almost all $a\in A_1^*$ among those with

conjecture | open | MathDB 页面 | 原文线索

V2-034677 | MATHDB-380632

Lück's homotopy-invariance conjecture for L2-Reidemeister torsion

摘录 / EXCERPT: Lück's homotopy-invariance conjecture. L^2-Reidemeister torsion is a homotopy invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-034678 | MATHDB-380641

Monotonicity conjecture for the length of real projective spaces

摘录 / EXCERPT: Let $\operatorname{\mathbb{R}P}^n$ denote real projective n-space, and let its length mean the invariant defined in the paper. Length monotonicity conjecture. The length of ... is...

conjecture | solved | MathDB 页面 | 原文线索

V2-034679 | MATHDB-380642

Witten's localization formula for reductions at regular co-adjoint orbits

摘录 / EXCERPT: For each $t\in\mathbb{R}$, $X\in\mathfrak{g}$, let $\Theta(M,t)(X):=\int_M e^{-itd_X\lambda^{M_{\mathcal O}}}\alpha(X)$, and let $\Theta_0^{\mathcal O}$ denote the generalized-func...

conjecture | open | MathDB 页面 | 原文线索

V2-034680 | MATHDB-380667

Homotopy hypothesis for strict non-unital n-groupoids

摘录 / EXCERPT: Homotopy hypothesis for snugroupoids. There are functors

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V2-034681 | MATHDB-380678

Corrected Feigin cohomology conjecture for twisted differential operators

摘录 / EXCERPT: Let $\lambda\in\mathbb C$, let ${\mathfrak{gl}}(\lambda)$ be embedded in Dif_1, and let ... be the first page of the associated Feigin Serre-Hochschild spectral sequence....

conjecture | open | MathDB 页面 | 原文线索

V2-034682 | MATHDB-380684

The generalized rosette knot cylinder-knot conjecture

摘录 / EXCERPT: Let s and k be positive integers, let $\varepsilon_i\in\{\pm1\}$ for $1\leq i\leq s-1$, and consider the closure of the braid ... A generalized rosette knot is this closed braid....

conjecture | open | MathDB 页面 | 原文线索

V2-034683 | MATHDB-380700

Kontsevich's quasi-isomorphism invariance conjecture for polyvector-field cohomology

摘录 / EXCERPT: Kontsevich's invariance conjecture. There should be an isomorphism of algebras

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V2-034684 | MATHDB-380701

Kontsevich's tangent-map conjecture for formality quasi-isomorphisms

摘录 / EXCERPT: Kontsevich's tangent-map conjecture. The induced map

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V2-034685 | MATHDB-380702

The Duflo formula for Q-manifolds

摘录 / EXCERPT: Duflo formula for Q-manifolds. 1. The map

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V2-034686 | MATHDB-380711

Virtual positive first Betti number conjecture for 3-manifolds

摘录 / EXCERPT: Virtual positive first Betti number conjecture. M has a finite cover $\tilde{M}$ with

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V2-034687 | MATHDB-380717

The Tor description conjecture for cohomology of Witt groups

摘录 / EXCERPT: Let F be a field with $|\cdot F/\cdot F^2|=2^n$, so that $E=E_n\cong (\mathbb Z/2)^n$, and let ${\mathcal G}_F$ be the associated W-group. Let...

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V2-034688 | MATHDB-380727

Infinitely many odd-superbridge knots conjecture

摘录 / EXCERPT: Odd-superbridge knots conjecture. There are infinitely many $(2n-1)$-superbridge knots for any positive integer $n\ge3$.

conjecture | open | MathDB 页面 | 原文线索

V2-034689 | MATHDB-380728

Superbridge index additivity under connected sum with a trefoil

摘录 / EXCERPT: Trefoil connected-sum conjecture. Every nontrivial knot K satisfies

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V2-034690 | MATHDB-380729

Full-twist conjecture for odd-superbridge torus braids

摘录 / EXCERPT: Full-twist odd-superbridge conjecture. For $n\ge3$ and $k\ge0$, the closure of this braid is a $(2n-1)$-superbridge knot.

conjecture | open | MathDB 页面 | 原文线索

V2-034691 | MATHDB-380733

Extra symmetry of transfer operators for the dilute A_3 model

摘录 / EXCERPT: Extra symmetry conjecture. As operators on these BRST cohomology groups,

conjecture | open | MathDB 页面 | 原文线索

V2-034692 | MATHDB-380734

Acyclicity and comparison conjectures for new globular and corner homology

摘录 / EXCERPT: Let $\mathcal{C}$ be an ω-category, let I^n denote the n-dimensional cube, let A and B be composable n-morphisms, and let $\widehat{\mathcal{C}}$ denote the modifi...

conjecture | open | MathDB 页面 | 原文线索

V2-034693 | MATHDB-380735

Thin corner-cycle conjecture for free globular omega-categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a free globular ω-category. For a natural number n, let $x_i\in\omega\operatorname{Cat}(I^n,\mathcal{C})^\pm$ and let λ_i be natural numbers i...

conjecture | open | MathDB 页面 | 原文线索

V2-034694 | MATHDB-380736

Thin-element decomposition conjecture for homotopic functors

摘录 / EXCERPT: Let $\mathcal{C}$ and $\mathcal{D}$ be free ω-categories, let $f,g:\mathcal{C}\to\mathcal{D}$ be homotopic non 1-contracting ω-functors, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034695 | MATHDB-380737

Corner homology of the tensor product with the interval

摘录 / EXCERPT: Let $\mathcal{C}$ be an ω-category and let I^1 be the interval in the biclosed monoidal structure on $\omega\operatorname{Cat}$. For either choice of sign, consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-034696 | MATHDB-380738

Corner homology of cubes, free morphism categories and globes

摘录 / EXCERPT: Let I^n be the n-cube, let 2_n be the free ω-category generated by an n-morphism, and let G_n for $n>0$ be the oriented n-globe, namely the free ω-categor...

conjecture | open | MathDB 页面 | 原文线索

V2-034697 | MATHDB-380739

Postnikov classification conjecture for semi-weak monoidal 2-categories

摘录 / EXCERPT: Let G be a group, H an abelian group, and R a commutative ring with unit group R^*. Let B_G be the classifying space of G, and for each cohomology class...

conjecture | open | MathDB 页面 | 原文线索

V2-034698 | MATHDB-380759

The universal locally free subgroup conjecture for finite-volume hyperbolic 3-manifolds

摘录 / EXCERPT: Universal locally free subgroup conjecture. The fundamental group $\pi_1(N)$ contains a subgroup which is locally free but not free.

conjecture | open | MathDB 页面 | 原文线索

V2-034699 | MATHDB-380760

The fibered-cover characterization of locally free subgroups

摘录 / EXCERPT: Fibered-cover characterization conjecture. The fundamental group $\pi_1(N)$ contains a subgroup which is locally free but not free if and only if N does not have a finite cover w...

conjecture | open | MathDB 页面 | 原文线索

V2-034700 | MATHDB-380761

Greuel-Lossen-Shustin asymptotic properness conjecture for the gamma invariant

摘录 / EXCERPT: Let $V_d^{\operatorname{irr}}(S_1,\dots,S_r)$ denote the family of reduced irreducible complex plane curves of degree d with isolated singular points of prescribed topological ty...

conjecture | open | MathDB 页面 | 原文线索

V2-034701 | MATHDB-380770

Schapira-Schneiders microlocal Euler class conjecture

摘录 / EXCERPT: Let X be a complex manifold of dimension d, and let ${\cal M}^\bullet$ be a complex of ${\cal D}_X$-modules with bounded good cohomology. Choose a filtration compatible with th...

conjecture | open | MathDB 页面 | 原文线索

V2-034702 | MATHDB-380772

Tsygan's generalized Gerstenhaber formality conjecture for chains

摘录 / EXCERPT: Tsygan's generalized Gerstenhaber formality conjecture. For any associative algebra A, the Hochschild chain complex $C_{\bullet}(A,A)$ is a G_{∞} module over the...

conjecture | open | MathDB 页面 | 原文线索

V2-034703 | MATHDB-380775

The conjectural form of the Poincaré-Betti series for A_n

摘录 / EXCERPT: Conjectural Poincaré-Betti series formula. The polynomial $q_n(t)$ satisfies $q_n(-1)\neq 0$, and

conjecture | open | MathDB 页面 | 原文线索

V2-034704 | MATHDB-380778

Witten's generalized conjecture for higher-spin curves

摘录 / EXCERPT: Let r-spin curves have moduli spaces $\overline{\mathcal{M}}_{g,n}^{1/r,\mathbf{m}}$, and let $\Psi(\mathbf{t})$ be the potential of the KdV_r hierarchy. Intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-034705 | MATHDB-380781

Vassiliev's stabilization conjecture for finite-type invariants

摘录 / EXCERPT: Let K be a knot, and let a Vassiliev (finite-type) invariant mean any finite-type invariant of knots. Vassiliev's stabilization conjecture. If every Vassiliev invariant of K is...

conjecture | open | MathDB 页面 | 原文线索

V2-034706 | MATHDB-380793

Makeev's concurrent-planes conjecture for regular simplices

摘录 / EXCERPT: Makeev's conjecture. There exists an $A\in SO(n)$ such that all $n(n+1)/2$ of these $(n-1)$-planes are concurrent. The conjecture is a reformulation of Makeev's universal-cover con...

conjecture | open | MathDB 页面 | 原文线索

V2-034707 | MATHDB-380812

Optimality of asymptotic Nielsen number bounds for trellis maps

摘录 / EXCERPT: Let T be a trellis and let f be a trellis map for T. Write $N_\infty(f)$ for the asymptotic Nielsen number and h_top for topological entropy. Optimality conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-034708 | MATHDB-380813

The natural-basis conjecture for homology of crepant resolutions of quotient singularities

摘录 / EXCERPT: Let $X=V/G$ be the quotient of a complex vector space V by a finite subgroup $G\subset SL(V)$, and assume that X admits a smooth crepant resolution $Y\to X$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-034709 | MATHDB-380816

The converse of the extension property proposition for diagram operads

摘录 / EXCERPT: Extension property converse. If every $(S,\underline D)$-diagram of ${\cal P}$-algebras can be extended to an S-diagram of ${\cal P}$-algebras, then the couple (S,D) has the...

conjecture | open | MathDB 页面 | 原文线索

V2-034710 | MATHDB-380837

Khovanov's presentation-independence conjecture for link-cobordism maps

摘录 / EXCERPT: Let S be a compact oriented surface and let $J,\widetilde J$ be representations of S by sequences of link diagrams, with endpoint diagrams J_0,J_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-034711 | MATHDB-380841

The asymptotic relation between degree-two and degree-three Vassiliev invariants of positive knots

摘录 / EXCERPT: Asymptotic Vassiliev-invariant conjecture. The numbers $\log_{v_2(K_i)}v_3(K_i)$, which are well-defined for $K_i\ne !3_1$, converge to 2 as $i\to\infty$. This predicts a precise...

conjecture | open | MathDB 页面 | 原文线索

V2-034712 | MATHDB-380863

Non-commutative Tkachenko-Uspenskij conjecture

摘录 / EXCERPT: Let X be a Tychonoff topological space, let $F_{\mathrm{SIN}}(X)$ be the free balanced topological group on X, and, for a Hilbert space ${\mathcal H}$, let $U({\mathcal H})_u$...

conjecture | open | MathDB 页面 | 原文线索

V2-034713 | MATHDB-380876

The bubble-separation conjecture for two intersecting spheres

摘录 / EXCERPT: Let two spheres in $\mathbb{R}^3$ be equipped with oriented normal bundles and intersect along a circle in the space $\tilde Y$, with the canonical orientation induced on the inter...

conjecture | open | MathDB 页面 | 原文线索

V2-034714 | MATHDB-380882

Boundary-path correspondence conjecture for compactified open-curve moduli spaces

摘录 / EXCERPT: Let X be the bordered holomorphic curve and let $\tilde M(f)$ be the compactified moduli space associated with a fixed homotopy class $[f]$. A simple path in X has no self-inte...

open_problem | open | MathDB 页面 | 原文线索

V2-034715 | MATHDB-380886

Paternain's real-analytic entropy conjecture for integrable geodesic flows

摘录 / EXCERPT: Let a closed manifold carry a geodesic flow that is analytically integrable, meaning integrable by real-analytic first integrals in the sense used in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034716 | MATHDB-380887

The exotic cosmetic surgery conjecture

摘录 / EXCERPT: A cosmetic surgery is a pair of Dehn fillings on a knot or 3-manifold exterior that produce homeomorphic filled manifolds; it is truly cosmetic when the homeomorphism preserves the...

conjecture | open | MathDB 页面 | 原文线索

V2-034717 | MATHDB-380890

The closed geodesic complement conjecture

摘录 / EXCERPT: Let N be a hyperbolic 3-manifold and let c_1 and c_2 be closed geodesics in N. Their complements are the manifolds obtained by removing the geodesics. Closed geodesic compl...

conjecture | open | MathDB 页面 | 原文线索

V2-034718 | MATHDB-380893

Stable secondary-class polynomiality conjecture for the Johnson kernel

摘录 / EXCERPT: Let ${\cal K}_g$ be the Johnson kernel, let $d_i\in H^{4i-3}({\cal K}_g;{\mathbb Q})$ be the higher secondary classes, and let ${\cal M}_g$ act on $H^*({\cal K}_g;{\mathbb Q})$. Se...

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V2-034719 | MATHDB-380898

Determinant de Rham cohomology formula for admissible connections

摘录 / EXCERPT: Determinant de Rham formula.

conjecture | open | MathDB 页面 | 原文线索

V2-034720 | MATHDB-380899

Epsilon-class formula for determinant de Rham cohomology

摘录 / EXCERPT: Epsilon-class conjecture. If ∇ is an integrable, admissible, absolute connection, then

conjecture | open | MathDB 页面 | 原文线索

V2-034721 | MATHDB-380916

The identification of the unframed invariant with the LMO invariant

摘录 / EXCERPT: Let M be an integer homology 3-sphere, let $\widetilde{I}_n(M)$ be the decoration-independent unframed invariant obtained by subtracting the framing correction from $I_n(M)$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-034722 | MATHDB-380917

The vanishing of the framing correction in higher degrees

摘录 / EXCERPT: Let $\delta_n(M)\in V_n$ be the framing correction for the invariant $I_n(M)$ of a homology 3-sphere M decorated with a framing or bundled bordism. Vanishing conjecture for the...

conjecture | open | MathDB 页面 | 原文线索

V2-034723 | MATHDB-380918

Weak reducibility conjecture for additive tunnel numbers

摘录 / EXCERPT: Let K_1 and K_2 be knots in S^3, and write $K=K_1\#K_2$. Let $E(K)$ denote the exterior of K, and let $t(K)$ denote its tunnel number. A Heegaard splitting is weakly reduci...

conjecture | open | MathDB 页面 | 原文线索

V2-034724 | MATHDB-380919

Fiendish knot conjecture on tunnel-number additivity

摘录 / EXCERPT: Let $K_1,K_2\subset S^3$, write $K=K_1\#K_2$, and let $E(K_i)$ denote the exterior of K_i. Let $t(K_i)$ denote the tunnel number, and call a Heegaard splitting minimal genus if i...

conjecture | open | MathDB 页面 | 原文线索

V2-034725 | MATHDB-380943

Ohtsuki's factorial integrality conjecture for homology 3-sphere invariants

摘录 / EXCERPT: Ohtsuki's factorial integrality conjecture. For every $n\geq 1$ and every integral homology 3-sphere M,

conjecture | open | MathDB 页面 | 原文线索

V2-034726 | MATHDB-380956

LMO's universality conjecture for Ohtsuki invariants

摘录 / EXCERPT: Let M be an integral homology three sphere. Let $+ default$ denote the generating series of the Ohtsuki invariants $ default$ and let $ default$ be the universal invariant of Le,...

conjecture | open | MathDB 页面 | 原文线索

V2-034727 | MATHDB-380962

The one-step integration conjecture for $T4T$ and TFI invariants

摘录 / EXCERPT: Let an invariant of chord diagrams satisfy the $T4T$ and TFI relations. An invariant is integrated one step when it is lifted to an invariant at the next diagrammatic level. The...

conjecture | open | MathDB 页面 | 原文线索

V2-034728 | MATHDB-380963

The graph-homology identification conjecture for diagram cohomology

摘录 / EXCERPT: For each m, let ${\cal D}^*_m$ be the dual diagram complex with differential ∂, and let $H^1_\partial({\cal D}^*_m)$ denote its first cohomology group. The graph-homolog...

conjecture | open | MathDB 页面 | 原文线索

V2-034729 | MATHDB-380964

The Lie-algebra construction conjecture for weight systems

摘录 / EXCERPT: Let ${\frak g}$ be a finite-dimensional Lie algebra with a metric and orthonormal basis $\{{\frak g}_a\}_{a=1}^{\dim{\frak g}}$, and let R be a finite-dimensional representation....

conjecture | open | MathDB 页面 | 原文线索

V2-034730 | MATHDB-380966

The wheeling conjecture for Chinese-character diagram products

摘录 / EXCERPT: Wheeling conjecture. Wheeling intertwines the two products on Chinese characters; more precisely,

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V2-034731 | MATHDB-380978

The effective categorification conjecture for A_{∞}-operads

摘录 / EXCERPT: Effective categorification conjecture. For any A_{∞}-operad ${\cal C}$, there is an operation $Cat _{{\cal C}}$ from $\widehat{{\cal C}}$-precats to $\widehat{{\cal C}}$-cat...

conjecture | open | MathDB 页面 | 原文线索

V2-034732 | MATHDB-380979

May's generalized Seifert-van Kampen conjecture for A_{∞}-groupoids

摘录 / EXCERPT: May's generalized Seifert-van Kampen conjecture. The loop space functor from spaces to grouplike $\widehat{{\cal C}}$-spaces takes pushouts of connected spaces to pushouts of...

conjecture | open | MathDB 页面 | 原文线索

V2-034733 | MATHDB-380981

The generalized pushout conjecture for little-cube categories

摘录 / EXCERPT: Generalized pushout conjecture. There is a notion of pushout of ${\cal C}$-categories such that the loop space functor from spaces to ${\cal C}$-groupoids takes pushouts to pushout...

conjecture | open | MathDB 页面 | 原文线索

V2-034734 | MATHDB-380983

Feigin's cohomology conjecture for the Lie algebra of differential operators

摘录 / EXCERPT: Let Dif_1 denote the Lie algebra of differential operators of order at most one on the circle, and let $\Psi_3,\widetilde\Psi_5,\widetilde\Psi_7,\widetilde\Psi_9,\dots$...

conjecture | open | MathDB 页面 | 原文线索

V2-034735 | MATHDB-380984

The lifting formula under conditions (i)--(iii)

摘录 / EXCERPT: Let $\mathfrak A$ be an associative algebra with a trace, and let $D_1,\dots,D_i$ be trace-preserving associative derivations satisfying conditions (i)--(iii): each D_j is trace-...

conjecture | open | MathDB 页面 | 原文线索

V2-034736 | MATHDB-380987

Bar-Natan's conjecture on Vassiliev invariants and quantum Lie group invariants

摘录 / EXCERPT: A Vassiliev invariant is a finite-type invariant of knots, and a quantum Lie group invariant is a knot invariant obtained from quantum-group representations. Bar-Natan's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-034737 | MATHDB-381019

The local model conjecture for homeomorphism groups of non-compact manifolds

摘录 / EXCERPT: Local model conjecture. The pair $(\mathcal H(M),\mathcal H_c(M))$ is locally homeomorphic to $(\square^\omega l_2,\boxdot^\omega l_2)$ at the identity id_M of M. In p...

conjecture | open | MathDB 页面 | 原文线索

V2-034738 | MATHDB-381050

Finiteness conjecture for irreducible simply connected 4-manifolds of fixed complexity

摘录 / EXCERPT: Let c be a natural number, and consider irreducible simply connected 4-manifolds of complexity c. Finiteness conjecture. For every natural number c there are finitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-034739 | MATHDB-381059

Leading-term conjecture for the Jones polynomial of alternating braids

摘录 / EXCERPT: Let L be a positive integer, let $a_1,\ldots,a_{2L}$ satisfy $a_i\geq 2$, and set $D=a_1+\cdots+a_{2L}$. For the braid word $x_1^{a_1}x_2^{a_2}\dots x_1^{a_{2L-1}}x_2^{a_{2L}}$,...

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V2-034740 | MATHDB-381065

The subexponential regulator conjecture for congruence arithmetic hyperbolic 3-manifolds

摘录 / EXCERPT: Subexponential regulator conjecture.

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V2-034741 | MATHDB-381067

Homotopy equivalence conjecture for ascending and descending links

摘录 / EXCERPT: Homotopy equivalence conjecture. The maps $l^\pm_V(p)$ are homotopy equivalences of ${\cal I}$-filtered spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-034742 | MATHDB-381091

Busemann's manifold conjecture for Busemann G-spaces

摘录 / EXCERPT: Let (X,d) be a Busemann G-space, meaning that it is a metric space satisfying Menger convexity, finite compactness, local extendibility, and uniqueness of extension. Suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-034743 | MATHDB-381094

Kawauchi's splitting conjecture for the Conway polynomial of achiral knots

摘录 / EXCERPT: Let K be an achiral knot, and let $C(z)$ denote its Conway polynomial. Kawauchi's conjecture. The polynomial $C(z)$ has the splitting property: there is a polynomial $F(z)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-034744 | MATHDB-381099

Higher-genus generalization of the Toeplitz operator result in TQFT

摘录 / EXCERPT: Let $\mathcal M(\Sigma,t)$ be the moduli space associated with a surface Σ and parameter t. In the higher-genus setting, model $\mathcal M(\Sigma,t)$ by a toric manifold,...

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V2-034745 | MATHDB-381120

Topological invariance of the state-integral invariants

摘录 / EXCERPT: Let M be a hyperbolic 3-manifold with a regular triangulation $\mathcal{T}$ and NeumannZagier datum, including a choice of quad type with $\det \mathbf{B}\neq 0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034746 | MATHDB-381121

Equivariant Dugundji characterization of metric G-ANE spaces

摘录 / EXCERPT: Let G be a compact group. A metric G-ANE space is a metric G-space $\Bbb X\in G$-ANE. An arbitrarily fine domination of $\Bbb X$ by G-CW-c...

conjecture | open | MathDB 页面 | 原文线索

V2-034747 | MATHDB-381144

The almost contact-to-contact existence conjecture

摘录 / EXCERPT: Let M be a $(2n+1)$-dimensional manifold equipped with an almost contact structure, meaning a codimension-1 cooriented distribution together with a conformally symplectic class...

conjecture | open | MathDB 页面 | 原文线索

V2-034748 | MATHDB-381177

Griffiths transversality for variational conservation laws of constant mean curvature surfaces

摘录 / EXCERPT: For a first-order deformation determined by the inhomogeneous Jacobi field $u=\mathrm{i}v_{\rho}+\mathrm{i}\delta v_0$, let ψ_j, $j=0,1,\ldots$, be the sequence of 1-forms o...

conjecture | open | MathDB 页面 | 原文线索

V2-034749 | MATHDB-381187

Torisu's unknotting-number-one conjecture for three-branched Montesinos knots

摘录 / EXCERPT: Let K be a Montesinos knot with three branches. Here $M(0; (p,r),(q,s),(2mn\pm 1, 2n^2))$ denotes the Montesinos knot with the displayed parameters. Torisu's conjecture. The knot...

conjecture | open | MathDB 页面 | 原文线索

V2-034750 | MATHDB-381198

Family Floer cohomology and mirror symmetry conjecture

摘录 / EXCERPT: Let X be the symplectic manifold under consideration. For each Lagrangian $L_i\subset X$, let $\mathcal L_i$ be the corresponding object in the category of α_X-twisted she...

conjecture | open | MathDB 页面 | 原文线索

V2-034751 | MATHDB-381237

Hyperbolic torsion polynomial conjecture for knot genus

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3, and let $b[?25l\cT_Kb[?25h \in \C[t^{\pm1}]$ be its hyperbolic torsion polynomial. The Seifert genus of K is denoted by $g(K)$. Hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-034752 | MATHDB-381238

Vanishing conjecture for homology below virtual cohomological dimension

摘录 / EXCERPT: Let $\Gamma_{n,s}$ be the graph automorphism group with parameters n and s, and let $2n+s-3$ be its virtual cohomological dimension. Below-vcd vanishing conjecture. All classes...

conjecture | open | MathDB 页面 | 原文线索

V2-034753 | MATHDB-381256

Bijection conjecture for models of two-point configuration spaces

摘录 / EXCERPT: Let A be a 1-connected Poincaré duality CDGA of formal dimension n odd. For each $\xi\in (A\otimes A)^{2n-2}$, let $C(\xi)$ be the associated CDGA, and let ... be the natural s...

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V2-034754 | MATHDB-381277

Conjecture on spheres with unique smooth structures

摘录 / EXCERPT: For an integer $n>4$, let S^n denote the n-dimensional sphere, and say that it has a unique smooth structure when every smooth manifold homeomorphic to S^n is diffeomorphic t...

conjecture | open | MathDB 页面 | 原文线索

V2-034755 | MATHDB-381284

Topological sliceness and ribbonness conjecture for even 3-strand pretzel knots

摘录 / EXCERPT: Let K be an even 3-strand pretzel knot, meaning a pretzel knot with an even parameter in the relevant parity convention. Topological sliceness and ribbonness conjecture. Then K...

conjecture | open | MathDB 页面 | 原文线索

V2-034756 | MATHDB-381293

Equality of sectional category and higher topological complexity for real projective spaces

摘录 / EXCERPT: Equality conjecture. An s-motion planning algorithm for $\operatorname{\mathbb{R}P}^m$ with $\operatorname{secat}(\pi_s)+1$ s-local rules can be constructed out of such a map....

conjecture | open | MathDB 页面 | 原文线索

V2-034757 | MATHDB-381323

Conjecture for the genus-zero free energy coefficient of multi-cut Toeplitz determinants

摘录 / EXCERPT: The large-n expansion of the Toeplitz determinant is organized by the remainder m of dividing n by $2r+1$, and let $F^{\{0\}}$ denote the corresponding $O(1)$ coefficient as...

conjecture | open | MathDB 页面 | 原文线索

V2-034758 | MATHDB-381376

The strong invertibility conjecture for L-space knots

摘录 / EXCERPT: Strong invertibility conjecture. Every L-space knot in S^3 is strongly invertible: there is an orientation-preserving involution of S^3 whose fixed-point set is a circle inters...

conjecture | open | MathDB 页面 | 原文线索

V2-034759 | MATHDB-381404

The bounded cohomology conjecture for smooth projective surfaces

摘录 / EXCERPT: Let X be a smooth projective surface, and let C be a curve on X. Write $h^i(\mathcal O_X(C))=\dim H^i(X,\mathcal O_X(C))$. The bounded cohomology conjecture asserts that ther...

conjecture | open | MathDB 页面 | 原文线索

V2-034760 | MATHDB-381444

Nontriviality conjecture for bucklings of spatial theta-graphs

摘录 / EXCERPT: Let G be a θ-graph in S^3. A belt B is an oriented annulus intersecting the edges of G in intervals, and $G(B)$ denotes the spatial graph obtained by buckling G al...

conjecture | open | MathDB 页面 | 原文线索

V2-034761 | MATHDB-381488

Associativity conjecture for the ring $(R_n,\star)$

摘录 / EXCERPT: Let R_n be the ring defined in the surrounding construction, with binary operation $\star$. Associativity conjecture. $(R_n,\star)$ is an associative ring. The supplied context d...

conjecture | open | MathDB 页面 | 原文线索

V2-034762 | MATHDB-381490

Hovey-Palmieri conjectures on Bousfield classes

摘录 / EXCERPT: Hovey-Palmieri conjectures. The following assertions hold:

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V2-034763 | MATHDB-381503

Lynch's cohomological-dimension conjecture for local rings

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a local ring and let I be an ideal of R. For an ideal I, write $\operatorname{cd}(I,R)$ for its cohomological dimension, and let $H^c_I(R)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-034764 | MATHDB-381518

Koike's local unitary-flatness conjecture for semi-positive line bundles

摘录 / EXCERPT: Let X be a complex manifold, let L be a semi-positive holomorphic line bundle on X, and let Y be a compact Kähler submanifold such that $L|_Y$ is topologically trivial. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-034765 | MATHDB-381519

Rational embeddability conjecture for circular spherical divisors

摘录 / EXCERPT: Let D_n be the circular spherical divisor with n components discussed above. A rational embedding means an embedding of D_n into a rational surface. Rational embeddability co...

conjecture | open | MathDB 页面 | 原文线索

V2-034766 | MATHDB-381540

Almost-sure scaling conjecture for Betti numbers in the Eden model

摘录 / EXCERPT: Let $\beta_i(t)$ be the ith Betti number of the d-dimensional Eden growth model at time t, with $1\leq i\leq d-1$. Almost-sure scaling conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-034767 | MATHDB-381586

The cyclic quandle cohomology suspension conjecture for R_p

摘录 / EXCERPT: The cyclic quandle cohomology suspension conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-034768 | MATHDB-381592

The metric thickening finite-support homotopy equivalence conjecture

摘录 / EXCERPT: Metric thickening finite-support homotopy equivalence conjecture. The inclusion induces a homotopy equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-034769 | MATHDB-381619

The filtration conjecture for quantum cohomology

摘录 / EXCERPT: The filtration conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-034770 | MATHDB-381620

The critical-eigenspace conjecture for completed symplectic cohomology

摘录 / EXCERPT: The critical-eigenspace conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-034771 | MATHDB-381621

The E2-degeneration conjecture for the symplectic spectral sequence

摘录 / EXCERPT: The E2-degeneration conjecture. The spectral sequence degenerates at the E_2 page.

conjecture | open | MathDB 页面 | 原文线索

V2-034772 | MATHDB-381622

The cut-and-join description conjecture for Chekhov-Eynard-Orantin topological recursion

摘录 / EXCERPT: The Chekhov-Eynard-Orantin topological recursion is a recursive procedure associated with data on a spectral curve; for simple ramification points, it can be related to cut-and-joi...

conjecture | open | MathDB 页面 | 原文线索

V2-034773 | MATHDB-381632

Conjectural slope bounds for finite-slope cohomology

摘录 / EXCERPT: Let $w\in{}^MW$, $\kappa\in X^\star(T^c)^{M_\mu,+}$, and let $\chi:T(\mathbb{Z}_p)\to {\overline{F}}^\times$ be a finite-order character. For a character λ of T^+ or...

conjecture | open | MathDB 页面 | 原文线索

V2-034774 | MATHDB-381633

Conjectural nonnegative slopes for locally analytic overconvergent cohomology

摘录 / EXCERPT: Fix $w\in{}^MW$, and let $\nu:T^c(\mathbb{Z}_p)\to\mathbb{C}_p^\times$ be a continuous character. For a character λ of T^+ or T^-, let $v(\lambda)$ denote its slope in...

conjecture | open | MathDB 页面 | 原文线索

V2-034775 | MATHDB-381636

The surgery-contactomorphism conjecture for Legendrian knots in the standard 3-sphere

摘录 / EXCERPT: Let L and L' be Legendrian knots in the standard contact 3-sphere (S^3,ξ_{st}). Write $L(-1)$ and $L(+1)$ for the contact manifolds obtained by Legendrian (-1)- and...

conjecture | open | MathDB 页面 | 原文线索

V2-034776 | MATHDB-381654

Sum's conjecture on the recursive construction of minimal generators for polynomial algebras

摘录 / EXCERPT: Sum's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-034777 | MATHDB-381741

The mathfrak{gl}(k) KhovanovRozansky categorification conjecture

摘录 / EXCERPT: Let $kge 2$, let $\beta\in \mathfrak{Br}_n$ be a braid, and let $L(\beta)$ denote its link closure. Write $\mathrm{H}_{0|k}(\beta)$ for the homology constructed in the paper and...

conjecture | open | MathDB 页面 | 原文线索

V2-034778 | MATHDB-381742

The $q\to -q/t$ symmetry conjecture for one-component links

摘录 / EXCERPT: Let β be a braid whose closure $L(\beta)$ has one connected component, written $\pi_0(L(\beta))=1$. Let $\mathrm{HH}_{m|m}(\beta)$ denote the corresponding homology, graded b...

conjecture | open | MathDB 页面 | 原文线索

V2-034779 | MATHDB-381802

HOMFLY-PT a-deformed Gukov-Manolescu conjecture

摘录 / EXCERPT: For every knot $K\subset S^3$, let $F_K^N(x,q)=F_K^{\mathfrak{sl}_N,\mathrm{sym}}(x,q)$ denote the series associated with symmetric representations, let $\Delta_K(x)$ be the Alexan...

conjecture | open | MathDB 页面 | 原文线索

V2-034780 | MATHDB-381813

G-dimension and complete-intersection-dimension conjecture for derivation modules

摘录 / EXCERPT: Derivation-module dimension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-034781 | MATHDB-381825

The chi-ribbon closure conjecture for alternating links

摘录 / EXCERPT: Chi-ribbon closure conjecture. Then L_1 is an alternating link. In particular, K_1 is an alternating knot in the knot case.

conjecture | open | MathDB 页面 | 原文线索

V2-034782 | MATHDB-381856

Giordano's generic cohomology conjecture for twisted polynomial superfunctors

摘录 / EXCERPT: Generic cohomology conjecture. For all $F,G\in\mathcal{P}$, there is a graded isomorphism, natural in F and G,

conjecture | open | MathDB 页面 | 原文线索

V2-034783 | MATHDB-381861

Relative cyclic open-closed map conjecture

摘录 / EXCERPT: Let X be a symplectic manifold and let $L\subset X$ be a null-homologous Lagrangian. Let $HC^-_*(Fuk(X),L)$ denote the relative cyclic homology associated to the object L, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034784 | MATHDB-381862

Flatness conjecture for the relative connection

摘录 / EXCERPT: Let $F:\mathcal{C}\rightarrow\mathcal{D}$ be an A_∞-functor, and let $HC^-_*(F)$ be its relative cyclic homology equipped with the relative connection ∇^F. Relative c...

conjecture | open | MathDB 页面 | 原文线索

V2-034785 | MATHDB-381879

The AJ conjecture for the colored Jones polynomial and the A-polynomial

摘录 / EXCERPT: Let K be a knot. Denote by $A_K(l,\alpha)$ its A-polynomial, by $A_q(K)(l,\alpha^2)$ the specialization at $q=1$ of the A_q-polynomial, and let ε be the evaluatio...

conjecture | open | MathDB 页面 | 原文线索

V2-034786 | MATHDB-381886

Coarse-convergence conjecture for Morse boundaries of hierarchically hyperbolic groups

摘录 / EXCERPT: Let G be a hierarchically hyperbolic group (HHG), and let $\partial_M G$ be its Morse boundary. Assume that G admits a largest acylindrical action. The topology on...

conjecture | open | MathDB 页面 | 原文线索

V2-034787 | MATHDB-381887

Visual-topology invariance for stabilized d_L-metrics

摘录 / EXCERPT: Let G act geometrically on two $\operatorname{CAT}(0)$ spaces Y and Y' whose d_L-metrics stabilize. Let $\partial_M^{\kappa}Y$ and $\partial_M^{\kappa}Y'$ denote their...

conjecture | open | MathDB 页面 | 原文线索

V2-034788 | MATHDB-381888

Visual-topology conjecture for sublinearly Morse boundaries of graph manifolds

摘录 / EXCERPT: Let G be the fundamental group of a non-positively curved graph manifold, let X be its universal cover, and let T_{∞} be the Bass-Serre tree of the graph-of-groups decom...

conjecture | open | MathDB 页面 | 原文线索

V2-034789 | MATHDB-381925

The standard-homology characterization of the small Heisenberg homology

摘录 / EXCERPT: Let $\mathcal{H}_g^{V(r)}$ be the small Heisenberg homology, let $\mathcal{H}_g^{V,\dagger}$ be the direct sum of the standard twisted homology groups, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-034790 | MATHDB-381926

Equality of the log mixed-motive categories

摘录 / EXCERPT: Let $({\rm {LM}}\flat)$ and $({\rm {LM}}\flat^*)$ be the categories with the same objects and with morphisms defined using, respectively, the Chern class constructions from K_0 an...

conjecture | open | MathDB 页面 | 原文线索

V2-034791 | MATHDB-381932

The norm-factorization homology comparison for compact Lie groups

摘录 / EXCERPT: Let G be a compact Lie group, let R be a cofibrant commutative algebra, and let U and U^{G} denote the relevant complete universes. Write $N_e^G R$ for the norm and...

conjecture | open | MathDB 页面 | 原文线索

V2-034792 | MATHDB-381933

Geometric fixed points of norms for positive-dimensional subgroups

摘录 / EXCERPT: Let G be a compact Lie group, let $K<G$ be a closed subgroup of positive dimension, and let X be a $\mathcal{D}^{d}$-algebra whose underlying orthogonal spectrum is cofibrant,...

conjecture | open | MathDB 页面 | 原文线索

V2-034793 | MATHDB-381934

Geometric fixed points of norms for finite subgroups with trivial adjoint action

摘录 / EXCERPT: Let G be a compact Lie group, let $K<G$ be finite, and let X be an input for the norm N_e^G. Assume that the adjoint action of K on T_eG is trivial, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-034794 | MATHDB-381935

Geometric fixed points of norms via the normalizer

摘录 / EXCERPT: Let G be a compact Lie group, let $K<G$ be finite, and let N be the normalizer of K in G. Let X be an input for the norm N_e^G. Normalizer geometric fixed-point conject...

conjecture | open | MathDB 页面 | 原文线索

V2-034795 | MATHDB-381936

Relative norm invariance under cofibrant weak equivalences

摘录 / EXCERPT: Let G be a compact Lie group, let $H<G$ be a closed subgroup, and let N_H^G be the point-set relative norm indexed on universes U_H and U. Suppose X' and X satisfy the...

conjecture | open | MathDB 页面 | 原文线索

V2-034796 | MATHDB-381937

The relative norm-induction comparison for equivariant commutative ring spectra

摘录 / EXCERPT: Let G be a compact Lie group, let $H<G$ be a closed subgroup, and let R be a cofibrant H-equivariant commutative ring orthogonal spectrum in the U_H-universe model structur...

conjecture | open | MathDB 页面 | 原文线索

V2-034797 | MATHDB-381938

Geometric fixed points of relative norms for normal subgroups

摘录 / EXCERPT: Let G be a compact Lie group, let $H<G$ be a closed subgroup, let $K\trianglelefteq G$ be normal, and let X be an H-equivariant $\mathcal{D}_V$-algebra satisfying the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-034798 | MATHDB-381939

Factorization-homology iteration for equivariant norms

摘录 / EXCERPT: Let G be a compact Lie group, let $K<H<G$, and let X be a K-equivariant little $T_{eK}(G/K)$-disk algebra. Let $D(T_{eH}(G/H))$ be the open unit disk and let i^* denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-034799 | MATHDB-381940

Iteration of relative norms through intermediate subgroups

摘录 / EXCERPT: Let G be a compact Lie group, let $K<H<G$, and let X be a K-equivariant little $T_{eK}(G/K)$-disk algebra whose underlying K-equivariant orthogonal spectrum is cofibrant in...

conjecture | open | MathDB 页面 | 原文线索

V2-034800 | MATHDB-381941

Symmetric monoidality of relative norms

摘录 / EXCERPT: Let G be a compact Lie group, let $H<G$ be a closed subgroup, and let X and Y be H-equivariant $\mathcal{D}_V$-algebras whose underlying H-equivariant orthogonal spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-034801 | MATHDB-381961

Strict inequality between the 13-crossing and 5-crossing numbers

摘录 / EXCERPT: Let K be a knot, and let $c_n(K)$ denote its n-crossing number, the minimum number of n-fold crossings in a projection of K. Strict inequality conjecture. For all knots K...

conjecture | open | MathDB 页面 | 原文线索

V2-034802 | MATHDB-381966

Equivariant positivity conjecture for CSM classes

摘录 / EXCERPT: Equivariant positivity conjecture. The coefficient $c_{v;w}(\alpha)$ is a polynomial in the positive roots α with non-negative coefficients. Numerical evidence supports this...

conjecture | open | MathDB 页面 | 原文线索

V2-034803 | MATHDB-382014

The topological analogue of the Slice-Ribbon Conjecture

摘录 / EXCERPT: A knot K is topologically slice if it bounds a topological slice disk Δ in B^4. It is topologically homotopy ribbon if it bounds such a disk for which the inclusion-indu...

conjecture | open | MathDB 页面 | 原文线索

V2-034804 | MATHDB-382016

The string homology-Legendrian contact homology correspondence for conormal bundles

摘录 / EXCERPT: String homology-Legendrian contact homology conjecture. For any compact oriented submanifold K in $\mathbb{R}^n$, $H^{\mathrm{string}}_*(\mathbb{R}^n,K)$ is isomorphic to the Leg...

conjecture | open | MathDB 页面 | 原文线索

V2-034805 | MATHDB-382019

SYZ mirror conjecture for Calabi-Yau manifolds

摘录 / EXCERPT: Let X be a Calabi-Yau manifold. A Lagrangian fibration is a map $\pi:X\to B$ onto a topological manifold B whose fibers are graded with respect to a holomorphic volume form...

conjecture | open | MathDB 页面 | 原文线索

V2-034806 | MATHDB-382022

Contractible-translate fundamental-domain conjecture for aspherical manifolds

摘录 / EXCERPT: Let M be an aspherical manifold, let $\widetilde M$ be its universal cover, and let $\pi_1(M)$ act on $\widetilde M$ by deck transformations. Contractible-translate conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-034807 | MATHDB-382023

Contractible-intersection fundamental-domain conjecture for aspherical closed manifolds

摘录 / EXCERPT: Let X be an aspherical closed manifold, let $\widetilde X$ be its universal cover, and let $\pi_1(X)$ act on $\widetilde X$ by deck transformations. A closed fundamental domain i...

conjecture | open | MathDB 页面 | 原文线索

V2-034808 | MATHDB-382032

The partial-conjugation relation conjecture for string links

摘录 / EXCERPT: Partial-conjugation relation conjecture. For each $1\leq i\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-034809 | MATHDB-382039

The persistence conjecture for first local cohomology of edge ideals

摘录 / EXCERPT: Let ${\Gamma}$ be a connected graph, let R be the polynomial ring associated to its vertices, let I be the edge ideal of ${\Gamma}$, and let ${\mathfrak m}$ be the homogeneous...

open_problem | open | MathDB 页面 | 原文线索

V2-034810 | MATHDB-382063

The ambient-monodromy realization conjecture for toric fibres

摘录 / EXCERPT: Ambient-monodromy realization conjecture. An isomorphism $\Phi\in\operatorname{Aut}H_2(X,T(x))$ can be realized as the ambient monodromy of a Hamiltonian diffeomorphism mapping...

conjecture | open | MathDB 页面 | 原文线索

V2-034811 | MATHDB-382118

Classification conjecture for regular homotopy classes of curves

摘录 / EXCERPT: Let $r(\alpha)$ be the integer that counts the number of rotations of the velocity vector along a curve α. Curves with $r(\alpha)=m$ include the curves γ_m in the fam...

conjecture | open | MathDB 页面 | 原文线索

V2-034812 | MATHDB-382126

The projective-dimension bound for binomial edge ideals of high regularity

摘录 / EXCERPT: Let G be a graph on $n\ge 7$ non-isolated vertices. Suppose that $\operatorname{reg}(J_G)=n-1$. Projective-dimension bound. Then ... The paper reports that this stronger bound is s...

conjecture | open | MathDB 页面 | 原文线索

V2-034813 | MATHDB-382127

The strong virtual Legendrian isotopy conjecture for sphere fibers

摘录 / EXCERPT: Strong virtual Legendrian isotopy conjecture. There exists no strong allowable positive, or non-constant non-negative, virtual Legendrian isotopy of the sphere fiber of ST^*M to...

conjecture | open | MathDB 页面 | 原文线索

V2-034814 | MATHDB-382128

The strong virtual Legendrian unlinking conjecture for cotangent links

摘录 / EXCERPT: Strong virtual Legendrian unlinking conjecture. There is no non-negative strong allowable virtual Legendrian isotopy of one component of L to the other.

conjecture | open | MathDB 页面 | 原文线索

V2-034815 | MATHDB-382129

The virtual Legendrian isotopy conjecture for spheres and complex projective spaces

摘录 / EXCERPT: Virtual Legendrian isotopy conjecture. Two Legendrian links in ST^*S^m or $ST^*\mathbb C P^n$ are Legendrian isotopic if and only if they are virtually Legendrian isotopic. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-034816 | MATHDB-382148

Exponential growth conjecture for compactly supported cohomology of moduli spaces of curves

摘录 / EXCERPT: Let g be the genus of a smooth proper curve and let k be a non-negative integer. Write $H^{2g+k}_c(\mathcal{M}_g)$ for the compactly supported cohomology of the moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-034817 | MATHDB-382202

Mixed Hodge structure conjecture for the Sullivan differential group of a Kähler manifold

摘录 / EXCERPT: Mixed Hodge structure conjecture. The coordinate ring and Lie algebra of G_M should have natural mixed Hodge structures such that the homomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-034818 | MATHDB-382209

The open-ball conjecture for full arc complexes of decorated once-punctured polygons

摘录 / EXCERPT: Let $n\geq2$, and let $\depu n$ be a decorated once punctured n-gon. Its full arc complex is the simplicial complex whose simplices consist of collections of pairwise compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-034819 | MATHDB-382215

Polynomial-time homotopy-type conjecture for independence complexes of circle graphs

摘录 / EXCERPT: Let G be a circle graph, and let $I(G)$ be its independence complex. The size of the input is measured by the number of vertices of G. Polynomial-time homotopy-type conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-034820 | MATHDB-382239

The contact-shell characterization of Reeb vector fields

摘录 / EXCERPT: Let v be a non-vanishing (traversing ?) vector field on a compact $(2n+1)$-dimensional manifold X, and let ξ be a $(2n)$-dimensional distribution normal to v that admits...

conjecture | open | MathDB 页面 | 原文线索

V2-034821 | MATHDB-382254

The critical COHA-deformed W-algebra isomorphism for Calabi-Yau threefolds

摘录 / EXCERPT: Let M be a Calabi-Yau threefold, let $H^*(M)$ denote its cohomology, and let $W^+(M)$ be the algebra generated by $T_n(\lambda)$ for $n{\geqslant} 0$ and $\lambda\in H^*(M)$, sub...

conjecture | open | MathDB 页面 | 原文线索

V2-034822 | MATHDB-382266

The conjecture on cohomology for minimal elliptic pairs

摘录 / EXCERPT: Let (T,U) be a minimal elliptic pair, and let $\chi\colon \mathcal{T}^{+}(\mathcal{O}_k)\to\overline{\mathbb{Q}}_\ell^{\times}$ be a smooth character. Let $s_\chi\geq 0$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-034823 | MATHDB-382285

Wang-Xu conjecture for very exotic spheres

摘录 / EXCERPT: Let a very exotic sphere mean a homotopy sphere that does not bound a parallelizable manifold. The preceding results show that very exotic spheres exist in at least 99 congruence...

conjecture | open | MathDB 页面 | 原文线索

V2-034824 | MATHDB-382291

Heyde's conjecture for locally compact Abelian groups without a 2-dimensional torus

摘录 / EXCERPT: Let X be a second countable locally compact Abelian group, let G be the subgroup of X generated by all elements of X of order 2, and let α be a topological automor...

conjecture | open | MathDB 页面 | 原文线索

V2-034825 | MATHDB-382331

Radial limit conjecture for GPPV invariants of rational homology spheres

摘录 / EXCERPT: Radial limit conjecture. For each $b\in\operatorname{Spin}^c(M)/\{\pm1\}$, there exist topological invariants $\Delta_b\in\mathbb{Q}$, $c\in\mathbb{Z}_{\ge0}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-034826 | MATHDB-382351

The HOMFLY-PT-Kauffman criterion for factorised HZ transforms of 2-component links

摘录 / EXCERPT: HOMFLY-PT-Kauffman factorisation criterion. If the HOMFLY-PT and Kauffman polynomials for a 2-component link $\mathcal{L}$ satisfy ... .

conjecture | open | MathDB 页面 | 原文线索

V2-034827 | MATHDB-382371

Halperin's conjecture on totally non-cohomologous to zero fibrations

摘录 / EXCERPT: Let F be an elliptic space with minimal model ΛV, and let $F\to X\to B$ be a fibration. The condition that $\dim V^{\mathrm{odd}}=\dim V^{\mathrm{even}}$ is imposed on t...

conjecture | open | MathDB 页面 | 原文线索

V2-034828 | MATHDB-382406

The converse between local compactness and local solidity of the unit filter

摘录 / EXCERPT: Local compactness-local solidity conjecture. X is locally compact if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-034829 | MATHDB-382444

Refined topological recursion conjecture for the multidifferentials \omega_{g,n}

摘录 / EXCERPT: Let $g\in\frac12\mathbb{Z}_{\geq0}$ and $n\in\mathbb{Z}_{\geq1}$ satisfy $2g-2+n\geq1$, and let $\omega_{g,n}$ be the multidifferential defined by Definition RTR. Refined topologic...

conjecture | open | MathDB 页面 | 原文线索

V2-034830 | MATHDB-382524

Seidel-Dostoglou decomposition conjecture for Floer cohomology of

摘录 / EXCERPT: Let Σ be a closed oriented surface of genus g, let $s^i\Sigma$ denote its i-th symmetric product, and let HF_g^* be the Floer cohomology ring associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-034831 | MATHDB-382535

Thomas's uniqueness conjecture for special Lagrangians under mean curvature flow

摘录 / EXCERPT: Let L be a graded Lagrangian submanifold of a Calabi-Yau n-fold X, with the phase of the holomorphic volume form chosen so that the cohomological phase satisfies $\phi(L)=0$....

conjecture | open | MathDB 页面 | 原文线索

V2-034832 | MATHDB-382542

The hyperpencil conjecture for integral symplectic forms

摘录 / EXCERPT: Let X be a closed $2n$-manifold and let ω be an integral symplectic form on X, meaning that its cohomology class lies in the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-034833 | MATHDB-382549

Conjecture on linking forms and Jones-Rong polynomial values for knots with signature four

摘录 / EXCERPT: Let K be a knot, let $\sg(K)$ denote its signature, let H_1 denote the first homology group of the double branched cover of K, and let Q_K be the associated polynomial valu...

conjecture | open | MathDB 页面 | 原文线索

V2-034834 | MATHDB-382557

Bernhard-Jablan conjecture on the unlinking number

摘录 / EXCERPT: Let L be a link. Its unlinking number $u(L)$ is the minimum number of crossing changes needed to transform L into an unlink, and $u_{BJ}(L)$ denotes its Bernhard-Jablan unlinki...

conjecture | open | MathDB 页面 | 原文线索

V2-034835 | MATHDB-382558

Simple homotopy conjecture for crosscut complexes of lattices

摘录 / EXCERPT: Let ${\mathcal L}$ be a lattice and let C be a crosscut of ${\mathcal L}$. Write $\Gamma(C,{\mathcal L})$ for the crosscut complex, $\operatorname{Bd}(\Gamma(C,{\mathcal L}))$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-034836 | MATHDB-382563

Polynomial invariants for unoriented links and framed links

摘录 / EXCERPT: Let T_n be a trivial link of n components, and let L_0,L_1,L_2,L_3 be a standard unoriented skein quadruple, with L_{i+1} obtained from L_i by a right-handed half-twist o...

conjecture | open | MathDB 页面 | 原文线索

V2-034837 | MATHDB-383165

Milnor's conjecture on fundamental groups

摘录 / EXCERPT: For every complete Riemannian manifold (M,g) with nonnegative Ricci curvature, $\operatorname{Ric}_g\ge 0$, the fundamental group $\pi_1(M)$ is finitely generated.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-034838 | MATHDB-383303

Gromov’s quantitative scalar-curvature-simplicial-volume conjecture

摘录 / EXCERPT: For every integer $n\geq 2$, there exists a constant $C_n>0$ such that, for every closed oriented n-manifold M, every Riemannian metric g on M, and every $\lambda\geq 0$, t...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-034839 | MATHDB-383535

Categorified Wrapping Number Conjecture

摘录 / EXCERPT: For every annular link L, the annular Khovanov homology of L is nonzero in the k-grading equal to the wrapping number of L:...

conjecture | partial | MathDB 页面 | 原文线索

V2-034840 | MATHDB-383818

Properties of subgroups of topological full groups

摘录 / EXCERPT: Try to understand relations, amenability and Liouville property of subgroups of topological full groups. In particular of $\left\langle \delta_a: a \in A \right\rangle$

open_problem | open | MathDB 页面 | 原文线索

V2-034841 | MATHDB-384345

Lipschitz homotopy groups of the Heisenberg group

摘录 / EXCERPT: If $\pi_k({\mathbb S}^n)$ is nontrivial, is the same true for $\pi_k^{\scriptstyle{Lip}}({\mathbb H}^n)$? In particular, is $\pi_3^{\scriptstyle{Lip}}({\mathbb H}^2) \ne 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-034842 | MATHDB-384657

Study the topological phase transition in the asymptotics of planar orthogonal polynomials with normal Random Matrix Theory.

摘录 / EXCERPT: Study the topological phase transition in the asymptotics of planar orthogonal polynomials with normal Random Matrix Theory.

open_problem | open | MathDB 页面 | 原文线索

V2-034843 | MATHDB-384672

Low Dimensional and Sparse Embeddings (problem 1)

摘录 / EXCERPT: Set $d(n)$ to be the minimal integer d with the property that every n-point metric space isometrically embeds into ℓ_∞^d. What are the asymptotics of $n - d(n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034844 | MATHDB-384673

Low Dimensional and Sparse Embeddings (problem 2)

摘录 / EXCERPT: For each $\alpha > 1, n \in \mathbb{N}$, let k^n_α denote the minimal k such that every n-point metric space embeds with distortion α into some k-dimensional no...

open_problem | open | MathDB 页面 | 原文线索

V2-034845 | MATHDB-384674

Low Dimensional and Sparse Embeddings (problem 3)

摘录 / EXCERPT: Does every n-point subset of ℓ_1 embed with average distortion $1+\varepsilon$ into ℓ_1^d, where $d = O_\varepsilon(\log n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034846 | MATHDB-384675

Low Dimensional and Sparse Embeddings (problem 4)

摘录 / EXCERPT: Fix a distortion $c<\infty$ and dimension $d<\infty$. Let M be an n-point subset of ℓ_1^d (or ℓ_∞^d). What is the least number of edges needed for a (weighted) g...

open_problem | open | MathDB 页面 | 原文线索

V2-034847 | MATHDB-385065

Group splittings and asymptotic topology (problem 1)

摘录 / EXCERPT: Let G be a hyperbolic group such that ∂G is not separated by a Cantor set. Is it true that ∂G is not separated by a simple path?

open_problem | open | MathDB 页面 | 原文线索

V2-034848 | MATHDB-385069

Group splittings and asymptotic topology (problem 5)

摘录 / EXCERPT: Let G be a finitely generated group such that there is a sequence of quasi-circles that separate its Cayley graph. Is G a virtually surface group?

open_problem | open | MathDB 页面 | 原文线索

V2-034849 | MATHDB-385071

Group splittings and asymptotic topology (problem 7)

摘录 / EXCERPT: Let G be a group with $asdim\,(G)\geq n$, $n>3$. Assume that a uniformly embedded copy of $\mathbb Z ^{n-2}$ separates coarsely the Cayley graph of G. Is it true then that G...

open_problem | open | MathDB 页面 | 原文线索

V2-034850 | MATHDB-385072

Group splittings and asymptotic topology (problem 8)

摘录 / EXCERPT: Let X be a locally connected homogeneous continuum of dimension ≥ 2. Is it true that X is not separated by an arc?

open_problem | open | MathDB 页面 | 原文线索

V2-034851 | MATHDB-385096

Find an example of a ... -action on a Cantor set such that the commutator subgroup of the topological full group is f.p.

摘录 / EXCERPT: Find an example of a $\mathbf{Z}^2$-action on a Cantor set such that the commutator subgroup of the topological full group is f.p.

open_problem | open | MathDB 页面 | 原文线索

V2-034852 | MATHDB-385132

End invariants and topological questions (problem 2)

摘录 / EXCERPT: Suppose (G,P) is relatively hyperbolic and consider rays in G for which there is a K such that no coset of a parabolic is visited for longer than K. Are any two such rays p...

open_problem | open | MathDB 页面 | 原文线索

V2-034853 | MATHDB-385134

End invariants and topological questions (problem 4)

摘录 / EXCERPT: If K is finite $K(\pi,1)$, is $\widetilde K$ inward tame?

open_problem | open | MathDB 页面 | 原文线索

V2-034854 | MATHDB-385135

End invariants and topological questions (problem 5)

摘录 / EXCERPT: Are any two Z-boundaries Z_1,Z_2 of a group cell-like equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-034855 | MATHDB-385175

Is there an analogue of the Geoghegan--Ontaneda result about Cech cohomology of the boundaries of CAT(0) groups for other boundaries?

摘录 / EXCERPT: Is there an analogue of the Geoghegan--Ontaneda result about Cech cohomology of the boundaries of CAT(0) groups for other boundaries? (c.f. Bestvina--Mess)

open_problem | open | MathDB 页面 | 原文线索

V2-034856 | MATHDB-385177

Miscellaneous (problem 12)

摘录 / EXCERPT: Does there exist a hyperbolic group G whose boundary is not a sphere, but is begin{enumerate} item[(i)] locally contractible item[(ii)] a homology n-manifold item[(iii)] satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-034857 | MATHDB-385208

Poisson boundary for random walks on free (semi)-groups with infinite moment

摘录 / EXCERPT: For any random walk on the free group or free semigroup, the Gromov boundary with the hitting measure is a topological model for the Poisson boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-034858 | MATHDB-385305

Mean Curvature Flow (problem 1)

摘录 / EXCERPT: Let $C\subset \mathbb{R}^2$ be the topologist's sine curve. More precisely, ... where Γ is a smooth curve connecting the origin and the point $(\frac{1}{2 \pi}, 0)$ that doesn...

open_problem | open | MathDB 页面 | 原文线索

V2-034859 | MATHDB-385374

Topological vs. Lagrangian arboreal singularities

摘录 / EXCERPT: What structure on a skeleton allows us to reconstruct uniquely its Weinstein neighborhood?

open_problem | open | MathDB 页面 | 原文线索

V2-034860 | MATHDB-385375

Can the structure of a topological arboreal space be globally upgraded to the structure of a Lagrangian arboreal space?

摘录 / EXCERPT: Can the structure of a topological arboreal space be globally upgraded to the structure of a Lagrangian arboreal space? And are all such Lagrangian arboreal realizations symplectom...

open_problem | open | MathDB 页面 | 原文线索

V2-034861 | MATHDB-385384

Is there a closed arboreal space whose sheaf category or symplectic cohomology is zero?

摘录 / EXCERPT: Is there a closed arboreal space whose sheaf category or symplectic cohomology is zero?

open_problem | open | MathDB 页面 | 原文线索

V2-034862 | MATHDB-385397

Prove without using holomorphic curves that an exotic parametrization of ... is not regularly homotopic to the standard embedding without dangerous self-tangencies.

摘录 / EXCERPT: Prove without using holomorphic curves that an exotic parametrization of $S^n\subset\mathbb{R}^{n+1}$ is not regularly homotopic to the standard embedding without dangerous self-ta...

open_problem | open | MathDB 页面 | 原文线索

V2-034863 | MATHDB-385403

Is there a reasonable notion of string topology on an arboreal space?

摘录 / EXCERPT: Is there a reasonable notion of string topology on an arboreal space?

open_problem | open | MathDB 页面 | 原文线索

V2-034864 | MATHDB-385431

Complex geometry (problem 4)

摘录 / EXCERPT: M K"ahler, θ big cohomology class. Can we define a family of distances generalizing the Finsler geometry of K"ahler potentials and compatible with weak Mabuchi geodesics?

open_problem | open | MathDB 页面 | 原文线索

V2-034865 | MATHDB-385529

Is there an alternate approach to homotopy groups that is adapted to Alexandrov spaces?

摘录 / EXCERPT: Is there an alternate approach to homotopy groups that is adapted to Alexandrov spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-034866 | MATHDB-385547

Manifolds of Cohomogeneity-one and Polar Actions (problem 4)

摘录 / EXCERPT: Compute topological invariants of cohomogeneity-one manifolds. Classify topologically the new candidates for positive curvature.

open_problem | open | MathDB 页面 | 原文线索

V2-034867 | MATHDB-385650

Scissors Congruence (problem 1)

摘录 / EXCERPT: Consider the sequence ... which is known to be exact. There are two known proofs: one uses homological methods (Dupont), the other uses more elementary methods (Jessen). How do they...

open_problem | open | MathDB 页面 | 原文线索

V2-034868 | MATHDB-385654

K-Theory and Algebraic Topology (problem 2)

摘录 / EXCERPT: Is there a K-theory spectrum that has homology groups $H_i(SL_n({\\mathbb{Z}});St({\\mathbb{Q}}))$?

open_problem | open | MathDB 页面 | 原文线索

V2-034869 | MATHDB-385655

K-Theory and Algebraic Topology (problem 3)

摘录 / EXCERPT: Develop projectivised scissors congruence K-theory. (For instance, can you prove polytopes with only vertices on the quadric generate the group?)

open_problem | open | MathDB 页面 | 原文线索

V2-034870 | MATHDB-385656

K-Theory and Algebraic Topology (problem 4)

摘录 / EXCERPT: Develop universal characterisation for K-theory for the following kinds of categories: begin{itemize} item 1) Monoidal Categories item 2) Exact Categories, CGW categories, Subtra...

open_problem | open | MathDB 页面 | 原文线索

V2-034871 | MATHDB-385657

K-Theory and Algebraic Topology (problem 5)

摘录 / EXCERPT: What is the relationship between total scissors congruence & discrete orthogonal K-Theory? Role of rank filtration of the associated $S_\bullet$-construction?

open_problem | open | MathDB 页面 | 原文线索

V2-034872 | MATHDB-385658

Steinberg Modules & Homology (problem 1)

摘录 / EXCERPT: What can we say about $H_\ast(O(3)^\delta$)? $H_\ast(PSL_2({\\mathbb{R}})^\delta)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034873 | MATHDB-385659

Steinberg Modules & Homology (problem 2)

摘录 / EXCERPT: Does $H_i(O(n)^\delta,St_n({\\mathbb{R}})\otimes {\\mathbb{Q}})$ vanish for sufficiently large n? Applications to homological stability?

open_problem | open | MathDB 页面 | 原文线索

V2-034874 | MATHDB-385660

Steinberg Modules & Homology (problem 3)

摘录 / EXCERPT: Compute homology of SK-automorphism groups of manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-034875 | MATHDB-385680

Can we describe and make sense of quantum cohomology of ... ?

摘录 / EXCERPT: Can we describe and make sense of quantum cohomology of D_{Cox}?

open_problem | open | MathDB 页面 | 原文线索

V2-034876 | MATHDB-386430

#12 Understanding the space of maps

摘录 / EXCERPT: How do we know how far apart two redistricting maps are? Can you capture the number of steps it takes in a Markov chain or political characteristics? What does the topology of the...

open_problem | open | MathDB 页面 | 原文线索

V2-034877 | MATHDB-386431

#13 Computational tools for exploring properties of a distribution over maps

摘录 / EXCERPT: Given a target measure, are there simple/computationally feasible tools to study the topology and metric properties of the space of graph partitions?

open_problem | open | MathDB 页面 | 原文线索

V2-034878 | MATHDB-386506

Mappings between balls (problem 5)

摘录 / EXCERPT: Let $f:\mathbb{B}^{n} \rightarrow \mathbb{B}^{N}$ be a proper rational map. Is f homotopic to a polynomial map p of the same degree? That is, $\forall t \in [0, 1]$, each...

open_problem | open | MathDB 页面 | 原文线索

V2-034879 | MATHDB-386507

Give a Nielsen-Thurston classification type theorem for big mapping classes.

摘录 / EXCERPT: Give a Nielsen-Thurston classification type theorem for big mapping classes.

open_problem | open | MathDB 页面 | 原文线索

V2-034880 | MATHDB-386509

Classification of elements of big mapping class groups (problem 3)

摘录 / EXCERPT: Given an (irreducible) big mapping class f and a simple closed curve α, do $f^n(\alpha)$ and $f^{-n}(\alpha)$ converge to a lamination?

open_problem | open | MathDB 页面 | 原文线索

V2-034881 | MATHDB-386511

Is there a dynamical description of an irreducible mapping class?

摘录 / EXCERPT: Is there a dynamical description of an irreducible mapping class? For instance, given two simple closed curves α and β, what can be said about the asymptotics of...

open_problem | open | MathDB 页面 | 原文线索

V2-034882 | MATHDB-386516

Classify surfaces whose mapping class groups act continuously by unbounded orbits on a (hyperbolic) connected locally finite complex.

摘录 / EXCERPT: Classify surfaces whose mapping class groups act continuously by unbounded orbits on a (hyperbolic) connected locally finite complex.

open_problem | open | MathDB 页面 | 原文线索

V2-034883 | MATHDB-386517

Among the surfaces which do have a hyperbolic complex on which their mapping classes act, classify the loxodromic elements.

摘录 / EXCERPT: Among the surfaces which do have a hyperbolic complex on which their mapping classes act, classify the loxodromic elements. Or, more simply, give explicit constructions of loxodrom...

open_problem | open | MathDB 页面 | 原文线索

V2-034884 | MATHDB-386518

Big mapping class groups acting on complexes (problem 5)

摘录 / EXCERPT: For S with no punctures and infinite genus, such as the Loch Ness Monster or the ladder surface, is there a hyperbolic complex on which $\operatorname{Map}(S)$ acts? Note that wh...

open_problem | open | MathDB 页面 | 原文线索

V2-034885 | MATHDB-386519

Classify surfaces with finite-invariance index 1, 2, and 3.

摘录 / EXCERPT: Classify surfaces with finite-invariance index 1, 2, and 3. (Note: finite invariance index is defined by Durham-Fanoni-Vlamis.)

open_problem | open | MathDB 页面 | 原文线索

V2-034886 | MATHDB-386520

Is there a good graph for surfaces with finite-invariance index 0?

摘录 / EXCERPT: Is there a good graph for surfaces with finite-invariance index 0?

open_problem | open | MathDB 页面 | 原文线索

V2-034887 | MATHDB-386521

When a given surface has multiple hyperbolic complexes associated to it, what relationships exist between those complexes?

摘录 / EXCERPT: When a given surface has multiple hyperbolic complexes associated to it, what relationships exist between those complexes? Are they quasi-isometric? Are there Lipschitz maps betwee...

open_problem | open | MathDB 页面 | 原文线索

V2-034888 | MATHDB-386522

Big mapping class groups acting on complexes (problem 9)

摘录 / EXCERPT: For a Cantor set C, $\operatorname{Map}(\mathbb{R}^2- C)$ is not acylindrically hyperbolic, and so it contains no WPDs. Are there any WWPDs elements? If so, describe them.

open_problem | open | MathDB 页面 | 原文线索

V2-034889 | MATHDB-386523

When a graph associated to a surface is hyperbolic, what can be said about its Gromov boundary?

摘录 / EXCERPT: When a graph associated to a surface is hyperbolic, what can be said about its Gromov boundary? For instance, can we describe the boundary of the ray/loop graph as space of laminat...

open_problem | open | MathDB 页面 | 原文线索

V2-034890 | MATHDB-386524

Big mapping class groups acting on complexes (problem 11)

摘录 / EXCERPT: Construct mapping classes that act parabolically on the ray graph of $\mathbb{R}^2-C$, where C is a cantor set.

open_problem | open | MathDB 页面 | 原文线索

V2-034891 | MATHDB-386525

Big mapping class groups acting on complexes (problem 12)

摘录 / EXCERPT: Consider the horseshoe map. After puncturing along its invariant Cantor set, is the resulting mapping class loxodromic on the ray graph? What about other "naturally occurring" home...

open_problem | open | MathDB 页面 | 原文线索

V2-034892 | MATHDB-386527

Teichmüller theory and other tools (problem 2)

摘录 / EXCERPT: If $f \in \operatorname{Map}(S)$ satisfies that there exists λ such that $\forall \alpha, \beta$ and $n \in \mathbb{N}$, $i(f^n(\alpha),\beta)< c \cdot \lambda^n$, is it tr...

open_problem | open | MathDB 页面 | 原文线索

V2-034893 | MATHDB-386531

Teichmüller theory and other tools (problem 6)

摘录 / EXCERPT: A possible counting problem is as follows. Fix a hyperbolic metric R on the blooming 3--pod surface. Fix a curve α and a basepoint $x \in R$, then does there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-034894 | MATHDB-386534

Algebraic and topological properties of big mapping class groups (problem 2)

摘录 / EXCERPT: For every S does there exist a countable group G such that G is not a subgroup of $\operatorname{Map}(S)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034895 | MATHDB-386536

Are there "nice" or "natural" generators for ... ?

摘录 / EXCERPT: Are there "nice" or "natural" generators for $\operatorname{Map}(S)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034896 | MATHDB-386537

Algebraic and topological properties of big mapping class groups (problem 5)

摘录 / EXCERPT: Can we see the topology of S in the algebraic structure of $\operatorname{Map}(S)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034897 | MATHDB-386538

Algebraic and topological properties of big mapping class groups (problem 6)

摘录 / EXCERPT: Does $\operatorname{Map}(S)$ have a canonical word metric?

open_problem | open | MathDB 页面 | 原文线索

V2-034898 | MATHDB-386543

Algebraic and topological properties of big mapping class groups (problem 11)

摘录 / EXCERPT: Do lattices map into $\operatorname{Map}(S)$ or $\overline {\operatorname{PMap}_c(S)}$? In particular, can higher-rank lattices map into these groups?

open_problem | open | MathDB 页面 | 原文线索

V2-034899 | MATHDB-386549

Which mapping classes are realized by affine automorphisms on some translation surface?

摘录 / EXCERPT: Which mapping classes are realized by affine automorphisms on some translation surface?

open_problem | open | MathDB 页面 | 原文线索

V2-034900 | MATHDB-386553

Infinite translation surfaces (problem 5)

摘录 / EXCERPT: Is the billiard flow on the triangle with side lengths 3, 4, and 5 ergodic?

open_problem | open | MathDB 页面 | 原文线索

V2-034901 | MATHDB-386555

Trisections of 4-manifolds (problem 1)

摘录 / EXCERPT: Let Σ be an embedded surface in X which bounds a handle body. Assume that the map $\pi_1(\Sigma)\rightarrow \pi_1(X)$ induced by the inclusion of Σ in X is surjec...

open_problem | open | MathDB 页面 | 原文线索

V2-034902 | MATHDB-386556

Trisections of 4-manifolds (problem 2)

摘录 / EXCERPT: Suppose T_1 and T_2 are balanced trisections of X, with the same core $\Sigma\subset X$. Is there an ambient isotopy $\phi:X\rightarrow X$ which maps T_1 to T_2 i.e....

open_problem | open | MathDB 页面 | 原文线索

V2-034903 | MATHDB-386557

Trisections of 4-manifolds (problem 3)

摘录 / EXCERPT: A trisection of S^4 is called emph{standard} if it is a stabilization of the genus zero trisection. Is there a non-standard balanced or unbalanced trisection of S^4?

open_problem | open | MathDB 页面 | 原文线索

V2-034904 | MATHDB-386558

Trisections of 4-manifolds (problem 4)

摘录 / EXCERPT: A balanced trisection is called emph{minimal} if its core has the smallest possible genus. Does there exist a non-minimal balanced trisection of X which is not a stabilization of...

open_problem | open | MathDB 页面 | 原文线索

V2-034905 | MATHDB-386559

Is the unbalanced trisection genus of ... equal to the balanced trisection genus?

摘录 / EXCERPT: Is the unbalanced trisection genus of X equal to the balanced trisection genus? Or equivalently, is there an example where unbalanced genus is strictly smaller than balanced genu...

open_problem | open | MathDB 页面 | 原文线索

V2-034906 | MATHDB-386561

Trisections of 4-manifolds (problem 7)

摘录 / EXCERPT: Given a smooth map (if necessary assume it is homotopy equivalence and degree one) $f:X\rightarrow Y$, does there exist trisections of X and Y such that f is homotopic to a m...

open_problem | open | MathDB 页面 | 原文线索

V2-034907 | MATHDB-386562

Trisections of 4-manifolds (problem 8)

摘录 / EXCERPT: Assume X is the double of a smooth 4-manifold with boundry Y. What can we say about the trisections of X? For example suppose Y is obtained from surgery on an embedded li...

open_problem | open | MathDB 页面 | 原文线索

V2-034908 | MATHDB-386563

Trisections of 4-manifolds (problem 9)

摘录 / EXCERPT: A Morse 2-function to a disk can be homotoped to a trisected Morse 2-function. What about Morse 2-functions to higher genus surfaces with boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-034909 | MATHDB-386564

Trisections of 4-manifolds (problem 10)

摘录 / EXCERPT: Suppose X is a homotopy S^4 together with a trisection such that X_1 is diffeomorphic to B^4. Is X diffeomorphic to S^4?

open_problem | open | MathDB 页面 | 原文线索

V2-034910 | MATHDB-386565

What Reimannian geometry conditions imply about trisections e.g.

摘录 / EXCERPT: What Reimannian geometry conditions imply about trisections e.g. Assume X is hyperbolic with large injectivity radius, does that imply the trisection genus of X is large?

open_problem | open | MathDB 页面 | 原文线索

V2-034911 | MATHDB-386566

Trisections of 4-manifolds (problem 12)

摘录 / EXCERPT: Translate 4-manifold operations e.g. rational blow downs, Gluck twist, corks,... to trisections. Find trisection diagrams for exotic pairs of 4-manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-034912 | MATHDB-386567

Find new invariants of smooth ... -manifolds using Heegaard Floer homology techniques.

摘录 / EXCERPT: Find new invariants of smooth 4-manifolds using Heegaard Floer homology techniques.

open_problem | open | MathDB 页面 | 原文线索

V2-034913 | MATHDB-386568

Find an algorithm to translate a Heegaard diagram of ... to a standard one via isotopies and handleslides.

摘录 / EXCERPT: Find an algorithm to translate a Heegaard diagram of $\#_k S^1\times S^2$ to a standard one via isotopies and handleslides. Find an invariant which gives a bound on the number of h...

open_problem | open | MathDB 页面 | 原文线索

V2-034914 | MATHDB-386570

Trisections of 4-manifolds (problem 16)

摘录 / EXCERPT: Given a Lefschetz fibration of X over a disk. Can you convert it to a trisection? Characterize which trisections arise from this construction. What about Lefschetz fibrations ove...

open_problem | open | MathDB 页面 | 原文线索

V2-034915 | MATHDB-386571

Develop a notion of trisection for higher dimensional manifolds.

摘录 / EXCERPT: Develop a notion of trisection for higher dimensional manifolds. What is the trisection for exotic 7-spheres?

open_problem | open | MathDB 页面 | 原文线索

V2-034916 | MATHDB-386572

Trisections of 4-manifolds (problem 18)

摘录 / EXCERPT: Study Spin bordism group. Find a trisection interpretation for a pair (X,F^2) where $[F^2]=w_2(X)$ or general bordiam (X,F^2).

open_problem | open | MathDB 页面 | 原文线索

V2-034917 | MATHDB-386573

Trisections of 4-manifolds (problem 19)

摘录 / EXCERPT: A (g,g)-balanced trisection is reducible and it is a diagram for a connected sum of copies of $S^1\times S^3$'s. If for a (g,k)-trisection we have that $\frac{k}{g}\approx 1$,...

open_problem | open | MathDB 页面 | 原文线索

V2-034918 | MATHDB-386574

Trisections of 4-manifolds (problem 20)

摘录 / EXCERPT: Let X be the double of the non-trivial disk bundle over $\mathbb{RP}^2$. Is $I(X)=7$? Find a closed smooth 4-manifold X such that $I(X)$ is between 0 and 7.

open_problem | open | MathDB 页面 | 原文线索

V2-034919 | MATHDB-386575

Show the following families of are standard: ...

摘录 / EXCERPT: Show the following families of are standard: begin{enumerate} item[(a)] $(b;1)$-bridge trisections (Remark: These surfaces have cyclic π_1 so interesting surfaces in this famil...

open_problem | open | MathDB 页面 | 原文线索

V2-034920 | MATHDB-386576

Find new invariants of knotted surfaces using Heegaard Floer homology techniques.

摘录 / EXCERPT: Find new invariants of knotted surfaces using Heegaard Floer homology techniques.

open_problem | open | MathDB 页面 | 原文线索

V2-034921 | MATHDB-386577

Find Khovanov type invariants coming from bridge trisections of knotted surfaces.

摘录 / EXCERPT: Find Khovanov type invariants coming from bridge trisections of knotted surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-034922 | MATHDB-386579

Find skein theoretic invariants of knotted surfaces.

摘录 / EXCERPT: Find skein theoretic invariants of knotted surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-034923 | MATHDB-386580

Can we use trisections to study knotted concordances?

摘录 / EXCERPT: Can we use trisections to study knotted concordances?

open_problem | open | MathDB 页面 | 原文线索

V2-034924 | MATHDB-386581

Classify trivial tangles which have two different plat closures to unlink.

摘录 / EXCERPT: Classify trivial tangles which have two different plat closures to unlink. (Remark: Generate examples of bridge trisections)

open_problem | open | MathDB 页面 | 原文线索

V2-034925 | MATHDB-386582

Relative trisections (problem 1)

摘录 / EXCERPT: Suppose X is a smooth, oriented 4-manifold with boundary. Consider an open book decomposition on $M=\partial X$ and let T be a relative trisection for X corresponding to th...

open_problem | open | MathDB 页面 | 原文线索

V2-034926 | MATHDB-386583

For ... -manifolds with boundary, how trisection genus changes under boundary connected sum?

摘录 / EXCERPT: For 4-manifolds with boundary, how trisection genus changes under boundary connected sum?

open_problem | open | MathDB 页面 | 原文线索

V2-034927 | MATHDB-386584

Use the theory of group trisections to find invariants of ... -manifold trisections.

摘录 / EXCERPT: Use the theory of group trisections to find invariants of 4-manifold trisections.

open_problem | open | MathDB 页面 | 原文线索

V2-034928 | MATHDB-386585

Define a notion of group trisection for trisections with boundary.

摘录 / EXCERPT: Define a notion of group trisection for trisections with boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-034929 | MATHDB-386586

Adapt group trisections to the setting of bridge trisections and use it to get invariants of knotted surfaces in ... .

摘录 / EXCERPT: Adapt group trisections to the setting of bridge trisections and use it to get invariants of knotted surfaces in S^4.

open_problem | open | MathDB 页面 | 原文线索

V2-034930 | MATHDB-386587

Group trisections (problem 4)

摘录 / EXCERPT: Accourding to a classical theorem by Wall, any two homotopoy equivalent, simply connected smooth closed 4-manifolds X and Y become diffeomorphic after stabilizing by taking c...

open_problem | open | MathDB 页面 | 原文线索

V2-034931 | MATHDB-386588

The extension problem (problem 1)

摘录 / EXCERPT: Given an Engel structure $\mathcal{D}$ on an open neighbourhood U of the 3-sphere $S^3\subset\mathbb{R}^4$, when does there exist an Engel structure $\hat{\mathcal{D}}$ on D^4...

open_problem | open | MathDB 页面 | 原文线索

V2-034932 | MATHDB-386589

Does a parametric version of the question in Problem 1.1 hold?

摘录 / EXCERPT: Does a parametric version of the question in Problem 1.1 hold? For example, if we have a 1-parametric family of germs of Engel structures $\mathcal{D}_\lambda$ on...

open_problem | open | MathDB 页面 | 原文线索

V2-034933 | MATHDB-386590

Do there exist two Engel structures that are formally homotopic but not homotopic?

摘录 / EXCERPT: Do there exist two Engel structures that are formally homotopic but not homotopic?

open_problem | open | MathDB 页面 | 原文线索

V2-034934 | MATHDB-386591

Can properties of the space of closed Engel curves be used to distinguish Engel structures?

摘录 / EXCERPT: Can properties of the space of closed Engel curves be used to distinguish Engel structures? In particular for the standard Engel structure on $\mathbb{R}^4$, is there a difference...

open_problem | open | MathDB 页面 | 原文线索

V2-034935 | MATHDB-386592

Is there an h-principle for transverse embedded surfaces in a given Engel structure?

摘录 / EXCERPT: Is there an h-principle for transverse embedded surfaces in a given Engel structure?

open_problem | open | MathDB 页面 | 原文线索

V2-034936 | MATHDB-386593

What are the constraints on the dynamics of the characteristic line field ... of an even contact structure imposed by an Engel structure?

摘录 / EXCERPT: What are the constraints on the dynamics of the characteristic line field $\mathcal{W}$ of an even contact structure imposed by an Engel structure? Otherwise said, which even conta...

open_problem | open | MathDB 页面 | 原文线索

V2-034937 | MATHDB-386594

Do Engel structures admit plugs in the sense of Wilson for ... ?

摘录 / EXCERPT: Do Engel structures admit plugs in the sense of Wilson for $\mathcal{W}$?

open_problem | open | MathDB 页面 | 原文线索

V2-034938 | MATHDB-386595

Does there exists an Engel structure on the 4-torus such that the characteristic foliation has exactly one closed leaf that is non-degenerate?

摘录 / EXCERPT: Does there exists an Engel structure on the 4-torus such that the characteristic foliation has exactly one closed leaf that is non-degenerate?

open_problem | open | MathDB 页面 | 原文线索

V2-034939 | MATHDB-386596

What can one say about the dynamics of the line field induced on a hypersurface on an Engel manifold?

摘录 / EXCERPT: What can one say about the dynamics of the line field induced on a hypersurface on an Engel manifold? For instance, are there non-trivial constraints of Bennequin type on the monod...

open_problem | open | MathDB 页面 | 原文线索

V2-034940 | MATHDB-386597

Is there an Engel structure with trivial automorphism group?

摘录 / EXCERPT: Is there an Engel structure with trivial automorphism group?

open_problem | open | MathDB 页面 | 原文线索

V2-034941 | MATHDB-386598

Is every compact holomorphic Engel manifold ... either a Cartan prolongation of a holomorphic contact 3 manifold or a Lorentz prolongation of a holomorphic conformal 3 manifold?

摘录 / EXCERPT: Is every compact holomorphic Engel manifold X^4 either a Cartan prolongation of a holomorphic contact 3 manifold or a Lorentz prolongation of a holomorphic conformal 3 manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-034942 | MATHDB-386599

Describe Engel structures that are realizable as complex line fields on compact complex surfaces.

摘录 / EXCERPT: Describe Engel structures that are realizable as complex line fields on compact complex surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-034943 | MATHDB-386600

Describe Engel structures that are realizable as Lagrangian, respectively symplectic, distributions on symplectic manifolds.

摘录 / EXCERPT: Describe Engel structures that are realizable as Lagrangian, respectively symplectic, distributions on symplectic manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-034944 | MATHDB-386601

Relations with other geometric structures (problem 4)

摘录 / EXCERPT: Is there a useful notion of a Goursat or similar cobordism/con-cor-dance between Engel structures?

open_problem | open | MathDB 页面 | 原文线索

V2-034945 | MATHDB-386602

Relations with other geometric structures (problem 5)

摘录 / EXCERPT: Let N be a contact 3 manifold with connected boundary, fix an inner product on ξ and let $\phi:N\rightarrow N$ be a contactomorphism that is the identity near ∂N. L...

open_problem | open | MathDB 页面 | 原文线索

V2-034946 | MATHDB-386603

Relations with other geometric structures (problem 6)

摘录 / EXCERPT: Let (N,ξ) be a contact 3 manifold and let $\Sigma_1,\Sigma_2\subset N$ be surfaces transversal to ξ. Which diffeomorphism $\psi:\Sigma_1\rightarrow\Sigma_2$ can be realized...

open_problem | open | MathDB 页面 | 原文线索

V2-034947 | MATHDB-386604

Are there canonical constructions on Engel structures that would yield geometric structures that obey a version of Moser's Theorem, i.e.

摘录 / EXCERPT: Are there canonical constructions on Engel structures that would yield geometric structures that obey a version of Moser's Theorem, i.e. admit no deformation or only a finite param...

open_problem | open | MathDB 页面 | 原文线索

V2-034948 | MATHDB-386605

What are cut-and-paste constructions of Engel structures?

摘录 / EXCERPT: What are cut-and-paste constructions of Engel structures? For example, are there branched cover or connected sum constructions? Are there emph{localizable} constructions?

open_problem | open | MathDB 页面 | 原文线索

V2-034949 | MATHDB-386606

Topological or cut-and-paste constructions (problem 2)

摘录 / EXCERPT: If $\pi:B\rightarrow S$ is a submersion (or a Lefschetz pencil) with B and S compact, B parallelisable, can $\ker(\pi_*)\subset TB$ be perturbed to an Engel structure?

open_problem | open | MathDB 页面 | 原文线索

V2-034950 | MATHDB-386609

Geography (problem 1)

摘录 / EXCERPT: begin{enumerate} item [(a)] Does there exist a simply-connected symplectic 4-manifold $X \neq \mathbb{C}P^2$ on the BMY line? item [(b)] Are there symplectic 4-manifolds on the BMY...

open_problem | open | MathDB 页面 | 原文线索

V2-034951 | MATHDB-386610

Geography (problem 2)

摘录 / EXCERPT: Consider a (possibly singular) surface $C \subset \mathbb{C}P^2$ homeomorphic to S^2 that either (i) is symplectic or (ii) satisfies the adjunction formula. begin{enumerate} item...

open_problem | open | MathDB 页面 | 原文线索

V2-034952 | MATHDB-386611

Symplectic BMY Inequality

摘录 / EXCERPT: If (X,ω) satisfies $c_1^2 \geq 0$, then $c_1^2 \leq 3 c_2$.

open_problem | open | MathDB 页面 | 原文线索

V2-034953 | MATHDB-386612

Does Miyaoka's inequality follow by showing that some homology class in the projectivization ... does not admit a symplectic representative?

摘录 / EXCERPT: Does Miyaoka's inequality follow by showing that some homology class in the projectivization $\mathbb{P}(T^*X)$ does not admit a symplectic representative?

open_problem | open | MathDB 页面 | 原文线索

V2-034954 | MATHDB-386613

Branched covers and braided surfaces (problem 1)

摘录 / EXCERPT: Let (X,ω) be symplectic but not complex (e.g. minimal but violating the Noether inequality), and let $f: X \to \mathbb{C}P^2$ be a branched covering with braided branch surf...

open_problem | open | MathDB 页面 | 原文线索

V2-034955 | MATHDB-386614

Is there a “universal” singular symplectic curve in ... over which all symplectic four-manifolds can be realized as branched covers?

摘录 / EXCERPT: Is there a “universal” singular symplectic curve in $\mathbb{C}P^2$ over which all symplectic four-manifolds can be realized as branched covers?

open_problem | open | MathDB 页面 | 原文线索

V2-034956 | MATHDB-386615

Branched covers and braided surfaces (problem 3)

摘录 / EXCERPT: Let $f: X \to \mathbb{C}P^2$ be a branched covering in the sense described above. In what ways can we modify the branch locus and preserve the symplectic four-manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-034957 | MATHDB-386616

Branched covers and braided surfaces (problem 4)

摘录 / EXCERPT: Let (X_1,ω_1) and (X_2,ω_2) be symplectic four-manifolds with the same c_1^2, $c_1 \cdot \omega$, c_2, and ω^2. begin{enumerate} item [(a)] Are X_1 and...

open_problem | open | MathDB 页面 | 原文线索

V2-034958 | MATHDB-386617

Branched covers and braided surfaces (problem 5)

摘录 / EXCERPT: Suppose that (X,ω) has vanishing canonical class $K=0$. Conjecturally, X is diffeomorphic to $K3$ or a T^2-bundle over T^2. What can we understand about X from the s...

open_problem | open | MathDB 页面 | 原文线索

V2-034959 | MATHDB-386618

Branched covers and braided surfaces (problem 6)

摘录 / EXCERPT: Let $X \to \mathbb{C}P^2$ be a symplectic branched covering (with smooth or singular branch locus). Can the Seiberg-Witten invariants of X be recovered from the topological data...

open_problem | open | MathDB 页面 | 原文线索

V2-034960 | MATHDB-386619

Efficient trisections of four-manifolds are well-behaved under branched covers.

摘录 / EXCERPT: Efficient trisections of four-manifolds are well-behaved under branched covers. Motivated by this, we ask: begin{enumerate} item [(a)] Does every simply-connected (smooth/symplecti...

open_problem | open | MathDB 页面 | 原文线索

V2-034961 | MATHDB-386620

Branched covers and braided surfaces (problem 8)

摘录 / EXCERPT: By work of Rudolph and Boileau-Orevkov, quasipositive knots in S^3 are precisely those knots which can be realized as positive transverse knots in $(S^3,\xi_{\operatorname{st}})$...

open_problem | open | MathDB 页面 | 原文线索

V2-034962 | MATHDB-386621

Branched covers and braided surfaces (problem 9)

摘录 / EXCERPT: Let (Y,ξ) be a contact three-manifold obtained as a cyclic branched cover of the standard contact S^3 along a quasipositive knot K. It is known that (Y,ξ) is Stein-fill...

open_problem | open | MathDB 页面 | 原文线索

V2-034963 | MATHDB-386622

Branched covers and braided surfaces (problem 10)

摘录 / EXCERPT: Suppose that Y be a three-manifold containing a rationally fibered knot K whose associated contact structure ξ is (Stein or weakly/strongly symplectically) fillable. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-034964 | MATHDB-386623

Branched covers and braided surfaces (problem 11)

摘录 / EXCERPT: Consider the relation of domination among symplectic 4-manifolds, writing $(X,\omega) \leq (X',\omega')$ if there is a symplectic branched covering $X' \to X$ in the sense of Aurou...

open_problem | open | MathDB 页面 | 原文线索

V2-034965 | MATHDB-386624

Embedding problems and configurations of curves (problem 1)

摘录 / EXCERPT: Consider the collections of algebraic, symplectic, and smooth surface knot groups in $\mathbb{C}P^2$, i.e. the groups $\pi_1(\mathbb{C}P^2 \setminus \Sigma)$ for Σ a surface...

open_problem | open | MathDB 页面 | 原文线索

V2-034966 | MATHDB-386625

Describe Stein structures on complements of symplectic surfaces in ... using Kirby diagrams.

摘录 / EXCERPT: Describe Stein structures on complements of symplectic surfaces in $\mathbb{C}P^2$ using Kirby diagrams.

open_problem | open | MathDB 页面 | 原文线索

V2-034967 | MATHDB-386626

Embedding problems and configurations of curves (problem 3)

摘录 / EXCERPT: [$\mathbb{Q}$-acyclic surfaces, plumbing trees]

open_problem | open | MathDB 页面 | 原文线索

V2-034968 | MATHDB-386630

Characterization of 0-bipolar knots

摘录 / EXCERPT: Can one characterize 0-positive, 0-negative, or 0-bipolar knots, either via the vanishing of invariants or geometrically?

open_problem | open | MathDB 页面 | 原文线索

V2-034969 | MATHDB-386631

Bipolar quotients

摘录 / EXCERPT: Does $\mathcal{T}_n/ \mathcal{T}_{n+1}$ contain a $\mathbb{Z}^{\infty}$-summand? What about a $(\mathbb{Z}/2\mathbb{Z})^{\infty}$ subgroup?

open_problem | open | MathDB 页面 | 原文线索

V2-034970 | MATHDB-386635

Torsion in the cokernel

摘录 / EXCERPT: Does $coker(\phi_{smooth})$ have torsion?

open_problem | open | MathDB 页面 | 原文线索

V2-034971 | MATHDB-386636

Concordance of knots in homology spheres.

摘录 / EXCERPT: Is every knot in every integer homology sphere topologically concordant in some integer homology cobordism to a knot in S^3? (i.e. is ϕ_{top} surjective?)

open_problem | open | MathDB 页面 | 原文线索

V2-034972 | MATHDB-386639

Small stable 4-genera

摘录 / EXCERPT: Find a knot K with $0< g_4^{st}(K)< 1/2$.

open_problem | open | MathDB 页面 | 原文线索

V2-034973 | MATHDB-386640

Stable 4-genus and 2-torsion

摘录 / EXCERPT: Find a knot K which has infinite order in $\mathcal{C}$ and yet which has $g_4^{st}(K)=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-034974 | MATHDB-386641

Injectivity of winding number 0 satellite operators

摘录 / EXCERPT: Is there a pattern P with winding number 0 such that $P(K)$ is slice if and only if K is slice, in either category?

open_problem | open | MathDB 页面 | 原文线索

V2-034975 | MATHDB-386643

Genus g traces and concordance

摘录 / EXCERPT: For a knot K in S^3, let $X^g(K)$ denote the 4-ball union $\Sigma_g \times D^2$, where Σ_g is a genus g surface with one boundary component, attached to B^4 along a...

open_problem | open | MathDB 页面 | 原文线索

V2-034976 | MATHDB-386645

Surgeries and concordance

摘录 / EXCERPT: Suppose K and J are knots in S^3 such that for all $p/q \in \mathbb{Q}$, the Dehn surgeries $S^3_{p/q}(K)$ and $S^3_{p/q}(J)$ are integer homology cobordant. Must K and J...

open_problem | open | MathDB 页面 | 原文线索

V2-034977 | MATHDB-386648

Difficult sliceness problems

摘录 / EXCERPT: Are any of the following slice? begin{enumerate} item The $(2,1)$ cable of the figure-eight knot. item The positive Whitehead double of any left-handed torus knot. item The positiv...

open_problem | open | MathDB 页面 | 原文线索

V2-034978 | MATHDB-386650

Understanding topological slice discs

摘录 / EXCERPT: Explicitly describe some non-smooth topologically slice disc.

open_problem | open | MathDB 页面 | 原文线索

V2-034979 | MATHDB-386651

Topological ribbon concordance

摘录 / EXCERPT: Define knots K and J to be topologically homotopy ribbon concordant, and write $K \leq_{thrc} J$, if K and J are topologically concordant via an annulus A such that the i...

open_problem | open | MathDB 页面 | 原文线索

V2-034980 | MATHDB-386652

Grope metric

摘录 / EXCERPT: Understand the grope metric d_1 defined by Cochran-Harvey-Powell.

open_problem | open | MathDB 页面 | 原文线索

V2-034981 | MATHDB-386655

Relative Homology

摘录 / EXCERPT: Given graphs Γ and Γ', where Γ is obtained from Γ' via an edge deletion, can we compute...

open_problem | open | MathDB 页面 | 原文线索

V2-034982 | MATHDB-386656

Poincar\'e Polynomial

摘录 / EXCERPT: What do Poincar'e polynomials of ordered or unordered configuration spaces look like? Do they have internal zeroes? Are the coefficients unimodal/log concave? Are they real rooted?...

open_problem | open | MathDB 页面 | 原文线索

V2-034983 | MATHDB-386657

Are braid groups CAT(0)?

摘录 / EXCERPT: Can we use this fact to determine if braid groups are CAT(0) by looking at some sort of limits of configuration spaces on graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-034984 | MATHDB-386658

Generation of $\pi_1(\operatorname{Conf}_n(\Gamma))$.

摘录 / EXCERPT: Can we generate graph braid group using Y-exchanges and 1-particle cycles? Specifically, for $\pi_1(\operatorname{Conf}_n(\Theta_3))$, if we quotient out the 1-particle cycles, a...

open_problem | open | MathDB 页面 | 原文线索

V2-034985 | MATHDB-386659

Verdier Duality for $\operatorname{Conf}_n(\Gamma)$

摘录 / EXCERPT: How does this compactified version relate to the original configuration space? Can we understand Verdier duality for $\operatorname{Conf}_n(\Gamma)$?

open_problem | open | MathDB 页面 | 原文线索

V2-034986 | MATHDB-386660

Genus 1 groups.

摘录 / EXCERPT: Is the 2-string braid group of a non-planar graph genus 1 and how can we tell if a group is genus 1 or not? Ko and Park showed that for if the 3 string braid group of a graph is ge...

open_problem | open | MathDB 页面 | 原文线索

V2-034987 | MATHDB-386661

Embedding genus of a graph.

摘录 / EXCERPT: Can we use the homology of configuration spaces of graphs to see if there can be an S_2-equivariant map $\operatorname{Conf}_n(K_7) \to \operatorname{Conf}_2(S^1 \times S^1)$? Ca...

open_problem | open | MathDB 页面 | 原文线索

V2-034988 | MATHDB-386662

Using the cohomology ring structure to detect embedding genus.

摘录 / EXCERPT: Can we compute the cup product structure or Steenrod operations on $H^*(\operatorname{Conf}_n(\Gamma))$? Can we use this structure to say anything about the min or max embedding ge...

open_problem | open | MathDB 页面 | 原文线索

V2-034989 | MATHDB-386663

Torsion in homology of configuration space of complete bipartite graph.

摘录 / EXCERPT: In general, is there $r+1$-torsion in $H_1(\overline{\operatorname{Conf}}_{r+1}(K_{r+2, r+2}))$?

open_problem | open | MathDB 页面 | 原文线索

V2-034990 | MATHDB-386664

Quasi-isometry type.

摘录 / EXCERPT: What can we say about the quasi-isometry type of configuration spaces of graphs? For example, all of the 2 string braid groups of trees are quasi-isometric. Can we classify graphs...

open_problem | open | MathDB 页面 | 原文线索

V2-034991 | MATHDB-386666

Non-k equals.

摘录 / EXCERPT: Can we compute the homology and homotopy groups for configuration spaces where collisions of k or fewer points are allowed?

open_problem | open | MathDB 页面 | 原文线索

V2-034992 | MATHDB-386667

Leray Spectral Sequence.

摘录 / EXCERPT: What does the Leray spectral sequence look like for the inclusion $\operatorname{Conf}_n(\Gamma) \hookrightarrow \Gamma^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-034993 | MATHDB-386668

Leray spectral sequence of embedding into a surface.

摘录 / EXCERPT: What does the Leray spectral sequence look like for a map $\operatorname{Conf}_n(\Gamma) \to \operatorname{Conf}_n(\Sigma_g)$ induced by an embedding $\Gamma \to \Sigma_g$.

open_problem | open | MathDB 页面 | 原文线索

V2-034994 | MATHDB-386669

Pseudo-manifolds.

摘录 / EXCERPT: The n-string ordered (or unordered) configuration space on K_{2n+1}, $K_{n+1,n+1}$, or $K_{n-1,n+1}$ is a manifold away from the codimension 3 skeleton. Can we investigate this...

open_problem | open | MathDB 页面 | 原文线索

V2-034995 | MATHDB-386670

Topological Complexity

摘录 / EXCERPT: What is LS category and topological complexity of $\operatorname{Conf}_n(\Gamma)$ or $\overline{\operatorname{Conf}}_n(\Gamma)$ (in particular when n is at least twice the number...

open_problem | open | MathDB 页面 | 原文线索

V2-034996 | MATHDB-386671

At what point do things stabilize?

摘录 / EXCERPT: Fix i and Γ. We know that for $n \gg 0$, $H_i(\overline{\operatorname{Conf}}_n(\Gamma))$ stabilizes. When does it stabilize? Alternatively, can we answer similar questions...

open_problem | open | MathDB 页面 | 原文线索

V2-034997 | MATHDB-386672

Module structure.

摘录 / EXCERPT: Can we explicitly compute $\bigoplus_n H_i(\overline{\operatorname{Conf}}_n(\Gamma))$ as a $Z[E]$-module?

open_problem | open | MathDB 页面 | 原文线索

V2-034998 | MATHDB-386673

Random graphs.

摘录 / EXCERPT: Given a random graph Γ (from your favorite model), what is the expected value of $\operatorname{rk}H_i(\operatorname{Conf}_n(\Gamma))$? When does a certain property occur wi...

open_problem | open | MathDB 页面 | 原文线索

V2-034999 | MATHDB-386674

Leading coefficient of the growth polynomial of the first betti number.

摘录 / EXCERPT: Given a graph Γ, is there any geometric meaning behind the growth polynomial for the first betti number of $\overline{\operatorname{Conf}}_n(\Gamma)$? Byung Hee An has a con...

open_problem | open | MathDB 页面 | 原文线索

V2-035000 | MATHDB-386675

Graph invariants that are functorial with respect to deletions and contractions.

摘录 / EXCERPT: What graph invariants are functorial with respect to deletions and contractions? In particular, can we find invariants that play nicely with the graph minor category?

open_problem | open | MathDB 页面 | 原文线索

V2-035001 | MATHDB-386676

Uniqueness of Joyce-Lee-Tsui translator

摘录 / EXCERPT: Let L_t be an ancient, almost calibrated, connected, exact LMCF. Assume that the tangent flow at $-\infty$ is equal to $P_1\cup P_2$ where P_1, P_2 are two Lagrangian planes in...

open_problem | open | MathDB 页面 | 原文线索

V2-035002 | MATHDB-386677

Weak LMCF

摘录 / EXCERPT: Find a good notion of weak solutions to LMCF.

open_problem | open | MathDB 页面 | 原文线索

V2-035003 | MATHDB-386678

Examples of non-compact Special Lagrangians

摘录 / EXCERPT: Find new examples of non-compact Special Lagrangians. (For example in $T^{\star}S^n$)

open_problem | open | MathDB 页面 | 原文线索

V2-035004 | MATHDB-386679

Generic singularities of LMCF

摘录 / EXCERPT: Is the Lawlor neck a generic singularity for LMCF?

open_problem | open | MathDB 页面 | 原文线索

V2-035005 | MATHDB-386680

Singularities of LMCF

摘录 / EXCERPT: Can we classify the singularities of LMCF in the equivariant setting?

open_problem | open | MathDB 页面 | 原文线索

V2-035006 | MATHDB-386681

Lagrangians in Landau-Ginzburg models

摘录 / EXCERPT: Are there good conditions for the existence of calibrated Lagrangians in Landau-Ginzburg models?

open_problem | open | MathDB 页面 | 原文线索

V2-035007 | MATHDB-386682

Characterize Translators for LMCF

摘录 / EXCERPT: Under what conditions will eternal solutions to LMCF be translators?

open_problem | open | MathDB 页面 | 原文线索

V2-035008 | MATHDB-386683

Characterize flat Lagrangian shrinkers

摘录 / EXCERPT: Under what conditions must a non-compact complete Lagrangian shrinker be flat? (e.g. is one end and simply connected enough?)

open_problem | open | MathDB 页面 | 原文线索

V2-035009 | MATHDB-386684

Classification of Lagrangian tori

摘录 / EXCERPT: Can we classify Lagrangian shrinking tori in $\mathbb{C}^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-035010 | MATHDB-386685

Blow-up of mean curvature at type II singularity

摘录 / EXCERPT: Does the mean curvature blow-up at a type II singularity? (Construct examples where it does not.)

open_problem | open | MathDB 页面 | 原文线索

V2-035011 | MATHDB-386686

Preserving HS conditions

摘录 / EXCERPT: Can we somehow couple the LMCF to the K"ahler-Ricci flow to preserve the Hamiltonian stationary condition?

open_problem | open | MathDB 页面 | 原文线索

V2-035012 | MATHDB-386687

LMCF and holomorphic curves

摘录 / EXCERPT: Relate the behaviour of LMCF to the existence of J-holomorphic curves.

open_problem | open | MathDB 页面 | 原文线索

V2-035013 | MATHDB-386688

Convergence of LMCF

摘录 / EXCERPT: Does the LMCF converge if the length of the path in the space of calibrated Lagrangian remains bounded?

open_problem | open | MathDB 页面 | 原文线索

V2-035014 | MATHDB-386689

Estimates for dHYM in the subcritical case

摘录 / EXCERPT: Find apriori estimate for the dHYM equation in the subcritical phase case.

open_problem | open | MathDB 页面 | 原文线索

V2-035015 | MATHDB-386690

Behaviour of line bundle MCF in the unstable case

摘录 / EXCERPT: Describe the behaviour of the Line Bundle Mean Curvature Flow in the unstable case. (Find singularity models and solitons)

open_problem | open | MathDB 页面 | 原文线索

V2-035016 | MATHDB-386691

Higher rank dHYM

摘录 / EXCERPT: Study the deformed Hermitian-Yang-Mills equation in higher rank. (eg. on non-split extension of TM and $\mathcal{O}_M$ on a Calabi-Yau surface)

open_problem | open | MathDB 页面 | 原文线索

V2-035017 | MATHDB-386692

Finite-dimensional interpretation of Bridgeland stability

摘录 / EXCERPT: Find or describe a finite-dimensional analogue of the dHYM equation and its relation to stability, similar to how finite-dimensional geometric invariant theory is formally similar...

open_problem | open | MathDB 页面 | 原文线索

V2-035018 | MATHDB-386693

Dimensional reduction of deformed Hermitian Yang--Mills

摘录 / EXCERPT: Understand the solvability of the dHYM equation on higher rank bundles in any examples. In particular: begin{itemize} item On complex surfaces with (special) Lagrangian torus fibra...

open_problem | open | MathDB 页面 | 原文线索

V2-035019 | MATHDB-386694

Toric mirror symmetry and asymptotically admissible Lagrangians

摘录 / EXCERPT: Provide an algebro-geometric or categorical interpretation of the central charge-type term ... appearing in the study of deformations of special Lagrangians in toric mirror symmetry...

open_problem | open | MathDB 页面 | 原文线索

V2-035020 | MATHDB-386695

Lagrangian mean curvature flow for K\"ahler--Einstein manifolds with definite first Chern class.

摘录 / EXCERPT: What does Lmcf look like for K"ahler--Einstein manifolds with definite first Chern class. In particular: begin{itemize} item What does Lmcf look like for $c_1(X)>0$. For example Ev...

open_problem | open | MathDB 页面 | 原文线索

V2-035021 | MATHDB-386696

Lagrangian mean curvature flow with dilations

摘录 / EXCERPT: What happens under Lmcf to Lagrangians in, for example, $\mathbb{R}^{2n} \cong T^* \mathbb{R}^n$ where one also allows dilations of the fibres (see work of Neves who shows that sin...

open_problem | open | MathDB 页面 | 原文线索

V2-035022 | MATHDB-386697

Definition of dHYM in higher rank for complexes of vector bundles

摘录 / EXCERPT: What is the right definition of the higher rank dHYM equation for complexes of vector bundles? Such a definition would need to satisfy certain desirable properties: begin{itemize}...

open_problem | open | MathDB 页面 | 原文线索

V2-035023 | MATHDB-386698

Deriving higher rank dHYM from Lagrangian multisections using mirror symmetry

摘录 / EXCERPT: How can one derive mathematically the higher rank dHYM equation from Lagrangian multisections using mirror symmetry. For example consider the case $T\mathbb{CP}^2$ relating to a mu...

open_problem | open | MathDB 页面 | 原文线索

V2-035024 | MATHDB-386699

Thomas--Yau--Joyce conjecture for the flat 2-torus

摘录 / EXCERPT: Prove a version of the Thomas--Yau--Joyce conjecture for curves in the flat 2-torus, or just on $\mathbb{R}^2$. Technicalities that occur include: begin{itemize} item One must rest...

open_problem | open | MathDB 页面 | 原文线索

V2-035025 | MATHDB-386700

Geometric structures on the space of complexified Kahler classes and of stability conditions

摘录 / EXCERPT: What are natural Weil--Peterson-type metrics on moduli of complexified K"ahler classes, and what can be said about the geometry of the moduli space using them. For example what are...

open_problem | open | MathDB 页面 | 原文线索

V2-035026 | MATHDB-386701

Examples of Joyce--Lee--Tsui translators in explicit cases

摘录 / EXCERPT: Study whether an almost-calibrated translator arises in Lmcf, for example in the simplest case ($\mathbb{C}^2$). The grim reaper translator is not allowed as this is not strictly a...

open_problem | open | MathDB 页面 | 原文线索

V2-035027 | MATHDB-386702

Study mirror symmetry in model Calabi-Yau metrics arising out of real Monge--Ampere equations near large volume limit.

摘录 / EXCERPT: Study the existence and flows for dHYM and special Lagrangians in semi-flat SYZ model Calabi--Yau metrics arising from real Monge--Ampere equations near the large volume limit.

open_problem | open | MathDB 页面 | 原文线索

V2-035028 | MATHDB-386703

Finite time singularities in dHYM flows and graphical Lagrangian mean curvature flow.

摘录 / EXCERPT: Study the possibility of finite-time singularities for dHYM flows (any which seem applicable to the problem, or new flows). Note that finite-time singularity formation does occur i...

open_problem | open | MathDB 页面 | 原文线索

V2-035029 | MATHDB-386704

Subsolutions of higher rank dHYM and Z-critical equation

摘录 / EXCERPT: Is a solution to the higher rank dHYM equation necessarily a subsolution in the sense of Dervan--McCarthy--Sektnan. The analogous condition for dHYM in rank 1 is an equivalence by...

open_problem | open | MathDB 页面 | 原文线索

V2-035030 | MATHDB-386705

Generating Lagrangians in the Fukaya category of P^2 relative to an elliptic curve

摘录 / EXCERPT: Do special Lagrangians generate the Fukaya--Seidel category $\mathrm{FS}(W)$ for a superpotential W mirror to $\mathbb{CP}^2$ relative to an elliptic curve E, $\mathbb{CP}^2/E$...

open_problem | open | MathDB 页面 | 原文线索

V2-035031 | MATHDB-386706

Chern number inequalities in higher rank dHYM

摘录 / EXCERPT: Does existence of solutions to the higher rank dHYM equation imply any interesting Chern number inequalities, for example a generalised Bogomolov--Gieseker inequality. For example...

open_problem | open | MathDB 页面 | 原文线索

V2-035032 | MATHDB-386707

Isotopy to holomorphic Landau--Ginzburg model

摘录 / EXCERPT: Suppose $W:X\to \mathbb{C}$ is a symplectic Landau--Ginzburg model. That is, there exists a compact $K\subset \mathbb{C}$ such that on $X\backslash W^{-1}(K)$ the fibers are symple...

open_problem | open | MathDB 页面 | 原文线索

V2-035033 | MATHDB-386708

Equivariant chromatic support of N_{∞}-rings

摘录 / EXCERPT: Given an N_{∞}-operad, and a R an algebra over it, what are the possible chromatic supports of it?

open_problem | open | MathDB 页面 | 原文线索

V2-035034 | MATHDB-386710

$\mathbb{E}_{\infty}$-total power operation maps for Lubin--Tate theories and Morava K-theories

摘录 / EXCERPT: Let $\tau_{\geq0}^{G}$ denote the connective cover in G-spectra which is the connective cover on each categorical fixed point. Let R be E_n or $K(n)$, and equip R with the...

open_problem | open | MathDB 页面 | 原文线索

V2-035035 | MATHDB-386711

Multiplicative structure on equivariant algebraic K-theory

摘录 / EXCERPT: How does equivariant algebraic K-theory behave with respective to multiplicative structure? For example, if I have an N_{∞}-algebra or an $\mathbb{E}_{V}$-algebra A in G...

open_problem | open | MathDB 页面 | 原文线索

V2-035036 | MATHDB-386712

Self-injectivity of equivariant ring spectra

摘录 / EXCERPT: Given R a G-ring spectrum, when is the RO_G-graded Mackey functor valued homotopy ring of R self-injective?

open_problem | open | MathDB 页面 | 原文线索

V2-035037 | MATHDB-386713

RO_G-graded homotopy via formal groups

摘录 / EXCERPT: Given a complex orientable G-ring spectrum R (in a suitable sense), is there an interpretation of the RO_G-graded homotopy groups π_VR in terms of formal groups?

open_problem | open | MathDB 页面 | 原文线索

V2-035038 | MATHDB-386714

A K-theoretic criterion for maps to even rings.

摘录 / EXCERPT: Given an $\mathbb{E}_{1}$-ring R with the unit nontrivial in rational K-theory, does R admit an $\mathbb{E}_{1}$ map to a nonzero even ring?

open_problem | open | MathDB 页面 | 原文线索

V2-035039 | MATHDB-386715

Spin orientation from equivariant homotopy theory

摘录 / EXCERPT: Can the spin orientation $\mathrm{MSpin} \to \mathrm{ko}$ be recovered from the map $\mathrm{MU}_{\mathbb{R}} \to \mathrm{ku}_{\mathbb{R}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-035040 | MATHDB-386716

Detecting Hopf invariant 1 elements in fixed points of Lubin--Tate theories

摘录 / EXCERPT: Do the Hopf invariant one elements that don't exist in the sphere also not exist in E_n^{hG} if the group G is sufficiently large?

open_problem | open | MathDB 页面 | 原文线索

V2-035041 | MATHDB-386717

Blueshift for $\mathbb{E}_{\infty}$-rings

摘录 / EXCERPT: Given an $\mathbb{E}_{\infty}$-ring R that is $K(n)$-acyclic, is R^{tC_p} $K(n-1)$-acyclic?

open_problem | open | MathDB 页面 | 原文线索

V2-035042 | MATHDB-386718

Multilinear relations for 1-semiadditive power operations

摘录 / EXCERPT: At height n, does the operation $\theta_{G,X}$ satisfy an $n+1$-multilinear equation with respect to products?

open_problem | open | MathDB 页面 | 原文线索

V2-035043 | MATHDB-386719

THH for structured G-ring-spectra

摘录 / EXCERPT: Is is possible to build a version of THH for G-ring-spectra that is G-symmetric monoidal?

open_problem | open | MathDB 页面 | 原文线索

V2-035044 | MATHDB-386730

a inverted Artin--Tate $\mathbb{R}$-motives

摘录 / EXCERPT: What category do you get when you invert the class a in ${\\textrm{Sp}}_{\mathbb{R}}^{AT}$?

open_problem | open | MathDB 页面 | 原文线索

V2-035045 | MATHDB-386732

Equivariant and synthetic

摘录 / EXCERPT: Can one profitably mix synthetic spectra with equivariant homotopy theory?

open_problem | open | MathDB 页面 | 原文线索

V2-035046 | MATHDB-386736

The Hurewicz image of equivariant truncated Brown-Peterson spectra

摘录 / EXCERPT: What is the Hurewicz image of the truncated Brown--Peterson spectrum $\mathrm{BP}^{((G))}\langle m \rangle$?

open_problem | open | MathDB 页面 | 原文线索

V2-035047 | MATHDB-386737

Understanding n-slices

摘录 / EXCERPT: Is there a way to get a good computational grasp of n-slices? For example understanding their Postnikov towers, homotopy groups etc.

open_problem | open | MathDB 页面 | 原文线索

V2-035048 | MATHDB-386739

The $G\cdot \overline{v}_n$-Bockstein spectral sequence for $\mathrm{BP}^{((G))}\langle n\rangle$

摘录 / EXCERPT: Can one understand the $G\cdot \overline{v}_n$-Bockstein spectral sequence for this quotient map?

open_problem | open | MathDB 页面 | 原文线索

V2-035049 | MATHDB-386741

Relations in $\mathrm{BP}^{((G))}$

摘录 / EXCERPT: Is $8\sigma = 0$ in the fixed points of $\mathrm{BP}^{((G))}$?

open_problem | open | MathDB 页面 | 原文线索

V2-035050 | MATHDB-386742

Recollement for $\mathbb{E}_V$-algebras

摘录 / EXCERPT: Given a G-representation V, can one similarly reconstruct the category of $\mathbb{E}_V$-algebras from Borel data and geometric fixed point data?

open_problem | open | MathDB 页面 | 原文线索

V2-035051 | MATHDB-386743

The structure on $\mathrm{BP}_{\mathbb{R}}\langle n \rangle$

摘录 / EXCERPT: What multiplicative structure can be put on $\mathrm{BP}_{\mathbb{R}}\langle n \rangle$? Is it $\mathbb{E}_{2\sigma+1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-035052 | MATHDB-386744

Improved quotients of even rings

摘录 / EXCERPT: Given an even $\mathbb{E}_\infty$-ring R and $f \in \pi_*R$, is $\mathrm{cof} (f^n)$ an $\mathbb{E}_{n+1}$-algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-035053 | MATHDB-386745

Quotients of $\mathbb{E}_V$-algebras

摘录 / EXCERPT: Given an $\mathbb{E}_V$-algebra, and an element f in its $RO(G)$-graded homotopy, are the quotients of powers (or norms) of f highly structured?

open_problem | open | MathDB 页面 | 原文线索

V2-035054 | MATHDB-386746

Multiplicative structure on equivariant quotients of E_m

摘录 / EXCERPT: What multiplicative structures can be supported on C_{p^n}-equivariant quotients of E_m, such as those studied in ?

open_problem | open | MathDB 页面 | 原文线索

V2-035055 | MATHDB-386747

Primitive pth root of unity in π_0

摘录 / EXCERPT: Does $\mathrm{KU}[\zeta_p]$ admit an $\mathbb{E}_2$-algebra structure?

open_problem | open | MathDB 页面 | 原文线索

V2-035056 | MATHDB-386750

Study resolution of the diagonal on symmetric products (equipped with potentials) and applications to fixed point Floer cohomology models for link invariants.

摘录 / EXCERPT: Study resolution of the diagonal on symmetric products (equipped with potentials) and applications to fixed point Floer cohomology models for link invariants.

open_problem | open | MathDB 页面 | 原文线索

V2-035057 | MATHDB-386751

Continuing the previous problem, understand explicitly the resolution of diagonals in Landau--Ginzburg models.

摘录 / EXCERPT: Continuing the previous problem, understand explicitly the resolution of diagonals in Landau--Ginzburg models.

open_problem | open | MathDB 页面 | 原文线索

V2-035058 | MATHDB-386752

Give a mathematical construction of a link invariant using the Fukaya category associated to potentials on ... and Mak--Smith's cylindrical model.

摘录 / EXCERPT: Give a mathematical construction of a link invariant using the Fukaya category associated to potentials on $\text{Hilb}^n(\mathbb{C}^* \times \mathbb{C}^*)_{hor}$ and Mak--Smith's...

open_problem | open | MathDB 页面 | 原文线索

V2-035059 | MATHDB-386753

Barmeier--Wang recently exhibited a nontrivial deformation of the extended arc algebra via Hochschild cohomology computations.

摘录 / EXCERPT: Barmeier--Wang recently exhibited a nontrivial deformation of the extended arc algebra via Hochschild cohomology computations. What is the symplectic meaning of this?

open_problem | open | MathDB 页面 | 原文线索

V2-035060 | MATHDB-386754

What is the B-model interpretation of the deformation introduced in the previous problem?

摘录 / EXCERPT: What is the B-model interpretation of the deformation introduced in the previous problem?

open_problem | open | MathDB 页面 | 原文线索

V2-035061 | MATHDB-386755

How can we explicitly understand the mirror symmetry equivalence between the A model and B model knot homology constructions in a way which allows one to compare (e.g., functorial) structures?

摘录 / EXCERPT: How can we explicitly understand the mirror symmetry equivalence between the A model and B model knot homology constructions in a way which allows one to compare (e.g., functorial)...

open_problem | open | MathDB 页面 | 原文线索

V2-035062 | MATHDB-386756

Can one adapt any of the symplectic Khovanov homology constructions to the case of knots in lens spaces?

摘录 / EXCERPT: Can one adapt any of the symplectic Khovanov homology constructions to the case of knots in lens spaces? Specifically, what about Aganagic's construction of knot Floer homology?

open_problem | open | MathDB 页面 | 原文线索

V2-035063 | MATHDB-386757

Can one establish a surgery formula (or exact triangle) in Aganagic's construction?

摘录 / EXCERPT: Can one establish a surgery formula (or exact triangle) in Aganagic's construction?

open_problem | open | MathDB 页面 | 原文线索

V2-035064 | MATHDB-386758

Can one introduce and then apply the computational tool of “nice diagrams" of to Aganagic's construction and use it to simplify computations?

摘录 / EXCERPT: Can one introduce and then apply the computational tool of “nice diagrams" of to Aganagic's construction and use it to simplify computations?

open_problem | open | MathDB 页面 | 原文线索

V2-035065 | MATHDB-386759

Asymptotics of spectral invariants

摘录 / EXCERPT: For a symplectic surface (Σ, ω) and a Hamiltonian $H \in C^{\infty}([0,1] \times \Sigma, \mathbb{R})$, investigate the subleading asymptotic of...

open_problem | open | MathDB 页面 | 原文线索

V2-035066 | MATHDB-386760

Further developments of quantitative Floer theory in low dimensions

摘录 / EXCERPT: What is the relationship between the link and PFH spectral invariants?

open_problem | open | MathDB 页面 | 原文线索

V2-035067 | MATHDB-386761

Can quantitative Heegaard Floer invariants be used to give quantitative invariants of contact forms on ... -manifolds related to ECH spectral invariants by analogy with the previous problem?

摘录 / EXCERPT: Can quantitative Heegaard Floer invariants be used to give quantitative invariants of contact forms on 3-manifolds related to ECH spectral invariants by analogy with the previous...

open_problem | open | MathDB 页面 | 原文线索

V2-035068 | MATHDB-386762

If one could provide a positive answer to the previous question, can it be used to better understand the subleading asymptotics of ECH spectral invariants?

摘录 / EXCERPT: If one could provide a positive answer to the previous question, can it be used to better understand the subleading asymptotics of ECH spectral invariants?

open_problem | open | MathDB 页面 | 原文线索

V2-035069 | MATHDB-386763

What more can the computational apparatus in Heegaard Floer theory tell us about dynamics?

摘录 / EXCERPT: What more can the computational apparatus in Heegaard Floer theory tell us about dynamics?

open_problem | open | MathDB 页面 | 原文线索

V2-035070 | MATHDB-386764

Miscellaneous

摘录 / EXCERPT: To what degree can link spectral invariants and their applications be generalized to higher dimensions?

open_problem | open | MathDB 页面 | 原文线索

V2-035071 | MATHDB-386765

Symplectic topology and dynamics (problem 1)

摘录 / EXCERPT: Given a closed symplectic manifold (M, ω), consider $\text{Ham}^{\text{aut}}(M, \omega) \subset \text{Ham}(M, \omega)$ the subset of autonomous Hamiltonian diffeomorphism gr...

open_problem | open | MathDB 页面 | 原文线索

V2-035072 | MATHDB-386766

Develop a toolkit for producing Lagrangian links in symplectic manifolds, and study its relationship to questions such as the Lagrangian packing problem and problems from dynamics.

摘录 / EXCERPT: Develop a toolkit for producing Lagrangian links in symplectic manifolds, and study its relationship to questions such as the Lagrangian packing problem and problems from dynamics.

open_problem | open | MathDB 页面 | 原文线索

V2-035073 | MATHDB-386767

Symplectic topology and dynamics (problem 3)

摘录 / EXCERPT: Is the volume preserving diffeomorphism group of $\mathbb{R}^2$ with respect to the standard volume form $dx \wedge dy$ perfect?

open_problem | open | MathDB 页面 | 原文线索

V2-035074 | MATHDB-386768

How about the perfectness for volume-preserving diffeomorphism groups of surfaces of infinite type?

摘录 / EXCERPT: How about the perfectness for volume-preserving diffeomorphism groups of surfaces of infinite type?

open_problem | open | MathDB 页面 | 原文线索

V2-035075 | MATHDB-386769

Symplectic topology and dynamics (problem 5)

摘录 / EXCERPT: Do existing techniques say anything about the structure of $\overline{\text{Ham}}(M, \omega)$ for $\text{dim} M \geq 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-035076 | MATHDB-386770

Symplectic topology and dynamics (problem 6)

摘录 / EXCERPT: Is the group of C^2 diffeomorphisms of S^1 simple?

open_problem | open | MathDB 页面 | 原文线索

V2-035077 | MATHDB-386771

Symplectic topology and dynamics (problem 7)

摘录 / EXCERPT: Can one better/explicitly understand the abelianization of the $\overline{\text{Ham}}(\Sigma, \omega)$ for (Σ, ω) a symplectic surface? More generally, how about quotie...

open_problem | open | MathDB 页面 | 原文线索

V2-035078 | MATHDB-386772

It's known that the group of volume-preserving diffeomorphisms on ... is perfect.

摘录 / EXCERPT: It's known that the group of volume-preserving diffeomorphisms on S^3 is perfect. Is the commutator length uniformly bounded?

open_problem | open | MathDB 页面 | 原文线索

V2-035079 | MATHDB-386773

Can Cristina Anghel's approach to knot polynomials (via intersections of Lagrangians in configuration spaces) be categorified in instances via Lagrangian Floer cohomology?

摘录 / EXCERPT: Can Cristina Anghel's approach to knot polynomials (via intersections of Lagrangians in configuration spaces) be categorified in instances via Lagrangian Floer cohomology?

open_problem | open | MathDB 页面 | 原文线索

V2-035080 | MATHDB-386774

Knot homology theories and categorification (problem 2)

摘录 / EXCERPT: Is it true that $\mathfrak{gl}_{n|m}$ and $\mathfrak{gl}_{m|n}$ knot homologies are isomorphic if one switches the role of t and q variable? In particular, what is happening fo...

open_problem | open | MathDB 页面 | 原文线索

V2-035081 | MATHDB-386775

Compute knot homology of a local' infinite twist in any of these cases.

摘录 / EXCERPT: Compute knot homology of a local' infinite twist in any of these cases.

open_problem | open | MathDB 页面 | 原文线索

V2-035082 | MATHDB-386776

What is a B-side construction of Heegaard knot Floer homology?

摘录 / EXCERPT: What is a B-side construction of Heegaard knot Floer homology?

open_problem | open | MathDB 页面 | 原文线索

V2-035083 | MATHDB-386778

How does Ekholm--Shende's “Skeins on Branes" story fit into a symplectic production of, e.g., (nil) Hecke algebras, and how can we better use it?

摘录 / EXCERPT: How does Ekholm--Shende's “Skeins on Branes" story fit into a symplectic production of, e.g., (nil) Hecke algebras, and how can we better use it?

open_problem | open | MathDB 页面 | 原文线索

V2-035084 | MATHDB-386779

Knot homology theories and categorification (problem 7)

摘录 / EXCERPT: Is there a direct proof that $HF^*(I \ \text{brane}, E \ \text{brane})$ is a knot invariant which does not pass through resolving E branes by T branes?

open_problem | open | MathDB 页面 | 原文线索

V2-035085 | MATHDB-386780

Does Aganagic's Khovanov homology construction have a Lee/Bar-Natan deformation?

摘录 / EXCERPT: Does Aganagic's Khovanov homology construction have a Lee/Bar-Natan deformation? Does this theory provide a geometric explanation for odd Khovanov homology?

open_problem | open | MathDB 页面 | 原文线索

V2-035086 | MATHDB-386781

Are there spectral refinements of KLRW and nil Hecke algebra?

摘录 / EXCERPT: Are there spectral refinements of KLRW and nil Hecke algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-035087 | MATHDB-386782

Are there spectral refinements of link spectral invariants and applications?

摘录 / EXCERPT: Are there spectral refinements of link spectral invariants and applications?

open_problem | open | MathDB 页面 | 原文线索

V2-035088 | MATHDB-386784

Loop spaces

摘录 / EXCERPT: Is there a good notion of a loop space of a digital space?

open_problem | open | MathDB 页面 | 原文线索

V2-035089 | MATHDB-386785

Topological realization

摘录 / EXCERPT: Is there a functor F from the category of digital images to that of topological spaces such that the digital invariants of a digital image X agree with the corresponding classi...

open_problem | open | MathDB 页面 | 原文线索

V2-035090 | MATHDB-386794

Singular homology of graphs

摘录 / EXCERPT: Compute $H^{sing}_*(\mathbb{Z}/n, c_k)$. Is it isomorphic to the homomology of the clique complex of $(\mathbb{Z}/n, c_k)$?

open_problem | open | MathDB 页面 | 原文线索

V2-035091 | MATHDB-386798

Equivariant discrete homotopy theory

摘录 / EXCERPT: Develop equivariant discrete homotopy theory.

open_problem | open | MathDB 页面 | 原文线索

V2-035092 | MATHDB-386808

Volume Conjecture (problem 9)

摘录 / EXCERPT: Find a shadow formula for Teichm"uller TQFT

open_problem | open | MathDB 页面 | 原文线索

V2-035093 | MATHDB-386813

Indecomposable SL_3 webs

摘录 / EXCERPT: For a triangulated punctured surface, find the indecomposable SL_3 webs for the Knutsen-Tao cone

open_problem | open | MathDB 页面 | 原文线索

V2-035094 | MATHDB-386817

Quantum trace for closed surfaces

摘录 / EXCERPT: For Bloomquist-Karuo-Le quantum trace for closed surface, what happens when we change the pants decomposition? (Hatcher-Thurston move)

open_problem | open | MathDB 页面 | 原文线索

V2-035095 | MATHDB-386818

Skein module for 3 manifold with $S(\Sigma)$ action

摘录 / EXCERPT: Let $S(\Sigma)$ act on N be a module of the Kauffman Bracket Skein Algebra. Can we determine if N is the skein module of a 3 manifold M with $S(\Sigma)$ action given by...

open_problem | open | MathDB 页面 | 原文线索

V2-035096 | MATHDB-386823

Categorification of CJP

摘录 / EXCERPT: Categorify Colored Jones Polynomial to study the volume conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-035097 | MATHDB-386824

Lift the recursion relation of the colored Jones Polynomial to categorified CJP.

摘录 / EXCERPT: Lift the recursion relation of the colored Jones Polynomial to categorified CJP.

open_problem | open | MathDB 页面 | 原文线索

V2-035098 | MATHDB-386827

c^k-approximations and contactomorphisms (problem 2)

摘录 / EXCERPT: For what k can we c^k-approximate the mapping tori of ergodic time-1 flows of contact vector fields by a contactomorphism whose mapping torus is convex?

open_problem | open | MathDB 页面 | 原文线索

V2-035099 | MATHDB-386829

Are all Liouville manifolds with half-dimensional homotopy type Liouville homotopic to Weinstein manifolds?

摘录 / EXCERPT: Are all Liouville manifolds with half-dimensional homotopy type Liouville homotopic to Weinstein manifolds? 1-parametric version?

open_problem | open | MathDB 页面 | 原文线索

V2-035100 | MATHDB-386830

What about in the special case where the skeleton is stratified by manifolds of dimension at most half?

摘录 / EXCERPT: What about in the special case where the skeleton is stratified by manifolds of dimension at most half?

open_problem | open | MathDB 页面 | 原文线索

V2-035101 | MATHDB-386832

Must every almost contact class admit a tight structure?

摘录 / EXCERPT: Must every almost contact class admit a tight structure?

open_problem | open | MathDB 页面 | 原文线索

V2-035102 | MATHDB-386835

Can Donaldson's theorem be proved on a closed symplectic manifolds via CHT techniques?

摘录 / EXCERPT: Can Donaldson's theorem be proved on a closed symplectic manifolds via CHT techniques?

open_problem | open | MathDB 页面 | 原文线索

V2-035103 | MATHDB-386837

Give a bypass description of convex surfaces foliating the region in ... between two Darboux balls.

摘录 / EXCERPT: Give a bypass description of convex surfaces foliating the region in $(S^5,\xi_k):=\text{OBD}(T^*S^2, \tau^{2k+1})$ between two Darboux balls.

open_problem | open | MathDB 页面 | 原文线索

V2-035104 | MATHDB-386838

Find other interesting explicit bypass decompositions of contact manifolds.

摘录 / EXCERPT: Find other interesting explicit bypass decompositions of contact manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-035105 | MATHDB-386840

Can triviality be detected via capacities on balls around the intersection point?

摘录 / EXCERPT: Can triviality be detected via capacities on balls around the intersection point?

open_problem | open | MathDB 页面 | 原文线索

V2-035106 | MATHDB-386843

Can we define an appropriate notion for overtwistedness for conformally symplectic manifolds?

摘录 / EXCERPT: Can we define an appropriate notion for overtwistedness for conformally symplectic manifolds? Two potential options: the existence of an embedded model, having a contactization whi...

open_problem | open | MathDB 页面 | 原文线索

V2-035107 | MATHDB-386844

Symplectic fillings of contact submanifolds (problem 1)

摘录 / EXCERPT: Given a Weinstein domain, can we construct a distinct Weinstein filling of its countact boundary and/or a contactomorphism of its boundary inducing the same augmentation.

open_problem | open | MathDB 页面 | 原文线索

V2-035108 | MATHDB-386848

Flexibility

摘录 / EXCERPT: Do properly embedded codimension 2 Weinstein domains satisfy an existence h-principle?

open_problem | open | MathDB 页面 | 原文线索

V2-035109 | MATHDB-386851

Integrable systems (problem 1)

摘录 / EXCERPT: Is there a topological problem (e.g. SFT(?)) giving rise to this?

open_problem | open | MathDB 页面 | 原文线索

V2-035110 | MATHDB-386852

Integrable systems (problem 2)

摘录 / EXCERPT: Let V be an odd dimensional manifold equipped with a stable Hamiltonian structure. Find conditions on V for the rational SFT homology algebra $(\mathcal{P},\{h,\cdot\})$ to hav...

open_problem | open | MathDB 页面 | 原文线索

V2-035111 | MATHDB-386900

Is strong homology consistently additive for closed subspaces of Euclidean space, or more generally for locally compact metric spaces?

摘录 / EXCERPT: Is strong homology consistently additive for closed subspaces of Euclidean space, or more generally for locally compact metric spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-035112 | MATHDB-386954

Is there a notion of Symmetry Protected Topological (SPT) phases protected by fusion categorical symmetry?

摘录 / EXCERPT: Is there a notion of Symmetry Protected Topological (SPT) phases protected by fusion categorical symmetry? Perhaps, they can be built using the cohomology groups of classifying spa...

open_problem | open | MathDB 页面 | 原文线索

V2-035113 | MATHDB-386975

Are there non-trivial links that can be created by only local operators from the vacuum and if so what topology can we get?

摘录 / EXCERPT: Are there non-trivial links that can be created by only local operators from the vacuum and if so what topology can we get?

open_problem | open | MathDB 页面 | 原文线索

V2-035114 | MATHDB-386979

List of all question (problem 37)

摘录 / EXCERPT: Given a MTC $\mathcal{C}$ and a condensable algebra A, how can one describe the A-action on $\mathcal{C}^{loc}_A$? Describe a cohomological theory of it.

open_problem | open | MathDB 页面 | 原文线索

V2-035115 | MATHDB-386980

What is the homotopy type of the space of braided tensor categories?

摘录 / EXCERPT: What is the homotopy type of the space of braided tensor categories?

open_problem | open | MathDB 页面 | 原文线索

V2-035116 | MATHDB-386983

Understand the topology of the classes of Topological Phases of Matter (TPMs).

摘录 / EXCERPT: Understand the topology of the classes of Topological Phases of Matter (TPMs).

open_problem | open | MathDB 页面 | 原文线索

V2-035117 | MATHDB-387007

Countable and finitary reducibility (problem 2)

摘录 / EXCERPT: Consider the class B defined by some $L_{\omega_1,\omega}$-sentence φ (where the topology is generated by the fragments of the language). What are the possible complexiti...

open_problem | open | MathDB 页面 | 原文线索

V2-035118 | MATHDB-387029

Does the superspace coinvariant ring defined as ... relate to link homology?

摘录 / EXCERPT: Does the superspace coinvariant ring defined as ... relate to link homology? What object does ... define in the trace of Hecke category?

open_problem | open | MathDB 页面 | 原文线索

V2-035119 | MATHDB-387033

Khovanov-Rozansky homology (problem 1)

摘录 / EXCERPT: Is the Khovanov-Rozansky homology functorial?

open_problem | open | MathDB 页面 | 原文线索

V2-035120 | MATHDB-387037

Khovanov-Rozansky homology (problem 5)

摘录 / EXCERPT: Describe annulus maps in Khovanov-Rozansky homology HHH.

open_problem | open | MathDB 页面 | 原文线索

V2-035121 | MATHDB-387039

Khovanov homology (problem 2)

摘录 / EXCERPT: Compute $JM_1^{t_1}\dots JM_n^{t_n}$ for Khovanov homology.

open_problem | open | MathDB 页面 | 原文线索

V2-035122 | MATHDB-387042

Compute endomorphism algebra (in the homotopy category) of projectors, and relate it to coordinate ring and of charts on Hilb ...

摘录 / EXCERPT: Compute endomorphism algebra (in the homotopy category) of projectors, and relate it to coordinate ring and of charts on Hilb$^n(\mathbb{C}^2)$

open_problem | open | MathDB 页面 | 原文线索

V2-035123 | MATHDB-387058

Is there an affine Springer fiber model for colored link homology?

摘录 / EXCERPT: Is there an affine Springer fiber model for colored link homology?

open_problem | open | MathDB 页面 | 原文线索

V2-035124 | MATHDB-387690

Research Problems in Function Theory - Problem 5.29

摘录 / EXCERPT: A G_δ set is a subset of a topological space that is a countable intersection of open sets. Let E be a G_δ set of measure zero on $|z|=1$. Then does there exist an...

open_problem | open | MathDB 页面 | 原文线索

V2-035125 | MATHDB-389679

Problem 1.20 - Justify the above arguments rigorously.

摘录 / EXCERPT: Justify the above arguments rigorously.

open_problem | open | MathDB 页面 | 原文线索

V2-035126 | MATHDB-389749

Problem 4.12 - Construct invariants of 3-manifolds via a linear skein theo ry based on the Kauffman skein module.

摘录 / EXCERPT: Construct invariants of 3-manifolds via a linear skein theo ry based on the Kauffman skein module.

open_problem | open | MathDB 页面 | 原文线索

V2-035127 | MATHDB-389776

Problem 7.2 - (S.K.

摘录 / EXCERPT: (S.K. Hansen, T. Takata) Find pairs of non-homeomorphic rational homology 3-spheres that can be distinguished by th eir quantum G invariants τ G r or their quantum P G invariants τ...

open_problem | open | MathDB 页面 | 原文线索

V2-035128 | MATHDB-389878

Problem 12.14 - (N.

摘录 / EXCERPT: (N. Askitas) Can a knot of 4-genus gs always be sliced (made into a slice knot) by gs crossing switches?

open_problem | open | MathDB 页面 | 原文线索

V2-035129 | MATHDB-390135

Problem 2.4 - Let Γ be a subgroup of finite index in the mapping class group Mod1,2 and let φ: Γ → Γ be an automorphism.

摘录 / EXCERPT: Let Γ be a subgroup of finite index in the mapping class group Mod1,2 and let φ: Γ → Γ be an automorphism. Is φ the restriction of an automorphism of Mod1,2? 1 In all situations we...

open_problem | open | MathDB 页面 | 原文线索

V2-035130 | MATHDB-390172

Question - Do there exist any other examples of non commuting homeomorphisms g andh which are not both Dehn twists and satisfy a...

摘录 / EXCERPT: Do there exist any other examples of non commuting homeomorphisms g andh which are not both Dehn twists and satisfy a braid relation ghg...[U+ED19] [U+ED18][U+ED17] [U+ED1A] mi,j =hgh...[U+ED19] [U+ED18][U+ED17] [U+ED1A] mi,j for some m...

open_problem | open | MathDB 页面 | 原文线索

V2-035131 | MATHDB-390245

Problem 12 - (Analog of Ratner theorem).

摘录 / EXCERPT: (Analog of Ratner theorem).

open_problem | open | MathDB 页面 | 原文线索

V2-035132 | MATHDB-391911

Grünbaum’s 1960 hyperplane equipartition conjecture; Ramos’ hyperplane equipartition conjecture

摘录 / EXCERPT: For every integer $n\ge 1$ and every finite, nonzero Borel measure μ on $\mathbb{R}^n$ that is absolutely continuous with respect to Lebesgue measure, there exist n affine hy...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035133 | MATHDB-391938

Conjecture 1.1 of Chen-Hussain-Yau-Zuo on generalized moduli algebras

摘录 / EXCERPT: Let $(V,0)$ be an isolated complex hypersurface singularity defined by $f\in\mathbb{C}\{x_1,\ldots,x_n\}$, with $n\ge 2$ and $\operatorname{mult}(f)\ge 3$. Let $A(V)$ be the moduli...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035134 | MATHDB-392110

Wei-Wylie question on Bakry-Émery curvature and Ricci curvature

摘录 / EXCERPT: Given a complete weighted Riemannian manifold (M,g,e^{-f}) with bounded potential f and nonnegative Bakry-Émery Ricci curvature, namely...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035135 | MATHDB-392113

Chernov-Maguire conjecture on Khovanov detection of Legendrian simple knots

摘录 / EXCERPT: For every smooth knot type K that is Legendrian simple-meaning that any two Legendrian representatives L_1,L_2 of K with $\operatorname{tb}(L_1)=\operatorname{tb}(L_2)$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035136 | MATHDB-392114

Guilbault-Tinsley question on Z-compactifiability and pseudo-collarability

摘录 / EXCERPT: Does there exist an open 4-manifold M that admits a finite-dimensional compact ANR $\mathcal{Z}$-compactification but is not pseudo-collarable?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035137 | MATHDB-392126

Peskine-Szpiro conjectures on modules of finite projective dimension

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a Noetherian local ring, and let M and N be finitely generated R-modules such that $\operatorname{pd}_R M<\infty$ and $\ell_R(M\otimes_RN)<\infty$. T...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035138 | MATHDB-392264

Sarnak’s Conjecture

摘录 / EXCERPT: For every dynamical system (X,T) on a compact metric space with zero topological entropy, every $f\in C(X)$, and every $x\in X$, one has...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035139 | MATHDB-392270

Geography problem of spin symplectic 4-manifolds

摘录 / EXCERPT: For every admissible lattice point (χ,c), does there exist a closed, simply connected, spin symplectic 4-manifold X such that $(\chi_h(X),c_1^2(X))=(\chi,c)$?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-035140 | MATHDB-392273

Kanalas' problem of continuously realising types

摘录 / EXCERPT: For every coherent theory $\mathcal{T}$, every topological space X, and every continuous assignment of model-theoretic types $p:X\to S(\mathcal{T})$, does there exist a sheaf mod...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035141 | MATHDB-392471

Fukaya-Yamaguchi and Kapovitch-Petrunin-Tuschmann conjectures

摘录 / EXCERPT: ...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-035142 | MATHDB-392775

Associativity and Hamiltonian-product comparison for preorbifold cohomology

摘录 / EXCERPT: Theorem 6. Let Y be an almost complex manifold with a T-action preserving the almost complex structure. Then $(PH^{*,\diamond}_T(Y),\wedge)$ is a graded, associative ring. If...

open_problem | open | MathDB 页面 | 原文线索

V2-035143 | MATHDB-392799

Transcendental holomorphic Morse inequality for nef classes

摘录 / EXCERPT: Let X be a compact complex manifold of dimension n. If $[U+007F]curl[U+007F]\alpha\u007f$ and $[U+007F]curl[U+007F]\beta\u007f$ are nef cohomology classes of type $(1,1)$ on X satisfying ... then the clas...

open_problem | open | MathDB 页面 | 原文线索

V2-035144 | MATHDB-392817

Multiplicative structures of crepant resolutions in dimension three

摘录 / EXCERPT: Completely determine the multiplicative structures in the cohomology or K-theory of crepant resolutions of quotient singularities $\mathbb{C}^3/G$, for finite subgroups...

open_problem | open | MathDB 页面 | 原文线索

V2-035145 | MATHDB-393025

Homological stabilization for mapping class groups of 3-manifolds

摘录 / EXCERPT: Let N be a compact, connected, oriented 3-manifold whose universal cover satisfies the Poincaré conjecture and which has no connected-sum summand homeomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-035146 | MATHDB-393033

Equivariant Borel homology and Floer homologies

摘录 / EXCERPT: Conjecturally, equivariant Borel homology of these spectra recovers the Floer homologies of Ozsváth-Szabó and the Seiberg-Witten-Floer homology.

open_problem | open | MathDB 页面 | 原文线索

V2-035147 | MATHDB-393043

Polynomial-growth cohomology versus ordinary group cohomology

摘录 / EXCERPT: For a given finitely generated discrete group G with a word length function ℓ, are the canonical maps ... isomorphisms for all $n\in\mathbb{N}$?

open_problem | open | MathDB 页面 | 原文线索

V2-035148 | MATHDB-393077

Finiteness of diffeomorphism classes under bounded sectional curvature and diameter

摘录 / EXCERPT: Given fixed $n\in\mathbb{N}$, $D>0$, and $c,C\in\mathbb{R}$, are there at most finitely many diffeomorphism classes of pairwise homotopy equivalent closed Riemannian n-dimensiona...

open_problem | open | MathDB 页面 | 原文线索

V2-035149 | MATHDB-393078

Minimal dimension for the Theorem A examples

摘录 / EXCERPT: Determine the minimal integer n for which there exist an infinite sequence $(M_k^n)_{k\\in\mathbb N}$ of pairwise homotopy equivalent but mutually non-homeomorphic closed simply...

open_problem | open | MathDB 页面 | 原文线索

V2-035150 | MATHDB-393079

Finiteness of diffeomorphism types under nonnegative curvature and diameter bounds

摘录 / EXCERPT: Given fixed $n\in\mathbb{N}$ and $D,C>0$, are there at most finitely many diffeomorphism types of pairwise homotopy equivalent closed Riemannian n-manifolds with sectional curvat...

open_problem | open | MathDB 页面 | 原文线索

V2-035151 | MATHDB-393226

Happel’s question on vanishing Hochschild cohomology

摘录 / EXCERPT: Let Λ be a finite-dimensional algebra over a field K. If $\operatorname{HH}^{n}(\Lambda)=0$ for all sufficiently large n, must the global dimension of Λ be fini...

open_problem | open | MathDB 页面 | 原文线索

V2-035152 | MATHDB-393232

Hochschild cohomology and infinitesimal deformations of derived categories

摘录 / EXCERPT: For a differential graded algebra A over a field of characteristic 0, one expects a canonical bijection between the second Hochschild cohomology $HH^2(A,A)$ and the equivalence...

open_problem | open | MathDB 页面 | 原文线索

V2-035153 | MATHDB-393235

Compact Kähler manifolds not having the homotopy type of a complex projective manifold

摘录 / EXCERPT: For every dimension $n \ge 4$, there exist compact Kähler manifolds which do not have the homotopy type of a complex projective manifold.

open_problem | open | MathDB 页面 | 原文线索

V2-035154 | MATHDB-393238

Equality of the mathematical and topological vertices for formal toric Calabi-Yau threefolds

摘录 / EXCERPT: The generating functions $\widetilde{C}_{\vec{\mu}}(\lambda;\mathbf{n})$ of formal relative Gromov-Witten invariants of indecomposable formal toric Calabi-Yau threefolds coincide w...

open_problem | open | MathDB 页面 | 原文线索

V2-035155 | MATHDB-393312

Projective-bundle push-forward formula for a multiplicative formal group law

摘录 / EXCERPT: Let k be any field and let A^* be an oriented weak cohomology theory on Sm/k with multiplicative formal group law ... Let X be an object of...

open_problem | open | MathDB 页面 | 原文线索

V2-035156 | MATHDB-393330

Homology-sphere decomposition of 2-CM complexes

摘录 / EXCERPT: Can the S_i’s in Theorem 7 be taken to be homology spheres?

open_problem | open | MathDB 页面 | 原文线索

V2-035157 | MATHDB-393399

Superpolynomial conjecture for Khovanov-Rozansky homology

摘录 / EXCERPT: For every knot K, there exists a finite polynomial $C_P(K)\in\mathbb{Z}[a^{\pm1},q^{\pm1},t^{\pm1}]$ such that ... for all sufficiently large N, where ...

conjecture | open | MathDB 页面 | 原文线索

V2-035158 | MATHDB-393400

Superpolynomial stabilization for unreduced Khovanov-Rozansky homology

摘录 / EXCERPT: For a knot K, there exists a finite polynomial $\overline{\mathcal P}(K)\in\mathbb{Z}[a^{\pm1},q^{\pm1},t^{\pm1}]$ such that ... for all sufficiently large N, where ...

open_problem | open | MathDB 页面 | 原文线索

V2-035159 | MATHDB-393401

Stabilization conjecture for reduced Khovanov-Rozansky homology

摘录 / EXCERPT: For a knot K, there exists a finite polynomial $\mathcal{P}(K)\in\mathbb{Z}_{\ge 0}[a^{\pm1},q^{\pm1},t^{\pm1}]$ such that ... for all sufficiently large N.

conjecture | open | MathDB 页面 | 原文线索

V2-035160 | MATHDB-393403

Uniqueness of smooth extensions of generalized cohomology theories

摘录 / EXCERPT: Let h be a multiplicative generalized cohomology theory, let R be a $ ... -graded algebra over $ ... $, and let $r:hto HR...

open_problem | open | MathDB 页面 | 原文线索

V2-035161 | MATHDB-393438

Topology of the set S associated with the matrix A

摘录 / EXCERPT: It is an important open problem to explore the topology of ... where $A=(\alpha_{ij})_{i,j=1}^{p}$ is given by ... for pairwise distinct points $a_1,\ldots,a_p\in\mathbb{R}^{3}$.

open_problem | open | MathDB 页面 | 原文线索

V2-035162 | MATHDB-393466

Topology of the resonant action-level intersection

摘录 / EXCERPT: The report presents as still partly conjectural the claim that, for a given action value vector a, there exist integers $p_1,\ldots,p_m$ with $\sum_{\ell=1}^{m}p_\ell=M$ such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-035163 | MATHDB-393523

Relation between symplectic and Khovanov-Rozansky link homologies

摘录 / EXCERPT: The authors conjecture that their symplectic link invariants are related to Khovanov-Rozansky homology of the mirror link $\kappa!$ by ...

open_problem | open | MathDB 页面 | 原文线索

V2-035164 | MATHDB-393703

Relation between the induced product and quantum cohomology for local \(\mathbb P^2\)

摘录 / EXCERPT: For $S=\mathbb P^2$ and $X=\mathcal O_{\mathbb P^2}(-3)$, the induced product structure should be related, as Bridgeland conjectured, to the quantum cohomology of $\mathbb P^2$; th...

open_problem | open | MathDB 页面 | 原文线索

V2-035165 | MATHDB-393723

Locally compact group topologies and orbit equivalence relations

摘录 / EXCERPT: Is it true that H does not have a locally compact second countable group topology including the original topology if and only if $E_0^{\d4a9}$ is Borel reducible to E_X^H for s...

open_problem | open | MathDB 页面 | 原文线索

V2-035166 | MATHDB-393752

Cohomological-index bound for odd-cycle Hom-complexes

摘录 / EXCERPT: Let G be a graph with an odd cycle and let $r\ge 1$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-035167 | MATHDB-393768

Trivialization of the D=11 supersymmetry four-cocycle and the underlying gauge supergroup

摘录 / EXCERPT: The report states that the standard $D=11$ supersymmetry-algebra cohomology four-cocycle ω_4, structurally equivalent to dA_3, is trivialized on every member...

open_problem | open | MathDB 页面 | 原文线索

V2-035168 | MATHDB-393847

Ravenel’s telescope conjecture (smashing-localization version)

摘录 / EXCERPT: A version of Ravenel’s telescope conjecture states that every smashing localization functor on the stable homotopy category $\mathcal S$-that is, a localization functor satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-035169 | MATHDB-393851

Algebraic models for rational equivariant cohomology theories

摘录 / EXCERPT: Conjecture. For a compact Lie group G there is an abelian category $A(G)$ and a Quillen equivalence ... such that:

open_problem | open | MathDB 页面 | 原文线索

V2-035170 | MATHDB-393855

Converse to the virtual-smallness theorem for complete intersections

摘录 / EXCERPT: Let R be a commutative noetherian local ring. The report proves that if R is a complete intersection, then every complex M of R-modules with finitely generated homology...

open_problem | open | MathDB 页面 | 原文线索

V2-035171 | MATHDB-393894

Lifting Chern classes to integral Deligne cohomology

摘录 / EXCERPT: Let X be a smooth manifold. Can the Chern class be lifted, in the same way as the Chern character, to a natural transformation with values in integral Deligne cohomology...

open_problem | open | MathDB 页面 | 原文线索

V2-035172 | MATHDB-393969

Conjecture 10: connection between projective-injective endomorphism algebras and Khovanov homology

摘录 / EXCERPT: Conjecture 10 ([11]). (1) For any natural number m, there is an isomorphism of algebras ... where H_m denotes Khovanov’s algebra. (2) The homological tangle invariant ... is Khovan...

conjecture | open | MathDB 页面 | 原文线索

V2-035173 | MATHDB-393980

Baby-Aleshin closure as a pro-\2 Sylow subgroup

摘录 / EXCERPT: The conjecture is that $G(B)$ is a pro-2-Sylow subgroup of $\mathrm{SL}_2(\mathbb{Z}_2)$, where $G(B)$ denotes the topological closure associated with the Baby-Aleshin automaton...

open_problem | open | MathDB 页面 | 原文线索

V2-035174 | MATHDB-394008

Baby Teichmüller space: cohomology and connected components

摘录 / EXCERPT: Consider the collection of all functions $f:\mathbb{Z}/n\mathbb{Z}\to\mathbb{RP}^1$ such that $f(i)\ne f(i+1)$ for any i (taking indices modulo n) and whose image contains at l...

open_problem | open | MathDB 页面 | 原文线索

V2-035175 | MATHDB-394027

Joint \\-convergence conditions for evolutionary problems

摘录 / EXCERPT: Let Q be equipped with a Hausdorff topology. Consider sequences of pairs of functionals (E_k,R_k) for dissipative systems, (E_k,D_k) for rate-independent systems, or...

open_problem | open | MathDB 页面 | 原文线索

V2-035176 | MATHDB-394038

Colliot-Thélène’s Hasse principle conjecture (Conjecture HP)

摘录 / EXCERPT: Let k be a perfect field of virtual cohomological dimension 2, let $\cOmega_k$ denote the space of orderings of k, and for each $v \cin \cOmega_k$ let k_v denote the real c...

conjecture | open | MathDB 页面 | 原文线索

V2-035177 | MATHDB-394050

Benson’s regularity conjecture for cohomology rings

摘录 / EXCERPT: Benson’s regularity conjecture: Castelnuovo-Mumford regularity is always zero for a cohomology ring.

conjecture | open | MathDB 页面 | 原文线索

V2-035178 | MATHDB-394052

Classification of maximal subgroups of low-dimensional almost simple classical groups

摘录 / EXCERPT: Classify the maximal subgroups of the almost simple classical groups in dimensions at most 12. The report states that dimensions $2,3,4,5,6,$ and 7 had been completed, while di...

open_problem | open | MathDB 页面 | 原文线索

V2-035179 | MATHDB-394074

Positive topological entropy for restricted-contact-type hypersurfaces

摘录 / EXCERPT: Let M be a closed energy-hyperbolic manifold. Let $\a3\subset T^*M$ be a hypersurface enclosing the zero-section 0_M, and choose a smooth Hamiltonian $H:T^*M\to[0,\infty)$ such...

open_problem | open | MathDB 页面 | 原文线索

V2-035180 | MATHDB-394081

Symplectic Floer homology and Nielsen fixed point theory for mapping classes

摘录 / EXCERPT: For any mapping class $g\in\Gamma=\pi_0(\operatorname{Diff}^+(M))$, there is a monotone representative $\phi\in\operatorname{Symp}^m(M,\omega)$, with respect to some ϕ-invaria...

open_problem | open | MathDB 页面 | 原文线索

V2-035181 | MATHDB-394092

Uniformly bounded numbers of closed geodesics in homotopy classes of a torus

摘录 / EXCERPT: Does there exist a Riemannian metric on T^2 and a constant $n_0\in\mathbb N$ such that, in each homotopy class, the number of closed geodesics is less than n_0?

open_problem | open | MathDB 页面 | 原文线索

V2-035182 | MATHDB-394110

A spectral sequence from HOMFLY homology to knot Floer homology

摘录 / EXCERPT: There should be a spectral sequence $E_k(0)$ relating the HOMFLY homology H to knot Floer homology.

open_problem | open | MathDB 页面 | 原文线索

V2-035183 | MATHDB-394114

Torsion in integer-coefficient Heegaard Floer homology

摘录 / EXCERPT: Are there any 3-manifolds for which there is torsion in any of the integer-coefficient Heegaard Floer groups? The report answers affirmatively: if Σ_g is a surface of genus...

open_problem | open | MathDB 页面 | 原文线索

V2-035184 | MATHDB-394122

Annulus Conjecture

摘录 / EXCERPT: Annulus Conjecture. If there is an embedded $S^{n-1}\times[0,1]$ in $\mathbb{R}^n$, then it is isotopic to the standard embedding, in the topological category. The report states th...

conjecture | open | MathDB 页面 | 原文线索

V2-035185 | MATHDB-394123

The manifold Hauptvermutung

摘录 / EXCERPT: The manifold Hauptvermutung asserts that every homeomorphism $h\colon L\to M$ of compact n-dimensional PL manifolds is homotopic to a PL homeomorphism. The report states that thi...

open_problem | open | MathDB 页面 | 原文线索

V2-035186 | MATHDB-394167

Kernel comparison for finite-dimensional topological and orthogonal groups

摘录 / EXCERPT: For all relevant dimensions n, the restriction maps ... have the same kernel.

open_problem | open | MathDB 页面 | 原文线索

V2-035187 | MATHDB-394168

Conjecture C on the rational nullhomotopy of the derivative map

摘录 / EXCERPT: Let n be odd, and let $\operatorname{reg}(D^n\times D^2,D^2)$ be the space of smooth regular (nonsingular) maps $f:D^n\times D^2\to D^2$ that agree with the projection near the b...

conjecture | open | MathDB 页面 | 原文线索

V2-035188 | MATHDB-394170

Topological equisingularity versus Milnor-number constancy in the surface case

摘录 / EXCERPT: Does the equivalence between topological equisingularity and constancy of the Milnor number for families of hypersurface singularities with isolated singularities extend to the sur...

open_problem | open | MathDB 页面 | 原文线索

V2-035189 | MATHDB-394171

Characterization of topological equisingularity by equisingularity at the critical set

摘录 / EXCERPT: Conjecture: equisingularity at the critical set characterizes topological equisingularity.

open_problem | open | MathDB 页面 | 原文线索

V2-035190 | MATHDB-394177

Connected sum problem for homotopy equivalences

摘录 / EXCERPT: Fix a closed manifold $X^n=X_1\#X_2$ with $n>2$ and write $\pi_1X=G_1*G_2$. Is every homotopy equivalence $h\colon M\to X$ splittable; that is, does there exist a connected-sum dec...

open_problem | open | MathDB 页面 | 原文线索

V2-035191 | MATHDB-394181

Finiteness of the cohomology of connected covers

摘录 / EXCERPT: Let X be a finite space with finite $\\pi_1 X$ and $n\ge 1$. Is $H^*(X\langle n\rangle;\mathbb{F}_p)$ finitely generated as an algebra over the Steenrod algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-035192 | MATHDB-394357

The g-conjecture for spheres

摘录 / EXCERPT: If Δ is a sphere (PL-sphere, homology sphere), then the g-vector of Δ is an M-vector.

conjecture | open | MathDB 页面 | 原文线索

V2-035193 | MATHDB-394362

Elementary proof of the homotopy type of a multiple mapping cylinder of torus projections

摘录 / EXCERPT: Consider the multiple mapping cylinder $MC(X\xrightarrow{f_1}Y_1,\ldots,X\xrightarrow{f_n}Y_n)$ formed as the quotient of the disjoint union...

open_problem | open | MathDB 页面 | 原文线索

V2-035194 | MATHDB-394410

Convergence theorem for the no-response test

摘录 / EXCERPT: Let G be a domain homotopic to a ball B_R, with $R>0$ sufficiently large, in the sense that there exists a continuous mapping $[0,1]\ni\lambda\mapsto G_\lambda$ with $G_0=B_R$,...

open_problem | open | MathDB 页面 | 原文线索

V2-035195 | MATHDB-394429

Dense topological graphs contain many pairwise intersecting edges

摘录 / EXCERPT: For any $\[1m\varepsilon>0\[0m$ and for any positive integer k, there are $\[1m\delta>0\[0m$ and n_0 such that every topological graph with $n\ge n_0$ vertices, at least...

open_problem | open | MathDB 页面 | 原文线索

V2-035196 | MATHDB-394513

Finite-total-curvature minimal surfaces with non-simply-connected topology in \(\mathbb H^2\times\mathbb R\)

摘录 / EXCERPT: Construct complete minimal surfaces of finite total curvature in $\mathbb H^2\times\mathbb R$ that are annuli, or that have more complicated topology.

open_problem | open | MathDB 页面 | 原文线索

V2-035197 | MATHDB-394518

Properness of finite-topology minimal surfaces

摘录 / EXCERPT: If $N\in\mathcal{N}$ has nonnegative scalar curvature and is not flat, then any surface $M\in\mathcal{M}_0(N)$ of finite topology is properly embedded in N.

open_problem | open | MathDB 页面 | 原文线索

V2-035198 | MATHDB-394607

Higher-codimension extensions of positive-scalar-curvature obstructions

摘录 / EXCERPT: Is it possible to work with submanifolds of even higher codimension, using more conditions on the ambient manifold, such as vanishing of higher homotopy groups?

open_problem | open | MathDB 页面 | 原文线索

V2-035199 | MATHDB-394618

Persistence of nonzero Lyapunov exponents under small perturbations

摘录 / EXCERPT: Does any sufficiently small perturbation, in the $C^{1+\alpha}$ topology, of a volume-preserving diffeomorphism with nonzero Lyapunov exponents possess nonzero exponents on a set o...

open_problem | open | MathDB 页面 | 原文线索

V2-035200 | MATHDB-394682

Equality of the Lusternik-Schnirelmann categories for closed 1-forms

摘录 / EXCERPT: For a finite CW-complex X and a cohomology class $\xi\in H^1(X;\mathbb{R})$, are the invariants $\operatorname{cat}(X,\xi)$ and $\operatorname{cat}_1(X,\xi)$ always equal?

open_problem | open | MathDB 页面 | 原文线索

V2-035201 | MATHDB-394683

Koszulity of the McCool-group cohomology ring

摘录 / EXCERPT: Determine if the cohomology ring $H^{*}(P\Sigma_{\ell};\mathbb{Q})$ is Koszul.

open_problem | open | MathDB 页面 | 原文线索

V2-035202 | MATHDB-394684

Existence of a noncommutative Alexander polynomial with Thurston-norm and Novikov-homology properties

摘录 / EXCERPT: Let M be a closed 3-manifold that is aspherical, and let $G=\pi_1(M)$. Does there exist an element $\widetilde{\Delta}(M)\in \mathbb Z[G]$ such that: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-035203 | MATHDB-394696

Topological-complexity inequality for a locally trivial fibration

摘录 / EXCERPT: If $F \to E \to B$ is a locally trivial fibration, is it true that $\operatorname{TC}(E) \leq \operatorname{TC}(F)\,\operatorname{TC}(B)$?

open_problem | open | MathDB 页面 | 原文线索

V2-035204 | MATHDB-394699

Topological complexity of closed 3-manifolds and their fundamental groups

摘录 / EXCERPT: Let M be a closed 3-manifold. Does $\operatorname{TC}(M)$ depend only on $\pi_1(M)$?

open_problem | open | MathDB 页面 | 原文线索

V2-035205 | MATHDB-394710

Existence of an L^2 rho-invariant associated to a degree-two cohomology class

摘录 / EXCERPT: An open question is whether it is possible to define an L^2 invariant $\rho^{[c]}_{(2)}(D)$ associated with the Dirac operator D and the cohomology class...

open_problem | open | MathDB 页面 | 原文线索

V2-035206 | MATHDB-394728

Exact sequence for absolute K-theory and cyclic homology of sub-harmonically stabilized algebras

摘录 / EXCERPT: For a complex algebra A and a sub-harmonic operator ideal J, the report states that there is an exact sequence ... Here K_* denotes algebraic K-theory and HC_* denotes alge...

open_problem | open | MathDB 页面 | 原文线索

V2-035207 | MATHDB-394751

Homotopy type of the open-closed cobordism category

摘录 / EXCERPT: The report states that there is a homotopy equivalence ... Here Y_{+} denotes Y with a disjoint basepoint added, and $Q(Y)=\operatorname*{colim}_{k}\Omega^{k}\Sigma^{k}Y$.

open_problem | open | MathDB 页面 | 原文线索

V2-035208 | MATHDB-394780

Nonvanishing of a filtered L^2-cohomology group and Abel-Jacobi cycles

摘录 / EXCERPT: For a Picard modular 3-fold X with universal family $f:A\to X$, assuming unipotent monodromy at infinity, if Γ is sufficiently small and $H^0(X,\Omega_X^1)$ contains two...

open_problem | open | MathDB 页面 | 原文线索

V2-035209 | MATHDB-394783

Dolbeault cohomology of nilmanifolds with left-invariant complex structure

摘录 / EXCERPT: For every nilmanifold $M=\Gamma\backslash G$ with left-invariant complex structure J, can the Dolbeault cohomology $H^{p,q}(M,J)$, for all bidegrees (p,q), be calculated using...

open_problem | open | MathDB 页面 | 原文线索

V2-035210 | MATHDB-394847

Global dimension of twisted group algebras

摘录 / EXCERPT: Let G be a torsion-free nilpotent group of finite cohomological dimension, with $\operatorname{cd}(G)=n$. Let $\alpha\in H^2(G,\mathbb{C}^{\times})$, and suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-035211 | MATHDB-394932

A cohomological proof of the failure of type FP_d for isotropic S-arithmetic groups

摘录 / EXCERPT: Assuming that Γ is of type FP_d, can one use the resolution $(*)$ and Borel-Serre's result to show that, for isotropic G, the top cohomology $\widetilde H^d_c(C;M)$ is n...

open_problem | open | MathDB 页面 | 原文线索

V2-035212 | MATHDB-394977

Perfection detected by relative Atiyah classes

摘录 / EXCERPT: Let A be a noetherian ring and let $X\in D(A)$ be a complex of right A-modules whose total homology $H(X)$ is a finitely generated graded A-module. Let $\rho:B\to A$ be a rin...

open_problem | open | MathDB 页面 | 原文线索

V2-035213 | MATHDB-395047

Homotopy equivalence of the Bigelow-generator complex

摘录 / EXCERPT: The complex with this differential is conjectured to be homotopy equivalent to the Seidel-Smith and Khovanov complexes.

open_problem | open | MathDB 页面 | 原文线索

V2-035214 | MATHDB-395054

Behavior of clock homology under Reidemeister moves

摘录 / EXCERPT: Study how the clock homology behaves under Reidemeister moves. This may help to modify the construction here to yield a topological invariant.

open_problem | open | MathDB 页面 | 原文线索

V2-035215 | MATHDB-395085

Dynamics of topological defects in nonlinear field theories: approximation by timelike Minkowski minimal surfaces

摘录 / EXCERPT: The report states and proves the following result. Let $N\ge 2$, $k\in\{1,2\}$ with $N>k$, and let u solve ... with initial data ... Here $q:\mathbb R\to\mathbb R$ solves...

open_problem | open | MathDB 页面 | 原文线索

V2-035216 | MATHDB-395208

Topological dimension and Z-stability: the Toms-Winter conjecture

摘录 / EXCERPT: Conjecture: For a separable, finite, nonelementary, simple, unital, nuclear C^*-algebra A, the following are equivalent: (α) A has finite decomposition rank. (β) A absorbs...

conjecture | open | MathDB 页面 | 原文线索

V2-035217 | MATHDB-395214

Homotopy invariants of automorphism groups of unital Kirchberg algebras

摘录 / EXCERPT: For a unital Kirchberg algebra D, compute the homotopy invariants of $ \operatorname{Aut}(D)$, including $\pi_n(\operatorname{Aut}(D))$ for $n\ge 1$,...

open_problem | open | MathDB 页面 | 原文线索

V2-035218 | MATHDB-395226

Topology bounds for real threefolds fibred by rational curves

摘录 / EXCERPT: Let $f\colon W\to X$ be a real smooth projective threefold fibred by rational curves over a geometrically rational real surface X, and suppose that $W(\mathbb{R})$ is orientable....

open_problem | open | MathDB 页面 | 原文线索

V2-035219 | MATHDB-395275

Realization of the adding machine on the two-torus

摘录 / EXCERPT: For the two-torus $\mathbb T^2$, is the adding-machine ergodic system measurably conjugate to a minimal topological system (M,λ,h) in the stated sense, where M is a compa...

open_problem | open | MathDB 页面 | 原文线索

V2-035220 | MATHDB-395279

Kuhn’s realization conjecture

摘录 / EXCERPT: Which abstract graded rings over the field $\mathbb{F}_p$, with a compatible action of the Steenrod algebra, are isomorphic to mod p cohomology rings of spaces?

conjecture | open | MathDB 页面 | 原文线索

V2-035221 | MATHDB-395284

Actions of $SAut(F_n)$ on generalized $[U+007F]\mathbb{Z}_2$-homology spheres

摘录 / EXCERPT: If $n\ge 3$ and $d<n-1$, then any action of $SAut(F_n)$ by homeomorphisms on a generalized d-sphere over $\mathbb{Z}_2$ is trivial, and hence $Aut(F_n)$ can act only via the dete...

open_problem | open | MathDB 页面 | 原文线索

V2-035222 | MATHDB-395287

Unbounded strong realization conjecture

摘录 / EXCERPT: Let X be a topological space, and regard $H^*(X;\mathbb F_2)$ as an unstable module over the mod-2 Steenrod algebra. If $H^*(X;\mathbb F_2)\notin U_0$, then there exists an int...

conjecture | open | MathDB 页面 | 原文线索

V2-035223 | MATHDB-395296

Threshold for vanishing integral homology in a random 2-dimensional complex

摘录 / EXCERPT: For the random two-dimensional simplicial-complex model $X(n,p)$ with full one-dimensional skeleton K_n, in which each triple is added independently with probability p, what is...

open_problem | open | MathDB 页面 | 原文线索

V2-035224 | MATHDB-395311

Linear-edge bound for endpoint-disjoint grids in topological graphs

摘录 / EXCERPT: For every positive integer ℓ, there exists a constant $c_\ell>0$ such that every topological graph with n vertices and more than $c_\ell n$ edges contains an ℓ-grid in...

open_problem | open | MathDB 页面 | 原文线索

V2-035225 | MATHDB-395344

Poincaré conjecture for simply connected closed 3-manifolds

摘录 / EXCERPT: Every simply connected, orientable, closed 3-manifold is topologically equivalent to S^3.

conjecture | open | MathDB 页面 | 原文线索

V2-035226 | MATHDB-395358

Equality in Gromov’s entropy inequality for unequal dynamical degrees

摘录 / EXCERPT: For a dominant meromorphic mapping $f:X\to X$ on a compact complex projective surface, with topological degree $d_t\ne\lambda$ (where d_t is the number of preimages of a generic...

open_problem | open | MathDB 页面 | 原文线索

V2-035227 | MATHDB-395359

Extension of the good-birational-model/degree-growth theory to small-topological-degree maps

摘录 / EXCERPT: Let $f:X\to X$ be a dominant meromorphic mapping of a compact projective complex surface, with topological degree d_t and first dynamical degree λ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-035228 | MATHDB-395360

Uniqueness of the maximal-entropy measure for small-topological-degree maps

摘录 / EXCERPT: Let $f:X\dashrightarrow X$ be a dominant meromorphic mapping on a compact complex projective surface with small topological degree, meaning that its first dynamical degree satisfie...

open_problem | open | MathDB 页面 | 原文线索

V2-035229 | MATHDB-395362

Topological characterization of postsingularly finite exponential maps

摘录 / EXCERPT: Every postsingularly finite topological exponential map, namely a covering map $f:S^2\setminus\{\infty\}\to S^2\setminus\{0,\infty\}$ for which the orbit of 0 is preperiodic, is...

open_problem | open | MathDB 页面 | 原文线索

V2-035230 | MATHDB-395482

Positive discrete curvature and the topology of triangulated manifolds

摘录 / EXCERPT: Let M^d be a triangulated d-manifold (simply connected, closed) with a discrete metric such that the discrete curvature (angle defect) along any $(d-2)$-simplex is positive. Gi...

open_problem | open | MathDB 页面 | 原文线索

V2-035231 | MATHDB-395566

Weak coercivity for unit-area metrics on the 2-sphere

摘录 / EXCERPT: Let M be a topological 2-sphere endowed with a Riemannian metric g of unit area, and let $ ... Equivalently, conformal factors of unit-area metrics on S^2 with a fixed first...

open_problem | open | MathDB 页面 | 原文线索

V2-035232 | MATHDB-395601

Independence of the characteristic homomorphism from the heart

摘录 / EXCERPT: Let d4d0 be the heart of a t-structure in a K-linear triangulated category $\d4d3$. If there is a characteristic homomorphism from the Hochschild cohomology of $\d4d0$ to the gra...

open_problem | open | MathDB 页面 | 原文线索

V2-035233 | MATHDB-395602

Characteristic homomorphism from the Hochschild cohomology of a heart

摘录 / EXCERPT: If A is the heart of a K-linear t-structure in a triangulated category $\mathcal{T}$, is there then already a characteristic homomorphism from the Hochschild cohomology of the...

open_problem | open | MathDB 页面 | 原文线索

V2-035234 | MATHDB-395613

Counterexample to finite generation of Hochschild cohomology modulo nilpotents

摘录 / EXCERPT: Let k be a field with $\operatorname{char} k=2$, and let E_0 be the small category with objects x,y, automorphisms of x generated by commuting involutions g,h (so...

open_problem | open | MathDB 页面 | 原文线索

V2-035235 | MATHDB-395650

Affine Yangian module structure of affine Laumon-space cohomology

摘录 / EXCERPT: Let M be the localized equivariant cohomology space of the affine Laumon spaces P_d, as defined in the report. The report conjectures that M is isomorphic to the universal Ve...

open_problem | open | MathDB 页面 | 原文线索

V2-035236 | MATHDB-395656

Affine geometric Satake correspondence for Uhlenbeck compactifications

摘录 / EXCERPT: Braverman and Finkelberg conjecture that intersection cohomology sheaves on the Uhlenbeck compactification of the framed moduli space of G_{cpt}-instantons on...

open_problem | open | MathDB 页面 | 原文线索

V2-035237 | MATHDB-395693

A topology-free proof of the cohomology coincidence for a perfect locality

摘录 / EXCERPT: Can the coincidence ... be proved without any topological consideration, where the inverse limit is taken over the contravariant functor sending an $\mathcal F$-selfcentralizing obje...

open_problem | open | MathDB 页面 | 原文线索

V2-035238 | MATHDB-395694

Generation of the graded center by Tate cohomology and almost vanishing elements

摘录 / EXCERPT: For which finite p-groups P is the graded center $Z^*(\operatorname{mod}(kP))$ generated by Tate cohomology and almost vanishing elements?

open_problem | open | MathDB 页面 | 原文线索

V2-035239 | MATHDB-395700

Generation of self-homotopy equivalences of a product

摘录 / EXCERPT: Let X and Y be pointed spaces. Define $E(X\times Y)$ as the group of based homotopy classes of based homotopy equivalences $X\times Y\to X\times Y$. Let $E_X(X\times Y)$ consis...

open_problem | open | MathDB 页面 | 原文线索

V2-035240 | MATHDB-395712

I-adic completion of homological sheaf models

摘录 / EXCERPT: For each i, the I-adic completion of $\widetilde{H}_{i,0}$ is equal to $\widehat{H}_i$.

open_problem | open | MathDB 页面 | 原文线索

V2-035241 | MATHDB-395820

Surjectivity of the bounded-to-Borel cohomology map

摘录 / EXCERPT: For semisimple Lie groups with finite center, is the natural map $H^*B_b(G,\mathbb{R})\to H^*B(G,\mathbb{R})$ onto?

open_problem | open | MathDB 页面 | 原文线索

V2-035242 | MATHDB-395821

Rosenberg’s conjecture on homotopy invariance of negative algebraic K-theory

摘录 / EXCERPT: For every integer $n\le 0$, the assignment $X\mapsto K_n(C(X))$, where X ranges over compact Hausdorff spaces and $C(X)$ is the commutative C^*-algebra of continuous complex-va...

conjecture | open | MathDB 页面 | 原文线索

V2-035243 | MATHDB-395828

Transfer theorem for cooperads up to homotopy

摘录 / EXCERPT: Theorem 2. Let C be a coaugmented cooperad in the category of chain complexes over a field k of characteristic zero. Then:

open_problem | open | MathDB 页面 | 原文线索

V2-035244 | MATHDB-395831

General differential cohomology in the ∞-topos of ∞-stacks

摘录 / EXCERPT: What is the general notion of differential cohomology in $\mathcal H$, the $(\infty,1)$-topos of ∞-stacks on Diff?

open_problem | open | MathDB 页面 | 原文线索

V2-035245 | MATHDB-395851

Galois-Suslin cohomology agrees with étale cohomology

摘录 / EXCERPT: Let k be a perfect field, let X be separated and of finite type over k, let G_k be the absolute Galois group of k, and let A be a continuous, finitely generated m-tor…

open_problem | open | MathDB 页面 | 原文线索

V2-035246 | MATHDB-395861

Rosenberg’s homotopy-invariance conjecture for negative K-theory

摘录 / EXCERPT: For every integer $n<0$, the functor from compact Hausdorff spaces and continuous maps to abelian groups, … where $C(X)$ is the algebra of continuous functions $X\to\mathbb C$, is…

conjecture | open | MathDB 页面 | 原文线索

V2-035247 | MATHDB-395863

Comparison of cohomological maps for SK_1 and SK_2 of division algebras

摘录 / EXCERPT: Compare the three Galois-cohomology maps constructed by the methods described in the report: maps of the form…

open_problem | open | MathDB 页面 | 原文线索

V2-035248 | MATHDB-395964

Relative Connes-Chern character with boundary information

摘录 / EXCERPT: Find explicit representations for the Connes-Chern character of the fundamental relative K-homology class $[D]\in K_\bullet(M,\partial M)$ that allow geometric information about th…

open_problem | open | MathDB 页面 | 原文线索

V2-035249 | MATHDB-395972

Topological T-duality for families of orbifold quotients of tori

摘录 / EXCERPT: Describe topological T-duality for families of orbifold quotients of tori.

open_problem | open | MathDB 页面 | 原文线索

V2-035250 | MATHDB-395991

Nash problem for surface singularities is topological

摘录 / EXCERPT: The report presents the result that the Nash Conjecture for normal surface singularities is topological, depending only on the resolution graph.

open_problem | open | MathDB 页面 | 原文线索

V2-035251 | MATHDB-396005

Homological Mirror Symmetry for Cusp Singularities

摘录 / EXCERPT: For a signature $A=(\alpha_1,\alpha_2,\alpha_3)$ of positive integers, set $X_A=\mathbb{P}^1_{\alpha_1,\alpha_2,\alpha_3}$ and, for $q\in\mathbb{C}^*$, set … Writing…

open_problem | open | MathDB 页面 | 原文线索

V2-035252 | MATHDB-396008

Keum-Naie surfaces and their moduli component

摘录 / EXCERPT: Theorem. Let S be a smooth complex projective surface which is homotopically equivalent to a Keum-Naie surface. Then S is a Keum-Naie surface. The connected component of the Gi…

open_problem | open | MathDB 页面 | 原文线索

V2-035253 | MATHDB-396017

BGG correspondence and the cohomology of compact Kähler manifolds

摘录 / EXCERPT: The report’s Theorem 2 states that the graded module $Q_X=H^*(X,\comega_X)$, together with its structure as a module over $E=\cLambda^*H^1(X,\cmathcal O_X)$, determines the project…

open_problem | open | MathDB 页面 | 原文线索

V2-035254 | MATHDB-396030

Finite Topology Conjecture I (Hoffman-Meeks)

摘录 / EXCERPT: Let $\mathcal C$ be the space of connected, complete, embedded minimal surfaces in $\mathbb{R}^3$. An orientable surface M of finite topology with genus g and k ends, where…

conjecture | open | MathDB 页面 | 原文线索

V2-035255 | MATHDB-396031

Finite Topology Conjecture II (Meeks-Rosenberg)

摘录 / EXCERPT: For every non-negative integer g, there exists a unique non-planar $M\in\mathcal{C}$ with genus g and one end.

conjecture | open | MathDB 页面 | 原文线索

V2-035256 | MATHDB-396043

Properness of complete embedded minimal surfaces of finite topology

摘录 / EXCERPT: Let Σ be a complete, minimally embedded surface in $\mathbb{R}^3$ of finite topology (i.e. diffeomorphic to a finitely punctured compact surface). Then Σ is properly…

open_problem | open | MathDB 页面 | 原文线索

V2-035257 | MATHDB-396070

Constructive foundations recovering classical and algorithmic results

摘录 / EXCERPT: To find new basic definitions allowing both tackling the problems of constructive homological algebra and recovering essential algorithmic results.

open_problem | open | MathDB 页面 | 原文线索

V2-035258 | MATHDB-396144

Twisting an Open Knotted Elastic Rod

摘录 / EXCERPT: Find the equilibria of an infinite, inextensible, unshearable elastic rod of prescribed trefoil or cinquefoil centerline topology, with clamped boundary conditions at both endpoint…

open_problem | open | MathDB 页面 | 原文线索

V2-035259 | MATHDB-396152

Higher homotopy groups of quotients by type-definable equivalence relations

摘录 / EXCERPT: Let X be a definable set and E a type-definable equivalence relation of bounded index such that X/E has the logic topology. Assume the strong forms (A1) and (A2) of the repor...

open_problem | open | MathDB 页面 | 原文线索

V2-035260 | MATHDB-396250

Asymptotic approximation in single-slip crystal plasticity with linear hardening

摘录 / EXCERPT: Let $p=2$ and let ... with the strong L^1 topology. For $\varepsilon>0$, define ... where $s,m\in S^1$ and $s\perp m$, and define $E_\varepsilon:X\to\mathbb R$ and $E:X\to\mathbb R$...

open_problem | open | MathDB 页面 | 原文线索

V2-035261 | MATHDB-396301

Tilting modules and perverse cohomology of parity sheaves under geometric Satake

摘录 / EXCERPT: In the setup of the report, let G be a simple, simply connected reductive group over k, let $\Lambda=X(T)^+$ be the dominant characters, and let $\mathcal E(\lambda)$ denote th...

open_problem | open | MathDB 页面 | 原文线索

V2-035262 | MATHDB-396322

Conjecture 1 on cohomological dimension of elementary amenable groups

摘录 / EXCERPT: Fix a commutative ring k. For a group G, define ... Then, for every elementary amenable group G, G has finite cohomological dimension over k if and only if: 1. G has fini...

conjecture | open | MathDB 页面 | 原文线索

V2-035263 | MATHDB-396323

Conjecture 5 on elementary amenable groups of cohomological dimension 2

摘录 / EXCERPT: Let G be an elementary amenable group. Then $\operatorname{cd}_k(G)=2$ if and only if G admits an action on a 2-dimensional contractible complex with stabilizers finite of or...

conjecture | open | MathDB 页面 | 原文线索

V2-035264 | MATHDB-396329

Bux-Wortman conjecture on low-dimensional fillings of S-arithmetic groups

摘录 / EXCERPT: Let K be a global field (number field or function field), let S be a finite set of inequivalent valuations of K containing one representative from each class of archimedean v...

conjecture | open | MathDB 页面 | 原文线索

V2-035265 | MATHDB-396353

Area minimizers in the two remaining nonzero mod-2 homology classes of the Tetra Cosm

摘录 / EXCERPT: In the Tetra Cosm, the two remaining nonzero homology classes (with $\[?m\mathbb{Z}_2\[0m$ coefficients) that do not admit embedded planar representatives are conjectured to have a...

open_problem | open | MathDB 页面 | 原文线索

V2-035266 | MATHDB-396395

Distribution of Koszul cohomology groups for a K3 surface

摘录 / EXCERPT: Let S be a K3 surface and let L be a globally generated line bundle on S. Let c denote the (constant) Clifford index of smooth irreducible curves associated with L. Suppo...

open_problem | open | MathDB 页面 | 原文线索

V2-035267 | MATHDB-396409

Topological information carried by analytic torsion on spaces with conical and edge singularities

摘录 / EXCERPT: What topological information does the analytic torsion carry in the presence of conical and edge singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-035268 | MATHDB-396432

Quantum-homological invariance of a monotone elementary Lagrangian cobordism

摘录 / EXCERPT: If $(V;L,L')$ is a monotone elementary Lagrangian cobordism, then, for an appropriately defined relative quantum homology, $QH(V,L)=0=QH(V,L')$. In particular, there is a ring isom...

open_problem | open | MathDB 页面 | 原文线索

V2-035269 | MATHDB-396433

Homological rigidity for Lagrangian cobordisms

摘录 / EXCERPT: Assume $(V;L,L')$ is a Lagrangian cobordism with L and L' connected. If $QH(L)\neq 0$, then the inclusion-induced maps $i:H_1(L;\mathbb Z_2)\to H_1(V;\mathbb Z_2)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-035270 | MATHDB-396484

Sharpness of the cohomological vanishing bound for symmetric groups

摘录 / EXCERPT: Does there exist a $\mathfrak{a}_n$-module M whose cohomology vanishes for $(n-1)^2$ consecutive degrees, but not for $(n-1)^2+1$ consecutive degrees?

open_problem | open | MathDB 页面 | 原文线索

V2-035271 | MATHDB-396485

First nonzero cohomology degree of a Young module

摘录 / EXCERPT: Given a partition $\lambda\vdash d$, find the first integer $i>0$ such that $H^i(\Sigma_d,Y^\lambda)\neq 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035272 | MATHDB-396492

Yoneda representatives for skyline-basis cohomology classes

摘录 / EXCERPT: Find Yoneda representatives of the cohomology classes represented by skyline diagrams.

open_problem | open | MathDB 页面 | 原文线索

V2-035273 | MATHDB-396494

Bockstein splitting conjecture

摘录 / EXCERPT: There is an additive basis for the two-local cohomology of symmetric groups consisting of skyline basis classes which are integral cycles and (higher) Bocksteins on skyline basis c...

conjecture | open | MathDB 页面 | 原文线索

V2-035274 | MATHDB-396508

Higher Mahler measure as a Massey product in Deligne cohomology

摘录 / EXCERPT: Give an interpretation of the higher Mahler measures $m_k(P)$ in terms of evaluations of Massey products in Deligne cohomology, analogous to Deninger’s interpretation of the classi...

open_problem | open | MathDB 页面 | 原文线索

V2-035275 | MATHDB-396511

Analogues of the Hida-Mazur representation and adjoint p-adic L-function in knot theory

摘录 / EXCERPT: Can we produce analogues of ρ_p and $L_p(T)$ in knot theory?

open_problem | open | MathDB 页面 | 原文线索

V2-035276 | MATHDB-396618

Isomorphism between Heegaard Floer homology and embedded contact homology

摘录 / EXCERPT: Let M be a closed, connected, oriented 3-manifold. There is an isomorphism ... where $\widehat{HF}$ and $\widehat{ECH}$ denote the hat versions of Heegaard Floer homology and emb...

open_problem | open | MathDB 页面 | 原文线索

V2-035277 | MATHDB-396623

Stable homology of diffeomorphism groups of highly connected manifolds

摘录 / EXCERPT: Let $W_g:=\mathop{\#}_{g}(S^n\times S^n)$, with $n\ne 2$. Let $\theta:BO(2n)[n+1,\infty)\to BO(2n)$ be the n-connected cover, and let $MT\theta$ be the Thom spectrum of the virtu...

open_problem | open | MathDB 页面 | 原文线索

V2-035278 | MATHDB-396624

Algebraic independence of generalized Miller-Morita-Mumford classes

摘录 / EXCERPT: Let M be a closed oriented smooth n-manifold, let $\operatorname{Diff}^{+}(M)$ be the topological group of orientation-preserving diffeomorphisms, and let $f\colon E\to B$ be a...

open_problem | open | MathDB 页面 | 原文线索

V2-035279 | MATHDB-396625

Finiteness of the homotopy p-rank of symmetry

摘录 / EXCERPT: For any finite simply connected CW-complex X, $\operatorname{h\text{-}rk}_p(X)<\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-035280 | MATHDB-396626

Algebraic models for rational torus-equivariant stable homotopy theories

摘录 / EXCERPT: For any torus T^n, there is an abelian category $\mathcal A(T^n)$ of sheaves over the space of closed subgroups of T^n and a Quillen equivalence of model categories ... Furthermo...

open_problem | open | MathDB 页面 | 原文线索

V2-035281 | MATHDB-396627

Local freeness of complemented quasi-bundles over a monoid-group algebra

摘录 / EXCERPT: Let X be a compact Hausdorff space and let $R:=\mathbb{C}[\mathbb{Z}^n\times\mathbb{N}^m]$ be equipped with the fine locally convex topology. An R-quasi-bundle is a continuous...

open_problem | open | MathDB 页面 | 原文线索

V2-035282 | MATHDB-396628

Homotopy invariance of negative algebraic K-theory of algebras of continuous functions

摘录 / EXCERPT: Rosenberg conjectured that the assignment $X\mapsto K_{-n}(C(X))$ is homotopy invariant as a functor on the category of compact Hausdorff topological spaces, where $C(X)$ is the ri...

open_problem | open | MathDB 页面 | 原文线索

V2-035283 | MATHDB-396631

Operadic L_∞-structure for deformations of arbitrary algebra diagrams

摘录 / EXCERPT: For a completely general diagram D of algebras over a fixed operad, understand GerstenhaberSchack cohomology from the operadic point of view well enough to equip the correspondin...

open_problem | open | MathDB 页面 | 原文线索

V2-035284 | MATHDB-396649

Walker’s conjecture for planar polygon spaces

摘录 / EXCERPT: Walker conjectured that the graded isomorphism type of the integral cohomology ring classifies the configuration spaces M_ℓ of planar polygonal linkages in the general case.

conjecture | open | MathDB 页面 | 原文线索

V2-035285 | MATHDB-396650

Conjectured bounds for higher symmetrized topological complexity

摘录 / EXCERPT: For every path-connected space X and every integer $n\ge 2$, one has ...

conjecture | open | MathDB 页面 | 原文线索

V2-035286 | MATHDB-396651

Higher symmetric topological complexity conjecture

摘录 / EXCERPT: For an ENR space X and $n\ge 2$, the report conjectures ...

conjecture | open | MathDB 页面 | 原文线索

V2-035287 | MATHDB-396653

Topological complexity of Euclidean configuration spaces

摘录 / EXCERPT: For integers $n,m\ge 2$, the topological complexity of the configuration space $F_n(\mathbb{R}^m)$ is ...

open_problem | open | MathDB 页面 | 原文线索

V2-035288 | MATHDB-396657

Topological complexity of odd-torsion low-torsion lens spaces

摘录 / EXCERPT: For a low-torsion lens space $L^{2n+1}(m)=S^{2n+1}/\mathbb Z_m$-that is, m divides $\binom{2n}{n}$-does the upper bound $\mathrm{TC}(L^{2n+1}(m))\le 4n$ hold when m is odd?

open_problem | open | MathDB 页面 | 原文线索

V2-035289 | MATHDB-396658

Complex-oriented cohomology of the two-point configuration space of real projective space

摘录 / EXCERPT: For a complex-oriented cohomology theory h^*-with the universal case MU suggested as sufficient-what can be said about the map ... where P_n is real projective n-space and...

open_problem | open | MathDB 页面 | 原文线索

V2-035290 | MATHDB-396659

Yuzvinsky’s conjecture for the topological complexity of hyperplane-arrangement complements

摘录 / EXCERPT: Let $\mathcal A\subset \mathbb C^\ell$ be a hyperplane arrangement. Yuzvinsky’s conjecture states that ... where $A^*(\mathcal A)$ is the Orlik-Solomon algebra of $\mathcal A$.

conjecture | open | MathDB 页面 | 原文线索

V2-035291 | MATHDB-396661

Topological complexity of hyperplane-arrangement complements

摘录 / EXCERPT: Let $A\subset \mathbb{C}^{\ell}$ be a hyperplane arrangement. Determine $\operatorname{TC}(M(A))$.

open_problem | open | MathDB 页面 | 原文线索

V2-035292 | MATHDB-396727

Local compactness of the distinguished character space

摘录 / EXCERPT: Is the subset $X_0\subseteq X$ corresponding to the characters in $\operatorname{Rep}_0(\mathcal L,\mathbb C)$ locally compact with respect to the relative topology inherited from...

open_problem | open | MathDB 页面 | 原文线索

V2-035293 | MATHDB-396735

Homotopy groups of rapidly decreasing mapping groups

摘录 / EXCERPT: Let m be a positive integer and let H be a Lie group modelled on a locally convex space. For every $k=1,2,\ldots$, the homotopy groups satisfy ...

open_problem | open | MathDB 页面 | 原文线索

V2-035294 | MATHDB-396753

The general-degree-\ cohomology of the Takhtajan\Zograf metric

摘录 / EXCERPT: Determine the degree- cohomology groups $H^*_{(2)}(M_{g,n},\omega_{\mathrm{TZ}})$ for general (g,n), where the moduli-space context in the report assumes $2g-2+n>0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035295 | MATHDB-396759

Characterization of homologically fibered knots

摘录 / EXCERPT: For a knot $K\subset S^3$ with a minimal-genus Seifert surface R of genus $g=g(K)$, its complementary sutured manifold becomes a homology cylinder after fixing a pair of boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-035296 | MATHDB-396760

Structure and topological meaning of mapping-class-group Johnson cokernels

摘录 / EXCERPT: Determine the structure of the cokernels $h_g(k)/\tau_k(M_{g,1}[k+1])$ and their topological meaning.

open_problem | open | MathDB 页面 | 原文线索

V2-035297 | MATHDB-396763

Homology of moduli spaces and unstable classes

摘录 / EXCERPT: Compute homology of moduli spaces $M_{g,n}$. More precisely, find unstable classes.

open_problem | open | MathDB 页面 | 原文线索

V2-035298 | MATHDB-396765

Cohomology theory of the stabilized arc-operad loop spectrum

摘录 / EXCERPT: What is the cohomology theory corresponding to the loop spectrum constructed by stabilizing the arc operad? Answer by calculating the homotopy groups of this spectrum.

open_problem | open | MathDB 页面 | 原文线索

V2-035299 | MATHDB-396768

Sullivan space and moduli-space compactifications

摘录 / EXCERPT: Is the Sullivan space homotopic to a moduli space with some compactification? True for $g=0,n,m=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035300 | MATHDB-396857

Weak selections and weak orderability

摘录 / EXCERPT: Is every topological space which admits a weak selection weakly orderable? Here a weak selection on a topological space X is a continuous function $ ... $ to X such that $ ... $ fo...

open_problem | open | MathDB 页面 | 原文线索

V2-035301 | MATHDB-396875

Existence of an aperiodic planar monotile

摘录 / EXCERPT: Does there exist a single prototile, usually meant to be a topological disk, whose copies tile the Euclidean plane without gaps or overlaps but admit no tiling invariant under any...

open_problem | open | MathDB 页面 | 原文线索

V2-035302 | MATHDB-396896

Topological equivalence of the BV and second-order TGV norms

摘录 / EXCERPT: Theorem 8. Let $\bOmega\subset\mathbb{R}^d$ be a bounded Lipschitz domain. Then there exist constants $0<c<C<\infty$ such that for each $u\in \mathrm{BGV}^2_\alpha(\Omega)$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-035303 | MATHDB-396915

Torsion in homology groups of independence complexes of triangle-free graphs

摘录 / EXCERPT: Is it possible to have torsion in the homology groups of the independence complex of a triangle-free graph?

open_problem | open | MathDB 页面 | 原文线索

V2-035304 | MATHDB-396922

Algebraic realization of the \b3-vector of a flag homology sphere

摘录 / EXCERPT: Let Δ be a flag homology sphere. Find an algebraic structure $A(\Delta)$ whose Hilbert series equals $\gamma(\Delta)$ that will prove Conjecture 4.

open_problem | open | MathDB 页面 | 原文线索

V2-035305 | MATHDB-396943

Miranda-Hamilton conjecture for quantization of higher-dimensional integrable systems with nondegenerate singularities

摘录 / EXCERPT: For a $2n$-dimensional compact integrable system whose momentum map has only nondegenerate singularities, ... Moreover, the cohomology $H^n(M;J)$ has contributions of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-035306 | MATHDB-397004

Grove’s finiteness question for rational homotopy types

摘录 / EXCERPT: Let $\mathcal M_{\le D}^{-1\le \sec\le +1}(n)$ denote the class of simply connected n-manifolds with diameter at most D and sectional curvature satisfying $-1\le \sec\le +1$. D...

open_problem | open | MathDB 页面 | 原文线索

V2-035307 | MATHDB-397007

Direct computation of dual string-topology operations from a Sullivan model

摘录 / EXCERPT: Calculate the following dual string-topology operations directly from the Sullivan model $\mathcal M_M$ of a closed oriented smooth manifold M of dimension m: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-035308 | MATHDB-397010

Topological obstructions to a symplectic manifold being Kählerian

摘录 / EXCERPT: There are two topological obstructions for a closed symplectic manifold not to be Kählerian: the Hard Lefschetz property and formality. Moreover, it is known that neither property...

open_problem | open | MathDB 页面 | 原文线索

V2-035309 | MATHDB-397015

Uniform bound for holomorphic unipotent factorizations

摘录 / EXCERPT: For every finite-dimensional reduced Stein space X of dimension m and every null-homotopic holomorphic mapping $f:X\to\mathrm{SL}_n(\mathbb{C})$, does there exist a natural num...

open_problem | open | MathDB 页面 | 原文线索

V2-035310 | MATHDB-397016

Holomorphic factorization of mappings into $SL_m(\mathbb{C})$

摘录 / EXCERPT: Let X be a finite-dimensional reduced Stein space and let $f:X\to SL_m(\mathbb{C})$ be a null-homotopic holomorphic mapping. Then there exist $K\in\mathbb{N}$ and holomorphic map...

open_problem | open | MathDB 页面 | 原文线索

V2-035311 | MATHDB-397036

Homology class of the non-invariance hypersurface for higher-genus real plane curves

摘录 / EXCERPT: For $g>0$, investigate the hypersurface ... in particular determining its homology class.

open_problem | open | MathDB 页面 | 原文线索

V2-035312 | MATHDB-397092

Classification of translation surfaces with homology not generated by periodic orbits

摘录 / EXCERPT: Describe the translation surfaces whose homology is not generated by the periodic orbits of their geodesic flow.

open_problem | open | MathDB 页面 | 原文线索

V2-035313 | MATHDB-397120

Homological stability for diffeomorphism groups of connected sums

摘录 / EXCERPT: For $M_g^{2d}=\#^g(S^d\times S^d)$ with an embedded disk $D^{2d}\subset M_g^{2d}$, does there exist a stability range of dimensions in which the stabilization homomorphism ... is an...

open_problem | open | MathDB 页面 | 原文线索

V2-035314 | MATHDB-397125

Hypothesis B on smooth maps to the plane

摘录 / EXCERPT: For all even integers $n\ge 4$, the map ∇ is rationally nullhomotopic with derived $O(2)$-invariance.

conjecture | open | MathDB 页面 | 原文线索

V2-035315 | MATHDB-397127

THH localization sequence for the Adams summand

摘录 / EXCERPT: The report establishes the conjectured topological Hochschild homology localization sequence ...

open_problem | open | MathDB 页面 | 原文线索

V2-035316 | MATHDB-397133

Reducibility of the Spivak normal fibre space

摘录 / EXCERPT: Let Γ be a virtual Poincaré duality group with periodic Farrell cohomology. Writing $k=\operatorname{cd}\Gamma$, let X be a finite Poincaré complex of dimension $n+k$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-035317 | MATHDB-397152

Almost-g homologically independent short loops on closed Riemannian surfaces

摘录 / EXCERPT: Let $\eta:\mathbb{N}\to\mathbb{N}$ satisfy ... Then there exists a constant C_λ such that every closed Riemannian surface M of genus g contains at least $\eta(g)$ homolog...

open_problem | open | MathDB 页面 | 原文线索

V2-035318 | MATHDB-397158

Analogue of Zucker’s conjecture for the moduli space of curves

摘录 / EXCERPT: Does the analogue of Zucker’s conjecture hold for the Siegel metric on $\mathcal M_g$: is the L^2-cohomology of $\mathcal M_g$, equipped with the pullback of the locally symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-035319 | MATHDB-397174

Finite generation of Galois cohomology for families of (φ,Γ)-modules

摘录 / EXCERPT: The Galois cohomology of families of (φ,Γ)-modules is finitely generated.

open_problem | open | MathDB 页面 | 原文线索

V2-035320 | MATHDB-397182

Cohomology of non-basic Rapoport-Zink spaces

摘录 / EXCERPT: Let X be a p-divisible group over $\bar{\mathbb F}_p$ with a decomposition $X=X_1\times\cdots\times X_r$ into isoclinic groups, with Newton slopes $\nu_i>\nu_{i+1}$. Let M be...

open_problem | open | MathDB 页面 | 原文线索

V2-035321 | MATHDB-397204

Katok’s approximation question for zero-entropy conservative systems

摘录 / EXCERPT: In low dimensions (2 for maps, 3 for flows), are all conservative dynamical systems with zero topological entropy a limit of integrable systems?

open_problem | open | MathDB 页面 | 原文线索

V2-035322 | MATHDB-397313

Algorithmic undecidability of topological d-representability

摘录 / EXCERPT: For every fixed integer $d\ge 5$, it is algorithmically undecidable, given a finite simplicial complex K abstractly as a collection of faces, whether K is topologically d-rep...

open_problem | open | MathDB 页面 | 原文线索

V2-035323 | MATHDB-397350

General index problem in K-homology

摘录 / EXCERPT: Let X be a finite CW complex. Given an element $\xi\in KK^0(C(X),\mathbb C)$, explicitly construct a K-cycle (M,E,φ) on X such that $\eta(M,E,\varphi)=\xi$. Here M is...

open_problem | open | MathDB 页面 | 原文线索

V2-035324 | MATHDB-397357

Invariant second cohomology of the dual of a Drinfeld-Jimbo quantum group

摘录 / EXCERPT: Let G be a simply connected semisimple compact Lie group, let G_q be its Drinfeld-Jimbo deformation for $q>0$, and let P and Q be the weight and root lattices of the Lie al...

open_problem | open | MathDB 页面 | 原文线索

V2-035325 | MATHDB-397361

Bivariant local cyclic cohomology of reduced group C^*-algebras of word-hyperbolic groups

摘录 / EXCERPT: For a word-hyperbolic group Γ, the report states the calculation ... where $\mathbb{C}\Gamma_{\mathrm{fin}}$ is the linear span of the torsion elements of Γ.

open_problem | open | MathDB 页面 | 原文线索

V2-035326 | MATHDB-397366

The Segal conjecture for topological Hochschild homology of complex cobordism

摘录 / EXCERPT: The map $\widehat{\Gamma}_1:\mathrm{THH}(\mathrm{MU})\longrightarrow \mathrm{THH}(\mathrm{MU})^{tC_p}$ is a p-adic equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-035327 | MATHDB-397367

Operads, modules and Goodwillie towers

摘录 / EXCERPT: Let $\mathcal C$ and $\mathcal D$ each be either the category of pointed spaces or the category of spectra, and let $F:\mathcal C\to\mathcal D$ be a pointed homotopy functor. What...

open_problem | open | MathDB 页面 | 原文线索

V2-035328 | MATHDB-397457

Realization of finitely presented rational Poincaré duality groups

摘录 / EXCERPT: Given a finitely presented group Γ satisfying $\mathbb Q$-Poincaré duality, is there a closed $\mathbb Q$-homology manifold M with $\pi_1M=\Gamma$ and $\widetilde M$...

open_problem | open | MathDB 页面 | 原文线索

V2-035329 | MATHDB-397458

Davis’s existence question for Poincaré duality groups

摘录 / EXCERPT: Is every torsion-free finitely presented group satisfying R-Poincaré duality the fundamental group of an aspherical closed R-homology n-manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-035330 | MATHDB-397459

Obstruction from torsion in uniform lattices

摘录 / EXCERPT: Let Γ be a uniform lattice in a semisimple Lie group containing p-torsion, for $p\ne 2$. Then there does not exist an ANR $\mathbb{Q}$-homology manifold X having...

open_problem | open | MathDB 页面 | 原文线索

V2-035331 | MATHDB-397468

Topological homology \(\widehat A\)-class for Witt spaces

摘录 / EXCERPT: Let X be a Witt space. Under which hypothesis can one define, topologically, the homology $\widehat A$-roof class $\widehat A_*(X)$? A sufficient condition should be the existenc...

open_problem | open | MathDB 页面 | 原文线索

V2-035332 | MATHDB-397469

Riemannian stratifolds with positive-scalar-curvature links as a model for connective KO-homology

摘录 / EXCERPT: Conjectural answer: Consider Riemannian stratifolds with all strata spin and all links with positive scalar curvature. Conjecture: This gives a homology theory that is equivalent t...

open_problem | open | MathDB 页面 | 原文线索

V2-035333 | MATHDB-397480

4-periodic rational cohomology with nonvanishing odd Betti numbers in dimensions congruent to 2 modulo 4

摘录 / EXCERPT: For an integer $n\equiv 2\pmod 4$ with $n\ge 10$, does there exist a closed, simply connected n-manifold M whose rational cohomology is 4-periodic and has a nonvanishing odd...

open_problem | open | MathDB 页面 | 原文线索

V2-035334 | MATHDB-397584

Pairwise disjointness of parabolic Deligne-Lusztig cohomology modules

摘录 / EXCERPT: Let G be a connected quasi-simple reductive group over the algebraic closure of a finite field of characteristic p, and let F be an isogeny such that G^F is finite. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-035335 | MATHDB-397613

Boundedness of continuous cohomology classes for simple Lie groups

摘录 / EXCERPT: For every connected simple Lie group G with finite centre, is the natural comparison map ... surjective? Equivalently, is every class in $H^\bullet_{\mathrm{c}}(G;\mathbb{R})$ repr...

open_problem | open | MathDB 页面 | 原文线索

V2-035336 | MATHDB-397678

Weak Lefschetz property for homology spheres

摘录 / EXCERPT: Every homology sphere has the weak Lefschetz property (WLP).

open_problem | open | MathDB 页面 | 原文线索

V2-035337 | MATHDB-397679

Topological or PL-ball property of the reconstructed complex

摘录 / EXCERPT: Let Δ be a homology $(d-1)$-sphere having the weak Lefschetz property over a field k of characteristic 0, with h-vector $(h_0,h_1,\ldots,h_d)$. Let $1\le r\le d/2$ be an...

open_problem | open | MathDB 页面 | 原文线索

V2-035338 | MATHDB-397735

Threshold conjecture for 1-equivariant wave maps in the zero topological class

摘录 / EXCERPT: The conjecture is that scattering to zero should hold for data in H_0 with energy up to $2E(Q)$, where Q is the 1-equivariant harmonic map.

conjecture | open | MathDB 页面 | 原文线索

V2-035339 | MATHDB-397740

Dynamics near the degree-one Schrödinger-map soliton

摘录 / EXCERPT: For Schrödinger-map solutions arising from the restricted class of homotopy-one maps $(r,\theta) \mapsto (\phi(r),\theta+\alpha(r))$, with $ \phi(0)=0$ and $ \phi(\infty)=\pi$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-035340 | MATHDB-397759

Nontrivial basepoint dependence of \f413HF in Heegaard Floer homology

摘录 / EXCERPT: The report conjectures that moving the basepoint can produce non-trivial automorphisms of the based Heegaard Floer group $\widehat{HF}(Y,p)$; equivalently, that $\widehat{HF}(Y,p)$...

open_problem | open | MathDB 页面 | 原文线索

V2-035341 | MATHDB-397767

Vanishing contact homology versus overtwistedness

摘录 / EXCERPT: Every closed contact 3-manifold has vanishing contact homology if and only if it is overtwisted.

open_problem | open | MathDB 页面 | 原文线索

V2-035342 | MATHDB-397771

Infinite-rank subgroup in the quotient of topologically slice knots

摘录 / EXCERPT: The quotient C_T^S/C_Δ contains a subgroup isomorphic to $\mathbb{Z}^{\infty}$, where C_T^S is the concordance group of topologically slice knots and...

open_problem | open | MathDB 页面 | 原文线索

V2-035343 | MATHDB-397806

Universal polynomial for curves with prescribed isolated singularities

摘录 / EXCERPT: For every collection of isolated analytic or topological singularity types $\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_{l(\alpha)})$, there exists a universal polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-035344 | MATHDB-397892

Extension of Thurston’s Lipschitz theorem to negatively curved surfaces

摘录 / EXCERPT: Let M_1 and M_2 be compact surfaces with negative curvature, and let $f:M_1\to M_2$ be a homotopy equivalence. If ... where the supremum is over all closed geodesics γ in...

open_problem | open | MathDB 页面 | 原文线索

V2-035345 | MATHDB-397986

Hirota equations for the non-perturbative topological-recursion partition function

摘录 / EXCERPT: Conjecture 1. The non-perturbative partition function $Z_{\hbar}$ satisfies formally Hirota equations with respect to an infinite number of deformation parameters of the spectral c...

open_problem | open | MathDB 页面 | 原文线索

V2-035346 | MATHDB-397987

Corrected topological-recursion formula for colored Jones asymptotics

摘录 / EXCERPT: Keeping the preceding notation, there exists a choice of characteristics-probably among the even half-integer characteristics-such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-035347 | MATHDB-397997

Homology torsion growth and hyperbolic volume

摘录 / EXCERPT: Let X be an irreducible, orientable 3-manifold whose boundary is either empty or a union of tori. Let $\pi=\pi_1(X)$, and for each finite-index normal subgroup...

open_problem | open | MathDB 页面 | 原文线索

V2-035348 | MATHDB-398000

Homological growth and L^2-torsion for aspherical closed manifolds

摘录 / EXCERPT: Let M be an aspherical closed manifold of dimension d, let $G=\pi_1(M)$, and let $\{G_i\}_{i\in I}$ be an inverse system of normal subgroups of G with finite index and trivia...

open_problem | open | MathDB 页面 | 原文线索

V2-035349 | MATHDB-398006

Grothendieck-Riemann-Roch in Deligne-Beĭlinson cohomology for general proper holomorphic morphisms

摘录 / EXCERPT: The Grothendieck-Riemann-Roch theorem in Deligne-Beĭlinson cohomology remains an open question for general proper holomorphic morphisms between complex manifolds.

open_problem | open | MathDB 页面 | 原文线索

V2-035350 | MATHDB-398024

Characterization of response matrices for graphs on surfaces

摘录 / EXCERPT: The question of characterizing which matrices L occur as a function of the topology of Σ remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-035351 | MATHDB-398038

Arone-Turchin model for spaces of long embeddings

摘录 / EXCERPT: For all positive integers m,n with $n>m+2$, there is a weak homotopy equivalence ... where B_m and B_n are the operads of little m- and n-balls, respectively, and...

open_problem | open | MathDB 页面 | 原文线索

V2-035352 | MATHDB-398046

Homological mirror symmetry compatibility for complete intersections in toric stacks

摘录 / EXCERPT: Generalize the known compatibility between homological mirror symmetry for projective space and for its Calabi-Yau hypersurface to, for example, complete intersections in toric sta...

open_problem | open | MathDB 页面 | 原文线索

V2-035353 | MATHDB-398067

Sharpness of the syzygy-order bound for equivariant cohomology

摘录 / EXCERPT: Let $T=(S^1)^r$ act on a rational Poincaré duality space X, and let $R=H^*(BT)$. The report states that if $H_T^*(X)$ is not free over R, then its syzygy order is at most...

open_problem | open | MathDB 页面 | 原文线索

V2-035354 | MATHDB-398076

Problem 8: Cohomology of open wonderful models of hyperplane arrangements

摘录 / EXCERPT: Let $\mathcal A$ be an arrangement of hyperplanes in $\mathbb P^\ell$. Let $\pi:X\to\mathbb P^\ell$ be the De Concini-Procesi wonderful model of $\mathcal A$ corresponding to the m...

open_problem | open | MathDB 页面 | 原文线索

V2-035355 | MATHDB-398101

Mean topological dimension and von Neumann-Lück dimension

摘录 / EXCERPT: For any countable amenable group Γ and any countable left $\mathbb{Z}\Gamma$-module M, the mean topological dimension of the induced algebraic action...

open_problem | open | MathDB 页面 | 原文线索

V2-035356 | MATHDB-398102

Metric realization of mean topological dimension

摘录 / EXCERPT: For every continuous action of a countable amenable group Gamma on a compact metrizable space X, is ... where the infimum is over all compatible metrics ρ on X?

open_problem | open | MathDB 页面 | 原文线索

V2-035357 | MATHDB-398239

Complex or algebraic structures on Tian’s topological compactifications

摘录 / EXCERPT: Do Tian’s topological compactifications of moduli spaces of slope-semistable vector bundles admit complex or even algebraic structures?

open_problem | open | MathDB 页面 | 原文线索

V2-035358 | MATHDB-398254

Stabilization of cohomology groups of moduli spaces of curves with symmetries

摘录 / EXCERPT: The cohomology groups of the moduli spaces $M_{g,G,\rho}$ stabilize for $g' \gg 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035359 | MATHDB-398324

Continuity of invariants under Benjamini-Schramm convergence

摘录 / EXCERPT: Which invariants are continuous in the Benjamini-Schramm topology?

open_problem | open | MathDB 页面 | 原文线索

V2-035360 | MATHDB-398369

Linear edge bound for fixed-[U+007F]t[U+007F]-quasi-planar topological graphs

摘录 / EXCERPT: For every integer $t\ge 2$, there is a constant c_t such that every t-quasi-planar topological graph on n vertices has at most $c_t n$ edges, where a topological graph is dra...

open_problem | open | MathDB 页面 | 原文线索

V2-035361 | MATHDB-398424

Proof of the Eilenberg-MacLane correspondence via the Eilenberg-Zilber vector field

摘录 / EXCERPT: Complete the not-yet-finished proof of the Eilenberg-MacLane conjecture (1953) concerning the correspondence between the topological and algebraic notions of classifying spaces, fo...

open_problem | open | MathDB 页面 | 原文线索

V2-035362 | MATHDB-398463

Cohomological-dimension conjecture for soluble groups

摘录 / EXCERPT: If G is a soluble group with $\\operatorname{cd}_k(G)<\infty$, then $|G|\leq\aleph_n$ for some $n<\omega$, and ...

conjecture | open | MathDB 页面 | 原文线索

V2-035363 | MATHDB-398481

Local-global injectivity for Galois cohomology via patching

摘录 / EXCERPT: With the notation above, the local-global map ... is injective for all integers m such that $\operatorname{char}(k)\nmid m$ and for all $n>1$. Here $F_1=\prod_{P\in\mathcal P}F_P$...

open_problem | open | MathDB 页面 | 原文线索

V2-035364 | MATHDB-398511

Rational homotopy-algebraic K-theory isomorphism conjecture with trace-class coefficients

摘录 / EXCERPT: Let G be a group, let $\\mathrm{Fin}$ be the family of finite subgroups of G, let $E(G,\mathrm{Fin})$ be the corresponding classifying space, and let $\mathcal{L}^{1}$ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-035365 | MATHDB-398517

Conjectural structure of the unstable \[mathbb A^1\]-homotopy sheaf of a punctured affine space

摘录 / EXCERPT: Let k be a perfect field of characteristic different from 2. For every integer $n\ge 4$, there is an exact sequence of sheaves ... The sequence becomes short exact after n-fold...

open_problem | open | MathDB 页面 | 原文线索

V2-035366 | MATHDB-398595

Inverse σ_k problem when the cone condition fails

摘录 / EXCERPT: Let (M,ω) be a compact Kähler manifold, let χ be a Kähler form with cohomology class $[\chi]$, and let $1\le k\le n$. For $k\ne n$, define ... while...

open_problem | open | MathDB 页面 | 原文线索

V2-035367 | MATHDB-398596

Existence of a solution to the inverse σ_k equation

摘录 / EXCERPT: Let (M,ω) be a compact Kähler manifold, let χ be another Kähler metric, and let $[\chi]$ be its cohomology class. For $1\le k\le n$, define ... Does there exist a metric...

open_problem | open | MathDB 页面 | 原文线索

V2-035368 | MATHDB-398603

Positive solutions of the supercritical Lane-Emden equation on topologically nontrivial domains

摘录 / EXCERPT: Is it true that, for every smoothly bounded domain $\Omega\subset\mathbb{R}^n$ with $n\ge 3$ and nontrivial topology-in particular, for domains sufficiently close to an annulus-the...

open_problem | open | MathDB 页面 | 原文线索

V2-035369 | MATHDB-398626

Entropy and the Fuglede-Kadison determinant for principal algebraic actions

摘录 / EXCERPT: Let Γ be a countable discrete amenable group and let f be a non-zero-divisor in $\mathbb{Z}\Gamma$. Then the topological entropy of the natural algebraic action of...

open_problem | open | MathDB 页面 | 原文线索

V2-035370 | MATHDB-398740

Numerical analysis of a direct Galerkin discretization of the curl eigenvalue problem

摘录 / EXCERPT: For a domain Ω topologically equivalent to a ball $(\beta=0)$, analyze the direct Galerkin discretization of the self-adjoint curl eigenvalue problem with closed traces: fin...

open_problem | open | MathDB 页面 | 原文线索

V2-035371 | MATHDB-398816

Automatic continuity for the isometry group of the Urysohn space

摘录 / EXCERPT: Does the group of isometries of the Urysohn space $\mathbb U$, equipped with the pointwise-convergence topology, have the automatic continuity property; that is, is every homomorph...

open_problem | open | MathDB 页面 | 原文线索

V2-035372 | MATHDB-398826

Tempered graded affine Hecke algebra modules with solvable central character

摘录 / EXCERPT: We conjecture that any tempered module with a solvable central character has non-zero Dirac cohomology.

open_problem | open | MathDB 页面 | 原文线索

V2-035373 | MATHDB-398828

Comparison Conjecture for Lie algebra homology

摘录 / EXCERPT: Let $\mathfrak h\subset\mathfrak g$ be a real spherical subalgebra and let V be a Harish-Chandra module with smooth completion V^∞. For the natural maps ... the Comparison C...

conjecture | open | MathDB 页面 | 原文线索

V2-035374 | MATHDB-398907

Torsion translates of cohomology jump loci for compact Kähler manifolds

摘录 / EXCERPT: Let X be a compact Kähler manifold, let $\operatorname{Char}(X)=\operatorname{Hom}(\pi_1(X),\mathbb{C}^*)$, and define ... Then every irreducible component of $\Sigma_m^k(X)$ is a...

open_problem | open | MathDB 页面 | 原文线索

V2-035375 | MATHDB-398910

Carlson-Toledo conjecture on the second cohomology of infinite Kähler groups

摘录 / EXCERPT: Is it true that every infinite Kähler group Γ has (virtually) $H^{2}(\Gamma,\mathbb{R})\neq 0$? Here a Kähler group means the fundamental group of a closed Kähler manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-035376 | MATHDB-398932

Relating the Kontsevich-Soibelman operad to Hopf-algebroid calculi

摘录 / EXCERPT: It is, at present, not obvious whether one can relate the singular homology operad of the Kontsevich-Soibelman operad, or a suitable generalisation thereof, to the pair...

open_problem | open | MathDB 页面 | 原文线索

V2-035377 | MATHDB-399083

Homological determinant conjecture for AS Gorenstein algebras

摘录 / EXCERPT: Let k be an algebraically closed field, and let A be an Artin-Schelter Gorenstein $\\mathbb{N}$-graded k-algebra. Then the homological determinant of its Nakayama automorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-035378 | MATHDB-399117

Classification of finite linear quotients that are topological manifolds

摘录 / EXCERPT: For a finite subgroup $G< O(n)$, the quotient space $\4\mathbb{R}^n/G$4 is a topological manifold if and only if ... where G_{ps} is a pseudoreflection group and each...

open_problem | open | MathDB 页面 | 原文线索

V2-035379 | MATHDB-399153

Plectic cohomology conjecture

摘录 / EXCERPT: Let ... be the respective stabilisers of the conjugacy class $[\mu]$ of the cocharacter attached to (G,X) in the groups...

conjecture | open | MathDB 页面 | 原文线索

V2-035380 | MATHDB-399177

Euclidean-type spectral multiplier theorem for arbitrary stratified groups

摘录 / EXCERPT: For an arbitrary stratified Lie group G of topological dimension d, a homogeneous sublaplacian L on G, and a Borel function $F:\mathbb{R}\to\mathbb{C}$ satisfying ... for som...

open_problem | open | MathDB 页面 | 原文线索

V2-035381 | MATHDB-399182

Sharp L^p regularity for the wave equation on Heisenberg groups

摘录 / EXCERPT: Let G be a Heisenberg group of topological dimension d, and let L be its usual sums-of-squares sub-Laplacian. The report states that, for $1<p<\infty$ and ... the wave propagat...

open_problem | open | MathDB 页面 | 原文线索

V2-035382 | MATHDB-399206

Stolz’s conjecture for positive scalar curvature and twisted spin cobordism

摘录 / EXCERPT: For a closed smooth manifold M of dimension $n>4$ whose universal cover is spin, the vanishing of the appropriate twisted connective K-homology obstruction (the twisted analogu...

conjecture | open | MathDB 页面 | 原文线索

V2-035383 | MATHDB-399217

Lens rigidity problem

摘录 / EXCERPT: The lens rigidity problem is whether the lens relation L (and possibly, ℓ) determine g (and the topology of M) up to an isometry as above.

open_problem | open | MathDB 页面 | 原文线索

V2-035384 | MATHDB-399253

Beauville-Voisin conjecture for projective hyper-Kähler manifolds

摘录 / EXCERPT: For every projective hyper-Kähler manifold X, the cycle class map from the Chow ring to cohomology is injective on the subalgebra generated by the first Chern classes of line bun...

conjecture | open | MathDB 页面 | 原文线索

V2-035385 | MATHDB-399284

Kühnel’s generalized Heawood inequality for skeletons of simplices

摘录 / EXCERPT: Conjecture 1. Let k,n,b be integers. If the k-dimensional skeleton $ ... Here homology and Betti numbers are taken with $mathbb Z2$ coefficients.

open_problem | open | MathDB 页面 | 原文线索

V2-035386 | MATHDB-399306

Three-sphere recognition

摘录 / EXCERPT: Given a three-dimensional triangulation T, decide whether the underlying topological space $M=|T|$ is homeomorphic to S^3.

open_problem | open | MathDB 页面 | 原文线索

V2-035387 | MATHDB-399348

Non-trivial contraction groups in compactly generated topologically simple t.d.l.c. groups

摘录 / EXCERPT: Does every compactly generated t.d.l.c. group that is topologically simple and non-discrete have at least one non-trivial contraction group?

open_problem | open | MathDB 页面 | 原文线索

V2-035388 | MATHDB-399350

Products of conjugacy classes in finitely generated free groups

摘录 / EXCERPT: For a finitely generated free group F_n, every product of finitely many conjugacy classes is closed in the profinite topology.

open_problem | open | MathDB 页面 | 原文线索

V2-035389 | MATHDB-399499

Description of the fiber at bad primes in the chromatic splitting conjecture

摘录 / EXCERPT: Give a plausible description of the homotopy type of ΣF_n, where $F_n\simeq F(L_{n-1}S,L_nS)$, at “bad” primes p with respect to n.

conjecture | open | MathDB 页面 | 原文线索

V2-035390 | MATHDB-399501

Rational injectivity for Whitehead groups under centralizer homology finiteness

摘录 / EXCERPT: Let G be a group. Assume that, for every finite cyclic subgroup $C\subseteq G$, the abelian groups $H_1(BZ_GC;\mathbb Z)$ and $H_2(BZ_GC;\mathbb Z)$ are finitely generated. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-035391 | MATHDB-399505

Stable geometric dimension and Mackey cohomological dimension

摘录 / EXCERPT: Let G be a discrete group. Define a stable decomposition of the sphere spectrum $\bb S$ in $\\mathrm{Ho}(\\mathrm{Sp}^G)$ to be a homotopy-colimit presentation...

open_problem | open | MathDB 页面 | 原文线索

V2-035392 | MATHDB-399627

Characterization of uniqueness of unstable Adams operations

摘录 / EXCERPT: For a p-local compact group G, can uniqueness up to homotopy of an unstable Adams operation of a given degree be completely characterized by properties of G?

open_problem | open | MathDB 页面 | 原文线索

V2-035393 | MATHDB-399637

Lie algebra structure of Hochschild cohomology for a stratifying triangular algebra

摘录 / EXCERPT: Let K be a field of characteristic 2. For the algebra ... with stratifying idempotent ... what is the precise Lie algebra structure of $\operatorname{HH}^*(B)$? Can the long exact...

open_problem | open | MathDB 页面 | 原文线索

V2-035394 | MATHDB-399641

Weak polynomiality of homology of general congruence groups

摘录 / EXCERPT: For any ring without unit I and any $d\in\mathbb{N}$, the functor $H_d(\mathrm{GL}(I))\colon \mathcal{S}(\mathbb{Z})\to\mathrm{Ab}$ is weakly polynomial of degree $\leq 2d$.

open_problem | open | MathDB 页面 | 原文线索

V2-035395 | MATHDB-399686

Helton’s theorem for group rings of cohomological dimension at most two

摘录 / EXCERPT: We conjecture that Helton’s theorem holds for the complex group rings of groups with cohomological dimension at most two.

open_problem | open | MathDB 页面 | 原文线索

V2-035396 | MATHDB-399688

Novikov conjecture on homotopy invariance of higher signatures

摘录 / EXCERPT: Let M be a closed oriented manifold with $\[1m\pi_1(M)=\Gamma$[0m, let $f:M\to B\Gamma$ be continuous, and let $[x]\in H^*(B\Gamma;\mathbb{Q})$. For every orientation-preserving...

conjecture | open | MathDB 页面 | 原文线索

V2-035397 | MATHDB-399701

Volume and homology growth of aspherical manifolds

摘录 / EXCERPT: For every $n\in\mathbb{N}$ and $V_0>0$ there is $C(n,V_0)>0$ with the following property: Let M be an n-dimensional closed connected aspherical Riemannian manifold such that ev...

open_problem | open | MathDB 页面 | 原文线索

V2-035398 | MATHDB-399709

Agol-Leininger-Margalit bounds for pseudo-Anosov stretch factors and homology

摘录 / EXCERPT: Let S_g be a closed orientable surface of genus $g\ge 2$. For a pseudo-Anosov homeomorphism $f:S_g\to S_g$, let $\kappa(f)=\dim\operatorname{Fix}(f_*)$, where f_* is the induce...

open_problem | open | MathDB 页面 | 原文线索

V2-035399 | MATHDB-399728

Generalized Hopf conjecture for the homotopy Euler characteristic

摘录 / EXCERPT: A closed manifold M^n of positive curvature satisfies ... A closed manifold M^n of non-negative curvature satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-035400 | MATHDB-399729

Homotopy Euler characteristic and positive curvature

摘录 / EXCERPT: Conjecture. A closed manifold M^n of positive curvature satisfies ... where ... Thus, the conjecture asserts that the homotopy Euler characteristic is 1 in odd dimensions and 0 i...

open_problem | open | MathDB 页面 | 原文线索

V2-035401 | MATHDB-399891

Monotonicity of optimal homotopies for curves bounding a planar triangulated graph

摘录 / EXCERPT: For two curves which bound an unweighted triangulated planar graph, is an optimal homotopy monotonic?

open_problem | open | MathDB 页面 | 原文线索

V2-035402 | MATHDB-399895

Counterexamples to the topological Tverberg conjecture in the range $d\ge 2r$

摘录 / EXCERPT: Let $r\ge 2$ and $d\ge 1$. A map $f\colon K\to\mathbb{R}^d$ is an almost r-embedding if $f(\sigma_1)\cap\cdots\cap f(\sigma_r)=\varnothing$ whenever $\sigma_1,\ldots,\sigma_r$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-035403 | MATHDB-399930

Explicit counterexample to the topological Tverberg conjecture for six faces

摘录 / EXCERPT: There exists a continuous map $f\colon \Delta^{100}\to\mathbb{R}^{19}$ such that, for any six pairwise disjoint faces $\sigma_1,\ldots,\sigma_6$ of Δ^{100}, ...

conjecture | open | MathDB 页面 | 原文线索

V2-035404 | MATHDB-399949

Homotopy equivalence of the proper part with the pseudo-join subposet

摘录 / EXCERPT: Given (P,c), the pseudo-join of a collection of atoms is the unique sink in $G(P,c)$ of the smallest face containing all of these atoms as well as the unique source in P. Is th...

open_problem | open | MathDB 页面 | 原文线索

V2-035405 | MATHDB-399950

Closed formula and generating function for cohomology weights after edge replacement

摘录 / EXCERPT: Given a graph G whose independence complex has exactly one cohomology generator and an edge e of G, consider the sequence of graphs G_k given by replacing e by a path on...

open_problem | open | MathDB 页面 | 原文线索

V2-035406 | MATHDB-400086

Vanishing of higher Hochschild-Witt homology over a field

摘录 / EXCERPT: For a k-vector space M, it is reasonable to expect that $WHH_i(k,M)=0$ for every $i\ge 1$.

open_problem | open | MathDB 页面 | 原文线索

V2-035407 | MATHDB-400095

Graded-commutativity of Hochschild cohomology of ring objects

摘录 / EXCERPT: If (R,μ,e) is a ring object in a monoidal category $(\mathcal C,\wedge,I,\alpha,\lambda,\rho)$ enriched over abelian groups, then its Hochschild cohomology ring $HH^*(R)$ is gr...

open_problem | open | MathDB 页面 | 原文线索

V2-035408 | MATHDB-400140

Computing the topological complexity of real projective spaces

摘录 / EXCERPT: Determine the topological complexity $\operatorname{TC}(P^n)$ of the real projective space P^n as a function of n.

open_problem | open | MathDB 页面 | 原文线索

V2-035409 | MATHDB-400141

Invariant topological complexity of a reflection action on a sphere

摘录 / EXCERPT: Determine the invariant topological complexity of the 4mathbb{Z}/2u0004$-action on S^n given by reflection along an $(n-1)$-dimensional hyperplane.

open_problem | open | MathDB 页面 | 原文线索

V2-035410 | MATHDB-400143

Product formula for topological complexity with a sphere

摘录 / EXCERPT: The analogue of Ganea’s conjecture for topological complexity asks whether, for every finite complex X and every $k\ge 1$, one has ...

open_problem | open | MathDB 页面 | 原文线索

V2-035411 | MATHDB-400144

Topological complexity of the planar ordered configuration space

摘录 / EXCERPT: Let $F(\mathbb{C},n)=\{(x_1,\ldots,x_n)\in\mathbb{C}^n\mid x_i\ne x_j\text{ if }i\ne j\}$. Then, for $n\ge 2$, $\operatorname{TC}(F(\mathbb{C},n))=2n-2$.

open_problem | open | MathDB 页面 | 原文线索

V2-035412 | MATHDB-400155

Attainment of topologically constrained exchange-energy minima for domain walls

摘录 / EXCERPT: For every $\theta\\in(0,\\pi/2]$ and $q\\in\\{-1,0,1\\}$, is ... attained, where Q denotes either the degree D or the winding number W? What happens for a general...

open_problem | open | MathDB 页面 | 原文线索

V2-035413 | MATHDB-400266

Nonvanishing of a tautological cohomology group

摘录 / EXCERPT: Show that $\mathrm{RH}^{4}(\mathcal{M}_{5},\mathbb{V}_{2222})\neq 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035414 | MATHDB-400325

Global Torelli-type question for smooth hypersurfaces

摘录 / EXCERPT: Let $X,X'\subset \mathbb{P}^{n+1}$ be smooth hypersurfaces of degree d. Are X and X' isomorphic if there exists a Hodge isometry ... between their primitive middle cohomologies...

open_problem | open | MathDB 页面 | 原文线索

V2-035415 | MATHDB-400363

Acyclicity under the fundamental class in Weil-Arakelov cohomology

摘录 / EXCERPT: The report conjectures the existence of Weil-Arakelov cohomology with compact support, denoted $R\Gamma_{\mathrm{ar},c}(-,A(n))$ and valued in $D^b(\mathrm{LCA})$, and asserts that...

open_problem | open | MathDB 页面 | 原文线索

V2-035416 | MATHDB-400369

Cyclic homology and rigid cohomology

摘录 / EXCERPT: Develop a version of periodic cyclic theory that reproduces Berthelot’s rigid cohomology for commutative algebras, but which is defined also for non-commutative V-algebras.

open_problem | open | MathDB 页面 | 原文线索

V2-035417 | MATHDB-400373

Analytic KH-theory and the generic fibre of continuous K-theory

摘录 / EXCERPT: Let K be a complete discretely valued field with ring of integers O_K, uniformizer $[U+007F]pi$, and residue field k of characteristic 0. Let A_0 be an algebra of topologically...

open_problem | open | MathDB 页面 | 原文线索

V2-035418 | MATHDB-400374

Explicit computation of the differential and relations in the elliptic-curve group-homology complex

摘录 / EXCERPT: Explicitly compute the differential ∂ and the relations R in the complex occurring in the exact sequence ... for an arbitrary field k, an elliptic curve E/k with fixed...

open_problem | open | MathDB 页面 | 原文线索

V2-035419 | MATHDB-400376

Bass’s homotopy-stability conjecture for commutative noetherian rings

摘录 / EXCERPT: Bass conjectured that for a commutative noetherian ring R, without assuming that R has many units, the relative homotopy groups satisfy ... The report states that this general co...

conjecture | open | MathDB 页面 | 原文线索

V2-035420 | MATHDB-400409

Stable twisted cohomology of \(\mathrm{Aut}(F_\infty)\)

摘录 / EXCERPT: The report states that, as a $\mathbb{Q}[\Sigma_q]$-module,...

open_problem | open | MathDB 页面 | 原文线索

V2-035421 | MATHDB-400410

Stable twisted cohomology of automorphism groups of free groups

摘录 / EXCERPT: For the stable automorphism group $\\operatorname{Aut}(F_\\infty)$, with $H=H_1(F_n;\\mathbb Z)$ and $H_{\\mathbb Q}=H\\otimes\\mathbb Q$, the report states that...

open_problem | open | MathDB 页面 | 原文线索

V2-035422 | MATHDB-400412

Modular cohomology classes of finite general linear groups

摘录 / EXCERPT: Let $N\ge 2$, and let n be the natural number satisfying $p^{n-1}<N\le p^n$. Then $H^*(GL_N(\mathbb{F}_{p^r});\mathbb{F}_p)$ has a nonzero element in degree $r(2p^n-2p^{n-1}-1)$....

open_problem | open | MathDB 页面 | 原文线索

V2-035423 | MATHDB-400443

Theorem 2 on overconvergent étale cohomology of a one-dimensional affinoid

摘录 / EXCERPT: Let Y be a connected one-dimensional affinoid defined over a finite extension K of $\mathbb{Q}_p$, and let $Y^{\dagger}$ be the associated overconvergent space after extending...

open_problem | open | MathDB 页面 | 原文线索

V2-035424 | MATHDB-400483

Fuglede-Kadison determinants and sofic entropy

摘录 / EXCERPT: Let $f\in M_{k,n}(\mathbb{Z}(G))$ and let G be a sofic group. Then: (1) the topological entropy of the action of G on X_f is finite if and only if f is injective as a convo...

open_problem | open | MathDB 页面 | 原文线索

V2-035425 | MATHDB-400488

Characterization of C*-simplicity by a topologically free boundary action

摘录 / EXCERPT: A discrete group is C^*-simple if and only if it has a topologically free boundary action.

open_problem | open | MathDB 页面 | 原文线索

V2-035426 | MATHDB-400517

Highest multiple-point order supporting the vanishing homology

摘录 / EXCERPT: Let $f:(\mathbb{C}^n,S)\to(\mathbb{C}^{n+1},0)$ be a germ of A_e-codimension 1 with $|S|<\infty$, and let f_t be a stable perturbation. If $\mu_I(f)=1$, then the unique nonze...

open_problem | open | MathDB 页面 | 原文线索

V2-035427 | MATHDB-400521

Equality of the pole-order and Hodge filtrations on first Milnor-fiber cohomology

摘录 / EXCERPT: Let $C:f=0$ be a reduced plane curve of degree $d\ge 3$ in $\mathbb{P}^2$, where $f\in S=\mathbb{C}[x,y,z]$ is homogeneous. Let $F=\{(x,y,z)\in\mathbb{C}^3\mid f(x,y,z)=1\}$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-035428 | MATHDB-400529

Equality between the exponential Bernstein-Sato zero locus and the multivariable cohomology-support locus

摘录 / EXCERPT: Let $F=(f_1,\dots,f_r):(\mathbb{C}^n,0)\to(\mathbb{C}^r,0)$ be a germ of a holomorphic map, put $f=\prod_i f_i$, and let $B_F\subset\mathbb{C}[s_1,\ldots,s_r]$ be the ideal of poly...

open_problem | open | MathDB 页面 | 原文线索

V2-035429 | MATHDB-400598

Serre’s conjecture II for semisimple simply connected groups

摘录 / EXCERPT: For every semisimple simply connected algebraic group G defined over a perfect field k of cohomological dimension $\operatorname{cd}(k)\le 2$, the Galois cohomology set...

conjecture | open | MathDB 页面 | 原文线索

V2-035430 | MATHDB-400609

Kernel of mod-2 cohomology for a quadratic Pfister function field

摘录 / EXCERPT: Let $Q=\langle 1,a_1\rangle\otimes\cdots\otimes\langle 1,a_m\rangle\otimes[1,b]$ be an anisotropic quadratic Pfister form over a field F of characteristic 2. Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-035431 | MATHDB-400610

Existence of a notion of K-orientable theories

摘录 / EXCERPT: Let K be a generalized cohomology theory, perhaps analogous to K_0. Does there exist a notion of K-orientable theories? Such theories should be presheaves of rings on smooth...

open_problem | open | MathDB 页面 | 原文线索

V2-035432 | MATHDB-400614

Low-dimensional mod-2 cohomology of the Janko group J_4

摘录 / EXCERPT: Compute the low-dimensional mod-2 cohomology of the Janko group J_4.

open_problem | open | MathDB 页面 | 原文线索

V2-035433 | MATHDB-400617

Explicit computation of the Borel-Serre cohomology exact sequence for arithmetic groups

摘录 / EXCERPT: Given an arithmetic group $\Gamma\subset G(\mathbb R)$, a Γ-module M finitely generated as a $\mathbb Z$-module, the symmetric space $X=G(\mathbb R)/K_\infty$, and the ass...

open_problem | open | MathDB 页面 | 原文线索

V2-035434 | MATHDB-400664

Putman’s stable cohomology conjecture for level subgroups

摘录 / EXCERPT: For level $\ell\ge 2$, the conjecture asserts that ...

conjecture | open | MathDB 页面 | 原文线索

V2-035435 | MATHDB-400665

Stability conjecture for the cohomology of level mapping class groups

摘录 / EXCERPT: For ... where $\ell\ge 2$, $\operatorname{Mod}_g(\ell)$ is the level-ℓ congruence subgroup, and $g\gg k$ is the report's notation for the stable range.

conjecture | open | MathDB 页面 | 原文线索

V2-035436 | MATHDB-400670

Number-theoretic properties of pseudo-Anosov stretch factors and mapping tori

摘录 / EXCERPT: How are the number theoretic properties of λ reflected in the topology and geometry of M_ψ?

open_problem | open | MathDB 页面 | 原文线索

V2-035437 | MATHDB-400677

The dark matter problem 1

摘录 / EXCERPT: For $g\ge 6$, find elements in the rational cohomology ring $H^*(\operatorname{Mod}_g;\mathbb{Q})$ which are not polynomials in the κ_i-classes.

open_problem | open | MathDB 页面 | 原文线索

V2-035438 | MATHDB-400680

Vanishing of high-codimension rational cohomology of mapping class groups

摘录 / EXCERPT: Conjecture 1 ([2]): $H^{(4g-5)-i}(\operatorname{Mod}_g;\mathbb{Q})=0$ for $g\gg i$.

open_problem | open | MathDB 页面 | 原文线索

V2-035439 | MATHDB-400699

Asymptotic behavior of local cohomology of thickenings of ideals

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a local ring of dimension d, let $I\subset R$ be an ideal, and let $(I_n)_{n\geq 1}$ be a decreasing sequence of ideals cofinal with $(I^n)_{n\geq 1}$. F...

open_problem | open | MathDB 页面 | 原文线索

V2-035440 | MATHDB-400703

Gromov’s question on virtual cohomological dimension

摘录 / EXCERPT: Is there a global bound for the virtual cohomological dimension of a right-angled hyperbolic Coxeter group?

open_problem | open | MathDB 页面 | 原文线索

V2-035441 | MATHDB-400746

Givental decomposition for irregular spectral curves

摘录 / EXCERPT: The relation between Givental decomposition and topological recursion generalises to irregular spectral curves, with the Airy partition function Z_A associated to regular branch...

open_problem | open | MathDB 页面 | 原文线索

V2-035442 | MATHDB-400747

Gluing local spectral curves into global curves

摘录 / EXCERPT: Can the local spectral curves produced by the stated topological-recursion procedure be glued into global curves? The report answers this affirmatively, at least in cases satisfyin...

open_problem | open | MathDB 页面 | 原文线索

V2-035443 | MATHDB-400847

Homological criteria for Ringel selfdual quasihereditary algebras

摘录 / EXCERPT: The report asks whether every connected Ringel selfdual quasihereditary algebra has a simple-preserving duality, dominant dimension at least 2, and a characteristic tilting modul...

open_problem | open | MathDB 页面 | 原文线索

V2-035444 | MATHDB-400925

Complete-minor characterization by homological dimension

摘录 / EXCERPT: Conjecture 3. For every positive integer d and every finite graph G, $K_{d+2}\prec G$ if and only if $\gamma(G)\ge d$. Here $\gamma(G)$ is the greatest integer r such that...

open_problem | open | MathDB 页面 | 原文线索

V2-035445 | MATHDB-400928

Homological Sperner-type theorem for rainbow simplices

摘录 / EXCERPT: Let K be a simplicial complex whose vertices are painted with the colors $I=\{1,\ldots,m\}$. For $S\subseteq I$, let K_S be the subcomplex of K generated by vertices whose co...

open_problem | open | MathDB 页面 | 原文线索

V2-035446 | MATHDB-400948

Homology of the nerve of a polyhedral cover

摘录 / EXCERPT: Let X be a polyhedron, let $\mathcal{F}=\{A_1,\dots,A_m\}$ be a polyhedral cover of X such that each A_i is nonempty but not necessarily connected, and let N be the nerve o...

open_problem | open | MathDB 页面 | 原文线索

V2-035447 | MATHDB-400955

Rank-function determination of semialgebraic multiparameter persistence modules

摘录 / EXCERPT: Prove that, for the fruit-fly wing application-and, more generally, for applications in which persistent homology is constructed geometrically from semialgebraic data in...

open_problem | open | MathDB 页面 | 原文线索

V2-035448 | MATHDB-400973

Computing Zariski cohomology of twists of loop torsors over Laurent polynomial rings

摘录 / EXCERPT: Let k be a field of characteristic zero, let $n\ge 1$, set $R_n=k[t_1^{\pm1},\ldots,t_n^{\pm1}]$, and let G be a linear algebraic group over k. For each loop G-torsor E o...

open_problem | open | MathDB 页面 | 原文线索

V2-035449 | MATHDB-400979

Equality of the cohomological Hall algebra with the Yangian up to central elements

摘录 / EXCERPT: The authors conjecture that $\mathrm{COHA}=Y_Q$ up to some central elements.

open_problem | open | MathDB 页面 | 原文线索

V2-035450 | MATHDB-401079

Nontrivial higher homotopy groups of the observer moduli space of positive Ricci metrics on spheres

摘录 / EXCERPT: For every $k\in\mathbb{N}$, there is an integer $N(k)$ such that, for every odd $n>N(k)$, the group $\pi_i\!\left(\mathcal{M}_{x_0}^{\mathrm{Ric}>0}(S^n),[g_0]\right)$ is non-trivi...

open_problem | open | MathDB 页面 | 原文线索

V2-035451 | MATHDB-401084

Homeomorphism of the metric parametrization in the Hölder-exponent-zero topology

摘录 / EXCERPT: For $\kappa\in\{0,1\}$, let $M_0=\mathbb C$, $M_1=S^2$, let g_κ be a complete metric of constant curvature κ, and let $D(M_\kappa)$ be the orientation-preserving diff...

open_problem | open | MathDB 页面 | 原文线索

V2-035452 | MATHDB-401085

Determine the sphere dimensions detecting rational homotopy of spaces of nonnegatively curved metrics

摘录 / EXCERPT: Given one of the manifolds U considered in the preceding result and an integer i for which that result guarantees nonzero groups for sufficiently large m, determine explicitl...

open_problem | open | MathDB 页面 | 原文线索

V2-035453 | MATHDB-401088

Homotopy equivalence of positive-scalar-curvature and Mishchenko-invertible metrics

摘录 / EXCERPT: Is the map $\operatorname{Riem}^{+}(M) \to \operatorname{Riem}^{\mathrm{inv}\infty}(M)$ a homotopy equivalence?

open_problem | open | MathDB 页面 | 原文线索

V2-035454 | MATHDB-401090

Topological symmetric spaces and affine symmetric spaces

摘录 / EXCERPT: Is every topological symmetric space that is homeomorphic to a finite-dimensional manifold necessarily an affine symmetric space?

open_problem | open | MathDB 页面 | 原文线索

V2-035455 | MATHDB-401091

The K(π,1)-question for Artin complexes

摘录 / EXCERPT: For the Artin complex $A_{S(L)}$ associated to a Coxeter system (W,S) with nerve L, is its aspherical realization $BA_{S(L)}$ aspherical, equivalently homotopy equivalent to a...

open_problem | open | MathDB 页面 | 原文线索

V2-035456 | MATHDB-401092

Action dimension of an Artin group when the top mod-2 homology of the nerve is nonzero

摘录 / EXCERPT: Suppose the $K(\pi,1)$-Question has a positive answer for the Artin complex $A_S(L)$, and let $d=\dim L$. If $H_d(L;\mathbb{Z}/2)\neq 0$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-035457 | MATHDB-401114

Measure of the non-minimal sets in rational asymptotic directions

摘录 / EXCERPT: Let g be a reversible Finsler metric on $T^2=\mathbb{R}^2/\mathbb{Z}^2$ with zero topological entropy. For a geodesic $c:\mathbb{R}\to\mathbb{R}^2$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-035458 | MATHDB-401133

Topological lower bound for Fourier coefficients

摘录 / EXCERPT: Assume $f\in C(\mathbb T)$ has n sign changes. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-035459 | MATHDB-401140

Smale conjecture for the 3-sphere

摘录 / EXCERPT: The inclusion $O(4)\hookrightarrow \operatorname{Diff}(S^3)$ is a homotopy equivalence, equivalently $\operatorname{Diff}(S^3)\simeq O(4)$.

conjecture | open | MathDB 页面 | 原文线索

V2-035460 | MATHDB-401143

Arnold-Novikov conjecture on constant-geodesic-curvature curves

摘录 / EXCERPT: It is conjectured that there exist closed embedded curves of any prescribed constant geodesic curvature on a topological two-sphere.

conjecture | open | MathDB 页面 | 原文线索

V2-035461 | MATHDB-401147

Topological invariance of the mass for ALE scalar-flat Kähler manifolds

摘录 / EXCERPT: The mass of an ALE scalar-flat Kähler manifold (M,g,J) is a topological invariant, determined entirely by the smooth manifold M, together with the first Chern class...

open_problem | open | MathDB 页面 | 原文线索

V2-035462 | MATHDB-401148

Order-two homotopy classes in spaces of metrics with invertible Dirac operator

摘录 / EXCERPT: Let M^m be a smooth closed spin manifold of dimension $m\ge 6$, and let $\mathcal{R}(M)$ be the space of Riemannian metrics on M with the C^∞-topology. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-035463 | MATHDB-401197

Even-degree concentration of triply graded knot homology

摘录 / EXCERPT: For which knots is HHH concentrated only in even degrees?

open_problem | open | MathDB 页面 | 原文线索

V2-035464 | MATHDB-401198

Homology of torus knots via Dyck paths

摘录 / EXCERPT: The Poincaré polynomial of the triply graded Khovanov-Rozansky homology HHH of the (m,n)-torus knot, up to a monomial factor, is given by ... where the sum is over (m,n)-Dyck p...

open_problem | open | MathDB 页面 | 原文线索

V2-035465 | MATHDB-401298

Spectral convergence under Chabauty-Fell convergence

摘录 / EXCERPT: For a finite alphabet A, let $I:=\{\Omega\subseteq A^{\mathbb Z^d}:\Omega\text{ is closed and invariant}\}$ with the Chabauty-Fell topology. For $w\in A^{\mathbb Z^d}$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-035466 | MATHDB-401300

Characterization of minimal mean-equicontinuous systems

摘录 / EXCERPT: A minimal topological dynamical system (X,T) is mean equicontinuous if and only if there exists an equicontinuous topological dynamical system (X',T') such that $(X,T,\bc)$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-035467 | MATHDB-401303

Positive boundary measure and topological entropy for irregular model sets

摘录 / EXCERPT: Does $|\partial W|>0$ imply positive topological entropy?

open_problem | open | MathDB 页面 | 原文线索

V2-035468 | MATHDB-401429

Computing symbolic extension entropy from internal dynamical data

摘录 / EXCERPT: For a zero-dimensional topological dynamical system (X,T), how can one compute the symbolic extension entropy $h_{\mathrm{sex}}(X,T)$ from internal properties of (X,T), without...

open_problem | open | MathDB 页面 | 原文线索

V2-035469 | MATHDB-401442

Localization-Topology Correspondence for gapped periodic magnetic Schrödinger operators

摘录 / EXCERPT: Under suitable technical hypotheses, consider a gapped periodic (magnetic) Schrödinger operator. It is always possible to construct a system $w=(w_1,\ldots,w_m)$ of composite Wanni...

open_problem | open | MathDB 页面 | 原文线索

V2-035470 | MATHDB-401459

Established properties of topological Hochschild homology

摘录 / EXCERPT: Corollary 16 states that for every stable $\b5$-category $\b6C$, $\b6THH(C)=\b6THH(\operatorname{id}_C)$ carries a canonical $\b8T$-action. For every functor...

open_problem | open | MathDB 页面 | 原文线索

V2-035471 | MATHDB-401469

Directional differentiability for elliptic quasi-variational inequalities

摘录 / EXCERPT: Let X be a locally compact topological space, countable at infinity, with Radon measure $[U+007F]\xi$, let $V\subset L^2(X;\xi)=:H$ be a Hilbert space continuously and densely embedded...

open_problem | open | MathDB 页面 | 原文线索

V2-035472 | MATHDB-401487

Geometricity of homotopy equivalences for strongly hyperbolic Artin neighbourhoods

摘录 / EXCERPT: For strongly hyperbolic Artin neighbourhoods X and Y, the retraction r of Theorem 1 is a bijection. Equivalently, all isomorphisms between their étale homotopy types in the s...

open_problem | open | MathDB 页面 | 原文线索

V2-035473 | MATHDB-401522

Weak polynomiality conjecture for the homology of IA-automorphisms

摘录 / EXCERPT: The functor $H_k(IA_\bullet):S(\mathbb{Z})\to\mathrm{Ab}$ is weak polynomial of degree $3k$.

conjecture | open | MathDB 页面 | 原文线索

V2-035474 | MATHDB-401535

Torsion-freeness of homology of configuration spaces of graphs

摘录 / EXCERPT: It is conjectured that $H_i(\operatorname{Conf}_n(G))$ is always torsion-free.

open_problem | open | MathDB 页面 | 原文线索

V2-035475 | MATHDB-401536

Finite generation for homology of configuration spaces of arbitrary graphs

摘录 / EXCERPT: Is the functor $G\mapsto H_q(\operatorname{Conf}_k(G))$ from the category of graphs and injective maps to abelian groups finitely generated for all $k,q\ge 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-035476 | MATHDB-401538

Homological stability for orthogonal-group inclusions

摘录 / EXCERPT: Let k be a maximal ordered field (for example, $k=\mathbb{R}$). For all n and i, consider the monomorphism of discrete orthogonal groups ... Does this map induce an isomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-035477 | MATHDB-401539

Does the cohomology algebra of a central toric arrangement determine its poset of layers?

摘录 / EXCERPT: Let $\mathcal A$ be a central toric arrangement in the torus T, let L be its poset of layers, and let $M(\mathcal A)$ be the complement in T. Consider the cohomology algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-035478 | MATHDB-401541

Cohomology of the moduli space of smooth quartics with a bitangent

摘录 / EXCERPT: Let $H_{4,2}$ be the moduli space of smooth plane quartic curves, and let $H_{4,2}(1)$ be the space of smooth quartics equipped with a bitangent line; the latter is a degree-28 c...

open_problem | open | MathDB 页面 | 原文线索

V2-035479 | MATHDB-401544

FI-like category for the cohomology of braid-group Milnor fibers

摘录 / EXCERPT: Is there a category C similar to FI such that $H^i(F_n,\mathbb{Q})$ becomes a finitely generated C-module?

open_problem | open | MathDB 页面 | 原文线索

V2-035480 | MATHDB-401545

Finite generation of twisted pure-braid-group homology as an FI-module

摘录 / EXCERPT: Let P_n be the pure braid groups, and let V_n be a sequence of polynomial coefficient systems for the braid groups. The collection $H_*(P_n,V_n)$ is an FI-module. Is it finitel...

open_problem | open | MathDB 页面 | 原文线索

V2-035481 | MATHDB-401546

Finite generation of twisted pure-braid-group cohomology as an FI-module

摘录 / EXCERPT: Let P_n be the pure braid groups. For a sequence $\{V_n\}$ of polynomial coefficient systems for the braid groups B_n, the collection $\{H^*(P_n,V_n)\}_n$ is an FI-module. Is t...

open_problem | open | MathDB 页面 | 原文线索

V2-035482 | MATHDB-401547

Lie algebra homology of a free Lie algebra tensored with a square-zero extension

摘录 / EXCERPT: Suppose E and V are finite-dimensional vector spaces over $\\mathbb{C}$. Define the commutative algebra $A:=\mathbb{C}\oplus E$, with all products of elements of E equal to z...

open_problem | open | MathDB 页面 | 原文线索

V2-035483 | MATHDB-401552

Convergence of discrete minimal-surface minimisers

摘录 / EXCERPT: Given a finite set $\Gamma=\{\Gamma_1,\Gamma_2,\Gamma_3,\ldots\}$ of closed embedded curves in $\mathbb{R}^m$, and surfaces of prescribed topology spanning Γ, do the discret...

open_problem | open | MathDB 页面 | 原文线索

V2-035484 | MATHDB-401566

Unramified degree-three cohomology for semisimple groups of mixed Dynkin types

摘录 / EXCERPT: Let G be a semisimple group of mixed Dynkin types (i.e., each component of the Dynkin diagram is of an arbitrary type) over an algebraically closed field. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-035485 | MATHDB-401567

Degree-3 invariants for a quotient of three spin groups of type B

摘录 / EXCERPT: Let F be an algebraically closed field, and let ... where $\mu=\langle(-1,-1,-1)\rangle\subset\mu_2^3$. Determine the degree-3 cohomological invariants of G (and its subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-035486 | MATHDB-401572

Classification of cohomological invariants of Witt classes

摘录 / EXCERPT: Any cohomological invariant α of I^n can be written as a unique combination ... with $a_d\in H^*(k,\mu_2)$, where $u^{(n)}_{nd}=e_{nd}\circ\pi^d_n$ and, in particular,...

open_problem | open | MathDB 页面 | 原文线索

V2-035487 | MATHDB-401574

Cohomological invariants of Weyl groups modulo 2

摘录 / EXCERPT: Let G be the Weyl group of a root system, and let k_0 be a field of characteristic different from 2. Write $I_G=\operatorname{Inv}_{k_0}(G)$ for the ring of mod-2 cohomolog...

open_problem | open | MathDB 页面 | 原文线索

V2-035488 | MATHDB-401591

Sequential weak* continuity of countably additive states

摘录 / EXCERPT: Given a state g on a von Neumann algebra that is countably additive on projections, prove in ZFC that for every sequence $x_n\to x$ in the weak topology, one has $g(x_n)\to g(x)$...

open_problem | open | MathDB 页面 | 原文线索

V2-035489 | MATHDB-401634

Theorem C on finite-adelic point groups and adele rings

摘录 / EXCERPT: Let K and L be number fields, and let G be a linear algebraic group over $\bb{Q}$ that is fertile for both K and L. Then there is a topological group isomorphism ... if and...

open_problem | open | MathDB 页面 | 原文线索

V2-035490 | MATHDB-401668

Infinitesimal characterization of topological manifolds

摘录 / EXCERPT: Let X be a locally compact space with an upper curvature bound in the sense of Alexandrov. Then the following are equivalent: (1) X is an n-dimensional topological manifold.…

open_problem | open | MathDB 页面 | 原文线索

V2-035491 | MATHDB-401721

Parametrization of supersingular pro-\p-Iwahori Hecke modules by the monodromy parameter

摘录 / EXCERPT: Can one parametrize the supersingular $H(1)_s$-modules appearing in the cohomology of $\hat B_s$ via the parameter t?

open_problem | open | MathDB 页面 | 原文线索

V2-035492 | MATHDB-401731

The Stable Cannon Conjecture

摘录 / EXCERPT: Let G be a hyperbolic 3-dimensional Poincaré duality group. Let N be any smooth, PL, or topological manifold, respectively, which is closed and whose dimension is $\2$. Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-035493 | MATHDB-401737

Relationship between hyperbolic geometry and Floer-homological invariants of hyperbolic 3-manifolds

摘录 / EXCERPT: For a hyperbolic three-manifold Y, is there any relationship between the topological invariants arising from the hyperbolic geometry of Y-for example, the volume, injectivity r...

open_problem | open | MathDB 页面 | 原文线索

V2-035494 | MATHDB-401741

Pairing delocalized continuous cyclic cocycles with the topological structure set

摘录 / EXCERPT: It is an open problem to compute the pairing of delocalized continuous cyclic cocycles with the topological structure set.

open_problem | open | MathDB 页面 | 原文线索

V2-035495 | MATHDB-401750

Brylinski's conjecture for Hochschild homology of transformation-groupoid convolution algebras

摘录 / EXCERPT: Let H act smoothly on a manifold M, let A be the smooth convolution algebra of the transformation groupoid $H\ltimes M$, and let $\Lambda^0(H\ltimes M)$ be its loop space. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-035496 | MATHDB-401872

Dubrovin-Zhang bi-Hamiltonian-structure conjecture

摘录 / EXCERPT: For a homogeneous cohomological field theory, the system of PDEs ... where $w^\alpha=\eta^{\alpha\mu}\frac{\partial^2F}{\partial t^1_0\,\partial t^\mu_0}$, $x=t^1_0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-035497 | MATHDB-401873

DR/DZ equivalence conjecture

摘录 / EXCERPT: The double ramification (DR) hierarchy associated to a cohomological field theory is related to the Dubrovin-Zhang system (1) by a polynomial change of variables.

conjecture | open | MathDB 页面 | 原文线索

V2-035498 | MATHDB-401874

Bi-Hamiltonian structure for the Dubrovin-Zhang hierarchy

摘录 / EXCERPT: For a homogeneous cohomological field theory in the semisimple setting in which system (1) is defined, the Dubrovin-Zhang system (1) possesses a bi-Hamiltonian structure.

open_problem | open | MathDB 页面 | 原文线索

V2-035499 | MATHDB-401886

Homology classes of saddle connections between two marked zeros

摘录 / EXCERPT: Let g,n be nonnegative integers with $2g-2+n>0$, and let $\mu=(m_1,\ldots,m_n)$ be a tuple of nonnegative integers satisfying $\sum_{i=1}^n m_i=2g-2$ and of length at least 2....

open_problem | open | MathDB 页面 | 原文线索

V2-035500 | MATHDB-401974

Topological recursion for fully simple maps after exchanging spectral-curve coordinates

摘录 / EXCERPT: Conjecture 1. For the double-trace models described in the report, $X_n^{(g)}$ continues analytically to $\mathbb{C}^n$, and is computed by the Eynard-Orantin topological recursion...

open_problem | open | MathDB 页面 | 原文线索

V2-035501 | MATHDB-402001

Tightness and moment convergence for the Lévy-Lorentz gas when $\alpha\in(0,1)$

摘录 / EXCERPT: Study of the regime of $\alpha\in(0,1)$: tightness under a suitable topology, moments convergence with respect to the quenched and annealed law.

open_problem | open | MathDB 页面 | 原文线索

V2-035502 | MATHDB-402024

Clustered Hajós conjecture for topological-minor-free graphs

摘录 / EXCERPT: For every positive integer t, the class of K_{t+1}-topological-minor-free graphs is clustered t-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-035503 | MATHDB-402046

Finite generation of cohomology for finite-dimensional Nichols algebras of diagonal type

摘录 / EXCERPT: The cohomology ring of a finite-dimensional Nichols algebra $\mathcal{B}(V)$ of diagonal type is finitely generated.

open_problem | open | MathDB 页面 | 原文线索

V2-035504 | MATHDB-402047

Finite generation of cohomology for finite-dimensional Hopf algebras

摘录 / EXCERPT: For every finite-dimensional Hopf algebra A over a field k, is the cohomology algebra $H^*(A,k)$ finitely generated as a k-algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-035505 | MATHDB-402077

Lie algebra of Picard groups of orders and Hochschild cohomology

摘录 / EXCERPT: In the setting where A is an $\mathcal{O}$-order in a separable K-algebra and $\operatorname{Pic}(A)$ is known to be an affine algebraic group over k, determine whether-and,...

open_problem | open | MathDB 页面 | 原文线索

V2-035506 | MATHDB-402154

Joyce’s multiplicity conjecture for special Lagrangian spherical objects

摘录 / EXCERPT: For a Lagrangian homology sphere L, Joyce conjectured that the quantity $w(L)=|H_1(L,\mathbb{Z})|$ is the correct multiplicity when counting special Lagrangian spherical objects.

conjecture | open | MathDB 页面 | 原文线索

V2-035507 | MATHDB-402164

Atiyah-Floer conjecture for integral homology 3-spheres

摘录 / EXCERPT: Let Y be an integral homology 3-sphere with a Heegaard splitting $Y=H_0\cup_{\Sigma_g}H_1$. Let $\chi(\Sigma_g)$ be the space of $SU(2)$-representations of $\pi_1(\Sigma_g)$ modu...

conjecture | open | MathDB 页面 | 原文线索

V2-035508 | MATHDB-402190

Formal convergence analysis of the MIPDECO trust-region algorithm

摘录 / EXCERPT: A formal convergence analysis of the algorithm based on topology optimization is an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-035509 | MATHDB-402197

Topology of a smooth proper morphism over a general log disc

摘录 / EXCERPT: Conjecture. Let $f:X\to Y=V_P$ be a smooth, proper, and exact morphism of fine saturated log analytic spaces, with v the origin of V_P. After possibly shrinking V_P around...

open_problem | open | MathDB 页面 | 原文线索

V2-035510 | MATHDB-402206

Algebraic analogue of relative topological Hochschild homology

摘录 / EXCERPT: What is the algebraic analogue of C_n-relative topological Hochschild homology?

open_problem | open | MathDB 页面 | 原文线索

V2-035511 | MATHDB-402222

Topology of open PIC1 manifolds

摘录 / EXCERPT: Conjecture 2. An open n-dimensional PIC1 Riemannian manifold, with $n \ge 3$, is diffeomorphic to $\mathbb{R}^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-035512 | MATHDB-402231

The homological Arnold conjecture for Hamiltonian homeomorphisms

摘录 / EXCERPT: The proposed extension of the homological Arnold conjecture asks whether every Hamiltonian homeomorphism of a closed, connected symplectic manifold (M,ω) must have at least...

conjecture | open | MathDB 页面 | 原文线索

V2-035513 | MATHDB-402244

Counterexamples to topology bounds in high-dimensional minimal hypersurfaces

摘录 / EXCERPT: In ambient dimensions $n+1\geq 8$, the authors conjecture that there are counterexamples to bounding the topology-measured by the total Betti number-of bounded-area minimal hypersu...

open_problem | open | MathDB 页面 | 原文线索

V2-035514 | MATHDB-402264

Stratified homotopy hypothesis for the exit-path construction

摘录 / EXCERPT: For every poset P, let $\operatorname{Top}^{\mathrm{ex}}_{/P}\subset \operatorname{Top}_{/P}$ be the full subcategory of P-stratified topological spaces T for which...

conjecture | open | MathDB 页面 | 原文线索

V2-035515 | MATHDB-402357

Generic Lefschetz theorem for PL rational homology spheres

摘录 / EXCERPT: Theorem 1 (A., [1]). Consider a PL $(d-1)$-sphere Σ, or more generally a PL rational homology sphere of that dimension, and the associated graded commutative face ring...

open_problem | open | MathDB 页面 | 原文线索

V2-035516 | MATHDB-402526

Alternating full groups versus derived subgroups

摘录 / EXCERPT: Let a group G act minimally on a Cantor set X. Define the topological full group of the action as the group of homeomorphisms $f:X\to X$ such that, for every $x\in X$, there is...

open_problem | open | MathDB 页面 | 原文线索

V2-035517 | MATHDB-402528

Topological finite presentation of the étale fundamental group

摘录 / EXCERPT: Let X be a smooth projective curve over a finite field. Then the étale fundamental group $\d70b_1^{\mathrm{ét}}(X)$ is topologically finitely presented. Moreover, it admits a top...

open_problem | open | MathDB 页面 | 原文线索

V2-035518 | MATHDB-402584

Generalized lower-bound inequality for balanced homology manifolds

摘录 / EXCERPT: Let Δ be a $(d-1)$-dimensional balanced simplicial complex. If Δ is an F-homology manifold without boundary, where F is a field, and $d\ge 4$, then ... Equivalently...

open_problem | open | MathDB 页面 | 原文线索

V2-035519 | MATHDB-402585

Balanced generalized lower bound inequality for homology manifolds

摘录 / EXCERPT: Let Δ be a $(d-1)$-dimensional balanced simplicial complex. If Δ is an F-homology manifold without boundary and $d\ge 4$, then ... Equivalently, ... Furthermore, if...

open_problem | open | MathDB 页面 | 原文线索

V2-035520 | MATHDB-402592

Breuil-Strauch-type conjecture for coherent cohomology of the Drinfeld tower

摘录 / EXCERPT: Conjecture 1.3. (a) For all $n\ge 0$, $\mathcal O(\Sigma_n)^*$ is an admissible locally analytic representation of G. (b) $\mathcal O(\Sigma_n)^*$ always has finite length.

conjecture | open | MathDB 页面 | 原文线索

V2-035521 | MATHDB-402593

Finite length of the coherent cohomology representations of the Drinfeld tower

摘录 / EXCERPT: For all $n\ge 0$, $\mathcal O(\Sigma_n)^\ast$ always has finite length.

open_problem | open | MathDB 页面 | 原文线索

V2-035522 | MATHDB-402635

Floer-cohomological recovery of quantum cohomology

摘录 / EXCERPT: Given a self-transverse immersion $\phi:L\to X$ with L Lagrangian, the report describes constructing the Fukaya algebra $A(\phi)$ and states that the quantum cohomology of X sh...

open_problem | open | MathDB 页面 | 原文线索

V2-035523 | MATHDB-402639

Conjecture 1: symplectic cohomology and the intrinsic mirror algebra

摘录 / EXCERPT: For any maximally degenerate (positive) Calabi-Yau pair (M,D), there is a ring isomorphism $SH^0(X)\cong A$.

conjecture | open | MathDB 页面 | 原文线索

V2-035524 | MATHDB-402683

Stabilization of approximating equivalence relations

摘录 / EXCERPT: Let X be a set, let $\sigma\subseteq\tau$ be topologies on X, and let $\alpha\leq\omega_1$. Let $(\tau_\xi)_{\xi<\alpha}$ be a filtration from σ to τ:...

open_problem | open | MathDB 页面 | 原文线索

V2-035525 | MATHDB-402684

CH clause of the rigidity conjecture for corona C*-algebras

摘录 / EXCERPT: Let A be a separable nonunital C^*-algebra. Does $\\mathrm{CH}$ imply that there exist $2^{\\aleph_1}$ automorphisms of $Q(A)$ that are not topologically trivial?

conjecture | open | MathDB 页面 | 原文线索

V2-035526 | MATHDB-402700

Analytic versus periodic cyclic homology in higher dimensions

摘录 / EXCERPT: Determine for which algebras the analytic cyclic homology theory described in the report is isomorphic to periodic cyclic homology; in particular, determine to what extent the isom...

open_problem | open | MathDB 页面 | 原文线索

V2-035527 | MATHDB-402727

Cohomology of p-adic analytic spaces: a comparison theorem

摘录 / EXCERPT: Let K be a field complete for the discretely valued valuation v_p, let C be the completion of an algebraic closure of K, let $\breve C=W(k_C)[1/p]$, where k_C is the resi...

open_problem | open | MathDB 页面 | 原文线索

V2-035528 | MATHDB-402739

Rational homotopy of diffeomorphism groups of even-dimensional discs in the unresolved bands

摘录 / EXCERPT: For which values of d in the bands $\displaystyle\bigcup_{r\ge 2}[2r(n-2)-1,2rn-1]$, with $2n\ge 6$ and $d\ge 4n-9$, is ... nonzero, and what are these rational homotopy groups?

open_problem | open | MathDB 页面 | 原文线索

V2-035529 | MATHDB-402742

Boundary isolation and finite topology of branched singularities for analytic boundaries

摘录 / EXCERPT: The report conjectures that, when the prescribed boundary curve $\Gamma \subset \mathbb{R}^{2+n}$ is analytic, the branched singular points for a minimizer T are isolated and, lo...

open_problem | open | MathDB 页面 | 原文线索

V2-035530 | MATHDB-402745

Existence of free boundary minimal surfaces of every topological type in the unit 3-ball

摘录 / EXCERPT: Does the unit ball $B^3\subset\mathbb{R}^3$ contain an embedded free boundary minimal surface of every prescribed topological type?

open_problem | open | MathDB 页面 | 原文线索

V2-035531 | MATHDB-402752

Tensor product property for cohomological support maps of small quantum Borel algebras

摘录 / EXCERPT: The Negron-Pevtsova conjecture that the cohomological support maps for small quantum Borel algebras associated with all finite-dimensional complex simple Lie algebras satisfy the t...

open_problem | open | MathDB 页面 | 原文线索

V2-035532 | MATHDB-402765

Guedj’s question on invariant foliations

摘录 / EXCERPT: Let h be a bimeromorphic map of a compact Kähler manifold which is not cohomologically hyperbolic. Is it true that h preserves a foliation?

open_problem | open | MathDB 页面 | 原文线索

V2-035533 | MATHDB-402875

Kähler representatives of Hermitian-symplectic Aeppli classes

摘录 / EXCERPT: Does every Hermitian-symplectic structure on a complex manifold admit a Kähler metric in its Aeppli cohomology class?

open_problem | open | MathDB 页面 | 原文线索

V2-035534 | MATHDB-402881

Conjecture 2.5 on anti-invariant cohomology

摘录 / EXCERPT: For an almost complex structure J on a compact 4-dimensional manifold M, if $h_J^- \ge 3$, then J is integrable, where h_J^- is the real dimension of the space of closed...

conjecture | open | MathDB 页面 | 原文线索

V2-035535 | MATHDB-402883

Question 3.23 on anti-invariant cohomology of almost complex 4-manifolds

摘录 / EXCERPT: Are there compact, 4-dimensional examples of non-integrable almost complex structures J with $h^-_J \ge 2$ other than the ones arising from [3, Proposition 3.21]?

open_problem | open | MathDB 页面 | 原文线索

V2-035536 | MATHDB-402928

Metric realization of the adapted topology of rank r

摘录 / EXCERPT: For every $r\ge 0$, provide a metric on the space $S(V)$ of discrete-time adapted processes evolving in a linear space V that induces the adapted topology of rank r, τ_r.

open_problem | open | MathDB 页面 | 原文线索

V2-035537 | MATHDB-402929

Metric or semimetric inducing the adapted topology of every rank

摘录 / EXCERPT: For every $r\ge 0$, the map ... is a semimetric on $\mathcal S(V)$ that induces the adapted topology of rank r, τ_r.

open_problem | open | MathDB 页面 | 原文线索

V2-035538 | MATHDB-402969

Whether scalability is an R-homotopy invariant

摘录 / EXCERPT: Is scalability an R-homotopy invariant?

open_problem | open | MathDB 页面 | 原文线索

V2-035539 | MATHDB-402984

Nontrivial homotopy groups of the diffeomorphism group of the 4-disk

摘录 / EXCERPT: For each $k\in\{1,4,8\}$, $\pi_k\bigl(\mathrm{Diff}_{\partial}(D^4)\bigr)\ne 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-035540 | MATHDB-402991

Algorithmic recognition of special cube complexes for triangulated 3-manifolds

摘录 / EXCERPT: Find an algorithm that takes as input a triangulated (hyperbolic?) 3-manifold M and decides whether M is homotopy equivalent to a special cube complex. Equivalently, the algori...

open_problem | open | MathDB 页面 | 原文线索

V2-035541 | MATHDB-403087

Low-dimensional manifold learning for high-resolution medical images

摘录 / EXCERPT: Fit low-dimensional manifolds to real high-dimensional medical-image data using deep-learning techniques, including autoencoders and GANs, to obtain low-dimensional representations...

open_problem | open | MathDB 页面 | 原文线索

V2-035542 | MATHDB-403098

Mazur’s homotopy-discreteness conjecture for derived Galois deformation rings

摘录 / EXCERPT: For the derived Galois deformation ring $\mathcal R^\rho_\Gamma$ representing the unrestricted derived deformation functor associated with an irreducible residual representation...

conjecture | open | MathDB 页面 | 原文线索

V2-035543 | MATHDB-403153

Topological recursion for spin Hurwitz numbers

摘录 / EXCERPT: For $s\ge 1$ and $r=2s$, the spectral curve ... produces, via Eynard-Orantin topological recursion, correlators $\omega_{g,n}$ satisfying ...

open_problem | open | MathDB 页面 | 原文线索

V2-035544 | MATHDB-403210

Homotopy type of spaces of positive-scalar-curvature metrics

摘录 / EXCERPT: Conjecture 3. Let M and N be two closed d-manifolds, $d \ge 5$, with the same normal 2-type. If both M and N admit positive-scalar-curvature metrics, then ... Here two ma...

open_problem | open | MathDB 页面 | 原文线索

V2-035545 | MATHDB-403251

Equality between a topological Lagrangian and its support

摘录 / EXCERPT: Let $L\in\widehat{\mathcal L}(T^*M)$ be a topological Lagrangian, meaning that $\operatorname{supp}(L)$ is a topological n-dimensional submanifold. Must...

open_problem | open | MathDB 页面 | 原文线索

V2-035546 | MATHDB-403276

Conjectural asymptotic conicality of one-ended finite-topology self-shrinkers

摘录 / EXCERPT: All one-ended self-shrinkers of finite topology are asymptotically conical.

open_problem | open | MathDB 页面 | 原文线索

V2-035547 | MATHDB-403277

Topology of low-entropy asymptotically conical self-shrinkers in \(\mathbb{R}^4\)

摘录 / EXCERPT: Suppose $M^3\subset\mathbb{R}^4$ is a smooth 2-sided asymptotically conical self-shrinker with entropy less than Λ_1, and k ends, where Λ_k is the Colding-Minicoz...

open_problem | open | MathDB 页面 | 原文线索

V2-035548 | MATHDB-403394

Efficient cobordisms between local knots in orientable 3-manifolds

摘录 / EXCERPT: Is there an example of the phenomenon of the theorem in an orientable 3-manifold? In other words, can we use the “additional topology” of an orientable 3-manifold for finding more...

open_problem | open | MathDB 页面 | 原文线索

V2-035549 | MATHDB-403422

Residual nilpotence of the augmentation ideal: topological generation of the free orthogonal quantum group

摘录 / EXCERPT: Does there exist a discrete group Γ which is residually torsion-free nilpotent and such that $\mathcal{O}(O_N^+)$ is topologically generated by SO_N and $\widehat{\Gamma}$...

open_problem | open | MathDB 页面 | 原文线索

V2-035550 | MATHDB-403431

Lifting the singular Hochschild cohomology morphism to a $B_\$-algebra morphism

摘录 / EXCERPT: Let A be a right noetherian algebra such that $A^e=A\otimes A^{\mathrm{op}}$ is also noetherian. With...

open_problem | open | MathDB 页面 | 原文线索

V2-035551 | MATHDB-403432

Comparison of cellular models with complicial sets for $n>2$

摘录 / EXCERPT: Determine whether the cellular model of (∞,n)-categories is equivalent, in a precise homotopical sense, to the model of n-complicial sets for $n>2$.

open_problem | open | MathDB 页面 | 原文线索

V2-035552 | MATHDB-403433

String diagrams and moduli spaces of Riemann surfaces with boundary

摘录 / EXCERPT: Conjecture: The space of string diagrams SD is homotopy equivalent to the moduli space of Riemann surfaces with boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-035553 | MATHDB-403439

Floer cohomology of Milnor monodromy and contact loci

摘录 / EXCERPT: Let $f:\mathbb{C}^n\to\mathbb{C}$ be a polynomial with an isolated singularity at 0. For $d\in\mathbb{N}$, let ϕ_f^d denote the d-th iterate of the Milnor monodromy and le...

open_problem | open | MathDB 页面 | 原文线索

V2-035554 | MATHDB-403440

Correspondence between the pair-of-pants product and contact-locus cohomology

摘录 / EXCERPT: What does the pair-of-pants product on the bi-graded group $\bigoplus_d HF^*(\phi_f^d)$ correspond to in $\bigoplus_d H_c^*(\chi_d(f))$?

open_problem | open | MathDB 页面 | 原文线索

V2-035555 | MATHDB-403444

Topological determination of polar data for special surface singularities

摘录 / EXCERPT: Question 2: Can we find special classes of surfaces $(X,0)$ for which the relative positions between the families of generic hyperplane sections and polar curves are determined by...

open_problem | open | MathDB 页面 | 原文线索

V2-035556 | MATHDB-403469

Koszulness for wonderful models with flag nested-set complexes

摘录 / EXCERPT: Is the cohomology algebra of a De Concini-Procesi wonderful model Koszul for any building set such that the corresponding nested set complex is a flag complex?

open_problem | open | MathDB 页面 | 原文线索

V2-035557 | MATHDB-403470

Koszulness of the cohomology ring of the moduli space of stable genus-zero curves

摘录 / EXCERPT: The cohomology ring $H^{\bullet}(\overline{\mathcal{M}}_{0,n+1},\mathbb{Q})$ is Koszul.

open_problem | open | MathDB 页面 | 原文线索

V2-035558 | MATHDB-403491

Completeness of intersection cohomology as primitive generators

摘录 / EXCERPT: For every dimension vector $d\in\mathbb{N}^{Q_0}$ such that there exists a simple d-dimensional Π_Q-module, the authors conjecture that $\operatorname{IC}(X_d)$ is a complete...

open_problem | open | MathDB 页面 | 原文线索

V2-035559 | MATHDB-403495

Center of the small quantum group and cohomology of an affine Springer fiber

摘录 / EXCERPT: Let $\check G$ be a quasi-simple, connected, simply connected linear group, and let $u_\zeta\subset U_\zeta$ be the small and big Lusztig quantum groups at a root of unity ζ...

open_problem | open | MathDB 页面 | 原文线索

V2-035560 | MATHDB-403496

Centers of quantum-group blocks and affine Springer-fiber cohomology

摘录 / EXCERPT: Conjecture: There is a canonical isomorphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-035561 | MATHDB-403579

Homological filling functions and the word problem

摘录 / EXCERPT: Does the homological filling function FA_H detect solvability of the word problem for groups H of type FH_2?

open_problem | open | MathDB 页面 | 原文线索

V2-035562 | MATHDB-403588

Interval property for first cohomology of Banach-Lamperti representations

摘录 / EXCERPT: Given a continuous homomorphism $\sigma : G \to \operatorname{Aut}(X,[\mu]) \ltimes L^0(X,\mu;\{-1,1\})$, is the set ... an interval?

open_problem | open | MathDB 页面 | 原文线索

V2-035563 | MATHDB-403627

BPS q-series and the mod-2 refinement of WRT/CGP invariants

摘录 / EXCERPT: Conjecture 1. Assume that Y is a rational homology sphere, that is, $b_1(Y)=0$. There exists a family of q-series $\hat Z_s(Y)\in\mathbb Z[[q]]$, indexed by...

open_problem | open | MathDB 页面 | 原文线索

V2-035564 | MATHDB-403628

Habiro-ring refinement of the BPS q-series conjecture for 3-manifold invariants

摘录 / EXCERPT: Assume that Y is a rational homology sphere, equivalently $b_1(Y)=0$. There exists a family of elements ... such that, for the relevant $ \omega$ and r, ... where $C_s(Y,\omega)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-035565 | MATHDB-403629

A four-dimensional TQFT sensitive to exotic smooth structure

摘录 / EXCERPT: Construct a four-dimensional topological quantum field theory in the sense of Atiyah-Segal that is sensitive to exotic smooth structure.

open_problem | open | MathDB 页面 | 原文线索

V2-035566 | MATHDB-403697

Costello’s conjecture on fermionization and the Yangian of \(\mathfrak{gl}(1|1)\)

摘录 / EXCERPT: Prove or refute the conjecture attributed to Kevin Costello that, for the noncommutative resolved-conifold case, the partially fermionized cohomological Hall algebra/Yangian obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-035567 | MATHDB-403703

Topological realization of the diagram of quasi-invariants

摘录 / EXCERPT: Given a compact connected Lie group G with maximal torus $T\subseteq G$ and Weyl group $W=W_G(T)$, construct a diagram of spaces $X_m(G,T)$ over the poset $M(W)$, ... together with...

open_problem | open | MathDB 页面 | 原文线索

V2-035568 | MATHDB-403704

Uniqueness of the topological realization of quasi-invariants

摘录 / EXCERPT: For every compact connected Lie group G with maximal torus T and Weyl group $W=W_G(T)$, do the axioms (P1)-(P5) for the diagram of spaces $X_m(G,T)$ over $M(W)$ characterize th...

open_problem | open | MathDB 页面 | 原文线索

V2-035569 | MATHDB-403744

Fixed-point cohomology under torus symmetry

摘录 / EXCERPT: If T^d, with $d\ge 5$, acts effectively by isometries on a connected, closed, orientable, positively curved Riemannian manifold, then every component of the fixed-point set has t...

open_problem | open | MathDB 页面 | 原文线索

V2-035570 | MATHDB-403746

Decomposition of topologically PSC manifolds by codimension-\ surgeries

摘录 / EXCERPT: Can all topologically PSC n-manifolds be built out of simple topologically PSC n-manifolds by performing codimension ≥ 3 surgeries?

open_problem | open | MathDB 页面 | 原文线索

V2-035571 | MATHDB-403747

Finite-orbifold-fundamental-group decomposition of PSC 4-manifolds

摘录 / EXCERPT: Let M be a closed, oriented, topologically PSC 4-manifold. Can one obtain M from a closed, oriented, topologically PSC 4-orbifold M' with isolated singularities and the p...

open_problem | open | MathDB 页面 | 原文线索

V2-035572 | MATHDB-403759

Positive entropy and non-diagonal asymptotic pairs for expansive algebraic actions

摘录 / EXCERPT: Let ... Here h denotes the common topological/Haar-measure entropy, and ρ is a compatible metric on X.

open_problem | open | MathDB 页面 | 原文线索

V2-035573 | MATHDB-403764

Characterization of CF-Nil(k) systems

摘录 / EXCERPT: For every minimal topological dynamical system (X,T) and every $k\ge 0$, (X,T) is a CF-Nil(k) system if and only if (X,T) is $(k+1)$-cube uniquely ergodic.

open_problem | open | MathDB 页面 | 原文线索

V2-035574 | MATHDB-403765

A closed Property (T) subgroup of the circle homeomorphism group

摘录 / EXCERPT: Does there exist a closed subgroup $G\leq \operatorname{Homeo}(S^1)$, where $\operatorname{Homeo}(S^1)$ has the compact-open topology, such that G has Property (T), acts non-elem...

open_problem | open | MathDB 页面 | 原文线索

V2-035575 | MATHDB-403815

C∗-stability obstruction from nonvanishing rational cohomology

摘录 / EXCERPT: Let Γ be a countable discrete MF-group that admits a γ-element. If $H^k(\Gamma,\mathbb{Q})\neq 0$ for some $k\ge 1$, then Γ is not C*-stable. In fact, under...

open_problem | open | MathDB 页面 | 原文线索

V2-035576 | MATHDB-403857

Global distance distributions and homotopy type of metric graphs

摘录 / EXCERPT: Let X and Y be metric graphs with $L_H(X,Y)=0$. Must X and Y be homotopy equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-035577 | MATHDB-403869

Duality for pro-étale cohomology of partially proper analytic spaces over a finite extension of $\mathbb{Q}_p$

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}_p$. Let Y_K be a partially proper analytic space (or a dagger space) defined over K, smooth, connected, and of dimension d. Then:...

open_problem | open | MathDB 页面 | 原文线索

V2-035578 | MATHDB-403870

Geometric Verdier duality for pro-étale cohomology over \(\mathbb C\)

摘录 / EXCERPT: Conjecture. Let Y be a partially proper analytic space (or a dagger space) defined over $\mathbb C$, smooth, connected, and of dimension d. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-035579 | MATHDB-404037

Homological versus homotopical filling length of graphs

摘录 / EXCERPT: Given a graph G and a collection of cycles $\mathcal C$, let $\lVert\mathcal C\rVert_1=\sum_{\gamma\in\mathcal C}|\gamma|$. Given a field $\mathbb F$ and $\mathcal C$, glue corre...

open_problem | open | MathDB 页面 | 原文线索

V2-035580 | MATHDB-404075

Cohomological dimension of fields of meromorphic functions on Stein compacta

摘录 / EXCERPT: Let X be a normal Stein space of dimension n and let $K\subset X$ be a connected compact Stein subset. Then the field $M(K)$ has cohomological dimension n.

open_problem | open | MathDB 页面 | 原文线索

V2-035581 | MATHDB-404089

Bobadilla-Kollár conjecture on smoothness from locally constant integral cohomology

摘录 / EXCERPT: Let $f:X\to Y$ be a proper morphism between complex manifolds. Then f is smooth if and only if $R^i f_*\mathbb{Z}$ is a local system for all i.

conjecture | open | MathDB 页面 | 原文线索

V2-035582 | MATHDB-404121

Projectivity and rank of the heart in analytic Lubin-Tate cohomology

摘录 / EXCERPT: In the setup above, for an open subgroup $U\subseteq\Gamma_L$ and an analytic (φ_L,Γ_L)-module M, the report expects that the heart ... is projective as a $D(U,K)$-mod...

open_problem | open | MathDB 页面 | 原文线索

V2-035583 | MATHDB-404122

Coherence of the cohomology of the \b[1m\(\Psi\)-complex for analytic Lubin\b[0m\b[1m\b[0m-\b[1m\((\varphi_L,\Gamma_L)\)-modules

摘录 / EXCERPT: For an analytic Lubinb[1mb[0m-b[1m(φ_L,Γ_L)-module M over the relative Robba ring $\mathcal R_K$, do the cohomology groups of the complex...

open_problem | open | MathDB 页面 | 原文线索

V2-035584 | MATHDB-404132

Homological instability for moduli spaces of smooth 4-manifolds

摘录 / EXCERPT: Let X be a simply-connected smooth closed 4-manifold and let $\mathring{X}=X\setminus\operatorname{int}(D^4)$ be its puncture. For every $k>0$, there exists a sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-035585 | MATHDB-404145

Upper bound on the proposed Hempel distance of a trisection

摘录 / EXCERPT: Suppose that X is a smooth 4-manifold containing a non-separating 3-manifold Y, and let $\a3$ be the central surface of a trisection of X. Use the topology of Y and the...

open_problem | open | MathDB 页面 | 原文线索

V2-035586 | MATHDB-404147

CTF-detection of slopes distance one from the longitudinal slope

摘录 / EXCERPT: Suppose that M_1 is a knot manifold which is an integer homology solid torus. Then any slope of distance 1 from its longitudinal slope is CTF-detected.

open_problem | open | MathDB 页面 | 原文线索

V2-035587 | MATHDB-404164

Existence of a physical measure for smooth surface diffeomorphisms

摘录 / EXCERPT: Let f be a C^{∞} diffeomorphism of a closed surface M with positive topological entropy. If ... has positive volume, then f admits a physical measure.

open_problem | open | MathDB 页面 | 原文线索

V2-035588 | MATHDB-404174

Positive entropy for symmetric homoclinic tangles involving multi-parameter families

摘录 / EXCERPT: The question of whether symmetric homoclinic tangles involving such families of periodic orbits always yield positive topological entropy remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-035589 | MATHDB-404179

Nonuniqueness of free boundary minimal surfaces with prescribed topology and symmetry

摘录 / EXCERPT: For any sufficiently large integer g, there exist in the unit ball of Euclidean $ \mathbb{R}^3$ two distinct, properly embedded, free boundary minimal surfaces having genus g,...

open_problem | open | MathDB 页面 | 原文线索

V2-035590 | MATHDB-404198

Higher homotopy groups of the h-cobordism moduli space

摘录 / EXCERPT: For a compact smooth manifold M, can the higher homotopy groups $\pi_i\mathcal H(M)$ of the moduli space of h-cobordisms on M be computed in terms of algebraic K-theory?

open_problem | open | MathDB 页面 | 原文线索

V2-035591 | MATHDB-404250

Topological structure of the spectrum under random perturbation of the Fibonacci Hamiltonian

摘录 / EXCERPT: Let H be the Fibonacci Hamiltonian (the discrete Schrödinger operator with the potential given by the Fibonacci substitution sequence). What can one say about the topological str...

open_problem | open | MathDB 页面 | 原文线索

V2-035592 | MATHDB-404251

Transversal Hölder regularity resolves the cohomology-limit conundrum

摘录 / EXCERPT: Let Ω be the tiling space under consideration, and let $S^r_\alpha(\Omega)$ be the space of functions described on the source pages. Let $H^*_{r,\alpha}(\Omega;\mathbb{R})$...

open_problem | open | MathDB 页面 | 原文线索

V2-035593 | MATHDB-404298

Hopkins-Wickelgren formality conjecture for mod-2 Galois cohomology

摘录 / EXCERPT: The mod-2 Galois cohomology algebra of every field is formal.

conjecture | open | MathDB 页面 | 原文线索

V2-035594 | MATHDB-404311

Examples with a strict inclusion between homotopy obstruction sets

摘录 / EXCERPT: Find examples where $X(\[?]A_K)^h \[?]\subsetneq X(\[?]A_K)^{h,2}$.[?]

open_problem | open | MathDB 页面 | 原文线索

V2-035595 | MATHDB-404315

Positive-density local-global principle for Galois sections

摘录 / EXCERPT: Let X be a smooth geometrically connected curve over a number field K, and consider its étale homotopy exact sequence $1\to\Delta_X\to\Pi_X\to G_K\to 1$. Let s and t be sec...

open_problem | open | MathDB 页面 | 原文线索

V2-035596 | MATHDB-404356

Annihilation of local cohomology for symbolic powers

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be a complete Cohen-Macaulay normal domain of arbitrary characteristic, of dimension d, and let $I\subseteq R$ be an ideal of pure height h. For each...

open_problem | open | MathDB 页面 | 原文线索

V2-035597 | MATHDB-404363

Avramov’s embedded deformation problem

摘录 / EXCERPT: Let R be a noetherian local ring, with homotopy Lie algebra $\[1m\pi_*(R)$. An embedded deformation of R is an expression $\widehat R\cong S/(f)$ for some quotient S of the r...

open_problem | open | MathDB 页面 | 原文线索

V2-035598 | MATHDB-404367

Vasconcelos’ rigidity conjecture for the first Koszul homology module

摘录 / EXCERPT: Let R be a commutative noetherian ring, let $I\subseteq R$ be an ideal of finite projective dimension over R, and set $S=R/I$. Then I is locally generated by a regular sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-035599 | MATHDB-404375

Topological criterion for mutability in a cosilting pair

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field K, let (Z,I) be a cosilting pair in $\[1m\operatorname{Mod} A\u001b[0m$, and let $M\in Z$. Then: (1) every module $I\in I$...

open_problem | open | MathDB 页面 | 原文线索

V2-035600 | MATHDB-404415

Smooth reconstruction of étale Lie groupoids from Cartan-pair data

摘录 / EXCERPT: Conversely, suppose we are given an étale Lie groupoid, i.e. the topological groupoid G has a smooth-manifold structure with $G^{(0)}$ a submanifold, the maps $r,s:G\to G^{(0)}$...

open_problem | open | MathDB 页面 | 原文线索

V2-035601 | MATHDB-404469

The $T\oplus T$ recurrence problem

摘录 / EXCERPT: If T is a recurrent continuous linear operator on a topological vector space X, does it follow that $T\oplus T$ is also recurrent?

open_problem | open | MathDB 页面 | 原文线索

V2-035602 | MATHDB-404482

Keller’s conjecture for singular Hochschild cohomology

摘录 / EXCERPT: Let A be a finite-dimensional algebra over a field k, let $D=D_{\rm dg}^b(A\text{-mod})$ be the bounded dg derived category, and assume that D is smooth. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-035603 | MATHDB-404483

Grothendieck-fibration reconstruction of enhancements

摘录 / EXCERPT: Let $\mathrm{Top\text{-}Cat}$ be the category of small categories enriched in topological spaces, with the stated model structure, and let $\mathrm{Pos}\subset\mathrm{Cat}$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-035604 | MATHDB-404500

Soft topological assumptions for collapsed Ricci limits

摘录 / EXCERPT: For Ricci limits $(X,d,\mu,p)\mathop{\longleftarrow}^{\mathrm{mpGH}}(M_i^n,g_i,d\mathrm{Vol}_{g_i}/\mathrm{Vol}_{g_i}(B_1(p_i)),p_i)$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-035605 | MATHDB-404505

Minimal topological volume of the 3-torus

摘录 / EXCERPT: Let $L\subset\mathbb T^3$ be the 3-component link depicted in Figure 2 of the report, consisting of three disjoint straight segments in the cubic fundamental domain, each joining a...

open_problem | open | MathDB 页面 | 原文线索

V2-035606 | MATHDB-404530

Finite presentation of the relative tame fundamental group

摘录 / EXCERPT: Is the relative tame fundamental group $\pi_1^t(X/K)$ topologically finitely presented?

open_problem | open | MathDB 页面 | 原文线索

V2-035607 | MATHDB-404561

Finite generation of cohomology for pointed Hopf algebras with abelian group-likes

摘录 / EXCERPT: The well-known finite-generation-of-cohomology conjecture of Etingof and Ostrik was positively answered in [2] for finite-dimensional pointed Hopf algebras with abelian group of gr...

open_problem | open | MathDB 页面 | 原文线索

V2-035608 | MATHDB-404565

Homeomorphism problem for the Balmer-cohomology map

摘录 / EXCERPT: Let A be a finite-dimensional Hopf algebra over a field k, and let ... be the natural continuous map. When is the map Ψ a homeomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-035609 | MATHDB-404685

Scalar-curvature preservation under convergence without a specified topology

摘录 / EXCERPT: Suppose a sequence (M^3_j,g_j) with $\\operatorname{Scal}\\ge 0$ converges in some sense to a smooth Riemannian manifold (M^3_0,g_0). Does (M^3_0,g_0) have...

open_problem | open | MathDB 页面 | 原文线索

V2-035610 | MATHDB-404783

Classifying finite presentations up to homotopy equivalence

摘录 / EXCERPT: Given a finitely presented group G, classify its finite presentations up to homotopy equivalence, where two finite presentations P and P' are homotopy equivalent when their a...

open_problem | open | MathDB 页面 | 原文线索

V2-035611 | MATHDB-404801

Effective computation of Hilbert series of Koszul modules

摘录 / EXCERPT: Given an n-dimensional complex vector space V and a subspace $K\subseteq\bigwedge^2V$, can one effectively compute the Hilbert series of the Koszul module $W(V,K)$, the homolog...

open_problem | open | MathDB 页面 | 原文线索

V2-035612 | MATHDB-404802

Integral Hodge classes of curves on intermediate Jacobians of very general cubic threefolds

摘录 / EXCERPT: Let $Y\subset \mathbb P^4_{\mathbb C}$ be a very general cubic hypersurface. Then the homology class of every curve $C\subset J_Y$ on its intermediate Jacobian is an even multiple...

open_problem | open | MathDB 页面 | 原文线索

V2-035613 | MATHDB-404813

Quantum-cohomological computation of the braid associated with a two-ray game

摘录 / EXCERPT: Conjecture 4. The braid appearing in Conjecture 3 can be computed as the monodromy of the eigenvalues of $c_1(X)$ acting on the small quantum cohomology ring $QH^*(X,\mathbb{C})$,...

open_problem | open | MathDB 页面 | 原文线索

V2-035614 | MATHDB-404820

Finite generation of the étale fundamental group of a proper rigid space

摘录 / EXCERPT: Let X be a rigid space proper over an algebraically closed field k. Is its étale fundamental group $\pi_1^{\mathrm{ét}}(X,\bar{x})$ topologically finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-035615 | MATHDB-404896

Conjugate sectors in topologically ordered models

摘录 / EXCERPT: Under what circumstances does a conjugate sector exist? It would be interesting to have a physically relevant criterion for the case of superselection sectors of topologically orde...

open_problem | open | MathDB 页面 | 原文线索

V2-035616 | MATHDB-404907

Contractibility of the automorphism group of the hyperfinite type III$_0$ factor

摘录 / EXCERPT: Determine whether the automorphism group $\operatorname{Aut}(R_{\mathrm{III}_0})$ of the hyperfinite type III$_0$ factor with separable predual is contractible in the u-topology.

open_problem | open | MathDB 页面 | 原文线索

V2-035617 | MATHDB-404916

Duality for condensed cohomology and suave duality

摘录 / EXCERPT: Is the duality for condensed cohomology of the Weil group of a p-adic field an instance of suave duality in the Heyer-Mann six-functor formalism on condensed anima?

open_problem | open | MathDB 页面 | 原文线索

V2-035618 | MATHDB-404919

Classification of \Z-algebra spectra with polynomial homotopy

摘录 / EXCERPT: Classify the Z-algebra spectra A such that $\\pi_*A\cong \u001a\mathbb{F}_p[s]$ for $|s|\ge 1$. Equivalently, via Koszul duality, classify $\\Z$-algebra spectra with homotopy an...

open_problem | open | MathDB 页面 | 原文线索

V2-035619 | MATHDB-404920

Solid higher local class field theory via Weil groups

摘录 / EXCERPT: For a d-local field F, define its Weil group W_F and its solid Milnor K-theory $K^{\\mathrm M,\\blacksquare}_d(F)$-obtained by incorporating the topology of F and replaci...

open_problem | open | MathDB 页面 | 原文线索

V2-035620 | MATHDB-404984

Functoriality of analytic lattice homology under flat deformations

摘录 / EXCERPT: For reduced curve singularities, the analytic lattice homology is conjectured to be functorial with respect to flat deformations.

open_problem | open | MathDB 页面 | 原文线索

V2-035621 | MATHDB-405086

Periodic geodesics in each homotopy class for half-Lie groups

摘录 / EXCERPT: Let (G,G) be a half-Lie group equipped with a strong right-invariant metric G satisfying the assumptions before (e) in [5, Thm. 7.7]. Assume in addition that $\pi_1(G)\ne 0$. T...

open_problem | open | MathDB 页面 | 原文线索

V2-035622 | MATHDB-405127

Riemannian topological expansion of compact locally symmetric spaces

摘录 / EXCERPT: For a simple noncompact Lie group G of real rank $d\ge 2$ with symmetric space $X=G/K$, the family of compact locally symmetric spaces $\{\Gamma\backslash X\}_{\Gamma}$ associate...

open_problem | open | MathDB 页面 | 原文线索

V2-035623 | MATHDB-405135

Noetherianity of the remixed twisted cohomology ring

摘录 / EXCERPT: Conjecture. Let G be a finite p-group. Define $\widetilde{B}(G)=B(G)\setminus\{k[1]\}$ and $\mathbb{N}^{\widetilde{B}(G)}=\{q:\widetilde{B}(G)\to\mathbb{N}\}$. For the...

open_problem | open | MathDB 页面 | 原文线索

V2-035624 | MATHDB-405141

Diffeomorphism of thickenings of homotopy-equivalent 2-complexes

摘录 / EXCERPT: For finite 2-complexes X and Y in the thickening construction $M(X)=\partial N(X)$, if $X\simeq Y$, are $M(X)$ and $M(Y)$ diffeomorphic?

open_problem | open | MathDB 页面 | 原文线索

V2-035625 | MATHDB-405142

Artemev’s conjecture

摘录 / EXCERPT: For every integer $p\geq 1$, the resonance transformations of the $(2,2p+1)$ minimal string coincide with the transformations induced by interchanging the two coordinates,...

conjecture | solved | MathDB 页面 | 原文线索

V2-035626 | MATHDB-405152

Johnson-McClendon conjecture on continuous orbit equivalence under bounded speedups

摘录 / EXCERPT: For every free minimal $\mathbb{Z}^d$-odometer T on a Cantor space and every minimal bounded speedup S of T (in particular, every minimal constant speedup), the actions S a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035627 | MATHDB-405156

Chas’s conjecture on exact curve counting

摘录 / EXCERPT: Let T be the once-punctured torus, let $\mathcal{C}(T)$ denote the set of closed curves on T, and let $\mathcal{O}$ be any mapping class group orbit in $\mathcal{C}(T)$. For ev...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035628 | MATHDB-405174

Lipshitz-Sarkar mirror-detection question

摘录 / EXCERPT: ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035629 | MATHDB-405189

Length conjecture for twist knots

摘录 / EXCERPT: ...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-035630 | MATHDB-405234

Li-Haldane conjecture on universal entanglement spectra

摘录 / EXCERPT: Let $|\Psi\rangle$ be a ground state in a two-dimensional topological phase, let A be a spatial subsystem, and let $\rho_A=\operatorname{Tr}_{\bar A}|\Psi\rangle\langle\Psi|$ be...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-035631 | MATHDB-405264

Hahn-Wilson conjecture

摘录 / EXCERPT: For each height n, the category of spectra of fp-type n is the thick subcategory generated by the truncated Brown-Peterson spectrum $\mathrm{BP}\langle n\rangle$; equivalently,...

conjecture | partial | MathDB 页面 | 原文线索

B07 分析 / Analysis | 2,727 条

V2-035632 | MATHDB-315468

Connes’s group-factor rigidity conjecture

摘录 / EXCERPT: Conjecture 1.1 (Connes Rigidity Conjecture). If G and H are ICC property (T) groups such that $\mathrm{L}(G)\cong\mathrm{L}(H)$, then $G\cong H$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035633 | MATHDB-315478

Toroidal Elton-Odell theorem

摘录 / EXCERPT: Does every infinite-dimensional complex normed space admit a uniformly separated sequence after quotienting unit vectors by unimodular scalars?

open_problem | open | MathDB 页面 | 原文线索

V2-035634 | MATHDB-315479

A unital Banach algebra that is not a Calkin algebra

摘录 / EXCERPT: Is there a unital Banach algebra that cannot occur as the Calkin algebra of any Banach space?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035635 | MATHDB-315480

Strict cosingularity and adjoints for separable range

摘录 / EXCERPT: Suppose X and Y are separable and $T:X\to Y$ is strictly cosingular. Is T^* strictly singular?

open_problem | open | MathDB 页面 | 原文线索

V2-035636 | MATHDB-315481

Weakly compact factorization through a space with a basis

摘录 / EXCERPT: Can the DFJP factorization space always be chosen to have a Schauder basis whenever the range space has one?

open_problem | open | MathDB 页面 | 原文线索

V2-035637 | MATHDB-315482

Primariness of $L_p(L_1)$

摘录 / EXCERPT: Is $L_p(L_1)$ primary for every $1<p\leq\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-035638 | MATHDB-315503

Carbery’s almost-orthogonality inequality in L^p

摘录 / EXCERPT: For p ≥ 2, does Carbery’s proposed many-function almost-orthogonality inequality hold with the pairwise overlap coefficients raised to the power 2? If not, what is the largest poss...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035639 | MATHDB-315521

Neural Jacobian Conjecture at width $N=n+1$

摘录 / EXCERPT: The cited paper states the Neural Jacobian Conjecture as follows (Conjecture 2.3): for every $F\in\mathcal{N}_{n,N}^{+}$, the map F is globally injective, where ... For the problem...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035640 | MATHDB-315607

First Proof Problem 2 - uniform Whittaker test vector

摘录 / EXCERPT: Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$. Let N_r denote the subgroup of $\mathrm{GL}_r(F)$ consisting of upper-triangular unipotent matrices...

open_problem | open | MathDB 页面 | 原文线索

V2-035641 | MATHDB-315624

Planar strictly convex hyperrigidity

摘录 / EXCERPT: Let K be a Hilbert space, let X,Y be commuting positive contractions on K, and let $P\in B(K)$ be an orthogonal projection such that PXP and PYP commute. Let...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035642 | MATHDB-315663

KLS conjecture for quadratic forms

摘录 / EXCERPT: Does the Kannan-Lovász-Simonovits variance inequality hold with a universal constant, and in particular for every quadratic form under an isotropic log-concave measure?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-035643 | MATHDB-315669

Weissler’s missing two-point inequality

摘录 / EXCERPT: Let $1\leq p\leq q<\infty$ and $z\in\mathbb{C}$. Weissler's conjecture asserts that the complex noise operator T_z on the Hamming cube has dimension-free norm from L^p to L^q...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035644 | MATHDB-315717

Brezis’s Fourier-degree problem below $\alpha=1/3$

摘录 / EXCERPT: For a fixed $0<\alpha\leq 1/3$, let $\left(\sigma_{n,\varepsilon}\right)_{n\in\mathbb{Z},\,0<\varepsilon<1}\subset\mathbb{C}$ be a summation process satisfying ... and ... Does there e...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035645 | MATHDB-315718

Lower bound for Korenblum’s constant

摘录 / EXCERPT: How large can the universal annular threshold in Korenblum’s maximum principle for the Bergman space $A^2(\mathbb D)$ be?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-035646 | MATHDB-315730

Lukic’s weighted-entropy decomposition conjecture

摘录 / EXCERPT: Is Lukic’s weighted entropy condition for finitely many critical points equivalent to decomposing the Verblunsky coefficients into components localized at those points?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035647 | MATHDB-315733

Glöckner-Neeb multiplication-growth question

摘录 / EXCERPT: Is the Glöckner--Neeb multiplication-growth condition automatic for every Mackey-complete continuous inverse algebra?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-035648 | MATHDB-315898

Fuglede's conjecture

摘录 / EXCERPT: For $\Omega \subset \mathbb{R}^{d}$ with positive finite Lebesgue measure, say that Ω is spectral if there exists a set $\Lambda \subset \mathbb{R}^{d}$ such that ... is an or...

conjecture | open | MathDB 页面 | 原文线索

V2-035649 | MATHDB-315899

Goodman's conjecture

摘录 / EXCERPT: Let p be a positive integer and let $\mathbb{D}=\{z\in\mathbb{C}:|z|<1\}$. Call a function f holomorphic on $\mathbb{D}$ p-valent if for every $w\in\mathbb{C}$ the equation...

conjecture | open | MathDB 页面 | 原文线索

V2-035650 | MATHDB-315901

Lehmer's conjecture

摘录 / EXCERPT: For a nonzero polynomial $P(x) = a_{0}\prod_{i=1}^{D}(x-\alpha_{i}) \in \mathbb{Z}[x]$ of degree D, factored over $\mathbb{C}$ with leading coefficient a_{0} and roots...

conjecture | open | MathDB 页面 | 原文线索

V2-035651 | MATHDB-315903

Pompeiu problem

摘录 / EXCERPT: Let $n \ge 2$, and let $M(n)$ denote the group of rigid motions of $\mathbb{R}^n$, i.e. the maps $\sigma(x) = \rho x + b$ with $\rho \in O(n)$ and $b \in \mathbb{R}^n$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-035652 | MATHDB-315905

Vitushkin's conjecture

摘录 / EXCERPT: For an open set $U\subseteq\mathbf{C}$, let $\mathcal{H}^{\infty}(U)$ denote the set of bounded analytic functions $f\colon U\to\mathbf{C}$, and for f analytic near ∞ put...

conjecture | open | MathDB 页面 | 原文线索

V2-035653 | MATHDB-316156

Mizohata-Takeuchi conjecture

摘录 / EXCERPT: In harmonic analysis, a branch of mathematics, the Mizohata-Takeuchi conjecture proposed a weighted L^{2} inequality for the Fourier extension operator associated with a smooth h...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035654 | MATHDB-316157

Kadison-Singer problem

摘录 / EXCERPT: Let ℓ^2 denote the Hilbert space of square-summable sequences of complex numbers, with standard orthonormal basis $(e_n)_{n\in\mathbb{N}}$. Let B be the C-algebra of all bou...

open_problem | open | MathDB 页面 | 原文线索

V2-035655 | MATHDB-316158

Ahlfors measure conjecture

摘录 / EXCERPT: In mathematics, the Ahlfors conjecture, now a theorem, states that the limit set of a finitely generated Kleinian group is either the whole Riemann sphere, or has measure zero.

conjecture | open | MathDB 页面 | 原文线索

V2-035656 | MATHDB-316159

Gradient conjecture

摘录 / EXCERPT: Let $U \subseteq \mathbb{R}^n$ be an open set, let $f : U \to \mathbb{R}$ be real analytic, and let $x_0 \in U$ be an isolated critical point of f, i.e. $\nabla f(x_0) = 0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-035657 | MATHDB-316247

The Lipschitz Kakeya maximal-function conjecture

摘录 / EXCERPT: Lipschitz Kakeya maximal-function conjecture. For some $1<p<2$ and some finite N, for all $0<\delta<1$ and all Lipschitz vector fields v, the operator...

conjecture | open | MathDB 页面 | 原文线索

V2-035658 | MATHDB-316248

The fixed-width Lipschitz Kakeya maximal-function conjecture

摘录 / EXCERPT: Fixed-width Lipschitz Kakeya conjecture. For some $1<p<2$ and some finite N, for all $0<\delta<1$, all Lipschitz vector fields v, and...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-035659 | MATHDB-316256

Asymptotic density of equal consecutive Collatz stopping times

摘录 / EXCERPT: For $m\geq2$, define ... The equal-stopping-time density conjecture. ... This predicts that equal total stopping times for consecutive integers occur with asymptotic density one-half....

conjecture | open | MathDB 页面 | 原文线索

V2-035660 | MATHDB-316272

Polynomial realization conjecture for zero Kobayashi-Royden metric in the spectral ball

摘录 / EXCERPT: Polynomial realization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-035661 | MATHDB-316343

Whole-sequence convergence conjecture for polynomial eigenfunctions

摘录 / EXCERPT: Let T_{λ} be a pencil of general type with polynomial eigenfunctions $p_{n,j}(z)$ and eigenvalues $\lambda_{n,j}$, and let Ψ_j be the limiting functions from Theorem....

conjecture | open | MathDB 页面 | 原文线索

V2-035662 | MATHDB-316345

Connected-complement conjecture for limiting measures

摘录 / EXCERPT: Let μ_j be the limiting measure associated with the jth sequence of polynomial eigenfunctions. Connected-complement conjecture. For each j, the set ... is connected. The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-035663 | MATHDB-316390

The small ball inequality conjecture for hyperbolic Haar sums over large rectangles

摘录 / EXCERPT: Small ball inequality conjecture. For every $d\ge3$, the displayed inequality should hold. It is a central conjecture in the paper, with consequences for irregularities of distribu...

conjecture | open | MathDB 页面 | 原文线索

V2-035664 | MATHDB-316391

The smooth small ball inequality conjecture

摘录 / EXCERPT: Smooth small ball inequality conjecture. The displayed inequality should hold, with an implied constant depending only on d and φ. This is a smooth analogue of the Haar s...

conjecture | open | MathDB 页面 | 原文线索

V2-035665 | MATHDB-316392

The discrepancy lower-bound conjecture in dimensions at least three

摘录 / EXCERPT: Let $\mathcal A_N$ be an arbitrary set of N points in $[0,1]^d$, and define its discrepancy function by ... where $[0,x)=\prod_{j=1}^d[0,x_j)$. Discrepancy lower-bound conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-035666 | MATHDB-316409

The zero-set characterization conjecture for the Kobayashi-Royden pseudometric of the spectral unit ball

摘录 / EXCERPT: Let Ω_n be the spectral unit ball, let $A\in\Omega_n$, and let $B\in\mathcal M_n$. Write $k_{\Omega_n}(A;B)$ for the Kobayashi-Royden pseudometric and C_A for the cone of...

conjecture | open | MathDB 页面 | 原文线索

V2-035667 | MATHDB-316428

Connectedness suffices for convergence of Faber zero counting measures

摘录 / EXCERPT: Let G be the domain under consideration, let $ u_n$ denote the counting measure associated with the nth Faber polynomial, and let bc_L be the equilibrium measure of L. Conv...

conjecture | open | MathDB 页面 | 原文线索

V2-035668 | MATHDB-316433

Minimal-idempotent reduction conjecture for the Kadison-Singer problem

摘录 / EXCERPT: Minimal-idempotent reduction conjecture. If a minimal idempotent has a unique state extension, then every point has a unique state extension. Thus, for any fixed minimal idempotent...

conjecture | open | MathDB 页面 | 原文线索

V2-035669 | MATHDB-316435

Non-recurrent-point range conjecture for the completely positive map

摘录 / EXCERPT: Non-recurrent-point range conjecture. If $\boxed{\textstyle\text{?}}$ is non-recurrent, then

conjecture | open | MathDB 页面 | 原文线索

V2-035670 | MATHDB-316532

Generalized discrete holomorphicity conjecture for random cluster models

摘录 / EXCERPT: Let $q\in[0,4]$, and consider the random cluster model on lattice domains. For the Ising model, let $F(z)$ denote the function in Proposition, which is discrete holomorphic and sat...

conjecture | open | MathDB 页面 | 原文线索

V2-035671 | MATHDB-316574

Conformal-radius convergence-rate conjecture for the Method of Fundamental Solutions

摘录 / EXCERPT: Let $ Omega$ be a simply connected domain with analytic boundary, and let ρ_z denote the conformal radius of z with respect to $ Omega$. Place the MFS points equally spaced...

conjecture | open | MathDB 页面 | 原文线索

V2-035672 | MATHDB-316575

Conformal-radius coefficient-growth conjecture for the Method of Fundamental Solutions

摘录 / EXCERPT: Let $ Omega$ be a simply connected domain with analytic boundary, and let $ Gamma_R=\{z:\rho_z=R\}$ be the curve of points with conformal radius $R>1$. Choose the MFS charge points...

conjecture | open | MathDB 页面 | 原文线索

V2-035673 | MATHDB-316576

Jordan-curve coefficient-growth conjecture for the Method of Fundamental Solutions

摘录 / EXCERPT: Let $ Omega$ be the domain of the Helmholtz boundary-value problem, and let $ Gamma$ be any Jordan curve enclosing $\overline{\Omega}$ with positive distance from $\partial\Omega$....

conjecture | open | MathDB 页面 | 原文线索

V2-035674 | MATHDB-316609

Random square DFT submatrix norm conjecture

摘录 / EXCERPT: Let n be a positive integer, let $\mathbf{F}$ be the normalized $n\times n$ DFT matrix, and let $\mathsfsl{P}_\delta$ and $\mathsfsl{P}_\delta'$ denote the coordinate projections...

conjecture | open | MathDB 页面 | 原文线索

V2-035675 | MATHDB-316634

Mikhlin-Hörmander multiplier conjecture for the Grushin operator

摘录 / EXCERPT: Mikhlin-Hörmander multiplier conjecture. The operator $m(G)$ is bounded on L_p for $1<p<\infty$ and is of weak type $(1,1)$ whenever $s>1$.

conjecture | open | MathDB 页面 | 原文线索

V2-035676 | MATHDB-316635

Higher-dimensional Grushin wave-estimate conjecture

摘录 / EXCERPT: Higher-dimensional Grushin wave-estimate conjecture. For every ${\mathscr{C}}_1>0$, $t>0$, $1\leq p\leq\infty$, and $\alpha>(d-1)|1/p-1/2|$, there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-035677 | MATHDB-316712

The BMV conjecture in algebraic form

摘录 / EXCERPT: BMV conjecture, algebraic form. The polynomial p has only nonnegative coefficients. This is an algebraic reformulation of the Bessis-Moussa-Villani conjecture, which originated i...

conjecture | open | MathDB 页面 | 原文线索

V2-035678 | MATHDB-316717

The linear adjoint cone restriction conjecture

摘录 / EXCERPT: Linear adjoint cone restriction conjecture. The inequality above holds with constants depending on S, n and p,q if and only if $q>\frac{2n}{n-1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-035679 | MATHDB-316742

The Spectral-Tile Conjecture

摘录 / EXCERPT: Let E be a measurable subset of ${\mathbb{R}}^n$. An additive tile is a measurable set that tiles ${\mathbb{R}}^n$ by translations from a discrete subset of ${\mathbb{R}}^n$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-035680 | MATHDB-316839

The integral almost-commuting matrices conjecture

摘录 / EXCERPT: The integral almost-commuting matrices conjecture. There exist $k\in\mathbb N$ and symmetric matrices $A',B'\in M_{nk}(\mathbb Z)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-035681 | MATHDB-316840

The integer-operator conjecture for group-measure-space rings

摘录 / EXCERPT: The integer-operator conjecture for group-measure-space rings. Every normal operator in this subring is an integer operator. This is the group-measure-space analogue of the conject...

conjecture | open | MathDB 页面 | 原文线索

V2-035682 | MATHDB-316859

Finiteness conjecture for crescents in Alt minimizers

摘录 / EXCERPT: Finiteness conjecture. Any Alt minimizer M for $Gamma$ has at most $C(Gamma)$ crescents, where $C(Gamma)$ depends only on the C^4 data of $Gamma$.

conjecture | open | MathDB 页面 | 原文线索

V2-035683 | MATHDB-316868

Jung-Ko-Pearcy convergence conjecture for iterated Aluthge transforms

摘录 / EXCERPT: Let T be a complex matrix, and let $T=U|T|$ be its polar decomposition. The Aluthge transform is ... and define its iterates by $\Delta^0(T)=T$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-035684 | MATHDB-316869

Joint continuity conjecture for iterated lambda-Aluthge transforms

摘录 / EXCERPT: Let $\lambda\in(0,1)$ and let $T\in\mathcal{M}_r(\mathbb{C})$ have polar decomposition $T=U|T|$. Define the λ-Aluthge transform by ... and let $\Delta_\lambda^\infty(T)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-035685 | MATHDB-316876

The rational dilation conjecture for compact planar domains

摘录 / EXCERPT: Let $X\subseteq\mathbb{C}$ be a compact domain with boundary $B=\partial X$, let H be a Hilbert space, and let $T\in\mathcal{B}(H)$ satisfy ... Assume that ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-035686 | MATHDB-316887

Asymptotic harmonic-mean conjecture for generalized Taylor means

摘录 / EXCERPT: Let a,b be positive real numbers, let $M_{p,m_{1},m_{2}}(a,b)$ denote the means defined in the paper, and set $n=m_{1}+m_{2}+1$. The harmonic mean is $H(a,b)=\tfrac{2ab}{a+b}$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-035687 | MATHDB-316890

Mean-energy equality conjecture for Brody curves in the projective line

摘录 / EXCERPT: For a Brody curve f in $\mathbb{C} P^1$, let $e(f)$ be its mean energy, and let $e(\mathbb{C} P^1)$ be the supremum of $e(f)$ over all Brody curves in $\mathbb{C} P^1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-035688 | MATHDB-316949

Semi-invertibility conjecture for extensions of the reduced C*-algebra of the free group

摘录 / EXCERPT: Let $C^*_r(\mathbb F_2)$ be the reduced C-algebra of the free group on two generators. An extension is a short exact sequence ... where ${\mathbb K}$ denotes the compact operators. S...

conjecture | open | MathDB 页面 | 原文线索

V2-035689 | MATHDB-316955

Convergence conjecture for kinetic functions of high-order equivalent equations

摘录 / EXCERPT: Convergence conjecture. As $q\to\infty$, the kinetic function $\psi^{\flat,q}_\alpha$ converges to the exact kinetic function $\varphi^\flat_\alpha$:

conjecture | open | MathDB 页面 | 原文线索

V2-035690 | MATHDB-317000

Almost-sure multifractal spectrum conjecture for SLE

摘录 / EXCERPT: Let $\bar\beta(t)$ denote the average spectrum of SLE, and let t_{min} and t_{max} be the two points such that a tangent to $\bar\beta(t)$ intersects the y-axis at -1. The...

conjecture | open | MathDB 页面 | 原文线索

V2-035691 | MATHDB-317019

Rigidity conjecture for constant-horizontal-normal sets in Carnot groups

摘录 / EXCERPT: Let $\mathbb G$ be a Carnot group, let $E\subset\mathbb G$ be a set with constant horizontal normal $\nu\in\mathbf S^{m-1}$, and let H be a vertical halfspace with the same horiz...

conjecture | open | MathDB 页面 | 原文线索

V2-035692 | MATHDB-317028

Emamirad-Laadnani trace-norm approximation conjecture

摘录 / EXCERPT: Let $U(t)$ be the Dirichlet-to-Neumann semigroup and let $V_{\gamma,\partial\Omega}(t/n)^n$ be its approximants, acting on the boundary Hilbert space $\partial\mathcal{H}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-035693 | MATHDB-317081

Hahn-Banach extension conjecture for ultra pseudo-normed modules

摘录 / EXCERPT: Let E be an ultra pseudo-normed $\widetilde{\mathbb K}$-module, let E' denote its continuous linear functionals, and let V be a submodule of E. For $\varphi\in V'$, the ext...

conjecture | open | MathDB 页面 | 原文线索

V2-035694 | MATHDB-317123

Uniqueness conjecture for harmonic unitaries on the noncommutative torus

摘录 / EXCERPT: Uniqueness conjecture. In each connected component of $PU(A_\theta^\infty)$, the functional E has a unique minimum, given by scalar multiples of U^nV^m. These are the only harm...

conjecture | open | MathDB 页面 | 原文线索

V2-035695 | MATHDB-317245

Parabolic-point measure-zero conjecture for exceptional minimal sets

摘录 / EXCERPT: Parabolic-point measure-zero conjecture. The set $K_{\mathcal P}$ has Lebesgue measure zero. The conjecture is equivalent, in the stated C^r setting, to the general measure-zero...

conjecture | open | MathDB 页面 | 原文线索

V2-035696 | MATHDB-317281

D'Angelo's conjecture for rigid pseudoconvex domains with monomial mixed terms

摘录 / EXCERPT: Let $\mathcal O_n$ be the ring of germs of holomorphic functions at the origin in $\mathbb C^n$, and let I be the ideal generated by $f_1,\dots,f_l\in\mathcal O_n$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-035697 | MATHDB-317284

The local spectral radius formula conjecture for the operators $\widetilde T_p$

摘录 / EXCERPT: Local spectral radius formula conjecture. Theorems for these settings, when read in the equivalent local-spectral form, are local spectral radius formulas:

conjecture | open | MathDB 页面 | 原文线索

V2-035698 | MATHDB-317297

K-theory conjectures for the 2-graphs associated with two-dimensional subshifts

摘录 / EXCERPT: Let $C^*(\Lambda)$ be the C^*-algebra associated with the 2-graph Λ, let δ_2 be the boundary map used in its K-theory computation, and let q, c_1, and c_2...

conjecture | open | MathDB 页面 | 原文线索

V2-035699 | MATHDB-317357

The minimal-element area and genus growth conjecture for CMC surfaces

摘录 / EXCERPT: Let $M\subset \mathbb{R}^3$ satisfy the hypotheses of the Minimal Element Theorem, and let $\Sigma\in\mathcal{T}(M)$ be a minimal element. For $n\in\{1,2,3\}$, consider the area an...

conjecture | open | MathDB 页面 | 原文线索

V2-035700 | MATHDB-317384

Amenability conjecture for centres of compact group algebras

摘录 / EXCERPT: Let G be a compact group, and let ... be the centre of the group algebra $\mathrm{L}^1(G)$. Amenability conjecture. The algebra $\mathrm{ZL}^1(G)$ is amenable if and only if G ad...

conjecture | open | MathDB 页面 | 原文线索

V2-035701 | MATHDB-317400

Periodization conjecture for the generalized sine-kernel Fredholm determinant

摘录 / EXCERPT: Let $I+V$ be the generalized sine-kernel operator (GSK), let x be the large asymptotic parameter, and let ν denote the parameter whose integer shifts restore the determinant'...

conjecture | open | MathDB 页面 | 原文线索

V2-035702 | MATHDB-317595

The $\pi/2$ lower-bound conjecture for nonnegative autoconvolutions

摘录 / EXCERPT: Let $g\in L^1(\mathbb R)\cap L^2(\mathbb R)$ be nonnegative and supported on an interval of length I. Write $g\ast g$ for its autoconvolution and $\|\cdot\|_r$ for the L^r norm...

conjecture | open | MathDB 页面 | 原文线索

V2-035703 | MATHDB-317596

The sharp L^2 improvement conjecture for nonnegative autoconvolutions

摘录 / EXCERPT: Let $g\in L^1(\mathbb R)\cap L^2(\mathbb R)$ be nonnegative, and let $g\ast g$ denote its autoconvolution. Write $\|\cdot\|_r$ for the L^r norm. The sharp autoconvolution L^2 i...

conjecture | open | MathDB 页面 | 原文线索

V2-035704 | MATHDB-317612

The converse chordality conjecture for positive definite function extensions

摘录 / EXCERPT: Converse chordality conjecture. For every $S\subset G$ such that $\Gamma(G,S)$ is not chordal, there exists a positive definite function

conjecture | open | MathDB 页面 | 原文线索

V2-035705 | MATHDB-317653

Endpoint Christ-Kiselev estimate for the derivative nonlinear Schrödinger equation

摘录 / EXCERPT: Let $S(t)$ be the linear propagator, let $\mathscr{A}f(t)=\int_{\mathbb{R}}S(t-\tau)f(\tau)\,d\tau$ be the associated full-time integral operator, and let $\Box_k$ be the frequency...

conjecture | open | MathDB 页面 | 原文线索

V2-035706 | MATHDB-317697

Nonconvex rectifiable curve unfolding conjecture

摘录 / EXCERPT: Let $\mathbf f:\mathbb R/2\pi\longrightarrow\mathbb R^2$ be a rectifiable simple closed curve that is not convex. Nonconvex unfolding conjecture. There exists an absolutely continu...

conjecture | open | MathDB 页面 | 原文线索

V2-035707 | MATHDB-317739

The periodic tiling conjecture for regions

摘录 / EXCERPT: Periodic tiling conjecture. If a region $\Omega\subset\mathbb{R}^n$ tiles $\mathbb{R}^n$ by translations, then it has a periodic tiling.

conjecture | open | MathDB 页面 | 原文线索

V2-035708 | MATHDB-317809

Three-dimensional multilinear restriction conjecture for transverse hypersurfaces

摘录 / EXCERPT: Three-dimensional multilinear restriction conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-035709 | MATHDB-317850

Steinlein's fixed point conjecture for eventually compact maps

摘录 / EXCERPT: Let $\mathcal{E}$ be a Banach space, let $\mathscr{C}\subset \mathcal{E}$ be a nonempty closed bounded convex set, and let $f\colon \mathscr{C}\to\mathscr{C}$ be a continuous map....

conjecture | open | MathDB 页面 | 原文线索

V2-035710 | MATHDB-317855

Smoothed-analysis conjecture on random perturbations and condition numbers

摘录 / EXCERPT: Let A be an arbitrary $n \times n$ matrix, and let M_n be a random $n \times n$ matrix. For an invertible matrix M, define its condition number by ... and set...

conjecture | open | MathDB 页面 | 原文线索

V2-035711 | MATHDB-317866

Asymptotic estimate for the interaction parameter in optical-lattice condensates

摘录 / EXCERPT: Let $\widehat{\tau}$, $\widehat{g}$, and τ denote the parameters introduced in the model, and let ϵ tend to zero. Asymptotic interaction-parameter conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-035712 | MATHDB-317916

Ding's global well-posedness conjecture for the one-dimensional Schrödinger map flow

摘录 / EXCERPT: A Schrödinger map is a time-dependent map from a one-dimensional domain into a compact Kähler manifold evolving according to the Schrödinger map flow. Ding's conjecture. The Schröd...

conjecture | open | MathDB 页面 | 原文线索

V2-035713 | MATHDB-318078

Non-degeneracy conjecture for the minima of the ground-state eigenvalue

摘录 / EXCERPT: Let k be a positive integer, let $Q(\alpha,1)$ be the operator whose lowest eigenvalue is $\lambda_0(\alpha,1)$, and let $\hat{\nu}$ denote the minimum value of...

conjecture | open | MathDB 页面 | 原文线索

V2-035714 | MATHDB-318227

The p-adic Corona conjecture for the coordinate ideal

摘录 / EXCERPT: Let $C\langle X_1,\ldots,X_d\rangle$ be the algebra of functions on the polydisc Δ^d, and let $g_1,\ldots,g_n$ belong to it. Write $(g_1,\ldots,g_n)$ and $(X_1,\ldots,X_d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-035715 | MATHDB-318249

Bergman projection boundedness conjecture for tube domains over symmetric cones

摘录 / EXCERPT: Let $T_\Omega=\mathbb{R}^n+i\Omega$ be a tube domain over an irreducible symmetric cone $\Omega\subset\mathbb{R}^n$ of rank r, with determinant function Δ, and let .....

conjecture | open | MathDB 页面 | 原文线索

V2-035716 | MATHDB-318251

The area-nonincreasing functional differential inequality conjecture

摘录 / EXCERPT: Area-nonincreasing functional conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-035717 | MATHDB-318263

Isometric multiplication operators on Bloch spaces of bounded homogeneous domains

摘录 / EXCERPT: Let D be a bounded homogeneous domain, let $H(D)$ denote the holomorphic functions on D, and let M_ψ be the multiplication operator induced by $\psi\in H(D)$ on the Bloch...

conjecture | open | MathDB 页面 | 原文线索

V2-035718 | MATHDB-318275

The Maslov-index criterion for local polynomial convexity near thin CR singularities

摘录 / EXCERPT: Maslov-index criterion. The surface $\mathcal{S}$ is locally polynomially convex at p if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-035719 | MATHDB-318311

Anderson-Cheeger-Colding-Tian manifold conjecture for three-dimensional Ricci limits

摘录 / EXCERPT: Anderson-Cheeger-Colding-Tian conjecture. The metric space (M,d) is itself a manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-035720 | MATHDB-318317

The Van Vleck zero correspondence conjecture

摘录 / EXCERPT: Let $\tilde v_{j,k}$ be the points uniquely defined, for $k=0,1,2$ and $j=0,\dots,M_k$, by ... where $M_k=[m_kn(1-\varepsilon)]$. A Van Vleck zero is a zero of a Van Vleck polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-035721 | MATHDB-318367

Embedded Calabi-Yau conjecture for minimal embeddings of open surfaces

摘录 / EXCERPT: Let M be an open surface. An end of M has infinite genus if every representative of the end has infinite genus. Let $\mathcal{D}_\infty$ be the bounded domain in $\mathbb{R}^3$...

conjecture | open | MathDB 页面 | 原文线索

V2-035722 | MATHDB-318390

The conjectural L^2 factorization equivalence for loops in $SU(2)$

摘录 / EXCERPT: The conjectural L^2 factorization equivalence. These three conditions should be equivalent, and the factorization should define a bijective correspondence between $\zeta\in l^2$...

conjecture | open | MathDB 页面 | 原文线索

V2-035723 | MATHDB-318464

Microlocality conjecture for the HUM control operator

摘录 / EXCERPT: Let Ω be the domain, let $\chi(t,x)$ be the control function, let Λ be the HUM control operator, and let the optical rays be the associated bicharacteristic rays. As...

conjecture | open | MathDB 页面 | 原文线索

V2-035724 | MATHDB-318483

Variance-growth conjecture for interval-graph estimators

摘录 / EXCERPT: Let $X(\infty)$ be the limiting estimator produced by the memoryless iterative process on an interval graph of length n, and let $Z(\infty)$ be the benchmark estimator used in th...

conjecture | open | MathDB 页面 | 原文线索

V2-035725 | MATHDB-318581

Prăjitură's dense escaping-orbit conjecture

摘录 / EXCERPT: Prăjitură's conjecture. If A_T is non-empty, then A_T is norm-dense in X. This conjecture concerns the possible size of the set of vectors whose orbits under an operator tend...

conjecture | open | MathDB 页面 | 原文线索

V2-035726 | MATHDB-318613

The strip-density classification conjecture

摘录 / EXCERPT: Let $\mathbb{R}^2$ have the strip density, equal to 1 in the strip $\{|y|\leq 1\}$ and equal to $\lambda>1$ outside it. The relevant candidate sets are classified as types (i), (...

conjecture | open | MathDB 页面 | 原文线索

V2-035727 | MATHDB-318614

The horizontal-or-vertical-line conjecture for $e^{-x^2-y^4}$

摘录 / EXCERPT: Let $\mathbb{R}^2$ carry the density ... An isoperimetric region minimizes weighted perimeter among regions of prescribed weighted volume. Horizontal-or-vertical-line conjecture. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-035728 | MATHDB-318645

Mori's exact constant conjecture for planar quasiconformal maps

摘录 / EXCERPT: Let QC_K be the family of all K-quasiconformal self-maps of the unit disk $\mathbf{B}^2$ that fix the origin, and let $M(2,K)$ be the least constant such that ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-035729 | MATHDB-318651

Reductive symmetry conjecture for recursively composed quasi-means

摘录 / EXCERPT: Reductive symmetry conjecture. All the symmetries of a quasi-mean obtained by recursive composition from G_2 are reductive symmetries, meaning that each can be formally proved us...

conjecture | open | MathDB 页面 | 原文线索

V2-035730 | MATHDB-318720

The characterization of convex domains by invariant metric equality

摘录 / EXCERPT: Let $\mathcal D$ be a domain in a complex Banach space X that is biholomorphically equivalent to a bounded domain. Here $C_{\mathcal D}$, $K_{\mathcal D}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-035731 | MATHDB-318741

Orthogonal-polynomial Christoffel function growth conjecture

摘录 / EXCERPT: Orthogonal-polynomial growth conjecture. If, for some $0\leq p<1$, γ_n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-035732 | MATHDB-318742

Orthogonal-polynomial harmonic-oscillation norm conjecture

摘录 / EXCERPT: Orthogonal-polynomial harmonic-oscillation conjecture. For every ω in the support $sp(a)$ of $a(\omega)$,

conjecture | open | MathDB 页面 | 原文线索

V2-035733 | MATHDB-318746

The ball conjecture for global minimizers of the three-dimensional local problem

摘录 / EXCERPT: Let $e_0^{\rm 3d}$ be the three-dimensional local minimization problem, and consider its global minimizers when they exist. Ball conjecture. All global minimizers of the problem...

conjecture | open | MathDB 页面 | 原文线索

V2-035734 | MATHDB-318768

Conjectured sharp constant for Schatten-norm commutator inequalities

摘录 / EXCERPT: Schatten-norm commutator conjecture. In the restricted case $p=q$,

conjecture | open | MathDB 页面 | 原文线索

V2-035735 | MATHDB-318808

The SRS boundary measure conjecture

摘录 / EXCERPT: Let $\mathbf{r}=(r_0,\ldots,r_{d-1})\in\operatorname{int}(\mathcal{D}_d)$ be a reduced parameter, and let $\mathcal{T}_{\mathbf{r}}(\mathbf{x})$ be the associated SRS tile for...

conjecture | open | MathDB 页面 | 原文线索

V2-035736 | MATHDB-318851

Conjecture on rank-two partial isometries being unitarily equivalent to complex symmetric matrices

摘录 / EXCERPT: Let T be a partial isometry on a 4-dimensional complex Hilbert space, and suppose that $\operatorname{Rank} T=2$. A matrix T is unitarily equivalent to a complex symmetric ma...

conjecture | open | MathDB 页面 | 原文线索

V2-035737 | MATHDB-318882

Multiplicity-one and codimension-three singular-set conjecture for tangent flows

摘录 / EXCERPT: Multiplicity-one and codimension-three conjecture. A time slice of any tangent flow of the mean curvature flow starting at M_0 has multiplicity one, and its singular set has dime...

conjecture | open | MathDB 页面 | 原文线索

V2-035738 | MATHDB-318885

Non-realizability of $A_{\aleph_0\aleph_1}$ as a graph CCR algebra

摘录 / EXCERPT: Let $\mathcal Z$ be the Jiang--Su algebra and let ... For a graph G, let $B(G)$ denote the associated CCR algebra. Non-realizability claim. The algebra $A_{\aleph_0\aleph_1}$ is no...

conjecture | open | MathDB 页面 | 原文线索

V2-035739 | MATHDB-319043

Vector-valued extension conjecture for rearrangement operators on UMD spaces

摘录 / EXCERPT: Let T be the scalar-valued rearrangement operator associated with a rearrangement of the dyadic Haar system, and let L^p_E denote the Bochner space of E-valued functions. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-035740 | MATHDB-319045

Finite-composition conjecture for Semenov-type shifts

摘录 / EXCERPT: Let $M=\{m_j:j\geq1\}$ be a shift, and set $N_j(M)$ for the associated Semenov counting quantity. A sequence is called decomposable when it satisfies the hypothesis of Definition 1...

conjecture | open | MathDB 页面 | 原文线索

V2-035741 | MATHDB-319046

Rosenberg's energy-minimization conjecture for rotation algebras

摘录 / EXCERPT: Rosenberg's conjecture. In this connected component, E attains its minimum exactly at the scalar multiples of U^mV^n.

conjecture | open | MathDB 页面 | 原文线索

V2-035742 | MATHDB-319108

The local orthogonality-preserver conjecture for full Hilbert C*-modules

摘录 / EXCERPT: Local orthogonality-preserver conjecture. If $\theta:E\to F$ is a $\mathbb C$-linear local map preserving orthogonality, namely

conjecture | open | MathDB 页面 | 原文线索

V2-035743 | MATHDB-319121

The rank conjecture for the p-adic valuation of the Fibonacci sequence

摘录 / EXCERPT: Let F_n denote the nth Fibonacci number, let $ u_p$ be the p-adic valuation, and let $\alpha(p)$ be the least positive integer n such that p divides F_n. For a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-035744 | MATHDB-319139

The affine-independence conjecture for holomorphic extension on domains

摘录 / EXCERPT: Affine-independence conjecture. Then $fcin A(cpartialcOmega)$.

conjecture | open | MathDB 页面 | 原文线索

V2-035745 | MATHDB-319142

Self-duality conjecture for finite Grassmann algebras

摘录 / EXCERPT: Self-duality conjecture. For every $f\in {\bf{L}}_{{\mathfrak{B}}_{{L},\mathrm{ev}}}({\mathfrak{B}}_{{L},\mathrm{od}}:{\mathfrak{B}}_{{L}})$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-035746 | MATHDB-319176

Lin's conjecture for almost commuting real symmetric matrices

摘录 / EXCERPT: Let A) and B be real, symmetric matrices, and let $[A,B]=AB-BA$ denote their commutator. For a matrix T, write $\lVert T\rVert$ for its operator norm. Lin's conjecture for re...

conjecture | open | MathDB 页面 | 原文线索

V2-035747 | MATHDB-319177

Lin's conjecture for almost commuting self-dual Hermitian matrices

摘录 / EXCERPT: Let A and B be self-dual, Hermitian matrices, and let $[A,B]=AB-BA$ denote their commutator. For a matrix T, write $\lVert T\rVert$ for its operator norm. Lin's conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-035748 | MATHDB-319178

Approximation conjecture for almost commuting real symmetric triples

摘录 / EXCERPT: Let H_1,H_2,H_3 be real, symmetric matrices, and let $[H_i,H_j]=H_iH_j-H_jH_i$ denote their commutators. Let I be the identity matrix. Real symmetric triple approximation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-035749 | MATHDB-319251

Fine's reconstruction conjecture for pure braid invariants

摘录 / EXCERPT: Let P_n denote the group of pure braids on n strands, let $\mathcal{P}_n$ be the L^2 space of square-summable functions $f:\mathcal{P}_n\to\mathbb{C}$, and let $Z_i(b)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-035750 | MATHDB-319319

Dimension-independent quasiconvex connected-set conjecture

摘录 / EXCERPT: Let $d\geq 2$. For every subset $K\subset \mathbb{R}^d$, the main theorem asserts the existence of a connected set $\tilde\Gamma\subset\mathbb{R}^d$ containing K, with controlled...

conjecture | open | MathDB 页面 | 原文线索

V2-035751 | MATHDB-319474

Furuta's matrix trace inequality conjecture

摘录 / EXCERPT: Furuta's conjecture. It was asked whether

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V2-035752 | MATHDB-319497

The nullspace characterization of Bloch eigenvalues

摘录 / EXCERPT: Let E be the linear operator in the layer-potential system for the unknown densities $[\eta;\xi]$, where $\eta=[\tau;-\sigma]$, and let (ω,a,b) denote the frequency and qu...

conjecture | open | MathDB 页面 | 原文线索

V2-035753 | MATHDB-319517

Intrinsic-flat noncancellation conjecture for three-manifolds with positive scalar curvature

摘录 / EXCERPT: Intrinsic-flat noncancellation conjecture. A converging sequence of three-dimensional Riemannian manifolds with positive scalar curvature, a uniform lower bound on volume, and no i...

conjecture | open | MathDB 页面 | 原文线索

V2-035754 | MATHDB-319540

RCBM's radial log-convex measure conjecture

摘录 / EXCERPT: Let $\mu=e^{v(r)}\,dx$ be a radially symmetric measure on $\mathbb{R}^d$, where v is a convex smooth potential and $d\geq 2$. The balls about the origin are $B_R=\{x:|x|\leq R\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-035755 | MATHDB-319552

The linear-independence conjecture for minimal intermediate subfactors

摘录 / EXCERPT: Linear-independence conjecture. There exist vectors $\xi_i,\xi\in N'\cap M_1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-035756 | MATHDB-319561

Unique choice is Weihrauch-equivalent to choice on Baire space

摘录 / EXCERPT: Let ${\rm UC}_X$ denote the unique closed choice operation on a represented Hausdorff space X, and let ${\rm C}_X$ denote closed choice on X. For Baire space...

conjecture | open | MathDB 页面 | 原文线索

V2-035757 | MATHDB-319606

Jorgensen-Pedersen dual spectral set conjecture

摘录 / EXCERPT: Let $\Gamma\subset\mathbb R$ be a set. A spectrum for a spectral set Ω is a set of frequencies whose normalized exponentials form an orthonormal basis of $L^2(\Omega)$. A pr...

conjecture | open | MathDB 页面 | 原文线索

V2-035758 | MATHDB-319645

A Bernstein-type inequality for translates of powers of a cosine bump

摘录 / EXCERPT: Bernstein-type conjecture. The function G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-035759 | MATHDB-319667

Connes' embedding conjecture for groups

摘录 / EXCERPT: Let G be a countable i.c.c. group, meaning that every conjugacy class other than the identity class is infinite, and let $VN(G)$ be its group von Neumann algebra. Let R be the...

conjecture | open | MathDB 页面 | 原文线索

V2-035760 | MATHDB-319668

Kirchberg's QWEP conjecture

摘录 / EXCERPT: A C^*-algebra A has the weak expectation property (WEP) if it is weakly completely positive complemented in $B(H)$ for a faithful representation $A\subseteq B(H)$. It has QWEP...

conjecture | open | MathDB 页面 | 原文线索

V2-035761 | MATHDB-319699

The level-set area bound for bounded domains

摘录 / EXCERPT: Let $U\subset\mathbb{R}^n$ be a bounded open set. A function $f:U\rightarrow\mathbb{R}$ assigns to each $y\in\mathbb{R}$ the level set $f^{-1}(y)$. Naive conjecture. There is such...

conjecture | open | MathDB 页面 | 原文线索

V2-035762 | MATHDB-319739

Boundary Forelli conjecture for bundles of complex lines

摘录 / EXCERPT: Let $B^n\subset\mathbb C^n$ be the unit ball, and let $a_1,\ldots,a_{n+1}\in B^n$ be vertices in general position. An $n+1$-bundle of complex lines consists of the complex lines co...

conjecture | open | MathDB 页面 | 原文线索

V2-035763 | MATHDB-319772

Feichtinger's conjecture for kernel functions

摘录 / EXCERPT: Let $\mathcal{H}$ be a reproducing kernel Hilbert space, and for each point z let $k_z^{\mathcal{H}}$ be its kernel function and $\widetilde{k}_z^{\mathcal{H}}$ the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-035764 | MATHDB-319862

Toeplitz singular Bohr-Sommerfeld conjecture

摘录 / EXCERPT: Let $\hat{J}$ and $\hat{H}$ be the semiclassical Toeplitz operators associated with a semitoric system near a focus-focus singularity. The singular Bohr-Sommerfeld formula describe...

conjecture | open | MathDB 页面 | 原文线索

V2-035765 | MATHDB-319890

The paving conjecture for zero-diagonal operators

摘录 / EXCERPT: Let ℓ_2^n be the n-dimensional Hilbert space, let T be a linear operator on ℓ_2^n whose matrix has zero diagonal, and for $A\triangleleft\{1,\ldots,n\}$ let Q_A denot...

conjecture | open | MathDB 页面 | 原文线索

V2-035766 | MATHDB-319962

Discrete logarithmic-derivative measure conjecture

摘录 / EXCERPT: Let $\mathcal{L}\text{-}\mathcal{P}_n$ denote the real polynomials of degree n with only real zeros, let $\mu(p)$ be the mesh size, and let m denote Lebesgue measure. Discrete...

conjecture | open | MathDB 页面 | 原文线索

V2-035767 | MATHDB-319988

Three-term idempotent counterexample conjecture

摘录 / EXCERPT: Let $\mathbb T:=\mathbb R/2\pi\mathbb Z$ be the one-dimensional torus, and write $e_k(x):=e^{2\pi i kx}$. An idempotent polynomial is an exponential sum with coefficients in...

conjecture | open | MathDB 页面 | 原文线索

V2-035768 | MATHDB-319998

The Schwartz-function version of the HRT conjecture

摘录 / EXCERPT: Let $(t_j,\xi_j)_{j=1}^n$ be $n\ge 2$ distinct points in the plane, and let $f:\mathbb R\to\mathbb C$ be a Schwartz function. A nontrivial linear dependence means that not all coef...

conjecture | open | MathDB 页面 | 原文线索

V2-035769 | MATHDB-320043

The Bourgain-Tzafriri restricted invertibility conjecture

摘录 / EXCERPT: Bourgain-Tzafriri conjecture. There is a universal constant $A>0$ such that, for every $B>1$, there is a natural number $r=r(B)$ with the following property: for every natural numb...

conjecture | open | MathDB 页面 | 原文线索

V2-035770 | MATHDB-320108

Wall's conjecture for maximal fusion subalgebras

摘录 / EXCERPT: Fusion-algebra version of Wall's conjecture. The number of maximal fusion subalgebras generated by a subset of the simple objects of ${\cal F}$ is less than n.

conjecture | open | MathDB 页面 | 原文线索

V2-035771 | MATHDB-320111

The tensor-product conjecture for minimal intermediate subfactors

摘录 / EXCERPT: Tensor-product conjecture. The number of minimal intermediate subfactors in $N_1\otimes N_2\subset M_1\otimes M_2$ not of either excluded form is at most

conjecture | open | MathDB 页面 | 原文线索

V2-035772 | MATHDB-320112

The differentiable support-point length conjecture for planar curves

摘录 / EXCERPT: Let $\gamma:[a,b]\rightarrow\mathbb{E}^2$ be an arbitrary continuous rectifiable parametric curve such that, for every $t\in(a,b)$, there is a non-zero derivative vector...

conjecture | open | MathDB 页面 | 原文线索

V2-035773 | MATHDB-320113

Uniform continuity of unit normals for weakly convex sets

摘录 / EXCERPT: Let E be a uniformly convex and smooth space, let $A\subset E$ and $\operatorname{cl}(E\backslash A)$ be closed and weakly convex with modulus $\gamma(\varepsilon)$, and assume...

conjecture | open | MathDB 页面 | 原文线索

V2-035774 | MATHDB-320147

The multifractal spectrum conjecture for self-similar measures

摘录 / EXCERPT: Multifractal spectrum conjecture. For all $t \in [t_{\min},t_{\max}]$,

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V2-035775 | MATHDB-320178

Topping's uniqueness conjecture for instantaneously complete Ricci flows

摘录 / EXCERPT: Let $(\mathcal{M}^2,g_0)$ be a smooth, possibly incomplete Riemannian surface, and let $g(t)$ be an instantaneously complete Ricci flow with $g(0)=g_0$. Topping's uniqueness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-035776 | MATHDB-320202

Bounded homomorphism characterization for Fourier algebras

摘录 / EXCERPT: Let G and H be locally compact groups. Suppose that G is amenable, and let $\Phi:{\mathrm{A}(G)}\to{\mathrm{B}(H)}$ be a bounded homomorphism. A continuous map...

open_problem | open | MathDB 页面 | 原文线索

V2-035777 | MATHDB-320258

Punctured-disk canonical volume form conjecture

摘录 / EXCERPT: Let ... be the punctured disk, and let $h_{\mathrm{can},\Delta^{*}}$ be the canonical singular hermitian metric on $K_{\Delta^{*}}$ constructed from pluricanonical forms. Punctured-d...

conjecture | open | MathDB 页面 | 原文线索

V2-035778 | MATHDB-320319

Refined conjecture for the Minkowski question mark function convolution

摘录 / EXCERPT: Refined conjecture. For every $\beta\in[0,2]\setminus\{1\}$, there exists a constant $C(\beta)\geq 0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-035779 | MATHDB-320340

The amenability-similarity conjecture for commutative operator algebras

摘录 / EXCERPT: Amenability-similarity conjecture. A commutative Banach subalgebra of $\mathfrak{B}(\mathfrak{H})$ is amenable if and only if it is similar to a C^*-algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-035780 | MATHDB-320468

Andersson's nonvanishing Mergelyan conjecture

摘录 / EXCERPT: Let K be a compact set with connected complement, and let f be continuous on K, analytic in its interior, and nonvanishing in the interior. Andersson's nonvanishing Mergelyan...

conjecture | open | MathDB 页面 | 原文线索

V2-035781 | MATHDB-320579

Borcea's univalence conjecture for polynomial critical points

摘录 / EXCERPT: Borcea's univalence conjecture. For $p\geq1$, F is not univalent in any closed disk centered at zero whose radius is larger than

conjecture | open | MathDB 页面 | 原文线索

V2-035782 | MATHDB-320580

Borcea's normal-matrix invertibility conjecture

摘录 / EXCERPT: Borcea's normal-matrix invertibility conjecture. If every eigenvalue of every B_ℓ lies outside the unit disk and

conjecture | open | MathDB 页面 | 原文线索

V2-035783 | MATHDB-320581

Borcea's Toeplitz reformulation of the 2-variance conjecture

摘录 / EXCERPT: Borcea's Toeplitz conjecture. The matrix B has at least one eigenvalue λ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-035784 | MATHDB-320582

Schmeisser's strengthening of Sendov's conjecture

摘录 / EXCERPT: Schmeisser's conjecture. For every $\zeta\in\operatorname{conv}(V(F))$, the closed disk centered at ζ with radius $\sigma_\infty(F)$ contains a critical point of F.

conjecture | open | MathDB 页面 | 原文线索

V2-035785 | MATHDB-320583

Borcea's normal-matrix spectral-distance conjecture

摘录 / EXCERPT: Borcea's normal-matrix spectral-distance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035786 | MATHDB-320584

Minimal-support variance conjecture for planar point-mass measures

摘录 / EXCERPT: Minimal-support variance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035787 | MATHDB-320629

The eigenfunction-space conjecture for spectral projections on Sierpinski fractafolds

摘录 / EXCERPT: Let Γ be the graph under consideration, let μ be the spectral measure, and let P_λ and $\tilde{P_\lambda}$ denote the spectral projection kernels introduced abov...

conjecture | open | MathDB 页面 | 原文线索

V2-035788 | MATHDB-320659

The strong openness conjecture for multiplier ideal sheaves

摘录 / EXCERPT: Strong openness conjecture. The following equality of sheaves holds:

conjecture | open | MathDB 页面 | 原文线索

V2-035789 | MATHDB-320691

The adelic Tau conjecture

摘录 / EXCERPT: For $f\in\mathbb{Z}[x_1]$, let $ \tau(f)$ denote its number of nonzero coefficients, and let the roots of f be counted in the indicated fields. Adelic Tau conjecture. There is an...

conjecture | open | MathDB 页面 | 原文线索

V2-035790 | MATHDB-320725

Formal asymptotic profile conjecture for degenerate Ricci flow neckpinches

摘录 / EXCERPT: Formal asymptotic profile conjecture. For every $n\geq 2$, every $k\geq 3$, and every $c>0$, there exist Ricci flow solutions $g(t)$ that develop at some finite time T a degenera...

conjecture | open | MathDB 页面 | 原文线索

V2-035791 | MATHDB-320792

Bounded conic intersection conjecture for sufficiently Euclidean Riemannian metrics

摘录 / EXCERPT: Let $\Phi(x,y)=\rho(x,y)$, where ρ is a Riemannian metric sufficiently close to the Euclidean metric. The bounded conic intersection conjecture. Then Φ satisfies the boun...

conjecture | open | MathDB 页面 | 原文线索

V2-035792 | MATHDB-320809

Finite-support minimizer conjecture for causal variational principles on the sphere

摘录 / EXCERPT: Finite-support minimizer conjecture. For any $\tau\geq 1$, there is a minimizer $\rho\in\mathfrak{M}$ of the variational problem on the sphere which is a weighted counting measure...

conjecture | open | MathDB 页面 | 原文线索

V2-035793 | MATHDB-320853

Muscalu-Tao-Thiele uniform Hausdorff-Young conjecture for the scattering transform

摘录 / EXCERPT: Let $f\in \mathrm{L}^p(\mathbb{R})$ and let $a(\xi)$ be the coefficient associated with the scattering transform of f, as above. Let p and q be conjugated exponents with...

conjecture | open | MathDB 页面 | 原文线索

V2-035794 | MATHDB-320878

Iwaniec-Martin conjecture for the Donaldson-Sullivan signature operator

摘录 / EXCERPT: Let S be the Donaldson-Sullivan signature operator on differential forms over $\mathbb{R}^n$, defined by ... For $1<p<\infty$, let $p^*=\max\{p,p/(p-1)\}$. Iwaniec-Martin conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-035795 | MATHDB-320879

Conjecture on Lévy-spherical multipliers

摘录 / EXCERPT: Let $\mathbb{S}$ be the unit sphere in $\mathbb{R}^n$, let $\varphi:\mathbb{S}\to\mathbb{C}$ satisfy $\|\varphi\|_{L^{\infty}}\leq 1$, let $0<r<\infty$ and $n\geq 2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-035796 | MATHDB-320884

Conjecture on the pointwise decay of Euler-Lagrange solutions

摘录 / EXCERPT: Let d be the dimension, let $p_0(d)$ be the exponent used in the generalized Euler-Lagrange equation, and let υ be the weight ... Let $\lambda\in\mathbb C$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-035797 | MATHDB-320893

The generic coincidence conjecture for approximated partial limits

摘录 / EXCERPT: Let $\left\{a_n^{(\omega)}\right\}_{n=1}^{\infty}$ range over all possible sequences, and let $p\tilde{\lim}a_n^{(\omega)}$ denote the set of approximated partial limits while the...

conjecture | open | MathDB 页面 | 原文线索

V2-035798 | MATHDB-320964

The type III factor realization conjecture for ICC groups

摘录 / EXCERPT: Let Γ be a non-amenable finitely presented ICC group. A finite presentation is written $\langle S\mid R\rangle$, and $R_{1},\ldots,R_{m}$ are admissible rules; let...

conjecture | open | MathDB 页面 | 原文线索

V2-035799 | MATHDB-320977

Strict monotonicity conjecture for the threshold m_0(n)

摘录 / EXCERPT: For $n\in\mathbb N$, let $m_0(n)$ be the least positive integer such that the associated Fock-Bargmann-Hartogs domain $D_{n,m}$ is a Lu Qi-Keng domain. Strict monotonicity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-035800 | MATHDB-320978

Upper-bound conjecture for the threshold m_0(n)

摘录 / EXCERPT: For $n\in\mathbb N$, let $m_0(n)$ be the least positive integer such that the associated Fock-Bargmann-Hartogs domain $D_{n,m}$ is a Lu Qi-Keng domain, and let $[x]$ denote the nea...

conjecture | open | MathDB 页面 | 原文线索

V2-035801 | MATHDB-320979

Limit conjecture for the largest root r_{n,m}

摘录 / EXCERPT: For $n,m\in\mathbb N$, let $r_{n,m}$ denote the largest real root of the polynomial associated with the Bergman kernel of the Fock-Bargmann-Hartogs domain $D_{n,m}$. Limit conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-035802 | MATHDB-321054

The Pełczyński decomposition conjecture for commutators

摘录 / EXCERPT: Let $\mathcal{X}$ be a Banach space such that ... where $1\leq p\leq\infty$ or $p=0$ (we say that such a space admits a Pełczyński decomposition). Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-035803 | MATHDB-321073

T-Winter conjecture on regularity properties of simple nuclear C*-algebras

摘录 / EXCERPT: Let A be a simple unital nuclear separable C^*-algebra. The properties of finite nuclear dimension, $\mathcal{Z}$-stability, and strict comparison of positive elements are the...

conjecture | open | MathDB 页面 | 原文线索

V2-035804 | MATHDB-321085

The kernel decomposition conjecture for braided Wick operators

摘录 / EXCERPT: Let $\mathcal{H}$ be the Hilbert space underlying a Wick algebra, let T be a braided operator on $\mathcal{H}^{\mathcal{otimes}2}$, and let R_m denote the corresponding Wick sy...

conjecture | open | MathDB 页面 | 原文线索

V2-035805 | MATHDB-321097

Schoen's curvature estimate conjecture for stable minimal hypersurfaces

摘录 / EXCERPT: Let M^4 be a Riemannian manifold, let $B_{r_0}=B_{r_0}(x)\subset M^4$, and let $\Sigma^3\subset B_{r_0}$ be an immersed 2-sided stable minimal hypersurface with...

conjecture | open | MathDB 页面 | 原文线索

V2-035806 | MATHDB-321123

Convergence conjecture for the coefficient-ratio sequences

摘录 / EXCERPT: For each $n\geq1$, write ... for integers D_n,E_n,F_n,G_n, and define the sequences $D=\\{D_{n+1}/D_n\\}_{n\geq1}$, $E=\\{E_{n+1}/E_n\\}_{n\geq1}$, $F=\\{F_{n+1}/F_n\\}_{n\geq1}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-035807 | MATHDB-321155

Embeddedness conjecture for finite critical tangent-point energy

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^n$ be an immersed m-dimensional C^1-manifold, and let $\mathcal{E}_{2m}(\Sigma)$ denote its tangent-point energy at the critical exponent. Embedded...

conjecture | open | MathDB 页面 | 原文线索

V2-035808 | MATHDB-321283

Three-region zero-separation conjecture for the electrostatic polynomial sequence

摘录 / EXCERPT: Zero-separation conjecture. For every positive integer d, the zeros of $G(d;z)$ form three groups separated by the sets A_1, A_2, and A_3, except that when 6 divides d...

conjecture | open | MathDB 页面 | 原文线索

V2-035809 | MATHDB-321284

Alternating-location conjecture for zeros near the unit circles

摘录 / EXCERPT: Alternating-location conjecture. If d is even, all the zeros of $G(d;z)$ in A_1 are the zeros on the unit circle $\mathcal C_0$ given in that proposition. If d is odd, the ze...

conjecture | open | MathDB 页面 | 原文线索

V2-035810 | MATHDB-321285

The fundamental conjecture for optimal Riesz energy on spheres

摘录 / EXCERPT: Let $d\geq 2$, let $-2<s<d+2$ with $s\neq 0,d$, and let $\mathbb{S}^d$ be the unit sphere in $\mathbb{R}^{d+1}$. Write $\mathcal{E}_s(\mathbb{S}^d;N)$ for the optimal discrete Ries...

conjecture | open | MathDB 页面 | 原文线索

V2-035811 | MATHDB-321290

General blowup conjecture for the L^2-critical boson star equation

摘录 / EXCERPT: Suppose that $u \in C^0([0,T);H^{1/2}(\mathbb{R}^3))$ solves the boson star equation with $m\geqslant 0$ and blows up at a finite time $0<T<+\infty$. Let $u^*\in L^2(\mathbb{R}^3)$...

conjecture | open | MathDB 页面 | 原文线索

V2-035812 | MATHDB-321393

Pisier's dual amenability conjecture for dual operator algebras

摘录 / EXCERPT: Let ${\mathcal{M}}$ be a unital dual operator algebra admitting a normal virtual diagonal u. Pisier's dual amenability conjecture. ${\mathcal{M}}$ should be similar to an injecti...

conjecture | open | MathDB 页面 | 原文线索

V2-035813 | MATHDB-321435

The sharp two-weight A_p-A_c0 inequality for truncated Calder3n\adZygmund operators

摘录 / EXCERPT: Sharp A_p-A_∞ conjecture. There is a constant $C_{T,p}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-035814 | MATHDB-321512

Uniform restriction conjecture for toral eigenfunctions

摘录 / EXCERPT: Let $d\geq 2$ and let $\Sigma\subset\mathbb T^d$ be a real analytic hypersurface. For an eigenfunction φ_λ of the Laplacian on $\mathbb T^d$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-035815 | MATHDB-321545

Continuity characterization by preservation of connected sets for arbitrary polynomials

摘录 / EXCERPT: Connected-set continuity conjecture. Condition (2) characterizes continuity: P is continuous if and only if $P(C)$ is connected for every connected subset C of E.

conjecture | open | MathDB 页面 | 原文线索

V2-035816 | MATHDB-321548

Extension of Strichartz's results without heat kernel estimates

摘录 / EXCERPT: Let K be the fractal and assume the conditions of Theorems and. Let the results of Strichartz referred to in the source be the stated spectral and fractal-analytic results, norma...

conjecture | open | MathDB 页面 | 原文线索

V2-035817 | MATHDB-321578

Friedgut-Kalai entropy/influence conjecture

摘录 / EXCERPT: Let $f:0,1n_{1/2}\to\{-1,1\}$ be a Boolean function on the discrete cube with uniform measure $\mu_{1/2}$. Write $\operatorname{Ent}_{1/2}(f)$ for the entropy of its Fourier-weight...

conjecture | open | MathDB 页面 | 原文线索

V2-035818 | MATHDB-321579

Biased-measure entropy/influence conjecture

摘录 / EXCERPT: Let μ_p be the product measure on the discrete cube $\{0,1\}^n$, and let $f:\{0,1\}^n_p\to\{-1,1\}$ be a Boolean function. Write $\operatorname{Ent}_p(f)$ for its spectral entr...

conjecture | open | MathDB 页面 | 原文线索

V2-035819 | MATHDB-321580

Fourier concentration conjecture for low-influence Boolean functions

摘录 / EXCERPT: Fix $0<p<1$. Let $f:\{-1,1\}^n_p\to\{-1,1\}$ be a Boolean function, let $\operatorname{I}(f)$ denote its total influence, and let $\hat f(S)$ be its Fourier coefficients. Fourier c...

conjecture | open | MathDB 页面 | 原文线索

V2-035820 | MATHDB-321603

Conformal-factorization conjecture for normalized symmetric gradients

摘录 / EXCERPT: Let $\Omega\subset \mathbb R^n$ be a bounded open connected domain with $n\geq 3$. Let $u\in W^{1,n}(\Omega)$ and $v\in W^{1,1}(\Omega)$ satisfy ... almost everywhere, and suppose th...

conjecture | open | MathDB 页面 | 原文线索

V2-035821 | MATHDB-321605

The projectional-skeleton characterization for scattered compact lines

摘录 / EXCERPT: Projectional-skeleton characterization. $C(K)$ has a projectional skeleton if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-035822 | MATHDB-321702

Finite-length conjecture for locally analytic vectors

摘录 / EXCERPT: Let G be a ${\mathbb Q}_p$-analytic group over L, and let Π be an absolutely irreducible admissible L-Banach space representation of G. Denote by $\Pi_{\rm an}$ the spa...

conjecture | open | MathDB 页面 | 原文线索

V2-035823 | MATHDB-321742

The comparison conjecture for Schatten-class and compact-operator K-theory

摘录 / EXCERPT: Let Γ be a group. Let K be the C^*-algebra of all compact operators on an infinite-dimensional separable Hilbert space, and let $C_r^*(\Gamma)$ be the reduced group...

conjecture | open | MathDB 页面 | 原文线索

V2-035824 | MATHDB-321796

Jarosz's renorming conjecture for prescribed isometry groups

摘录 / EXCERPT: Let G be a group and let X be a real Banach space with $\dim X \geqslant |G|$. An equivalent renorming of X is a norm equivalent to its original norm, and...

conjecture | open | MathDB 页面 | 原文线索

V2-035825 | MATHDB-321797

Near-triviality conjecture for isometries of complex HI spaces

摘录 / EXCERPT: Let X be a complex HI space, meaning a hereditarily indecomposable Banach space, and let the group of isometries of X act on X. Near-triviality conjecture. The group of isome...

conjecture | open | MathDB 页面 | 原文线索

V2-035826 | MATHDB-321882

The pure-state separation conjecture for Hilbert C^*-modules

摘录 / EXCERPT: Let E be a Hilbert C^*-module over a C^*-algebra $\sA$, let $L\subset E$ be a closed convex subset, and let $x_0\in E\setminus L$. For each state $\go$ on $\sA$, let $E^\go$...

conjecture | open | MathDB 页面 | 原文线索

V2-035827 | MATHDB-321883

The pure-state separation conjecture for closed $\sA$-convex sets

摘录 / EXCERPT: Let $\sA$ be a C^*-algebra, let $\sA_+$ denote its positive cone, and let $A\subset\sA_+$ be a closed $\sA$-convex subset. A subset is $\sA$-convex if it is closed under finite e...

conjecture | open | MathDB 页面 | 原文线索

V2-035828 | MATHDB-321884

The implication from positive-set separation to submodule separation

摘录 / EXCERPT: Let Conjecture c:one be the assertion that pure states separate the localized image of an $\sA$-submodule from a point outside it, and let Conjecture c:two be the assertion that ev...

conjecture | open | MathDB 页面 | 原文线索

V2-035829 | MATHDB-321899

Positive-curvature flow convergence conjecture

摘录 / EXCERPT: Let M be a closed Riemannian manifold with positive sectional curvature. Consider the flow defined in the paper, denoted by ... converges uniformly to a nonzero Killing vector fiel...

conjecture | open | MathDB 页面 | 原文线索

V2-035830 | MATHDB-321935

Meromorphic extension conjecture for pluripolar curves

摘录 / EXCERPT: Let γ be a pluripolar curve, let $\hat{\gamma}$ denote its projective hull, and let Σ be a set contained in $\hat{\gamma}$. Let ϕ be the function associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-035831 | MATHDB-321956

Theta-line illumination conjecture

摘录 / EXCERPT: Theta-line illumination conjecture. For every such f, θ, and P, one has

conjecture | open | MathDB 页面 | 原文线索

V2-035832 | MATHDB-322055

Lau-Ülger-Pym conjecture on factorization in Banach algebras

摘录 / EXCERPT: Lau-Ülger-Pym conjecture. There is no faithful infinite-dimensional non-unital weakly sequentially complete Banach algebra A with a BAI such that

conjecture | open | MathDB 页面 | 原文线索

V2-035833 | MATHDB-322060

Dimension bound for the nonuniqueness locus of tangent cones

摘录 / EXCERPT: Nonuniqueness-locus conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035834 | MATHDB-322061

Dimension bound for the locus of tangent cones with differing homeomorphism types

摘录 / EXCERPT: Homeomorphism-type conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035835 | MATHDB-322100

The capacity-deficit asymptotic conjecture for planar shrinking condensers

摘录 / EXCERPT: Let Δ be the unit disc in the complex plane, let $\mathbb T=\partial\Delta$, and let $E\subseteq\mathbb T$ be closed. For $0<r<1$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-035836 | MATHDB-322108

The unconditional convergence and Bessel-sequence rescaling conjecture for multipliers

摘录 / EXCERPT: Let ${\cal H}$ be a Hilbert space and let $M_{m,\Phi,\Psi}$ be a multiplier associated with scalar sequence m and sequences (ϕ_n) and (ψ_n) in ${\cal H}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-035837 | MATHDB-322115

The immediate-successor conjecture for the ideal generated by the formal identity

摘录 / EXCERPT: Let $d_{w,p}$ be the Lorentz sequence space, and let $\mathcal K(d_{w,p})$ denote the ideal of compact operators on it. Let $j\colon\ell_p\to d_{w,p}$ be the formal identity, and w...

conjecture | open | MathDB 页面 | 原文线索

V2-035838 | MATHDB-322191

Convex-hull conjecture for Orthomin(1)

摘录 / EXCERPT: Let $\zeta_1,\dots,\zeta_d\in\mathbb{C}$ be the spectral points used in the finite-dimensional Orthomin analysis, and let $\operatorname{Hull}(\zeta_1,\dots,\zeta_d)$ denote their...

conjecture | open | MathDB 页面 | 原文线索

V2-035839 | MATHDB-322264

The laminate-sequence conjecture for the Burkholder function

摘录 / EXCERPT: Let p and τ be parameters, let $c_{\mathcal M}$ be the relevant martingale constant, and let $\widetilde{U}_{c_{\mathcal M}}$ be the associated Burkholder function. A lamina...

conjecture | open | MathDB 页面 | 原文线索

V2-035840 | MATHDB-322309

Ilman's asymptotic-end conjecture for embedded self-shrinkers

摘录 / EXCERPT: Ilman's asymptotic-end conjecture. There exists $R>0$ such that $\Sigma\setminus B_R$ decomposes into a finite number of ends U_j, and for each j either:

conjecture | open | MathDB 页面 | 原文线索

V2-035841 | MATHDB-322310

Ilman's rigidity conjecture for cylindrical self-shrinking ends

摘录 / EXCERPT: Ilman's cylindrical rigidity conjecture. If one end of Σ is asymptotic to a cylinder, then Σ is isometric to the self-shrinking cylinder.

conjecture | open | MathDB 页面 | 原文线索

V2-035842 | MATHDB-322394

Homogeneity criterion for agreement between population TDMI estimators

摘录 / EXCERPT: Let $\bar{I}(\tau)$ and $\hat{I}(\tau)$ denote the two population time-delayed mutual information estimators, and define their difference by ... A population is statistically homogen...

conjecture | open | MathDB 页面 | 原文线索

V2-035843 | MATHDB-322421

The S^{p}-dimension conjecture for noncommutative L^{p} spaces

摘录 / EXCERPT: Let Γ be an $\mathcal{R}^{\omega}$-embeddable group, fix $2<p<\infty$, and consider the multiplication action of Γ on $L^{p}(L(\Gamma),\tau_{\Gamma})^{\oplus n}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-035844 | MATHDB-322423

Fourier lower-bound conjecture for sofic I^{p} dimension

摘录 / EXCERPT: The Fourier lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035845 | MATHDB-322424

The left-ideal lower-bound conjecture for S^{p} dimension

摘录 / EXCERPT: The left-ideal lower-bound conjecture. For the action of Γ by left multiplication,

conjecture | open | MathDB 页面 | 原文线索

V2-035846 | MATHDB-322468

Fourier coefficient lower-bound conjecture for rank one transformations

摘录 / EXCERPT: Let T be a rank one transformation with a fixed sequence of tower partitions (ξ_n) such that ξ_{n+1} refines ξ_n and every measurable set A can be approximated by a...

conjecture | open | MathDB 页面 | 原文线索

V2-035847 | MATHDB-322495

The coefficient-vanishing conjecture for the determinant expansion

摘录 / EXCERPT: Coefficient-vanishing conjecture. For $j=1,\ldots,np(m)$, the coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-035848 | MATHDB-322503

Zwegers's radial smoothness and asymptotic matching conjecture for mock theta functions

摘录 / EXCERPT: Let f be a mock theta function in Ramanujan's sense, and let ξ be a root of unity. Radial limits are taken along the line through ξ, either inside or outside the unit cir...

conjecture | open | MathDB 页面 | 原文线索

V2-035849 | MATHDB-322507

Uniform restricted X-ray transform estimates for polynomial curves

摘录 / EXCERPT: Let $d\geq 3$ and let $P:{\bf R}\to{\bf R}^{d-1}$ be a polynomial of degree N. Write ... for affine arclength measure, and let X^P denote the restricted X-ray transform associate...

conjecture | open | MathDB 页面 | 原文线索

V2-035850 | MATHDB-322512

Goldberg's constant-sign conjecture for trace polynomials

摘录 / EXCERPT: Let m_i and n_j be the parameters in the trace polynomial $a_{1,1}+a_{2,2}$ of the matrix product $A^{m_1}B^{-n_1}\ldots A^{m_k}B^{-n_k}$. Substitute ... with an arbitrary combin...

conjecture | open | MathDB 页面 | 原文线索

V2-035851 | MATHDB-322606

Sharp heat and Poisson kernel estimates for all Fourier-Bessel type indices

摘录 / EXCERPT: Final conjecture. The estimates from the heat-kernel theorem and the Poisson-kernel theorem hold for all $\nu>-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-035852 | MATHDB-322681

Higher-dimensional Young-varifold representation conjecture

摘录 / EXCERPT: Let Ω be the domain, let $u\in{\mathrm{BV}}(\Omega)$ satisfy $\overline{F}(u)<\infty$, and let $\mathbb{V}(u)$ denote the class of Young varifolds associated with u. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-035853 | MATHDB-322683

Upper estimate conjecture for the coderivative of a differential inclusion's reachable map

摘录 / EXCERPT: Let F define a differential inclusion on $[0,T]$ and let $R:\mathbb{R}^{n}\rightrightarrows\mathbb{R}^{n}$ be the reachable map, with reference points $(\bar{x},\bar{y})$. For th...

conjecture | open | MathDB 页面 | 原文线索

V2-035854 | MATHDB-322718

Winter-Tikuisis regularity conjecture for simple nuclear C*-algebras

摘录 / EXCERPT: Let A be a simple, unital, separable, nuclear, infinite-dimensional C*-algebra. Winter-Tikuisis regularity conjecture. The following are equivalent: 1. A has finite nuclear...

conjecture | open | MathDB 页面 | 原文线索

V2-035855 | MATHDB-322751

Spectral naturality under quasi-containment

摘录 / EXCERPT: Let G be a locally compact group, and let τ_ρ denote the representation associated with a representation ρ. A representation ρ is spectrally natural when...

conjecture | open | MathDB 页面 | 原文线索

V2-035856 | MATHDB-322883

The complete heat-content asymptotic expansion for singular data

摘录 / EXCERPT: Let M be a compact Riemannian manifold with smooth boundary ∂M, let δ denote the geodesic distance to the boundary, let D be an operator of Laplace type with Dir...

conjecture | open | MathDB 页面 | 原文线索

V2-035857 | MATHDB-322884

The logarithmic heat-content asymptotic at the critical singularity

摘录 / EXCERPT: Let M be a compact Riemannian manifold with smooth boundary ∂M, let δ denote the distance to the boundary, let D be an operator of Laplace type with Dirichlet...

conjecture | open | MathDB 页面 | 原文线索

V2-035858 | MATHDB-322888

Vanishing-order conjecture for the umbilicity tensor and Weyl curvature

摘录 / EXCERPT: Let g) be a smooth Riemannian metric defined on the unit half n-ball B^+_1, where $4\leq n\leq24$. Suppose there is a sequence of positive solutions $\{u_i\}$ of ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-035859 | MATHDB-322891

Zhang-Lin-Liu eigenvalue conjecture for a scaled trigonometric matrix

摘录 / EXCERPT: Zhang-Lin-Liu eigenvalue conjecture. $P(\omega)$ has one positive eigenvalue λ_+ and one negative eigenvalue λ_-, both of which have multiplicity 2, are independe...

conjecture | open | MathDB 页面 | 原文线索

V2-035860 | MATHDB-322915

Hanner inequality for Schatten norms of operator tuples

摘录 / EXCERPT: Let $A_0,A_1,\ldots,A_{n-1}$ be operators on a separable Hilbert space, and let p,q satisfy $1/p+1/q=1$. Operator Hanner-type conjecture. For $2\leq p<\infty$, ... whereas for...

conjecture | open | MathDB 页面 | 原文线索

V2-035861 | MATHDB-322916

Smale's mean-value conjecture for polynomial critical points

摘录 / EXCERPT: Let p be a polynomial of degree n with $p(0)=0$ and $p'(0)\neq0$. Smale's conjecture. At least one root μ_j of p' satisfies ... The source lists this among open conjectures...

conjecture | open | MathDB 页面 | 原文线索

V2-035862 | MATHDB-322955

Huang-Ji-Yin gap-rigidity conjecture for proper rational maps between balls

摘录 / EXCERPT: Let $n\ge 3$ be a positive integer. Write $K(n)$ for the largest positive integer m such that ... For each k with $1\le k\le K(n)$, let $\mathcal{I}_k$ be the collection of integ...

conjecture | open | MathDB 页面 | 原文线索

V2-035863 | MATHDB-322972

Extension of the block-matrix majorization inequality to normal off-diagonal blocks

摘录 / EXCERPT: Let a positive semidefinite matrix in $\mathbb{M}_{2n}^{+}$ be written in blocks of the same size, with normal off-diagonal blocks. Normal-block majorization conjecture. The inequa...

conjecture | open | MathDB 页面 | 原文线索

V2-035864 | MATHDB-323008

Monotonicity conjecture for the function F_q

摘录 / EXCERPT: Monotonicity conjecture. F_q is decreasing on $[0,\frac{\pi}{2}]$ for $q>2$. For $q\leq 2$, F_q is nondecreasing on $[0,\frac{\pi}{2}]$ when $\left[\frac{(n+1)q}{2}\right]$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-035865 | MATHDB-323044

Rakhmanov-Saff-Zhou logarithmic energy asymptotic conjecture for the sphere

摘录 / EXCERPT: Let $\mathcal{E}_{\mathrm{log}}(\mathbb{S}^2;N)$ denote the optimal logarithmic energy of N points on the unit sphere, and let $C_{\mathrm{log},2}$ and $D_{\mathrm{log},2}$ be co...

conjecture | open | MathDB 页面 | 原文线索

V2-035866 | MATHDB-323045

Next-order Riesz energy asymptotic conjecture for spheres

摘录 / EXCERPT: Let $d\geq 2$, let $\mathbb{S}^d$ be the unit sphere, let $\mathcal{H}_d(\mathbb{S}^d)$ denote its d-dimensional Hausdorff measure, and let $V_s(\mathbb{S}^d)$ denote its continu...

conjecture | open | MathDB 页面 | 原文线索

V2-035867 | MATHDB-323046

Critical Riesz energy asymptotic conjecture on spheres

摘录 / EXCERPT: Let $d\geq1$, let $\mathbb{S}^d$ be the unit sphere, and let $\mathcal{H}_d(\mathbb{B}^d)$ and $\mathcal{H}_d(\mathbb{S}^d)$ denote the relevant measures of the unit ball and spher...

conjecture | open | MathDB 页面 | 原文线索

V2-035868 | MATHDB-323068

The squeezing-function characterization of the unit ball

摘录 / EXCERPT: Let $D\subset\subset\mathbb{C}^n$ with $n>1$ be a strongly pseudoconvex domain with smooth boundary. The squeezing function s_D measures how closely D can be biholomorphically...

conjecture | open | MathDB 页面 | 原文线索

V2-035869 | MATHDB-323082

Hadamard lacunary extension of the non-maximal-summability theorem

摘录 / EXCERPT: Let Q be a sequence that is lacunary in Hadamard sense. The non-maximal-summability theorem, and perhaps even the PLA plus uniform-lacunarity theorem, should hold for Q. This i...

conjecture | open | MathDB 页面 | 原文线索

V2-035870 | MATHDB-323092

Vector-valued Hopf-Dunford-Schwartz conjecture for l^q-valued functions

摘录 / EXCERPT: Let $(T_t)_{tgeqslant0}$ be a strongly continuous semi-group satisfying the contraction property, and let $1<q<+infty$. For a sequence-valued function f, write $\overline{M_T}f$...

conjecture | open | MathDB 页面 | 原文线索

V2-035871 | MATHDB-323096

Uniformity conjecture for measures with frame measures

摘录 / EXCERPT: Let μ be a finite Borel measure on $\mathbb{R}^d$. Say that μ is translationally absolutely continuous if, for every Borel set F in the support of μ with $\mu(F)>0$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-035872 | MATHDB-323181

Borwein-Sims basin-of-attraction conjecture for the sphere-line Douglas-Rachford iteration

摘录 / EXCERPT: Let X be an N-dimensional Hilbert space, let $S:=\{x\in X\colon\|x\|=1\}$ be the unit sphere, and let $L:=\{x=\lambda a+\alpha b\colon\lambda\in\mathbb{R}\}$, where a,b are o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-035873 | MATHDB-323194

The injectivity conjecture for finitely separated graph C*-algebras

摘录 / EXCERPT: Injectivity conjecture. The natural map

conjecture | open | MathDB 页面 | 原文线索

V2-035874 | MATHDB-323195

The ordered K-theory conjecture for finitely separated graph C*-algebras

摘录 / EXCERPT: Ordered K-theory conjecture. This homomorphism of partially pre-ordered abelian groups is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-035875 | MATHDB-323204

Hexagonal conjecture for the boundary length of minimal spectral equipartitions

摘录 / EXCERPT: Hexagonal boundary-length conjecture. The normalized boundary length satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-035876 | MATHDB-323242

Finite-dimensionality conjecture for weak and strong vector-valued Hardy spaces

摘录 / EXCERPT: Let $1<p<\infty$ and let F be a Banach space. Write $H^p(\mathbb{D};F)$ for the space of F-valued holomorphic functions on the unit disk with strong H^p boundary behavior, an...

conjecture | open | MathDB 页面 | 原文线索

V2-035877 | MATHDB-323277

The conjecture on the sharp norm of the Bergman projection

摘录 / EXCERPT: Sharp Bergman projection norm conjecture. The upper quantity is expected to attain the lower bound:

conjecture | open | MathDB 页面 | 原文线索

V2-035878 | MATHDB-323292

Logarithmic bounds conjecture for maximal directional singular integrals

摘录 / EXCERPT: Let $T_{\mathbf V}$ be the maximal directional singular integral associated with a finite set of N directions ${\mathbf V}$, as defined in the source. Here p' denotes the Hölde...

conjecture | open | MathDB 页面 | 原文线索

V2-035879 | MATHDB-323293

The largest-entry-free determinant bound for positive diagonally balanced symmetric matrices

摘录 / EXCERPT: Let J be a positive, diagonally balanced symmetric matrix, and let S denote the comparison matrix used in the determinant bound. Largest-entry-free determinant conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-035880 | MATHDB-323342

Prăjitură's orbit-growth conjecture for Banach-space operators

摘录 / EXCERPT: Prăjitură's conjecture. A_R is either empty or dense.

conjecture | open | MathDB 页面 | 原文线索

V2-035881 | MATHDB-323405

Ideal rigidity conjecture for uncountable free Banach lattices

摘录 / EXCERPT: Let $\textswab{a}$ be an uncountable set, let X be a projective Banach lattice, and suppose that X contains a closed ideal isomorphic to the free Banach lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-035882 | MATHDB-323406

Atomic determination conjecture for order continuous functionals on projective Banach lattices

摘录 / EXCERPT: Let X be a projective Banach lattice, and let an order continuous functional mean a continuous linear functional preserving limits of decreasing nets to zero in order. The atoms...

conjecture | open | MathDB 页面 | 原文线索

V2-035883 | MATHDB-323410

Conjectural refined asymptotic for the generating function of restricted partitions

摘录 / EXCERPT: For each integer $k\geq2$, let $p_k(n)$ denote the number of partitions of n with no sequence of k consecutive positive integers, and define its generating function by ... For re...

conjecture | open | MathDB 页面 | 原文线索

V2-035884 | MATHDB-323495

The spectral equality conjecture for weighted composition operators on Banach lattice subspaces

摘录 / EXCERPT: Let X be a Banach lattice with Stonean compact K, let Y be the subspace and T the weighted composition operator from Proposition 1, and let S be the corresponding weighte...

conjecture | open | MathDB 页面 | 原文线索

V2-035885 | MATHDB-323496

The crossed-product factorization conjecture for Orlicz spaces

摘录 / EXCERPT: The crossed-product factorization conjecture. For any Orlicz function ψ,

conjecture | open | MathDB 页面 | 原文线索

V2-035886 | MATHDB-323505

The classification conjecture for subfactors of index between 5 and $3+\sqrt{5}$

摘录 / EXCERPT: Classification conjecture. Any subfactor with index in $(5,3+\sqrt{5})$ has principal graph A_∞, $\Gamma(\mathcal A)$, or $\Gamma(\mathcal B)$.

conjecture | open | MathDB 页面 | 原文线索

V2-035887 | MATHDB-323530

Demailly-Kollár openness conjecture for jumping numbers

摘录 / EXCERPT: Demailly-Kollár jumping-number openness conjecture. If $c_x^{\mathfrak q}(\varphi)<\infty$, then $|\mathfrak q|^2\exp(-2c_x^{\mathfrak q}(\varphi)\varphi)$ is not locally integrabl...

conjecture | open | MathDB 页面 | 原文线索

V2-035888 | MATHDB-323597

Bourgain's conjecture for trigonometric polynomials on the Bohr compactification of $\mathbb R$

摘录 / EXCERPT: Bourgain's conjecture in the Bohr compactification. For any increasing sequence $(\omega_j)_{j\in\mathbb N}$ of real numbers,

conjecture | open | MathDB 页面 | 原文线索

V2-035889 | MATHDB-323605

Characterization of the Rohlin property for flows on injective factors

摘录 / EXCERPT: The Rohlin-property characterization. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-035890 | MATHDB-323622

Douglas's boundedness conjecture for effective potentials

摘录 / EXCERPT: Let M be a compact n-dimensional manifold, let $g=e^{2\varphi}g_0$ range over a conformal class of metrics, let v be a positive warp factor, and let $F^{(p)}$ be the field st...

conjecture | open | MathDB 页面 | 原文线索

V2-035891 | MATHDB-323629

Holomorphic-observable convergence conjecture for self-avoiding walks

摘录 / EXCERPT: Let $\Omega\subsetneq\mathbb{C}$ be a simply connected domain, let $z\in\Omega$, and let a,b be distinct boundary points, with the boundary smooth near b. For $\delta>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-035892 | MATHDB-323642

Growth rate conjecture for Chern-Simons asymptotics

摘录 / EXCERPT: Assume the asymptotic expansion conjecture holds for a closed oriented 3-manifold M. Let $\mathcal{M}(M)$ be the moduli space of flat $\operatorname{SU}(N)$-connections, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-035893 | MATHDB-323676

Endpoint weak-type conjecture for the quartile operator

摘录 / EXCERPT: Endpoint weak-type conjecture. For each $p_1,p_2>1$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-035894 | MATHDB-323685

Strictness conjecture for connected choice in dimension two versus interval choice

摘录 / EXCERPT: Let $\text{\rm\sffamily CC}_2$ be connected choice for non-empty connected closed subsets of $[0,1]^2$, and let $\text{\rm\sffamily C}_{[0,1]}$ be closed choice on the unit interva...

conjecture | open | MathDB 页面 | 原文线索

V2-035895 | MATHDB-323686

Non-idempotency conjecture for connected choice in dimension two

摘录 / EXCERPT: Let $\text{\rm\sffamily CC}_2$ denote connected choice for non-empty connected closed subsets of $[0,1]^2$. For a multi-valued operation f, write $f\times f$ for two parallel ins...

conjecture | open | MathDB 页面 | 原文线索

V2-035896 | MATHDB-323687

Cylinder conjecture for connected choice in dimension two

摘录 / EXCERPT: Let $\text{\rm\sffamily CC}_2$ denote connected choice for non-empty connected closed subsets of $[0,1]^2$. For a multi-valued operation f, write $\operatorname{id}\times f$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-035897 | MATHDB-323704

Conjecture on the convergence rate of discretized Pickands constants

摘录 / EXCERPT: Convergence-rate conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-035898 | MATHDB-323720

Sharp constants for the strong multilinear Bohnenblust-Hille inequality

摘录 / EXCERPT: Assume that the strong multilinear Bohnenblust-Hille inequality holds, so that a universal constant $K_{\mathbb{K}}$ exists for each scalar field...

conjecture | open | MathDB 页面 | 原文线索

V2-035899 | MATHDB-323793

Dubrovin's pole-location conjecture for the tritronquée solution

摘录 / EXCERPT: Let $y(\lambda)$ be the tritronquée solution of Painlevé I, and let $\alpha\in\mathbb{C}$ be a pole of this solution. Dubrovin's conjecture. If α is a pole of the tritronqué...

conjecture | open | MathDB 页面 | 原文线索

V2-035900 | MATHDB-323823

Generic existence and uniqueness of means in tree shape spaces

摘录 / EXCERPT: Generic means conjecture. For a generic set of points $x_1,\ldots,x_n$ in $\bar{X}$, $\bar{\bar{X}}$, $\bar{Z}$, or $\bar{\bar{Z}}$, means exist and are unique.

conjecture | open | MathDB 页面 | 原文线索

V2-035901 | MATHDB-323824

Local triple derivation conjectures for JB*-triples

摘录 / EXCERPT: Local triple derivation conjectures. The following statements are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-035902 | MATHDB-323853

Mockenhaupt's three-term idempotent conjecture

摘录 / EXCERPT: Let $\mathbb T:=\mathbb R/\mathbb Z$, let $e(t):=e^{2\pi i t}$, and write $e_j(x):=e(jx)$. For $k\in\mathbb N$, define the three-term idempotent polynomials ... Let $2k<p<2k+2$. Mock...

conjecture | open | MathDB 页面 | 原文线索

V2-035903 | MATHDB-323859

Roe-Strichartz theorem for iterated Laplacian sequences on symmetric spaces

摘录 / EXCERPT: Let X be the Riemannian symmetric space under consideration, let K/M be its Furstenberg boundary, and let $q' $ satisfy $1/q+1/q'=1$. Fix $q\in(1,2)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-035904 | MATHDB-323917

The spine ideal and Clifford spectrum conjecture for Fourier-Stieltjes algebras

摘录 / EXCERPT: Let G be a locally compact group, and let $vn(G)=\ell^\infty\text{-}\bigoplus_{\pi\in\hat{G}}\mathcal{B}(\mathcal{H}_\pi)$ be the von Neumann algebra dual of $a(G)$. The spine id...

conjecture | open | MathDB 页面 | 原文线索

V2-035905 | MATHDB-323997

Fong-Tsui conjecture for bounded operators

摘录 / EXCERPT: Fong-Tsui conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-035906 | MATHDB-324002

Prolate quadrature conjecture on the relative accuracy of coefficient coordinates

摘录 / EXCERPT: Relative-accuracy conjecture. The coordinates of $\beta^{(n)}$ satisfy, for every $1\leq k\leq N$,

conjecture | open | MathDB 页面 | 原文线索

V2-035907 | MATHDB-324003

Prolate quadrature error conjecture for lower-index prolate functions

摘录 / EXCERPT: Quadrature-error conjecture. The quadrature error satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-035908 | MATHDB-324004

Prolate quadrature conjecture for exponential-function errors

摘录 / EXCERPT: Exponential quadrature-error conjecture. The quadrature error for the exponential function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-035909 | MATHDB-324035

Colding-Ilmanen-Minicozzi-White entropy conjecture for closed hypersurfaces

摘录 / EXCERPT: Let $M^n\subset \mathbb{R}^{n+1}$ be a closed hypersurface, and let λ denote entropy. The cited theorem asserts that, for a closed self-shrinker, the round sphere minimizes...

conjecture | open | MathDB 页面 | 原文线索

V2-035910 | MATHDB-324036

Colding-Ilmanen-Minicozzi-White low-entropy self-shrinker conjecture

摘录 / EXCERPT: Let $\Sigma^n\subset\mathbb{R}^{n+1}$ be a self-shrinker, and call it non-flat if it is not a hyperplane. Let λ denote entropy. Colding-Ilmanen-Minicozzi-White low-entropy...

conjecture | open | MathDB 页面 | 原文线索

V2-035911 | MATHDB-324037

Generic tangent-flow singular-set dimension conjecture for mean curvature flow

摘录 / EXCERPT: Let $M_0\subset\mathbb{R}^{n+1}$ be a smooth closed embedded hypersurface, and consider its mean curvature flow. A time slice of a tangent flow is a hypersurface obtained by parabo...

conjecture | open | MathDB 页面 | 原文线索

V2-035912 | MATHDB-324038

Cylindrical self-shrinker rigidity conjecture

摘录 / EXCERPT: Let $\Sigma^n\subset\mathbb{R}^{n+1}$ be a smooth embedded self-shrinker with entropy at most α, where $\alpha>1$. Let B_R be the Euclidean ball of radius R, and let H...

conjecture | open | MathDB 页面 | 原文线索

V2-035913 | MATHDB-324039

Canonical neighborhood conjecture at cylindrical singularities

摘录 / EXCERPT: Let M_t be a mean curvature flow of smooth closed hypersurfaces in $\mathbb{R}^{n+1}$. A cylindrical singularity is a singularity whose tangent flow is a generalized cylinder. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-035914 | MATHDB-324043

The cotype-2 threshold-embedding conjecture

摘录 / EXCERPT: Let X be a Banach space of cotype 2, and let $S\subseteq X$. The subset S threshold-embeds into Hilbert space if there is a threshold embedding of S into a Hilbert space. C...

conjecture | open | MathDB 页面 | 原文线索

V2-035915 | MATHDB-324083

Strong-type endpoint conjecture for finite-field k-plane transforms

摘录 / EXCERPT: Let F_q be a finite field, let $dgeq 2$ and $1leq kleq d-1$ be integers, and let T_{Pi_k} denote the k-plane transform on the collection Pi_k of all k-planes in F_q^d,...

conjecture | open | MathDB 页面 | 原文线索

V2-035916 | MATHDB-324108

The discrepancy coefficient limsup-liminf conjecture

摘录 / EXCERPT: Let d_N denote the coefficient of the leading term of the two-dimensional L_1 discrepancy for N-point sets. Discrepancy coefficient limsup-liminf conjecture. The following st...

conjecture | open | MathDB 页面 | 原文线索

V2-035917 | MATHDB-324122

The regularity conjecture for simple nuclear C*-algebras

摘录 / EXCERPT: Let A be a simple nuclear separable unital infinite-dimensional C^*-algebra. The properties of having finite nuclear dimension, being $\mathcal{Z}$-stable, and having strict co...

conjecture | open | MathDB 页面 | 原文线索

V2-035918 | MATHDB-324123

Gaudin-type determinantal norm conjecture for periodic Macdonald spherical functions

摘录 / EXCERPT: Gaudin-type determinantal norm conjecture. For any $\mu\in P^{\vee,+}_c$, this quadratic norm is

conjecture | open | MathDB 页面 | 原文线索

V2-035919 | MATHDB-324318

Blackadar's semiprojectivity conjecture for split extensions

摘录 / EXCERPT: Let $0 \to A \to B \to \mathbb{C} \to 0$ be a split exact sequence of separable C^*-algebras. A C^*-algebra is semiprojective if every $*$-homomorphism into a quotient by the c...

conjecture | open | MathDB 页面 | 原文线索

V2-035920 | MATHDB-324425

Wall's conjecture for intermediate subfactors

摘录 / EXCERPT: Let $N\subset M$ be an irreducible subfactor, meaning that $N'\cap M=\mathbb{C}$, with finite index. Let $M\subset M_1$ be the Jones basic construction, so that $N'\cap M_1$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-035921 | MATHDB-324426

Wall's conjecture for intermediate conformal subnets

摘录 / EXCERPT: Suppose that ${\cal A}\subset {\cal B}$ are conformal subnets with finite index. Let the vacuum representation of ${\cal B}$ carry the action of ${\cal A}$, and let V be the spac...

conjecture | open | MathDB 页面 | 原文线索

V2-035922 | MATHDB-324430

Eckhoff's conjecture on maximal-accuracy reconstruction of jump locations

摘录 / EXCERPT: Eckhoff's conjecture. The jump locations of f can be reconstructed from its first $2M+1$ Fourier coefficients, with accuracy $O\left(M^{-d-2}\right)$, by solving the perturbed no...

conjecture | open | MathDB 页面 | 原文线索

V2-035923 | MATHDB-324491

Compactness conjecture for manifolds satisfying a generalized curvature-dimension inequality

摘录 / EXCERPT: Let $\mathbb M$ be the contact manifold considered in the paper, with curvature-dimension parameters ρ_1, ρ_2, ρ_3 and κ. Compactness conjecture. If ... then th...

conjecture | open | MathDB 页面 | 原文线索

V2-035924 | MATHDB-324529

Kadec-Pełczyński dichotomy for preduals of JBW*-triples

摘录 / EXCERPT: Kadec-Pełczyński dichotomy conjecture. There exist a subsequence $(\phi_{\tau(n)})$ and a sequence of mutually orthogonal tripotents (e_n) in W such that

conjecture | open | MathDB 页面 | 原文线索

V2-035925 | MATHDB-324581

Kadison-Kastler conjecture for von Neumann algebras

摘录 / EXCERPT: Let $\mathcal H$ be a Hilbert space, let $\mathcal B(\mathcal H)$ denote the bounded operators on $\mathcal H$, and let $d(M,N)$ be the Hausdorff distance between the unit balls of...

conjecture | open | MathDB 页面 | 原文线索

V2-035926 | MATHDB-324602

Bond-Łaba-Volberg structural conjecture for vanishing cyclotomic divisors

摘录 / EXCERPT: Bond-Łaba-Volberg conjecture. For any $A(x)$ as above, there is a divisor Q of N such that s does not divide Q for every $s\in S_A$, and

conjecture | open | MathDB 页面 | 原文线索

V2-035927 | MATHDB-324608

Hytönen-Lacey mixed A_p-A_infty estimate with one supremum

摘录 / EXCERPT: Let $1<p<finfty$, let T be a Calderon-Zygmund operator, and let $w otin A_p$. Write $p'=p/(p-1)$ and let $\sigma=w^{-1/(p-1)}$ be the dual weight. For mixed constants,...

conjecture | open | MathDB 页面 | 原文线索

V2-035928 | MATHDB-324632

The determinant bound for positive diagonally balanced symmetric matrices

摘录 / EXCERPT: Let J be an entrywise positive, diagonally balanced symmetric real $n\times n$ matrix, where $n\geq 2$. Diagonal balance means that, for every $i=1,\ldots,n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-035929 | MATHDB-324633

Coarse total vertical perimeter conjecture on the Heisenberg group

摘录 / EXCERPT: Coarse total vertical perimeter conjecture. For every measurable $A\subseteq\mathbb H(\mathbb R)$ and every $\varepsilon\in(0,1/2)$,

conjecture | open | MathDB 页面 | 原文线索

V2-035930 | MATHDB-324634

Sharp cl_1 distortion conjecture for Heisenberg balls

摘录 / EXCERPT: Let $\mathbb H$ be the discrete Heisenberg group, let B_n be its radius-n ball with the word metric d_W, and let $c_1(B_n,d_W)$ denote the least distortion of an embedding of...

conjecture | open | MathDB 页面 | 原文线索

V2-035931 | MATHDB-324664

Perturbative stability conjecture for computed Maass-form eigenpairs

摘录 / EXCERPT: Let $\tilde\lambda$ be a fixed trial eigenvalue, let η be a small complex parameter, and let $(h,\vec\alpha)_{|\tilde\lambda}$ and $(h_\eta,\vec\alpha_\eta)_{|\tilde\lambda}$...

conjecture | open | MathDB 页面 | 原文线索

V2-035932 | MATHDB-324687

Curvature type-I conjecture for finite-time Kähler-Ricci flow singularities

摘录 / EXCERPT: Let $(M,\omega(t))$ be a solution of the Kähler-Ricci flow on its maximal time interval $[0,T)$, and let Rm denote its curvature tensor. If $T<\infty$, then curvat...

conjecture | open | MathDB 页面 | 原文线索

V2-035933 | MATHDB-324727

Operator Kantorovich inequality for 2-positive linear maps

摘录 / EXCERPT: Let $0<m\le A\le M$, and let X,Y be partial isometries on a Hilbert space $\mathcal{H}$ whose final spaces are orthogonal to each other. Let Φ be a 2-positive linear map....

conjecture | open | MathDB 页面 | 原文线索

V2-035934 | MATHDB-324750

Nonsingularity conjecture for the Riccati-derived M-matrices

摘录 / EXCERPT: Let K be a regular singular M-matrix and let u_1,u_2,v_1,v_2 be the vectors associated with K as in the source. The scalar products $u_1^Tv_1$ and $u_2^Tv_2$ compare the co...

conjecture | open | MathDB 页面 | 原文线索

V2-035935 | MATHDB-324757

Leander's generalized Helleseth conjecture for permutations

摘录 / EXCERPT: Leander's generalized Helleseth conjecture. If f is a permutation of L, then

conjecture | open | MathDB 页面 | 原文线索

V2-035936 | MATHDB-324758

Sarwate's conjecture on Fourier coefficients of power permutations

摘录 / EXCERPT: Sarwate's conjecture. If $[L:{\mathbb F}_2]$ is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-035937 | MATHDB-324762

The graph C*-algebra classification conjecture by filtered ordered K-theory

摘录 / EXCERPT: Let E be a graph and let $C^*(E)$ be its graph C^*-algebra. Suppose that $C^*(E)$ has finitely many ideals, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-035938 | MATHDB-324783

Reflexivity characterization by quasihyperbolic geodesics in open half spaces

摘录 / EXCERPT: Reflexivity conjecture. The Banach space X is reflexive if and only if each of its open half spaces is geodesic in the quasihyperbolic metric.

conjecture | open | MathDB 页面 | 原文线索

V2-035939 | MATHDB-324785

Common-solvability conjecture for systems of linear matrix inequalities

摘录 / EXCERPT: Common-solvability conjecture. The k LMIs

conjecture | open | MathDB 页面 | 原文线索

V2-035940 | MATHDB-324801

Conjecture on monotonicity and floating-point accuracy of PSWF quadrature errors

摘录 / EXCERPT: Let $c>1$ be real, let $n>2c/\pi$ be an integer, and let S_n be the stated quadrature rule. For even integers m with $0\leq m<n$, let $\delta_n(\psi_m)$ be the quadrature error...

conjecture | open | MathDB 页面 | 原文线索

V2-035941 | MATHDB-324803

Conjecture on the decay threshold for prolate spheroidal eigenvalues

摘录 / EXCERPT: Let $c>1$ and $0<\varepsilon<1$ be real numbers, and let $n>0$ be an integer. Let λ_n denote the prolate spheroidal eigenvalue. Eigenvalue-decay conjecture. If ... then ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-035942 | MATHDB-324824

The 2-adic binomial reciprocal-sum valuation conjecture

摘录 / EXCERPT: The 2-adic binomial reciprocal-sum valuation conjecture. If $0\le k<2^e$, then

conjecture | open | MathDB 页面 | 原文线索

V2-035943 | MATHDB-324825

The odd-argument 2-adic binomial reciprocal-sum valuation conjecture

摘录 / EXCERPT: The odd-argument 2-adic valuation conjecture. If $0\le k<2^{e-1}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-035944 | MATHDB-324856

The square-root commutator conjecture for positive contractions

摘录 / EXCERPT: Let H and A belong to a C^{*}-algebra, with H a positive contraction satisfying $0\leq H\leq1$ and $\left\Vert A\right\Vert\leq1$. For the square-root function...

conjecture | open | MathDB 页面 | 原文线索

V2-035945 | MATHDB-324885

Restriction conjecture for measurably continuous functions

摘录 / EXCERPT: Let (X,μ) be a measure space, and consider a symmetric measurably continuous function of two variables on $(X\times X,\mu\times\mu)$. A measurably continuous function is one th...

conjecture | open | MathDB 页面 | 原文线索

V2-035946 | MATHDB-324956

Fourier-compatibility conjecture for Lie algebra smooth transfer

摘录 / EXCERPT: Let $\mathfrak s$ and $\mathfrak s'$ be the tangent spaces of the relevant symmetric spaces, equipped with the smooth-transfer relation defined by the transfer factor κ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-035947 | MATHDB-325017

Conjecture on limit points of zeros of exponential-integral sections

摘录 / EXCERPT: Let F be an exponential integral function as defined in the paper, and let s_nF denote its normalized sections. Let $D_{a,b}$ and D_imag be the curves introduced in th...

conjecture | open | MathDB 页面 | 原文线索

V2-035948 | MATHDB-325020

Equality of intrinsic and extrinsic reinforced spaces on the cube

摘录 / EXCERPT: Let Q be the unit cube in $\mathbb{R}^n$ for $n\geq 2$. For $1\leq p<\infty$, $0<q<\infty$, and $s>\sigma_{p,q}$, let $F_{p,q}^{s,\operatorname{rinf}}(Q)$ denote the reinforced f...

conjecture | open | MathDB 页面 | 原文线索

V2-035949 | MATHDB-325056

Global converse for intrinsic supersmoothness

摘录 / EXCERPT: Global converse conjecture. The curve γ is smooth at P if and only if one can glue two continuously differentiable functions along γ as above so that F is not dif...

conjecture | open | MathDB 页面 | 原文线索

V2-035950 | MATHDB-325057

Ros's conjecture on properly embedded minimal surfaces in hyperbolic 3-space

摘录 / EXCERPT: An open surface is a non-compact surface without boundary. Let S be an open, connected, orientable surface. Ros's conjecture. Every such surface can be properly and minimally emb...

conjecture | open | MathDB 页面 | 原文线索

V2-035951 | MATHDB-325080

Plancherel-Rotach edge asymptotic conjecture for orthonormal polynomials

摘录 / EXCERPT: Let $\{\widehat P_k(x)\}_{k=1}^n$ be a sequence of polynomials orthonormal with respect to a weight function $w(x)$ on an unbounded interval, and let t_n be the large transition...

conjecture | open | MathDB 页面 | 原文线索

V2-035952 | MATHDB-325086

Dimension conservation conjecture for non-degenerate projections

摘录 / EXCERPT: Let $B\subset\mathbb{R}^{3}$ be an analytic set, and let $(\rho_{\theta})_{\theta\in U}$ and $(\pi_{\theta})_{\theta\in U}$ be non-degenerate families of projections onto lines and...

conjecture | open | MathDB 页面 | 原文线索

V2-035953 | MATHDB-325108

The Gaussian-isoperimetric stability conjecture for hyperplanes

摘录 / EXCERPT: Equip $\mathbb{R}^n$ with the Gaussian metric. A hyperplane is a codimension-one affine subspace, and it is regarded as a solution of the isoperimetric problem when it separates th...

conjecture | open | MathDB 页面 | 原文线索

V2-035954 | MATHDB-325174

Negative critical-exponent conjecture for two-weight Riesz potentials

摘录 / EXCERPT: Let $0<\alpha<n$, let $1<p<\infty$, let (u,σ) be a pair of weights, and let I_α and M_α denote the Riesz potential and fractional maximal operator, respectivel...

conjecture | open | MathDB 页面 | 原文线索

V2-035955 | MATHDB-325175

Cruz-Moen weak-type bump conjecture at the critical exponent

摘录 / EXCERPT: Let $0<\alpha<n$, let $1<p<\infty$, let $\bar{\Psi}\in B_{p'}$, and suppose $(u,\sigma)\in A_{p',p',\Psi}^\alpha$. Weak-type critical-exponent bump conjecture. Then ... This is the e...

conjecture | open | MathDB 页面 | 原文线索

V2-035956 | MATHDB-325176

Cruz-Moen strong-type bump conjecture at the critical exponent

摘录 / EXCERPT: Strong-type critical-exponent bump conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-035957 | MATHDB-325208

The critical exponent conjecture for powers of doubly nonnegative matrices

摘录 / EXCERPT: Critical exponent conjecture. For all integers $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-035958 | MATHDB-325306

Sarason's conjecture for the Bergman space

摘录 / EXCERPT: Sarason's conjecture. The operator $T_fT_g^*$ is bounded on $A^2(\mathbb{D})$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-035959 | MATHDB-325307

Sarason's conjecture for the Hardy space

摘录 / EXCERPT: Sarason's conjecture. The operator $\mathcal{T}_f\mathcal{T}_g^*$ is bounded on $H^2(\mathbb{T})$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-035960 | MATHDB-325379

Full-dimensional projection conjecture for non-degenerate families in three dimensions

摘录 / EXCERPT: Full-dimensional projection conjecture. For almost every $\theta\in J$,

conjecture | open | MathDB 页面 | 原文线索

V2-035961 | MATHDB-325472

Equivalence of metric regularity and strong metric regularity of the subdifferential

摘录 / EXCERPT: Equivalence of metric regularity and strong metric regularity. Under these hypotheses, metric regularity and strong metric regularity of the subdifferential are the same notions.

conjecture | open | MathDB 页面 | 原文线索

V2-035962 | MATHDB-325586

Higher regularity conjecture for quasisymmetric maps between relative Schottky sets

摘录 / EXCERPT: Higher regularity conjecture. The map f is conformal at every point $p\in S$ and belongs to $C^{\infty}(S)$; that is, derivatives of all orders exist on S in the sense of the d...

conjecture | open | MathDB 页面 | 原文线索

V2-035963 | MATHDB-325595

Borchardt mean conjecture for genus-two theta constants

摘录 / EXCERPT: Let $\mathcal{H}_2$ be the Siegel upper half-space of degree 2, let $\mathcal{F}_2$ denote the relevant fundamental domain, and let B_2 be the Borchardt mean applied to four sq...

conjecture | open | MathDB 页面 | 原文线索

V2-035964 | MATHDB-325611

The definable Lebesgue measure conjecture for the unit interval

摘录 / EXCERPT: Let $\mathbb{K}$ be a definably complete expansion of an ordered field, and let μ be the outer measure defined by infima of sums of lengths of definable open-interval covers in...

conjecture | open | MathDB 页面 | 原文线索

V2-035965 | MATHDB-325612

Commutativity of definable sums over discrete sets

摘录 / EXCERPT: Let $\mathbb{K}$ be an expansion of an ordered field, let $D \subseteq K_{\geq 0}$ be a definable closed discrete subset, let $h:D\to K_{\geq 0}$ be definable, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-035966 | MATHDB-325675

Lasoń's pattern-repetition conjecture for measurable colorings of Euclidean space

摘录 / EXCERPT: Let $d\geq 1$, and let a measurable k-coloring of $\mathbb{R}^d$ mean a measurable partition into k color classes. Two axis-aligned d-dimensional cubes are measure indiscerni...

conjecture | open | MathDB 页面 | 原文线索

V2-035967 | MATHDB-325676

Existence of measures distinguishing disjoint cuboids in Euclidean space

摘录 / EXCERPT: Let $d\geq 1$. A finite continuous measure on $\mathbb{R}^d$ has no atoms, and two axis-aligned cuboids are measure discernible for a collection of measures when their measure vect...

conjecture | open | MathDB 页面 | 原文线索

V2-035968 | MATHDB-325720

The L^1 discrepancy conjecture in dimensions at least three

摘录 / EXCERPT: Let $\mathcal P_N$ be a collection of N points in $[0,1]^d$, and let D_N denote its discrepancy function. For a measurable function f on $[0,1]^d$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-035969 | MATHDB-325776

Classification conjecture for actions of discrete Kac algebras via invariants

摘录 / EXCERPT: Let α be an action of $\widehat{\mathbb{G}}$ on an injective factor M. Define the $\widehat{\mathbb{G}}^{\mathrm{opp}}$-action γ^α on ... by ... Let θ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-035970 | MATHDB-325801

Pole-freeness conjecture for the tritronquée solutions of P_{I}^2

摘录 / EXCERPT: Let $U_0(x,t)$ and $V_0(x,t)$ be the solutions of P_{I}^2 with uniform algebraic asymptotics in the sectors specified by ... , respectively. Pole-freeness conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-035971 | MATHDB-325834

Density version of Lerman's L^2 curve learning theorem

摘录 / EXCERPT: Let μ be a locally finite Borel measure on $\mathbb{R}^n$, let $Q_1\in\vec{\Delta}(\mathbb{R}^n)$, and let $\lambda>1$ be the constant from Lerman's theorem. Define ... Suppose t...

conjecture | open | MathDB 页面 | 原文线索

V2-035972 | MATHDB-325909

Closure conjecture for the linear hull of exponential solutions

摘录 / EXCERPT: Let the linear hull in Proposition be the span of the relevant exponential solutions, and let its closure be taken in the function space under consideration. Closure conjecture. Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-035973 | MATHDB-326052

Coarse index product formula for Hilbert module bundle coefficients

摘录 / EXCERPT: Let M_1 and M_2 be complete Riemannian manifolds with Hilbert A_1-module and A_2-module bundles $E_1\to M_1$ and $E_2\to M_2$, respectively. The external tensor product def...

conjecture | open | MathDB 页面 | 原文线索

V2-035974 | MATHDB-326061

Douglas's strengthened essential normality conjecture

摘录 / EXCERPT: Let K be a finite dimensional Hilbert space, let H^2_d denote Drury-Arveson space, and let $L\subseteq H^2_d\otimes K$ be a graded submodule. Write $\dim(L)$ for the dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-035975 | MATHDB-326078

The Rohlin-property characterization conjecture for flows on AFD factors

摘录 / EXCERPT: Rohlin-property characterization conjecture. The following three conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-035976 | MATHDB-326133

Isometric-isomorphic universality conjecture for Banach spaces

摘录 / EXCERPT: Let ... , with either isomorphic or isometric embeddings. Isometric-isomorphic universality conjecture. The universality number of the class of Banach spaces of density ... with isomet...

conjecture | open | MathDB 页面 | 原文线索

V2-035977 | MATHDB-326145

Rarity of Pareto-dominated fictitious-play orbits

摘录 / EXCERPT: Consider the space of $n\times n$ bimatrix games with entries in $[0,1]$. Restrict attention to games with a unique, completely mixed Nash equilibrium and to typical fictitious-pla...

conjecture | open | MathDB 页面 | 原文线索

V2-035978 | MATHDB-326151

The high-dimensional maximal surface area conjecture for radial measures

摘录 / EXCERPT: Let $\varphi(t)$ be any real-valued function on the positive semi-axis, and let X be a random vector in $\mathbb{R}^n$ with density $e^{-\varphi(|X|)}$. Write $\mathcal{K}_n$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-035979 | MATHDB-326198

Bourgain's sharp logarithmic Strichartz bound on the two-dimensional torus

摘录 / EXCERPT: Let P_N denote the frequency projection onto frequencies $\lesssim N$, and let $e^{it\Delta_{\mathbb{T}^2}}$ be the Schrödinger propagator on $\mathbb{T}^2$. For all sufficiently...

conjecture | open | MathDB 页面 | 原文线索

V2-035980 | MATHDB-326199

Character classification for stable simple AF algebras

摘录 / EXCERPT: Let A be a stable simple AF algebra not isomorphic to $\mathbb{K}$. Write $TW(A)$ for the tracial state space associated with A, $U_\to(A)$ for the relevant infinite-dimensiona...

open_problem | open | MathDB 页面 | 原文线索

V2-035981 | MATHDB-326203

Fundamental singularity conjecture for minimal laminations

摘录 / EXCERPT: Let $S\subset\mathbb{R}^3$ be a closed set whose one-dimensional Hausdorff measure is zero, and let $\mathcal{L}$ be a minimal lamination of $\mathbb{R}^3-S$. Fundamental singulari...

conjecture | open | MathDB 页面 | 原文线索

V2-035982 | MATHDB-326226

Isaev's embedding conjecture for the restriction of $\widetilde F$ to M_t^3

摘录 / EXCERPT: Let M_t^3 be the real hypersurface with parameter $t>1$, and let $\widetilde F$ be the map constructed and analyzed in the paper. Isaev's embedding conjecture. The restriction of...

conjecture | open | MathDB 页面 | 原文线索

V2-035983 | MATHDB-326373

Extension of Flett's mean value theorem to improper derivatives

摘录 / EXCERPT: Let f be a function on the interval $\langle a,b\rangle$ having a proper or improper derivative at each point. Assume that the tangents at the endpoints are parallel. Flett's the...

conjecture | open | MathDB 页面 | 原文线索

V2-035984 | MATHDB-326385

Optimal error estimate for the optimization-based atomistic-to-continuum method

摘录 / EXCERPT: Let $\bar{u}^{\rm atc}$ be the atomistic-to-continuum approximation, let $\bar{u}$ be the exact solution, let $\mathcal{L}$ be the computational lattice, let $\mathcal{L}_{\rm c}$...

conjecture | open | MathDB 页面 | 原文线索

V2-035985 | MATHDB-326395

The odd-order and generation conjecture for the differential-operator algebra of a matrix weight

摘录 / EXCERPT: Let W be the matrix weight under consideration, and let $\mathcal D(W)$ denote its algebra of differential operators. The paper constructs four differential operators...

conjecture | open | MathDB 页面 | 原文线索

V2-035986 | MATHDB-326406

Sokal's simple-zero conjecture for the entire function F

摘录 / EXCERPT: Let ... where $q\in\mathbb{C}$ satisfies $0<|q|\leq 1$. Sokal's conjecture. The entire function $F(z;q)$ can have only simple zeros. The conjecture concerns the multiplicity of the z...

conjecture | open | MathDB 页面 | 原文线索

V2-035987 | MATHDB-326407

Polynomial Sokal conjecture for the roots of P_N

摘录 / EXCERPT: For $N>0$, let ... For $|q|<1$, the polynomial Sokal conjecture. The polynomial $P_N(z;q)$ can have only simple roots, separated in modulus by at least the factor $|q|$. This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-035988 | MATHDB-326442

Weinberger-Yu's finite K-theory conjecture for group C*-algebras

摘录 / EXCERPT: Weinberger-Yu's conjecture. The group $\mathcal M_{\gamma_1,\ldots,\gamma_n}$ has rank n, and every nonzero element of it lies outside the image of the assembly map

conjecture | open | MathDB 页面 | 原文线索

V2-035989 | MATHDB-326444

The hyperplane conjecture for convex bodies

摘录 / EXCERPT: Hyperplane conjecture. There exists a constant $C>0$ such that for every n and every convex body $K\subset\mathbb{R}^n$ of volume 1 and barycenter at the origin, there is a dir...

conjecture | open | MathDB 页面 | 原文线索

V2-035990 | MATHDB-326511

The separated bumps conjecture for Calderón-Zygmund operators

摘录 / EXCERPT: Separated bumps conjecture. For all Calderón-Zygmund operators T, there holds

conjecture | open | MathDB 页面 | 原文线索

V2-035991 | MATHDB-326515

Solyanik asymptotic conjecture for the uncentered Hardy-Littlewood maximal operator

摘录 / EXCERPT: Let $C_{\operatorname{HL}, b}(\alpha)$ denote the Tauberian constant associated with the uncentered Hardy-Littlewood maximal operator in $\mathbb{R}^n$. Solyanik asymptotic conject...

conjecture | open | MathDB 页面 | 原文线索

V2-035992 | MATHDB-326516

Solyanik limit conjecture for convex density bases

摘录 / EXCERPT: Solyanik limit conjecture. The associated Tauberian constants satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-035993 | MATHDB-326517

Exact-level-set criterion for Solyanik failure in convex density bases

摘录 / EXCERPT: Exact-level-set criterion. Then there exist a set E_γ and a constant $c(\gamma)>1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-035994 | MATHDB-326529

Conjecture on the $O(N^{-2})$ accuracy of WKB eigenvalues

摘录 / EXCERPT: Let $\lambda_{N,k}^{\mathrm{wkb}}$ be the WKB eigenvalues given by the Bohr-Sommerfeld quantization condition, and let $\lambda_{N,k}$ denote the true eigenvalues of the Zakharov-S...

conjecture | open | MathDB 页面 | 原文线索

V2-035995 | MATHDB-326604

Zalcman space conjecture for complex affine spaces

摘录 / EXCERPT: Zalcman space conjecture. $\mathbb C^n$ is a Zalcman space for each $n\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-035996 | MATHDB-326632

The hyperplane conjecture for time flat surfaces in Minkowski spacetime

摘录 / EXCERPT: Let Σ be a closed surface in $\mathbb{R}^{3,1}$ that is contained in a complete, spacelike hypersurface. A surface is time flat when its associated connection one-form has v...

conjecture | open | MathDB 页面 | 原文线索

V2-035997 | MATHDB-326634

The monotonicity conjecture for LGL interval quotients

摘录 / EXCERPT: Let $\xi^N_0<\xi^N_1<\cdots<\xi^N_N$ be the LGL points, let $\Delta^N_k=[\xi^N_k,\xi^N_{k+1}]$, and define $q_k^N=|\Delta^N_k|/|\Delta^N_{k-1}|$ for the relevant indices. The famil...

conjecture | open | MathDB 页面 | 原文线索

V2-035998 | MATHDB-326648

The optimal removability conjecture for bounded quasiregular mappings

摘录 / EXCERPT: A closed set $E\subset \mathbb{R}^n$ is removable under bounded K-quasiregular mappings if every bounded K-quasiregular mapping defined on an open set minus E extends to a...

conjecture | open | MathDB 页面 | 原文线索

V2-035999 | MATHDB-326726

Dyson's crank limiting-shape conjecture

摘录 / EXCERPT: Dyson's crank limiting-shape conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-036000 | MATHDB-326749

Hitchin WKB local-estimate conjecture

摘录 / EXCERPT: Let $\gamma:[0,1]\to\widetilde X^{\ast}$ be a noncritical path, meaning that the pullbacks $\gamma^{\ast}\operatorname{Re}\phi_i$ are distinct for every $t\in[0,1]$, and order them...

conjecture | open | MathDB 页面 | 原文线索

V2-036001 | MATHDB-326789

Lacey's two-weight Hilbert transform conjecture with common point masses

摘录 / EXCERPT: Let ... mathbb{R}$. For each interval I, define the Poisson integral by ... Let b_I be the unique, if it exists, point such that ... Let $\widetilde\sigma_I:=\sigma-\sigma\{b_I\}\del...

conjecture | open | MathDB 页面 | 原文线索

V2-036002 | MATHDB-326850

Infinite-SNR asymptotic MMSE limit

摘录 / EXCERPT: Let $D(R,\beta)$ denote the asymptotic MMSE at measurement rate R and inverse noise variance (SNR) parameter β, and let m_a be the a-priori magnetization. In the low-nois...

conjecture | open | MathDB 页面 | 原文线索

V2-036003 | MATHDB-326880

Sharp error-term conjecture for discrete harmonic propagation of smallness

摘录 / EXCERPT: Sharp error-term conjecture. For all $\varepsilon>0$, $n_0 a0in N$, $C>0$ and $d a0geq 2$, there exists a harmonic function $u: \mathbb{Z}^d \to \mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036004 | MATHDB-326894

The Muckenhoupt-Wheeden two-weight conjecture for Calderón-Zygmund operators

摘录 / EXCERPT: Let $1<p<\infty$, let $p'=p/(p-1)$, let u and v be weights, and let T be a Calderón-Zygmund operator. Write $M:L^p(u)\to L^p(v)$ for boundedness of the maximal operator betwe...

conjecture | open | MathDB 页面 | 原文线索

V2-036005 | MATHDB-326896

The maximal Kakeya operator conjecture

摘录 / EXCERPT: Maximal Kakeya operator conjecture. For all $n\geq 2$, the displayed inequality holds for every such family of tubes.

conjecture | open | MathDB 页面 | 原文线索

V2-036006 | MATHDB-326947

Second-order L^{1} convergence of the finite element approximation

摘录 / EXCERPT: Let $H=L^{1}([x_{min},x_{max}],\mathbb{R}_{+})$, let H^{N} be the piecewise-constant approximation space generated by the basis functions β_i^N, and let $f^{N}(t,x)$ solve...

conjecture | open | MathDB 页面 | 原文线索

V2-036007 | MATHDB-326998

The Marr conjecture on uniqueness from wavelet zero sets

摘录 / EXCERPT: Let M_σ denote the wavelet transform kernel at positive scale σ, and let f be a locally integrable function of sufficiently rapid decay. For a sequence of positive...

conjecture | open | MathDB 页面 | 原文线索

V2-036008 | MATHDB-327023

General discrete restriction conjecture

摘录 / EXCERPT: Let $n\ge 2$, let S be either the sphere S^{n-1} or the truncated paraboloid P^{n-1}, and let $e(a)=e^{2\pi i a}$. Let $\Lambda\subset S$ be a $\delta^{1/2}$-separated set, l...

conjecture | open | MathDB 页面 | 原文线索

V2-036009 | MATHDB-327024

L^2 weak decoupling conjecture for the sphere and paraboloid

摘录 / EXCERPT: Let $P^{n-1}=\{(\xi_1,\ldots,\xi_{n-1},\xi_1^2+\cdots+\xi_{n-1}^2):|\xi_i|\le 1/2\}$, let S be either P^{n-1} or S^{n-1}, and let ${\mathcal N}_\delta$ be the δ-neighb...

conjecture | open | MathDB 页面 | 原文线索

V2-036010 | MATHDB-327025

Limit discrete restriction conjecture for the sphere and paraboloid

摘录 / EXCERPT: Let $n\ge 2$, let S be either the truncated paraboloid P^{n-1} or the sphere S^{n-1}, let $\Lambda\subset S$ be a $\delta^{1/2}$-separated set, let $B_R\subset\mathbb R^n$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-036011 | MATHDB-327045

Rademacher's conjecture on restricted partition partial-fraction coefficients

摘录 / EXCERPT: Let $C_{hk\ell}(N)$ denote the coefficient associated with the root of unity $e^{2\pi i h/k}$ in the Nth partial-fraction approximation to the restricted partition generating fun...

conjecture | open | MathDB 页面 | 原文线索

V2-036012 | MATHDB-327046

Asymptotic conjecture for the coefficient C_{011}(N)

摘录 / EXCERPT: Let $\operatorname{Li_2}(z)$ be the dilogarithm, let w_0 be the unique solution of … with $w_0\approx 0.916198+0.182459i$, and define … Thus $w_0=1-e^{-2\pi i z_0}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036013 | MATHDB-327047

Asymptotic conjecture for the coefficient C_{121}(N)

摘录 / EXCERPT: Let $w_0\approx 0.916198+0.182459i$ be the unique solution of $\operatorname{Li_2}(w)+2\pi i\log(w)=0$, and define … Then $w_0=1-e^{-2\pi i z_0}$ and $1<\operatorname{Re}(z_0)<2$....

conjecture | open | MathDB 页面 | 原文线索

V2-036014 | MATHDB-327054

Existence of suitable decay for cube-based CIS interpolation in higher dimensions

摘录 / EXCERPT: The cube-CIS recovery conjecture. There exists $\omega(n)>0$ such that the interpolation operator I_α associated to g_α admits the use of CISs for cubes in...

conjecture | open | MathDB 页面 | 原文线索

V2-036015 | MATHDB-327073

The scatteredness-subprojectivity characterization for separable C*-algebras

摘录 / EXCERPT: Scatteredness-subprojectivity conjecture. A separable C^*-algebra is scattered if and only if it is subprojective.

conjecture | open | MathDB 页面 | 原文线索

V2-036016 | MATHDB-327089

Break-up geometry conjecture for semiclassical DS II

摘录 / EXCERPT: Consider rapidly decreasing smooth initial data in $L_{2}(\mathbb{R}^{2})$ with a single maximum. Let the defocusing and focusing variants of the semiclassical DS II equation be th...

conjecture | open | MathDB 页面 | 原文线索

V2-036017 | MATHDB-327090

Scaling and long-time behavior conjecture for semiclassical DS II

摘录 / EXCERPT: Consider rapidly decreasing smooth initial data in $L_{2}(\mathbb{R}^{2})$ with a single maximum, and let t_c be the critical time of the corresponding semiclassical DS II soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-036018 | MATHDB-327091

Two-point gradient-catastrophe conjecture for focusing semiclassical DS II

摘录 / EXCERPT: Consider generic rapidly decreasing initial data with a single hump for the focusing semiclassical DS II system, and let x_c and y_c denote the spatial coordinates of gradient-...

conjecture | open | MathDB 页面 | 原文线索

V2-036019 | MATHDB-327200

Convergence conjecture for eventually periodic 2-adic arguments

摘录 / EXCERPT: Let $x\in\mathbb Z_2$ have binary digits $x_i\in\{0,1\}$, so that $x=\sum_{i\ge0}x_i2^i$. Suppose that, for some $d\ge2$ and $i_0\ge0$, $x_{i+d}=x_i$ for all $i\ge i_0$. The eventu...

conjecture | open | MathDB 页面 | 原文线索

V2-036020 | MATHDB-327201

Quantitative convergence conjecture for partial Stirling functions

摘录 / EXCERPT: Let $x\in\mathbb Z_2$ be a finite element, and let i_0 and d be positive integers such that $x_{i_0}=0$ and $x_{i+d}=x_i$ for all $i\ge i_0$, with $2^{i+d}\le x$. Define ... the...

conjecture | open | MathDB 页面 | 原文线索

V2-036021 | MATHDB-327202

Negative-argument congruence conjecture for partial Stirling functions

摘录 / EXCERPT: Let $P_n(x)$ denote the partial Stirling function, let ν be the 2-adic valuation, and let $\operatorname{o}(n)=n/2^{\nu(n)}$ be the odd part of n. For integers e and...

conjecture | open | MathDB 页面 | 原文线索

V2-036022 | MATHDB-327203

One-zero congruence-class conjecture for partial Stirling functions

摘录 / EXCERPT: Let $5\le\Delta\le8$, let $z_{\Delta,\ell}\in\mathbb Z_2$ be the two zeros of P_Δ indexed by $\ell\in\{0,1\}$, and let $\varepsilon_{\Delta,\ell}$ be 2 for...

conjecture | open | MathDB 页面 | 原文线索

V2-036023 | MATHDB-327204

Difference-valuation conjecture for partial Stirling functions

摘录 / EXCERPT: Let $5\le\Delta\le8$, $0\le m<2^{e-1}$, and $e\ge4$. If ... then the difference-valuation conjecture. For every $d\ge0$, ... This is presented as a reduction step toward the one-zero c...

conjecture | open | MathDB 页面 | 原文线索

V2-036024 | MATHDB-327205

Auxiliary valuation conjecture for partial Stirling functions

摘录 / EXCERPT: Let $5\le\Delta\le8$ and $e\ge5$. For $j\ge0$, define ... For $j>0$, $j\ne2^{e-1}$, and any integer m, define ... Auxiliary valuation conjecture. The quantities satisfy all the follo...

conjecture | open | MathDB 页面 | 原文线索

V2-036025 | MATHDB-327222

Divergence conjecture for non-equidistant sampling in system approximation

摘录 / EXCERPT: Let $t_k\}_{k\in\mathbb{Z}}\subset\mathbb{R}$ be an ordered complete interpolating sequence for $\mathcal{PW}_{\pi}^{2}$, let ϕ_k be as defined in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036026 | MATHDB-327224

Logarithmic lower-bound conjecture for complete interpolating sequences

摘录 / EXCERPT: Let $\{t_k\}_{k\in\mathbb{Z}}\subset\mathbb{R}$ be an ordered complete interpolating sequence for $\mathcal{PW}_{\pi}^{2}$. Logarithmic lower-bound conjecture. There exists a posit...

conjecture | open | MathDB 页面 | 原文线索

V2-036027 | MATHDB-327294

Asymptotic zero distribution conjecture for Gonchar polynomials of the third kind

摘录 / EXCERPT: Let Γ be the set consisting of the boundary of the union of the two unit disks centered at -1 and 0, together with the line segment connecting their intersection points....

conjecture | open | MathDB 页面 | 原文线索

V2-036028 | MATHDB-327303

Matrix weighted square function conjecture

摘录 / EXCERPT: Let W be a $d\times d$ matrix weight in A_2, and let S_W denote the associated square function on $L^2(W)$. Matrix weighted square function conjecture. ... The scalar analogue...

conjecture | open | MathDB 页面 | 原文线索

V2-036029 | MATHDB-327314

The Lipschitz multiplier conjecture for Fourier-constrained measures

摘录 / EXCERPT: Lipschitz multiplier conjecture. Suppose that ϕ is Lipschitz and its image contains n linearly independent points. Then

conjecture | open | MathDB 页面 | 原文线索

V2-036030 | MATHDB-327363

Carathéodory completeness conjecture for smooth bounded pseudoconvex domains

摘录 / EXCERPT: Let $D\subset\mathbb{C}^n$ be a bounded pseudoconvex domain with C^∞ boundary. A domain D is c-complete when its Carathéodory pseudodistance c_D is a complete metric....

conjecture | open | MathDB 页面 | 原文线索

V2-036031 | MATHDB-327371

Formal compactness and lim-inf conjecture for the three-dimensional bilayer membrane model

摘录 / EXCERPT: Formal compactness and lim-inf conjecture. There exists a subsequence, not relabelled, such that

conjecture | open | MathDB 页面 | 原文线索

V2-036032 | MATHDB-327396

Extension from singular hypersurfaces for plurisubharmonic weights

摘录 / EXCERPT: Let W be the hypersurface and let φ be a plurisubharmonic weight for which the upper density $D^+_{\varphi}(W)$ is defined. The theorem asserting the relevant Bargmann-Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-036033 | MATHDB-327493

Converse to the invariance of asymptotic centers and radii under zero distance

摘录 / EXCERPT: Let ${\bar x}$ and ${\bar y}$ be bounded sequences in a Banach space X. For a bounded sequence ${\bar z}=(z_n)$, write ... and let $d({\bar x},{\bar y})$ denote the distance betwee...

conjecture | open | MathDB 页面 | 原文线索

V2-036034 | MATHDB-327541

Sharp interior norm estimates for layer potentials in strictly convex domains

摘录 / EXCERPT: Strictly convex layer-potential conjecture. There is a constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-036035 | MATHDB-327543

Noguchi's extension conjecture for holomorphic maps across polar sets

摘录 / EXCERPT: Let X be a Kobayashi hyperbolic compact manifold of dimension n. Let ${\mathbb B}$ be the unit ball in $\mathbb C^p$, and let E be a closed polar subset of ${\mathbb B}$. Nog...

conjecture | open | MathDB 页面 | 原文线索

V2-036036 | MATHDB-327551

Complex Gr"unbaum lower-bound conjecture for regular fans

摘录 / EXCERPT: Let m be a positive integer, let $k\geq 1$, and let $q_1,\ldots,q_k>2$ be integers. Set ... Here $\Delta_{\mathbb{C}}(m;q_1,\ldots,q_k)$ is the minimum dimension d such that any...

conjecture | open | MathDB 页面 | 原文线索

V2-036037 | MATHDB-327614

Essential self-adjointness criterion for one-dimensional graph Laplacians

摘录 / EXCERPT: Let $G=(V,c,E)$ be the one-dimensional weighted graph considered above, with edge conductances $c_{n,n-1}=a_n$ and energy Hilbert space $\mathscr{H}_{E}$. Let Δ denote the c...

conjecture | open | MathDB 页面 | 原文线索

V2-036038 | MATHDB-327670

Hellerstein-Sheil-Small-Wiman conjecture for real meromorphic functions

摘录 / EXCERPT: Hellerstein-Sheil-Small-Wiman conjecture. The function f satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036039 | MATHDB-327684

A boundary growth conjecture for harmonic mappings into linearly connected domains

摘录 / EXCERPT: Let $f=h+\bar{g}\in\mathcal{S}_{H}^{0}$ be a normalized locally univalent harmonic mapping, and let $\Omega=f(\mathbb{D})$ be an M-linearly connected domain. Write Λ_f fo...

conjecture | open | MathDB 页面 | 原文线索

V2-036040 | MATHDB-327690

Subsequence boundedness conjecture for Hardy-space reconstruction

摘录 / EXCERPT: Q2. For each $f\in\mathcal{H}^1$, there exists a subsequence $\{N_l\}_{l\in\mathbb{N}}$ such that the corresponding $\limsup$ of the supremum norm remains bounded for that f. In...

conjecture | open | MathDB 页面 | 原文线索

V2-036041 | MATHDB-327767

Fargues's strong noetherianity conjecture for the Banach algebras B_I

摘录 / EXCERPT: Fargues's conjecture. The Banach algebra B_I is strongly noetherian: for every $n\geq 1$, the Tate algebra

conjecture | open | MathDB 页面 | 原文线索

V2-036042 | MATHDB-327768

Large-Riesz-parameter limit conjecture for polygonal equilibrium radii

摘录 / EXCERPT: Fix $n\geq3$, and let $r(n,\beta)$ be a radius corresponding to an equilibrium position different from the origin for the system of positive unit point charges at the vertices of a...

conjecture | open | MathDB 页面 | 原文线索

V2-036043 | MATHDB-327791

Monotonicity conjecture for sublevel volumes of the pluricomplex Green function

摘录 / EXCERPT: Monotonicity conjecture. If $Omega$ is pseudoconvex in $mathbb C^n$ then this function is non-decreasing on $(-infty,0]$. The conjecture is proved in dimension one; in arbitrary di...

conjecture | open | MathDB 页面 | 原文线索

V2-036044 | MATHDB-327792

Logarithmic convexity conjecture for sublevel volumes of the pluricomplex Green function

摘录 / EXCERPT: Logarithmic convexity conjecture. If $Omega$ is pseudoconvex in $mathbb C^n$ then this function is convex on $(-infty,0]$. The conjecture would imply the preceding monotonicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-036045 | MATHDB-327793

Balanced-domain characterization of equality in the Bergman-Kobayashi indicatrix inequality

摘录 / EXCERPT: Equality characterization conjecture. $F_{Omega}(w)=1$ if and only if there exists a balanced domain $Omega'$ (not necessarily convex) and a biholomorphic mapping $H:OmegatoOmega'$...

conjecture | open | MathDB 页面 | 原文线索

V2-036046 | MATHDB-327819

The Banzhaf form of Penrose's limit theorem

摘录 / EXCERPT: Let $W\subseteq\mathbb{R}_{\ge 0}$ be a set of weights, let $N^{(0)}\subsetneq N^{(1)}\subsetneq\dots$ be an infinite increasing chain of finite non-empty sets, and put...

conjecture | open | MathDB 页面 | 原文线索

V2-036047 | MATHDB-327926

Ozawa's nuclear embedding conjecture for exact C*-algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be an exact C-algebra, meaning that its reduced C-algebra is exact. Let $I(\mathcal{A})$ denote the injective envelope of $\mathcal{A}$. Ozawa's conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-036048 | MATHDB-327940

The entropy gap conjecture for closed hypersurfaces in dimensions at most six

摘录 / EXCERPT: Let $M^n\subset\mathbb{R}^{n+1}$ be a closed hypersurface with $n\leq 6$, and let $\lambda(M)$ denote its entropy. Let $\lambda(\mathbb{S}^n)$ be the entropy of the round sphere. E...

conjecture | open | MathDB 页面 | 原文线索

V2-036049 | MATHDB-327941

The Gelfand-transform criterion for elementary generation in Banach algebras

摘录 / EXCERPT: Gelfand-transform criterion. $F\in SL_n(R)$ belongs to $E_n(R)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-036050 | MATHDB-327942

Elementary generation in Hardy algebras of the polydisc and ball

摘录 / EXCERPT: Hardy-algebra elementary-generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036051 | MATHDB-327946

Finiteness conjecture for components of the singular set of mean curvature flow

摘录 / EXCERPT: Finiteness conjecture. The space-time singular set has only finitely many components.

conjecture | open | MathDB 页面 | 原文线索

V2-036052 | MATHDB-327955

The linear Kakeya conjecture

摘录 / EXCERPT: For $0<\delta\ll1$, let $\mathbb{T}$ be a collection of rectangular δ-tubes in $\mathbb{R}^d$, each with $d-1$ sides of length δ and one side of length 1, whose ori...

conjecture | open | MathDB 页面 | 原文线索

V2-036053 | MATHDB-327956

The bilinear restriction conjecture for positively curved hypersurfaces

摘录 / EXCERPT: Let S_1 and S_2 be compact smooth $(d-1)$-dimensional submanifolds of $\mathbb{R}^d$, with induced measures $d\sigma_1$ and $d\sigma_2$. They are transversal if the wedge produ...

conjecture | open | MathDB 页面 | 原文线索

V2-036054 | MATHDB-327958

The strengthened multilinear restriction conjecture

摘录 / EXCERPT: Let $S_1,\ldots,S_d$ be transversal smooth codimension-one submanifolds of $\mathbb{R}^d$, with induced measures $d\sigma_j$. Let p' be the Hölder conjugate of p. Strengthened...

conjecture | open | MathDB 页面 | 原文线索

V2-036055 | MATHDB-327959

The nonlinear Brascamp-Lieb conjecture

摘录 / EXCERPT: Let $(\mathbf{L},\mathbf{p})$ be a Brascamp-Lieb datum with finite Brascamp-Lieb constant $\mathbf{BL}(\mathbf{L},\mathbf{p})$. For each $1\leq j\leq k$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-036056 | MATHDB-328018

Unique nondegenerate minimum conjecture for the magnetic wedge edge eigenvalue

摘录 / EXCERPT: Let $\alpha_{0}\in(0,\pi)$, and let $\mu^\mathsf{e}(s,\eta)$ denote the lowest eigenvalue of the Neumann realization of ... on the sector $\mathcal{S}_{\alpha_{0}}$. Unique-minimum c...

conjecture | open | MathDB 页面 | 原文线索

V2-036057 | MATHDB-328082

Balanced algebraic-function conjecture for positive motherbody measures

摘录 / EXCERPT: Balanced algebraic-function conjecture. Every such $P(\mathcal C,z)$ admits a positive motherbody measure.

conjecture | open | MathDB 页面 | 原文线索

V2-036058 | MATHDB-328205

The all-dimensional L^2 curvature bound conjecture

摘录 / EXCERPT: All-dimensional L^2 curvature bound conjecture. There exists $C=C(n,{\rm v})>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036059 | MATHDB-328231

Factorisation conjecture for pre-Hilbert-Schmidt operators

摘录 / EXCERPT: Soient E et F deux espaces de Banach. Un opérateur pré-Hilbert-Schmidt est un opérateur $u:E\to F$ de cette classe; un espace de Hilbert-Schmidt est l'espace à travers lequel u...

conjecture | open | MathDB 页面 | 原文线索

V2-036060 | MATHDB-328240

Mendel-Naor heat-smoothing conjecture for tail functions

摘录 / EXCERPT: Let $1<p<\infty$. Let $f\colon\{-1,1\}^{n}\to\mathbb{R}$ satisfy ... for every $S\subseteq\{1,\ldots,n\}$ with $|S|<k$, where P_t is the heat semigroup and L is its generator. Me...

conjecture | open | MathDB 页面 | 原文线索

V2-036061 | MATHDB-328254

Bonami-Bernicot bilinear decomposition conjecture for products of BMO and H^1 functions

摘录 / EXCERPT: Let ${\mathcal X}$ be the underlying space of homogeneous type, and let $BMO({\mathcal X})$, $H^1({\mathcal X})$, $L^1({\mathcal X})$, and $H^{\log}({\mathcal X})$ denote the space...

conjecture | open | MathDB 页面 | 原文线索

V2-036062 | MATHDB-328267

The one-dimensional formula conjecture for the Ginzburg-Landau energy

摘录 / EXCERPT: Let $E(L)$ be the limiting energy defined by ... and let λ be the function introduced in the source. For $b>0$, define $E^{\rm 1D}(b)=\mathfrak{b}(\alpha_0,b)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-036063 | MATHDB-328336

Linnik-Ibragimov universality conjecture for meromorphic Dirichlet series

摘录 / EXCERPT: Let $f(s)=\sum_{n=1}^{\infty}a_n e^{-\lambda_n s}$ be a Dirichlet series that is meromorphically continuable to the left of its half-plane of absolute convergence. Linnik-Ibragimov...

conjecture | open | MathDB 页面 | 原文线索

V2-036064 | MATHDB-328360

Positivity conjecture for Faber-polynomial error norms at corners

摘录 / EXCERPT: Positivity conjecture for Faber-polynomial error norms. If L has at least one corner with exterior angle different from 0, π, and 2π, then

conjecture | open | MathDB 页面 | 原文线索

V2-036065 | MATHDB-328518

The closed-sets extension conjecture for unital C^*-algebras

摘录 / EXCERPT: Let $\mathscr{T}: \mathbf{KHaus} \rightarrow \mathbf{CMSLat}$ assign to a compact Hausdorff space its lattice of closed sets and to a continuous map the corresponding image homomor...

conjecture | open | MathDB 页面 | 原文线索

V2-036066 | MATHDB-328519

The unitary-invariance conjecture for partial ideals of C^*-algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C^*-algebra. A partial ideal is a choice of a closed ideal I_V in each unital commutative sub-C^*-algebra $V\subset\mathcal{A}$ such that, whene...

conjecture | open | MathDB 页面 | 原文线索

V2-036067 | MATHDB-328520

The invariant partial ideal characterization conjecture for C^*-algebras

摘录 / EXCERPT: Invariant partial ideal conjecture. A partial ideal of a C^*-algebra arises from a total ideal if and only if it is an invariant partial ideal. Consequently, the map...

conjecture | open | MathDB 页面 | 原文线索

V2-036068 | MATHDB-328545

Classification conjecture for stable capillary hypersurfaces in a Euclidean ball

摘录 / EXCERPT: Let M^n, with $n\geq 3$, be a capillary hypersurface in the unit Euclidean ball B^{n+1}. Classification conjecture. The only stable such hypersurfaces are a totally geodesic hy...

conjecture | open | MathDB 页面 | 原文线索

V2-036069 | MATHDB-328550

Positive semidefiniteness conjecture for even-order symmetric quasi-double B0-tensors

摘录 / EXCERPT: Let $\mathcal{B}=(b_{i_1\cdots i_m})$ be a tensor of order m and dimension n. For $i,j\in N$ with $i\neq j$, suppose that ... Here, $\mathcal{B}$ is a quasi-double B_0-tensor w...

conjecture | open | MathDB 页面 | 原文线索

V2-036070 | MATHDB-328553

White's building-dimension characterization for closed cones

摘录 / EXCERPT: Let C be a closed cone in $\mathbb{R}^n$. Its building dimension is ... where $\operatorname{diff}(S):=\{x-y:x,y\in S\}$. White's building-dimension conjecture. The building dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-036071 | MATHDB-328554

White's sharp Hausdorff-dimension conjecture for cone stratifications

摘录 / EXCERPT: Let $\mathcal{C}$ be a compact family of closed cones. Suppose that at each point of a set S, each blow-up of S is contained in some element of $\mathcal{C}$. Let d be the la...

conjecture | open | MathDB 页面 | 原文线索

V2-036072 | MATHDB-328557

The polynomial-power conjecture on Taylor-polynomial intersection roots

摘录 / EXCERPT: Suppose that $f(z)=z^n$, where $n\in\mathbb{N}$ and $n\geq r+1$, and let P_c denote the Taylor polynomial of f of order r at $x=c$. Polynomial-power conjecture. Every pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-036073 | MATHDB-328570

Convergence conjecture for the normalized CR Yamabe flow

摘录 / EXCERPT: Let $(M^3,J,\mathring{\theta})$ be a closed spherical CR 3-manifold with positive Tanaka-Webster scalar curvature and vanishing torsion. Consider the normalized CR Yamabe flow st...

conjecture | open | MathDB 页面 | 原文线索

V2-036074 | MATHDB-328574

White-Wang uniqueness conjecture for translating soliton blowup limits

摘录 / EXCERPT: A hypersurface $M^n\subset\mathbb{R}^{n+1}$ is a translating soliton if its mean curvature H and unit normal vector ν satisfy ... for some nonzero $V\in\mathbb{R}^{n+1}$. Such...

conjecture | open | MathDB 页面 | 原文线索

V2-036075 | MATHDB-328604

Uniform QTT-rank conjecture for scaled matrix exponentials of the one-dimensional Laplacian

摘录 / EXCERPT: Let Δ_1 be the one-dimensional discrete Laplacian on a grid of size N, and let $a,b>0$ be fixed. For $t_k>0$, indexed by $k=-M,\ldots,M$, consider the family of matrix exp...

conjecture | open | MathDB 页面 | 原文线索

V2-036076 | MATHDB-328637

The l2 parametrization conjecture for absolutely continuous factorization measures

摘录 / EXCERPT: Let Σ be the nondecreasing lift associated with the factorization map, and write $d\Sigma=\Sigma' d\theta$ when its measure is absolutely continuous with respect to Lebesgue...

conjecture | open | MathDB 页面 | 原文线索

V2-036077 | MATHDB-328674

Ebenfelt-Huang analyticity conjecture for CR-diffeomorphisms

摘录 / EXCERPT: Ebenfelt-Huang analyticity conjecture. Any C^{∞}-smooth CR (local) diffeomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-036078 | MATHDB-328753

Dimension-free quantitative Gaussian isoperimetric inequality

摘录 / EXCERPT: Let $E\subset\mathbb{R}^n$ be an open set with Lipschitz boundary and Gaussian measure $\gamma(E)=\phi(s)$, where ... Its Gaussian perimeter is ... and define the Gaussian isoperimetri...

conjecture | open | MathDB 页面 | 原文线索

V2-036079 | MATHDB-328778

Hamilton's Type-III asymptotic soliton conjecture

摘录 / EXCERPT: Hamilton's Type-III asymptotic soliton conjecture. Type-III solutions asymptotically approach soliton metrics that are locally homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-036080 | MATHDB-328780

The algebraic soliton linear stability conjecture

摘录 / EXCERPT: An algebraic soliton is a Ricci soliton arising algebraically, including the solvsolitons and nilsolitons considered in the paper. The algebraic soliton linear stability conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-036081 | MATHDB-328874

Conjecture on the mass curve of cubic-quintic ground states

摘录 / EXCERPT: Let P_ω denote the ground state soliton for the cubic-quintic nonlinear Schrödinger equation on $\mathbb{R}^3$, and let $M(P_\omega)$ be its mass. Mass-curve conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-036082 | MATHDB-328875

Conjecture on injectivity of the ground-state beta ratio

摘录 / EXCERPT: Injectivity conjecture. The mapping $\omega\mapsto\beta(\omega)$ is injective; equivalently, ground state solitons are uniquely identified by the ratio $\beta(\omega)$. The claim i...

conjecture | open | MathDB 页面 | 原文线索

V2-036083 | MATHDB-328904

Willmore's conjecture for tori in the three-sphere

摘录 / EXCERPT: Let $\mathcal W(\Sigma)=\int_{\Sigma}(1+H^2)\\,d\mu$ denote the Willmore energy of a compact surface $\Sigma\subset S^3$, where H is its mean curvature. Willmore's conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-036084 | MATHDB-328931

Optimal second-order SSP coefficient conjecture for fully implicit Runge-Kutta methods

摘录 / EXCERPT: Let an s-stage fully implicit Runge-Kutta method have order conditions ... and let $r>0$ satisfy the absolute-monotonicity conditions ... with $I+rA$ invertible and...

conjecture | open | MathDB 页面 | 原文线索

V2-036085 | MATHDB-328943

Nearly unperforated Cuntz semigroups of Z-stable C*-algebras

摘录 / EXCERPT: Let A be a $\mathcal{Z}$-stable C^*-algebra, meaning that $A\cong A\otimes\mathcal{Z}$. Let $\operatorname{Cu}(A)$ be its Cuntz semigroup, and recall that a positively ordered...

conjecture | open | MathDB 页面 | 原文线索

V2-036086 | MATHDB-328998

The exact Carleson constant formula for postorder-rearranged dyadic collections

摘录 / EXCERPT: Let $N\in\mathbb{N}$ with $N\geq 2$ and let $0\leq \ell\leq N-2$. For the dyadic collection ${\mathcal E}^N_{\ell,0}$ and the postorder rearrangement operator τ_N, write...

conjecture | open | MathDB 页面 | 原文线索

V2-036087 | MATHDB-329082

Power-improved two-weight Bergman inequality

摘录 / EXCERPT: Power-improved two-weight conjecture. If $(\mu_1^r,\mu_2^r)\in A_p^+(\mathbb{R}_{+}^2)$ for some $r>1$, then the displayed inequality holds for some $C>0$. This is presented as a v...

conjecture | open | MathDB 页面 | 原文线索

V2-036088 | MATHDB-329178

The dimension formula for multi-scale constructions

摘录 / EXCERPT: Let $A(y)$ be the almost periodic filter represented by a polynomial P, let $M(A)=M(P)$ be its Mahler measure, and let λ be the scaling parameter. For a Borel measure f...

conjecture | open | MathDB 页面 | 原文线索

V2-036089 | MATHDB-329181

Trace-norm commutator conjecture for matrix functions

摘录 / EXCERPT: Trace-norm commutator conjecture. For certain functions f still to be determined, there exists a constant c independent of d such that

conjecture | open | MathDB 页面 | 原文线索

V2-036090 | MATHDB-329230

Non-injectivity conjecture for average-distance minimizers below the critical exponent

摘录 / EXCERPT: Let μ be a measure and, for $\lambda>0$ and $1\leq p<2$, consider the average-distance minimization problem whose global minimizers are parameterized curves. Non-injectivity co...

conjecture | open | MathDB 页面 | 原文线索

V2-036091 | MATHDB-329236

The Dixmier-Douady vanishing conjecture for subquotients of exponential Lie group C^*-algebras

摘录 / EXCERPT: Let G be an exponential Lie group and let $C^*(G)$ be its C^*-algebra. A continuous-trace subquotient is a quotient $\mathcal J_2/\mathcal J_1$ of $C^*(G)$ by ideals...

conjecture | open | MathDB 页面 | 原文线索

V2-036092 | MATHDB-329254

Equilateral-mesh Jacobian spectral conjecture

摘录 / EXCERPT: Equilateral-mesh spectral conjecture. The Jacobian matrix of Θ at x has eigenvalues of absolute value <1 except for exactly four. In particular, the group orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-036093 | MATHDB-329335

Bellissard's gap labeling conjecture for quasicrystals without magnetic field

摘录 / EXCERPT: Let Λ be the quasicrystal with transversal Ω_{trans}, and let $\mathcal{A}_{\theta=0}$ be its twisted groupoid C^*-algebra for zero magnetic field. The trace induc...

conjecture | open | MathDB 页面 | 原文线索

V2-036094 | MATHDB-329378

Popa-Vaes app-paving conjecture and universal paving-size bound

摘录 / EXCERPT: Let $A\subset M$ be a MASA in a von Neumann algebra, and suppose first that there is a normal conditional expectation from M onto A. The app-paving conjecture. Every MASA in a...

conjecture | open | MathDB 页面 | 原文线索

V2-036095 | MATHDB-329430

Curvilinear-triangle conjecture for spectral and Painleve zero distributions

摘录 / EXCERPT: Curvilinear-triangle conjecture. As $n\to\infty$, the sequences

conjecture | open | MathDB 页面 | 原文线索

V2-036096 | MATHDB-329471

Conjecture on Liouville transformation methods for complex-valued coefficients

摘录 / EXCERPT: Liouville transformation conjecture. Theorem holds, and the proposed method works under these weaker conditions on p and r.

conjecture | open | MathDB 页面 | 原文线索

V2-036097 | MATHDB-329477

The real-analytic boundary conjecture for martingale connectedness

摘录 / EXCERPT: Let $w\subset\mathbb{R}^d$ be a bounded closed connected set with non-empty interior, and suppose that its boundary is a real-analytic submanifold of $\mathbb{R}^d$. The real-analy...

conjecture | open | MathDB 页面 | 原文线索

V2-036098 | MATHDB-329501

Asymptotic zero distribution conjecture for exceptional orthogonal polynomials

摘录 / EXCERPT: Let an exceptional orthogonal polynomial family have weight ... where $W_0(x)$ is a classical orthogonal-polynomial weight and $\eta(x)$ is a polynomial whose degree equals the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-036099 | MATHDB-329629

The Chern-Yamabe conjecture for Hermitian conformal structures

摘录 / EXCERPT: Let X be a compact complex manifold of complex dimension n, and let $\\{\omega\} \in \mathcal{C}^H_X$ be a Hermitian conformal structure on X. For a representative...

conjecture | open | MathDB 页面 | 原文线索

V2-036100 | MATHDB-329662

A supercongruence for truncated hypergeometric series and the p-adic Gamma function

摘录 / EXCERPT: The truncated hypergeometric supercongruence. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036101 | MATHDB-329701

Fractional Lieb-Thirring coupling conjecture for the optimal constant

摘录 / EXCERPT: Fractional Lieb-Thirring coupling conjecture. The optimal constant also satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036102 | MATHDB-329735

The limiting-hyperbola conjecture for domination roots of generalized friendship graphs

摘录 / EXCERPT: Let $F_{4,n}$ be the generalized friendship graph with parameter n, and consider the complex zeros of its domination polynomial $D(F_{4,n},x)$ as n varies. Limiting-hyperbola c...

conjecture | open | MathDB 页面 | 原文线索

V2-036103 | MATHDB-329794

The finite singular times conjecture for mean curvature flow

摘录 / EXCERPT: Let M_t be the evolving hypersurface described above. Finite singular times conjecture. For M_t as above, the evolving hypersurface is completely smooth except at finitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-036104 | MATHDB-329895

System singularity and system pole conjecture for Hermite-Padé approximants

摘录 / EXCERPT: Let ${\bf f}=(f_1,\ldots,f_d)$ and ${\bf m}=(m_1,\ldots,m_d)\in\mathbb{Z}_+^d\setminus\{{\bf 0}\}$, and suppose that the denominator $Q_{n,\bf m}$ is unique for all sufficiently la...

conjecture | open | MathDB 页面 | 原文线索

V2-036105 | MATHDB-329936

Drori-Teboulle conjecture for the smooth convex gradient method

摘录 / EXCERPT: Drori-Teboulle conjecture. Every sequence of iterates generated in this way satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036106 | MATHDB-329937

Strongly convex gradient-norm worst-case conjecture

摘录 / EXCERPT: Strongly convex gradient-norm conjecture. Every sequence of iterates generated in this way satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036107 | MATHDB-329964

Asymptotic equivalence of uniform Riemann sums and infinite-limit Riemann sums

摘录 / EXCERPT: Let a continuous curve be given, and consider a uniform Riemann sum for it together with a Riemann sum whose limit is infinite. Asymptotic equivalence conjecture. A uniform Riemann...

conjecture | open | MathDB 页面 | 原文线索

V2-036108 | MATHDB-329966

The iterated-logarithm infinity conjecture

摘录 / EXCERPT: Let $f(x)$ be a function such that ... and let $k=d4a$ denote an infinite iteration index. Iterated-logarithm infinity conjecture. When x reaches infinity before k, one has ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-036109 | MATHDB-329967

The L-function comparison conjecture

摘录 / EXCERPT: L-function comparison conjecture. Every L-function at infinity is ultimately continuous and monotonic. If f and g are L-functions, then in

conjecture | open | MathDB 页面 | 原文线索

V2-036110 | MATHDB-329968

The infinite-interval L-function comparison conjecture

摘录 / EXCERPT: Infinite-interval comparison conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036111 | MATHDB-329969

The infinite-sequence monotonicity determination claim

摘录 / EXCERPT: Let $a_n|_{n=d4a}$ be a sequence evaluated at infinity, and let z denote a relation in $*G$. Infinite-sequence monotonicity claim. The monotonicity of $a_n|_{n=d4a}$ can be deter...

conjecture | open | MathDB 页面 | 原文线索

V2-036112 | MATHDB-329970

The infinite-integer ratio transfer conjecture

摘录 / EXCERPT: Let $\mathbb{N}_{\infty}$ denote the infinite integers. Infinite-integer ratio transfer conjecture. There exist ratios of infinite integers of the form ... that can be transferred to...

conjecture | open | MathDB 页面 | 原文线索

V2-036113 | MATHDB-329996

Suita's conjecture on the curvature of the fundamental metric

摘录 / EXCERPT: Suita's conjecture. The Gaussian curvature of the fundamental metric of X is at most -4. Moreover, it is exactly -1 if and only if X is the unit disk.

conjecture | open | MathDB 页面 | 原文线索

V2-036114 | MATHDB-330051

Sharp Poisson transform inequality on the Heisenberg group

摘录 / EXCERPT: Sharpness and optimizer conjecture. There exists a sharp constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036115 | MATHDB-330054

Generic bumpy boundary conjecture for minimal submanifolds

摘录 / EXCERPT: Let G be a finite set of isometries of $\mathbf{R}^n$. A smooth, closed, G-invariant, embedded submanifold of $\mathbf{R}^n$ is one whose boundary-bounded minimal submanifolds...

conjecture | open | MathDB 页面 | 原文线索

V2-036116 | MATHDB-330056

Extension of the lifted operator theorem to matrix Muckenhoupt weights

摘录 / EXCERPT: Let $1<p<\infty$ and let $W=(w_{ij})\colon \mathbb{R}^n\times \mathbb{R}^m\to\mathbb{C}^{N\times N}$ be a matrix weight in the set of Muckenhoupt weights...

conjecture | open | MathDB 页面 | 原文线索

V2-036117 | MATHDB-330064

The Real Polarization Problem

摘录 / EXCERPT: Let $d\geq 2$, let $x_1,\ldots,x_d$ be unit vectors in $\mathbb{R}^d$, and let v be a unit vector in $\mathbb{R}^d$. The dth linear polarization constant $c_d(\mathcal{H})$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-036118 | MATHDB-330101

Saff-Varga width conjecture for zeros of partial sums

摘录 / EXCERPT: Saff-Varga's width conjecture. There exists an infinite sequence of positive integers N such that no region $S_\theta(\tau)$ is devoid of all zeros of all partial sums $p_n(z)$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-036119 | MATHDB-330102

Edrei-Saff-Varga modified width conjecture

摘录 / EXCERPT: Let f be an entire function of positive, finite order λ, and let $p_n(z)$ denote its n^th partial sum. Edrei-Saff-Varga's modified width conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-036120 | MATHDB-330126

The strong Novikov conjecture

摘录 / EXCERPT: Let $B\pi$ be the classifying space of a discrete group π, and consider the assembly map ... Here $C^*(\pi)$ is the group C^*-algebra. Strong Novikov conjecture. The map...

conjecture | open | MathDB 页面 | 原文线索

V2-036121 | MATHDB-330173

Existence of bounded analytic functions that are not Cowen-Douglas functions

摘录 / EXCERPT: Let $m\in\mathbb{N}$. An m-folder complex analytic function is a function $\phi(z)\in\mathcal{H}^{\infty}(\mathbb{D})$ with the stated multiplicity m, where $\mathbb{D}$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-036122 | MATHDB-330278

The nonintegrability conjecture for PV refinable distributions

摘录 / EXCERPT: Let λ be a Pisot-Vijayaraghavan (PV) number, let f be a nonzero refinable distribution for dilation by λ, and suppose its translations lie in ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-036123 | MATHDB-330280

The noncommutative Brouwer retract conjecture for C*-algebras

摘录 / EXCERPT: Noncommutative Brouwer retract conjecture. There is no $*$-homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-036124 | MATHDB-330369

Non-concentrated directions conjecture for projections of planar sets

摘录 / EXCERPT: Assume that $K \subset B(0,1) \subset \mathbb{R}^{2}$ is a set with $\mathcal{H}^{1}_{\infty}(K) \sim 1$. Let $E \subset S^{1}$ be a δ-separated set of directions with cardi...

conjecture | open | MathDB 页面 | 原文线索

V2-036125 | MATHDB-330406

Asymptotic structure conjecture for two-dimensional self-shrinkers

摘录 / EXCERPT: Asymptotic structure conjecture. For each U_i, either, as $\lambda\to\infty$, $\lambda^{-1}U_i$ converges locally smoothly to a cone smooth except at the origin $\mathbf{0}$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-036126 | MATHDB-330407

Plane-or-cylinder asymptotic conjecture for self-shrinkers

摘录 / EXCERPT: Plane-or-cylinder asymptotic conjecture. As $\lambda\to\infty$, $\Sigma-\lambda\mathbf{v}_0$ converges locally smoothly to a plane or a self-shrinking cylinder of multiplicity one.

conjecture | open | MathDB 页面 | 原文线索

V2-036127 | MATHDB-330424

The loose extension is equivalent to the adjunction matrix

摘录 / EXCERPT: Let $\Phi:{\mathbb A}\looparrowright{\mathbb B}$ be a matrix. Its loose extension is the matrix ... with objects (a,b) and structure maps...

conjecture | open | MathDB 页面 | 原文线索

V2-036128 | MATHDB-330425

Minimal bicompletion conjecture for tight extensions of matrices

摘录 / EXCERPT: Let $\Phi:{\mathbb A}\xrightarrow{}{\mathbb B}$ be a matrix. Its tight extension is the matrix ... obtained from the loose extension by retaining the morphisms that are mono and whos...

conjecture | open | MathDB 页面 | 原文线索

V2-036129 | MATHDB-330455

The factor-valued pinching conjecture for type-II-infinity and type-III factors

摘录 / EXCERPT: Factor-valued pinching conjecture. If $A\in{\mathfrak{R}}$ satisfies $W_e^{\mathfrak{R}}(A)\supset{\mathcal{D}}$ and $\{X_i\}_{i=1}^{\infty}\subset{\mathfrak{R}}$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-036130 | MATHDB-330474

Characterization of the maximal igregularity class by \mathcal N

摘录 / EXCERPT: Let $\mathcal N$ be the class of compact sets $A\subset\mathbb{R}^d$ for which there exists a monotone sequence of compact smooth domains ... with the masses of the normal cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-036131 | MATHDB-330475

Almost-everywhere sublevel sets of MA functions belong to \mathcal N

摘录 / EXCERPT: Let $\operatorname{MA}(\mathbb{R}^d)$ denote the class used in the source, and let $\mathcal N$ be the class of compact sets admitting decreasing smooth-domain approximations with...

conjecture | open | MathDB 页面 | 原文线索

V2-036132 | MATHDB-330498

The Schwartz-algebra conjecture for Bernstein components

摘录 / EXCERPT: Let F be a non-archimedean local field, G a connected reductive algebraic group over F, and ${\mathfrak s}=[L,\omega]_G$ an inertial equivalence class. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036133 | MATHDB-330538

Separate subharmonicity conjecture

摘录 / EXCERPT: Separate subharmonicity conjecture. If F is subharmonic with respect to every pair (x_j,x_{j+d}), then F is non-negative. This geometric statement is intended to provide the...

conjecture | open | MathDB 页面 | 原文线索

V2-036134 | MATHDB-330539

Endpoint asymptotic conjecture for anisotropic differential operators

摘录 / EXCERPT: Endpoint asymptotic conjecture. If T_1 cannot be expressed as a linear combination of the other T_j, then

conjecture | open | MathDB 页面 | 原文线索

V2-036135 | MATHDB-330540

Geometric lower-bound conjecture for separately convex homogeneous functions

摘录 / EXCERPT: Let $F:\mathbb{R}^d\to\mathbb{R}$ be a Lipschitz separately convex function that is p-positively homogeneous, meaning ... Suppose that $F(x)\leqslant |x|^p$. Geometric lower-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-036136 | MATHDB-330569

The multilinear Hilbert transform boundedness conjecture

摘录 / EXCERPT: Multilinear Hilbert transform boundedness conjecture. For $k\geq 1$ and exponents $1<p_1,\dots,p_k,p<\infty$ satisfying $\frac{1}{p}=\frac{1}{p_1}+\dots+\frac{1}{p_k}$, the truncat...

conjecture | open | MathDB 页面 | 原文线索

V2-036137 | MATHDB-330581

W-cyclicity conjecture for distributive subfactor planar algebras

摘录 / EXCERPT: W-cyclicity conjecture. Every distributive subfactor planar algebra is w-cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-036138 | MATHDB-330614

The higher-dimensional fractional $T1$ conjecture

摘录 / EXCERPT: Let $\mathbf{T}^{\alpha,n}$ denote an elliptic vector of standard α-fractional singular integrals in $\mathbb{R}^{n}$, and let σ and ω be measures on...

conjecture | open | MathDB 页面 | 原文线索

V2-036139 | MATHDB-330642

Nazarov-Peller weak type Lipschitz conjecture

摘录 / EXCERPT: Let H be a Hilbert space, let $L_1(H)$ denote the trace-class operators on H, and let $L_{1,cinfty}(H)$ denote the weak-L_1 space of compact operators T whose singular valu...

conjecture | open | MathDB 页面 | 原文线索

V2-036140 | MATHDB-330680

The hullizability conjecture for joint spectra of commuting semiclassical unitary operators

摘录 / EXCERPT: Consider pairwise commuting unitary semiclassical operators $U_1(\hbar), \ldots, U_d(\hbar)$ with joint principal symbol $F=(f_0^1,\ldots,f_0^d)$, and assume that $F(M)$ is closed....

conjecture | open | MathDB 页面 | 原文线索

V2-036141 | MATHDB-330701

Relative Novikov and Baum-Connes conjectures

摘录 / EXCERPT: Let $Y\subseteq X$ be aspherical compact spaces, and let ... be the maximal relative Baum-Connes map. If $j\colon \pi_1(Y)\to \pi_1(X)$ is injective, let ... be the reduced relative Ba...

conjecture | open | MathDB 页面 | 原文线索

V2-036142 | MATHDB-330723

Computable fixed-point selection for nonexpansive mappings on two-dimensional Banach spaces

摘录 / EXCERPT: Let E be a uniformly convex, smooth, computable Banach space of dimension two, and let $K\subseteq E$ be bounded, convex, and located, meaning that its distance function is compu...

conjecture | open | MathDB 页面 | 原文线索

V2-036143 | MATHDB-330724

Projection invariance of fixed points for sunny nonexpansive retractions

摘录 / EXCERPT: Let E be a smooth and uniformly convex Banach space of dimension two. Let $K\subseteq E$ be nonempty, closed, bounded, and convex, and let $f\colon K\to E$ be nonexpansive with...

open_problem | open | MathDB 页面 | 原文线索

V2-036144 | MATHDB-330788

Rigorous Wilson loop observable existence conjecture

摘录 / EXCERPT: Existence conjecture. $\operatorname{WLO}_{rig}(L)$ is well-defined. In particular, all limits involved exist.

conjecture | open | MathDB 页面 | 原文线索

V2-036145 | MATHDB-330795

Lee-Lerario-Lundberg zero-count bound for harmonic polynomial equations

摘录 / EXCERPT: Lee-Lerario-Lundberg bound. The number of solutions of this equation is bounded by

conjecture | open | MathDB 页面 | 原文线索

V2-036146 | MATHDB-330811

Greene-Krantz finite-type conjecture for pseudoconvex domains

摘录 / EXCERPT: Let Ω be a bounded pseudoconvex domain with C^∞ boundary, and let $\mathcal{L}(\Omega)$ denote the limit set of its automorphism group. A point...

conjecture | open | MathDB 页面 | 原文线索

V2-036147 | MATHDB-330831

Assem's stability conjecture for nilpotent orbital integrals of quasi-split unitary groups

摘录 / EXCERPT: Assem's stability conjecture. For every stable nilpotent orbit $\mathcal{O}$ of $U_n(E/F)$, there should be, up to constant, a unique linear combination of orbital integrals among...

conjecture | open | MathDB 页面 | 原文线索

V2-036148 | MATHDB-330855

Equality of regular and generic non-tangential singular sets

摘录 / EXCERPT: Let μ be the measure under consideration, and let $\mathcal{S}_{nt}^{reg}(\mu)$ and $\mathcal{S}_{nt}^{gen}(\mu)$ denote the regular and generic parts of its non-tangential sin...

conjecture | open | MathDB 页面 | 原文线索

V2-036149 | MATHDB-330856

Meagreness of the non-tangential singular set

摘录 / EXCERPT: Let μ be a measure under consideration, and let $\mathcal{S}_{nt}^{sing}(\mu)$ denote its non-tangential singular set. Meagreness conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-036150 | MATHDB-330857

Boundary characterization at regular isolated-set points

摘录 / EXCERPT: Let $\mathcal{S}_{nt}^{iso}(\mu)$ denote the isolated-point part of the non-tangential singular set, let $\mathcal{E}$ be the relevant edge set, and let $\partial\mathcal{L}(x)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-036151 | MATHDB-330859

Permutation lower-bound conjecture for partial Sylvester sums

摘录 / EXCERPT: Permutation lower-bound conjecture. For every positive integer n and every permutation $\sigma:\{1,2,\ldots,2^n\}\to\{1,2,\ldots,2^n\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-036152 | MATHDB-330876

Long-time existence conjecture for the $\bar{\partial}$-harmonic map heat flow

摘录 / EXCERPT: Let M be a compact almost Hermitian manifold and N an almost Hermitian manifold. Suppose that the sectional curvature of N is non-positive, that ∇J_N is bounded, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036153 | MATHDB-330893

Leinster-Willerton planar magnitude conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^2$ be a compact set suitable for the perimeter and classical Euler characteristic to be defined. Here $|tX|$ denotes the magnitude of the dilation of X...

conjecture | open | MathDB 页面 | 原文线索

V2-036154 | MATHDB-330913

The non-frame conjecture for singular counterexamples to the B-spline Gabor frame property

摘录 / EXCERPT: Non-frame conjecture. The Gabor system $\mathcal{G}(B_2,a_0,b_0)$ is not a frame. Furthermore, it is not a frame along the hyperbolas

conjecture | open | MathDB 页面 | 原文线索

V2-036155 | MATHDB-330969

The adjoint restriction conjecture for the paraboloid

摘录 / EXCERPT: Let S be a non-empty smooth compact subset of the paraboloid ... where $n\geq1$, and let $\mathrm{d}\sigma$ be the pull-back of n-dimensional Lebesgue measure under the projectio...

conjecture | open | MathDB 页面 | 原文线索

V2-036156 | MATHDB-330977

Solynin's area-growth conjecture for hyperbolic cells

摘录 / EXCERPT: Let $D_n^{(s)}$ and $(D_n^*)^{(s)}$ be the objects defined before, with $s\geq 0$ and $n\geq 3$. Solynin's area-growth conjecture. One has ... Equality holds only if D_n is a rotat...

conjecture | open | MathDB 页面 | 原文线索

V2-036157 | MATHDB-331002

Sills-Zeilberger asymptotic conjecture for $C_{011}(N)$

摘录 / EXCERPT: Let $C_{011}(N)$ denote the coefficient of $(q-1)^{-1}$ in the partial-fraction decomposition of $\prod_{j=1}^N(1-q^j)^{-1}$. Let $w_0\approx0.916-0.182i$ be the specified solution...

conjecture | open | MathDB 页面 | 原文线索

V2-036158 | MATHDB-331003

Generalized asymptotic conjecture for the coefficients $C_{01\ell}(N)$

摘录 / EXCERPT: Generalized asymptotic conjecture. For every positive integer m,

conjecture | open | MathDB 页面 | 原文线索

V2-036159 | MATHDB-331008

Conjecture on the optimal set for the quantitative isoperimetric functional

摘录 / EXCERPT: Let $\mathcal{F}$ be the quantitative isoperimetric functional considered in the paper, and let Ω_0 denote an optimal set for $\mathcal{F}$. Suppose that Ω_0 is conne...

conjecture | open | MathDB 页面 | 原文线索

V2-036160 | MATHDB-331009

Numerical optimality conjecture for the mask set

摘录 / EXCERPT: Let M_0 be the mask-shaped set described in the paper, with parameters α, θ, and x_0. The quantitative isoperimetric functional is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-036161 | MATHDB-331054

The magnetic gap-labelling conjecture for minimal actions

摘录 / EXCERPT: Magnetic gap-labelling conjecture. The magnetic gap-labelling group is contained in the subgroup of $\mathbb R$ given by

conjecture | open | MathDB 页面 | 原文线索

V2-036162 | MATHDB-331073

The aliasing-norm formula for consistent sampling from oblique decompositions

摘录 / EXCERPT: Aliasing-norm conjecture. The formula in Proposition should hold for the consistent sampling corresponding to the oblique decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-036163 | MATHDB-331124

Bañuelos's sharp bound conjecture for stable Lévy multipliers

摘录 / EXCERPT: Bañuelos's conjecture. For every $1<p<\infty$, T_{m_r} is bounded on $L^p(\mathbb{R}^n)$ and

conjecture | open | MathDB 页面 | 原文线索

V2-036164 | MATHDB-331148

The positivity conjecture for derivatives of the sum of squared logarithms

摘录 / EXCERPT: Let $e=(e_1,\ldots,e_n)\in\mathbb{R}_+^n$, and let $f\circ\varphi$ be the composition defined in the paper, where φ parametrizes the roots associated with e. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036165 | MATHDB-331195

The maximal function conjecture for the two-dimensional Schrödinger equation

摘录 / EXCERPT: Let $A(1)=\{\xi:|\xi|\sim 1\}$. For $R\geq 1$, let $B(0,R)$ denote the ball of radius R centered at the origin, and let $e^{it\Delta}f$ be the free Schrödinger evolution of a fun...

conjecture | open | MathDB 页面 | 原文线索

V2-036166 | MATHDB-331200

Chalker-Mehlig conjecture for the two-point eigenvector overlap function

摘录 / EXCERPT: Chalker-Mehlig conjecture. (i) For any two points z_1,z_2 such that $|z_1|<1$, $|z_2|<1$ and $|z_1-z_2|>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-036167 | MATHDB-331235

Newman's L^1-flatness conjecture for Littlewood polynomials

摘录 / EXCERPT: A Littlewood polynomial is an analytic polynomial whose coefficients are all 7. For a polynomial P of degree n, write ... for its L^1 norm on the unit circle. Newman's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-036168 | MATHDB-331303

The analytic zero-divisor conjecture for torsion-free groups

摘录 / EXCERPT: Analytic zero-divisor conjecture. If $0 \neq \alpha \in \mathbb{C}G$ and $0 \neq f \in \ell^2(G)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036169 | MATHDB-331362

Hilbert-space Erdős discrepancy conjecture

摘录 / EXCERPT: Let $x_1,x_2,\dots$ be a sequence of unit vectors in a real or complex Hilbert space. Hilbert-space Erdős discrepancy conjecture. For every C there exist n and d such that .....

conjecture | open | MathDB 页面 | 原文线索

V2-036170 | MATHDB-331386

Mean value decoupling conjecture for two-dimensional polynomial manifolds

摘录 / EXCERPT: Mean value decoupling conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-036171 | MATHDB-331419

Marshall's asymptotic tail variance conjecture for Bloch functions

摘录 / EXCERPT: Marshall's conjecture. For every $\varphi\in\mathcal{S}$,

conjecture | open | MathDB 页面 | 原文线索

V2-036172 | MATHDB-331422

The sharp trilinear inequality for antipodally symmetric functions on the circle

摘录 / EXCERPT: The trilinear inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-036173 | MATHDB-331423

Hypomonotonicity-to-quadraticity conjecture for proximally smooth sets

摘录 / EXCERPT: Hypomonotonicity-to-quadraticity conjecture. Then A belongs to $\Omega_N^{\psi_1}(r)$, where

conjecture | open | MathDB 页面 | 原文线索

V2-036174 | MATHDB-331425

Vanishing translation parameters for spectral almost-Parseval-frame towers

摘录 / EXCERPT: Let (N_j,B_j) be the almost-Parseval-frame tower defined in Theorem 0.1, and let bc be its associated measure. The measure bc is spectral if there exists a set...

conjecture | open | MathDB 页面 | 原文线索

V2-036175 | MATHDB-331471

The dimension conjecture for real hypersurfaces in complex two-space

摘录 / EXCERPT: Let M be a 3-dimensional connected real-analytic CR-manifold of hypersurface type, locally embedded in ${\mathbb C}^2$, and let $\mathop{\mathfrak {hol}}\nolimits(M,p)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-036176 | MATHDB-331507

The dyadic Bellman-function concave-envelope conjecture

摘录 / EXCERPT: Let $Q\subset\mathbb{R}^d$ be a cube, let $\mathrm{BMO}_{\varepsilon}^{\mathrm{dyad}}(Q)$ denote the dyadic BMO class with parameter ε, and define ... An extension...

conjecture | open | MathDB 页面 | 原文线索

V2-036177 | MATHDB-331533

Strictness of the weak-product inclusion for Dirichlet-series spaces

摘录 / EXCERPT: Strict weak-product inclusion conjecture. The inclusion between the standard weak product and its skew counterpart is strict:

conjecture | open | MathDB 页面 | 原文线索

V2-036178 | MATHDB-331558

The conjecture relating $B(\chi,[1,b])$ to the p-adic local Langlands correspondence

摘录 / EXCERPT: Local Langlands conjecture for $B(\chi,[1,b])$. Up to a twist by a character, $B(\chi,[1,b])$ is isomorphic to $\Pi(V_{\chi,[1,b]}^\vee)$ as a Banach space representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-036179 | MATHDB-331619

Brown-Shields conjecture on cyclic functions in the Dirichlet space

摘录 / EXCERPT: Let D_1 denote the Dirichlet space, and let $f\in D_1$. A function is cyclic in D_1 if the polynomials multiplied by it are dense in D_1. A function $f\in D_1$ is outer if it...

conjecture | open | MathDB 页面 | 原文线索

V2-036180 | MATHDB-331629

The linear weighted bound for rough homogeneous singular integrals

摘录 / EXCERPT: Let T_{Ω} be the rough homogeneous singular integral associated with an angular function Ω, let $w\in A_2$ be a Muckenhoupt weight, and let $[w]_{A_2}$ denote its cla...

conjecture | open | MathDB 页面 | 原文线索

V2-036181 | MATHDB-331630

Linear weighted bounds for the Beurling-type operators B^m

摘录 / EXCERPT: Let B^m denote the mth operator in the family considered in the paper, and let $w\in A_2$ with classical A_2 characteristic $[w]_{A_2}$. Linear A_2 conjecture for B^m. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-036182 | MATHDB-331673

Endpoint multilinear k_j-restriction conjecture

摘录 / EXCERPT: Endpoint multilinear k_j-restriction conjecture. When the U_j are sufficiently small,

conjecture | open | MathDB 页面 | 原文线索

V2-036183 | MATHDB-331674

Endpoint Brascamp-Lieb type restriction conjecture

摘录 / EXCERPT: Endpoint Brascamp-Lieb type restriction conjecture. When the U_j are sufficiently small,

conjecture | open | MathDB 页面 | 原文线索

V2-036184 | MATHDB-331730

Desalvo-Pak ratio conjecture for the partition function

摘录 / EXCERPT: Let $p(n)$ denote the number of partitions of n. Desalvo-Pak's conjecture. For every integer $n\geq45$, ... The coefficient $\frac{\pi}{\sqrt{24}}$ is asymptotically best possible,...

conjecture | open | MathDB 页面 | 原文线索

V2-036185 | MATHDB-331751

The similarity-degree conjecture for Fourier algebras

摘录 / EXCERPT: Let G be a locally compact group. Write $\widetilde{\mathrm{A}}(G)_1$ for the unit ball of the unitization of the Fourier algebra $\mathrm{A}(G)$, let $\mathcal{C}_0(G)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-036186 | MATHDB-331831

Uniform singular-vector bound conjecture for powers of the difference matrix

摘录 / EXCERPT: Singular-vector bound conjecture. There exists a constant c such that, for any r, the singular vectors V of D^{-r} satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-036187 | MATHDB-331874

Hytönen-Lacey one-supremum conjecture for Calderón-Zygmund operators

摘录 / EXCERPT: Hytönen-Lacey one-supremum conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-036188 | MATHDB-331882

The signed small ball conjecture for multiparameter Haar functions

摘录 / EXCERPT: Let $d\ge 2$. Let $\mathcal D^d$ be the dyadic rectangles in $[0,1)^d$, and let h_R be the associated multiparameter Haar function. For each $n\in\mathbb N_0$, consider rectangle...

conjecture | open | MathDB 页面 | 原文线索

V2-036189 | MATHDB-331919

Approximate structure conjecture for low-degree Boolean functions on the symmetric group

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathcal{A} \subset S_n$, and let $t \in \mathbb{N}$. Let f be the characteristic function of $\mathcal{A}$, and let U_t be the degree-...

conjecture | open | MathDB 页面 | 原文线索

V2-036190 | MATHDB-331956

Multilinear Marcinkiewicz integral extrapolation conjecture

摘录 / EXCERPT: Multilinear Marcinkiewicz integral extrapolation conjecture. If μ is bounded from $L^{q_1}({\mathbb R}^n)\times\cdots\times L^{q_m}({\mathbb R}^n)$ to $L^q({\mathbb R}^n)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-036191 | MATHDB-332034

Amenability of rank algebras implies hyperfiniteness

摘录 / EXCERPT: Amenable-rank hyperfiniteness conjecture. The rank function rk_ρ is hyperfinite.

conjecture | open | MathDB 页面 | 原文线索

V2-036192 | MATHDB-332035

Embedding countable amenable rank algebras into the continuous algebra

摘录 / EXCERPT: Continuous-algebra embedding conjecture. Every countable-dimensional amenable rank algebra embeds into M_K. Conversely, every countable-dimensional rank subalgebra of M_K is am...

conjecture | open | MathDB 页面 | 原文线索

V2-036193 | MATHDB-332048

Discrete-spectrum base-change conjecture for complex symmetric spaces

摘录 / EXCERPT: Let G be a real reductive group and let π be an irreducible unitary representation of $G(\mathbb{C})$. Discrete-spectrum base-change conjecture. There is at most one pure inn...

conjecture | open | MathDB 页面 | 原文线索

V2-036194 | MATHDB-332123

Pierce's discrete quadratic Carleson conjecture on \(\ell^2(\mathbb Z)\)

摘录 / EXCERPT: Let $f\in\ell^2(\mathbb Z)$ and write $e(t)=e^{2\pi i t}$. Pierce's discrete quadratic Carleson conjecture. The inequality ... holds. This is a discrete analogue of Stein's maximal q...

conjecture | open | MathDB 页面 | 原文线索

V2-036195 | MATHDB-332140

Non-invariance of regularized Riesz energies for even-dimensional submanifolds

摘录 / EXCERPT: Let M be a closed submanifold of Euclidean space, let $m=\dim M$, and let $E_M(z)$ denote its regularized Riesz energy. The scale-anomaly term $R_M(-2m)$ measures the failure of...

conjecture | open | MathDB 页面 | 原文线索

V2-036196 | MATHDB-332265

The multilinear restriction conjecture

摘录 / EXCERPT: Multilinear restriction conjecture. Under these assumptions, there is a constant C, depending on finitely many derivatives of the Σ_i, the sets U_i, ν, and n, such...

conjecture | open | MathDB 页面 | 原文线索

V2-036197 | MATHDB-332294

Stabilization conjecture for the spaces ${}_{\Sigma}L_p^k$

摘录 / EXCERPT: Stabilization conjecture. There exists l such that

conjecture | open | MathDB 页面 | 原文线索

V2-036198 | MATHDB-332295

Density conjecture for annihilators of ${}_{\Sigma}L_p$

摘录 / EXCERPT: Let Σ be a smooth manifold with the L_p restriction property, and let Π_{Σ} denote the restriction map to Σ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-036199 | MATHDB-332296

Uncertainty conjecture for distributions supported on a surface

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^d$ be the surface under consideration, let $\mathcal{S}'(\mathbb{R}^d)$ be the space of tempered distributions, and let $\mathcal{F}^{-1}$ denote the i...

conjecture | open | MathDB 页面 | 原文线索

V2-036200 | MATHDB-332297

Bochner-Riesz regularity conjecture for ${}_{\mathrm{S}^{d-1}}L_p^k$

摘录 / EXCERPT: Let ${}_{\mathrm{S}^{d-1}}L_p^k$ be the Hardy-type space associated with the unit sphere, and let the Bochner-Riesz operator of order $\lambda>0$ be given as in the source. Bochner...

conjecture | open | MathDB 页面 | 原文线索

V2-036201 | MATHDB-332298

Interpolation stability conjecture for the scale ${}_{\Sigma}L_p$

摘录 / EXCERPT: Let ${}_{\Sigma}L_p$ denote the family of spaces associated with the surface Σ, and let $[\,\]_{\theta}$ and $(\,\)_{\theta,r}$ denote the complex and real interpolation m...

conjecture | open | MathDB 页面 | 原文线索

V2-036202 | MATHDB-332307

Wu's second asymptotic conjecture for binomial sums

摘录 / EXCERPT: Let m be a positive integer, and let γ denote the Euler-Mascheroni constant. Wu's second asymptotic conjecture. ... This is one of three asymptotic conjectures used to reduc...

conjecture | open | MathDB 页面 | 原文线索

V2-036203 | MATHDB-332318

The weighted Bhatia-Kittaneh conjecture for positive semidefinite matrices

摘录 / EXCERPT: Let A and B be $n\times n$ positive definite semidefinite matrices. For $t\in[0,1]$, let λ_j denote the eigenvalues and σ_j the singular values, indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-036204 | MATHDB-332354

Calderón sum formula conjecture for discrete wavelet frames

摘录 / EXCERPT: Let A be an invertible matrix, let Γ be a lattice in $\mathbb{R}^n$, let $B=A^T$, and let $\Psi=\{\psi_1,\dots,\psi_L\}$ generate the wavelet system...

conjecture | open | MathDB 页面 | 原文线索

V2-036205 | MATHDB-332478

Fourier-critical-point conjecture for the auxiliary Bessel derivative function

摘录 / EXCERPT: Let $n\in\mathbb{N}_0$, let $\nu\in\mathbb{R}$, and define the real entire function ... A Fourier critical point of a real entire function is a point at which some derivative has a c...

conjecture | open | MathDB 页面 | 原文线索

V2-036206 | MATHDB-332493

A trace inequality for four orthogonal projections

摘录 / EXCERPT: Projection trace inequality conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-036207 | MATHDB-332498

Linnell's L^2-property conjecture

摘录 / EXCERPT: Let Γ be a torsion-free discrete group, let $\mathbb C\Gamma$ be its complex group ring, and let $\ell^2(\Gamma)$ be the Hilbert space of square-summable functions on...

conjecture | open | MathDB 页面 | 原文线索

V2-036208 | MATHDB-332515

Giusto-Simpson conjecture on the reverse-mathematical strength of strong Tietze extension

摘录 / EXCERPT: Giusto-Simpson conjecture. The following are equivalent over RCA_0:

conjecture | open | MathDB 页面 | 原文线索

V2-036209 | MATHDB-332534

Bang's 1-codimensional cylinder covering conjecture

摘录 / EXCERPT: Let $\operatorname{cyl}_1(\mathbf{K})$ be the 1-st cylindrical covering parameter of a convex body $\mathbf{K}$ in $\mathbb{E}^3$, obtained by infimizing the total relative cylin...

conjecture | open | MathDB 页面 | 原文线索

V2-036210 | MATHDB-332672

Capacity convergence conjecture for the second Hermite approximant

摘录 / EXCERPT: Let f belong to the Laguerre class $\mathscr L$, represented by ... where $\alpha_j\in\mathbb R\setminus\mathbb Z$, $e_j\in\mathbb R$, and $e_1<\dots<e_{2q}$. Assume that $1,f,f^2$...

conjecture | open | MathDB 页面 | 原文线索

V2-036211 | MATHDB-332675

Kye's decomposability conjecture for 2-positive maps on $M_3(\mathbb{C})$

摘录 / EXCERPT: Kye's conjecture. Every 2-positive, respectively 2-copositive, linear map in $B(M_3(\mathbb{C}),M_3(\mathbb{C}))$ is decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-036212 | MATHDB-332685

Newman-Byrnes merit-factor conjecture for Littlewood polynomials

摘录 / EXCERPT: Let $\mathcal{L}=\bigcup_{N=1}^{+\infty}\{+1,-1\}^N$ be the class of Littlewood sequences, and let $U_N(z)$ denote the associated L^2-normalized analytic polynomial for a sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-036213 | MATHDB-332700

The NTV conjecture for the Hilbert transform

摘录 / EXCERPT: Let b and b be weights on b^n, let H be the Hilbert transform, and let b_Q denote the two-tailed auxiliary function associated with a cube Q. Write b_Q and b_Q for...

conjecture | open | MathDB 页面 | 原文线索

V2-036214 | MATHDB-332701

The general $T1$ conjecture for elliptic fractional singular integrals

摘录 / EXCERPT: Let $b^{\alpha,n}$ denote an elliptic vector of standard b-fractional singular integrals in b^n, and let b and b be weights. The notation $\mathcal{A}_2^\alpha$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-036215 | MATHDB-332702

The energy necessity conjecture for elliptic fractional singular integrals

摘录 / EXCERPT: Let $b^{\alpha,n}$ denote an elliptic vector of standard b-fractional singular integrals in b^n, acting between weighted spaces $L^2(\mathbb{R}^n;\sigma)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036216 | MATHDB-332706

The duality conjecture for covering numbers

摘录 / EXCERPT: Covering-number duality conjecture. There are universal constants $c,C>0$ such that, for every dimension n and every pair of symmetric convex bodies $K,L\subset\mathbb R^n$,

open_problem | open | MathDB 页面 | 原文线索

V2-036217 | MATHDB-332716

No-unwanted-roots conjecture for the limiting sphere-packing functions

摘录 / EXCERPT: Let f and $\widehat f$ be the limiting functions from the convergence conjecture, and call the prescribed zeros of the construction the forced roots. No-unwanted-roots conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-036218 | MATHDB-332718

Rational quadratic-coefficient conjecture for the limiting functions

摘录 / EXCERPT: Normalize the limiting functions f and $\widehat f$ so that $f(0)=\widehat f(0)=1$, and consider their Taylor expansions at 0. Rational quadratic-coefficient conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036219 | MATHDB-332719

Mellin midpoint-value conjecture for the dimension-8 limiting functions

摘录 / EXCERPT: Let ... and define $M_{\widehat f}$ analogously for the Fourier transform $\widehat f$. Normalize the limiting functions by $f(0)=\widehat f(0)=1$. Mellin midpoint-value conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-036220 | MATHDB-332721

Single-root Fourier-eigenfunction convergence conjecture

摘录 / EXCERPT: For $\varepsilon\in\{0,1\}$, let ... where $r^\varepsilon_{n,k}$ has degree at most $2k+\varepsilon$, is nonzero, vanishes at $2,4,\ldots,2k$, and is a linear combination of the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-036221 | MATHDB-332722

Explicit formulas conjecture for single-root Fourier eigenfunctions

摘录 / EXCERPT: Let g^ε_n be the limiting Fourier eigenfunctions from the single-root convergence conjecture. Explicit formulas conjecture. If the scaling is chosen appropriately, then...

conjecture | open | MathDB 页面 | 原文线索

V2-036222 | MATHDB-332728

Cohn-Miller rationality conjecture for optimal radial functions

摘录 / EXCERPT: Let f and $\widehat{f}$ be optimal radial functions for the Euclidean linear programming sphere-packing bound, normalized so that ... Their quadratic Taylor coefficients are the co...

conjecture | open | MathDB 页面 | 原文线索

V2-036223 | MATHDB-332740

The alternative p-adic condition for adelic Beltrami differentials

摘录 / EXCERPT: Alternative p-adic condition conjecture. In the p-adic case, the renormalized average series condition in the definition of an adelic Beltrami differential can be replaced by the c...

conjecture | open | MathDB 页面 | 原文线索

V2-036224 | MATHDB-332748

Minkowski's measure power-law Jacobi convergence conjecture

摘录 / EXCERPT: Minkowski's measure power-law Jacobi convergence conjecture. There exist two positive constants A,B such that

conjecture | open | MathDB 页面 | 原文线索

V2-036225 | MATHDB-332749

Minkowski's orthogonal-polynomial zero convergence conjecture

摘录 / EXCERPT: Minkowski's orthogonal-polynomial zero convergence conjecture. As j tends to infinity, the zeros ζ_l^j converge uniformly to the zeros θ_l^j; moreover, there exist p...

conjecture | open | MathDB 页面 | 原文线索

V2-036226 | MATHDB-332750

Minkowski's zero-distribution discrepancy conjecture

摘录 / EXCERPT: Minkowski's zero-distribution discrepancy conjecture. As j tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-036227 | MATHDB-332751

Minkowski's ratio-measure power-law convergence conjecture

摘录 / EXCERPT: Minkowski's ratio-measure power-law convergence conjecture. There exist two positive constants A,B such that

conjecture | open | MathDB 页面 | 原文线索

V2-036228 | MATHDB-332754

Level-difference conjecture for Fisher-KPP fronts

摘录 / EXCERPT: Level-difference conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036229 | MATHDB-332755

Refined Fisher-KPP front expansion conjecture under a third-moment condition

摘录 / EXCERPT: Refined Fisher-KPP expansion conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036230 | MATHDB-332777

Asymptotic binomial-sum conjecture for even moments

摘录 / EXCERPT: Even-moment asymptotic conjecture. There is a constant $c(k)>0$ such that, as $j\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-036231 | MATHDB-332795

Foschi's Gaussian maximizer conjecture for the paraboloid extension inequality

摘录 / EXCERPT: Let $d\geq 3$. The quantity $\mathcal S_d$ denotes the supremum in the relevant paraboloid extension inequality over functions ψ on $\mathbb{R}^d$. Foschi's Gaussian maximizer...

conjecture | open | MathDB 页面 | 原文线索

V2-036232 | MATHDB-332863

Lipschitz continuity conjecture for Riemannian max-flow solution sets

摘录 / EXCERPT: Let $g_{ij}(\tau)$ be a continuous family of metrics on a Riemannian manifold $\left(M^d,\partial\right)$ whose boundary has non-outward-pointing mean curvature. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036233 | MATHDB-333000

Fredholm range and kernel conjecture for the coagulation-fragmentation operator

摘录 / EXCERPT: Let $\operatorname{R}(W_{\phi})$ and $\operatorname{N}(W_{\phi})$ denote the range and null space of the operator W_{ϕ} on $L^1([0,L])$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036234 | MATHDB-333019

Triangular truncation conjecture for positive semidefinite matrices

摘录 / EXCERPT: Let A be an $n\times n$ positive semidefinite matrix whose diagonal entries are all L_1. Define ... For some constant C_1, assume that ... Let Γ be the upper triangular pa...

conjecture | open | MathDB 页面 | 原文线索

V2-036235 | MATHDB-333025

The Khinchin extremizer conjecture below the critical exponent

摘录 / EXCERPT: Let $p_0\cong1.8474$ be the threshold referred to in the paper, and let (e_i) be the standard unit vectors. For a scalar sequence $(a_j)_{j=1}^{\infty}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-036236 | MATHDB-333062

Huang et al.'s boundary uniqueness conjecture for holomorphic mappings

摘录 / EXCERPT: Huang et al.'s conjecture. Under these assumptions, f vanishes identically.

conjecture | open | MathDB 页面 | 原文线索

V2-036237 | MATHDB-333153

Ruscheweyh-Salinas-Sugawa conjecture on universal convexity of the Ramanujan sequence function

摘录 / EXCERPT: Let $\Lambda=\mathbb{C}\setminus[1,\infty)$, and let $\operatorname{Hol}_1(\Lambda)$ be the functions analytic on Λ satisfying $f(0)=f'(0)-1=0$. The Ramanujan sequence is...

conjecture | open | MathDB 页面 | 原文线索

V2-036238 | MATHDB-333228

The quantitative nonlinear Brascamp-Lieb conjecture

摘录 / EXCERPT: Let U be a neighbourhood of $x_0\in\mathbb{R}^n$, let $B_j:U\to\mathbb{R}^{n_j}$ be local submersions, and let $L_j:\mathbb{R}^n\to\mathbb{R}^{n_j}$ satisfy $dB_j(x_0)=L_j$. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-036239 | MATHDB-333233

The first collapsing-function identity for worms and spiders

摘录 / EXCERPT: Let $\uppsi_0$ be the ordinal collapsing function and let $\cond_{\top}$ denote the corresponding worm/spider collapsing operation. The first collapsing-function identity. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-036240 | MATHDB-333235

Nonvanishing of the numerator factor near the denominator roots

摘录 / EXCERPT: Suppose $0<\gamma<\gamma_*$, and let $1<p^1_{\gamma}<p^2_{\gamma}$ be the roots of $g_0(p;\gamma)$. Define $M_{\delta}(p;\gamma)$ as in the source. The numerator noncancellation co...

conjecture | open | MathDB 页面 | 原文线索

V2-036241 | MATHDB-333237

ALR-busting source conjecture for compactly supported sources

摘录 / EXCERPT: ALR-busting source conjecture. There exist $\delta_{\gamma}>0$ and $C_{\gamma}>0$ such that $E_{\delta}(a)\le C_{\gamma}$ for all $0<\delta\le\delta_{\gamma}$. The conjecture propo...

conjecture | open | MathDB 页面 | 原文线索

V2-036242 | MATHDB-333337

Bejenaru's multilinear restriction conjecture for curved transversal hypersurfaces

摘录 / EXCERPT: Let $n\geq 1$, let $U_i\subset\mathbb{R}^n$ be bounded parameter domains, and let $S_i\subset\mathbb{R}^{n+1}$ be smooth, compact hypersurfaces parametrized by smooth maps...

conjecture | open | MathDB 页面 | 原文线索

V2-036243 | MATHDB-333476

Strong ratio-limit conjecture for subcritical and critical heat kernels

摘录 / EXCERPT: Strong ratio-limit conjecture. Locally uniformly for $(x,y)\in M\times M$,

conjecture | open | MathDB 页面 | 原文线索

V2-036244 | MATHDB-333524

Conjecture on universal rational-function limits of cluster-expansion terms

摘录 / EXCERPT: Universal-limit conjecture. The limits obtained using E_1, E_2, or E are equal, and

conjecture | open | MathDB 页面 | 原文线索

V2-036245 | MATHDB-333526

Scaling conjecture for modified cluster-expansion limits

摘录 / EXCERPT: Modified scaling conjecture. The limits obtained using E_1, E_2, or E are equal and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-036246 | MATHDB-333530

Strict mass loss for exotic Jacobi semigroups

摘录 / EXCERPT: Strict mass-loss conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036247 | MATHDB-333552

Asymptotic growth conjecture for square-ice counts

摘录 / EXCERPT: Let $a_\ell(n)$ be the number of configurations with n removed oranges, where ℓ is the size parameter. Asymptotic growth conjecture. For $\ell\ll n^{1/3}$, ... as $n\to\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-036248 | MATHDB-333624

The higher-dimensional energy condition conjecture for strongly gradient elliptic singular integrals

摘录 / EXCERPT: Let T be a vector of standard singular integrals in higher dimensions, and suppose that T is both strongly elliptic and strongly gradient elliptic. The energy conditions are th...

conjecture | open | MathDB 页面 | 原文线索

V2-036249 | MATHDB-333734

Poincaré-type stability conjecture for weighted Einstein manifolds

摘录 / EXCERPT: Let $(M^n,g,v,m,0)$ be a closed weighted Einstein manifold with $P_\phi^m=\lambda g>0$ and scale $\kappa>0$. Let $I_1,I_2\colon C^\infty(M)\to\mathbb{R}$ be the functionals defined...

conjecture | open | MathDB 页面 | 原文线索

V2-036250 | MATHDB-333737

The logarithmic Hilbertian norm conjecture for normed spaces

摘录 / EXCERPT: Let $(X,\|\cdot\|_X)$ be an n-dimensional normed space, let $q\in[1,\infty)$, and let $I_q(X)$ and $M(X)$ denote the quantities defined in the paper for a Hilbertian norm...

conjecture | open | MathDB 页面 | 原文线索

V2-036251 | MATHDB-333764

The completion conjecture in L-theory for real group algebras

摘录 / EXCERPT: Completion conjecture in L-theory. This map is an isomorphism. The conjecture is the additional comparison statement needed in the source to identify the real Baum-Connes and alg...

conjecture | open | MathDB 页面 | 原文线索

V2-036252 | MATHDB-333860

The weighted congruence deduction conjecture

摘录 / EXCERPT: Weighted congruence deduction conjecture. Every weighted congruence can be deduced from Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-036253 | MATHDB-333921

Complex relative entropy principle for imaginary arguments

摘录 / EXCERPT: Complex relative entropy principle. For each positive subsequence, the limit points of the normalized critical complex entropies are contained in the real parts of the critical val...

conjecture | open | MathDB 页面 | 原文线索

V2-036254 | MATHDB-333922

Even-odd subsequence selection conjecture for imaginary arguments

摘录 / EXCERPT: Even-odd subsequence selection conjecture. The even-N limit is a piecewise real-analytic function of y, with analyticity defects where the curves

conjecture | open | MathDB 页面 | 原文线索

V2-036255 | MATHDB-333943

Ext-duality conjecture for Spanier-Whitehead K-duals and Paschke duals

摘录 / EXCERPT: Let A) be a separable C^*-algebra in the bootstrap category with finitely generated K-theory. Let DA be a Spanier-Whitehead K-dual of A, let ${\mathbb{P}}(A)$ be its Pa...

conjecture | open | MathDB 页面 | 原文线索

V2-036256 | MATHDB-333972

Day-Nordlander conjecture for the moduli gamma-minus and gamma-plus

摘录 / EXCERPT: Day-Nordlander conjecture. The displayed inequalities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-036257 | MATHDB-333973

Hilbert-space bounds for the duality-map moduli d-minus and d-plus

摘录 / EXCERPT: Conjecture for the duality-map moduli. The following inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-036258 | MATHDB-333974

Milman-moduli characterization of the zeta-minus and zeta-plus moduli

摘录 / EXCERPT: Milman-moduli conjecture. For positive ε,

conjecture | open | MathDB 页面 | 原文线索

V2-036259 | MATHDB-334067

Lin's discreteness conjecture for harmonic-map energies on the two-sphere

摘录 / EXCERPT: Lin's discreteness conjecture. The subset of critical values of $\mathscr{E}_{g,h}$ is closed and discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-036260 | MATHDB-334109

L^p estimates for semi-degenerate simplex multipliers

摘录 / EXCERPT: A multi-index $\vec{\alpha} \in \mathbb{R}^n$ is called semi-degenerate when it satisfies the semi-degeneracy conditions defined in the paper, and $C^{\vec{\alpha}}$ denotes the co...

conjecture | open | MathDB 页面 | 原文线索

V2-036261 | MATHDB-334164

Cubical area growth conjecture for finite-genus constant-mean-curvature surfaces

摘录 / EXCERPT: Let M be a complete surface of finite genus embedded in $\mathbb{R}^3$ with constant mean curvature. Say that M has at most cubical area growth if there is a constant C depen...

conjecture | open | MathDB 页面 | 原文线索

V2-036262 | MATHDB-334180

Peterson's unique maximal amenable extension conjecture for free group factors

摘录 / EXCERPT: Let $L(\mathbb{F}_n)$ be a free group factor and let $B\subset L(\mathbb{F}_n)$ be a diffuse amenable subalgebra. A maximal amenable extension of B is an amenable subalgebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-036263 | MATHDB-334226

Sufficiency of the bow-tie parameter region for maximum step-size coefficient

摘录 / EXCERPT: Maximum step-size conjecture. All methods in this family with $(\alpha,\beta)\in{\cal{B}}$ have maximum step-size coefficient $\gamma=1$. This claim is intended to characterize the...

conjecture | open | MathDB 页面 | 原文线索

V2-036264 | MATHDB-334243

The type I conjecture for boundary-transitive groups acting on trees

摘录 / EXCERPT: Type I conjecture. Then G is a type ${\rm I}$ group.

conjecture | open | MathDB 页面 | 原文线索

V2-036265 | MATHDB-334271

The radial Fourier multiplier conjecture

摘录 / EXCERPT: Radial Fourier multiplier conjecture. For every $d\geq 2$, the multiplier operator T_m is bounded on $L^p(\mathbb{R}^d)$ if and only if $K\in L^p(\mathbb{R}^d)$, throughout the r...

conjecture | open | MathDB 页面 | 原文线索

V2-036266 | MATHDB-334427

Existence conjecture for rPD twin surfaces

摘录 / EXCERPT: Let $\Sigma_{\tau,\delta}$ denote the minimal twin surface associated with the rPD configuration, where τ is the neck-size parameter and δ is the relevant geometric par...

conjecture | open | MathDB 页面 | 原文线索

V2-036267 | MATHDB-334481

The multinomial projection subgaussianity conjecture

摘录 / EXCERPT: Multinomial projection subgaussianity conjecture. This random variable is $O\left(\frac{m}{\alpha_1+\dotsm+\alpha_k}\right)$-subgaussian. The conjecture is part of a family of conc...

conjecture | open | MathDB 页面 | 原文线索

V2-036268 | MATHDB-334501

Weak Zalcman conjecture via the functionals (B_t)

摘录 / EXCERPT: Let S be the class of normalized univalent functions in the unit disk, with … and, for $t\in[0,1]$, let (B_t) denote the statement … for every $f\in S$ and every $n\ge 2$. Firs...

conjecture | open | MathDB 页面 | 原文线索

V2-036269 | MATHDB-334502

Weak Zalcman conjecture via the functionals (C_r)

摘录 / EXCERPT: Let S be the class of normalized univalent functions in the unit disk, with … and, for $r\in[0,1]$, let (C_r) denote the statement … for every $f\in S$ and every $n\ge 2$. Seco...

conjecture | open | MathDB 页面 | 原文线索

V2-036270 | MATHDB-334542

Asymptotic distribution conjecture for pure imaginary zeros of modified Bessel functions

摘录 / EXCERPT: Asymptotic distribution conjecture. The asymptotic location of the n-th pure imaginary zero is determined by

conjecture | open | MathDB 页面 | 原文线索

V2-036271 | MATHDB-334615

Genus-zero singular points of convergence conjecture

摘录 / EXCERPT: Let N be a flat 3-torus, and let M_n be a sequence of closed constant mean curvature surfaces converging to a limit surface M_∞. Let $q\in N$ be a singular point of co...

conjecture | open | MathDB 页面 | 原文线索

V2-036272 | MATHDB-334696

The amenability necessity conjecture for crossed-product transference

摘录 / EXCERPT: Amenability necessity conjecture. Under these assumptions, the action θ is amenable. This conjecture would show that amenability is necessary for the transference results co...

conjecture | open | MathDB 页面 | 原文线索

V2-036273 | MATHDB-334755

The soap bubble conjecture for planar clusters

摘录 / EXCERPT: Let N be a positive integer, let $\boldsymbol a\in\mathbb{R}^N_+$ be a vector of prescribed positive areas, and let $\mathcal M(\boldsymbol a)$ denote the set of perimeter-minimi...

conjecture | open | MathDB 页面 | 原文线索

V2-036274 | MATHDB-334812

Reproducing-kernel conjecture for asymptotics of Chebyshev polynomials

摘录 / EXCERPT: Reproducing-kernel conjecture. Asymptotics for Chebyshev polynomials in a complex domain ${\mathcal{D}}$ can be given in terms of the reproducing kernels of a suitable Hilbert spac...

conjecture | open | MathDB 页面 | 原文线索

V2-036275 | MATHDB-334913

Bombieri's conjecture for odd-even coefficient pairs below the line $n=(4m+2)/5$

摘录 / EXCERPT: Bombieri's odd-even refinement. If

conjecture | open | MathDB 页面 | 原文线索

V2-036276 | MATHDB-334945

Fourier-transform asymptotic expansion conjecture for the Painlevé I tau function

摘录 / EXCERPT: Let $q(t\,|\,\boldsymbol\nu)$ be a Painlevé I function with Stokes parameters $\boldsymbol\nu$, and let $\mathcal R_k$ be a ray on which $t\to\infty$. Define ... Let $G(z)$ denote th...

conjecture | solved | MathDB 页面 | 原文线索

V2-036277 | MATHDB-334958

Conjecture on asymptotic stability of steady shocks for relativistic Burgers flow

摘录 / EXCERPT: Let $M>0$, let $v=v(t,r)\in[-1,1]$ solve the relativistic Burgers equation on the Schwarzschild exterior $r>2M$, and let the initial data be a compactly perturbed steady shock. Asy...

conjecture | open | MathDB 页面 | 原文线索

V2-036278 | MATHDB-334959

Godefroy-Kalton-Lancien conjecture on Banach spaces with summable Szlenk index

摘录 / EXCERPT: Let X be a Banach space with summable Szlenk index, and suppose that its dual X^{*} is isomorphic to ℓ_{1}. Godefroy-Kalton-Lancien conjecture. Then X is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-036279 | MATHDB-335014

Liu's exact norm conjecture for the Bergman projection

摘录 / EXCERPT: Let P be the Bergman projection on the unit disk. For $1<p<\infty$, let $q:=p/(p-1)$ be the conjugate exponent and let $\|P\|_p$ denote the operator norm of P on...

conjecture | open | MathDB 页面 | 原文线索

V2-036280 | MATHDB-335016

Classification conjecture for rank-two elliptic quadratic operator cones

摘录 / EXCERPT: Let $Q:{\mathbb R}^n\to{\mathbb R}^m$ be an elliptic quadratic operator with $\textup{rg}Q=2$, and let ${\overline K}'_Q$ denote its associated cone. Let C_3 be a three-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-036281 | MATHDB-335030

Closedness of the inverse coordinate map

摘录 / EXCERPT: Let K be the fractal space, let $\mathcal{H}$ be the Hilbert space of 1-forms, and let π be the injective map described above, so that its densely defined inverse...

conjecture | open | MathDB 页面 | 原文线索

V2-036282 | MATHDB-335070

The maximum principle conjecture for the Riesz transform

摘录 / EXCERPT: Maximum principle conjecture. There is a constant $C=C(d,s)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036283 | MATHDB-335113

Phase-linearity characterization of inner-product spaces

摘录 / EXCERPT: Let $(X,\|\cdot\|)$ be a complex normed space, and let $\langle\cdot\mid\cdot\rangle$ denote its product from the defining construction. Phase-linearity characterization. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-036284 | MATHDB-335114

The equality conjecture for the classes of univalent harmonic mappings

摘录 / EXCERPT: Equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036285 | MATHDB-335148

Chen's short-time regularization conjecture for the Calabi flow

摘录 / EXCERPT: Chen's short-time regularization conjecture. Given any initial $C^{1,1}$ Kähler potential, the Calabi flow admits a short-time solution and it immediately becomes smooth when $t>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-036286 | MATHDB-335169

The conjecture on Salem sets of prescribed non-integer size

摘录 / EXCERPT: Prescribed-size Salem-set conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036287 | MATHDB-335175

The radial-measure simplex conjecture

摘录 / EXCERPT: Let μ be a radially symmetric measure on the ambient Euclidean space with positive density $\rho(r)$ that is monotone decreasing in the radius. Let $B_0(r)$ be the ball of radi...

conjecture | open | MathDB 页面 | 原文线索

V2-036288 | MATHDB-335179

The HRT conjecture for Schwartz functions

摘录 / EXCERPT: Schwartz-function sub-conjecture. For every nonzero $g\in S(\mathbb{R})$ and every such Λ, the system $\mathcal{G}(g,\Lambda)$ is linearly independent in $L^2(\mathbb{R})$.

conjecture | open | MathDB 页面 | 原文线索

V2-036289 | MATHDB-335180

Heil's four-point HRT conjecture for a specific configuration

摘录 / EXCERPT: Heil's four-point conjecture. The system $\mathcal{G}(g,\Lambda)$ is linearly independent.

conjecture | open | MathDB 页面 | 原文线索

V2-036290 | MATHDB-335241

The distributive biprojection lattice conjecture for subfactor planar algebras

摘录 / EXCERPT: Distributive biprojection lattice conjecture. Every irreducible subfactor planar algebra with a distributive biprojection lattice is w-cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-036291 | MATHDB-335252

Circulant symmetric expectation conjecture for almost Hadamard matrices

摘录 / EXCERPT: Let $O(N)$ be the real orthogonal group, and let $\Phi(U,B)$ be the quantity defined earlier in the paper through the second variation of the 1-norm. Suppose that $U\in O(N)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-036292 | MATHDB-335253

Strict almost Hadamard and isolated complex Hadamard equivalence conjecture

摘录 / EXCERPT: Let $U\in\sqrt N U(N)$, where $U(N)$ is the unitary group. A strict almost Hadamard matrix is a strict local maximizer in the paper's sense, an isolated complex Hadamard matrix is...

conjecture | open | MathDB 页面 | 原文线索

V2-036293 | MATHDB-335281

The first-wave asymptotic expansion conjecture

摘录 / EXCERPT: First-wave asymptotic conjecture. For every positive integer m, one has

conjecture | open | MathDB 页面 | 原文线索

V2-036294 | MATHDB-335282

The second-wave asymptotic expansion conjecture

摘录 / EXCERPT: Second-wave asymptotic conjecture. For every positive integer m, one has

conjecture | open | MathDB 页面 | 原文线索

V2-036295 | MATHDB-335283

The asymptotic equivalence conjecture for the first exceptional waves

摘录 / EXCERPT: Asymptotic equivalence conjecture. The quantities $\mathcal{E}_1(N,-\lambda N)$ and $3\mathcal{C}_2(N,-\lambda N)$ have the same asymptotic expansion as $N\to\infty$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-036296 | MATHDB-335384

Sharp restricted averaging conjecture for cones in dimensions divisible by four

摘录 / EXCERPT: Restricted averaging conjecture for cones. We have $A_C(p\to r)\lesssim 1$ if and only if $\left(\frac{1}{p},\frac{1}{r}\right)$ lies in the convex hull of the points $(0,0)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036297 | MATHDB-335414

Gaussian extremizer conjecture for the Schrödinger Strichartz estimate

摘录 / EXCERPT: Gaussian extremizer conjecture. A function $f(\boldsymbol{x})$ maximizes the Strichartz estimate if and only if it has the form

conjecture | open | MathDB 页面 | 原文线索

V2-036298 | MATHDB-335447

Vanishing conjecture for tempered holomorphic functions on relatively compact subanalytic pseudo-convex opens

摘录 / EXCERPT: Let $X=\mathbb{C}^n$, and let U be a relatively compact subanalytic pseudo-convex open subset of X. The derived global sections of the sheaf of tempered holomorphic functions,...

conjecture | open | MathDB 页面 | 原文线索

V2-036299 | MATHDB-335463

Sawyer's extrapolation conjecture for mixed weak-type maximal estimates

摘录 / EXCERPT: Sawyer's extrapolation conjecture. There is a finite constant c, depending on the A_1 constant of u and on c_v, such that

conjecture | open | MathDB 页面 | 原文线索

V2-036300 | MATHDB-335464

Sawyer's mixed weak-type conjecture for two A_1 weights

摘录 / EXCERPT: Let M be the Hardy-Littlewood maximal operator on $\mathbb{R}^n$, and let $u,v\in A_1$. Sawyer's mixed weak-type conjecture. There exists a dimensional constant c_n such that ……

conjecture | open | MathDB 页面 | 原文线索

V2-036301 | MATHDB-335521

Boundary conjecture for cubical covers

摘录 / EXCERPT: Let $E\subset \mathbb{R}^n$ be compact, and suppose that its boundary ∂E is $C^{1,1}$. For each integer d, let $\mathcal{C}^E_d$ denote the cubical cover of E at scal...

conjecture | open | MathDB 页面 | 原文线索

V2-036302 | MATHDB-335559

Nonvanishing Euler characteristic for irreducible subfactor planar algebras

摘录 / EXCERPT: Nonvanishing Euler-characteristic conjecture. The Euler characteristic $\chi(\mathcal{P})$ is nonzero.

conjecture | open | MathDB 页面 | 原文线索

V2-036303 | MATHDB-335560

Nonvanishing Euler characteristic for Eulerian biprojection intervals

摘录 / EXCERPT: Eulerian-interval conjecture. If $[e_1,\operatorname{id}]$ is Eulerian, then $\varphi(\mathcal{P})$ is nonzero, and consequently $\mathcal{P}$ is w-cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-036304 | MATHDB-335576

The Hilbert-space conjecture for irreducible energy forms

摘录 / EXCERPT: Let E be an irreducible energy form with domain $D(E)$ and kernel $\ker E$. Hilbert-space conjecture. The quotient $(D(E)/\ker E,E)$ is a Hilbert space. This conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-036305 | MATHDB-335608

The Nitsche inequality conjecture for (ρ,n)-harmonic mappings between annuli

摘录 / EXCERPT: Let $A(1,R)$ and $A(1,R_\ast)$ be annuli, and let ρ be the weight defining (ρ,n)-harmonicity. For an increasing radial diffeomorphism between them, write ... and set...

conjecture | open | MathDB 页面 | 原文线索

V2-036306 | MATHDB-335613

Meeks' classification conjecture for minimal surfaces with two-plane area growth

摘录 / EXCERPT: Meeks' classification conjecture. Σ is either a catenoid or a Scherk example.

conjecture | open | MathDB 页面 | 原文线索

V2-036307 | MATHDB-335614

Meeks' uniqueness conjecture for tangent cones at infinity

摘录 / EXCERPT: Meeks' uniqueness conjecture. Σ has a unique tangent cone at infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-036308 | MATHDB-335616

Dimension version of Falconer's distance set conjecture

摘录 / EXCERPT: Dimension version of Falconer's distance set conjecture. If $\dim F>d/2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036309 | MATHDB-335644

Large-parameter positivity conjecture for the transmission eigenvalue variation

摘录 / EXCERPT: Large-parameter positivity conjecture. There exists $y_1>0$ such that $\delta^T(y)>0$ on $[y_1,+\infty)$. The conjecture concerns the eventual monotonicity of the real part of the...

conjecture | open | MathDB 页面 | 原文线索

V2-036310 | MATHDB-335646

Weak differential subordination estimate for finite-dimensional Banach spaces

摘录 / EXCERPT: Weak differential subordination conjecture. There exists $C_p\geq 1$ such that, for every pair of continuous martingales $M,N:\mathbb R_+\times\Omega\to X$ with N weakly differen...

conjecture | open | MathDB 页面 | 原文线索

V2-036311 | MATHDB-335763

The uniqueness conjecture for spectral values of the partial theta function in the half-disk

摘录 / EXCERPT: Uniqueness conjecture. The spectral values $\tilde{q}_1$ and $v_{\pm}$ are the only spectral values of θ for

conjecture | open | MathDB 页面 | 原文线索

V2-036312 | MATHDB-335790

Finite redundancy implies atomicity for reproducing pairs

摘录 / EXCERPT: Atomicity conjecture. If $R(\Phi)<\infty$, then (X,μ) is atomic.

conjecture | open | MathDB 页面 | 原文线索

V2-036313 | MATHDB-335915

The real-rank conjecture for uniform Roe algebras

摘录 / EXCERPT: Let X be a bounded geometry metric space, and let $C^*_u(X)$ denote its uniform Roe algebra. The real-rank conjecture. If X has asymptotic dimension at least two, then...

conjecture | open | MathDB 页面 | 原文线索

V2-036314 | MATHDB-335919

The sharp constant in the average absolute inner-product inequality

摘录 / EXCERPT: For each dimension n and every $x\in\mathbb{R}^n$, define the average absolute inner product over the sign vectors $v_i\in\{-1,1\}^n$ by ... The largest absolute constant α...

conjecture | open | MathDB 页面 | 原文线索

V2-036315 | MATHDB-335942

Multiplicative separability of the positive linear functional contrast

摘录 / EXCERPT: Multiplicative separability conjecture. For all positive definite attributes A, whenever U belongs to $\mathcal{G}$, the contrast is multiplicatively separable and

conjecture | open | MathDB 页面 | 原文线索

V2-036316 | MATHDB-335968

Younsi-Ransford monotonicity conjecture for analytic capacity ratios

摘录 / EXCERPT: Younsi-Ransford conjecture. For all $z_1,\dots,z_n,w_1,\dots,w_m$, the function $R(r)$ is decreasing on $0<r<\delta/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-036317 | MATHDB-335996

The large-exponent conjecture for complex shifted powers

摘录 / EXCERPT: Let s be a positive integer and let ... be a family of s complex shifted powers. Large-exponent conjecture. There are absolute constants a and b such that, for all sufficient...

conjecture | open | MathDB 页面 | 原文线索

V2-036318 | MATHDB-336010

Generalized Cesàro subordination threshold conjecture

摘录 / EXCERPT: Generalized Cesàro threshold conjecture. For $\rho\in(0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-036319 | MATHDB-336011

Largest-parameter positivity conjecture for generalized Cesàro means

摘录 / EXCERPT: Largest-parameter positivity conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-036320 | MATHDB-336079

Local norm-attainment conjecture for Banach spaces

摘录 / EXCERPT: Local norm-attainment conjecture. If there exists a weakly open subset U of X^* such that

conjecture | open | MathDB 页面 | 原文线索

V2-036321 | MATHDB-336086

Faithful Cuntz semigroup embedding conjecture for stably finite C*-algebras

摘录 / EXCERPT: Let A be a separable, stably finite C-algebra. A faithful Cuntz semigroup embedding conjecture asserts that there is a faithful Cu-morphism ... If true, this would...

conjecture | open | MathDB 页面 | 原文线索

V2-036322 | MATHDB-336158

Bing Wang's entropy lower-bound conjecture for four-dimensional shrinking Ricci solitons

摘录 / EXCERPT: Let (M,g,f) be a complete non-compact four-dimensional shrinking Ricci soliton, and let $\bar{E}>0$. Its Euler characteristic $\chi(M)$ is finite. Assume $|\chi(M)|\leq \bar{E}$....

conjecture | open | MathDB 页面 | 原文线索

V2-036323 | MATHDB-336159

Uniform scalar-curvature conjecture for four-dimensional shrinking Ricci solitons

摘录 / EXCERPT: Let $\bar{E}>0$, and let (M,g,f) be a four-dimensional shrinking Ricci soliton whose Euler characteristic satisfies $|\chi(M)|\leq \bar{E}$. Write $\mathcal{R}_g$ for the scalar...

conjecture | open | MathDB 页面 | 原文线索

V2-036324 | MATHDB-336174

The thin-annuli formulation of the Bochner-Riesz conjecture

摘录 / EXCERPT: Let χ be a Schwartz function satisfying ... and let S_τ be the Fourier multiplier with symbol $\chi((|\xi|-1)/\tau)$, where $0<\tau<1$. Thin-annuli formulation of the Bochn...

conjecture | open | MathDB 页面 | 原文线索

V2-036325 | MATHDB-336248

Burbea's coefficient inequality conjecture for Hardy spaces

摘录 / EXCERPT: Burbea's conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-036326 | MATHDB-336249

Weak-type level-set conjecture for normalized Hardy functions

摘录 / EXCERPT: Weak-type level-set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036327 | MATHDB-336250

Contractive coefficient inequality for Hardy spaces in the dual range

摘录 / EXCERPT: Contractive dual coefficient conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-036328 | MATHDB-336345

Entropy-reduction conjecture for block maps

摘录 / EXCERPT: Let x be an element in the setting of a finite-dimensional subfactor planar algebra, with Fourier transform denoted by $\mathfrak{F}$, and let B_{λ} be the associated blo...

conjecture | open | MathDB 页面 | 原文线索

V2-036329 | MATHDB-336394

Static approximate Gauss-Lucas conjecture for rational functions

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be bounded and convex, and let $\epsilon>0$ be given. There is some constant $C>0$ such that if $k/(n-k)>C$, then K satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-036330 | MATHDB-336395

Rational-function extension of the static approximate Gauss-Lucas conjecture

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be convex, and let $\epsilon>0$ be given. There exists some constant $C>0$ for which the following holds. Let $g,h\in\mathbb{C}(z)$ be rational functions,...

conjecture | open | MathDB 页面 | 原文线索

V2-036331 | MATHDB-336477

Strong SL-algebra generation problem for conformal observables

摘录 / EXCERPT: Let the algebras in Corollary 4.10 and Theorem 4.6 be the scaling-limit algebras constructed in the paper. A strong SL-algebra is the strong scaling-limit algebra structure defined...

open_problem | open | MathDB 页面 | 原文线索

V2-036332 | MATHDB-336516

The multilinear Kakeya conjecture

摘录 / EXCERPT: For $j=1,\ldots,n$, let $\mathcal{T}_j$ be a family of δ-tubes in $\mathbb{R}^n$. Call the families transversal when, for each j, every tube in $\mathcal{T}_j$ points in a...

conjecture | open | MathDB 页面 | 原文线索

V2-036333 | MATHDB-336603

Sierakowski's ideal-separation conjecture for reduced crossed products

摘录 / EXCERPT: Sierakowski's conjecture. If the action of Γ on $\widehat{A}$ is essentially free, then A separates the ideals of $A\rtimes_r\Gamma$.

conjecture | open | MathDB 页面 | 原文线索

V2-036334 | MATHDB-336634

Extension theorem without the deformable pseudoeffectivity hypothesis

摘录 / EXCERPT: Let X be a compact Kähler manifold, let $H\to X$ be a holomorphic line bundle, and let $Z\subset X$ be a smooth hypersurface. Let h be a singular Hermitian metric on H satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-036335 | MATHDB-336657

Critical-exponent conjecture for discontinuous Heaviside diffusivity

摘录 / EXCERPT: Consider the diffusion-absorption model with parameter $\lambda>0$ and discontinuous Heaviside diffusivity having jump ϵ. Let α denote the time-decay parameter in t...

conjecture | open | MathDB 页面 | 原文线索

V2-036336 | MATHDB-336725

The conjecture that the quadratic supremum equals the sharp lower bound

摘录 / EXCERPT: Let G be the energy function and let $M_2(\lambda,c_0)$ denote the corresponding supremum, with parameters $\lambda>0$ and $c_0>0$. Quadratic-supremum conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-036337 | MATHDB-336726

The conjectured formula for the cubic supremum at large cutoff

摘录 / EXCERPT: Let $M_3(\lambda,c_0)$ be the cubic supremum, and let $c_1(\lambda)$ be a threshold such that $M_3(\lambda,c_0)=M_3(\lambda,c_1)$ for all $c_0\geq c_1(\lambda)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036338 | MATHDB-336777

Positive-influence conjecture for simply-rooted Boolean functions

摘录 / EXCERPT: Positive-influence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036339 | MATHDB-336778

The antisymmetry lower Hausdorff dimension conjecture for bundle measures

摘录 / EXCERPT: Let ϕ be a smooth, nonconstant line bundle on $\mathbb{R}^{n}\setminus\{0\}$, and let $a(\phi)$ be the least $k\in\{0,1,\ldots,n-1\}$ such that, for every $(k+1)$-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-036340 | MATHDB-336779

The Rogers-Wang Fourier-independence dimension conjecture for bundle measures

摘录 / EXCERPT: Let μ be a bundle measure, and suppose its Fourier transform contains n linearly independent vectors. Let $\mu\in M_{\phi}(\mathbb{R}^{n})$ for some line bundle ϕ. Roger...

conjecture | open | MathDB 页面 | 原文线索

V2-036341 | MATHDB-336780

Faulhuber's rotational invariance conjecture for Weyl-Heisenberg frames

摘录 / EXCERPT: Rotational invariance conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-036342 | MATHDB-336852

Fässler-Orponen Marstrand-type projection conjecture for curved families in three dimensions

摘录 / EXCERPT: Let $J\subset\mathbb{R}$ be a bounded open interval, let $\gamma\colon J\to S^2$ be a $\mathcal{C}^2$ curve satisfying ... and let $\rho_\theta\colon\mathbb{R}^3\to\mathbb{R}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-036343 | MATHDB-337039

Equality of Bernoulli double sums for multiplicative inverses

摘录 / EXCERPT: Let z and N be integers with $\gcd(z,N)=1$. Let B_1 and B_2 denote the first and second Bernoulli polynomials, respectively, and interpret $z(k-\ell)\bmod N$ as the residue...

conjecture | open | MathDB 页面 | 原文线索

V2-036344 | MATHDB-337067

Garbit-Mustapha-Raschel conjecture on critical exponents of quadrant walks

摘录 / EXCERPT: Let ${\mathcal S}\subset\mathbb{Z}^2$ be a non-singular step set, let $Q(1,1;t)$ be the corresponding generating function, and let ... Let (x^*,y^*) be the unique point minimizing...

conjecture | open | MathDB 页面 | 原文线索

V2-036345 | MATHDB-337107

Subexponential Lebesgue constants for pseudo Leja sequences

摘录 / EXCERPT: Let E be a regular compact set in $\mathbb{C}$, let $(a_n)_{n=0}^{\infty}$ be a pseudo Leja sequence of bounded Edrei growth in $\partial \widehat{E}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036346 | MATHDB-337144

Random Roth fusion lemma for generalized random rational functions

摘录 / EXCERPT: Let K_1, K_2, and k be random compact subsets of $\mathbb C$ with $K_1(\omega)\cap K_2(\omega)=\emptyset$ for every ω. Assume ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036347 | MATHDB-337222

Polynomial-growth conjecture for minimal intermediate subfactors

摘录 / EXCERPT: Let $N\subseteq M$ be an irreducible subfactor with finite index, and let the number of minimal intermediate subfactors be counted over the inclusion $N\subseteq M$. Polynomial-gro...

conjecture | open | MathDB 页面 | 原文线索

V2-036348 | MATHDB-337268

The generalized Andérsen-Lempert theorem

摘录 / EXCERPT: Generalized Andérsen-Lempert theorem. There exists a sequence of entire univalent maps which converges to f uniformly on compact subsets of ${\mathbb{B}^n}$. The source presents...

conjecture | open | MathDB 页面 | 原文线索

V2-036349 | MATHDB-337270

The Fatou-Bieberbach realization conjecture for non-Runge ball domains

摘录 / EXCERPT: Fatou-Bieberbach realization conjecture. There exists a Fatou-Bieberbach domain $\Omega\subseteq{\mathbb{C}^n}$ with $D\subseteq\Omega$ such that (D,Ω) is a Runge pair. This...

conjecture | open | MathDB 页面 | 原文线索

V2-036350 | MATHDB-337271

The Fatou-Bieberbach approximation conjecture for restrictions of univalent maps

摘录 / EXCERPT: Fatou-Bieberbach approximation conjecture. For every $r\in(0,1)$, there exists a domain $\Omega_r\subseteq{\mathbb{C}^n}$ biholomorphic to ${\mathbb{C}^n}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036351 | MATHDB-337272

The strong Fatou-Bieberbach approximation conjecture

摘录 / EXCERPT: Strong Fatou-Bieberbach approximation conjecture. For each $r\in(0,1)$, there exist a Fatou-Bieberbach domain Ω_r and a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-036352 | MATHDB-337313

WKB accuracy conjecture for the direct spectral transform

摘录 / EXCERPT: Suppose that A and $S-S_\infty$ are Schwartz-class functions for some constant $S_\infty\in\mathbb{R}$, and that $k\in\mathbb{C}\setminus\{0\}$ admits a global classical solution...

conjecture | open | MathDB 页面 | 原文线索

V2-036353 | MATHDB-337376

Uniform asymptotic expansion conjecture for eigenvalues of higher-order Toeplitz symbols

摘录 / EXCERPT: Let $g_m(x)=\left(2\sin\frac{x}{2}\right)^{2m}$ for an integer $m\geq 3$, and let $\lambda_{n,j}$ be the eigenvalues of the Toeplitz matrices generated by g_m. Let d_k denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-036354 | MATHDB-337446

Equality of convex functions from equal minimal subgradient norms

摘录 / EXCERPT: Equal minimal subgradient norm conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-036355 | MATHDB-337482

Optimal asymptotic estimate for unimodular multilinear forms on sequence spaces

摘录 / EXCERPT: Optimal asymptotic estimate. There exist constants $B_d,C_d<\infty$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036356 | MATHDB-337502

The L^{n-1}-curvature conjecture for two-convex mean curvature flows

摘录 / EXCERPT: Let $\{M_t\subset\mathbb{R}^{n+1}\}_{t\in [0,T)}$ be a mean curvature flow of two-convex closed embedded hypersurfaces. Here $\lvert A\rvert$ denotes the norm of the second fundame...

conjecture | open | MathDB 页面 | 原文线索

V2-036357 | MATHDB-337503

The bounded diameter conjecture for two-convex mean curvature flows

摘录 / EXCERPT: Let $\{M_t\subset\mathbb{R}^{n+1}\}_{t\in [0,T)}$ be a mean curvature flow of two-convex closed embedded hypersurfaces, and let d_t be the intrinsic distance induced on M_t. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-036358 | MATHDB-337530

Conjecture on the dimension of directions spanned by planar sets

摘录 / EXCERPT: Let $K\subset\mathbb{R}^{2}$ be a Borel set, not contained on a line, and define the set of unit vectors spanned by K by ... Here $\dim_{\mathrm{H}}$ denotes Hausdorff dimension. D...

conjecture | open | MathDB 页面 | 原文线索

V2-036359 | MATHDB-337554

Fuglede-Putnam conjecture for unbounded operators

摘录 / EXCERPT: Let H be a complex Hilbert space. Let A be an operator on H, densely defined and closed if necessary, and let $B\in B(H)$ be normal. Fuglede-Putnam conjecture. The inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-036360 | MATHDB-337643

The positive-curvature restriction conjecture under geometric control

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold and let $\boldsymbol{\Sigma}$ be an interior hypersurface with positive definite second fundamental form satisfying the transversal geometric c...

conjecture | open | MathDB 页面 | 原文线索

V2-036361 | MATHDB-337707

Breuil-Schneider lattice separation conjecture

摘录 / EXCERPT: Let $G=GL_2(F)$), let C be as in the source with ring of integers $\mathcal{O}_C$, and let $\Theta_{\underline{k},a}$ be the $\mathcal{O}_C[G]$-lattice in $\Pi_{\underline{k},a}$...

conjecture | open | MathDB 页面 | 原文线索

V2-036362 | MATHDB-337742

Asymptotic conjecture for the best Markov constant

摘录 / EXCERPT: Let $c_n(\alpha)$ denote the best constant in the Markov L_2-inequality, with n the polynomial degree and α the parameter. Asymptotic conjecture. The best Markov constan...

conjecture | open | MathDB 页面 | 原文线索

V2-036363 | MATHDB-337746

Conjecture on bounds for normalized Bessel eigenfunction constants

摘录 / EXCERPT: Bessel normalization conjecture. For every $p\in\mathbb{N}^*$,

conjecture | open | MathDB 页面 | 原文线索

V2-036364 | MATHDB-337817

The rational-matrix Fuglede-Gabor conjecture

摘录 / EXCERPT: Let $K\subset \mathbb R^d$ and let B be a rational matrix with $\det(B)=1$. Let s be the least common multiple of the denominators of the entries b_{ij}. Rational-matrix Fugl...

conjecture | open | MathDB 页面 | 原文线索

V2-036365 | MATHDB-337840

Parametrized asymptotic expansion conjecture for colored Jones polynomials

摘录 / EXCERPT: Parametrized asymptotic expansion conjecture. The family determines the asymptotic expansion

conjecture | open | MathDB 页面 | 原文线索

V2-036366 | MATHDB-337844

Uniqueness conjecture for the holomorphic central-charge-24 conformal net

摘录 / EXCERPT: A local conformal net is holomorphic when its only irreducible representation is the vacuum representation. Consider a holomorphic local conformal net with central charge $c=24$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-036367 | MATHDB-337923

Saitoh's conjecture for conjugate Hardy H^{2} kernels

摘录 / EXCERPT: Let D be a planar regular region whose boundary consists of n analytic Jordan curves. Let $\hat{R}(z,\bar{w})$ be its conjugate Hardy H^{2} kernel and $B(z,\bar{w})$ its Berg...

conjecture | open | MathDB 页面 | 原文线索

V2-036368 | MATHDB-337926

Umehara's exception conjecture for Kähler immersions into complex hyperbolic space and Hilbert space

摘录 / EXCERPT: Let (M,g) be a Kähler-Einstein manifold. A Kähler immersion is an isometric holomorphic immersion into a Kähler manifold. The spaces $\mathbb{C}\mathrm{H}^{\infty}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036369 | MATHDB-337936

Asymptotic equality conjecture for positive and negative Fourier eigenfunctions

摘录 / EXCERPT: For $s\in\{+1,-1\}$ and dimension d, let $\mathrm{A}_s(d)$ denote the corresponding optimal radius in the Fourier-eigenfunction uncertainty principle. Asymptotic equality conject...

conjecture | open | MathDB 页面 | 原文线索

V2-036370 | MATHDB-338032

Weaver's KS_2 conjecture

摘录 / EXCERPT: Let d,m be positive integers, and let $w_1,\dots,w_m\in\mathbb{C}^d$ satisfy $\lVert w_i\rVert\leq 1$ for every i. Suppose there are universal constants $\eta\geq 2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036371 | MATHDB-338043

The essential norm formula for the Leray transform on strongly complex-convex hypersurfaces

摘录 / EXCERPT: Essential norm formula for the Leray transform. The essential L^2-norm of $\bm{L}_{{\mathcal S}}$ is the supremum of the corresponding model norms over points of ${\mathcal S}$....

conjecture | open | MathDB 页面 | 原文线索

V2-036372 | MATHDB-338059

Ardehali's general Romelsberger-index asymptotics conjecture

摘录 / EXCERPT: Let $\mathcal{I}(b,\beta)$ be a general Romelsberger index, and let the hyperbolic limit mean $\beta\to0^+$ with $b\in]0,\infty[$ fixed. The estimate in question is understood with...

conjecture | open | MathDB 页面 | 原文线索

V2-036373 | MATHDB-338105

Calabi-Chen global existence conjecture for Calabi flow

摘录 / EXCERPT: Calabi-Chen conjecture. Initiating from any smooth Kähler potential, the Calabi flow always exists globally.

conjecture | open | MathDB 页面 | 原文线索

V2-036374 | MATHDB-338124

Conjecture on finite expected condition numbers for random rank-one tuples

摘录 / EXCERPT: Let n_1,n_2,n_3 satisfy $n_1 \ge n_2 \ge n_3 \ge 3$, and let random rank-$1$ tuples of length r in $\mathbb{R}^{n_1 \times n_2 \times n_3}$ have their expected condition number...

conjecture | open | MathDB 页面 | 原文线索

V2-036375 | MATHDB-338145

Injectivity conjecture for Fréchet means of covariance operators

摘录 / EXCERPT: Let $\mathcal H$ be the underlying Hilbert space, and let $\Sigma_1,\dots,\Sigma_n$ be covariance operators on $\mathcal H$. Assume that Σ_1 is injective, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036376 | MATHDB-338249

The duality conjecture for the Dirichlet series $\varphi_{1/p}$ on $\mathcal{H}^p$

摘录 / EXCERPT: Let $0<p\leq2$, and let $\varphi_{1/p}$ be the Dirichlet series defined in the paper by the referenced formula. The space $\mathcal{H}^p$ denotes the corresponding Hardy space of D...

conjecture | open | MathDB 页面 | 原文线索

V2-036377 | MATHDB-338287

Fejes Tóth's continuous energy conjecture on the sphere

摘录 / EXCERPT: Fejes Tóth's continuous energy conjecture. The energy integral is maximized by ν_{ONB}:

conjecture | open | MathDB 页面 | 原文线索

V2-036378 | MATHDB-338324

Conjecture on the radius of convergence for generalized Lambert W inverse series

摘录 / EXCERPT: Radius-of-convergence conjecture. There exist integers $l_1,\ldots,l_m$ such that the radius of convergence of this series is

conjecture | open | MathDB 页面 | 原文线索

V2-036379 | MATHDB-338371

Existence of nonsingular complete holomorphic foliations on Stein manifolds

摘录 / EXCERPT: Let X be a Stein manifold of dimension $n\ge 2$ equipped with a Riemannian metric. A nonsingular holomorphic foliation by smooth complete closed complex hypersurfaces is a foliat...

open_problem | open | MathDB 页面 | 原文线索

V2-036380 | MATHDB-338448

Blackadar's conjecture on semiprojectivity of commutative C*-algebras

摘录 / EXCERPT: Let X be a compact metric space. An absolute neighborhood retract is a space that is a neighborhood retract whenever embedded as a closed subset of a suitable metric space. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-036381 | MATHDB-338457

Integral relation for periodic functions and their derivatives

摘录 / EXCERPT: Let $w:\mathbb{R}\to\mathbb{R}$ be a 2π-periodic function of class $C^2(\mathbb{R})$. Integral relation. The function w satisfies ... The relation is presented as a generalizat...

conjecture | open | MathDB 页面 | 原文线索

V2-036382 | MATHDB-338462

The Kähler Plateau correspondence conjecture in complex dimension two

摘录 / EXCERPT: Kähler Plateau correspondence conjecture. Every index-one submanifold of M is a complex submanifold. Consequently, every causally convex Plateau solution of real dimension 2,...

conjecture | open | MathDB 页面 | 原文线索

V2-036383 | MATHDB-338463

The intersection conjecture for compact immersed minimal currents

摘录 / EXCERPT: Minimal-current intersection conjecture. The currents V_n and W_n must intersect.

conjecture | open | MathDB 页面 | 原文线索

V2-036384 | MATHDB-338464

The finite-variation stability conjecture

摘录 / EXCERPT: Finite-variation stability conjecture. To determine whether the solution is stable over a compact domain, it suffices to examine derivatives of only finitely many orders.

conjecture | open | MathDB 页面 | 原文线索

V2-036385 | MATHDB-338465

The uniform finite-order stability conjecture

摘录 / EXCERPT: Uniform finite-order stability conjecture. There exists a constant $K(\lambda,n,R)$ such that stability for the local length-minimization problem in every such U can be determine...

conjecture | open | MathDB 页面 | 原文线索

V2-036386 | MATHDB-338512

Waldspurger's endoscopic transfer and Fourier-transform compatibility conjecture

摘录 / EXCERPT: Waldspurger's conjecture. (1) For every $f\in C_c^\infty(\mathfrak{g})$, there exists a transfer $f^H\in C_c^\infty(\mathfrak{h})$. (2) For compatible Fourier transforms...

conjecture | open | MathDB 页面 | 原文线索

V2-036387 | MATHDB-338560

Liouville conjecture for the perturbed Paneitz equation on the three-sphere

摘录 / EXCERPT: Liouville conjecture. u must be a constant function.

conjecture | open | MathDB 页面 | 原文线索

V2-036388 | MATHDB-338611

Conjectural L^2 convergence rate for multidimensional convection-diffusion collocation

摘录 / EXCERPT: Multidimensional convergence conjecture. There exists a constant $C=C(R,k,m,|u|_{H^{m+2}(R_0)})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036389 | MATHDB-338757

Semiclassical asymptotics conjecture for scalar products of eigen-half-densities

摘录 / EXCERPT: Let M be a $2n$-dimensional Poisson manifold with a real polarization P and two integrable systems given by Lagrangian fibrations $\pi_i:M\to B_i$, with generically transverse...

conjecture | open | MathDB 页面 | 原文线索

V2-036390 | MATHDB-338761

Uniqueness up to permutation of best tuples in cyclic adaptive Fourier decomposition

摘录 / EXCERPT: Let $n>1$, and let $E(\mathbf{z})$ be the energy function in the Blaschke-form approximation of degree n for a function $f\in H^2$, defined as in. A best n-tuple is a point of...

conjecture | open | MathDB 页面 | 原文线索

V2-036391 | MATHDB-338780

Weightless functions and invariant distributions conjecture

摘录 / EXCERPT: Let G be a p-adic group, let $\mathcal K\subset\mathcal H(G)$ be the space of weightless functions, let $\mathcal E\subset\mathcal D^G$ be the distinguished space of invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-036392 | MATHDB-338786

Baernstein's norm-ratio conjecture for polynomials with unimodular roots

摘录 / EXCERPT: Let $Q(z)=1+z^n$, where n is a positive integer, and let $\|\cdot\|_s$ denote the corresponding quasi-norm on the unit circle. Baernstein's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-036393 | MATHDB-339027

Magyar-Stein-Wainger maximal inequality in four dimensions

摘录 / EXCERPT: Magyar-Stein-Wainger conjecture. If $p>2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036394 | MATHDB-339046

Sendov's reformulated conjecture for an interior zero

摘录 / EXCERPT: Let ... with $a\in(0,1)$ and $|z_j|\leq 1$ for $j=1,\ldots,n-1$. Sendov's reformulated conjecture. The disk ... contains a critical point of $P(z)$. This is a reformulation obtained us...

conjecture | open | MathDB 页面 | 原文线索

V2-036395 | MATHDB-339090

Passage conjecture for L-theory from rational to real group rings and reduced real group C*-algebras

摘录 / EXCERPT: Passage conjecture. The maps j_2 and j_3 appearing in the diagram are bijective. The preceding discussion establishes several related comparison maps and split-injectivity resu...

conjecture | open | MathDB 页面 | 原文线索

V2-036396 | MATHDB-339260

The Heil-Ramanathan-Topiwala conjecture on time-frequency shifts

摘录 / EXCERPT: Let $f\in L^2(\operatorname{\mathbb{R}})$ be nonzero, and let $\Lambda\subset\operatorname{\mathbb{R}}^2$ be a finite set. For $\lambda=(x,\omega)\in\operatorname{\mathbb{R}}^2$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-036397 | MATHDB-339274

Complex-coefficient extension of the Koebe radius conjecture

摘录 / EXCERPT: Let F be the polynomial defined by ... For a real parameter α, consider the rotations $e^{-i\alpha}F(ze^{i\alpha})$. Complex-coefficient Koebe conjecture. The Koebe-radius c...

conjecture | open | MathDB 页面 | 原文线索

V2-036398 | MATHDB-339408

Generalized accuracy conjecture for the near-field limit of high-order rogue waves

摘录 / EXCERPT: Let $\{K_n\subset\mathbb{R}^2\}_{n=1}^\infty$ be a sequence of compact sets satisfying ... Let ψ_{2n} and ψ_{2n-1} denote the even- and odd-order finite rogue waves, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-036399 | MATHDB-339465

The invariant partial ideal conjecture for C*-algebras

摘录 / EXCERPT: The invariant partial ideal conjecture. A partial ideal of $\mathcal{A}$ arises from a total ideal if and only if it is an invariant partial ideal. Consequently, the map...

conjecture | open | MathDB 页面 | 原文线索

V2-036400 | MATHDB-339572

Koumandos-Ruscheweyh weaker conjecture for partial sums

摘录 / EXCERPT: Koumandos-Ruscheweyh weaker conjecture. For every $z\in\mathbb{D}$, $n\in\mathbb{N}$, and $0<\mu\leq\mu^{\ast}(\rho)$,

conjecture | open | MathDB 页面 | 原文线索

V2-036401 | MATHDB-339573

Harmonic-mean conjecture for Koumandos-Ruscheweyh parameters

摘录 / EXCERPT: Harmonic-mean conjecture. If the main conjecture and the weaker conjecture hold for $0<\rho_1,\rho_2\leq1$, then both conjectures also hold for their harmonic mean...

conjecture | solved | MathDB 页面 | 原文线索

V2-036402 | MATHDB-339671

Approximation of full Lagrange functions by local Lagrange functions

摘录 / EXCERPT: Let $G=(\mathcal{V},E)$ be the graph underlying the neighborhood $B(v_0,K)$, let $\widetilde{\mathcal{V}}\subset\mathcal{V}$ be the interpolation set, and let $\chi(\cdot,v_0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036403 | MATHDB-339700

Kuijlaars-Milson conjecture on the asymptotics of exceptional polynomial zeros

摘录 / EXCERPT: Let $\{p_n(x)\}_{n=m}^{\infty}$ be a family of exceptional orthogonal polynomials with weight function ... where $W_0(x)$ is the weight function of a classical orthogonal polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-036404 | MATHDB-339708

Semiclassical mean-field limit of the Levy-Lieb functional

摘录 / EXCERPT: Semiclassical mean-field limit conjecture. For every function f satisfying $f\geqslant 0$, $\sqrt{f}\in H^1(\mathbb{R}^d)$ and $\int_{\mathbb{R}^d}f=1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-036405 | MATHDB-339744

The one-way crossing conjecture for complex zeros of the partial theta function

摘录 / EXCERPT: Let $\theta(q,x)$ be the partial theta function. For $q\in(0,1)$, the zeros are complex-conjugation invariant, so nonreal zeros occur in complex conjugate pairs; for each...

conjecture | open | MathDB 页面 | 原文线索

V2-036406 | MATHDB-339761

Beloshapka's maximum conjecture for totally nondegenerate CR models

摘录 / EXCERPT: Let $M\subset\mathbb C^{n+k}$ be a totally nondegenerate CR model of length $\rho\geq 3$, with graded infinitesimal CR automorphism algebra ... Here, rigidity means that the positive...

conjecture | open | MathDB 页面 | 原文线索

V2-036407 | MATHDB-339788

The characterization of Fourier transformable measures by positive definiteness

摘录 / EXCERPT: Fourier-transformability characterization. The measure μ is Fourier transformable if and only if it is a linear combination of positive definite measures.

open_problem | open | MathDB 页面 | 原文线索

V2-036408 | MATHDB-339922

Improved time control with equality on initial ball for higher-dimensional Ricci flow

摘录 / EXCERPT: Let $nin \mathbb{N}$ with $n\geq 3$. There are constants $\mathcal{A}=\mathcal{A}(n)>0$, $c=c(n)>0$, and $\mathcal{R}=\mathcal{R}(n)>0$ such that, whenever $R\geq \mathcal{R}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036409 | MATHDB-339969

Littlewood's flat polynomial conjecture

摘录 / EXCERPT: Littlewood's conjecture. There exist positive constants c_1 and c_2 such that, for every n, one can choose p_n satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-036410 | MATHDB-340083

Voronoi-cell criterion for selecting a root permutation

摘录 / EXCERPT: Let the roots of the corresponding univariate polynomials be all distinct, and let $\mathbf{a}$ be a vector whose entries are points in the complex plane. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036411 | MATHDB-340154

The general Maxwellian long-time asymptotics conjecture for collision-dominated homoenergetic flows

摘录 / EXCERPT: Let B be a collision cross-section satisfying condition, and let γ denote its homogeneity. Let $Q(t)$ and L_0 be the matrix-valued quantities appearing in the homoenerge...

conjecture | open | MathDB 页面 | 原文线索

V2-036412 | MATHDB-340155

The cylindrical-dilatation Maxwellian asymptotics conjecture

摘录 / EXCERPT: Let B be a collision cross-section satisfying condition with homogeneity $\gamma<\gamma_{\mathrm{crit}}=-\frac32$, and let $g(t,w)$ be a weak solution of the cylindrical-dilatati...

conjecture | open | MathDB 页面 | 原文线索

V2-036413 | MATHDB-340170

Uncertainty conjecture for supports in the finite affine plane

摘录 / EXCERPT: Finite-plane uncertainty conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036414 | MATHDB-340177

Divisor function monotonicity conjecture

摘录 / EXCERPT: For $r>1$, let C_r denote the number of connected components of the closure of the image of the regular divisor function $[?]$. Divisor function monotonicity conjecture. C_r is...

conjecture | open | MathDB 页面 | 原文线索

V2-036415 | MATHDB-340324

The multilinear good-fibration Brascamp-Lieb inequality

摘录 / EXCERPT: Let V be a manifold and let $\Gamma_1,\dots,\Gamma_n$ be a transverse family of good fibrations of V, with corresponding maps $\pi_j:V\to U_j$ and forms ω_j. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-036416 | MATHDB-340411

Local removability conjecture

摘录 / EXCERPT: Local removability conjecture. A compact set is conformally removable if and only if it is locally conformally removable.

conjecture | open | MathDB 页面 | 原文线索

V2-036417 | MATHDB-340413

Non-removability conjecture for planar compact sets containing a Cantor strip

摘录 / EXCERPT: Let C be the middle-thirds Cantor set. A planar compact set K is said to contain a homeomorphic copy of $C\times[0,1]$ if there is a subset of K homeomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-036418 | MATHDB-340420

Braverman-Kazhdan's basic-function Fourier-eigenfunction conjecture

摘录 / EXCERPT: Let G be a reductive group over F, let ρ be a finite-dimensional representation of ${}^{L}G$, let $l=2\langle\rho_B,\lambda\rangle$ with λ the highest weight of...

conjecture | open | MathDB 页面 | 原文线索

V2-036419 | MATHDB-340471

The L^p-exotic group C*-algebra conjecture

摘录 / EXCERPT: For a locally compact group G and $1\leq p\leq\infty$, let $C^*_{L^p}(G)$ and $C^*_{L^{p+}}(G)$ denote the corresponding exotic group C-algebras, and let $B_{L^p}(G)$ be the asso...

conjecture | open | MathDB 页面 | 原文线索

V2-036420 | MATHDB-340472

Improved remainder estimate under strong convexity

摘录 / EXCERPT: Let A^0 be a scalar operator and let ... be a strongly convex surface, meaning that ... for all $\xi\in\Sigma^0_\lambda$ and all η satisfying ... where the sign depends on the con...

conjecture | open | MathDB 页面 | 原文线索

V2-036421 | MATHDB-340533

Dimension bound for genuine boundary singularities of mass-minimizing currents

摘录 / EXCERPT: Dimension conjecture. The Hausdorff dimension of the set of genuine boundary singularities is at most $m-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-036422 | MATHDB-340534

Vanishing boundary-singular measure for mass-minimizing currents

摘录 / EXCERPT: Boundary-measure conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036423 | MATHDB-340535

Dimension bound for genuine singularities of linearized half-valued minimizers

摘录 / EXCERPT: Linearized dimension conjecture. The Hausdorff dimension of the set of genuine singularities is at most $m-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-036424 | MATHDB-340536

Null boundary-singular set for linearized half-valued minimizers

摘录 / EXCERPT: Linearized measure conjecture. The boundary singular set is a $\mathcal{H}^{m-1}$-null set.

conjecture | open | MathDB 页面 | 原文线索

V2-036425 | MATHDB-340537

Discreteness conjecture for analytic interfaces in the two-dimensional linearized problem

摘录 / EXCERPT: Linearized discreteness conjecture. The singular set is discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-036426 | MATHDB-340538

White's discreteness conjecture for analytic boundaries in dimension two

摘录 / EXCERPT: White's discreteness conjecture. The union of the boundary and interior singular sets is discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-036427 | MATHDB-340539

Uniqueness of tangent cones at analytic-boundary points

摘录 / EXCERPT: Tangent-cone uniqueness conjecture. There is a unique tangent cone to T at p.

open_problem | open | MathDB 页面 | 原文线索

V2-036428 | MATHDB-340576

Lipschitz-interface conjecture for isoperimetric subsets of convex domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded, convex domain, and let $E\subset\Omega$ be an open isoperimetric subset. Lipschitz-interface conjecture. The interface...

conjecture | open | MathDB 页面 | 原文线索

V2-036429 | MATHDB-340611

GILP's absolute continuity conjecture for distance measures

摘录 / EXCERPT: GILP's absolute continuity conjecture. If $s>n/2$, then for almost all $g \otin\mathbb{O}(n)$, ν_g is absolutely continuous with respect to Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-036430 | MATHDB-340612

The maximal tubular incidence conjecture for direction-separated lines

摘录 / EXCERPT: Maximal tubular incidence conjecture. Let ${l_i}_{i\in\{1,\dots,L\}}$ be L unit line segments in $\mathbb{R}^3$ whose directions are δ-separated. Then there is a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-036431 | MATHDB-340634

Weak form of Gaiotto-Moore-Neitzke's conjecture for the $SU(n)$ Hitchin moduli space

摘录 / EXCERPT: Let $({\overline{\partial}}_E,\varphi)$ be a Higgs bundle in the regular locus $\mathcal M'$, and let g_{L^2} and g_{sf} denote Hitchin's L^2-metric and the semiflat...

conjecture | open | MathDB 页面 | 原文线索

V2-036432 | MATHDB-340722

Burns-Krantz conjecture for finite-type pseudoconvex domains

摘录 / EXCERPT: Let $\Omega \subset \mathbb{C}^d$ be a pseudoconvex domain of finite type and let $\xi_0 \in \partial \Omega$. A holomorphic self-map is a map $f:\Omega\to\Omega$ that is holomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-036433 | MATHDB-340723

Convex-domain boundary rigidity conjecture

摘录 / EXCERPT: Let $\Omega \subset \mathbb{C}^d$ be a bounded convex domain and let $\xi_0\in\partial\Omega$. A holomorphic self-map is a holomorphic map $f:\Omega\to\Omega$. The tangent cone of...

conjecture | open | MathDB 页面 | 原文线索

V2-036434 | MATHDB-340741

Failure of smooth boundary regularity without genericity

摘录 / EXCERPT: Assume that $\mathcal L$ is a small perturbation satisfying the partial symmetry hypothesis of a tuple $\mathcal L^0$ satisfying the full symmetry hypothesis, and let $\mathbf E$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-036435 | MATHDB-340753

Generalized convergence in law for non-pluripolar background measures

摘录 / EXCERPT: Let dV be the background measure in the convergence theorem, and suppose that the associated pluripotential-theoretic objects are defined. Non-pluripolar background-measure conje...

conjecture | open | MathDB 页面 | 原文线索

V2-036436 | MATHDB-340793

Gromov's Hölder embedding conjecture for the Heisenberg group

摘录 / EXCERPT: Let Ω be an open subset of $\mathbb{R}^k$, let n be a positive integer, and let $C^{\gamma}(\Omega,\mathbb{H}_n)$ denote the maps that are γ-Hölder with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-036437 | MATHDB-340833

Negative-order Bochner-Riesz conjecture

摘录 / EXCERPT: Negative-order Bochner-Riesz conjecture. The operator $\mathcal R^{-\alpha}$ is bounded from $L^p(\mathbb R^d)$ to $L^q(\mathbb R^d)$ if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-036438 | MATHDB-340850

Infinite-dimensional quasi-isometric geometry conjecture for the symplectic Banach-Mazur space of the sphere

摘录 / EXCERPT: Let S^2 be the two-dimensional sphere, let $\bar{\mathcal C}_{S^2}$ be the relevant completion or quotient space of persistence-module objects associated with S^2, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036439 | MATHDB-340869

The graph extension conjecture for unital graph C^*-algebras

摘录 / EXCERPT: Graph extension conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-036440 | MATHDB-340905

The UMD$^-_{\rm PW}$ conjecture for Musielak-Orlicz spaces

摘录 / EXCERPT: The UMD$^-_{\rm PW}$ conjecture. If $\Phi\in\Delta_2$, then $L^{\Phi}(T)$ is UMD$^-_{\rm PW}$.

conjecture | open | MathDB 页面 | 原文线索

V2-036441 | MATHDB-340939

Conjecture on the L^p-boundedness of the generalized spherical maximal operator

摘录 / EXCERPT: Let $n\ge 2$ and $1< p < \infty$. Let M^{β}_* denote the generalized spherical maximal operator on $\mathbb{R}^n$, and let condition (cnd18) be ... with the last inequality wea...

conjecture | open | MathDB 页面 | 原文线索

V2-036442 | MATHDB-340945

Bauer's geodesic-completeness conjecture for uniformly stronger weak Riemannian metrics

摘录 / EXCERPT: Let G^1 and G^2 be smooth, weak Riemannian metrics on an infinte-dimensional Banach manifold $\mathcal H$. Assume that G^2 is uniformly stronger than G^1, both metrics admi...

conjecture | open | MathDB 页面 | 原文线索

V2-036443 | MATHDB-340958

The one-sided Calderón-Zygmund weighted estimate conjecture

摘录 / EXCERPT: Let T be a one-sided Calderón-Zygmund operator: an operator bounded on L^2 with a kernel $K(x,y)$ satisfying ... ... for fixed $0<\alpha\leq 1$, and $K(x,y)=0$ when $x<y$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036444 | MATHDB-340965

The third author's rigidity conjecture for order-type limits

摘录 / EXCERPT: Let cmu be a measure on $cmathbf{R}^2$ that charges no line, and suppose that its support has Hausdorff dimension strictly greater than 1. Let $cell_cmu$ be the order-type limi...

conjecture | open | MathDB 页面 | 原文线索

V2-036445 | MATHDB-341073

Existence of infinite-dimensional crossed-product algebras for the Higson-Kasparov-Trout algebra

摘录 / EXCERPT: Existence conjecture. Two C^*-algebras playing the roles of $\mathscr{SC}(U)\rtimes U$ and $\mathscr{SC}(U)\rtimes_\tau U$ can be defined, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-036446 | MATHDB-341090

Denjoy's multivariate Peano differential conjecture

摘录 / EXCERPT: Let Q^d be the d-dimensional cube, and let f be a function on Q^d. For an integer $k\geq 2$, write $D_x^k f$ for the Peano k-th differential at x and $\partial_x^k f$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-036447 | MATHDB-341138

Unbounded commutator conjecture for self-adjoint operators

摘录 / EXCERPT: Let A be an unbounded self-adjoint operator on a Hilbert space. Unbounded commutator conjecture. There is always a bounded self-adjoint operator B such that $AB-BA$ is unbounde...

conjecture | open | MathDB 页面 | 原文线索

V2-036448 | MATHDB-341139

Mortad's conjecture on trivial domains of powers and adjoints

摘录 / EXCERPT: Let $n\in\mathbb{N}$. For a closed, densely defined operator T, write $D(T^k)$ for the domain of its k-th power and $D[T^{*k}]$ for the domain of the k-th power of its adjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-036449 | MATHDB-341159

Connes's characterization of approximately inner automorphisms

摘录 / EXCERPT: Connes's conjecture. The automorphism θ is approximately inner if and only if it is weakly inner and $\mathrm{mod}(\theta)$ is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-036450 | MATHDB-341211

Pure rough path signature-tail asymptotics conjecture

摘录 / EXCERPT: Pure rough path signature-tail asymptotics conjecture. For every pure m-rough path, the tail asymptotics equals its local m-variation, namely

conjecture | open | MathDB 页面 | 原文线索

V2-036451 | MATHDB-341323

Levitina-Sukochev-Zanin's interpolation characterization for symmetric sequence spaces

摘录 / EXCERPT: Let E be a quasi-Banach symmetric sequence space. For functions or sequences f and g with nonincreasing rearrangements f^* and g^*, write $f\triangleright g$ when ... The s...

conjecture | open | MathDB 页面 | 原文线索

V2-036452 | MATHDB-341350

Upper Banach density conjecture for product-free subsets of free groups

摘录 / EXCERPT: Upper Banach density conjecture. No product-free subset $S\subset\mathbf{F}_{\mathscr{A}}$ has upper Banach density exceeding

conjecture | open | MathDB 页面 | 原文线索

V2-036453 | MATHDB-341380

Asymptotic form conjecture for partition ideals with finite basis

摘录 / EXCERPT: Let $X=F_Z$ be a partition ideal with finite basis Z. Let $p(n,X)$ denote its counting function, and let C be the constant appearing in the exponential term. Asymptotic form co...

conjecture | open | MathDB 页面 | 原文线索

V2-036454 | MATHDB-341433

Stability conjecture for Wick algebra C^*-algebras

摘录 / EXCERPT: Let $T: \mathcal{H}^{\otimes 2} \rightarrow \mathcal{H}^{\otimes 2}$ be a self-adjoint braided operator with $\lVert T\rVert < 1$. The associated C^*-algebra is...

open_problem | open | MathDB 页面 | 原文线索

V2-036455 | MATHDB-341489

Minicozzi-Sogge local smoothing conjecture for compact manifolds

摘录 / EXCERPT: Local smoothing conjecture for compact manifolds. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-036456 | MATHDB-341490

Signature-refined local smoothing conjecture for Fourier integral operators

摘录 / EXCERPT: Signature-refined local smoothing conjecture. There is $1/p$ local smoothing for $\mathcal{F}$, meaning that the local smoothing estimate holds for every $\sigma<1/p$, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-036457 | MATHDB-341499

Generalized Strohmer-Beaver conjecture for Gaussian Gabor frames

摘录 / EXCERPT: Let $g_0(t)=2^{1/4}e^{-\pi t^2}$ be the standard Gaussian, let $\delta>1$, and define ... For each Λ in this class, let A and B denote the lower and upper frame bounds of...

conjecture | open | MathDB 页面 | 原文线索

V2-036458 | MATHDB-341500

Separable Gaussian Gabor frame conjecture for the square lattice

摘录 / EXCERPT: Let $g_0(t)=2^{1/4}e^{-\pi t^2}$ be the standard Gaussian, let $\delta>1$, and define ... For each separable lattice in this class, let A and B denote the lower and upper frame b...

conjecture | open | MathDB 页面 | 原文线索

V2-036459 | MATHDB-341508

Rademacher's conjecture for universal coverings of punctured planes

摘录 / EXCERPT: Let $\Gamma\subset\mathbb{C}$ be a discrete subset, let $f:\mathbb{D}\to\mathbb{C}\setminus\Gamma$ be a universal covering map, let $r(f)$ denote the relevant covering radius, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036460 | MATHDB-341559

Gaussian multi-bubble conjecture

摘录 / EXCERPT: Let $me3$, and let $a_1,dots,a_m>0$ satisfy ... Let $\Omega_1,\dots,\Omega_m\subseteq\mathbb{R}^{n+1}$ be a minimizing partition for the Gaussian multi-bubble problem: the sets cover...

conjecture | open | MathDB 页面 | 原文线索

V2-036461 | MATHDB-341571

The Bernstein-Nikolskii limit conjecture for finite q

摘录 / EXCERPT: Bernstein-Nikolskii limit conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-036462 | MATHDB-341600

The second-derivative characterization of h-convexity

摘录 / EXCERPT: Second-derivative characterization. The function f is h-convex if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-036463 | MATHDB-341608

The zero-strip-decreasing conjecture for complex finite difference operators

摘录 / EXCERPT: Zero-strip-decreasing conjecture. The operator $\Delta_{\theta,h}$ defined on $\mathbb{C}[z]$ is zero strip decreasing.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-036464 | MATHDB-341631

Daws's cotype conjecture for coefficient spaces

摘录 / EXCERPT: Let Γ be a discrete group, let $1\leq p\leq 2$, and let q be its conjugate exponent, so that ... Let $B_p(\Gamma)$ denote the coefficient space associated with isometric rep...

conjecture | open | MathDB 页面 | 原文线索

V2-036465 | MATHDB-341691

Meziane-Mortad conjecture on intertwining with a normal operator

摘录 / EXCERPT: Let H be a Hilbert space, let T be an operator on H, densely defined and closed if necessary, and let $B\in B(H)$ be normal. Meziane-Mortad's conjecture. If ... then ... This con...

conjecture | open | MathDB 页面 | 原文线索

V2-036466 | MATHDB-341695

Spectrality and equi-positivity for Hadamard triple towers

摘录 / EXCERPT: Let $\mu(N_n,B_n)$ be the Cantor-Moran measure associated with a sequence $\{(N_n,B_n,L_n)\}$ forming a Hadamard triple tower, and let $\nu_{>n}$ denote the pulled-back tail measur...

conjecture | open | MathDB 页面 | 原文线索

V2-036467 | MATHDB-341699

Diagonal matrices maximize log-minor variance

摘录 / EXCERPT: Let $\mathcal M_{\kappa}$ be the set of positive-definite $n\times n$ matrices with condition number κ. For $k<n$, κ, and $M\in\mathcal M_{\kappa}$, define $Y_k(M)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-036468 | MATHDB-341713

Density conjecture for fixed-isoperimetric-ratio subdivision surfaces

摘录 / EXCERPT: Fixed-ratio density conjecture. For every $v\in(0,1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-036469 | MATHDB-341744

Finite-scale structure conjecture for free boundaries with finitely many curves

摘录 / EXCERPT: Let u be a classical solution in the unit disk $\mathbb{D}$, and let its free boundary $F(u)$ consist of at most N curves. Finite-scale structure conjecture. There are absolute...

conjecture | open | MathDB 页面 | 原文线索

V2-036470 | MATHDB-341756

Trautman's local realizability conjecture for three-dimensional CR manifolds

摘录 / EXCERPT: Let M^3 be a three-dimensional CR manifold, and let its canonical bundle be the bundle of CR $(1,0)$-forms. A section is closed when its exterior derivative vanishes, and it is n...

conjecture | open | MathDB 页面 | 原文线索

V2-036471 | MATHDB-341821

Chamberland-Meisters eigenvalue conjecture for injectivity

摘录 / EXCERPT: Chamberland-Meisters conjecture. Then F is injective. This conjecture is presented as a sufficient condition for injectivity after the real Jacobian conjecture was disproved; its...

conjecture | open | MathDB 页面 | 原文线索

V2-036472 | MATHDB-341827

Stein's Hilbert transform conjecture for Lipschitz vector fields

摘录 / EXCERPT: Stein's conjecture. The operator H_{Γ} is bounded on $L^p(\mathbb{R}^2)$ for every $1<p<\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-036473 | MATHDB-341861

Failure of radial Schwartz interpolation on the density-one hexagonal lattice radii

摘录 / EXCERPT: Let $r_1,r_2,\dots$ be the positive real numbers of the form ... where j and k are integers. Radial Schwartz functions $f\colon{\mathbb R}^2\to{\mathbb R}$ are considered through...

conjecture | open | MathDB 页面 | 原文线索

V2-036474 | MATHDB-341902

Lempert's stable embedding conjecture for three-dimensional CR manifolds

摘录 / EXCERPT: Let (X, HX, J) be a three dimensional strongly pseudoconvex CR manifold. A stable embedding is a CR embedding $f:(X, HX, J)\rightarrow\mathbb C^k$ such that every sufficiently sm...

conjecture | open | MathDB 页面 | 原文线索

V2-036475 | MATHDB-341921

Boundedness of the harmonic Fock projection in arbitrary dimension

摘录 / EXCERPT: Let $n\geq 2$, let $d\mu_{\beta}$ be the Gaussian measure on $\mathbb{R}^{n}$ with parameter $\beta>0$, and let P_{α} denote the harmonic Fock-space projection for...

conjecture | open | MathDB 页面 | 原文线索

V2-036476 | MATHDB-341970

Simplex Calderón-Zygmund form conjecture

摘录 / EXCERPT: Simplex Calderón-Zygmund form conjecture. Then

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V2-036477 | MATHDB-341989

Conjecture on contour deformation for the general excitation sectors

摘录 / EXCERPT: Assume that the minimal structure property holds. Let ${\boldsymbol{n}}$ range over ${\frak{S}}_n$, and let ${\mathcal{I}}_{{\mathrm{tot}}}^{(n)}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036478 | MATHDB-342027

Conjecture on nonreal cyclotomic coincidence roots

摘录 / EXCERPT: Nonreal-root modulus conjecture. If $z\in\mathbb{C}\setminus\mathbb{R}$ and $\Phi_m(z)=\Phi_n(z)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036479 | MATHDB-342032

Kapaev-Kitaev asymptotic expansion conjecture for the generic Painlevé IV tau function

摘录 / EXCERPT: Let $\sigma(s)$ be a Painlevé IV function and define its Painlevé IV tau function $\uptau(s)$ by ... The function $\uptau(s)$ is entire, and its zeros correspond to simple poles of...

conjecture | open | MathDB 页面 | 原文线索

V2-036480 | MATHDB-342033

Matching conjecture for symmetric Painlevé IV vortex-filament solutions

摘录 / EXCERPT: Let a and ε be the parameters of the symmetric Painlevé IV solutions $\sigma_{\pm a,\varepsilon}(s)$, let $\mathsf C(s)$ denote the curvature scaling function, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-036481 | MATHDB-342038

Exponential smallness of discrete fractal uncertainty exponents

摘录 / EXCERPT: Exponential-smallness conjecture. Fix $\delta\in(1/2,1)$. There exists a sequence of pairs $(M_j,\mathcal A_j)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036482 | MATHDB-342039

Generic dilation conjecture for the discrete fractal uncertainty principle

摘录 / EXCERPT: Generic dilation conjecture. There exists $\beta>\max(0,\frac12-\delta)$ depending only on δ such that, for a generic choice of $\alpha\in[1,M]$,

open_problem | open | MathDB 页面 | 原文线索

V2-036483 | MATHDB-342040

Higher-dimensional logarithmic-phase fractal uncertainty conjecture

摘录 / EXCERPT: Higher-dimensional logarithmic-phase FUP conjecture. For each $\nu>0$ there exists $\beta=\beta(\nu)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036484 | MATHDB-342041

Exceptional-configuration conjecture for the discrete higher-dimensional FUP

摘录 / EXCERPT: Let $\mathcal A,\mathcal B\subset\{0,\ldots,M-1\}^2$ be digit sets, let $\mathcal C_{k,\mathcal A}$ and $\mathcal C_{k,\mathcal B}$ be the associated discrete product Cantor sets,...

conjecture | open | MathDB 页面 | 原文线索

V2-036485 | MATHDB-342071

Low-entropy shrinker classification in dimensions at most four

摘录 / EXCERPT: Low-entropy cylinder conjecture. There exists $\epsilon>0$ such that, for $n\leq 4$ and in any codimension, the only shrinkers with entropy $<\lambda({\boldsymbol S}^1)+\epsilon$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-036486 | MATHDB-342092

Uniqueness conjecture for closed CAMC hypersurfaces under conditions (I), (II), or (III)

摘录 / EXCERPT: Let $\gamma:S^n\to {\mathbb R}_{>0}$ be a C^2 function, and let $X:M\to {\mathbb R}^{n+1}$ be a closed CAMC hypersurface. For each $r=1,\ldots,n$, assume that the r-th anisotro...

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V2-036487 | MATHDB-342093

Genus-zero uniqueness conjecture for anisotropic self-similar shrinkers

摘录 / EXCERPT: Let $\gamma:S^2\to {\mathbb R}_{>0}$ be a C^2 function, and let $\xi_\gamma:S^2\to {\mathbb R}^{3}$ be its Cahn-Hoffman map. Consider closed embedded self-similar shrinking solut...

conjecture | open | MathDB 页面 | 原文线索

V2-036488 | MATHDB-342109

Spectrally reasonable measures conjecture for locally compact Abelian groups

摘录 / EXCERPT: Spectrally reasonable measures conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036489 | MATHDB-342178

Decay conjecture for spectral coefficients of smooth mesh functions

摘录 / EXCERPT: Let the assumptions for the Fokker-Planck operator hold, and let an orthogonal decomposition based on such an operator be given. For smooth functions...

conjecture | open | MathDB 页面 | 原文线索

V2-036490 | MATHDB-342231

Conjecture on non-isolated Eguchi-Hanson singularities in Ricci flow

摘录 / EXCERPT: Let an Eguchi-Hanson singularity arise as in Theorem ... mathbb{R}^4/mathbb{Z}2. ... is not isolated, and all four blow-up limits (i)-(iv) occur. In particular, it is accompanied by a...

conjecture | open | MathDB 页面 | 原文线索

V2-036491 | MATHDB-342232

Conjectural blow-up limits for U(2)-invariant Ricci flow singularities

摘录 / EXCERPT: Let $(M_k,g(t))$ be a $U(2)$-invariant Ricci flow encountering a singularity at the tip of M_k, as the area of the two-sphere S^2_o decreases to zero. A blow-up limit at...

conjecture | open | MathDB 页面 | 原文线索

V2-036492 | MATHDB-342233

Variation conjectures for nested fractals

摘录 / EXCERPT: Nested-fractal variation conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-036493 | MATHDB-342234

Critical weak Bakry-Émery conjecture for generalized Sierpinski carpets

摘录 / EXCERPT: Generalized Sierpinski carpet criticality conjecture. For generalized Sierpinski carpets and similar fractals,

conjecture | open | MathDB 页面 | 原文线索

V2-036494 | MATHDB-342267

The radial projection exceptional-set conjecture

摘录 / EXCERPT: Radial projection exceptional-set conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-036495 | MATHDB-342277

The triangular-ratio metric conjecture for the inverse hyperbolic tangent

摘录 / EXCERPT: For $z_1,z_2 e z_3$ in the unit disk $ \mathbb{D}$, let $s_{\mathbb{D}}(z_1,z_2)$ denote the triangular ratio metric. The inverse-hyperbolic-tangent conjecture. The function ... sati...

conjecture | open | MathDB 页面 | 原文线索

V2-036496 | MATHDB-342379

Higher-dimensional generalization of Ohsawa's holomorphic convexity theorem

摘录 / EXCERPT: Higher-dimensional Ohsawa conjecture. Then Z is holomorphically convex.

conjecture | open | MathDB 页面 | 原文线索

V2-036497 | MATHDB-342380

Boundary-kernel characterization of modifications of Stein spaces

摘录 / EXCERPT: Modification characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036498 | MATHDB-342389

The Fourier completeness criterion for affine contractive IFS measures

摘录 / EXCERPT: Fourier completeness criterion. The Fourier frequencies $\{e_n\}_{n\in\mathbb{N}_0^d}$ are total in $L^2(\mu)$ if and only if v is non-constant, namely, not proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-036499 | MATHDB-342440

Zero-distribution conjecture for Painlevé V expansion polynomials

摘录 / EXCERPT: Zero-distribution conjecture. Every zero of every polynomial $a_k^1(a_1^1)$ lies inside the circle of radius R centered at the origin of the complex plane. Numerical studies sugg...

conjecture | open | MathDB 页面 | 原文线索

V2-036500 | MATHDB-342484

The equal-exponent conjecture for the simplex Hilbert form

摘录 / EXCERPT: Equal-exponent conjecture. There exists a constant C such that for all tuples of Schwartz functions $(F_j)_{j=1}^h$ the form is bounded by

conjecture | open | MathDB 页面 | 原文线索

V2-036501 | MATHDB-342499

Bounded-constant conjecture for small values of bounded-spectrum functions

摘录 / EXCERPT: Bounded-constant conjecture. In the estimate above, C_n can be replaced by a bounded sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-036502 | MATHDB-342546

The Koebe radius conjecture for polynomials of degree N

摘录 / EXCERPT: Let p_N be a polynomial of degree N in the normalized univalent class $\mathcal S$, and define its Koebe radius by ... The Koebe radius is the infimum of the boundary modulus amo...

conjecture | open | MathDB 页面 | 原文线索

V2-036503 | MATHDB-342555

The ACCT conjecture on the manifold structure of Ricci limit spaces

摘录 / EXCERPT: Let X be a noncollapsed Ricci limit space of dimension n, with singular set ${{\cal S}}$. For each $k\in\{0,\ldots,n-1\}$, let ${{\cal S}}_k$ be the subset of ${{\cal S}}$ cons...

conjecture | open | MathDB 页面 | 原文线索

V2-036504 | MATHDB-342556

The open problem on complete Ricci-flow continuation from nonnegative Ricci curvature

摘录 / EXCERPT: Continuation problem. Does there exist a smooth complete Ricci flow continuation starting from (M,g_0)? This is described in the source as an intriguing long-standing open proble...

open_problem | open | MathDB 页面 | 原文线索

V2-036505 | MATHDB-342576

Time-frequency shift invariance conjecture for Gabor spaces with an S_0-generator

摘录 / EXCERPT: Let $g\in S_0({\mathbb R})$ and let $\Lambda \subset {\mathbb R}^2$ be a lattice such that (g,Λ) is a Riesz basis for its closed linear span $\mathcal G(g,\Lambda)$. Time-f...

conjecture | open | MathDB 页面 | 原文线索

V2-036506 | MATHDB-342616

Linnell's analytic zero divisor conjecture for torsion-free groups

摘录 / EXCERPT: Let G be a group, let $\mathbb C G$ be its complex group algebra, and let $\ell^2(G)$ be the Hilbert-space completion of $\mathbb C G$ in the 2-norm. Linnell's analytic zero divi...

conjecture | open | MathDB 页面 | 原文线索

V2-036507 | MATHDB-342620

The square-root-oscillation conjecture for radial Toeplitz operators

摘录 / EXCERPT: Let n be the fixed positive integer defining the polyanalytic Bargmann-Segal-Fock setting. A bounded square-root-oscillating sequence is a bounded sequence with the square-root o...

conjecture | open | MathDB 页面 | 原文线索

V2-036508 | MATHDB-342621

The scalar-limit conjecture for radial Toeplitz operators

摘录 / EXCERPT: Scalar-limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036509 | MATHDB-342623

Average embedding conjecture for finite metric spaces into iterated ll_p spaces

摘录 / EXCERPT: Let $(\mathcal{M},d_\mathcal{M})$ be an n-point metric space. For $p,D\geqslant 1$, a bi-Lipschitz distortion-D embedding of $\mathcal{M}$ into a metric space means an embeddin...

conjecture | open | MathDB 页面 | 原文线索

V2-036510 | MATHDB-342624

Snowflake embedding conjecture for finite-dimensional normed spaces

摘录 / EXCERPT: Finite-dimensional snowflake embedding conjecture. For every $\upomega\in(0,\tfrac12)$, there exists $\mathsf{C}_\upomega>0$ such that, for every integer $k\geqslant 2$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-036511 | MATHDB-342715

Strohmer and Beaver's conjecture on optimal Gaussian Gabor lattice geometry

摘录 / EXCERPT: Strohmer and Beaver's conjecture. Among Gaussian Gabor systems with fixed lattice density greater than 1, the condition number $\operatorname{cond}(S_{g_0,\Lambda})=B/A$ is minim...

conjecture | open | MathDB 页面 | 原文线索

V2-036512 | MATHDB-342716

Stronger frame-bound conjecture for optimal Gaussian Gabor lattices

摘录 / EXCERPT: Let $g_0(t)=2^{1/4}e^{-\pi t^2}$ be the Gaussian, let $\Lambda\subset\mathbb{R}^2$ be a lattice of fixed density δ, and let A and B denote the sharp lower and upper fram...

conjecture | open | MathDB 页面 | 原文线索

V2-036513 | MATHDB-342727

Kusner's classification conjecture for the min-max decomposition

摘录 / EXCERPT: Let ${\vec{\Phi}}_1,\ldots,{\vec{\Phi}}_p,\vec{\Psi}_1,\ldots,\vec{\Psi}_q$ be the Willmore spheres given by the decomposition in equation (12), where the preceding discussion conc...

conjecture | open | MathDB 页面 | 原文线索

V2-036514 | MATHDB-342798

Strict monotonicity conjecture for $C(k,p)$

摘录 / EXCERPT: For $0<p<1$, let $C(k,p)$ be the coefficient-estimate constants considered in the paper. Strict monotonicity conjecture. The sequence $C(k,p)$ is strictly increasing. This is prese...

conjecture | open | MathDB 页面 | 原文线索

V2-036515 | MATHDB-342822

Shadow-boundary finiteness conjecture for convex bodies

摘录 / EXCERPT: Shadow-boundary finiteness conjecture. For every $k\leq d-1$, there exists a k-dimensional linear subspace L of $\mathbb{R}^d$ such that $\Gamma(B,L)$ has finite $(d-k-1)$-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-036516 | MATHDB-342825

Log-harmonic coefficient conjecture

摘录 / EXCERPT: Let $f(z)=zh(z)\overline{g(z)}\in\mathcal{S}_{Lh}$, where h and g are given by the coefficient expansions in the source. Thus, for $n\geq 1$, a_n and b_n denote the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-036517 | MATHDB-342930

Asymptotic decomposition conjecture for the XX-chain time-like correlation ratio

摘录 / EXCERPT: Let $r(m,t,h):=(-1)^m\langle \sigma^-_{1}(0)\sigma^+_{m+1}(t)\rangle_{-h,T}/I_m(t)$ be the ratio formed from the time-like correlation function and its leading asymptotic factor. F...

conjecture | open | MathDB 页面 | 原文线索

V2-036518 | MATHDB-343072

Conjecture on uniform spectral separation implying injectivity

摘录 / EXCERPT: Let $F:\mathbb{R}^n\rightarrow\mathbb{R}^n$ be a C^1 map. Suppose there exists an $\varepsilon>0$ such that $|\lambda|\geqslant\varepsilon$ for every eigenvalue λ of...

conjecture | open | MathDB 页面 | 原文线索

V2-036519 | MATHDB-343093

Perelman's classification conjecture for compact ancient non-collapsed Ricci flows on S^3

摘录 / EXCERPT: Let $(S^3,g(t))$ be a compact ancient κ-noncollapsed solution to the three-dimensional Ricci flow ... existing for $t\in(-\infty,T)$ and shrinking to a round point at T. Per...

conjecture | open | MathDB 页面 | 原文线索

V2-036520 | MATHDB-343106

Lavrentiev-Andersson zero-free polynomial approximation conjecture

摘录 / EXCERPT: Let K be a compact set with connected complement, and let f be a continuous function on K that is analytic in the interior K° and zero-free on K°. Lavrentiev-An...

conjecture | solved | MathDB 页面 | 原文线索

V2-036521 | MATHDB-343153

Continuous extension conjecture for complex geodesics of strictly convex domains

摘录 / EXCERPT: Continuous extension conjecture. Every complex geodesic of Ω extends continuously to $\overline{\mathbb{D}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-036522 | MATHDB-343156

Rectifiability conjecture for the boundary of non-collapsed RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf d,\mathcal{H}^{N})$ be a non-collapsed $\mathsf{RCD}(K,N)$ space. Let $\boldsymbol\partial^{*}X$ denote its reduced boundary. Boundary rectifiability conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-036523 | MATHDB-343208

The degree-uniform improving conjecture for polynomial averages

摘录 / EXCERPT: Let $d\geq 2$ be an integer, let $p(x)$ be any polynomial of degree d mapping the integers to the integers, and let $N\geq 1$. Define ... For the interval $I=[0,N^d]$, let f be s...

conjecture | open | MathDB 页面 | 原文线索

V2-036524 | MATHDB-343257

The optimal uniform L^p resolvent region conjecture on the torus

摘录 / EXCERPT: Optimal resolvent-region conjecture. For every $\varepsilon > 0$, there is a constant C_{ε} such that, whenever $z \in \mathcal{R}_{\mathrm{opt}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-036525 | MATHDB-343258

The discrete restriction conjecture for lattice points on spheres

摘录 / EXCERPT: Let $n \geq 3$, let $\lambda \geq 1$, and let S^{n-1} denote the unit sphere in $\mathbb{R}^n$. For a function f on $\mathbb{T}^n$, write $\widehat f(k)$ for its Fourier coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-036526 | MATHDB-343302

Burago-Zalgaller-Ros conjecture on isoperimetric regions in projective spaces

摘录 / EXCERPT: Let $\mathbb{K}\in\{\mathbb{R},\mathbb{C},\mathbb{H}\}$, and let $\mathbb{P}^{n+1}(\mathbb{K})$ be the corresponding projective space. A projective subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-036527 | MATHDB-343307

Generalized distance formula for projections of arbitrary rank

摘录 / EXCERPT: Let $\mathbb{M}_n(\mathbb{C})$ be the algebra of $n\times n$ complex matrices, let $\mathcal{N}(\mathbb{C}^n)$ be the set of nilpotent operators, and let $\nu_{r,n}$ denote the dis...

conjecture | open | MathDB 页面 | 原文线索

V2-036528 | MATHDB-343331

Almost-everywhere completeness conjecture for the triple-junction singular basis

摘录 / EXCERPT: Completeness conjecture. There exists a measure-zero set S such that, for every $(\theta_{1},\theta_{2},a,b)\in Y\setminus S$, there are constants $p_{i,j}$, $i=0,1,\ldots,N$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036529 | MATHDB-343380

The mass law for polynomial-tuple expansions

摘录 / EXCERPT: Let $\mathcal{F}=\{\mathcal{S}_j\}_{j=1}^{\infty}$ be a collection of tuples in $\mathbb{R}[x]$, and let $\mathcal{S}\in\mathcal{F}$ satisfy $\operatorname{deg}(\mathcal{S})>1$. Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-036530 | MATHDB-343444

The bilinear sparse square-function domination conjecture

摘录 / EXCERPT: Bilinear sparse square-function domination conjecture. There exist $\eta=\eta(d,T)\in(0,1)$ and an η-sparse family $\mathcal{S}$ of cubes in $\mathbb{R}^d$, depending on...

conjecture | open | MathDB 页面 | 原文线索

V2-036531 | MATHDB-343448

Conjecture on positive boundary value solutions under Ricci and convexity bounds

摘录 / EXCERPT: Let M^n be a compact Riemannian manifold with $Ric\geq 0$ and second fundamental form $\Pi\geq 1$ on ∂M. Let u be a positive solution of ... where $\lambda>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036532 | MATHDB-343489

Engel et al.'s antichain Hausdorff-measure conjecture

摘录 / EXCERPT: Let $[0,1]^n$ be equipped with the coordinatewise partial order, and let an antichain be a subset $A\subset[0,1]^n$ containing no two distinct comparable points. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-036533 | MATHDB-343538

The lower bounded k-th mean curvature characterization of balls

摘录 / EXCERPT: Let $K\subseteq\mathbf{R}^{n+1}$ be a convex body. For $k=1,\ldots,n$, let $H_k(K,\cdot)$ denote its pointwise k-th mean curvature, and write $\mathscr{H}^{n}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036534 | MATHDB-343565

Square-cap decoupling conjecture for the cone

摘录 / EXCERPT: Let ${\mathcal C}^2=\{(\xi_1,\xi_2,\sqrt{\xi_1^2+\xi_2^2}):\frac14\leq\xi_1^2+\xi_2^2\leq4\}$, and let ${\mathcal N}_\delta({\mathcal C}^2)$ denote its δ-neighborhood. Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-036535 | MATHDB-343604

Uniform asymptotics for normalized correlator sums

摘录 / EXCERPT: Uniform normalized-correlator conjecture. For every fixed $C<2$,

conjecture | open | MathDB 页面 | 原文线索

V2-036536 | MATHDB-343605

Uniform individual-correlator asymptotics in logarithmically growing dimension

摘录 / EXCERPT: Uniform individual-correlator conjecture. For every $C<2$,

conjecture | open | MathDB 页面 | 原文线索

V2-036537 | MATHDB-343622

Mean value conjecture for the moment curve

摘录 / EXCERPT: For integers $n\geq 2$, parameters $0\leq \beta\leq n-1$, and $s\geq 1$, let $e(t)=e^{2\pi i t}$ and integrate over $[0,1]^{n-1}\times[0,N^{-\beta}]$. Mean value conjecture for the...

conjecture | open | MathDB 页面 | 原文线索

V2-036538 | MATHDB-343623

Small cap decoupling conjecture for the cone

摘录 / EXCERPT: Let ${{\mathbb C}}o^2=\\{(\xi_1,\xi_2,\sqrt{\xi_1^2+\xi_2^2}):1\leq \xi_1^2+\xi_2^2\leq 2\\}$ be the truncated cone, and let $\Gamma(\delta)$ partition its δ-neighborhood in...

conjecture | open | MathDB 页面 | 原文线索

V2-036539 | MATHDB-343630

The basic spectral decomposition conjecture for analytic Langlands eigenfunctions

摘录 / EXCERPT: Basic spectral decomposition conjecture. The Hilbert space has an orthogonal decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-036540 | MATHDB-343680

Necessity of Hardy-space membership for generalized integration operators

摘录 / EXCERPT: Let $0<q<p<\infty$, let $a\in\mathbb{C}^{n-1}$, and define s by ... For an analytic symbol g, let $T_{g,a}:H^p\to H^q$ be the generalized integration operator ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-036541 | MATHDB-343752

Amenability, exactness, and crossed products of the stable Higson corona

摘录 / EXCERPT: Amenability-exactness conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-036542 | MATHDB-343791

The flat-point Hausdorff-dimension conjecture for area-minimizing currents mod p

摘录 / EXCERPT: Flat-point conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036543 | MATHDB-343792

The even-modulus regularity conjecture for area-minimizing currents

摘录 / EXCERPT: Even-modulus regularity conjecture. The conclusions of Theorem hold for p even as well; that is, when p is even, $\operatorname{Sing}(T)$ is $(m-1)$-rectifiable and, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-036544 | MATHDB-343820

UMD openness conjecture

摘录 / EXCERPT: Let X be an r-intermediate UMD space for some $r\in(2,\infty)$. Equivalently, there exist a UMD space Y and a Hilbert space H forming a compatible couple such that X is i...

conjecture | open | MathDB 页面 | 原文线索

V2-036545 | MATHDB-343823

Kato's maximal strip-width conjecture for positive commutators

摘录 / EXCERPT: Let $f\in K_a$ and $g\in K_b$, where K_a and K_b are maximal for f and g, respectively; the functions are non-constant, and P and Q denote the momentum and position ope...

conjecture | open | MathDB 页面 | 原文线索

V2-036546 | MATHDB-343889

The monotonicity conjecture for the spherical restriction functional

摘录 / EXCERPT: Monotonicity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-036547 | MATHDB-343894

Microlocal trace formula conjecture for the bulk-edge invariant

摘录 / EXCERPT: Let ρ_1,ρ_2 be the ranks of the bulk Bloch bundles $\pazocal{E}_+$ and $\pazocal{E}_-$, and assume ... Let $\pazocal{I}_e(P)$ denote the edge invariant, let $\mathbb{P}_h$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-036548 | MATHDB-343958

Huisken's Penrose inequality conjecture for asymptotically flat support surfaces

摘录 / EXCERPT: Let S be an asymptotically flat support surface with mean curvature $H^S\geq 0$. Let $S'\subset S$ be an exterior surface with free boundary minimal surface Σ and exterior...

conjecture | open | MathDB 页面 | 原文线索

V2-036549 | MATHDB-343983

Characterization of bounded iterated commutators of the Hilbert transform

摘录 / EXCERPT: Let b_1,b_2 satisfy the same assumptions as in Theorems and. Let H denote the Hilbert transform, let A,B,C be Young functions, let $\bar A,\bar B,\bar C$ be their complementa...

conjecture | open | MathDB 页面 | 原文线索

V2-036550 | MATHDB-344013

Richards' static Gauss-Lucas conjecture

摘录 / EXCERPT: For a bounded convex set $K\subset\mathbb{C}$, an $\epsilon>0$, and a polynomial p, let K_ϵ be the ϵ-neighborhood of K and let $Z(p,K)$ denote the number of zero...

conjecture | open | MathDB 页面 | 原文线索

V2-036551 | MATHDB-344039

The selfsimilarity conjecture for limit flows of mean-convex surfaces

摘录 / EXCERPT: Let $M\subset\mathbb{R}^3$ be a smooth closed embedded mean-convex surface, and let $\mathcal{M}=\{M_t\}_{t\geq 0}$ be its mean curvature flow. For a spacetime point...

conjecture | open | MathDB 页面 | 原文线索

V2-036552 | MATHDB-344056

Conjecture on the potential kernel for the radial Dunkl Laplacian

摘录 / EXCERPT: Potential-kernel conjecture. For $x\in B^+$ and $y\in S^+$,

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V2-036553 | MATHDB-344080

Resampling prevents false convergence of the iterative ensemble Kalman filter

摘录 / EXCERPT: Let σ_t denote the scalar ensemble standard deviation of the parameter $\boldsymbol{\theta}$ at iteration t, and let...

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V2-036554 | MATHDB-344201

Oppositely transformed matrices can be made hollow and almost hollow

摘录 / EXCERPT: Hollowisation conjecture. There exists an orthogonal matrix $V\in\mathbb{R}^{n\times n}$ such that V^TAV is hollow and VBV^T is almost hollow.

conjecture | open | MathDB 页面 | 原文线索

V2-036555 | MATHDB-344226

Schwarz singularities at focal points for analytic curves

摘录 / EXCERPT: Let Γ be an analytic curve, and let R denote the radius of curvature at a point of highest curvature on Γ. A Schwarz singularity is a singularity of the analytic co...

conjecture | open | MathDB 页面 | 原文线索

V2-036556 | MATHDB-344262

Dorsey et al.'s linear preserver conjecture for semipositive matrices

摘录 / EXCERPT: Dorsey et al.'s conjecture. If $L:M_{m,n}\to M_{m,n}$ is invertible and is an into preserver of $S(\mathbb{R}^n_+,\mathbb{R}^m_+)$, then there exist matrices X and Y such that

conjecture | open | MathDB 页面 | 原文线索

V2-036557 | MATHDB-344325

The conjecture on the minimum eigenvalue derivative for block-Toeplitz symbols

摘录 / EXCERPT: Let $\deg>0$, $j>0$, and $z>0$. Let $\hat{f}_{z,j}$ be the symbol appearing in the paper, and let $\lambda”_{\rm min}(\hat{f}_{z,j})|_{0}$ denote the second derivative at 0 of it...

conjecture | open | MathDB 页面 | 原文线索

V2-036558 | MATHDB-344336

Flat skew-symmetric Littlewood polynomial conjecture

摘录 / EXCERPT: Let $\mathcal L_n$ be the family of Littlewood polynomials of degree n. A polynomial $f(z)\in\mathcal L_n$ is skew-symmetric if its coefficients satisfy the skew-symmetry conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-036559 | MATHDB-344357

The conjecture for the discrete polynomial fractional integral estimate

摘录 / EXCERPT: Let P be a polynomial of degree d, and let $\mathcal I(P)$ denote the set of all triples (p,q,λ) for which the inequality ... holds if and only if ... This conjecture chara...

conjecture | open | MathDB 页面 | 原文线索

V2-036560 | MATHDB-344470

Four step-like solutions conjecture for the KdV similarity equation

摘录 / EXCERPT: Let $u(x)$) and $y(x)$ be functions satisfying equation ... possesses four step-like solutions for any $H_0>-4$ and $H_1>0$. Two of them have the asymptotic behaviour ... and another t...

conjecture | open | MathDB 页面 | 原文线索

V2-036561 | MATHDB-344506

Uniqueness of unimodal maximizers for the improvement dynamics

摘录 / EXCERPT: Let b be a kernel satisfying Assumption MainKernel, let the nonlinearity satisfy Assumption MainNonl, and let $\mathcal{C}_K$ be the admissible class on which the functional...

conjecture | open | MathDB 页面 | 原文线索

V2-036562 | MATHDB-344546

Equality of self-extension constants for real \ell_1 and its finite-dimensional subspaces

摘录 / EXCERPT: Let $se(X)$ denote the self-extension constant of a Banach space X, namely the infimum of the constants controlling norm-preserving extensions of operators from closed subspaces...

conjecture | open | MathDB 页面 | 原文线索

V2-036563 | MATHDB-344635

Elliptic extension conjecture above rectangles

摘录 / EXCERPT: Elliptic extension conjecture. For N sufficiently large, $\varepsilon_0>0$ sufficiently large, and $1\leq p,q\leq\infty$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-036564 | MATHDB-344654

Discrete restriction conjecture for the curve (x,x^3)

摘录 / EXCERPT: Let $N\in\mathbb{N}$ and let $a\in\ell^2(\mathbb{Z})$. Define the truncated extension operator ... where $(\alpha,\beta)\in\mathbb{T}^2$ and $e(t)=e^{2\pi i t}$. Discrete restriction...

conjecture | open | MathDB 页面 | 原文线索

V2-036565 | MATHDB-344655

Sharp discrete restriction conjecture for two polynomial phases

摘录 / EXCERPT: Let $\phi_1,\phi_2\in\mathbb{Z}[n]$ be polynomials with respective degrees k_1 and k_2, and define ... where $a\in\ell^2(\mathbb{Z})$ and $(\alpha_1,\alpha_2)\in\mathbb{T}^2$. As...

conjecture | open | MathDB 页面 | 原文线索

V2-036566 | MATHDB-344656

Sharp discrete restriction conjecture for a univariate polynomial

摘录 / EXCERPT: Let $\phi\in\mathbb{Z}[n]$ be a polynomial of degree $k\geq3$, and let $a\in\ell^2(\mathbb{Z})$. Define the extension operator ... where $\alpha\in\mathbb{T}$ and $e(t)=e^{2\pi i t}$...

conjecture | open | MathDB 页面 | 原文线索

V2-036567 | MATHDB-344710

The invariant little Grothendieck conjecture for invariant function spaces

摘录 / EXCERPT: Let G be a compact abelian group, let $Z(G)$ be an invariant function space, and let $T:H_1\to H_2$ be an operator between Hilbert spaces. Say that T factorizes invariantly thr...

conjecture | open | MathDB 页面 | 原文线索

V2-036568 | MATHDB-344735

Strichartz's conjecture on the Poisson transform over noncompact symmetric spaces

摘录 / EXCERPT: Strichartz's conjecture. (i) If $\lambda\in\mathfrak{a}_{\mathrm{reg}}^*$ and $F\in\mathcal{E}_{\lambda}(X)$, then $F=P_\lambda f$ for some $f\in L^2(B)$ if and only if

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V2-036569 | MATHDB-344821

Purely magnetic tunneling conjecture for elliptical domains

摘录 / EXCERPT: Let Ω be an ellipse with boundary Γ of length $2L$, and let s denote the curvilinear coordinate. Let $\lambda_1(h)$ and $\lambda_2(h)$ be the first two eigenvalues,...

conjecture | open | MathDB 页面 | 原文线索

V2-036570 | MATHDB-344867

Uniqueness conjecture for subcritical nonlinear boundary problems on the ball

摘录 / EXCERPT: Uniqueness conjecture. If $1<q<n/(n-2)$ and $0<a\leq 1/(q-1)$, then u is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-036571 | MATHDB-344911

Global persistence conjecture for simple eigenvector solutions

摘录 / EXCERPT: Let $(x_*,0,\lambda_*)$ be a simple solution of the eigenvalue problem, meaning that the first three hypotheses stated in the paper hold. Suppose that G and H are separable, an...

conjecture | open | MathDB 页面 | 原文线索

V2-036572 | MATHDB-345001

Breuil-Schneider non-vanishing conjecture for $B_{\xi,\zeta}(G)$

摘录 / EXCERPT: Let ${\bf G}$ be an F-split connected reductive algebraic group defined over F, let ${\bf T}$ be a maximal F-split torus, and let ξ and ζ be the parameters used to...

conjecture | open | MathDB 页面 | 原文线索

V2-036573 | MATHDB-345032

Half finite approximation conjecture

摘录 / EXCERPT: Assume Hilbert spaces are separable with complex scalars. Let $\mathscr{X}$ be a Hilbert space, let $\rho:\mathbb{F}\times\mathbb{F}\to\mathrm{U}(\mathscr{X})$ be a unitary represe...

conjecture | open | MathDB 页面 | 原文线索

V2-036574 | MATHDB-345033

Low energy extension conjecture for matrix-valued positive definite functions

摘录 / EXCERPT: Let $\mathbb{F}$ be the free group, let $\mathbb{B}_r$ be the radius-r ball, and let $\mathrm{NSPD}(r,d)$ denote the class of normalized strictly positive definite $d\times d$ ma...

conjecture | open | MathDB 页面 | 原文线索

V2-036575 | MATHDB-345036

Low-entropy mean-convexity conjecture for translating solitons

摘录 / EXCERPT: Low-entropy mean-convexity conjecture. Then Σ is mean convex.

conjecture | open | MathDB 页面 | 原文线索

V2-036576 | MATHDB-345037

Stable translating soliton classification conjecture

摘录 / EXCERPT: Stable translating soliton classification conjecture. Then Σ is mean convex and, up to an ambient isometry, is one of the following translating solitons:

conjecture | open | MathDB 页面 | 原文线索

V2-036577 | MATHDB-345084

Atomic approximation conjecture for biparameter martingale filtrations

摘录 / EXCERPT: Let X) be a separable Banach space. For integers $N,M\geq 0$, let $(f_{n,m}:0\leq n\leq N,0\leq m\leq M)$ be an X-valued martingale on $(\Omega,\mathcal{F},\mathbb{P})$ with re...

conjecture | open | MathDB 页面 | 原文线索

V2-036578 | MATHDB-345106

Characterization of envelopes of moving rotational cones

摘录 / EXCERPT: Envelope characterization. The surface Φ contains the envelope of the family C_t if and only if one of the following conditions holds:

conjecture | open | MathDB 页面 | 原文线索

V2-036579 | MATHDB-345117

The surrogate maximum principle for tensor-product weights on n-trees

摘录 / EXCERPT: Surrogate maximum principle. Every such weight satisfies this principle with $\kappa=1/n$ and $C=C(n)$ independent of w. This is known for $n=1,2,3$; the conjecture concerns the...

conjecture | open | MathDB 页面 | 原文线索

V2-036580 | MATHDB-345118

The n-tree energy estimate for positive measures

摘录 / EXCERPT: Let T^n be the n-tree, let μ be a positive measure on T^n, let $\delta>0$, and let $\mathbf{V}_{\delta}^{\mu}$ and $\mathcal{E}[\mu]$ denote the associated truncated pote...

conjecture | open | MathDB 页面 | 原文线索

V2-036581 | MATHDB-345227

Conjecture on the semiclassical ground-state limit for non-degenerate minima

摘录 / EXCERPT: Let M be a connected manifold and let V be a potential whose minima are non-degenerate in the sense of Shubin's Condition C. Let λ_t denote the relevant analytic eigenv...

conjecture | open | MathDB 页面 | 原文线索

V2-036582 | MATHDB-345252

Inversion transformation formula for Dwork's p-adic hypergeometric functions

摘录 / EXCERPT: Let $s\geq 1$ be an integer and let $a\in\mathbb{Z}_p$. Dwork's p-adic hypergeometric function is ... where the numerator and denominator are hypergeometric power series. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-036583 | MATHDB-345255

Figà-Talamanca-Picardello conjecture on the radicalizer of the Fourier algebra

摘录 / EXCERPT: Figà-Talamanca-Picardello conjecture. $A_r(G)$ is dense in $B_0(G)$, equivalently

conjecture | open | MathDB 页面 | 原文线索

V2-036584 | MATHDB-345296

Regularity conjecture for generators of T-continuous semigroups

摘录 / EXCERPT: Let X be a Banach space, let $d4a5=\left\{F_t\right\}_{t\ge0}\subset C^k(\mathcal{D})$ be a semigroup on a domain $\mathcal{D}\subset X$, and let T-continuity mean continuity i...

conjecture | open | MathDB 页面 | 原文线索

V2-036585 | MATHDB-345387

Albuquerque-Rezende's norm estimate conjecture for unimodular multilinear forms

摘录 / EXCERPT: Albuquerque-Rezende's conjecture. There exist constants $B_{m},C_{m}>0$, depending only on m, such that

conjecture | open | MathDB 页面 | 原文线索

V2-036586 | MATHDB-345391

Optimality conjecture for the weight classes S for Schrödinger operators

摘录 / EXCERPT: Optimality conjecture. There exist $c_1,c_2>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036587 | MATHDB-345392

Weaker optimality conjecture for the weight classes H for Schrödinger operators

摘录 / EXCERPT: Weaker optimality conjecture. There exist $c_1,c_2,m_1,m_2>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036588 | MATHDB-345442

The asymptotic growth conjecture for Golomb-like systems

摘录 / EXCERPT: Asymptotic growth conjecture. Any solution to this system grows asymptotically like

conjecture | open | MathDB 页面 | 原文线索

V2-036589 | MATHDB-345449

Banach-algebra linearity criterion for holomorphic reflexivity

摘录 / EXCERPT: Banach-algebra linearity conjecture. The condition

conjecture | open | MathDB 页面 | 原文线索

V2-036590 | MATHDB-345459

Sharpness and convergence conjecture for two-dimensional Vilenkin-Fourier sums

摘录 / EXCERPT: Let $H_p^{\square}(G_m^2)$ denote the two-dimensional martingale Hardy space and let M_k be the associated scale sequence. For f in these spaces, write...

conjecture | open | MathDB 页面 | 原文线索

V2-036591 | MATHDB-345460

Strong summability conjecture for two-dimensional Vilenkin-Fourier series

摘录 / EXCERPT: Strong summability conjecture. There exists an absolute constant c such that

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V2-036592 | MATHDB-345469

Contractive Hardy-Littlewood inequality

摘录 / EXCERPT: Contractive Hardy-Littlewood inequality. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036593 | MATHDB-345497

Binding conjecture for the nonlinear Schrödinger equation with orthonormal functions

摘录 / EXCERPT: Binding conjecture. For every $N\geq2$ and every $1<p<\min\left(2,1+\frac{2}{d}\right)$, the binding inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-036594 | MATHDB-345498

Absence of binding in one dimension for the nonlinear Schrödinger equation

摘录 / EXCERPT: Absence-of-binding conjecture. For all $2\leq p<3$ and all $N\geq2$, $J(N)$ has no minimiser and

conjecture | open | MathDB 页面 | 原文线索

V2-036595 | MATHDB-345542

Real minimizer conjecture for nearest Hurwitz-stable matrices

摘录 / EXCERPT: Real minimizer conjecture. The problem has a real solution $B\in\mathbb{R}^{n\times n}$. The conjecture concerns whether minimizing over all complex matrices can always be achieved...

conjecture | open | MathDB 页面 | 原文线索

V2-036596 | MATHDB-345581

Bernoulli foliation conjecture for asymptotically flat 3-manifolds

摘录 / EXCERPT: Let (M,g) be an asymptotically flat 3-manifold with nonnegative scalar curvature and positive ADM mass. A Bernoulli surface is a surface $S=\partial A$ for a compact set A such...

conjecture | open | MathDB 页面 | 原文线索

V2-036597 | MATHDB-345582

Large-volume capacity minimizer conjecture

摘录 / EXCERPT: Let (M,g) be an asymptotically flat 3-manifold with nonnegative scalar curvature and positive ADM mass, and assume that ∂M is empty or minimal. For $V>0$ sufficiently la...

conjecture | open | MathDB 页面 | 原文线索

V2-036598 | MATHDB-345608

Generic uniqueness conjecture for sparse periodic phase retrieval

摘录 / EXCERPT: Let N be even, let K be the sparsity, and let x be a K-sparse generic signal whose periodic auto-correlation is a_x. An intrinsic symmetry is an element of the group D...

conjecture | open | MathDB 页面 | 原文线索

V2-036599 | MATHDB-345611

Generic uniqueness conjecture for sparse phase retrieval on finite abelian groups

摘录 / EXCERPT: Let A be a finite abelian group, let V be the vector space of functions $A\to\mathbb K$, and let L_S be the subspace of functions supported in $S\subseteq A$. The periodic au...

conjecture | open | MathDB 页面 | 原文线索

V2-036600 | MATHDB-345641

Moduli-space conjecture for the canonical embedding of holomorphic automorphisms

摘录 / EXCERPT: Moduli-space conjecture. $\operatorname{Ran}({\frak{X}}^n)$ is a moduli space for all $n\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-036601 | MATHDB-345676

The multilinear restriction conjecture for curved hypersurfaces

摘录 / EXCERPT: Let $S_i\subset\mathbb{R}^n$, $i=1,\ldots,k$, be smooth, compact hypersurfaces, where $1\leq k\leq n$. For functions f_i supported in neighborhoods U_i of S_i, write...

conjecture | open | MathDB 页面 | 原文线索

V2-036602 | MATHDB-345688

Local Stein restriction conjecture for positively curved surfaces in three dimensions

摘录 / EXCERPT: Let S be a smooth compact surface in $\mathbb{R}^3$ with positive definite second fundamental form, and let $d\sigma$ be its induced Lebesgue measure. Define the Fourier extensio...

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V2-036603 | MATHDB-345714

Euclidean volume growth conjecture for the escape rate

摘录 / EXCERPT: Let (M,x) be an open manifold with $\operatorname{Ric}\ge 0$, let $\widetilde{M}$ be its Riemannian universal cover, and let $E(M,x)$ denote the escape rate. Euclidean volume gro...

conjecture | open | MathDB 页面 | 原文线索

V2-036604 | MATHDB-345715

Escape rate gap conjecture for virtually abelian fundamental groups

摘录 / EXCERPT: Let n be a positive integer, let (M,x) be an open n-manifold with $\operatorname{Ric}\ge 0$, and let $E(M,x)$ denote its escape rate. Escape rate gap conjecture. There is a u...

conjecture | open | MathDB 页面 | 原文线索

V2-036605 | MATHDB-345731

Sparse form domination implies worse sparse form domination

摘录 / EXCERPT: Sparse form domination conjecture. The same estimate should also hold after replacing q_0 by any $q\in(0,q_0]$.

conjecture | open | MathDB 页面 | 原文线索

V2-036606 | MATHDB-345797

Sawyer-type weak inequality for the multilinear maximal operator with rectangular weights

摘录 / EXCERPT: Let $1\leq p_1,\dots,p_m<\infty$ and define p by ... Let $w_1,\dots,w_m$ and ν be weights such that $(w_1,\dots,w_m,\nu)\in A_{\vec P}^{\mathcal R}$, and let v be a weight sa...

conjecture | open | MathDB 页面 | 原文线索

V2-036607 | MATHDB-345916

K-theory conjecture for products of odometers

摘录 / EXCERPT: Let $(\mathbb{Z},\Lambda)$ be a product of odometers with $k\geq 2$. Write $g\Lambda=\gcd\{n_i-1:1\leq i\leq k\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036608 | MATHDB-345944

The linear-programming and sign uncertainty constants conjecture

摘录 / EXCERPT: Linear-programming and sign uncertainty conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036609 | MATHDB-345945

The one-dimensional positive sign uncertainty conjecture

摘录 / EXCERPT: Positive sign uncertainty conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036610 | MATHDB-345946

The discrete Fourier-transform feasibility conjecture

摘录 / EXCERPT: Discrete Fourier-transform feasibility conjecture. If (k,q) is s-feasible, then $(k+1,q)$ and $(k,q-1)$ are s-feasible. The function $q\mapsto\mathbb{A}_s^{\textup{disc}}(q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-036611 | MATHDB-346038

The interval representation conjecture for fusible numbers

摘录 / EXCERPT: Interval representation conjecture. Every fusible number in $I_{x,n}$, for $n\geq 1$, can be written as

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V2-036612 | MATHDB-346048

Unimodality and ordering conjecture for even generalized sextic Freud recurrence coefficients

摘录 / EXCERPT: Let $\beta_n(t;\lambda)$ be the recurrence coefficients of the orthogonal polynomials for the generalized sextic Freud weight ... For each relevant nonnegative integer n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-036613 | MATHDB-346064

Small-eigenvalue asymptotics at the positive threshold

摘录 / EXCERPT: Let a be a sufficiently regular kernel, and define ... Let ξ_σ be the rescaled support parameter, $\overline{w}_\sigma$ the rescaled profile, $\overline{\chi}$ the indicat...

conjecture | open | MathDB 页面 | 原文线索

V2-036614 | MATHDB-346111

Uniform asymptotic exactness of Witten correlator approximations

摘录 / EXCERPT: Uniform asymptotic exactness conjecture. For any strictly positive constant C,

conjecture | open | MathDB 页面 | 原文线索

V2-036615 | MATHDB-346121

The bounded diameter conjecture for mean curvature flow

摘录 / EXCERPT: Let $\mathcal{M}=\{M_{t}\}_{t\in [0,T)}$ be a family of closed embedded hypersurfaces evolving under mean curvature flow, with first singular time T. Let d_t be the geodesic di...

conjecture | open | MathDB 页面 | 原文线索

V2-036616 | MATHDB-346122

The no cylinder conjecture for embedded shrinking solitons

摘录 / EXCERPT: Let Σ be an embedded shrinking soliton in $\mathbb{R}^3$, and suppose that Σ is not a round cylinder. No cylinder conjecture. Then Σ cannot have an end asympto...

conjecture | open | MathDB 页面 | 原文线索

V2-036617 | MATHDB-346159

The Markov group conjecture for Markovian semigroups on \ell^1

摘录 / EXCERPT: Markov group conjecture. Then T has bounded generator.

conjecture | open | MathDB 页面 | 原文线索

V2-036618 | MATHDB-346170

Higher-dimensional polyhedral singularity conjecture for the Monge-Ampère equation

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^n$ be a compact convex polytope, and let Γ_k denote its k-skeleton. A convex function is a function whose epigraph is convex. Higher-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-036619 | MATHDB-346247

Sharp third Hankel determinant bounds for Ozaki close-to-convex function classes

摘录 / EXCERPT: Sharp-value conjecture. If $f\in\mathcal F$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036620 | MATHDB-346261

Logarithmic rank-growth conjecture for interpolative decomposition butterfly factorization

摘录 / EXCERPT: Let A be an odd or even associated Legendre transform matrix of size $N\times N$, and let $r(N)$ be the least upper bound for the ranks of the relevant non-oscillatory blocks and...

conjecture | open | MathDB 页面 | 原文线索

V2-036621 | MATHDB-346325

Lower semicontinuity conjecture for convex integral functionals of matricial measures

摘录 / EXCERPT: Let $\mathfrak{H}_{+,m}$ denote the cone of positive semidefinite Hermitian $m\times m$ matrices, let $\mathbb T^d$ be the d-dimensional torus, and let $\mathrm{d}M$ be a nonnega...

conjecture | open | MathDB 页面 | 原文线索

V2-036622 | MATHDB-346480

Weak Johnson conjecture for unconditional bases

摘录 / EXCERPT: Let X be a Johnson space, meaning a separable Banach space with only countably many non-isomorphic subspaces. Weak Johnson conjecture. Every Johnson space has an unconditional ba...

conjecture | open | MathDB 页面 | 原文线索

V2-036623 | MATHDB-346482

Nonexistence of simultaneously d_2-minimal and d_2-HI Banach spaces

摘录 / EXCERPT: For a finite-dimensional Banach space F, let $d_2(F)$ denote its Banach-Mazur distance to the Euclidean space of the same dimension. A Banach space is d_2-minimal if it has the...

conjecture | open | MathDB 页面 | 原文线索

V2-036624 | MATHDB-346484

Existence of a tropical Fermat-Weber point in tropical principal component analysis

摘录 / EXCERPT: Existence conjecture. There exists a tropical Fermat-Weber point

conjecture | open | MathDB 页面 | 原文线索

V2-036625 | MATHDB-346582

Maximal perturbations conjecture

摘录 / EXCERPT: Let $\mathcal{S}_{even}(\mathbb{R})$ denote the space of real-valued even Schwartz functions on $\mathbb{R}$, and let $\widehat{f}$ be the Fourier transform of f. For a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-036626 | MATHDB-346606

Mixed-norm Strichartz conjecture for integral quadratic forms

摘录 / EXCERPT: Let $(\cdot,\cdot)$ be a positive-definite quadratic form of integral coefficients in r variables, let B be a bounded region in $\mathbb{R}^r$, and let I be a bounded interva...

conjecture | open | MathDB 页面 | 原文线索

V2-036627 | MATHDB-346607

Optimal Strichartz range for compact Lie groups

摘录 / EXCERPT: The optimal Strichartz-range conjecture. The estimate should hold on any compact Lie group for every $p>2+\frac{4}{d}$. The paper provides evidence for this assertion for class fun...

conjecture | open | MathDB 页面 | 原文线索

V2-036628 | MATHDB-346608

Optimal eigenfunction bounds on compact Lie groups

摘录 / EXCERPT: The optimal eigenfunction-bound conjecture. This estimate should hold for every $p>2+\frac{4}{d-2}$, with an ε-loss if $2\leq r\leq 4$. The paper establishes evidence f...

conjecture | open | MathDB 页面 | 原文线索

V2-036629 | MATHDB-346609

Bourgain's conjectured torus eigenfunction bounds

摘录 / EXCERPT: Let B be a bounded region in $\mathbb{R}^r$, let $r\geq 3$, and let $(\cdot,\cdot)$ be the positive-definite integral quadratic form used to define the frequency sphere. For coef...

conjecture | open | MathDB 页面 | 原文线索

V2-036630 | MATHDB-346617

Semiclassical equivalence conjecture for geometric quantizations

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold. A geometric quantization is a sequence of maps $\{T_k:C^\infty(M)\to\mathcal{L}(H_k)\}_{k\in\mathbb{N}}$ satisfying the geometric...

conjecture | open | MathDB 页面 | 原文线索

V2-036631 | MATHDB-346640

Sharp two-dimensional maximal estimate for the generalized Zakharov-Kuznetsov equation

摘录 / EXCERPT: Let $U(t)$ denote the linear evolution operator and let u_0 be initial data, with the mixed-norm estimate ... Here d is the spatial dimension, p is the spatial integrability ex...

conjecture | open | MathDB 页面 | 原文线索

V2-036632 | MATHDB-346673

Froese's resonance counting conjecture

摘录 / EXCERPT: Let V be a super-exponentially decreasing potential, so that its Fourier-Laplace transform $\widehat V$ is entire, and let F be its Froese function, ... Assume that $\widehat V$...

conjecture | refuted | MathDB 页面 | 原文线索

V2-036633 | MATHDB-346718

Interlacing property for the matrices C(m)

摘录 / EXCERPT: Let the skew-symmetric matrix $C(m)\in\mathbb{R}^{m\times m}$ be defined by ... where ... and ... with $\epsilon(m)=(m-1)^{-1}(1-m^{-1/2})$. The interlacing conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-036634 | MATHDB-346720

Characterization conjecture for the unique pseudo-expectation property of regular inclusions

摘录 / EXCERPT: Unique pseudo-expectation characterization conjecture. 1. $({\mathcal{C}},{\mathcal{D}})$ has the unique pseudo-expectation property. 2. $({\mathcal{C}},{\mathcal{D}})$ is a coveri...

conjecture | open | MathDB 页面 | 原文线索

V2-036635 | MATHDB-346747

Impossibility of exact attainment for the asymptotic rate-error lower bound

摘录 / EXCERPT: An asymptotically scale-invariant centered MRQ is a family of centered multi-resolution quantizers $\{Q_s\}_{s>0}$ whose asymptotic distribution function is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-036636 | MATHDB-346844

Negative covariance conjecture for scrambled $(0,m,s)$-nets

摘录 / EXCERPT: Let $\tilde P_n$ be a scrambled $(0,m,s)$-net in base b, and let $f\in L^2([0,1)^s)$ be a function whose base b Walsh series decomposition satisfies ... where $a\in[0,1]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036637 | MATHDB-346848

The projective-space characterization of the Birkhoff-James property for linear operators

摘录 / EXCERPT: Let $\mathbb{X}$ be an n-dimensional real Banach space, let $S_{\mathbb{X}}$ be its unit sphere, and let T be a linear operator on $\mathbb{X}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-036638 | MATHDB-346875

The bounded-base tube-domain polydisc conjecture

摘录 / EXCERPT: Let $d\geq 1$, let $C\subset\operatorname{\mathbb{R}}^d$ be a bounded convex domain, and let $\Omega=C+i\operatorname{\mathbb{R}}^d$ be its tube domain. Bounded-base tube-domain co...

conjecture | open | MathDB 页面 | 原文线索

V2-036639 | MATHDB-346876

The Gromov-hyperbolic tube-domain ball conjecture

摘录 / EXCERPT: Let $d\geq 1$, let $C\subset\operatorname{\mathbb{R}}^d$ be a convex domain, and let ... be its tube domain. Assume that (Ω,K_Ω) is Gromov hyperbolic; equivalently, C i...

conjecture | open | MathDB 页面 | 原文线索

V2-036640 | MATHDB-346888

The type II ring-isomorphism conjecture for locally measurable operator algebras

摘录 / EXCERPT: Let M and N be von Neumann algebras of type II. Let $LS(M)$ and $LS(N)$ denote their algebras of locally measurable operators. A ring isomorphism is a bijective ring homomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-036641 | MATHDB-346932

The duality conjecture for the BPF norm on b-classes

摘录 / EXCERPT: Let $\mathcal{X}$ be the space of smooth birational models, let $\operatorname{BPF}^k(\mathcal{X})$ be the base-point-free cone, let...

conjecture | open | MathDB 页面 | 原文线索

V2-036642 | MATHDB-346933

The injectivity conjecture for the canonical map of the Fujita space

摘录 / EXCERPT: Let $\mathcal{X}$ be the space of smooth birational models, let $\operatorname{c\text{-}N}^1(\mathcal{X})$ be the Cartier numerical classes, let $\operatorname{BPF}^{d-2}(\mathcal{X})$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-036643 | MATHDB-346934

The model-approximation conjecture for the Fujita space

摘录 / EXCERPT: Let $\mathcal{X}$ be the space of smooth birational models, let $\operatorname{\mathcal{F}_{\Sigma}}(\mathcal{X})$ be the Fréchet space of classes defined in the source, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-036644 | MATHDB-346945

Incomparability conjecture for non-Heisenberg uncertainty principles

摘录 / EXCERPT: Incomparability conjecture. The image of F_q is all of $\mathbb{R}_{>0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-036645 | MATHDB-347058

Conjectured lower semicontinuity of the singular density

摘录 / EXCERPT: Let $(M,g(t))$ be a Ricci flow maximally defined on $[0,T)$, with singular density function $\Theta\colon M\to[0,+\infty]$ defined using an optimal pair (α,β) by ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-036646 | MATHDB-347059

Liminf characterization conjecture for Ricci singularities

摘录 / EXCERPT: Let $(M,g(t))$ be a Ricci flow maximally defined on $[0,T)$, and let Σ_I and Σ_{II} denote the Type I and Type II singular sets. Theorem characterizes these sets usin...

conjecture | open | MathDB 页面 | 原文线索

V2-036647 | MATHDB-347102

Frisch-Parisi conjecture for prescribed multifractal spectra

摘录 / EXCERPT: Frisch-Parisi conjecture. In that Baire functional space, any typical element f obeys some multifractal formalism and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036648 | MATHDB-347199

Cone restriction conjecture for the wave equation

摘录 / EXCERPT: Let u solve the wave equation on $\mathbb{R}\times\mathbb{R}^n$ with initial data $u(0,x)=0$ and $u_t(0,x)=f(x)$. The solution is expressed through the half-wave oscillatory inte...

conjecture | open | MathDB 页面 | 原文线索

V2-036649 | MATHDB-347224

Conjecture on completely bounded isometries of noncommutative Lp-spaces

摘录 / EXCERPT: Let M be a finite von Neumann algebra and let N be a semifinite von Neumann algebra. Suppose $1{\leqslant} p<\infty$ with $p\ne 2$, and let ... be a not necessarily surjective is...

conjecture | open | MathDB 页面 | 原文线索

V2-036650 | MATHDB-347225

Conjecture on complete boundedness of the Jordan component in Yeadon factorizations

摘录 / EXCERPT: Let M be a finite von Neumann algebra and let N be a semifinite von Neumann algebra. Suppose $1{\leqslant} p<\infty$ with $p\ne 2$, and let ... be an isometry with a Yeadon facto...

conjecture | open | MathDB 页面 | 原文线索

V2-036651 | MATHDB-347286

Low-conformal-volume isotopy conjecture for minimal surfaces in hyperbolic three-space

摘录 / EXCERPT: Low-conformal-volume isotopy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-036652 | MATHDB-347426

The norm-attainment equivalence conjecture for free Banach lattices

摘录 / EXCERPT: Norm-attainment equivalence conjecture. One has

conjecture | refuted | MathDB 页面 | 原文线索

V2-036653 | MATHDB-347480

Conjecture on the complete range of elliptic hyperboloid extension estimates

摘录 / EXCERPT: Let d be the spatial dimension, let Σ be the elliptic hyperboloid equipped with its Lorentz-invariant measure μ, and let $\mathcal{E}$ denote the associated extension...

conjecture | open | MathDB 页面 | 原文线索

V2-036654 | MATHDB-347566

The sharp L^1 inequality for sup-convolution

摘录 / EXCERPT: Let $C\subset\mathbb{R}^k$ be a compact convex domain and let $f:C\to\mathbb{R}$ be bounded and measurable. Write $f^{\ast n}$ for the n-fold sup-convolution of f, let...

conjecture | open | MathDB 页面 | 原文线索

V2-036655 | MATHDB-347597

Fillability conjecture for tall infinite Jordan curves

摘录 / EXCERPT: Let Γ be an infinite, connected, tall Jordan curve in $\partial_\infty (\mathbb{H}^2\times \mathbb{R})$. A curve is fat at infinity when it satisfies the fatness condition a...

conjecture | open | MathDB 页面 | 原文线索

V2-036656 | MATHDB-347637

KPV spherical extension conjecture for finite-field spheres

摘录 / EXCERPT: KPV spherical extension conjecture. The following bounds give the sharp $L^p\to L^4$ estimates:

conjecture | open | MathDB 页面 | 原文线索

V2-036657 | MATHDB-347795

The quantitative scalar-curvature decay conjecture near a hypersurface

摘录 / EXCERPT: Quantitative scalar-curvature decay conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-036658 | MATHDB-347837

Non-archimedean Broyden method Q-superlinear convergence conjecture

摘录 / EXCERPT: Let the hypotheses and notation be those of Theorem, in the general non-archimedean setting considered in the paper. Non-archimedean Broyden convergence conjecture. Broyden's metho...

conjecture | open | MathDB 页面 | 原文线索

V2-036659 | MATHDB-347838

Non-archimedean Broyden convergence conjecture over formal power series

摘录 / EXCERPT: Formal-power-series Broyden convergence conjecture. Broyden's method with $\nu_n=\frac{s_n}{s_n^t\cdot s_n}$ has locally Q-superlinear convergence. This is proposed as an adaptatio...

conjecture | open | MathDB 页面 | 原文线索

V2-036660 | MATHDB-347849

Chui's equispaced-poles conjecture in the Bergman space

摘录 / EXCERPT: Let $\mathbb D$ be the unit disk and let $L^1(\mathbb D)$ denote the space of integrable functions on $\mathbb D$ with respect to normalized planar Lebesgue measure. For a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-036661 | MATHDB-347882

Hamilton's gradient shrinking soliton conjecture for finite-time Ricci-flow singularities

摘录 / EXCERPT: Hamilton's conjecture. Finite-time singularities of the Ricci flow are “mostly” (or at least somewhere) modeled on a gradient shrinking soliton.

conjecture | open | MathDB 页面 | 原文线索

V2-036662 | MATHDB-347907

The Banach-space characterization conjecture for perfect earth mover's correlation

摘录 / EXCERPT: Let X and Y be Banach-space-valued random variables, and let f be a similarity transformation of the Banach space. The Banach-space perfect-correlation conjecture. The axiom...

conjecture | open | MathDB 页面 | 原文线索

V2-036663 | MATHDB-347928

Chamberland-Meisters injectivity conjecture for uniformly nonvanishing Jacobian eigenvalues

摘录 / EXCERPT: Let $F:\mathbb{R}^n\to\mathbb{R}^n$ be a C^1 map. Chamberland-Meisters' injectivity conjecture. Suppose there exists an $\varepsilon>0$ such that ... for every eigenvalue λ...

conjecture | open | MathDB 页面 | 原文线索

V2-036664 | MATHDB-347954

The characterization of positive functionals admitting Haar unitaries

摘录 / EXCERPT: Haar-unitary conjecture. The functional φ admits a Haar unitary if and only if there is no ideal $I\subseteq A$ such that A/I is scattered and

conjecture | open | MathDB 页面 | 原文线索

V2-036665 | MATHDB-347996

The matrix inverse realization conjecture for noncommutative rational functions

摘录 / EXCERPT: Matrix inverse realization conjecture. Each entry of $\mathfrak A^{-1}$ is a noncommutative rational expression in the entries of $\mathfrak A$: there exist noncommutative rational...

conjecture | open | MathDB 页面 | 原文线索

V2-036666 | MATHDB-347997

The polynomial inverse-factorization alternative for invertible noncommutative polynomial matrices

摘录 / EXCERPT: Let $\mathbb K\langle\underline{x}\rangle$ denote the free noncommutative polynomial algebra, let $\mathfrak r=(p_{ij})\in\mathbb K\langle\underline{x}\rangle^{L\times L}$ have non...

conjecture | open | MathDB 页面 | 原文线索

V2-036667 | MATHDB-348031

Characterization of continuous maps triple derivable at orthogonal pairs on JB*-triples

摘录 / EXCERPT: Orthogonal-pair derivability conjecture. Every continuous linear mapping $T:E\to E$ which is triple derivable at orthogonal pairs must be of the form $T=\delta+S$, where δ i...

conjecture | open | MathDB 页面 | 原文线索

V2-036668 | MATHDB-348044

Wiegerinck's conjecture on Bergman spaces of Stein domains

摘录 / EXCERPT: Let D be a Stein domain in $\mathbb C^n$, and let its Bergman space be the space of holomorphic functions on D that belong to $L^2(D)$ with respect to Lebesgue measure. Wiegeri...

conjecture | open | MathDB 页面 | 原文线索

V2-036669 | MATHDB-348045

Wiegerinck-type conjecture for L^2 holomorphic sections on compact Riemann surfaces

摘录 / EXCERPT: Let M be a compact, connected Riemann surface, let $E\rightarrow M$ be a holomorphic vector bundle with a smooth Hermitian metric h, and let dV be a smooth volume form. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-036670 | MATHDB-348046

Arveson's essential normality conjecture for quotient Bergman modules

摘录 / EXCERPT: Let $\Omega\subseteq\mathbb{C}^m$ be a bounded strongly pseudoconvex domain with smooth boundary, let A be the algebra of analytic functions under consideration, and let I be a...

conjecture | open | MathDB 页面 | 原文线索

V2-036671 | MATHDB-348063

The dimension-reduction conjecture for four-dimensional steady Ricci soliton singularity models

摘录 / EXCERPT: Dimension-reduction conjecture. If (M,g,f) is a 4-dimensional steady gradient Ricci soliton singularity model, then it dimension reduces to 3-manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-036672 | MATHDB-348079

Capacity rigidity conjecture for Riemannian manifolds

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold of dimension $n\geq 3$. For $p\in M$ and sufficiently small $r>0$, let $\operatorname{cap}_g(r,\lambda r)$ denote the relative capacity of the...

conjecture | open | MathDB 页面 | 原文线索

V2-036673 | MATHDB-348090

Rigidity conjecture for proper maps between type-I bounded symmetric domains

摘录 / EXCERPT: Let $D^{\mathrm{I}}_{p,q}$ denote the irreducible type-I bounded symmetric domain, and let $F:D^{\mathrm{I}}_{p,q}\to D^{\mathrm{I}}_{p',q'}$ be a proper holomorphic map, where...

conjecture | open | MathDB 页面 | 原文线索

V2-036674 | MATHDB-348134

The Rado-minor conjecture for M-natural-concave valuations

摘录 / EXCERPT: Let $v:2^{\mathcal G}\to\mathbb R$ be a valuation. An endowment operation with respect to $T\subseteq\mathcal G$ produces the valuation $v'(X)=v(X\cup T)-v(T)$, and a Rado minor va...

conjecture | open | MathDB 页面 | 原文线索

V2-036675 | MATHDB-348142

Nonvanishing normal derivative conjecture for Green potentials on the Sierpinski gasket

摘录 / EXCERPT: Nonvanishing normal derivative conjecture. For every integer $t\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-036676 | MATHDB-348313

Dimension-free estimates for discrete maximal functions of q-balls

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and let $B^q(d)\subset\mathbb{R}^d$ be a q-ball. The discrete restricted maximal function is denoted by $\mathcal{M}_{\ast,>D}^{G}$, and $\mathcal{C}(G,p)$ d...

open_problem | open | MathDB 页面 | 原文线索

V2-036677 | MATHDB-348335

Near second-order Markov property of MA-filtered first-order Markov processes

摘录 / EXCERPT: Near second-order Markov conjecture. The MA filtering of the first-order Markov process generates nearly a second-order Markov process.

conjecture | open | MathDB 页面 | 原文线索

V2-036678 | MATHDB-348380

The Lipschitz-rate and constant-density conjecture for collapsed Ricci limits

摘录 / EXCERPT: Volume-density conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-036679 | MATHDB-348401

Uniform curvature bound conjecture for inverse mean curvature flow in Euclidean three-space

摘录 / EXCERPT: Uniform curvature bound conjecture. The curvature remains uniformly bounded up to the maximal time:

conjecture | open | MathDB 页面 | 原文线索

V2-036680 | MATHDB-348444

The general v-adic convergence conjecture for Hayes-module exponentials and logarithms

摘录 / EXCERPT: General v-adic convergence conjecture. The exponential series $e_\rho(z)$ converges in $\mathbb{C}_v$ whenever

conjecture | open | MathDB 页面 | 原文线索

V2-036681 | MATHDB-348478

Euclidean local smoothing conjecture for wave equations

摘录 / EXCERPT: Suppose $M=\mathbb R^d$ has the flat metric, u solves the Cauchy problem ... and define $s_p=\frac{d-1}{2}-\frac{d-1}{p}$ and $p_d=\frac{2d}{d-1}$. Euclidean local smoothing conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-036682 | MATHDB-348498

Hélein's bounded Coulomb-frame conjecture for maps into the sphere

摘录 / EXCERPT: Hélein's conjecture. There exist an orthonormal moving frame $(\mathbf e_1,\mathbf e_2)$ and a function f such that

conjecture | open | MathDB 页面 | 原文线索

V2-036683 | MATHDB-348530

Amenability characterization by completely bounded Fourier multipliers

摘录 / EXCERPT: Amenability characterization. Γ is amenable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-036684 | MATHDB-348574

The sharp maximal-variation inequality on the integer lattice

摘录 / EXCERPT: Let $p e(1/2,1]$ and let $f:\mathbb{Z}\to\mathbb{R}$ be a function in $\ell^{p}(\mathbb{Z})$. Write ${\rm Var}_p f$ for the p-variation of f, and let Mf denote the centered di...

open_problem | open | MathDB 页面 | 原文线索

V2-036685 | MATHDB-348633

The Nitsche-threshold conjecture for minimizing harmonic diffeomorphisms of annuli

摘录 / EXCERPT: Nitsche-threshold conjecture. One has $r_\diamond<R$, and for every $r>r_\diamond$ there exists a minimizing ρ-harmonic diffeomorphism from $\mathbb{A}(r,1)$ onto...

conjecture | open | MathDB 页面 | 原文线索

V2-036686 | MATHDB-348674

Kakeya conjecture for Hausdorff, packing and Minkowski dimension

摘录 / EXCERPT: A Besicitch set is a subset $B\subset\mathbb{R}^n$ containing a unit line segment in every direction. For such a set, let ${\mathrm{dim}}\xspace_H B$, ${\mathrm{dim}}\xspace_P B$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036687 | MATHDB-348698

Yu and Chen's uniqueness conjecture for the Canham model

摘录 / EXCERPT: Let g denote the genus and let $\boldsymbol{\iota}_0\in(0,1]$ be the reduced area difference parameter for the Canham model. Let ... A solution is considered up to homothetic trans...

conjecture | open | MathDB 页面 | 原文线索

V2-036688 | MATHDB-348699

Positivity conjecture for the Canham-model recurrence sequence

摘录 / EXCERPT: Let (d_n) be the unique sequence of rational numbers with initial terms ... satisfying the explicit order-seven linear recurrence ... where each $r_i(n)\in\mathbb{Z}[n]$ is defined b...

conjecture | open | MathDB 页面 | 原文线索

V2-036689 | MATHDB-348739

The completely bounded unital contraction conjecture for operator algebras

摘录 / EXCERPT: Let $A\subset B(\mathcal H)$ be a unital operator algebra. Let $\theta:A\to M_n(\mathbb{C})$ be a unital completely bounded homomorphism, and let $\alpha:A\to A$ be a unital antili...

conjecture | open | MathDB 页面 | 原文线索

V2-036690 | MATHDB-348811

The circularly slit disk formula for the squeezing function

摘录 / EXCERPT: Let $\Omega\subset\mathbb C$ be an m-connected domain without degenerate boundary components. For each $z\in\Omega$ and $j=0,\ldots,m-1$, let $f_{z,j}$ be the unique conformal ma...

conjecture | open | MathDB 页面 | 原文线索

V2-036691 | MATHDB-348867

Ash-Catoiu conjecture on directional and relative Lipschitz continuity

摘录 / EXCERPT: Ash-Catoiu conjecture. The set F is a null set.

conjecture | open | MathDB 页面 | 原文线索

V2-036692 | MATHDB-348885

Multiplicity invariance under left-definite theory

摘录 / EXCERPT: Let ${\bf A}$ be a semi-bounded self-adjoint operator in a Hilbert space $\mathcal{H}$, and let λ be an eigenvalue of ${\bf A}$ with multiplicity m. Let ${\bf A}_r$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-036693 | MATHDB-348908

Derivative-order conjecture for residues of closed submanifolds

摘录 / EXCERPT: Let M be a closed submanifold of dimension m, and let $j\ge2$. Its ν-weighted residue at $z=-m-2j$ is determined locally from a graph representation of M. Derivative-orde...

conjecture | open | MathDB 页面 | 原文线索

V2-036694 | MATHDB-348911

Invariants conjecture for overconvergent substitutions in the Robba ring

摘录 / EXCERPT: Invariants conjecture. If φ is overconvergent, then

conjecture | open | MathDB 页面 | 原文线索

V2-036695 | MATHDB-348912

Convergence conjecture for substitutions in the Robba ring

摘录 / EXCERPT: Convergence conjecture. If $h\in\mathcal{R}$ is such that $\varphi(h)$ is convergent on the annulus

conjecture | open | MathDB 页面 | 原文线索

V2-036696 | MATHDB-348913

Invariant field conjecture for substitutions in the Robba ring

摘录 / EXCERPT: Invariant field conjecture. In general,

conjecture | open | MathDB 页面 | 原文线索

V2-036697 | MATHDB-348923

Conjecture on an n-independent convergence rate for Kalantari's bound

摘录 / EXCERPT: Let $p(z)$ be a polynomial of degree n, let $\rho_{\min}:=\min_{\rho:p(\rho)=0}|\rho|$, and let L_m be the lower bound in the inequality $L_m/\rho_{\min}\geq 1-\epsilon$. n-ind...

conjecture | open | MathDB 页面 | 原文线索

V2-036698 | MATHDB-348966

Eigenvalue realization conjecture for locally singular matrices in \(\mathcal{S}^{n,k}\)

摘录 / EXCERPT: Let $H(e^n_k)$ denote the hyperbolicity cone of the elementary symmetric polynomial e^n_k, and let $DG(n,k)D$ be a diagonal congruence of the matrix $G(n,k)$ by a diagonal matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-036699 | MATHDB-349019

Subcritical p-improving estimate conjecture for discrete polynomial curves

摘录 / EXCERPT: Let $d\geq 1$ and let $\gamma(n)=(P_1(n),\ldots,P_d(n))$ be a polynomial curve in $\mathbb{Z}^d$, where $P_1,\ldots,P_d\in\mathbb{Z}[X]$ have separated degrees, meaning...

conjecture | open | MathDB 页面 | 原文线索

V2-036700 | MATHDB-349048

The conjecture on the number of maximum modulus curves for magic entire functions

摘录 / EXCERPT: Let f be an entire function of the form specified in the paper, and let μ_f denote its inner degree. Write J_f for the index set parametrizing the components of the maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-036701 | MATHDB-349073

Asymptotic inadequacy of bounded-degree Gaussian polynomials for $\mathcal{C}^+(A)$

摘录 / EXCERPT: For $A>1$, let $\mathcal{C}^+(A)$ denote the relevant Fourier optimization problem, let $J_A^+(F)$ be its objective functional, and let $\mathcal{F}_n$ be the class of nonzero func...

conjecture | open | MathDB 页面 | 原文线索

V2-036702 | MATHDB-349087

Characterization of C-approximate symmetry of Birkhoff-James orthogonality

摘录 / EXCERPT: The conjecture. Birkhoff-James orthogonality is C-approximately symmetric in X if and only if the local property (P) holds for each $x\in\operatorname{Ext} B_X$.

conjecture | open | MathDB 页面 | 原文线索

V2-036703 | MATHDB-349106

Remizov's first-order convergence conjecture for Chernoff approximations

摘录 / EXCERPT: Let $(e^{tL})_{t\geq 0}$ be a C_0-semigroup in a Banach space $\mathcal{F}$ with generator $(L,\operatorname{dom}(L))$, and let G be a Chernoff function for L. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-036704 | MATHDB-349108

Remizov's dense second-order approximation-subspace conjecture

摘录 / EXCERPT: Remizov's dense second-order approximation-subspace conjecture. The function G has a dense approximation subspace $A^{[0,t_0)}_{\zeta}$ of order $\zeta(n)=1/n^2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036705 | MATHDB-349145

The multiplier conjecture for spectral balls of positive Rockland operators

摘录 / EXCERPT: Let $\mathbb{G}$ be a graded Lie group with $\dim(\mathbb{G})\geq 2$, let $\mathcal{G}$ be a positive Rockland operator on $\mathbb{G}$, and let χ be the characteristic functi...

conjecture | open | MathDB 页面 | 原文线索

V2-036706 | MATHDB-349158

Mass-based asymptotic selection for perturbed unstable cubic-quintic ground states

摘录 / EXCERPT: For $\omega<\omega_c$, consider initial data of the form ... Here Q_ω is a cubic-quintic ground state, M denotes mass, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036707 | MATHDB-349197

Monotonicity and ordering conjecture for generalised Airy recurrence coefficients

摘录 / EXCERPT: Let $\alpha_n(t;\lambda)$ and $\beta_n(t;\lambda)$ be the recurrence coefficients appearing in the generalised Airy polynomial system, and let t_n^* denote the location of the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-036708 | MATHDB-349263

Conjecture on polynomial asymptotic structures of PT-symmetric vector rational rogue waves

摘录 / EXCERPT: Let $\mathcal{PT}$-symmetric vector rational rogue waves be solutions of the multi-component nonlinear Schrödinger equations, and let their asymptotic structures refer to the patte...

conjecture | open | MathDB 页面 | 原文线索

V2-036709 | MATHDB-349295

Continuity of the Levy-Lieb and Lieb functionals

摘录 / EXCERPT: Continuity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-036710 | MATHDB-349411

Confluent exponentiation conjecture for conformal blocks of type G

摘录 / EXCERPT: Confluent exponentiation conjecture for type $\mathcal G$. The confluent conformal block exponentiates semiclassically:

conjecture | open | MathDB 页面 | 原文线索

V2-036711 | MATHDB-349434

Existence of non-normal EP operators with prescribed range

摘录 / EXCERPT: Existence conjecture. There exists a non-normal EP operator T on $\mathcal H$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036712 | MATHDB-349464

The L^{12} square-root cancellation conjecture for nondegenerate curves in ${\mathbb R}^4$

摘录 / EXCERPT: Let ϕ_3,ϕ_4 be real analytic functions on $(0,3)$ satisfying the derivative bounds and nondegeneracy conditions ... ... For a finite interval $I\subset {\mathbb Z}$, define .....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-036713 | MATHDB-349465

The L^p square-root cancellation conjecture for nondegenerate curves in ${\mathbb R}^4$

摘录 / EXCERPT: Square-root cancellation in L^p. For $11\le p\le12$ and $\alpha\ge\beta\ge0$ satisfying $\alpha+\beta=p/2-3$, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-036714 | MATHDB-349564

Equicontinuity conjecture for derivative nonlinear Schrödinger flows

摘录 / EXCERPT: Equicontinuity conjecture. For any $Q\subseteq\mathcal{S}$ that is L^2-bounded and equicontinuous, the set Q_* is also L^2-equicontinuous.

conjecture | open | MathDB 页面 | 原文线索

V2-036715 | MATHDB-349565

Equicontinuity conjecture for flows conserving the perturbation determinant

摘录 / EXCERPT: Perturbation-determinant equicontinuity conjecture. For any L^2-bounded and equicontinuous $Q\subseteq\mathcal{S}$, the set Q_* is also L^2-equicontinuous.

conjecture | open | MathDB 页面 | 原文线索

V2-036716 | MATHDB-349598

Existence of unitary orthonormal bases for finite-index regular subfactors

摘录 / EXCERPT: Let $N\subset M$ be a finite-index regular inclusion of factors of type II_1. A unitary orthonormal basis conjecture asserts that M/N has a unitary orthonormal basis. The quest...

conjecture | open | MathDB 页面 | 原文线索

V2-036717 | MATHDB-349604

A structured numerical-range realization conjecture

摘录 / EXCERPT: Let $B\in \mathbb{C}^{d \times d}$, and suppose that its numerical range $\mathcal{W}(B)$ is invariant under multiplication by $e^{2\pi i/n}$. Let $\mathcal{C}_{\mathbb{C}}(n,d)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-036718 | MATHDB-349605

An arbitrary-size cyclic weighted-shift conjecture for higher numerical ranges

摘录 / EXCERPT: Let $F_B\in\mathbb{C}[t,x,y]^{C_n}_d$ be the homogeneous polynomial associated with a matrix $B\in\mathbb{C}^{d\times d}$, and let $\mathcal{W}_k(B)$ denote its k-th numerical ra...

conjecture | open | MathDB 页面 | 原文线索

V2-036719 | MATHDB-349608

Nonexistence conjecture for nonnegative solutions under integral volume conditions

摘录 / EXCERPT: Nonexistence conjecture. If either of these integral conditions is satisfied on M, then every nonnegative solution of the equation referred to as $$ is identically zero.

conjecture | open | MathDB 页面 | 原文线索

V2-036720 | MATHDB-349644

Longest minimal length partition conjecture

摘录 / EXCERPT: Longest minimal length partition conjecture. Given $c\in(0,1)$, the set Ω maximizing $L(\Omega,c)$ under the constraint $|\Omega|=v_d$ is the ball. Given $n>1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036721 | MATHDB-349648

Global-minimum conjecture for the Fourier discrepancy function

摘录 / EXCERPT: Let N be the parameter appearing in the definition ... and let $d^*=\frac{N}{2}$, which is a local minimum of $g:[0,N]\to\mathbb{R}$. Global-minimum conjecture. The point...

conjecture | open | MathDB 页面 | 原文线索

V2-036722 | MATHDB-349664

Scalar-curvature entropy conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Scalar-curvature entropy conjecture. If the scalar curvature of g is at least -6, then

conjecture | open | MathDB 页面 | 原文线索

V2-036723 | MATHDB-349717

Asymptotic strengthened convexity conjecture for d-th power partitions

摘录 / EXCERPT: Let $d\in\mathbb{N}_{\geq 2}$, and let $p_d(n)$ denote the number of partitions of n into d-th powers, defined by ... A sequence $(a_n)_{n\in\mathbb{N}}$ is convex when...

conjecture | open | MathDB 页面 | 原文线索

V2-036724 | MATHDB-349718

Asymptotic strengthened log-concavity conjecture for d-th power partitions

摘录 / EXCERPT: Let $d\in\mathbb{N}_{\geq 2}$, and let $p_d(n)$ denote the number of partitions of n into d-th powers, defined by ... A sequence of positive real numbers $(a_n)_{n\in\mathbb{N}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-036725 | MATHDB-349722

Vector-valued Maclaurin inequality

摘录 / EXCERPT: Fix vectors $v_1,\dots,v_m\in\mathbb{R}^d$ with $1\le d\le m$. For integers k satisfying $2\le k\le d$, let $|v_{i_1}\wedge\cdots\wedge v_{i_k}|$ denote the Euclidean norm of the...

conjecture | open | MathDB 页面 | 原文线索

V2-036726 | MATHDB-349755

Balanced-refinement core conjecture for Lipschitz selections

摘录 / EXCERPT: Let $({\mathcal M},\rho)$ be a pseudometric space, let X be a Banach space, and let m be a fixed positive integer. Set ... For a set-valued mapping...

conjecture | open | MathDB 页面 | 原文线索

V2-036727 | MATHDB-349756

Stabilization conjecture for iterated balanced refinements

摘录 / EXCERPT: Stabilization conjecture. There exist $\ell+1$ numbers $\lambda_1,\ldots,\lambda_{\ell+1}\ge1$ such that, for every F satisfying the finiteness-principle hypothesis that each res...

conjecture | open | MathDB 页面 | 原文线索

V2-036728 | MATHDB-349760

Existence conjecture for global minimal-surface-type solutions asymptotic to Lawson cones

摘录 / EXCERPT: Let C_{kl} be the Lawson cone and let Φ be an integrand constructed as in Theorem Main. An elliptic extension of Φ is an extension to $\mathbb{R}^{k+l+3}$ that remains...

conjecture | open | MathDB 页面 | 原文线索

V2-036729 | MATHDB-349764

Threshold conjecture for propagation and blocking in funnel-shaped domains

摘录 / EXCERPT: Propagation and blocking threshold conjecture. For every $R>0$, there is $\alpha_R\in(0,\pi/2]$ such that complete propagation holds for all $\alpha\in(0,\alpha_R)$, while blocking...

conjecture | open | MathDB 页面 | 原文线索

V2-036730 | MATHDB-349790

Cohn-Kumar-Miller-Radchenko-Viazovska conjecture on hexagonal-lattice interpolation

摘录 / EXCERPT: Let $r_1,r_2,\ldots$ be the positive real numbers of the form ... where j,k are integers. A radial Schwartz function is a function $f:\mathbf{R}^2\to\mathbf{R}$ invariant under rot...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-036731 | MATHDB-349904

Barceló-Bennett-Carbery-Rogers conjecture on wave-equation divergence sets

摘录 / EXCERPT: Let $d\geq 3$ and, for $s>0$, define ... where ... and $\dim_H$ denotes Hausdorff dimension. Barceló-Bennett-Carbery-Rogers conjecture. ... The quantity $\mathcal D_d(s)$ measures the la...

conjecture | open | MathDB 页面 | 原文线索

V2-036732 | MATHDB-349944

Khavinson's conjecture for hyperbolic harmonic functions on the unit ball

摘录 / EXCERPT: Khavinson's conjecture. Let $p\in(1,\infty]$, $n\geq3$, and $x\in{\mathbb B}^n\setminus\{0\}$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-036733 | MATHDB-349988

A priori error estimate for the approximate C^1 isogeometric discretization

摘录 / EXCERPT: Let $\varphi_h\in\widetilde{\mathcal{V}}^1_{h,0}$ solve the discrete problem on the two-patch domain, and let $p_1^{(S)}\geq 2$, $p_2\geq 3$, and $r_1,r_2,\widetilde{r}\geq 1$. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-036734 | MATHDB-350105

Gauge-coaction normality conjecture for reductions to amenable groups

摘录 / EXCERPT: Gauge-coaction normality conjecture. Every gauge coaction on a representation of Λ is normal. Consequently, the other results in the relevant section should require only th...

conjecture | open | MathDB 页面 | 原文线索

V2-036735 | MATHDB-350170

The endpoint fourth-moment conjecture for quadratic Weyl sums

摘录 / EXCERPT: Endpoint fourth-moment conjecture. The conjecture is

conjecture | open | MathDB 页面 | 原文线索

V2-036736 | MATHDB-350178

Conjectured Kolmogorov-number lower bound for quasi-Banach Schatten embeddings

摘录 / EXCERPT: Let S_p^N and S_q^N be Schatten classes of $N\times N$ matrices, and let d_n denote the nth Kolmogorov number of the embedding. For $0<p\leq\infty$, $0<q\leq1$, $q\leq p$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036737 | MATHDB-350215

The class-number asymptotic conjecture for \(\Gamma(2)'\)

摘录 / EXCERPT: Let L be the ambient subgroup under consideration, and let $h(t)$ be the number of $\operatorname{SL}_2(\mathbb{Z})$-conjugacy classes of hyperbolic matrices in L with trace...

conjecture | open | MathDB 页面 | 原文线索

V2-036738 | MATHDB-350218

The average word-length conjecture for trace classes

摘录 / EXCERPT: Let L be the ambient subgroup, let $h(t)$ count the $\operatorname{SL}_2(\mathbb{Z})$-conjugacy classes of hyperbolic matrices in L with trace t, and let $\ell([g])$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-036739 | MATHDB-350239

The small-or-perfect geodesic minimizer conjecture for G-alpha

摘录 / EXCERPT: Let G_α, with $-1\leq\alpha\leq 1$, be the homogeneous Riemannian 3-manifold equipped with its canonical left-invariant metric. For a geodesic segment γ of length T...

conjecture | open | MathDB 页面 | 原文线索

V2-036740 | MATHDB-350240

Grayson's balanced figure-eight convergence conjecture

摘录 / EXCERPT: A balanced figure-eight is a figure-eight curve whose lobes bound regions of equal area, and the curve-shortening flow is the evolution of a plane curve by curvature. Grayson's con...

conjecture | open | MathDB 页面 | 原文线索

V2-036741 | MATHDB-350241

Volume entropy monotonicity conjecture for the G-alpha family

摘录 / EXCERPT: For $-1\leq\alpha\leq 1$, let G_α be the homogeneous Riemannian manifold in the interpolating family, and let $h(G_\alpha)$ denote its volume entropy, defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-036742 | MATHDB-350242

Period-function monotonicity and limiting-value conjecture for G-alpha

摘录 / EXCERPT: Let $P(\beta)$ be the period function in G_α, with parameter α in the interpolating family, and let β be its period-function parameter. Period-function conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-036743 | MATHDB-350243

Limit conjecture for the half-period coordinate in G-alpha

摘录 / EXCERPT: For $\alpha\in(0,1]$, define ... where $P(x_0,\alpha)$ is the period and $b_{x_0,\alpha}$ is the corresponding coordinate along the geodesic flow. Limit conjecture for $L(\alpha)$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-036744 | MATHDB-350268

A liminf upper bound for the radii of univalence of harmonic derivatives

摘录 / EXCERPT: Let $f(z)=\sum_{k=1}^{\infty}a_k z^k+\overline{\sum_{k=1}^{\infty}b_kz^k}\in \mathcal{S}_{H}$. Let R_f be the radius of convergence of f, let R_n be the radius of univalence...

conjecture | open | MathDB 页面 | 原文线索

V2-036745 | MATHDB-350284

De Lellis-Otto rectifiability and jump-structure conjecture for Eikonal entropy productions

摘录 / EXCERPT: Let $[...].$

conjecture | open | MathDB 页面 | 原文线索

V2-036746 | MATHDB-350303

Khavinson-Shapiro conjecture for planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}$ be a bounded domain whose boundary consists of finitely many non-intersecting Jordan curves. For each polynomial boundary datum on $\partial\Omega$, c...

conjecture | open | MathDB 页面 | 原文线索

V2-036747 | MATHDB-350382

The sector cross-ratio supremum conjecture

摘录 / EXCERPT: The sector cross-ratio supremum conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036748 | MATHDB-350383

The sharp quotient bounds conjecture for the Möbius metric in sectors

摘录 / EXCERPT: Let $Q(k,\theta)$ denote the quotient studied in the paper, with $0<k<1$ and $0<\theta<2\pi$. The sharp quotient bounds conjecture. For all such k and θ, ... and ... The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-036749 | MATHDB-350384

The convex polygonal-domain Möbius-hyperbolic inequality conjecture

摘录 / EXCERPT: Let $G\subsetneq\mathbb{R}^2$ be a bounded convex polygonal domain, and let ρ_G and δ_G denote the Möbius and hyperbolic metrics in G. The convex polygonal-domain ine...

conjecture | open | MathDB 页面 | 原文线索

V2-036750 | MATHDB-350385

The regular polygon convergence conjecture for Möbius and hyperbolic metrics

摘录 / EXCERPT: Let G_k be the regular k-gon described in the paper, and let $x,y\in\bigcap_k G_k$ be distinct points. Let ρ_{G_k} and δ_{G_k} denote the Möbius and hyperbolic metr...

conjecture | open | MathDB 页面 | 原文线索

V2-036751 | MATHDB-350398

Regularity of maximizers of the weighted Cheeger functional on polygons

摘录 / EXCERPT: Let $\mathcal{J}$ be the functional on the class $\mathcal{P}_n$ of simple polygons used in the paper. For $n\in\{3,4\}$, the maximizers of $\mathcal{J}$ over $\mathcal{P}_n$ are r...

conjecture | open | MathDB 页面 | 原文线索

V2-036752 | MATHDB-350399

Regularity of critical polygons for the weighted Cheeger functional

摘录 / EXCERPT: Let $\mathcal{J}$ be the functional on the class $\mathcal{P}_n$ of simple polygons used in the paper. A polygon is a critical point of $\mathcal{J}$ when the limit in the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-036753 | MATHDB-350447

Maximal asymmetric alpha-stability conjecture

摘录 / EXCERPT: Let ρ satisfy $0\leq\rho\leq1$, let $b1\in[1,9]$, and let $\mathbf{MaxStab}_{\alpha}(1/2)$ denote the maximal asymmetric $b1$-stability at mean $1/2$. A dictator function is a...

conjecture | open | MathDB 页面 | 原文线索

V2-036754 | MATHDB-350472

The Bochner-Riesz conjecture for Hermite expansions

摘录 / EXCERPT: The Bochner-Riesz conjecture for Hermite expansions. The uniform estimate

conjecture | open | MathDB 页面 | 原文线索

V2-036755 | MATHDB-350488

Conjecture on signed measures with prescribed vanishing moments

摘录 / EXCERPT: Let $a,b\in \mathbb{R}$ with $0<a<b$. Let $(n_k)_{k \in \mathbb{N}}$ be a strictly increasing sequence of positive integers such that ... A non-trivial signed measure with prescribed...

conjecture | open | MathDB 页面 | 原文线索

V2-036756 | MATHDB-350498

Local moment conjecture for exponential sums

摘录 / EXCERPT: Local moment conjecture. For every $j\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-036757 | MATHDB-350500

Sphere exponential-sum restriction conjecture

摘录 / EXCERPT: Let $d\geq 2$, $N\geq 1$, and define ... Let $\mathcal{M}$ be a real analytic hypersurface in $\mathbb{T}^d$ with nonnegative curvature and surface measure σ. Sphere restricti...

conjecture | open | MathDB 页面 | 原文线索

V2-036758 | MATHDB-350501

Paraboloid exponential-sum restriction conjecture

摘录 / EXCERPT: Let $d\geq 2$, $N\geq 1$, and define ... ... Let $\mathcal{M}$ be a real analytic hypersurface in $\mathbb{T}^d$ with nonnegative curvature and surface measure σ. Paraboloid res...

conjecture | open | MathDB 页面 | 原文线索

V2-036759 | MATHDB-350543

L^p smallness conjecture for three-dimensional Dirac operators

摘录 / EXCERPT: L^p smallness conjecture. There exists a positive constant $d4c$ independent of V such that

conjecture | open | MathDB 页面 | 原文线索

V2-036760 | MATHDB-350544

Positive weights at the minimal exact discretization

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a compact subset, let μ be a finite measure on Ω, and let X_N be an N-dimensional subspace of the real space $L_2(\Omega,\mu)$ who...

conjecture | open | MathDB 页面 | 原文线索

V2-036761 | MATHDB-350565

Zigzag-front peak-angle convergence conjecture

摘录 / EXCERPT: Let θ be the direction of a zigzag front, and let $\theta_{\mathrm{m}}(t)$ and $\theta_{\mathrm{p}}(t)$ determine the two angles forming a peak, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-036762 | MATHDB-350584

Tsfasman-Vladuts generalized Brauer-Siegel conjecture

摘录 / EXCERPT: Tsfasman-Vladuts conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036763 | MATHDB-350734

BMS conjecture on L^2-positivity preservation

摘录 / EXCERPT: Let (M,g) be a complete Riemannian manifold. It is L^2-positivity preserving if, for every $u\in L^2(M)$, the implication ... holds. BMS conjecture. If (M,g) is geodesically co...

conjecture | open | MathDB 页面 | 原文线索

V2-036764 | MATHDB-350780

Hellerstein-Sheil-Small-Wittich classification conjecture for real meromorphic functions

摘录 / EXCERPT: Hellerstein-Sheil-Small-Wittich conjecture. Then f satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036765 | MATHDB-350785

Lusin cotype characterization of martingale cotype for Schrödinger semigroups

摘录 / EXCERPT: Let $\{T_t\}_{t>0}$ be the heat semigroup generated by a Schrödinger operator as above, and let $\{P_t\}_{t>0}$ be the Poisson semigroup subordinated to $\{T_t\}_{t>0}$. Let X be...

conjecture | open | MathDB 页面 | 原文线索

V2-036766 | MATHDB-350807

The desingularization conjecture for singular spherical and hyperbolic orbifolds

摘录 / EXCERPT: Let $(M_o,\mathbf{g}_o)$ be a singular spherical or hyperbolic compact orbifold. A sequence of smooth Einstein metrics is said to converge to $(M_o,\mathbf{g}_o)$ in the Gromov-Hau...

conjecture | open | MathDB 页面 | 原文线索

V2-036767 | MATHDB-350811

Lemos-Soares singular-value log-majorization conjecture

摘录 / EXCERPT: Lemos-Soares conjecture. If $A,B\geq0$, $r,s\in\mathbb{R}$, and $0\leq t\leq1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036768 | MATHDB-350812

Generalized eigenvalue log-majorization conjecture for geometric means

摘录 / EXCERPT: Generalized geometric-mean conjecture. If $0\leq t\leq1$, $p\geq1$, and either $r,s\geq1$ or $r,s\leq0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036769 | MATHDB-350813

Complementary norm inequality for arithmetic and geometric means

摘录 / EXCERPT: Complementary norm conjecture. For $1\leq p\leq\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-036770 | MATHDB-350824

Angenent-Daskalopoulos-Sesum uniqueness conjecture for symmetric ancient ovals

摘录 / EXCERPT: Let an ancient oval be an ancient compact noncollapsed mean curvature flow that is not self-similar, and let $1\leq k\leq n$. An ancient oval is...

conjecture | open | MathDB 页面 | 原文线索

V2-036771 | MATHDB-350933

The converse characterization of convex Lipschitzian functions

摘录 / EXCERPT: Let X be the domain of a Lipschitzian function $F:X\to\mathbb{R}$, and let $\nabla F(\boldsymbol{x})$ denote the set of subgradients of F at $\boldsymbol{x}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036772 | MATHDB-350994

Maximal left ideal classification conjecture for Hermitian locally compact groups

摘录 / EXCERPT: Let G be a Hermitian locally compact group. For a representation π of G and a unit vector ξ in its representation space $\mathcal{H}_\pi$, define ... and, for $f\in L^1(G)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036773 | MATHDB-350995

Vanishing-at-infinity criterion for maximal left ideals

摘录 / EXCERPT: Let G be a Hermitian locally compact group, let $\widehat{G}$ denote its irreducible representations, and for $\pi\in\widehat{G}$ and a unit vector $\xi\in\mathcal{H}_\pi$ define...

conjecture | open | MathDB 页面 | 原文线索

V2-036774 | MATHDB-351034

The Fubini property conjecture for unions of affine subspaces

摘录 / EXCERPT: Fubini property conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-036775 | MATHDB-351205

The Nikodym maximal-function conjecture

摘录 / EXCERPT: Nikodym maximal-function conjecture. For $1<p<n-d+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-036776 | MATHDB-351206

The off-diagonal Nikodym maximal-function conjecture

摘录 / EXCERPT: For $1\leq d<n$, let $\mathcal N_\delta$ denote the (d,n)-Nikodym maximal function. For $1\leq p\leq n-d+1$, set $q=p'(n-d)$, where p' is the Hölder conjugate of p. Off-diago...

conjecture | open | MathDB 页面 | 原文线索

V2-036777 | MATHDB-351212

The finite-dimensional limiting dimension conjecture for finitely strictly singular adjustment

摘录 / EXCERPT: Let X be a Banach space, and let $(M_n)_{\mathbb{N}'}$ and $(P_n)_{\mathbb{N}'}$ be sequences of subspaces of X, with $(P_n)_{\mathbb{N}'}$ finite-dimensional. For $\lambda<1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036778 | MATHDB-351272

Forest conjecture for supports of asymptotic derivative measures

摘录 / EXCERPT: Let P be a polynomial and let $\x03bc_{\x03b1,P}$ be the asymptotic root-counting measure from Theorem. Forest conjecture. For $0<\x03b1<\deg P$, the support of...

conjecture | open | MathDB 页面 | 原文线索

V2-036779 | MATHDB-351307

Kindler-Safra conjecture for balanced Boolean functions on the symmetric group

摘录 / EXCERPT: Let S_n be the symmetric group, and let $f\colon S_n\to\{0,1\}$ be a Boolean function. Say that f is ϵ-close to degree d if its distance from the space of degree-d...

conjecture | open | MathDB 页面 | 原文线索

V2-036780 | MATHDB-351325

Wigderson-Wigderson conjecture on the range of Fourier norm ratios

摘录 / EXCERPT: Wigderson-Wigderson conjecture. For all $q\neq 2$, the image of

conjecture | open | MathDB 页面 | 原文线索

V2-036781 | MATHDB-351340

Generalized Mazya conjecture for weakly cancelling Fourier multipliers

摘录 / EXCERPT: Let $\mathrm{T}\colon S^{d-1}\to\mathbb{R}^{\ell}$ be sufficiently smooth, at least Hölder continuous, and weakly cancelling: ... For $\alpha\in(0,d)$, define ... let...

open_problem | open | MathDB 页面 | 原文线索

V2-036782 | MATHDB-351388

The Fourier-reflexivity characterization of hierarchical posets

摘录 / EXCERPT: Fourier-reflexivity characterization. If $|H_i|\geqslant2$ for all $i\in\Omega$ and $\mathcal{Q}(\boldsymbol{H},\boldsymbol{P})$ is Fourier-reflexive, then $\boldsymbol{P}$ is hier...

conjecture | open | MathDB 页面 | 原文线索

V2-036783 | MATHDB-351413

Asymptotic limit conjecture for Bloch-to-Bergman inclusion norms

摘录 / EXCERPT: Asymptotic limit conjecture. For any $\alpha>-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-036784 | MATHDB-351425

Quaternionic quotient-field conjecture for slice-preserving semiregular functions

摘录 / EXCERPT: Let D be a symmetric slice domain in the quaternions, and let $\mathcal{O}_s(D)$ denote the integral domain of slice-preserving regular functions on D, with quotient field...

conjecture | open | MathDB 页面 | 原文线索

V2-036785 | MATHDB-351450

Uniform decoupling conjecture for the parabola below scale N^2

摘录 / EXCERPT: Let $s\in[1,2)$ and $2\le p\le 2+2s$. Write $D(N,N^s,p)$ for the decoupling constant associated with the parabola at spatial scale N^s and exponent p. Uniform decoupling conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-036786 | MATHDB-351451

Hamilton's prescribed cross curvature conjecture on the three-sphere

摘录 / EXCERPT: Let $\mathbb S^3$ be the three-sphere, let Y be a positive-definite symmetric $(0,2)$-tensor field on $\mathbb S^3$, and let $X(g)$ denote the cross curvature tensor of a Riemann...

conjecture | open | MathDB 页面 | 原文线索

V2-036787 | MATHDB-351535

Forward stability conjecture for the Schur canonical form

摘录 / EXCERPT: Let $A_0=U_0T_0U_0^*\in\mathbb{C}^{n\times n}$, where U_0 is unitary and T_0 is upper triangular. A factorization of A as $A=UTU^*$ is a Schur decomposition when U is unita...

conjecture | open | MathDB 页面 | 原文线索

V2-036788 | MATHDB-351570

Rigidity conjecture for round cylinders in Ricci shrinkers

摘录 / EXCERPT: Round-cylinder rigidity conjecture. There exists a small constant $\epsilon=\epsilon(n,N)>0$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-036789 | MATHDB-351624

Uniqueness conjecture for fair-cut centers

摘录 / EXCERPT: Let M be a compact convex manifold, and let a fair-cut center mean a point maximizing the fair-cut function φ (equivalently, attaining the fair-cut index $\Phi(M)$). Uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-036790 | MATHDB-351626

Dimension-independent lower-bound conjecture for the fair-cut index

摘录 / EXCERPT: Let M be a compact convex manifold with non-positive sectional curvature, and let $\Phi(M)$ denote its fair-cut index. Fair-cut index lower-bound conjecture. The sharp inequality...

conjecture | open | MathDB 页面 | 原文线索

V2-036791 | MATHDB-351679

Banach's conjecture on affinely equivalent hyperplane sections

摘录 / EXCERPT: Let B be a convex body in $\mathbb{R}^{n+1}$ with the origin as an interior point. Two hyperplane sections through the origin are affinely equivalent if an affine isomorphism map...

conjecture | open | MathDB 页面 | 原文线索

V2-036792 | MATHDB-351680

Banach's conjecture in dimension four

摘录 / EXCERPT: Let B be a 4-dimensional convex body with the origin as an interior point. Two hyperplane sections through the origin are affinely equivalent if an affine isomorphism maps one...

conjecture | open | MathDB 页面 | 原文线索

V2-036793 | MATHDB-351755

Failure of the pointwise-convergence hypothesis in the non-compact sufficiency theorem

摘录 / EXCERPT: Pointwise-convergence failure conjecture. The conclusion that p is a Birkhoff-James extension fails if the assumption that $p(x_n,\,\rule{3mm}{0.15mm})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036794 | MATHDB-351845

Endpoint trace inequality characterization for divergence-free charges

摘录 / EXCERPT: Let F be an $\mathbb{R}^d$-valued charge with $\operatorname*{div}F=0$, let $ u$ be a measure on $\mathbb{R}^d$, and let $K_1,K_2,\ldots,K_d$ be sufficiently smooth functions on...

conjecture | open | MathDB 页面 | 原文线索

V2-036795 | MATHDB-351861

Dimension-free irrelevance of low-influence directions for convex sets

摘录 / EXCERPT: Let $K \subseteq \mathbb{R}^n$ be symmetric and convex, and let $v \in \mathbb{S}^{n-1}$ satisfy $\operatorname{Inf}_v[K] \leq \varepsilon$. A symmetric convex set L can be compa...

open_problem | open | MathDB 页面 | 原文线索

V2-036796 | MATHDB-351868

The hierarchy conjecture for polynomials preserving nonnegative matrices

摘录 / EXCERPT: For $n\geq 3$, let $\mathscr{P}_n$ denote the set of polynomials that preserve entrywise nonnegative $n\times n$ matrices under polynomial evaluation. Hierarchy conjecture. ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-036797 | MATHDB-351894

Iwaniec's L^p norm conjecture for the Ahlfors-Beurling operator

摘录 / EXCERPT: Let T be the Ahlfors-Beurling operator on $L^p(\mathbb C)$, let $p^*=\max\{p,p/(p-1)\}$, and write $\|T\|_p$ for its operator norm. Iwaniec's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-036798 | MATHDB-351912

Asymptotic formula for the meromorphic 3D-index

摘录 / EXCERPT: Let M be a one-cusped hyperbolic 3-manifold, let $\hat M$ be its end compactification, and let $\mathcal{X}^{\mathrm{geo}}$ be the relevant set of conjugacy classes of geometric-...

conjecture | open | MathDB 页面 | 原文线索

V2-036799 | MATHDB-351915

The local smoothing conjecture for fractional Schrödinger equations

摘录 / EXCERPT: Local smoothing conjecture. Fix $\alpha\in(0,1)\cup(1,\infty)$. For $p>\frac{2n}{n-1}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-036800 | MATHDB-351916

The restriction type conjecture for the operator E_R

摘录 / EXCERPT: Let $n\geq 3$, let $\alpha\in(0,1)\cup(1,\infty)$, and define the restriction-type operator ... where $e(t):=e^{2\pi it}$, ψ is compactly supported and smooth, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036801 | MATHDB-351958

The full numerator criterion for pure stable polynomials

摘录 / EXCERPT: Full numerator criterion. For every $f\in\mathbb{C}[z_1,z_2]$, f/p is locally H^{∞} if and only if $f\in\mathcal{I}$.

conjecture | open | MathDB 页面 | 原文线索

V2-036802 | MATHDB-351968

Anisotropic cancellation conjecture for fractional integration

摘录 / EXCERPT: Let $a\in\mathbb{R}^d$ have positive coordinates satisfying $\sum_{j=1}^d a_j=d$. Let K_a be the anisotropic kernel defined by ... where S^{d-1} is the unit sphere. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-036803 | MATHDB-351984

Realization conjecture for rational inner functions of matrix argument

摘录 / EXCERPT: Let $\mathrm B_m$ and $\mathrm B_\alpha$ denote the matrix balls, and let $\mathop{\mathrm {Inn}}\nolimits(m,\alpha)$ be the space of all rational inner maps...

conjecture | open | MathDB 页面 | 原文线索

V2-036804 | MATHDB-351989

Continuous Zauner's conjecture for equiangular tight continuous frames

摘录 / EXCERPT: Let (Ω,μ) be a measure space, let $d\in\mathbb{N}$, and let $\{\tau_\alpha\}_{\alpha\in\Omega}$ be a continuous frame for the Hilbert space $\mathbb{C}^d$. The frame is...

conjecture | open | MathDB 页面 | 原文线索

V2-036805 | MATHDB-351998

Uniform diameter bound for finite-time Kähler-Ricci flow singularities

摘录 / EXCERPT: Let $\omega=\omega(t)$ be a solution of the Kähler-Ricci flow on a compact manifold X, defined on its maximal time interval $[0,T)$. Here ${\rm diam}(X,\omega(t))$ denotes the di...

conjecture | open | MathDB 页面 | 原文线索

V2-036806 | MATHDB-352011

Conjectured asymptotics for type II Hermite-Padé polynomials on a three-sheeted Riemann surface

摘录 / EXCERPT: Let $f\in\mathbb C(z,w)$, let w_∞ be the specified element of the function w, and let $f_\infty\in\mathscr H(\infty)$ be the corresponding element of f. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-036807 | MATHDB-352020

The orthonormal wavelet implies wavelet-set conjecture

摘录 / EXCERPT: Orthonormal wavelet implies wavelet-set conjecture. For each pair (B,Γ^*) such that there exists a (B,Γ^*) orthonormal wavelet, there exists an (A,Γ) wavelet s...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-036808 | MATHDB-352036

Weighted strong Poincaré inequality for biparametric rectangles

摘录 / EXCERPT: Let w be a weight in $A_{p,\mathfrak{R}}$ for $p>1$. For a rectangle $R=I_1\times I_2$, let $\pi_R(f)$ denote the projection used in the Poincaré inequality, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036809 | MATHDB-352126

Stein's density conjecture for Jacobian operators of entire mappings

摘录 / EXCERPT: Stein's conjecture. If $J(F)=1$, then $\Delta_i^F(E_n)$ is dense in E_n for $n-1$ different indices $i\in\{1,\ldots,n\}$. The conjecture, stated for $n=2$ in Stein's work, would...

conjecture | open | MathDB 页面 | 原文线索

V2-036810 | MATHDB-352154

The HRT conjecture on linear independence of finite Gabor systems

摘录 / EXCERPT: Let $\Lambda\subset\mathbb{R}^{2}$ be a fixed finite set, let $f\in L^{2}(\mathbb{R})$ be nonzero, and write ... Here $\mathcal{G}(f,\Lambda)$ is the associated finite Gabor system....

conjecture | open | MathDB 页面 | 原文线索

V2-036811 | MATHDB-352195

Cordoba's Fourier-series norm equivalence conjecture

摘录 / EXCERPT: Cordoba's conjecture. For $alpha<2k$, there exists a constant C_α such that

conjecture | open | MathDB 页面 | 原文线索

V2-036812 | MATHDB-352208

Eligibility characterization for absolute continuity in types C and D

摘录 / EXCERPT: Let C_n and D_n denote the classical compact Lie groups of types C and D. For an element x in one of these groups, let S_x be defined from its dominant type, and call a...

conjecture | open | MathDB 页面 | 原文线索

V2-036813 | MATHDB-352225

The peripheral-spectrum conjecture for unital completely positive maps

摘录 / EXCERPT: Let $\Phi:{\mathfrak A}\to{\mathfrak A}$ be a unital completely positive map acting on the unital C^*-algebra ${\mathfrak A}$, and let $\sigma(\Phi)$ denote its spectrum. Periphe...

conjecture | open | MathDB 页面 | 原文线索

V2-036814 | MATHDB-352227

Uniform large-genus asymptotics conjecture for normalized psi-class intersections

摘录 / EXCERPT: For nonnegative integers $g,n,d_1,\dots,d_n$ with $2g-2+n>0$ and $d_1+\cdots+d_n=3g-3+n$, let $\mathcal{G}_{d_1,\dots,d_n}(g)$ be the normalized psi-class intersection numbers. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036815 | MATHDB-352262

Bourgain-Demeter-Kemp decoupling conjecture for analytic surfaces in three dimensions

摘录 / EXCERPT: Bourgain-Demeter-Kemp decoupling conjecture. There is a partition $\mathcal P_\delta=\mathcal P_\delta(\phi)$ of $[-1,1]^2$ into δ-flat subsets S such that, for every f...

conjecture | open | MathDB 页面 | 原文线索

V2-036816 | MATHDB-352325

The relative Novikov conjecture for manifolds with boundary

摘录 / EXCERPT: Let $(M,partial M)$ be a compact oriented smooth manifold with boundary, and let G and Γ be groups associated to the boundary and manifold through a homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-036817 | MATHDB-352336

White's conjecture on analytic Plateau solutions

摘录 / EXCERPT: White's conjecture. The union of the boundary and interior singular points of Σ is discrete. In particular, the overall singular set is finite, Σ has finite genus g...

conjecture | open | MathDB 页面 | 原文线索

V2-036818 | MATHDB-352344

Derived Hom-colimit conjecture for Fréchet spaces of compact type

摘录 / EXCERPT: Derived Hom-colimit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036819 | MATHDB-352354

The general multilinear Fourier extension conjecture for the paraboloid

摘录 / EXCERPT: Let $k\geq2$, and let $U_1,\ldots,U_k$ parametrize transversal caps of the paraboloid $x\mapsto|x|^2$ in $\mathbb{R}^{d+1}$. For functions g_j on U_j, write p' for the Hölder...

conjecture | open | MathDB 页面 | 原文线索

V2-036820 | MATHDB-352355

The restriction-Brascamp-Lieb conjecture for smooth submanifolds

摘录 / EXCERPT: For each $1\leq j\leq m$, let $\Sigma_j:U_j\to\mathbb{R}^n$ be a smooth parametrization of an n_j-dimensional submanifold S_j by a neighborhood U_j of the origin in...

conjecture | open | MathDB 页面 | 原文线索

V2-036821 | MATHDB-352356

Lieb-Seiringer reformulation of the Bessis-Moussa-Villani conjecture

摘录 / EXCERPT: Let A and B be $n \times n$ positive semidefinite matrices, and let m and r be integers with $m \geq r \geq 0$. Lieb-Seiringer's conjecture. The coefficient of t^r in ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-036822 | MATHDB-352448

Conjecture on contour-domain edges for elliptic and hyperbolic domains

摘录 / EXCERPT: Let G be either the elliptic domain ... or the hyperbolic domain ... For $0<t<1$, define the contour domain ... After a similarity transformation, assume that the center point is $z=0$...

conjecture | open | MathDB 页面 | 原文线索

V2-036823 | MATHDB-352451

Sharp Michael-Simon inequality conjecture for higher mean curvatures

摘录 / EXCERPT: Let M be a compact hypersurface in $\mathbb{R}^{n+1}$, possibly with boundary ∂M, and let f be a positive smooth function on M. For $1\leq k\leq n-1$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-036824 | MATHDB-352625

Conjecture on minimal algebraic Bergman-kernel degree in higher dimensions

摘录 / EXCERPT: Let $G\subset \mathbb{C}^n$ with $n\geq 2$ be a smoothly bounded pseudoconvex domain, and let K be the Bergman kernel of G. The minimal-degree conjecture. If K is algebraic,...

conjecture | open | MathDB 页面 | 原文线索

V2-036825 | MATHDB-352635

The variance conjecture for symmetric convex bodies

摘录 / EXCERPT: Let $K\subset\mathbb{R}^d$ be a symmetric convex body, let X be uniformly distributed on K, and let Σ be the covariance matrix of X, with entries ... Write $\|\Sigma\|$...

conjecture | open | MathDB 页面 | 原文线索

V2-036826 | MATHDB-352637

Uniform iteration conjecture for essentially linear fractional self-maps of the disk

摘录 / EXCERPT: Let $\mathbb{D}$ be the open unit disk, and let φ be an essentially linear fractional self-map of $\mathbb{D}$ with exactly one fixed point w in $\overline{\mathbb{D}}$....

conjecture | open | MathDB 页面 | 原文线索

V2-036827 | MATHDB-352676

The frequency-one-over-k conjecture for Widom domains

摘录 / EXCERPT: Let $\Omega=\mathbb{C}\setminus\mathsf{E}$ be a Widom domain with a minimal DCT violation, where E satisfies the gap-endpoint conditions denoted in the source by (33sept...

conjecture | open | MathDB 页面 | 原文线索

V2-036828 | MATHDB-352688

The L^2-boundedness conjecture for Grushin pseudo-multipliers

摘录 / EXCERPT: Let G be the Grushin operator, let $\boldsymbol{L}$ and $\boldsymbol{U}$ denote the associated operator-valued spectral variables, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036829 | MATHDB-352704

Holomorphic stability characterization for weighted Bergman spaces

摘录 / EXCERPT: Holomorphic stability conjecture. For any boundary-accessable measure μ on $\mathbb{D}$, the pair $(B^2(\mathbb{D},\mu),h^1(\mathbb{D}))$ is holomorphically stable if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-036830 | MATHDB-352745

The spherical-function growth conjecture for Weyl-invariant Dunkl analysis

摘录 / EXCERPT: Spherical-function growth conjecture. If $\lambda,X\in\overline{\mathfrak{a}^+}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036831 | MATHDB-352746

Uniform discrete derivative bound for the function p_3

摘录 / EXCERPT: Uniform discrete derivative bound. There is a uniform bound

conjecture | open | MathDB 页面 | 原文线索

V2-036832 | MATHDB-352750

Barrett-Edholm essential norm conjecture for the Leray transform

摘录 / EXCERPT: Let ${\mathcal S} \subset \mathbb C^2$ be a smoothly bounded, strongly $\mathbb C$-convex hypersurface, let $\beta_{\mathcal S}(\zeta)$ be its projective invariant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036833 | MATHDB-352762

Hopf's Gaussian curvature conjecture for minimal graphs

摘录 / EXCERPT: Hopf's Gaussian curvature conjecture. The optimal constant is

conjecture | open | MathDB 页面 | 原文线索

V2-036834 | MATHDB-352799

Besicovitch's almost-everywhere line intersection conjecture

摘录 / EXCERPT: Let $E\subset\mathbb{R}^2$ be $\mathcal{H}^1$ measurable and purely 1-unrectifiable, with $\mathcal{H}^1(E)<\infty$. Besicovitch's almost-everywhere line intersection conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-036835 | MATHDB-352842

Blackadar-Kirchberg conjecture on quasidiagonality of stably finite nuclear algebras

摘录 / EXCERPT: Let A be a C^*-algebra. Blackadar-Kirchberg conjecture. If A is separable, stably finite and nuclear, then A is quasidiagonal. This conjecture concerns the converse to the...

conjecture | refuted | MathDB 页面 | 原文线索

V2-036836 | MATHDB-352843

Quasidiagonality of extensions of quasidiagonal C^*-algebras

摘录 / EXCERPT: Let I, A, and B be C^*-algebras in a short exact sequence ... Assume that I and B are separable, nuclear, and quasidiagonal. Extension quasidiagonality conjecture. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-036837 | MATHDB-352868

Macroscopic curvature and codimension 2 width

摘录 / EXCERPT: Macroscopic curvature and codimension 2 width. There exist positive dimensional constants δ_n, C_n, and R_n such that, if every ball of radius R_n in M has volume at...

conjecture | open | MathDB 页面 | 原文线索

V2-036838 | MATHDB-352906

Jørgensen-Schmitt-Werner stability conjecture for Wick algebra envelopes

摘录 / EXCERPT: For a family of coefficients T_{ij}^{kl}, let $W(T)$ be the Wick $*$-algebra generated by a_j,a_j^*, and let $T\colon\mathcal H^{\otimes 2}\to\mathcal H^{\otimes 2}$ be its ope...

conjecture | open | MathDB 页面 | 原文线索

V2-036839 | MATHDB-352918

Forster's conjecture on proper holomorphic embeddings of open Riemann surfaces

摘录 / EXCERPT: An open connected Riemann surface is a connected one-dimensional Stein manifold. Forster's conjecture. Every open connected Riemann surface embeds properly holomorphically into...

conjecture | open | MathDB 页面 | 原文线索

V2-036840 | MATHDB-352943

The multidimensional BFECC interpolation accuracy conjecture

摘录 / EXCERPT: Let d be a positive integer, and consider a uniform rectangular grid in d dimensions with mesh sizes ... Let the grid be shifted along any vector, and apply BFECC interpolation f...

conjecture | open | MathDB 页面 | 原文线索

V2-036841 | MATHDB-352953

Orponen's radial projection dimension conjecture

摘录 / EXCERPT: Let $X\subset\mathbb{R}^d$ be a Borel set, with $\operatorname{dim_H}(X)\leq d-1$, that is not contained in a hyperplane. For $y\in X$, let ... be the radial projection of X from...

conjecture | open | MathDB 页面 | 原文线索

V2-036842 | MATHDB-352989

Conjecture on the refined reconstruction formula for discrete Radon transform data

摘录 / EXCERPT: Let x_0 be a generic point, let H_0 have well-behaved level sets in the sense that any open interval of length $L\ge L_0$ contains at most ρL points of any level set...

conjecture | open | MathDB 页面 | 原文线索

V2-036843 | MATHDB-353009

Super-exponential decay conjecture for the orthogonality parameter

摘录 / EXCERPT: Let $sigma_T(kappa,H,ell)>0$ be the parameter satisfying ... Here $sigma_T$ measures the quasi-orthogonality of g on Y, and depends on the wavenumber through the wavenumber-depen...

conjecture | open | MathDB 页面 | 原文线索

V2-036844 | MATHDB-353020

Ferenczi-Lopez-Abad conjecture on separable Fraïssé spaces

摘录 / EXCERPT: A Banach space X is Fraïssé when for every $\varepsilon>0$ and every $k\in\mathbb{N}$ there is a $\delta>0$ such that for every k-dimensional subspace E of X and every...

conjecture | open | MathDB 页面 | 原文线索

V2-036845 | MATHDB-353023

Nitsche's uniqueness conjecture for free boundary minimal annuli

摘录 / EXCERPT: Let $\mathbb{B}^3$ be the unit ball in Euclidean three-space, and let a free boundary minimal annulus be an embedded minimal annulus whose boundary lies on $\partial\mathbb{B}^3$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-036846 | MATHDB-353068

The hyperbolic Horn conjecture for lpm-polynomials

摘录 / EXCERPT: Let P be any degree k lpm-polynomial, and let $\lambda,\mu\in\mathbb{R}^k$ with λ majorizing μ. Hyperbolic Horn conjecture. There exists a symmetric matrix X such...

conjecture | open | MathDB 页面 | 原文线索

V2-036847 | MATHDB-353102

Zero-freeness conjecture for optimal polynomial approximants in Hardy spaces

摘录 / EXCERPT: Let $1<p<\infty$, $f\in H^p$, and $n\in\mathbb{N}$. If $f(0)\neq 0$, then the optimal polynomial approximant $q_{n,p}[f,1]$ is the polynomial of degree at most n that best approx...

conjecture | open | MathDB 页面 | 原文线索

V2-036848 | MATHDB-353127

Kohn's conjecture on finite generation of the trivial multiplier

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}^n$ be a bounded pseudoconvex domain with smooth boundary of finite type. A q-multiplier is a germ of a smooth function whose product with a $(0,q)$-f...

conjecture | open | MathDB 页面 | 原文线索

V2-036849 | MATHDB-353128

Effective Kohn's conjecture on the order of subellipticity

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}^n$ be a bounded pseudoconvex domain with smooth boundary of finite type, and let $f=1$ be the trivial multiplier generated by Kohn's procedures. The or...

conjecture | open | MathDB 页面 | 原文线索

V2-036850 | MATHDB-353157

Modular Zauner's conjecture for Hilbert C*-modules

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra, let $d\in\mathbb{N}$, and let $\mathcal{A}^d$ denote the standard Hilbert C-module. A unit inner product frame has...

conjecture | open | MathDB 页面 | 原文线索

V2-036851 | MATHDB-353158

Strong modular Zauner's conjecture for Hilbert C*-modules

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra with the invariant basis number property or let $\mathcal{A}$ be a W-algebra. For $d\in\mathbb{N}$, consider unit inner product frames in th...

conjecture | open | MathDB 页面 | 原文线索

V2-036852 | MATHDB-353161

The quadratic convergence conjecture for the decreasing threshold sequence

摘录 / EXCERPT: Fix $\kappa\geq4$ with $\kappa\in\mathbb{N}_e$, and let $\{\mathcal{E}_n\}_{n=0}^{\infty}$ be the strictly decreasing sequence from the stated theorem, with limit...

conjecture | open | MathDB 页面 | 原文线索

V2-036853 | MATHDB-353167

The decreasing-limit conjecture for even-indexed upside-down wells

摘录 / EXCERPT: Let $\kappa\geq4$ be even and let $2\leq j\leq\lfloor\kappa/2\rfloor$ be even. Let $\{\mathcal{E}_n\}$ be the sequence constructed in the two stated theorems. Upside-down-well limi...

conjecture | open | MathDB 页面 | 原文线索

V2-036854 | MATHDB-353181

Bochner-Riesz multiplier conjecture for curved hypersurfaces

摘录 / EXCERPT: Multiplier conjecture. For $1\leq p\leq\infty$, the estimate

conjecture | open | MathDB 页面 | 原文线索

V2-036855 | MATHDB-353182

Sharp uniform resolvent estimate outside the uniform boundedness range

摘录 / EXCERPT: Sharp resolvent conjecture. If $(\frac1p,\frac1q)\in\mathcal R_0\setminus([B,E]\cup[B',E']\cup[D,C)\cup[D',C))$, then

conjecture | open | MathDB 页面 | 原文线索

V2-036856 | MATHDB-353344

Hörmander-Mikhlin conjecture for stratified Lie groups

摘录 / EXCERPT: Let G be an n-dimensional stratified Lie group with homogeneous dilation weights $(\ell_1,\ell_2,\ldots,\ell_n)$, rescaled so that $\min(\ell_j)=1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036857 | MATHDB-353412

Entropy bounds for double cones with product symmetry

摘录 / EXCERPT: Let $\mathcal{C}_{n,p}$ denote the double cone with $O(p+1)\times O(n-p+1)$ symmetry, and let λ denote entropy. Solomon's conjecture. Any double cone with...

conjecture | open | MathDB 页面 | 原文线索

V2-036858 | MATHDB-353557

Monotonicity conjecture for the binomial-logarithm sequence

摘录 / EXCERPT: Monotonicity conjecture. The sequence a_p is decreasing in p. The preceding asymptotic result gives $a_p\to 0$ as $p\to\infty$, so this conjecture would describe the finite-p...

conjecture | open | MathDB 页面 | 原文线索

V2-036859 | MATHDB-353690

Hard Lefschetz and Hodge-Riemann conjecture for smooth valuations

摘录 / EXCERPT: Let n be a positive integer, let $0\leq r\leq\lfloor\frac n2\rfloor$, and let $\mathcal K(\mathbb R^n)$ denote the space of convex bodies in $\mathbb R^n$. For $i=0,\dots,n-2r$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036860 | MATHDB-353705

The sharp Gaussian decay conjecture for Fourier uncertainty classes

摘录 / EXCERPT: Sharp Gaussian decay conjecture. If both quantities are finite, then

conjecture | open | MathDB 页面 | 原文线索

V2-036861 | MATHDB-353723

Parametric mod-2 isoperimetric inequality for relative cycles

摘录 / EXCERPT: Let $F:X\rightarrow\mathcal{Z}_k([0,1]^n,\partial[0,1]^n;\mathbb{Z}_2)$ be a contractible family, and let $I_{k+1}([0,1]^n;\mathbb{Z}_2)$ denote the $(k+1)$-chains with mod-2 coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-036862 | MATHDB-353724

Parametric localized isoperimetric conjecture for domains with corners

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a connected domain whose piecewise smooth boundary has θ-corners, with $\theta\in(0,\pi)$, and let $0\leq k<n$. Let X^p be a p-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-036863 | MATHDB-353725

Parametric Federer-Fleming filling conjecture for Euclidean cycles

摘录 / EXCERPT: Let $F:X^p\rightarrow\mathcal{Z}_k(\mathbb{R}^n;\mathbb{Z})$ be a continuous family of cycles. Parametric Euclidean filling conjecture. There should exist a family of chains...

conjecture | open | MathDB 页面 | 原文线索

V2-036864 | MATHDB-353726

Boundary modification conjecture for localized sweepouts

摘录 / EXCERPT: Let $n>k\geq1$, let $F:X^p\rightarrow\mathcal{Z}_k([0,1]^n,\partial[0,1]^n;\mathbb{Z}_2)$ be a continuous family, and let $r\in(0,\tfrac12)$. There should exist a family...

conjecture | open | MathDB 页面 | 原文线索

V2-036865 | MATHDB-353727

Boundary localization conjecture for sweepouts in domains with corners

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a connected domain with piecewise smooth boundary having θ-corners, where $\theta\in(0,\pi)$. Fix $\eta>0$. For all $p\geq p_0(\Omega)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036866 | MATHDB-353749

Tensor-product Fourier dual-certificate conjecture for Hadamard matrices

摘录 / EXCERPT: Let $U(d)$ be the unitary group, let ν be normalized Haar measure on $U(d)$, let e be its identity, and let $\mu_{[\bm H]}$ be the smoothed measure associated with the equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-036867 | MATHDB-353825

Thurston's $K=2$ conjecture for conformal maps to domes

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}$ be a simply connected domain. Define $K(\Omega)$ as the infimum of the maximal dilatations of quasiconformal maps in $\mathcal{F}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036868 | MATHDB-353827

Harmonic-map projection conjecture for pinched Hadamard manifolds

摘录 / EXCERPT: Let M be a pinched Hadamard manifold, meaning a simply connected, complete Riemannian manifold whose sectional curvature is bounded between two negative constants. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036869 | MATHDB-353864

Lipman-Rosenberg conjecture on Dixmier-Douady invariants of nilpotent Lie groups

摘录 / EXCERPT: Let G be a connected nilpotent Lie group, and let A be a continuous trace subquotient of $C^*(\mathsf{G})$, meaning a subquotient whose spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-036870 | MATHDB-353867

Minimal-integrability conjecture for weak estimates of rough singular integral operators

摘录 / EXCERPT: Let Ω be homogeneous of degree zero on $\mathbb{R}^d$, satisfy the vanishing moment condition, and let A be a function on $\mathbb{R}^d$ whose derivatives of order one bel...

conjecture | open | MathDB 页面 | 原文线索

V2-036871 | MATHDB-353876

Fejes Tóth's real Hilbert-space zone conjecture

摘录 / EXCERPT: Let $v_1,v_2,\ldots,v_n$ be unit vectors in a real Hilbert space $\mathcal{H}$. Fejes Tóth's real Hilbert-space zone conjecture. There exists a unit vector $u\in\mathcal{H}$ such t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-036872 | MATHDB-353908

Commutative C*-algebraic Sendov's conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital commutative C-algebra whose group of invertible elements $G(\mathcal{A})$ is dense in $\mathcal{A}$. Let $n\in\mathbb{N}\setminus\{1\}$ and ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-036873 | MATHDB-353909

C*-algebraic Sendov's conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra, let $n\in\mathbb{N}\setminus\{1\}$, and let ... with $a_1,\dots,a_n\in\overline{\mathbb{D}}^*(0,1)$. Define ... where the hatted factor is omit...

conjecture | open | MathDB 页面 | 原文线索

V2-036874 | MATHDB-353960

Global optimizer conjecture for confined elasticae

摘录 / EXCERPT: Global optimizer conjecture. There is a number $\xi(j)\in(0,1)$ such that the globally least bent confined elastica is a j-$(j+1)$-clew if $r_L/R<j+\xi$, while the global optimiz...

conjecture | open | MathDB 页面 | 原文线索

V2-036875 | MATHDB-353968

Jørgensen-Schmitt-Werner conjecture on Wick algebra envelopes

摘录 / EXCERPT: Let T be the operator of coefficients of a Wick algebra $W(T)$ on $d4c=\mathbb C^n$, and let $\mathcal{W}(T)=C^*(W(T))$ be its universal C^*-envelope. The operator T is self-...

conjecture | open | MathDB 页面 | 原文线索

V2-036876 | MATHDB-354008

Fourier extension conjecture for the truncated paraboloid

摘录 / EXCERPT: Let $P^{n-1} = \{\xi_n = \sum_{j=1}^{n-1} \xi_j^2,\ |\xi_j| \leq 1,\ \forall 1 \leq j \leq n-1\}$ be the truncated unit paraboloid, let $\mathrm{d}\sigma$ be its natural surface me...

conjecture | open | MathDB 页面 | 原文线索

V2-036877 | MATHDB-354050

Ginzburg-Landau global-minimizer conjecture for the constant- and variable-curvature solutions

摘录 / EXCERPT: Let $(\psi\equiv 0,a=a^{b_r})$ be the constant-curvature solution (1.3), let (ψ,a) be the solution (1.7), and let h_r be the relevant Hermitian metric. Let b_r and...

conjecture | open | MathDB 页面 | 原文线索

V2-036878 | MATHDB-354156

Bounds conjecture for the Stirling error continued-fraction approximants

摘录 / EXCERPT: Bounds conjecture. The functions $g_m(n)$ give lower bounds for r_n when m is even and upper bounds when m is odd; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-036879 | MATHDB-354186

Hellerstein-Shen-Williamson conjecture on real meromorphic functions

摘录 / EXCERPT: Let f be a real transcendental meromorphic function in the plane with at least one pole, meaning that $f(\mathbb R)\subseteq\mathbb R\cup\{\infty\}$. Assume that all zeros and po...

conjecture | open | MathDB 页面 | 原文线索

V2-036880 | MATHDB-354246

Lipschitz structure conjecture for branch points of minimizers

摘录 / EXCERPT: Let u be a minimizer of the functional in on some open set $\Omega\subset\mathbb R^n$, and let $\Gamma_{\mathrm{BP}}(u)$ denote its branch-point set. For a relatively compact set...

conjecture | open | MathDB 页面 | 原文线索

V2-036881 | MATHDB-354282

Conjecture on the exceptional set of Seshadri functions in Picard number two

摘录 / EXCERPT: Exceptional-set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-036882 | MATHDB-354312

Haagerup-Størmer's conjecture on pointwise inner automorphisms of type III_1 factors

摘录 / EXCERPT: Let M be a type ${\rm III}_1$ factor with separable predual. An automorphism $\Theta\in\operatorname{Aut}(M)$ is pointwise inner if, for every normal state $\psi\in M_*$, there i...

conjecture | open | MathDB 页面 | 原文线索

V2-036883 | MATHDB-354333

Sub-affine-critical translators conjecture

摘录 / EXCERPT: Let $n\geq 2$ and let $\alpha<\frac{1}{n+2}$. An α-translator is a translating soliton in $\mathbb{R}^{n+1}$ for the power of Gauss curvature specified in the paper, and an...

conjecture | open | MathDB 页面 | 原文线索

V2-036884 | MATHDB-354467

Full generic regularity I for area-minimizing hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be a closed smooth manifold. A Riemannian metric g is called Baire C^∞-generic if it belongs to a Baire generic subset of the space of smooth Riemannian metr...

conjecture | open | MathDB 页面 | 原文线索

V2-036885 | MATHDB-354498

The Fourier restriction conjecture for Frostman measures on the parabola

摘录 / EXCERPT: Fourier restriction conjecture for Frostman measures on the parabola. For every $0\leq s\leq 1$ and $\epsilon>0$, there exists $p=p(\epsilon,s)\geq 1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036886 | MATHDB-354508

Uniqueness conjecture for curve shortening flow on the flat plane

摘录 / EXCERPT: Uniqueness conjecture. CSF is unique on the flat plane.

conjecture | open | MathDB 页面 | 原文线索

V2-036887 | MATHDB-354509

Uniqueness conjecture for curve shortening flow and blooming at infinity

摘录 / EXCERPT: Uniqueness conjecture. CSF is unique on $(\mathbb{R}^2,g)$ if and only if g does not allow blooming at infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-036888 | MATHDB-354516

Planar analytic-boundary conjecture for harmonic-function quotients

摘录 / EXCERPT: Let $n=2$. Let Ω be a one-sided NTA domain, and let u and v be as above, with u/v agreeing on $B(z,r)\cap\partial\Omega$ with a real-analytic function R. Assume more...

conjecture | solved | MathDB 页面 | 原文线索

V2-036889 | MATHDB-354517

Proportionality conjecture for regular Cantor boundaries

摘录 / EXCERPT: Let $n=2$. Let Ω be a one-sided NTA domain, and let u and v be as above, with u/v agreeing on $B(z,r)\cap\partial\Omega$ with a real-analytic function R. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-036890 | MATHDB-354518

Green-function decay and boundary regularity conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^n$ be a one-sided NTA domain whose Green function with some pole p satisfies, for some $\delta>0$ and constants $c_1,c_2>0$, bounds of the form ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-036891 | MATHDB-354519

Dimension drop for harmonic measure of conformal self-similar sets

摘录 / EXCERPT: Let J be a conformal self-similar set of dimension $\delta<1$, and let $\omega_{\mathbb C\setminus J}$ denote harmonic measure for its complement. Dimension-drop conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-036892 | MATHDB-354586

Finite-size correction conjecture for symmetric longest increasing subsequences

摘录 / EXCERPT: Let $l_N^{\boxbslash}$ and $l_N^{\boxslash}$ denote the lengths of the two symmetric longest increasing subsequences considered in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036893 | MATHDB-354637

Radial projection exceptional-set conjecture above codimension one

摘录 / EXCERPT: Let $E\subset \mathbb{R}^d$ be a Borel set, let $\dim_H(E)>d-1$, and let $s\leq d-1$ be a real number. Radial projection exceptional-set conjecture. ... This conjecture strengthens t...

conjecture | open | MathDB 页面 | 原文线索

V2-036894 | MATHDB-354638

Liu-Orponen radial projection conjecture

摘录 / EXCERPT: Let $E\subset \mathbb{R}^d$ be a Borel set with $\dim_H(E)\in(k-1,k]$ for some $k\in\{0,1,\ldots,d-1\}$. Liu-Orponen's radial projection conjecture. ... This strengthens the exceptio...

conjecture | open | MathDB 页面 | 原文线索

V2-036895 | MATHDB-354665

Conjecture on regularity of J_p-harmonic maps

摘录 / EXCERPT: Let M and N be hyperbolic surfaces, let $p>2$, and let $u_p:M\rightarrow N$ satisfy the J_p-Euler-Lagrange equations. Regularity conjecture. The map u_p belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-036896 | MATHDB-354666

Piecewise description conjecture for the radial infinity-harmonic Dirichlet solution

摘录 / EXCERPT: Let $L>1$, and let R_p be the solutions to the p-approximation of the Dirichlet problem, with $R=\lim_{p\rightarrow\infty}R_p$. For $R_j=R(s_j)$ and $R'_j=dR/ds(s_j)$, suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-036897 | MATHDB-354674

Pointwise convergence conjecture for recurrence-defined functions

摘录 / EXCERPT: Let $\{F_n\}$, $\{G_n\}$ and $\{H_n\}$ be sequences of real functions on $\{1,\ldots,n\}$ satisfying ... Define $g_n(x):=nG_n(\lfloor nx\rfloor)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036898 | MATHDB-354675

Endpoint-limit conjecture for recurrence-defined functions

摘录 / EXCERPT: Let $\{F_n\}$, $\{G_n\}$ and $\{H_n\}$ be the sequences of real functions and let g_n, h_n, g, h and Θ be as in the preceding pointwise convergence conjecture. Assum...

conjecture | open | MathDB 页面 | 原文线索

V2-036899 | MATHDB-354676

The coefficient relation between Ramanujan exponential approximations

摘录 / EXCERPT: Let ψ_r and ρ_r denote the coefficients in the asymptotic expansions of $\Psi_n(0)$ and $\theta_n(0)$, respectively. Coefficient relation. For all $r \geqslant 0$, ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-036900 | MATHDB-354681

Wright's conjecture on phase retrieval by position, Fourier, and unitary measurements

摘录 / EXCERPT: Let $L^2(\mathbb{R})$ be the Hilbert space of square-integrable functions, let $\mathcal{F}$ denote the Fourier transform, and let ... A tuple of measurement operators does phase ret...

conjecture | open | MathDB 页面 | 原文线索

V2-036901 | MATHDB-354784

Gromov's generalized Weyl law conjecture for widths of cycles

摘录 / EXCERPT: Let M^n be a closed n-dimensional manifold with $n\geq 2$, and let $\omega_p^1(M^n,g)$ denote its 1-dimensional p-width with respect to a Riemannian metric g. The volume...

conjecture | open | MathDB 页面 | 原文线索

V2-036902 | MATHDB-354785

The conjectured inner amenability characterization by decomposable Fourier multipliers

摘录 / EXCERPT: Inner amenability and decomposable multipliers conjecture. The group G is inner amenable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-036903 | MATHDB-354786

The free-group conjecture on Fourier multipliers outside the Fourier-Stieltjes closure

摘录 / EXCERPT: Free-group Fourier multiplier conjecture. There exists a completely bounded Fourier multiplier

conjecture | open | MathDB 页面 | 原文线索

V2-036904 | MATHDB-354852

Regularity conjecture for the Hofstadter H sequence limit density

摘录 / EXCERPT: Let mu be the absolutely continuous limit measure of the numbers $alpha H(n)\mod 1$, where $alpha$ is the real root of $x^3+x=1$ and H is Hofstadter's H sequence. Regularity...

conjecture | open | MathDB 页面 | 原文线索

V2-036905 | MATHDB-354877

Super-exponential decay conjecture for the localized normal derivative

摘录 / EXCERPT: Let $cvarphi^{\rm loc}$ be the local function on the patch $comega$, let $cgamma_{\partial_n}$ denote the normal-derivative trace on $cpartial\omega$, and let $cell$ be the localiz...

conjecture | open | MathDB 页面 | 原文线索

V2-036906 | MATHDB-354929

Zagier's conjecture on the 2-adic valuations of Laurent coefficients

摘录 / EXCERPT: Let $m=2^n m_0$, where m_0 is odd, and let $ u= u_2$ denote the 2-adic valuation. Define $ u((k)!)$ by the 2-adic valuation of the factorial. Zagier's conjecture. For the Laurent...

conjecture | open | MathDB 页面 | 原文线索

V2-036907 | MATHDB-354930

Asymptotic slope conjecture for the 2-adic valuations of Taylor coefficients

摘录 / EXCERPT: Let a_m be the Taylor coefficients of $\Theta(z)=1/\Psi(1/z)$, let $m=2^n m_0$ with m_0 odd, and fix $n=\overline{n}$. Write $ u= u_2$ for the 2-adic valuation. Asymptotic slop...

conjecture | open | MathDB 页面 | 原文线索

V2-036908 | MATHDB-354931

Zagier-type conjecture on the 2-adic valuations of Taylor coefficients

摘录 / EXCERPT: Let $m=2^n m_0$, where m_0 is odd, and let $ u= u_2$ denote the 2-adic valuation. Let a_m be the Taylor coefficients of the reciprocal Riemann mapping function. Zagier-type con...

conjecture | open | MathDB 页面 | 原文线索

V2-036909 | MATHDB-354940

Ivrii's conjecture on singly generated invariant subspaces of the Bergman space

摘录 / EXCERPT: Let A^2_1 be the Bergman space ... A closed subspace of A^2_1 is singly generated if it is the minimal closed invariant subspace containing one function, where invariance is with...

conjecture | open | MathDB 页面 | 原文线索

V2-036910 | MATHDB-354956

Fourier multiplier characterization on the product Hardy space

摘录 / EXCERPT: Fourier multiplier conjecture. The function $\mathcal{F}(\lambda)$ is a Fourier multiplier from $H^1(\mathbb{R}_+^2\times\mathbb{R}_+^2)$ into L^1 if and only if there exists a c...

conjecture | open | MathDB 页面 | 原文线索

V2-036911 | MATHDB-355008

C*-algebraic Schoenberg conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a C-algebra, let $d\in\mathbb{N}\setminus\{1\}$, and let ... be a polynomial over $\mathcal{A}$ with $a_1,\ldots,a_d\in\mathcal{A}$. Suppose that its formal deri...

conjecture | open | MathDB 页面 | 原文线索

V2-036912 | MATHDB-355009

C*-algebraic de Bruin-Sharma conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a C-algebra, let $d\in\mathbb{N}\setminus\{1\}$, and let ... be a polynomial over $\mathcal{A}$ with $a_1,\ldots,a_d\in\mathcal{A}$. Suppose that ... on...

conjecture | open | MathDB 页面 | 原文线索

V2-036913 | MATHDB-355072

Zauner's maximal equiangular tight frame conjecture

摘录 / EXCERPT: For each dimension m, let ρ_1,ρ_2,ρ_3 denote the three quantities defined in the paper, and let $\rho(m)$ be the associated complex projection constant. Zauner's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-036914 | MATHDB-355105

Conjectured higher-order L^2 error estimate for the nonconforming hybrid method

摘录 / EXCERPT: Let u be the exact solution of the two-dimensional vector Laplacian problem and u_h its approximation by the order-k nonconforming primal hybrid finite element method. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036915 | MATHDB-355142

The family of fractal Fourier restriction estimates

摘录 / EXCERPT: The family of fractal Fourier restriction estimates. For every $\epsilon>0$ there is a constant C_ϵ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036916 | MATHDB-355211

Liquid-drop minimizer conjecture

摘录 / EXCERPT: Liquid-drop minimizer conjecture. The energy $E^{\rm G}(m)$ has a minimizer if and only if $m\leqslant m_*$. Moreover, whenever a minimizer exists, it is a ball. The assertion refl...

conjecture | open | MathDB 页面 | 原文线索

V2-036917 | MATHDB-355226

Dashti-Stuart conjecture on MAP estimators for Banach-space Gaussian inverse problems

摘录 / EXCERPT: Dashti-Stuart conjecture. Assume that the conditions in Assumption 4 hold. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-036918 | MATHDB-355244

Separation conjecture for polynomial roots and isodynamic points

摘录 / EXCERPT: Separation conjecture. No circle or line in $\mathbb C$ separates the zero locus of $P(z)$ from $\mathcal I_P$.

conjecture | open | MathDB 页面 | 原文线索

V2-036919 | MATHDB-355254

Fässler-Orponen restricted Marstrand conjecture for space curves

摘录 / EXCERPT: Let $I\subset\mathbb{R}$ be a compact interval, and let $\gamma:I\to S^2$ be a C^2 curve satisfying the escaping great circle condition ... for every $\theta\in I$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036920 | MATHDB-355322

Conjecture on divergence of classical Jacobi polynomial collocations

摘录 / EXCERPT: Divergence conjecture. Such collocations will likely diverge as the mesh size grows large. For Gauss/Gauss-Radau discretizations, the divergence is attributed to the divergence est...

conjecture | open | MathDB 页面 | 原文线索

V2-036921 | MATHDB-355334

Schoen-Yau positive mass conjecture for arbitrary asymptotically flat ends

摘录 / EXCERPT: Schoen-Yau's conjecture. The ADM mass of $\mathcal{E}$ is non-negative.

conjecture | open | MathDB 页面 | 原文线索

V2-036922 | MATHDB-355335

Positive mass conjecture for asymptotically hyperbolic manifolds with arbitrary ends

摘录 / EXCERPT: Positive mass conjecture for arbitrary asymptotically hyperbolic ends. The mass functional of the end $\mathcal{E}$ is timelike future-directed or zero.

conjecture | open | MathDB 页面 | 原文线索

V2-036923 | MATHDB-355422

Yamada's conjecture on logarithmic capacity and reproducing kernels

摘录 / EXCERPT: Let D be a bounded planar domain with analytic boundary and connectivity $n>1$. For $z e w$ in D, let $G_D(w,z)$ be the Green function and define the logarithmic capacity by .....

conjecture | open | MathDB 页面 | 原文线索

V2-036924 | MATHDB-355423

Generalized Yang's conjecture on periodic differential polynomials

摘录 / EXCERPT: Generalized Yang's conjecture. If $f(z)^n f^{(k)}(z)$ is a periodic function, then $f(z)$ is also a periodic function.

conjecture | open | MathDB 页面 | 原文线索

V2-036925 | MATHDB-355452

Characterization conjecture for weighted composition operators on Bloch spaces

摘录 / EXCERPT: Let D be a bounded homogeneous domain in $\mathbb{C}^n$, let ψ be a holomorphic function on D, and let φ be a holomorphic self-map of D. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-036926 | MATHDB-355465

The two-dimensional Schatten-to-L_p isometric embedding conjecture

摘录 / EXCERPT: Two-dimensional Schatten embedding conjecture. For any $p\geq 1$ and $A,B\in\mathcal{S}_p$, the real space

conjecture | open | MathDB 页面 | 原文线索

V2-036927 | MATHDB-355466

All-order trace derivative positivity conjecture

摘录 / EXCERPT: All-order trace derivative positivity conjecture. The function $t\mapsto\operatorname{tr} f(A+tB)$ has non-negative kth derivative.

conjecture | open | MathDB 页面 | 原文线索

V2-036928 | MATHDB-355486

Higher-derivative local Fock-space contraction conjecture

摘录 / EXCERPT: Let $\mathcal{F}_\alpha^2$ be the Fock space, let $f^{(n)}$ denote the nth derivative of f, let $\mu_{n,\alpha}$ be the measure used to define the weighted derivative norm, and...

conjecture | open | MathDB 页面 | 原文线索

V2-036929 | MATHDB-355487

Global derivative contraction conjecture for Fock space

摘录 / EXCERPT: The global derivative contraction conjecture. For every $f\in\mathcal{F}_\alpha^2$,

conjecture | open | MathDB 页面 | 原文线索

V2-036930 | MATHDB-355578

T-symmetric Koebe radius conjecture for schlicht polynomials

摘录 / EXCERPT: Let T and N be positive integers, and let $F^{(0)}$ be the polynomial proposed as the extremizer for the T-symmetric univalent-polynomial optimization problem. A schlicht pol...

conjecture | open | MathDB 页面 | 原文线索

V2-036931 | MATHDB-355580

Compactness characterization for weighted composition operators on bounded homogeneous domains

摘录 / EXCERPT: Let D be a bounded homogeneous domain in $\mathbb{C}^n$, let $\psi\in H(D)$, and let φ be a holomorphic self-map of D. Write $\mathcal{B}(D)$ for the Bloch space and...

conjecture | open | MathDB 页面 | 原文线索

V2-036932 | MATHDB-355603

Energy conjecture for polyhedral cones with reduced dihedral angles

摘录 / EXCERPT: Let $\boldsymbol{C}$ and $\bar{\boldsymbol{C}}$ be k-faced polyhedral cones, and let $\bar P$ be a fixed k-gonal pyramid over $\bar{\boldsymbol{C}}$. For $\xi\in\mathbb{S}^2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-036933 | MATHDB-355641

Modular Johnson-Lindenstrauss flattening conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra, let $0<\varepsilon<1$, and let $M,N\in\mathbb{N}$ and $\mathbf{x}_1,\dots,\mathbf{x}_M\in\mathcal{A}^N$. Modular Johnson-Lindenstrauss flat...

conjecture | open | MathDB 页面 | 原文线索

V2-036934 | MATHDB-355649

Benítez-Sarantopoulos-Tonge conjecture on the linear polarization constant of Euclidean space

摘录 / EXCERPT: Benítez-Sarantopoulos-Tonge conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-036935 | MATHDB-355723

Restriction conjecture for the flat disk over finite fields

摘录 / EXCERPT: Let $n=2d\ge 4$ be even, let $\mathcal{F}$ be the flat disk in $\mathbb{F}_q^n$, and let $R^*_{\mathcal{F}}(p\to r)$ denote its restriction operator estimate. If $(1/p,1/r)$ lies i...

conjecture | open | MathDB 页面 | 原文线索

V2-036936 | MATHDB-355728

Curvature-controlled Fourier decay conjecture for convex planar sets

摘录 / EXCERPT: Curvature-controlled Fourier decay conjecture. There exists an absolute constant $C_1>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036937 | MATHDB-355737

Decomposition conjecture for finite Blaschke factors in rotationally symmetric Hardy spaces

摘录 / EXCERPT: Let H^{α} be the Hardy space associated with a continuous rotationally symmetric norm α, let B be a finite Blaschke factor of degree n, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-036938 | MATHDB-355738

Decomposition conjecture for normalized gauge-norm Hardy spaces

摘录 / EXCERPT: Let α be a continuous $\|\cdot\|_1$-dominating normalized gauge norm, and let H^{α} and $M_{\alpha}(z^n)$ be the associated Hardy space and invariant subspace. Decomp...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-036939 | MATHDB-355856

Strong self-absorption criterion for coronas

摘录 / EXCERPT: Let A and D be separable and unital C^*-algebras, with D strongly self-absorbing. For a C^*-algebra B, write $\mathcal Q(B)=\mathcal M(B)/B$ for its c...

conjecture | open | MathDB 页面 | 原文线索

V2-036940 | MATHDB-355873

Distinct asymptotic directions for the components of a finite point-charge electrical field

摘录 / EXCERPT: Distinct-asymptotic-directions conjecture. The asymptotic directions of the points in the zero sets $X=0$ and $Y=0$, in the Type I and Type II domains, are distinct. Hence, outside...

conjecture | open | MathDB 页面 | 原文线索

V2-036941 | MATHDB-355887

Uniqueness conjecture for the GJMS equation on the sphere

摘录 / EXCERPT: Let $n\geq 2$, let $\mathbb{S}^n$ be the round sphere, let $P_{\mathbb{S}^n}$ denote its GJMS operator, let $Q_{g_{\mathbb{S}^n}}=(n-1)!/2$ be its constant Q-curvature, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-036942 | MATHDB-356008

Holomorphic equivalence conjecture for isometric Bergman spaces

摘录 / EXCERPT: Holomorphic equivalence conjecture. If there is a linear isometry between $A^p(D_1)$ and $A^p(D_2)$, then D_1 and D_2 are holomorphically equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-036943 | MATHDB-356030

Strohmer's conjecture on Gaussian Gabor frame-condition asymptotics

摘录 / EXCERPT: Strohmer's conjecture. As the critical density is approached from above, $\alpha\to1^+$,

conjecture | open | MathDB 页面 | 原文线索

V2-036944 | MATHDB-356031

Universal Gaussian Gabor frame-condition asymptotics and hexagonal optimality

摘录 / EXCERPT: Universal Gaussian Gabor asymptotic conjecture. There exists a constant C_Λ, depending on the geometry of $\Lambda(1)$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-036945 | MATHDB-356034

Product spectrality conjecture

摘录 / EXCERPT: Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be bounded, measurable sets, and let $\Omega=A\times B$. Product spectrality conjecture. The product Ω is spectral...

conjecture | open | MathDB 页面 | 原文线索

V2-036946 | MATHDB-356094

Non-Archimedean functional Zauner conjecture

摘录 / EXCERPT: Non-Archimedean functional Zauner conjecture. For every $d\in\mathbb{N}$, there exist such vectors and functionals satisfying $f_j(\tau_j)=1$ for all $1\leq j\leq d^2$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-036947 | MATHDB-356224

The kernel-growth conjecture for optimal OGF coefficients

摘录 / EXCERPT: Kernel-growth conjecture. An optimal ordinary generating function $\mathcal{C}(q)$ with integer-valued coefficients, $\mathcal{C}(0)\neq 0$, that attains the theoretical maximum of...

conjecture | open | MathDB 页面 | 原文线索

V2-036948 | MATHDB-356262

Brevig-Ortega-Cerdà-Seip-Zhao conjecture on the Riesz projection

摘录 / EXCERPT: For $1\leq p<\infty$, let $p'=\frac{p}{p-1}$, let P_+ denote the Riesz projection on $L^{p'}(\mathbb T)$, and let $H^{4/p}$ be the Hardy space with exponent $4/p$. Brevig-Ortega-...

conjecture | open | MathDB 页面 | 原文线索

V2-036949 | MATHDB-356299

Generic isolatedness conjecture for mean curvature flow

摘录 / EXCERPT: Let M_t be a mean curvature flow with generic initial data. Generic isolatedness conjecture. The flow has only isolated spacetime singularities. This conjecture addresses whether...

conjecture | open | MathDB 页面 | 原文线索

V2-036950 | MATHDB-356300

Colding-Minicozzi's thin-ring conjecture for mean curvature flow

摘录 / EXCERPT: Let $\gamma\subset\mathbb{R}^3$ be any C^1 curve. A thin-ring mean curvature flow is a mean convex flow whose singular set is prescribed by such a curve. Colding-Minicozzi's thin...

conjecture | open | MathDB 页面 | 原文线索

V2-036951 | MATHDB-356301

Generic nonexistence conjecture for firecracker singularities

摘录 / EXCERPT: Let a firecracker singularity be a degenerate singularity that is the limit of a sequence of nondegenerate spacetime singularities. Generic nonexistence conjecture for firecracker...

conjecture | open | MathDB 页面 | 原文线索

V2-036952 | MATHDB-356330

Conjecture on affiliation of the incoming wave operator for the radial part of $SL(2,\mathbb{R})$

摘录 / EXCERPT: Let $H_{\mu,\nu}$ be the operator parametrized by $\mu,\nu\geq0$, let $H_{\rm D}$ be the Dirichlet reference operator, and let $W_-(H_{\mu,\nu},H_{\rm D})$ denote the incoming wave...

conjecture | open | MathDB 页面 | 原文线索

V2-036953 | MATHDB-356352

The Hermite spectral projection estimate on the full exponent square

摘录 / EXCERPT: Hermite spectral projection estimate. For $(1/p,1/q)\in {\mathlarger{\square}}$, the estimate

conjecture | open | MathDB 页面 | 原文线索

V2-036954 | MATHDB-356359

Peterson's conjecture on diagonal embeddings into ultrapowers of Haagerup factors

摘录 / EXCERPT: Let N be a non-Gamma $\rm II_1$ factor with the Haagerup property. For an ultrafilter $\mathcal{U}$ on a set I, let $N^\mathcal{U}$ be the corresponding ultrapower, and let the...

conjecture | open | MathDB 页面 | 原文线索

V2-036955 | MATHDB-356377

Harshitha's asymptotic conjectures for alternating trigonometric sums

摘录 / EXCERPT: Harshitha's conjectures. The limits satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-036956 | MATHDB-356397

Universal principal-minor Fourier permutation conjecture

摘录 / EXCERPT: Let $N\in\mathbb{N}$, let $\mathbb{Z}_N$ denote the residues modulo N, and let $\boldsymbol{\mathrm{S}}(\mathbb{Z}_N)$ be their permutation group. For...

conjecture | open | MathDB 页面 | 原文线索

V2-036957 | MATHDB-356412

Nonsingular intertwining conjecture for maxima of translates functions

摘录 / EXCERPT: Let $n\in\mathbb{N}$, let $\nu_1,\ldots,\nu_n>0$, let K be a strictly concave and strictly monotone kernel function, and let J be an upper semicontinuous n-field function. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-036958 | MATHDB-356415

The xz-conjecture for Laurent polynomials with polynomial coefficients

摘录 / EXCERPT: Let h be an admissible function on $\mathbb{T}^k\times[0,1]^I$ of the form ... where $\mathbf m=(m_1,\ldots,m_k)$ is an integer multi-index, each $c_{\mathbf m}$ is a complex polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-036959 | MATHDB-356416

The Mathieu xz-conjecture for moments of admissible functions

摘录 / EXCERPT: Let g and h be admissible functions on $\mathbb{T}^k\times[0,1]^I$, with admissibility as in the xz-conjecture. Mathieu xz-conjecture. If ... for every positive integer P, then...

conjecture | open | MathDB 页面 | 原文线索

V2-036960 | MATHDB-356423

Non-Archimedean Zauner conjecture

摘录 / EXCERPT: Non-Archimedean Zauner conjecture. For every $d\in\mathbb{N}$, there exist such vectors satisfying $\langle\tau_j,\tau_j\rangle=1$ for all $1\leq j\leq d^2$, with S_τ diagonal...

conjecture | open | MathDB 页面 | 原文线索

V2-036961 | MATHDB-356499

The sticky Kakeya conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be the set of affine lines in $\mathbb{R}^n$, with packing dimension used for subsets of this line space. A compact set $K\subset\mathbb{R}^n$ is a sticky Kakeya...

conjecture | open | MathDB 页面 | 原文线索

V2-036962 | MATHDB-356580

Injectivity conjecture for convolution with the unit-ball indicator

摘录 / EXCERPT: Let $n\ge 2$. For $1 \le p \le \frac{2n}{n-1}$, consider the convolution operator ... where B_1 is the unit ball in $\mathbb{R}^n$. Injectivity conjecture. The convolution operator...

conjecture | open | MathDB 页面 | 原文线索

V2-036963 | MATHDB-356590

Degree conjecture for degenerate sphere maps

摘录 / EXCERPT: Let $H:\mathbb{S}^{2n-1}\rightarrow\mathbb{S}^{2N-1}$ be a sphere map with linearly independent components, and let $s>0$. Degree conjecture. If H is s-degenerate, then H has...

conjecture | open | MathDB 页面 | 原文线索

V2-036964 | MATHDB-356610

Admissible-solution parity conjecture for the convolution equation

摘录 / EXCERPT: Let n be an integer and let an admissible solution mean a solution of the convolution equation defined earlier in the paper satisfying the stated admissibility conditions. Parity...

conjecture | open | MathDB 页面 | 原文线索

V2-036965 | MATHDB-356667

Existence conjecture for the Gauss image problem under the weak Aleksandrov condition

摘录 / EXCERPT: Let μ and λ be Borel measures on S^{n-1}, with λ absolutely continuous. Say that μ is weak Aleksandrov related to λ when it satisfies the weak Ale...

conjecture | open | MathDB 页面 | 原文线索

V2-036966 | MATHDB-356722

Extension conjecture for residue functions and adjoint ideals

摘录 / EXCERPT: Conjecture 1.1.3. Under the curvature assumption given in Theorem, there exists a sufficiently large constant $\ell\geq e$, depending only on ψ and δ, such that for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-036967 | MATHDB-356770

Logarithmic sparse-domination conjecture for Banach function spaces

摘录 / EXCERPT: Let T satisfy the full-range sparse domination bound referred to in the source. Let X be a Banach function space such that $M:X\to X$ and $M:X'\to X'$ are bounded. Logarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-036968 | MATHDB-356771

Multilinear extrapolation conjecture for Banach function spaces

摘录 / EXCERPT: Let $p_1,\ldots,p_m\in(1,\infty]$ with $\frac{1}{p}=\sum_{j=1}^m\frac{1}{p_j}>0$, and let $w=\prod_{j=1}^m w_j$. Suppose T is well-defined on all m-tuples...

conjecture | open | MathDB 页面 | 原文线索

V2-036969 | MATHDB-356816

Ohsawa's Bergman-kernel restriction conjecture

摘录 / EXCERPT: Let M be a Stein manifold of complex dimension n, let S be a closed complex submanifold, and let $G(,S)$ be the upper envelope of the polar functions associated with S. Let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-036970 | MATHDB-356817

Yamada's extended Suita conjecture

摘录 / EXCERPT: Let Ω be an open Riemann surface admitting a nontrivial Green function G_Ω, let u be harmonic on Ω, and set $\rho=e^{-2u}$. Define $B_{\Omega,\rho}$ from an...

conjecture | solved | MathDB 页面 | 原文线索

V2-036971 | MATHDB-356822

The SL_2 Kakeya conjecture

摘录 / EXCERPT: Let $\mathcal L_{SL_2}$ be the set of lines in $\varmathbb{R}^3$ of the form $\ell_{(a,b,c,d)}=(a,b,0)+\operatorname{span}(c,d,1)$ with $ad-bc=1$, and let $\mathcal L_{SL_2}^*$ als...

conjecture | open | MathDB 页面 | 原文线索

V2-036972 | MATHDB-356823

Relative biexactness conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Relative biexactness conjecture. G is biexact relative to $\{H_i\}_{i=1}^{n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-036973 | MATHDB-356904

The maximality conjecture for Gaussian Riemann derivatives

摘录 / EXCERPT: Maximality conjecture for Gaussian Riemann derivatives. For all functions f and points x:

conjecture | open | MathDB 页面 | 原文线索

V2-036974 | MATHDB-356952

Coulhon-Duong conjecture on the Riesz transform for complete manifolds

摘录 / EXCERPT: Coulhon-Duong conjecture. The Riesz transform is bounded on $L^p(M)$ for every $1<p<2$.

conjecture | open | MathDB 页面 | 原文线索

V2-036975 | MATHDB-356959

Short-time existence conjecture for three-dimensional complete Ricci flows with non-negative Ricci curvature

摘录 / EXCERPT: Short-time existence conjecture. There exists $T>0$ and a smooth family of complete metrics $g(t)$ on $M\times[0,T)$ such that $g(0)=g_0$ and $g(t)$ solves the Ricci flow equation...

conjecture | open | MathDB 页面 | 原文线索

V2-036976 | MATHDB-356971

Reciprocal conjecture for Banach spaces of holomorphic functions

摘录 / EXCERPT: Let Ω be a domain and let $B(\Omega)$ be the Banach space of holomorphic functions on Ω. Reciprocal conjecture. If $\psi\in B(\Omega)$ and there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-036977 | MATHDB-356972

Multiplier reciprocal conjecture for Banach spaces of holomorphic functions

摘录 / EXCERPT: Let Ω be a domain, let $B(\Omega)$ be a Banach space of holomorphic functions on Ω, and let $\mathcal{M}(B(\Omega))$ denote its algebra of multipliers. Multiplier rec...

conjecture | open | MathDB 页面 | 原文线索

V2-036978 | MATHDB-357006

Existence of compact sets with a prescribed hierarchy of vanishing ideals

摘录 / EXCERPT: Let $R(K)$ denote the uniform algebra of rational functions with poles off a compact set $K\subset\mathbb{C}$. For $x\in K$, write M_x for the maximal ideal of functions in...

conjecture | open | MathDB 页面 | 原文线索

V2-036979 | MATHDB-357021

Nonconstant fixed-point conjecture for the uncentered Hardy-Littlewood maximal operator

摘录 / EXCERPT: Nonconstant fixed-point conjecture. Under these assumptions, there is no non-constant fixed point $f\in L^{1}_{\operatorname{loc}}(\nu)$ of M_ν.

conjecture | open | MathDB 页面 | 原文线索

V2-036980 | MATHDB-357029

Asymptotic expansion conjecture for extreme eigenvalues of non-Hermitian Toeplitz matrices

摘录 / EXCERPT: Extreme-eigenvalue expansion conjecture. There exists an integer $k\geqslant1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-036981 | MATHDB-357039

The continuous periodic tiling conjecture

摘录 / EXCERPT: Continuous periodic tiling conjecture. Let Σ be a bounded measurable subset of $\mathbb{R}^d$ of positive measure. Then the tiling equation

conjecture | open | MathDB 页面 | 原文线索

V2-036982 | MATHDB-357061

Logarithmic-density conjecture for binary partition values that are sums of two squares

摘录 / EXCERPT: Let $\mathcal{T}(x)$ count the integers $n\leq x$ such that $b(n)$ is a sum of two squares, where $b(n)$ is the binary partition function. Logarithmic-density conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-036983 | MATHDB-357073

Quantitative Besicovitch projection conjecture for AD-regular planar sets

摘录 / EXCERPT: Let $s\in (0,1)$ and $C_0\in (1,\infty)$. Let $E\subset \mathbb{R}^2$ be a bounded AD-regular set with constant C_0, meaning that E is closed and, for every $x\in E$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-036984 | MATHDB-357154

Existence conjecture for convex weak refractors with general target sets

摘录 / EXCERPT: Convex refractor existence conjecture. Under these hypotheses, such a convex weak solution exists for the target set and data described above.

conjecture | open | MathDB 页面 | 原文线索

V2-036985 | MATHDB-357181

The finite-union conjecture for bounded below approximate Schauder frames

摘录 / EXCERPT: Finite-union conjecture. Every bounded below ASF can be partitioned as a finite union of ARSs. The source's statement uses “ARBs,” but the preceding definition introduces ARSs; thi...

conjecture | open | MathDB 页面 | 原文线索

V2-036986 | MATHDB-357205

Weak conjecture on pre-quasi Gelfand triples of Hilbert spaces

摘录 / EXCERPT: Weak conjecture. Every pre-quasi Gelfand triple of Hilbert spaces is a quasi Gelfand triple.

conjecture | open | MathDB 页面 | 原文线索

V2-036987 | MATHDB-357206

Strong conjecture on pre-quasi Gelfand triples

摘录 / EXCERPT: Strong conjecture. Every pre-quasi Gelfand triple is a quasi Gelfand triple.

conjecture | open | MathDB 页面 | 原文线索

V2-036988 | MATHDB-357295

The tracial non-isomorphism conjecture for q-Araki-Woods von Neumann algebras

摘录 / EXCERPT: Let $q,q'\in(-1,1)$ with $q\neq 0$. Let $(U_t)_{t\in\mathbb R}$ be a group of orthogonal transformations of ℓ^2, let $\mathsf{H}_{\mathbb R}$ be a finite-dimensional real Hilb...

conjecture | open | MathDB 页面 | 原文线索

V2-036989 | MATHDB-357359

Kenig's spectral radius conjecture for the double-layer operator

摘录 / EXCERPT: Kenig's conjecture. The spectral radius of D'_{Γ} on $L^2_0(\Gamma)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-036990 | MATHDB-357485

The Kisyński-space conjecture for non-autonomous second-order Cauchy problems

摘录 / EXCERPT: Let X be the underlying Banach space, let D denote the common domain appearing in the definition of the evolutionary fundamental solution, and let ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-036991 | MATHDB-357548

The NTV two-weight conjecture for the Hilbert transform

摘录 / EXCERPT: The NTV conjecture. If $\mathcal{A}_{2}(\sigma,\omega)$ is finite and both testing characteristics $\mathfrak{T}_{H}(\sigma,\omega)$ and $\mathfrak{T}_{H}(\omega,\sigma)$ are finit...

conjecture | open | MathDB 页面 | 原文线索

V2-036992 | MATHDB-357564

Symmetric-lens conjecture for the two-dimensional strip-to-disk transition state

摘录 / EXCERPT: Let $\omega=(4\pi-3\sqrt{3})/18\approx0.4095$ be the fixed volume fraction, and let c and K be the phase-field parameters. A transition state is the saddle point along the mini...

conjecture | open | MathDB 页面 | 原文线索

V2-036993 | MATHDB-357566

Frank's continuation conjecture for the three-dimensional Bohr-Wheeler branch

摘录 / EXCERPT: Let χ be the fissility parameter. The Bohr-Wheeler branch is the bifurcation branch of three-dimensional saddle points, and a saddle point is the transition state along the co...

conjecture | open | MathDB 页面 | 原文线索

V2-036994 | MATHDB-357620

Kotěšovec's asymptotic conjecture for the coefficients of a partition function

摘录 / EXCERPT: Kotěšovec's conjecture. The coefficients of $P(z)$ have the asymptotic growth rate

conjecture | open | MathDB 页面 | 原文线索

V2-036995 | MATHDB-357688

Uniqueness of simultaneously stabilizable and stabilized means

摘录 / EXCERPT: Let K, M, and N be means, and let G denote the geometric mean. For means A and B, write $A\otimes B$ for their indicated mean composition. Uniqueness conjecture for sta...

conjecture | open | MathDB 页面 | 原文线索

V2-036996 | MATHDB-357993

Dimension-independent constants conjecture for Banach-space fillings

摘录 / EXCERPT: Let m be a positive integer, and let $c_1(m)$ and $c_2(m)$ be the constants from Theorem 1.5, which applies to compact subsets X of Banach spaces with nonzero filling content...

conjecture | open | MathDB 页面 | 原文线索

V2-036997 | MATHDB-357994

The nonlinear sandwich extension conjecture

摘录 / EXCERPT: Let X be a vector space over the scalar structure associated with $\mathbf{C}$, let $\mathbb{V}$ be the ordered codomain, let $H:X\to \mathbb{V}$ be $\mathbf{C}$-sublinear and mo...

conjecture | open | MathDB 页面 | 原文线索

V2-036998 | MATHDB-358008

Delecroix-Goujard-Zograf-Zorich frequency-ratio conjecture for simple closed geodesics

摘录 / EXCERPT: Delecroix-Goujard-Zograf-Zorich conjecture. The ratio of frequencies of separating over nonseparating simple closed geodesics admits the uniform asymptotics

conjecture | open | MathDB 页面 | 原文线索

V2-036999 | MATHDB-358050

Maximizing-pattern conjecture for numerical ranges of cyclic shift matrices

摘录 / EXCERPT: Let $S(a_1,\dots,a_n)$ denote the $n\times n$ cyclic shift matrix with weights $a_1,\dots,a_n$. For $n\geq 2$, let $0\leq a_1\leq\cdots\leq a_n$ and set ... For a permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-037000 | MATHDB-358055

The perturbation-limit conjecture for the minimal surface boundary value problem

摘录 / EXCERPT: Perturbation-limit conjecture. Up to addition of a constant, the smooth solution u is the suitable limit of the sequence of solutions w_ε as $\varepsilon\to0$.

conjecture | open | MathDB 页面 | 原文线索

V2-037001 | MATHDB-358109

Conjectured mesoscopic asymptotics for the inverse Kasteleyn matrix

摘录 / EXCERPT: Let $x \in \mathtt{W}_{\varepsilon_1}$ and $y \in \mathtt{B}_{\varepsilon_2}$ with $\varepsilon_1,\varepsilon_2 \in \\{0,1\\}$. Let $K_0(\alpha)$ and $K_1(\alpha)$ denote modified...

conjecture | open | MathDB 页面 | 原文线索

V2-037002 | MATHDB-358110

Conjectured asymptotics for the correction integrals in the inverse Kasteleyn matrix

摘录 / EXCERPT: Let $n=4m$, $0<a<1$, and take $x=(x_1,x_2)\in\mathtt{W}_{\varepsilon_1}$ and $y=(y_1,y_2)\in\mathtt{B}_{\varepsilon_2}$ with $\varepsilon_1,\varepsilon_2\in\\{0,1\\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037003 | MATHDB-358118

Minimal measure doubling conjecture for simple Lie groups

摘录 / EXCERPT: Minimal measure doubling conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037004 | MATHDB-358119

Nonabelian Brunn-Minkowski conjecture for simply connected simple Lie groups

摘录 / EXCERPT: Nonabelian Brunn-Minkowski conjecture. For every pair of compact sets $A,B\subseteq G$,

conjecture | open | MathDB 页面 | 原文线索

V2-037005 | MATHDB-358200

Purity and compact-support conjecture for self-similar states

摘录 / EXCERPT: Let (A,L) satisfy the requirements of the strictly contractive dual iterated function system, and let $\phi_0\in\mathcal{S}_{\mathrm{c}}(A)$. Purity and compact-support conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-037006 | MATHDB-358201

Compact-support conjecture for self-similar states

摘录 / EXCERPT: Let (A,L) satisfy the requirements of the strictly contractive dual iterated function system, let $\phi_0\in\mathcal{S}_{\mathrm{c}}(A)$, and let ϕ_π be the self-similar s...

conjecture | open | MathDB 页面 | 原文线索

V2-037007 | MATHDB-358224

Coincidence of Aronszajn null sets and heat kernel null sets

摘录 / EXCERPT: Let G be a stratified Banach-Lie group with finite-dimensional commutator subgroup. An Aronszajn null set is a set null on the relevant affine lines in the Aronszajn sense, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-037008 | MATHDB-358228

Random alternant-code filtration conjecture

摘录 / EXCERPT: Random alternant-code filtration conjecture. We expect that

conjecture | open | MathDB 页面 | 原文线索

V2-037009 | MATHDB-358230

Zygmund's conjecture I on maximal operators for parametrized rectangle bases

摘录 / EXCERPT: Zygmund's conjecture I. The maximal operator $M_{B_{\boldsymbol{\phi}}}$ is bounded from $L^1\left(1+\log^+L^1\right)^{k-1}$ to $L^{1,\infty}$.

conjecture | refuted | MathDB 页面 | 原文线索

V2-037010 | MATHDB-358231

Stokolos's reformulation of Zygmund's conjecture for translation-invariant rectangle bases

摘录 / EXCERPT: Stokolos's reformulation of Zygmund's conjecture. There exists an integer k with $1\leq k\leq n$ such that $M_{\mathcal{B}}$ is sharply bounded from...

conjecture | open | MathDB 页面 | 原文线索

V2-037011 | MATHDB-358309

The uniqueness conjecture for zeros of diagonal Laguerre functions

摘录 / EXCERPT: Uniqueness conjecture for diagonal Laguerre zeros. The only $(j,n)\in\mathbb{N}\times\mathbb{Z}^*$ for which $l_j(n)=0$ is $(j,n)=(2,2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-037012 | MATHDB-358317

The low- and high-degree Fourier-level conjecture for symmetric hypercube connections

摘录 / EXCERPT: Let $\sigma:\{\pm1\}^d\to[0,1]$ be a symmetric connection, and let $\widehat{\sigma}(S)$ denote its Fourier coefficient indexed by $S\subseteq[d]$. The low- and high-degree Fourier...

conjecture | open | MathDB 页面 | 原文线索

V2-037013 | MATHDB-358330

Polynomial convexity of totally real discs in singular Levi-flat hypersurfaces

摘录 / EXCERPT: Second conjecture. The disc K is polynomially convex if

conjecture | open | MathDB 页面 | 原文线索

V2-037014 | MATHDB-358345

The monotonicity and continuity conjecture for generalized Constantin-Lax-Majda fixed points

摘录 / EXCERPT: Monotonicity and continuity conjecture. For any $a_1\leq a_2\leq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037015 | MATHDB-358434

The arrival conjecture for Collatz-invariant measures

摘录 / EXCERPT: Let $\mathrm{Col}:\mathbb N\to\mathbb N$ be the Collatz map, and let μ be a finitely additive Col-invariant measure on $\mathbb N$, with every singleton measurable....

conjecture | open | MathDB 页面 | 原文线索

V2-037016 | MATHDB-358439

Generalized Maximum Modulus Principle for cyclic groups

摘录 / EXCERPT: Generalized Maximum Modulus Principle. There exists $C(d,N)>0$, independent of n, such that for every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037017 | MATHDB-358441

Budaghyan et al.'s differential spectrum conjecture for a power function over finite fields

摘录 / EXCERPT: Budaghyan et al.'s differential spectrum conjecture. The equation has q^2 solutions for one value of b, has q^2-q solutions for q values of b, and has at most 2 solutio...

conjecture | open | MathDB 页面 | 原文线索

V2-037018 | MATHDB-358474

Kisil's nonemptiness conjecture for Clifford spectra

摘录 / EXCERPT: Let $A_{1},\dots,A_{d}$ be Hermitian matrices in $\boldsymbol{M}_{n}(\mathbb{C})$, and let $\Lambda(A_{1},\dots,A_{d})$ denote their Clifford spectrum. Kisil's conjecture. The Clif...

conjecture | open | MathDB 页面 | 原文线索

V2-037019 | MATHDB-358476

No one-dimensional manifold Clifford spectrum conjecture

摘录 / EXCERPT: Let $A_{1},\dots,A_{d}$ be Hermitian matrices in $\boldsymbol{M}_{n}(\mathbb{C})$, and let $\Lambda(A_{1},\dots,A_{d})$ denote their Clifford spectrum. No-one-manifold conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-037020 | MATHDB-358489

The discreteness conjecture for Lipschitz quotient mappings

摘录 / EXCERPT: Discreteness conjecture. The mapping f is discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-037021 | MATHDB-358496

The equivalence of nuclearity, symmetric amenability, strong amenability and unitary-group amenability

摘录 / EXCERPT: Amenability equivalence conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-037022 | MATHDB-358497

Amenability of the component group of the unitary group

摘录 / EXCERPT: Component-group amenability conjecture. The discrete group

conjecture | open | MathDB 页面 | 原文线索

V2-037023 | MATHDB-358498

Pisier's Grothendieck-pair conjecture with the bounded approximation property

摘录 / EXCERPT: Let $\mathcal{E}=(E_n)_{n\geqslant 1}$ be a sequence of finite-dimensional Banach spaces with $\dim E_n=n$, and let F be a Banach space with the bounded approximation property. A...

conjecture | open | MathDB 页面 | 原文线索

V2-037024 | MATHDB-358499

Pisier's K-convexity conjecture under finite cotype

摘录 / EXCERPT: Let X be an infinite-dimensional Banach space with the bounded approximation property. For a Banach space X, write $q(X)=\inf\{q:X\text{ is of cotype }q\}$. Pisier's K-convexit...

conjecture | open | MathDB 页面 | 原文线索

V2-037025 | MATHDB-358500

Pisier's finite-cotype conjecture for nontrivial type

摘录 / EXCERPT: Let X be an infinite-dimensional Banach space. For a Banach space X, write $p(X)=\sup\{p:X\text{ is of type }p\}$ and $q(X)=\sup\{q:X\text{ is of cotype }q\}$. Pisier's finite-...

conjecture | open | MathDB 页面 | 原文线索

V2-037026 | MATHDB-358511

The periodic tiling conjecture for translational tiles

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^d$ be a set of finite, positive measure. A translational tiling of $\mathbb R^d$ is a collection of translates of Ω covering almost every point e...

conjecture | refuted | MathDB 页面 | 原文线索

V2-037027 | MATHDB-358515

Héra's conjecture on the dimension of unions of k-planes

摘录 / EXCERPT: Let $0<k\le d<n$ be integers and let $\beta\in[0,k+1]$. Let $\mathcal{V}\subset A(k,n)$ be a set of k-planes in $\mathbb{R}^n$, with ... Here, $A(k,n)$ denotes the affine Grassmann...

conjecture | open | MathDB 页面 | 原文线索

V2-037028 | MATHDB-358531

The visibility conjecture for visible parts of compact sets

摘录 / EXCERPT: Let $E\subset\mathbb{R}^d$ be compact. For $\theta\in\mathbb{S}^{d-1}$, let ... be the closed half-line spanned by θ, and define the visible part of E in direction θ...

conjecture | open | MathDB 页面 | 原文线索

V2-037029 | MATHDB-358584

Boundary-continuous rigidity conjecture for proper maps between balls

摘录 / EXCERPT: Let $\mathbb{B}^m\subset\mathbb{C}^m$ and $\mathbb{B}^M\subset\mathbb{C}^M$ be unit balls, and let $f:\mathbb{B}^m\rightarrow\mathbb{B}^M$ be a proper holomorphic map extending to...

conjecture | open | MathDB 页面 | 原文线索

V2-037030 | MATHDB-358593

Partial-moment conjecture for secondary Gauss-Legendre nodes

摘录 / EXCERPT: Let $\bar z_i$ be the secondary intermediate nodes and let the first-order partial moments be defined by $\alpha_0=0$ and $\alpha_i=\alpha_{i-1}-2x_iw_i$. Partial-moment conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-037031 | MATHDB-358681

Local finiteness conjecture for the synthetic curvature measure

摘录 / EXCERPT: Let $X\in\text{Alex}^n(-1)$, and let ... be the synthetic curvature measure, where ${\operatorname R}$, $\mathcal{K}$, and θ are the corresponding density functions. Local fin...

conjecture | open | MathDB 页面 | 原文线索

V2-037032 | MATHDB-358687

The vector-valued Borell inequality conjecture

摘录 / EXCERPT: Vector-valued Borell inequality. If $0<\rho<1$ and

conjecture | open | MathDB 页面 | 原文线索

V2-037033 | MATHDB-358737

Conjecture on isometric transitivity of norming order units

摘录 / EXCERPT: Isometric transitivity conjecture. If $u\in V$ with $\lVert u\rVert=1$ is a norming order unit, then there exists a surjective linear isometry

conjecture | open | MathDB 页面 | 原文线索

V2-037034 | MATHDB-358746

Real-rank-zero conjecture for amenable group C^*-algebras

摘录 / EXCERPT: Real-rank-zero conjecture. Then G is locally finite.

conjecture | open | MathDB 页面 | 原文线索

V2-037035 | MATHDB-358748

Boundary conjecture for the analytic filtration

摘录 / EXCERPT: Let $\mathfrak A=\{\mathfrak A_\alpha\}_{\alpha\le1}$ be the analytic filtration, and let F_α denote the function used in its definition. For each $\alpha\le1$, consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-037036 | MATHDB-358749

Boundary conjecture for the hyperbolic filtration

摘录 / EXCERPT: Boundary conjecture. The function f belongs to $\partial\mathfrak H_\alpha$ whenever $\alpha\in(0,2/3]$. The source notes that several other boundary functions are known, while t...

conjecture | open | MathDB 页面 | 原文线索

V2-037037 | MATHDB-358750

No boundary repelling fixed point conjecture for the generalized filtration

摘录 / EXCERPT: Let $\mathfrak G^{(0)}=\{\mathfrak G^{(0)}_\alpha\}_{\alpha\in[\alpha_*,1]}$ be the generalized filtration, and let $f\in\mathfrak G^{(0)}_\alpha$. The function f generates a sem...

conjecture | open | MathDB 页面 | 原文线索

V2-037038 | MATHDB-358751

Strictness conjecture for the filtration of classes $\mathfrak L_\alpha$

摘录 / EXCERPT: Let $\{\mathfrak L_\alpha\}_{\alpha\in[0,1]}$ be the filtration of $\mathcal G_0$ defined in the paper. The family is called strict when its inclusions are strict as the parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-037039 | MATHDB-358859

Tsai-Tsui-Wang area-nonincreasing rigidity conjecture

摘录 / EXCERPT: Let $(\mathbb S^n,g_{std})$ and $(\mathbb{S}^m,g_{std})$ be round spheres, and let $(\mathbb C \mathbb P ^n,g_{FS})$ and $(\mathbb C \mathbb P ^m,g_{FS})$ be complex projective spa...

conjecture | open | MathDB 页面 | 原文线索

V2-037040 | MATHDB-358882

Uniqueness conjecture for energy ground states on the \b1-metric graph

摘录 / EXCERPT: Let $p\in(2,6)$ and let $\nu>0$ be a mass. An energy ground state is a positive solution of the NLS variational problem on the $\mathcal{T}$-metric graph that minimizes the energy...

conjecture | open | MathDB 页面 | 原文线索

V2-037041 | MATHDB-358892

Spectral projector bounds on narrow windows for the two-dimensional torus

摘录 / EXCERPT: Spectral projector conjecture. If $p\geq2$ and $\kappa>0$, then, whenever $\delta>\lambda^{-1+\kappa}$,

conjecture | open | MathDB 页面 | 原文线索

V2-037042 | MATHDB-358947

Spectral extension property for unitizations of semisimple Banach algebras

摘录 / EXCERPT: Let $(A,\|\cdot\|)$ be a non-unital, semisimple Banach algebra, and let A_e denote its unitization. Say that A has the spectral extension property (SEP) when every norm on A...

conjecture | open | MathDB 页面 | 原文线索

V2-037043 | MATHDB-358988

Geometric sign conjecture for the phase-two extrapolation parameters

摘录 / EXCERPT: Assume $f obreak\text{ belongs to } C^{1,1}_\nu(\mathbb{R}^n)$. Let $\hat{f}$ be the linear function that interpolates f at any set of $n+1$ affinely independent vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-037044 | MATHDB-359048

Isometric decomposition conjecture for von Neumann factors

摘录 / EXCERPT: Let $\mathfrak{R}$ be a type-II_{∞} or type-III factor acting on a separable Hilbert space, and let $W_e^{\mathfrak{R}}(A)$ denote its essential numerica...

conjecture | open | MathDB 页面 | 原文线索

V2-037045 | MATHDB-359049

Essentially Hermitian conjecture for block-matrix norm inequalities

摘录 / EXCERPT: Let $X\in\mathbb{M}_n$. Say that X has the universal block-matrix inequality if, for every positive block matrix with X as its off-diagonal block, ... Essentially Hermitian conje...

conjecture | open | MathDB 页面 | 原文线索

V2-037046 | MATHDB-359083

The semifinite embedding conjecture for nonassociative cL^p-spaces

摘录 / EXCERPT: Let $\mathcal{M}$ be a JW^*-algebra equipped with a normal semifinite faithful trace τ, and let $1{\leqslant} p<\infty$. Let $\mathrm{L}^p(\mathcal{M},\tau)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-037047 | MATHDB-359102

Uniformly separated bounded sets are ell^1-bounded

摘录 / EXCERPT: Uniform separation conjecture. If M is bounded and uniformly separated, then M is ell^1-bounded. This is posed as an open question in the paper; no resolution is supplied the...

open_problem | open | MathDB 页面 | 原文线索

V2-037048 | MATHDB-359159

Topping's existence conjecture for Ricci flow on complete noncompact 3-manifolds

摘录 / EXCERPT: Topping's Ricci-flow existence conjecture. The Ricci flow equation

conjecture | open | MathDB 页面 | 原文线索

V2-037049 | MATHDB-359170

Trace inequality conjecture for stable spherical Voronoi clusters

摘录 / EXCERPT: Let Ω be a 3-Plateau, stable, perpendicular spherical Voronoi q-cluster on $\mathbb{S}^n$ with equatorial cells and $q\leq n+1$. Let $\mathbf{F}$ be its conformal-to-vol...

conjecture | open | MathDB 页面 | 原文线索

V2-037050 | MATHDB-359171

Bounded conformal Jacobi fields conjecture

摘录 / EXCERPT: Let Ω be a 3-Plateau, stable, perpendicular spherical Voronoi q-cluster with equatorial cells and $q\leq n+1$ in $\mathbb{S}^n$. Let $\Omega_t=\Phi_t(\Omega)$ be the PCF...

conjecture | open | MathDB 页面 | 原文线索

V2-037051 | MATHDB-359172

No-new-interfaces conjecture for Gram perturbations of minimizers

摘录 / EXCERPT: Let Ω be a spherical Voronoi q-cluster on $\mathbb{S}^n$ with $q\leq n+1$ that is an isoperimetric minimizer. Let $\{\Omega_t\}$ be its Gram perturbation, and let $t_e>0$....

conjecture | open | MathDB 页面 | 原文线索

V2-037052 | MATHDB-359295

Classification conjecture for finite depth objects in bicommutant categories

摘录 / EXCERPT: Let $\mathcal{B}$ be a bicommutant category admitting an absorbing object whose endomorphism algebra is a hyperfinite II_∞ or III_1 factor. Assume that ……

conjecture | open | MathDB 页面 | 原文线索

V2-037053 | MATHDB-359326

Saitoh's strict inequality conjecture for conjugate Hardy and Bergman kernels

摘录 / EXCERPT: Saitoh's conjecture. If D is not simply connected, then

conjecture | open | MathDB 页面 | 原文线索

V2-037054 | MATHDB-359367

Generalized inf-sup conjecture for Rhie-Chow stabilization

摘录 / EXCERPT: Let $\Omega=[-1,1]^d$ with $d=2,3$, let $\mathcal{V}_h$ be the discrete velocity or pressure space as appropriate, and let $\Pi_h q_h$ be the projection used in the Rhie-Chow formu...

conjecture | open | MathDB 页面 | 原文线索

V2-037055 | MATHDB-359375

Non-uniqueness conjecture for quotient-neck singularities in 4d Ricci flow

摘录 / EXCERPT: Non-uniqueness conjecture. For every $k\geq 3$ there exists a Ricci flow on $S^1\times S^3/\mathbb{Z}_k$ that forms a quotient-neck singularity with tangent flow...

conjecture | open | MathDB 页面 | 原文线索

V2-037056 | MATHDB-359459

Flatness conjecture for stable nonlocal minimal hypercones in dimensions three through seven

摘录 / EXCERPT: Let $s\in(0,1)$, and consider stable s-minimal hypercones in $\mathbb{R}^n\setminus\{0\}$. Flatness conjecture. For every integer n with $3\leqslant n\leqslant 7$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-037057 | MATHDB-359463

Adapted-function reduction for quadratic-manifold restriction estimates

摘录 / EXCERPT: Let $fbf{Q}(fxi)=(Q_1(fxi),\ldots,Q_n(fxi))$ be an n-tuple of real quadratic forms in d variables, and let ... For a rectangle $B\subset[0,1]^d$, call f adapted to B if...

conjecture | open | MathDB 页面 | 原文线索

V2-037058 | MATHDB-359464

Core-carrier conjecture for analytic weighted Bergman spaces

摘录 / EXCERPT: Let Gcolon $(0,\infty)\to(0,\infty)$ be the function occurring in the measure structure defined by, let $w\in\mathcal{L}^1(\mathbb{T})$ be a non-negative weight, and let ... for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-037059 | MATHDB-359465

Cyclicity conjecture for singular inner functions in analytic weighted spaces

摘录 / EXCERPT: Let G and $w\in\mathcal{L}^1(\mathbb{T})$ be as in the core-carrier conjecture, let μ have the form, and let $\operatorname{core}(w)$ denote the core of w. For a singular m...

conjecture | open | MathDB 页面 | 原文线索

V2-037060 | MATHDB-359466

Coefficient-growth conjecture for the Cauchy transform

摘录 / EXCERPT: Coefficient-growth conjecture. The conclusion of the uncertainty theorem referred to as thref{UncertThmRSD} can be reached if merely

conjecture | open | MathDB 页面 | 原文线索

V2-037061 | MATHDB-359510

The limiting-shape conjecture for optimal radar codes

摘录 / EXCERPT: Let $T>0$ denote the noise-temperature parameter, and let $epsilon_T(t)$ be an optimal radar code for each T. Limiting-shape conjecture. The limit ... defines a well-defined limiti...

conjecture | open | MathDB 页面 | 原文线索

V2-037062 | MATHDB-359602

Classical zero-curvature trilinear Hilbert transform conjecture

摘录 / EXCERPT: Let $\gamma(t)=(a_1t,a_2t,a_3t)$, where a_1,a_2,a_3 are nonzero and pairwise distinct, so that γ is a nondegenerate line in $\mathbb{R}^3$. For $\vec f=(f_1,f_2,f_3)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037063 | MATHDB-359652

Nonexistence of category-(c) normalizable frames

摘录 / EXCERPT: Let H be a Hilbert space, and let category (c) be the class of frames described in Theorem of the source. A frame is normalizable when normalizing each nonzero vector produces a...

conjecture | open | MathDB 页面 | 原文线索

V2-037064 | MATHDB-359665

The Cantor-normal-form conjecture for logics of ordinal beta spaces

摘录 / EXCERPT: Let $gamma$ be an ordinal with Cantor normal form, and write $alpha_1$ for its leading exponent. Let $beta(gamma)$ be the corresponding beta space, $sf L(beta(gamma))$ its logic, a...

conjecture | open | MathDB 页面 | 原文线索

V2-037065 | MATHDB-359668

Higher-dimensional Monge-Ampère decomposition conjecture

摘录 / EXCERPT: Higher-dimensional decomposition conjecture. For every $D\in\mathcal{C}^{m,\gamma}(\bar\omega,\mathbb{R}^{d\times d}_{\mathrm{sym}})$,

conjecture | open | MathDB 页面 | 原文线索

V2-037066 | MATHDB-359693

Smale's mean value conjecture

摘录 / EXCERPT: Smale's mean value conjecture.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037067 | MATHDB-359705

Relative bicentralizer conjecture for maximal abelian subalgebras with expectation

摘录 / EXCERPT: Let $N \subset M$ be an irreducible inclusion of factors with expectation and with separable preduals, meaning that $N' \cap M=\mathbb{C}$ and there is a faithful normal conditiona...

conjecture | open | MathDB 页面 | 原文线索

V2-037068 | MATHDB-359706

Relative bicentralizer formula for the continuous core

摘录 / EXCERPT: Let $N \subset M$ be an inclusion of von Neumann algebras with expectation, and let $\varphi \in N_*$ be a faithful normal state. Write $c(N)=N \rtimes_{\sigma^\varphi} \mathbb{R}$...

conjecture | open | MathDB 页面 | 原文线索

V2-037069 | MATHDB-359707

Bicentralizer conjecture for inclusions with expectations

摘录 / EXCERPT: Let $N \subset M$ be an inclusion of von Neumann algebras with expectations. For a strictly semifinite weight $\varphi \in \mathcal{P}(N)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-037070 | MATHDB-359791

Extension of the main theorem to endpoint dimensions

摘录 / EXCERPT: Let m, q, and s_P be the parameters in Theorem. Endpoint extension conjecture. Theorem also holds when $s_P=m-1$, $s_P=m$, or $s_P=q$. This would extend the theorem to the en...

conjecture | open | MathDB 页面 | 原文线索

V2-037071 | MATHDB-359792

Prescribing arbitrary Hausdorff and packing dimensions of doubling measures

摘录 / EXCERPT: Prescribing-dimensions conjecture. Theorem also holds without the requirement that $d_*=q\Leftrightarrow D_*=q$ and $d^*=q\Leftrightarrow D^*=q$. The paper's construction does not...

conjecture | open | MathDB 页面 | 原文线索

V2-037072 | MATHDB-359807

Generalised Erdős-Lax inequality for powers of polynomials

摘录 / EXCERPT: Let ... where $c\in\mathbb{C}$, $m\in\mathbb{N}$, $s_k\geq 1$, $ms_k\in\mathbb{N}$, and $|a_k|\geq 1$. For any branch $f(z)=P(z)^{1/m}$, write $\|\cdot\|_{\mathbb{T}}$ for the maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-037073 | MATHDB-359837

Conjecture on extending the eigenvalue formula for $L_{n,p,k}$ to all p and n

摘录 / EXCERPT: Let $L_{n,p,k}$ be the matrix whose scaled eigenvalues and grids are described by Theorem, with $1\le p\le100$, $0\le k\le\min(1,p-1)$, and the grids...

conjecture | open | MathDB 页面 | 原文线索

V2-037074 | MATHDB-359853

The Karpelevic polygon conjecture for Farey-neighbor arcs

摘录 / EXCERPT: Let $n>3$, and let p/q and r/s be Farey neighbors. Let $\a0\lambda$ be a function satisfying the properties listed in the second step of the main proposition, and let E_n den...

conjecture | open | MathDB 页面 | 原文线索

V2-037075 | MATHDB-359892

Exactness of the k-dependent terms in Poisson order-k median asymptotics

摘录 / EXCERPT: Exactness conjecture for the median asymptotics. The terms in these two formulas that depend only on k are exact.

conjecture | open | MathDB 页面 | 原文线索

V2-037076 | MATHDB-359896

Haagerup and Størmer's conjecture for pointwise inner automorphisms

摘录 / EXCERPT: Let M be a type $\rm III_1$ factor with separable predual. An automorphism $\theta\in\operatorname{Aut}(M)$ is pointwise inner if, for every positive linear functional...

conjecture | open | MathDB 页面 | 原文线索

V2-037077 | MATHDB-360049

Decoupling conjecture for equally distributed points on a monomial curve

摘录 / EXCERPT: Let $d\geq 3$ be an integer, let $\Gamma=\{(\xi,\xi^d)\in\mathbb R^2:1\leq|\xi|\leq2\}$, and let $P_j=(\xi_j,\xi_j^d)$ for $j=1,\ldots,N$ be equally distributed in Γ. For ea...

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V2-037078 | MATHDB-360065

Conjecture on biholomorphic Loewner chains in reflexive Banach spaces

摘录 / EXCERPT: Biholomorphic-chain conjecture. If $k_+(A)<2m(A)$ and $f(z,t)$ is the A-normalized univalent subordination chain given by the displayed limit, then $f(\cdot,t)$ is biholomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-037079 | MATHDB-360189

Lemvig and Nielsen's obstruction conjecture for second-order B-spline Gabor frames

摘录 / EXCERPT: Lemvig and Nielsen's conjecture. The Gabor system $\mathcal{G}(Q_2,a,b)$ is not a frame along the hyperbolas

conjecture | open | MathDB 页面 | 原文线索

V2-037080 | MATHDB-360190

Nielsen's obstruction conjecture for second-order B-spline Gabor frames

摘录 / EXCERPT: Let $Q_2=Q_1*Q_1$, where $Q_1(x):=\chi_{[-1/2,1/2]}(x)$, and let $\mathcal{G}(Q_2,a,b)$ denote the associated Gabor system. Nielsen's conjecture. The Gabor system...

conjecture | open | MathDB 页面 | 原文线索

V2-037081 | MATHDB-360200

Continuous Kraus conjecture for continuous Parseval frames

摘录 / EXCERPT: Let (Ω,μ) and (Δ,ν) be measure spaces, let $\mathcal{H}$ be a Hilbert space, and let $\{\tau_\alpha\}_{\alpha\in\Omega}$ and $\{\omega_\beta\}_{\beta\in\Delta}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-037082 | MATHDB-360313

Uniqueness conjecture for semigraphical translators

摘录 / EXCERPT: Let w be a width, and let Σ_w denote the corresponding semigraphical translator: a helicoid when $w<\pi$ and a pitchfork when $w\geqslant\pi$. Uniqueness conjecture. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-037083 | MATHDB-360385

Esterle's Hilbert-space contraction conjecture

摘录 / EXCERPT: Let E be a closed subset of the unit circle $\mathbb{T}$ of Lebesgue measure zero. A Hilbert-space contraction is a contraction acting on a Hilbert space, and T is unitary when...

conjecture | open | MathDB 页面 | 原文线索

V2-037084 | MATHDB-360386

Generalized partial-estimate conjecture for operator-valued inner functions

摘录 / EXCERPT: Let E be a closed subset of the unit circle $\mathbb{T}$ such that $|E|=0$. Let $\mathcal{F}$ and $\mathcal{F}'$ be separable Hilbert spaces, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037085 | MATHDB-360607

Conjecture on the monotonicity of Riemann sums for reciprocal quadratics

摘录 / EXCERPT: Let $f_b(x):=\frac{1}{1-bx+x^2}$, and let $\beta^+=\frac{3+\sqrt{13}}{4}$. Monotonicity conjecture. 1. $L_n(f_b)$ is neither increasing nor decreasing for all...

conjecture | open | MathDB 页面 | 原文线索

V2-037086 | MATHDB-360613

Singular-set conjecture for locally least-gradient functions

摘录 / EXCERPT: Singular-set conjecture. Suppose that $d\geq 8$. Then every function of locally least gradient $u:M\to\mathbf{R}$ has a singular set S_u of codimension 8, such that on...

conjecture | open | MathDB 页面 | 原文线索

V2-037087 | MATHDB-360719

Weighted maximal operator conjecture for Walsh-Fourier partial sums in H_1

摘录 / EXCERPT: Weighted maximal operator conjecture. For every $f\in H_1$, the operator $S^{\ast,\nabla}$ is bounded from H_1 to L_1. Moreover, if $\sup_{k\in\mathbb{N}}|A_{n_k}|=\infty$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037088 | MATHDB-360867

The hook-number ratio conjecture for odd-part and distinct-part partitions

摘录 / EXCERPT: The hook-number ratio conjecture. For fixed $h\geq 2$, there exists an integer $N_h>0$ such that $a_h(n)\geq b_h(n)$ for all $n>N_h$, and there exists a constant $\gamma_h>1$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-037089 | MATHDB-360868

The hook-number ratio conjecture for self-conjugate and distinct-odd partitions

摘录 / EXCERPT: Let $a_h^*(n)$ and $b_h^*(n)$ denote the numbers of hooks of length h among self-conjugate partitions of n and partitions of n into distinct odd parts, respectively. The hook...

conjecture | open | MathDB 页面 | 原文线索

V2-037090 | MATHDB-360894

Generalized Chern-Hamilton conjecture for closed contact 3-manifolds

摘录 / EXCERPT: Let (M,ζ) be a closed contact 3-manifold, where ζ is a contact distribution, and consider the Chern-Hamilton energy functional on the space of compatible metrics for...

conjecture | open | MathDB 页面 | 原文线索

V2-037091 | MATHDB-360898

Kaczynski's vanishing-area conjecture for unrestricted trapezoids

摘录 / EXCERPT: Kaczynski's vanishing-area conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037092 | MATHDB-360899

Kaczynski's logarithmic-area conjecture for unrestricted trapezoids

摘录 / EXCERPT: For the smallest unrestricted trapezoid area ... where T^n_σ is the union of the parallelograms associated with σ, Kaczynski's logarithmic-area conjecture. there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-037093 | MATHDB-360900

Kaczynski's monotonicity conjecture for unrestricted trapezoid areas

摘录 / EXCERPT: For each positive integer n, let ... where T^n_σ is the smallest-area unrestricted trapezoid associated with a permutation of n parallelograms. Kaczynski's monotonicity co...

conjecture | open | MathDB 页面 | 原文线索

V2-037094 | MATHDB-360901

Bagemihl's three-segment property conjecture

摘录 / EXCERPT: Let $\mathbb{D}$ be the open unit disk. A disk function has the three-segment property at a boundary point z if the cluster sets along three Jordan arcs that are unions of rectil...

conjecture | open | MathDB 页面 | 原文线索

V2-037095 | MATHDB-360961

The sharp third-moment threshold for centroidal closed curves

摘录 / EXCERPT: Sharp third-moment threshold conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-037096 | MATHDB-361020

The n/k to n norm conjecture for the maximal operator ${\mathcal N}^k$

摘录 / EXCERPT: The n/k to n norm conjecture for ${\mathcal N}^k$. There is a constant C_n, depending on n and ${\mathbb F}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-037097 | MATHDB-361032

Clunie-Eremenko-Rossi conjecture on zeros of sums of Cauchy kernels

摘录 / EXCERPT: Clunie-Eremenko-Rossi conjecture. The function $f(z)$ has infinitely many zeros.

conjecture | open | MathDB 页面 | 原文线索

V2-037098 | MATHDB-361063

Level set Crouzeix conjecture for finite Blaschke products

摘录 / EXCERPT: Level set Crouzeix conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-037099 | MATHDB-361064

Pseudohyperbolic disk conjecture for unicritical Blaschke products

摘录 / EXCERPT: Let $t\in[0,1)$, and let Θ_t be of the unicritical form referred to in the source. Let $W(\Theta_t)$ denote the associated numerical range. Pseudohyperbolic disk conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-037100 | MATHDB-361070

The non-\lineability conjecture for functions vanishing at infinity with a unique maximum

摘录 / EXCERPT: Let $n\geq 2$. Denote by $\hat{\mathcal{C}}_0(\mathbb{R}^n)$ the set of functions vanishing at infinity that have a unique maximum. A subset of a vector space is k-lineable if it...

conjecture | open | MathDB 页面 | 原文线索

V2-037101 | MATHDB-361111

Shapiro's shadow-domain conjecture for derivatives of polynomial powers

摘录 / EXCERPT: Shapiro's shadow conjecture. The set Υ_P is a closed domain in the convex hull of the roots of P; all critical points of P lie on its boundary; its boundary has no inf...

conjecture | open | MathDB 页面 | 原文线索

V2-037102 | MATHDB-361125

The Fourier-Min-Entropy-Influence conjecture

摘录 / EXCERPT: Fourier-Min-Entropy-Influence conjecture. There exists a constant $c>0$ such that, for every Boolean function $f:\\{-1,1\\}^n\to\\{-1,1\\}$,

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V2-037103 | MATHDB-361141

The large-l counting conjecture for Gaudin Bethe solutions

摘录 / EXCERPT: Let $d_0=d_1$ and let $\operatorname{Part}(d_0)$ denote the number of partitions of d_0. Large-l Gaudin counting conjecture. For generic l,m, the Gaudin Bethe Ansatz system h...

conjecture | open | MathDB 页面 | 原文线索

V2-037104 | MATHDB-361145

The large-l Gaudin counting conjecture for unequal degrees

摘录 / EXCERPT: Let $r=d_0-d_1$ and use the polynomials W_r,U_r and the Hermite-Wronskian polynomial V_μ defined in the source. Unequal-degree large-l conjecture. For generic l,m, the Ga...

conjecture | open | MathDB 页面 | 原文线索

V2-037105 | MATHDB-361268

Generation conjecture for sparse two-factor analysis model ideals

摘录 / EXCERPT: A sparse two-factor analysis model is the model associated with a graph G having two latent variables, with $M_{\leq 1}(G)$ denoting the set of model polynomials indexed by subse...

conjecture | open | MathDB 页面 | 原文线索

V2-037106 | MATHDB-361288

Herremans, Huybrechs and Trefethen's lightning plus polynomial approximation conjecture

摘录 / EXCERPT: Herremans, Huybrechs and Trefethen's conjecture. There exist coefficients $\{a_j\}_{j=1}^{N_1}$ and a polynomial P_{N_2} with $N_2=\mathcal{O}(\sqrt{N_1})$ such that

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V2-037107 | MATHDB-361323

Wigderson's range conjecture for the Fourier norm functional

摘录 / EXCERPT: Wigderson's conjecture. The image of $\mathcal{F}_{p,2}^{(1)}$ is all of $\mathbb{R}_{>0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-037108 | MATHDB-361342

Global maximality of the FCC lattice for truncated density of unit ball packings

摘录 / EXCERPT: FCC truncated-density conjecture. For all >0, among packings of unit balls in ^{3}, the $$ -truncated density has a maximum at the corresponding FCC lattice.

open_problem | open | MathDB 页面 | 原文线索

V2-037109 | MATHDB-361367

The principal-curvature conjecture for Fourier dimension of hypersurfaces

摘录 / EXCERPT: A hypersurface M in Euclidean space has principal curvatures, the eigenvalues of its Weingarten map, at each point. Its Fourier dimension, denoted by $\dim_F(M)$, measures the st...

conjecture | open | MathDB 页面 | 原文线索

V2-037110 | MATHDB-361368

The Fourier-dimension conjecture for cones and cylinders generated by arbitrary sets

摘录 / EXCERPT: Cone-and-cylinder Fourier-dimension conjecture. Then

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V2-037111 | MATHDB-361393

The explicit contraction factor conjecture for Chebyshev subdivision

摘录 / EXCERPT: Let $0<\alpha<1$ and let $0\leq\beta\leq 1-\alpha$. Write $\tau_{\alpha,\beta}$ for the asymptotic degree ratio associated with approximating a Chebyshev polynomial after the affin...

conjecture | open | MathDB 页面 | 原文线索

V2-037112 | MATHDB-361419

Aliasing estimate for ultraspherical rectangular collocation

摘录 / EXCERPT: Aliasing estimate. There exists $c(\lambda)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037113 | MATHDB-361450

Choksi-Peletier liquid drop minimizer conjecture

摘录 / EXCERPT: Choksi-Peletier conjecture. For $V\leq V_*$, the round ball of volume V uniquely minimizes $\mathcal E(\cdot)$ among measurable sets $\Omega\subset\mathbb{R}^3$ with $|\Omega|=V$...

conjecture | open | MathDB 页面 | 原文线索

V2-037114 | MATHDB-361460

Audenaert-Kittaneh's Clarkson inequalities conjecture for Schatten operators

摘录 / EXCERPT: Audenaert-Kittaneh's conjecture. For $1<p\le 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-037115 | MATHDB-361499

Eskenazis-Ivanisvili's dimension-free Laplacian estimate for low-degree functions

摘录 / EXCERPT: Let $(X,\|\cdot\|_X)$ be a K-convex Banach space. For $n\in\mathbb{N}$, let $\operatorname{Deg}_{\le d}$ denote the space of functions $f:\{-1,1\}^n\to X$ whose Hamming cube Four...

conjecture | open | MathDB 页面 | 原文线索

V2-037116 | MATHDB-361510

Bourgain-Chang bilinear estimate conjecture over finite fields

摘录 / EXCERPT: Bourgain-Chang bilinear estimate conjecture. For every $\varepsilon>0$, there exists C_ε such that, for all sufficiently large primes p,

conjecture | open | MathDB 页面 | 原文线索

V2-037117 | MATHDB-361588

One-dimensional null-space conjecture for the Laplace-Beltrami integral equation

摘录 / EXCERPT: Let Γ be a smooth surface, and let $\mathcal I+\mathcal R_{\text{LB}}$ denote the integral operator in, where $\mathcal R_{\text{LB}}$ is the composition of the Laplace-Belt...

conjecture | open | MathDB 页面 | 原文线索

V2-037118 | MATHDB-361624

Conjecture on the smallest positive zero of the Mittag-Leffler function with parameter beta equal to alpha

摘录 / EXCERPT: Smallest-zero conjecture for $E_{\alpha,\alpha}$. For any $\alpha\in(1,2]$, $E_{\alpha,\alpha}(-z^\alpha)>0$ for all $z\in[0,2.93785]$. Moreover, as $\alpha\to1+$,

conjecture | open | MathDB 页面 | 原文线索

V2-037119 | MATHDB-361626

Conjecture on the smallest positive zero of the Mittag-Leffler function with parameter beta equal to 1

摘录 / EXCERPT: Smallest-zero conjecture for $E_{\alpha,1}$. For any $\alpha\in(1,2]$, $E_{\alpha,1}(-z^\alpha)>0$ for all $z\in[0,1.559]$. Moreover, as $\alpha\to1+$,

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V2-037120 | MATHDB-361666

Approximation-extension conjecture for Banach-valued holomorphic functions on the polydisk maximal ideal space

摘录 / EXCERPT: Extension conjecture. An X-valued holomorphic function f on U admits a continuous extension to $\widehat U$ if and only if $f(V)\Subset X$ for every subset...

conjecture | open | MathDB 页面 | 原文线索

V2-037121 | MATHDB-361697

Periodic Schrödinger maximal-function level set conjecture

摘录 / EXCERPT: Let N be a positive integer, let $(a_n)_{n=1}^N$ satisfy ... and let λ satisfy $N^{\frac14}\leq\lambda\leq N^{\frac12}$. Periodic Schrödinger maximal-function level set con...

conjecture | open | MathDB 页面 | 原文线索

V2-037122 | MATHDB-361725

Four-value CM conjecture for infinite-order meromorphic functions

摘录 / EXCERPT: Let f and g be two distinct transcendental meromorphic functions of infinite order, and let a_1,a_2,a_3,a_4 be four distinct values in the extended complex plane. Four-value...

conjecture | open | MathDB 页面 | 原文线索

V2-037123 | MATHDB-361726

Nonexistence conjecture for four-value sharing by infinite-order meromorphic functions

摘录 / EXCERPT: Let f and g be two distinct non-constant meromorphic functions of infinite order, and let a_1,a_2,a_3,a_4 be four distinct values in the extended complex plane. Nonexistence...

conjecture | open | MathDB 页面 | 原文线索

V2-037124 | MATHDB-361728

The restriction conjecture for the paraboloid

摘录 / EXCERPT: Restriction conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-037125 | MATHDB-361758

Generic multiplicity-one convergence conjecture for mean curvature flow

摘录 / EXCERPT: Let N be a Riemannian manifold and let $(M(t))_{t>0}$ be a mean curvature flow starting from a generic closed hypersurface. A long-time limit has multiplicity 1 if it occurs wi...

conjecture | open | MathDB 页面 | 原文线索

V2-037126 | MATHDB-361789

Universal-cover conjecture for operator-algebraic strict deformation quantization

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold, and let $C^{\infty}_c(M)$ denote the compactly supported smooth functions on M. A strict deformation quantization is understood in the...

conjecture | open | MathDB 页面 | 原文线索

V2-037127 | MATHDB-361811

Monotonicity conjecture for the convergence-rate quotients of the canonical iterates

摘录 / EXCERPT: Monotonicity conjecture. For all $m\in\mathbb N$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037128 | MATHDB-361833

Manifold-structure conjecture for higher-degree Bézier spline spaces

摘录 / EXCERPT: Let M be a d-dimensional Riemannian manifold and $U\subseteq M$ a normal convex neighborhood. For each nonnegative integer k, let $\mathcal{B}^1_k(U)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037129 | MATHDB-361864

Uniqueness conjecture for Schauder-Orlicz decompositions in ℓ_1 and c_0

摘录 / EXCERPT: A Schauder-Orlicz decomposition is a Schauder decomposition whose associated coefficient space is an Orlicz sequence space. Fix the dimensions of its component subspaces. Uniquenes...

conjecture | open | MathDB 页面 | 原文线索

V2-037130 | MATHDB-361870

The multidimensional critical exponent conjecture for the Riesz projection

摘录 / EXCERPT: Critical exponent conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-037131 | MATHDB-361872

The monotonicity and high-dimensional limit conjecture for Riesz projection exponents

摘录 / EXCERPT: For $d\geq1$, let $\mathfrak{p}_d(q)$ be the critical exponent of the Riesz projection on $\mathbb{T}^d$, and let $\mathfrak{q}(d)$ be its minimal admissible exponent. Monotonicity...

conjecture | open | MathDB 页面 | 原文线索

V2-037132 | MATHDB-361882

Measure classification from monotonicity of boundary content under Minkowski addition

摘录 / EXCERPT: Measure-classification conjecture. Then μ is a constant multiple of Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-037133 | MATHDB-361902

Handelman's integrability conjecture for log-convex functions

摘录 / EXCERPT: Let $h:(0,\infty)\rightarrow(0,\infty)$ be a log-convex function satisfying ... Assume that $h^2(x)/h(2x)$ is integrable on $(0,\infty)$. Handelman's integrability conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-037134 | MATHDB-361929

The critical gradient extrapolation conjecture

摘录 / EXCERPT: Consider gradient descent run for a fixed number N of steps with constant stepsize $h\in(0,1]$, starting from x_0 with $\lVert x_0-x_\star\rVert=D$. For simple extrapolation by...

conjecture | open | MathDB 页面 | 原文线索

V2-037135 | MATHDB-361943

Low-frequency spectral asymptotics for the Poincaré operator on ellipsoids

摘录 / EXCERPT: Low-frequency spectral asymptotics.

conjecture | open | MathDB 页面 | 原文线索

V2-037136 | MATHDB-361955

Ultra-flatness conjecture at roots of unity

摘录 / EXCERPT: Let $n>4$, and let ... be a polynomial with coefficients $a_i\in\{-1,1\}$. Ultra-flatness conjecture at roots of unity. There exists $\varepsilon_0>0$ such that ... This is a sampling...

conjecture | open | MathDB 页面 | 原文线索

V2-037137 | MATHDB-362014

Contraction conjecture for the modified interval iteration

摘录 / EXCERPT: Let Z_A, Z_B, χ, and $\breve{F}$ be the modified variables and iteration introduced for the case $\mathbb{K}_A \ne \mathbb{K}_B$, and let χ denote the modified log-od...

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V2-037138 | MATHDB-362015

Binomial Fisher limiting prior bounds

摘录 / EXCERPT: For $0<x_A<x_B<1$, let $P_N^*(x_A,x_B)$ denote the prior of a Binomial Fisher game with size parameter N. Binomial Fisher limiting prior bounds. The prior does not converge as...

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V2-037139 | MATHDB-362016

First-order asymptotics conjecture for the Binomial Bayesian prior

摘录 / EXCERPT: Let $0<x_A<x_B<1$, and let $\vartheta_N^*(x_A,x_B)$ be the Binomial Bayesian equilibrium log-odds. Let $\vartheta^{*,\varphi}_{\loopedsquare}(x_A,x_B)$ be its conjectured periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-037140 | MATHDB-362154

Mathieu-type admissible-function conjecture for $Sp(N)$

摘录 / EXCERPT: Let $N\in\mathbb{N}$, and let S^* denote the unit circle. A function ... is $\frac{1}{N}$-admissible if it is a finite sum $\sum_{\vec m}c_{\vec m}(x)z^{\vec m}$, where the exponen...

conjecture | open | MathDB 页面 | 原文线索

V2-037141 | MATHDB-362155

Mathieu-type admissible-function conjecture for G_2

摘录 / EXCERPT: Let S^* denote the unit circle, and let $\tilde J_{G_2}$ be the weight function associated with the paper's Euler-angle parametrization of G_2. A function ... is $\frac{1}{4}$-ad...

conjecture | open | MathDB 页面 | 原文线索

V2-037142 | MATHDB-362202

The large-set planar unit-ball decoupling conjecture

摘录 / EXCERPT: Large-set planar conjecture. If $|Y|\geq R^{1/2}$, then is true with

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V2-037143 | MATHDB-362211

Kalai-Wilson conjecture on the largest orthogonality-free subset of the sphere

摘录 / EXCERPT: Let $A\subseteq S^2$ be a measurable subset containing no two orthogonal directions. Its measure is normalized by $|S^2|=1$. Kalai-Wilson's conjecture. The maximal measure of such...

conjecture | open | MathDB 页面 | 原文线索

V2-037144 | MATHDB-362212

Maximal non-Kochen-Specker set conjecture

摘录 / EXCERPT: Let a non-Kochen-Specker set in dimension d be a measurable subset $A\subseteq S^{d-1}$ admitting a valuation map $v:A\to\{0,1\}$ such that $v(-n)=v(n)$, every mutually orthogona...

conjecture | open | MathDB 页面 | 原文线索

V2-037145 | MATHDB-362229

Halmos' reducible-operator conjecture for separable factors

摘录 / EXCERPT: Let $\mathcal{M}$ be a separable factor. An operator $T\in\mathcal{M}$ is reducible if there is a projection $P\in\mathcal{M}$ with $P\ne 0,I$ and $PT=TP$. Halmos' conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-037146 | MATHDB-362230

Approximate bicommutant conjecture for separable factors

摘录 / EXCERPT: Let $\mathcal{M}$ be a separable factor, and let $\mathcal{A}\subseteq\mathcal{M}$ be a separable unital C^*-subalgebra. The relative approximate bicommutant of $\mathcal{A}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-037147 | MATHDB-362251

Isoperimetric conjecture on the three-dimensional cube

摘录 / EXCERPT: Isoperimetric conjecture on the three-dimensional cube. For every $\bar v\in(0,1)$, at least one of the following sets or its complement is an isoperimetric minimizer in...

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V2-037148 | MATHDB-362330

Holomorphic continuation conjecture for stationary-measure solutions in the Schottky case

摘录 / EXCERPT: Let μ be a countably supported measure whose support generates a Fuchsian group $\Gamma \subset PSU(1,1)$ of second type. Let a solution be a function satisfying the equation r...

conjecture | open | MathDB 页面 | 原文线索

V2-037149 | MATHDB-362394

The conormal blow-up conjecture for self-similar soliton solutions

摘录 / EXCERPT: Let α be sufficiently close to zero, let $\mathfrak{M}_\alpha$ be a blow-up of the compactification of Minkowski space, and let W denote the soliton profile. A function is...

conjecture | open | MathDB 页面 | 原文线索

V2-037150 | MATHDB-362479

Fässler-Orponen projection conjecture for non-degenerate curves on the sphere

摘录 / EXCERPT: Let $\gamma:[0,1]\to\mathbb S^2$ be a C^2 curve satisfying ... For each θ, let $p_{\gamma(\theta)}$ denote projection onto the line in direction $\gamma(\theta)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037151 | MATHDB-362495

High-dimensional p decoupling conjecture for smooth hypersurfaces

摘录 / EXCERPT: Let $\phi:\mathbb{R}^{n-1}\to\mathbb{R}$ be smooth, and let $\mathcal{M}_{\phi}=\{(\xi,\phi(\xi)):\xi\in[0,1]^{n-1}\}$. A set $S\subset[0,1]^{n-1}$ is (ϕ,δ)-flat when it...

conjecture | open | MathDB 页面 | 原文线索

V2-037152 | MATHDB-362558

The right-triangle stability conjecture for the constant $C_{\rm st,2}$

摘录 / EXCERPT: Let T be a triangle with maximal interior angle ω, and let $C_{\rm st,2}$ denote the stability constant associated with the hybrid high-order method. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-037153 | MATHDB-362609

Gilbert's conjecture on Real Hardy spaces for Clifford modules

摘录 / EXCERPT: Gilbert's conjecture. There exist η and $\mathfrak{H}_0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037154 | MATHDB-362611

The absence of infinitesimals above 1 in maximal Hardy fields

摘录 / EXCERPT: Maximal Hardy-field conjecture. If H is maximal, then there is no $y\in\mathcal{C}^1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037155 | MATHDB-362646

Commutativity conjecture for asymptotic partition ratios with interleaved arithmetic progressions

摘录 / EXCERPT: Commutativity conjecture. The limits satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-037156 | MATHDB-362745

Sawyer-type inequality for the Hardy-Littlewood maximal operator with rearrangement weights

摘录 / EXCERPT: Fix exponents $q3e=1$ and $\varsigma>0$ (or $03c\varsigma3c=1$), and write ... Let u and v be positive, measurable functions such that $u^{\varsigma}\in A_{\varrho}^{\mathcal R}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-037157 | MATHDB-362802

Conjectured moduli of the quarter-sphere quadrilateral and its conjugate

摘录 / EXCERPT: Conjectured modulus values.

conjecture | open | MathDB 页面 | 原文线索

V2-037158 | MATHDB-362815

Boundedness conjecture for wavelet-type Malmquist-Takenaka operators with bounded hyperbolic gaps

摘录 / EXCERPT: Let $D>0$ be a constant, and let $\mathbf{a}=(a_n)_{n=1}^{\infty}$ be a sequence such that the hyperbolic distance between a_n and a_{n+1} is at most D for every...

conjecture | open | MathDB 页面 | 原文线索

V2-037159 | MATHDB-362886

Quantitative Besicovitch conjecture for planar sets

摘录 / EXCERPT: Quantitative Besicovitch conjecture. There exist positive constants ε and δ such that, whenever $E\subset\mathbb R^2$ is compact and

conjecture | open | MathDB 页面 | 原文线索

V2-037160 | MATHDB-362927

Twin-volume conjecture for constant mean curvature hypersurfaces

摘录 / EXCERPT: Let (M^{n+1},g) be a closed Riemannian manifold with $3\leq n+1\leq 7$, and let $v\in(0,\operatorname{Vol}(M))$. Twin-volume conjecture. The manifold M admits at least two dist...

conjecture | open | MathDB 页面 | 原文线索

V2-037161 | MATHDB-362963

Banach's affine section conjecture for convex bodies

摘录 / EXCERPT: Let V be a finite-dimensional real vector space, let $2\leq k<\dim V$, and let $K\subset V$ be a convex body. Suppose that all k-dimensional sections of K through a fixed int...

conjecture | open | MathDB 页面 | 原文线索

V2-037162 | MATHDB-362964

Pointwise convergence and maximal cone multiplier conjecture

摘录 / EXCERPT: Pointwise convergence and maximal cone multiplier conjecture. The range of p for which

conjecture | open | MathDB 页面 | 原文线索

V2-037163 | MATHDB-362967

The Germain-Myerson spectral projection conjecture for tori

摘录 / EXCERPT: The Germain-Myerson conjecture. If $\delta>\lambda^{-1+\kappa}$ for some fixed $\kappa\in(0,1]$, then there is a constant $C=C(n,p,\kappa,L^*)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037164 | MATHDB-362970

The rigid-transformation packing conjecture

摘录 / EXCERPT: Rigid-transformation packing conjecture. Under this dimension hypothesis, the set obtained by applying the transformations in Θ to E has positive d-dimensional Lebesgue...

conjecture | open | MathDB 页面 | 原文线索

V2-037165 | MATHDB-363005

Algebraic criterion for stability of a viscous hyperbolic operator

摘录 / EXCERPT: Algebraic stability conjecture. The operator $\Omega(k)$ has negative eigenvalues if and only if $\lambda_{\ell,1}<0$ for every $\ell=1,\ldots,N$.

conjecture | open | MathDB 页面 | 原文线索

V2-037166 | MATHDB-363021

Kaad's pure-state separation conjecture for Hilbert C^*-modules

摘录 / EXCERPT: Let $\mathscr A$ be a C^*-algebra, let $\mathscr E$ be a Hilbert C^*-module over $\mathscr A$, and let $\mathscr L$ be a closed $\mathscr A$-submodule of $\mathscr E$. For a st...

conjecture | open | MathDB 页面 | 原文线索

V2-037167 | MATHDB-363077

Widom's conjecture for discontinuous pseudodifferential symbols and spectral projectors

摘录 / EXCERPT: Assume the conditions denoted by Assumptions. Let $\Pi_{\hbar,\mu}$ be the spectral projector defined in, and let ... Let $g:[0,1]\to\mathbb{R}$ be continuous with $g(0)=g(1)=0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-037168 | MATHDB-363078

Equivariance implies admissibility for groupoid Busby homomorphisms

摘录 / EXCERPT: Equivariance-implies-admissibility conjecture. If μ is G-equivariant, then μ is G-admissible.

conjecture | open | MathDB 页面 | 原文线索

V2-037169 | MATHDB-363086

Asymptotic conjecture for the balanced model metric function

摘录 / EXCERPT: Asymptotic conjecture. There exists a constant C_β such that

conjecture | open | MathDB 页面 | 原文线索

V2-037170 | MATHDB-363092

Structural instability of critical quasicircle maps

摘录 / EXCERPT: Let $f:\mathcal{H}_q\to\mathcal{H}_q$ be a critical quasicircle map with irrational rotation number, and let $\mathcal{N}$ be the Banach neighborhood of f consisting of nearby un...

conjecture | open | MathDB 页面 | 原文线索

V2-037171 | MATHDB-363121

Paszkiewicz's conjecture on products of decreasing positive contractions

摘录 / EXCERPT: Paszkiewicz's conjecture. The sequence (S_n) converges strongly.

conjecture | open | MathDB 页面 | 原文线索

V2-037172 | MATHDB-363341

Maximal-operator duality for Musielak-Orlicz spaces

摘录 / EXCERPT: Let ϕ be a generalized Φ-function, let ϕ^* be its complementary generalized Φ-function, and let M denote the Hardy-Littlewood maximal operator. Assume that bo...

conjecture | open | MathDB 页面 | 原文线索

V2-037173 | MATHDB-363342

Weighted Musielak-Orlicz maximal-operator boundedness conjecture

摘录 / EXCERPT: Let ϕ be a generalized Φ-function, let ϕ^* be its complementary function, and define ... Assume that ϕ^* satisfies the Δ_2 condition and that ϕ satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-037174 | MATHDB-363343

Maximal-operator duality for weighted Morrey and block spaces

摘录 / EXCERPT: Morrey-block duality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037175 | MATHDB-363344

Global strong Muckenhoupt characterization for weighted Morrey spaces

摘录 / EXCERPT: Let $1<p\leq q<\infty$, let w be a weight, and let M denote the Hardy-Littlewood maximal operator. Let T range over all Calderón-Zygmund operators and let R_j,...

conjecture | open | MathDB 页面 | 原文线索

V2-037176 | MATHDB-363348

Kernel inclusion for the WKB-deformed central charge

摘录 / EXCERPT: Kernel inclusion conjecture. There is an inclusion of kernels

conjecture | open | MathDB 页面 | 原文线索

V2-037177 | MATHDB-363395

Analytic extension conjecture for semigroups generated by nonlinear resolvents

摘录 / EXCERPT: Let G_r be the nonlinear resolvent associated with $r>0$, and let q and γ_r be the parameters defined in the paper. The semigroup generated by G_r is considered with r...

conjecture | open | MathDB 页面 | 原文线索

V2-037178 | MATHDB-363452

The COMB strategy's global optimality range

摘录 / EXCERPT: COMB optimality conjecture. The COMB strategy is globally asymptotically optimal on $\mathbb{S}_n$ only for $n\leq 4$ experts. The paper reports strong numerical evidence against g...

conjecture | open | MathDB 页面 | 原文线索

V2-037179 | MATHDB-363453

The unique globally optimal non-COMB strategy for five experts

摘录 / EXCERPT: Non-COMB uniqueness conjecture. The non-COMB strategy $(1,0,1,0,0)$ is the only globally asymptotically optimal strategy for $n=5$ experts. The claim is supported by strong numeric...

conjecture | open | MathDB 页面 | 原文线索

V2-037180 | MATHDB-363454

Nonexistence of constant globally optimal adversary strategies for at least six experts

摘录 / EXCERPT: Constant-strategy nonexistence conjecture. For $n\geq 6$ experts, there is no globally asymptotically optimal adversary strategy that is constant on $\mathbb{S}_n$. The paper repor...

conjecture | open | MathDB 页面 | 原文线索

V2-037181 | MATHDB-363499

Quadratic curvature decay is unnecessary for uniqueness on average in surfaces

摘录 / EXCERPT: Let M be the surface setting of the main theorem, so $n=2$, and let the theorem's other hypotheses and notation remain in force. Surface-case conjecture. Theorem 1.1 holds withou...

open_problem | open | MathDB 页面 | 原文线索

V2-037182 | MATHDB-363537

The exact dimension formula for line sets of trapped sets

摘录 / EXCERPT: Exact dimension formula. One has

conjecture | open | MathDB 页面 | 原文线索

V2-037183 | MATHDB-363552

The weighted Muckenhoupt conjecture for the Carleson-Dunkl operator

摘录 / EXCERPT: Let $\alpha\geq -1/2$, $p\in(1,\infty)$, and let $w\colon\mathbb{R}\to[0,\infty)$ satisfy ... where the supremum is over all intervals $B\subset\mathbb{R}$ and $1/p+1/p'=1$. The weig...

conjecture | open | MathDB 页面 | 原文线索

V2-037184 | MATHDB-363635

Rearrangement conjecture for characteristic functions and quadratic spectral concentration

摘录 / EXCERPT: Characteristic-function rearrangement conjecture. For every measurable A with $|A|<\infty$ and every $W>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-037185 | MATHDB-363636

Alternating-exponential integral conjecture

摘录 / EXCERPT: Let n be a positive integer and let $\alpha_1,\ldots,\alpha_{2n}$ be real numbers satisfying ... Consider the alternating exponential sum $\sum_{j=1}^{2n}(-1)^j e^{i\alpha_jt}$. Al...

conjecture | open | MathDB 页面 | 原文线索

V2-037186 | MATHDB-363669

Breuil-Schneider conjecture for condensed generalized z-adic Banach representations

摘录 / EXCERPT: Let X be a finite extension of a local nonarchimedean field, let $G(X)$ be a z-adic reductive group, and let $\mathrm{Gal}_{X,2}$ denote the two-fold covering of the Galois gro...

conjecture | open | MathDB 页面 | 原文线索

V2-037187 | MATHDB-363687

Tightness conjecture for gradient descent with variable mid-range stepsizes

摘录 / EXCERPT: Let $f\in\mathcal{F}_{\mu,L}$ with $\mu\in(-\infty,0)$, and consider N gradient-descent iterations from x_0 with stepsizes satisfying $\gamma_iL\in[1,\gbari[1]]$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-037188 | MATHDB-363748

Exponential concentration implies a finite Poincaré constant

摘录 / EXCERPT: Let $g(t)=\exp(t-e^t)$, and let $\gamma(x,\xi)=\exp(-a|x|-b|\xi|)$ with $a,b>0$. Suppose that $f\in L^2(\mathbb{R})$ has exponential time-frequency concentration, meaning ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037189 | MATHDB-363771

Colding-Minicozzi entropy lower-bound conjecture for closed self-shrinkers

摘录 / EXCERPT: Let $\Sigma\subset \mathbb R^N$ be a closed n-dimensional self-shrinker. The Colding-Minicozzi entropy should satisfy ... This conjecture concerns the entropy-minimizing closed sel...

conjecture | open | MathDB 页面 | 原文线索

V2-037190 | MATHDB-363773

Iwaniec-Martin removability conjecture for bounded quasiregular mappings

摘录 / EXCERPT: Iwaniec-Martin removability conjecture. Sets of Hausdorff d-measure zero, where

conjecture | open | MathDB 页面 | 原文线索

V2-037191 | MATHDB-363797

Bellettini's compactness conjecture for branched stable minimal immersions

摘录 / EXCERPT: Consider the class of branched two-sided stable minimal n-dimensional immersions whose singular sets have locally finite $(n-2)$-measure. Bellettini's compactness conjecture. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-037192 | MATHDB-363843

Weak-type spherical Carleson conjecture

摘录 / EXCERPT: Spherical Carleson conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-037193 | MATHDB-363860

Centered maximal variation conjecture with endpoint improvement

摘录 / EXCERPT: Let M be the centered Hardy--Littlewood maximal operator on fR, and let $f\in\operatorname{BV}(fR)$ have limits $f(\infty)$ and $f(-\infty)$ at the two ends. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-037194 | MATHDB-363955

Initial value problem of Oshida-Rule

摘录 / EXCERPT: Let m be a positive integer, let e denote the distinguished initial vector, and let h, h', f_1, and $H_{245}=L\circ S$ be the maps defined above, with...

conjecture | open | MathDB 页面 | 原文线索

V2-037195 | MATHDB-363970

Strengthened minimax optimal constant-stepsize conjecture for gradient descent

摘录 / EXCERPT: Let $\mathcal{F}_{L,D}$ be the class of pairs (f,x_0) where f is L-smooth and convex, has a minimizer $x_\star$ with $\|x_0-x_\star\|\leq D$, and x_N is obtained by N ste...

conjecture | open | MathDB 页面 | 原文线索

V2-037196 | MATHDB-364046

Optimal variance conjecture for Gaussian regularized Shannon sampling

摘录 / EXCERPT: Optimal variance conjecture. The parameter specified by ... with the Gaussian function ... .

conjecture | open | MathDB 页面 | 原文线索

V2-037197 | MATHDB-364076

Order-n Turán-type oscillation conjecture for convex domains

摘录 / EXCERPT: Let $K\Subset\mathbb C$ be a compact convex domain, let $0<q<\infty$, and let $\mathcal P_n(K)$ denote the polynomials of degree at most n whose zeros lie in K. For a polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-037198 | MATHDB-364077

Uniform geometric Turán-type oscillation conjecture for convex domains

摘录 / EXCERPT: Let $K\Subset\mathbb C$ be a compact convex domain with width w and diameter d, and let $\mathcal P_n(K)$ denote the polynomials of degree at most n whose zeros lie in K. F...

conjecture | open | MathDB 页面 | 原文线索

V2-037199 | MATHDB-364106

Characterization of real Hilbert spaces by smoothness of norm-one operators

摘录 / EXCERPT: Real Hilbert-space characterization conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-037200 | MATHDB-364163

Tao's maximal Bochner-Riesz conjecture

摘录 / EXCERPT: Tao's conjecture. For any $1\leq p<2$, the operator $T^\lambda_\ast$ extends boundedly on $L^p(\mathbb R^n)$ whenever

conjecture | open | MathDB 页面 | 原文线索

V2-037201 | MATHDB-364164

The planar $L^{3/2}$ maximal-operator conjecture

摘录 / EXCERPT: The planar $L^{3/2}$ conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-037202 | MATHDB-364171

Internal characterization of non-horocyclically convex domains in the unit disk

摘录 / EXCERPT: Let $\mathbb{D}$ denote the unit disk, and let G be a simply connected subdomain of $\mathbb{D}$. Let $\mathcal{C}$ be the class of domains $\Omega\subset\mathbb{D}$ of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-037203 | MATHDB-364197

The optimality conjecture for the $2 \times n$ bad science matrix

摘录 / EXCERPT: Optimality conjecture. An optimal $2 \times n$ matrix is given by

conjecture | open | MathDB 页面 | 原文线索

V2-037204 | MATHDB-364308

Monotonicity conjecture for Gaussian and inverse multiquadric RBF condition numbers

摘录 / EXCERPT: Condition-number monotonicity conjecture. For the Gaussian or inverse multiquadric RBF,

conjecture | open | MathDB 页面 | 原文线索

V2-037205 | MATHDB-364369

The coarse Novikov conjecture for discrete metric spaces

摘录 / EXCERPT: Coarse Novikov conjecture. The induced map

conjecture | open | MathDB 页面 | 原文线索

V2-037206 | MATHDB-364370

The Strong Novikov conjecture for finitely presented groups

摘录 / EXCERPT: Strong Novikov conjecture. The induced map

conjecture | open | MathDB 页面 | 原文线索

V2-037207 | MATHDB-364573

The Abrams-Tomforde conjecture for Leavitt path algebras and graph C^*-algebras

摘录 / EXCERPT: Let E and F be graphs and let k be a field. Consider the Leavitt path algebras $L_\mathsf{k}(E)$ and $L_\mathsf{k}(F)$, and the graph C^*-algebras $C^*(E)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037208 | MATHDB-364686

Typical dense sets are strongly non-Rado in strictly convex Banach spaces with separable dual

摘录 / EXCERPT: Strong non-Rado conjecture. Typical countable dense sets in X are strongly non-Rado.

conjecture | open | MathDB 页面 | 原文线索

V2-037209 | MATHDB-364687

Extension conjecture for step-isometries on dense subsets of Banach spaces

摘录 / EXCERPT: Step-isometry extension conjecture. The map T can be extended to a step-isometry on the whole of X.

conjecture | open | MathDB 页面 | 原文线索

V2-037210 | MATHDB-364702

Nonconvergent-scaling conjecture for approximate multi-solitons

摘录 / EXCERPT: Fix distinct soliton speeds, and let f_{start} denote the starting error whose leading behavior at one soliton is t^{-1}. Let W_a be ground-state solitons, let...

conjecture | open | MathDB 页面 | 原文线索

V2-037211 | MATHDB-364708

Rough local smoothing conjecture for spherical averages

摘录 / EXCERPT: Let $E\subset [1,2]$ satisfy $\dim_M E=\dim_A E=\beta$. For each scale j, let $\mathcal{Z}_j(E)$ index a minimal collection of δ-separated points t_ν whose intervals...

conjecture | open | MathDB 页面 | 原文线索

V2-037212 | MATHDB-364751

Optimal regularity conjecture for K-finite matrix coefficients on compact symmetric spaces

摘录 / EXCERPT: Optimal regularity conjecture. Any K-finite matrix coefficient of a unitary representation of U is in $C^{(r,\delta)}(U_r)$, and this regularity is optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-037213 | MATHDB-364784

Weak discretization error conjecture for the particle system

摘录 / EXCERPT: Let $g:\mathbb{R}\to\mathbb{R}$ be smooth with bounded derivatives, let $G^P(x)=\frac{1}{P}\sum_{j=1}^P g(x_j)$, and let $\bm X_N^P$ be a realization of the particle system with co...

conjecture | open | MathDB 页面 | 原文线索

V2-037214 | MATHDB-364796

Saitoh's conjecture for multiply connected planar regions

摘录 / EXCERPT: Let D be a planar regular region with finitely many boundary components that are analytic Jordan curves. Let $c_{\beta}(z)$ denote the logarithmic capacity and let $B(z)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-037215 | MATHDB-364852

Higher-dimensional projection dimension conjecture

摘录 / EXCERPT: Let $E\subset\mathbb{R}^d$ and let $\mathcal{V}\subset G(d,d-1)$ be Borel sets such that $\dim_{\mathcal{H}}\mathcal{V}>0$ and $\mathcal{V}^{\perp}$ is not contained in a great cir...

conjecture | open | MathDB 页面 | 原文线索

V2-037216 | MATHDB-364942

Twisted coarse Baum-Connes conjecture with coefficients

摘录 / EXCERPT: Let X be a metric space with bounded geometry, let $P_d(X)$ be its Rips complex at scale d, and let $\Gamma(X,\mathcal A)$ be a coarse X-algebra. The twisted assembly map is...

conjecture | open | MathDB 页面 | 原文线索

V2-037217 | MATHDB-364943

Twisted coarse Novikov conjecture with coefficients

摘录 / EXCERPT: Let X be a metric space with bounded geometry and let $\Gamma(X,\mathcal A)$ be a coarse X-algebra. Denote by ... the twisted assembly map associated with X and $\mathcal A$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-037218 | MATHDB-365008

The Brazilian Dream Problem for uniqueness of quasiconformal extensions

摘录 / EXCERPT: The Brazilian Dream Problem. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037219 | MATHDB-365317

The weighted Laurent-polynomial Mathieu conjecture

摘录 / EXCERPT: Weighted Laurent-polynomial Mathieu conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037220 | MATHDB-365328

The quantitative coarse Baum-Connes conjecture with coefficients

摘录 / EXCERPT: Quantitative coarse Baum-Connes conjecture with coefficients. For every $k\in\mathbb{N}$, there exists $0<\varepsilon<1/4$ such that, for every $r>0$, there exists $k'\geq k$ for w...

conjecture | open | MathDB 页面 | 原文线索

V2-037221 | MATHDB-365330

The uniformly quantitative coarse Baum-Connes conjecture with coefficients

摘录 / EXCERPT: Uniformly quantitative coarse Baum-Connes conjecture with coefficients. For every $k\in\mathbb{N}$, there exists $0<\varepsilon<1/4$ such that, for every $r>0$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-037222 | MATHDB-365393

Isodiametric conjecture for Riesz capacity

摘录 / EXCERPT: Let $n\geq 2$, let $q(n)\leq n-2$, and let $K\subset\mathbb R^n$ be compact. Write $\operatorname{diam}(K)$ for its diameter and $\operatorname{Cap}_q(K)$ for its Riesz capacity. I...

conjecture | open | MathDB 页面 | 原文线索

V2-037223 | MATHDB-365405

Proposed norm inequalities for extending the radius estimate to $1<p\leq2$

摘录 / EXCERPT: Let $1<p<2$, set $r=2p/(2-p)$, and let $i_{p2}\colon\ell_p\to\ell_2$ be the formal identity map. Let $\pi_{r,2}(\cdot)$ denote the $(r,2)$-absolutely summing norm. For a linear map...

conjecture | open | MathDB 页面 | 原文线索

V2-037224 | MATHDB-365414

The conjecture on ancient Ricci flows from orbifold cylinders

摘录 / EXCERPT: Orbifold-cylinder conjecture. There exist ancient Ricci flows from orbifold cylinders.

conjecture | open | MathDB 页面 | 原文线索

V2-037225 | MATHDB-365416

The uniqueness conjecture for ancient Ricci flows constructed in Theorem mainthm1

摘录 / EXCERPT: Uniqueness conjecture. Up to time reparametrization and diffeomorphism, the ancient solutions to Ricci flow from Theorem $$ are unique among smooth ancient solutions to Ricci flow...

conjecture | open | MathDB 页面 | 原文线索

V2-037226 | MATHDB-365417

The higher-dimensional conjecture on Ricci flows with Calabi bubbles

摘录 / EXCERPT: Higher-dimensional Calabi-bubble conjecture. In any dimension $2n\geq 6$, there exist:

conjecture | open | MathDB 页面 | 原文线索

V2-037227 | MATHDB-365419

The conjecture on obstructions to desingularizing Ricci solitons

摘录 / EXCERPT: Ricci-soliton obstruction conjecture. Such (M_o,g_o) cannot be the limit of smooth Ricci solitons in the pointed Gromov--Hausdorff sense.

conjecture | open | MathDB 页面 | 原文线索

V2-037228 | MATHDB-365454

Nonnegativity conjecture for the truncated Bessel kernel

摘录 / EXCERPT: Nonnegativity conjecture. For all $0\leqslant\alpha\leqslant\frac{1}{2}$, $M\in\mathbb{N}$, and $x\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-037229 | MATHDB-365481

Carleson's ε^2-conjecture for tangent points of Jordan domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a Jordan domain, and let $\partial\Omega$ denote its boundary. For $x\in\partial\Omega$, let $\varepsilon(x,r)$ be the Carleson ε-fun...

conjecture | open | MathDB 页面 | 原文线索

V2-037230 | MATHDB-365496

Equality of the union and indexed annihilator families

摘录 / EXCERPT: Let V be a Banach algebra, and let X be a Banach space. Let $\varphi\!:V\to X$ be a bounded linear map. For $z\in V$, write ${\mathfrak{C}}_{-1}(z)$ for the set of correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-037231 | MATHDB-365497

Local characterization of the cap family by cup families

摘录 / EXCERPT: Let V be a Banach algebra, let Y be a Banach space, and let $\varphi\!:V\to Y$ be a bounded linear map. For $z\in V$, let ${\mathfrak{S}}_\cap^\varphi(z)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037232 | MATHDB-365564

Zhang's unitary-orbit n-tuple Schatten inequality conjecture

摘录 / EXCERPT: Let $A_1,\ldots,A_n\in\mathbb{M}_m$, and let $|A|$ denote the positive operator $(A^*A)^{1/2}$. Zhang's conjecture. For $p>2$, there exist $\frac{n(n-1)}{2}+1$ unitary matrices U...

conjecture | open | MathDB 页面 | 原文线索

V2-037233 | MATHDB-365565

Noncommutative Clarkson inequality for Schatten norms

摘录 / EXCERPT: Let $\mathscr{H}$ be a Hilbert space and let $\mathbb{B}_p(\mathscr{H})$ denote the Schatten p-class on $\mathscr{H}$. For $A,B\in\mathbb{B}_p(\mathscr{H})$, the noncommutative C...

conjecture | open | MathDB 页面 | 原文线索

V2-037234 | MATHDB-365566

Strengthened Fourier-type n-operator Clarkson inequality conjecture

摘录 / EXCERPT: Let $\mathscr{H}$ be a Hilbert space, let $\mathbb{B}_p(\mathscr{H})$ denote the Schatten p-class, and define $\omega_j=e^{ji\pi/n}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037235 | MATHDB-365585

Rational approximation conjecture for ×(2,3)-circular Carathéodory functions

摘录 / EXCERPT: Let $\mathbb{D}$ be the unit disk, and let $\operatorname{Cara}(\mathbb{D};2,3)$ be the set of Carathéodory functions that are ×2- and ×3-circular, where ×n-circular means...

conjecture | open | MathDB 页面 | 原文线索

V2-037236 | MATHDB-365586

Periodic equidistribution conjecture for rational extreme Carathéodory functions

摘录 / EXCERPT: Let $\mathbb{D}$ be the unit disk, and let $\operatorname{Cara}(\mathbb{D};2,3)$ be the set of ×2- and ×3-circular Carathéodory functions; call an element rational extreme when it...

conjecture | open | MathDB 页面 | 原文线索

V2-037237 | MATHDB-365587

Conditional periodic equidistribution conjecture for tracial states

摘录 / EXCERPT: Let $C^*(\mathbb{Z}[1/6])$ be regarded as a subalgebra of $C^*(\mathfrak{F}_{2,3})$, let $\operatorname{TSt}(C^*(\mathfrak{F}_{2,3}))$ denote the tracial states, call a state condi...

conjecture | open | MathDB 页面 | 原文线索

V2-037238 | MATHDB-365593

Uniqueness conjecture for minimal hypersurfaces near quadratic cones

摘录 / EXCERPT: Let $\mathcal C_1$ be a quadratic cone in B_1 and let Σ be its link. For $g\in C^{2,\alpha}(\Sigma)$ with $|g|$ sufficiently small, write...

conjecture | open | MathDB 页面 | 原文线索

V2-037239 | MATHDB-365601

The real Hilbert space factorization property conjecture

摘录 / EXCERPT: Let X and Y be real Banach spaces. The pair (X,Y) has the Hilbert space factorization property (HFP) when every bounded operator $\phi:X\to Y$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-037240 | MATHDB-365602

The planar convex-body factorization conjecture

摘录 / EXCERPT: Planar ellipse conjecture. Then K or L is an ellipse.

conjecture | open | MathDB 页面 | 原文线索

V2-037241 | MATHDB-365626

Positive boundedness conjecture for case iv singular Brascamp-Lieb data

摘录 / EXCERPT: Positive boundedness conjecture. Every such datum $\mathbf{H}_\mathbf{M}$ is $\mathbf{p}$-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-037242 | MATHDB-365627

Rectified Baum-Connes conjecture for transformation groupoids

摘录 / EXCERPT: Let Γ be a finitely generated discrete group acting on a metrizable finite-dimensional compact space X, and let ${\mathcal G}=X\rtimes\Gamma$. Write $RK_i(\Gamma,X)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-037243 | MATHDB-365629

Constantin-Strauss-Varvaruca conjecture on the periodic Hilbert transform kernel

摘录 / EXCERPT: Let $d>0$ and let β_d be the kernel of the periodic Hilbert transform on the strip of depth d, defined for $s\in\mathbb{R}\setminus 2\pi\mathbb{Z}$ by the kernel formulas i...

conjecture | open | MathDB 页面 | 原文线索

V2-037244 | MATHDB-365649

Variable-coefficient Mizohata-Takeuchi conjecture

摘录 / EXCERPT: Let $\mathcal T^\lambda$ be the operator defined earlier in the paper, let B_λ be its spatial domain, and let $\{\Gamma_{\theta,v}^\lambda\}_{\theta,v}$ be the associated c...

conjecture | open | MathDB 页面 | 原文线索

V2-037245 | MATHDB-365675

Optimal reproducing-filter norm conjecture for shift-invariant subspaces

摘录 / EXCERPT: Optimal reproducing-filter norm conjecture. Every s-dimensional SIS admits a reproducing filter $\varphi\in\mathbb{C}_n(\mathbb{Z})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037246 | MATHDB-365703

The l^p restriction conjecture for uniformly bounded orthonormal systems

摘录 / EXCERPT: Let $2\le q<p<\infty$, and let Φ be a finite, uniformly bounded orthonormal system. For a finite collection Θ of functions, define $K_p^*(\Theta)$ to be the smallest co...

conjecture | open | MathDB 页面 | 原文线索

V2-037247 | MATHDB-365704

The discretized three-dimensional Kakeya conjecture

摘录 / EXCERPT: Let $\varepsilon>0$. A family L of lines is directionally δ-separated when its line directions are separated at scale δ, and let $E\subset[0,1]^3$ be a union of...

conjecture | open | MathDB 页面 | 原文线索

V2-037248 | MATHDB-365747

Schmeisser's weighted generalization of Sendov's conjecture

摘录 / EXCERPT: Schmeisser's conjecture. For every such choice of weights,

conjecture | open | MathDB 页面 | 原文线索

V2-037249 | MATHDB-365748

Pawlowski's asymptotic conjecture for the critical-point radius

摘录 / EXCERPT: Let F be a complex polynomial of degree $n\geq 2$ with zeros $z_1,\dots,z_n$ satisfying $|z_k|\leq 1$, and critical points $w_1,\dots,w_{n-1}$. Let γ_n be the radius of th...

conjecture | open | MathDB 页面 | 原文线索

V2-037250 | MATHDB-365749

Schoenberg's inequality for polynomial zeros and critical points

摘录 / EXCERPT: Let F be a complex polynomial of degree $n\geq 2$ with zeros $z_1,\dots,z_n$ and critical points $w_1,\dots,w_{n-1}$. Schoenberg's conjecture. ... with equality if and only if all...

conjecture | open | MathDB 页面 | 原文线索

V2-037251 | MATHDB-365750

Generalized Borcea variance conjecture for convex combinations of zeros

摘录 / EXCERPT: Generalized Borcea variance conjecture. For every $p\geq 1$ and every such choice of weights,

conjecture | open | MathDB 页面 | 原文线索

V2-037252 | MATHDB-365800

Local Fourier restriction conjecture

摘录 / EXCERPT: Let $S\subset\mathbb{R}^n$ be a compact C^2 hypersurface, possibly with boundary, with strictly positive second fundamental form. Let E_S be its Fourier extension operator. Loc...

conjecture | open | MathDB 页面 | 原文线索

V2-037253 | MATHDB-365801

Local restriction conjecture for the normalized extension operator

摘录 / EXCERPT: Normalized local restriction conjecture. If $p\geq\frac{2n}{n-1}$, then for every $\varepsilon>0$ and $R>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037254 | MATHDB-365806

The weak-type constants conjecture for Λ_m and Λ_m^*

摘录 / EXCERPT: For $m\in\mathbb{N}$, let Λ_m and Λ_m^* be the operators considered in the paper, and let $\mathcal{E}_m$, $\mathcal{F}_m$, $\mathcal{E}_m^*$, and $\mathcal{F}_m^*$...

conjecture | open | MathDB 页面 | 原文线索

V2-037255 | MATHDB-365856

Fredholm conjecture for the three-body Helmholtz operator on anisotropic Hilbert spaces

摘录 / EXCERPT: Let $P=\Delta+V-\lambda^2$ be the Helmholtz operator on $\mathbb{R}^n$, where $\lambda^2>0$ and $V=\sum_{\alpha\in I}\pi_\alpha^*(V^\alpha)$ is a three-body potential with...

conjecture | open | MathDB 页面 | 原文线索

V2-037256 | MATHDB-365931

Reality conjecture for the limiting spectrum of real block Toeplitz matrices

摘录 / EXCERPT: Let $B(z)$ be a real $k\times k$ matrix-valued symbol. Define ... where $f(z,\lambda)$ is the characteristic function of B, and let $\Lambda_0(B)$ denote the set of limiting spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-037257 | MATHDB-365956

Polynomial convexity conjecture for the domain ${\mathcal C}$

摘录 / EXCERPT: Polynomial convexity conjecture. The domain ${\mathcal C}$ is polynomially convex.

conjecture | open | MathDB 页面 | 原文线索

V2-037258 | MATHDB-365957

Certifying determinantal representation conjecture for polynomials on ${\mathcal T}_{27}$

摘录 / EXCERPT: Certifying determinantal representation conjecture. There exists a polynomial $q(w_1,\ldots,w_{27})$ such that pq has a certifying determinantal representation: there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-037259 | MATHDB-365970

The smooth Alpert trilinear Fourier extension conjecture for the paraboloid

摘录 / EXCERPT: Let $0<\delta<1$ and $\kappa\in\mathbb{N}$ with $\kappa>\frac{20}{\delta}$. For $1<q<\infty$, $0<\varepsilon,\nu<1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-037260 | MATHDB-365971

The strong Kakeya maximal operator conjecture in three dimensions

摘录 / EXCERPT: Let $0<\varepsilon<1$, and let $\mathcal{K}^{\ast}(\otimes_1L^{\infty}\rightarrow L^{\frac{3}{2}};\varepsilon)$ be the statement that for every $0<\delta<1$ and every family...

conjecture | open | MathDB 页面 | 原文线索

V2-037261 | MATHDB-366091

Non-integrability of one-point functions in p,q-adic analysis

摘录 / EXCERPT: Let p and q be primes, let $\mathbb{Z}_p$ be the ring of p-adic integers, and let $\mathbf{1}_{\mathfrak a}$ be the one-point function at $\mathfrak a\in\mathbb{Z}_p$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-037262 | MATHDB-366092

Quasi-integrability of the multidimensional numen in the noncommutative case

摘录 / EXCERPT: Let H be as in the source's theorem giving the multidimensional $\mathcal{F}$-series for χ_H, and suppose that $\mathbf I_d-\alpha_H(\mathbf 0)$ is invertible. Let ψ_H...

conjecture | open | MathDB 页面 | 原文线索

V2-037263 | MATHDB-366103

Spherical cap conjecture for compact CMC surfaces bounded by a circle

摘录 / EXCERPT: Let Σ be a compact surface in $\mathbb{R}^3$ with non-empty circular boundary and nonzero constant mean curvature. Spherical cap conjecture. If either (i) Σ has genus...

conjecture | open | MathDB 页面 | 原文线索

V2-037264 | MATHDB-366207

Suita's conjecture on the Bergman kernel and logarithmic capacity

摘录 / EXCERPT: Let Ω be an open Riemann surface admitting a nontrivial Green function G_{Ω}. For $z_0\in\Omega$, let $c_{\beta}(z_0)$ denote its logarithmic capacity and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037265 | MATHDB-366242

Li-Solovej convex integral inequality for holomorphic functions on the complex ball

摘录 / EXCERPT: Let ${\mathbb B}_n$ be the unit ball in $\mathbb C^n$, with Lebesgue measure $dv(z)$, and define the hyperbolic measure by ... For $0<p<\infty$ and $\alpha>n$, let A^p_α be th...

conjecture | open | MathDB 页面 | 原文线索

V2-037266 | MATHDB-366337

Uniqueness and sharp existence bound for invariant CMC tubes

摘录 / EXCERPT: Let c be a screw motion geodesic in a homogeneous space with parameters $c\kappa$ and $c\tau$, and let a tube invariant by translations along c have constant mean curvature H...

conjecture | open | MathDB 页面 | 原文线索

V2-037267 | MATHDB-366347

Hypoellipticity and invertibility of the Vladimirov sub-Laplacian

摘录 / EXCERPT: Vladimirov sub-Laplacian conjecture. The Vladimirov sub-Laplacian of order $\alpha>0$ is a hypoelliptic operator on G and is invertible on the space of mean-zero functions.

conjecture | open | MathDB 页面 | 原文线索

V2-037268 | MATHDB-366368

Classification conjecture for four-dimensional normed spaces of type IV

摘录 / EXCERPT: Let $K^{\lor}$ be the spherical completion of the underlying valued field K, let V_K denote the relevant multiplicative value subgroup, and let $\sim$ denote the equivalence re...

conjecture | open | MathDB 页面 | 原文线索

V2-037269 | MATHDB-366389

Pinned dot product measure conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}^n$ be a Borel set, and for $a,x\in\mathbb{R}^n$ define the pinned dot product set ... In particular, $\Pi_0^a(A)$ is the dot product set pinned at a. Pinned...

conjecture | open | MathDB 页面 | 原文线索

V2-037270 | MATHDB-366409

Wulff inequality for minimal submanifolds

摘录 / EXCERPT: Wulff inequality for minimal submanifolds. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-037271 | MATHDB-366483

Stein-Iosevich-Sawyer maximal estimate conjecture from Fourier decay

摘录 / EXCERPT: Let S be a smooth hypersurface in ${\mathbb R}^d$ given by the graph of a smooth function, let σ_1 be the associated measure, and define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-037272 | MATHDB-366484

The discrete distributed spectral pair classification conjecture

摘录 / EXCERPT: Let $({\mathcal A},{\mathcal B})$ be a distributed spectral pair in $\{0,1,\ldots,M-1\}$. A distributed spectral pair classification conjecture. The pair...

conjecture | open | MathDB 页面 | 原文线索

V2-037273 | MATHDB-366520

Characterization of Hilbert spaces by minimal K-sets

摘录 / EXCERPT: Hilbert-space characterization conjecture. The following conditions are equivalent: $\mathbb{X}$ is a Hilbert space; and a nonempty set $A\subset S_{\mathbb{X}}$ is a minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-037274 | MATHDB-366589

The finiteness conjecture for SU(3)-invariant complete AC shrinkers

摘录 / EXCERPT: Consider complete AC shrinkers with $\textup{SU}(3)$ symmetry on the relevant cohomogeneity-one $\textup{SU}(3)$-invariant geometry. SU(3)-invariant shrinker-finiteness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-037275 | MATHDB-366590

The SU(3)-invariant quadratic-exponential expander-family conjecture

摘录 / EXCERPT: Let $\Lambda^2_-\mathbb{CP}^2$ denote the anti-self-dual bundle of the complex projective plane. A complete expander has quadratic-exponential volume growth when its volume grows a...

conjecture | open | MathDB 页面 | 原文线索

V2-037276 | MATHDB-366591

The explicit Sp(2)-invariant AC shrinker uniqueness conjecture

摘录 / EXCERPT: Let $\Lambda^2_-\mathbb{S}^4$ be the anti-self-dual bundle of the 4-sphere, and let the explicit AC shrinker be the complete $\textup{Sp}(2)$-invariant shrinker described in Exampl...

conjecture | open | MathDB 页面 | 原文线索

V2-037277 | MATHDB-366592

The finite SU(3)-invariant AC shrinker conjecture for fixed negative parameter

摘录 / EXCERPT: Fix $\lambda<0$ and consider the 4-dimensional space of local $\textup{SU}(3)$-invariant shrinkers, together with the complete AC shrinkers among them on $\Lambda^2_-\mathbb{CP}^2$...

conjecture | open | MathDB 页面 | 原文线索

V2-037278 | MATHDB-366595

Ilmanen's smooth-blowup conjecture for embedded hypersurfaces

摘录 / EXCERPT: Let an embedded hypersurface evolve by mean curvature flow in $\mathbb{R}^{n+1}$, with $n \leq 6$, and consider blow-ups at singularities. Ilmanen's smooth-blowup conjecture. The b...

conjecture | open | MathDB 页面 | 原文线索

V2-037279 | MATHDB-366614

Zograf's large-genus asymptotic conjecture for Weil-Petersson volumes

摘录 / EXCERPT: Zograf's conjecture. For any fixed $n\geq 0$, as $g\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-037280 | MATHDB-366615

Kimura's large-genus boundary-volume asymptotic conjecture

摘录 / EXCERPT: Kimura's conjecture. For any given $n\geq 0$ and boundary lengths $\mathbf{L}=(L_1,\ldots,L_n)$, as $g\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-037281 | MATHDB-366629

Brue-Pigati-Semola local biHölder regularity conjecture for RCD spaces

摘录 / EXCERPT: Brue-Pigati-Semola conjecture. (X,d) is locally biHölder homeomorphic to a smooth, complete Riemannian manifold (M^3,g).

conjecture | open | MathDB 页面 | 原文线索

V2-037282 | MATHDB-366650

Self-similar halo modulation problem for weakly localized solutions

摘录 / EXCERPT: Consider weakly localized solutions and their stability analysis by the method of modulation equations. Let the halo part denote the component outside the core or localized part of...

open_problem | open | MathDB 页面 | 原文线索

V2-037283 | MATHDB-366655

The GPRS conjecture for large-genus super Weil-Petersson volumes

摘录 / EXCERPT: Let $V^{\Theta}_{g,n}(L_1,\ldots,L_n)$ denote the super Weil-Petersson volume with n boundary lengths, and let $[\prod_{i=1}^n\tau_{d_i}]^{\Theta}_g$ denote the corresponding int...

conjecture | open | MathDB 页面 | 原文线索

V2-037284 | MATHDB-366656

The conjecture that the universal super Weil-Petersson constant equals $\pi^{-9/2}$

摘录 / EXCERPT: Constant conjecture. The constant C should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-037285 | MATHDB-366657

Brendle-Guan-Li Alexandrov-Fenchel conjecture for k-convex star-shaped hypersurfaces in the sphere

摘录 / EXCERPT: Let M be a smooth, closed, connected, embedded, k-convex, star-shaped hypersurface in the sphere $\mathbb{S}^{n+1}$ enclosing a bounded domain Ω. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037286 | MATHDB-366854

Bozkurt's conjecture on mixed norms of Cauchy-Toeplitz matrices

摘录 / EXCERPT: Let T_n be the Cauchy-Toeplitz matrix ... For $1\leq p,q\leq\infty$, let $\|T_n\|_{p,q}$ denote its $\ell_{p,q}$ norm. Bozkurt's conjecture. The inequalities ... and ... are valid. The...

conjecture | open | MathDB 页面 | 原文线索

V2-037287 | MATHDB-366855

The strict Weihrauch separation between finite and infinite iterations of NON

摘录 / EXCERPT: Let NON be the problem discussed in the paper, and let NON^ω and NON^∞ denote its countable and inverse-limit iterations, respectively. T...

conjecture | open | MathDB 页面 | 原文线索

V2-037288 | MATHDB-366943

Nuclearity conjecture for pseudo-Cartan inclusions and their Cartan envelopes

摘录 / EXCERPT: Nuclearity conjecture. ${\mathcal{C}}$ is nuclear if and only if ${\mathcal{A}}$ is nuclear.

conjecture | open | MathDB 页面 | 原文线索

V2-037289 | MATHDB-366959

Gaussian operator norm conjecture for arbitrary matrices

摘录 / EXCERPT: Let p and q satisfy $p\le 2\le q$, and let $A=(a_{ij})_{i\le m,j\le n}$ be a deterministic $m\times n$ matrix. Write p^* for the Hölder conjugate of p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037290 | MATHDB-366967

Strong spherical maximal function conjecture

摘录 / EXCERPT: Strong spherical maximal function conjecture. For all $n\geq 2$ and $p>\frac{n+1}{n-1}$, there exists a constant $C_{n,p}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037291 | MATHDB-367019

Weak equivalence preserves exponents of p-adic differential modules

摘录 / EXCERPT: Let I be an open polysegment, let P be a ∇-module over $A^n_K(I)$ satisfying the Robba condition, and let A be an exponent of P. A multisubset B of...

conjecture | open | MathDB 页面 | 原文线索

V2-037292 | MATHDB-367020

A weaker weight condition for the John-Nirenberg implication

摘录 / EXCERPT: Let v be a weight on the unit disk and let p be the parameter appearing in Proposition 1. Assume that the function $x\mapsto v(1-x)x^{\frac{1}{p}-\epsilon}$ is almost increasin...

conjecture | open | MathDB 页面 | 原文线索

V2-037293 | MATHDB-367021

A counterexample to conformal invariance for increasing radial weights

摘录 / EXCERPT: Let $\mathbb D$ denote the unit disk, let $v\colon\mathbb D\to(0,\infty)$ be an increasing radial weight that is not admissible, let $\hat\phi\in\operatorname{Aut}(\mathbb D)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-037294 | MATHDB-367023

Boundedness of composition operators induced by analytic polynomial symbols

摘录 / EXCERPT: Let v be an admissible weight, and let $\phi\colon\mathbb D\to\mathbb D$ be an analytic polynomial symbol. The polynomial-symbol conjecture. The composition operator ... is bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-037295 | MATHDB-367041

Monotonicity conjecture for the optimal regulation measure

摘录 / EXCERPT: Let $\bar{\eta}_1$ and $\bar{\eta}_2$ be cumulative-effort constraints satisfying $0<\bar{\eta}_1<\bar{\eta}_2$, and let $\mu_{\alpha_{opt},\bar{\eta}_i}$ be the measures associate...

conjecture | open | MathDB 页面 | 原文线索

V2-037296 | MATHDB-367057

The endpoint absolute summability conjecture for the inclusion $X_p\subset L^p$

摘录 / EXCERPT: Endpoint absolute summability conjecture. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-037297 | MATHDB-367058

The absolute summability conjecture for exact interpolation spaces

摘录 / EXCERPT: Absolute summability conjecture for exact interpolation spaces. The embedding

conjecture | open | MathDB 页面 | 原文线索

V2-037298 | MATHDB-367135

Euclidean distortion conjecture for finite subsets of L_p

摘录 / EXCERPT: For $n\to\finity$ and $p>2$, let $\mathsf{c}_2^n(L_p)$ denote the maximum, over all n-point metric spaces, of the least distortion with which the space embeds into L_p and then...

conjecture | open | MathDB 页面 | 原文线索

V2-037299 | MATHDB-367157

Universality conjecture for Gaussian time-frequency localization eigenvalues

摘录 / EXCERPT: Universality conjecture. For every such Ω,

conjecture | open | MathDB 页面 | 原文线索

V2-037300 | MATHDB-367174

The Specht property for finitely generated generalized operator algebras

摘录 / EXCERPT: Let A be a W-algebra over a field of characteristic zero. A class of algebras has the Specht property if every T-ideal of the class is finitely generated. The Specht conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-037301 | MATHDB-367183

Minimum Fourier entropy-influence conjecture

摘录 / EXCERPT: Minimum Fourier entropy-influence conjecture. There exists a constant $c>0$ such that, for every n and every Boolean function f,

conjecture | open | MathDB 页面 | 原文线索

V2-037302 | MATHDB-367188

The restriction conjecture for the sphere

摘录 / EXCERPT: The restriction conjecture for the sphere. This estimate should hold when

conjecture | open | MathDB 页面 | 原文线索

V2-037303 | MATHDB-367198

Uniform operator-norm conjecture for the linear BGK spectral approximation

摘录 / EXCERPT: Let $N\in\mathbb{N}$ and let X_N be the finite-dimensional polynomial space with orthogonal projection Π_{X_N}. Let Ω, ∂_x, and ∂_x^* be the operator...

conjecture | open | MathDB 页面 | 原文线索

V2-037304 | MATHDB-367268

The Tube Doubling Conjecture

摘录 / EXCERPT: Let $n\geq 2$ and $\varepsilon>0$. For sufficiently small $\delta>0$, let $\mathbb{T}$ be a set of δ-tubes in $\mathbb{R}^n$, and let $\widetilde T$ denote the 2-fold dilate...

conjecture | open | MathDB 页面 | 原文线索

V2-037305 | MATHDB-367269

Keleti's line segment extension conjecture

摘录 / EXCERPT: Let L be a set of line segments in $\mathbb{R}^n$. For each line segment ℓ, let $\widetilde\ell$ denote the line containing ℓ. Keleti's line segment extension conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-037306 | MATHDB-367291

R. Yang's Hilbert-Schmidt conjecture for finitely generated submodules

摘录 / EXCERPT: Let $H^2(\mathbb{D}^2)$ be the Hardy space on the bidisk, and let M be a submodule. Define ... where R_z and R_w are multiplication by the coordinate functions on M. The subm...

conjecture | open | MathDB 页面 | 原文线索

V2-037307 | MATHDB-367292

R. Yang's uniform invariant conjecture for rank-one submodules

摘录 / EXCERPT: Let $H^2(\mathbb{D}^2)$ be the Hardy space on the bidisk, and for a submodule M define ... where R_z and R_w are multiplication by the coordinate functions on M. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037308 | MATHDB-367348

The smooth approximation conjecture for stationary varifolds

摘录 / EXCERPT: Smooth approximation conjecture. There is a smooth classical minimal m-dimensional graph $\mathcal{M}$ in some neighborhood of x_0 such that

conjecture | open | MathDB 页面 | 原文线索

V2-037309 | MATHDB-367349

The codimension-one conjecture for stationary integer rectifiable varifolds

摘录 / EXCERPT: Let V be a stationary integer rectifiable m-dimensional varifold, and let its interior singular set consist of the points of ${\rm spt}(V)$ that are not regular. The codimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-037310 | MATHDB-367350

The measure-zero conjecture for stationary varifolds

摘录 / EXCERPT: Let V be a stationary integer rectifiable m-dimensional varifold, and let its interior singular set be the nonregular points in the interior. The measure-zero conjecture. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-037311 | MATHDB-367351

The high-order contact conjecture for stationary varifolds

摘录 / EXCERPT: Let V and x_0 satisfy the assumptions of the flat tangent-cone regularity conjecture. The high-order contact conjecture. There is a smooth classical minimal m-dimensional gra...

conjecture | open | MathDB 页面 | 原文线索

V2-037312 | MATHDB-367352

The unique-continuation conjecture for stationary varifolds

摘录 / EXCERPT: Let V be a stationary varifold, let $x_0\in{\rm spt}(V)\cap U$, and let $\mathcal{M}$ be a classical smooth minimal m-dimensional surface such that ... The unique-continuation co...

conjecture | open | MathDB 页面 | 原文线索

V2-037313 | MATHDB-367456

Kakeya L^p conjecture

摘录 / EXCERPT: Fix $d\geq2$. For each $1/R$-cap θ on S^{d-1}, let θ^* be the corresponding radius-one, length-R tube, and let x_θ be arbitrary. Kakeya L^p conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-037314 | MATHDB-367460

The absolute-continuity conjecture for multiplicative convolutions of self-similar measures

摘录 / EXCERPT: Multiplicative-convolution conjecture. The measure $\mu_1\cdot \mu_2 \dotsb \mu_k$ is absolutely continuous with respect to Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-037315 | MATHDB-367461

The Fourier-decay conjecture for non-degenerate analytic images of self-similar measures

摘录 / EXCERPT: Fourier-decay conjecture. There exists $\sigma>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037316 | MATHDB-367520

Stein's dimension-free maximal inequality conjecture for convex symmetric bodies

摘录 / EXCERPT: Stein's conjecture. For every $p\in(1,\infty]$ there is a constant $C_p>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037317 | MATHDB-367524

The Furstenberg dimension conjecture on Riemannian surfaces

摘录 / EXCERPT: Let $E\subset\Omega$ be an (s,t)-Furstenberg set on a Riemannian surface: there is a family Γ of local geodesics with $\dim_{\mathcal H}\Gamma\geq t$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-037318 | MATHDB-367525

The maximal Kakeya conjecture

摘录 / EXCERPT: For $0<\delta<1$, let $K_\delta f(\omega)$ be the Kakeya maximal function, defined using averages over the δ-neighborhoods of unit line segments in direction...

conjecture | open | MathDB 页面 | 原文线索

V2-037319 | MATHDB-367534

Conjectured polynomial properties of the coefficients $\tilde{c}_{2k-1,m}$

摘录 / EXCERPT: Polynomial-properties conjecture. First, $a=0$ is a first-order zero of $P_{2n-1,m}(a,\tilde{c}_{-1,3})$, while $P_{2n,m}(0,\tilde{c}_{-1,3})\neq0$. Second, when $m\geqslant k+3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-037320 | MATHDB-367535

Conjectured generating-function structure of the coefficients $\tilde{c}_{2k-1,m}$

摘录 / EXCERPT: Generating-structure conjecture. For $m\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-037321 | MATHDB-367574

Restriction conjecture for the hyperbolic paraboloid

摘录 / EXCERPT: Let $d\geq 2$ and let $\vec{v}=(v_1,\ldots,v_{d-1})\in\{-1,1\}^{d-1}$. Consider the hyperbolic paraboloid ... For f on $[0,1]^{d-1}$, define its extension operator by ... where...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037322 | MATHDB-367594

Sogge's local smoothing conjecture for the wave equation

摘录 / EXCERPT: Let u solve the wave equation Cauchy problem ... with $u(x,0)=f(x)$ and $\partial_tu(x,0)=0$ on $\mathbb{R}^d\times\mathbb{R}$. For $s_p=\frac{d-1}{2}-\frac{d-1}{p}$, Sogge's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-037323 | MATHDB-367598

The strong-frame conjecture for interpolation frames of the Schwartz space

摘录 / EXCERPT: Strong-frame conjecture. Every $\mathfrak s$-frame for $\mathcal S$ induced by a pair of sequences as in Theorem 2 of KNS is a strong frame.

conjecture | open | MathDB 页面 | 原文线索

V2-037324 | MATHDB-367610

Conjecture on the Bohr radius for bounded functions with boundary restriction

摘录 / EXCERPT: Bohr-radius conjecture. If $a\in(a_1,1]$, then

conjecture | open | MathDB 页面 | 原文线索

V2-037325 | MATHDB-367618

Uniqueness conjecture for critical rotational annuli in spherical and hyperbolic balls

摘录 / EXCERPT: Let $\mathbb{S}^3_+$ denote the round hemisphere and let $\mathbb{H}^3$ denote hyperbolic three-space. A free boundary minimal annulus is a compact minimal annulus in a geodesic ba...

conjecture | open | MathDB 页面 | 原文线索

V2-037326 | MATHDB-367739

Phase retrieval conjecture for the harmonic oscillator Schrödinger equation

摘录 / EXCERPT: Harmonic-oscillator phase-retrieval conjecture. If $|u(t,x)|=|v(t,x)|$ for every $t\in\mathbb{R}$ and every $x\in\mathbb{R}^d$, then there exists $c\in\mathbb{R}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-037327 | MATHDB-367785

Lipschitz characterization of Fock spaces via the Fock distance

摘录 / EXCERPT: Let F^p_α be the Fock space on ${\mathbb C}^n$, let $L^p({\mathbb C}^n,dv)$ be Lebesgue L^p space, and let $d_\alpha(z,w)$ be the distance function associated with the Foc...

conjecture | open | MathDB 页面 | 原文线索

V2-037328 | MATHDB-367809

The XZ conjecture for $SU(N)$

摘录 / EXCERPT: Let $N\geq 2$, and let $f:[0,1]^{\frac{N(N-1)}{2}}\times \mathbb{T}^{\frac{N(N+1)}{2}-1}\to\mathbb{C}$ be an admissible function, meaning that it has a finite expansion in the toru...

conjecture | open | MathDB 页面 | 原文线索

V2-037329 | MATHDB-367810

The XZ conjecture for G_2

摘录 / EXCERPT: The XZ conjecture for G_2. If

conjecture | open | MathDB 页面 | 原文线索

V2-037330 | MATHDB-367828

Analytic embedding conjecture for Bergman spaces on non-compact Riemann surfaces

摘录 / EXCERPT: Analytic Bergman-space conjecture. The statement of the smooth affine algebraic-curve theorem holds for M: for every open subset $U\subseteq M$, the Bergman space $A^2(U,\sigma)$...

conjecture | open | MathDB 页面 | 原文线索

V2-037331 | MATHDB-367888

The asymptotic expansion conjecture for minimal Riesz 2-energy on the sphere

摘录 / EXCERPT: Let $\mathbb{S}^2$ be the unit sphere and let $\mathcal{E}_2(\mathbb{S}^2;N)$ denote the minimal Riesz 2-energy of N points on $\mathbb{S}^2$. Let $\gamma_n(\alpha)$ be the gen...

conjecture | open | MathDB 页面 | 原文线索

V2-037332 | MATHDB-367933

Invertibility conjecture for Gram matrices of monomial-potential orthogonal polynomials

摘录 / EXCERPT: Let d be a positive integer and $a\geq0$, and consider the hermitian inner product with density ... For each nonnegative integer N, let $G_N=((z^n,z^m))_{n,m=0}^N$ be its Gram ma...

conjecture | open | MathDB 页面 | 原文线索

V2-037333 | MATHDB-367945

Strong Bergman asymptotics for vector-valued polynomial spaces

摘录 / EXCERPT: Strong Bergman asymptotics. The measures satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-037334 | MATHDB-368075

Baire-measurable characterization of the Pedersen-Baire envelope

摘录 / EXCERPT: Baire-measurable characterization. The vertical right arrow identifies $\mathcal{A}^{\infty}$ with those bounded affine functions on $\mathsf{States}(\mathcal{A})$ that are Baire m...

conjecture | open | MathDB 页面 | 原文线索

V2-037335 | MATHDB-368136

Decoupling conjecture for smooth surfaces of vanishing Gaussian curvature

摘录 / EXCERPT: Let S be a compact piece of a smooth surface in $\mathbb{R}^n$, and let $N_\delta(M)$ denote its δ-neighborhood. A family of parallelograms has width at least δ if...

conjecture | open | MathDB 页面 | 原文线索

V2-037336 | MATHDB-368137

Trace-class approximation by direct sums of irreducible operators for type II_1 factors

摘录 / EXCERPT: Type II$_1$ factor approximation conjecture. Suppose that T is an operator in $\mathcal{B}(\mathcal{H})$ such that $W^*(T)$ is a type II_1 factor. Then for every...

conjecture | open | MathDB 页面 | 原文线索

V2-037337 | MATHDB-368192

The nonempty weighted frequent universality conjecture for polar compact sets

摘录 / EXCERPT: Let $K\subset {\mathbb{C}}\setminus \overline{{\mathbb{D}}}$ be a polar compact set. A weight α is understood in the sense defined earlier in the paper, and...

conjecture | open | MathDB 页面 | 原文线索

V2-037338 | MATHDB-368193

The bounded-annulus obstruction conjecture for frequent universality

摘录 / EXCERPT: Let $1\leq r<R<\infty$, let $A(r,R):=\{z\in {\mathbb{C}}:r\leq |z|\leq R\}$, and let ${\mathcal{M}}(r,R)$ be the set of polynomial compact sets in ${\mathcal{M}}$ contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-037339 | MATHDB-368245

Asymptotic cylinder-shape conjecture for magnetic ground-state minimizers

摘录 / EXCERPT: Let $B\geq 0$ be the strength of a constant magnetic field. Let $\Omega^*(B)$ be a minimizing domain, and let $h(\Omega^*(B))$ and $R(\Omega^*(B))$ denote its height and radius. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-037340 | MATHDB-368280

The computable Scott sentence conjecture

摘录 / EXCERPT: Computable Scott sentence conjecture. For each even $n\geq 2$, there is a computable structure $\mathcal{M}$ with a Π_n Scott sentence but no computable Σ_{2n} Scott sen...

conjecture | open | MathDB 页面 | 原文线索

V2-037341 | MATHDB-368303

GLT closure under geometric mean

摘录 / EXCERPT: Let $\{A_n\}_n$ and $\{B_n\}_n$ be Hermitian positive definite GLT matrix-sequences with matrix-valued GLT symbols ... Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-037342 | MATHDB-368342

Stability-barrier conjecture for Hermite-Volterra discretizations

摘录 / EXCERPT: Let L and R denote the numbers of left- and right-sided Hermite degrees of freedom in a semi-discretized Hermite-Volterra (HV) method for an advection equation. Stability-barri...

conjecture | open | MathDB 页面 | 原文线索

V2-037343 | MATHDB-368368

The qNCC-I quantitative Nonlinear Carleson Conjecture

摘录 / EXCERPT: Let $\boldsymbol{\textstyle\gamma}=\\{\gamma_n\}_{n\in\mathbb{Z}_+}$ be the sequence of Verblunsky coefficients associated with the measure σ, and let $M(\xi,\sigma)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-037344 | MATHDB-368390

Local smoothing conjecture for curve averages

摘录 / EXCERPT: Local smoothing conjecture. The displayed inequality holds for all $\sigma < \sigma(p,n)$. This conjecture is solved affirmatively for $n=2$; for $n \geq 3$, it is known when...

conjecture | open | MathDB 页面 | 原文线索

V2-037345 | MATHDB-368391

L^p boundedness conjecture for geometric maximal operators of curve averages

摘录 / EXCERPT: Maximal-operator conjecture. The operator M_γ maps $L^p(\mathbb R^n)$ to $L^p(\mathbb R^n)$ boundedly if and only if $p>n$. This conjecture concerns the L^p theory of geom...

conjecture | open | MathDB 页面 | 原文线索

V2-037346 | MATHDB-368392

Extension of commutator-calculus identities to non-symmetric matrices

摘录 / EXCERPT: Let d be a positive integer, let $\mathbf{G}\in\mathbb{C}^{d\times d}$, and let $\mathbf{A}\in\mathbb{C}^{d\times d}$ be such that $\log\mathbf{A}$ exists. Suppose Definition is...

conjecture | open | MathDB 页面 | 原文线索

V2-037347 | MATHDB-368403

Asymptotic L^2-mass conjecture for pullbacks of Saito-Kurokawa lifts

摘录 / EXCERPT: Let N tend to infinity through odd square-free integers. For an SK lift $F_f\in \mathrm{SK}_{k+1}^{new}(N)$, let F_f° be its pullback and let $N(F_f)$ be its normalized...

conjecture | open | MathDB 页面 | 原文线索

V2-037348 | MATHDB-368426

Principal-pair spectral conjecture for orthorecursive coefficients

摘录 / EXCERPT: Let $(c_n)_{n\geq 0}$ be the orthorecursive coefficients, and let g^* be the function whose zero with smallest positive real part is ... where $\alpha_1\approx1.34652$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037349 | MATHDB-368484

Syntactic Radon-Nikodym conjecture

摘录 / EXCERPT: Let S_n be a syntactic fractal space with a family of definable sets cF_n, and let ν_n and μ_n be S_n-measures. Write $\nu_n \ll \mu_n$ when $\mu_n(A)=0$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-037350 | MATHDB-368536

Classical-limit conjecture for multiple chordal SLE(0) partition functions

摘录 / EXCERPT: Let $\mathcal{Z}_{\alpha}(\boldsymbol{z},u)$ and $\mathcal{Z}_{\alpha}(\boldsymbol{x})$ be pure partition functions, and let $\Phi(\boldsymbol{x},\boldsymbol{\xi},u)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037351 | MATHDB-368572

Non-vanishing conjecture for the polynomials $Q_{m,n}$ on the unit circle

摘录 / EXCERPT: Let $m>n$ be positive integers with $\gcd(m,n)=1$, and let $Q_{m,n}$ be the degree $2m-2n$ palindromic polynomial associated with this pair. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-037352 | MATHDB-368602

The fine-property conjecture for functions in deviatoric bounded deformation

摘录 / EXCERPT: Fine-property conjecture. For all $u\in \mathrm{BD}_{\mathrm{dev}}(\Omega)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037353 | MATHDB-368669

Lee-Rogers-Seeger maximal Schrödinger conjecture

摘录 / EXCERPT: Let I be a compact interval. For $p,q\in[1,\infty]$, write $L^q_xL^\infty_t(\mathbb{R}\times I)$ for the corresponding mixed-norm space. For $\alpha\in\mathbb{R}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-037354 | MATHDB-368692

The absence of unbounded d-weakly convergent sequences in normed linear spaces

摘录 / EXCERPT: The absence conjecture. There is no normed linear space where an unbounded sequence converges d-weakly.

conjecture | open | MathDB 页面 | 原文线索

V2-037355 | MATHDB-368718

The epsilon-loss local Mizohata-Takeuchi conjecture

摘录 / EXCERPT: Let B_R be the ball of radius R centered at the origin, let $\mathbb{T}$ be the relevant family of tubes, and write $w(T)$ for the weight of a tube. Under the hypotheses of the...

conjecture | open | MathDB 页面 | 原文线索

V2-037356 | MATHDB-368725

Permutation rigidity conjecture for a diagonal stationarity condition

摘录 / EXCERPT: Let $\Lambda_1\in\mathbb{R}^{m\times n}$ and $\Lambda_2\in\mathbb{R}^{n\times n}$ be diagonal matrices, with $\Lambda_2\succ0$ having distinct diagonal entries, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037357 | MATHDB-368785

Rakočević's conjecture on continuity of the Moore-Penrose inverse

摘录 / EXCERPT: Let A be a unital Banach algebra, and let A_{MP} denote the set of Moore-Penrose invertible elements of A. For $a\in A_{\mathrm{MP}}$, write $a^\dag$ for its Moore-P...

conjecture | open | MathDB 页面 | 原文线索

V2-037358 | MATHDB-368806

Haag duality conjecture for potentially non-unitary Wightman conformal field theories

摘录 / EXCERPT: Let $(\cF,\cD,U,\Omega)$ be a potentially non-unitary Wightman conformal field theory on S^1 with invariant nondegenerate Hermitian form. Let $\cQ(I)=\cP(I)''$ be the associated a...

conjecture | open | MathDB 页面 | 原文线索

V2-037359 | MATHDB-368850

A quadratic refinement of Sendov's conjecture

摘录 / EXCERPT: Let β be a real number in $[0,1]$. For a polynomial P of degree at least 2 with complex coefficients, all roots in the closed unit disk, and at least one root at β,...

conjecture | open | MathDB 页面 | 原文线索

V2-037360 | MATHDB-368852

The two-critical-point refinement conjecture for Sendov's conjecture

摘录 / EXCERPT: Let $\beta\in[0,1]$, let $S(\beta)$ be the set of polynomials of degree at least 2 with complex coefficients, all roots in the closed unit disk, and at least one root at β,...

conjecture | open | MathDB 页面 | 原文线索

V2-037361 | MATHDB-368873

Lang's general Flat Chain Conjecture via the comparison map

摘录 / EXCERPT: Lang's general Flat Chain Conjecture. For every $d\in\mathbb{N}$ and $k\in\mathbb{N}\cup\{0\}$, if $T\in\mathcal{D}_k(\boldsymbol{R}^d)$ has compact support, then

conjecture | open | MathDB 页面 | 原文线索

V2-037362 | MATHDB-368898

Domingo-Salazar-Lacey-Rey weighted weak-type conjecture for the intrinsic square function

摘录 / EXCERPT: Let S_{int} be the intrinsic square function and let $w\in A_2$ be a Muckenhoupt weight, with characteristics $[w]_{A_2}$ and $[w]_{A_{\infty}}$. The weighted weak...

conjecture | open | MathDB 页面 | 原文线索

V2-037363 | MATHDB-368899

Reverse Hölder self-improvement conjecture

摘录 / EXCERPT: Reverse Hölder self-improvement conjecture. If $w\in\operatorname{RH}_p$ and $1<p<\infty$, then for

conjecture | open | MathDB 页面 | 原文线索

V2-037364 | MATHDB-368961

Best Stein-Tomas conjecture for quadratic manifolds

摘录 / EXCERPT: Let $\mathbf{Q}$ be a quadratic manifold, and let “well-curved” and “best Stein-Tomas” denote the geometric and Fourier-analytic conditions defined in the paper. Best Stein-Tomas c...

conjecture | open | MathDB 页面 | 原文线索

V2-037365 | MATHDB-369032

Lipschitz magnitude conjecture for tractable subspaces

摘录 / EXCERPT: Lipschitz magnitude conjecture. Let $T\subseteq L_1$ be tractable. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037366 | MATHDB-369039

Dual form of the Kakeya maximal operator conjecture in three dimensions

摘录 / EXCERPT: Let $0<\delta<1$. A δ-tube is a tube of length 1 and cross-sectional diameter δ, and a family $\mathbb{T}$ of δ-tubes is δ-separated when distinct tub...

conjecture | open | MathDB 页面 | 原文线索

V2-037367 | MATHDB-369040

Disjoint trilinear dual Kakeya maximal operator conjecture in three dimensions

摘录 / EXCERPT: Let $0<\delta\leq\nu<1$. Three families $\mathbb{T}_{1},\mathbb{T}_{2},\mathbb{T}_{3}$ of δ-tubes are ν-disjoint when their orientations lie in spherical patches of diam...

conjecture | open | MathDB 页面 | 原文线索

V2-037368 | MATHDB-369041

Modulated single-scale Fourier square function disjoint trilinear extension conjecture

摘录 / EXCERPT: Let $U_{1},U_{2},U_{3}\subset U$ be a ν-disjoint triple of squares, meaning that each has side length comparable to ν and distinct squares are separated by at least ν....

conjecture | open | MathDB 页面 | 原文线索

V2-037369 | MATHDB-369053

Conjectured structure of the top stratum in the no-sign obstacle problem

摘录 / EXCERPT: Let u be a solution of the no-sign obstacle problem ... and let Σ_{n-1} denote the top stratum of the free boundary, with $\Sigma_{n-1,a}$ its anomalous part. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-037370 | MATHDB-369068

The tubular geometric control conjecture for L^p-Logvinenko-Sereda sets on spheres

摘录 / EXCERPT: Let S^m be the standard sphere, let $1\le p<\frac{2m}{m-1}$, and let $\mathcal{A}=\{A_\lambda\}_{\lambda\ge1}$ be a family of subsets of S^m. An L^p-Logvinenko-Sereda family...

conjecture | open | MathDB 页面 | 原文线索

V2-037371 | MATHDB-369069

The converse L^2-optimality conjecture for strongly upper semi-continuous functions

摘录 / EXCERPT: Let $D\subset\mathbb{C}^n$ be a domain, and let φ be a strongly upper semi-continuous function on D, meaning that for every $x\in D$ and every subset $S\subset D$ of meas...

conjecture | open | MathDB 页面 | 原文线索

V2-037372 | MATHDB-369070

Griffiths semipositivity conjecture for continuous L^2-optimal Hermitian metrics

摘录 / EXCERPT: Let $D\subset\mathbb{C}^n$ be a domain, let E be a trivial holomorphic vector bundle, and let h be a continuous L^2-optimal Hermitian metric on E, meaning that for every sm...

conjecture | open | MathDB 页面 | 原文线索

V2-037373 | MATHDB-369084

The avoidable-singularity conjecture for the Willmore flow

摘录 / EXCERPT: Let $f_0:\mathbb{S}^2\to\mathbb{R}^3$ be a smooth immersion with $W(f_0)\leq 12\pi$. An avoidable-singularity conjecture asserts that there exists such an f_0 for which the Willm...

conjecture | open | MathDB 页面 | 原文线索

V2-037374 | MATHDB-369110

Support conjecture for tight-frame spectra

摘录 / EXCERPT: Support conjecture for tight-frame spectra. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037375 | MATHDB-369111

Non-spectrality conjecture for piecewise smooth closed curves

摘录 / EXCERPT: Closed-curve non-spectrality conjecture. The measure mu is not spectral.

conjecture | open | MathDB 页面 | 原文线索

V2-037376 | MATHDB-369113

Late-time decay-rate conjecture for nonlinear wave modes

摘录 / EXCERPT: Consider the nonlinear wave equation ... with $p\in\mathbb{N}$ and $p\geqslant3$. Expand the field in spherical harmonics, ... At fixed finite radius r, the modes are expected to dec...

conjecture | open | MathDB 页面 | 原文线索

V2-037377 | MATHDB-369133

Conjecture that the crossed-product algebra is no longer type III

摘录 / EXCERPT: Let $\mathcal{M}$ be a local von Neumann factor, let σ_s be its modular action, and let $\mathcal{R}(\mathcal{M},\sigma_s)$ be the algebra generated by $\mathcal{M}$ togethe...

conjecture | open | MathDB 页面 | 原文线索

V2-037378 | MATHDB-369141

Uniform, degenerate, and sublevel set decoupling for polynomial graphs

摘录 / EXCERPT: Let $n,d\ge 1$. For each n, let $U(n)$ denote uniform decoupling for the polynomial class $\mathcal P_{n,d}$, let $Z(n)$ denote uniform degenerate decoupling for...

conjecture | open | MathDB 页面 | 原文线索

V2-037379 | MATHDB-369147

Lee's conjecture on the Frobenius norm

摘录 / EXCERPT: Lee's conjecture. For all $A,B\in M_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-037380 | MATHDB-369171

Zero-free Taylor polynomial conjecture for square roots of truncated geometric series

摘录 / EXCERPT: Let $d>2$ and define the coefficients $b_k^{(d)}$ by ... For each $n\geq 0$, the corresponding partial sum is the polynomial $\sum_{k=0}^n b_k^{(d)}z^k$. Zero-free Taylor polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-037381 | MATHDB-369172

Positivity conjecture for trigonometric polynomials from square-root coefficients

摘录 / EXCERPT: Positivity conjecture. For $d\geq 2$, the trigonometric polynomials $\mathcal T_n^{(d)}(t)$ are positive for every $t\in\mathbb R$. Positivity of these polynomials is suggested as...

conjecture | open | MathDB 页面 | 原文线索

V2-037382 | MATHDB-369180

Strong stability conjecture under double commutation on the synthesis kernel

摘录 / EXCERPT: Let H be a Hilbert space, let $T_1,T_2\in B(H)$ commute, and suppose that $\\{T_1^iT_2^j\varphi\\}_{i,j\geq 0}$ is an overcomplete frame for H. Let R_1 and R_2 be the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-037383 | MATHDB-369188

Reverse trace inequality for powers greater than one

摘录 / EXCERPT: Let $A,B\geq 0$. Let f be a nonnegative and nondecreasing function on an interval $\mathcal{J}$ such that $\operatorname{spec}(A)\subseteq\mathcal{J}$. Reverse trace inequality....

conjecture | open | MathDB 页面 | 原文线索

V2-037384 | MATHDB-369222

Interior variant of Falconer's conjecture for pinned chain distance sets

摘录 / EXCERPT: Let $E\subset \mathbb{R}^d$ be compact with Hausdorff dimension $\dim_{\rm H}(E)>\frac{d}{2}$. For $n\in\mathbb{N}$ and $x\in E$, let $\Delta_x^n(E)$ denote the pinned n-chain se...

conjecture | open | MathDB 页面 | 原文线索

V2-037385 | MATHDB-369265

Larson's conjecture on wavelet sets in Fourier supports

摘录 / EXCERPT: Larson's conjecture. The support of the Fourier transform of a dyadic, orthonormal wavelet contains a wavelet set.

conjecture | open | MathDB 页面 | 原文线索

V2-037386 | MATHDB-369267

Weak Larson conjectures on measurable subsets of supports

摘录 / EXCERPT: Let ψ be a dyadic orthonormal wavelet, let $f:\mathbb{R}\to\mathbb{R}$ tile $\obreak\mathbb{R}$ by translations and dilations, and let μ denote Lebesgue measure. Define a...

conjecture | open | MathDB 页面 | 原文线索

V2-037387 | MATHDB-369337

de Branges' conjecture that (H1) implies (H2)

摘录 / EXCERPT: A Hilbert space of entire functions satisfies axiom (H1), which asserts that, whenever F belongs to the space and w is not real, the function ... belongs to the space with no lar...

conjecture | open | MathDB 页面 | 原文线索

V2-037388 | MATHDB-369414

Biswas-Pal-Tomar's rigidity conjecture for the hexablock

摘录 / EXCERPT: Let $\mathbb{H}$ denote the hexablock, let $G(\mathbb{H})$ be the group of explicitly given automorphisms generated by the maps described above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037389 | MATHDB-369433

Backwards uniqueness conjecture past singularities for mean curvature flow in R^3

摘录 / EXCERPT: Let $M^i\subset\mathbb{R}^3$, $i=1,2$, be two smooth, connected, compact hypersurfaces in $\mathbb{R}^3$. Denote by M^i_t the outermost mean curvature flows emanating from M^i....

conjecture | open | MathDB 页面 | 原文线索

V2-037390 | MATHDB-369492

Ilmanen's genus reduction conjecture for mean curvature flow

摘录 / EXCERPT: Let M_t be any mean curvature flow in $\mathbb{R}^3$. A genus reduction conjecture asserts that the genus of M_t strictly decreases at the time of a singularity, unless the sin...

conjecture | open | MathDB 页面 | 原文线索

V2-037391 | MATHDB-369499

Visibility characterization by cluster sets of non-visible geodesics

摘录 / EXCERPT: The cluster-set conjecture. If $p,q\in\Gamma(\{\gamma_k\})$ and $p\ne q$, then either

conjecture | open | MathDB 页面 | 原文线索

V2-037392 | MATHDB-369500

Complex visibility characterization for bounded convex domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb C^N$ be a bounded convex domain. It is complex visible if for every distinct $p,q\in\partial\Omega$ there exist a compact set $K\Subset\Omega$ and open ne...

conjecture | open | MathDB 页面 | 原文线索

V2-037393 | MATHDB-369501

Geodesic equicontinuity characterization of visibility

摘录 / EXCERPT: Let $\Omega\subset\mathbb C^N$ be a bounded convex domain, let $z_0\in\Omega$, and let $\mathcal G_{z_0}$ be the family of all geodesic segments and geodesic rays starting from...

conjecture | open | MathDB 页面 | 原文线索

V2-037394 | MATHDB-369502

Denjoy-Wolff characterization by sequential horospheres

摘录 / EXCERPT: Let $\Omega\subset\mathbb C^N$ be a bounded convex domain whose boundary contains no non-trivial analytic discs. A sequential horosphere is a set of the form ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-037395 | MATHDB-369503

Coordinate convergence conjecture for fixed-point-free polydisc self-maps

摘录 / EXCERPT: Let $N\ge1$ and let $F:\mathbb D^N\to\mathbb D^N$ be holomorphic and without fixed points. Write $F^{\circ n}$ for the n-fold iterate and let $\pi_j(z)=z_j$ be projection onto th...

conjecture | open | MathDB 页面 | 原文线索

V2-037396 | MATHDB-369521

Iwaniec's diffeomorphic minimizer conjecture for the disk

摘录 / EXCERPT: Let $f_0:\overline{\mathbb{D}}\to\overline{\mathbb{D}}$ be the prescribed boundary homeomorphism, let $1<p<\infty$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037397 | MATHDB-369569

The conjecture that the logarithmic loss in the general estimate is technical

摘录 / EXCERPT: The technical-loss conjecture. For every $k_0>0$ and $s\in\mathbb R$, there is $C>0$ such that, for all $N>0$, $k>k_0$, and $0<\epsilon<1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037398 | MATHDB-369589

Conjugate unitary hollowization conjecture for traceless Hermitian operators

摘录 / EXCERPT: Let $L,M\in\mathbb{C}^{n\times n}$ be traceless Hermitian matrices. A matrix is hollow when all its diagonal entries are zero. Conjugate unitary hollowization conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-037399 | MATHDB-369672

HRT conjecture for finite Gabor systems

摘录 / EXCERPT: Let $\Lambda=\{(x_k,y_k):1\leq k\leq N\}\subset \mathbb{R}^d\times\mathbb{R}^d$ be a finite set of distinct points, and let $0\neq f\in L^2(\mathbb{R}^d)$. HRT conjecture. The fini...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037400 | MATHDB-369675

The single-base representation conjecture for sums of exponential terms

摘录 / EXCERPT: Single-base representation conjecture. For every $a_n,b,c_n,d\in\mathbb{R}$ and $n\in\mathbb{N}$, there exist $u_n,{\Greekmath 010D}\in\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037401 | MATHDB-369692

Bourgain's eigenfunction bounds on the three-dimensional torus

摘录 / EXCERPT: Let u be an eigenfunction of the Laplacian $-\Delta$ on $\mathbb{T}^3$ with eigenvalue λ^2. Bourgain's eigenfunction conjecture. One should have ... The loss...

conjecture | open | MathDB 页面 | 原文线索

V2-037402 | MATHDB-369702

Andersson-Gauthier conjecture on zero-free polynomial approximation

摘录 / EXCERPT: Let K be a compact subset of the complex plane, and write K° for its interior. A continuous function $f:K\to\mathbb C$ is zero-free on K° when $f(z)\neq 0$ for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-037403 | MATHDB-369742

The flat chain conjecture for metric currents

摘录 / EXCERPT: Let $k\in\mathbb{N}$, let X be a metric space, and let $\mathbf{M}_k(\mathrm{X})$ denote the space of metric k-currents on X. Normal k-currents are metric...

conjecture | open | MathDB 页面 | 原文线索

V2-037404 | MATHDB-369779

Bshouty-Hengartner's maximal valence conjecture for logharmonic polynomials

摘录 / EXCERPT: Let p and q be holomorphic polynomials with $n:=\deg p$ and $m:=\deg q$, and suppose that p is not a constant multiple of q. A logharmonic polynomial is a function of the f...

conjecture | open | MathDB 页面 | 原文线索

V2-037405 | MATHDB-369834

Matrix-analysis inequality implying the tree real-part conjecture

摘录 / EXCERPT: Matrix-analysis conjecture. Under these hypotheses,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037406 | MATHDB-369835

Self-conjugate-grid pointwise interpolation-error conjecture

摘录 / EXCERPT: Self-conjugate-grid pointwise conjecture. For every $0\leqslant k\leqslant n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037407 | MATHDB-369837

Uniform interpolation-error conjecture for self-conjugate grids

摘录 / EXCERPT: Self-conjugate-grid uniform conjecture. For every $0\leqslant k\leqslant n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037408 | MATHDB-369960

Higher-codimension borderline minimality conjecture for elastic curves

摘录 / EXCERPT: Let $n\geq 2$ and let $\gamma\in W^{2,2}_{loc}(\mathbf{R};\mathbf{R}^n)$ satisfy ... A curve has a self-intersection if there exist distinct parameters with the same image point. Hig...

conjecture | open | MathDB 页面 | 原文线索

V2-037409 | MATHDB-370035

The asymptotic principle for area-minimizing flat chains modulo n

摘录 / EXCERPT: Consider the direct-limit coefficient group $\mathbb{Q}/\mathbb{Z}$ obtained from the groups $\mathbb{Z}/n\mathbb{Z}$ under the multiplication maps, equipped with the norm induced...

conjecture | open | MathDB 页面 | 原文线索

V2-037410 | MATHDB-370072

The limiting alternative-measure conjecture for sparse polynomials

摘录 / EXCERPT: Let $c(P)$ be the alternative measure of a polynomial, and let $k\geq 3$ with successive limits taken in the displayed order. The preceding discussion considers the sparse polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-037411 | MATHDB-370075

The three-dimensional anisotropic Kakeya--Nikodym conjecture

摘录 / EXCERPT: Let $n=3$, let Σ be a strictly convex hypersurface, and let $\mathop{\mathrm{MKNE}}$ denote the maximal Kakeya--Nikodym estimate referred to in the source as equation (MKNE)...

conjecture | open | MathDB 页面 | 原文线索

V2-037412 | MATHDB-370110

Stable Allen-Cahn level-set curvature conjecture in dimensions at most seven

摘录 / EXCERPT: Stable Allen-Cahn regularity conjecture. If $n \leq 7$, then the curvatures of the level sets of u_{ε_j} in these interfacial regions are uniformly bounded along the se...

conjecture | open | MathDB 页面 | 原文线索

V2-037413 | MATHDB-370177

Maggi's conjecture on anisotropic critical sets

摘录 / EXCERPT: Maggi's conjecture. Wulff shapes are the unique such sets.

conjecture | open | MathDB 页面 | 原文线索

V2-037414 | MATHDB-370178

Anisotropic Serrin-type conjecture for rough domains

摘录 / EXCERPT: Anisotropic Serrin-type conjecture for rough domains. Wulff shapes are the unique such sets satisfying the overdetermined system.

conjecture | open | MathDB 页面 | 原文线索

V2-037415 | MATHDB-370198

Gauthier-Seshadri conjecture on rigidity of Kobayashi isometries

摘录 / EXCERPT: Let M and M' be Kobayashi hyperbolic complex manifolds. Let $f:(M,K_M)\to (M',K_{M'})$ be a Kobayashi isometry. If M and M' are not biholomorphic to a product of complex ma...

conjecture | open | MathDB 页面 | 原文线索

V2-037416 | MATHDB-370224

Higher-order Hecke L-function moment conjecture with switching behavior

摘录 / EXCERPT: Higher-order switching conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037417 | MATHDB-370245

Schur stability conjecture for image reconstruction operators

摘录 / EXCERPT: Schur stability conjecture. The operators $P(t)$ and $R(t)$ are stable, meaning that

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037418 | MATHDB-370374

The sharp zero-constant conjecture for the non-local interaction-energy bound

摘录 / EXCERPT: Sharp zero-constant conjecture. Proposition should remain true with $C=0$; that is,

conjecture | open | MathDB 页面 | 原文线索

V2-037419 | MATHDB-370397

The reflexive Banach space fixed point property conjecture

摘录 / EXCERPT: Reflexive-space fixed point conjecture. Every reflexive Banach space has the fixed point property for nonexpansive maps. Equivalently, if X is reflexive and $C\subset X$ is close...

conjecture | open | MathDB 页面 | 原文线索

V2-037420 | MATHDB-370497

The matricial-range conjecture for unital self-adjoint preserving maps on the operator system of J_3

摘录 / EXCERPT: Let J_3 denote the 3×3 nilpotent Jordan block, let $\mathcal{H}$ be a Hilbert space, and let $\mathcal{OS}(J_3)$ and $\mathcal{OS}(X)$ be the operator systems generated b...

conjecture | open | MathDB 页面 | 原文线索

V2-037421 | MATHDB-370529

The relative solidity conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Let G be a group that is hyperbolic relative to a finite family of subgroups $\{H_1,\ldots,H_n\}$. Denote by $\mathcal M=\mathcal L(G)$ the von Neumann algebra corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-037422 | MATHDB-370541

Fractional-power numerical-radius conjecture for accretive matrices

摘录 / EXCERPT: Let $A\in\Gamma_n$, where Γ_n denotes the class of accretive $n\times n$ matrices, and let t satisfy $0<t<1$. Fractional-power numerical-radius conjecture. ... This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-037423 | MATHDB-370601

The coarse Baum-Connes and coarse Novikov conjectures

摘录 / EXCERPT: Coarse Baum-Connes and coarse Novikov conjectures. The coarse Baum-Connes conjecture claims that μ is an isomorphism, while the coarse Novikov conjecture claims that μ is i...

conjecture | open | MathDB 页面 | 原文线索

V2-037424 | MATHDB-370602

The relative coarse Baum-Connes and relative coarse Novikov conjectures

摘录 / EXCERPT: Relative coarse Baum-Connes and relative coarse Novikov conjectures. The relative coarse Baum-Connes conjecture for (X,Y) claims that $\mu_{Y,\infty}$ is an isomorphism, while th...

conjecture | open | MathDB 页面 | 原文线索

V2-037425 | MATHDB-370609

Uniform-integrability closure conjecture for Schrödinger-wave observability

摘录 / EXCERPT: For each $d\geq 1$, let Y be the normed function space introduced in the paper for Schrödinger-wave observability, and let Y_0 denote the closure of continuous functions in Y...

conjecture | open | MathDB 页面 | 原文线索

V2-037426 | MATHDB-370610

Time-independent closure conjecture for Schrödinger-wave observability

摘录 / EXCERPT: For each $d\geq 1$, let Y_0 be the closure space for Schrödinger waves introduced in the paper, and let $\tilde{Y}_0\subset Y_0$ be its time-independent subspace. Time-independen...

conjecture | open | MathDB 页面 | 原文线索

V2-037427 | MATHDB-370654

Robins' conjecture on Fourier transforms of convex centrally symmetric bodies

摘录 / EXCERPT: Robins' conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037428 | MATHDB-370698

Classification conjecture for ancient asymptotically cylindrical 4-dimensional Ricci flows

摘录 / EXCERPT: Consider an ancient asymptotically cylindrical 4-dimensional Ricci flow, meaning a Ricci flow existing for all sufficiently negative times and asymptotic at negative infinity to a...

conjecture | open | MathDB 页面 | 原文线索

V2-037429 | MATHDB-370880

Shapiro's Conjecture 12 for real polynomials of even degree

摘录 / EXCERPT: Let $p(x)$ be a real polynomial of even degree n. For a real polynomial q, write $\sharp_r q$ for the number of its real zeros. Shapiro's Conjecture 12. ... The conjecture concer...

conjecture | open | MathDB 页面 | 原文线索

V2-037430 | MATHDB-370896

Characterization conjecture for maximal monotone operators with unique representation

摘录 / EXCERPT: Characterization conjecture. The operator A must either be the subdifferential ∂f of a proper convex lower semicontinuous function f of the form described in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-037431 | MATHDB-370907

Li and Yang's conjecture on sharing small functions with a derivative

摘录 / EXCERPT: Let f be a non-constant entire function, let k be an integer, and let a_1 and a_2 be two distinct small functions of f. Li and Yang's conjecture. If f and $f^{(k)}$ sha...

conjecture | open | MathDB 页面 | 原文线索

V2-037432 | MATHDB-370909

Product formula conjecture for Dunkl-Bessel functions

摘录 / EXCERPT: Let $N\geq 1$, let $\mathcal{R}\subset\mathbb{R}^N$ be a finite root system, and let $\theta\in\theta(\mathcal{R})$ be nonnegative. For $a_1,a_2\in\mathbb{R}^N$, consider the Dunkl...

conjecture | open | MathDB 页面 | 原文线索

V2-037433 | MATHDB-370977

Existence of a continuous function nowhere differentiable along every unit speed curve

摘录 / EXCERPT: A unit speed curve is a curve parametrized by arclength, and a C^1 curve has a continuous first derivative. Existence conjecture. There is a continuous function ... whose restricti...

conjecture | open | MathDB 页面 | 原文线索

V2-037434 | MATHDB-370978

Generic nowhere differentiability along unit speed curves

摘录 / EXCERPT: Let U be a bounded open set in ${\mathbb R}^d$. Write $C(\overline U)$ for the space of continuous real-valued functions on $\overline U$, and let G_δ denote a countable i...

conjecture | open | MathDB 页面 | 原文线索

V2-037435 | MATHDB-370986

The equioscillating-spline bound for quadrature weights

摘录 / EXCERPT: Let $\bm{x}$ be a set of unisolvent interpolation points, let $c(\bm{x})$ denote the equioscillating spline associated with these points, let $I(1)$ denote the integral of the cons...

conjecture | open | MathDB 页面 | 原文线索

V2-037436 | MATHDB-370990

Higher-dimensional bounded-diameter conjecture for Type I Ricci flows

摘录 / EXCERPT: Let M be a compact manifold and $g(t)$, $t\in [0,T)$ be a smooth Ricci flow on M, which may become singular at $t=T$. A Ricci flow is Type I if, for some constant $C_I<+\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-037437 | MATHDB-371029

Restricted Marstrand projection conjecture for analytic families

摘录 / EXCERPT: Let $P_t:\mathbb{R}^n\rightarrow\mathbb{R}^m$ be an analytic family of linear maps, meaning that its coordinate functions are analytic functions of t. Assume that for every subsp...

conjecture | open | MathDB 页面 | 原文线索

V2-037438 | MATHDB-371030

Generalized restricted Marstrand conjecture for analytic linear families

摘录 / EXCERPT: Let $P_t:\mathbb{R}^n\rightarrow\mathbb{R}^m$ be an analytic family of linear maps. Define $f:[0,n]\rightarrow\mathbb{R}$ as follows: for an integer $x\in[0,n]$, let $f(x)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-037439 | MATHDB-371031

The closed-form conjecture for the optimal Schatten norm constant for m matrices

摘录 / EXCERPT: Let $M_n(\mathbb C)=\mathbb C^{n\times n}$, and for $A\in M_n(\mathbb C)$ write $|A|=(A^*A)^{1/2}$. For $p>1$ and $m\geq2$, let $c_p(m)$ be the smallest number such that ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-037440 | MATHDB-371034

Smale's mean value conjecture for complex polynomials

摘录 / EXCERPT: Let P be a complex polynomial of degree $n\ge2$ satisfying $P(0)=0$ and $P'(0)=1$. A point ζ is a critical point of P when $P'(\zeta)=0$. Smale's mean value conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-037441 | MATHDB-371035

Dual Smale's mean value conjecture for complex polynomials

摘录 / EXCERPT: Let P be a complex polynomial of degree $n\ge2$ satisfying $P(0)=0$ and $P'(0)=1$. A point ζ is a critical point of P when $P'(\zeta)=0$. Dual Smale's mean value conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-037442 | MATHDB-371048

Schur critical-index conjecture for simple Lie groups and lattices

摘录 / EXCERPT: Schur critical-index conjecture. The critical index satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-037443 | MATHDB-371071

The conjecture that the critical dimension in the Alt-Caffarelli problem is six

摘录 / EXCERPT: Critical-dimension conjecture. The critical dimension d^* is 6. In particular, for any minimizer u of the Alt-Caffarelli problem in $\Omega\subset\mathbb{R}^d$ with $d\leq 6$...

conjecture | open | MathDB 页面 | 原文线索

V2-037444 | MATHDB-371116

Sharp $L^2\to L^6$ extension inequality conjecture on the circle

摘录 / EXCERPT: Sharp $L^2\to L^6$ extension conjecture. For all $f\in L^2(\mathbb{S}^1)\setminus\{0\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-037445 | MATHDB-371179

Conjecture on consecutive logarithmic coefficients of univalent functions

摘录 / EXCERPT: Let $f\in\mathcal{S}$ be a univalent function with logarithmic coefficients γ_n, defined by the logarithmic expansion associated with f. Conjecture on consecutive logarith...

conjecture | open | MathDB 页面 | 原文线索

V2-037446 | MATHDB-371229

Rectifiability and backward uniqueness conjecture for the top Ricci flow singular stratum

摘录 / EXCERPT: Let $(Z,d_Z,\mathfrak{t})$ be a noncollapsed Ricci flow limit space arising as the pointed Gromov--Hausdorff limit of a sequence in $\mathcal{M}(n,Y,T)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037447 | MATHDB-371263

The one-dimensional Mizohata-Takeuchi conjecture for convex curves

摘录 / EXCERPT: Let Γ be a convex C^2 function on $[0,1]$, let E_Γ be its extension operator, and let $\omega:B_R\to[0,\infty)$ be a one-dimensional weight, meaning ... for every bal...

conjecture | open | MathDB 页面 | 原文线索

V2-037448 | MATHDB-371264

The one-dimensional Mizohata-Takeuchi conjecture for the truncated parabola

摘录 / EXCERPT: Let E be the extension operator for the truncated parabola, corresponding to $\Gamma(\xi)=\xi^2$, and let $\omega:B_R\to[0,\infty)$ be a one-dimensional weight, so that ... for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-037449 | MATHDB-371313

Carbery-Wright oscillatory integral conjecture

摘录 / EXCERPT: Carbery-Wright conjecture. There exists an absolute constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037450 | MATHDB-371314

Small-variance conjecture for projected normals on Stiefel manifolds

摘录 / EXCERPT: Let $\mathbb{K}$ be the underlying scalar field, let $\operatorname{M}_{n,k}(\mathbb{K})$ be the ambient matrix space, and let $\operatorname{St}_{n,k}(\mathbb{K})$ be the Stiefel...

conjecture | open | MathDB 页面 | 原文线索

V2-037451 | MATHDB-371403

The non-resonant boundedness conjecture for the curved trilinear Hilbert transform and curved n-linear maximal operator

摘录 / EXCERPT: The non-resonant boundedness conjecture. Up to end-points, both operators have the boundedness range

conjecture | open | MathDB 页面 | 原文线索

V2-037452 | MATHDB-371412

Friedgut-Kalai Fourier-Entropy-Influence conjecture for the uniform cube

摘录 / EXCERPT: Let $f:(\{0,1\}^n,\mu_{1/2}^n)\to\{\pm1\}$ be Boolean-valued, with spectral entropy ... and total influence ${\rm I}^{(1/2)}[f]=\sum_{i=1}^n{\rm Inf}_i^{(1/2)}[f]$. Friedgut-Kalai's...

conjecture | open | MathDB 页面 | 原文线索

V2-037453 | MATHDB-371413

Keller-Mossel-Schlank Fourier-Entropy-Influence conjecture for the biased cube

摘录 / EXCERPT: Let $0<p<1$, let n be positive, and let $f:(\{0,1\}^n,\mu_p^n)\to\{\pm1\}$ be Boolean-valued. Write ... and ${\rm I}^{(p)}[f]=\sum_{i=1}^n{\rm Inf}_i^{(p)}[f]$. Keller-Mossel-Schla...

conjecture | open | MathDB 页面 | 原文线索

V2-037454 | MATHDB-371414

Binary-entropy lower-bound conjecture for Fourier entropy on the biased cube

摘录 / EXCERPT: Let $f:(\{0,1\}^n,\mu_p^n)\to\{\pm1\}$ be Boolean-valued and let $0<p<1$. Define ... and let $h(q)=-q\log q-(1-q)\log(1-q)$ be the binary entropy function. Binary-entropy lower-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-037455 | MATHDB-371416

Strict inner-function inequality conjecture in Hardy spaces

摘录 / EXCERPT: Let H^p denote the Hardy space on the unit disk, let $d54b$ be the unit circle, and let $f\in H^p$ satisfy $f(0)\neq 0$. An inner function is a bounded analytic function on the u...

conjecture | open | MathDB 页面 | 原文线索

V2-037456 | MATHDB-371498

Sibony's conjecture on generating Ahlfors currents by one entire curve

摘录 / EXCERPT: Let X be a compact complex manifold, and let an Ahlfors current mean a current obtained as an asymptotic limit from the concentric holomorphic discs associated to an entire curve...

conjecture | open | MathDB 页面 | 原文线索

V2-037457 | MATHDB-371552

Conjecture on asymptotic growth of cluster scattering wall-function coefficients

摘录 / EXCERPT: Let $b,c>0$ with $bc>4$, and let r lie in the Badlands interval ... For positive integers i,j, let $\tau(i,j)$ be the corresponding wall-function coefficient. Asymptotic growth c...

conjecture | open | MathDB 页面 | 原文线索

V2-037458 | MATHDB-371560

Infinitely many spectral eigenvalues for singular continuous spectral pairs

摘录 / EXCERPT: Let (μ,Λ) be a singular continuous spectral pair in $\mathbb{R}$. A real number $t\in\mathbb{R}$ is called a spectral eigenvalue of (μ,Λ) if $(\mu,t\Lambda)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-037459 | MATHDB-371568

The tightness conjecture for the learning coefficient in factor analysis

摘录 / EXCERPT: Tightness conjecture. The bound from Theorem bound-alternate is tight for all such p, k, and r; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-037460 | MATHDB-371599

Worst-case performance conjecture for gradient descent on linear-operator compositions

摘录 / EXCERPT: Let g be a mu_g-strongly convex and L_g-smooth function, and let M belong to the class of linear operators with singular values in $[mu_M,L_M]$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-037461 | MATHDB-371629

Characterization of measure sequences defining a density

摘录 / EXCERPT: Measure-sequence characterization. The sequences of measures defining a density in this sense are precisely those satisfying

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-037462 | MATHDB-371687

Morita equivalence of continuous-trace subquotients of exponential solvable Lie group C^*-algebras

摘录 / EXCERPT: Let G) be an exponential solvable Lie group, and let ${\mathcal J}_1\subseteq{\mathcal J}_2$ be closed two-sided ideals of $C^*(G)$ such that ${\mathcal J}_2/{\mathcal J}_1$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-037463 | MATHDB-371768

The openness and smooth-witness conjecture for repeated integrals

摘录 / EXCERPT: The (P_n) conjecture. (P_n) is true for all nonnegative integers n.

conjecture | open | MathDB 页面 | 原文线索

V2-037464 | MATHDB-371770

A weighted Paley-Wiener characterization for entire functions

摘录 / EXCERPT: Weighted Paley-Wiener conjecture. The following equivalences hold:

conjecture | open | MathDB 页面 | 原文线索

V2-037465 | MATHDB-371889

The Riemannian isoperimetric inequality for black-hole surfaces

摘录 / EXCERPT: Let Σ be a compact surface in an asymptotically flat manifold with nonnegative scalar curvature $R\geq0$. Let M be the mass and define the Schwarzschild radius by $r_H=2M$...

conjecture | open | MathDB 页面 | 原文线索

V2-037466 | MATHDB-371910

Gaussian maximizer conjecture for paraboloid Strichartz inequalities

摘录 / EXCERPT: Gaussian maximizer conjecture. The maximizers for the paraboloid Strichartz inequality are exactly the Gaussian-type functions $g\in\mathcal{G}$. This conjecture has been confirmed...

conjecture | open | MathDB 页面 | 原文线索

V2-037467 | MATHDB-371929

The local Mizohata-Takeuchi conjecture

摘录 / EXCERPT: Local Mizohata-Takeuchi conjecture. For every $\alpha>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037468 | MATHDB-371957

Naor's lattice quotient embedding conjecture

摘录 / EXCERPT: Naor's lattice quotient conjecture. There exists a constant $c>0$ such that for every $r\in\mathbb{N}$ and every lattice $\Lambda\subset\mathbb{R}^r$,

conjecture | open | MathDB 页面 | 原文线索

V2-037469 | MATHDB-371968

The discretized Kakeya conjecture

摘录 / EXCERPT: Discretized Kakeya conjecture. The following two assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-037470 | MATHDB-372038

Erdős-Herzog-Piranian maximal lemniscate length conjecture

摘录 / EXCERPT: Let $n \geq 1$, and let $p\colon\mathbb{C}\to\mathbb{C}$ be a monic polynomial of degree n. For $R\geq 0$, define ... and ... The Erdős-Herzog-Piranian conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-037471 | MATHDB-372042

The critical-volume conjecture for the nonlocal isoperimetric problem in Euclidean space

摘录 / EXCERPT: Critical-volume conjecture. For $m\leq m_{*}$, the ball of volume m uniquely minimizes $\mathcal{E}(\cdot)$ among measurable sets $F\subset\mathbb{R}^n$ with $|F|=m$, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-037472 | MATHDB-372049

The prevalent singularity spectrum conjecture for inhomogeneous Besov spaces

摘录 / EXCERPT: Prevalent singularity spectrum conjecture. For every $f\in\widetilde{b}^{\mu,\Psi}_{p,q}([0,1]^d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-037473 | MATHDB-372053

Fau's graph-algebra nowhere-scatteredness conjecture

摘录 / EXCERPT: Fau's graph-algebra conjecture. $C^*(E)$ is nowhere scattered if and only if $C^*(E)$ is $\mathcal{Z}$-stable, if and only if E has Condition (K) and distinct detours.

conjecture | open | MathDB 页面 | 原文线索

V2-037474 | MATHDB-372054

The Toms-Winter conjecture

摘录 / EXCERPT: The Toms-Winter conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-037475 | MATHDB-372170

Rathmair's conjecture on local stability and Cheeger constants for the Gabor transform

摘录 / EXCERPT: Rathmair's conjecture. The local stability and Cheeger constants of f should be finite and non-zero.

conjecture | open | MathDB 页面 | 原文线索

V2-037476 | MATHDB-372184

Uniform boundedness conjecture for the Fourier interpolation basis functions

摘录 / EXCERPT: Let $f_n(x)$ denote the basis functions of the Fourier interpolation considered in the paper, for $n\in\mathbb{Z}_{\ge0}$ and $x\ge0$. Uniform boundedness conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-037477 | MATHDB-372187

Logarithmic square-norm conjecture for the Fourier interpolation basis functions

摘录 / EXCERPT: Let $f_n(x)$ be the Fourier interpolation basis functions. Logarithmic square-norm conjecture. For $n>1$, one has ... The lower bound is already known from the result cited in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-037478 | MATHDB-372188

Fourier uniqueness-pair conjecture for the square-root sequence

摘录 / EXCERPT: Let $\mathcal{H}_{\operatorname{e}}$ be the function space defined in the paper, and let ... A pair (Λ_0,Λ_0) is a Fourier uniqueness pair for...

conjecture | open | MathDB 页面 | 原文线索

V2-037479 | MATHDB-372210

The Nevanlinna-Pick norm conjecture for commutative Banach algebras

摘录 / EXCERPT: Let A be a commutative Banach algebra with the Nevanlinna-Pick property represented by $A\in NP_{\infty}$. Nevanlinna-Pick norm conjecture. If $A\in NP_{\infty}$, then A is a...

conjecture | open | MathDB 页面 | 原文线索

V2-037480 | MATHDB-372211

The Nevanlinna-Pick norm conjecture for uniform algebras

摘录 / EXCERPT: Let K be a compact space and let A be a closed subalgebra of $C(K)$, with the Nevanlinna-Pick property represented by $A\in NP_{\infty}$. Nevanlinna-Pick norm conjecture for un...

conjecture | open | MathDB 页面 | 原文线索

V2-037481 | MATHDB-372364

Monotonicity conjecture for the first Steklov eigenvalue of an eccentric annulus

摘录 / EXCERPT: Monotonicity conjecture. The first non-zero Steklov eigenvalue $\sigma_1(\Omega_{\epsilon})$ is a monotone decreasing function of ϵ on $[0,1-\delta)$.

conjecture | open | MathDB 页面 | 原文线索

V2-037482 | MATHDB-372409

Iosevich-Sawyer-Stein conjecture for maximal operators

摘录 / EXCERPT: Let $\rho>0$ and let ρ-Fourier decay mean that a measure $ u$ satisfies the decay inequality ... Suppose $Gamma$ is a smooth hypersurface and a measure mu defined on $Gamma$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-037483 | MATHDB-372410

Reformulated Iosevich-Sawyer-Stein conjecture for analytic hypersurfaces

摘录 / EXCERPT: Let $p_{cr}\ge 2$ and let Γ be a compact analytic hypersurface. Let M_Γ denote its associated maximal operator, let μ be the measure on Γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037484 | MATHDB-372446

Kaplansky's conjecture for Type II$_1$ AW*-factors

摘录 / EXCERPT: Kaplansky's conjecture. Every Type II$_1$ AW*-factor is a W*-factor.

conjecture | open | MathDB 页面 | 原文线索

V2-037485 | MATHDB-372447

The 2-quastrace conjecture

摘录 / EXCERPT: The 2-quastrace conjecture. Every 2-quasitrace on a unital C*-algebra is a trace.

conjecture | open | MathDB 页面 | 原文线索

V2-037486 | MATHDB-372448

The W*-Pedersen characterization by maximal abelian self-adjoint subalgebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C*-algebra. A MASA is a maximal abelian C*-subalgebra of $\mathcal{A}$, and a W*-algebra is a C*-algebra admitting a Banach-space predual....

conjecture | open | MathDB 页面 | 原文线索

V2-037487 | MATHDB-372496

Weak majorization of higher-power coefficients in the exponential product formula

摘录 / EXCERPT: Let $A,B\in\mathbb{H}_n$ be Hermitian matrices, and let H and R_k denote the operators defined in the paper from the coefficients of $e^{(A+B)t}$ and $e^{At}e^{Bt}$, respective...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037488 | MATHDB-372530

The HRT conjecture for time-frequency translates

摘录 / EXCERPT: HRT conjecture. The functions g_k, $1\le k\le N$, are linearly independent in $L^2(\mathbb R,\mathrm{d}x)$. This is a central open problem in time-frequency analysis, also called...

conjecture | open | MathDB 页面 | 原文线索

V2-037489 | MATHDB-372531

The Fock-space formulation of the HRT conjecture

摘录 / EXCERPT: HRT conjecture. The functions

conjecture | open | MathDB 页面 | 原文线索

V2-037490 | MATHDB-372537

Oberlin's conjecture on Hausdorff dimensions of unions of affine lines

摘录 / EXCERPT: Oberlin's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037491 | MATHDB-372551

Nonexistence of exotic ancient ovals

摘录 / EXCERPT: Let $M_t\subset\mathbb{R}^{n+1}$ be an ancient oval whose tangent flow at $-\infty$ is $\mathbb{R}^{k}\times S^{n-k}$. A k-oval is an ancient oval with this tangent flow whose hi...

conjecture | open | MathDB 页面 | 原文线索

V2-037492 | MATHDB-372664

The positive-area conjecture for non-removable curves

摘录 / EXCERPT: Let $\textrm{CH}(\gamma)$ be the set of homeomorphisms of the Riemann sphere that are conformal off a curve $c(cgamma)$. A curve is conformally removable if every element of...

conjecture | refuted | MathDB 页面 | 原文线索

V2-037493 | MATHDB-372695

Linear smoothing conjecture for Fourier integral operators

摘录 / EXCERPT: Linear smoothing conjecture for FIOs. For any $p\geq\overline{p}_d$ and $\sigma<\frac{1}{p}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037494 | MATHDB-372708

Arveson's hyperrigidity conjecture for operator systems

摘录 / EXCERPT: A separable operator system S is a unital, $*$-closed subspace of a C^*-algebra, and a unital completely positive map has the unique extension property (u.e.p.) if every unital...

conjecture | open | MathDB 页面 | 原文线索

V2-037495 | MATHDB-372726

The critical-index boundary conjecture for Shimorin-type operators

摘录 / EXCERPT: Let ν be a finite positive Borel measure on $[0,1]$, and let $T_\nu:L^p(\mathbb{D})\to L^q(\mathbb{D})$ be the associated Shimorin-type operator. Define ... where c' is the dua...

conjecture | open | MathDB 页面 | 原文线索

V2-037496 | MATHDB-372727

Endpoint boundedness conjecture for Shimorin-type operators

摘录 / EXCERPT: Let ν be a finite positive Borel measure on $[0,1]$, let $T_\nu:L^p(\mathbb{D})\to L^q(\mathbb{D})$ be the associated Shimorin-type operator, and let $\mathcal{C}$ be the bound...

conjecture | open | MathDB 页面 | 原文线索

V2-037497 | MATHDB-372761

Asymptotic conjecture for near-perfect powers of partition numbers

摘录 / EXCERPT: The asymptotic conjecture for $M_k(d)$. For $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-037498 | MATHDB-372765

Asymptotic conjecture for near-perfect powers of overpartition numbers

摘录 / EXCERPT: Let $\overline{p}(n)$ denote the overpartition function, and define $M_{\overline{p},k}(d)$ analogously to $M_k(d)$ as the largest index whose overpartition number lies within d...

conjecture | open | MathDB 页面 | 原文线索

V2-037499 | MATHDB-372793

Nieraeth's maximal-operator duality conjecture for s-concave Banach function spaces

摘录 / EXCERPT: Nieraeth's conjecture. If M is bounded on X, then M is bounded on X'.

conjecture | open | MathDB 页面 | 原文线索

V2-037500 | MATHDB-372844

Berger-Coburn conjecture on the borderline heat transform criterion

摘录 / EXCERPT: Let $H^2(\mathbb C^n,d\mu)$ be the Bargmann-Fock space, let g be measurable, and let k_a denote the normalized reproducing kernel at $a\in\mathbb C^n$. For $t>0$, define the he...

conjecture | open | MathDB 页面 | 原文线索

V2-037501 | MATHDB-372987

The symmetry-rigidity conjecture for proper holomorphic maps between balls

摘录 / EXCERPT: Let $f: \operatorname{\mathbb{B}}^m \rightarrow \operatorname{\mathbb{B}}^M$ be a proper holomorphic map, and let $\operatorname{\mathsf{G}}_f$ denote its symmetry group. Suppose t...

conjecture | open | MathDB 页面 | 原文线索

V2-037502 | MATHDB-372989

The lattice-symmetry rigidity conjecture for proper holomorphic maps between balls

摘录 / EXCERPT: Let $f: \operatorname{\mathbb{B}}^m \rightarrow \operatorname{\mathbb{B}}^M$ be a proper holomorphic map that extends to a Hölder-continuous map...

conjecture | open | MathDB 页面 | 原文线索

V2-037503 | MATHDB-372997

Holomorphic potential conjecture under the operator conjecture

摘录 / EXCERPT: Holomorphic potential conjecture. Assuming the limiting-measure and operator conjecture, there exists a set of full ν_κ-measure

conjecture | open | MathDB 页面 | 原文线索

V2-037504 | MATHDB-373002

The coarse-geometric rigidity conjecture for uniform Roe algebras

摘录 / EXCERPT: Uniform Roe algebra rigidity conjecture. The C^*-algebras $C^*_u(X)$ and $C^*_u(Y)$ are Morita equivalent (respectively, isomorphic) if and only if (X,d) and (Y,∂) are...

conjecture | open | MathDB 页面 | 原文线索

V2-037505 | MATHDB-373043

Multilinear Fourier extension conjecture for transversal curved hypersurfaces

摘录 / EXCERPT: Let $k\geq 2$. Let $S_1,\ldots,S_k\subseteq\mathbb{R}^{d}$ be smooth hypersurfaces equipped with measures $\sigma_1,\ldots,\sigma_k$, respectively. They are transversal if there is...

conjecture | open | MathDB 页面 | 原文线索

V2-037506 | MATHDB-373060

Fyodorov-Keating supercritical moment asymptotic conjecture for CβE

摘录 / EXCERPT: Let $\beta>0$, $k\in\mathbb{N}$, and $s\in\mathbb{R}_+$. Let $\mathsf{M}^{(\beta)}_N(k;s)$ denote the corresponding CβE field moment, and let $\mathfrak{c}^{(\beta)}(k;s)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-037507 | MATHDB-373099

Asymptotic growth conjecture for homogeneous Golombic and Levine sequences

摘录 / EXCERPT: Asymptotic growth conjecture. The golombic sequence γw is either bounded or

conjecture | open | MathDB 页面 | 原文线索

V2-037508 | MATHDB-373115

Connectedness conjecture for Anosov metrics on negatively curved 3-manifolds

摘录 / EXCERPT: Connectedness conjecture. The space of Anosov metrics on Σ is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-037509 | MATHDB-373117

The quotient-manifold criterion for Banach-Lie subgroups

摘录 / EXCERPT: Quotient-manifold criterion. H is a Banach-Lie subgroup if and only if G/H carries the structure of a Banach manifold for which q has a surjective differential at each point...

conjecture | open | MathDB 页面 | 原文线索

V2-037510 | MATHDB-373137

The unimodular Spencer conjecture for complex vectors

摘录 / EXCERPT: Let $n\geq 1$ be an integer, and let $a_1,\ldots,a_n\in\mathbb{C}^n$ satisfy ... A complex number of modulus 1 is unimodular, and a vector in $\mathbb{C}^n$ is a unimodular vector...

conjecture | open | MathDB 页面 | 原文线索

V2-037511 | MATHDB-373138

The complex Banach-Mazur distance conjecture for ℓ_1 and ℓ_∞

摘录 / EXCERPT: For $n\geq 1$, let $\ell_1^n(\mathbb{C})$ and $\ell_{\infty}^n(\mathbb{C})$ denote the n-dimensional complex sequence spaces, and let d_{BM} denote the Banach-Mazur distance be...

conjecture | open | MathDB 页面 | 原文线索

V2-037512 | MATHDB-373139

The complex cross-2 Banach-Mazur distance conjecture

摘录 / EXCERPT: The complex cross-2 distance conjecture. For every integer $n\geq 1$ and all $1\leq q\leq 2\leq p\leq\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-037513 | MATHDB-373182

Matkowski's piecewise-linear conjecture for continuous quasi graph-additive functions

摘录 / EXCERPT: Matkowski's conjecture. The function f solves this equation if and only if there exist $a,b\in\mathbb{R}$ with $a\neq b$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037514 | MATHDB-373189

Calabi's unboundedness conjecture for complete minimal hypersurfaces

摘录 / EXCERPT: Let $n\geq 2$, and let $\Sigma^{n}\subset\mathbb{R}^{n+1}$ be a complete minimal hypersurface. Calabi's conjecture. The hypersurface Σ^{n} must be unbounded. This is the un...

conjecture | open | MathDB 页面 | 原文线索

V2-037515 | MATHDB-373204

The p-adic Schanuel conjecture

摘录 / EXCERPT: Suppose $z_1,\ldots,z_n\in\mathbb{D}_p\subset\mathbb{C}_p$ are $\mathbb{Q}$-linearly independent, where $\mathbb{D}_p$ is the domain of the p-adic exponential function. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-037516 | MATHDB-373235

One-node branching conjecture for dyadic Walsh--Hadamard truncations

摘录 / EXCERPT: Let $\mathcal{C}$ be a set of columns of a dyadic-truncated Walsh--Hadamard matrix W_Φ with exactly one node, such that each branch is complete above the node. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037517 | MATHDB-373236

Optimality of the standard truncation among two-branch matrices

摘录 / EXCERPT: For fixed N, let $B_{N-1,K}$ and $W_N^{\rm opt}$ denote the two-branch and standard dyadic Walsh--Hadamard truncation matrices described in the paper, with $K=0,\dots,N$. Two-bra...

conjecture | open | MathDB 页面 | 原文线索

V2-037518 | MATHDB-373256

Indicator-element realization conjecture for Choquet decompositions of invariant means

摘录 / EXCERPT: Let Γ be a discrete abelian group, let $m\in\mathcal M_{\tau}(\Gamma)$ be an invariant mean, let σ be a Choquet decomposition of m, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037519 | MATHDB-373278

Non-trivial type characterization for no-dimensional colorful Tverberg convergence

摘录 / EXCERPT: Let X be a Banach space, let $k\ge 2$ be fixed, and let $n\ge 2$. For n-point sets $Q_1,\dots,Q_n\subset X$, define $\operatorname{Tv}(X,k,n)$ as the smallest radius factor suc...

conjecture | open | MathDB 页面 | 原文线索

V2-037520 | MATHDB-373318

Infinite-dimensional operator invariant-subspace conjecture

摘录 / EXCERPT: Let $f_1,\ldots,f_l,g\in\mathbb{C}\langle\underline{x}\rangle$. For a complex Hilbert space $\mathcal{H}$, let $\mathcal{B}(\mathcal{H})$ denote the bounded operators on...

conjecture | open | MathDB 页面 | 原文线索

V2-037521 | MATHDB-373323

The relative index comparison conjecture of Chang, Weinberger and Yu

摘录 / EXCERPT: Relative index comparison conjecture. There exists a map

conjecture | open | MathDB 页面 | 原文线索

V2-037522 | MATHDB-373441

Zhan's convergence conjecture for iterated Aluthge transforms

摘录 / EXCERPT: Let $A\in\mathbb{M}_n(\mathbb{C})$, and let $\Delta(A)=P^{1/2}UP^{1/2}$ be its Aluthge transform, where $A=UP$ is a polar decomposition with $P=|A|$. Write Δ^m for the m-f...

conjecture | open | MathDB 页面 | 原文线索

V2-037523 | MATHDB-373491

Rump's 100 Euro Conjecture for matrices

摘录 / EXCERPT: Rump's 100 Euro Conjecture. There exists $x\in\mathbb{R}^n\setminus\{0\}$ such that

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037524 | MATHDB-373492

B1nger--Seeger's 200 Euro Conjecture

摘录 / EXCERPT: B1nger--Seeger's 200 Euro Conjecture. There exists $x\in\mathbb{R}^n\setminus\{0\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037525 | MATHDB-373493

The weak union of uniform closed balls conjecture

摘录 / EXCERPT: Weak union of uniform closed balls conjecture. There exists $r'>0$ such that S is the union of closed balls with common radius r'.

conjecture | open | MathDB 页面 | 原文线索

V2-037526 | MATHDB-373549

The discrete restriction conjecture for toral eigenfunctions

摘录 / EXCERPT: Let $p\geq 2$ and $d\geq 2$. Let $\mathbb{T}^d=\mathbb{R}^d/\mathbb{Z}^d$ be the square torus, and let e_N be an eigenfunction of $\sqrt{-\Delta}$ with eigenvalue $2\pi N$. The d...

conjecture | open | MathDB 页面 | 原文线索

V2-037527 | MATHDB-373554

Damase's continuity approximation conjecture for positive definite functions on spheres

摘录 / EXCERPT: Let S^{n-1} be the unit sphere, and let a positive definite function on S^{n-1} be understood in the usual sense for functions on the sphere. Damase's continuity approximation...

conjecture | open | MathDB 页面 | 原文线索

V2-037528 | MATHDB-373599

Localization conjecture for strongly self-absorbing equivariant \C^*-algebras

摘录 / EXCERPT: Localization conjecture. D satisfies the localization condition with respect to every separable compact G-C^*-algebra B.

conjecture | open | MathDB 页面 | 原文线索

V2-037529 | MATHDB-373632

Koebe's conjecture on conformal uniformization by circle domains

摘录 / EXCERPT: A domain in the Riemann sphere $\widehat{\mathbb C}=\mathbb C\cup\{\infty\}$ is a connected open set. A circle domain is a domain whose complementary components are points or close...

conjecture | open | MathDB 页面 | 原文线索

V2-037530 | MATHDB-373633

He-Schramm rigidity conjecture for circle domains

摘录 / EXCERPT: A circle domain $\Omega\subset\widehat{\mathbb C}$ is a domain whose boundary components are points or circles. It is conformally rigid if every conformal map of...

conjecture | open | MathDB 页面 | 原文线索

V2-037531 | MATHDB-373638

Cross-ratio rigidity conjecture for surfaces in \mathcal{S}

摘录 / EXCERPT: Let $\mathcal{R}$ and $\mathcal{R}'$ be two surfaces in $\mathcal{S}$. Suppose the cell of $\mathcal{R}$ at position (i,j) has cross ratio $k_{i,j}$, while the cell of...

conjecture | open | MathDB 页面 | 原文线索

V2-037532 | MATHDB-373647

Linear growth conjecture for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $Q(n)$ denote the sequence defined by the perturbed Hofstadter recursion. Linear growth conjecture. The sequence satisfies ... Numerical experiments strongly indicate slope $1/2$...

conjecture | open | MathDB 页面 | 原文线索

V2-037533 | MATHDB-373648

Sublinear fluctuations conjecture for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $E(n)$ denote the fluctuation term in the decomposition of $Q(n)$ into its main term and fluctuation. Sublinear fluctuations conjecture. The fluctuation term satisfies ... This i...

conjecture | open | MathDB 页面 | 原文线索

V2-037534 | MATHDB-373649

Log-periodic modulation conjecture for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $Q(n)$ be the sequence, let $\log_2 n$ denote the base-two logarithm, and let $\rho(n)$ be a smaller correction term. Log-periodic modulation conjecture. There appears to exist...

conjecture | open | MathDB 页面 | 原文线索

V2-037535 | MATHDB-373650

Dyadic renormalization conjecture for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $E(n)$ denote the fluctuation term and let $\epsilon(n)$ denote a bounded perturbation. Dyadic renormalization conjecture. The fluctuation term approximately satisfies ... This h...

conjecture | open | MathDB 页面 | 原文线索

V2-037536 | MATHDB-373651

Dyadic clock structure conjecture for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $Q(n)$ be the sequence and define the clock sequences ... Let ϕ_1 and ϕ_2 be bounded or slowly varying functions. Dyadic clock structure conjecture. There appear to exi...

conjecture | open | MathDB 页面 | 原文线索

V2-037537 | MATHDB-373668

Four-domain additive inequality for the hyperbolic metric

摘录 / EXCERPT: Let $D_1,D_2\subsetneq\mathbb{C}$ be simply connected domains, let $z\in D_1\cap D_2$, and assume that $D_1\cup D_2$ is hyperbolic. The hyperbolic densities satisfy ... This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-037538 | MATHDB-373675

The limiting achievability-region conjecture for block-structured coded retrieval

摘录 / EXCERPT: Limiting achievability-region conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037539 | MATHDB-373739

Polynomial coercivity conjecture for the Ginzburg-Landau Hessian

摘录 / EXCERPT: Let $E”(u)$ be the Hessian of the Ginzburg-Landau energy at a minimizer u, and let $\rho(\kappa)^{-1}$ denote its coercivity constant. Assume the general hypotheses of the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-037540 | MATHDB-373771

Kovrizhkin's lacunary Logvinenko-Sereda conjecture

摘录 / EXCERPT: Kovrizhkin's conjecture. The Logvinenko-Sereda theorem holds for this lacunary set F^c.

conjecture | open | MathDB 页面 | 原文线索

V2-037541 | MATHDB-373774

Identity conjecture for the unitary-conjugation descent construction

摘录 / EXCERPT: Identity conjecture for the unitary-conjugation descent construction. The resulting class coincides with the original class $[u]$. In particular, the lift map is injective and its...

conjecture | open | MathDB 页面 | 原文线索

V2-037542 | MATHDB-373790

Nicola-Riccardi-Tilli convex-order conjecture for weighted Bergman functions

摘录 / EXCERPT: Let $\mathbb{B}_n$ be the Bergman ball, let $p\in(0,\infty)$ and $\alpha>n$, and let $A^p_\alpha(\mathbb{B}_n)$ denote the weighted Bergman space of holomorphic functions with norm...

conjecture | open | MathDB 页面 | 原文线索

V2-037543 | MATHDB-373797

The small-ball to Fourier-decay conjecture for polynomial images

摘录 / EXCERPT: Let $X\sim\mu$, where μ is a log-concave measure on $\mathbb{R}^n$, and let $f:\mathbb{R}^n\to\mathbb{R}$ be a polynomial of degree d. For a Borel measure ν on...

open_problem | open | MathDB 页面 | 原文线索

V2-037544 | MATHDB-373821

Cantor-weighted discrete restriction conjecture for the logarithmic phase

摘录 / EXCERPT: Let $\widetilde\nu$ be the Cantor measure pushed forward to $(0,1)$, let $H_{M-1}=\sum_{n=1}^{M-1}n^{-1}$ be the harmonic number, and set $M=\lfloor\sqrt t\rfloor$. Cantor-weighted...

conjecture | open | MathDB 页面 | 原文线索

V2-037545 | MATHDB-373835

Separation conjecture for Hilbert C*-submodules by bounded modular functionals

摘录 / EXCERPT: Separation conjecture. There does not exist any non-trivial bounded A-linear map from $\mathcal N$ to A that is equal to the zero map on $\mathcal M$.

conjecture | open | MathDB 页面 | 原文线索

V2-037546 | MATHDB-373890

Zero-measure conjecture for the sum of the $p=2$ value-set closure

摘录 / EXCERPT: Zero-measure conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037547 | MATHDB-373973

Normal-derivative Green-function conjecture for C-GNP domains

摘录 / EXCERPT: Green-function normal-derivative conjecture. As $4y4$ approaches $4x4$ along the normal,

conjecture | open | MathDB 页面 | 原文线索

V2-037548 | MATHDB-373976

The conjecture that the harmonic univalence threshold of the unit disk is $3/2$

摘录 / EXCERPT: Let ${\mathbb D}$ be the unit disk, let $\sigma_H({\mathbb D})$ denote the harmonic univalence threshold of ${\mathbb D}$, and write $\|\cdot\|$ for the Schwarzian norm on...

conjecture | open | MathDB 页面 | 原文线索

V2-037549 | MATHDB-373977

The conjecture that the harmonic threshold $3/2$ implies global homeomorphic extension

摘录 / EXCERPT: Let ${\mathbb D}$ be the unit disk, let f be a harmonic mapping in ${\mathbb D}$, and let $\|S_f\|$ denote its Schwarzian norm. A mapping f admits a homeomorphic extension to...

conjecture | open | MathDB 页面 | 原文线索

V2-037550 | MATHDB-373990

Twin Bubble Conjecture for constant-mean-curvature hypersurfaces

摘录 / EXCERPT: Let (M^{n+1},g) be a closed Riemannian manifold with $3\le n+1\le 7$, and let $c>0$. A closed hypersurface has constant mean curvature c if its mean curvature is equal to c a...

conjecture | open | MathDB 页面 | 原文线索

V2-037551 | MATHDB-373995

Existence of a symmetry-axis compatible background blow-up profile

摘录 / EXCERPT: There exist a time $T>0$ and smooth functions ... with smooth initial data ... for smooth f_1,f_2,f_3 and a smooth even angular profile $\phi(\theta)$ adapted to...

conjecture | open | MathDB 页面 | 原文线索

V2-037552 | MATHDB-374020

Sharp exponent conjecture for the bilinear Heisenberg tube estimate

摘录 / EXCERPT: Let the bilinear tube estimate in Theorem 1 be understood with its stated L^p exponent, and let the sharp exponent mean the smallest exponent for which that estimate holds. Sha...

conjecture | open | MathDB 页面 | 原文线索

V2-037553 | MATHDB-374034

The Kakeya maximal-function volume conjecture

摘录 / EXCERPT: Let $\mathbb{T}$ be a set of length-one δ-tubes in $\mathbb{R}^n$, and let $U(\mathbb{T})$ denote their union. Assume that the tubes have δ-separated directions. Kake...

conjecture | open | MathDB 页面 | 原文线索

V2-037554 | MATHDB-374105

Abate's weak Wolff-Denjoy conjecture for taut acyclic manifolds

摘录 / EXCERPT: Let $\mathscr{X}$ be a taut, acyclic complex manifold, meaning that $H_j(\mathscr{X},\mathbb{Z})=0$ for all $j\in\mathbb{N}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037555 | MATHDB-374170

Do-He-Robertson's coefficient-gap conjecture for large-genus Hurwitz numbers

摘录 / EXCERPT: Do-He-Robertson's coefficient-gap conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-037556 | MATHDB-374176

Yau's conjecture on infinitely many minimal surfaces

摘录 / EXCERPT: Let M be a closed three-dimensional manifold. Yau's conjecture. M contains infinitely many immersed minimal surfaces. This conjecture has been resolved: Song proved the asserti...

conjecture | open | MathDB 页面 | 原文线索

V2-037557 | MATHDB-374177

Equivariant Yau conjecture for infinitely many invariant minimal hypersurfaces

摘录 / EXCERPT: Let M^{n+1} be a closed Riemannian manifold equipped with a compact Lie group G acting by isometries, satisfying ... A closed embedded minimal hypersurface is G-invariant when...

conjecture | open | MathDB 页面 | 原文线索

V2-037558 | MATHDB-374192

Buttazzo-Gavitońe conjecture on the sharp upper bound for normalized torsional rigidity

摘录 / EXCERPT: Buttazzo-Gavitońe conjecture. The sharp upper bound for $\mathcal F$ in arbitrary dimension is

conjecture | open | MathDB 页面 | 原文线索

V2-037559 | MATHDB-374195

Interpolation conjecture for two-indexed Schatten quasi-norms

摘录 / EXCERPT: Interpolation conjecture. The quasi-Banach spaces defined in the paper are complex interpolation spaces, completely isometrically, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-037560 | MATHDB-374211

Semiclassical Weyl law for the Baouendi-Grushin example

摘录 / EXCERPT: On $\mathbb{T}^2$, let $X=\partial_x$, $Y=2(1-\cos x)\partial_y$, and let $\Delta_H=X^2+Y^2$. The singular set is $S=\{0\}\times\mathbb{T}$, and $c>0$ is the constant in the singul...

conjecture | open | MathDB 页面 | 原文线索

V2-037561 | MATHDB-374276

Higher-order finite-horizon correction conjecture for quantile Kelly

摘录 / EXCERPT: Fix $\alpha\in(0,1)$, and let $W^{\mathrm K}$ denote the ordinary Kelly optimizer. Suppose the local chamber/stratum geometry is nondegenerate near $W^{\mathrm K}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037562 | MATHDB-374277

Pointwise character bound extension to spherical functions on compact symmetric spaces

摘录 / EXCERPT: Let X be an irreducible symmetric space of compact type, with spectral parameter $\mu\in \Lambda^+\subset i\mathfrak{t}^*$ and spatial parameter $H\in\mathfrak{t}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-037563 | MATHDB-374317

Affine Fourier-spectrum threshold conjecture for distance sets

摘录 / EXCERPT: Affine Fourier-spectrum threshold conjecture. If, for some $\theta\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-037564 | MATHDB-374327

Exterior-domain rigidity conjecture for the discrete-time overdetermined heat problem

摘录 / EXCERPT: Exterior-domain rigidity conjecture. Complements of closed balls are the only exterior domains whose solution satisfies the discrete-time overdetermined boundary condition.

conjecture | open | MathDB 页面 | 原文线索

V2-037565 | MATHDB-374531

Volumetric zone conjecture on spherical zone unions

摘录 / EXCERPT: Let $d\geq 2$, let $u_1,\dots,u_n\in\mathbb{S}^{d-1}$, and let $\alpha\in[0,\pi/2]$. For a unit vector u, define the spherical zone ... For $j=1,\dots,n$, set ... Let σ_{d-1}...

conjecture | open | MathDB 页面 | 原文线索

V2-037566 | MATHDB-374557

Krzyż's conjecture for Taylor coefficients of zero-free disk maps

摘录 / EXCERPT: Krzyż's conjecture. For every $f\in\Omega$ and every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037567 | MATHDB-374595

Holevo-Utkin conjecture on sharp ℓ_p norms for zero-sum vectors

摘录 / EXCERPT: Holevo-Utkin conjecture. For $0<p\le 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037568 | MATHDB-374709

Extension beyond compact resolvent via regularization

摘录 / EXCERPT: Let $X_1,\ldots,X_r$ be spectral operators in the sense of Dunford on separable Hilbert spaces $\mathcal{H}_1,\ldots,\mathcal{H}_r$. Assume there are positive self-adjoint operator...

conjecture | open | MathDB 页面 | 原文线索

V2-037569 | MATHDB-374711

Non-Brjuno divergence conjecture for Bishop quadrics

摘录 / EXCERPT: Let μ be a complex number that is not a root of unity and $|\mu|=1$. A number μ is Brjuno when it satisfies the Brjuno condition referred to in the source. Let Q_{γ}...

open_problem | open | MathDB 页面 | 原文线索

V2-037570 | MATHDB-374763

Necessity of the conditions in the sufficiency theorem for vector-valued Runge-Kutta methods

摘录 / EXCERPT: Let a vector-valued explicit or implicit Runge-Kutta method have order p, and let the conditions in the sufficiency theorem referred to in the source be imposed on such a method....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037571 | MATHDB-374842

Conjectured stability for unequal block sizes

摘录 / EXCERPT: Let the conditions of the stability theorem hold, but now for block sizes n_i that are not necessarily equal. Define the mean block size and variance by ... If the relative standar...

conjecture | open | MathDB 页面 | 原文线索

V2-037572 | MATHDB-374900

Volume-ratio conjecture for linear images of the Schatten-1 ball

摘录 / EXCERPT: Let $M:\mathbb{R}^{d\times d}\to\boxed{?}$ be a linear map, and let ... be the Schatten-1 ball. Volume-ratio conjecture. Is it true that ... A positive answer would give an...

open_problem | open | MathDB 页面 | 原文线索

V2-037573 | MATHDB-374926

Akutagawa's radiality conjecture for Yamabe constants on products with Euclidean space

摘录 / EXCERPT: Akutagawa's radiality conjecture. This infimum is achieved by a function radial in the $\mathbb{R}^{m_2}$ variable.

conjecture | open | MathDB 页面 | 原文线索

V2-037574 | MATHDB-374938

Pal's permanent asymptotic conjecture for symmetric cost functions

摘录 / EXCERPT: Let $\mathcal{C}:[0,1]\times[0,1]\to\mathbb{R}$ be twice continuously differentiable up to the boundary, symmetric in its two variables, zero on the diagonal, and satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-037575 | MATHDB-374988

Ball-Frenkel strong polarization conjecture

摘录 / EXCERPT: Strong polarization conjecture. The inequality above holds for every set of n vectors in $\mathbb{S}^{d-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-037576 | MATHDB-375000

Finite-infinite separation conjecture for interval discrepancy

摘录 / EXCERPT: Let $\mathcal{S}=(\mathcal{I}_1,\mathcal{I}_2,\dots)$ be an infinite-length strategy, and define ... Here $\operatorname{disc}(n)$ is the optimal discrepancy for finite strategies of...

conjecture | open | MathDB 页面 | 原文线索

V2-037577 | MATHDB-375036

Balazs's weighted lower semi-frame conjecture for complete families

摘录 / EXCERPT: Let $\mathcal{F}$ be a family in a separable Hilbert space $\mathcal{H}$. The family is complete if its closed linear span is $\mathcal{H}$, and it is a weighted lower semi frame i...

conjecture | open | MathDB 页面 | 原文线索

V2-037578 | MATHDB-375037

Logarithmic biorthogonal-norm estimate for the Gaussian Gabor system

摘录 / EXCERPT: For $(m,n)\in\mathbb{Z}^2$, let $\gamma_{m,n,0,0}$ denote the biorthogonal element associated with the Gaussian Gabor atom indexed by (m,n), and let $c_{m,n}>0$. The Gaussian Gab...

open_problem | open | MathDB 页面 | 原文线索

V2-037579 | MATHDB-375039

Generic nondegenerate singularity conjecture for mean curvature flow

摘录 / EXCERPT: Generic nondegeneracy conjecture. For a generic compact embedded initial hypersurface L_0, the mean curvature flow L_t admits only spherical and nondegenerate cylindrical singu...

conjecture | open | MathDB 页面 | 原文线索

V2-037580 | MATHDB-375068

Clouâtre-Ostermann-Ransford conjecture on contractive homomorphisms

摘录 / EXCERPT: Clouâtre-Ostermann-Ransford conjecture. (i) If $\|\Lambda\|\leq 2$, then $\|\theta\|\leq 2$. (ii) If α is a complete contraction, θ is completely bounded, and...

conjecture | open | MathDB 页面 | 原文线索

V2-037581 | MATHDB-375120

The visibility conjecture for visible parts of planar compact sets

摘录 / EXCERPT: Let $K \subset \mathbb{R}^{2}$ be compact. For $\sigma \in[-1,1]$, let $\operatorname{vis}_{\sigma}(K)$ denote the points of K that are not followed by another point of K along...

conjecture | open | MathDB 页面 | 原文线索

V2-037582 | MATHDB-375121

Exceptional-set estimate for visible parts

摘录 / EXCERPT: Let $K \subset \mathbb{R}^{2}$ be compact, let $\alpha \in[0,1]$, and let $\operatorname{vis}_{\sigma}(K)$ be the visible part of K in direction σ. Exceptional-set estimat...

conjecture | open | MathDB 页面 | 原文线索

V2-037583 | MATHDB-375148

Harmonic-radius bound for epsilon-acK manifolds

摘录 / EXCERPT: Let $\epsilon\in(0,a]$, and consider ϵ-acK manifolds with parameters (n,d,k,i_0,a). Harmonic-radius conjecture. The $W^{2,p}$ harmonic radius is uniformly bounded below...

conjecture | open | MathDB 页面 | 原文线索

V2-037584 | MATHDB-375155

The direct-summand conjecture for norm-preserving matrix powers

摘录 / EXCERPT: Direct-summand conjecture. If

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037585 | MATHDB-375156

The foci conjecture for elliptical numerical ranges of block matrices

摘录 / EXCERPT: Foci conjecture. If $\operatorname{cl}(W(T))$ is an elliptical disk, then its foci are contained in $\{0,1,-1\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-037586 | MATHDB-375163

Friedgut-Kahn-Kalai-Keller spectral correlation conjecture

摘录 / EXCERPT: Friedgut-Kahn-Kalai-Keller spectral correlation conjecture. For any increasing Boolean functions $f,g:\{0,1\}^n\to\{0,1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-037587 | MATHDB-375188

Hamilton's scalar-curvature blow-up conjecture for Ricci flow

摘录 / EXCERPT: Let $(M,g(t))_{t\in[0,T)}$ be a solution to Ricci flow on a closed smooth n-dimensional Riemannian manifold, with T its maximal time. Hamilton's scalar-curvature blow-up conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-037588 | MATHDB-375228

The characterization of 3-Hilbert spaces by inductive fixed-point data

摘录 / EXCERPT: Let $\fX\in 3\mathsf{Vect}$ be an $O(3)$-fixed point. For every $F\colon 2\mathsf{Vect}\to\fX$, suppose that F is equipped with an $O(2)$-fixed point structure, and that...

conjecture | open | MathDB 页面 | 原文线索

V2-037589 | MATHDB-375295

Sharp Gaussian curvature inequality for noncompact self-shrinkers

摘录 / EXCERPT: Let $M^n\subset\mathbb{R}^{n+1}$ be a connected, complete, embedded self-shrinker with polynomial volume growth, and let A be its second fundamental form. Sharp Gaussian curvatur...

conjecture | open | MathDB 页面 | 原文线索

V2-037590 | MATHDB-375337

Unitization by folding finitely many infinite paths in a graph

摘录 / EXCERPT: Let a graph have all but finitely many vertices lying on finitely many infinite paths. Fold each of these paths so that it produces countably infinitely many edges between two dist...

conjecture | open | MathDB 页面 | 原文线索

V2-037591 | MATHDB-375341

Infinite-dimensional submajorization conjecture for spectral suprema

摘录 / EXCERPT: Let $\mathcal{M}$ be a semifinite von Neumann algebra endowed with a faithful normal semifinite trace τ, and let $\mathcal{P}$ be the positive cone of the space of all τ-...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037592 | MATHDB-375342

Dimension-free maximal symmetric modulus conjecture

摘录 / EXCERPT: Maximal symmetric modulus conjecture. For every dimension d, one may choose $c_{\vee}(d)=1$. The assertion would settle the corresponding finite-dimensional question in its stron...

conjecture | open | MathDB 页面 | 原文线索

V2-037593 | MATHDB-375343

Zhang's quasisymmetric modulus conjecture

摘录 / EXCERPT: For $Z\in\mathbb{M}_d$, let $|Z|_{\mathrm{qsym}}$ denote the quasisymmetric modulus used in Zhang's conjecture. For $A,B\in\mathbb{M}_d$, there are unitaries $U,V\in\mathbb{M}_d$....

conjecture | open | MathDB 页面 | 原文线索

V2-037594 | MATHDB-375352

Discrete Strichartz conjecture for the periodic linear KdV equation

摘录 / EXCERPT: Let $N \ge 1$ be an integer and let $\varepsilon > 0$. Suppose $u_0 \in L^2(\mathbb{T})$ is frequency-localized as ... Let $u(x,t)$ be the corresponding solution to the periodic line...

conjecture | open | MathDB 页面 | 原文线索

V2-037595 | MATHDB-375369

Stein's restriction conjecture for smooth surfaces with nonzero Gaussian curvature

摘录 / EXCERPT: Let ... be the graph of a smooth function h, and define the extension operator ... Write $q'=\frac{q}{q-1}$. Stein's restriction conjecture. If the Gaussian curvature of $\mathcal S$...

conjecture | open | MathDB 页面 | 原文线索

V2-037596 | MATHDB-375478

Uniform boundedness conjecture for bilinear quadratic functionals over disjoint strips

摘录 / EXCERPT: Uniform boundedness conjecture. For every $\epsilon>0$, uniformly over all collections ${\mathcal S}$ of disjoint strips,

conjecture | open | MathDB 页面 | 原文线索

V2-037597 | MATHDB-375503

The Green's function mass conjecture for the fractional Yamabe problem

摘录 / EXCERPT: Let (X,g_+) be a conformally compact Einstein manifold with compactification $(\overline X,\overline g)$ and boundary N, and suppose that $\Lambda^\gamma(N,[\overline g|_N])>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-037598 | MATHDB-375546

Yang's periodicity conjecture for products of entire functions and derivatives

摘录 / EXCERPT: Let f be a transcendental entire function and let k be a positive integer. For $k\geq 2$, suppose that either $ff^{(k)}$ is periodic or ... for some entire function h. Yang's p...

conjecture | open | MathDB 页面 | 原文线索

V2-037599 | MATHDB-375575

Brauchart-Hardin-Saff conjecture for the minimal Riesz 2-energy on the sphere

摘录 / EXCERPT: Brauchart-Hardin-Saff conjecture. The following asymptotic expansion holds:

conjecture | open | MathDB 页面 | 原文线索

V2-037600 | MATHDB-375597

Mizohata-Takeuchi conjecture for Fourier extension operators

摘录 / EXCERPT: Mizohata-Takeuchi conjecture. If Σ is convex, then for all $f\in L^2(\Sigma)$ and measurable weights $w\colon\mathbb{R}^n\to[0,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-037601 | MATHDB-375606

Second-order Chernoff approximation error conjecture

摘录 / EXCERPT: Let $(e^{tL})_{t\geq 0}$ be a C_0-semigroup in a Banach space $\mathcal{F}$ with generator $(L,D(L))$. Let G be a Chernoff function for L satisfying condition (N) or (N'). Fi...

conjecture | open | MathDB 页面 | 原文线索

V2-037602 | MATHDB-375702

The Margulis-type regular subalgebra conjecture for II$_1$ factors

摘录 / EXCERPT: Let M be a separable II$_1$ factor, and let $P\subset M$ be a regular von Neumann subalgebra, meaning that the normalizer of P in M generates M. Margulis-type conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-037603 | MATHDB-375713

Symmetry efficiency conjecture

摘录 / EXCERPT: Let $G_{\mathrm{data}} \subseteq G$ act on M, and let Γ be a continuous boundary whose symmetry group contains G_{data}. For a target precision $\varepsilon>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-037604 | MATHDB-375783

Conjecture on the location of zeros of the Padé denominator

摘录 / EXCERPT: Consider the $[P/P]$-Padé approximant $R_P(z)=T_P(z)/Q_P(z)$, where T_P and Q_P have degree P and are computed from the first $2P+1$ Taylor coefficients of f. Let t denot...

conjecture | open | MathDB 页面 | 原文线索

V2-037605 | MATHDB-375934

Nielsen-Zhou conjecture on Schauder-basis failure of Daubechies wavelet packets

摘录 / EXCERPT: Let a Daubechies filter have length at least 4, and let its associated basic wavelet packets be the functions obtained from the corresponding low-pass/high-pass filter pair. For...

conjecture | open | MathDB 页面 | 原文线索

V2-037606 | MATHDB-375959

The rational Robba-ring admissibility conjecture

摘录 / EXCERPT: Let $\mathcal R_{\mathbb Q}=\bigcup_N\mathcal R_{1/N}$ be the rational Robba ring over $\mathbb C_p$, completed under the Gauss norm. Rational Robba-ring admissibility conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-037607 | MATHDB-375962

Fourier-frame spectrality conjecture for non-overlapping Bernoulli convolutions

摘录 / EXCERPT: Let $\mu_{\rho,d}$ be the Bernoulli convolution with contraction parameter $0<\rho\le1/2$ and scale parameter $d>0$. A measure is spectral when its associated L^2 space admits an...

conjecture | open | MathDB 页面 | 原文线索

V2-037608 | MATHDB-376051

Logarithmic upper-bound conjecture for the gap in minimal isogeny degrees

摘录 / EXCERPT: Logarithmic-gap conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-037609 | MATHDB-376114

Lipschitz extension spectral synthesis conjecture for virtual persistence diagram groups

摘录 / EXCERPT: Let (X,d,A) be a separable metric pair, let E be a complex Banach space, and let $f\colon K(X,A)\to E$ be Lipschitz. Lipschitz extension spectral synthesis conjecture. Every su...

conjecture | open | MathDB 页面 | 原文线索

V2-037610 | MATHDB-376115

Conjecture on polynomial convexity and analytic discs for the surfaces M_t

摘录 / EXCERPT: Polynomial-convexity and analytic-disc conjecture. The surface M_t is locally polynomially convex if $t>1$. The local polynomial hull of M_t contains an analytic disc for $t<1$...

conjecture | open | MathDB 页面 | 原文线索

V2-037611 | MATHDB-376126

Near-isometric duality conjecture for Hardy norms of matrices

摘录 / EXCERPT: Near-isometric duality conjecture. For every $n\times n$ matrix A with $n>2$,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037612 | MATHDB-376229

Strichartz-Tse conjecture on integrability of harmonic energy-measure densities

摘录 / EXCERPT: Let K be the Sierpiński gasket. For a nonconstant harmonic function h on K, let $u_h=\mu_h/\mu_h(K)$ be its normalized energy measure. For nonconstant harmonic functions...

conjecture | open | MathDB 页面 | 原文线索

V2-037613 | MATHDB-376238

Strong Ramadanov conjecture for strictly pseudoconvex domains

摘录 / EXCERPT: Strong Ramadanov conjecture. If $\rho \in C^{n+2}(\overline{D})$, then ∂D is locally spherical.

conjecture | open | MathDB 页面 | 原文线索

V2-037614 | MATHDB-376323

Dyadic one-function form of Zygmund's conjecture

摘录 / EXCERPT: Let $\Phi:\mathbb{Z}^{d-1}\to\mathbb{Z}$ be monotone increasing in each argument, and let $\mathcal{Z}_d$ be the collection of dyadic rectangles in $\mathbb{R}^d$ whose sidelengths...

conjecture | open | MathDB 页面 | 原文线索

V2-037615 | MATHDB-376324

Antichain k-wise intersection conjecture

摘录 / EXCERPT: Let $d\geq2$, and let $\mathcal{E}$ be a $\frac12$-sparse antichain of d-dimensional dyadic rectangles. Let $\operatorname{sh}(\mathcal{E})$ be the shadow of $\mathcal{E}$. Antic...

conjecture | open | MathDB 页面 | 原文线索

V2-037616 | MATHDB-376369

David-Mayboroda conjecture on the weak geometric lemma for uniform domains

摘录 / EXCERPT: David-Mayboroda conjecture. Under these hypotheses, the Weak Geometric Lemma implies the Bilateral Weak Geometric Lemma, and consequently uniform rectifiability of E.

conjecture | open | MathDB 页面 | 原文线索

V2-037617 | MATHDB-376396

Locality conjecture for the Schwartz space on an L-monoid

摘录 / EXCERPT: Let G be a reductive group over F, let ρ be the representation defining the reductive monoid X_ρ, and let $C^(X_\rho(F))$ be the space of locally constant functions o...

conjecture | open | MathDB 页面 | 原文线索

V2-037618 | MATHDB-376397

Compactly supported functions conjecture for Schwartz spaces on L-monoids

摘录 / EXCERPT: Let G be a reductive group over F, let ρ define the reductive monoid X_ρ, and let $C^\infty_c(X_\rho(F))$ be the space of compactly supported locally constant functio...

open_problem | open | MathDB 页面 | 原文线索

V2-037619 | MATHDB-376398

Euler-Schwartz inclusion conjecture for L-monoids

摘录 / EXCERPT: Let G be a reductive group over F, let ρ define the reductive monoid X_ρ, and write $\mathcal{S}_{\mathrm{ES}}(X_\rho(F)):=C^\infty_c(X_\rho(F))$. View these function...

conjecture | open | MathDB 页面 | 原文线索

V2-037620 | MATHDB-376420

The symmetric Pisier estimate for the $ ell$-functional

摘录 / EXCERPT: Let $K\subset\mathbb R^n$ be an origin-symmetric convex body, let GL_n denote the group of invertible linear transformations of $\mathbb R^n$, and let $\ell(K)$ be the product fu...

conjecture | open | MathDB 页面 | 原文线索

V2-037621 | MATHDB-376528

Log-convexity conjecture for admissible bundles

摘录 / EXCERPT: Let $\bar{M}$ be the curve under consideration, and let $\cal{L}$ and $\cal{M}$ be two bundles on $\bar{M}$ given by gluing conditions. Log-convexity conjecture. If both...

conjecture | open | MathDB 页面 | 原文线索

V2-037622 | MATHDB-376602

The smallest singular value conjecture for Gaussian matrix perturbations

摘录 / EXCERPT: Let $\widetilde{A}$ be an arbitrary square matrix in $\mathbb{R}^{n\times n}$, and let A be a Gaussian perturbation of $\widetilde{A}$ with variance σ^2. Smallest singular...

conjecture | open | MathDB 页面 | 原文线索

V2-037623 | MATHDB-376603

The conjectured polynomial-exponential tail bound for Gaussian LU growth with partial pivoting

摘录 / EXCERPT: Let $\widetilde{A}$ be an $n\times n$ matrix with $\|\widetilde{A}\|_2\leq 1$, and let A be a Gaussian perturbation of $\widetilde{A}$ with variance $\sigma^2\leq1$. Let U be t...

conjecture | open | MathDB 页面 | 原文线索

V2-037624 | MATHDB-376604

The conjectured optimal smoothed tail bound for the condition number

摘录 / EXCERPT: Let $\widetilde{A}$ be an $n\times n$ matrix satisfying $\|\widetilde{A}\|_2\leq\sqrt{n}$, and let A be a Gaussian perturbation of $\widetilde{A}$ with variance $\sigma^2\leq1$....

conjecture | open | MathDB 页面 | 原文线索

V2-037625 | MATHDB-376609

The clique function's maximal flat-spectrum conjecture for the \{I,H\}^n transform set

摘录 / EXCERPT: Let n be a positive integer, let $\{I,H\}^n$ denote the transform set consisting of n choices between the identity transform I and the Hadamard transform H, and let the cli...

conjecture | open | MathDB 页面 | 原文线索

V2-037626 | MATHDB-376653

Existence conjecture for harmonic extensions of admissible boundary maps

摘录 / EXCERPT: Harmonic extension conjecture. The admissible boundary map can be extended to a harmonic map from X to Y with bounded energy density if and only if the inverse image under the...

conjecture | open | MathDB 页面 | 原文线索

V2-037627 | MATHDB-376719

The largest-unitization conjecture for the Moyal plane

摘录 / EXCERPT: Let $\mathcal{A}_1=\mathcal{D}_{L^2}(\mathbb{R}^{2N})$ be the algebra of Moyal multipliers acting on the relevant Hilbert space, let $\mathcal{B}(\mathbb{R}^{2N})$ denote the propo...

conjecture | open | MathDB 页面 | 原文线索

V2-037628 | MATHDB-376774

Exponentials generate holomorphic functions on simply connected convex regions

摘录 / EXCERPT: Let R be a simply connected convex region of a discrete Riemann surface, and let the associated discrete exponentials be holomorphic functions on R. Exponential-generation conj...

conjecture | open | MathDB 页面 | 原文线索

V2-037629 | MATHDB-376775

Ramification bound and attainment for discrete polynomial maps

摘录 / EXCERPT: Let f be a polynomial of degree k on a discrete Riemann surface, and let the ramification number of f at a cycle γ be its winding number around the origin, when define...

conjecture | open | MathDB 页面 | 原文线索

V2-037630 | MATHDB-376807

Agranovsky-Quinto conjecture on higher-dimensional non-injectivity sets

摘录 / EXCERPT: Let $S\subset\mathbb{R}^n$ and let $C_c(\mathbb{R}^n)$ denote the compactly supported continuous functions on $\mathbb{R}^n$. A rigid motion of $\mathbb{R}^n$ is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-037631 | MATHDB-376820

One-dimensional Maxwell bound for critical points of point-charge potentials

摘录 / EXCERPT: One-dimensional Maxwell bound. For any values of the charges, the function $V_{\alpha}^*(x)$ has at most $2l-1$ real critical points. This is a stronger one-dimensional version of...

conjecture | open | MathDB 页面 | 原文线索

V2-037632 | MATHDB-376832

Asymptotic cosine conjecture for elliptic Faulhaber polynomials

摘录 / EXCERPT: Let $\mathcal{F}_m(\lambda)$ denote the elliptic Faulhaber polynomial and let $\mathcal{B}_{2m-2}$ denote the corresponding elliptic Bernoulli number. For real x in a finite inte...

conjecture | open | MathDB 页面 | 原文线索

V2-037633 | MATHDB-376974

The characterization of non-strictly singular embeddings into L_1

摘录 / EXCERPT: Embedding conjecture. The natural embedding $E\subset L_1([0,1])$ is not strictly singular if and only if G embeds continuously into E.

conjecture | open | MathDB 页面 | 原文线索

V2-037634 | MATHDB-377000

The AF-embedding conjecture for separable exact quasidiagonal C*-algebras

摘录 / EXCERPT: Let A be a separable exact quasidiagonal C^*-algebra. AF-embedding conjecture. A is isomorphic to a subalgebra of an AF algebra. The preceding theorem gives strong evidence f...

conjecture | open | MathDB 页面 | 原文线索

V2-037635 | MATHDB-377036

The Kakeya conjecture in maximal-function form

摘录 / EXCERPT: Fix $n\geq 2$ and $0<\delta\ll 1$. Let Ω be a maximal δ-separated subset of directions on S^{n-1}, and for each $\omega\in\Omega$ let T_ω be a...

conjecture | open | MathDB 页面 | 原文线索

V2-037636 | MATHDB-377158

Non-embedding conjecture for complete pluriminimal immersions of complex two-space

摘录 / EXCERPT: Let $f:\mathbb{C}^2\to\mathbb{R}^6$ be a complete pluriminimal immersion. Non-embedding conjecture. The immersion f is not an embedding. The claim is motivated by the observation...

conjecture | open | MathDB 页面 | 原文线索

V2-037637 | MATHDB-377203

Automatic continuity and spatial form of biseparating maps between C*-algebras

摘录 / EXCERPT: Let $\mathcal{A}_{1}$ and $\mathcal{A}_{2}$ be $C^{\ast}$-algebras, and let $T:\mathcal{A}_{1}\to\mathcal{A}_{2}$ be a biseparating map, meaning that both T and T^{-1} preserve...

conjecture | open | MathDB 页面 | 原文线索

V2-037638 | MATHDB-377218

The real rank versus completely positive rank conjecture for C*-algebras

摘录 / EXCERPT: Real-rank inequality conjecture. For every C^*-algebra A, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037639 | MATHDB-377220

The coefficient ratio conjecture for complete Nevanlinna-Pick kernels

摘录 / EXCERPT: Coefficient ratio conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-037640 | MATHDB-377221

Generic asymptotic eigenvalue-counting conjecture for piecewise continuous elliptic systems

摘录 / EXCERPT: Let $L f(x)=A(x)f'(x)$ on $L^2((\alpha,\beta),\mathbf C^n)$, where $A(x)$ is invertible and diagonalizable for each $x\in(\alpha,\beta)$, with boundary conditions $f(\alpha)=u$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-037641 | MATHDB-377222

Weakly real-constructible ind-sheaves are closed under ind-internal Hom and tensor product

摘录 / EXCERPT: Let X be a real analytic manifold, let $F\in D^b_{{\rm w-}\mathbb{R}\text{-c}}(\mathrm{I}[k_X])$, and let $G\in D^b_{{\rm w-}\mathbb{R}\text{-c}}(k_X)$. Closure conjecture. The o...

conjecture | open | MathDB 页面 | 原文线索

V2-037642 | MATHDB-377255

The self-dual relaxed lattice equality conjecture

摘录 / EXCERPT: Self-dual relaxed lattice conjecture. In every dimension, the largest possible value of r_1 among self-dual relaxed lattices equals the smallest value of r possible in Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-037643 | MATHDB-377274

Conditional expectation conjecture for contractive projections on rearrangement-invariant spaces

摘录 / EXCERPT: Let X be a rearrangement-invariant function space on $[0,1]$, so that X contains the constant functions, and let P be a contractive projection on X satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-037644 | MATHDB-377275

Block-structure conjecture for norm-one projections with unconditional bases

摘录 / EXCERPT: Let X be a strictly monotone Banach space with a 1-unconditional basis, sufficiently different from a Hilbert space. Block-structure conjecture. Every norm-one projection on...

conjecture | open | MathDB 页面 | 原文线索

V2-037645 | MATHDB-377276

Finite-dimensional characterization conjecture for spaces with complemented hyperplanes

摘录 / EXCERPT: Let X be a finite-dimensional Banach space, with $n=\dim X$. Finite-dimensional characterization conjecture. (a) If every one-dimensional subspace of X can be represented as an...

conjecture | open | MathDB 页面 | 原文线索

V2-037646 | MATHDB-377331

Bellissard's gap-labelling conjecture

摘录 / EXCERPT: Bellissard's gap-labelling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037647 | MATHDB-377472

Carleson-Hunt estimate for the scattering transform

摘录 / EXCERPT: Let F be the scattering data and let $a(x)$ denote the upper-left entry of the associated matrix G as a function of the parameter k, with x ranging over the spatial variabl...

conjecture | open | MathDB 页面 | 原文线索

V2-037648 | MATHDB-377553

Connectedness conjecture for minimizing double bubbles in the three-torus

摘录 / EXCERPT: Connectedness conjecture. An area-minimizing double bubble in T^3 has connected regions and complement.

conjecture | open | MathDB 页面 | 原文线索

V2-037649 | MATHDB-377554

Small-volume standard double-bubble conjecture in the three-torus

摘录 / EXCERPT: Small-volume standard double-bubble conjecture. For two small volumes the standard double bubble is optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-037650 | MATHDB-377555

Slab Lens conjecture for one very small volume

摘录 / EXCERPT: Slab Lens conjecture. For one very small volume and two moderate volumes, the Slab Lens is optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-037651 | MATHDB-377556

Delaunay-chain first-transition conjecture for equal small volumes

摘录 / EXCERPT: Delaunay-chain transition conjecture. The first phase transition as small equal, or close to equal, volumes grow is from the standard double bubble to a chain of two bubbles bounde...

conjecture | open | MathDB 页面 | 原文线索

V2-037652 | MATHDB-377559

No periodic three-volume foam conjecture for the cubic three-torus

摘录 / EXCERPT: No periodic three-volume foam conjecture. There are no least-area divisions of T^3 into three volumes that lift to a foam in $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-037653 | MATHDB-377560

Complete classification conjecture for double bubbles in the three standard tori

摘录 / EXCERPT: Complete-classification conjecture. The double bubbles of Figure 1 together with the Hexagonal Honeycomb of Figure 4 comprise the complete set of area-minimizing double bubbles for...

conjecture | open | MathDB 页面 | 原文线索

V2-037654 | MATHDB-377561

Small-volume triple-bubble attachment conjecture in the cubic three-torus

摘录 / EXCERPT: Small-volume triple-bubble conjecture. The minimizers will look like the double bubbles of Figure 2 with a small ball attached, and the phase diagram will look just like Figure 3.

conjecture | open | MathDB 页面 | 原文线索

V2-037655 | MATHDB-377572

Conjecture on the distance formula and strong exposedness of the functions f_β

摘录 / EXCERPT: Let A^1 be the Bergman space, let $(A^1)^\perp$ denote its annihilator in the relevant dual space, let C denote the subspace of constant functions, and let φ_β and...

conjecture | open | MathDB 页面 | 原文线索

V2-037656 | MATHDB-377625

A characterization of interpolating sequences for multipliers of weighted Dirichlet spaces

摘录 / EXCERPT: Let w_n be given by equation (15), let ${\mathcal H}_{\mathbf w}$ be the corresponding Hilbert space, and let $Mult({\mathcal H}_{\mathbf w})$ denote its multiplier algebra. A...

open_problem | open | MathDB 页面 | 原文线索

V2-037657 | MATHDB-377626

Montgomery-Smith-Semenov hypothesis on signed sums in the integral-uniform norm

摘录 / EXCERPT: Montgomery-Smith-Semenov hypothesis. There exist signs $\theta_i\in\{-1,1\}$ and a constant $c_0>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037658 | MATHDB-377627

Weaker sufficient estimate for systems satisfying condition (d)

摘录 / EXCERPT: Let $\{f_i\}_{i=1}^n$ be a system of functions satisfying condition (d) with parameter $p\in[0,1/2)$. For each fixed $x\in X$, let $f_k^*(x)$ denote the decreasingly ordered values...

open_problem | open | MathDB 页面 | 原文线索

V2-037659 | MATHDB-377895

Relative perturbation theory criterion for homogeneous polynomials

摘录 / EXCERPT: Let $f(x_1,\ldots,x_n)$ be a homogeneous polynomial and let $\mathcal D$ be as in the preceding conjecture. Thus condition (A) means that f can be written as ... where each f_m...

conjecture | open | MathDB 页面 | 原文线索

V2-037660 | MATHDB-377960

Patterson-Wiedemann conjecture on the minimum Fourier amplitude of Boolean functions

摘录 / EXCERPT: Patterson-Wiedemann conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-037661 | MATHDB-377966

The lacunary-sided polygon restricted maximal conjecture

摘录 / EXCERPT: Restricted lacunary-sided polygon conjecture. For $2\le p<\infty$, the maximal operator $f\mapsto\sup_{k\in\mathbb N}|T_kf|$ maps $L^p(\mathbb R^2)$ into itself. The source calls t...

conjecture | open | MathDB 页面 | 原文线索

V2-037662 | MATHDB-377972

Connectedness conjecture for least-perimeter disk partitions

摘录 / EXCERPT: Let a minimizing graph be a least-perimeter partition of the unit disk into regions of prescribed areas. Connectedness conjecture. A minimizing graph separates the disk into connec...

conjecture | open | MathDB 页面 | 原文线索

V2-037663 | MATHDB-378016

Conjecture on phase portraits of minimizing double bubbles on flat tori

摘录 / EXCERPT: Let $A_0\geq A_2\geq A_1$ be areas, and let a flat torus have area $A_0+A_1+A_2$. A non-tiling minimizing double bubble is a perimeter-minimizing double bubble enclosing areas...

conjecture | open | MathDB 页面 | 原文线索

V2-037664 | MATHDB-378034

Spectral variation conjecture for normal operators and differentiator compressions

摘录 / EXCERPT: Let A be a normal operator on a complex Hilbert space of dimension $n\ge 2$. Let $\rho(A)$ denote the radius of the smallest closed disk containing the spectrum of A, let A'...

conjecture | open | MathDB 页面 | 原文线索

V2-037665 | MATHDB-378139

Miller's quadratic refinement conjecture for Sendov's problem

摘录 / EXCERPT: Let $S(\beta)$ be the set of polynomials of degree at least 2 with complex coefficients, all roots in the unit disk and at least one root at β. For $P\in S(\beta)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-037666 | MATHDB-378162

The majorization conjecture for twisted root maps

摘录 / EXCERPT: Majorization conjecture. If $T\in\mathcal{A}'$ and $n\ge 1$, then for every $P\in\mathcal{C}_n$ at least one of the relations

conjecture | open | MathDB 页面 | 原文线索

V2-037667 | MATHDB-378272

The vanishing-power conjecture for smooth functions on projective space

摘录 / EXCERPT: Let k and ℓ be positive integers. Vanishing-power conjecture. There exists a smooth nonconstant function v on $\mathbb{P}^k$ such that ... if and only if ... The proposed thre...

conjecture | open | MathDB 页面 | 原文线索

V2-037668 | MATHDB-378309

Diederich-Pinchuk extension conjecture for proper holomorphic mappings

摘录 / EXCERPT: Let $D,D'\subset\mathbb{C}^n$, with $n\geq 2$, be bounded domains whose boundaries are real analytic and geometrically smooth, and let $h:D\to D'$ be a proper holomorphic mapping....

conjecture | open | MathDB 页面 | 原文线索

V2-037669 | MATHDB-378363

Higher-order differentiability conjecture for spectral functions

摘录 / EXCERPT: Higher-order differentiability conjecture. The spectral function $f\circ\lambda$ is k times continuously differentiable at X if and only if f is k times continuously differ...

conjecture | open | MathDB 页面 | 原文线索

V2-037670 | MATHDB-378378

Paley-Wiener conjecture for the Dunkl transform

摘录 / EXCERPT: Let G be a finite reflection group. For a subset $S\subset\mathfrak a$, let ${\mathcal D}(S)$ denote the smooth compactly supported functions with support contained in S. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-037671 | MATHDB-378472

Baugher's integral identity for the Morse-index coefficient

摘录 / EXCERPT: Let m be the complex dimension, let $m\leq q\leq 2m$, and let Y_{2m-q}, $\Delta(\lambda)$, $\Delta(\xi)$ and the integration variables be as in the source's preceding definitio...

conjecture | open | MathDB 页面 | 原文线索

V2-037672 | MATHDB-378480

Minimal condition number conjecture for lattices

摘录 / EXCERPT: Let $R_{g,\Lambda}$ denote the matrix associated with the Gaussian g and a lattice Λ, and let ${x_1}(\Lambda)$ denote the lattice parameter used in the source. Minimal co...

conjecture | open | MathDB 页面 | 原文线索

V2-037673 | MATHDB-378481

Weighted decay conjecture for integral operators

摘录 / EXCERPT: Weighted decay conjecture. There exists $\delta>0$ such that $h_L\in\mathcal L^1_{\tau_\delta}$. If true, this would support identifying $\mathcal L^1$ with a union of the weighted...

conjecture | open | MathDB 页面 | 原文线索

V2-037674 | MATHDB-378482

Decay preservation for quotient spaces of integral-operator algebras

摘录 / EXCERPT: Quotient-space decay conjecture. The spaces $\mathcal L^{1'}_v$ and $\mathcal L'_v$ possess the same decay properties as $\mathcal L^1_v$ and $\mathcal L_v$, respectively. The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-037675 | MATHDB-378519

Boundary convergence conjecture for type II parabolic self-maps

摘录 / EXCERPT: Let ϕ be an analytic self-map of the unit disk $\mathbb{D}$, and write $\phi_n=\phi^{\circ n}$ for its n-th iterate. Suppose that ϕ is type II parabolic, meaning that i...

conjecture | open | MathDB 页面 | 原文线索

V2-037676 | MATHDB-378542

Approximate symmetry conjecture for real Painlevé I solutions

摘录 / EXCERPT: Let $y_0\in\mathbb R$ and $y_l\leq y_0$ define a first solution of the first Painlevé equation, and let the second solution be defined by the data $-y_l$ and $-y_0$. Let Δ b...

conjecture | open | MathDB 页面 | 原文线索

V2-037677 | MATHDB-378575

The AW*-factor problem for type II$_1$ factors

摘录 / EXCERPT: Let an AW-algebra be a unital C-algebra whose left annihilator of every nonempty subset is a principal right ideal generated by a projection. An AW-factor is an AW-algebra that is...

open_problem | open | MathDB 页面 | 原文线索

V2-037678 | MATHDB-378591

The quasitrace conjecture for arbitrary C*-algebras

摘录 / EXCERPT: Quasitrace conjecture. Quasitraces on arbitrary C-algebras are traces.

conjecture | open | MathDB 页面 | 原文线索

V2-037679 | MATHDB-378630

Interlacing conjecture for polynomial recurrences on the equimodular discriminant

摘录 / EXCERPT: Let $p_{n+1}(z)=\sum_{i=1}^{k}Q_i(z)p_{n-i}(z)$ be a finite recurrence with fixed polynomial coefficients, and suppose $\deg p_n(z)=n$ for $n\in\mathbb N$. Let Ξ_Q be the equim...

conjecture | open | MathDB 页面 | 原文线索

V2-037680 | MATHDB-378712

Noncommutative Q-Gauss summations for basic hypergeometric series

摘录 / EXCERPT: Let A, B, and C be noncommutative parameters in a Banach algebra, with Q commuting with each of A, B, and C. Assume that $BC^{-1}A$ commutes with each of A, B, an...

conjecture | open | MathDB 页面 | 原文线索

V2-037681 | MATHDB-378818

Unique limit tangent cone conjecture for minimal surfaces

摘录 / EXCERPT: Let M be a properly immersed minimal surface in ${\mathbb R}^3$ with quadratic area growth, meaning that its area growth constant $A(M)$ is finite. A limit tangent cone at infini...

conjecture | open | MathDB 页面 | 原文线索

V2-037682 | MATHDB-378833

Uniqueness conjecture for the square grid Z^c

摘录 / EXCERPT: Let Z^c be the square grid circle pattern with $0<c<2$, whose circles are labeled by ... and which covers the infinite sector with angle $c\pi/2$, with the centers of the border ci...

conjecture | open | MathDB 页面 | 原文线索

V2-037683 | MATHDB-378912

Nonpositive-curvature local diffeomorphism conjecture for cusp metrics

摘录 / EXCERPT: Nonpositive-curvature local diffeomorphism conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037684 | MATHDB-378926

The sheaf conjecture for continuous valuations

摘录 / EXCERPT: Let X be a manifold, and for each open subset $U\subset X$ let ${\cal S}_i(U)$ be the space of continuous valuations on U. These spaces form a presheaf ${\cal S}_{i,X}$ on X....

conjecture | open | MathDB 页面 | 原文线索

V2-037685 | MATHDB-378935

Kalai's transitive-symmetric low-level Fourier-weight conjecture

摘录 / EXCERPT: For $d\geq1$, let C_n be the collection of odd, transitive-symmetric functions $f:\{-1,1\}^n\to\{-1,1\}$. Let $\widehat f(S)$ denote Fourier coefficients, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037686 | MATHDB-378996

Eventual periodic representative conjecture for asymptotically periodic sequences

摘录 / EXCERPT: Let $S({\bf C}^{N})^{\infty}_{\mathrm{AP}}$ denote the asymptotically periodic sequences and let $S({\bf C}^{N})^{\infty}_{\mathrm{EP}}$ denote the eventually periodic sequences. W...

conjecture | open | MathDB 页面 | 原文线索

V2-037687 | MATHDB-379032

The asymptotic spherical-distribution conjecture for the angular sum

摘录 / EXCERPT: Let $\mathcal{Q}_N$ denote the configuration space of N points in three-dimensional space, and let F_N be the angular sum defined in the paper. The asymptotic spherical-distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-037688 | MATHDB-379079

Scalar-curvature volume-entropy conjecture for closed 3-manifolds

摘录 / EXCERPT: Volume-entropy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037689 | MATHDB-379083

The superreflexive fixed-point conjecture for higher rank groups and lattices

摘录 / EXCERPT: Let $G=\prod_{i=1}^{m}\mathbf{G}_{i}(k_{i})$, where each k_i is a local field and each $\mathbf{G}_{i}(k_i)$ is the group of k_i-points of a Zariski connected simple k_i-alge...

conjecture | open | MathDB 页面 | 原文线索

V2-037690 | MATHDB-379116

Hillar's commuting-minimizers conjecture for the Bessis-Moussa-Villani problem

摘录 / EXCERPT: Hillar's commuting-minimizers conjecture. Such matrices A and B commute.

conjecture | open | MathDB 页面 | 原文线索

V2-037691 | MATHDB-379171

Quasi-free Hilbert-module p-reductivity conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ be a finite-rank, quasi-free, p-reductive Hilbert module over $A(\Omega)$, and let $\mathcal{S}$ be a submodule such that $\mathcal{S}^{\bot}$ is spanned by gen...

conjecture | open | MathDB 页面 | 原文线索

V2-037692 | MATHDB-379176

The coarse geometric Novikov conjecture

摘录 / EXCERPT: The coarse geometric Novikov conjecture. The map Ind is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-037693 | MATHDB-379228

Iversen-type conjecture for planar harmonic functions

摘录 / EXCERPT: Iversen-type conjecture. If the critical set of f is nowhere dense and the cluster set of f has empty interior, then every cluster point of f is an asymptotic value. In parti...

conjecture | open | MathDB 页面 | 原文线索

V2-037694 | MATHDB-379256

The linear restriction conjecture for curved hypersurfaces

摘录 / EXCERPT: Let $d\geq 2$, let U be a compact neighbourhood of the origin in $\mathbb{R}^{d-1}$, and let $\Sigma:U\to\mathbb{R}^d$ be a smooth parametrisation of a $(d-1)$-dimensional subman...

conjecture | open | MathDB 页面 | 原文线索

V2-037695 | MATHDB-379272

Linear-eliminant ideal-membership conjecture for factor-analysis invariants

摘录 / EXCERPT: Linear-eliminant membership conjecture. If $m\geq 2$, $p\geq 2m+2$, and $R\cup C\neq\bar R\cup\bar C$, then

conjecture | open | MathDB 页面 | 原文线索

V2-037696 | MATHDB-379273

Minimal generator conjecture for the two-factor model

摘录 / EXCERPT: Two-factor minimal-generation conjecture. The ideal $I_{p,2}$ is minimally generated by

conjecture | open | MathDB 页面 | 原文线索

V2-037697 | MATHDB-379384

The nearly symmetric word conjecture for positive definite letters

摘录 / EXCERPT: Nearly symmetric word conjecture. A word has positive trace for every pair of positive definite letters if and only if the word is nearly symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-037698 | MATHDB-379412

Subsequential limit conjecture for generalized spiral polynomials

摘录 / EXCERPT: Generalized spiral limit conjecture. The rescaled functions $p_k\left(z_0+\frac{z}{\sqrt{k}}\right)$ have a subsequence converging to a normalization of P_λ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-037699 | MATHDB-379441

Hypergeometric asymptotic expansion conjecture for $F(k,-\rho;k+n/2;z)$

摘录 / EXCERPT: Let n be the dimension parameter, let $\rho=(n-1)/2$, and let $F(a,b;c;z)$ denote the hypergeometric function. For the function $F_k(z)=F(k,-\rho;k+n/2;z)$, with complex variable...

conjecture | open | MathDB 页面 | 原文线索

V2-037700 | MATHDB-379449

Mixed-norm bound for the higher-dimensional Kakeya maximal operator

摘录 / EXCERPT: Higher-dimensional Kakeya maximal-operator conjecture. For $2\leq k<d$, $p>\frac{d}{k}$, and $1\leq q\leq kp$,

conjecture | open | MathDB 页面 | 原文线索

V2-037701 | MATHDB-379450

Mixed-norm bound for the higher-dimensional Kakeya maximal operator at small p

摘录 / EXCERPT: Higher-dimensional Kakeya maximal-operator conjecture. For $k\geq1$, $p<\frac{d}{k}$, and $q\leq(d-k)p'$, where p' is the Hölder conjugate of p,

conjecture | open | MathDB 页面 | 原文线索

V2-037702 | MATHDB-379488

Uniform oscillation inequality for the sharply truncated Hilbert transform

摘录 / EXCERPT: Let $J_H(y)=\boldsymbol{1}_{[-1,1]}(y)y^{-1}$ be the Hilbert transform kernel with a sharp cutoff, and let $\operatorname{Osc}_n(J_H;f)$ denote the corresponding oscillation operat...

conjecture | open | MathDB 页面 | 原文线索

V2-037703 | MATHDB-379511

Borel-series zero conjecture

摘录 / EXCERPT: Let $\{a_i\}_{i\in\mathbb{N}}$ be positive with finite sum, and let $\{z_i\}_{i\in\mathbb{N}}$ be distinct complex numbers such that ... for some $\rho>0$. Borel-series zero conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-037704 | MATHDB-379512

Operator-theoretic Borel-series zero conjecture

摘录 / EXCERPT: Let $\{z_n\}_{n\in\mathbb{N}}$ be distinct points in the open unit disk converging to 1. Let $\mathcal{H}$ be a separable infinite-dimensional complex Hilbert space with orthonor...

conjecture | open | MathDB 页面 | 原文线索

V2-037705 | MATHDB-379513

Operator-theoretic Clunie-Eremenko-Rossi conjecture

摘录 / EXCERPT: Let $\{\alpha_n\}_{n\in\mathbb{N}}$ be unimodular complex numbers and let $\{\beta_n\}_{n\in\mathbb{N}}$ be positive numbers converging to 0, with the ratios...

conjecture | open | MathDB 页面 | 原文线索

V2-037706 | MATHDB-379514

Complex-charge equilibrium zero conjecture

摘录 / EXCERPT: Complex-charge equilibrium zero conjecture. The meromorphic function

conjecture | open | MathDB 页面 | 原文线索

V2-037707 | MATHDB-379522

Miller's conjecture on maximal polynomials

摘录 / EXCERPT: Let $\beta\in\bar{D}$ and let $S(n,\beta)$ be the set of monic complex polynomials in S_n having at least one zero at β. For $\alpha\in\mathbf{C}$ and $p\in S(n,\beta)$, de...

conjecture | open | MathDB 页面 | 原文线索

V2-037708 | MATHDB-379523

Global maximality conjecture for the function d on S(n,0)

摘录 / EXCERPT: Let $S(n,0)$ be the set of monic complex polynomials of degree $n\ge 2$ whose zeros lie in the closed unit disk and which vanish at 0. For $p\in S_n$, define ... The paper has dete...

conjecture | open | MathDB 页面 | 原文线索

V2-037709 | MATHDB-379536

Conjecture on zeros of meromorphic logarithmic potentials in the plane

摘录 / EXCERPT: Conjecture on infinitely many zeros. The function f has infinitely many zeros in $\mathbb C$.

conjecture | open | MathDB 页面 | 原文线索

V2-037710 | MATHDB-379537

Borcea's conjecture on zeros in a disk

摘录 / EXCERPT: Borcea's conjecture. The function f has infinitely many zeros in $|z|<\rho$.

conjecture | open | MathDB 页面 | 原文线索

V2-037711 | MATHDB-379538

Equivalent half-plane formulation of Borcea's conjecture

摘录 / EXCERPT: Equivalent half-plane formulation. The function F has infinitely many zeros in $\operatorname{Re}w>\frac12$.

conjecture | open | MathDB 页面 | 原文线索

V2-037712 | MATHDB-379599

Stable-completeness characterization of bases for exponential families

摘录 / EXCERPT: Let $1<p<\infty$, and let $U=\{u_k\}$ be an exponential-function family in $L^p(0,1)$, with the distance between normalized exponential-function families defined by the displayed q...

conjecture | open | MathDB 页面 | 原文线索

V2-037713 | MATHDB-379729

Modified Newton distance conjecture for oscillatory integral operators

摘录 / EXCERPT: Modified Newton distance conjecture. If $S\in\mathfrak S^m\mathbb R^{n_X+n_Z}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-037714 | MATHDB-379749

Conjecture on the Haagerup graphs Γ_k as principal graphs

摘录 / EXCERPT: Let Γ_k denote the graphs in the paper's Figure 2. Haagerup graph principal-graph conjecture. For every $k>1$, the graph Γ_k is not a principal graph of a subfactor....

conjecture | open | MathDB 页面 | 原文线索

V2-037715 | MATHDB-379760

The logarithmic lower-bound conjecture for Fourier norms on compact Abelian groups

摘录 / EXCERPT: Logarithmic Fourier-norm conjecture. If, for every finite subgroup $V\leq\widehat G$ with $|V|\leq M$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-037716 | MATHDB-379796

Many derivative-root circles conjecture for Newton basins

摘录 / EXCERPT: Let p be a degree-n univariate complex polynomial. Let its DR-circles be the circles centered at the roots of p' with radii ρ_j, where ... For each root of p, consider i...

conjecture | open | MathDB 页面 | 原文线索

V2-037717 | MATHDB-379851

Twisted Bost conjecture for unconditional twisted group-algebra completions

摘录 / EXCERPT: Twisted Bost conjecture. The map $\mu_\sigma^{\mathcal A}$ is an isomorphism for $j=0,1$. The conjecture is presented as an unconditional analogue of the twisted Baum-Connes conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-037718 | MATHDB-379927

The short interval L^4 conjecture for square frequencies

摘录 / EXCERPT: The short interval conjecture. For every $\alpha<1$, every such trigonometric polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-037719 | MATHDB-379956

Gukov's asymptotic expansion conjecture for colored Jones polynomials

摘录 / EXCERPT: Asymptotic expansion conjecture. As $N\to\infty$ and $k\to\infty$ with u fixed,

conjecture | open | MathDB 页面 | 原文线索

V2-037720 | MATHDB-379975

A continuity-based criterion for failure of property (iv)

摘录 / EXCERPT: The continuity-based criterion. Under these assumptions, A does not have property (iv). The conjecture proposes that the conclusion of Theorem 2 remains valid when the relevant g...

conjecture | open | MathDB 页面 | 原文线索

V2-037721 | MATHDB-380015

The asymptotic constant conjecture for the operator determinant inequality

摘录 / EXCERPT: Asymptotic constant conjecture. There exists a constant $1\leq c\leq\sqrt e$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-037722 | MATHDB-380030

Hausdorff-content comparability under principal quasiconformal mappings

摘录 / EXCERPT: Let $d\in(0,2]$, let $E\subset\mathbb C$ be compact, and let h be a principal K-quasiconformal mapping conformal on $\mathbb C\setminus E$. Here $\mathcal M^d$ denotes d-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-037723 | MATHDB-380032

The (n,k) maximal-function conjecture

摘录 / EXCERPT: (n,k) maximal-function conjecture. For all $\epsilon>0$ and $0<\delta\ll1$,

conjecture | open | MathDB 页面 | 原文线索

V2-037724 | MATHDB-380070

The L^p discrepancy conjecture below the endpoint

摘录 / EXCERPT: The subunit-L^p discrepancy conjecture. For every dimension $d\geq2$ and every $0<p<1$, there is a constant depending only on d and p such that

conjecture | open | MathDB 页面 | 原文线索

V2-037725 | MATHDB-380148

The flat-neck volume conjecture

摘录 / EXCERPT: Let $H^+\subseteq\mathbb{R}^{(a,b)}$ be the positive pseudosphere and let $N^+\subset H^+$ be a neck, with Riemannian volume $\operatorname{Vol}N^+$. Flat-neck volume conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-037726 | MATHDB-380181

Classification conjecture for time-frequency representations on finite Abelian groups

摘录 / EXCERPT: Let G be a finite Abelian group, let $V_g f$ denote the time-frequency representation appearing in the paper, let $\widehat f$ denote the Fourier transform of f, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-037727 | MATHDB-380239

Ivrii's generic spectral asymptotics conjecture for even-dimensional magnetic Schrödinger operators

摘录 / EXCERPT: Ivrii's conjecture. The main part of the spectral asymptotics is given by ${\mathcal E}^{\rm {MW}}$, while: (i) for fixed (g^{jk}) and general $(V_j),V$, the remainder is...

conjecture | open | MathDB 页面 | 原文线索

V2-037728 | MATHDB-380281

The BMV conjecture

摘录 / EXCERPT: Let A and B be positive definite matrices, and let n be a positive integer. The BMV conjecture. The polynomial in t ... has all positive coefficients. The source presents thi...

conjecture | open | MathDB 页面 | 原文线索

V2-037729 | MATHDB-380287

The three-matrix coefficient-minor stability conjecture

摘录 / EXCERPT: Let D_1,D_2,D_3 be positive definite matrices, and let M be the matrix whose entries are the coefficients of y and z in ... A minor means the determinant of any finite square...

conjecture | open | MathDB 页面 | 原文线索

V2-037730 | MATHDB-380299

The R_ϵ-conjecture for unit norm Riesz basic sequences

摘录 / EXCERPT: R_ϵ-conjecture. Every unit norm Riesz basic sequence is a finite union of ϵ-Riesz basic sequences.

conjecture | open | MathDB 页面 | 原文线索

V2-037731 | MATHDB-380300

The quantitative R_ϵ-conjecture for finite Riesz basic sequences

摘录 / EXCERPT: Quantitative R_ϵ-conjecture. There is a natural number $r=r(\epsilon,A,B)$ such that, for every $n\in\mathbb N$ and every such sequence, there is a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-037732 | MATHDB-380365

Vector-spread majorization conjecture for Ritz value errors

摘录 / EXCERPT: Let ${\mathcal X}$ and ${\mathcal Y}$ be subspaces of a Hilbert space ${\mathcal H}$ with ${{\rm dim}} \mathcal X={{\rm dim}} \mathcal Y$, and let A be Hermitian. For a vector...

conjecture | open | MathDB 页面 | 原文线索

V2-037733 | MATHDB-380371

Dimension-free linear weighted bound for the Cauchy transform

摘录 / EXCERPT: Dimension-free Cauchy-transform conjecture. For every $w\in A_2$ and the Cauchy transform C in $\mathbb C^n$, the displayed estimate holds with an implied constant independent of...

open_problem | open | MathDB 页面 | 原文线索

V2-037734 | MATHDB-380378

The stable-rank-two conjecture for the real disc algebra

摘录 / EXCERPT: Let $A_\mathbb{R}(\mathbb{D})$ denote the real disc algebra, consisting of functions analytic on $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and satisfying the relevant re...

conjecture | open | MathDB 页面 | 原文线索

V2-037735 | MATHDB-380389

The PSD block-sum Schatten maximization conjecture

摘录 / EXCERPT: Let $Q^{(i)}=[Q^{(i)}_{(jk)}]_{j,k=1}^2$ be arbitrary positive semidefinite 2×2 block matrices, and let $q^{(i)}_k$ be arbitrary non-negative numbers. For a matrix A, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-037736 | MATHDB-380478

Révész-Sarantopoulos conjecture on the real linear polarization constants

摘录 / EXCERPT: Let $c_n(X)$ denote the nth linear polarization constant of a normed space X, and let ${\bf R}^n$ be the real Euclidean space. Révész-Sarantopoulos conjecture. ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-037737 | MATHDB-380494

The quotient-manifold characterization of Banach-Lie subgroups

摘录 / EXCERPT: Let G be a Banach-Lie group and let H be a closed subgroup. The quotient set G/H is formed from the left cosets gH, and the quotient map is ... A closed subgroup H of a Ban...

conjecture | open | MathDB 页面 | 原文线索

V2-037738 | MATHDB-380498

The strict isoperimetric-ratio conjecture for conformal manifolds with boundary

摘录 / EXCERPT: Let $n\geq 3$, let (M^n,g) be a smooth compact Riemannian manifold with nonempty boundary, and suppose that $\lambda_1(L_g)>0$. For a conformal metric $\widetilde g\in[g]$ with z...

conjecture | open | MathDB 页面 | 原文线索

V2-037739 | MATHDB-380545

Extension of Theorem 3 to negative mean curvature

摘录 / EXCERPT: Extension conjecture. Theorem 3 should hold without the hypothesis $H\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-037740 | MATHDB-380610

Hadamard-type conjecture on graph and level-set concavity

摘录 / EXCERPT: Hadamard-type conjecture. (i) $\mathcal G$ is concave if and only if $G_v\geq 0$ for every $x\in\Omega$. (ii) v is $-\infty$-concave, meaning that $\mathcal G$ has convex level s...

conjecture | open | MathDB 页面 | 原文线索

V2-037741 | MATHDB-380647

The sharp restriction conjecture

摘录 / EXCERPT: The sharp restriction conjecture. $R^*(p\to q)$ holds whenever

conjecture | open | MathDB 页面 | 原文线索

V2-037742 | MATHDB-380648

Bilinear restriction range conjecture

摘录 / EXCERPT: Bilinear restriction range conjecture. If $n\geq2$, then $R^*(p\times p\to q)$ holds whenever

conjecture | open | MathDB 页面 | 原文线索

V2-037743 | MATHDB-380649

Bilinear Kakeya range conjecture

摘录 / EXCERPT: Bilinear Kakeya range conjecture. $K^*(q'\times q'\to p'/2)$ holds if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-037744 | MATHDB-380652

The generalized triple bubble conjecture with enlarged prescribed areas

摘录 / EXCERPT: Let A_1, A_2, and A_3 be positive real numbers. Consider planar graphs enclosing and separating finite areas B_1, B_2, and B_3 satisfying ... A standard triple bubble is...

conjecture | open | MathDB 页面 | 原文线索

V2-037745 | MATHDB-380692

Energy minimization conjecture for the directed volume of K-plus

摘录 / EXCERPT: Energy conjecture. The quantity $Q(\operatorname{\stackrel{\longrightarrow}{Vol}}K^+)$ is uniquely minimized when K is an ellipsoid.

conjecture | open | MathDB 页面 | 原文线索

V2-037746 | MATHDB-380715

The GL2 characterization by finite-cotype lattice embeddings

摘录 / EXCERPT: Let X be a K-convex Banach space. The property GL_2 is the Gordon-Lewis property with exponent 2, and a Banach lattice has finite cotype in the usual Banach-space...

conjecture | open | MathDB 页面 | 原文线索

V2-037747 | MATHDB-380820

The C^1 unipotent Jacobian univalence conjecture

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be a C^1 map, and let $J(f)$ denote its Jacobian matrix. A matrix is unipotent when all its eigenvalues are 1. The C^1 unipotent Jacobian...

conjecture | open | MathDB 页面 | 原文线索

V2-037748 | MATHDB-380823

The two-fixed-point Jacobian eigenvalue conjecture

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be a C^1 map, and let $J(f)$ denote its Jacobian matrix. A point x is a fixed point if $f(x)=x$. The two-fixed-point conjecture. If f has...

conjecture | open | MathDB 页面 | 原文线索

V2-037749 | MATHDB-380827

Uniqueness of the diffusively dominant circle packing

摘录 / EXCERPT: A circle packing is a packing of congruent circles in the plane, and a packing is diffusively dominant if its mass distribution diffusively dominates that of every other packing. C...

conjecture | open | MathDB 页面 | 原文线索

V2-037750 | MATHDB-380897

Globevnik-Stout conjecture on holomorphic extension from tangent complex lines

摘录 / EXCERPT: Let $\Omega\subset\subset D$ be convex domains with C^2 boundaries in $\mathbb{C}^n$, and let f be continuous on bD. Globevnik-Stout conjecture. If f extends continuously t...

conjecture | open | MathDB 页面 | 原文线索

V2-037751 | MATHDB-380913

The dual spectral-set conjecture for spectra and tiling sets

摘录 / EXCERPT: Let $L\subset\mathbb{R}^{d}$. A spectrum is a set L for which there exists a set Ω such that (Ω,L) is a spectral pair, and a tiling set is a set L for which there...

conjecture | open | MathDB 页面 | 原文线索

V2-037752 | MATHDB-380914

The cube spectral-tiling conjecture

摘录 / EXCERPT: Let I^{d} denote the d-dimensional unit cube, and let $L\subset\mathbb{R}^{d}$. The cube spectral-tiling conjecture. (I^{d},L) is a spectral pair if and only if (I^{d},L) i...

conjecture | open | MathDB 页面 | 原文线索

V2-037753 | MATHDB-381008

The vanishing conjecture for Hessian nilpotent polynomials

摘录 / EXCERPT: Let $z=(z_1,z_2,\dots,z_n)$ be free commutative variables, let $D_i=\frac{\partial}{\partial z_i}$, and let ... be the Laplace operator on $\mathbb C[z]$. A polynomial $P(z)$ is Hess...

conjecture | open | MathDB 页面 | 原文线索

V2-037754 | MATHDB-381073

Schoen's Weyl-vanishing conjecture for Yamabe blow-up points

摘录 / EXCERPT: Schoen's Weyl-vanishing conjecture. If $\overline{y}$ is a blow-up point, then the Weyl tensor W_{g_0} of g_0 vanishes at $\overline{y}$ to order $\left[\frac{n-6}{2}\right]$;...

conjecture | open | MathDB 页面 | 原文线索

V2-037755 | MATHDB-381081

Conjecture on convergence and continuity of the scattering transform for integrable functions

摘录 / EXCERPT: Scattering convergence and continuity conjecture. Condition $$ holds for all $f\in {\bf L}^1({{\mathbb{R}}}^d)$. Moreover, if $f\in {\bf L}^1({{\mathbb{R}}}^d)$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-037756 | MATHDB-381128

Uniqueness conjecture for bounded symmetric domains with the same harmonic measure distribution

摘录 / EXCERPT: Uniqueness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-037757 | MATHDB-381132

Separated bump conjecture for strong-type inequalities of Riesz potentials

摘录 / EXCERPT: Separated bump conjecture. Given $0<\alpha<n$ and $1<p\leq q<\infty$, the strong-type inequality for the Riesz potential holds for every pair of weights (u,σ) satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-037758 | MATHDB-381174

Polynomial upper-bound conjecture for the smallest singular value of submatrices

摘录 / EXCERPT: Let $F=[g_1,g_2,\dots,g_{n+k}]$ be an $n\times(n+k)$ matrix, and define ... Assume that $\|g_j\|\leq1$ for every j. Polynomial upper-bound conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-037759 | MATHDB-381201

Symmetric basis conjecture for eigenvectors of generators on \(\ell_p\)

摘录 / EXCERPT: Let A be the generator of a C_0-group on ℓ_p, with $p>1$, whose eigenvalues, counted with multiplicity, are $\{\lambda_n\}_{n=1}^{\infty}$ and whose corresponding normaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-037760 | MATHDB-381215

Lee-Sormani intrinsic flat stability conjecture for asymptotically flat manifolds

摘录 / EXCERPT: Let M_j be asymptotically flat n-dimensional Riemannian manifolds with nonnegative scalar curvature and no interior closed minimal surfaces, with either no boundary or boundary...

conjecture | refuted | MathDB 页面 | 原文线索

V2-037761 | MATHDB-381226

Two-weight Bergman inequality without the pointwise weight comparison condition

摘录 / EXCERPT: Consider the upper half-plane $\mathbb{R}_{+}^2=\{z\in\mathbb{C}:\operatorname{Im}z>0\}$, two weights μ_1 and μ_2 on $\mathbb{R}_{+}^2$, and the two-weight inequality ... Her...

open_problem | open | MathDB 页面 | 原文线索

V2-037762 | MATHDB-381258

Conformal polynomial model conjecture for holomorphic functions on the disk

摘录 / EXCERPT: Let f be holomorphic on a neighborhood of the closure of the unit disk $\mathbb{D}$. An injective holomorphic function is a holomorphic map that is one-to-one. Conformal polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-037763 | MATHDB-381333

The converse characterization of extreme points in PRP_2

摘录 / EXCERPT: Converse characterization conjecture. If f is an extreme point of PRP_2, then p is $\mathbb{T}^2$-saturated and $p-\widetilde{p}$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-037764 | MATHDB-381353

Killing-form convergence conjecture for Hermitian curvature flow on complex simple groups

摘录 / EXCERPT: Let G be a complex Lie group whose Lie algebra is the complexification $\mathfrak g=\mathfrak g_{\mathbb R}\otimes\mathbb C$ of a simple compact real Lie algebra...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037765 | MATHDB-381372

Random Runge approximation conjecture for holomorphic functions on compact sets

摘录 / EXCERPT: Random Runge approximation conjecture. There is a sequence $R_k(\omega,z)$ of random rational functions such that, for each ω, $R_k(\omega,\cdot)\rightarrow f(\omega,\cdot)$...

conjecture | open | MathDB 页面 | 原文线索

V2-037766 | MATHDB-381409

Uniqueness and convexity conjecture for optimal densities

摘录 / EXCERPT: Let $Omega$ and k satisfy the assumptions in Section 1. Suppose also that $Omega$ is convex and $rho_->0$. An optimal-density uniqueness and convexity conjecture asserts that the...

conjecture | open | MathDB 页面 | 原文线索

V2-037767 | MATHDB-381441

The finite-difference stability conjecture for stable polynomials

摘录 / EXCERPT: Finite-difference stability conjecture. For any stable polynomial p, the polynomials

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037768 | MATHDB-381460

Classicality criterion for the odd-indexed Painlevé V solution

摘录 / EXCERPT: Let l be an odd positive integer, and let a_l^l denote the coefficient in the expansion of the solution referred to as Expansion. Classicality criterion. The solution with odd...

conjecture | open | MathDB 页面 | 原文线索

V2-037769 | MATHDB-381474

Path connectedness of the regular part of non-collapsed RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf d,\mathcal{H}^{N})$ be a non-collapsed $\mathsf{RCD}(K,N)$ space, and let $({\mathcal R}_{N})_{\varepsilon}$ denote its ε-regular set. Then the interior...

conjecture | open | MathDB 页面 | 原文线索

V2-037770 | MATHDB-381480

MacDonald's conjecture on closest projections to nilpotent operators

摘录 / EXCERPT: Let $\mathcal{N}(\mathbb{C}^n)$ denote the set of nilpotent operators on $\mathbb{C}^n$, and let δ_n be the distance from the set of non-zero projections in...

conjecture | open | MathDB 页面 | 原文线索

V2-037771 | MATHDB-381521

Nuclearity of induced morphisms between colimit Banach rings

摘录 / EXCERPT: For $0<r_2<r_1<1$, let ... be the Banach ring constructed as the non-expanding colimit of the rings $\mathbb{Z}\{(x/r)^{1/n}\}$ under the morphisms $\alpha_{n,m}$. Nuclearity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-037772 | MATHDB-381551

Weak-type weighted inequality characterization for the maximal operator

摘录 / EXCERPT: Let X be a space of homogeneous type, let M be the uncentered Hardy-Littlewood maximal operator, and let $p(39)\in\mathrm{LH}$ satisfy $p_+<\infty$. Let w be a weight, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-037773 | MATHDB-381609

Conjecture on semiclassical transport of localized data along curved interfaces

摘录 / EXCERPT: Semiclassical transport conjecture. There exists $\beta<3/4$ such that, under the assumptions in, the solution Ψ_t of

conjecture | open | MathDB 页面 | 原文线索

V2-037774 | MATHDB-381735

Blackadar-Handelman conjecture on dimension functions

摘录 / EXCERPT: Let A be a unital C^*-algebra. Define the classical Cuntz semigroup $\mathrm{W}(A)$ from positive elements in $M_\infty(A)$, and let $\mathrm{DF}(A)$ be the set of normalized d...

conjecture | open | MathDB 页面 | 原文线索

V2-037775 | MATHDB-381798

Optimal cell-average decomposition equality for tensor-product and total-degree polynomial spaces

摘录 / EXCERPT: Let $\theta\in[-1,1]$ and let $\mathbb V^k$ denote either the tensor-product polynomial space $\mathbb Q^k$ or the total-degree polynomial space $\mathbb P^k$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-037776 | MATHDB-381799

Squared critical positive polynomial conjecture for optimal cell-average decomposition

摘录 / EXCERPT: Let $\theta\in[-1,1]$, let $\mathbb V^k$ be either $\mathbb Q^k$ or $\mathbb P^k$, and let $\mathbb V^k_+$ denote the corresponding cone of positive polynomials. For...

conjecture | open | MathDB 页面 | 原文线索

V2-037777 | MATHDB-381836

Conjectural asymptotics for the moment generating function of linear statistics

摘录 / EXCERPT: Fix $\beta>0$. Let $g:\mathbb{R}\to\mathbb{R}$ be 2π-periodic and belong to $C^{1,q}$ for some $0<q\leq 1$, let $t\in\mathbb{R}$, and let I denote the integral whose normaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-037778 | MATHDB-382028

The folk conjecture on von Neumann subalgebras of uniform Roe and quasi-local algebras

摘录 / EXCERPT: Folk conjecture. The only von Neumann algebras that can embed into a uniform Roe algebra or a quasi-local algebra associated to a uniformly locally finite metric space are those of...

conjecture | open | MathDB 页面 | 原文线索

V2-037779 | MATHDB-382029

The refined embedding conjecture for von Neumann algebras in uniform Roe and quasi-local algebras

摘录 / EXCERPT: Refined embedding conjecture. The only von Neumann algebras that can embed into

conjecture | open | MathDB 页面 | 原文线索

V2-037780 | MATHDB-382068

Conjecture on decay of mixed derivatives of the symmetric coordinates

摘录 / EXCERPT: Let u_r be the symmetric homogeneous polynomial coordinates in N variables constructed in the main theorem. For a partition $\sigma\vdash r$ and a differential operator of orde...

conjecture | open | MathDB 页面 | 原文线索

V2-037781 | MATHDB-382117

Contour-integral bound conjecture on a V-shaped domain

摘录 / EXCERPT: Contour-integral bound conjecture. The remaining contour integral satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-037782 | MATHDB-382131

Modular Dvoretzky conjecture for Banach modules over C*-algebras

摘录 / EXCERPT: Let $\mathcal{A}$ be a unital C-algebra with the invariant basis number property, and let $\mathcal{E}$ be an n-rank Banach module over $\mathcal{A}$. For $k\in\mathbb{N}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-037783 | MATHDB-382158

Abdulhadi-Hengartner conjecture on the valence of logharmonic polynomials

摘录 / EXCERPT: Let $f(z)=p(z)\overline{q(z)}$ be a logharmonic polynomial, where p and q are analytic polynomials of degrees n and m, respectively, with p not a constant multiple of q...

conjecture | open | MathDB 页面 | 原文线索

V2-037784 | MATHDB-382159

The spheres-tubes-slabs conjecture for the relative isoperimetric problem in the cube

摘录 / EXCERPT: Let $[0,1]^n$ be the unit cube in $\mathbb{R}^n$ with the Euclidean metric. For a measurable set $U\subset [0,1]^n$ of locally finite perimeter, let $\operatorname{Vol}(U)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-037785 | MATHDB-382160

The cubical minimizers conjecture for the relative isoperimetric problem

摘录 / EXCERPT: Let $[0,1]^n$ be the unit cube in $\mathbb{R}^n$, and let a cubical subset be a compact subset of $[0,1]^n$ whose boundary is contained in finitely many hyperplanes with outer norm...

conjecture | open | MathDB 页面 | 原文线索

V2-037786 | MATHDB-382166

Uniqueness conjecture for bounded-girth ancient Ricci flows on spheres

摘录 / EXCERPT: Let $n\ge 4$. An ancient Ricci flow is a Ricci flow defined for all sufficiently negative times. Consider the solution on S^n constructed in the paper's main existence theorem, a...

conjecture | open | MathDB 页面 | 原文线索

V2-037787 | MATHDB-382204

Strong Breuillard-Green conjecture for \(\mathrm{SO}(3,\mathbb{R})\)

摘录 / EXCERPT: Let $A\subseteq \mathrm{SO}(3,\mathbb{R})$ be open, and let μ denote normalized Haar measure on $\mathrm{SO}(3,\mathbb{R})$. For a unit vector $u\in\mathbb{R}^3$, define the an...

conjecture | open | MathDB 页面 | 原文线索

V2-037788 | MATHDB-382245

Conjecture on weighted frame representations of invertible unconditionally convergent multipliers

摘录 / EXCERPT: Let ${\cal H}$ be a Hilbert space, let $\Phi=(\varphi_n)$ and $\Psi=(\psi_n)$ be sequences in ${\cal H}$, and let $m=(m_n)$ be a complex sequence. The multiplier $M_{m,\Phi,\Psi}$...

conjecture | open | MathDB 页面 | 原文线索

V2-037789 | MATHDB-382252

Asymptotic growth conjecture for the balanced embedding solution

摘录 / EXCERPT: Let $f^\beta(x)$ be the unique solution of equation (2) with $f^\beta(0)=1$. Asymptotic growth conjecture. As $x\to+\infty$, the function f^β should be asymptotic to ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-037790 | MATHDB-382370

Regularity from a unique flat tangent cone and density concentration

摘录 / EXCERPT: Let V be a stationary integer rectifiable varifold, let x_0 be a point with a unique flat tangent cone, let $\Theta(V,x_0)$ denote its density at x_0, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-037791 | MATHDB-382382

Conjectured validity of confidence sets for the mean of random functions

摘录 / EXCERPT: Let $\boldsymbol{t}\feature$?

conjecture | open | MathDB 页面 | 原文线索

V2-037792 | MATHDB-382396

Nesterov acceleration asymptotic conjecture

摘录 / EXCERPT: Let $\rho\in\mathbb{N}$, and assume (A1) obtained from the preceding lemma. Let $(a_k)_{k\geqslant 0}$ be the sequence defined by the two Nesterov equations referred to in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-037793 | MATHDB-382401

Uniqueness conjecture II for the length vector of AF$_\beta$ gravitational instantons

摘录 / EXCERPT: In the setting of Theorem, let $\bm{l}=\bm{l}(\beta)$ be the length vector associated with the construction of $\operatorname{AF_\beta}$ gravitational instantons. Uniqueness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-037794 | MATHDB-382483

Naive bilinear Kakeya conjecture for transversal Heisenberg tubes

摘录 / EXCERPT: Naive bilinear Kakeya conjecture. Let $\varepsilon>0$. For every $\delta\in(0,1)$ and every pair of transversal families of Heisenberg δ-tubes...

conjecture | open | MathDB 页面 | 原文线索

V2-037795 | MATHDB-382503

Regularity assumption for parallel overlapping Maxwell domain decomposition

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be the cylindrical domain ... where $\Omega_p\subset\mathbb{R}^2$ is a Lipschitz polygon. Suppose that each partition-of-unity function χ_ℓ de...

conjecture | open | MathDB 页面 | 原文线索

V2-037796 | MATHDB-382541

Unique solvability of symmetric word equations

摘录 / EXCERPT: Let A and B be real positive definite matrices, and let a symmetric word equation be an equation obtained by evaluating a symmetric word in A and B at a prescribed matrix....

conjecture | open | MathDB 页面 | 原文线索

V2-037797 | MATHDB-382543

Conjecture on least-perimeter partitions of the disk into six regions

摘录 / EXCERPT: Least-perimeter six-region partition conjecture. The least-perimeter way of dividing the unit disk into six regions of prescribed areas is given by the configuration shown in Figur...

conjecture | open | MathDB 页面 | 原文线索

V2-037798 | MATHDB-382871

Brin’s quarter-pinching conjecture

摘录 / EXCERPT: Conjecture 1.3 (Brin--Karcher, 1984). Let M be a compact connected manifold with a C^3 Riemannian metric of negative variable sectional curvature K, satisfying...

conjecture | partial | MathDB 页面 | 原文线索

V2-037799 | MATHDB-382874

Generalized Chang-Yang conjecture

摘录 / EXCERPT: Conjecture 1 (Generalized Chang--Yang conjecture). For every integer $N\geq 1$ and every $\alpha\geq\frac{1}{2}$, let

conjecture | solved | MathDB 页面 | 原文线索

V2-037800 | MATHDB-383241

Tang-Zhang Schatten norm conjecture

摘录 / EXCERPT: For $m\ge 2$ and $1<p<\infty$, define the all-dimensional best constant $c_p(m)$ by ... The Tang-Zhang conjecture asserts that $c_p(m)$ equals the explicit closed-form expression pro...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037801 | MATHDB-383291

Stein’s vector Riesz transform weak-type problem

摘录 / EXCERPT: Let $\mathcal{R}f=(R_1f,\ldots,R_nf)$ be the vector Riesz transform on $\mathbb{R}^n$, where R_j is the Fourier multiplier with symbol $-i\xi_j/|\xi|$. Does there exist a univers...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037802 | MATHDB-383295

Yang’s monotonicity conjecture

摘录 / EXCERPT: For every finitely generated polynomial submodule $M\subset H^2(\mathbb D^2)$, Yang's numerical invariants satisfy $\Sigma_0(M)\geq \Sigma_1(M)\geq \Sigma_2(M)\geq\cdots$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037803 | MATHDB-383320

Zariski-density conjecture for singular hyperbolic metrics

摘录 / EXCERPT: For every non-hyperbolic Riemann surface X and every singular hyperbolic metric g on X in the sense of potential theory, the associated monodromy group...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037804 | MATHDB-383321

Petrunin’s question on real projective spaces

摘录 / EXCERPT: Let $m\ge 2$, let g be a Riemannian metric on $\mathbb{R}\mathbb{P}^{m}$, and let $\operatorname{sys}(g)$ denote the length of the shortest loop representing the nontrivial eleme...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-037805 | MATHDB-383350

Arveson’s hyperrigidity conjecture

摘录 / EXCERPT: Let A be a unital operator algebra generating the C^*-algebra $B=C^*(A)$. If every irreducible $*$-representation $\pi:B\to B(H_\pi)$ has the unique completely positive extensi...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037806 | MATHDB-383363

Guerra-Tione Question 5.5 on gradient mappings

摘录 / EXCERPT: For each dimension $n\ge 2$, determine whether the following implication holds: for every domain $\Omega\subset\mathbb{R}^{n}$, every $u\in W^{2,n}_{\mathrm{loc}}(\Omega)$, and eve...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-037807 | MATHDB-383403

Crouzeix conjecture

摘录 / EXCERPT: For every integer $n\ge 1$, every matrix $A\in\mathbb{C}^{n\times n}$, and every polynomial $p\in\mathbb{C}[z]$, one has $\|p(A)\|_2\le 2\sup_{z\in W(A)}|p(z)|$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037808 | MATHDB-383404

Gautschi’s conjecture

摘录 / EXCERPT: For every integer $n\ge 1$, every $-1<\alpha<\beta$, and every $0<c\le 1$, let π_n be the monic degree-n polynomial orthogonal on $[-c,c]$ with respect to the Jacobi weight...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-037809 | MATHDB-383502

Kadison’s orthonormal bases of unitaries problem

摘录 / EXCERPT: Does every type II_1 factor M, equipped with its normalized trace τ, admit an orthonormal basis of $L^2(M,\tau)$ consisting of unitaries in M? Equivalently, doe...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-037810 | MATHDB-383503

Brezis Open Problem 9.3

摘录 / EXCERPT: Let $N\ge 1$, $p\in[1,\infty)$, $\gamma>0$, and let $\Omega\subset\mathbb{R}^N$ be a bounded open interval if $N=1$ or a bounded Lipschitz domain if $N\ge 2$. For $\lambda>0$, defi...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-037811 | MATHDB-383509

Bangert’s question on codimension-one stable norms

摘录 / EXCERPT: For $n\ge 3$, let g be a smooth Riemannian metric on the torus $\mathbb{T}^n$. Suppose that every primitive class $\alpha\in H_{n-1}(\mathbb{T}^n;\mathbb{Z})$ is represented by a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-037812 | MATHDB-383555

Banach’s isometric conjecture

摘录 / EXCERPT: Let X be a complex normed space and let n satisfy $2\le n<\dim_{\mathbb C}X$. If every pair of n-dimensional complex subspaces of X is isometric as metric spaces, then ther...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037813 | MATHDB-383637

Kitaev determinant trivialization conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a complex Hilbert space and let $U,V\in\mathcal{B}(\mathcal{H})$ be invertible. If $(U-\mathbf{1})(V-\mathbf{1})$ and $(V-\mathbf{1})(U-\mathbf{1})$ belong to...

conjecture | solved | MathDB 页面 | 原文线索

V2-037814 | MATHDB-383678

Schäffer’s matrix inequality problem

摘录 / EXCERPT: For each $n\geq 1$, define S_n to be the least constant such that, for every n-dimensional complex Banach space X and every invertible operator $T\in\mathcal{L}(X)$,...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-037815 | MATHDB-383679

Endpoint Nevanlinna defect problem

摘录 / EXCERPT: Determine whether, for every countable family of hyperplanes $\{H_j\}_{j\in\mathbb{N}}$ in general position in $\mathbb{P}^m$ and every linearly nondegenerate holomorphic curve...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037816 | MATHDB-383731

Generalized hyperbolicity conjecture for shadowing

摘录 / EXCERPT: For every Banach space X and every invertible bounded linear operator $T\in\mathcal{B}(X)$, if T has the shadowing property, then T is generalized hyperbolic. In particular,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-037817 | MATHDB-384340

Dependence on parameters

摘录 / EXCERPT: Do canonical Lipschitz extensions exist which depend nicely on parameters?

open_problem | open | MathDB 页面 | 原文线索

V2-037818 | MATHDB-384342

Absolute Lipschitz retract constructions

摘录 / EXCERPT: If Y is an absolute Lipschitz retract, is the same true of the space ${\mathbb Q}_Q(Y)$?

open_problem | open | MathDB 页面 | 原文线索

V2-037819 | MATHDB-384343

Banach space pairs with the Lipschitz extension property

摘录 / EXCERPT: Does the pair (L^2,L^1) have the Lipschitz extension property?

open_problem | open | MathDB 页面 | 原文线索

V2-037820 | MATHDB-384344

Absolute Lipschitz retracts

摘录 / EXCERPT: Are all Hadamard manifolds absolute Lipschitz retracts?

open_problem | open | MathDB 页面 | 原文线索

V2-037821 | MATHDB-384347

Deterministic proofs of bi-Lipschitz embedding theorems

摘录 / EXCERPT: Give a deterministic proof of Naor--Neiman's improved Assouad embedding theorem.

open_problem | open | MathDB 页面 | 原文线索

V2-037822 | MATHDB-384359

Quasiconformal mappings of the plane which destroy the rectifiability of uncountably many disjoint lines

摘录 / EXCERPT: Is there a quasiconformal map f of the plane and an uncountable set $E\subset{\mathbb R}$ so that $f(E\times{\mathbb R})$ contains no rectifiable curves?

open_problem | open | MathDB 页面 | 原文线索

V2-037823 | MATHDB-384361

Product quasiconformal mappings of the plane

摘录 / EXCERPT: Let f be a quasiconformal map of the plane. Suppose at a.e. point, the differential of f either preserves both the x-direction and the y-direction, or switches the two dire...

open_problem | open | MathDB 页面 | 原文线索

V2-037824 | MATHDB-384363

Bi-Lipschitz embeddings of the Sierpi\'nski carpet into itself

摘录 / EXCERPT: Is every bi-Lipschitz embedding of the $\tfrac13$ Sierpi'nski carpet S_3 into itself the restriction of an affine mapping?

open_problem | open | MathDB 页面 | 原文线索

V2-037825 | MATHDB-384364

Quasisymmetric rigidity and non-rigidity of positive area Sierpi\'nski carpets

摘录 / EXCERPT: Which positive area Sierpi'nski carpets are quasisymmetrically rigid? Which are nonrigid?

open_problem | open | MathDB 页面 | 原文线索

V2-037826 | MATHDB-384366

Hausdorff measures in ℓ^p_n

摘录 / EXCERPT: Let P be a polyhedral m-cycle in ℓ^p_n, $p\ne 2$, with faces $F_1,\ldots,F_k$. Is ${\mathcal H}^m(F_1) \le \sum_{j=2}^k {\mathcal H}^m(F_j)$?

open_problem | open | MathDB 页面 | 原文线索

V2-037827 | MATHDB-384367

Compact deformations of minimizing sets

摘录 / EXCERPT: An m-rectifiable closed set $S \subset V = \ell^\infty_n$ is called {it minimizing} if for every Lipschitz map $f:V\to V$ such that f is the identity off a cube C and...

open_problem | open | MathDB 页面 | 原文线索

V2-037828 | MATHDB-384368

Isoperimetric inequalities in metric measure spaces

摘录 / EXCERPT: Can the perimeter measure P on the right hand side of the relative isoperimetric inequality ... where $E \subset X$ is a Borel set and $B \subset X$ is a ball of radius r, be rep...

open_problem | open | MathDB 页面 | 原文线索

V2-037829 | MATHDB-384374

Matrix Theory (problem 4)

摘录 / EXCERPT: Any real polynomial can be written as $\det(A-\lambda I),$ where A has the (tridiagonal) form ...

open_problem | open | MathDB 页面 | 原文线索

V2-037830 | MATHDB-384377

Log Concavity (problem 2)

摘录 / EXCERPT: If $p\in\mathcal{L}\text{-}\mathcal{P}^+$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-037831 | MATHDB-384382

Zeros of derivatives (problem 1)

摘录 / EXCERPT: Let $p\in\mathbb{R}[x]$, $\deg(p)=n>2$. There are at least $n-1$ points of extreme curvature $\kappa(p)$ (more generally where $\kappa'(p)=0$).

open_problem | open | MathDB 页面 | 原文线索

V2-037832 | MATHDB-384384

Orthogonal polynomials (problem 1)

摘录 / EXCERPT: There are no common zeros between L_m^α and L_n^α, $|m-n|>1$, for α rational.

open_problem | open | MathDB 页面 | 原文线索

V2-037833 | MATHDB-384387

Riemann Ξ-function (problem 1)

摘录 / EXCERPT: Investigate the zeros of P_n and $Q_n.$

open_problem | open | MathDB 页面 | 原文线索

V2-037834 | MATHDB-384388

Riemann Ξ-function (problem 2)

摘录 / EXCERPT: Are there any non-real zeros of ...

open_problem | open | MathDB 页面 | 原文线索

V2-037835 | MATHDB-384389

Zeros of specific functions (problem 1)

摘录 / EXCERPT: Let ... where, $n=2,3,4,\ldots.$ Each p_n has no real zeros for even n, and exactly two real zeros for odd n.

open_problem | open | MathDB 页面 | 原文线索

V2-037836 | MATHDB-384390

Zeros of specific functions (problem 2)

摘录 / EXCERPT: Consider the function ... For which values of a is f in $\mathcal{L}\text{-}\mathcal{P}$?

open_problem | open | MathDB 页面 | 原文线索

V2-037837 | MATHDB-384391

Zeros of specific functions (problem 3)

摘录 / EXCERPT: For all $|w|>1$, all the zeros of $P_n(\cdot,w)$ are simple and separated in modulus (by a factor of at least $|w|^2$).

open_problem | open | MathDB 页面 | 原文线索

V2-037838 | MATHDB-384392

Zeros of specific functions (problem 4)

摘录 / EXCERPT: Generalize conjecture to $Q_n(z,w)=\sum_{k=0}^n a_kz^kw^{k(n-k)}$. For what class of $\{a_n\}$ does the conjecture still hold?

open_problem | open | MathDB 页面 | 原文线索

V2-037839 | MATHDB-384393

Miscellaneous Problems (problem 1)

摘录 / EXCERPT: Let be $K\subset\mathbb{C}$ be compact, let $\mathbb{C}\setminus K$ be connected, and let f be a continuous function on K which is holomorphic and non-zero on the interior of...

open_problem | open | MathDB 页面 | 原文线索

V2-037840 | MATHDB-384394

Miscellaneous Problems (problem 2)

摘录 / EXCERPT: Let be $K\subset\mathbb{C}$ be compact, $\mathbb{C}\setminus K$ connected, and f be a continuous function on K which is holomorphic and non-zero on the interior of K. Suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-037841 | MATHDB-384396

Miscellaneous Problems (problem 4)

摘录 / EXCERPT: Let $p(z)\in\mathbb{C}[z]$, $\deg(p)=n$. Suppose $p(z)\neq 0$ for all $|z|\le 1$. Define $p^*(z):=z^n\overline{p\left(\frac{1}{\bar{z}}\right)}$. Then it is known that ... (A is emp...

open_problem | open | MathDB 页面 | 原文线索

V2-037842 | MATHDB-384397

Find more examples and attempt to characterize Markov processes preserving stability.

摘录 / EXCERPT: Find more examples and attempt to characterize Markov processes preserving stability. Explicitly, let $\psi:\mathbb{R}_{MA}[x,y,z]\to\mathbb{R}_{MA}[x,y,z]$ be a linear operator. W...

open_problem | open | MathDB 页面 | 原文线索

V2-037843 | MATHDB-384398

Estimates for $\overline{\partial}$ (problem 1)

摘录 / EXCERPT: Let Ω be a pseudoconvex domain in $\mathbb{C}^n$ that contains 0 and let $\phi\in L^1_{loc}(\Omega)$ be a weight function on Ω. Let H be a hyperplane through 0...

open_problem | open | MathDB 页面 | 原文线索

V2-037844 | MATHDB-384399

Estimates for $\overline{\partial}$ (problem 2)

摘录 / EXCERPT: Let $\mathcal{W}$ be the smooth Diederich--Fornaess worm domain in $\mathbb{C}^2$ (see) and ρ be a defining function for $\mathcal{W}$. Let g_{ρ} be the K"ahler metric o...

open_problem | open | MathDB 页面 | 原文线索

V2-037845 | MATHDB-384401

Estimates for $\overline{\partial}$ (problem 4)

摘录 / EXCERPT: Let $\mathcal{W}$ be the smooth Diederich--Fornaess worm domain in $\mathbb{C}^2$ and $\mathbf{B}_{\mathcal{W}}$ denote the Bergman projection operator on $\mathcal{W}$. Prove or d...

open_problem | open | MathDB 页面 | 原文线索

V2-037846 | MATHDB-384404

Obstruction to Compactness (problem 2)

摘录 / EXCERPT: Let $\Omega_2\subset \Omega_1$ be two smooth bounded pseudoconvex domains that share a boundary point p. If the restriction map from $A^2(\Omega_1)$ into $A^2(\Omega_2)$ is not c...

open_problem | open | MathDB 页面 | 原文线索

V2-037847 | MATHDB-384406

Obstruction to Compactness (problem 4)

摘录 / EXCERPT: Is there a smooth bounded pseudoconvex domain Ω in $\mathbb{C}^n~ (n\geq 3$) that contains a non-trivial analytic disc in $b\Omega$ and yet the $\overline{\partial}$-Neumann...

open_problem | open | MathDB 页面 | 原文线索

V2-037848 | MATHDB-384408

Mappings (problem 2)

摘录 / EXCERPT: Let $R(z,\overline{z})$ be a bihomogeneous polynomial. Find necessary and sufficient conditions such that there exists an integer N with ... Here $p_j(z)$'s are linearly independen...

open_problem | open | MathDB 页面 | 原文线索

V2-037849 | MATHDB-384409

Mappings (problem 3)

摘录 / EXCERPT: Let $R(z,\overline{z})$ be a Hermitian polynomial that is a sum of squares (SOS) ... where $p_j(z)$'s are linearly independent holomorphic polynomials. Suppose that...

open_problem | open | MathDB 页面 | 原文线索

V2-037850 | MATHDB-384410

Mappings (problem 4)

摘录 / EXCERPT: Let D_1 and D_2 be two bounded symmetric domains such that neither is a ball. Let $F:D_1\to D_2$ be a proper holomorphic map. Is F a textit{trivial map}, that is...

open_problem | open | MathDB 页面 | 原文线索

V2-037851 | MATHDB-384412

Estimates for Bergman Kernel (problem 1)

摘录 / EXCERPT: Let M be a compact complex manifold without boundary and L be a holomorphic line bundle over M with positive curvature. Let $B_k(z,w)$ be the Bergman kernel associated to...

open_problem | open | MathDB 页面 | 原文线索

V2-037852 | MATHDB-384413

Determine the boundary behavior of the Bergman kernel on Reinhardt domains.

摘录 / EXCERPT: Determine the boundary behavior of the Bergman kernel on Reinhardt domains. In particular, assume that the domain is smooth, complete and finite type. Start with the domains...

open_problem | open | MathDB 页面 | 原文线索

V2-037853 | MATHDB-384418

Spectrum of $\overline{\partial}$-Neumann Laplacian (problem 4)

摘录 / EXCERPT: Let $\phi:\mathbb{C}^n\to\mathbb{R}$ be a C^2 plurisubharmonic function. Determine the relation between the eigenvalues of the complex Hesssian of ϕ and the compactness of t...

open_problem | open | MathDB 页面 | 原文线索

V2-037854 | MATHDB-384419

Painlev\'e Property (problem 1)

摘录 / EXCERPT: Given an equation with a Painlev'e property, how does one identify which linear system it is?

open_problem | open | MathDB 页面 | 原文线索

V2-037855 | MATHDB-384420

Painlev\'e Property (problem 2)

摘录 / EXCERPT: Can one solve the above for discrete Painlev'e? (If one has a nonlinear difference equation that one suspects may be Painlev'e, can one work out which it is?)

open_problem | open | MathDB 页面 | 原文线索

V2-037856 | MATHDB-384421

Painlev\'e Property (problem 3)

摘录 / EXCERPT: How does one check the Painlev'e property?

open_problem | open | MathDB 页面 | 原文线索

V2-037857 | MATHDB-384422

Painlev\'e Property (problem 4)

摘录 / EXCERPT: In general, is there a classification of determinants that matches that for Painlev'e?

open_problem | open | MathDB 页面 | 原文线索

V2-037858 | MATHDB-384423

Painlev\'e Property (problem 5)

摘录 / EXCERPT: Are the spacing distributions of the exceptional groups related to Painlev'e?

open_problem | open | MathDB 页面 | 原文线索

V2-037859 | MATHDB-384424

Is there an integrable structure related to general ... -ensembles?

摘录 / EXCERPT: Is there an integrable structure related to general β-ensembles?

open_problem | open | MathDB 页面 | 原文线索

V2-037860 | MATHDB-384425

Expectations of Characteristic Polynomials (problem 1)

摘录 / EXCERPT: Identify whether or not there exists an integrable system (by the usual accepted criteria) underlying ... where $Z=\det(I-\lambda A)$.

open_problem | open | MathDB 页面 | 原文线索

V2-037861 | MATHDB-384426

Expectations of Characteristic Polynomials (problem 2)

摘录 / EXCERPT: Consider the expectation given in 2.1. For $h=k$, formulate a conjecture for $k\in\mathbb{R}$

open_problem | open | MathDB 页面 | 原文线索

V2-037862 | MATHDB-384427

Expectations of Characteristic Polynomials (problem 3)

摘录 / EXCERPT: For $h=k$, is there a Painlev'e equation related to the second term?

open_problem | open | MathDB 页面 | 原文线索

V2-037863 | MATHDB-384428

Expectations of Characteristic Polynomials (problem 4)

摘录 / EXCERPT: Now for $h=0$, is there a connection with Painlev'e?

open_problem | open | MathDB 页面 | 原文线索

V2-037864 | MATHDB-384429

How would the conclusions change if one replaces ... by an average over the other classical groups?

摘录 / EXCERPT: How would the conclusions change if one replaces $\mathbb{E}_{A\in U(N)}$ by an average over the other classical groups?

open_problem | open | MathDB 页面 | 原文线索

V2-037865 | MATHDB-384434

Prove the connection between these two equations - it is claimed that one is derivative of the other.

摘录 / EXCERPT: Prove the connection between these two equations - it is claimed that one is derivative of the other.

open_problem | open | MathDB 页面 | 原文线索

V2-037866 | MATHDB-384435

Is there a formulation of the Riemann Hypothesis in Function Fields in terms of Hankel determinants?

摘录 / EXCERPT: Is there a formulation of the Riemann Hypothesis in Function Fields in terms of Hankel determinants?

open_problem | open | MathDB 页面 | 原文线索

V2-037867 | MATHDB-384438

Polynomials and Painlev\'e (problem 2)

摘录 / EXCERPT: What is the precise shape of the boundary of the region containing the roots for Painlev'e V? What about for the location of the zeros?

open_problem | open | MathDB 页面 | 原文线索

V2-037868 | MATHDB-384439

Polynomials and Painlev\'e (problem 3)

摘录 / EXCERPT: Consider the polynomials related to rational solutions of Painlev'e equations. Are they exceptional orthogonal polynomials?

open_problem | open | MathDB 页面 | 原文线索

V2-037869 | MATHDB-384440

Conformal Field Theory (problem 1)

摘录 / EXCERPT: Can one understand the connection between Painlev'e and Conformal Field Theory (conformal blocks)?

open_problem | open | MathDB 页面 | 原文线索

V2-037870 | MATHDB-384441

Conformal Field Theory (problem 2)

摘录 / EXCERPT: Can one understand the τ-functions for Painlev'e VI as Freidholm determinants?

open_problem | open | MathDB 页面 | 原文线索

V2-037871 | MATHDB-384442

Other (problem 1)

摘录 / EXCERPT: Given a function whose logarithmic derivative satisfies a third order ODE of Chazy type, what can one say about the zeros of its Fourier transform? For example, are they all real?

open_problem | open | MathDB 页面 | 原文线索

V2-037872 | MATHDB-384444

Is there a Borodin-Okounkov formula describing transition asymptotics?

摘录 / EXCERPT: Is there a Borodin-Okounkov formula describing transition asymptotics?

open_problem | open | MathDB 页面 | 原文线索

V2-037873 | MATHDB-384447

Entanglement entropy for Schrödinger operators (problem 3)

摘录 / EXCERPT: Entanglement impurity asymptotics in $d=1$: at which order of the asymptotic expansion of the entanglement entropy can the impact of a local perturbation (e.g. V short-range pote...

open_problem | open | MathDB 页面 | 原文线索

V2-037874 | MATHDB-384448

Closure of the asymptotics: up to which order can known (full) asymptotics for both symbol and test function smooth be closed for non-smooth test functions such as ... ?

摘录 / EXCERPT: Closure of the asymptotics: up to which order can known (full) asymptotics for both symbol and test function smooth be closed for non-smooth test functions such as...

open_problem | open | MathDB 页面 | 原文线索

V2-037875 | MATHDB-384449

The weak coupling asymptotics of the ground state energy-density for the delta Fermi gas with frozen spin leads to the asymptotic analysis of the Gaudin integral equation.

摘录 / EXCERPT: The weak coupling asymptotics of the ground state energy-density for the delta Fermi gas with frozen spin leads to the asymptotic analysis of the Gaudin integral equation. For sect...

open_problem | open | MathDB 页面 | 原文线索

V2-037876 | MATHDB-384450

For the Fermi gas with delta pair interaction, is there a characterization of the grounds state correlation functions in terms of a two(?)-particle operator, similar to the determinantal description i

摘录 / EXCERPT: For the Fermi gas with delta pair interaction, is there a characterization of the grounds state correlation functions in terms of a two(?)-particle operator, similar to the determi...

open_problem | open | MathDB 页面 | 原文线索

V2-037877 | MATHDB-384451

Double scaling limits for (e)merging singularities (problem 1)

摘录 / EXCERPT: (E)merging singularites at singular points: consider a symbol f_t, $t>0$, with a fixed Fisher-Hartwig singularity at $z=1$ such that, for $t\to 0$, additional singularites (e)mer...

open_problem | open | MathDB 页面 | 原文线索

V2-037878 | MATHDB-384452

Spectral density (clustering asyptotics) for non-selfadjoint Toeplitz operators and the study of related double scaling asymptotics.

摘录 / EXCERPT: Spectral density (clustering asyptotics) for non-selfadjoint Toeplitz operators and the study of related double scaling asymptotics.

open_problem | open | MathDB 页面 | 原文线索

V2-037879 | MATHDB-384453

Existence of global weak solutions

摘录 / EXCERPT: The modified SQG equation reads begin{aligned} partialttheta +ucdot nablatheta &=0, u &= mathcal{R}^perp|nabla|^{-varepsilon}theta, end{aligned} where...

open_problem | open | MathDB 页面 | 原文线索

V2-037880 | MATHDB-384454

Towards strong global solutions ?

摘录 / EXCERPT: Is it possible to describe the formation of singularities in ?

open_problem | open | MathDB 页面 | 原文线索

V2-037881 | MATHDB-384456

Well-posedness issues

摘录 / EXCERPT: For which values of $\alpha<1$ is it possible to find global solutions to ... in the energy space $H^{\alpha /2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-037882 | MATHDB-384457

Other modifications of NLS

摘录 / EXCERPT: Can you find solutions to other modified NLS equations, such as ... where dissipation is involved ?

open_problem | open | MathDB 页面 | 原文线索

V2-037883 | MATHDB-384458

About the kinetic timescale

摘录 / EXCERPT: Writing ... the averaged amplitude $\mathbb{E}(|a_k|^2)$ solves the kinetic wave equation up to the so-called kinetic timescale $\tau \sim \frac {L^{2d}} {\lambda^4}$.

open_problem | open | MathDB 页面 | 原文线索

V2-037884 | MATHDB-384459

Validity of the RPA

摘录 / EXCERPT: In the preceding setting, up to what time can we assume that the phases remain independent ?

open_problem | open | MathDB 页面 | 原文线索

V2-037885 | MATHDB-384461

Well-posedness theory

摘录 / EXCERPT: Find the optimal space(s) in which the kinetic wave equation is locally well-posed, or at least locally well-posed around the Zakharov solution.

open_problem | open | MathDB 页面 | 原文线索

V2-037886 | MATHDB-384462

Large time behavior

摘录 / EXCERPT: What can be said about the formation of condensates ? Is there a self-similar mechanism to be understood ?

open_problem | open | MathDB 页面 | 原文线索

V2-037887 | MATHDB-384463

Investigation of the Crouzeix-Palencia proof (problem 1)

摘录 / EXCERPT: Better estimate for fg than $\|fg\|\leq \|f\|\|g\|$. Modifications of the Cauchy integral, especially concerning its behavior on constant functions.

open_problem | open | MathDB 页面 | 原文线索

V2-037888 | MATHDB-384464

Investigation of the Crouzeix-Palencia proof (problem 2)

摘录 / EXCERPT: Let θ be a continuous homomorphism from a uniform algebra A into a C^{*}-algebra B and $\alpha:A\rightarrow A$ be antilinear and such that begin{itemize} item[]...

open_problem | open | MathDB 页面 | 原文线索

V2-037889 | MATHDB-384465

Investigation of the Crouzeix-Palencia proof (problem 3)

摘录 / EXCERPT: Does it help to replace λ^2 in the proof by $p(\lambda)$ for polynomial other than λ^2?

open_problem | open | MathDB 页面 | 原文线索

V2-037890 | MATHDB-384466

Investigation of the Crouzeix-Palencia proof (problem 4)

摘录 / EXCERPT: Let f be such that $|f(z)|=1$ for $z\in\partial\Omega$. Then ... How to prove that $C(\bar{f},z)|\leq 1$? (what about Blaschke products?)

open_problem | open | MathDB 页面 | 原文线索

V2-037891 | MATHDB-384467

Investigation of the Crouzeix-Palencia proof (problem 5)

摘录 / EXCERPT: Let f be such that $|f(z)|=1$ for $z\in\partial\Omega$. Then ... How to prove that $C(\bar{f},z)|\leq 1$ directly?

open_problem | open | MathDB 页面 | 原文线索

V2-037892 | MATHDB-384468

Investigation of the Crouzeix-Palencia proof (problem 6)

摘录 / EXCERPT: Can anything be said about $\delta,\hat{\delta}$, in particular ...

open_problem | open | MathDB 页面 | 原文线索

V2-037893 | MATHDB-384469

In what situations (other than in Caldwell-Greenbaum-Li) does the estimate ... hold?

摘录 / EXCERPT: In what situations (other than in Caldwell-Greenbaum-Li) does the estimate ... hold? Here, $f=B\circ \phi$ is such that B is a Blaschke product, ϕ is a conformal mapping from...

open_problem | open | MathDB 页面 | 原文线索

V2-037894 | MATHDB-384470

Investigation of the Crouzeix-Palencia proof (problem 8)

摘录 / EXCERPT: What if Ω consists of a union of disjoint regions? It still holds that ... but how does is this related to $\|f(A)\|$? What is the optimal f, like in the case of the union o...

open_problem | open | MathDB 页面 | 原文线索

V2-037895 | MATHDB-384471

3×3 matrix and dilation theory (problem 1)

摘录 / EXCERPT: For which $A \in \mathbb{C}^{3\times 3}$ holds that $W(T) \subseteq W(A)$ implies that $T = X^*(I\otimes A)X$ for some $X^*X = I$, i.e., $I\otimes A$ is a dilation of T? More gen...

open_problem | open | MathDB 页面 | 原文线索

V2-037896 | MATHDB-384472

3×3 matrix and dilation theory (problem 2)

摘录 / EXCERPT: As $\Omega\to$ line segment, does $C_{\Omega}\to 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-037897 | MATHDB-384473

Barycentric approximation to solve the conjecture (problem 1)

摘录 / EXCERPT: Try barycentric approximation for more general $W(A)$ in order to show that $W(f(A))\subset \frac{5}{4}\mathbb{D}$ for $\|f\|_{\infty,W(A)}=1$.

open_problem | open | MathDB 页面 | 原文线索

V2-037898 | MATHDB-384474

Lemniscates = Ω (problem 1)

摘录 / EXCERPT: Numerical investigations of Blaschke products

open_problem | open | MathDB 页面 | 原文线索

V2-037899 | MATHDB-384475

Lemniscates = Ω (problem 2)

摘录 / EXCERPT: Minimum residual/Arnoldi lemniscates vs. K-spectral sets.

open_problem | open | MathDB 页面 | 原文线索

V2-037900 | MATHDB-384476

Similarity transformation approach (problem 1)

摘录 / EXCERPT: Analytic approach to find X to give "good" XAX^{-1}.

open_problem | open | MathDB 页面 | 原文线索

V2-037901 | MATHDB-384477

Ideas that do not seem to work (problem 1)

摘录 / EXCERPT: Given f with $\|f\|_{\infty,W(A)}=1$, can we always find $\tilde{f}$ defined on a ball D enclosing $W(A)$ such that $\tilde{f}\approx f$ on $W(A)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-037902 | MATHDB-384478

Related open problems (problem 1)

摘录 / EXCERPT: Fix Ω. How does C_{Ω} depend on Ω? Continuously?

open_problem | open | MathDB 页面 | 原文线索

V2-037903 | MATHDB-384479

What is the spectrum of the double layer potential operator on ... ?

摘录 / EXCERPT: What is the spectrum of the double layer potential operator on $\partial\Omega$?

open_problem | open | MathDB 页面 | 原文线索

V2-037904 | MATHDB-384480

Optimal constant dependence for Ritt-operators (problem 1)

摘录 / EXCERPT: Is this C-dependence optimal? What about finite dimensions? In the examples I know the dependence is linear.

open_problem | open | MathDB 页面 | 原文线索

V2-037905 | MATHDB-384481

A possible counterexample via FOM

摘录 / EXCERPT: To extend the above inequality to a counterexample of Crouzeix's conjecture, one may try among other ideas to (1) find matrices A where (most of) the prescribed Ritz values lie o...

open_problem | open | MathDB 页面 | 原文线索

V2-037906 | MATHDB-384482

K-spectral sets for $\Omega\not\subset W(A)$ (problem 1)

摘录 / EXCERPT: Use of lemniscates to bound K-spectral sets.

open_problem | open | MathDB 页面 | 原文线索

V2-037907 | MATHDB-384483

Joint numerical ranges (problem 1)

摘录 / EXCERPT: For what types of polynomials $p(x,y)$ does it hold that ... A next step is then to consider the question for three matrices A_{1},A_{2},A_{3}.

open_problem | open | MathDB 页面 | 原文线索

V2-037908 | MATHDB-384484

What does a low (or high) degree tell us about the matrix?

摘录 / EXCERPT: What does a low (or high) degree tell us about the matrix?

open_problem | open | MathDB 页面 | 原文线索

V2-037909 | MATHDB-384487

A_2 conjecture for rough singular integrals.

摘录 / EXCERPT: Prove or disprove: ...

open_problem | open | MathDB 页面 | 原文线索

V2-037910 | MATHDB-384488

Linear bound for the Beurling operator

摘录 / EXCERPT: Prove or disprove: ...

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V2-037911 | MATHDB-384489

A_2 for Commutators

摘录 / EXCERPT: Consider the commutator defined by ... Then we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-037912 | MATHDB-384490

Boundedness of T_{Ω} on L^p

摘录 / EXCERPT: Prove or disprove ...

open_problem | open | MathDB 页面 | 原文线索

V2-037913 | MATHDB-384491

Weak-type estimates

摘录 / EXCERPT: For all the problems 1.1 to 1.4, it would be interesting to find a sparse approach to obtain sharp weak-type estimates.

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V2-037914 | MATHDB-384492

A generalized related problem

摘录 / EXCERPT: Determine, by using sparse operators, if the following inequality is true for $r \in [1, 2)$. ...

open_problem | open | MathDB 页面 | 原文线索

V2-037915 | MATHDB-384493

Vector valued bounds for maximal operator

摘录 / EXCERPT: Is the operator A bounded on $\ell^p (\ell^r)$ for $1 < p, r < \infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-037916 | MATHDB-384494

Discrete operators (problem 2)

摘录 / EXCERPT: Find a sparse control over the operator A, that is, find $s \geq 1$, $0 < \varepsilon \leq 1$ and a sparse colection $\mathcal{S}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-037917 | MATHDB-384495

Variation of spherical averages

摘录 / EXCERPT: Find $\varepsilon > 0$ and a sparse colection $\mathcal{S}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-037918 | MATHDB-384496

Generalized spherical averages

摘录 / EXCERPT: Find r,s and a sparse collection $\mathcal{S}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-037919 | MATHDB-384497

Bochner-Riesz operator

摘录 / EXCERPT: Prove the existence of an $\varepsilon > 0$ such that the following weighted inequality holds ...

open_problem | open | MathDB 页面 | 原文线索

V2-037920 | MATHDB-384498

End point sparse bounds

摘录 / EXCERPT: Find domination by sparse operators at the end points for the lacunary spherical maximal average ...

open_problem | open | MathDB 页面 | 原文线索

V2-037921 | MATHDB-384499

Sparse adapted to the geometry

摘录 / EXCERPT: Find sparse bounds for the lacunary spherical maximal average, using sparse forms given by sets better adapted to the geometry of the problem.

open_problem | open | MathDB 页面 | 原文线索

V2-037922 | MATHDB-384500

Radial weights

摘录 / EXCERPT: Find sparse bounds adapted to estimates with respect to radial weights and/or radial functions.

open_problem | open | MathDB 页面 | 原文线索

V2-037923 | MATHDB-384501

Bilinear spherical maximal operator

摘录 / EXCERPT: Find a correct way to define a sparse bound for the bilinear spherical maximal averaging operator.

open_problem | open | MathDB 页面 | 原文线索

V2-037924 | MATHDB-384502

Lower bound for the square function

摘录 / EXCERPT: Find a sparse domination proof for the lower bound ...

open_problem | open | MathDB 页面 | 原文线索

V2-037925 | MATHDB-384503

Matrix A_2 conjecture

摘录 / EXCERPT: Prove or disprove: ...

open_problem | open | MathDB 页面 | 原文线索

V2-037926 | MATHDB-384504

Extrapolation results

摘录 / EXCERPT: Is it possible to develop some extrapolation theory for matrix weights?

open_problem | open | MathDB 页面 | 原文线索

V2-037927 | MATHDB-384505

Sparse operators

摘录 / EXCERPT: It is necessary to find the correct notion of a sparse operator in a two or more parameters setting.

open_problem | open | MathDB 页面 | 原文线索

V2-037928 | MATHDB-384506

Exponential results

摘录 / EXCERPT: Determine if we have the following ...

open_problem | open | MathDB 页面 | 原文线索

V2-037929 | MATHDB-384507

Zygmund dilations

摘录 / EXCERPT: Is there a notion of sparse domination for this kind of operators?

open_problem | open | MathDB 页面 | 原文线索

V2-037930 | MATHDB-384508

Representation theorem

摘录 / EXCERPT: Find a representation theorem for Calder'on-Zygmund operators with Zygmund dilations that allows to produce an sparse bound by dominating the Haar shifts.

open_problem | open | MathDB 页面 | 原文线索

V2-037931 | MATHDB-384509

L^p boundedness for strong maximal function

摘录 / EXCERPT: Considering the weighted strong maximal function $\mathcal{M}_s^{w}$. Is it possible to obtain by means of sparse domination, a sharp dependence of...

open_problem | open | MathDB 页面 | 原文线索

V2-037932 | MATHDB-384510

Separated bump conjecture

摘录 / EXCERPT: Determine if the conditions ... imply that I_{α} is bounded from $L^p(\sigma)$ to $L^p(u)$.

open_problem | open | MathDB 页面 | 原文线索

V2-037933 | MATHDB-384511

A_2 conjecture in non-homogeneous setting

摘录 / EXCERPT: Find a sparse domination result, to prove (one of) the following conjectures: begin{enumerate} item $\displaystyle \| T \|_{L^2(w) \rightarrow L^2(w)} \lesssim [w]_{A_2^0(\mu)}$ it...

open_problem | open | MathDB 页面 | 原文线索

V2-037934 | MATHDB-384512

Control of a maximal function

摘录 / EXCERPT: What is the relation between $Z_p(\mu)$ and $A_2^{\eta}(\mu)$ defined in problem 7.1.

open_problem | open | MathDB 页面 | 原文线索

V2-037935 | MATHDB-384513

Sparse domination of $\widetilde{\mathcal{M}}$

摘录 / EXCERPT: Can we find a sparse operator that controls the maximal function $\widetilde{\mathcal{M}}$ related to the Tolsa $Z_p(\mu)$ condition.

open_problem | open | MathDB 页面 | 原文线索

V2-037936 | MATHDB-384514

Representative class

摘录 / EXCERPT: Find a representative class for Tolsa's theorem. That is, find a subfamily of operators (for example, Riesz transforms) such that the boundedness on $L^p(w)$ of the elements of thi...

open_problem | open | MathDB 页面 | 原文线索

V2-037937 | MATHDB-384515

Structure theory

摘录 / EXCERPT: Develop some structure theory for Z_p weights. This includes, explore extrapolation for such class of weights.

open_problem | open | MathDB 页面 | 原文线索

V2-037938 | MATHDB-384516

Different characterizations of $\widetilde{\mathcal{M}}$

摘录 / EXCERPT: Find another characterization for the boundedness of $\widetilde{\mathcal{M}}$ on $L^p(w)$ similar to the work of Jawerth, in terms of an A_p condition and boundedness of two wei...

open_problem | open | MathDB 页面 | 原文线索

V2-037939 | MATHDB-384517

$T1$ theorem

摘录 / EXCERPT: Prove some version of the $T1$ theorem for the maximal function $\widetilde{\mathcal{M}}$ or for the Calder'on-Zygmund operators.

open_problem | open | MathDB 页面 | 原文线索

V2-037940 | MATHDB-384518

Are there nonspherical, compact, nondegenerate CR manifolds which admit essential conformal vector fields?

摘录 / EXCERPT: Are there nonspherical, compact, nondegenerate CR manifolds which admit essential conformal vector fields?

open_problem | open | MathDB 页面 | 原文线索

V2-037941 | MATHDB-384519

Spectral problems (problem 1)

摘录 / EXCERPT: Let $L=\partial_b + f\bar\partial_b$ on the standard CR three-sphere. Does L have closed range if and only if the Paneitz operator is nonnegative?

open_problem | open | MathDB 页面 | 原文线索

V2-037942 | MATHDB-384520

Spectral problems (problem 2)

摘录 / EXCERPT: Consider the domains ... and ... (a) Can one compute the spectrum of the Kohn Laplacian on $(0,1)$-forms in either case? (b) Can one distinguish Ω_k from U_k using the spectru...

open_problem | open | MathDB 页面 | 原文线索

V2-037943 | MATHDB-384521

CR maps (problem 1)

摘录 / EXCERPT: Let $f\colon S^{2n-1}\to S^{2N-1}$ and $g\colon S^{2n-1}\to S^{2N-1}$ be two CR maps with N minimal; i.e. f and g are not CR maps into a lower dimensional sphere. How can one...

open_problem | open | MathDB 页面 | 原文线索

V2-037944 | MATHDB-384523

Foliations (problem 1)

摘录 / EXCERPT: Let $(M^{2n+d},\mathcal{F}^d)$ be a transversely holomorphic foliation. There is a "complexification" $\tilde M^{2n+2d}$ of $(M,\mathcal{F})$. Does $\tilde M$ have a complex struct...

open_problem | open | MathDB 页面 | 原文线索

V2-037945 | MATHDB-384524

Can one understand the existence, regularity, and stability properties of exponentially subelliptic harmonic maps?

摘录 / EXCERPT: Can one understand the existence, regularity, and stability properties of exponentially subelliptic harmonic maps? Specifically: (a) What is the first variation of the exponential...

open_problem | open | MathDB 页面 | 原文线索

V2-037946 | MATHDB-384525

Chains and pseudo-circles (problem 1)

摘录 / EXCERPT: What do pseudo-circles really mean? More precisely: (a) Is there a variational formulation of chains and pseudo-circles? For example, is there a Finsler structure on M^3 which yi...

open_problem | open | MathDB 页面 | 原文线索

V2-037947 | MATHDB-384526

Iterated commutators and derivatives of the Levi form (problem 1)

摘录 / EXCERPT: Let M be a pseudoconvex CR manifold of hypersurface type. Let L be a (1,0) vector field. Let $\lambda(L,\bar L)$ denote the Levi form on L. Consider two notions: the type of L...

open_problem | open | MathDB 页面 | 原文线索

V2-037948 | MATHDB-384527

Convergence of the polygonal Steklov spectrum to the spectrum of a smooth domain

摘录 / EXCERPT: Let $\mathbb{D}$ be a disk with P_n a regular n-gon inscribed in $\mathbb{D}$ or circumscribing $\mathbb{D}$. Estimate the difference between their kth Steklov eigenvalues ……

open_problem | open | MathDB 页面 | 原文线索

V2-037949 | MATHDB-384528

Oscillating behaviour of eigenfunctions

摘录 / EXCERPT: Capture geometrically the oscillatory behaviour of eigenfunctions in higher dimension. Can we factor out the oscillatory behaviour from the eigenfunctions like it is done in dimens...

open_problem | open | MathDB 页面 | 原文线索

V2-037950 | MATHDB-384529

Steklov asymptotics for generalised cylinders and cones

摘录 / EXCERPT: Calculate Steklov asymptotics for domains with corners, or specifically for cylinders and cones in term of the Laplace spectrum of the base manifold?

open_problem | open | MathDB 页面 | 原文线索

V2-037951 | MATHDB-384530

The Steklov spectrum of a Helmholtz problem

摘录 / EXCERPT: For a bounded function ρ, and fixed λ, consider the problem ... a) What is the relation between ρ and the number of negative eigenvalues? b) What are the asymptotic...

open_problem | open | MathDB 页面 | 原文线索

V2-037952 | MATHDB-384531

Heat trace asymptotics for the Steklov problem on orbifolds, polygons and polyhedra

摘录 / EXCERPT: a) Prove existence of heat trace asymptotics for orbifolds, polygons or polyhedrons. b) Compute the heat trace coefficient in those settings.

open_problem | open | MathDB 页面 | 原文线索

V2-037953 | MATHDB-384532

Weyl law for Lipschitz domain

摘录 / EXCERPT: Prove Weyl's law for the Steklov problem on domains with Lipschitz boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-037954 | MATHDB-384533

Properties of the single and double layer operators

摘录 / EXCERPT: Explore the relation between the spectral properties of the Dirichlet-to-Neumann map, the single layer and the double layer operators in the presence of corners. Does inverse simpl...

open_problem | open | MathDB 页面 | 原文线索

V2-037955 | MATHDB-384534

Can one find the same relations for the Helmholtz case?

摘录 / EXCERPT: Can one find the same relations for the Helmholtz case? For the Steklov-Maxwell case?

open_problem | open | MathDB 页面 | 原文线索

V2-037956 | MATHDB-384535

Steklov spectrum for the Lamé operator

摘录 / EXCERPT: On a domain with corners, find the behaviour in κ of the Steklov problem for the Lamé operator, ...

open_problem | open | MathDB 页面 | 原文线索

V2-037957 | MATHDB-384536

What is the Steklov-Maxwell problem?

摘录 / EXCERPT: Find the most natural formulation of the Steklov-Maxwell problem.

open_problem | open | MathDB 页面 | 原文线索

V2-037958 | MATHDB-384537

Eigenfunctions on the ellipse

摘录 / EXCERPT: Calculate explicitly the eigenfunctions of the Steklov problem on the ellipse

open_problem | open | MathDB 页面 | 原文线索

V2-037959 | MATHDB-384538

Obtaining the domain from polynomial eigenfunctions

摘录 / EXCERPT: If all Steklov eigenfunctions are polynomials on a planar domain then that domain is a disk.

open_problem | open | MathDB 页面 | 原文线索

V2-037960 | MATHDB-384539

Obtaining the domain from analytic continuation

摘录 / EXCERPT: Suppose that all Steklov eigenfunctions of a smooth planar domain can be continued harmonically across the boundary to the whole plane. Does it follow that this domain is a disk?

open_problem | open | MathDB 页面 | 原文线索

V2-037961 | MATHDB-384540

Homogeneisation of the Steklov-Neumann problem

摘录 / EXCERPT: Let Ω be a domain in $\mathbb R^2$ and fix a fraction of the perimeter. Consider partitions of its boundary into $\Gamma = \Gamma_N \cup \Gamma_S$. Put Neumann boundary cond...

open_problem | open | MathDB 页面 | 原文线索

V2-037962 | MATHDB-384541

Symmetry of Steklov optimisers

摘录 / EXCERPT: Show analytically that the actual maximisers to the Steklov problem do exhibit those symmetries.

open_problem | open | MathDB 页面 | 原文线索

V2-037963 | MATHDB-384542

Eigenvalue degeneration for Riemannian metrics

摘录 / EXCERPT: In dimension $d = 3$, can one find a sequence of Riemannian metrics such that both the boundary and interior volume stay bounded but such that $\sigma_1 \to \infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-037964 | MATHDB-384543

Eigenvalue optimisation in dimension 2

摘录 / EXCERPT: a) Is the 4π bound sharp? If not, find a sharp bound. b) Find the maximising domain or show that such a domain does not exist.

open_problem | open | MathDB 页面 | 原文线索

V2-037965 | MATHDB-384544

Shape optimisation in the Riemannian setting under volume constraint.

摘录 / EXCERPT: Let $\mathbb B^d$ be a ball of unit volume with metric g in the conformal class of the Euclidean metric. Is it true that the Euclidean ball maximises σ_1? Is σ_1 bo...

open_problem | open | MathDB 页面 | 原文线索

V2-037966 | MATHDB-384545

Pólya-Szegö conjecture for the Steklov problem

摘录 / EXCERPT: Do regular n-gons maximise σ_1 among all n-gons for given area or perimeter?

open_problem | open | MathDB 页面 | 原文线索

V2-037967 | MATHDB-384546

Eigenvalue multiplicity in dimension two

摘录 / EXCERPT: a) Prove that the spectrum in the sloshing problem is always simple (Conjecture of Kozlov-Kuznetsov-Motygin) b) Does there exist a simply connected planar domain with a Steklov eig...

open_problem | open | MathDB 页面 | 原文线索

V2-037968 | MATHDB-384547

Spectrum of infinite domains

摘录 / EXCERPT: a) Does the analytic continuation of the resolvent have poles? If so, characterize them. b) What is the convergence of the spectrum for truncated domains converging to this infinit...

open_problem | open | MathDB 页面 | 原文线索

V2-037969 | MATHDB-384548

Inverse problems for polygons

摘录 / EXCERPT: Reconstruct a polygon from Steklov spectral asymptotics modulo $o(1)$.

open_problem | open | MathDB 页面 | 原文线索

V2-037970 | MATHDB-384549

Can one hear the genus of a surface from its Steklov spectrum

摘录 / EXCERPT: a) Given a Riemannian surface with boundary, can one obtain its genus from the Steklov spectrum? b)? Given a closed surface embedded in $\mathbb{R}^3$, can one determine the genus...

open_problem | open | MathDB 页面 | 原文线索

V2-037971 | MATHDB-384550

Inhomogeneous Landau Equation

摘录 / EXCERPT: Let us consider the inhomogeneous Landau equation $\partial_f+v\cdot\nabla_xf=Q(f,f)$, where $f=f(x,v,t)$ and ... with $\Pi(z)=\frac{1}{8\pi}(id-\frac{z\otimes z}{|z|^2})$. Is it pos...

open_problem | open | MathDB 页面 | 原文线索

V2-037972 | MATHDB-384551

Homogeneous Landau equation

摘录 / EXCERPT: Let us now look at the homogeneous Landau equation, i.e. $\partial_tf=Q(f,f)$, where Q is as the previous problem. Can we prove that $f\in L_t^{\infty}L_v^p$, for $p>1$? Another...

open_problem | open | MathDB 页面 | 原文线索

V2-037973 | MATHDB-384552

Isotropic Landau equation

摘录 / EXCERPT: What is the long-time decay of the solution toward the steady state $f_{\infty}=0$?

open_problem | open | MathDB 页面 | 原文线索

V2-037974 | MATHDB-384553

Aggregation equation on manifolds

摘录 / EXCERPT: The aggregation equation $\partial_t\rho-\nabla\dot(\rho\nabla K*\rho)=0$ has been succesfully studied by several authors in ${\\mathbb{R}}^d$ and in manifolds...

open_problem | open | MathDB 页面 | 原文线索

V2-037975 | MATHDB-384554

Fractional parabolic Keller-Segel

摘录 / EXCERPT: Consider the following type of Keller-Segel equation ... where $s\in[0,1]$. The main issue with this problem is the lack of coercivity, which is needed to control the nonlinear terms...

open_problem | open | MathDB 页面 | 原文线索

V2-037976 | MATHDB-384555

Long time behavior of aggregation-diffusion equation

摘录 / EXCERPT: Let us consider the aggregation-diffusion equation $\partial_t\rho=\epsilon\Delta\rho+\nabla\cdot(\rho\nabla W*\rho)$, where $W\in L^1$ is a bounded attractive interaction potentia...

open_problem | open | MathDB 页面 | 原文线索

V2-037977 | MATHDB-384556

Particle approximation for aggregation equation

摘录 / EXCERPT: Let us consider the aggregation equation $\partial_t\rho=\nabla\cdot(\rho\nabla W*\rho)$ and the energy functional associated $E[\rho]=\frac{1}{2}\int_{\mathbb{R}^d}W*\rho\,d\rho$,...

open_problem | open | MathDB 页面 | 原文线索

V2-037978 | MATHDB-384557

"Fractional thin film" equation

摘录 / EXCERPT: Let us consider the aggregation equation $\partial_t\rho=\nabla\cdot(\rho\nabla W*\rho)$, where $W(x)=-\frac{|x|^\alpha}{\alpha}$ for $\alpha\le d$. What can we say about the well-...

open_problem | open | MathDB 页面 | 原文线索

V2-037979 | MATHDB-384558

After blow-up

摘录 / EXCERPT: Let us consider the Keller-Segel equation $\partial_t\rho=\partial_{xx}\rho+\partial_x(\rho\partial_x(\log|x|*\rho))$ in 1D. Could we derive a generalized KS model for which we can...

open_problem | open | MathDB 页面 | 原文线索

V2-037980 | MATHDB-384561

Numerics for Landau collision operator

摘录 / EXCERPT: Among the deterministic numerical methods for the Landau equation, it is known that the Fourier spectral method can produce very accurate results and is efficient to implement. How...

open_problem | open | MathDB 页面 | 原文线索

V2-037981 | MATHDB-384562

Numerical methods for nonlinear equation with gradient flow structure

摘录 / EXCERPT: Numerical methods for nonlinear equations with a gradient flow structure are expected to conserve mass, preserve positivity, dissipate the energy, and capture the steady states in...

open_problem | open | MathDB 页面 | 原文线索

V2-037982 | MATHDB-384563

Clustering in biology

摘录 / EXCERPT: Let us assume there are some dividing cells communicating with each other and moving around in $\mathbb{R}^3$ or $\mathbb{R}^2$ with both cell growth and not necessarily totally co...

open_problem | open | MathDB 页面 | 原文线索

V2-037983 | MATHDB-384564

Gang Dynamics

摘录 / EXCERPT: Let us consider two opponent gangs of graffiti (A and B). We can derive a model describing their behavior from the lattice model. Indeed, if g_A, g_B are the densities correspo...

open_problem | open | MathDB 页面 | 原文线索

V2-037984 | MATHDB-384570

Problems on Holomorphic Function Spaces (problem 6)

摘录 / EXCERPT: Let $b \in H^\infty (\Delta)$ with $|| b ||_{H^\infty} \leq 1$. Define $H(b):=(I-T_b T_{\overline{b}})^{1/2} H^2(\Delta)$, where T_b denotes the multiplication operator by b. b...

open_problem | open | MathDB 页面 | 原文线索

V2-037985 | MATHDB-384572

Problems on Holomorphic Function Spaces (problem 8)

摘录 / EXCERPT: Define...

open_problem | open | MathDB 页面 | 原文线索

V2-037986 | MATHDB-384576

Problems on Holomorphic Function Spaces (problem 12)

摘录 / EXCERPT: Can region of attractions for non-polynomial automorphisms of $\mathbb{C}^2$ have C^1 boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-037987 | MATHDB-384578

Problems on Holomorphic Function Spaces (problem 14)

摘录 / EXCERPT: For a domain $\Omega \subset \mathbb{C}$, define ... Does log-capacity$(\mathbb{C}\setminus \Omega) > 0$ imply ${\rm dim}\, H^2 (\Omega) > 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-037988 | MATHDB-384582

Problems on Complex Dynamics (problem 2)

摘录 / EXCERPT: Investigate the dynamics of maps of the form ... where $p(x)$ is a degree 2 rational map.

open_problem | open | MathDB 页面 | 原文线索

V2-037989 | MATHDB-384584

Study hedgehogs for gems of automorphism of ... with both elliptic eigenvalues.

摘录 / EXCERPT: Study hedgehogs for gems of automorphism of $\mathbb{C}^2$ with both elliptic eigenvalues.

open_problem | open | MathDB 页面 | 原文线索

V2-037990 | MATHDB-384585

Problems on Complex Dynamics (problem 5)

摘录 / EXCERPT: Let C be the cubic surface defined by ... where D is a complex parameter. Let $s_x: C \rightarrow C$ be defined by ... The maps s_y, s_z are defined similarly. Consider, for exam...

open_problem | open | MathDB 页面 | 原文线索

V2-037991 | MATHDB-384586

Monotonicity of the spectral gap

摘录 / EXCERPT: For a bounded convex domain $\Omega \subset \mathbb{R}^n$, the spectral gap $\lambda_2(\Omega, \alpha)-\lambda_1(\Omega, \alpha)$ is strictly increasing as a function of $\alpha>0$...

open_problem | open | MathDB 页面 | 原文线索

V2-037992 | MATHDB-384587

Bareket's conjecture

摘录 / EXCERPT: Let $\lambda_1(\Omega, \alpha)$ be the lowest Robin eigenvalue on a bounded domain Ω with a negative boundary parameter α. begin{enumerate} item In two dimensions, th...

open_problem | open | MathDB 页面 | 原文线索

V2-037993 | MATHDB-384588

Discrete Bareket's conjecture

摘录 / EXCERPT: Among all triangles of a given area $\lambda_1(\Omega, \alpha)$ is: begin{enumerate} item minimized by the equilateral triangle when $\alpha>0$. item maximized by the equilateral t...

open_problem | open | MathDB 页面 | 原文线索

V2-037994 | MATHDB-384589

Negative Robin eigenvalues on a locally deformed half-space

摘录 / EXCERPT: Prove or disprove that there are always negative eigenvalues below $-\alpha^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-037995 | MATHDB-384590

Multiplicity of optimal Dirichlet eigenvalues

摘录 / EXCERPT: Let Ω_k^* be a minimizer of the shape optimization problem ... Prove that $\lambda_{k-1}(\Omega_k^*)=\lambda_k(\Omega_k^*)$.

open_problem | open | MathDB 页面 | 原文线索

V2-037996 | MATHDB-384591

Mahler inequality for the principal Dirichlet eigenvalue

摘录 / EXCERPT: A hypercube in $\mathbb{R}^n$ is an attainer for the maximization problem ...

open_problem | open | MathDB 页面 | 原文线索

V2-037997 | MATHDB-384592

Discrete Faber-Krahn inequality

摘录 / EXCERPT: Among all polygons of a given area and having not more than N edges the lowest Dirichlet eigenvalue $\lambda_1(\Omega)$ is minimized by the regular N-gon.

open_problem | open | MathDB 页面 | 原文线索

V2-037998 | MATHDB-384593

van den Berg's conjecture

摘录 / EXCERPT: There is a dimensional constant C such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-037999 | MATHDB-384594

Concavity of the principal Dirichlet eigenfunction

摘录 / EXCERPT: The best concavity exponent $\alpha(\Omega)$ is maximized by the ball.

open_problem | open | MathDB 页面 | 原文线索

V2-038000 | MATHDB-384595

Optimizers for Neumann eigenvalues among convex sets of a fixed perimeter

摘录 / EXCERPT: Prove the existence of optimizers for ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-038001 | MATHDB-384596

Maximizing the ratio of Neumann eigenvalues among convex domains

摘录 / EXCERPT: For any convex $\Omega\subset \mathbb{R}^n$, it holds that begin{enumerate}[label=textup{(alph)}] item $\mu_2(\Omega)/\mu_1(\Omega)\leq 4$. item...

open_problem | open | MathDB 页面 | 原文线索

V2-038002 | MATHDB-384597

Hot spots conjecture

摘录 / EXCERPT: The first non-trivial Neumann eigenfunction does not have global extrema in the interior of Ω if begin{enumerate} item $\Omega\subset \mathbb{R}^2$ is simply connected. item...

open_problem | open | MathDB 页面 | 原文线索

V2-038003 | MATHDB-384598

Directional hot spots conjecture

摘录 / EXCERPT: Can one always find a direction $\mathbf{e}\in\mathbb{R}^2$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-038004 | MATHDB-384599

Non-existence of a maximizer for the first Steklov eigenvalue with perimeter constraint

摘录 / EXCERPT: Prove that there is no $\Omega\subset \mathbb{R}^n$ which maximizes the quantity $P(\Omega)^{1/(n-1)}\sigma_1(\Omega)$.

open_problem | open | MathDB 页面 | 原文线索

V2-038005 | MATHDB-384601

Hermite-Hadamard inequality in higher dimensions

摘录 / EXCERPT: Find the optimal constants in the inequalities.

open_problem | open | MathDB 页面 | 原文线索

V2-038006 | MATHDB-384602

Asymptotic partitioning problems

摘录 / EXCERPT: Find conditions for problems involving optimal N-partitions of a planar domain such that one observes in the limit $N\to \infty$ a packing of hexagons.

open_problem | open | MathDB 页面 | 原文线索

V2-038007 | MATHDB-384603

The ovals of Benguria and Loss

摘录 / EXCERPT: The following inequality $\lambda_1(H_\Gamma) \ge 1$ holds.

open_problem | open | MathDB 页面 | 原文线索

V2-038008 | MATHDB-384604

Faber-Krahn for the buckling problem

摘录 / EXCERPT: Prove that the lowest buckling eigenvalue $\Lambda_1(\Omega)$ is minimized by the disk among all simply connected sets of equal area.

open_problem | open | MathDB 页面 | 原文线索

V2-038009 | MATHDB-384606

Nodal structure of the k-th Neumann eigenfunction for collapsing domains

摘录 / EXCERPT: Under what assumptions can one prove that the nodal domains are in a sense ordered?

open_problem | open | MathDB 页面 | 原文线索

V2-038010 | MATHDB-384608

Study the bifurcation problem of Taylor--Couette instability.

摘录 / EXCERPT: Study the bifurcation problem of Taylor--Couette instability.

open_problem | open | MathDB 页面 | 原文线索

V2-038011 | MATHDB-384611

Prove sharp estimates of the ... norm of solutions to the 1D periodic viscous Burger's equation.

摘录 / EXCERPT: Prove sharp estimates of the $H ^1$ norm of solutions to the 1D periodic viscous Burger's equation. It is proven the enstrophy...

open_problem | open | MathDB 页面 | 原文线索

V2-038012 | MATHDB-384612

Prove the global existence of the 2D Burger's equation with hyperviscosity.

摘录 / EXCERPT: Prove the global existence of the 2D Burger's equation with hyperviscosity. ...

open_problem | open | MathDB 页面 | 原文线索

V2-038013 | MATHDB-384613

Prove the global existence of strong solutions to the 1D surface growth model.

摘录 / EXCERPT: Prove the global existence of strong solutions to the 1D surface growth model. ... Show the ill-posedness of the inviscid version of this model.

open_problem | open | MathDB 页面 | 原文线索

V2-038014 | MATHDB-384615

Long-time dynamics (problem 2)

摘录 / EXCERPT: Doering conjecture: for a transport equation ... show that ...

open_problem | open | MathDB 页面 | 原文线索

V2-038015 | MATHDB-384617

A stronger form of Aaronson-Ambainis conjecture

摘录 / EXCERPT: Show that there exists $C>0$ such that for all $n\ge 1$ and for all $f:\{-1,1\}^n\to \mathbb{R}$ we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-038016 | MATHDB-384618

Some special cases

摘录 / EXCERPT: Show that the above Aaronson-Ambainis conjecture and its stronger form are valid for begin{enumerate} item all homogeneous functions on $\{-1,1\}^n$; item all functions...

open_problem | open | MathDB 页面 | 原文线索

V2-038017 | MATHDB-384619

Heat-smoothing conjecture

摘录 / EXCERPT: Fix $d\ge 1$ and $1< p < \infty$. Show that there exists $c_p>0$ such that for all $n\ge 1$ and all $f:\{-1,1\}^n\to \mathbb{R}$ of the form $f=\sum_{|S|\ge d}\widehat{f}(S)\chi_S$...

open_problem | open | MathDB 页面 | 原文线索

V2-038018 | MATHDB-384621

Gaussian heat-smoothing

摘录 / EXCERPT: Fix $d\ge 1$ and $1< p < \infty$. Show that there exists $c_p>0$ such that for all $n\ge 1$ and all polynomials $f\in \text{Lin}\{x^k,k\ge d\}$, we have ... where $P_t=e^{-tN}$ is th...

open_problem | open | MathDB 页面 | 原文线索

V2-038019 | MATHDB-384622

Gaussian heat-smoothing: a weaker form

摘录 / EXCERPT: Fix $d\ge 1$ and $1< p < \infty$. Show that there exists $c_p>0$ such that for all $n\ge 1$ and all polynomials $f\in \text{Lin}\{x^k,k\ge d\}$, we have ... where $N=\Delta-x\nabla$...

open_problem | open | MathDB 页面 | 原文线索

V2-038020 | MATHDB-384623

Reverse heat-smoothing

摘录 / EXCERPT: Fix $d\ge 1$ and $1< p < \infty$. Show that there exists $c_p>0$ such that for all $n\ge 1$, begin{enumerate} item and for all polynomials f on $\{-1,1\}^n$ of degree at most d...

open_problem | open | MathDB 页面 | 原文线索

V2-038021 | MATHDB-384624

Best constants in Bohnenblust-Hille inequalities for hypercubes

摘录 / EXCERPT: Fix $d\ge 1$. Show that there exists $C>0$ such that for all $n\ge 1$ and all polynomials $f=\sum_{|S|\le d}\widehat{f}(S)\chi_S$ on $\{-1,1\}^n$ of degree at most d, we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-038022 | MATHDB-384625

Homogeneous case

摘录 / EXCERPT: Fix $d\ge 1$. Show that there exists $C>0$ such that for all $n\ge 1$ and all d-homogeneous polynomials $f=\sum_{|S|= d}\widehat{f}(S)\chi_S$ on $\{-1,1\}^n$, we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-038023 | MATHDB-384626

The sensitivity theorem

摘录 / EXCERPT: The sensitivity theorem says that for all $n\ge 1$ and all Boolean function $f:\{-1,1\}^n\to \{-1,1\}$, we have ... More proofs of this result? Is it possible to formulate a reasonab...

open_problem | open | MathDB 页面 | 原文线索

V2-038024 | MATHDB-384627

Moment comparison

摘录 / EXCERPT: Consider the following problems: begin{enumerate} item Find the best $C>0$ such that for all $f:\{-1,1\}^n\to \mathbb{R}$, ... item For $1< p < q \le 2$, find the best $C_{p,q}>0$ su...

open_problem | open | MathDB 页面 | 原文线索

V2-038025 | MATHDB-384628

Pathological sets (problem 1)

摘录 / EXCERPT: Assume CH and fix $s \in [0, 1)$. Is there a set $E \subseteq {\\mathbb{R}}^2$ such that $\text{dim}_H(E) = 1 + s$ and for every continuous function...

open_problem | open | MathDB 页面 | 原文线索

V2-038026 | MATHDB-384629

Pathological sets (problem 2)

摘录 / EXCERPT: Assume CH and fix $s \in [0, 1]$. Is there a set $E \subseteq 2^\omega$ such that $\text{dim}_H(E) = s$ and for every Borel $f \colon 2^\omega \to 2^\omega$,...

open_problem | open | MathDB 页面 | 原文线索

V2-038027 | MATHDB-384630

Pathological sets (problem 3)

摘录 / EXCERPT: Let $Z \in 2^\omega$ be arbitrary and fix $s \in (0, 1)$. Is there some $X \in 2^\omega$ such that $\text{dim}^Z(X) = s$ and for all $Y \leq_T X'$, $\text{dim}^{0'}(Y) \leq s$?

open_problem | open | MathDB 页面 | 原文线索

V2-038028 | MATHDB-384631

Pathological sets (problem 4)

摘录 / EXCERPT: Does the existence of a set E such that E is not σ-finite in dimension 1 but for every superlinear gauge function φ, $H_\varphi(E) = 0$ imply that there is a c...

open_problem | open | MathDB 页面 | 原文线索

V2-038029 | MATHDB-384632

Most constructions of sets with pathological behavior related to Hausdorff dimension require some assumption beyond ZFC, i.e.

摘录 / EXCERPT: Most constructions of sets with pathological behavior related to Hausdorff dimension require some assumption beyond ZFC, i.e. CH, Martin's axiom, the failure of the Borel conjectur...

open_problem | open | MathDB 页面 | 原文线索

V2-038030 | MATHDB-384633

Positive results (problem 1)

摘录 / EXCERPT: For what s can we prove that if $E \subseteq {\\mathbb{R}}^2$ is of Hausdorff dimension greater than 1 then $\text{dim}_H(\Delta(E)) \geq s$?

open_problem | open | MathDB 页面 | 原文线索

V2-038031 | MATHDB-384634

Positive results (problem 2)

摘录 / EXCERPT: Marstrand's theorem for works when $\text{dim}_H(E) = \text{dim}_P(E)$---i.e. for such an E and for almost every direction e,...

open_problem | open | MathDB 页面 | 原文线索

V2-038032 | MATHDB-384635

Is there an effective proof of Marstrand's second projection theorem?

摘录 / EXCERPT: Is there an effective proof of Marstrand's second projection theorem? Also, is the following effective version of the theorem true: if $x \in {\\mathbb{R}}^2$ is such that...

open_problem | open | MathDB 页面 | 原文线索

V2-038033 | MATHDB-384636

Positive results (problem 4)

摘录 / EXCERPT: Is there a set $E \subseteq [0,1]^2$ which is Π^0_1 and of Hausdorff dimension > 1 such that for every non-Schnorr random direction e, $\text{proj}_e(E)$ has measure 0? O...

open_problem | open | MathDB 页面 | 原文线索

V2-038034 | MATHDB-384637

Are there geometric properties that imply (or are equivalent to) a set having a Hausdorff optimal oracle?

摘录 / EXCERPT: Are there geometric properties that imply (or are equivalent to) a set having a Hausdorff optimal oracle? More generally, is the definition of Hausdorff optimal oracle “right”? Is...

open_problem | open | MathDB 页面 | 原文线索

V2-038035 | MATHDB-384639

Positive results (problem 7)

摘录 / EXCERPT: For which f do we have $(2^k\cdot x)_{k \in {\\mathbb{N}}}$ equidistributed mod 1 whenever x is f-normal?

open_problem | open | MathDB 页面 | 原文线索

V2-038036 | MATHDB-384643

Toeplitz Determinants and Toeplitz Operators (problem 4)

摘录 / EXCERPT: (Berger-Coburn 1994): On the Fock/Bergman space, define the projection $\prod:L^2(\mathbb{C},e^{-|z|^2}dA) \rightarrow \mathscr{F}_{\mathbb{C}}$. Take...

open_problem | open | MathDB 页面 | 原文线索

V2-038037 | MATHDB-384648

Riemann-Hilbert Problems (problem 4)

摘录 / EXCERPT: For planar orthogonal polynomials, we know that ... and ... Is there an analogue of small norm theory of such $\overline{\partial}$-problems (i.e....

open_problem | open | MathDB 页面 | 原文线索

V2-038038 | MATHDB-384649

Riemann-Hilbert Problems (problem 5)

摘录 / EXCERPT: Analyze the condition given by ... where $Q=|z|^2+2t\text{Re}(z^2)$ and where $\overline{H_n}$ is Hermite.

open_problem | open | MathDB 页面 | 原文线索

V2-038039 | MATHDB-384650

Riemann-Hilbert Problems (problem 6)

摘录 / EXCERPT: Consider ... for constants a_i, b_i. Suppose Wiener-Hopf factorization has no partial indices. Study the behavior as $N \rightarrow \infty$. The asymptotics are known if the a_i'...

open_problem | open | MathDB 页面 | 原文线索

V2-038040 | MATHDB-384651

Riemann-Hilbert Problems (problem 7)

摘录 / EXCERPT: Consider ... Study Z_N as $N \rightarrow \infty$, where Γ is a smooth loop on the Riemann sphere.

open_problem | open | MathDB 页面 | 原文线索

V2-038041 | MATHDB-384652

The discrete Bessel process is connected to Poissonized Plancherel measure.

摘录 / EXCERPT: The discrete Bessel process is connected to Poissonized Plancherel measure. Consider the distance of the largest particle: $\mathbb{P}[Q_{\text{max}}\leq x]=q(x)$, where q solves...

open_problem | open | MathDB 页面 | 原文线索

V2-038042 | MATHDB-384654

Spectrum of Hilbert Matrices (problem 1)

摘录 / EXCERPT: The $\mathscr{l}^2$-spectrum for the Hilbert matrix $H_{m n}=\frac{1}{m+n+1}$ is known. Find the spectrum on spaces of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-038043 | MATHDB-384655

Spectrum of Hilbert Matrices (problem 2)

摘录 / EXCERPT: Define ... for $\alpha >0$ and $|\xi|=1$. Given ξ_1, ξ_2, ξ_3 and $\alpha_1, \alpha_2, \alpha_3$, can we approximate ...

open_problem | open | MathDB 页面 | 原文线索

V2-038044 | MATHDB-384658

Embedding Unions (problem 1)

摘录 / EXCERPT: Suppose A and B are subsets of some metric space such that both A and B biLipschitzly embed into L_1. Does $A \cup B$ biLipschitzly embed into L_1?

open_problem | open | MathDB 页面 | 原文线索

V2-038045 | MATHDB-384659

Embedding Unions (problem 2)

摘录 / EXCERPT: Fix $D<\infty$. Suppose $A_1, A_2, \dots A_k$ are subsets of some metric space such that $c_2(A_i) \leq D$ for each i. What are the asymptotics (in terms of k) of...

open_problem | open | MathDB 页面 | 原文线索

V2-038046 | MATHDB-384660

Uniform and Coarse Embeddings (problem 1)

摘录 / EXCERPT: Suppose X is a Banach space that coarsely contains every locally finite metric space. Does X have trivial cotype?

open_problem | open | MathDB 页面 | 原文线索

V2-038047 | MATHDB-384661

Does every metric space of bounded geometry admit a coarse embedding into a (possibly infinite dimensional) CAT(0) space?

摘录 / EXCERPT: Does every metric space of bounded geometry admit a coarse embedding into a (possibly infinite dimensional) CAT(0) space? Does every finitely generated group admit such an embeddin...

open_problem | open | MathDB 页面 | 原文线索

V2-038048 | MATHDB-384662

Uniform and Coarse Embeddings (problem 3)

摘录 / EXCERPT: Let $q \in (2,\infty)$. Does L_q admit a coarse or uniform embedding into ℓ_q?

open_problem | open | MathDB 页面 | 原文线索

V2-038049 | MATHDB-384663

Uniform and Coarse Embeddings (problem 4)

摘录 / EXCERPT: Let $q \in (2,\infty)$. Does $(L_q,\|\cdot\|^{\frac12})$ admit a biLipschitz embedding into L_q?

open_problem | open | MathDB 页面 | 原文线索

V2-038050 | MATHDB-384664

Is there a bounded geometry space that coarsely contains every bounded geometry space?

摘录 / EXCERPT: Is there a bounded geometry space that coarsely contains every bounded geometry space? Is there a bounded geometry space with Property A that coarsely contains every bounded geomet...

open_problem | open | MathDB 页面 | 原文线索

V2-038051 | MATHDB-384665

Uniform and Coarse Embeddings (problem 6)

摘录 / EXCERPT: Is it true that for every bounded degree graph G, there exists a finitely generated group that coarsely contains G?

open_problem | open | MathDB 页面 | 原文线索

V2-038052 | MATHDB-384666

Does the unit ball of every superreflexive Banach space uniformly embed into some Hilbert space?

摘录 / EXCERPT: Does the unit ball of every superreflexive Banach space uniformly embed into some Hilbert space?

open_problem | open | MathDB 页面 | 原文线索

V2-038053 | MATHDB-384667

Is every Banach space with Markov type 2 superreflexive?

摘录 / EXCERPT: Is every Banach space with Markov type 2 superreflexive? More generally, is every Banach space with nontrivial Markov type superreflexive?

open_problem | open | MathDB 页面 | 原文线索

V2-038054 | MATHDB-384668

Is Markov type 2 equivalent to maximal Markov type 2 for general metrics spaces?

摘录 / EXCERPT: Is Markov type 2 equivalent to maximal Markov type 2 for general metrics spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-038055 | MATHDB-384669

Does there exist a sequence of ... -regular graphs with unbounded girth that serve as test spaces for superreflexivity?

摘录 / EXCERPT: Does there exist a sequence of k-regular graphs with unbounded girth that serve as test spaces for superreflexivity?

open_problem | open | MathDB 页面 | 原文线索

V2-038056 | MATHDB-384670

Does there exist a bounded degree graph that is a nonlinear expander with respect to every Banach space of nontrivial cotype?

摘录 / EXCERPT: Does there exist a bounded degree graph that is a nonlinear expander with respect to every Banach space of nontrivial cotype? Is every bounded degree expander a nonlinear expander...

open_problem | open | MathDB 页面 | 原文线索

V2-038057 | MATHDB-384671

Does there exist a locally finite transitive graph whose balls (or some subsequence of balls) are expanders?

摘录 / EXCERPT: Does there exist a locally finite transitive graph whose balls (or some subsequence of balls) are expanders?

open_problem | open | MathDB 页面 | 原文线索

V2-038058 | MATHDB-384676

Transportation Cost and Wasserstein Spaces (problem 1)

摘录 / EXCERPT: Does there exist a function $f: [0,\infty) \to [0,\infty)$ such that, for every finite metric space M, the distortion of M into convex combinations of dominating tree metrics i...

open_problem | open | MathDB 页面 | 原文线索

V2-038059 | MATHDB-384677

Transportation Cost and Wasserstein Spaces (problem 2)

摘录 / EXCERPT: Let $M = \{1,\dots \Delta\}^d$ equipped with the ℓ_1 or ℓ_2 metric. Let Y the subset of $W_1(M)$ consisting of uniform distributions over s-point subsets of M. What...

open_problem | open | MathDB 页面 | 原文线索

V2-038060 | MATHDB-384678

Transportation Cost and Wasserstein Spaces (problem 3)

摘录 / EXCERPT: Let $\{G_n\}_{n=1}^\infty$ be a sequence of bounded degree expanders. Does the ℓ_2-sum $(\oplus_n \mathrm{TC}(G_n))_2$ have trivial cotype?

open_problem | open | MathDB 页面 | 原文线索

V2-038061 | MATHDB-384679

Transportation Cost and Wasserstein Spaces (problem 4)

摘录 / EXCERPT: Let $\{1,\dots n\}^d$ denote the d-dimensional grid graph of side length $n-1$. What are the asymptotics of $c_1({\rm TC}(\{1,\dots n\}^d))$ for $d \geq 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-038062 | MATHDB-384680

BiLipschitz Embeddings of Groups (problem 1)

摘录 / EXCERPT: Does the planar lamplighter group $\mathbb{Z}_2 \wr \mathbb{Z}^2$ biLipschitzly embed into L_1?

open_problem | open | MathDB 页面 | 原文线索

V2-038063 | MATHDB-384681

BiLipschitz Embeddings of Groups (problem 2)

摘录 / EXCERPT: Does $\mathrm{SL}_3(\mathbb{Z})$ biLipschitzly embed into L_1?

open_problem | open | MathDB 页面 | 原文线索

V2-038064 | MATHDB-384683

Online Embeddings (problem 1)

摘录 / EXCERPT: Find a nontrivial lower bound (better than $\log n$) for online embedding an n-point metric space into ℓ_1.

open_problem | open | MathDB 页面 | 原文线索

V2-038065 | MATHDB-384684

Euclidean Distortion (problem 1)

摘录 / EXCERPT: For an n-point subset M of a d-dimensional normed space, equipped with the $1/2$-snowflake metric, find, in polynomial time, a nonexpansive embedding of M into ℓ_2 wit...

open_problem | open | MathDB 页面 | 原文线索

V2-038066 | MATHDB-384685

Euclidean Distortion (problem 2)

摘录 / EXCERPT: Suppose M is an n-point metric space that embeds with distortion $D<\infty$ into a convex combination of dominating tree metrics. Is $c_2(M) = o_D(\sqrt{\log n})$?

open_problem | open | MathDB 页面 | 原文线索

V2-038067 | MATHDB-384686

Finding a Planted Vector (problem 1)

摘录 / EXCERPT: Suppose we have n vectors in the unit sphere of $\ell_2^{O(\log n)}$ drawn uniformly at random. We place an extra vector at distance 0.1 from one of the random vectors. Can thi...

open_problem | open | MathDB 页面 | 原文线索

V2-038068 | MATHDB-384687

Computing Spread Constants (problem 1)

摘录 / EXCERPT: Can the spread constant of an n-point metric space be approximated within a constant factor in $O(\text{poly } } n)$ time?

open_problem | open | MathDB 页面 | 原文线索

V2-038069 | MATHDB-384688

Finding Subsets of Vectors with Large Average Correlation (problem 1)

摘录 / EXCERPT: For $\varepsilon,\delta>0$ and unit vectors $v_1,v_2,\ldots,v_n\in \ell_2$, consider the following two properties: begin{enumerate} item[(a)]...

open_problem | open | MathDB 页面 | 原文线索

V2-038070 | MATHDB-385007

Calkin algebra (problem 3)

摘录 / EXCERPT: Is it consistent that, for a nonseparable Hilbert space $\mathcal{H}$, $\mathcal{B}\left(\mathcal{H}\right) \left/ \mathcal{K}\left(\mathcal{H}% \right) \right. $ has an outer au...

open_problem | open | MathDB 页面 | 原文线索

V2-038071 | MATHDB-386937

Validating numerics for analysis

摘录 / EXCERPT: How can we produce rigorous numerics with an emphasis on questions arising in mathematical analysis?

open_problem | open | MathDB 页面 | 原文线索

V2-038072 | MATHDB-387480

Research Problems in Function Theory - Problem 1.4

摘录 / EXCERPT: Let $f(z)$ be an entire function of finite order ρ, and let $n_1(r,a)$ denote the number of simple zeros of the equation $f(z)=a$. If ... where $a\neq b, c<\rho$, is it true tha...

open_problem | open | MathDB 页面 | 原文线索

V2-038073 | MATHDB-387482

Research Problems in Function Theory - Problem 1.6

摘录 / EXCERPT: Arakelyan has proved that, given $\rho>\frac{1}{2}$ and a countable set E, there exists an entire function $f(z)$ of order ρ, for which all the points of E are deficient....

open_problem | open | MathDB 页面 | 原文线索

V2-038074 | MATHDB-387486

Research Problems in Function Theory - Problem 1.11

摘录 / EXCERPT: If $f(z)$ is a meromorphic function of finite order with at least one finite deficient value, does the conclusion of Problem 1.10 hold?

open_problem | open | MathDB 页面 | 原文线索

V2-038075 | MATHDB-387487

Research Problems in Function Theory - Problem 1.12

摘录 / EXCERPT: Edrei, Fuchs and Hellerstein ask if $f(z)$ is an entire function of infinite order with real zeros, is $\delta(0,f)>0$? More generally, is $\delta(0,f)=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-038076 | MATHDB-387488

Research Problems in Function Theory - Problem 1.13

摘录 / EXCERPT: If $f(z)$ is an entire function of finite order ρ and lower order λ with real zeros, find the best possible bound $B=B(\rho,\lambda)$ such that $\delta(0,f)\geq B$. Fr...

open_problem | open | MathDB 页面 | 原文线索

V2-038077 | MATHDB-387491

Research Problems in Function Theory - Problem 1.17

摘录 / EXCERPT: (Paley's conjecture) For any entire function $f(z)$ of finite order ρ in the plane, we have ... where $C(\rho)$ depends on ρ only. This follows very simply from Hayman . It...

open_problem | open | MathDB 页面 | 原文线索

V2-038078 | MATHDB-387492

Research Problems in Function Theory - Problem 1.21

摘录 / EXCERPT: If $f(z)$ is non-constant in the plane, it is known (see Hayman) that ... These inequalities are sharp. It is not known whether ... can be greater than one, or even infinite. It is kn...

open_problem | open | MathDB 页面 | 原文线索

V2-038079 | MATHDB-387493

Research Problems in Function Theory - Problem 1.22

摘录 / EXCERPT: The defect relation ([source label: 1.2]) is a consequence of the inequality (see Hayman), which is called the second fundamental theorem'', ... which holds for any distinct numbers...

open_problem | open | MathDB 页面 | 原文线索

V2-038080 | MATHDB-387495

Research Problems in Function Theory - Problem 1.24

摘录 / EXCERPT: If f is meromorphic in the plane, can $n(r,a)$ be compared in general with its average value ... in the same sort of way that $N(r,a)$ can be compared with $T(r)$? In particular, i...

open_problem | open | MathDB 页面 | 原文线索

V2-038081 | MATHDB-387496

Research Problems in Function Theory - Problem 1.25

摘录 / EXCERPT: In the opposite direction to Problem 1.24, does there exist a meromorphic function such that for every pair of distinct values a, b, we have ... Note, of course, that either of the...

open_problem | open | MathDB 页面 | 原文线索

V2-038082 | MATHDB-387499

Research Problems in Function Theory - Problem 1.28

摘录 / EXCERPT: Are there upper bounds of any kind on the set of asymptotic values of a meromorphic function of finite order? (D. Drasin and A. Weitsman)

open_problem | open | MathDB 页面 | 原文线索

V2-038083 | MATHDB-387503

Research Problems in Function Theory - Problem 1.33

摘录 / EXCERPT: Let f be a meromorphic function of finite order ρ. Does the condition ... imply that 2ρ is an integer? (A. Eremenko)

open_problem | open | MathDB 页面 | 原文线索

V2-038084 | MATHDB-387504

Research Problems in Function Theory - Problem 1.34

摘录 / EXCERPT: Let $n_1(r,a,f)$ denote the number of simple zeros of $f(z)-a$ in $\{|z|\leq r\}$. Selberg has shown that if: [(a)] ; f is a meromorphic function of finite order ρ, and ;...

open_problem | open | MathDB 页面 | 原文线索

V2-038085 | MATHDB-387512

Research Problems in Function Theory - Problem 1.42

摘录 / EXCERPT: Let f be meromorphic in $\mathbb{C}$, and suppose that the function ... is non-constant, where $k\geq3$ and the coefficients a_j are polynomials. Characterise those functions f...

open_problem | open | MathDB 页面 | 原文线索

V2-038086 | MATHDB-387514

Research Problems in Function Theory - Problem 2.2

摘录 / EXCERPT: Produce a general method for constructing an entire function of finite order, and in fact, minimal growth, which tends to different asymptotic values $w_1, w_2, \ldots, w_k$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-038087 | MATHDB-387516

Research Problems in Function Theory - Problem 2.4

摘录 / EXCERPT: Suppose that $f(z)$ is a meromorphic function in the plane, and that for some θ, $0\leq\theta<2\pi$, $f(z)$ assumes every value infinitely often, with at most two exceptions...

open_problem | open | MathDB 页面 | 原文线索

V2-038088 | MATHDB-387517

Research Problems in Function Theory - Problem 2.5

摘录 / EXCERPT: What can we say about the set E of values a which an entire function $f(z)$ assumes infinitely often in every angle? Simple examples show that E may be the whole open plane,...

open_problem | open | MathDB 页面 | 原文线索

V2-038089 | MATHDB-387519

Research Problems in Function Theory - Problem 2.7

摘录 / EXCERPT: If $f(z)$ of finite order, can anything be asserted about the length of Γ_∞, which is the path on which $f(z)$ tends to ∞, or the part of it in $|z|\leq r$?

open_problem | open | MathDB 页面 | 原文线索

V2-038090 | MATHDB-387529

Research Problems in Function Theory - Problem 2.18

摘录 / EXCERPT: Consider the function $b(r)$ of Problem 2.17. Since $\log M(r,f)$ is an analytic function of r, except for isolated points, $b(r)$ exists except at isolated points where the righ...

open_problem | open | MathDB 页面 | 原文线索

V2-038091 | MATHDB-387531

Research Problems in Function Theory - Problem 2.20

摘录 / EXCERPT: If $f(z)$ is an entire function, the iterates $f_n(z), n=1,2,\ldots$ are defined inductively by ... A point z satisfying the equation $f_n(z)=z$, but such that $f_k(z)\neq z$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-038092 | MATHDB-387532

Research Problems in Function Theory - Problem 2.21

摘录 / EXCERPT: If, in the terminology of Problem 2.20, z_0 is a fixed point of exact order n for $f(z)$, the fixed point is called repelling if $|{f_n}'(z_0)|>1$. It is a problem of Fatou (se...

open_problem | open | MathDB 页面 | 原文线索

V2-038093 | MATHDB-387533

Research Problems in Function Theory - Problem 2.22

摘录 / EXCERPT: With the terminology of Problem 2.20, denote by $\mathcal{F}(f)$ the set of points where the sequence $\{f_n(z)\}$ is not normal. Fatou asks if there is an entire function $f(z)$ f...

open_problem | open | MathDB 页面 | 原文线索

V2-038094 | MATHDB-387534

Research Problems in Function Theory - Problem 2.23

摘录 / EXCERPT: Baker has proved that if $f(z)$ is a transcendental entire function, then $\mathcal{F}(f)$ is not restricted to a straight line in the plane. This implies (see Problem 2.22) that,...

open_problem | open | MathDB 页面 | 原文线索

V2-038095 | MATHDB-387535

Research Problems in Function Theory - Problem 2.24

摘录 / EXCERPT: Can an entire function have all its zeros and ones on two distinct straight lines, having infinitely many on each line? Edrei has proved (unpublished) that if l, m are intersecti...

open_problem | open | MathDB 页面 | 原文线索

V2-038096 | MATHDB-387536

Research Problems in Function Theory - Problem 2.25

摘录 / EXCERPT: If f, g are linearly independent entire functions of order ρ, which is not a positive multiple of $\frac{1}{2}$, can $fg'-gf'$ have order less than ρ? This is possible...

open_problem | open | MathDB 页面 | 原文线索

V2-038097 | MATHDB-387539

Research Problems in Function Theory - Problem 2.28

摘录 / EXCERPT: A meromorphic function $f(z)$ in the plane, is said to be of bounded value distribution (b.v.d.) if, for every positive r, there exists a fixed constant $C(r)$ such that the equa...

open_problem | open | MathDB 页面 | 原文线索

V2-038098 | MATHDB-387540

Research Problems in Function Theory - Problem 2.29

摘录 / EXCERPT: Is it possible to give an analogous characterisation of the solutions of ([source label: 2.5]) in the case where the $f_\nu(z)$ are polynomials? (P. Tur'an)

open_problem | open | MathDB 页面 | 原文线索

V2-038099 | MATHDB-387541

Research Problems in Function Theory - Problem 2.30

摘录 / EXCERPT: Let $S_k, k=1, 2, \ldots$ be sets which have no finite limit points. Does there exist a sequence n_k and an entire function $f(z)$, so that whenever $z\in S_k$ we have...

open_problem | open | MathDB 页面 | 原文线索

V2-038100 | MATHDB-387542

Research Problems in Function Theory - Problem 2.31

摘录 / EXCERPT: Let A, B be two countable dense sets in the plane. Does there exist an entire function $f(z)$, so that $f(z)\in B$, if and only if $z\in A$? If the answer is negative, it would b...

open_problem | open | MathDB 页面 | 原文线索

V2-038101 | MATHDB-387543

Research Problems in Function Theory - Problem 2.32

摘录 / EXCERPT: Let $f(z)=\sum^\infty_{n=0} a_nz^n$ be a transcendental entire function where $a_n\geq0$ for $n\geq0$, and set ... Then ... In addition, if $f(z)=e^z=\sum^\infty_{n=0} z^n/n!$, we have...

open_problem | open | MathDB 页面 | 原文线索

V2-038102 | MATHDB-387546

Research Problems in Function Theory - Problem 2.35

摘录 / EXCERPT: If Γ is a continuum that recedes to ∞, it is known (see Hayman) that as $z\to\infty$ on Γ, ... where A is an absolute constant. Is it true that $A=1$? This is...

open_problem | open | MathDB 页面 | 原文线索

V2-038103 | MATHDB-387548

Research Problems in Function Theory - Problem 2.37

摘录 / EXCERPT: Let r_n be a sequence of P'olya peaks (as defined by Edrei) of order ρ. Then Edrei showed that there exists $K=K(\alpha,\rho)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-038104 | MATHDB-387549

Research Problems in Function Theory - Problem 2.38

摘录 / EXCERPT: It was shown by Kjellberg that if $0<\alpha<1$ and ... then ... where $0<\beta\leq\infty$. If $\alpha=1$, it was shown by Hayman that unless $f(z)=Ae^{Bz}$, the corresponding result ho...

open_problem | open | MathDB 页面 | 原文线索

V2-038105 | MATHDB-387551

Research Problems in Function Theory - Problem 2.40

摘录 / EXCERPT: Let $f(z)$ be a non-constant entire function, and assume that for some constant c the plane measure of the set $E(c)$ where $|f(z)|>c$ is finite. What is the minimum growth rate...

open_problem | open | MathDB 页面 | 原文线索

V2-038106 | MATHDB-387552

Research Problems in Function Theory - Problem 2.42

摘录 / EXCERPT: Let $f(z)$ be an entire function (of sufficiently high order) with l, $l\geq2$ different asymptotic values a_k, $k=1,\ldots, l$. Suppose that γ_k is a path such that...

open_problem | open | MathDB 页面 | 原文线索

V2-038107 | MATHDB-387554

Research Problems in Function Theory - Problem 2.44

摘录 / EXCERPT: For $f(z)$ entire of order ρ, and non-constant, let $\nu(r)$ be the number of points on $|z|=r$ where $|f(z)|=1$. Is it true that ... If one replaces $\nu(r)$ by the number of p...

open_problem | open | MathDB 页面 | 原文线索

V2-038108 | MATHDB-387562

Research Problems in Function Theory - Problem 2.52

摘录 / EXCERPT: What is the best function $g(\sigma)$, $\sigma\geq 0$ such that, for a non-constant entire function $f(z)$ with maximum and minimum modulus $M(r,f)$ and $m_0(r,f)$ respectively, th...

open_problem | open | MathDB 页面 | 原文线索

V2-038109 | MATHDB-387567

Research Problems in Function Theory - Problem 2.57

摘录 / EXCERPT: If f is an entire function such that $\log M(r,f)=O(\log r)^2$ as $r\to\infty$, then Hayman has shown that $\log |f(re^{i\theta})|\sim\log M(r,f)$, as $r\to\infty$, for...

open_problem | open | MathDB 页面 | 原文线索

V2-038110 | MATHDB-387568

Research Problems in Function Theory - Problem 2.58

摘录 / EXCERPT: Suppose that f is entire with a non-zero Picard exceptional value α. Then f has α as an asymptotic value. It can be shown that $f\to\alpha$ along a level curve...

open_problem | open | MathDB 页面 | 原文线索

V2-038111 | MATHDB-387574

Research Problems in Function Theory - Problem 2.65

摘录 / EXCERPT: Since the knowledge of the zeros of an entire function f leaves an unknown factor, e^h say, in the Hadamard product for f, one can ask if f is determined by the zeros of...

open_problem | open | MathDB 页面 | 原文线索

V2-038112 | MATHDB-387585

Research Problems in Function Theory - Problem 2.76

摘录 / EXCERPT: Let Ω be a component of the normal set of an entire function (under iteration). Is $\dim(\partial\Omega) > 1$? Or is $\partial\Omega$ a circle/line? (D. Hamilton)

open_problem | open | MathDB 页面 | 原文线索

V2-038113 | MATHDB-387595

Research Problems in Function Theory - Problem 2.86

摘录 / EXCERPT: Let the function $f(z)$, $f(z) = \lambda(e^z-1)$ with $|\lambda| = 1$, have a Siegel singular disc S_λ that contains zero. [(a)] ; Prove that there exists some number...

open_problem | open | MathDB 页面 | 原文线索

V2-038114 | MATHDB-387601

Research Problems in Function Theory - Problem 3.1

摘录 / EXCERPT: If $u(z)$ is harmonic in the plane, and not a polynomial, does there exist a path Γ_n for every positive integer n, such that ... as $z\to\infty$ along Γ_n. Does ther...

open_problem | open | MathDB 页面 | 原文线索

V2-038115 | MATHDB-387602

Research Problems in Function Theory - Problem 3.2

摘录 / EXCERPT: If $u(x)$ is harmonic and not constant in space of 3 or more dimensions, is it true that there exists a path Γ such that $u(x)\to+\infty$ as $x\to\infty$ along Γ? T...

open_problem | open | MathDB 页面 | 原文线索

V2-038116 | MATHDB-387603

Research Problems in Function Theory - Problem 3.3

摘录 / EXCERPT: Suppose that $u(z)$ is subharmonic and $u(z)<0$ in the half-plane $|\theta|<\pi/2$, where $z=re^{i\theta}$. Suppose also that ... Is it true then that ... The result $u(r)\leq-K/3$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-038117 | MATHDB-387606

Research Problems in Function Theory - Problem 3.6

摘录 / EXCERPT: It follows from a result of Wolf , that if ... where ... then $u(z)$ is bounded above, and so is constant. What is the 3-dimensional analogue of this result?

open_problem | open | MathDB 页面 | 原文线索

V2-038118 | MATHDB-387607

Research Problems in Function Theory - Problem 3.7

摘录 / EXCERPT: Problem 1.17 can be reformulated for subharmonic functions, if we replace $\log M(r,f)$ by a general subharmonic function $u(z)$. The same positive theorems hold, and the same conc...

open_problem | open | MathDB 页面 | 原文线索

V2-038119 | MATHDB-387610

Research Problems in Function Theory - Problem 3.11

摘录 / EXCERPT: If $u(x)$ is a homogeneous harmonic polynomial of degree n in $\mathbb{R}^m$, what are the upper and lower bounds of ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-038120 | MATHDB-387616

Research Problems in Function Theory - Problem 3.17

摘录 / EXCERPT: Let D, D' be Lipschitz domains in $\mathbb{R}^n$, $n\geq 3$ with $D' \subset D$ and $\partial D'\cap\partial D$ lying compactly in the interior of a set Γ in ∂D;...

open_problem | open | MathDB 页面 | 原文线索

V2-038121 | MATHDB-387618

Research Problems in Function Theory - Problem 3.19

摘录 / EXCERPT: Let C_0 be a tangential path in $\mathbb{D}$ which ends at $z=1$, and let C_θ be any rotation of C_0. Littlewood showed that there exists a function $u(z)$, harmonic and...

open_problem | open | MathDB 页面 | 原文线索

V2-038122 | MATHDB-387620

Research Problems in Function Theory - Problem 3.21

摘录 / EXCERPT: Let α be a continuum in the closure of the unit disc $\mathbb{D}$, and let $\omega(z) = \omega(z; \mathbb{D}; \alpha)$ be the harmonic measure at z of α with respec...

open_problem | open | MathDB 页面 | 原文线索

V2-038123 | MATHDB-387622

Research Problems in Function Theory - Problem 3.23

摘录 / EXCERPT: Determine whether or not there exists a function $g(r)$, defined for $r\geq0$, with $g(r)\to0$ as $r\to\infty$, such that the following holds: if u is any Green potential in...

open_problem | open | MathDB 页面 | 原文线索

V2-038124 | MATHDB-387623

Research Problems in Function Theory - Problem 3.24

摘录 / EXCERPT: For which positive p does there exist a function u, $u\not\equiv0$ harmonic on $\mathbb{R}^3$ and vanishing on the cone $x^2_1+x^2_2=px^2_3$? (H. S. Shapiro)

open_problem | open | MathDB 页面 | 原文线索

V2-038125 | MATHDB-387624

Research Problems in Function Theory - Problem 3.25

摘录 / EXCERPT: Is there a harmonic polynomial $P(x_1, x_2, x_3)$, $P\not\equiv 0$ that is divisible by $x^4_1+x^4_2+x^4_3$? (H. S. Shapiro)

open_problem | open | MathDB 页面 | 原文线索

V2-038126 | MATHDB-387625

Research Problems in Function Theory - Problem 3.26

摘录 / EXCERPT: Given $n, n\geq4$, find a continuous function f on $(0,1)$ such that the following statement is true: if u is a subharmonic function in the unit ball B of $\mathbb{R}^n$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-038127 | MATHDB-387626

Research Problems in Function Theory - Problem 3.27

摘录 / EXCERPT: Let D be an unbounded domain in $\mathbb{R}^n$, $n\geq2$. Is there a positive continuous function $\varepsilon(|x|)$ such that, if u is harmonic in D and...

open_problem | open | MathDB 页面 | 原文线索

V2-038128 | MATHDB-387631

Research Problems in Function Theory - Problem 3.32

摘录 / EXCERPT: Let Ω be an open ball in $\mathbb{R}^n$, $n\geq 2$. It is shown by Armitage that $V\in L^p(\Omega)$ for any positive superharmonic function V on Ω and any p in...

open_problem | open | MathDB 页面 | 原文线索

V2-038129 | MATHDB-387635

Research Problems in Function Theory - Problem 4.1

摘录 / EXCERPT: Let $\{z_n\}, 1\leq n<\infty$ be an infinite sequence such that $|z_n|=1$. Define ... Is it true that $\limsup_{n\to\infty}A_n=\infty$, and if so, how quickly must A_n tend to infi...

open_problem | open | MathDB 页面 | 原文线索

V2-038130 | MATHDB-387636

Research Problems in Function Theory - Problem 4.2

摘录 / EXCERPT: Let $p(z)=a_0+a_1z+\ldots+a_nz^n$ be a polynomial, all of whose zeros are on $|z|=1$. If ... is $M\geq2A$?

open_problem | open | MathDB 页面 | 原文线索

V2-038131 | MATHDB-387642

Research Problems in Function Theory - Problem 4.8

摘录 / EXCERPT: Assume that $E^{(n)}_f$ is connected. Is it true that ... Pommerenke proved this with $\frac{1}{2}en^2$ instead of $\frac{1}{2}n^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-038132 | MATHDB-387651

Research Problems in Function Theory - Problem 4.19

摘录 / EXCERPT: Littlewood conjectured that if $n_1, n_2, \ldots, n_k$ are distinct positive integers then ... The best result known in this direction is due to Davenport who proved ([source label:...

open_problem | open | MathDB 页面 | 原文线索

V2-038133 | MATHDB-387659

Research Problems in Function Theory - Problem 4.28

摘录 / EXCERPT: Suppose that P is a non-linear polynomial with real coefficients. Show that $P^2(z)+P'(z)$ has non-real zeros. We conjecture that the lower bound for the number of non-real zeros...

open_problem | open | MathDB 页面 | 原文线索

V2-038134 | MATHDB-387660

Research Problems in Function Theory - Problem 4.29

摘录 / EXCERPT: Yang claims to prove the following: let $P(z), Q(z)$ be monic polynomials such that (i) $P(z)=0 \iff Q(z)=0$, and (ii) $P'(z)=0 \iff Q'(z)=0$. Then there exist positive integer...

open_problem | open | MathDB 页面 | 原文线索

V2-038135 | MATHDB-387667

Research Problems in Function Theory - Problem 5.5

摘录 / EXCERPT: If $f(z)=u+iv$ assumes only values in the right half-plane, then subordination shows that ... It is of interest to ask what other hypotheses on the values assumed by $f(z)$ result in...

open_problem | open | MathDB 页面 | 原文线索

V2-038136 | MATHDB-387669

Research Problems in Function Theory - Problem 5.7

摘录 / EXCERPT: If c_k is a sequence of positive numbers such that ... and n_k is an arbitrary sequence of positive integers, then ... is bounded in $\mathbb{D}$, and so takes no values outside a...

open_problem | open | MathDB 页面 | 原文线索

V2-038137 | MATHDB-387673

Research Problems in Function Theory - Problem 5.11

摘录 / EXCERPT: $f(z)$ meromorphic in $\mathbb{D}$, $f(z)\neq0, f^{(l)}(z)\neq1$, where $l\geq1$.

open_problem | open | MathDB 页面 | 原文线索

V2-038138 | MATHDB-387674

Research Problems in Function Theory - Problem 5.13

摘录 / EXCERPT: $f(z)$ meromorphic in $\mathbb{D}$, $f'(z)f(z)^n\neq1$, for $n\geq3$.

open_problem | open | MathDB 页面 | 原文线索

V2-038139 | MATHDB-387675

Research Problems in Function Theory - Problem 5.14

摘录 / EXCERPT: $f'-f^n\neq a$, where a is some complex number, and $n\geq 5$ if f is meromorphic, $n\geq3$ if f is entire. The corresponding results for functions in the place are proved by...

open_problem | open | MathDB 页面 | 原文线索

V2-038140 | MATHDB-387676

Research Problems in Function Theory - Problem 5.15

摘录 / EXCERPT: Is it possible to remove the restriction that D^* is simply connected in (a) and (b) above? It might be possible to start with the case when D and D^* are both doubly-con...

open_problem | open | MathDB 页面 | 原文线索

V2-038141 | MATHDB-387685

Research Problems in Function Theory - Problem 5.24

摘录 / EXCERPT: Is the intersection of two finitely generated ideals in H^∞ finitely generated? (L. A. Rubel)

open_problem | open | MathDB 页面 | 原文线索

V2-038142 | MATHDB-387689

Research Problems in Function Theory - Problem 5.28

摘录 / EXCERPT: Let f be continuous in $\overline{\mathbb{D}}$ and analytic in $\mathbb{D}$. Let ... ... Is it true that ... The Ruben, Shields and Taylor (unpublished) have shown that there is an abs...

open_problem | open | MathDB 页面 | 原文线索

V2-038143 | MATHDB-387694

Research Problems in Function Theory - Problem 5.33

摘录 / EXCERPT: Let L be a regular triangular lattice in the plane. Let $f(z)$ map $\mathbb{D}$ onto the universal covering surface over the complement of L. Is it true that the coefficients...

open_problem | open | MathDB 页面 | 原文线索

V2-038144 | MATHDB-387701

Research Problems in Function Theory - Problem 5.40

摘录 / EXCERPT: Suppose that $f(z) = \sum^\infty_0 a_nz^n$ and that $F(z)$ is analytic in $\mathbb{D}$, with $f\prec F$. What non-trivial conditions on F imply that ... In particular, is ([source...

open_problem | open | MathDB 页面 | 原文线索

V2-038145 | MATHDB-387710

Research Problems in Function Theory - Problem 5.50

摘录 / EXCERPT: A function f analytic in $\mathbb{D}$, is said to be annular if there exists a sequence $\{J_n\}$ of Jordan curves in $\mathbb{D}$ such that [(a)] ; J_n lies in the inside of...

open_problem | open | MathDB 页面 | 原文线索

V2-038146 | MATHDB-387721

Research Problems in Function Theory - Problem 5.61

摘录 / EXCERPT: Let $w(z)$ be analytic in $\mathbb{D}$ with $w(0)=0$. If mbox{$|w(z)+zw'(z)|<1$}, for $|z|<1$, then a simple application of Schwarz's lemma shows that $|w(z)|<1$, for $|z|<1$. Mill...

open_problem | open | MathDB 页面 | 原文线索

V2-038147 | MATHDB-387725

Research Problems in Function Theory - Problem 5.65

摘录 / EXCERPT: Does there exist a non-constant function f in the disc algebra such that $f(e^{i\theta})\in f(\mathbb{D})$ for almost all θ? Caution: the Rudin-Carleson theorem (see Bisho...

open_problem | open | MathDB 页面 | 原文线索

V2-038148 | MATHDB-387749

Research Problems in Function Theory - Problem 6.11

摘录 / EXCERPT: If $f(z)$, $g(z)$ are convex functions in S, is it true that for $0<\lambda<1$, $\lambda f+(1-\lambda)g$ is star-like and univalent? A function $w=f(z)$ in S is said to be star...

open_problem | open | MathDB 页面 | 原文线索

V2-038149 | MATHDB-387754

Research Problems in Function Theory - Problem 6.16

摘录 / EXCERPT: Let S^* be the class of all star-like functions $f(z)$ in S. Marx conjectured that for each fixed z_0, $|z_0|<1$, the set of all numbers $f'(z_0)$ for $f\in S^*$ coincides wi...

open_problem | open | MathDB 页面 | 原文线索

V2-038150 | MATHDB-387774

Research Problems in Function Theory - Problem 6.36

摘录 / EXCERPT: Suppose that f in S and define ... What can be said about bounds for $a_{n,p}$? If $|a_{n,1}|\leq Kn$ for all n and fixed K, then, for integral p, ... but it might be easier...

open_problem | open | MathDB 页面 | 原文线索

V2-038151 | MATHDB-387776

Research Problems in Function Theory - Problem 6.38

摘录 / EXCERPT: With the notation of Problem 6.37, is it true that ... where $\beta=(\sqrt{2}-1)^2$? (K. W. Lucas)

open_problem | open | MathDB 页面 | 原文线索

V2-038152 | MATHDB-387779

Research Problems in Function Theory - Problem 6.41

摘录 / EXCERPT: Let $K(\alpha)$ and $S^*(\alpha)$ be those subsets of S consisting of the class of functions convex in $\mathbb{D}$ of order α i.e. ... and star-like of order α in...

open_problem | open | MathDB 页面 | 原文线索

V2-038153 | MATHDB-387782

Research Problems in Function Theory - Problem 6.44

摘录 / EXCERPT: Let f, g be formal power series ... respectively, and define ... Let S_R denote the class of functions in S with real coefficients. Prove (or disprove) that $f,g\in S_R$ implie...

open_problem | open | MathDB 页面 | 原文线索

V2-038154 | MATHDB-387783

Research Problems in Function Theory - Problem 6.45

摘录 / EXCERPT: Let $S^*(\alpha)$ be the class of α-strongly-star-like functions f, that is, those f in S for which ... where $0 < \alpha < 1$. [(a)] ; Prove (or disprove) that...

open_problem | open | MathDB 页面 | 原文线索

V2-038155 | MATHDB-387784

Research Problems in Function Theory - Problem 6.46

摘录 / EXCERPT: Suppose that f in S and is star-like. Is it true that ... This is certainly true if $\lim_{r\to1} (1-r)M(r,f) > 0$, (D. A. Brannan, unpublished). T. Sheil-Small (uncited) has obt...

open_problem | open | MathDB 页面 | 原文线索

V2-038156 | MATHDB-387791

Research Problems in Function Theory - Problem 6.53

摘录 / EXCERPT: Hentgartner and Schobe and Goodman and Saff have shown that if $f(z) = z + a_2z^2 +\ldots$ maps $\mathbb{D}$ univalently onto a domain G_1 that is convex in the direction of the...

open_problem | open | MathDB 页面 | 原文线索

V2-038157 | MATHDB-387796

Research Problems in Function Theory - Problem 6.59

摘录 / EXCERPT: Let D be a plane domain containing ∞. Let there be given a continuous assignment of numbers (thought of as angles) to the components of $\mathbb{C}\setminus D$. Consider c...

open_problem | open | MathDB 页面 | 原文线索

V2-038158 | MATHDB-387803

Research Problems in Function Theory - Problem 6.66

摘录 / EXCERPT: Describe the extreme points of the class Σ_0 consisting of all functions g in Σ with constant term $b_0=0$. Springer (see Pommerenke) showed that every g in...

open_problem | open | MathDB 页面 | 原文线索

V2-038159 | MATHDB-387804

Research Problems in Function Theory - Problem 6.67

摘录 / EXCERPT: Let f be univalent in $\mathbb{D}$ and let $f(\mathbb{D})$ be a Jordan domain. Does the condition ... imply that f has a quasi-conformal extension over the unit circle? It is kno...

open_problem | open | MathDB 页面 | 原文线索

V2-038160 | MATHDB-387806

Research Problems in Function Theory - Problem 6.69

摘录 / EXCERPT: Let B be the Banach space of analytic functions ϕ in $\{|z|>1\}$ with finite norm ... Let S and T be the subsets defined by $S:=\{f''/f':f\in \sum\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-038161 | MATHDB-387807

Research Problems in Function Theory - Problem 6.70

摘录 / EXCERPT: Is every extreme point of S a support point? Is every support point an extreme point? (P. L. Duren)

open_problem | open | MathDB 页面 | 原文线索

V2-038162 | MATHDB-387810

Research Problems in Function Theory - Problem 6.73

摘录 / EXCERPT: Let $f(z)=z+\sum^\infty_{n=2}a_nz^n$ be in S. Is it true that ... Hamilton has proved that this is true for odd functions in S, functions of maximal growth in S, and spiral-lik...

open_problem | open | MathDB 页面 | 原文线索

V2-038163 | MATHDB-387816

Research Problems in Function Theory - Problem 6.79

摘录 / EXCERPT: Let $S_k(\infty)$ denote the class of all analytic and univalent functions $f(z)=z+a_2z^2+\ldots$ defined in $\mathbb{D}$ which admit a k-quasiconformal extension $(0<k<1)$ to th...

open_problem | open | MathDB 页面 | 原文线索

V2-038164 | MATHDB-387822

Research Problems in Function Theory - Problem 6.85

摘录 / EXCERPT: Each function f in S that maximises $\text{Re}\, \{L(g) :g \in S\}$ for some continuous linear functional L must map the unit disc onto the complement of an arc Γ that i...

open_problem | open | MathDB 页面 | 原文线索

V2-038165 | MATHDB-387861

Research Problems in Function Theory - Problem 7.9

摘录 / EXCERPT: Suppose that f is a plane K-quasiconformal mapping of the unit disc $\mathbb{D}$ onto itself. Show that there exists a finite constant $b = b(K)$ such that ... for each measurabl...

open_problem | open | MathDB 页面 | 原文线索

V2-038166 | MATHDB-387863

Research Problems in Function Theory - Problem 7.11

摘录 / EXCERPT: Show that each quasiconformal mapping of $\mathbb{R}^n$ onto $\mathbb{R}^n$ has a quasiconformal extension to $\mathbb{R}^{n+1}$. This has been established by Ahlfors when $n = 2$...

open_problem | open | MathDB 页面 | 原文线索

V2-038167 | MATHDB-387864

Research Problems in Function Theory - Problem 7.12

摘录 / EXCERPT: Suppose that f is an n-dimensional K-quasi-analytic function. Show that the partial derivatives of f are locally L-integrable for $n\leq p\leq n + c$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-038168 | MATHDB-387866

Research Problems in Function Theory - Problem 7.14

摘录 / EXCERPT: (Boundary values of Cauchy integrals). Let γ be a C^1 curve in the plane and let f be a continuous function on γ. Put ... Does F have non-tangential boundary valu...

open_problem | open | MathDB 页面 | 原文线索

V2-038169 | MATHDB-387869

Research Problems in Function Theory - Problem 7.17

摘录 / EXCERPT: Let $f(z)$ be analytic and bounded for $\Re z> 0$. Suppose that $|\alpha|<\frac{1}{2}\pi$ and that (r_n) is a sequence of positive integers with $\sum\frac{1}{r_n}=\infty$. Show...

open_problem | open | MathDB 页面 | 原文线索

V2-038170 | MATHDB-387874

Research Problems in Function Theory - Problem 7.22

摘录 / EXCERPT: Suppose that A and B are disjoint linked Jordan curves in $\mathbb{R}^3$ which lie at a distance 1 from each other. Show that the length of A is at least 2π. The corres...

open_problem | open | MathDB 页面 | 原文线索

V2-038171 | MATHDB-387882

Research Problems in Function Theory - Problem 7.30

摘录 / EXCERPT: Let $u(z)$ be a real bounded continuous function on $U = \{|z|< 1\}$, and suppose that to each $z \in U$ there corresponds a real number $r(z)$ with $0<r(z)< 1-|z|$ such that ... Mus...

open_problem | open | MathDB 页面 | 原文线索

V2-038172 | MATHDB-387884

Research Problems in Function Theory - Problem 7.32

摘录 / EXCERPT: Let $\mu(t)$ be a continuous monotonic increasing function of t for $t\in[0,1]$ and let $\omega_1(h,\mu)$, $\omega_2(h,\mu)$ denote its modulus of continuity and modulus of smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-038173 | MATHDB-387892

Research Problems in Function Theory - Problem 7.40

摘录 / EXCERPT: Let $D(z, r)=\{w:|w-z|\leq r\}$. Given a sequence $\{D(z_j, r)\}^N_{j=1}$ of disjoint closed balls all contained in $|z|\leq\frac{1}{2}$, put ... Let h be the function in...

open_problem | open | MathDB 页面 | 原文线索

V2-038174 | MATHDB-387893

Research Problems in Function Theory - Problem 7.41

摘录 / EXCERPT: Let $\{z_\nu\}$ be a sequence of distinct points in the unit disc $\mathbb{D}$ such that $\sum(1-|z_\nu|)<\infty$. Let B be the Blaschke product corresponding to this sequence. I...

open_problem | open | MathDB 页面 | 原文线索

V2-038175 | MATHDB-387895

Research Problems in Function Theory - Problem 7.43

摘录 / EXCERPT: A bounded simply-connected domain D is said to be conformally rigid if there is some $\varepsilon>0$ such that if f is a conformal self-map of D satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-038176 | MATHDB-387896

Research Problems in Function Theory - Problem 7.44

摘录 / EXCERPT: Let U be an open set in the plane and λ^U_a be the harmonic measure at the point a with respect to U. Then O{}ksendal showed that λ^U_a is singular with respe...

open_problem | open | MathDB 页面 | 原文线索

V2-038177 | MATHDB-387897

Research Problems in Function Theory - Problem 7.45

摘录 / EXCERPT: Let D be the unit disc cut along p radial slits from the outer boundary, all of the same given length. Let u be the harmonic measure of $\{|z|=1\}$ in D. Find the configura...

open_problem | open | MathDB 页面 | 原文线索

V2-038178 | MATHDB-387898

Research Problems in Function Theory - Problem 7.46

摘录 / EXCERPT: (A Universal' Phr'agmen-Lindel"of Theorem) Let D be an arbitrary unbounded plane domain. Suppose that $f(z)$ is analytic on D and continuous on $\overline{D}$. If $|f(z)|\leq1$...

open_problem | open | MathDB 页面 | 原文线索

V2-038179 | MATHDB-387908

Research Problems in Function Theory - Problem 7.56

摘录 / EXCERPT: Let Γ be a closed Jordan curve in the extended plane, and suppose that $\infty\in\Gamma$. Let f_1, f_2 map the upper and lower half-planes, respectively, onto the two diff...

open_problem | open | MathDB 页面 | 原文线索

V2-038180 | MATHDB-387911

Research Problems in Function Theory - Problem 7.59

摘录 / EXCERPT: Let (P, Q) denote a pair of polynomials with the following property: ... If (P, Q) has the property ([source label: J7.59]), is it necessarily true that (Q, P) also has the pro...

open_problem | open | MathDB 页面 | 原文线索

V2-038181 | MATHDB-387912

Research Problems in Function Theory - Problem 7.60

摘录 / EXCERPT: Let P be a polynomial of degree m in which the coefficient of z^m_1 is non-zero, and let $Q(z) = z^m_1$. [(a)] ; Does the pair (P, Q) have the property ([source label: J7.5...

open_problem | open | MathDB 页面 | 原文线索

V2-038182 | MATHDB-387915

Research Problems in Function Theory - Problem 7.63

摘录 / EXCERPT: For $n\geq 2$ and $\alpha > 0$, let ... where $R > 0$ and $f\in\mathcal{S}(\mathbb{R}^n)$. Is it true, for all $f\in L^{2n/(n+1)}(\mathbb{R}^n)$ and all $\alpha > 0$, that...

open_problem | open | MathDB 页面 | 原文线索

V2-038183 | MATHDB-387930

Research Problems in Function Theory - Problem 7.78

摘录 / EXCERPT: Does there exist a sequence $\{z_n\}^\infty_1$ of distinct complex numbers such that ... for all $z\in\mathbb{C}$? This has the following physical interpretation. If we imagine elect...

open_problem | open | MathDB 页面 | 原文线索

V2-038184 | MATHDB-387934

Research Problems in Function Theory - Problem 7.82

摘录 / EXCERPT: Let $E\subset \mathbb{C}$, and the function $F:E\times \mathbb{D}\to\mathbb{C}$ satisfy the following conditions: [(a)] ; F is injective on E, for all $z\in \mathbb{D}$; ; F...

open_problem | open | MathDB 页面 | 原文线索

V2-038185 | MATHDB-387937

Research Problems in Function Theory - Problem 8.2

摘录 / EXCERPT: Given functions $f_1,\ldots,f_N\in H^\infty$, let $I = I(f_1,\ldots,f_N)$ be the ideal of H^∞ generated by $f_1,\ldots,f_N$ and let $J = J(f_1,\ldots,f_N)$ denote the set of...

open_problem | open | MathDB 页面 | 原文线索

V2-038186 | MATHDB-387955

Research Problems in Function Theory - Problem 8.20

摘录 / EXCERPT: Let $f_1,\ldots,f_n$ and g be H^∞ functions on $\mathbb{D}$. If $|g(z)| \leq |f_1(z)| + \ldots + |f_n(z)|$, are there necessarily functions $g_1,\ldots,g_n\in H^\infty$ su...

open_problem | open | MathDB 页面 | 原文线索

V2-038187 | MATHDB-387957

Research Problems in Function Theory - Problem 8.22

摘录 / EXCERPT: Do there exist inner functions in any strictly pseudo-convex domain of $\mathbb{C}^n$, $n\geq2$? Alexandrov (no citation) and independently L{o}w (no citation) and Hakim and Sibony...

open_problem | open | MathDB 页面 | 原文线索

V2-038188 | MATHDB-387964

Research Problems in Function Theory - Problem 9.3

摘录 / EXCERPT: [(a)] ; Suppose that $f, f_1, f_2,\ldots,f_n\in H^\infty$, and ... Do there necessarily exist $h_1,h_2,\ldots,h_n\in H^\infty$ such that ... (If $|f_1|+|f_2|+\ldots+|f_n|\geq\delta>0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-038189 | MATHDB-390594

Bao-Wulan multiplier problem for QK spaces

摘录 / EXCERPT: For the Möbius-invariant space Q_K on the unit disc, determine whether the pointwise multiplier algebra satisfies $\mathcal{M}(Q_K)=H^\infty\cap Q_K^{\log}$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038190 | MATHDB-391918

Altun-Hancer-Ateş question on enriched P-contractions

摘录 / EXCERPT: Let X be a normed linear space, let $C\subseteq X$ be nonempty, and let $T:C\to C$. Is it true that if there exists $b\geq 0$ such that...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038191 | MATHDB-391919

Yau's conjecture on smooth Reinhardt domains

摘录 / EXCERPT: For every integer $n\ge 1$, if a Reinhardt domain $\Omega\subset\mathbb{C}^n$ has smooth boundary and its Bergman metric is complete and Einstein, then Ω is biholomorphic to...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038192 | MATHDB-391920

Escobar's conjecture

摘录 / EXCERPT: For every integer $n\ge 3$, every compact n-dimensional Riemannian manifold (M^n,g) with nonnegative Ricci curvature, $\operatorname{Ric}_g\ge 0$, and every principal curvature...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038193 | MATHDB-391932

Douglas question for functions of the form z̄+h

摘录 / EXCERPT: Let $\mathbb{D}=\{z\in\mathbb{C}:|z|<1\}$, let $A(\mathbb{D})$ denote the disk algebra, and let $A^2(\mathbb{D})$ be the Bergman space. For $h\in A(\mathbb{D})$, define the Bergman...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038194 | MATHDB-391936

Vuorinen survey Problems 3.38(a)-(b): Newton iteration for the inverse Grötzsch modulus

摘录 / EXCERPT: Let $\mu:(0,1)\to(0,\infty)$ be the Grötzsch modulus function $\mu(r)=\frac{\pi}{2}\frac{\mathcal{K}(\sqrt{1-r^2})}{\mathcal{K}(r)}$, where $\mathcal{K}$ is the complete elliptic i...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038195 | MATHDB-392089

Kolountzakis-Papageorgiou Question 1 on common tiling functions

摘录 / EXCERPT: For a fixed dimension $d\ge 2$, do there exist arbitrarily large integers N and lattices $\Lambda_1,\ldots,\Lambda_N\subset\mathbb{R}^d$ such that...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038196 | MATHDB-392094

Nevanlinna’s half-plane problem

摘录 / EXCERPT: Let $\mathbb{H}=\{z\in\mathbb{C}:\operatorname{Im}z>0\}$, and let $N(\mathbb{H})$ be the class of meromorphic functions on $\mathbb{H}$ that are quotients of two bounded analytic f...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038197 | MATHDB-392099

Semmes’s open problem on conformal welding of chord-arc curves

摘录 / EXCERPT: Let Γ be a closed Jordan curve, let h be its conformal welding, let C_h be the pull-back operator on BMO, and let A_h be the analytic projection of C_h. D...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038198 | MATHDB-392128

Dembo-Cover-Thomas conjecture and related projection inequalities

摘录 / EXCERPT: For every integer $n\ge 2$, every pair of full-dimensional zonoids $K,L\subset\mathbb{R}^n$, and every $u\in S^{n-1}$, one has ... where $P_{u^\perp}$ denotes orthogonal projection o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038199 | MATHDB-392141

Generalized Geroch conjecture; spectral Geroch conjecture

摘录 / EXCERPT: For every integer $n\ge 3$, every n-manifold N, and every parameter γ with $0\le \gamma<\frac{2n}{n-1}$, the connected sum $\mathbb{T}^n\# N$ admits no complete Riemanni...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038200 | MATHDB-392271

Barría-Halmos problem on the Cesàro operator

摘录 / EXCERPT: Let $H^2(\mathbb D)$ be the Hardy space of the unit disk. Define the Cesàro operator $C:H^2(\mathbb D)\to H^2(\mathbb D)$ by...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038201 | MATHDB-392297

Nevanlinna’s omitted-values and Blaschke-divisor question

摘录 / EXCERPT: Let $F\colon\mathbb{C}\to\widehat{\mathbb{C}}$ be a nonconstant meromorphic function such that $F^{-1}(\{0,1,\infty\})\subseteq\mathbb{R}$, equivalently, F omits 0, 1, and...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038202 | MATHDB-392307

Choi-Wang-Reilly boundary eigenvalue conjecture and Hui-Hsien Wang boundary-area conjecture

摘录 / EXCERPT: For every integer $n\ge 2$, if (M^{n+1},g) is a compact Riemannian manifold with boundary satisfying $\operatorname{Ric}_g\ge n g$ and $\operatorname{II}_g\ge 0$ on ∂M,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038203 | MATHDB-392309

Boundedness characterization of Volterra-type operators on area Nevanlinna spaces

摘录 / EXCERPT: For each $1\le p<\infty$ and $\alpha>-1$, determine exactly which analytic symbols g make the Volterra-type operators J_g and I_g metrically bounded on the area Nevanlinna sp...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038204 | MATHDB-392465

Complete Crouzeix conjecture

摘录 / EXCERPT: For every $n,m\ge 1$, every matrix $A\in M_n(\mathbb{C})$, and every matrix-valued polynomial $P(z)=\sum_{k=0}^d P_k z^k$ with $P_k\in M_m(\mathbb{C})$, one has...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038205 | MATHDB-392479

Dual Mean Value Conjecture

摘录 / EXCERPT: For every integer $d\ge 2$ and every polynomial $P\in\mathbb{C}[z]$ of degree d satisfying $P(0)=0$ and $P(1)=1$, there exists a critical point $\zeta\ne 0$ with $P'(\zeta)=0$ su...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038206 | MATHDB-392483

Pólya's conjecture and the Weyl-Berry conjecture for the Vladimirov-Taibleson operator

摘录 / EXCERPT: Let $\Omega\subset\mathbb{Q}_p^d$ be an admissible bounded domain, let D_{Ω}^{α} denote the Dirichlet realization of the Vladimirov-Taibleson operator, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038207 | MATHDB-392689

Nonlinear Fredholm alternative for the p-Laplacian

摘录 / EXCERPT: Let X and Y be Banach spaces, let $J:X\to Y$ be an odd, $\[1m\tau\[0m$-homogeneous homeomorphism with $[J]_a>0$, and let $F:X\to Y$ be odd, $\[1m\tau\[0m$-homogeneous, and comp...

open_problem | open | MathDB 页面 | 原文线索

V2-038208 | MATHDB-392705

Tight frames on ellipsoids defined by compact perturbations of the identity

摘录 / EXCERPT: For an ellipsoidal surface $E_A=A S^1$ in an infinite-dimensional real or complex Hilbert space, with $A-I$ compact, determine necessary and sufficient conditions for E_A to cont...

open_problem | open | MathDB 页面 | 原文线索

V2-038209 | MATHDB-392710

Approximation of Frames by Normalized Tight Ones

摘录 / EXCERPT: Let $\{x_i\}_i$ be a frame of a Hilbert subspace $K\subseteq H$ of a given separable Hilbert space H, with frame transform θ and frame operator $S=(\theta^*\theta)^{-1}$....

open_problem | open | MathDB 页面 | 原文线索

V2-038210 | MATHDB-392716

Characterization of type B positive trace-class operators

摘录 / EXCERPT: Let H be an infinite-dimensional separable Hilbert space, and let $[U+007F]e_n[U+007F]_n$ be an orthonormal basis for H. For $v\in H$, define ... which may be infinite. For a positive trace-cl...

open_problem | open | MathDB 页面 | 原文线索

V2-038211 | MATHDB-392937

Maximal L^p-regularity for a non-autonomous parabolic problem

摘录 / EXCERPT: Let X be a UMD Banach space, let $1<p<\infty$, and let $(A(t),D(A(t)))_{t\in[0,T]}$ be a family of operators satisfying (RA) and (AT), where ... Then (NACP) has L^p-maximal regul...

open_problem | open | MathDB 页面 | 原文线索

V2-038212 | MATHDB-392941

Whether every UMD space has the stated interpolation-subspace representation

摘录 / EXCERPT: Is every UMD Banach space isomorphic, as a Banach space, to a subspace of an intermediate space $[X_0,X_1]_t$ arising from a compatible couple (X_0,X_1) for Calderón’s complex in...

open_problem | open | MathDB 页面 | 原文线索

V2-038213 | MATHDB-392942

Existence of Banach spaces without frequently hypercyclic operators

摘录 / EXCERPT: Conjecture: There exists an infinite-dimensional separable Banach space X such that no bounded operator $T:X\to X$ is frequently hypercyclic.

open_problem | open | MathDB 页面 | 原文线索

V2-038214 | MATHDB-392944

R-sectoriality and maximal regularity for elliptic boundary value problems

摘录 / EXCERPT: Let E be a UMD Banach space, let $1<p<\infty$, and let $G\subset\mathbb{R}^{n+1}$ be an open connected set with compact C^{2m}-boundary. Consider ... with $m_j<2m$, and assume th...

open_problem | open | MathDB 页面 | 原文线索

V2-038215 | MATHDB-392948

Characterization of Hilbert spaces by vector-valued Bloch coefficient multipliers

摘录 / EXCERPT: Let X be a complex Banach space. Does ... imply that X is isomorphic to a Hilbert space?

open_problem | open | MathDB 页面 | 原文线索

V2-038216 | MATHDB-392949

Taylor coefficients of vector-valued Bloch functions

摘录 / EXCERPT: For which complex Banach spaces X does the following implication hold? If ... then $(x_n)_{n\ge 0}\in\ell(2,\infty,X)$? Equivalently, for which X does ... hold, where...

open_problem | open | MathDB 页面 | 原文线索

V2-038217 | MATHDB-392989

Complex interpolation of compact operators

摘录 / EXCERPT: Let A_0 and A_1 be compatible Banach spaces, i.e. form a Banach pair, and let B_0 and B_1 likewise be compatible Banach spaces. Suppose that $T:A_0+A_1\to B_0+B_1$ is a lin...

open_problem | open | MathDB 页面 | 原文线索

V2-038218 | MATHDB-392991

Complex interpolation of compact operators under compactness at both endpoints

摘录 / EXCERPT: Let (A_0,A_1) and (B_0,B_1) be compatible Banach couples. Suppose that $T:A_0+A_1\to B_0+B_1$ is linear, that $T:A_0\to B_0$ is compact, and that $T:A_1\to B_1$ is also compact...

open_problem | open | MathDB 页面 | 原文线索

V2-038219 | MATHDB-393050

Proper affine isometric actions of $SL(n,\a0\mathbb Z)$ on uniformly convex Banach spaces

摘录 / EXCERPT: Does $SL(n;\mathbb Z)$ admit a proper affine isometric action on some uniformly convex Banach space for every integer $n\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-038220 | MATHDB-393211

The Big Z: optimal deterministic programming of POVMs

摘录 / EXCERPT: For a system of dimension d, a fixed ancilla Hilbert-space dimension, and a fixed number of outcomes $N=|Z|=|P|$, find the joint observables Z that are most efficient for progr...

open_problem | open | MathDB 页面 | 原文线索

V2-038221 | MATHDB-393359

Transitivity of automorphisms on normalized units of arbitrary Arveson systems

摘录 / EXCERPT: For arbitrary Arveson systems (that is, stationary continuous products of Hilbert spaces), is the group of automorphisms transitive on the normalized units? The report answers no:...

open_problem | open | MathDB 页面 | 原文线索

V2-038222 | MATHDB-393531

Translation-invariant subspaces of weighted sequence spaces

摘录 / EXCERPT: For every sequence of positive numbers $\omega=(\omega(n))_{n\in\mathbb Z}$ for which there exists $c>0$ such that ... does the Hilbert space ... have a nontrivial translation-invarian...

open_problem | open | MathDB 页面 | 原文线索

V2-038223 | MATHDB-393533

Schaeffer's conjecture on determinant-inverse norm inequalities

摘录 / EXCERPT: For every invertible operator (matrix) $T:X\to X$ on an N-dimensional Banach space, Schaeffer proved that ... His conjecture that the constants k_N could be bounded independently...

conjecture | open | MathDB 页面 | 原文线索

V2-038224 | MATHDB-393534

Comparison of quotient norms for three analytic function algebras

摘录 / EXCERPT: For a function Banach algebra A on $\[?]D$, define ... The comparison problem is to determine the growth of $k_n(A)$. The report considers $A=B^1_{1,1}$, W, and $B^0_{\infty,1}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-038225 | MATHDB-393539

Equivalence of invariant-subspace index and boundary-limit conditions under point-dependent lower bounds

摘录 / EXCERPT: Let H be a Hilbert space of analytic functions on $\mathbb D$ such that $\mathopen\|M_\zeta\mathclose\|\le 1$ and $\dim(H/\zeta H)=1$. Assume the weaker pointwise condition that,...

open_problem | open | MathDB 页面 | 原文线索

V2-038226 | MATHDB-393540

Index one from a single function with nontangential limits

摘录 / EXCERPT: For a Bergman-type Hilbert space H of analytic functions on the unit disc and a closed M_z-invariant subspace $M\subset H$, if M contains one nonzero function having nontange...

open_problem | open | MathDB 页面 | 原文线索

V2-038227 | MATHDB-393557

Primitivity of twists from unitary representations

摘录 / EXCERPT: Let G be a compact semisimple Lie group, let V be a finite-dimensional unitary representation of G, and let LG act on the polarized Hilbert space $L^2(S^1,V)$. The resultin...

open_problem | open | MathDB 页面 | 原文线索

V2-038228 | MATHDB-393558

LeBrun positive frequency conjecture

摘录 / EXCERPT: Let S be a smooth submanifold of the Banach manifold $M^{k,\alpha}$ of $C^{k,\alpha}$ metrics on S^3 near the round metric $g_{(0)}$, containing $g_{(0)}$ and tangent at...

conjecture | open | MathDB 页面 | 原文线索

V2-038229 | MATHDB-393893

Baum-Connes-type right-hand side and assembly map for loop groups

摘录 / EXCERPT: For a loop group (or Kac-Moody group) G, does there exist a suitable reduced C^*-algebra-possibly an l^1-algebra or another associated Banach algebra-whose K-theory provide...

open_problem | open | MathDB 页面 | 原文线索

V2-038230 | MATHDB-393929

Nonperturbative analysis of warped-product sigma-models

摘录 / EXCERPT: Investigate nonperturbative aspects of the functional-integral formulation of the two-dimensional nonlinear sigma-models with warped-product target spaces of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-038231 | MATHDB-393934

Rigorous nonperturbative confirmation of infrared asymptotic freedom in two-dimensional noncompact sigma models

摘录 / EXCERPT: Perturbative Renormalization Group analysis predicts for 2 dimensions asymptotic freedom in the infrared. Whether this is confirmed by a rigorous nonperturbative analysis or turns...

open_problem | open | MathDB 页面 | 原文线索

V2-038232 | MATHDB-393936

Existence characterization for a singular quadratic regulator problem

摘录 / EXCERPT: In the report’s setting, let X,U,Y be Hilbert spaces, let A generate an analytic semigroup on X, let $B:U\to X$ and $G:U\to Y$ be bounded linear operators with...

open_problem | open | MathDB 页面 | 原文线索

V2-038233 | MATHDB-394129

Necessity of the B_0 condition for supercyclic subspaces

摘录 / EXCERPT: Let E be an infinite-dimensional separable complex Banach space and let $T\in\mathcal L(E)$. Suppose that T satisfies the Supercyclicity Criterion for a sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-038234 | MATHDB-394250

Finding Moyal quantizers

摘录 / EXCERPT: Given a phase space X, a Liouville measure μ on X, and the Hilbert space H associated to (X,μ), find a Moyal quantizer: a map $\Omega:X\to$ the selfadjoint operators...

open_problem | open | MathDB 页面 | 原文线索

V2-038235 | MATHDB-394269

Banach-Mazur distance for polytopes with few vertices

摘录 / EXCERPT: Let $P_n\subset B^d$ be a polytope with $n=d+k$ vertices, where $k=1,\ldots,d$ (equivalently, $d+1\le n\le 2d$). Conjecture: there is an absolute constant $\tilde c>0$ such that ……

open_problem | open | MathDB 页面 | 原文线索

V2-038236 | MATHDB-394727

Karoubi’s conjecture for compact-operator stabilization

摘录 / EXCERPT: If $J=L^{\infty}$ is the ideal of all compact operators and L is a unital Banach algebra, then $K_{*}(L\widehat\otimes J)=K^{\mathrm{top}}_{*}(L\widehat\otimes J)$; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-038237 | MATHDB-394926

Every Banach space contains a copy of c_0 or $\\\\\ell_p$

摘录 / EXCERPT: Does every Banach space contain a copy of c_0 or ℓ_p, $1\le p<\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-038238 | MATHDB-394955

Direct sums of hypercyclic operators

摘录 / EXCERPT: Let X be a separable Banach space and let T be a bounded hypercyclic operator on X. Must $T\oplus T$ be hypercyclic on $X\oplus X$?

open_problem | open | MathDB 页面 | 原文线索

V2-038239 | MATHDB-395138

Stability of matter for Coulomb molecules with fermionic electrons

摘录 / EXCERPT: For a molecule with M nuclei of equal charge $+Z$ at positions $R_1,\c\c,R_M\in\mathbb{R}^3$ and N electrons, let ... where ... ... With the electronic Hilbert space restricted to an...

open_problem | open | MathDB 页面 | 原文线索

V2-038240 | MATHDB-395195

Convergence of the inexact Uzawa method

摘录 / EXCERPT: Let V and Q be Hilbert spaces with inner products $a_0(\cdot,\cdot)$ and $(\cdot,\cdot)$, and induced norms $|\cdot|_V$ and $\|\cdot\|_Q$. Let $A:V\to V^*$ and $C:Q\to Q^*$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-038241 | MATHDB-395220

Commutant of a nonatomic MASA in a maximal tensor product

摘录 / EXCERPT: Let H be a Hilbert space with $H\cong\ell^{2}(\mathbb{N})$, and let C be a MASA of $B(H)$ isomorphic to $L^{\infty}(0,1)$. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-038242 | MATHDB-395505

Abstract variational model problem and its optimization-based reformulation

摘录 / EXCERPT: Let V and H be Hilbert spaces, let V^* be the dual of V, and let $f\in V^*$. The model problem is to seek $u\in V$ such that ... where $Q:V\times V\to\mathbb{R}$ is continuou...

open_problem | open | MathDB 页面 | 原文线索

V2-038243 | MATHDB-395824

Hilbert space compression of Grigorchuk’s group

摘录 / EXCERPT: What is the compression of Grigorchuk’s group of intermediate growth?

open_problem | open | MathDB 页面 | 原文线索

V2-038244 | MATHDB-395825

Hilbert compression of amenable groups

摘录 / EXCERPT: Is there an amenable group of Hilbert space compression value $<\frac{1}{2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-038245 | MATHDB-396189

Existence for an abstract Hilbert-space initial-value problem

摘录 / EXCERPT: Let H be a Hilbert space and let $h:H\times H\to\mathbb{R}$ be smooth, with $h(x,\cdot)$ uniformly convex and growing superlinearly at infinity, uniformly in x. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-038246 | MATHDB-396197

Rigorous error analysis of CRBC corner conditions

摘录 / EXCERPT: Rigorous Error Analysis Including the Corner Conditions: One possibility is to write down an equivalent integral operator and try to adapt work on integral equations on nonsmooth d...

open_problem | open | MathDB 页面 | 原文线索

V2-038247 | MATHDB-396248

Perturbations of separable nuclear C*-algebras

摘录 / EXCERPT: Let A and B be C-algebras acting on a separable Hilbert space H. If A is separable and nuclear and their Kadison-Kastler distance satisfies $\lVert A-B\rVert<10^{-11}$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-038248 | MATHDB-396253

Rigorous homogenization of partially high-contrast periodic media

摘录 / EXCERPT: Develop a systematic mathematical theory and rigorous analysis of homogenization for PDEs with partially high-contrast periodic inclusions. The theory should appropriately modify t...

open_problem | open | MathDB 页面 | 原文线索

V2-038249 | MATHDB-396635

Banach frames for shearlet coorbit spaces

摘录 / EXCERPT: Assume the hypotheses of Theorem 5: the irreducible unitary representation π is w-integrable; $\psi\in L^{2}(\mathbb{R}^{n})$ is appropriately normalized and satisfies ... choo...

open_problem | open | MathDB 页面 | 原文线索

V2-038250 | MATHDB-396674

Feichtinger conjecture

摘录 / EXCERPT: Every frame in a Hilbert space is a finite union of Riesz basis sequences.

conjecture | open | MathDB 页面 | 原文线索

V2-038251 | MATHDB-396677

Optimality of the growth condition for cyclicity of bicyclic operators

摘录 / EXCERPT: Conjecture: In Theorem 2, let X be a complex separable Banach space and let T be a bounded invertible bicyclic operator on X. Assume that there is a nonnegative integer k s...

open_problem | open | MathDB 页面 | 原文线索

V2-038252 | MATHDB-396683

The Rε-Conjecture

摘录 / EXCERPT: For every $\varepsilon>0$, every unit norm Riesz basic sequence in a Hilbert space can be written as a finite union of ε-Riesz basic sequences.

conjecture | open | MathDB 页面 | 原文线索

V2-038253 | MATHDB-396684

The Harmonic Analysis Conjecture

摘录 / EXCERPT: For every measurable set $E\subseteq[0,1]$ and every $\varepsilon>0$, there exists a finite partition $(A_j)_{j=1}^r$ of $\mathbb{Z}$ such that, for every $j=1,2,\ldots,r$ and ever...

conjecture | open | MathDB 页面 | 原文线索

V2-038254 | MATHDB-396694

Corona theorem for multiplier algebras of RKHSs with 1-positive square

摘录 / EXCERPT: If the reproducing kernel Hilbert space $H(\Omega)$ has 1-positive square, does the corona theorem hold for its multiplier algebra $M(H(\Omega))$?

open_problem | open | MathDB 页面 | 原文线索

V2-038255 | MATHDB-396696

Invertibility bounds and thresholds for H^{∞} trace algebras

摘录 / EXCERPT: Given a sequence $\sigma=\{\lambda_j\}$ in the unit disk $\mathbb D=\{z\in\mathbb C:|z|<1\}$, consider all normalized matrices A with $\|A\|\le 1$ (or Hilbert-space operators) su...

open_problem | open | MathDB 页面 | 原文线索

V2-038256 | MATHDB-396698

A resolvent bound with the factor $(|z|-1)^{\alpha}$

摘录 / EXCERPT: Let T be an operator on a finite-dimensional Banach space satisfying $P(T)=\sup_{k\ge 0}\|T^k\|<\infty$, let $R(z,T)=(zI-T)^{-1}$, and let $0<\alpha<1$. Is there a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-038257 | MATHDB-396699

Possible extension of the classical Kreiss resolvent condition

摘录 / EXCERPT: Let T be an operator on a finite-dimensional Banach space satisfying $P(T)=\sup_{k\ge 0}\|T^k\|<\infty$, let $r(T)$ be its spectral radius, and let $R(z,T)=(zI-T)^{-1}$. For ever...

open_problem | open | MathDB 页面 | 原文线索

V2-038258 | MATHDB-396867

Existence of nontrivial isometries on Banach spaces

摘录 / EXCERPT: Does every Banach space contain a non-trivial isometry?

open_problem | open | MathDB 页面 | 原文线索

V2-038259 | MATHDB-396868

Equivalent almost transitive norms on superreflexive Banach spaces

摘录 / EXCERPT: Does every superreflexive Banach space admit an equivalent almost transitive norm?

open_problem | open | MathDB 页面 | 原文线索

V2-038260 | MATHDB-396869

Mazur’s rotation problem

摘录 / EXCERPT: Any separable Banach space X whose norm is transitive-that is, for every two points x,y on the unit sphere of X, there exists a surjective linear isometry $T:X\to X$ such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-038261 | MATHDB-396870

Wood’s maximal-norm problem for Banach spaces

摘录 / EXCERPT: Does every Banach space admit an equivalent maximal norm?

open_problem | open | MathDB 页面 | 原文线索

V2-038262 | MATHDB-396871

Nonexistence of equivalent maximal norms

摘录 / EXCERPT: Let X be a separable, complex, reflexive, hereditarily indecomposable Banach space without a Schauder basis. Then X admits no equivalent maximal norm and hence $GL(X)$ contains...

open_problem | open | MathDB 页面 | 原文线索

V2-038263 | MATHDB-396900

Formal analysis of locality, sparsity, and local maxima for bounded biharmonic weights

摘录 / EXCERPT: For the bounded biharmonic weights [U+007F]wj[U+007F] defined as minimizers of the Laplacian energy subject to the handle interpolation, cage-face linearity, partition-of-unity, and bounds in eq...

open_problem | open | MathDB 页面 | 原文线索

V2-038264 | MATHDB-396990

Norm-controlled inversion for polynomial off-diagonal decay matrices

摘录 / EXCERPT: Let $B=B(\ell^2(\mathbb Z))$ and, for $s>1$, let J_s be the Banach algebra of matrices $A=(A_{k,l})_{k,l\in\mathbb Z}$ with norm ... Jaffard's and Baskakov's results show that J_s...

open_problem | open | MathDB 页面 | 原文线索

V2-038265 | MATHDB-396992

Deriving the Gabor duality principle from R-duals

摘录 / EXCERPT: Can the duality principle in Gabor analysis actually be derived from the theory of R-duals?

open_problem | open | MathDB 页面 | 原文线索

V2-038266 | MATHDB-396994

Nonlinear least-squares subspace approximation (Problem 1)

摘录 / EXCERPT: Let H be a Hilbert space, let $\mathcal{F}=\{f_1,\dots,f_m\}\subset H$ be a finite set, let $\mathcal{C}$ be a family of closed subspaces of H, and fix an integer $l\ge 1$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-038267 | MATHDB-396995

Density and independence for complementary Parseval-frame subsets

摘录 / EXCERPT: Proposition 1. Let H be a Hilbert space with orthonormal basis $\\{e_j\}_{j\in S}$, let P be the orthogonal projection onto a closed subspace of H, and let $B\subseteq S$. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-038268 | MATHDB-396997

Partitioning an equal-norm Parseval frame into independent spanning sets

摘录 / EXCERPT: Let $\{f_i\}_{i\in I}$ be an equal-norm Parseval frame for an N-dimensional Hilbert space H_N, with $|I|=rN+k$ and $0\le k<N$. Then there is a partition $\{I_i\}_{i=1}^{r+1}$ o...

open_problem | open | MathDB 页面 | 原文线索

V2-038269 | MATHDB-397136

General analysis and adaptive methods for a convection-diffusion eigenproblem

摘录 / EXCERPT: For the model convection-diffusion eigenproblem ... where $(\lambda,u)\in\mathbb C\times\bigl(H_0^1(\Omega;\mathbb C)\cap H^2_{\mathrm{loc}}(\Omega;\mathbb C)\bigr)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-038270 | MATHDB-397349

Coresets for the minimum-volume enclosing ellipsoid

摘录 / EXCERPT: For a subset $S\subseteq\mathbb{R}^d$, let $\operatorname{MEE}(S)$ denote the ellipsoid of minimum volume containing S, and let DBM denote the Banach-Mazur dista...

open_problem | open | MathDB 页面 | 原文线索

V2-038271 | MATHDB-397360

Intermediate algebra for the higher Godbillon-Vey index pairing

摘录 / EXCERPT: In the foliated-bundle-with-cylindrical-ends setup of the report, there exists a holomorphically closed intermediate Banach subalgebra $\mathfrak J$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-038272 | MATHDB-397443

Convergence analysis for the surface Taylor-Hood discretization

摘录 / EXCERPT: Determine whether the planar convergence analysis for the quadratic-linear Taylor-Hood element (P2-P1) extends to the corresponding discretization on immersed hypersurfaces for the...

open_problem | open | MathDB 页面 | 原文线索

V2-038273 | MATHDB-397542

Convergence analysis for the regularized inverse-scattering iteration

摘录 / EXCERPT: It is an open problem to extend the convergence analysis to the modification of iteration (10) in which S is replaced by $S_n:=N_{\rho,\alpha_n}\circ R$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-038274 | MATHDB-397775

Optimal union of subspaces for subspace segmentation

摘录 / EXCERPT: Let B be a Banach space and let $W=\{w_1,\ldots,w_N\}\subset B$ be a finite set. For $i=1,\ldots,M$, let $\mathcal C_i$ be a family of closed subspaces of B, each containing...

open_problem | open | MathDB 页面 | 原文线索

V2-038275 | MATHDB-397776

Existence of optimal unions of subspaces for $p\ne 2$

摘录 / EXCERPT: Let B be a Banach space, let $W=\{w_1,\ldots,w_N\}\subset B$ be finite, let $0<p\le\infty$, and let $M\ge1$. For each $i=1,\ldots,M$, let $\mathcal C_i$ be a family of closed sub...

open_problem | open | MathDB 页面 | 原文线索

V2-038276 | MATHDB-397777

Efficient inversion of phaseless frame measurements

摘录 / EXCERPT: Let H be a finite-dimensional real or complex Hilbert space and let $\mathcal{F}=\{f_1,\ldots,f_m\}$ be a frame for H. With the equivalence relation $x\sim y$ if $y=zx$ for a s...

open_problem | open | MathDB 页面 | 原文线索

V2-038277 | MATHDB-397789

Characterization of strictly scalable frames

摘录 / EXCERPT: For a frame $\Phi=(\varphi_j)_{j\in J}$ for a separable Hilbert space H, characterize strict scalability. The report answers this by proving that Φ is strictly scalable if a...

open_problem | open | MathDB 页面 | 原文线索

V2-038278 | MATHDB-398121

Recognition of perfectoid algebras from their adic spectra

摘录 / EXCERPT: Let A be a uniform Banach algebra over $\mathbb{Q}_p$ equipped with its spectral norm. Suppose that $\operatorname{Spa}(A,A^\circ)$ is a perfectoid adic space. Must A be a per...

open_problem | open | MathDB 页面 | 原文线索

V2-038279 | MATHDB-398137

Consistency-error estimate for the theoretical $\alpha=0$ DPG analysis

摘录 / EXCERPT: In the theoretical case $\alpha=0$, demonstrate that the consistency error resulting from the weak enforcement of interelement continuity is of order $\omega(\omega h)^r$, where th...

open_problem | open | MathDB 页面 | 原文线索

V2-038280 | MATHDB-398173

Complete the unresolved normal-form analysis of the front dynamics

摘录 / EXCERPT: Complete the computation of the normal-form coefficients for the center-manifold reduction governing the front dynamics in the minimally nonlinear FitzHugh-Nagumo/Allen-Cahn-type s...

open_problem | open | MathDB 页面 | 原文线索

V2-038281 | MATHDB-398274

Mathematical analysis of the multiscale sound-proof atmospheric-flow problem

摘录 / EXCERPT: Analyze the asymptotic three-time-scale problem for compressible atmospheric flow and obtain mathematical insight into the remaining two-time-scale sound-proof models when only the...

open_problem | open | MathDB 页面 | 原文线索

V2-038282 | MATHDB-398590

Comparison of restricted and classical Kolmogorov widths

摘录 / EXCERPT: Let K be a compact subset of a Hilbert space H. Define the classical Kolmogorov width ... where F ranges over subspaces of H. Define the restricted width ... where F is requi...

open_problem | open | MathDB 页面 | 原文线索

V2-038283 | MATHDB-398630

Amenable operator algebras and similarity to C*-algebras

摘录 / EXCERPT: Does every separable amenable closed subalgebra of $L(H)$, for a Hilbert space H, have to be similar to a C^*-algebra?

open_problem | open | MathDB 页面 | 原文线索

V2-038284 | MATHDB-398646

Strong finite embeddability of countable discrete groups into Hilbert space

摘录 / EXCERPT: Is every countable discrete group Γ strongly finitely embeddable into a Hilbert space H? Here, strong finite embeddability means that for every $n\in\mathbb{N}$ and every...

open_problem | open | MathDB 页面 | 原文线索

V2-038285 | MATHDB-398652

Invariant subspace problem for separable infinite-dimensional Hilbert spaces (version I)

摘录 / EXCERPT: Let H be an infinite-dimensional, separable Hilbert space. Is there, for every $T\in B(H)$, a projection $p\in B(H)$ such that $p\notin\{0,1\}$ and $Tp=pTp$?

open_problem | open | MathDB 页面 | 原文线索

V2-038286 | MATHDB-398653

Invariant subspace problem relative to a von Neumann algebra

摘录 / EXCERPT: Let H be an infinite-dimensional separable Hilbert space. Is there, for arbitrary $T\in B(H)\setminus \mathbb{C}I$, a projection p belonging to the von Neumann algebra generate...

open_problem | open | MathDB 页面 | 原文线索

V2-038287 | MATHDB-398742

Variational characterization for a projected and restricted self-adjoint operator

摘录 / EXCERPT: The report states the following theorem. Let A_0 be self-adjoint in a Hilbert space H, let P be a closed subspace of H with associated orthogonal projection (also denoted...

open_problem | open | MathDB 页面 | 原文线索

V2-038288 | MATHDB-398855

Invariant-subspace characterization in the Hilbert space of Dirichlet series

摘录 / EXCERPT: Define E_2 as the space of Dirichlet series $D(s)=\sum_{n=1}^{\infty}a_n n^{-s}$ such that the series converges for some s, and $D(s)/s$ has an analytic continuation $F(s)$ to...

open_problem | open | MathDB 页面 | 原文线索

V2-038289 | MATHDB-398857

Bohr absolute-convergence strips for Banach-valued Hardy Dirichlet series

摘录 / EXCERPT: The report answers its natural question affirmatively: in the Banach-space-valued Dirichlet-series setup, with $H_p(X)$ and the abscissas $\c3_a^X$, $\c3_p^X$ defined analogously t...

open_problem | open | MathDB 页面 | 原文线索

V2-038290 | MATHDB-398860

Bohr’s absolute convergence problem in Banach spaces

摘录 / EXCERPT: For a complex Banach space X, let ... where the supremum is over all X-valued Dirichlet series $D(s)=\sum_{n=1}^{\infty}a_n n^{-s}$, and $\sigma_a(D)$ and $\sigma_u(D)$ are respe...

open_problem | open | MathDB 页面 | 原文线索

V2-038291 | MATHDB-398861

Cotype versus hypercontractive polynomial cotype

摘录 / EXCERPT: The report asks whether the notion of cotype q and the notion of hypercontractive polynomial cotype q coincide. Here, for $2\le q<\infty$, a Banach space X has hypercontracti...

open_problem | open | MathDB 页面 | 原文线索

V2-038292 | MATHDB-398874

Characterization of Banach sequence spaces with uniformly positive Bohr radii

摘录 / EXCERPT: Let X be a Banach sequence space such that $ ... If $inf{ninmathbb{N}}K(B{Xn})0$, is it necessarily true that $X=ell1$ as sets?

open_problem | open | MathDB 页面 | 原文线索

V2-038293 | MATHDB-398970

Lions’ problem for maximal regularity of non-autonomous symmetric forms

摘录 / EXCERPT: Let V and H be Hilbert spaces with V continuously and densely embedded in H, and let $a:[0,T]\times V\times V\to\mathbb{R}$ be a bounded, coercive, non-autonomous form. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-038294 | MATHDB-398971

Crouzeix’s numerical-range spectral-set conjecture with constant $Q=2$

摘录 / EXCERPT: Let $A\in L(D(A),H)$ be a linear operator on a complex (or complexified) Hilbert space H, and let ... be its numerical range. The conjecture is that $W(A)$ is a 2-spectral set fo...

conjecture | open | MathDB 页面 | 原文线索

V2-038295 | MATHDB-399053

Commutativity criterion via the joint point spectrum

摘录 / EXCERPT: Suppose $A=(A_1, \ldots,A_n)$ is an n-tuple of compact self-adjoint operators on a separable Hilbert space H. Then the operators in A pairwise commute if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-038296 | MATHDB-399105

Rigorous asymptotics for flat ASEP fluctuations

摘录 / EXCERPT: Make rigorous the formal critical-point analysis for the flat ASEP Fredholm-Pfaffian formula: under the appropriate scaling, prove that the Fredholm Pfaffian converges to a limitin...

open_problem | open | MathDB 页面 | 原文线索

V2-038297 | MATHDB-399223

Bound on the Banach indicatrix for injective-registration deformations

摘录 / EXCERPT: The report asks whether one can establish a bound of the form ... where N_y is the Banach indicatrix $N_y(x,E)=\operatorname{card}(y^{-1}(x)\cap E)$, R_M is the motion regulariza...

open_problem | open | MathDB 页面 | 原文线索

V2-038298 | MATHDB-399315

Extension of the robust LOD theory to wider classes of high-contrast coefficients

摘录 / EXCERPT: Can the localizable orthogonal decomposition (LOD) theory, including its robust localization and convergence analysis, be extended to wider classes of coefficients- in particular a...

open_problem | open | MathDB 页面 | 原文线索

V2-038299 | MATHDB-399317

Systematic and principled data summaries for object-oriented data analysis

摘录 / EXCERPT: The open problem proposed here is to develop a systematic and principled approach to finding data summaries-including cluster centers and indications of local variation in each clu...

open_problem | open | MathDB 页面 | 原文线索

V2-038300 | MATHDB-399425

Integrability of the infinitesimal representation associated with reflection positivity

摘录 / EXCERPT: Given a symmetric pair (G,H) with involution τ, a reflection-positive Hilbert space (E,E_{+},θ), and a reflection-positive unitary representation π satisfying (RP...

open_problem | open | MathDB 页面 | 原文线索

V2-038301 | MATHDB-399820

Ul’yanov-Type Problem

摘录 / EXCERPT: Let $\{T_N\}_{N\in\mathbb N}\subset L(B_1,B_2)$ be an approximation method of $T\in L(B_1,B_2)$, where B_1 and B_2 are Banach spaces. Does there exist a strictly increasing seq...

open_problem | open | MathDB 页面 | 原文线索

V2-038302 | MATHDB-399953

Cayley Transform Problem on Hilbert Spaces

摘录 / EXCERPT: Let X be a Hilbert space and let A be the generator of a bounded C_0-semigroup $(e^{tA})_{t\ge 0}$ on X. Is the Cayley transform ... power bounded, i.e. does ... hold?

open_problem | open | MathDB 页面 | 原文线索

V2-038303 | MATHDB-399985

Computational barrier for preventing false discovery in interactive data analysis

摘录 / EXCERPT: If one-way functions exist, then there is no procedure that runs in time $\operatorname{poly}(n,\log|X|)$ per query and answers significantly more than n^2 interactive queries.

open_problem | open | MathDB 页面 | 原文线索

V2-038304 | MATHDB-399999

Convergence of ESFEM for a regularized solution-driven surface with linear finite elements

摘录 / EXCERPT: For the regularized velocity law ... where $\alpha>0$ is fixed and g is a given smooth scalar-valued function, establish a complete stability and convergence analysis for the evolv...

open_problem | open | MathDB 页面 | 原文线索

V2-038305 | MATHDB-400280

Uniqueness of the signature of a rough path up to tree-like equivalence

摘录 / EXCERPT: For a rough path X over some Banach space, the signature of X is trivial if and only if X can be realized as a loop in some real tree. In particular, a rough path is uniquely...

open_problem | open | MathDB 页面 | 原文线索

V2-038306 | MATHDB-400284

Large-field analysis for the subcritical Bose gas

摘录 / EXCERPT: Complete the rigorous analysis of the large-field problem in the coherent-state/block-spin construction for the three-dimensional lattice Bose gas in the regime...

open_problem | open | MathDB 页面 | 原文线索

V2-038307 | MATHDB-400425

Extension of the operator-algebraic construction to the sine-Gordon and Thirring models

摘录 / EXCERPT: Extend the construction described in the report to the sine-Gordon and Thirring models, comparing the construction with the Euclidean methods in reference [1].

open_problem | open | MathDB 页面 | 原文线索

V2-038308 | MATHDB-400444

Nonvanishing of the patched Banach representation

摘录 / EXCERPT: In the specialization to $F=\mathbb{Q}_p$ and $n=2$, is $\Pi_y\neq 0$ for every $y\in\operatorname{m\text{-}Spec}R_\infty[1/p]$?

open_problem | open | MathDB 页面 | 原文线索

V2-038309 | MATHDB-400467

Fixed-point property for superreflexive Banach spaces

摘录 / EXCERPT: Higher-rank connected simple real Lie groups have property (F_X) for every superreflexive Banach space X.

open_problem | open | MathDB 页面 | 原文线索

V2-038310 | MATHDB-400497

Does Yu’s property A imply coarse embeddability into Hilbert space for groups?

摘录 / EXCERPT: Does Yu’s property A imply that a group admits a coarse embedding into a Hilbert space? The report states that the answer is negative: there exist finitely generated groups with Yu...

open_problem | open | MathDB 页面 | 原文线索

V2-038311 | MATHDB-400511

Analysis of the TD-LSM for obstacles touching waveguide walls

摘录 / EXCERPT: Analyze the Time Domain Linear Sampling Method (TD-LSM) for the waveguide problem when the obstacle touches the walls of the pipe.

open_problem | open | MathDB 页面 | 原文线索

V2-038312 | MATHDB-400532

Rigorous analysis of the seamless multilevel ETPF scaling

摘录 / EXCERPT: Provide a rigorous analysis that confirms or refutes the reported $\b2=2$ variance-scaling behavior of the seamless multilevel ensemble transform particle filter for higher-dimensi...

open_problem | open | MathDB 页面 | 原文线索

V2-038313 | MATHDB-400613

Systematic analysis of a ferroelectric phase-diagram model

摘录 / EXCERPT: A systematic analysis of the proposed model for the phase diagram of a solid solution of lead titanate and lead zirconate remains open. The model is described as similar to a rando...

open_problem | open | MathDB 页面 | 原文线索

V2-038314 | MATHDB-400710

Reconstructing the global-field dynamics from a non-involutive Banach algebra

摘录 / EXCERPT: Is it possible to reconstruct the dynamics $I_K\curvearrowright X_K$ from the non-involutive Banach algebra which is the closure of $C(X_K)\rtimes_{\mathrm{alg}} I_K$ in its natura...

open_problem | open | MathDB 页面 | 原文线索

V2-038315 | MATHDB-400742

General effective integration rules for Galerkin-based isogeometric analysis

摘录 / EXCERPT: The development of a general and effective solution for Galerkin-based isogeometric analysis methods, consisting of efficient integration rules that reduce the high array-formation...

open_problem | open | MathDB 页面 | 原文线索

V2-038316 | MATHDB-400849

Geometry, analysis, and numerics for infinite-dimensional rough-volatility models

摘录 / EXCERPT: Develop geometric, analytic, and numerical theories for infinite-dimensional models with components involving fractional Brownian motions or, more generally, rough paths.

open_problem | open | MathDB 页面 | 原文线索

V2-038317 | MATHDB-400891

Analysis Problem for a finite-dimensional discrete-time linear system

摘录 / EXCERPT: Let ... where the state, input, and output spaces are finite-dimensional. Internal stability holds if and only if $\sigma(A)\subset\mathbb D$, equivalently if and only if there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-038318 | MATHDB-400967

Multivariate extension of minimum-volume optimal designs

摘录 / EXCERPT: Extend the analysis of minimum-volume optimal designs from the univariate Fourier and weighted-polynomial regression settings to models with more than one explanatory variable, and...

open_problem | open | MathDB 页面 | 原文线索

V2-038319 | MATHDB-401020

Convergence analysis for multidimensional spectral estimation of diffusion coefficients

摘录 / EXCERPT: Let $D\subset\mathbb{R}^d$ be a bounded domain, and suppose a stationary diffusion has invariant density μ and generator in divergence form ... with positive-definite matrix-valu...

open_problem | open | MathDB 页面 | 原文线索

V2-038320 | MATHDB-401463

Infinite-horizon bilinear optimal control problem

摘录 / EXCERPT: Consider ... subject to ... ... Here, $V\subset Y\subset V^*$ is a Gelfand triple of real Hilbert spaces, $A:D(A)\subset Y\to Y$ is the infinitesimal generator of an analytic C_0-semig...

open_problem | open | MathDB 页面 | 原文线索

V2-038321 | MATHDB-401464

Derivation and analysis of multiscale models for collective-behavior control

摘录 / EXCERPT: The derivation of multiscale (MS) models for the control of collective behavior and their analysis.

open_problem | open | MathDB 页面 | 原文线索

V2-038322 | MATHDB-401465

Constrained Pontryagin maximum principle in an infinite-dimensional state-constrained control problem

摘录 / EXCERPT: Let X be an infinite-dimensional separable Banach space, $I=[0,1]$, and consider the mild solutions of ... subject to ... where u is a measurable selection of a measurable set-valu...

open_problem | open | MathDB 页面 | 原文线索

V2-038323 | MATHDB-401499

Expander graphs and equi-coarse embeddings into super-reflexive Banach spaces

摘录 / EXCERPT: Let Y be an arbitrary super-reflexive Banach space and $(G_n)_{n\in\mathbb N}$ be a sequence of expander graphs. Is it true that $(G_n)_{n\in\mathbb N}$ does not equi-coarsely em...

open_problem | open | MathDB 页面 | 原文线索

V2-038324 | MATHDB-401500

Uniform homeomorphism between the unit spheres of a separable super-reflexive Banach space and \(\ell_2\)

摘录 / EXCERPT: Let X be a separable super-reflexive Banach space. Is the unit sphere of X uniformly homeomorphic to the unit sphere of ℓ_2?

open_problem | open | MathDB 页面 | 原文线索

V2-038325 | MATHDB-401501

Problem 8: Expander graphs and asymptotically Hilbertian Banach spaces

摘录 / EXCERPT: Let Y be an asymptotically Hilbertian Banach space and $(G_n)_{n\in\mathbb N}$ be a sequence of expander graphs. Is it true that $(G_n)_{n\in\mathbb N}$ does not equi-coarsely em...

open_problem | open | MathDB 页面 | 原文线索

V2-038326 | MATHDB-401585

Conjecture 5: interpolation characterization of symmetric quasi-Banach sequence spaces

摘录 / EXCERPT: Let $E\subset \ell^2$ be a symmetric quasi-Banach sequence space. The following conditions are equivalent: (1) If $\mu_2(x)\prec\mu_2(y)$ and $x\in E$, then $y\in E$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-038327 | MATHDB-401813

HHR property of convex bounded domains in infinite-dimensional Hilbert spaces

摘录 / EXCERPT: Is a convex bounded domain in an infinite-dimensional Hilbert space HHR?

open_problem | open | MathDB 页面 | 原文线索

V2-038328 | MATHDB-401977

Interpreting traffic independence via modified free independence

摘录 / EXCERPT: Can we interpret traffic independence as some modification of free independence in order to use free analysis theory?

open_problem | open | MathDB 页面 | 原文线索

V2-038329 | MATHDB-402057

Conditions for an E-weak mode to be a strong mode

摘录 / EXCERPT: Let X be a real separable Banach space, let μ be a measure on X, and let $\E\subsetneq X$ be a dense subspace. For $x\in X$, an E-weak mode is characterized by ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-038330 | MATHDB-402071

Symmetry of the completed representation-ring Banach *-algebra

摘录 / EXCERPT: Is it true in general that the completion $\widehat{a}(G)$ of the complexified representation ring $a_{\mathbb C}(G)$ is a symmetric Banach $*$-algebra? Here, a Banach $*$-algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-038331 | MATHDB-402525

Fixed-point property for uniformly convex Banach spaces

摘录 / EXCERPT: Every lattice in a connected simple Lie group of rank at least two has property (F_X) for every uniformly convex Banach space X.

open_problem | open | MathDB 页面 | 原文线索

V2-038332 | MATHDB-402823

The remaining real case of Banach’s isometric conjecture for $n=133$

摘录 / EXCERPT: Let V be a real Banach space of dimension m, with $1<133<m$, such that all 133-dimensional subspaces of V are isometric to one another. Must V be a Hilbert space?

conjecture | open | MathDB 页面 | 原文线索

V2-038333 | MATHDB-402824

Banach’s geometric hypothesis in the exceptional case $n=133$

摘录 / EXCERPT: Does every finite-dimensional real Banach space V whose 133-dimensional subspaces are all mutually isometric necessarily have an ellipsoidal unit ball, and hence be a Hilbert s...

conjecture | open | MathDB 页面 | 原文线索

V2-038334 | MATHDB-402942

Three-dimensional overlapping FEM-BEM implementation and analysis

摘录 / EXCERPT: Develop and analyze an overlapping FEM-BEM implementation for the three-dimensional case using the framework in reference [4].

open_problem | open | MathDB 页面 | 原文线索

V2-038335 | MATHDB-402944

Rigorous analysis of the stabilized BEM shape-gradient formula

摘录 / EXCERPT: Give a rigorous theoretical justification for the expectation that, after the singular terms in expression (7) have been suitably rearranged and stabilized, substituting boundary-e...

open_problem | open | MathDB 页面 | 原文线索

V2-038336 | MATHDB-403135

Dissipativity-based turnpike analysis with active constraints

摘录 / EXCERPT: Despite substantial progress on the dissipativity-based analysis of optimal control problems, the vast majority of available results assume that inequality constraints on states an...

open_problem | open | MathDB 页面 | 原文线索

V2-038337 | MATHDB-403294

Removal of the loss of 1 in Kolmogorov-width decay under holomorphic mappings

摘录 / EXCERPT: For a holomorphic map F between Banach spaces V_1 and V_2, with F holomorphic in a neighborhood of a compact set $K_1\subset V_1$ and $K_2=F(K_1)$, the report proves that .....

open_problem | open | MathDB 页面 | 原文线索

V2-038338 | MATHDB-404081

Converse to Shalit’s stable division criterion for essential normality

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and let $I\subset\mathbb{C}[z_1,\ldots,z_d]$ be a radical homogeneous ideal. Let H_d^2 be the Hilbert space obtained by completing the polynomial ring with r...

open_problem | open | MathDB 页面 | 原文线索

V2-038339 | MATHDB-404391

Classification of unitary Yang-Baxter solutions up to braid-group-character equivalence

摘录 / EXCERPT: Classify the quotient $\mathcal R(V)/\sim$, where V is a finite-dimensional Hilbert space and $\mathcal R(V)$ is the collection of unitary endomorphisms...

open_problem | open | MathDB 页面 | 原文线索

V2-038340 | MATHDB-404464

Extension of disjoint frequently hypercyclic tuples

摘录 / EXCERPT: For every integer $N\ge 1$, every Banach space X supporting a hereditarily frequently hypercyclic operator, and every $T_1,\ldots,T_N\in L(X)$ such that $(T_1,\ldots,T_N)$ is d-f...

open_problem | open | MathDB 页面 | 原文线索

V2-038341 | MATHDB-404496

Existence of an infinite-dimensional Wishart process for nonintegral shape parameter

摘录 / EXCERPT: Let H be an infinite-dimensional separable real Hilbert space, and consider the Wishart equation ... Here $\alpha\in\mathbb R$, $A:D(A)\subset H\to H$ is the generator of a C_0-s...

open_problem | open | MathDB 页面 | 原文线索

V2-038342 | MATHDB-404626

Banach’s section-equivalence problem

摘录 / EXCERPT: Does a convex body $K\subset\mathbb{R}^n$ have to be an ellipsoid if all k-dimensional sections through a fixed interior point are linearly (affinely) equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-038343 | MATHDB-404747

Analysis of the amalgam in Part II of the Meierfrankenfeld program

摘录 / EXCERPT: The investigation of the amalgam $\langle L,H\rangle$ by means of suitable uniqueness subgroups has yet to be carried out.

open_problem | open | MathDB 页面 | 原文线索

V2-038344 | MATHDB-405154

Schoen’s conjecture on scalar-curvature singularities

摘录 / EXCERPT: Let (M^n,g_0) be a closed Einstein manifold, and let $\Sigma\subset M$ be a singular set. Suppose that g is a continuous (or, in the weaker formulation, L^∞) Riemannian...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038345 | MATHDB-405158

Lin’s interior mass-bound problem

摘录 / EXCERPT: Determine whether, for every dimension n and every algebraic projection multiplicity Q, there exists a constant $C(n)$ such that every confined area-minimizing rectifiable curr...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038346 | MATHDB-405165

Gromov’s mean-of-the-mean-curvature conjecture

摘录 / EXCERPT: For every closed Riemannian n-manifold (M,g) and every real number λ, there exists a constant $C(M,g,\lambda)$ such that, for every compact Riemannian $(n+1)$-manifold...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038347 | MATHDB-405176

Unique predual problem for Lipschitz spaces

摘录 / EXCERPT: Determine whether, for every complete pointed metric space $(M,d,0)$, the Banach space $\operatorname{Lip}_0(M)$ of real-valued Lipschitz functions vanishing at 0 has a strongly...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038348 | MATHDB-405177

Every bounded operator on an infinite-dimensional Banach space has norm attainment

摘录 / EXCERPT: Does there exist an infinite-dimensional Banach space X such that every bounded linear operator $T\in\mathcal{L}(X,X)$ attains its norm; that is, for every $T\in\mathcal{L}(X,X)$...

open_problem | open | MathDB 页面 | 原文线索

V2-038349 | MATHDB-405191

Lyubarskii-Nes conjecture on the frame set of the first Hermite function

摘录 / EXCERPT: Let $h_1(t)=Cte^{-pi t^2}$ be the first Hermite function, where $C\neq 0$, and for $a,b>0$ define the Gabor system...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038350 | MATHDB-405220

Nick Higham’s open problem on conditioning of Sylvester and Lyapunov equation solutions

摘录 / EXCERPT: Given matrices A, B, and C, determine conditions under which the solution X of the Sylvester equation $AX-XB=C$ is well-conditioned, rather than merely existing uniquely or...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038351 | MATHDB-405227

Olikier-Waldspurger open question on proximal-gradient cluster points

摘录 / EXCERPT: Let $F=f+g$ be a possibly nonconvex, nonsmooth objective, and let (x_k) be a sequence generated by a proximal-gradient method for minimizing F, with one component locally Lipsc...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038352 | MATHDB-405235

Kaplansky’s idempotent conjecture for integral quandle rings

摘录 / EXCERPT: For every finite Latin quandle Q, let $\mathbb{Z}[Q]$ be the free abelian group with basis Q, equipped with the bilinear multiplication extending the quandle operation $x*y$ on...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038353 | MATHDB-405236

Stein’s weak-type (1,1) conjecture for rough maximal operators

摘录 / EXCERPT: For $n\ge 2$ and every measurable angular kernel $\Omega\in L^1(\mathbb S^{n-1})$, define...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038354 | MATHDB-405238

Chern’s conjecture

摘录 / EXCERPT: For each integer $n\ge 2$, let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038355 | MATHDB-405239

Gromov’s critical rate of decay conjecture

摘录 / EXCERPT: Let $n\ge 3$ and let (M^n,g) be a complete, connected, orientable, noncompact Riemannian manifold with positive scalar curvature. If, for some (equivalently any) $x_0\in M$,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038356 | MATHDB-405241

Clouâtre-Ostermann-Ransford conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a uniform algebra, let H be a Hilbert space, let $\pi:\mathcal A\to\mathcal B(H)$ be a bounded unital homomorphism, and let $\alpha:\mathcal A\to\mathcal A$ b...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038357 | MATHDB-405256

Yu's nonempty-interior conjectures for radial projections and division sets

摘录 / EXCERPT: Yu's conjectures assert, for the relevant pairs of Cantor sets and their radial projections or division sets, that the corresponding sets have nonempty interior. The supplied sourc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038358 | MATHDB-405269

De Giorgi conjecture on the regularity of minimizers of Cartesian area in one dimension and codimension one

摘录 / EXCERPT: For every pair of positive integers n,m, every bounded open set $\Omega\subset\mathbb{R}^n$, and admissible data $g:\Omega\to\mathbb{R}^m$, consider minimizers...

conjecture | claimed_progress | MathDB 页面 | 原文线索

B08 偏微分方程 / Partial Differential Equations | 710 条

V2-038359 | MATHDB-315529

Equilibria of the Vlasov-Maxwell-Landau system

摘录 / EXCERPT: Under the paper’s smoothness, positivity, Schwartz-decay, and score assumptions, are all steady solutions of the Coulomb Vlasov-Maxwell-Landau system necessarily spatially uniform...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038360 | MATHDB-315594

Unique invariant measure for the skew stochastic heat equation

摘录 / EXCERPT: For the stochastic heat equation $\partial_tu=\tfrac12\partial_x^2u+\delta_0(u)+\dot W$ on $[0,1]$ with Dirichlet boundary conditions, does its Markov semigroup have at most one in...

open_problem | open | MathDB 页面 | 原文线索

V2-038361 | MATHDB-315775

Lower bounds for advection-diffusion

摘录 / EXCERPT: For mean-zero solutions of advection-diffusion equations on the two-dimensional torus, derive explicit constructive lower bounds that rule out excessively fast mixing in three regi...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038362 | MATHDB-316344

Support-Stokes-line conjecture for asymptotic measures

摘录 / EXCERPT: Let $\mu_1,\ldots,\mu_k$ be the asymptotic measures associated with the polynomial eigenfunctions of the differential operator $\sum_{i=0}^k Q_i(z)\frac{d^i}{dz^i}$. The global Sto...

conjecture | open | MathDB 页面 | 原文线索

V2-038363 | MATHDB-316693

The second-prolongation invariant-count conjecture for differential equations defined by vector fields

摘录 / EXCERPT: Second-prolongation invariant-count conjecture. For any value n of independent variables, the number of functionally independent invariants of the second prolongation of G is

conjecture | open | MathDB 页面 | 原文线索

V2-038364 | MATHDB-316718

Short-term a priori bound conjecture for periodic Navier-Stokes

摘录 / EXCERPT: Short-term a priori bound for periodic Navier-Stokes. There exists a non-decreasing function $G:{\mathbf{R}}^+\to{\mathbf{R}}^+$ such that, for every $0<T\leq1$ and every such smoo...

conjecture | open | MathDB 页面 | 原文线索

V2-038365 | MATHDB-316774

Chudnovsky-Chudnovsky conjecture on globally nilpotent indigenous bundles

摘录 / EXCERPT: Let $\overline{C}$ be a smooth projective curve defined over a number field. Let $({\mathcal E}, \nabla)$ be an indigenous bundle on $\overline{C}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038366 | MATHDB-316775

Dwork's conjecture on globally nilpotent rank-two bundles

摘录 / EXCERPT: Let $\overline{C}$ be a smooth projective curve of genus 0 defined over a number field, and let $({\mathcal E}, \nabla)$ be a flat vector bundle of rank 2 which is globally nil...

conjecture | open | MathDB 页面 | 原文线索

V2-038367 | MATHDB-316804

Critical Sobolev heat-kernel limit and hyperfunction measure classes

摘录 / EXCERPT: Let ν_t^{s,m_0} be the heat kernels associated with the W^s norm on loops, viewed as measures on $\operatorname{Hyp}(S^1,G)$, where $s>1/2$, $t>0$, and $m_0>0$. Let μ_l b...

conjecture | open | MathDB 页面 | 原文线索

V2-038368 | MATHDB-317027

Hislop-Lutzer exponential decay conjecture for gamma-harmonic liftings

摘录 / EXCERPT: Let $\{\phi_k\}_{k=1}^{\infty}$ be eigenfunctions of the Dirichlet-to-Neumann operator, with corresponding eigenvalues λ_k, and let v_{ϕ_k} denote the γ-harmoni...

conjecture | open | MathDB 页面 | 原文线索

V2-038369 | MATHDB-317121

The exterior-square CY(5) conjecture

摘录 / EXCERPT: Let ${\cal P}={\cal P}(\theta,z)$ be a CY(4)-operator, and let ${\cal Q}$ be the fifth-order differential operator constructed from ${\cal P}$ by requiring that, for any two linear...

conjecture | open | MathDB 页面 | 原文线索

V2-038370 | MATHDB-317321

Heteroclinic-orbit conjecture for eternal solutions

摘录 / EXCERPT: Let u be an eternal solution of the semilinear parabolic equation $\$, with coefficients a_i that decay sufficiently fast as $|x|\to\infty$. Heteroclinic-orbit conjecture. All...

conjecture | open | MathDB 页面 | 原文线索

V2-038371 | MATHDB-317326

Finiteness conjecture for smooth global solutions

摘录 / EXCERPT: Consider the differential equation ... . This finiteness would justify the use of a numerical solver to determine the number of global solutions for a given M-shaped ϕ, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-038372 | MATHDB-317396

Generic simplicity of eigenvalues along heteroclinic orbits

摘录 / EXCERPT: Generic simplicity conjecture. There is a generic subset (a Baire subset) of choices for ϕ in $$ such that, if u is a heteroclinic orbit, all of the eigenvalues of $H(t)$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-038373 | MATHDB-317568

Dichotomy conjecture for negative-Schwarzian delay equations

摘录 / EXCERPT: Assume that $a\geq 0$ and that f is an S-map. Let K be the unique fixed point of f. Equation (1) has an equilibrium at K; a global attractor attracts the phase space, and...

conjecture | open | MathDB 页面 | 原文线索

V2-038374 | MATHDB-317574

Low-regularity well-posedness conjecture for the modified Benjamin-Ono equation

摘录 / EXCERPT: Let H^s denote the Sobolev space of order s, let H^∞ denote the smooth-data space, and let S_T^∞ be the solution map for the real-valued modified Benjamin-Ono equ...

conjecture | open | MathDB 页面 | 原文线索

V2-038375 | MATHDB-317811

Polynomial-structure conjecture for finite-dimensional semisimple Lie algebras

摘录 / EXCERPT: Let $(A_{s s'})$ be a Cartan matrix for a finite-dimensional semisimple Lie algebra $\mathcal G$. Let $H_s(z)>0$ satisfy the master equations ... with boundary conditions ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-038376 | MATHDB-317844

Sverak's conjecture on isolated singularities of stationary Navier-Stokes flows

摘录 / EXCERPT: Let u be a solution of the stationary Navier-Stokes equations with zero force in $\mathbb{R}^3\setminus\{0\}$ satisfying the bound ... for some $C_*>0$. A Landau-solution conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-038377 | MATHDB-317956

The Hasse-Arf theorem for mixed-characteristic ramification filtrations

摘录 / EXCERPT: Let K be a complete discretely valued field of mixed characteristic $(0,p)$, and let $\rho:G_K\rightarrow GL(V_\rho)$ be a representation with finite local monodromy. The Artin a...

conjecture | open | MathDB 页面 | 原文线索

V2-038378 | MATHDB-318009

The global boundedness and scattering conjecture for energy-supercritical NLS

摘录 / EXCERPT: Global boundedness and scattering conjecture. The solution u is global and there exists a function $C:[0,\infty)\to[0,\infty)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038379 | MATHDB-318019

Nonvanishing imaginary-part conjecture for reduced Beltrami solutions

摘录 / EXCERPT: Nonvanishing imaginary-part conjecture. Either $\operatorname{Im}(f_z)$ is constant, or

conjecture | open | MathDB 页面 | 原文线索

V2-038380 | MATHDB-318021

Iwaniec-Martin-Šverák conjecture on mappings of integrable inner distortion

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a domain, let $f\in W^{1,n}_{\operatorname{loc}}(\Omega,\mathbb{R}^n)$, and let $K_I(\cdot,f)$ denote the inner distortion of f. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-038381 | MATHDB-318188

Parity conjecture for logarithmic singularities in the Suslov problem

摘录 / EXCERPT: Parity conjecture. Singular points s_1 and s_2 are non-logarithmic for any odd integer p and logarithmic for any even $p\in\mathbb{Z}^{\ast}$.

conjecture | open | MathDB 页面 | 原文线索

V2-038382 | MATHDB-318349

Chudnovsky and Chudnovsky's conjecture on globally nilpotent Lamé equations

摘录 / EXCERPT: Consider Lamé equations with parameter $n=-\frac{1}{2}$ defined over $\overline{\mathbb{Q}}$, and the four classes numbered $1,2,3,4$ above. Chudnovsky and Chudnovsky's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-038383 | MATHDB-318404

Spectral-measure convergence conjecture for higher Lamé operators

摘录 / EXCERPT: Let $\mathfrak d(z)=\sum_{i=1}^k Q_i(z)\frac{d^i}{dz^i}$ be a non-degenerate higher Lamé operator, and let $r=\max_{i=1,\ldots,k}(\deg Q_i-i)$. Let Pol_r be the space of monic po...

conjecture | open | MathDB 页面 | 原文线索

V2-038384 | MATHDB-318405

Unbounded-root conjecture for degenerate Lamé operators

摘录 / EXCERPT: Let $\mathfrak d(z)$ be a degenerate Lamé operator, meaning that its leading coefficient satisfies $\deg Q_k<k+r$, and let N_0 be any positive integer. Unbounded-root conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-038385 | MATHDB-318406

Root-interlacing conjecture for eigenpolynomials of exactly solvable operators

摘录 / EXCERPT: Let $\mathfrak d(z)$ be an exactly solvable operator, and let $\{S_n(z)\}_{n\in\mathbb N,\,n\geq n_0}$ be its eigenpolynomials with $\deg S_n(z)=n$. Let $\mathfrak S_\mu$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-038386 | MATHDB-318418

The limiting Van Vleck measure conjecture for generalized Lamé equations

摘录 / EXCERPT: Let $Q(z)$ and $P(z)$ be polynomials with $\deg Q(z)=l\geq 2$ and $\deg P(z)\leq l-1$, and let $\mathcal V_n$ be the finite measure on the space Pol_{l-2} of monic polynomials of...

conjecture | open | MathDB 页面 | 原文线索

V2-038387 | MATHDB-318787

Criticality of the Ladyzhenskaya-Lions exponent for hyper-dissipative Navier-Stokes equations

摘录 / EXCERPT: Let $n\geq 3$ and let $\alpha_{\rm \text{\tiny \sc l}}(n)$ denote the Ladyzhenskaya-Lions exponent for the hyper-dissipative Navier-Stokes equations. The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-038388 | MATHDB-318823

Null geodesic solution conjecture for distributional wave equations

摘录 / EXCERPT: Let M be a manifold and choose an atlas A'. Let $g_{\mu\nu}\in D^{0}_{2 S(A' o)}(M)$ admit $A_s(g_{\mu\nu})$, a Lorentzian-signature metric tensor field. Let Ω_{Ch} be a...

conjecture | open | MathDB 页面 | 原文线索

V2-038389 | MATHDB-318825

Global regularity conjecture for wave maps

摘录 / EXCERPT: Let ${\mathcal S}$ be the space of classical data (ϕ_0,ϕ_1), where $\phi_0:{\mathbf R}^2\to{\mathbf H}$ is Schwartz modulo constants and...

conjecture | open | MathDB 页面 | 原文线索

V2-038390 | MATHDB-319195

Conjecture on the Pearcey process transition probability with inliers

摘录 / EXCERPT: Let $p>2$ and let ${\mathbb Q}(\tau_1,\ldots,\tau_m;u_1,\ldots,u_{p-2};{\mathbb E})$ be the logarithm of the transition probability that the Pearcey process with inliers...

conjecture | open | MathDB 页面 | 原文线索

V2-038391 | MATHDB-319610

Katz's conjectural description of the generic Galois group

摘录 / EXCERPT: Let ${\mathcal M}=(M,\nabla)$ be a differential module over $\mathbb Q(x)$, let $\operatorname{Gal}({\mathcal M})$ be its generic Galois group, and let ψ_p denote the p-curv...

conjecture | open | MathDB 页面 | 原文线索

V2-038392 | MATHDB-319722

Regularity and minimal-growth conjecture for Fuchsian singularities

摘录 / EXCERPT: Let X) be the domain in Eq. (1), let $\hat X$ be its compactification, and let $\zeta\in\partial\hat X$. Assume that Eq. (1) has a Fuchsian type singularity at ζ and admits...

conjecture | open | MathDB 页面 | 原文线索

V2-038393 | MATHDB-319723

Strong comparison principle for p-Laplacian type equations

摘录 / EXCERPT: Strong comparison principle. Either $u<v$ throughout Ω', or $u=v$ throughout Ω'. The weak comparison principle supplies the non-strict inequality $u\leq v$; the conje...

conjecture | open | MathDB 页面 | 原文线索

V2-038394 | MATHDB-319731

Cantrell-Cosner-Lou concentration conjecture for positive local maxima

摘录 / EXCERPT: Cantrell-Cosner-Lou concentration conjecture. u concentrates precisely on the set of positive local maximum points of $m(x)$ as $\alpha\rightarrow\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-038395 | MATHDB-320044

Existence of a fully invariant critical functional for the Trudinger-Moser nonlinearity

摘录 / EXCERPT: Let $H_0^1({\mathbb D})$ be the Sobolev space on the unit disk, let η_ζ denote the disk automorphism associated with $\zeta\in{\mathbb D}$, and let h_s be the radial dil...

conjecture | open | MathDB 页面 | 原文线索

V2-038396 | MATHDB-320401

Continuity and consistency of moment-based matching operators

摘录 / EXCERPT: Let $\left(\overline{\mathcal R_{[L]}}\right)_{L=1,2,\dots}$ be a sequence of restriction operators whose macroscopic state variables are the first L centralized moments, with...

conjecture | open | MathDB 页面 | 原文线索

V2-038397 | MATHDB-320435

Khavinson-Shapiro Schwarz potential singularity conjecture

摘录 / EXCERPT: Let Γ be a nonsingular analytic surface, and consider the Cauchy problem for Laplace's equation on Γ with real-entire data. Let u be its solution, and let w be th...

conjecture | open | MathDB 页面 | 原文线索

V2-038398 | MATHDB-320455

Equality of multiplicity bounds for polynomial and piecewise linear coefficients

摘录 / EXCERPT: Let $\mu_{1}(m,n)$ be the maximum multiplicity of $z=0$ when $B(t)$ and $A(t)$ are polynomial functions of degrees m and n, respectively. Let $\mu_{2}(m,n)$ be the maximum mult...

conjecture | open | MathDB 页面 | 原文线索

V2-038399 | MATHDB-320666

Existence of Cartesian vector fields for constrained Lagrangian systems

摘录 / EXCERPT: Let $\textsc{Q}$ be the configuration space, let λ_k be the functions defining the system, and let ... for $j=M+1,\ldots,N$. A vector field is called Cartesian if it generate...

conjecture | open | MathDB 页面 | 原文线索

V2-038400 | MATHDB-321030

Quasi-Monte Carlo convergence conjecture for mapped $(0,2)$-sequences on the sphere

摘录 / EXCERPT: Convergence conjecture. The squared worst-case error in the Sobolev space $\mathcal{H}^{3/2}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038401 | MATHDB-321048

General comparison with measurable ingredients for integro-differential operators

摘录 / EXCERPT: General comparison with measurable ingredients conjecture. Under these assumptions, the conclusion of the comparison proposition holds with M^+_{CS} in place of M^+_A.

conjecture | open | MathDB 页面 | 原文线索

V2-038402 | MATHDB-321312

Optimal exponent conjecture for fully nonlinear uniformly elliptic equations

摘录 / EXCERPT: Let λ and Λ be the ellipticity constants in Proposition 2.1, and let ε denote the exponent in its $W^{2,\varepsilon}$ estimate. Optimal exponent conject...

conjecture | open | MathDB 页面 | 原文线索

V2-038403 | MATHDB-321374

The approximation-rate conjecture for Wendland functions

摘录 / EXCERPT: Approximation-rate conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-038404 | MATHDB-321407

Sobolev-norm limit conjecture for regularized stochastic Allen-Cahn solutions

摘录 / EXCERPT: Let u_N be the solution to the regularized problem, and let $\|\cdot\|_s$ denote the $H^s(\mathbb{T}^2)$ norm. For every $t>0$, the Sobolev-norm limit conjecture. ... This predicts...

conjecture | open | MathDB 页面 | 原文线索

V2-038405 | MATHDB-321520

Divergence with probability one of the multilevel Monte Carlo Euler method

摘录 / EXCERPT: Let $T\in(0,\infty)$, let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space with a normal filtration $(\mathcal{F}_t)_{t\in[0,T]}$, let $W^{l,k}$ be a family of independent...

conjecture | open | MathDB 页面 | 原文线索

V2-038406 | MATHDB-321541

Zero-mean conjecture for irregular delay-equation solutions

摘录 / EXCERPT: Zero-mean conjecture. For every such ϕ and every $\beta\in I$, $\mu(\beta;\phi)=0$. The authors state that this is unproved for irregular solutions; it holds for symmetric per...

conjecture | open | MathDB 页面 | 原文线索

V2-038407 | MATHDB-321542

Variance-scaling conjecture for deterministic Brownian motion

摘录 / EXCERPT: Let I be the parameter interval for which the delay equation has no stable steady state or stable periodic solution. For $\phi\in C([-1,0],\mathbb{R})$, let $v_\beta(t;\phi)$ sol...

conjecture | open | MathDB 页面 | 原文线索

V2-038408 | MATHDB-321641

The one-ended non-negative Ricci curvature conjecture for the Riesz transform

摘录 / EXCERPT: Let M be a Riemannian manifold satisfying a Sobolev inequality, with non-negative Ricci curvature outside a compact set and only one end. The one-ended Riesz transform conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-038409 | MATHDB-321703

Existence conjecture for complete descriptions of nonholonomic systems

摘录 / EXCERPT: Let $\mathbf{v}$ be the vector field given by formula. Consider the equations and the system, together with the linear constraints ... for $k=1,\ldots,M$. Existence conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-038410 | MATHDB-321771

Conjecture on stability of the stalemate equilibrium in Lanchester theory

摘录 / EXCERPT: Let the stalemate equilibrium be the equilibrium defined by Eqs. ... for the system of differential equations studied in the paper. Stability conjecture. The stalemate equilibrium is...

conjecture | open | MathDB 页面 | 原文线索

V2-038411 | MATHDB-321850

Null condition conjecture for strongly tame spherically symmetric Einstein-matter systems

摘录 / EXCERPT: Null condition conjecture. If a spherically symmetric Einstein-matter system satisfies a suitable “null condition”, then the system is strongly tame.

conjecture | open | MathDB 页面 | 原文线索

V2-038412 | MATHDB-321997

Global regularity for homogeneous Schwartz data

摘录 / EXCERPT: Let $(u_0,0,T)$ be a homogeneous Schwartz set of data. Global regularity for homogeneous Schwartz data. There exists a smooth finite energy solution $(u,p,u_0,0,T)$ with the indica...

conjecture | open | MathDB 页面 | 原文线索

V2-038413 | MATHDB-321998

Global regularity for periodic data

摘录 / EXCERPT: Let (u_0,f,T) be a smooth periodic set of data. Global regularity for periodic data. There exists a smooth periodic solution (u,p,u_0,f,T) with the indicated data. This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-038414 | MATHDB-321999

Global regularity for periodic data with normalised pressure

摘录 / EXCERPT: Let (u_0,f,T) be a smooth periodic set of data. A periodic solution has normalised pressure when its pressure satisfies the normalisation used in the periodic Navier-Stokes formu...

conjecture | open | MathDB 页面 | 原文线索

V2-038415 | MATHDB-322000

A priori periodic H^1 bound

摘录 / EXCERPT: Let (u,p,u_0,f,T,L) be a smooth periodic, normalised-pressure solution, with periodic H^1 data norm ${\mathcal H}^1(u_0,f,T,L)$ finite. A priori periodic H^1 bound. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-038416 | MATHDB-322001

Global almost regularity for homogeneous H^1 data

摘录 / EXCERPT: Let $(u_0,0,T)$ be a smooth homogeneous H^1 set of data. An almost smooth finite energy solution is smooth for positive times and has the stated continuous spatial and time-deriv...

conjecture | open | MathDB 页面 | 原文线索

V2-038417 | MATHDB-322002

Global well-posedness in homogeneous H^1

摘录 / EXCERPT: Let (u_0,f,T) be H^1 data and let an H^1 mild solution be a solution with $u\in L^\infty_tH^1_x\cap L^2_tH^2_x$ and the corresponding pressure and Duhamel formulation. Global...

conjecture | open | MathDB 页面 | 原文线索

V2-038418 | MATHDB-322003

A priori global homogeneous H^1 bound

摘录 / EXCERPT: Let $(u,p,u_0,0,T)$ be a smooth homogeneous H^1 solution, with $\|u_0\|_{H^1_x(\mathbf{R}^3)}\leq A<\infty$. A priori global homogeneous H^1 bound. There exists a function...

conjecture | open | MathDB 页面 | 原文线索

V2-038419 | MATHDB-322004

Global well-posedness from spatially smooth H^1 data

摘录 / EXCERPT: Let (u_0,f,T) be an H^1 set of data such that, for every multi-index α and every compact $K\subset\mathbf{R}^3$, ... and ... Global well-posedness from spatially smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-038420 | MATHDB-322005

A priori homogeneous periodic H^1 bound

摘录 / EXCERPT: Let $(u,p,u_0,0,T,L)$ be a smooth periodic, homogeneous normalised-pressure solution, with $0<T<T_0<\infty$ and periodic data norm ${\mathcal H}^1(u_0,0,T,L)\leq A<\infty$. A prior...

conjecture | open | MathDB 页面 | 原文线索

V2-038421 | MATHDB-322006

Global regularity for homogeneous periodic data with normalised pressure

摘录 / EXCERPT: Let $(u_0,0,T)$ be a smooth periodic set of data. Global regularity for homogeneous periodic data with normalised pressure. There exists a smooth periodic solution $(u,p,u_0,0,T)$...

conjecture | open | MathDB 页面 | 原文线索

V2-038422 | MATHDB-322015

The higher-dimensional Sobolev space-time estimate for the Schrödinger operator

摘录 / EXCERPT: Higher-dimensional space-time estimate. Then

conjecture | open | MathDB 页面 | 原文线索

V2-038423 | MATHDB-322029

Roitman's folding polynomial conjecture

摘录 / EXCERPT: Let $S=\{s_1,\dots,s_\nu\}\subset\mathbb{C}$. A polynomial q is a folding polynomial for S if $q(S)\subset\mathbb{R}$ and q has only real critical values. Roitman's folding p...

conjecture | open | MathDB 页面 | 原文线索

V2-038424 | MATHDB-322438

The general-order linear ODE symmetry-generator conjecture for the group G_c

摘录 / EXCERPT: General-order linear ODE symmetry-generator conjecture. The vector field X^0 is the infinitesimal generator of G_c.

conjecture | open | MathDB 页面 | 原文线索

V2-038425 | MATHDB-322477

Uniqueness conjecture for radial decreasing quasi-linear Schrödinger solutions

摘录 / EXCERPT: Uniqueness conjecture. Then $lambda_1=lambda_2$ and $r_2=r_1(\cdot+\xi)$ for some $xi\in\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-038426 | MATHDB-322821

Small-energy existence and convergence conjecture for the four-dimensional L^2 flow

摘录 / EXCERPT: Let $C_S(g)$ denote the Sobolev constant of a compact four-dimensional Riemannian manifold (M^4,g), and let $\mathcal F(g)$ be the L^2 curvature functional. Small-energy existe...

conjecture | open | MathDB 页面 | 原文线索

V2-038427 | MATHDB-322965

Boundary-hitting conjecture for Wishart stochastic differential equations

摘录 / EXCERPT: Let β and Q be arbitrary $d\times d$ matrices, let $p<\frac{d+1}{2}$, and let X_t be a solution of the Wishart stochastic differential equation with initial condition...

conjecture | open | MathDB 页面 | 原文线索

V2-038428 | MATHDB-323153

Conjecture on the monodromy matrix around the critical point

摘录 / EXCERPT: Monodromy-matrix conjecture. The monodromy matrix around $z=z_c$ is

conjecture | open | MathDB 页面 | 原文线索

V2-038429 | MATHDB-323438

The generalized native-space inclusion conjecture

摘录 / EXCERPT: Generalized native-space inclusion conjecture. The native space $\mathcal{N}_G^{\mathscr{P}}(\mathbb{R}^2)$ associated with G and $\mathscr{P}$ is a subspace of...

conjecture | open | MathDB 页面 | 原文线索

V2-038430 | MATHDB-323490

The CR sphere's k-Yamabe classification conjecture

摘录 / EXCERPT: CR sphere k-Yamabe conjecture. If the pseudohermitian structure $\widetilde{\theta}=e^{2u}\hat{\theta}\in[\hat{\theta}]_{+}$ has constant pseudohermitian k-curvature, then...

conjecture | open | MathDB 页面 | 原文线索

V2-038431 | MATHDB-323491

The CR k-Yamabe existence conjecture below the spherical constant

摘录 / EXCERPT: CR k-Yamabe existence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-038432 | MATHDB-323509

Bizoń et al.'s large-data blow-up conjecture for equivariant wave maps

摘录 / EXCERPT: Bizoń et al.'s large-data blow-up conjecture. For initial data ... blow up in finite time in this sense.

conjecture | open | MathDB 页面 | 原文线索

V2-038433 | MATHDB-323594

Criterion for existence of non-monotone wavefronts in the delayed KPP-Fisher equation

摘录 / EXCERPT: Let $\mathfrak{D}_{n}$ denote the set of parameter pairs $(\tau,c)\in\mathbb{R}_+^2$ for which equation (17) admits a non-monotone wavefront, and let $c^*(\tau)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-038434 | MATHDB-323623

Local continuation conjecture for warped compactification solutions

摘录 / EXCERPT: Fix $(M,g_0,T_{\text{st}}^{g_0})$ and let $(\varphi,\mathcal{F})$ depend smoothly on a real parameter t. Assume that at $t=0$ the data satisfy one of the situations governed by t...

conjecture | open | MathDB 页面 | 原文线索

V2-038435 | MATHDB-323807

Intrinsic norm conjecture for the maximal Dirac trace space

摘录 / EXCERPT: Intrinsic norm conjecture. The $\check{R}$-norm on $R(\operatorname{dom} D_{\rm max})$ is equivalent to the $\check{H}$-norm. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-038436 | MATHDB-323894

Feinberg's persistence conjecture for weakly reversible reaction systems

摘录 / EXCERPT: Let N be a confined, weakly reversible mass-action ordinary differential equation system, and let $\mathfrak M^\ell(N)$ denote its associated differential inclusion. A differenti...

conjecture | open | MathDB 页面 | 原文线索

V2-038437 | MATHDB-323988

The one-dimensional Sobolev gradient bound for infimal convolution

摘录 / EXCERPT: Let L satisfy the assumptions given in the introduction, let u be the Sobolev function under consideration on $\mathbb{R}$, and let u_t denote its infimal convolution with th...

conjecture | open | MathDB 页面 | 原文线索

V2-038438 | MATHDB-324124

Equivalence of the induced norm with the Sobolev norm

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be a Lipschitz domain, let $\emptyset\neq\Gamma\subset\partial\Omega$ be relatively open, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038439 | MATHDB-324162

Inter-critical NLS critical Sobolev boundedness conjecture

摘录 / EXCERPT: Let $d\geq 1$ and $p>0$ satisfy ... Let $u:I\times\mathbb{R}^d\to\mathbb{C}$ be a maximal-lifespan solution to the nonlinear Schrödinger equation, with...

conjecture | open | MathDB 页面 | 原文线索

V2-038440 | MATHDB-324181

Sibuya-type conjecture for the third-order conformal-limit equation

摘录 / EXCERPT: Let $M>0$, and consider the third-order equation ... Here $p(x,E)$ is the potential in the conformal-limit equation, and x is understood on a suitable cover of the punctured comple...

conjecture | open | MathDB 页面 | 原文线索

V2-038441 | MATHDB-324183

Differentiability conjecture for mixed transport-diffusion quantum graph semigroups

摘录 / EXCERPT: Differentiability conjecture. If $\athsf{E}_d\not=\emptyset$, then the semigroup is differentiable for every $t>T$; equivalently, for every $x\in X$, the map

conjecture | open | MathDB 页面 | 原文线索

V2-038442 | MATHDB-324190

Uniqueness for solutions with different vanishing orders

摘录 / EXCERPT: The uniqueness conjecture. Such a solution $u(t)$ cannot possess another solution $v(t)$ satisfying these initial conditions.

conjecture | open | MathDB 页面 | 原文线索

V2-038443 | MATHDB-324629

Conjecture on the omega-limit set for an extended dissipative system

摘录 / EXCERPT: Let $u: \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ solve the parabolic equation with initial data u_0 satisfying ... where $(b_n)_{n\in\mathbb{N}}$ is strictly increasing,...

conjecture | open | MathDB 页面 | 原文线索

V2-038444 | MATHDB-324906

Universality of singularity reduction for isomonodromy systems

摘录 / EXCERPT: An isomonodromy system is a differential system whose monodromy data remain constant under deformation, and a singularity reduction is a gauge transformation that regularizes the s...

conjecture | open | MathDB 页面 | 原文线索

V2-038445 | MATHDB-325151

Duke's conjecture on Betti number and minimum genus

摘录 / EXCERPT: Let G be a graph, with Betti number $\beta(G)$ and minimum genus $\gamma(G)$. Duke's conjecture. ... for all graphs G. This strengthens the general Euler-equation bound and would...

conjecture | open | MathDB 页面 | 原文线索

V2-038446 | MATHDB-325200

Log-Lipschitz Mané projector conjecture for dissipative PDE attractors

摘录 / EXCERPT: Let H be the phase space, let H^2 denote the more regular space, let F be the non-linearity satisfying the reasonable assumptions considered in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038447 | MATHDB-325261

Zagier-type conjecture for integral differential-equation solutions

摘录 / EXCERPT: Consider the differential equation ... An integral solution means a solution of this equation whose coefficients are integral. Zagier-type conjecture. Any integral solution to this d...

conjecture | open | MathDB 页面 | 原文线索

V2-038448 | MATHDB-325285

Local analytic Cauchy problem for the parabolic monogenic equation

摘录 / EXCERPT: Local extension conjecture. There is a unique parabolic monogenic spinor Ψ converging on a neighborhood of $\mathcal V$ in $\mathcal U$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038449 | MATHDB-325600

The forbidden physical mass interval conjecture for small-data global solvability

摘录 / EXCERPT: For the scalar semilinear equation, let n denote the spatial dimension, m the physical mass, and let the nonlinearity exponent satisfy $\alpha\in(0,\infty)$. Forbidden physical...

conjecture | open | MathDB 页面 | 原文线索

V2-038450 | MATHDB-325663

Positive Sobolev functions on one-ended groups of polynomial growth

摘录 / EXCERPT: Let G be a one-ended group of polynomial growth, let $1\leq p<\infty$, and let f be a positive function in the Sobolev-type space $\mathscr H^p(G)$. The positive-function conje...

conjecture | open | MathDB 页面 | 原文线索

V2-038451 | MATHDB-326266

The refinement-process obstruction to property (PM)

摘录 / EXCERPT: The refinement-process obstruction to property (PM). Under the stated approximation condition, no fixed pair $(\boldsymbol{a},\boldsymbol{b})$ can make property (PM) hold through...

conjecture | open | MathDB 页面 | 原文线索

V2-038452 | MATHDB-326597

Relations between geometric invariants and ell-numbers of Calabi-Yau differential equations

摘录 / EXCERPT: Let y_0 be the holomorphic solution of a fourth-order Calabi-Yau differential equation with geometric invariants (H^3,c_2H,c_3). Write ... and, for coefficients C_n, define ... S...

conjecture | open | MathDB 页面 | 原文线索

V2-038453 | MATHDB-326964

Dwork's log-growth Newton polygon conjecture

摘录 / EXCERPT: Dwork's conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-038454 | MATHDB-326965

Dwork's specialization conjecture for log-growth Newton polygons

摘录 / EXCERPT: Dwork's specialization conjecture. The polygon $\mathrm{NP}_{\log,0}(D)$ is above $\mathrm{NP}_{\log,t}(D_t)$.

conjecture | solved | MathDB 页面 | 原文线索

V2-038455 | MATHDB-327037

Peakedness conjecture for extremal Sobolev functions

摘录 / EXCERPT: Peakedness conjecture. Within the allowable range of exponents,

conjecture | open | MathDB 页面 | 原文线索

V2-038456 | MATHDB-327195

Navier-Stokes global regularity conjecture

摘录 / EXCERPT: Navier-Stokes global regularity conjecture. For every $\nu>0$ and every such initial data u_0, such a global smooth finite-energy solution exists.

conjecture | open | MathDB 页面 | 原文线索

V2-038457 | MATHDB-327272

Isett's generic energy-profile regularity conjecture for Hölder Euler flows

摘录 / EXCERPT: Generic energy-profile regularity conjecture. A generic solution with regularity at most $1/3$ fails to conserve energy and has an energy profile of minimal regularity. More precis...

conjecture | open | MathDB 页面 | 原文线索

V2-038458 | MATHDB-327289

The zero-location and asymptotic conjecture for the pantograph equation

摘录 / EXCERPT: Zero-location and asymptotic conjecture. The set $\mathbb{A}$ is countably infinite and contained in $\mathbb{R}$. Its elements can be enumerated as ${x_n}_{n\geq1}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-038459 | MATHDB-327342

Conjecture on improved flux approximation by a point source

摘录 / EXCERPT: Let $c^*(t)$ denote the quantity measuring the flux difference between the full problem and the reduced point-source problem, and let c^* be its upper bound. The estimate used to...

conjecture | open | MathDB 页面 | 原文线索

V2-038460 | MATHDB-327556

Numerical blow-up profile conjecture for the rescaling method

摘录 / EXCERPT: Consider the semilinear parabolic problem denoted by equation ... z=frac{x}{sqrt{(T-t)|log(T-t)|}}. ... bar{fbeta}(z)=left(p-1+b(beta)|z|^2right)^{-frac{1}{p-1}}, ... has a solution...

conjecture | open | MathDB 页面 | 原文线索

V2-038461 | MATHDB-327649

The product decomposition conjecture for the completion of smooth functions

摘录 / EXCERPT: Let $0<p<1$, let $\mathcal{M}=(M_k)_{k\geq 0}$ be the weight sequence defining the quasi-norm ... and let $\widehat{\mathcal{C}}$ be the completion of C^{∞} with respect to th...

conjecture | open | MathDB 页面 | 原文线索

V2-038462 | MATHDB-327650

The product decomposition conjecture for the Carleman-Sobolev completion

摘录 / EXCERPT: Let $0<p<1$, let $\mathcal{M}=(M_k)_{k\geq 0}$ be the weight sequence defining the quasi-norm ... and let $\widehat{\mathcal{S}}$ be the completion of the subclass $\mathcal{S}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-038463 | MATHDB-327817

The Calabi-type conjecture for generalized Kähler structures

摘录 / EXCERPT: Let $(M^{2n},g,J_\pm)$ be a generalized Kähler structure satisfying $[J_+,J_-]=0$. Let $\mathcal H$ be the space of formally generalized Kähler classes, and let P be the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-038464 | MATHDB-327872

Liouvillian first-integral conjecture for the quadratic differential system

摘录 / EXCERPT: Let the differential system denoted by ... admits Liouvillian first integrals. The surrounding discussion concerns polynomial and hypergeometric differential equations and the search...

conjecture | open | MathDB 页面 | 原文线索

V2-038465 | MATHDB-328085

The distinct-symbol conjecture for quadratic differential-operator embeddings

摘录 / EXCERPT: Let k,l be positive integers, let σ and τ be real parameters, and let $W_2^{\alpha,\beta}(\mathbb{R}^2)$ denote the anisotropic Sobolev space appearing in the quadrat...

conjecture | open | MathDB 页面 | 原文线索

V2-038466 | MATHDB-328905

The conjecture that slower-than-$C\nu^{1/4}$ convergence invalidates Prandtl theory

摘录 / EXCERPT: Let u solve the two-dimensional Navier-Stokes equations with viscosity $ u$, and let $\overline{u}$ denote the corresponding Euler solution. Suppose the vanishing viscosity limit...

conjecture | open | MathDB 页面 | 原文线索

V2-038467 | MATHDB-329029

Ferromagnetic comparison conjecture for the log-Sobolev constant

摘录 / EXCERPT: Ferromagnetic comparison conjecture. The measure μ_{Λ} satisfies a log-Sobolev inequality with constant ϱ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038468 | MATHDB-329632

Existence of a cohomologous k-positive form satisfying the subsolution inequality

摘录 / EXCERPT: Geometric subsolution conjecture. There exists a k-positive form $'\text{ in }[]$ satisfying the inequality from the paper with $'$ in place of ...

conjecture | open | MathDB 页面 | 原文线索

V2-038469 | MATHDB-329739

Generic absence of response solutions near resonances

摘录 / EXCERPT: Let ${\mathcal F}_{\varepsilon}$ be the family of analytic operators under consideration, let a resonance be a resonant value of ε, and equip the families with the topo...

conjecture | open | MathDB 页面 | 原文线索

V2-038470 | MATHDB-329740

Generic singularities near resonances of the linearized operator

摘录 / EXCERPT: Consider the problem referred to in the paper as equation ... is not defined in the supplied span.

conjecture | open | MathDB 页面 | 原文线索

V2-038471 | MATHDB-329741

Generic failure of analyticity on quadratic resonance neighborhoods

摘录 / EXCERPT: Let N_ε be the operator in the response-solution problem, let ε_0 be a resonance for N_ε, and let a generic family mean a family in the class co...

conjecture | open | MathDB 页面 | 原文线索

V2-038472 | MATHDB-329870

The supersymmetric symmetry criterion for integrability

摘录 / EXCERPT: Let $\mathcal{L}_1$ be a maximal finite-dimensional superalgebra of Lie point symmetries of a nonlinear system of partial differential equations, and let $\mathcal{L}_2$ be a maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-038473 | MATHDB-330568

Pearcey asymptotics conjecture for dissipative dKP near gradient catastrophe

摘录 / EXCERPT: Pearcey asymptotics conjecture. Near the first singularity, the solution has the asymptotic expansion

conjecture | open | MathDB 页面 | 原文线索

V2-038474 | MATHDB-330665

Conjecture on extinction of the disease in the stochastic SIS model

摘录 / EXCERPT: Extinction conjecture. If $R_0^S<1$ and

conjecture | open | MathDB 页面 | 原文线索

V2-038475 | MATHDB-330882

Endpoint complex interpolation conjecture for Sobolev spaces on quantum tori

摘录 / EXCERPT: Endpoint complex interpolation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-038476 | MATHDB-330943

Strong maximum and minimum principle in backwards light cones

摘录 / EXCERPT: Let U_T be the spacetime domain, let $U_{(x,t)}$ denote the backwards light cone ... and let $\widetilde{Q}$ be the spatial operator in the relativistic heat equation. Strong maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-038477 | MATHDB-330944

Tangency principle in backwards light cones for the relativistic heat equation

摘录 / EXCERPT: Let U_T be the spacetime domain, let $U_{(x,t)}$ be the backwards light cone of $(x,t)\in U_T$, and let $\widetilde{Q}$ be the spatial operator in the relativistic heat equation....

conjecture | open | MathDB 页面 | 原文线索

V2-038478 | MATHDB-331001

Fractional-power domain conjecture for the operators A_α

摘录 / EXCERPT: Let Ω be the spatial domain, let Γ be its boundary, and let A_α denote the operators associated with the strongly damped wave equation for $\alpha\in[0,1]$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-038479 | MATHDB-331071

Uniqueness conjecture for the multi-layer limiting configuration

摘录 / EXCERPT: Let $(\beta_1,\ldots,\beta_{k-1})$ be the configuration appearing in the k-layer limit problem of Theorem defining the corresponding multi-layer radial solution. Uniqueness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-038480 | MATHDB-331441

Uniqueness conjecture for Trudinger-Moser maximizers

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a ball, and consider the Trudinger-Moser variational problem ... A maximizer is a function attaining this supremum. Uniqueness conjecture. Maximize...

conjecture | open | MathDB 页面 | 原文线索

V2-038481 | MATHDB-331632

Bounded roots conjecture for eigenpolynomials of exactly solvable operators

摘录 / EXCERPT: Consider a fixed order-k linear ordinary differential operator family with monic polynomial leading coefficient $Q_k(z)$ of degree k. An operator is exactly solvable if its coe...

conjecture | open | MathDB 页面 | 原文线索

V2-038482 | MATHDB-331764

Existence of wake-asymptotic solutions for two-dimensional Navier-Stokes flow

摘录 / EXCERPT: Consider the steady two-dimensional Navier-Stokes equations with boundary conditions $\boldsymbol{u}^{*}$ and source term $\boldsymbol{f}$, and let $\boldsymbol{F}$ denote the net...

conjecture | open | MathDB 页面 | 原文线索

V2-038483 | MATHDB-331813

The free Log-Sobolev inequality at coefficient one-half for the zero potential

摘录 / EXCERPT: Zero-potential Log-Sobolev conjecture. For $Q\equiv0$, $LSI(1/2)$ is true. The paper proves that $LSI((1+\delta)/4)$ holds for some $\delta>0$, and conjectures the stronger coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-038484 | MATHDB-332001

The redundancy of the curve-intersection assumption in the intrinsic Lipschitz lemma

摘录 / EXCERPT: Let $f:{\mathbb{R}}^d\longrightarrow {\mathbb{R}}^m$ be continuous and piecewise Lipschitz with exceptional set Θ. Suppose that f is intrinsically Lipschitz with respect t...

conjecture | open | MathDB 页面 | 原文线索

V2-038485 | MATHDB-332277

The Sobolev sampling-recovery conjecture in the diagonal case

摘录 / EXCERPT: Let $\varrho_m(\mathbf W^r_p,L_p)$ denote the optimal sampling-recovery error from the Sobolev class $\mathbf W^r_p$ in the L_p norm, using m samples. Diagonal Sobolev sampling...

conjecture | open | MathDB 页面 | 原文线索

V2-038486 | MATHDB-332284

Logarithmic Sobolev inequality for the noncommutative two-torus

摘录 / EXCERPT: Let A_{θ}^{∞} be the smooth noncommutative two-torus with canonical trace τ, generated by unitaries U,V, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038487 | MATHDB-332464

A posteriori KAM existence conjecture for whiskered tori in the Boussinesq equation

摘录 / EXCERPT: A posteriori KAM existence conjecture. There exist three explicit functions $a,b_d,b_a:\mathbb R^+\to\mathbb R^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038488 | MATHDB-332465

KAM existence conjecture for whiskered tori in the Boussinesq system

摘录 / EXCERPT: KAM existence conjecture. There exist three explicit functions $a,b_d,b_a:\mathbb R^+\to\mathbb R^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038489 | MATHDB-332467

Forbidden mass interval conjecture for semilinear wave equations in de Sitter space

摘录 / EXCERPT: Forbidden mass interval conjecture. The interval

conjecture | open | MathDB 页面 | 原文线索

V2-038490 | MATHDB-332562

Equality of the bifurcation parameters in the one-dimensional Liouville type equation

摘录 / EXCERPT: Let $(\lambda(\alpha),U(x;\alpha))$ and the constants α_1, α_2, and α_3 be as in Theorem, where $\alpha_*<\alpha_1\leq\alpha_2\leq\alpha_3$ describe the degene...

conjecture | open | MathDB 页面 | 原文线索

V2-038491 | MATHDB-332739

Blow-up and critical-profile conjecture for radial bifurcation branches

摘录 / EXCERPT: Let $N\geqslant 3$, let $i\geqslant 2$, and assume $2+oubles$?

conjecture | open | MathDB 页面 | 原文线索

V2-038492 | MATHDB-332752

Sobolev smoothness and invertibility conjecture for the operator Ψ^L

摘录 / EXCERPT: Smoothness and invertibility conjecture. For each $k>\frac{d}{2}+1$, the operator Ψ^L depends smoothly on the immersion $f\in\operatorname{Imm}^k(M,N)$ and is invertible as a...

conjecture | open | MathDB 页面 | 原文线索

V2-038493 | MATHDB-332798

Sharpness of the Yudovich regularity-loss estimate

摘录 / EXCERPT: Let $\omega_0\in H^1(\mathbb{T}^2)\cap L^\infty(\mathbb{T}^2)$ be mean-zero, and let the corresponding solution of the two-dimensional incompressible Euler equations be a Yudovich...

conjecture | open | MathDB 页面 | 原文线索

V2-038494 | MATHDB-332799

Audounet-Matignon-Montseny conjecture on scalar nonlinear fractional stability

摘录 / EXCERPT: Audounet-Matignon-Montseny conjecture. The local stability of $x^{*}=0$ is governed by the global stability of the linearized system as follows: (i) $x^{*}=0$ is locally asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-038495 | MATHDB-332976

Classification conjecture for positive viscosity solutions of the conformally invariant equation

摘录 / EXCERPT: Let (f,Γ) satisfy conditions (01)--(04), and let $0<v\in C^0_{\mathrm{loc}}(\mathbb{R}^n)$ be a viscosity solution of the equation (B0). Write the family referred to as (vfo...

conjecture | open | MathDB 页面 | 原文线索

V2-038496 | MATHDB-333010

Peres-Tetali curvature-to-MLSI conjecture

摘录 / EXCERPT: Peres-Tetali conjecture. The modified log-Sobolev constant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038497 | MATHDB-333058

Uniform particle-distance bound for the Newtonian shock-wave model

摘录 / EXCERPT: Uniform particle-distance conjecture. There exist $T>0$ and $a<b$, depending only on W, ϕ_0^x, and ϕ_0^τ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-038498 | MATHDB-333106

Treves' conjecture on analytic hypoellipticity of sums of squares

摘录 / EXCERPT: Treves' conjecture. The operator P is analytic hypoelliptic if and only if every stratum in the above-described Poisson-Treves stratification is symplectic.

conjecture | open | MathDB 页面 | 原文线索

V2-038499 | MATHDB-333451

Fractional Triebel-Lizorkin regularity conjecture for principal Beltrami solutions

摘录 / EXCERPT: Let $0<s<2$ and $1<p<\infty$ satisfy $s<\frac{2}{p}$. Let $\mu,\nu\in F^s_{p,q}\cap L^\infty_c$ satisfy the ellipticity condition, and let f be the principal solution to the Belt...

conjecture | open | MathDB 页面 | 原文线索

V2-038500 | MATHDB-333540

Uniqueness conjecture for Rumanov's beta-six auxiliary system

摘录 / EXCERPT: Rumanov auxiliary-system uniqueness conjecture. The system has a unique smooth solution (q_2,α) satisfying these conditions as $t\to+\infty$. Such uniqueness would make the...

conjecture | open | MathDB 页面 | 原文线索

V2-038501 | MATHDB-333572

The constant-delay expansion conjecture for steady-state bifurcations

摘录 / EXCERPT: Let $u(t)=0$ be the steady state of a state-dependent delay differential equation, and suppose its expansion has terms through a given order involving only constant delays, with al...

conjecture | open | MathDB 页面 | 原文线索

V2-038502 | MATHDB-333881

Dynamic representation conjecture for OCEs in jump-diffusion models

摘录 / EXCERPT: Consider a Markovian claim $X=X^y$ driven by a jump diffusion Y, with drift b, volatility $sigma$, jump coefficients $gamma^{(i)}$, compensator measures nu^i, running loss...

conjecture | open | MathDB 页面 | 原文线索

V2-038503 | MATHDB-334373

Long-time convergence of the Physarum continuum dynamics to Monge-Kantorovich transport

摘录 / EXCERPT: Long-time Monge-Kantorovich convergence conjecture. The solution tends as $t\to\infty$ to a solution $(\mu^*,u^*)\in L^{\infty}(\Omega)\times Lip_1(\Omega)$, where $Lip_1(\Omega)$...

conjecture | open | MathDB 页面 | 原文线索

V2-038504 | MATHDB-334383

Convergence of the asynchronous PDE solver's higher-order remainder

摘录 / EXCERPT: Convergence conjecture. The higher-order remainder satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038505 | MATHDB-334539

Surjectivity conjecture for the Jacobian on bounded Lipschitz domains

摘录 / EXCERPT: Bounded-domain Jacobian conjecture. The Jacobian maps

conjecture | open | MathDB 页面 | 原文线索

V2-038506 | MATHDB-334624

The D-progressive trajectory conjecture for scalar electrons

摘录 / EXCERPT: Let $\hat{x}(\circ,\bullet)$ be a D-progressive four-dimensional stochastic process as defined by the forward and backward stochastic differential equations, with diffusion paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-038507 | MATHDB-334763

Gaussian concentration implies modified logarithmic Sobolev inequalities under Ricci curvature bounds

摘录 / EXCERPT: Let $(\mathcal{X},Q,\pi)$ be a Markov chain with invariant measure π, and suppose that a concentration property with respect to the distance $d_{\mathcal{W}}$ holds with profil...

conjecture | open | MathDB 页面 | 原文线索

V2-038508 | MATHDB-334809

The differential equation conjecture for weighted minimal d-Hurwitz generating functions

摘录 / EXCERPT: Let d be a positive integer. For a partition α of n, let $h^d(\alpha)$ denote the minimal d-Hurwitz number, let $\mu^d(\alpha)$ denote its minimal factorization length...

conjecture | open | MathDB 页面 | 原文线索

V2-038509 | MATHDB-335081

Linearization sufficiency conjecture for oscillation-free nonlinear PDE methods

摘录 / EXCERPT: Let a nonlinear partial differential equation be discretized by a nonlinear numerical method, and let a consistent linearized form of that method be given. Linearization sufficienc...

conjecture | open | MathDB 页面 | 原文线索

V2-038510 | MATHDB-335359

Uniqueness and asymptotics of the Toda differential-equation solutions

摘录 / EXCERPT: Let $M>1/2$, let k be any value, and let $\psi^{(1)}_{[k]}(z,\overline{z};\theta)$ denote the Symanzik-rescaled solution of the differential equations for $\psi^{(1)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-038511 | MATHDB-335632

Strict monotonicity conjecture for the mass of the TFDW Euler-Lagrange solution

摘录 / EXCERPT: Let $c_{TF}>0$, and for $0<\mu<\frac{15}{64c_{TF}}$ let Q_μ be the unique positive solution of the associated Euler-Lagrange equation, with mass $M(\mu)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038512 | MATHDB-335946

The no-isola problem for slowly oscillating periodic solutions of Wright's equation

摘录 / EXCERPT: Let $\alpha>\tfrac{\pi}{2}$, and consider the family of slowly oscillating periodic solutions (SOPS) of Wright's equation ... An isola is a connected component of the SOPS family tha...

open_problem | open | MathDB 页面 | 原文线索

V2-038513 | MATHDB-336722

High-order accuracy of the DPM with implicit geometry

摘录 / EXCERPT: High-order accuracy conjecture. The error in the computed solution has second- and fourth-order accuracy in the maximum norm for the single- and composite-domain parabolic problems...

conjecture | open | MathDB 页面 | 原文线索

V2-038514 | MATHDB-336781

Continuity and piecewise real-analyticity of the infimum function

摘录 / EXCERPT: Continuity and real-analyticity conjecture. The function $\mathfrak{I}_{\mathcal{G}}(\sigma)$ is continuous in σ and is real-analytic except at integers and half-integers.

conjecture | open | MathDB 页面 | 原文线索

V2-038515 | MATHDB-337020

Necessary and sufficient Kolchin-polynomial criterion for partial differential Chow forms

摘录 / EXCERPT: Necessary and sufficient Kolchin-polynomial criterion. The Δ-Chow form of V exists if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-038516 | MATHDB-337084

Metric completion conjecture for the outer metric on embeddings

摘录 / EXCERPT: Let M be a smooth, compact manifold without boundary with $\dim M=m$, and let $\operatorname{Emb}(M,\mathbb R^d)$ be the space of smooth embeddings equipped with the outer metric...

conjecture | open | MathDB 页面 | 原文线索

V2-038517 | MATHDB-337085

Trace and extension conjecture for Sobolev embeddings

摘录 / EXCERPT: Let M be a smooth, compact manifold without boundary with $\dim M=m$, and let $\mathcal E^{s'}(M,\mathbb R^d)$ be the space of $H^{s'}$-embeddings. For $s>d/2$, set ... The trace o...

conjecture | open | MathDB 页面 | 原文线索

V2-038518 | MATHDB-337330

The gradient estimate for the critical vector-field exponent

摘录 / EXCERPT: Let $(\sigma_k)_{k\in\mathbb{Z}^2_0}$ be the family of vector fields on $\mathbb{T}^2$ defined, for $\gamma\geq 2$, by ... Assume that the associated gradient estimate is the estimat...

open_problem | open | MathDB 页面 | 原文线索

V2-038519 | MATHDB-337413

The critical-time conjecture for stationary interfaces of the parabolic p-Laplace equation

摘录 / EXCERPT: Let u solve the one-dimensional p-parabolic boundary-value problem ... on $(-1,1)\times(0,T)$ with boundary values $u(-1,t)=0$, $u(1,t)=1$, and initial value...

conjecture | open | MathDB 页面 | 原文线索

V2-038520 | MATHDB-337480

Large-shear conjecture for positivity of the top Lyapunov exponent

摘录 / EXCERPT: Consider the random dynamical system induced by the stochastic differential equation in the paper, and fix $a>0$ and $\beta\in\mathbb{R}$. Let $\lambda_{\rm top}$ denote its top Ly...

conjecture | open | MathDB 页面 | 原文线索

V2-038521 | MATHDB-337535

The K41-Onsager conjecture for vanishing-viscosity Navier-Stokes flows

摘录 / EXCERPT: K41-Onsager conjecture for Navier-Stokes. There exists a finite open interval I and a sequence of suitable weak solutions v_{ν_j} with $\nu_j\to0$ that is uniformly bounded i...

conjecture | open | MathDB 页面 | 原文线索

V2-038522 | MATHDB-337849

Liouville-type conjecture for the critical p-Laplacian on a half-space

摘录 / EXCERPT: Liouville-type conjecture. Every weak solution $u\in\mathcal D^{1,p}_0({\mathbb R}_+^N)$ of this equation is identically zero:

conjecture | open | MathDB 页面 | 原文线索

V2-038523 | MATHDB-337938

The concavifying-function conjecture for Lovelock symmetric functions

摘录 / EXCERPT: Let $\overrightarrow{a}=(a_0,a_1,\ldots,a_p)\in(\mathbb{R}_+)^{p+1}$ and define ... Here Γ_n denotes the Gårding cone. Concavifying-function conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-038524 | MATHDB-337957

Non-density of Lipschitz mappings in jet space Carnot-group Sobolev spaces

摘录 / EXCERPT: Let $n,k\geq 1$ and let $J^k(\mathbb R^n)$ be the k-th jet space Carnot group. For $n\leq p<n+1$, consider the Sobolev space $W^{1,p}(\mathbb B^{n+1},J^k(\mathbb R^n))$ and the c...

conjecture | open | MathDB 页面 | 原文线索

V2-038525 | MATHDB-338323

Generalized Wright conjecture on the global attractor

摘录 / EXCERPT: Let $\alpha>0$ and let $U(\alpha)$ be the forward extension by the semiflow of a local unstable manifold at zero for Wright's equation. Write ... Let $\overline{U(\alpha)}$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-038526 | MATHDB-338343

Frames for Sobolev spaces of differential forms

摘录 / EXCERPT: Frames conjecture. The sets $B^J_{k,p}$ and $\widetilde B_{k,p}$ are frames for H^1_k, for every $k\in\{1,\ldots,m\}$ and $p\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-038527 | MATHDB-338416

Center-manifold conjecture for the modified curve-shortening flow

摘录 / EXCERPT: Center-manifold conjecture. The PDE has an n-dimensional center manifold on which its flow is approximated by this ODE system, in the sense that the error terms of solutions are...

conjecture | open | MathDB 页面 | 原文线索

V2-038528 | MATHDB-338466

The uniqueness and minimal-solution representation conjecture

摘录 / EXCERPT: Uniqueness and representation conjecture. The field ψ is uniquely determined in U by f, its boundary values, and those two equations. Moreover, ψ can be expressed in...

conjecture | open | MathDB 页面 | 原文线索

V2-038529 | MATHDB-338467

The exterior-determination conjecture for nonunique solutions

摘录 / EXCERPT: Exterior-determination conjecture. The field ψ is then determined precisely inside U.

conjecture | open | MathDB 页面 | 原文线索

V2-038530 | MATHDB-338602

Conjecture on solutions of the Hirzebruch functional equation and the associated differential equation

摘录 / EXCERPT: Solution conjecture. Every solution $f(z)$ of the Hirzebruch functional equation with $n>2$ and these initial conditions is a solution of the associated differential equation.

conjecture | open | MathDB 页面 | 原文线索

V2-038531 | MATHDB-338610

Character identification for the admissible affine VOA $V_{-3/2}(G_2)$

摘录 / EXCERPT: Let $V=V_{-3/2}(G_2)$ be the affine vertex operator algebra at admissible level $-3/2$, and let f_0 and $7f_{4/5}$ be the solutions of the fourth-order modular linear differentia...

conjecture | open | MathDB 页面 | 原文线索

V2-038532 | MATHDB-338675

Existential closedness for reducts of differentially closed fields with the j-function

摘录 / EXCERPT: Let $\mathfrak{L}_{j}$ denote the language of the reduct associated with the differential equation of the modular j-function, and let $\emph{EC}$ denote the existential-closednes...

conjecture | open | MathDB 页面 | 原文线索

V2-038533 | MATHDB-338855

De Giorgi's conjecture for minimizers of the hyperbolic variational functional

摘录 / EXCERPT: De Giorgi's conjecture. There exists a limit

conjecture | open | MathDB 页面 | 原文线索

V2-038534 | MATHDB-338859

Regularity conjecture for minimizers of the electrostatic Born-Infeld energy

摘录 / EXCERPT: Let N be the ambient dimension, let $q>N$, let $\rho\in L^q_{loc}(\mathbb{R}^N)\cap\mathcal{X}^*$, and let u_ρ be the minimizer associated with ρ. Regularity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-038535 | MATHDB-338941

The weak null condition conjecture for quasilinear wave equations

摘录 / EXCERPT: Let u solve a quasilinear wave system of the form ... with the weak null condition (WNC) defined by the existence of a global asymptotic-PDE solution for the ansatz ... whose derivat...

conjecture | open | MathDB 页面 | 原文线索

V2-038536 | MATHDB-339055

Non-removability conjecture for Sierpiński carpets

摘录 / EXCERPT: Sierpiński carpet non-removability conjecture. Every Sierpiński carpet is non-removable for quasiconformal maps and for $W^{1,p}$ functions, for

conjecture | open | MathDB 页面 | 原文线索

V2-038537 | MATHDB-339221

Equality of critical thresholds for diffeomorphism groups

摘录 / EXCERPT: For a group of diffeomorphisms, define the distance threshold by ... the smoothness threshold by ... and the Fredholm threshold by ... Here d_s is the H^s geodesic distance,...

conjecture | open | MathDB 页面 | 原文线索

V2-038538 | MATHDB-339222

Complete distance characterization for volume-preserving diffeomorphisms

摘录 / EXCERPT: Let M be a manifold of dimension at least 3, and let $\operatorname{Diff}_\mu(M)$ denote its group of volume-preserving diffeomorphisms equipped with the right-invariant H^s...

conjecture | open | MathDB 页面 | 原文线索

V2-038539 | MATHDB-339280

Regularity via minors for odd k

摘录 / EXCERPT: Regularity via minors (odd k). If $\bigwedge^k df\in\widetilde H_{>k}$ is smooth, then f is smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-038540 | MATHDB-339306

Bergeron-Favreau-Krob conjecture on differential equations for bounded-height tableau generating functions

摘录 / EXCERPT: Bergeron-Favreau-Krob conjecture. For each k, there are polynomials $p_m(t)$ of degree at most $\lfloor\frac{k}{2}\rfloor$ such that $Y_k(t)$ is a solution of a linear differenti...

conjecture | open | MathDB 页面 | 原文线索

V2-038541 | MATHDB-339307

Armstrong-Silvestre-Smart's optimal exponent conjecture for viscosity supersolutions

摘录 / EXCERPT: Armstrong-Silvestre-Smart's conjecture. The optimal exponent in this estimate is

conjecture | open | MathDB 页面 | 原文线索

V2-038542 | MATHDB-339484

The order-1/d exponential ergodicity conjecture for generalized reflected Brownian motion

摘录 / EXCERPT: Let $\operatorname{GRBM}(R,\mu,\Gamma,U)$ be the generalized reflected Brownian motion defined by the strong solution of the stochastic differential equation in the paper, with tra...

conjecture | open | MathDB 页面 | 原文线索

V2-038543 | MATHDB-339883

Conjecture on singular points in trajectories of planar fractional-order systems

摘录 / EXCERPT: Singular-point conjecture. There exist singular points in the trajectory if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-038544 | MATHDB-340190

Existence of delay-induced switched states near the balance point

摘录 / EXCERPT: Let $f\in C^{1}(\mathbb{R})$ satisfy conditions (H1)-(H3), and let β be the balance point. Consider initial histories satisfying $x(\theta)\in[\chi^{-},\chi^{+}]$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-038545 | MATHDB-340250

The weak null conjecture

摘录 / EXCERPT: Consider a system of wave equations and its asymptotic system, obtained by neglecting the terms expected to have better behaviour, such as terms involving good derivatives. Let r...

conjecture | open | MathDB 页面 | 原文线索

V2-038546 | MATHDB-340336

Liouville conjecture for entire 2-convex solutions of the 2-Hessian equation

摘录 / EXCERPT: Let $u\in C^4(\mathbb{R}^n)$ be an entire 2-convex function, meaning that its Hessian eigenvalues lie in the closure of the 2-Hessian ellipticity cone, and suppose that ... through...

conjecture | open | MathDB 页面 | 原文线索

V2-038547 | MATHDB-340956

The rectifiability-dimension conjecture for measures satisfying linear PDE constraints

摘录 / EXCERPT: Let ... -free measure, and let ... ... satisfies ... This would improve the known lower bound involving ... . The conjecture was also advanced by Raita and appears in related work cited by t...

conjecture | open | MathDB 页面 | 原文线索

V2-038548 | MATHDB-340984

One-dimensionality conjecture for minimizers of the one-phase free boundary energy

摘录 / EXCERPT: Assume (A1)-(A2): β is non-negative and C^1 with $\textrm{supp}\,\beta=[0,1]$ and $\int_0^1\beta=1$, while $u\geq0$ satisfies $\|\nabla u\|_{L^\infty(\mathbb{R}^n)}<\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-038549 | MATHDB-340985

Non-one-dimensional minimizer conjecture in dimension seven

摘录 / EXCERPT: Assume (A1): β is non-negative and C^1, with $\textrm{supp}\,\beta=[0,1]$ and $\int_0^1\beta=1$. Let ... where $\Phi'=\beta$. Dimension-seven non-one-dimensional minimizer co...

conjecture | open | MathDB 页面 | 原文线索

V2-038550 | MATHDB-341080

Lü-Li-Yang conjecture on shared polynomials and derivatives of powers of entire functions

摘录 / EXCERPT: Let f be a transcendental entire function and let n be a positive integer. Let k be a positive integer, and let Q_{1} and Q_{2} be polynomials with...

conjecture | open | MathDB 页面 | 原文线索

V2-038551 | MATHDB-341104

The cancellation conjecture for Riesz potentials on constrained function spaces

摘录 / EXCERPT: Let d and ℓ be positive integers, let S^{d-1} be the unit sphere in $\mathbb{R}^d$, and let $G(\ell,k)$ denote the Grassmannian of k-dimensional linear subspaces of...

conjecture | open | MathDB 页面 | 原文线索

V2-038552 | MATHDB-341105

The Besov--Lorentz strengthening of the cancellation conjecture

摘录 / EXCERPT: With $\mathfrak{W}$, Ω, d, ℓ, k, and S^{d-1} as above, for $\alpha\in(0,d)$, the Besov--Lorentz strengthening. The Riesz potential $\operatorname{I}_{\alpha}$ map...

conjecture | open | MathDB 页面 | 原文线索

V2-038553 | MATHDB-341337

The conjecture on determinants of fundamental matrices for reflected systems

摘录 / EXCERPT: Determinant conjecture. The determinant $|X(t)|$ can be obtained as a component of the solution of a linear system of differential equations with constant coefficients, whose coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-038554 | MATHDB-341345

The L_1 Sobolev cubature lower-bound conjecture

摘录 / EXCERPT: Let $\mathbf W^r_{1,\alpha}$ be the weighted Sobolev-type function class, and let $\kappa_m(\mathbf W^r_{1,\alpha})$ denote the minimal cubature error over formulas with m knots....

conjecture | open | MathDB 页面 | 原文线索

V2-038555 | MATHDB-341346

The L_∞ Sobolev cubature lower-bound conjecture

摘录 / EXCERPT: Let $\mathbf W^r_{\infty,\alpha}$ be the weighted Sobolev-type function class, and let $\kappa_m(\mathbf W^r_{\infty,\alpha})$ denote the minimal cubature error over formulas with...

conjecture | open | MathDB 页面 | 原文线索

V2-038556 | MATHDB-341450

Continuous dependence conjecture for weak solutions of Problem P

摘录 / EXCERPT: Let $T>0$) and $R>0$. Let $\mathcal{X}_{01}=(U_{01},U_{11})$ and $\mathcal{X}_{02}=(U_{02},U_{12})$ belong to $\mathcal{H}_0$, with $\mathcal{X}_{01}$ and $\mathcal{X}_{02}$ having...

conjecture | open | MathDB 页面 | 原文线索

V2-038557 | MATHDB-341512

Lions' small-time global null controllability conjecture for the Navier-Stokes equation

摘录 / EXCERPT: Let Ω be the rectangular domain described above, and let $L^2_{\operatorname{div}}(\Omega)$ denote the space of square-integrable divergence-free vector fields tangent to...

conjecture | open | MathDB 页面 | 原文线索

V2-038558 | MATHDB-341745

Caffarelli Perron-solution regularity conjecture in dimension two

摘录 / EXCERPT: Let a strict supersolution be understood in the sense of the Perron process for the one-phase free-boundary problem, and consider solutions obtained as infima of such strict supers...

conjecture | open | MathDB 页面 | 原文线索

V2-038559 | MATHDB-341746

Flat-implies-Lipschitz conjecture with a narrow cylindrical hole

摘录 / EXCERPT: Set ... Let u be a classical solution to the one-phase free-boundary problem in $\mathbb{B}_1\setminus\mathcal{C}_\delta$, with contractible positive and zero phases. Flat-implies-...

conjecture | open | MathDB 页面 | 原文线索

V2-038560 | MATHDB-341747

Flat-implies-Lipschitz conjecture with a small concentric hole

摘录 / EXCERPT: Let u be a classical solution to the one-phase free-boundary problem in the punctured ball $\mathbb{B}_1\setminus\mathbb{B}_\delta$, with contractible positive and zero phases. F...

conjecture | open | MathDB 页面 | 原文线索

V2-038561 | MATHDB-341748

One-dimensional rigidity conjecture for semilinear solutions with a topological one-dimensional model

摘录 / EXCERPT: Let $f\in C_0^\infty(\mathbb{R})$ satisfy $f\ge0$. Let u solve ... and suppose there is a homeomorphism $\Phi:\mathbb{R}^n\to\mathbb{R}^n$ and a one-dimensional solution U of the...

conjecture | open | MathDB 页面 | 原文线索

V2-038562 | MATHDB-341749

Half-space rigidity conjecture for semilinear solutions in dimension three

摘录 / EXCERPT: Let $f\in C_0^\infty(\mathbb{R})$ satisfy $f\ge0$, and let u solve ... Assume that u has finitely many critical points and that $u(x)>0$ implies $x_1>0$. Half-space rigidity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-038563 | MATHDB-341750

Quantitative intrinsic connectivity conjecture for semilinear positive phases

摘录 / EXCERPT: Let $f\in C_0^\infty(\mathbb{R})$ satisfy $f\ge0$ and $f(s)=0$ for all $s\le0$. Let u solve ... and suppose that the blow-down limit ... exists and has connected positive phase...

conjecture | open | MathDB 页面 | 原文线索

V2-038564 | MATHDB-341783

Conjecture on nonconstant proportionality factors admitting Beltrami fields

摘录 / EXCERPT: Beltrami-field proportionality conjecture. Nonconstant proportionality factors f admitting a nonzero Beltrami field have the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-038565 | MATHDB-341817

Gromov's Sobolev-type stability conjecture for almost nonnegative scalar curvature

摘录 / EXCERPT: Let X be an n-manifold with Riemannian metric g, and let R_g denote its scalar curvature. A Sobolev-type weak metric is a weak metric on the space of n-manifolds. Gromov'...

conjecture | open | MathDB 页面 | 原文线索

V2-038566 | MATHDB-341845

Global convergence conjecture for polyflow approximations of stable systems

摘录 / EXCERPT: Consider the system $\dot{x}=f(x)$, and suppose that $x=0$ is an asymptotically stable equilibrium. Let $\psi(x_0,t)$ denote its solution with initial condition x_0, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038567 | MATHDB-341947

Upper bound conjecture for zeros of Wronskians of disconjugate equations

摘录 / EXCERPT: Let ... be a linear ordinary homogeneous differential equation of order n with real-valued continuous coefficients, disconjugate on an interval I. For a positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-038568 | MATHDB-342018

Harmonicity conjecture for the limiting solution outside the support of the datum

摘录 / EXCERPT: Let $cOmega$ be a bounded domain, let $comegasubsetsubsetcOmega$ be the region where the nonnegative datum is supported, and let u be the function given by Theorem 1 with...

conjecture | open | MathDB 页面 | 原文线索

V2-038569 | MATHDB-342019

Higher-dimensional convergence conjecture for the transformed solutions

摘录 / EXCERPT: Higher-dimensional convergence conjecture. The sequence cv_n is bounded in $W_0^{1,2}(\Omega)\cap L^\infty(\Omega)$ and therefore converges, up to subsequences, to a bounded nonn...

conjecture | open | MathDB 页面 | 原文线索

V2-038570 | MATHDB-342020

Caputo, Dai Pra and Posta's dimension-free modified log-Sobolev conjecture for inhomogeneous zero-range processes

摘录 / EXCERPT: Consider an inhomogeneous zero-range process with n sites, rate functions r_i, particle number m, and site weights $p=(1/n,\ldots,1/n)$. Assume that there are constants...

conjecture | open | MathDB 页面 | 原文线索

V2-038571 | MATHDB-342088

Vacuum constraint extension conjecture for Bartnik boundary data

摘录 / EXCERPT: Vacuum constraint extension conjecture. If (M,g) is an extension of (γ,H) in $\mathcal P$, then there exists an extension in $\mathcal C_{(\gamma,H)}$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-038572 | MATHDB-342214

The sharp Sobolev injectivity conjecture for Schrödinger potentials

摘录 / EXCERPT: Let $d/2\leq p<\infty$, let Ω be a bounded domain with Lipschitz boundary, and let $H_c^{s,p}(\Omega)$ denote the compactly supported Sobolev space of potentials. For a pote...

conjecture | open | MathDB 页面 | 原文线索

V2-038573 | MATHDB-342266

Optimal-rate conjecture for general quasilinear Skorohod SDEs

摘录 / EXCERPT: Let X solve the quasilinear Skorohod SDE ... where $X_0\in L^p(\Omega)$ for some $p>2$, a satisfies the global Lipschitz and boundedness condition ... for all $t\in[0,1]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-038574 | MATHDB-342493

The lower-bound conjecture for transformed SDE approximation under (A1)-(A3)

摘录 / EXCERPT: Let $x_0\in\mathbb{R}$, let $\mu,\sigma\colon\mathbb{R}\to\mathbb{R}$ satisfy (A1)-(A3), and let X be the solution of the corresponding SDE. For $n\in\mathbb{N}$, observations co...

conjecture | open | MathDB 页面 | 原文线索

V2-038575 | MATHDB-342812

Bounded critical Sobolev norm conjecture for defocusing NLS and NLW

摘录 / EXCERPT: For a defocusing nonlinear Schrödinger equation or nonlinear wave equation, let u be a solution whose critical Sobolev norm remains bounded in time. Bounded critical Sobolev norm...

conjecture | open | MathDB 页面 | 原文线索

V2-038576 | MATHDB-342862

Conjecture on scaling limits selecting viscosity solutions of Euler equations

摘录 / EXCERPT: Let Q_{N_i,ν_i} be probability laws associated with sequences $\nu_i\to 0$ and $N_i\to\infty$, and suppose that $\{Q_{N_i,\nu_i}\}_i$ is weakly convergent to a probability meas...

conjecture | open | MathDB 页面 | 原文线索

V2-038577 | MATHDB-342951

Conjecture on singularities from three formal solutions of a third-order ODE

摘录 / EXCERPT: Let a,b,c,d be analytic functions at $x_0 otin\mathbb R$, and consider the linear homogeneous ordinary differential equation ... Suppose that this equation admits three linearly in...

conjecture | open | MathDB 页面 | 原文线索

V2-038578 | MATHDB-342997

Continuity and propagation conjecture for transition densities of Lévy-driven SDEs

摘录 / EXCERPT: Continuity and propagation conjecture. The process X has a continuous transition density function $p(t,x,y)$. Moreover, if

conjecture | open | MathDB 页面 | 原文线索

V2-038579 | MATHDB-343014

Generic density of hyperbolic periodic solutions for scalar parabolic equations

摘录 / EXCERPT: Generic density conjecture. For any $A>0$ and K, ${\cal G}_{2}(3A/2,K)$ is dense in ${\cal G}_{3/2}(3A/2,K)\cap {\cal G}_{2}(A,K)$.

conjecture | open | MathDB 页面 | 原文线索

V2-038580 | MATHDB-343015

Local hyperbolization of simple periodic solutions

摘录 / EXCERPT: Local hyperbolization conjecture. There exists a function $\tilde f\in\mathcal{N}$ such that p is a hyperbolic periodic solution of the equation with nonlinearity $\tilde f$.

conjecture | open | MathDB 页面 | 原文线索

V2-038581 | MATHDB-343175

Short-time shock continuation conjecture for asymptotically self-similar Euler shocks

摘录 / EXCERPT: For ρ the density and $u=(u_r,u_\uptheta)$ the velocity solving the two-dimensional isentropic compressible Euler equations, suppose the solution constructed in Theorem genera...

conjecture | open | MathDB 页面 | 原文线索

V2-038582 | MATHDB-343233

Order-independent termination of the reversible-realization heuristic

摘录 / EXCERPT: Let a kinetic differential equation be represented by a polynomial vector field, and consider the heuristic that repeatedly pairs monomial reactant vectors with reaction vectors to...

conjecture | open | MathDB 页面 | 原文线索

V2-038583 | MATHDB-343234

Uniqueness conjecture for weakly reversible deficiency-zero realizations

摘录 / EXCERPT: A reaction network is a directed network of complexes whose mass-action kinetics induces a kinetic differential equation; it is weakly reversible when every reaction lies in a stro...

conjecture | open | MathDB 页面 | 原文线索

V2-038584 | MATHDB-343694

Circularity conjecture for bounded Euler flows in a disk

摘录 / EXCERPT: Let D be an open non-empty disk and let $z\in D$. Let v be a $C^2(\overline{D}\!\setminus\!\{z\})$ and bounded flow solving the Euler equations and satisfying $v\cdot n=0$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-038585 | MATHDB-344309

Global-attractor conjecture for anisotropic micropolar fluids

摘录 / EXCERPT: Let λ and ν be the two eigenvalues of the planar microinertia block, with the microstructure called inertially oblong when $\lambda>\nu$ and inertially oblate when...

conjecture | open | MathDB 页面 | 原文线索

V2-038586 | MATHDB-344343

Sharp fully nonlinear Sobolev trace inequality on the ball

摘录 / EXCERPT: Sharp Sobolev trace inequality. One has

conjecture | open | MathDB 页面 | 原文线索

V2-038587 | MATHDB-344373

Nonexistence conjecture for radially symmetric backward self-similar solutions

摘录 / EXCERPT: Consider the compressible Navier-Stokes equations referred to as, and backward self-similar solutions with radial symmetry. Nonexistence conjecture. There are no backward self-simi...

conjecture | open | MathDB 页面 | 原文线索

V2-038588 | MATHDB-344392

The BD jump-set conjecture

摘录 / EXCERPT: Let $u\in\operatorname{BD}_{\mathrm{loc}}(\mathbb{R}^d)$, with S_u its approximate discontinuity set and J_u its jump set. BD jump-set conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-038589 | MATHDB-344709

The invariant little Grothendieck conjecture for Sobolev spaces

摘录 / EXCERPT: Let W^{1}_{1} denote the Sobolev space on the relevant torus, and let T be an operator between Hilbert spaces that factorizes through W^{1}_{1}. Invariant little Grothendieck...

conjecture | open | MathDB 页面 | 原文线索

V2-038590 | MATHDB-345010

Monotonicity and nonexistence conjecture for the Emden-Fowler equation

摘录 / EXCERPT: Let $N\ge 3$, let $p>p_{\rm S}$, and let $\Theta(\Gamma)$ denote the relevant solution-branch parameter for the Emden-Fowler equation ... has no regular solution for...

conjecture | open | MathDB 页面 | 原文线索

V2-038591 | MATHDB-345011

Limit behavior of distinguished angles at the Sobolev critical exponent

摘录 / EXCERPT: Let $\overline{\Theta}$ be the angle from Corollary B(iv), and let Θ^* be the angle from Theorem C, for the Emden-Fowler problem on a spherical cap. Critical-exponent limit...

conjecture | open | MathDB 页面 | 原文线索

V2-038592 | MATHDB-345271

Uniform decay for quadratic Hecke sums

摘录 / EXCERPT: Let $(\varphi_k)_k$ be the sequence of holomorphic Hecke eigenforms under consideration, with Hecke eigenvalues $\lambda_{\varphi_k}(n)$. Let $\mathcal{S}_N$ denote the Sobolev nor...

conjecture | open | MathDB 页面 | 原文线索

V2-038593 | MATHDB-345277

Infinite-order resolution ODE convergence conjecture

摘录 / EXCERPT: Infinite-order resolution ODE conjecture. Under certain regularity conditions on $g(z,s)$ and s, including, for example, that $g(z,s)$ is infinitely differentiable and that s i...

conjecture | open | MathDB 页面 | 原文线索

V2-038594 | MATHDB-345681

Fifth-order differential equation conjecture for the degree-of-symmetry generating function of Dyck paths

摘录 / EXCERPT: Differential-equation conjecture. The generating function $D(s,z)$ is D-finite in z but not algebraic. Specifically, it satisfies a fifth-order linear differential equation with...

conjecture | open | MathDB 页面 | 原文线索

V2-038595 | MATHDB-345888

Bressan's mixing conjecture for Sobolev velocity fields

摘录 / EXCERPT: Let $b\in L^{\infty}([0,\infty),W^{1,1}(\mathbb{T}^d,\mathbb{R}^d))$ be divergence free, and let ρ denote the mixing rate associated with an initial datum u_0. Bressan's mix...

conjecture | open | MathDB 页面 | 原文线索

V2-038596 | MATHDB-345889

Poon-type logarithmic lower bound for Sobolev advection diffusion

摘录 / EXCERPT: Fix $p\in[1,+\infty)$. Let $b\in C^{\infty}([0,T]\times\mathbb{T}^d)$ be divergence free and let $u_0\in W^{1,2}(\mathbb{T}^d)$. If u^ν is a solution of the advection diffusion...

conjecture | open | MathDB 页面 | 原文线索

V2-038597 | MATHDB-345902

S-shape multiplicity conjecture for the case III semilinear elliptic problem

摘录 / EXCERPT: S-shape conjecture. The interval $(\underline{\lambda},\overline{\lambda})$ should be expandable so that problem ... should be removable.

conjecture | open | MathDB 页面 | 原文线索

V2-038598 | MATHDB-345903

Bifurcation-diagram conjecture for the case VIII semilinear elliptic problem

摘录 / EXCERPT: Case VIII bifurcation conjecture. The branches of solutions to ... .

conjecture | open | MathDB 页面 | 原文线索

V2-038599 | MATHDB-346117

Exponential turnpike conjecture for semilinear control with local control

摘录 / EXCERPT: Exponential turnpike conjecture. There are constants $K>0$ and $\mu>0$, independent of T, such that

conjecture | open | MathDB 页面 | 原文线索

V2-038600 | MATHDB-346119

Homeomorphism conjecture for neohookean energy minimizers

摘录 / EXCERPT: Homeomorphism conjecture. Every minimizer $h_\circ\in\mathscr M^p(\overline{\mathbb{X}},\overline{\mathbb{Y}})$ is a homeomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-038601 | MATHDB-346152

Filmus-O'Donnell-Wu log-Sobolev conjecture for the multislice

摘录 / EXCERPT: Let $\kappa=(\kappa_1,\ldots,\kappa_L)$ be the parameter defining the multislice, let $\tau_{\textsc{ls}}(\kappa)$ denote the log-Sobolev constant of its transposition walk, and se...

conjecture | open | MathDB 页面 | 原文线索

V2-038602 | MATHDB-346541

Local PDE and PDI sufficiency conjecture for neural networks

摘录 / EXCERPT: Let $\mathcal{N}$ be a feedforward neural network with inputs labeled by the coordinate functions $x_1,\dots,x_n$, and let $F=F(x_1,\dots,x_n)$ be smooth near a point...

conjecture | open | MathDB 页面 | 原文线索

V2-038603 | MATHDB-346653

Optimal integrability conjecture for extension distortions and their inverses

摘录 / EXCERPT: Let $\mathcal G_s(K)$ be the class defined in the paper by, and let $\mathcal F_s(g)$ be the class defined by. For a map f, write K_f and $K_{f^{-1}}$ for the distortions of...

conjecture | open | MathDB 页面 | 原文线索

V2-038604 | MATHDB-346813

Landis's qualitative conjecture for Schrödinger equations

摘录 / EXCERPT: “Qualitative” Landis conjecture. Then $u\equiv 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-038605 | MATHDB-347125

Eigenvalue blow-up conjecture for finite-time Navier-Stokes singularities

摘录 / EXCERPT: Let $u\in C\left(\left[0,T_{max}\right);\dot{H}^1_{df}\right)$ be a smooth solution of the Navier-Stokes equation with finite maximal existence time $T_{max}<+\infty$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-038606 | MATHDB-347128

The regularity and minimal-growth solution conjecture for Fuchsian singularities

摘录 / EXCERPT: Regularity and minimal-growth solution conjecture. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-038607 | MATHDB-347422

Finite-time blow-up conjecture for the damped compressible Euler equations

摘录 / EXCERPT: Let w solve the equation denoted by ... lambda=1,quad 0<muleq 2,qquadtext{or}qquad lambda1,quad mu0. ... will blow up in finite time for a family of smooth initial data...

conjecture | open | MathDB 页面 | 原文线索

V2-038608 | MATHDB-347667

Conjecture on concentration for sign-changing radial solutions as beta increases to one

摘录 / EXCERPT: Let $\Lambda^*<\lambda_*<\Lambda_1$ and set $\beta_*=1$. Let $S_{1,\lambda,\beta}$ denote the class of radial sign-changing solutions with the index specified in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038609 | MATHDB-347716

Optimal logarithmic power conjecture for Lorentz-Sobolev multiplier conditions

摘录 / EXCERPT: Optimal logarithmic power conjecture. If $K'<\infty$, then there is a constant $C=C(s_1,\dots,s_n,p,n,d,\psi)$ such that every $f\in\mathscr C_0^\infty(\mathbb R^n)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038610 | MATHDB-347777

Uniform smoothness conjecture for soliton addition and removal maps

摘录 / EXCERPT: Uniform smoothness conjecture. The maps B^N_+ and B^N_- are uniformly smooth on bounded sets in H^s.

conjecture | open | MathDB 页面 | 原文线索

V2-038611 | MATHDB-347779

Global uniform smoothness conjecture for soliton dressing and undressing maps

摘录 / EXCERPT: Global uniform smoothness conjecture. Identifying $\mathbf S_U^N$ and $\mathbf M_U^N$, the soliton addition and removal maps are uniformly smooth globally,

conjecture | open | MathDB 页面 | 原文线索

V2-038612 | MATHDB-347946

Supercritical nonuniqueness and non-Leray-Hopf conjecture for the Navier-Stokes equations

摘录 / EXCERPT: Supercritical nonuniqueness and non-Leray-Hopf conjecture. Then there exist two weak solutions $u,v\in X^{p,q}([0,T];\mathbb{T}^d)$ of the Navier-Stokes equations with

conjecture | open | MathDB 页面 | 原文线索

V2-038613 | MATHDB-348018

Difference quotient characterization conjecture for the Radon-Nikodým property of Banach function spaces

摘录 / EXCERPT: Let $ X(Omega)$ be a Banach function space, and let the difference quotient criterion be the condition in which the estimate referenced as equation DQC1:1 characterizes membership...

conjecture | open | MathDB 页面 | 原文线索

V2-038614 | MATHDB-348484

Improved two-grid approximation conjecture for fractional PDE-constrained optimization

摘录 / EXCERPT: Let ${\rm dom}$ be a convex domain, let s satisfy $0<s<1$, and let ${\mathcal H}_h^s$ and ${\mathcal G}_h^s$ denote the fine-grid and two-grid operators, respectively, with mesh...

conjecture | open | MathDB 页面 | 原文线索

V2-038615 | MATHDB-348526

Convergence-rate conjecture for the randomized truncated Milstein method

摘录 / EXCERPT: Convergence-rate conjecture. There is a constant H independent of Δ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038616 | MATHDB-348581

The positivity conjecture for nonzero eigenvalues of $\mathfrak{L}_0$

摘录 / EXCERPT: Let $a>1$, and let $\mathfrak{L}_0$ be the linear operator arising from the $n=0$ mode of the linearized problem. Positivity conjecture for $\mathfrak{L}_0$. For all $a>1$, all non...

conjecture | open | MathDB 页面 | 原文线索

V2-038617 | MATHDB-348582

The positive-real-part conjecture for $\mathfrak{L}_1$

摘录 / EXCERPT: Let $a>1$ and $\sigma>0$, and let $\mathfrak{L}_1$ be the linear operator for the first nonzero Fourier mode after reducing the eigenvalue problem from $\mathfrak{L}_n$ to $n=1$. P...

conjecture | open | MathDB 页面 | 原文线索

V2-038618 | MATHDB-348646

Satisfiability of the Poincaré assumption for smooth domains

摘录 / EXCERPT: The paper considers Assumption ... can be satisfied for smooth domains. The preceding discussion establishes satisfiability for rectangular domains; the conjecture proposes the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-038619 | MATHDB-348693

Approximation conjecture for solutions of the inner-function curvature equation

摘录 / EXCERPT: Approximation conjecture. Any solution u of this equation such that $u\leq u_{I',\infty}$ can be approximated uniformly on compact subsets by solutions u_k that are bounded abo...

conjecture | open | MathDB 页面 | 原文线索

V2-038620 | MATHDB-349119

Iwaniec's continuous right-inverse conjecture for the Jacobian

摘录 / EXCERPT: For $p\text{ in }[1,\text{infinity})$, consider the Jacobian operator ... where $\mathscr{H}^p(\mathbb{R}^n)$ is the real Hardy space and $\dot{W}^{1,np}(\mathbb{R}^n,\mathbb{R}^n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-038621 | MATHDB-349125

The conjecture on decay of oscillatory solutions below Lillo's constant

摘录 / EXCERPT: Consider the delay differential equation (1), with coefficient $c(t)$ and delay function $\Greekmath 011C (t)$ satisfying the assumptions in (10a). Assume that $c(t)\leq 0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-038622 | MATHDB-349126

The conjecture on asymptotics of rapidly oscillating solutions to delay equations

摘录 / EXCERPT: Let $s\in[1,2]$ be arbitrary. Consider the delay differential equation (1), with coefficient $c(t)$ and delay function $\Greekmath 011C (t)$ satisfying the assumptions in (10a), an...

conjecture | open | MathDB 页面 | 原文线索

V2-038623 | MATHDB-349595

Uniqueness of localized quasimonochromatic breather profiles

摘录 / EXCERPT: Uniqueness conjecture. There exist $\gamma,m,z\in\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038624 | MATHDB-349596

Monotone-profile classification for localized quasimonochromatic breathers

摘录 / EXCERPT: Monotone-profile conjecture. There exist $\gamma,m>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038625 | MATHDB-349754

Symmetry conjecture for least energy nodal solutions on squares

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a square, and let p be close to 2. A least energy nodal solution (l.e.n.s.) is a sign-changing solution with minimum energy. Symmetry conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-038626 | MATHDB-349791

Sobolev improving conjecture for non-degenerate curves

摘录 / EXCERPT: Non-degenerate curve Sobolev improving conjecture. If γ is non-degenerate and $2n-2<p<\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-038627 | MATHDB-349792

Finite-type Sobolev improving conjecture

摘录 / EXCERPT: Finite-type Sobolev improving conjecture. The operator A_{γ} maps L^p to L^p_{α} for all $p\geq 2$ and

conjecture | open | MathDB 页面 | 原文线索

V2-038628 | MATHDB-349895

Existence of holomorphic-function quads for almost all constant parameters

摘录 / EXCERPT: Existence conjecture. For almost all values of the constant parameters, there exists a discrete set of quads of functions p,q,r,s such that all three systems of equations are ful...

conjecture | open | MathDB 页面 | 原文线索

V2-038629 | MATHDB-350048

Conjecture on vanishing IVIS for Ginibre system matrices

摘录 / EXCERPT: Let A be a system matrix sampled from the Ginibre ensemble, and let $\mathbf{x}_{0}$ be an initial condition whose coordinates $x_{0,i}$ have second-order moments bounded by a co...

conjecture | open | MathDB 页面 | 原文线索

V2-038630 | MATHDB-350222

The multitype $b[Lambda]b[0m$-coalescent differential-equation speed conjecture

摘录 / EXCERPT: Multitype coalescent speed conjecture. Provided the conditions of the theorem establishing coming down from infinity hold, the ordinary differential equation

conjecture | open | MathDB 页面 | 原文线索

V2-038631 | MATHDB-350320

Hang-Yang conjecture on positive critical points of the perturbed Paneitz problem

摘录 / EXCERPT: Hang-Yang conjecture. Every such solution $u_epsilon$ is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-038632 | MATHDB-350509

Two-term asymptotic conjecture for the buckling counting function

摘录 / EXCERPT: Let Ω be a bounded domain in $\mathbb R^d$ with smooth boundary. Let $N(z)$ denote the counting function for the eigenvalues of the buckling problem, and let B_k denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-038633 | MATHDB-350868

Parfenov-Fedotov approximation conjecture for strongly elliptic operators

摘录 / EXCERPT: Let $\mathcal L\in\operatorname{\mathcal S\mathcal E}$, and let X be a compact set in $\mathbb C$. The spaces $P_{\mathcal L}(X)$ and $C_{\mathcal L}(X)$ denote, respectively, th...

conjecture | open | MathDB 页面 | 原文线索

V2-038634 | MATHDB-350869

Regularity conjecture for strongly elliptic operators

摘录 / EXCERPT: Let $\mathcal L\in\operatorname{\mathcal S\mathcal E}$, and let $G\subset\mathbb C$ be a bounded simply connected domain. The domain G is $\mathcal L$-regular when the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-038635 | MATHDB-350892

Differential equations for the exponential generating functions of two vincular avoidance classes

摘录 / EXCERPT: Conjectures. For $E=E[\overline{123}]$ and for $E=E[\overline{213}]$, respectively,

conjecture | open | MathDB 页面 | 原文线索

V2-038636 | MATHDB-350898

Strong minimality conjecture for generic differential equations

摘录 / EXCERPT: Let $f(x)$ be a generic differential polynomial of fixed order $h>1$ and fixed degree $d>1$, meaning that its coefficients are independent differentially transcendental elements an...

conjecture | open | MathDB 页面 | 原文线索

V2-038637 | MATHDB-350899

Geometric triviality conjecture for generic differential equations

摘录 / EXCERPT: Let $f(x)$ be a generic differential equation of fixed order $h>1$ and fixed degree $d>1$, and let a and b be two solutions of f. Geometric triviality conjecture. Any two sol...

conjecture | open | MathDB 页面 | 原文线索

V2-038638 | MATHDB-350902

The elliptic-cone higher-integrability conjecture for PDE-constrained measures

摘录 / EXCERPT: The elliptic-cone higher-integrability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-038639 | MATHDB-350987

CR analogue of the rigidity theorem for subcritical semilinear equations

摘录 / EXCERPT: Let M^{2m+1} be a closed pseudohermitian manifold with vanishing pseudohermitian torsion $A_{\alpha\beta}=0$ and Webster Ricci tensor satisfying ... Suppose a positive function f...

conjecture | open | MathDB 页面 | 原文线索

V2-038640 | MATHDB-351004

Global dynamics conjecture for the competition system with delay

摘录 / EXCERPT: Consider the competition system ... with positive initial data, basic reproduction numbers $\mathcal{R}_0^{(1)}>1$ and $\mathcal{R}_0^{(2)}>1$, boundary equilibria E_1,E_2, and coe...

conjecture | open | MathDB 页面 | 原文线索

V2-038641 | MATHDB-352051

The apparent-singularity conjecture for the associated third-order ODE

摘录 / EXCERPT: Apparent-singularity conjecture. The point 0 is an apparent singularity if and only if $B=D_0=D=0$; equivalently, this polynomial system has only the trivial solution $B=D_0=D=0$...

conjecture | open | MathDB 页面 | 原文线索

V2-038642 | MATHDB-352245

Bi-Lipschitz reduction conjecture for generalized outward cuspidal domains

摘录 / EXCERPT: Bi-Lipschitz reduction conjecture. For every generalized outward cuspidal domain U_ψ^n, there exists a Lipschitz cuspidal function $\hat\psi$, a Lipschitz base domain $\hat U$...

conjecture | open | MathDB 页面 | 原文线索

V2-038643 | MATHDB-352566

Sharp Sobolev regularity conjecture for restricted X-ray transforms

摘录 / EXCERPT: Sharp Sobolev regularity conjecture. The operator $\mathfrak R$ boundedly maps L^p to L_α^p for $\alpha \le \alpha(p)$. The sharp L^p regularity estimate is open in high...

conjecture | open | MathDB 页面 | 原文线索

V2-038644 | MATHDB-352736

The max-mid matrix gap conjecture for the strain constraint space

摘录 / EXCERPT: Max-mid matrix gap conjecture. There are no nontrivial max-mid matrices in the strain constraint space:

conjecture | open | MathDB 页面 | 原文线索

V2-038645 | MATHDB-352899

Equality of particle-system and mean-field log-Sobolev constants

摘录 / EXCERPT: Log-Sobolev constant limit conjecture. Under assumptions A1 and A2,

conjecture | open | MathDB 页面 | 原文线索

V2-038646 | MATHDB-353204

Quadratic convergence conjecture for planar diagonal diffusion matrices with constant trace

摘录 / EXCERPT: Let Ω be the domain and let u^{ε} and u denote, respectively, the solution of the rapidly oscillating elliptic equation and its homogenized limit, with data...

conjecture | open | MathDB 页面 | 原文线索

V2-038647 | MATHDB-353220

Ancient-solution triviality conjecture below the Sobolev exponent

摘录 / EXCERPT: Let (M,g) be a complete, n-dimensional Riemannian manifold with bounded geometry. An ancient, positive solution is a positive solution u of ... that is defined for all past tim...

conjecture | open | MathDB 页面 | 原文线索

V2-038648 | MATHDB-353266

Local uniform Sobolev inequality near an isolated analytic singularity

摘录 / EXCERPT: Local uniform Sobolev inequality conjecture. There exist a neighbourhood U of o in N and a constant C, independent of δ, such that every positive $\mathcal C^1$ func...

conjecture | open | MathDB 页面 | 原文线索

V2-038649 | MATHDB-353270

Uniform solvability at one for relative (φ_2,∇_2)-modules

摘录 / EXCERPT: Let $I=\{1,2\}$, let M be a Φ-equivariant F-isocrystal $(M,\varphi,\varphi_1,\varphi_2)$ in the category...

conjecture | open | MathDB 页面 | 原文线索

V2-038650 | MATHDB-353670

Regularity conjecture for solutions of a singular fractional differential equation

摘录 / EXCERPT: Regularity conjecture. If $h\in C[0,1]$ or $h\in L^1(0,1)$, then the solution u belongs to $C^1_{2-\alpha}[0,1]$. Moreover, if $h\in C(0,1]$ satisfies (H), then the solution...

conjecture | open | MathDB 页面 | 原文线索

V2-038651 | MATHDB-353820

Equivalence of the optimal fractional Poincaré constants

摘录 / EXCERPT: Equivalence of the optimal fractional Poincaré constants. For $t>s\geq 0$, the optimal Poincaré constant is

conjecture | open | MathDB 页面 | 原文线索

V2-038652 | MATHDB-354120

The minimal-intensity conjecture for finite-time Euler singularities

摘录 / EXCERPT: Let $T_*<\infty$ and let $\omega\in C^\infty_c(\mathbb{R}^3\times[0,T_*))$ be a smooth solution of the 3D Euler equation that becomes singular as $t\to T_*$. Minimal-intensity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-038653 | MATHDB-354122

The Hou-Luo finite-time singularity conjecture for axisymmetric Euler flow

摘录 / EXCERPT: Let (r,z) belong to the cylindrical domain $\{(r,z):0\leq r\leq1\}$. Hou-Luo conjecture. There exists a smooth solution to the axisymmetric Euler equation on this cylindrical dom...

conjecture | open | MathDB 页面 | 原文线索

V2-038654 | MATHDB-354164

Non-uniqueness conjecture for the Sobolev divergence chain rule

摘录 / EXCERPT: Non-uniqueness conjecture. Under the stated integrability condition and suitable assumptions on β and T, such a density ρ and incompressible Sobolev vector field u e...

conjecture | open | MathDB 页面 | 原文线索

V2-038655 | MATHDB-354277

The kinetic Krylov-Safonov conjecture

摘录 / EXCERPT: Kinetic Krylov-Safonov conjecture. There exist constants $\alpha>0$ and C, depending only on the dimension and λ,Λ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-038656 | MATHDB-354278

The kinetic integro-differential Krylov-Safonov conjecture

摘录 / EXCERPT: Kinetic integro-differential Krylov-Safonov conjecture. There exist constants $\alpha>0$ and C, depending only on the dimension, λ, and Λ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-038657 | MATHDB-354286

Backwards conjecture for slow manifolds of non-autonomous wave PDEs

摘录 / EXCERPT: Consider the non-autonomous wave PDE ... on a domain $\mathcal X$, with the stated edge conditions and assumptions on f and g. Partition the space into N elements, let...

conjecture | open | MathDB 页面 | 原文线索

V2-038658 | MATHDB-354411

Minimal spectral abscissa conjecture for scalar linear systems with two delays

摘录 / EXCERPT: Two-delay minimal spectral abscissa conjecture. For arbitrary $a_1,a_2\in\mathbb{R}$, the minimal spectral abscissa is

conjecture | open | MathDB 页面 | 原文线索

V2-038659 | MATHDB-354424

Finite-time blowup conjecture for axisymmetric swirl-free Euler flow in dimensions at least four

摘录 / EXCERPT: Let $u^0 ot\equiv 0$ belong to $H^s_{as\&df}(\mathbb{R}^d)$, where $d\geq 4$ and $s>1+\frac{d}{2}$, and let ω^0 be its associated vorticity. Assume that ω^0 is odd in...

conjecture | open | MathDB 页面 | 原文线索

V2-038660 | MATHDB-354439

Conjecture on critical and subcritical solutions for discrete p-biharmonic equations and Lane-Emden systems

摘录 / EXCERPT: Let G be a graph of polynomial growth with homogeneous dimension $N\geq 3$. For the discrete p-biharmonic equation ... where $1<p<N/2$, and for the discrete Lane-Emden system ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-038661 | MATHDB-354876

Solvability conjecture for invariant differential operators on homogeneous bundles

摘录 / EXCERPT: Let Xconjectures be the manifold under consideration, let $\mathbb{E}_\gamma$, $\mathbb{E}_\tau$, and $\mathbb{F}$ be homogeneous vector bundles on X, and let ... be invariant di...

conjecture | open | MathDB 页面 | 原文线索

V2-038662 | MATHDB-354916

Tran-Dritschel conjecture on maximum enstrophy dissipation

摘录 / EXCERPT: Tran-Dritschel conjecture. The enstrophy dissipation satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038663 | MATHDB-355058

The DNLS well-posedness threshold conjecture

摘录 / EXCERPT: Let $u=u(t,x)$ solve the derivative nonlinear Schrödinger equation on $\mathbb{R}$, ... For $s\in\mathbb{R}$, let $H^s(\mathbb{R})$ denote the Sobolev space of regularity s. DNLS w...

conjecture | open | MathDB 页面 | 原文线索

V2-038664 | MATHDB-355943

Constant-exponent conjecture for regular relative connections

摘录 / EXCERPT: Let I be open, let X be connected and affinoid with a finite Shilov boundary, and let $\mathcal{E}$ be a regular connection of rank n. Suppose there are $x\in X$, a closed su...

conjecture | open | MathDB 页面 | 原文线索

V2-038665 | MATHDB-355944

Relative regularization conjecture for integrable connections on polyannuli

摘录 / EXCERPT: Let K be the coefficient field, let m be a positive integer, and let $\epsilon\in(0,1)$. Let $A_K[\epsilon,1)^m$ denote the corresponding polyannulus, and let $\mathcal{E}$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-038666 | MATHDB-356167

Conjecture on trivial differential Galois groups for the inhomogeneous equation

摘录 / EXCERPT: The trivial-Galois-group conjecture. However large α, β, and r are, the Galois group of the associated third-order homogeneous equation is trivial in the category of...

conjecture | open | MathDB 页面 | 原文线索

V2-038667 | MATHDB-356202

Strict convexity conjecture for the general inverse σ_k equation

摘录 / EXCERPT: Strict convexity conjecture. If $r_f(x)$ is strictly right-Noetherian, then the level set $\{f=0\}$ is strictly convex; equivalently, the Hessian matrix

conjecture | open | MathDB 页面 | 原文线索

V2-038668 | MATHDB-356378

Convergence to endemic equilibrium in the stochastic epidemic model

摘录 / EXCERPT: Convergence conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-038669 | MATHDB-356379

Uniqueness of positive bounded entire solutions for the limiting epidemic equations

摘录 / EXCERPT: Uniqueness conjecture. The limiting system above has a unique positive and bounded solution on $\mathbb{R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-038670 | MATHDB-356637

Asymptotic classification conjecture for positive singular solutions of poly-harmonic equations

摘录 / EXCERPT: Asymptotic classification conjecture. If $p\in(1,2_m^*-1)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-038671 | MATHDB-356679

Non-localized data focusing lifespan conjecture

摘录 / EXCERPT: Let a one-dimensional dispersive flow on the real line have a cubic nonlinearity, and let its initial data have size $epsilon$ in a suitable Sobolev norm. Non-localized data focusi...

conjecture | open | MathDB 页面 | 原文线索

V2-038672 | MATHDB-356806

Relatively algebraic non-characteristic derived Cauchy problem

摘录 / EXCERPT: Relatively algebraic non-characteristic derived Cauchy problem. This map is an isomorphism. The conjecture extends the non-characteristic Cauchy problem from linear to nonlinear de...

conjecture | open | MathDB 页面 | 原文线索

V2-038673 | MATHDB-356807

Analytic non-characteristic derived Cauchy problem

摘录 / EXCERPT: Analytic non-characteristic derived Cauchy problem. This map is an isomorphism of sheaves of spaces over Y. The statement is proposed as a general nonlinear derived Cauchy-Kowale...

conjecture | open | MathDB 页面 | 原文线索

V2-038674 | MATHDB-356887

Shnirelman's compact-orbit conjecture for generic 2D Euler limit sets

摘录 / EXCERPT: Let (ω,u) be a solution of the two-dimensional incompressible Euler equation on $\mathbb{T}^2$, and consider its weak limit set as $t\to\pm\infty$. A solution lies on a comp...

conjecture | open | MathDB 页面 | 原文线索

V2-038675 | MATHDB-356896

Precompactness conjecture for critical radial solutions of the Lin-Ni-Takagi problem

摘录 / EXCERPT: Let $B_R\subset\mathbb{R}^N$ be the ball of radius R, let $2^*=\frac{2N}{N-2}$, and let $(u_k)_{k\geq 0}$ be radial solutions of the critical Lin-Ni-Takagi problem with exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-038676 | MATHDB-356993

The Sobolev gradient convolution trace inequality conjecture

摘录 / EXCERPT: Let $K\colon \mathbb{R}^d\to\mathbb{R}^d$ be homogeneous of order $1-d$ and smooth outside the origin. Let ν be a measure satisfying ... For $f\in C_0^\infty(\mathbb{R}^d)$, cons...

conjecture | open | MathDB 页面 | 原文线索

V2-038677 | MATHDB-357018

The zero-product conjecture for Toeplitz operators on Fock-Sobolev spaces

摘录 / EXCERPT: Let $m\neq 0$, let $f,g,h,k\in\mathcal{A}_1$, and define ... Here T_u and T_v are the Toeplitz operators with symbols u and v on the Fock-Sobolev space $F^{2,m}$. Zero-produc...

conjecture | open | MathDB 页面 | 原文线索

V2-038678 | MATHDB-357109

The conjecture that diffusion-equation solution computation is #P-complete

摘录 / EXCERPT: Let $x_0,x_1,y_0\in\mathbb{R}_c\cap[0,+\infty)$ and consider the initial-boundary value problems and solution operators described in Theorems 1 and 2, including the stated polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-038679 | MATHDB-357112

De Giorgi's continuity conjecture for singular elliptic equations

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a domain and consider weak solutions of ... Assume that the coefficient matrix satisfies ... for $\xi\in\mathbb{R}^N$ and almost every $x\in\Omega$,...

conjecture | open | MathDB 页面 | 原文线索

V2-038680 | MATHDB-357225

BMS conjecture on positivity preservation of geodesically complete Riemannian manifolds

摘录 / EXCERPT: BMS conjecture. Assume that (M,g) is geodesically complete. Then (M,g) is L^{2}-positivity preserving.

conjecture | open | MathDB 页面 | 原文线索

V2-038681 | MATHDB-357419

The algebraic-independence characterization of general type equations

摘录 / EXCERPT: General type characterization conjecture. A first order differential equation over k (respectively, an autonomous differential equation over C) is not of general type if and on...

conjecture | open | MathDB 页面 | 原文线索

V2-038682 | MATHDB-357465

Long-time dynamics conjecture for generic solutions of the 2D Euler equation

摘录 / EXCERPT: Let solutions of the two-dimensional Euler equation evolve as $t\to\pm\infty$. A solution is called generic in the informal sense intended by the conjecture, and its weak limit set...

conjecture | open | MathDB 页面 | 原文线索

V2-038683 | MATHDB-357469

Nakayama's local epsilon conjecture for de Rham (φ,Γ)-modules

摘录 / EXCERPT: Nakayama's local epsilon conjecture. There exists an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-038684 | MATHDB-357494

Strong well-posedness conjecture for measure-valued drifts driven by fractional Brownian motion

摘录 / EXCERPT: Let b be a finite signed measure on $\mathbb{R}$, let $H\in(0,1/2)$, and consider the stochastic differential equation ... where B^H is fractional Brownian motion with Hurst para...

conjecture | open | MathDB 页面 | 原文线索

V2-038685 | MATHDB-357542

Fernández-Real-Ros-Oton conjecture on one-dimensional minimizers

摘录 / EXCERPT: Fernández-Real-Ros-Oton conjecture. Suppose that $u:\mathbb{R}^n\to(0,\infty)$ minimizes I_1 locally. If $n\leq n_e^*-1$, then u is one-dimensional: in a suitable Euclidean coo...

conjecture | open | MathDB 页面 | 原文线索

V2-038686 | MATHDB-357885

The conjecture on the structure of the test space for reactive transport

摘录 / EXCERPT: The test-space structure conjecture. The test space $\mathcal{Y}$ is isometrically isomorphic to $H^1(\vec{b},\Omega)$.

conjecture | open | MathDB 页面 | 原文线索

V2-038687 | MATHDB-357963

Landis-type uniqueness-at-infinity conjecture for periodic elliptic equations

摘录 / EXCERPT: Let u be a super-exponentially decaying solution of a second-order periodic, or more generally an appropriate class of periodic elliptic, equation. Landis-type conjecture. Such a...

conjecture | open | MathDB 页面 | 原文线索

V2-038688 | MATHDB-358022

Christol-André conjecture on globally bounded D-finite power series

摘录 / EXCERPT: Let $y(x)=\sum_{k\geq 0}a_kx^k\in\mathbb{Q}[[x]]$ be D-finite. Assume that the sequence $(a_k)_{k\geq 0}$ has at most geometric growth, that there exists $N\in\mathbb{N}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-038689 | MATHDB-358023

Grothendieck's conjecture on algebraic solutions and reductions modulo primes

摘录 / EXCERPT: Let $\mathscr{L}\in\mathbb{Q}(x)\langle\partial_x\rangle$ be the differential operator attached to the paper's linear differential equation, and let $\mathscr{L}_p$ denote its redu...

conjecture | open | MathDB 页面 | 原文线索

V2-038690 | MATHDB-358115

Higher Sobolev boundedness conjecture for global cubic-wave solutions

摘录 / EXCERPT: Let u solve the cubic wave equation on $[0,\infty)$ with initial data $(u_0,u_1)\in\mathcal{H}^{1/2}\cap\mathcal{H}^1$, where $\boldsymbol{u}(t)=(u(t),\partial_tu(t))$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-038691 | MATHDB-358117

Non-uniqueness conjecture for 2D Euler solutions with L^p vorticity

摘录 / EXCERPT: Let $2<p<\infty$. Consider the two-dimensional Euler equation with initial velocity $v^\circ\in L^2$ and initial vorticity $\omega^\circ\in L^1\cap L^p$. A weak solution has veloci...

conjecture | open | MathDB 页面 | 原文线索

V2-038692 | MATHDB-358121

MLDE conjecture for twisted characters of DC-series affine vertex algebras

摘录 / EXCERPT: Let V be a simple affine vertex algebra associated with one of the Lie algebras in the DC series ... at level $-h^{\vee}/6-1$. Let $\chi^{\mathrm{twi}}(q)$ denote a norm...

conjecture | open | MathDB 页面 | 原文线索

V2-038693 | MATHDB-358418

Transport-noise well-posedness conjecture for the 2D Euler equations

摘录 / EXCERPT: Let W be a noise satisfying Assumption ... {rm d} omega + operatorname{curl}^{-1} omegacdot nabla omega, {rm d} t + circ {rm d} Wcdot nabla omega = 0 ... such that these stochastic 2...

conjecture | open | MathDB 页面 | 原文线索

V2-038694 | MATHDB-358619

Hamel-Nadirashvili's punctured-disk circular-flow conjecture

摘录 / EXCERPT: Hamel-Nadirashvili's conjecture. If $|\mathbf{v}|>0$ in $\overline{D}\backslash\{z\}$, then z is the origin and $\mathbf{v}$ is a circular flow.

conjecture | open | MathDB 页面 | 原文线索

V2-038695 | MATHDB-358620

Hamel-Nadirashvili's one-stagnation-point circular-flow conjecture

摘录 / EXCERPT: Hamel-Nadirashvili's weaker conjecture. If $z\in D$ is the only stagnation point of $\mathbf{v}$ in $\overline{D}$, meaning that $|\mathbf{v}(z)|=0$ and $|\mathbf{v}|>0$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-038696 | MATHDB-358630

Sharpness conjecture for the local well-posedness regularity range

摘录 / EXCERPT: Consider the cubic problems denoted by ... . Sharpness conjecture. Generically, the problem ... is ill-posed in H^s for $s<2$. The theorem establishes local well-posedness above thes...

conjecture | open | MathDB 页面 | 原文线索

V2-038697 | MATHDB-358770

Beck-Flandoli-Gubinelli-Maurelli's conjecture on failure of regularisation beyond the LPS condition

摘录 / EXCERPT: Beck-Flandoli-Gubinelli-Maurelli's conjecture. If the LPS condition does not hold, namely if

conjecture | open | MathDB 页面 | 原文线索

V2-038698 | MATHDB-358781

Uniqueness conjecture for extended Bogomolny equations under weaker asymptotic conditions

摘录 / EXCERPT: The uniqueness conjecture. Then $u=u_3$. The inequalities need not be uniform in z.

conjecture | open | MathDB 页面 | 原文线索

V2-038699 | MATHDB-358809

The conjecture on strong convergence of the renormalized sequence

摘录 / EXCERPT: Strong-convergence conjecture. The renormalized sequence converges strongly:

conjecture | open | MathDB 页面 | 原文线索

V2-038700 | MATHDB-358864

Finiteness conjecture for virtually solvable telescoper Galois groups

摘录 / EXCERPT: Let T be a minimal-order telescoper for an integral ... where a is algebraic in x and h. Finiteness conjecture. If the Galois group of T is virtually solvable, then it is f...

conjecture | open | MathDB 页面 | 原文线索

V2-038701 | MATHDB-358870

Tilt-stable log-Sobolev conjecture for log-concave measures

摘录 / EXCERPT: Let μ be a log-concave probability measure on $\mathbb R^n$. It is β-tilt-stable if, for every $h\in\mathbb R^n$, the covariance operator of the tilted measure...

conjecture | open | MathDB 页面 | 原文线索

V2-038702 | MATHDB-358974

Existence of a periodic-orbit cylinder in the generalized Chazy differential system

摘录 / EXCERPT: Periodic-cylinder conjecture. For $k=q+1$ and q a positive integer, the generalized Chazy differential system has an invariant cylinder foliated by periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-038703 | MATHDB-358983

Hamel-Nadirashvili's rigidity conjecture for steady Euler flows in a punctured disk

摘录 / EXCERPT: Let D be an open disk centered at the origin. Let $z\in D$ and let $\mathbf{v}\in C^2(\overline{D}\backslash \\{z\\})$ be a bounded flow solving the Euler equations on...

conjecture | open | MathDB 页面 | 原文线索

V2-038704 | MATHDB-359019

Grothendieck's p-curvature conjecture for linear differential equations

摘录 / EXCERPT: Grothendieck's p-curvature conjecture. If $L_py=0$ has a basis of $\mathbb{F}_p[[x^p]]$-linearly independent solutions in $\mathbb{F}_p[[x]]$ for almost all prime numbers p, th...

conjecture | open | MathDB 页面 | 原文线索

V2-038705 | MATHDB-359020

Bézivin's conjecture on integral power-series solutions

摘录 / EXCERPT: Bézivin's conjecture. If $Ly=0$ has a $\mathbb{Q}$-linearly independent basis of solutions in $\mathbb{Z}[[x]]$, then these solutions are algebraic over $\mathbb{Q}(x)$.

conjecture | open | MathDB 页面 | 原文线索

V2-038706 | MATHDB-359058

Conjecture on the unique blowup point for Fourier-restricted Euler and hypodissipative Navier-Stokes

摘录 / EXCERPT: Let $u\in C\left([0,T_{max}); \dot{H}^s_{\mathcal{M}}\right)$, with $s>\frac{5}{2}$, be an odd, permutation symmetric, σ-mirror symmetric solution of the Fourier-restricted...

conjecture | open | MathDB 页面 | 原文线索

V2-038707 | MATHDB-359074

Conjecture on exceptional Gaussian measures for Euler-flow non-invariance

摘录 / EXCERPT: Exceptional-support conjecture. Part (i) of the theorem on invariant measures already contains essentially all counterexamples to non-invariance: for...

conjecture | open | MathDB 页面 | 原文线索

V2-038708 | MATHDB-359075

Optimal average Sobolev-norm growth bound for generic Gaussian Euler data

摘录 / EXCERPT: Optimal growth conjecture. There exists $C(s)>0$ such that, for every $t\in[0,t_0)$ and every $s\in(0,\sigma]$,

conjecture | open | MathDB 页面 | 原文线索

V2-038709 | MATHDB-359328

Weak-type weighted Sobolev conjecture for the nonlinear operator N

摘录 / EXCERPT: Weak-type weighted Sobolev conjecture. For such w and f,

conjecture | open | MathDB 页面 | 原文线索

V2-038710 | MATHDB-359357

Peres-Tetali conjecture on coarse Ricci curvature and modified log-Sobolev inequalities

摘录 / EXCERPT: Let Ω be a finite unweighted graph and consider the stochastic matrix P associated with the simple random walk on this graph, $\tilde{P} = \frac{P+I}{2}$ associated to the...

conjecture | open | MathDB 页面 | 原文线索

V2-038711 | MATHDB-359388

Mazzeo's weighted X-ray transform isomorphism conjecture

摘录 / EXCERPT: Let (M,g) be a simple Riemannian manifold with boundary defining function ρ, and fix $\gamma>-1$. An α-boundary defining function (α-bdf) for ∂_+SM is...

conjecture | open | MathDB 页面 | 原文线索

V2-038712 | MATHDB-359505

Time of first blowup for the 3D relativistic Euler equations with vorticity and entropy

摘录 / EXCERPT: Time-of-first-blowup conjecture. The perturbed solution forms a shock at a time that is a perturbation of T_{Shock}; its first blowup time is the smallest value of t a...

conjecture | open | MathDB 页面 | 原文线索

V2-038713 | MATHDB-359506

Structure of the crease and singular boundary for the 3D relativistic Euler equations

摘录 / EXCERPT: Crease-and-singular-boundary conjecture. The perturbed solution has a crease and singular boundary that are perturbations of the crease and singular boundary of the background solu...

conjecture | open | MathDB 页面 | 原文线索

V2-038714 | MATHDB-359507

Structure of the Cauchy horizon for the 3D relativistic Euler equations with vorticity and entropy

摘录 / EXCERPT: Cauchy-horizon conjecture. The perturbed solution has a Cauchy horizon that is a perturbation of the Cauchy horizon of the background solution.

conjecture | open | MathDB 页面 | 原文线索

V2-038715 | MATHDB-359716

Inviscid-limit convergence conjecture for steady Navier-Stokes flows

摘录 / EXCERPT: Let $M\subset\mathbb{R}^2$ be any simply connected domain such that the constant-vorticity Euler solution u_* on M has a single eddy, meaning that its streamfunction has a sing...

conjecture | open | MathDB 页面 | 原文线索

V2-038716 | MATHDB-359744

Nonattainment conjecture for the one-dimensional Bianchi-Egnell quotient

摘录 / EXCERPT: Let $d=1$ and $s\in(0,\frac{1}{2})$. For $f\in\dot{H}^s(\mathbb{R})\setminus\mathcal B$, let $\mathcal Q(f)$ denote the Bianchi-Egnell quotient, let $c_{BE}(s)$ be its infimum, let...

conjecture | open | MathDB 页面 | 原文线索

V2-038717 | MATHDB-359913

Unitary inequivalence of representations from orthogonal vacua

摘录 / EXCERPT: Let $|0\rangle_p$ and $|0\rangle_q$ be the vacuum states in the Hilbert spaces $L^2_p({\cal F})$ and $L^2_q({\cal F})$, respectively, associated with the Sobolev parameters p and...

conjecture | open | MathDB 页面 | 原文线索

V2-038718 | MATHDB-360088

The H^1 strong cosmic censorship conjecture for the Einstein-Maxwell-charged-Klein-Gordon system

摘录 / EXCERPT: Let Σ be a regular, compact spacelike hypersurface without boundary. Prescribe generic initial data in the spherically symmetric class for the Einstein-Maxwell-charged-Klein...

conjecture | open | MathDB 页面 | 原文线索

V2-038719 | MATHDB-360914

Wang's classification conjecture for semilinear subelliptic equations on CR manifolds

摘录 / EXCERPT: Wang's conjecture. If $1<\alpha\leqslant\frac{n+2}{n}$ and $\lambda\leqslant\frac{n}{2(\alpha-1)}$, then the only positive solution is $u\equiv\lambda^{\frac{1}{\alpha-1}}$. Otherw...

conjecture | open | MathDB 页面 | 原文线索

V2-038720 | MATHDB-360987

Dancer's stable solution conjecture

摘录 / EXCERPT: Consider the semilinear elliptic equation ... where f is a C^1 function, and let u be a bounded stable C^2 solution, meaning that ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-038721 | MATHDB-361082

Sharp balanceability conjecture for the exterior derivative

摘录 / EXCERPT: Let d and q be integers with $1\leq q\leq d-1$, and consider the exterior derivative acting on $(d-q)$-forms. The operator is k-balanceable when it satisfies the balanceabili...

conjecture | open | MathDB 页面 | 原文线索

V2-038722 | MATHDB-361340

Finiteness conjecture for quasi-geometric connections with fixed Riemann symbol

摘录 / EXCERPT: Let S be a finite subset of $\boldsymbol{P}^1$ and let P be a quasiunipotent Riemann symbol for S, meaning the list of eigenvalues of monodromies around points of S. A conn...

conjecture | open | MathDB 页面 | 原文线索

V2-038723 | MATHDB-361854

Generalized symmetry algebra conjecture for the fine Kolmogorov backward equation

摘录 / EXCERPT: Generalized symmetry algebra conjecture. Up to equivalence of generalized symmetries and after neglecting the Lie symmetries associated with the linear superposition of solutions,...

conjecture | open | MathDB 页面 | 原文线索

V2-038724 | MATHDB-361898

Unramified exponential-type conjecture for quantum connections of Kähler manifolds

摘录 / EXCERPT: Let X be a Kähler manifold, and let $\nabla_{\frac{d}{du}}$ be its associated quantum connection. Unramified exponential-type conjecture. The connection $\nabla_{\frac{d}{du}}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-038725 | MATHDB-361919

Dimension equality conjecture for W_3 conformal-block solutions at $c=2$

摘录 / EXCERPT: Let $\mathcal C^{(c=2)}_\varsigma$ be the constructed space spanned by the relevant Specht-polynomial solutions, and let $\mathcal S^{(c=2)}_\varsigma$ be the full solution space o...

conjecture | open | MathDB 页面 | 原文线索

V2-038726 | MATHDB-362196

Effective diffusivity conjecture for critical fractional stochastic Navier-Stokes equations

摘录 / EXCERPT: Effective diffusivity conjecture. The sequence u^N converges weakly to the stationary solution u of

conjecture | open | MathDB 页面 | 原文线索

V2-038727 | MATHDB-362556

High-order k-Hessian bifurcation conjecture

摘录 / EXCERPT: Let Ω be the domain and let f be the nonlinearity in the k-Hessian problem ... has at least two convex solutions for every $\lambda>\lambda_{*}$, exactly one convex soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-038728 | MATHDB-362557

High-order k-Hessian solution multiplicity conjecture

摘录 / EXCERPT: Let Ω be the domain and let f be the nonlinearity in the k-Hessian problem ... has no nontrivial solution for every $\lambda\in(0,\lambda_{*})$ and at least two nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-038729 | MATHDB-362751

Generic sharpness conjecture for local well-posedness thresholds

摘录 / EXCERPT: Let s denote the Sobolev regularity exponent, and let Theorems and be the local well-posedness results stated in the source for the two-dimensional and n-dimensional cubic prob...

conjecture | open | MathDB 页面 | 原文线索

V2-038730 | MATHDB-362752

Large-data local well-posedness under nontrapping conjecture

摘录 / EXCERPT: Let s denote the Sobolev regularity exponent, and let Theorems and be the stated local well-posedness results for the two-dimensional and n-dimensional cubic problems. Large-da...

conjecture | open | MathDB 页面 | 原文线索

V2-038731 | MATHDB-362980

Ren-Wang's concavity conjecture for elementary symmetric functions

摘录 / EXCERPT: Ren-Wang's concavity conjecture. For every $\xi=(\xi_1,\ldots,\xi_n)\in\mathbb{R}^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-038732 | MATHDB-363241

Sampling-width rate for weighted mixed Sobolev spaces

摘录 / EXCERPT: Let $d\geq2$, let $r\in\mathbb N$, let λ be an even integer with $\lambda>4$, and let $\varrho_n(\boldsymbol W^r_2(\mathbb R^d;\mu),L_2(\mathbb R^d;\mu))$ denote the sampli...

conjecture | open | MathDB 页面 | 原文线索

V2-038733 | MATHDB-363254

Modified Gamma conjecture I, strong form

摘录 / EXCERPT: Let X be a Fano manifold. Let $S(z)$ be the quantum differential-equation fundamental solution, μ the grading operator, c_1 act by quantum multiplication, and...

conjecture | open | MathDB 页面 | 原文线索

V2-038734 | MATHDB-363283

The converse to the analytic characterization of convexity

摘录 / EXCERPT: Converse convexity conjecture. Inequality is fulfilled for solutions of problem, provided μ is sufficiently large.

conjecture | open | MathDB 页面 | 原文线索

V2-038735 | MATHDB-363364

Log-Sobolev KLS conjecture

摘录 / EXCERPT: Log-Sobolev KLS conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-038736 | MATHDB-363588

Pathwise uniqueness conjecture for Lagrangian trajectories of three-dimensional Leray solutions

摘录 / EXCERPT: Let $u_0\in L^2_\sigma$ be divergence-free initial data, let u be an associated Leray solution of the three-dimensional Navier-Stokes equations, and consider the stochastic diffe...

conjecture | open | MathDB 页面 | 原文线索

V2-038737 | MATHDB-363590

Benign non-convexity conjecture for dimensional logarithmic Sobolev optimization

摘录 / EXCERPT: Let $U_r\in\mathbb{R}^{d\times r}$ be an orthonormal matrix, and let $\mathrm{Gr}(d,r)$ denote the Grassmann manifold of r-dimensional subspaces of $\mathbb{R}^d$. Define ... as in...

conjecture | open | MathDB 页面 | 原文线索

V2-038738 | MATHDB-363789

Finite-time blowup conjecture for the Navier-Stokes strain equation without advection

摘录 / EXCERPT: Let S be a mild solution of the Navier-Stokes strain equation without advection, with $S otag\in C\left([0,T_{max});L^2_{st}\right)$, where L^2_{st} denotes the space of square...

conjecture | open | MathDB 页面 | 原文线索

V2-038739 | MATHDB-363810

High-differentiability characteristic gluing conjecture

摘录 / EXCERPT: Characteristic gluing conjecture. The data x_1 can be $C^k_u C^\infty_{(r,x^A)}$-glued to data induced on a deformation of $\mathbf S_2$ within a nearby member of $\mathcal F$.

conjecture | open | MathDB 页面 | 原文线索

V2-038740 | MATHDB-363829

Minimal travelling-wave selection conjecture for the free-boundary PDE

摘录 / EXCERPT: Let (u,L) solve the free-boundary problem, and define ... Let $\Pi_{\min}(x):=\int_x^{\infty}\pi_{\min}(y)\,dy$ denote the integrated minimal travelling wave. Minimal travelling-wa...

conjecture | open | MathDB 页面 | 原文线索

V2-038741 | MATHDB-364652

Invariant bisolution basis conjecture for de Sitter space

摘录 / EXCERPT: Let d be the spacetime dimension, let $u\in\mathbb{C}$ satisfy ... and let G_0^{sym}, $G_0^{\mathrm{sym},A}$, G^{PJ}, and $G^{\mathrm{PJ},A}$ be the invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-038742 | MATHDB-364906

Scattering-data conjecture for Kerr-de Sitter spacetimes

摘录 / EXCERPT: Fix Kerr-de Sitter parameters b. Let $R^\alpha H_{\mathrm{b}}^\infty$ and $R^\alpha H_{\mathrm{b}}^d$ denote the indicated weighted b-Sobolev spaces, and suppose that fo...

conjecture | open | MathDB 页面 | 原文线索

V2-038743 | MATHDB-365054

Landau-solution classification conjecture for critical stationary Navier-Stokes flows

摘录 / EXCERPT: Landau-solution classification conjecture. Then u is one of the Landau solutions, which are minus-one homogeneous and axisymmetric without swirl.

conjecture | open | MathDB 页面 | 原文线索

V2-038744 | MATHDB-365337

The Parisi PDE conjecture for the enriched bipartite free energy

摘录 / EXCERPT: Parisi PDE conjecture. The enriched free energy F_N should converge to a function $f:\mathbb{R}_+\times\mathcal Q^2\to\mathbb{R}$ solving

conjecture | open | MathDB 页面 | 原文线索

V2-038745 | MATHDB-365650

Painlevé property conjecture for the full constrained 3D system

摘录 / EXCERPT: Consider the full constrained 3D system studied in the paper, rather than either of its restrictions to an invariant hypersurface. The Painlevé property means that its solutions ha...

conjecture | open | MathDB 页面 | 原文线索

V2-038746 | MATHDB-365818

Kenig-Merle critical norm conjecture for nonlinear Schrödinger equations

摘录 / EXCERPT: Kenig-Merle critical norm conjecture. If u has bounded critical Sobolev norm in this sense, then u is global and scatters to a free solution.

conjecture | open | MathDB 页面 | 原文线索

V2-038747 | MATHDB-365938

Dwork's conjecture on asymptotic growth of horizontal sections

摘录 / EXCERPT: Let K be a complete discrete valuation field of mixed characteristic $(0,p)$ with non-trivial valuation, let M be a finite free differential module over $K[\![t]\!]_0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038748 | MATHDB-366223

Alvino-Ferone-Trombetti conjecture on extremal functions for Hardy-Sobolev-Maz'ya inequalities

摘录 / EXCERPT: Alvino-Ferone-Trombetti's conjecture. Every extremal function u for inequality (1.1) is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-038749 | MATHDB-366536

Solvability conjecture for positive Hessian quotient equations

摘录 / EXCERPT: Let M be a connected, closed Riemannian manifold with metric g, and let χ be a symmetric $(2,0)$ tensor on M. Suppose that there is an admissible function $v\in C^4(M)$...

conjecture | open | MathDB 页面 | 原文线索

V2-038750 | MATHDB-366537

C2 estimate conjecture for general Hessian quotient equations

摘录 / EXCERPT: Let $1\leq l<k$ and consider the general Hessian quotient equation ... Let u be a solution on a closed Riemannian manifold (M,g) endowed with a symmetric $(2,0)$ tensor χ. C...

conjecture | open | MathDB 页面 | 原文线索

V2-038751 | MATHDB-366711

Asymptotic stability conjecture for line solitary waves of the Amick-Schonbek system

摘录 / EXCERPT: Asymptotic stability conjecture. The line solitary waves of the Amick-Schonbek system are asymptotically stable.

conjecture | open | MathDB 页面 | 原文线索

V2-038752 | MATHDB-366712

Finite-time gradient-catastrophe conjecture for non-cavitating-violating initial data in one dimension

摘录 / EXCERPT: Gradient-catastrophe conjecture. Initial data violating the non-cavitation condition in one point in 1D lead to a gradient catastrophe in finite time.

conjecture | open | MathDB 页面 | 原文线索

V2-038753 | MATHDB-366713

Dispersion conjecture for localized data in the two-dimensional Amick-Schonbek system

摘录 / EXCERPT: Dispersion conjecture. Localised smooth initial data satisfying the non-cavitation condition do not lead to stable structures localised in both spatial dimension. The solutions exi...

conjecture | open | MathDB 页面 | 原文线索

V2-038754 | MATHDB-366753

Algebraicity, integrality, and prime-wise integrality of rational differential-equation solutions

摘录 / EXCERPT: Let $g\in \mathbb{Q}(z,y_0,\ldots,y_{n-1})$ be a rational function, let $t_0,\ldots,t_{n-1}\in\mathbb{Q}$ be initial values for which $g(0,t_0,\ldots,t_{n-1})$ is defined, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038755 | MATHDB-366756

Algebraicity or infinite local monodromy for integral solutions at singular points

摘录 / EXCERPT: Let $X/\mathbb{C}$ be a smooth projective curve, let $D\subset X$ be a reduced effective divisor, and let $(\mathscr{E},\nabla:\mathscr{E}\to\mathscr{E}\otimes \Omega_X^1(\log D))$...

conjecture | open | MathDB 页面 | 原文线索

V2-038756 | MATHDB-366767

Compactness-rate conjecture for weighted Sobolev embeddings

摘录 / EXCERPT: Compactness-rate conjecture. There exist $C>0$ and $N\in\mathbb{N}$ such that, for every $m\geq N$ and every $u\in\overline{\operatorname{Span}\{q_j\}_{j>m}}^{H^1(\nu)}$,

conjecture | open | MathDB 页面 | 原文线索

V2-038757 | MATHDB-366833

Weak involutivity conjecture for linear gauge invariant PDE systems

摘录 / EXCERPT: Let a system of linear gauge invariant PDEs be given in the form of system. A system is doubly weakly involutive if it has the corresponding weak involutivity property for both the...

conjecture | open | MathDB 页面 | 原文线索

V2-038758 | MATHDB-366940

The MLDE conjecture for characters of the vertex algebras $\mathcal{C}_n$

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the vertex algebra under consideration, and let ... be its character. An MLDE is a modular linear differential equation of the form described using Serre deriv...

conjecture | open | MathDB 页面 | 原文线索

V2-038759 | MATHDB-367381

Ollivier-Ricci curvature and the discrete log-Sobolev inequality

摘录 / EXCERPT: Ollivier-Ricci curvature and LSI. There exists a universal constant $c<\infty$ such that whenever the Ollivier-Ricci curvature $\tilde{\kappa}$ is positive, we have

conjecture | refuted | MathDB 页面 | 原文线索

V2-038760 | MATHDB-367532

The one-third regularity threshold for Euler solutions with anomalous diffusion

摘录 / EXCERPT: Let $d\in\{2,3\}$. Let $u\in C^0([0,1];C^\alpha(\mathbb T^d))\cap C^\alpha([0,1];C^0(\mathbb T^d))$ be a weak solution of the incompressible Euler equations, for some Hölder expone...

conjecture | open | MathDB 页面 | 原文线索

V2-038761 | MATHDB-367564

Characterization of mutual singularity of Sobolev spaces on Laakso-type fractals

摘录 / EXCERPT: Let $p_1,p_2\in(1,\infty)$, let $\mathscr{F}_{p_1}$ and $\mathscr{F}_{p_2}$ be the corresponding Sobolev spaces, and let $U_{p_1,+}$ and $U_{p_2,+}$ be their optimal potential func...

conjecture | open | MathDB 页面 | 原文线索

V2-038762 | MATHDB-367575

The BC method's unification conjecture for inverse boundary value problems

摘录 / EXCERPT: The BC method concerns the recovery of information in inverse boundary value problems (BVPs), including hyperbolic, elliptic, and parabolic problems. BC method unification conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-038763 | MATHDB-367595

The $L^p\to L^p$ local smoothing conjecture for spherical averages

摘录 / EXCERPT: Let A_t denote the spherical averaging operator, let ∂_t^γ be the fractional time derivative, and let $L^p_s(\mathbb{R}^d)$ denote the inhomogeneous Sobolev space....

conjecture | open | MathDB 页面 | 原文线索

V2-038764 | MATHDB-367617

Strictness characterization for divergence-free advection-diffusion flows

摘录 / EXCERPT: Strictness characterization.

conjecture | open | MathDB 页面 | 原文线索

V2-038765 | MATHDB-367767

Potential blow-up scenario for the 3D Navier-Stokes equations

摘录 / EXCERPT: Let $\alpha(\{\delta_j\}_{j=1}^n)$ be the exponent associated with the multi-level logarithmically improved criteria, and let s denote the Sobolev regularity exponent. Potential...

conjecture | open | MathDB 页面 | 原文线索

V2-038766 | MATHDB-367850

Markus-Yamabe conjecture

摘录 / EXCERPT: Let $X=(P_1,\dots,P_n)$ be an autonomous C^1-vector field on $\mathbb{R}^n$ with a unique singularity at the origin. For each $x\in\mathbb{R}^n$, let $DX(x)$ denote its Jacobian...

conjecture | open | MathDB 页面 | 原文线索

V2-038767 | MATHDB-367936

Sharpness conjecture for the local well-posedness Sobolev range

摘录 / EXCERPT: Let s denote the Sobolev regularity exponent in Theorem. Sharpness conjecture for the local well-posedness Sobolev range. The range of s in Theorem is sharp for generic problem...

conjecture | open | MathDB 页面 | 原文线索

V2-038768 | MATHDB-367975

Salez-Youssef conjecture on Ollivier curvature and log-Sobolev inequalities

摘录 / EXCERPT: Salez-Youssef conjecture. There is a universal constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038769 | MATHDB-368121

Gao's conjecture on entire solutions of a degenerate Fermat-type differential equation

摘录 / EXCERPT: Gao's conjecture. If $9ab^2=-4$, then every entire solution is either

conjecture | open | MathDB 页面 | 原文线索

V2-038770 | MATHDB-368122

Gao's conjecture on entire solutions of a Fermat-type differential equation

摘录 / EXCERPT: Gao's conjecture. This equation has no entire solution.

conjecture | open | MathDB 页面 | 原文线索

V2-038771 | MATHDB-368372

Onsager's conjecture for the 3D incompressible Euler equations

摘录 / EXCERPT: Consider the 3D incompressible Euler equations on $[0,T]\times \mathbb{T}^3$. A weak solution is understood in the usual distributional sense, and C^{β} denotes the Hölder sp...

conjecture | open | MathDB 页面 | 原文线索

V2-038772 | MATHDB-368409

Nonsingularity and odd-order initial conditions for weighted polyharmonic minimizers

摘录 / EXCERPT: Let m and α be the parameters defining the weighted Sobolev quotient $\mathcal{S}(m,2,\alpha,\alpha,\infty)$, and suppose that this quotient admits a 0-th nonsingular mi...

conjecture | open | MathDB 页面 | 原文线索

V2-038773 | MATHDB-368733

Generic non-precompactness of flow maps for divergence-free Sobolev velocities

摘录 / EXCERPT: Generic flow-map non-precompactness conjecture. For a residual subset of $\mathcal{U}_p$, the associated flow maps $\{\Phi_t\}_{t\geq 0}$ are not precompact in L^1.

conjecture | open | MathDB 页面 | 原文线索

V2-038774 | MATHDB-368864

Mean-field weight-noise invariance conjecture

摘录 / EXCERPT: Let the assumptions of Theorem hold, with the dynamics of $(\bm{X}_i(t))_{i\in[N]}$ and $(\bm{w}_{i,j}(t))_{i,j\in[N]}$ governed by the SDEs with weight-noise intensity $\eta>0$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-038775 | MATHDB-368865

Singular-limit conjecture for adaptive weight dynamics

摘录 / EXCERPT: Let the assumptions of Theorem hold. For $\epsilon>0$, let $(\bm{X}^{(\epsilon)}(t),\bm{w}^{(\epsilon)}(t))$ solve, and let $(\bm{X}^{(0)}(t))$ solve the limiting system with the s...

conjecture | open | MathDB 页面 | 原文线索

V2-038776 | MATHDB-369015

Chen-Véron conjecture on decaying fundamental solutions of the logarithmic Laplacian

摘录 / EXCERPT: Chen-Véron conjecture. In dimensions 1 and 2, such a fundamental solution exists, is radial and locally integrable, and satisfies the displayed bound.

conjecture | open | MathDB 页面 | 原文线索

V2-038777 | MATHDB-369186

Continuity conjecture for bounded-domain nonlocal Poisson equations

摘录 / EXCERPT: Let Ω be the bounded domain in the nonlocal Poisson problem, let u solve that problem, let $f\in L^2(\Omega)$, and let β be the kernel exponent. Continuity conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-038778 | MATHDB-369673

Bases as solutions to HODEs

摘录 / EXCERPT: Let $(\mathcal{F}_{\mathbb{M}}^{\mathbb{M}},{\Greekmath 011A})$ be a Hilbert function space. Assume that $\mathcal{E}=\{e_i\}_{i\in I}\subset \mathcal{F}_{\mathbb{M}}^{\mathbb{M}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-038779 | MATHDB-369687

Nonexistence of spiral-like blocking strategies below the critical speed

摘录 / EXCERPT: Let $\sigma>0$ be the barrier speed in the isotropic fire problem, and let $\bar\sigma=2.6144\ldots$ denote the critical value obtained from the spectral analysis of the retarded d...

conjecture | open | MathDB 页面 | 原文线索

V2-038780 | MATHDB-369725

Threshold classification conjecture for the normalized time-delay system

摘录 / EXCERPT: Consider the normalized delay differential equation referred to as, with solution $u(t;\beta,w_0)$ for parameter β and initial velocity w_0. Let the comparison properties b...

conjecture | open | MathDB 页面 | 原文线索

V2-038781 | MATHDB-369736

Lin's nodal and singular set measure conjecture for elliptic equations

摘录 / EXCERPT: Let u solve the divergence-form elliptic equation ... in $B(0,1)$, with ... and ... Let $Z(u)$ be the nodal set, $S(u)$ the singular set, and β_u the frequency quantity used in t...

conjecture | open | MathDB 页面 | 原文线索

V2-038782 | MATHDB-369756

Ballistic aggregation's stochastic differential approximation conjecture

摘录 / EXCERPT: Stochastic differential approximation conjecture. The process $\mathcal{X}_n$ behaves like a continuous-time process X_t satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-038783 | MATHDB-369976

Diagonal Strichartz conjecture for orthonormal systems of wave equations

摘录 / EXCERPT: Let $n\geq2$, $q\geq \frac{2(n+1)}{n-1}$, and let $(f_j)_j$ be an orthonormal system in the homogeneous Sobolev space $\dot{H}^s(\mathbb{R}^n)$. For a sequence of coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-038784 | MATHDB-369985

Convergence of the log-Sobolev constant in the central limit theorem

摘录 / EXCERPT: Let $\mu\in\mathcal{P}(\mathbb{R}^d)$ be an isotropic distribution, meaning that ... Let ν_n denote the normalized n-fold convolution sequence from the central limit theorem, s...

conjecture | open | MathDB 页面 | 原文线索

V2-038785 | MATHDB-370031

Attracting-face convergence conjecture for degenerate diffusions

摘录 / EXCERPT: Let $\mathbb{X}^{\circ}$ denote the interior of the state cube, let $\mathbb{F}$ be the collection of faces, and let $F\in\mathbb{F}$ be an attracting face supporting the invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-038786 | MATHDB-370111

Wang-Wei classification conjecture for stable Allen-Cahn solutions

摘录 / EXCERPT: Wang-Wei classification conjecture. If $n\leq 7$, then u is one-dimensional: either $u\equiv \pm1$, or

conjecture | open | MathDB 页面 | 原文线索

V2-038787 | MATHDB-370324

Li and Yang's exponential-type conjecture for entire solutions

摘录 / EXCERPT: Let k be a positive integer, let $a,b\in\mathbb{C}$ with $a\neq b$, let $a_i\in\mathbb{C}$ for $i=0,1,\ldots,k$ with $a_k=1$, and let φ be an entire function. For a funct...

conjecture | open | MathDB 页面 | 原文线索

V2-038788 | MATHDB-370428

Symmetry conjecture for the Robin torsion overdetermined system

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded Lipschitz domain, let ν denote the unit outward normal on $\partial\Omega$, and let $\beta>0$. Consider the overdetermined system .....

conjecture | open | MathDB 页面 | 原文线索

V2-038789 | MATHDB-370611

The Schiffer conjecture for bounded Lipschitz domains

摘录 / EXCERPT: Schiffer conjecture. If this problem admits a non-trivial solution, then $\operatorname{\Omega_S}$ is a ball. The conjecture is a well-known open problem in geometric analysis and,...

conjecture | open | MathDB 页面 | 原文线索

V2-038790 | MATHDB-370897

Linear dependence of the Sobolev gap on the weight parameter

摘录 / EXCERPT: Weight-dependence conjecture. There exists a constant $C=C_G>0$ depending only on G such that

conjecture | open | MathDB 页面 | 原文线索

V2-038791 | MATHDB-370898

Uniform finiteness conjecture for Sobolev gaps over the unitary dual

摘录 / EXCERPT: Uniform finiteness conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-038792 | MATHDB-371293

Nonexistence conjecture for singular self-similar profiles with intermediate exponent

摘录 / EXCERPT: Nonexistence conjecture. The equation does not have solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-038793 | MATHDB-371446

Weak convergence conjecture for the surviving species in the stochastic Rosenzweig-MacArthur model

摘录 / EXCERPT: Let M_+ be the interior state space of the stochastic Rosenzweig-MacArthur model, let $x_1^x(t)$ denote the first component of the solution started from $x\in M_+$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038794 | MATHDB-371718

Test vector conjecture for local triple product periods

摘录 / EXCERPT: Test vector conjecture for local triple product periods. There exist unit vectors $\varphi_i\in\pi_i$ such that: (i) the Sobolev norms of φ_3 are controlled by M, namely,...

conjecture | open | MathDB 页面 | 原文线索

V2-038795 | MATHDB-371763

Conjecture on joined energy curves and bifurcation from zero and infinity

摘录 / EXCERPT: Let the curves in Figure be energy curves for the problem, with solutions parametrized by their energy and with parameter $\lambda\in\mathbb{R}$. The conjectured joined diagram is...

conjecture | open | MathDB 页面 | 原文线索

V2-038796 | MATHDB-371767

Fixed-order MLDE conjecture for generalized Schur partition functions

摘录 / EXCERPT: Fixed-order MLDE conjecture. For any given 4d ${\cal N}=2$ superconformal field theory, ${\cal I}_k(q)$ solves a modular linear differential equation of fixed order and Wronskian i...

conjecture | open | MathDB 页面 | 原文线索

V2-038797 | MATHDB-371823

Batchelor Scale Conjecture for dissipation in stochastic and hyperviscous Navier-Stokes flows

摘录 / EXCERPT: Let u be a velocity field given by either a two-dimensional stochastic Navier-Stokes equation or a three-dimensional hyperviscous Navier-Stokes equation. For the passive scalar a...

conjecture | open | MathDB 页面 | 原文线索

V2-038798 | MATHDB-371955

Zagier's modularity conjecture for integral differential-equation solutions

摘录 / EXCERPT: Let $P(t)$ be a non-degenerate quadratic polynomial, and consider the differential equation ... An integral solution is a solution at $t=0$ of the form ... with $y_n\in\mathbb{Z}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-038799 | MATHDB-372080

The weighted Sobolev decay conjecture for the Green kernel of the p-Laplacian

摘录 / EXCERPT: Let M be a metric-measure space, let $u>p>1$, and let $\Omega\subset M$ be a connected open set containing a fixed origin o. Write $r(x)=\operatorname{dist}(x,o)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-038800 | MATHDB-372112

Yang-Li's conjecture on meromorphic solutions of a polynomial differential equation

摘录 / EXCERPT: Yang-Li's conjecture. The equation has no transcendental meromorphic solution when P_1/P_3 is a non-zero constant and P_2/P_3 is not the square of any rational function.

conjecture | open | MathDB 页面 | 原文线索

V2-038801 | MATHDB-372124

Lane-Emden conjecture for the subcritical Lane-Emden system

摘录 / EXCERPT: Lane-Emden conjecture. If (p,q) is subcritical, then the system admits no positive solution in $\left(C^2(\mathbb{R}^N)\right)^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-038802 | MATHDB-372125

Hénon-Lane-Emden conjecture for the subcritical weighted system

摘录 / EXCERPT: Hénon-Lane-Emden conjecture. If (p,q,a,b) is subcritical, then the system admits no positive solutions in $\left(C^2(\mathbb{R}^N\setminus\{0\})\cap C(\mathbb{R}^N)\right)^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-038803 | MATHDB-372419

Higher-time regularity conjecture for the Augmented formula

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let $T_{\varepsilon,\kappa,R}$ denote the time scale introduced for the Augmented formula. The solution takes values in the product of the Sobolev space...

conjecture | open | MathDB 页面 | 原文线索

V2-038804 | MATHDB-372557

Generic boundedness conjecture for inviscid limits in the critical Besov class

摘录 / EXCERPT: Let $u_0\in L^2$ be initial data, and consider sequences of Leray-Hopf weak solutions of the Navier-Stokes equations whose inviscid limits are denoted by u. The critical Besov cl...

conjecture | open | MathDB 页面 | 原文线索

V2-038805 | MATHDB-372640

Bihamiltonian coefficient-recursion conjecture for the local DR hierarchy

摘录 / EXCERPT: Consider the formal PDE referred to in the source as equation ... mathcal{P}1^{[0]}neq0,qquad mathcal{P}2^{[0]}neqlambdamathcal{P}1^{[0]}quadtext{for every }lambdainmathbb{C}. ... biha...

conjecture | open | MathDB 页面 | 原文线索

V2-038806 | MATHDB-372656

Liouville conjecture for stationary Navier-Stokes fields

摘录 / EXCERPT: Let v be a vector field on $\mathbb R^{3}$, with pressure p, satisfying the stationary Navier-Stokes system ... where $\nu>0$, together with ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-038807 | MATHDB-372657

Entire-function Liouville conjecture for stationary Navier-Stokes fields

摘录 / EXCERPT: Let v be a vector field satisfying the assumptions of the Liouville conjecture for stationary Navier-Stokes fields. In addition, suppose that v is the restriction to...

conjecture | open | MathDB 页面 | 原文线索

V2-038808 | MATHDB-373184

Sharpness conjecture for the enstrophy dissipation rate of measure vorticities

摘录 / EXCERPT: Let $\delta\in(0,1)$ and $p>1$. On the two-dimensional torus $\mathbb{T}^2$, let u_0 be a divergence-free vector field in $L^2(\mathbb{T}^2)$ with vorticity...

conjecture | open | MathDB 页面 | 原文线索

V2-038809 | MATHDB-373448

The A-infinity weighted Sobolev conjecture

摘录 / EXCERPT: The $\textup{A}_\infty$ weighted Sobolev conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-038810 | MATHDB-373485

Critical-parameter dichotomy conjecture for the nonlocal parabolic MEMS equation

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$, with $1\leq N\leq 3$, be a bounded domain with smooth boundary, and let $u_0\in H^2\cap H_0^1(\Omega)$. Consider equation (1.1), its solution...

conjecture | open | MathDB 页面 | 原文线索

V2-038811 | MATHDB-373486

Monotonicity conjecture for solutions of the nonlocal parabolic MEMS equation

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$, with $1\leq N\leq 3$, be a bounded domain with smooth boundary, and let $u_0\in H^2\cap H_0^1(\Omega)$. Let λ^{*} be the critical parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-038812 | MATHDB-373487

Critical-threshold conjecture for the hyperbolic MEMS equation

摘录 / EXCERPT: Assume (B,T), (u_{0},u_{1}), d, and Ω are given, and let u be the unique maximal solution to the hyperbolic problem. Critical-threshold conjecture. There exist criti...

conjecture | open | MathDB 页面 | 原文线索

V2-038813 | MATHDB-373587

Kumbhakar-Roy-Srinivasan classification conjecture for first-order differential equations

摘录 / EXCERPT: Let (k,δ) be a differential field of characteristic zero with field of constants C. Consider a first-order differential equation over k, and, respectively, an autonomous...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038814 | MATHDB-373697

The Dimension Conjecture for fewer differential equations than variables

摘录 / EXCERPT: Dimension Conjecture. If $m<n$, then every irreducible component Σ_1 of Σ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-038815 | MATHDB-373720

Variable-exponent extension of the multiplicity and singular existence results

摘录 / EXCERPT: Variable-exponent conjecture. Theorems concerning multiplicity for ... should remain valid when q is replaced by a variable exponent q.

conjecture | open | MathDB 页面 | 原文线索

V2-038816 | MATHDB-373721

Metric-measure and Heisenberg-group extension of the Neumann problem result

摘录 / EXCERPT: Metric-measure and Heisenberg-group conjecture. Problem $$ should yield the same result when considered over a metric measure space or a Heisenberg group.

conjecture | open | MathDB 页面 | 原文线索

V2-038817 | MATHDB-373799

Generalized convex-growth existence and uniqueness conjecture for regime-switching HJB systems

摘录 / EXCERPT: Generalized convex-growth existence and uniqueness conjecture. Under these assumptions, the system admits a unique solution $u\in\mathcal{A}_g^k$. This conjecture proposes extendin...

conjecture | open | MathDB 页面 | 原文线索

V2-038818 | MATHDB-373800

Comparison-principle conjecture for regime-switching HJB systems with common convex growth

摘录 / EXCERPT: Comparison-principle conjecture. Under these assumptions, the comparison principle of the paper's system comparison theorem extends to solutions in $\mathcal{A}_g^k$. This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-038819 | MATHDB-373813

Spherical truncation incidence-growth conjecture

摘录 / EXCERPT: Let Λ_N^{sph} denote the spherical truncation and let $S^{\mathrm{sph}}(N)$ be its associated incidence quantity. Spherical truncation incidence-growth conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-038820 | MATHDB-374002

Chewi-Stromme conjecture on low-temperature log-Sobolev asymptotics

摘录 / EXCERPT: Chewi-Stromme conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-038821 | MATHDB-374112

Tao's finite-time singularity conjecture for incompressible Euler flow

摘录 / EXCERPT: Let (M,g) be a closed, compact Riemannian manifold without boundary, and let (u,p) be a smooth solution of the incompressible Euler equations on M. Tao's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-038822 | MATHDB-374515

Iwaniec-Martin conjecture on weakly quasiregular mappings

摘录 / EXCERPT: Let $\cuOmega\textbackslashsubset\textbackslashmathbb R^n$ be a domain, let $K\textbackslashge 1$, and let $p\textbackslashge 1$. A mapping...

conjecture | open | MathDB 页面 | 原文线索

V2-038823 | MATHDB-375119

Unconditional local well-posedness threshold for the periodic generalized KdV equation

摘录 / EXCERPT: Let $k\geq 3$ and consider the k-generalized Korteweg-de Vries equation on the periodic domain, with initial data in the Sobolev space $H^s(\mathbb T)$. Unconditional well-posedn...

conjecture | open | MathDB 页面 | 原文线索

V2-038824 | MATHDB-375133

Non-uniqueness conjecture for axially symmetric swirl-free Navier-Stokes solutions

摘录 / EXCERPT: Let u solve the incompressible Navier-Stokes equations on $[0,1]\times\mathbb{R}^3$, ... For any $\alpha\in[0,1/2)$, consider initial data $u_0\in H^\alpha\cap L^{3,\infty}$ and so...

conjecture | open | MathDB 页面 | 原文线索

V2-038825 | MATHDB-375149

Classical randomized simulation conjecture for divergence-free quadratic ODEs

摘录 / EXCERPT: Let $\beta\geq 5/3$ and define ... Assume there is an efficient procedure for sampling from the probability distribution induced by the normalized tensor c/J. Consider the quadratic...

conjecture | open | MathDB 页面 | 原文线索

V2-038826 | MATHDB-375150

Conjecture on polynomial-time classical simulation with strong dissipation

摘录 / EXCERPT: Let u_0 be an observable depending only on the first k coordinates, so that ... Let λ_i be dissipation rates satisfying $\lambda_i\propto i^p$ for an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-038827 | MATHDB-375380

Global stability of the natural equilibrium in the persistence component of the SIT-free model

摘录 / EXCERPT: Global-stability conjecture. The natural equilibrium X_+^{**} is globally asymptotically stable relative to $\mathcal P_+$: $\mathcal P_+$ is positively invariant, X_+^{**} is...

conjecture | open | MathDB 页面 | 原文线索

V2-038828 | MATHDB-375456

Polynomial solvability conjecture for the formal asymptotic expansion

摘录 / EXCERPT: Let $P(t)=\sum_{j=0}^d\pi_jt^j$ be a polynomial satisfying ... and set $p_0(x)=x$. Let p_k denote polynomials whose coefficients are determined recursively by the formal expansion...

conjecture | open | MathDB 页面 | 原文线索

V2-038829 | MATHDB-375457

Nonnegative-coefficient conjecture for the formal asymptotic polynomials

摘录 / EXCERPT: Let $P(t)=\sum_{j=0}^d\pi_jt^j$ satisfy $\pi_0=0$ and $\pi_1=\pi_2$, and let $p_0(x)=x$. Assume that the formal expansion has polynomial coefficients p_k as in the polynomial sol...

conjecture | open | MathDB 页面 | 原文线索

V2-038830 | MATHDB-375458

Convergence conjecture for the formal asymptotic expansion

摘录 / EXCERPT: Let $P(t)=\sum_{j=0}^d\pi_jt^j$ satisfy $\pi_0=0$ and $\pi_1=\pi_2$, and set $p_0(x)=x$. Consider the formal expansion with recursively defined polynomial coefficients p_k, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-038831 | MATHDB-375757

Spectral-tree and mother-body conjecture for high-order Heun operators

摘录 / EXCERPT: Spectral-tree and mother-body conjecture. The measures μ_n converge weakly to a probability measure μ_Q depending only on Q_k. Its support is a finite planar tree contain...

conjecture | open | MathDB 页面 | 原文线索

V2-038832 | MATHDB-375758

Mother-body conjecture for the spectral limit of high-order Heun operators

摘录 / EXCERPT: Mother-body conjecture. The root-counting measures of Sp_n converge weakly to a positive measure μ_{Q_k} depending only on Q_k. Its support is a finite planar tree...

conjecture | open | MathDB 页面 | 原文线索

V2-038833 | MATHDB-375759

WKB Stokes-period conjecture for the spectral tree of high-order Heun operators

摘录 / EXCERPT: WKB Stokes-period conjecture. The support Γ_{Q_k} is

conjecture | open | MathDB 页面 | 原文线索

V2-038834 | MATHDB-375939

Perelman's nonexistence conjecture for rotated backward self-similar Navier-Stokes profiles

摘录 / EXCERPT: Perelman's conjecture. Every smooth rotated backward self-similar profile with $\alpha\neq 0$ and the stated decay is trivial.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038835 | MATHDB-376013

Multiplicity conjecture for solutions with finitely many modes

摘录 / EXCERPT: Multiplicity conjecture. For each $M\in\mathbb{N}_0$, there exists at least one distinct solution u_M of problem $$ having at most M modes.

conjecture | open | MathDB 页面 | 原文线索

V2-038836 | MATHDB-376169

The local characterization conjecture for P-Runge pairs of smooth Whitney jets

摘录 / EXCERPT: Let $P(D)$ be a hypoelliptic partial differential operator on $\mathbb{R}^d$ which is orthogonally degenerate on its linear characteristic cone ... Moreover, let $F_1\subset F_2$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-038837 | MATHDB-376583

The five-solution conjecture for 1-periodic solutions of bistable ODEs

摘录 / EXCERPT: Consider an ODE of the form of equation with a bistable function $f(y)$. Five-solution conjecture. Such an ODE has at most five 1-periodic solutions. This conjecture concerns the...

conjecture | open | MathDB 页面 | 原文线索

V2-038838 | MATHDB-376783

Darboux characterization conjecture for polynomial solutions of bilinear hypergeometric equations

摘录 / EXCERPT: Darboux characterization conjecture. Any polynomial solutions to the bilinear hypergeometric equations of propositions and are given by ... , respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-038839 | MATHDB-376858

Smooth representative conjecture for finite-energy homotopy sectors

摘录 / EXCERPT: Smooth representative conjecture. Every 2-homotopy sector of finite-energy maps contains a smooth representative; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-038840 | MATHDB-376859

Admissible representative conjecture for finite-energy lifts

摘录 / EXCERPT: Let M be a manifold, G a Lie group, $X=G/H$, and let φ be smooth. An admissible lift is a map u whose gauge potential $u^{-1}du$ satisfies the admissibility condition...

conjecture | open | MathDB 页面 | 原文线索

V2-038841 | MATHDB-376860

Non-Abelian gauge-fixing conjecture for L^p potentials

摘录 / EXCERPT: Non-Abelian gauge-fixing conjecture. There exists a gauge transformation $\lambda\in W^{1,p}(\operatorname{Ad}P)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038842 | MATHDB-376882

Strong convergence conjecture for fully nonlinear KdV-type equations

摘录 / EXCERPT: Consider the scalar conservation law ... Let u solve the fully nonlinear fourth-order approximation ... where f is strictly convex, g is concave, and $g(0)=0$. Strong convergence...

conjecture | open | MathDB 页面 | 原文线索

V2-038843 | MATHDB-377045

Eyink's anomalous-dissipation conjecture for viscosity solutions of 2D Euler

摘录 / EXCERPT: Let $\boldsymbol{u}^ u,\boldsymbol{ abla}\boldsymbol{\times}\boldsymbol{u}^ u=\boldsymbol{ abla}\times\boldsymbol{u}^ u$ be a sequence of 2D Navier-Stokes solutions satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-038844 | MATHDB-377096

Necessity conjecture for the CYBE solutions in the constructed ansatz

摘录 / EXCERPT: Necessity conjecture. All solutions of the classical Yang-Baxter equation are described by the preceding ansatz and differential-equation condition. The statement is presented as t...

conjecture | open | MathDB 页面 | 原文线索

V2-038845 | MATHDB-377165

Compactness and existence conjecture for conformally invariant curvature equations

摘录 / EXCERPT: Compactness and existence conjecture. If $A_g\in\Gamma_k^+$, then there exists a conformal deformation $\tilde{g}=e^{2u}g$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-038846 | MATHDB-377319

Generalized Wright conjecture for equations with negative Schwarzian

摘录 / EXCERPT: Consider the Wright delay differential equation ... Assume that all conditions (H) on f are satisfied. Generalized Wright conjecture. If $-f'(0)<\pi/2$, then ... for every solution...

conjecture | open | MathDB 页面 | 原文线索

V2-038847 | MATHDB-377327

Smith and Wright's global stability conjecture for scalar delay differential equations

摘录 / EXCERPT: Consider the scalar delay differential equation ... under conditions (W), with the trivial solution and its local asymptotic stability understood as in the paper. Smith and Wright's...

conjecture | open | MathDB 页面 | 原文线索

V2-038848 | MATHDB-377353

Local Sobolev regularity conjecture for Schrödinger operators under Assumption A

摘录 / EXCERPT: Let M be a Riemannian manifold, let E be a Hermitian vector bundle over M, and let V be a potential satisfying Assumption A: $V=V_{+}+V_{-}$, with $V_{+}(x)8e0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-038849 | MATHDB-377775

Self-similar solution-count conjecture for the nonlinear parabolic system

摘录 / EXCERPT: Self-similar solution-count conjecture. For fixed $\kappa>0$ and $n>4$, this boundary-value problem has $\nu(n)-2$ non-trivial solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-038850 | MATHDB-377881

The below-scaling ill-posedness conjecture for semilinear wave equations

摘录 / EXCERPT: For a semilinear wave equation, let $H^s \times H^{s-1}$ denote the Sobolev data space, and let s_0 be the scaling index determined by the equation's scaling. Below-scaling ill-p...

conjecture | open | MathDB 页面 | 原文线索

V2-038851 | MATHDB-377906

The cubic homogeneous nilpotent-Jacobian formulation of the Jacobian conjecture

摘录 / EXCERPT: Cubic nilpotent-Jacobian conjecture. If $JH(z)$ is nilpotent, then $N_t(z)$ is a polynomial solution in both t and z. This is presented as an equivalent formulation of the Jaco...

conjecture | open | MathDB 页面 | 原文线索

V2-038852 | MATHDB-377916

Grothendieck-Katz and Bombieri-Dwork conjectures for differential systems

摘录 / EXCERPT: Grothendieck-Katz and Bombieri-Dwork conjectures. The Lie algebra of the differential Galois group of D is minimal with the property that, for almost all primes $\mathfrak p$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-038853 | MATHDB-378541

Uniqueness conjecture for the solution attaining the extremal minimum $\Xi_{min}(x_0)$

摘录 / EXCERPT: Uniqueness conjecture. For any x_0, the solution with a pole at x_0 and minimum at $\Xi_{min}(x_0)$ is unique. The claim is presented as a numerical and heuristic extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-038854 | MATHDB-378543

Pointwise ordering conjecture for the maximal and minimal Painlevé I solutions

摘录 / EXCERPT: Let $y_{max}(x_0;x)$ and $y_{min}(x_0;x)$ be the two real solutions with pole at $x_0\leq0$, with x in the interval of existence of $y_{min}(x_0;x)$. Ordering conjecture. ... The i...

conjecture | open | MathDB 页面 | 原文线索

V2-038855 | MATHDB-378544

Comparison conjecture for the extremal locations X and X_{min}

摘录 / EXCERPT: Let $x_1<x_2<0$, and let $X(x)$ and $X_{min}(x)$ denote the corresponding extremal location functions for real solutions of the first Painlevé equation. Comparison conjecture. ... Th...

open_problem | open | MathDB 页面 | 原文线索

V2-038856 | MATHDB-378612

S.G. Krein's conjecture on dissipative two-point boundary value problems

摘录 / EXCERPT: A two-point boundary value problem (BVP) for an ordinary differential operator is called dissipative when its associated operator is dissipative, and Birkhoff-regular when its boun...

conjecture | open | MathDB 页面 | 原文线索

V2-038857 | MATHDB-378973

Minimality conjecture for the ground state of a critical parabolic operator

摘录 / EXCERPT: Let P be a critical operator in a Riemannian manifold $\mathcal{M}$, and let φ denote its ground state. Let $\mathcal{H}_P(\mathcal{M}\times\mathbb{R})$ be the cone of po...

conjecture | open | MathDB 页面 | 原文线索

V2-038858 | MATHDB-379363

Persistence of global hypoellipticity failure for worm domains

摘录 / EXCERPT: Let D be a worm domain whose $\overline\partial$-Neumann problem is not globally hypoelliptic. Let A be its boundary annulus, let $V\subset\mathbb C^2$ be an arbitrarily small...

conjecture | open | MathDB 页面 | 原文线索

V2-038859 | MATHDB-379398

Conjecture on unique solvability and guided cycles

摘录 / EXCERPT: Unique-solvability conjecture. The boundary value problem is uniquely solvable if and only if $\mathcal{N}^\Omega_\zeta=\emptyset$.

conjecture | open | MathDB 页面 | 原文线索

V2-038860 | MATHDB-379858

A priori bounds for the almost-complex Calabi-Yau equation

摘录 / EXCERPT: Let X be a compact 4-manifold, let Ω be a symplectic form on X, and let $\mathcal{P}$ be a constraint manifold defined by an almost-complex structure tamed by Ω...

conjecture | open | MathDB 页面 | 原文线索

V2-038861 | MATHDB-379859

A priori bounds for asymptotic unimodular constraint manifolds

摘录 / EXCERPT: Let X be a compact 4-manifold, let Ω be a symplectic form on X, and let $\mathcal{P}$ be a unimodular constraint manifold with negative chords that is asymptotic to a...

conjecture | open | MathDB 页面 | 原文线索

V2-038862 | MATHDB-380002

Necessary and sufficient condition for convergence of the Magnus series

摘录 / EXCERPT: Let $Y(t;\kappa)$ denote the solution of ... Denote the eigenvalues of $Y(t;\kappa)$ by $\lambda_n(t;\kappa)$, ordered so that each λ_n is continuous as a function of t. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-038863 | MATHDB-380276

Local well-posedness conjecture for cubic NLS below L^2

摘录 / EXCERPT: Consider the cubic nonlinear Schrödinger equation referred to as, with initial data in the Sobolev space H^s. Local well-posedness conjecture. The cubic NLS equation is locally w...

conjecture | open | MathDB 页面 | 原文线索

V2-038864 | MATHDB-380459

Nonexistence of centers for polynomial differential equations without linear terms

摘录 / EXCERPT: Let $N\geq 3$, let $A_j(t)$, for $j=2,3,\ldots,N-1$, be polynomial functions, and consider the differential equation ... A solution is a center at the origin if nearby solutions have...

conjecture | open | MathDB 页面 | 原文线索

V2-038865 | MATHDB-380699

Small-data critical well-posedness conjecture for one-dimensional wave maps

摘录 / EXCERPT: Small-data critical well-posedness conjecture. The Cauchy problem is well-posed for small data in $\dot H^{\frac{n}{2}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-038866 | MATHDB-381025

Compactness conjecture for heteroclinic orbits modulo time translation

摘录 / EXCERPT: Let $\mathcal{H}$ be the space of heteroclinic orbits of the semilinear parabolic equation, and let Y_λ denote the ambient space in which these orbits are considered. Ident...

conjecture | open | MathDB 页面 | 原文线索

V2-038867 | MATHDB-381170

Asymptotic expansion conjecture for stationary Navier-Stokes flows with non-zero net force

摘录 / EXCERPT: Asymptotic expansion conjecture. For a large class of boundary conditions $\mathbf{u}^{*}$, respectively source terms $\boldsymbol{f}$, there exists a solution of the body problem,...

conjecture | open | MathDB 页面 | 原文线索

V2-038868 | MATHDB-381250

Nonexistence conjecture for the self-similar boundary value problem

摘录 / EXCERPT: Let g satisfy the self-similar boundary value problem referred to as, with the boundary conditions described in the preceding discussion, including $g'(0)<0$. Nonexistence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-038869 | MATHDB-381257

Large-domain decay conjecture for mixed normal-superconducting states

摘录 / EXCERPT: Let Ω be the domain, let B_n be the normal-state magnetic field, and let Proposition provide the conditions under consideration. For a compact set...

conjecture | open | MathDB 页面 | 原文线索

V2-038870 | MATHDB-381314

Supercritical degenerate Hopf-bifurcation conjecture for car-following wavefronts

摘录 / EXCERPT: Supercritical degenerate Hopf-bifurcation conjecture. At the parameter value $h=h_H$ and solution $w=w^{c_1(h_H),d}$, the canonical equation, and hence the original car-following e...

conjecture | open | MathDB 页面 | 原文线索

V2-038871 | MATHDB-381348

Jones' uniqueness conjecture for slowly oscillating periodic solutions of Wright's equation

摘录 / EXCERPT: Let ... with parameter $\alpha>\tfrac{\pi}{2}$. A slowly oscillating periodic solution (SOPS) is a periodic solution $y(t)$ for which, up to time translation, there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-038872 | MATHDB-381356

Existence of a nontrivial slowly oscillating periodic solution for the cyclic delay system

摘录 / EXCERPT: Periodic-solution conjecture. Under these assumptions, system $$ has a nontrivial slowly oscillating periodic solution.

conjecture | open | MathDB 页面 | 原文线索

V2-038873 | MATHDB-381381

Inviscid K41-Onsager conjecture for globally dissipative Euler flows

摘录 / EXCERPT: Let I be an open interval, and let (v,p) be a weak solution of the incompressible Euler equations on $I \times \mathbb{T}^3$. Assume that $v \in L_t^\infty C_x^{1/3}$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-038874 | MATHDB-381392

Monotonicity conjecture for the period length of SOPS to Wright's equation

摘录 / EXCERPT: Let x be a slowly oscillating periodic solution (SOPS) of Wright's equation ... with parameter α and period length $L(\alpha)$. Monotonicity conjecture. The period length of...

conjecture | open | MathDB 页面 | 原文线索

V2-038875 | MATHDB-381414

The stability conjecture for Allen-Cahn equations

摘录 / EXCERPT: Stability conjecture. If $n\leq 7$, then all level sets of u are hyperplanes.

conjecture | open | MathDB 页面 | 原文线索

V2-038876 | MATHDB-381492

Nonexistence conjecture for the Lane-Emden system in strictly starshaped domains

摘录 / EXCERPT: Let Ω be a bounded domain, strictly starshaped with respect to $0\in\Omega$, and suppose the condition ... . Consider the problem with $u=v=0$ on $\partial\Omega$. Lane-Emden...

conjecture | open | MathDB 页面 | 原文线索

V2-038877 | MATHDB-381526

Multiplicity and branch-shape conjecture for the semilinear Dirichlet problem in case VI

摘录 / EXCERPT: Multiplicity and branch-shape conjecture. The branches of solutions behave as depicted in the case-VI bifurcation diagram: the problem has at least two solutions for every...

conjecture | open | MathDB 页面 | 原文线索

V2-038878 | MATHDB-381527

Branch-shape and uniqueness conjecture for the semilinear Dirichlet problem in case VII

摘录 / EXCERPT: Branch-shape and uniqueness conjecture. The solution branches in cases VII2-1 and VII2-2 of Theorem VII1 have the same behaviour as the branches depicted in the case-IIb and case-V...

conjecture | open | MathDB 页面 | 原文线索

V2-038879 | MATHDB-381542

The nonexistence conjecture for proper complex-valued $(p,1)$-harmonic morphisms

摘录 / EXCERPT: Let $p\ge 2$ and let $\phi:U\to\mathbb{C}$ be a complex-valued $(p,1)$-harmonic morphism defined locally on the standard Euclidean space $\mathbb{R}^{2p-1}$. The nonexistence conje...

conjecture | open | MathDB 页面 | 原文线索

V2-038880 | MATHDB-381590

Large-shear chaos conjecture for stochastically perturbed Hopf systems

摘录 / EXCERPT: Consider the random dynamical system induced by the stochastic differential equation, and fix $a>0$ and $\beta\in\mathbb{R}$. Let Λ_1 denote its largest Lyapunov exponent....

conjecture | open | MathDB 页面 | 原文线索

V2-038881 | MATHDB-381701

Uniqueness conjecture for Trudinger-Moser maximizers on balls

摘录 / EXCERPT: Uniqueness conjecture. If $\alpha=4\pi$ and Ω is a ball, then the maximizers of the Trudinger-Moser inequality are unique.

conjecture | open | MathDB 页面 | 原文线索

V2-038882 | MATHDB-381711

Hypoellipticity characterization for symmetric degenerate elliptic operators

摘录 / EXCERPT: Hypoellipticity characterization. The operator L is locally C^∞-hypoelliptic if and only if both L_1 and L_2 satisfy the superlogarithmic estimate.

conjecture | open | MathDB 页面 | 原文线索

V2-038883 | MATHDB-381759

Kinetic Poincaré inequality for higher-order Kolmogorov equations

摘录 / EXCERPT: Let $k\geq 1$, $d\geq 1$, and let $u=u(t,x_1,\dots,x_k,v)$ be a nonnegative weak subsolution of ... Set ... ... and define...

conjecture | open | MathDB 页面 | 原文线索

V2-038884 | MATHDB-381800

Hörmander nondegeneracy conjecture for irreversible drift

摘录 / EXCERPT: Hörmander nondegeneracy conjecture. If the parabolic Hörmander condition holds everywhere, then it cannot be the case that $b(x)\in\operatorname{Im}\sigma(x)$ for any x; equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-038885 | MATHDB-381833

Uniqueness conjecture for strong-KPP steady states on bounded Lipschitz domains

摘录 / EXCERPT: Uniqueness conjecture. The problem admits at most one solution.

conjecture | open | MathDB 页面 | 原文线索

V2-038886 | MATHDB-381946

Conjecture on the Kotake-Narasimhan theorem for ultradifferentiable structures

摘录 / EXCERPT: Let $\mathcal{U}$ be an ultradifferentiable structure that is closed under derivation and invariant under composition with mappings of class $\mathcal{U}$. The Kotake-Narasimhan th...

conjecture | open | MathDB 页面 | 原文线索

V2-038887 | MATHDB-381995

Linearizability conjecture for the Riccati chain

摘录 / EXCERPT: Linearizability conjecture. For $n\geq 3$, the equation is linearizable if and only if $F_y=0$; in that case, it is already a linear equation. The claim extends the verified analys...

conjecture | open | MathDB 页面 | 原文线索

V2-038888 | MATHDB-382184

Pointwise positivity conjecture for the second- and fourth-order curvature quantities

摘录 / EXCERPT: Pointwise positivity conjecture. The estimates

conjecture | open | MathDB 页面 | 原文线索

V2-038889 | MATHDB-382270

Convergence of inflated dynamic Laplacian minimisers to material solutions

摘录 / EXCERPT: Let M be the spatial manifold, let $[0,\tau]\times M$ be the spacetime manifold $\mathbb{M}_0$, and for each $a\geq 0$ let S_a and H_a denote the Sobolev and Cheeger constant...

conjecture | open | MathDB 页面 | 原文线索

V2-038890 | MATHDB-382294

The Dirichlet-Neumann eigenvalue gap conjecture for bounded Euclidean domains

摘录 / EXCERPT: Let Ω be a bounded domain in $\mathbb{R}^{n}$, and let $\lambda_k(\Omega)$ and $\mu_k(\Omega)$ denote its Dirichlet and Neumann eigenvalues, respectively. For positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-038891 | MATHDB-382319

Strong monotonicity conjecture for the glacier surface motion map

摘录 / EXCERPT: Let $\textnormal{r}>2$ be such that the Lipschitz continuity conjecture holds, set $\mathcal{X}=W^{1,\textnormal{r}}(\Omega)$, fix $b\in\mathcal{X}$, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-038892 | MATHDB-382459

Positivity criterion for the Green's function of the reflected piecewise-constant problem

摘录 / EXCERPT: Positivity conjecture. The Green's function $H_{m,M}$ is positive if and only if $M\in(-m,\lambda_1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-038893 | MATHDB-382525

On blowup for large data in equivariant wave maps

摘录 / EXCERPT: On blowup for large data. For initial data with sufficiently large energy, the solutions of equation blow up in finite time in the sense that the derivative $u_r(t,0)$ diverges as...

conjecture | open | MathDB 页面 | 原文线索

V2-038894 | MATHDB-382559

Non-existence conjecture for regular solutions of the system $\mathcal{E}^{\gamma,\delta}_{k,\ell}$

摘录 / EXCERPT: Let γ and δ be real analytic functions with $\gamma_4\neq0$, let k and ℓ be integers, and let $\mathcal{E}^{\gamma,\delta}_{k,\ell}$ denote the partial differe...

conjecture | open | MathDB 页面 | 原文线索

V2-038895 | MATHDB-382872

Chan-Wei radiality conjecture

摘录 / EXCERPT: For every admissible integer $n\ge 12$ and every exponent p satisfying $p_{\mathrm{JL}}(n)<p<p_{\mathrm{JL}}(n-1)$, every positive smooth entire solution...

conjecture | solved | MathDB 页面 | 原文线索

V2-038896 | MATHDB-383167

3D Euler regularity and singularity problem

摘录 / EXCERPT: For the incompressible Euler equations on boundary-free three-dimensional space, $\partial_t u+(u\cdot\nabla)u=-\nabla p$, $\nabla\cdot u=0$ on $\mathbb{R}^3\times[0,\infty)$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-038897 | MATHDB-383222

Pólya's conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{d}$ be a bounded domain, and let $\lambda_k(\Omega)$ and $\mu_k(\Omega)$ denote its k-th Dirichlet and Neumann Laplacian eigenvalues, respectively....

conjecture | special_case | MathDB 页面 | 原文线索

V2-038898 | MATHDB-383226

Brezis Open Problem 2.2

摘录 / EXCERPT: Let $B_1\subset\mathbb{R}^2$ be the unit disk, let $\varepsilon>0$, and define the Ginzburg--Landau energy...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038899 | MATHDB-383243

Delaunay-solution and singular-classification problems for the critical boundary equation

摘录 / EXCERPT: Let $N\ge 3$, let $\mathbb{R}^N_+=\{x\in\mathbb{R}^N:x_N>0\}$, and let $0\in\partial\mathbb{R}^N_+$. Consider positive solutions of the critical problem ... with an isolated singular...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038900 | MATHDB-383246

Close the strip: type-I Navier-Stokes blowup and the direction-regularity exponent in [-1/2, 0]

摘录 / EXCERPT: Setting. For a Navier-Stokes velocity field with vorticity ω, direction $\xi = \omega/|\omega|$, and peak set $P(t)$ (the high-vorticity region), define the scale-invariant...

open_problem | open | Justin Hill | MathDB 页面 | 原文线索

V2-038901 | MATHDB-383262

Open $C^{1,\alpha}$ regularity problem for singular fractional p-Laplacian equations

摘录 / EXCERPT: Let $n\ge 2$, $1<p<2$, and let $\alpha_{\rm loc}(n,p)>0$ be an admissible interior Hölder exponent for gradients of local p-harmonic functions. Is it true that, for every...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038902 | MATHDB-383388

Magnanini's conjecture on critical points of the torsion function

摘录 / EXCERPT: For every bounded simply-connected domain $\Omega\subset\mathbb{R}^2$, let u be its torsion function, defined by $-\Delta u=1$ in Ω and $u=0$ on $\partial\Omega$, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038903 | MATHDB-383498

Levinson cavity asymptotics problem

摘录 / EXCERPT: For the three-dimensional axially symmetric steady incompressible cavity-flow free-boundary problem, determine whether the cavity solutions whose existence was established by Garab...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038904 | MATHDB-384522

Sobolev inequalities (problem 1)

摘录 / EXCERPT: Assuming they exist, how do the level sets of the extremals of the Sobolev inequality ... vary as p varies?

open_problem | open | MathDB 页面 | 原文线索

V2-038905 | MATHDB-384607

Rigorous derivation of a Hopf bifurcation in Reynolds number for 2D incompressible Navier--Stokes flow or Gross-Pitaevskii flow passing a (cylindrical) obstacle.

摘录 / EXCERPT: Rigorous derivation of a Hopf bifurcation in Reynolds number for 2D incompressible Navier--Stokes flow or Gross-Pitaevskii flow passing a (cylindrical) obstacle. Can it be proven w...

open_problem | open | MathDB 页面 | 原文线索

V2-038906 | MATHDB-384610

Study the inviscid limit of the non-unique Leray solution to the forced Navier--Stokes equation.

摘录 / EXCERPT: Study the inviscid limit of the non-unique Leray solution to the forced Navier--Stokes equation.

open_problem | open | MathDB 页面 | 原文线索

V2-038907 | MATHDB-384614

Study the large-time dynamics of 2D Navier--Stokes with high modes forcing.

摘录 / EXCERPT: Study the large-time dynamics of 2D Navier--Stokes with high modes forcing.

open_problem | open | MathDB 页面 | 原文线索

V2-038908 | MATHDB-384616

Investigate global asymptotic behavior of Magnetic relaxation equations (MRE): ... with connections to the Magnetohydrodynamics (MHD) equation and the Euler equation.

摘录 / EXCERPT: Investigate global asymptotic behavior of Magnetic relaxation equations (MRE): ... with connections to the Magnetohydrodynamics (MHD) equation and the Euler equation.

open_problem | open | MathDB 页面 | 原文线索

V2-038909 | MATHDB-385422

Monge-Ampère equations (problem 5)

摘录 / EXCERPT: (textit{Conjecture of Nadirashvili-Tkachev-Vlu{a}duc{t}}) A uniformly elliptic equation in divergence form admits a solution in $W^{1,p}$ for some $p>1$.

open_problem | open | MathDB 页面 | 原文线索

V2-038910 | MATHDB-386168

Sampling high dimension distributions

摘录 / EXCERPT: Find a measure μ supported on a hypersurface $S \subset \mathbb{R}^n$ and a nonlinear, degenerate, parabolic equation for which solutions converge to μ. Remark: This should...

open_problem | open | MathDB 页面 | 原文线索

V2-038911 | MATHDB-386174

Global well-posedness for 3D incompressible axisymmetric Navier-Stokes equation

摘录 / EXCERPT: Prove global well-posedness for 3D incompressible axisymmetric Navier-Stokes for large initial data.

open_problem | open | MathDB 页面 | 原文线索

V2-038912 | MATHDB-386334

How do we use the Wasserstein geometry on RMBs to approximation PDE solutions?

摘录 / EXCERPT: How do we use the Wasserstein geometry on RMBs to approximation PDE solutions?

open_problem | open | MathDB 页面 | 原文线索

V2-038913 | MATHDB-386941

2D Euler equation - From graph-like interface to splash

摘录 / EXCERPT: Find a computer-assisted proof that the 2D Euler equation can develop a splash when starting from a graph-like structure.

open_problem | open | MathDB 页面 | 原文线索

V2-038914 | MATHDB-387579

Research Problems in Function Theory - Problem 2.70

摘录 / EXCERPT: Let H be an entire function, let f_1, f_2 be linearly independent solutions of the differential equation $w'' + Hw = 0$, and let $E = f_1 f_2$. Clearly f_1, f_2 and E are...

open_problem | open | MathDB 页面 | 原文线索

V2-038915 | MATHDB-387581

Research Problems in Function Theory - Problem 2.72

摘录 / EXCERPT: Let $\{f_1,\ldots,f_n\}$ be a fundamental system for the differential equation ... where $a_o,...,a_{n-1}$ are polynomials. Frank has proved that each function $f_1,\ldots,f_n$ has f...

open_problem | open | MathDB 页面 | 原文线索

V2-038916 | MATHDB-390504

Verón's problem for p-Laplace-type equations

摘录 / EXCERPT: Let (M^n,g) be a closed Riemannian manifold with $\operatorname{Ric}_g\geq (n-1)g$. For $1<p<n$, $p<q<p^*:=\frac{np}{n-p}$, and $\lambda>0$, determine when every positive solutio...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038917 | MATHDB-391848

Troesch problem

摘录 / EXCERPT: For a given sensitivity parameter $\lambda>0$, determine a function $u\in C^2([0,1])$ satisfying the nonlinear boundary-value problem $u''(x)=\lambda\sinh(\lambda u(x))$ for...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038918 | MATHDB-391900

Kwaśnicki's uniform eigenfunction bound conjecture and the Kaleta-Kwaśnicki-Małecki conjectural remainder

摘录 / EXCERPT: Let $\alpha\in(0,2)$, and let $\{\varphi_n^{(\alpha)}\}_{n\geq 1}$ be an $L^2((-1,1))$-normalized sequence of eigenfunctions of the fractional Laplacian on $(-1,1)$ with the approp...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038919 | MATHDB-391908

Girela-Peláez conjecture

摘录 / EXCERPT: The supplied sources identify the Girela-Peláez conjecture as a characterization problem for function classes induced by the Möbius-invariant Laplacian operator on the unit ball, w...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038920 | MATHDB-391915

Stability of generic inflow wave patterns for the one-dimensional Navier-Stokes-Fourier system

摘录 / EXCERPT: For the one-dimensional Navier-Stokes-Fourier system on the half-line with inflow boundary data in the subsonic region, consider the composite wave consisting of a degenerate bound...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038921 | MATHDB-391921

Rivière's regularity question for n-Laplace systems

摘录 / EXCERPT: For $n\geq 2$ and $m\geq 1$, let $u\in W^{1,n}(\mathbb{B}^n,\mathbb{R}^m)$ be a weak solution of...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038922 | MATHDB-391934

Critical orthonormal Strichartz estimate problem

摘录 / EXCERPT: For $d\ge 2$ and parameters q,r,s in the scaling range $\frac{2}{q}+\frac{d}{r}=\frac{d}{2}-s$ with $2<q<\frac{r}{d'}$ and $d'=\frac{d}{d-1}$, determine whether there is a consta...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-038923 | MATHDB-392096

Global minimality and rigidity of the constraint map vortex

摘录 / EXCERPT: Let $3\le n\le 6$ and $0<a<1$. Consider the variational problem...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038924 | MATHDB-392098

Minimax spectral-density estimation under local differential privacy for dependent Gaussian data

摘录 / EXCERPT: Let $X_1,\ldots,X_N$ be observations from a centered stationary Gaussian process with spectral density f belonging to a Sobolev-type class $\mathcal F_s$, and let the observation...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038925 | MATHDB-392111

Rivière’s regularity question for critical n-Laplace systems with antisymmetric potentials

摘录 / EXCERPT: For every $n>2$, let $U\in W^{1,n}(B^n,\mathbb{R}^{n+2})$ and let $\Omega\in L^n(B^n,\mathfrak{so}(n+2)\otimes\mathbb{R}^n)$ satisfy...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038926 | MATHDB-392118

Strong maximum principle conjecture for higher-order GJMS operators

摘录 / EXCERPT: For every closed Riemannian manifold (M^n,g) with $n\ge 7$, if $Q_g^{(2)}>0$, $Q_g^{(4)}>0$, $Q_g^{(6)}>0$, and the sixth-order GJMS operator $P_{6,g}$ is strictly positive as a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038927 | MATHDB-392123

Cazenave’s open problem for energy-critical complex Ginzburg-Landau equations

摘录 / EXCERPT: Determine whether the energy-critical complex Ginzburg--Landau initial-value problem admits global regularity in the focusing and defocusing undamped aligned cases in dimensions...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038928 | MATHDB-392282

Bernstein problem for the affine maximal type equation

摘录 / EXCERPT: For $n\ge 2$, let $f\in C^\infty(\mathbb{R}^n)$ be locally uniformly convex, so that D^2f is positive definite, and suppose its graph is complete with respect to the induced Eucl...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-038929 | MATHDB-392467

Higher-Dimensional Nitsche Conjecture

摘录 / EXCERPT: For every $n\ge 3$, let $0<r<1$ and let $h:\mathbb{A}(r,1)\to\mathbb{A}(R,1)\subset\mathbb{R}^n$ be an onto homeomorphism whose coordinate functions are harmonic, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-038930 | MATHDB-392480

Sharp-interface limit of coupled Allen-Cahn equations

摘录 / EXCERPT: For a suitable vector-valued coupled Allen-Cahn evolution with diffuse-interface parameter $\varepsilon\to0$, prove that the associated diffuse interfaces converge globally and unc...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038931 | MATHDB-392486

Maz’ya’s Problem 8 on the capacitary-distance Hardy inequality

摘录 / EXCERPT: Let $n\ge 3$. For every open set $\Omega\subset\mathbb{R}^n$, set $F:=\mathbb{R}^n\setminus\Omega$ and, for $0<\alpha\le 1$ and $x\in\Omega$, define...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-038932 | MATHDB-392618

MEMS problem

摘录 / EXCERPT: Let $1\le n\le 6$, let $\Omega\subset\mathbb{R}^n$ be a domain, and let $f:[0,1)\to(0,+\infty)$ be nondecreasing and convex, with $\lim_{t\to1^-}f(t)=+\infty$ and...

open_problem | partial | MathDB 页面 | 原文线索

V2-038933 | MATHDB-392659

Renormalized entropy solutions for quasilinear anisotropic degenerate parabolic equations

摘录 / EXCERPT: For the Cauchy problem ... with $(t,x)\in(0,T)\times\mathbb{R}^d$, fixed $T>0$, data $u_0\in L^1(\mathbb{R}^d)$ and $F\in L^1((0,T)\times\mathbb{R}^d)$, diffusion matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-038934 | MATHDB-392662

Global well-posedness of periodic KdV in low-regularity Sobolev spaces

摘录 / EXCERPT: For every ... is globally [1mC^0[0m-well-posed on [1m$H^\alpha(\mathbb{T})$[0m: for every [1mq[0m the solution exists for all time, and the solution map [1m...

open_problem | open | MathDB 页面 | 原文线索

V2-038935 | MATHDB-393127

Criterion for the Liouville theorem for periodic elliptic equations

摘录 / EXCERPT: Is it possible to determine for a given periodic elliptic equation whether the Liouville theorem holds?

open_problem | open | MathDB 页面 | 原文线索

V2-038936 | MATHDB-393336

Feedback stabilization of the two-dimensional Navier-Stokes equation by a Riccati-based boundary feedback

摘录 / EXCERPT: For a bounded, connected domain $\Omega\subset\mathbb{R}^{2}$ with regular boundary, an unstable stationary solution (w,χ), and a boundary restriction operator M localizing...

open_problem | open | MathDB 页面 | 原文线索

V2-038937 | MATHDB-393363

Uniqueness of the invariant measure under full mode propagation

摘录 / EXCERPT: If $Z_1=\\mathbb{Z}^2_0$, then the two-dimensional stochastic Navier-Stokes equation (1) has a unique invariant measure in L^2.

open_problem | open | MathDB 页面 | 原文线索

V2-038938 | MATHDB-393377

Uniqueness theorem for 2BSDEs and fully nonlinear parabolic PDEs

摘录 / EXCERPT: Let ... with terminal condition ... Suppose that this nonlinear PDE and its terminal condition satisfy comparison in the sense of viscosity solutions. Assume further that f is ellipt...

open_problem | open | MathDB 页面 | 原文线索

V2-038939 | MATHDB-393430

Forcing stationary braid classes in uniformly parabolic dynamics

摘录 / EXCERPT: Given a braid v fixed by a particular uniformly parabolic PDE, which other classes of braids u are forced to exist as stationary curves?

open_problem | open | MathDB 页面 | 原文线索

V2-038940 | MATHDB-393515

Stability of stationary profiles for plane Kolmogorov flow

摘录 / EXCERPT: The stability of the profiles with respect to the PDE problem (1) is not understood yet.

open_problem | open | MathDB 页面 | 原文线索

V2-038941 | MATHDB-393940

Minimisation conditions versus phase conditions for warped MPDAE initial-boundary value problems

摘录 / EXCERPT: For initial-boundary value problems for warped multirate partial differential-algebraic equations, can conditions based on minimisation compete with smooth phase conditions?

open_problem | open | MathDB 页面 | 原文线索

V2-038942 | MATHDB-394029

Attainability of variational solutions for degenerate elliptic equations

摘录 / EXCERPT: Let $\rho\ge 0$ on $\mathbb R^d$, with $\rho,\rho^{-1}\in L^1_{\mathrm{loc}}(\mathbb R^d)$, and let A be a measurable symmetric matrix satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-038943 | MATHDB-394224

Existence and behavior of weakly regular vortex-sheet solutions

摘录 / EXCERPT: Determine whether solutions of the Birkhoff-Rott equation exist in spaces less regular than $C^{1+\nu}(\mathbb{R})$, for $\nu>0$, or than Sobolev spaces $H^s(\mathbb{R})$, for...

open_problem | open | MathDB 页面 | 原文线索

V2-038944 | MATHDB-394231

Dispersive blow-up and rogue-wave formation for the full Euler equations

摘录 / EXCERPT: Does the dispersive-blow-up theory established for the KdV equation and Boussinesq systems, which relies on their unbounded group and phase velocities, pertain to the development o...

open_problem | open | MathDB 页面 | 原文线索

V2-038945 | MATHDB-394256

Critical Sobolev-limit behavior of heat kernel measures on loop groups

摘录 / EXCERPT: In the setup with $X=S^1$, K a simply connected compact Lie group, G its complexification, and LG the formal completion of the G-loop space, conjecturally ... where the inner...

open_problem | open | MathDB 页面 | 原文线索

V2-038946 | MATHDB-394257

Critical Sobolev-degree limit of heat kernel measures on loop groups

摘录 / EXCERPT: For the Sobolev inclusion $H=W^s(S^1,K)\subset G=C_0(S^1,K)$, with K a simply connected compact Lie group, conjecturally $\nu^{H\subset G}_t\to\widetilde\mu_l$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-038947 | MATHDB-394737

Small-data global well-posedness for KP-II in the critical homogeneous Sobolev space

摘录 / EXCERPT: There exists a space $\dot{Z}^{-\frac12}([0,\infty))\subset C([0,\infty);\dot{H}^{-\frac12,0}(\mathbb{R}^2))$ and $\delta>0$ such that, for every initial datum...

open_problem | open | MathDB 页面 | 原文线索

V2-038948 | MATHDB-394741

Global well-posedness for the one-dimensional defocusing quintic NLS at and below the energy regularity

摘录 / EXCERPT: For the defocusing quintic nonlinear Schrödinger Cauchy problem ... is the problem globally well-posed in time for every initial datum in the Sobolev spaces $H^s(\mathbb R)$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-038949 | MATHDB-394830

Existence of time-periodic motions of a rigid body in a Navier-Stokes liquid

摘录 / EXCERPT: Consider a mechanical system $\{S,L\}$ consisting of a rigid body S moving in a Navier-Stokes liquid L that fills the three-dimensional region exterior to S. Assume that the...

open_problem | open | MathDB 页面 | 原文线索

V2-038950 | MATHDB-394831

Well-posedness of a fish-like swimming fluid-solid system

摘录 / EXCERPT: The initial and boundary value problem obtained by coupling the Navier-Stokes equations for the fluid to Newton’s law for the creature is well-posed in Sobolev-type spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-038951 | MATHDB-394832

Existence of time-periodic weak motions of a rigid body in a Navier-Stokes liquid

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be the domain exterior to a rigid body S. Let $F,M\in C([0,T];\mathbb{R})$ satisfy $F(0)=F(T)$ and $M(0)=M(T)$. Then there exists at least one T...

open_problem | open | MathDB 页面 | 原文线索

V2-038952 | MATHDB-395157

Characterization of Segal-Bargmann images of Sobolev spaces on compact symmetric spaces

摘录 / EXCERPT: Let $X=U/K$ be a compact Riemannian symmetric space, let $X_C=U_C/K_C$ be its complexification, and let H_t denote the heat-kernel (Segal-Bargmann) transform. For every integer...

open_problem | open | MathDB 页面 | 原文线索

V2-038953 | MATHDB-395343

Possible monodromy and differential Galois groups of accessory-parameter-free differential equations

摘录 / EXCERPT: Which kinds of monodromy groups or differential Galois groups can appear among irreducible differential equations that are free from accessory parameters and are obtained from one-...

open_problem | open | MathDB 页面 | 原文线索

V2-038954 | MATHDB-395446

Global existence of weak solutions for multidimensional isentropic Euler equations

摘录 / EXCERPT: Establish global existence of weak solutions for the multidimensional system of isentropic Euler equations.

open_problem | open | MathDB 页面 | 原文线索

V2-038955 | MATHDB-395499

Infinite-horizon zonal temperature LQR control problem

摘录 / EXCERPT: Find a control u minimizing ... where $R>0$, $r(t)$ is the desired average temperature, and ... subject to the thermal PDE ... For the discussion in the report, $r(t)=0$. Under suitabl...

open_problem | open | MathDB 页面 | 原文线索

V2-038956 | MATHDB-395585

Invariant manifolds of Euler equations

摘录 / EXCERPT: For the Euler equation on a bounded domain ... with phase space ... let $v_0\in H^{k+4}(\Omega)$ be a steady state and let L be the linearized Euler operator at v_0. If there are c...

open_problem | open | MathDB 页面 | 原文线索

V2-038957 | MATHDB-395913

Leray’s problem for stationary Navier-Stokes equations with inhomogeneous boundary data

摘录 / EXCERPT: For the stationary Navier-Stokes equations in three-dimensional bounded domains whose boundary consists of several connected surfaces, existence is known when the prescribed bounda...

open_problem | open | MathDB 页面 | 原文线索

V2-038958 | MATHDB-396614

Local well-posedness for the KdV equation in generalized Sobolev spaces

摘录 / EXCERPT: The Cauchy problem for the KdV equation is locally well-posed in $\widehat H_s^r(\mathbb R)$ if $1<r<2$ and ... where $\frac1r+\frac1{r'}=1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-038959 | MATHDB-396790

Composition Conjecture for the Abel differential equation

摘录 / EXCERPT: Let $A(\theta)$ and $B(\theta)$ be trigonometric polynomials, and consider ... If this Abel differential equation has a center, then there exist a periodic function $u(\theta)$ and f...

conjecture | open | MathDB 页面 | 原文线索

V2-038960 | MATHDB-396891

Necessity of the integral representation for a non-constant-coefficient parabolic equation

摘录 / EXCERPT: Is the explicit integral representation of solutions to the ill-posed linear parabolic equation with non-constant coefficients also necessary, in addition to being sufficient?

open_problem | open | MathDB 页面 | 原文线索

V2-038961 | MATHDB-396985

Unbounded Sobolev-norm growth for the cubic defocusing NLS on the torus

摘录 / EXCERPT: For the cubic defocusing nonlinear Schrödinger equation $i\partial_tu=-\Delta u+\varepsilon|u|^2u$ with $u(t,x)\in\mathbb C$, $x\in\mathbb T^d$, and $\varepsilon>0$ small, it is co...

open_problem | open | MathDB 页面 | 原文线索

V2-038962 | MATHDB-397186

Sobolev regularity of L^2 best rank-r tensor-product approximations

摘录 / EXCERPT: Let $f\in H^s(\mathbb{R}^{n+m+l})$ and suppose that $\operatorname{rank}f\ge r$. Assume that, for $k=1,2,\ldots,r$, $u_k\in L^2(\mathbb{R}^n)$, $v_k\in L^2(\mathbb{R}^m)$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-038963 | MATHDB-397378

Small ramification of the quantum differential equation of a Fano manifold

摘录 / EXCERPT: Conjecture. If X is a Fano manifold, then $\\operatorname{Sol}\\widehat{Q}_X$ has small ramification. More precisely, ... where $P_X^{n/2,n/2}$ is the primitive cohomology of Hodge...

open_problem | open | MathDB 页面 | 原文线索

V2-038964 | MATHDB-397451

Onsager’s conjecture for weak incompressible Euler flows

摘录 / EXCERPT: For weak solutions of the incompressible Euler equations on the torus, the conjecture is that the solutions preserve the energy if they are Hölder-continuous with exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-038965 | MATHDB-397727

Local well-posedness of the classical Stefan problem under the Taylor sign condition

摘录 / EXCERPT: For the classical Stefan problem described by (1), the weighted high-order Sobolev-type energy is coercive under the Taylor sign condition $\partial_n p_0<0$ on the initial interfa...

open_problem | open | MathDB 页面 | 原文线索

V2-038966 | MATHDB-397733

Asymptotic rotational symmetry of geometric-flow singularities

摘录 / EXCERPT: Do singularities of geometric PDE become asymptotically rotationally symmetric? In other words, is rotational symmetry stable in a suitable geometric sense?

open_problem | open | MathDB 页面 | 原文线索

V2-038967 | MATHDB-397737

Almost-sure global weak solutions for randomized 3D periodic Navier-Stokes data

摘录 / EXCERPT: Fix $T>0$ and $0<\alpha<\frac13$. Let $\vec f\in (H^{-\alpha}(\mathbb T^3))^3$ be divergence-free and of mean zero. Then there exists a set $\Sigma\subset\Omega$ of probability 1...

open_problem | open | MathDB 页面 | 原文线索

V2-038968 | MATHDB-397900

Sobolev regularity for solutions of the Monge-Ampère equation

摘录 / EXCERPT: For convex solutions of ... with $0<\lambda\le f\le\Lambda$, does one have at least $u\in W^{2,1}_{\mathrm{loc}}(\Omega)$, or even $u\in W^{2,1+\varepsilon}_{\mathrm{loc}}(\Omega)$ f...

open_problem | open | MathDB 页面 | 原文线索

V2-038969 | MATHDB-397924

Past stability of the FLRW Big Bang under near-FLRW perturbations

摘录 / EXCERPT: For initial data on the spacelike hypersurface $\Sigma'_1=\{t=1\}$ that are sufficiently close to the FLRW data in a Sobolev norm, the maximal globally hyperbolic development conta...

open_problem | open | MathDB 页面 | 原文线索

V2-038970 | MATHDB-397925

Existence of a CMC hypersurface for near-FLRW Einstein-stiff-fluid data

摘录 / EXCERPT: For initial data for the Einstein-stiff-fluid system posed on the spacelike hypersurface $\Sigma'_1=\{t=1\}$, sufficiently near the FLRW data in a Sobolev norm, the maximal globall...

open_problem | open | MathDB 页面 | 原文线索

V2-038971 | MATHDB-397948

Dissipative Hölder-continuous Euler-flow existence theorem

摘录 / EXCERPT: Let $\bar e:[0,1]\to\mathbb{R}$ be a positive smooth function and let $\theta<1/10$. Then there is a weak solution (v,p) of the incompressible Euler equations on...

open_problem | open | MathDB 页面 | 原文线索

V2-038972 | MATHDB-397953

Leray’s problem for stationary Navier-Stokes equations

摘录 / EXCERPT: For the stationary nonhomogeneous Navier-Stokes problem ... where $\Omega\subset\mathbb{R}^n$ has a multiply connected boundary $\partial\Omega=\bigcup_{j=1}^N\Gamma_j$, and the boun...

open_problem | open | MathDB 页面 | 原文线索

V2-038973 | MATHDB-397981

Intrinsic characterization of solutions to the rough HJB equation

摘录 / EXCERPT: For the controlled rough differential equation setup in the report, solutions of the rough HJB equation ... are defined as limits of viscosity solutions corresponding to smooth appro...

open_problem | open | MathDB 页面 | 原文线索

V2-038974 | MATHDB-398122

Continuity and finiteness of the convergence Newton polygon of a p-adic differential equation

摘录 / EXCERPT: Let $(K,|\cdot|)$ be a complete valued ultrametric field of characteristic 0, let X be a quasi-smooth K-analytic curve in the sense of Berkovich, and let (M,∇) be a di...

open_problem | open | MathDB 页面 | 原文线索

V2-038975 | MATHDB-398230

Simultaneous moving-horizon estimation and nonlinear model predictive control for PDE models

摘录 / EXCERPT: A logical further step is the development of a simultaneous MHE and NMPC method “in one step”. This requires both the theoretical investigation of joint sensitivity, conditioning a...

open_problem | open | MathDB 页面 | 原文线索

V2-038976 | MATHDB-398258

Convexity of the maximum set for a singular elliptic equation

摘录 / EXCERPT: For the zero-boundary minimizer u of $F(u)=\int_\Omega |Du|+\frac1p\int_\Omega |Du|^p-\int_\Omega au$, with $a>0$, $p\in(1,\infty)$, and $\Omega\subset\mathbb{R}^n$ a bounded dom...

open_problem | open | MathDB 页面 | 原文线索

V2-038977 | MATHDB-398266

Gradient potential estimate for quasilinear elliptic equations

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be open, let μ be a Radon measure with compact support, and let $u\in C^1(\Omega)$ solve ... where $a\in C^1(\mathbb{R}^n;\mathbb{R}^n)$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-038978 | MATHDB-398267

Boundary estimate for non-negative solutions of a parabolic equation

摘录 / EXCERPT: The report states the following established theorem. Let H be the operator in (1), with the structural conditions (2) in the report and $p=2$. Let $\Omega_T=\Omega\times(0,T)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-038979 | MATHDB-398275

L^1-uniqueness for the Kolmogorov operator of the 2D stochastic Navier-Stokes equation with space-time white noise

摘录 / EXCERPT: For the two-dimensional stochastic Navier-Stokes equations on the torus with space-time white noise, L^1-uniqueness of the associated Kolmogorov operator has been obtained for su...

open_problem | open | MathDB 页面 | 原文线索

V2-038980 | MATHDB-398277

Sobolev inviscid limit and Prandtl expansion for the half-plane Navier-Stokes equations

摘录 / EXCERPT: For the two-dimensional incompressible Navier-Stokes system (NS_ν) in the half-plane with no-slip boundary conditions, can the formal Prandtl asymptotic expansion be rigorously...

open_problem | open | MathDB 页面 | 原文线索

V2-038981 | MATHDB-398279

Large-time behavior of two-dimensional exterior-domain Navier-Stokes solutions

摘录 / EXCERPT: For the two-dimensional exterior-domain problem ... with ... clarify the large-time behavior, as $t\to\infty$, of solutions to its initial-value problem.

open_problem | open | MathDB 页面 | 原文线索

V2-038982 | MATHDB-398477

Stability of non-constant equilibria for the compressible Navier-Stokes equations

摘录 / EXCERPT: For the non-isothermal compressible Navier-Stokes equations in bounded domains, with initial data sufficiently close to equilibria, the report states that global existence and uniq...

open_problem | open | MathDB 页面 | 原文线索

V2-038983 | MATHDB-398538

Unbounded growth of Sobolev norms for the cubic defocusing NLS

摘录 / EXCERPT: For the cubic defocusing nonlinear Schrödinger equation (0.1) with $V=0$ and periodic boundary conditions in dimension 2 or higher, do there exist solutions whose H^s-norm has...

open_problem | open | MathDB 页面 | 原文线索

V2-038984 | MATHDB-398540

Growth of Sobolev norms for the cubic nonlinear Schrödinger equation with a convolution potential

摘录 / EXCERPT: Fix $s>1$. Let $s_0>70s/17$ and take $V\in H^{s_0}(\mathbb T^2)$ with real Fourier coefficients. Then there exists $c>0$ with the following property: for any small $\mu\ll1$ and la...

open_problem | open | MathDB 页面 | 原文线索

V2-038985 | MATHDB-398589

Weak greedy parameter selection for transport-dominated reduced basis methods

摘录 / EXCERPT: For the current reduced basis X_n, find a parameter $\bc_{n+1}$ such that ... where u_μ is the solution of the parametric PDE and $c\ge 0$ is a constant.

open_problem | open | MathDB 页面 | 原文线索

V2-038986 | MATHDB-398965

Adaptation to both Sobolev ellipsoid parameters

摘录 / EXCERPT: Is sharp minimax testing adaptable simultaneously to both the smoothness parameter β and the ellipsoid-size parameter M?

open_problem | open | MathDB 页面 | 原文线索

V2-038987 | MATHDB-398973

Nonlinear Landau damping in Sobolev spaces for the Vlasov-HMF model

摘录 / EXCERPT: Let $m_0>1/2$ and define ... For the Vlasov-HMF equation with $P(x)=\cos x$, write $f(t,x,v)=\eta(v)+\varepsilon r(t,x,v)$ and $g(t,x,v)=r(t,x+tv,v)$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-038988 | MATHDB-398977

Maximal Sobolev regularity for a Neumann problem for gradient systems

摘录 / EXCERPT: Let $O=\{x\in X:G(x)<0\}$, where G is convex, $G\in C^1(X)\cap\bigcap_{p>1}W^{2,p}_0(X,\mu)$, $DG\ne0$ on ∂O, and $1/\|Q^{1/2}DG\|\in\bigcap_{p>1}L^p(X,\mu)$. For...

open_problem | open | MathDB 页面 | 原文线索

V2-038989 | MATHDB-399055

Carleson measures for Besov-Sobolev spaces on the complex ball

摘录 / EXCERPT: For the analytic Besov-Sobolev space $B^{\sigma}_2(\mathbb B_{2d})$, characterize the positive Borel measures μ on $\mathbb B_{2d}$ such that ... The report identifies the diffic...

open_problem | open | MathDB 页面 | 原文线索

V2-038990 | MATHDB-399056

Carleson measures for Besov-Sobolev spaces in the difficult range

摘录 / EXCERPT: Characterize the positive Borel measures μ on the unit ball $B_2^d\subset\mathbb C^d$ for which ... in the range $\tfrac12<\sigma<\tfrac d2$.

open_problem | open | MathDB 页面 | 原文线索

V2-038991 | MATHDB-399172

Exterior higher-order PDE for the magnitude extremal problem

摘录 / EXCERPT: For every $m\in\mathbb N$ and every convex compact set $X\subseteq\mathbb R^n$, there exists a unique $h\in H^m(\mathbb R^n)$ such that ... Moreover, ... where ν is the unit normal...

open_problem | open | MathDB 页面 | 原文线索

V2-038992 | MATHDB-399173

Endpoint Sobolev boundedness of the centered Hardy-Littlewood maximal operator

摘录 / EXCERPT: Is the operator $f\mapsto \nabla Mf$ bounded from $W^{1,1}(\mathbb{R}^{d})$ into $L^{1}(\mathbb{R}^{d})$? Equivalently, does Mf have a weak derivative satisfying, for some consta...

open_problem | open | MathDB 页面 | 原文线索

V2-038993 | MATHDB-399394

Appropriate methods for dynamic gas-network optimization

摘录 / EXCERPT: Develop appropriate methods and algorithms for solving gas-network optimization problems in the general dynamical setting, where binary decision variables and partial differential...

open_problem | open | MathDB 页面 | 原文线索

V2-038994 | MATHDB-399576

Ill-posedness of the Kaup system in L^2-based Sobolev spaces

摘录 / EXCERPT: For the admissible choice $a=\frac13$ and $b=c=d=0$, the $abcd$-system becomes the Kaup system ... Despite its inverse-scattering formalism, this model is ill-posed in L^2-based So...

open_problem | open | MathDB 页面 | 原文线索

V2-038995 | MATHDB-399590

Stability of quarkonial frames in negative-order function spaces

摘录 / EXCERPT: Can quarkonial frames be made stable in negative-order Sobolev or Besov spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-038996 | MATHDB-399803

Double-exponential vorticity-gradient growth in the bulk of the fluid

摘录 / EXCERPT: Is double-exponential growth of the vorticity gradient possible in the bulk of the fluid for a solution of the two-dimensional Euler equation, rather than only at the boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-038997 | MATHDB-399804

Double-exponential growth of the vorticity gradient for 2D Euler flow on the unit disk

摘录 / EXCERPT: Consider the two-dimensional Euler equation on a unit disk D. There exists smooth initial data $\\omega_0$ with ... such that the corresponding solution $\omega(x,t)$ satisfies ... f...

open_problem | open | MathDB 页面 | 原文线索

V2-038998 | MATHDB-399805

Weak limits of approximate solutions and the incompressible Euler equations

摘录 / EXCERPT: Given an approximate sequence of solutions for the incompressible Euler equations-for example, Leray-Hopf solutions of the Navier-Stokes equations with vanishing viscosity-does its...

open_problem | open | MathDB 页面 | 原文线索

V2-038999 | MATHDB-399809

Energy conservation for physically realizable weak solutions of 2D Euler

摘录 / EXCERPT: Let $u\in C(0,T;L^{2}(\mathbb{T}^{2}))$ be a physically realizable weak solution of the incompressible 2D Euler equations. Suppose that $u_{0}\in L^{2}$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-039000 | MATHDB-399810

Planar case of Leray’s problem for stationary Navier-Stokes equations

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{2}$ be a bounded domain with C^{2}-smooth boundary $\partial\Omega=\bigcup_{j=0}^{N}\Gamma_{j}$, where the Γ_j are disjoint boundary componen...

open_problem | open | MathDB 页面 | 原文线索

V2-039001 | MATHDB-399811

Uniqueness of 2D incompressible Euler weak solutions in the Delort class

摘录 / EXCERPT: Given initial data $u_0\in L^2_{\mathrm{loc}}$ with $\omega_0=\operatorname{curl}u_0\in BM_+$, is the global weak solution of the two-dimensional incompressible Euler equations in...

open_problem | open | MathDB 页面 | 原文线索

V2-039002 | MATHDB-399814

Local existence and uniqueness for the rotational 3D free-boundary Euler equations

摘录 / EXCERPT: Let $\delta>0$. Consider the incompressible free-boundary Euler equations in Lagrangian coordinates on $\Omega=\mathbb{R}^2\times(0,1)$, with period-one periodicity in x_1,x_2, f...

open_problem | open | MathDB 页面 | 原文线索

V2-039003 | MATHDB-399815

Determining modes for the 3D Navier-Stokes equations in an average sense

摘录 / EXCERPT: Can determining modes for the three-dimensional Navier-Stokes equations be obtained in an average sense-for example, through a time-dependent determining wavenumber whose time aver...

open_problem | open | MathDB 页面 | 原文线索

V2-039004 | MATHDB-399817

BMO norm of the 3D Navier-Stokes vector potential as a blowup criterion

摘录 / EXCERPT: The report conjectures that, in three dimensions, blowup at $t=t^*$ implies ...

open_problem | open | MathDB 页面 | 原文线索

V2-039005 | MATHDB-399818

Finite-time blow-up versus global regularity for the 3D incompressible Euler equations

摘录 / EXCERPT: Do solutions of the 3D incompressible Euler equations remain globally regular, or can a finite-time singularity (blow-up) occur?

open_problem | open | MathDB 页面 | 原文线索

V2-039006 | MATHDB-400279

Optimality of the stochastic Sobolev regularity exponent for a scalar conservation law

摘录 / EXCERPT: For quasi-solutions of $dv+\frac12\partial_xv^2\circ d\beta_t=0$ on the one-dimensional torus $\mathbb T$, the report proves that $v(t)\in W^{\alpha,1}(\mathbb T)$ for every...

open_problem | open | MathDB 页面 | 原文线索

V2-039007 | MATHDB-400292

Onsager’s conjecture, dissipative direction

摘录 / EXCERPT: For every $\theta<\frac{1}{3}$, there exist weak solutions $v\in C^\theta$ of the incompressible Euler equations that do not conserve the energy.

conjecture | open | MathDB 页面 | 原文线索

V2-039008 | MATHDB-400293

Kolmogorov-Onsager conjecture: non-conservative Euler solutions with critical Sobolev regularity

摘录 / EXCERPT: There exists a weak solution $v \in L^\infty([0,1];H^{1/3-\varepsilon}(\mathbb{T}^3))$ of the incompressible Euler equations (1) that does not conserve kinetic energy.

conjecture | open | MathDB 页面 | 原文线索

V2-039009 | MATHDB-400294

Kolmogorov-Onsager conjecture for weak Euler solutions

摘录 / EXCERPT: For any $\theta<\frac{1}{3}$, there exist weak solutions $v\in C([0,1];L^2(\mathbb T^3))$ to the Euler equations (1), belonging to the Besov space...

conjecture | open | MathDB 页面 | 原文线索

V2-039010 | MATHDB-400450

Completeness of Sobolev metrics on Sobolev immersions

摘录 / EXCERPT: Let $n\ge 2$ and let G_n be a reparametrization-invariant Sobolev metric on $\operatorname{Imm}(S^1,\mathbb{R}^d)$ of order n with constant coefficients. Define ... Then...

open_problem | open | MathDB 页面 | 原文线索

V2-039011 | MATHDB-400655

Describe the stochastic process associated with a regular representation of a non-regular weighted Sobolev Dirichlet form

摘录 / EXCERPT: Let $d\ge 2$ and let $\rho:\mathbb{R}^{d}\to[0,\infty)$ be measurable with $\rho,\rho^{-1}\in L^{1}_{\mathrm{loc}}(\mathbb{R}^{d})$. Define ... let $\mathcal{H}$ be the completion of...

open_problem | open | MathDB 页面 | 原文线索

V2-039012 | MATHDB-400855

Existence and uniqueness for the calibrated PDV McKean equation

摘录 / EXCERPT: Existence and uniqueness of a solution to the McKean stochastic differential equation that describes the calibrated model.

open_problem | open | MathDB 页面 | 原文线索

V2-039013 | MATHDB-401014

Could noise regularize 2D Euler equations?

摘录 / EXCERPT: Could noise regularize 2D Euler equations?

open_problem | open | MathDB 页面 | 原文线索

V2-039014 | MATHDB-401015

Could noise regularize 2D Euler equations at the critical exponent $\alpha=1$?

摘录 / EXCERPT: For the stochastic 2D Euler equations on $\mathbb{T}^2=\mathbb{R}^2/\mathbb{Z}^2$ with transport noise ... where ... does the noise produce a regularization for the case $\alpha=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-039015 | MATHDB-401127

Pointwise convergence for Schrödinger solutions at the conjectured Sobolev threshold

摘录 / EXCERPT: The convergence holds for $f\in H^s(\mathbb{R}^n)$ provided that $s>\dfrac{n}{2(n+1)}$; that is, $\lim_{t\to 0}e^{it\Delta}f(x)=f(x)$ almost everywhere.

conjecture | open | MathDB 页面 | 原文线索

V2-039016 | MATHDB-401216

Completely computable a posteriori discretization-error estimates for discrete POD optimal control models

摘录 / EXCERPT: For the linear-quadratic optimal control problem governed by a parabolic equation with pointwise control bounds described in the report, can one obtain completely computable a post...

open_problem | open | MathDB 页面 | 原文线索

V2-039017 | MATHDB-401395

Pointwise estimate for the one-dimensional compressible Navier-Stokes problem

摘录 / EXCERPT: For the system (1), with initial data $(v_0-1,u_0)=\bigl(\varepsilon e^{-x}\chi(x),0\bigr)$ and sound speed $c=1$, there exist $\varepsilon_0>0$, $C_0>0$, and $K_0>0$ such that, fo...

open_problem | open | MathDB 页面 | 原文线索

V2-039018 | MATHDB-401441

Selecting physical solutions of the interacting-electron Green-function equations

摘录 / EXCERPT: What could be a general strategy (analytical and/or numerical) to solve the set of functional differential equations (2) and to select the physical solution?

open_problem | open | MathDB 页面 | 原文线索

V2-039019 | MATHDB-401577

Complete logarithmic Sobolev inequality for selfadjoint Lindblad generators

摘录 / EXCERPT: Does every selfadjoint Lindblad generator satisfy λ-CLSI for some $\lambda(L)$?

open_problem | open | MathDB 页面 | 原文线索

V2-039020 | MATHDB-401831

Extension to the full PDE chemical model

摘录 / EXCERPT: Extend the model in [6] to the full PDE chemical model proposed in [10, 11].

open_problem | open | MathDB 页面 | 原文线索

V2-039021 | MATHDB-402178

Incurable ill-posedness of the two-dimensional Euler equations

摘录 / EXCERPT: Conjecture. The 2-dimensional Euler equations (both incompressible and compressible) are incurably ill-posed on any space that contains $H^1(\mathbb{R}^2)$, where “incurably ill-po...

open_problem | open | MathDB 页面 | 原文线索

V2-039022 | MATHDB-402183

Existence of stationary martingale solutions satisfying weak anomalous dissipation

摘录 / EXCERPT: For the stochastically forced three-dimensional Navier-Stokes equations on $\mathbb{T}^3$, ... with mean-zero, white-in-time, spatially colored Gaussian forcing...

open_problem | open | MathDB 页面 | 原文线索

V2-039023 | MATHDB-402210

Singular limit for Navier-Stokes equations in a curved thin domain

摘录 / EXCERPT: For a closed, connected, oriented, smooth surface $\Gamma\subset\mathbb{R}^3$, smooth thickness functions g_0,g_1 with $g=g_1-g_0\ge c>0$, and the curved thin domains...

open_problem | open | MathDB 页面 | 原文线索

V2-039024 | MATHDB-402249

Boundedness of stable solutions to semilinear elliptic equations in dimensions at most 9

摘录 / EXCERPT: Let $ Omega\subset\mathbb{R}^n$ be a bounded domain and let $f:\mathbb{R}\to\mathbb{R}$ be positive, nondecreasing, convex, and superlinear at $+\infty$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-039025 | MATHDB-402253

Vortex filament conjecture for the three-dimensional Euler equations

摘录 / EXCERPT: The big open problem is whether one can find solutions of the Euler equations for which the vorticity remains close for a significant period of time to a filament evolving by binor...

conjecture | open | MathDB 页面 | 原文线索

V2-039026 | MATHDB-402293

Onsager conjecture for incompressible Euler equations

摘录 / EXCERPT: Weak solutions of the incompressible Euler equations conserve the energy as long as they belong to the Hölder space $C^{0,\alpha}$ with $\alpha>\frac{1}{3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-039027 | MATHDB-402295

Vanishing-viscosity limit for Navier-Stokes flow with boundaries

摘录 / EXCERPT: Study the limit, or possible limits, as $\nu\to0$, of solutions of the Navier-Stokes equations in a fluid domain with boundaries, with no-slip (Dirichlet) boundary condition $u=0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-039028 | MATHDB-402300

Global existence for the half-space free-boundary Navier-Stokes problem

摘录 / EXCERPT: For the free-boundary incompressible Navier-Stokes system (1.1) in the half-space setting with no surface tension or gravity, initial domain $\Omega_0=\mathbb{R}^3_+$, and sufficie...

open_problem | open | MathDB 页面 | 原文线索

V2-039029 | MATHDB-402336

Consistent thin-layer approximation of PDEs on moving hypersurfaces

摘录 / EXCERPT: Assuming that interfacial phenomena are described by partial differential equations on moving hypersurfaces, how can these equations be consistently approximated when the hypersurf...

open_problem | open | MathDB 页面 | 原文线索

V2-039030 | MATHDB-402405

Whether the Neumann and Dirichlet wave problems on the fractal tree differ

摘录 / EXCERPT: Do the solutions of the Neumann problem (N) and Dirichlet problem (D) differ; in particular, do the weighted Sobolev spaces satisfy $H^1_{\\bc,0}(T)=H^1_{\\bc}(T)$?

open_problem | open | MathDB 页面 | 原文线索

V2-039031 | MATHDB-402910

Rigorous derivation of the Vlasov-Navier-Stokes limit for sedimenting particles

摘录 / EXCERPT: Starting from the microscopic system of many spherical particles coupled to the incompressible fluid equations, give a rigorous derivation, under a suitable scaling as the number o...

open_problem | open | MathDB 页面 | 原文线索

V2-039032 | MATHDB-402922

Locally Lipschitz continuity of the Sobolev rough-path solution map

摘录 / EXCERPT: The solution map is locally Lipschitz continuous with respect to an inhomogeneous Sobolev rough-path distance.

open_problem | open | MathDB 页面 | 原文线索

V2-039033 | MATHDB-402924

Non-uniqueness for reflected rough differential equations

摘录 / EXCERPT: The general uniqueness question for reflected rough differential equations driven by rough signals is refuted by the following counterexample. Let ... let e_1,e_2 be the canonical...

open_problem | open | MathDB 页面 | 原文线索

V2-039034 | MATHDB-403040

Other branching reference processes for unbalanced optimal transport

摘录 / EXCERPT: When the reference law R is the law of a branching Brownian motion with $p_0=p_2=1/2$, and $\nu\to0$ with $\lambda\sim \nu^{-1}e^{-1/\nu}$, the Γ-convergence at the PDE le...

open_problem | open | MathDB 页面 | 原文线索

V2-039035 | MATHDB-403053

Higher-order entropy-stable discretisations for the relaxed Navier-Stokes-Korteweg system

摘录 / EXCERPT: The generalisation to higher-order methods like the ones used in [1, 3, 5] remains an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-039036 | MATHDB-403090

The Batchelor power spectrum of passive scalar turbulence at fixed Reynolds number

摘录 / EXCERPT: Under the two-dimensional stochastically forced Navier-Stokes/passive-scalar model (2)-(5), for every admissible viscosity ν and forcing coefficient sequence $\{q_k\}$, there e...

open_problem | open | MathDB 页面 | 原文线索

V2-039037 | MATHDB-403097

Rational conserved integrals and inverse polynomial volume forms for polynomial ODEs

摘录 / EXCERPT: For a polynomial ordinary differential equation (ODE) $\frac{dx}{dt}=f(x)$, what rational conserved integral(s) and (inverse) polynomial volume form (if any) does the ODE possess?

open_problem | open | MathDB 页面 | 原文线索

V2-039038 | MATHDB-403187

Conforming finite element methods for general-order PDEs in arbitrary dimensions

摘录 / EXCERPT: How to construct conforming finite element methods for general m and d for $2m$-th-order partial differential equations in $\mathbb{R}^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-039039 | MATHDB-403291

Optimal-cost adaptive algorithms for fully discrete parametric elliptic PDE approximations

摘录 / EXCERPT: Whether an adaptive algorithm can be constructed that finds such fully discrete approximations at optimal computational cost, that is, using $O(N)$ operations.

open_problem | open | MathDB 页面 | 原文线索

V2-039040 | MATHDB-403357

Time-uniform Sobolev composition inequality for Robinson-Trautman equations

摘录 / EXCERPT: Let $G:\mathbb R\to\mathbb R$ be smooth with $G(0)=0$, and let $s>1$. Is there a constant $C_2=C_2(G,s)$ such that, for every time-dependent function $\phi(t)$ satisfying ... one has...

open_problem | open | MathDB 页面 | 原文线索

V2-039041 | MATHDB-403546

Fully intertwined numerical methods for risk-averse PDE-constrained optimization

摘录 / EXCERPT: A fully intertwined numerical method linking inexact calculation of the Newton steps, adaptive sampling (where necessary), and hierarchical methods or adaptive finite elements with...

open_problem | open | MathDB 页面 | 原文线索

V2-039042 | MATHDB-403727

Invariant measure for 3D stochastic Navier-Stokes equations

摘录 / EXCERPT: Does the three-dimensional Navier-Stokes system with periodic boundary conditions driven by space-time white noise admit an invariant measure?

open_problem | open | MathDB 页面 | 原文线索

V2-039043 | MATHDB-403728

Global existence and non-uniqueness for stochastic 3D Navier-Stokes equations

摘录 / EXCERPT: For the three-dimensional Navier-Stokes system with periodic boundary conditions ... where B is a cylindrical Wiener process on a stochastic basis...

open_problem | open | MathDB 页面 | 原文线索

V2-039044 | MATHDB-403734

Existence of minimizers for the critical Sobolev variational problem in all dimensions

摘录 / EXCERPT: For every dimension $n\ge 2$, every $p\in(1,n)$, every open bounded domain $\Omega\subset\mathbb{R}^n$ with boundary of class C^2, and every $T>0$, does the variational problem .....

open_problem | open | MathDB 页面 | 原文线索

V2-039045 | MATHDB-403735

Rigidity in the optimal Sobolev comparison theorem

摘录 / EXCERPT: For the comparison theorem $\Phi_{\Omega}(T)\geq \Phi_B(T)$ in (4), does rigidity hold?

open_problem | open | MathDB 页面 | 原文线索

V2-039046 | MATHDB-403778

Fractal pointwise convergence for the Schrödinger equation

摘录 / EXCERPT: For a fixed dimension d and $d/2<\alpha\le d$, determine the critical Sobolev exponent ...

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V2-039047 | MATHDB-403860

Planar Sobolev extension domains with a prescribed exponent range

摘录 / EXCERPT: In the planar case, given $1<q<2$, can one construct a domain that has the Sobolev extension property precisely for $1\le p<q$ or $p>2$?

open_problem | open | MathDB 页面 | 原文线索

V2-039048 | MATHDB-403896

Differential equation for the Rubinstein-Keating-Rodgers-Roditty-Gershon-Rudnick generating function

摘录 / EXCERPT: Let $G_{k,N}(x)$ be the function defined in the report by the random-matrix integral associated with the k-fold divisor function, and set $F=xG'_{k,N}/G_{k,N}$. The problem is to...

open_problem | open | MathDB 页面 | 原文线索

V2-039049 | MATHDB-403903

Concave maximization formulation for the incompressible Euler equations

摘录 / EXCERPT: For the kinetic linear program described on the report’s preceding page, convex duality gives the concave maximization problem ... where A is divergence-free, tangent to...

open_problem | open | MathDB 页面 | 原文线索

V2-039050 | MATHDB-403904

Free-boundary incompressible Navier-Stokes equations in critical Besov spaces

摘录 / EXCERPT: For $n\ge 2$, let u_0 and η_0 be initial velocity and boundary data, and define ... The report considers the free-boundary problem ... where ... For $n/2<p<2n-1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-039051 | MATHDB-403970

Analysis of a higher-order space-time CutFEM approach for Navier-Stokes equations on evolving domains

摘录 / EXCERPT: The combination of higher-order space-time finite element discretizations with the fictitious-domain approach, spatial extension, ghost-penalty stabilization, and integration over...

open_problem | open | MathDB 页面 | 原文线索

V2-039052 | MATHDB-404135

Geometric coordinate conditions guaranteeing convergence of the center of mass

摘录 / EXCERPT: Can one find geometric, Poincaré-equivariant PDE conditions selecting asymptotic coordinate charts on every asymptotically Euclidean relativistic initial data set 8IDStt with...

open_problem | open | MathDB 页面 | 原文线索

V2-039053 | MATHDB-404136

Geometric asymptotic-coordinate conditions ensuring convergence of center of mass

摘录 / EXCERPT: Find geometric, Poincaré-equivariant (PDE) conditions selecting asymptotic coordinates (x^i) on an arbitrary asymptotically Euclidean initial data set with...

open_problem | open | MathDB 页面 | 原文线索

V2-039054 | MATHDB-404323

Error, reliability, and accuracy analysis for the Sobolev spectral method

摘录 / EXCERPT: A consistent study of the error, reliability, and accuracy of the Sobolev spectral method remains an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-039055 | MATHDB-404337

Time-varying control and stabilization of traffic-flow PDE/ODE systems

摘录 / EXCERPT: For the traffic-flow PDE/ODE model (4)-(6), can the existence theory established for constant controls be extended to time-varying controls $u(t)$, and, a fortiori, can the associa...

open_problem | open | MathDB 页面 | 原文线索

V2-039056 | MATHDB-404401

Global existence of weak solutions for the multidimensional isentropic Euler equations

摘录 / EXCERPT: For the Cauchy problem ... in $[0,\infty)\times\mathbb R^d$, with initial data $(\varrho,u)(0,\cdot)=(\bar\varrho,\bar u)$, and for polytropic gases ... determine whether global-in-tim...

open_problem | open | MathDB 页面 | 原文线索

V2-039057 | MATHDB-404463

Approximation rates for shallow ReLU^k neural networks on Sobolev and Besov spaces

摘录 / EXCERPT: Determine the correct approximation rates, in terms of the number of parameters, for approximating functions from Sobolev and Besov spaces by shallow ReLU^k neural networks in th...

open_problem | open | MathDB 页面 | 原文线索

V2-039058 | MATHDB-404524

Converse to Eisenstein’s last theorem for nonlinear differential equations

摘录 / EXCERPT: Let $f(z)=\sum a_nz^n\in\mathbb{Q}[[z]]$ be a solution to a possibly nonlinear differential equation ... with $F\in\mathbb{Q}(z,y_0,\ldots,y_{n-1})$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-039059 | MATHDB-404666

Complete PDE analysis of the two-dimensional network flow

摘录 / EXCERPT: “Concerning the 2-dimensional version, I aim to a complete analysis of the network flow in a PDE framework, from well-posedness, to the study of long-time behaviour and asymptotic...

open_problem | open | MathDB 页面 | 原文线索

V2-039060 | MATHDB-404863

Multilinear Sobolev smoothing (multilinear Weyl) estimate

摘录 / EXCERPT: Let $1<p_1,\ots,p_k<\infty$ and let p satisfy ... For all $C_1,C_2\ge 1$, there exists $c\in(0,1)$, possibly depending on k, $\mathcal P=\{P_1,\dots,P_k\}$, $p_1,\dots,p_k$, p,...

open_problem | open | MathDB 页面 | 原文线索

V2-039061 | MATHDB-405087

Rigorous dynamical metastability of the mean-field PDE

摘录 / EXCERPT: A rigorous proof of dynamical metastability of the mean-field PDE (2), in the sense of Otto-Reznikoff, remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-039062 | MATHDB-405148

Compactness of the ∂-Neumann operator versus analytic discs in the boundary

摘录 / EXCERPT: For every $n\ge 2$ and every bounded pseudoconvex domain $\Omega\subset\mathbb{C}^n$ with smooth boundary, does the existence of a nonconstant analytic disc in the boundary, רמולכ,...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-039063 | MATHDB-405187

Neumann analogue of the Pólya-Szegő polygonal eigenvalue conjecture

摘录 / EXCERPT: For every integer $n\ge 3$ and every convex planar n-gon P of area A, let $\lambda_1^N(P)$ denote the first nonzero eigenvalue of the Laplace operator with Neumann boundary c...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-039064 | MATHDB-405192

Interior C^2 estimate problem for the graphical scalar-curvature equation

摘录 / EXCERPT: For every dimension $n\ge 3$, let $\Omega\subset\mathbb{R}^{n}$ be a domain and let $u\in C^{3}(\Omega)$ be admissible, meaning that the principal-curvature vector of its graph...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-039065 | MATHDB-405196

Holopainen’s L^{p-1}-Liouville conjecture

摘录 / EXCERPT: Let M be a complete Riemannian manifold satisfying the geometric hypotheses in Holopainen’s conjecture. Is every nonnegative p-harmonic function v on M such that...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-039066 | MATHDB-405216

Fast Dynamo Conjecture of Zeldovich and Sakharov

摘录 / EXCERPT: Does there exist an integer $k\ge 1$ and a nonempty C^k-open family of smooth, autonomous, divergence-free velocity fields $\mathbf u\colon\mathbb T^3\to\mathbb R^3$ such that ev...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-039067 | MATHDB-405228

McKean’s conjecture for the Landau-Coulomb equation

摘录 / EXCERPT: Let f_t be a sufficiently regular solution of the spatially homogeneous Landau equation with Coulomb interaction, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-039068 | MATHDB-405270

Width of the Lie algebra of vector fields on a manifold

摘录 / EXCERPT: Let M be a smooth manifold. Is it true that any vector field on M is a Lie bracket of two other vector fields? Open even in the case $M=S^2$.

open_problem | open | Nikolay Sheshko | MathDB 页面 | 原文线索

B09 动力系统 / Dynamical Systems | 1,971 条

V2-039069 | MATHDB-315751

Spectral edge of the quartic SYK model

摘录 / EXCERPT: Determine the leading asymptotic of the largest eigenvalue of the N-Majorana quartic SYK Hamiltonian as $N\to\infty$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-039070 | MATHDB-315777

Quantitative two-chaos fourth-moment theorem

摘录 / EXCERPT: Can the qualitative fourth-moment theorem for sums of two Wiener-Itô integrals of different parity be strengthened to an explicit total-variation rate depending only on the fourth...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-039071 | MATHDB-315915

Arnold-Givental conjecture

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold of dimension $2n$, and let $\tau:M\to M$ be an anti-symplectic involution, i.e. a diffeomorphism with ... Assume that the fixed poin...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-039072 | MATHDB-315917

Collatz conjecture

摘录 / EXCERPT: Define $f:\mathbb{Z}_{>0}\to\mathbb{Z}_{>0}$ by ... and for each $n\in\mathbb{Z}_{>0}$ let $a_0=n$ and $a_i=f(a_{i-1})$ for $i\ge 1$, so that $a_i=f^{\,i}(n)$ is the i-th iterate o...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-039073 | MATHDB-315918

Eden's conjecture

摘录 / EXCERPT: Let $(\{\varphi^{t}\}_{t\ge 0},U)$ be a dynamical system on an open set U of a finite-dimensional Euclidean space (or, more generally, of a Hilbert space), where each map...

conjecture | open | MathDB 页面 | 原文线索

V2-039074 | MATHDB-315919

Fatou conjecture

摘录 / EXCERPT: For $c \in \mathbb{C}$ let f_c be the quadratic polynomial ... regarded as a holomorphic self-map of the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039075 | MATHDB-315920

Kaplan-Yorke conjecture

摘录 / EXCERPT: Let $n \ge 2$ and let f be a smooth dynamical system on $\mathbb{R}^n$ (a diffeomorphism, or the time-one map of a smooth flow) possessing a compact attractor A that carries an...

conjecture | open | MathDB 页面 | 原文线索

V2-039076 | MATHDB-315921

Hilbert-Arnold problem

摘录 / EXCERPT: Fix $k\in\mathbb{N}$ and let $B\subset\mathbb{R}^{k}$ be a compact set. Consider k-parameter families $\{v_{\varepsilon}\}_{\varepsilon\in B}$ of smooth vector fields on the two-...

open_problem | open | MathDB 页面 | 原文线索

V2-039077 | MATHDB-315923

open problems

摘录 / EXCERPT: For a compact convex set $P \subset \mathbb{R}^2$ with nonempty interior, and a point x in the exterior $\mathbb{R}^2 \setminus P$, call a point $m \in \partial P$ a right suppor...

open_problem | open | MathDB 页面 | 原文线索

V2-039078 | MATHDB-315924

Weinstein conjecture

摘录 / EXCERPT: Let M be a closed (compact, without boundary) smooth manifold of odd dimension $2n+1$ with $n \ge 1$, and let α be a contact form on M, i.e. a smooth 1-form such that...

conjecture | open | MathDB 页面 | 原文线索

V2-039079 | MATHDB-316167

Zimmer's conjecture

摘录 / EXCERPT: Let G be a connected simple Lie group whose real rank is $r=\operatorname{rank}_{\mathbb{R}}(G)\ge 2$ (that is, a maximal $\mathbb{R}$-split torus in G has dimension r), and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-039080 | MATHDB-316254

A logarithmic upper bound for Collatz total stopping time

摘录 / EXCERPT: For $n\geq2$, let $\sigma_{\infty}(n)$ be the least $k\in\mathbb{N}\cup\{\infty\}$ such that $T^{(k)}(n)=1$, and let $\delta(n)$ be a slowly varying function, possibly eventually c...

conjecture | open | MathDB 页面 | 原文线索

V2-039081 | MATHDB-316292

Periodicity conjecture for integer sequences with bounded affine recurrence

摘录 / EXCERPT: Let λ be a real number with $|\lambda|<2$. Consider an integer sequence $(a_k)_{k\in\mathbb Z}$ satisfying ... for every $k\in\mathbb Z$. Periodicity conjecture. Every such s...

conjecture | open | MathDB 页面 | 原文线索

V2-039082 | MATHDB-316425

The accumulation conjecture for foliations with ample normal bundle

摘录 / EXCERPT: Let X be a compact connected complex manifold of dimension $n\ge 3$, and let ${\mathcal F}$ be a codimension one holomorphic foliation on X. Its normal bundle $N_{\mathcal F}$...

conjecture | open | MathDB 页面 | 原文线索

V2-039083 | MATHDB-316434

Non-recurrent-point unique extension conjecture

摘录 / EXCERPT: Non-recurrent-point conjecture. If $\boxed{\textstyle\text{?}}$ is non-recurrent, then $s_{\boxed{\textstyle\text{?}}}$ extends uniquely to $B(\ell^2(G))$.

conjecture | open | MathDB 页面 | 原文线索

V2-039084 | MATHDB-316436

Wandering-point unique extension conjecture

摘录 / EXCERPT: Wandering-point conjecture. If $\boxed{\textstyle\text{?}}$ is wandering for the G-action on G^*, then the state corresponding to evaluation at $\boxed{\textstyle\text{?}}$ ext...

conjecture | open | MathDB 页面 | 原文线索

V2-039085 | MATHDB-316456

Invariant-distribution conjecture for volume-preserving vector fields

摘录 / EXCERPT: Let M be a closed, connected, orientable manifold, and let X be a volume-preserving vector field on M. Write $\mathcal I_X(M)$ for the space of all X-invariant distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-039086 | MATHDB-316542

Furstenberg-Katok-Spatzier-Margulis measure homogeneity conjecture

摘录 / EXCERPT: Let $n\geq 3$, let X_n be the space of lattices of rank n, and let H be a maximal $\mathbb{R}$-split torus in $\mathrm{PGL}_n(\mathbb{R})$. Let μ be an ergodic H-invari...

conjecture | open | MathDB 页面 | 原文线索

V2-039087 | MATHDB-316584

Adiabatic vanishing conjecture for the transient current

摘录 / EXCERPT: Let H be the coupled Hamiltonian, let $P^{d}(H)$ be the projection onto the subspace generated by the discrete eigenfunctions of H, and let ${\varrho}_{\alpha}(t)$ denote the t...

conjecture | open | MathDB 页面 | 原文线索

V2-039088 | MATHDB-316621

Global phase-locking conjecture for homogeneous Kuramoto oscillators

摘录 / EXCERPT: Let N phase oscillators evolve according to the homogeneous Kuramoto system ... where $\theta_0\in\mathbb{R}^N$ and $R(\theta)$ is the magnitude of the order parameter. Global phas...

conjecture | open | MathDB 页面 | 原文线索

V2-039089 | MATHDB-316637

Hamiltonian-cycle conjecture for graphs of non-degenerate classical transportation polytopes

摘录 / EXCERPT: Let $P_c=\{x\mid Bx=c,\ x\geq 0\}$ be a non-degenerate $m\times n$ classical transportation polytope, and let its graph have the vertices and edges of P_c. Hamiltonian-cycle conj...

conjecture | open | MathDB 页面 | 原文线索

V2-039090 | MATHDB-316689

Density conjecture for diffeomorphisms of the circle with non-abelian centralizer

摘录 / EXCERPT: Let S^1 be the circle, let $\operatorname{Diff}^1(S^1)$ denote the space of C^1 diffeomorphisms of S^1, and for $f\in\operatorname{Diff}^1(S^1)$ let $C(f)$ be its centralizer...

conjecture | open | MathDB 页面 | 原文线索

V2-039091 | MATHDB-316690

Density conjecture for C^1-conjugacy classes in $\hat D_{\alpha,\beta}$

摘录 / EXCERPT: Let $\alpha>1$ and $0<\beta<1$, and let $\hat D_{\alpha,\beta}\subset\operatorname{Diff}^1([0,1])$ be the set of diffeomorphisms whose fixed points are precisely 0 and 1, with...

conjecture | open | MathDB 页面 | 原文线索

V2-039092 | MATHDB-316695

The dimension conjecture for divergent trajectories of non-quasi-unipotent flows

摘录 / EXCERPT: Let G/Γ be a finite-volume, noncompact homogeneous space and let (g_t) be a non-quasi-unipotent flow on it. Define $D\subset G/\Gamma$ to be the set of points whose trajec...

conjecture | open | MathDB 页面 | 原文线索

V2-039093 | MATHDB-316720

Nearly periodic slow motion in deterministic fully coupled averaging

摘录 / EXCERPT: Let $w_{-}(r)$ and $w_{+}(r)$ be strictly increasing and decreasing functions, respectively, satisfying ... Suppose that $w_{-}(\frac{}{\lambda})=w_{+}(\lambda)=w^*$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-039094 | MATHDB-316796

Existence of an L_{0}-local Hamiltonian for L_{0}-regular spin chains

摘录 / EXCERPT: Existence conjecture. For any L_{0}-regular spin chain there always exists a Hamiltonian which is L_{0}-local.

conjecture | open | MathDB 页面 | 原文线索

V2-039095 | MATHDB-316895

Tits-type conjecture for automorphism groups of compact Kähler manifolds

摘录 / EXCERPT: Tits-type conjecture. The quotient $G/N(G)$ is a free abelian group of rank at most $n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-039096 | MATHDB-316897

The two-fixed-points-or-high-periodic-points conjecture for torus homeomorphisms

摘录 / EXCERPT: Two-fixed-points-or-high-periodic-points conjecture. Either F has two fixed points or F has periodic points of arbitrarily high periods.

conjecture | open | MathDB 页面 | 原文线索

V2-039097 | MATHDB-317063

Conjecture on concentration of first-eigenfunction limits on maximal recurrent subsets

摘录 / EXCERPT: Concentration conjecture. There exists a function, not necessarily differentiable, such that every possible limit of this sequence is a measure in

conjecture | open | MathDB 页面 | 原文线索

V2-039098 | MATHDB-317181

Integrability classification conjecture for the minimally coupled scalar-field system

摘录 / EXCERPT: Let ... be the Hamiltonian of the minimally coupled scalar-field system. Assume $\Lambda\ne0$, let n be an integer satisfying ... and consider a generic energy level $E\ne0$. Integra...

conjecture | open | MathDB 页面 | 原文线索

V2-039099 | MATHDB-317196

Rigidity of Jacobians along Lyapunov foliations

摘录 / EXCERPT: Jacobian-rigidity conjecture. Jacobians along Lyapunov foliations for α are rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-039100 | MATHDB-317197

Arithmeticity of Lyapunov exponents and entropy

摘录 / EXCERPT: Arithmeticity conjecture. The values of the Lyapunov exponents, and hence of the entropy, are logarithms of algebraic integers.

conjecture | open | MathDB 页面 | 原文线索

V2-039101 | MATHDB-317198

Local rigidity of measure and Lyapunov exponents

摘录 / EXCERPT: Local measure-rigidity conjecture. The action α' has an ergodic invariant measure μ' satisfying the same assumptions; μ' can be chosen so that $\mu'\to\mu$ in the we...

conjecture | open | MathDB 页面 | 原文线索

V2-039102 | MATHDB-317199

Existence of higher-dimensional actions satisfying Main Theorem (1) assumptions

摘录 / EXCERPT: Existence conjecture. An action satisfying the assumptions of Main Theorem (1) exists on every compact manifold of dimension three or higher.

conjecture | open | MathDB 页面 | 原文线索

V2-039103 | MATHDB-317219

Nonfaithfulness conjecture for C^1 actions of finite-index mapping class groups

摘录 / EXCERPT: Let S be a connected orientable surface with finitely many punctures, finitely many boundary components, and genus at least 2, and let G be a finite-index subgroup of the map...

conjecture | open | MathDB 页面 | 原文线索

V2-039104 | MATHDB-317226

Equality of the Hofer and virtual indices for Hamiltonian circle actions

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold, and let γ be a Hamiltonian circle action generated by a Morse Hamiltonian. The Hofer index $I^{H}(\gamma)$ is the maximal d...

conjecture | open | MathDB 页面 | 原文线索

V2-039105 | MATHDB-317232

Transversality conjecture for the first nullcline intersection

摘录 / EXCERPT: Let $\bar\gamma_{\bar\mu}$ be the special solution of the rescaled system, and consider its intersections with the nullcline $\bar y=\bar x^2$. Transversality conjecture. The least...

conjecture | open | MathDB 页面 | 原文线索

V2-039106 | MATHDB-317233

Primary-canard and exponentially narrow transition conjecture

摘录 / EXCERPT: Let $\bar\mu(\varepsilon)$ be a parameter function for the canonical three-dimensional system, and let $\bar\mu_0=-\frac{a(a+b)}{2b^2}$. Primary-canard transition conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-039107 | MATHDB-317239

The diameter-barycenter norm conjecture for generalized Sierpinski bodies

摘录 / EXCERPT: Diameter-barycenter norm conjecture. For every basis ${\cal E}$ of ${\mathbb R}^{n+1}$ there exist constants A and B such that

conjecture | open | MathDB 页面 | 原文线索

V2-039108 | MATHDB-317242

Positive entropy conjecture from nonzero residual secondary classes

摘录 / EXCERPT: Let ${\mathcal F}$ be a C^2-foliation. Let $h_I\wedge c_J$ be a residual class with $|J|=q$, and suppose ... is nonzero for some $\ell>1$. Positive entropy conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-039109 | MATHDB-317243

Hector's measure-zero conjecture for exceptional minimal sets

摘录 / EXCERPT: Let ${\mathcal F}$ be a codimension-one, C^2-foliation of a closed manifold M. An exceptional minimal set is a minimal set that is neither a compact leaf nor the whole manifold...

conjecture | open | MathDB 页面 | 原文线索

V2-039110 | MATHDB-317265

Invariance of Lyapunov exponents under limits of tidy perturbations

摘录 / EXCERPT: Let f be the diffeomorphism under consideration, and let $g_i=h_i f h_i^{-1}$ be tidy perturbations of f converging to $g=hfh^{-1}$, where the diffeomorphisms h_i converge to...

conjecture | open | MathDB 页面 | 原文线索

V2-039111 | MATHDB-317300

Renault's essential freeness conjecture for reduced crossed products

摘录 / EXCERPT: Let (A,G) be a C-dynamical system with G discrete. The action of G on the spectrum $\widehat{A}$ is called essentially free if, for every closed invariant subset...

conjecture | open | MathDB 页面 | 原文线索

V2-039112 | MATHDB-317304

Higher-regularity conjecture for conjugacies near irreducible toral automorphisms

摘录 / EXCERPT: Let L be a hyperbolic automorphism of $\mathbb T^d$, $d\geq 3$, with simple real spectrum and irreducible characteristic polynomial over $\mathbb Z$. Let $\mathcal U$ be the neig...

conjecture | open | MathDB 页面 | 原文线索

V2-039113 | MATHDB-317306

Local counterexample conjecture to smooth conjugacy from coinciding periodic data

摘录 / EXCERPT: Let $\mathcal U$ be the neighborhood of the hyperbolic toral automorphism considered in the source. Local counterexample conjecture. For any $f\in\mathcal U$ there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-039114 | MATHDB-317348

Spectral bounded isometry conjecture

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold, let $\operatorname{Symp}_0(M,\omega)$ be its identity component, and let $\operatorname{Ham}(M,\omega)$ be the Hamiltonian subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039115 | MATHDB-317403

Perturbation conjecture for Hamiltonian identity cycles

摘录 / EXCERPT: Let $\mathcal{F}^n=\{df=0\}$ be a degree n Hamiltonian foliation, and let δ be a Hamiltonian identity cycle, meaning an identity cycle of this foliation. A degree n pert...

conjecture | open | MathDB 页面 | 原文线索

V2-039116 | MATHDB-317570

Finite orbit-tile covering conjecture near dyadic Veech points

摘录 / EXCERPT: Finite orbit-tile covering conjecture. For $k=3,4,5,\ldots$, no neighborhood of V_{2^k} has a finite covering by orbit tiles.

conjecture | open | MathDB 页面 | 原文线索

V2-039117 | MATHDB-317597

Conjecture on variance and ergodic capacity in canonical and Kronecker channels

摘录 / EXCERPT: Variance-smoothing conjecture. For a large class of channels in the medium- to high-SNR regime,

conjecture | open | MathDB 页面 | 原文线索

V2-039118 | MATHDB-317644

The Dynamical Lehmer conjecture

摘录 / EXCERPT: Dynamical Lehmer conjecture. If $\alpha\in\overline K$ is not preperiodic for φ, then there is a constant $C=C(\varphi,K)$ not depending on α such that

conjecture | open | MathDB 页面 | 原文线索

V2-039119 | MATHDB-317706

Holomorphic mass equidistribution conjecture for Hecke eigencuspforms

摘录 / EXCERPT: Let $\mathbb{H}$ be the upper half plane with hyperbolic measure $d\mu z=y^{-2}dxdy$, let $\Gamma=\operatorname{SL}_2(\mathbb{Z})$, and set $X=\Gamma\backslash\mathbb{H}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039120 | MATHDB-317707

Quantum unique ergodicity conjecture for Hecke-Maass cusp forms

摘录 / EXCERPT: Let X be the quotient space above, let $\mathcal{L}(X)$ denote the class of functions under consideration, and let $\{u_j\}$ be an orthonormal basis of Hecke-Maass cusp forms on...

conjecture | open | MathDB 页面 | 原文线索

V2-039121 | MATHDB-317708

Classification conjecture for rational maps with equal dynamical degrees

摘录 / EXCERPT: Classification conjecture. Either F is holomorphic in some model, or F preserves a rational fibration, or F is conjugated to a monomial map.

conjecture | open | MathDB 页面 | 原文线索

V2-039122 | MATHDB-317714

Uniqueness conjecture for analytic families with continuous modulus representatives

摘录 / EXCERPT: Uniqueness conjecture. The only families with representative $\psi_{\epsilon}^{0,\infty}$ of the modulus which are analytic in ϵ are the ones presented in the theorem on t...

conjecture | open | MathDB 页面 | 原文线索

V2-039123 | MATHDB-317799

Ghys's alternative conjecture for diffeomorphism groups

摘录 / EXCERPT: Let M be a compact manifold and let Γ be a finitely generated group of smooth diffeomorphisms of M. Ghys's alternative conjecture. Either Γ contains a free group...

conjecture | open | MathDB 页面 | 原文线索

V2-039124 | MATHDB-317834

Power stability of good sequences for multiple ergodic convergence

摘录 / EXCERPT: Let (s_n) be a sequence of integers, and let $\boldsymbol{\text{good for }\ell\text{-convergence}}$ mean that the sequence has the corresponding multiple ergodic convergence prop...

conjecture | open | MathDB 页面 | 原文线索

V2-039125 | MATHDB-317835

Power transfer from mean $(k\ell)$-convergence to ℓ-convergence

摘录 / EXCERPT: Let (s_n) be a sequence of integers, and let $k,\ell\in\mathbb{N}$. Power-transfer conjecture. If (s_n) is good for mean $(k\ell)$-convergence, then (s_n^k) is good for...

conjecture | open | MathDB 页面 | 原文线索

V2-039126 | MATHDB-317836

Nilmanifold characterization of good sequences for multiple convergence

摘录 / EXCERPT: Let (s_n) be a sequence of integers. An ℓ-step nilsystem is a measure-preserving system arising from a translation on a compact homogeneous space G/Γ, where G is an...

conjecture | open | MathDB 页面 | 原文线索

V2-039127 | MATHDB-317974

Conjecture on the limiting process with a growing number of light particles

摘录 / EXCERPT: The limiting-process conjecture. The limiting process $\mathbf{Q}(\tau)$ is (a) straight motion if $\beta=\alpha+1/2-\epsilon$; (b) the integral of an Ornstein-Uhlenbeck process,

conjecture | open | MathDB 页面 | 原文线索

V2-039128 | MATHDB-317989

Derivative-criterion conjecture for Bernoulli abstract linked-twist maps

摘录 / EXCERPT: Let M be a smooth manifold of dimension 2, let C_i be cylinders, let $E_i:C_i\to A\subset M$ be embeddings, and let $T_i:C_i\to C_i$ be twist maps, with associated linked-twi...

conjecture | open | MathDB 页面 | 原文线索

V2-039129 | MATHDB-317991

The Lehmer-type height conjecture for polarized dynamical systems

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system over a number field K, with canonical height $\hat h_{\mathscr L}$, and let $x\in X(\overline{\mathbb Q})$ be a non-preperi...

conjecture | open | MathDB 页面 | 原文线索

V2-039130 | MATHDB-317992

The uniform boundedness problem for rational preperiodic points

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system defined over a number field K, with weight d, dimension $\dim X$, and polarization degree $c_1(\mathscr L)^{\dim X}$. Uni...

open_problem | open | MathDB 页面 | 原文线索

V2-039131 | MATHDB-317994

The equidistribution conjecture for small points in polarized dynamical systems

摘录 / EXCERPT: Let $(X,f,\mathscr L)$ be a polarized dynamical system over a number field K. For $x\in X(\overline{\mathbb Q})$, let δ_x be the probability measure on $X(\mathbb C)$ obta...

open_problem | open | MathDB 页面 | 原文线索

V2-039132 | MATHDB-318016

All-orders Hamiltonian conjecture for rotational water waves

摘录 / EXCERPT: Let ϵ and δ be the perturbation parameters, and let $\mathcal{H}_{nm}$ denote the Hamiltonian contribution of bidegree (n,m) in the grading with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-039133 | MATHDB-318069

Mishchenko-Fomenko conjecture for infinitely dimensional analytic integrals

摘录 / EXCERPT: Let M be a real-analytic symplectic manifold of dimension $2n$, and consider a Hamiltonian system ... Assume that the system is completely integrable by an infinitely dimensional n...

conjecture | open | MathDB 页面 | 原文线索

V2-039134 | MATHDB-318128

Persistence conjecture for compact normally AS laminations

摘录 / EXCERPT: A compact lamination $(L,\mathcal L)$ embedded in a manifold M is preserved by a diffeomorphism f and is normally AS if there exists $\epsilon>0$ such that the saturated nonw...

conjecture | open | MathDB 页面 | 原文线索

V2-039135 | MATHDB-318142

The generic failure of persistence through the focus-node bifurcation

摘录 / EXCERPT: Let t^{**} be the anti-bump and let t^*_μ be the corresponding bump, with these quantities defined by ... Assume hypotheses HFN1 through HFN5. The focus-node persistence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-039136 | MATHDB-318143

No persistence along the slow curve after the focus-node bifurcation

摘录 / EXCERPT: Consider the system ... Let R denote the relief function and let a and b be the parameters appearing in the focus-node bifurcation, with b the focus-node bifurcation point. T...

conjecture | open | MathDB 页面 | 原文线索

V2-039137 | MATHDB-318144

Equality of the asymptotic expansions at the focus-node bifurcation

摘录 / EXCERPT: Let $X_-(b)$ and $X_+(b)$ be the two values of the trajectory quantity at the focus-node bifurcation point b, obtained from the two distinguished trajectories considered in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-039138 | MATHDB-318175

Extension of Bourgain's Return Times theorem below $p=2$

摘录 / EXCERPT: Return Times conjecture. For every function $g\in L^q(X)$ there is a universal set $X_0\subseteq X$ with $\mu(X_0)=1$ such that, for every second dynamical system...

conjecture | open | MathDB 页面 | 原文线索

V2-039139 | MATHDB-318185

Ergodicity and degenerate limit shape for multiplicative statistics with rad_1<1

摘录 / EXCERPT: Let $rad_1<1$ in the setting of Theorem, and suppose that f has a pole at rad_1. Let $mu^{(n)}$ denote the corresponding measures and let $alpha^{(n)}$ be their scaling functio...

conjecture | open | MathDB 页面 | 原文线索

V2-039140 | MATHDB-318247

Parry's conjecture on absolutely continuous spectral types

摘录 / EXCERPT: A measure-preserving transformation with simple spectrum has a reduced maximal spectral type. Parry's conjecture. If the reduced maximal spectral type of some transformation with s...

conjecture | open | MathDB 页面 | 原文线索

V2-039141 | MATHDB-318310

The transcendence-degree conjecture for periodic solutions of the circular double Sitnikov problem

摘录 / EXCERPT: Let a periodic solution of the circular double Sitnikov problem have period $\tau=2n\pi$ for some $n\in\mathbb N$, and let $k^*\in\mathbb R\setminus\mathbb A$. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-039142 | MATHDB-318341

The Luo-Sarnak variance formula for Maass-Hecke eigenfunctions

摘录 / EXCERPT: Luo-Sarnak variance conjecture. The displayed asymptotic holds for the relevant Hecke eigenfunctions. Luo and Sarnak proved the analogous statement for holomorphic Hecke eigenforms...

conjecture | open | MathDB 页面 | 原文线索

V2-039143 | MATHDB-318347

Reduced smoothness conjecture for cascades in continuous families

摘录 / EXCERPT: Let $f:\mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ be a continuous family and let $U \subset \mathbb{R} \times \mathbb{R}^n$ be such that the periodic orbits in U are contai...

conjecture | open | MathDB 页面 | 原文线索

V2-039144 | MATHDB-318356

Eigenstate characterization conjecture for cuspidal nilpotent Lax orbits

摘录 / EXCERPT: For a higher group with more Hamiltonians, consider Lax operators whose orbits are cuspidal nilpotent orbits: their operators have all vanishing eigenvalues and are not diagonaliza...

conjecture | open | MathDB 页面 | 原文线索

V2-039145 | MATHDB-318382

The strengthened billiard-trajectory conjecture for rounded disk-polygons

摘录 / EXCERPT: Rounded disk-polygon billiard conjecture. Any of the shortest generalized billiard trajectories in $D(\varepsilon)$ is a 2-periodic one for all ε at most as large as...

conjecture | open | MathDB 页面 | 原文线索

V2-039146 | MATHDB-318411

Hamiltonicity conjecture for minimum-degree-three random graphs

摘录 / EXCERPT: Let $G_{n,cn}$ be the random graph on n vertices with cn edges, conditioned on having minimum degree at least 3. Hamiltonicity conjecture. For any $c\geq 3/2$, $G_{n,cn}$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-039147 | MATHDB-318454

Birman-Williams braid conjugacy conjecture for isotopic Lorenz orbits

摘录 / EXCERPT: Let b and b' be Birman-Williams braids associated with two isotopic periodic orbits of the Lorenz flow. If t is the common period of these orbits, then the Birman-Williams br...

conjecture | open | MathDB 页面 | 原文线索

V2-039148 | MATHDB-318477

Palis's finiteness and prevalence conjecture for ergodic attractors

摘录 / EXCERPT: Let M be a compact manifold and let ${\rm Diff}^r(M)$ denote the space of C^r diffeomorphisms of M. An ergodic attractor is an attractor supporting an ergodic invariant proba...

conjecture | open | MathDB 页面 | 原文线索

V2-039149 | MATHDB-318478

Hurley's conjecture on quasi-attractors of generic points

摘录 / EXCERPT: Let M be a compact manifold, let ${\rm Diff}^r(M)$ be the space of C^r diffeomorphisms of M, and call a point generic when it belongs to the full-measure generic set under co...

conjecture | open | MathDB 页面 | 原文线索

V2-039150 | MATHDB-318551

Almost Reducibility Conjecture for analytic cocycles

摘录 / EXCERPT: Let (α,A) be an analytic cocycle, and let A_ϵ denote its complexified cocycle at imaginary shift ϵ. Assume that $L(\alpha,A_\epsilon)=0$ for $a<\epsilon<b$...

conjecture | open | MathDB 页面 | 原文线索

V2-039151 | MATHDB-318576

Hamiltonicity hitting-time conjecture for edge-disjoint Hamilton cycles in random geometric graphs

摘录 / EXCERPT: Hamiltonicity hitting-time conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-039152 | MATHDB-318634

Downarowicz and Newhouse's symbolic extension conjecture for smooth diffeomorphisms

摘录 / EXCERPT: Downarowicz and Newhouse's conjecture. Every $f\in\mathrm{Diff}^r(M)$ has a symbolic extension.

conjecture | open | MathDB 页面 | 原文线索

V2-039153 | MATHDB-318800

The return-time dimension conjecture for a large class of dynamical systems

摘录 / EXCERPT: Return-time dimension conjecture. For a large class of dynamical systems,

conjecture | open | MathDB 页面 | 原文线索

V2-039154 | MATHDB-318824

Finite-product independence conjecture for measure-preserving dynamical systems

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of subsets of the positive integers, and let a measure-preserving dynamical system (m.d.s.) be called $\mathcal{F}$-independent when it has the indepe...

conjecture | open | MathDB 页面 | 原文线索

V2-039155 | MATHDB-318872

Thomassen's resilience conjecture for Hamilton cycles in regular tournaments

摘录 / EXCERPT: Let G be a regular tournament on $2k+1$ vertices, and let A be any set of at most $k-1$ edges of G. Thomassen's conjecture. The graph $G-A$ has a Hamilton cycle. This is a we...

conjecture | open | MathDB 页面 | 原文线索

V2-039156 | MATHDB-318873

Nash-Williams's minimum-degree conjecture for Hamilton cycle packings

摘录 / EXCERPT: Let G be a graph on n vertices with minimum degree at least $n/2$. Nash-Williams's conjecture. The graph G contains $n/8+o(n)$ edge-disjoint Hamilton cycles. The conjecture c...

conjecture | open | MathDB 页面 | 原文线索

V2-039157 | MATHDB-318886

Monotonicity conjecture for the stability range of the delayed logistic map

摘录 / EXCERPT: Let K be the carrying capacity, r the growth parameter, and τ a nonnegative integer time delay. Consider the delayed logistic recurrence ... For a fixed delay, write the two...

conjecture | open | MathDB 页面 | 原文线索

V2-039158 | MATHDB-318945

Nonextendability of the dominated splitting at the singularities

摘录 / EXCERPT: Let Λ be the singular attractor containing the hyperbolic singularities s_0 and s_1 of different indexes. On the part of Λ away from the equilibria, suppose a d...

conjecture | open | MathDB 页面 | 原文线索

V2-039159 | MATHDB-318946

Positive Lyapunov exponents for quotient maps outside the sink basin

摘录 / EXCERPT: Let X be the vector field from Theorem 3.2, let $\mathcal{U}$ be its neighborhood in $\mathfrak{X}^2(M)$, and for $Y\in\mathcal{U}$ let g_Y be the quotient map. Let $p(Y)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-039160 | MATHDB-318947

Physical hyperbolic measure for quotient maps away from the sink basin

摘录 / EXCERPT: Let X be the vector field under consideration and let Y be a vector field C^2 close to X. Let g_Y be the quotient map, let μ_Y denote a candidate physical measure, an...

conjecture | open | MathDB 页面 | 原文线索

V2-039161 | MATHDB-318954

Kurlberg-Rudnick's limiting distribution conjecture for Hecke matrix elements

摘录 / EXCERPT: Kurlberg-Rudnick's conjecture. As $N\to\infty$ through primes, the limiting distribution of the normalized matrix elements $F_j^{(N)}$ is the distribution of

conjecture | open | MathDB 页面 | 原文线索

V2-039162 | MATHDB-318955

Kurlberg-Rudnick's IID conjecture for quadratic-form-indexed Hecke matrix elements

摘录 / EXCERPT: Quadratic-form IID conjecture. For every $\nu\in\mathbb{Z}$, $Y_\nu(\psi)$ has limiting distribution equal to that of $\operatorname{tr}(U_\nu)$, where U_ν is Haar-random in...

conjecture | open | MathDB 页面 | 原文线索

V2-039163 | MATHDB-319006

Mañé's strong conjecture on generic minimizing measures

摘录 / EXCERPT: Mañé's strong conjecture. The set $\mathcal{O}_1(L)$ is residual in $C^{\infty}(M)$. Equivalently, for a generic potential perturbation, there is a unique minimizing measure and it...

conjecture | open | MathDB 页面 | 原文线索

V2-039164 | MATHDB-319008

Mañé's Aubry-set conjecture for generic potential perturbations

摘录 / EXCERPT: Mañé's Aubry-set conjecture. The set $\mathcal{O}_3(L)$ is residual in $C^{\infty}(M)$. The source states that this conjecture is equivalent to the strong conjecture on generic min...

conjecture | open | MathDB 页面 | 原文线索

V2-039165 | MATHDB-319020

Reformulated Wright conjecture on the principal branch and isolas of slowly oscillating periodic solutions

摘录 / EXCERPT: Consider Wright's equation ... Let $\mathcal{F}_0$ be the branch of slowly oscillating periodic solutions (SOPS) that bifurcates forward in α at $\alpha=\frac{\pi}{2}$. A fold...

conjecture | open | MathDB 页面 | 原文线索

V2-039166 | MATHDB-319082

The conjectured sharp threshold for connectivity and Hamiltonicity in random intersection graphs

摘录 / EXCERPT: Connectivity and Hamiltonicity conjecture. Under these assumptions, $\mathcal{G}(n,m,p)$ is k-connected for every fixed constant k and contains a Hamilton cycle with high proba...

conjecture | open | MathDB 页面 | 原文线索

V2-039167 | MATHDB-319094

The 3x+1 tree growth constant conjecture

摘录 / EXCERPT: 3x+1 tree growth constant conjecture. For every integer $a\not\equiv0\pmod 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-039168 | MATHDB-319095

Furstenberg's conjecture on projections of ×2- and ×3-invariant sets

摘录 / EXCERPT: Let $T_m:[0,1]\to[0,1]$ be the map $x\mapsto mx\bmod 1$, and let $\Pi_{2,1}$ be the space of orthogonal projections from $\mathbb{R}^2$ to one-dimensional subspaces. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039169 | MATHDB-319116

The GMY structure characterization conjecture

摘录 / EXCERPT: A dynamical system is a system to which the notions of a Gibbs-Markov-Young (GMY) structure and nonuniform hyperbolicity apply. GMY structure characterization conjecture. A dynamic...

conjecture | open | MathDB 页面 | 原文线索

V2-039170 | MATHDB-319208

The simultaneous preperiodicity conjecture for 0 and 1 in the quadratic family

摘录 / EXCERPT: For $c\in\mathbb{C}$, let $f_c(z)=z^2+c$, and let $S_{0,1}$ denote the set of parameters c for which both 0 and 1 are preperiodic under iteration of f_c. Simultaneous prepe...

conjecture | open | MathDB 页面 | 原文线索

V2-039171 | MATHDB-319213

Dissipativity conjecture for products of conservative interval maps without acips

摘录 / EXCERPT: Dissipativity conjecture. The product system (I^2,f_2,λ_2) is dissipative.

conjecture | open | MathDB 页面 | 原文线索

V2-039172 | MATHDB-319254

Asymptotic local resilience of binomial random graphs for Hamiltonicity

摘录 / EXCERPT: Let $\mathcal{G}(n,p)$ be the binomial random graph on n vertices, let $\mathcal{HAM}$ denote the property of being Hamiltonian, and let $r_\ell(G,\mathcal{HAM})$ denote the loca...

conjecture | open | MathDB 页面 | 原文线索

V2-039173 | MATHDB-319258

Positive-coupling real-spectrum conjecture for inhomogeneous spin chains

摘录 / EXCERPT: Let S be the spin, let γ be the deformation parameter, and let $\boldsymbol{H}$ be the Hamiltonian of a spin chain with inhomogeneous coupling constants a_n. Positive-co...

conjecture | open | MathDB 页面 | 原文线索

V2-039174 | MATHDB-319337

Generalized local entropy estimate for ^{r} maps

摘录 / EXCERPT: Let $r>1$, let M be a compact manifold of dimension d, and let $T:M\to M$ be a ^{r} map. For nonintegral r, assume $T\in \mathcal{C}^{\lfloor r\rfloor}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039175 | MATHDB-319349

Bergelson-Leibman polynomial ergodic averages convergence conjecture

摘录 / EXCERPT: Let $(X,\mathcal{X},\mu)$ be a probability space, let $T_1,\ldots,T_\ell\colon X\to X$ be commuting, invertible measure-preserving transformations, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039176 | MATHDB-319352

The generic dichotomy between zero exponents and nonuniform Anosov ergodicity

摘录 / EXCERPT: Let M be the compact manifold equipped with volume measure m, and let $\mathrm{Diff}_m^1(M)$ denote the space of C^1 volume-preserving diffeomorphisms. A diffeomorphism is no...

conjecture | open | MathDB 页面 | 原文线索

V2-039177 | MATHDB-319353

Smale's density conjecture for hyperbolic surface diffeomorphisms

摘录 / EXCERPT: Soit S une surface compacte, et notons $\operatorname{Diff}^1(S)$ l'espace des difféomorphismes de classe C^1 de S. Un difféomorphisme est hyperbolique s'il satisfait la défi...

conjecture | open | MathDB 页面 | 原文线索

V2-039178 | MATHDB-319354

Palis's density conjecture for hyperbolicity and dynamical obstructions

摘录 / EXCERPT: Soit M une variété compacte, et notons $\operatorname{Diff}^k(M)$ l'espace des difféomorphismes de classe C^k de M. Une tangence homocline est une intersection non transverse...

conjecture | open | MathDB 页面 | 原文线索

V2-039179 | MATHDB-319355

Bonatti's finiteness conjecture for chain-recurrence classes away from homoclinic tangencies

摘录 / EXCERPT: Soit M une variété compacte, et considérons des difféomorphismes de classe C^1 dans $\operatorname{Diff}^1(M)$. Un difféomorphisme est loin des tangences homoclines s'il appart...

conjecture | open | MathDB 页面 | 原文线索

V2-039180 | MATHDB-319356

Bonatti-Díaz hyperbolicity conjecture for robust heterodimensional cycles

摘录 / EXCERPT: Soit M une variété compacte, et notons $\operatorname{Diff}^1(M)$ l'espace des difféomorphismes de classe C^1. Un cycle hétérodimensionnel robuste est un cycle hétérodimensionn...

conjecture | open | MathDB 页面 | 原文线索

V2-039181 | MATHDB-319357

Smale's density conjecture for hyperbolic diffeomorphisms on surfaces

摘录 / EXCERPT: Soit M une variété compacte de dimension 2, et notons $\operatorname{Diff}^1(M)$ l'espace des difféomorphismes de classe C^1. Un difféomorphisme est hyperbolique s'il satisfa...

conjecture | open | MathDB 页面 | 原文线索

V2-039182 | MATHDB-319358

Palis's hyperbolicity conjecture with homoclinic and heterodimensional obstructions

摘录 / EXCERPT: Soit M une variété compacte, et notons $\operatorname{Diff}^1(M)$ l'espace des difféomorphismes de classe C^1. Une tangence homocline est une intersection non transverse entre...

conjecture | open | MathDB 页面 | 原文线索

V2-039183 | MATHDB-319359

Bonatti's finiteness conjecture away from homoclinic tangencies

摘录 / EXCERPT: Soit M une variété compacte, et soit $\mathcal{T}$ l'ensemble des difféomorphismes présentant une tangence homocline. Considérons le complémentaire...

conjecture | open | MathDB 页面 | 原文线索

V2-039184 | MATHDB-319363

Spectral decomposition conjecture for typical one-frequency Schrödinger operators

摘录 / EXCERPT: Let v, α, and θ be typical parameters for the one-frequency Schrödinger operator $H_{v,\alpha,\theta}$. The operator is a one-frequency Schrödinger operator, and “l...

conjecture | open | MathDB 页面 | 原文线索

V2-039185 | MATHDB-319373

Primary piecewise Fagnano orbit chronology conjecture for the Koch snowflake billiard

摘录 / EXCERPT: Let KS be the Koch snowflake billiard, and let $pp\mathscr{F}$ denote the proposed primary piecewise Fagnano orbit of KS. Its chronology is the order in which the billiard traj...

conjecture | open | MathDB 页面 | 原文线索

V2-039186 | MATHDB-319374

Fractal flat surface and flow conjecture for the Koch snowflake billiard

摘录 / EXCERPT: Let KS_n be the prefractal polygonal rational billiards converging to the Koch snowflake billiard KS, and let $\mathcal{S}_{KS_n}$ be their associated flat surfaces. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039187 | MATHDB-319402

Bidaut-Véron-Szulkin conjecture for the Hamiltonian system

摘录 / EXCERPT: Bidaut-Véron-Szulkin conjecture. System (SH) with $a=b=0$ admits no radial or nonradial global solutions if and only if (δ,μ) lies under the hyperbola

conjecture | open | MathDB 页面 | 原文线索

V2-039188 | MATHDB-319406

Common critical subsolution conjecture for commuting time-periodic Tonelli Hamiltonians

摘录 / EXCERPT: Common critical subsolution conjecture. If $[H_1,H_2]=0$, then there exists at least one common $C^{1,1}$ subsolution for

conjecture | open | MathDB 页面 | 原文线索

V2-039189 | MATHDB-319438

The unstable-manifold conditional-measure conjecture

摘录 / EXCERPT: Recall that a strongly simple action has one-dimensional coarse Lyapunov distributions, which integrate to Lyapunov foliations, and that some Lyapunov foliations have conditional m...

conjecture | open | MathDB 页面 | 原文线索

V2-039190 | MATHDB-319439

Uniqueness of large invariant measures for infranilmanifold actions

摘录 / EXCERPT: Let N be a simply connected nilpotent Lie group, let A be a group of affine transformations acting freely with a finite-index translation subgroup that is a lattice in N, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039191 | MATHDB-319495

Relative uniqueness conjecture for maximal-pressure measures

摘录 / EXCERPT: Let (X,Y,π) be a factor triple, let ν be a fully supported ergodic measure on Y, and let c_π denote the class degree of the factor map. For a function...

conjecture | open | MathDB 页面 | 原文线索

V2-039192 | MATHDB-319757

Residual ergodicity and vanishing new Lyapunov exponents for area-preserving surface homeomorphisms

摘录 / EXCERPT: Let S be the surface under consideration, let $\lambdaup$ denote Lebesgue measure, and let $\text{Hom}_\lambdaup(S)$ be the space of area-preserving homeomorphisms of S. A Ly...

conjecture | open | MathDB 页面 | 原文线索

V2-039193 | MATHDB-319758

Three-dimensional zero-exponent approximation conjecture for area-preserving homeomorphisms

摘录 / EXCERPT: Let M be a compact, connected, boundaryless, smooth three-dimensional manifold, let $\lambdaup$ be its Lebesgue measure, and let $\text{Hom}_\lambdaup(M)$ denote the space of a...

conjecture | open | MathDB 页面 | 原文线索

V2-039194 | MATHDB-319800

The star-system hyperbolicity conjecture

摘录 / EXCERPT: A star system is a dynamical system for which all sufficiently small perturbations have only hyperbolic closed orbits; its non-wandering set is denoted by Ω. Star-system hyp...

open_problem | open | MathDB 页面 | 原文线索

V2-039195 | MATHDB-319818

Nakada-Natsui full-measure conjecture for the matching set

摘录 / EXCERPT: Let $\{T_\alpha\}_{\alpha\in(0,1]}$ be the family of α-continued fraction transformations. A parameter interval $I\subset(0,1]$ is a matching interval if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-039196 | MATHDB-319836

Asymptotic nilpotency conjecture for cellular automata in all dimensions

摘录 / EXCERPT: Asymptotic nilpotency conjecture. Any asymptotically nilpotent d-dimensional cellular automaton $(A^{\mathbb{M}^d},F)$, with $d\in\mathbb{N}_+$, is nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-039197 | MATHDB-319849

Classification conjecture for integrable bi-Hamiltonian equations

摘录 / EXCERPT: An integrable bi-Hamiltonian equation in one dependent variable u is considered up to contact transformations. The relevant model hierarchies are the linear, KdV, and HD hierarch...

conjecture | open | MathDB 页面 | 原文线索

V2-039198 | MATHDB-319852

Local nondegeneracy conjecture for the frequency map of ellipsoid billiards

摘录 / EXCERPT: Let Q be a nondegenerate ellipsoid, let $\boldsymbol{\boldsymbol{a}}=(a_1,\ldots,a_n)$ be its parameters, let Λ be the nonsingular caustic space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039199 | MATHDB-319975

Generalized quantum unique ergodicity conjecture for ergodic flows and maps

摘录 / EXCERPT: Let Φ^t_H be an ergodic Hamiltonian flow on an energy shell $\mathcal{E}_E$, with quantum Hamiltonian $\hat H_\hbar$ and eigenstates $\psi_{\hbar,j}$ of energies...

conjecture | open | MathDB 页面 | 原文线索

V2-039200 | MATHDB-319976

Zelditch's nodal-measure equidistribution conjecture

摘录 / EXCERPT: Let (M,g) be a compact smooth Riemannian manifold with ergodic geodesic flow, and let $(\psi_j)_{j\geq1}$ be any orthonormal basis of eigenfunctions. For an eigenfunction, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039201 | MATHDB-319978

Durrett-Rogers conjecture on sublinear growth of repulsive self-interacting motion

摘录 / EXCERPT: Suppose that $f:\mathbb{R}\rightarrow\mathbb{R}$ is an odd function of compact support such that $xf(x)\ge 0$. Let X be the process defined by the Durrett-Rogers equation. Durret...

conjecture | open | MathDB 页面 | 原文线索

V2-039202 | MATHDB-319990

The general structural conjecture for joinings of partially pro-nilsystem actions

摘录 / EXCERPT: General Structural Conjecture. Depending only on these data, there are finite families of pairs $(\Lambda_{i,j},m_{i,j})$ for $i=1,2,\ldots,r$, with $m_{i,j}\geq 0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039203 | MATHDB-319992

Bang-Jensen, Gutin and Li's dominated-pair conjecture

摘录 / EXCERPT: Bang-Jensen, Gutin and Li's conjecture. If $d(x)+d(y)\geq 2n-1$ for every dominated pair of non-adjacent vertices in G, then G has a Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-039204 | MATHDB-320008

Frieze-Krivelevich Hamilton decomposition conjecture for random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the binomial random graph and let $\delta(G_{n,p})$ denote its minimum degree. Frieze-Krivelevich conjecture. For any p, with high probability, $G_{n,p}$ contain...

conjecture | open | MathDB 页面 | 原文线索

V2-039205 | MATHDB-320056

Ergodicity conjecture for large skewed-stadium parameter

摘录 / EXCERPT: Fix a parameter $b<1$ and let c be the second parameter of the paper's two-parameter family of convex billiard tables. The billiard dynamics refer to the dynamics of the billiard...

conjecture | open | MathDB 页面 | 原文线索

V2-039206 | MATHDB-320073

The characterization conjecture for C^1-inverse stable endomorphisms

摘录 / EXCERPT: Characterization conjecture. The C^1-inverse stable endomorphisms are exactly those which satisfy axiom A and the strong transversality condition. The preceding results establish...

conjecture | open | MathDB 页面 | 原文线索

V2-039207 | MATHDB-320180

Quasi-flat minimizer conjecture for spectral length

摘录 / EXCERPT: Let M be a symplectic manifold with symplectic form ω, and let $\widetilde{\phi}\in\widetilde{\operatorname{Ham}}(M,\omega)$ be Floer non-degenerate. A generalized path fr...

conjecture | open | MathDB 页面 | 原文线索

V2-039208 | MATHDB-320181

Pseudo-injectivity radius conjecture for the symplectic two-sphere

摘录 / EXCERPT: Let (S^2,ω_{st}) be the symplectic two-sphere normalized by ... Let $inj(M,\omega)$ be the pseudo-injectivity radius, namely the supremum of the radii r such that...

conjecture | open | MathDB 页面 | 原文线索

V2-039209 | MATHDB-320334

Generic maximality conjecture for scaling sequences

摘录 / EXCERPT: Let T be an ergodic automorphism, and let a T-admissible Lipschitz metric be a metric satisfying the admissibility and Lipschitz conditions for T. A metric is T-maximal whe...

conjecture | open | MathDB 页面 | 原文线索

V2-039210 | MATHDB-320398

The cellular Markov partition conjecture for expanding Thurston maps

摘录 / EXCERPT: Cellular Markov partition conjecture. Every expanding Thurston map admits a cellular Markov partition.

conjecture | open | MathDB 页面 | 原文线索

V2-039211 | MATHDB-320469

Feingold-Peres-type conjecture for Patterson-Sullivan quantum variances

摘录 / EXCERPT: Let ${\mathbf X_{\Gamma}}$ be a compact hyperbolic surface. For a zeroth-order pseudo-differential operator with principal symbol a_0, let $PS_a(\lambda)$ denote the variance of...

conjecture | open | MathDB 页面 | 原文线索

V2-039212 | MATHDB-320473

Ergodicity conjecture for convergence of the information state

摘录 / EXCERPT: Ergodicity conjecture. The conditions of that theorem can be weakened to the case when the underlying chain is only ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-039213 | MATHDB-320623

Conjecture on resonant cyclotron acceleration in the original dynamical system

摘录 / EXCERPT: Let $I(t)=(F(t)-|a(t)|)/2$ and $\varphi(t)=bt+\phi(t)$ be the action-angle variables, where $F(t)$ and $\phi(t)$ are the variables of the original dynamical system, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039214 | MATHDB-320639

Conjecture on limit cycles in regions 1, 2, 4-8, and 12

摘录 / EXCERPT: Let (ρ,ϕ) be the parameters of the collective Parrondo game, and let the parameter plane be divided into the regions shown in Figure. For a finite word w in the game symb...

conjecture | open | MathDB 页面 | 原文线索

V2-039215 | MATHDB-320640

Conjecture on limit cycles for adjacent even cycle lengths

摘录 / EXCERPT: Let $n\ge4$ be even. Let ρ and ϕ be the game parameters, let π_0 and b_{n-2} be the quantities defined for the stationary dynamics, and let $G_{n,n-2}$, E_{n-2},...

conjecture | open | MathDB 页面 | 原文线索

V2-039216 | MATHDB-320643

Hofbauer's converse conjecture for stable dynamics in bimatrix games

摘录 / EXCERPT: Let (A,B) be a bimatrix game with a unique Nash equilibrium point in the interior $\mathring\Sigma$ of the strategy space Σ. Suppose this equilibrium is stable under eithe...

conjecture | open | MathDB 页面 | 原文线索

V2-039217 | MATHDB-320655

Strong Palis conjecture for robust heterodimensional cycles

摘录 / EXCERPT: Let M be a closed manifold, and let $\operatorname{Diff}^1(M)$ be the space of C^1-diffeomorphisms of M. A robust heterodimensional cycle is a heterodimensional cycle that pe...

conjecture | open | MathDB 页面 | 原文线索

V2-039218 | MATHDB-320660

Generic symplectic Oseledets splitting and ergodicity conjecture

摘录 / EXCERPT: Generic symplectic Oseledets splitting conjecture. For a generic $f\in \operatorname{Diff}^1_\omega(M)$ either all Lyapunov exponents vanish almost everywhere, or else the Oseledet...

conjecture | open | MathDB 页面 | 原文线索

V2-039219 | MATHDB-320667

Hamiltonian stability conjecture for compact semisimple Lagrangian orbits in complex projective space

摘录 / EXCERPT: Let G be a compact (semi)simple subgroup of $\SU{N}$ for some N, and let $\mathcal O$ be a Lagrangian orbit of G in $\CP^{N-1}$. Hamiltonian stability conjecture. If G admi...

conjecture | open | MathDB 页面 | 原文线索

V2-039220 | MATHDB-320700

Treglown's oriented graph analogue of Pósa's conjecture

摘录 / EXCERPT: Let G be an oriented graph, meaning a directed graph with no pair of opposite directed edges, on n vertices. Write $\delta^+(G)$ and $\delta^-(G)$ for its minimum outdegree and...

conjecture | open | MathDB 页面 | 原文线索

V2-039221 | MATHDB-320751

Solitonic behaviour conjecture for discrete dynamical systems from general whurl relations

摘录 / EXCERPT: The loops around the cylinder are called whurls, and a whurl relation is the transformation R on the parameters of adjacent whurls that preserves boundary measurements. In the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-039222 | MATHDB-320825

De Sole-Kac-Wakimoto conjecture on normal forms of Hamiltonian operators

摘录 / EXCERPT: De Sole-Kac-Wakimoto conjecture. There exists a contact transformation that brings H to either a quasiconstant coefficient skew-adjoint differential operator, or to a linear comb...

conjecture | open | MathDB 页面 | 原文线索

V2-039223 | MATHDB-320826

Conjecture on reducing Hamiltonian operators by contact transformations and potentiation

摘录 / EXCERPT: Potentiation normal-form conjecture. H can be transformed into a quasiconstant coefficient skew-adjoint differential operator using either a contact transformation or a compositi...

conjecture | open | MathDB 页面 | 原文线索

V2-039224 | MATHDB-320827

Conjecture on reducing Hamiltonian operators to the Gardner operator

摘录 / EXCERPT: Gardner normal-form conjecture. H can be transformed into the Gardner operator D by either a transformation from the potentiation normal-form conjecture or a composition thereo...

conjecture | open | MathDB 页面 | 原文线索

V2-039225 | MATHDB-320828

No-elliptic-islands conjecture for the universal area-preserving map

摘录 / EXCERPT: Let F_* be the universal area-preserving map, let $\mathcal{D}^\infty$ be its domain of analytic continuation, and let $\tilde{\mathcal{D}}^\infty$ denote the corresponding invar...

conjecture | open | MathDB 页面 | 原文线索

V2-039226 | MATHDB-320829

Disjoint complex-root conjecture for the polynomials S_n

摘录 / EXCERPT: For each integer $n>1$, define ... The complex roots of S_n are its roots other than $x=\pm1$. Disjoint-root conjecture. If $m,n>1$ are distinct integers, then the sets of complex...

conjecture | open | MathDB 页面 | 原文线索

V2-039227 | MATHDB-320830

Tangent-equation disjointness conjecture

摘录 / EXCERPT: Tangent-equation conjecture. These two equations have no common solutions in $(0,\pi/2)$. The source states this as an equivalent formulation of the disjoint-root conjecture for th...

conjecture | open | MathDB 页面 | 原文线索

V2-039228 | MATHDB-320888

Polynomiality conjecture for the second Hamiltonian structure of the principal hierarchy

摘录 / EXCERPT: Let $P_2^{[0]}$ be the second Hamiltonian structure of the principal hierarchy, and let P_2 denote its image under the quasi-Miura transformation ... Here a Hamiltonian structure h...

conjecture | open | MathDB 页面 | 原文线索

V2-039229 | MATHDB-320897

Arbitrary-regularity approximation conjecture for Lyapunov vector fields

摘录 / EXCERPT: Arbitrary-regularity approximation conjecture. In any C^r neighborhood of X, there exists a vector field that coincides with X in a neighborhood of X and satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-039230 | MATHDB-320945

Noise sensitivity of crossing events under nearest-neighbor dynamics

摘录 / EXCERPT: Nearest-neighbor noise-sensitivity conjecture. If $t_n \gg (n^2\alpha_4(n))^{-1} \approx n^{-3/4}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-039231 | MATHDB-320965

Robust exponential decay for smooth Lorenz attractors

摘录 / EXCERPT: Let a Lorenz attractor be generated by a flow of regularity $C^{1+\alpha}$. The Lorenz-attractor conjecture. $C^{1+\alpha}$ Lorenz attractors have robust exponential decay of corre...

conjecture | open | MathDB 页面 | 原文线索

V2-039232 | MATHDB-320966

Robust decay for higher-dimensional sectionally-hyperbolic singular flows

摘录 / EXCERPT: Consider smooth singular flows in dimensions higher than three that are sectionally-hyperbolic, meaning that they belong to the higher-dimensional class described in the source, ge...

conjecture | open | MathDB 页面 | 原文线索

V2-039233 | MATHDB-321023

Aperiodic symplectic forms on four-manifolds with positive b-plus

摘录 / EXCERPT: Aperiodicity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-039234 | MATHDB-321077

Symbolic concatenation conjecture for periodic directions in the double pentagon

摘录 / EXCERPT: Let a,A and b,B be two pairs of cyclic symbolic orbits corresponding to two periodic directions joined by a side of an ideal pentagon of some generation. A cyclic symbolic orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-039235 | MATHDB-321078

Symbolic orbit splitting conjecture for periodic directions in the double pentagon

摘录 / EXCERPT: Let β and $\dots,\gamma_{-2},\gamma_{-1},\gamma_0,\gamma_1,\gamma_2,\dots$ denote the periodic directions introduced in the paper's preceding period theorem. A short symbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-039236 | MATHDB-321084

Equality of the ergodic control eigenvalues

摘录 / EXCERPT: Let u^* be the convex, $C^{1,1}(\mathbb{R}^n)$ solution associated to the eigenvalue λ^*, and define ... where the infimum is over the relevant $C^2(\mathbb{R}^n)$ supersol...

conjecture | open | MathDB 页面 | 原文线索

V2-039237 | MATHDB-321091

Made's strong genericity conjecture for minimizing measures

摘录 / EXCERPT: Made's strong conjecture. For a generic Lagrangian, there is a unique minimizing measure, and this measure is supported on a periodic orbit or a fixed point of the Euler-Lagrange f...

conjecture | open | MathDB 页面 | 原文线索

V2-039238 | MATHDB-321181

Boundary-orbit conjecture for atoms of the compactified outer billiard map

摘录 / EXCERPT: Recall that the sets $\widehat A$ and $\widehat B$ are partitioned into atoms on whose interiors the first-return map of $\widehat\Psi$ is defined. Boundary-orbit conjecture. No po...

conjecture | open | MathDB 页面 | 原文线索

V2-039239 | MATHDB-321187

Almost-everywhere unique ergodicity along lines and analytic curves

摘录 / EXCERPT: The lines-and-curves conjecture. (Lines) If $\mathbf{a}$ is uniquely ergodic and ℓ is a line in $\mathbb{R}^d_+$ passing through $\mathbf{a}$, then there is a neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-039240 | MATHDB-321188

Unique ergodicity under almost-everywhere relative deformation

摘录 / EXCERPT: Relative unique-ergodicity conjecture. There is a neighborhood $\mathcal{U}$ of 0 in REL such that $\mathbf{a}+\mathcal{U}\subset\mathbb{R}^d_+$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039241 | MATHDB-321233

Polynomial multiple ergodic averages conjecture

摘录 / EXCERPT: Polynomial multiple ergodic averages conjecture. These averages converge in $L^2(\mu)$ for every choice of $f_1,f_2,\ldots,f_k\in L^\infty(\mu)$. This would extend the known L^2-...

conjecture | open | MathDB 页面 | 原文线索

V2-039242 | MATHDB-321240

Bonatti's finite chain-recurrence classes conjecture

摘录 / EXCERPT: Let M be the compact manifold underlying ${\operatorname{Diff}}^1(M)$, and let $\overline{\mathrm{HT}}$ denote the closure of the set of diffeomorphisms with homoclinic tangencies. Bo...

conjecture | open | MathDB 页面 | 原文线索

V2-039243 | MATHDB-321280

Pseudo-stable behavior conjecture for SBC and SBI systems

摘录 / EXCERPT: Let $x(t)\in\mathbb{R}^n$ be a trajectory of an SBC or SBI system, and let a trajectory have pseudo-stable behavior after a time τ when the agents split into non-empty fixed a...

conjecture | open | MathDB 页面 | 原文线索

V2-039244 | MATHDB-321379

Sarnak's quantum unique ergodicity conjecture on locally symmetric spaces

摘录 / EXCERPT: Let Y be the locally symmetric space under consideration, and let $\bar\mu_n$ be the probability measures associated with a sequence of Laplace-Beltrami eigenfunctions whose eige...

conjecture | open | MathDB 页面 | 原文线索

V2-039245 | MATHDB-321428

Generic spectral-radius conjecture for complex transfer operators

摘录 / EXCERPT: Let I be the underlying expanding dynamical system, let τ be a cocycle, and let $\boldsymbol{G}=?$. In the one-dimensional torus case $\boldsymbol{G}=?$, for $q e 0$ define...

conjecture | open | MathDB 页面 | 原文线索

V2-039246 | MATHDB-321433

Conjecture on optimal weighted beamforming vectors for the two-user broadcast channel

摘录 / EXCERPT: Let ${\bf\Sigma}_1$ and ${\bf\Sigma}_2$ denote the channel covariance matrices, let ρ be the signal-to-noise ratio, and let $\mathcal R_{\sf weighted}$ be the weighted ergodic...

conjecture | open | MathDB 页面 | 原文线索

V2-039247 | MATHDB-321469

Schweiger's absorbing-set conjecture for the invariant set D

摘录 / EXCERPT: Schweiger's conjecture. For every $a\geq 1$ and $b\geq 2$, the set D is absorbing for $T_{a,b}$.

conjecture | open | MathDB 页面 | 原文线索

V2-039248 | MATHDB-321470

Schweiger's ergodicity conjecture for the projected subtractive algorithm

摘录 / EXCERPT: Let $a\geq 1$ and $b\geq 2$. Define ... and let $S_{a,b}:B\to B$ be the map induced by $T_{a,b}$ under the projection ... Let D also denote the projection of the invariant set...

conjecture | open | MathDB 页面 | 原文线索

V2-039249 | MATHDB-321592

The effective stability conjecture for integrable Hamiltonians

摘录 / EXCERPT: Effective stability conjecture. An integrable Hamiltonian is effectively stable if and only if it is rationally steep. The paper notes that effective stability implies rational ste...

conjecture | open | MathDB 页面 | 原文线索

V2-039250 | MATHDB-321648

The volume entropy bound for Hilbert geometries

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}\mathbb{P}^n$ be a convex proper open set, and let $h_{vol}(\Omega,d_{\Omega})$ denote the volume entropy of its Hilbert geometry, defined using the Bus...

conjecture | open | MathDB 页面 | 原文线索

V2-039251 | MATHDB-321655

Ergodic scaling conjectures for thinning-invariant gap distributions

摘录 / EXCERPT: Let $W=(W_1,W_2,\ldots)$ have a thinning-invariant gap distribution, let $T_1,T_2,\ldots$ be independent exponential random variables of mean one, and let M be the almost-surely...

conjecture | open | MathDB 页面 | 原文线索

V2-039252 | MATHDB-321724

Asymptotic semi-periodicity of interspike intervals for periodic inputs

摘录 / EXCERPT: Asymptotic semi-periodicity conjecture. If f is periodic, then for every t the sequence $\eta_n(t)$ is asymptotically semi-periodic. If f is almost periodic, then for every...

conjecture | open | MathDB 页面 | 原文线索

V2-039253 | MATHDB-321744

Palis's approximation conjecture for robustly non-hyperbolic vector fields

摘录 / EXCERPT: Let a vector field be robustly non-hyperbolic if it belongs to the robustly non-hyperbolic class considered in the conjecture, and let C^r denote the relevant differentiability t...

conjecture | open | MathDB 页面 | 原文线索

V2-039254 | MATHDB-321746

The strong Palis conjecture for vector fields

摘录 / EXCERPT: Let X be a vector field. A singular Axiom A vector field without cycle has only finitely many chain recurrent classes, each singular hyperbolic; a homoclinic tangency is a non-tr...

conjecture | open | MathDB 页面 | 原文线索

V2-039255 | MATHDB-321747

Bonatti's periodic-orbit conjecture for singular chain recurrent classes

摘录 / EXCERPT: Let X be a generic vector field, and let $C(\sigma)$ be a non-trivial chain recurrent class containing a hyperbolic singularity σ. A chain recurrent class is non-trivial w...

conjecture | open | MathDB 页面 | 原文线索

V2-039256 | MATHDB-321748

The singular Axiom A or robust-cycle approximation conjecture

摘录 / EXCERPT: Let X be a vector field. A singular Axiom A vector field without cycle has finitely many chain recurrent classes, each singular hyperbolic; a robustly heterodimensional cycle is...

conjecture | open | MathDB 页面 | 原文线索

V2-039257 | MATHDB-321749

The finite chain-recurrent-class conjecture away from homoclinic tangencies

摘录 / EXCERPT: Let M be the underlying compact manifold, let ${\cal X}^1(M)$ denote the space of C^1 vector fields, and let $X\in{\cal X}^1(M)$. A vector field is approximable by homoclinic t...

conjecture | open | MathDB 页面 | 原文线索

V2-039258 | MATHDB-321751

Hammel-Yorke-Grebogi finite Hölder shadowing conjecture

摘录 / EXCERPT: Hammel-Yorke-Grebogi conjecture. A typical dissipative map $f:\mathbb{R}^2\to\mathbb{R}^2$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-039259 | MATHDB-321803

Classical motion conjecture for continuously observed quantum blobs

摘录 / EXCERPT: A quantum blob is a phase-space set denoted by $\mathcal{QB}^{2n}$. Let f_t^H be the classical Hamilton flow associated with a Hamiltonian H, and let t be the observation tim...

conjecture | open | MathDB 页面 | 原文线索

V2-039260 | MATHDB-321825

Hardy-Petrovitch-Hutchinson conjecture on iteration of hyperbolic polynomials

摘录 / EXCERPT: Iteration-convergence conjecture. If $0<q\leq\widetilde q$, then $\{f_j(x)\}$ converges uniformly on $[-1,0]$, together with all derivatives, to

conjecture | open | MathDB 页面 | 原文线索

V2-039261 | MATHDB-321919

Carleson's conjecture on the density of elliptic islands in the Standard family

摘录 / EXCERPT: Let f_κ denote the Standard Map with coupling constant κ, and consider the set of parameter values for which f_κ has at least one elliptic island. Carleson's c...

conjecture | open | MathDB 页面 | 原文线索

V2-039262 | MATHDB-321921

Zhang's weak-resonance obstruction conjecture for hyperbolic diffeomorphisms

摘录 / EXCERPT: Zhang's conjecture. The diffeomorphism φ is not embedded in a C^{∞} (respectively, analytic) flow.

conjecture | refuted | MathDB 页面 | 原文线索

V2-039263 | MATHDB-321941

Analytic graph conjecture for the Kuramoto global attractor

摘录 / EXCERPT: Let q_h be the heteroclinic solution of the Kuramoto equation connecting the incoherent equilibrium to the synchronized equilibrium, and let $W^{\frac{1}{2\pi}}$ denote the globa...

conjecture | open | MathDB 页面 | 原文线索

V2-039264 | MATHDB-321942

Kuramoto global attractor convergence to the Ott-Antonsen manifold

摘录 / EXCERPT: Let $\Phi_\mathcal H$ be the graph parametrizing the two-dimensional global attractor $W^u(\frac{1}{2\pi})$, let $q_{OA}(z)$ denote the corresponding state on the Ott-Antonsen mani...

conjecture | open | MathDB 页面 | 原文线索

V2-039265 | MATHDB-321943

Time-rescaled dynamical convergence to the no-diffusion Kuramoto manifold

摘录 / EXCERPT: Let $W^u(\frac{1}{2\pi})$ be the unstable manifold of the incoherent equilibrium for the diffusive Kuramoto equation, and let $\mathcal M_{0A}$ be the corresponding invariant manif...

conjecture | open | MathDB 页面 | 原文线索

V2-039266 | MATHDB-322128

The uniform exceptional-linear-subspace conjecture in dynamical Mordell-Lang

摘录 / EXCERPT: Let $\f:{\mathbb P}^n\to{\mathbb P}^n$ be a morphism of degree $d\ge2$, let $P\in{\mathbb P}^n({\mathbb C})$ have Zariski-dense forward orbit ... and let $r\ge1$. An $(r-1)$-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-039267 | MATHDB-322144

Pressure criterion for a spectral gap in damped quantum chaotic systems

摘录 / EXCERPT: Let $P(\hbar)$ be the damped quantum operator, let K be the set of least damped trajectories, let Φ^t denote the relevant flow, and let φ^{+} be the unstable Jacobia...

conjecture | open | MathDB 页面 | 原文线索

V2-039268 | MATHDB-322168

Arithmetic ergodicity dichotomy for countable Borel equivalence relations

摘录 / EXCERPT: Let E be a countable Borel equivalence relation, and let $\equiv_A$ denote arithmetic equivalence equipped with the arithmetic cone measure. Arithmetic ergodicity conjecture. Exa...

conjecture | open | MathDB 页面 | 原文线索

V2-039269 | MATHDB-322181

Completeness conjecture for equivalence relations of weakly mixing actions

摘录 / EXCERPT: Completeness conjecture. Conjugacy, orbit equivalence, and von Neumann equivalence in $A^{\star}_{\rm wm}(\Gamma,X,\mu)$ are complete analytic equivalence relations, viewed as subs...

conjecture | open | MathDB 页面 | 原文线索

V2-039270 | MATHDB-322193

Herman's local instability conjecture for the planetary problem

摘录 / EXCERPT: Herman's local instability conjecture. If the masses of the planets are small enough, the wandering set accumulates on the set of circular, coplanar, Keplerian motions.

conjecture | open | MathDB 页面 | 原文线索

V2-039271 | MATHDB-322205

Quantum ergodicity conjecture for resonant states on cusp surfaces

摘录 / EXCERPT: Quantum ergodicity conjecture for resonant states. Under these assumptions, there exists a density-one subsequence of resonant states in every strip...

conjecture | open | MathDB 页面 | 原文线索

V2-039272 | MATHDB-322270

Frieze-Krivelevich conjecture on Hamilton-cycle packings in random graphs

摘录 / EXCERPT: Frieze-Krivelevich conjecture. Asymptotically almost surely, G contains

conjecture | open | MathDB 页面 | 原文线索

V2-039273 | MATHDB-322299

The generalized lock inequality for positive words

摘录 / EXCERPT: Let w be a positive word of the form ... Suppose that $R(w,r,s)=p/q$, where p/q is reduced. Generalized lock inequality. Then ... For the linear representation with $a=R_r$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039274 | MATHDB-322300

The rationality conjecture for maximal rotation numbers in the commutator subgroup

摘录 / EXCERPT: Let F be the free group generated by a and b, and let $w\in [F,F]$. Commutator-subgroup rationality conjecture. Then ... This is a rationality conjecture for the maximum of...

conjecture | open | MathDB 页面 | 原文线索

V2-039275 | MATHDB-322342

The Bohigas-Giannoni-Schmit conjecture on chaotic quantum spectra

摘录 / EXCERPT: Bohigas-Giannoni-Schmit conjecture. If the classical problem is a sufficiently chaotic system, then the local statistics of the E_n are generically those of EGO or EGU, according...

conjecture | open | MathDB 页面 | 原文线索

V2-039276 | MATHDB-322387

The star-flow sectional-Axiom A conjecture

摘录 / EXCERPT: Let $n\geq 3$ and let X be a C^1-generic star flow on a closed n-manifold. The star-flow sectional-Axiom A conjecture. X is sectional-Axiom A. This conjecture seeks the hig...

conjecture | open | MathDB 页面 | 原文线索

V2-039277 | MATHDB-322388

The sectional-Axiom A dichotomy for generic vector fields

摘录 / EXCERPT: Let X be a C^1-generic vector field on a closed manifold. A point is accumulated by hyperbolic periodic orbits of different Morse indices if it lies in the accumulation set of...

conjecture | open | MathDB 页面 | 原文线索

V2-039278 | MATHDB-322467

The symbolic extension conjecture for C^2 maps

摘录 / EXCERPT: A map is called a symbolic extension of a dynamical system (M,f) if it is a subshift (Y,g) over a finite alphabet together with a continuous surjection $\pi:Y\to M$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-039279 | MATHDB-322485

Conjecture on the thermodynamic-limit excitation spectrum of an interacting Fermi gas

摘录 / EXCERPT: Let H^L be the finite-volume Hamiltonian of an interacting Fermi gas in a periodic box, with ground-state energy E^L. For momentum ${\bf k}$, let $\epsilon^\pm({\bf k})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-039280 | MATHDB-322486

Transversality conjecture for tridiagonal Fibonacci Hamiltonians

摘录 / EXCERPT: Let $\boldsymbol{\theta}=(\theta_1,\theta_2)$ denote the parameters of the tridiagonal Fibonacci Hamiltonian, with $\theta_1 eq 0$ and $\boldsymbol{\theta} eq(1,0)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039281 | MATHDB-322487

Analyticity conjecture for the Hausdorff dimension of tridiagonal Fibonacci spectra

摘录 / EXCERPT: Let $\alpha(t)=(\mathfrak{p}(t),\mathfrak{q}(t))$ be an analytic curve in $\mathbb{R}^2\setminus\{(1,0)\}$ such that $\mathfrak{p}(t)\neq 0$ for all t, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039282 | MATHDB-322488

Monotonicity conjecture for the Hausdorff dimension of diagonal Fibonacci spectra

摘录 / EXCERPT: Let $\mathfrak{q}\in[0,\infty)$, and let $\Sigma_{\mathfrak{q}}$ denote the spectrum of the diagonal Fibonacci operator with parameter $\mathfrak{q}$. Monotonicity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-039283 | MATHDB-322602

Arithmetic degree conjecture for dominant rational self-maps of projective space

摘录 / EXCERPT: Let $\varphi:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map defined over $\overline{\mathbb{Q}}$. For a point $P\in\mathbb{P}^N(\overline{\mathbb{Q}})$ whose o...

conjecture | open | MathDB 页面 | 原文线索

V2-039284 | MATHDB-322603

Polynomial correction conjecture for degree growth

摘录 / EXCERPT: Polynomial correction conjecture. The infimum ℓ_φ exists and is an integer satisfying $0\leq\ell_\varphi\leq N$.

conjecture | open | MathDB 页面 | 原文线索

V2-039285 | MATHDB-322604

Positive canonical height conjecture for dense orbits

摘录 / EXCERPT: Let $\varphi:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map defined over $\overline{\mathbb{Q}}$, with dynamical degree $\delta_\varphi>1$. Let ℓ_φ...

conjecture | open | MathDB 页面 | 原文线索

V2-039286 | MATHDB-322673

The three-part decomposition conjecture for the nonwandering set

摘录 / EXCERPT: Let ϕ_λ be the square billiard map with parameter $0<\lambda<1$, and let Ω_λ be its nonwandering set. Let P be the parabolic attractor, let A_λ de...

conjecture | open | MathDB 页面 | 原文线索

V2-039287 | MATHDB-322674

Homoclinic relation conjecture for the periodic points q_n

摘录 / EXCERPT: Let q_n be the periodic points of the billiard map, let $W^s(q_n)$ and $W^u(q_n)$ denote their stable and unstable manifolds, and let c_n and λ_1 be the bifurcation par...

conjecture | open | MathDB 页面 | 原文线索

V2-039288 | MATHDB-322675

Two homoclinic classes conjecture for the periodic points p_n

摘录 / EXCERPT: Let p_n be the periodic points of the billiard map, let C_0 and C_1 denote homoclinic classes, and let a_n, b_n, and b_{17} be the parameters appearing in the paper. Tw...

conjecture | open | MathDB 页面 | 原文线索

V2-039289 | MATHDB-322676

Transverse homoclinic connection conjecture for periodic points p_n

摘录 / EXCERPT: Let ϕ_λ be the billiard map, let Ω_λ be its nonwandering set, let P be the parabolic attractor, let p_λ be the fixed point, and let p_n be the p...

conjecture | open | MathDB 页面 | 原文线索

V2-039290 | MATHDB-322735

Plaque expansivity conjecture for partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Plaque expansivity conjecture. Any partially hyperbolic, dynamically coherent diffeomorphism is plaque expansive.

conjecture | open | MathDB 页面 | 原文线索

V2-039291 | MATHDB-322744

Uniform norm-ratio conjecture for iterates of the linearized renormalization operator

摘录 / EXCERPT: Let $\mathcal{L}_\omega$ be the map defined by the source, let $\omega_0\in\Omega$, and choose nonzero vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-039292 | MATHDB-322747

Uniform contraction conjecture for the linearized quasi-periodic renormalization map

摘录 / EXCERPT: Let $\mathcal{B}_1'$ be the space in which the linearized renormalization dynamics is considered, let L_1 be the map given by the referenced equation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039293 | MATHDB-322748

Norm-ratio boundedness conjecture for the linearized renormalization operator

摘录 / EXCERPT: Let $\mathcal{R}\mathcal{H}(\mathbb{W}_{\rho})$ be the space appearing in the definition of the linearized operator, let $\mathcal{L}_{\omega}$ be the map from ... to itself given by...

conjecture | open | MathDB 页面 | 原文线索

V2-039294 | MATHDB-322756

Bondy's conjecture on long circuits generating the cycle space

摘录 / EXCERPT: Bondy's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-039295 | MATHDB-322778

Convex hull conjecture for non-wandering sets of open billiards

摘录 / EXCERPT: For $i\ne j$, let $(p_{ij},p_{ji})\in K_i\times K_j$ be a pair realizing the minimum distance between K_i and K_j. Let ... be their convex hull. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039296 | MATHDB-322787

Positive-curvature sphere conjecture for left-handed geodesic flows

摘录 / EXCERPT: Let Σ be a 2-sphere with a not necessarily constant positive curvature, and let Φ_{Σ} denote its geodesic flow. Positive-curvature sphere conjecture. The geodesi...

conjecture | open | MathDB 页面 | 原文线索

V2-039297 | MATHDB-322808

The generic wickedness conjecture for chain recurrent maps

摘录 / EXCERPT: Generic wickedness conjecture. A generic element of $\mathcal{C}$ is wicked.

conjecture | open | MathDB 页面 | 原文线索

V2-039298 | MATHDB-322935

Singular-limit convergence of C-shaped homoclinic bifurcation curves

摘录 / EXCERPT: Let $A=(p^*,0)$, $B=(p_-,0)$, and $C=(p_-,s^*)$, where $s^*\approx 1.50815$. Consider the C-shaped homoclinic bifurcation curves in (p,s)-parameter space for the FitzHugh-Nagumo...

conjecture | open | MathDB 页面 | 原文线索

V2-039299 | MATHDB-322952

Exponential closeness conjecture for the FitzHugh-Nagumo homoclinic C-curve

摘录 / EXCERPT: Let $C_{l,\epsilon}$ denote the left equilibrium of the FitzHugh-Nagumo system, let q be the relevant saddle equilibrium, and let $W^u(C_{l,\epsilon})$ and $W^s(q)$ be their unst...

conjecture | open | MathDB 页面 | 原文线索

V2-039300 | MATHDB-322970

Canonical-height finiteness and arithmetic-degree conjecture for plane polynomial maps

摘录 / EXCERPT: Let $f:{\mathbf{A}}^2\to{\mathbf{A}}^2$ be a dominant polynomial mapping over $\overline{{\mathbf{Q}}}$ with first dynamical degree $\lambda_1>1$ and second dynamical degree...

conjecture | open | MathDB 页面 | 原文线索

V2-039301 | MATHDB-322993

Katok's polynomial growth conjecture for generalized diagonals in polygonal billiards

摘录 / EXCERPT: Let M be a polygon, let P_n denote the number of generalized diagonals of M whose discrete length is at most n, and let $\epsilon>0$. Katok's conjecture. For every polygon...

conjecture | open | MathDB 页面 | 原文线索

V2-039302 | MATHDB-323135

Hamilton decomposition conjecture for regular tripartite tournaments

摘录 / EXCERPT: Tripartite tournament conjecture. Every regular tripartite tournament has a Hamilton decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-039303 | MATHDB-323140

Connected four-reflection approximation conjecture for random reflectors

摘录 / EXCERPT: Let M_n be sets of mirrors and let Λ denote the measure on light rays. The sets M_n are required to satisfy the conclusions of Theorem 16.2 except condition (b), namely...

conjecture | open | MathDB 页面 | 原文线索

V2-039304 | MATHDB-323155

Smooth extension conjecture for integrable billiards

摘录 / EXCERPT: Let Ω be a smooth integrable billiard domain, and let $\mathcal E$ be an ellipse with the same perimeter and the same Lazutkin perimeter. Consider the conjugacy between the...

conjecture | open | MathDB 页面 | 原文线索

V2-039305 | MATHDB-323240

Avila-Bochi conjecture on generic non-uniform hyperbolicity

摘录 / EXCERPT: Avila-Bochi conjecture. Either all Lyapunov exponents of f vanish almost everywhere, or the Pesin region $\operatorname{Nuh}(f)$ has full measure, f is ergodic, and the Oselede...

conjecture | open | MathDB 页面 | 原文线索

V2-039306 | MATHDB-323241

Central-support conjecture for orthogonal measures

摘录 / EXCERPT: Let ${\mathfrak A}$ be a unital C^*-algebra, and let $\omega\in{\mathcal S}({\mathfrak A})$ have central support in the bidual algebra: ... Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-039307 | MATHDB-323259

Conjecture on convergence of approximating densities for Bernoulli convolutions

摘录 / EXCERPT: Let ν_{β} be the Bernoulli convolution associated with β, and let f_n denote the sequence of approximating functions introduced in the paper. Convergence conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-039308 | MATHDB-323365

Generic graph separatrix conjecture for trigonometric Hamiltonians

摘录 / EXCERPT: Generic graph separatrix conjecture. For a generic trigonometric polynomial V, the associated Hamiltonian system has a hyperbolic critical point whose invariant manifolds coincid...

conjecture | open | MathDB 页面 | 原文线索

V2-039309 | MATHDB-323373

Lapidus-Niemeyer conjecture on elusive limit points of Koch snowflake billiards

摘录 / EXCERPT: Let $K\!S$ be the Koch snowflake. For a sequence of compatible periodic hybrid orbits, let $x_0^{k_0}$ denote its sequence of basepoints, and suppose the basepoints are dynamically...

conjecture | open | MathDB 页面 | 原文线索

V2-039310 | MATHDB-323412

Existence of adapted metrics for sectional-hyperbolic sets

摘录 / EXCERPT: A sectional-hyperbolic set is an invariant set with a partially hyperbolic splitting whose contracted bundle is uniformly contracted and whose complementary bundle is sectional-exp...

conjecture | open | MathDB 页面 | 原文线索

V2-039311 | MATHDB-323432

The nonexistence of wandering domains for twisted tent maps

摘录 / EXCERPT: Let f be the twisted tent map and let K be its filled-in invariant set. A wandering domain is a Fatou domain whose forward images are pairwise disjoint and are not eventually p...

conjecture | open | MathDB 页面 | 原文线索

V2-039312 | MATHDB-323433

The escape-time atlas conjecture for twisted tent maps

摘录 / EXCERPT: Let c be a parameter in the parameter plane, let $K(c)$ be the associated filled-in set, and let the polygonal locus denote the parameter-space locus discussed in the source. An...

conjecture | open | MathDB 页面 | 原文线索

V2-039313 | MATHDB-323434

The coded-coloring atlas conjecture for twisted tent maps

摘录 / EXCERPT: Let c be a parameter in the parameter plane, let $K(c)$ be the associated filled-in set, and let a coded-coloring be the coloring procedure described in the source using symbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-039314 | MATHDB-323435

The coded-coloring similarity conjecture for twisted tent maps

摘录 / EXCERPT: Let c be a parameter, let $K(c)$ be its filled-in set, and let a coded-coloring be the symbolic coloring procedure used in the source. Let the test point be nonzero. Coded-colori...

conjecture | open | MathDB 页面 | 原文线索

V2-039315 | MATHDB-323436

The rational-angle conjecture for flares in the parameter plane

摘录 / EXCERPT: Let the centered unit disk in the parameter plane be the disk centered at 0 of radius 1, and let a flare be one of the sun-flare-like parameter sets arising in the coded-colori...

conjecture | open | MathDB 页面 | 原文线索

V2-039316 | MATHDB-323467

Strong resolvent convergence of the scaled Pauli-Fierz Hamiltonians

摘录 / EXCERPT: Strong resolvent convergence conjecture. In the sense of strong convergence of resolvents,

conjecture | open | MathDB 页面 | 原文线索

V2-039317 | MATHDB-323485

Hamiltonian instability conjecture for non-minimal Lagrangians in indefinite complex space

摘录 / EXCERPT: Let $\mathcal L$ be an H-minimal, non-minimal, Lagrangian submanifold of complex space $\mathbb C^n$ equipped with the pseudo-Kähler structure determined by the Hermitian form ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-039318 | MATHDB-323506

Complexified-geodesic nodal-zero equidistribution conjecture

摘录 / EXCERPT: Let (M^2,g) be a real analytic Riemannian surface with ergodic geodesic flow, and let $\gamma_{x,\xi}$ be a geodesic satisfying the QER hypothesis. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039319 | MATHDB-323574

Categorical quantization commutes with reduction

摘录 / EXCERPT: Let $(X,\omega_X,G,\mathbf J)$ be a Hamiltonian G-space for G a reductive group, and let $\mathcal X_{red}$ be the derived Marsden-Weinstein quotient associated with a coadjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-039320 | MATHDB-323598

The density conjecture for real limits of generalized autoconjugacies

摘录 / EXCERPT: Let $T_{m,r}$ be the generalized Collatz map, let $\Omega_{k,m,r}(x)$ be the truncated generalized autoconjugacy, and define $\hat{\Omega}_{m,r}(x)$ as the real limit of...

conjecture | open | MathDB 页面 | 原文线索

V2-039321 | MATHDB-323616

Identification of the near-critical limit with the derivative-martingale measure

摘录 / EXCERPT: Let M^{γ} be the subcritical chaos measures for $\gamma^2<2d$, and let M^c be the positive random measure obtained in Proposition 3.1 as a limit...

conjecture | open | MathDB 页面 | 原文线索

V2-039322 | MATHDB-323617

Glassy-phase convergence to a stable random measure

摘录 / EXCERPT: For $\gamma^2>2d$, let ... set $\alpha=\frac{\sqrt{2d}}{\gamma}\in(0,1)$, and let M' be the derivative-martingale measure. Let N_α be an independently scattered random measu...

conjecture | open | MathDB 页面 | 原文线索

V2-039323 | MATHDB-323636

Eigenvalue multiplicity scaling conjecture for cubic Julia sets

摘录 / EXCERPT: Eigenvalue multiplicity scaling conjecture. For $p=3$,

conjecture | open | MathDB 页面 | 原文线索

V2-039324 | MATHDB-323706

Shift-ergodicity conjecture for the invariant measure

摘录 / EXCERPT: Let F be the cellular automaton under consideration and let μ_F be its measure. Shift-ergodicity conjecture. The measure μ_F is shift-ergodic. The paper has established t...

conjecture | open | MathDB 页面 | 原文线索

V2-039325 | MATHDB-323709

The conjecture that the BCZ map is an adic transformation

摘录 / EXCERPT: Let the BCZ map denote the Boca-Cobeli-Zaharescu map arising as a Poincaré return map for horocycle flow on the space of lattices. An adic transformation is a measurable transforma...

conjecture | open | MathDB 页面 | 原文线索

V2-039326 | MATHDB-323724

Higher-spin XXZ identification at the root of unity

摘录 / EXCERPT: Let h' be the modified local Hamiltonian and let $h_{\mathrm{XXZ}_{\ell/2}}$ be the spin-$\ell/2$ higher-spin XXZ Hamiltonian density defined from the quadratic Casimir construct...

conjecture | open | MathDB 页面 | 原文线索

V2-039327 | MATHDB-323725

Spectral symmetry under exchange of couplings

摘录 / EXCERPT: Let $H_N(x,y)$ be the spin-chain Hamiltonian with twist angle ϕ, depending on couplings x and y. The spectral exchange conjecture. For twist angle $\phi=\pi$, the spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-039328 | MATHDB-323726

Existence of representative zero-energy states

摘录 / EXCERPT: Let $H_N(x,y)$ be the Hamiltonian on N sites, with twist angle ϕ, and let S_N^3 denote the total third spin component. The representative singlet conjecture. For every N...

conjecture | open | MathDB 页面 | 原文线索

V2-039329 | MATHDB-323729

Additional zero-energy states at the elliptic supersymmetric point

摘录 / EXCERPT: Let H_N be the Fateev-Zamolodchikov spin-chain Hamiltonian, with twist angle ϕ and total third spin component S_N^3. The elliptic zero-mode conjecture. For odd $N=2n+1$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-039330 | MATHDB-323737

Menzin's conjecture for bicycle monodromy

摘录 / EXCERPT: Let F be a front-wheel path whose trajectory is a simple closed curve bounding a region of area A. For a bicycle of length ℓ, let ... be the bicycle monodromy, which maps an...

conjecture | open | MathDB 页面 | 原文线索

V2-039331 | MATHDB-323749

Power-law spectrum conjecture for matrix-valued Gaussian multiplicative chaos

摘录 / EXCERPT: Power-law spectrum conjecture. The moments of M satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-039332 | MATHDB-323868

The global-attractor conjecture for the Hartree equation

摘录 / EXCERPT: Let $u(t)$) be the unique solution to the Hartree equation for some $u_0\in H^1(\mathbb{R}^3)$, and let $\mathscr{A}_Z$ denote the global attractor. Take any sequence of times...

conjecture | open | MathDB 页面 | 原文线索

V2-039333 | MATHDB-323983

Singularity conjecture for the first exceptional time infinite cluster

摘录 / EXCERPT: Let FETIC be the law of the configuration at the first exceptional time, and let IIC denote the incipient infinite cluster measure. Singularity conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-039334 | MATHDB-323984

Existence and identification of the limiting local-time measure

摘录 / EXCERPT: For a fixed interval $[a,b]$, let $\mathsf{\overline{\mu}}_r[a,b]$ be the finite-radius approximations to the local-time measure and let $\mathsf{\overline{\mu}}[a,b]$ denote their...

conjecture | open | MathDB 页面 | 原文线索

V2-039335 | MATHDB-323985

Full-support conjecture for the dynamical percolation local time

摘录 / EXCERPT: Let $\mathsf{\mathscr{E}}$ be the exceptional time set and let $\mathsf{\mu}$ be the associated local-time measure, which is supported inside $\mathsf{\mathscr{E}}$. Full-support c...

conjecture | open | MathDB 页面 | 原文线索

V2-039336 | MATHDB-324045

Finite-occurrence conjecture for non-maximal dynamical Galois groups

摘录 / EXCERPT: Fix a critical point $\gamma\in\mathbb{Q}$, and let $f_{\gamma,t}=f_t$ be the corresponding quadratic polynomial with t an indeterminate. Let $V_{n,\gamma}=V_n$ be the union of t...

conjecture | open | MathDB 页面 | 原文线索

V2-039337 | MATHDB-324074

The return-times ergodic Hilbert transform conjecture

摘录 / EXCERPT: Return-times ergodic Hilbert transform conjecture. For every $x\in X_f$, and for every $(Y,\mathcal{G},\nu,S)$ and $g\in L^q(\nu)$, this limit exists for ν-almost every y. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-039338 | MATHDB-324157

Mu3 conjecture for hypohamiltonian snarks

摘录 / EXCERPT: Let G be a snark, and let $\mu_3(G)$ be the number of edges not covered by the union of three 1-factors, minimized over all choices of three 1-factors. A graph is hypohamiltonian...

conjecture | open | MathDB 页面 | 原文线索

V2-039339 | MATHDB-324227

Ergodicity conjecture for generalized Boole type transformations

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{+}$, let Σ_n be the symmetric group on n elements, and let $\sigma\in\Sigma_n$. Define the generalized Boole type transformation ... on...

conjecture | open | MathDB 页面 | 原文线索

V2-039340 | MATHDB-324240

Dettmann's maximal-horizon asymptotic conjecture for periodic Lorentz processes

摘录 / EXCERPT: Let $\mathcal L$ be a lattice defining a periodic Lorentz process, let $F(t)$ be the free-path tail probability, and let $\mathbb H$ denote the set of maximal non-incipient horizon...

conjecture | open | MathDB 页面 | 原文线索

V2-039341 | MATHDB-324241

Dettmann's incipient-principal-horizon tail conjecture for periodic Lorentz processes

摘录 / EXCERPT: Let $\mathcal L$ be a lattice defining a periodic Lorentz process, let d be the dimension, and let $F(t)$ be the free-path tail probability. A principal horizon has the highest p...

conjecture | open | MathDB 页面 | 原文线索

V2-039342 | MATHDB-324242

Dettmann's truncated-correlation conjecture for periodic Lorentz processes

摘录 / EXCERPT: Let $\mathcal L$ be a lattice defining a periodic Lorentz process with phase space M, invariant measure μ, free-flight time τ, and flow Φ_t. Let $f,g:M\to\mathbb R$...

conjecture | open | MathDB 页面 | 原文线索

V2-039343 | MATHDB-324277

Flow-equivalence conjecture for irreducible shifts of finite type and renewal systems

摘录 / EXCERPT: An irreducible shift of finite type (SFT) is an irreducible symbolic dynamical system defined by finitely many forbidden words, and a renewal system is a shift generated by concate...

conjecture | open | MathDB 页面 | 原文线索

V2-039344 | MATHDB-324330

Four-boson critical-coupling conjecture on infinitely many bound states

摘录 / EXCERPT: Let H be the Hamiltonian of $N=4$ particles in the setting of $(1.08;3)$--$(1.08;4)$, with all pair potentials $v_{ik}\leq 0$ bounded and of finite range. Suppose ... that the pair...

conjecture | open | MathDB 页面 | 原文线索

V2-039345 | MATHDB-324340

Stability of the regular n-gon at the bifurcation point

摘录 / EXCERPT: Let n be the number of vortices and consider the regular n-gon at the bifurcation point, with stability determined by the positivity of the relevant quadratic form. Stability c...

conjecture | open | MathDB 页面 | 原文线索

V2-039346 | MATHDB-324398

Stojaković-Szabó threshold-bias conjecture for the random-graph Hamiltonicity game

摘录 / EXCERPT: Stojaković-Szabó conjecture. There exists a constant C such that for every

conjecture | open | MathDB 页面 | 原文线索

V2-039347 | MATHDB-324415

The conjecture on postcritically-finite rational maps and dynamically independent critical points

摘录 / EXCERPT: Conjecture on postcritically-finite rational maps. The map f_t is postcritically finite for a Zariski-dense subset of $t\in V$ if and only if there are at most N dynamically in...

conjecture | open | MathDB 页面 | 原文线索

V2-039348 | MATHDB-324480

Hamiltonianity conjecture for symplectic circle actions with isolated fixed points

摘录 / EXCERPT: Let (X^6,ω) be a closed symplectic six-manifold equipped with a symplectic S^1 action, and suppose that its fixed-point set contains at least one isolated fixed point. Ham...

conjecture | open | MathDB 页面 | 原文线索

V2-039349 | MATHDB-324575

Strict ordering conjecture for billiard winding numbers in ellipsoids

摘录 / EXCERPT: Let $m_0,\ldots,m_{n-1}$ be the winding numbers of a periodic billiard trajectory inside an ellipsoid, as defined by the oscillations of its elliptic coordinates. Winding-number or...

conjecture | open | MathDB 页面 | 原文线索

V2-039350 | MATHDB-324577

Winding-number and signature correspondence conjecture for elliptic billiards

摘录 / EXCERPT: Let $\tau=(\tau_1,\ldots,\tau_n)$ be a signature associated with a nonsingular periodic billiard trajectory, and let $\widetilde m_0,\ldots,\widetilde m_{n-1}$ be its elliptic wind...

conjecture | open | MathDB 页面 | 原文线索

V2-039351 | MATHDB-324647

The conjecture on an open set of helical asymptotic orbits

摘录 / EXCERPT: Consider the Heisenberg-group dynamical system described above, with initial conditions and orbits under its Hamiltonian flow. An orbit is asymptotic to a helix if it approaches a...

conjecture | open | MathDB 页面 | 原文线索

V2-039352 | MATHDB-324760

Hohlweg-Labbé-Ripoll conjecture on accumulation points of normalized Coxeter roots

摘录 / EXCERPT: Let W be an infinite Coxeter group with simple roots $Delta$ acting on a vector space V with bilinear form B. Set ... Let E be the set of accumulation points of normalized ro...

conjecture | open | MathDB 页面 | 原文线索

V2-039353 | MATHDB-324799

Bohigas-Giannoni-Schmit conjecture for chaotic spectral statistics

摘录 / EXCERPT: Bohigas-Giannoni-Schmit conjecture. The eigenvalues are distributed like the eigenvalues of Hermitian random matrices, with the following limiting ensembles: without time-reversal...

conjecture | open | MathDB 页面 | 原文线索

V2-039354 | MATHDB-324890

Strong quantitative quantum ergodicity conjecture for Maass-Hecke cusp forms

摘录 / EXCERPT: Let ϕ be a Maass-Hecke cusp form with spectral parameter t_ϕ, Fourier coefficients $\rho_\phi(n)$, and let $\delta_{0,m}$ denote the Kronecker delta. For...

conjecture | open | MathDB 页面 | 原文线索

V2-039355 | MATHDB-324940

Palis's density conjecture for dynamical systems

摘录 / EXCERPT: Palis's conjecture. Every system can be approximated either by Morse-Smale systems or by systems exhibiting a horseshoe (non-trivial hyperbolic basic set).

conjecture | open | MathDB 页面 | 原文线索

V2-039356 | MATHDB-325023

Small Incremental Entangling conjecture

摘录 / EXCERPT: Bravyi's Small Incremental Entangling conjecture. There is a universal constant c such that, for all ancilla dimensions, all states $|\Psi\rangle$, and all such Hamiltonians,

conjecture | open | MathDB 页面 | 原文线索

V2-039357 | MATHDB-325101

The Kaplan-Yorke conjecture for typical dynamical systems

摘录 / EXCERPT: Let M be a locally compact subset of $\mathbb R^d$, and let $F:M\to M$ be a piecewise C^1 dynamical system with an ergodic invariant physical measure μ. Write $D_1(\mu)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-039358 | MATHDB-325156

The entropy threshold conjecture for cat-map quantum limits

摘录 / EXCERPT: Entropy threshold conjecture. The entropy of μ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-039359 | MATHDB-325157

The logarithmic quasimode threshold conjecture for QUE on hyperbolic surfaces

摘录 / EXCERPT: Logarithmic quasimode threshold conjecture. Then the sequence $\{\psi_j\}$ satisfies Quantum Unique Ergodicity.

conjecture | open | MathDB 页面 | 原文线索

V2-039360 | MATHDB-325177

Geometric rigidity conjecture for smooth critical circle maps

摘录 / EXCERPT: Geometric rigidity conjecture. Any two C^3 critical circle maps with the same irrational rotation number of bounded type and the same odd criticality are conjugate to each other...

conjecture | open | MathDB 页面 | 原文线索

V2-039361 | MATHDB-325194

The periodic-attractor bound for perturbations of one-dimensional maps

摘录 / EXCERPT: Periodic-attractor bound. There exists a neighbourhood ${\cal F} \subset C^3({\cal N})$ of f such that for any $g\in {\cal F}$ the number of periodic attractors of g is bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-039362 | MATHDB-325248

Katok's conjecture on unstable homogeneous flows

摘录 / EXCERPT: Katok's conjecture. Every homogeneous flow on a compact homogeneous space that fails to be stable projects onto a Liouvillean linear flow on a torus. In this case, the flow is stil...

conjecture | open | MathDB 页面 | 原文线索

V2-039363 | MATHDB-325292

Global rigidity conjecture for higher-rank Anosov actions on infranilmanifolds

摘录 / EXCERPT: Infranilmanifold global rigidity conjecture. All irreducible Anosov genuine $\mathbb{Z}^r$-actions on compact infranilmanifolds should be C^∞-conjugate to actions by affine...

conjecture | open | MathDB 页面 | 原文线索

V2-039364 | MATHDB-325377

Quasi-stationary distribution conjecture for N-BBM and N-BRW

摘录 / EXCERPT: Consider N-BBM and N-BRW, with empirical measures viewed under their invariant measure λ^N. Quasi-stationary distribution conjecture. Both N-BBM and N-BRW are ergod...

conjecture | open | MathDB 页面 | 原文线索

V2-039365 | MATHDB-325378

Fleming-Viot invariant-measure conjecture for Yaglom limits

摘录 / EXCERPT: Let $Z=(Z_t,t\geq0)$ be a Markov driving process on $\Lambda\cup\{0\}$, with 0 absorbing, and let τ be its absorption time. For $N\geq2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039366 | MATHDB-325407

Unique maximization of the invariant for logarithmic pentagram spirals

摘录 / EXCERPT: Uniqueness conjecture. For any (n,k), the invariant Z is uniquely maximized, and has a unique critical point, at the point representing the logarithmic pentagram spiral.

conjecture | open | MathDB 页面 | 原文线索

V2-039367 | MATHDB-325408

Total integrability of the pentagram map on spiral configuration spaces

摘录 / EXCERPT: Integrability conjecture. The map $T_{n,k}$ acting on ${\cal C}(n,k)$ is a discrete totally integrable system.

conjecture | open | MathDB 页面 | 原文线索

V2-039368 | MATHDB-325409

Non-ergodicity and Li-Yorke chaos for the family W_alpha

摘录 / EXCERPT: Let S^2 be the two-dimensional simplex, and let $W_\alpha:S^2\to S^2$ be the quadratic stochastic operator defined in the source by equation W_α. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-039369 | MATHDB-325489

Karlsson-Nussbaum boundary-limit conjecture for fixed-point-free maps

摘录 / EXCERPT: Karlsson-Nussbaum conjecture. There exists a convex set Λ in $\partial\Omega$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-039370 | MATHDB-325544

Non-degeneracy of the symplectic form on connected normally hyperbolic submanifolds

摘录 / EXCERPT: Symplectic non-degeneracy conjecture. ω_N is non-degenerate on the whole of N.

conjecture | open | MathDB 页面 | 原文线索

V2-039371 | MATHDB-325568

Palis' conjecture for surface diffeomorphisms

摘录 / EXCERPT: Let M be a real compact surface, and let $k\geq 1$. For $f\in\mathrm{Diff}^k(M)$, a homoclinic tangency means a tangency between the stable and unstable manifolds of some saddle...

conjecture | open | MathDB 页面 | 原文线索

V2-039372 | MATHDB-325591

Finite ergodic decomposition conjecture for the piecewise affine square map

摘录 / EXCERPT: Let $F=F_\theta$ be the map of the square ${\mathcal S}$, with parameter $\theta\in(0,\frac{\pi}{4})$. An ergodic decomposition conjecture asserts that, for Lebesgue almost all...

conjecture | open | MathDB 页面 | 原文线索

V2-039373 | MATHDB-325650

Solutions conjecture for the lognormal star equation

摘录 / EXCERPT: Let $alpha e0$ and let $gamma^2\triangleleftteq2d$ describe the parameters in the lognormal star-scale invariant setting; M denotes a solution random measure. Solutions conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-039374 | MATHDB-325651

Derivative-chaos law conjecture for the maximum of log-correlated Gaussian fields

摘录 / EXCERPT: Let $(X_n)_n$ be a cutoff approximation of a centered Gaussian field on $[0,1]^d$ with logarithmic covariance, and set $c_n={\rm Var}(X_n)$. Let M' be the derivative chaos and le...

conjecture | open | MathDB 页面 | 原文线索

V2-039375 | MATHDB-325653

Vortex-free ground-state conjecture for ladder Hamiltonians

摘录 / EXCERPT: Let H be the Hamiltonian in the paper's equation (Hamiltonian), and let $\widetilde{H}$ be its extended Hamiltonian in equation (HamiltonianExtended). A ladder may be closed or o...

conjecture | open | MathDB 页面 | 原文线索

V2-039376 | MATHDB-325665

Positivity conjecture for the sixth cumulant of multiple Wiener integrals

摘录 / EXCERPT: Let X be a multiple Wiener--Itô integral of order $p\geq 2$, and let $\kappa_6(X)$ denote its sixth cumulant. Sixth-cumulant positivity conjecture. For every such X, one has .....

conjecture | open | MathDB 页面 | 原文线索

V2-039377 | MATHDB-325694

Weak variant of Mayer's conjecture in dimension one

摘录 / EXCERPT: Weak variant of Mayer's conjecture. All eigenvalues of $\mathcal{L}$ with sufficiently small modulus should be real.

conjecture | refuted | MathDB 页面 | 原文线索

V2-039378 | MATHDB-325708

The decay-of-correlations summability claim for non-uniformly expanding maps

摘录 / EXCERPT: Decay-of-correlations summability claim. A non-uniformly expanding map having a sufficiently fast rate of decay of correlations satisfies the summability condition above. This cond...

conjecture | open | MathDB 页面 | 原文线索

V2-039379 | MATHDB-325712

Young's geodesic QUE conjecture for restricted eigenfunctions

摘录 / EXCERPT: Let u_j be even Hecke-Maass forms normalized by the paper's L^2 normalization, and let f be an L^2-normalized holomorphic Hecke cusp form of weight k. If...

conjecture | open | MathDB 页面 | 原文线索

V2-039380 | MATHDB-325714

Young's horocycle QUE conjecture for restricted eigenfunctions

摘录 / EXCERPT: Let $U(z)$ be either $y^{k/2}f(z)$ or $u_j(z)$, and let $\psi:\mathbb{Z}\backslash\mathbb{R}\to\mathbb{R}$ be smooth. For fixed $y>0$, Young's horocycle QUE conjecture. ... as the we...

conjecture | open | MathDB 页面 | 原文线索

V2-039381 | MATHDB-325726

Conjecture on Frobenius-structure eigenvalues and Hamiltonian spectral numbers

摘录 / EXCERPT: Let H be a non-degenerate Hamiltonian on M, let $\tilde U_{\xi}$ be the associated space, and let $\star:T U\rightarrow \operatorname{End}(E)$ be the Frobenius structure over...

conjecture | open | MathDB 页面 | 原文线索

V2-039382 | MATHDB-325787

Convergence to a point or interval conjecture for graph-based generalized Pólya urns

摘录 / EXCERPT: Let G be a finite connected graph, and let $x(n)$ denote the associated urn process. Convergence to a point or interval conjecture. There exists either a point $v(G)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-039383 | MATHDB-325792

Palis's conjecture on arithmetic differences of affine Cantor sets

摘录 / EXCERPT: A regular Cantor set K is called affine if its defining map has constant derivative on every interval K_i. For two affine Cantor sets K and K', consider their arithmetic di...

conjecture | open | MathDB 页面 | 原文线索

V2-039384 | MATHDB-325812

Dynamic comparison implies strict comparison for transformation group C*-algebras

摘录 / EXCERPT: Let (X,h) be a minimal dynamical system, and assume that (X,h) has the dynamic comparison property. Write $C^{*}({\mathbb Z},X,h)$ for the transformation group C^{*}-algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-039385 | MATHDB-325833

The E4 Weyl group symmetry and invariant tiling conjecture

摘录 / EXCERPT: Let $g=(A_1A_2)^2$ with s irrational, and let the associated double lattice PET from the compactification theorem be defined on a compactified space with invariant foliation. Den...

conjecture | open | MathDB 页面 | 原文线索

V2-039386 | MATHDB-325876

Semirandom pointwise averaging conjecture

摘录 / EXCERPT: Semirandom pointwise averaging conjecture. Then the random sequence $R^\omega=\{r_1<r_2<r_3<\dots\}$ satisfies, in every dynamical system, for bounded functions F_i,

conjecture | open | MathDB 页面 | 原文线索

V2-039387 | MATHDB-325925

Periodic spectral-radius attainment for linear cocycles with closing by periodic orbits

摘录 / EXCERPT: Let $T\colon W\rightarrow W$ have the closing by periodic orbits property, and let $\boldsymbol{C}\in\operatorname{C}(W,\mathbb{R}^{d\times d})$ be a continuous matrix-valued funct...

open_problem | open | MathDB 页面 | 原文线索

V2-039388 | MATHDB-325929

Rodriguez Hertz-Rodriguez Hertz-Ures conjecture on dynamical coherence

摘录 / EXCERPT: Let M be a compact connected 3-manifold without boundary, and let $f:M\to M$ be a partially hyperbolic diffeomorphism, meaning that there are $N>0$ and a Df-invariant continu...

conjecture | open | MathDB 页面 | 原文线索

V2-039389 | MATHDB-325930

Pujals-Bonatti-Wilkinson classification conjecture for transitive partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let M be a 3-manifold and let $f:M\to M$ be a transitive partially hyperbolic diffeomorphism. Two dynamically coherent partially hyperbolic diffeomorphisms are leaf conjugate i...

conjecture | open | MathDB 页面 | 原文线索

V2-039390 | MATHDB-325943

General projection conjecture for irrational $(\times m,\times n)$-invariant measures

摘录 / EXCERPT: Let $T_{m,n}:\mathbb{T}^2\to\mathbb{T}^2$ denote the coordinatewise action of T_m and T_n, let $\Pi_{2,1}$ be the set of orthogonal projections from $\mathbb{R}^2$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-039391 | MATHDB-325947

Hamiltonian equations for agreeable Lagrangians

摘录 / EXCERPT: Hamiltonian equations for agreeable Lagrangians. For an agreeable Lagrangian L, the Hamiltonian equations of motion for first class functions are given by

conjecture | open | MathDB 页面 | 原文线索

V2-039392 | MATHDB-325984

The supercritical Gaussian multiplicative chaos renormalization conjecture

摘录 / EXCERPT: Supercritical renormalization conjecture. As $\varepsilon\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-039393 | MATHDB-325985

The inner phase II complex Gaussian multiplicative chaos conjecture

摘录 / EXCERPT: Inner phase II conjecture. The following convergence in law holds:

conjecture | open | MathDB 页面 | 原文线索

V2-039394 | MATHDB-325986

The triple-point complex Gaussian multiplicative chaos conjecture

摘录 / EXCERPT: Triple-point conjecture. As $\varepsilon\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-039395 | MATHDB-326001

The Bernoulli-shift conjecture for nonzero-entropy translations on homogeneous spaces

摘录 / EXCERPT: The Bernoulli-shift conjecture. If g has an eigenvalue λ with $|\lambda|\neq 1$-equivalently, if the entropy of T_g is nonzero-then T_g is measurably isomorphic to a...

conjecture | open | MathDB 页面 | 原文线索

V2-039396 | MATHDB-326002

The conjecture that Anosov manifolds are virtually nilmanifolds

摘录 / EXCERPT: The conjecture from the 1960s. If there is an Anosov diffeomorphism f on M, then some finite cover of M is a nilmanifold. Furthermore, lifting f to the finite cover yields...

conjecture | open | MathDB 页面 | 原文线索

V2-039397 | MATHDB-326041

Grant's total-degree conjecture for anti-directed Hamiltonian cycles

摘录 / EXCERPT: Let D be a directed graph on $2n$ vertices, and let $\delta(D)$ denote its minimum total degree. Grant's conjecture. If ... then D contains an anti-directed Hamiltonian cycle. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-039398 | MATHDB-326043

Conjecture on semi-degree thresholds for non-anti-directed cycle orientations

摘录 / EXCERPT: Let D be a directed graph on n vertices, and let $\vec C$ be an orientation of a cycle on n vertices that is not anti-directed. Non-anti-directed cycle conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-039399 | MATHDB-326147

Average-payoff dominance conjecture for fictitious play

摘录 / EXCERPT: Consider bimatrix games with a unique, completely mixed Nash equilibrium and their typical fictitious-play orbits. Average-payoff dominance conjecture. For most such games, typical...

conjecture | open | MathDB 页面 | 原文线索

V2-039400 | MATHDB-326225

Hamilton-cycle bound for circular distance-two labelings

摘录 / EXCERPT: Let (G,H) be a pair of graphs with $H\subset G$, let μ be an integer, and let $\Delta=\Delta(H)\geq 1$. A $\mu(G,H)$-labeling is a (G,H)-labeling using labels from...

conjecture | open | MathDB 页面 | 原文线索

V2-039401 | MATHDB-326227

The resist-impurities conjecture for pairs of 2×2 matrices

摘录 / EXCERPT: The resist-impurities conjecture. The set of pairs $(H,R)\in\mathcal{H}\times\mathcal{E}$ that resist impurities has full Lebesgue measure in $\mathcal{H}\times\mathcal{E}$. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-039402 | MATHDB-326298

Nonvanishing of the order-five obstruction outside exceptional parameter values

摘录 / EXCERPT: Let $K_{4,36}$ denote the order-five variational-equation entry obtained by transporting the jet along the cycle $\gamma=\gamma_-^{-1}\gamma_+^{-1}\gamma_-\gamma_+$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039403 | MATHDB-326388

The uniqueness and non-degeneracy conjecture for the lowest band-function minimum

摘录 / EXCERPT: Let $\zeta_{1}(\tau)$ be the lowest band function associated with the three-dimensional magnetic Hamiltonian with axisymmetric potential. The preceding analysis shows that...

conjecture | open | MathDB 页面 | 原文线索

V2-039404 | MATHDB-326416

Arnold's uniform quasi-periodic motion conjecture for the spatial three-body problem

摘录 / EXCERPT: Let the spatial three-body problem be considered with mass parameter μ and planetary inclinations, in the notation of the corresponding planar theorem. The planar theorem asser...

conjecture | open | MathDB 页面 | 原文线索

V2-039405 | MATHDB-326543

Almost-everywhere asymptotic periodicity for dissipative outer billiards of regular polygons

摘录 / EXCERPT: For a regular polygon and a parameter $\lambda\in(0,1)$, let T_λ denote its dissipative outer billiard map. Asymptotic periodicity conjecture. For almost every...

conjecture | open | MathDB 页面 | 原文线索

V2-039406 | MATHDB-326650

Symmetric-norm minimizer conjecture for slow entropy

摘录 / EXCERPT: Let χ_i be the linear forms associated with a Cartan action, and let q range over the norms occurring in the minimization problem for the slow-entropy estimate. The minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-039407 | MATHDB-326674

Box-dimension equality conjecture for leaves near resonant complex saddles

摘录 / EXCERPT: Box-dimension equality conjecture. The equality holds in the displayed lower-bound formula: $\dim_B(L_a)$ equals the corresponding value in the cases listed there.

conjecture | open | MathDB 页面 | 原文线索

V2-039408 | MATHDB-326782

Akiyama's finiteness-property conjecture for shift radix systems

摘录 / EXCERPT: Akiyama's finiteness-property conjecture. For every $d\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-039409 | MATHDB-326812

The quasi-ergodic hypothesis for typical Hamiltonian systems

摘录 / EXCERPT: Consider a Hamiltonian system with Hamiltonian $H(p,q)=h(p)+\epsilon f(q,p)$ on $T^*\mathbb{T}^n$, and a typical energy surface. Quasi-ergodic hypothesis. For a typical Hamiltonian...

conjecture | open | MathDB 页面 | 原文线索

V2-039410 | MATHDB-326929

Circle-intersection Devron conjecture for polygons

摘录 / EXCERPT: Circle-intersection Devron conjecture. After $2n-6$ steps, one reaches another polygon for which circle intersection takes half of the vertices to a common point.

conjecture | open | MathDB 页面 | 原文线索

V2-039411 | MATHDB-326930

Khesin-Soloviev higher-dimensional pentagram Devron conjecture in dimension three

摘录 / EXCERPT: Khesin-Soloviev higher-dimensional pentagram Devron conjecture. If $B=F^{n-3}(A)$, then $B_1,B_3,\ldots,B_{2n-1}$ are coplanar, and $B_2,B_4,\ldots,B_{2n}$ are also coplanar.

conjecture | open | MathDB 页面 | 原文线索

V2-039412 | MATHDB-326931

Schubert-flip Devron conjecture for configurations of lines

摘录 / EXCERPT: Schubert-flip Devron conjecture. Apply Schubert flips in a bipartite manner, beginning with the even lines. After $2n-7$ steps, one reaches another configuration in which half of t...

conjecture | open | MathDB 页面 | 原文线索

V2-039413 | MATHDB-326986

Conjectures on Hamilton cycles in regular directed and oriented graphs

摘录 / EXCERPT: Conjectures on Hamilton cycles in regular directed and oriented graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-039414 | MATHDB-327072

Nash-Williams's Hamilton cycle packing conjecture

摘录 / EXCERPT: Nash-Williams's conjecture. Suppose that G is a graph on n vertices with minimum degree

conjecture | open | MathDB 页面 | 原文线索

V2-039415 | MATHDB-327109

Random mapping-torus volume ergodic and central-limit conjecture

摘录 / EXCERPT: Fix a generating set Γ of the mapping class group of a surface of genus g. Let ϕ_N be a random word of length N, and let $V(\phi_N)$ be the volume of its mapping to...

conjecture | open | MathDB 页面 | 原文线索

V2-039416 | MATHDB-327125

Direct-observation conjecture for filter stability

摘录 / EXCERPT: Let $(X_k,Y_k)_{k\,in\mathbb{Z}}$ be a stationary infinite-dimensional hidden Markov model as in Section 3, with $X_k\in\{-1,1\}^{\mathbb{Z}}$ and $Y_k\in\{-1,1\}^{\mathbb{Z}}$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-039417 | MATHDB-327193

Uniform periodic approximation conjecture for continued-fraction means

摘录 / EXCERPT: Uniform periodic approximation conjecture. For each $c\in(0,1]$ and almost all α, the limit $F(c)$ exists and

conjecture | open | MathDB 页面 | 原文线索

V2-039418 | MATHDB-327247

Unboundedness of polynomial ergodic averages' variation operator on L^p

摘录 / EXCERPT: Let $(X,\mathcal{B},\mu,T)$ be the measure-preserving system under consideration, let $P(n)$ be an integer-valued polynomial, and for $N\geq 1$ write ... For a sequence (a_N), let...

conjecture | open | MathDB 页面 | 原文线索

V2-039419 | MATHDB-327259

Self-replicating tiling conjecture for irreducible Pisot substitutions

摘录 / EXCERPT: Let σ be an irreducible Pisot substitution, and let K_σ be its associated representation space. Self-replicating tiling conjecture. Every irreducible Pisot substituti...

conjecture | open | MathDB 页面 | 原文线索

V2-039420 | MATHDB-327308

Kühn-Lapinskas-Osthus exact Hamilton-cycle packing conjecture

摘录 / EXCERPT: Let G be a graph on n vertices with minimum degree $\delta(G)\ge n/2$. Kühn-Lapinskas-Osthus conjecture. The graph G contains $\operatorname{reg}_{\operatorname{even}}(G)/2$...

conjecture | open | MathDB 页面 | 原文线索

V2-039421 | MATHDB-327309

Bermond's line-graph Hamilton-decomposition conjecture

摘录 / EXCERPT: Let G be a graph with a Hamilton decomposition, and let $L(G)$ denote its line graph. Bermond's conjecture. The line graph $L(G)$ has a Hamilton decomposition. The conjecture is...

conjecture | open | MathDB 页面 | 原文线索

V2-039422 | MATHDB-327310

Random-regular-graph Hamilton-decomposition conjecture

摘录 / EXCERPT: Let $d=d(n)$ satisfy $d\to\infty$ and $d=o(n)$. Random-regular-graph decomposition conjecture. Asymptotically almost surely, a random d-regular graph on n vertices has a decomp...

conjecture | open | MathDB 页面 | 原文线索

V2-039423 | MATHDB-327311

Random-subcube Hamiltonicity conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional cube, and let $Q_{n,p}$ be the random subgraph obtained by retaining each edge independently with probability p. Random-subcube Hamiltonicity con...

conjecture | open | MathDB 页面 | 原文线索

V2-039424 | MATHDB-327312

Random-graph square Hamilton-cycle conjecture

摘录 / EXCERPT: Let $G_{n,p}$ be the binomial random graph. The square of a Hamilton cycle is obtained from a Hamilton cycle by adding edges between all vertices at distance at most two on the cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-039425 | MATHDB-327378

Uniform boundedness conjecture for stable periodic solutions

摘录 / EXCERPT: Consider the stable periodic solutions of the system under discussion, parametrized by sufficiently small $epsilon$, with stability understood in the sense established above. Unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-039426 | MATHDB-327394

Pósa's conjecture on the square of a Hamiltonian cycle

摘录 / EXCERPT: Let G be a graph on n vertices. The square of a Hamiltonian cycle is C_n^2, where vertices at cyclic distance at most two are adjacent. Pósa's conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-039427 | MATHDB-327397

The almost-sure existence conjecture for increasing Hamiltonian paths

摘录 / EXCERPT: Increasing Hamiltonian path conjecture. With probability tending to 1 as $n\to\infty$, K_n contains an f-increasing Hamiltonian path.

conjecture | open | MathDB 页面 | 原文线索

V2-039428 | MATHDB-327436

The no-limit-cycles conjecture for the wound-healing angiogenesis system

摘录 / EXCERPT: Let (u,w) denote the phase-plane variables and let (α,β,c) be the system parameters in system ... possesses no limit cycles under the parameter regimes Case 1 or Case 2....

conjecture | open | MathDB 页面 | 原文线索

V2-039429 | MATHDB-327437

The Case 2 heteroclinic-orbit conjecture for the wound-healing angiogenesis system

摘录 / EXCERPT: Let (u_{C_1},w_{C_1}), (u_{C_2},w_{C_2}), and (u_W,w_W) be the equilibria of system ... possesses heteroclinic orbits connecting (u_{C_2},w_{C_2}) to (u_{C_1},w_{C_1}) and...

conjecture | open | MathDB 页面 | 原文线索

V2-039430 | MATHDB-327452

Metric adaptation conjecture for positive center-unstable Lyapunov exponents

摘录 / EXCERPT: Metric adaptation conjecture. If f satisfies the positive-Lyapunov-exponent condition along E^{cu} on H, then there exists a Riemannian metric on M for which f satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-039431 | MATHDB-327485

Conjecture on the non-deformability of a partially hyperbolic diffeomorphism

摘录 / EXCERPT: Let E_α be the total space of the smoothly non-trivial bundle constructed in Section, and let $f\colon E_\alpha\to E_\alpha$ be the constructed partially hyperbolic diffeomo...

conjecture | open | MathDB 页面 | 原文线索

V2-039432 | MATHDB-327590

Stable-law and logarithmic central limit conjecture for symmetric observables

摘录 / EXCERPT: Let $\gamma\in(\frac12,1)$, let $v:X\to\mathbb{R}^d$ and $h:X\to G$ be the observable and cocycle from the system, and let $\operatorname{Fix} h(0)=\{w\in\mathbb{R}^d:h(0)w=w\}$. W...

conjecture | open | MathDB 页面 | 原文线索

V2-039433 | MATHDB-327604

Convexity conjecture for the positive total Hofer norm

摘录 / EXCERPT: Let (M,ω) be the symplectic manifold under consideration and let $f\in \operatorname{Ham}(M,\omega)$. Define the positive total Hofer norm as the function assigning to each...

conjecture | open | MathDB 页面 | 原文线索

V2-039434 | MATHDB-327705

Möbius disjointness conjecture for exact C*-algebras

摘录 / EXCERPT: Let $\mathfrak{A}$ be a unital exact C^*-algebra and let α be an endomorphism of $\mathfrak{A}$. The pair $(\mathfrak{A},\alpha)$ is a noncommutative flow. The Möbius func...

conjecture | open | MathDB 页面 | 原文线索

V2-039435 | MATHDB-327723

Baker-DeMarco's dynamical André-Oort conjecture for quadratic polynomials

摘录 / EXCERPT: Let $C\subset {\mathbb A}^2$ be a complex plane curve. A map is postcritically finite (PCF) if every critical point has a finite forward orbit under iteration. Baker-DeMarco's dyna...

conjecture | open | MathDB 页面 | 原文线索

V2-039436 | MATHDB-327728

Universal pentagram maps conjectures on Hamiltonianity and integrability

摘录 / EXCERPT: Let $T_{\mathcal I,\mathcal J}$ be a universal pentagram map on d-tuples of twisted n-gons in ${\mathbb {RP}}^d$, determined by matrices $\mathcal I$ and $\mathcal J$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039437 | MATHDB-327730

Torus symplectic quotient multiplicity conjecture

摘录 / EXCERPT: Let T be a torus acting Hamiltonianly on a symplectic manifold X, with moment map $\mu:X\to\mathfrak{t}^*$. Let $\mathfrak{F}(X)$ be the Fukaya category of X, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039438 | MATHDB-327731

Non-abelian symplectic reduction multiplicity conjecture

摘录 / EXCERPT: Let G act Hamiltonianly on a compact symplectic manifold X, with moment map μ, and let μ be a regular value. Let $\mathfrak{X}$ denote the representation associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-039439 | MATHDB-327802

Avila-Crovisier-Wilkinson conjecture on stable ergodicity and dominated splittings

摘录 / EXCERPT: Let M be a compact manifold, let ω be a volume form, and let $\operatorname{Diff}^r_\omega(M)$ denote the space of C^r volume-preserving diffeomorphisms, with $r>1$. A d...

conjecture | open | MathDB 页面 | 原文线索

V2-039440 | MATHDB-327803

Volume-hyperbolicity dichotomy conjecture for diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold, and let $\operatorname{Diff}^1(M)$ be the space of C^1 diffeomorphisms. A chain-recurrence class is a sink when it is represented by an attracting...

conjecture | open | MathDB 页面 | 原文线索

V2-039441 | MATHDB-327833

The dynamical Manin-Mumford conjecture for complex polynomial automorphisms

摘录 / EXCERPT: Let f be a complex polynomial automorphism of Hénon type of the affine plane, and let C be an irreducible algebraic curve containing infinitely many periodic points of f. An...

conjecture | open | MathDB 页面 | 原文线索

V2-039442 | MATHDB-327834

Uniform periodic-point bound for Hénon-type polynomial automorphisms

摘录 / EXCERPT: Let f be a complex polynomial automorphism of Hénon type such that $|\operatorname{Jac}(f)|\ne 1$. For an algebraic curve C, write $\deg(C)$ for its degree, and write $\deg(f)$...

conjecture | open | MathDB 页面 | 原文线索

V2-039443 | MATHDB-327835

Common-iterate conjecture for Hénon-type polynomial automorphisms

摘录 / EXCERPT: Suppose f and g are two complex polynomial automorphisms of Hénon type sharing infinitely many periodic points. Common-iterate conjecture. There exist non-zero integers n and...

conjecture | open | MathDB 页面 | 原文线索

V2-039444 | MATHDB-327837

The p-stable random-walk scenery entropy formula

摘录 / EXCERPT: Let $(\mathbf{Y},\sigma)$ be the system and cocycle corresponding to a p-stable random walk on $\mathbb{Z}$ for some $p\in(1,2)$, let $\mathbf{X}$ be a Bernoulli flow, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039445 | MATHDB-327838

The vanishing two-sided information conjecture for planar random walks

摘录 / EXCERPT: Let $\mathbf{e}_1,\mathbf{e}_2$ be the usual basis of $\mathbb{Z}^2$. Let $\mathbf{Y}=(\{\pm\mathbf{e}_1,\pm\mathbf{e}_2\}^{\mathbb{Z}},\nu,S)$ be the Bernoulli shift with...

conjecture | open | MathDB 页面 | 原文线索

V2-039446 | MATHDB-327927

Braga-Mello conjecture on arbitrary numbers of limit cycles

摘录 / EXCERPT: Braga-Mello conjecture. Given $n\in\mathbb{N}$, there is a piecewise linear system with two zones in the plane with exactly n limit cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-039447 | MATHDB-327943

The approximation conjecture for three-dimensional flows

摘录 / EXCERPT: Approximation conjecture. Every three-dimensional flow can be C^1 approximated by a flow exhibiting a homoclinic tangency or by a singular-Axiom A flow without cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-039448 | MATHDB-328044

Convergence to equilibrium for strongly reinforced Pólya urns

摘录 / EXCERPT: Convergence to equilibrium. For any WARM with $\alpha>1$, there exists a random vector $\vec{X}=(X_1,\dots,X_n)$, supported on the set of linearly-stable and critical equilibria, s...

conjecture | open | MathDB 页面 | 原文线索

V2-039449 | MATHDB-328154

Mixing conjecture for Bratteli-diagram staircase suspensions

摘录 / EXCERPT: Mixing conjecture. The vertical flow on $S(\mathcal{B},w^\pm, \leq_{r,s})$ is mixing.

conjecture | open | MathDB 页面 | 原文线索

V2-039450 | MATHDB-328156

Gauss-map diffeomorphism conjecture for Sturmian spectra

摘录 / EXCERPT: Let $d706>0$ and $d6fc\in(0,1)\setminus\mathbb{Q}$, and let $G(\alpha)$ denote the Gauss map applied to $d6fc$. Write $d6a9_{\lambda,\alpha}$ and $d6a9_{\lambda,G(\alpha)}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-039451 | MATHDB-328183

Lalonde-Polterovich bounded isometry conjecture

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold. Write $\operatorname{Symp}_0(M)$ for the identity component of the symplectomorphism group, and let ${\rm BI}(M)$ be the normal subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-039452 | MATHDB-328184

Higher-order bounded isometry conjecture

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold. For each integer $k\geq 0$, let ${\rm BI}_{k,0}(M)$ be the subgroup of k-bounded symplectomorphisms in $\operatorname{Symp}_0(M)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039453 | MATHDB-328217

The disk obstruction conjecture for traversing boundary concave vector fields

摘录 / EXCERPT: Disk obstruction conjecture. The standard $(n+1)$-disk D^{n+1} does not admit a traversing boundary concave vector field.

conjecture | open | MathDB 页面 | 原文线索

V2-039454 | MATHDB-328260

Superfluity of the linear growth assumption for multiple correlation sequences

摘录 / EXCERPT: Growth-assumption conjecture. In Theorems 4 and $4'$, the growth assumption on (r_n) is superfluous.

conjecture | open | MathDB 页面 | 原文线索

V2-039455 | MATHDB-328270

Integrable complex structure conjecture for Hamiltonian symplectic Fano actions

摘录 / EXCERPT: Let (Z,ω) be a six-dimensional symplectic Fano manifold with a Hamiltonian S^1-action. An integrable S^1-invariant complex structure compatible with ω is an integ...

conjecture | open | MathDB 页面 | 原文线索

V2-039456 | MATHDB-328274

Standard Koszulity conjecture for geometric category O

摘录 / EXCERPT: Let $\mathcal{O}_{\!\operatorname{g}}$ be the geometric category $\mathcal{O}$ associated with a conical symplectic resolution and a Hamiltonian torus action. Standard Koszulity co...

conjecture | open | MathDB 页面 | 原文线索

V2-039457 | MATHDB-328299

AMP-UD recovery conjecture for stationary ergodic signals

摘录 / EXCERPT: Consider a linear inverse problem in which an input signal ${\bf x}$, generated by a stationary ergodic source, is estimated from noisy measurements ${\bf y}$ and measurement matri...

conjecture | open | MathDB 页面 | 原文线索

V2-039458 | MATHDB-328343

Smooth geometric linearization conjecture for integrable dynamical systems

摘录 / EXCERPT: In a smooth integrable dynamical system, let the system be defined near a nondegenerate singular point, meaning a singular point satisfying the paper's nondegeneracy condition for...

conjecture | open | MathDB 页面 | 原文线索

V2-039459 | MATHDB-328445

The expanding-map lifting conjecture for infra-nilmanifolds

摘录 / EXCERPT: The expanding-map lifting conjecture. The infra-nilmanifold $\Gamma \backslash G$ admits an expanding map if and only if the nilmanifold $N \backslash G$ admits an expanding map.

conjecture | open | MathDB 页面 | 原文线索

V2-039460 | MATHDB-328559

Ratner's non-divergence conjecture for unipotent flows

摘录 / EXCERPT: Let G be a Lie group, let Γ be a discrete subgroup of G, let $\{u_t\}_{t\in\mathbb{R}}\subseteq G$ be a unipotent one-parameter subgroup, and let $x=g\Gamma\in G/\Gamma$...

conjecture | open | MathDB 页面 | 原文线索

V2-039461 | MATHDB-328572

Realization conjecture for continua in the interior of torus rotation sets

摘录 / EXCERPT: Let $f\in{\mathrm{Homeo}}_0(\mathbb{T}^2)$ have a rotation set $\rho(F)$ with non-empty interior, and let $C\subseteq{\mathrm{int}}(\rho(F))$ be a continuum. A realization conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-039462 | MATHDB-328581

Strong Terminating Conjecture for M&m sequences

摘录 / EXCERPT: Starting from the normalized initial values $0<x<1$, let $(x_4,x_5,\ldots)$ be the associated M&m sequence, and let $L(x)$ denote the minimum integer k such that $x_n=x_k$ for al...

conjecture | open | MathDB 页面 | 原文线索

V2-039463 | MATHDB-328700

Density conjecture for geodesically traversally generic metrics

摘录 / EXCERPT: Geodesic genericity conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-039464 | MATHDB-328821

Universality of the Hodge hierarchy for deformations of the Riemann hierarchy

摘录 / EXCERPT: Let a tau-symmetric integrable Hamiltonian deformation of the Riemann hierarchy be given by a hierarchy of scalar Hamiltonian evolutionary PDEs with Hamiltonian densities and Hamil...

conjecture | open | MathDB 页面 | 原文线索

V2-039465 | MATHDB-328822

Canonical-form conjecture for tau-symmetric deformations of the Riemann hierarchy

摘录 / EXCERPT: Let a tau-symmetric integrable Hamiltonian deformation of the Riemann hierarchy be considered modulo normal Miura-type transformations. Write $w_k=\partial_x^k w$. The canonical de...

conjecture | open | MathDB 页面 | 原文线索

V2-039466 | MATHDB-328909

Transversal local rigidity conjecture for parabolic \(\mathbb Z^2\) actions

摘录 / EXCERPT: Transversal local rigidity conjecture. Every sufficiently transversal and sufficiently small perturbation $\{\tilde\rho_\lambda\}$ of the family $\{\rho_\lambda\}$ in a neighborhoo...

conjecture | open | MathDB 页面 | 原文线索

V2-039467 | MATHDB-328941

The S-adic Pisot conjecture

摘录 / EXCERPT: Let S be a finite or infinite set of unimodular substitutions over the finite alphabet $\mathcal{A}$, and let $\boldsymbol{\sigma}\in S^\mathbb{N}$ be a primitive, algebraically...

conjecture | open | MathDB 页面 | 原文线索

V2-039468 | MATHDB-328944

Arbieto's conjecture on sectional-hyperbolic decomposition of generic star flows

摘录 / EXCERPT: Arbieto's conjecture. The nonwandering set of a C^1 generic star flow is the disjoint union of finitely many transitive sets which are positively or negatively sectional-hyperbol...

conjecture | open | MathDB 页面 | 原文线索

V2-039469 | MATHDB-328954

Karlsson-Nussbaum conjecture for finite-dimensional Hilbert metric spaces

摘录 / EXCERPT: Karlsson-Nussbaum conjecture. If (Σ,δ) is finite-dimensional, then there exists a convex set $\Omega\subseteq\partial\Sigma$ such that, for every $x\in\Sigma$, all accu...

conjecture | open | MathDB 页面 | 原文线索

V2-039470 | MATHDB-328968

Turaev's Newhouse-region conjecture for homoclinic tangencies

摘录 / EXCERPT: Turaev's conjecture. There is a Newhouse region C^r-close to a homoclinic tangency if and only if that tangency is not volume-hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-039471 | MATHDB-328969

Generic mechanical non-domination conjecture for chain-recurrence classes

摘录 / EXCERPT: Generic mechanical non-domination conjecture. There is a residual subset $\mathcal R$ of C^r-diffeomorphisms f such that if $\mathcal C(P,f)$ is not dominated of index i, the...

conjecture | open | MathDB 页面 | 原文线索

V2-039472 | MATHDB-328970

Generic mechanical non-volume-hyperbolicity conjecture

摘录 / EXCERPT: Generic mechanical non-volume-hyperbolicity conjecture. For any integer $r\geq 1$, there is a residual subset $\mathcal R$ of C^r-diffeomorphisms f such that if...

conjecture | open | MathDB 页面 | 原文线索

V2-039473 | MATHDB-328971

Generic sink-or-source characterization of non-volume-hyperbolic homoclinic classes

摘录 / EXCERPT: Generic sink-or-source conjecture. For any integer $r\geq 1$, for any diffeomorphism f in a residual subset $\mathcal R\subset\operatorname{Diff}^r(M)$, a homoclinic class...

conjecture | open | MathDB 页面 | 原文线索

V2-039474 | MATHDB-328972

Preperiodicity conjecture for invariant measures

摘录 / EXCERPT: Preperiodicity conjecture. For $r\geq 1$, there exists a C^r-residual set of diffeomorphisms for which any invariant measure supported in a chain-recurrent set is C^r-preperiod...

conjecture | open | MathDB 页面 | 原文线索

V2-039475 | MATHDB-328973

Extension of the Lyapunov-map theorem to heterodimensional cycles

摘录 / EXCERPT: Heterodimensional-cycle extension conjecture. Theorem on possible limit Lyapunov maps for a homoclinic tangency still works if one replaces the homoclinic tangency by a heterodimen...

conjecture | open | MathDB 页面 | 原文线索

V2-039476 | MATHDB-329063

Reduced integrability conjecture for diffusion-invariant stochastic dynamical systems

摘录 / EXCERPT: Let $X = X_0 + \sum X_i \circ \dfrac{dB_t^i}{dt}$ be an SDS on a manifold M. Suppose that X is integrable and diffusion invariant with respect to an action of a compact Lie gro...

conjecture | open | MathDB 页面 | 原文线索

V2-039477 | MATHDB-329133

QUE conjecture for quasimodes on compact hyperbolic surfaces

摘录 / EXCERPT: QUE conjecture for quasimodes. Any sequence of $o(1/\log r)$-quasimodes satisfies QUE.

conjecture | open | MathDB 页面 | 原文线索

V2-039478 | MATHDB-329179

Generic maximal-growth trajectory conjecture for switching systems

摘录 / EXCERPT: Maximal-growth trajectory conjecture. For a generic discrete-time system, there exists a trajectory $x(k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-039479 | MATHDB-329229

Minimality and unique ergodicity of the casse-briques for irrational slope

摘录 / EXCERPT: Let φ be the casse-briques dynamical system and let its limit set be ... The parameter α is the slope parameter of the system. Irrational-slope conjecture. If α...

conjecture | open | MathDB 页面 | 原文线索

V2-039480 | MATHDB-329255

Global contraction conjecture for geometric mesh smoothing

摘录 / EXCERPT: Global contraction conjecture. For any $4\leq N<\infty$ and any connectivity C with cardinality N, there exists a metric $\|\;\|_X$ on $\tilde{X}_C$ such that $\tilde{\Theta}$...

conjecture | open | MathDB 页面 | 原文线索

V2-039481 | MATHDB-329269

Spectral correspondence between Frobenius eigenvalues and Hamiltonian spectral numbers

摘录 / EXCERPT: Let H be a non-degenerate Hamiltonian on M, let U be the base of the associated Frobenius structure on a vector bundle E, and let X_E be its Euler vector field. The 'eige...

conjecture | open | MathDB 页面 | 原文线索

V2-039482 | MATHDB-329303

Infinitely many minimal subsystems in universal symbolic systems

摘录 / EXCERPT: A universal symbolic system is a symbolic dynamical system capable of simulating every effective symbolic system. Universality conjecture. Every universal symbolic system has infin...

conjecture | open | MathDB 页面 | 原文线索

V2-039483 | MATHDB-329384

Biran and Cornea's Lagrangian suspension conjecture

摘录 / EXCERPT: Let $(W;L_0,L_1)$ be an exact Lagrangian cobordism, and let a Lagrangian suspension denote the suspension associated to a Hamiltonian isotopy. Biran and Cornea's conjecture. Any ex...

conjecture | open | MathDB 页面 | 原文线索

V2-039484 | MATHDB-329514

The horocycle-flow obstruction conjecture for variable negative curvature

摘录 / EXCERPT: Horocycle-flow obstruction conjecture. The natural analogues of those theorems should hold for the horocycle flow, with an infinite countable family of obstructions.

conjecture | open | MathDB 页面 | 原文线索

V2-039485 | MATHDB-329546

Novikov-Maltsev conjecture on the Hausdorff dimension of chaotic regimes

摘录 / EXCERPT: Novikov-Maltsev conjecture. For every fixed M, chaotic regimes form a subset of Hausdorff dimension strictly smaller than 2 in $\mathcal{R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-039486 | MATHDB-329556

Unbounded Hofer distance from autonomous Hamiltonian diffeomorphisms

摘录 / EXCERPT: Unboundedness conjecture. For every closed symplectic manifold (M,ω),

conjecture | open | MathDB 页面 | 原文线索

V2-039487 | MATHDB-329589

Hertz-Hertz-Ures weak ergodicity conjecture

摘录 / EXCERPT: Consider an orientable 3-manifold and a conservative partially hyperbolic diffeomorphism on it. Hertz-Hertz-Ures weak ergodicity conjecture. The only orientable 3-manifolds admitti...

conjecture | open | MathDB 页面 | 原文线索

V2-039488 | MATHDB-329590

Hertz-Hertz-Ures weak dynamical coherence conjecture

摘录 / EXCERPT: Consider orientable 3-manifolds supporting partially hyperbolic dynamics. Hertz-Hertz-Ures weak dynamical coherence conjecture. The only orientable 3-manifolds supporting non-dynam...

conjecture | open | MathDB 页面 | 原文线索

V2-039489 | MATHDB-329667

Unique ergodic SRB measure conjecture for obtuse triangular billiards

摘录 / EXCERPT: Let Δ be an obtuse triangle, let $f\in\mathcal B$ be a contracting reflection law, and let $\lambda(f)$ denote its contraction parameter. The billiard map is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-039490 | MATHDB-329669

Isolation conjecture for parabolic periodic points in positive characteristic

摘录 / EXCERPT: Let f be an analytic map over an ultrametric field of positive characteristic, and let ζ_0 be a parabolic periodic point, meaning that for its minimal period n, the multi...

conjecture | open | MathDB 页面 | 原文线索

V2-039491 | MATHDB-329795

Converse torus conjecture for dynamical coherence

摘录 / EXCERPT: Let M^3 be a 3-dimensional manifold and let $f:M^3\rightarrow M^3$ be a partially hyperbolic diffeomorphism. A cu-torus is a 2-torus tangent to the center-unstable bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-039492 | MATHDB-329841

The arbitrarily long periodic-orbit conjecture for triangle tilings

摘录 / EXCERPT: Triangle periodic-length conjecture. There exist triangle tilings with periodic trajectories of arbitrarily large length.

conjecture | open | MathDB 页面 | 原文线索

V2-039493 | MATHDB-329842

The $4n+2$ period conjecture for triangle tilings

摘录 / EXCERPT: $4n+2$ period conjecture. In a triangle tiling, every periodic trajectory has a period of the form

conjecture | open | MathDB 页面 | 原文线索

V2-039494 | MATHDB-329843

The no-spiraling conjecture for triangle tilings

摘录 / EXCERPT: Triangle no-spiraling conjecture. Trajectories on triangle tilings never spiral.

conjecture | open | MathDB 页面 | 原文线索

V2-039495 | MATHDB-329852

Simon's subshift conjecture for Verblunsky coefficients

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite alphabet of Verblunsky coefficients, and let $\Omega\subset \mathcal{A}^{\mathbb{Z}}$ be a minimal subshift that is not periodic. The associated measu...

conjecture | open | MathDB 页面 | 原文线索

V2-039496 | MATHDB-329947

Conjecture on the connection of periodic-orbit families in case VI_3^ϵ

摘录 / EXCERPT: Consider the regularized planar piecewise smooth system in case VI_3^ϵ, with the two families of periodic orbits obtained respectively in Proposition and Proposition. Peri...

conjecture | open | MathDB 页面 | 原文线索

V2-039497 | MATHDB-329953

Existence of a basin point for every finite cycle

摘录 / EXCERPT: Consider the three difference equations referred to in the source as Eqs. (1), (8) and (9), including the equation ... A finite-length cycle means a cycle of finite period, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-039498 | MATHDB-329973

The type A quantum Hamiltonian reduction by stages conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra of type A, and let e and e' be nilpotent elements such that e' covers e in the dominance ordering. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039499 | MATHDB-329974

The Slodowy-slice identification for type A Hamiltonian reductions

摘录 / EXCERPT: Let $e_1,e_2\in\mathfrak{g}$ be nilpotent elements, and let $\mathfrak{m}_1,\mathfrak{m}_2\subseteq\mathfrak{g}$ be the subalgebras constructed for a covering pair of nilpotent orb...

conjecture | open | MathDB 页面 | 原文线索

V2-039500 | MATHDB-329975

The type A reduced W-algebra identification conjecture

摘录 / EXCERPT: Reduced W-algebra identification conjecture. The reduced algebra is isomorphic to the W-algebra $U(\mathfrak{g},e_2)$:

conjecture | open | MathDB 页面 | 原文线索

V2-039501 | MATHDB-330073

The Morse-Smale broken-trajectory lower-bound conjecture

摘录 / EXCERPT: Broken-trajectory lower-bound conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-039502 | MATHDB-330180

Eigenstate thermalisation hypothesis

摘录 / EXCERPT: Let A be a few-body observable, and let $|E_k\rangle$ be an eigenvector of the Hamiltonian of a large interacting many-body system with energy E_k. Eigenstate thermalisation hy...

conjecture | open | MathDB 页面 | 原文线索

V2-039503 | MATHDB-330214

The Morse-Smale trajectory-type and broken-trajectory correspondence conjecture

摘录 / EXCERPT: Let $f:M\to\mathbb R$ be a Morse function on a closed manifold M of dimension $n+1$, and let v be its gradient-like field satisfying the Morse-Smale transversality condition. F...

conjecture | open | MathDB 页面 | 原文线索

V2-039504 | MATHDB-330272

The intrinsic Palis conjecture for homoclinic classes

摘录 / EXCERPT: Let M be a compact manifold and let $\operatorname{Diff}^1(M)$ be the space of C^1-diffeomorphisms. A residual subset is a countable intersection of open dense subsets, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-039505 | MATHDB-330330

The index-vanishing conjecture for commuting vector fields on 3-manifolds

摘录 / EXCERPT: Index-vanishing conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-039506 | MATHDB-330344

Divergence conjecture for the generalized Rogers-Ramanujan continued fraction at roots of unity

摘录 / EXCERPT: Let $a=\zeta_k^j=e^{2\pi i\frac{j}{k}}$ and $x=\zeta_m^l=e^{2\pi i\frac{l}{m}}$ be roots of unity, with $\gcd(j,k)=\gcd(l,m)=1$. Assume $5\mid m$ and $k\mid m$, and define the maps...

conjecture | open | MathDB 页面 | 原文线索

V2-039507 | MATHDB-330351

Hamiltonicity conjecture for non-simple Venn diagrams

摘录 / EXCERPT: Hamiltonicity conjecture. Every non-simple Venn diagram has a Hamiltonian associated planar graph.

conjecture | open | MathDB 页面 | 原文线索

V2-039508 | MATHDB-330373

Percival's conjecture for mixed dynamical systems

摘录 / EXCERPT: Let M be a compact Riemannian manifold with flow domain $\mathcal{D}$, and suppose that $\mathcal{D}$ is the disjoint union of two invariant subsets U and...

conjecture | open | MathDB 页面 | 原文线索

V2-039509 | MATHDB-330401

Asymptotic length-spectrum conjecture for generic algebraic billiard curves

摘录 / EXCERPT: Asymptotic length-spectrum conjecture. If Q is a generic axisymmetric algebraic curve, then

conjecture | open | MathDB 页面 | 原文线索

V2-039510 | MATHDB-330403

Dynamical power non-isotriviality conjecture for polynomial maps

摘录 / EXCERPT: Let K be the function field and let $\bar K$ be an algebraic closure of K. For a polynomial $\phi\in K[x]$, write $\mathcal{O}_\phi(\gamma)$ for the forward orbit of γ u...

conjecture | open | MathDB 页面 | 原文线索

V2-039511 | MATHDB-330602

Rubio-Massegu's conjecture on openness of the good set

摘录 / EXCERPT: Let $f:D\to\mathbb{C}$ be a rational function, where D is its natural domain, and assume that f depends effectively on its last variable. Consider the associated difference equ...

conjecture | open | MathDB 页面 | 原文线索

V2-039512 | MATHDB-330636

Bidirectional tunnel classification for the simplest bomb shape

摘录 / EXCERPT: A bidirectional tunnel is a tunnel whose motion occurs in two directly opposite directions. For the simplest bomb shape, only the wall hit by the particle is erased, and the partic...

conjecture | open | MathDB 页面 | 原文线索

V2-039513 | MATHDB-330637

Periodic tunnels for all non-corner starting positions at the slopes s_n

摘录 / EXCERPT: Let $s_1=3$ and define a sequence by ... A particle starts at a position in the square grid with slope s_n. Periodic-tunnel conjecture. For s_n as above, any starting position wh...

conjecture | open | MathDB 页面 | 原文线索

V2-039514 | MATHDB-330638

Bounded-distance wall visitation in billiards with bombs

摘录 / EXCERPT: A particle moves on the square grid, bouncing from walls and erasing them according to the bomb rule; a starting condition specifies its initial position and direction. Bounded-dis...

conjecture | open | MathDB 页面 | 原文线索

V2-039515 | MATHDB-330639

Complete wall visitation for slope-2 particles

摘录 / EXCERPT: A particle moves on the square grid, and its slope is the slope of its direction of movement. A starting position in the square is non-corner-hitting if the resulting trajectory do...

conjecture | open | MathDB 页面 | 原文线索

V2-039516 | MATHDB-330640

Divisibility criterion for tunneling with wedge-shaped bombs

摘录 / EXCERPT: Let $s>1$ be an integer slope, let n be the size of a wedge-shaped bomb of either type, and choose a starting point in the square that does not cause the particle to hit a corner...

conjecture | open | MathDB 页面 | 原文线索

V2-039517 | MATHDB-330646

The quasi-isomorphism conjecture for arithmetic graphs and the plaid model

摘录 / EXCERPT: Let p/q be the rational parameter, let $Gamma(p/q)$ be its arithmetic landscape, and let $T_{p/q}$ be the affine map carrying the plaid model's fundamental domain to the arithmet...

conjecture | open | MathDB 页面 | 原文线索

V2-039518 | MATHDB-330657

Shear constraint on virtual eigenvalues of ${\mathbb Z}^2$ subshifts

摘录 / EXCERPT: Let Ξ be a minimal, aperiodic and uniquely ergodic ${\mathbb Z}^2$ subshift, and let Ω be its suspension. Let $\lambda=(\lambda_x,\lambda_y)$ be a virtual eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-039519 | MATHDB-330681

Fractal-solution conjecture for the kinetic Boltzmann equation of Saturn's ring

摘录 / EXCERPT: Let the particles of Saturn's Ring be modeled by a kinetic Boltzmann equation associated with the many-body system referred to as system (15). A fractal solution means a solution e...

conjecture | open | MathDB 页面 | 原文线索

V2-039520 | MATHDB-330777

Chirikov-Lochak conjecture for quasi-periodic time-dependent perturbations

摘录 / EXCERPT: Let H be the quasi-periodically time-dependent Hamiltonian ... where $(\theta,I)\in\mathbb T^n\times B$, $(\varphi,J)\in\mathbb T^m\times\mathbb R^m$, $\alpha\in\mathbb R^m$ is non...

conjecture | open | MathDB 页面 | 原文线索

V2-039521 | MATHDB-330785

Goodness of generic one-parameter families on the two-sphere

摘录 / EXCERPT: Good-family conjecture. Generic one-parameter families of vector fields in the two-sphere are good.

conjecture | open | MathDB 页面 | 原文线索

V2-039522 | MATHDB-330786

Goodness of generic two-parameter families on the two-sphere

摘录 / EXCERPT: Good-family conjecture for two parameters. Generic two-parameter families of vector fields in the two-sphere are good.

conjecture | open | MathDB 页面 | 原文线索

V2-039523 | MATHDB-330806

Crovisier's central-bundle conjecture for aperiodic classes

摘录 / EXCERPT: Crovisier's conjecture. The bundle E^c has dimension at least two and admits no non-trivial dominated splitting.

conjecture | open | MathDB 页面 | 原文线索

V2-039524 | MATHDB-330807

The contracted-bundle conjecture for Lyapunov stable aperiodic classes

摘录 / EXCERPT: Contracted-bundle conjecture. The bundle E is contracted.

conjecture | open | MathDB 页面 | 原文线索

V2-039525 | MATHDB-330808

The no-splitting conjecture for bi-Lyapunov stable aperiodic classes

摘录 / EXCERPT: No-splitting conjecture. The class Λ admits no non-trivial dominated splitting.

conjecture | open | MathDB 页面 | 原文线索

V2-039526 | MATHDB-330860

The higher-dimensional singular value pressure inequality

摘录 / EXCERPT: Higher-dimensional singular value pressure conjecture. For each $d\geq 1$, there exist an integer $n(d)$ and a continuous function $\hat{K}_d\colon(0,+\infty)\to(0,+\infty)$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-039527 | MATHDB-330954

The Hamiltonicity bounds for once-subdivided multigraphs

摘录 / EXCERPT: Let M be a multigraph, and let H be obtained from M by subdividing each edge of M at least once. For a positive integer k, let $h_k(H)$ denote the least number of colors...

conjecture | open | MathDB 页面 | 原文线索

V2-039528 | MATHDB-331013

Díaz-Gorodetski global non-hyperbolic measure conjecture

摘录 / EXCERPT: Let M be a compact smooth manifold without boundary, and let $\operatorname{Diff}^r(M)$ denote the space of C^r diffeomorphisms of M, where $r\geq 1$. An invariant measure is...

conjecture | open | MathDB 页面 | 原文线索

V2-039529 | MATHDB-331014

Díaz-Gorodetski homoclinic-class non-hyperbolic measure conjecture

摘录 / EXCERPT: Let M be a compact smooth manifold without boundary, and let $\operatorname{Diff}^1(M)$ be the space of C^1 diffeomorphisms of M. For a periodic point p, its homoclinic cla...

conjecture | open | MathDB 页面 | 原文线索

V2-039530 | MATHDB-331015

Bonatti-Díaz local heterodimensional-cycle conjecture

摘录 / EXCERPT: Let M be a compact smooth manifold without boundary, and let $\operatorname{Diff}^1(M)$ be the space of C^1 diffeomorphisms of M. A heterodimensional cycle is a pair of hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-039531 | MATHDB-331055

The magnetic gap-labelling conjecture for strongly minimal actions

摘录 / EXCERPT: Magnetic gap-labelling conjecture for strongly minimal actions. If the action is strongly minimal, then the containment in the magnetic gap-labelling conjecture for minimal actions...

conjecture | open | MathDB 页面 | 原文线索

V2-039532 | MATHDB-331067

Multicomponent EMM conjecture

摘录 / EXCERPT: Let $\mathcal M$ be a closed $GL(2,\mathbb R)$-invariant subset of a stratum of multicomponent surfaces, and assume $\mathcal M$ consists entirely of surfaces all of whose componen...

conjecture | open | MathDB 页面 | 原文线索

V2-039533 | MATHDB-331168

Connectedness conjecture for tertiary classes of convex bodies

摘录 / EXCERPT: Connectedness conjecture. Every equivalence class in the family of convex bodies defined by the tertiary classification system is connected: for any two convex bodies K_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-039534 | MATHDB-331169

Physicality conjecture for tertiary edges

摘录 / EXCERPT: Physicality conjecture. All tertiary edges of $\mathcal G$ are physical.

conjecture | open | MathDB 页面 | 原文线索

V2-039535 | MATHDB-331183

The approximate-point-spectrum equality for the adding-machine operator

摘录 / EXCERPT: Let S be the operator under consideration, and let E and $\sigma_a(S)$ denote the sets defined in the paper. Approximate-spectrum conjecture. ... This identifies the dynamically...

conjecture | open | MathDB 页面 | 原文线索

V2-039536 | MATHDB-331184

The compactness and connected-complement conjecture for the set E

摘录 / EXCERPT: Let $(p_i)_{i\geqslant 1}$ be the parameter sequence and let E be the set defined in the paper. Compactness and connectedness conjecture. The set E is compact and...

conjecture | open | MathDB 页面 | 原文线索

V2-039537 | MATHDB-331194

The Space-Bounded Church-Turing thesis for computing asymptotic stochastic behavior

摘录 / EXCERPT: Let $\mathcal{S}$ be a closed, discrete-time stochastic system over a state space $\mathcal{X}$, and let its memory be ... where the inner supremum ranges over all distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-039538 | MATHDB-331239

Pesin's density conjecture for non-uniformly hyperbolic conservative diffeomorphisms

摘录 / EXCERPT: Pesin's density conjecture. Arbitrarily close to f in $\operatorname{Diff}^r_m(M)$, there is a diffeomorphism $g\in \operatorname{Diff}^r_m(M)$ without zero Lyapunov exponents on...

conjecture | open | MathDB 页面 | 原文线索

V2-039539 | MATHDB-331240

Herman's dominated-splitting conjecture for conservative diffeomorphisms

摘录 / EXCERPT: Herman's conjecture. Under these assumptions, f admits a dominated splitting.

conjecture | open | MathDB 页面 | 原文线索

V2-039540 | MATHDB-331292

Arnold's instability conjecture for generic elliptic equilibria

摘录 / EXCERPT: Let H be an analytic Hamiltonian system with an elliptic equilibrium point, having n degrees of freedom. Assume that $n\geq 3$ and that the quadratic part of the Hamiltonian at...

conjecture | open | MathDB 页面 | 原文线索

V2-039541 | MATHDB-331364

Dense-mod-1 conjecture for finite sets of irrational numbers

摘录 / EXCERPT: Dense-mod-1 conjecture. The sums $s_{n,d}$ are dense modulo 1.

conjecture | open | MathDB 页面 | 原文线索

V2-039542 | MATHDB-331478

Bochi-Fayad conjecture on resistant pairs of matrices

摘录 / EXCERPT: Bochi-Fayad conjecture. The set of all resistant pairs $(H,R)\in\mathcal{H}\times\mathcal{R}$ has full Lebesgue measure. Partial results are known, and explicit resistant pairs can...

conjecture | open | MathDB 页面 | 原文线索

V2-039543 | MATHDB-331538

Brough-Eick exceptional-isomorphism conjecture for Hamiltonian and special algebras

摘录 / EXCERPT: Let $H(4,(m_1,m_2,m_3,1))$ and $S(3,(m_1,m_2,m_3))$ denote the indicated graded Lie algebras of Cartan type in characteristic 2. Brough-Eick's exceptional-isomorphism conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-039544 | MATHDB-331539

Brough-Eick conjecture on contact and Hamiltonian algebra quotients

摘录 / EXCERPT: Let $mathbb F$ be a field of characteristic 2. Let $n\in\mathbb N$ be odd, $m\in\mathbb N^n$, and set $K=K(n,m)$. Write K' for the derived algebra and $N(K')$ for its relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-039545 | MATHDB-331540

Centralizer conjecture for sofic entropy

摘录 / EXCERPT: Centralizer conjecture. If the centralizer of T is ergodic, then the sofic entropy and quenched sofic entropy coincide; if it is weakly mixing, then the sofic entropy and doubly...

conjecture | open | MathDB 页面 | 原文线索

V2-039546 | MATHDB-331578

Rich restrictions imply non-stable minimality for interval exchange transformations

摘录 / EXCERPT: Rich-restriction instability conjecture. If $\boldsymbol{\boldsymbol{\test}}$ is rich with respect to $\boldsymbol{\boldsymbol{\test}}$ and $T_{\boldsymbol{\boldsymbol{\test}}}$ sa...

conjecture | open | MathDB 页面 | 原文线索

V2-039547 | MATHDB-331604

Hamiltonian non-displaceability conjecture for compact minimal Lagrangian submanifolds

摘录 / EXCERPT: Let L be a compact connected minimal Lagrangian submanifold of an irreducible Hermitian symmetric space of compact type. Hamiltonian non-displaceability conjecture. The submanifo...

conjecture | open | MathDB 页面 | 原文线索

V2-039548 | MATHDB-331655

Conjecture on the structure of extended tongues

摘录 / EXCERPT: Extended-tongue structure conjecture. The connected components of ET_τ are disjoint. The boundary of every connected component consists of two disjoint connected components: a...

conjecture | open | MathDB 页面 | 原文线索

V2-039549 | MATHDB-331656

Conjecture on tangential accumulation of parabolic curves

摘录 / EXCERPT: Parabolic-curve accumulation conjecture. Parabolic curves accumulate on the unit circle at isolated points and tangentially to it. Moreover, if a parabolic curve accumulates on a p...

conjecture | open | MathDB 页面 | 原文线索

V2-039550 | MATHDB-331782

Period-two conjecture for the complex rational difference equation

摘录 / EXCERPT: Let (z_n) be a solution of the difference equation ... with arbitrary initial values and parameters α and β satisfying $\beta=\alpha+1$. Period-two conjecture. All sol...

conjecture | open | MathDB 页面 | 原文线索

V2-039551 | MATHDB-331783

Conjecture on the parameter condition for chaotic solutions

摘录 / EXCERPT: Consider the difference equation ... Here α and β are complex parameters, and a solution is called chaotic in the sense used by the paper's numerical dynamical analysis....

conjecture | open | MathDB 页面 | 原文线索

V2-039552 | MATHDB-331806

The stochastic diffusive behavior conjecture for the generalized Arnold example

摘录 / EXCERPT: The stochastic diffusive behavior conjecture. For a generic perturbation $\varepsilon H_1(\cdot)$, there are smooth functions $b(I)$ and $\sigma(I)>0$, depending only on H_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-039553 | MATHDB-331810

Stationarity and convergence conjecture for the spatial queue process

摘录 / EXCERPT: A state is a configuration $\mathbf{x}=(x_0,x_1,x_2,\ldots)$ satisfying ... and the spatial queue process is the discrete-time Markov chain $\mathbf{X}(t)$ on the space of such confi...

conjecture | open | MathDB 页面 | 原文线索

V2-039554 | MATHDB-331865

Orbit-restrictedness conjecture for cyclic SDS maps over \mathbb Z/m\mathbb Z

摘录 / EXCERPT: Orbit-restrictedness conjecture. The collections $\{F_{m,n}\}_{n\geq 3}$ and $\{G_{m,n}\}_{n\geq 3}$ are each orbit-restricted.

conjecture | open | MathDB 页面 | 原文线索

V2-039555 | MATHDB-331867

Period-divisibility conjecture for cyclic SDS maps over powers of two

摘录 / EXCERPT: Period-divisibility conjecture. If $F_{m,n}$ has a periodic orbit of period r, then

conjecture | open | MathDB 页面 | 原文线索

V2-039556 | MATHDB-331932

Rigidity conjecture for multicritical circle maps with the same signature

摘录 / EXCERPT: Rigidity conjecture. If f and g have the same signature, then they are conjugate by a C^1 diffeomorphism. Moreover, if their common rotation number is of bounded type, then t...

conjecture | open | MathDB 页面 | 原文线索

V2-039557 | MATHDB-331964

Unique non-Markov measure maximizing ascendancy on the golden mean shift

摘录 / EXCERPT: Let the golden mean shift be the shift space on two symbols in which the block 11 is forbidden, and let ν range over all invariant probability measures on it. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-039558 | MATHDB-331972

Generalized-polynomial extension of the integer-part polynomial convergence theorems

摘录 / EXCERPT: Generalized-polynomial convergence conjecture. Theorems 1 and 3 hold if $p_{i,j}$ are generalized polynomials.

conjecture | open | MathDB 页面 | 原文线索

V2-039559 | MATHDB-331980

Oh's conjecture on Hamiltonian volume minimization of product tori

摘录 / EXCERPT: Oh's conjecture. The Lagrangian torus $T(\boldsymbol{b})$ in $\mathbb C^n$ is Hamiltonian volume minimizing.

conjecture | open | MathDB 页面 | 原文线索

V2-039560 | MATHDB-332016

Arithmetic dynamical Mordell-Lang conjecture for $\mathbb{P}^1$

摘录 / EXCERPT: Let $X=\mathbb{P}^1$, and let Y be a curve defined over a field K of characteristic zero. Suppose that $\lambda:Y\to X$ is a finite K-morphism and that $\phi:X\to X$ is a mor...

conjecture | open | MathDB 页面 | 原文线索

V2-039561 | MATHDB-332025

The non-linear limit theorem for ordinal numbers in Pascal automorphism

摘录 / EXCERPT: Let $x=\{x_n,n\in\textbf N\}\in\textbf Z_2$ be a point, and let $t_n(x)$ be the ordinal number of its length-n initial fragment among the binary sequences of length n having th...

conjecture | open | MathDB 页面 | 原文线索

V2-039562 | MATHDB-332157

The Erdős-Hajós conjecture for cylindric billiard flows

摘录 / EXCERPT: Let $\mathbf{Q}=\mathbb{T}^d\setminus(C_1\cup\dots\cup C_k)$ be the configuration space of a cylindric billiard, with cylindric scatterers C_i as described above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039563 | MATHDB-332159

The chaos and NP-hardness conjecture for nonlinear systems

摘录 / EXCERPT: Let $({\cal P}_0)$ be the discretized nonlinear algebraic problem associated with a nonlinear dynamical system, and let $({\cal P})$ be the corresponding optimization problem with...

conjecture | open | MathDB 页面 | 原文线索

V2-039564 | MATHDB-332179

The four-bar linkage pop-reachability conjecture

摘录 / EXCERPT: Four-bar linkage pop-reachability conjecture. For almost every four-bar linkage, the neighborhood of any configuration that can be reached by smooth motion can also be reached by a...

conjecture | open | MathDB 页面 | 原文线索

V2-039565 | MATHDB-332229

Even-length diagonal-twist ground-state conjecture

摘录 / EXCERPT: Let N be the length of the spin-one chain and let x be its anisotropy parameter. Consider the Hamiltonian with diagonal twist and its spectrum, including eigenvalue degeneracie...

conjecture | open | MathDB 页面 | 原文线索

V2-039566 | MATHDB-332230

Odd-length diagonal-twist spectral conjecture

摘录 / EXCERPT: Let N be the length of the spin-one chain and let x be its anisotropy parameter. Consider the Hamiltonian with diagonal twist and its eigenvalue degeneracies. Odd-length diagon...

conjecture | open | MathDB 页面 | 原文线索

V2-039567 | MATHDB-332231

Anti-diagonal-twist ground-state conjecture

摘录 / EXCERPT: Let N be the length of the spin-one chain and let x be its anisotropy parameter. Consider the Hamiltonian with anti-diagonal twist. Anti-diagonal-twist conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-039568 | MATHDB-332246

Microlocal quantum unique ergodicity for paths on the Bruhat-Tits graph

摘录 / EXCERPT: Fix $n\in\mathbb{Z}_{\geqslant 0}$ and let $N\geqslant n$ tend to infinity. Let $\mathbf{Y}_{-N..N}$ be the space of paths used above, let $\mathcal{F}_{-N..N}$ be the specified fa...

conjecture | open | MathDB 页面 | 原文线索

V2-039569 | MATHDB-332332

The genericity, freedom and velocity conjectures for Arnold diffusion

摘录 / EXCERPT: Arnold diffusion conjecture. For sufficiently large $r\in{\mathbb N}\cup\{\infty,\omega\}$ and an open dense set of C^r perturbations: (A) diffusion exists; (B) the projection...

conjecture | open | MathDB 页面 | 原文线索

V2-039570 | MATHDB-332431

Dynamics of the projectively natural polygon iteration on projective polygon space

摘录 / EXCERPT: Let $n \geq 5$ and let H_{λ} be the polygon iteration on the projective moduli space $\mathfrak{P}_n = (\mathbb{R}P^2)^n/PGL(3,\mathbb{R})$ of projective equivalence classe...

conjecture | open | MathDB 页面 | 原文线索

V2-039571 | MATHDB-332432

Affine dynamics of the polygon iteration on ordinary polygon space

摘录 / EXCERPT: Let $n \geq 5$, let H_{λ} be the polygon iteration on the space $\mathcal{P}_n=(\mathbb{R}P^2)^n$ of n-gons, and let $\lambda_0=2\cos(\pi/n)$. Affine dynamics conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-039572 | MATHDB-332454

Symbolic Furstenberg conjecture for Jewett-Krieger models

摘录 / EXCERPT: Let $S,T\colon[0,1)\to[0,1)$ be the maps $Sx=px\pmod 1$ and $Tx=qx\pmod 1$, where $p,q\geq 1$ are multiplicatively independent integers. Let μ be a Borel probability measure on...

conjecture | open | MathDB 页面 | 原文线索

V2-039573 | MATHDB-332488

Multiple recurrence conjecture for computable commuting transformations

摘录 / EXCERPT: Let (X,μ) be a computable probability space. Let $T_1,\ldots,T_k$ be computable measure-preserving transformations that commute pairwise. Let P be a Π^0_1 class with...

conjecture | open | MathDB 页面 | 原文线索

V2-039574 | MATHDB-332558

The triangulation-to-Liouville quantum gravity convergence conjecture

摘录 / EXCERPT: Let $\mathcal{T}_{N}$ be the set of triangulations of the sphere with N faces, and let $\mathcal{T}_{N,3}$ be the set of triangulations with N faces and three marked faces. Equ...

conjecture | open | MathDB 页面 | 原文线索

V2-039575 | MATHDB-332597

Conjecture on fractal-like and fractal-unlike chaotic trajectories

摘录 / EXCERPT: Let ... be the rational map associated with parameters α, β, γ, and δ. A chaotic trajectory is called fractal-like or fractal-unlike according to the obser...

conjecture | open | MathDB 页面 | 原文线索

V2-039576 | MATHDB-332609

Liouville mushroom conjecture

摘录 / EXCERPT: Liouville mushroom conjecture. For sufficiently well-approximable MUPO-free ρ,

conjecture | open | MathDB 页面 | 原文线索

V2-039577 | MATHDB-332610

No-incipient-MUPOs conjecture

摘录 / EXCERPT: No-incipient-MUPOs conjecture. The above equation for r is never satisfied without the existence of another real MUPO.

conjecture | open | MathDB 页面 | 原文线索

V2-039578 | MATHDB-332641

The mixing conjecture for matrix equilibrium states

摘录 / EXCERPT: Mixing conjecture. If for some $s>0$ the equilibrium state of (A,s) is ergodic with respect to σ^d, then it is mixing with respect to σ. If there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-039579 | MATHDB-332642

The Bernoulli property conjecture for totally ergodic matrix equilibrium states

摘录 / EXCERPT: Bernoulli property conjecture. The natural extension of every totally ergodic matrix equilibrium state has the Bernoulli property.

conjecture | open | MathDB 页面 | 原文线索

V2-039580 | MATHDB-332643

The \overline{d}-continuity conjecture for equilibrium states under parameter variation

摘录 / EXCERPT: overline{d}-continuity conjecture. The measure μ_s depends $\overline{d}$-continuously on s.

conjecture | open | MathDB 页面 | 原文线索

V2-039581 | MATHDB-332644

The local \overline{d}-continuity conjecture for mixing matrix equilibrium states

摘录 / EXCERPT: Local overline{d}-continuity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-039582 | MATHDB-332661

Attracting limit cycle converging to the singular cycle

摘录 / EXCERPT: Let α and ξ be parameters with $\alpha>\xi$, let $\varepsilon>0$ be sufficiently small, and let Γ_0 denote the singular cycle described by the concatenation of th...

conjecture | open | MathDB 页面 | 原文线索

V2-039583 | MATHDB-332683

Nonexistence conjecture for Morse cocycles with simple Lebesgue component

摘录 / EXCERPT: A Morse cocycle is a cocycle arising from a Morse rank-one construction, and a simple Lebesgue component is a multiplicity-one Lebesgue component in the spectrum of the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-039584 | MATHDB-332794

Franks-Manning conjecture on Anosov diffeomorphisms

摘录 / EXCERPT: Let M be a closed manifold and let $f\colon M\to M$ be an Anosov diffeomorphism, meaning a diffeomorphism with a hyperbolic splitting of the tangent bundle. A hyperbolic infra-ni...

conjecture | open | MathDB 页面 | 原文线索

V2-039585 | MATHDB-332871

Bounded number of stable sets for n-expansive homeomorphisms

摘录 / EXCERPT: Bounded stable-set conjecture. There exists $\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-039586 | MATHDB-332916

The generalized admissible triple convergence conjecture for graded graphs

摘录 / EXCERPT: Let Γ be a graded graph and let μ be an ergodic central measure on the path space $T(\Gamma)$. Let $\rho=\rho_0$ be a metric on $T(\Gamma)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-039587 | MATHDB-332917

The absolute-boundary conjecture for distributive-lattice graded graphs

摘录 / EXCERPT: Let Y be a poset, let $L(Y)$ be the distributive lattice of all finite ideals of Y, and let Γ be the Hasse diagram of $L(Y)$. Consider the space of all minimal infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-039588 | MATHDB-333071

The conjecture on the maximum of the GFF on the circle

摘录 / EXCERPT: Let V_N be the maximum of the discretized Gaussian free field on the circle with singularity parameter $\alpha\geq 0$. Let $M_{(\tau=1,\alpha,\alpha)}$ denote the critical Morris...

conjecture | open | MathDB 页面 | 原文线索

V2-039589 | MATHDB-333072

The freezing hypothesis for the exponential functional of the GFF

摘录 / EXCERPT: Let X be the normalized exponential functional introduced in the source, and let q be in the range where the displayed moments exist. Freezing hypothesis. For $\beta>1$, ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-039590 | MATHDB-333166

The homoclinic-orbit simplicity conjecture

摘录 / EXCERPT: Let G be the group under consideration, and call a real-analytic subset of G simple if it is contained in a finite union of sets of the form ... with $\chi\in\widehat G$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039591 | MATHDB-333189

Log-normal limit conjecture for increasing Hamiltonian paths

摘录 / EXCERPT: For each n, let X be the number of Hamiltonian paths that are increasing in a uniformly random ordering of the edges of K_n. Log-normal limit conjecture. As $n\to\infty$, the...

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V2-039592 | MATHDB-333199

The unbounded autonomous Hofer distance conjecture

摘录 / EXCERPT: Let X be a closed symplectic manifold. Define ... where d_H is Hofer's metric and $\operatorname{Aut}(X)$ denotes the autonomous Hamiltonian diffeomorphisms. Unbounded autonomous...

conjecture | open | MathDB 页面 | 原文线索

V2-039593 | MATHDB-333206

The scaling conjecture for radius of comparison under powers

摘录 / EXCERPT: Let X be a compact metric space, let $n\in\mathbb{Z}_{>0}$, and let $h\colon X\to X$ be a homeomorphism such that h^n is minimal. The scaling conjecture. ... This conjecture is p...

conjecture | open | MathDB 页面 | 原文线索

V2-039594 | MATHDB-333209

Weak baby multicorns dynamically distinct conjecture

摘录 / EXCERPT: Let d be the degree of a multicorn, let ω denote the rotational symmetry used in the parameter space, and let H_1 and H_2 be hyperbolic components of odd periods k_1...

conjecture | open | MathDB 页面 | 原文线索

V2-039595 | MATHDB-333285

Shortest forbidden-pattern conjecture for symmetric tent maps

摘录 / EXCERPT: Shortest forbidden-pattern conjecture. The shortest μ-forbidden pattern is of the form σ_n. Equivalently, if n is the length of the shortest μ-forbidden pattern, t...

conjecture | open | MathDB 页面 | 原文线索

V2-039596 | MATHDB-333286

Monotonicity conjecture for allowed patterns of symmetric tent maps

摘录 / EXCERPT: Let T_μ and T_ν be symmetric tent maps with $\frac{1}{2}<\mu<\nu\leq 1$, and let h_μ and h_ν be their commuter functions. Monotonicity conjecture. ... Consequently, t...

conjecture | open | MathDB 页面 | 原文线索

V2-039597 | MATHDB-333291

Infinite cell-boundary crossings for almost all crossing trajectories

摘录 / EXCERPT: Let a trajectory of the ABC flow cross the boundary of a cell. Infinite-crossing conjecture. Almost all trajectories that cross the boundary of a cell will cross cell boundaries in...

conjecture | open | MathDB 页面 | 原文线索

V2-039598 | MATHDB-333292

Open sets of monotone ballistic trajectories in ABC flows

摘录 / EXCERPT: Consider initial conditions for the ABC flow in the non-KAM region. Principal-vortex conjecture. There exists an open set of positive measure of initial conditions such that y (o...

conjecture | open | MathDB 页面 | 原文线索

V2-039599 | MATHDB-333293

KAM trapping and recurrent cell crossing in ABC flows

摘录 / EXCERPT: Fix an initial value $z(0)=z_0$. The xy-plane is divided into disjoint KAM and non-KAM regions, and a KAM region is one of the regions in which the associated trajectories are tr...

conjecture | open | MathDB 页面 | 原文线索

V2-039600 | MATHDB-333382

Positivity of Goncharov-Kenyon Hamiltonians in upper cluster algebras

摘录 / EXCERPT: Let Q be a quiver in the classification under consideration, and let H_i denote its Goncharov-Kenyon Hamiltonians. The positivity conjecture. The Hamiltonians H_i are element...

conjecture | open | MathDB 页面 | 原文线索

V2-039601 | MATHDB-333420

McMullen's Arithmetic Chaos conjecture

摘录 / EXCERPT: McMullen's Arithmetic Chaos conjecture. There is a compact subset $\mathcal{Y}\subset\mathcal{X}$ such that, for every real quadratic field K, the set of closed geodesics defined...

conjecture | open | MathDB 页面 | 原文线索

V2-039602 | MATHDB-333421

The continued-fraction formulation of Arithmetic Chaos

摘录 / EXCERPT: Arithmetic Chaos continued-fraction conjecture. There is an $A<\infty$ such that, for every real quadratic field K, the set

conjecture | open | MathDB 页面 | 原文线索

V2-039603 | MATHDB-333565

Uniqueness of the ascent-maximizing Markov measure on the golden mean shift

摘录 / EXCERPT: Let (X,σ) be the golden mean shift, let α be the partition used to define the ascent rate, and let $r\geq1$. An r-step Markov measure is a shift-invariant measure w...

conjecture | open | MathDB 页面 | 原文线索

V2-039604 | MATHDB-333566

Distinct ascent-maximizing Markov measures on the golden mean shift

摘录 / EXCERPT: Let (X,T) be the golden mean shift, let α be the partition used to define ascent, and let μ_1 and μ_2 denote the maximizing one-step and two-step Markov measures,...

conjecture | open | MathDB 页面 | 原文线索

V2-039605 | MATHDB-333567

Non-Markovian uniqueness of the ascent maximizer on the golden mean shift

摘录 / EXCERPT: Let (X,T) be the golden mean shift, let α be the partition used to define ascent, and let $\mu_{\max}$ denote the measure of maximal entropy. Non-Markovian ascent-maximize...

conjecture | open | MathDB 页面 | 原文线索

V2-039606 | MATHDB-333568

Two information-maximizing one-step Markov measures on the full two-shift

摘录 / EXCERPT: Let (X,T) be the full two-shift, let α be the partition used to define information, and let the dualizing map interchange the symbols 0 and 1. Two-maximizers conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-039607 | MATHDB-333569

A fully supported local information maximum on the full two-shift

摘录 / EXCERPT: Let (X,T) be the full two-shift, let α be the partition used to define information, and consider the space of one-step Markov measures on X. A measure is fully supported...

conjecture | open | MathDB 页面 | 原文线索

V2-039608 | MATHDB-333571

Uniqueness conjecture for continuous solutions of the Furstenberg function equations

摘录 / EXCERPT: The function-equation conjecture. The only such function is $f(x)=x$. The conjecture is presented as a sufficient condition for Furstenberg's conjecture on continuous ergodic...

conjecture | open | MathDB 页面 | 原文线索

V2-039609 | MATHDB-333604

Dynamical unlikely intersection conjecture for two points

摘录 / EXCERPT: Let $k={\mathbb C}(X)$, where X is a compact Riemann surface, and let $f\in k(z)$ be non-isotrivial. For $P,Q\in{\mathbb P}^1(k)$, define ... Assume that neither P nor Q is per...

conjecture | open | MathDB 页面 | 原文线索

V2-039610 | MATHDB-333686

Generalized middle levels conjecture for hypercube level subgraphs

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, and let $Q_{n,[a,b]}$ denote the subgraph induced by the vertices whose levels lie in $[a,b]$. For integers $k\geq 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039611 | MATHDB-333807

Finite-horizon approximation conjecture for extremum-seeking systems

摘录 / EXCERPT: Finite-horizon approximation conjecture. For any $\delta>0$, any compact set $K \subset \mathbb{R}^n$, and any $t_0,T \in \mathbb{R}_{\geq 0}$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-039612 | MATHDB-333825

Higher-Hamiltonian eigenvalue conjecture for generalized Macdonald polynomials

摘录 / EXCERPT: Higher-Hamiltonian eigenvalue conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-039613 | MATHDB-333832

The ergodic coboundary non-degeneracy conjecture

摘录 / EXCERPT: Ergodic coboundary non-degeneracy conjecture. Then $\delta=0$ ν-almost everywhere.

conjecture | open | MathDB 页面 | 原文线索

V2-039614 | MATHDB-333908

Maximality conjecture for the noncommutative spectral-curve Hamiltonians

摘录 / EXCERPT: Let A be the free associative algebra of the system, let $[A,A]$ be its commutator subspace, and write the coefficients of the noncommutative spectral curve as ... The induced Lie...

conjecture | open | MathDB 页面 | 原文线索

V2-039615 | MATHDB-333911

Conjecture on ideal-free reducible strategies in source-sink landscapes

摘录 / EXCERPT: Assume $1\le k<n$, assumptions $\mathrm{(A2)}$-$\mathrm{(A4)}$ hold, (d_{ij}) is ideal-free on patches $1\le i\le k$ and satisfies $d_{ij}=0$ whenever $i>k$ or $j>k$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039616 | MATHDB-333914

The Fredholm index conjecture for formal quantization of b-symplectic manifolds

摘录 / EXCERPT: Let M be an integral b-symplectic manifold equipped with a Hamiltonian T-action with nonzero modular weight, and let $Q(M)$ denote its formal geometric quantization. Fredholm...

conjecture | open | MathDB 页面 | 原文线索

V2-039617 | MATHDB-333933

Buchstaber-Tertychnyi garland conjecture for phase-lock areas

摘录 / EXCERPT: For every $r\in\mathbb Z$, let L_r denote the phase-lock area with rotation number r, and set ... Let $\mathcal A_{r,1},\mathcal A_{r,2},\ldots$ denote the adjacencies in the lin...

conjecture | open | MathDB 页面 | 原文线索

V2-039618 | MATHDB-333934

Small-frequency limit conjecture for the upper part of phase-lock areas

摘录 / EXCERPT: Small-frequency limit conjecture. As $\omega\to0$, the maximal distance from a point of $L_{r,1}$ to the ray $\{A\geq1\}$ on the A-axis tends to zero. This describes the limiting...

conjecture | open | MathDB 页面 | 原文线索

V2-039619 | MATHDB-333935

Nonintersection conjecture for phase-lock areas and vertical lines

摘录 / EXCERPT: Nonintersection conjecture. The phase-lock area L_{r+1} does not intersect Λ_r. This conjecture is discussed as a possible consequence of a monodromy/Riccati statement, w...

conjecture | open | MathDB 页面 | 原文线索

V2-039620 | MATHDB-333936

Shifted nonintersection conjecture for phase-lock areas

摘录 / EXCERPT: Shifted nonintersection conjecture. The phase-lock area L_r does not intersect the line Λ_{r-2}. The source notes that this conjecture follows from the preceding noninter...

conjecture | open | MathDB 页面 | 原文线索

V2-039621 | MATHDB-333937

Non-tangency conjecture for phase-lock adjacencies

摘录 / EXCERPT: Let $\omega>0$, and let $(B_0,A_0)\in\mathbb R_+^2$ be an adjacency of a phase-lock area. Consider the two branches of the boundary of that phase-lock area meeting at (B_0,A_0)....

conjecture | open | MathDB 页面 | 原文线索

V2-039622 | MATHDB-333992

Uniform dynamical-degree lower-bound conjecture for dominant rational maps

摘录 / EXCERPT: Uniform dynamical-degree lower-bound conjecture. There exists a constant $\gamma_N>0$ such that, for all dominant rational maps $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$,

open_problem | open | MathDB 页面 | 原文线索

V2-039623 | MATHDB-333993

Conjecture on the exceptional set of nondifferentiability of the entropy function

摘录 / EXCERPT: Let H be the entropy function and let ${\mathcal{E}}$ denote the exceptional set introduced in the paper. By Theorem 2, H is differentiable inside each entropy plateau...

conjecture | open | MathDB 页面 | 原文线索

V2-039624 | MATHDB-333994

Smoothness conjecture for the exponential attractor under finite dissipation

摘录 / EXCERPT: Let $f=f_B$, and let k be sufficiently large depending on the internal parameters $ Omega$, L, b, and p. There is a proper fractal exponential attractor ... with finite fract...

conjecture | open | MathDB 页面 | 原文线索

V2-039625 | MATHDB-334074

Rudnick-Sarnak's quantum unique ergodicity conjecture

摘录 / EXCERPT: Let N be a compact Riemannian manifold with negative sectional curvature, let $(\Psi_n)_{n=0}^{\infty}$ be an orthonormal basis of Laplace eigenfunctions on N, and let μ_n...

conjecture | open | MathDB 页面 | 原文线索

V2-039626 | MATHDB-334090

No-wild-historic-behavior Lyapunov-exponent conjecture

摘录 / EXCERPT: No-wild-historic-behavior conjecture. Absence of wild historic points implies that every invariant probability measure is either atomic or has only zero Lyapunov exponents. An anal...

conjecture | open | MathDB 页面 | 原文线索

V2-039627 | MATHDB-334091

Approximate homoclinic relation from periodic historic measures

摘录 / EXCERPT: Homoclinic-approximation conjecture. If there exist x and $a,b\geq0$ with $a+b>1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-039628 | MATHDB-334128

Exponential conditional mutual information decay for Gibbs states on lattices

摘录 / EXCERPT: Let ρ be a Gibbs state of a short-ranged Hamiltonian defined on a D-dimensional spin lattice. Let A, B, and C be three regions, with B shielding A from C, and le...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-039629 | MATHDB-334147

Existence of an escaping orbit for every rational standard-map parameter

摘录 / EXCERPT: Let $b1=p/q$ be rational, and let $y_0=a/b$ be an initial level for the original map. An orbit is escaping if it is unbounded. Escaping-orbit existence conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-039630 | MATHDB-334148

Quadratic growth conjecture for the length of the escaping orbit

摘录 / EXCERPT: Let $b1=1/q$ and $y_0=0$, and let $ell(q)$ denote the unique odd-length period under f on L_o. Quadratic escaping-orbit length conjecture. The length of the escaping orbit grow...

conjecture | open | MathDB 页面 | 原文线索

V2-039631 | MATHDB-334149

Linear bound conjecture for bounded trajectories of the standard map

摘录 / EXCERPT: Let q be the denominator of the rational parameter in the transformation f, and consider its bounded trajectories. Bounded-trajectory length conjecture. The maximum length of t...

conjecture | open | MathDB 页面 | 原文线索

V2-039632 | MATHDB-334150

Coincidence conjecture for large-period distributions at even denominators

摘录 / EXCERPT: Let q range over even denominators, and consider the periodic trajectories of the transformation f. Large-period distribution conjecture. The distributions of large periods of...

conjecture | open | MathDB 页面 | 原文线索

V2-039633 | MATHDB-334165

Furstenberg's intersection conjecture for multiplicatively independent dynamics

摘录 / EXCERPT: Let $p,q\ge 2$ be integers with $p\nsim q$, where $a\nsim b$ means that $\log a/\log b\notin\mathbb Q$. Let $T_m:[0,1]\rightarrow[0,1]$ be the map $x\mapsto mx\mod 1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039634 | MATHDB-334199

Smale's generic Axiom A conjecture

摘录 / EXCERPT: Let M be a compact manifold. A diffeomorphism of M is Baire generic if it belongs to a countable intersection of open dense sets. Smale's conjecture. A Baire generic diffeomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-039635 | MATHDB-334200

Arnold-Kolmogorov typicality conjecture for finiteness of attractors

摘录 / EXCERPT: Let M be a compact manifold. A property is Arnold-Kolmogorov typical if it holds for Lebesgue almost every small parameter in a Baire generic set of smooth families of dynamics....

conjecture | open | MathDB 页面 | 原文线索

V2-039636 | MATHDB-334201

Sup-polynomial emergence conjecture

摘录 / EXCERPT: Let M be the compact manifold under consideration, and let $\operatorname{Diff}(M)$ denote its diffeomorphism space. Sup-polynomial emergence conjecture. There exists an open set...

conjecture | open | MathDB 页面 | 原文线索

V2-039637 | MATHDB-334210

Selberg distribution conjecture for Bacry-Muzy GMC on the interval

摘录 / EXCERPT: Selberg distribution conjecture. Its distribution has Mellin transform

conjecture | open | MathDB 页面 | 原文线索

V2-039638 | MATHDB-334211

Morris distribution conjecture for Bacry-Muzy GMC on the circle

摘录 / EXCERPT: Morris distribution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-039639 | MATHDB-334212

Infinite divisibility conjecture for logarithmic GMC total mass

摘录 / EXCERPT: Infinite divisibility conjecture. The distribution of

conjecture | open | MathDB 页面 | 原文线索

V2-039640 | MATHDB-334213

Self-duality conjecture for the Bacry-Muzy GMC Mellin transform

摘录 / EXCERPT: Self-duality conjecture. Under $\tau\mapsto1/\tau$ and $q\mapsto q/\tau$, it satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-039641 | MATHDB-334214

Covariance perturbation invariance conjecture for GMC regularizations

摘录 / EXCERPT: Let $\widetilde{\omega}_{\mu,\varepsilon}(s)$ be a Gaussian process whose covariance agrees with the covariance in Eq. up to $O(\varepsilon)$ as $\varepsilon\to0$, and whose mean a...

conjecture | open | MathDB 页面 | 原文线索

V2-039642 | MATHDB-334215

Centered GMC measure moment-limit conjecture

摘录 / EXCERPT: Let $\omega_{\mu,\varepsilon}(u)$ be the regularized Gaussian field and define its centered version by ... For $-1/\mu-1/2<\operatorname{Re}(q)<2/\mu$, and for a test function...

conjecture | open | MathDB 页面 | 原文线索

V2-039643 | MATHDB-334356

Hohenberg-Kohn density-potential uniqueness conjecture

摘录 / EXCERPT: Let $H=T+V_{\mathrm{ee}}+V_{\mathrm{ext}}$ be the Hamiltonian of a bounded many-electron system, with external potential v and one-particle density n in a non-degenerate ground...

conjecture | open | MathDB 页面 | 原文线索

V2-039644 | MATHDB-334393

Conjecture on longest orbit lengths for ambiguous and non-ambiguous modular elements

摘录 / EXCERPT: Let $g\in\operatorname{PSL}_{2}(\mathbb{Z})$ be hyperbolic. Let $L(g;p)$ denote the longest orbit of g on $\mathbb{Y}_{-2}(\mathbb{F}_{p})$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-039645 | MATHDB-334403

Non-approximability conjecture for convex combinations of hyperbolic ergodic measures

摘录 / EXCERPT: Non-approximability conjecture. For every pair of hyperbolic ergodic measures μ_1 and μ_2 with

conjecture | open | MathDB 页面 | 原文线索

V2-039646 | MATHDB-334421

The conjecture extending the sign pattern of h_1 to every h_n

摘录 / EXCERPT: Sign-pattern conjecture. Lemma (b) is true for every h_n, $n\in\mathbb{N}$; that is, for $q\in Q_k$ and $0<t<1/2$, one has $(-1)^{k-1}h_n'(t)>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-039647 | MATHDB-334537

Finitary simplicity of the limiting dynamical symmetry Lie algebra

摘录 / EXCERPT: Let $\Phi[\sigma;1;f]$ be the limiting object considered in the paper, and let $(\mathfrak l_{k_h})_h$ be the system of dynamical symmetry Lie algebras associated with the embeddin...

conjecture | open | MathDB 页面 | 原文线索

V2-039648 | MATHDB-334718

Dense periodic points imply periodicity for monotone maps with polyhedral order cones

摘录 / EXCERPT: Dense periodicity conjecture. If $\mathcal{P}(T)$ is dense in X, then T is periodic: there is a positive integer k such that $X=\mathcal{P}_k(T)$.

conjecture | open | MathDB 页面 | 原文线索

V2-039649 | MATHDB-334774

Weinstein's symplectic conjecture on closed characteristics

摘录 / EXCERPT: Weinstein conjecture. Any closed contact-type hypersurface in a symplectic manifold admits a closed characteristic.

conjecture | open | MathDB 页面 | 原文线索

V2-039650 | MATHDB-334787

Conjecture on the second asymmetric invariant set $\mathcal{A}_2$

摘录 / EXCERPT: Second asymmetric-set conjecture. There exists such a polyhedron P_3 for which $\mathcal{A}_2$ is symmetric with respect to S_1,S_2 and S_4, asymmetric with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-039651 | MATHDB-334788

Conjecture on symmetric invariant sets at the values ε_n

摘录 / EXCERPT: Symmetric invariant-set conjecture. At every ε_n, new symmetric invariant sets appear; these sets are contained in $\mathcal{S}$ and are symmetric. The conjecture is mo...

conjecture | open | MathDB 页面 | 原文线索

V2-039652 | MATHDB-334802

Fyodorov-Keating conjecture on moments of regularized counting statistics

摘录 / EXCERPT: Fyodorov-Keating conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-039653 | MATHDB-334810

Instability persistence under little-o perturbations

摘录 / EXCERPT: Let $g: \mathbb{R}^d \to \mathbb{R}^d$ be continuous and linearly homogeneous. Let $f: \mathbb{R}^d \to \mathbb{R}^d$ be continuous and satisfy ... A fixed point is Lyapunov stable i...

conjecture | open | MathDB 页面 | 原文线索

V2-039654 | MATHDB-334993

Quasiconvexity conjecture for radially symmetric effective Hamiltonians

摘录 / EXCERPT: Assume that $H(p)=\varphi(|p|)$, where $\varphi:[0,\infty)\to\mathbb{R}$ satisfies (H7), and let ... be the associated sequence, with $M_i=\varphi(s_{2i-1})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039655 | MATHDB-335020

Minimality conjecture for the remaining affine interval exchange transformations

摘录 / EXCERPT: Let F_t be the affine interval exchange transformation associated with the parameter t, and let $\Lambda_{\Gamma}\subset S^1$ be the Cantor set of remaining parameters, defined...

conjecture | open | MathDB 页面 | 原文线索

V2-039656 | MATHDB-335024

Robust fast growth for generic families of diffeomorphisms

摘录 / EXCERPT: Let M be the manifold under consideration, let $r\in\{1,\ldots,\infty,\omega\}$, and write $\operatorname{Diff}^r(M)$ for the space of C^r diffeomorphisms of M. A family...

conjecture | open | MathDB 页面 | 原文线索

V2-039657 | MATHDB-335094

Stable ergodicity is open and dense for volume-preserving diffeomorphism tuples

摘录 / EXCERPT: Let M be a compact Riemannian manifold, let r be sufficiently large, and let $m\geq 2$ be an integer. An m-tuple of C^r volume-preserving diffeomorphisms of M is called...

conjecture | open | MathDB 页面 | 原文线索

V2-039658 | MATHDB-335103

Conjecture on Heisenberg cosets of regular reductions of W-algebras

摘录 / EXCERPT: Let n,m be positive integers and let p satisfy ... Let $\mathcal{W}_{n,m}$ be the family of W-algebras defined by quantum Hamiltonian reduction, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039659 | MATHDB-335264

Existence conjecture for stationary measures of the nonlinear process

摘录 / EXCERPT: Let $\delta\geq0$, and let λ be the interaction strength. Consider the nonlinear process X defined by the rates in the source. Stationary-measure existence conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-039660 | MATHDB-335335

Palis's three-dimensional dichotomy conjecture for vector fields

摘录 / EXCERPT: Let $r\geq 1$ and let M be a three-dimensional manifold. A vector field belongs to ${\cal X}^r(M)$, the space of C^r vector fields on M, and a homoclinic tangency is a nontra...

conjecture | open | MathDB 页面 | 原文线索

V2-039661 | MATHDB-335336

Three-dimensional density of singular hyperbolicity

摘录 / EXCERPT: Let M be a three-dimensional manifold, let ${\cal X}^1(M)$ denote the space of C^1 vector fields on M, and call a vector field singular hyperbolic when its chain-recurrent se...

conjecture | open | MathDB 页面 | 原文线索

V2-039662 | MATHDB-335356

Boundedness conjecture for simply connected period-two shell components

摘录 / EXCERPT: Let Λ be a dynamically natural slice for a family of meromorphic maps $\{f_\lambda,\ \lambda \in \Lambda\}$ in $\mathcal{M}_\infty$ in the strict sense. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-039663 | MATHDB-335419

Smale's classification conjecture for Anosov diffeomorphisms

摘录 / EXCERPT: Let M be a closed manifold and let $f:M\to M$ be an Anosov diffeomorphism. A hyperbolic affine infra-nil-automorphism is an affine automorphism of an infra-nil-manifold whose lin...

conjecture | open | MathDB 页面 | 原文线索

V2-039664 | MATHDB-335424

Palis-Smale structural stability conjecture for diffeomorphisms

摘录 / EXCERPT: Let $1\le r\le\infty$ or let $\mathcal C=\mathcal H$, and let $\mathcal C$ be the corresponding category of maps. A $\mathcal C$-diffeomorphism is structurally stable if every...

conjecture | open | MathDB 页面 | 原文线索

V2-039665 | MATHDB-335427

Przytycki's strong-transversality conjecture for coverings

摘录 / EXCERPT: Let f be a covering map. It satisfies axiom A and the strong transversality condition when its non-wandering set is locally maximal, its non-wandering set is the union of a hyper...

conjecture | open | MathDB 页面 | 原文线索

V2-039666 | MATHDB-335428

Berger-Rovella inverse-limit stability conjecture for endomorphisms

摘录 / EXCERPT: Let f be an endomorphism, with inverse limit space $\overleftarrow M_f$ and natural extension $\overleftarrow f$. It is C^1-inverse limit stable if every C^1 perturbation...

conjecture | open | MathDB 页面 | 原文线索

V2-039667 | MATHDB-335446

Expected dimension conjecture for portrait-level dynamical moduli spaces

摘录 / EXCERPT: Let $N\ge1$, and let $\mathcal P=(\mathcal W,\mathcal V,\Phi,\epsilon)$ be a preperiodic portrait. Write $\operatorname{wt}(\mathcal P)$ for its weight, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039668 | MATHDB-335471

Floquet stability conjecture for the reduced periodic orbit

摘录 / EXCERPT: Let $ u$ be the reduced dynamical system ... associated with the planar dynamics in the paper, and let γ_0 be the periodic orbit whose existence is established by Lemma. Floqu...

conjecture | open | MathDB 页面 | 原文线索

V2-039669 | MATHDB-335479

The instability-gap conjecture for generic area-preserving twist diffeomorphisms

摘录 / EXCERPT: Let Δ_0 and Δ_1 denote the instability quantities associated with an area-preserving twist diffeomorphism, and let V be a C^2 potential. Instability-gap conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-039670 | MATHDB-335480

Edrei's conjecture on weak local contractions

摘录 / EXCERPT: Let X be a compact metric space, and let $f:X\to X$ be a weak local contraction, meaning that for every $x\in X$ there exists an $r_x>0$ such that $d(x,y)<r_x$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-039671 | MATHDB-335549

Finiteness conjecture for torsion-free orbit depth and orbit depth

摘录 / EXCERPT: Finiteness conjecture. For any γ, the torsion-free orbit depth k and the orbit depth κ are finite.

conjecture | open | MathDB 页面 | 原文线索

V2-039672 | MATHDB-335550

Abundance of strange attractors in the EBM family

摘录 / EXCERPT: Strange-attractor abundance conjecture. For each natural number n there exists an interval $I_n\subset (1/\sqrt{2},t_n)$ such that Λ_t displays, at least, 2^{n-1} diffe...

conjecture | open | MathDB 页面 | 原文线索

V2-039673 | MATHDB-335656

Spectral instability conjecture for periodic homographic motions of planar strong-force systems

摘录 / EXCERPT: Consider the planar N-body problem with masses $m_k>0$ under a $1/r^\alpha$ potential, where $0<\alpha<2$, and let q be a central configuration. A periodic homographic motion i...

conjecture | open | MathDB 页面 | 原文线索

V2-039674 | MATHDB-335660

Ergodicity conjecture for the super-Apollonian systems

摘录 / EXCERPT: The systems are defined on the complex projective line $\mathbb P^1(\mathbb C)$ with transformations T_A and T_B and invariant measures μ_A and μ_B, respectively. Ergodicit...

conjecture | open | MathDB 页面 | 原文线索

V2-039675 | MATHDB-335661

The generalized Kähler Calabi-Yau existence, uniqueness, and rigidity conjecture

摘录 / EXCERPT: Let (M^{4n}, g, I, J) be a nondegenerate generalized Kähler manifold, and consider its Ω-Hamiltonian deformation class. Generalized Kähler Calabi-Yau conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-039676 | MATHDB-335730

Geometric counting conjecture for homogeneous spaces

摘录 / EXCERPT: Homogeneous-space counting conjecture. There exists an exhausting compact family $\mathcal{B}^Z=(B_R^Z)_{R>0}$ of Z such that the quadruple...

open_problem | open | MathDB 页面 | 原文线索

V2-039677 | MATHDB-335767

The finiteness conjecture for homoclinic classes of Venice masks

摘录 / EXCERPT: Let X be a Venice mask on a compact manifold, and let $M(X)$ denote its maximal invariant set. A homoclinic class is the homoclinic class associated with a periodic orbit. The fi...

conjecture | open | MathDB 页面 | 原文线索

V2-039678 | MATHDB-335776

Lax-type representation conjecture for state-dependent Hamilton-Jacobi equations

摘录 / EXCERPT: Let $H(x,p,t)$ be a Hamiltonian satisfying $H(x,p,t)\in C^2$ and convexity with respect to p, and let (A5) denote the assumption referenced in the source. Let φ be the vi...

conjecture | open | MathDB 页面 | 原文线索

V2-039679 | MATHDB-335786

McMullen's arithmetic chaos conjecture for quadratic fields

摘录 / EXCERPT: Let $\mathfrak C_m$ be the set of continued fractions of height at most m, and for a real quadratic field $\mathbb{Q}(\sqrt d)$ define ... McMullen's arithmetic chaos conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-039680 | MATHDB-335850

The total form domain intersection conjecture for uniform dynamical decoupling

摘录 / EXCERPT: Total form domain intersection conjecture. If dynamical decoupling works uniformly, then

conjecture | open | MathDB 页面 | 原文线索

V2-039681 | MATHDB-336026

Individual ergodic convergence conjecture for mean-bounded positive invertible operators

摘录 / EXCERPT: Mean-bounded ergodic convergence conjecture. The sequence $(A_nx)_{n\geq1}$ converges bilaterally almost uniformly to Px for every $x\in L_p(\mathcal{M})$. If additionally...

conjecture | open | MathDB 页面 | 原文线索

V2-039682 | MATHDB-336027

The Dynamical Pink-Zilber conjecture for split endomorphisms of $(\mathbb{P}^1)^n$

摘录 / EXCERPT: Let $n\in\operatorname{N}$, let $f_1,\dots,f_n\in\mathbb{C}(x)$ be non-exceptional rational functions of degrees >1, and let ... be their coordinatewise action ... For each...

conjecture | open | MathDB 页面 | 原文线索

V2-039683 | MATHDB-336089

Virtual matrix distributions classify nonstandard ergodic filtrations

摘录 / EXCERPT: Let $c4$ and $c4'$ be finitely isomorphic ergodic filtrations. Assume that both filtrations are nonstandard and that both have virtual matrix distributions for the sequences of ass...

conjecture | open | MathDB 页面 | 原文线索

V2-039684 | MATHDB-336090

Positive-entropy adic realizations are nonstandard and have Poulsen limiting simplices

摘录 / EXCERPT: Consider a graph providing an adic realization of an automorphism with positive entropy, together with its associated limiting simplex. Positive-entropy adic graph conjecture. Such...

conjecture | open | MathDB 页面 | 原文线索

V2-039685 | MATHDB-336222

Local Birkhoff conjecture near ellipses

摘录 / EXCERPT: For an integer $q_0\geq 2$, a billiard domain Ω is q_0-rationally integrable if its billiard map admits integrable rational caustics with rotation number p/q for every...

conjecture | open | MathDB 页面 | 原文线索

V2-039686 | MATHDB-336273

HHU Problem 16 on the structure and regularity of accessibility classes

摘录 / EXCERPT: Let M be a manifold and let $f:M\to M$ be a partially hyperbolic diffeomorphism. For $x\in M$, the accessibility class $AC(x)$ is the set of points that can be joined to x by a...

open_problem | open | MathDB 页面 | 原文线索

V2-039687 | MATHDB-336451

Aperiodic multiplicative orthogonality conjecture for homogeneous ergodic averages

摘录 / EXCERPT: Let $\bm \nu$ be an aperiodic bounded multiplicative function and let $l\geq 2$ be a positive integer. Let $T_1,\ldots,T_l$ be commuting measure-preserving transformations acting o...

conjecture | open | MathDB 页面 | 原文线索

V2-039688 | MATHDB-336459

The dynamical RLL conjecture for elliptic L-operators

摘录 / EXCERPT: Let $U_{q,p}([...??]$ be the elliptic quantum algebra with half currents assembled into the L-operator $[...??]$. For two auxiliary tensor components, write $[...??]$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039689 | MATHDB-336482

The Sullivan-type invariant-measure conjecture for nonorientable surfaces

摘录 / EXCERPT: Sullivan-type invariant-measure conjecture. There exists a unique, up to scaling, ergodic $\operatorname{Map}(S)$-invariant Radon measure supported on $\mathcal{ML}^+(S)$, and it i...

conjecture | open | MathDB 页面 | 原文线索

V2-039690 | MATHDB-336561

Generic rigidity conjecture for periodic potentials

摘录 / EXCERPT: Let m denote the number of Fourier modes in the periodic potentials, let n be the ambient dimension, and let V_1,V_2 be smooth periodic functions. The rigidity result referre...

conjecture | open | MathDB 页面 | 原文线索

V2-039691 | MATHDB-336748

Conjecture on Furstenberg systems of multiplicative functions

摘录 / EXCERPT: Let $f\colon\mathbb N\to[-1,1]$ be a multiplicative function. A Furstenberg system for f is a measure-preserving dynamical system generated by the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-039692 | MATHDB-336749

Bernoulli-odometer dichotomy for Furstenberg systems

摘录 / EXCERPT: Let $f\colon\mathbb N\to\{-1,1\}$ be a multiplicative function. Its Furstenberg system is the measure-preserving dynamical system generated by the sequence $(f(n))_{n\in\mathbb N}$...

conjecture | open | MathDB 页面 | 原文线索

V2-039693 | MATHDB-336797

Localization transition for generalized directed polymers

摘录 / EXCERPT: Let the Hamiltonian be convex and let its associated Lagrangian have superlinear growth. A generalized directed polymer is the polymer process associated with the corresponding ran...

conjecture | open | MathDB 页面 | 原文线索

V2-039694 | MATHDB-336800

Generation of Darboux polynomials by linear polynomials in the oligopoly dynamical system

摘录 / EXCERPT: Consider the oligopoly dynamical system with parameters $alpha_i$, $epsilon_i$, and the associated Darboux polynomials, including the linear Darboux polynomials obtained from the d...

conjecture | open | MathDB 页面 | 原文线索

V2-039695 | MATHDB-336809

Birkhoff's sphere conjecture for real-analytic area-preserving diffeomorphisms

摘录 / EXCERPT: Let f be an orientation-preserving, real-analytic, Lebesgue measure-preserving diffeomorphism of the 2-sphere $\mathbb{S}^2$ with only two periodic points, which are necessarily...

conjecture | open | MathDB 页面 | 原文线索

V2-039696 | MATHDB-336832

Finite Hamilton-cover conjecture for two-ended transitive graphs

摘录 / EXCERPT: Let Γ be a two-ended transitive graph. A finite Hamilton cover is a finite collection of Hamiltonian spanning subgraphs whose union covers the relevant graph structure, as i...

conjecture | open | MathDB 页面 | 原文线索

V2-039697 | MATHDB-336952

Arnold-Givental conjecture for real Lagrangian fixed point sets

摘录 / EXCERPT: Arnold-Givental conjecture. Assume that (M,ω,ρ) is a closed real symplectic manifold and ϕ_H is a Hamiltonian diffeomorphism such that L and $\phi_H(L)$ intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-039698 | MATHDB-336953

Weak Arnold-Givental conjecture on Hamiltonian nondisplaceability

摘录 / EXCERPT: Weak Arnold-Givental conjecture. The Lagrangian submanifold L is Hamiltonianly nondisplaceable.

conjecture | open | MathDB 页面 | 原文线索

V2-039699 | MATHDB-337076

Almost-everywhere density conjecture for infinite subsets under interval exchange transformations

摘录 / EXCERPT: Let $I=[0,1]$, let X be an infinite subset of I, and let T be an interval exchange transformation (IET) of I. Almost-everywhere density conjecture. For almost every IET T...

conjecture | open | MathDB 页面 | 原文线索

V2-039700 | MATHDB-337077

Quantitative almost-everywhere density conjecture for infinite subsets under interval exchanges

摘录 / EXCERPT: Let $I=[0,1]$, let X be an infinite subset of I, and let T be an interval exchange transformation (IET) of I. Quantitative density conjecture. For almost every IET T, ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-039701 | MATHDB-337125

The PEPS bulk-boundary gap and short-range conjecture

摘录 / EXCERPT: Let a PEPS have a bulk parent Hamiltonian and an associated one-dimensional boundary Hamiltonian. Bulk-boundary gap conjecture. The bulk parent Hamiltonian of the PEPS is gapped if...

conjecture | open | MathDB 页面 | 原文线索

V2-039702 | MATHDB-337126

Quasi-local Araki conjecture

摘录 / EXCERPT: Quasi-local Araki conjecture. The displayed approximation and norm bounds hold for quasi-local Hamiltonian interactions.

conjecture | open | MathDB 页面 | 原文线索

V2-039703 | MATHDB-337127

PEPS boundary-state locality necessity conjecture

摘录 / EXCERPT: Let Λ be the underlying lattice, let $A\subset\Lambda$ be any rectangular region, and let H_A denote the corresponding parent Hamiltonian restricted to A. A boundary st...

conjecture | open | MathDB 页面 | 原文线索

V2-039704 | MATHDB-337189

Bermond's conjecture on Cartesian products of Hamilton-decomposable graphs

摘录 / EXCERPT: Let G_1 and G_2 be finite graphs, and let $G_1\square G_2$ denote their Cartesian product: vertices (g,h) and (g',h') are adjacent exactly when either $g=g'$ and h is adj...

conjecture | open | MathDB 页面 | 原文线索

V2-039705 | MATHDB-337236

No-self-intersection conjecture for the ad hoc periodgon

摘录 / EXCERPT: Let the ad hoc periodgon be the polygon formed from the periodgon edges in the specified clockwise order, for parameters in the interior of the slit disk. The ad hoc periodgon no-s...

conjecture | open | MathDB 页面 | 原文线索

V2-039706 | MATHDB-337237

Conjectural description of the non-convexity region of the ad hoc periodgon

摘录 / EXCERPT: Let the ad hoc periodgon be the polygon constructed from the eigenvalue-period edges of the vector field family. The non-convexity-region conjecture states that the only region in...

conjecture | open | MathDB 页面 | 原文线索

V2-039707 | MATHDB-337291

Competitive modes conditions for chaos in nonlinear dynamical systems

摘录 / EXCERPT: Consider a nonlinear autonomous system ... whose second-order form is ... so that the functions g_i act as oscillator frequencies and the functions h_i are the remaining terms. Com...

conjecture | open | MathDB 页面 | 原文线索

V2-039708 | MATHDB-337344

Conjecture on the number of ergodic equilibrium states for singular value potentials

摘录 / EXCERPT: Let $(A_1,\ldots,A_N)\in GL_d(\mathbb{R})$ and let $s\in(0,d)\smallsetminus\mathbb{Z}$. For a tuple of matrices, let φ^s denote Falconer's singular value function, and call...

conjecture | open | MathDB 页面 | 原文线索

V2-039709 | MATHDB-337416

Dual degenerate insertion moment and beta-factorization conjecture

摘录 / EXCERPT: Dual degenerate insertion conjecture. For every real $p<\frac{4}{\gamma^2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-039710 | MATHDB-337475

Realisation of formal invariant curves for algebraically isolated vector fields

摘录 / EXCERPT: Let X be a smooth germ of vector field at the origin of $\mathbb{R}^{n}$ with an algebraically isolated singularity, and suppose it has a formal invariant curve...

conjecture | open | MathDB 页面 | 原文线索

V2-039711 | MATHDB-337476

Smooth realisation of formal subgroups of formal diffeomorphisms

摘录 / EXCERPT: Let $\hat{G}\subset\widehat{\operatorname{Diff}}(\mathbb{R}^{n}_{0})$ be a subgroup of formal diffeomorphisms. Realisation conjecture. There exists a smooth realisation...

conjecture | open | MathDB 页面 | 原文线索

V2-039712 | MATHDB-337477

Smooth realisation of formal integrable one-forms

摘录 / EXCERPT: Let $\hat{\omega}\in\hat{\Omega}_{n}^{1}$ be a formal integrable one-form. Realisation conjecture. There exists an integrable one-form $\omega\in\Omega_{n}^{1}$ such that ... Logarit...

conjecture | open | MathDB 页面 | 原文线索

V2-039713 | MATHDB-337553

Birkhoff's strong conjecture on integrable planar billiards

摘录 / EXCERPT: Let a billiard table be a strictly convex planar domain, and call its billiard dynamics integrable when its phase space admits a one-dimensional foliation by invariant curves. Birk...

conjecture | open | MathDB 页面 | 原文线索

V2-039714 | MATHDB-337622

The equivariant symplectic Fano conjecture in dimension six

摘录 / EXCERPT: Equivariant symplectic Fano conjecture. M is diffeomorphic to a complex projective Fano 3-fold.

conjecture | open | MathDB 页面 | 原文线索

V2-039715 | MATHDB-337644

Restricted QUE conjecture for Hecke-Maass cusp forms

摘录 / EXCERPT: Let u_j be a Hecke-Maass cusp form with Laplace eigenvalue $1/4+T_j^2$, Hecke eigenvalues $\lambda_{u_j}(m)$, parity $u_j(-x+iy)=\delta_j u_j(x+iy)$ with $\delta_j\in\{-1,1\}$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-039716 | MATHDB-337790

Large deviations for sectional-hyperbolic attracting sets

摘录 / EXCERPT: Consider a C^2 flow with a general sectional-hyperbolic attracting set and a neighborhood of that attracting set. Let Lebesgue measure be compared with the physical measures, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039717 | MATHDB-337791

Persistence of singular-hyperbolic large deviations for C^1 flows

摘录 / EXCERPT: Consider the statements of Theorem and the sectional-hyperbolic attracting-set conjecture for C^2 flows. C^1 regularity conjecture. The statements of Theorem and Conjecture sho...

conjecture | open | MathDB 页面 | 原文线索

V2-039718 | MATHDB-337878

Typically periodic optimization conjecture

摘录 / EXCERPT: Typically periodic optimization (TPO) conjecture. The set V_{Per} contains an open dense subset of V.

conjecture | open | MathDB 页面 | 原文线索

V2-039719 | MATHDB-337895

Persistence of Cat's eyes for all rotation parameters

摘录 / EXCERPT: Persistence-of-Cat's-eyes conjecture. The conclusion of Theorem holds for all $s>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-039720 | MATHDB-337956

Gevrey regularity conjecture for the parametrization of invariant circles

摘录 / EXCERPT: Let u_{ε} denote the parametrization of the quasi-periodic invariant circle for the dissipative standard map, viewed as a function of the complex parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-039721 | MATHDB-337971

Univalence or typical reality of the multiplier function

摘录 / EXCERPT: Univalence conjecture. For any N and T there is a choice of γ such that the function $\Phi(z)$ is univalent or typically real in $\mathbb D$. The largest value of...

conjecture | open | MathDB 页面 | 原文线索

V2-039722 | MATHDB-337972

Optimality of the constructed polynomials for $J_N^{(T)}$

摘录 / EXCERPT: Optimality conjecture. For any N and T the polynomials $q(z)$ with $\sigma=\tau=1$ give the solutions to the optimization problem $J_N^{(T)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-039723 | MATHDB-337978

Dyatlov-Zworski trace formula conjecture for smooth Anosov flows

摘录 / EXCERPT: Let M be a compact manifold and let $\varphi_t:M\to M$ be a $\mathcal{C}^\infty$ Anosov flow. Write $P=-iV$, where V is the generator of φ_t. For each periodic orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-039724 | MATHDB-338013

Affine indicial roots conjecture for admissible triples

摘录 / EXCERPT: Affine indicial roots conjecture. The roots of an admissible triple are always affine, and, when $\operatorname{Re}s=0$, no root lies on the imaginary axis.

conjecture | open | MathDB 页面 | 原文线索

V2-039725 | MATHDB-338020

Almost-sure convergence of delayed random walk on ergodic unimodular random graphs

摘录 / EXCERPT: Delayed random-walk convergence conjecture. The distribution of (G,X_t) converges to the initial unimodular measure almost surely.

open_problem | open | MathDB 页面 | 原文线索

V2-039726 | MATHDB-338033

Triviality conjecture for jointly T_a- and analytic f-invariant sets

摘录 / EXCERPT: Let T_a be the fixed transformation, and let f be a real-analytic function on either the real line or the circle: ... Assume that f is not affine. A set is jointly T_a- and...

conjecture | open | MathDB 页面 | 原文线索

V2-039727 | MATHDB-338257

The cluster-chart conjecture for mirror Landau-Ginzburg models

摘录 / EXCERPT: Cluster-chart conjecture. The set of cluster charts $(\mathbb{C}^*)^n\subset\check X$ is in bijection with the set of monotone Lagrangian tori in X up to Hamiltonian isotopy. If...

conjecture | open | MathDB 页面 | 原文线索

V2-039728 | MATHDB-338342

The Texan conjecture on dimension spectra of conformal iterated function systems

摘录 / EXCERPT: Texan conjecture. For every compact and perfect subset K of $[0,+\infty)$, there exists a conformal iterated function system $\mathcal{S}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-039729 | MATHDB-338445

Quantized conductance conjecture for the almost Mathieu operator

摘录 / EXCERPT: Let H_0 be the Hamiltonian whose potential on the sites n is the quasi-periodic function $v(n)=U\cos(2\pi\omega n)$, where $-2<U<2$ and ω is irrational. Assume that the...

conjecture | open | MathDB 页面 | 原文线索

V2-039730 | MATHDB-338451

Bonatti's finiteness conjecture for chain-recurrence classes

摘录 / EXCERPT: Let a diffeomorphism be generic and far from homoclinic tangencies, and let chain-recurrence classes be the equivalence classes defined by mutual chain recurrence. Bonatti's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-039731 | MATHDB-338543

Brunella's conjecture on leaf accumulation for foliations with ample normal bundle

摘录 / EXCERPT: Brunella's conjecture. Every leaf of ${\mathscr{F}}$ accumulates to the singular set ${\rm Sing}({\mathscr{F}})$.

conjecture | open | MathDB 页面 | 原文线索

V2-039732 | MATHDB-338544

Generalized Brunella's conjecture for foliations with ample determinant of the normal bundle

摘录 / EXCERPT: Generalized Brunella's conjecture. If $n\geq 2r+1$, then every leaf of ${\mathscr{F}}$ accumulates to the singular set ${\rm Sing}({\mathscr{F}})$.

conjecture | open | MathDB 页面 | 原文线索

V2-039733 | MATHDB-338613

Variance bound for energy differences between incongruent states

摘录 / EXCERPT: Let $H_{\Lambda,J}$ be the finite-volume Hamiltonian in a box Λ with $|\Lambda|=L^d$, and let σ^1 and σ^2 be chosen as in Theorem. Write Var...

conjecture | open | MathDB 页面 | 原文线索

V2-039734 | MATHDB-338712

Conjecture on Hamiltonian simple subalgebras and special Hamiltonian algebras

摘录 / EXCERPT: Let $L=H(2n;\underline{1})^{(1)}$ be a Lie algebra of Hamiltonian type, with $n\geq3$, over a field of characteristic 2. The Hamiltonian-special Hamiltonian conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-039735 | MATHDB-338765

Stochastic homogenization conjecture for the Hamiltonian H_{ℓ}

摘录 / EXCERPT: Stochastic homogenization conjecture. The Hamiltonian $H_{\ell}(p,x,\omega)$ is stochastically homogenizable.

conjecture | open | MathDB 页面 | 原文线索

V2-039736 | MATHDB-338814

Minimality conjecture for interval exchange transformations with flips

摘录 / EXCERPT: Let $\mathbb{S}^1$ be the circle, let n be any positive integer, and let $F\in\operatorname{CET}_n$ be an interval exchange transformation with flips whose flip parameter is...

conjecture | open | MathDB 页面 | 原文线索

V2-039737 | MATHDB-338897

Odd-even polynomial-integral conjecture for natural systems on the torus

摘录 / EXCERPT: Let T^2 be the two-dimensional torus, and let a natural Hamiltonian system on it have a nonconstant potential. A linear, quadratic, or higher-degree polynomial integral is an int...

conjecture | open | MathDB 页面 | 原文线索

V2-039738 | MATHDB-338898

Birkhoff's conjecture for planar convex billiards

摘录 / EXCERPT: Let $b2b2\text{?}$ be a billiard table $b2b2\text{?}b2b2$ in $b2b2\text{?}b2b2$ with boundary curve $b2b2\text{?}b2b2$. A caustic is an invariant curve tangent to the billiard traj...

conjecture | open | MathDB 页面 | 原文线索

V2-039739 | MATHDB-338899

Tabachnikov's projective-involution conjecture for convex foliations

摘录 / EXCERPT: Let $b2b2\text{?}b2b2\text{?}b2b2$ be a smooth closed strictly convex curve in $b2b2\text{?}b2b2$, and let $b2b2\text{?}b2b2$ be a smooth foliation by convex closed curves in an ex...

conjecture | open | MathDB 页面 | 原文线索

V2-039740 | MATHDB-338900

Kaloshin's rationally integrable billiard conjecture

摘录 / EXCERPT: Let $q_0\geq 2$ be an integer. A billiard table is q_0-rationally integrable if it possesses p/q-rational caustics for every $0<p/q<1/q_0$. Kaloshin's conjecture. Every q_0-r...

conjecture | open | MathDB 页面 | 原文线索

V2-039741 | MATHDB-338901

Bialy's rotational-invariant-circle conjecture for plane billiards

摘录 / EXCERPT: A rotational invariant circle of a billiard map is an invariant circle consisting of periodic points. Bialy's conjecture. If a plane billiard has a family of rotational invariant c...

conjecture | open | MathDB 页面 | 原文线索

V2-039742 | MATHDB-338920

Generic finite-parameter families of smooth vector fields have finiteness properties

摘录 / EXCERPT: Generic-family conjecture. Smooth vector fields met in a generic finite-parameter family belong to $Vect^*\,S^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-039743 | MATHDB-338945

Quantum Shannon-McMillan-Breiman conjecture for quantum Martin-Löf random states

摘录 / EXCERPT: Quantum ShannonfMcMillanfBreiman conjecture. If μ is an ergodic computable state on $M_\e1$ and ρ is quantum Martin-Lf random with respect to μ, then

conjecture | open | MathDB 页面 | 原文线索

V2-039744 | MATHDB-338993

Polynomiality and monodromy invariance of the Novikov-algebra coordinates

摘录 / EXCERPT: Polynomiality and monodromy-invariance conjecture. Under these hypotheses, every function $u^i(\boldsymbol{v})$ belongs to the invariant polynomial ring...

conjecture | open | MathDB 页面 | 原文线索

V2-039745 | MATHDB-339017

Joint Furstenberg-system structure conjecture for bounded multiplicative functions

摘录 / EXCERPT: Joint Furstenberg-system structure conjecture. The functions $f_1,\ldots,f_\ell$ have a unique joint Furstenberg system, which is ergodic and isomorphic to the direct product of a...

conjecture | open | MathDB 页面 | 原文线索

V2-039746 | MATHDB-339018

Ergodic-component structure conjecture for joint Furstenberg systems

摘录 / EXCERPT: Ergodic-component structure conjecture. Every joint Furstenberg system of $f_1,\ldots,f_\ell$ has ergodic components isomorphic to direct products of procyclic systems and Bernoull...

conjecture | open | MathDB 页面 | 原文线索

V2-039747 | MATHDB-339066

Exponential memory decay for deterministic walks in random environment

摘录 / EXCERPT: Let $z=(z(n))_{n\geq 0}$ be a path in the admissible path space, let $z^k=z(1),\ldots,z(k-1)$ denote its past, and let $\widehat z_m(k)=z(m+k)-z(m)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-039748 | MATHDB-339083

The ergodicity conjecture for the 3x+1 conjugacy map

摘录 / EXCERPT: Let Q be the 3x+1 conjugacy map on $\mathbb Z_2$, and let the ergodic domain of Q be the union of the invariant sets on which Q is ergodic. Ergodicity Conjecture. The ergodic...

conjecture | open | MathDB 页面 | 原文线索

V2-039749 | MATHDB-339084

The odd cycles conjecture for the 3x+1 conjugacy map

摘录 / EXCERPT: Let Q be the 3x+1 conjugacy map on $\mathbb Z_2$, and call a periodic orbit an odd cycle when all its elements are odd. Odd Cycles Conjecture. The function Q has exactly two od...

conjecture | open | MathDB 页面 | 原文线索

V2-039750 | MATHDB-339091

Existence of continuous non-trivial embeddings of minimal interval exchange transformations

摘录 / EXCERPT: Existence conjecture. For any $d\geq 3$, there is a minimal d-IET (I,f) which admits a continuous, non-trivial embedding into a d-PWI.

conjecture | open | MathDB 页面 | 原文线索

V2-039751 | MATHDB-339122

Elliptic Ruijsenaars-Schneider and quantum K-theory spectral correspondence

摘录 / EXCERPT: Let $\mathscr E_r^{(nN)}$, for $r=1,\dots,nN$, denote the eigenvalues of the elliptic Ruijsenaars-Schneider Hamiltonians, and let $\mathscr U$ be the universal bundle over the modu...

conjecture | open | MathDB 页面 | 原文线索

V2-039752 | MATHDB-339131

The Lorenz renormalization conjecture on post-critical attractors

摘录 / EXCERPT: Let f be a Lorenz map, and let $w=(w_0,w_1)$ be a renormalization type. A map is infinitely w-renormalizable when every successive renormalization is w-renormalizable. Denote...

conjecture | open | MathDB 页面 | 原文线索

V2-039753 | MATHDB-339132

The Lorenz renormalization conjecture on rigidity classes

摘录 / EXCERPT: Let $\mathcal{T}_w$ be the set of infinitely w-renormalizable Lorenz maps, and let $\mathcal{L}$ denote the space of Lorenz maps. The Lorenz rigidity conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-039754 | MATHDB-339133

The Lorenz renormalization conjecture on rigidity and renormalization alternatives

摘录 / EXCERPT: Let $\mathcal{T}_w$ be the set of infinitely w-renormalizable Lorenz maps. A rigidity class of $f\in\mathcal{T}_w$ is the set of maps in $\mathcal{T}_w$ smoothly conjugate to f...

conjecture | open | MathDB 页面 | 原文线索

V2-039755 | MATHDB-339163

The interval conjecture for consecutive constrictions of phase-lock areas

摘录 / EXCERPT: The interval conjecture. For every $r\in\mathbb Z$ and $j\in\mathbb N$,

conjecture | open | MathDB 页面 | 原文线索

V2-039756 | MATHDB-339164

The right-of-axis conjecture for phase-lock areas

摘录 / EXCERPT: The right-of-axis conjecture. Each phase-lock area L_r lies to the right of Λ_{r-1}, namely

conjecture | open | MathDB 页面 | 原文线索

V2-039757 | MATHDB-339165

The uniqueness conjecture for positive simple intersections

摘录 / EXCERPT: The uniqueness conjecture. For every $r\in\mathbb Z\setminus\{0\}$, the simple intersection with positive ordinate is unique.

conjecture | open | MathDB 页面 | 原文线索

V2-039758 | MATHDB-339166

The positivity conjecture for constrictions

摘录 / EXCERPT: The positivity conjecture. Every constriction is positive.

conjecture | open | MathDB 页面 | 原文线索

V2-039759 | MATHDB-339167

The vertical-ray conjecture for phase-lock areas

摘录 / EXCERPT: The vertical-ray conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-039760 | MATHDB-339168

The triangular-term sign conjecture for constrictions

摘录 / EXCERPT: The triangular-term sign conjecture. The ratio of these triangular terms is negative, in the source's notation meaning

conjecture | open | MathDB 页面 | 原文线索

V2-039761 | MATHDB-339169

The no-queer-constrictions conjecture

摘录 / EXCERPT: The no-queer-constrictions conjecture. There are no queer constrictions.

conjecture | open | MathDB 页面 | 原文线索

V2-039762 | MATHDB-339343

Evenness conjecture for the Hurwitz continued-fraction invariant density

摘录 / EXCERPT: Let h be the invariant density, viewed as a function of $x+y\mathbb{i}$ with real coordinates x,y. Consider the points $\pm.5\pm.5\mathbb{i}$, $\pm.5$, and $\pm.5\mathbb{i}$...

conjecture | open | MathDB 页面 | 原文线索

V2-039763 | MATHDB-339344

Vanishing odd-derivative conjecture for the $V_{3,1}$ Hurwitz density

摘录 / EXCERPT: Let $h_{m,n}^{3,1}(x,y)$ denote the coefficients or derivatives used for the invariant density associated with the region $V_{3,1}$, evaluated at the center (x,y). Vanishing odd-...

conjecture | open | MathDB 页面 | 原文线索

V2-039764 | MATHDB-339451

Mean-dimension embedding conjecture for real flows

摘录 / EXCERPT: Mean-dimension embedding conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-039765 | MATHDB-339490

Two-sided total variation estimate for centered Gamma approximation in the second Wiener chaos

摘录 / EXCERPT: Let $\nu>0$, and let $F=I_2(f)$ belong to the second Wiener chaos with ... Let $G(\nu)\sim\operatorname{CenteredGamma}(\nu)$. Two-sided total variation conjecture. There exist consta...

conjecture | open | MathDB 页面 | 原文线索

V2-039766 | MATHDB-339539

Asymptotic-average replacement conjecture for negative growth

摘录 / EXCERPT: Asymptotic-average replacement conjecture. In the discrete setting, the conclusions of Theorems 1 and 2 remain valid when the assumptions $A^-(x)<0$ or $\tilde A^-(x)<0$ are replac...

conjecture | open | MathDB 页面 | 原文线索

V2-039767 | MATHDB-339540

Derivative-free validity conjecture for the sink theorem

摘录 / EXCERPT: Derivative-free validity conjecture. The result of Theorem 1 remains valid without any extra assumptions on the derivative.

conjecture | open | MathDB 页面 | 原文线索

V2-039768 | MATHDB-339541

Bowen-type saddle-connection conjecture for vector fields

摘录 / EXCERPT: Bowen-type saddle-connection conjecture. Then G exhibits saddle connections similar to the saddle connections in Example 1.

conjecture | open | MathDB 页面 | 原文线索

V2-039769 | MATHDB-339542

Open-and-dense sink/source conjecture for flows and semiflows

摘录 / EXCERPT: Open-and-dense sink/source conjecture. There exists an open and dense family of continuous flows or smooth semiflows on manifolds for which sectional asymptotic expansion or contra...

conjecture | open | MathDB 页面 | 原文线索

V2-039770 | MATHDB-339618

Palis's conjecture for flows with singularities

摘录 / EXCERPT: Let a flow, equivalently a vector field, be given on a manifold. A flow is hyperbolic if its chain recurrent dynamics are hyperbolic; a singular cycle is a cycle involving singular...

conjecture | open | MathDB 页面 | 原文线索

V2-039771 | MATHDB-339619

Strong Palis conjecture for singular flows

摘录 / EXCERPT: Let a flow, equivalently a vector field, be given on a manifold. A singular hyperbolic flow is one whose invariant sets admit the corresponding singular-hyperbolic structure; a hom...

conjecture | open | MathDB 页面 | 原文线索

V2-039772 | MATHDB-339620

Singular hyperbolicity and finiteness of chain recurrence classes for star flows

摘录 / EXCERPT: Let X be a vector field and let its chain recurrent set be the set of points returning to themselves by arbitrarily small pseudo-orbits. A vector field is a star vector field if...

conjecture | open | MathDB 页面 | 原文线索

V2-039773 | MATHDB-339653

Strong terminating conjecture for the mean-median map

摘录 / EXCERPT: For an initial condition $x\in[0,1]$, let (x_n) be the orbit generated by the mean-median map from the initial set $[0,x,1]$. The strong terminating conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-039774 | MATHDB-339654

Stabilisation conjecture for normal form orbits

摘录 / EXCERPT: For every odd integer $t\geqslant 5$, let (y_n) be the normal form orbit of order t. The stabilisation conjecture for normal form orbits. The orbit (y_n) stabilises. Normal f...

conjecture | open | MathDB 页面 | 原文线索

V2-039775 | MATHDB-339655

Upper-bound conjecture for normal form limit values

摘录 / EXCERPT: For every odd integer $t\geqslant 9$, let N_t be the index ... where ... and let m_t be the limit of the normal form orbit and y_{N_t} its term at index N_t. The upper-bound co...

conjecture | open | MathDB 页面 | 原文线索

V2-039776 | MATHDB-339681

Exhaustion of coupling operators by decomposed generator products

摘录 / EXCERPT: For $i \in \{1,\ldots,n\}$, let $(\mathcal{L}_{i},\mathcal{D}(\mathcal{L}_i))$ be generators of ergodic Feller semigroups. Let A_k^i, B^i, and $\mathcal{J}$ arise from decompos...

conjecture | open | MathDB 页面 | 原文线索

V2-039777 | MATHDB-339884

Arnold diffusion conjecture for typical perturbations

摘录 / EXCERPT: Consider a near-integrable Hamiltonian system with slow action variables y and perturbations of the form ... where $x\in\mathbb{T}^n$, $y\in\mathbb{R}^n$, $t\in\mathbb{T}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039778 | MATHDB-339885

Arnold diffusion conjecture for a priori unstable systems

摘录 / EXCERPT: Consider the near-integrable, time-periodic a priori unstable Hamiltonian ... where $y\in\overline{\mathcal D}\subset\mathbb{R}^n$, $x\in\mathbb{T}^n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-039779 | MATHDB-339974

The converse conjecture for deleting one leak in linear compartmental models

摘录 / EXCERPT: Let $\widetilde{\mathcal M}$ be a linear compartmental model that is strongly connected and has at least one input and exactly one leak. Let $\mathcal M$ be the model obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-039780 | MATHDB-339983

Gaussian marginal conjecture for the pure-noise Gibbs measure

摘录 / EXCERPT: Let $\mathcal{S}^N$ be the sphere, let H_0 be the pure-noise Hamiltonian, and for $\beta>0$ let $\pi_{0,\beta}$ be the Gibbs measure with density proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-039781 | MATHDB-340130

Continuous-measure recurrence conjecture for recurrence sets

摘录 / EXCERPT: Let $S\subseteq\mathbb Z\setminus\{0\}$ be a set of recurrence. A Borel probability measure σ on $\mathbb T$ is continuous if $\sigma(\{a\})=0$ for every $a\in\mathbb T$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-039782 | MATHDB-340171

The uniqueness conjecture for invariants of four-parameter unfoldings

摘录 / EXCERPT: Let $v\in {\mathbf{TH}}_{N}$ be a generic vector field, and let v_{0} denote the reference member of a generic 4-parameter unfolding. Let $\Lambda _{e}(v_{0})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039783 | MATHDB-340247

The full-dimensionality conjecture for multi-rotation orbits

摘录 / EXCERPT: Let $k\geq 1$ be an integer and let $\Lambda=\{\alpha_1,\dots,\alpha_k\}$ be a set of k rationally independent numbers. Let $\omega\in\Lambda^\mathbb{N}$ be an arbitrary sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-039784 | MATHDB-340300

Stein universality lemma for second-Wiener-chaos targets

摘录 / EXCERPT: Stein universality lemma. For every $h\in\mathcal{H}$, this equation admits a bounded solution f_h that is d times differentiable, with

conjecture | open | MathDB 页面 | 原文线索

V2-039785 | MATHDB-340301

Gamma-to-cumulant conjecture for iterated Malliavin Gamma operators

摘录 / EXCERPT: Let F_∞ be the target random variable in the second Wiener chaos of the form referenced in the source, and let $F=I_q(f)$ be a chaotic random variable in the qth Wiener ch...

conjecture | open | MathDB 页面 | 原文线索

V2-039786 | MATHDB-340359

Generic positive exponent conjecture for random products in higher dimensions

摘录 / EXCERPT: Higher-dimensional random-product conjecture. There exist a C^1 dense and C^r open set of volume-preserving diffeomorphisms such that the random product has at least one positi...

conjecture | open | MathDB 页面 | 原文线索

V2-039787 | MATHDB-340360

The explicit subaction conjecture for the Cantor-distance potential

摘录 / EXCERPT: Let K be the set used in the source, let $d(x,K)$ denote the distance from x to K, and set $A(x)=-d(x,K)$. Let $T(x)=2x\pmod 1$ on $[0,1]$, with inverse branch...

conjecture | open | MathDB 页面 | 原文线索

V2-039788 | MATHDB-340361

The explicit subaction conjecture associated with the period-two maximizing orbit

摘录 / EXCERPT: Let A and $T(x)=2x\pmod 1$ be as in the preceding construction, with inverse branches $\tau_1(x)=x/2$ and $\tau_2(x)=(x+1)/2$. Define $\eta=\tau_2\circ\tau_1$ and ... A subaction i...

conjecture | open | MathDB 页面 | 原文线索

V2-039789 | MATHDB-340455

Higher-degree Hamiltonian conjecture for affine Gaudin models

摘录 / EXCERPT: Let E_+ be the set of positive exponents of the underlying Lie algebra, let $\varphi(z)$ be the twist function, and let $h^\vee$ be the dual Coxeter number. The quadratic Hamilto...

conjecture | open | MathDB 页面 | 原文线索

V2-039790 | MATHDB-340456

Oper formula for higher-degree affine Gaudin Hamiltonian eigenvalues

摘录 / EXCERPT: Let d be a positive exponent with $d>1$, let γ be a Pochhammer contour, and let $\Psi^{\mathrm{on}}_{\bm c}(\bm w)$ be an on-shell Bethe state in $H_{\bm\lambda}$. Associa...

conjecture | open | MathDB 页面 | 原文线索

V2-039791 | MATHDB-340466

Pointwise convergence of double Birkhoff averages for \b7R^m-systems

摘录 / EXCERPT: Double Birkhoff averages conjecture. Every $\mathbb{R}^{m}$-system is good for double Birkhoff averages: for every $f_1,f_2\in L^{\infty}(\mu)$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-039792 | MATHDB-340486

The escaping-trajectory conjecture for Rauzy-gasket triangle tilings

摘录 / EXCERPT: Rauzy-gasket escaping-trajectory conjecture. Every trajectory through the circumcenter is escaping, and trajectories passing increasingly close to the circumcenter, with...

conjecture | open | MathDB 页面 | 原文线索

V2-039793 | MATHDB-340487

The $4n+2$ period conjecture for periodic trajectories on triangle tilings

摘录 / EXCERPT: $4n+2$ conjecture. Every periodic trajectory on a triangle tiling has period $4n+2$ for some positive integer n.

conjecture | open | MathDB 页面 | 原文线索

V2-039794 | MATHDB-340497

Conjecture on dominant harmonics after a Hamiltonian-Hopf bifurcation

摘录 / EXCERPT: Dominant-harmonic splitting conjecture. For $\nu\in(\nu_0,\nu_1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-039795 | MATHDB-340569

Arnold's Lyapunov instability conjecture for elliptic equilibria

摘录 / EXCERPT: Consider an analytic Hamiltonian system with an elliptic equilibrium point, and let n denote the number of degrees of freedom. Say that the quadratic part of the Hamiltonian func...

conjecture | open | MathDB 页面 | 原文线索

V2-039796 | MATHDB-340570

Herman's accumulation conjecture for elliptic equilibria and KAM tori

摘录 / EXCERPT: An elliptic equilibrium is an elliptic fixed point of an analytic Hamiltonian system, and a KAM torus is an invariant quasi-periodic torus of such a system satisfying the KAM condi...

conjecture | open | MathDB 页面 | 原文线索

V2-039797 | MATHDB-340691

Matsumoto's minimality conjecture for horocycle flows on Riemannian foliations

摘录 / EXCERPT: Let $\mathcal{F}$ be a minimal Riemannian foliation by hyperbolic surfaces, and let h_s^+ denote its horocycle flow. Matsumoto's conjecture. The horocycle flow h_s^+ is minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-039798 | MATHDB-340705

Alternate normal form conjecture for generic unfoldings of saddle-nodes

摘录 / EXCERPT: Alternate normal form conjecture. Any such unfolding is orbitally conjugate to an unfolding of the form

conjecture | open | MathDB 页面 | 原文线索

V2-039799 | MATHDB-340852

Asymmetric periodic trajectory period conjecture for the pentagon

摘录 / EXCERPT: Asymmetric-period conjecture. Every even number other than 2, 12, 14, and 18 occurs as the period of an asymmetric periodic billiard trajectory on the pentagon.

conjecture | open | MathDB 页面 | 原文线索

V2-039800 | MATHDB-340853

Aperiodic bounce-word complexity conjecture for the regular pentagon

摘录 / EXCERPT: Aperiodic-word complexity conjecture. For n sufficiently large, the complexity of each individual aperiodic word is

conjecture | open | MathDB 页面 | 原文线索

V2-039801 | MATHDB-340854

Cubic asymptotic complexity conjecture for the regular pentagon billiard language

摘录 / EXCERPT: Pentagon-language complexity conjecture. The normalized complexity has the limit

conjecture | open | MathDB 页面 | 原文线索

V2-039802 | MATHDB-340875

Shifted Chvátal-resilience conjecture for Hamilton cycles in random graphs

摘录 / EXCERPT: Shifted Chvátal-resilience conjecture. For every $\varepsilon>0$, there exists $C>0$ such that, for $p\geq C\log n/n$, a.a.s. the random graph $G_{n,p}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-039803 | MATHDB-340928

The full-dimensional Newhouse lamination conjecture

摘录 / EXCERPT: Newhouse lamination conjecture. Every d-dimensional unfolding of a map with a strong homoclinic tangency contains a codimension 2 Newhouse lamination with Hausdorff dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-039804 | MATHDB-340957

Scaling-limit conjecture for dynamical supercritical oriented percolation

摘录 / EXCERPT: Let Γ_s^ϵ be the set of right-most infinite open paths at dynamical time s. Let $S_{\alpha(p),\sigma(p),\epsilon}$ denote the corresponding space-time rescaling, an...

conjecture | open | MathDB 页面 | 原文线索

V2-039805 | MATHDB-340966

Sharpness of the entropy threshold for infinitely many ergodic measures of maximal entropy

摘录 / EXCERPT: Let f be a C^r diffeomorphism of a closed surface, where $1<r<\infty$, and let $\lambda_{\min}(f)$ denote the minimum expansion rate appearing in the entropy estimates. A measu...

conjecture | open | MathDB 页面 | 原文线索

V2-039806 | MATHDB-341068

Ergodicity conjecture for degraded oversampled adaptive sensing distortions

摘录 / EXCERPT: Ergodicity conjecture. As $N\uparrow\infty$, $\{\hat{d}_n(\mathbf{Y}_1):n\in[N]\}$ is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-039807 | MATHDB-341171

The Novikov conjecture on exceptional directions

摘录 / EXCERPT: Let $F\to oalign{\vskip\baselineskip} $ be a generic smooth function on the three-torus, and let ${\cal E}_F$ denote the exceptional set of directions associated with its quasiperi...

conjecture | open | MathDB 页面 | 原文线索

V2-039808 | MATHDB-341176

Dynnikov's connectedness conjecture for chaotic sections

摘录 / EXCERPT: Let M_g^2 be a surface in ${\mathbb T}^3$, let B be a direction, and let ${\cal F}_B(M_g^2)$ denote the corresponding foliation or section system. Call ${\cal F}_B(M_g^2)$ chao...

conjecture | open | MathDB 页面 | 原文线索

V2-039809 | MATHDB-341181

Furstenberg's intersection problem for invariant Cantor sets

摘录 / EXCERPT: Furstenberg intersection conjecture. Every point of $A_2\cap A_3$ is rational. This is posed as a strengthening of known dimension and sparsity results for intersections of...

conjecture | open | MathDB 页面 | 原文线索

V2-039810 | MATHDB-341182

The target-hitting estimate for generic isometries

摘录 / EXCERPT: Let X be a compact metric space with finite upper box dimension, let $G\subset\operatorname{Iso}(X)$ be a compact subgroup with Haar measure μ_G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039811 | MATHDB-341183

The long-arithmetic-progression conjecture for digit positions

摘录 / EXCERPT: For $\beta>1$, let $A_k(\beta)$ denote the set of positions of digit 1 in the binary expansion of β^k. Long-arithmetic-progression conjecture. If $\log\beta/\log2$ is irrat...

conjecture | open | MathDB 页面 | 原文线索

V2-039812 | MATHDB-341186

Conical singularities of momentum-section zero loci for proper Hamiltonian groupoids

摘录 / EXCERPT: Conical-singularity conjecture. The zero locus of the momentum section has only conical singularities.

conjecture | open | MathDB 页面 | 原文线索

V2-039813 | MATHDB-341199

Full box-dimension conjecture for alpha-beta sequences

摘录 / EXCERPT: Let α and β be real numbers such that $1,\alpha,\beta$ are independent over the field of rational numbers. An $\alpha\beta$-sequence is a sequence $\{x_n\}_{n\geq 1}$...

open_problem | open | MathDB 页面 | 原文线索

V2-039814 | MATHDB-341340

Ergodicity conjecture for geodesic flows on surfaces with nonpositive curvature

摘录 / EXCERPT: Let (M,g) be a smooth, connected and closed surface of genus $d524\geq 2$ with nonpositive curvature. Let SM be its unit tangent bundle, let Λ denote the set of flat ge...

conjecture | open | MathDB 页面 | 原文线索

V2-039815 | MATHDB-341348

Periodgon bifurcation-ray conjecture

摘录 / EXCERPT: Let k be the parameter in the vector-field family, let $s\in(0,1)$, and let θ be the angular parameter. The periodgon records the arrangement of the sides associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-039816 | MATHDB-341452

Kählerness conjecture for six-dimensional monotone Hamiltonian circle manifolds

摘录 / EXCERPT: Kählerness conjecture. Then (M,ω) is S^1-equivariantly symplectomorphic to some Kähler manifold (X,ω_X,J) with some holomorphic Hamiltonian S^1-action.

conjecture | open | MathDB 页面 | 原文线索

V2-039817 | MATHDB-341487

The alpha-limit conjecture for generic real polynomial maps

摘录 / EXCERPT: Alpha-limit conjecture. There is some non-empty open subset $A\subset f(\mathbb{R}\mathrm{P}^2)$ such that $\alpha_{N_f}(x)$ is equal to the set of non-regular points of the bounda...

conjecture | open | MathDB 页面 | 原文线索

V2-039818 | MATHDB-341488

Simple-dynamics conjectures for Newton maps of real polynomial maps

摘录 / EXCERPT: Simple-dynamics conjectures. The following properties hold:

conjecture | open | MathDB 页面 | 原文线索

V2-039819 | MATHDB-341511

Conjecture on atomic limits of outlier measures

摘录 / EXCERPT: Atomic-limit conjecture. There exists a sequence of atomic ergodic measures converging to a measure with a positive atomic part.

conjecture | open | MathDB 页面 | 原文线索

V2-039820 | MATHDB-341611

Diffusive propagation and disorder-scaling conjecture for ergodic Schrödinger potentials

摘录 / EXCERPT: Diffusive propagation and disorder-scaling conjecture. For every bounded potential u and every $\lambda>0$, these constants are positive and finite, with...

conjecture | open | MathDB 页面 | 原文线索

V2-039821 | MATHDB-341645

Full-dimensional parameter set for the CS property

摘录 / EXCERPT: Let $U\subset\mathbb R^k$ be the parameter space of the iterated function systems Φ_u, and let $\mathfrak m$ be a slowly decaying σ-invariant ergodic probability measur...

conjecture | open | MathDB 页面 | 原文线索

V2-039822 | MATHDB-341672

Measured-generic Morse-Smale or asymptotically periodic conjecture for transversely affine foliations

摘录 / EXCERPT: Measured-generic Morse-Smale or asymptotically periodic conjecture. A transversely affine foliation is generically, in the measured sense, Morse-Smale or asymptotically periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-039823 | MATHDB-341681

Chaos conjecture for periodically switched reversible vector fields

摘录 / EXCERPT: Let $X_{\lambda}(x,y)$ be a fixed type of normal form for a one-parameter family of codimension 1 reversible vector fields. Let λ_1 and λ_2 be two distinct real v...

conjecture | open | MathDB 页面 | 原文线索

V2-039824 | MATHDB-341684

Sharp phase transition for Hamilton cycles and near perfect matchings in the KPKVB model

摘录 / EXCERPT: Let $G(n;\alpha,\nu)$ denote the KPKVB random graph, where n is the number of vertices, α controls the radial distribution, and ν controls the angular range. A near pe...

conjecture | open | MathDB 页面 | 原文线索

V2-039825 | MATHDB-341781

The combined parameterization conjecture for Hamiltonian Cycle

摘录 / EXCERPT: Let G be a graph on n vertices, and let k be a nonnegative integer. Suppose that at least $n-k$ vertices of G have degree at least $n/2-k$. Combined parameterization conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-039826 | MATHDB-341847

Novikov-Maltsev conjecture on the Hausdorff dimension of the Rauzy gasket

摘录 / EXCERPT: Let R be the Rauzy gasket, the unique non-empty compact subset of the standard 2-simplex satisfying ... Here the maps $f_j\colon\Delta\to\Delta_j$ are the projectivizations of th...

conjecture | open | MathDB 页面 | 原文线索

V2-039827 | MATHDB-341927

Effective SMB theorem for measures

摘录 / EXCERPT: Let $\mathbb A^\infty$ be the space of one-sided infinite sequences over the alphabet A, with shift operator T. Let μ be a computable ergodic T-invariant measure, and def...

conjecture | open | MathDB 页面 | 原文线索

V2-039828 | MATHDB-341969

Polynomial Wiener-Wintner singular integral conjecture

摘录 / EXCERPT: Polynomial Wiener-Wintner singular integral conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-039829 | MATHDB-342125

Strong growth conjecture for Fibonacci-polynomial cycles with m = 8 + 12(61 + 64d)

摘录 / EXCERPT: Let $m=8+12q$ with $q=61+64d$ for some nonnegative integer d. Define the sequence $\{g_l\}_{l\ge 10}$ as in the preceding definition, with g_{10} and g_{10}' selected accordi...

conjecture | open | MathDB 页面 | 原文线索

V2-039830 | MATHDB-342126

Fluid-limit trajectories conjecture for stable mean field game equilibria

摘录 / EXCERPT: Let N be the number of players, and let $n^N(t)$ denote the number of players in state 0 at time t under the unique symmetric Markov perfect equilibrium for the $N+1$-player...

conjecture | open | MathDB 页面 | 原文线索

V2-039831 | MATHDB-342131

Thom's conjecture on invariant functions of generic vector fields

摘录 / EXCERPT: Let M be a compact connected manifold, let X be a C^1-generic vector field on M, and let $f:M\to\mathbb{R}$ be a function that is either C^1 or C^0. Call f invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-039832 | MATHDB-342164

Quantum-walk conjecture for reversible Markov chains

摘录 / EXCERPT: Let $\mathcal{P}$ be a reversible, ergodic Markov chain with stationary distribution $\uppi$, and let M be a set of marked states. Define ... Assume that $p_M<0.5$, and consider Al...

conjecture | open | MathDB 页面 | 原文线索

V2-039833 | MATHDB-342216

Boundary Gibbs-state conjecture for the spectral gap of PEPS parent Hamiltonians

摘录 / EXCERPT: Boundary Gibbs-state conjecture. The parent Hamiltonian of the PEPS is gapped if and only if the boundary states can be written as Gibbs states of one-dimensional short-range Hamil...

conjecture | open | MathDB 页面 | 原文线索

V2-039834 | MATHDB-342227

The soft-core Gibbs-to-non-Gibbs transition-time conjecture

摘录 / EXCERPT: Soft-core transition-time conjecture. There exists a time $t_{NG}(\beta)$ such that the time-evolved measure is Gibbs for all $t<t_{NG}(\beta)$ and non-Gibbs for all...

conjecture | open | MathDB 页面 | 原文线索

V2-039835 | MATHDB-342287

Weak-strong consistency conjecture for stationary ergodic distributions

摘录 / EXCERPT: Let H_0 and H_1 be hypotheses consisting of stationary ergodic distributions. A test is weakly asymmetrically consistent (respectively, weakly asymptotically consistent) when t...

conjecture | open | MathDB 页面 | 原文线索

V2-039836 | MATHDB-342335

The classification of satisfiable condition sets for elementary slopes

摘录 / EXCERPT: Condition-set classification conjecture. The 19 condition sets given in the table below are the only condition sets that can be satisfied.

conjecture | open | MathDB 页面 | 原文线索

V2-039837 | MATHDB-342375

Injectivity conjecture for harmonic models of semi-lopsided algebraic actions

摘录 / EXCERPT: Let G be a countable discrete group with a left-invariant partial order $\preceq$. Let $f\in \mathbb{Z}(G)$ be semi-lopsided and satisfy ... Set $m=\tau(f)$. Suppose either that...

conjecture | open | MathDB 页面 | 原文线索

V2-039838 | MATHDB-342407

Planck-scale mass equidistribution conjecture for dihedral Maaß newforms

摘录 / EXCERPT: Let $g\to\frac12+it_g$ be a sequence of newforms in $\mathcal{B}_0^{\ast}(q,\chi)$, and let R denote the radius of a hyperbolic ball. The volume-level shrinking-ball equidistribu...

conjecture | open | MathDB 页面 | 原文线索

V2-039839 | MATHDB-342522

Ergodicity invariance under isomorphism of Volterra evolution algebras

摘录 / EXCERPT: Let ${\mathcal E}_1$ and ${\mathcal E}_2$ be two n-dimensional isomorphic Volterra evolution algebras, and let the corresponding Volterra quadratic stochastic operators be the op...

conjecture | open | MathDB 页面 | 原文线索

V2-039840 | MATHDB-342544

Fractal exponential attractor conjecture for the flow-plate evolution

摘录 / EXCERPT: Fractal exponential attractor conjecture. The evolution $(T_t,\mathbf H)$ has a fractal exponential attractor, and

conjecture | open | MathDB 页面 | 原文线索

V2-039841 | MATHDB-342545

Finite determining set conjecture for the entire flow-plate dynamics

摘录 / EXCERPT: Finite determining set conjecture. There exists a set $\mathbb L$ which is a finite determining set for the entire dynamics (S_t,Y).

conjecture | open | MathDB 页面 | 原文线索

V2-039842 | MATHDB-342563

Hamiltonian displacement conjecture for Vianna's tori and real projective planes

摘录 / EXCERPT: Let $T_{a,b,c}\subset\mathbb{C}P^2$ be Vianna's torus associated with a Markov triple (a,b,c), let $\mathbb{R}P^2\subset\mathbb{C}P^2$ be the real projective plane, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-039843 | MATHDB-342567

Mixing conjecture for the KdV flow

摘录 / EXCERPT: Let $U(t)$ denote the Koopman operator associated with the KdV flow on the underlying probability space, acting on observables by composition with the flow, and let the projection...

conjecture | open | MathDB 页面 | 原文线索

V2-039844 | MATHDB-342634

Mirzakhani's conjecture on higher-rank orbit closures of translation surfaces

摘录 / EXCERPT: Let an orbit closure be an orbit closure of the $SL(2,\bR)$ action on a stratum of translation surfaces, and let its rank be the rank associated with the corresponding affine invar...

conjecture | open | MathDB 页面 | 原文线索

V2-039845 | MATHDB-342655

Periodic-factor conjecture for noncorrelated binary pattern sequences

摘录 / EXCERPT: The periodic-factor conjecture. The sequence a is the product of a periodic sequence and a dilation-invariant noncorrelated binary pattern sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-039846 | MATHDB-342679

Stable ergodicity from positive metric entropy

摘录 / EXCERPT: Let M be a smooth compact manifold, let m be a smooth volume measure, and let $\operatorname{Diff}^{1}_{m}(M)$ denote the volume-preserving C^{1} diffeomorphisms. Stable ergo...

conjecture | open | MathDB 页面 | 原文线索

V2-039847 | MATHDB-342680

Generic zero-exponents or minimal invariant foliation conjecture in dimension 3

摘录 / EXCERPT: Let M^{3} be a three-dimensional smooth compact manifold, let m be a smooth volume measure, and let $\operatorname{Diff}^{1}_{m}(M^{3})$ denote the volume-preserving C^{1} di...

conjecture | open | MathDB 页面 | 原文线索

V2-039848 | MATHDB-342768

Hertz-Hertz-Ures ergodic conjecture for three-dimensional partially hyperbolic systems

摘录 / EXCERPT: Let f be a conservative partially hyperbolic diffeomorphism of a 3-manifold. Hertz-Hertz-Ures ergodic conjecture. If f is non-ergodic, then there is a 2-torus tangential to...

conjecture | open | MathDB 页面 | 原文线索

V2-039849 | MATHDB-342805

Essential one-to-one conjecture for generalized Bernoulli algebraic actions

摘录 / EXCERPT: Let Γ be a countable group and let $[\,\cdot\,]\colon\Gamma\to\mathbb{Z}$ be a homomorphism. Suppose that ... where $M>\sum_{s\in I}f_s$, and $[s]\geqslant1$ and $f_s>0$ for e...

conjecture | open | MathDB 页面 | 原文线索

V2-039850 | MATHDB-342861

Anantharaman-Nonnenmacher entropy conjecture for Anosov flows

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold, let φ^t be its geodesic flow on S^*M, let $\mathcal M(\Delta_g)$ be the set of semiclassical measures of Laplace eigenfunctions, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039851 | MATHDB-342983

Factorization conjecture for nonstandard slow-fast vector fields

摘录 / EXCERPT: Let $H(z,\varepsilon)$ be a vector field satisfying Assumptions --, with slow manifold S_{ε} and linear fast fibers $\mathcal{N}_{\varepsilon}(p)$ at basepoints...

conjecture | open | MathDB 页面 | 原文线索

V2-039852 | MATHDB-343055

The ergodicity conjecture for invariant states on symplectic abelian groups

摘录 / EXCERPT: Let $({\mathcal G},\sigma)$ be a ${\mathbb T}$-symplectic abelian group, let ${\mathcal W}_{\mathcal G}$ be its associated algebra, and let ${\rm Sp}({\mathcal G})$ act on...

conjecture | open | MathDB 页面 | 原文线索

V2-039853 | MATHDB-343056

The diagonalizable irrational case of the ergodicity conjecture for invariant states

摘录 / EXCERPT: Let $({\mathcal G},\sigma)$ be an irrational ${\mathbb T}$-symplectic abelian group, let ${\mathcal W}_{\mathcal G}$ be its associated algebra, and let ${\rm Sp}({\mathcal G})$ act...

conjecture | open | MathDB 页面 | 原文线索

V2-039854 | MATHDB-343058

Katok's conjecture on intermediate entropies of diffeomorphisms

摘录 / EXCERPT: Katok's conjecture. The set of entropies satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-039855 | MATHDB-343065

Commutativity conjecture for the quantum DELL Hamiltonians

摘录 / EXCERPT: Consider the quantum double-elliptic (DELL) system with $N-1$ degrees of freedom and Hamiltonians ... where the operators $\widehat{\mathcal{O}}_a$ are Fourier modes of the DELL curr...

conjecture | open | MathDB 页面 | 原文线索

V2-039856 | MATHDB-343133

The density property for quadrilateral tiling billiards

摘录 / EXCERPT: Consider quadrilateral tiling billiards and their trajectories. Density property. The density property holds for quadrilateral tiling billiards. In the surrounding discussion, dens...

conjecture | open | MathDB 页面 | 原文线索

V2-039857 | MATHDB-343134

The Winding Conjecture for fully flipped interval exchange maps

摘录 / EXCERPT: Let $\CETn$ denote the family of maps used in the source, and consider periodic trajectories of a map $F\in\CETn$. Winding Conjecture. The winding number of a periodic trajectory d...

conjecture | open | MathDB 页面 | 原文线索

V2-039858 | MATHDB-343230

Decorated rotation number determines smooth rigidity classes for circle diffeomorphisms with breaks

摘录 / EXCERPT: Rigidity conjecture. There is a full-measure set of decorated rotation numbers γ such that, whenever T_1 and T_2 are circle diffeomorphisms with breaks having breaks of...

conjecture | open | MathDB 页面 | 原文线索

V2-039859 | MATHDB-343231

Hyperbolicity of the renormalisation operator on the attractor

摘录 / EXCERPT: Renormalisation hyperbolicity conjecture. The renormalisation operator $\mathcal{R}$, restricted to $\mathcal{E}_{\mathbf{c}}$, is hyperbolic when restricted to its attractor.

conjecture | open | MathDB 页面 | 原文线索

V2-039860 | MATHDB-343379

Boundary-separation conjecture for expansivity phases

摘录 / EXCERPT: Let $\{\mathcal{S}_j\}_{j=1}^{\infty}$ be the collection of all tuples in $\mathbb{R}[x]$, and let $\mathcal{Z}[(\gamma^{-1}\circ\beta\circ\gamma\circ\nabla)^n(\mathcal{S}_j)]$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-039861 | MATHDB-343384

Convexity conjecture for the pendulum time map

摘录 / EXCERPT: Let $T_B(z,\phi)$ be the time map defined in the paper for parameters z and ϕ, with $z\in(0,2)$ and $\phi\in(0,\frac{\pi}{2})$. Convexity conjecture. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-039862 | MATHDB-343441

The integer-coefficient supremum conjecture for maximal eigenvalue multiplicities

摘录 / EXCERPT: For $N\in\mathbb N^*$, let $\mathfrak M(N)$ be the family of admissible arrays $\mathfrak A$, and for $k\in\mathbb N^*$ let $\mathfrak m_k(N)$ be the supremum of the multiplicity o...

conjecture | open | MathDB 页面 | 原文线索

V2-039863 | MATHDB-343514

Charged quantum-system chaos bound with an effective inverse temperature

摘录 / EXCERPT: Let H be the Hamiltonian, Q a conserved charge, and define ... For operators that can create states with arbitrarily large values of Q, consider the regularized out-of-time-ord...

conjecture | open | MathDB 页面 | 原文线索

V2-039864 | MATHDB-343579

The universal tightness conjecture for contact structures with nowhere-invariant complex structure

摘录 / EXCERPT: Let (M^3,ξ) be a closed contact manifold admitting a contact form α and a complex structure J on ξ such that ... everywhere, where X_α is the Reeb field of...

conjecture | open | MathDB 页面 | 原文线索

V2-039865 | MATHDB-343600

Constriction abscissa conjecture for phase-lock areas

摘录 / EXCERPT: Let L_ρ be the phase-lock area with rotation number ρ, and call the separation points between consecutive bounded domains in its chain constrictions when they do not lie...

conjecture | open | MathDB 页面 | 原文线索

V2-039866 | MATHDB-343601

Monotonicity conjecture for phase-lock area portraits

摘录 / EXCERPT: Let $\omega_0>0$, $l\in\mathbb N$, $\mu>0$, $B_0=l\omega_0$, and $A_0=2\mu\omega_0$. Let $s\in\mathbb Z_{\geqslant0}$, and suppose $(B_0,A_0)\in\partial L_{s,\pm}(\omega_0)$ for an...

conjecture | open | MathDB 页面 | 原文线索

V2-039867 | MATHDB-343619

The positive-entropy stable ergodicity conjecture

摘录 / EXCERPT: Let M be a compact manifold, let m denote Lebesgue measure, and let $\operatorname{Diff}^{1}_{m}(M)$ be the space of C^1 Lebesgue-measure-preserving diffeomorphisms. A diffeo...

conjecture | open | MathDB 页面 | 原文线索

V2-039868 | MATHDB-343620

The minimal expanding-or-contracting foliation conjecture in dimension three

摘录 / EXCERPT: Let M^3 be a three-dimensional manifold, let m denote Lebesgue measure, and let $\operatorname{Diff}^{1}_{m}(M^3)$ be the space of C^1 Lebesgue-measure-preserving diffeomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-039869 | MATHDB-343621

The minimal foliation criterion for stable Bernoulli dynamics

摘录 / EXCERPT: Let M be a compact manifold, let m denote Lebesgue measure, and let $\operatorname{Diff}^{1}_{m}(M)$ be the space of C^1 Lebesgue-measure-preserving diffeomorphisms. An invar...

conjecture | open | MathDB 页面 | 原文线索

V2-039870 | MATHDB-343698

Meagreness conjecture for Dirac physical measures on saddle-type hyperbolic periodic points

摘录 / EXCERPT: Let M be the compact manifold underlying the space of diffeomorphisms, and let $\operatorname{Diff}^1(M)$ denote the space of C^1 diffeomorphisms of M. A saddle-type hyperbol...

conjecture | open | MathDB 页面 | 原文线索

V2-039871 | MATHDB-343704

Extension of the nondivergence proposition to other rational subgroups

摘录 / EXCERPT: Let $\mathbf{H}$ be a $\mathbb{Q}$-subgroup, let $S_{\mathbf{H}}$ be the subgroup used to define the corresponding set $\Omega_{g,\varepsilon}\subseteq\operatorname{Lie}(S_{\mathbf{H}})$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039872 | MATHDB-343772

Conjecture on weak omega-limit sets of Volterra quadratic stochastic operators

摘录 / EXCERPT: Let $V\in\tilde{\mathcal V}^-$ and let $\frak{f}(V)=(a_{ki})$. For ${\mathbf{x}}\in S$, write $\operatorname{supp}({\mathbf{x}})$ for its support, let ${\mathbf e}_j$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-039873 | MATHDB-343780

Conjecture on BV stability for the general self-consistent system

摘录 / EXCERPT: BV stability conjecture. The uniform density is stable with respect to the BV-norm for all $\varepsilon>0$, while every other acim is unstable with respect to the BV-norm.

conjecture | open | MathDB 页面 | 原文线索

V2-039874 | MATHDB-343800

Classification of BATS tip solutions by steady tip growth

摘录 / EXCERPT: Let Y_μ be an open parameter set for which the tip solutions $x_\alpha(\,\cdot\,;\mu)$ are uniquely defined, and let $A_\mu(Y)$, $B_\mu(Y)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-039875 | MATHDB-343871

Birkhoff's conjecture for Riemannian billiards

摘录 / EXCERPT: Let an annulus carry a Riemannian metric, let one boundary component be a strictly convex curve Γ, and consider the billiard system near Γ. Birkhoff's conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-039876 | MATHDB-343899

Infinitely many connected components in a Master Teapot slice

摘录 / EXCERPT: Let Ξ_λ denote the λ-slice of Thurston's Master Teapot, and let $\overline{\mathbb{D}}$ be the closed unit disk. Connected-component conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-039877 | MATHDB-343977

Greenfield-Wallach conjecture on globally hypoelliptic vector fields

摘录 / EXCERPT: Let M be a closed, connected, orientable manifold, and let X be a vector field on M. The vector field X is globally hypoelliptic when every distributional solution of...

conjecture | open | MathDB 页面 | 原文线索

V2-039878 | MATHDB-344006

Commutativity of QM transformation and discretization for QP systems

摘录 / EXCERPT: Let a QP differential system be given, and consider the multiplicative and additive discretizations ... [U+0915][U+0947] and ... where each φ_i is differentiable. Commutativity conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-039879 | MATHDB-344228

Quantization commutes with reduction for b-symplectic manifolds

摘录 / EXCERPT: Let (M,Z,ω,μ) be a compact connected Hamiltonian G-space, where G is compact and connected, M is prequantizable, and $Q_G(M,Z,\omega)\in R(G)$ is its equivariant qua...

conjecture | open | MathDB 页面 | 原文线索

V2-039880 | MATHDB-344249

Hamiltonian invariance conjecture for Plücker-torus potentials

摘录 / EXCERPT: Let $d4c$ be a Plücker sequence of type (k,n), and let $L_\mathfrak{s}\subset\operatorname{Gr}(k,n)$ be its associated Plücker torus. The Laurent polynomial $W_\mathfrak{s}$ is d...

conjecture | open | MathDB 页面 | 原文线索

V2-039881 | MATHDB-344263

The positive entropy conjecture for typical symplectic surface diffeomorphisms

摘录 / EXCERPT: A symplectic surface diffeomorphism is said to have positive metric entropy if it leaves invariant an ergodic physical probability measure with positive entropy. Positive entropy c...

conjecture | open | MathDB 页面 | 原文线索

V2-039882 | MATHDB-344264

The high-emergence conjecture for differentiable dynamics

摘录 / EXCERPT: High-emergence conjecture. Dynamics of high emergence are typical in many senses and contexts. This is stated as the first aim of the authors' program on emergence; the source leav...

conjecture | open | MathDB 页面 | 原文线索

V2-039883 | MATHDB-344272

Robust expansiveness for sectional hyperbolic attracting sets without strong dissipativeness

摘录 / EXCERPT: Robust expansiveness conjecture. Theorem is still true for all C^1 vector fields exhibiting a sectional hyperbolic attracting set.

conjecture | open | MathDB 页面 | 原文线索

V2-039884 | MATHDB-344340

Continuity of the top Lyapunov exponent under a single uniform holonomy

摘录 / EXCERPT: Let $\hat{A}_k$ and $\hat{A}$ be continuous linear cocycles over the same hyperbolic system. Suppose that each cocycle is equipped with a uniform unstable holonomy $H^{u,k}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-039885 | MATHDB-344441

Unimodality conjecture for Hamiltonian circle actions with isolated fixed points

摘录 / EXCERPT: Let (M,ω) be a $2n$-dimensional closed symplectic manifold equipped with a Hamiltonian S^1-action whose fixed points are all isolated. Write b_j for the j-th Betti num...

conjecture | open | MathDB 页面 | 原文线索

V2-039886 | MATHDB-344539

Invariance of sign stability under mutation-loop representations

摘录 / EXCERPT: Let $b3_i:t_i\to t'_i$ $(i=1,2)$ be edge paths in the exchange graph $b4_I$ representing the same mutation loop $d\rm[b3_1]_{b2}=d\rm[b3_2]_{b2}$. A path is sign-stable when it has...

conjecture | open | MathDB 页面 | 原文线索

V2-039887 | MATHDB-344579

Finite transit-time conjecture for mean-median orbits

摘录 / EXCERPT: Finite transit-time conjecture. For every initial set, the sequence of new numbers generated by iterating the mean-median map is eventually constant.

conjecture | open | MathDB 页面 | 原文线索

V2-039888 | MATHDB-344657

Unique fixed-point convergence for monotone-function code ensembles

摘录 / EXCERPT: Let $q^{\mathrm{BEC}}_\ell(x)$ denote the state after ℓ iterations of the BEC dynamical system for a code ensemble generated by a monotone function, initialized at x. Unique...

conjecture | open | MathDB 页面 | 原文线索

V2-039889 | MATHDB-344658

Soft-information bounds for monotone-function code ensembles

摘录 / EXCERPT: Let $\alpha=C/R$, and let q^{BEC}_t and q^{BSC}_t be the dynamical systems initialized at x_0 by ... and ... Here ι^{BP}_ℓ is the soft informa...

conjecture | open | MathDB 页面 | 原文线索

V2-039890 | MATHDB-344686

Positive controllability diameter conjecture for bilinear Schrödinger systems

摘录 / EXCERPT: Controllability-diameter conjecture. For every such Hamiltonian, the implication

conjecture | open | MathDB 页面 | 原文线索

V2-039891 | MATHDB-344727

The symmetric chain decomposition extension conjecture for hypercubes

摘录 / EXCERPT: A symmetric chain decomposition (SCD) of the hypercube Q_n is a partition of its vertices into symmetric chains, where a symmetric chain is a path $(x_k,x_{k+1},\ldots,x_{n-k})$...

conjecture | open | MathDB 页面 | 原文线索

V2-039892 | MATHDB-344775

Uniqueness of the upper transition attractor for bounded affine families

摘录 / EXCERPT: Let F_t be a bounded one-parameter affine family with attractors A_t, and let t_0 be the threshold at which attractors cease to exist. A compact set A^* is an upper transit...

open_problem | open | MathDB 页面 | 原文线索

V2-039893 | MATHDB-344839

Algebraic Stein operator characterization for Gaussian polynomials

摘录 / EXCERPT: Let $X\sim N(0,1)$), and let $Y=h(X)$, where $h\in\mathbb{K}[X]$ is a polynomial with $\deg(h)\geq 3$ and ${\rm I\kern-0.16em E}[h(X)]=0$. For $p_0\in\mathbb{K}_0[Y]$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-039894 | MATHDB-344881

Uniqueness of the energetically stable q=1 ring in the large-D limit

摘录 / EXCERPT: Large-D stability conjecture. In the limit $D\rightarrow\infty$, the only energetically stable equilibrium is the $q=1$ ring.

conjecture | open | MathDB 页面 | 原文线索

V2-039895 | MATHDB-344964

Periodicity conjecture for Hexhouses

摘录 / EXCERPT: A Hexhouse is the lattice polygon obtained by placing a trapezoid on top of a square, with its moduli described by three integral parameters. Hexhouse periodicity conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-039896 | MATHDB-344965

Periodicity conjecture for special octagons

摘录 / EXCERPT: A special octagon is a lattice octagon whose opposite sides are parallel with slopes 0, ±1, and ∞, and which has an axis of symmetry parallel to a pair of sides. Speci...

conjecture | open | MathDB 页面 | 原文线索

V2-039897 | MATHDB-344968

Ye's conjecture on invariant lines of planar polynomial systems

摘录 / EXCERPT: Let $P_n(x,y)$ and $Q_n(x,y)$ be polynomials of degree n, defining the planar polynomial system ... An invariant line is a straight line preserved by the flow of this system. Ye's...

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V2-039898 | MATHDB-345057

Kähler realization conjecture for six-dimensional monotone Hamiltonian circle manifolds

摘录 / EXCERPT: Let (M,ω) be a six-dimensional closed monotone symplectic manifold equipped with an effective Hamiltonian circle action. A Kähler realization conjecture asserts that...

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V2-039899 | MATHDB-345085

Averaging conjecture for non-vacuum LRS Bianchi III solutions

摘录 / EXCERPT: Let $(H(t),\mathbf x(t))$ be the solution of the quasi-standard system with arbitrary non-vacuum initial data satisfying $\Omega>0$, the Hamiltonian constraint, and $H>0$. Let...

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V2-039900 | MATHDB-345107

Birkhoff-section existence conjecture for nondegenerate Reeb vector fields

摘录 / EXCERPT: Let M be a closed 3-manifold and let R be a nondegenerate Reeb vector field on M. A Birkhoff section is a surface with boundary whose interior is embedded and transverse to...

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V2-039901 | MATHDB-345267

Typicality of super-polynomial emergence in dynamical systems

摘录 / EXCERPT: Let f be a dynamical system with emergence function $\mathcal E_f(\epsilon)$. When ... the system is said to have super-polynomial emergence. Super-polynomial emergence conjecture....

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V2-039902 | MATHDB-345268

Locally dense wandering Fatou components with high emergence

摘录 / EXCERPT: Let $\operatorname{Aut}(\mathbb C^2)$ denote the space of polynomial automorphisms of $\mathbb C^2$. A dynamical system has a wandering Fatou component when it displays a Fatou com...

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V2-039903 | MATHDB-345270

Holomorphic quantum unique ergodicity for quaternionic Shimura curves

摘录 / EXCERPT: Holomorphic mass equidistribution conjecture. For every such Ψ,

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V2-039904 | MATHDB-345274

Extension of invariant manifold theorems to normally hyperbolic compact invariant sets

摘录 / EXCERPT: Let A^+ and A^- be general normally hyperbolic compact invariant sets of the respective limit systems. Invariant manifold extension conjecture. The statements for equilibria in...

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V2-039905 | MATHDB-345353

The multisingular partial hyperbolicity dichotomy for vector fields

摘录 / EXCERPT: A vector field X is called multisingular partially hyperbolic if its chain-recurrence set can be split into finitely many compact invariant sets, each admitting a multisingular p...

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V2-039906 | MATHDB-345381

Uniqueness of the equal-eigenvalue balanced configuration in the 3-body problem in four dimensions

摘录 / EXCERPT: Let a balanced configuration of the three-body problem in $\mathbb{R}^4$ have angular momentum L and eigenvalues μ_1 and μ_2. Uniqueness conjecture. For any choice of mas...

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V2-039907 | MATHDB-345382

Unique critical points on balanced families in the 3-body problem in four dimensions

摘录 / EXCERPT: Consider a balanced family in the three-body problem in $\mathbb{R}^4$, with functions h and k defined along the family. Critical-point conjecture. On each balanced family, the...

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V2-039908 | MATHDB-345549

Iommi-Kiwi conjecture on inflections of two-branch Lyapunov spectra

摘录 / EXCERPT: Let f be a two-branch expanding map, and let L denote its Lyapunov spectrum. Iommi-Kiwi conjecture. The Lyapunov spectrum L has at most two points of inflection. This conject...

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V2-039909 | MATHDB-345588

Invariant-measure periodic-orbit conjecture for geodesic flows

摘录 / EXCERPT: Invariant-measure periodic-orbit conjecture. If $\mu(S^*H)>0$, then $\mu|_{S^*H}$ is supported on a union of periodic geodesics γ such that $\gamma\cap S^*H$ has positive ar...

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V2-039910 | MATHDB-345639

Uniform distribution conjecture for powers under self-similar measures

摘录 / EXCERPT: Let μ be a non-atomic self-similar measure with support contained in $[1,\infty)$. A sequence $(y_n)_{n=1}^{\infty}$ is uniformly distributed modulo one if, for every interval...

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V2-039911 | MATHDB-345684

Buzzi-Crovisier-Sarig existence conjecture for measures of maximal entropy

摘录 / EXCERPT: Let $r\in(1,+\infty)$, let f be a C^r diffeomorphism of a surface, and define ... where ... A measure of maximal entropy (m.m.e.) is an invariant probability measure whose metric e...

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V2-039912 | MATHDB-345685

Uniqueness conjecture for measures of maximal entropy near the standard map

摘录 / EXCERPT: Let $f_k:\mathbb{T}^2\to\mathbb{T}^2$ be the standard map ... A C^2-diffeomorphism g is sufficiently C^2-close to f_k when it lies in a sufficiently small C^2 neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-039913 | MATHDB-345690

Diagonal monotonicity conjecture for kicked Harper rotation sets

摘录 / EXCERPT: Diagonal monotonicity conjecture. If $0\leq\alpha\leq\tilde\alpha$, then

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V2-039914 | MATHDB-345703

Generic symmetry-reduced integrability conjecture for point-vortex Hamiltonian systems

摘录 / EXCERPT: Generic symmetry-reduced integrability conjecture. A generic G-invariant Hamiltonian system on P^N_M is integrable only when these geometric conditions are such that symplectic...

conjecture | open | MathDB 页面 | 原文线索

V2-039915 | MATHDB-345807

Circular-arc conjecture for minimizers of the Gamma-limit functional

摘录 / EXCERPT: Assume that $\mathcal{K}$ is not central. Let $M ein\mathbb{N}$ be as in the symmetry condition, and let $(\Delta\theta)_{\min}$ be the minimum total angle compatible with...

conjecture | open | MathDB 页面 | 原文线索

V2-039916 | MATHDB-345861

The conjecture on thermodynamic equivalence of the Bogoliubov Hamiltonian to the perfect Bose gas

摘录 / EXCERPT: Let H_{Λ}^{B} be the Bogoliubov Hamiltonian, and let $p^{B}({\Greekmath 010C},{\Greekmath 0116})$ and $p^{PBG}({\Greekmath 010C},{\Greekmath 0116})$ denote, respectively, t...

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V2-039917 | MATHDB-345891

Noise-induced order from positive Lyapunov exponents

摘录 / EXCERPT: Let I be the phase space and let $T:I\to I$ be a C^1, nonsingular dynamical system admitting a unique absolutely continuous invariant measure with positive Lyapunov exponent. L...

conjecture | open | MathDB 页面 | 原文线索

V2-039918 | MATHDB-345911

Sinai's positive entropy conjecture for the standard map

摘录 / EXCERPT: Sinai's conjecture. For every parameter $\lambda\ne 0$, the metric entropy of f_λ is positive. This is presented as another famous positive-entropy conjecture; the source d...

conjecture | open | MathDB 页面 | 原文线索

V2-039919 | MATHDB-345933

The Lyapunov genus bound for vector fields with finitely many zeros

摘录 / EXCERPT: Let X be a compact manifold and let v be a smooth vector field that is of gradient type in a neighborhood of its finite zero set $Z(v)$. Let $\mathsf{Lyap}(v)$ be the least car...

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V2-039920 | MATHDB-346004

Buff's conjecture on continuous dependence of invariant measures on the multiplier

摘录 / EXCERPT: Let $\mathbb{U}_X = \{\,\exp(2i\pi x)\,;x\in X\,\}$ and let $\mathcal{B}$ denote the set of Brjuno numbers. The function ... has a continuous extension to $\overline{\mathbb{D}}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-039921 | MATHDB-346083

Uniform upper-bound conjecture for acip densities of piecewise expanding unimodal maps

摘录 / EXCERPT: Let $\mathfrak{X}$ be the class of maps under consideration, and for each $f\in\mathfrak{X}$ let the acip denote its absolutely continuous invariant probability measure, with densi...

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V2-039922 | MATHDB-346116

Linear starting-point conjecture for boundary roots of Newton's method

摘录 / EXCERPT: Let p be a polynomial of degree d, rescaled so that all its roots lie in the unit disk. For $r>1$, let $\partial D_r(0)$ be the circle of radius r centered at the origin, and...

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V2-039923 | MATHDB-346141

Uniform boundedness conjecture for the pathwise critical coupling strength

摘录 / EXCERPT: Let N be the number of particles, let Θ^0 be an initial phase vector, and let $\Omega=(\nu_1,\ldots,\nu_N)$ be the vector of natural frequencies. Write $D(\Omega)$ for the...

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V2-039924 | MATHDB-346142

Global basin conjecture for the Kuramoto phase-locked state

摘录 / EXCERPT: Consider the finite-particle all-to-all Kuramoto model with natural-frequency vector Ω and frequency diameter $D(\Omega)$. Let Θ^0 be an initial phase vector, and let...

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V2-039925 | MATHDB-346172

Axis-ratio invariance for Circumellipses in Moses' pencil

摘录 / EXCERPT: Let a triangle vary through a family of 3-periodic orbits, and consider the Circumellipses in Moses' pencil, namely the Circumellipses whose centers lie on the Feuerbach Circumhype...

conjecture | open | MathDB 页面 | 原文线索

V2-039926 | MATHDB-346177

Classical completeness and quantum completeness for formally self-adjoint differential operators

摘录 / EXCERPT: Classical-quantum completeness conjecture. The Hamiltonian flow of p is complete if and only if P is essentially self-adjoint.

conjecture | open | MathDB 页面 | 原文线索

V2-039927 | MATHDB-346187

Continuum many bounded displacement non-equivalences for substitution tiling spaces

摘录 / EXCERPT: Continuum-many BD classes conjecture. If

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V2-039928 | MATHDB-346200

Symmetry of coarse conjugacy

摘录 / EXCERPT: Let $f:X\to X$ and $g:Y\to Y$ be maps between metric spaces. Two maps are close if there is a constant $K>0$ such that $d(\varphi(x),\varphi'(x))\le K$ for every x, and a map is...

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V2-039929 | MATHDB-346201

Iteration invariance of coarse conjugacy

摘录 / EXCERPT: Let $f:X\to X$ and $g:Y\to Y$ be maps between metric spaces, and let coarse conjugacy mean that there are coarse equivalences in both directions intertwining the maps up to bounded...

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V2-039930 | MATHDB-346252

Niţică-Rodriguez Hertz generic stable ergodicity problem

摘录 / EXCERPT: Let M be a compact manifold with volume measure m, and let $\operatorname{Diff}^{1}_{m}(M)$ be the space of volume-preserving C^1-diffeomorphisms of M. For...

open_problem | open | MathDB 页面 | 原文线索

V2-039931 | MATHDB-346296

Moses' pencil axis-ratio invariance conjecture for 3-periodic circumellipses

摘录 / EXCERPT: Moses' pencil conjecture. Over the family of 3-periodics, all circumellipses in Moses' pencil conserve the ratio of their axes.

conjecture | open | MathDB 页面 | 原文线索

V2-039932 | MATHDB-346297

Excentral X_3-inconic axis-ratio conjecture for 3-periodics

摘录 / EXCERPT: Excentral X_3-inconic conjecture. The ratio μ'_3/μ_3 is invariant over all 3-periodics and is given by

conjecture | open | MathDB 页面 | 原文线索

V2-039933 | MATHDB-346331

Undecidability conjecture for the generalized Collatz problem $qn+1$

摘录 / EXCERPT: Let q be an odd integer with $q\geq 5$, let x be a positive integer in $\mathbb Z_{cq}$, and let $F_q(x)$ be the modified Collatz general function. Undecidability conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-039934 | MATHDB-346332

Isomorphism conjecture for the transformations T_{θ} and T_N

摘录 / EXCERPT: Isomorphism conjecture. The transformations T_{θ} and T_N are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-039935 | MATHDB-346415

Characterization of equilibria in point cloud registration dynamics

摘录 / EXCERPT: Let ${\bm s}$ denote the state of the dynamical system and let ${\cal F}$ be its vector field, so that equilibria satisfy ... Besides the equilibrium corresponding to the global mini...

conjecture | open | MathDB 页面 | 原文线索

V2-039936 | MATHDB-346416

The BKL scattering conjecture for the Einstein-scalar field system

摘录 / EXCERPT: BKL scattering conjecture. Scattering at $tau to infty$ from the dynamical system defined by $\mathcal{A}$ to that defined by $\mathcal{E}_Phi$ defines a smooth map

conjecture | open | MathDB 页面 | 原文线索

V2-039937 | MATHDB-346481

Weak ergodicity conjecture for non-ergodic Banach spaces

摘录 / EXCERPT: Let X be a separable Banach space. A space is non-ergodic when it is not ergodic, and a Banach space is Hilbertian when it is isomorphic to a Hilbert space. Weak ergodicity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-039938 | MATHDB-346560

The singular Hamiltonian Seifert conjecture

摘录 / EXCERPT: Let (M,ω) be a b^m-symplectic manifold, and let $H\in C^\infty(M)$ be a proper smooth Hamiltonian. A singular periodic orbit is a periodic orbit in the singular Hamiltonia...

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V2-039939 | MATHDB-346699

Erdős-Taylor conjecture on the supremum of planar random-walk local times

摘录 / EXCERPT: For $x\in\mathbb{Z}^2$ and $N\geq 1$, let ℓ_x^N be the total number of visits to x by a planar simple random walk started at the origin before exiting $[-N,N]^2$. Define the...

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V2-039940 | MATHDB-346792

Frantzikinakis's logarithmic divergence conjecture for Hardy-field averages

摘录 / EXCERPT: Let $f_1,\ldots,f_k$ be functions from a Hardy field, and define ... Let $(X,\mathcal{B},\mu,T)$ be an ergodic measure-preserving system and let $h_1,\ldots,h_k\in L^\infty(X)$. Fran...

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V2-039941 | MATHDB-346793

Totally ergodic convergence conjecture for power functions

摘录 / EXCERPT: Let $f_1,\ldots,f_k$ be linearly independent functions of the form ... Let $(X,\mathcal{B},\mu,T)$ be a totally ergodic measure-preserving system and let...

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V2-039942 | MATHDB-346794

Nilsequence structure conjecture for Hardy-field multicorrelations

摘录 / EXCERPT: Nilsequence structure conjecture. There exist a nilmanifold $Y=G/\Gamma$, a continuous function $F\in C(Y)$, a point $y\in Y$, elements $a_1,\ldots,a_k\in G$, and a sequence...

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V2-039943 | MATHDB-346795

Weak compatibility conjecture for weighted Hardy-field averages

摘录 / EXCERPT: Let $\mathcal{H}$ be a Hardy field, let $f_1,\ldots,f_k,W\in\mathcal{H}$, and define ... Property (WP) is that, for every $f\in\operatorname{span}^*(f_1,\ldots,f_k)$ and...

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V2-039944 | MATHDB-346814

Conjectured dynamic convergence for mixed finite elements for Mindlin plates

摘录 / EXCERPT: Dynamic convergence conjecture. Assuming a smooth solution to the Mindlin plate problem, the error estimates

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V2-039945 | MATHDB-346815

Conjectured convergence for weakly symmetric mixed finite elements for Mindlin plates

摘录 / EXCERPT: Weak-symmetry convergence conjecture. Assuming a smooth solution to the Mindlin plate problem, the error estimates

conjecture | open | MathDB 页面 | 原文线索

V2-039946 | MATHDB-346835

Conjecture on Theorems 2 and 3 for non-quasi-analytic weights of moderate growth

摘录 / EXCERPT: Conjecture. Theorem 2 and Theorem 3 hold true for any such weight φ.

conjecture | open | MathDB 页面 | 原文线索

V2-039947 | MATHDB-346841

Dynamical spectral rigidity of ellipses among axially symmetric smooth convex domains

摘录 / EXCERPT: Let an ellipse be regarded as a smooth convex domain in the class of axially symmetric smooth convex domains, and let dynamical spectral rigidity mean that every C^1-deformation...

conjecture | open | MathDB 页面 | 原文线索

V2-039948 | MATHDB-346842

Dynamical spectral rigidity of low-eccentricity ellipses among smooth $b2_{2}$-symmetric convex domains

摘录 / EXCERPT: Let e denote the eccentricity of an ellipse, and consider smooth convex domains invariant under the $b2_{2}$-symmetry. A domain is dynamically spectrally rigid if every C^1-def...

conjecture | open | MathDB 页面 | 原文线索

V2-039949 | MATHDB-346869

Ergodicity of the uniform transverse measure

摘录 / EXCERPT: Let ν_u be the limiting transverse measure obtained from a convergent sequence of uniform probability measures on finite sphere triangulations, and let G be the associated gr...

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V2-039950 | MATHDB-346870

Veech's conjecture on the Pinsker orthocomplement of Möbius orbit measures

摘录 / EXCERPT: Veech's conjecture. For any $\eta\in\mathcal{I}_S(\bm \mu)$,

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V2-039951 | MATHDB-346871

Conjecture on spectral isomorphism of the Pinsker systems

摘录 / EXCERPT: Spectral-isomorphism conjecture. For any $\eta\in\mathcal{I}_S(\bm\mu)$, the dynamical systems

conjecture | open | MathDB 页面 | 原文线索

V2-039952 | MATHDB-346913

Generalised Alspach conjecture for decompositions into double-rays and Hamiltonian circles

摘录 / EXCERPT: Generalised Alspach conjecture. If G satisfies condition P_i, then G has a decomposition into i Hamiltonian double-rays and $k-i$ Hamiltonian circles. Here a Hamiltonian do...

conjecture | open | MathDB 页面 | 原文线索

V2-039953 | MATHDB-346929

Blanc-Van Santen's Perron conjecture for polynomial maps

摘录 / EXCERPT: Let K be a field of characteristic 0, let d be a positive integer, and let $f\colon \mathbb{A}^d_K\to\mathbb{A}^d_K$ be a polynomial map. Recall that a real number...

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V2-039954 | MATHDB-346930

Blanc-Van Santen's Perron conjecture for polynomial automorphisms

摘录 / EXCERPT: Let K be a field of characteristic 0, let d be a positive integer, and let $f\colon \mathbb{A}^d_K\to\mathbb{A}^d_K$ be a polynomial automorphism. Blanc-Van Santen's conjectu...

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V2-039955 | MATHDB-346995

Hutchings's Ruelle-invariant conjecture for ECH error terms

摘录 / EXCERPT: Hutchings's Ruelle-invariant conjecture. If (X,ω) is generic, then

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V2-039956 | MATHDB-347030

Well-posedness hypothesis for the regularity-class wave equation

摘录 / EXCERPT: Well-posedness hypothesis. There exists a unique solution $\boldsymbol z\in\mathcal C^1([0,\infty);\mathcal X_\kappa)\cap\mathcal C([0,\infty);\mathcal Z_\kappa)$ with...

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V2-039957 | MATHDB-347036

Alexandrov's spectral-radius conjecture for global mappings

摘录 / EXCERPT: Alexandrov's conjecture. The image $f(\mathbb{R}^n)$ is a convex domain in $\mathbb{R}^n$, and f is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-039958 | MATHDB-347037

Chamberland's injectivity conjecture for mappings with bounded inverse spectral radius

摘录 / EXCERPT: Chamberland's conjecture. The mapping f is injective.

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V2-039959 | MATHDB-347038

The C^1 fixed point conjecture

摘录 / EXCERPT: C^1 fixed point conjecture. The origin 0 is the unique fixed point of f.

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V2-039960 | MATHDB-347046

Monotonicity of throughputs for queueing-inventory systems

摘录 / EXCERPT: Consider three exponential ergodic queueing-inventory systems with common arrival rate λ, service rate μ, replenishment rate ν, and individual ageing rate γ...

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V2-039961 | MATHDB-347060

Local C^∞-closing property for standard symplectic vector spaces

摘录 / EXCERPT: Local C^∞-closing property. The standard symplectic vector space $({\mathbb R}^{2n},\Omega_{\textrm{standard}})$ has the C^∞-closing property for every $n\geq 2$. In...

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V2-039962 | MATHDB-347070

Conjecture on the growth of the DC pipes region

摘录 / EXCERPT: DC pipes growth conjecture. The wedge formed by the “DC pipes” triples will grow larger and wider as R_1 and R_2 increase.

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V2-039963 | MATHDB-347071

Conjecture on the volumetric extent of additional Turing-like sub-structures

摘录 / EXCERPT: Volumetric extent conjecture. The structures “areas”, “UC pipes”, and “UC spheres” will gain volumetric extent when the values of R_1 and R_2 are increased further.

conjecture | open | MathDB 页面 | 原文线索

V2-039964 | MATHDB-347120

Biased-normal characterization of attractor-generating sequences

摘录 / EXCERPT: Attractor-generation conjecture. Given a hyperbolic IFS with probabilities, a sequence σ is an element of S-that is, σ generates the attractor of the IFS as describ...

conjecture | open | MathDB 页面 | 原文线索

V2-039965 | MATHDB-347139

The almost-spanning cycle conjecture for sparse random hypercube subgraphs

摘录 / EXCERPT: Let $\mathcal{Q}^n$ be the n-dimensional hypercube, and let $\mathcal{Q}^n_p$ be the random subgraph obtained by retaining each edge independently with probability $p=p(n)$. The...

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V2-039966 | MATHDB-347141

The approximate packing conjecture for Hamilton cycles in random hypercube subgraphs

摘录 / EXCERPT: Let $\mathcal{Q}^n$ be the n-dimensional hypercube, let $\mathcal{Q}^n_p$ be its random subgraph obtained by retaining each edge independently with probability p, and let...

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V2-039967 | MATHDB-347150

Threshold-equality conjecture for commensurate fractional-order chaotic systems

摘录 / EXCERPT: Consider a commensurate fractional-order system ... with threshold value α_t, and let $\bar{\alpha}$ be the maximum stability bound $\alpha_*(E)$ over the equilibrium points a...

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V2-039968 | MATHDB-347296

Kolyada-Misiurewicz-Snoha conjecture on closed special alpha-limit sets

摘录 / EXCERPT: Let $f:[0,1]\to[0,1]$ be continuous. A special α-limit set $s\alpha(x)$ is the union of the accumulation-point sets of all backward orbit branches of x. Kolyada-Misiurewic...

conjecture | open | MathDB 页面 | 原文线索

V2-039969 | MATHDB-347297

The period-three coexistence conjecture for special alpha-limit sets

摘录 / EXCERPT: Let $f:[0,1]\to[0,1]$ be continuous, let $x\in[0,1]$, and let $s\alpha(x)$ be the special α-limit set of x. Period-three coexistence conjecture. If $s\alpha(x)$ contains a...

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V2-039970 | MATHDB-347340

Belyakov-Devaney bifurcation conjecture for triple Copenhagen homoclinics

摘录 / EXCERPT: Let $\gamma_{1,2,3}$ denote the short triple Copenhagen homoclinic orbits and let $\mathfrak{D}$ be the critical curve in parameter space. Belyakov-Devaney conjecture. Any paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-039971 | MATHDB-347341

Non-continuation conjecture for the gamma-1 family

摘录 / EXCERPT: Let γ_1 be the triple Copenhagen homoclinic family and let $\mathfrak{D}$ be the critical curve. Gamma-1 non-continuation conjecture. The γ_1 family never continues t...

conjecture | open | MathDB 页面 | 原文线索

V2-039972 | MATHDB-347342

Critical-curve continuation conjecture for the gamma-2 family

摘录 / EXCERPT: Let γ_2 be the triple Copenhagen homoclinic family, let $\mathfrak{D}$ be the critical curve, and let L_0 and L_2 be the relevant libration points. Gamma-2 critical-curv...

conjecture | open | MathDB 页面 | 原文线索

V2-039973 | MATHDB-347343

Separating-parameter conjecture for the gamma-3 family

摘录 / EXCERPT: Let γ_3 be the homoclinic family, let $\mathfrak{D}$ be the critical curve, and let m_1,m_2,m_3 be the mass parameters. Gamma-3 separation conjecture. The γ_3 famil...

conjecture | open | MathDB 页面 | 原文线索

V2-039974 | MATHDB-347424

W*-rigidity conjecture for stationary stable random-field properties

摘录 / EXCERPT: Let $\mathcal{G}$ be a class of countable groups, let $\mathbb{F}_m$ denote the free group on m generators, and let $W^\ast_m$-rigidity and $W^\ast_R$-rigidity refer to the rigid...

conjecture | open | MathDB 页面 | 原文线索

V2-039975 | MATHDB-347425

Ergodicity conjecture for the Furstenberg-Poisson-boundary stable random field

摘录 / EXCERPT: Let $\mathbb{F}_m$ act on its Furstenberg-Poisson boundary $\partial\mathbb{F}_m$, and let the stationary $S\alpha S$ random field associated with this boundary action be the one c...

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V2-039976 | MATHDB-347436

Matsuzawa-Matsumoto-Sano-Zhang conjecture on bounded-degree points with non-maximal arithmetic degree

摘录 / EXCERPT: Matsuzawa-Matsumoto-Sano-Zhang conjecture. The set $Z_f(d)$ is not Zariski dense in X. This concerns the distribution of algebraic points of bounded degree whose arithmetic degre...

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V2-039977 | MATHDB-347460

Ergodic product-disintegration conjecture

摘录 / EXCERPT: Ergodic product-disintegration conjecture. A T-invariant measure on $S^{\mathbb N}$ is T-ergodic exactly when it admits a representation as a mixture of product measures whose...

conjecture | open | MathDB 页面 | 原文线索

V2-039978 | MATHDB-347516

Transverse super-homoclinic orbit conjecture for coupled nonlinear Schrödinger equations

摘录 / EXCERPT: Let the coupled nonlinear Schrödinger equations be the system given by ... possess transverse super-homoclinic orbits. The conjecture asserts that the global stable and unstable inva...

conjecture | open | MathDB 页面 | 原文线索

V2-039979 | MATHDB-347641

Dockery's slowest-disperser conjecture for multi-species communities

摘录 / EXCERPT: Let $N\geq 3$ species have distinct positive dispersal rates $0<d_1<\cdots<d_N$, and let E_i denote the equilibrium at which only species i persists, with E_0 the extinction...

conjecture | open | MathDB 页面 | 原文线索

V2-039980 | MATHDB-347722

The Hamiltonian no-torsion uniruledness conjecture

摘录 / EXCERPT: Hamiltonian no-torsion uniruledness conjecture. Each closed symplectic manifold with non-trivial Hamiltonian torsion must be uniruled.

conjecture | open | MathDB 页面 | 原文线索

V2-039981 | MATHDB-347750

Unitarity criterion for partitioned Bose-Hubbard Hamiltonians

摘录 / EXCERPT: Let the (M,L)-partitioned Hamiltonians be the Hamiltonians referred to as ... have a spectrum. In the leading-order approximation, the partitioned-Hamiltonian unitarity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-039982 | MATHDB-347806

Orbit-closure conjecture for times two and three on tori

摘录 / EXCERPT: Let $d\geq 2$ be an integer. Let $A\subset\mathbb{T}^d$ be an orbit closure under the $\times 2,\times 3$ actions. Orbit-closure conjecture. Then A is a union of finitely many (p...

conjecture | open | MathDB 页面 | 原文线索

V2-039983 | MATHDB-347843

The double-hendecagon trace-field direction conjecture

摘录 / EXCERPT: On the double-hendecagon translation surface, a direction is called hyperbolic or parabolic according to the corresponding type for the surface's Veech group, and the trace field i...

conjecture | open | MathDB 页面 | 原文线索

V2-039984 | MATHDB-347852

Tao's nilmanifold formulation of the logarithmic Sarnak conjecture

摘录 / EXCERPT: Let $s\in\mathbb{N}$, let G/Γ be an s-step nilmanifold, where G is a connected, simply connected s-step nilpotent Lie group and Γ is a cocompact discrete subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-039985 | MATHDB-347937

Ergodicity of the multidimensional Lorentz gas with spherical scatterers

摘录 / EXCERPT: The Lorentz gas is the billiard system in ${\mathbb R}^d$ with $d\geq 2$ and identical, fixed spherical scatterers in an otherwise arbitrary configuration satisfying the stated gen...

conjecture | open | MathDB 页面 | 原文线索

V2-039986 | MATHDB-347938

Ergodicity of the aperiodic Ehrenfest wind-tree gas

摘录 / EXCERPT: Consider an aperiodic Ehrenfest wind-tree gas whose scatterer configuration satisfies the conditions of the wind-tree flow cited in the source. Ergodicity conjecture. The aperiodic...

conjecture | open | MathDB 页面 | 原文线索

V2-039987 | MATHDB-347939

Ergodicity of the periodic wind-tree gas

摘录 / EXCERPT: Consider the periodic wind-tree gas with rectangular scatterers, and assume that their shape parameter satisfies the conditions stated in the cited work. Ergodicity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-039988 | MATHDB-347940

The complexity hypothesis for typical three-dimensional billiards

摘录 / EXCERPT: Let K_n be the upper bound on the number of smooth pieces into which the nth iterate of a smooth convex expanding submanifold is chopped by the billiard map. Complexity hypothe...

conjecture | open | MathDB 页面 | 原文线索

V2-039989 | MATHDB-347941

Exponential correlation decay for planar Sinai billiards with corner points

摘录 / EXCERPT: Consider planar Sinai billiard flows whose boundary may have corner points, with the angle between the intersecting boundary pieces at each corner nonzero. Corner-point correlation...

conjecture | open | MathDB 页面 | 原文线索

V2-039990 | MATHDB-348259

Spectral duality conjecture for mutation-loop matrices

摘录 / EXCERPT: Let γ be a path representing a mutation loop, and let $w \in \mathcal{X}_{(v_0)}^{\mathrm{uf}}(\mathbb{R}^{\mathrm{trop}})$ have strict sign...

conjecture | open | MathDB 页面 | 原文线索

V2-039991 | MATHDB-348261

Uniform sign-stability and North-South dynamics conjecture for seed patterns

摘录 / EXCERPT: Let $\boldsymbol{s}$ be any seed pattern and let $\phi \in \Gamma_{\boldsymbol{s}}$. Uniform sign-stability and North-South dynamics conjecture. The mutation loop ϕ is uniform...

conjecture | open | MathDB 页面 | 原文线索

V2-039992 | MATHDB-348315

Periodic correction conjecture for Lindstedt-series coefficient growth

摘录 / EXCERPT: Periodic correction conjecture. For $k\gg 1$, $A_\rho(k)$ approximately satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-039993 | MATHDB-348516

The dynamical-independence conjecture for integral PCF points in algebraic families

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k including all the archimedean places, and let $d\geq 2$ and $N\geq 1$. Let (X,f_x) be an N-dimensional algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-039994 | MATHDB-348531

Structural equivalence conjecture for dynamical and static Curie-Weiss Potts rate functionals

摘录 / EXCERPT: Structural equivalence conjecture. There exist maps ψ_1 and ψ_2 with the stated diffeomorphism properties such that, for every $(\alpha,\beta,t)\in D$ and $m\in\Delta^2$,

conjecture | open | MathDB 页面 | 原文线索

V2-039995 | MATHDB-348694

The spectral-curve uniqueness conjecture for generalized simple intersections

摘录 / EXCERPT: Fix $\ell\in\mathbb Z_{\neq 0}$ and $\omega>0$. Let $\Gamma_\ell\subset\mathbb R^2_{(\lambda,\mu)}$ be the real spectral curve of the special double confluent Heun equations, and i...

conjecture | solved | MathDB 页面 | 原文线索

V2-039996 | MATHDB-348695

The constriction-curve and phase-lock-area garland conjecture

摘录 / EXCERPT: Consider the three-dimensional parameter space $\mathbb R^2_{B,A}\times(\mathbb R_+)_{\omega}$. For $r\in\mathbb Z_{\neq0}$, the growth points of the phase-lock areas form the r-...

conjecture | open | MathDB 页面 | 原文线索

V2-039997 | MATHDB-348771

Linearity of the logarithmic dominant eigenvalue on the feasible set

摘录 / EXCERPT: Let $\mathbf{D}=\operatorname{diag}(d_1,\ldots,d_n)$ with $d_i>0$, and suppose the conditions of Theorem 22 hold: $\lambda^*(\mathbf{D}^{-1}\mathbf{Q}_m)>1$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-039998 | MATHDB-348772

Unique maximizer of the logarithmic dominant eigenvalue

摘录 / EXCERPT: Let $\mathbf{D}=\operatorname{diag}(d_1,\ldots,d_n)$ with $d_i>0$, and suppose the conditions of Theorem 22 hold. For each $\tau\geq 0$, let ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-039999 | MATHDB-348940

Invariance of the perimeter of focus-inversive polygons

摘录 / EXCERPT: A focus-inversive polygon is obtained from an N-periodic billiard polygon by the focus-inversive construction described in the paper. Its perimeter is denoted by $L^\dagger$. Per...

conjecture | open | MathDB 页面 | 原文线索

V2-040000 | MATHDB-348941

Arnold's diffusion conjecture for nearly integrable Hamiltonian systems

摘录 / EXCERPT: Let h be an integrable Hamiltonian on the action space, let P be a generic perturbation, and consider the nearly integrable Hamiltonian ... For any two points y' and y'' on th...

conjecture | open | MathDB 页面 | 原文线索

V2-040001 | MATHDB-348943

Generic Schwarzschild diffusion conjecture on bounded phase space

摘录 / EXCERPT: Define the bounded-motion region ... Here E is the particle energy, L its angular momentum, V the effective potential, and H the Hamiltonian. Generic Schwarzschild bounded-ph...

conjecture | open | MathDB 页面 | 原文线索

V2-040002 | MATHDB-348970

Generic Morse-Smale conjecture for directional foliations on dilation surfaces

摘录 / EXCERPT: Generic Morse-Smale conjecture. The collection of directions $m\in\mathbb{RP}^1$ for which the directional foliation of X is Morse-Smale has full measure.

conjecture | open | MathDB 页面 | 原文线索

V2-040003 | MATHDB-349058

Chaos conjecture for the subcritical Kasner circle eIFS

摘录 / EXCERPT: Let $v\in(0,1/2)$, and let the Kasner circle eIFS be the iterated function system consisting of the maps ${\cal K}_{\mu\nu\zeta}:\mathrm{K}^{\ocircle}\to\mathrm{K}^{\ocircle}$ defi...

conjecture | open | MathDB 页面 | 原文线索

V2-040004 | MATHDB-349059

Stable-set conjecture for period-two heteroclinic cycles

摘录 / EXCERPT: Consider Bianchi type VIII and IX models in the supercritical case $v\in(1/2,1)$, with time parameter satisfying $\tau_-\to\infty$. Period-two stable-set conjecture. Each heterocli...

conjecture | open | MathDB 页面 | 原文线索

V2-040005 | MATHDB-349060

Stable-set conjecture for supercritical Cantor-set heteroclinic chains

摘录 / EXCERPT: Consider Bianchi type VIII and IX models in the supercritical case $v\in(1/2,1)$, let C be the Cantor set associated with infinite Bianchi type II heteroclinic chains, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040006 | MATHDB-349061

Supercritical Kasner-circle attractor conjecture

摘录 / EXCERPT: Consider Bianchi type VIII and IX models in the supercritical case $v\in(1/2,1)$, with stable set $S\subset\mathrm{K}^{\ocircle}$, attractor ${\cal A}_-$, and $\tau_-\to\infty$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-040007 | MATHDB-349062

Subcritical and critical Kasner-circle attractor conjecture

摘录 / EXCERPT: Consider Bianchi type VIII and IX models in the subcritical and critical cases $v\in(0,1/2]$, and let ${\cal A}_-$ denote the attractor as $\tau_-\to\infty$. Define the invariant s...

conjecture | open | MathDB 页面 | 原文线索

V2-040008 | MATHDB-349068

Conjecture that iterative graph normalization always converges

摘录 / EXCERPT: Let G be a node-weighted graph, and let iterative graph normalization (IGN) denote the iteration of graph normalization, with a non-linear activation applied after each normaliza...

conjecture | open | MathDB 页面 | 原文线索

V2-040009 | MATHDB-349069

Conjecture on activated iterative graph normalization for maximal independent sets

摘录 / EXCERPT: Let G be a node-weighted graph with all node weights distinct. Let iterative graph normalization be applied with a suitable non-linear activation. Activated IGN maximal-independe...

conjecture | open | MathDB 页面 | 原文线索

V2-040010 | MATHDB-349088

Furstenberg's dimension conjecture for multiplicatively independent orbits

摘录 / EXCERPT: Let p and q be two multiplicatively independent natural numbers, and let $x\in[0,1)$ be a real number. Define the forward orbit ... where T_q is the map on...

conjecture | open | MathDB 页面 | 原文线索

V2-040011 | MATHDB-349131

Marchal's conjecture on the continuation class of the Lagrange triangle and figure eight

摘录 / EXCERPT: Marchal's conjecture. The three-body equilateral triangle of Lagrange and the three-body figure eight are in the same continuation class.

conjecture | open | MathDB 页面 | 原文线索

V2-040012 | MATHDB-349192

Existence of exponential mixing skew-product semiflows without constant stable-leaf return times

摘录 / EXCERPT: Let an expanding semiflow be given together with its stable leaves and an original flow to which a skew-product model may be semiconjugated. Exponential mixing skew-product conject...

conjecture | open | MathDB 页面 | 原文线索

V2-040013 | MATHDB-349193

Uniform non-integrability for singular-hyperbolic attracting sets

摘录 / EXCERPT: Let an ergodic physical probability measure be supported on a non-trivial singular-hyperbolic attracting set. Uniform non-integrability conjecture. The Uniform Non-Integrability (U...

conjecture | open | MathDB 页面 | 原文线索

V2-040014 | MATHDB-349194

Exponential mixing for equilibrium states of singular flows

摘录 / EXCERPT: Let a singular flow admit equilibrium states associated to Hölder continuous potentials. Equilibrium-state mixing conjecture. The techniques of Tsujii and Zhang can be adapted to s...

conjecture | open | MathDB 页面 | 原文线索

V2-040015 | MATHDB-349195

Exponential mixing for the multidimensional Lorenz attractor

摘录 / EXCERPT: The multidimensional Lorenz attractor is the sectional-hyperbolic attractor introduced by Bonatti, Pumariño and Viana. Multidimensional Lorenz mixing conjecture. The multidimension...

conjecture | open | MathDB 页面 | 原文线索

V2-040016 | MATHDB-349228

Infinitely many period cycles in the quadratic family near the homoclinic parameter

摘录 / EXCERPT: Let $F_c(x)=x^2+\dfrac{c}{x}$ with $c<0$, and consider its period-n cycles as the parameter c varies. The homoclinic case is the parameter at which the critical value and the f...

conjecture | open | MathDB 页面 | 原文线索

V2-040017 | MATHDB-349229

Escape intervals between successive period cycles in the quadratic family

摘录 / EXCERPT: Let $F_c(x)=x^2+\dfrac{c}{x}$ with $c<0$, and let the critical point be the point whose orbit is being iterated as the parameter c varies. Consider successive period-n cycles i...

conjecture | open | MathDB 页面 | 原文线索

V2-040018 | MATHDB-349260

Berger's blender conjecture for entropy-dominated hyperbolic basic sets

摘录 / EXCERPT: Berger's conjecture. There exists a C^r-neighborhood $\mathcal{U}$ of $\mathcal{F}$ and an infinite-codimensional subset $\mathcal{N}\subset\mathcal{U}$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-040019 | MATHDB-349336

Admissibility characterization for pair large intersections

摘录 / EXCERPT: Pair characterization conjecture. The pair $\{\varphi,\psi\}$ has the large intersections property if and only if it is admissible.

conjecture | open | MathDB 页面 | 原文线索

V2-040020 | MATHDB-349337

Non-ergodic counterexample to large intersections for admissible pairs

摘录 / EXCERPT: Non-ergodic failure conjecture. There exist a necessarily non-ergodic measure-preserving system $(X,\mathcal B,\mu,(T_g)_{g\in G})$, a set $A\in\mathcal B$ with $\mu(A)>0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040021 | MATHDB-349339

Failure of large intersections for quadruple recurrence

摘录 / EXCERPT: Quadruple-recurrence conjecture. If $k\ge4$, then $\{\varphi_1,\dots,\varphi_k\}$ does not have the large intersections property.

conjecture | open | MathDB 页面 | 原文线索

V2-040022 | MATHDB-349340

Parallelogram characterization of large intersections for admissible triples

摘录 / EXCERPT: Parallelogram conjecture. The admissible triple (r,s,t) has the large intersections property if and only if $(0,r,s,t)$ forms a parallelogram.

conjecture | open | MathDB 页面 | 原文线索

V2-040023 | MATHDB-349342

Multiplicative recurrence along Pythagorean parametrizations

摘录 / EXCERPT: A measure preserving $(\mathbb{N},\times)$-system is a tuple $(X,\mathcal{B},\mu,T)$ in which $(X,\mathcal{B},\mu)$ is a probability space and T is a measure-preserving action of...

conjecture | open | MathDB 页面 | 原文线索

V2-040024 | MATHDB-349355

Oscillation conjecture for zero-energy Kepler-Heisenberg orbits

摘录 / EXCERPT: Let H be the Kepler-Heisenberg Hamiltonian, and let $z(t)$ denote the vertical coordinate of a zero-energy orbit. An orbit avoids the z-axis if it never meets the set where the...

conjecture | open | MathDB 页面 | 原文线索

V2-040025 | MATHDB-349356

Bifurcation conjecture for zero-energy orbits meeting the Heisenberg z-axis

摘录 / EXCERPT: Let J denote the dilational momentum and consider a zero-energy orbit meeting the z-axis. Bifurcation conjecture. If ... then the orbit has self-similarity like a generic $H=0$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-040026 | MATHDB-349369

Lyapunov instability transfer conjecture for boundary equilibria in Filippov systems

摘录 / EXCERPT: Let f be a Filippov system with boundary equilibrium $x^*={\bf 0}$, and let F be its reduced system. Assume the setting and hypotheses of the paper, including $c_1<0$. Lyapunov...

conjecture | open | MathDB 页面 | 原文线索

V2-040027 | MATHDB-349438

Symmetric rigidity conjecture for uniformly quasisymmetric circle endomorphisms

摘录 / EXCERPT: Let f and g be uniformly quasisymmetric circle endomorphisms of the same degree $d\geq 2$ with $f(1)=g(1)=1$, and suppose both preserve Lebesgue measure on the circle. Let h...

conjecture | open | MathDB 页面 | 原文线索

V2-040028 | MATHDB-349440

Continuity conjecture for Lyapunov exponents of volume-preserving Anosov diffeomorphisms on the three-torus

摘录 / EXCERPT: Continuity conjecture. The Lyapunov exponents vary continuously as functions of f in the space of $\mathcal{C}^2$ volume-preserving Anosov diffeomorphisms on $\mathbb T^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-040029 | MATHDB-349570

Periodicity of cubic vectors without Preliminary Stage 2

摘录 / EXCERPT: Periodicity conjecture without Preliminary Stage 2. Excluding Preliminary Stage 2 from the sin-squared algorithm will not affect periodicity for cubic vectors.

conjecture | open | MathDB 页面 | 原文线索

V2-040030 | MATHDB-349617

The generic Conley conjecture for Hamiltonian diffeomorphisms

摘录 / EXCERPT: Generic Conley conjecture. On every closed symplectic manifold, almost all Hamiltonian diffeomorphisms have infinitely many simple contractible periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-040031 | MATHDB-349745

Polterovich's lower-bound conjecture for the Poisson bracket invariant

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold. For a finite open cover $\mathcal{U}=\{U_i\}$ of M, let $e(U_i)$ denote the displacement energy of U_i and set ... The Poisson b...

conjecture | open | MathDB 页面 | 原文线索

V2-040032 | MATHDB-349801

The Flux Conjecture

摘录 / EXCERPT: Flux Conjecture. The subgroup $\operatorname{Ham}(M,\omega)$ is C^1 close in $\operatorname{Symp}(M,\omega)_0$; equivalently, it is C^1-closed there.

conjecture | open | MathDB 页面 | 原文线索

V2-040033 | MATHDB-349822

Novikov-Maltsev Hausdorff-dimension conjecture for chaotic sections

摘录 / EXCERPT: Let M be a fixed surface, and let H be a covector whose corresponding sections of the level surfaces associated with M may exhibit chaotic behavior. Novikov-Maltsev conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-040034 | MATHDB-349823

Tree conjecture for cyclic quadrilateral tiling billiards

摘录 / EXCERPT: Let γ be a periodic trajectory in a cyclic quadrilateral tiling, and let Ω_{γ} be the domain bounded by it. Tree conjecture for cyclic quadrilateral tiling billi...

conjecture | open | MathDB 页面 | 原文线索

V2-040035 | MATHDB-350097

All non-periodic points satisfy the Poisson limit law under exponential mixing

摘录 / EXCERPT: Exponential-mixing conjecture. If f is exponentially mixing, then the conclusion of Proposition (a) holds for all non-periodic points x.

conjecture | open | MathDB 页面 | 原文线索

V2-040036 | MATHDB-350109

Minimality criterion for Veech N-example flows and interval exchanges

摘录 / EXCERPT: Let X_N be the Veech N-example surface, with directional flow ϕ_t^{θ}, and let T_f and T_{1-f} be the associated interval exchange transformations. Write the slop...

conjecture | open | MathDB 页面 | 原文线索

V2-040037 | MATHDB-350151

Invariance of cosine sums for bicentric pedal polygons

摘录 / EXCERPT: A bicentric pedal polygon is obtained from a bicentric polygon by taking its pedal polygon with respect to a limiting point, denoted by ℓ_1 or ℓ_2. Cosine-sum invariance...

conjecture | open | MathDB 页面 | 原文线索

V2-040038 | MATHDB-350186

The carrying-capacity conjecture for balanced multi-patch logistic systems

摘录 / EXCERPT: Carrying-capacity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-040039 | MATHDB-350209

Hamiltonian structure conjecture for antisymmetric L-functions

摘录 / EXCERPT: Hamiltonian structure conjecture. There exist an energy $\mathcal{E}:\mathcal{Z}\to\mathbb{R}$ and a skew-symmetric operator...

conjecture | open | MathDB 页面 | 原文线索

V2-040040 | MATHDB-350257

The finite Riemann surface two-branch-point covering conjecture

摘录 / EXCERPT: Two-branch-point covering conjecture. On every finite Riemann surface U, there is a branched covering

conjecture | open | MathDB 页面 | 原文线索

V2-040041 | MATHDB-350287

Monotonicity conjecture for the principal Floquet exponent in dispersal-induced growth

摘录 / EXCERPT: Let $\Lambda(m,\omega)$ denote the relevant Floquet exponent for dispersal rate $m>0$ and temporal frequency $\omega>0$ in the time-periodic two-patch model, under $\bar{r}_1<0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-040042 | MATHDB-350288

Concavity conjecture for the dispersal-induced growth transition curve

摘录 / EXCERPT: Assume $\bar{r}_1<0$, $\bar{r}_2<0$, and $\chi>0$, and let $m^*>0$ be the threshold from the sink-sink result. For $m\in(0,m^*)$, let $\omega_{crit}(m)$ denote the transition frequ...

conjecture | open | MathDB 页面 | 原文线索

V2-040043 | MATHDB-350289

Infinite-slope conjecture for the dispersal-induced growth transition curve

摘录 / EXCERPT: Assume $\bar{r}_1<0$, $\bar{r}_2<0$, and $\chi>0$, and let $\omega_{crit}(m)$ be the transition frequency for $m\in(0,m^*)$. The curve is understood to extend to the origin with...

conjecture | open | MathDB 页面 | 原文线索

V2-040044 | MATHDB-350328

Dippolito's transverse-measure conjecture for exceptional minimal sets

摘录 / EXCERPT: Let $(M,\mathcal F)$ be a codimension-one foliation of a closed manifold that is transversely C^2, with exceptional minimal set Λ. Dippolito's conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-040045 | MATHDB-350329

Dippolito's semi-exceptional-leaf holonomy conjecture

摘录 / EXCERPT: Let $(M,\mathcal F)$ be a codimension-one foliation of a closed manifold that is transversely C^2, with exceptional minimal set Λ. A semi-exceptional leaf is a boundary l...

conjecture | open | MathDB 页面 | 原文线索

V2-040046 | MATHDB-350334

Jenkinson-Morris conjecture on periodic Lyapunov-minimizing measures

摘录 / EXCERPT: Jenkinson-Morris conjecture. For a generic $T\in\mathcal{E}^k$, the Lyapunov minimizing measure is unique and supported on a periodic orbit of T.

conjecture | open | MathDB 页面 | 原文线索

V2-040047 | MATHDB-350335

Generic periodic and full-support measures for upper Lyapunov exponents

摘录 / EXCERPT: Upper Lyapunov exponent maximizing-measure conjecture. If $k\geq 2$, then for a C^k-generic expanding self-map T, the upper Lyapunov exponent has a unique maximizing measure, s...

conjecture | open | MathDB 页面 | 原文线索

V2-040048 | MATHDB-350336

Existence and uniqueness conjecture for minimizing upper-Lyapunov measures

摘录 / EXCERPT: Minimizing upper-Lyapunov measure conjecture. For a generic expanding self-map T on a manifold M, the minimizing measure of the upper Lyapunov exponent λ_1 exists, is u...

conjecture | open | MathDB 页面 | 原文线索

V2-040049 | MATHDB-350386

The vanishing-density conjecture for pentagram-map dynamical domains

摘录 / EXCERPT: Vanishing-density conjecture. The dynamical domain has asymptotic density zero over the tower of finite fields:

conjecture | open | MathDB 页面 | 原文线索

V2-040050 | MATHDB-350400

Central limit conjecture for the Sudler product along Euler's number

摘录 / EXCERPT: Euler Sudler-product central limit conjecture. For every $t\in\mathbb R$,

conjecture | open | MathDB 页面 | 原文线索

V2-040051 | MATHDB-350405

Conjecture on infinitely many periodic magnetic geodesics near zero speed

摘录 / EXCERPT: Let (g,b) be a magnetic system on S^2 that is resonant at zero speed. A sequence of speeds ... should exist such that, for every $n\in\mathbb{N}$, there are infinitely many perio...

conjecture | open | MathDB 页面 | 原文线索

V2-040052 | MATHDB-350443

Slow-thermalization conjecture at the dynamical transition

摘录 / EXCERPT: Slow-thermalization conjecture. Starting from the uniform measure, Langevin dynamics takes exponential time to reach equilibrium for every $T<T_{sh}$: there is a constant $c>0$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-040053 | MATHDB-350465

Arnold-Thom conjecture for gradient flow lines

摘录 / EXCERPT: Let f be a real analytic function on an open set $U\subset\mathbb{R}^n$, and let $x(t)$ be a gradient flow line of f with a limit point $x_0\in U$. Arnold-Thom conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-040054 | MATHDB-350491

Finite decomposition complexity implies finite dynamical complexity for Cantor actions

摘录 / EXCERPT: Cantor-action complexity conjecture. If Γ has finite decomposition complexity, then $X\rtimes\Gamma$ has finite dynamical complexity for every free action of Γ on the...

conjecture | open | MathDB 页面 | 原文线索

V2-040055 | MATHDB-350525

Hakimi-Schmeichel-Thomassen conjecture on Hamiltonian cycles in planar triangulations

摘录 / EXCERPT: Let G be a 4-connected planar triangulation on n vertices. The Hakimi-Schmeichel-Thomassen conjecture. G has at least ... hamiltonian cycles, with equality if and only if G...

conjecture | open | MathDB 页面 | 原文线索

V2-040056 | MATHDB-350536

Jarque-Villadelprat conjecture on isochronous centers of real polynomial Hamiltonian systems

摘录 / EXCERPT: Let n be the degree parameter of the real polynomial Hamiltonian system ... has no isochronous centers. At present, this conjecture remains open; recent developments are reported b...

conjecture | open | MathDB 页面 | 原文线索

V2-040057 | MATHDB-350636

The inversion number of the Hamiltonian-path reversal tournament

摘录 / EXCERPT: Hamiltonian-path reversal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-040058 | MATHDB-350716

The spectral-radius conjecture for mutation-loop presentation matrices

摘录 / EXCERPT: Let $w \in \mathcal{X}_{(t_0)}(\mathbb{R}^{\mathrm{trop}})$, and let γ be a path representing a mutation loop. For the sign sequence $\boldsymbol{\epsilon}_\gamma(w)$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-040059 | MATHDB-350717

Ishibashi-Kano's sign-stability and North-South dynamics conjecture

摘录 / EXCERPT: Let $\boldsymbol{s}$ be any seed pattern and let ϕ be a mutation loop. Ishibashi-Kano's conjecture. The loop ϕ is uniformly sign-stable with cluster stretch factor greate...

conjecture | open | MathDB 页面 | 原文线索

V2-040060 | MATHDB-350718

Pseudo-Anosovness of uniformly sign-stable mutation loops

摘录 / EXCERPT: Let $t \in \mathbb{T}_I$, and let ϕ be a mutation loop that is uniformly sign-stable with cluster stretch factor $\lambda^{(t)}_\phi>1$ at t. Pseudo-Anosov conjecture. The f...

conjecture | open | MathDB 页面 | 原文线索

V2-040061 | MATHDB-350719

Herman's analytic positive entropy conjecture

摘录 / EXCERPT: Let the disk carry its standard analytic symplectic structure, and let an analytic symplectic perturbation of the identity mean an analytic symplectic diffeomorphism sufficiently c...

conjecture | open | MathDB 页面 | 原文线索

V2-040062 | MATHDB-350720

The analytic perturbation conjecture for elliptic symplectic surface maps

摘录 / EXCERPT: Let (S,Ω) be an analytic closed symplectic surface, and let $f\in \operatorname{Diff}^{\omega}_{\Omega}(S)$ be an analytic symplectic diffeomorphism displaying an elliptic p...

conjecture | open | MathDB 页面 | 原文线索

V2-040063 | MATHDB-350776

The Monodromy Conjecture for shift-polynomial braid groups

摘录 / EXCERPT: Let ${\mathcal S}_d$ be the parameter space of degree-d shift polynomials, let $\pi_1({\mathcal S}_d)$ be its fundamental group, and let $\hat{B}_d$ denote the image of the monod...

conjecture | open | MathDB 页面 | 原文线索

V2-040064 | MATHDB-350857

Big components big ends conjecture for tautological laminations

摘录 / EXCERPT: Let $mathcal S_q$ be the shift locus in a one-dimensional slice, and let $mathbb T_q$ be the tautological tree whose ends correspond to components of the complement of...

conjecture | open | MathDB 页面 | 原文线索

V2-040065 | MATHDB-350858

Big ends eventually periodic conjecture

摘录 / EXCERPT: Let $mathbb T_q$ be the tautological tree, whose ends determine sequences of ell-values and associated F-sequences. An end is big when ... and it is of type S when its associat...

conjecture | open | MathDB 页面 | 原文线索

V2-040066 | MATHDB-350878

The exact Hausdorff dimension formula for trajectory sets of equivalent g-templates

摘录 / EXCERPT: Exact dimension conjecture. For every $\Lambda\in X_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-040067 | MATHDB-350949

The type A enumeration conjecture for Coxeter Pop-Tsack torsing

摘录 / EXCERPT: Let A_{n-1} be the type A Coxeter system, let $4frak S_n$ be its permutation group, and let Pop_T be the Coxeter Pop-Tsack map. A permutation requires exactly k it...

conjecture | open | MathDB 页面 | 原文线索

V2-040068 | MATHDB-350962

Universality conjecture for nodal surplus distributions on large metric graphs

摘录 / EXCERPT: Let $\left\{\Gamma^{(\beta)}\right\}_{\beta\nearrow\infty}$ be any sequence of standard graphs, labeled by their first Betti numbers. For each $\Gamma^{(\beta)}$, choose arbitrary...

conjecture | open | MathDB 页面 | 原文线索

V2-040069 | MATHDB-350982

Mean-median map stabilization conjecture

摘录 / EXCERPT: Let $[x_1,\ldots,x_{n_0}]$ be a finite non-empty real multiset, and let the mean-median map adjoin x_{n+1} so that the arithmetic mean of the enlarged multiset equals the median...

conjecture | open | MathDB 页面 | 原文线索

V2-040070 | MATHDB-350983

Mean-median map median convergence conjecture

摘录 / EXCERPT: Let $[x_1,\ldots,x_{n_0}]$ be a finite non-empty real multiset, and let $(x_n)_{n=1}^{\infty}$ be its mean-median map orbit. Write $\mathcal{M}_n$ for the median of...

conjecture | open | MathDB 页面 | 原文线索

V2-040071 | MATHDB-350984

Unbounded transit-time conjecture for the mean-median map

摘录 / EXCERPT: For an initial set of size three, use affine invariance to reduce to the initial set $[0,x,1]$ with $x\in\left[\frac{1}{2},\frac{2}{3}\right]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040072 | MATHDB-350985

Continuity conjecture for the mean-median map limit function

摘录 / EXCERPT: For an initial set of size three, use affine invariance to reduce to the initial set $[0,x,1]$ with $x\in\left[\frac{1}{2},\frac{2}{3}\right]$. Let $m(x)\in\mathbb{R}$ be the limit...

conjecture | open | MathDB 页面 | 原文线索

V2-040073 | MATHDB-350996

Formal-group isogeny conjecture for stable p-adic dynamical systems

摘录 / EXCERPT: Formal-group isogeny conjecture. There exists a formal group F with coefficients in $\mathcal{O}_K$, two endomorphisms f_F and u_F of F, and a nonzero power series h such...

conjecture | open | MathDB 页面 | 原文线索

V2-040074 | MATHDB-351011

Almost-everywhere matching for odd-partial-quotient -continued fractions

摘录 / EXCERPT: Let g denote the parameter introduced in the paper, and consider the family of α-continued fraction maps with odd partial quotients for $\alpha\in[0,g]$. Almost-everywhere...

conjecture | refuted | MathDB 页面 | 原文线索

V2-040075 | MATHDB-351014

The C^r closing lemma for area-preserving diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold equipped with a symplectic form ω, and let $\operatorname{Diff}^{r}(M,\omega)$ be the space of C^{r} diffeomorphisms preserving ω, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-040076 | MATHDB-351102

Virasoro symmetry conjecture for deformations of the Principal Hierarchy

摘录 / EXCERPT: Let the Principal Hierarchy be associated to a semisimple Frobenius manifold, and consider a deformation of its bihamiltonian structure with constant central invariants. Denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-040077 | MATHDB-351215

Normality conjecture for self-similar measures

摘录 / EXCERPT: Let $\{\varphi_i(x)=\lambda_i x+t_i\}_{i\in \mathcal{I}}$ be an iterated function system of contracting similarities, and let $b\geq 2$ be an integer. Suppose that ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-040078 | MATHDB-351228

Hamiltonian lcs nonsqueezing conjecture

摘录 / EXCERPT: Hamiltonian lcs nonsqueezing conjecture. There is no compactly supported Hamiltonian lcs map

conjecture | open | MathDB 页面 | 原文线索

V2-040079 | MATHDB-351231

Equidistribution formulation of the dynamical Mordell-Lang conjecture

摘录 / EXCERPT: Dynamical Mordell-Lang equidistribution conjecture. If $x\in X_f(\mathbf{k})$ and $O_f(x)$ is Zariski dense in X, then

conjecture | open | MathDB 页面 | 原文线索

V2-040080 | MATHDB-351303

Conjecture on popular four-term polynomial configurations over rings of integers

摘录 / EXCERPT: Let K be a number field with ring of integers O_K. Let $r,s\thisattacksim$ be distinct and nonzero. For an O_K-system $(X,\mathcal{B},\mu,T)$, an O_K-valued intersective po...

conjecture | open | MathDB 页面 | 原文线索

V2-040081 | MATHDB-351308

The Shroer-Sauer-Ott-Yorke predictability conjecture for time-delay embeddings

摘录 / EXCERPT: Let $T\colon X \to X$ be a smooth diffeomorphism of a compact Riemannian manifold X, let $\Lambda \subset X$ be an attractor of T with a natural measure μ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-040082 | MATHDB-351309

Weak mixing sequence conjecture for functions satisfying the correlation condition

摘录 / EXCERPT: Let $(X,\mathscr{B},\mu,T)$ be a measure-preserving system, let U denote its Koopman operator, and let $f\in L^2(X,\mu)$ satisfy ... for every $g\in L^2(X,\mu)$. A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-040083 | MATHDB-351363

The one-symbol difference conjecture for codimension-one bifurcations

摘录 / EXCERPT: Consider periodic orbits continued from the anti-integrable limit of the three-dimensional quadratic map. Associate to each orbit its symbol sequence. One-symbol difference conject...

conjecture | open | MathDB 页面 | 原文线索

V2-040084 | MATHDB-351422

The hyperbolic closed geodesic conjecture for strict dilation surfaces

摘录 / EXCERPT: The hyperbolic closed geodesic conjecture. Every strict dilation surface contains a hyperbolic closed geodesic.

conjecture | open | MathDB 页面 | 原文线索

V2-040085 | MATHDB-351423

The density conjecture for hyperbolic closed geodesic directions on dilation surfaces

摘录 / EXCERPT: The density conjecture for hyperbolic closed geodesic directions. The set of directions of hyperbolic closed geodesics of X forms a dense subset of $\mathbb{S}^{1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-040086 | MATHDB-351424

The Morse-Smale directional-flow conjecture for strict dilation surfaces

摘录 / EXCERPT: The Morse-Smale directional-flow conjecture. There is an open dense subset U of $\mathbb{S}^{1}$ such that the directional flow of X along any direction of U is Morse-Smale.

conjecture | open | MathDB 页面 | 原文线索

V2-040087 | MATHDB-351492

Existence of a generalized Hopf bifurcation in the predator-prey model

摘录 / EXCERPT: Consider the predator-prey model given by Equation (Main), and suppose it admits a Hopf bifurcation as in the stated Hopf-bifurcation theorem for a given set of parameters. The coe...

conjecture | open | MathDB 页面 | 原文线索

V2-040088 | MATHDB-351493

Existence of a homoclinic bifurcation under predator harvesting

摘录 / EXCERPT: Consider the harvested predator-prey model given by Equation (HarvestingEqn), with all parameters fixed except the harvesting effort $q\geq 0$. A homoclinic loop is a trajectory th...

conjecture | open | MathDB 页面 | 原文线索

V2-040089 | MATHDB-351494

Harvesting-induced recovery from prey extinction

摘录 / EXCERPT: Consider the predator-prey model given by Equation (HarvestingEqn), and let q^* denote the harvesting threshold appearing in the conjecture. Here q is the predator-harvesting e...

conjecture | open | MathDB 页面 | 原文线索

V2-040090 | MATHDB-351623

Nonautonomous weak Markus-Yamabe conjecture

摘录 / EXCERPT: Let $f\colon\mathbb{R}^{+}\times\mathbb{R}^{n}\to\mathbb{R}^{n}$ be as in the system $\dot{x}=f(t,x)$, and assume that it satisfies (G1) and (G3*). Here (G3*) requires th...

conjecture | open | MathDB 页面 | 原文线索

V2-040091 | MATHDB-351699

Centralizer conjecture for uniquely ergodic Heisenberg nilflows

摘录 / EXCERPT: Let $H_3(\mathbb R)$ be the three-dimensional Heisenberg group, let $\Gamma< H_3(\mathbb R)$ be a cocompact lattice, and let ... Let $\phi_{\mathbb R}$ be a uniquely ergodic, or Diop...

conjecture | open | MathDB 页面 | 原文线索

V2-040092 | MATHDB-351772

The Mahler-measure-zero conjecture for finite-measure rank one maps

摘录 / EXCERPT: Mahler-measure-zero conjecture. The Mahler measure of the spectrum of T is zero.

conjecture | open | MathDB 页面 | 原文线索

V2-040093 | MATHDB-351843

The central-invariant criterion for linearizable Virasoro symmetries

摘录 / EXCERPT: Let a semisimple Frobenius manifold have an associated Principal Hierarchy, and let (P_0,P_1) be a tau-symmetric bihamiltonian deformation of that hierarchy. The deformation has...

conjecture | open | MathDB 页面 | 原文线索

V2-040094 | MATHDB-351951

Nonlinear Perron-Frobenius conjecture for statistical Hamiltonians

摘录 / EXCERPT: Nonlinear Perron-Frobenius conjecture. There exists a value $E^\star(\boldsymbol{\gamma})$ such that: for $E>E^\star(\boldsymbol{\gamma})$, all orbits tend towards the system bound...

conjecture | open | MathDB 页面 | 原文线索

V2-040095 | MATHDB-351961

Parity-dependent asymptotic domination of contractible Hamiltonian cycles

摘录 / EXCERPT: Let $h_m^c(n)$ and $h_m^{nc}(n)$ denote, respectively, the numbers of contractible and non-contractible Hamiltonian cycles in the grid cylinder graph $P_{m+1}\times C_n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040096 | MATHDB-351963

Multiple ergodic convergence for squared fractional powers

摘录 / EXCERPT: Let $(X,\mathcal{B},\mu,T)$ be a measure-preserving system, let $c_1,\ldots,c_k$ be distinct positive non-integers, and let $f_1,\ldots,f_k\in L^{\infty}(\mu)$. The multiple ergodi...

open_problem | open | MathDB 页面 | 原文线索

V2-040097 | MATHDB-351982

Liouville-component conjecture for semiclassical measures

摘录 / EXCERPT: Let (M,g) be a compact manifold with Anosov geodesic flow, let μ be a semiclassical measure on S^*M, and let μ_L denote the Liouville measure. Liouville-component conje...

conjecture | open | MathDB 页面 | 原文线索

V2-040098 | MATHDB-352046

Common-measure conjecture for rational maps

摘录 / EXCERPT: Common-measure conjecture. Either $d707$ is discrete, or Q and R share the same measure of maximal entropy.

conjecture | open | MathDB 页面 | 原文线索

V2-040099 | MATHDB-352057

Furstenberg's slicing conjecture

摘录 / EXCERPT: Let $X,Y\subset [0,1)$ be closed and invariant under T_a,T_b respectively, where $T_m(x)=mx\bmod 1$ is multiplication by m on the circle. Assume that $\log a/\log b$ is irratio...

conjecture | open | MathDB 页面 | 原文线索

V2-040100 | MATHDB-352151

Hooke billiard classification conjecture for smooth connected walls

摘录 / EXCERPT: A Hooke billiard is a mechanical billiard in $\mathbb{R}^2$ with Hooke potential $U=f r^2$, where r is the distance from a fixed center. Let its reflection wall be smooth and con...

conjecture | open | MathDB 页面 | 原文线索

V2-040101 | MATHDB-352203

Full-measure conjecture for closed geodesics on real affine structures

摘录 / EXCERPT: Let Σ be a closed surface, and consider the finite-dimensional parameter space of real affine structures on Σ. A real affine structure carries a closed geodesic if it...

conjecture | open | MathDB 页面 | 原文线索

V2-040102 | MATHDB-352280

Rigidity conjecture for circle maps of an oval

摘录 / EXCERPT: Let γ be an oval. For directions u and v, let $f_u,f_v:\gamma\to\gamma$ be the involutions induced by the family of parallel lines in those directions, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-040103 | MATHDB-352281

Möbius-conjugacy rigidity conjecture for an oval

摘录 / EXCERPT: Let γ be an oval. For points P,Q, let $f_P,f_Q:\gamma\to\gamma$ be the involutions associated with the points, and set $F_{P,Q}=f_P\circ f_Q$. Consider pairs for which eit...

conjecture | open | MathDB 页面 | 原文线索

V2-040104 | MATHDB-352290

Boutet de Monvel's conjecture on complexified Hamiltonian flows and Poisson kernels

摘录 / EXCERPT: Boutet de Monvel's conjecture. There is a maximal $\epsilon_0>0$ such that, for every $\epsilon\in(0,\epsilon_0)$, the extended flows define a real-analytic diffeomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-040105 | MATHDB-352359

Rolland Rigidity conjecture for self-diffeomorphisms of locally symmetric spaces

摘录 / EXCERPT: Suppose M is a locally symmetric space with natural Riemannian metric ρ and Levi-Civita connection $ abla$. Let f be a self-diffeomorphism of M, and let $(g_{1,b})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040106 | MATHDB-352462

Dichotomy conjecture for Hamiltonian conservation in Euler-α

摘录 / EXCERPT: Euler-α Hamiltonian dichotomy conjecture. Weak solutions obey the following dichotomy: if $s>1$, every weak solution in $C_t^0H_x^1\cap L_t^3B_{3,\infty}^s$ conserves...

conjecture | open | MathDB 页面 | 原文线索

V2-040107 | MATHDB-352486

The period-set containment conjecture for max-type equations

摘录 / EXCERPT: Period-set containment conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-040108 | MATHDB-352487

Grove-Ladas non-equilibrium limit conjecture for a max-ratio recurrence

摘录 / EXCERPT: Grove-Ladas conjecture. No positive non-equilibrium solution has a limit; the question is also to extend and generalize this assertion.

conjecture | open | MathDB 页面 | 原文线索

V2-040109 | MATHDB-352501

Positive-cone and optimality conjecture for the general first Physarum ansatz

摘录 / EXCERPT: Let $X(t)$ be the trajectory of the general first ansatz for the semidefinite program, and write $X(0)\succ F$ when $X(0)-F$ is positive definite. Let F be a feasible solution of...

conjecture | open | MathDB 页面 | 原文线索

V2-040110 | MATHDB-352614

Higher-dimensional Lyapunov-exponent conjecture for diffeomorphisms

摘录 / EXCERPT: Let M be a compact manifold of any dimension, let $f:M\circlearrowleft$ be a C^∞ diffeomorphism, and for k define ... where $\Lambda^k df$ is the action induced by f on...

conjecture | open | MathDB 页面 | 原文线索

V2-040111 | MATHDB-352692

Isolation conjecture for invariant subsets in the Apollonian orbifold

摘录 / EXCERPT: Let M_A be the Apollonian orbifold, let F^*M_A denote its relevant frame space, let H be the acting subgroup, and let $\mathcal{S}\subset F^*M_A$ be an H-invariant subset....

conjecture | open | MathDB 页面 | 原文线索

V2-040112 | MATHDB-352722

Local strong Birkhoff conjecture for almost every ellipse

摘录 / EXCERPT: Fix an integer $q_0\geq 3$ and $\delta>0$. There exist $m=m(q_0,\delta)$ and $n=n(q_0,\delta)$ in $\mathbb{N}$ such that, for every $0<e<1-\delta$ and $c\geq 0$, there is an...

conjecture | open | MathDB 页面 | 原文线索

V2-040113 | MATHDB-352723

Finite-parameter local strong Birkhoff conjecture for almost every ellipse

摘录 / EXCERPT: Fix an integer $q_0\geq 3$ and $\delta>0$. There exist $m=m(q_0,\delta)$ and $n=n(q_0,\delta)$ in $\mathbb{N}$ such that, for each but a finite number of pairs with $0<e<1-\delta$...

conjecture | open | MathDB 页面 | 原文线索

V2-040114 | MATHDB-352866

Natural-extension conjecture for simple alternating N-expansions

摘录 / EXCERPT: Natural-extension conjecture. The restriction $\overline{\mathcal{T}_T}:=\mathcal{T}_T|_X$ is bijective Lebesgue almost everywhere, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040115 | MATHDB-352913

Tabachnikov's projective Birkhoff conjecture

摘录 / EXCERPT: Let C be a strictly convex closed curve equipped with a projective billiard structure, and call the billiard Birkhoff integrable when its boundary and closed caustics form a foli...

conjecture | open | MathDB 页面 | 原文线索

V2-040116 | MATHDB-352914

Tabachnikov's dual Birkhoff conjecture for projective billiards

摘录 / EXCERPT: Let $\gamma\subset\mathbb{RP}^2$ be a strictly convex closed curve. A dual billiard structure on γ is a family of non-trivial projective involutions on the projective tangen...

conjecture | open | MathDB 页面 | 原文线索

V2-040117 | MATHDB-352925

Ellipses characterized by four-cusp caustics

摘录 / EXCERPT: Let γ be a planar billiard table that is not an ellipse. For a light source chosen inside γ, let Γ_n be the caustic by reflection after n reflections, with...

conjecture | open | MathDB 页面 | 原文线索

V2-040118 | MATHDB-352978

Global stability conjecture for the equilibrium of the delayed quadratic-rational equation

摘录 / EXCERPT: Consider the delayed quadratic-rational difference equation referred to as Eq., with parameter p, and let $\bar{y}$ denote its equilibrium point. Global stability conjecture. Man...

conjecture | open | MathDB 页面 | 原文线索

V2-040119 | MATHDB-353052

Accessibility implies ergodicity for partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let f be a C^{2} partially hyperbolic, volume-preserving diffeomorphism, and suppose that f has the essential accessibility property, meaning that every measurable set satura...

conjecture | open | MathDB 页面 | 原文线索

V2-040120 | MATHDB-353309

Nonconvergence conjecture for the double bracket flow in infinite spin systems

摘录 / EXCERPT: Consider a d-dimensional hypercubic lattice with translation-invariant Hamiltonians H and V. For a sequence of lattices of increasing size, let $H(B)$ be the solution of the...

conjecture | open | MathDB 页面 | 原文线索

V2-040121 | MATHDB-353347

Arithmetic oscillation disjointness conjecture for zero-entropy torus flows

摘录 / EXCERPT: Let ${\bf c}=(c_n)_{n\in\mathbb N}$ be a sequence of complex numbers. For every real-coefficient polynomial P of degree at most d and every pair of integers $0\leq l<k$, suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-040122 | MATHDB-353397

Relative dynamical Bogomolov conjecture for split polarized endomorphisms

摘录 / EXCERPT: Relative dynamical Bogomolov conjecture. There exists $\epsilon=\epsilon(\mathbf{F},\mathbf{V})>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040123 | MATHDB-353440

The L1 convergence conjecture for destinations of flip processes

摘录 / EXCERPT: Let $\mathcal{R}$ be a flip process, let U be a graphon, and let W be a destination of U, meaning that $\Phi^{t}U \xrightarrow{\square} W$ as $t\to\infty$. Destination conver...

conjecture | open | MathDB 页面 | 原文线索

V2-040124 | MATHDB-353533

Zero-rotation locus conjecture for the circumcenter map of regular polygons

摘录 / EXCERPT: Let $\mathcal{P}$ be a regular n-gon, with $n\geq 3$, and let $\mathcal{P}'=\mathcal{C}_M^n(\mathcal{P})$. Write α for the angle of rotation in the similarity taking...

conjecture | open | MathDB 页面 | 原文线索

V2-040125 | MATHDB-353534

Regularity maximizes area-contracting regions for the circumcenter map

摘录 / EXCERPT: Let $\mathcal{Q}$ be a simple n-gon, meaning that it has no self-intersections, and consider the regions of points M for which the circumcenter map produces area contraction. R...

conjecture | open | MathDB 页面 | 原文线索

V2-040126 | MATHDB-353543

Non-degenerate unfolding conjecture for homoclinic tangencies of Beltrami fields

摘录 / EXCERPT: Let $X\in\mathscr B(\mathbb R^3)$ be a Beltrami field, and let a periodic saddle orbit of X have a homoclinic tangency. A homoclinic tangency is non-degenerately unfolded in...

conjecture | open | MathDB 页面 | 原文线索

V2-040127 | MATHDB-353581

Conjecture on the perturbed threshold for the third power of a Hamilton cycle at density one quarter

摘录 / EXCERPT: Threshold conjecture. The perturbed threshold satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-040128 | MATHDB-353589

Existence of disentangling QCAs for bulk-trivial commuting projector Hamiltonians

摘录 / EXCERPT: Disentangling-QCA conjecture. For every such Hamiltonian H, there exists a QCA that disentangles its ground state. A pair of disentangling QCAs are equivalent if and only if the...

conjecture | open | MathDB 页面 | 原文线索

V2-040129 | MATHDB-353641

Strong Generalised Weinstein Conjecture for contact foliations

摘录 / EXCERPT: Strong Generalised Weinstein Conjecture. Every contact foliation has a closed leaf, that is, a leaf homeomorphic to a torus $\mathbb{T}^q$.

conjecture | open | MathDB 页面 | 原文线索

V2-040130 | MATHDB-353693

Quantum Sinai theorem conjecture for environmental entropy

摘录 / EXCERPT: Let $\mathcal{W}:\mu\mapsto\mu'$ be a quantum evolutionary process, with quantum relative fitness operator $U:=\frac{1}{\overline{W}}\mathcal{W}^\dagger(\operatorname{Id}')$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040131 | MATHDB-353717

Orbifold tangent-cone conjecture for convex bodies with continuous billiard evolution

摘录 / EXCERPT: Orbifold tangent-cone conjecture. If K admits a continuous billiard evolution, or a quasigeodesic flow, then all its tangent cones are orbifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-040132 | MATHDB-353739

Conjecture on approximate invariant manifolds and chaotic dynamics in the three-body problem

摘录 / EXCERPT: Let ${\cal M}({\cal E})=\{({\rm r},{\rm G},{\rm g}):{\rm E}({\rm r},{\rm G},{\rm g})={\cal E}\}$ be the lifted level manifolds of the Euler integral, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040133 | MATHDB-353765

Second-order ground-state energy conjecture for the Gross-Pitaevskii three-body Bose gas

摘录 / EXCERPT: Let $H_N^{\rm GP}$ be the Gross-Pitaevskii Hamiltonian, let $E_{\rm GP}(N)$ denote its ground-state energy, and let $e_{\rm GP}$ be the minimum of the corresponding Gross-Pitaevski...

conjecture | open | MathDB 页面 | 原文线索

V2-040134 | MATHDB-353767

Gross-Pitaevskii limit conjecture for combined two- and three-body interactions

摘录 / EXCERPT: Let $H_N^{\rm Comb}$ be the Hamiltonian with two-body potential W at scaling parameter $\beta=1$ and three-body potential V. Let $e^{\rm Comb}_{\rm GP}(b_1,b_2)$ be the minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-040135 | MATHDB-353797

Periodic striped ground-state conjecture for the two-dimensional dipolar Ising model

摘录 / EXCERPT: Consider the two-dimensional Ising model on the periodic torus ... with nearest-neighbor ferromagnetic coupling J, Hamiltonian $\mathcal H_L$, periodic striped configurations...

conjecture | open | MathDB 页面 | 原文线索

V2-040136 | MATHDB-353888

Dynamical uniform boundedness conjecture for function fields

摘录 / EXCERPT: Let k be a field, let K be the function field of an integral curve over k, and let $\operatorname{Rat}_d(\overline K)$ denote the degree-d rational functions over an algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-040137 | MATHDB-353989

Arc-disjoint hamiltonian paths in Cartesian products of directed cycles

摘录 / EXCERPT: Arc-disjoint hamiltonian paths conjecture. The Cartesian product $C_1\mathbin{\Box}C_2\mathbin{\Box}\cdots\mathbin{\Box}C_r$ has two arc-disjoint hamiltonian paths.

conjecture | open | MathDB 页面 | 原文线索

V2-040138 | MATHDB-353991

Alspach's decomposition conjecture for abelian Cayley graphs

摘录 / EXCERPT: Alspach's conjecture. The Cayley graph $\mathop{\mathsf{Cay}}(G;S)$ can be decomposed into hamiltonian cycles, together with a 1-factor if its valency is odd.

conjecture | open | MathDB 页面 | 原文线索

V2-040139 | MATHDB-354013

Conjectured optimal convergence criterion for Hardy-field functions

摘录 / EXCERPT: Conjectured optimal convergence criterion. Under this assumption, the averages

conjecture | open | MathDB 页面 | 原文线索

V2-040140 | MATHDB-354034

The shrinking-target conjecture for the doubling map

摘录 / EXCERPT: Let $T_2:\mathbb{R}/\mathbb{Z}\to\mathbb{R}/\mathbb{Z}$ be the doubling map, let μ denote Lebesgue measure on $\mathbb{R}/\mathbb{Z}$, let $\psi:\mathbb{N}\to[0,\infty)$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-040141 | MATHDB-354042

Bounded Lagrangians are spectrally bounded

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold and let L be an exact Lagrangian. For any Hamiltonianly isotopic exact Lagrangian L_1 such that $\gamma(L,L_1)$ is sufficiently small,...

conjecture | open | MathDB 页面 | 原文线索

V2-040142 | MATHDB-354048

Computational hardness of constructing Type III dynamics for games

摘录 / EXCERPT: Let g be a normal-form game. A game is degenerate if it has a non-isolated set of Nash equilibria, and a dynamical system is of Type III as defined in the paper. Computational ha...

conjecture | open | MathDB 页面 | 原文线索

V2-040143 | MATHDB-354054

Katok-Spatzier conjecture on algebraic models for higher-rank Anosov actions

摘录 / EXCERPT: Katok-Spatzier conjecture. Up to finite cover, the action is C^∞ conjugate to an algebraic action.

conjecture | open | MathDB 页面 | 原文线索

V2-040144 | MATHDB-354074

Boston-Jones model conjecture outside exceptional orbit types

摘录 / EXCERPT: Boston-Jones exceptional-orbit conjecture. The distribution of the factorization process of f converges to limiting distributions arising from the Boston-Jones model unless f h...

conjecture | open | MathDB 页面 | 原文线索

V2-040145 | MATHDB-354196

The smoothability conjecture for finitely generated PL groups

摘录 / EXCERPT: Let $G\le \operatorname{\mathsf{PL}}([0,1])$ be a finitely generated group of piecewise linear homeomorphisms of $[0,1]$. Its standard action is assumed to be conjugate to a...

conjecture | open | MathDB 页面 | 原文线索

V2-040146 | MATHDB-354232

Generic bounded-complexity conjecture for dispersing billiard tables

摘录 / EXCERPT: Bounded-complexity conjecture. For typical finite horizon dispersing billiard configurations, the sequence of complexities K_n is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-040147 | MATHDB-354309

Finite-order conjecture for reciprocal and twinning involutions

摘录 / EXCERPT: Let the reciprocal and twinning operations be the two involutions on tetrahedra defined in the surrounding discussion. Their compositions in either order are not themselves involut...

conjecture | open | MathDB 页面 | 原文线索

V2-040148 | MATHDB-354322

Brin's conjecture on ergodicity of the frame flow

摘录 / EXCERPT: Brin's conjecture. If (M,g) is $delta$-pinched for some $delta>0.25$, then the frame flow is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-040149 | MATHDB-354359

Upper semicontinuity conjecture for singular physical measures

摘录 / EXCERPT: Let $\mathcal{V}$ be the open family of C^2 vector fields admitting a singular-hyperbolic attracting set Λ. For each $G\in\mathcal{V}$, let $s(G)\in\mathbb{Z}_0^+$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-040150 | MATHDB-354360

Generic strict inequality for physical measures and Lorenz-like singularities

摘录 / EXCERPT: Consider the family of 3-vector fields G of class C^2 exhibiting a singular-hyperbolic attracting set containing some Lorenz-like singularity. Write s for the number of erg...

conjecture | open | MathDB 页面 | 原文线索

V2-040151 | MATHDB-354361

Generalized Lorenz-like extension to sectional-hyperbolic attracting sets

摘录 / EXCERPT: Let a sectional-hyperbolic attracting set be an attracting set with stable and center-unstable dimensions satisfying $d_s\geq 1$ and $d_{cu}\geq 2$, and replace Lorenz-like singula...

conjecture | open | MathDB 页面 | 原文线索

V2-040152 | MATHDB-354362

Morales theorem extension for sectional-hyperbolic attractors

摘录 / EXCERPT: Let a sectional-hyperbolic attracting set be C^1 close to a sectional-hyperbolic attractor, with no dimensional restrictions; equivalently, allow any combination of stable and ce...

conjecture | open | MathDB 页面 | 原文线索

V2-040153 | MATHDB-354363

Generic C^1 bounds for physical and SRB measures

摘录 / EXCERPT: Consider generic C^1 vector fields having a sectional-hyperbolic attracting set. Let the bounds referred to in Theorem and Corollary be the corresponding bounds for the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-040154 | MATHDB-354469

Uniqueness and terminal attraction of upper transition attractors

摘录 / EXCERPT: Let F_t be a one-parameter iterated function system satisfying properties (H1), (H2), and (H3) of Section 4. If $A_{\bullet}=A^{\bullet}$ for some upper transition attractor of...

conjecture | open | MathDB 页面 | 原文线索

V2-040155 | MATHDB-354535

Weak non-uniform sectional expansion conjecture for physical and SRB measures

摘录 / EXCERPT: Let G be a C^2 vector field with a partially hyperbolic attracting set $\Lambda=\Lambda_G(U)$, and let $\Omega\subset U$ be a positive-volume subset on which the weak non-unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-040156 | MATHDB-354593

Hasselblatt-Schmeling dimension conjecture for hyperbolic sets

摘录 / EXCERPT: Let a hyperbolic set have stable and unstable slices, and interpret “fractal dimension” as either Hausdorff dimension or upper box dimension. Hasselblatt-Schmeling conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-040157 | MATHDB-354626

Bogatyi-Shavgulidze covering-system periodic point conjecture

摘录 / EXCERPT: Bogatyi-Shavgulidze conjecture. For every covering system $(I_1,\ldots,I_k;f)$, there exists $x_0\in I_1\cup\cdots\cup I_k$ and an integer $l\leq k$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040158 | MATHDB-354716

Island accumulation conjecture for the frontier of the parameter region

摘录 / EXCERPT: Island accumulation conjecture. Every point in the frontier of $\mathcal R$ is the limit of islands in $\mathcal W$ of diameter going to zero.

conjecture | open | MathDB 页面 | 原文线索

V2-040159 | MATHDB-354793

The BKK-bound growth conjecture for second-order averaged functions

摘录 / EXCERPT: Let $H_2(n,m)$ denote the quantity of second-order averaged functions under consideration, and let $ u_{n,m}$ be its BKK bound. BKK-bound growth conjecture. ... This conjecture is mo...

conjecture | open | MathDB 页面 | 原文线索

V2-040160 | MATHDB-354885

Unique ergodicity conjecture for the Jeandel-Rao subshift

摘录 / EXCERPT: Let ... is uniquely ergodic. The preceding discussion states this as an unresolved conjecture, concerning the pattern frequencies of configurations even though the shift is not minim...

conjecture | open | MathDB 页面 | 原文线索

V2-040161 | MATHDB-354897

Global convergence conjecture for the dynamic Kuramoto model

摘录 / EXCERPT: Let the dynamic Kuramoto model be given by with zero natural frequencies, and let the phase vector start from a deterministic point outside the set of spurious fixed points...

conjecture | open | MathDB 页面 | 原文线索

V2-040162 | MATHDB-354923

The extension of the Fregier circle area proposition to all longer orbits

摘录 / EXCERPT: Let n be an integer with $n>3$, and consider the proposition immediately preceding this statement, which asserts that the sum of the areas of the Frégier circles associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-040163 | MATHDB-355010

The four-dimensional discretized rotation periodicity conjecture

摘录 / EXCERPT: Let $c_1=\alpha_1+\alpha_2$ and $c_2=\alpha_1\alpha_2+2$, where $\alpha_1,\alpha_2\in(-2,2)$ and $\alpha_1\ne\alpha_2$. For an integer sequence (k_n) satisfying ... choose any init...

conjecture | open | MathDB 页面 | 原文线索

V2-040164 | MATHDB-355011

The almost-everywhere origin periodicity conjecture for four-dimensional discretized rotations

摘录 / EXCERPT: Let $c_1=\alpha_1+\alpha_2$ and $c_2=\alpha_1\alpha_2+2$, with $(\alpha_1,\alpha_2)\in(-2,2)^2$, and let (k_n) satisfy ... Start the recurrence at the origin,...

conjecture | open | MathDB 页面 | 原文线索

V2-040165 | MATHDB-355030

Convergence to steady states in symmetric nonlocal advection-diffusion systems

摘录 / EXCERPT: Steady-state convergence conjecture. The solution $\mathbf{u}$ converges towards a steady state.

conjecture | open | MathDB 页面 | 原文线索

V2-040166 | MATHDB-355042

The intermediate-regime conjecture for Cramér distance discretizations

摘录 / EXCERPT: Let $\tilde \mu_1,\dots,\tilde \mu_M$ be probability measures on $\mathbf{S}^1$, with the density of $\tilde\mu_m$ given by $a_m/\int_{\mathbf{S}^1}a_m$. Let $K\in\mathbf{N}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-040167 | MATHDB-355068

Conditional mixing conjecture for Lozi maps

摘录 / EXCERPT: Let $f(x,y)=(1+y-a|x|,bx)$ be a Lozi map with parameters (a,b) in the chaotic regime described above, and let $\rho(\cdot\mid x_0)$ denote the conditional SRB measure on the leve...

conjecture | open | MathDB 页面 | 原文线索

V2-040168 | MATHDB-355069

Conditional mixing conjecture for analytic Anosov surface diffeomorphisms

摘录 / EXCERPT: Let f be an analytic Anosov diffeomorphism of a compact surface, and let H be an analytic function on the surface whose zero level set has no critical points. Conditional measu...

conjecture | open | MathDB 页面 | 原文线索

V2-040169 | MATHDB-355070

Conditional mixing conjecture for slice measures of piecewise hyperbolic maps

摘录 / EXCERPT: Let f be the piecewise hyperbolic map under consideration, let ρ be its SRB measure, and let $\rho_{\mathcal S}$ be the slice measure on the singular set. Write $\mathcal M$...

conjecture | open | MathDB 页面 | 原文线索

V2-040170 | MATHDB-355075

Fried's genus-one Birkhoff section conjecture for Anosov flows

摘录 / EXCERPT: Let ϕ be an Anosov flow whose stable and unstable foliations are orientable. A Birkhoff section is a compact surface transverse to the flow in its interior, with boundary cons...

conjecture | open | MathDB 页面 | 原文线索

V2-040171 | MATHDB-355084

Dynamical Lang conjecture for rational maps

摘录 / EXCERPT: Fix an embedding of the moduli space $\mathscr{M}_d$ in projective space, with associated height $h_{\mathscr{M}_d}$. Let K be a number field and $d\geq2$. Dynamical Lang conject...

conjecture | open | MathDB 页面 | 原文线索

V2-040172 | MATHDB-355179

Ih's dynamical S-integral preperiodic points conjecture

摘录 / EXCERPT: Let K be a number field, let S be a finite set of places containing all the archimedean ones, let $f\colon \mathbb{P}^1\to\mathbb{P}^1$ be a rational function of degree at leas...

conjecture | open | MathDB 页面 | 原文线索

V2-040173 | MATHDB-355206

The ionization conjecture for atoms

摘录 / EXCERPT: Let H_N be the N-electron atomic Hamiltonian with nuclear charge $Z\in\mathbb{N}$, and let $E_N=\inf\sigma(H_N)$. Denote by $N_c=N_c(Z)$ the largest number of electrons for whi...

conjecture | open | MathDB 页面 | 原文线索

V2-040174 | MATHDB-355207

Strong ionization conjecture for atomic Hamiltonians

摘录 / EXCERPT: Let H_N be the N-electron atomic Hamiltonian with nuclear charge $Z\in\mathbb{N}$. Strong ionization conjecture. There is a universal constant $C>0$ such that, whenever $N>Z+C$...

conjecture | open | MathDB 页面 | 原文线索

V2-040175 | MATHDB-355208

Dynamical critical-mass conjecture for the Hartree equation

摘录 / EXCERPT: Dynamical Hartree conjecture. For every initial state $u_0\in H^1(\mathbb{R}^3)$ and every $R>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-040176 | MATHDB-355209

Dynamical neutrality conjecture for Thomas-Fermi theory

摘录 / EXCERPT: Thomas-Fermi dynamical neutrality conjecture. For every initial state (φ_0,ρ_0) satisfying $0\leqslant\rho_0\in L^1(\mathbb{R}^3)\cap L^{5/3}(\mathbb{R}^3)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040177 | MATHDB-355210

Universality conjecture for many-body atomic densities

摘录 / EXCERPT: Let ρ_Z be the one-body density of a ground state of the many-body atomic Hamiltonian with nuclear charge Z and $N=N_c(Z)$ electrons. Universality conjecture. Along subseque...

conjecture | open | MathDB 页面 | 原文线索

V2-040178 | MATHDB-355353

Hamilton-cycle time-and-budget trade-off conjecture

摘录 / EXCERPT: Let a (t,b)-strategy B of Builder be a strategy that observes at most t arriving edges and purchases at most b of them. A graph is Hamiltonian if it contains a Hamilton cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-040179 | MATHDB-355357

Continuity conjecture for the Rényi entropy rate at order one

摘录 / EXCERPT: Continuity conjecture. For every stationary ergodic process Z,

conjecture | open | MathDB 页面 | 原文线索

V2-040180 | MATHDB-355358

Non-strict sparse coupling conjecture for weak Markov blankets

摘录 / EXCERPT: Let J be the Jacobian of a multiscale, coupled random dynamical system, with state components $\boldsymbol{\eta}$ and $\boldsymbol{\mu}$, and let H denote its Hessian form. A w...

conjecture | open | MathDB 页面 | 原文线索

V2-040181 | MATHDB-355424

Continuity of thermal operations with respect to inverse temperature

摘录 / EXCERPT: Let H_S be a Hermitian Hamiltonian on an n-dimensional system, so that $H_S\in i\mathfrak u(n)$. For each temperature $T\in(0,\infty]$, let $\overline{\mathsf{TO}(H_S,T)}$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-040182 | MATHDB-355439

The higher-order recurrence conjecture for Bohr-Hamming balls

摘录 / EXCERPT: For $r\in\mathbb N$, let $BH:=BH(\bm\beta,\bm y,k,\varepsilon)$ be a proper Bohr-Hamming Ball, with $\varepsilon>0$ and $y\in\mathbb T^r$. Define ... A set is (δ,k)-recurrent...

conjecture | open | MathDB 页面 | 原文线索

V2-040183 | MATHDB-355508

Exponential complexity conjecture for local maxima of mixed p-spin Hamiltonians

摘录 / EXCERPT: Let H_N be a mixed p-spin Hamiltonian on the sphere, let $\mathcal{N}(D)$ denote the number of its local maxima on S_{N-1} whose normalized radial derivative lies in D, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040184 | MATHDB-355517

Additional-food predator explosion conjecture for general predator-pest models

摘录 / EXCERPT: Consider the general predator-pest model ... where x is the pest population, y is the predator population, ξ is the quantity of additional food, and f is a pest-dependent f...

conjecture | open | MathDB 页面 | 原文线索

V2-040185 | MATHDB-355547

The quotient groupoid identification for non-periodic Deaconu systems

摘录 / EXCERPT: Quotient groupoid identification. If T is non-periodic, then the map

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V2-040186 | MATHDB-355724

Conjecture on periodic orbits in no-slip billiards with external force

摘录 / EXCERPT: Periodic-orbit conjecture. For a no-slip billiard with an external force, 2-periodic non-path-reversing orbits exist between any two accessible points on an arbitrary table, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040187 | MATHDB-355725

Conjecture on stable periodic orbits in forced no-slip billiards

摘录 / EXCERPT: Stable-orbit conjecture. Stable periodic orbits exist between circular scatterers for no-slip billiards with force, creating nontrivial invariant regions.

conjecture | open | MathDB 页面 | 原文线索

V2-040188 | MATHDB-355726

Launch-angle dependence conjecture for forced no-slip periodic orbits

摘录 / EXCERPT: Launch-angle dependence conjecture. The stability of path-reversing 2-periodic orbits in no-slip billiards with force depends on the launch angle.

conjecture | open | MathDB 页面 | 原文线索

V2-040189 | MATHDB-355774

Locally conformally symplectic non-squeezing conjecture

摘录 / EXCERPT: Locally conformally symplectic non-squeezing conjecture. If $R\geq 1$, there is no compactly supported Hamiltonian lcs map

conjecture | open | MathDB 页面 | 原文线索

V2-040190 | MATHDB-355854

Analytic-conjugacy rigidity conjecture for three-scatterer billiards

摘录 / EXCERPT: Analytic-conjugacy rigidity conjecture. Any two 3-scatterer systems that are analytically conjugate to one another in a neighborhood of the homoclinic orbit must be isometric. Eq...

conjecture | open | MathDB 页面 | 原文线索

V2-040191 | MATHDB-355917

The quadric threefold dynamical-degree and periodic-point conjecture

摘录 / EXCERPT: Let $F:\mathbb{C}^4\rightarrow\mathbb{C}^4$ be the birational map ... let $\phi(x_1,x_2,x_3,x_4)=x_1x_4-x_2x_3$, and for generic $c\in\mathbb{C}$ let $X_c\subset\mathbb{P}^4$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-040192 | MATHDB-355970

Bellow-Furstenberg pointwise convergence conjecture for multi-parameter polynomial averages

摘录 / EXCERPT: Bellow-Furstenberg conjecture. For μ-almost every $x\in X$, these averages converge as $\min\{M_1,\ldots,M_k\}\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-040193 | MATHDB-356029

Number operator representation conjecture for quantized time evolution

摘录 / EXCERPT: Let $f:\mathbb{C}^{m}\to\mathbb{R}$ be a classical Hamiltonian, let $\mathcal{E}_{b}(f(\vec{Z}))$ be its quantization, and define ... Let N_b be the number operator and E_b the p...

conjecture | open | MathDB 页面 | 原文线索

V2-040194 | MATHDB-356062

The spectral-gap obstruction conjecture for non-uniformly hyperbolic local diffeomorphisms

摘录 / EXCERPT: Let $f:M\rightarrow M$ be a C^{2} local diffeomorphism on a Riemannian manifold M. For a potential ϕ, define the transfer operator by ... A suitable Banach space may consist...

conjecture | open | MathDB 页面 | 原文线索

V2-040195 | MATHDB-356081

Planar outer billiard trajectories characterization of symplectic balls

摘录 / EXCERPT: Planar outer billiard conjecture. All outer billiard trajectories of K are planar if and only if K is a ball up to an affine symplectomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-040196 | MATHDB-356086

Fyodorov-Keating conjecture on thick points of CUE characteristic polynomials

摘录 / EXCERPT: Let U_N be a random matrix distributed according to the Haar measure on the group of $N\times N$ unitary matrices, and let $p_N(\theta)=\det(1-U_N e^{-i\theta})$ be its character...

conjecture | open | MathDB 页面 | 原文线索

V2-040197 | MATHDB-356098

Approximation conjecture for steady states in the sigmoidal bounded-confidence model

摘录 / EXCERPT: Approximation conjecture. There exist $\mathbf{x}'\in C_\delta(\mathbf{x})$, a sequence $\{\gamma^{(\ell)}\}_\ell$, and a sequence $\{\mathbf{x}^{(\ell)}\}_\ell$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040198 | MATHDB-356109

The lower bound for the Mina-margin endpoint

摘录 / EXCERPT: Let λ be the constant for which the set of Mina-margin values of positive ABMN solutions is $[\lambda,\lambda^{-1}]$, with the established upper bound...

conjecture | open | MathDB 页面 | 原文线索

V2-040199 | MATHDB-356112

Coexistence conjecture for quasi-periodic motions and hyperbolic tori

摘录 / EXCERPT: In the outer, retrograde configuration, let $\mathcal{D}_{{{{{{{{{\rm t} t} t} t} t} t} t} t} s}$ be the 8-dimensional phase-space region almost completely filled with a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-040200 | MATHDB-356188

Donoso-Koutsogiannis-Sun joint ergodicity conjecture for polynomial iterates

摘录 / EXCERPT: Let $(X,\mathcal X,\mu)$ be a probability space, let $T_1,\ldots,T_\ell$ be commuting measure-preserving transformations, and let $p_1,\ldots,p_\ell\in\mathbb Z[n]$. A sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-040201 | MATHDB-356205

Weyl's conjecture for the Dirichlet Laplace spectrum

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a bounded domain with C^∞ boundary. Let $N(\lambda)=|\{\lambda_i\leq\lambda\}|$ count the Dirichlet Laplace eigenvalues, and let...

conjecture | solved | MathDB 页面 | 原文线索

V2-040202 | MATHDB-356211

Khinchin's Cesàro convergence question for periodic indicator functions

摘录 / EXCERPT: Let $E\subset(0,1)$ be a measurable set, and define $f(x)=\chi_E(\{x\})$, where $\{x\}$ denotes the fractional part of x. Khinchin's question. Is it true that for almost every...

open_problem | open | MathDB 页面 | 原文线索

V2-040203 | MATHDB-356219

Equality of real and complex Markov-Bernstein constants

摘录 / EXCERPT: The equality conjecture. For all $\ell,n\in{\mathbb N}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-040204 | MATHDB-356286

The C^0-flux conjecture for Hamiltonian diffeomorphisms

摘录 / EXCERPT: Let (M,ω) be a closed connected symplectic manifold. Denote by $\operatorname{Symp}(M,\omega)$ the group of symplectomorphisms, by $\operatorname{Symp}_0(M,\omega)$ its iden...

conjecture | open | MathDB 页面 | 原文线索

V2-040205 | MATHDB-356319

Nonuniform nonautonomous Markus-Yamabe conjecture

摘录 / EXCERPT: Nonuniform nonautonomous Markus-Yamabe conjecture. Under these conditions, the trivial solution of the nonlinear system is globally nonuniformly asymptotically stable.

conjecture | open | MathDB 页面 | 原文线索

V2-040206 | MATHDB-356357

Birkhoff rigidity conjecture for analytic symplectic pseudo-rotations

摘录 / EXCERPT: Let S be an orientable analytic surface. An analytic symplectomorphism of the sphere with only two fixed points and no other periodic point, or of a compact cylinder without peri...

conjecture | open | MathDB 页面 | 原文线索

V2-040207 | MATHDB-356478

Positive critical infection-rate conjecture for dynamical long-range percolation

摘录 / EXCERPT: Positive critical-rate conjecture. For every $\gamma>0$ and $q\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-040208 | MATHDB-356546

Arnold's transition-chain conjecture for general Hamiltonian systems

摘录 / EXCERPT: A transition chain is a chain of hyperbolic invariant tori whose stable and unstable manifolds can be shadowed by drifting orbits in a near-integrable Hamiltonian system. Arnold's...

conjecture | open | MathDB 页面 | 原文线索

V2-040209 | MATHDB-356547

Herman's global instability conjecture for the N-body problem

摘录 / EXCERPT: Consider the spatial N-body problem with $N\geq 3$. Fix the center of mass at the origin. For each energy level e, let $H_e^{-1}(0)$ denote the corresponding energy surface, an...

conjecture | open | MathDB 页面 | 原文线索

V2-040210 | MATHDB-356585

Smoothness conjecture for the Poincaré problem on convex simple domains

摘录 / EXCERPT: Smoothness conjecture. The solution $u(t)$ of the Poincaré problem is smooth.

conjecture | open | MathDB 页面 | 原文线索

V2-040211 | MATHDB-356802

One Force -- One Solution principle for the open KPZ equation

摘录 / EXCERPT: Fix $u,v\in\R$. Let $h_0,\tilde h_0\in\operatorname{C}([0,1])$ be arbitrary random functions on the same probability space, and let ξ be white noise on $\mathbb{R}\times[0,1]$....

conjecture | open | MathDB 页面 | 原文线索

V2-040212 | MATHDB-356859

Gishboliner-Krivelevich-Michaeli oriented discrepancy conjecture

摘录 / EXCERPT: Let G be an oriented graph on $n\geq 3$ vertices, and let $\delta(G)$ denote its minimum degree. For a Hamilton cycle C in G, let $\sigma_{\max}(C)$ be the larger of the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-040213 | MATHDB-356920

The crossing conjecture for longest Hamiltonian cycles

摘录 / EXCERPT: Let S be a point set in the plane in general position, and let a longest Hamiltonian cycle be a Hamiltonian cycle on S whose total edge length is maximal. Longest-cycle crossin...

conjecture | open | MathDB 页面 | 原文线索

V2-040214 | MATHDB-356974

The zero-entropy conjecture for wild automorphisms of compact Kähler spaces

摘录 / EXCERPT: Let X be a compact Kähler space, and let $\sigma\in\operatorname{Aut}(X)$ be a wild automorphism. Zero-entropy conjecture. The automorphism σ has zero entropy. The stateme...

conjecture | open | MathDB 页面 | 原文线索

V2-040215 | MATHDB-357031

Splitting-from-singularity-filling conjecture for long-range free-fermion chains

摘录 / EXCERPT: Let H_{BDI} be an open chain of size L, whose related bulk Hamiltonian has winding number $\omega>0$ and is described by ... Define ... Suppose that $m(z)$ has singu...

conjecture | open | MathDB 页面 | 原文线索

V2-040216 | MATHDB-357102

Infinite-volume scaling limit conjecture for the uniform infinite meandric system

摘录 / EXCERPT: Let $(\mathcal M,P,\Gamma)$ be the infinite random planar map decorated by a bi-infinite Hamiltonian path and the loops and possibly bi-infinite paths associated to the UIMS. Unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-040217 | MATHDB-357104

Pointed half-plane meandric-system scaling limit conjecture

摘录 / EXCERPT: Let $(\mathcal M^\circ,P^\circ,\Gamma^\circ)$ be the infinite random planar map decorated by a bi-infinite Hamiltonian path and loops and possibly infinite paths associated with th...

conjecture | open | MathDB 页面 | 原文线索

V2-040218 | MATHDB-357113

Arnold conjecture for compact symplectic manifolds

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold, and let $H:\mathbb{R}\times M \to \mathbb{R}$ be a time-dependent Hamiltonian function. Suppose that the solutions of period 1...

conjecture | open | MathDB 页面 | 原文线索

V2-040219 | MATHDB-357202

Mokhov's conjecture on the Doyle-Potemin form of local Poisson operators

摘录 / EXCERPT: Let $P=\sum_{e=1}^{d+2} P^{ij}_e {\partial}_x^{d+2-e}$ be a local operator of homogeneous differential order $d+2$, meaning that ... where $d\geq 0$. Assume that P defines a Poisso...

conjecture | open | MathDB 页面 | 原文线索

V2-040220 | MATHDB-357330

Nonexistence of infinitely many phase transitions for surface diffeomorphisms

摘录 / EXCERPT: Let S be a compact surface, let $f\in \operatorname{Diff}^{1}(S)$ be a diffeomorphism, and let the potential be Holder continuous. A phase transition is a failure of analyticity...

conjecture | open | MathDB 页面 | 原文线索

V2-040221 | MATHDB-357479

Entropy thermalization in quantum chaotic systems with energy conservation

摘录 / EXCERPT: Let a quantum system evolve under a time-independent Hamiltonian, and let A be a subsystem. Entropy thermalization means that, after long-time evolution, the entropy of A agree...

conjecture | open | MathDB 页面 | 原文线索

V2-040222 | MATHDB-357487

Passage-time decorrelation conjecture in the chaotic regime

摘录 / EXCERPT: Let T_0 and T_t be the passage times at perturbation times 0 and t, and write $\operatorname{Var}(T)$ for the variance of the passage time and...

conjecture | open | MathDB 页面 | 原文线索

V2-040223 | MATHDB-357507

Conjecture on the classification of final motions by super-hyperbolic building blocks

摘录 / EXCERPT: Final-motion classification conjecture. For the N-body problem with $N\geq 4$, every logically possible combination of

conjecture | open | MathDB 页面 | 原文线索

V2-040224 | MATHDB-357508

Growth-rate classification conjecture for sub-superhyperbolic N-body motions

摘录 / EXCERPT: Growth-rate classification conjecture. Every such orbit is a combination of $\mathcal P$, $\mathcal B$, and $\mathcal{OS}$. Conversely, every logically possible combination of...

conjecture | open | MathDB 页面 | 原文线索

V2-040225 | MATHDB-357509

Generic-motion conjecture for noncollision and super-hyperbolic N-body orbits

摘录 / EXCERPT: Generic-motion conjecture. For generic initial condition, the solution of the N-body problem is globally defined on $\mathbb R$. As $t\to\pm\infty$, each body approaches either a...

conjecture | open | MathDB 页面 | 原文线索

V2-040226 | MATHDB-357571

The conjecture that every triangle-free planar graph is POSH

摘录 / EXCERPT: A planar graph is POSH (partial one-sided Hamiltonian) if it is a spanning subgraph of a graph admitting a plane embedding with a one-sided Hamiltonian cycle. Triangle-free planar...

conjecture | open | MathDB 页面 | 原文线索

V2-040227 | MATHDB-357572

The conjecture that every 5-connected planar triangulation is POSH

摘录 / EXCERPT: A planar triangulation is a planar graph that admits a plane embedding in which every face is a triangle, and a graph is POSH (partial one-sided Hamiltonian) if it is a spanning su...

conjecture | open | MathDB 页面 | 原文线索

V2-040228 | MATHDB-357588

Furstenberg's Hausdorff convergence conjecture for times p and times q

摘录 / EXCERPT: Furstenberg's Hausdorff convergence conjecture. If F is an infinite closed T_p-invariant subset, then the iterates $T_q^n(F)$ converge in the Hausdorff distance to...

conjecture | open | MathDB 页面 | 原文线索

V2-040229 | MATHDB-357589

Furstenberg's weak-star convergence conjecture for times p-invariant measures

摘录 / EXCERPT: Furstenberg's weak-star convergence conjecture. If μ is a continuous Borel probability measure on ${\mathbb T}$ that is T_p-invariant, then $T_q^n\mu$ converges in the weak-s...

conjecture | open | MathDB 页面 | 原文线索

V2-040230 | MATHDB-357680

Bergelson-Tao-Ziegler conjecture on polynomial generation of ergodic systems

摘录 / EXCERPT: Let $\mathbb{F}_p^\omega=\varinjlim\mathbb{F}_p^n$ act measure-preservingly on a countably generated probability space (X,μ), and call the resulting system an...

conjecture | solved | MathDB 页面 | 原文线索

V2-040231 | MATHDB-357692

Rapaport Strasser's conjecture on Hamilton cycles in Cayley graphs

摘录 / EXCERPT: A Cayley graph is a graph constructed from a group and a generating set; a graph is Hamiltonian if it contains a Hamilton cycle. Rapaport Strasser's conjecture. Every connected Cay...

conjecture | open | MathDB 页面 | 原文线索

V2-040232 | MATHDB-357693

Pak-Radoičić conjecture on random Cayley graphs

摘录 / EXCERPT: Let G be a group of order n, let $d=C\log n$ for a sufficiently large absolute constant C, and choose a set $A\subseteq G$ of size d uniformly at random. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-040233 | MATHDB-357716

Non-shadowing conjecture for chain-recurrent sets with attached singularities

摘录 / EXCERPT: Let M be a closed manifold and let X be a C^1 vector field on M. Let $\Lambda\subset M$ be a chain-recurrent set with an attached singularity. Attached-singularity non-shad...

conjecture | open | MathDB 页面 | 原文线索

V2-040234 | MATHDB-357717

Non-shadowing conjecture for chain-recurrent sets with attached hyperbolic singularities

摘录 / EXCERPT: Let M be a closed manifold and let X be a C^1 vector field on M. Let $\Lambda\subset M$ be a chain-recurrent set with an attached hyperbolic singularity. Hyperbolic attache...

conjecture | open | MathDB 页面 | 原文线索

V2-040235 | MATHDB-357756

The IIP conjecture beyond moment conditions and ergodicity

摘录 / EXCERPT: Let $d\ge1$, let $\Omega^\star$ and $\Omega_{p,q}$ denote the sets of environments defined in the paper, and let IIP denote the invariance invariance principle establish...

conjecture | open | MathDB 页面 | 原文线索

V2-040236 | MATHDB-357757

The long-range conductance IIP conjecture

摘录 / EXCERPT: Let $d\ge2$ and let $p,q>d/2$ satisfy $1/p+1/q<2/d$. Consider symmetric long-range conductance configurations $\{c(x,y)=c(y,x):x,y\in\mathbb Z^d\}$ that are averaging and ergodic i...

conjecture | open | MathDB 页面 | 原文线索

V2-040237 | MATHDB-357774

Calta-Kraaikamp-Schmidt natural-extension conjecture for triangle-group continued fractions

摘录 / EXCERPT: Calta-Kraaikamp-Schmidt conjecture. For all $n\geq 3$ and all $\alpha\in(0,1)$, the natural extension of the first pointwise expansive power of $T_{n,\alpha}$ is given by the first...

conjecture | open | MathDB 页面 | 原文线索

V2-040238 | MATHDB-357803

Zero-entropy extension of the multiple ergodic averages theorem

摘录 / EXCERPT: Let $(X,\mathscr{B},\mu)$ be a probability space and let $T,R,S:X\rightarrow X$ be measure-preserving automorphisms. Suppose that R and S commute, S is weakly mixing, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040239 | MATHDB-357915

Local survival is equivalent to metastability for the contact process

摘录 / EXCERPT: Let $I_n(t)$ denote the expected number of infected vertices at time t for the contact process on the dynamical graph with initially all vertices infected. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-040240 | MATHDB-357995

Subsequence Sarnak conjecture for aperiodic multiplicative functions

摘录 / EXCERPT: Subsequence Sarnak conjecture. There exists a subsequence $N_k\to\infty$ such that, for every $w\colon \mathbb{N}\to\mathbb{U}$ that is completely deterministic for Cesàro averages...

conjecture | open | MathDB 页面 | 原文线索

V2-040241 | MATHDB-358036

Classification conjecture for Furstenberg systems of non-pretentious multiplicative functions

摘录 / EXCERPT: Let $f:\mathbb{N}\to\{-1,0,+1\}$ be a non-pretentious multiplicative function. Write $1_{f(n)=0}$ for the multiplicative function that equals 1 when $f(n)=0$ and 0 otherwise. C...

conjecture | open | MathDB 页面 | 原文线索

V2-040242 | MATHDB-358087

The characterization of star-shaped trajectories with rotation number two-fifths

摘录 / EXCERPT: Let D be the domain under consideration, and let $\triangle P_1P_2P_3$ be a triangle in D. Write $\psi_{\triangle P_1P_2P_3}$ for its associated billiard map,...

conjecture | open | MathDB 页面 | 原文线索

V2-040243 | MATHDB-358149

The singular reduction conjecture for Hamiltonian circle actions

摘录 / EXCERPT: Let X be a Liouville manifold equipped with a Hamiltonian S^1 action such that the fixed locus X^{S^1} has codimension four and there are no finite non-trivial stabiliser gro...

conjecture | open | MathDB 页面 | 原文线索

V2-040244 | MATHDB-358199

Mean ergodic conjecture for locally semi-consistent extended spectral metric spaces

摘录 / EXCERPT: Let (A,L) be a locally semi-consistent extended complete spectral metric space, and let $f:A\to M(A)$ be a relatively proper $*$-homomorphism. Write T for the induced map and l...

conjecture | open | MathDB 页面 | 原文线索

V2-040245 | MATHDB-358234

Sabok's completeness conjecture for conjugacy of automorphisms

摘录 / EXCERPT: Sabok's conjecture. The action of $\operatorname{Aut}(X,\mu)$ on itself by conjugation is a complete orbit equivalence relation.

conjecture | open | MathDB 页面 | 原文线索

V2-040246 | MATHDB-358237

Eliasson's Hausdorff-dimension conjecture for reducible cocycles

摘录 / EXCERPT: Let β be a parameter in a generic one-parameter family of analytic quasi-periodic $\mathrm{GL}(m,\mathbb C)$-cocycles, and consider the set of parameters for which the cocycl...

conjecture | open | MathDB 页面 | 原文线索

V2-040247 | MATHDB-358297

Pointwise convergence conjecture for polynomial multilinear ergodic averages

摘录 / EXCERPT: Pointwise convergence conjecture. For every such system, polynomial family, and m-tuple $f=(f_1,\dots,f_m)$ with $f_j\in L^\infty(\mathbb X)$ for $j\in[m]$, the limit

conjecture | open | MathDB 页面 | 原文线索

V2-040248 | MATHDB-358367

Convergence of numerically determined anti-integrability regions

摘录 / EXCERPT: Let ${\cal R}_n^+$ and ${\cal R}_n^-$ be the numerically determined parameter regions, and let ${\cal R}_{\mathcal A}^+$ and ${\cal R}_{\mathcal A}^-$ be the analytically determine...

conjecture | open | MathDB 页面 | 原文线索

V2-040249 | MATHDB-358368

Period-doubling symbol-sequence conjecture for quadratic diffeomorphisms

摘录 / EXCERPT: Consider the symbol sequences of periodic orbits of the map in the paper, with a period-doubled orbit obtained from a parent periodic orbit. When the sequences are ordered appropri...

conjecture | open | MathDB 页面 | 原文线索

V2-040250 | MATHDB-358560

The standard-flow simplex conjecture for symmetric convex ancient solutions

摘录 / EXCERPT: Standard-flow simplex conjecture. The invariant set

conjecture | open | MathDB 页面 | 原文线索

V2-040251 | MATHDB-358576

Crossing-cycle persistence conjecture for tropical polynomial systems

摘录 / EXCERPT: Let a tropical dynamical system have a hyperbolic crossing cycle γ_0. Crossing-cycle persistence conjecture. For all $0<\epsilon\ll 1$, the associated smooth system has a li...

conjecture | open | MathDB 页面 | 原文线索

V2-040252 | MATHDB-358577

Tropical singularity persistence conjecture for smooth planar systems

摘录 / EXCERPT: Let a tropical dynamical system have a tropical singularity at (u_0,v_0) of sink, source, or saddle type. Let X be a neighborhood of (u_0,v_0). Tropical singularity persisten...

conjecture | open | MathDB 页面 | 原文线索

V2-040253 | MATHDB-358594

Zeeman's monotonicity conjecture for the Lyness rotation number

摘录 / EXCERPT: Consider the Lyness map with parameter $\alpha>0$ and let ρ^v_α denote its rotation number as a function of the level parameter v. Zeeman's conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-040254 | MATHDB-358800

Non-singularity of the reduced Information Bottleneck operator

摘录 / EXCERPT: Non-singularity conjecture. The Jacobian matrix

conjecture | open | MathDB 页面 | 原文线索

V2-040255 | MATHDB-358886

Finite-time predator blow-up conjecture for the delayed predator-prey model

摘录 / EXCERPT: Finite-time predator blow-up conjecture. For the delayed system, the predator population explodes to infinity in finite time.

conjecture | open | MathDB 页面 | 原文线索

V2-040256 | MATHDB-358920

Avila's almost reducibility conjecture

摘录 / EXCERPT: A cocycle (α,A) consists of a frequency α and an analytic matrix-valued map A. It is subcritical when its Lyapunov exponent vanishes in a neighborhood of the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-040257 | MATHDB-359036

Keller's invariant-measure representation conjecture for regular Toeplitz $free$ systems

摘录 / EXCERPT: Keller's conjecture. For every $nuin\ta{P}(X_varphi)$, there exists $rhoin\ta{P}(Hxtimes\ta{0,1\ta}^{\mathbb{Z}},Rxtimessigma)$ such that, for every measurable $Asubseteq X_varphi$...

conjecture | open | MathDB 页面 | 原文线索

V2-040258 | MATHDB-359099

Quantum unique ergodicity conjecture for Liouville quantum gravity

摘录 / EXCERPT: Fix $\gamma\in(0,2)$, let μ_γ be the Liouville measure on a bounded domain D, and let $(\mathbf{f}_n)$ be eigenfunctions normalised to have unit $L^2(\mu_\gamma)$ norm....

conjecture | open | MathDB 页面 | 原文线索

V2-040259 | MATHDB-359101

Existence conjecture for endemic equilibria in the multigroup prion system

摘录 / EXCERPT: Let System ... admits at least one endemic equilibrium when $\mathcal{R}_0>1$. This conjecture is motivated by extensive numerical simulations; the stated existence result is not est...

conjecture | open | MathDB 页面 | 原文线索

V2-040260 | MATHDB-359115

Dynamical nonperiodicity for automorphisms with small spectral radius

摘录 / EXCERPT: Let ${\mathbf{k}}$ be an algebraically closed field of characteristic 0, let $X=\mathcal{M}(A)$ be the affinoid space under consideration, and let $f:X\to X$ be an automorphism w...

conjecture | open | MathDB 页面 | 原文线索

V2-040261 | MATHDB-359181

The refined de Sitter conjecture

摘录 / EXCERPT: Let $\Phi=(\phi^1,\dots,\phi^N)$ be scalar fields with potential $V(\Phi)$ in a low-energy effective field theory coupled to dynamical D-dimensional spacetime. Let ∇_i den...

conjecture | open | MathDB 页面 | 原文线索

V2-040262 | MATHDB-359183

The swampland distance conjecture

摘录 / EXCERPT: Let $\Phi=(\phi^1,\dots,\phi^N)$ be fields in a low-energy effective field theory coupled to dynamical D-dimensional spacetime, with vacuum state $|\Omega\rangle$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040263 | MATHDB-359241

Wieczorek-Xie-Jones conjecture for normally hyperbolic compact invariant sets

摘录 / EXCERPT: The compactified system is obtained from a nonautonomous system with autonomous future and past limit systems, and the equilibria of those limit systems correspond to equilibria of...

conjecture | open | MathDB 页面 | 原文线索

V2-040264 | MATHDB-359294

The conjecture that all fractals are dynamical fractals

摘录 / EXCERPT: Dynamical-fractal universality conjecture. All fractals existing in our world are dynamical fractals.

conjecture | open | MathDB 页面 | 原文线索

V2-040265 | MATHDB-359382

Prevalence of cyclic vectors for ergodic transformations with discrete spectrum

摘录 / EXCERPT: Prevalence conjecture. If the ergodic transformation $T\in\mathrm{MPT}$ has discrete spectrum, then $\mathcal{U}=\mathcal{U}_T$ has a prevalent set of cyclic vectors in L^2_μ.

conjecture | open | MathDB 页面 | 原文线索

V2-040266 | MATHDB-359415

Hodge mapping universality conjecture for the abstract local RH hierarchy

摘录 / EXCERPT: Consider the abstract local RH hierarchy and its hamiltonian perturbations possessing a τ-structure; the abstract local Hodge mapping hierarchy is the hierarchy constructed fr...

conjecture | open | MathDB 页面 | 原文线索

V2-040267 | MATHDB-359428

Conjecture on the asymptotic landing points of constriction curves

摘录 / EXCERPT: Let $\beta_{\ell,k}=(0,s_{\ell,k})$ be the points on the s-axis, where $s_{\ell,1},s_{\ell,2},\dots$ are the positive zeros of the Bessel function J_ℓ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040268 | MATHDB-359482

Existence of a zero-Hopf bifurcation in the eco-epidemiological model

摘录 / EXCERPT: Consider the three-dimensional autonomous eco-epidemiological model, with all parameters fixed except the pair (k_2,d_0). A zero-Hopf bifurcation is a bifurcation involving an eq...

conjecture | open | MathDB 页面 | 原文线索

V2-040269 | MATHDB-359483

Multiplicity of zero-Hopf bifurcations in the eco-epidemiological model

摘录 / EXCERPT: Consider the three-dimensional autonomous eco-epidemiological model, with all parameters fixed except the pair (k_2,K). A zero-Hopf bifurcation is a bifurcation involving an equi...

conjecture | open | MathDB 页面 | 原文线索

V2-040270 | MATHDB-359484

Multiplicity of saddle-node transcritical bifurcations in the eco-epidemiological model

摘录 / EXCERPT: Consider the three-dimensional autonomous eco-epidemiological model, with all parameters fixed except the pair (k_2,K). A saddle-node transcritical bifurcation is a codimension-t...

conjecture | open | MathDB 页面 | 原文线索

V2-040271 | MATHDB-359626

Straight-line motion conjecture for heat-flow zeros

摘录 / EXCERPT: Let $T_t:\mathbb{C}\to\mathbb{C}$ be the deterministic transport maps associated with the hydrodynamic limit, let t_{sing} be the first singular time, and let $z_j(t)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-040272 | MATHDB-359879

Eventual stability of party trees for ultimate leader pairs

摘录 / EXCERPT: Ultimate-leader-pair stability conjecture. For every ultimate leader pair, there exists a step k after which the party tree of the ultimate leader pair is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-040273 | MATHDB-359898

The approximate periodicity conjecture for long-time Floquet properties

摘录 / EXCERPT: Let $H(t)$ be a time-dependent Hamiltonian with period 1, let $U(0,t)$ be its time-evolution operator, and let $|\psi(t)\rangle=U(0,t)|\psi(0)\rangle$ be the state evolved from a...

conjecture | open | MathDB 页面 | 原文线索

V2-040274 | MATHDB-359899

The conjecture on exponentially long-lived discrete time-crystalline behavior

摘录 / EXCERPT: Consider the Floquet system defined by the Hamiltonian model in equation, with parameters $(\mathbf h,\mathbf J)\in[-20,20]^{\times(3N-1)}$. Discrete time-crystal stability conject...

conjecture | open | MathDB 页面 | 原文线索

V2-040275 | MATHDB-359928

Damjanović-Wilkinson-Xu affine rigidity conjecture with centralizer isomorphism

摘录 / EXCERPT: Let G be a Lie group, let $X=G/\Gamma$ be a homogeneous space with $\Gamma<G$ cocompact and discrete, and let $f_0=L_a\circ\Psi$ be an affine diffeomorphism, where $a\in G$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040276 | MATHDB-359947

Spectral-gap and convergence conjecture for SU(2)-symmetric random circuits

摘录 / EXCERPT: Let ... be the Hamiltonian associated with a one-dimensional local random circuit, where $T_k^{(i,i+1)}$ acts on neighboring sites. Let the circuit have open boundary conditions, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040277 | MATHDB-360013

Jackson's Hamiltonicity conjecture for regular oriented graphs

摘录 / EXCERPT: Jackson's conjecture. For each $d>2$, every d-regular oriented graph on $n\leq 4d+1$ vertices has a Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-040278 | MATHDB-360075

The Hofer-Zehnder conjecture on infinitely many periodic orbits

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold, and let a Hamiltonian map on M have more fixed points than are necessarily required by the V. Arnold conjecture. Hofer-Zehnder...

conjecture | open | MathDB 页面 | 原文线索

V2-040279 | MATHDB-360152

Non-ergodicity conjecture for Nevanlinna functions with only prepole asymptotic values

摘录 / EXCERPT: Non-ergodicity conjecture. When $K=N$, the action of f on its Julia set $\widehat{\mathbb{C}}$ is not ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-040280 | MATHDB-360271

Higher-order Lazutkin coefficient structure conjecture

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^2$ be a smooth, bounded, strictly convex billiard table, and let $\Gamma \subset \Omega$ be a convex caustic of length $|\Gamma|$ with Lazutkin param...

conjecture | open | MathDB 页面 | 原文线索

V2-040281 | MATHDB-360299

Bochi-Katok-Rodriguez Hertz flexibility conjecture for Lyapunov exponents

摘录 / EXCERPT: Bochi-Katok-Rodriguez Hertz flexibility conjecture. There exists an ergodic diffeomorphism $f \in C$ such that ξ_i, for $i=1,\ldots,d$, are the Lyapunov exponents of f with r...

conjecture | open | MathDB 页面 | 原文线索

V2-040282 | MATHDB-360317

Bomfim-Carneiro conjecture on spectral gaps for nonhyperbolic local diffeomorphisms

摘录 / EXCERPT: Let M be a compact Riemannian manifold and let $f:M\rightarrow M$ be a C^{2} local diffeomorphism. A transfer operator associated to a complex continuous potential...

conjecture | open | MathDB 页面 | 原文线索

V2-040283 | MATHDB-360384

The W-state signature conjecture for Lifshitz transitions

摘录 / EXCERPT: The W-state signature conjecture. If such a translation-invariant Hamiltonian is tuned so that the zero-mode $|W_1\rangle$-like state and the product state...

conjecture | open | MathDB 页面 | 原文线索

V2-040284 | MATHDB-360434

The Collatz chain descent conjecture

摘录 / EXCERPT: Let $C_n(L)$ denote the Collatz chain associated with an odd integer L in the interval under discussion, and let $\alpha(C_n)$ and $\beta(C_n)$ denote the numbers of applications...

conjecture | open | MathDB 页面 | 原文线索

V2-040285 | MATHDB-360439

Homoclinic or heteroclinic termination conjecture for Rössler periodic trajectories

摘录 / EXCERPT: Termination conjecture. For every periodic trajectory $T\in C$, there exists a parameter $v\in P$ such that, as the Rössler system is perturbed from p to v through parameter sp...

conjecture | open | MathDB 页面 | 原文线索

V2-040286 | MATHDB-360534

Flaminio-Katok conjecture on smooth stable and unstable foliations

摘录 / EXCERPT: Let $f:M\to M$ be a C^k-smooth Anosov diffeomorphism. Denote its stable and unstable foliations by $\mathcal{F}^s_f$ and $\mathcal{F}^u_f$, respectively. Flaminio-Katok conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-040287 | MATHDB-360537

Low-moment star graph spectral-gap conjecture

摘录 / EXCERPT: For a star graph $G_{\star}$ with $n_{\star}$ vertices, let $\Delta(G_{\star},n_{\star},k)$ denote the spectral gap of its Hamiltonian at moment k. Low-moment star graph conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-040288 | MATHDB-360561

The singular Weinstein conjecture for b-contact manifolds

摘录 / EXCERPT: Let (M,Z,α) be a compact b^m-contact manifold, with critical set Z. A singular periodic orbit is an integral curve of the Reeb b-vector field in $M\setminus Z$ whose l...

conjecture | open | MathDB 页面 | 原文线索

V2-040289 | MATHDB-360562

The generalized Weinstein conjecture for eventually singular contact manifolds

摘录 / EXCERPT: Let M be an eventually singular contact manifold, and let Z denote its critical set. An orbit in $M\setminus Z$ is quasi-closed if it is either closed or its α- and...

conjecture | open | MathDB 页面 | 原文线索

V2-040290 | MATHDB-360596

Hamiltonian-reduction chart conjecture for framed torsion-free sheaves

摘录 / EXCERPT: Let Z be the affine bundle constructed from the extension data, let G be the acting group, and let ... be the resulting moment-map analogue. For $\zeta\in HH_0(\mathscr C)$, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-040291 | MATHDB-360634

The directional weak-mixing conjecture for translation surfaces

摘录 / EXCERPT: Directional weak-mixing conjecture. For every translation surface that does not factor over the circle, the directionally typical translation flow is weakly mixing.

conjecture | open | MathDB 页面 | 原文线索

V2-040292 | MATHDB-360635

The almost-everywhere ergodicity and weak-mixing conjecture for polygonal billiards

摘录 / EXCERPT: Polygonal-billiard conjecture. For almost all simply connected polygonal tables, the billiard flow is ergodic and weakly mixing.

conjecture | open | MathDB 页面 | 原文线索

V2-040293 | MATHDB-360637

Ruskey's Hamiltonian-path conjecture for the graph of linear extensions

摘录 / EXCERPT: Let P be a finite sign-balanced poset, and let $\mathcal G(P)$ be the graph whose vertices are the linear extensions of P and whose edges correspond to adjacent switches. Ruske...

conjecture | open | MathDB 页面 | 原文线索

V2-040294 | MATHDB-360761

The ergodicity-tracking conjecture for unexpectedness

摘录 / EXCERPT: Let x be a situation, let $C_W(x)$ denote its causal or world complexity, and let $C_D(x)$ denote its description complexity. In the minimal frequentist core of Simplicity Theory...

conjecture | open | MathDB 页面 | 原文线索

V2-040295 | MATHDB-360823

Braak's G-conjecture for shifted quantum Rabi eigenvalues

摘录 / EXCERPT: Let H be the quantum Rabi Hamiltonian with parity classes having eigenvalues E_n^+ and E_n^-. For a fixed parity class, consider the shifted eigenvalues $E_n^\pm g^2$ and the...

conjecture | open | MathDB 页面 | 原文线索

V2-040296 | MATHDB-360824

Braak-Nguyen-Reyes-Bustos-Wakayama spacing conjecture for the quantum Rabi model

摘录 / EXCERPT: Let $\{E_k\}$ be the ordered eigenvalues of the quantum Rabi Hamiltonian, including both parity classes, with $E_1\leq E_2\leq\cdots$, and define the nearest-neighbor spacings...

conjecture | open | MathDB 页面 | 原文线索

V2-040297 | MATHDB-360851

Asymptotic stability conjecture for the equilibria of the fractional-order dark-sector system

摘录 / EXCERPT: Let $p>0$ and let $X_{1,2}^*$ denote the two equilibria of the fractional-order system, given by ... where $P=\sqrt{p^2+4}$. Let q be the fractional order. Asymptotic stability con...

conjecture | open | MathDB 页面 | 原文线索

V2-040298 | MATHDB-360883

Dirichlet entropy formula conjecture for contact 3-manifolds

摘录 / EXCERPT: Let (M,α) be an arbitrary contact 3-manifold. Let X_α be the Reeb vector field generated by α, let $\mathcal{M}(\alpha)$ denote the space of contact metrics...

conjecture | open | MathDB 页面 | 原文线索

V2-040299 | MATHDB-360890

The folklore conjecture on endomorphisms of Fano manifolds of Picard number 1

摘录 / EXCERPT: Folklore conjecture. If X admits a non-isomorphic surjective endomorphism, then X is a projective space.

conjecture | open | MathDB 页面 | 原文线索

V2-040300 | MATHDB-360934

Space-like asymptotics conjecture for transverse thermal two-point functions of the XXZ chain

摘录 / EXCERPT: Let $T>0$ be fixed, let m denote the spatial separation, and let t denote time. For the transverse thermal dynamical two-point function of the XXZ spin-$1/2$ chain, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040301 | MATHDB-360956

The Beltrami-welding conjecture for an independent construction of the SLE loop

摘录 / EXCERPT: Beltrami-welding conjecture. The above Beltrami welding yields an SLE loop measure and thus an independent second construction of the SLE loop in Theorem 1.3 of Ang (2021).

conjecture | open | MathDB 页面 | 原文线索

V2-040302 | MATHDB-361100

Ergodicity conjecture for additive one-dimensional nearest-neighbor interacting particle systems

摘录 / EXCERPT: Let an additive interacting particle system (IPS) evolve on a one-dimensional lattice with homogeneous, one-sided, nearest-neighbor interaction and positive rates. Additive ergodic...

conjecture | open | MathDB 页面 | 原文线索

V2-040303 | MATHDB-361101

Ordering conjecture for multiplying bifurcations of the symmetric periodic orbit

摘录 / EXCERPT: Ordering conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-040304 | MATHDB-361254

The kneading-invariant conjecture for the Rössler system near trefoil parameters

摘录 / EXCERPT: Let P be the parameter space of the Rössler system, let $p\in P$ be a trefoil parameter, and let $K(v)\in\{1,2\}^{\mathbf{N}}$ be the sequence obtained by recording whether the s...

conjecture | open | MathDB 页面 | 原文线索

V2-040305 | MATHDB-361325

Elliptic Ruijsenaars-Schneider eigenfunction conjecture

摘录 / EXCERPT: Let $\mathcal{H}^{e\text{RS}}_k$ be the elliptic Ruijsenaars-Schneider Hamiltonians for $k=1,\dots,N-1$, and let $\mathcal{Z}^{\text{RS}}(\textbf{a},\textbf{x})$ be a function of t...

conjecture | open | MathDB 页面 | 原文线索

V2-040306 | MATHDB-361346

Equidistribution conjecture for unstable periodic points of heterochaos baker maps

摘录 / EXCERPT: Let $(a,b)\in\Delta$, let f be the corresponding heterochaos baker map on $[0,1]^3$, and let μ_α and μ_β be the two ergodic measures of maximal entropy. Denote b...

conjecture | open | MathDB 页面 | 原文线索

V2-040307 | MATHDB-361347

Linear response conjecture for ergodic maximal-entropy measures

摘录 / EXCERPT: Let Δ be the parameter domain for the heterochaos baker maps $f_{a,b}$ on $[0,1]^3$, and let $\mu_{\alpha,a,b}$ and $\mu_{\beta,a,b}$ be their two ergodic measures of maxima...

conjecture | open | MathDB 页面 | 原文线索

V2-040308 | MATHDB-361348

Bernoulli property at the mostly neutral parameter

摘录 / EXCERPT: Let M be the parameter appearing in the family of heterochaos baker maps F_a on $[0,1]^d$, and let Leb denote Lebesgue measure. At the mostly neutral parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-040309 | MATHDB-361351

Ghazouani's Morse-Smale conjecture for dilation surfaces

摘录 / EXCERPT: Let S be a dilation surface that is not a translation surface, and let $\theta\in S^1$ be a direction. A directional foliation is Morse-Smale if it has finitely many closed leave...

conjecture | open | MathDB 页面 | 原文线索

V2-040310 | MATHDB-361404

Global convergence to a fixed point of the SUCPA-map

摘录 / EXCERPT: Let the SUCPA-map be $\mathbf{f}=[f_1,\ldots,f_K]:\mathbb{R}^K\to\mathbb{R}^K$, and let $\omega(\boldsymbol{\beta}^{[0]})$ denote the ω-limit set of an initial condition...

conjecture | open | MathDB 页面 | 原文线索

V2-040311 | MATHDB-361405

Uniqueness of the fixed-point line for the SUCPA-map

摘录 / EXCERPT: Let $\mathbf{f}=[f_1,\ldots,f_K]:\mathbb{R}^K\to\mathbb{R}^K$ be the SUCPA-map. For a fixed point $\boldsymbol{\beta}^*$ of $\mathbf{f}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040312 | MATHDB-361457

The orbit-depth dichotomy for polynomial Hamiltonians and Melnikov-function length

摘录 / EXCERPT: Orbit-depth and Melnikov-length conjectures. (i) For any polynomial F and any non-trivial cycle γ of F, either the depth is unbounded, or it is 1, or 2. (ii) For any...

conjecture | open | MathDB 页面 | 原文线索

V2-040313 | MATHDB-361458

Parallel-curve characterizations of orbit-group solvability and orbit-depth boundedness

摘录 / EXCERPT: Parallel-curve characterization conjecture. (i) The non-solvability of the group $\pi_1^{\mathcal{O}\text{-ab}}$ is characterized by the presence of a pair of parallel curves in th...

conjecture | open | MathDB 页面 | 原文线索

V2-040314 | MATHDB-361502

The strong coincidence conjecture for irreducible Pisot substitutions

摘录 / EXCERPT: Let σ be an irreducible Pisot substitution, and let the associated substitution dynamical system and its strong coincidence condition be understood. Strong coincidence conje...

conjecture | open | MathDB 页面 | 原文线索

V2-040315 | MATHDB-361582

Finite-graph conjecture for gradient reaction-diffusion systems

摘录 / EXCERPT: Finite-graph conjecture. The graph of this system has finitely many nodes, and every node is a fixed point, that is, an equilibrium solution.

conjecture | open | MathDB 页面 | 原文线索

V2-040316 | MATHDB-361641

Schroer-Sauer-Ott-Yorke conjecture on prediction-error decay

摘录 / EXCERPT: Let $T\colon M \to M$ be a smooth diffeomorphism of a compact Riemannian manifold M with a compact T-invariant attractor $X \subset M$ and a natural measure μ on X of inf...

conjecture | open | MathDB 页面 | 原文线索

V2-040317 | MATHDB-361660

The bounded-interval conjecture for the real Collatz map

摘录 / EXCERPT: Let Col_R denote the real analogue of the Collatz map considered in the paper, and let r be a real number satisfying $r>9/4$. Bounded-interval conjecture. There...

open_problem | open | MathDB 页面 | 原文线索

V2-040318 | MATHDB-361661

The real Collatz co-null liminf conjecture

摘录 / EXCERPT: Let C denote the real Collatz map used in the paper, and let $x\in(3/4,\infty)$. A subset A of a measurable set $B\subseteq\mathbb{R}$ is Lebesgue-co-null in B when...

conjecture | open | MathDB 页面 | 原文线索

V2-040319 | MATHDB-361667

Noble-number conjecture for the local minima of the Brjuno function

摘录 / EXCERPT: Let B_1 denote the classical Brjuno function, let $ \mathcal{M}$ be the set of its local minima, and let $\mathcal{N}$ be the set of noble numbers, namely the numbers whose regul...

conjecture | open | MathDB 页面 | 原文线索

V2-040320 | MATHDB-361678

Fried-Ghys genus one Birkhoff section conjecture for Anosov flows

摘录 / EXCERPT: Let ϕ^t be a transitive Anosov flow with orientable stable and unstable foliations. A genus one Birkhoff section is a Birkhoff section whose underlying surface has genus one....

conjecture | open | MathDB 页面 | 原文线索

V2-040321 | MATHDB-361800

Full-support conjecture for eigenfunction limits on negatively curved manifolds

摘录 / EXCERPT: Let (M,g) be a compact connected Riemannian manifold of negative sectional curvature. A weak limit $\tilde\mu$ is a weak- limit of the probability measures...

conjecture | open | MathDB 页面 | 原文线索

V2-040322 | MATHDB-361801

Full-support conjecture for semiclassical measures on complex hyperbolic quotients

摘录 / EXCERPT: Let M be a compact complex hyperbolic quotient, and let μ be a semiclassical measure, meaning the semiclassical limit of a sequence of Laplacian eigenfunctions on M. Its su...

conjecture | open | MathDB 页面 | 原文线索

V2-040323 | MATHDB-361826

Sub-square-root growth conjecture for point vortex systems

摘录 / EXCERPT: Sub-square-root growth conjecture. Every solution has global existence and grows as $o(\sqrt{t})$.

conjecture | open | MathDB 页面 | 原文线索

V2-040324 | MATHDB-361931

The conjecture that Ecalle's proof can be made fully detailed

摘录 / EXCERPT: Ecalle's proof concerns the injection of accelero-summable transseries into germs of functions near infinity, as used to establish the quasianalyticity needed in the analysis of po...

conjecture | open | MathDB 页面 | 原文线索

V2-040325 | MATHDB-362013

Six-periodic isosceles-triangle orbit conjecture

摘录 / EXCERPT: Let q_t and r_t be the maps used to describe the two symmetry-related images of a triangle, and let a triangle be non-equilateral when it is not equilateral. Six-periodic isosc...

conjecture | open | MathDB 页面 | 原文线索

V2-040326 | MATHDB-362222

Pre-compactness of pentagram-map iterates modulo affine transformations

摘录 / EXCERPT: Pre-compactness conjecture. The sequence $\{P^{\ell}\}$ is pre-compact modulo affine transformations. That is, this sequence has a convergent subsequence which converges to another...

conjecture | open | MathDB 页面 | 原文线索

V2-040327 | MATHDB-362326

Dirichlet infinite-linking convergence conjecture

摘录 / EXCERPT: Dirichlet infinite-linking convergence conjecture. The constants and minimisers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-040328 | MATHDB-362414

Nonsplit quantum unique ergodicity conjecture for Bianchi Maaß forms

摘录 / EXCERPT: Nonsplit Bianchi quantum unique ergodicity conjecture. If $H\in C_b(\Gamma\backslash\mathbb{H})$ is fixed, then

conjecture | open | MathDB 页面 | 原文线索

V2-040329 | MATHDB-362449

The fixed-point conjecture for first-step primitive elements in PL homeomorphism groups

摘录 / EXCERPT: Fixed-point conjecture. There exists a generating set S of Γ such that every first-step primitive element with respect to S fixes a point in I, while there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-040330 | MATHDB-362462

Generalized rainbow linear-forest coloring conjecture

摘录 / EXCERPT: Let H be a subgraph of K_{2n+1} with n edges, and suppose its edges are colored so that each color class induces a linear forest. A coloring extension is an extension of this...

conjecture | open | MathDB 页面 | 原文线索

V2-040331 | MATHDB-362490

The finite metric-space convergence conjecture for Markov distribution mappings

摘录 / EXCERPT: Let S be a finite space with distinct distances, and let $\pi_{j,k}$ be the distribution mapping defined from those distances. For an initial distribution θ on S, write...

conjecture | open | MathDB 页面 | 原文线索

V2-040332 | MATHDB-362491

The planar \(\pi_{1,2}\) point-mass convergence conjecture

摘录 / EXCERPT: Let S be a finite subset of the plane equipped with Euclidean distance, and let $\pi_{1,2}$ be the corresponding distribution mapping. For an initial distribution on S, conside...

conjecture | open | MathDB 页面 | 原文线索

V2-040333 | MATHDB-362506

Convergence of adaptive filter roots to rotation and island-period points

摘录 / EXCERPT: Polynomial-root convergence conjecture. There exist $M_*>0$ and a sequence of roots $z_{K,n}$ of q_K such that

conjecture | open | MathDB 页面 | 原文线索

V2-040334 | MATHDB-362536

Many-body ground states conjecture for frustration-free FBI Hamiltonians

摘录 / EXCERPT: Let a frustration-free FBI Hamiltonian be a many-body Hamiltonian whose ground-state energy is attained when each positive-semidefinite local term is minimized, and let its Hartree...

conjecture | open | MathDB 页面 | 原文线索

V2-040335 | MATHDB-362623

Hamilton-cycle profile conjecture for the hypercube

摘录 / EXCERPT: Let Q^n be the n-dimensional hypercube, and let the profile of a subgraph be the tuple recording the number of its edges in each coordinate direction. A tuple is even if every...

conjecture | open | MathDB 页面 | 原文线索

V2-040336 | MATHDB-362695

Weak Ergodicity Conjecture for partially hyperbolic diffeomorphisms on 3-manifolds

摘录 / EXCERPT: Let M^3 be an orientable closed 3-manifold and let $f:M^3\to M^3$ be a C^r ($r>1$) non-ergodic partially hyperbolic diffeomorphism. Weak Ergodicity Conjecture. Then M^3 must...

conjecture | open | MathDB 页面 | 原文线索

V2-040337 | MATHDB-362696

Strong Ergodicity Conjecture for partially hyperbolic diffeomorphisms on 3-manifolds

摘录 / EXCERPT: Let f be a C^r ($r>1$) conservative partially hyperbolic diffeomorphism on a 3-manifold, with invariant bundles E^s and E^u. Strong Ergodicity Conjecture. If f is non-erg...

conjecture | open | MathDB 页面 | 原文线索

V2-040338 | MATHDB-362779

Coboundary characterization of nonfluctuating irrational rotation averages

摘录 / EXCERPT: Let $\theta otin\boldsymbol{Q}$ and let $4\mathsf{x}={x_n}_n$ with $x_n=n\theta$. For a continuous function f, write ... We say that these averages fluctuate when they are infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-040339 | MATHDB-362780

Generic fluctuation for dissipative ergodic averages

摘录 / EXCERPT: Let $(X,\mathcal{B},\mathbb{P},T)$ be a measure-preserving system, and let $\mu_N^T f=\sum_{n\in\mathbb{Z}}\mu_N(n)f\circ T^n$ be a dissipative sequence of operators. Assume that .....

conjecture | open | MathDB 页面 | 原文线索

V2-040340 | MATHDB-362840

Viterbo's spectral bound conjecture for cotangent bundles

摘录 / EXCERPT: Let (M,g) be a closed Riemannian manifold. Define ... and let 0_M denote the image of the zero section of T^*M. For a compactly supported Hamiltonian diffeomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-040341 | MATHDB-362852

Invariant-measure encoding conjecture for computational hardness

摘录 / EXCERPT: A constraint satisfaction or optimization problem is a computational problem whose feasible assignments or candidate solutions can be evaluated by its constraints or objective func...

conjecture | open | MathDB 页面 | 原文线索

V2-040342 | MATHDB-362971

The conjectural formula for the degree of difficulty

摘录 / EXCERPT: Conjectural formula. The number of degree of difficulty is

conjecture | open | MathDB 页面 | 原文线索

V2-040343 | MATHDB-362981

Unimodality conjecture for symplectically rational 8-manifolds

摘录 / EXCERPT: Unimodality conjecture. If (M,ω) is a closed symplectically rational 8-manifold admitting a Hamiltonian S^1-action, then the even Betti numbers of M are unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-040344 | MATHDB-362985

Conservation conjecture for truncated modified Hamiltonians in symplectic Euler

摘录 / EXCERPT: Let $\mathcal{P},\mathcal{Q}\subseteq\mathbb{R}^{d}$ be closed and convex, and let $F:\mathcal{P}\rightarrow\mathbb{R}$ and $G:\mathcal{Q}\rightarrow\mathbb{R}$ be L-smooth of or...

conjecture | open | MathDB 页面 | 原文线索

V2-040345 | MATHDB-363112

Generic existence of regular points and Birkhoff averages

摘录 / EXCERPT: Let a dynamical system have a positive-volume subset of its ambient space. A point is $p\omega$-regular if it has the regularity property denoted by $p\omega$ in the source; forwar...

conjecture | open | MathDB 页面 | 原文线索

V2-040346 | MATHDB-363113

Uniqueness of the physical measure for the maps $\varphi_{\alpha,b}$

摘录 / EXCERPT: Let $\varphi_{\alpha,b}$ be one of the maps in the setting of Theorem; an SRB/physical measure is a measure with the corresponding SRB or physical property. Uniqueness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-040347 | MATHDB-363114

Statistical stability of physical measures in $\mathcal{U}$

摘录 / EXCERPT: Let $\mathcal{U}$ be the open family of maps considered in the paper, and let f_n be C^∞ dynamics in $\mathcal{U}$ with physical measures μ_n. Suppose that f_n conve...

conjecture | open | MathDB 页面 | 原文线索

V2-040348 | MATHDB-363330

Mixed-equilibrium conjecture for multiple pathogens

摘录 / EXCERPT: Consider the multi-species SLIR system for V pathogens and P species, with pathogens indexed by $a=1,dots,k$ satisfying $\mathcal{R}_{0a}<1$ and pathogens indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-040349 | MATHDB-363332

Quantum GIT conjecture for Fukaya categories

摘录 / EXCERPT: Let X be a monotone almost complex manifold with a Hamiltonian action of a group G, moment map μ, and symplectic quotient $X/\!/G=\mu^{-1}(0)/G$. Let $\mathcal{F}(X)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-040350 | MATHDB-363334

Critical-limit conjecture for random multiplicative functions

摘录 / EXCERPT: Let $\alpha(n)$ be the random completely multiplicative function appearing in the paper, let μ be the Möbius function, and let $m_{x,\infty}$ be the random measure defined in t...

conjecture | open | MathDB 页面 | 原文线索

V2-040351 | MATHDB-363535

Local resilience conjecture for Hamiltonicity in random geometric graphs

摘录 / EXCERPT: Local resilience conjecture. For every $\epsilon\in (0,1/2]$ and integer $d\geq1$, there exists a constant $C=C(d,\epsilon)>0$ such that, for $r\geq C(\log n/n)^{1/d}$, asymptotica...

conjecture | open | MathDB 页面 | 原文线索

V2-040352 | MATHDB-363554

Four-cusp conjecture for caustics by reflection in ellipses

摘录 / EXCERPT: Let C be an ellipse and let O be an interior point that is not a focus of C. For each $n\geq 1$, the n-th caustic by reflection from O is the envelope of the family of bi...

conjecture | open | MathDB 页面 | 原文线索

V2-040353 | MATHDB-363555

Non-ellipses have caustics with more than four cusps

摘录 / EXCERPT: Let C be an oval, meaning a smooth strictly convex closed curve in the plane, and let an n-th caustic by reflection from an interior point O be the envelope of billiard traje...

conjecture | open | MathDB 页面 | 原文线索

V2-040354 | MATHDB-363565

Strong rigidity of periodic points outside equivariant torus quotients

摘录 / EXCERPT: Let F be a field of characteristic zero and let S_F be a normal affine surface over F. Suppose that $f,g\rbrace\in\operatorname{Aut}(S_F)$ are loxodromic automorphisms. Stron...

conjecture | refuted | MathDB 页面 | 原文线索

V2-040355 | MATHDB-363659

Ivrii's conjecture on periodic billiard points

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a plane region with smooth boundary, and consider the billiard in Ω. The periodic points are the points of the billiard phase space corres...

conjecture | open | MathDB 页面 | 原文线索

V2-040356 | MATHDB-363680

The bounded orbit conjecture for plane homeomorphisms

摘录 / EXCERPT: Let $f:\mathbb R^2\rightarrow\mathbb R^2$ be a homeomorphism, and suppose that every orbit of f is bounded. Bounded orbit conjecture. Then f has a fixed point. This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-040357 | MATHDB-363702

Jimbo-Miwa-Ueno differential conjecture for hbar-deformed meromorphic connections

摘录 / EXCERPT: Jimbo-Miwa-Ueno differential conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-040358 | MATHDB-363704

Asymptotic threshold for consensus discovery in structured coordination games

摘录 / EXCERPT: Let $\Gamma_{n,N(n)}$ be an Erdős-Rényi random graph, let $Q_{\Gamma_{n,N(n)}}^0$ denote the trivial consensus partition, and let $\phi(Q_{\Gamma_{n,N(n)}}^0)$ be the probability t...

conjecture | open | MathDB 页面 | 原文线索

V2-040359 | MATHDB-363982

Wojtkowski's falling particle conjecture

摘录 / EXCERPT: Let $n\ge 2$ point particles of masses $m_1\ge m_2\ge\dots\ge m_n>0$ move on the vertical half line $q\ge 0$, with positions $0\le q_1\le q_2\le\dots\le q_n$, under constant gravit...

conjecture | open | MathDB 页面 | 原文线索

V2-040360 | MATHDB-364056

Birkhoff-Poritsky conjecture for integrable billiards

摘录 / EXCERPT: Let Ω be a strictly convex domain with smooth boundary, and let its billiard map act on the cylinder parametrized by...

conjecture | open | MathDB 页面 | 原文线索

V2-040361 | MATHDB-364057

Polynomiality conjecture for compatible Ferapontov and Doyle--Potëmín operators

摘录 / EXCERPT: Polynomiality conjecture. If P and R are compatible, meaning that their Schouten bracket vanishes, $[P,R]=0$, then every entry of $Q^{\alpha\beta}$ is a polynomial of degree tw...

conjecture | open | MathDB 页面 | 原文线索

V2-040362 | MATHDB-364059

Alekseev's dense collision-orbit conjecture

摘录 / EXCERPT: Alekseev's dense collision-orbit conjecture. Is there an open subset $\mathcal{U}$ of the phase space such that for a dense subset of initial conditions the associated trajectories...

open_problem | open | MathDB 页面 | 原文线索

V2-040363 | MATHDB-364065

Existence of an endemic equilibrium for adaptive SIS systems with $R_0>1$

摘录 / EXCERPT: Consider system ... admits at least one EE when $R_0>1$. This conjecture is based on extensive numerical simulations; the provided text does not establish the claim analytically or i...

conjecture | open | MathDB 页面 | 原文线索

V2-040364 | MATHDB-364169

The Stuck--Zimmer conjecture

摘录 / EXCERPT: Let G be a semisimple Lie group of rank at least 2, and let a p.m.p. action of G be irreducible, meaning that every factor of G acts ergodically. An action is essentially f...

conjecture | open | MathDB 页面 | 原文线索

V2-040365 | MATHDB-364170

The co-amenable subgroup conjecture for higher-rank semisimple groups

摘录 / EXCERPT: Let G be a semisimple Lie group without compact factors and of rank at least two. A discrete subgroup $\Lambda\le G$ is co-amenable when the Hilbert space $L^2(G/\Lambda)$ admits...

conjecture | open | MathDB 页面 | 原文线索

V2-040366 | MATHDB-364209

Lower-period completion conjecture for positively oriented periodic polygons

摘录 / EXCERPT: Let a disjoint set of positively oriented periodic polygons be given. Lower-period completion conjecture. There exists an invariant lamination such that all additional periodic pol...

conjecture | open | MathDB 页面 | 原文线索

V2-040367 | MATHDB-364210

Dimension conjecture for co-landing loci of finite dynamical laminations

摘录 / EXCERPT: Let an FDL be a finite dynamical lamination for degree d polynomials, and let its free criticality be the number of critical polygons, counted with multiplicity, that fit outside...

conjecture | open | MathDB 页面 | 原文线索

V2-040368 | MATHDB-364211

Connectedness conjecture for co-landing loci

摘录 / EXCERPT: Let an FDL be a finite dynamical lamination, and let its co-landing locus be intersected with the connectedness locus. Connectedness conjecture for co-landing loci. This intersecti...

conjecture | open | MathDB 页面 | 原文线索

V2-040369 | MATHDB-364218

Bellon-Viallet integrality conjecture for dynamical degrees

摘录 / EXCERPT: Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map, and let ... be its dynamical degree. Bellon-Viallet conjecture. The dynamical degree is an algebraic integ...

conjecture | open | MathDB 页面 | 原文线索

V2-040370 | MATHDB-364221

Dynamical Lang height conjecture

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a number field, let $\mathcal{M}_d^N$ be the moduli space of degree-d endomorphisms of $\mathbb{P}^N$, and let $h_{\mathcal{M}}$ be an ample height on...

conjecture | open | MathDB 页面 | 原文线索

V2-040371 | MATHDB-364282

Conjecture on intersections of independent orbit-dimension level sets

摘录 / EXCERPT: For integers $p,q\geq2$, let T_p and T_q be the corresponding transformations on $[0,1)$, and define ... Assume that p and q are multiplicatively independent, written...

conjecture | open | MathDB 页面 | 原文线索

V2-040372 | MATHDB-364298

Funk-Jackson-Labbate-Sheehan conjecture on 2-factor Hamiltonian bipartite graphs

摘录 / EXCERPT: Let G be a 2-factor Hamiltonian k-regular bipartite graph, meaning that every 2-factor of G is a Hamiltonian circuit. Let $K_{3,3}$ and H_0 be the two specified cubic b...

conjecture | open | MathDB 页面 | 原文线索

V2-040373 | MATHDB-364299

Sheehan's conjecture on 2-factor Hamiltonian regular bipartite graphs

摘录 / EXCERPT: Let G be a 2-factor Hamiltonian k-regular bipartite graph, meaning that every 2-factor of G is a Hamiltonian circuit. Sheehan's conjecture. There are no 2-factor Hamilt...

conjecture | open | MathDB 页面 | 原文线索

V2-040374 | MATHDB-364300

Aldred-Funk-Jackson-Labbate-Sheehan conjecture on 2-factor isomorphic graphs

摘录 / EXCERPT: Let G be a connected k-regular bipartite graph. A graph is 2-factor isomorphic when all its 2-factors are isomorphic, and it is 2-factor Hamiltonian when every 2-factor is...

conjecture | open | MathDB 页面 | 原文线索

V2-040375 | MATHDB-364301

Abreu-Diwan-Jackson-Labbate-Sheehan 3-edge-connected conjecture

摘录 / EXCERPT: Let G be a 3-edge-connected 2-factor isomorphic cubic bipartite graph.
Abreu-Diwan-Jackson-Labbate-Sheehan's conjecture. Then G is a 2-factor Hamiltonian cubic bipartite graph....

conjecture | open | MathDB 页面 | 原文线索

V2-040376 | MATHDB-364302

Abreu-Aldred-Funk-Jackson-Sheehan conjecture on 4-regular non-bipartite graphs

摘录 / EXCERPT: Let K_5 denote the complete graph on five vertices, and let a graph be 2-factor Hamiltonian when every 2-factor is a Hamiltonian circuit.
Abreu-Aldred-Funk-Jackson-Sheehan's co...

conjecture | open | MathDB 页面 | 原文线索

V2-040377 | MATHDB-364341

The positive-characteristic dynamical Mordell-Lang conjecture for polynomial product maps

摘录 / EXCERPT: Positive-characteristic dynamical Mordell-Lang conjecture. The return set R is a union of finitely many infinite arithmetic progressions together with finitely many sets of the f...

conjecture | open | MathDB 页面 | 原文线索

V2-040378 | MATHDB-364401

Root compatibility conjecture for abelian mirror branes

摘录 / EXCERPT: Let Y be a Hamiltonian G-manifold with maximal torus T, Weyl group W, and root system. Let $(Y^\vee,f)$ be its two-dimensional mirror Landau-Ginzburg model, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040379 | MATHDB-364507

Non-local exponentiality conjecture for semidirect products of smooth functions

摘录 / EXCERPT: Non-local exponentiality conjecture. The Lie group $C^\infty(M,{\mathbb{R}}) \rtimes_\alpha {\mathbb{R}}$ is not locally exponential. More specifically, there is a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-040380 | MATHDB-364609

Ergodicity conjecture for the inhibited discrete-time Hawkes process

摘录 / EXCERPT: Let $a,b,c\in\mathbb{R}$ be parameters of the Markov chain (X_n). Let P be the polynomial associated with these parameters, and write $\operatorname{Disc}(P)$ for its discrimin...

conjecture | open | MathDB 页面 | 原文线索

V2-040381 | MATHDB-364820

Solenoidal injectivity conjecture for locally symmetric metrics

摘录 / EXCERPT: Let g_0 be a locally symmetric metric, and let $D_{g_0}^*$ be the operator appearing in the paper. Denote by $\Pi_{\operatorname{Ker}(D_{g_0}^*)}$ the projection onto...

conjecture | open | MathDB 页面 | 原文线索

V2-040382 | MATHDB-364838

Generic diffeomorphisms are not robustly Turing-universal

摘录 / EXCERPT: Generic non-universality conjecture. For any compact computable manifold M (possibly with boundary), there is a C^∞ generic set $\mathcal{S}$ of diffeomorphisms $f:M\to M$...

conjecture | open | MathDB 页面 | 原文线索

V2-040383 | MATHDB-364846

Existence of unitary input encoding for the nonstationary ESP

摘录 / EXCERPT: Existence of unitary input encoding for the nonstationary ESP. If $\rho(W)<1$, then there always exists a non-trivial unitary input encoding $\hat{R}$ such that Eq. $$ is satisfied...

conjecture | open | MathDB 页面 | 原文线索

V2-040384 | MATHDB-364992

Absolute continuity conjecture for contracting-on-average self-similar measures

摘录 / EXCERPT: Let G be the group acting on $\mathbb{R}^d$ in the self-similar setting, and let ν be the associated stationary measure of a finitely supported, contracting on average and ir...

conjecture | open | MathDB 页面 | 原文线索

V2-040385 | MATHDB-365089

The full-measure conjecture for conditions (I') and (II')

摘录 / EXCERPT: Full-measure conjecture. There is a subset of $\triangle$ of full Lebesgue measure on which condition (I') or condition (II') is satisfied.

conjecture | open | MathDB 页面 | 原文线索

V2-040386 | MATHDB-365090

The almost-everywhere entry conjecture for the region U

摘录 / EXCERPT: Almost-everywhere entry conjecture. For Lebesgue-almost every point in $\triangle$, the region U can be reached after finitely many iterations of the $\mathbb{R}^3(e,13,e)$ TRIP...

conjecture | open | MathDB 页面 | 原文线索

V2-040387 | MATHDB-365134

The Hamiltonian-path conjecture for non-conflicting flows

摘录 / EXCERPT: Let G be a bridgeless cubic graph containing a Hamiltonian path, and let $\overline{F}$ denote a 2-factor of G. A non-conflicting nowhere-zero $Z_2\times Z_2$-flow with respe...

conjecture | open | MathDB 页面 | 原文线索

V2-040388 | MATHDB-365219

Bellissard's almost-everywhere constancy conjecture for Sturmian spectrum dimension

摘录 / EXCERPT: Let $H_{\lambda,\alpha,\omega}$ be the Sturmian Hamiltonian with coupling $\lambda>0$, irrational frequency $\alpha\in(0,1)$, and phase $\omega\in S^1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040389 | MATHDB-365242

Generic stable ergodicity for tuples of volume-preserving diffeomorphisms

摘录 / EXCERPT: Let M be a closed manifold with a volume, let $k\ge 1$, and let $\operatorname{Diff}^k_{\operatorname{vol}}(M)$ denote the space of C^k volume-preserving diffeomorphisms of M...

conjecture | open | MathDB 页面 | 原文线索

V2-040390 | MATHDB-365243

Generic exponential mixing for volume-preserving partially hyperbolic systems

摘录 / EXCERPT: Consider volume-preserving partially hyperbolic systems, meaning systems preserving a volume and admitting a partially hyperbolic invariant splitting. Generic exponential mixing co...

conjecture | open | MathDB 页面 | 原文线索

V2-040391 | MATHDB-365290

Basmajian-Hakobyan-Pandazis-Šarić strong Kahn-Marković conjecture

摘录 / EXCERPT: Let $X(\{\ell_n\},\{t_n\})$ be a flute surface whose cuff lengths form a non-decreasing sequence $\{\ell_n\}$, and let each twist parameter satisfy $t_n\in\{0,1/2\}$. Strong Kahn-M...

conjecture | open | MathDB 页面 | 原文线索

V2-040392 | MATHDB-365292

Global structure conjecture for the stable manifold of plane-wave equilibria

摘录 / EXCERPT: The dynamical system has plane-wave fixed points $\mathrm{PW}^{\pm}$, a fixed point source RT, a fixed point PW^{0}, a fixed point T_3, and variabl...

conjecture | open | MathDB 页面 | 原文线索

V2-040393 | MATHDB-365293

Kasner-Bianchi II attractor conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ denote the attractor describing the generic behaviour near the initial singularity, let $\mathcal{K}$ be the Bianchi type I Kasner subset, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040394 | MATHDB-365294

Single-transition attractor conjecture for Bianchi type VI$_{-1/9}$

摘录 / EXCERPT: Let $\mathcal{A}$ denote the attractor describing generic behaviour near the initial singularity, and let K°, $\mathcal{T}_{R_1}$, $\mathcal{T}_{R_3}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-040395 | MATHDB-365316

Generic degeneracies in two-parameter families of vector fields on surfaces

摘录 / EXCERPT: Generic two-parameter degeneracy conjecture. In generic two-parameter families of vector fields on a surface, only the following degeneracies may occur: one or two simultaneous deg...

conjecture | open | MathDB 页面 | 原文线索

V2-040396 | MATHDB-365384

Hamiltonicity threshold conjecture via linear forests

摘录 / EXCERPT: Hamiltonicity threshold conjecture. The sharp d-threshold for $H_n\cup G(n,p)$ to be Hamiltonian coincides with the sharp threshold for $G(n-d,p)$ to contain a linear forest with...

conjecture | open | MathDB 页面 | 原文线索

V2-040397 | MATHDB-365385

Hamiltonicity threshold conjecture for randomly perturbed directed graphs

摘录 / EXCERPT: Directed Hamiltonicity threshold conjecture. The d-threshold for Hamiltonicity in randomly perturbed directed graphs is η/n^2.

conjecture | open | MathDB 页面 | 原文线索

V2-040398 | MATHDB-365421

Hamiltonian actions in Floer theory

摘录 / EXCERPT: Assume the setting of the Lie monoid action conjecture in Floer theory, with the actions Hamiltonian. Let $\mathcal{W}(T^*G)$ and $\mathcal{W}(T^*H)$ denote the wrapped Fukaya cate...

conjecture | open | MathDB 页面 | 原文线索

V2-040399 | MATHDB-365449

Thomas-Yau conjecture for special Lagrangian representatives

摘录 / EXCERPT: Thomas-Yau conjecture. There exists a unique special Lagrangian in the Hamiltonian isotopy class of L if and only if L is stable. This conjecture proposes a stability criterion...

conjecture | open | MathDB 页面 | 原文线索

V2-040400 | MATHDB-365582

Furstenberg's ×(2,3) conjecture for invariant measures

摘录 / EXCERPT: Let $\mathbb{T}$ be the unit circle, let $\operatorname{Prob}(\mathbb{T};2,3)$ denote the set of measures invariant under the maps $\omega\mapsto\omega^2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040401 | MATHDB-365583

Periodic approximation conjecture for ×(2,3)-invariant measures

摘录 / EXCERPT: Let $\mathbb{T}$ be the unit circle, let a measure be ×(2,3)-invariant when it is invariant under $\omega\mapsto\omega^2$ and $\omega\mapsto\omega^3$, and let vague convergence mea...

conjecture | open | MathDB 页面 | 原文线索

V2-040402 | MATHDB-365584

Rationality conjecture for extreme ×(2,3)-circular Carathéodory functions

摘录 / EXCERPT: Let $\mathbb{D}$ be the unit disk, and let $\operatorname{Cara}(\mathbb{D};2,3)$ be the set of holomorphic functions $\psi:\mathbb{D}\to\mathbb{C}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-040403 | MATHDB-365598

Viana's conjecture on nonzero Lyapunov exponents and SRB measures

摘录 / EXCERPT: Viana's conjecture. If f has only nonzero Lyapunov exponents at Lebesgue almost every point, then it admits some SRB measure.

conjecture | open | MathDB 页面 | 原文线索

V2-040404 | MATHDB-365613

Existence of hamiltonian 2-form metrics on direct sum bundles

摘录 / EXCERPT: Existence conjecture. The conclusions of Theorem 1 hold for M, depending on the sign of $c_1(\det E\otimes K_B^{\vee})$. If $d_\ell=0$ and $\det E\otimes K_B^{\vee}\to B$ is triv...

conjecture | open | MathDB 页面 | 原文线索

V2-040405 | MATHDB-365658

Facet-Hamiltonicity of type B permutahedra

摘录 / EXCERPT: A type B permutahedron is the Coxeter permutahedron associated with the type B reflection group; a polytope is facet-Hamiltonian when its skeleton contains a cycle that visits...

conjecture | open | MathDB 页面 | 原文线索

V2-040406 | MATHDB-365663

Alekseev's conjecture on the measure of oscillatory motions

摘录 / EXCERPT: Alekseev's conjecture. The set OM has Lebesgue measure zero.

conjecture | open | MathDB 页面 | 原文线索

V2-040407 | MATHDB-365680

McMullen's expansion conjecture for arithmetic Veech groups

摘录 / EXCERPT: For a translation surface in the stratum $\mathcal{H}(2)$, the associated Veech group is arithmetic when its orbit gives rise to the graphs under consideration. McMullen's expansio...

conjecture | open | MathDB 页面 | 原文线索

V2-040408 | MATHDB-365684

The multidimensional unique-active-transition idempotency conjecture

摘录 / EXCERPT: Unique-active-transition idempotency conjecture. The automaton τ is not idempotent if and only if there exists $t\in S$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040409 | MATHDB-365716

Two-cluster convergence conjecture for causal transformer dynamics with simple dominant eigendirection

摘录 / EXCERPT: Let Q and K be arbitrary matrices, and let V be diagonalizable with d different positive real eigenvalues. Denote its largest eigenvalue by $\lambda_{\max}$, and let ξ...

conjecture | open | MathDB 页面 | 原文线索

V2-040410 | MATHDB-365717

One-cluster convergence conjecture for causal attention with a multidimensional dominant eigenspace

摘录 / EXCERPT: Let Q and K be arbitrary matrices, and let V be a matrix whose largest eigenvalue $\lambda_{\max}>0$ is real, with eigenspace L satisfying $\mathop{\rm dim} L\geq2$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-040411 | MATHDB-365744

The conjecture that the connectedness locus has no isolated points

摘录 / EXCERPT: Let $\mathcal{N}$ be the connectedness locus for pairs of affine maps on $\mathbb{R}^2$. A point of $\mathcal{N}$ is isolated if it has a neighborhood in the ambient parameter spac...

conjecture | open | MathDB 页面 | 原文线索

V2-040412 | MATHDB-365780

Boundary-of-chaos conjecture for Hénon maps

摘录 / EXCERPT: Let $\mathcal H$ be the Hénon family $F_{a,b}(x,y)=(x^2+a-by,x)$ with $a,b\in\mathbb R$, and let $\partial_{\operatorname{chaos}}$ denote the boundary in $\mathcal H$ of the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-040413 | MATHDB-365787

The singular-hyperbolicity exclusion conjecture for heteroclinic knots

摘录 / EXCERPT: Let F be a smooth vector field on S^3 generating a heteroclinic knot H, and suppose that $Ess(F)$ with respect to H is finite, possibly empty. An invariant set...

conjecture | open | MathDB 页面 | 原文线索

V2-040414 | MATHDB-365788

The ordered-chaotic-mixed classification conjecture for invariant one-dimensional sets

摘录 / EXCERPT: Let F be a smooth vector field on S^3 with fixed points $\{x_1,\ldots,x_n\}$, and let H be a one-dimensional flow-invariant set such that each component C of...

conjecture | open | MathDB 页面 | 原文线索

V2-040415 | MATHDB-365789

The transversality conjecture for essential periodic-orbit neighborhoods

摘录 / EXCERPT: Let G be an ordered vector field, let H be the fixed invariant one-dimensional set, and let $T_1,\ldots,T_k$ be the periodic orbits in $Ess(G)$. Choose tubular neighborhoods...

conjecture | open | MathDB 页面 | 原文线索

V2-040416 | MATHDB-365790

The Chaotic Hypothesis for attractors

摘录 / EXCERPT: Let F be a smooth vector field on $\mathbf{R}^3$ generating a chaotic attractor A. Two flows are orbitally equivalent when their trajectories correspond up to a reparametrizati...

conjecture | open | MathDB 页面 | 原文线索

V2-040417 | MATHDB-365791

The no-all-Möbius periodic-orbit suspension conjecture

摘录 / EXCERPT: Let F be a C^k vector field on S^3, with $k\geq 3$. Let Λ be a compact connected invariant set and S a cross-section for Λ such that the first-return map...

conjecture | open | MathDB 页面 | 原文线索

V2-040418 | MATHDB-365792

The dense hyperbolic periodic-orbit suspension problem

摘录 / EXCERPT: Let D be a disc, let $\sigma:\{1,2\}^{\mathbf{Z}}\to\{1,2\}^{\mathbf{Z}}$ be the double-sided shift, and let $f:D\to D$ be an orientation-preserving diffeomorphism. Suppose there...

open_problem | refuted | MathDB 页面 | 原文线索

V2-040419 | MATHDB-365793

Finiteness conjecture for finite orbits when the Poncelet group is trivial

摘录 / EXCERPT: Finiteness conjecture. If $\mathrm{F}(p)$ is trivial, then there are only finitely many finite orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-040420 | MATHDB-365899

Finite-type boundary conjecture for stable regions

摘录 / EXCERPT: Let a stable region be a connected region in the parameter space of interval translation maps on which the maps are stable, and let its boundary consist of the maps obtained at the...

conjecture | open | MathDB 页面 | 原文线索

V2-040421 | MATHDB-365900

Eventually periodic maps conjecture for stable-region closures

摘录 / EXCERPT: Let an interval translation map be eventually periodic when its critical points eventually become periodic, and let a stable region be a region of stable maps in the parameter spac...

conjecture | open | MathDB 页面 | 原文线索

V2-040422 | MATHDB-365988

The degree-two bound for irreducible polynomial integrals on the two-torus

摘录 / EXCERPT: Let the geodesic flow be the Hamiltonian flow of a Riemannian metric on the 2-torus. A polynomial first integral is a polynomial in the momenta that Poisson-commutes with the Hamil...

conjecture | open | MathDB 页面 | 原文线索

V2-040423 | MATHDB-366013

Stably C^1 or stably fractal conjecture for partially hyperbolic systems

摘录 / EXCERPT: Let f be a C^r partially hyperbolic diffeomorphism on a compact manifold M, with $r\geq 2$. The distribution E^s_f is stably C^1 if, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-040424 | MATHDB-366014

Stably C^1 or stably fractal conjecture for Anosov systems

摘录 / EXCERPT: Let $r\geq 2$ and consider C^r Anosov diffeomorphisms, with their stable and unstable distributions. A distribution is stably C^1 if it remains C^1 under every sufficiently...

conjecture | open | MathDB 页面 | 原文线索

V2-040425 | MATHDB-366068

Facet-Hamiltonian conjecture for B-permutahedra

摘录 / EXCERPT: Facet-Hamiltonian conjecture. The B-permutahedron of any dimension $d\geq 2$ has a facet-Hamiltonian cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-040426 | MATHDB-366093

The Fourier-transform criterion for periodic and divergent trajectories

摘录 / EXCERPT: Let $H:\mathbb{Z}\to\mathbb{Z}$ be a contracting, semi-basic p-Hydra map, and let $\widehat{\chi}_H$ be a Fourier transform of its numen. Fourier-transform dynamical criterion. (...

conjecture | open | MathDB 页面 | 原文线索

V2-040427 | MATHDB-366117

Nonlinear orbital stability conjecture for modulated traveling waves

摘录 / EXCERPT: Nonlinear stability conjecture. Under these assumptions, the modulated traveling wave is exponentially and asymptotically orbitally stable.

conjecture | open | MathDB 页面 | 原文线索

V2-040428 | MATHDB-366205

Typical weak-mixing conjecture for Bruin-Troubetzkoy interval translation maps

摘录 / EXCERPT: Let $d49c$ be the measure obtained in the theorem on unique ergodicity, and consider Bruin-Troubetzkoy interval translation maps of infinite type. Weak-mixing conjecture. Almost al...

conjecture | open | MathDB 页面 | 原文线索

V2-040429 | MATHDB-366244

Onsager-type conjecture for active scalar equations

摘录 / EXCERPT: Let θ be a weak solution of the active scalar equation, let δ be the parameter appearing in the equation, and set ... The Hamiltonian is ... with T_0 the Fourier multip...

conjecture | open | MathDB 页面 | 原文线索

V2-040430 | MATHDB-366248

Accumulation-point conjecture for quotient values of binary sequences

摘录 / EXCERPT: Let ${\bf u}=(u_n)_{n\in\mathbb{N}}$ be a binary sequence that is not eventually periodic. Suppose that the set of return lengths $\mathcal{R}(\mathbf{u})$ is finite, and that ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-040431 | MATHDB-366349

Infinitely many bi-normal trajectories near the primaries in the spatial circular restricted three-body problem

摘录 / EXCERPT: Let $q(t)=(q_1(t),q_2(t),q_3(t))$ be a trajectory in the spatial Circular Restricted Three-Body Problem, in the low-energy range and near the primaries. A trajectory is bi-normal t...

conjecture | open | MathDB 页面 | 原文线索

V2-040432 | MATHDB-366416

Hyperbolic continuation of high-entropy homoclinic classes

摘录 / EXCERPT: Let f be a C^r diffeomorphism of a compact surface. For $0<t\le h_{\rm top}(f)$, let $\mathcal H(f,t)$ be the set of homoclinic classes of measures C satisfying $h(f,C)\ge t$...

conjecture | open | MathDB 页面 | 原文线索

V2-040433 | MATHDB-366417

Upper semicontinuity of high-entropy SRB measures

摘录 / EXCERPT: Let f be a C^∞ surface diffeomorphism and let $b>0$. Let $N^{\mathrm{SRB}}_g(b)$ denote the number of SRB measures of g with metric entropy larger than b. SRB-measure...

conjecture | open | MathDB 页面 | 原文线索

V2-040434 | MATHDB-366607

Ore-type oriented Hamilton cycle discrepancy conjecture

摘录 / EXCERPT: Let D be an oriented graph on $n\ge 3$ vertices with minimum degree δ. For vertices u,v, write $d(u)$ and $d(v)$ for their degrees, and call them non-adjacent if neither...

conjecture | open | MathDB 页面 | 原文线索

V2-040435 | MATHDB-366608

Generalized oriented Hamilton cycle discrepancy conjecture via s-star

摘录 / EXCERPT: Let D be an oriented graph on $n\ge 3$ vertices. For distinct vertices u,v, call them non-adjacent if neither directed arc joins them, and define ... when D has a pair of non-a...

conjecture | open | MathDB 页面 | 原文线索

V2-040436 | MATHDB-366768

Global phase portrait conjecture for the predator-prey system in region II-b

摘录 / EXCERPT: Global phase portrait conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-040437 | MATHDB-366792

Existence of a dense wave front in every convex billiard

摘录 / EXCERPT: Let a convex billiard table be a convex planar domain, and let $W_t(P)$ denote the billiard wave front at time t from a point P in the table. The wave front becomes dense if it...

conjecture | open | MathDB 页面 | 原文线索

V2-040438 | MATHDB-366793

Convex polygon billiard wave-front density conjecture

摘录 / EXCERPT: Let a convex polygon billiard be a billiard in a convex polygonal table, and let $W_t(P)$ denote the billiard wave front at time t from a point P in the table. The wave front b...

conjecture | open | MathDB 页面 | 原文线索

V2-040439 | MATHDB-366797

Non-periodic connection points on the double regular heptagon

摘录 / EXCERPT: Heptagon connection-point conjecture. There exist non-periodic connection points on the double regular heptagon.

conjecture | open | MathDB 页面 | 原文线索

V2-040440 | MATHDB-366798

Non-periodic connection points on double regular n-gons for odd n at least 9

摘录 / EXCERPT: Non-periodic connection-point conjecture. For $n\geq 9$ odd, there are no non-periodic connection points on the double regular n-gon.

conjecture | open | MathDB 页面 | 原文线索

V2-040441 | MATHDB-366857

Ferenczi-Rosendal conjecture on non-Hilbertian Banach spaces

摘录 / EXCERPT: Ferenczi-Rosendal conjecture. Every non-Hilbertian separable Banach space is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-040442 | MATHDB-366924

The inverse-limit characterization of uniformly rigid dynamical systems

摘录 / EXCERPT: Inverse-limit characterization. A dynamical system is uniformly rigid if and only if it is a factor of an inverse limit of odometers.

conjecture | open | MathDB 页面 | 原文线索

V2-040443 | MATHDB-366925

Nonexistence of large paradoxical Collatz sequences

摘录 / EXCERPT: Let T be the Collatz map. A sequence $n,T(n),\ldots,T^j(n)$ is paradoxical when $C_j(n)<1$ and $T^j(n)\geq n$. Nonexistence conjecture for large paradoxical sequences. There is n...

conjecture | open | MathDB 页面 | 原文线索

V2-040444 | MATHDB-366926

Logarithmic delay and polynomial maximum-excursion bounds for Collatz sequences

摘录 / EXCERPT: Let $d_{\text{T}}(n)$ denote the delay and $M_{\text{T}}(n)$ the maximum excursion of the Collatz sequence starting at the positive integer n. Delay and maximum-excursion bound c...

conjecture | open | MathDB 页面 | 原文线索

V2-040445 | MATHDB-366936

Birkhoff's conjecture on integrable billiards

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded convex domain whose boundary $\partial\Omega$ is a C^2-smooth convex curve. Suppose that a neighborhood of $\partial\Omega$ is foliat...

conjecture | open | MathDB 页面 | 原文线索

V2-040446 | MATHDB-366961

Generalized quantum Zernike superintegrability and polynomial symmetry conjecture

摘录 / EXCERPT: Let N be a positive integer and let γ_k be coefficients satisfying $1\leq k\leq N$. Let $\hat{\mathcal{H}}_N$ be the quantum Hamiltonian, let $\hat{\mathcal{C}}$ be the qu...

conjecture | open | MathDB 页面 | 原文线索

V2-040447 | MATHDB-366966

Necessary sampling-condition conjecture for tensor dynamical sampling

摘录 / EXCERPT: Let $\mathcal{F} \in \mathbb{C}^{m \times p \times n}$ be a signal, let Ψ denote its samples under the stated tensor-product dynamical evolution, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040448 | MATHDB-366980

Lagrangian Poincaré recurrence conjecture

摘录 / EXCERPT: Let (X,ω) be a symplectic manifold, let $\psi\in\operatorname{Ham}(X,\omega)$, and let $L\subset(X,\omega)$ be a closed Lagrangian submanifold. Lagrangian Poincaré recurrenc...

conjecture | open | MathDB 页面 | 原文线索

V2-040449 | MATHDB-367193

Measure classification conjecture for stationary measures of symplectic diffeomorphism groups

摘录 / EXCERPT: Measure classification conjecture. The measure ν is a μ-invariant measure on a smooth submanifold M of Q, where the case $M=Q$ is allowed.

conjecture | open | MathDB 页面 | 原文线索

V2-040450 | MATHDB-367228

Finite-Hecke-operator arithmetic quantum unique ergodicity conjecture

摘录 / EXCERPT: Let F be the number field in the paper, let $\mathbb{X}_F$ be the associated quotient, and let $\bar{\operatorname{G}}(F)\backslash\bar{\operatorname{G}}(\mathbb{A})$ carry the v...

conjecture | open | MathDB 页面 | 原文线索

V2-040451 | MATHDB-367280

Eckmann-Ruelle conjecture on pointwise dimensions of hyperbolic measures

摘录 / EXCERPT: Eckmann-Ruelle conjecture. For any hyperbolic measure μ of a $C^{1+\alpha}$ diffeomorphism f, the pointwise dimension exists almost everywhere and is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-040452 | MATHDB-367298

Invariance of connection matrices under compatible functions

摘录 / EXCERPT: Let f_1 and f_2 be functions compatible with the same input vector field, and fix the orders of simplices within each Morse set. Connection-matrix invariance conjecture. The co...

conjecture | open | MathDB 页面 | 原文线索

V2-040453 | MATHDB-367372

Rational invariants as invariants of isogenies of interval exchange systems

摘录 / EXCERPT: Let $(X,\operatorname{\sigma})$ be an interval exchange system (IES) with an invariant decomposition into ergodic probability measures, and let its rational invariants consist of t...

conjecture | open | MathDB 页面 | 原文线索

V2-040454 | MATHDB-367373

Isogeny classification conjecture for Denjoy systems

摘录 / EXCERPT: Let $(\mathcal{D}_{F_k},F_k)$, for $k\in\{0,1\}$, be Denjoy systems with invariants (ρ_k,Q_k). For $A\subset\mathbb{T}$, write $\mathbb{Q}A\subset\mathbb{T}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-040455 | MATHDB-367380

Takens-linearization conjecture for periodic heteroclinic chains

摘录 / EXCERPT: Let $u=[a_{0},a_{1},\ldots]$ be an infinite periodic continued fraction with minimal period $p\ge 3$. The corresponding heteroclinic chain is the dynamical object associated with t...

conjecture | open | MathDB 页面 | 原文线索

V2-040456 | MATHDB-367517

Mean-field limit for the complex Euler-Monge-Ampère equation

摘录 / EXCERPT: Let $x_1(t),\ldots,x_N(t)$ be the trajectories of the Hamiltonian flow generated by $E^{(N)}$, and let $\delta_N((x_1(t),\ldots,x_N(t)))$ be their empirical measure. Let ρ_t a...

conjecture | open | MathDB 页面 | 原文线索

V2-040457 | MATHDB-367563

Explicit complete Lyapunov function conjecture for the synchronisation map on the torus

摘录 / EXCERPT: Let G be the synchronisation diffeomorphism of the torus considered in the paper, and let S and R be the open basins of attraction of its corresponding asymptotically stable...

conjecture | open | MathDB 页面 | 原文线索

V2-040458 | MATHDB-367570

HAW conjecture for bounded geodesics on locally symmetric spaces

摘录 / EXCERPT: Let M be a locally symmetric space of non-compact type with finite Riemannian volume. For $p\in M$, let $S_p(M)$ denote the unit tangent sphere at p, and let $E^+(p,\infty)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-040459 | MATHDB-367571

HAW conjecture for bounded diagonal rays on homogeneous spaces

摘录 / EXCERPT: HAW conjecture. For any $x\in X$, the set $E^+(x,\infty)$ is hyperplane absolute winning (HAW) on $\mathbf{P}^+(\mathfrak{g})$. This is the homogeneous-space counterpart of the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-040460 | MATHDB-367586

Mean convergence conjecture for multiplicative actions and linearly factorable rational polynomials

摘录 / EXCERPT: Mean convergence conjecture. These averages converge in $L^2(\mu)$ as $N\to\infty$. Furthermore, if all the rational polynomials have degree 0, then the conclusion holds for all...

conjecture | open | MathDB 页面 | 原文线索

V2-040461 | MATHDB-367587

Multiple recurrence conjecture for multiplicative actions and linearly factorable rational polynomials

摘录 / EXCERPT: Multiple recurrence conjecture. This liminf is positive. Furthermore, if all the rational polynomials have degree 0 and $R_j(m_0,n_0)=1$ for $j=1,\ldots,\ell$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-040462 | MATHDB-367648

Typical periodic optimization for open expanding multi-valued dynamical systems

摘录 / EXCERPT: Let (X,T) be an open expanding multi-valued dynamical system, and let $\operatorname{Lip}(X)$ denote the Banach space of Lipschitz real-valued functions on X. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040463 | MATHDB-367802

Structural stability implies hyperbolicity and alpha-expansion for piecewise Möbius transformations

摘录 / EXCERPT: Let F be a piecewise Möbius transformation (PMT), and let structural stability, hyperbolicity, and $b1$-expansion have their meanings defined in the paper. Structural-stability c...

conjecture | open | MathDB 页面 | 原文线索

V2-040464 | MATHDB-367803

Structural stability characterization for piecewise Möbius transformations

摘录 / EXCERPT: Let F be a piecewise Möbius transformation with component transformations f_k, and let structural stability, hyperbolicity, and $b1$-expansion have their meanings defined in th...

conjecture | open | MathDB 页面 | 原文线索

V2-040465 | MATHDB-367886

The n-dimensional classification conjecture for bounded omega-limit sets

摘录 / EXCERPT: Let Tcolon $\mathbb{R}^n\to\mathbb{R}^n$ be an nD map as given in Definition 1, and let $\mathcal{A}$ be a bounded ω-limit set different from the fixed point O or from r...

conjecture | open | MathDB 页面 | 原文线索

V2-040466 | MATHDB-367893

Conjecture on stability and convergence of equilibria in the microbial virus model

摘录 / EXCERPT: Assume that $r\kappa\mu_1\neq v\sigma$ and that the $(\operatorname{Coex}_{2,3})$-condition holds, which in particular implies $mv>r+v$. Let $(n_{1a}^*,n_{1i}^*,n_3^*)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040467 | MATHDB-368011

The weak non-uniform hyperbolicity conjecture for physical measures

摘录 / EXCERPT: Let f be a diffeomorphism and let the attracting set under consideration have a dominated splitting. Weak non-uniform expansion means that ... on a positive-volume subset of points...

conjecture | open | MathDB 页面 | 原文线索

V2-040468 | MATHDB-368012

The robust polynomial mixing conjecture for dominated splittings

摘录 / EXCERPT: Let $r\ge1$. Consider diffeomorphisms with a dominated splitting, together with non-uniform expansion and non-uniform contraction, and with neither a uniformly expanding nor a unif...

conjecture | open | MathDB 页面 | 原文线索

V2-040469 | MATHDB-368013

The stable-direction tail dependence conjecture for mixing rates

摘录 / EXCERPT: Consider smooth diffeomorphisms with hyperbolic physical measures. A stable leaf has uniform size on a full-volume subset if there is a cylinder in the ambient space on which that...

conjecture | open | MathDB 页面 | 原文线索

V2-040470 | MATHDB-368014

The ergodic-versus-geometric basin conjecture for hyperbolic cu-Gibbs states

摘录 / EXCERPT: Let f be a smooth diffeomorphism with a hyperbolic cu-Gibbs state. Its geometric basin is the set of points whose empirical measures accumulate on that state, while its ergodic...

conjecture | open | MathDB 页面 | 原文线索

V2-040471 | MATHDB-368051

The Hausdorff-dimension-zero conjecture for exceptional bigeodesic times

摘录 / EXCERPT: Fix a dynamical last passage percolation model, and let $\mathscr{T}$ be the set of times at which non-trivial bigeodesics exist. Hausdorff-dimension-zero conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-040472 | MATHDB-368055

Rosenblatt-Wierdl conjecture on divergence for zero-density ergodic averages

摘录 / EXCERPT: Rosenblatt-Wierdl conjecture. does not converge almost everywhere. This conjecture concerns divergence of sparse ergodic averages at the L^1 endpoint; the paper studies quantitat...

conjecture | open | MathDB 页面 | 原文线索

V2-040473 | MATHDB-368181

Generalized orthogonality of polynomial correlation sequences

摘录 / EXCERPT: Generalized orthogonality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-040474 | MATHDB-368182

General-position conjecture for measurable polynomial recurrence

摘录 / EXCERPT: Let $P=\{p_1,\ldots,p_k\}$ and $Q=\{q_1,\ldots,q_l\}$ be sets of polynomials. Let $\textup{\textsf{{WP}}}(P)$ and $\textup{\textsf{{WP}}}(Q)$ denote their associated Weyl-polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-040475 | MATHDB-368183

Leibman's conjecture on polynomial subgroups of nilpotent Lie groups

摘录 / EXCERPT: Let $(X=G/\Gamma,\mu_X,T)$ be a connected nilsystem, with T induced by left translation by $\tau\in G$. Write $G=G_0\Gamma$, where G_0 is the identity component, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-040476 | MATHDB-368297

The digit-determining conjecture for non-strongly recurrent quadratic laminations

摘录 / EXCERPT: Let $\alpha\in\mathbb{T}$ be a parameter, and let $GCS_n(\alpha)$ denote the generalized cylinder set associated with the first $n+1$ letters of the itinerary $I(\alpha)$. Say that...

conjecture | open | MathDB 页面 | 原文线索

V2-040477 | MATHDB-368300

The billiard decomposition conjecture for planar domains

摘录 / EXCERPT: Let $\Omega\subset \mathbb{R}^2$ be a bounded domain with smooth boundary, and let $p=1,2,\dots$. A billiard trajectory is a trajectory in Ω satisfying the billiard reflecti...

conjecture | open | MathDB 页面 | 原文线索

V2-040478 | MATHDB-368494

Fine-Panov's diffeomorphism conjecture for six-dimensional positive monotone Hamiltonian manifolds

摘录 / EXCERPT: Let (M,ω) be a positive monotone symplectic manifold of dimension six admitting a Hamiltonian circle action. A Fano manifold is a compact complex manifold whose anticanonica...

conjecture | open | MathDB 页面 | 原文线索

V2-040479 | MATHDB-368538

The black hole settling-down conjecture

摘录 / EXCERPT: Black hole settling-down conjecture. At $\Sigma_{t'}$, with $t'\gg t$, the spacetime will be described by a stationary metric. This is a dynamical black-hole statement: after black...

conjecture | open | MathDB 页面 | 原文线索

V2-040480 | MATHDB-368562

Maximal abelianness conjecture for frame representations of commutative semigroups

摘录 / EXCERPT: Let $\mathcal{S}$ be a unital commutative semigroup, possibly with the cancellation property, and let π be a frame representation of $\mathcal{S}$. Let $\mathcal{A}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-040481 | MATHDB-368626

HHU ergodicity conjecture for conservative partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let $f: M^3\rightarrow M^3$ be a C^r ($r > 1$) conservative partially hyperbolic diffeomorphism of a closed 3-manifold that fails to be ergodic. Here, E^s and E^u denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-040482 | MATHDB-368863

The rigidity conjecture for partially rigid motions in the n-body problem

摘录 / EXCERPT: Rigidity conjecture. There are no partially rigid solutions of the n-body problem in $\mathbb{\mathbb{R}}^d$. Equivalently, the only solutions such that one of the mutual distanc...

conjecture | refuted | MathDB 页面 | 原文线索

V2-040483 | MATHDB-369003

Conjecture on polynomial patterns in relatively totally ergodic systems

摘录 / EXCERPT: Polynomial-pattern conjecture. For any integer-valued polynomial $P(x)\in\mathbb{Q}[x]$ such that both $P(x)$ and $P(x)+x$ are nonconstant, there exists $t\in\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040484 | MATHDB-369016

The degenerate Arnol'd conjecture for Hamiltonian-isotopic Lagrangians

摘录 / EXCERPT: Degenerate Arnol'd conjecture. The intersection satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-040485 | MATHDB-369021

Logarithmic joint ergodicity criterion for logarithmic iterates

摘录 / EXCERPT: Let $(X,\mathcal{X},\mu,T_1,T_2)$ be a measure-preserving system. Joint ergodicity for logarithmic averages means convergence of the corresponding logarithmically weighted multiple...

conjecture | open | MathDB 页面 | 原文线索

V2-040486 | MATHDB-369042

Classification conjecture for abelian Cayley graphs with one symmetry class of Hamiltonian cycles

摘录 / EXCERPT: Let Γ be a finite abelian group with generating set S, and let $G=\operatorname{Cay}(\Gamma,S)$. Let $\mathcal{H}$ denote the class of graphs having a unique Hamiltonian c...

conjecture | open | MathDB 页面 | 原文线索

V2-040487 | MATHDB-369082

The converse contraction conjecture for non-exceptional Hurwitz bisets

摘录 / EXCERPT: Let B be a non-exceptional Hurwitz biset, and let B^* denote its associated semigroup of elements. An element of B^* is unobstructed if it has no Thurston obstruction. Conver...

conjecture | open | MathDB 页面 | 原文线索

V2-040488 | MATHDB-369145

The periodic-orbit Lyapunov exponent rigidity conjecture for Anosov geodesic flows

摘录 / EXCERPT: Periodic-orbit Lyapunov rigidity conjecture. If the unstable Lyapunov exponents are constant across all periodic orbits, then M has constant negative sectional curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-040489 | MATHDB-369169

Häggkvist's degree-sum conjecture for Hamilton cycles in oriented graphs

摘录 / EXCERPT: Häggkvist's conjecture. If

conjecture | solved | MathDB 页面 | 原文线索

V2-040490 | MATHDB-369226

Continuity conjecture for conservative matrix field limits

摘录 / EXCERPT: Let $v_0\in\mathbb{Z}^d\setminus\{0\}$, and suppose $\mathcal{L}^{p,q}_{x,v_0}(n)$ converges to $l\in\mathbb{R}$ with non-trivial convergence rate. An angular domain...

conjecture | open | MathDB 页面 | 原文线索

V2-040491 | MATHDB-369264

Convergence of expanding twist-torus measures for Veech surfaces

摘录 / EXCERPT: Let (M,ω) be a horizontally periodic Veech surface, let $\mathcal{M}=\overline{\mathrm{SL}_2(\mathbb{R})\cdot\mathbb{T}(\omega)}$, and let $\mu_\mathbb{T}$ be a fully suppor...

conjecture | open | MathDB 页面 | 原文线索

V2-040492 | MATHDB-369286

The length-complexity conjecture for dynamical bordisms

摘录 / EXCERPT: Length-complexity conjecture. Let $f:\mathbb{N}\dashrightarrow\mathbb{N}$ be a partial recursive function. Given a Turing machine $\mathcal{G}$, for any $n\in\mathbb{N}$ let $T(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-040493 | MATHDB-369294

Katok's intermediate entropy property conjecture

摘录 / EXCERPT: Let f be a C^2-diffeomorphism on a compact Riemannian manifold of dimension greater than one. The map f has the intermediate entropy property when, for every number...

conjecture | open | MathDB 页面 | 原文线索

V2-040494 | MATHDB-369295

Generic entropy flexibility conjecture for diffeomorphisms and flows

摘录 / EXCERPT: Entropy flexibility is the property that, for each admissible entropy level, there is an invariant compact set whose restricted system has that entropy. A property is C^1-generic...

conjecture | open | MathDB 页面 | 原文线索

V2-040495 | MATHDB-369308

Conjecture on localizable genericity of high emergence

摘录 / EXCERPT: Let high emergence mean that the number of probability measures needed to describe a dynamical system up to precision ε grows super-polynomially in ε^{-1}....

open_problem | open | MathDB 页面 | 原文线索

V2-040496 | MATHDB-369309

Arnold's conjecture on exponential growth of periodic points in typical families

摘录 / EXCERPT: Let f be a C^∞ mapping, and let $P_f(n)$ denote the number of isolated points of period n. In a typical sufficiently parameterized family (f_a), interpret “almost all”...

conjecture | open | MathDB 页面 | 原文线索

V2-040497 | MATHDB-369456

Böhm-Lafuente's dynamical Alekseevskii conjecture

摘录 / EXCERPT: Dynamical Alekseevskii conjecture. If a homogeneous space has an immortal Ricci flow, then its universal cover is diffeomorphic to $\mathbb{R}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-040498 | MATHDB-369467

Fredkin-chain conjecture on annihilation of non-cyclic invariant eigenstates

摘录 / EXCERPT: Let $\Sigma^\pm$ be the raising and lowering operators for the periodic Fredkin chain, let C be the cyclic shift operator, and let H be its Hamiltonian. An eigenstate...

conjecture | open | MathDB 页面 | 原文线索

V2-040499 | MATHDB-369596

The two-or-infinitely-many periodic Reeb orbits conjecture

摘录 / EXCERPT: Two-or-infinitely-many periodic Reeb orbits conjecture. The associated Reeb flow has either exactly two or infinitely many periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-040500 | MATHDB-369598

J.-E. Chang's conjectures on the period-energy map

摘录 / EXCERPT: J.-E. Chang's conjectures. If $\lambda\geq 0$, then $E\mapsto\Theta_\lambda(E)$ is strictly monotonically decreasing. If $\lambda<0$, there is a value $V_\lambda^\ast>V_\lambda^0$...

conjecture | open | MathDB 页面 | 原文线索

V2-040501 | MATHDB-369611

Abbondandolo's contact type conjecture for exact magnetic systems

摘录 / EXCERPT: Let M be the configuration manifold, let L be the corresponding magnetic Lagrangian, and let Σ_{κ} be the energy hypersurface at energy κ. The lowest Mañé cr...

conjecture | open | MathDB 页面 | 原文线索

V2-040502 | MATHDB-369667

Fisher-Melnick classification conjecture for lattice actions on closed n-manifolds

摘录 / EXCERPT: Fisher-Melnick classification conjecture. For every smooth action $\rho:\Gamma\rightarrow\operatorname{Diff}(M)$, one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-040503 | MATHDB-369697

Gatzouras-Peres conjecture on measures of maximal dimension

摘录 / EXCERPT: Gatzouras-Peres conjecture. There exists a unique ergodic T-invariant measure with the same Hausdorff dimension as K. Moreover, this measure is mixing for T and is possibly m...

conjecture | open | MathDB 页面 | 原文线索

V2-040504 | MATHDB-369738

The finite constant-solution conjecture for deterministic state-space systems

摘录 / EXCERPT: Finite constant-solution conjecture. If $\#\Sc^{\mathrm{det}}(\Seq{u})$ is constant on $\US^-$ and this constant is finite, then the state-space system has the FMP.

conjecture | open | MathDB 页面 | 原文线索

V2-040505 | MATHDB-369739

The generic constancy conjecture for causal stochastic solutions

摘录 / EXCERPT: Generic constancy conjecture. Suppose that $\Pc_{\mathrm{in}}^-$ and $\Pc_{\mathrm{state}}^-$ are Polish and that...

conjecture | open | MathDB 页面 | 原文线索

V2-040506 | MATHDB-369753

Positive Rates Conjecture for one-dimensional interacting particle systems

摘录 / EXCERPT: Positive Rates Conjecture. Every IPS on a one-dimensional lattice with homogeneous interactions of bounded range and positive rates is ergodic.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-040507 | MATHDB-369754

Positive Rates Conjecture for simple interacting particle systems

摘录 / EXCERPT: PRC for simple IPS. Every simple IPS with positive rates is ergodic.

conjecture | open | MathDB 页面 | 原文线索

V2-040508 | MATHDB-369768

Ergodic variant of the quadratic Rado conjecture

摘录 / EXCERPT: Ergodic variant of the quadratic Rado conjecture. There exist distinct $x,y,z\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040509 | MATHDB-369780

The q-analog $3x+1$ orbit conjecture for T_q

摘录 / EXCERPT: Let $\mathbb{Z}_2$ denote the 2-adic integers, let $F_2[[q]]$ be the ring of formal power series over F_2, and let T and T_q be the maps in the source. A conjugacy is a bij...

conjecture | open | MathDB 页面 | 原文线索

V2-040510 | MATHDB-369781

The polynomial-correspondence conjecture for $T_{1,1+q^2}$

摘录 / EXCERPT: Let $\mathbb{Z}_2$ denote the 2-adic integers, let $F_2[[q]]$ be the ring of formal power series over F_2, and let T and $T_{1,1+q^2}$ be the maps in the source. A conjugacy...

conjecture | open | MathDB 页面 | 原文线索

V2-040511 | MATHDB-369796

Conjecture on coincidence of pathwise and phase-locking critical coupling strengths

摘录 / EXCERPT: Let $ u_1,\ldots, u_N$ be fixed natural frequencies, and let $ u=(u_1,\ldots, u_N)$. Consider the inertial Kuramoto system $\$, with coupling strength κ, phase initial data...

conjecture | open | MathDB 页面 | 原文线索

V2-040512 | MATHDB-369804

Lacoin-Rhodes-Vargas left-tail conjecture for derivative Gaussian multiplicative chaos

摘录 / EXCERPT: Let $D\subset\mathbb{R}^d$ and let a log-correlated Gaussian field on D admit a smooth white noise decomposition. Denote its derivative Gaussian multiplicative chaos by...

conjecture | open | MathDB 页面 | 原文线索

V2-040513 | MATHDB-369821

Monotonicity conjecture for the random billiard walk variance

摘录 / EXCERPT: Let $b\in S^{d-1}$ be a direction, let $p\in(0,1)$ be the parameter, and let $\sigma_{b,p}^2$ denote the variance of the random billiard walk in direction b. Monotonicity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-040514 | MATHDB-369822

Discontinuity conjecture for the random billiard walk variance

摘录 / EXCERPT: Let $p\in(0,1)$ be fixed, let $b\in S^{d-1}$ be a direction, and let $\sigma_{b,p}$ denote the standard deviation associated with the random billiard walk. A direction is called ra...

conjecture | open | MathDB 页面 | 原文线索

V2-040515 | MATHDB-369823

Invariance conjecture for the random refraction billiard walk

摘录 / EXCERPT: Invariance conjecture. For every $R_0\in\mathbb{R}^d$, $b\in S^{d-1}$ and $p\in(0,1)$, the rescaled trajectories $R_t^{(n)}$, $t\in[0,1]$, converge to isotropic Brownian motion in...

conjecture | open | MathDB 页面 | 原文线索

V2-040516 | MATHDB-369907

Pisot-condition thresholds for classical multidimensional continued fraction algorithms

摘录 / EXCERPT: Consider the Brun, Jacobi--Perron, and Selmer multidimensional continued fraction algorithms in dimension d. The convergence-threshold conjecture. The Brun algorithm satisfies th...

conjecture | open | MathDB 页面 | 原文线索

V2-040517 | MATHDB-369989

Minkowski dimension conjecture for degenerate-focus spirals

摘录 / EXCERPT: Minkowski dimension conjecture. Any such spiral trajectory has Minkowski dimension

conjecture | open | MathDB 页面 | 原文线索

V2-040518 | MATHDB-370249

Analytic-function extension of the polynomial Koksma-Weyl theorem

摘录 / EXCERPT: Let $g:\mathbb{R}\to(1,\infty)$ be a nonconstant analytic function, let $b:\mathbb{N}\to\mathbb{R}$ tend to infinity and be scattered, let $k\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040519 | MATHDB-370333

Shub's conjecture on eigenvalue products and Grassmannian distortion

摘录 / EXCERPT: Shub's conjecture. There is a constant $c_{d,k}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040520 | MATHDB-370359

Low-temperature ordering conjecture for weakly coupled mixed-chaotic Hamiltonian systems

摘录 / EXCERPT: Let a Hamiltonian dynamical system exhibit mixed chaos, with chaotic trajectories and islands of regularity in its phase space. Suppose the system is weakly coupled to a thermal ba...

conjecture | open | MathDB 页面 | 原文线索

V2-040521 | MATHDB-370380

Antihydra's odd-even frequency conjecture

摘录 / EXCERPT: Consider the map $H:^+\to^+$ defined by ... Starting from $x=8$, classify each iterate as odd or even. Antihydra's odd-even frequency conjecture. At no point in the iteration are the...

conjecture | open | MathDB 页面 | 原文线索

V2-040522 | MATHDB-370406

Unique Ferapontov Hamiltonian-operator conjecture for first-order WDVV systems

摘录 / EXCERPT: Let A_1 be a first-order Hamiltonian operator of Ferapontov type for a first-order WDVV system, compatible with the third-order Hamiltonian operator supplied by the Hamiltonian t...

conjecture | open | MathDB 页面 | 原文线索

V2-040523 | MATHDB-370407

The metastable random-walk conjecture for the dynamical sine-Gordon model

摘录 / EXCERPT: Metastable random-walk conjecture. As $\epsilon\to0$, the rescaled process approaches a symmetric simple random walk on

conjecture | open | MathDB 页面 | 原文线索

V2-040524 | MATHDB-370427

Favre's algebraic-integrality conjecture for dynamical degrees

摘录 / EXCERPT: Let k be a field, let n be a positive integer, and let $f\in\operatorname{End}(\mathbb{A}^n_k)$. Its dynamical degree is ... where $\deg(f)$ is the maximum of the degrees of the...

conjecture | open | MathDB 页面 | 原文线索

V2-040525 | MATHDB-370531

Fried's surgery conjecture for suspension pseudo-Anosov flows

摘录 / EXCERPT: Let Φ_1 and Φ_2 be suspension pseudo-Anosov flows. A Dehn-Goodman-Fried surgery is a surgery performed on a periodic orbit of a pseudo-Anosov flow to produce another pseu...

conjecture | open | MathDB 页面 | 原文线索

V2-040526 | MATHDB-370532

The Markovian-family conjecture for pseudo-Anosov flows

摘录 / EXCERPT: Let Φ be a pseudo-Anosov flow, let $\mathcal{P}$ be its bifoliated plane, and let $\mathcal{R}$ be a Markovian family for Φ. A reduced Markov partition of Φ is a Mar...

conjecture | open | MathDB 页面 | 原文线索

V2-040527 | MATHDB-370600

Uniqueness conjecture for homogeneous quasi-morphisms on Hamiltonian groups of complete Liouville manifolds

摘录 / EXCERPT: Let M be a complete Liouville manifold, and consider its Hamiltonian group or its universal cover. A homogeneous quasi-morphism is a quasi-morphism $ u$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-040528 | MATHDB-370731

Kotzig's perfect 1-factorisation conjecture for complete graphs

摘录 / EXCERPT: Let $n\geq 2$ be an even integer. A perfect 1-factorisation of a graph is a partition of its edge set into perfect matchings such that the union of any two distinct perfect match...

conjecture | open | MathDB 页面 | 原文线索

V2-040529 | MATHDB-370751

Uniqueness conjecture for expanding-attractor flows on the cyclic branched cover

摘录 / EXCERPT: Let $p:\widetilde{M}^3\to\mathbb{S}^3$ be a 2-fold cyclic branched covering map branched over the figure-eight knot l, and let $p^{-1}(l)$ be its preimage knot. Let f^t_* be...

conjecture | open | MathDB 页面 | 原文线索

V2-040530 | MATHDB-370762

Diffusion conjecture for random Hamiltonian diffeomorphisms

摘录 / EXCERPT: Diffusion conjecture. For every $\varepsilon>0$, there exists a law-defining datum $\mathcal{D}$ such that the push-forward measure $(ev_p)_*\mu_{\operatorname{Ham}}^{\mathcal{D}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-040531 | MATHDB-370763

Crofton-type conjecture for random Hamiltonian diffeomorphisms

摘录 / EXCERPT: Crofton-type conjecture. There exist a constant $C>0$ and a smooth function $\rho:M\to\mathbb{R}_{>0}$, depending only on $\mathcal{D}$ and L and satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-040532 | MATHDB-370804

Sabok's conjecture on the complexity of isomorphism of ergodic measure-preserving transformations

摘录 / EXCERPT: Let EMPTs denote ergodic measure-preserving transformations, and consider their isomorphism relation under measure-preserving conjugacy. An equivalence relation is Borel reducible...

conjecture | open | MathDB 页面 | 原文线索

V2-040533 | MATHDB-370811

Lipschitz right-inverse conjecture for cocycle operators

摘录 / EXCERPT: Lipschitz right-inverse conjecture. There exists $L=L(\lambda,C)$ and a not necessarily linear function $S_{\alpha,A}:X\to X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040534 | MATHDB-370842

Three-parameter Cartesian-product conjecture for disconnected large bifurcation supports

摘录 / EXCERPT: Let a generic 3-parameter family have a disconnected large bifurcation support. Three-parameter Cartesian-product conjecture. The family is equivalent either to a Cartesian produ...

conjecture | open | MathDB 页面 | 原文线索

V2-040535 | MATHDB-370843

ET-to-SET conjecture for generic glocal families

摘录 / EXCERPT: Let a glocal family have the extension triviality property (ET), meaning that its glocal extensions are moderately equivalent near the relevant large bifurcation support to trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-040536 | MATHDB-370845

Tameness conjecture for generic codimension-one degeneracies

摘录 / EXCERPT: Let $\v \in \operatorname{Vect}^{*}_k(\Omega)$, where Ω is an open subset of S^2 or S^2 itself, and suppose that v contains only a generic degeneracy of codimension...

conjecture | open | MathDB 页面 | 原文线索

V2-040537 | MATHDB-370846

Basic-list conjecture for disconnected large bifurcation supports

摘录 / EXCERPT: Let a generic two-parameter glocal family of vector fields have a disconnected large bifurcation support. The basic list consists of the six unavoidable generic one-parameter degen...

conjecture | open | MathDB 页面 | 原文线索

V2-040538 | MATHDB-370864

The Weinstein conjecture for periodic orbits on contact-type energy levels

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold and let $H:M\to\mathbb{R}$ be a Hamiltonian. An energy level is a level set $H^{-1}(c)$, and a hypersurface $Y\subset M$ is of contact type...

conjecture | open | MathDB 页面 | 原文线索

V2-040539 | MATHDB-370865

Hofer-Wysocki-Zehnder's two-or-infinity conjecture

摘录 / EXCERPT: Let Y be a star-shaped hypersurface in $\mathbb{R}^4$, and call a periodic orbit simple if it is not a multiple cover. Hofer-Wysocki-Zehnder's two-or-infinity conjecture. A Hamil...

conjecture | open | MathDB 页面 | 原文线索

V2-040540 | MATHDB-370875

Jumps of the roundness function

摘录 / EXCERPT: Let ρ be a parameter, let Λ_ρ be the associated set, and let $R_\rho(\Lambda_\rho)$ denote its roundness. For each $k\in\mathbb{N}\setminus\{0,1\}$, consider the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-040541 | MATHDB-370910

The vertical amplitude conjecture for horizontal unitracks

摘录 / EXCERPT: Vertical amplitude conjecture. The sequence $V(\gamma_n)$ is unbounded. The preceding theorem establishes analogous linear bounds and unboundedness for the horizontal amplitudes...

conjecture | open | MathDB 页面 | 原文线索

V2-040542 | MATHDB-370911

The self-intersection conjecture for horizontal unitracks

摘录 / EXCERPT: Let γ_0 be an initial curve in the horizontal unitrack construction, and let γ_n denote the successive curves. Call γ_0 trivial when it is the trivial initial...

conjecture | open | MathDB 页面 | 原文线索

V2-040543 | MATHDB-370912

The unitrack self-intersection conjecture

摘录 / EXCERPT: Let γ_0 be an initial curve in the horizontal unitrack construction, and let γ_n denote the successive curves for $n\in\mathbb{N}$. The unitrack is the union of these...

conjecture | open | MathDB 页面 | 原文线索

V2-040544 | MATHDB-370922

Robust transitivity with robust singularities near a product map

摘录 / EXCERPT: Let M and N be closed manifolds of arbitrary finite dimension. Suppose that M supports an expanding map F. Product-map conjecture. The product $M\times N$ supports a C^1...

conjecture | open | MathDB 页面 | 原文线索

V2-040545 | MATHDB-370945

Existence of 3-periodic orbits for polygonal outer length billiards

摘录 / EXCERPT: Let Q be a polygon, and consider its outer length billiard, including its periodic orbits and their lengths. 3-periodic orbit conjecture. Every polygonal outer length billiard ad...

conjecture | open | MathDB 页面 | 原文线索

V2-040546 | MATHDB-370946

Hausdorff dimension and dense-orbit conjecture for polygonal outer length billiards

摘录 / EXCERPT: Let Q be a polygon, and let its singularity set be the set of points where the outer length billiard is singular. Let an orbit be a sequence of points generated by the billiard,...

conjecture | open | MathDB 页面 | 原文线索

V2-040547 | MATHDB-370947

Open ball of diverging orbits around a square in outer length billiards

摘录 / EXCERPT: Consider the outer length billiard associated with a square, and call an orbit diverging if it is unbounded. Diverging-orbit conjecture. There exists an open ball of diverging orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-040548 | MATHDB-370952

Pointwise Bernoulli Furstenberg-system conjecture for superpolynomial sequences

摘录 / EXCERPT: Let $(X,\mathcal{B},\mu,T)$ be a measure-preserving system, let $0<c<1/2$, and let $f\in L^{\infty}(\mu)$. For $x\in X$, define ... A Furstenberg system associated to a sequence is a...

conjecture | open | MathDB 页面 | 原文线索

V2-040549 | MATHDB-370953

Averaging and recurrence conjecture for zero-entropy systems along superpolynomial times

摘录 / EXCERPT: Let $(X,\mathcal{B},\mu)$ be a probability space, and let $T,S:X\to X$ be two not necessarily commuting measure-preserving transformations. Let $0<c<1/2$, and suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-040550 | MATHDB-370974

Pugh-Shub conjecture on Kolmogorov-typical finiteness of attractors

摘录 / EXCERPT: Pugh-Shub conjecture. For every $d\geq 1$, a d-Kolmogorov typical diffeomorphism displays finitely many sinks and other attractors.

conjecture | open | MathDB 页面 | 原文线索

V2-040551 | MATHDB-370975

Conjecture on local Kolmogorov typicality of generically localizable properties

摘录 / EXCERPT: Localizability conjecture. There are many spaces of dynamical systems for which any generically localizable property is locally Kolmogorov typical.

conjecture | open | MathDB 页面 | 原文线索

V2-040552 | MATHDB-370976

Berger's conjecture on local typicality of high emergence

摘录 / EXCERPT: Berger's high-emergence conjecture. In many spaces of dynamical systems, high emergence is locally typical, in many senses.

conjecture | open | MathDB 页面 | 原文线索

V2-040553 | MATHDB-371008

Existence of nonlocal Lie symmetries for the integrable maps

摘录 / EXCERPT: The integrable map corresponding to the linear equation, respectively the nonlinear equation, is denoted by ... . A nonlocal diffeomorphism is a diffeomorphism whose action is nonloc...

conjecture | open | MathDB 页面 | 原文线索

V2-040554 | MATHDB-371009

Lie algebra correspondence for nonlocal symmetries

摘录 / EXCERPT: Let $\mathcal{G}$ be the Lie group of nonlocal symmetry diffeomorphisms associated with the integrable map ... . Let its Lie algebra of generators be the Lie algebra of the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-040555 | MATHDB-371143

Arnold's conjecture for non-degenerate Hamiltonians

摘录 / EXCERPT: Arnold's conjecture for non-degenerate Hamiltonians.

conjecture | open | MathDB 页面 | 原文线索

V2-040556 | MATHDB-371189

Subcritical Gaussian multiplicative chaos Fourier coefficient limit conjecture

摘录 / EXCERPT: Let $\gamma\in(1/\sqrt{2},\sqrt{2})$, let μ_γ be the Gaussian multiplicative chaos measure on $[0,1]$ with parameter γ, and let c_n be its n-th Fourier coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-040557 | MATHDB-371251

Bergelson-Leibman-Lesigne conjecture on jointly intersective polynomial recurrence

摘录 / EXCERPT: Let $p_1,\ldots,p_\ell\in\mathbb{Z}[t]$ be jointly intersective, meaning that for every $r\in\mathbb{N}$ there exists $n\in\mathbb{N}$ such that r divides each of...

conjecture | open | MathDB 页面 | 原文线索

V2-040558 | MATHDB-371265

Pugh-Shub accessibility conjecture for partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let M be a closed Riemannian manifold, and let $f:M\to M$ be a partially hyperbolic diffeomorphism, meaning that TM admits a Df-invariant splitting ... where E^s is uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-040559 | MATHDB-371299

Ergodic polynomial odd-recurrence conjecture

摘录 / EXCERPT: Ergodic polynomial odd-recurrence conjecture. The set R has nonempty intersection with every infinite arithmetic progression of step size k.

conjecture | open | MathDB 页面 | 原文线索

V2-040560 | MATHDB-371302

Boundary equilibrium bifurcation under variation of an ablation parameter

摘录 / EXCERPT: Consider the piecewise glacial cycles system ... exhibits a boundary equilibrium bifurcation. The claim concerns parameter-driven changes in the type of equilibrium, but the supplied...

conjecture | open | MathDB 页面 | 原文线索

V2-040561 | MATHDB-371305

Conjugacy conjecture for Lorenz-attractor first-return maps

摘录 / EXCERPT: Let P be the parameter space, let R_0 and R_1 be the two components of the cross-section R, and let $h_v:R_0\cup R_1\to\mathbb{R}^2$ be the map specified in Fig. S. For...

conjecture | open | MathDB 页面 | 原文线索

V2-040562 | MATHDB-371307

Generic no-cut-and-paste conjecture for Lorenz maps

摘录 / EXCERPT: Let O be the collection of vector fields under consideration, let $\mathcal{L}$ be the collection of Lorenz maps that form factor maps for vector fields $V\in O$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040563 | MATHDB-371404

Transverse intersection conjecture for dimension functions of quadratic Sturmian spectra

摘录 / EXCERPT: For each quadratic irrationality α, let $\Sigma_{\alpha,\lambda}$ be the spectrum of the corresponding Sturmian Hamiltonian at coupling constant λ. Transverse inters...

conjecture | open | MathDB 页面 | 原文线索

V2-040564 | MATHDB-371405

Non-differentiability conjecture for dimensions of bounded-type Sturmian spectra

摘录 / EXCERPT: Let $\Sigma_{\alpha,\lambda}$ be the spectrum of the Sturmian Hamiltonian with frequency α and coupling constant λ. Bounded-type non-differentiability conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-040565 | MATHDB-371419

Palis's conjecture on Morse-Smale systems and transversal homoclinic orbits

摘录 / EXCERPT: A diffeomorphism f is a diffeomorphism of a compact manifold. Palis's conjecture. Any diffeomorphism can be C^r approximated by a Morse-Smale one or by one exhibiting a transve...

conjecture | open | MathDB 页面 | 原文线索

V2-040566 | MATHDB-371431

Dedieu-Shub random-deterministic exponent conjecture for real matrices

摘录 / EXCERPT: Dedieu-Shub conjecture. There are constants $c_{d,k}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-040567 | MATHDB-371440

Palis's finitude of attractors conjecture for interval maps

摘录 / EXCERPT: Let $C^r([0,1])$ denote the space of C^r interval maps. For a generic k-parameter family $(f_t)_{t\in[-1,1]^k} \subset C^r([0,1])$, where $t\in[-1,1]^k$ is equipped with Lebesg...

conjecture | open | MathDB 页面 | 原文线索

V2-040568 | MATHDB-371488

Baker-DeMarco's dynamical André-Oort conjecture

摘录 / EXCERPT: Dynamical André-Oort conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-040569 | MATHDB-371497

High-entropy finiteness of maximal-entropy measures for finite-regularity local diffeomorphisms

摘录 / EXCERPT: Fix $r>1$, and let M be the underlying compact manifold and $f:M\rightarrow M$ a $\mathcal{C}^r$ local diffeomorphism. Write $\lVert f\rVert_{\mathcal{C}^1}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-040570 | MATHDB-371501

Conjecture on arguments of complex Markov triples

摘录 / EXCERPT: Let (z_1,z_2,z_3) be an initial seed triple of roots of unity, and let n be the minimal positive integer such that $z_i^n=1$ for each i. A triple of rational numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-040571 | MATHDB-371591

Shnirelman-Šverák relaxation conjecture for two-dimensional Euler flows

摘录 / EXCERPT: Shnirelman-Šverák relaxation conjecture. The set $\mathcal{A}_\infty$ consists of precisely those $f\in L^\infty(\Omega)$ for which $\{\mathcal{S}_t(f)\}_{t\in\mathbb{R}}$ is pre-c...

conjecture | open | MathDB 页面 | 原文线索

V2-040572 | MATHDB-371643

ASE implies non-uniform sectional expansion

摘录 / EXCERPT: ASE-to-NU2SE conjecture. A partially hyperbolic compact invariant set for a smooth flow satisfying the ASE condition (pASE) also satisfies the NU2SE (NUpSE) condition.

conjecture | open | MathDB 页面 | 原文线索

V2-040573 | MATHDB-371644

Physical/SRB measures without slow recurrence

摘录 / EXCERPT: Physical/SRB without slow recurrence conjecture. In Theorem and Theorem the slow recurrence conditions are superfluous.

conjecture | open | MathDB 页面 | 原文线索

V2-040574 | MATHDB-371645

Unique physical measure for the constructed p-sectional expanding attractors

摘录 / EXCERPT: Physical-measure conjecture. The attractors constructed in Section admit a unique physical/SRB measure with a full volume ergodic basin.

conjecture | open | MathDB 页面 | 原文线索

V2-040575 | MATHDB-371670

The property $(T_n)_{L^1}$ Zimmer conjecture

摘录 / EXCERPT: Let Γ be a group with property $(T_n)_{L^1}$, meaning the L^1-coefficient form of higher property T used in the source. The $(T_n)_{L^1}$ Zimmer conjecture. Every diffeomo...

conjecture | open | MathDB 页面 | 原文线索

V2-040576 | MATHDB-371675

Stuck-Zimmer's IRS rigidity conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be a center-free irreducible higher-rank lattice. Stuck-Zimmer's IRS rigidity conjecture. For every ergodic probability-measure-preserving action of Γ on a prob...

conjecture | open | MathDB 页面 | 原文线索

V2-040577 | MATHDB-371730

Avoidance conjecture for weakly strict saddle points

摘录 / EXCERPT: Avoidance conjecture. If a compact connected component C of $\operatorname{Crit}(f)$ consists of weakly strict saddle points, then $W^s_-(C)$ has measure zero. Moreover, if M i...

conjecture | open | MathDB 页面 | 原文线索

V2-040578 | MATHDB-371748

Hotspot theorem conjecture for deterministic dynamically generated basic sequences

摘录 / EXCERPT: Let $Q=(q_n)_{n\geq 1}$ be a deterministic dynamically generated basic sequence. Let the strongest hotspot theorem refer to the assertion that the hotspot hypotheses imply membersh...

conjecture | open | MathDB 页面 | 原文线索

V2-040579 | MATHDB-371749

Normality equivalence conjecture for deterministic dynamically generated basic sequences

摘录 / EXCERPT: Let $Q=(q_n)_{n\geq 1}$ be a deterministic dynamically generated basic sequence. Write $\mathcal{N}(Q)$ for the set of Q-normal numbers, $\mathcal{UN}(Q)$ for the uniformly Q-n...

conjecture | open | MathDB 页面 | 原文线索

V2-040580 | MATHDB-371750

Descriptive-complexity conjecture for nondeterministic dynamically generated basic sequences

摘录 / EXCERPT: Let $Q=(q_n)_{n\geq 1}$ be a nondeterministic dynamically generated basic sequence. Let $\mathcal{N}(Q)$, $\mathcal{DN}(Q)$, and $\mathcal{UN}(Q)$ denote the corresponding normalit...

conjecture | open | MathDB 页面 | 原文线索

V2-040581 | MATHDB-371803

Effective SPR conjecture for surface diffeomorphisms

摘录 / EXCERPT: Effective SPR conjecture. For every $1>\alpha_1>\alpha_2>0$, there exist $\beta_0>0$ and $C>0$ such that, for every ergodic measure μ of f satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-040582 | MATHDB-371967

El Sahili's arc-deletion conjecture for oriented Hamiltonian paths

摘录 / EXCERPT: Let T be a strong tournament of order $n\geq 8$, let e be an arc of T, and let P be an oriented path of order n. El Sahili's conjecture. The tournament $T-e$ contains P...

conjecture | open | MathDB 页面 | 原文线索

V2-040583 | MATHDB-371988

Khesin-Soloviev's dynamical degree conjecture for truly skew pentagram maps

摘录 / EXCERPT: Let $T_{a,b,c,d}$ be a truly skew pentagram map, with parameters (a,b,c,d) as in the paper, and let its dynamical degree be denoted by $\rho(T_{a,b,c,d})$. Khesin-Soloviev's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-040584 | MATHDB-372009

Cyclicity conjecture for the highest Hamiltonian of the quadratic Poisson structure

摘录 / EXCERPT: Cyclicity conjecture. The Hamiltonian $h^{(N)}$ is the cyclic variable of the quadratic Poisson structure.

conjecture | open | MathDB 页面 | 原文线索

V2-040585 | MATHDB-372052

Bistability conjecture for a three-class structured population model

摘录 / EXCERPT: Bistability conjecture. There exists a choice of parameters such that $u^\star$ is locally asymptotically stable but not globally asymptotically stable. The ω-limit set also...

conjecture | open | MathDB 页面 | 原文线索

V2-040586 | MATHDB-372067

Orbit classification conjecture for points on K3 surfaces

摘录 / EXCERPT: Let X be a complex K3 surface, let $\operatorname{Aut}(X)$ be its automorphism group, and let $\Gamma\subseteq \operatorname{Aut}(X)$ be a nonelementary subgroup, meaning that it...

conjecture | open | MathDB 页面 | 原文线索

V2-040587 | MATHDB-372068

Zero-Lyapunov-exponent measure conjecture for K3 automorphisms

摘录 / EXCERPT: Let X be a complex K3 surface, let $\Gamma\subseteq \operatorname{Aut}(X)$ be a nonelementary subgroup, and let μ be a probability measure on a generating set of Γ. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-040588 | MATHDB-372069

Symplectic orbit-closure and stationary-measure stratification conjecture

摘录 / EXCERPT: Let Q be a compact symplectic manifold, and let μ be supported on a group Γ_μ of smooth symplectomorphisms of Q. Assume that μ has uniform expansion on every i...

open_problem | open | MathDB 页面 | 原文线索

V2-040589 | MATHDB-372108

The upper bound conjecture for ergodic invariant measures of FIETs

摘录 / EXCERPT: Upper bound conjecture. A FIET can have at most

conjecture | open | MathDB 页面 | 原文线索

V2-040590 | MATHDB-372221

The infinite spectral-diameter conjecture for symplectic manifolds

摘录 / EXCERPT: Let (M,ω) be a compact symplectic manifold, and let γ be the Floer-theoretic pseudometric on the universal cover of $\mathrm{Ham}(M)$. Assume that ω vanishes o...

conjecture | open | MathDB 页面 | 原文线索

V2-040591 | MATHDB-372398

Universal pathwise critical coupling conjecture for the Winfree model

摘录 / EXCERPT: Let $\Omega=\\{\omega_1,\cdots,\omega_N\\}\in \mathbb{R}^N$ be a frequency vector, and let $\Theta^0=\\{\theta_1^0,\cdots,\theta_N^0\\}$ be an initial phase vector. Define the crit...

conjecture | open | MathDB 页面 | 原文线索

V2-040592 | MATHDB-372401

The PYY conjecture on singular star flows and aperiodic classes

摘录 / EXCERPT: A singular star flow is a star flow with singularities, and a singular aperiodic class is a chain-recurrent class containing singularities but no periodic orbit. A homoclinic class...

conjecture | open | MathDB 页面 | 原文线索

V2-040593 | MATHDB-372412

Dry Ten Martini conjecture for type-I operators

摘录 / EXCERPT: Let α be irrational and let $v\in C^\omega(\mathbb{T},\mathbb{R})$. Write $\Sigma_{v,\alpha}$ for the spectrum and $N_{v,\alpha}$ for the integrated density of states. An en...

conjecture | open | MathDB 页面 | 原文线索

V2-040594 | MATHDB-372440

The deterministic limiting velocity conjecture for stationary ergodic random environments

摘录 / EXCERPT: Let $(X_n)_{n\geq 0}$ be a random walk in a stationary ergodic environment on $\mathbb{Z}^d$. A deterministic limiting velocity is a vector v such that ... for almost every realisa...

conjecture | open | MathDB 页面 | 原文线索

V2-040595 | MATHDB-372441

Kalikow's 0-1 conjecture for directional escape in random environments

摘录 / EXCERPT: Kalikow's 0-1 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-040596 | MATHDB-372533

The Morse-Smale telic-problem tractability conjecture

摘录 / EXCERPT: Morse-Smale tractability conjecture. Every telic problem coming from the system is contained in $\textsf{P}$.

conjecture | open | MathDB 页面 | 原文线索

V2-040597 | MATHDB-372534

The sensitivity conjecture for intractable telic problems

摘录 / EXCERPT: Sensitivity conjecture. If (X,T) admits an intractable telic problem, then there is a closed subset $E\subset X$ on which the system has sensitivity to initial conditions.

conjecture | open | MathDB 页面 | 原文线索

V2-040598 | MATHDB-372604

Silverman's dynamical Lang conjecture for canonical heights

摘录 / EXCERPT: Let $d\geq 2$, let $f:\mathbb{P}^1\to\mathbb{P}^1$ be a degree-d dynamical system, let $h_{\mathcal{M}}$ be a height on the moduli space $\mathcal{M}_d$ of degree-d dynamical s...

conjecture | open | MathDB 页面 | 原文线索

V2-040599 | MATHDB-372638

Standard-form classification conjecture for tau-symmetric RH perturbations

摘录 / EXCERPT: Let a τ-symmetric Hamiltonian perturbation of the Riemann--Hopf hierarchy be written in generalized standard form, with first Hamiltonian ... Here $\mathcal{P}_k$ denotes the se...

conjecture | open | MathDB 页面 | 原文线索

V2-040600 | MATHDB-372648

Quasiconformal universality conjecture for Basilica limit sets

摘录 / EXCERPT: Let K_i, for $i\in\{1,2\}$, be Basilica limit sets of geometrically finite conformal dynamical systems, such as limit sets of geometrically finite Kleinian groups or Julia sets o...

conjecture | open | MathDB 页面 | 原文线索

V2-040601 | MATHDB-372649

David hierarchy conjecture for Basilica limit sets

摘录 / EXCERPT: Let K_i, for $i\in\{1,2\}$, be Basilica limit sets of geometrically finite conformal dynamical systems. A map is weakly type-preserving if it sends parabolic points to parabolic...

conjecture | open | MathDB 页面 | 原文线索

V2-040602 | MATHDB-372650

Quasiconformal universality conjecture for conformal-dimension-one limit sets

摘录 / EXCERPT: Let K_i, for $i\in\{1,2\}$, be limit sets of hyperbolic conformal dynamical systems, such as limit sets of convex co-compact Kleinian groups or Julia sets of hyperbolic rational...

conjecture | open | MathDB 页面 | 原文线索

V2-040603 | MATHDB-372847

Conjecture on dynamically stable solutions for curve diffusion, elastic, and ideal flows

摘录 / EXCERPT: Stability classification conjecture. The only dynamically stable solutions are circles for the curve diffusion flow; multiply-covered lemniscates of Bernoulli and multiply-covered...

conjecture | open | MathDB 页面 | 原文线索

V2-040604 | MATHDB-373106

Furstenberg-Bergelson-Leibman pointwise convergence conjecture

摘录 / EXCERPT: Furstenberg-Bergelson-Leibman conjecture. These averages converge for μ-almost every $x\in X$ as $\min\\{N_1,\ldots,N_k\\}\to\infty$. When $N_1=\cdots=N_k=N$, write the average...

conjecture | open | MathDB 页面 | 原文线索

V2-040605 | MATHDB-373213

Falconer's conjecture on stable intersections of regular Cantor sets

摘录 / EXCERPT: Let $K\subset\mathbb{R}^d$ be a regular Cantor set, and write $\operatorname{dim_{H}}(K)$ for its Hausdorff dimension. A pair of sets has a C^r-stable intersection if this inters...

conjecture | open | MathDB 页面 | 原文线索

V2-040606 | MATHDB-373358

Conjecture on stability and convergence of the coordinate-wise positive equilibrium

摘录 / EXCERPT: Stability and convergence conjecture. (A) If

conjecture | open | MathDB 页面 | 原文线索

V2-040607 | MATHDB-373501

The phase-transition conjecture for minima of $sigma$-Brjuno functions

摘录 / EXCERPT: Let $B_\sigma:[0,1]\to\mathbb R$ be the σ-Brjuno function, and let $\eta_n=(\sqrt{n^2+4}-n)/2$ be the n-th fixed point of the Gauss map. For each $n\in\mathbb N$, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-040608 | MATHDB-373511

Random subgraph Hamiltonicity conjecture for Cayley graphs

摘录 / EXCERPT: Let G be a Cayley graph on n vertices with degree d, and let G_p be the random graph obtained by retaining each edge of G independently with probability p. Random Cayle...

conjecture | open | MathDB 页面 | 原文线索

V2-040609 | MATHDB-373523

Bialy's conjecture on spectral rigidity among ellipses

摘录 / EXCERPT: Let ρ_0 and ρ_1 be two distinct rotation numbers in $(0,1/2]$. For ellipses $\mathscr E$ and $\mathscr E'$, let $\beta_{\mathscr E}(\rho)$ denote the Mather beta function...

conjecture | open | MathDB 页面 | 原文线索

V2-040610 | MATHDB-373556

The Orbit Equidistribution Conjecture for Collatz orbits

摘录 / EXCERPT: Let T denote the odd-to-odd Syracuse map, let n_0 be odd, and define $a_i=T^i(n_0)\bmod 2^M$. Let $\mu_{\mathrm{orb},N}=\frac{1}{N}\sum_{i=1}^N\delta_{a_i}$ and let μ_U be...

conjecture | open | MathDB 页面 | 原文线索

V2-040611 | MATHDB-373557

The observable-specific weak-mixing conjecture for phantom gain discrepancies

摘录 / EXCERPT: Let T be the odd-to-odd Syracuse map, let n_0 be odd, and let h_K be the phantom gain observable at depth K, with reference mean $R(K)$. Define ... Observable-specific weak-m...

conjecture | open | MathDB 页面 | 原文线索

V2-040612 | MATHDB-373558

The depthwise discrepancy recurrence conjecture for Collatz orbits

摘录 / EXCERPT: Let $\delta_K(n_0)$ denote the total-variation discrepancy at modular depth K for the orbit starting at odd n_0. Depthwise discrepancy recurrence conjecture. There exist consta...

conjecture | open | MathDB 页面 | 原文线索

V2-040613 | MATHDB-373559

The gain-observable recurrence conjecture for Collatz orbits

摘录 / EXCERPT: Let $\eta_K(n_0)$ be the phantom gain discrepancy at depth K for the odd-to-odd Syracuse orbit starting at odd n_0. Gain-observable recurrence conjecture. There exist constants...

conjecture | open | MathDB 页面 | 原文线索

V2-040614 | MATHDB-373560

The summable alignment-renewal conjecture for Collatz shadows

摘录 / EXCERPT: Let $\eta_K^{\mathrm{renew}}(n_0)$ denote the total discrepancy contribution from renewed depth-K alignment events for an orbit of the odd-to-odd Syracuse map starting at odd...

conjecture | open | MathDB 页面 | 原文线索

V2-040615 | MATHDB-373561

The spectral diffusion conjecture for Collatz orbit measures

摘录 / EXCERPT: Let μ_K be the orbit measure on $\mathbb{Z}/2^K\mathbb{Z}$, let $\operatorname{hw}(\xi)$ denote Hamming weight, and define the spectral content at weight w by ... Spectral diff...

conjecture | open | MathDB 页面 | 原文线索

V2-040616 | MATHDB-373562

The orbit covariance summability conjecture for Collatz drift increments

摘录 / EXCERPT: Let Δ_i be the odd-skeleton drift increments along a Collatz orbit, and let $\operatorname{Cov}_{\mathrm{orbit}}(\Delta_0,\Delta_h)$ denote their orbit covariance at lag h...

conjecture | open | MathDB 页面 | 原文线索

V2-040617 | MATHDB-373592

Mallée's parabolic-triangle isobilliard conjecture

摘录 / EXCERPT: Let B be the unit disk in the plane, and for a convex shape $K\subset\mathbb{R}^2$ let $c(K\times B)$ be the length of the shortest closed generalized billiard orbit in K. Mall...

conjecture | open | MathDB 页面 | 原文线索

V2-040618 | MATHDB-373596

Ryjáček et al.'s minimum-degree conjecture for forbidden-star graphs

摘录 / EXCERPT: Ryjáček et al.'s conjecture. Every 4-connected ${K_{1,4},K_{1,4}+e}$-free graph with minimum degree at least 6 is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-040619 | MATHDB-373597

Ryjáček et al.'s line-graph minimum-degree conjecture

摘录 / EXCERPT: Ryjáček et al.'s conjecture. Every 4-connected line graph with minimum degree at least 5 is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-040620 | MATHDB-373708

Conjecture on the sharp weak-noise threshold for spatial ergodicity

摘录 / EXCERPT: Let $ u_\theta$ denote the annealed invariant measure obtained from the constant initial profile $\theta\ri\in\mathbb{R}$, and let and denote, respectively, the stronger and baseli...

conjecture | open | MathDB 页面 | 原文线索

V2-040621 | MATHDB-373709

Conjecture on representation of annealed time-ergodic invariant measures

摘录 / EXCERPT: Consider an annealed time-ergodic invariant measure for the SPDE discussed in the paper. A random field is weakly spatially stationary when it has the weak spatial stationarity pro...

conjecture | open | MathDB 页面 | 原文线索

V2-040622 | MATHDB-373734

Dynamical strong Lehmer conjecture for Parry powers

摘录 / EXCERPT: Dynamical strong Lehmer conjecture. If $4\beta<\tau_{10}4$ is a Perron number, then $4\beta^n4$ is a Parry number for only finitely many integers $4n4$.

conjecture | open | MathDB 页面 | 原文线索

V2-040623 | MATHDB-373750

Persistence of saddle-center homoclinics in the planar elliptic restricted three-body problem

摘录 / EXCERPT: Persistence conjecture. For every small $e>0$ there exists $\mu(e)$ for which the orbit L_1 has a homoclinic orbit. Moreover, there is a neighborhood U of the curve...

conjecture | open | MathDB 页面 | 原文线索

V2-040624 | MATHDB-373833

Completeness conjecture for obtuse rational lattice triangles

摘录 / EXCERPT: Completeness conjecture. The known list of obtuse rational lattice triangles is complete: every obtuse rational lattice triangle is either a member of one of the two infinite famil...

conjecture | open | MathDB 页面 | 原文线索

V2-040625 | MATHDB-373842

Geometric localization conjecture for Hopf, Bogdanov-Takens, and Neimark-Sacker bifurcations

摘录 / EXCERPT: Geometric localization conjecture. Every coexistence equilibrium at which a Hopf or Bogdanov-Takens bifurcation occurs has prey coordinate in $(x_{\min},x_{\max})$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-040626 | MATHDB-373906

Fried-Christy-Ghys conjecture on almost equivalence of transitive Anosov flows

摘录 / EXCERPT: Fried-Christy-Ghys conjecture. Any two transitive Anosov flows with orientable stable and unstable foliations are almost equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-040627 | MATHDB-373913

Discrete stable self-consistent kernel conjecture

摘录 / EXCERPT: Let $\mathcal{K}$ be the space of kernels, let $\mathcal{S}^*(k)$ be the optimized path entropy for a kernel held fixed at k, and let a self-consistent kernel k^* be a fixed po...

conjecture | open | MathDB 页面 | 原文线索

V2-040628 | MATHDB-373914

RG critical-exponent conjecture for MaxCal kernel dynamics

摘录 / EXCERPT: Let $R_\Lambda:\mathcal{K}\to\mathcal{K}$ be the renormalization-group map, let $\beta(k)$ be its beta function, and let k^* be an RG fixed point satisfying $\beta(k^*)=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040629 | MATHDB-373938

Slow-dispersion conjecture for Floquet-Dirac Hamiltonians

摘录 / EXCERPT: Slow-dispersion conjecture. For every $\varepsilon>0$, there exists a choice of $\nu(t)$ such that the $L^1\to L^\infty$ decay rate is no faster than $t^{-\varepsilon}$; more preci...

conjecture | open | MathDB 页面 | 原文线索

V2-040630 | MATHDB-373975

Smooth Markov partition conjecture for expanding toral endomorphisms

摘录 / EXCERPT: Smooth Markov partition conjecture. The induced toral endomorphism f admits a smooth, in fact linear, Markov partition if and only if for some $k\in\mathbb{N}$, A^k is diagonal...

conjecture | open | MathDB 页面 | 原文线索

V2-040631 | MATHDB-373979

Optimality of the exponent one-half in convergence rates for dynamic random Hamilton-Jacobi equations

摘录 / EXCERPT: Consider the setting of Theorems --, with the convergence rates established there. Optimality conjecture. The convergence rates obtained therein are optimal up to slowly varying fa...

conjecture | open | MathDB 页面 | 原文线索

V2-040632 | MATHDB-373987

Strongly negative cases for Hamiltonicity in graphon random graphs

摘录 / EXCERPT: Let $W:\Omega^2\to[0,1]$ be a graphon, and let $\mathbb{G}(n,W)$ denote the associated inhomogeneous random graph. The four conditions in Proposition are: W is not a connected gr...

conjecture | open | MathDB 页面 | 原文线索

V2-040633 | MATHDB-373988

Hamilton powers conjecture for inhomogeneous random graphs

摘录 / EXCERPT: Let $W:\Omega^2\to[0,1]$ be a graphon, let $k\ge 2$ be fixed, and let $\mathbb{G}(n,W)$ be the associated inhomogeneous random graph. A K_{k+1}-fractional cover of W is a measu...

conjecture | open | MathDB 页面 | 原文线索

V2-040634 | MATHDB-374066

Eigenvalue spectrum conjecture for type j,k saddle-node ghosts

摘录 / EXCERPT: Let $\dot{x}=f(x,\rho)$, with $x\in\mathbb{R}^n$, $f\in\mathcal{C}^\omega$, and parameters $\rho=(\rho_1,\dots,\rho_m)^T\in\mathbb{R}^m$. Suppose that $x_g\in\mathcal{A}$ is a ghos...

conjecture | open | MathDB 页面 | 原文线索

V2-040635 | MATHDB-374125

Minimality conjecture for periodic spectral projections

摘录 / EXCERPT: Let $\eqp$ be an irreducible ergodic quantum process defined by $\seq{\theta, \phi}$, and let $\alpha\in\PerSpecEQP$. For the disordered quantum process defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-040636 | MATHDB-374127

Maximal-period invariants conjecture for irreducible quantum processes

摘录 / EXCERPT: Let $\opL$ be an irreducible object with associated set $\Lambda_\opL$, preorder $\preceq$, period N_α, and invariant ς_α for each $\alpha\in\Lambda_\opL$. Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-040637 | MATHDB-374168

Smilansky's nodal universality conjecture for quantum graphs

摘录 / EXCERPT: Nodal universality conjecture. The variance has linear growth

conjecture | open | MathDB 页面 | 原文线索

V2-040638 | MATHDB-374205

Special Tavis-Cummings systems for higher-order singularities

摘录 / EXCERPT: Special Tavis-Cummings conjecture. For each $N\geq 1$, there exists a special Tavis-Cummings system of order N, that is, there exist parameter configurations such that the polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-040639 | MATHDB-374248

Efficient Lie-algebraic simulation from polynomial dynamical Lie algebra dimension

摘录 / EXCERPT: Working conjecture. There exists a basis or representation of $\mathfrak{g}$ in which the preprocessing steps required by $\mathfrak{g}{\text -}{\rm sim}$-basis construction and s...

conjecture | open | MathDB 页面 | 原文线索

V2-040640 | MATHDB-374262

The all-α extension of the super-4 theorem

摘录 / EXCERPT: Let $calpha>2$ be real, and let $c_{n,k}$ and the associated quantities be those defined in the paper. All-α conjecture. The statement of Theorem cref{t:super4} holds true for ever...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-040641 | MATHDB-374350

The universality conjecture for irrational two-step affine systems

摘录 / EXCERPT: Let ${\boldsymbol{u}}$ denote the relevant dynamical system, let $A_{2,\alpha}$ be the associated system parametrized by an irrational parameter α, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040642 | MATHDB-374379

Deterministic sign-balance conjecture for Laplace eigenfunctions

摘录 / EXCERPT: Let $\mathcal{M}$ be a chaotic smooth compact d-manifold, and let $\{(\varphi_{j},\lambda_{j})\}_{j\ge 1}$ be the corresponding sequence of Laplace eigenfunctions and eigenvalues...

conjecture | open | MathDB 页面 | 原文线索

V2-040643 | MATHDB-374494

Boshernitzan-Kornfeld conjecture on the prevalence of finite type interval translation maps

摘录 / EXCERPT: Let $r\geqslant 2$, and let $\mathrm{ITM}(r)$ denote the space of interval translation maps on r intervals. An interval translation map is of infinite type when, for its attracto...

conjecture | open | MathDB 页面 | 原文线索

V2-040644 | MATHDB-374495

Aubry's completeness conjecture for twist maps

摘录 / EXCERPT: Let F be a twist map, let $c_1<c_2$, and let $\Lambda_{[c_1,c_2]}$ be the set of orbits corresponding to minimal configurations with $c\in[c_1,c_2]$. Let α denote the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-040645 | MATHDB-374517

The R_0-independent pathwise oscillator-death conjecture for the inertial Winfree model

摘录 / EXCERPT: Let N oscillators have initial data $(\{\theta_i^0\}_{i=1}^N,\{\omega_i^0\}_{i=1}^N)$ and system parameters $(\{\nu_i\}_{i=1}^N,\kappa,m)$. Write $\mathcal{V}=(\nu_i)_{i=1}^N$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-040646 | MATHDB-374518

The large-inertia and large-initial-velocity oscillator-death conjecture for the inertial Winfree model

摘录 / EXCERPT: Let $\mathcal{V}\in\mathbb{R}^N$ be the vector of intrinsic velocities, and let $\kappa_c(\mathcal{V})$ denote the critical coupling strength of the Winfree system. Let initial dat...

conjecture | open | MathDB 页面 | 原文线索

V2-040647 | MATHDB-374542

Kawahigashi's uniqueness conjecture for almost periodic outer flows on the hyperfinite type II_1 factor

摘录 / EXCERPT: Kawahigashi's uniqueness conjecture. The following assertions are equivalent: α is cocycle conjugate to σ^{∞}; α satisfies the Rokhlin property; α...

conjecture | open | MathDB 页面 | 原文线索

V2-040648 | MATHDB-374597

The asymptotic Lyapunov exponent conjecture for the generalized Gauss map

摘录 / EXCERPT: Let $T_{r,\alpha}$ be the generalized Gauss continued-fraction map, and let $\lambda(r,\alpha)$ be the Lyapunov exponent of the orbit of $x_0\in[0,1]$ under $T_{r,\alpha}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-040649 | MATHDB-374604

Vanishing of KAM islands in the thermodynamic limit

摘录 / EXCERPT: Let V_N denote the relative phase-space volume of the union of KAM islands for an N-particle system at fixed energy density and fixed forcing parameters. Vanishing-of-islands c...

conjecture | open | MathDB 页面 | 原文线索

V2-040650 | MATHDB-374613

Flower-support conjecture for maximizing measures of odd trigonometric polynomials

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{>0}$ be odd. Let $\mathbb{T}=\mathbb{R}/\mathbb{Z}$ be the circle, and let $E_2:\mathbb{T}\to\mathbb{T}$ be the doubling map. For observables ... consider measur...

conjecture | open | MathDB 页面 | 原文线索

V2-040651 | MATHDB-374681

Conjecture on long-time concentration in a dominant eigenspace of the interaction matrix

摘录 / EXCERPT: Let B and V be weight matrices satisfying Assumption, with $V\neq\mathrm{Id}$, and let $(\rho_t^\beta)_{t\geq 0}$ be the token-distribution evolution on ${\mathbb S}^{d-1}$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-040652 | MATHDB-374762

The weak-mixing Poisson conjecture for ASL-invariant point processes

摘录 / EXCERPT: Let Λ be an $\operatorname{ASL}_d(\mathbb R)$-invariant point process with distribution μ. Let $\mathcal{M}(\mathbb R^d)$ denote the space on which the point-process di...

conjecture | open | MathDB 页面 | 原文线索

V2-040653 | MATHDB-374829

Bounded-type regularity conjecture for Siegel-disk boundaries

摘录 / EXCERPT: Bounded-type regularity conjecture. If $b1$ is of bounded type, then the map Γ_{b1} is at least C^1.

conjecture | open | MathDB 页面 | 原文线索

V2-040654 | MATHDB-374830

Bounded-type nonsmoothness conjecture for Siegel-disk boundaries

摘录 / EXCERPT: Let $b1$ be a rotation number of bounded type, and let $\Gamma_{b1}(\theta)$ denote the parametrization of the boundary of the corresponding Siegel disk. Bounded-type nonsmoothness...

conjecture | open | MathDB 页面 | 原文线索

V2-040655 | MATHDB-374973

Semi-linearity conjecture for recurrence sets of rational maps

摘录 / EXCERPT: Let $f,g\in\mathbb{C}(z)\setminus\mathbb{C}$ be such that $\deg(f)\geq2$ or $\deg(g)\geq2$, and let $c\in\mathbb{C}(z)$. Define ... Also let $\hat{S}_{f,g,c}^2$ denote the correspond...

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V2-040656 | MATHDB-375078

Bor-Tabachnikov conjecture on cusps of elliptical billiard caustics

摘录 / EXCERPT: Let $\gamma\subset\mathbb{R}^2$ be an ellipse, let O be a light source inside γ and different from a focus, and for each $n\geq 1$ let Γ_n be the envelope of rays f...

conjecture | open | MathDB 页面 | 原文线索

V2-040657 | MATHDB-375130

Espuny Díaz, Lichev, and Wesolek's local-resilience conjecture for cyclic Cayley graphs

摘录 / EXCERPT: Let n and k be positive integers. Consider the Cayley graph on $4\mathbb{Z}/n\mathbb{Z}4$ with generating set $\{-k,\ldots,k\}\setminus\{0\}$. Espuny Díaz, Lichev, and Wesolek'...

conjecture | open | MathDB 页面 | 原文线索

V2-040658 | MATHDB-375131

Espuny Díaz, Lichev, and Wesolek's local-resilience conjecture for random geometric graphs

摘录 / EXCERPT: For integers $d\geq 1$, let $T_d(n,r)$ be the random graph obtained by sampling n random points of the d-dimensional torus $(\mathds{R}/\mathds{Z})^d$ and joining two vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-040659 | MATHDB-375132

The random sub-sampling local-resilience conjecture for powers of cycles

摘录 / EXCERPT: Let C_n^k denote the kth power of the cycle on n vertices, and let $\varepsilon>0$. The random sub-sampling local-resilience conjecture. There exists k large enough such th...

conjecture | open | MathDB 页面 | 原文线索

V2-040660 | MATHDB-375176

Polynomial Fourier decay for nonlinear self-conformal measures on analytic curves

摘录 / EXCERPT: Let $ u$ be a self-conformal measure on $\mathbb{C}$ that is not conjugate to a linear system and is supported on an analytic curve that is not a line. Polynomial Fourier decay con...

conjecture | open | MathDB 页面 | 原文线索

V2-040661 | MATHDB-375243

Rudnick-Sarnak Quantum Unique Ergodicity conjecture

摘录 / EXCERPT: Let (M,g) be a compact manifold of negative curvature without boundary, and let $(u_n)_{n\geq 1}$ be any orthogonal eigenbasis of the Laplacian. A semiclassical measure is a weak...

conjecture | open | MathDB 页面 | 原文线索

V2-040662 | MATHDB-375316

Ergodic decomposition of invariant measures on structure spaces

摘录 / EXCERPT: Let X be a set, let $\operatorname{\mathscr{L}}$ be a language, and let κ be a cardinal. Let μ be an invariant measure on...

conjecture | open | MathDB 页面 | 原文线索

V2-040663 | MATHDB-375333

Arnol'd's chord conjecture for closed Legendrians

摘录 / EXCERPT: For every tame contact manifold (Y,ξ), every contact form α with $\ker\alpha=\xi$, and every compact Legendrian submanifold $\Lambda\subset Y$, there exist $T>0$ and a s...

conjecture | solved | MathDB 页面 | 原文线索

V2-040664 | MATHDB-375340

Abdenur-Bonatti-Díaz conjecture on non-wandering sets with nonempty interior

摘录 / EXCERPT: Let M be the compact manifold underlying $Diff^1(M)$, let $Diff^1(M)$ denote the space of C^1 diffeomorphisms of M, and let $\Omega(f)$ be the non-wandering set of f. A sub...

conjecture | refuted | MathDB 页面 | 原文线索

V2-040665 | MATHDB-375416

Flexibility above the Besson-Courtois-Gallot threshold

摘录 / EXCERPT: Flexibility above the Besson-Courtois-Gallot threshold. For every

conjecture | open | MathDB 页面 | 原文线索

V2-040666 | MATHDB-375526

Sharkovskii-type conjecture for bifurcations on manifolds

摘录 / EXCERPT: Let M be a smooth, orientable, metrizable manifold of dimension at least 2. A bifurcation ordering conjecture. There exists a class of bifurcations $\{B_\alpha\}_\alpha$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-040667 | MATHDB-375534

The dynamical Ax-Schanuel conjecture for Böttcher coordinates

摘录 / EXCERPT: Let f be a non-exceptional polynomial of degree $d\ge 2$. Let $Z\subset \mathbb{D}_R^n\times U_\infty(f)^n$ be an analytic subset of dimension $0\le m\le n$ defined by the parame...

conjecture | open | MathDB 页面 | 原文线索

V2-040668 | MATHDB-375711

The polycyclic cyclic escape conjecture

摘录 / EXCERPT: Polycyclic cyclic escape conjecture. Every infinite finitely generated polycyclic group has the cyclic escape property.

conjecture | open | MathDB 页面 | 原文线索

V2-040669 | MATHDB-375780

Conjecture on one-dimensional Möbius presentations of Cha-Nguyen-Tauber systems

摘录 / EXCERPT: The Cha-Nguyen-Tauber procedure associates a solution generating system to a homogeneous quadratic equation over $\mathbb{R}^3$. A 1-D piecewise Möbius transformation is a one-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-040670 | MATHDB-375788

Palis-type conjecture for convex hypersurfaces and robust heterodimensional cycles

摘录 / EXCERPT: Palis-type conjecture. The space of closed oriented hypersurfaces in a contact manifold (Y,ξ) decomposes into two open and C^2-dense sets: the set of convex hypersurfaces and...

conjecture | open | MathDB 页面 | 原文线索

V2-040671 | MATHDB-375814

Ergodic Pythagorean recurrence conjecture

摘录 / EXCERPT: Ergodic Pythagorean recurrence conjecture. There exists a solution $(x,y,z)\in\mathbb{N}^3$ of $x^2+y^2=z^2$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-040672 | MATHDB-375821

Bergelson-Leibman conjecture on polynomial ergodic averages for nilpotent actions

摘录 / EXCERPT: Bergelson-Leibman conjecture. The limit of these averages exists in L^2-norm and almost everywhere. The conjecture concerns pointwise and norm convergence of polynomial multiple...

conjecture | open | MathDB 页面 | 原文线索

V2-040673 | MATHDB-375933

The \b7IP\b7rat-characteristic factor conjecture for rationally independent polynomial systems

摘录 / EXCERPT: Let $P=(p_1,\b7dots,p_r)$ be a system of rationally independent polynomials, and let $Z_k(X)$ denote the factor appearing in the definition of an $\b7IP\b7rat$-characteristic facto...

conjecture | open | MathDB 页面 | 原文线索

V2-040674 | MATHDB-375942

Periodic orbit-closure conjecture for exact-cluster tilings

摘录 / EXCERPT: Periodic orbit-closure conjecture. The orbit closure

conjecture | open | MathDB 页面 | 原文线索

V2-040675 | MATHDB-375954

Finiteness for pseudo-Anosov flows without negative lozenges

摘录 / EXCERPT: Let M be a closed 3-manifold. A pseudo-Anosov flow without negative lozenges is a pseudo-Anosov flow having no negative lozenges. Finiteness conjecture. There are finitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-040676 | MATHDB-375963

Global orientation principle for reset distributions

摘录 / EXCERPT: Let $\textnormal{(S1)--(S4)}$ be the standing assumptions, let Δ_{m-1}° be the interior of the distribution simplex, let π^* be a reset-neutral distribution, let...

conjecture | open | MathDB 页面 | 原文线索

V2-040677 | MATHDB-376019

Twenty-one-ball total-clearance existence conjecture in snooker

摘录 / EXCERPT: Let Ω be the shot-parameter space, let c_0^{ε} denote the initial cue-ball configuration with ball radius or clearance parameter ε, and let...

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V2-040678 | MATHDB-376027

Frantzikinakis--Kuca conjecture on seminorm control along arithmetic progressions

摘录 / EXCERPT: Let $a\colon\mathbb N\to\mathbb Z$ be strictly increasing. The notions of being good for seminorm control and for degree-$(\ell+1)$ seminorm control along ℓ-term arithmetic pr...

conjecture | open | MathDB 页面 | 原文线索

V2-040679 | MATHDB-376028

Frantzikinakis--Kuca pointwise vanishing conjecture for nilsystems

摘录 / EXCERPT: Let $(G/\Gamma,\mu_{G/\Gamma},R_\alpha)$ be an ergodic nilsystem, and let $k_1,\ldots,k_\ell\in\mathbb Z$ be nonzero and distinct. For functions...

conjecture | open | MathDB 页面 | 原文线索

V2-040680 | MATHDB-376029

Leibman's conjecture on polynomial nilmanifold orbit averages

摘录 / EXCERPT: Let $W=N/\Lambda$ be a connected nilmanifold, let $Y=\pi(H)$ be a connected subnilmanifold of W, where H is a connected closed subgroup of N and $\pi:N\to W$ is the quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-040681 | MATHDB-376034

Seymour's conjecture on powers of Hamilton cycles

摘录 / EXCERPT: Seymour's conjecture. For positive integers n and k with $n\geq k+1$ and $k\geq2$, if

conjecture | open | MathDB 页面 | 原文线索

V2-040682 | MATHDB-376057

The Cantor-set structure conjecture for entropy-maximizing primitive majors

摘录 / EXCERPT: Let $PM(d)$ be the space of primitive majors for degree d, and let $\Pi(s_1,\ldots,s_r)$ be a stratum other than $\Pi(d)$ and $\Pi(1,1,\ldots,1)$. Consider the set of primitive m...

conjecture | open | MathDB 页面 | 原文线索

V2-040683 | MATHDB-376184

Bedert-Dră ganić-Müyesser-Pavez-Signé random Cayley-graph Hamilton-cycle conjecture

摘录 / EXCERPT: Bedert-Dră ganić-Müyesser-Pavez-Signé conjecture. There is an absolute constant C such that, for every connected Cayley graph G of order n and degree d, the condition

conjecture | open | MathDB 页面 | 原文线索

V2-040684 | MATHDB-376202

Non-contractibility conjecture for non-Hamiltonian subcircle actions

摘录 / EXCERPT: Non-contractibility conjecture. Then for every non-Hamiltonian subcircle action of T, the orbits of the circle action and all of their iterates are non-contractible.

conjecture | open | MathDB 页面 | 原文线索

V2-040685 | MATHDB-376215

Burns-Climenhaga-Todd conjectures on Lyapunov exponent level sets

摘录 / EXCERPT: Let (T^1M,f^t) be the geodesic flow on a compact rank 1 surface of nonpositive curvature. For $\alpha\geq 0$, let $L(\alpha)$ be the level set of Lyapunov exponents, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-040686 | MATHDB-376328

Hamiltonicity conjecture for regular sublinear expanders

摘录 / EXCERPT: For $\varepsilon>0$, let an (ε,d)-expander be the sublinear-expansion notion used in the source. Hamiltonicity conjecture for regular sublinear expanders. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-040687 | MATHDB-376381

The VIP-system nice recurrence conjecture

摘录 / EXCERPT: VIP-system recurrence conjecture. The VIP-system v is good for nice recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-040688 | MATHDB-376382

Bergelson-McCutcheon IP-limit positivity conjecture for VIP-systems

摘录 / EXCERPT: Bergelson-McCutcheon IP-limit positivity conjecture. For every $f\in\mathcal H$,

conjecture | open | MathDB 页面 | 原文线索

V2-040689 | MATHDB-376418

Joint ergodicity conjecture in the abelian setting

摘录 / EXCERPT: Joint ergodicity conjecture. The polynomials $p_1,\ldots,p_\ell$ are jointly ergodic for $(X,\mathcal{X},\mu,U)$ if and only if: (1) the product action...

conjecture | open | MathDB 页面 | 原文线索

V2-040690 | MATHDB-376551

Odd-length XXZ spin-chain ground-state energy conjecture

摘录 / EXCERPT: Let H be the Hamiltonian referred to as ${\rm (H1)}$, with an odd number N of sites, and let E and S_z denote an energy eigenvalue and the total spin in the z-direction....

conjecture | open | MathDB 页面 | 原文线索

V2-040691 | MATHDB-376568

Twisted XXZ even-chain ground-state conjecture

摘录 / EXCERPT: Let $H^{(N)}$ be the XXZ Hamiltonian with twisted boundary conditions, with N an even number of sites, twisting angle ϕ, anisotropy Δ, and total spin z-axis project...

conjecture | open | MathDB 页面 | 原文线索

V2-040692 | MATHDB-376751

The asymptotic second Lyapunov exponent conjecture for strata of holomorphic differentials

摘录 / EXCERPT: Let ${\cal M}^{\cal H}_{(2g-2)}$ and ${\cal M}^{\cal H}_{(g-1,g-1)}$ denote the hyperelliptic components of the corresponding moduli spaces of holomorphic differentials, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040693 | MATHDB-376764

The homogeneous Hamiltonian torus-action conjecture

摘录 / EXCERPT: Consider the cotangent bundle $T^*(\mathbb{R}^n/\mathbb{Z}^n)$ with its standard symplectic structure. A homogeneous Hamiltonian torus action is an action compatible with the cotan...

conjecture | open | MathDB 页面 | 原文线索

V2-040694 | MATHDB-376777

Continuity conjecture for the non-deterministic density of states measure

摘录 / EXCERPT: Continuity conjecture. The measure $\mu-\mu^{ND}$ is continuous, i.e. it does not charge any point.

conjecture | open | MathDB 页面 | 原文线索

V2-040695 | MATHDB-376778

Absence of pure point spectrum under dynamical avoidance

摘录 / EXCERPT: Pure point spectrum conjecture. If $d_\infty=N$ and this condition holds for every real λ, then

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V2-040696 | MATHDB-376786

Nekhoroshev stability criterion for elliptic critical manifolds

摘录 / EXCERPT: Let ${\mathfrak C}_i\subset{\mathfrak C}_{gen}$ be a connectivity component of the critical manifold, and let $a\in{\mathfrak C}_i$. Suppose the nonzero spectrum of the vertical li...

conjecture | open | MathDB 页面 | 原文线索

V2-040697 | MATHDB-376794

Berry-Tabor conjecture on level statistics of integrable systems

摘录 / EXCERPT: The quantum mechanical system is desymmetrized with respect to all its unitary symmetries, eigenvalues are considered on the scale of the mean level spacing, and generic degeneraci...

conjecture | open | MathDB 页面 | 原文线索

V2-040698 | MATHDB-376796

Conjecture on density of non-degenerate Picard-group eigenvalues

摘录 / EXCERPT: Let the Picard group be the arithmetic group underlying the spectrum of the Laplacian, and consider the sequence of its non-degenerate eigenvalues. Conjecture on density of non-deg...

conjecture | open | MathDB 页面 | 原文线索

V2-040699 | MATHDB-376810

Berry's Poisson distribution conjecture for integrable quantum systems

摘录 / EXCERPT: Let H be a quantum Hamiltonian whose classical counterpart is integrable, and consider the point spectrum of H (or its normalized energy differences). Berry's conjecture. The p...

conjecture | open | MathDB 页面 | 原文线索

V2-040700 | MATHDB-376825

Global maximality of the circular leaky loop

摘录 / EXCERPT: Let $H_{\alpha,\Gamma}$ be the planar Hamiltonian with attractive interaction of strength $\alpha>0$ supported by a loop Γ of fixed length L, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040701 | MATHDB-376829

Conjecture on the common complexity of matrix transformations

摘录 / EXCERPT: Common-complexity conjecture. The complexity of K for patterns (G), (S), and (C) is the same. Its common value is the inverse of the modulus of the smaller root of

conjecture | open | MathDB 页面 | 原文线索

V2-040702 | MATHDB-376835

Kurlberg-Rudnick conjecture for normalized exponential sums

摘录 / EXCERPT: For a finite field $\mathbb{F}_{q}$, let $\mathcal{C}$ be the multiplicative group $\mathbb{F}_{q}^{*}$ or the group of norm-one elements in a quadratic extension, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040703 | MATHDB-376840

Power-law diagonal variance conjecture

摘录 / EXCERPT: Let Ω be the billiard domain, let $\hat{A}$ be multiplication by a test function A, and define its spatial average by ... For eigenfunctions ϕ_n of the Laplacian with e...

conjecture | open | MathDB 页面 | 原文线索

V2-040704 | MATHDB-376841

Feingold-Peres diagonal variance conjecture

摘录 / EXCERPT: Let Ω be a billiard domain and let $\hat{A}$ be multiplication by a test function A. Define the local diagonal variance $V_A(E)$ as the mean square of...

conjecture | open | MathDB 页面 | 原文线索

V2-040705 | MATHDB-376842

Quantum unique ergodicity conjecture for billiards

摘录 / EXCERPT: Let ϕ_n be Laplacian eigenfunctions on an ergodic billiard, and let a quantum limit mean a measure to which $|\phi_n|^2$ converges weakly. Quantum unique ergodicity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-040706 | MATHDB-376843

Feingold-Peres off-diagonal variance conjecture

摘录 / EXCERPT: Feingold-Peres off-diagonal variance conjecture. For ergodic flow, as $E\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-040707 | MATHDB-376852

Dubrovin's universality conjecture for Hamiltonian perturbations

摘录 / EXCERPT: Dubrovin's universality conjecture. There exists a solution $u=u(x,t;\epsilon)$ to the perturbed equation on the same domain with

conjecture | open | MathDB 页面 | 原文线索

V2-040708 | MATHDB-376863

The conjecture on triple-collision shape curves and their first eclipses

摘录 / EXCERPT: Let $\Gamma^{\ast}(s)$, $s\geq 0$, denote a triple-collision shape curve, and let the equator circle be the locus representing the first eclipse. Triple-collision shape-curve conje...

open_problem | open | MathDB 页面 | 原文线索

V2-040709 | MATHDB-376873

Genericity of periodic Hamiltonians

摘录 / EXCERPT: Fix a finite spectral interval $\Lambda=(a,b)$. A periodic Hamiltonian H is called generic on Λ if, for every band edge λ_0 occurring inside Λ, the band f...

conjecture | open | MathDB 页面 | 原文线索

V2-040710 | MATHDB-376879

Quantum-classical convergence of energy-velocity measures

摘录 / EXCERPT: Let ${\cal P}:=T^*{\bf T}$ be the classical phase space, with Hamiltonian $H(p,q):=\frac{1}{2}p^2+V(q)$ and Liouville measure λ. Let $\bar v:{\cal P}\to{\bf R}^d$ be the as...

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V2-040711 | MATHDB-376898

Small-voter-fraction nonergodicity conjecture for the hybrid process

摘录 / EXCERPT: Small-voter-fraction nonergodicity conjecture. For every $p<1/2$, there exists a number $\beta_0(p)>0$ such that, whenever $\beta<\beta_0(p)$, the hybrid process $\tilde\eta_t$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-040712 | MATHDB-376925

Hamiltonian Seifert conjecture for prescribed aperiodic energy values

摘录 / EXCERPT: Let H_0 be the proper Hamiltonian on ${\mathbb R}^{2n}$ under discussion, let $K\subset(0,\infty)$ be a compact set of measure zero, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-040713 | MATHDB-376926

Magnus conjecture for twisted geodesic flows

摘录 / EXCERPT: Let M be a manifold, let σ be a symplectic form on M, and let H be the twisted geodesic Hamiltonian on T^*M. Magnus conjecture. For every M and any symplectic form...

conjecture | open | MathDB 页面 | 原文线索

V2-040714 | MATHDB-377005

The uniqueness conjecture for positive-entropy domino-tiling measures

摘录 / EXCERPT: Let $\mu_{s,t}$ denote the measure on domino tilings of the plane associated with tilt (s,t). A measure is conditionally uniform when, given a tiling outside any finite region, t...

conjecture | open | MathDB 页面 | 原文线索

V2-040715 | MATHDB-377113

The invariant-measure classification conjecture for sphere crossed products

摘录 / EXCERPT: Let g and h be minimal diffeomorphisms of S^m and S^n, respectively, with $m,n\geq 3$ and odd. Let the associated transformation group C-algebras be $C^*(\mathbf{Z},S^m,g)$...

conjecture | open | MathDB 页面 | 原文线索

V2-040716 | MATHDB-377131

Mae9's generic uniqueness conjecture for minimizing measures

摘录 / EXCERPT: Man'e's conjecture. For a generic Lagrangian L on a closed manifold M, there exist a dense open set U_0 of $H^1(M,\mathbb{R})$ such that for every $\omega\in U_0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-040717 | MATHDB-377245

Higher-dimensional almost homoclinic sequence conjecture

摘录 / EXCERPT: Let $f:M\rightarrow M$ be a C^{2} diffeomorphism with a compact basic set Λ which has associated an almost homoclinic sequence. Let $p\in\Lambda$ be a hyperbolic...

conjecture | open | MathDB 页面 | 原文线索

V2-040718 | MATHDB-377282

Hamiltonian volume conjecture for vacuum data on hyperbolic 3-manifolds

摘录 / EXCERPT: Hamiltonian volume conjecture. The rescaled volume satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-040719 | MATHDB-377434

Dynamical Lefschetz trace formula for foliated spaces

摘录 / EXCERPT: Let X be a compact manifold with a one-codimensional foliation ${\mathcal F}$ and an ${\mathcal F}$-compatible flow ϕ. Assume that the fixed points and periodic orbits are n...

conjecture | open | MathDB 页面 | 原文线索

V2-040720 | MATHDB-377455

Thomassen's Hamiltonian graph conjecture

摘录 / EXCERPT: Let G be a hamiltonian loopless graph with n vertices, and let $P(G,t)$ denote its chromatic polynomial. Thomassen's conjecture. ... The conjecture extends the known zero-free in...

conjecture | open | MathDB 页面 | 原文线索

V2-040721 | MATHDB-377470

Bellissard's Gap Labeling Conjecture for Cantor actions

摘录 / EXCERPT: Let Σ be a Cantor set, meaning a compact, totally disconnected metric space without isolated points, and let $\mathbb{Z}^{n}$ act freely and minimally on Σ, with inva...

conjecture | open | MathDB 页面 | 原文线索

V2-040722 | MATHDB-377548

The real-rank-zero criterion for crossed products by uniquely ergodic minimal diffeomorphisms

摘录 / EXCERPT: Let M be a connected compact smooth manifold with $\dim(M)>0$, and let $h\colon M\to M$ be a uniquely ergodic minimal diffeomorphism. Let...

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V2-040723 | MATHDB-377609

Algebraic integrality of the first dynamical degree

摘录 / EXCERPT: Let $\f:\mathbb{P}^2\dashrightarrow\mathbb{P}^2$ be a rational map, and let $\lambda_1=\lim_{n\to\infty}\operatorname{deg}(\f^n)^{1/n}$ be its first dynamical degree. The algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-040724 | MATHDB-377648

Ergodicity conjecture for full-limit-set Kleinian groups

摘录 / EXCERPT: Let Γ be a finitely generated Kleinian group, with limit set Λ in the Riemann sphere. Ergodicity conjecture. If ... then Γ acts ergodically on ... This is describ...

conjecture | open | MathDB 页面 | 原文线索

V2-040725 | MATHDB-377765

Total expansion characterizes exponential decay of correlations

摘录 / EXCERPT: Let $f:M\to M$ be a non-uniformly expanding map with an absolutely continuous invariant measure μ. The map is totally expanding when every invariant measure has all Lyapunov ex...

conjecture | open | MathDB 页面 | 原文线索

V2-040726 | MATHDB-377771

Existence of formal twist quantizations for modified dynamical r-matrices

摘录 / EXCERPT: Existence conjecture. Any modified dynamical r-matrix admits a formal twist quantization.

conjecture | open | MathDB 页面 | 原文线索

V2-040727 | MATHDB-377786

Conjecture on the limiting distribution of matrix elements for the quantum cat map

摘录 / EXCERPT: Limiting-distribution conjecture. As $N\to\infty$ through primes, the limiting distribution of the normalized matrix elements $F_j^{(N)}$ is that of the random variable

conjecture | open | MathDB 页面 | 原文线索

V2-040728 | MATHDB-377826

Density of periodic points for Moser generic diffeomorphisms

摘录 / EXCERPT: Let $f:S^2\to S^2$ be an area-preserving diffeomorphism that is Moser generic. Write $\operatorname{Per}(f)$ for the set of periodic points of f. Density conjecture. The set of p...

conjecture | open | MathDB 页面 | 原文线索

V2-040729 | MATHDB-377889

Pujals-Sambarino-type description for codimension-one dominated splittings

摘录 / EXCERPT: Let f be a C^2-diffeomorphism and let K be a compact locally maximal invariant set of f admitting a dominated splitting ... where F has dimension 1 and E^s is uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-040730 | MATHDB-377890

Tameness of one-dimensional partially hyperbolic diffeomorphisms in dimension three

摘录 / EXCERPT: Let M be a compact 3-manifold, and let ${\cal P}{\cal H}(M)$ be the C^1-open set of partially hyperbolic diffeomorphisms of M admitting a dominated splitting ... where all th...

conjecture | open | MathDB 页面 | 原文线索

V2-040731 | MATHDB-377930

Quantization conjecture for Poisson dynamical algebras

摘录 / EXCERPT: Let L be an $\ffrak{h}$-base manifold, let P be an $\ffrak{h}$-manifold, and let π be a Poisson dynamical bracket on P over L. A quantization of the Poisson dynamical...

conjecture | open | MathDB 页面 | 原文线索

V2-040732 | MATHDB-377965

Campbell-Petersen Wiener-Wintner conjecture for measure-preserving systems

摘录 / EXCERPT: Campbell-Petersen's conjecture. For every measure-preserving system (X,μ,T) and every $f\in L^2(X)$,

conjecture | open | MathDB 页面 | 原文线索

V2-040733 | MATHDB-378003

Spectral transition conjecture for the family of analytic Markov maps

摘录 / EXCERPT: Let $\mathcal{P}$ be the operator associated with the one-parameter family of analytic Markov maps, acting on the Hilbert space $\mathcal{H}$, and let $r\in[0,1]$ denote the parame...

conjecture | open | MathDB 页面 | 原文线索

V2-040734 | MATHDB-378041

The Hofer-distance conjecture for commutator-length strata

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold, let ${\rm Diff}_{\rm Ham}(M)$ denote the Hamiltonian diffeomorphism group, and let ${\rm Diff}_{\rm Ham}^k(M)$ be the subspace of diffeom...

conjecture | open | MathDB 页面 | 原文线索

V2-040735 | MATHDB-378181

Two-cycle count conjectures for the discrete logarithm map

摘录 / EXCERPT: Two-cycle counting conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-040736 | MATHDB-378183

Two-cycle count conjectures from the auxiliary heuristic

摘录 / EXCERPT: Two-cycle count conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-040737 | MATHDB-378184

Relative-primitivity scaling conjectures for two-cycle counts

摘录 / EXCERPT: Relative-primitivity scaling conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-040738 | MATHDB-378209

Sturmfels-Zworski conjecture on reconstruction from cyclic resultants

摘录 / EXCERPT: Sturmfels-Zworski conjecture. A generic monic polynomial $f(x)\in\mathbb C[x]$ of degree d is determined by its first $d+1$ cyclic resultants. Moreover, if f is non-monic recip...

conjecture | open | MathDB 页面 | 原文线索

V2-040739 | MATHDB-378250

Margulis's algebraicity conjecture for invariant measures

摘录 / EXCERPT: Let G, Γ, and H be as in Margulis's orbit-closure conjecture, and let μ be an H-invariant, H-ergodic measure on $\Gamma\backslash G$. Suppose that for every...

conjecture | open | MathDB 页面 | 原文线索

V2-040740 | MATHDB-378251

Margulis's almost-everywhere algebraicity conjecture for invariant measures

摘录 / EXCERPT: Let G, Γ, and H be as in Margulis's orbit-closure conjecture, and let μ be an H-invariant, H-ergodic measure on $\Gamma\backslash G$. Suppose that for μ-almo...

conjecture | open | MathDB 页面 | 原文线索

V2-040741 | MATHDB-378296

Conjecture on ergodic Gibbs measures and free energy for quadri-tilings

摘录 / EXCERPT: Fix $(s,t,p,q)\in{\mathbb R}^4$. Let ${\cal Q}_n^{(s,t,p,q)}$ be the quadri-tilings of ${\cal Q}_n$ with first height change $(\lfloor ns\rfloor,\lfloor nt\rfloor)$ and second heig...

conjecture | open | MathDB 页面 | 原文线索

V2-040742 | MATHDB-378412

The dimension bound for compact manifolds with standard lattice actions

摘录 / EXCERPT: Dimension bound conjecture. The dimension of M must be at least n.

conjecture | open | MathDB 页面 | 原文线索

V2-040743 | MATHDB-378604

Exponential decay of correlations and Lyapunov exponents

摘录 / EXCERPT: Let f be a non-uniformly expanding map. An orbit has a zero Lyapunov exponent if its Lyapunov exponent is zero, and f has exponential decay of correlations if its correlations...

conjecture | open | MathDB 页面 | 原文线索

V2-040744 | MATHDB-378605

Persistence of nonuniform expansivity under small perturbations

摘录 / EXCERPT: Let f be a nonuniformly expanding map. Perturbation-persistence conjecture. Sufficiently small perturbations of f have positive probability of also being nonuniformly expanding...

conjecture | open | MathDB 页面 | 原文线索

V2-040745 | MATHDB-378606

Largeness of acip parameters in the quadratic family

摘录 / EXCERPT: Let the quadratic family be the parameterized family of quadratic maps, and call an invariant probability measure an acip when it is absolutely continuous with respect to Lebesgue...

conjecture | refuted | MathDB 页面 | 原文线索

V2-040746 | MATHDB-378633

The domain-of-attraction inclusion conjecture for $D_a(0)$

摘录 / EXCERPT: Let $D_a(0)$ and N_p^c be the sets defined in the paper, with $p\geq 2$ and $c>0$. Domain-of-attraction inclusion conjecture. For any $x\in D_a(0)$ there exists $p\geq 2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040747 | MATHDB-378648

Nonexistence conjecture for analytic first integrals of even Liénard systems

摘录 / EXCERPT: Nonexistence conjecture. There is no global analytic first integral for this system on $\mathbb{R}^2$. The claim concerns global analytic integrability of planar Liénard systems; t...

conjecture | open | MathDB 页面 | 原文线索

V2-040748 | MATHDB-378765

Quantization conjecture for formal classical dynamical r-matrices

摘录 / EXCERPT: Let $\mathfrak h\to\mathfrak g$ be an inclusion of Lie algebras, let $Z\to(\mathfrak{h}^{\wedge}3\mathfrak g)^{\mathfrak g}$, and let a formal classical dynamical r-matrix mean...

conjecture | open | MathDB 页面 | 原文线索

V2-040749 | MATHDB-378840

The almost one-to-one coding conjecture for Pisot automorphisms

摘录 / EXCERPT: Almost one-to-one coding conjecture. A Pisot automorphism with associated Pisot number β admits an almost 1-to-1 coding by S_β.

conjecture | open | MathDB 页面 | 原文线索

V2-040750 | MATHDB-378909

Poincaré's homoclinic density conjecture

摘录 / EXCERPT: Let M be a compact surface, let $\operatorname{Diff}_\mu^r(M)$ denote the space of C^r area-preserving diffeomorphisms of M, and let $W^s(p)$ and $W^u(p)$ be the stable and u...

conjecture | open | MathDB 页面 | 原文线索

V2-040751 | MATHDB-379028

Conjecture on ergodic real nodal hypersurfaces

摘录 / EXCERPT: Let (M,g) be a real analytic Riemannian manifold with ergodic geodesic flow, and let $\{\varphi_j\}$ be a density-one sequence of ergodic eigenfunctions. For a test function f,...

conjecture | open | MathDB 页面 | 原文线索

V2-040752 | MATHDB-379056

The billiard-flow deviation conjecture for rational-angle Euclidean polygons

摘录 / EXCERPT: Billiard-flow deviation conjecture. For all rational-angle Euclidean polygons, the deviation of ergodic averages for the billiard flow is the same for almost all directions, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-040753 | MATHDB-379180

The smoothness conjecture on orbit structure in Case 4

摘录 / EXCERPT: Smoothness conjecture. Orbits of points in Case 4 will consist of a countable number of copies of an interval I, the union of a number of subintervals J_n, in each of which the...

conjecture | open | MathDB 页面 | 原文线索

V2-040754 | MATHDB-379181

The generator-number conjecture for orbit levels

摘录 / EXCERPT: Generator-number conjecture. The highest possible level of an orbit is $n+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-040755 | MATHDB-379239

The transcendentality-multiplier group conjecture for quasiperiodic flows

摘录 / EXCERPT: Transcendentality-multiplier group conjecture. A quasiperiodic flow ψ is transcendental if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-040756 | MATHDB-379250

The recurrence criterion for uniqueness of invariant measures in random dynamical systems

摘录 / EXCERPT: Recurrence criterion. The random dynamical system has a unique invariant Borel probability measure if and only if its associated fundamental Markov system is recurrent.

conjecture | open | MathDB 页面 | 原文线索

V2-040757 | MATHDB-379341

Dynamical generalization of Siegel's theorem for integral points

摘录 / EXCERPT: Dynamical Siegel conjecture. For any nonpreperiodic point $\beta\in\mathbb{P}^1_{\mathfrak{o}_K}(\overline{K})$, there are at most finitely many preperiodic points of φ in...

conjecture | open | MathDB 页面 | 原文线索

V2-040758 | MATHDB-379366

The ergodicity dichotomy for separable Banach spaces

摘录 / EXCERPT: A separable Banach space is a Banach space with a countable dense subset, and a Banach space is ergodic when the isomorphism relation between its infinite-dimensional subspaces is...

conjecture | open | MathDB 页面 | 原文线索

V2-040759 | MATHDB-379478

The spectral gap conjecture for random isometries of the sphere

摘录 / EXCERPT: Let $n\geq 2$, let $\phi_1,\ldots,\phi_n$ be a collection of elements of $SU(2)$, and let $L^2_0(SU(2))$ denote the orthogonal complement of the constant functions in $L^2(SU(2))$....

conjecture | open | MathDB 页面 | 原文线索

V2-040760 | MATHDB-379480

The Effros-Hahn-induction conjecture for separable dynamical systems

摘录 / EXCERPT: Let (A,G,α) be a separable dynamical system, where A is a C^*-algebra, G is a locally compact group, and α is an action of G on A. The system satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-040761 | MATHDB-379498

Universally bad zero-Banach-density sequences conjecture

摘录 / EXCERPT: Zero-Banach-density sequences conjecture. There exists some $f\in L^1(\mu)$ such that these averages do not converge almost everywhere.

conjecture | open | MathDB 页面 | 原文线索

V2-040762 | MATHDB-379528

Hamiltonian cycle game's threshold conjecture

摘录 / EXCERPT: Let ${\cal H}$ be the set of Hamiltonian cycles in K_n. For a family of winning sets ${\cal F}$, let $b_{\cal F}^p$ denote the smallest bias for which Breaker wins the random gam...

conjecture | open | MathDB 页面 | 原文线索

V2-040763 | MATHDB-379529

Sharp threshold conjectures for Maker's matching and Hamiltonian cycle games

摘录 / EXCERPT: Let G be the evolving random graph process, and write $\tau(\mathcal E)$ for the hitting time at which event $\mathcal E$ first occurs. Let ${\cal M}$ be the perfect matching gam...

conjecture | open | MathDB 页面 | 原文线索

V2-040764 | MATHDB-379555

Measure uniqueness and measurable rigidity for Cartan actions on tori

摘录 / EXCERPT: Measure uniqueness and measurable rigidity conjecture. The set $\mathcal M$ consists of a single measure, and the semiconjugacy h is a measurable isomorphism between the actions...

conjecture | open | MathDB 页面 | 原文线索

V2-040765 | MATHDB-379700

Computable-modulus conjecture for the conformal-radius function

摘录 / EXCERPT: Computable-modulus conjecture. The function f has a computable modulus of continuity.

conjecture | open | MathDB 页面 | 原文线索

V2-040766 | MATHDB-379895

Extended Il'Yashenko category conjecture for non-integrable SR spheres

摘录 / EXCERPT: Let an SR sphere mean a sphere for a sub-Riemannian structure, and call the case non-integrable when the associated geodesic dynamics is not integrable. Il'Yashenko's category is t...

conjecture | open | MathDB 页面 | 原文线索

V2-040767 | MATHDB-379915

Finiteness conjecture for periodic diagonal orbits in compact sets

摘录 / EXCERPT: Let $n \geq 3$ and let H be the diagonal subgroup acting on $\operatorname{PGL}_n(\mathbb{Z}) \backslash \operatorname{PGL}_n(\mathbb{R})$. Finiteness conjecture for periodic orb...

conjecture | open | MathDB 页面 | 原文线索

V2-040768 | MATHDB-379916

Polynomial-volume equidistribution conjecture for periodic torus orbits

摘录 / EXCERPT: Fix $\rho > 0$. Let $G=\mathbf{G}(\mathbb{R})$ be an $\mathbb{R}$-split real algebraic group, let $\Gamma < G$ be an arithmetic lattice, and let H be a maximal $\mathbb{R}$-split...

conjecture | open | MathDB 页面 | 原文线索

V2-040769 | MATHDB-379952

Lalonde-McDuff c-splitting conjecture for Hamiltonian fiber bundles

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold, and let $(M,\omega)\rightarrow P\rightarrow B$ be a Hamiltonian fiber bundle, meaning that its structural group can be reduced to...

conjecture | open | MathDB 页面 | 原文线索

V2-040770 | MATHDB-380172

Conjecture on generic non-absolute continuity of central foliations

摘录 / EXCERPT: A volume-preserving partially hyperbolic diffeomorphism is a diffeomorphism preserving a volume measure and admitting a partially hyperbolic invariant splitting; when its center fo...

conjecture | open | MathDB 页面 | 原文线索

V2-040771 | MATHDB-380263

Extension of the vector-representation identification to twisted face weights

摘录 / EXCERPT: Let ${\bf g}$ be one of the twisted affine Lie algebra types $A_{2n}^{(2)}$ or $A_{2n-1}^{(2)}$, or let the type be $G_2^{(1)}$. Let ${\cal R}(\lambda)$ denote the universal dynami...

conjecture | open | MathDB 页面 | 原文线索

V2-040772 | MATHDB-380339

Piecewise conjugacy conjecture for tensor algebras of multivariable systems

摘录 / EXCERPT: Let $(X,\f8sigma)$ and $(Y,\f8tau)$ be multivariable dynamical systems, and let ${\mathcal{A}}(X,\sigma)$ and ${\mathcal{A}}(Y,\tau)$ denote their tensor algebras. Piecewise conjug...

conjecture | open | MathDB 页面 | 原文线索

V2-040773 | MATHDB-380465

The symplectic-singularities conjecture for untwisted nilpotent models

摘录 / EXCERPT: Symplectic-singularities conjecture. Every fiber Y of $\widetilde{\psi}_{G,X}/\!/G$ has symplectic singularities.

conjecture | open | MathDB 页面 | 原文线索

V2-040774 | MATHDB-380626

Conjecture on the caustics of iterates of off-center reflections

摘录 / EXCERPT: Caustics of iterates conjecture. For $0<r\leq 1/3$, the caustic of R_r^{2m+1} is a C^∞ curve with exactly four cusp singularities, with two of them occurring at...

conjecture | open | MathDB 页面 | 原文线索

V2-040775 | MATHDB-380750

Tensorial criterion for flat Kronecker bihamiltonian structures

摘录 / EXCERPT: Let a homogeneous bihamiltonian structure have type $\left(2k_{1}-1,2k_{2}-1,\dots,2k_{l}-1\right)$. Tensorial criterion for Kronecker structures. There exist $N>0$ and natural ten...

conjecture | open | MathDB 页面 | 原文线索

V2-040776 | MATHDB-380751

Flatness of factors of a flat product of bihamiltonian structures

摘录 / EXCERPT: Let M_{1} and M_{2} be manifolds equipped with bihamiltonian structures, and equip $M_{1}\times M_{2}$ with their product bihamiltonian structure. Product-flatness conjecture....

open_problem | open | MathDB 页面 | 原文线索

V2-040777 | MATHDB-380752

Generic Kronecker conjecture for Volterra, Toda, Euler-top, and regular Lie-coalgebra systems

摘录 / EXCERPT: The odd-dimensional open Volterra system, the even-dimensional periodic Volterra system, the full Toda lattice, the multidimensional Euler top, and the regular case of the cited bi...

conjecture | open | MathDB 页面 | 原文线索

V2-040778 | MATHDB-380753

Web-rigidity conjecture for homogeneous bihamiltonian structures

摘录 / EXCERPT: Let M and M' carry homogeneous bihamiltonian structures ... For small open subsets $U\subset M$ and $U'\subset M'$, let ${\cal B}_{U}$ and ${\cal B}_{U'}$ be the corresponding we...

conjecture | open | MathDB 页面 | 原文线索

V2-040779 | MATHDB-380754

Casimir-family criterion for generic Kronecker bihamiltonian structures

摘录 / EXCERPT: Let M be a manifold with compatible Poisson structures $\{,\}_{1}$ and $\{,\}_{2}$. Let L be a finite set with r elements, and let $F_{I,\lambda}$, for $I\in L$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-040780 | MATHDB-380821

The C^1 stability conjecture

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be a C^1 map, and let $J(f)$ denote its Jacobian matrix. The C^1 stability conjecture. If the eigenvalues of $J(f)$ have strictly negative r...

conjecture | open | MathDB 页面 | 原文线索

V2-040781 | MATHDB-380831

The local classification conjecture for micro-Kronecker bihamiltonian structures

摘录 / EXCERPT: Local classification conjecture. There is a local isomorphism of bihamiltonian structures between M near m and $\Phi(\cal W)$ near m'.

conjecture | open | MathDB 页面 | 原文线索

V2-040782 | MATHDB-380834

The anti-involution conjecture for micro-Kronecker bihamiltonian structures

摘录 / EXCERPT: Anti-involution conjecture. There are a neighborhood $U\ni m$ and an anti-involution α of the bihamiltonian structure on U such that

conjecture | open | MathDB 页面 | 原文线索

V2-040783 | MATHDB-380859

The equality for the period of a Mersenne-indexed quotient ring

摘录 / EXCERPT: Mersenne-period conjecture. For all $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-040784 | MATHDB-380930

The conjecture on semiclassical measures of long-living open-baker eigenstates

摘录 / EXCERPT: Open-baker semiclassical-measure conjecture. All long-living semiclassical measures are supported on $\overline{\Gamma_+}$, and almost all long-living eigenstates are non-diffracti...

conjecture | open | MathDB 页面 | 原文线索

V2-040785 | MATHDB-380938

Integrability conjecture for the tabulated homogeneous Newton systems

摘录 / EXCERPT: Integrability conjecture. These equations are integrable in the Jacobi sense with polynomial first integrals I_1 and I_2, where

conjecture | open | MathDB 页面 | 原文线索

V2-040786 | MATHDB-381087

Devil's-staircase conjecture for quasiperiodically forced Arnold circle maps

摘录 / EXCERPT: Devil's-staircase conjecture. The set Λ_∞ can be chosen so that it is contained in the closure of the union of non-collapsed mode-locking plateaus. The same is true fo...

conjecture | open | MathDB 页面 | 原文线索

V2-040787 | MATHDB-381092

Conjecture on non-Gaussian and almost-everywhere Gaussian fluctuations for substitutions

摘录 / EXCERPT: Let σ be a primitive substitution of constant length d with eigenvalue 1 and eigenvector γ. Write $S_{\lfloor d^n t\rfloor}(\gamma)$ for the associated ergodic su...

conjecture | open | MathDB 页面 | 原文线索

V2-040788 | MATHDB-381124

Relative independence of joinings with ergodic nilsystems

摘录 / EXCERPT: Let (X,μ,T) be an ergodic system satisfying the spectral hypothesis of Corollary, and let (Y,ν,S) be an ergodic nilsystem. A joining of these systems is a joining of the me...

conjecture | open | MathDB 页面 | 原文线索

V2-040789 | MATHDB-381164

Canonical extension conjecture for the characteristic map

摘录 / EXCERPT: The characteristic map is a homeomorphism on the invariant set, with values in a cylinder. Canonical extension conjecture. The map $\widetilde{T}$ can be canonically extended to a...

conjecture | open | MathDB 页面 | 原文线索

V2-040790 | MATHDB-381167

Barbashin-type characterization of uniform exponential stability

摘录 / EXCERPT: Let X be the Banach space underlying a linear discrete-time system, and let A_m^k denote its associated evolution operator from time k to time m. The system is uniformly ex...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-040791 | MATHDB-381190

Essential boundedness conjecture for geometric ergodicity of i-cSMC

摘录 / EXCERPT: Let $T\in\mathbb{N}$, let π_t denote the time-t marginal of the target distribution, and let G_t be the corresponding potential function. Write P_N for the iterated condi...

conjecture | open | MathDB 页面 | 原文线索

V2-040792 | MATHDB-381204

Conjecture on almost-everywhere limit sets of differentially positive systems

摘录 / EXCERPT: Limit-set conjecture. Under the assumptions of Theorem $$, for almost every $\xi\in\mathcal{X}$, the ω-limit set $\omega(\xi)$ is given by either a fixed point, a limit cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-040793 | MATHDB-381228

Full-measure conjecture for dominant products of matrix families

摘录 / EXCERPT: Let ${\cal{A}}=\{A_1,\ldots,A_m\}$ be a family of m real $d\times d$ matrices, and let ${\cal{Q}}_m$ denote the space of all such families, identified with ${\mathbb R}^{m d^2}$....

conjecture | open | MathDB 页面 | 原文线索

V2-040794 | MATHDB-381232

Finite verification conjecture for rational discrete equidecomposability of polygons

摘录 / EXCERPT: Finite verification conjecture. Checking condition (3) can be reduced to a finite number of verifications. In particular, there exists a finite procedure for determining whether...

conjecture | open | MathDB 页面 | 原文线索

V2-040795 | MATHDB-381272

The measure-zero conjecture for minimal restricted interval exchange transformations

摘录 / EXCERPT: Measure-zero conjecture. The subset $M_{\pi,\mathscr R}$ has zero Lebesgue measure in $X_{\pi,\mathscr R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-040796 | MATHDB-381288

Isolation of parameter values admitting heteroclinic orbits in cyclic competition games

摘录 / EXCERPT: Let (ϵ_X,ϵ_Y) be a pair of game parameters, and let Z^a,Z^b,Z^c,Z^d,Z^e,Z^f denote the equilibria appearing in the cyclic competition bimatrix game. A heteroclini...

open_problem | open | MathDB 页面 | 原文线索

V2-040797 | MATHDB-381310

The collinearity conjecture for polynomial Heun eigenfunction points

摘录 / EXCERPT: For $l\in\mathbb N$, let $\mathcal P_l\in\{B=l\omega\}$ be the point with maximal A-coordinate for which the corresponding Heun equation has a polynomial solution. Collinearity c...

conjecture | open | MathDB 页面 | 原文线索

V2-040798 | MATHDB-381311

The Riccati-eigenfunction rotation-number conjecture

摘录 / EXCERPT: Let $l\in\mathbb N$, and let the parameter μ satisfy one of the equations governing the relevant monodromy cases according as l is odd or even. Consider the corresponding mon...

conjecture | open | MathDB 页面 | 原文线索

V2-040799 | MATHDB-381312

Numerical validity of the island hypotheses in dark-gray parameter regions

摘录 / EXCERPT: Consider the repulsive one-dimensional Gross-Pitaevskii equation with cosine potential, parameters ω and A, and its associated Poincaré map T. Let $\Delta_n^\pm$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-040800 | MATHDB-381330

Uniqueness conjecture for the positive zeros of the twisted-crown function

摘录 / EXCERPT: Let F be the function defined in the paper's equation for the twisted $(2,n)$-crowns, with $n\geq 4$. Its positive zeros are denoted by z_1 and z_2. Uniqueness conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-040801 | MATHDB-381342

Non-recurrence conjecture for the residual intersection of homoclinic classes

摘录 / EXCERPT: Let X be a Venice mask supported on a compact 3-manifold M. Let H_1 and H_2 be two different homoclinic classes in $M(X)$), let $\Lambda=H_1\cap H_2$, and suppose that th...

conjecture | open | MathDB 页面 | 原文线索

V2-040802 | MATHDB-381343

Containment conjecture for intersections of homoclinic classes

摘录 / EXCERPT: Let X be a Venice mask supported on a compact 3-manifold M, and let H_1 and H_2 be two different homoclinic classes in $M(X)$. Set $\Lambda=H_1\cap H_2$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-040803 | MATHDB-381374

The bifurcation-locus conjecture for the periodgon

摘录 / EXCERPT: Bifurcation-locus conjecture. The bifurcation locus of the periodgon of this family is precisely Σ.

conjecture | open | MathDB 页面 | 原文线索

V2-040804 | MATHDB-381379

Bodine et al.'s conjecture on irreducible sign patterns requiring $\mathbb{H}_n$

摘录 / EXCERPT: Let $n\ge 3$, let $\mathbb{H}_n=\{(0,n,0,0),(0,n-2,0,2),(2,n-2,0,0)\}$, and let an $n\times n$ sign pattern ${\cal A}$ require $\mathbb{H}_n$ when ... A sign pattern is irreducible i...

conjecture | open | MathDB 页面 | 原文线索

V2-040805 | MATHDB-381459

Uniqueness conjecture for the annihilated Boltzmann hierarchy with hard-sphere interactions

摘录 / EXCERPT: Uniqueness conjecture. The sequence $bm{nu}^{infty}={nu_{k}}_{k}$ is the unique solution in $L^{infty}([0,infty),\mathcal{X})$ to the annihilated Boltzmann hierarchy in the sense o...

conjecture | open | MathDB 页面 | 原文线索

V2-040806 | MATHDB-381466

Global dynamics conjecture for the Langford vector field

摘录 / EXCERPT: Let α be the parameter of the Langford vector field, and let $\alpha_1\approx0.697144898322973$, $\alpha_2\approx0.823$, $\alpha_3\approx0.925$,...

conjecture | open | MathDB 页面 | 原文线索

V2-040807 | MATHDB-381481

Conjectural hardness of certification below the spectral threshold for the SK Hamiltonian

摘录 / EXCERPT: Let $\mathsf{M}(\bm W)$ denote the Sherrington-Kirkpatrick Hamiltonian instance with random matrix $\bm W$, and let $c(\bm W)$ be the approximation ratio certified by an algorithm....

conjecture | open | MathDB 页面 | 原文线索

V2-040808 | MATHDB-381491

Extension of the longest-period distribution theorem to $sigma>r$

摘录 / EXCERPT: Extension conjecture. Theorem 1 holds in the same form for $\sigma>r$; in particular, for any fixed $\sigma\geq 1$ and $r\geq 1$ with $\sigma>r$, $X_{\sigma,n}/n^{\sigma/2}$ conver...

conjecture | open | MathDB 页面 | 原文线索

V2-040809 | MATHDB-381538

Ferenczi-Rosendal conjecture on ergodicity of non-Hilbertian Banach spaces

摘录 / EXCERPT: Let X be a separable Banach space that is not isomorphic to a Hilbert space. Recall that X is ergodic if the equivalence relation $\mathbf{E}_0$ is Borel-reducible to the isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-040810 | MATHDB-381570

Gevrey regularity conjecture for the Lindstedt series of the dissipative standard map

摘录 / EXCERPT: Let $ u$ be a parameter and let $\boldsymbol{\mathcal D}(u,1)$ denote the corresponding set of frequencies. For $\omega\in\mathcal D(\nu,1)$, consider the Lindstedt series ... of qu...

conjecture | open | MathDB 页面 | 原文线索

V2-040811 | MATHDB-381631

Non-displaceability conjecture for fibers of symplectic cluster 4-manifolds

摘录 / EXCERPT: Let X be a symplectic cluster 4-manifold equipped with a cluster presentation $\{\pi,L_1,\dots,L_n\}$. A fiber means a fiber of π over a point of the base. Non-displaceabil...

conjecture | open | MathDB 页面 | 原文线索

V2-040812 | MATHDB-381657

Hypothesis A on vanishing shadow volume for Euclidean obstacles

摘录 / EXCERPT: Let $K\subset\mathbb E^n$ be a smooth manifold of codimension at least 2 with a regular embedding, and let $\mathcal W(\epsilon)$ denote the volume of the trapping locus...

conjecture | open | MathDB 页面 | 原文线索

V2-040813 | MATHDB-381683

Margulis's measure-rigidity conjecture for the diagonal action

摘录 / EXCERPT: Let $G={\rm SL}(n,\mathbb{R})$, $\Gamma={\rm SL}(n,\mathbb{Z})$, and $X=G/\Gamma$ for $n\geq 3$. Let $A<G$ be the group of positive diagonal matrices with determinant one. A Borel...

conjecture | open | MathDB 页面 | 原文线索

V2-040814 | MATHDB-381903

Furstenberg-Johnson-Lesigne-Wierdl conjecture on mean convergence along arithmetic progressions

摘录 / EXCERPT: Let $a\colon {\mathbb N}\to {\mathbb Z}$ be a sequence and let $\ell\in {\mathbb N}$. A sequence is good for mean convergence along ℓ-term arithmetic progressions if the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-040815 | MATHDB-381904

Furstenberg-Lesigne-Wierdl recurrence conjecture for arithmetic progressions

摘录 / EXCERPT: Let $a\colon {\mathbb N}\to {\mathbb Z}$ be a sequence. A sequence is good for ℓ-step irrational equidistribution if it satisfies the corresponding irrational equidistribution...

conjecture | open | MathDB 页面 | 原文线索

V2-040816 | MATHDB-381905

Seminorm-control conjecture for arithmetic progressions

摘录 / EXCERPT: Let $a\colon {\mathbb N}\to {\mathbb Z}$ be a strictly increasing sequence. For a system $(X,{\mathcal X},\mu,T)$, say that a is good for seminorm control along ℓ-term arith...

conjecture | open | MathDB 页面 | 原文线索

V2-040817 | MATHDB-381906

The Hölder upper-bound conjecture for metric mean dimension on the interval

摘录 / EXCERPT: Let $\phi:[0,1]\rightarrow [0,1]$ be an α-Hölder continuous map. The metric mean dimension $\operatorname{mdim}_{\operatorname{M}}([0,1],|\cdot|,\phi)$ is defined using the...

conjecture | open | MathDB 页面 | 原文线索

V2-040818 | MATHDB-381907

Nonexistence of positive-Hölder maps with full metric mean dimension

摘录 / EXCERPT: Let $\phi:[0,1]\rightarrow [0,1]$ be an α-Hölder continuous map, with $\alpha>0$. The metric mean dimension $\operatorname{mdim}_{\operatorname{M}}([0,1],|\cdot|,\phi)$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-040819 | MATHDB-381908

Vanishing metric mean dimension for Hölder maps on the interval

摘录 / EXCERPT: Let $\phi:[0,1]\rightarrow [0,1]$ be an α-Hölder continuous map for some $\alpha\in(0,1)$. The metric mean dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-040820 | MATHDB-381993

The geometric realization conjecture for Markovian families of transitive Anosov flows

摘录 / EXCERPT: Let Φ be a transitive Anosov flow on a 3-manifold M, let $\widetilde{\Phi}$ be its lift to $\widetilde{M}=\mathbb{R}^3$, and let $\mathcal{P}$ be the bifoliated plane asso...

conjecture | open | MathDB 页面 | 原文线索

V2-040821 | MATHDB-382006

Schroer-Sauer-Ott-Yorke predictability conjecture

摘录 / EXCERPT: Let $T\colon X \to X$ be a smooth diffeomorphism of a compact Riemannian manifold X with a natural measure μ of information dimension $\operatorname{ID}(\mu)=D$. An observabl...

conjecture | open | MathDB 页面 | 原文线索

V2-040822 | MATHDB-382045

Cantor families of periodic solutions for spherical capillary water waves

摘录 / EXCERPT: The spherical capillary water waves system is considered on the sphere, with small-amplitude time-periodic solutions as the objects of interest. Periodic-solution conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-040823 | MATHDB-382112

Dynamical André-Oort conjecture for curves

摘录 / EXCERPT: Let $(f_t)_{t \in \Lambda}$ be a non-isotrivial algebraic family of rational maps with degree $d\geq 2$, parametrized by an algebraic curve Λ over $\mathbb C$. A rational ma...

conjecture | open | MathDB 页面 | 原文线索

V2-040824 | MATHDB-382142

Density conjecture for generic orbits of the bicycle map

摘录 / EXCERPT: Fix a degree d and consider the torus $\mathcal H_d$ of hedgehogs whose support functions are trigonometric polynomials of degree d, with the amplitudes of each harmonic fixed....

conjecture | open | MathDB 页面 | 原文线索

V2-040825 | MATHDB-382180

Generic unbounded distortion for non-Dini moduli of continuity

摘录 / EXCERPT: Generic unbounded-distortion conjecture. For every non-Dini-integrable $\omega\in K$, unbounded distortion is C^1-generic in $\Gamma_{\omega}^{1}(\mathbb S^1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-040826 | MATHDB-382222

Conjecture on multiple limit cycles in a predator-prey model with group defence

摘录 / EXCERPT: Let the predator-prey model described in the paper be given, with its interior equilibrium point and model parameters as defined there. Multiple-limit-cycle conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-040827 | MATHDB-382223

Conjecture on cyclical fold bifurcations in a predator-prey model with group defence

摘录 / EXCERPT: Let the predator-prey model described in the paper be given, with its parameters and equilibria as defined there. Cyclical-fold conjecture. There exist parameter regimes under whic...

conjecture | open | MathDB 页面 | 原文线索

V2-040828 | MATHDB-382236

Uniform convergence conjecture for weak competition in an allelopathic phytoplankton model

摘录 / EXCERPT: Let (u,v) solve the reaction-diffusion system of allelopathic phytoplankton with fear function $k(x)$, subject to the parameter restrictions ... where $\mathbf{k}$ is either...

conjecture | open | MathDB 页面 | 原文线索

V2-040829 | MATHDB-382348

Hamiltonicity conjecture for the face lattice of every polytope

摘录 / EXCERPT: Hamiltonicity conjecture. The graph $G(L(P))$ has a Hamiltonian cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-040830 | MATHDB-382357

Ergodicity conjecture for anti-symmetric skew products over locally Hamiltonian flows

摘录 / EXCERPT: Let $\psi_\mathbb{R}$ be a locally Hamiltonian flow on a compact surface M for which Theorem holds. Let M' be one of its minimal components of genus $g\geq 1$, and set ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-040831 | MATHDB-382430

The MPT-like transition conjecture for the forced glacial cycles model

摘录 / EXCERPT: Let the system of equations be forced by time-dependent functions $Q=Q(t)$ and $s_2=s_2(t)$, and let $T_0(t)$ be the specified ramp function. MPT-like transition conjecture. The so...

conjecture | open | MathDB 页面 | 原文线索

V2-040832 | MATHDB-382431

The triggering-of-deglaciation conjecture in the glacial cycles model

摘录 / EXCERPT: Consider the autonomous system of equations and its equilibria, classified as regular, boundary, or virtual according to their location relative to the switching boundary. Triggeri...

conjecture | open | MathDB 页面 | 原文线索

V2-040833 | MATHDB-382432

The skipped-obliquity conjecture for the forced glacial cycles model

摘录 / EXCERPT: Let T_0 be the critical temperature and let $T_1(t)$ denote the other time-dependent critical temperature in the model. Skipped-obliquity conjecture. When $|T_0-T_1(t)|$ is suffi...

conjecture | open | MathDB 页面 | 原文线索

V2-040834 | MATHDB-382433

The two-attracting-equilibria conjecture for the glacial cycles model

摘录 / EXCERPT: Let ... For the piecewise-defined system, call an equilibrium regular if it lies in its associated domain, boundary if it lies on the boundary between domains and is an equilibrium o...

conjecture | open | MathDB 页面 | 原文线索

V2-040835 | MATHDB-382434

The attracting-periodic-orbit conjecture for equilibrium types

摘录 / EXCERPT: Consider the piecewise-defined system of equations and classify its attracting equilibria as regular, boundary, or virtual according to their locations relative to the switching do...

conjecture | open | MathDB 页面 | 原文线索

V2-040836 | MATHDB-382511

Radial time-average conjecture for the equal-vortex model

摘录 / EXCERPT: Let $z=(z_1,\ldots,z_N)$ be a configuration of N equal vortices, each of vorticity 1, and define the pre-shape set ... For almost every $z\in\Gamma_N$, the time average of the em...

conjecture | open | MathDB 页面 | 原文线索

V2-040837 | MATHDB-382875

Bosch-Simó strange-attractor conjecture

摘录 / EXCERPT: For a dissipative dynamical system undergoing a Shilnikov-Hopf bifurcation, the Bosch-Simó conjecture predicts that, in the parameter unfolding, chaotic and periodic regimes are in...

conjecture | solved | MathDB 页面 | 原文线索

V2-040838 | MATHDB-383809

Centralizer of a diffeomorphism

摘录 / EXCERPT: Which ergodic affine maps $f:X\to X$ of a homogeneous space X have the property that if g is a perturbation of f such that $Z_{\textrm{Diff}(X)}(g)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-040839 | MATHDB-383810

Fixed point proportion of dynamical exceptional polynomials

摘录 / EXCERPT: 1. What can we say about the fixed-point proportion of groups acting on rooted trees? 2. Fixing the tree T, can we find a family of groups such that its fixed-point proportion is...

open_problem | open | MathDB 页面 | 原文线索

V2-040840 | MATHDB-383811

Analogous of paths on iterated monodromy groups for iterated Galois groups of dynamical maps

摘录 / EXCERPT: 1. Can we algebraically construct "nice" paths to describe G_∞^{geom} by more explicit wreath recursions? More precisely, given $z\in f^{-1}(t)$, can we find explicit endomo...

open_problem | open | MathDB 页面 | 原文线索

V2-040841 | MATHDB-384445

Universal upper bounds on local entanglement entropy for Schrödinger operators: under which condition is it true that for local Schrödinger operators such as ... with bounded (not necessarily ergodic) p

摘录 / EXCERPT: Universal upper bounds on local entanglement entropy for Schrödinger operators: under which condition is it true that for local Schrödinger operators such as $-\Delta +V$ with boun...

open_problem | open | MathDB 页面 | 原文线索

V2-040842 | MATHDB-384446

Which effect do cusps/corners have on the leading/lower order of trace asymptotics for ergodic Scrödinger operators?

摘录 / EXCERPT: Which effect do cusps/corners have on the leading/lower order of trace asymptotics for ergodic Scrödinger operators?

open_problem | open | MathDB 页面 | 原文线索

V2-040843 | MATHDB-385415

A non-Hermitian Operator from Physics

摘录 / EXCERPT: What is the distribution of eigenvalues of the non-Hermitian Hamiltonian ... Here a and $a^\dagger$ are the raising and lowering operators respectively and ω, χ,...

open_problem | open | MathDB 页面 | 原文线索

V2-040844 | MATHDB-385485

Extending the Burns-Climenhaga-Fisher-Thompson Technology (problem 1)

摘录 / EXCERPT: Let S be a surface which has a flat cylinder of parallel periodic orbits. In, a criterion for the existence and uniqueness of equilibrium states includes a pressure gap,...

open_problem | open | MathDB 页面 | 原文线索

V2-040845 | MATHDB-386159

The Maslow index allow you to prove stability results for symmetric pulses in Hamiltonian systems.

摘录 / EXCERPT: The Maslow index allow you to prove stability results for symmetric pulses in Hamiltonian systems. Can this be done also for non-symmetric pulses?

open_problem | open | MathDB 页面 | 原文线索

V2-040846 | MATHDB-386204

Stochastic analysis

摘录 / EXCERPT: Consider the multi-layer extension of the stochastic heat equation introduced by O'Connell and Warren. It is defined via chaos series -- show that it can also be defined by spatial...

open_problem | open | MathDB 页面 | 原文线索

V2-040847 | MATHDB-386234

Spin glass with i.i.d. couplings

摘录 / EXCERPT: Consider the spin glass model on $\mathbb{Z}_n^d$ with i.i.d. couplings, textit{i.e.} with Hamiltonian given by ... where $J=(J_{ij})$ is a random symmetric matrix with i.i.d....

open_problem | open | MathDB 页面 | 原文线索

V2-040848 | MATHDB-386352

Non-Gaussian multiplicative chaoses

摘录 / EXCERPT: Let $P_n(z) = \sum_{0\le k\le n} \frac1{\sqrt{k}} \xi_k z^k$ with ξ_k iid centered random variables of variance one, and consider the random measures μ_n on S^1 with dens...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-040849 | MATHDB-386411

#1 ReCom ergodicity

摘录 / EXCERPT: - (a) Given a map, take the two districts with the largest shared perimeter, then redraw to minimize boundary. Does this process eventually terminate in a simple, Voronoi-like part...

open_problem | open | MathDB 页面 | 原文线索

V2-040850 | MATHDB-386416

#34 Cycle walk variant on splittable trees

摘录 / EXCERPT: Suppose we run the up-down walk but restrict to trees that are k-splittable. Is this Markov chain ergodic, and if so, does it mix rapidly?

open_problem | open | MathDB 页面 | 原文线索

V2-040851 | MATHDB-386857

SFT (problem 1)

摘录 / EXCERPT: Completeness of the SFT Hamiltonian system i.e. do the Hamiltonians $h_{\alpha,i}$ form a maximal commuting set in $H_*(\mathcal{P})?$ In particular, in the case of symplectic mapp...

open_problem | open | MathDB 页面 | 原文线索

V2-040852 | MATHDB-386859

Computation of the commuting Hamiltonians for subcritical Stein-fillable manifolds using Legendrian SFT.

摘录 / EXCERPT: Computation of the commuting Hamiltonians for subcritical Stein-fillable manifolds using Legendrian SFT. Particular example: ... where $f(x_1,..,x_n)$ is a weighted homogenous polyno...

open_problem | open | MathDB 页面 | 原文线索

V2-040853 | MATHDB-386960

Find the eigenspace decomposition of the Hilbert space for the golden chain Hamiltonian.

摘录 / EXCERPT: Find the eigenspace decomposition of the Hilbert space for the golden chain Hamiltonian. Are the eigenspaces 1-dimensional.

open_problem | open | MathDB 页面 | 原文线索

V2-040854 | MATHDB-386973

Can we always introduce gapped boundary for the ... D commuting Hamiltonian?

摘录 / EXCERPT: Can we always introduce gapped boundary for the 2D commuting Hamiltonian?

open_problem | open | MathDB 页面 | 原文线索

V2-040855 | MATHDB-388322

Mañé's last theorem

摘录 / EXCERPT: In the space of area-preserving C^1 diffeomorphisms of a compact manifold, is it generic that the dynamics is either hyperbolic or has zero Lyapunov exponents?

open_problem | open | MathDB 页面 | 原文线索

V2-040856 | MATHDB-388361

Stable cellular-automaton limit sets

摘录 / EXCERPT: Characterize the stable limit sets of one-dimensional cellular automata.

open_problem | open | MathDB 页面 | 原文线索

V2-040857 | MATHDB-391896

Eulerian walker range conjecture of Priezzhev, Dhar, Dhar, and Krishnamurthy

摘录 / EXCERPT: Let each site of $\mathbb{Z}^2$ be assigned an independent uniformly random initial direction among its four outgoing directions. Starting at the origin, the Eulerian walker turns...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-040858 | MATHDB-392456

Avigad-Rute open question on norm variation of multiple ergodic averages

摘录 / EXCERPT: Let $n\ge 2$, let $(X,\mathcal{B},\mu)$ be a probability space, and let $T_1,\ldots,T_n$ be commuting measure-preserving transformations of X. For $f_1,\ldots,f_n\in L^\infty(X)$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-040859 | MATHDB-392470

Margulis’s conjecture on boundary ergodicity

摘录 / EXCERPT: Let G be a connected real semisimple Lie group of real rank at least 2, let $P<G$ be a minimal parabolic subgroup, and let $\Gamma<G$ be a discrete subgroup. If the action...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-040860 | MATHDB-392658

Infinitely many multi-troughed solitary waves in a water-wave parameter regime

摘录 / EXCERPT: For a parameter regime in which the four-dimensional invariant manifold is governed by the Hamiltonian equation $u^{(4)}+Pu''+u-u^2=0$, with $P\\in(-2,-2+\\epsilon)$, there are inf...

open_problem | open | MathDB 页面 | 原文线索

V2-040861 | MATHDB-392802

Universal period statistics for R-reversible polynomial automorphisms

摘录 / EXCERPT: For non-integrable polynomial automorphisms that induce permutations of $\mathbb{F}_p^2$ and possess the specified R-reversibility, does the appropriately scaled periodic-orbit p...

open_problem | open | MathDB 页面 | 原文线索

V2-040862 | MATHDB-393013

Conjectural dynamical critical exponent for Kawasaki dynamics

摘录 / EXCERPT: At inverse critical temperature (betac), does the critical dynamical structure function have the scaling form ... where ϕ vanishes at infinity and $\phi(0)=1$, with dynamical cr...

open_problem | open | MathDB 页面 | 原文线索

V2-040863 | MATHDB-393015

Ergodic/non-ergodic transition in the Kob-Andersen model

摘录 / EXCERPT: For the Kob-Andersen kinetically constrained lattice gas on the three-dimensional cubic lattice with constraint parameter $m=3$, numerical simulations conjectured an ergodic/non-er...

open_problem | open | MathDB 页面 | 原文线索

V2-040864 | MATHDB-393130

Spectral asymptotics near the lowest Landau level

摘录 / EXCERPT: Let H_0 be the Landau Hamiltonian on $L^2(\mathbb{R}^2)$ with magnetic-field strength $B>0$, let $V\ge 0$ satisfy $V\in L^1(\mathbb{R}^2)$ and have compact support...

open_problem | open | MathDB 页面 | 原文线索

V2-040865 | MATHDB-393262

A power-based Brayton-Moser description of standard mechanical systems

摘录 / EXCERPT: The report states the following result. For a standard conservative mechanical system with Hamiltonian equations $\dot q=H_p(q,p)$, $\dot p=-H_q(q,p)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-040866 | MATHDB-393263

Behavioral equivalence of AR-MA-X representations

摘录 / EXCERPT: In the setting of zero-mean Gaussian stationary ergodic vector processes on $\a0\mathbb{Z}$, consider AR-MA-X representations $A(\sigma)R(\sigma)w=M(\sigma)\varepsilon$, where A...

open_problem | open | MathDB 页面 | 原文线索

V2-040867 | MATHDB-393464

Arnold diffusion in a priori unstable Hamiltonian systems

摘录 / EXCERPT: Let ... where $(p,q)\in\mathbb R\times\mathbb T$, $(x,y)\in\mathbb T^n\times\mathbb R^n$, P is time-1-periodic and sufficiently small, and $H\in C^r$ with $r\ge 3$. Assume: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-040868 | MATHDB-393465

Arnold diffusion in Hamiltonian systems: a priori unstable case

摘录 / EXCERPT: Let ... where $u=(q,x)$, $v=(p,y)$, $(p,q)\in\mathbb R\times\mathbb T$, $(x,y)\in\mathbb T^n\times\mathbb R^n$, and P is time-1-periodic. Assume $H\in C^r$, $r\ge 3$, and: -...

open_problem | open | MathDB 页面 | 原文线索

V2-040869 | MATHDB-393467

Hypothetical conjugacy for smooth Hamiltonians

摘录 / EXCERPT: Let $\mathcal N=\bigcup_{\alpha\in\mathbb R^p}\mathcal N_\alpha=\{\alpha\cdot r+O(r^2)\}_{\alpha\in\mathbb R^p}$. For every $N^o\in\mathcal N$, there is a non-unique germ of a...

open_problem | open | MathDB 页面 | 原文线索

V2-040870 | MATHDB-393645

Absorbing nonlocal Hamiltonian time ordering into a propagator

摘录 / EXCERPT: For the DFR effective nonlocal theory with interaction Hamiltonian $H_I(t)=g\int dx_1\cdots dx_n\,K_t(x_1,\ldots,x_n):\!\phi(x_1)\cdots\phi(x_n)\!:$, the fact that Hamiltonian time...

open_problem | open | MathDB 页面 | 原文线索

V2-040871 | MATHDB-393683

Inverse spectral reconstruction for finite-gap orthogonal polynomials on the unit circle

摘录 / EXCERPT: For a stationary ergodic sequence of Verblunsky coefficients $\{\alpha_n\}\subset\mathbb D$ with $\sup_n|\alpha_n|<1$, represented as $\alpha_n(\omega)=g(\tau^{n-1}(\omega))$ on a...

open_problem | open | MathDB 页面 | 原文线索

V2-040872 | MATHDB-393684

Non-uniform continuous cocycles over uniquely ergodic minimal systems

摘录 / EXCERPT: Does every uniquely ergodic minimal dynamical system admit a non-uniform continuous $SL(2,\mathbb{R})$ cocycle?

open_problem | open | MathDB 页面 | 原文线索

V2-040873 | MATHDB-393745

Upper bound for Hamiltonian cycles in tournaments

摘录 / EXCERPT: Conjecture: For every tournament T on n vertices, the number $\\Psi(T)$ of Hamiltonian cycles satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-040874 | MATHDB-393775

Energy density entering the Eshelby stress

摘录 / EXCERPT: What is the energy to be considered in the Eshelby stress? In many practical problems one distinguishes between different thermodynamical situations, among these, isothermal evolut...

open_problem | open | MathDB 页面 | 原文线索

V2-040875 | MATHDB-393861

Density of the Ruelle-Sullivan image and aperiodicity

摘录 / EXCERPT: Let P be a repetitive Delone set of finite local complexity. Suppose that the translation action on its hull is uniquely ergodic. Let $\bc$ be the ergodic measure. Then P is ap...

open_problem | open | MathDB 页面 | 原文线索

V2-040876 | MATHDB-394003

Negative convergence of stretch lines with uniquely ergodic stump

摘录 / EXCERPT: Let $t\mapsto g_t^\mu$, $t\in\mathbb{R}$, be a stretch line directed by a complete geodesic lamination μ and passing through $g\in\mathcal{T}(\Sigma)$. If μ has a non-empty...

open_problem | open | MathDB 页面 | 原文线索

V2-040877 | MATHDB-394083

Infinite Hofer diameter of Hamiltonian groups

摘录 / EXCERPT: For every closed symplectic manifold (M,ω), does the metric space $\operatorname{Ham}(M,\omega)$, equipped with the Hofer norm $\|\cdot\|$, have infinite diameter?

open_problem | open | MathDB 页面 | 原文线索

V2-040878 | MATHDB-394084

The C^0-closedness of the Hamiltonian diffeomorphism group

摘录 / EXCERPT: For a closed symplectic manifold (M,ω), is ${\rm Ham}(M,\omega)$ always C^0-closed in ${\rm Diff}_0(M)$?

open_problem | open | MathDB 页面 | 原文线索

V2-040879 | MATHDB-394093

Embeddings of finitely generated groups into Hamiltonian diffeomorphism groups of surfaces

摘录 / EXCERPT: Does there exist a finitely generated group which admits a monomorphism to $\operatorname{Ham}(\Sigma_g)$ but not to $\operatorname{Ham}(\Sigma_h)$ where $g>h$? In the opposite dir...

open_problem | open | MathDB 页面 | 原文线索

V2-040880 | MATHDB-394096

Closed geodesic representatives of elements of the Hamiltonian group’s fundamental group

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold and let $G=\operatorname{Ham}(M,\omega)$. Is every element of $\pi_1(G)$ represented by a closed geodesic with respect to Hofer’s m...

open_problem | open | MathDB 页面 | 原文线索

V2-040881 | MATHDB-394098

Entropy values for higher-rank actions

摘录 / EXCERPT: Conjecture: Let $k\ge 2$, let α be a $C^{1+\varepsilon}$ action of $\mathbb{Z}^k$ on a $(k+1)$-dimensional manifold, where $\varepsilon>0$, and let μ be an ergodic...

open_problem | open | MathDB 页面 | 原文线索

V2-040882 | MATHDB-394128

Devaney chaos for a polynomial shift on \(\ell^p\)

摘录 / EXCERPT: For the particular case $\lambda_p(A)=\ell^p$, let q be a fixed complex polynomial of degree $d>1$ with $q(0)=0$, and define $P:\ell^p\to\ell^p$ by...

open_problem | open | MathDB 页面 | 原文线索

V2-040883 | MATHDB-394163

Exceptional times in the Brownian dynamical circle-covering model

摘录 / EXCERPT: Let C be the circle of circumference 1, and let $(\ell_n)_{n\ge 1}$ be a decreasing sequence of positive numbers tending to 0 that satisfies ... Consider the Brownian model, in...

open_problem | open | MathDB 页面 | 原文线索

V2-040884 | MATHDB-394193

Coexistence of invariant curves of periodic billiard trajectories

摘录 / EXCERPT: Can invariant curves consisting of periodic trajectories coexist for different periods, or for the same period but different rotation numbers, in a billiard other than a circle or...

open_problem | open | MathDB 页面 | 原文线索

V2-040885 | MATHDB-394219

Pósa-Seymour Hamilton-power conjecture

摘录 / EXCERPT: At the threshold [4m[0m$\delta(G)\ge (r-1)n/r$, a graph G on n vertices contains a copy of the $(r-1)$-st power of a Hamiltonian cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-040886 | MATHDB-394225

Full homogenization of long-wave water waves over a random bottom

摘录 / EXCERPT: For free-surface water waves in the long-wave KdV regime, with the fluid bottom given by a realization of a stationary ergodic process having a sufficiently strong mixing property,...

open_problem | open | MathDB 页面 | 原文线索

V2-040887 | MATHDB-394435

A five-regular graph without a cycle-independent dominating set

摘录 / EXCERPT: Does there exist a five-regular graph G with a Hamiltonian cycle C that does not contain any C-independent dominating set, where a C-independent dominating set is an indepe...

open_problem | open | MathDB 页面 | 原文线索

V2-040888 | MATHDB-394556

Dynamical stability of equilibria in a binary-gas phase-transition model

摘录 / EXCERPT: “A very relevant question to be addressed in this context is the dynamical stability of the equilibrium solutions and its relation with the thermodynamic stability expressed by the...

open_problem | open | MathDB 页面 | 原文线索

V2-040889 | MATHDB-394604

Universal tightness implies dynamical hyperbolicity

摘录 / EXCERPT: Every universally tight contact structure on a hyperbolic three-manifold is dynamically hyperbolic, meaning that every Reeb vector field associated with the contact structure has a...

open_problem | open | MathDB 页面 | 原文线索

V2-040890 | MATHDB-394609

The Conley conjecture for Hamiltonian diffeomorphisms of a torus

摘录 / EXCERPT: A Hamiltonian diffeomorphism of a torus has infinitely many periodic points; more precisely, if such a diffeomorphism has finitely many fixed points, then it has simple periodic po...

conjecture | open | MathDB 页面 | 原文线索

V2-040891 | MATHDB-394610

Generalized Conley conjecture for closed symplectically aspherical manifolds

摘录 / EXCERPT: Every Hamiltonian diffeomorphism of a closed symplectically aspherical manifold with finitely many fixed points has simple periodic points of arbitrarily large period.

conjecture | open | MathDB 页面 | 原文线索

V2-040892 | MATHDB-394614

Periodic-orbit lower bound for non-length-minimizing Hamiltonian paths

摘录 / EXCERPT: Let H be a nondegenerate Hamiltonian on a closed symplectic manifold (M,ω). If the path generated by H does not minimize the positive or negative Hofer length in its hom...

open_problem | open | MathDB 页面 | 原文线索

V2-040893 | MATHDB-394715

Deriving Fourier’s law from microscopic Hamiltonian dynamics

摘录 / EXCERPT: Can Fourier’s law be derived from microscopic Hamiltonian dynamics for Fermi-Pasta-Ulam chains of coupled anharmonic oscillators, with Hamiltonian ...

open_problem | open | MathDB 页面 | 原文线索

V2-040894 | MATHDB-394785

Quantization of a polynomial completely integrable Hamiltonian system

摘录 / EXCERPT: Let $R=\mathbb{C}[p_1,\ldots,p_n,q_1,\ldots,q_n]$, with its canonical Poisson bracket, and let $I_1,\ldots,I_n\in R$ be functionally independent and satisfy $\{I_i,I_j\}=0$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-040895 | MATHDB-394904

Pósa-Seymour conjecture on powers of Hamiltonian cycles

摘录 / EXCERPT: For sufficiently large n, a graph G on n vertices with minimum degree $\delta(G)\ge \frac{r-1}{r}n$ contains a copy of the $(r-1)$st power of a Hamiltonian cycle. Here the...

conjecture | open | MathDB 页面 | 原文线索

V2-040896 | MATHDB-395151

Almost-everywhere convergence of bilinear ergodic averages for two arbitrary commuting transformations

摘录 / EXCERPT: Let $(X,\text{probability space})$ be a probability space, let S and T be commuting but otherwise arbitrary measure-preserving transformations of X, and let f,g be bounded...

open_problem | open | MathDB 页面 | 原文线索

V2-040897 | MATHDB-395152

Quantitative lattice-point counting under mean ergodic and regularity hypotheses

摘录 / EXCERPT: Let G be a locally compact second-countable group, let $\93\subset G$ be a lattice, and let (B_t) be a growing family of domains in G. Assume: - A: the (stable) mean ergodic...

open_problem | open | MathDB 页面 | 原文线索

V2-040898 | MATHDB-395163

Uniform lattice-point counting from a quantitative mean ergodic bound

摘录 / EXCERPT: Let $G=\mathrm{SL}_2(\mathbb{R})$, let Γ be any lattice in G, let $\Delta\subset\Gamma$ be a finite-index subgroup, and fix any norm on $M_2(\mathbb{R})$. For...

open_problem | open | MathDB 页面 | 原文线索

V2-040899 | MATHDB-395248

Uniformity in the spin magnitude for low-temperature quantum phase transitions

摘录 / EXCERPT: Assume a quantum spin model satisfies conditions (i)-(iii) stated in the report: its quantum Hamiltonian is reflection positive for every spin magnitude...

open_problem | open | MathDB 页面 | 原文线索

V2-040900 | MATHDB-395369

Relative property (T) for subequivalence relations of the SL$_2(\mathbb Z)$ action on the 2-torus

摘录 / EXCERPT: Let $\mathcal S$ be the equivalence relation induced by the action $\mathrm{SL}_2(\mathbb Z)\curvearrowright\mathbb T^2$. Then every ergodic subequivalence relation $\mathcal R$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-040901 | MATHDB-395428

High-diffusivity asymptotics for the parabolic Anderson model with voter-dynamics catalyst

摘录 / EXCERPT: For the catalytic medium (c), namely simple symmetric voter dynamics with either the Bernoulli product initial distribution or the ergodic invariant initial distribution, both of d...

open_problem | open | MathDB 页面 | 原文线索

V2-040902 | MATHDB-395429

High-dimension large-diffusion asymptotics for the parabolic Anderson model in voter-dynamics media

摘录 / EXCERPT: For the catalytic medium (c), namely simple symmetric voter dynamics with votes 0 and 1, started from either the Bernoulli product measure or the ergodic invariant measure, and for...

open_problem | open | MathDB 页面 | 原文线索

V2-040903 | MATHDB-395437

Uniqueness for planar transport with a weak Sard Hamiltonian

摘录 / EXCERPT: Let $b\in L^\infty(\mathbb R^2;\mathbb R^2)$ have compact support and satisfy $\operatorname{div}b=0$. Let $H\in\operatorname{Lip}_c(\mathbb R^2)$ satisfy ... and the weak Sard prope...

open_problem | open | MathDB 页面 | 原文线索

V2-040904 | MATHDB-395453

Direct derivation of macroscopic Lagrangian/Hamiltonian structures

摘录 / EXCERPT: Can the macroscopic Lagrangian and Hamiltonian structures associated with the reduced equations (3) and (8) be derived directly from the microscopic Lagrangian/Hamiltonian structur...

open_problem | open | MathDB 页面 | 原文线索

V2-040905 | MATHDB-395455

Average speed of generalized traveling waves in spatially ergodic media

摘录 / EXCERPT: For the equation $u_t=u_{xx}+b(x)f(u)$, where $f(0)=f(1)=0$ and $b(x)$ is a positive smooth function on $\mathbb{R}$, the generalized traveling wave has an average speed if $b(x)$...

open_problem | open | MathDB 页面 | 原文线索

V2-040906 | MATHDB-395543

Minimizing the bottom of the spectrum in the random displacement model

摘录 / EXCERPT: For the random displacement Hamiltonian $H_\omega=-\Delta+V_\omega$ on $L^2(\mathbb R^d)$, with $V_\omega(x)=\sum_{i\in\mathbb Z^d}q(x-i-\omega_i)$, where q is reflection-symmetr...

open_problem | open | MathDB 页面 | 原文线索

V2-040907 | MATHDB-395614

Smale’s periodic-orbit question for nonvanishing vector fields on the solid 3-torus

摘录 / EXCERPT: Does every nonvanishing smooth vector field X on the solid 3-torus have a periodic orbit?

open_problem | open | MathDB 页面 | 原文线索

V2-040908 | MATHDB-395620

Choice of expansion point in rational Krylov model reduction

摘录 / EXCERPT: For a stable linear dynamical system with impulse response expanded in the scaled Laguerre basis as $h(t)=\sum_{i=0}^{\infty} f_i\,\varphi_i^{\alpha}(t)$, choose the Laguerre param...

open_problem | open | MathDB 页面 | 原文线索

V2-040909 | MATHDB-395629

Observability of quantized nonlinear systems

摘录 / EXCERPT: Let $x(t+1)=f(x(t))$ be a nonlinear discrete-time dynamical system with $x(t)\in\mathbb{R}^n$ and known map $f:\mathbb{R}^n\to\mathbb{R}^n$. Let $P_1,\ldots,P_N\subset\mathbb{R}^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-040910 | MATHDB-395630

Global Observability of Spaces of Negative Curvature

摘录 / EXCERPT: Consider the observed dynamical system $\dot{x}=f(x)$, $y=h(x)$ on an n-dimensional manifold, with $x(0)=x_0$ and output $y\in\mathbb{R}^p$. For some $T>0$, suppose that the loca...

open_problem | open | MathDB 页面 | 原文线索

V2-040911 | MATHDB-395637

Pointwise finiteness of the shrinking-target exponent for uniquely ergodic IETs

摘录 / EXCERPT: Let $J=[0,1)=\mathbb{R}/\mathbb{Z}$, let $T:J\to J$ be an interval exchange transformation, and define ... where $\langle a\rangle=\min_{k\in\mathbb{Z}}|a-k|$. If T is uniquely erg...

open_problem | open | MathDB 页面 | 原文线索

V2-040912 | MATHDB-395868

Abundance of one-dimensional non-uniformly hyperbolic attractors for surface endomorphisms

摘录 / EXCERPT: For every B of C^2-norm b sufficiently small, there exists a subset $ ... preserves a physical, ergodic, hyperbolic SRB probability measure $mu...

open_problem | open | MathDB 页面 | 原文线索

V2-040913 | MATHDB-395869

Existence of a leaf-wise intersection below the Reeb-period bound

摘录 / EXCERPT: Let Σ be a hypersurface of restricted contact type in an exact symplectic manifold M that is convex at infinity, and let ϕ be a Hamiltonian diffeomorphism. If the Hof...

open_problem | open | MathDB 页面 | 原文线索

V2-040914 | MATHDB-395871

Infinitely many periodic orbits for area-preserving disc maps

摘录 / EXCERPT: Let $\varphi:D\to D$ be a C^∞-smooth, area-preserving diffeomorphism of the closed 2-disc such that, on the boundary circle, φ coincides with a rotation by an irra...

open_problem | open | MathDB 页面 | 原文线索

V2-040915 | MATHDB-395872

Herman’s weak quasi-ergodic question for nearly integrable Hamiltonians

摘录 / EXCERPT: Can one find an example of a C^∞-Hamiltonian H in a C^r-small neighborhood of $H_0(I)=\sum_i I_i^2/2$ such that on the unit energy surface there is a dense trajectory?

open_problem | open | MathDB 页面 | 原文线索

V2-040916 | MATHDB-395876

Minimal average length of cyclically related billiard trajectories

摘录 / EXCERPT: Let $\Pi\subset\mathbb D$ be an ideal hyperbolic polygon with k counterclockwise-enumerated sides, and let Π_0 be a regular ideal hyperbolic k-gon, also equipped with a cou...

open_problem | open | MathDB 页面 | 原文线索

V2-040917 | MATHDB-395878

Billiards with small holes - Question C

摘录 / EXCERPT: If infinitely many particles start on the table according to some distribution μ_0, then at any given time, how much of the initial mass remains?

open_problem | open | MathDB 页面 | 原文线索

V2-040918 | MATHDB-395960

Critical-density formula for the simplified spatial-permutation model

摘录 / EXCERPT: For the simplified spatial-random-permutation model with Hamiltonian H_2 and cycle weights ... the report conjectures that the critical density for the occurrence of infinite cycle...

open_problem | open | MathDB 页面 | 原文线索

V2-040919 | MATHDB-395961

Strict lowering of the Be-like $J=1$ multiconfiguration energy below the ${}^{3}P_{1}$ sector

摘录 / EXCERPT: For the four-electron Be-like Hamiltonian $H=\sum_{i=1}^{4}\left(-\frac{\Delta_i}{2}-\frac{4}{|x_i|}\right)+\sum_{1\le i<j\le4}\frac{1}{|x_i-x_j|}$, consider the $J=1$ multiconfigu...

open_problem | open | MathDB 页面 | 原文线索

V2-040920 | MATHDB-395971

Phase transition theorem for the Toeplitz algebra of the affine semigroup over \(\mathbb{N}\)

摘录 / EXCERPT: For the C-dynamical system $(\mathcal{T}(\mathbb{N}\rtimes\mathbb{N}^{\times}),\sigma)$ described in the report, there are no KMS$_\beta$ states for $\beta<1$. For $\beta\in[1,2]$,...

open_problem | open | MathDB 页面 | 原文线索

V2-040921 | MATHDB-395973

Type and ergodicity of the Connes-Marcolli ${\rm GL}_2$-system action

摘录 / EXCERPT: For every $\beta\in(1,2]$, the action of ${\rm GL}_2(\mathbb Q)$ on $\bigl({\rm PGL}_2(\mathbb R)\times {\rm Mat}_2(\mathbb A_f),\,\mu_\infty\times\mu_{\beta,f}\bigr)$ is amenable,...

open_problem | open | MathDB 页面 | 原文线索

V2-040922 | MATHDB-396277

Cusp billiard with finite-time reflection accumulation

摘录 / EXCERPT: Give an explicit example of a billiard with a cusp on the boundary such that reflections accumulate at the tip in finite time.

open_problem | open | MathDB 页面 | 原文线索

V2-040923 | MATHDB-396278

Bounding the impacts of glancing billiard trajectories

摘录 / EXCERPT: For a planar billiard table with smooth closed convex border, under the non-existence of smooth convex caustics, Mather established the existence of infinitely many billiard trajec...

open_problem | open | MathDB 页面 | 原文线索

V2-040924 | MATHDB-396279

Ivrii’s conjecture on two-dimensional families of periodic billiard trajectories

摘录 / EXCERPT: There are no convex planar billiard domains with 2-dimensional families of n-periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-040925 | MATHDB-396282

Asymptotics of periodic billiard lengths near the boundary

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a bounded strictly convex domain with smooth boundary. For winding numbers satisfying $0<m/n<1/2$, let $\Gamma(m,n)$ be the collection of periodic...

open_problem | open | MathDB 页面 | 原文线索

V2-040926 | MATHDB-396283

Existence of the quadratic asymptotic for periodic billiard orbits

摘录 / EXCERPT: For every rational polygon P, does the limit $\displaystyle\lim_{L\to\infty}\frac{N_P(L)}{L^2}$ exist?

open_problem | open | MathDB 页面 | 原文线索

V2-040927 | MATHDB-396284

Periodic orbit in every triangle

摘录 / EXCERPT: Does every triangle have a periodic orbit?

open_problem | open | MathDB 页面 | 原文线索

V2-040928 | MATHDB-396286

Escape to infinity for dual billiards on general polygons

摘录 / EXCERPT: For the dual billiard map associated with a general convex polygon Q, can some orbits escape to infinity?

open_problem | open | MathDB 页面 | 原文线索

V2-040929 | MATHDB-396435

Osin’s conjecture on dynamical and geometric relative quasiconvexity

摘录 / EXCERPT: Osin’s conjecture: for relatively hyperbolic groups, dynamical quasiconvexity is equivalent to geometric relative quasiconvexity.

conjecture | open | MathDB 页面 | 原文线索

V2-040930 | MATHDB-396436

Measure-zero quadrangular trajectories in planar billiards

摘录 / EXCERPT: In any piecewise-smooth planar billiard, with sufficiently smooth pieces of boundary, the set of quadrangular orbits has measure zero.

open_problem | open | MathDB 页面 | 原文线索

V2-040931 | MATHDB-396472

Weak KAM existence theorem for the Hamilton-Jacobi equation

摘录 / EXCERPT: Let M be a compact smooth manifold without boundary and let $H:T^*M\to\mathbb{R}$ be a smooth Hamiltonian. If the matrix $(\partial_{p_i}\partial_{p_j}H)$ is positive definite an...

open_problem | open | MathDB 页面 | 原文线索

V2-040932 | MATHDB-396709

Diffusion in a weakly coupled quantum Hamiltonian system

摘录 / EXCERPT: Prove diffusion, in the long-time limit $t\to\infty$, for a quantum Hamiltonian model consisting of a large-mass particle with a finite-dimensional internal degree of freedom, movi...

open_problem | open | MathDB 页面 | 原文线索

V2-040933 | MATHDB-396731

Moment levels are orbits for symplectic Howe pairs

摘录 / EXCERPT: Let commuting Hamiltonian proper actions of the connected Lie groups G_1 and G_2 on the symplectic manifold (M,ω) be given, with equivariant moment maps...

open_problem | open | MathDB 页面 | 原文线索

V2-040934 | MATHDB-396876

Monotonicity and weak-coupling asymptotics of Fibonacci-spectrum thickness

摘录 / EXCERPT: The thickness of the Fibonacci-Hamiltonian spectrum Σ_λ is conjectured to be a monotonically decreasing function of λ and to behave like $1/\lambda$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-040935 | MATHDB-396933

Generalized quantisation commutes with reduction at zero

摘录 / EXCERPT: Let a Lie group G act prequantisably and Hamiltonianly on a symplectic manifold (M,ω), with M/G compact. Let $Q_G(M,\omega):=\mu_M^G([\not\!D_M])\in K_0(C^*G)$ be the ge...

open_problem | open | MathDB 页面 | 原文线索

V2-040936 | MATHDB-396935

Guillemin-Sternberg quantization-commutes-with-reduction equality

摘录 / EXCERPT: Let (M,ω) be a symplectic manifold with a Hamiltonian action of a Lie group G and moment map $\Phi:M\to\mathfrak g^*$. Suppose 0 is a regular value of Φ, so that...

open_problem | open | MathDB 页面 | 原文线索

V2-040937 | MATHDB-396941

Spin-quantization commutes with reduction

摘录 / EXCERPT: Let (M,ω,Φ) be a compact Spin-prequantized Hamiltonian K-manifold, where K is a compact connected Lie group. Let σ be the smallest face of the closed Weyl cham...

open_problem | open | MathDB 页面 | 原文线索

V2-040938 | MATHDB-396986

Direct Lagrangian-side proof for general variational integrators

摘录 / EXCERPT: To obtain a general statement for any variational integrator, a proof directly on the Lagrangian side (rather than switching to the Hamiltonian side) is desirable.

open_problem | open | MathDB 页面 | 原文线索

V2-040939 | MATHDB-397088

Ergodic properties of extensions of typical locally Hamiltonian flows

摘录 / EXCERPT: What are the ergodic properties of extensions $(\Phi_t^f)_{t\in\mathbb R}$ of a typical locally Hamiltonian flow?

open_problem | open | MathDB 页面 | 原文线索

V2-040940 | MATHDB-397089

Hausdorff-dimension dichotomy for non-arithmetic Veech surfaces

摘录 / EXCERPT: Does the dichotomy for the Hausdorff dimension of the set of non-ergodic directions or the set of divergent directions still hold if T is replaced with a non-arithmetic Veech sur...

open_problem | open | MathDB 页面 | 原文线索

V2-040941 | MATHDB-397098

Existence of a well-defined Koch snowflake billiard

摘录 / EXCERPT: Does there exist a well-defined billiard $\Omega(KS)$ with boundary equal to the Koch snowflake KS?

open_problem | open | MathDB 页面 | 原文线索

V2-040942 | MATHDB-397206

Blowing up a non-periodic orbit in a maximal-rank toral action

摘录 / EXCERPT: For a maximal-rank $\mathbb{Z}^{d-1}$-action on $\mathbb{T}^d$, can one blow up a non-periodic orbit as a construction of a new action? If such a construction is possible, what deg...

open_problem | open | MathDB 页面 | 原文线索

V2-040943 | MATHDB-397210

Conley conjecture under vanishing first Chern class or cotangent-bundle hypotheses

摘录 / EXCERPT: Let M be a symplectic manifold and let $H:S^1\times M\to\mathbb{R}$ be a Hamiltonian, with time-one map φ_H. Assume that φ_H has only finitely many fixed points a...

conjecture | open | MathDB 页面 | 原文线索

V2-040944 | MATHDB-397283

Strongly consistent learning of the best prediction for Gaussian processes

摘录 / EXCERPT: Let $\{Y_n\}_{n=-\infty}^{\infty}$ be a stationary, ergodic Gaussian process, and let the predictor be a sequence of functions $g=\{g_i\}_{i=1}^{\infty}$. Is there a prediction rul...

open_problem | open | MathDB 页面 | 原文线索

V2-040945 | MATHDB-397286

Strong universal consistency in static forecasting under mere integrability

摘录 / EXCERPT: Does there exist a sequence of static forecasting estimates $m_n(X_0,D_{-1}^{-n})$ such that, for every stationary and ergodic sequence $((X_n,Y_n))_{n\in\mathbb Z}$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-040946 | MATHDB-397398

Persistence of strong scars without arithmetic symmetries

摘录 / EXCERPT: For Laplacian eigenmodes of two-dimensional chaotic billiards without the relevant arithmetic symmetries, does the enhancement of probability density near periodic orbits persist i...

open_problem | open | MathDB 页面 | 原文线索

V2-040947 | MATHDB-397400

Quantum ergodicity for ergodic billiards

摘录 / EXCERPT: Let (M,g) be a compact smooth Riemannian manifold with piecewise smooth boundary. Let Δ_g denote the Dirichlet Laplacian induced by the metric, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-040948 | MATHDB-397401

Ergodicity of the geodesic frame flow under strict quarter-pinching

摘录 / EXCERPT: It is conjectured that the geodesic frame flow is ergodic whenever the manifold is strictly $\frac{1}{4}$-pinched, that is, whenever all sectional curvatures lie in the interval...

open_problem | open | MathDB 页面 | 原文线索

V2-040949 | MATHDB-397403

Arithmetic quantum unique ergodicity for compact congruence surfaces

摘录 / EXCERPT: Let $M=\Gamma\backslash\mathbb H$, where Γ is a congruence lattice over $\mathbb Q$. If M is compact, then the only arithmetic quantum limit is the normalized volume measu...

open_problem | open | MathDB 页面 | 原文线索

V2-040950 | MATHDB-397405

Kurlberg-Rudnick estimates for quantum unique ergodicity of a Hecke basis

摘录 / EXCERPT: Let d4f6 be the Anosov toral automorphism considered in the report, let d4a2=mathbb{Z}[alpha] for its largest eigenvalue α, let I be the corresponding $\mathcal O$-ideal,...

open_problem | open | MathDB 页面 | 原文线索

V2-040951 | MATHDB-397406

Conjectured rate of quantum ergodicity

摘录 / EXCERPT: For the quantum-ergodic subsequence, if ... then the decay exponent is conjectured to be ...

conjecture | open | MathDB 页面 | 原文线索

V2-040952 | MATHDB-397408

Existence of the integrated density of states for uniquely ergodic Delone-Anderson systems

摘录 / EXCERPT: Assume that the Delone dynamical system X_D is uniquely ergodic. Then, for every energy $E\in\mathbb{R}$, the normalized finite-volume eigenvalue counting function ... has a limit...

open_problem | open | MathDB 页面 | 原文线索

V2-040953 | MATHDB-397491

Real-analyticity of the Hausdorff dimension of radial Julia sets

摘录 / EXCERPT: Suppose that $f:\mathbb{C}\to\widehat{\mathbb{C}}$ is a dynamically regular meromorphic function of divergence type which belongs to class S. If $\Lambda\subseteq\mathbb{C}$ is o...

open_problem | open | MathDB 页面 | 原文线索

V2-040954 | MATHDB-397506

Zero speed under a logarithmic moment condition for random conductances

摘录 / EXCERPT: Let $Q\in M_1(\Omega_2)$ be invariant and ergodic under spatial shifts in $\mathbb{Z}^2$. If, for some $\alpha>1$, $E_Q[\log^\alpha\omega(x,y)]<\infty$ for a nearest-neighbor edge...

open_problem | open | MathDB 页面 | 原文线索

V2-040955 | MATHDB-397520

Ergodic strengthening for non-Martin-Löf-random points

摘录 / EXCERPT: Is the following true? If x is not Martin-Löf random, then there is a computable ergodic transformation T and a computably enumerable set A such that the ergodic average...

open_problem | open | MathDB 页面 | 原文线索

V2-040956 | MATHDB-397624

Powers of Hamiltonian cycles in pseudorandom graphs

摘录 / EXCERPT: An (n,d,λ)-graph is a d-regular graph on n vertices whose second-largest adjacency-matrix eigenvalue in absolute value is at most λ. There exist a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-040957 | MATHDB-397726

Construction of dynamical vacuum black holes without symmetries

摘录 / EXCERPT: Given suitable smooth scattering data on the event horizon $\mathcal{H}^{+}$ and future null infinity $\mathcal{I}^{+}$, asymptoting to the induced Kerr geometry with parameters...

open_problem | open | MathDB 页面 | 原文线索

V2-040958 | MATHDB-397738

Homogenization and asymptotics for small transaction costs

摘录 / EXCERPT: The report states the following Main Theorem. With $z=x+y$, define ... If $\bar a$ is the ergodic-control quantity defined earlier and $a:=\eta v_z\bar a$, then, as $\varepsilon\to0$...

open_problem | open | MathDB 页面 | 原文线索

V2-040959 | MATHDB-397863

Which continuous functions generate continuous Hamiltonian isotopies?

摘录 / EXCERPT: Given a symplectic manifold (M,ω), which continuous functions H on $[0,1]\times M$ generate a continuous Hamiltonian isotopy in the following sense: there exist smooth Ham...

open_problem | open | MathDB 页面 | 原文线索

V2-040960 | MATHDB-397923

Dynamical instability of Schwarzschild-AdS and Kerr-AdS

摘录 / EXCERPT: We conjecture that Schwarzschild-AdS and Kerr-AdS are dynamically unstable.

open_problem | open | MathDB 页面 | 原文线索

V2-040961 | MATHDB-397976

Hamiltonian cycles in the unit disk with bounded squared edge-length sum

摘录 / EXCERPT: Let B be the disk of radius one in the plane. Is it true that any finite point set $P\subset B$ can be spanned by a Hamiltonian cycle with the sum of the squares of its edge leng...

open_problem | open | MathDB 页面 | 原文线索

V2-040962 | MATHDB-398081

Convergence of multiple ergodic averages along Hardy-field sequences

摘录 / EXCERPT: Fix $k\ge 1$. Let $a_i\in\mathcal H$ for $1\le i\le k$ have at most polynomial growth, and consider the averages ... Is it true that these averages converge if and only if, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-040963 | MATHDB-398089

Classification of transformation-group crossed products by the Elliott invariant

摘录 / EXCERPT: Let (X,G,α) be a dynamical system (with, say, X compact metrizable and G discrete, countable, and amenable). To what extent is the crossed-product C^*-algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-040964 | MATHDB-398395

Hamiltonian stability of Gauss images of nonhomogeneous isoparametric hypersurfaces

摘录 / EXCERPT: Study the Hamiltonian stability of the Gauss images of compact nonhomogeneous isoparametric hypersurfaces of OT-FKM type embedded in spheres with $g=4$.

open_problem | open | MathDB 页面 | 原文线索

V2-040965 | MATHDB-398414

Quenched functional CLT under a finite-mean stationary ergodic environment

摘录 / EXCERPT: For $d\ge 2$, does the quenched functional central limit theorem hold for the variable-speed random walk in every stationary ergodic random-conductance environment...

open_problem | open | MathDB 页面 | 原文线索

V2-040966 | MATHDB-398533

Arnold diffusion and symbolic dynamics for small generic perturbations

摘录 / EXCERPT: For a generic nearly integrable Hamiltonian system subject to a sufficiently small perturbation, do there exist trajectories whose action coordinate changes by a positive constant...

open_problem | open | MathDB 页面 | 原文线索

V2-040967 | MATHDB-398535

Hyperbolic periodic orbits force infinitely many periodic orbits

摘录 / EXCERPT: Let M be a closed symplectic manifold. Its minimal Chern number N is defined by $\langle c_1(TM),\pi_2(M)\rangle=N\mathbb Z$, where $N\ge 0$. The manifold M is strictly monot...

open_problem | open | MathDB 页面 | 原文线索

V2-040968 | MATHDB-398536

The HZ-conjecture for Hamiltonian diffeomorphisms of \(\mathbb{C}P^n\)

摘录 / EXCERPT: The HZ-conjecture predicts that a Hamiltonian diffeomorphism of $\mathbb{C}P^n$ with more than the expected threshold of $n+1$ fixed points has infinitely many periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-040969 | MATHDB-398542

C^0-integrable Tonelli Hamiltonians that are not C^1-integrable

摘录 / EXCERPT: Does there exist a C^0-integrable Tonelli Hamiltonian that is not C^1-integrable?

open_problem | open | MathDB 页面 | 原文线索

V2-040970 | MATHDB-398544

Strong Arnold diffusion along a prescribed resonant path

摘录 / EXCERPT: Let H_0 be a smooth strictly convex integrable Hamiltonian on $B^2\subset\mathbb R^2$, and consider $H_\varepsilon(\theta,p,t)=H_0(p)+\varepsilon H_1(\theta,p,t)$ on...

open_problem | open | MathDB 页面 | 原文线索

V2-040971 | MATHDB-398545

Quasiergodic hypothesis

摘录 / EXCERPT: A quasiergodic hypothesis states that a generic Hamiltonian system on a generic energy surface has a dense orbit.

conjecture | open | MathDB 页面 | 原文线索

V2-040972 | MATHDB-398578

Dynamical stability of compact Ricci-flat metrics

摘录 / EXCERPT: Let $(M,\hat g)$ be a compact Ricci-flat manifold. If $\hat g$ is a local maximizer of Perelman's λ-functional, then for every $C^{k,\alpha}$-neighborhood U of $\hat g$,...

open_problem | open | MathDB 页面 | 原文线索

V2-040973 | MATHDB-398654

Role of dynamical regularity properties in structure theory

摘录 / EXCERPT: Do the dynamic regularity properties play a role for the structure theory of dynamical systems similar to that of nuclear C^*-algebras?

open_problem | open | MathDB 页面 | 原文线索

V2-040974 | MATHDB-398657

Extensive adiabatic invariant for the Klein-Gordon chain in the thermodynamic limit

摘录 / EXCERPT: The report states that, for the Klein-Gordon Hamiltonian $H(x,y)=\frac12\sum_{j=1}^{N}\left(y_j^2+x_j^2+a(x_j-x_{j-1})^2+\frac12x_j^4\right)$ with periodic boundary conditions...

open_problem | open | MathDB 页面 | 原文线索

V2-040975 | MATHDB-398710

Strict positivity of the Green-Kubo conductivity with positive coupling and noise

摘录 / EXCERPT: For the chain of Hamiltonian cells perturbed by an energy-preserving noise of intensity $\varsigma>0$ and coupled by a smooth nearest-neighbor potential of strength ε,...

open_problem | open | MathDB 页面 | 原文线索

V2-040976 | MATHDB-398749

Propagation of chaos for the 2D viscous vortex model

摘录 / EXCERPT: Let $w_0\ge 0$ satisfy ... Let $X^N=(X_t^{N,1},\ldots,X_t^{N,N})_{t\ge0}$ be the vortex system with i.i.d. initial positions of density w_0, ... where $\nu>0$, the B^i are independ...

open_problem | open | MathDB 页面 | 原文线索

V2-040977 | MATHDB-398881

Goldman's ergodicity conjecture for non-maximal Euler-class components

摘录 / EXCERPT: When S_g is a closed surface of genus $g\ge 2$, Goldman conjectured that the action of $\mathrm{MCG}(S_g)$ on the components of $\mathcal X(S_g)$ with non-maximal Euler class is...

conjecture | open | MathDB 页面 | 原文线索

V2-040978 | MATHDB-398918

Bounded Negativity for Dynamically Minimal Positive-Entropy Automorphisms

摘录 / EXCERPT: Prove the Bounded Negativity Conjecture for complex projective surfaces admitting dynamically minimal automorphisms with positive entropy.

open_problem | open | MathDB 页面 | 原文线索

V2-040979 | MATHDB-399158

Analysis of a modified least-action principle for dissipative Hamiltonian dynamics

摘录 / EXCERPT: For a closed bounded subset S of a Euclidean space, define ... The associated Hamiltonian ODE is ... With $d_0K[X]$ denoting the element of minimal Euclidean norm in the subdifferent...

open_problem | open | MathDB 页面 | 原文线索

V2-040980 | MATHDB-399412

Absolute continuity of diffraction relative to the maximal dynamical spectral measure

摘录 / EXCERPT: The diffraction spectrum is absolutely continuous relative to the maximal spectral measure of the dynamical spectrum.

open_problem | open | MathDB 页面 | 原文线索

V2-040981 | MATHDB-399448

Uniqueness of an $n\times n$ Hamiltonian from its monodromy matrix

摘录 / EXCERPT: Let $B=B^*\in\mathbb C^{n\times n}$ and let ... be the polar decomposition of the matrix $-iB$. Let $H(\cdot)$ and $\widetilde H(\cdot)$ be nonnegative Hamiltonians admitting represe...

open_problem | open | MathDB 页面 | 原文线索

V2-040982 | MATHDB-399491

Commuting billiard maps

摘录 / EXCERPT: Let two nested convex domains in the plane have smooth boundaries. Their billiard ball maps, T_1 and T_2, act on the oriented lines that intersect both domains. Assume that the...

open_problem | open | MathDB 页面 | 原文线索

V2-040983 | MATHDB-399679

Strictly ergodic Cantor models with prescribed subgroup restrictions

摘录 / EXCERPT: Let $G\curvearrowright (X,\mu)$ be a free ergodic probability-measure-preserving action of a countable amenable discrete group, and let $\{H_i\}$ be a countable totally ordered col...

open_problem | open | MathDB 页面 | 原文线索

V2-040984 | MATHDB-399760

No minimal energy level for a proper Hamiltonian on \(\mathbb{R}^4\)

摘录 / EXCERPT: If H is a smooth and proper Hamiltonian on $\mathbb{R}^4$, then no energy level of H is minimal.

open_problem | open | MathDB 页面 | 原文线索

V2-040985 | MATHDB-399762

Full-measure ergodicity for non-rational polygonal billiards

摘录 / EXCERPT: Is there, for any number of edges, a set of full Lebesgue measure consisting of ergodic non-rational polygons?

open_problem | open | MathDB 页面 | 原文线索

V2-040986 | MATHDB-399864

Universality of the critical scaling theory for interacting two-dimensional Ising models

摘录 / EXCERPT: For the Ising models with Hamiltonian $H_\Lambda=-J\sum_{\langle x,y\rangle}\sigma_x\sigma_y-\lambda\sum_{x,y}\sigma_xv(x-y)\sigma_y$, where $J>0$, $\Lambda\subset\mathbb Z^2$ is a...

open_problem | open | MathDB 页面 | 原文线索

V2-040987 | MATHDB-399881

Conjugacy classification for ergodic transformations with discrete spectrum

摘录 / EXCERPT: For ergodic measure-preserving transformations with discrete spectrum, the subgroup of the unit circle K formed by the eigenvalues is a complete conjugacy invariant: two such tra...

open_problem | open | MathDB 页面 | 原文线索

V2-040988 | MATHDB-399998

Dynamic formation of polymerization shocks from smooth data

摘录 / EXCERPT: Can these phenomena also result dynamically from a suitable class of smooth initial data?

open_problem | open | MathDB 页面 | 原文线索

V2-040989 | MATHDB-400246

Tail bounds for the number of Hamiltonian cycles

摘录 / EXCERPT: Let $G(n,m)$ denote the random graph on n vertices and m edges, where $m=\\Theta(n^2)$. Let H be the number of Hamiltonian cycles in the graph, and let $\\mu=\\mathbb{E}H$. E...

open_problem | open | MathDB 页面 | 原文线索

V2-040990 | MATHDB-400351

General isothermal two-phase Riemann problem for thermodynamically consistent equations of state

摘录 / EXCERPT: For the one-dimensional isothermal Euler system $\partial_t\rho+\partial_x(\rho u)=0$, $\partial_t(\rho u)+\partial_x(\rho u^2+p)=0$, with liquid and vapor phases distinguished by...

open_problem | open | MathDB 页面 | 原文线索

V2-040991 | MATHDB-400426

Many-body localization for a disordered interacting quantum spin chain

摘录 / EXCERPT: For the random-field, random-transverse-field, random-exchange Ising Hamiltonian ... on $\Lambda=[-K,K]\cap\mathbb Z$, acting on $\mathcal H=\bigotimes_{i\in\Lambda}\mathbb C^2$, whe...

open_problem | open | MathDB 页面 | 原文线索

V2-040992 | MATHDB-400481

Bowen's conjecture on ergodic hyperfinite primitive subrelations

摘录 / EXCERPT: Every ergodic p.m.p. treeable equivalence relation contains an ergodic hyperfinite primitive subrelation.

conjecture | open | MathDB 页面 | 原文线索

V2-040993 | MATHDB-400658

Quenched mixing-time upper bound for random walk on supercritical dynamical percolation

摘录 / EXCERPT: Let $p>p_c(\mathbb{Z}^d)$ with $\theta(p)>1/2$. Then there exists $a>0$ such that, for all $\varepsilon>0$, all sufficiently large n, and every starting environment η_0, ... w...

open_problem | open | MathDB 页面 | 原文线索

V2-040994 | MATHDB-400711

Dynamical reconstruction from operator-space envelopes

摘录 / EXCERPT: For a pair of unital operator spaces A_1,A_2, do there exist suitable one-parameter groups of automorphisms $\sigma_i:\mathbb{R}\to\operatorname{Aut}(C^*_{\mathrm{env}}(A_i))$ su...

open_problem | open | MathDB 页面 | 原文线索

V2-040995 | MATHDB-400827

Anti-classification theorem for measure-preserving diffeomorphisms of the 2-torus

摘录 / EXCERPT: Let $\mathcal{G}$ be the collection of diffeomorphisms $T\in \operatorname{Diff}^{\infty}(\mathbb{T}^{2},\lambda)$ such that: (1) T is ergodic; and (2) T is isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-040996 | MATHDB-401067

Derivation of the Landau-Pekar equations from a microscopic polaron model

摘录 / EXCERPT: It is an open problem to derive the Landau-Pekar equations from a microscopic model of a polaron based on the Fröhlich Hamiltonian.

open_problem | open | MathDB 页面 | 原文线索

V2-040997 | MATHDB-401069

Variational minimization for degenerate KdV compactons

摘录 / EXCERPT: For the Hamiltonian $H[u]$ and mass constraint $\|u\|_{L^2}^2=M_0$, where $M_0>0$, the minimization problem ... admits a minimizer for $2<p<8$. Every minimizer is, up to translatio...

open_problem | open | MathDB 页面 | 原文线索

V2-040998 | MATHDB-401104

Non-KAM-stable elliptic equilibria for smooth Hamiltonians

摘录 / EXCERPT: For every dimension parameter $d\ge 3$, examples of smooth Hamiltonians are presented having elliptic equilibria with arbitrary frequency that are not KAM-stable.

open_problem | open | MathDB 页面 | 原文线索

V2-040999 | MATHDB-401109

Dynamically convex contact forms on S^3 violating the Viterbo bound

摘录 / EXCERPT: For every $\varepsilon>0$ there exists a dynamically convex contact form α on S^3 such that $\rho(S^3,\alpha)>2-\varepsilon$; here dynamically convex means that every...

open_problem | open | MathDB 页面 | 原文线索

V2-041000 | MATHDB-401117

Localized C^∞ closing lemma for Hamiltonian surface diffeomorphisms

摘录 / EXCERPT: Let (S,ω) be a closed symplectic two-manifold, let φ be a Hamiltonian diffeomorphism of (S,ω), and let U be a nonempty open subset of S. Can the sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-041001 | MATHDB-401230

Strong arithmetic quantum unique ergodicity for Hecke-Maass cusp forms

摘录 / EXCERPT: Let $M:=\mathrm{SL}_2(\mathbb Z)\backslash\mathbb H$. Let φ range over Hecke-Maass cusp forms on M, with $\Delta\varphi=-\lambda\varphi$ and $\lambda\to\infty$, and defin...

open_problem | open | MathDB 页面 | 原文线索

V2-041002 | MATHDB-401305

Decomposition of homeomorphisms between FLC tiling spaces

摘录 / EXCERPT: Let Ω and Ω' be aperiodic, minimal, finite-local-complexity tiling spaces, with Ω uniquely ergodic. For every homeomorphism $h:\Omega\to\Omega'$, there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-041003 | MATHDB-401586

Noncommutative maximal ergodic theorem for ball averages under doubling and asymptotic invariance

摘录 / EXCERPT: Let G be a locally compact group with right Haar measure m, and let d be an invariant measurable metric such that m is a Radon Borel measure with respect to (G,d). For...

open_problem | open | MathDB 页面 | 原文线索

V2-041004 | MATHDB-401865

Quenched invariance principle in one-dimensional dynamical random conductances

摘录 / EXCERPT: Let $d=1$ and assume the conductances satisfy the stated Assumption: they are non-negative, obey $a_t(x,y)=a_t(y,x)$, and their law $[U+007F]P$ is stationary and ergodic under space-time...

open_problem | open | MathDB 页面 | 原文线索

V2-041005 | MATHDB-401869

Analyticity of the McCoy-Wu free energy away from the critical point

摘录 / EXCERPT: For the two-dimensional Ising model with columnar disorder on $\Lambda_N=(-N,N)^2\cap\mathbb Z^2$, with Hamiltonian ... where $E_1>0$, $E_2(j)=E_2+\Delta\eta(j)$, $E_2>0$, $\Delta>0$...

open_problem | open | MathDB 页面 | 原文线索

V2-041006 | MATHDB-402215

Special Lagrangian representatives in Hamiltonian isotopy classes

摘录 / EXCERPT: In a Calabi-Yau manifold, which Hamiltonian isotopy classes of compact zero Maslov Lagrangians admit a special Lagrangian representative?

open_problem | open | MathDB 页面 | 原文线索

V2-041007 | MATHDB-402225

Sarnak’s spectral-rigidity conjecture for convex planar domains

摘录 / EXCERPT: Every convex planar domain Omega subset mathbb{R}^2 with smooth boundary is dynamically spectrally rigid: any deformation preserving the length of every periodic billiard orbit ins...

conjecture | open | MathDB 页面 | 原文线索

V2-041008 | MATHDB-402233

Barcode endpoint bound for fixed points of Hamiltonian homeomorphisms

摘录 / EXCERPT: Let (M,ω) be a closed, connected, and symplectically aspherical manifold. Let $\varphi\in\operatorname{Ham}(M,\omega)$ be a Hamiltonian homeomorphism. If...

open_problem | open | MathDB 页面 | 原文线索

V2-041009 | MATHDB-402288

Realizing the K-theory of an infinite compact metric space by a minimal dynamical system

摘录 / EXCERPT: For a given infinite compact metric space X, can one find a second space Y admitting a minimal homeomorphism and having the same K-theory as X?

open_problem | open | MathDB 页面 | 原文线索

V2-041010 | MATHDB-402342

Gross-Pitaevskii limit of a homogeneous Bose gas at positive temperature

摘录 / EXCERPT: For a nonnegative interaction potential v with finite scattering length, consider the N-boson Hamiltonian in a three-dimensional periodic box of side length L with interactio...

open_problem | open | MathDB 页面 | 原文线索

V2-041011 | MATHDB-402376

Geometric interpretation of automorphic L-values via Hamiltonian duality

摘录 / EXCERPT: Conjecture: When $\mathfrak{o}=k[[t]]$, with k a finite field or $\mathbb{C}$, the derived category of $G(\mathfrak{o})$-equivariant constructible sheaves on $X(F)$ is equivalent...

open_problem | open | MathDB 页面 | 原文线索

V2-041012 | MATHDB-402377

Involutivity of the Hamiltonian duality assignment

摘录 / EXCERPT: We conjecture that, if the derived category of constructible sheaves on $X(F)$ is viewed as a quantization of the Hamiltonian space $M=T^*X$, then the assignment $M\mapsto\check M$...

open_problem | open | MathDB 页面 | 原文线索

V2-041013 | MATHDB-402624

Rapid mixing of Glauber dynamics for colorings at the ergodicity threshold

摘录 / EXCERPT: A well-known conjecture is that Glauber dynamics mixes in polynomial time (rapid mixing) for every $k \ge \Delta + 2$ (and hence that there exists a polynomial-time approximation s...

open_problem | open | MathDB 页面 | 原文线索

V2-041014 | MATHDB-402773

Quantum ergodicity for compact quotients of $\mathrm{SL}_d(\mathbb{R})/\mathrm{SO}(d)$ in the level aspect

摘录 / EXCERPT: Let $d\ge 3$, $S=\mathrm{SL}_d(\mathbb{R})/\mathrm{SO}(d)$, and let $\Gamma_n\le \mathrm{SL}_d(\mathbb{R})$ be a sequence of uniformly discrete, torsion-free, cocompact lattices su...

open_problem | open | MathDB 页面 | 原文线索

V2-041015 | MATHDB-402793

Metric and geodesic structure for generalized Wasserstein costs with matrix-valued mobility

摘录 / EXCERPT: Find conditions on a general matrix-valued function $K(\rho,u,t)\in\mathbb{R}^{d\times d}$ under which the dynamical description ... subject to ... provides a well-defined metric struc...

open_problem | open | MathDB 页面 | 原文线索

V2-041016 | MATHDB-403132

General data-driven identification of dynamical systems from imperfect measurements

摘录 / EXCERPT: Given a dynamical system with high-dimensional state $x\in X_1\subset\mathbb{R}^n$ and measurements $y_k\in\mathbb{R}^m$, where $n\gg1$ and $m\ll n$, modeled by ... with dynamics f...

open_problem | open | MathDB 页面 | 原文线索

V2-041017 | MATHDB-403134

Representations for nonlinear port-Hamiltonian systems

摘录 / EXCERPT: Finding appropriate representations for nonlinear port-Hamiltonian systems remains a largely open research field.

open_problem | open | MathDB 页面 | 原文线索

V2-041018 | MATHDB-403226

Extension of the Calabi invariant to Hamiltonian homeomorphisms of the disk

摘录 / EXCERPT: Does the Calabi invariant admit an extension to $\mathrm{Ham}(D,\omega)$?

open_problem | open | MathDB 页面 | 原文线索

V2-041019 | MATHDB-403230

Existence of non-isometric billiard germs with identical Birkhoff normal forms

摘录 / EXCERPT: There exist two non-isometric domain germs around bouncing-ball orbits for which the Birkhoff normal forms of the billiard maps around the orbits are the same.

open_problem | open | MathDB 页面 | 原文线索

V2-041020 | MATHDB-403233

Two-or-infinitely-many periodic orbits for nondegenerate Reeb vector fields

摘录 / EXCERPT: Let M be a closed oriented 3-manifold. If M is not the sphere or a lens space, then every nondegenerate Reeb vector field on M has infinitely many simple periodic orbits. I...

open_problem | open | MathDB 页面 | 原文线索

V2-041021 | MATHDB-403236

Tabachnikov’s local rigidity conjecture for rational caustics

摘录 / EXCERPT: In a sufficiently small C^r ($r=2,\ldots,\infty,w$) neighborhood of a circle, there are no billiard tables other than circles that admit both a $(1,2)$-rational caustic and a...

conjecture | open | MathDB 页面 | 原文线索

V2-041022 | MATHDB-403242

Lagrangian products, toric domains, and Minkowski billiards

摘录 / EXCERPT: For which domains $K,L\subset\mathbb{R}^n$ is the Lagrangian product $K\times_L L$ a toric domain, and how is the integrability of the corresponding Minkowski billiard system relat...

open_problem | open | MathDB 页面 | 原文线索

V2-041023 | MATHDB-403243

Lagrangian Poincaré recurrence

摘录 / EXCERPT: Let (M,ω) be a closed symplectic manifold and let $L\subset M$ be a displaceable closed Lagrangian submanifold. For every Hamiltonian diffeomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-041024 | MATHDB-403250

Convergence for Hamiltonians singular along nowhere \b3-coisotropic sets

摘录 / EXCERPT: In the report's setting $M=T^*N$, let $V\subset M$ be compact, let $H\in C^0(M\setminus V)$, and let $(H_k)_{k\ge 1}\subset C^\infty(M)$ converge uniformly on compact subsets of...

open_problem | open | MathDB 页面 | 原文线索

V2-041025 | MATHDB-403375

Flatness of ergodic compact Cauchy horizons

摘录 / EXCERPT: For a smooth, time-orientable vacuum spacetime of dimension $3+1$ with a smooth, compact, non-degenerate Cauchy horizon C, if every null generator of C densely fills C (case...

open_problem | open | MathDB 页面 | 原文线索

V2-041026 | MATHDB-403688

Erdős's conjecture on pancyclicity of Hamiltonian graphs

摘录 / EXCERPT: There exists a sufficiently large absolute constant c such that every Hamiltonian graph G with independence number $\b1(G)\le k$ and at least ck^2 vertices is pancyclic; that...

conjecture | open | MathDB 页面 | 原文线索

V2-041027 | MATHDB-403969

Mathematical proof of the predicted asymptotics for Hamiltonian cycles in planar bicubic maps

摘录 / EXCERPT: Let H_{2n} be the number of edge-rooted Hamiltonian cycles in genus-0 planar bicubic maps with $2n$ vertices, where the maps are bipartite and trivalent. The report states the...

open_problem | open | MathDB 页面 | 原文线索

V2-041028 | MATHDB-404061

Quantum algorithms for non-pure-point-spectrum ergodic flows

摘录 / EXCERPT: Whether similar quantum algorithms can be constructed for measure-preserving ergodic flows that are not of pure-point-spectrum type, particularly weak-mixing or mixing systems wher...

open_problem | open | MathDB 页面 | 原文线索

V2-041029 | MATHDB-404173

Existence of all combinations of Chazy asymptotic motions in the restricted isosceles three-body problem

摘录 / EXCERPT: Consider the Hamiltonian system H_G defined in (3). Denote by X^+ (respectively Y^-) either H^+, P^+, B^+, or OS^+ (respectively H^-, P^-, B^-, or OS^-), acco...

open_problem | open | MathDB 页面 | 原文线索

V2-041030 | MATHDB-404247

The Dry Ten Martini Problem for Sturmian dynamical systems

摘录 / EXCERPT: For every irrational $\b1\in[0,1]\setminus\mathbb{Q}$ and every $V\in\mathbb{R}\setminus\{0\}$, let $H_{\alpha,V}:\ell^2(\mathbb{Z})\to\ell^2(\mathbb{Z})$ be defined by ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-041031 | MATHDB-404324

Rigorous justification of the Bogoliubov prediction for strongly interacting Fermi gases

摘录 / EXCERPT: For spinless fermions on $(\mathbb R/\mathbb Z)^{3N}$ with Hamiltonian ... where the wave function is antisymmetric, $\alpha<\tfrac13$, and the Fourier coefficients satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-041032 | MATHDB-404678

Lower boundedness of the Hamiltonian mass for physically relevant two-dimensional ALH initial data

摘录 / EXCERPT: For all physically relevant, globally well-behaved initial data sets with matter satisfying suitable positivity conditions, the Hamiltonian mass should be bounded from below under...

open_problem | open | MathDB 页面 | 原文线索

V2-041033 | MATHDB-404846

Reeb Hofer-Zehnder and multiplicity conjecture

摘录 / EXCERPT: The Hamiltonian flow on a star-shaped energy level in a $2n$-dimensional symplectic vector space has either exactly n or infinitely many periodic orbits.

conjecture | open | MathDB 页面 | 原文线索

V2-041034 | MATHDB-404849

Global area spectral rigidity for symplectic billiards

摘录 / EXCERPT: For strictly convex planar domains 4Omegau00024 and 4Omega'u00024 of the same area, does 4A(Omega)=A(Omega')4 imply that 4Omegau00024 and 4Omega'u00024 are equal up to a unitary af...

open_problem | open | MathDB 页面 | 原文线索

V2-041035 | MATHDB-404850

Poincaré-Birkhoff theorem for C^0-Hamiltonian twist maps on cotangent domains

摘录 / EXCERPT: Let Q be a closed, orientable manifold, and let $f:(W,\omega)\to(W,\omega)$ be a C^0-Hamiltonian twist map on a fiber-wise starshaped domain $W\subset T^*Q$. Then either f ha...

open_problem | open | MathDB 页面 | 原文线索

V2-041036 | MATHDB-404857

Quadratic growth of periodic points for asymptotically linear Hamiltonian diffeomorphisms

摘录 / EXCERPT: The report conjectures that, if an asymptotically linear Hamiltonian diffeomorphism ϕ has infinitely many non-trivial periodic points, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-041037 | MATHDB-404936

Deterministic conductance models: invariance principle for averaging ergodic conductances

摘录 / EXCERPT: Let $0<a\le b<\infty$, let $\Omega=[a,b]^{E(\mathbb Z^d)}$, and let τ_x denote spatial shifts. An $\omega\in\Omega$ is called averaging if, for every $f\in C(\Omega)$, the lim...

open_problem | open | MathDB 页面 | 原文线索

V2-041038 | MATHDB-405035

Macroscopic particle transport for long-range bosonic systems

摘录 / EXCERPT: For bosons on an arbitrary D-dimensional lattice with Hamiltonian $H_t=-\sum_{i\ne j\in\Lambda}J_{ij}(t)\hat b_i^\dagger\hat b_j+\sum_{Z\subseteq\Lambda}h_Z(t)$, where $h_Z(t)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-041039 | MATHDB-405205

Two-centre Birkhoff-Poritsky conjecture

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded domain with real-analytic boundary containing the segment joining the two centres, and let B_h denote the billiard map on the fixed-e...

conjecture | claimed_solved | MathDB 页面 | 原文线索

B10 组合数学 / Combinatorics | 5,906 条

V2-041040 | MATHDB-315465

Binary-code exponent beyond the MRRW bound

摘录 / EXCERPT: Can the optimized MRRW upper bound for asymptotic binary-code rates be improved at every fixed relative distance?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041041 | MATHDB-315466

Spherical-code exponent beyond Kabatianskii-Levenshtein

摘录 / EXCERPT: Can the optimized Kabatianskii--Levenshtein exponent for high-dimensional spherical codes be improved for every fixed inner-product threshold?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041042 | MATHDB-315489

Elementary symmetric-polynomial bounds for centered vectors and matrices

摘录 / EXCERPT: How sharply can elementary symmetric polynomials be bounded under vector or matrix centering constraints, and what quantitative consequences follow for permutation mixtures and fin...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041043 | MATHDB-315502

Graffiti Conjecture 284

摘录 / EXCERPT: If G has order at least three and girth at least five, then ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041044 | MATHDB-315512

Hamilton decompositions of the directed 3-torus

摘录 / EXCERPT: For $D_3(m)=\vec C_m\square\vec C_m\square\vec C_m$, can the full arc set be partitioned into three directed Hamilton cycles for every integer $m\geq3$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041045 | MATHDB-315514

Ramsey numbers for book graphs

摘录 / EXCERPT: Does every n admit a graph on $4n-2$ vertices containing no book B_{n-1} whose complement contains no B_n, and hence prove $R(B_{n-1},B_n)=4n-1$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041046 | MATHDB-315516

Diagonal Ramsey upper-bound constant

摘录 / EXCERPT: Within the stated diagonal-Ramsey ansatz, choose a correction polynomial and auxiliary functions satisfying all sufficient inequalities while minimizing the resulting constant...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041047 | MATHDB-315526

Written on the Wall II, Graph Conjecture 2

摘录 / EXCERPT: Let G be a finite simple connected graph. For each $v\in V(G)$, let $N_G(v)$ be the open neighborhood of v and define ... where $\alpha(H)$ is the independence number of H, wit...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041048 | MATHDB-315527

Ben Green’s Open Problem 57

摘录 / EXCERPT: For an abelian group G, let $\Phi(G)$ be the space of all functions on G that are convex combinations of functions of the form ... where $\|f_i\|_\infty\le 1$. Let $\Phi'(G)$ be...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041049 | MATHDB-315532

Record compositions of alternating permutations

摘录 / EXCERPT: The two open problems stated in Section 1.4 of the cited paper are: 1. Is there a refinement of the record-partition enumeration to record compositions? Equivalently, for compositi...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041050 | MATHDB-315533

Reciprocals of partition polynomials

摘录 / EXCERPT: For $\operatorname{sp}(\lambda,x)=\prod_i(1+x^{\lambda_i})$, what divisibility, coprimality, recurrence, irreducibility, and coefficient-shape properties hold after reducing sums o...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041051 | MATHDB-315597

Hook coefficients of universal Hilbert series

摘录 / EXCERPT: For the universal Schur coefficients $c_\lambda(n)$ in the multigraded Hilbert series of the coinvariant algebra $R_n^{(m)}$, give a combinatorial interpretation when λ is...

open_problem | open | MathDB 页面 | 原文线索

V2-041052 | MATHDB-315610

Graffiti Conjecture 143

摘录 / EXCERPT: Graffiti Conjecture 143. If G is connected and not a tree, then

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041053 | MATHDB-315611

Graffiti Conjecture 154 under the standard-deviation reading

摘录 / EXCERPT: For every connected graph, is the deviation of its adjacency eigenvalues at most its order divided by its average distance?

open_problem | open | MathDB 页面 | 原文线索

V2-041054 | MATHDB-315612

TxGraffiti-Davila Conjecture 9

摘录 / EXCERPT: If G is connected, cubic, and diamond-free, must $Z(G)\leq\gamma(G)+2$?

open_problem | open | MathDB 页面 | 原文线索

V2-041055 | MATHDB-315613

Pandey parity conjecture for generalized Petersen graphs

摘录 / EXCERPT: Let $\operatorname{GP}(n,k)$ be the generalized Petersen graph, and let $\operatorname{I}(\operatorname{GP}(n,k),x)$ denote its independence polynomial. For all integers...

open_problem | open | MathDB 页面 | 原文线索

V2-041056 | MATHDB-315618

Koch-Narayan Conjecture 1

摘录 / EXCERPT: Let G be a bipartite graph without isolated vertices, of order n, domination number γ, and with a unique minimum dominating set. If $\gamma \geq 2$ and $n \geq 3\gamma$,...

open_problem | open | MathDB 页面 | 原文线索

V2-041057 | MATHDB-315619

Elizalde-Luo $\{1132,3312\}$ pattern-avoidance conjecture

摘录 / EXCERPT: Is the number of nonnesting multiset permutations avoiding 1132 and 3312 equal to $3^n-3\cdot2^{n-1}+1$?

open_problem | open | MathDB 页面 | 原文线索

V2-041058 | MATHDB-315620

Kurkov’s Fubini-number sum conjecture

摘录 / EXCERPT: For the Fubini numbers $a(n)$, with $a(0)=1$, Kurkov's conjecture states that ... for every integer $n>0$.

open_problem | open | MathDB 页面 | 原文线索

V2-041059 | MATHDB-315625

Levit-Mandrescu very-well-covered graph conjecture

摘录 / EXCERPT: For every very well-covered graph G, the coefficient sequence of its independence polynomial $I(G;x)$ is unimodal.

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041060 | MATHDB-315635

Antipodal cut code of K_{11}

摘录 / EXCERPT: For the antipodal cut code $\widehat C_m= C_{\mathrm{cut}}(K_m)\cup(\mathbf1+C_{\mathrm{cut}}(K_m))$, must $\rho(\widehat C_m)=\rho(C_{m-1})$ hold for every $m\geq7$?

open_problem | open | MathDB 页面 | 原文线索

V2-041061 | MATHDB-315636

Bondy’s minimum-degree longest-cycle conjecture

摘录 / EXCERPT: Let $k\geq 1$ be an integer, and let G be a k-connected graph on n vertices. If $\delta(G)\geq\frac{n+k(k-1)}{k+1}$, then for every longest cycle C of G, every path in...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041062 | MATHDB-315666

Rota’s matroid-flat unimodality conjecture

摘录 / EXCERPT: Let M be a matroid of rank r. For $0 \leq i \leq r$, let W_i denote the number of rank-i flats of M. Is the sequence $(W_0,W_1,\dots,W_r)$ unimodal for every matroid? Equ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041063 | MATHDB-315679

Powers of the Vandermonde determinant are eventually non-SNP

摘录 / EXCERPT: For every integer $k\geq 1$, there exists an integer N_k such that, for every $n\geq N_k$, the kth power $a_{\delta_n}^{k}$ of the Vandermonde determinant is non-SNP; that is,...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041064 | MATHDB-315682

Gold Partition Conjecture through fourteen elements

摘录 / EXCERPT: Does every finite non-chain poset satisfy the Gold Partition property, and hence the $1/3$-$2/3$ balance bound?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041065 | MATHDB-315688

Stanley’s rankwise lower-bound conjecture for differential posets

摘录 / EXCERPT: Must every r-differential poset P satisfy ... Here Y^r denotes the r-fold Cartesian power of Young's lattice.

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041066 | MATHDB-315690

Stanley’s weighted-multichain rationality problem

摘录 / EXCERPT: For every differential poset P and every fixed positive integer k, does there exist a rational function $R(q)$ such that ... Here $F_P(q)$ is the rank generating series, and ... Th...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041067 | MATHDB-315692

Written on the Wall II, Graph Conjecture 103

摘录 / EXCERPT: WOWII Conjecture 103. For every simple connected graph G, $a(G) \leq \left\lfloor b(G) - \ln\left(\frac{1}{|V(G)|}\sum_{v\in V(G)}\operatorname{ecc}(v)\right) \right\rfloor$, whe...

open_problem | open | MathDB 页面 | 原文线索

V2-041068 | MATHDB-315693

Written on the Wall II, Graph Conjecture 109

摘录 / EXCERPT: If G is a simple connected graph, then $a(G) \le \left\lfloor\frac{\operatorname{residue}(G)+2b(G)}{3}\right\rfloor$, where $a(G)$ is the independence number and $b(G)$ is the or...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041069 | MATHDB-315694

Written on the Wall II, Graph Conjecture 143

摘录 / EXCERPT: For every finite simple connected non-tree graph G, let $t(G)$ denote the largest order of an induced tree, let $g(G)$ denote the girth of G, and let $\delta'(G)$ denote the se...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041070 | MATHDB-315713

Optimal exponent relating sumsets and difference sets

摘录 / EXCERPT: Let A be a finite subset of a commutative additive group Z, and define $\sigma(A)=\frac{|A+A|}{|A|}$ and $\delta(A)=\frac{|A-A|}{|A|}$. The cited paper asks whether the exponen...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041071 | MATHDB-315715

Nikiforov’s spectral consecutive-cycle constant

摘录 / EXCERPT: What is the largest constant C forcing cycles of every length up to $(C-o(1))n$ from the spectral condition $\rho(G)>\sqrt{\lfloor n^2/4\rfloor}$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041072 | MATHDB-315731

WOW-284 distance-spectrum conjecture

摘录 / EXCERPT: Conjecture (WOW-284). If G has order at least three and girth at least five, then $\delta^*(G)\leq-\lambda_{\min}(D(G))$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041073 | MATHDB-315734

Kazhdan-Lusztig polynomial shape conjectures for matroids

摘录 / EXCERPT: For every matroid M, the coefficients of its Kazhdan--Lusztig polynomial $P_M(t)$ are non-negative (Conjecture 2.3). Moreover, the coefficient sequence of $P_M(t)$ is log-concave...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041074 | MATHDB-315740

Integral tangent classes for matroid wonderful models

摘录 / EXCERPT: Can the tangent bundle class of a realizable wonderful compactification be lifted to a canonical integral combinatorial tangent class for every matroid and building set?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041075 | MATHDB-315744

Monical’s SNP conjecture for Schur-positive chromatic functions

摘录 / EXCERPT: Let X_G be the chromatic symmetric function of a graph G, and let $X_G(x_1,\ldots,x_k)$ denote its specialization to k variables. A polynomial is SNP (has a saturated Newton...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-041076 | MATHDB-315906

1/3-2/3 conjecture

摘录 / EXCERPT: Let $P=(X,\le)$ be a finite partially ordered set, so that $\le$ is a reflexive, antisymmetric, transitive binary relation on the finite set X. A linear extension of P is a tot...

conjecture | open | MathDB 页面 | 原文线索

V2-041077 | MATHDB-315907

Dittert conjecture

摘录 / EXCERPT: For an $n \times n$ matrix $A = [a_{ij}]$ with real entries, let ... where the sum is over all permutations σ in the symmetric group S_n, and define ... Let J_n denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-041078 | MATHDB-315908

Problems in Latin squares

摘录 / EXCERPT: In mathematics, the theory of Latin squares is an active research area with many open problems. As in other areas of mathematics, such problems are often made public at professiona...

open_problem | open | MathDB 页面 | 原文线索

V2-041079 | MATHDB-315910

Map folding

摘录 / EXCERPT: Let $n\ge 1$, and let S_n be an ideal zero-thickness strip of n congruent rigid rectangular panels, joined consecutively by hinges along the $n-1$ common edges. A flat folded s...

open_problem | open | MathDB 页面 | 原文线索

V2-041080 | MATHDB-315912

Rudin's conjecture

摘录 / EXCERPT: For positive integers N,q,a, let ... that is, the number of perfect squares among the N terms $a,\ q+a,\ 2q+a,\ \ldots,\ (N-1)q+a$ of the arithmetic progression $qn+a$, and set ……

conjecture | open | MathDB 页面 | 原文线索

V2-041081 | MATHDB-315913

sunflower conjecture

摘录 / EXCERPT: For a set U, a collection W of subsets of U is called a sunflower (or Δ-system) if there is a set $S \subseteq U$ such that $A \cap B = S$ for all distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-041082 | MATHDB-315925

Hales-Jewett theorem

摘录 / EXCERPT: In mathematics, the Hales-Jewett theorem is a fundamental combinatorial result of Ramsey theory, named after Alfred W. Hales and Robert I. Jewett, that concerns the degree to which...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-041083 | MATHDB-315965

big-line-big-clique conjecture

摘录 / EXCERPT: For a finite set S of points in the Euclidean plane, say that two distinct points $p,q\in S$ are mutually visible (with respect to S) if no point of $S\setminus\{p,q\}$ lies on...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041084 | MATHDB-315966

Dirac-Motzkin conjecture

摘录 / EXCERPT: For a finite set S of points in the Euclidean plane, call a line ℓ an ordinary line of S if ℓ contains exactly two points of S. For an integer $n\geq 3$, let ... the...

conjecture | open | MathDB 页面 | 原文线索

V2-041085 | MATHDB-315969

Kalai's 3d conjecture

摘录 / EXCERPT: Let $d \ge 1$ and let $P \subset \mathbb{R}^d$ be a convex polytope of dimension d that is centrally symmetric, i.e. $P = -P$ after a suitable translation. For $0 \le i \le d$ le...

conjecture | open | MathDB 页面 | 原文线索

V2-041086 | MATHDB-315970

Kobon triangle problem

摘录 / EXCERPT: For an integer $k \ge 3$, let $\mathcal{A}$ be an arrangement of k distinct straight lines in the Euclidean plane $\mathbb{R}^2$. Call a set $T \subset \mathbb{R}^2$ a triangle o...

open_problem | open | MathDB 页面 | 原文线索

V2-041087 | MATHDB-315972

McMullen problem

摘录 / EXCERPT: For an integer $d\ge 1$, a finite set of points in $\mathbb{R}^d$ is said to be in general position if no $d+1$ of its points lie in a common hyperplane, and in convex position if...

open_problem | open | MathDB 页面 | 原文线索

V2-041088 | MATHDB-315973

Opaque forest problem

摘录 / EXCERPT: For a set $S\subseteq\mathbb{R}^2$, let its coverage be ... and for a set $K\subseteq\mathbb{R}^2$ call S an opaque set for K if $K\subseteq C(S)$, i.e. if every line that meets...

open_problem | open | MathDB 页面 | 原文线索

V2-041089 | MATHDB-316148

Unknotting problem

摘录 / EXCERPT: Fix the following encoding conventions. A knot diagram D is a connected 4-valent plane graph, given combinatorially (say, as a planar embedding together with the cyclic order o...

open_problem | open | MathDB 页面 | 原文线索

V2-041090 | MATHDB-316160

Erdős sumset conjecture

摘录 / EXCERPT: In additive combinatorics, the Erdős sumset conjecture is a conjecture which states that if a subset A of the natural numbers $\mathbb {N}$ has a positive upper density then ther...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041091 | MATHDB-316161

McMullen's g-conjecture

摘录 / EXCERPT: For a finite simplicial complex Δ of dimension d let $f_i = f_i(\Delta)$ denote the number of i-dimensional faces of Δ for $0 \le i \le d$, with the convention...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041092 | MATHDB-316162

Hirsch conjecture

摘录 / EXCERPT: Let $P \subseteq \mathbb{R}^d$ be a d-dimensional convex polytope, i.e. a bounded intersection of finitely many closed half-spaces whose affine hull is d-dimensional. A facet o...

conjecture | open | MathDB 页面 | 原文线索

V2-041093 | MATHDB-316163

Gessel's lattice path conjecture

摘录 / EXCERPT: Ira Martin Gessel is an American mathematician, known for his work in combinatorics. He is a long-time faculty member at Brandeis University and resides in Arlington, Massachusetts...

conjecture | open | MathDB 页面 | 原文线索

V2-041094 | MATHDB-316164

Stanley-Wilf conjecture

摘录 / EXCERPT: For a permutation σ of $\{1,2,\dots,n\}$ and a permutation β of $\{1,2,\dots,k\}$, say that σ contains β as a pattern if there are indices...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041095 | MATHDB-316166

Cameron-Erdős conjecture

摘录 / EXCERPT: In combinatorics, the Cameron-Erdős conjecture (now a theorem) is the statement that the number of sum-free sets contained in $[N]=\{1,\ldots,N\}$ is $O{\big (}{2^{N/2}}{\big)}.$ T...

conjecture | open | MathDB 页面 | 原文线索

V2-041096 | MATHDB-316169

angel problem

摘录 / EXCERPT: The angel problem is a question in combinatorial game theory proposed by John Horton Conway. The game is commonly referred to as the angels and devils game. The game is played by t...

open_problem | open | MathDB 页面 | 原文线索

V2-041097 | MATHDB-316244

Conjecture on fixed-length complete sums in cyclic groups

摘录 / EXCERPT: Fixed-length complete-sums conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041098 | MATHDB-316257

The polynomial-form conjecture for the functions \(\Psi_g\)

摘录 / EXCERPT: Let $g\geq 0$, and let u, Y_0, $Y_1,\ldots,Y_{2g}$, and $Z_1,\ldots,Z_{2g+1}$ be the variables occurring in the functions Ψ_g. Write ... Here f_g is a polynomial in its ar...

conjecture | open | MathDB 页面 | 原文线索

V2-041099 | MATHDB-316268

Minimal difference set embedding conjecture for abelian difference sets

摘录 / EXCERPT: Minimal difference set embedding conjecture. The difference set D contains a minimal difference set embedded in it: there exists a subgroup S of G such that

conjecture | open | MathDB 页面 | 原文线索

V2-041100 | MATHDB-316353

The exponential-formula conjecture for primitive ideals in quantum matrices

摘录 / EXCERPT: Exponential enumeration conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-041101 | MATHDB-316354

The asymptotic proportion conjecture for primitive ideals in quantum matrices

摘录 / EXCERPT: Primitive-proportion conjecture. As $n\rightarrow\infty$, the proportion of $\mathcal{H}$-primes that are primitive tends to

conjecture | open | MathDB 页面 | 原文线索

V2-041102 | MATHDB-316370

Branson's extremal determinant conjecture for conformally covariant operators on even-dimensional spheres

摘录 / EXCERPT: Branson's conjecture. The quantities

conjecture | open | MathDB 页面 | 原文线索

V2-041103 | MATHDB-316439

The subdivision-tropical oriented matroid correspondence

摘录 / EXCERPT: Let Δ_{n-1} and Δ_{d-1} be simplices, and let a tropical oriented matroid have parameters (n,d). A subdivision is a polyhedral subdivision of the product...

conjecture | open | MathDB 页面 | 原文线索

V2-041104 | MATHDB-316440

The topological representation theorem for tropical oriented matroids

摘录 / EXCERPT: A tropical pseudohyperplane is a subset of $\mathbb{TP}^{d-1}$ that is PL-homeomorphic to a tropical hyperplane, and an arrangement is a collection of such tropical pseudohyperplan...

conjecture | open | MathDB 页面 | 原文线索

V2-041105 | MATHDB-316442

The generic rigidity conjecture for k-triangulations

摘录 / EXCERPT: A k-triangulation is a maximal set of pairwise noncrossing diagonals of a convex polygon in which no $k+1$ diagonals mutually cross. A graph is generically minimally rigid in dim...

conjecture | open | MathDB 页面 | 原文线索

V2-041106 | MATHDB-316495

Power-word extremality conjecture in symmetric two-groups

摘录 / EXCERPT: Let W_n be the symmetric 2-group, let a denote a generator, and let w be a word of length 2^m. Power-word extremality conjecture. ... The conjecture says that among words o...

conjecture | open | MathDB 页面 | 原文线索

V2-041107 | MATHDB-316498

The Davenport-constant formula for weighted zero-sums of length the group order

摘录 / EXCERPT: Conjecture 1. For any finite abelian group G with exponent n and any non-empty subset A of $]n[$,

conjecture | open | MathDB 页面 | 原文线索

V2-041108 | MATHDB-316499

Ten-round-zero conjecture for singular extremal ternary sextics

摘录 / EXCERPT: Let $P_{3,6}$ be the cone of nonnegative ternary sextic forms, let $\Delta_{3,6}$ be its relevant cone of forms under consideration, and let $E(P_{3,6})$ denote the set of extremal...

conjecture | open | MathDB 页面 | 原文线索

V2-041109 | MATHDB-316525

Gowers inverse conjecture for the Boolean cube

摘录 / EXCERPT: Let $d\geq 1$ and let $\varepsilon>0$. For each $n\geq 1$, write $\mathbf{F}_2^n$ for the Boolean vector space, let $f:\mathbf{F}_2^n\to[-1,1]$, let $\|f\|_{U^d(\mathbf{F}_2^n)}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-041110 | MATHDB-316554

Eventual unimodality of the coefficients of the q-Catalan numbers

摘录 / EXCERPT: Let $C_n(q)=\sum_{k=0}^{n(n-1)}m_n(k)q^k$ denote the q-Catalan numbers and let $m_n(k)$ be their coefficients. Eventual unimodality conjecture. The sequence ... is unimodal when...

conjecture | open | MathDB 页面 | 原文线索

V2-041111 | MATHDB-316555

Eventual log-concavity of the coefficients of the q-Catalan numbers

摘录 / EXCERPT: Let $C_n(q)=\sum_{k=0}^{n(n-1)}m_n(k)q^k$ denote the q-Catalan numbers and let $m_n(k)$ be their coefficients. Log-concavity conjecture. There exists an integer t such that, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-041112 | MATHDB-316557

Very short sets conjecture for the internal zonotopal space

摘录 / EXCERPT: Let X be a finite list of vectors of rank n. Define the set of very short X-sets by ... Let ${\cal P}_-(X)$ be the internal ${\cal P}$-space. Very short sets conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-041113 | MATHDB-316615

Associated Hermite matching-generating-function conjecture

摘录 / EXCERPT: Associated Hermite matching-generating-function conjecture. The integral

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041114 | MATHDB-316616

The missing-face embeddability conjecture for triangulated spheres

摘录 / EXCERPT: Missing-face embeddability conjecture. The complex K does not embed in the $2d$-sphere.

conjecture | open | MathDB 页面 | 原文线索

V2-041115 | MATHDB-316623

Nonexistence conjecture for four-dimensional faces of permutation polytopes

摘录 / EXCERPT: Nonexistence conjecture. There are no four-dimensional faces of permutation polytopes having the combinatorial type of Q_1 or Q_2.

conjecture | open | MathDB 页面 | 原文线索

V2-041116 | MATHDB-316624

Strong embedding conjecture for permutation polytopes

摘录 / EXCERPT: Strong embedding conjecture. There exists a permutation group $G\leq S_{2d}$ such that $P(G)$ is combinatorially equivalent, or more strongly lattice equivalent, to P.

conjecture | open | MathDB 页面 | 原文线索

V2-041117 | MATHDB-316626

Birkhoff-polytope uniqueness conjecture for permutation groups

摘录 / EXCERPT: Birkhoff-polytope uniqueness conjecture. If $P(G)$ is combinatorially equivalent to B_n for some n, then G is effectively equivalent to S_n.

conjecture | open | MathDB 页面 | 原文线索

V2-041118 | MATHDB-316627

Affine-automorphism conjecture for abelian permutation groups

摘录 / EXCERPT: Affine-automorphism conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041119 | MATHDB-316628

Free-sum conjecture for centrally symmetric faces of permutation polytopes

摘录 / EXCERPT: Free-sum conjecture. The free sum of centrally symmetric faces of permutation polytopes can be combinatorially realized as a face of a permutation polytope.

conjecture | open | MathDB 页面 | 原文线索

V2-041120 | MATHDB-316629

Wedge conjecture for faces of permutation polytopes

摘录 / EXCERPT: Wedge conjecture. The wedge over F' can be combinatorially realized as a face of a permutation polytope if, or only if, the complement of F' in F is a face of F.

conjecture | open | MathDB 页面 | 原文线索

V2-041121 | MATHDB-316644

Faithful Laurent polynomial conjecture for lifted quadratic recurrences

摘录 / EXCERPT: Let $T(n,k)$ satisfy the two-dimensional recurrence ... with initial terms $T(i,j)=x_{i,j}$ for all $i<0$. Consider the recurrence families conjectured to produce integer sequences i...

conjecture | open | MathDB 页面 | 原文线索

V2-041122 | MATHDB-316645

Uniqueness conjecture for extremal lattice-constrained bodies

摘录 / EXCERPT: Let L be a lattice in $\mathbb R^d$, let $V\in (0,2^d\det L]$, and let $K_L(V)$ be the intersection of the d-dimensional ball centered at the origin with radius $r_L(V)$ and th...

conjecture | open | MathDB 页面 | 原文线索

V2-041123 | MATHDB-316646

The extremal 3/4 conjecture for fix-free code extension constants

摘录 / EXCERPT: Let P_γ denote the property that every relevant partial fix-free code whose Kraftsum is at most γ can be extended in the manner defined in the survey; let γ^*...

conjecture | open | MathDB 页面 | 原文线索

V2-041124 | MATHDB-316710

Real-rootedness and positivity conjecture for checkerboard triangulation polynomials

摘录 / EXCERPT: Real-rootedness and positivity conjecture. (i) The polynomials $Q_0(u,1),Q_1(u,1),\dots\in\mathbb N[u]$ have only real roots, all of which are non-positive, and the non-zero roots...

conjecture | open | MathDB 页面 | 原文线索

V2-041125 | MATHDB-316719

Module-support conjecture for maximum 2-intersecting permutation families

摘录 / EXCERPT: Let $P_t(n)$ be the graph whose vertices are the permutations of an n-set, with two vertices adjacent when the corresponding permutations agree on no more than t points. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-041126 | MATHDB-316726

The approximate polynomial growth conjecture for locally compact groups

摘录 / EXCERPT: Let G be a locally compact group, let $A\subset G$ be a compact neighborhood, and let μ be Haar measure on G. For $n\geqslant 1$, write A^n for the n-fold product set o...

conjecture | open | MathDB 页面 | 原文线索

V2-041127 | MATHDB-316746

Caro's weighted Erdős-Ginzburg-Ziv conjecture

摘录 / EXCERPT: Let $G=\mathbb{Z}/n\mathbb{Z}$ be a finite cyclic group, and let $\mathcal{F}(G)$ denote the free abelian monoid of sequences over G. For sequences $S,W\in\mathcal{F}(G)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-041128 | MATHDB-316760

Higher-codimension Nathanson height asymptotics conjecture

摘录 / EXCERPT: Higher-codimension Nathanson height conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041129 | MATHDB-316761

Perimeter-preserving bijection for two-staircase unimodal polygons

摘录 / EXCERPT: Let a unimodal polygon have two staircase regions with identical sides, possibly differing widths, and let the two boxes delimiting the staircase region be identified as in the pro...

conjecture | open | MathDB 页面 | 原文线索

V2-041130 | MATHDB-316764

Multidimensional polynomial Szemerédi property and joint intersectivity

摘录 / EXCERPT: Let $p_1,\ldots,p_r:\mathbb Z^m\to\mathbb Z^k$ be polynomial mappings. For a set $E\subseteq\mathbb Z^k$ of positive upper Banach density, define ... The family has the SPSZ property...

conjecture | open | MathDB 页面 | 原文线索

V2-041131 | MATHDB-316801

Gowers-Wolf conjecture on the true complexity of linear systems

摘录 / EXCERPT: Gowers-Wolf conjecture. The true complexity of $\mathcal{L}$ is equal to the smallest k such that the functions $L_1^{k+1},\dots,L_m^{k+1}$ are linearly independent. This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-041132 | MATHDB-316808

Nested q,t-Catalan symmetry and specialization conjecture

摘录 / EXCERPT: Nested q,t-Catalan conjecture. The polynomial $N_n(q,t)$ is symmetric in q and t, and

conjecture | open | MathDB 页面 | 原文线索

V2-041133 | MATHDB-316821

Property B for zero-sum free subsets of $\mathbb{Z}_n^2$

摘录 / EXCERPT: Let n be a positive integer. A subset $A\subseteq\mathbb{Z}_n^2$ is zero-sum free if no nonempty subset of A has sum equal to 0 in $\mathbb{Z}_n^2$. We say that n satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-041134 | MATHDB-316857

Panyushev's orbit conjecture for antichains above the simple roots

摘录 / EXCERPT: Let Δ be a reduced irreducible root system with positive roots Δ^+, simple roots Π, Coxeter number h, and reverse operator $\mathfrak X_0$ on the antichains of...

conjecture | open | MathDB 页面 | 原文线索

V2-041135 | MATHDB-316875

Weighted generalization of Gao's $n+D-1$ theorem

摘录 / EXCERPT: Weighted zero-sum conjecture. There are a nonempty subset $I\subset [1,k]$ and an injection $f:I\mapsto [1,n]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041136 | MATHDB-316908

Sufficiency of the NR-facet condition for parsimonious relaxations

摘录 / EXCERPT: Let P_n be the Graphical Traveling Salesman Polyhedron, let $\mathcal G_B$ be the graph associated with a relaxation $\mathcal R_B$, and call a facet of P_n an NR-facet when it h...

conjecture | open | MathDB 页面 | 原文线索

V2-041137 | MATHDB-316913

The g-conjecture for simplicial spheres

摘录 / EXCERPT: Let L be a simplicial sphere, with g-vector defined from its h-vector by $g_0=h_0$ and $g_i=h_i-h_{i-1}$ for $i\geq 1$. An M-sequence is the Hilbert function of a standard...

conjecture | open | MathDB 页面 | 原文线索

V2-041138 | MATHDB-316945

The CAPP conjecture for weighted matroid measures

摘录 / EXCERPT: Let a weighted matroid measure be obtained by assigning positive external-field weights to the independent sets of a matroid. For a measure on $2k$ variables, define the antipodal...

conjecture | open | MathDB 页面 | 原文线索

V2-041139 | MATHDB-316951

Ito-Terwilliger parameter-array existence and uniqueness conjecture for sharp tridiagonal systems

摘录 / EXCERPT: Let d be a nonnegative integer, let $\mathbb{K}$ be a field, and let ... be a sequence of scalars in $\mathbb{K}$. A sharp tridiagonal system is a tridiagonal system whose associat...

conjecture | open | MathDB 页面 | 原文线索

V2-041140 | MATHDB-316962

The cardioid limit-shape conjecture for reflective rotor-router aggregation

摘录 / EXCERPT: Let A_n be the region formed by rotor-router aggregation started with n particles at the origin in $\mathbb{Z}^2$, with rotors on the positive x-axis reflecting particles bac...

conjecture | open | MathDB 页面 | 原文线索

V2-041141 | MATHDB-316964

Representative-with-first-coin-two conjecture for non-empty +/- classes

摘录 / EXCERPT: Let a $+/-$-class be a class of orderly currencies determined by a specified $+/-$ pattern, and suppose that this class is non-empty. Representative-with-first-coin-two conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-041142 | MATHDB-316974

The half-integrality conjecture for peg-solitaire linear tests

摘录 / EXCERPT: Let $\mathfrak{S}$ be the set of points of a peg-solitaire board. Let $V(\mathfrak{S},\mathbb{Z})$ be the integer span of legal-move vectors and let $V^+(\mathfrak{S},\mathbb{Q})$...

conjecture | open | MathDB 页面 | 原文线索

V2-041143 | MATHDB-316975

The doubled-basis conjecture for peg-solitaire move spaces

摘录 / EXCERPT: Let $\mathfrak{S}$ be a peg-solitaire board, let $\mathscr{D}(\mathfrak{S})$ be its set of legal-move vectors, and let $\mathcal{B}\subset\mathscr{D}(\mathfrak{S})$ be a basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-041144 | MATHDB-316980

Higher-dimensional Balinski-type connectivity conjecture for cell complexes

摘录 / EXCERPT: Higher-dimensional Balinski-type connectivity conjecture. The graph ${\mathcal G}_k({\mathcal C})$ is

conjecture | open | MathDB 页面 | 原文线索

V2-041145 | MATHDB-316987

Plancherel-Hecke longest-subsequence fluctuation conjecture

摘录 / EXCERPT: Fluctuation conjecture. If $0<\alpha<\frac12$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041146 | MATHDB-316995

Unimodular regular triangulation conjecture for uniform clutters

摘录 / EXCERPT: Let $\mathcal C$ be a uniform clutter with incidence vectors $v_1,\ldots,v_q$. Its max-flow min-cut property means that the associated max-flow and min-cut optimization problems ha...

conjecture | open | MathDB 页面 | 原文线索

V2-041147 | MATHDB-317012

Conjecture on sharp polynomial bounds for triangles in pseudo-line arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be an affine arrangement or a projective arrangement of n pseudo-lines. The quantities $a_3(\mathcal{A})$ and $p_3(\mathcal{A})$ denote the numbers of triangula...

conjecture | open | MathDB 页面 | 原文线索

V2-041148 | MATHDB-317029

The existence conjecture for k-symmetric permutations

摘录 / EXCERPT: Let $4\mathfrak{S}_n4$ be the symmetric group on n symbols. For $4\tau\in\mathfrak{S}_k4$ and $4\sigma\in\mathfrak{S}_n4$, let $4\mathbf{X}^{\tau}(\sigma)4$ count the occurrences...

conjecture | open | MathDB 页面 | 原文线索

V2-041149 | MATHDB-317032

Non-realizable extension of the nonnegativity corollary

摘录 / EXCERPT: Let ${\bf X}=(X,\operatorname{rk})$ be a polymatroid, and let ${\mathcal H}[{\bf X}]$ and ${\mathcal P}[{\bf X}]$ denote the invariants appearing in Corollary ... should remain true...

conjecture | open | MathDB 页面 | 原文线索

V2-041150 | MATHDB-317033

Specialization of the polymatroid invariant to Speyer's invariant

摘录 / EXCERPT: Let ${\mathcal G}$ be the invariant of polymatroids defined in the paper, and let $g_{\bf X}(t)$ be Speyer's invariant of a matroid ${\bf X}$. Specialization conjecture. The invari...

conjecture | open | MathDB 页面 | 原文线索

V2-041151 | MATHDB-317034

Universality of the polymatroid quasi-symmetric invariant

摘录 / EXCERPT: Let $\operatorname{PolyMat}$ be the vector space of polymatroids, let QSym be the quasi-symmetric-function space, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-041152 | MATHDB-317036

Eventual quasi-polynomiality for profiles with bounded signature or finite kernel

摘录 / EXCERPT: Let R be a relational structure with profile φ_R. Assume that R has bounded signature or finite kernel, and that $\varphi_R(n)$ is bounded by some polynomial in n. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-041153 | MATHDB-317039

Matroid determination of the Khovanov differential

摘录 / EXCERPT: Let K and K' be knot diagrams such that their colored graphic matroids satisfy $M(K)\cong M(K')$. Let $\mathcal{C}(K)$ and $\mathcal{C}(K')$ be the Khovanov complexes generated...

conjecture | open | MathDB 页面 | 原文线索

V2-041154 | MATHDB-317069

Asymptotic probability conjecture for substitution matrices

摘录 / EXCERPT: Asymptotic probability conjecture. The effect of the range selection should vanish as n tends to infinity; more precisely,

conjecture | open | MathDB 页面 | 原文线索

V2-041155 | MATHDB-317103

Hardy-sequence Szemerédi conjecture for polynomial-growth sequences

摘录 / EXCERPT: Let $\mathcal{H}$ denote the Hardy field under consideration, let $[x]$ be the integer part of x, and let $\bar d(\Lambda)$ denote the upper density of $\Lambda\subset\mathbb{Z}$...

conjecture | open | MathDB 页面 | 原文线索

V2-041156 | MATHDB-317118

Generalized lower bound conjecture for triangulated spheres

摘录 / EXCERPT: Let d and k be integers with $d\geq 2k+1$, and let X be a triangulated d-sphere with face-vector $(f_0,\ldots,f_d)$. Generalized lower bound conjecture. For every j, the f...

conjecture | open | MathDB 页面 | 原文线索

V2-041157 | MATHDB-317125

Hou's linear Kneser conjecture for field extensions

摘录 / EXCERPT: Let $K\subset L$ be a commutative field extension, and let $A,B\subset L$ be nonzero finite-dimensional K-subspaces. Let $\langle AB\rangle$ denote the K-linear span of the pro...

conjecture | open | MathDB 页面 | 原文线索

V2-041158 | MATHDB-317160

Donaldson's existence conjecture for extremal metrics on toric manifolds

摘录 / EXCERPT: Let $P\subset \mathbb{R}^{n}$ be a Delzant polytope, let A be a smooth function on $\overline{P}$, and let L_A be the associated linear functional on convex functions. An admis...

conjecture | open | MathDB 页面 | 原文线索

V2-041159 | MATHDB-317168

The Cohen-Macaulay conjecture for unmixed admissible clutters

摘录 / EXCERPT: The Cohen-Macaulay conjecture. If ${\mathcal C}$ is an admissible clutter and ${\mathcal C}$ is unmixed, then ${\mathcal C}$ is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-041160 | MATHDB-317183

Polynomial bound conjecture for unavoidable tournaments

摘录 / EXCERPT: Let $T(k,\epsilon)$ be the minimum positive integer such that every tournament T with $n\geq T(k,\epsilon)$ vertices that is ϵ-far from being transitive contains the tou...

conjecture | open | MathDB 页面 | 原文线索

V2-041161 | MATHDB-317236

The Genocchi descent pattern conjecture for packed words

摘录 / EXCERPT: Let I and J be compositions of n, and let $W”(I,J)$ be the set of packed words w such that ... Here $\operatorname{totg}(w)$ denotes the number of occurrences of the patterns...

conjecture | open | MathDB 页面 | 原文线索

V2-041162 | MATHDB-317237

The permutation form of the Genocchi descent pattern conjecture

摘录 / EXCERPT: Let I and J be compositions of n, and let $P”(I,J)$ be the set of permutations σ such that ... Here $\operatorname{tot}(\sigma)$ denotes the number of occurrences of the...

conjecture | open | MathDB 页面 | 原文线索

V2-041163 | MATHDB-317254

Conjecture on the cardinality spectrum for q congruent to 3 modulo 4

摘录 / EXCERPT: Cardinality-spectrum conjecture. There exist $l_q,r_q\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041164 | MATHDB-317255

Conjecture on unbounded minimum cardinalities of maximal integral point sets

摘录 / EXCERPT: Unbounded-minimum conjecture. For each $w\in\mathbb{N}$ there exists $q_w\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041165 | MATHDB-317260

Erdős-Noll conjecture on the number of non-isomorphic clusters

摘录 / EXCERPT: An n_m-cluster is an integral point set in general position consisting of n points in the integer grid $\mathbb{Z}^m$. Erdős-Noll conjecture. For any $m>1$ and $n>1$, there exi...

conjecture | open | MathDB 页面 | 原文线索

V2-041166 | MATHDB-317262

The circle conjecture for plane integral point sets in semi-general position

摘录 / EXCERPT: Circle conjecture. The points of a plane integral point set in semi-general position are situated on a circle.

conjecture | open | MathDB 页面 | 原文线索

V2-041167 | MATHDB-317294

The $\mathbb{P}_4$ characterization conjecture for matroid representability

摘录 / EXCERPT: $\mathbb{P}_4$ characterization conjecture. A matroid M is representable over all finite fields with at least 4 elements if and only if M is representable over $\mathbb{P}_4$...

conjecture | open | MathDB 页面 | 原文线索

V2-041168 | MATHDB-317295

Idempotence conjecture for the lift of a partial field

摘录 / EXCERPT: Let $\mathbb{P}$ be a partial field, and let $\mathbb{L}\mathbb{P}$ denote its lift. Lift idempotence conjecture. ... The conjecture is related to whether the canonical homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-041169 | MATHDB-317296

Uniform partial-field characterization conjecture for matroids over extensions of $\operatorname{GF}(2)$

摘录 / EXCERPT: Uniform partial-field characterization conjecture. A matroid is representable over $\operatorname{GF}(2^k)$ for all $k>1$ if and only if it is representable over...

conjecture | open | MathDB 页面 | 原文线索

V2-041170 | MATHDB-317332

Cotterill's enumeration conjecture for labeled trees with a given indegree sequence

摘录 / EXCERPT: Let $\lambda=1^{e_1}2^{e_2}\ldots$ be a partition of $n-1$, and let $e_0=n-\ell(\lambda)$. For the set $\mathcal{T}_{n,\lambda}$ of labeled trees with local indegree sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-041171 | MATHDB-317342

Piecewise linearity of the Freiman dimension function and constancy of the covering function

摘录 / EXCERPT: Let $F(K)$ and $G(K)$ be the functions introduced earlier in the paper for measuring the linear structure of finite binary sets with doubling parameter K. The piecewise-linearity...

conjecture | open | MathDB 页面 | 原文线索

V2-041172 | MATHDB-317343

The extremal rotation conjecture for braid representations

摘录 / EXCERPT: Extremal rotation conjecture. Any rotation that transforms an extremum to another extremum of the same braid must take the form

conjecture | open | MathDB 页面 | 原文线索

V2-041173 | MATHDB-317349

Kalai's lower-bound conjecture for the second g-number of triangulated manifolds

摘录 / EXCERPT: Let K) be a connected triangulated d-manifold with $d\geq 3$, and let $g_2(K)$ denote its second g-number and $\beta_1(K;\mathbb{Q})$ its first Betti number over $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-041174 | MATHDB-317350

Tight-neighborly triangulations are tight

摘录 / EXCERPT: Let K be a triangulation of a connected sum of sphere bundles $S^{(d-1)}\times S^1$ and $S^{(d-1)}\text{$timeshspace{-1.62ex}hspace{-.4ex}hspace{.7ex}$}S^1$ that is tight-neighbo...

conjecture | open | MathDB 页面 | 原文线索

V2-041175 | MATHDB-317370

The parameter-count conjecture for even-dimensional combinatorial torus cube packings

摘录 / EXCERPT: Even-dimensional parameter-count conjecture. There exist non-extensible combinatorial torus cube packings with $n+2$ cubes and

conjecture | open | MathDB 页面 | 原文线索

V2-041176 | MATHDB-317428

Liu-Wang's Narayana transformation conjecture on log-convexity

摘录 / EXCERPT: Let $N(n,k)=\frac{1}{n}\binom{n}{k}\binom{n}{k+1}$ be the Narayana numbers. For a sequence $(a_k)_{k\geq 0}$ of positive real numbers, define its Narayana transform by ... A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-041177 | MATHDB-317443

The three-dimensional poset poly-antimatroid staircase conjecture

摘录 / EXCERPT: Three-dimensional staircase conjecture. Every three-dimensional poset poly-antimatroid is a step staircase.

conjecture | open | MathDB 页面 | 原文线索

V2-041178 | MATHDB-317444

The convex-dimension bound for poset poly-antimatroids

摘录 / EXCERPT: Convex-dimension bound conjecture. The convex dimension of an n-dimensional poset poly-antimatroid is at most n.

conjecture | open | MathDB 页面 | 原文线索

V2-041179 | MATHDB-317461

Wastlund's half-\bound conjecture for sorting permutations by block transpositions

摘录 / EXCERPT: Wastlund's conjecture. If $n\neq 13$ and $n\neq 15$, then every n-permutation can be sorted by at most

conjecture | open | MathDB 页面 | 原文线索

V2-041180 | MATHDB-317465

Crystallographic q-specialization conjecture for q,t-Fuss-Catalan numbers

摘录 / EXCERPT: Let W be a crystallographic reflection group, and let $\operatorname{Cat}^{(m)}(W,q)$ be the combinatorial q-Fuss-Catalan number obtained by summing $q^{\operatorname{coh}(R)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-041181 | MATHDB-317508

The fundamental-elements conjecture for universal partial fields

摘录 / EXCERPT: Let N be a matroid and let ${\mathbb{P}}_N$ denote its universal partial field. A ${\mathbb{P}}_N$-representation of N is a representation over ${\mathbb{P}}_N$, and two such r...

conjecture | open | MathDB 页面 | 原文线索

V2-041182 | MATHDB-317509

The 3-connected stabilizer conjecture for universal partial fields

摘录 / EXCERPT: Let N be a 3-connected matroid. A matroid is ${\mathbb{P}}_N$-representable when it has a representation over the universal partial field ${\mathbb{P}}_N$. The matroid N is a...

conjecture | open | MathDB 页面 | 原文线索

V2-041183 | MATHDB-317510

The level partial-field universality conjecture

摘录 / EXCERPT: Let ${\mathbb{P}}$ be a partial field, and let ${\cal M}$ be a class of matroids. Say that ${\mathbb{P}}$ is ${\cal M}$-universal if, for every partial field ${\mathbb{P}}'$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-041184 | MATHDB-317511

The fixed-element settlement conjecture for representable matroids

摘录 / EXCERPT: Let M be a representable matroid and let e be an element of M. The deletion $M{-} e$ settles M when representations of $M{-} e$ determine representations of M in the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-041185 | MATHDB-317512

The circuit-relation ideal conjecture for 3-connected matroid extensions

摘录 / EXCERPT: Let ${\mathbb{P}}$ be a partial field, let M be a 3-connected ${\mathbb{P}}$-representable matroid, and let $N=M{-} e$ be a 3-connected deletion that settles M. Let $I_{N,M}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-041186 | MATHDB-317514

Consecutive-block conjecture for high-multiplicity sumsets

摘录 / EXCERPT: Let $k,l\geqslant 1$ and $n\geqslant 3$ be integers, and write ... Suppose that $A\subseteq[0,l]$ is an integer set with $0,l\in A$, $\gcd(A)=1$, and $|A|=n$. Consecutive-block conje...

conjecture | open | MathDB 页面 | 原文线索

V2-041187 | MATHDB-317521

Hirsch's shade conjecture for t-intersecting families

摘录 / EXCERPT: Hirsch's shade conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041188 | MATHDB-317522

Asymptotic shade conjecture for cross-t-intersecting families

摘录 / EXCERPT: Asymptotic cross-shade conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-041189 | MATHDB-317558

Canonical embedding isomorphism conjecture for spherical latin bitrades

摘录 / EXCERPT: Let (W,B) be a spherical latin bitrade, and let $\mathcal{A}_W^*$ and $\mathcal{A}_B^*$ denote the canonical abelian groups into which the white and black trades embed, respectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-041190 | MATHDB-317562

The jumping-sequence characterization of the infinite admissible set

摘录 / EXCERPT: Jumping-sequence characterization. The set $\mathcal{A}_\infty$ is composed of the elements of the integer sequence $d(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-041191 | MATHDB-317571

The Pascal descent polynomial twin and anti-twin conjecture

摘录 / EXCERPT: Pascal descent polynomial conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041192 | MATHDB-317592

Green's removal-property conjecture for homogeneous linear systems

摘录 / EXCERPT: Green's conjecture. Every system of homogeneous linear equations $Mx=0$ has the removal property.

conjecture | open | MathDB 页面 | 原文线索

V2-041193 | MATHDB-317594

The logarithmic density bound for progression-free subsets of $\mathbb{Z}_4^n$

摘录 / EXCERPT: Logarithmic density conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-041194 | MATHDB-317603

The t-Hamming intersection conjecture for permutations

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let a k-permutation on $[n]$ be an ordered sequence of k distinct elements of $[n]$. For two such permutations P and Q, write $|P\cap_H Q|$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-041195 | MATHDB-317604

The t-intersecting multiset conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let a k-multiset on $[n]$ be a multiset of cardinality k whose elements lie in $[n]$. Two multisets are t-intersecting when their intersection h...

conjecture | open | MathDB 页面 | 原文线索

V2-041196 | MATHDB-317617

The extremal-family conjecture for shades of t-intersecting families

摘录 / EXCERPT: The shade extremal-family conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041197 | MATHDB-317618

The asymptotic Erdős-Ko-Rado conjecture for cross-t-intersecting families

摘录 / EXCERPT: The cross-t Erdős-Ko-Rado conjecture. For fixed k,l,t, one has

conjecture | open | MathDB 页面 | 原文线索

V2-041198 | MATHDB-317619

The Ahlswede-Khachatrian conjecture for cross-t-intersecting families

摘录 / EXCERPT: The cross-intersecting Ahlswede-Khachatrian conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041199 | MATHDB-317620

The extremal-shade conjecture for cross-t-intersecting families

摘录 / EXCERPT: Let $C(n,k,l,t)$ be the collection of cross-t-intersecting pairs $(\mathcal A,\mathcal B)$, and define ... For $0\le i+j\le n-t$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-041200 | MATHDB-317621

The asymptotic vanishing conjecture for cross-intersecting shades

摘录 / EXCERPT: The cross-shade asymptotic conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-041201 | MATHDB-317628

The revised SIAM Review conjecture on extremal distributions

摘录 / EXCERPT: Let $X_1,\ldots,X_n$ be iid random variables supported on $[0,1]$, with mean m satisfying $0<m<1$, and write $S_n=X_1+\cdots+X_n$. Define ... For $n\geq2$ and $0\leq t<mn$, call a...

conjecture | open | MathDB 页面 | 原文线索

V2-041202 | MATHDB-317648

The determinant-character conjecture for finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group, and let $A=\{a_1,\ldots,a_k\}$ and $B=\{b_1,\ldots,b_k\}$ be two k-subsets of G. Let K be any field containing an element of multiplicative...

conjecture | open | MathDB 页面 | 原文线索

V2-041203 | MATHDB-317704

Simion's unimodality conjecture for lattice-path numbers

摘录 / EXCERPT: Simion's conjecture. For each integer ℓ and each partition λ, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-041204 | MATHDB-317740

The finite spectral-set tiling conjecture in the integers

摘录 / EXCERPT: Finite spectral-set conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-041205 | MATHDB-317741

The Coven-Meyerowitz tiling criterion for finite subsets of the integers

摘录 / EXCERPT: Coven-Meyerowitz conjecture. The set A tiles $\mathbb{Z}$ by translations if and only if:

conjecture | open | MathDB 页面 | 原文线索

V2-041206 | MATHDB-317775

Central-frequency conjecture for MSTD sets

摘录 / EXCERPT: For a finite set $A\subset[1,n]$, define … Here an MSTD set is a finite set satisfying $|A+A|>|A-A|$. Central-frequency conjecture. For every fixed constant $0<\alpha<1/2$, as…

conjecture | open | MathDB 页面 | 原文线索

V2-041207 | MATHDB-317823

Conjecture on universal gaps in cyclic-group exponent sets

摘录 / EXCERPT: Universal-gap conjecture. For every positive integer k, there exist integers a and b such that

conjecture | open | MathDB 页面 | 原文线索

V2-041208 | MATHDB-317837

Strong-generator sparsity conjecture for numerical semigroups

摘录 / EXCERPT: For each genus g, let S_g be the number of strong generators and let W_g be the number of weak generators, counted over all numerical semigroups of genus g. Strong-generato…

conjecture | open | MathDB 页面 | 原文线索

V2-041209 | MATHDB-317851

Existence conjecture for k-homogeneous Latin bitrades

摘录 / EXCERPT: Let k-homogeneous Latin bitrades be Latin bitrades in which every symbol occurs k times, and let the volume denote the number of entries in the bitrade. Existence conjecture. F…

conjecture | open | MathDB 页面 | 原文线索

V2-041210 | MATHDB-317856

Conjecture that the monochromatic-box matrix is positive semidefinite

摘录 / EXCERPT: Positive-semidefiniteness conjecture. M_r is positive semidefinite for $r \geq 3$. The matrix is positive semidefinite in the cases considered computationally, enabling convex pr…

conjecture | open | MathDB 页面 | 原文线索

V2-041211 | MATHDB-317862

Kerov's positivity conjecture for character polynomials

摘录 / EXCERPT: For each permutation π, let K_{π} be the universal Kerov character polynomial defined by … For a cycle of length k, write K_k for the corresponding polynomial. Kerov's…

conjecture | open | MathDB 页面 | 原文线索

V2-041212 | MATHDB-317975

The Sprouts winning-strategy congruence conjecture

摘录 / EXCERPT: In the Sprouts game, a two-player game beginning with p spots in which the player making the last move wins, a player has a winning strategy for exactly one of the two players. S…

conjecture | open | MathDB 页面 | 原文线索

V2-041213 | MATHDB-317976

The nimber pattern conjecture for Sprouts on orientable surfaces

摘录 / EXCERPT: Consider a Sprouts starting position with n spots on an orientable compact surface, and let its nimber be the least excluded nonnegative integer among the nimbers of the position…

conjecture | open | MathDB 页面 | 原文线索

V2-041214 | MATHDB-317978

Terai's conjecture on h-vectors of Buchsbaum complexes

摘录 / EXCERPT: A vector $h=(1,h_1,\dots,h_d) \in \mathbb{Z}^{d+1}$ is the h-vector of a $(d-1)$-dimensional Buchsbaum complex Δ such that $\beta_k(\Delta)=0$ for $k \leq d-3$. A vector…

conjecture | open | MathDB 页面 | 原文线索

V2-041215 | MATHDB-317985

Coefficientwise q-unimodality conjecture for refined Eulerian polynomials

摘录 / EXCERPT: Let λ be a partition of n with exactly k parts of size 1, and write $A_{\lambda}^{{\rm maj},{\rm des},{\rm exc}}(q,p,t)$ as a polynomial in p and t. Coefficientwi…

conjecture | open | MathDB 页面 | 原文线索

V2-041216 | MATHDB-317999

The asymptotic containment conjecture for tuples of standard Young tableaux

摘录 / EXCERPT: Let $A_1,\ldots,A_k$ be standard Young tableaux with $\operatorname{sh}(A_i)=\alpha_i\vdash a_i$ for $i\in[k]$. Let $\mathcal{T}_n(A_1,\ldots,A_k)$ be the set of k-tuples…

conjecture | open | MathDB 页面 | 原文线索

V2-041217 | MATHDB-318003

Bialostocki's zero-sum permutation conjecture

摘录 / EXCERPT: Bialostocki's conjecture. There exists a permutation $\sigma\in S_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041218 | MATHDB-318048

Infinite log-concavity conjecture for the coefficients $d_{l,m}$

摘录 / EXCERPT: For each fixed $m\in\mathbb{N}$, let $\{d_{l,m}:0\leq l\leq m\}$ be the coefficient sequence discussed above. For a sequence $\{a_j\}$ of positive numbers, define … A nonnegative s…

conjecture | open | MathDB 页面 | 原文线索

V2-041219 | MATHDB-318093

The converse generation conjecture for order-invariant measures

摘录 / EXCERPT: Let $P=(Z,<)$ be a causal set, let Ω be its space of natural extensions, and let μ be an order-invariant measure on the associated measurable space. An element…

conjecture | open | MathDB 页面 | 原文线索

V2-041220 | MATHDB-318107

The finitely forcible solution conjecture for graphon constraints

摘录 / EXCERPT: Let $F_1,\dots,F_k$ be finite graphs, let $a_1,\dots,a_k$ be prescribed values, and let $t(F_i,W)$ denote the homomorphism density of F_i in a graphon W. Finitely forcible solu...

conjecture | open | MathDB 页面 | 原文线索

V2-041221 | MATHDB-318132

Automatic regularity of Butson matrices

摘录 / EXCERPT: Let $H_n(l)$ be the set of $n\times n$ Hadamard matrices whose entries are l-th roots of unity. A Butson matrix is regular when the scalar product of each pair of rows decomposes...

conjecture | open | MathDB 页面 | 原文线索

V2-041222 | MATHDB-318171

Mitra-Pyatov conjecture for the number of HSFPL diagrams

摘录 / EXCERPT: Let an FPL diagram of size L have size $(L-1)\times L/2$ when L is even and size $L\times (L-1)/2$ when L is odd. Let d be its depth, and let $S(L,d)$ denote the total numb...

conjecture | open | MathDB 页面 | 原文线索

V2-041223 | MATHDB-318172

Pyatov's conjecture for the HSFPL nest generating function

摘录 / EXCERPT: For half-turn symmetric FPL diagrams with $L=2n$ external terminals, let $P(L,m)$ count diagrams with $m+1$ nests and define ... Let $Z_{\rm HTSFPL}(2n)$ be the total number of half-...

conjecture | open | MathDB 页面 | 原文线索

V2-041224 | MATHDB-318194

The well-quasi-ordering conjecture for finite-field-representable matroids

摘录 / EXCERPT: Well-quasi-ordering conjecture. Any infinite set of $\mathrm{GF}(q)$-representable matroids contains two matroids, one of which is isomorphic to a minor of the other.

conjecture | open | MathDB 页面 | 原文线索

V2-041225 | MATHDB-318195

Efficient oracle recognition of binary internally 4-connected matroids

摘录 / EXCERPT: Binary-recognition conjecture. There is an efficient oracle algorithm for deciding whether an internally 4-connected matroid is binary.

conjecture | open | MathDB 页面 | 原文线索

V2-041226 | MATHDB-318196

Efficient oracle recognition of proper minor-closed classes of binary matroids

摘录 / EXCERPT: Subclass-recognition conjecture. There is an efficient oracle algorithm for deciding whether an internally 4-connected matroid belongs to $\mathcal{M}$.

conjecture | open | MathDB 页面 | 原文线索

V2-041227 | MATHDB-318221

Inverse conjecture for polynomially concentrated bilinear forms

摘录 / EXCERPT: Let A be an $n\times n$ matrix of nonzero entries, and let x,y be chosen independently and uniformly from $\{-1,1\}^n$. For a matrix decomposition $A=A_1+A_2+A_3$, say that...

conjecture | open | MathDB 页面 | 原文线索

V2-041228 | MATHDB-318228

Bonnafé-Geck-Iancu-Lam conjecture on left cells in type B_n

摘录 / EXCERPT: Bonnafé-Geck-Iancu-Lam conjecture. If $s\notin\mathbb{N}$, the Kazhdan-Lusztig left cells coincide with the irreducible combinatorial left cells of rank $\lfloor s\rfloor$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-041229 | MATHDB-318284

Skew blow-up conjecture for triangle densities

摘录 / EXCERPT: Skew blow-up conjecture. There is a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041230 | MATHDB-318285

Logarithmic balanced-clique conjecture for graphs with fixed triangle density

摘录 / EXCERPT: Logarithmic balanced-clique conjecture. There is an absolute constant $c>0$ such that G contains a copy of $K_{t,t,t}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-041231 | MATHDB-318307

The positive-proportion conjecture for rigid permutations in layered classes

摘录 / EXCERPT: Let $\mathcal{I}_k$ denote the class of permutations formed by the direct sum of at most k decreasing sequences, and call a permutation in $\mathcal{I}_k$ k-rigid when it satis...

conjecture | open | MathDB 页面 | 原文线索

V2-041232 | MATHDB-318314

Joint equidistribution of odd descent-tops, odd pairs, and increasing subsequences

摘录 / EXCERPT: Let S_{10} be the number of odd descent-tops, let S_{12} be the number of (odd,odd) pairs, and let S_{17} be the maximal subsequence of the form $1 2\ldots i$...

conjecture | open | MathDB 页面 | 原文线索

V2-041233 | MATHDB-318315

Refined joint equidistribution of permutation statistics

摘录 / EXCERPT: Let S_{10} be the number of odd descent-tops, S_{12} the number of (odd,odd) pairs, and S_{17} the maximal subsequence of the form $1 2\ldots i$ in a permutat...

conjecture | open | MathDB 页面 | 原文线索

V2-041234 | MATHDB-318342

Girard's conjecture on zero-sum subsequence lengths modulo the smallest invariant factor

摘录 / EXCERPT: Let $G \simeq C_{n_1} \oplus \dots \oplus C_{n_r}$, with $1<n_1\mid\dots\mid n_r\in\mathbb{N}$, be a finite Abelian group, and let S be a sequence over G with...

conjecture | open | MathDB 页面 | 原文线索

V2-041235 | MATHDB-318343

Conjecture on bounded-length zero-sum subsequences

摘录 / EXCERPT: Let G be a finite Abelian group of exponent m, and let $\eta_{\ell m}(G)$ be the smallest positive integer t such that every sequence over G of length at least t contains...

conjecture | open | MathDB 页面 | 原文线索

V2-041236 | MATHDB-318348

The extremal entire-function conjecture for the Bernstein problem

摘录 / EXCERPT: Extremal entire-function conjecture. The function F has the displayed form, where ϕ maps the upper half-plane onto the region in the upper half-plane above γ, satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-041237 | MATHDB-318353

Asymptotic constant for the discrepancy of the lex-least de Bruijn sequence

摘录 / EXCERPT: Discrepancy asymptotic constant conjecture. There is some constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-041238 | MATHDB-318394

Sufficiency conjecture for doubly refined alternating sign matrix numbers

摘录 / EXCERPT: Sufficiency conjecture. The linear equations, near-symmetry equations and special values determine the numbers $\hat{A}_{n,i,j}$ uniquely.

conjecture | open | MathDB 页面 | 原文线索

V2-041239 | MATHDB-318395

Linear-equations conjecture for d-refined alternating sign matrix enumeration

摘录 / EXCERPT: Linear-equations conjecture. The extension can be chosen so that, for $1\le i_1,i_2,\ldots,i_d\le n$,

conjecture | open | MathDB 页面 | 原文线索

V2-041240 | MATHDB-318400

Generalized lattice-path trigonometric positivity conjecture

摘录 / EXCERPT: Generalized positivity conjecture. The first sum is a polynomial in $1+\cos(x)$ with nonnegative integral coefficients, and the second satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041241 | MATHDB-318479

Eventual polynomiality conjecture for generalised stretched Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $\lambda,\mu,\nu,\lambda',\mu',\nu'$ be partitions, and define ... Assume that $f(n)=c(n\lambda;n\mu,n\nu)$ is a polynomial of degree d, and that $c(\lambda';\mu',\nu')\neq 0$....

conjecture | open | MathDB 页面 | 原文线索

V2-041242 | MATHDB-318494

Near-quadratic growth conjecture for similar copies in general position

摘录 / EXCERPT: Near-quadratic growth conjecture. The bound above should hold for every such m and P. A proof cannot follow from the paper's main theorem and would require a different construc...

conjecture | open | MathDB 页面 | 原文线索

V2-041243 | MATHDB-318495

Near-linear upper-bound conjecture for parallelogram-free patterns

摘录 / EXCERPT: Near-linear upper-bound conjecture. The function $S_{P}^{\nparallel}(n)$ should satisfy this bound for every parallelogram-free pattern P. The paper suggests that the constructio...

conjecture | open | MathDB 页面 | 原文线索

V2-041244 | MATHDB-318498

Extremal metric decomposition conjecture for projective bundles over curves

摘录 / EXCERPT: Extremal metric decomposition conjecture. The projective bundle $(M,J)=P(E)$ admits an extremal Kähler metric in some Kähler class if and only if E decomposes as a direct sum of...

conjecture | open | MathDB 页面 | 原文线索

V2-041245 | MATHDB-318499

Donaldson-type convexity criterion for extremal metrics on compatible classes

摘录 / EXCERPT: Convexity criterion conjecture. The following conditions should be equivalent: Ω admits an extremal Kähler metric; Ω admits a compatible extremal Kähler metric; and

conjecture | open | MathDB 页面 | 原文线索

V2-041246 | MATHDB-318537

Whittle's decomposition conjecture for near-regular matroids

摘录 / EXCERPT: Whittle's decomposition conjecture. Every near-regular matroid can be obtained from signed-graphic matroids, their duals, and members of some finite set by applying 1-, 2-, and...

conjecture | open | MathDB 页面 | 原文线索

V2-041247 | MATHDB-318538

Updated decomposition conjecture for near-regular matroids

摘录 / EXCERPT: Updated decomposition conjecture. Every near-regular matroid can be obtained from signed-graphic matroids, the duals of signed-graphic matroids, or members of a finite set...

conjecture | open | MathDB 页面 | 原文线索

V2-041248 | MATHDB-318556

The arbitrary-target distribution conjecture for pebbling

摘录 / EXCERPT: Arbitrary-target distribution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041249 | MATHDB-318575

The least-word conjecture for the Rudin-Shapiro orbit closure

摘录 / EXCERPT: Let ${\bf w}$ be the Rudin-Shapiro word and let ${\mathpzc O}_w$ be its orbit closure, the set of right infinite words all of whose finite prefixes are factors of ${\bf w}$. Order...

conjecture | open | MathDB 页面 | 原文线索

V2-041250 | MATHDB-318599

The higher-dimensional face-number conjecture for simplicial complexes of large 1-girth

摘录 / EXCERPT: The higher-dimensional face-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041251 | MATHDB-318600

The higher-girth face-number conjecture for simplicial complexes

摘录 / EXCERPT: The higher-girth face-number conjecture. There is a constant $C_{p,r,i}$ depending only on p,r,i such that

conjecture | open | MathDB 页面 | 原文线索

V2-041252 | MATHDB-318635

Balanced-run maximization conjecture for non-crossing pairings

摘录 / EXCERPT: Balanced-run maximization conjecture. For every such bit string $\mathbf{S}$,

conjecture | open | MathDB 页面 | 原文线索

V2-041253 | MATHDB-318636

Tree-polynomial ordering conjecture for non-crossing pairings

摘录 / EXCERPT: Tree-polynomial ordering conjecture. For every permutation $\sigma\in S_r$, there exists a re-ordering $\tau:\operatorname{Tr}\to\operatorname{Tr}$ such that $p_{T,\sigma}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-041254 | MATHDB-318643

The cardinality bound for minimal rank-determining sets of metric graphs

摘录 / EXCERPT: Let Γ be a metric graph of genus g. A minimal rank-determining set is a rank-determining set of Γ that has no proper subset that is rank-determining. Cardinality bo...

conjecture | open | MathDB 页面 | 原文线索

V2-041255 | MATHDB-318647

The weak Polynomial inverse conjecture for the U^3 norm over $\mathbb{Z}/N\mathbb{Z}$

摘录 / EXCERPT: Let $f:\mathbb{Z}/N\mathbb{Z}\to\mathbb{C}$ be a K-approximate quadratic. An elementary 2-step nilsequence is a sequence $F(g^nx_0)$, where $F:G/\Gamma\to\mathbb{C}$ is Lipschi...

conjecture | open | MathDB 页面 | 原文线索

V2-041256 | MATHDB-318654

Extended doubly-refined enumeration formula for alternating sign matrices

摘录 / EXCERPT: Extended enumeration conjecture. The explicit formula for the doubly-refined enumeration numbers also holds for the extended numbers:

conjecture | open | MathDB 页面 | 原文线索

V2-041257 | MATHDB-318690

Dukes-Herke conjecture on large orders of bases for finite cyclic groups

摘录 / EXCERPT: Let $\mathbb{Z}_n$ be the finite cyclic group of integers modulo n. For a subset $A\subseteq\mathbb{Z}_n$, write $\rho_n(A)$ for the least positive integer h such that every el...

conjecture | open | MathDB 页面 | 原文线索

V2-041258 | MATHDB-318722

Front crossing-nesting symmetry conjecture for set partitions

摘录 / EXCERPT: Let Π_n be the set of set partitions of $[n]$, and let $\operatorname{FCR}(\pi)$ and $\operatorname{FNE}(\pi)$ denote respectively the front crossing and front nesting statisti...

conjecture | open | MathDB 页面 | 原文线索

V2-041259 | MATHDB-318754

The planar graph braid group commutator presentation conjecture

摘录 / EXCERPT: Planar graph braid group commutator presentation conjecture. There exists a presentation for $B_2\Gamma$ such that the relators are all commutators corresponding to pairs of disjoi...

conjecture | open | MathDB 页面 | 原文线索

V2-041260 | MATHDB-318759

Extremal-size conjecture for percolating sets in the hypercube

摘录 / EXCERPT: Extremal percolating-set conjecture. For $r \in \mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-041261 | MATHDB-318760

Yang's conjecture on near-normal sequences

摘录 / EXCERPT: Yang's conjecture. $NN(n)\ne\emptyset$ for all positive even integers n.

conjecture | open | MathDB 页面 | 原文线索

V2-041262 | MATHDB-318847

The local limit conjecture for the Mahonian distribution on words

摘录 / EXCERPT: Local limit conjecture. Uniformly for all choices of $a_1,\dots,a_m$ and all integers k,

conjecture | open | MathDB 页面 | 原文线索

V2-041263 | MATHDB-318853

Conjecture on admissible pictures and Littlewood-Richardson crystals

摘录 / EXCERPT: Let λ, μ, and ν be Young diagrams with $|\lambda|+|\mu|=|\nu|$. Let A be an admissible order on the skew Young diagram $\nu\setminus\lambda$, and let A' be an a...

conjecture | open | MathDB 页面 | 原文线索

V2-041264 | MATHDB-318855

Ito-Terwilliger classification conjecture for sharp tridiagonal systems

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, let d be a nonnegative integer, and let $(\{\theta_i\}_{i=0}^d;\{\theta_i^*\}_{i=0}^d;\{\zeta_i\}_{i=0}^d)$ be a sequence of scalars in $\mathbb{F}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-041265 | MATHDB-318862

The Green-Tao conjecture for affine spaces of irreducible polynomials

摘录 / EXCERPT: Green-Tao conjecture for function fields. For every finite field $\mathbf{F}$, the monic irreducible polynomials in $\mathbf{F}[t]$ contain affine spaces of arbitrarily high dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-041266 | MATHDB-318876

The rational-strategy conjecture for lattice games

摘录 / EXCERPT: Rational-strategy conjecture. Every lattice game possesses a rational strategy; equivalently, the generating function for its P-positions is a ratio of polynomials with integer coe...

conjecture | open | MathDB 页面 | 原文线索

V2-041267 | MATHDB-318877

The affine-stratification conjecture for lattice games

摘录 / EXCERPT: Affine-stratification conjecture. Every lattice game possesses an affine stratification.

conjecture | open | MathDB 页面 | 原文线索

V2-041268 | MATHDB-318894

Logarithmic odd Dyson constant-term conjecture

摘录 / EXCERPT: Let k and m be positive integers. For variables $x_1,\ldots,x_{2k+1}$, let $\operatorname{CT}_{x_1,\ldots,x_{2k+1}}$ denote the constant term in their Laurent expansion. Then t...

conjecture | open | MathDB 页面 | 原文线索

V2-041269 | MATHDB-318905

Applegate-Jacobson-Sleator conjecture for misère Sprouts

摘录 / EXCERPT: Let S_p^- denote the misère version of Sprouts starting with p spots. Applegate-Jacobson-Sleator conjecture. The first player has a winning strategy in S_p^- if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-041270 | MATHDB-318906

Period-six conjecture for misère Sprouts

摘录 / EXCERPT: Let S_p^- denote the misère version of Sprouts starting with p spots. Period-six conjecture for misère Sprouts. The first player has a winning strategy in S_p^- if and only i...

conjecture | open | MathDB 页面 | 原文线索

V2-041271 | MATHDB-318969

Saturation number of $T_3^{\le 2}$

摘录 / EXCERPT: Let $T_3^{\le 2}$ denote the specified binary matrix family, and let $\operatorname{sat}(n,F)$ be the minimum number of columns in an n-row F-saturated matrix. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-041272 | MATHDB-318971

Characteristic-independence conjecture for formal characters of Khovanov-Lauda-Rouquier modules

摘录 / EXCERPT: Let $(I,\cdot)$ be a Cartan datum of finite type, let Q_+ be its positive root lattice, let $\alpha\in Q_+$, and let R_α be the corresponding Khovanov-Lauda-Rouquier algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-041273 | MATHDB-318981

Weighted Gao conjecture for the cyclic group

摘录 / EXCERPT: Let n be a positive integer and let A be a non-empty subset of the integers. For the cyclic group $G=\mathbb{Z}/n\mathbb{Z}$, let $D_A(n)$ be the least natural number k such...

conjecture | open | MathDB 页面 | 原文线索

V2-041274 | MATHDB-318983

Lemke-Kleitman conjecture on indices of sequences over cyclic groups

摘录 / EXCERPT: Let Z_n be the cyclic group of n elements, let d and n be positive integers with $d\mid n$, and let S be a sequence of n elements in Z_n. For a subsequence T of S...

conjecture | open | MathDB 页面 | 原文线索

V2-041275 | MATHDB-318984

Pomerance conjecture on four-term minimal zero-sum sequences

摘录 / EXCERPT: Pomerance conjecture. If n is a positive integer with $(n,6)=1$ and S is a minimal zero-sum sequence in Z_n with $|S|=4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041276 | MATHDB-318994

Openness conjecture for extremal norms and characteristic words

摘录 / EXCERPT: Let $r,d \in \mathbb{N}$, let $\mathcal{I}_r(\mathbb{K}^d)$ be the ambient class of r-tuples under consideration, and let Ω_r^n denote the words of length n. For...

conjecture | open | MathDB 页面 | 原文线索

V2-041277 | MATHDB-319024

The non-flowerable coin graph extremal equality conjecture

摘录 / EXCERPT: Non-flowerable extremal equality conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-041278 | MATHDB-319026

Donaldson's relative K-stability conjecture for labeled polytopes

摘录 / EXCERPT: Donaldson's relative K-stability conjecture. There is a solution $G\in\mathcal{S}(\Delta,u)$ to the extremal Kähler equation

conjecture | open | MathDB 页面 | 原文线索

V2-041279 | MATHDB-319041

The dual colorful Tverberg theorem in the plane

摘录 / EXCERPT: Let $3n$ straight lines in general position be given in the plane, and partition them into three color classes of n lines each. The dual colorful Tverberg theorem. The lines can...

conjecture | open | MathDB 页面 | 原文线索

V2-041280 | MATHDB-319044

Supporting-tree decomposition conjecture for dyadic rearrangements

摘录 / EXCERPT: Let $\t$ be a rearrangement of the dyadic intervals, and write $\cF_1,\dots,\cF_N$ for subcollections of the dyadic intervals. Suppose there is a constant C such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-041281 | MATHDB-319062

Keller-Pilpel's influence conjecture for monotone Boolean functions

摘录 / EXCERPT: Keller-Pilpel's conjecture. The total influence of f should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-041282 | MATHDB-319101

Finite-limit conjecture for normalized numerical-semigroup counts

摘录 / EXCERPT: Finite-limit conjecture. As $g \to \infty$, the normalized count $n_g\varphi^{-g}$ converges to a finite limit.

conjecture | open | MathDB 页面 | 原文线索

V2-041283 | MATHDB-319102

Convergence conjecture for the auxiliary numerical-semigroup sum

摘录 / EXCERPT: Convergence conjecture. The sum in the displayed series converges to a finite value.

conjecture | open | MathDB 页面 | 原文线索

V2-041284 | MATHDB-319103

Asymptotic concentration conjecture for numerical semigroups with Frobenius bound

摘录 / EXCERPT: Asymptotic concentration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041285 | MATHDB-319113

Existence of planar minimal graphs with arbitrarily many vertices

摘录 / EXCERPT: Existence conjecture. There exist minimal graphs attached to some fixed finite set of points in the plane with arbitrarily large numbers of vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-041286 | MATHDB-319117

Conjecture on the third Erdős-Szekeres-type function at $(6,1)$

摘录 / EXCERPT: Let $h(n,k)$ be the least positive integer such that every planar set ${\cal X}$ in general position with at least $h(n,k)$ points contains n points forming a convex n-gon whos...

conjecture | open | MathDB 页面 | 原文线索

V2-041287 | MATHDB-319161

The squarefree Veronese Stanley depth formula conjecture

摘录 / EXCERPT: Let K be a field, let $S=K[x_1,\dots,x_n]$, and let $I_{n,d}$ be the ideal generated by all squarefree monomials of degree d in S. For positive integers d and n with...

conjecture | open | MathDB 页面 | 原文线索

V2-041288 | MATHDB-319187

A lower bound for the ratio of the polynomials R_d and P_d

摘录 / EXCERPT: Let $R_d(x)$ and $P_d(x)$ be the polynomials whose ratio is under consideration. Lower-bound conjecture. If $d\ge5$ and $x\ge\frac12(d+\sqrt{d+2})$, then ... This conjecture is prese...

conjecture | open | MathDB 页面 | 原文线索

V2-041289 | MATHDB-319189

The combinatorial Sylvester four-point conjecture

摘录 / EXCERPT: Combinatorial Sylvester four-point conjecture. For every $n\geq 4$, the probability that $\mathsf{v}\in X$ is $1/4$.

conjecture | open | MathDB 页面 | 原文线索

V2-041290 | MATHDB-319215

Martin-O'Bryant's limiting MSTD probability conjecture

摘录 / EXCERPT: Let S be chosen uniformly from the subsets of $\{0,1,\dots,n\}$, and let ρ_n denote the probability that S is a more sums than differences (MSTD) set, meaning...

conjecture | open | MathDB 页面 | 原文线索

V2-041291 | MATHDB-319218

Completeness conjecture for honeycomb arrays

摘录 / EXCERPT: A honeycomb array is an arrangement of points on the triangular lattice satisfying the defining conditions described in the paper; two arrays are considered equivalent up to the sy...

conjecture | open | MathDB 页面 | 原文线索

V2-041292 | MATHDB-319227

Infinite log-concavity conjecture for Euler's difference-table sequences

摘录 / EXCERPT: Infinite log-concavity conjecture. The sequence $\{d_n^k\}_{0\leq k\leq n}$ is infinitely log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-041293 | MATHDB-319259

Razumov-Stroganov conjecture for fully packed loop connectivities

摘录 / EXCERPT: Razumov-Stroganov conjecture. For every link pattern pi of size $2n$,

conjecture | open | MathDB 页面 | 原文线索

V2-041294 | MATHDB-319287

Triangularity conjecture for the sharp-transformed S basis of FQSym

摘录 / EXCERPT: Triangularity conjecture. The matrix of ${{\bf S}^\sigma}^\sharp$ in the ${\bf S}$ basis is triangular. Its diagonal values are 1 for elements of X_n and 0 for elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-041295 | MATHDB-319288

Freeness conjecture for the sharp transform of the planar binary tree algebra

摘录 / EXCERPT: Freeness conjecture. The algebra ${\bf PBT}^\sharp$ is free over the set ${\bf P}_T^\sharp$, where T runs over trees with at least two nodes and such that the right subtree of th...

conjecture | open | MathDB 页面 | 原文线索

V2-041296 | MATHDB-319291

Monotonicity conjecture for card values in two-person whist

摘录 / EXCERPT: Consider two-person whist without the assumption of symmetry, where a player may force the opponent to discard by leading a suit in which the opponent is void. Cards are compared w...

conjecture | open | MathDB 页面 | 原文线索

V2-041297 | MATHDB-319292

Sign-alternation conjecture for the flat Bernoulli game coefficients

摘录 / EXCERPT: Let a_n be the coefficient of $t^n/n!$ in $1/(1-\ln(1+t))$, so that for $n\geq 1$, ... The first 11 entries are positive, while a_{12} is negative and the subsequent entries ap...

conjecture | open | MathDB 页面 | 原文线索

V2-041298 | MATHDB-319301

Infinite log-convexity conjecture for Apéry, Catalan-Larcombe-French, and related sequences

摘录 / EXCERPT: Let $A=a_i_{0\leq i\leq\infty}$ be a sequence, and define ... A sequence is k-log-convex if $\mathcal{L}^j(A)$ is log-convex for $j=0,1,\ldots,k-1$, and infinitely log-convex if...

conjecture | open | MathDB 页面 | 原文线索

V2-041299 | MATHDB-319309

The convex-combination version of Freiman's theorem for planar blockers

摘录 / EXCERPT: Let P be a set of points in the plane with at most $\frac{1}{2}|P|$ points collinear. Suppose that a set B blocks P, meaning that for every two distinct points $x,y\in P$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-041300 | MATHDB-319313

The monochromatic-line conjecture for coloured planar point sets

摘录 / EXCERPT: Let P be a finite set of points in the plane, and colour each point with one of k colours. A monochromatic line means a maximal set of collinear points in P whose points all...

conjecture | open | MathDB 页面 | 原文线索

V2-041301 | MATHDB-319341

Haglund-Haiman-Loehr formula criterion for Kostka-Macdonald polynomials

摘录 / EXCERPT: HaglundHaimanLoehr formula criterion. The formula is valid if and only if at least one of the following conditions holds: (i) $mu_1\leq 3$ and $mu_2\leq 2$; or (ii) $lambda$ is a h...

conjecture | open | MathDB 页面 | 原文线索

V2-041302 | MATHDB-319348

Bandlow-Schilling-Thiéry's uniqueness conjecture for affine promotion

摘录 / EXCERPT: Bandlow-Schilling-Thiéry's conjecture. There is a unique promotion operator, and hence a unique good affine structure of type $A_n^{(1)}$, on tensor products of type A_n crystal...

conjecture | open | MathDB 页面 | 原文线索

V2-041303 | MATHDB-319361

Cordovil-Moreira cyclic ordering conjecture for two matroid bases

摘录 / EXCERPT: Let M be a finite loopless matroid of rank r, and let $B=\{b_1,\dots,b_r\}$ and $B'=\{b'_1,\dots,b'_r\}$ be two bases of M. A cyclic ordering is an ordering considered cyclic...

conjecture | open | MathDB 页面 | 原文线索

V2-041304 | MATHDB-319362

Kajitani's cyclic arboricity conjecture for matroids

摘录 / EXCERPT: Let M be a finite loopless matroid with ground set E and rank function $r(A)$ for subsets $A\subseteq E$. A cyclic ordering of E is an ordering considered cyclically, and a b...

conjecture | open | MathDB 页面 | 原文线索

V2-041305 | MATHDB-319377

Generalized Yau-Tian-Donaldson conjecture for extremal metrics

摘录 / EXCERPT: Let M be a compact polarised manifold and let L be an ample line bundle on M. An extremal metric in the class $c_1(L)$ is a Kähler metric whose scalar curvature has holomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-041306 | MATHDB-319395

The rectangular peg conjecture

摘录 / EXCERPT: Let $r>0$ be a real number, and let $\gamma:S^1\to\mathbb{R}^2$ be a smooth embedding of the circle into the plane. A rectangle has aspect ratio r when the ratio of its side len...

conjecture | open | MathDB 页面 | 原文线索

V2-041307 | MATHDB-319396

The crosspolytopal peg conjecture

摘录 / EXCERPT: Let $d\geq 2$, and let $\gamma:S^{d-1}\to\mathbb{R}^d$ be a smooth embedding of the $(d-1)$-sphere into d-dimensional Euclidean space. A regular d-dimensional cross-polytope i...

conjecture | open | MathDB 页面 | 原文线索

V2-041308 | MATHDB-319398

Holroyd-Talbot-Borg strict EKR conjecture for simplicial complexes

摘录 / EXCERPT: HolroydfTalbotfBorg's conjecture. If Δ has minimal facet cardinality k, then Δ is r-EKR for $r\leq\frac{k}{2}$, and strictly r-EKR for $r<\frac{k}{2}$. The stri...

conjecture | open | MathDB 页面 | 原文线索

V2-041309 | MATHDB-319437

The one-row orthogonal Weingarten second-order conjecture

摘录 / EXCERPT: Let n and k be nonnegative integers with $n>k$. Let $G^{k+1}_{(k)}(n)$ denote the coefficient used in the orthogonal Weingarten expansion, and let Cat_k denote...

conjecture | open | MathDB 页面 | 原文线索

V2-041310 | MATHDB-319449

Equivelar triangulation existence conjecture for surfaces

摘录 / EXCERPT: Let M be a surface of Euler characteristic $\chi<0$, and let $n\geq\Bigl\lceil\tfrac{1}{2}(7+\sqrt{49-24\chi})\Bigr\rceil$. Set $q=6-\frac{6\chi}{n}$ and suppose that q is an i...

conjecture | open | MathDB 页面 | 原文线索

V2-041311 | MATHDB-319465

Approximate homomorphism testing conjecture

摘录 / EXCERPT: Let $f:\mathbb{F}_2^n \to \mathbb{F}_2^m$ be a function, and let ϵ measure the probability that it satisfies the additive homomorphism equation on uniformly sampled...

conjecture | open | MathDB 页面 | 原文线索

V2-041312 | MATHDB-319466

Structured approximate homomorphism conjecture

摘录 / EXCERPT: Structured approximate homomorphism conjecture. If $|\Delta f|\le K$, then there is a linear map $\ell:\mathbb{F}_2^n\to\mathbb{F}_2^m$ and a function e such that

conjecture | open | MathDB 页面 | 原文线索

V2-041313 | MATHDB-319467

Polynomial inverse Gowers conjecture for the U^3 norm

摘录 / EXCERPT: Let $f:\mathbb{F}_2^n\to\mathbb{F}_2$ be a Boolean function, and let $\|f\|_{U^3}$ denote its Gowers U^3 norm. Polynomial inverse Gowers conjecture for the U^3 norm. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-041314 | MATHDB-319471

A uniform cut-product bound for random Fibonacci words

摘录 / EXCERPT: Uniform cut-product conjecture. There is a constant C such that for all $n\geq 3$ and $1\leq k\leq f_n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-041315 | MATHDB-319502

Superlinear blocker-number conjecture

摘录 / EXCERPT: Let $b(n)$ be the minimum integer such that some set of n points in the plane in general position is blocked by a set of $b(n)$ points. Superlinear blocker-number conjecture. ... K...

conjecture | open | MathDB 页面 | 原文线索

V2-041316 | MATHDB-319512

Ball-homeomorphism conjecture for the twindragons B and C

摘录 / EXCERPT: The twindragons $\mathcal B$ and $\mathcal C$ are self-affine tiles in the plane. Ball-homeomorphism conjecture. The twindragons $\mathcal B$ and $\mathcal C$ are homeomorphic to a...

conjecture | open | MathDB 页面 | 原文线索

V2-041317 | MATHDB-319541

Conjectured enumeration formulas for affine permutations avoiding patterns in S_4

摘录 / EXCERPT: Enumeration conjecture. The following equalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-041318 | MATHDB-319543

The $2k+1$ threshold conjecture for maximal-chain trace extremal families

摘录 / EXCERPT: Let $W(n,k,r)$ be the maximum size of a family $\mathcal{A}\subset [n]^{(k)}$ whose trace on every r-subset of $[n]$ contains no maximal chain, and let $n_0(k,r)$ denote a thresh...

conjecture | open | MathDB 页面 | 原文线索

V2-041319 | MATHDB-319544

The complete extremal-family conjecture for maximal-chain traces

摘录 / EXCERPT: Let $W(n,k,r)$ denote the maximum size of a family $\mathcal{A}\subset [n]^{(k)}$ whose trace on every r-subset of $[n]$ contains no maximal chain. The complete extremal-family c...

conjecture | open | MathDB 页面 | 原文线索

V2-041320 | MATHDB-319545

Gowers's true complexity conjecture for systems of linear forms

摘录 / EXCERPT: Let $L_1,\dots,L_m$ be a system of linear forms in d variables over $\mathbb{F}_p^n$. Its true complexity is the least integer k such that, for every $\eta>0$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-041321 | MATHDB-319546

Gowers' linear forms and uniformity conjecture

摘录 / EXCERPT: Gowers' conjecture. The two averages are close whenever $\\|f-g\\|_{U^k}$ is small.

conjecture | open | MathDB 页面 | 原文线索

V2-041322 | MATHDB-319565

The good-word conjecture for simply-laced Coxeter groups

摘录 / EXCERPT: Let W be a Coxeter group with simple reflections, and let $u\leqslant v$ in the Bruhat order. Choose a reduced decomposition $v=s_1\cdots s_n$, where $n=l(v)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041323 | MATHDB-319583

Even-row lattice-path model for bounded-height standard Young tableaux

摘录 / EXCERPT: Even-row lattice-path conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041324 | MATHDB-319584

Knaster's problem for points on a sphere

摘录 / EXCERPT: Let S^{d-1} be a unit sphere in $\mathbb R^d$. Suppose we are given d points $x_1, \ldots, x_d\in S^{d-1}$ and a continuous function $f: S^{d-1}\to \mathbb R$. Knaster's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-041325 | MATHDB-319588

Real-root conjecture for fully packed loop polynomials

摘录 / EXCERPT: Let π be a matching, let $A_\pi(t)$ be its polynomial, let $d(\pi)$ denote the relevant degree, and let $m_p(\pi)$ denote the multiplicity assigned to the integer p. Real-roo...

conjecture | open | MathDB 页面 | 原文线索

V2-041326 | MATHDB-319589

Negative-integer factorization conjecture for fully packed loop polynomials

摘录 / EXCERPT: Let π be a matching and let p be an integer with $1\leq p\leq|\pi|-1$ such that $m_p(\pi)=0$. Write $\pi=\alpha\circ\beta$, where $|\alpha|=p$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-041327 | MATHDB-319590

Sum rules for the special values of fully packed loop polynomials

摘录 / EXCERPT: For a matching π, define $G_\pi=A_\pi(-|\pi|)$; let A_n and A_n^V denote the enumerative quantities used in the source, and let ^{n} denote the empty matching with n nest...

conjecture | open | MathDB 页面 | 原文线索

V2-041328 | MATHDB-319591

Coefficient-positivity conjecture for fully packed loop polynomials

摘录 / EXCERPT: Let π be a matching and let $A_\pi(t)$ be its polynomial. Coefficient-positivity conjecture. Every coefficient of $A_\pi(t)$ is nonnegative. The conjecture is supported by comp...

conjecture | open | MathDB 页面 | 原文线索

V2-041329 | MATHDB-319603

The polynomial complexity conjecture for graphon constraint sets

摘录 / EXCERPT: Let $F_1,F_2,\dots,F_n$ be finite graphs, let $t_1,t_2,\dots,t_m$ be real numbers in $[0,1]$, and let $\mathcal{S}$ be the set of graphons W satisfying ... A graphon W has polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-041330 | MATHDB-319656

Cameron-Praeger conjecture on block-transitive 6-designs

摘录 / EXCERPT: Cameron-Praeger conjecture. There are no non-trivial block-transitive 6-designs.

conjecture | open | MathDB 页面 | 原文线索

V2-041331 | MATHDB-319711

Symmetry of the algorithmic shuffling weight function

摘录 / EXCERPT: Let φ be the shuffling weight function produced by the algorithm, and call a shuffling weight function symmetric if $\varphi(i)+\varphi(31-i)$ is independent of i. Symmet...

conjecture | open | MathDB 页面 | 原文线索

V2-041332 | MATHDB-319712

The Weak Anti-Freiman conjecture

摘录 / EXCERPT: Weak Anti-Freiman conjecture. Then A contains either a $B_2[g]$ set or a $B^\circ_2[g]$ set of size at least $\delta'|A|$, where g and δ' depend only on δ.

conjecture | open | MathDB 页面 | 原文线索

V2-041333 | MATHDB-319714

Enumeration of matchings with no 2-left-nestings

摘录 / EXCERPT: Let a matching on $[2n]$ be a set of n disjoint arcs joining pairs of vertices. Assume that $i<j<k<\ell$, and call the arcs (i,ℓ) and (j,k) m-left-nesting when...

conjecture | open | MathDB 页面 | 原文线索

V2-041334 | MATHDB-319715

Second-order Eulerian distribution of left-nestings in matchings

摘录 / EXCERPT: Let $\mathcal{M}_{2n}$ be the set of all matchings on $[2n]$, and let $\operatorname{lne}(M)$ denote the number of left-nestings of a matching $M\in\mathcal{M}_{2n}$. The second-or...

conjecture | open | MathDB 页面 | 原文线索

V2-041335 | MATHDB-319760

The quadrilateral lower-bound conjecture for slicings of combinatorial 3-manifolds

摘录 / EXCERPT: Quadrilateral lower-bound conjecture. For every such slicing,

conjecture | open | MathDB 页面 | 原文线索

V2-041336 | MATHDB-319783

TBR adjacency implies BME adjacency

摘录 / EXCERPT: Let $\mathcal{T}_n$ be the space of trees on n leaves. For $T,T'\in\mathcal{T}_n$, let $d_{TBR}(T,T')$ denote the minimum number of tree-bisection-regrafting moves needed to tran...

conjecture | open | MathDB 页面 | 原文线索

V2-041337 | MATHDB-319798

Disjoint crossing families conjecture for geometric graphs

摘录 / EXCERPT: A geometric graph is a graph drawn in the plane with vertices represented by points and edges as straight-line segments. For integers $k,l\geq 1$, a (k,l)-crossing family is a pa...

conjecture | open | MathDB 页面 | 原文线索

V2-041338 | MATHDB-319799

Ackerman's linear extremal conjecture for forbidden circle-graph matchings

摘录 / EXCERPT: A geometric graph is a graph drawn in the plane with vertices represented by points and edges as straight-line segments. A circle graph F is the intersection graph of chords on a...

conjecture | open | MathDB 页面 | 原文线索

V2-041339 | MATHDB-319883

The golden-ratio limit conjecture for generalized diagonal Wythoff Nim

摘录 / EXCERPT: Fix $p,q\bin \mathbb{N}$, and let $a=a(p,q)$ and $b=b(p,q)$ denote the associated sequences. A pair (p,q) is a splitting pair if it is a Wythoff pair or a dual Wythoff pair. The...

conjecture | open | MathDB 页面 | 原文线索

V2-041340 | MATHDB-319884

The splitting conjecture for generalized diagonal Wythoff Nim

摘录 / EXCERPT: Fix $p,q\in\mathbb{N}$, let $\pi=\pi_{p,q}$, and let $a=a(p,q)$ and $b=b(p,q)$ be the associated sequences. Let U and $\mathbb{N}_0$ have the meanings fixed earlier in the paper,...

conjecture | open | MathDB 页面 | 原文线索

V2-041341 | MATHDB-319885

The two-limit conjecture for the $(1,2)$ and $(2,3)$ games

摘录 / EXCERPT: Fix $(p,q)=(1,2)$ or $(p,q)=(2,3)$, and let (a_n,b_n) be the associated sequences. Two sequences (l_i) and (u_i) are complementary when they are increasing and together parti...

conjecture | open | MathDB 页面 | 原文线索

V2-041342 | MATHDB-319932

Even-length alternating-permutation equivalence for eight patterns

摘录 / EXCERPT: Let $A_m(p)$ denote the set of alternating permutations of length m that avoid the pattern p. Two patterns are equivalent for even-length alternating permutations when...

conjecture | open | MathDB 页面 | 原文线索

V2-041343 | MATHDB-319933

Odd-length alternating-permutation equivalence of the six remaining patterns

摘录 / EXCERPT: Let $A_m(p)$ denote the set of alternating permutations of length m that avoid the pattern p. Two patterns are equivalent for odd-length alternating permutations when their avo...

conjecture | open | MathDB 页面 | 原文线索

V2-041344 | MATHDB-319936

Bukh's bounded-error conjecture for sums of dilates

摘录 / EXCERPT: Bukh's bounded-error conjecture. There is an absolute constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-041345 | MATHDB-319943

The polynomial-denominator conjecture for generalized factor order

摘录 / EXCERPT: Let $\mathbb{P}^*$ be the set of finite words over the positive integers, let $S(u;t,x)$ be the associated generating function, and let an increasing/decreasing factorization mean...

conjecture | open | MathDB 页面 | 原文线索

V2-041346 | MATHDB-319954

The simplex conjecture for random simplex volumes

摘录 / EXCERPT: Let K be a convex body in $\mathbb{R}^d$, let T^d be a d-dimensional simplex, let $p\geqslant 1$, and let $n\geqslant d+1$. Write $\mathbb{E}^p_*(K)$ and $\mathbb{E}^p_n(K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-041347 | MATHDB-319971

Volume-partition enumeration conjecture for a permutation pattern

摘录 / EXCERPT: Regard two permutations as equivalent if they give the same volume partition. Let ... Let $\widetilde{{\mathfrak A}}_n(\mathbf{p})$ denote the corresponding set of equivalence-class...

conjecture | open | MathDB 页面 | 原文线索

V2-041348 | MATHDB-319979

Graphic-cographic conjecture on NBC bases

摘录 / EXCERPT: Let M be a matroid on ground set $E(M)$. A standard ordering of $E(M)$ is an ordering satisfying the standard-ordering conditions defined for the matroid's broken-circuit complex...

conjecture | open | MathDB 页面 | 原文线索

V2-041349 | MATHDB-319986

The shifted-sum conjecture for pure O-sequences

摘录 / EXCERPT: Let $h=(1,h_1,\dotsc,h_e)$ and $h'=(1,h'_1,\dotsc,h'_{e-1})$ be pure O-sequences. For a vector $H=(1,H_1,H_2,\dots,H_t)$, write $\Delta H=(1,H_1-1,H_2-H_1,\dots,H_t-H_{t-1})$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-041350 | MATHDB-319987

The interval conjecture for pure O-sequences

摘录 / EXCERPT: Let $(1,h_1,\dotsc,h_e)$ be a sequence of natural numbers, and fix an index i. Suppose that for some positive integer α, both ... and ... are pure O-sequences. Interval conj...

conjecture | open | MathDB 页面 | 原文线索

V2-041351 | MATHDB-320007

Kitaev-Remmel conjecture on the generating function for $(2+2)$-free posets

摘录 / EXCERPT: Kitaev-Remmel conjecture. The generating function $P(t,z)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041352 | MATHDB-320015

The high-branch-width non-fragility conjecture

摘录 / EXCERPT: Let N be a $\operatorname{GF}(q)$-representable matroid, and let $\operatorname{bw}(M)$ denote the branch width of a matroid M. High-branch-width non-fragility conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-041353 | MATHDB-320016

The partial-field inequivalent-representation chain conjecture

摘录 / EXCERPT: Let $\mathbb{P}$ be a partial field, and let $\operatorname{\mathcal{M}}(\mathbb{P})$ denote the class of $\mathbb{P}$-representable matroids. Partial-field inequivalent-representa...

conjecture | open | MathDB 页面 | 原文线索

V2-041354 | MATHDB-320028

Narayana distribution conjecture for path lengths of 231-avoiding permutations

摘录 / EXCERPT: Narayana distribution conjecture. The path lengths have the Narayana distribution:

conjecture | open | MathDB 页面 | 原文线索

V2-041355 | MATHDB-320030

Polynomial Diameter Conjecture for polytope graphs

摘录 / EXCERPT: Polynomial Diameter Conjecture. Is there a polynomial function $f(n,d)$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-041356 | MATHDB-320031

The d-step Conjecture for polytope diameters

摘录 / EXCERPT: The d-step Conjecture. For every $d \geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-041357 | MATHDB-320032

Non-revisiting Conjecture for simple polytopes

摘录 / EXCERPT: Non-revisiting Conjecture. There is a path from u to v which at every step enters a different facet of P.

conjecture | open | MathDB 页面 | 原文线索

V2-041358 | MATHDB-320040

White's conjecture for lattice path polymatroids

摘录 / EXCERPT: Let Γ be a lattice path polymatroid. Its base ring is the polynomial ring $\mathbb{C}[Y_m: m$ is a basis of $\Gamma]$, and its toric ideal is the kernel of the map ... defined...

conjecture | open | MathDB 页面 | 原文线索

V2-041359 | MATHDB-320045

Hähnle's linear-in-dimension Hirsch conjecture

摘录 / EXCERPT: Let P be a d-dimensional polytope with n facets, and let its diameter be the maximum length of a shortest edge path between two vertices of P. Hähnle's conjecture. The diam...

conjecture | open | MathDB 页面 | 原文线索

V2-041360 | MATHDB-320046

Hähnle's sharpened linear Hirsch conjecture

摘录 / EXCERPT: Let P be a d-dimensional polytope with n facets, and let its diameter be the maximum length of a shortest edge path between two vertices of P. Hähnle's sharpened conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-041361 | MATHDB-320051

Fang's extremal metric conjecture for Dolbeault determinants on the sphere

摘录 / EXCERPT: Fang's conjecture. The upper bound is achieved precisely by the Fubini-Study metric on L, up to scaling.

conjecture | open | MathDB 页面 | 原文线索

V2-041362 | MATHDB-320065

The symmetric Galois group conjecture for Tutte polynomials of connected matroids

摘录 / EXCERPT: Let M be a finite connected matroid with positive rank $n=r(M)$, and let K be a field of characteristic zero. The symmetric Galois group conjecture. The Galois group of the Tut...

conjecture | open | MathDB 页面 | 原文线索

V2-041363 | MATHDB-320085

Critical-threshold conjecture for the splitting model

摘录 / EXCERPT: Critical-threshold conjecture. There exists a critical value $h_c=h_c(d)$ such that, for all $h<h_c$, the model is robust, while for all $h\geq h_c$, the model is explosive. Moreov...

conjecture | open | MathDB 页面 | 原文线索

V2-041364 | MATHDB-320116

The asymptotic limit conjecture for higher Davenport constants

摘录 / EXCERPT: Asymptotic limit conjecture. For every positive integer j, the limit

conjecture | open | MathDB 页面 | 原文线索

V2-041365 | MATHDB-320131

Classical conjecture on the systolic extremality of the round real projective 3-space

摘录 / EXCERPT: Let $(\mathbb{R} \mathbb{P}^3,sph)$ denote real projective 3-space endowed with its metric of constant curvature. A metric is extremal for the isosystolic inequality if it realizes...

conjecture | open | MathDB 页面 | 原文线索

V2-041366 | MATHDB-320132

Systolic extremality conjecture for the singular manifold (B_2,g_0)

摘录 / EXCERPT: Let (B_2,g_0) be the singular Riemannian manifold considered in the source. A metric is extremal for the isosystolic inequality if it realizes the optimal systolic constant in it...

conjecture | open | MathDB 页面 | 原文线索

V2-041367 | MATHDB-320140

The converse symmetry conjecture for skew quasisymmetric Schur functions

摘录 / EXCERPT: Let $\gamma\negthinspace\sslash\negthinspace\beta$ be a skew composition shape, and let $\mathcal{S}_{\gamma\negthinspace\sslash\negthinspace\beta}$ denote its skew quasisymmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-041368 | MATHDB-320145

Odd-cardinality simple wedge conjecture

摘录 / EXCERPT: Let V be a set of n points in the plane, where n is odd, and suppose that V is not contained in a line. A simple line is a line spanned by V that contains exactly two poi...

conjecture | open | MathDB 页面 | 原文线索

V2-041369 | MATHDB-320174

Little's conjecture on dual Edelman-Greene equivalence

摘录 / EXCERPT: Let $R(\infty)$ be the set of all reduced words of permutations. Two reduced words are dual EG-equivalent when their recording tableaux under Edelman-Greene insertion agree, that...

conjecture | open | MathDB 页面 | 原文线索

V2-041370 | MATHDB-320175

Postnikov-Stanley conjecture on the finite Fomin-Stanley algebra

摘录 / EXCERPT: Let W be a Weyl group, let R^+ be its positive roots ordered by $\alpha\prec\beta$ when $\beta-\alpha$ is a positive sum of simple roots, and let $J(R^+,\prec)$ be the set of u...

conjecture | open | MathDB 页面 | 原文线索

V2-041371 | MATHDB-320190

Conjecture on eventual completion of simple partial multi-latin squares

摘录 / EXCERPT: Let k and m be positive integers. A simple partial k-latin square is a partial k-latin square in which no symbol is repeated within a cell. Eventual embedding conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-041372 | MATHDB-320196

Asymptotic formula for the number of solutions when m equals 2

摘录 / EXCERPT: Let $p(2,n)$ denote the number of solutions of the multiple permutation problem with parameter $m=2$ and size n. Asymptotic conjecture. ... The paper presents this as an additional...

conjecture | open | MathDB 页面 | 原文线索

V2-041373 | MATHDB-320197

Unimodality conjecture for the μ-sequence of a solution

摘录 / EXCERPT: Unimodality conjecture. There must exist $i\leqslant n$ such that $\mu_i=\left\lfloor\frac{1}{2}(n+3)\right\rfloor$, and

open_problem | open | MathDB 页面 | 原文线索

V2-041374 | MATHDB-320198

Symmetry and central-circuit conjectures for ({1,2,3},6)- and ({2,3},6)-spheres

摘录 / EXCERPT: Symmetry and circuit conjecture. (i) A z- or c-knotted $(\{1,2,3\},6)$-sphere, except Trifolium C_{3v}, has symmetry C_1, C_2, C_3, D_2 or D_3. (ii) A $(\{2,3\},6)$...

conjecture | open | MathDB 页面 | 原文线索

V2-041375 | MATHDB-320199

Conjectural extremal central-circuit counts for ({1,2,3},6)-spheres

摘录 / EXCERPT: Extremal central-circuit conjecture. (i) $f_2(v)=v+1$, realized exactly by one sphere for each $v\geq1$ of symmetry C_{2h} with $c=1^v,(2v)_{0,v}$. (ii) For odd $v\geq3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-041376 | MATHDB-320221

Conjectural linear system for refined alternating sign matrix enumerations

摘录 / EXCERPT: Linear-system conjecture. The numbers $A(n;-;i_1,\ldots,i_d)$, for $1\le i_1,i_2,\ldots,i_d\le n$, are uniquely determined by the five families of linear equations given in the sou...

conjecture | solved | MathDB 页面 | 原文线索

V2-041377 | MATHDB-320239

Uniqueness of the extremal periodic configuration in the discrete random displacement model

摘录 / EXCERPT: Let $d\geq 2$, let $q\neq 0$ be sign-definite, and let $\omega^{\ast}$ denote the periodic configuration associated with the extremal spectral values E_{min} and E_{max}. A per...

conjecture | open | MathDB 页面 | 原文线索

V2-041378 | MATHDB-320265

Ross's sub-lattice conjecture for periodic framework rigidity

摘录 / EXCERPT: Let $\tilde{G}(\mathbf{p},\mathbf{L})$ be an infinitesimally rigid periodic framework with periodic graph $(\tilde{G},\varphi)$. Let $\Lambda<\mathbb{Z}^2$ be any sub-lattice, and...

conjecture | open | MathDB 页面 | 原文线索

V2-041379 | MATHDB-320282

The PBW basis parametrization conjecture for symplectic Lie algebras

摘录 / EXCERPT: PBW basis parametrization conjecture. The lattice points in $\mathcal{C}(P,A)_{(\lambda,0)}$ parametrize a PBW basis of $V(\lambda)$ for the symplectic Lie algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-041380 | MATHDB-320294

Dimension formula conjecture for multilinear right-commutative monomials

摘录 / EXCERPT: Let FRC_n be the multilinear subspace of degree n in the free right-commutative algebra on n generators. An ordered basis of FRC_n consists of the distinct right-commutativ...

conjecture | open | MathDB 页面 | 原文线索

V2-041381 | MATHDB-320321

Transitivity of arrows between impartial misère-game positions

摘录 / EXCERPT: Let α, β, and γ be impartial positions. An arrow $\delta\to\varepsilon$ exists when Left moving second can win $\varepsilon+\delta^{\circ}$, or under a similar...

conjecture | open | MathDB 页面 | 原文线索

V2-041382 | MATHDB-320322

Replacement conjecture for equivalent impartial misère-game positions

摘录 / EXCERPT: Replacement conjecture. One can replace $*$ by ξ in any position having $*$ as an option without changing the resulting misère monoid. This would provide a substitution princip...

conjecture | open | MathDB 页面 | 原文线索

V2-041383 | MATHDB-320380

No-common-edge shortest paths conjecture for multitriangulations

摘录 / EXCERPT: No-common-edge shortest paths conjecture. 1. A shortest path in $G_{n,k}$ between T and T' never flips a common edge of T and T'. 2. If f is a k-relevant edge of T,...

conjecture | open | MathDB 页面 | 原文线索

V2-041384 | MATHDB-320412

Conjecture on Fibonacci-numeration P-positions of Wythoff star

摘录 / EXCERPT: Conjecture on Fibonacci-numeration P-positions. For all $n\ge 3$ and all i satisfying these conditions, both displayed positions are P-positions of the game under consideration...

conjecture | open | MathDB 页面 | 原文线索

V2-041385 | MATHDB-320413

Conjecture on diagonal positions in Wythoff star

摘录 / EXCERPT: Diagonal Wythoff-star conjecture. The position (i,i) belongs to $\mathcal{P}(\mathrm{W}^\star)$ if $i\in(S_1\cup S_4)\setminus(S_2\cup S_3)$, and belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-041386 | MATHDB-320414

Conjecture on periodicity of complementary Beatty-sequence games

摘录 / EXCERPT: Beatty-periodicity conjecture. The set $\mathcal{P}(G)$ is periodic if and only if the modulus α of (a_n) is rational. This proposes a complete rational-versus-irrational...

conjecture | open | MathDB 页面 | 原文线索

V2-041387 | MATHDB-320416

The uniqueness conjecture for the coloring c_{2,n} of E(2,n)

摘录 / EXCERPT: The uniqueness conjecture. For $n>2$, $c_{2,n}$ is the only n-coloring, up to isomorphism, of the nonzero rational numbers without a monochromatic solution to $E(2,n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-041388 | MATHDB-320417

The multiplicative decomposition conjecture for minimal coloring counts

摘录 / EXCERPT: The multiplicative decomposition conjecture. If a, b, and n are integers, $n>2$, $a>1$, $|b|>1$, and $\gcd(a,b)=1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041389 | MATHDB-320418

The finite-or-continuum conjecture for minimal coloring counts

摘录 / EXCERPT: The finite-or-continuum conjecture. For every nonregular finite system $\mathcal{L}$ of linear homogeneous equations, either $\Delta(\mathcal{L};\mathbb{Q})$ is finite or...

conjecture | open | MathDB 页面 | 原文线索

V2-041390 | MATHDB-320426

Property of inner tableau translation for the weak order

摘录 / EXCERPT: Property of inner tableau translation. If

conjecture | open | MathDB 页面 | 原文线索

V2-041391 | MATHDB-320427

Cover-preserving inner tableau translation conjecture for the weak order

摘录 / EXCERPT: Inner tableau translation conjecture. If $S\lessdot_{weak}T$ is a covering relation in SYT_n^R and R' is obtained from R by applying a single dual Knuth relation, then

conjecture | open | MathDB 页面 | 原文线索

V2-041392 | MATHDB-320456

Freiman's small-doubling conjecture for finite subsets of torsion-free groups

摘录 / EXCERPT: Let A be a finite subset of a torsion-free group, with $|A|\geq 4$. A progression is a finite geometric progression in the group. Freiman's conjecture. If ... then A is covered b...

conjecture | open | MathDB 页面 | 原文线索

V2-041393 | MATHDB-320464

The general lower-bound conjecture for uniform partition integrals

摘录 / EXCERPT: Let $\rho\renewcommand{\arraystretch}{1.2}\in\mathcal{P}(n)$ be a partition of n elements with r blocks of sizes ... Let $K_{\rho,u}$ denote the quantity associated with the part...

conjecture | open | MathDB 页面 | 原文线索

V2-041394 | MATHDB-320487

Conjecture on the extremal process of branching Brownian motion

摘录 / EXCERPT: Extremal-process conjecture. In the limit of large times, the distributions of $ Xi(t)$ and $ Pi_r$ coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-041395 | MATHDB-320510

The critical probability conjecture for bootstrap percolation on the binary hypercube

摘录 / EXCERPT: Let r be fixed, and let $p_c([2]^d,r)$ denote the critical probability for r-neighbor bootstrap percolation on the d-dimensional binary hypercube $[2]^d$. The critical probab...

conjecture | open | MathDB 页面 | 原文线索

V2-041396 | MATHDB-320575

Diagonal Ramsey bound for planar graphs

摘录 / EXCERPT: Let H be a planar graph on n vertices, and let $R(H,H)$ denote its diagonal two-colour Ramsey number. The planar Ramsey conjecture. For every sufficiently large n and every p...

conjecture | open | MathDB 页面 | 原文线索

V2-041397 | MATHDB-320578

Phelps-Rodrigues extremal conjecture for Borcea's variance inequality

摘录 / EXCERPT: Phelps-Rodrigues extremal conjecture. If $p>1$, equality in Borcea's variance inequality occurs if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-041398 | MATHDB-320590

The diamond-free Lubell-function extremal conjecture

摘录 / EXCERPT: Diamond-free Lubell-function conjecture. Every such family satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041399 | MATHDB-320607

Observed relations between up-step generating functions for s-ary paths

摘录 / EXCERPT: Let m be a positive integer, and let $G^{(i,m)}$ and $G^{(j,m)}$ denote the generating functions introduced in the paper for the corresponding up-step statistics on m-ary paths...

conjecture | open | MathDB 页面 | 原文线索

V2-041400 | MATHDB-320610

Finiteness conjecture for non-singular unsatisfiable hitting clause-sets

摘录 / EXCERPT: Finiteness conjecture. For every deficiency $k\in\mathbb{N}$, there are only finitely many isomorphism types of non-singular unsatisfiable hitting clause-sets.

conjecture | open | MathDB 页面 | 原文线索

V2-041401 | MATHDB-320611

Dejean's conjecture on distant repetitions in c-ary sequences

摘录 / EXCERPT: Let $c\geq 5$. An infinite c-ary sequence is a sequence over an alphabet of c symbols. Two identical blocks of length n are separated by the number of terms between the block...

conjecture | open | MathDB 页面 | 原文线索

V2-041402 | MATHDB-320620

Triviality of higher-order compositions on coordinate-harmonic functions

摘录 / EXCERPT: Let A_k denote the space of fields with k components, and let $(\ast)$ denote the family of differential operations and directional derivatives considered in the paper. A funct...

conjecture | open | MathDB 页面 | 原文线索

V2-041403 | MATHDB-320628

Farber's basis pair graph connectivity conjecture

摘录 / EXCERPT: Farber's conjecture. The basis pair graph of any matroid is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-041404 | MATHDB-320697

Random-set Szemerédi threshold conjecture

摘录 / EXCERPT: Random-set Szemerédi threshold conjecture. For every $\delta>0$ and every integer $k\geq 3$, there exist positive constants c and C such that

conjecture | open | MathDB 页面 | 原文线索

V2-041405 | MATHDB-320730

Small-boundary reconstruction conjecture for discrete tomography

摘录 / EXCERPT: Let row sums be $\mathcal{R}=(r_1,r_2,\ldots,r_m)$ and column sums be $\mathcal{C}=(c_1,c_2,\ldots,c_n)$, where ... and ... Assume that the line sums are consistent. Small-boundary rec...

conjecture | open | MathDB 页面 | 原文线索

V2-041406 | MATHDB-320753

Pfaffian evaluations for Motzkin, Delannoy, Schröder and Narayana sequences

摘录 / EXCERPT: Let $M_n=\sum_{k=0}^n\binom{n}{2k}C_k$ be the Motzkin numbers, $D_n=\sum_{k=0}^n\binom{n}{k}\binom{n+k}{k}$ the central Delannoy numbers, $S_n=\sum_{k=0}^n\binom{n+k}{2k}C_k$ the S...

conjecture | open | MathDB 页面 | 原文线索

V2-041407 | MATHDB-320754

Pfaffian conjecture for the vertically symmetric alternating-sign-matrix sequence

摘录 / EXCERPT: Let $a_n=\frac{1}{2n+1}\binom{3n}{n}=\frac{1}{3n+1}\binom{3n+1}{n}$. Vertically symmetric alternating-sign-matrix Pfaffian conjecture. For every integer $n\geq1$, ... The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-041408 | MATHDB-320780

Conjectured lower bounds for the second nonlinear syzygy stage

摘录 / EXCERPT: Let G be a simple graph, let I_G be its edge ideal, and let $[k;a_1,\ldots,a_{k-1}]$ be the sequence associated to G. The lower-bound conjecture. The sequence satisfies ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-041409 | MATHDB-320801

Chari's shellable M-vector conjecture for matroid complexes

摘录 / EXCERPT: A matroid complex is the simplicial complex of independent sets of a matroid. For a $(d-1)$-dimensional simplicial complex Δ, its h-vector $h(\Delta)=(h_0,h_1,\ldots,h_d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-041410 | MATHDB-320804

The product conjecture for finite transitive sets

摘录 / EXCERPT: Product conjecture. For every positive integer k, there exists a positive integer n such that some scaling of X^n is k-Ramsey for X.

conjecture | open | MathDB 页面 | 原文线索

V2-041411 | MATHDB-320805

The block-permutation conjecture

摘录 / EXCERPT: Block-permutation conjecture. For every pair of positive integers m and k, there exist positive integers n and d such that every k-colouring of $[m]^n$ contains a block p...

conjecture | open | MathDB 页面 | 原文线索

V2-041412 | MATHDB-320806

The template conjecture for monochromatic block sets

摘录 / EXCERPT: Template conjecture. For every pair of positive integers m and k, and every template τ over $[m]$, there exist positive integers n and d such that every k-colouring...

conjecture | open | MathDB 页面 | 原文线索

V2-041413 | MATHDB-320807

The non-subtransitive cyclic-quadrilateral conjecture

摘录 / EXCERPT: Cyclic-quadrilateral conjecture. There exists a cyclic quadrilateral that is not subtransitive.

conjecture | open | MathDB 页面 | 原文线索

V2-041414 | MATHDB-320851

Erdős-Shelah conjecture for maximum union-free subfamilies

摘录 / EXCERPT: Erdős-Shelah conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041415 | MATHDB-320852

Barat-Füredi-Kantor-Kim-Patkos conjecture on a-union-free subfamilies

摘录 / EXCERPT: Barat-Füredi-Kantor-Kim-Patkos conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041416 | MATHDB-320868

The Boolean-algebra subfamily extremal conjecture

摘录 / EXCERPT: Let $m=2^n$, and consider families of m sets. A subfamily forms a Boolean algebra of dimension d if it is isomorphic to the Boolean lattice $2^{[d]}$. Boolean-algebra subfamily...

conjecture | open | MathDB 页面 | 原文线索

V2-041417 | MATHDB-320885

The conjecture that a cyclic quadrilateral is not Ramsey

摘录 / EXCERPT: The cyclic-quadrilateral non-Ramsey conjecture. This cyclic quadrilateral is not Ramsey. The paper proves that this quadrilateral does not embed into any finite transitive set, and...

conjecture | open | MathDB 页面 | 原文线索

V2-041418 | MATHDB-320886

Chan-Jensen-Rubei's tropical basis conjecture for matrix minors

摘录 / EXCERPT: Let d, n, and r be integers, and consider the $(r+1)\times(r+1)$ minors of a d-by-n matrix. A collection of tropical polynomials is a tropical basis when its tropical zer...

conjecture | open | MathDB 页面 | 原文线索

V2-041419 | MATHDB-320937

The inverse conjecture for the Gowers norm over finite fields

摘录 / EXCERPT: Inverse conjecture $\operatorname{GI}(s)$. Let $\delta>0$ and $s\geqslant 0$. Then there exists an $\varepsilon=\varepsilon_{\delta,s,\mathbb{F}}>0$ such that for every finite-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-041420 | MATHDB-320959

The planar graph braid-group Theta-four obstruction conjecture

摘录 / EXCERPT: Let Γ be a planar graph, let $B_n\Gamma$ be its braid group, and let $P_3\Gamma$ be its pure braid group on three points. A group is simple-commutator-related if it has a pr...

conjecture | open | MathDB 页面 | 原文线索

V2-041421 | MATHDB-320991

The asymptotic bound for families forbidding distance one

摘录 / EXCERPT: Distance-one extremal conjecture. Every family $\mathcal{A}\subset\mathcal{P}[n]$ that forbids distance one satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041422 | MATHDB-320998

Nonexistence of a finite planar fair-win goal set

摘录 / EXCERPT: In the planar achievement game, a finite point set $G\subset\mathbb{R}^2$ is used as the goal set: players alternately place points, and Player 1 seeks to build a congruent copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-041423 | MATHDB-321004

Clonal Ding-Oporowski-Oxley-Vertigan-type minor conjecture

摘录 / EXCERPT: Let m be a positive integer, and let M be a 3-connected clonal matroid. The matroids $U_{2,2m}$ and $U_{2m-2,2m}$ are uniform matroids, while Λ_m and Δ_m denot...

conjecture | open | MathDB 页面 | 原文线索

V2-041424 | MATHDB-321006

Conjecture on complete subsets in cyclic groups of order pq

摘录 / EXCERPT: Complete-subset conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041425 | MATHDB-321007

Characterization of extremal incomplete sets in cyclic groups of order pq

摘录 / EXCERPT: Extremal-set conjecture. There exists an element $g\in G$ of order pq such that

conjecture | open | MathDB 页面 | 原文线索

V2-041426 | MATHDB-321046

Polynomial-dimension conjecture for high additive energy in F_3^N

摘录 / EXCERPT: Let $N>0$ be a large parameter, with constants independent of N. Let $B,C\subseteq F_3^N$ satisfy $|B|,|C|\leq N^{100}$. Suppose that the number of quadruples...

conjecture | open | MathDB 页面 | 原文线索

V2-041427 | MATHDB-321047

Additive-structure conjecture for non-smoothing subsets of F_3^N

摘录 / EXCERPT: Let $\Delta\subseteq F_3^N$ be N^{σ} additively non-smoothing, and assume $|\Delta|\leq N^{100}$. Additive-structure conjecture. Then Δ admits an additive structure a...

conjecture | open | MathDB 页面 | 原文线索

V2-041428 | MATHDB-321060

Sparla's Euler-characteristic conjecture for centrally symmetric manifolds

摘录 / EXCERPT: Sparla's conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-041429 | MATHDB-321098

The divisibility conjecture for odd binomial residue counts modulo 16

摘录 / EXCERPT: Let $a_{16,r}(n)$ denote the number of binomial coefficients in row n of Pascal's triangle that are congruent to r modulo 16. Fix n and odd r. Divisibility conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-041430 | MATHDB-321134

Elizalde-Noy maximality conjecture for consecutive pattern avoidance

摘录 / EXCERPT: Elizalde-Noy maximality conjecture. The increasing pattern σ is the maximal pattern, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-041431 | MATHDB-321135

Maximality conjecture for triples of consecutive patterns

摘录 / EXCERPT: Maximality conjecture for triples. The pattern set B is the maximal pattern set among sets of three patterns, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-041432 | MATHDB-321174

Stationary-distribution integrality conjecture for the symmetric-group Markov chain

摘录 / EXCERPT: Let $W=S_n$, let w_0 be the longest element of W, and let ζ be the stationary distribution of the Markov chain on W defined by the transition probabilities in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-041433 | MATHDB-321179

Vanishing of the cotangent cohomology for flag combinatorial spheres

摘录 / EXCERPT: Vanishing conjecture. If $\operatorname{lk}(f,\mathcal{K})$ is a 4-gon or a 5-gon for every codimension-2 face f, then

conjecture | open | MathDB 页面 | 原文线索

V2-041434 | MATHDB-321204

Maximal-coefficient conjecture for Stern polynomials

摘录 / EXCERPT: Let ... and define ... Then the following equality holds: ... where k is equal to $n/2$ if n is even and $(n-1)/2$ if n is odd. This conjecture arose from investigations of the seq...

conjecture | open | MathDB 页面 | 原文线索

V2-041435 | MATHDB-321212

Ahmed-Kullmann-Snevily's double-zero conjecture for good partitions

摘录 / EXCERPT: For $t\geqslant 3$, let $S(t)$ be the set of good partitions considered in the paper, and for $B\in S(t)$ let $n_{00}(B)$ denote the number of occurrences of 00 in B. Double-ze...

conjecture | open | MathDB 页面 | 原文线索

V2-041436 | MATHDB-321213

Ahmed-Kullmann-Snevily's one-count bound for good partitions

摘录 / EXCERPT: For $t\geqslant 3$, let $S(t)$ be the set of good partitions considered in the paper, and for $B\in S(t)$ let $n_1(B)$ denote the number of 1's in B. One-count bound conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-041437 | MATHDB-321215

Currie-Simpson conjecture on nonrepetitive sequences modulo k

摘录 / EXCERPT: Let $k\geqslant 1$ be a fixed positive integer, and let $M(k)$ denote the minimum number of symbols needed to create arbitrarily long sequences whose subsequences along arithmetic...

conjecture | open | MathDB 页面 | 原文线索

V2-041438 | MATHDB-321228

van Rees' conjecture on latin squares with many subsquares of order 3

摘录 / EXCERPT: van Rees' conjecture. If L is a van Rees latin square, then the order of L is a power of 3.

conjecture | open | MathDB 页面 | 原文线索

V2-041439 | MATHDB-321231

Conjecture for asymptotic enumeration of regular graphs with loops

摘录 / EXCERPT: Regular loopy-graph enumeration conjecture. Uniformly as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-041440 | MATHDB-321255

Pattern-free sets conjecture for the six-term configuration A_6

摘录 / EXCERPT: A set of integers is A_6-free if it contains no subset of the form ... for some $a,b>0$ with $a\neq b$. Let $r(n,A_6)$ be the maximum size of an A_6-free subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-041441 | MATHDB-321311

Gács et al.'s prescribed-range polynomial conjecture

摘录 / EXCERPT: Gács et al.'s conjecture. Let $k<\sqrt{p}$. If no polynomial with range M has degree less than $q-k$, then M contains an element of multiplicity at least $q-k$.

conjecture | open | MathDB 页面 | 原文线索

V2-041442 | MATHDB-321317

The list-version of Thue's theorem for nonrepetitive sequences

摘录 / EXCERPT: Let $n\geqslant 1$ and let $L_{1},\ldots,L_{n}$ be sets of symbols, each of size 3. A sequence $s_{1}\ldots s_{n}$ is chosen from lists $L_{1},\ldots,L_{n}$ if $s_i\in L_i$ for e...

conjecture | open | MathDB 页面 | 原文线索

V2-041443 | MATHDB-321318

Borg's sum conjecture for cross-intersecting hereditary subfamilies

摘录 / EXCERPT: Borg's sum conjecture. If $k\geq n+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041444 | MATHDB-321319

Borg's two-configuration sum conjecture for cross-intersecting hereditary families

摘录 / EXCERPT: Borg's stronger sum conjecture. (i) If

conjecture | open | MathDB 页面 | 原文线索

V2-041445 | MATHDB-321320

Borg's product conjecture for cross-intersecting hereditary families

摘录 / EXCERPT: Borg's product conjecture. The product

conjecture | open | MathDB 页面 | 原文线索

V2-041446 | MATHDB-321375

König's extremal-kernel conjecture for Grothendieck's inequality

摘录 / EXCERPT: For a measure-space formulation of Grothendieck's inequality, let $K:\mathbb R^n\times\mathbb R^n\to\mathbb R$ be the oscillatory Gaussian kernel ... and let ... Define...

conjecture | open | MathDB 页面 | 原文线索

V2-041447 | MATHDB-321415

Hersh's shellability conjecture for skeleta of finite lattices

摘录 / EXCERPT: Hersh's shellability conjecture. The $(r-2)$-skeleton of L is pure and shellable.

conjecture | open | MathDB 页面 | 原文线索

V2-041448 | MATHDB-321441

Ané-Eulenstein conjecture on maximal groves

摘录 / EXCERPT: Let $\mathcal{S}$ be a set of taxon sets, and let a grove be a collection of taxon sets with the potential to construct informative supertrees. A grove is maximal in $\mathcal{S}$...

conjecture | open | MathDB 页面 | 原文线索

V2-041449 | MATHDB-321442

Conjecture on unions of intersecting strictly informative groves

摘录 / EXCERPT: Let two groves be strictly informative if each satisfies the strict informativeness criterion for determining supertrees. Let the groves be collections of taxon sets, with intersec...

conjecture | open | MathDB 页面 | 原文线索

V2-041450 | MATHDB-321466

Conjecture that complete graphs are strongly GT

摘录 / EXCERPT: Strongly GT conjecture for complete graphs. For $q\geq 3$, the complete graph K_q is strongly GT.

conjecture | open | MathDB 页面 | 原文线索

V2-041451 | MATHDB-321467

Surjective homomorphism swapping conjecture for simple graphs

摘录 / EXCERPT: Surjective homomorphism swapping conjecture. For every positive integer i,

conjecture | open | MathDB 页面 | 原文线索

V2-041452 | MATHDB-321476

Main conjectural inequality for the maximum permanent of completely indecomposable three-regular matrices

摘录 / EXCERPT: Main conjectural inequality (MCI). For $n_1,n_2\geq 3$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-041453 | MATHDB-321477

Homotopy-equivalent topological representations detect isomorphisms of matroids

摘录 / EXCERPT: Let X be a finite CW complex that is not contractible, and let (M,l) and (N,l') be ρ-immersed matroids such that ... Suppose there exists a surjective weak map...

conjecture | open | MathDB 页面 | 原文线索

V2-041454 | MATHDB-321478

Rota's unimodality conjecture for Whitney numbers of the first kind

摘录 / EXCERPT: Let L be a geometric lattice of rank r. The Whitney numbers of the first kind are the integers $w_i(L)$ for $0\leq i\leq r$. Rota's unimodality conjecture. For some...

conjecture | open | MathDB 页面 | 原文线索

V2-041455 | MATHDB-321479

Rota's unimodality conjecture for Whitney numbers of the second kind

摘录 / EXCERPT: Let L be a geometric lattice of rank r. The Whitney numbers of the second kind are the integers $W_i(L)$ for $0\leq i\leq r$. Rota's unimodality conjecture. For some...

conjecture | open | MathDB 页面 | 原文线索

V2-041456 | MATHDB-321488

Treglown's conjecture for squares of Hamiltonian cycles in oriented graphs

摘录 / EXCERPT: Let G be an oriented graph, meaning a directed graph with no 2-cycles, on n vertices. Let $\delta^0(G)$ denote its minimum semi-degree, the minimum of its minimum in-degree and...

conjecture | open | MathDB 页面 | 原文线索

V2-041457 | MATHDB-321538

The middle 4-cube covering conjecture

摘录 / EXCERPT: Let n be even, and consider the 4-dimensional subcubes of the hypercube whose layers range from $n/2-2$ to $n/2+2$; call these the middle 4-cubes. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-041458 | MATHDB-321539

The middle even-cube covering conjecture

摘录 / EXCERPT: Let d be fixed. For even n, call a $2d$-dimensional subcube of the hypercube a middle $2d$-cube when its layers range from $n/2-d$ to $n/2+d$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-041459 | MATHDB-321561

Conjecture on Betti cones of hypersurface rings

摘录 / EXCERPT: Betti-cone conjecture. The cone $\mathrm{B}_{\mathbb{Q}}(Q)$ is an $(n+1)$-dimensional cone, and its closure is defined by the following $(n+2)$ extremal rays:

conjecture | open | MathDB 页面 | 原文线索

V2-041460 | MATHDB-321583

Runners-up conjecture for the Erdős-Littlewood-Offord bound

摘录 / EXCERPT: Let λ range over integer partitions with exactly k parts and greatest common divisor 1, and let r_λ be the associated Erdős-Littlewood-Offord probability. Runne...

conjecture | open | MathDB 页面 | 原文线索

V2-041461 | MATHDB-321584

Polynomiality conjecture for right descents of 2-clumped permutations

摘录 / EXCERPT: Let S_n be the symmetric group on n elements. A right descent of a permutation is a pair of adjacent entries in which the left entry is larger than the right entry, and a permu...

conjecture | open | MathDB 页面 | 原文线索

V2-041462 | MATHDB-321604

Eventual form conjecture for exponentially dense minor-closed matroid classes

摘录 / EXCERPT: Eventual form conjecture. There exist an integer $k\geq 0$ and an integer d with $$

conjecture | open | MathDB 页面 | 原文线索

V2-041463 | MATHDB-321630

Rational-strategy conjecture for squarefree lattice games

摘录 / EXCERPT: Rational-strategy conjecture. Every squarefree lattice game possesses a rational strategy.

conjecture | open | MathDB 页面 | 原文线索

V2-041464 | MATHDB-321631

Affine stratification conjecture for squarefree lattice games

摘录 / EXCERPT: Affine stratification conjecture. Every squarefree lattice game possesses an affine stratification.

conjecture | open | MathDB 页面 | 原文线索

V2-041465 | MATHDB-321642

The symmetry conjecture for determinants of punctured-hexagon enumeration matrices

摘录 / EXCERPT: Let $a=b$ and $\alpha=\beta$, so the punctured hexagon $H_{a,a,c,\alpha,\alpha,\gamma}$ is symmetric. Let A, C, and M be the parameters associated with this hexagon, let...

conjecture | open | MathDB 页面 | 原文线索

V2-041466 | MATHDB-321656

Wójcik's extremal characterization of minimum-density union-closed families

摘录 / EXCERPT: Let $\mathscr{F}$ be a finite union-closed family of finite sets, let $n=|\bigcup\mathscr{F}|$, and define ... with s_n the minimum of $D(\mathscr{F})$ over all union-closed famili...

conjecture | open | MathDB 页面 | 原文线索

V2-041467 | MATHDB-321671

Universal extremal-size conjecture for t-set-intersecting families of permutations

摘录 / EXCERPT: Let S_n be the symmetric group. A family $\mathcal{A}\subset S_n$ is t-set-intersecting if for every $\sigma,\pi\in\mathcal{A}$ there is a t-element subset $x\subseteq[n]$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-041468 | MATHDB-321672

Double-translate conjecture for maximum t-intersecting families of permutations

摘录 / EXCERPT: Double-translate conjecture. A maximum-sized t-intersecting family in S_n must be of the form $\pi M_i\tau$ for some i in this range and some $\pi,\tau\in S_n$. The source st...

conjecture | open | MathDB 页面 | 原文线索

V2-041469 | MATHDB-321673

Exponential bound conjecture for reversing families of permutations

摘录 / EXCERPT: Let S_n be the symmetric group. A family $\mathcal{A}\subset S_n$ is reversing if for every $\sigma,\pi\in\mathcal{A}$ there is a pair $\\{i,j\\}\subseteq[n]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-041470 | MATHDB-321716

Multilinear representation conjecture for skew partial fields

摘录 / EXCERPT: A skew partial field is a partial field whose multiplicative group need not be commutative. Let $\mathbb{P}(n,\mathbb{F})$ denote the partial field associated with $n\times n$ matr...

conjecture | open | MathDB 页面 | 原文线索

V2-041471 | MATHDB-321717

Ingleton inequality conjecture for skew-partial-field matroids

摘录 / EXCERPT: Let M be a matroid representable over a skew partial field, and let a quadruple of subsets of M be given. Ingleton inequality conjecture. Ingleton's Inequality is satisfied by...

conjecture | open | MathDB 页面 | 原文线索

V2-041472 | MATHDB-321718

Generalized parallel connection conjecture for skew partial fields

摘录 / EXCERPT: Let $\mathbb{P}$ be a skew partial field, and let M_1 and M_2 be $\mathbb{P}$-representable matroids having a common flat N and representations that agree on N. If N is a...

conjecture | open | MathDB 页面 | 原文线索

V2-041473 | MATHDB-321719

Finite cross-ratio conjecture for quaternionic unimodular matroids

摘录 / EXCERPT: Let $\mathbb{H}$ be the quaternions, with norm $|\cdot|$, and let $p,q,r\in\mathbb{H}$ satisfy ... for all distinct $i,j\in\{0,1,p,q,r\}$. Let M be a quaternionic unimodular matroi...

conjecture | open | MathDB 页面 | 原文线索

V2-041474 | MATHDB-321720

2-uniform matroid conjecture for quaternionic unimodular matroids

摘录 / EXCERPT: A 2-uniform matroid is a matroid in the class of matroids represented over the 2-uniform partial field, and a QU matroid is a quaternionic unimodular matroid. 2-uniform matroid con...

conjecture | open | MathDB 页面 | 原文线索

V2-041475 | MATHDB-321721

Polynomial-time basis-counting conjecture for 2-uniform matroids

摘录 / EXCERPT: Let M be a 2-uniform matroid, given by a representation over the 2-uniform partial field. Basis-counting conjecture. There is a polynomial-time algorithm to count the number of b...

conjecture | open | MathDB 页面 | 原文线索

V2-041476 | MATHDB-321722

Cauchy-Binet conjecture for determinant-like maps on skew partial fields

摘录 / EXCERPT: Let $\mathbb{P}=(R,G)$ be a skew partial field, let n,r,s be positive integers with $s\geq r$, define ... and let A be an $X\times E$ weak $\mathbb{P}$-matrix. A map...

conjecture | open | MathDB 页面 | 原文线索

V2-041477 | MATHDB-321726

The homothety conjecture for infinity-shaped extremal curves

摘录 / EXCERPT: Homothety conjecture. All infinity-shaped extremal curves are homothetic to the curve γ_α for this choice of α, where $\xi\approx.90890856$.

conjecture | open | MathDB 页面 | 原文线索

V2-041478 | MATHDB-321727

Exact exceptional values conjecture for sum-free-set r-values

摘录 / EXCERPT: Let $1\neq s\in\mathbb{N}$. For $S\subseteq[1,N]$ with $|S|=s$, let $r(S)$ denote the associated r-value, and define ... Let $f_{s,N}$ and $g_{s,N}$ be the smallest and largest eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-041479 | MATHDB-321737

Jungers-Protasov-Blondel conjecture on the overlap-free-word growth exponents

摘录 / EXCERPT: Let A_1,A_2 be the special 20×20 matrices with nonnegative integer entries associated with overlap-free words, and let $\mathcal{M}=\{A_1,A_2\}$. Let β be the upper...

conjecture | open | MathDB 页面 | 原文线索

V2-041480 | MATHDB-321741

Combinatorial tropical gonality conjecture for lattice polygons

摘录 / EXCERPT: Let $\Delta\subset\mathbb{R}^2$ be a two-dimensional lattice polygon, and let $\Delta^{(1)}$ denote the convex hull of the interior lattice points of Δ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-041481 | MATHDB-321798

Polynomiality conjecture for zero weight spaces of tensor arrays

摘录 / EXCERPT: Let k and $n_1,n_2,\dots,n_k$ be positive integers, and let d be the degree. Consider the zero weight space in degree d for the $n_1\times\cdots\times n_k$ array representati...

conjecture | open | MathDB 页面 | 原文线索

V2-041482 | MATHDB-321820

Borg's conjecture on stars in signed-set families

摘录 / EXCERPT: Let $\mathcal{F}$ be a family and let $k\geq2$. The signed-set family $\mathcal{S}_{\mathcal{F},k}$ consists of sets obtained by assigning one of k labels to each element of a me...

conjecture | open | MathDB 页面 | 原文线索

V2-041483 | MATHDB-321821

Borg's large-label conjecture for t-intersecting signed sets

摘录 / EXCERPT: Let $\mathcal{F}$ be a family, let $k\geq2$, and let $\mathcal{S}_{\mathcal{F},k}$ be its signed-set family. A family has the t-star property if a largest t-intersecting subfam...

conjecture | open | MathDB 页面 | 原文线索

V2-041484 | MATHDB-321822

Brunk-Huczynska conjecture on t-intersecting partial permutations

摘录 / EXCERPT: Brunk-Huczynska conjecture. (i) One of $\mathcal{A}_0,\ldots,\mathcal{A}_p$ is a largest t-intersecting subfamily of $\mathcal{S}_{[n],k}^*$. (ii) Every largest t-intersecting...

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V2-041485 | MATHDB-321846

Sub-extremality conjecture for black holes in the Einstein-Maxwell-Klein-Gordon system

摘录 / EXCERPT: Sub-extremality conjecture. Among all the data admissible from Theorem 1, there exists a generic subclass such that, if the maximal future development has $$, then the black hole...

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V2-041486 | MATHDB-321860

The secretary problem bound for posets with a fixed number of maximal elements

摘录 / EXCERPT: Secretary problem conjecture. There is an algorithm for the secretary problem on $(P,\prec)$ that is successful with probability at least p_k.

conjecture | open | MathDB 页面 | 原文线索

V2-041487 | MATHDB-321878

Butler-Graham conjecture on marking coordinate lines

摘录 / EXCERPT: Let $[k]=\{1,\ldots,k\}$, and consider the grid $[k]^n$. A coordinate line is the set of k points obtained by fixing all but one coordinate, and a marking of the lines designates...

conjecture | open | MathDB 页面 | 原文线索

V2-041488 | MATHDB-321905

The minimum monochromatic 4-term progression density for orders not divisible by 4

摘录 / EXCERPT: The authors' conjecture. The upper bound $1/12$ is attained as the infimum over group orders not divisible by 4. This complements the known lower bound $7/96$ and narrows the gap...

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V2-041489 | MATHDB-321908

The integrality conjecture for the coefficients in the L-polynomials

摘录 / EXCERPT: Integrality conjecture. Every coefficient of these polynomials is an integer.

conjecture | open | MathDB 页面 | 原文线索

V2-041490 | MATHDB-321938

The two-dimensional sumset lower-bound conjecture for valid multisets

摘录 / EXCERPT: The two-dimensional sumset lower-bound conjecture. Then

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V2-041491 | MATHDB-321939

The sumset lower-bound conjecture for valid multisets in finite vector spaces

摘录 / EXCERPT: The finite-vector-space sumset lower-bound conjecture. The following bounds should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-041492 | MATHDB-321947

Hurlbert et al.'s monotonicity conjecture for the pebbling threshold

摘录 / EXCERPT: Let $\mathcal{G}=(G_1,G_2,\dots,G_n,\dots)$ and $\mathcal{H}=(H_1,H_2,\dots,H_n,\dots)$ be graph sequences. For each graph G_n or H_n, let π denote its pebbling number, and...

conjecture | open | MathDB 页面 | 原文线索

V2-041493 | MATHDB-321953

Fakhruddin's conjecture on the \mathfrak{sl}_2 conformal block cone of \overline{M}_{0,6}

摘录 / EXCERPT: Let $\operatorname{Nef}(\overline{M}_{0,6})$ be the nef cone. A Fakhruddin ray is an extremal ray spanned either by $D(\mathfrak{sl}_2,1,(1,1,1,1,1,1))$ or by one of the 127 pullba...

conjecture | open | MathDB 页面 | 原文线索

V2-041494 | MATHDB-321957

Mubayi-Rödl conjecture on M-intersecting families with short integer intervals

摘录 / EXCERPT: Let $M\subset [n]:=\{0,\ldots,n\}$, and let $I(M)$ denote the maximum number of consecutive integers contained in M. Let $\mathcal{A}\subset 2^V$ be an M-intersecting family, m...

conjecture | open | MathDB 页面 | 原文线索

V2-041495 | MATHDB-321985

The bounded-bias conjecture for ordered chain games on wedges

摘录 / EXCERPT: Bounded-bias wedge conjecture. There exists a constant k such that for every dimension d, Maker wins the $(d-k)$-biased ordered chain game on $4\mathcal{W}^{d}$.

conjecture | open | MathDB 页面 | 原文线索

V2-041496 | MATHDB-322008

Baernstein's extremal exit-time conjecture for schlicht functions

摘录 / EXCERPT: Let ${\cal S}$ be the class of normalized univalent functions on the unit disk, let ${\cal K}$ denote the Koebe functions, and let K be the Koebe function. For $f\in {\cal S}$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-041497 | MATHDB-322014

The strongly stratified and locally conical reflection arrangement conjecture

摘录 / EXCERPT: Let W be a well-generated reflection group acting on a complex vector space V. A frame Λ is the auxiliary structure used to define the properties strongly stratified an...

conjecture | open | MathDB 页面 | 原文线索

V2-041498 | MATHDB-322028

Folklore conjecture on the final size of random greedy triangle packing

摘录 / EXCERPT: Folklore conjecture. With high probability,

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V2-041499 | MATHDB-322031

Conjectured face numbers of the Tutte polytopes

摘录 / EXCERPT: Let n be a positive integer, let $0<q<1$ and $t>0$, and let ${{\text{\hskip.03cm\bf{T}}}}_n(q,t)$ denote the Tutte polytope defined in the source. Conjectured face numbers of the...

conjecture | open | MathDB 页面 | 原文线索

V2-041500 | MATHDB-322037

Sheffield's uniqueness conjecture for extremal Brownian Gibbs line ensembles

摘录 / EXCERPT: Let $\mathcal{A}$ be an $\mathbb{N}$-indexed line ensemble. Say that $\mathcal{A}$ is x-invariant if $\mathcal{A}(s+cdot)$ has the same distribution as $\mathcal{A}$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-041501 | MATHDB-322052

Grünbaum's conjecture on unstretchable simplicial arrangements

摘录 / EXCERPT: Grünbaum's conjecture. Up to isomorphism, there are only 5 simplicial unstretchable arrangements of 15 or 16 pseudolines.

conjecture | open | MathDB 页面 | 原文线索

V2-041502 | MATHDB-322053

Grünbaum's classification conjecture for simplicial line arrangements

摘录 / EXCERPT: Grünbaum's classification conjecture. There are no simplicial arrangements of straight lines besides the three infinite families and 90 sporadic arrangements listed in Grünbaum's...

conjecture | open | MathDB 页面 | 原文线索

V2-041503 | MATHDB-322063

The characterization of Ramsey-perfect numbers by Ramsey twins

摘录 / EXCERPT: Let $R(3,k)$ denote the Ramsey number for triangles versus independent sets of size k, and let a Ramsey twin be a pair of consecutive Ramsey numbers. A natural number is Ramsey-p...

conjecture | open | MathDB 页面 | 原文线索

V2-041504 | MATHDB-322064

The classification of Ramsey twins

摘录 / EXCERPT: A Ramsey twin is a pair of consecutive Ramsey numbers. Ramsey-twin classification conjecture. The only Ramsey twins are $\{3,6\}$ and $\{6,9\}$. The conjecture is motivated by the...

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V2-041505 | MATHDB-322077

The universal-word conjecture for demipotents in the 0-Hecke algebra

摘录 / EXCERPT: Universal-word conjecture. The word $u_N=(1,2,\ldots,N-2,N-1,N-2,\ldots,2,1)$ is universal. This conjecture proposes a uniform, simple expression whose signed versions produce demi...

conjecture | open | MathDB 页面 | 原文线索

V2-041506 | MATHDB-322079

Uniform error conjecture for symmetric integer matrices with prescribed row sums

摘录 / EXCERPT: Uniform error conjecture. For $n\geq 5$ and $\ell\geq 1$, one has

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V2-041507 | MATHDB-322085

Ruzsa's exponential bound conjecture for the Freiman-Ruzsa theorem

摘录 / EXCERPT: Let $F(r,K)$ denote the smallest bound such that, for every abelian group G in which every element has order at most r, every finite subset $A\subseteq G$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-041508 | MATHDB-322088

Sign-reversing involution conjecture for plethysm coefficients

摘录 / EXCERPT: Sign-reversing cancellation conjecture. There are a subset $S_0\subset\operatorname{SSTab}(\lambda[\mu])$ and a bijective map

conjecture | open | MathDB 页面 | 原文线索

V2-041509 | MATHDB-322096

Wilson's joint asymptotic enumeration conjecture for Steiner triple systems and 1-factorizations

摘录 / EXCERPT: Let $STS(n)$ be the number of Steiner triple systems on n points and let $F(n)$ be the number of 1-factorizations of the complete graph K_n. Wilson's joint asymptotic conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-041510 | MATHDB-322106

Bousquet-Mélou-Xin conjecture on non-D-finiteness of noncrossing partition generating functions

摘录 / EXCERPT: Let k-noncrossing set partitions be set partitions whose arc diagrams contain no k-crossing, and let their ordinary generating function count them by size. Bousquet-Mélou-Xin c...

conjecture | open | MathDB 页面 | 原文线索

V2-041511 | MATHDB-322107

Non-D-finiteness conjecture for nonnesting permutation generating functions

摘录 / EXCERPT: For each integer k, let k-nonnesting permutations be permutations whose corresponding arc diagrams contain no k-nesting, and let the ordinary generating function count these...

conjecture | open | MathDB 页面 | 原文线索

V2-041512 | MATHDB-322109

Geodesic containment conjecture for the Tamari lattice

摘录 / EXCERPT: Let T_1 and T_2 be trees with an upper bound in the Tamari lattice $\mathscr{C}_n$, and let $\mathscr{R}_n$ be the rotation graph on these trees. A shortest path in...

conjecture | open | MathDB 页面 | 原文线索

V2-041513 | MATHDB-322125

Neighborliness conjecture for permutation polytopes of cyclic groups

摘录 / EXCERPT: Let $G=\langle g\rangle$ be the cyclic permutation group of order d, acting with t orbits, and for $I\subseteq [t]$ let d_I denote the relevant order associated with the unio...

conjecture | open | MathDB 页面 | 原文线索

V2-041514 | MATHDB-322138

Polynomial partition-value limit conjecture

摘录 / EXCERPT: Let $\lambda=(a_1,\ldots,a_t)$ be a partition of n, written $\lambda\vdash n$, and let f be a polynomial of degree d with $d\geq 2$. Polynomial partition-value limit conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-041515 | MATHDB-322149

The induced-subposet conjecture for weighted games

摘录 / EXCERPT: Let J_n be the poset of linear games on n voters, and let $\mathcal{W}_n$ be the weighted games poset, consisting of all weighted voting systems on n voters with the partial...

conjecture | open | MathDB 页面 | 原文线索

V2-041516 | MATHDB-322213

Combinatorial R-matrix compatibility of the rigged-configuration bijection

摘录 / EXCERPT: Let $(\nu,J)\in\operatorname{RC}(B)$, where $B=B_1\otimes B_2$ is a two-fold tensor product of Kirillov--Reshetikhin crystals, and let $B'=B_2\otimes B_1$ be the tensor product wit...

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V2-041517 | MATHDB-322226

Conservativity of an ultrafilter of IP sets over \a0${\mathbf{ATR_0}}$

摘录 / EXCERPT: An IP set is a set of natural numbers containing all finite sums of an infinite sequence of natural numbers. Let $\mathfrak{U}$ be an ultrafilter on $\mathbb{N}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041518 | MATHDB-322227

Conservativity of ultrafilters with the P-point and Ramsey properties

摘录 / EXCERPT: Let T be one of ... An ultrafilter $\mathfrak{U}$ is a P-point if, for every partition $\mathbb{N}=S_0\cup S_1\cup\cdots$ with $S_n\notin\mathfrak{U}$ for every n, there is an...

conjecture | open | MathDB 页面 | 原文线索

V2-041519 | MATHDB-322252

Linearity conjecture for the first present sum in a sumset

摘录 / EXCERPT: First-present-sum linearity conjecture. For large k,

conjecture | open | MathDB 页面 | 原文线索

V2-041520 | MATHDB-322253

Linearity conjecture for the left fringe of a sumset

摘录 / EXCERPT: Let $A\subseteq[0,n-1]$ and define ... Let $W_n(A)$ denote the left endpoint of an interval of consecutive present sums in $A+A$, where $W_n(A)$ is the quantity defined in the surrou...

conjecture | open | MathDB 页面 | 原文线索

V2-041521 | MATHDB-322269

The modified uniform K-stability conjecture for toric extremal metrics

摘录 / EXCERPT: Modified uniform K-stability conjecture. The pair (M,ω) has a metric within the class $[\omega]$ that solves the Abreu equation if and only if (Δ,A) is uniformly K-st...

conjecture | open | MathDB 页面 | 原文线索

V2-041522 | MATHDB-322281

Huang-Rota's column Latin square conjecture

摘录 / EXCERPT: For a positive even integer m, let $CELS(m)$ and $COLS(m)$ denote respectively the sets of column-even and column-odd Latin (m,m)-squares. Huang-Rota's column Latin square conj...

conjecture | open | MathDB 页面 | 原文线索

V2-041523 | MATHDB-322301

Burton's completability conjecture for Latin-square t-plexes

摘录 / EXCERPT: Let a t-plex of a $M \times M$ Latin Square be the partially filled array described in the surrounding construction, with t entries in each row, column, and symbol class. Burto...

conjecture | open | MathDB 页面 | 原文线索

V2-041524 | MATHDB-322322

Stanley's toric g-vector conjecture for Gorenstein* meet-semilattices

摘录 / EXCERPT: Let P be a Gorenstein meet-semilattice, meaning a meet-semilattice with minimum element whose order complex after removing the minimum is Gorenstein. Denote its toric g-vector...

conjecture | open | MathDB 页面 | 原文线索

V2-041525 | MATHDB-322325

The growth-rate asymptotic conjecture for tree-pattern avoidance

摘录 / EXCERPT: Growth-rate asymptotic conjecture. For every set of labels X, there exists an integer d such that, for every pattern $P\in\mathcal{LT}(X)$ with at least d internal vertices,...

conjecture | open | MathDB 页面 | 原文线索

V2-041526 | MATHDB-322326

The easiest and hardest tree-pattern avoidance conjecture for combs

摘录 / EXCERPT: Comb avoidance extremality conjecture. The patterns $LC^<_n$, $LC^>_n$, and RC_n are the easiest to avoid, while H_n is the hardest to avoid.

conjecture | open | MathDB 页面 | 原文线索

V2-041527 | MATHDB-322332

Conjecture on the roots of the greedy Galois-game polynomials

摘录 / EXCERPT: Root-distribution conjecture. The sequence α_k is increasing and

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V2-041528 | MATHDB-322348

The Condorcet generating-function conjecture for the coefficients $b_{k,\ell}$

摘录 / EXCERPT: The Condorcet generating-function conjecture. (i) The series $B(x,P)$ converges for $0\leq x<\frac12$ and $x\leq P\leq\frac12$; (ii) $B(x,P)\geq 0$ on the same region; and (iii)

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V2-041529 | MATHDB-322352

Serial symmetric exchange conjecture for matroid bases

摘录 / EXCERPT: Serial symmetric exchange conjecture. For every subset $A_{1}\subset A$, there exists a subset $B_{1}\subset B$ such that A_{1} and B_{1} form a serial symmetric exchange relat...

conjecture | open | MathDB 页面 | 原文线索

V2-041530 | MATHDB-322353

Stones-Wanless conjecture on reduced Latin squares

摘录 / EXCERPT: Let $rels(n)$ and $rols(n)$ denote the numbers of reduced even and reduced odd Latin squares of order n, respectively. Stones-Wanless conjecture. ... for all positive n. The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-041531 | MATHDB-322354

Weak Rota basis conjecture for scrambled systems in odd dimensions

摘录 / EXCERPT: Let $S_1,\ldots,S_n$ be a scrambled system of bases of $\mathbb{R}^n$: each S_i has size n, and $\bigcup S_i$ is the multiset union of n bases. An independent transversal con...

conjecture | open | MathDB 页面 | 原文线索

V2-041532 | MATHDB-322380

The freeness conjecture for the Hopf monoid of acyclic orientations

摘录 / EXCERPT: Let $\mathbf{g}$ be a positive graphical species, and let $\overrightarrow{\mathbf{A}}$ denote the graphical species of acyclic orientations. Write $\circ$ for the composition prod...

conjecture | open | MathDB 页面 | 原文线索

V2-041533 | MATHDB-322395

The Kolakoski sequence digit distribution conjecture

摘录 / EXCERPT: Let $K=(K_n)_{n=1}^{\infty}$ be the classical Kolakoski sequence on the alphabet $\{\mathtt{1},\mathtt{2}\}$, and let $o_n=|\{i:K_i=\mathtt{1},\ 1\leq i\leq n\}|$ be the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-041534 | MATHDB-322500

Universal rotor-router classification conjecture

摘录 / EXCERPT: A rotor type is a finite sequence of directions; a rotor type is palindromic if it reads the same forwards and backwards, and block-repetitive if it is a concatenation of uniform b...

conjecture | open | MathDB 页面 | 原文线索

V2-041535 | MATHDB-322501

Universality conjecture for rotor classes $\mathbf{C}(p,b)$

摘录 / EXCERPT: Let $p\in\{0,1\}$ and $b\in\mathbb{N}$. Define $\mathbf{C}(p,b)$ to be the class of rotor types that are palindromic if and only if $p=1$ and are block-repetitive with block length...

conjecture | open | MathDB 页面 | 原文线索

V2-041536 | MATHDB-322502

Compressor universality conjecture for rotor types

摘录 / EXCERPT: Let $\mathbf{C}(p,b)$ be the class of rotor types that are palindromic if and only if $p=1$ and block-repetitive with block length b, but no larger block length, where...

conjecture | open | MathDB 页面 | 原文线索

V2-041537 | MATHDB-322530

Generalized Cambrian denominator-vector theorem

摘录 / EXCERPT: Let B be an exchange matrix, and let Theorem denote the source's theorem identifying the combinatorial Cambrian data with denominator vectors of cluster variables. Generalized Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-041538 | MATHDB-322531

Conjecture on monotone triangle evaluations and vertically symmetric alternating sign matrices

摘录 / EXCERPT: Let $\alpha(n;k_1,\ldots,k_n)$ denote the number of monotone triangles with bottom row $(k_1,\ldots,k_n)$, extended to arbitrary integer bottom rows by the associated evaluation. F...

conjecture | open | MathDB 页面 | 原文线索

V2-041539 | MATHDB-322541

Zhao's density conjecture for numerical semigroups with small Frobenius number

摘录 / EXCERPT: Let n_g be the number of numerical semigroups of genus g, and let t_g be the number of numerical semigroups Λ of genus g satisfying $f(\Lambda)<3m(\Lambda)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-041540 | MATHDB-322571

The extremal consecutive-level conjecture for trace-Sperner families

摘录 / EXCERPT: Let k and l' be positive integers with $l'<k$. An $(n-l')$-trace k-Sperner family is a family ${\mathcal F}\subseteq 2^{[n]}$ whose trace on every $(n-l')$-set is k-Sperner...

conjecture | open | MathDB 页面 | 原文线索

V2-041541 | MATHDB-322572

The small-parameter trace-Sperner asymptotic conjecture

摘录 / EXCERPT: Let $f(n,k,l)$ be the maximum size of an l-trace k-Sperner family ${\mathcal F}\subseteq 2^{[n]}$. For a family, being l-trace k-Sperner means that the trace on every l-s...

conjecture | open | MathDB 页面 | 原文线索

V2-041542 | MATHDB-322588

Laplacian estimate for the Chen-LeBrun-Weber extremal Kähler metric

摘录 / EXCERPT: Let k be the extremal Kähler metric on $\mathbb{C} P^2\sharp 2\overline{\mathbb{C} P^2}$ such that $s^{-2}k$ is the Chen-LeBrun-Weber metric with Einstein constant equal to 3,...

conjecture | open | MathDB 页面 | 原文线索

V2-041543 | MATHDB-322621

Special-values conjecture for coefficients of cap-shaped link patterns

摘录 / EXCERPT: Let $\mathcal{D}_n$ be the set of link patterns of size n, and let $c_{\alpha\pi}$ be the coefficients introduced in the paper. Write link patterns as binary words, with...

conjecture | open | MathDB 页面 | 原文线索

V2-041544 | MATHDB-322622

Wieland-type chord-diagram identity for fully packed loops

摘录 / EXCERPT: Let $i,j,\ell,m\geq 0$, and let A_X denote the number associated with a link pattern X in the paper's chord-diagram notation. Wieland-type chord-diagram conjecture. The equalit...

conjecture | open | MathDB 页面 | 原文线索

V2-041545 | MATHDB-322662

Singularity probability conjecture for random fixed-row-sum matrices

摘录 / EXCERPT: Let n be even, and let Q_n be a random n by n $(0,1)$ matrix whose rows are independent vectors with exactly $n/2$ zero components. Fixed-row-sum singularity probability co...

conjecture | open | MathDB 页面 | 原文线索

V2-041546 | MATHDB-322663

Gao-Ruzsa-Thangadurai conjecture on Olson's constant in finite-dimensional vector spaces

摘录 / EXCERPT: Let $\mathcal OL(\mathbf F_p^d)$ denote the Olson constant of the additive group $\mathbf F_p^d$, namely the smallest integer such that no subset of that group of the given cardina...

conjecture | open | MathDB 页面 | 原文线索

V2-041547 | MATHDB-322664

Asymptotic upper-bound conjecture for Olson's constant in finite-dimensional vector spaces

摘录 / EXCERPT: Let $\mathcal OL(\mathbf F_p^d)$ denote the Olson constant of the additive group $\mathbf F_p^d$. Fix an integer $d\ge 2$ and a real number $\gamma>0$. Asymptotic Olson-constant co...

conjecture | open | MathDB 页面 | 原文线索

V2-041548 | MATHDB-322668

The parity conjecture for cyclic triangulations of $S^2 \times S^1$

摘录 / EXCERPT: Parity conjecture. M has an even number of vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-041549 | MATHDB-322670

The asymptotic constant conjecture for maximum-weight monochromatic cliques

摘录 / EXCERPT: Let $r(n)$ be the diagonal two-color Ramsey number and assume that ... exists. For a red-blue coloring of K_n, let $f(n)$ be the largest real number such that every coloring contai...

conjecture | open | MathDB 页面 | 原文线索

V2-041550 | MATHDB-322684

Conjecture on left-crossing statistics in matchings and permutations

摘录 / EXCERPT: Conjecture on left-crossing statistics. The sets $M(n,A,B,C,D)$ and $P(n,A,B,C,D)$ are equinumerous for every positive integer n and all $A,B,C,D\subseteq[n]$.

conjecture | open | MathDB 页面 | 原文线索

V2-041551 | MATHDB-322701

The missing lemma for the modulo 23 Schröter-Russell-Ramanujan identity

摘录 / EXCERPT: The missing lemma. Condition (i) of Theorem holds for these parameters.

conjecture | open | MathDB 页面 | 原文线索

V2-041552 | MATHDB-322704

The ordinal Ramsey bound conjecture for well-founded trees

摘录 / EXCERPT: A well-founded tree is a prefix-closed set of finite sequences, and its order type is the ordinal order type used in the paper. A coloring of such a tree assigns each node one of t...

open_problem | open | MathDB 页面 | 原文线索

V2-041553 | MATHDB-322705

Exact critical threshold for simple games

摘录 / EXCERPT: Let $c_\mathcal{S}(n)$ denote the maximum critical threshold value among simple games on n voters. Simple-game threshold conjecture. For every $n\ge 4$, ... The paper proves the di...

conjecture | open | MathDB 页面 | 原文线索

V2-041554 | MATHDB-322706

Theta-square-root growth of complete-game critical thresholds

摘录 / EXCERPT: Let $c_\mathcal{C}(n)$ denote the maximum critical threshold value of a complete simple game on n voters. Complete-game growth conjecture. ... This is the asymptotic form of the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-041555 | MATHDB-322716

Nonnegative coefficient conjecture for three-residue partition products

摘录 / EXCERPT: Three-residue coefficient conjecture. If z does not divide y, then all coefficients $a(L,y,n,x,z,m)$ are non-negative; if z divides y, then only finitely many of these coef...

conjecture | open | MathDB 页面 | 原文线索

V2-041556 | MATHDB-322775

The continuous face-vector representation conjecture for flag complexes

摘录 / EXCERPT: Continuous face-vector representation conjecture. There are such numbers satisfying, for every $1\leq j\leq d$,

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V2-041557 | MATHDB-322776

The scaled face-vector conjecture for flag complexes

摘录 / EXCERPT: Scaled face-vector conjecture. There is an integer q such that $(1,qc_1,q^2c_2,\dots,q^dc_d)$ is the face vector of a flag complex.

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V2-041558 | MATHDB-322777

The equal-block minimizer conjecture for multipartite flag-complex constructions

摘录 / EXCERPT: Equal-block minimizer conjecture. The unique choice of constants minimizing

conjecture | open | MathDB 页面 | 原文线索

V2-041559 | MATHDB-322796

General-region period-two scale invariance conjecture for CIS-Nim

摘录 / EXCERPT: Let F be a forbidden set defining an instance of CIS-Nim, and let $S ameq\subseteq\mathbb{R}^3$ be open. For each k, let $\pi(S,k)$ denote the number of P-positions of the fo...

conjecture | open | MathDB 页面 | 原文线索

V2-041560 | MATHDB-322797

Labeled-pile and arbitrary-heap period-two scale invariance conjecture for CIS-Nim

摘录 / EXCERPT: Let m be any positive integer, let F be a set of positions in m-heap Nim with labeled piles, and let $S\subseteq\mathbb{R}^m$ be open. For each k, let $\pi(S,k)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-041561 | MATHDB-322828

Kerr-de Sitter Cauchy-horizon extendibility conjecture

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be the maximal Cauchy development for the Einstein vacuum equations with positive cosmological constant of sufficiently small perturbations of subextremal Ker...

conjecture | open | MathDB 页面 | 原文线索

V2-041562 | MATHDB-322829

Near-extremal Reissner-Nordström-de Sitter regularity conjecture

摘录 / EXCERPT: Let the initial data be sufficiently close to a Reissner-Nordström-de Sitter solution whose parameters are sufficiently close to extremal but still subextremal, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041563 | MATHDB-322830

Near-extremal Kerr-de Sitter failure of $L^2_{\rm loc}$ strong cosmic censorship

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be the maximal Cauchy development of sufficiently small perturbations of Kerr-de Sitter data with n black-hole regions, initial topology...

conjecture | open | MathDB 页面 | 原文线索

V2-041564 | MATHDB-322856

Period formula conjecture for subtraction games with FES sets of size 3

摘录 / EXCERPT: Period formula conjecture. If there exists a multiple m of $2a$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-041565 | MATHDB-322860

A four-point multiplicative configuration conjecture in finite groups

摘录 / EXCERPT: Let G be a finite group, and let $S\subset G\times G$. For a fixed density parameter $\delta>0$, consider sets satisfying $|S|\geq \delta|G|^2$. Four-point configuration conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-041566 | MATHDB-322897

The conjecture on URC-manifolds associated with indecomposable simple cells

摘录 / EXCERPT: Let P be a simple cell, and let $\mathcal{R}_P$ denote the associated manifold. A simple cell is indecomposable in the relevant sense if it is not combinatorially equivalent to a...

conjecture | open | MathDB 页面 | 原文线索

V2-041567 | MATHDB-322907

The square-root agreement-subtree conjecture for balanced phylogenetic trees

摘录 / EXCERPT: Let T_1 and T_2 be balanced, rooted, binary trees of height m. An agreement subtree is a common subtree of the two trees on a shared set of leaves. The square-root agreement-...

conjecture | open | MathDB 页面 | 原文线索

V2-041568 | MATHDB-322961

Asymptotic positivity for self-conjugate core partitions

摘录 / EXCERPT: Asymptotic positivity conjecture for self-conjugate core partitions. As n tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-041569 | MATHDB-323004

Cyclically distinct composition interpretation for the leafed-cycle cone

摘录 / EXCERPT: Cyclically distinct composition conjecture. The number of integer points $\lambda\in\mathcal{C}_n$ with $\lambda_0=m$ is equal to the number of compositions of m into n parts t...

conjecture | open | MathDB 页面 | 原文线索

V2-041570 | MATHDB-323010

Linear singular-index conjecture for unsatisfiable hitting clause-sets

摘录 / EXCERPT: Let $\mathcal{U}\mathcal{HIT}_{\!\delta=k}$ be the class of unsatisfiable hitting clause-sets of deficiency k. Let $\operatorname{singind}(F)$ denote the singular index of F, a...

conjecture | open | MathDB 页面 | 原文线索

V2-041571 | MATHDB-323013

Non-expandability conjecture for ultimately bipartite subtraction games

摘录 / EXCERPT: Let S be the subtraction set of an ultimately bipartite subtraction game, meaning that its nim-sequence is eventually periodic with period two and alternates between 0 and 1....

conjecture | open | MathDB 页面 | 原文线索

V2-041572 | MATHDB-323014

Period-equality conjecture for nim-sequences and zero-position sequences

摘录 / EXCERPT: Period-equality conjecture. For every subtraction game, the sequences $\{\mathcal{G}(n)\}$ and $\{\mathcal{V}(n)\}$ have the same period p. Since subtraction-game nim-sequences a...

conjecture | open | MathDB 页面 | 原文线索

V2-041573 | MATHDB-323032

Packing/covering conjecture for infinite matroid families

摘录 / EXCERPT: Let $(M_k\mid k\in K)$ be a family of matroids on the same ground set E. A packing consists of pairwise disjoint spanning sets, one for each M_k, and a covering consists of ind...

conjecture | open | MathDB 页面 | 原文线索

V2-041574 | MATHDB-323033

Unhindered-family covering conjecture

摘录 / EXCERPT: Let $(M_k\mid k\in K)$ be a family of matroids. A wave is a set together with disjoint spanning sets for the restrictions of the matroids to that set; a family is unhindered when i...

conjecture | open | MathDB 页面 | 原文线索

V2-041575 | MATHDB-323034

Packing conjecture for infinite matroid families

摘录 / EXCERPT: Let $(M_k\mid k\in K)$ be a family of matroids on the same ground set E. A covering for $(M_k.Y\mid k\in K)$ is a family of independent sets whose union covers Y, and a packing...

conjecture | open | MathDB 页面 | 原文线索

V2-041576 | MATHDB-323048

Period-$2m$ conjecture for arithmetic-progression Max-Welter positions

摘录 / EXCERPT: Let $a\geq 0$, $m\geq 1$, and $k\geq 1$. Then the sequence ... is ultimately periodic with period $p=2m$. This conjectures periodicity of the Sprague-Grundy values for positions whos...

conjecture | open | MathDB 页面 | 原文线索

V2-041577 | MATHDB-323058

David-Felsner-Rödl conjecture on maximal antichains and independent sets

摘录 / EXCERPT: David-Felsner-Rödl conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-041578 | MATHDB-323062

Periodicity conjecture for weighted finite taking-no-breaking games

摘录 / EXCERPT: Let $O=(n_1n_2\dots n_t,p_1p_2\dots p_t)$ and $P=(m_1m_2\dots m_l,q_1q_2\dots q_l)$ be two finite taking-no-breaking octal games. Assume that at least one n_s is neither 0 nor...

conjecture | open | MathDB 页面 | 原文线索

V2-041579 | MATHDB-323063

Eventual periodicity for finite taking-no-breaking octal games under the disjunctive sum

摘录 / EXCERPT: Disjunctive-sum periodicity conjecture. For every m, there exists N such that

conjecture | open | MathDB 页面 | 原文线索

V2-041580 | MATHDB-323064

Common eventual period under the selective sum

摘录 / EXCERPT: Selective-sum period conjecture. If $\mathcal{G}_s(n+_{\ell}m)$ eventually has period p, then $\mathcal{G}_s(n\triangledown m)$ also eventually has period p.

conjecture | open | MathDB 页面 | 原文线索

V2-041581 | MATHDB-323065

Common eventual period under the sequential join

摘录 / EXCERPT: Sequential-join period conjecture. If $\mathcal{G}_s(n+_{\ell}m)$ eventually has period p, then $\mathcal{G}_s(n\rhd m)$ also eventually has period p.

conjecture | open | MathDB 页面 | 原文线索

V2-041582 | MATHDB-323069

Borg's EKR conjecture for simplicial complexes

摘录 / EXCERPT: Let $t\leq r$ be two natural numbers. A family of faces is t-intersecting if every pair of faces in it has intersection of cardinality at least t. Let Δ be a simplicial...

conjecture | open | MathDB 页面 | 原文线索

V2-041583 | MATHDB-323073

A small-sumset counting conjecture

摘录 / EXCERPT: For positive integers n, let $[n]=\{1,\ldots,n\}$, and for a finite set S write $S+S=\{a+b:a,b\in S\}$. Small-sumset counting conjecture. For every $\delta>0$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-041584 | MATHDB-323085

Main conjecture on the recursive structure of fusible numbers

摘录 / EXCERPT: Let ${\mathcal{F}}$ be the set of fusible numbers. For $a\in{\mathcal{F}}$, let $s(a)$ be its successor, let $m(a)$ denote the relevant minimum gap, and write ... as before. Main con...

conjecture | open | MathDB 页面 | 原文线索

V2-041585 | MATHDB-323129

Asymptotic parity-class distribution conjecture for reduced Latin squares

摘录 / EXCERPT: Asymptotic parity-class distribution conjecture. If n is odd, then for all $k,m<n/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-041586 | MATHDB-323164

The closed-flow enumeration conjecture for forks

摘录 / EXCERPT: Let $\mathtt{Frk}_{i,n-i}$ be the fork with n vertices and stem of size i, and let $\#\mathbb{F}(\mathtt{Frk}_{i,n-i},k,0)$ denote the number of closed flows of size k on thi...

conjecture | open | MathDB 页面 | 原文线索

V2-041587 | MATHDB-323165

The saturated-flow formula for the PreLie quotient series

摘录 / EXCERPT: Let $\mathscr{Y}=\mathscr{E}\circ\mathscr{D}^{-1}$ be the quotient series in the group $\mathsf{G}_{\operatorname{PreLie}}$, let $\mathscr{Y}_T$ be the coefficient of a rooted tree...

conjecture | open | MathDB 页面 | 原文线索

V2-041588 | MATHDB-323179

Diagonal uniqueness conjecture for Sprague-Grundy values of F-Wythoff

摘录 / EXCERPT: Diagonal uniqueness conjecture. For all nonnegative integers a and g, there exists a unique integer b such that

conjecture | open | MathDB 页面 | 原文线索

V2-041589 | MATHDB-323219

Integrality conjecture for the alpha-triangle

摘录 / EXCERPT: Let $\alpha_{m,z}$ be defined by ... with $\alpha_{m,0}=1$ for $m\geq0$, $\alpha_{m,2m}=0$ for $m>0$, and $\alpha_{m,z}=0$ for $z>2m$. The resulting triangular array is called the...

conjecture | open | MathDB 页面 | 原文线索

V2-041590 | MATHDB-323220

Alpha-beta triangle decomposition conjecture

摘录 / EXCERPT: Let $\alpha_{m,z}$ be the entries of the α-triangle, and let $\beta_{z,y}$ be entries of a triangular array called the β-triangle. Alpha-beta decomposition conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-041591 | MATHDB-323221

Meeussen's Hermite-polynomial conjecture for the beta-triangle

摘录 / EXCERPT: Let $\beta_{n,0}$ denote the entries in the first column of the β-triangle, and let $H_n(x)$ be the n-th Hermite polynomial. Meeussen's conjecture. ... This conjecture specif...

conjecture | open | MathDB 页面 | 原文线索

V2-041592 | MATHDB-323222

Hanna's matrix conjecture for plane-partition numbers

摘录 / EXCERPT: Let $p_d(n)$ denote the relevant sequence of partition numbers. Hanna's conjecture. There exists a lower-triangular matrix $T=(\tau_{ij})$, with $i,j=0,1,2,\ldots$, whose entries a...

conjecture | open | MathDB 页面 | 原文线索

V2-041593 | MATHDB-323223

Hanna-Meeussen polynomiality conjecture for the T-matrix

摘录 / EXCERPT: Let $T=(\tau_{ij})$ be the lower-triangular integer matrix with ones on its diagonal associated with the sequence $p_d(n)$. Hanna-Meeussen conjecture. For the relevant indices m...

conjecture | open | MathDB 页面 | 原文线索

V2-041594 | MATHDB-323237

Monotonicity conjecture for ordinarization-number counts

摘录 / EXCERPT: Ordinarization-count monotonicity conjecture. For every $g,r\in{\mathbb N}_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-041595 | MATHDB-323244

The asymptotic covering constant for permutations

摘录 / EXCERPT: Let S_n be the set of permutations of $\{1,\ldots,n\}$, and let κ_n denote the minimum number of $(n+1)$-permutations needed to cover every n-permutation. Asymptotic cov...

conjecture | open | MathDB 页面 | 原文线索

V2-041596 | MATHDB-323245

Uniqueness of Ext multiplicities for basic highest weight categorifications

摘录 / EXCERPT: Multiplicity invariance conjecture. These numbers depend only on the combinatorial type of $\mathcal{C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-041597 | MATHDB-323266

Nonnegative-coefficient conjecture for rank enumerators of Rothe diagrams

摘录 / EXCERPT: Let $w\in\mathfrak{S}_n$ be a permutation, and let R_w be its Rothe diagram. For a finite field with q elements, let $\operatorname{mat}_q(n,R_w,r)$ count rank-r matrices who...

conjecture | open | MathDB 页面 | 原文线索

V2-041598 | MATHDB-323267

Poincaré-polynomial upper-bound conjecture for Rothe diagrams

摘录 / EXCERPT: Fix a permutation $w\in\mathfrak{S}_n$, and let R_w be its Rothe diagram. Let $\operatorname{inv}(w)$ denote the inversion number of w, and let $P_w(q)$ be the Poincaré polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-041599 | MATHDB-323268

Upper-bound conjecture via the northwest-justified diagram of a permutation

摘录 / EXCERPT: Fix a permutation $w\in\mathfrak{S}_n$, let R_w be its Rothe diagram, and let $H_L(w)$ be the associated northwest-justified diagram. Write $\#H_L(w)$ for the number of cells of...

conjecture | open | MathDB 页面 | 原文线索

V2-041600 | MATHDB-323274

The Haglund-Haiman-Loehr-Remmel-Ulyanov conjecture for diagonal coinvariants

摘录 / EXCERPT: Haglund-Haiman-Loehr-Remmel-Ulyanov conjecture. We have an identity

conjecture | open | MathDB 页面 | 原文线索

V2-041601 | MATHDB-323289

The toroidal no-three-in-line conjecture involving the divisor function

摘录 / EXCERPT: Let $a(n)$ be the smallest multiple of the number of divisors of n that is greater than or equal to n. On an $n\times n$ discrete torus, points are required to have no three in...

conjecture | open | MathDB 页面 | 原文线索

V2-041602 | MATHDB-323299

The flag-polytope conjecture for URC-manifolds

摘录 / EXCERPT: Let P be an n-dimensional flag simple polytope, meaning that every set of pairwise intersecting facets has nonempty total intersection, and suppose that P is not combinatoria...

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V2-041603 | MATHDB-323303

The conjecture that the line-intersection bound is tight

摘录 / EXCERPT: Let $d\ge 2$ and $n\ge 2$. Let $f_d(n)$ denote the largest volume of a d-dimensional open set with n connected components and no pair of points at an integral distance. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-041604 | MATHDB-323308

The quadratic growth conjecture for colored radial orderings

摘录 / EXCERPT: Let $g_\mathrm{col}(n)$ denote the relevant extremal quantity for colored radial orderings of bichromatic sets of $2n$ points. Quadratic colored-ordering conjecture. ... The paper no...

conjecture | open | MathDB 页面 | 原文线索

V2-041605 | MATHDB-323328

PFR conjecture in $\mathbb{Z}_m^n$

摘录 / EXCERPT: PFR conjecture. If $|A-A|\leq\lambda|A|$, then there is a subgroup $H\leq\mathbb{Z}_m^n$ with $|H|\leq|A|$ such that A is covered by $\lambda'=\lambda^{c_m}$ translates of H, w...

conjecture | open | MathDB 页面 | 原文线索

V2-041606 | MATHDB-323374

Second-order asymptotics for the critical probability of r-neighbor bootstrap percolation

摘录 / EXCERPT: Let $d\geq r\geq 2$, and let $p_c([n]^d,r)$ denote the critical probability for r-neighbor bootstrap percolation on the d-dimensional grid. Write $\lambda(d,r)$ for the leading...

conjecture | open | MathDB 页面 | 原文线索

V2-041607 | MATHDB-323383

The missing axiom conjecture for matroid representability in MSOL

摘录 / EXCERPT: Let a finite matroid be a finite set $E^{\mathcal{M}}$ equipped with a rank function ... Let MSOL denote monadic second-order logic for matroids, whose structures use finite ground s...

conjecture | open | MathDB 页面 | 原文线索

V2-041608 | MATHDB-323384

The MSOL strengthening of the paper's main theorem

摘录 / EXCERPT: Let the paper's main theorem be the assertion referred to as Theorem $\mathrm{main2}$, concerning the limitations of finite axiomatization in M-logic. MSOL strengthening conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-041609 | MATHDB-323413

Conjecture on decreasing limiting missing-difference distributions

摘录 / EXCERPT: Decreasing limiting-distribution conjecture. For all integers $m\geq 0$, this limiting function of m is decreasing.

conjecture | open | MathDB 页面 | 原文线索

V2-041610 | MATHDB-323414

Conjecture on decreasing missing-sum distributions

摘录 / EXCERPT: Missing-sum monotonicity conjecture. For $m\geq m^*$, the following assertions hold:

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V2-041611 | MATHDB-323420

Kotlar-Ziv conjecture on independent partial transversals in matroidal Latin squares

摘录 / EXCERPT: Let M be a matroid, and let an MLS of degree n be a matroidal Latin square of degree n, whose rows and columns are bases of M. An independent partial transversal is an inde...

conjecture | open | MathDB 页面 | 原文线索

V2-041612 | MATHDB-323422

Nonvanishing parity difference for reduced Latin squares

摘录 / EXCERPT: Let a Latin square be reduced if its first row and first column are the identity permutation. Let R_n^{E} and R_n^{O} denote the numbers of even and odd reduced L...

conjecture | open | MathDB 页面 | 原文线索

V2-041613 | MATHDB-323423

Bukh's asymptotic conjecture for the extremal number of finite posets

摘录 / EXCERPT: Bukh's conjecture. For all finite posets P,

conjecture | open | MathDB 页面 | 原文线索

V2-041614 | MATHDB-323428

Parity conjecture for the number of square rectangulations

摘录 / EXCERPT: Let t_m denote the total number of rectangulations of a square with m tiles, and write n for a positive integer. Parity conjecture. ... if $n=8k+1$ or $n=8k+4$; otherwise, ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-041615 | MATHDB-323480

Griggs-Lu-Li conjecture on the asymptotic Lubell function of poset-free families

摘录 / EXCERPT: Griggs-Lu-Li conjecture. For every finite poset $\mathcal{P}$, the limit $\pi(\mathcal{P})$ exists and is an integer.

conjecture | open | MathDB 页面 | 原文线索

V2-041616 | MATHDB-323487

Felsner-Krawczyk-Micek crossing-vector extremal conjecture

摘录 / EXCERPT: For positive integers k and w, let $f(k,w)$ be the maximum size of a subset of $\mathbb{Z}^w$ in which every two vectors are 1-crossing but no two vectors are k-crossing. F...

conjecture | open | MathDB 页面 | 原文线索

V2-041617 | MATHDB-323488

Felsner-Krawczyk-Micek conjecture for poset antichains

摘录 / EXCERPT: For a positive integer k, let $\mathbf{k}+\mathbf{k}$ be the poset consisting of two disjoint k-element chains with no comparabilities between distinct chains. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-041618 | MATHDB-323489

Felsner-Krawczyk-Micek generalized crossing-vector conjecture

摘录 / EXCERPT: For $w\ge 1$ and integers $1\le k_1\le\cdots\le k_w$, say that two vectors $A,B\in\mathbb{Z}^w$ are $(k_1,\ldots,k_w)$-crossing if there are coordinates i,j such that...

conjecture | open | MathDB 页面 | 原文线索

V2-041619 | MATHDB-323508

The conjectured swap-distance bounds for graphical degree sequences

摘录 / EXCERPT: For a degree sequence $\mathbf{d}=\{d_1,d_2,\ldots,d_n\}$, let $m=(\sum d_i)/2$ be the number of edges in any realization, and let ... For realizations G_1 and G_2 of this degree...

conjecture | open | MathDB 页面 | 原文线索

V2-041620 | MATHDB-323541

The framing conjecture for iterated balancing of energy-maximal exponent tuples

摘录 / EXCERPT: Framing conjecture. Suppose that all $\Lambda^0(d),\ldots,\Lambda^j(d)$ exist and that $\Lambda^j(d)$ is unframed. Then $\Lambda^0(d),\ldots,\Lambda^{j-1}(d)$ are all framed.

conjecture | open | MathDB 页面 | 原文线索

V2-041621 | MATHDB-323542

Additive-shift multiplicative-structure conjecture for product sets

摘录 / EXCERPT: Additive-shift multiplicative-structure conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041622 | MATHDB-323566

Castravet-Tevelev's hypertree conjecture for extremal divisors

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{0,n}$ be the moduli space of stable n-pointed genus-zero curves, and let D_Γ be the divisor associated with an irreducible hypertree Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-041623 | MATHDB-323604

Borsuk's conjecture on partitioning sets of bounded diameter

摘录 / EXCERPT: Let S be a set of diameter $d>0$ in Euclidean n-space $\mathbb R^n$. Borsuk's conjecture. Every such set is the union of $n+1$ sets, each of diameter less than d. The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-041624 | MATHDB-323606

Kahle's homology vanishing conjecture for random clique complexes

摘录 / EXCERPT: Kahle's homology vanishing conjecture. For each positive integer k and each $\varepsilon>0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-041625 | MATHDB-323610

The minor inequality facet conjecture for circulant set covering polyhedra

摘录 / EXCERPT: Let C_n^k be a circulant matrix, let $W\subset\mathbb Z_n$ define a relevant circulant minor isomorphic to $C_{n'}^{k'}$, and let $Q(C_n^k)$ denote the associated set covering po...

conjecture | open | MathDB 页面 | 原文线索

V2-041626 | MATHDB-323613

Conjecture on the exact value of the weighted Hajnal-Szemerédi threshold

摘录 / EXCERPT: For integers $r\geq 2$ and $t\in(0,1]$, let $\delta(r,t)$ denote the infimum of the minimum weighted-degree density guaranteeing a partition into r-cliques of total weight at lea...

conjecture | open | MathDB 页面 | 原文线索

V2-041627 | MATHDB-323656

Elekes's general-position rich-lines conjecture

摘录 / EXCERPT: Let $A,B\subset \mathbb{R}$ with $|A|=|B|=n$, and say that a line is cn-rich on $A\times B$ if it contains at least cn points of this Cartesian product. A collection of lines i...

conjecture | open | MathDB 页面 | 原文线索

V2-041628 | MATHDB-323661

Kahn's five-value range conjecture for Hamming-cube homomorphisms

摘录 / EXCERPT: Kahn's five-value range conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041629 | MATHDB-323671

Asymptotic normality of the number of independent-set classes in forests

摘录 / EXCERPT: Let $(F^{c(n)}_n)_{n geq 0}$ be a sequence of forests, where $F^{c(n)}_n$ has n vertices and $c(n)$ components. Let $X^{c(n)}_n$ be the number of classes in a uniformly chosen pa...

conjecture | open | MathDB 页面 | 原文线索

V2-041630 | MATHDB-323673

The asymptotic minimum regularity of maximal (2,3)-antichains

摘录 / EXCERPT: A maximal $(2,3)$-antichain on n points is a family of subsets in which every member has size 2 or 3, no member contains another, and which is maximal under these conditions.…

conjecture | open | MathDB 页面 | 原文线索

V2-041631 | MATHDB-323694

Conjecture on the asymptotic dominant orbit of p by 2 by 2 tensors over F_2

摘录 / EXCERPT: Asymptotic-orbit conjecture. As p becomes arbitrarily large, the set of $p \times 2 \times 2$ tensors over $\mathbb{F}_2$ contains a single orbit $\mathcal{O}$ of rank 4 which co...

conjecture | open | MathDB 页面 | 原文线索

V2-041632 | MATHDB-323730

Sum rules for the three singlets in the cyclic model

摘录 / EXCERPT: For odd $N=2n+1$, consider the cyclic spin-chain Hamiltonian with parameters λ,ζ and its three translation-invariant zero-energy singlets Ψ_N^a, indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-041633 | MATHDB-323736

Asymptotic slope conjecture for upper P-positions in Maharaja Nim

摘录 / EXCERPT: Let $k,l\in\mathbb{N}$ with $k<l$, and consider (k,l)-Maharaja Nim, obtained from Wythoff Nim by adjoining moves of the form (k,l) and (l,k). An upper P-position is a losing...

conjecture | open | MathDB 页面 | 原文线索

V2-041634 | MATHDB-323752

Kaneko-Noro-Tsurumaki conjecture on generating multiple zeta value relations

摘录 / EXCERPT: Let $\mathcal{H}^1$ and $\mathcal{H}^2$ be the word algebras used for multiple zeta values, with shuffle product denoted by $?$ and stuffle product by $*$. For an integer n, take...

conjecture | open | MathDB 页面 | 原文线索

V2-041635 | MATHDB-323758

Conjecture on almost-tree, minimal and obstructing clutters

摘录 / EXCERPT: Let d be a positive integer, and let $\mathscr{C}^{\rm a.tree}_d$, $\mathscr{C}^{\min}_d$, and $\mathscr{C}^{\rm obs}_d$ denote, respectively, the classes of d-uniform clutters...

conjecture | open | MathDB 页面 | 原文线索

V2-041636 | MATHDB-323790

Conjecture on minimizers of the number of Lyndon factors

摘录 / EXCERPT: Let $\ell(n)$ be the minimum number of Lyndon factors among Lyndon words of length n: ... A minimizer conjecture. If w is a Lyndon word with $|w|\neq 6$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-041637 | MATHDB-323795

Fischer-Riegler conjecture for vertically symmetric alternating-sign matrices

摘录 / EXCERPT: Let $\alpha(n;k_1,\ldots,k_n)$ denote the generalized monotone-triangle enumeration, and let $n\geq1$. An alternating-sign matrix (ASM) is a square matrix with entries in...

conjecture | open | MathDB 页面 | 原文线索

V2-041638 | MATHDB-323796

Even-argument reciprocity conjecture for generalized monotone triangles

摘录 / EXCERPT: Let $\alpha(n;k_1,\ldots,k_n)$ denote the generalized monotone-triangle enumeration, and let $n\geq1$. Even-argument reciprocity conjecture. The identity ... should hold. The paper l...

conjecture | open | MathDB 页面 | 原文线索

V2-041639 | MATHDB-323797

Block-repetition identities for alternating-sign-matrix numbers

摘录 / EXCERPT: Block-repetition conjecture. The identities

conjecture | open | MathDB 页面 | 原文线索

V2-041640 | MATHDB-323798

One-descent conjecture for generalized monotone triangles

摘录 / EXCERPT: Let $A_n:=\alpha(n;1,2,\ldots,n)$, and let $n\geq2$ and $i=1,\ldots,n-1$. One-descent conjecture. The identity ... should hold. This is a computational conjecture related to a broade...

conjecture | open | MathDB 页面 | 原文线索

V2-041641 | MATHDB-323799

Reverse-duplication conjecture for generalized monotone triangles

摘录 / EXCERPT: Let $A_n:=\alpha(n;1,2,\ldots,n)$, and let $n\geq1$, $i=1,\ldots,n-k+1$, and $k=1,\ldots,n$. Reverse-duplication conjecture. The identity ... should hold. In words, the consecutive s...

conjecture | open | MathDB 页面 | 原文线索

V2-041642 | MATHDB-323800

Hole-one-descent conjecture for generalized monotone triangles

摘录 / EXCERPT: Let $A_{n,j}$ denote the number of alternating-sign matrices of size n whose unique 1 in the first row is in column j, and let $n\geq1$ and $i=1,\ldots,n-1$. Hole-one-descent...

conjecture | open | MathDB 页面 | 原文线索

V2-041643 | MATHDB-323801

A formula for a repeated-entry generalized monotone-triangle enumeration

摘录 / EXCERPT: Let $A_{n-1}:=\alpha(n-1;1,2,\ldots,n-1)$ and let $n\geq4$. Repeated-entry formula conjecture. The identity ... should hold. The formula was obtained from computational experiments a...

conjecture | open | MathDB 页面 | 原文线索

V2-041644 | MATHDB-323802

Polynomial-ratio conjecture for repeated-entry generalized monotone triangles

摘录 / EXCERPT: Let α be the generalized monotone-triangle enumeration, let $A_{n-1}:=\alpha(n-1;1,2,\ldots,n-1)$, and let $n\geq k\geq4$. Polynomial-ratio conjecture. There exist polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-041645 | MATHDB-323826

Generalised lower bound conjecture for triangulated manifolds

摘录 / EXCERPT: Generalised lower bound conjecture for triangulated manifolds. The displayed inequality holds, and equality for some $l<\frac{d-1}{2}$ holds if and only if $M\in {\cal K}_l(d)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-041646 | MATHDB-323828

Kühnel and Lutz's tightness conjecture on strong minimality

摘录 / EXCERPT: Let X be a simplicial complex. Call X $\mathbb{F}$-tight when it has the tightness property over a field $\mathbb{F}$, and call it strongly minimal when it achieves the compone...

conjecture | open | MathDB 页面 | 原文线索

V2-041647 | MATHDB-323830

Extremal Betti table generation conjecture for finite-length bigraded modules

摘录 / EXCERPT: Extremal Betti table generation conjecture. All extremal Betti tables of the cone of bigraded T-modules with finite length are generated by Betti tables of modules M that satis...

conjecture | open | MathDB 页面 | 原文线索

V2-041648 | MATHDB-323844

Modified Calabi-flow convergence conjecture for extremal metrics

摘录 / EXCERPT: Let X be a Kähler manifold with Kähler class $[\omega]$. Suppose that there exists an extremal metric $\omega_0\in[\omega]$. Let $\omega_1\in[\omega]$ be a Kähler metric invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-041649 | MATHDB-323845

Apostolov-Calderbank-Gauduchon-Tønnesen-Friedman projective-bundle conjecture

摘录 / EXCERPT: Let $(X,J)=P(E)$ be a projective bundle over a compact curve Σ of genus at least 2. A subbundle is stable in the usual slope-stability sense. Apostolov-Calderbank-Gauducho...

conjecture | open | MathDB 页面 | 原文线索

V2-041650 | MATHDB-323852

Nonexistence conjecture for simple Venn diagrams with dihedral symmetry

摘录 / EXCERPT: Dihedral-symmetry nonexistence conjecture. There is no simple Venn diagram with dihedral symmetry if $n>3$.

conjecture | open | MathDB 页面 | 原文线索

V2-041651 | MATHDB-323877

Walk enumeration conjecture for square non-zero recurrent configurations

摘录 / EXCERPT: Square non-zero configuration conjecture. The cardinality satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041652 | MATHDB-323878

Symmetry conjecture for q,t-Narayana polynomials of parallelogram polyominoes

摘录 / EXCERPT: q,t-Narayana symmetry conjecture. For all positive integers m,n, the distribution is symmetric:

conjecture | open | MathDB 页面 | 原文线索

V2-041653 | MATHDB-323879

Transpose symmetry conjecture for q,t-Narayana polynomials

摘录 / EXCERPT: Transpose symmetry conjecture. For all positive integers m,n,

conjecture | open | MathDB 页面 | 原文线索

V2-041654 | MATHDB-323905

Factorial growth conjecture for hatted pattern-avoiding permutations

摘录 / EXCERPT: Factorial growth conjecture. The growth rate of $|S_n(\hat\tau_{(i)})|$ is factorial for every hatted pattern $\hat\tau_{(i)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-041655 | MATHDB-323941

PSpace-completeness conjecture for Pirates and Treasure

摘录 / EXCERPT: Consider the decision problem Pirates and Treasure, whose input is a graph G and whose outcome is determined by the game's scoring-play rules. PSpace-completeness conjecture. Pir...

conjecture | open | MathDB 页面 | 原文线索

V2-041656 | MATHDB-323942

Nonzero equivalence conjecture for negative scoring positions

摘录 / EXCERPT: Let $ \mathcal{PT}_{-x}$ be the set of scoring-play game positions in which every node has value $x>0$ and the initial score is 0. Write $G\cong 0$ for equivalence to the zero ga...

conjecture | open | MathDB 页面 | 原文线索

V2-041657 | MATHDB-323948

The primitive sigma-LFSR enumeration conjecture

摘录 / EXCERPT: Let m and n be positive integers. A primitive $c3$-LFSR of order n over ${\mathbb F}_{q^m}$ is a word-oriented feedback shift register whose associated sequence has maximal p...

conjecture | open | MathDB 页面 | 原文线索

V2-041658 | MATHDB-323949

Brlek-Reutenauer conjecture on factor and palindromic complexity

摘录 / EXCERPT: Brlek-Reutenauer conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041659 | MATHDB-323975

The monomial-count conjecture for eigenvectors of Markov matrices

摘录 / EXCERPT: Let A be the $n\times n$ matrix given by equation (44), and let $\mathbf{v}$ be the eigenvector given by equation (29). Express each coordinate of $\mathbf{v}$ in terms of the va...

conjecture | open | MathDB 页面 | 原文线索

V2-041660 | MATHDB-323992

Modified Nivat conjecture in terms of nonexpansive subspaces

摘录 / EXCERPT: Let $\eta\in\mathcal{A}^{\mathbb{Z}^2}$, let $X_{\eta}:=\overline{\mathcal{O}(\eta)}$ be its translation orbit closure, and for a finite set $\mathcal{S}\subseteq\mathbb{Z}^2$ defi...

conjecture | open | MathDB 页面 | 原文线索

V2-041661 | MATHDB-324007

Existence of maxtame extensions for closed or finitely connected open n-tame sets

摘录 / EXCERPT: Maxtame extension conjecture. Then U is contained in a maxtame subset of S^{n-1}, where a maxtame set is an open, maximal n-tame subset of S^{n-1}.

conjecture | open | MathDB 页面 | 原文线索

V2-041662 | MATHDB-324014

Concavity conjecture for normalized Bell numbers

摘录 / EXCERPT: For $x>0$, define ... so that $B(n)=B_n$ for nonnegative integers n, where B_n is the n-th Bell number. Bell-function concavity conjecture. The function $B(x)^{1/x}$ is log-con...

conjecture | open | MathDB 页面 | 原文线索

V2-041663 | MATHDB-324022

The consecutive monotone pattern conjecture

摘录 / EXCERPT: Let $m\geq 1$ and let $\sigma\in\mathcal{S}_m$ be a consecutive pattern. Define ... where $\alpha_n(\sigma)$ is the number of permutations in $\mathcal{S}_n$ avoiding σ as a c...

conjecture | open | MathDB 页面 | 原文线索

V2-041664 | MATHDB-324028

The conjecture on cycle-free partial transversals of Latin squares

摘录 / EXCERPT: A Latin square of order n is an $n\times n$ array in which every symbol occurs exactly once in each row and column. A cycle-free partial transversal is a partial transversal cont...

conjecture | open | MathDB 页面 | 原文线索

V2-041665 | MATHDB-324088

The Growth Rate Conjecture for minor-closed classes of matroids

摘录 / EXCERPT: Let $a\geq 1$ be an integer, and let $\mathcal{M}$ be a minor-closed class of matroids. For a matroid M, write $r(M)$ for its rank and let $\tau_a(M)$ denote the minimum number o...

conjecture | solved | MathDB 页面 | 原文线索

V2-041666 | MATHDB-324109

Albert-Atkinson conjecture on layered patterns maximizing avoidance

摘录 / EXCERPT: Let $S_n(q)$ denote the number of permutations of length n that avoid a pattern q. A pattern is layered if it consists of decreasing subsequences whose entries increase from on...

conjecture | open | MathDB 页面 | 原文线索

V2-041667 | MATHDB-324110

Maximal avoidance conjecture for the patterns M_{2m-1} and M_{2m}

摘录 / EXCERPT: Let $S_n(q)$ denote the number of permutations of length n that avoid a pattern q. For $m\geq 2$, define M_{2m} to be the pattern $132\cdots(2m-1)(2m-2)2m$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-041668 | MATHDB-324112

The three conjectural sporadics in the entropic order on a parabolic quotient of S_6

摘录 / EXCERPT: Three sporadic entropic-order conjectures. The following three relations hold:

conjecture | open | MathDB 页面 | 原文线索

V2-041669 | MATHDB-324113

The optimality conjecture for two-colourings avoiding three-term progressions

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, and consider colourings of $[n]$ with two colours. A monochromatic three-term arithmetic progression is a three-term arithmetic progression whose three...

conjecture | open | MathDB 页面 | 原文线索

V2-041670 | MATHDB-324114

The conjecture that random colourings can always be improved for monochromatic progressions

摘录 / EXCERPT: For integers $k\geq 3$ and $r\geq 2$, let $[n]=\{1,2,\ldots,n\}$ and consider colourings of $[n]$ with r colours. A monochromatic progression of length k is an arithmetic progr...

conjecture | open | MathDB 页面 | 原文线索

V2-041671 | MATHDB-324143

The flip characterization of faces of the triangle-measure cone

摘录 / EXCERPT: Flip face conjecture. A collection Col of triangles having no internal vertices spans a face of $fmathfrak K(S)$ if and only if, for every pair of triangles in Col forming a fl...

conjecture | open | MathDB 页面 | 原文线索

V2-041672 | MATHDB-324144

The oriented-matroid conjecture for the positive polygonal-measure cone

摘录 / EXCERPT: Oriented-matroid conjecture. The combinatorial structure of $\mathfrak K_{pos}(S)$ depends only on the oriented matroid associated with S.

conjecture | open | MathDB 页面 | 原文线索

V2-041673 | MATHDB-324182

Cameron-Ku conjecture on maximum non-centred intersecting families

摘录 / EXCERPT: Cameron-Ku conjecture. If $n\geq6$ and $\mathcal{A}\subset S_n$ is a non-centred intersecting family, then

conjecture | open | MathDB 页面 | 原文线索

V2-041674 | MATHDB-324193

Gessel's conjecture on the two-sided Eulerian polynomial

摘录 / EXCERPT: Let $A_n(s,t)$ denote the two-sided Eulerian polynomial. For $n\geq 1$, the polynomial is considered in the basis ... with indices satisfying $0\leq i-1$, j, and $j+2i\leq n+1$. Ge...

conjecture | open | MathDB 页面 | 原文线索

V2-041675 | MATHDB-324194

The quantum Ahlswede connection between zero-error capacity and deterministic strong subspace capacity

摘录 / EXCERPT: Let ${\mathcal N}\in\mathcal C({\mathcal H},{\mathcal K})$ have a decomposition ... where each ${\mathcal N}_s$ is extremal in ${\mathcal C}({\mathcal H},{\mathcal K})$ and $q(s)>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-041676 | MATHDB-324208

The $2d-1$ threshold conjecture for collapsing unit vectors

摘录 / EXCERPT: Let $\overline{\mathcal{C}}(k,d)$ denote the extremal quantity for k-collapsing families of unit vectors in d-dimensional normed spaces. The $2d-1$ threshold conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-041677 | MATHDB-324209

The $2d$ extremal-value conjecture for large collapsing parameters

摘录 / EXCERPT: Let $\overline{\mathcal{C}}(k,d)$ denote the extremal quantity for k-collapsing families of unit vectors in d-dimensional normed spaces. The $2d$ extremal-value conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-041678 | MATHDB-324221

Uniqueness conjecture for tight neighborly triangulated manifolds

摘录 / EXCERPT: Let $d\geq 3$, and let X be a triangulated d-manifold with d^2+5d+5 vertices. Write $\beta_1(X;\mathbb{Z}_2)$ for its first Betti number with coefficients in $\mathbb{Z}_2$....

open_problem | open | MathDB 页面 | 原文线索

V2-041679 | MATHDB-324246

Conjecture on eventual equality of card-visit counts

摘录 / EXCERPT: Let $c_i(t)$ denote the number of times card i has been seen by time t. Eventual equality conjecture. For any card i, there exists t such that ... This would make precise the...

conjecture | open | MathDB 页面 | 原文线索

V2-041680 | MATHDB-324247

Conjecture on the quadratic asymptotic constant in flashcard games

摘录 / EXCERPT: Let $T_1(n)$ be the first time card 1 is seen for the nth time. Quadratic asymptotic conjecture. We have ... for $c\approx 0.85\ldots$. This is suggested by numerical experiments...

conjecture | open | MathDB 页面 | 原文线索

V2-041681 | MATHDB-324248

Novikoff et al.'s bounded-increment conjecture for flashcard games

摘录 / EXCERPT: Let $T_i(n)$ denote the first time card i is seen for the nth time. Novikoff et al.'s bounded-increment conjecture. We have ... for all i and n. The paper states that this co...

conjecture | open | MathDB 页面 | 原文线索

V2-041682 | MATHDB-324249

Conjecture on the limiting curve of flashcard-game tableaux

摘录 / EXCERPT: Limiting-curve conjecture. There exists a curve Γ such that the area below Γ in the first quadrant is equal to 1 and $\frac{(n,k)}{\sqrt{T_n(k)}}$ lies outside of...

conjecture | open | MathDB 页面 | 原文线索

V2-041683 | MATHDB-324283

Conjecture on arbitrarily long zero gaps in the privileged complexity of the Thue-Morse word

摘录 / EXCERPT: Let t be the Thue-Morse word, and let $\mathpzc{A}_n(t)$ denote the number of privileged factors of t of length n. A zero gap is a consecutive interval of lengths on which...

conjecture | open | MathDB 页面 | 原文线索

V2-041684 | MATHDB-324284

The oriented-matroid determination conjecture for polytopal measure cones

摘录 / EXCERPT: Let S be a finite spanning set, and let $\mathfrak M_\mathbb Z(S)$ be the lattice generated by standard measures of simplices with vertices in S. Define the rational cone...

conjecture | open | MathDB 页面 | 原文线索

V2-041685 | MATHDB-324285

Bijection conjecture for fixed points of the involution on beta(0,1)-trees

摘录 / EXCERPT: Bijection conjecture. There is a bijection between beta$(0,1)$-trees on n nodes fixed under g and beta$(0,1)$-trees on $\lfloor n/2\rfloor+1$ nodes with one distinguished exces...

conjecture | open | MathDB 页面 | 原文线索

V2-041686 | MATHDB-324286

Joint equidistribution conjecture for statistics on beta(0,1)-trees

摘录 / EXCERPT: Joint equidistribution conjecture. The pairs

conjecture | open | MathDB 页面 | 原文线索

V2-041687 | MATHDB-324302

Linear stability conjecture for unions of t-cosets in S_n

摘录 / EXCERPT: Let $\mathcal{A} subset S_n$, and let $t in \mathbb{N}$. Let f denote the characteristic function of $\mathcal{A}$, and let f_t denote the orthogonal projection of f onto...

conjecture | open | MathDB 页面 | 原文线索

V2-041688 | MATHDB-324316

Cornuéjols-Guenin-Margot blocking-number conjecture for ideal minimally non-packing clutters

摘录 / EXCERPT: Let ${\mathscr C}$ be an ideal minimally non-packing clutter, meaning an ideal clutter that does not have the packing property but whose proper minors have the packing property. It...

conjecture | open | MathDB 页面 | 原文线索

V2-041689 | MATHDB-324317

Affine-plane ideal-solution-clutter conjecture

摘录 / EXCERPT: Let ${\mathscr C}$ be a clutter of a combinatorial affine plane, and suppose that its blocking number is at least 3. An ideal solution clutter is an ideal clutter associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-041690 | MATHDB-324339

Existence of a nonrepetitive palindrome-free sequence avoiding two subsequences

摘录 / EXCERPT: Sequence existence conjecture. There exists an infinite sequence on four letters, A,B,C and D such that the sequence is nonrepetitive, palindrome-free and avoids the subsequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-041691 | MATHDB-324360

Nakamura's extremal conjecture for non-overlapping patterns

摘录 / EXCERPT: Let $ath]σ\in\mathcal S_m$ be a non-overlapping pattern, and let $\alpha_n(σ)$ be the number of permutations of length n avoiding $ath]σ$. Nakamura's extremal conjecture. For eve...

conjecture | open | MathDB 页面 | 原文线索

V2-041692 | MATHDB-324370

Károlyi's inverse Erdős-Heilbronn conjecture for non-nilpotent groups

摘录 / EXCERPT: Károlyi's inverse Erdős-Heilbronn conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-041693 | MATHDB-324395

The eventual EKR threshold conjecture

摘录 / EXCERPT: Eventual EKR threshold conjecture. If X is eventually EKR and $n>\varphi^2r$, then X is EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-041694 | MATHDB-324441

Currie-Saari recurrence conjecture for unbordered factors of the Thue-Morse sequence

摘录 / EXCERPT: Let $\mathbf{t}=t_0t_1t_2\cdots$ be the Thue-Morse sequence, defined by $t_0=0$, $t_{2n}=t_n$, and $t_{2n+1}=1-t_n$ for $n\geq 0$. Let $f(n)$ denote the number of unbordered factor…

conjecture | open | MathDB 页面 | 原文线索

V2-041695 | MATHDB-324452

The square-root growth conjecture for distinct distance subsets

摘录 / EXCERPT: Let P be a set of N points in the plane, and let $delta(N)$ denote the largest size of a subset of P whose pairwise distances are all distinct. For every $epsilon>0$, conside…

conjecture | open | MathDB 页面 | 原文线索

V2-041696 | MATHDB-324462

The volume formula for interval-vector polytopes

摘录 / EXCERPT: Let n and i be parameters for the interval-vector polytope $\mathcal{P}_n(\mathcal{I}_{1,n-i})$, where $\mathcal{I}_{1,n-i}$ denotes the corresponding family of intervals and…

conjecture | open | MathDB 页面 | 原文线索

V2-041697 | MATHDB-324464

The Haiman-Loehr conjecture for higher q,t-Catalan polynomials

摘录 / EXCERPT: Let $C^{(m)}_n(q,t)$ denote the combinatorial higher q,t-Catalan polynomial, and let $AC^{(m)}_n(q,t)$ be the Garsia-Haiman expression … where the sum is over partitions of n a…

conjecture | open | MathDB 页面 | 原文线索

V2-041698 | MATHDB-324466

Non-embeddability conjecture for finite thick buildings

摘录 / EXCERPT: A finite thick building is a finite building in which every panel is contained in at least three chambers, and its dimension is denoted by d. Non-embeddability conjecture. No d…

conjecture | open | MathDB 页面 | 原文线索

V2-041699 | MATHDB-324469

The finite-group formulation of the Lonely Runner Conjecture

摘录 / EXCERPT: The finite-group formulation. There exist a natural number n and an element $x\in\mathbb{Z}_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041700 | MATHDB-324474

The extremal-distance characterization of regular polygons

摘录 / EXCERPT: Regular polygon conjecture. Then $A_1,A_2,\ldots,A_n$ are the vertices of a regular polygon inscribed in a circle concentric to Γ.

conjecture | open | MathDB 页面 | 原文线索

V2-041701 | MATHDB-324495

The completability-transition conjecture for random bipartite masks

摘录 / EXCERPT: Fix a rank r and let $G(n,n,c/n)$ be the random bipartite graph with edge density parameter c/n. Let t_r be the supremum of the constants t for which $G(n,n,t/n)$ is r-in…

conjecture | open | MathDB 页面 | 原文线索

V2-041702 | MATHDB-324496

The regular-mask conjecture for generic matrix completability

摘录 / EXCERPT: Fix a rank r and let $G(n,n,2r,2r)$ denote a uniformly random $2r$-regular bipartite graph with n vertices in each part. A mask is completable when its observed entries determi…

conjecture | open | MathDB 页面 | 原文线索

V2-041703 | MATHDB-324540

Supersolvability-Koszulness conjecture for Orlik-Solomon algebras

摘录 / EXCERPT: Let M be a matroid, or let $\mathcal{A}$ be a hyperplane arrangement, and let $A(M)$, respectively $A(\mathcal{A})$, denote its Orlik-Solomon algebra. A matroid or arrangement is…

conjecture | open | MathDB 页面 | 原文线索

V2-041704 | MATHDB-324555

Cocircuit-rank conjecture for non-graphic binary matroids

摘录 / EXCERPT: Cocircuit-rank conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041705 | MATHDB-324556

Dual-rank characterization of graphicness for binary matroids

摘录 / EXCERPT: Graphicness characterization conjecture. The following assertions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-041706 | MATHDB-324568

Finite-color extremal distribution conjecture

摘录 / EXCERPT: Finite-color extremal distribution conjecture. For every nonnegative integer n, among all probability distributions on $n+1$ colors, the maximum value of $D({\bf p})$ is achieved…

conjecture | open | MathDB 页面 | 原文线索

V2-041707 | MATHDB-324569

Unique finite-color extremal distribution conjecture

摘录 / EXCERPT: Let $D({\bf p})$ be the discrepancy for a probability distribution ${\bf p}$ on $n+1$ colors. For $n>0$ and $x\in[1/(n+1),1)$, define … and let x_n be a maximizer of…

conjecture | open | MathDB 页面 | 原文线索

V2-041708 | MATHDB-324570

Equality of finite- and countable-support discrepancy maxima

摘录 / EXCERPT: Support-maximization conjecture. The global maximum exists, equals the supremum of the finite-support maxima, and is strictly less than 1:

conjecture | open | MathDB 页面 | 原文线索

V2-041709 | MATHDB-324578

Multiplicity-dependence conjecture for separation probabilities of long-cycle products

摘录 / EXCERPT: Multiplicity-dependence conjecture. The quantity

conjecture | open | MathDB 页面 | 原文线索

V2-041710 | MATHDB-324588

Bounded first-player difference conjecture for the tau-game

摘录 / EXCERPT: Let $\tau_M(n,k)$ and $\tau_m(n,k)$ denote the terminal covering numbers in the τ-game when Maximizer or minimizer, respectively, makes the first move. Bounded first-player di...

conjecture | open | MathDB 页面 | 原文线索

V2-041711 | MATHDB-324605

Marton's conjecture

摘录 / EXCERPT: Marton's conjecture. A is $\exp(O(\log K))$-covered by a subgroup M of size at most $\exp(O(\log K))|A|$.

conjecture | open | MathDB 页面 | 原文线索

V2-041712 | MATHDB-324606

Polynomial $U^3(\mathbb F_2^n)$-inverse conjecture

摘录 / EXCERPT: Polynomial $U^3(\mathbb F_2^n)$-inverse conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041713 | MATHDB-324607

Rudin's conjecture that the squares form a $\Lambda(4)$-set

摘录 / EXCERPT: Rudin's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041714 | MATHDB-324615

Brockman-Kay conjecture on intersecting families of multisets

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $M(n,k)$ be the $k\times n$ rectangle representation of k-multisets, and let $\mathcal{M}(n,k,t)$ denote the families of t-intersecting k-multis...

conjecture | open | MathDB 页面 | 原文线索

V2-041715 | MATHDB-324616

Meagher-Purdy conjecture on intersecting families of multisets

摘录 / EXCERPT: Let k,n,t be positive integers with $t\leq k$, let $M(n,k)$ be the rectangle representation of k-multisets, and let $\mathcal{M}(n,k,t)$ denote the families of t-intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-041716 | MATHDB-324642

Asymptotic monotonicity conjecture for pattern-avoiding ordered set partitions

摘录 / EXCERPT: Let S_m be the symmetric group on m letters, let $\rho\in S_m$ be a permutation, and let $\operatorname{op}_{n,k}(\rho)$ denote the number of ordered set partitions of $[n]$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-041717 | MATHDB-324645

Lewis's even-length alternating-pattern enumeration conjecture

摘录 / EXCERPT: Let $A_n(\sigma)$ be the set of permutations of $[n]$ that avoid the pattern σ and are alternating. Lewis's conjecture. For $n\geq 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-041718 | MATHDB-324646

Lewis's odd-length alternating-pattern enumeration conjecture for 2134, 4312, 3214 and 4123

摘录 / EXCERPT: Let $A_n(\sigma)$ be the set of permutations of $[n]$ that avoid the pattern σ and are alternating. Lewis's conjecture. For $n\geq 0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-041719 | MATHDB-324673

Bounded arithmetic progressions in ranges of rational polynomials

摘录 / EXCERPT: Let $g(x)$ be a polynomial over the rationals, and write $g(\mathbb{Q})$ for its range on rational inputs. An arithmetic progression is a finite sequence of the form $a+id$ with fi...

conjecture | open | MathDB 页面 | 原文线索

V2-041720 | MATHDB-324674

Uniformly bounded geometric progressions in translated integer polynomial ranges

摘录 / EXCERPT: Let g be a polynomial over the integers. For each integer a, define … A geometric progression of length b is a sequence of b terms with a common ratio. Geometric-progressio...

conjecture | open | MathDB 页面 | 原文线索

V2-041721 | MATHDB-324691

Extremal-ray conjecture for homological data of squarefree complexes

摘录 / EXCERPT: Let $C(\operatorname{SqFree},n)$ be the positive rational cone generated by homological data triplets of free squarefree complexes over $\Bbbk[x_1,\ldots,x_n]$. A pure free squaref...

conjecture | open | MathDB 页面 | 原文线索

V2-041722 | MATHDB-324700

The conjecture that every subspace hereditary property is testable

摘录 / EXCERPT: An affine-invariant property $\mathcal P$ is subspace hereditary if, whenever $f:\mathbb{F}^n\to [R]$ satisfies $\mathcal P$, the restriction of f to every affine subspace of...

conjecture | open | MathDB 页面 | 原文线索

V2-041723 | MATHDB-324705

Completion conjecture for Foulis-Randall products

摘录 / EXCERPT: Let $H_1,\ldots,H_n$ be contextuality scenarios, let $\overline{H_1\otimes\ldots\otimes H_n}$ denote the completion of their n-fold product scenario, let $\bar H_i$ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-041724 | MATHDB-324712

The immunity bound conjecture for simply intersecting $\mathbb{B}_3$-spaces

摘录 / EXCERPT: The immunity bound conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-041725 | MATHDB-324738

Uniqueness conjecture for maximal tail-length starting sequences

摘录 / EXCERPT: Uniqueness conjecture. The starting sequence S_0 is unique and has properties P2, P3 and P4.

conjecture | open | MathDB 页面 | 原文线索

V2-041726 | MATHDB-324739

Nonexistence conjecture for doubly rotten sequences

摘录 / EXCERPT: Nonexistence conjecture. Doubly rotten sequences do not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-041727 | MATHDB-324771

Erdős-Kahn-Komlós-Szemerédi conjecture on singular Bernoulli matrices

摘录 / EXCERPT: Let M_n be an $n\times n$ random matrix with iid Bernoulli entries, and let $p_n={\mathbf P}(M_n\text{ is singular})$. Erdős-Kahn-Komlós-Szemerédi conjecture. ... This is described...

conjecture | open | MathDB 页面 | 原文线索

V2-041728 | MATHDB-324810

Conjecture on consecutive-layer extremal families avoiding a fixed poset

摘录 / EXCERPT: Consecutive-layer conjecture. For any fixed poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-041729 | MATHDB-324823

Generalized Alon-Saks-Seymour conjecture of order t

摘录 / EXCERPT: Let G be a graph and t a positive integer. A t-biclique covering is a collection of complete bipartite graphs covering every edge of G at least once and at most t times;...

conjecture | open | MathDB 页面 | 原文线索

V2-041730 | MATHDB-324829

Shephard-Kalai conjecture on subpolytopes of stacked polytopes

摘录 / EXCERPT: Let $d\ge 0$. A subpolytope of a polytope is the convex hull of a subset of its vertices, and a stacked polytope is a polytope obtained by starting with a simplex and successively...

conjecture | open | MathDB 页面 | 原文线索

V2-041731 | MATHDB-324835

The Fundamental Conjecture of Superposition

摘录 / EXCERPT: Let a superposition be a product of mutually compatible slices, and let a slice be a component obtained by a slicing algorithm from the initial syntax of a superposition. An order-...

conjecture | open | MathDB 页面 | 原文线索

V2-041732 | MATHDB-324838

Combinatorial computability of Legendrian contact homology for products of Legendrian knots

摘录 / EXCERPT: Legendrian knots are Legendrian submanifolds in the contactization of a one-dimensional base, and their Legendrian contact homology is encoded by a differential graded algebra gene...

conjecture | open | MathDB 页面 | 原文线索

V2-041733 | MATHDB-324844

Color count conjecture for highly colorable maps

摘录 / EXCERPT: Let $n\geq 2$. Define $m_1(n),m_2(n),m_3(n),m_4(n)$ to be the four largest possible numbers of valid color vectors, modulo the action of S_3 on the colors, among tree pairs corre...

conjecture | open | MathDB 页面 | 原文线索

V2-041734 | MATHDB-324888

Extremal ghost numbers for group algebras of p-groups

摘录 / EXCERPT: Ghost-number extremality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041735 | MATHDB-324895

The top-twisted equality conjecture for Jack characters

摘录 / EXCERPT: Let π be a partition, let λ be a Young diagram, and let $\alpha>0$. Denote by $\operatorname{Ch}_\pi^{(\alpha)}$ the Jack character and by...

conjecture | open | MathDB 页面 | 原文线索

V2-041736 | MATHDB-324912

Almost-homomorphism formulation of the Polynomial Freiman-Ruzsa conjecture

摘录 / EXCERPT: Almost-homomorphism formulation of the Polynomial Freiman-Ruzsa conjecture. Is it true that $f=\widetilde f+g$, where $\widetilde f:\mathbb{F}_2^n\rightarrow\mathbb{F}_2^{n'}$ is l...

open_problem | open | MathDB 页面 | 原文线索

V2-041737 | MATHDB-324916

The K-Kohnert rule for dual stable Grothendieck polynomials

摘录 / EXCERPT: Let α be a weak composition, let $J_{\alpha}^{(\beta)}$ be the generating function for diagrams produced from $\operatorname{Skyline}(\alpha)$ by the K-Kohnert rule, with...

conjecture | open | MathDB 页面 | 原文线索

V2-041738 | MATHDB-324986

The lexicographic minimization conjecture for noisy Boolean functions

摘录 / EXCERPT: Let X^n be i.i.d. Bernoulli$(1/2)$, let Z^n be i.i.d. Bernoulli(α) independently of X^n, where $0\leq\alpha\leq 1/2$, and let $Y^n=X^n\oplus Z^n$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-041739 | MATHDB-325013

Conjecture on the threshold function for nonnegative k-sums

摘录 / EXCERPT: Threshold conjecture. One has $f(k)=N_k$, and hence

conjecture | open | MathDB 页面 | 原文线索

V2-041740 | MATHDB-325063

Extremal gamma-vector conjecture for even-dimensional flag spheres

摘录 / EXCERPT: Let $d\geq 4$ be even, and let Δ be a flag simplicial $(d-1)$-sphere on n vertices. Let $f_{i-1}(r,m)$ denote the number of i-cliques in the complete r-partite Turán g...

conjecture | open | MathDB 页面 | 原文线索

V2-041741 | MATHDB-325064

Gabow-Kajitani-Ueno-Miyano cyclic ordering conjecture for matroids

摘录 / EXCERPT: Cyclic ordering conjecture. It is possible to place the elements of the ground set of M on a circle so that every $r(M)$ consecutive elements form a basis of M.

conjecture | open | MathDB 页面 | 原文线索

V2-041742 | MATHDB-325072

The one-third-two-thirds conjecture for antimatroids

摘录 / EXCERPT: The one-third-two-thirds conjecture for antimatroids. Every antimatroid $\mathcal{A}$ with more than one basic word satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-041743 | MATHDB-325116

Congruence pattern for the function S(a) modulo 3

摘录 / EXCERPT: Congruence-pattern conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041744 | MATHDB-325145

Configuration-type conjecture for three nonattacking pieces

摘录 / EXCERPT: Let $\mathbb P$ be a piece with set of basic moves $\mathbf M$, and consider nonattacking configurations of three pieces. Configuration-type conjecture. The number of nonattacking...

conjecture | open | MathDB 页面 | 原文线索

V2-041745 | MATHDB-325147

Triangular interval conjecture for exactly-coloured infinite subgraphs

摘录 / EXCERPT: Let $\Delta:\mathbb{N}^{(2)}\twoheadrightarrow[k]$ be a surjective k-colouring, and let $\mathcal{F}_{\Delta}$ denote the set of numbers of colours appearing on infinite induced...

conjecture | open | MathDB 页面 | 原文线索

V2-041746 | MATHDB-325148

Linear excess conjecture for the exact-colour spectrum

摘录 / EXCERPT: Let $\psi(k)$ denote the minimum possible size of $\mathcal{F}_{\Delta}$ over surjective k-colourings $\Delta:\mathbb{N}^{(2)}\twoheadrightarrow[k]$. Linear excess conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-041747 | MATHDB-325168

Falgas-Ravry-Vaughan conjecture for induced directed stars

摘录 / EXCERPT: Let $\vec{S}_k$ be the directed star on k vertices, with one center and $k-1$ edges oriented away from it. For an integer $k\geq 3$, let $\pi_{\vec{S}_k}(\emptyset)$ denote the m...

conjecture | open | MathDB 页面 | 原文线索

V2-041748 | MATHDB-325179

Hirsch's bounded Hirsch conjecture for polytopes

摘录 / EXCERPT: Bounded Hirsch conjecture. The diameter of the facet-ridge graph of any $(d+1)$-polytope on n vertices is at most $n-(d+1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-041749 | MATHDB-325182

The enumeration conjecture for Type B non-decreasing parking functions

摘录 / EXCERPT: The Type B non-decreasing parking functions are the elements of the submonoid of NDPF_{2N} generated by $\mu_i=\pi_i\pi_{2N-i}$ for $1\leq i\leq N$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-041750 | MATHDB-325185

The extremal conjecture for separated list chromatic number

摘录 / EXCERPT: For integers $c,n\ge 1$, let G be a graph with n vertices, and let $\chi_l(G,c)$ denote its list chromatic number with separation c. Extremal conjecture. ... This conjecture as...

conjecture | open | MathDB 页面 | 原文线索

V2-041751 | MATHDB-325187

Conjecture on the exact value of f(q+2)

摘录 / EXCERPT: Let q be the order of the finite projective plane, and let $f(n)$ denote the minimum number of lines whose odd-intersection point set has size n. Conjecture. ... The paper has es...

conjecture | open | MathDB 页面 | 原文线索

V2-041752 | MATHDB-325188

The minimal superpermutation length conjecture

摘录 / EXCERPT: Minimal superpermutation length conjecture. The length of the minimal superpermutation on n symbols is

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041753 | MATHDB-325189

The uniqueness conjecture for minimal superpermutations

摘录 / EXCERPT: Uniqueness conjecture for minimal superpermutations. The minimal superpermutation on n symbols is unique up to interchanging the roles of the symbols.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-041754 | MATHDB-325257

Structural conjectures for the tatami polynomial $P_n(z)$

摘录 / EXCERPT: Let $P_n(z)$ be the polynomial associated with the tatami-covering generating polynomial, and write $\langle z^i\rangle Q(z)$ for the coefficient of z^i in a polynomial $Q(z)$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-041755 | MATHDB-325278

The two-piece monochromatic path and cycle covering conjecture for 3-colored complete graphs

摘录 / EXCERPT: Let K_n be a complete graph whose edges are colored with three colors. A monochromatic path or cycle is a path or cycle all of whose edges have one color, and the two pieces are...

conjecture | open | MathDB 页面 | 原文线索

V2-041756 | MATHDB-325280

The monochromatic tree-cover averaging conjecture

摘录 / EXCERPT: Let t be a positive integer, let s satisfy $1\le s\le t-1$, and let K_n be a complete graph with its edges colored using t colors. A monochromatic tree is a tree all of who...

conjecture | open | MathDB 页面 | 原文线索

V2-041757 | MATHDB-325289

Folklore separation conjectures for Ramsey numbers

摘录 / EXCERPT: Folklore separation conjectures. (a) For every integer $ell\findent 3$, $R(ell-1,n)=o(R(ell,n))$ as $n\toinfty$. (b) For every integer $ell\findent 3$, there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-041758 | MATHDB-325312

Computational complexity conjecture for the alpha-permanent

摘录 / EXCERPT: Computational complexity conjecture. Computation of the α-permanent is $\#P$-complete, except for

conjecture | open | MathDB 页面 | 原文线索

V2-041759 | MATHDB-325313

Brown-Freedman conjecture on double 3-term arithmetic progressions

摘录 / EXCERPT: Brown-Freedman conjecture. There are two consecutive intervals I,J of the same length such that

conjecture | open | MathDB 页面 | 原文线索

V2-041760 | MATHDB-325314

Conjecture on double van der Waerden numbers

摘录 / EXCERPT: Double van der Waerden conjecture. The number $w^*(3,3)$ exists, whereas the number $w^*(2,4)$ does not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-041761 | MATHDB-325328

Uniqueness conjecture for triangulated disk configurations

摘录 / EXCERPT: Uniqueness conjecture. The uniqueness theorem for thin triangulated disk configurations remains true without the thinness requirement: every triangulated disk configuration is dete...

conjecture | open | MathDB 页面 | 原文线索

V2-041762 | MATHDB-325339

Rational Catalan q,t-symmetry refinement

摘录 / EXCERPT: Let $Y_{m/n}$ be the set of m/n Dyck paths, let $h_{+}(D)$ be the stated statistic, and define ... Let $\widetilde{P}_{n,m}$ be the transformed DAHA element, let h_n be the compl...

conjecture | open | MathDB 页面 | 原文线索

V2-041763 | MATHDB-325385

The logarithmic tail conjecture for Kadanoff sandpile configurations

摘录 / EXCERPT: Logarithmic tail conjecture. For a general parameter D and all N, there exists a column n in $O(\log N)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041764 | MATHDB-325386

A basis conjecture for the Partition Temperley-Lieb algebra

摘录 / EXCERPT: Let ${\rm PTL}_n$ be the Partition Temperley-Lieb algebra, let P_n denote the set-partition monoid on $\{1,\ldots,n\}$, and let F be a reduced word in the generators u...

conjecture | open | MathDB 页面 | 原文线索

V2-041765 | MATHDB-325405

Three-triangle matroid minor conjecture

摘录 / EXCERPT: A triangle in a matroid is a circuit of size 3. A matroid admits another matroid as a minor if that minor is obtained by deletions and contractions. Three-triangle matroid minor...

conjecture | open | MathDB 页面 | 原文线索

V2-041766 | MATHDB-325406

Binomial-count conjecture for parity differences of hit-huggers

摘录 / EXCERPT: Let n and c be odd integers satisfying ... Consider the even and odd hit-huggers beginning with 1 in S_n. A hit-hugger is a permutation subject to the hit-hugging relation co...

conjecture | open | MathDB 页面 | 原文线索

V2-041767 | MATHDB-325429

The higher-dimensional bootstrapping conjecture for lattice walks

摘录 / EXCERPT: Let ${\mathfrak{S}}$ be a set of small steps in $\mathbb{Z}^d$ for some dimension d, and let R be an open region in $\mathbb{Z}^d$ bounded by rational hyperplanes. Define ... Her...

conjecture | open | MathDB 页面 | 原文线索

V2-041768 | MATHDB-325430

The symmetry conjecture for exponential growth of quarter-plane walks

摘录 / EXCERPT: Let ${\mathfrak{S}}$ be a step set symmetric about an axis L, where L passes through the first quadrant, potentially along a boundary. Let $L^\perp$ be the line perpendicular t...

conjecture | open | MathDB 页面 | 原文线索

V2-041769 | MATHDB-325434

Geelen's conjecture on preserving two matroid connectivities

摘录 / EXCERPT: For a matroid M on ground set $E(M)$, let $Q,R,S,T\subseteq E(M)$ satisfy $Q\cap R=S\cap T=\emptyset$. Write ... Let $c:\mathbb{N}^2\to\mathbb{N}$ be a function. Geelen's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-041770 | MATHDB-325435

Weak intersecting partition conjecture

摘录 / EXCERPT: Let P_n be the set of all integer partitions of n, and let $P_n(1)$ be the set of partitions having 1 as a part. Two partitions intersect if they have at least one common par...

conjecture | open | MathDB 页面 | 原文线索

V2-041771 | MATHDB-325436

Strong intersecting partition conjecture

摘录 / EXCERPT: Let $P_{n,k}$ be the set of partitions of n of length k, and let $P_{n,k}(1)$ be the set of such partitions having 1 as a part. Two partitions intersect if they have at least...

conjecture | open | MathDB 页面 | 原文线索

V2-041772 | MATHDB-325449

The regular-word existence and uniqueness conjecture for generalized Thue-Morse sequences

摘录 / EXCERPT: Let $r\geqslant 2$ and let n be the length of a word over a two-letter alphabet. An r-regular word is a word satisfying the corresponding regularity condition, and...

conjecture | open | MathDB 页面 | 原文线索

V2-041773 | MATHDB-325464

The codegree decomposition conjecture for point configurations

摘录 / EXCERPT: Codegree decomposition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041774 | MATHDB-325466

The lattice join projection conjecture

摘录 / EXCERPT: Lattice join projection conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-041775 | MATHDB-325475

The equal-slope conjecture for Cookie Monster losing positions

摘录 / EXCERPT: Let p^0_i,q^0_i and p^1_i,q^1_i denote the coordinates of the losing positions of the forms $\{0,p^0_i,q^0_i\}$ and $\{1,p^1_i,q^1_i\}$, respectively. Here, the pairs are compa...

conjecture | open | MathDB 页面 | 原文线索

V2-041776 | MATHDB-325476

The bounded-difference conjecture for Cookie Monster losing positions

摘录 / EXCERPT: Let p^0_i,q^0_i and p^1_i,q^1_i enumerate, respectively, all losing positions of the forms $\{0,p^0_i,q^0_i\}$ and $\{1,p^1_i,q^1_i\}$, with each family arranged in increasing...

conjecture | open | MathDB 页面 | 原文线索

V2-041777 | MATHDB-325486

Infinite logconcavity conjecture for the sequence $\{d_{\ell}(m)\}$

摘录 / EXCERPT: Let $\{d_{\ell}(m)\}$ be the sequence described above, and define the operator ... A sequence is infinitely logconcave if $\mathfrak{L}^j(\{x_k\})$ is nonnegative for every...

conjecture | open | MathDB 页面 | 原文线索

V2-041778 | MATHDB-325490

The zero-sum Ramsey equality conjecture for intersecting families and hyperstars

摘录 / EXCERPT: Let G be a finite abelian group, let $k\geq 1$ and $r\geq 2$ be positive integers, and suppose that ... Here, $\mathcal{I}_{k\exp(G)}^{(r)}$ denotes the family of intersecting r-...

conjecture | open | MathDB 页面 | 原文线索

V2-041779 | MATHDB-325506

The product-dimension conjecture for forbidden submatrices

摘录 / EXCERPT: Let F be a fixed $(0,1)$-matrix. For matrices M and N, write $[M\,|\,N]$ for their horizontal concatenation and let $k\cdot M$ denote the concatenation of k copies of M....

conjecture | open | MathDB 页面 | 原文线索

V2-041780 | MATHDB-325518

Cubic formula conjecture for the minimum number of two-by-two spectral radii

摘录 / EXCERPT: For a finite set $A\subset\mathbb{R}^+$, define ... For $k\geq 1$, define ... A cubic minimum conjecture asserts ... where $T(k)=k(k+1)/2$ is the k-th triangular number; moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-041781 | MATHDB-325519

Strict exclusion conjecture for adjoining zero to positive sets

摘录 / EXCERPT: Strict exclusion conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-041782 | MATHDB-325527

Generalized rainbow matching conjecture for families of bipartite matchings

摘录 / EXCERPT: Generalized rainbow matching conjecture. Every family of n matchings, each of size n, in a bipartite graph has a rainbow matching of size $n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-041783 | MATHDB-325531

Pairwise inequivalence conjecture for seven Dickson-polynomial skew Hadamard difference sets

摘录 / EXCERPT: Let m be an odd integer with $m>7$ and $m\not\equiv 0 \pmod{3}$. In the additive group $({\mathbb F}_{3^m},+)$, let ${\rm P}$, ${\rm DY}(1)$, ${\rm DY}(-1)$, ${\rm RT}(1)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-041784 | MATHDB-325545

Asymptotic conjecture for the optimal cleanup threshold

摘录 / EXCERPT: Optimal cleanup threshold conjecture. For $n\geqslant 5$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-041785 | MATHDB-325546

Lower-bound conjecture for the optimal cleanup threshold

摘录 / EXCERPT: Approximate-threshold conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-041786 | MATHDB-325590

Bérczi's polynomial quotient conjecture for Thom-polynomial coefficients

摘录 / EXCERPT: Bérczi's neighbouring-coefficient conjecture. The quotient

conjecture | open | MathDB 页面 | 原文线索

V2-041787 | MATHDB-325601

The finite-field product-set arithmetic progression conjecture

摘录 / EXCERPT: Finite-field product-set conjecture. There is an absolute constant $c>0$ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-041788 | MATHDB-325619

Rudolph's popularity conjecture for 132-avoiding patterns

摘录 / EXCERPT: Let τ and μ be permutation patterns, and let $A_n(\tau)$ and $A_n(\mu)$ denote their total numbers of occurrences among permutations in $S_n(132)$. Associate to each patte...

conjecture | open | MathDB 页面 | 原文线索

V2-041789 | MATHDB-325644

Rota's matroid decomposition conjecture

摘录 / EXCERPT: Let ${\mathcal M}$ be a matroid, and let $F_1,\ldots,F_k$ be independent sets in ${\mathcal M}$ with $|F_i|=k$ for every i. An ${\mathcal M}$-SR is a system of representatives wh...

conjecture | open | MathDB 页面 | 原文线索

V2-041790 | MATHDB-325661

Decomposability conjecture for conjugation-free line arrangements and groups

摘录 / EXCERPT: Let $\mathcal{L}$ be a line arrangement, and let G be a group. They are conjugation-free when their defining relations have no conjugations; they are decomposable when their asso...

conjecture | open | MathDB 页面 | 原文线索

V2-041791 | MATHDB-325669

Bollobás-Leader conjecture on extremal \ell-balls in uniform set systems

摘录 / EXCERPT: Let n, k, r, and ℓ be parameters with $\mathcal{A}_{r,\ell}=\{A\in\binom{[n]}{k}:|A\cap[r]|\ge\ell\}$. These are the ell-balls referred to in the source; when $\ell=1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-041792 | MATHDB-325685

Geelen's Growth Rate Conjecture for minor-closed matroid classes

摘录 / EXCERPT: Growth Rate Conjecture. For every integer $a ge 1$, there is an integer $c>0$ such that at least one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-041793 | MATHDB-325689

Daykin-Häggkvist conjecture on completing dense partial Latin squares

摘录 / EXCERPT: Let P be an $n\times n$ partial Latin square in which each symbol, row, and column contains at most $\frac{1}{4}n$ nonblank cells. Daykin-Häggkvist conjecture. Every such partial...

conjecture | open | MathDB 页面 | 原文线索

V2-041794 | MATHDB-325700

Quadratic synchronizing-word bound conjecture

摘录 / EXCERPT: Let ${\cal M}\subset\{0,1\}^{n\times n}$ be a synchronizing automaton, with synchronizing words defined by products of matrices from ${\cal M}$ that have the form $e e_i^T$. Quadra...

conjecture | open | MathDB 页面 | 原文线索

V2-041795 | MATHDB-325703

Anti-pencil conjecture for the maximum number of non-negative-sum subsets

摘录 / EXCERPT: Anti-pencil conjecture. If $n\geq 4k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041796 | MATHDB-325729

Upper-bound conjecture for lines in rank-three matroids in \mathcal{U}(\ell)

摘录 / EXCERPT: Rank-three line-count conjecture. If ℓ is a sufficiently large integer and $M\in\mathcal{U}(\ell)$ has rank 3, then

conjecture | open | MathDB 页面 | 原文线索

V2-041797 | MATHDB-325777

GCD divisibility conjecture for drop functions of the Novelli-Pak-Stoyanovskii algorithm

摘录 / EXCERPT: Let $n\in{\mathbb N}$, let λ be a partition of n, and let $U\in\operatorname{SYT}(\lambda)$ be such that the corresponding Novelli-Pak-Stoyanovskii algorithm is uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-041798 | MATHDB-325783

The conjecture on polynomial tails for generic Kerr perturbations

摘录 / EXCERPT: Let smooth vacuum Cauchy data be generic and sufficiently near data corresponding to a subextremal Kerr metric, and let $(\mathcal{M},g)$ be the resulting vacuum spacetime with sca...

conjecture | open | MathDB 页面 | 原文线索

V2-041799 | MATHDB-325785

Hyatt's unimodality conjecture for the type B polynomial $T_k(x)$

摘录 / EXCERPT: Let ... where $B_m(x)$ is the type B Eulerian polynomial. Hyatt's unimodality conjecture. The polynomial $T_k(x)$ is unimodal for every integer $k\geq 0$. This conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-041800 | MATHDB-325786

The 4-adjacent Gray code conjecture for descent sequences in co-RGC order

摘录 / EXCERPT: 4-adjacent Gray code conjecture. The resulting list is a 4-adjacent Gray code.

conjecture | open | MathDB 页面 | 原文线索

V2-041801 | MATHDB-325818

The gap conjecture for vertices of non-degenerate 3-way transportation polytopes

摘录 / EXCERPT: Let $p \times q \times s$ be a non-degenerate 3-way transportation polytope specified by 2-margin matrices U,V,W, and let $f_0(M(U,V,W))$ denote its number of vertices. The g...

conjecture | open | MathDB 页面 | 原文线索

V2-041802 | MATHDB-325819

The diameter range conjecture for transportation polytopes

摘录 / EXCERPT: For positive integers p and q, consider the graph diameter of a $p \times q$ transportation polytope. The diameter range conjecture. All integer numbers between 1 and $p+q-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-041803 | MATHDB-325827

Godbole et al.'s eventual monotonicity conjecture for ordered set partitions

摘录 / EXCERPT: Godbole et al.'s conjecture. For each pattern p and each integer k, there exists a positive integer $n_0(k,p)$ such that for every $n\geq n_0(k,p)$,

conjecture | open | MathDB 页面 | 原文线索

V2-041804 | MATHDB-325860

Extremal real-part zeros attained by fibred stable alternating knots

摘录 / EXCERPT: For $n\geq1$, let Γ_{2n} be the set of Alexander polynomials of alternating knots of genus n, and let δ_{2n} be the maximum of the maximal real part of a zero among...

conjecture | open | MathDB 页面 | 原文线索

V2-041805 | MATHDB-325862

Hof-Knill-Simon conjecture on palindromic fixed points and conjugates in class P

摘录 / EXCERPT: Hof-Knill-Simon conjecture. ${\bf u}$ is palindromic if and only if there exists a morphism $\varphi\neq\operatorname{Id}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041806 | MATHDB-325924

The linear disjoint-matching conjecture for angularly monotone topological graphs

摘录 / EXCERPT: Angularly monotone matching conjecture. Every simple angularly monotone complete topological graph on n vertices contains a disjoint matching of size

conjecture | open | MathDB 页面 | 原文线索

V2-041807 | MATHDB-325938

The unistructurality conjecture for cluster algebras

摘录 / EXCERPT: Unistructurality conjecture. Every cluster algebra is unistructural.

conjecture | open | MathDB 页面 | 原文线索

V2-041808 | MATHDB-325944

The one-third approval-ratio conjecture for double-interval societies

摘录 / EXCERPT: One-third approval-ratio conjecture. For all pairwise-intersecting double-interval societies S, the approval ratio

conjecture | open | MathDB 页面 | 原文线索

V2-041809 | MATHDB-325977

Hähnle's connected layer multicomplex diameter conjecture

摘录 / EXCERPT: Let $H_{\operatorname{c.l.m.}}(n,d)$ be the maximum length of a connected layer multicomplex of rank d on n elements. Hähnle's conjecture. For all n and d, ... This is a mult...

conjecture | open | MathDB 页面 | 原文线索

V2-041810 | MATHDB-325990

The upper-bound comparison conjecture for Ramsey numbers with dropped cliques

摘录 / EXCERPT: Upper-bound comparison conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041811 | MATHDB-325991

The exact-overlaps conjecture for self-similar sets

摘录 / EXCERPT: Exact-overlaps conjecture. A strict inequality in the preceding box-dimension bound can occur only in the presence of exact overlaps.

conjecture | open | MathDB 页面 | 原文线索

V2-041812 | MATHDB-326022

Eventual constancy of large automorphic conjugacy class counts

摘录 / EXCERPT: Let F_2 be the free group on two generators. For each n, consider automorphic conjugacy classes whose minimal words have length n, and let the size of a class mean the cardin...

conjecture | open | MathDB 页面 | 原文线索

V2-041813 | MATHDB-326025

Weight-four size-one non-root class enumeration conjecture

摘录 / EXCERPT: Let $F_2=\langle a,b\rangle$ be the free group on two generators. An automorphic conjugacy class is a class under the action of $\operatorname{Aut}F_2$, its size is the number of v...

conjecture | open | MathDB 页面 | 原文线索

V2-041814 | MATHDB-326031

The generalized Cauchy-Davenport inequality for sums of non-empty subsets

摘录 / EXCERPT: Generalized Cauchy-Davenport conjecture. Let n be a positive integer and $X_1,\ldots,X_n$ be non-empty subsets of A. If $\mathbb A$ is cancellative, then

conjecture | open | MathDB 页面 | 原文线索

V2-041815 | MATHDB-326033

Maximal-length conjecture for inscribed kiss-free snakes

摘录 / EXCERPT: Let b and k be positive integers, and consider kiss-free snakes inscribed in a rectangle of size $b\times k$. The length of an inscribed partially directed snake is...

conjecture | open | MathDB 页面 | 原文线索

V2-041816 | MATHDB-326047

The bivariate symmetry conjecture for core partitions

摘录 / EXCERPT: Bivariate symmetry conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041817 | MATHDB-326090

Monochromatic subdiagram cover-dimension conjecture

摘录 / EXCERPT: Let P be a poset, let D be its order diagram, let E be the edge set of D, and let $\phi:E\to\{1,2,\dots,r\}$ be an r-coloring of the edges. For a subdiagram D' with ass...

conjecture | open | MathDB 页面 | 原文线索

V2-041818 | MATHDB-326098

The maximal-singular-locus conjecture for K-orbit closures

摘录 / EXCERPT: Let Y_{γ} be a K-orbit closure and let $\mathcal O_{\alpha}$ be a K-orbit appearing in the maximal singular locus of Y_{γ}. Maximal-singular-locus conjecture. All...

conjecture | open | MathDB 页面 | 原文线索

V2-041819 | MATHDB-326136

Formation width conjecture for $abc(acb)^t abc$

摘录 / EXCERPT: Let t be a nonnegative integer, and let $\mathit{fw}(u)$ denote the formation width of a sequence u. Formation width conjecture. ... The paper states that this would imply nearly...

conjecture | open | MathDB 页面 | 原文线索

V2-041820 | MATHDB-326138

Gao-Wang's conjecture for the K1 constant

摘录 / EXCERPT: Gao-Wang's conjecture. The equality $\mathsf{K}_{1}(G)=\mathsf{K}_{1}^{*}(G)$ holds for all finite abelian groups G. This is the analogous expected-value conjecture for the...

conjecture | open | MathDB 页面 | 原文线索

V2-041821 | MATHDB-326146

Ostrovski's transition-combinatorics conjecture for fictitious play

摘录 / EXCERPT: Ostrovski's conjecture. Under these conditions, the Nash equilibrium does not Pareto dominate typical fictitious-play orbits. This conjecture gives a sufficient transition-combinat...

conjecture | open | MathDB 页面 | 原文线索

V2-041822 | MATHDB-326165

Bounded combinatorial dimension along an admissible subsequence

摘录 / EXCERPT: Let k be a field and let V be a finite-dimensional k-vector space. An infinite sequence is admissible if all its truncations are admissible; for an admissible sequence s, w...

conjecture | open | MathDB 页面 | 原文线索

V2-041823 | MATHDB-326166

A direct-system conjecture for bounded fundamental combinatorial dimension

摘录 / EXCERPT: Let k be a field and let V be a finite-dimensional k-vector space. An infinite sequence is admissible if all its truncations are admissible. For an infinite subsequence t o...

conjecture | open | MathDB 页面 | 原文线索

V2-041824 | MATHDB-326178

The regular-simplex conjecture for unions of spherical caps

摘录 / EXCERPT: Regular-simplex conjecture. The maximizing configuration is always the one in which the $n+1$ centers are the vertices of a regular simplex in $\mathbb{R}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-041825 | MATHDB-326220

Quasigroup formulation of the Brown-Erdős-Sós conjecture

摘录 / EXCERPT: Quasigroup formulation of the Brown-Erdős-Sós conjecture. For every $c>0$, there is a threshold n_0 such that if Q has order $n\geq n_0$, then for every such subset S, whenev...

conjecture | open | MathDB 页面 | 原文线索

V2-041826 | MATHDB-326221

Parker's improved conjecture on type-I complementary arrays

摘录 / EXCERPT: Let n be a positive integer. A type-I complementary array over the alphabet ${0,1,-1}$ is an array represented by a function $f_{n-1,0}({\bf{x}})$, with the indices and parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-041827 | MATHDB-326259

The extremal completion conjecture for bordered-permutation matrices

摘录 / EXCERPT: Let E_m be the $m\times m$ $(0,-1)$-matrix with -1s in positions $(2,m),(3,m-1),\ldots,(m,2)$, and let F_m be the $m\times m$ $(0,-1)$-matrix with -1s in positions...

conjecture | open | MathDB 页面 | 原文线索

V2-041828 | MATHDB-326288

Manickam-Singhi conjecture for vector spaces

摘录 / EXCERPT: Let V be an n-dimensional vector space over the finite field $\mathbb{F}_{q}$, and let ... be a weighting of the 1-dimensional subspaces such that ... For a subspace $S\subset V$...

conjecture | open | MathDB 页面 | 原文线索

V2-041829 | MATHDB-326317

Classification conjecture for primitive regular hexagonal tilings of the integer lattice

摘录 / EXCERPT: Classification conjecture. If $\mathcal{T}$ is a primitive, regular, hexagonal tiling of $\mathbb{Z}^2$ by 90°-symmetric tiles, then

conjecture | open | MathDB 页面 | 原文线索

V2-041830 | MATHDB-326320

Salzborn's basis-set conjecture for normalised families

摘录 / EXCERPT: Salzborn's conjecture. Every normalised family $mathcal N eq\{emptyset\}$ contains a basis set B with

conjecture | open | MathDB 页面 | 原文线索

V2-041831 | MATHDB-326321

Poonen's power-set conjecture for separating families

摘录 / EXCERPT: Poonen's conjecture. Unless $mathcal A$ is a power set, it contains an element that appears in strictly more than half of its member-sets.

conjecture | open | MathDB 页面 | 原文线索

V2-041832 | MATHDB-326322

The extremal-frequency form of Frankl's conjecture

摘录 / EXCERPT: Frankl's extremal-frequency conjecture. For every integer $n\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-041833 | MATHDB-326325

Enayati and Green's asymptotic conjecture for binary subword complexity sequences

摘录 / EXCERPT: Let $a_k(n)$ denote the number of distinct subword complexity sequences of length $n \ge 1$ over a k-letter alphabet. Enayati and Green's conjecture. ... This conjecture predicts t...

conjecture | open | MathDB 页面 | 原文线索

V2-041834 | MATHDB-326326

Asymptotic conjecture for subword complexity sequences over a k-letter alphabet

摘录 / EXCERPT: Let $a_k(n)$ denote the number of distinct subword complexity sequences of length $n \ge 1$ over a k-letter alphabet. Asymptotic conjecture. ... The conjecture is motivated by nume...

conjecture | open | MathDB 页面 | 原文线索

V2-041835 | MATHDB-326327

Conjectured recurrence for differences of subword complexity sequence counts

摘录 / EXCERPT: Let $a_k(n)$ denote the number of distinct subword complexity sequences of length $n \ge 1$ over a k-letter alphabet. Recurrence conjecture. There exists a function $f(k)$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-041836 | MATHDB-326342

The connected-heavy-edge conjecture for T_5-factors

摘录 / EXCERPT: Let M be a standard multigraph on n vertices, where n is divisible by 3. Let $\delta(M)$ denote its minimum degree, let H_M be its heavy-edge graph, and let a T_5-facto...

conjecture | open | MathDB 页面 | 原文线索

V2-041837 | MATHDB-326345

Extremality of towers of cubes for maximal chains

摘录 / EXCERPT: Let $X=\{1,2,\dots,n\}$, let $\mathcal{P}(X)$ be its power set, and suppose that $n=tk$ with $t,k\in\mathbb{N}$. For $1\leq i\leq k$, set $X_i=\{(i-1)t+1,\dots,it\}$, and define th...

conjecture | open | MathDB 页面 | 原文线索

V2-041838 | MATHDB-326346

Extremality of generalized towers of cubes for maximal chains

摘录 / EXCERPT: Let $X=\{1,2,\dots,n\}$ and let $\mathcal{P}(X)$ be its power set. A generalized tower of cubes is a family ... where $X_1,\dots,X_k$ are pairwise disjoint subsets of X whose union...

conjecture | open | MathDB 页面 | 原文线索

V2-041839 | MATHDB-326348

Inverse concentration conjecture for random multilinear forms

摘录 / EXCERPT: Inverse concentration conjecture. If $\rho_{\xi,\beta}(A)\ge n^{-B}$ for some generic ξ and small β, then most entries of A can be β-approximated by coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-041840 | MATHDB-326350

Closed-form enumeration conjecture for maximal dominant weights of affine sl(n)-modules

摘录 / EXCERPT: Enumeration conjecture. For fixed $n\geq 2$, the number of maximal dominant weights of $V(k\Lambda_0)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-041841 | MATHDB-326365

The structural conjecture for vertically 4-connected matroids with a modular 4-point line

摘录 / EXCERPT: Structural conjecture. Either M is ternary, for some $e\in L$ the deletion $M\backslash e$ is binary, or M has an F_7-minor and either an F_7^-- or $(F_7^-)^*$-minor.

conjecture | open | MathDB 页面 | 原文线索

V2-041842 | MATHDB-326394

Frankl-Akiyama conjecture on antichains in convex families

摘录 / EXCERPT: Let $\mathcal{B}_n$ be the Boolean lattice of subsets of $[n]$. For a family $\mathcal{F}\subseteq\mathcal{B}_n$, define its convex closure by ... A family is convex when...

conjecture | open | MathDB 页面 | 原文线索

V2-041843 | MATHDB-326402

The extremal incomparable-copy conjecture for the three-element poset V

摘录 / EXCERPT: Let B_n denote the Boolean lattice of subsets of an n-element set, and let V be the three-element poset with elements a,b,c and relations $a<b$ and $a<c$. Let $M_1(V,n)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-041844 | MATHDB-326412

De Loera-Lee-Margulies-Miller conjecture on minor-minimal non-weakly-orientable matroids

摘录 / EXCERPT: Minor-minimality conjecture. For all integers $n\geq 0$, $M\!I_n$ are minor-minimal with respect to being non-weakly-orientable. Hence, weak-orientability cannot be described by a...

conjecture | open | MathDB 页面 | 原文线索

V2-041845 | MATHDB-326456

Extremality of the small-rainbow colouring for infinite graphs

摘录 / EXCERPT: Small-rainbow extremality conjecture. If n is a natural number and $k>\binom{n}{2}+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-041846 | MATHDB-326464

Gromov's filling area conjecture for all genera

摘录 / EXCERPT: Let a membrane have genus $p\ge 0$ and a prescribed boundary metric, and let an Ahlfors map mean the extremal conformal map used to compute its area through isothermic coordinates....

conjecture | open | MathDB 页面 | 原文线索

V2-041847 | MATHDB-326465

Ahlfors extremals realizing the gonality

摘录 / EXCERPT: Let a compact bordered Riemann surface have gonality equal to the least degree of a conformal map to the disc, and let $f_{a,b}$ denote an Ahlfors extremal function associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-041848 | MATHDB-326530

da Silva's conjecture on realizability of positively oriented matroids

摘录 / EXCERPT: da Silva's conjecture. Every positively oriented matroid is realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-041849 | MATHDB-326540

Benevides-Griffiths-Przykucki conjecture on maximum percolation time

摘录 / EXCERPT: Benevides-Griffiths-Przykucki conjecture. For all fixed $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-041850 | MATHDB-326555

Nonexistence of a CMDRR$(5,3)$

摘录 / EXCERPT: A complete mixed doubles round robin tournament, abbreviated CMDRR(n,k), is a tournament on $2n$ players with the parameter k specifying the required number of games or rounds...

conjecture | open | MathDB 页面 | 原文线索

V2-041851 | MATHDB-326558

Savage-Visontai equidistribution conjecture for signed multiset permutations

摘录 / EXCERPT: Let $s=(1,4,3,8,5,12,\ldots)$, with $s_{2i}=4i$ and $s_{2i-1}=2i-1$ for $i\geq 1$. Let I_n be the set of s-inversion sequences of length n, and let P_n be the set of signed...

conjecture | open | MathDB 页面 | 原文线索

V2-041852 | MATHDB-326568

The proposed formula for the maximum number of directed 2-factors in odd tournaments

摘录 / EXCERPT: Odd-tournament maximum conjecture. For every odd positive integer $n>7$,

conjecture | open | MathDB 页面 | 原文线索

V2-041853 | MATHDB-326596

Extension condition for strict gammoids to satisfy the maximality axiom

摘录 / EXCERPT: Let (D,B_0) be a dimaze, and let $M_L(D,B_0)$ be its linkability system. For every $I\in M_L(D,B_0)$ and every $B\subseteq B_0$, suppose there is a maximal independent set in...

conjecture | open | MathDB 页面 | 原文线索

V2-041854 | MATHDB-326598

Elekes's uniform statistical conjecture for small image sets of lines

摘录 / EXCERPT: Let $\mathcal{L}$ denote the set of lines under consideration. For sets $\Phi\subseteq\mathcal{L}$ and $H\subseteq\mathbb{R}$ of size at least n, let $E\subseteq\Phi\times H$ def...

conjecture | open | MathDB 页面 | 原文线索

V2-041855 | MATHDB-326642

Ruzsa-Matolcsi conjecture on triangulations of planar sum sets

摘录 / EXCERPT: Let A and B be finite non-collinear point sets in $\mathbb{R}^2$. For a finite non-collinear set S, let b_S and i_S denote the numbers of points of S on the boundary an...

conjecture | open | MathDB 页面 | 原文线索

V2-041856 | MATHDB-326665

The extremal-function conjecture for generalized Davenport-Schinzel sequences

摘录 / EXCERPT: Let $t_1,t_2,\dots,t_s$ be integers and let $t=\max(t_1,t_2,\dots,t_s)$. For positive integers r,k,n, let $\operatorname{Ex}_r(12^{t_1}3^{t_2}\dots s^{t_{s-1}}(s+1)^{t_s}1,k,n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-041857 | MATHDB-326690

Interior optimal-barrier conjecture for the square

摘录 / EXCERPT: Interior optimal-barrier conjecture. An optimal barrier for the square is interior.

conjecture | open | MathDB 页面 | 原文线索

V2-041858 | MATHDB-326691

Optimal exterior-barrier length conjecture for the unit square

摘录 / EXCERPT: Exterior-barrier length conjecture. The length of an optimal exterior barrier for the unit square is 3.

conjecture | open | MathDB 页面 | 原文线索

V2-041859 | MATHDB-326729

Inverse large sieve stability conjecture for finite sets

摘录 / EXCERPT: Let $X_0\in\mathbb{N}$, let $\rho>0$, and let $X\in\mathbb{N}$ be sufficiently large in terms of X_0 and ρ. Suppose that $A,B\subset[X]$ satisfy ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-041860 | MATHDB-326730

Symmetric inverse large sieve conjecture for sets of integers

摘录 / EXCERPT: Let $\mathscr{A}$ be a set of positive integers satisfying ... for all sufficiently large primes p. Symmetric inverse large sieve conjecture. Either there is a rational quadratic...

conjecture | open | MathDB 页面 | 原文线索

V2-041861 | MATHDB-326731

Quantitative inverse Goldbach conjecture for dense sets

摘录 / EXCERPT: Let $\delta>0$. Dense-set inverse Goldbach conjecture. If X is sufficiently large in terms of δ and $A,B\subset[X]$ satisfy ... then $A+B$ contains a composite number. This...

conjecture | open | MathDB 页面 | 原文线索

V2-041862 | MATHDB-326734

Biró-Füredi-Jahanbekam stability conjecture for standard examples in posets

摘录 / EXCERPT: Let P be a finite poset, let $|P|$ denote its number of elements, and let $\dim(P)$ denote its order dimension. For a positive integer m, let S_m be the standard example on...

conjecture | open | MathDB 页面 | 原文线索

V2-041863 | MATHDB-326747

The fixed intersection property for uniform subsets

摘录 / EXCERPT: Fixed intersection conjecture. If $n\ne 2p$, then $[n]^{(p)}$ has the fixed intersection property.

conjecture | open | MathDB 页面 | 原文线索

V2-041864 | MATHDB-326761

Positivity conjecture for fused multi-state ASEP update matrices

摘录 / EXCERPT: Let $M_{i,i+1}^{(\ell)}$ be the update matrix ... Normalize $b_r^{(\ell)}=\begin{bmatrix}\ell\\r\end{bmatrix}\widetilde b_r^{(\ell)}$, where ... Because the update matrix is symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-041865 | MATHDB-326774

Jockusch's edge lower-bound conjecture for cubical polytopes

摘录 / EXCERPT: Let K be a cubical polytope of dimension $d\geq 3$, with V vertices and E edges. Jockusch's conjecture. ... The source attributes this conjecture to Jockusch and discusses part...

conjecture | open | MathDB 页面 | 原文线索

V2-041866 | MATHDB-326836

Solymosi's sparse removal conjecture

摘录 / EXCERPT: Let H be a graph with exponent α, meaning that its Turán number satisfies ... A graph is said to have the unique-copy property for H if every edge lies in exactly one copy...

conjecture | open | MathDB 页面 | 原文线索

V2-041867 | MATHDB-326851

The equality of quadratic-generation classes for matroid toric ideals

摘录 / EXCERPT: Let ${\mathcal M}_{{\mathcal Q}{\mathcal G}}$ be the class of matroids whose toric ideal J_M has a quadratic Gröbner basis, let ${\mathcal M}_{\mathcal Q}$ be the class of matroi...

conjecture | open | MathDB 页面 | 原文线索

V2-041868 | MATHDB-326852

The quadratic conjecture for 3-connected matroids

摘录 / EXCERPT: Let M be a 3-connected matroid, and let J_M be its toric ideal. 3-connected quadratic conjecture. For any 3-connected matroid M, 1. J_M is generated by quadratic binomi...

conjecture | open | MathDB 页面 | 原文线索

V2-041869 | MATHDB-326895

Li et al.'s rational sign-pattern minimum-rank conjecture

摘录 / EXCERPT: Li et al.'s conjecture. If $r\geqslant t-2$, then the minimum rank of S over the rationals is also r.

conjecture | open | MathDB 页面 | 原文线索

V2-041870 | MATHDB-326906

Alon and Tarsi's Latin square conjecture

摘录 / EXCERPT: Let n be a positive integer. A Latin square of order n is an $n\times n$ array with entries in $[n]=\{1,\ldots,n\}$, in which every element occurs precisely once in each row an...

conjecture | open | MathDB 页面 | 原文线索

V2-041871 | MATHDB-326918

Kadell's q-analogue of his constant-term hypothesis

摘录 / EXCERPT: Let n be a positive integer, let $M\subset\{1,2,\dots,n\}$, and for each $s\in M$ choose an index r_s such that $r_s\notin M$. Let $\boldsymbol{a}=(a_1,\dots,a_n)$ be nonnegati...

conjecture | open | MathDB 页面 | 原文线索

V2-041872 | MATHDB-326991

Mayhew-Newman-Whittle finite rank-axiom conjecture for representable matroids

摘录 / EXCERPT: A representable matroid is a matroid satisfying the rank axioms of a matroid and representable over a field. Mayhew-Newman-Whittle conjecture. It is impossible to characterise the...

conjecture | open | MathDB 页面 | 原文线索

V2-041873 | MATHDB-326992

Kinser class non-duality conjecture

摘录 / EXCERPT: For each integer $n>5$, let $\mathcal{K}_n$ be the class of matroids satisfying the n-th Kinser inequality, and let $\mathcal{K}_n^*$ denote its dual class. Kinser class non-dual...

conjecture | open | MathDB 页面 | 原文线索

V2-041874 | MATHDB-326993

Kinser-minus separation conjecture

摘录 / EXCERPT: For an integer $n>5$, let ${\mathrm{Kin}(n)}^-$ denote the matroid referred to in the source as ${\mathrm{Kin}(n)}^-$. Let $\mathcal{K}_n$ be the class of matroids satisfying the...

conjecture | open | MathDB 页面 | 原文线索

V2-041875 | MATHDB-326994

Kinser hierarchy dual-containment conjecture

摘录 / EXCERPT: For each integer $n>4$, let $\mathcal{K}_n$ be the class of matroids satisfying the n-th Kinser inequality, and let $\mathcal{K}_n^*$ denote its dual class. Kinser hierarchy dual...

conjecture | open | MathDB 页面 | 原文线索

V2-041876 | MATHDB-326995

Infinite Kinser class self-duality conjecture

摘录 / EXCERPT: Let $\mathcal{K}_\infty$ denote the class obtained from the hierarchy of Kinser classes, and let $\mathcal{K}_\infty^*$ denote its dual class. Infinite Kinser class self-duality co...

conjecture | open | MathDB 页面 | 原文线索

V2-041877 | MATHDB-326996

Dowling geometry Kinser-inequality conjecture

摘录 / EXCERPT: Let G be an abelian group, and let a Dowling geometry be the matroid constructed from the corresponding labelled graph whose circuits are the positive cycles and minimal connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-041878 | MATHDB-326997

Skew partial-field Kinser-inequality conjecture

摘录 / EXCERPT: Let $\mathbb{P}=(R,G)$ be a skew partial field, and let M be a matroid that is $\mathbb{P}$-representable, meaning that it arises from a $\mathbb{P}$-chain group as in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-041879 | MATHDB-327009

Classification conjecture for weighted type compositions of complete simple games

摘录 / EXCERPT: Classification conjecture. Each type composition $\widehat{n}$ is either weighted or there exists a type composition $\widehat{m}$ in this list with $\widehat{n}\ge\widehat{m}$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-041880 | MATHDB-327010

The transfinite columns property conjecture

摘录 / EXCERPT: Transfinite columns property conjecture. If A has bounded row sums, then A has the transfinite columns property.

conjecture | open | MathDB 页面 | 原文线索

V2-041881 | MATHDB-327032

Conjecture on contributions in simple matrices

摘录 / EXCERPT: Let a simple $m\times n$ $(0,1)$ matrix be a matrix with pairwise distinct columns, and let a contribution mean the configuration defined earlier in the paper. Contribution conject...

conjecture | open | MathDB 页面 | 原文线索

V2-041882 | MATHDB-327103

The doubling conjecture for the forbidden set $\{ababa,abcacbc\}$

摘录 / EXCERPT: Doubling conjecture. In general, it is not true that

conjecture | open | MathDB 页面 | 原文线索

V2-041883 | MATHDB-327104

The nonlinear forbidden-sequence containment conjecture

摘录 / EXCERPT: Nonlinear containment conjecture. Every nonlinear sequence σ contains $ababa$, $abcacbc$, or some sequence morally equivalent to $abcacbc$.

conjecture | open | MathDB 页面 | 原文线索

V2-041884 | MATHDB-327110

Random Heegaard-splitting volume central-limit conjecture

摘录 / EXCERPT: Let N_ϕ^3 be a random Heegaard splitting defined by a random gluing map ϕ of combinatorial length n, and let $V(N_\phi^3)$ be its hyperbolic volume. Heegaard-splitting...

conjecture | open | MathDB 页面 | 原文线索

V2-041885 | MATHDB-327128

Left Gog-GOGAm trapezoid equienumeration conjecture

摘录 / EXCERPT: For $k\leqslant n$, a left Gog or GOGAm trapezoid of shape (n,k) is obtained from the corresponding triangle by retaining the k leftmost diagonals. Left Gog-GOGAm trapezoid con...

conjecture | open | MathDB 页面 | 原文线索

V2-041886 | MATHDB-327129

Equidistribution conjecture for the alpha, beta and gamma statistics

摘录 / EXCERPT: For Gog, Magog and GOGAm triangles, define the statistics α, β, and γ as in the paper; in particular, $\alpha_{\mathrm{Gog}}(X)$ counts indices k with...

conjecture | open | MathDB 页面 | 原文线索

V2-041887 | MATHDB-327130

Refined left Gog-GOGAm trapezoid equienumeration conjecture

摘录 / EXCERPT: For each n,k, consider left Gog and GOGAm trapezoids of shape (n,k), and let $X_{1,1}=l$ denote their bottom entry. Refined left Gog-GOGAm conjecture. For each n,k,l, the...

conjecture | open | MathDB 页面 | 原文线索

V2-041888 | MATHDB-327131

The Polynomial Freiman-Ruzsa conjecture for subsets of ${\mathbb F}_2^n$

摘录 / EXCERPT: Let $S\subseteq{\mathbb F}_2^n$, and write its doubling constant as $|S+S|/|S|$. A translate of a subspace is a set of the form $x+V$, where V is a subspace. Polynomial Freiman-R...

conjecture | open | MathDB 页面 | 原文线索

V2-041889 | MATHDB-327132

The near-optimal linear compression structural conjecture

摘录 / EXCERPT: Let $S\subseteq{\mathbb F}_2^n$. Its linear compression dimension is the minimum dimension of a linear map that is injective on S, while its nonlinear compression dimension is...

conjecture | open | MathDB 页面 | 原文线索

V2-041890 | MATHDB-327154

The $G(\rho)<1+2\rho$ conjecture for addable minor-closed classes

摘录 / EXCERPT: Let $\mathcal{G}$ be an addable, minor-closed class of graphs, let $G(z)$ be its exponential generating function, and let ρ be the radius of convergence of $G(z)$....

conjecture | open | MathDB 页面 | 原文线索

V2-041891 | MATHDB-327186

The index conjecture for minimal zero-sum sequences of length four

摘录 / EXCERPT: The index conjecture. Every minimal zero-sum sequence S over G of length $|S|=4$ has

conjecture | open | MathDB 页面 | 原文线索

V2-041892 | MATHDB-327233

Hivert-Mallet-Mallet vectorial conjecture for irreducible k-shapes and surjective pistols

摘录 / EXCERPT: For $k\geq 3$, let IS_k be the set of irreducible k-shapes. For an irreducible k-shape λ, let $\overrightarrow{fr}(\lambda)=(t_1,\ldots,t_{k-2})\in\{0,1\}^{k-2}$ reco...

conjecture | open | MathDB 页面 | 原文线索

V2-041893 | MATHDB-327262

Boundedness conjecture for extremal classes on birational irreducible holomorphic symplectic manifolds

摘录 / EXCERPT: Boundedness conjecture. Let Y be an irreducible holomorphic symplectic manifold birational to X. The set

conjecture | open | MathDB 页面 | 原文线索

V2-041894 | MATHDB-327266

The exact-value conjecture for multicoloured triangles in connected colourings

摘录 / EXCERPT: Let $f(k)$ denote the minimum number of multicoloured triangles in a connected k-colouring of a complete graph. The conjecture concerns the exact value of this function for every...

conjecture | open | MathDB 页面 | 原文线索

V2-041895 | MATHDB-327276

Moshkovitz-Shapira's lower-bound conjecture for bipartite saturation

摘录 / EXCERPT: Let $1\le s\le t$ be integers. A bipartite graph is $K_{s,t}$-saturated if it contains no copy of $K_{s,t}$, but adding any missing edge between its color classes creates a copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-041896 | MATHDB-327299

Existence of arbitrarily long S-crucial permutations with respect to squares

摘录 / EXCERPT: Let an S-crucial permutation with respect to squares be a permutation that is crucial for the property of avoiding squares, with S denoting the relevant permutation property. S-c...

conjecture | open | MathDB 页面 | 原文线索

V2-041897 | MATHDB-327301

Wilf-equivalence of 2143 and 246135 with 2413

摘录 / EXCERPT: For a permutation pattern τ, let τ-avoidance mean avoidance of τ, and for sets of patterns let $\beta ot i\text{ any pattern in }B$ be understood through the permuta...

conjecture | open | MathDB 页面 | 原文线索

V2-041898 | MATHDB-327324

The maximal-move-range piece conjecture for coefficient periods

摘录 / EXCERPT: Maximal-move-range piece conjecture. Among all pieces with $|\mathbb P|\leq\lambda$, $\mathbb P^{\max}_\lambda$ maximizes the period of every coefficient γ_i, for fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-041899 | MATHDB-327325

The γ_3 period conjecture for rider pieces

摘录 / EXCERPT: Let $\mathbf{M}$ be the move set of a piece $\mathbb P$, with basic moves (c_r,d_r), and write ... Let γ_3 denote the third coefficient of the quasipolynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-041900 | MATHDB-327326

The γ_4 period divisibility conjecture

摘录 / EXCERPT: Let $\mathbf{M}$ be the move set of a piece $\mathbb P$, with basic moves (c_r,d_r), and set ... Let γ_4 denote the fourth coefficient of the quasipolynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-041901 | MATHDB-327327

The lcm-maximal piece conjecture for coefficient periods

摘录 / EXCERPT: For a piece $\mathbb P$ with move set $\mathbf{M}$, write ... For a positive integer λ, let $\mathbb P^{\operatorname{lcm}}_\lambda$ be the lcm-maximal piece whose move set c...

conjecture | open | MathDB 页面 | 原文线索

V2-041902 | MATHDB-327328

The monotonicity and divisibility conjecture for coefficient periods

摘录 / EXCERPT: Let $u_{\mathbb P}(q;n)$ be the quasipolynomial for a piece $\mathbb P$, and write p_i for the period of its coefficient γ_i. Coefficient-period monotonicity conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-041903 | MATHDB-327329

The period-denominator equality conjecture for inside-out polytopes

摘录 / EXCERPT: Let $\mathscr{A}_{\mathbb P}$ be the inside-out arrangement associated with the piece $\mathbb P$, and let $D([0,1]^{2q},\mathscr{A}_{\mathbb P})$ denote the denominator of the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-041904 | MATHDB-327340

The shifted polygon setup perimeter-to-area conjecture

摘录 / EXCERPT: A shifted (k,n)-setup is a configuration of k regular polygons with n vertices and unit side length, where the polygons are shifted as in the paper. Let k and n be coprim...

conjecture | open | MathDB 页面 | 原文线索

V2-041905 | MATHDB-327350

Conjecture on small sumsets of iterated product sets

摘录 / EXCERPT: Iterated product-set sum conjecture. The k-fold sumset of $A^{(k)}$ should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-041906 | MATHDB-327351

Threefold product-of-sums growth conjecture

摘录 / EXCERPT: Threefold product-of-sums conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-041907 | MATHDB-327359

Mayhew-Royle's conjecture on the number of bases of matroids

摘录 / EXCERPT: Let (n,r,b) be integers with $0<r\leq n$ and $1\leq b\leq\binom{n}{r}$. An (n,r,b)-matroid is a matroid with n elements, rank r, and exactly b bases. Mayhew-Royle's conje...

conjecture | solved | MathDB 页面 | 原文线索

V2-041908 | MATHDB-327360

Conjecture on the existence of linear matroids with prescribed numbers of bases

摘录 / EXCERPT: Let (n,r,b) be integers with $0<r\leq n$ and $1\leq b\leq\binom{n}{r}$. A linear (n,r,b)-matroid is a matroid with n elements, rank r, and exactly b bases that is represe...

conjecture | open | MathDB 页面 | 原文线索

V2-041909 | MATHDB-327375

The maximum-point conjecture for maximal intersecting families

摘录 / EXCERPT: Let $N(k)$ denote the maximum number of points in a maximal intersecting family of k-sets. Maximum-point conjecture. For $k\geq 2$, ... Hanson and Toft proved the equality for...

conjecture | open | MathDB 页面 | 原文线索

V2-041910 | MATHDB-327376

Tuza's conjecture for the numbers n(k,t)

摘录 / EXCERPT: Let $n(k,t)$ be the quantity defined for the transversal problem on uniform intersecting families. Tuza's conjecture. For $k\geq t+2$, ... The paper attributes this precise conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-041911 | MATHDB-327381

Blum's conjecture on tilings of hexagonal dungeons

摘录 / EXCERPT: Blum's conjecture. The number of tilings of $HD_{a,2a,b}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-041912 | MATHDB-327385

The symmetry conjecture for the rational combinatorial Hilbert series

摘录 / EXCERPT: The combinatorial Hilbert-series symmetry conjecture. The series is symmetric in q and t for all m and n:

conjecture | open | MathDB 页面 | 原文线索

V2-041913 | MATHDB-327386

The termination conjecture for the affine inverse algorithm

摘录 / EXCERPT: Affine inverse-algorithm termination conjecture. For the resulting U, there exists N such that for every $i\ge N$ and every $t\in\mathbb{N}$,

open_problem | open | MathDB 页面 | 原文线索

V2-041914 | MATHDB-327389

Fischer-Riegler multivariate Laurent polynomial inversion conjecture

摘录 / EXCERPT: Fischer-Riegler Laurent polynomial inversion conjecture. If $s\leq t$, then for every $i\in\{1,\ldots,s+t-1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-041915 | MATHDB-327405

Douglas-Reinbacher-Yau conjecture on extremal bundles

摘录 / EXCERPT: Let X be a simply-connected Calabi-Yau threefold, and let V be a stable vector bundle of rank n with $c_1(V)=0$. Prescribe $c_2(V)$ and $c_3(V)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041916 | MATHDB-327416

The critical-number conjecture for dense triangle-free binary matroids

摘录 / EXCERPT: Let M be a simple triangle-free binary matroid, let $r(M)$ denote its rank, and let $|M|$ denote its number of elements. The critical number of M is the minimum codimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-041917 | MATHDB-327442

Lickorish's collapsibility conjecture for subdivisions of simplices

摘录 / EXCERPT: Lickorish's conjecture. Any simplicial subdivision of a d-simplex is collapsible.

conjecture | open | MathDB 页面 | 原文线索

V2-041918 | MATHDB-327443

Shephard's dimension-preserving subpolytope conjecture

摘录 / EXCERPT: Shephard's conjecture. For every $d\ge 0$, every combinatorial type of d-dimensional polytope can be realized using subpolytopes of d-dimensional stacked polytopes.

conjecture | open | MathDB 页面 | 原文线索

V2-041919 | MATHDB-327455

Hirsch conjecture for dual graphs of polytope boundary ideals

摘录 / EXCERPT: Hirsch conjecture. If Δ is the boundary of a convex polytope, then I_Δ is Hirsch.

conjecture | open | MathDB 页面 | 原文线索

V2-041920 | MATHDB-327489

Product-form conjecture for hexagonal dungeons with defects

摘录 / EXCERPT: Let $HH_{a,b}$ be the hexagonal dungeon with jagged contour side-lengths $a,a+1,b,a,a+1,b$, after removing $a-1$ triangles from each of the west and east sides. For a tiling...

open_problem | open | MathDB 页面 | 原文线索

V2-041921 | MATHDB-327495

Hajek's multinomial maximum conjecture

摘录 / EXCERPT: Let k,n be positive integers. For each $j\in\{1,\ldots,k\}$, let S_j be the collection of subsets $L\subseteq[n]^k$ such that $|L|=n$ and no two distinct elements of L have t...

conjecture | open | MathDB 页面 | 原文线索

V2-041922 | MATHDB-327532

Conjecture on sums of any number of points in finite fields

摘录 / EXCERPT: Let $E\subset \mathbb F_q^d$, with $d\geq 2$, and for every integer $k\geq 2$ define the k-resultant magnitude set ... where $\|\mathbf{v}\|=v_1^2+\cdots+v_d^2$. Conjecture on sums...

conjecture | open | MathDB 页面 | 原文线索

V2-041923 | MATHDB-327569

Ben-Sasson-Ron-Zewi approximate duality conjecture

摘录 / EXCERPT: Let $A,B\subset\mathbb{F}_2^r$ be sets. Define their approximate duality measure by ... and call them ε-approximate dual when this measure is at least ε. Appr...

conjecture | open | MathDB 页面 | 原文线索

V2-041924 | MATHDB-327571

The Graßmann-Tamari lattice conjecture

摘录 / EXCERPT: Let $\mathcal{T}_{k,n}$ denote the Graßmann-Tamari poset on the relevant Graßmannian tableaux. The usual Tamari order is recovered when $k=2$. Graßmann-Tamari lattice conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-041925 | MATHDB-327573

The Polynomial Freiman-Ruzsa conjecture for approximate homomorphisms

摘录 / EXCERPT: Polynomial Freiman-Ruzsa conjecture. There is a linear map $\tilde f:V\rightarrow V$ and a set $\tilde S$ with $|\tilde S|\ll |S|^C$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041926 | MATHDB-327591

Green-Tao nilsequence inverse conjecture for the Gowers norms

摘录 / EXCERPT: Inverse conjecture for the Gowers norms. The stated correlation with a bounded Lipschitz nilsequence should hold for every s, δ, N, and f satisfying the hypotheses.

conjecture | solved | MathDB 页面 | 原文线索

V2-041927 | MATHDB-327621

Fox-Mahdian-Radoičić's classification conjecture for rainbow regular matrices

摘录 / EXCERPT: Let A be a matrix with integer entries. The rows of A are linearly independent when no nontrivial linear combination of them is zero, and a vector in $\ker(A)$ has distinct pos...

conjecture | open | MathDB 页面 | 原文线索

V2-041928 | MATHDB-327641

Distinct prefix normal words have distinct extension-count sequences

摘录 / EXCERPT: Let $v,w e 0$ be distinct prefix normal binary words, with extension counts $\textit{ext}(v,m)$ and $\textit{ext}(w,m)$ denoting the numbers of prefix normal extensions of length...

conjecture | open | MathDB 页面 | 原文线索

V2-041929 | MATHDB-327642

Asymptotic conjecture for prefix normal words and extension-critical words

摘录 / EXCERPT: For each n, let $\textit{pnw}(n)$ be the number of prefix normal binary words of length n, and let $\textit{crit}(n)$ be the number of extension-critical words of length n. P...

conjecture | open | MathDB 页面 | 原文线索

V2-041930 | MATHDB-327643

Linear average-case performance conjecture for MemberPN

摘录 / EXCERPT: Let w be a binary word, and let ${\sc MemberPN}(w)$ be the proposed membership tester that applies the two linear-time rejection tests followed by a quadratic-time prefix-normali...

conjecture | open | MathDB 页面 | 原文线索

V2-041931 | MATHDB-327666

A lower-bound conjecture for the regularity of long linear equations

摘录 / EXCERPT: For a positive integer r, let an equation be r-regular when every coloring of the positive integers with r colors has a monochromatic positive-integer solution. Lower-bound c...

conjecture | open | MathDB 页面 | 原文线索

V2-041932 | MATHDB-327682

A reconstruction claim for binomial partial Steiner triple systems

摘录 / EXCERPT: Reconstruction claim. There are two nonisomorphic $\binkonf(n,1)$-configurations, each with m complete subgraphs K_n, such that the corresponding complementary subconfiguration...

conjecture | open | MathDB 页面 | 原文线索

V2-041933 | MATHDB-327696

The harmonic-bundle conjecture for four-covector flats of vee-systems

摘录 / EXCERPT: Harmonic-bundle conjecture. The corresponding four lines form a harmonic bundle with two orthogonals.

conjecture | open | MathDB 页面 | 原文线索

V2-041934 | MATHDB-327697

The uniqueness conjecture for irreducible vee-systems

摘录 / EXCERPT: Uniqueness conjecture. An irreducible $\vee$-system $\mathcal{A}$ is uniquely determined, modulo the linear group $GL(V^*)$, by its matroid M and the corresponding ν-function...

conjecture | open | MathDB 页面 | 原文线索

V2-041935 | MATHDB-327698

The extension conjecture for vee-systems and their nu-functions

摘录 / EXCERPT: Extension conjecture. The values of the ν-functions on the corresponding flats of an irreducible $\vee$-system and its extension are proportional.

conjecture | open | MathDB 页面 | 原文线索

V2-041936 | MATHDB-327726

Matroid packing/covering conjecture

摘录 / EXCERPT: Let M and N be tame matroids on the same ground set E. A packing for a pair of matroids on a set is a pair of disjoint spanning sets, one for each matroid, and a covering is...

conjecture | open | MathDB 页面 | 原文线索

V2-041937 | MATHDB-327764

Lauritzen's score matching estimability conjecture

摘录 / EXCERPT: Lauritzen's conjecture. The graph G is n-estimable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-041938 | MATHDB-327770

The logarithmic upper-bound conjecture for anti-van der Waerden numbers

摘录 / EXCERPT: Logarithmic upper-bound conjecture. There exists a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-041939 | MATHDB-327771

The power-of-three anti-van der Waerden conjecture

摘录 / EXCERPT: Power-of-three anti-van der Waerden conjecture. For every nonnegative integer m,

conjecture | open | MathDB 页面 | 原文线索

V2-041940 | MATHDB-327808

The first index-theoretic alternative form of the rational cuspidal curve conjecture

摘录 / EXCERPT: Let $C\subset\mathbb{C}P^2$ be a rational cuspidal curve of degree d with ν cusps and let F be the function defined from the semigroup counting functions of its singulariti...

conjecture | open | MathDB 页面 | 原文线索

V2-041941 | MATHDB-327810

Sheffield's extremal Brownian Gibbs line ensemble conjecture

摘录 / EXCERPT: Sheffield's extremal Brownian Gibbs line ensemble conjecture. As a convex set, the extremal points of $\mathscr{G}_{\Theta}$ are exactly $\{\mathcal{L}^c\mid c\in\mathbb{R}\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-041942 | MATHDB-327818

Graver-basis edge-connectivity conjecture for fiber graphs

摘录 / EXCERPT: Graver-basis edge-connectivity conjecture. In all Graver fiber graphs of A, the edge-connectivity equals the minimal degree.

conjecture | open | MathDB 页面 | 原文线索

V2-041943 | MATHDB-327828

Manickam-Singhi conjecture for sets

摘录 / EXCERPT: Manickam-Singhi conjecture. If $n\geq 4k$, then the number of k-element subsets $S\subseteq X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-041944 | MATHDB-327839

Linial-Morgenstern conjecture on minimizing 4-cycles in tournaments

摘录 / EXCERPT: Let T be a tournament on n vertices, and let c_3 and c_4 denote its normalized numbers of cyclic triples and cyclic 4-cycles, respectively. For a transitive tournament...

conjecture | open | MathDB 页面 | 原文线索

V2-041945 | MATHDB-327844

Primitivity conjecture for groups generated by a supersimple design

摘录 / EXCERPT: Let $\mathcal{D}=(\Omega,\mathcal{B})$ be a supersimple 2-$(n,4,\lambda)$ design, and let $\pi_\infty(\mathcal{D})$ denote the group associated with the point $\infty\in\Omega$....

conjecture | open | MathDB 页面 | 原文线索

V2-041946 | MATHDB-327845

Eventual symmetric-or-alternating-group conjecture for supersimple designs

摘录 / EXCERPT: Let $\mathcal{D}=(\Omega,\mathcal{B})$ be a supersimple 2-$(n,4,\lambda)$ design, and let $\pi_\infty(\mathcal{D})$ be the permutation group associated with the point...

conjecture | open | MathDB 页面 | 原文线索

V2-041947 | MATHDB-327848

Eventual infinite log-concavity conjecture for powers of positive polynomials

摘录 / EXCERPT: Eventual infinite log-concavity conjecture. There exists N such that, for every $\lambda\geqslant N$, the coefficients of $p(x)^\lambda$ are ∞-log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-041948 | MATHDB-327850

Li's unimodality conjecture for bounded-composition largest parts

摘录 / EXCERPT: Let $a(k,n,q)$ denote the number of compositions of k with n parts whose largest part is q, and let $\binom{n,q}{k}$ denote the coefficient of x^k in...

conjecture | open | MathDB 页面 | 原文线索

V2-041949 | MATHDB-327855

Armstrong-Hanusa-Jones average-size conjecture for core partitions

摘录 / EXCERPT: Armstrong-Hanusa-Jones conjecture. The average size in each of these two sets is

conjecture | open | MathDB 页面 | 原文线索

V2-041950 | MATHDB-327884

Hirsch bound conjecture for the circuit diameter of polyhedra

摘录 / EXCERPT: Circuit diameter bound. For any n-dimensional polyhedron with f facets, the circuit diameter is bounded above by

conjecture | open | MathDB 页面 | 原文线索

V2-041951 | MATHDB-327907

The domination conjecture for symmetric biplanes

摘录 / EXCERPT: Biplane domination conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-041952 | MATHDB-327908

The residual-design domination conjecture

摘录 / EXCERPT: Residual-design domination conjecture. The domination numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-041953 | MATHDB-327917

Kim's symmetry conjecture for crossing and nesting numbers of set partitions

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, and let P be a set partition of $[n]$. Its front representation is the linear representation obtained by joining consecutive elements in each block. F...

conjecture | open | MathDB 页面 | 原文线索

V2-041954 | MATHDB-327918

de Mier's symmetry conjecture for crossing and nesting numbers of linked partitions

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, and let P be a linked partition of $[n]$. In its linear representation, arcs are pairs (i,j) with $i<j$. A k-crossing consists of arcs satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-041955 | MATHDB-327929

The difference conjecture for disjoint Golomb rulers

摘录 / EXCERPT: Let $H(I,J)$ denote the least positive integer n such that there exist I disjoint Golomb rulers, each a J-subset of $\\{1,2,\ldots,n\\}$. Difference conjecture. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-041956 | MATHDB-327930

The persistence conjecture for regular disjoint Golomb rulers

摘录 / EXCERPT: Let $H(I,J)$ be the least positive integer n such that there exist I disjoint Golomb rulers, each a J-subset of $\\{1,2,\ldots,n\\}$. A DGR is regular when $H(I,J)=IJ$. Persi...

conjecture | open | MathDB 页面 | 原文线索

V2-041957 | MATHDB-327931

The regularity conjecture for $H(I,I+2)$

摘录 / EXCERPT: Let $H(I,J)$ be the least positive integer n such that there exist I disjoint Golomb rulers, each a J-subset of $\\{1,2,\ldots,n\\}$. Regularity conjecture. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-041958 | MATHDB-327948

The conjectural generating-series formula for minimal transitive cycle factorizations with three cycles

摘录 / EXCERPT: Conjectural formula for Ψ_3. With this notation,

conjecture | open | MathDB 页面 | 原文线索

V2-041959 | MATHDB-327991

The COVP characterization conjecture for the feasible $(4,2)$-AP

摘录 / EXCERPT: COVP characterization conjecture. A feasible instance of the $(4,2)$-AP with cost array C of size $n\neq 3$ has the COVP if and only if C is sum-decomposable with parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-041960 | MATHDB-328000

The independent set conjecture for triangle-free 3-connected matroids

摘录 / EXCERPT: Independent set conjecture. If M is a triangle-free 3-connected matroid with a simple 3-connected minor N, then M has an independent set I of N-contractible elements...

conjecture | open | MathDB 页面 | 原文线索

V2-041961 | MATHDB-328049

Enumeration of minimal factors of the form xxx^R

摘录 / EXCERPT: Minimal-xxx^R enumeration conjecture. For $n=F_{3k+1}$ there are 4 such words of length n. For $n=F_{3k+1}\pm F_{3k-2}$ there are 2 such words. Otherwise there are none. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-041962 | MATHDB-328051

Friedland's Lower Matching Conjecture

摘录 / EXCERPT: Let G be a d-regular bipartite graph on $v(G)=2n$ vertices, and let $m_k(G)$ denote the number of matchings of size k. Put $p=\frac{k}{n}$. Friedland's Lower Matching Conject...

conjecture | open | MathDB 页面 | 原文线索

V2-041963 | MATHDB-328061

The general sweep-map bijectivity conjecture

摘录 / EXCERPT: Let $A=x_1,\ldots,x_k\}$ be an alphabet and $\operatorname{wt}:A\rightarrow\mathbb{Z}$ a weight function. For nonnegative integers $n_1,\ldots,n_k$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-041964 | MATHDB-328073

Butterfly supersaturation conjecture up to the packing number

摘录 / EXCERPT: Butterfly supersaturation conjecture. Let $E=E(n)\le K(n,\lceil n/2\rceil+1)$. If n is large enough, then the minimum number of butterflies a family $\mathcal F\subset 2^{[n]}$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-041965 | MATHDB-328074

Butterfly-free family stability conjecture

摘录 / EXCERPT: Butterfly-free stability conjecture. Let $\mathcal F\subseteq 2^{[n]}$ be butterfly-free with $\emptyset,[n]\notin\mathcal F$. If $|\mathcal M|$ is not too large, then

conjecture | open | MathDB 页面 | 原文线索

V2-041966 | MATHDB-328075

Consecutive-level minimizer conjecture for tree-shaped posets

摘录 / EXCERPT: Tree-poset consecutive-level minimizer conjecture. There exists a constant $c=c(T)$ with $0<c\le1$ such that, if $E\le c\binom{n}{\lfloor n/2\rfloor}$, then some family...

conjecture | open | MathDB 页面 | 原文线索

V2-041967 | MATHDB-328076

Nevo's generalized lower bound conjecture for homology spheres

摘录 / EXCERPT: For integers $d,i\geq 1$, let $\mathcal{HS}(i,d)$ be the family of $(d-1)$-dimensional homology spheres without missing faces of dimension greater than i. For...

conjecture | open | MathDB 页面 | 原文线索

V2-041968 | MATHDB-328086

Vanishing proportion conjecture for MSTD subsets of lattice polytopes

摘录 / EXCERPT: Let P be a polytope in $\mathbb{R}^D$ with vertices in $\mathbb{Z}^D$, and let $L(P)=P\cap\mathbb{Z}^D$. Call $L(P)$ balanced when it has the balance property defined in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-041969 | MATHDB-328102

Critical-number strengthening for dense triangle-free binary matroids

摘录 / EXCERPT: Let M be a simple triangle-free binary matroid, with ground-set size $|M|$ and rank $r(M)$. Critical-number strengthening conjecture. If ... then $\chi(M)\in\{1,2\}$. The preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-041970 | MATHDB-328103

Critical thresholds for critical-number-two binary matroids

摘录 / EXCERPT: Let $\mathcal{N}$ be the class of simple binary matroids of critical number 2. For $\delta\in\{0,\tfrac14,\tfrac12\}$, let $\mathcal{N}_\delta$ be the three subclasses defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-041971 | MATHDB-328104

Critical threshold formula for higher critical number

摘录 / EXCERPT: Let N be a simple binary matroid of critical number $c\geq 2$. A k-codimensional subspace of N is obtained by intersecting its ground set, in a projective-geometry representa...

conjecture | open | MathDB 页面 | 原文线索

V2-041972 | MATHDB-328117

Real-rootedness of the Harer-Zagier coefficient polynomial

摘录 / EXCERPT: Real-rootedness conjecture. For any fixed g, the polynomial $K_g(x)$ has only real zeros located in $(-\frac14,0]$. Consequently, the sequence $\\{\kappa_{g,t}\\}_{t=0}^g$ is inf...

conjecture | open | MathDB 页面 | 原文线索

V2-041973 | MATHDB-328133

Regularity conjecture for the rational functions $A_{g,r}(z)$

摘录 / EXCERPT: Regularity conjecture. The rational function $A_{g,r}(z)$ is regular at every nontrivial rth root of unity.

conjecture | open | MathDB 页面 | 原文线索

V2-041974 | MATHDB-328208

The higher-dimensional universality conjecture for bootstrap percolation update families

摘录 / EXCERPT: Higher-dimensional universality conjecture. begin{enumerate} item If $\mathcal{U}$ is subcritical, then

conjecture | open | MathDB 页面 | 原文线索

V2-041975 | MATHDB-328246

The upper matching conjecture for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph on $2n$ vertices, with $d\mid 2n$, and let ${\mathcal M}(G)$ be its collection of all matchings. Write $K(n,d)$ for the disjoint union of n/d copie...

conjecture | open | MathDB 页面 | 原文线索

V2-041976 | MATHDB-328247

The upper matching conjecture for matchings of each size

摘录 / EXCERPT: Let G be a d-regular graph on $2n$ vertices, with $d\mid 2n$, and let ${\mathcal M}_t(G)$ denote its matchings of size t. Write $K(n,d)$ for the disjoint union of n/d copie...

conjecture | open | MathDB 页面 | 原文线索

V2-041977 | MATHDB-328249

The Widom-Rowlinson extremal conjecture for regular graphs

摘录 / EXCERPT: Let $H_{\rm WR}$ be the complete fully looped graph on three vertices with one non-loop edge removed, equivalently the complete looped path on three vertices. Let G be an n-ver...

conjecture | open | MathDB 页面 | 原文线索

V2-041978 | MATHDB-328268

Achievement-game nimber conjecture for symmetric and alternating groups

摘录 / EXCERPT: For $n\geq 5$, consider the symmetric group S_n and alternating group A_n, and let $\text{\sf GEN}(S_n)$ and $\text{\sf GEN}(A_n)$ denote their achievement games, with nim-numb...

conjecture | open | MathDB 页面 | 原文线索

V2-041979 | MATHDB-328300

Liu-Wang's log-convexity conjecture for the Eulerian transformation

摘录 / EXCERPT: Let $\{x_n\}_{n\geq 0}$ be a sequence of nonnegative numbers, and define $z_n=\sum_{k=0}^n A_{n,k}x_k$ using the Eulerian triangle $\{A_{n,k}\}_{0\leq k\leq n}$. A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-041980 | MATHDB-328314

Extension conjecture for Nim sequences under the recurrence $b_i=(m+1)b_{i-1}-(m-1)b_{i-2}$

摘录 / EXCERPT: Let $(a_i)_{i=0}^n$ with $n\geq2$ be a finite representing sequence with representation word w, and let m be an integer with $n\geq2$. Define (b_i) by ... and let v be the re...

conjecture | open | MathDB 页面 | 原文线索

V2-041981 | MATHDB-328325

The second-moment lower-bound conjecture for lonely runners

摘录 / EXCERPT: Let X be the random variable used in the paper to count the relevant loneliness events for a set of distinct speeds $v_1,\dots,v_n$, and let $\delta<1$. Second-moment lower-bound...

conjecture | open | MathDB 页面 | 原文线索

V2-041982 | MATHDB-328330

The residue-table meta-conjecture for k-Schur functions

摘录 / EXCERPT: Let k be a positive integer, let $ u$ be a k-bounded partition, and let $R(u)$ be its residue table. The theory also involves k-bounded partitions and k-Schur functions, i...

conjecture | open | MathDB 页面 | 原文线索

V2-041983 | MATHDB-328338

Hansen and Zwick's Fibonacci conjecture for Order-Regular matrices

摘录 / EXCERPT: Let an Order-Regular matrix be a binary matrix satisfying the Order-Regularity constraints arising from Policy Iteration on cube acyclic Unique Sink Orientations. For an n-column...

conjecture | open | MathDB 页面 | 原文线索

V2-041984 | MATHDB-328416

Boundedness conjecture for axisymmetric scalar waves in extremal Kerr interiors

摘录 / EXCERPT: Let $(\mathcal M,g)$ be extremal Kerr spacetime with mass M and angular momentum per unit mass a, where $M=|a|\neq 0$. Let ϕ be an axisymmetric solution of ... arising from...

conjecture | open | MathDB 页面 | 原文线索

V2-041985 | MATHDB-328421

Las Vergnas's single-reorientation-class conjecture for hypercube matroids

摘录 / EXCERPT: Let C_d be the matroid obtained from the affine d-dimensional hypercube. A single-reorientation-class conjecture asserts that C_d has a single reorientation class for every...

conjecture | open | MathDB 页面 | 原文线索

V2-041986 | MATHDB-328422

Las Vergnas's simplex conjecture for uniform oriented matroids

摘录 / EXCERPT: A uniform oriented matroid is an oriented matroid whose underlying matroid is uniform, and a mutation is the operation referred to in the source for such oriented matroids. Las Ver...

conjecture | open | MathDB 页面 | 原文线索

V2-041987 | MATHDB-328491

Hitting clause-sets attain the minimum variable-degree of minimally unsatisfiable clause-sets

摘录 / EXCERPT: For $k\in\mathbb{N}$, let $\operatorname{minnonmer}(k)$ be the minimum variable-degree over minimally unsatisfiable clause-sets of deficiency k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041988 | MATHDB-328492

Sharp lower bound for the minimum variable-degree of minimally unsatisfiable clause-sets

摘录 / EXCERPT: For $k\in\mathbb{N}$, let $\operatorname{minnonmer}(k)$ denote the minimum variable-degree among minimally unsatisfiable clause-sets of deficiency k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041989 | MATHDB-328493

Full clauses attain the non-Mersenne lower bound up to one

摘录 / EXCERPT: For $k\in\mathbb{N}$, let $\operatorname{maxsmar}(k)$ be the maximum number of full clauses in a minimally unsatisfiable clause-set of deficiency k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-041990 | MATHDB-328497

Exact linear bound for nonsingular unsatisfiable hitting clause-sets

摘录 / EXCERPT: For $k\ge2$, let $\mathcal{U}\mathcal{HIT}_{\delta=k}^{\mathrm{ns}}$ be the class of nonsingular unsatisfiable hitting clause-sets of deficiency k, and let $n(F)$ denote the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-041991 | MATHDB-328500

The finite Turán threshold conjecture for posets

摘录 / EXCERPT: Finite Turán threshold conjecture. Every poset has a finite Turán threshold, that is, $\pi^{*}(P)<\infty$ for every finite poset P.

conjecture | open | MathDB 页面 | 原文线索

V2-041992 | MATHDB-328501

The finite Lubell threshold conjecture for posets

摘录 / EXCERPT: Finite Lubell threshold conjecture. Every poset has a finite Lubell threshold, that is, $\lambda^{*}(P)<\infty$ for every finite poset P.

conjecture | open | MathDB 页面 | 原文线索

V2-041993 | MATHDB-328511

Grósz-Methuku-Tompkins conjecture on forbidden subposet densities

摘录 / EXCERPT: Grósz-Methuku-Tompkins conjecture. For every poset P, the limit $\pi(P)$ exists and

conjecture | open | MathDB 页面 | 原文线索

V2-041994 | MATHDB-328512

Induced forbidden subposet density conjecture

摘录 / EXCERPT: Induced forbidden subposet density conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-041995 | MATHDB-328532

Preservation of the arithmetic hierarchy for cohesiveness and Erdős-Moser principles

摘录 / EXCERPT: Preservation conjecture. Both $coperatorname{COH}$ and $coperatorname{EM}$ admit preservation of the arithmetic hierarchy.

conjecture | open | MathDB 页面 | 原文线索

V2-041996 | MATHDB-328537

Nonnegativity of the beta parameter for regular Stanley sequences

摘录 / EXCERPT: Let $S(A)=\{a_n\}$ be a regular Stanley sequence. For sufficiently large k, write ... where $\sigma(A)$ is the shift index and $\beta(A)$ is the associated rational parameter. Nonn...

conjecture | open | MathDB 页面 | 原文线索

V2-041997 | MATHDB-328538

Type 2 growth for irregular Stanley sequences

摘录 / EXCERPT: Type 2 growth conjecture. Every irregular Stanley sequence follows Type 2 growth. The paper proves that regular Stanley sequences follow Type 1 growth and identifies this assertion...

conjecture | open | MathDB 页面 | 原文线索

V2-041998 | MATHDB-328539

Faithfulness of regular Stanley sequences to their cores

摘录 / EXCERPT: Let $S(A)=\{a_n\}$ and $S(A')=\{a'_n\}$ be Stanley sequences. Say that $S(A)$ is faithful to $S(A')$ if, for every $a'_n<\omega(A')$, there is an index m such that $a_m=a'_n$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-041999 | MATHDB-328540

Character range conjecture for Stanley sequences

摘录 / EXCERPT: For a regular Stanley sequence $S(A)$, let $\lambda(A)$ denote its character. The range of the character function is the set of values attained by $\lambda(A)$ over regular Stanley...

conjecture | open | MathDB 页面 | 原文线索

V2-042000 | MATHDB-328541

Generalized shifted-sequence construction for regular Stanley sequences

摘录 / EXCERPT: Generalized shift conjecture. Then $S_k(c,A)$ is defined and is a regular Stanley sequence with core $S(A)$. This would strengthen the preceding construction theorem for shifted St...

conjecture | open | MathDB 页面 | 原文线索

V2-042001 | MATHDB-328542

Infinitely many deletable elements in dependent Stanley sequences

摘录 / EXCERPT: Let $S(A)$ be a dependent Stanley sequence, meaning a regular Stanley sequence that is not independent. An element of $S(A)$ is deletable if deleting it produces another dependent...

conjecture | open | MathDB 页面 | 原文线索

V2-042002 | MATHDB-328564

The Schur superpolynomial identification conjecture

摘录 / EXCERPT: A superpartition is denoted by $[?]$; let $s^c_\Lambda(x,\theta)$ be the combinatorially defined Schur superpolynomial and let $s_\Lambda(x,\theta)$ be the specialization at...

conjecture | open | MathDB 页面 | 原文线索

V2-042003 | MATHDB-328609

Katona-Lu-Milans induced forbidden subposet bound

摘录 / EXCERPT: Katona-Lu-Milans conjecture. For every poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-042004 | MATHDB-328610

Lu-Milans bounded Lubell function conjecture for induced-free families

摘录 / EXCERPT: Lu-Milans conjecture. For every poset P, there is a constant C such that every induced P-free family $\mathcal{F}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042005 | MATHDB-328612

Shifted Lubell conjecture in terms of middle-level parameters

摘录 / EXCERPT: Shifted Lubell conjecture. For every P-free family $\mathcal{F}$,

conjecture | open | MathDB 页面 | 原文线索

V2-042006 | MATHDB-328618

First-singularity conjecture for nonlinear waves on extremal Reissner-Nordström spacetimes

摘录 / EXCERPT: First-singularity conjecture. Spherically symmetric solutions of this equation have their first singularities on the event horizon $\mathcal{H}^{+}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042007 | MATHDB-328621

The two-type growth conjecture for Stanley sequences

摘录 / EXCERPT: A Stanley sequence $S(A)=\{a_n\}$ is generated by the greedy construction from a finite 3-free set A of nonnegative integers. The two-type growth conjecture. Every Stanley sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-042008 | MATHDB-328632

The Vassiliev degree bound for far combinatorial cohomology classes of long knots

摘录 / EXCERPT: Let $H_{d,\text{far}}^i$ denote the submodule of i-th cohomology classes in degree d represented by cocycles in the far subcomplex, and let $H^i(\mathcal{K}\setminus\Sigma)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042009 | MATHDB-328719

The exact shortest-cycle conjecture for oriented 3-graphs

摘录 / EXCERPT: Let $f(n)$ denote the minimum possible length of the shortest cycle in a tournament on n vertices. Exact shortest-cycle conjecture. For $n\ge 4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042010 | MATHDB-328720

The asymptotic shortest-cycle conjecture for oriented 3-graphs

摘录 / EXCERPT: Let $f(n)$ denote the minimum possible length of the shortest cycle in a tournament on n vertices. Asymptotic shortest-cycle conjecture. For all $n\ge 4$, ... This is the asymptoti...

conjecture | open | MathDB 页面 | 原文线索

V2-042011 | MATHDB-328722

Hereditary algebraicity of sum-closures of ordered binary structures

摘录 / EXCERPT: Let $\mathfrak D$ be a hereditary class of indecomposable ordered binary structures. Suppose that $\mathfrak D$ is hereditary well-quasi-ordered and hereditary algebraic. Sum-closu...

conjecture | open | MathDB 页面 | 原文线索

V2-042012 | MATHDB-328747

Contractibility conjecture for the clique complex of the Buchberger graph

摘录 / EXCERPT: Let I be a monomial ideal, let ${\mathfrak{BuG}}(I)$ be its Buchberger graph, and let $\Cl({\mathfrak{BuG}}(I))$ denote the simplicial complex whose simplices are the subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-042013 | MATHDB-328748

The full Erdős-Ko-Rado conjecture for t-intersecting perfect matchings

摘录 / EXCERPT: Let K_{2n} be the complete graph on $2n$ vertices, and let $\mathcal{F}$ be a family of its perfect matchings. The family is t-intersecting if $|m\cap m'|>t$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-042014 | MATHDB-328757

Bruhat-order conjecture for the first column of Zhao's table

摘录 / EXCERPT: Let $Q=(u,u-r)$ with $r\ge3$, and let $P_{k,1}$ denote the entries in the first column of the table $\mathcal T(Q)$. Recall that for partitions $P=(p_1\ge p_2\ge\cdots)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042015 | MATHDB-328766

The squarefree-triangulation conjecture for symmetric doubly-stochastic matrices

摘录 / EXCERPT: Let S_n be the polytope of symmetric doubly-stochastic matrices, and let $h^*(S_n)$ denote its Ehrhart h^*-polynomial. A regular unimodular triangulation of S_n should exist,...

conjecture | open | MathDB 页面 | 原文线索

V2-042016 | MATHDB-328767

The connected-decomposition conjecture for Gröbner basis elements of I_{S_n}

摘录 / EXCERPT: Let S_n be the polytope of symmetric doubly-stochastic matrices, let I_{S_n} be its toric ideal, and let $\mathscr{G}$ be the reduced Gröbner basis of I_{S_n}. For $g eq 0$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-042017 | MATHDB-328769

The unimodality conjecture for P_n

摘录 / EXCERPT: Let $mathit{Sigma}_n$ be the polytope considered in the source, and let P_n be the convex hull of the lattice points in $mathit{Sigma}_n$. Let $h^*(P_n)$ denote the Ehrhart h^*...

conjecture | open | MathDB 页面 | 原文线索

V2-042018 | MATHDB-328789

Points-Lines-Planes conjecture for Whitney numbers of rank-four matroids

摘录 / EXCERPT: Let M be a rank-four matroid, and let W_k denote the number of flats of M of rank k. Points-Lines-Planes conjecture. ... This is a special case of conjectured log-concavity i...

conjecture | open | MathDB 页面 | 原文线索

V2-042019 | MATHDB-328838

Convex-hull characterization of the \(\mathbb{F}_{2^2}\) relaxed code polytope

摘录 / EXCERPT: Let $\mathcal{E}$ be the set defined by the stated integer constraint lemma for $\mathbb{F}_{2^2}$ and by the all-ones check of length d. Let $\mathbb{U}_4$ be the relaxed code p...

conjecture | open | MathDB 页面 | 原文线索

V2-042020 | MATHDB-328877

Konyagin-Lev conjecture on Freiman isomorphisms into short intervals

摘录 / EXCERPT: Let $A\subset \mathbb{Z}$ be a finite set with $|A|=m$. A Freiman isomorphism is a map preserving the additive relations of A; write $[N]=\{1,\ldots,N\}$. Konyagin-Lev conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042021 | MATHDB-328890

The inverse sieve conjecture

摘录 / EXCERPT: Inverse sieve conjecture. At least one of the following holds: $|X|\ll_{\epsilon}N^{\epsilon}$, or there exists a polynomial $f(x)\in\mathbb{Q}[x]$ of degree $d\in[2,C]$ and height...

conjecture | open | MathDB 页面 | 原文线索

V2-042022 | MATHDB-328895

Extremal Picard-number conjecture for non-S_3-splittable toric Fano manifolds

摘录 / EXCERPT: Non-S_3-splittable extremal conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042023 | MATHDB-328896

Extremal Picard-number conjecture for non-splittable toric Fano manifolds

摘录 / EXCERPT: Non-splittable extremal conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042024 | MATHDB-328912

Triple sumset conjecture for unit vectors

摘录 / EXCERPT: Let V be a set of n unit vectors in Euclidean space, with at most $n^{o(1)}$ endpoints of V on any plane. Triple sumset conjecture. Then ... The source presents this as an inde...

conjecture | open | MathDB 页面 | 原文线索

V2-042025 | MATHDB-328917

A generalized Prouhet-Thue-Morse identity for products of binary sums

摘录 / EXCERPT: The generalized Prouhet-Thue-Morse identity. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042026 | MATHDB-328918

A generalized Prouhet-Thue-Morse divisibility conjecture

摘录 / EXCERPT: Generalized Prouhet-Thue-Morse divisibility conjecture. If $k=2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042027 | MATHDB-328930

Higher-dimensional generalization of the good-spacing conjecture

摘录 / EXCERPT: Let $A\subseteq[1,n]\times[1,n]$ satisfy $|A|=\delta n^2$. Higher-dimensional good-spacing conjecture. There exist constants $c_1,c_2,c_3>0$ depending only on δ and a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-042028 | MATHDB-328932

The tightness conjecture for antipodal n-fold covers of spheres

摘录 / EXCERPT: Tightness conjecture. For $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-042029 | MATHDB-328955

The quotient-of-sumsets small-doubling conjecture

摘录 / EXCERPT: Quotient-of-sumsets small-doubling conjecture. There exist absolute constants c and c' such that

conjecture | open | MathDB 页面 | 原文线索

V2-042030 | MATHDB-328986

Holmes-Plummer enumeration conjecture for cds-sortable permutations

摘录 / EXCERPT: Holmes-Plummer conjecture. The number of cds-sortable elements in $\mathfrak{S}_{2k+1}$ equals $(k+1)(2k)!$. This conjecture proposes a closed formula for the odd-degree terms in t...

conjecture | open | MathDB 页面 | 原文线索

V2-042031 | MATHDB-329032

Collapsibility conjecture for nerve complexes of circular arcs

摘录 / EXCERPT: Collapsibility conjecture. The complex $\mathcal{N}({n},{k})$ simplicially collapses to $\partial C_{2l+2}(n)$.

open_problem | open | MathDB 页面 | 原文线索

V2-042032 | MATHDB-329047

2-primitivity conjecture for Conway groupoids

摘录 / EXCERPT: Let $\mathcal{D}$ be a supersimple $2-(n,4,\lambda)$ design other than $\mathbb{P}_3$, let ∞ be a point in $\mathcal{D}$, and suppose that the hole stabilizer...

conjecture | open | MathDB 页面 | 原文线索

V2-042033 | MATHDB-329092

Near-minimum-ratio existence conjecture for incomplete pairwise balanced designs

摘录 / EXCERPT: Let $K \subseteq \mathbb{Z}_{\ge 2}$. For a real number $\epsilon>0$, an incomplete pairwise balanced design $\operatorname{IPBD}((v;w),K)$ is a pairwise balanced design on v poi...

conjecture | open | MathDB 页面 | 原文线索

V2-042034 | MATHDB-329099

Equitable partition conjecture for prescribed block sizes

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $s^{n,k}=\frac{1}{k}\binom{n+1}{2}$ be an integer, and let a partition of $[n]$ implement $\{p_i\}_{i=1}^k$ when its blocks can be bijectively matched...

conjecture | open | MathDB 页面 | 原文线索

V2-042035 | MATHDB-329121

Big line, big clique conjecture for planar visibility graphs

摘录 / EXCERPT: Let X be a finite set of points in the Euclidean plane. For $u,v\in X$, let (uv) denote the open line segment with endpoints u and v, and let the visibility graph G_X be...

conjecture | open | MathDB 页面 | 原文线索

V2-042036 | MATHDB-329132

Factorization and squarefreeness conjecture for dimer packings on odd square lattices

摘录 / EXCERPT: Let a_N denote the number of dimer packings of the odd lattice $(2k+1) \times (2k+1)$, and let c_k be the integer in the factorization below. Factorization and squarefreeness c...

conjecture | open | MathDB 页面 | 原文线索

V2-042037 | MATHDB-329135

Richness of binary projections of episturmian words

摘录 / EXCERPT: Projection richness conjecture. The word $\zeta(\mathbf u)$ is rich.

conjecture | open | MathDB 页面 | 原文线索

V2-042038 | MATHDB-329156

Column-sign Latin square conjecture

摘录 / EXCERPT: Let an $n\times n$ array whose rows and columns are permutations of $[n]=\{1,\ldots,n\}$ be a Latin square. The product of the signs of its columns is its column sign; call the squ...

conjecture | open | MathDB 页面 | 原文线索

V2-042039 | MATHDB-329184

Conjecture on combinatorial and affine symmetries of representation polytopes

摘录 / EXCERPT: Let $G\leqslant \operatorname{GL}(d,\mathbb{R})$ be a representation of a finite group, and let $P(G)$ denote its representation polytope. The combinatorial symmetry group consists...

conjecture | open | MathDB 页面 | 原文线索

V2-042040 | MATHDB-329208

The boundary conjecture for extremal Lagrangian tori in the ball

摘录 / EXCERPT: Let $B^{2n}(1)$ be the unit ball, and call a Lagrangian torus extremal if it maximizes $A_{\rm min}$, so that $A_{\rm min}(L)=c_L(B^{2n}(1))$. The extremal-torus boundary conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-042041 | MATHDB-329222

The (I3) conjecture for path-transversal systems

摘录 / EXCERPT: A path-transversal system is a system arising from path transversals in a dimaze. The (I3) conjecture. Any path-transversal system satisfies (I3). The preceding discussion notes th...

conjecture | open | MathDB 页面 | 原文线索

V2-042042 | MATHDB-329223

The duality conjecture for C^A-free gammoids

摘录 / EXCERPT: A C^A-free gammoid is a gammoid avoiding the excluded configuration C^A. The duality conjecture. The class of C^A-free gammoids is closed under duality. This is presented as...

conjecture | open | MathDB 页面 | 原文线索

V2-042043 | MATHDB-329232

Myers' asymptotic monotone-subsequence conjecture

摘录 / EXCERPT: Let k be a positive integer. For a permutation τ of $[n]=\{1,\dots,n\}$, let $m_k(\tau)$ denote the number of monotone subsequences of length $k+1$. Myers' asymptotic conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-042044 | MATHDB-329239

Monomial positivity conjecture for products of Schur functions

摘录 / EXCERPT: Let Y_n denote the set of Young diagrams with n boxes, and let s_λ be the Schur symmetric function indexed by λ. Let $\lambda,\hat\lambda\in Y_n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042045 | MATHDB-329250

Kleiner's CLP-to-Loewner conjecture for hyperbolic group boundaries

摘录 / EXCERPT: Let Z be a compact metric space that is quasi-Möbius homeomorphic to the boundary of a hyperbolic group, and suppose that Z satisfies the combinatorial Loewner property (CLP)....

conjecture | open | MathDB 页面 | 原文线索

V2-042046 | MATHDB-329251

A flag Schur identity in the KR stable quotient

摘录 / EXCERPT: Let $m,n_1,\ldots,n_t,x,y_1,\ldots,y_t$ be integers satisfying ... ... Set $\alpha=(a,a,a+1,a+2,\ldots,a+t)$, and let J_α denote the function and $[r]$ the interval notation use...

conjecture | open | MathDB 页面 | 原文线索

V2-042047 | MATHDB-329260

Alon-Frankl conjecture on comparable pairs above the square-root threshold

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathcal{F}$ be a family of m subsets of $[n]$, and let $c(n,m)$ be the maximum number of comparable pairs in a family of m subsets of $[n]$. Alo...

conjecture | open | MathDB 页面 | 原文线索

V2-042048 | MATHDB-329272

Erdős-Szemerédi's sunflower conjecture in the Boolean cube

摘录 / EXCERPT: A 3-sunflower is a collection of three sets having the same pairwise intersections, and $[n]=\{1,\ldots,n\}$. Erdős-Szemerédi's Boolean-cube sunflower conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-042049 | MATHDB-329273

Alon-Shpilka-Umans' sunflower conjecture over ${\mathbb Z}_D^n$

摘录 / EXCERPT: Let ${\mathbb Z}_D=\{1,\ldots,D\}$. Vectors $v_1,v_2,v_3\in{\mathbb Z}_D^n$ form a 3-sunflower if, in every coordinate, their entries are either all equal or all distinct. Alon-S...

conjecture | open | MathDB 页面 | 原文线索

V2-042050 | MATHDB-329274

Alon-Shpilka-Umans' weak sunflower conjecture over ${\mathbb Z}_D$

摘录 / EXCERPT: Let ${\mathbb Z}_D=\{1,\ldots,D\}$, and call three vectors in ${\mathbb Z}_D^n$ a 3-sunflower when their entries in each coordinate are either all equal or all distinct. Alon-Shp...

conjecture | open | MathDB 页面 | 原文线索

V2-042051 | MATHDB-329291

Factor-poset size bound for tight frames

摘录 / EXCERPT: Let H_n be an n-dimensional Hilbert space, let I be an index set with $|I|=k=mn$ for some $m\in\mathbb{N}$, and let $F=\{f_i\}_{i\in I}\subseteq H_n\setminus\{0\}$ be a tight...

conjecture | open | MathDB 页面 | 原文线索

V2-042052 | MATHDB-329390

The general existence conjecture for Heffter arrays

摘录 / EXCERPT: Heffter array existence conjecture. There exists a Heffter array $H(m,n;s,t)$ for all m,n,s,t with $s,t\geq 3$ and $ms=nt$. If the conditions of the necessary integer-sum lemma a...

conjecture | open | MathDB 页面 | 原文线索

V2-042053 | MATHDB-329391

The full Heffter array conjecture

摘录 / EXCERPT: Full Heffter array conjecture. There exist Heffter arrays $H(m,n)$ for all $m,n\geq 3$. Moreover, they are integer Heffter arrays when $mn\equiv 0,3\pmod{4}$, and are shiftable whe...

conjecture | open | MathDB 页面 | 原文线索

V2-042054 | MATHDB-329394

The signed sumset minimum conjecture for finite abelian groups

摘录 / EXCERPT: Let G be an abelian group of order n and type $(n_1,\dots,n_r)$, and let $\rho_{\pm}(G,m,h)$ denote the minimum size of the h-fold signed sumset of an m-subset of G. Defi...

conjecture | open | MathDB 页面 | 原文线索

V2-042055 | MATHDB-329395

The converse to the sufficient condition for equality of signed and ordinary sumset minima

摘录 / EXCERPT: The converse conjecture. If $2\leq h\leq p-1$ and $m>(c+1)p^k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042056 | MATHDB-329400

Linear Beer-index bound for p-componentwise simply Δ-connected sets

摘录 / EXCERPT: Call a set $S\subseteq\mathbb{R}^2$ p-componentwise simply Δ-connected if, for every triangle T with $\partial T\subseteq S$, one has $T\subseteq S$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-042057 | MATHDB-329401

Boundary convexity conjecture for the Beer index

摘录 / EXCERPT: Let $S\subseteq\mathbb{R}^2$ be a measurable set, and let $\operatorname{b}(S)$ denote its Beer index and λ planar Lebesgue measure. Boundary convexity conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-042058 | MATHDB-329402

Skeleton conjecture for higher-order Beer indices

摘录 / EXCERPT: For a set $X\subseteq\mathbb{R}^d$ and an integer $k\geq 0$, define ... where $\binom{X}{k+1}$ is the set of $(k+1)$-element subsets of X. Let $\operatorname{b}_k(S)$ be the k-th...

conjecture | open | MathDB 页面 | 原文线索

V2-042059 | MATHDB-329403

Contractible p-component conjecture for the $(d-1)$-st Beer index

摘录 / EXCERPT: Let $d\geq 2$, let $S\subseteq\mathbb{R}^d$ be a set whose $(d-1)$-st Beer index $\operatorname{b}_{d-1}(S)$ is defined, and assume that every p-component of S is contractible. L...

conjecture | open | MathDB 页面 | 原文线索

V2-042060 | MATHDB-329407

The multidimensional permanent upper-bound conjecture

摘录 / EXCERPT: Let A be a d-dimensional $(0,1)$-matrix of order n. For each i, let r_i be the number of ones in the ith hyperplane of A. Multidimensional permanent upper-bound conje...

conjecture | open | MathDB 页面 | 原文线索

V2-042061 | MATHDB-329435

The q-analogue identity for rook placements and the Harer-Zagier formula

摘录 / EXCERPT: Let $Y(n,s)$ be the set of all Young diagrams with n rows of lengths between s and $n+s$. For a rook placement C on a diagram $[?]$ in $Y(n,s)$, let $\operatorname{inv}(C)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-042062 | MATHDB-329446

Equality conjecture for the h-critical bound in Theorem cor h=3

摘录 / EXCERPT: Let n and m be the parameters occurring in Theorem cor h=3, whose bound concerns the quantity $\rho(\mathbb{Z}_n,m,3)$ and its upper estimate in terms of $u(n,m,3)$. Equality c...

conjecture | open | MathDB 页面 | 原文线索

V2-042063 | MATHDB-329447

Equality conjecture for the h=3 restricted critical number

摘录 / EXCERPT: Let n be an integer, and let Proposition h=3 rest crit denote the proposition concerning the $h=3$ restricted critical number. Equality conjecture. For all values of $n\geq 11$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042064 | MATHDB-329461

The asymptotic lower bound for n-splitting families

摘录 / EXCERPT: Let $[k]=\{1,\ldots,k\}$, and let an n-splitting family be a family of subsets of $[k]$ such that every collection of n subsets of $[k]$ has a splitter in the family. The minim...

conjecture | open | MathDB 页面 | 原文线索

V2-042065 | MATHDB-329462

Disjointness minimizes the number of simultaneous splitters

摘录 / EXCERPT: Let $[k]=\{1,\ldots,k\}$, and let a splitter of a collection of subsets be a subset that splits every member of the collection. Disjoint-hardest conjecture. Let $B_1,\ldots,B_n$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042066 | MATHDB-329472

Parity condition for simple Hurwitz numbers

摘录 / EXCERPT: Let $g\ge 0$ and let μ be a partition with $|\mu|\ge 3$. Write $h_{g,\mu}$ for the corresponding simple Hurwitz number, $r(g,\mu)$ for its associated ramification number, and...

conjecture | open | MathDB 页面 | 原文线索

V2-042067 | MATHDB-329506

The tail-end inversion conjecture for rectangular tableaux

摘录 / EXCERPT: Let $n\geq 1$ and $m\geq 2$. Consider the m-row shapes ... For each integer i, let $S_i(\lambda)$ and $S_{i-m+1}(\widetilde{\lambda})$ denote the corresponding sets of tableaux w...

conjecture | open | MathDB 页面 | 原文线索

V2-042068 | MATHDB-329523

Andrews-Curtis conjecture for finite contractible 2-complexes

摘录 / EXCERPT: Let L be a finite, contractible 2-complex. A 3-deformation is the operation, described in the source, of successively gluing 3-balls along open 2-cells and performing inverse ope...

conjecture | open | MathDB 页面 | 原文线索

V2-042069 | MATHDB-329524

Sum Rule II for canonical-basis component sums

摘录 / EXCERPT: Sum Rule II. For all $L\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-042070 | MATHDB-329526

Kazhdan-Lusztig positivity conjecture for matroids

摘录 / EXCERPT: Let M be a matroid, and let $P_M(t)\in\mathbb{Z}[t]$ be its Kazhdan-Lusztig polynomial, characterized by the recursive conditions in the paper. Kazhdan-Lusztig positivity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-042071 | MATHDB-329527

Positivity conjecture for structure coefficients of the matroid Kazhdan-Lusztig basis

摘录 / EXCERPT: Structure-coefficient positivity conjecture. For all $F,G,H\in L(M)$, $C_{FG}^H(q)\in\mathbb{N}[q]$. The conjecture was disproved by examples in rank 4 and higher, including a un...

conjecture | open | MathDB 页面 | 原文线索

V2-042072 | MATHDB-329532

The sum-rule conjecture for two-boundary Temperley-Lieb ground states

摘录 / EXCERPT: Let si be the ground-state vector, and let $S^{\mathrm{X}}_{N}(q,Q)$ denote the sum of all its components on the Kazhdan-Lusztig basis of boundary type X, where...

conjecture | open | MathDB 页面 | 原文线索

V2-042073 | MATHDB-329533

The coefficient-polynomial conjecture for Type A and Type BIII sums

摘录 / EXCERPT: At $q=1$, write the Type A and Type BIII sums as ... with nonnegative integer coefficients $S_{N,i}$ and $S'_{N,i}$. Coefficient-polynomial conjecture. The coefficients satisfy ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-042074 | MATHDB-329534

The Type A symmetric-binary-matrix coefficient conjecture

摘录 / EXCERPT: Let $\mathcal{A}_N$ be the set of symmetric binary $N\times N$ matrices with no row sum greater than one. For $a=(a_{i,j})$, define ... Let $S_{N,i}$ be the coefficient of Q^i in t...

conjecture | open | MathDB 页面 | 原文线索

V2-042075 | MATHDB-329535

The Type BIII signed-permutation coefficient conjecture

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the set of signed permutation matrices of size $2(n-1)\times2(n-1)$ invariant under diagonal and anti-diagonal reflections and avoiding the pattern $(-2,-1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042076 | MATHDB-329536

The Type BII first-coefficient conjecture

摘录 / EXCERPT: At $q=1$, write the Type BII sum as ... with $S_{N,j}\in\mathbb{N}$. Type BII first-coefficient conjecture. ... The paper reports verification through $N=20$, but no general proof or r...

conjecture | open | MathDB 页面 | 原文线索

V2-042077 | MATHDB-329537

The Type BII coefficient-polynomial conjecture

摘录 / EXCERPT: At $q=1$, write the Type BII sum as ... Type BII coefficient-polynomial conjecture. For the relevant coefficient index i, ... where ... with $p_{i,k}\in\mathbb{Z}$. The source reports...

conjecture | open | MathDB 页面 | 原文线索

V2-042078 | MATHDB-329538

The Type A component symmetric-matrix formula

摘录 / EXCERPT: Let $\mathcal{S}$ be the set of symmetric binary matrices with no row sum greater than one. For $s\in\mathcal{S}$, let $A(s)$ be the set of upper-triangular positions containing a...

conjecture | open | MathDB 页面 | 原文线索

V2-042079 | MATHDB-329539

The Type BIII component signed-permutation formula

摘录 / EXCERPT: Let $\mathcal{C}_{N+1}$ be the set of signed permutation matrices of size $2N\times2N$ invariant under diagonal and anti-diagonal reflections and avoiding $(-2,-1)$. Let $B(c)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042080 | MATHDB-329557

Conjecture on lines in metric spaces with finitely many distances

摘录 / EXCERPT: Let (V,ρ) be an n-point metric space with $n\geq 2$, and suppose that the number of distinct distances occurring in the metric is bounded by an absolute constant. Conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042081 | MATHDB-329574

White's three equivalence-relation conjectures for matroid bases

摘录 / EXCERPT: White's TE conjecture. All matroids satisfy each of the three properties:

conjecture | open | MathDB 页面 | 原文线索

V2-042082 | MATHDB-329579

Király-Nagy-Pálvölgyi-Visontai conjecture on weakly cross intersecting set pair systems

摘录 / EXCERPT: Király-Nagy-Pálvölgyi-Visontai conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042083 | MATHDB-329596

The delta-process tree probability conjecture

摘录 / EXCERPT: The delta-process tree probability conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042084 | MATHDB-329597

The probabilistic determinant formula for the delta process

摘录 / EXCERPT: The probabilistic determinant formula. For every $D\subseteq[k-1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-042085 | MATHDB-329603

First-elongation characterization for uniformly coloured spherical frameworks

摘录 / EXCERPT: First-elongation conjecture. Suppose $(G=(V,E),\chi)$ is uniformly coloured. Then (G,χ) is isostatic on $r\mathbb{S}^d$ if and only if G is simple and E is a base of the f...

conjecture | open | MathDB 页面 | 原文线索

V2-042086 | MATHDB-329610

Existence of an extremizer for the proposed Mahler-volume lower bound

摘录 / EXCERPT: Existence conjecture. There is a symmetric convex set K in $\mathbb{R}^n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042087 | MATHDB-329612

The lower-bound conjecture for coefficients of Ehrhart polynomials

摘录 / EXCERPT: Let ${\mathcal P} \subset {\mathbb R}^d$ be a lattice polytope with $d \geq 3$. Write $g_r({\mathcal P})$ for the coefficients of its Ehrhart polynomial, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042088 | MATHDB-329645

Alphabet-independence conjecture for the sets R(n,m)

摘录 / EXCERPT: Alphabet-independence conjecture. The set $R(n,m)$ depends only on n.

conjecture | open | MathDB 页面 | 原文线索

V2-042089 | MATHDB-329665

The extremal eigenvalue conjecture for the double layer potential

摘录 / EXCERPT: Let K be the double layer potential on the boundary of a smooth region in $\boldsymbol{R}^3$, and let ... where $\sigma_p(K)$ denotes the point spectrum. Extremal eigenvalue conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-042090 | MATHDB-329687

The one-away multiline-queue enumeration conjecture

摘录 / EXCERPT: Let $N\ge n>k\ge 1$ and let $0\le b_1<\dots<b_n\le N-1$. For the permutation s_kw_0 obtained by applying the simple reflection s_k to the longest permutation w_0, let A_k b...

conjecture | open | MathDB 页面 | 原文线索

V2-042091 | MATHDB-329688

The multi-away multiline-queue enumeration conjecture

摘录 / EXCERPT: Let $N\ge n$, let $\mathbf{k}=(k_1\ge\cdots\ge k_r\ge 1)$ be a partition, and let $\mathbf{k}'$ be its conjugate. For $S\subseteq [r]$, let $\mathbf{k}(S)$ be the partition consist...

conjecture | open | MathDB 页面 | 原文线索

V2-042092 | MATHDB-329714

Symmetric joint distribution of left-to-right and right-to-left maxima

摘录 / EXCERPT: Symmetric-distribution conjecture. For any n, LRmax and RLmax have symmetric joint distribution on $\operatorname{Av}_n(132)$.

conjecture | open | MathDB 页面 | 原文线索

V2-042093 | MATHDB-329760

The partial-sum sequencing conjecture for subsets of cyclic groups

摘录 / EXCERPT: Let $A=a_1,\dots,a_k$ be a subset of the nonzero elements of ${\mathbb Z}_n$. For an ordering $(a_1,\dots,a_k)$ of A, define the partial sums by ... with arithmetic in...

conjecture | open | MathDB 页面 | 原文线索

V2-042094 | MATHDB-329762

Restricted zero-sum sequencing conjecture

摘录 / EXCERPT: Let A be a subset of ${\mathbb Z}_n$ satisfying the zero-total condition ... A sequencing of A is an ordering whose partial sums are all distinct. Restricted zero-sum sequencing...

conjecture | open | MathDB 页面 | 原文线索

V2-042095 | MATHDB-329763

No-opposite-elements conjecture for zero-sum sequencings

摘录 / EXCERPT: Let $A=\{a_1,\dots,a_k\}$ be a subset of ${\mathbb Z}_n$ satisfying the zero-total condition from the restricted zero-sum sequencing conjecture. Assume that no two elements of A...

conjecture | open | MathDB 页面 | 原文线索

V2-042096 | MATHDB-329764

Small-density zero-sum sequencing conjecture

摘录 / EXCERPT: Let $A=\{a_1,\dots,a_k\}$ be a subset of ${\mathbb Z}_n$ satisfying the hypotheses of the no-opposite-elements conjecture. Small-density zero-sum sequencing conjecture. If n is...

conjecture | open | MathDB 页面 | 原文线索

V2-042097 | MATHDB-329765

Linear-density partial-sum sequencing problem

摘录 / EXCERPT: Let A be a k-element subset of the nonzero elements of ${\mathbb Z}_n$, and let the partial-sum sequencing conjecture be the assertion that A has an ordering with pairwise di...

open_problem | open | MathDB 页面 | 原文线索

V2-042098 | MATHDB-329766

Fixed-size partial-sum sequencing problem

摘录 / EXCERPT: Let k be fixed, and let A be a k-element subset of the nonzero elements of ${\mathbb Z}_n$. Fixed-size sequencing problem. Determine whether there exists a constant N such...

open_problem | open | MathDB 页面 | 原文线索

V2-042099 | MATHDB-329769

Gnedin-Olshanski conjecture on the Martin boundary of the zigzag graph

摘录 / EXCERPT: Let $\mathcal{Z}$ be the zigzag graph, let $\partial_{M}\mathcal{Z}$ and $\partial_{\min}\mathcal{Z}$ denote its Martin and minimal entrance boundaries, and let $\mathcal{U}^{(2)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-042100 | MATHDB-329770

The constant-boundary enumeration conjecture for tree sl_n webs

摘录 / EXCERPT: Let $R_{p,r}(k)$ denote the generalized Catalan number associated with parameters p,r,k. An sl_n web is a diagrammatic object for the representation theory of $U_q(sl_n)$; a we...

conjecture | open | MathDB 页面 | 原文线索

V2-042101 | MATHDB-329771

The non-constant-boundary enumeration conjecture for tree sl_n webs

摘录 / EXCERPT: Let $R_{p,r}(k)$ denote the generalized Catalan number associated with parameters p,r,k. For $n \geq 3$, let an sl_n web have a boundary string consisting of one boundary edge...

conjecture | open | MathDB 页面 | 原文线索

V2-042102 | MATHDB-329772

Polytopality conjecture for compatibility fans of graphical nested complexes

摘录 / EXCERPT: A graphical nested complex is associated with a graph, and its primal and dual compatibility fans are the fans constructed from the corresponding compatibility data. Polytopality c...

conjecture | open | MathDB 页面 | 原文线索

V2-042103 | MATHDB-329777

Coefficient and unimodality conjectures for specialized Grothendieck enumerators

摘录 / EXCERPT: Set $d:=\gamma-1$. For $w\in\mathbb{S}_n$, let $L_n(d)$ and $L_{n,1}(\alpha,\beta,d)$ be the specialized enumerators defined from the generalized Grothendieck polynomials. Coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-042104 | MATHDB-329796

Diagram matroid basis-count conjecture

摘录 / EXCERPT: Let $w\in S_n$, and let DM_w be the diagram matroid of w, obtained from a generic matrix $[A\mid I_n]$ whose entries satisfy $A_{ij}=0$ whenever $(i,j)\in D(w)$, where ... For a...

conjecture | open | MathDB 页面 | 原文线索

V2-042105 | MATHDB-329797

Tutte-polynomial equality for diagram matroids and positroids

摘录 / EXCERPT: Let $w\in S_n$. Let DM_w be the diagram matroid associated to the permutation diagram of w, and let $\mathscr{P}_w$ be the corresponding positroid. Tutte-polynomial equality co...

conjecture | open | MathDB 页面 | 原文线索

V2-042106 | MATHDB-329807

Even tetrahedron count conjecture for tetrahedral link complements

摘录 / EXCERPT: Evenness conjecture. Every tetrahedral link complement has an even number of tetrahedra; equivalently, a corresponding combinatorial tetrahedral tessellation has an even number of...

conjecture | open | MathDB 页面 | 原文线索

V2-042107 | MATHDB-329851

Optimal-colouring conjecture for fixed-length de Bruijn cycle decompositions

摘录 / EXCERPT: Let $\mathcal{E}(q,k,\ell)$ denote the maximum number of eBugs in an ℓ-valid colouring with q colours and k LEDs per eBug. An optimal colouring is one attaining the upper...

conjecture | solved | MathDB 页面 | 原文线索

V2-042108 | MATHDB-329855

The existence conjecture for difference covering arrays satisfying P1 and P2

摘录 / EXCERPT: Let p be a positive integer with $p\geq 3$. A difference covering array $\operatorname{DCA}(4,2p+1;2p)$ is an array with the indicated parameters, and P1 and P2 denote the two pr...

conjecture | open | MathDB 页面 | 原文线索

V2-042109 | MATHDB-329858

Pach-Tardos conjecture for forest matrices

摘录 / EXCERPT: Given a $0/1$ matrix M and an $n\times n$ matrix A, say that A contains the pattern M if the 1-entries of M occur in A on increasing choices of rows and columns. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-042110 | MATHDB-329897

The winning-strategy conjecture for Building Nim

摘录 / EXCERPT: Let $\operatorname{BN}(2n,\ell)$ denote the Building Nim game with parameters $2n$ and ℓ, and let P1 and P2 be its two players. Winning-strategy conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042111 | MATHDB-329900

Entropicity conjecture for polymatroids satisfying Ingleton and strong union inequalities

摘录 / EXCERPT: Let h be a polymatroid function on a lattice. The entropicity conjecture. If h satisfies all Ingleton inequalities and all strong union inequalities, then h is entropic. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042112 | MATHDB-329921

Ramsey expansion conjecture for finite-signature homogeneous structures

摘录 / EXCERPT: Let Γ be a homogeneous structure with a finite relational signature. An expansion of Γ is a structure obtained by adding relations to its signature; its age is the cl...

conjecture | open | MathDB 页面 | 原文线索

V2-042113 | MATHDB-329939

Pseudomodularity conjecture for Type 1 Stanley sequences

摘录 / EXCERPT: Pseudomodularity conjecture. A Stanley sequence follows Type 1 growth if and only if it is pseudomodular.

conjecture | open | MathDB 页面 | 原文线索

V2-042114 | MATHDB-329940

Completion conjecture for 3-free sets

摘录 / EXCERPT: Completion conjecture. Every 3-free set has a completion.

conjecture | open | MathDB 页面 | 原文线索

V2-042115 | MATHDB-329941

Density conjecture for modular Stanley sequences

摘录 / EXCERPT: Density conjecture. There exists a modular sequence $S(A)=(a_n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042116 | MATHDB-329942

Non-power cardinality conjecture for modular p-free sets

摘录 / EXCERPT: Non-power cardinality conjecture. For every $n\in\mathbb{N}\cup\{0\}$, there exists a modular p-free set A such that

conjecture | open | MathDB 页面 | 原文线索

V2-042117 | MATHDB-329943

Density conjecture for modular p-Stanley sequences

摘录 / EXCERPT: p-density conjecture. There exists a modular p-Stanley sequence $S_p(A)=(a_n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042118 | MATHDB-329985

Sharp universal bounds for maximal sum-free subsets of finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group of order n, and let $f_{\max}(G)$ denote the number of maximal sum-free subsets of G. Universal bounds conjecture. For every such group, ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-042119 | MATHDB-329986

Conjecture on an infinite family with few maximal sum-free subsets

摘录 / EXCERPT: Let $\mu(G)$ denote the relevant group parameter used in the paper, and let $f_{\max}(G)$ denote the number of maximal sum-free subsets of a finite abelian group G. Few maximal s...

conjecture | open | MathDB 页面 | 原文线索

V2-042120 | MATHDB-329992

A shifted-coset formula for rational Catalan polynomials

摘录 / EXCERPT: Coset age conjecture. There is an age function ι on the cosets $\mathfrak{c}\in\boldsymbol{\Lambda}/\boldsymbol{\Lambda}_T$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042121 | MATHDB-329993

Residue-class unimodality conjecture for rational Catalan polynomials

摘录 / EXCERPT: Residue-class unimodality conjecture. For every residue class r, $0\leq r<a$, the coefficients of q^{ak+r} in $\mathbf{Cat}_{a,b}(q)$ are unimodal. The conjecture refines ordin...

conjecture | open | MathDB 页面 | 原文线索

V2-042122 | MATHDB-330016

Periodic-limit conjecture for normalized curve-diagram counts

摘录 / EXCERPT: Let $g_{n,k}$ denote the number of curve diagrams of genus n and complexity k, as in the paper. Let $n\geq 2$ be an integer. Periodic-limit conjecture. There exists a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-042123 | MATHDB-330052

Dvir-Gopi rich-lines conjecture

摘录 / EXCERPT: Let $r\geq 2$, and let $\mathcal P\subset\mathbb{C}^d$ be a set of n points. Write $\mathcal{L}_r(\mathcal P)$ for the set of lines incident to at least r points of...

conjecture | open | MathDB 页面 | 原文线索

V2-042124 | MATHDB-330082

Pfender's recurrence conjecture for critical clique edge densities

摘录 / EXCERPT: For each integer $k\ge 2$, let $d_k=d_k(K_k)$ denote the critical edge density of the complete graph K_k. Pfender's recurrence conjecture. The critical edge densities satisfy ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-042125 | MATHDB-330097

Carousel homomorphism conjecture for odd carousel tournaments

摘录 / EXCERPT: Carousel homomorphism conjecture. For every $n\in\mathbb{N}$, the carousel homomorphism maximizes the density of R_{2n+1}:

conjecture | open | MathDB 页面 | 原文线索

V2-042126 | MATHDB-330098

Uniqueness conjecture for the carousel homomorphism maximizer

摘录 / EXCERPT: Carousel maximizer uniqueness conjecture. For every $n\geq2$, a homomorphism $\phi\in\operatorname{Hom}^+({\mathcal{A}}^0,\mathbb{R})$ maximizes the density of R_{2n+1} if and on...

conjecture | open | MathDB 页面 | 原文线索

V2-042127 | MATHDB-330114

Hirsch's diameter conjecture for polyhedra

摘录 / EXCERPT: Let $f > d \geq 2$. Let P be a d-dimensional polyhedron with f facets. Hirsch's conjecture. The combinatorial diameter of P is at most $f-d$. The conjecture is false in gen...

conjecture | open | MathDB 页面 | 原文线索

V2-042128 | MATHDB-330115

Circuit Hirsch conjecture

摘录 / EXCERPT: Let P be a d-dimensional polyhedron with f facets. Its circuit diameter is the largest circuit distance between any two vertices, where circuit walks generalize edge walks by...

conjecture | open | MathDB 页面 | 原文线索

V2-042129 | MATHDB-330142

Gog-Magog trapezoid conjecture

摘录 / EXCERPT: An (m,n,k)-Gog trapezoid is the array defined by the row, column, diagonal, and rightmost-column-bound conditions in the source; an (m,n,k)-Magog trapezoid is the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-042130 | MATHDB-330147

Kleitman-Pinchasi conjecture on bichromatic lines

摘录 / EXCERPT: Let P be a set of n red points and either n or $n-1$ blue points in the Euclidean plane. A bichromatic line is a line containing at least one red and one blue point, and P...

conjecture | open | MathDB 页面 | 原文线索

V2-042131 | MATHDB-330175

Kano and Suzuki's colorful simplex partition conjecture

摘录 / EXCERPT: Let $r \ge d\ge 3$ and $n\ge 1$ be integers. Let $X_1, X_2, \dots, X_r$ be r disjoint point sets in $\mathbb{R}^d$. Assume that no $d+1$ points of...

conjecture | open | MathDB 页面 | 原文线索

V2-042132 | MATHDB-330219

Coordinate-star extremal conjecture for intersecting families under product measures

摘录 / EXCERPT: Coordinate-star extremal conjecture. Assume

conjecture | open | MathDB 页面 | 原文线索

V2-042133 | MATHDB-330220

The asymptotic exponent conjecture for sets free of prescribed sumsets

摘录 / EXCERPT: Let $r\ge 2$ and let $2\le \ell_1\le \cdots \le \ell_r$ be integers. Write $F(n,\mathcal L^{(r)}_{\ell_1,\dots,\ell_r})$ for the maximum size of a subset of $\\{1,\dots,n\\}$ conta...

conjecture | open | MathDB 页面 | 原文线索

V2-042134 | MATHDB-330222

Zauner's conjecture on maximal complex equiangular tight frames

摘录 / EXCERPT: Let $M\geq 2$. An equiangular tight frame (ETF) in $\mathbb{C}^M$ with N vectors is denoted by $\operatorname{ETF}(M,N)$; the maximal possible number of vectors in the complex ca...

conjecture | open | MathDB 页面 | 原文线索

V2-042135 | MATHDB-330232

Conjecture on bounded doubling of large-spectrum sets

摘录 / EXCERPT: Bounded-doubling spectrum conjecture. There exists a subset $A'\subseteq A$ with $|A'|\geq |A|/C$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042136 | MATHDB-330246

Weighted average-size conjecture for self-conjugate core partitions

摘录 / EXCERPT: Weighted self-conjugate core-average conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042137 | MATHDB-330302

Klostermeyer et al.'s self-similarity conjecture for the Pascal rhombus modulo 2

摘录 / EXCERPT: Let $R[n,k]$ denote the entry in row n and column k of the Pascal rhombus, reduced modulo 2. For $n\ge 1$, consider the triangle whose corners are $[1,0]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042138 | MATHDB-330303

Klostermeyer et al.'s row-count formula conjecture for the Pascal rhombus

摘录 / EXCERPT: Let I_n be the number of ones in row $m=2^n$ of the Pascal rhombus, with entries reduced modulo 2. Klostermeyer et al.'s row-count formula conjecture. Then ... The formula predic...

conjecture | open | MathDB 页面 | 原文线索

V2-042139 | MATHDB-330304

Klostermeyer et al.'s diagonal-periodicity conjecture for the Pascal rhombus

摘录 / EXCERPT: For each $k\ge 1$, define diagonal D_k of the Pascal rhombus modulo 2 to consist of the cells $[n,-n+k+1]$ for $n\ge (k+1)/2$. Let D_0 be the diagonal consisting of the cells...

conjecture | open | MathDB 页面 | 原文线索

V2-042140 | MATHDB-330305

Klostermeyer et al.'s odd-entry density conjecture for the Pascal rhombus

摘录 / EXCERPT: Let G_n and H_n be the numbers of odd and even entries, respectively, among the first n rows of the Pascal rhombus. Klostermeyer et al.'s odd-entry density conjecture. Then .....

conjecture | open | MathDB 页面 | 原文线索

V2-042141 | MATHDB-330314

Extremal transition conjecture for quantum cohomology of toric degenerations

摘录 / EXCERPT: Extremal transition conjecture. The following hold: (1) the structure constants of the small quantum product of $X_{\rm res}$ are polynomials in $q_1,\dots,q_l$ with coefficients r...

conjecture | open | MathDB 页面 | 原文线索

V2-042142 | MATHDB-330357

The power-of-two conjecture for the Sprague-Grundy function of Tetris Nim

摘录 / EXCERPT: Let $x=(x_0,x_1,x_2)$ be a position with $x_0\in\mathbb{Z}_{\geq}$ and $x_2\geq x_1=2^k$ for some $k\in\mathbb{Z}_{\geq}$. Write $\ell(x)=x_0+(x_1\mathbin{\oplus}x_2)$ for the lowe...

conjecture | open | MathDB 页面 | 原文线索

V2-042143 | MATHDB-330358

The power-of-two interval conjecture for the Sprague-Grundy function

摘录 / EXCERPT: Let $x=(x_0,x_1,x_2)$ be a position with $x_1\leq x_2$, and let $u(x)=x_0+x_1+x_2$ denote the upper bound for its Sprague-Grundy value. The power-of-two interval conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-042144 | MATHDB-330359

The multiple-of-a-power-of-two conjecture for the Sprague-Grundy function

摘录 / EXCERPT: Let $x=(x_0,x_1,x_2)$ be a position satisfying, for some $k\in\mathbb{Z}_{\geq}$, $2^{k-1}<x_1<x_0+x_1<2^k$, and assume $x_1\leq x_2$. Let $u(x)=x_0+x_1+x_2$ be the upper bound. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-042145 | MATHDB-330360

The eventual upper-bound conjecture for Tetris Nim

摘录 / EXCERPT: Let $x=(x_0,x_1,x_2)$ be a position satisfying, for some $k\in\mathbb{Z}_{\geq}$, $2^{k-1}<x_1<x_0+x_1<2^k$. Let $u(x)=x_0+x_1+x_2$ be the upper bound. The eventual upper-bound con...

conjecture | open | MathDB 页面 | 原文线索

V2-042146 | MATHDB-330361

The eventual periodicity conjecture for the exceptional even case

摘录 / EXCERPT: Let $x^*_0=1$ and let x^*_1 be even with $2^{k-1}<x^*_1<2^k$ for some $k\in\mathbb{Z}_{\geq}$. Define the defect...

conjecture | open | MathDB 页面 | 原文线索

V2-042147 | MATHDB-330364

Finiteness conjecture for tame integral positive \(\operatorname{SL}_3\)-friezes of height 4

摘录 / EXCERPT: Let a tame integral positive SL_3-frieze be a tame frieze pattern whose entries are nonnegative integers and whose adjacent 3×3 minors have determinant 1;...

conjecture | open | MathDB 页面 | 原文线索

V2-042148 | MATHDB-330365

Asymptotic growth conjecture for central-polynomial codimensions

摘录 / EXCERPT: Let A be a PI algebra, and let $\delta_n(A)$ denote the nth codimension sequence associated with its central polynomials. For ordinary codimensions, Giambruno and Zaicev proved...

conjecture | open | MathDB 页面 | 原文线索

V2-042149 | MATHDB-330366

Bollobás-Leader directed edge-disjoint paths conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, identified with the power set $\mathcal{P}[n]$. For disjoint subsets $A,B\subseteq \mathcal{P}[n]$, let $\overrightarrow{p}_e(A,B)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-042150 | MATHDB-330375

Periodicity conjecture for fusion sequences of finite groups

摘录 / EXCERPT: Let G be a finite group, and let its fusion sequence denote the sequence of fusion numbers associated with the poset of Occam characteristic functions of G. Periodicity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-042151 | MATHDB-330380

Geelen-Gerards-Whittle polynomial-time girth conjecture for proper minor-closed classes

摘录 / EXCERPT: Let $\mathcal{M}$ be a proper minor-closed class of binary matroids. Geelen-Gerards-Whittle's conjecture. There is a polynomial-time algorithm for computing the girth of matroids i...

conjecture | open | MathDB 页面 | 原文线索

V2-042152 | MATHDB-330405

Combinatorial discriminant conjecture for complete-intersection torus fibrations

摘录 / EXCERPT: Combinatorial discriminant conjecture. After choosing suitable triangulation data of the dual polytope, the image of each component of $C^{(i)}$ retracts to a codimension-one subse...

conjecture | open | MathDB 页面 | 原文线索

V2-042153 | MATHDB-330428

The minimal 2-design conjecture

摘录 / EXCERPT: Let a 2-design in dimension d be a finite projective 2-design, and let a SIC denote a symmetric informationally complete measurement while an MUB denotes a complete set of mutual...

conjecture | open | MathDB 页面 | 原文线索

V2-042154 | MATHDB-330435

Unimodality of rectangular Narayana numbers

摘录 / EXCERPT: Let $N(n,m;k\mid 1)$ denote the rectangular Narayana numbers, indexed by $0\le k\le (n-1)(m-1)$. The sequence is symmetric about its midpoint. Rectangular Narayana unimodality conj...

conjecture | open | MathDB 页面 | 原文线索

V2-042155 | MATHDB-330436

Conjugation symmetry for stretched Kostka coefficients

摘录 / EXCERPT: Let λ and μ be partitions of the same size. Write $a(\lambda,\mu)$ and $b(\lambda,\mu)$ for the associated invariants introduced in the paper, with $b(\lambda,\mu)$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-042156 | MATHDB-330454

The bounds relating non-Mersenne numbers and Smarandache parameters

摘录 / EXCERPT: The non-Mersenne-Smarandache bounds conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042157 | MATHDB-330477

Existence of the limiting expected weight of k edge-disjoint spanning trees

摘录 / EXCERPT: Existence conjecture. For every positive integer k, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042158 | MATHDB-330487

Continuity of the projection function $proj$

摘录 / EXCERPT: Let $Erdahl(n)$ denote the space used in the paper, and for $f\in Erdahl(n)$ let $proj(f)=g$, where (g,h) is the unique extremal pair associated with f. Continuity conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-042159 | MATHDB-330490

Hurlbert-Mykkeltveit-Parker conjecture on de Bruijn arrays

摘录 / EXCERPT: Hurlbert-Mykkeltveit-Parker conjecture. There exists an $(r,v;n,m)_d$-array whenever

conjecture | open | MathDB 页面 | 原文线索

V2-042160 | MATHDB-330520

Witsenhausen's conjecture on large spherical sets without orthogonal pairs

摘录 / EXCERPT: Let S^n be the unit sphere with its rotation-invariant volume, and let $A\subseteq S^n$ be measurable. Suppose that A contains no two orthogonal vectors. Witsenhausen's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-042161 | MATHDB-330524

Schur's conjecture on faces of diameter complexes

摘录 / EXCERPT: Let A be a set of n points in $\mathbb R^d$, and let its diameter complex be the simplicial complex whose faces are subsets of A with diameter equal to the diameter of A. S...

conjecture | open | MathDB 页面 | 原文线索

V2-042162 | MATHDB-330528

Frankl-Rödl/Frankl-Wilson conjecture with sum restrictions

摘录 / EXCERPT: For $S\subset[n]$, write $\lVert S\rVert=\sum_{s\in S}s$. Let $\alpha_1,\alpha_2,\beta_1,\beta_2$ satisfy $0<\alpha_1,\alpha_2<1$ and $0<\beta_1,\beta_2<1$, and let $\mathcal G$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042163 | MATHDB-330547

Virtual combinatorial R-matrix conjecture for KR crystals

摘录 / EXCERPT: Let $B^{r,s}$ and $B^{r',s'}$ be KR crystals of type $D_4^{(3)}$, and let the virtual combinatorial R-matrix be induced by restriction of the type $D_4^{(1)}$ combinatorial R-m...

conjecture | open | MathDB 页面 | 原文线索

V2-042164 | MATHDB-330574

Noonan-Zeilberger conjecture on P-recursiveness of pattern-avoidance sequences

摘录 / EXCERPT: Let S_k be the symmetric group on k letters. For a fixed finite set of patterns $\subset S_k$, let $C_n(\mathcal{F})$ be the number of permutations in S_n avoiding every patt...

conjecture | open | MathDB 页面 | 原文线索

V2-042165 | MATHDB-330590

Limit-game conjecture for the Manickam-Miklós-Singhi problem

摘录 / EXCERPT: Fix $p\in(0,1)$. In the limit game, an admissible strategy is a countable sequence of coefficients, with all but finitely many coefficients zero in the stated candidate solutions;...

conjecture | open | MathDB 页面 | 原文线索

V2-042166 | MATHDB-330591

Finite-game conjecture for the Manickam-Miklós-Singhi problem

摘录 / EXCERPT: Let $n,k\in\mathbb N$, and consider the finite game in which the coefficients satisfy the zero-sum normalization from the Manickam-Miklós-Singhi problem. Two solutions are equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-042167 | MATHDB-330592

Kikuta-Ruckle conjecture

摘录 / EXCERPT: Let $n,k\in\mathbb N$ and $d\in(0,1)$. In the KR-Problem, choose nonnegative numbers $a_1,\ldots,a_n$ with $\sum_{i=1}^n a_i=1$ to maximize the probability that the sum over a unif...

conjecture | open | MathDB 页面 | 原文线索

V2-042168 | MATHDB-330593

Refined Kikuta-Ruckle conjecture

摘录 / EXCERPT: Let $n,k\in\mathbb N$ and $d\in(0,1)$. In the KR-Problem, an optimal solution has equal nonzero coordinates as in the Kikuta-Ruckle conjecture. Define $\lfloor x\rfloor$ and the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-042169 | MATHDB-330594

Sharpness conjecture for the discrete double-limit caching game

摘录 / EXCERPT: Let $v(k,j,n,h)$ be the value of the discrete caching game and let $v^*(k,j,h)$ be the corresponding limit-game value. The source states the upper bound ... For $k=j=2$, the transfor...

conjecture | open | MathDB 页面 | 原文线索

V2-042170 | MATHDB-330596

Sharpness conjecture for the two-nut discrete double-limit bound

摘录 / EXCERPT: Assume the two-nut case $k=j=2$. Let the value of the game be bounded above by the quantity in the source theorem. Two-nut sharpness conjecture. The bound ... is sharp if ... and eithe...

conjecture | open | MathDB 页面 | 原文线索

V2-042171 | MATHDB-330597

Large-depth value conjecture for the two-nut caching game

摘录 / EXCERPT: Assume the two-nut caching game with parameters n and h, and let the value denote the optimal winning probability. Large-depth value conjecture. If ... then the value of the game...

conjecture | open | MathDB 页面 | 原文线索

V2-042172 | MATHDB-330598

Sharpness conjecture for the discrete-limit upper bound

摘录 / EXCERPT: Assume the two-nut caching game and let the discrete-limit theorem provide its stated upper bound for h. Discrete-limit sharpness conjecture. If ... then the best upper bound suppl...

conjecture | open | MathDB 页面 | 原文线索

V2-042173 | MATHDB-330599

Exact-value conjecture for the multi-nut caching game

摘录 / EXCERPT: Multi-nut exact-value conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-042174 | MATHDB-330600

Integral-depth multi-nut exact-value conjecture

摘录 / EXCERPT: Consider the caching game with $k=j\ge2$, in which the hider caches k nuts and the searcher aims to find all of them. Integral-depth multi-nut conjecture. If ... then the value of...

conjecture | open | MathDB 页面 | 原文线索

V2-042175 | MATHDB-330604

Haws' regular unimodular triangulation conjecture for matroid polytopes

摘录 / EXCERPT: Let M be a matroid with set of bases $\mathcal{B}$, and let ... where e_B is the sum of the standard basis vectors indexed by B. A triangulation is regular unimodular when it i...

conjecture | open | MathDB 页面 | 原文线索

V2-042176 | MATHDB-330605

Haws' unimodular-subsimplex covering conjecture for matroid polytopes

摘录 / EXCERPT: Let M be a matroid and let $P(M)$ be its matroid polytope. A unimodular subsimplex is a unimodular simplex contained in $P(M)$. Haws' conjecture. For any matroid M, the polytop...

conjecture | open | MathDB 页面 | 原文线索

V2-042177 | MATHDB-330619

Linear-growth conjecture for distinct slopes of extreme points of S_r

摘录 / EXCERPT: Let S_r be the polytope in the slope variables $(s_0,s_1,\dots,s_{r+1})$ arising from the discretized search, and let s_i denote the slope values. Linear-growth conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042178 | MATHDB-330621

The corner-peeling conjecture for extremal classes

摘录 / EXCERPT: Let C be an extremal class. A concept $c\in C$ is a corner of C if $C\setminus\{c\}$ is extremal. Corner-peeling conjecture. Every nonempty extremal class C has at least one...

conjecture | open | MathDB 页面 | 原文线索

V2-042179 | MATHDB-330648

The denominator conjecture for even moments of uniform random walks

摘录 / EXCERPT: Denominator conjecture. For every $\nu \geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-042180 | MATHDB-330649

The cluster monomial basis conjecture for finite-type LP algebras

摘录 / EXCERPT: An LP algebra is an algebra defined from seeds and mutations, and a cluster monomial is a product of powers of variables $X_1^{a_1}\dots X_n^{a_n}$ whose variables $X_1,\dots,X_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-042181 | MATHDB-330693

Baxter's conjecture on switch-multiplicity and hesitating walks

摘录 / EXCERPT: Let $q(n,m)$ be the number of Q-hesitating excursions of length $2n$ with switch-multiplicity m, and let $a(n,m)$ be the number of W_2-hesitating walks of length $2n$ from...

conjecture | open | MathDB 页面 | 原文线索

V2-042182 | MATHDB-330694

Baxter's stronger symmetry conjecture for hesitating walks

摘录 / EXCERPT: Let $a(n;i,j,m)$ denote the number of W_2-hesitating walks of length $2n$ from $(0,0)$ to (i,j) with m marked steps each leaving the diagonal $x=y$. Baxter's stronger symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-042183 | MATHDB-330695

Conjecture that binary words maximize the number of abelian square factors

摘录 / EXCERPT: For a finite word, an abelian square is a factor formed by concatenating two anagrammatic words. Binary-word maximality conjecture. If a word of length n contains k distinct ab...

conjecture | open | MathDB 页面 | 原文线索

V2-042184 | MATHDB-330696

Conjecture on the number of inequivalent abelian squares

摘录 / EXCERPT: Two abelian squares are inequivalent when they have different Parikh vectors.
Inequivalent-abelian-square bound conjecture. Every word of length n contains $O(n\sqrt{n})$ inequiv...

conjecture | open | MathDB 页面 | 原文线索

V2-042185 | MATHDB-330763

Equality of the absolute Galois group with the dessin automorphism subgroup

摘录 / EXCERPT: Let $\mathcal{L}$ be the lattice of biased dessins, let $\Gamma' = \operatorname{Aut}(\mathcal{L})$, and let Γ'_0 be the subgroup of Γ' that fixes the three vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-042186 | MATHDB-330791

Bounded-row conjecture for three-pile Sharing Nim nim-sequences

摘录 / EXCERPT: Bounded-row conjecture. Given $a\geq 1$, the nim-sequence $(\mathcal{G}(0,a,n))_{n\geq a}$ is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-042187 | MATHDB-330792

Non-periodicity conjecture for rows of the three-pile Sharing Nim nim-value table

摘录 / EXCERPT: Non-periodicity conjecture. Rows of Table 1 are not ultimately periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-042188 | MATHDB-330793

Support bound for higher nim-values in three-pile Sharing Nim

摘录 / EXCERPT: Support-bound conjecture. If $\mathcal{G}(0,a,b)=g\geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042189 | MATHDB-330849

Smoothness conjecture for complete pointsets in gauged linear sigma-models

摘录 / EXCERPT: Smoothness conjecture. If a gauged linear σ-model is defined by pointsets ${\Scr A}$ and ${\Scr B}$ which are both ${\Scr A}$-complete and ${\Scr B}$-complete, then this mod...

conjecture | open | MathDB 页面 | 原文线索

V2-042190 | MATHDB-330894

Minimal-denominator conjecture for relative extremal projectors

摘录 / EXCERPT: Let ${\mathfrak g}$ be a reductive Lie algebra, ${\mathfrak l}$ a Levi subalgebra, and $P({\mathfrak g},{\mathfrak l})$ the relative extremal projector. Let ... be the lower-bound de...

conjecture | open | MathDB 页面 | 原文线索

V2-042191 | MATHDB-330940

The cherry-count conjecture for random tanglegrams

摘录 / EXCERPT: A cherry is a pair of leaves joined to a common parent in a binary tree. Let a tanglegram of size n be chosen uniformly at random, and consider its left binary tree. Cherry-count...

conjecture | open | MathDB 页面 | 原文线索

V2-042192 | MATHDB-330941

The generator-count asymptotic conjecture for random tanglegrams

摘录 / EXCERPT: Let B_n denote the set of binary trees with n leaves. For a binary tree T, let $l(T)$ be its number of leaves and $k(T)$ its number of symmetries, meaning vertices with ident...

conjecture | open | MathDB 页面 | 原文线索

V2-042193 | MATHDB-330959

Ingleton's critical-graph excluded-minor conjecture

摘录 / EXCERPT: Ingleton's critical-graph conjecture. If Δ is a critical graph with no obstructions, then $M(\Delta)$ is an excluded minor for $\mathcal{BO}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042194 | MATHDB-330960

Ingleton's obstruction excluded-minor conjecture for base-orderability

摘录 / EXCERPT: Obstruction excluded-minor conjecture. If Δ has an obstruction, then $M(\Delta)$ is an excluded minor for $\mathcal{BO}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042195 | MATHDB-330970

The fiber representation conjecture for COMs

摘录 / EXCERPT: Fiber representation conjecture. Every COM is a fiber of some OM.

conjecture | open | MathDB 页面 | 原文线索

V2-042196 | MATHDB-330971

The realizable fiber representation conjecture for locally realizable COMs

摘录 / EXCERPT: Realizable fiber representation conjecture. Every locally realizable COM is a fiber of a realizable OM.

conjecture | open | MathDB 页面 | 原文线索

V2-042197 | MATHDB-330984

Subset preservation conjecture for multiple monotone submodular functions

摘录 / EXCERPT: Subset preservation conjecture. Under these hypotheses, such a set X exists. The claim would provide a simultaneous value guarantee for a constant number of monotone submodular o...

conjecture | open | MathDB 页面 | 原文线索

V2-042198 | MATHDB-331016

Batyrev's bounded primitive collections conjecture for complete nonsingular fans

摘录 / EXCERPT: Let Σ be an n-dimensional complete nonsingular fan, let $G(\Sigma)$ be the set of generators of its rays, and let ℓ be the Picard number of Σ. A primitive coll...

conjecture | open | MathDB 页面 | 原文线索

V2-042199 | MATHDB-331022

Gross et al.'s weak synchronicity conjecture under convolution

摘录 / EXCERPT: Let A, B, and C be three log-concave nonnegative sequences without internal zeros. Write $A*C$ and $B*C$ for their convolution sequences, and let $A\sim_{\mathrm{w}} B$ mean...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042200 | MATHDB-331024

Keleti's perimeter-to-area conjecture for unions of unit squares

摘录 / EXCERPT: Let U be the union of finitely many unit squares in a Euclidean plane, and let $P(U)$ and $A(U)$ denote its perimeter and area, respectively. Keleti's perimeter-to-area conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-042201 | MATHDB-331028

Asymptotic bound for equiangular lines with angle arccos 1/(2r-1)

摘录 / EXCERPT: Let $N_{1/(2r-1)}(d)$ denote the maximum number of equiangular lines in $\mathbb{R}^d$ with common angle $\arccos \frac{1}{2r-1}$, where $r\geq 2$ is an integer. Asymptotic equiang...

conjecture | open | MathDB 页面 | 原文线索

V2-042202 | MATHDB-331029

Polynomial bound for spherical codes with a negative interval and k positive inner products

摘录 / EXCERPT: Polynomial spherical-code conjecture. Suppose $\alpha_1,\dotsc,\alpha_k$ are any k real numbers. Then

conjecture | open | MathDB 页面 | 原文线索

V2-042203 | MATHDB-331033

Strong B-rigidity conjecture for flag Gorenstein* complexes

摘录 / EXCERPT: A simplicial sphere K is strongly B-rigid if, whenever a simplicial sphere L satisfies ... then K and L are combinatorially equivalent. Strong B-rigidity conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-042204 | MATHDB-331040

The Knutson-Vakil puzzle conjecture for equivariant Grassmannian K-theory

摘录 / EXCERPT: Let $\Delta_{\lambda,\mu,\nu}$ be the triangular puzzle region whose left, right, and bottom boundary labels are respectively λ, μ, and ν. A KV-puzzle is a filling...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042205 | MATHDB-331064

The anti-concentration norm-sum conjecture for inhomogeneous random walks

摘录 / EXCERPT: Anti-concentration norm-sum conjecture. There exist consecutive indices $i_0,\dots,i_0+n^{1-\varepsilon}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042206 | MATHDB-331068

Maximal even $(\bar p,\bar q)$-core partition conjecture

摘录 / EXCERPT: Let p and q be distinct odd primes. A partition is a $(\bar p,\bar q)$-core partition if it is simultaneously a $\bar p$-core and a $\bar q$-core, and it is even when its corre...

conjecture | open | MathDB 页面 | 原文线索

V2-042207 | MATHDB-331152

Conjectured sharp Rogers-Shephard inequality for entropy

摘录 / EXCERPT: Let X and X' be independent random vectors in ${\mathbb R}^n$ drawn from the same log-concave distribution. Let h denote differential entropy and define the entropy power by...

conjecture | open | MathDB 页面 | 原文线索

V2-042208 | MATHDB-331165

Ratio conjecture for independent sets in regular graphs

摘录 / EXCERPT: Let G be a d-regular graph on n vertices, and let $H_{d,n}$ denote the disjoint union of $n/(2d)$ copies of $K_{d,d}$, assuming that $2d$ divides n. For each k, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042209 | MATHDB-331205

The conjecture on the minimum negative difference set size

摘录 / EXCERPT: Negative difference-set conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-042210 | MATHDB-331216

Erdős's lower-bound conjecture for representation functions of asymptotic bases

摘录 / EXCERPT: Let $h\geqslant 2$, let $\mathcal{S}$ be the space of subsets of the nonnegative integers, and for an asymptotic h-basis A let $r_h(n;A)$ denote its representation function. Er...

conjecture | open | MathDB 页面 | 原文线索

V2-042211 | MATHDB-331217

Erdős's minimal-growth conjecture for regular asymptotic bases

摘录 / EXCERPT: Let $h\geqslant 2$ and let f be a function for which the notion of a $\Theta(f)$-regular asymptotic h-basis is defined. Adapted Erdős 1956 conjecture. There are no $\Theta(f)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-042212 | MATHDB-331265

Geelen and Nelson's critical threshold conjecture for matroids

摘录 / EXCERPT: Let N be a matroid, let $\chi(N)$ denote its critical number, and let its critical threshold be the infimum of the density thresholds forcing bounded critical number. Geelen and...

conjecture | open | MathDB 页面 | 原文线索

V2-042213 | MATHDB-331339

The quasi-plane graph linear density conjecture

摘录 / EXCERPT: Call a topological graph k-quasi-plane if it has no k pairwise crossing edges. The quasi-plane graph linear density conjecture. For any integer $k \geq 2$ there is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-042214 | MATHDB-331343

Cubical-distance conjecture for largest minors

摘录 / EXCERPT: Let $\mathcal{J}\subset {[n]\choose k}$ be an arrangement of largest minors, and let $W\in {[n]\choose k}$. Say that W is $(\geq t,\mathcal{J})$-largest if, for every arrangement...

conjecture | open | MathDB 页面 | 原文线索

V2-042215 | MATHDB-331352

The regular-vector reconstruction conjecture for biplanes

摘录 / EXCERPT: Let ${\cal B}$ be a biplane, let $a_1,a_2,\ldots,a_m\in{\cal \grave{M}}^v$ be vectors obeying the regular-vector property $|f_{a,j}^{-1}(2)|=q(a)$ for $j\in Q(a)$, and let ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-042216 | MATHDB-331370

The ordered-partition expansion conjecture for basic sequences

摘录 / EXCERPT: Let k be a field, let $\{\mathbf e_i\}_{1\leq i\leq \ell}$ be the standard basis of k^ℓ, and let $\mathbf B=(B_1,\ldots,B_\ell)$ be a delta ℓ-tuple with...

conjecture | open | MathDB 页面 | 原文线索

V2-042217 | MATHDB-331387

Non-property-PP conjecture for the Tribonacci word

摘录 / EXCERPT: Let ${\bf PP}$ denote the property that, for infinitely many n, every factor of length n is a product of two palindromes. Let the Tribonacci word be the standard infinite word...

conjecture | open | MathDB 页面 | 原文线索

V2-042218 | MATHDB-331415

Dafermos-Luk conjecture on weak null singularities in subextremal Kerr interiors

摘录 / EXCERPT: Let $\mathcal{H}^+\cup \widetilde{\mathcal{H}}^+$ be a bifurcate null hypersurface and let $\mathcal{CH}^+$ denote the bifurcate Cauchy horizon in the maximal development of charac...

conjecture | open | MathDB 页面 | 原文线索

V2-042219 | MATHDB-331416

Conjecture on C^0 extendibility beyond the Cauchy horizon of extremal Kerr

摘录 / EXCERPT: Consider axisymmetric characteristic initial data for the vacuum Einstein equations on a hypersurface $\mathcal{H}^+$ with future-affine complete null generators and on a hypersurf...

conjecture | open | MathDB 页面 | 原文线索

V2-042220 | MATHDB-331439

The piecewise syndetic {x+y,xy} conjecture

摘录 / EXCERPT: Let $E\subset\mathbb{N}$ be additively and multiplicatively piecewise syndetic, and suppose that E has positive lower density. The piecewise syndetic $\{x+y,xy\}$ conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042221 | MATHDB-331440

The DC-set {x+y,xy} conjecture

摘录 / EXCERPT: Let $E\subset\mathbb{N}$ be a DC set, meaning a set with the central-set property referred to in the source. The DC-set $\{x+y,xy\}$ conjecture. The set E contains a configur...

conjecture | open | MathDB 页面 | 原文线索

V2-042222 | MATHDB-331445

BV-alpha positivity conjecture for regular permutohedra

摘录 / EXCERPT: BV-alpha positivity conjecture. Every regular permutohedron is BV-α-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-042223 | MATHDB-331446

All-two tuple conjecture for maximal lower-central-series containments

摘录 / EXCERPT: Let A_n be the free associative algebra on n generators, and let $j=(2,\dots,2)\in\mathbb{N}^k$. All-two tuple conjecture. ... The formula is motivated by combinatorial considera...

conjecture | open | MathDB 页面 | 原文线索

V2-042224 | MATHDB-331481

Dimension-free stability conjecture for the uniform cover inequality

摘录 / EXCERPT: Let $d\in\mathbb{N}$ with $d\geq 2$, let $S\subset\mathbb{Z}^d$ be finite, and let ${\cal G}$ be the uniform cover family with projection maps π_g, covering multiplicity...

conjecture | open | MathDB 页面 | 原文线索

V2-042225 | MATHDB-331491

The matroidal Merino-Welsh conjectures

摘录 / EXCERPT: Let M be a matroid without loops or coloops, and let $T(M;x,y)$ be its Tutte polynomial. Matroidal Merino-Welsh conjectures. The following inequalities hold: ... ... and ... These conj...

conjecture | open | MathDB 页面 | 原文线索

V2-042226 | MATHDB-331492

Kron-Knuth strengthening of the noncommutative super Schur formula

摘录 / EXCERPT: Let $\mathcal{A}$ be the ordered colored alphabet, let $\mathcal{U}$ be the associated noncommutative algebra, let $\mathfrak{J}_\nu(\mathbf{u})$ be the noncommutative super Schur...

conjecture | open | MathDB 页面 | 原文线索

V2-042227 | MATHDB-331493

LLT-type Schur expansion conjecture for the stable Kronecker quotient

摘录 / EXCERPT: Let $\mathcal{U}_\varnothing$ be the subalgebra generated by the uncolored variables, let $\mathfrak{J}^\varnothing_\nu$ be the corresponding noncommutative Schur function for a pa...

conjecture | open | MathDB 页面 | 原文线索

V2-042228 | MATHDB-331496

Courtiel's signed peak-count conjecture for shuffles of horizontal and vertical loops

摘录 / EXCERPT: Let $\mathcal{H}$ (respectively, $\mathcal{V}$) be a path beginning and ending at the origin of half-length i (respectively, j) on the alphabet $\{\rightarrow, \leftarrow\}$ (r...

conjecture | open | MathDB 页面 | 原文线索

V2-042229 | MATHDB-331497

Albert and Bousquet-Mélou's $(x+1)$-positivity conjecture for quarter-planar loops

摘录 / EXCERPT: A quarter-planar loop is a path in the quarter plane, and its peak-count polynomial is the generating function whose coefficient of x^k counts loops with k peak-count. A polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-042230 | MATHDB-331498

Albert and Bousquet-Mélou's $(x+1)$-positivity conjecture for balanced path projections

摘录 / EXCERPT: Let $\mathcal{V}$ be a positive vertical path with u up-steps and d down-steps, let Q be the set of quarter-planar paths with r right-steps, l left-steps, and vertical pr...

conjecture | open | MathDB 页面 | 原文线索

V2-042231 | MATHDB-331501

The toggle-buildability conjecture for signed peak-count generating functions

摘录 / EXCERPT: Let G_2 be the signed peak-count generating function for the quarter-planar paths under consideration. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-042232 | MATHDB-331531

Classification conjecture for maximum (d,t)-EKR sets in dual polar graphs

摘录 / EXCERPT: Let Y be a maximum-size (d,t)-EKR set of generators in a polar space of rank d and type parameter e, meaning that any two members of Y meet in dimension at least $d-t$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-042233 | MATHDB-331543

The chromatic-number bound for Latin square graphs

摘录 / EXCERPT: The chromatic-number conjecture. Every Latin square $\mathbf L$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042234 | MATHDB-331570

Baker-Wantz conjecture on the 2-rank of unitals

摘录 / EXCERPT: Let $\mathcal U_\theta$ be the unital in the shift plane $\Pi(f)$ constructed from a planar function f on $\mathbb F_{q^2}$, where q is odd, and let $C_2(\mathcal U_\theta)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042235 | MATHDB-331608

Set-coloured complete-graph tree-cover conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. In an (r,k)-colouring, each edge receives a k-element subset of a set of r colours; let $\operatorname{tc}_{r,k}(K_n)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-042236 | MATHDB-331609

Chen-Fujita-Gyárfás-Lehel-Tóth bipartite tree-cover conjecture

摘录 / EXCERPT: Let $K_{n,m}$ be a complete bipartite graph, and let $\operatorname{tc}_{r,1}(K_{n,m})$ denote the minimum number of monochromatic trees needed to cover every r-colouring of its...

conjecture | open | MathDB 页面 | 原文线索

V2-042237 | MATHDB-331646

Canonical-form conjecture for reduced switches

摘录 / EXCERPT: Canonical-form conjecture. For all $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-042238 | MATHDB-331661

The exact formula conjecture for the topological Tverberg threshold

摘录 / EXCERPT: Exact formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042239 | MATHDB-331662

Asymptotic stability conjecture for the Kerr family

摘录 / EXCERPT: Asymptotic stability of the Kerr family. The maximal Cauchy development of a small perturbation of sub-extremal Kerr initial data possesses a black hole and exterior region, and in...

conjecture | open | MathDB 页面 | 原文线索

V2-042240 | MATHDB-331667

Multiplicative reduction conjecture for the toroidal covering parameter

摘录 / EXCERPT: Let n be an odd positive integer, and let $\xi(n)$ denote the toroidal covering parameter. Multiplicative reduction conjecture. ... This conjecture proposes an exact reduction of t...

conjecture | open | MathDB 页面 | 原文线索

V2-042241 | MATHDB-331668

Asymptotic density conjecture for odd toroidal covering parameters

摘录 / EXCERPT: Let $\xi(n)$ denote the toroidal covering parameter, and let n range over positive integers. Asymptotic density conjecture. ... The conjecture predicts the limiting density of the...

conjecture | open | MathDB 页面 | 原文线索

V2-042242 | MATHDB-331677

The big-O conjecture for union-free families

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, and let $M(n)$ denote the maximum size of a union-free family of subsets of $[n]$. Let $q(n)$ be the lower-bound function for $M(n)$ constructed in the...

conjecture | open | MathDB 页面 | 原文线索

V2-042243 | MATHDB-331679

The strict ordering of non-one-block patterns

摘录 / EXCERPT: Strict ordering conjecture. If $k\geq 4$, $\tau\in\Pi_k$, and $\tau\neq\beta_k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042244 | MATHDB-331685

The Symmetric Gaussian Problem

摘录 / EXCERPT: Symmetric Gaussian Problem. If $\rho>0$, then either $(B(0,r_a),B(0,r_b)^c)$ or $(B(0,r_a')^c,B(0,r_b'))$ achieves

conjecture | open | MathDB 页面 | 原文线索

V2-042245 | MATHDB-331686

The quadratic Symmetric Gaussian Problem

摘录 / EXCERPT: Quadratic Symmetric Gaussian Problem. Either $B(0,r_a)$ or $B(0,r_a')^c$ achieves

conjecture | open | MathDB 页面 | 原文线索

V2-042246 | MATHDB-331740

Rota-Heron-Welsh log-concavity conjecture for matroid characteristic coefficients

摘录 / EXCERPT: Let M be a matroid on a finite set E, and write $r+1$ for its rank. Let $\chi_{\mathrm{M}}(\lambda)$ be its characteristic polynomial, and let $w_k(\mathrm{M})$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-042247 | MATHDB-331741

Welsh-Mason log-concavity conjecture for independent-set numbers of matroids

摘录 / EXCERPT: Let M be a matroid on a finite set E, write $r+1$ for its rank, and call a subset of E independent if no element belongs to the closure of the other elements. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-042248 | MATHDB-331752

Gyárfás' staircase conjecture for binary matrices

摘录 / EXCERPT: Gyárfás' conjecture. Every 0-1 matrix M of size $n\times n$ contains a homogeneous staircase of size at least $n-1$, equivalently, if $st(M)$ is the maximum size of a homogen...

conjecture | open | MathDB 页面 | 原文线索

V2-042249 | MATHDB-331759

The extremal classification conjecture for multiply union families

摘录 / EXCERPT: The extremal classification conjecture. Let $r\geq2$ and let s be a positive integer. If $A\subseteq{\mathbb N}^n$ is r-wise s-union, then

conjecture | open | MathDB 页面 | 原文线索

V2-042250 | MATHDB-331763

Finiteness conjecture for d-configurations with large characteristic number

摘录 / EXCERPT: Finiteness conjecture for large characteristic numbers. For every $\varepsilon>0$ and fixed $d\geqslant2$, there exist only finitely many d-configurations $\mathcal{C}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042251 | MATHDB-331820

Pach's conjecture on touchings among pairwise-intersecting pseudo-segments

摘录 / EXCERPT: Let a family of pseudo-segments be a family of Jordan arcs in which any pair of curves intersects at most once apart from endpoints. Let $\mathcal{C}$ be such a family, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-042252 | MATHDB-331821

L.-Z.'s corner enumeration conjecture for tree-like tableaux

摘录 / EXCERPT: L.-Z.'s corner enumeration conjecture. The number of corners in the set of tree-like tableaux of size n is

conjecture | open | MathDB 页面 | 原文线索

V2-042253 | MATHDB-331822

The (a,b)-analogue conjecture for non-occupied corners of tree-like tableaux

摘录 / EXCERPT: The (a,b)-analogue conjecture. For $n\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-042254 | MATHDB-331824

Polynomial enumeration conjecture for non-occupied corners in tree-like tableaux

摘录 / EXCERPT: Polynomial enumeration conjecture. For $n\geq 3$,

conjecture | solved | MathDB 页面 | 原文线索

V2-042255 | MATHDB-331825

Polynomial enumeration conjecture for non-occupied corners in symmetric tree-like tableaux

摘录 / EXCERPT: Symmetric polynomial enumeration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042256 | MATHDB-331943

Lazzarini's boundary conjecture for extremal Lagrangian tori

摘录 / EXCERPT: Let D^{2n} be the $2n$-dimensional unit disc with the standard symplectic form $ \omega_0$, and let $S^{2n-1}=\partial D^{2n}$. A Lagrangian torus $L\subset D^{2n}$ is extremal w...

conjecture | open | MathDB 页面 | 原文线索

V2-042257 | MATHDB-331944

Monotonicity conjecture for extremal Lagrangian tori in complex projective space

摘录 / EXCERPT: Let $\mathbb{C}P^n$ carry the Fubini-Study symplectic form $\omega_{\operatorname{FS}}$. A Lagrangian torus $L\subset \mathbb{C}P^n$ is extremal when it realizes the relevant Lagra...

conjecture | open | MathDB 页面 | 原文线索

V2-042258 | MATHDB-331947

The f-conjecture for union-closed families

摘录 / EXCERPT: Let $f(n,a)$ denote the optimization quantity defined in the paper for union-closed families on n ground-set elements with maximum element frequency at most a. f-conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042259 | MATHDB-331954

Multicolor sunflower product conjecture

摘录 / EXCERPT: For integers n and k, let ${\cal F}(n,k)$ be the collection of k-tuples of families ${\cal A}_i\subset 2^{[n]}$ that contain no multicolor sunflower with k petals, and defi...

conjecture | open | MathDB 页面 | 原文线索

V2-042260 | MATHDB-331962

The projective Bollobás-type conjecture for affine subspaces

摘录 / EXCERPT: Projective Bollobás-type conjecture. Every cross-intersecting sequence of pairs of projective subspaces satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042261 | MATHDB-331963

Sun's ratio monotonicity conjecture for the sequence S_n

摘录 / EXCERPT: Sun's ratio monotonicity conjecture. The sequence ${S_{n+1}/S_n}_{n\geq 3}$ is strictly increasing to the limit 9, and the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-042262 | MATHDB-332004

Conjecture on uniform cutset sizes in the truncated Boolean lattice

摘录 / EXCERPT: Uniform cutset conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042263 | MATHDB-332005

The minimum-width conjecture for cutsets in the truncated Boolean lattice

摘录 / EXCERPT: Minimum-width conjecture. For $n\gg m$,

conjecture | open | MathDB 页面 | 原文线索

V2-042264 | MATHDB-332006

The equality conjecture for level-width and cutset-width in the truncated Boolean lattice

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$ and let ${\cal B}_n(m,l)$ be the truncated Boolean lattice consisting of all subsets of $[n]$ of sizes from m through l. A cutset is a collection mee...

conjecture | open | MathDB 页面 | 原文线索

V2-042265 | MATHDB-332007

Conjectures on the asymptotic independence numbers of cyclic groups

摘录 / EXCERPT: For a positive integer n and t, let $s(\mathbb{Z}_n,t)$ denote the maximum cardinality of a t-independent subset of the cyclic group $\mathbb{Z}_n$. The authors propose two c...

conjecture | open | MathDB 页面 | 原文线索

V2-042266 | MATHDB-332008

Conjectural formula for restricted sumsets in cyclic groups

摘录 / EXCERPT: Let $\mathbb{Z}_n$ be the cyclic group of order n. For an m-subset of $\mathbb{Z}_n$, let $\rho\hat{\;}(\mathbb{Z}_n,m,h)$ denote the minimum size of its restricted h-fold s...

conjecture | open | MathDB 页面 | 原文线索

V2-042267 | MATHDB-332012

Bound on the number of exceptional restricted double-sumset sizes

摘录 / EXCERPT: Let G be an abelian group of order n, and let $\rho_{\pm}(G,m,2)$ and $\rho(G,m,2)$ denote the minimum signed and unrestricted double sumset sizes of an m-subset of G. Bajn...

conjecture | open | MathDB 页面 | 原文线索

V2-042268 | MATHDB-332013

Conjectural formula for unrestricted-length restricted sumsets in cyclic groups

摘录 / EXCERPT: Let n be a positive integer and let $m\leq n$. For an m-subset A of $\mathbb{Z}_n$, write $\Sigma A=\bigcup_{h=0}^{\infty}h\hat{\;}A$ for its restricted sumset over all numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-042269 | MATHDB-332028

The radial subtiling conjecture for monohedral disk tilings

摘录 / EXCERPT: Radial subtiling conjecture. Any monohedral tiling of the disk is a subtiling of a radially generated tiling.

conjecture | open | MathDB 页面 | 原文线索

V2-042270 | MATHDB-332029

The classification conjecture for subtilings of radially generated tilings

摘录 / EXCERPT: Subtiling classification conjecture. A subtiling of a radially generated tiling is one of the following:

conjecture | open | MathDB 页面 | 原文线索

V2-042271 | MATHDB-332030

The complete classification conjecture for monohedral disk tilings

摘录 / EXCERPT: Complete classification conjecture. The full classification of monohedral disk tilings is present in this paper.

conjecture | open | MathDB 页面 | 原文线索

V2-042272 | MATHDB-332037

The combinatorial injection conjecture for dominance order on compositions

摘录 / EXCERPT: Let A,a,B,b be integers with $A \geq B \geq 0$ and $a \geq b \geq 0$. For $m,n \in \mathbb{N}$, let $\mathcal{U}^m_n$ denote the set of compositions of n with m non-negative...

conjecture | open | MathDB 页面 | 原文线索

V2-042273 | MATHDB-332039

Almost-everywhere optimal empty-box conjecture for Latin squares

摘录 / EXCERPT: For a d-dimensional permutation A, let $\mathcal T\subseteq[n]^{d+1}$ be a box and write $A(\mathcal T)$ for the number of 1-entries of A in it. An empty box is a box with...

conjecture | open | MathDB 页面 | 原文线索

V2-042274 | MATHDB-332040

Conjecture that non-extremal acoustic determinant implies a non-extremal quadratic form

摘录 / EXCERPT: Let $f(\boldsymbol{\xi})=\boldsymbol{\xi}C\boldsymbol{\xi}^T$ be a quasiconvex quadratic form, where $\boldsymbol{\xi}\in\mathbb R^{3\times 3}$ and $C\in(\mathbb R^3)^4$ is a fourt...

conjecture | open | MathDB 页面 | 原文线索

V2-042275 | MATHDB-332066

Sum-product conjecture

摘录 / EXCERPT: Let A be a finite subset of real numbers. Define the sumset and productset by ... Sum-product conjecture. For arbitrarily small positive ϵ, ... This is a central conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042276 | MATHDB-332071

Elimination pattern conjecture for admissible colorings

摘录 / EXCERPT: Let K be a combinatoric, let g be the genus of $\abs{K}$, and let $K^\ast$ be its dual complex. An admissible coloring of $K^\ast$ is a coloring satisfying the condition that e...

conjecture | open | MathDB 页面 | 原文线索

V2-042277 | MATHDB-332084

The Erdős-Ko-Rado-type conjecture for L-intersecting families

摘录 / EXCERPT: Let $0<s\leq k\leq n$ be positive integers. Let $L=\{\ell_1,\ldots,\ell_s\}$ be a set of s positive integers such that $0<\ell_1<\cdots<\ell_s$. Suppose that $n>k^2-k+1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-042278 | MATHDB-332099

The nonexistence conjecture for Ulrich partitions with two large blocks

摘录 / EXCERPT: Let (α,β,γ) be the type of an Ulrich partition. Nonexistence conjecture. There are no Ulrich partitions of type (α,β,γ) if $\beta\geq3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042279 | MATHDB-332104

Low-dimensional subset conjecture for sets with small doubling

摘录 / EXCERPT: Low-dimensional subset conjecture. There is a subset $X \subseteq A$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042280 | MATHDB-332116

The small-set avoidance conjecture for matroid antichains

摘录 / EXCERPT: Let M be a matroid and let $\mathcal{V}(M)$ denote its associated antichain of subsets. Fix an integer k. Small-set avoidance conjecture. Asymptotically almost all antichains...

conjecture | open | MathDB 页面 | 原文线索

V2-042281 | MATHDB-332144

The diameter bound and lattice-vector realization for primitive zonotopes

摘录 / EXCERPT: Diameter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042282 | MATHDB-332154

Generic Kerr wave instability conjecture for compactly supported data

摘录 / EXCERPT: Let Σ_0 be a Cauchy hypersurface in a subextremal rotating Kerr spacetime, and let ψ solve the linear wave equation ... Assume the initial data on Σ_0 are smooth a...

conjecture | open | MathDB 页面 | 原文线索

V2-042283 | MATHDB-332199

Extremal Steklov domain symmetry and uniqueness conjecture

摘录 / EXCERPT: Extremal Steklov domain conjecture. The maximizer $\Omega^{p\star}$ of $\Lambda_p(\Omega)$ is unique up to dilations and rigid transformations, has p-fold symmetry and an axis of...

conjecture | open | MathDB 页面 | 原文线索

V2-042284 | MATHDB-332200

The zero-partition maximization conjecture

摘录 / EXCERPT: For $ \in\mathbb{N}$ and $\mathbf{x}\in\mathbb{N}^{n}$, a zero partition of $\mathbf{x}$ is a choice of signs $\sigma\in\{-1,1\}^{n}$ such that $\langle\mathbf{x},\sigma\rangle=0$....

conjecture | open | MathDB 页面 | 原文线索

V2-042285 | MATHDB-332243

Additivity of nonnegative rank under Cartesian products

摘录 / EXCERPT: Let P_1 and P_2 be polytopes. Their nonnegative rank is the smallest number of nonnegative factors in a nonnegative factorization of a slack matrix of the polytope. Product add...

conjecture | open | MathDB 页面 | 原文线索

V2-042286 | MATHDB-332244

Generic maximal nonnegative rank for hypersimplices

摘录 / EXCERPT: Generic maximal-rank conjecture. The combinatorial hypersimplices of nonnegative rank $2n$ form a dense open subset of $\mathcal{R}_{n,k}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042287 | MATHDB-332273

Analytic-content extremal-domain conjecture

摘录 / EXCERPT: Analytic-content extremal-domain conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042288 | MATHDB-332279

Monotonicity of the function F_k

摘录 / EXCERPT: Monotonicity conjecture for F_k. The function F_k is monotonically increasing; equivalently, if $q<q'$, then $F_k(q)\leq F_k(q')$.

conjecture | open | MathDB 页面 | 原文线索

V2-042289 | MATHDB-332319

Almkvist's conjecture on the unimodality of Gaussian polynomials

摘录 / EXCERPT: Almkvist's conjecture. The polynomial $f_{n,r}(q)$ is an ${\rm A}$-polynomial if either r is even and $n\ge 1$, or r is odd and $n\ge 11$. This conjecture gives sufficient cond...

conjecture | open | MathDB 页面 | 原文线索

V2-042290 | MATHDB-332329

Brenti's real-rootedness conjecture for labeled-poset Eulerian polynomials

摘录 / EXCERPT: Brenti's conjecture. For any labeled poset (P,ω), the polynomial $W(P,\omega;t)$ has only real zeros as a polynomial in t. This is the Poset Conjecture, also called the Ne...

conjecture | open | MathDB 页面 | 原文线索

V2-042291 | MATHDB-332339

Solymosi's rational expansion conjecture

摘录 / EXCERPT: Let $f\in\mathbb Q[x,y]$ be a polynomial of degree d, and let $A\subset\mathbb Q$. Solymosi's conjecture. There is a constant $c>0$ such that ... unless $f(x,y)=g(ax+by)$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-042292 | MATHDB-332342

Binary single-\diamond universal partial word existence conjecture

摘录 / EXCERPT: Let $A={0,1}$, let $n\binom{}{}\geq 1$, and consider linear upwords for A^n containing a single $\Diamond$ at position k. The cases covered by Theorem are $k=n$ for $n\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042293 | MATHDB-332355

Finite extremal conjecture for copies of rooted binary trees

摘录 / EXCERPT: Let E_n denote the rooted binary tree with n leaves, and let $c(F,T)$ be the number of copies of a rooted binary tree F in a rooted binary tree T. For intege...

conjecture | open | MathDB 页面 | 原文线索

V2-042294 | MATHDB-332357

The power-of-two conjecture for odd core partitions into distinct parts

摘录 / EXCERPT: Let s be odd, and consider $(s,s+2)$-core partitions into distinct parts. Power-of-two conjecture. Their number equals ... The paper presents this as a proposed extension beyond th...

conjecture | open | MathDB 页面 | 原文线索

V2-042295 | MATHDB-332370

The extremal-point conjecture for the support of the invariant measure

摘录 / EXCERPT: Extremal-point conjecture. The set T is a singleton $\{z_0\}$, and the support of the invariant measure that is not supported by one of the two axes is

conjecture | open | MathDB 页面 | 原文线索

V2-042296 | MATHDB-332371

Common-basis conjecture for partitions of a matroid ground set

摘录 / EXCERPT: Common-basis conjecture. There exist partitions of $E\setminus\{x\}$ and $E\setminus\{y\}$ into k pairwise disjoint bases that share a common basis.

conjecture | open | MathDB 页面 | 原文线索

V2-042297 | MATHDB-332392

The contractive-grid Cauchy-sequence conjecture

摘录 / EXCERPT: The contractive-grid conjecture. Every such grid contains a 1-way Cauchy sequence. The conjecture is introduced as a formulation capturing the essence of Austin's common fixed poin...

conjecture | open | MathDB 页面 | 原文线索

V2-042298 | MATHDB-332396

The extremal length conjecture for tournaments and the cyclic tournament

摘录 / EXCERPT: The extremal length conjecture. For every $n\geq 3$, $r\in[n-1]$, and $T\in\mathrm{Tour}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-042299 | MATHDB-332407

Conjectured decomposition of level-k Dyck vectors

摘录 / EXCERPT: Conjectured decomposition. There exist possibly identical subsets $\mathcal{DV}_{n,\mu}$ and $\overline{\mathcal{DV}}_{n,\mu}$ of $\mathcal{DV}_{n,k}$ such that: (a) each of the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-042300 | MATHDB-332408

Conjectured chain decomposition of Dyck partitions

摘录 / EXCERPT: Conjectured chain decomposition. There exists a sequence of Dyck partitions $\{\gamma_{\lambda,i}:i\in\mathbb{Z}_{>0}\}$ with strictly increasing sizes satisfying all five stated c...

conjecture | open | MathDB 页面 | 原文线索

V2-042301 | MATHDB-332420

Characterization of fixed ordering configurations by formal expansion fixedness

摘录 / EXCERPT: Let $\mathcal{E}$ be a set of point labels and $\mathcal{B}$ a set of coordinate indices. A linear ordering configuration $\mathcal{C}$ on $(\mathcal{E},\mathcal{B})$ is fixed when...

conjecture | open | MathDB 页面 | 原文线索

V2-042302 | MATHDB-332427

Local evacuation-shuffle algorithm for arbitrary skew tableaux

摘录 / EXCERPT: Let T be any (semi)standard skew tableau and let $mathbox$ be an inner co-corner of T. Local evacuation-shuffle algorithm conjecture. There exists a local algorithm for computi...

conjecture | open | MathDB 页面 | 原文线索

V2-042303 | MATHDB-332446

Conjectural structure of lines in GC_n sets

摘录 / EXCERPT: Conjecture concerning GC_n sets. There is an integer s with $2k-n-1\leq s\leq k$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042304 | MATHDB-332461

The Mieczkowska-Sileikis conjecture for maximal i.i.d. tail probabilities

摘录 / EXCERPT: Let $m_k(x)$ denote the supremum of $\mathbb{P}(X_1+\cdots+X_k\geq 1)$ over nonnegative, independent and identically distributed random variables with $\mathbb{E}(X_i)\leq x$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-042305 | MATHDB-332468

The asymmetric Ramsey threshold conjecture for graphs

摘录 / EXCERPT: For any two graphs G and H, let $m_2(G)$ denote the 2-density of G, and define ... where ... Assume that G and H are not forests and that $m_2(G)\leq m_2(H)$. The asymmetric...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042306 | MATHDB-332480

Mixed cubic identity for Catalan triangle numbers

摘录 / EXCERPT: Mixed cubic identity. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-042307 | MATHDB-332495

Diameter conjecture for pointihedra

摘录 / EXCERPT: Let P be a convex n-gon with $n\geq 13$, and let $P^{\text{\raisebox{1px}{\scalebox{0.7}{$star$}}}}$ be the resulting polygon with a puncture placed in its interior. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-042308 | MATHDB-332499

The nonrepresentability conjecture for almost all matroids

摘录 / EXCERPT: Nonrepresentability conjecture. Asymptotically almost every matroid is not representable over any field.

conjecture | open | MathDB 页面 | 原文线索

V2-042309 | MATHDB-332500

The sparse paving minor conjecture

摘录 / EXCERPT: Sparse paving minor conjecture. Let N be a fixed sparse paving matroid. Asymptotically almost every matroid has an N-minor.

conjecture | open | MathDB 页面 | 原文线索

V2-042310 | MATHDB-332501

The sparse paving minor conjecture for almost all sparse paving matroids

摘录 / EXCERPT: Sparse paving minor conjecture for sparse paving matroids. Let N be a fixed sparse paving matroid. Asymptotically almost every sparse paving matroid has an N-minor.

conjecture | open | MathDB 页面 | 原文线索

V2-042311 | MATHDB-332502

The connectivity dichotomy for matroids without a uniform minor

摘录 / EXCERPT: Connectivity dichotomy conjecture. Let $k,l\in\mathbb{N}$. Asymptotically almost all matroids without a $U_{k,2k}$-minor are either vertically l-connected or cyclically l-conne...

conjecture | open | MathDB 页面 | 原文线索

V2-042312 | MATHDB-332508

The Catalan-Larcombe-French sequence is infinitely log-convex

摘录 / EXCERPT: Let $\{P_n\}_{n\geq 0}$ be the Catalan-Larcombe-French sequence. A sequence is infinitely log-convex if every sequence obtained by repeatedly applying the operator...

conjecture | open | MathDB 页面 | 原文线索

V2-042313 | MATHDB-332511

The highly vertically connected matroid minor conjecture

摘录 / EXCERPT: Let M be a matroid. For an integer k, M is vertically k-connected if, for every $A\subseteq E(M)$ with $\lambda_M(A)<k-1$, either A or $E(M)-A$ is spanning in M. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-042314 | MATHDB-332519

Supersaturation conjecture for ISP-systems

摘录 / EXCERPT: Let N and n be positive integers, and let $\mathcal{F}\subset \mathcal{P}(N)\times\mathcal{P}(N)$ have size ... Assume that every $(A_j,B_j)\in\mathcal{F}$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-042315 | MATHDB-332620

Kohayakawa-Łuczak-Rödl conjecture on random Turán thresholds

摘录 / EXCERPT: Let F be a graph with at least one edge. For every subgraph F' of F with at least one edge, define ... and set ... For a graph H, let $\operatorname{ex}(H,F)$ denote the maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-042316 | MATHDB-332625

The combinatorial Loewner property implies quasi-Möbius equivalence to a Loewner space

摘录 / EXCERPT: CLP-to-Loewner conjecture. If Z satisfies the CLP, then Z is quasi-Möbius equivalent to a Loewner space.

conjecture | open | MathDB 页面 | 原文线索

V2-042317 | MATHDB-332626

Conjecture on the maximum length of rich square-free words

摘录 / EXCERPT: Let $r(n)$ be the maximum length of a rich square-free word over an alphabet of size n, and let w_n be the recursively constructed rich square-free word over an alphabet of siz...

conjecture | open | MathDB 页面 | 原文线索

V2-042318 | MATHDB-332639

The trace coefficient formula for web-colouring matrices

摘录 / EXCERPT: Trace coefficient conjecture. For all $n>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-042319 | MATHDB-332680

Modified Orlik's conjecture for characteristic-polynomial divisibility

摘录 / EXCERPT: Let ${\mathcal{A}}$ be an ℓ-arrangement and let $H\in{\mathcal{A}}$. Write ${\mathcal{A}}'={\mathcal{A}}\setminus\{H\}$, and let $\pi({\mathcal{A}};t)$ denote the Poincaré pol...

conjecture | open | MathDB 页面 | 原文线索

V2-042320 | MATHDB-332681

The conjecture that the lower bound is sharp for all integer intervals

摘录 / EXCERPT: Sharp lower-bound conjecture. Corollary 1 holds for any $k\in\mathbb N$; equivalently, whenever every integer in $[1,D(I_k)]$ divides t, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042321 | MATHDB-332761

Hähnle's normal-complex diameter conjecture

摘录 / EXCERPT: Let d and n be positive integers. A normal pure $(d-1)$-complex is a pure simplicial complex in which the link of every face of codimension at least two is connected. Its diame...

conjecture | open | MathDB 页面 | 原文线索

V2-042322 | MATHDB-332778

The paving matroid conjecture

摘录 / EXCERPT: A rank-r matroid is paving if all its circuits have cardinality at least r. For each positive integer n, consider matroids on an n-element ground set. The paving matroid co...

conjecture | open | MathDB 页面 | 原文线索

V2-042323 | MATHDB-332805

Kalai-Wigderson's Euler characteristic conjecture for continuous polytopes

摘录 / EXCERPT: Suppose that K is a convex body in $\mathbb{R}^d$ and let $F_k(K)$ be the space of all k-dimensional faces of K, with the topology induced by the Hausdorff metric. Assume tha...

conjecture | open | MathDB 页面 | 原文线索

V2-042324 | MATHDB-332881

The exact-variable Finiteness Conjecture for nonsingular unsatisfiable hitting clause-sets

摘录 / EXCERPT: Exact-variable Finiteness Conjecture. For every $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-042325 | MATHDB-332892

The Cauchy-Davenport constant inequality for cancellative semigroups

摘录 / EXCERPT: Let $\mathbb A$ be a cancellative semigroup with underlying set A, and let $X_1,\ldots,X_n\subseteq A$. For each tuple, let $\gamma(X_1,\ldots,X_n)$ denote its Cauchy-Davenport c...

conjecture | open | MathDB 页面 | 原文线索

V2-042326 | MATHDB-332914

Flawlessness conjecture for broken circuit complex h-vectors

摘录 / EXCERPT: Let M be a loopless matroid, and let $(h_0,h_1,\ldots,h_s)$ be the h-vector of its broken circuit complex $BC(M)$, where s is the largest index j such that $h_j\ne0$. Flawl...

conjecture | open | MathDB 页面 | 原文线索

V2-042327 | MATHDB-332933

Straub's largest-size conjecture for odd $(s,s+2)$-core partitions

摘录 / EXCERPT: Let s be an odd positive integer. An $(s,s+2)$-core partition is a partition that is simultaneously an s-core and an $(s+2)$-core, and its parts are required to be distinct. St...

conjecture | open | MathDB 页面 | 原文线索

V2-042328 | MATHDB-332980

Friedgut's stability conjecture for t-intersecting families

摘录 / EXCERPT: Friedgut's stability conjecture. If β is non-singular and $|\mathcal{F}|\geq(1-\epsilon)|\mathcal{F}(n,k,t,r^*)|$, then there exists $B\subset[n]$ with $|B|=t+2r^*$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042329 | MATHDB-333005

Maximal m-palindromic density conjecture

摘录 / EXCERPT: Let $m\in\mathbb{N}$, and let $A\in\mathbb{N}^{\mathbb{N}_0}$ be an m-palindromic sequence that is not eventually periodic. Let $d_m([A])$ denote its m-palindromic density. For...

conjecture | open | MathDB 页面 | 原文线索

V2-042330 | MATHDB-333006

Noncontractibility conjecture for coset complexes of Boolean intervals

摘录 / EXCERPT: Let $[H,G]$ be an interval of finite groups, and let $C(H,G)$ denote the associated coset poset with order complex $\Delta(C(H,G))$. An interval $[H,G]$ is Boolean when its subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-042331 | MATHDB-333040

Even-grid upper-bound conjecture for triangle-free point constellations

摘录 / EXCERPT: Let $T(n,k)$ be the largest number of points in an n by k regular grid such that no three selected points form an isosceles triangle of nonzero area. Even-grid upper-bound conj...

conjecture | open | MathDB 页面 | 原文线索

V2-042332 | MATHDB-333041

Exact three-column constellation conjecture

摘录 / EXCERPT: Let $T(n,k)$ be the largest number of points in an n by k regular grid such that no three selected points form an isosceles triangle of nonzero area. Three-column exact-value c...

conjecture | open | MathDB 页面 | 原文线索

V2-042333 | MATHDB-333045

Cayley conjecture for lattice polytopes of large dimension

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ be a lattice polytope of dimension d and degree s. A Cayley polytope is a lattice polytope expressible as a Cayley sum of lattice polytopes. Cayley c...

conjecture | open | MathDB 页面 | 原文线索

V2-042334 | MATHDB-333067

The 4-cycle obstruction conjecture for κ-matroidal connectivity functions

摘录 / EXCERPT: Let E be a finite set, and let $ u$ be the connectivity function on $\\{a,b,c,d\\}$ defined by ... ... and with all other values determined by symmetry. A connectivity function is un...

conjecture | open | MathDB 页面 | 原文线索

V2-042335 | MATHDB-333068

Finite-obstruction conjecture for κ-matroidal connectivity functions

摘录 / EXCERPT: Let E be a finite set, let μ be a unitary connectivity function on E, and let $\mu\downarrow X$ denote the connectivity-function minor obtained by the operation defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-042336 | MATHDB-333076

Equivariant Kazhdan-Lusztig positivity conjecture for matroids

摘录 / EXCERPT: Equivariant Kazhdan-Lusztig positivity conjecture. For every equivariant matroid $W\curvearrowright M$,

conjecture | open | MathDB 页面 | 原文线索

V2-042337 | MATHDB-333096

Gaussian-width conjecture for arithmetic-progression counts

摘录 / EXCERPT: Gaussian-width conjecture. For any fixed $k \ge 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-042338 | MATHDB-333097

Polylogarithmic convex-hull conjecture for gradients of arithmetic-progression counts

摘录 / EXCERPT: Polylogarithmic convex-hull conjecture. For every fixed $k \ge 4$, one can choose an admissible set S satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-042339 | MATHDB-333104

Simpliciality conjecture for bipartite biCambrian fans

摘录 / EXCERPT: Simpliciality conjecture. The fan $\operatorname{biCamb}(W,c)$ is simplicial. Equivalently, its dual polytope is simple.

conjecture | open | MathDB 页面 | 原文线索

V2-042340 | MATHDB-333110

The asymptotic enumeration conjecture for representable matroids

摘录 / EXCERPT: The asymptotic enumeration conjecture. Asymptotically almost all representable matroids on $[n]$ have rank in $\{\lfloor n/2\rfloor,\lceil n/2\rceil\}$ and exactly $d(n)-1$ nonbase...

conjecture | open | MathDB 页面 | 原文线索

V2-042341 | MATHDB-333111

Upper growth rates are realized by sum closed classes

摘录 / EXCERPT: Let an upper growth rate of a permutation class mean a real number that occurs as the upper exponential growth rate of that class, and let a sum closed class be a permutation class...

conjecture | open | MathDB 页面 | 原文线索

V2-042342 | MATHDB-333191

Bárány's topological Tverberg conjecture

摘录 / EXCERPT: Let $d\geq 1$ and $r\geq 2$ be integers, let $N=(d+1)(r-1)$, and let $f:\Delta_N\to\mathbb{R}^d$ be continuous. A collection of faces is pairwise disjoint if the faces have no comm...

conjecture | open | MathDB 页面 | 原文线索

V2-042343 | MATHDB-333194

Monotonicity and CM-monotonicity for valuations

摘录 / EXCERPT: Monotonicity conjecture. $d719$ is monotone if and only if $d719$ is CM-monotone.

conjecture | open | MathDB 页面 | 原文线索

V2-042344 | MATHDB-333227

The GLS combinatorial criterion for square-irreducibility

摘录 / EXCERPT: Let $\mathfrak{m}=\Delta_1+\dots+\Delta_k$ be a multisegment, let $C_\mathfrak{m}$ be the vector space of compatible coordinates, and let $X_\mathfrak{m}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042345 | MATHDB-333231

Strong cross-star conjecture for lifted set families

摘录 / EXCERPT: Let $t\geq 1$, let $\mathcal{F}$ be a family, and let m be a positive integer. The notation $\mathcal{S}_{\mathcal{F},m}$ denotes the associated lifted set family, and the strong...

conjecture | open | MathDB 页面 | 原文线索

V2-042346 | MATHDB-333232

Strong cross-star conjecture for hereditary families

摘录 / EXCERPT: Let $1\leq t\leq r$, let $\emptyset\neq S\subseteq [r]$, and let $\mathcal{H}$ be a hereditary family with covering number $\mu(\mathcal{H})\geq (t+1)(r-t+1)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-042347 | MATHDB-333247

The bellows conjecture for flexible polyhedra in Euclidean space

摘录 / EXCERPT: A flexor is a flexible polyhedron in $\mathbb{R}^3$, and a flexion is a continuous deformation preserving its combinatorial type and the shapes of its faces. The bellows conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042348 | MATHDB-333279

The triangle-free average-to-maximum independent set ratio conjecture

摘录 / EXCERPT: Average-to-maximum ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042349 | MATHDB-333280

The fugacity-uniform triangle-free average-to-maximum ratio conjecture

摘录 / EXCERPT: Fugacity-uniform ratio conjecture. For every $4\epsilon>04$, there exists $4\lambda>04$ such that, for every triangle-free graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-042350 | MATHDB-333298

Existence of the limiting distribution of the truncated extremal process

摘录 / EXCERPT: Let $\varrho_{n,n-k}$ denote the point measure of particles at generation n whose most recent common ancestor with the minimum particle is at least $n-k$ generations from the roo...

conjecture | open | MathDB 页面 | 原文线索

V2-042351 | MATHDB-333343

Tao's asymptotic tightness conjecture for colored product-free subsets

摘录 / EXCERPT: Let G be a finite group, let $\mathbb{F}$ be a field, and let $X_1,\dots,X_k$ be subspaces of the group algebra $\mathbb{F}[G]$ satisfying ... Define $M_k(G)$ to be the minimum, ov...

conjecture | open | MathDB 页面 | 原文线索

V2-042352 | MATHDB-333349

Alternating expansion conjecture for Young quasisymmetric Schur functions of distinct-part partitions

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\lambda_2,\ldots,\lambda_k)$ be a partition whose k parts are all distinct. For $\sigma\in S_k$, define ... and let $\ell(\sigma)$ be the length of σ...

conjecture | open | MathDB 页面 | 原文线索

V2-042353 | MATHDB-333362

The period characterization conjecture for three-element subtraction games

摘录 / EXCERPT: Period characterization conjecture. In Case I,

conjecture | open | MathDB 页面 | 原文线索

V2-042354 | MATHDB-333364

The augmented-weighing conjecture for counterfeit coin strategies

摘录 / EXCERPT: Let $t=fk+r$, where k and r are positive integers, $0<r<f$, and $k\geq 4$. Split the coins into groups $A_1,\ldots,A_f$, $B_1,\ldots,B_f$, and $C_1,\ldots,C_f$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-042355 | MATHDB-333370

The complementary-signature conjecture for weak c-Wilf equivalence

摘录 / EXCERPT: For $k\geq 2$, let $\sigma\in\{+,-\}^k$ be a signature, and define its complementary signature $\sigma_c=\sigma_c^0\sigma_c^1\ldots\sigma_c^{k-1}$ by setting $\sigma_c^i=+$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-042356 | MATHDB-333371

The alternating-signature conjecture for c-Wilf equivalence

摘录 / EXCERPT: For $k\geq 2$, an alternating signature of size k is a signature σ such that either $\sigma^i=+$ for all even i and $\sigma^i=-$ for all odd i, or $\sigma^i=+$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-042357 | MATHDB-333380

Hausel-Rodriguez-Villegas genus-one combinatorial identity

摘录 / EXCERPT: For a partition $lambda$, let $\mathcal{H}_{lambda}(z,w)$ be the genus-one hook function, and let $\vertlambdarvert$ denote the size of $lambda$. Hausel-Rodriguez-Villegas genus-on...

conjecture | open | MathDB 页面 | 原文线索

V2-042358 | MATHDB-333402

Cubic mixing-time conjecture for the building-block sampler

摘录 / EXCERPT: Let n denote the width parameter and let $\mathcal{M}_{\mathit{blocks}}$ be the Markov-chain sampler on building block sequences described in the paper. Cubic mixing-time conject...

conjecture | open | MathDB 页面 | 原文线索

V2-042359 | MATHDB-333418

The MMS star conjecture for Latin squares of order 6

摘录 / EXCERPT: MMS star conjecture for order 6. Any Latin square of order 6 has the MMS star property.

conjecture | open | MathDB 页面 | 原文线索

V2-042360 | MATHDB-333419

The strict MMS star conjecture for Latin squares of order 7

摘录 / EXCERPT: Strict MMS star conjecture for order 7. Any Latin square of order 7 has the strict MMS star property.

conjecture | open | MathDB 页面 | 原文线索

V2-042361 | MATHDB-333426

Bukh's conjecture on spherical codes with separated negative inner products

摘录 / EXCERPT: Let $\beta\in(0,1)$ be fixed, let $\alpha_1,\dots,\alpha_k$ be any k real numbers, and let a spherical L-code be a finite non-empty set of unit vectors in $\mathbb{R}^n$ whose...

conjecture | open | MathDB 页面 | 原文线索

V2-042362 | MATHDB-333427

The heavy-handed Twyst-off conjecture

摘录 / EXCERPT: Heavy-handed Twyst-off conjecture. Heavy-handed Twyst-off has the same P positions as Twyst-off on three stacks.

conjecture | open | MathDB 页面 | 原文线索

V2-042363 | MATHDB-333440

The eventual constancy recurrence for phylogenetic complexity

摘录 / EXCERPT: Eventual constancy recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-042364 | MATHDB-333441

The phylogenetic complexity conjecture for the Klein four-group

摘录 / EXCERPT: Klein four-group phylogenetic complexity conjecture. The phylogenetic complexity of G is

conjecture | open | MathDB 页面 | 原文线索

V2-042365 | MATHDB-333444

Conjectured exponential stability bound for the permanent near maximal row norm

摘录 / EXCERPT: Exponential stability conjecture. There is some constant $C>0$ and some polynomial $f(n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042366 | MATHDB-333446

Krattenthaler-Müller's average-to-worst-case conjecture for the NPS algorithm

摘录 / EXCERPT: Krattenthaler-Müller's average-to-worst-case conjecture. The average case complexity $C(\lambda^{(n)})$ is asymptotically one half of the worst case complexity $W(\lambda^{(n)})$.

conjecture | open | MathDB 页面 | 原文线索

V2-042367 | MATHDB-333447

Krattenthaler-Müller's two-row formula conjecture for NPS average complexity

摘录 / EXCERPT: Krattenthaler-Müller's two-row formula conjecture. The average case complexity is

conjecture | open | MathDB 页面 | 原文线索

V2-042368 | MATHDB-333478

Asymptotically optimal colorful Tverberg tolerance conjecture

摘录 / EXCERPT: Let r,d be fixed positive integers. A colorful partition of M families of r points each in $\mathds{R}^d$ is a partition into r sets $A_1,\ldots,A_r$ assigning one point fr...

conjecture | open | MathDB 页面 | 原文线索

V2-042369 | MATHDB-333483

The tournament Erdős-Hajnal conjecture

摘录 / EXCERPT: Tournament Erdős-Hajnal conjecture. For every tournament S, there exists a constant $\delta(S)>0$ such that every S-free tournament T satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042370 | MATHDB-333487

The positive EH-coefficient conjecture for tournaments

摘录 / EXCERPT: Positive EH-coefficient conjecture. Every tournament has a positive EH-coefficient.

conjecture | open | MathDB 页面 | 原文线索

V2-042371 | MATHDB-333510

Sunflower conjecture in $(\mathbb Z/D\mathbb Z)^n$

摘录 / EXCERPT: Let $k\leq D$, and let $A\subset(\mathbb Z/D\mathbb Z)^n$. A collection of vectors is a k-sunflower if, in every coordinate, its entries are either all different or all equal;...

conjecture | open | MathDB 页面 | 原文线索

V2-042372 | MATHDB-333519

Burton's conjecture on intersections of normal-surface-theory fans

摘录 / EXCERPT: For $d\geq2$, let $U(d,k)$ denote the maximum number of inclusion-maximal cones in the intersection of k balanced fans in general position in $\mathbb{R}^d$, where each fan consi...

conjecture | open | MathDB 页面 | 原文线索

V2-042373 | MATHDB-333520

Wegner-style collapse conjecture for avoiding complexes

摘录 / EXCERPT: Let C be a centered colorful configuration in relative general position, and let $\mathcal{A}(C)$ be its avoiding complex. A complex d-collapses onto a subcomplex if it can be...

conjecture | open | MathDB 页面 | 原文线索

V2-042374 | MATHDB-333537

Nonvanishing conjecture for two-partition descent polynomials

摘录 / EXCERPT: Let $bbda=(bbda_1,bbda_2)$ be a partition with $bbda_1>bbda_2$. Write $C_{bbda}(x,q)$ for the associated orbit q-Eulerian polynomial, $P_{bbda_2}^{(0,bbda_1-bbda_2)}$ for the Jac...

conjecture | open | MathDB 页面 | 原文线索

V2-042375 | MATHDB-333538

Rectangularity conjecture for unitary factors of orbit q-Eulerian polynomials

摘录 / EXCERPT: Let $bbda=(bbda_1,dots,bbda_m)$ be a partition, and let $C_{bbda}(x,q)$ be its orbit q-Eulerian polynomial. A factor is unitary when it is a factor of the form $1+xq^j$. Rectangu...

conjecture | open | MathDB 页面 | 原文线索

V2-042376 | MATHDB-333546

The asymptotic connectivity conjecture for matroid complexes

摘录 / EXCERPT: Asymptotic connectivity conjecture. For any $k\geq 1$ there exists a function $f(k)$ such that, if $b(M)\geq f(k)$, then M^{*k}_{Δ} is $(k\rho(V)-2)$-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-042377 | MATHDB-333550

Polynomial-time approximation conjecture for two-row Buttons & Scissors boards

摘录 / EXCERPT: Two-row Buttons & Scissors conjecture. There is a polynomial-time algorithm that removes all but s buttons from any full $2\times n$ board with $C=2$ colors, for some constant...

conjecture | open | MathDB 页面 | 原文线索

V2-042378 | MATHDB-333551

Polynomial expansion conjecture for square-ice counts

摘录 / EXCERPT: Let $a_\ell(n)$ denote the number of configurations with n removed oranges for fixed $[?]$. For fixed n and $\ell\geq \lceil n/2\rceil$, the quantity $a_\ell(n)$ is a polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-042379 | MATHDB-333576

The planar strip occupancy conjecture

摘录 / EXCERPT: Let N be a positive integer and let $x_1,\ldots,x_N$ be points in the unit disk in the plane. A strip of width $1/N$ is a planar strip with that width. Planar strip occupancy con...

open_problem | open | MathDB 页面 | 原文线索

V2-042380 | MATHDB-333577

The variable-width spherical strip occupancy conjecture

摘录 / EXCERPT: Let N be a positive integer, let $x_1,\ldots,x_N$ be points on the unit sphere $\mathbb S^{2}$, and let $w>0$. For a unit vector v, consider the strip...

open_problem | open | MathDB 页面 | 原文线索

V2-042381 | MATHDB-333609

The ternary code support 2-design conjecture

摘录 / EXCERPT: Let $m\geq 3$ be odd, let $v=(3^m-1)/2$, and let $\mathcal{C}_m$ be either of the ternary cyclic codes described in the paper. Let A_k denote the number of codewords of Hamming w...

conjecture | open | MathDB 页面 | 原文线索

V2-042382 | MATHDB-333610

The weak closure property for matroids over hyperfields

摘录 / EXCERPT: Weak Closure Property. The Weak Closure Property holds for all matroids over hyperfields. This would characterize a broad class of tract-like coefficient systems for which vector s...

conjecture | open | MathDB 页面 | 原文线索

V2-042383 | MATHDB-333619

Selig's conjecture on the addition-distribution independence of stochastic sandpile stationarity

摘录 / EXCERPT: Selig's conjecture. The stationary distribution for this SSM does not depend on bc.

conjecture | open | MathDB 页面 | 原文线索

V2-042384 | MATHDB-333621

Conjecture on the asymptotic ratios for anti-power prefixes of the Thue-Morse word

摘录 / EXCERPT: Let ${\bf t}$ be the Thue-Morse word, and let $\gamma(k)$ denote the least positive integer n such that the prefix of ${\bf t}$ of length nk is a k-anti-power. Then the asymp...

conjecture | open | MathDB 页面 | 原文线索

V2-042385 | MATHDB-333628

Halfspace meta-conjecture for extremal families with bounded matching number

摘录 / EXCERPT: Halfspace meta-conjecture. For any n,s, the maximum defining $e(n,s)$ is attained by $\mathcal F(\boldsymbol\alpha)$ for a suitable $\boldsymbol\alpha\in\mathbb R^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-042386 | MATHDB-333629

Hilton-Milner-type conjecture for nontrivial Erdős matching families

摘录 / EXCERPT: Hilton-Milner-type conjecture. If $\nu(\mathcal F)=s$ and $\tau(\mathcal F)>s$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042387 | MATHDB-333630

Özkahya-Young anti-Ramsey matching conjecture

摘录 / EXCERPT: Let $ar(n,k,s)$ be the least number of nonempty color classes in a coloring (partition) of ${[n]\choose k}$ that guarantees a rainbow s-matching, meaning s pairwise disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-042388 | MATHDB-333633

The combinatorial formula conjecture for arrangement monodromy

摘录 / EXCERPT: Combinatorial monodromy conjecture. The characteristic polynomial is determined by this formula. The preceding results establish the formula in several cases, including arrangement...

conjecture | open | MathDB 页面 | 原文线索

V2-042389 | MATHDB-333643

Extremal-distance distribution conjecture for first-passage local sets

摘录 / EXCERPT: First-passage extremal-distance conjecture. The extremal distance $\operatorname{ED}(\Lambda_a^\Sigma,B_2(\Sigma))$ is distributed as the last visit time of level a by a Brownian...

conjecture | open | MathDB 页面 | 原文线索

V2-042390 | MATHDB-333648

Extremal-pair existence conjecture for moduli spaces on a quadric surface

摘录 / EXCERPT: Extremal-pair conjecture. Every ξ above the Rudakov δ-surface has an extremal pair, and every controlling exceptional bundle of an extremal pair is contained in two extr...

conjecture | open | MathDB 页面 | 原文线索

V2-042391 | MATHDB-333657

Multivariate alternating-sum partition identity for modulus m

摘录 / EXCERPT: Let $m\geq 2$. For a partition λ, let $\Sigma_i(\lambda)$ denote the ith component of its alternating sum type, and let $l_i(\mu)$ denote the number of parts of μ con...

conjecture | open | MathDB 页面 | 原文线索

V2-042392 | MATHDB-333709

The two-round minimal percolating set conjecture

摘录 / EXCERPT: Let Π_q be a finite projective plane of order q, let r be the infection rate, and let $t_r(A)$ denote the percolation time of a percolating set A. Two-round minimal perco...

conjecture | open | MathDB 页面 | 原文线索

V2-042393 | MATHDB-333710

The monotonicity conjecture for percolation time in finite projective planes

摘录 / EXCERPT: Let Π_q be a finite projective plane of order q, let $T_r(\Pi_q)$ denote the maximum percolation time among percolating sets at infection rate r, and let r_1 and r_2 be...

conjecture | open | MathDB 页面 | 原文线索

V2-042394 | MATHDB-333744

Bloom-Saracino's easiest-to-avoid partition conjecture

摘录 / EXCERPT: Let τ be a set partition of $[k]$, and let $A_n(\tau)$ denote the number of set partitions of $[n]$ avoiding τ. The one-block partition $(12\cdots k)$ consists of all ele...

conjecture | open | MathDB 页面 | 原文线索

V2-042395 | MATHDB-333745

Gunby's permeability formula conjecture for set-partition avoidance

摘录 / EXCERPT: Let τ be a set partition of $[n]$. Its permeability $pm(\tau)$ is the minimum k such that $[n]$ can be partitioned into $k+1$ intervals of consecutive integers, each contain...

conjecture | open | MathDB 页面 | 原文线索

V2-042396 | MATHDB-333746

Logarithmic convergence conjecture for set-partition avoidance proportions

摘录 / EXCERPT: Let $F_n(\tau)$ be the avoidance proportion and let c be chosen so that ... with $c\geq2$. Logarithmic convergence conjecture. For the appropriate c, ... This conjecture seeks a qu...

conjecture | open | MathDB 页面 | 原文线索

V2-042397 | MATHDB-333756

Guba-Sapir closure conjecture for diagram groups

摘录 / EXCERPT: Let $\mathcal{K}$ be a directed 2-complex, let u be a 1-path, and let H be a subgroup of the diagram group $\mathrm{DG}(\mathcal{K},u)$. A subgroup is closed for components...

conjecture | open | MathDB 页面 | 原文线索

V2-042398 | MATHDB-333773

Ellis and Narayanan's strengthening for symmetric 3-wise intersecting families

摘录 / EXCERPT: Ellis and Narayanan's strengthening. If $\mathcal{A}\subset\mathcal{P}_n$ is a symmetric 3-wise intersecting family, then

conjecture | open | MathDB 页面 | 原文线索

V2-042399 | MATHDB-333793

Multicomplex realization conjecture for QE, QI and FBP complexes

摘录 / EXCERPT: Multicomplex realization conjecture. There exists a map $\cal F$ from the class of ordered simplicial complexes in $QE\cap QI\cap FBP$ to the class of multicomplexes such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-042400 | MATHDB-333796

Eigenvalue-centralization extremal conjecture for bipartite graphs

摘录 / EXCERPT: Let $\mathcal{B}(n_0,n_1)$ be the class of all bipartite graphs with bipartition P_0 and P_1, where $|P_i|=n_i$ for $i\in\{0,1\}$. Let $H(v,n_0,n_1)$ denote the rooted depth-...

conjecture | open | MathDB 页面 | 原文线索

V2-042401 | MATHDB-333800

Continuous higher-dimensional thrackle conjecture

摘录 / EXCERPT: Let a $(d-1)$-thrackle be a higher-dimensional thrackle with m facets and n ridges. Continuous higher-dimensional thrackle conjecture. ... This is proposed as the continuous anal...

conjecture | open | MathDB 页面 | 原文线索

V2-042402 | MATHDB-333833

The effective-weight upper-bound conjecture for numerical semigroups

摘录 / EXCERPT: Let S be a numerical semigroup of genus g, and let $\operatorname{ewt}(S)$ denote its effective weight. Effective-weight upper-bound conjecture. One has ... The authors describe...

conjecture | open | MathDB 页面 | 原文线索

V2-042403 | MATHDB-333870

The associahedron conjecture for finite-type mutation matrices

摘录 / EXCERPT: Associahedron conjecture. All $\widetilde{\sf M}$-associahedra of a given finite type have the same combinatorial type, meaning that they have the same face lattice. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-042404 | MATHDB-333904

Average-distance conjecture for touching Jordan curves

摘录 / EXCERPT: Average-distance conjecture. For any k-touching family $\mathcal{F}$ of Jordan curves, the average distance in $\mathcal{F}$ is at most

conjecture | open | MathDB 页面 | 原文线索

V2-042405 | MATHDB-333931

McFarland's conjecture on reversible difference sets

摘录 / EXCERPT: Let D be a reversible difference set, meaning a difference set equivalent to its inverse under the relevant group operation. A Hadamard difference set is a difference set with pa...

conjecture | open | MathDB 页面 | 原文线索

V2-042406 | MATHDB-333941

Failure of automorphism invariance for Andrews-Curtis equivalence

摘录 / EXCERPT: Let F_2 be the free group on x,y, let ${\mathcal B}_2$ be the set of balanced presentations of the trivial group with two generators, and let $\operatorname{Aut}(F_2)$ act comp...

conjecture | open | MathDB 页面 | 原文线索

V2-042407 | MATHDB-333949

The period-one classification for partial chess pieces

摘录 / EXCERPT: Let a partial queen $\mathbb Q^{hk}$ be a chess piece with h horizontal or vertical basic moves and k diagonal basic moves, where $h,k\leq 2$. Period-one classification conject...

conjecture | open | MathDB 页面 | 原文线索

V2-042408 | MATHDB-333981

The cube conjecture for pattern-avoiding polytopes $P_n(132,213)$

摘录 / EXCERPT: Let $P_n(132,213)$ be the pattern-avoiding polytope associated with the patterns 132 and 213. A cube conjecture asserts that, for every n, $P_n(132,213)$ is a combinatorial c...

conjecture | open | MathDB 页面 | 原文线索

V2-042409 | MATHDB-333996

Kleitman's conjecture on minimizing the number of k-chains

摘录 / EXCERPT: Kleitman's conjecture. Let $n,M>0$ and $k\geq 2$ be integers. Among families $\mathcal{F}\subseteq\mathcal{P}(n)$ of size M, the number of k-chains in $\mathcal{F}$ is minimize...

conjecture | open | MathDB 页面 | 原文线索

V2-042410 | MATHDB-333997

Centered-set conjecture for k-chains in grid posets

摘录 / EXCERPT: Centered-set conjecture. Given m, there exists a number $d_0(m)$ such that if $d\geq d_0(m)$, then the sets minimizing the number of k-chains among subsets of $[m]^d$ of any pr...

conjecture | open | MathDB 页面 | 原文线索

V2-042411 | MATHDB-334042

Van Kampen obstruction criterion for simplicial maps of trees

摘录 / EXCERPT: Van Kampen obstruction conjecture. The van Kampen obstruction vanishes if and only if φ is approximable by embeddings.

open_problem | open | MathDB 页面 | 原文线索

V2-042412 | MATHDB-334046

Extrapolated tiling counts for $2s\times(2s+2)$ rectangles

摘录 / EXCERPT: Let $T_{n\times m}(s,k)$ denote the number of ways to tile an $n\times m$ rectangle using $s\times s$ squares and 1×1 squares, with exactly k squares of size $s\times s$....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-042413 | MATHDB-334058

Stable Terao conjecture for hyperplane arrangements

摘录 / EXCERPT: Stable Terao conjecture. Stable freeness is combinatorially determined: if arrangements ${\mathcal A}_1$ and ${\mathcal A}_2$ have isomorphic lattices and ${\mathcal A}_1$ is stabl...

conjecture | open | MathDB 页面 | 原文线索

V2-042414 | MATHDB-334075

Asymptotic asymmetry conjecture for matroids

摘录 / EXCERPT: Asymmetry conjecture. Asymptotically almost all matroids are asymmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-042415 | MATHDB-334076

Transposition-generated symmetry conjecture for matroids

摘录 / EXCERPT: Transposition conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042416 | MATHDB-334078

Central-rank conjecture for matroids

摘录 / EXCERPT: Central-rank conjecture. Asymptotically almost all matroids satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-042417 | MATHDB-334079

Transposition generation conjecture for symmetric matroids

摘录 / EXCERPT: Transposition-generation conjecture. Asymptotically almost all symmetric matroids have an automorphism group that is generated by a transposition.

conjecture | open | MathDB 页面 | 原文线索

V2-042418 | MATHDB-334108

Polynomial Ramsey growth conjecture for binary formulas in NIP structures

摘录 / EXCERPT: Let $\mathcal{M}$ be an NIP structure, and let $\varphi(x_1,x_2;z)$ be a formula. The quantity $R^*_{\varphi}(n)$ denotes the Ramsey-growth function associated with φ. Poly...

conjecture | open | MathDB 页面 | 原文线索

V2-042419 | MATHDB-334112

Optimality of the partition-matching exponent for set-partition avoidance

摘录 / EXCERPT: Partition-matching exponent conjecture.

conjecture | refuted | MathDB 页面 | 原文线索

V2-042420 | MATHDB-334124

Riehl's basketball-walk enumeration conjecture for increasing unary-binary trees

摘录 / EXCERPT: Riehl's conjecture. The number of basketball walks of length n starting at the origin and ending at altitude 1 that never touch or pass below the x-axis equals the number of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042421 | MATHDB-334125

Stanley-Wilf-like conjecture for pattern avoidance in increasing trees

摘录 / EXCERPT: Stanley-Wilf-like conjecture. There is a constant C, depending on the pattern and on ϕ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-042422 | MATHDB-334172

Conjecture that the compatibility density of delta-matroids tends to zero

摘录 / EXCERPT: For each $n\geq 2$, let Γ_n be the proportion defined in the paper for compatible pairs of labelled delta-matroids. The sequence $(\Gamma_n)_{n\geq 2}$ is decreasing and bou...

conjecture | open | MathDB 页面 | 原文线索

V2-042423 | MATHDB-334186

The $F(n)$ upper-bound conjecture for 2-increasing sequences

摘录 / EXCERPT: The $F(n)$ upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042424 | MATHDB-334187

The power-bound conjecture for Condition 4 labellings

摘录 / EXCERPT: Condition 4 power-bound conjecture. There exists $\epsilon>0$ such that every such labelling has at most

conjecture | open | MathDB 页面 | 原文线索

V2-042425 | MATHDB-334189

The decomposition conjecture for extremal 2-increasing sequences

摘录 / EXCERPT: Decomposition conjecture for extremal 2-increasing sequences.

conjecture | open | MathDB 页面 | 原文线索

V2-042426 | MATHDB-334190

The classification conjecture for extremal 2-increasing sequences

摘录 / EXCERPT: Classification conjecture for extremal 2-increasing sequences.

conjecture | open | MathDB 页面 | 原文线索

V2-042427 | MATHDB-334228

The diameter conjecture for lattice polytopes

摘录 / EXCERPT: Let $\delta(d,k)$ denote the maximum possible diameter of a lattice (d,k)-polytope, where d is the dimension and k is the maximum number of lattice points on any edge. A Mink...

conjecture | open | MathDB 页面 | 原文线索

V2-042428 | MATHDB-334250

Disjoint-triangles conjecture for decompositions of complete geometric graphs

摘录 / EXCERPT: Let K_n be a complete geometric graph and let $[\mathsf{K}_n,P]$ be a decomposition. Disjoint-triangles conjecture. If P contains at least ... triangles, then P contai...

conjecture | open | MathDB 页面 | 原文线索

V2-042429 | MATHDB-334342

The simple-polytope lower-interval conjecture for the face lattice of a phase tropical hypersurface

摘录 / EXCERPT: Simple-polytope lower-interval conjecture. For each element $(\sigma,J)\in\mathcal W$, the lower interval $\mathcal W_{\preceq(\sigma,J)}$ is isomorphic to the face lattice of a si...

conjecture | open | MathDB 页面 | 原文线索

V2-042430 | MATHDB-334347

Equivariant Kazhdan-Lusztig polynomial formula for thagomizer matroids

摘录 / EXCERPT: Equivariant Kazhdan-Lusztig polynomial formula. For all $n>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-042431 | MATHDB-334348

The conjecture that the reconstruction gap is unbounded

摘录 / EXCERPT: Let $G(n)$ denote the largest integer k such that every partition of n is uniquely determined by its multiset of k-minors, with the precise definition of $G(n)$ as in the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-042432 | MATHDB-334350

The equality of the generalized sums $\tilde{S}_{k,j}(n)$ and $S_{k,j}(n)$

摘录 / EXCERPT: The generalized-sums equality. For any $k\geq 0$ and $j\geq 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042433 | MATHDB-334351

The polynomial form of $S_{k,j}(n)$

摘录 / EXCERPT: Polynomial-factorization conjecture. There exist polynomials $P_k(t,x)$ in x of degree k, with coefficients rationally depending on t, such that

conjecture | open | MathDB 页面 | 原文线索

V2-042434 | MATHDB-334353

The polynomiality conjecture for $S_{k,j}(n)$

摘录 / EXCERPT: Polynomiality conjecture. For the indicated integers k and j, $S_{k,j}(n)$ is expressed as a polynomial in

conjecture | open | MathDB 页面 | 原文线索

V2-042435 | MATHDB-334354

The Bernoulli-number formula for the terminal coefficients $p_{k,k-1}(n)$

摘录 / EXCERPT: Terminal-coefficient conjecture. The polynomials and constants are given by

conjecture | open | MathDB 页面 | 原文线索

V2-042436 | MATHDB-334374

Finiteness conjecture for irreducible polyhedral graphs with fixed rigid vertices

摘录 / EXCERPT: Let ${\mathcal{P}}_k^\text{irr}$ be the set of all irreducible polyhedral graphs with k rigid vertices. Finiteness conjecture. Each set ${\mathcal{P}}_k^\text{irr}$ is finite. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-042437 | MATHDB-334463

Geelen-Nelson critical-threshold conjecture for binary matroids

摘录 / EXCERPT: Let N be a binary matroid with critical number $\bchi(N)=c$, and let $\btheta(N)$ denote its critical threshold. Geelen-Nelson's conjecture. ... where $i=2$ if and only if no...

conjecture | open | MathDB 页面 | 原文线索

V2-042438 | MATHDB-334464

Regularity conjecture for intersected polynomial factors

摘录 / EXCERPT: Let $V=\mathbb{F}_2^n$. For any nondecreasing function $r':\mathbb{Z}_{\geq 0}\to\mathbb{Z}_{\geq 0}$ and parameters δ,η,r,d, there exists a constant K such that a poly...

conjecture | open | MathDB 页面 | 原文线索

V2-042439 | MATHDB-334509

Classification of unavoidable Tverberg partitions

摘录 / EXCERPT: Classification conjecture. For every Tverberg type $\mathcal I$, if $\mathcal I$ is colorful, then $\operatorname{Tv}_{\mathcal I}$ is unavoidable; otherwise,...

conjecture | open | MathDB 页面 | 原文线索

V2-042440 | MATHDB-334538

Lin's extremal-preimage conjecture for the Jacobian on the Hardy space

摘录 / EXCERPT: Lin's conjecture. For every $h\in\mathcal{H}^1(\mathbb C)$, there exists $u\in\dot{W}^{1,2}(\mathbb C,\mathbb C)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042441 | MATHDB-334564

Large matching with bounded representation conjecture

摘录 / EXCERPT: For every positive integer m, let $c(m)$ be a number. Let G be a bipartite graph and let $E_1,\ldots,E_m$ be arbitrary edge sets. Bounded representation conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-042442 | MATHDB-334581

Carraher et al.'s odd-even density conjecture for three-element distance sets

摘录 / EXCERPT: Carraher et al.'s odd-even density conjecture. If $k\geqslant 3j$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042443 | MATHDB-334582

Haralambis' equality conjecture for three-element distance sets

摘录 / EXCERPT: Haralambis' equality conjecture. Every three-element D-set satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042444 | MATHDB-334592

Bijection conjecture for hesitating walks and excursions in the octant and quadrant

摘录 / EXCERPT: Bijection conjecture. The following families are in bijection: hesitating excursions of length $2n$ in the quadrant; hesitating axis-walks of length $2n$ in the octant; and hesitat...

conjecture | open | MathDB 页面 | 原文线索

V2-042445 | MATHDB-334603

Iterated difference-product growth conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}$ be a finite set, and for a positive integer k write ... Here $X\gg_{k,l}Y$ means that $X\geq c_{k,l}Y$ for a positive constant depending only on k and...

conjecture | open | MathDB 页面 | 原文线索

V2-042446 | MATHDB-334604

Small-product-set difference containment conjecture

摘录 / EXCERPT: Let $X\subset\mathbb{R}$ be a finite real set, and let $XX=\{xy:x,y\in X\}$. Let $A\subset\mathbb{R}$ be a set with difference set $A-A=\{a-a':a,a'\in A\}$. Small-product-set diffe...

conjecture | open | MathDB 页面 | 原文线索

V2-042447 | MATHDB-334608

The 6n complexity conjecture for binary generalized pseudostandard words

摘录 / EXCERPT: Let $\mathbf u$ be a binary generalized pseudostandard word, and let $\mathcal C_{\mathbf u}(n)$ denote its complexity, namely the number of factors of length n in $\mathbf u$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-042448 | MATHDB-334626

Unimodality and palindromicity conjecture for the flag descent polynomial

摘录 / EXCERPT: Let $\mathcal{F}_n^\alpha(x)$ be the flag descent polynomial associated with the quotient $\mathbb{Z}_{\alpha}\wr \mathfrak{S}_n/H$. Unimodality and palindromicity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042449 | MATHDB-334667

The symmetric three-curve area inequality

摘录 / EXCERPT: Symmetric area inequality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042450 | MATHDB-334668

The combinatorial non-crossing-sums inequality

摘录 / EXCERPT: Combinatorial formulation conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-042451 | MATHDB-334672

Non-degeneracy conjecture for connected regular matroids

摘录 / EXCERPT: Non-degeneracy conjecture. Every connected regular matroid is non-degenerate.

conjecture | open | MathDB 页面 | 原文线索

V2-042452 | MATHDB-334673

Interlacing conjecture for Kazhdan-Lusztig polynomials of matroids

摘录 / EXCERPT: Interlacing conjecture. If M and M/e are both non-degenerate, then $Q_M(t)$ interlaces $Q_{M/e}(t)$.

conjecture | open | MathDB 页面 | 原文线索

V2-042453 | MATHDB-334709

Extension-language separation conjecture for prefix normal words

摘录 / EXCERPT: Let $v,w\in 1\{0,1\}^*$ be distinct prefix normal binary words. For each $m\geq 1$, let $\textit{ext}(v,m)$ and $\textit{ext}(w,m)$ denote the numbers of length-m extensions that...

conjecture | open | MathDB 页面 | 原文线索

V2-042454 | MATHDB-334710

Empirical asymptotics for extension-critical and prefix normal word counts

摘录 / EXCERPT: Let $\Sigma=\{0,1\}$, let $\mathcal L_{\mathrm{PN1}}$ be the language of prefix normal words, and let $\textit{ecrit}(n)$ count the extension-critical words in...

conjecture | open | MathDB 页面 | 原文线索

V2-042455 | MATHDB-334721

Extremal restraint conjecture for odd cycles

摘录 / EXCERPT: Extremal restraint conjecture. If $r\in R_{\operatorname{max}}(C_n,1)$, then $r\cong r^*$.

conjecture | open | MathDB 页面 | 原文线索

V2-042456 | MATHDB-334725

The intrinsic volumes conjecture for weakly smoothable Alexandrov spaces

摘录 / EXCERPT: Let X be a compact weakly smoothable Alexandrov space, meaning that it is a Gromov-Hausdorff limit of closed smooth connected Riemannian manifolds with uniformly bounded above di...

conjecture | open | MathDB 页面 | 原文线索

V2-042457 | MATHDB-334769

Asymptotic conjecture for parabolic double cosets in symmetric groups

摘录 / EXCERPT: Let S_n be the symmetric group, and let p_n denote the total number of distinct parabolic double cosets in S_n, equivalently ... where c_w is the number of parabolic double c...

conjecture | open | MathDB 页面 | 原文线索

V2-042458 | MATHDB-334805

McDiarmid and Scott's conjecture on the diameter of block-stable random graphs

摘录 / EXCERPT: Let C_n be a uniform random graph from a block class, with each block receiving weight 0 or 1, and let t_n be a sequence tending to infinity. McDiarmid and Scott's...

conjecture | open | MathDB 页面 | 原文线索

V2-042459 | MATHDB-334813

The finite-size component conjecture for the supersymmetric open XXZ chain

摘录 / EXCERPT: Finite-size component conjecture. For each $n\geqslant 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-042460 | MATHDB-334819

Strong polynomial Freiman-Ruzsa conjecture via generalized arithmetic progressions

摘录 / EXCERPT: Strong polynomial Freiman-Ruzsa conjecture. There exists an absolute constant $c>0$ such that the following holds. If $|A+A|\leq K|A|$, then there exist a d-dimensional GAP...

conjecture | open | MathDB 页面 | 原文线索

V2-042461 | MATHDB-334829

Nonexistence conjecture for balanced neighborly homology spheres with 6k vertices

摘录 / EXCERPT: Let $k\geq 2$. A balanced k-neighborly homology $(2k-1)$-sphere is a balanced simplicial homology sphere of dimension $2k-1$ in which every set of at most k vertices with disti...

conjecture | open | MathDB 页面 | 原文线索

V2-042462 | MATHDB-334848

Hoppen-Lefmann-Odermann conjecture for vector-space colourings

摘录 / EXCERPT: Hoppen-Lefmann-Odermann conjecture. For (r,t)-colourings of the k-dimensional subspaces of $\mathbb{F}_q^n$:

conjecture | open | MathDB 页面 | 原文线索

V2-042463 | MATHDB-334889

Quadratic zero-sum block conjecture for two-valued sequences

摘录 / EXCERPT: Let r, s and k be positive integers such that $(r+s)$ divides k. A function $f:[n]\to\{-r,s\}$ has total sum $f([n])=0$ when the sum of its values over $[n]$ is zero. A ZS...

conjecture | open | MathDB 页面 | 原文线索

V2-042464 | MATHDB-334895

The sharp stability conjecture for the edge-isoperimetric inequality in the discrete cube

摘录 / EXCERPT: The sharp stability conjecture. If $\mathcal{F}\subset\mathcal{P}([n])$ and $\mathcal{L}\subset\mathcal{P}([n])$ is as above, then there exists a family...

conjecture | open | MathDB 页面 | 原文线索

V2-042465 | MATHDB-334900

The odd-configuration inversion conjecture

摘录 / EXCERPT: Let $n=2m+1$, and let $\widetilde{\Delta^n}$ denote the stable configurations reachable from Δ^n by chip-firing. In the odd case, these configurations may be identified with...

conjecture | open | MathDB 页面 | 原文线索

V2-042466 | MATHDB-334902

Borovik-Lubotzky-Myasnikov generalized Andrews-Curtis conjecture

摘录 / EXCERPT: Let G be a finitely generated group, let $w(G)$ be its weight, and let $G_{ab}=G/[G,G]$ be its abelianization. For $n\geq\max(w(G),2)$, an ordered n-tuple is normally generatin...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-042467 | MATHDB-334903

Burns-Macedonska generalized Andrews-Curtis conjecture

摘录 / EXCERPT: Let G be a finitely generated group of rank n. An ordered n-tuple is normally generating when its components normally generate G, and its recalcitrance is the least number...

conjecture | open | MathDB 页面 | 原文线索

V2-042468 | MATHDB-334924

Edge-cover conjecture for the intersection of two matroids

摘录 / EXCERPT: Let $\mathcal P$ and $\mathcal Q$ be matroids on the same ground set. For a simplicial complex $\mathcal C$, let $\beta(\mathcal C)$ denote the minimum number of members of...

conjecture | open | MathDB 页面 | 原文线索

V2-042469 | MATHDB-334925

Almost fair representation conjecture for dimatroids

摘录 / EXCERPT: Let $\mathcal P$ and $\mathcal Q$ be matroids on the same ground set V, let $\mathcal D=\mathcal P\cap\mathcal Q$, and set ... A set S represents a part A almost α-fairl...

conjecture | open | MathDB 页面 | 原文线索

V2-042470 | MATHDB-334927

Balancing conjecture for pairs of dimatroid sets

摘录 / EXCERPT: Let $\mathcal D$ be a dimatroid, meaning the intersection of two matroids on a common ground set, and let $C,D\in\mathcal D$. Balancing conjecture. There exist $C',D'\in\mathcal D$...

conjecture | open | MathDB 页面 | 原文线索

V2-042471 | MATHDB-334940

Asymptotic growth conjecture for right-angle-free subsets of finite vector spaces

摘录 / EXCERPT: Let $R(n,q)$ denote the maximum cardinality of a subset of the vector space $4\mathbb{F}_q^n4$ containing no right angles. The notation $4\Theta4$ refers to asymptotic equality up...

conjecture | open | MathDB 页面 | 原文线索

V2-042472 | MATHDB-334949

Povolotsky's recurrence conjecture for perfect lattice paths

摘录 / EXCERPT: Let $\mathcal{I}_n(n)$ denote the number of the relevant perfect lattice paths in an $n\times n$ table. Povolotsky's recurrence conjecture. For positive integers n, the following...

conjecture | open | MathDB 页面 | 原文线索

V2-042473 | MATHDB-334950

Coefficient-matrix determinant conjecture for lattice-path tables

摘录 / EXCERPT: Let T be an $m\times n$ table with $2n\geqslant m$. Let $\mathcal{C}(T)$ be the coefficient matrix whose entries are the coefficients arising when the relevant quantities...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042474 | MATHDB-334957

The Rearrangement Conjecture for generalized factor order

摘录 / EXCERPT: Rearrangement Conjecture. If u and v are Wilf equivalent, then they are rearrangements of each other; that is, they have the same multiset of letters. The converse is false: fo...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042475 | MATHDB-334962

Group divisible design conjecture for the triples (X,W_2,W_k)

摘录 / EXCERPT: Group divisible design conjecture. For all $m\geq k\geq 3$, the triple (X,W_2,W_k) is a $\operatorname{GDD}(2^m-2,2,k)$ with

conjecture | solved | MathDB 页面 | 原文线索

V2-042476 | MATHDB-334964

Akbari-Alipour's unbounded gap conjecture for Latin arrays

摘录 / EXCERPT: Let $\ell(n)$ be the least positive integer such that $\ell(n)\geqslant n$ and every Latin array of order n with at least $\ell(n)$ distinct symbols contains a transversal. Akbar...

conjecture | open | MathDB 页面 | 原文线索

V2-042477 | MATHDB-334981

Random-list colorability threshold conjecture for graphs of growing maximum degree

摘录 / EXCERPT: Random-list colorability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042478 | MATHDB-334986

Binary-alphabet extremal conjecture for abelian-square factors

摘录 / EXCERPT: For a finite word, an abelian-square factor is a factor formed by concatenating two anagrams of one another. Let n be a positive integer, and let k be the number of distinct ab...

conjecture | open | MathDB 页面 | 原文线索

V2-042479 | MATHDB-334987

Extremal bound conjecture for inequivalent abelian squares

摘录 / EXCERPT: Two abelian squares are inequivalent when they have different Parikh vectors, where the Parikh vector records the number of occurrences of each letter. Let n be a positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-042480 | MATHDB-334992

Noy's characterization conjecture for subcritical minor-closed graph classes

摘录 / EXCERPT: Noy's conjecture. An addable, minor-closed class of labelled graphs is subcritical if and only if it has a planar forbidden minor.

conjecture | open | MathDB 页面 | 原文线索

V2-042481 | MATHDB-335012

The bar-list correspondence conjecture for strongly stable monomial ideals

摘录 / EXCERPT: Strongly stable-ideal bar-list conjecture. There is a biunivocal correspondence between $\mathcal{S}_{\pi}(p_1,\ldots,p_n)$ and $\mathcal{B}_{(p_1,\ldots,p_n)}$. This would provide...

conjecture | open | MathDB 页面 | 原文线索

V2-042482 | MATHDB-335051

Extremal contranominal-scale conjecture for formal contexts

摘录 / EXCERPT: A formal context consists of a set of objects, a set of attributes, and an incidence relation between them. A contranominal scale of size j is a context of the form ... Fix integer...

conjecture | open | MathDB 页面 | 原文线索

V2-042483 | MATHDB-335060

Lovász-Szegedy finite-dimensionality conjecture for finitely forcible graphons

摘录 / EXCERPT: Lovász-Szegedy finite-dimensionality conjecture. The space of typical vertices of every finitely forcible graphon has finite dimension.

conjecture | refuted | MathDB 页面 | 原文线索

V2-042484 | MATHDB-335064

Necessary-condition sufficiency conjecture for quantum lens-space equivalence

摘录 / EXCERPT: Let $r=2^t p_1^{\alpha_1}\cdots p_k^{\alpha_k}$, where $t\in\{0,1\}$ and the p_i are distinct odd primes. Let $\overline m,\overline m'\in(\mathbb Z_r)^n$ be parameter vectors, a...

conjecture | open | MathDB 页面 | 原文线索

V2-042485 | MATHDB-335065

Uniform-fiber conjecture for quantum lens-space equivalence classes

摘录 / EXCERPT: Let $S_{r,n}$ be the parameter set for the quantum lens spaces under consideration, and let $\sim$ be the equivalence relation on it. Uniform-fiber conjecture. All equivalence clas...

conjecture | open | MathDB 页面 | 原文线索

V2-042486 | MATHDB-335069

Classification conjecture for primitive morphism fixed points among binary generalized pseudostandard words

摘录 / EXCERPT: Classification conjecture. Let $\mathbf u$ be an aperiodic binary generalized pseudostandard word, not standard Sturmian, being a fixed point of a primitive morphism. Then

conjecture | open | MathDB 页面 | 原文线索

V2-042487 | MATHDB-335075

Counting conjecture for sum-free subsets

摘录 / EXCERPT: Let $\mathcal{L}$ denote the equation $x+y=z$. A subset of $[n]$ is $(\mathcal{L},2)$-free if it admits a 2-colouring with no monochromatic solution to $\mathcal{L}$. Counting conj...

conjecture | open | MathDB 页面 | 原文线索

V2-042488 | MATHDB-335088

Full-level extremality conjecture for posets of unequal heights

摘录 / EXCERPT: Let $l(P)$ denote the height of a poset P, namely the length of its longest chain. For posets P,Q with $l(P)>l(Q)$, let $c(Q,\mathcal F_n)$ be the number of copies of Q in a...

conjecture | open | MathDB 页面 | 原文线索

V2-042489 | MATHDB-335091

Simplex and cube extremality conjecture for the Monge-Ampère eigenvalue

摘录 / EXCERPT: Simplex and cube extremality conjecture. (i) Among all bounded open convex sets in $\mathbb R^n$ having a fixed positive volume, the n-dimensional regular simplex-the interior of...

conjecture | open | MathDB 页面 | 原文线索

V2-042490 | MATHDB-335096

Derangement excedance conjecture for the Chow groups of the generic Specht matroid

摘录 / EXCERPT: Let $M(2,1^{n-1})$ be the Specht matroid associated with the partition $(2,1^{n-1})$, and let $A^k(M(2,1^{n-1}))$ denote its degree-k Chow group. Derangement excedance conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042491 | MATHDB-335097

The convex-extension conjecture for COM tope graphs

摘录 / EXCERPT: Convex-extension conjecture. Every $G\in\mathcal{G}_{\rm COM}$ is a convex subgraph of a graph $G'\in\mathcal{G}_{\rm OM}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042492 | MATHDB-335120

Strong Dirac conjecture for planar point sets

摘录 / EXCERPT: Let $\mathcal{P}$ be a set of n non-collinear points in the plane, and let $\mathcal{L}(\mathcal{P})$ be the set of lines determined by pairs of points in $\mathcal{P}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-042493 | MATHDB-335139

Gomory-Johnson's facet conjecture for the infinite group problem

摘录 / EXCERPT: Let $f\in(0,1)$ be fixed, and consider the infinite group problem with variables $s_r\in\mathbb{Z}_+$ for $r\in[0,1)$, finite support, and the equality constraint ... where additions...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042494 | MATHDB-335151

A q-super Catalan Laurent-polynomial conjecture

摘录 / EXCERPT: Let m, n, s and t be positive integers. Let r be a non-negative integer with $r+s+t\equiv 1\pmod{2}$, and let j be an integer. Let $B_{m,k}(q)$ and $B_{n,k}(q)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-042495 | MATHDB-335162

Erdős's productset conjecture for sets of positive Banach density

摘录 / EXCERPT: Erdős's productset conjecture. If $\operatorname{BD}(A)>0$, then there are infinite sets $B,C\subseteq\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042496 | MATHDB-335185

Asymptotic conjecture for decomposing the complete graph product $E(K_4)\times E(K_n)$

摘录 / EXCERPT: Asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042497 | MATHDB-335211

Web-combinatorics conjecture for the cluster structure of Gr(3,n)

摘录 / EXCERPT: Let $\mathbb{C}[\operatorname{Gr}^{\sim}(3,n)]$ be the coordinate ring of the affine cone over the Grassmannian. A web invariant is called indecomposable, non-elliptic, or arboriza...

conjecture | open | MathDB 页面 | 原文线索

V2-042498 | MATHDB-335214

Asymptotic rarity of high chromatic index in Steiner triple systems

摘录 / EXCERPT: Asymptotic rarity conjecture. The proportion of Steiner triple systems of order v whose chromatic index is at least $m(v)+2$ tends to 0 as $v\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-042499 | MATHDB-335217

Csóka's weak caching-game value conjecture

摘录 / EXCERPT: Let $v_A(n,d,k)$ denote the value of Alpern's Caching Game with n doors, d treasures, and searcher power k. Csóka's conjecture. If k is an integer and $n\geq dk$, then ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-042500 | MATHDB-335218

Csóka's strong caching-game value conjecture

摘录 / EXCERPT: Let $v_A(n,d,k)$ denote the value of Alpern's Caching Game with n doors, d treasures, and searcher power k. Csóka's conjecture. If $k\geq 1+\frac{2}{d-1}$ and $n\geq dk$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-042501 | MATHDB-335221

Armstrong's characteristic-polynomial conjecture for the Tesler poset

摘录 / EXCERPT: Armstrong's conjecture. The characteristic polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042502 | MATHDB-335222

The lower-bound conjecture for generalized Tesler matrices

摘录 / EXCERPT: Generalized Tesler-matrix lower-bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042503 | MATHDB-335223

Symplectic-product growth conjecture for point sets

摘录 / EXCERPT: Symplectic-product growth conjecture. Suppose $|P|\leq p$. Either $\omega(P)$ is empty or, for every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-042504 | MATHDB-335236

The generalized polynomial Wronskian conjecture

摘录 / EXCERPT: Let $P_1,\dots,P_m$ be m polynomials of arbitrary degrees $0\leq d_1,\dots,d_m\leq m-1$ in $(\mathbb{F}[\alpha_1,\dots,\alpha_n])[\alpha]$. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-042505 | MATHDB-335255

Finite-commutative-ring product-of-difference-sets conjecture

摘录 / EXCERPT: Finite-ring product-of-difference-sets conjecture. There exist N and k, depending only on α, such that for every finite commutative ring R with $|R|\ge N$ and every se...

conjecture | open | MathDB 页面 | 原文线索

V2-042506 | MATHDB-335262

The ultrafilter forcing conjecture for topological Ramsey spaces

摘录 / EXCERPT: Topological Ramsey space forcing conjecture. Every ultrafilter which satisfies some partition relation and is forced by some $c3$-closed forcing is actually forced by some topologi...

conjecture | open | MathDB 页面 | 原文线索

V2-042507 | MATHDB-335263

The Solovay-model genericity conjecture for Ramsey filters

摘录 / EXCERPT: Solovay-model genericity conjecture. Suppose that there is a supercompact cardinal in V. If $\mathcal{U}$ is Ramsey, then $\mathcal{U}$ is generic for the forcing...

conjecture | open | MathDB 页面 | 原文线索

V2-042508 | MATHDB-335271

The odd-order edge lower-bound conjecture for k-in-out graphs

摘录 / EXCERPT: Odd-order edge lower-bound conjecture. Any k-in-out graph of order $2k+1$ for odd k has at least $4k-3$ edges.

conjecture | open | MathDB 页面 | 原文线索

V2-042509 | MATHDB-335272

The four-colour conjecture for t-perfect graphs

摘录 / EXCERPT: Let G be a t-perfect graph, meaning that its stable set polytope is determined by non-negativity, edge, and odd-cycle inequalities. Four-colour conjecture for t-perfect graph...

conjecture | open | MathDB 页面 | 原文线索

V2-042510 | MATHDB-335295

Metastable Local Disjunction conjecture

摘录 / EXCERPT: Let $D=D^{s_1}\sqcup\ldots\sqcup D^{s_r}$ be a disjoint union of r disks, let $f:D\to B^d$ be a proper PL map such that...

conjecture | open | MathDB 页面 | 原文线索

V2-042511 | MATHDB-335296

Metastable Mabillard-Wagner conjecture

摘录 / EXCERPT: Let $s,d,r\ge2$ be integers satisfying ... and let K be a finite s-dimensional simplicial complex. An almost r-embedding is a map $f:K\to\mathbb R^d$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-042512 | MATHDB-335297

Kiss-Rozgonyi-Sándor conjecture on higher-order representation functions

摘录 / EXCERPT: Kiss-Rozgonyi-Sándor conjecture. The equality $R_{h,C}^{(1)}(n)=R_{h,D}^{(1)}(n)$ from some point onward holds if and only if there exist positive integers n_0 and M, and finit...

conjecture | open | MathDB 页面 | 原文线索

V2-042513 | MATHDB-335298

Chen-Lev conjecture on Hilbert-cube pairs

摘录 / EXCERPT: Chen-Lev conjecture. If $R_C(n)=R_D(n)$ for every positive integer n, then there exist positive integers $d_{i_1}<d_{i_2}<\cdots$ belonging to D such that

conjecture | open | MathDB 页面 | 原文线索

V2-042514 | MATHDB-335299

Chen-Lev conjecture on periodic intersections of representation-equivalent sets

摘录 / EXCERPT: Chen-Lev conjecture. If $R_C(n)=R_D(n)$ for every positive integer n, then there exists an integer $l\ge1$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-042515 | MATHDB-335300

Chen-Lev strengthened conjecture on explicit Hilbert-cube pairs

摘录 / EXCERPT: Chen-Lev strengthened conjecture. If $R_C(n)=R_D(n)$ for every positive integer n, then there exists an integer $l\ge1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042516 | MATHDB-335307

Sturmfels's binomial conjecture for the maximum number of Gaussian-mixture modes

摘录 / EXCERPT: Sturmfels's conjecture. For all integers $d,k\geq 1$, the maximum number of modes is

conjecture | open | MathDB 页面 | 原文线索

V2-042517 | MATHDB-335312

The AP conjecture on Tverberg prescribable dimensions

摘录 / EXCERPT: Let $r\geq 2$ and $d\geq 1$ be integers. An r-tuple $(d_1,\dots,d_r)$ is Tverberg admissible if ... and $\sum_i d_i=(r-1)d$. AP conjecture. Every Tverberg admissible r-tuple is T...

conjecture | open | MathDB 页面 | 原文线索

V2-042518 | MATHDB-335313

The colorful Tverberg partition conjecture

摘录 / EXCERPT: Fix integers $r\geq 2$ and $d\geq 1$. A partition $X_1\sqcup\cdots\sqcup X_r$ of $\{1,2,\dots,(r-1)(d+1)+1\}$ is colorful if, for every $1\leq k\leq d+1$, the block ... contains exac...

conjecture | open | MathDB 页面 | 原文线索

V2-042519 | MATHDB-335328

The generating-function conjecture for capacity-two multiplex juggling cards

摘录 / EXCERPT: Generating-function conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042520 | MATHDB-335333

Morris's conjecture on regular generators of non-FC families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $U(\mathcal{S})$ denote the union of the sets in $\mathcal{S}$. A minimal generator of a union-closed family $\mathcal{F}$ is a family whose genera...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042521 | MATHDB-335361

Carpenter-Tonchev rank conjecture for designs from maximal arcs

摘录 / EXCERPT: Let $t\geq 2$, and let $\mathcal{D}$ be a $2-(2^{2t-1}-2^{t-1},2^{t-1},1)$ design with incidence matrix A. Carpenter-Tonchev's rank conjecture. The binary rank of A satisfies .....

conjecture | open | MathDB 页面 | 原文线索

V2-042522 | MATHDB-335365

Ultimate periodicity of Sprague-Grundy values along diagonals

摘录 / EXCERPT: Let a_1 and a_2 be nonnegative integers, and let $\mathcal{G}(a,b)$ denote the Sprague-Grundy value of the position (a,b) in the game under consideration. Diagonal ultimate-p...

conjecture | open | MathDB 页面 | 原文线索

V2-042523 | MATHDB-335386

Chen-Liang-Wang's total-positivity conjecture for the type-A Narayana triangle

摘录 / EXCERPT: Let $NA(n,k)=\frac{1}{k+1}\binom{n+1}{k}\binom{n}{k}$ be the Narayana numbers, and let ... be the Narayana triangle of type A. Chen-Liang-Wang's conjecture. The Narayana triangle...

conjecture | open | MathDB 页面 | 原文线索

V2-042524 | MATHDB-335387

Total positivity of Fuss-Narayana triangles and strict positivity of their squares

摘录 / EXCERPT: Fuss-Narayana conjecture. For any $m\geq 1$, the Fuss-Narayana triangles are totally positive and the Fuss-Narayana squares are strictly totally positive. The source introduces thi...

conjecture | open | MathDB 页面 | 原文线索

V2-042525 | MATHDB-335388

Strict total positivity conjecture for the Eulerian square

摘录 / EXCERPT: Eulerian-square conjecture. The Eulerian square $A^{\ulcorner}$ is strictly totally positive: all its minors are positive. The conjecture is motivated in the source by the strict t...

conjecture | open | MathDB 页面 | 原文线索

V2-042526 | MATHDB-335392

The maximal-orbit conjecture for $C_{1,n}$

摘录 / EXCERPT: Maximal-orbit conjecture for $C_{1,n}$. The orbit of t under the map $C_{1,n}(1,-)$ has length 2^j. The general orbit bound proved in the source shows that this is the maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-042527 | MATHDB-335393

The negative-alphabet maximal-orbit conjecture for $C_{-1,n}$

摘录 / EXCERPT: Let n be an even integer, possibly nonpositive, and let $j>0$. Set ... Let t be either the sequence $(-1)v^{j-1}$ or the sequence $v^{j-1}n$; both have length $2j-1$. The map...

conjecture | open | MathDB 页面 | 原文线索

V2-042528 | MATHDB-335400

The Binomial Approximation Conjecture for random graph degree sequences

摘录 / EXCERPT: Let $\mathcal{B}_p(n)$ be the model consisting of n independent random variables with distribution $\operatorname{Bin}(n-1,p)$, and let $\mathcal{D}(\mathcal{G}(n,p))$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-042529 | MATHDB-335416

The truncated theta function has no polynomial-length short generating function

摘录 / EXCERPT: Let r be a positive integer and define the truncated theta function by ... For a short generating function f, let $\operatorname{index}(f)$ be the maximum number of denominator f...

conjecture | open | MathDB 页面 | 原文线索

V2-042530 | MATHDB-335417

The arithmetic-progression property for short generating functions

摘录 / EXCERPT: Fix $c>0$ and $k\geq3$. A short GF g has the (c,k)-property if either ... or $\operatorname{supp}(g)$ contains a k-term arithmetic progression, denoted AP_k. L...

conjecture | open | MathDB 页面 | 原文线索

V2-042531 | MATHDB-335436

Two-part Hilton-Milner conjecture for non-trivially intersecting families

摘录 / EXCERPT: Let X_1 and X_2 be disjoint sets with sizes n_1 and n_2, and let ${X_1, X_2\choose k,\ell}$ denote the family of sets F satisfying $|F\cap X_1|=k$ and $|F\cap X_2|=\ell$....

conjecture | open | MathDB 页面 | 原文线索

V2-042532 | MATHDB-335442

Covert-Koh-Pham even-dimensional k-resultant modulus conjecture

摘录 / EXCERPT: Let $E_1,\ldots,E_k\subset\mathbb F_q^d$, with $k\geq 2$, and define ... Let γ_k be the smallest exponent such that $\prod_{j=1}^k|E_j|\geq Cq^{\gamma_k}$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-042533 | MATHDB-335483

The polynomial-exponent enumeration conjecture for 2-level polytopes

摘录 / EXCERPT: Let $\ell(d)$ be the number of combinatorially distinct 2-level polytopes in dimension d. Enumeration conjecture for 2-level polytopes. The number $\ell(d)$ of combinatorially...

conjecture | open | MathDB 页面 | 原文线索

V2-042534 | MATHDB-335484

The asymptotic suspension conjecture for 2-level polytopes

摘录 / EXCERPT: Let $\ell(d)$ and $s(d)$ respectively denote the numbers of combinatorially distinct 2-level polytopes and 2-level suspensions in dimension d. Suspension conjecture for 2-lev...

conjecture | open | MathDB 页面 | 原文线索

V2-042535 | MATHDB-335504

Reverse-center equidistribution conjecture for $\mathcal{G}^X$-parking functions

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geq 3$, and let $X\subseteq(1,n)$. For the hyperplane-arrangement family $\mathcal{G}^X$, consider its parking functions and the reverse center statist...

conjecture | open | MathDB 页面 | 原文线索

V2-042536 | MATHDB-335508

Support conjecture for key polynomials

摘录 / EXCERPT: Let α be a composition, let t_{ij} swap positions i and j, and let $m_{ij}(\alpha)=\alpha+e_i-e_j$. Define $\beta<_\kappa\alpha$ when β can be generated from...

conjecture | open | MathDB 页面 | 原文线索

V2-042537 | MATHDB-335527

Cherlin's conjecture on almost all triangle-free oriented graphs

摘录 / EXCERPT: A T_3-free oriented graph is an oriented graph containing no transitive tournament on three vertices. Cherlin's conjecture. Almost all T_3-free oriented graphs are tripartite....

conjecture | open | MathDB 页面 | 原文线索

V2-042538 | MATHDB-335553

The regularity conjecture for the k-abelian complexity of automatic sequences

摘录 / EXCERPT: Automatic-sequence regularity conjecture. The k-abelian complexity of any ℓ-automatic sequence is an ℓ-regular sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-042539 | MATHDB-335571

All plausible tau-parities are eventually actual

摘录 / EXCERPT: Actuality conjecture. For any fixed k and all sufficiently large N, all plausible τ-parities are actual for (k,N).

conjecture | open | MathDB 页面 | 原文线索

V2-042540 | MATHDB-335574

Bounded branch-width conjecture for excluded minors of frame matroids

摘录 / EXCERPT: A matroid M is a frame matroid if there is a matroid M' with a basis V such that ... and, for each $e\in E(M)$, the unique circuit in $V\cup\{e\}$ has size at most 3. An excl...

conjecture | open | MathDB 页面 | 原文线索

V2-042541 | MATHDB-335578

Conjecture on recurrence orders and coefficient polynomials for power sums

摘录 / EXCERPT: Recurrence and coefficient conjecture. The linear recurrence corresponding to k has order $\left\lfloor k/2\right\rfloor+3$, and each coefficient $c_j(q)$ is a linear polynomial...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042542 | MATHDB-335584

Noel-Scott-Sudakov-Balogh-Wagner centeredness conjecture for product posets

摘录 / EXCERPT: Let k be a positive integer, and let n_0 be an integer. Equip the product poset $\{0,1,\ldots,k\}^n$ with the coordinatewise order. A family is centered when its elements are c...

conjecture | open | MathDB 页面 | 原文线索

V2-042543 | MATHDB-335586

The asymptotic sharpness conjecture for Betti numbers of complexes with highly connected links

摘录 / EXCERPT: Let k and ℓ be integers with $0\leq \ell<k$, and let $B_{n,k,\ell}$ be the bound defined in the paper. For a simplicial complex X with ... write $\lambda_{\ell,k-\ell-2}(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042544 | MATHDB-335587

Anstee-Sali conjecture on forbidden configuration growth

摘录 / EXCERPT: Let F be a $k\times \ell$ matrix with $F\ne \begin{bmatrix}0\\ 1\end{bmatrix}$. Let I_k be the $k\times k$ identity matrix, I_k^c its (0,1)-complement, and T_k the...

conjecture | open | MathDB 页面 | 原文线索

V2-042545 | MATHDB-335588

Nonexistence conjecture for 5-rowed minimal cubic configurations

摘录 / EXCERPT: A minimal cubic configuration is a minimal configuration associated with the cubic forbidden-configuration problem; a configuration is minimal when deleting any row or column destr...

conjecture | open | MathDB 页面 | 原文线索

V2-042546 | MATHDB-335621

Limit-shape conjecture for recurrent configurations of well-behaved Bulgarian solitaires

摘录 / EXCERPT: Let a sequence of well-behaved $csigma_n$-solitaires have stable configurations with limit shape $cphi$. Limit-shape conjecture. If $cphi$ is a limit shape of the stable configurat...

conjecture | open | MathDB 页面 | 原文线索

V2-042547 | MATHDB-335622

Triangular limit-shape conjecture for sparse random proportion Bulgarian solitaire

摘录 / EXCERPT: Let p_n and q_n be the parameters of the p_n-random q_n-proportion Bulgarian solitaire, and consider the asymptotic regime in which ... as $n\to\infty$. Triangular limit-shap...

conjecture | open | MathDB 页面 | 原文线索

V2-042548 | MATHDB-335623

Piecewise-linear limit-shape conjecture for the critical random proportion Bulgarian solitaire regime

摘录 / EXCERPT: Let p_n and q_n be the parameters of the p_n-random q_n-proportion Bulgarian solitaire, and consider the critical asymptotic regime in which ... for some constant $C>0$. Piec...

conjecture | open | MathDB 页面 | 原文线索

V2-042549 | MATHDB-335655

The characterization of connected combinatorial metrics satisfying the MacWilliams extension property

摘录 / EXCERPT: Let $\boldsymbol{ obreak\text{$boldsymbol{ obreak ewcommand$}}}$

conjecture | open | MathDB 页面 | 原文线索

V2-042550 | MATHDB-335684

The conjecture on the uniform rigged configuration bijection

摘录 / EXCERPT: Let B be a tensor product of Kirillov--Reshetikhin crystals of affine type. Let $\operatorname{RC}(B)$ be the corresponding set of rigged configurations, let η interchange r...

conjecture | open | MathDB 页面 | 原文线索

V2-042551 | MATHDB-335701

Conjecture on the sharpness of the agreement-number bound for toroidal societies

摘录 / EXCERPT: Sharpness conjecture. The lower bound

conjecture | open | MathDB 页面 | 原文线索

V2-042552 | MATHDB-335720

Generalized Zimin divisibility conjecture for formulas with reversal

摘录 / EXCERPT: Generalized Zimin divisibility conjecture. The formula phi is unavoidable if and only if it divides $Z_{m,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042553 | MATHDB-335721

Deletion conjecture for unavoidable formulas with reversal

摘录 / EXCERPT: Deletion conjecture. If phi is unavoidable, then $delta_{X\cup X^R}(phi)$ is unavoidable.

conjecture | open | MathDB 页面 | 原文线索

V2-042554 | MATHDB-335732

Newman's conjecture on the maximal codensity of four-element sets

摘录 / EXCERPT: Newman's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042555 | MATHDB-335756

Kløve's generalized formula for permutation-code sphere polynomials

摘录 / EXCERPT: Kløve's generalized conjecture. For every positive integer d,

conjecture | open | MathDB 页面 | 原文线索

V2-042556 | MATHDB-335759

The serpent nest conjecture for hollow quadrangulations

摘录 / EXCERPT: Serpent nest conjecture. For any hollow quadrangulation $\mathrm{Q}_\circ$, there is a bijection between the serpent nests of $\mathrm{Q}_\circ$ and the maximal $\mathrm{Q}_\circ$-...

conjecture | open | MathDB 页面 | 原文线索

V2-042557 | MATHDB-335760

Optimal transport conjectures for generalized symmetric partitions

摘录 / EXCERPT: Let $k,n\in\mathbb N$, let $\sigma\in Perm(k+1)$, and let $\pi\in\mathscr P_k(n)$. Write $T_\sigma:\mathbb R^{k+1}\to\mathbb R^{k+1}$ for the linear map whose matrix in the standar...

conjecture | open | MathDB 页面 | 原文线索

V2-042558 | MATHDB-335764

The isosceles right triangle conjecture for fixed-area right triangles

摘录 / EXCERPT: Isosceles right triangle conjecture. Among all right triangles with area $A>0$, the one with largest torsional rigidity is the isosceles right triangle with area A.

conjecture | open | MathDB 页面 | 原文线索

V2-042559 | MATHDB-335769

Karzanov-Lomonosov conjecture on the size of cross-free families

摘录 / EXCERPT: Karzanov-Lomonosov conjecture. Let $k\geq 2$ and let n be positive integers. If $\mathcal{F}\subset 2^{[n]}$ is a k-cross-free family, then

conjecture | open | MathDB 页面 | 原文线索

V2-042560 | MATHDB-335796

The Lipschitz-surface characterization conjecture for d-controlling sequences

摘录 / EXCERPT: The Lipschitz-surface characterization conjecture. The sequence $(x_i)_{i\in I}$ is d-controlling if and only if there exist a Lipschitz map $g:\mathbb{R}^d\to\mathbb{R}^m$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-042561 | MATHDB-335852

Whitney-flip invariance conjecture for extended graph permanents

摘录 / EXCERPT: Whitney-flip invariance conjecture. If two graphs differ by a Whitney flip, then they have equal extended graph permanents.

conjecture | open | MathDB 页面 | 原文线索

V2-042562 | MATHDB-335862

Reimer's condition conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let ${\mathcal A}\subseteq 2^{[n]}$, and suppose ${\mathcal A}$ satisfies Condition 1: there is a filter ${\mathcal F}\subseteq 2^{[n]}$ and a bijection...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-042563 | MATHDB-335872

The weighted vertical projection realization conjecture for alternating sign matrices

摘录 / EXCERPT: Let n be a positive integer, and let $(c_1,c_2,\ldots,c_n)$ be a vector of positive integers such that ... Here, an alternating sign matrix (ASM) is an $n\times n$ matrix with entr...

conjecture | open | MathDB 页面 | 原文线索

V2-042564 | MATHDB-335878

Kepler's conjecture for asymptotic sphere-packing density

摘录 / EXCERPT: Let Γ be a container, let $\rho(1,k\Gamma)$ denote the optimal density of unit-sphere packings in the k-fold magnification $k\Gamma$, and define ... For a large class of con...

conjecture | open | MathDB 页面 | 原文线索

V2-042565 | MATHDB-335882

The OTE characterization of sticky matroids

摘录 / EXCERPT: Let M be a matroid. A modular cut is a set of flats closed upward and closed under intersections of modular pairs; M is OTE (only trivially extendable) if every modular cut oth...

conjecture | open | MathDB 页面 | 原文线索

V2-042566 | MATHDB-335891

Equality of two quadrant marked mesh pattern polynomials

摘录 / EXCERPT: Equality conjecture. For all $k\geq 1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-042567 | MATHDB-335893

Euclidean ball extremality conjecture for finite-facet volume approximation

摘录 / EXCERPT: Fix $n\in\mathbb{N}$ with $n\geq 2$ and $N\geq n+1$, and let K be a convex body in $\mathbb{R}^n$. For a polytope P with at most N facets, write $\Delta_v(P,K)$ for the symme...

conjecture | open | MathDB 页面 | 原文线索

V2-042568 | MATHDB-335900

Conjecture on extremal Betti numbers of Cartwright-Sturmfels ideals

摘录 / EXCERPT: Let S be the polynomial ring with its ${\mathbb Z}^n$-grading, let I be a Cartwright-Sturmfels (CS) ideal, and let J be its ${\mathbb Z}^n$-graded generic initial ideal. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042569 | MATHDB-335931

Distinct symplectic-form values conjecture for non-collinear planar point sets

摘录 / EXCERPT: Let $E\in\mathbb{R}^2$ be a non-collinear point set, and let $\omega[E]$ denote the set of values of the symplectic form ω on all ordered pairs of points of E. Equivalentl...

conjecture | open | MathDB 页面 | 原文线索

V2-042570 | MATHDB-335932

Balog-Roche-Newton-Zhelezov growth conjecture for difference sets

摘录 / EXCERPT: Let A be a set of scalars with at least three elements, and put $D=A-A$. For a positive integer m, write D^m for the m-fold product set of D. Balog-Roche-Newton-Zhelezov...

conjecture | open | MathDB 页面 | 原文线索

V2-042571 | MATHDB-335957

A small-doubling structural conjecture for finite sets of integers

摘录 / EXCERPT: Let A be a finite set of integers, and write $A-A=\{a-a':a,a'\in A\}$. Small-doubling structural conjecture. There exists a natural number K such that, whenever $|A|\ge K$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042572 | MATHDB-335958

Enumeration conjecture for r-wise intersecting families

摘录 / EXCERPT: Let $\binom{[n]}{k}$ be the family of k-subsets of $[n]$. A family is intersecting if it contains no two disjoint sets, and it is r-wise intersecting if it can be partitioned i...

conjecture | open | MathDB 页面 | 原文线索

V2-042573 | MATHDB-335976

Fraenkel's conjecture on complementary Beatty sequences

摘录 / EXCERPT: Let $d\geq 3$, let $\alpha_1<\alpha_2<\ldots<\alpha_d$, and let $\gamma_1,\ldots,\gamma_d$ be real numbers. Suppose that the vectors ... for $n\in\mathbb{N}$, split the positive inte...

conjecture | open | MathDB 页面 | 原文线索

V2-042574 | MATHDB-335991

Bárány-Kalai-Meshulam conjecture on connectivity of deleted joins of matroids

摘录 / EXCERPT: Bárány-Kalai-Meshulam conjecture. There exists an integer $n_k\geq 1$ depending only on k such that, whenever M has at least n_k disjoint bases, the complex M^{*k}_Δ i...

conjecture | open | MathDB 页面 | 原文线索

V2-042575 | MATHDB-336019

The minimal red progression length in Euclidean Ramsey theory

摘录 / EXCERPT: For integers $n,k\geq 1$, let $\mathbb{E}^n$ denote n-dimensional Euclidean space, and let ℓ_s denote a set of s equally spaced collinear points. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-042576 | MATHDB-336029

The Grid-associahedron as a facet reduction of a polytope

摘录 / EXCERPT: Let λ be a shape, let $\mathcal{A}_{\lambda}=\{H_s:s\in\operatorname{Seg}(\lambda)\}$ be the arrangement associated with its segments, and let $\mathcal{F}_{\lambda}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-042577 | MATHDB-336030

The F-triangle and M-triangle identity for Grid-associahedra

摘录 / EXCERPT: Let λ be a shape, let $\Psi=\Psi^f(\lambda)$ be its shard intersection order, and let r be its rank. Define ... Let $F(x,y)$ be the F-triangle associated with the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-042578 | MATHDB-336034

Asymptotic upper-bound conjecture for toroidal n-queens configurations

摘录 / EXCERPT: Asymptotic upper-bound conjecture. For every integer n not divisible by 2 or 3,

conjecture | open | MathDB 页面 | 原文线索

V2-042579 | MATHDB-336075

Defant's asymptotic conjecture for odd antipower prefix lengths

摘录 / EXCERPT: Defant's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042580 | MATHDB-336076

Conjecture on the growth of the Thue-Morse antipower threshold

摘录 / EXCERPT: Growth conjecture. There exists a sequence (a_i) converging to 0 such that, whenever n lies between $3\cdot 2^i$ and $2^{i+1}(1-a_i)$,

conjecture | open | MathDB 页面 | 原文线索

V2-042581 | MATHDB-336091

The asymptotic robustness limit conjecture for matchings

摘录 / EXCERPT: Let G range over graphs. For each $K\in\mathbb{N}$, let α_K denote the smallest K-asymptotic robustness ratio over all graphs, and let $\alpha^*(G)$ denote the randomize...

conjecture | open | MathDB 页面 | 原文线索

V2-042582 | MATHDB-336115

The coloop-extension conjecture for powerful sets

摘录 / EXCERPT: Coloop-extension conjecture. If T is a powerful set with at least one vector of weight 1, then T is a coloop extension of some powerful set S.

conjecture | open | MathDB 页面 | 原文线索

V2-042583 | MATHDB-336116

The coordinate-deletion conjecture for powerful sets

摘录 / EXCERPT: Coordinate-deletion conjecture. If S is a powerful set, then one can find a coordinate such that deleting this coordinate from all the elements of S yields the set...

conjecture | open | MathDB 页面 | 原文线索

V2-042584 | MATHDB-336153

The quadratic sumset covering conjecture for difference bases

摘录 / EXCERPT: Quadratic covering conjecture. There is an integer l such that whenever $A\subset\mathbb{Z}_q$ and $A-A=\mathbb{Z}_q$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042585 | MATHDB-336156

Facet-defining conjecture for clique inequalities in the two-level graph partitioning polytope

摘录 / EXCERPT: Let $G=(V,E)$ be the graph and let k,k' be the partition parameters, with r, r', and R as defined for the corresponding clique inequalities. For a clique C, write $y(C)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042586 | MATHDB-336161

The automaticity conjecture for generalised-polynomial sequences

摘录 / EXCERPT: Let $f\colon\mathbb{N}_0\to\Omega$ be a sequence, where Ω is a finite alphabet. Call f automatic if it is generated by a finite automaton, and call it a generalised-polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-042587 | MATHDB-336164

The finite-graph INT_f-minimizing partition conjecture

摘录 / EXCERPT: Finite-graph INT_f-minimizing partition conjecture. For every continuous and strictly concave function $f:[0,1]\rightarrow\mathbb R$ and every $\epsilon>0$, there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-042588 | MATHDB-336181

The logarithmic Lubell-mass conjecture for poset-free subsets of grids

摘录 / EXCERPT: Logarithmic Lubell-mass conjecture. There exists a constant $C(P)$ such that for every positive integer k and every integer n satisfying $n\geq\dim P$, if $S\subset [k]^n$ does...

conjecture | open | MathDB 页面 | 原文线索

V2-042589 | MATHDB-336196

List local antimagic probability conjecture

摘录 / EXCERPT: Let G be a graph with m edges, none of which is an isolated edge, and assign to each edge a list of m positive integers. A feasible assignment is an injective map...

conjecture | open | MathDB 页面 | 原文线索

V2-042590 | MATHDB-336229

Einstein's threshold conjecture for matrices of subspaces

摘录 / EXCERPT: Let $cell_1,cell_2,\ldots,cell_k$ be positive integers with $cell_i\ge t$ for every i. Let U be a linear space over a field $\mathbb F$, and let U_{ij} be subspaces arranged...

conjecture | open | MathDB 页面 | 原文线索

V2-042591 | MATHDB-336237

The compositional shuffle conjecture

摘录 / EXCERPT: The compositional shuffle conjecture. For any composition α,

conjecture | open | MathDB 页面 | 原文线索

V2-042592 | MATHDB-336252

Shuffle-compatibility of the up-down run and peak statistics

摘录 / EXCERPT: Let a permutation statistic assign an invariant to each permutation, and call it shuffle-compatible if, for any two disjoint permutations, the multiset of its values on their shuff...

conjecture | open | MathDB 页面 | 原文线索

V2-042593 | MATHDB-336274

Trace extension conjecture for forbidden posets

摘录 / EXCERPT: Let P be a poset, let $e(P)$ be the largest integer k such that, for every j and n, the family ... is P-free, and let $Tr_{n-k}(n,P)$ denote the maximum size of a family wh...

conjecture | open | MathDB 页面 | 原文线索

V2-042594 | MATHDB-336275

Butterfly trace conjecture

摘录 / EXCERPT: Let B be the butterfly poset on four elements a,b,c,d, with $a,b<_B c,d$. For a family $\mathcal F\subseteq 2^{[n]}$, let $Tr_{n-1}(n,B)$ denote the maximum size of a family wh...

conjecture | open | MathDB 页面 | 原文线索

V2-042595 | MATHDB-336276

The random-interval sequence k-width scaling conjecture

摘录 / EXCERPT: Let $k\geq 2$, and let R_n be a sequence of n random intervals. Write $E[k\operatorname{-wd}(R_n)]$ for the expected k-width of R_n. Random-interval sequence scaling conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-042596 | MATHDB-336277

The random-interval set k-width scaling conjecture

摘录 / EXCERPT: Let $k\geq 2$, and let W_n be a set of n random intervals. Write $E[k\operatorname{-wd}(W_n)]$ for the expected k-width of W_n. Random-interval set scaling conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-042597 | MATHDB-336301

Alon-Spencer's clique-packing conjecture for random graphs

摘录 / EXCERPT: Alon-Spencer's conjecture. The trivial upper bound gives the true order of magnitude of the expected packing number:

conjecture | open | MathDB 页面 | 原文线索

V2-042598 | MATHDB-336308

Closure-size conjecture for three-dimensional line-sparse sets

摘录 / EXCERPT: Closure-size conjecture. There are constants $c_1,c_2>0$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-042599 | MATHDB-336309

The communication-complexity equivalence conjecture for partition and exactness

摘录 / EXCERPT: For positive integers n and k, let $Part_{n,k}$ denote the problem of determining whether pairwise disjoint sets $S_1,\ldots,S_k$ form a partition of $[n]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042600 | MATHDB-336312

The crystal-bijection and cocharge conjecture for rigged configurations

摘录 / EXCERPT: Crystal-bijection and cocharge conjecture. The map Φ is a classical crystal isomorphism and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042601 | MATHDB-336313

The permutation-equivariance conjecture for the rigged-configuration bijection

摘录 / EXCERPT: Permutation-equivariance conjecture. The diagram relating $\operatorname{RC}(B)$ to $\operatorname{RC}(B')$ by the identity and B to B' by R, with vertical maps given by...

conjecture | open | MathDB 页面 | 原文线索

V2-042602 | MATHDB-336336

Asymptotic density conjecture for multidimensional boustrophedons

摘录 / EXCERPT: Asymptotic density conjecture. For every $1\leq\alpha\leq 6$, the limit $p_{\infty}^{(\alpha)}=\lim_{n\rightarrow\infty}p_n^{(\alpha)}$ exists and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042603 | MATHDB-336341

The missing-edge rigidity conjecture for simplicial 2-spheres

摘录 / EXCERPT: Let Γ be a 2-dimensional simplicial sphere or a simplicial manifold, and let $\{\{u_1,w_1\},\ldots,\{u_m,w_m\}\}$ be a collection of pairwise disjoint missing edges of...

conjecture | open | MathDB 页面 | 原文线索

V2-042604 | MATHDB-336365

Two-slope extremality conjecture for continuous classical DFFs

摘录 / EXCERPT: Two-slope extremality conjecture. If the function has no uncovered components, then it is extreme.

conjecture | open | MathDB 页面 | 原文线索

V2-042605 | MATHDB-336392

The homological dimension conjecture for complete graph minors

摘录 / EXCERPT: Let G be a graph, let $\therefore$ denote the graph-minor relation, and let $b3(G)$ be its homological dimension. For a positive integer d, consider the complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-042606 | MATHDB-336393

The surface (p,q) piercing conjecture

摘录 / EXCERPT: Let p and q be integers with $p\geq q\geq 3$, let S be a surface, and let $\mathcal{F}$ be a finite family of open connected subsets of S. The (p,q) property means that a...

conjecture | open | MathDB 页面 | 原文线索

V2-042607 | MATHDB-336396

Odlyzko-Stanley dichotomy conjecture for Stanley sequence growth

摘录 / EXCERPT: Let $S(A)=(a_n)$ be a Stanley sequence, meaning the greedy sequence generated from a finite 3-free set A of nonnegative integers. Let α denote the constant appearing in th...

conjecture | open | MathDB 页面 | 原文线索

V2-042608 | MATHDB-336397

Near-modular set conjecture for realizing all even independent Stanley characters

摘录 / EXCERPT: A set A is near-modular modulo $3^{\ell+1}$ if it has the near-modularity property used in the paper. Near-modular set conjecture. For every $\ell\in\mathbb{Z}^{+}$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-042609 | MATHDB-336402

The interlacing conjecture for Z-polynomials of nice matroid families

摘录 / EXCERPT: Let $\{M_d\mid d\geq 0\}$ be a nice family of matroids, and let $Z_d(t)$ denote the Z-polynomial associated with M_d. For polynomials with real roots, say that $f(t)$ interlace...

conjecture | open | MathDB 页面 | 原文线索

V2-042610 | MATHDB-336403

The strong equivariant log-concavity conjecture for equivariant matroid Z-polynomials

摘录 / EXCERPT: Let M be a matroid equipped with an action of a group Γ, and let $Z_M^\Gamma(t)=\bigoplus_i V_i t^i$ be its equivariant Z-polynomial, viewed as a graded virtual represen...

conjecture | open | MathDB 页面 | 原文线索

V2-042611 | MATHDB-336415

Motion-family k-centerpoint conjecture

摘录 / EXCERPT: Let P be a set of n points in $\mathbb{R}^d$, and let $0\leq k\leq d$. For a point q, let F_q be a $(d-k-1)$-flat through q, and let F_o be such a flat disjoint from...

conjecture | open | MathDB 页面 | 原文线索

V2-042612 | MATHDB-336436

The three-strand braid group conjecture for the nodal cubic

摘录 / EXCERPT: Let Z be the nodal cubic, and let Z' be the space from Example, namely the union of the unit sphere in $\mathbb{R}^3$ and a line segment connecting its poles. Write P_3 and...

conjecture | open | MathDB 页面 | 原文线索

V2-042613 | MATHDB-336443

Hibi and Ohsugi's reflexive IDP polytope unimodality conjecture

摘录 / EXCERPT: Let $\mathcal P$ be a lattice polytope. It is reflexive if its polar dual is also a lattice polytope, and it satisfies the integer decomposition property if every lattice point in...

conjecture | open | MathDB 页面 | 原文线索

V2-042614 | MATHDB-336450

Binomial-transform conjecture for classical and enhanced k-crossing-avoiding partitions

摘录 / EXCERPT: For $k\geq2$, let $C_k(n)$ be the number of partitions of $[n]$ avoiding classical k-crossings, and let $E_k(n)$ be the number avoiding enhanced k-crossings. Binomial-transform...

conjecture | open | MathDB 页面 | 原文线索

V2-042615 | MATHDB-336494

Minimum clues for $5 \times n$, $6 \times n$, and $7 \times n$ Numbrix puzzles

摘录 / EXCERPT: For an $m \times n$ Numbrix puzzle, call a set of clues defining exactly one puzzle a defining clue set, and let the minimum number of clues be the smallest size of such a set. Min...

conjecture | open | MathDB 页面 | 原文线索

V2-042616 | MATHDB-336509

The length-restricted antipalindrome sum conjecture

摘录 / EXCERPT: The length-restricted antipalindrome sum conjecture. Every even integer of length n, for n odd and $n\geq 9$, is the sum of at most 10 antipalindromes of length $n-3$.

conjecture | open | MathDB 页面 | 原文线索

V2-042617 | MATHDB-336510

The four-antipalindrome conjecture

摘录 / EXCERPT: Four-antipalindrome conjecture. Every even natural number is the sum of at most 4 antipalindromes.

conjecture | open | MathDB 页面 | 原文线索

V2-042618 | MATHDB-336511

The exceptional three-antipalindrome conjecture

摘录 / EXCERPT: Exceptional three-antipalindrome conjecture. Every even natural number outside E is the sum of at most 3 antipalindromes.

conjecture | open | MathDB 页面 | 原文线索

V2-042619 | MATHDB-336558

Little-o upper-bound conjecture for equal-weight separating hash families

摘录 / EXCERPT: Little-o upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042620 | MATHDB-336565

The 3-design conjecture for the trace-cubic design

摘录 / EXCERPT: The trace-cubic 3-design conjecture. If $m\not\equiv 0\pmod{4}$, then the pair $({\mathcal{P}},{\mathcal{B}}_2)$ is a 3-design.

conjecture | open | MathDB 页面 | 原文线索

V2-042621 | MATHDB-336566

The odd-dimensional weight-enumerator conjecture for the design code

摘录 / EXCERPT: Let $m\geq 3$ be odd. Let $\mathbb{D}=({\mathcal{P}},{\mathcal{B}}_2)$ be the 2-design described in the paper, and let ${\mathcal{C}}_2(\mathbb{D})$ denote its binary code. The o...

conjecture | open | MathDB 页面 | 原文线索

V2-042622 | MATHDB-336567

The even-dimensional weight-distribution conjecture for the design codes

摘录 / EXCERPT: Let $m\geq 4$ be even. Let $\mathbb{D}=({\mathcal{P}},{\mathcal{B}}_2)$ be the 2-design in Theorem 5, or let $\mathbb{D}=({\mathcal{P}},{\mathcal{B}})$ be the 2-design in Theor...

conjecture | open | MathDB 页面 | 原文线索

V2-042623 | MATHDB-336574

Characterization conjecture for morphic authentic PWWF substitutions

摘录 / EXCERPT: Morphic-mode characterization conjecture. The substitution σ is morphic if and only if its generated mode, up to permutation of the letters, is an instance of a morphic mode...

conjecture | open | MathDB 页面 | 原文线索

V2-042624 | MATHDB-336578

Cherlin's metric homogeneity classification conjecture

摘录 / EXCERPT: Cherlin's metric homogeneity classification conjecture. The countable metrically homogeneous graphs are the following: those of diameter $\delta\leq 2$, namely the homogeneous grap...

conjecture | open | MathDB 页面 | 原文线索

V2-042625 | MATHDB-336597

Conjectured convolution inverse for the Tutte q-polynomial transform

摘录 / EXCERPT: Conjectured convolution-inverse formula. For the relevant indices a,b,c,d,

conjecture | open | MathDB 页面 | 原文线索

V2-042626 | MATHDB-336611

The perfect Skolem set conjecture

摘录 / EXCERPT: Let $A = \{a_1,a_2,\dots,a_n\}$ be a set of positive integers with $a_1 < a_2 < \dots < a_n$. A Skolem set is a set for which $\{1,\dots,2n\}$ can be partitioned into pairs whose d...

conjecture | open | MathDB 页面 | 原文线索

V2-042627 | MATHDB-336638

Sharp-threshold conjecture for reconstructibility of random jigsaws

摘录 / EXCERPT: Let $J(n,q)$ be the random (n,q)-jigsaw, and let reconstructibility mean that the jigsaw is uniquely determined by its deck. The critical number of colours is conjectured to be a...

conjecture | open | MathDB 页面 | 原文线索

V2-042628 | MATHDB-336642

Sharpness of the antichain Hausdorff measure bound

摘录 / EXCERPT: Sharpness conjecture. For every $n\geq 3$, there exists an n-cube antichain S such that

conjecture | open | MathDB 页面 | 原文线索

V2-042629 | MATHDB-336649

Terao-type conjecture for the invariant ν of line arrangements

摘录 / EXCERPT: Terao-type conjecture. The invariant $\nu(C)$ is combinatorially determined.

conjecture | open | MathDB 页面 | 原文线索

V2-042630 | MATHDB-336708

The Tu-Deng conjecture on binary Hamming weights

摘录 / EXCERPT: Tu-Deng conjecture. For every such k and t, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042631 | MATHDB-336773

Purity of mutation-closed domains in pure oriented matroids

摘录 / EXCERPT: Purity conjecture for mutation-closed domains. If ${\mathcal{M}}$ is a pure oriented matroid, then every mutation-closed domain ${{\mathcal{D}}}$ for ${\mathcal{M}}$ is an...

conjecture | open | MathDB 页面 | 原文线索

V2-042632 | MATHDB-336774

Six-element-minor characterization of pure oriented matroids

摘录 / EXCERPT: Six-element-minor conjecture. The oriented matroid ${\mathcal{M}}$ is pure if and only if all of its six-element minors are pure. Equivalently, ${\mathcal{M}}$ is pure if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-042633 | MATHDB-336775

Restriction of maximal separated collections to mutation-closed domains

摘录 / EXCERPT: Restriction conjecture. If $\mathcal{S}$ is a maximal-by-size ${\mathcal{M}}$-separated collection inside 2^E and ${{\mathcal{D}}}$ is a mutation-closed domain, then...

conjecture | open | MathDB 页面 | 原文线索

V2-042634 | MATHDB-336786

Rolnick's growth-rate conjecture for Stanley sequences

摘录 / EXCERPT: Let $S(A)=(a_n)$ be a Stanley sequence, meaning the sequence generated greedily from a finite 3-free set $A\subset\mathbb N_0$. Rolnick's growth-rate conjecture. For all n larg...

conjecture | open | MathDB 页面 | 原文线索

V2-042635 | MATHDB-336787

Rolnick's character-range conjecture for independent Stanley sequences

摘录 / EXCERPT: Let $S(A)=(a_n)$ be an independent Stanley sequence: there are constants $\lambda=\lambda(A)$ and $\kappa=\kappa(A)$ such that for all $k\ge\kappa$ and $0\le i<2^k$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042636 | MATHDB-336789

Eventual partition-count inequality for bounded part sets

摘录 / EXCERPT: Let L and s be integers. Define $\mathcal{C}_{L,s,1}$ to be the set of partitions whose smallest part is s, whose parts are all at most $L+s$, and in which $L+s-1$ does not a...

conjecture | open | MathDB 页面 | 原文线索

V2-042637 | MATHDB-336790

Eventual partition-count inequality with a fixed forbidden part

摘录 / EXCERPT: Let $L\geq s+1$ and let $\mathcal{C}^*_{L,s,1}$ be the set of partitions whose smallest part is s, whose parts are all at most $L+s$, and in which L does not appear. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-042638 | MATHDB-336791

The extremal bound for almost-equidistant diameter sets

摘录 / EXCERPT: Let an almost-equidistant diameter set be a set of points in $\mathbb R^d$ of diameter 1 in which every three points contain a pair at distance 1. The extremal bound conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-042639 | MATHDB-336792

Schur's conjecture on intersecting unit simplices

摘录 / EXCERPT: Let S_1 and S_2 be two unit simplices in $\mathbb R^d$ forming a set of diameter 1, with k and m vertices respectively, and assume $k\geq m$. Schur's conjecture. The simp...

conjecture | solved | MathDB 页面 | 原文线索

V2-042640 | MATHDB-336793

Strong crossing-to-touching conjecture for Jordan arcs

摘录 / EXCERPT: Strong crossing-to-touching conjecture. The ratio of the number of intersection points to the number of touching points should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-042641 | MATHDB-336826

Sign-balance conjecture for nonzero symmetric-group character values

摘录 / EXCERPT: Let S_n be the symmetric group, and let $chi(\mu)$ denote the value of an irreducible character chi on the conjugacy class corresponding to a partition mu of n. Choose an e...

conjecture | open | MathDB 页面 | 原文线索

V2-042642 | MATHDB-336855

FindStat's double-deficiency refinement for Dyck paths

摘录 / EXCERPT: Double-deficiency refinement. For a Dyck path D, an index k is in $\mathcal F$ but not in $\mathcal N\cap\mathcal D$ if and only if it is a double deficiency in the 321-avoid...

conjecture | open | MathDB 页面 | 原文线索

V2-042643 | MATHDB-336866

Infinite-existence conjecture for non-rigid tight Euclidean designs

摘录 / EXCERPT: Let e be a positive integer, let ε_S denote the parameter used in the source, and let $[x]$ denote the integer-part notation in the source. Consider a tight Euclidean...

conjecture | open | MathDB 页面 | 原文线索

V2-042644 | MATHDB-336898

Round matroid hyperplane bound conjecture

摘录 / EXCERPT: Round matroid hyperplane bound conjecture. If M is round, has sufficiently large rank, and has no $U_{2,\ell+2}$-minor, then M has at most

conjecture | open | MathDB 页面 | 原文线索

V2-042645 | MATHDB-336900

The zero defect conjecture for aperiodic primitive morphism fixed points

摘录 / EXCERPT: Zero defect conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042646 | MATHDB-336901

Class P' conjecture for primitive morphism fixed points

摘录 / EXCERPT: Class P' conjecture. There exists a morphism of class P' such that its fixed point has the same language as ${\mathbf u}$.

conjecture | open | MathDB 页面 | 原文线索

V2-042647 | MATHDB-336904

Symmetry criterion for facet-defining building block classes

摘录 / EXCERPT: Let $\mathcal T^{\bm m}$ be a valid building block class for the constructed inequalities. Symmetry criterion. The class $\mathcal T^{\bm m}$ is facet-defining if and only if it is...

conjecture | open | MathDB 页面 | 原文线索

V2-042648 | MATHDB-336909

Molloy-Reed conjecture on the chromatic neighborhood theorem

摘录 / EXCERPT: Molloy-Reed conjecture. The theorem holds for every Δ; equivalently, one can take $\Delta_1=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-042649 | MATHDB-336910

The characterization of diameter-Ramsey simplices by their circumcentres

摘录 / EXCERPT: Diameter-Ramsey simplex conjecture. A simplex is diameter-Ramsey if and only if its circumcentre is contained in its convex hull.

conjecture | open | MathDB 页面 | 原文线索

V2-042650 | MATHDB-336929

The EKR-module conjecture for 2-transitive groups

摘录 / EXCERPT: Let G be a finite 2-transitive permutation group. An intersecting set of permutations is a subset of G in which any two elements agree on at least one point, and the EKR-modu...

conjecture | open | MathDB 页面 | 原文线索

V2-042651 | MATHDB-336936

The maximal-energy-concentration obstruction to Poissonian pair correlations

摘录 / EXCERPT: Let $(a(n))_{n\geq1}$ be the sequence of integers used to form $A_N=\{a(1),\ldots,a(N)\}$, and let $E(A_N)$ denote its additive energy. For a real number α, consider the seq...

conjecture | open | MathDB 页面 | 原文线索

V2-042652 | MATHDB-336937

The quasi-arithmetic obstruction to Poissonian pair correlations

摘录 / EXCERPT: Let $(a(n))_{n\geq1}$ be a strictly increasing sequence of positive integers. Call it quasi-arithmetic if it has the finite-dimensional arithmetic-progression structure defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-042653 | MATHDB-336967

Imaginary-part bound for roots of the box polynomial

摘录 / EXCERPT: Let $B_{m,n}(x)$ be the box polynomial, and consider its complex roots. The notation $O(m\cdot\sqrt{n})$ denotes an asymptotic upper bound in the parameters m and n. Imaginary-...

conjecture | open | MathDB 页面 | 原文线索

V2-042654 | MATHDB-336969

The equal-parameter Armstrong conjecture for simultaneous bicores

摘录 / EXCERPT: Let s be a positive integer and let a,b satisfy $0\leq a,b<s$. For parameters r and c, write $\mathcal{C}_{(r\,|\,0,c)}$ for the corresponding set of bipartitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042655 | MATHDB-336975

The algebraic and symmetric shifting Betti-number conjecture

摘录 / EXCERPT: Algebraic and symmetric shifting conjecture. For all i,j,

conjecture | open | MathDB 页面 | 原文线索

V2-042656 | MATHDB-336977

Extremal-ray generation conjecture for the algebra of conformal blocks

摘录 / EXCERPT: Let $\mathrm{M}_{\mathbf{p}}(r,0)$ be the moduli space of parabolic bundles, let $\mathrm{Eff}(\mathrm{M}_{\mathbf{p}}(r,0))$ denote its effective cone, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042657 | MATHDB-337017

Affine semigroup conjecture for the f-vector set of 4-polytopes under connected sum addition

摘录 / EXCERPT: Let $\mathcal{F}(\mathcal{P}^4)$ denote the set of f-vectors of all 4-polytopes, and let $\mathbin{{\scriptstyle\boxplus}”}$ be the modified addition realized by connected sum...

conjecture | open | MathDB 页面 | 原文线索

V2-042658 | MATHDB-337019

The non-acyclic graph conjecture on average instability

摘录 / EXCERPT: Let D be an interaction graph and let $\mathrm{F}[D,q]$ denote the finite dynamical systems over an alphabet of size q with interaction graph D. For $j\in\{0,1,2\}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042659 | MATHDB-337023

Combinatorial reciprocity conjecture for the determinants $D_{s,t}(n)$

摘录 / EXCERPT: Let $D_{s,t}(n)$ denote the determinant associated with the corresponding family of cyclically symmetric rhombus-tiling regions, with the parity of $s-t$ distinguishing ordinary ti...

conjecture | open | MathDB 页面 | 原文线索

V2-042660 | MATHDB-337030

Enumeration conjecture for maximal chains without plus-full-sets in Tamari lattices

摘录 / EXCERPT: Enumeration conjecture. For all $i\geq-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-042661 | MATHDB-337040

Finiteness and strict growth conjecture for compact packings with n radii

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let Π_n denote the set of compact packings of the plane with n distinct circle radii, represented by radius ratios $(r_{n-1},r_{n-2},\ldots,r_1)$....

conjecture | open | MathDB 页面 | 原文线索

V2-042662 | MATHDB-337045

Leth's conjecture on dense almost progressions

摘录 / EXCERPT: Fix $t\in\mathbb{N}$ and $c>0$. A set $A\subseteq\mathbb{N}$ nearly contains a (t,d,w)-progression on $[1,n]$ when it contains the almost-progression configuration defined earlie...

conjecture | open | MathDB 页面 | 原文线索

V2-042663 | MATHDB-337052

Serra-Zémor's $3k-4$ conjecture for subsets of $\mathbb{Z}_p$

摘录 / EXCERPT: Serra-Zémor's conjecture. Then A is included in an arithmetic progression of length $|A| + r$.

conjecture | open | MathDB 页面 | 原文线索

V2-042664 | MATHDB-337053

The r-critical pair conjecture in $\mathbb{Z}_p$

摘录 / EXCERPT: The r-critical pair conjecture. There exist intervals $I,J,K\subset \mathbb{Z}_p$ and $n\in\mathbb{Z}_p\setminus\{0\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042665 | MATHDB-337054

The symmetric three-set r-critical pair conjecture in $\mathbb{Z}_p$

摘录 / EXCERPT: Symmetric three-set r-critical pair conjecture. There exist intervals $I_1,I_2,I_3\subset \mathbb{Z}_p$ and $n\in\mathbb{Z}_p\setminus\{0\}$ such that $n\cdot A_j\subset I_j$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042666 | MATHDB-337066

Genericity conjecture for effective minimal-critical-point assumptions

摘录 / EXCERPT: Let $F({\mathbf z})\in\mathbb{Z}(z_1,\ldots,z_n)$ be a rational function, and let (A5) denote that the Jacobian matrix of the relevant system of equations (GenSys1)-(GenSys5) has fu...

conjecture | open | MathDB 页面 | 原文线索

V2-042667 | MATHDB-337068

The gap conjecture for gapful bigraphs

摘录 / EXCERPT: Gap conjecture. A bigraph is gapful if and only if it contains a cyclic bigraph.

conjecture | open | MathDB 页面 | 原文线索

V2-042668 | MATHDB-337069

Babson-Kozlov connectivity conjecture for Hom complexes

摘录 / EXCERPT: Let G be a graph with maximal degree d, and let K_n be the complete graph on n vertices. Babson-Kozlov's conjecture. The complex $\operatorname{Hom}(G,K_n)$ is at least...

conjecture | open | MathDB 页面 | 原文线索

V2-042669 | MATHDB-337108

Weak Gotsman-Linial conjecture on polynomial threshold function influence

摘录 / EXCERPT: Let $f:\{-1,1\}^n\to\{-1,1\}$ be a degree-d polynomial threshold function, and let $\operatorname{\mathbf{I}}[f]$ denote its total influence. Weak Gotsman-Linial conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-042670 | MATHDB-337121

Brown-Graham-Landman 2-large conjecture

摘录 / EXCERPT: Let $D\subseteq\mathbb{Z}^+$, and let $r\in\mathbb{Z}^+$. An arithmetic progression $a,a+d,a+2d,\ldots,a+(k-1)d$ with $d\in D$ is a D-arithmetic progression (or D-ap). The set...

conjecture | open | MathDB 页面 | 原文线索

V2-042671 | MATHDB-337136

The exact Davenport constant conjecture for powers of cyclic groups

摘录 / EXCERPT: Let C_n denote the cyclic group of order n, and let C_n^r be the direct sum of r copies of C_n. For a finite Abelian group G, let $\mathsf{D}(G)$ be the smallest intege...

conjecture | open | MathDB 页面 | 原文线索

V2-042672 | MATHDB-337196

The PFR conjecture over \d8352\d8357 and \d8357\d8357: GAP formulation

摘录 / EXCERPT: PFR conjecture, GAP formulation. There should exist such a progression P, with $a_i \in G$, rank $d = O(\log 2K)$ and size $O(K^{O(1)})|A|$, together with a set $X \subseteq G$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-042673 | MATHDB-337197

The PFR conjecture over \d8352\d8357 and \d8357\d8357: ellipsoid progression formulation

摘录 / EXCERPT: Let $G = \mathbb{R}^m$ or $\mathbb{Z}^m$, and let $A \subseteq G$ be finite with $|A+A| \leq K|A|$. An ellipsoid progression is a set ... where $a_0,\dots,a_d \in G$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042674 | MATHDB-337205

The combinatorial rigidity conjecture for unicritical and exponential families

摘录 / EXCERPT: Let ${\cal F}$ be the exponential family or a family of unicritical polynomials. For $f\in{\cal F}$, its orbit portrait is the collection of all tuples of rays landing at parabolic...

conjecture | open | MathDB 页面 | 原文线索

V2-042675 | MATHDB-337206

The quasiconformal formulation of combinatorial rigidity

摘录 / EXCERPT: Let ${\cal F}$ be the exponential family or a family of unicritical polynomials. Two maps in ${\cal F}$ are combinatorially equivalent when they have the same combinatorial data, s...

conjecture | open | MathDB 页面 | 原文线索

V2-042676 | MATHDB-337217

Solymosi's general-position conjecture for rich lines in grids

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, let $0<\alpha\leq 1$, and let a line ℓ in $\mathbb{F}^2$ be α-rich in a Cartesian product $Y\times Y$ if ... A set of lines is in general pos...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042677 | MATHDB-337219

The extremal matching conjecture for almost alternating threshold graphs

摘录 / EXCERPT: Let $k\geq 0$. Let A be an almost alternating threshold graph, and let G be a threshold graph with the same number of vertices and edges as A. Write $m_k(H)$ for the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-042678 | MATHDB-337258

The asymptotic enumeration conjecture for Ingleton matroids

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let an Ingleton matroid be a matroid whose rank function satisfies Ingleton's inequality for every quadruple of subsets of its ground set. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-042679 | MATHDB-337260

Conjecture on the growth of AP_k-covering sequences

摘录 / EXCERPT: Growth conjecture for AP_k-covering sequences. (i) For every integer $k\geq 3$, there exists an AP_k-covering sequence A such that

conjecture | open | MathDB 页面 | 原文线索

V2-042680 | MATHDB-337288

The pure-EKR conjecture for generalized cluster complexes

摘录 / EXCERPT: Generalized cluster complex conjecture. Every generalized cluster complex $\mathcal C(m,\Phi)$ is pure-EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-042681 | MATHDB-337290

The x not 1 modulo k-1 conjecture for prographs and set-valued tableaux

摘录 / EXCERPT: Assume $x\equiv a\pmod{k-1}$, where $2\leq a\leq k-1$. Let ... and let $\operatorname{PC}_x^k(n,m)$ denote the relevant prographs and $\operatorname{\mathbb{S}}(\lambda/\mu,\rho)$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-042682 | MATHDB-337298

Kuzmin-Warmuth extension conjecture for maximum classes

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\mathcal{F}\subseteq 2^{[n]}$. The Vapnik-Chervonenkis dimension $\operatorname{dim}_{VC}(\mathcal{F})$ is the size of the largest set shattered b...

conjecture | solved | MathDB 页面 | 原文线索

V2-042683 | MATHDB-337299

Equivalent Sperner-family formulation of the s-extremal extension conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathcal{S}\subseteq 2^{[n]}$ be a Sperner family, meaning that no member of $\mathcal{S}$ contains another, and let $h:\mathcal{S}\to 2^{[n]}$ satis...

conjecture | open | MathDB 页面 | 原文线索

V2-042684 | MATHDB-337309

Grigoriev-Koshevoy's exponential lower-bound conjecture for skew-Schur polynomials

摘录 / EXCERPT: Grigoriev-Koshevoy's conjecture. The tropical semiring complexity of $\operatorname{Trop}(s_{\lambda/\mu})$ is at least exponential.

conjecture | open | MathDB 页面 | 原文线索

V2-042685 | MATHDB-337314

The maximum stable-state count conjecture for neighborhood grids

摘录 / EXCERPT: Let P be a point configuration on n^2 points, and let $f^{(n,\ldots,n)}$ denote the number of standard Young tableaux of the square partition $(n,\ldots,n)$, equivalently the n...

conjecture | open | MathDB 页面 | 原文线索

V2-042686 | MATHDB-337323

Characterization of equality in the four-quotient lower bound

摘录 / EXCERPT: Let S be a numerical semigroup of multiplicity m, with quotient $q=4$. Let P, L, S_1, X_1, X_2, X_3, X_4, D, and ρ denote the objects defined in the paper,...

conjecture | open | MathDB 页面 | 原文线索

V2-042687 | MATHDB-337367

The asymptotic forbidden fork-pair conjecture for induced subposets

摘录 / EXCERPT: Forbidden fork-pair conjecture. For all $k \ge 2$ and $r \ge 2$, there is an $n_0=n_0(k,r)$ such that if $n\ge n_0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042688 | MATHDB-337385

Uniform asymptotic bound for skew standard Young tableaux

摘录 / EXCERPT: Let $\lambda \vdash n$ and $\mu \vdash k$ be balanced partitions. The quantity $A_{\lambda/\mu}$ measures the deviation associated with the skew shape λ/μ. Uniform bound...

conjecture | open | MathDB 页面 | 原文线索

V2-042689 | MATHDB-337410

The incremental conjecture for set multipartite Ramsey numbers

摘录 / EXCERPT: Let $s\geq 2$ and $n\geq 2$ be integers. For positive integers s,n, let $M_s(n)$ denote the set multipartite Ramsey number for C_4 versus the star $K_{1,n}$, namely the least...

conjecture | open | MathDB 页面 | 原文线索

V2-042690 | MATHDB-337425

Conjecture that every gaussoid is compatible with all quadrics in J_n

摘录 / EXCERPT: All-quadrics compatibility conjecture. Every gaussoid is compatible with all quadrics in J_n, not just trinomials.

conjecture | open | MathDB 页面 | 原文线索

V2-042691 | MATHDB-337453

Uniqueness conjecture for Laman graphs maximizing the number of realizations

摘录 / EXCERPT: Let $M_2(n)$ and $M_3(n)$ denote the maximum numbers of realizations in dimensions 2 and 3, respectively, among Laman graphs with n vertices. For each $n\geqslant2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042692 | MATHDB-337456

Miller-Reiner Smith normal form conjecture for differential posets

摘录 / EXCERPT: Let P be a differential poset, let p_n denote the size of its rank-n set, and let $U_{n-1}D_n$ be the composite of the down operator D_n from rank n to rank $n-1$ with th...

conjecture | open | MathDB 页面 | 原文线索

V2-042693 | MATHDB-337470

The Griggs-Lu conjecture on asymptotic poset-free families

摘录 / EXCERPT: Let P be a finite poset. Write ... and ... Define $\pi(P)=\lim_{n\to\infty}La(n,P)/\binom{n}{\lfloor n/2\rfloor}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042694 | MATHDB-337483

Cyclic calibrated representation conjecture for cluster variables

摘录 / EXCERPT: Cyclic calibrated representation conjecture. If S is finite, then there is a non-initial cluster variable z such that

conjecture | open | MathDB 页面 | 原文线索

V2-042695 | MATHDB-337487

P1-win conjecture for infinitely many disjoint complete boards

摘录 / EXCERPT: Let G be the target graph from the surrounding discussion, and let $\bigsqcup_\omega K_n$ denote a countable disjoint union of copies of the complete graph K_n. In the strong R...

conjecture | open | MathDB 页面 | 原文线索

V2-042696 | MATHDB-337505

Ur-equivalence determined by the original and deleted posets

摘录 / EXCERPT: Let P and Q be posets, with elements $x\in P$ and $y\in Q$. Write $P[x\to\varnothing]$ and $Q[y\to\varnothing]$ for the posets obtained by applying the Ur-operation at x and...

conjecture | open | MathDB 页面 | 原文线索

V2-042697 | MATHDB-337512

Gallai-Ramsey formula for disjoint unions of paths on three vertices

摘录 / EXCERPT: Let P_3 denote the path on three vertices, and let $gr_k(K_3:n_1P_3, \ldots,n_kP_3)$ be the corresponding multicolor Gallai-Ramsey number. For positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-042698 | MATHDB-337514

General bipartite graph conjecture for Gallai-Ramsey numbers

摘录 / EXCERPT: Let H be a connected bipartite graph, let $R(H,H)=R$ be its two-color Ramsey number, and let $s(H)$ denote the parameter used in the source. The general bipartite Gallai-Ramsey c...

conjecture | open | MathDB 页面 | 原文线索

V2-042699 | MATHDB-337547

The interval decomposition conjecture for Ulam sequences U(1,n)

摘录 / EXCERPT: Interval decomposition conjecture. There exist integer coefficients m_i,p_i,k_i,r_i such that, for every integer $n\geq 4$,

conjecture | solved | MathDB 页面 | 原文线索

V2-042700 | MATHDB-337548

Rigidity Conjecture for generalized Ulam sets

摘录 / EXCERPT: Rigidity Conjecture. For every positive integer a, there exists an integer $L\geq1$ such that, for every $c\in(\mathbb{Z}/L\mathbb{Z})^\times$, there exists a positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-042701 | MATHDB-337549

Numerical rigidity conjecture for the intervals of U(1,n)

摘录 / EXCERPT: Numerical rigidity conjecture. There exists an integer $N_0>1$ such that, for every $n\geq N_0$, the endpoints are given by

conjecture | open | MathDB 页面 | 原文线索

V2-042702 | MATHDB-337551

Stein's equi-n-square conjecture

摘录 / EXCERPT: Stein's equi-n-square conjecture. Every such array S has a partial transversal of size $n-1$.

conjecture | refuted | MathDB 页面 | 原文线索

V2-042703 | MATHDB-337584

The EW-tableaux Eulerian distribution conjecture

摘录 / EXCERPT: EW-tableaux Eulerian distribution conjecture. The number of EW-tableaux of size n with k columns containing a 0 equals

conjecture | open | MathDB 页面 | 原文线索

V2-042704 | MATHDB-337604

The even-cycle multigraph host conjecture

摘录 / EXCERPT: Even-cycle multigraph conjecture. For $k\geq 6$,

conjecture | open | MathDB 页面 | 原文线索

V2-042705 | MATHDB-337649

Griesmer's inverse conjecture for Kneser's inequality in compact groups

摘录 / EXCERPT: Griesmer's conjecture. Under these hypotheses, such compact subsets A' and B' exist. This conjecture proposes an inverse theorem for near equality in Kneser's inequality that e...

conjecture | open | MathDB 页面 | 原文线索

V2-042706 | MATHDB-337664

Finite-strategy conjecture for normal quasicones

摘录 / EXCERPT: Let $\mathfrak{g}$ be an affine Kac-Moody algebra, let V be a weight module, and let $\mathfrak{C}$ be the set of normal quasicones arising as annihilators of vectors in V. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-042707 | MATHDB-337665

Finite-obstruction meta-conjecture for packing sparse graphs

摘录 / EXCERPT: Finite-obstruction meta-conjecture. If there is no simple obstruction to packing $\mathcal{G}$ into H, then a packing exists; equivalently, there is a finite list of obstructions...

conjecture | open | MathDB 页面 | 原文线索

V2-042708 | MATHDB-337666

The characterization of P-positions for triangular chocolate bar games when k is 1 modulo 4

摘录 / EXCERPT: Let k be a positive integer, and let (x,y,z) denote a position in the triangular chocolate bar game, with $x,y,z\in Z_{\geq 0}$. A position is a P-position if the next player h...

conjecture | open | MathDB 页面 | 原文线索

V2-042709 | MATHDB-337667

Bezrukavnikov's transpose symmetry conjecture for wall-crossing algorithms

摘录 / EXCERPT: Bezrukavnikov's conjecture. For every partition λ of n and every such interval I, one has

conjecture | open | MathDB 页面 | 原文线索

V2-042710 | MATHDB-337671

Minimal-degree conjecture for intervals in the Tamari lattice

摘录 / EXCERPT: Let $\operatorname{\mathbf{Tam}}_n$ be the Tamari lattice, and let $\mathbb{D}_n$ be its refined interval polynomial in variables $x,y,\overline{x},\overline{y}$. An interval has d...

conjecture | open | MathDB 页面 | 原文线索

V2-042711 | MATHDB-337672

Motzkin-number conjecture for doubly minimal-degree Tamari intervals

摘录 / EXCERPT: Let $\operatorname{\mathbf{Tam}}_n$ be the Tamari lattice, and consider its intervals. For an interval, measure degree separately in the variable pairs (x,y) and...

conjecture | open | MathDB 页面 | 原文线索

V2-042712 | MATHDB-337675

The signed-shift pattern enumeration conjecture

摘录 / EXCERPT: Signed-shift pattern enumeration conjecture. For every such n and σ,

conjecture | open | MathDB 页面 | 原文线索

V2-042713 | MATHDB-337677

Polylogarithmic bilinear Bogolyubov conjecture

摘录 / EXCERPT: Let V be the ambient vector space, let $P\subset V\times V$ have density $\delta>0$, and let $c(\delta)$ be the constant in Theorem 1.1, which asserts that there are subspaces...

conjecture | open | MathDB 页面 | 原文线索

V2-042714 | MATHDB-337684

Authors' classification conjecture for non-bipartite distance-regular graphs

摘录 / EXCERPT: Authors' classification conjecture. When D is large enough, Γ is one of the following graphs: the odd polygons; folded $(2D+1)$-cubes; the odd graphs O_k; the Hamming gr...

conjecture | open | MathDB 页面 | 原文线索

V2-042715 | MATHDB-337690

Flag lower-bound conjecture for edges of simplicial spheres

摘录 / EXCERPT: Fix positive integers n and m, and consider flag triangulations of the $(n-1)$-dimensional sphere with m vertices. Let the $(n-2)$-fold suspension of the one-dimensional sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-042716 | MATHDB-337692

Equivariant Gal conjecture for crystallographic Coxeter complexes

摘录 / EXCERPT: Let W be a finite crystallographic Coxeter group, and let $\operatorname{Cox}(W)$ be its Coxeter complex. Equivariant Gal conjecture. The pair $(\operatorname{Cox}(W),W)$ exhibit...

conjecture | open | MathDB 页面 | 原文线索

V2-042717 | MATHDB-337756

Donald's existence conjecture for stable toric triples

摘录 / EXCERPT: Let $(\Delta,\mathbf L)$ be the momentum polytope of a smooth compact toric Kähler manifold (X,ω_0), let F be a fixed union of facets of Δ, and let $Z\subset X$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-042718 | MATHDB-337759

Universal non-Poissonian pair correlation conjecture for sequences of maximal additive energy

摘录 / EXCERPT: Let $\left(a_n\right)_{n\geq 1}$ be a sequence of distinct integers, and let A_N denote its first N elements. The additive energy of a finite set A of reals is ... where the su...

conjecture | open | MathDB 页面 | 原文线索

V2-042719 | MATHDB-337760

Quasi-arithmetic sequences have no Poissonian pair correlation conjecture

摘录 / EXCERPT: Let $\left(a_n\right)_{n\geq 1}$ be a strictly increasing sequence of positive integers. It is quasi-arithmetic of degree d if there exist constants $C,K>0$ and a strictly increa...

conjecture | open | MathDB 页面 | 原文线索

V2-042720 | MATHDB-337764

Four-square representation conjecture for the middle-thirds Cantor set

摘录 / EXCERPT: Let C denote the middle-thirds Cantor set. For a real number $u\in[0,1]$, consider representations of u as a sum of four squares of elements of C. Four-square representation...

conjecture | open | MathDB 页面 | 原文线索

V2-042721 | MATHDB-337785

Nonnegative Demazure-character expansion for Kohnert polynomials of Demazure diagrams

摘录 / EXCERPT: Let D be a demazure diagram, meaning a finite cell diagram satisfying the demazure condition described in the paper, and let ${\mathfrak K}_D$ denote its Kohnert polynomial. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042722 | MATHDB-337786

Nonnegative Schubert expansion conjecture for skew polynomials

摘录 / EXCERPT: Let a be a weak composition, let $\mathbb S(a)$ be the associated skew diagram, and let ${\mathfrak K}_{\mathbb S(a)}$ be its skew polynomial. The skew-polynomial expansion conje...

conjecture | refuted | MathDB 页面 | 原文线索

V2-042723 | MATHDB-337797

Maximal determinant conjecture for 2-consecutive-ones matrices

摘录 / EXCERPT: Let $A\in\{0,1\}^{n\times n}$ be a matrix with the 2-consecutive ones property, meaning that the ones in each row occur in at most two blocks in some ordering of the columns. 2-con...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-042724 | MATHDB-337807

Reiner-Stanton conjecture on unimodality of symmetric q-binomial differences

摘录 / EXCERPT: Let k, m, and b be integers with $k\ge 2$, $m\ge k$, and, for $k\ge 3$, $k-2\le b\le \frac{km-4k+4}{k-2}$, where $k(m-b)$ is even. Define ... Here $\binom{m}{k}_q$ denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-042725 | MATHDB-337814

Bao Shou's same-fingerprint symbol-invariant conjecture

摘录 / EXCERPT: Let rigid semisimple operators be labeled by pairs of partitions $(\lambda^{'},\lambda^{”})$. Fix a fingerprint invariant, and consider operators satisfying different conditions in...

conjecture | open | MathDB 页面 | 原文线索

V2-042726 | MATHDB-337832

Conjecture on quantum D-modules for primitive extremal transitions of Calabi-Yau threefolds

摘录 / EXCERPT: A primitive extremal transition is a transition between Calabi-Yau 3-folds satisfying the hypotheses specified in the paper. The quantum D-module conjecture. The statement of The...

conjecture | open | MathDB 页面 | 原文线索

V2-042727 | MATHDB-337870

The large-intersection conjecture for hypercube subspaces

摘录 / EXCERPT: Let $H(\infty,k)$ denote the set of intersection sizes arising from linear subspaces of dimension k with the hypercube, and let $\mathbb{N}$ denote the natural numbers. Large-int...

conjecture | open | MathDB 页面 | 原文线索

V2-042728 | MATHDB-337871

The interval conjecture for hypercube intersection sizes

摘录 / EXCERPT: Let $H(\infty,k)$ denote the set of intersection sizes arising from linear subspaces of dimension k with the hypercube. For a positive integer r, write $[r]=\{1,2,\ldots,r\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-042729 | MATHDB-337876

The NonKrupp subarrangement conjecture for pseudocircle arrangements

摘录 / EXCERPT: NonKrupp subarrangement conjecture. If every triple of pseudocircles which forms a triangle in ${\cal A}$ is NonKrupp, then ${\cal A}$ contains ${\cal N}_6^\Delta$ as a subarrangem...

conjecture | open | MathDB 页面 | 原文线索

V2-042730 | MATHDB-337877

Connectivity conjecture for the triangle flip-graph of pseudocircle arrangements

摘录 / EXCERPT: Triangle flip-graph connectivity conjecture. The triangle flip-graph on the set of all intersecting digon-free arrangements of n pseudocircles is connected for every...

conjecture | open | MathDB 页面 | 原文线索

V2-042731 | MATHDB-337882

The mod-4 relation between transversals and minimal covers

摘录 / EXCERPT: Let L be a Latin square of even order n. Let t be the number of transversals of L, and let $q_{\min}$ be the number of minimal $(n+1)$-covers of L. The mod-4 conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042732 | MATHDB-337898

Polyanskii's conjecture on sums of balanced unit-vector sets

摘录 / EXCERPT: Polyanskii's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042733 | MATHDB-337932

The unimodality conjecture for Euclid polynomials

摘录 / EXCERPT: Let $E_k(\lambda)$ be the Euclid polynomials. A polynomial $p(\lambda)=\sum_{j=0}^{n}a_j\lambda^j$ is unimodal when its coefficient vector $[a_0,a_1,\ldots,a_n]$ consists of positi...

conjecture | open | MathDB 页面 | 原文线索

V2-042734 | MATHDB-337937

Root-location conjecture for an extremizer in dimension 28

摘录 / EXCERPT: Let $\mathcal{A}_+(28)$ denote the class of radial Schwartz functions on $\mathbb{R}^{28}$ satisfying the positive Fourier-eigenfunction uncertainty conditions. Dimension-28 root-l...

conjecture | open | MathDB 页面 | 原文线索

V2-042735 | MATHDB-338007

Geelen-Gerards-Whittle's matroid structure theorem

摘录 / EXCERPT: Matroid structure theorem. The stated tree-decomposition exists.

conjecture | open | MathDB 页面 | 原文线索

V2-042736 | MATHDB-338008

Geelen-Gerards-Whittle's perturbation conjecture for represented matroids

摘录 / EXCERPT: Perturbation conjecture. There exist $k,t\in\mathbb Z_+$ such that each vertically k-connected member of $\mathcal M$ is a rank-$(\leq t)$ perturbation of an $\mathbb F$-represen...

conjecture | open | MathDB 页面 | 原文线索

V2-042737 | MATHDB-338009

Grace-van Zwam's frame-template structure conjecture

摘录 / EXCERPT: Frame-template conjecture. There exist $k\in\mathbb Z_+$ and frame templates $\Phi_1,\ldots,\Phi_s,\Psi_1,\ldots,\Psi_t$ such that $\mathcal{M}$ contains each class...

conjecture | open | MathDB 页面 | 原文线索

V2-042738 | MATHDB-338016

The equal-slit-length conjecture for doubly-intruded hexagons

摘录 / EXCERPT: Consider the scaling regime in which $x+y=y+t=z=N$, all fern lobe sizes are equal to 1, both ferns have an even number of lobes, and their combined length is $m+n=M$. Let the reg...

conjecture | open | MathDB 页面 | 原文线索

V2-042739 | MATHDB-338038

High-girth Ramsey graph conjecture for ordered forests

摘录 / EXCERPT: Let H and H' be ordered forests, and let t be a positive integer. Write $R_<(H,H')$ for the family of Ramsey graphs of (H,H'), and let $\operatorname{girth}(F)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-042740 | MATHDB-338041

Forest Ramsey-graph conjecture for finite ordered forest pairs

摘录 / EXCERPT: Let H and H' be ordered forests, and let $R_<(H,H')$ denote the family of Ramsey graphs of (H,H'). Forest Ramsey-graph conjecture. If (H,H') is Ramsey finite, then...

conjecture | open | MathDB 页面 | 原文线索

V2-042741 | MATHDB-338042

Characterization of finite pairs of chi-unavoidable connected ordered graphs

摘录 / EXCERPT: Let H and H' be chi-unavoidable connected ordered graphs, each with at least two edges. Connected ordered-graph finiteness conjecture. The pair (H,H') is Ramsey finite if a...

conjecture | open | MathDB 页面 | 原文线索

V2-042742 | MATHDB-338060

The exponential-profile conjecture for deep extremal level sets

摘录 / EXCERPT: Exponential-profile conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-042743 | MATHDB-338084

Eggleton-Erdős conjecture on subset sums of zero-sum free subsets

摘录 / EXCERPT: Eggleton-Erdős conjecture. The set of subset sums satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042744 | MATHDB-338114

EPPA and maximal amenability conjecture for the class of finite F-graphs

摘录 / EXCERPT: Let F be the structure defining the class $\mathcal{G}_F$, let O_F be the corresponding ordered structure, and let M_F be its Fraïssé limit. Write $\operatorname{Aut}(O_F)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-042745 | MATHDB-338157

The conjectural norm formula for Jack superpolynomials

摘录 / EXCERPT: The norm conjecture. The superpolynomial ${P}^{(\alpha)}_\Lambda$ has norm, with respect to the combinatorial scalar product, given by

conjecture | open | MathDB 页面 | 原文线索

V2-042746 | MATHDB-338164

Hähnle's monomial-family bound conjecture

摘录 / EXCERPT: Let $g(d,n)$ be the maximum number t of pairwise-disjoint nonempty families $F_1,F_2,\dots,F_t$ of degree-d monomials in n variables satisfying: for every $1\le i<j<k\le t$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042747 | MATHDB-338210

The arbitrary-group induced removal conjecture

摘录 / EXCERPT: Arbitrary-group induced removal conjecture. For every $0 < \epsilon < 1/2$, if $A \subseteq G$ is ϵ-far from bi-induced-F-free, then

conjecture | open | MathDB 页面 | 原文线索

V2-042748 | MATHDB-338224

Minimum complexity conjecture for square-free ternary words

摘录 / EXCERPT: Let ${\mathbf T}$ be the ternary Thue word, the fixed point of the morphism $0\mapsto012$, $1\mapsto02$, and $2\mapsto1$. A ternary infinite word is square-free if it contains no f...

conjecture | open | MathDB 页面 | 原文线索

V2-042749 | MATHDB-338225

Minimum critical exponent conjecture for ternary words of complexity at most $2n+1$

摘录 / EXCERPT: Let ${\mathbf G}$ be the ternary infinite word defined earlier in the paper, and let the subword complexity of an infinite word be the number of its distinct factors of each length...

conjecture | open | MathDB 页面 | 原文线索

V2-042750 | MATHDB-338226

Shur's Restivo-Salemi conjecture for power-free languages

摘录 / EXCERPT: Let Σ_k be a k-letter alphabet, and let $L_{k,\alpha}$ denote the language of α-power-free finite words over Σ_k. For a language L, write ${\sf ext}(L)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-042751 | MATHDB-338227

Erdős-Kleitman minimum saturated-family conjecture

摘录 / EXCERPT: Erdős-Kleitman's conjecture. Every s-saturated family satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-042752 | MATHDB-338278

Pitman-Stanley flow-polytope volume conjecture for the net flow vector $(1,0,1,0,\ldots,0)$

摘录 / EXCERPT: Let $\textup{PS}_{n+1}$ be the Pitman-Stanley graph on $n+1$ vertices, and let $\mathcal{F}_{\textup{PS}_{n+1}}(1,0,1,0,\ldots,0)$ denote its flow polytope with the indicated net f...

conjecture | open | MathDB 页面 | 原文线索

V2-042753 | MATHDB-338279

Caracol flow-polytope volume conjecture for three initial parameters

摘录 / EXCERPT: Let $\textup{Car}_{n+1}$ be the Caracol graph on $n+1$ vertices, and let $\mathcal{F}_{\textup{Car}_{n+1}}(a,b,c,\ldots,c)$ denote its flow polytope with the indicated net flow vec...

conjecture | open | MathDB 页面 | 原文线索

V2-042754 | MATHDB-338280

Caracol graph Ehrhart polynomial product conjecture

摘录 / EXCERPT: Let $\textup{Car}_{n+1}$ be the Caracol graph on $n+1$ vertices, and let $E_{\textup{Car}_{n+1}}(x)$ denote the Ehrhart polynomial associated with its flow polytope. Caracol Ehrhar...

conjecture | open | MathDB 页面 | 原文线索

V2-042755 | MATHDB-338297

The cluster-number ordering conjecture for equal endpoint differences

摘录 / EXCERPT: Let $cpi,\tau\binom{}{}$ be permutations in $cmathfrak{S}_m$, and call a permutation in standard form when $cpi_1<cpi_m$ and $cpi_1+cpi_m\boxtimes m+1$. Let $r_{n,k}^{cpi}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-042756 | MATHDB-338298

Condition (I) conjecture for the elementary divisors of quasihomogeneous singularities

摘录 / EXCERPT: Let $f\in\mathbb{C}[x_1,\ldots,x_n]$ be a quasihomogeneous singularity. Define the finite sets $M_1\supset\cdots\supset M_{\nu_{max}}$ from the multiplicities of the cyclotomic fac...

conjecture | open | MathDB 页面 | 原文线索

V2-042757 | MATHDB-338320

Fractal sum-product expansion conjecture

摘录 / EXCERPT: Let $\alpha,\beta,\delta \in (0,1)$. Assume that $A,B,E \subset [0,1]$ are compact sets with $\dim A \leq \alpha$, $\dim B \geq \beta$, and $\dim B + \dim E \geq \dim A + \delta$....

conjecture | open | MathDB 页面 | 原文线索

V2-042758 | MATHDB-338330

Balog's conjecture on the size of $AA+A$

摘录 / EXCERPT: Balog's conjecture. For every such set A,

conjecture | open | MathDB 页面 | 原文线索

V2-042759 | MATHDB-338331

The diagonal asymptotic density conjecture for higher-dimensional rooks

摘录 / EXCERPT: Let $a_{k,\ell}$ denote the asymptotic packing density parameter for ℓ-rooks in k dimensions. Diagonal density conjecture. For integers $k\ge 2$, ... This is the most surprisi...

conjecture | open | MathDB 页面 | 原文线索

V2-042760 | MATHDB-338332

The asymptotic covering density conjecture for higher-dimensional rooks

摘录 / EXCERPT: Let $c_{k,\ell}$ denote the asymptotic covering density parameter for ℓ-rooks in k dimensions, with positive integers $k\ge \ell\ge 2$. Covering density conjecture. For posi...

conjecture | open | MathDB 页面 | 原文线索

V2-042761 | MATHDB-338344

Balanced diagonal conjecture for frequency squares

摘录 / EXCERPT: Let L be a frequency square of type $F(m\lambda;\lambda^m)$. A balanced diagonal conjecture asserts that if $(m-1)\lambda$ is even, then L contains a balanced diagonal. The pap...

conjecture | open | MathDB 页面 | 原文线索

V2-042762 | MATHDB-338345

Plex-existence conjecture for Latin squares

摘录 / EXCERPT: Plex-existence conjecture. If n is even, every Latin square of order n has a $2k$-plex for each $k\in N(n/2)$. If n is odd, every Latin square of order n has a k-plex for...

conjecture | open | MathDB 页面 | 原文线索

V2-042763 | MATHDB-338378

Locality conjecture for cross ratios and quasi-Plücker coordinates

摘录 / EXCERPT: Let N be a matroid, let H be a skew hyperfield, and let $cr:CR_N\rightarrow H$ be a map assigning cross-ratio values. The map is called consistent if it arises from quasi-Plück...

conjecture | open | MathDB 页面 | 原文线索

V2-042764 | MATHDB-338379

Algebraic-group characterization from a unit cross ratio

摘录 / EXCERPT: Let a,b,c,d be elements in an algebraic matroid, and let $cr(a,b,c,d)$ denote their cross ratio. Algebraic-group characterization conjecture. ... in an algebraic matroid if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-042765 | MATHDB-338386

Almost-everywhere obstruction conjecture for Poissonian pair correlations

摘录 / EXCERPT: Almost-everywhere obstruction conjecture. If for almost all α the pair correlations of $((\lbrace a_n\alpha\rbrace)_{n\in\mathbb N})$ are not Poissonian, then the pair corre...

conjecture | open | MathDB 页面 | 原文线索

V2-042766 | MATHDB-338395

Istrate's scaling conjecture for heapable subsequences

摘录 / EXCERPT: Istrate's scaling conjecture. For every $k\boldsymbol{$ there exists $_k>0$ such that, as $n\to\boldsymbol{\frac{E[MHS_k]}{(n)}$ converges to $_k$. Moreover, $_2=\frac{1+{5}}{2}$ i...

conjecture | solved | MathDB 页面 | 原文线索

V2-042767 | MATHDB-338400

Gedeon-Proudfoot-Young negative-root conjecture for matroid Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let M be a matroid, and let $P_M(t)$ denote its Kazhdan-Lusztig polynomial. Gedeon-Proudfoot-Young's conjecture. The polynomial $P_M(t)$ has only negative zeros for every matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-042768 | MATHDB-338401

Gedeon-Proudfoot-Young interlacing conjecture for matroid Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let M be a matroid and let e be an element of its ground set. Let M/e be the contraction of M at e. A matroid is non-degenerate if its rank is 0 or its Kazhdan-Lusztig...

conjecture | open | MathDB 页面 | 原文线索

V2-042769 | MATHDB-338402

Proudfoot-Xu-Young negative-root conjecture for matroid Z-polynomials

摘录 / EXCERPT: Proudfoot-Xu-Young's conjecture. The polynomial $Z_M(t)$ has only negative zeros for every matroid M. The source gives no resolution of this conjecture for arbitrary matroids.

conjecture | open | MathDB 页面 | 原文线索

V2-042770 | MATHDB-338409

Arity-dependence conjecture for inducibility of nonbinary trees

摘录 / EXCERPT: Arity-dependence conjecture. Among all d-ary trees D, only binary trees satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-042771 | MATHDB-338417

Full-dimension Erdős semi-group conjecture

摘录 / EXCERPT: Full-dimension Erdős semi-group conjecture. If E is an ESG with full Hausdorff dimension, then

conjecture | open | MathDB 页面 | 原文线索

V2-042772 | MATHDB-338418

A modified order-complex and Chow-ring identity for uniform matroids

摘录 / EXCERPT: Let $U_{n,r}$ be the uniform matroid of rank r on n elements, let $L(U_{n,r})$ be its lattice of flats, and let $\Delta(L(U_{n,r}))$ be the order complex of this lattice. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-042773 | MATHDB-338421

Face nonrevisiting conjecture for lattice Hasse diagrams of simple polytopes

摘录 / EXCERPT: Let P be a simple polytope and let ${\bf c}$ be a generic cost vector such that $G(P, {\bf c})$ is the Hasse diagram of a lattice. A directed path in $G(P,{\bf c})$ is said to re...

conjecture | open | MathDB 页面 | 原文线索

V2-042774 | MATHDB-338427

Fici-Mignosi binary reduction conjecture for distinct abelian squares

摘录 / EXCERPT: Let a word of length n contain k distinct abelian square factors. Fici-Mignosi's conjecture. There exists a binary word of length n containing at least k distinct abelian s...

conjecture | open | MathDB 页面 | 原文线索

V2-042775 | MATHDB-338428

Kucherov-Rytter-Rytter-Wendlandt conjecture on non-equivalent abelian squares

摘录 / EXCERPT: For a binary word of length n, let $XNL(n)$ denote the maximum number of non-equivalent abelian squares it contains. Kucherov et al.'s conjecture. ... The conjecture proposes a pol...

conjecture | open | MathDB 页面 | 原文线索

V2-042776 | MATHDB-338429

Fraenkel-Paterson-Simpson minimum non-equivalent abelian-square conjecture

摘录 / EXCERPT: For a binary linear word of length n, let $MNL(n)$ be the minimum number of non-equivalent abelian squares it contains. Fraenkel-Paterson-Simpson's conjecture. For every positive...

conjecture | open | MathDB 页面 | 原文线索

V2-042777 | MATHDB-338430

Fici-Saarela minimum distinct abelian-square conjecture

摘录 / EXCERPT: For a binary linear word of length n, let $MDL(n)$ be the minimum number of distinct abelian squares it contains. Fici-Saarela's conjecture. ... and the only words of length $4k+3$...

conjecture | open | MathDB 页面 | 原文线索

V2-042778 | MATHDB-338431

Minimum distinct abelian squares in circular binary words conjecture

摘录 / EXCERPT: For a circular binary word of length n, let $MDC(n)$ denote the minimum number of distinct abelian squares it contains. The circular minimum conjecture. The minimum number is ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-042779 | MATHDB-338444

The Gallai-Ramsey conjecture for books

摘录 / EXCERPT: Gallai-Ramsey conjecture for books. For $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-042780 | MATHDB-338459

Connected-sum superadditivity conjecture for Thompson's width

摘录 / EXCERPT: Let M_1 and M_2 be n-dimensional simplicial manifolds, and choose a connected sum $M_1 \# M_2$. Here $\Omega(M)$ denotes Thompson's width of a simplicial manifold M. Connec...

conjecture | open | MathDB 页面 | 原文线索

V2-042781 | MATHDB-338460

Lower-bound conjecture for width under connected sums of odd-dimensional manifolds

摘录 / EXCERPT: Let M_1 and M_2 be closed odd-dimensional triangulated manifolds, and let $M_1 \# M_2$ be any connected sum. Let $\Omega(M)$ denote Thompson's width. Odd-dimensional connected-...

conjecture | open | MathDB 页面 | 原文线索

V2-042782 | MATHDB-338477

Bean-Tannock-Ulfarsson maximal staircase-encoding conjecture

摘录 / EXCERPT: Let $J(\ell,k)$ denote the number of 132-avoiding permutations of length ℓ whose staircase encoding has k nonzero boxes, and let C_i denote the i^{th} Catalan n...

conjecture | open | MathDB 页面 | 原文线索

V2-042783 | MATHDB-338486

Motion conjecture for primitive geometric distance-regular graphs

摘录 / EXCERPT: Let X be a primitive geometric distance-regular graph of diameter $d\geq 3$. Its motion is the minimum number of vertices moved by a nonidentity automorphism. Motion conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042784 | MATHDB-338510

The Sárközy-Sós conjecture for multivariate linear forms

摘录 / EXCERPT: Sárközy-Sós conjecture. There exists some infinite set of positive integers $\mathcal{A}$ such that $r_{\mathcal{A}}(n;k_1,\dots,k_d)$ is constant for n large enough if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-042785 | MATHDB-338525

Decorated Sierpinski graph conjecture for the edge-isoperimetric profile

摘录 / EXCERPT: Decorated Sierpinski graph conjecture. For every ℓ with $0\leq\ell\leq m^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-042786 | MATHDB-338526

Subadditivity-plus-sigma conjecture for lexicographic edge boundaries

摘录 / EXCERPT: Subadditivity-plus-sigma conjecture. For all $n,m,\ell_a,\ell_b\in\mathbb N$ with $m^n\geq\ell_a\geq\ell_b>0$, if $\ell_a+\ell_b\leq m^n$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042787 | MATHDB-338527

The ultimate periodicity conjecture for super-monochromatic factorisations

摘录 / EXCERPT: Let x be an infinite word over an alphabet $\mathcal{A}$. A super-monochromatic factorisation of a suffix of x is a factorisation into finite words such that all factors and ev...

conjecture | open | MathDB 页面 | 原文线索

V2-042788 | MATHDB-338528

Vanishing containment probability conjecture for mesh patterns

摘录 / EXCERPT: Let m be a mesh pattern with at least one shaded box, and let a random mesh pattern of length n be chosen uniformly. Vanishing containment probability conjecture. The probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-042789 | MATHDB-338529

Conway's bounded octal game periodicity conjecture

摘录 / EXCERPT: An octal game is a game played with tokens divided into heaps, where a move either removes some or all tokens in one heap, or removes some but not all tokens from a heap and divide...

conjecture | open | MathDB 页面 | 原文线索

V2-042790 | MATHDB-338530

Periodicity conjecture for star-Kayles games

摘录 / EXCERPT: Let $STARKAYLES_{k,n}$ be the graph game in which the underlying graph is obtained by starting with a star graph on k vertices and extending one branch to a path of edge-length...

conjecture | open | MathDB 页面 | 原文线索

V2-042791 | MATHDB-338531

Periodicity conjecture for $KAYLES_n\boxplus(*1)$

摘录 / EXCERPT: Let $KAYLES_n$ denote the Kayles graph game on a path of edge-length n, and let $\boxplus$ denote the selective compound operation used in the paper; write $*1$ for the one-point...

conjecture | open | MathDB 页面 | 原文线索

V2-042792 | MATHDB-338589

Zeta-polynomial conjecture for m-divisible odd noncrossing partitions

摘录 / EXCERPT: For $m\geq 1$, let $C=(x_1,x_2,\ldots,x_m)$ be an m-multichain in $\mathcal{O\!N\C}_{2n+1}$, with extended delta sequence $\delta_o(C)=(d_0;d_1,d_2,\ldots,d_m)$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-042793 | MATHDB-338590

The generalized Sidon set counting conjecture

摘录 / EXCERPT: An α-generalized Sidon set in $[n]=\{1,\ldots,n\}$ is a set containing at most α Sidon 4-tuples. Let $N_n(\alpha)$ denote the number of α-generalized Sidon set...

conjecture | open | MathDB 页面 | 原文线索

V2-042794 | MATHDB-338594

Equality of the honest-effort and attainable tournament regions

摘录 / EXCERPT: Let $\mathcal{A}_n$ denote the set of score vectors attainable in tournaments on n players, and let $\mathcal{A}^{*}_n$ denote the region generated by the tournament configuratio...

conjecture | open | MathDB 页面 | 原文线索

V2-042795 | MATHDB-338625

Arithmetic periodicity conjecture for partition games excluding \{1,2\}

摘录 / EXCERPT: Arithmetic periodicity conjecture. Every instance $\mathcal C$ of CUT for which $\{1,2\}\not\subset \mathcal C$ is ultimately arithmetic-periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-042796 | MATHDB-338629

Product formula conjecture for centrally symmetric tilings of $R^{\odot}$-type regions

摘录 / EXCERPT: Let $R^{\odot}_{x,yz}(\textbf{a};\ \textbf{c};\ \textbf{b})$ be the region described in the paper, and consider its tilings invariant under 180° rotations. Product formula...

conjecture | open | MathDB 页面 | 原文线索

V2-042797 | MATHDB-338640

Minimality conjecture for the constructed bicommutative system

摘录 / EXCERPT: Let $n=4k+7$, $m=2k+3$, and $k\geq 2$. Let $\mathcal{A}_m$ be the constructed m-bicommutative system in $\mathcal{P}_n(m)$. Bicommutative-system minimality conjecture. Does...

open_problem | open | MathDB 页面 | 原文线索

V2-042798 | MATHDB-338641

Asymptotic size conjecture for the constructed cone

摘录 / EXCERPT: Let $n=4k+7$ with $k\geq 2$, and let $\operatorname{Cone}=\operatorname{Cone}(\operatorname{Cone}_{2k+1}\cup\operatorname{Cone}_{2k+3})$. Cone asymptotic-size conjecture. Is ... Nume...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-042799 | MATHDB-338642

Asymptotic size conjecture for the constructed system D

摘录 / EXCERPT: Let $n=4k+7$ and let $\mathcal{D}$ be the constructed maximal commutative algebraic system in $\mathcal{P}_n(*)$. $\mathcal{D}$ asymptotic-size conjecture. Is ... Numerical checks th...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-042800 | MATHDB-338643

Global bound for the constructed maximal commutative system

摘录 / EXCERPT: Let $n=4k+9$ with $k\geq 2$, and define ... where $\mathcal{C}$ is the constructed maximal commutative algebraic system. K_n bound conjecture. Is $K_n<1$ for all $n=4k+9$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-042801 | MATHDB-338657

Linear-size conjecture for bisection-closed families

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, and let $\mathcal{F}$ be a bisection-closed family, meaning a fractional L-intersecting family of subsets of $[n]$ with $L=\left\{\frac{1}{2}\right\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-042802 | MATHDB-338672

Difference-product and difference-quotient sum-product conjectures

摘录 / EXCERPT: Let $A\subset\mathbb{R}$ be finite, and define ... with the convention $0^{-1}=0$. Also let $AA=\{ab:a,b\in A\}$. Difference-product and difference-quotient conjecture. For every fix...

conjecture | open | MathDB 页面 | 原文线索

V2-042803 | MATHDB-338674

The conjecture for even-clique Ramsey-Turán densities

摘录 / EXCERPT: Let $R(3,s)$ be the smallest integer N such that every red-blue colouring of K_N contains a red K_3 or a blue K_s. Let $\varrho(K_3,K_{2s})$ denote the multicolour Ramsey-T...

conjecture | open | MathDB 页面 | 原文线索

V2-042804 | MATHDB-338685

Dimca-Sticlaru conjecture on the combinatorial freeness defect of line arrangements

摘录 / EXCERPT: Dimca-Sticlaru conjecture. The invariant $\nu(C)$ is combinatorially determined.

conjecture | open | MathDB 页面 | 原文线索

V2-042805 | MATHDB-338686

The universal upper-bound conjecture for proportional cake-cutting with different entitlements

摘录 / EXCERPT: Universal upper-bound conjecture. For every n, every set of n value measures, and every entitlement vector t, there exists a t-proportional allocation with at most $2n-2$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-042806 | MATHDB-338717

Conjecture on polynomial-time enumeration of regular graphs

摘录 / EXCERPT: Regular-graph enumeration conjecture. The sequence a_n can be computed in polynomial time in n.

conjecture | open | MathDB 页面 | 原文线索

V2-042807 | MATHDB-338718

Conjecture on hard square tiling enumeration

摘录 / EXCERPT: Square-tiling hardness conjecture. There exists a finite set of tiles ${\bf T}$ such that counting the number of tilings of $[n\times n]$ with ${\bf T}$ is $\#\mathrm{EXP}$-complet...

conjecture | open | MathDB 页面 | 原文线索

V2-042808 | MATHDB-338721

Wilfian-formula conjecture for self-avoiding walks

摘录 / EXCERPT: Self-avoiding-walk Wilfian conjecture. The sequence a_n has no Wilfian formula of type (W1).

conjecture | open | MathDB 页面 | 原文线索

V2-042809 | MATHDB-338722

Combinatorial-interpretation conjecture for the Gessel sequence

摘录 / EXCERPT: Gessel-sequence interpretation conjecture. The sequence b_n has a combinatorial interpretation of type (C1).

conjecture | open | MathDB 页面 | 原文线索

V2-042810 | MATHDB-338732

The minimum-volume conjecture for simple matroids

摘录 / EXCERPT: Let M be a simple matroid of rank r on n elements. Write $U_{a,b}$ for the uniform matroid of rank a on b elements, and let $\oplus$ denote matroid direct sum. Minimum-vo...

conjecture | open | MathDB 页面 | 原文线索

V2-042811 | MATHDB-338733

Equality of connective constants for knot and knotoid shadows

摘录 / EXCERPT: Let sk_n be the number of n-crossing knotoid shadows, let μ_I be the connective constant for knot shadows, and let μ_K be the connective constant for knotoid shadows. E...

conjecture | open | MathDB 页面 | 原文线索

V2-042812 | MATHDB-338734

Asymptotic growth conjecture for knotoid diagrams

摘录 / EXCERPT: Let $s\kappa_n$ be the number of n-crossing knotoid diagrams, let sk_n be the number of n-crossing knotoid shadows, and let μ_K be the knotoid-shadow connective constant....

conjecture | open | MathDB 页面 | 原文线索

V2-042813 | MATHDB-338747

Complex ETF existence conjecture beyond twice the dimension

摘录 / EXCERPT: Let D and N be positive integers with $N>2D>2$, and let an $\operatorname{ETF}(D,N)$ denote a complex equiangular tight frame of N vectors in dimension D. Call (D,N) posi...

conjecture | open | MathDB 页面 | 原文线索

V2-042814 | MATHDB-338753

Berend et al.'s balanced knight coloring conjecture

摘录 / EXCERPT: Let n be a positive integer and consider an $n\times n$ chessboard. For each position (i,j), let $N[i,j]$ be the closed neighborhood of the corresponding knight-move graph vert...

conjecture | open | MathDB 页面 | 原文线索

V2-042815 | MATHDB-338758

The Gallai-Ramsey conjecture for even cycles and paths

摘录 / EXCERPT: Let $n\ge3$ and $k\ge2$. For each $i\in\{0,1,\ldots,n-2\}$, let $G_i=P_{2i+3}$, and let G_{n-1} be either C_{2n} or P_{2n+1}. Choose integers $i_1,\ldots,i_k$ satisfying ... He...

conjecture | open | MathDB 页面 | 原文线索

V2-042816 | MATHDB-338785

The asymptotic upper-bound conjecture for the weighted Davenport constant

摘录 / EXCERPT: Let $n=n_1n_2$ under the same assumptions as in Theorem, and let $D_{\{1,s\}}(\mathbb Z_n)$ denote the $\{1,s\}$-weighted Davenport constant of the cyclic group $\mathbb Z_n$. Asym...

conjecture | open | MathDB 页面 | 原文线索

V2-042817 | MATHDB-338790

The central zonotopal algebra conjecture for connected graphs

摘录 / EXCERPT: Let G_1 and G_2 be connected graphs. Their central zonotopal algebras are denoted by $4\mathcal{C}_{G_1}^{\mathcal{C}}4$ and $4\mathcal{C}_{G_2}^{\mathcal{C}}4$. For a graph, i...

conjecture | open | MathDB 页面 | 原文线索

V2-042818 | MATHDB-338791

The central zonotopal algebra conjecture for matrices

摘录 / EXCERPT: Let $A_1\in\mathbb{R}^{n_1\times m_1}$ and $A_2\in\mathbb{R}^{n_2\times m_2}$ have ranks n_1 and n_2, respectively. A column is a bridge-column if deleting it decreases the ran...

conjecture | open | MathDB 页面 | 原文线索

V2-042819 | MATHDB-338804

Anthony's conjecture on the number of polynomial threshold functions

摘录 / EXCERPT: Let $T(n,d)$ be the number of n-variable polynomial threshold functions of degree d, and let ... Here $\binom{n}{\le d}$ denotes the number of monomials in at most d variables'...

conjecture | open | MathDB 页面 | 原文线索

V2-042820 | MATHDB-338809

Conjecture that LS-function combinatoric polytopes are simple

摘录 / EXCERPT: Let an LS function determine a combinatoric polytope obtained from its associated scattering-form or kinematic-space construction. Simplicity conjecture. All combinatoric polytopes...

conjecture | open | MathDB 页面 | 原文线索

V2-042821 | MATHDB-338838

The even Williamson conjecture

摘录 / EXCERPT: A Williamson matrix is a matrix arising from Williamson's construction; its order is the number of rows and columns. The even Williamson conjecture. Williamson matrices exist in ev...

conjecture | open | MathDB 页面 | 原文线索

V2-042822 | MATHDB-338849

The combinatorial Stiefel map determines linkage matching fields

摘录 / EXCERPT: Stiefel-map determination conjecture. A linkage matching field is determined by the collection of transversal matroids associated to its maximal tope linkage covectors by the combi...

conjecture | open | MathDB 页面 | 原文线索

V2-042823 | MATHDB-338851

The asymptotic number of copies of $\wedge_s$ in $\{\wedge_k,\vee_l\}$-free families

摘录 / EXCERPT: Let $\wedge_r$ be the poset on elements $a,b_1,\ldots,b_r$ with $a<b_i$ for every i, and let $\vee_r$ be its dual. For a family of posets $\mathcal P$ and a poset Q, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042824 | MATHDB-338875

Parabolic Coxeter-Catalan numbers for types A, B, H and I

摘录 / EXCERPT: Let (W,S) be a Coxeter system with $W\in\{A_n,B_n,H_3,I_2(m)\}$, and let $J\subseteq S$. The sets $\operatorname{Align}(W^J,c)$, $\operatorname{NC}(W^J,c)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042825 | MATHDB-338906

Schmidt's conjecture on perfect binary sequences with parameter $d=1$

摘录 / EXCERPT: Schmidt's conjecture. There do not exist perfect binary sequences with

conjecture | open | MathDB 页面 | 原文线索

V2-042826 | MATHDB-338908

Bergelson-Host-Kra conjecture on monochromatic arithmetic progressions

摘录 / EXCERPT: Let $0<\alpha<1$, let $\epsilon>0$, and let $k=3$ or 4. For a positive integer N, write $[N]=\{1,2,\dots,N\}$. A length-k arithmetic progression in $[N]$ has a common differe...

conjecture | open | MathDB 页面 | 原文线索

V2-042827 | MATHDB-338917

Gurvits's inner-product conjecture for homogeneous real stable polynomials

摘录 / EXCERPT: Let $p,q \in \mathbb{R}_+[x_1,\ldots,x_n]$ be homogeneous real stable polynomials of total degree d. For a multi-index μ with $|\mu|=d$, write p_μ and q_μ for the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-042828 | MATHDB-338928

Weak Foulkes conjecture for multiset partitions

摘录 / EXCERPT: Let a and b be positive integers with $a<b$, and let M be a multiset of size ab. A multiset partition divides M into an unordered collection of submultisets. Weak Foulkes...

conjecture | open | MathDB 页面 | 原文线索

V2-042829 | MATHDB-338977

The detachable-pair classification conjecture for 3-connected matroids

摘录 / EXCERPT: Detachable-pair classification conjecture. Either M has a detachable pair; M is a spike; $M \cong M(K_{3,k})$ or $M \cong M^*(K_{3,k})$ for some $k \ge 5$; M is the cycle mat...

conjecture | open | MathDB 页面 | 原文线索

V2-042830 | MATHDB-338990

Generalised spike-structure conjecture for matroids with the $(t_1,2t_1,t_2,2t_2)$-property

摘录 / EXCERPT: Generalised spike-structure conjecture. There exists a function $f(t_1,t_2)$ such that if M is a matroid with the $(t_1,2t_1,t_2,2t_2)$-property and

conjecture | open | MathDB 页面 | 原文线索

V2-042831 | MATHDB-338991

Chain theorem conjecture for highly connected matroids and t-spikes

摘录 / EXCERPT: Chain theorem conjecture. There exists a function $f(t)$ such that if M is a $(2t-1)$-connected matroid with no circuits or cocircuits of size $2t-1$ and

conjecture | open | MathDB 页面 | 原文线索

V2-042832 | MATHDB-338992

Krattenthaler's Gog-Magog trapezoid correspondence conjecture

摘录 / EXCERPT: For non-negative integers m,n,k with $k\leq n$, an (m,n,k)-Gog trapezoid and an (m,n,k)-Magog trapezoid are the corresponding generalized trapezoid arrays; a minimum and maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-042833 | MATHDB-338996

The finite-character-subset conjecture for perfect phylogenies

摘录 / EXCERPT: Finite-character-subset conjecture. For each positive integer r, there exists an integer $f(r)$ such that, for any finite set X and any set C of r-state characters on X,...

conjecture | open | MathDB 页面 | 原文线索

V2-042834 | MATHDB-339010

Deloera-Haws-Köppe conjecture for matroid polytopes

摘录 / EXCERPT: Deloera-Haws-Köppe conjecture. For any matroid M, the matroid polytope $P(M)$ is both Ehrhart positive and Ehrhart unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-042835 | MATHDB-339050

Linear-union conjecture for sunflower-free set systems

摘录 / EXCERPT: Linear-union conjecture. There exists a constant $D>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042836 | MATHDB-339051

Bounded subcover conjecture for sunflower-free set systems

摘录 / EXCERPT: Bounded subcover conjecture. There exist $F_1,\ldots,F_{2k}\in\cal F$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042837 | MATHDB-339070

Kupavskii's diversity conjecture for the families \(\mathcal{D}_r\)

摘录 / EXCERPT: Kupavskii's conjecture. Fix $n\ge2k$. If, for some $r\in\mathbb{Z}_{\ge0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-042838 | MATHDB-339071

The even-order non-uniform diversity conjecture

摘录 / EXCERPT: Even-order diversity conjecture. If k is not a power of 2, then

conjecture | open | MathDB 页面 | 原文线索

V2-042839 | MATHDB-339075

The q,t Catalan difference divisibility conjecture

摘录 / EXCERPT: Let $\mathcal{C}_n(q,t)$ and $\mathcal{C}'_n(q,t)$ be the two q,t-Catalan polynomials introduced in the source. Divisibility conjecture. We have ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-042840 | MATHDB-339076

The two-shifted q-Motzkin determinant factorization conjecture

摘录 / EXCERPT: Factorization conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-042841 | MATHDB-339077

The three-shifted q-Motzkin determinant conjecture

摘录 / EXCERPT: Three-shift conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-042842 | MATHDB-339078

Positivity of even-shifted q-Motzkin Hankel determinants

摘录 / EXCERPT: Positivity conjecture. For any $k,n\in\mathbb{Z}_{>0}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-042843 | MATHDB-339106

The Z-independence conjecture for amplituhedron combinatorics

摘录 / EXCERPT: Fix $Z i\operatorname{Gr}_{>0}(k+m,n)$. Let $f,f_1,\dots,f_N$ be affine permutations in $\tilde{S_n}(-k,n-k)$, and let g be another such affine permutation. The notions of Z-ad...

conjecture | open | MathDB 页面 | 原文线索

V2-042844 | MATHDB-339185

Geelen-Gerards-Whittle conjecture on branch-width under circuit-hyperplane relaxation

摘录 / EXCERPT: Let M_1 be a matroid with a circuit-hyperplane H, and let M_2 be obtained from M_1 by relaxing H. If both M_1 and M_2 are representable over a finite field...

conjecture | open | MathDB 页面 | 原文线索

V2-042845 | MATHDB-339198

Three-dimensional Polytope Translation hardness conjecture

摘录 / EXCERPT: Let $P e\varnothing$ be a rational polytope in $\mathbb{R}^3$. The Polytope Translation hardness conjecture. The Polytope Translation problem for rational polytopes...

conjecture | open | MathDB 页面 | 原文线索

V2-042846 | MATHDB-339199

Fixed-dimensional simplex Polytope Translation hardness conjecture

摘录 / EXCERPT: For some fixed positive integer n, let $\Delta\subset \mathbb{R}^n$ be a rational simplex. The simplex Polytope Translation hardness conjecture. For some fixed n, the Polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-042847 | MATHDB-339210

The extremal symmetric-sum conjecture for perfect k-hashing

摘录 / EXCERPT: Let $k\geq 3$, let $x=(x_1,\dots,x_k)$ have nonnegative coordinates satisfying ... and let $S_h^{k-1}(x)$ denote the h-th elementary symmetric sum of $x_1,\dots,x_{k-1}$. The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-042848 | MATHDB-339218

Kronecker-product supermultiplicativity conjecture for principal Schubert evaluations

摘录 / EXCERPT: Kronecker-product supermultiplicativity conjecture. For every $w\in S_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-042849 | MATHDB-339228

The partition-weight conjecture in dimensions divisible by four

摘录 / EXCERPT: Let $d\geq4$ with $d\equiv0\pmod4$, and let $\pi=\{\pi_{i_1\ldots i_{d-1}}\}_{i_1,\ldots,i_{d-1}\geq1}$ be a $(d-1)$-partition. Let w_π be the signed fixed-point weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-042850 | MATHDB-339236

Almost-all Cartesian bi-magicness of three-dimensional boards

摘录 / EXCERPT: Let (p,q,r)-boards be the three-dimensional boards considered in the paper, and let a board be Cartesian bi-magic when it admits a Cartesian bi-magic labeling. Cartesian bi-magic...

conjecture | open | MathDB 页面 | 原文线索

V2-042851 | MATHDB-339243

Growth bounds for cyclic permutations avoiding a pattern of length three

摘录 / EXCERPT: Growth-bounds conjecture. For each pattern q of length 3 and every $n \geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-042852 | MATHDB-339246

Cross-intersection conjecture for levels of hereditary families

摘录 / EXCERPT: Cross-intersection conjecture for hereditary families. If $\mu(\mathcal{H})\geq 2r$ and $\mathcal{A},\mathcal{B}\subseteq\mathcal{H}^{(r)}$ are cross-intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-042853 | MATHDB-339247

Kirshner-Samorodnitsky conjecture on additive energy on Hamming spheres

摘录 / EXCERPT: Kirshner-Samorodnitsky conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-042854 | MATHDB-339273

Univalence conjecture for the extremal Chebyshev polynomial

摘录 / EXCERPT: Univalence conjecture. The polynomial F is univalent in the unit disk $\mathbb D$. Numerical evidence supports the assertion for all N, while it is established in the cases...

conjecture | open | MathDB 页面 | 原文线索

V2-042855 | MATHDB-339275

Parameterized Chebyshev-polynomial univalence conjecture

摘录 / EXCERPT: Parameterized univalence conjecture. The polynomial $F(q,z)$ is univalent for every $q=1,\ldots,N$ and every positive integer N. The case $q=1$ recovers the preceding extremal po...

conjecture | open | MathDB 页面 | 原文线索

V2-042856 | MATHDB-339276

Odd conjugate-polynomial extremal-value conjecture

摘录 / EXCERPT: Let ... where the coefficients a_j are real and $a_1=1$. Odd conjugate-polynomial conjecture. One has ... and the solution is unique, namely ... This is an extremal problem for conjuga...

conjecture | open | MathDB 页面 | 原文线索

V2-042857 | MATHDB-339277

The Brualdi-Ryser conjecture for partial transversals in latin squares

摘录 / EXCERPT: Brualdi-Ryser conjecture. Every latin square L of order n possesses a near transversal, and if n is odd then L possesses a transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-042858 | MATHDB-339278

The Candy Nim pile-splitting conjecture

摘录 / EXCERPT: Candy Nim pile-splitting conjecture. There exist such j and $a_1,\ldots,a_j$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-042859 | MATHDB-339279

The extremal Windsor-move conjecture for Candy Nim games

摘录 / EXCERPT: Extremal Windsor-move conjecture. There exist, not necessarily distinct, games G_1 and G_2 with $N(G_1)=N(G_2)=N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042860 | MATHDB-339294

The projective-plane core conjecture for minimally nonideal matrices

摘录 / EXCERPT: A minimally nonideal matrix is a clutter matrix that is not ideal but whose proper minors are all ideal. Its core is the submatrix consisting of its minimum-weight rows. A non-dege...

conjecture | open | MathDB 页面 | 原文线索

V2-042861 | MATHDB-339295

The cubic minimally nonideal matrix order conjecture

摘录 / EXCERPT: A cubic minimally nonideal matrix is a minimally nonideal matrix whose Lehman core is cubic, and an $n\times n$ matrix is square with n rows and n columns. Cubic mni order conj...

conjecture | open | MathDB 页面 | 原文线索

V2-042862 | MATHDB-339332

Sacrifice-size conjecture for cumulative subtraction games

摘录 / EXCERPT: Let a cumulative subtraction game be played with a finite action set, and suppose optimal play includes sacrifices by both players. Sacrifice-size conjecture. In such a game, Negat...

conjecture | open | MathDB 页面 | 原文线索

V2-042863 | MATHDB-339333

Two-pile periodicity conjecture for cumulative subtraction games

摘录 / EXCERPT: Consider cumulative subtraction on two piles, where on each turn the active player chooses a pile and plays the single-pile game on that pile. Two-pile periodicity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042864 | MATHDB-339334

Diagonal periodicity conjecture for two-pile cumulative subtraction games

摘录 / EXCERPT: Consider cumulative subtraction on two piles, and fix an integer k. A k-diagonal is a diagonal half-line of positions of the form $(x,k+x)$. Diagonal periodicity conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-042865 | MATHDB-339337

Finite-bound conjecture for independent three-variable word-equation systems

摘录 / EXCERPT: Let a system of equations be a set of word equations, and call it independent if it is not equivalent to any proper subset. A system is constant-free when its equations contain onl...

conjecture | open | MathDB 页面 | 原文线索

V2-042866 | MATHDB-339365

Forbidden-substring conjecture for greedy universal cycles

摘录 / EXCERPT: Let $\textbf{T}(n,k)$ denote the set of strings of length n over the alphabet of symbols used in the paper. Let $\gamma\in\textbf{T}(m,k)$ be a string containing a symbol...

conjecture | open | MathDB 页面 | 原文线索

V2-042867 | MATHDB-339366

Suffix-language characterization conjecture for FKM universal cycles

摘录 / EXCERPT: Let $\textbf{T}(n,k)$ denote the set of strings of length n over the paper's alphabet, and let $\textbf{S}\subseteq\textbf{T}(n,k)$ be closed under rotations. Given a string...

conjecture | open | MathDB 页面 | 原文线索

V2-042868 | MATHDB-339377

Herzog's extremal Betti numbers conjecture for square-free initial ideals

摘录 / EXCERPT: Let S be a standard graded polynomial ring, let I be a homogeneous ideal of S, and let $<$ be a term order on S. Suppose that the initial ideal $\operatorname{in}(I)$ is sq...

conjecture | open | MathDB 页面 | 原文线索

V2-042869 | MATHDB-339412

Symmetric log-concavity conjecture for peak-polynomial coefficients

摘录 / EXCERPT: Peak-polynomial coefficient conjecture. The sequence $(b_k)_{k\in\mathbb N}$ consists of non-negative integers and is symmetric and log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-042870 | MATHDB-339420

Rainbow circuit conjecture for regular matroids

摘录 / EXCERPT: Let M be a simple matroid of rank $n-1$, and let c be a colouring of its ground set $E(M)$ with n colours. A circuit is rainbow when no two of its elements have the same colo...

conjecture | open | MathDB 页面 | 原文线索

V2-042871 | MATHDB-339431

Holroyd-Talbot conjecture for levels of hereditary families

摘录 / EXCERPT: Let $\mathcal{H}$ be a hereditary family, let $\mu(\mathcal{H})$ denote the size of a smallest base of $\mathcal{H}$, and let $\mathcal{H}^{(r)}$ be its rth level. The rth leve...

conjecture | open | MathDB 页面 | 原文线索

V2-042872 | MATHDB-339432

Borg's strong form for extremal cross-intersecting levels

摘录 / EXCERPT: Let $\mathcal{H}$ be a hereditary family, with $\mathcal{H}^{(r)}$ and $\mathcal{H}^{(s)}$ its rth and sth levels. For cross-intersecting subfamilies, let...

conjecture | open | MathDB 页面 | 原文线索

V2-042873 | MATHDB-339433

The empty-family extremal conjecture for hereditary levels

摘录 / EXCERPT: Let $\mathcal{H}$ be a hereditary family, let $\mathcal{H}^{(r)}$ and $\mathcal{H}^{(s)}$ be its levels, and suppose $\mathcal{A}\subseteq\mathcal{H}^{(r)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-042874 | MATHDB-339445

Adjacent-swap connectivity conjecture for binary tropical matrix identities

摘录 / EXCERPT: Let Σ be a two-letter alphabet, let $w,v\in\Sigma^+$ with $w\neq v$, and let $n\geq 2$. Write $w\leftrightarrow v$ when w and v differ by one adjacent swap, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042875 | MATHDB-339446

Local-isolation characterization of binary tropical matrix isoterms

摘录 / EXCERPT: Let $n\geq 2$. For a word w over a two-letter alphabet, call w locally isolated in $\mathcal{UT}_n(\mathbb{T})$ if every word w' satisfying $w'\leftrightarrow w$ is inequival...

conjecture | open | MathDB 页面 | 原文线索

V2-042876 | MATHDB-339447

Decay of the isoterm fraction in binary tropical matrix identities

摘录 / EXCERPT: Let $W(\ell r,\ell(1-r))$ denote the set of binary words with the indicated numbers of the two letters. Fix $r\in(0,1)$ and $n\geq 2$, and let $i(n,\ell,r)$ be the fraction of...

conjecture | open | MathDB 页面 | 原文线索

V2-042877 | MATHDB-339448

Asymptotic equivalence of tropical matrix identities for random binary words

摘录 / EXCERPT: Fix $r\in(0,1)$ and consider words in $W(\ell r,\ell(1-r))$. Choose w uniformly from this set, choose u uniformly from the $\sim_2$-class of w, and choose v uniformly from...

conjecture | open | MathDB 页面 | 原文线索

V2-042878 | MATHDB-339449

Decay of the isoterm-class ratio for binary tropical matrix identities

摘录 / EXCERPT: Fix $r\in(0,1)$. For binary words in $W(\ell r,\ell(1-r))$, let $e(\ell,r)$ be the ratio of the number of $\mathcal{UT}_2(\mathbb{T})$ isoterms to the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-042879 | MATHDB-339450

Generalised Catalan conjecture for largest binary tropical matrix equivalence classes

摘录 / EXCERPT: Let $W(\ell_a,\ell_a)$ be the set of binary words containing ℓ_a occurrences of each letter, and consider its $\mathcal{UT}_2(\mathbb{T})$ equivalence classes. Generalised Cat...

conjecture | open | MathDB 页面 | 原文线索

V2-042880 | MATHDB-339459

Miana-Ohtsuka-Romero's Catalan triangle cubic-sum identity

摘录 / EXCERPT: Let $B_{n,k}:=\frac{k}{n}{2n\choose n-k}$ for $n\geqslant k\geqslant 0$, and let $r=\min\{n,m\}$ and $s=\max\{n,m\}$. Miana-Ohtsuka-Romero's conjecture. For all positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-042881 | MATHDB-339466

Universal combinatorial list-decodability for finite simple groups

摘录 / EXCERPT: Universal finite-simple-group conjecture. The class of finite simple groups is universally CombEcon.

conjecture | open | MathDB 页面 | 原文线索

V2-042882 | MATHDB-339470

Narayanan's conjecture on the growth of the extremal function away from triangular boundaries

摘录 / EXCERPT: Narayanan's conjecture. For every $C>0$, there exists a positive integer ℓ such that, for every $m\in\mathbb{N}$ satisfying the displayed inequalities,

conjecture | open | MathDB 页面 | 原文线索

V2-042883 | MATHDB-339471

Narayanan's sublinear-growth conjecture for the extremal function

摘录 / EXCERPT: Narayanan's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-042884 | MATHDB-339472

Threshold conjecture for simple connectivity in random 2-complexes

摘录 / EXCERPT: Let $Y_2(n,p)$ be the Linial-Meshulam random 2-complex on n vertices, where each possible 2-simplex is included independently with probability p, and let $\gamma:=4^4/3^3$....

conjecture | open | MathDB 页面 | 原文线索

V2-042885 | MATHDB-339477

Unbounded rank gaps in differential posets

摘录 / EXCERPT: Unbounded rank-gap conjecture. In any differential poset,

conjecture | open | MathDB 页面 | 原文线索

V2-042886 | MATHDB-339480

The inflation structure conjecture for cyclic matroids

摘录 / EXCERPT: Inflation structure conjecture. If t is even, then M can be obtained from a spike or a swirl by a sequence of inflations. If t is odd, then M can be obtained from a wheel o...

conjecture | open | MathDB 页面 | 原文线索

V2-042887 | MATHDB-339487

The K-theoretic coefficient-sum conjecture for glides and kaons

摘录 / EXCERPT: A weak composition is a finite sequence of nonnegative integers. For weak compositions a and b, let $M^a_b(\beta)$ be the coefficient of the glide polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-042888 | MATHDB-339488

The kaon-glide positivity conjecture

摘录 / EXCERPT: A weak composition is a finite sequence of nonnegative integers. For weak compositions a and b, let $\overline{\mathfrak{P}}_a$ denote the kaon associated with a, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042889 | MATHDB-339497

Binary matroid conjecture excluding an independent set and a triangle

摘录 / EXCERPT: Let s be a positive integer. A simple binary matroid is a binary matroid with no loops or parallel elements; an induced I_s-restriction and an induced $\operatorname{PG}(1,2)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-042890 | MATHDB-339514

The switch map's dinv and fixed-point conjecture

摘录 / EXCERPT: Let $\mathcal{P}_{3,n}$ be the set of $(3,n)$-parking functions, let $\operatorname{pides}(\pi)=2^a1^b$, let Φ be the involution whose fixed points are the fixed parking funct...

conjecture | open | MathDB 页面 | 原文线索

V2-042891 | MATHDB-339518

Hamada's conjecture on p-ranks of geometric designs

摘录 / EXCERPT: Let D be a design with the parameters of a geometric design G in $\operatorname{PG}(n,q)$ or $\operatorname{AG}(n,q)$, where $q=p^m$. Let the p-rank denote the rank of the in...

conjecture | open | MathDB 页面 | 原文线索

V2-042892 | MATHDB-339557

Square-root recurrence conjecture for extremal reciprocal algebraic integers

摘录 / EXCERPT: Let $\mathrm{mr}(d)$ be the minimum of the houses of reciprocal algebraic integers of degree d that are not roots of unity. An algebraic integer attaining $\mathrm{mr}(d)$ is cal...

conjecture | open | MathDB 页面 | 原文线索

V2-042893 | MATHDB-339559

Composite-degree extremal reciprocal polynomial conjecture

摘录 / EXCERPT: Composite-degree extremal reciprocal polynomial conjecture. If $R_{2p_i}(x)$ is the minimal polynomial of the extremal reciprocal algebraic integer of degree $2p_i$, then

conjecture | open | MathDB 页面 | 原文线索

V2-042894 | MATHDB-339565

Conjecture on exclusion from the sets S^{TFFF}(n) and S^{FFFF}(n)

摘录 / EXCERPT: Let $S^{TFFF}(n)$ and $S^{FFFF}(n)$ be the classes of cyclically presented groups defined in the source, and let $\Gamma_n(1,n/2-1)$ be the corresponding group. Exclusion conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-042895 | MATHDB-339568

Cuckler-Yuster conjecture on cyclic triangle factors in regular tournaments

摘录 / EXCERPT: Cuckler-Yuster conjecture. If n is an odd positive multiple of 3, then every regular tournament on n vertices has a cyclic triangle factor.

conjecture | open | MathDB 页面 | 原文线索

V2-042896 | MATHDB-339589

Affine evacuation weight-vector conjecture

摘录 / EXCERPT: Let w be an affine permutation with AMBC data ... where P and Q are tabloids of shape λ, and let $r(w)$ denote the rotation of w. For a vector, call its dominant repr...

conjecture | open | MathDB 页面 | 原文线索

V2-042897 | MATHDB-339635

Conjecture on diagonal and off-diagonal online random Ramsey numbers

摘录 / EXCERPT: Let $m,n\be 3$ and let $p\be(0,1)$. The online random Ramsey number $\tilde{r}_{\operatorname{rand}}(m,n)$ is the maximum, over p, of the number of turns needed by Builder to win...

conjecture | open | MathDB 页面 | 原文线索

V2-042898 | MATHDB-339636

Conjectured asymptotics for the Subgraph Query Problem on cliques

摘录 / EXCERPT: For $p\in(0,1)$, let $f(H,p)$ be the minimum number of turns, over Builder strategies, needed to build a copy of H with probability at least $1/2$ when each queried edge succeeds...

conjecture | open | MathDB 页面 | 原文线索

V2-042899 | MATHDB-339674

The d-sequence conjecture for coefficients of uniform-matroid Kazhdan-Lusztig polynomials

摘录 / EXCERPT: For integers m and d, and $0\leq i\leq d$, define ... A finite sequence of real numbers is an n-sequence when the associated polynomial has real zeros of one common sign. The...

conjecture | open | MathDB 页面 | 原文线索

V2-042900 | MATHDB-339710

Arc-complexity conjecture for uniform matroids

摘录 / EXCERPT: Let $r,n\to\mathbb{N}$ with $n\geq r$. The uniform matroid of rank r on n elements is $U_{r,n}=({\{1,2,\ldots,n\}},\mathcal{I}_{r,n})$, where ... Let $T=\{1,2,\ldots,r\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-042901 | MATHDB-339711

Unbounded arc-complexity conjecture for gammoids

摘录 / EXCERPT: Let $\mathrm{C}_A(M)$ denote the arc complexity of a gammoid $M=(E,\mathcal{I})$, where $|E|$ is its ground-set size. Unbounded arc-complexity conjecture for gammoids. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-042902 | MATHDB-339714

The generalized Erdős-Ginzburg-Ziv constant conjecture for cyclic groups

摘录 / EXCERPT: Let C_n be the cyclic group of order n, and let $\mathsf s^{(k)}(C_n)$ denote its generalized Erdős-Ginzburg-Ziv constant: the least integer ℓ such that every sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-042903 | MATHDB-339721

Nunge's unimodality conjecture for segmented permutation statistics

摘录 / EXCERPT: Nunge's unimodality conjecture. For every positive integer n, the sequences

conjecture | open | MathDB 页面 | 原文线索

V2-042904 | MATHDB-339728

The four-term sum-product conjecture for fields of characteristic different from 2 and 3

摘录 / EXCERPT: Four-term sum-product conjecture. There are $a,b,c,d\in F$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-042905 | MATHDB-339753

Erdős-Faudree-Rousseau-Schelp conjecture on cycle-clique Ramsey numbers

摘录 / EXCERPT: Let C_n be the cycle on n vertices, K_m the complete graph on m vertices, and let $R(G,H)$ denote the two-colour Ramsey number of graphs G and H. Erdős-Faudree-Rousseau...

conjecture | open | MathDB 页面 | 原文线索

V2-042906 | MATHDB-339772

The refined pipe-dream and quarter-plane walk bijection conjecture

摘录 / EXCERPT: Fix integers n and k. Consider pipe dreams P with $\omega_P\in\mathfrak{S}_n$ whose factorization into atomics consists of k identity permutations, and quarter-plane walks...

conjecture | open | MathDB 页面 | 原文线索

V2-042907 | MATHDB-339773

The Steep-Bounce bijection conjecture

摘录 / EXCERPT: Fix integers n and k with $k\le n$. A Dyck path is a lattice path of semilength n from $(0,0)$ to (n,n) that stays weakly on one side of its diagonal; a nested pair...

conjecture | open | MathDB 页面 | 原文线索

V2-042908 | MATHDB-339801

Bounded covering radius conjecture for lattices generated by unit vectors

摘录 / EXCERPT: Let $\mathcal{L}$ be a lattice generated by vectors of Euclidean norm 1, and let $\rho_\infty(\mathcal{L})$ denote its covering radius with respect to the ℓ_∞ norm. Uni...

conjecture | open | MathDB 页面 | 原文线索

V2-042909 | MATHDB-339809

Polynomial counting conjecture for square-board chess pieces

摘录 / EXCERPT: Let $u_{\mathbb P}(q;n)$ denote the counting function for q copies of a piece $\mathbb P$ on the square board. Polynomial counting conjecture. The rook, semirook, semibishop, and...

conjecture | open | MathDB 页面 | 原文线索

V2-042910 | MATHDB-339811

Largest-denominator conjecture for semiqueens

摘录 / EXCERPT: Let F_k be the Fibonacci numbers, and let $\mathbb Q^{21}$ denote the semiqueen. For q semiqueens, consider all vertices of the associated inside-out polytope. Semiqueen maxima...

conjecture | open | MathDB 页面 | 原文线索

V2-042911 | MATHDB-339813

Kotěšovec's Fibonacci-period conjecture for queens

摘录 / EXCERPT: Let F_q be the qth Fibonacci number, let $[F_q]=\{1,2,\ldots,F_q\}$, and let $u_{\mathbb Q}(q;n)$ be the counting quasipolynomial for q queens. Kotěšovec's conjecture. The pe...

conjecture | open | MathDB 页面 | 原文线索

V2-042912 | MATHDB-339814

Queen inside-out-polytope denominator conjecture

摘录 / EXCERPT: Let F_q be the qth Fibonacci number and let $[F_q]=\{1,2,\ldots,F_q\}$. For the inside-out polytope associated with q queens, let D denote its denominator. Queen denominato...

conjecture | open | MathDB 页面 | 原文线索

V2-042913 | MATHDB-339820

Nakamura's strong c-Wilf equivalence conjecture

摘录 / EXCERPT: Let $\mathfrak{S}_m$ denote the symmetric group on m letters, and let $\pi,\tau\in\mathfrak{S}_m$. Write $\pi\sim\tau$ for ordinary Wilf-equivalence and...

conjecture | open | MathDB 页面 | 原文线索

V2-042914 | MATHDB-339831

The Generalized Delta conjecture

摘录 / EXCERPT: Let D be a partially labelled decorated Dyck path of size $n+m$ with m zero labels, n nonzero labels, and k decorated rises. Its area word is $a(D)=a_1(D)\cdots a_{n+m}(D)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042915 | MATHDB-339832

Positivity for the C_α refinement

摘录 / EXCERPT: Let Λ be the ring of symmetric functions, let ∇ and the Delta operators act on it, and let ϕ be defined by ... For a composition...

conjecture | open | MathDB 页面 | 原文线索

V2-042916 | MATHDB-339833

Oda's strong factorization conjecture for smooth fans

摘录 / EXCERPT: Let Σ_1 and Σ_2 be smooth fans with the same support. A smooth star subdivision is the subdivision of a fan obtained by inserting the ray through the sum of the primi...

conjecture | open | MathDB 页面 | 原文线索

V2-042917 | MATHDB-339851

The insertion monotonicity conjecture for right palindromic length

摘录 / EXCERPT: Let Σ be an alphabet, let $w\in\Sigma^+$ be a nonempty word, and let $I_R(w)$ denote its right palindromic length. A letter $x\in\Sigma$ can be inserted into w to obtain a...

conjecture | open | MathDB 页面 | 原文线索

V2-042918 | MATHDB-339852

The vanishing proportion conjecture for square-free M-unambiguous ternary and larger alphabets

摘录 / EXCERPT: Let Σ be an ordered alphabet with $|\Sigma|\ge 3$. A word $w\in\Sigma^*$ is square-free-M-unambiguous when it is not M-equivalent to any other distinct square-free word....

conjecture | open | MathDB 页面 | 原文线索

V2-042919 | MATHDB-339853

Large-pillar conjecture for equiangular sets

摘录 / EXCERPT: Let an equiangular set have angle α and base size K, and let a pillar be one of the structures in the paper's pillar decomposition. Large-pillar conjecture. There is a con...

conjecture | open | MathDB 页面 | 原文线索

V2-042920 | MATHDB-339889

Alphabet-independence conjecture for the threshold avoiding powers and anti-powers

摘录 / EXCERPT: Let $N_\alpha(k,k)$ denote the quantity defined in the paper as the relevant threshold for words over an alphabet of size α, avoiding k-powers and k-anti-powers. Alphabe...

conjecture | open | MathDB 页面 | 原文线索

V2-042921 | MATHDB-339902

Reiner-Tenner-Yong's barely set-valued tableau enumeration conjecture

摘录 / EXCERPT: A barely set-valued semistandard Young tableau is a set-valued semistandard Young tableau in which exactly one square is assigned two integers and every other square is assigned a...

conjecture | open | MathDB 页面 | 原文线索

V2-042922 | MATHDB-339903

Reiner-Tenner-Yong's Fomin-Kirilov polynomial ratio conjecture

摘录 / EXCERPT: Let w be a dominant permutation, meaning a 132-avoiding permutation, whose Lehmer code $\lambda=(c_1(w),\ldots,c_n(w))$ is the rectangular staircase shape $\lambda=\delta_d(b^a)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042923 | MATHDB-339954

Extremal monotonicity conjecture for the side length function of regular polygon stars

摘录 / EXCERPT: Let P_n be a regular polygon with n sides, let s_{2l+1} denote the side length function for inscribed $(2l+1)$-pointed stars, and call a point of P_n a midpoint crossing or...

conjecture | open | MathDB 页面 | 原文线索

V2-042924 | MATHDB-339956

The inclusion conjecture for decomposable clutters

摘录 / EXCERPT: Let $4\mathfrak{C}_d4$ denote the class of chordal d-uniform clutters and let $4\mathfrak{C}'_d4$ denote the class of decomposable d-uniform clutters. The inclusion conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042925 | MATHDB-339966

Exponential density conjecture for matroids excluding an induced I_t-restriction

摘录 / EXCERPT: Exponential density conjecture. If $M=(E,G)$ is a full-rank, r-dimensional matroid with no induced I_t-restriction, then

conjecture | open | MathDB 页面 | 原文线索

V2-042926 | MATHDB-339967

Alternating-string conjecture for minimal autocorrelation

摘录 / EXCERPT: Alternating-string conjecture. The alternating string $x=\texttt{1010...}$ minimizes the autocorrelation coefficient κ^2.

conjecture | open | MathDB 页面 | 原文线索

V2-042927 | MATHDB-339978

Füredi's higher-dimensional Erdős-Szekeres conjecture

摘录 / EXCERPT: Let $ES_d(n)$ be the minimum integer N such that every set of N points in $\mathbb{R}^d$ in general position contains n points in convex position. Füredi's conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-042928 | MATHDB-339984

Conjecture on vanishing values of seaweed composition counts

摘录 / EXCERPT: Let $C_{p,q}(n,k)$ denote the composition-counting function studied in the paper. Vanishing conjecture. ... for integers $s,p,k\geq 0$ with $k=2^p$. The values of $n_{k,4}$ suggest t...

conjecture | open | MathDB 页面 | 原文线索

V2-042929 | MATHDB-339985

Conjectured first diagonal formula for $C_{2,t}$

摘录 / EXCERPT: Let $C_{2,t}(n,k)$ be the composition-counting function. First-diagonal conjecture. If t is even, then ... for integers $t>2$ and $s\geq 0$. This describes one of the first nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-042930 | MATHDB-339987

Conjectured recurrence for the red cells of $C_{4,4}$

摘录 / EXCERPT: Let $C_{4,4}(n,k)$ be the composition-counting function. Red-cell recurrence conjecture. ... This conjecturally describes the red cells in Table 4; no proof or resolution is supplied...

conjecture | open | MathDB 页面 | 原文线索

V2-042931 | MATHDB-340002

Engbers-Galvin conjecture on colorings of the Hamming cube

摘录 / EXCERPT: Engbers-Galvin conjecture. For each fixed q,

conjecture | open | MathDB 页面 | 原文线索

V2-042932 | MATHDB-340003

The crown graph extremal representation-number conjecture

摘录 / EXCERPT: Let $H_{n,n}$ be the crown graph obtained from the complete bipartite graph on two parts of size n by deleting a perfect matching. The crown graph extremal conjecture. $H_{n,n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-042933 | MATHDB-340005

The incompatible intersection conjecture for 3-wise and 2-wise intersecting families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $2^{[n]}$ denote the family of all subsets of $[n]$, and let $\mathcal F\subset 2^{[n]}$. The family is r-wise t-intersecting if every r members...

conjecture | open | MathDB 页面 | 原文线索

V2-042934 | MATHDB-340006

The shadow-doubling conjecture for 3-wise intersecting families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $2^{[n]}$ denote the family of all subsets of $[n]$, and let $\mathcal F\subset 2^{[n]}$. Define the immediate shadow by ... and set...

conjecture | open | MathDB 页面 | 原文线索

V2-042935 | MATHDB-340015

Infinitely many cyclic groups with the weak acyclic matching property

摘录 / EXCERPT: Weak acyclic matching infinitude conjecture. There are infinitely many n for which $\mathbb{Z}/n\mathbb{Z}$ has the weak acyclic matching property.

conjecture | open | MathDB 页面 | 原文线索

V2-042936 | MATHDB-340020

Andrews-Gordon-type q-series identities for the Kanade-Russell conjectures

摘录 / EXCERPT: Kanade-Russell q-series conjectures. The following identities are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-042937 | MATHDB-340021

The anti-power doubling conjecture for shifted Thue-Morse words

摘录 / EXCERPT: Let b be the Thue-Morse word, and let $AP_j(\mathbf{t},k)$ denote the set of lengths of anti-power j-fixes of order k in b. For fixed $j,k\in\mathbb{Z}^{\geq 0}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-042938 | MATHDB-340022

The asymptotic anti-power conjecture for shifted Thue-Morse words

摘录 / EXCERPT: Let $\gamma_j(k)$ be the function measuring the least relevant anti-power length of order k for the shift parameter j in the Thue-Morse word. For each nonnegative integer j,...

conjecture | open | MathDB 页面 | 原文线索

V2-042939 | MATHDB-340043

Asymptotic sparsity of composites in finite-group semirings

摘录 / EXCERPT: Let G be a finite group, let $\mathbb{N}[G]_n$ denote the n-th grade of the group semiring, and let $\operatorname{Comp}(\mathbb{N}[G])_n$ denote its composite elements in that...

conjecture | open | MathDB 页面 | 原文线索

V2-042940 | MATHDB-340050

Possible circular critical exponents for Rudin-Shapiro factors

摘录 / EXCERPT: Let the Rudin-Shapiro sequence be viewed as an infinite word, and consider each nonempty factor as a circular word. Its circular critical exponent is the largest exponent of a repe...

conjecture | open | MathDB 页面 | 原文线索

V2-042941 | MATHDB-340076

The Erdős-Ko-Rado conjecture for t-intersecting families of perfect matchings

摘录 / EXCERPT: Let K_{2n} be the complete graph on $2n$ vertices, and let $\mathcal{F}$ be a family of perfect matchings of K_{2n}. The family is t-intersecting if any two of its members sh...

conjecture | open | MathDB 页面 | 原文线索

V2-042942 | MATHDB-340078

Rationality conjecture for generating functions of anchored bounded-gap permutations

摘录 / EXCERPT: A permutation of $1,2,\ldots,n$ is k-bounded if every pair of consecutive entries differs by no more than k, and it is anchored if its first entry is 1 and its last entry is...

conjecture | open | MathDB 页面 | 原文线索

V2-042943 | MATHDB-340088

The nested matching ordered Ramsey number conjecture

摘录 / EXCERPT: Let NM_k be the ordered matching on $[2k]$ whose edges are (i,j) with $i+j=2k+1$, and let $r_<(G,H)$ denote the off-diagonal ordered Ramsey number. Nested matching conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-042944 | MATHDB-340094

Jelínek's equidistribution bijection conjecture for restricted permutations and Fishburn matrices

摘录 / EXCERPT: Jelínek's bijection conjecture. For every n, there is a bijection α between $Sn(!raisebox{-0.3em}{ begin{tikzpicture}[line width=0.7pt, scale=0.15] tikzstyle{disc} = [circ...

conjecture | open | MathDB 页面 | 原文线索

V2-042945 | MATHDB-340095

Jelínek's symmetric joint distribution conjecture for LRmax and RLmax

摘录 / EXCERPT: Jelínek's distribution conjecture. For any n, LRmax and RLmax have symmetric joint distribution on $Sn(!raisebox{-0.3em}{ begin{tikzpicture}[lin...

conjecture | open | MathDB 页面 | 原文线索

V2-042946 | MATHDB-340096

The Billera-Lee upper-bound conjecture for spheres

摘录 / EXCERPT: Billera-Lee upper-bound conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-042947 | MATHDB-340097

The Billera-Lee upper-bound conjecture for triangulated spheres

摘录 / EXCERPT: Billera-Lee upper-bound conjecture. For every integer i with $0\leq i\leq d$,

conjecture | open | MathDB 页面 | 原文线索

V2-042948 | MATHDB-340099

Benjamini-Häggström-Mossel range conjectures for graph-indexed random walks

摘录 / EXCERPT: Benjamini-Häggström-Mossel range conjectures. The weak conjecture asserts

conjecture | open | MathDB 页面 | 原文线索

V2-042949 | MATHDB-340104

The root-difference conjecture for free hyperplane arrangements

摘录 / EXCERPT: Root-difference conjecture. Then $d=d_i$ for some i, or $d>d_\ell$. The source notes that this conclusion was established under the additional assumption that...

conjecture | open | MathDB 页面 | 原文线索

V2-042950 | MATHDB-340108

The weak acyclic matching property for abelian groups

摘录 / EXCERPT: Let G be an abelian group. A pair of finite subsets $A,B\subseteq G$ with $\#A=\#B$ and $A\cap(A+B)=\emptyset$ has an acyclic matching if there is a matching $f:A\to B$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-042951 | MATHDB-340109

The terminal acyclicity-sequence conjecture for matchings

摘录 / EXCERPT: Let G be an abelian group, and let $A,B\subseteq G$ be finite sets with $\#A=\#B$. Let $\mathcal{M}(A,B)$ be the set of matchings from A to B, and let $F^{(t)}_{(A,B)}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-042952 | MATHDB-340110

The acyclic matching conjecture with unit acyclicity sequence

摘录 / EXCERPT: Let G be an abelian group, and let $A,B\subseteq G$ be finite subsets with $A\cap(A+B)=\emptyset$. Write $\mathcal{M}(A,B)$ for the set of matchings from A to B. An acyclicit...

conjecture | open | MathDB 页面 | 原文线索

V2-042953 | MATHDB-340111

The unit acyclicity-sequence criterion for strong acyclic matching

摘录 / EXCERPT: Let A and B be finite subsets of an abelian group, and let $\mathcal{M}(A,B)$ be the set of matchings from A to B. Say that A is acyclically matched to B if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-042954 | MATHDB-340115

Equivalent quasi-transversal conjecture for Latin squares

摘录 / EXCERPT: Quasi-transversal conjecture. Every Latin square has a quasi-transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-042955 | MATHDB-340116

Rodney's conjecture on 2-transversals in Latin squares

摘录 / EXCERPT: Rodney's conjecture. Every Latin square has a 2-transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-042956 | MATHDB-340121

Positive proportion conjecture for partitions into MSTD subsets

摘录 / EXCERPT: A k-decomposition into MSTD subsets is a partition of $I_r=\{1,2,\ldots,r\}$ into k subsets, each having more pairwise sums than pairwise differences. Positive proportion conje...

conjecture | open | MathDB 页面 | 原文线索

V2-042957 | MATHDB-340151

Delannoy face-number conjecture for state polytopes of P(l,0,...,0)

摘录 / EXCERPT: Let n and l be positive integers, and let ... For the polytope $P(\ell)$, write f_k for the number of its faces of dimension k. Delannoy face-number conjecture. The state pol...

conjecture | open | MathDB 页面 | 原文线索

V2-042958 | MATHDB-340166

The growth-rate conjecture for the number of Hadamard matrices

摘录 / EXCERPT: Let $H(n)$ denote the number of Hadamard matrices of order n. If a Hadamard matrix of order n exists, permuting its rows and columns already gives at least $(n!)^2$ distinct Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-042959 | MATHDB-340167

The Deneanu-Vu conjecture on normal sign matrices

摘录 / EXCERPT: A real matrix is normal if it commutes with its adjoint; for real matrices this means ... Let a sign matrix be a matrix whose entries belong to $\{\pm1\}$. Since every real symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-042960 | MATHDB-340192

Freiman's volume conjecture with dimension

摘录 / EXCERPT: Let $k,T,d\in \mathbb{N}$, and let c,b satisfy ... where ... Here $\operatorname{vol}(k,T,d)$ is the maximum volume of a set of integers of cardinality k, doubling T, and additi...

conjecture | open | MathDB 页面 | 原文线索

V2-042961 | MATHDB-340193

Volume bound for one-dimensional sets composed of few segments

摘录 / EXCERPT: Let A be a 1-dimensional set, meaning a set with additive dimension 1, and let s be the minimum number of disjoint segments into which it can be decomposed. If...

conjecture | open | MathDB 页面 | 原文线索

V2-042962 | MATHDB-340195

The balanced cyclic composition proportion conjecture

摘录 / EXCERPT: Balanced cyclic composition proportion conjecture. For even n,

conjecture | open | MathDB 页面 | 原文线索

V2-042963 | MATHDB-340202

Hamidoune's atom cardinality conjecture for torsion-free groups

摘录 / EXCERPT: Let G be a torsion-free group, let C be a finite subset of G containing the identity element, and let n be a positive integer. For a finite subset $X\subseteq G$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-042964 | MATHDB-340205

Frid's Fibonacci-word palindromic-length lower-bound conjecture

摘录 / EXCERPT: Let $\boldsymbol{f}$ be the Fibonacci word, namely the fixed point of the morphism $0\mapsto 01$, $1\mapsto 0$. Let τ denote the golden ratio. Frid's Fibonacci-word lower-boun...

conjecture | open | MathDB 页面 | 原文线索

V2-042965 | MATHDB-340211

Erdős's weak Dirac conjecture

摘录 / EXCERPT: Let $\mathcal{P}\subset \mathbb{P}^{2}_{\mathbb{C}}$ be a finite set of mutually distinct n points, and let $\mathcal{L}(\mathcal{P})$ be the set of lines determined by pairs of...

conjecture | open | MathDB 页面 | 原文线索

V2-042966 | MATHDB-340229

The realization conjecture for edge-preserving symmetries of 3-polytopes

摘录 / EXCERPT: Realization conjecture. For every convex 3-dimensional polytope P, there exists a convex 3-dimensional polytope Q combinatorially equivalent to P such that the lengths of eac...

conjecture | open | MathDB 页面 | 原文线索

V2-042967 | MATHDB-340263

Finiteness conjecture for simple connected ordered matroids with prescribed broken-circuit topology

摘录 / EXCERPT: Let d,k be positive integers. For an ordered matroid M, write $\overline{BC_<(M)}$ for its reduced broken circuit complex, and let $h_{d-1}(\overline{BC_<(M)})$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-042968 | MATHDB-340276

Las Vergnas's second Tutte polynomial conjecture for binary matroids

摘录 / EXCERPT: Let M be a binary matroid, let $T_M(x,y)$ denote its Tutte polynomial, and let z be an integer. Las Vergnas's second conjecture. The value ... is an odd integer for every binary...

conjecture | open | MathDB 页面 | 原文线索

V2-042969 | MATHDB-340290

Nonnegative-coefficient property of the parabolic F-triangle

摘录 / EXCERPT: F-triangle conjecture. The rational function $F_{\alpha}(x,y)$ is a polynomial with nonnegative integer coefficients if and only if α has at most one part exceeding 1.

conjecture | open | MathDB 页面 | 原文线索

V2-042970 | MATHDB-340291

Davoodi-Javadi-Omoomi lower-bound conjecture for sigma clique coverings

摘录 / EXCERPT: Let $K_t(d)$ be the complete t-partite graph with each part of size d. Davoodi-Javadi-Omoomi's conjecture. There exists a function f and a constant $c>0$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-042971 | MATHDB-340296

Precise anchored rectangle packing conjecture

摘录 / EXCERPT: Let $m(n)$ be the minimum, over all sets of n distinct points in the unit square $U=[0,1]^2$ containing the origin, of the maximum total area of an anchored rectangle packing. Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-042972 | MATHDB-340297

Symmetry conjecture for self-inverse point-order permutations

摘录 / EXCERPT: Let π be a permutation of $\{1,2,\ldots,n-1\}$, and let $M(\pi)$ denote the minimum possible maximal anchored-packing area among point sets satisfying π. A permutation is s...

conjecture | open | MathDB 页面 | 原文线索

V2-042973 | MATHDB-340298

Mountain-end anchored packing conjecture

摘录 / EXCERPT: Consider a mountain permutation on the points, and suppose it starts with m and ends with $(m-1,m-2,\ldots,1)$ for some $m\leq n$. Let k be the proportion of the rectangular re...

conjecture | open | MathDB 页面 | 原文线索

V2-042974 | MATHDB-340333

Archdeacon-Dinitz-Mattern-Stinson distinct partial sums conjecture

摘录 / EXCERPT: Archdeacon-Dinitz-Mattern-Stinson conjecture. For any cyclic group $\mathbb{Z}_n$ and any subset $A\subseteq\mathbb{Z}_n\setminus\{0\}$, there is an ordering of the elements of A...

conjecture | open | MathDB 页面 | 原文线索

V2-042975 | MATHDB-340373

Las Vergnas's conjecture on simplicial topes

摘录 / EXCERPT: Las Vergnas's conjecture. Every simple oriented matroid ${\cal M}$ of rank r has an acyclic reorientation $ _{-F}{\cal M}$ that has exactly r extreme points, or equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-042976 | MATHDB-340374

Mutation connectivity conjecture for uniform matroid polytopes of rank 4

摘录 / EXCERPT: Mutation connectivity conjecture for uniform matroid polytopes. Every pair of uniform matroid polytopes of rank 4 on the same ground set can be transformed into each other by a f...

open_problem | open | MathDB 页面 | 原文线索

V2-042977 | MATHDB-340387

The optimal singularity-probability conjecture for random symmetric Bernoulli matrices

摘录 / EXCERPT: Let M_n be a random $n\times n$ symmetric matrix whose upper-diagonal entries are independent Bernoulli random variables taking the values 1 and -1 with probability $1/2$ eac...

conjecture | open | MathDB 页面 | 原文线索

V2-042978 | MATHDB-340398

Linear average-length conjecture for the Zeckendorf Game

摘录 / EXCERPT: For a uniformly random Zeckendorf Game on n initial copies of 1, let M_n be the number of moves until the Zeckendorf decomposition of n is reached, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-042979 | MATHDB-340399

Deterministic Game is Best for Tribonacci

摘录 / EXCERPT: Let the Tribonacci game be the game on representations of an integer in which legal moves are those prescribed by the Tribonacci recurrence, and let the deterministic game be the g...

conjecture | open | MathDB 页面 | 原文线索

V2-042980 | MATHDB-340474

Added good box conjecture for combinatorial wall-crossing

摘录 / EXCERPT: Let two Young diagrams be related by adding one box to the first row, with the corresponding parameter in the interval $(0,1/n)$. Added good box conjecture. At every step of the ab...

conjecture | open | MathDB 页面 | 原文线索

V2-042981 | MATHDB-340478

Ajjanagadde's entropy conjecture for sums from Sidon sets

摘录 / EXCERPT: Let A be a Sidon set in an ambient abelian group, and let X and Y be independent random variables taking values in A. Let H denote Shannon entropy. Ajjanagadde's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-042982 | MATHDB-340479

Recurrence conjecture for the coefficients of $W_d(t)$

摘录 / EXCERPT: Let $W_d(t)$ be a power series, and write $W_d[t^k]$ for the coefficient of t^k in $W_d(t)$. Let $k\in\mathbb{N}$ and $d\geq 2k$. Recurrence conjecture. For fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-042983 | MATHDB-340485

Asymptotic density conjecture for cutoffs in take-away games

摘录 / EXCERPT: Let a cutoff be a rational number at which the associated take-away-game structure changes, and let $\gamma(n)$ denote the number of cutoffs up to n. Asymptotic density conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-042984 | MATHDB-340502

Cubic-curve extremal conjecture for blocking points

摘录 / EXCERPT: Cubic-curve extremal conjecture. If $|B|=n$, then all points in $P\cup B$ lie on a cubic curve.

conjecture | open | MathDB 页面 | 原文线索

V2-042985 | MATHDB-340513

Indecomposability conjecture for matroids arising from finite projective spaces

摘录 / EXCERPT: A matroid arising from a projective space over a finite field is a matroid constructed from the points and lines of that projective space. Indecomposability conjecture. All matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-042986 | MATHDB-340529

The simpliciality conjecture for arrangements with fewer than half double points

摘录 / EXCERPT: Let $\mathcal{A}$ be an arrangement consisting of n pseudolines, and let $t^\mathcal{A}_2$ denote its number of double points. Its characteristic polynomial $\chi(\mathcal{A},t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-042987 | MATHDB-340530

Erdős' asymptotic conjecture for bounded-multiplicity line arrangements

摘录 / EXCERPT: Let $k\in\mathbb{N}$ with $k\geq 5$, and let $\mathfrak{A}_k$ be the set of isomorphism classes of line arrangements in $\mathbb{P}^2(\mathbb{R})$ having multiplicity at most k....

conjecture | open | MathDB 页面 | 原文线索

V2-042988 | MATHDB-340531

Rota-Welsh conjecture on the Whitney numbers of matroids

摘录 / EXCERPT: Soit M un matroïde et soit $r=\operatorname{rk}(M)$. Pour chaque entier k, $W_k(M)$ désigne le nombre de plats de rang k de M. Conjecture de Rota-Welsh. ... est une suite log...

conjecture | open | MathDB 页面 | 原文线索

V2-042989 | MATHDB-340584

The structural conjecture for η-closed subsets of tensor products

摘录 / EXCERPT: Let d be a positive integer and let $ obreak\boldsymbol{\text{?}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-042990 | MATHDB-340613

Refined ballot-odd-order equinumerosity conjecture

摘录 / EXCERPT: Let $B(n)$ be the set of ballot permutations of size n, and let $b(n,d)$ be the number of elements of $B(n)$ with exactly d descents. Let $P(n)$ be the set of odd order permuta...

conjecture | open | MathDB 页面 | 原文线索

V2-042991 | MATHDB-340616

The sparse simplicial-complex index conjecture

摘录 / EXCERPT: Let $\mathcal H\subset 2^{[n]}$ be a simplicial complex, and let $\mathcal H^{*2}_{\Delta}$ be its deleted join. Its $\mathbb Z_2$-index, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-042992 | MATHDB-340617

Grytczuk's $(k+2)$-conjecture for k-Thue sequences

摘录 / EXCERPT: A sequence is k-Thue when every arithmetic subsequence with common difference d, for $1\leq d\leq k$, is non-repetitive, meaning that it contains no two consecutive identical b...

conjecture | open | MathDB 页面 | 原文线索

V2-042993 | MATHDB-340618

Samotij-Sudakov's Kleitman analog for projective cubes in b\mathbb{Z}_{2^n}d

摘录 / EXCERPT: Let $n\geq 2$ and let M be an integer. A family $\mathcal{F}\subseteq \mathbb{Z}_{2^n}$ has size $|\mathcal{F}|=M$. A family is centred if there is an $i\in[n+1]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-042994 | MATHDB-340619

Samotij's theorem analog for projective cubes in $\mathbb{Z}_{2^n}$

摘录 / EXCERPT: Let $n\geq k\geq 2$ and let M be an integer. A family $\mathcal{F}\subseteq\mathbb{Z}_{2^n}$ has size $|\mathcal{F}|=M$, and centred families are defined by taking initial layers...

conjecture | open | MathDB 页面 | 原文线索

V2-042995 | MATHDB-340620

Optimality conjecture for the largest d-cube-free subsets of $\mathbb{Z}_{2^n}$

摘录 / EXCERPT: Let d,n be positive integers with $d\leq n$, and let $\mathcal{C}_d\subseteq\mathbb{Z}_{2^n}$ be the construction defined in the paper. Optimality conjecture for $\mathcal{C}_d$....

conjecture | open | MathDB 页面 | 原文线索

V2-042996 | MATHDB-340621

The 3-cube-free density conjecture in $\mathbb{Z}_{2^n}$

摘录 / EXCERPT: Let n be a positive integer, and consider subsets of $\mathbb{Z}_{2^n}$ that contain no 3-cube. The 3-cube-free density conjecture. The largest 3-cube-free family in...

conjecture | open | MathDB 页面 | 原文线索

V2-042997 | MATHDB-340644

Early's winding-number formula for the Ehrhart h*-vector of the hypersimplex

摘录 / EXCERPT: Early's conjecture. The number of hypersimplicial decorated ordered set partitions of type (k,n) with winding number d is $h^{*}_d(\Delta_{k,n})$.

conjecture | open | MathDB 页面 | 原文线索

V2-042998 | MATHDB-340645

Early's winding-number formula for the Ehrhart h*-vector of generic hypercube cross sections

摘录 / EXCERPT: Early's conjecture. The number of r-hypersimplicial decorated ordered set partitions of type (k,n) with winding number d is $h_{d}^{*}(I_{r,k}^{n})$.

conjecture | open | MathDB 页面 | 原文线索

V2-042999 | MATHDB-340658

Positive-proportion missing-values conjecture for hypercube intersections

摘录 / EXCERPT: Let $H(\infty,k)=\bigcup_{n\geq k}H(n,k)$ be the set of all intersection sizes of a k-dimensional linear subspace with the vertices of some n-dimensional hypercube, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043000 | MATHDB-340661

The CGT-value criterion for Left wins in simultaneous Hackenbush

摘录 / EXCERPT: Let G be a Hackenbush tree, and let $o_S(G)$ denote its outcome under simultaneous play. If the Conway game-theoretic value of G is greater than $2/3$, then the CGT-value conje...

conjecture | open | MathDB 页面 | 原文线索

V2-043001 | MATHDB-340676

Symmetry, unimodality, and factorial sum for diagonal numerators of the blade array

摘录 / EXCERPT: Let $T^B_{n,k}$ be the array whose entries count blades of dimension $n-k$ in the canonical basis of graduated blades, with diagonal generating functions formed from the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-043002 | MATHDB-340689

Kräuter's rank bound conjecture for permanents of sign matrices

摘录 / EXCERPT: Let $\mathbb{F}$ be a field of zero characteristics. For positive integers k,n, let $M_{k,n}(\mathbb{F})$ be the set of k-by-n matrices over $\mathbb{F}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043003 | MATHDB-340720

The upper-bound conjecture for the density of zero Möbius values

摘录 / EXCERPT: Upper-bound conjecture. The values d_n are bounded from above by 0.6040.

conjecture | open | MathDB 页面 | 原文线索

V2-043004 | MATHDB-340736

Generic maximizers are not tuples of ellipsoids

摘录 / EXCERPT: Let $({\mathcal L},{\mathbf e})$ satisfy the appropriate nondegeneracy and admissibility hypotheses. Non-ellipsoidal maximizer conjecture. For generic $({\mathcal L},{\mathbf e})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043005 | MATHDB-340737

Partial-symmetry symmetrization conjecture for maximizers

摘录 / EXCERPT: Let $({\mathcal L},{\mathbf e})$ satisfy the appropriate nondegeneracy and admissibility hypotheses, and suppose that $\mathcal L$ satisfies the partial symmetry hypothesis. A maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-043006 | MATHDB-340738

Uniqueness of symmetrized maximizers under partial symmetry

摘录 / EXCERPT: Let $({\mathcal L}^0,\mathbf e)$ be nondegenerate and strictly admissible. For generic admissible $({\mathcal L},\mathbf e)$ satisfying the partial symmetry hypothesis, with...

conjecture | open | MathDB 页面 | 原文线索

V2-043007 | MATHDB-340739

Convexity of maximizers beyond the perturbative regime

摘录 / EXCERPT: Let $\mathcal L$ satisfy the partial symmetry hypothesis. The component sets of a maximizer are sets E_i occurring in a maximizing tuple $\mathbf E$. Global convexity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-043008 | MATHDB-340740

Existence and convexity without partial symmetry

摘录 / EXCERPT: Let $\mathcal L$ be generic nondegenerate data, without assuming the partial symmetry hypothesis. The relevant maximization problem has maximizers, and their component sets are con...

conjecture | open | MathDB 页面 | 原文线索

V2-043009 | MATHDB-340750

The cross-intersection product criterion for uniform families

摘录 / EXCERPT: Let n,k,l be positive integers with $k+l<n<\max\{2k,2l\}$. For each $j\geq 0$, let ... and ... Let $M(n,k,l)$ be the maximum of $|\mathcal A||\mathcal B|$ over cross-intersecting fam...

conjecture | open | MathDB 页面 | 原文线索

V2-043010 | MATHDB-340751

The equality characterization for the biased cross-intersection theorem

摘录 / EXCERPT: Let Ω be the parameter domain used in Theorem, and let μ_α and μ_β denote the corresponding biased measures on $2^{[n]}$. Let $\mathcal A_j$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043011 | MATHDB-340855

Extremality of the automata $\mathcal{E}_n$, $\mathcal{E}'_n$, $\mathcal{O}_n$ and $\mathcal{O}'_n$ in the family $\mathcal{A}_{ij}$

摘录 / EXCERPT: Let $\mathcal{E}_n$, $\mathcal{E}'_n$, $\mathcal{O}_n$ and $\mathcal{O}'_n$ be the explicitly constructed automata in the paper, and let $\{\mathcal{A}_{ij}\}_{i\neq j}$ be the ind...

conjecture | open | MathDB 页面 | 原文线索

V2-043012 | MATHDB-340873

Sufficiency of admissibility for enclosings at the extremal order

摘录 / EXCERPT: Let n and m be positive integers such that $m=2n-2$, and let $r\geq 2$. Suppose that r, μ, and λ are positive integers independent of n, with $\mu>\lambda$. A d...

conjecture | open | MathDB 页面 | 原文线索

V2-043013 | MATHDB-340919

Nonnegativity conjecture for the K-theoretic Tutte polynomial of flag matroids

摘录 / EXCERPT: Nonnegativity conjecture. The coefficients of the K-theoretic Tutte polynomial of a flag matroid are always nonnegative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-043014 | MATHDB-340935

Schreyer's conjecture on extremal canonical syzygies

摘录 / EXCERPT: Let C be a curve of genus g and non-maximal gonality $3\leq k \leq \frac{g+1}{2}$. Assume $W^1_k(C)=\{A\}$ is a reduced single point and A is the unique line bundle of degree...

conjecture | solved | MathDB 页面 | 原文线索

V2-043015 | MATHDB-340936

Daykin's minimum-degree conjecture for r-wise union families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let ${\mathcal F}\subset 2^{[n]}$ be r-wise union if the union of any r members has size at most $n-t$ for some t. The minimum degree of...

conjecture | open | MathDB 页面 | 原文线索

V2-043016 | MATHDB-340941

Wu-Zhang's conjecture on extremal reciprocal algebraic integers

摘录 / EXCERPT: Let α be a reciprocal algebraic integer that is not a root of unity, and let $d=\deg(\alpha)\geq 2$. Let $\mathrm{mr}^d(d)$ denote the relevant degree-d extremal reciproca...

conjecture | open | MathDB 页面 | 原文线索

V2-043017 | MATHDB-340942

The primitivity conjecture for extremal reciprocal polynomials of composite half-degree

摘录 / EXCERPT: Let d be an even natural number, and let $P_d(x)$ be the minimal polynomial of an extremal reciprocal algebraic integer of degree d. A polynomial is primitive in the sense used...

conjecture | open | MathDB 页面 | 原文线索

V2-043018 | MATHDB-340943

The octanomial exclusion conjecture for extremal reciprocal algebraic integers

摘录 / EXCERPT: Let an extremal reciprocal primitive be an extremal reciprocal algebraic integer that is primitive, and let an octanomial be a polynomial with eight nonzero monomials. Octanomial e...

conjecture | open | MathDB 页面 | 原文线索

V2-043019 | MATHDB-340946

The asymptotic cardinality conjecture for systems of simple loops

摘录 / EXCERPT: Asymptotic cardinality conjecture. The maximum cardinality of a k-system of simple loops on S_g is

conjecture | open | MathDB 页面 | 原文线索

V2-043020 | MATHDB-340989

Cycle-free orientation counting conjecture for directed cycles

摘录 / EXCERPT: Cycle-free orientation counting conjecture. With high probability as $n\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-043021 | MATHDB-340996

Iterated blow-up conjecture for inducibility of directed paths

摘录 / EXCERPT: Iterated blow-up conjecture. The number of induced copies of $\vec{P}_k$ over all oriented graphs on n vertices is maximized by an iterated balanced blow-up of $\vec{C}_{k+1}$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-043022 | MATHDB-340997

Directed-path inducibility conjecture for transitive-tournament-free oriented graphs

摘录 / EXCERPT: Transitive-tournament-free inducibility conjecture. The number of induced copies of $\vec{P}_k$ over all $\vec{T}_3$-free oriented graphs on n vertices is maximized by a balanced...

conjecture | open | MathDB 页面 | 原文线索

V2-043023 | MATHDB-341003

Strict-inclusion conjecture for classes of free arrangements

摘录 / EXCERPT: Class-inclusion conjecture. The following strict inclusion and non-inclusion relations should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-043024 | MATHDB-341049

Alon-Balogh-Morris-Samotij conjecture on sets with small doubling in [n]

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let s and K be parameters. A set $J\subset[n]$ has doubling constant at most K when $|J+J|\leq K|J|$.
Alon-Balogh-Morris-Samotij conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043025 | MATHDB-341050

Conjecture on groups with exponentially few maximal sum-free sets

摘录 / EXCERPT: For an abelian group G, let $\mu(G)$ denote the size of a largest sum-free subset of G, and let $f_{\max}(G)$ denote the number of maximal sum-free subsets of G. Conjecture o...

conjecture | open | MathDB 页面 | 原文线索

V2-043026 | MATHDB-341051

Conjecture on maximal sum-free sets in type I groups

摘录 / EXCERPT: Let G be a finite abelian group of type I, and let $\mu(G)$ be the size of a largest sum-free subset of G, while $f_{\max}(G)$ is the number of maximal sum-free subsets of G....

conjecture | open | MathDB 页面 | 原文线索

V2-043027 | MATHDB-341079

Mammoliti-Britz conjecture on nonintersecting d-cluster-free families

摘录 / EXCERPT: Let $[n]$ denote the set $\{1,\ldots,n\}$, and let $\binom{[n]}{k}$ be the collection of all k-sets of $[n]$. A family is d-cluster-free if it contains no collection of d dif...

conjecture | open | MathDB 页面 | 原文线索

V2-043028 | MATHDB-341096

Conjectures on successive threshold values for equal-product partitions

摘录 / EXCERPT: For integers $r\geq2$ and $n\geq3$, let $s_r^*(n)$ be the smallest positive integer, if it exists, such that every integer $s\geq s_r^*(n)$ has at least r different n-partition...

conjecture | open | MathDB 页面 | 原文线索

V2-043029 | MATHDB-341098

Ellis et al.'s extremal conjecture for intersecting families of permutations

摘录 / EXCERPT: Let $d4d5$ be the set of permutations in $d4d5_{n,n,n}$, and call $d4d5$ t-intersecting when any two members agree on at least t points. For each admissible i, let $d4a2_i$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-043030 | MATHDB-341102

OEIS conjecture for the distribution of mesh pattern 3

摘录 / EXCERPT: Distribution conjecture for mesh pattern 3. The distribution of mesh pattern Nr. 3 is the triangle, read by rows, given by

conjecture | open | MathDB 页面 | 原文线索

V2-043031 | MATHDB-341115

Stanley's conjecture on decompositions of k-fold acyclic complexes

摘录 / EXCERPT: Stanley's conjecture. If Δ is a k-fold acyclic simplicial complex, then its face poset can be decomposed into disjoint rank k boolean intervals whose minimal faces toget...

conjecture | open | MathDB 页面 | 原文线索

V2-043032 | MATHDB-341121

The generalized Turán additive-approximation hardness conjecture

摘录 / EXCERPT: Let T be a graph, let $\mathcal{F}$ be a family of graphs such that no $F\in\mathcal{F}$ is a subgraph of a blow-up of T, and let $\epsilon>0$. For a graph G on n vertices,...

conjecture | open | MathDB 页面 | 原文线索

V2-043033 | MATHDB-341143

Same-set hitting conjecture for correlated spaces

摘录 / EXCERPT: Same-set hitting conjecture. Every correlated space with $\operatorname{diag}(\Omega)\subseteq\operatorname{supp}(\mathcal{P})$ is same-set hitting.

conjecture | open | MathDB 页面 | 原文线索

V2-043034 | MATHDB-341146

Divisibility conjecture for optimal hyperplane-cover weights

摘录 / EXCERPT: Divisibility conjecture. Every entry $\lambda_{i,j}$ is an integer multiple of δ.

conjecture | open | MathDB 页面 | 原文线索

V2-043035 | MATHDB-341149

Order-two essential-weight conjecture for extremal matrices

摘录 / EXCERPT: Order-two essential-weight conjecture. The values k_{α}, for $\alpha\in\operatorname{supp}(K)$, are essential weights of an optimal hyperplane cover for some multidimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-043036 | MATHDB-341161

Fibonacci lower-growth conjecture for bounded-depth numerical semigroups

摘录 / EXCERPT: Let $\mathcal F(g,q\leq d)$ denote the set of numerical semigroups of genus g and depth at most d. Bounded-depth growth conjecture. For all $d,g\geq 2$, ... For $d=3$ this extend...

conjecture | open | MathDB 页面 | 原文线索

V2-043037 | MATHDB-341162

Bounded-depth growth conjecture for the truncated numerical-semigroup counts

摘录 / EXCERPT: Let n'_g denote the number of numerical semigroups of genus g and depth at most 3, and let n_g denote the total number of numerical semigroups of genus g. Bounded-depth r...

conjecture | open | MathDB 页面 | 原文线索

V2-043038 | MATHDB-341163

Monotonic-level conjecture for infinite numerical-semigroup subtrees

摘录 / EXCERPT: Let $\mathcal T$ be the tree of numerical semigroups, and let S be a numerical semigroup such that the subtree $\mathcal T(S)$ rooted at S is infinite. Infinite-subtree growth...

conjecture | open | MathDB 页面 | 原文线索

V2-043039 | MATHDB-341169

Polynomial dependence conjecture for the restricted-doubling structure theorem

摘录 / EXCERPT: Let $A,B\subset \mathbb Z$ be finite sets of equal size, and let $\Gamma\subset A\times B$ be a set of pairs defining the restricted sumset $A+_{\Gamma}B$. Suppose the hypotheses of...

conjecture | open | MathDB 页面 | 原文线索

V2-043040 | MATHDB-341192

Conjecture on extremal copies of even d-ary trees

摘录 / EXCERPT: Let $d\geq 2$ and $k\geq 1$ be fixed positive integers. Let E_n^d denote the even d-ary tree with n leaves, let $c(E_k^d,T)$ denote the number of copies of E_k^d in a d-a...

conjecture | open | MathDB 页面 | 原文线索

V2-043041 | MATHDB-341197

Asymptotic conjecture for uniform sets with restricted intersections

摘录 / EXCERPT: Let $l\geqslant 1$ be an integer. For an L-intersecting n-family of subsets of $[n]$, where $L=\{0,1,\ldots,l\}$, let $\kappa_{\leqslant l}(n,n)$ denote the maximum possible...

conjecture | open | MathDB 页面 | 原文线索

V2-043042 | MATHDB-341204

The rank-three paving matroid enumeration conjecture

摘录 / EXCERPT: Let $p(n,r)$ denote the number of paving matroids on an n-element ground set of rank r, and let $p_k(n,r)$ denote the number of paving matroids without hyperplanes of cardinali...

conjecture | open | MathDB 页面 | 原文线索

V2-043043 | MATHDB-341220

The asymptotic conjecture for minimum splitters of splittable set families

摘录 / EXCERPT: Let a splittable n-set family be a family of n sets admitting at least one splitter, and let k denote the size of the underlying ground set. The asymptotic splitter conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-043044 | MATHDB-341225

Generic solution-count conjecture for the quantum KdV oper system

摘录 / EXCERPT: Let $P_n(N)$ be the number of n-coloured partitions of N, let $n=\operatorname{rank}\mathfrak{g}$, and let $\operatorname{systemonc}$ denote the displayed system of equations f...

conjecture | open | MathDB 页面 | 原文线索

V2-043045 | MATHDB-341264

The Fishburn-Wilf equivalence conjecture for patterns 2413, 2431 and 3241

摘录 / EXCERPT: Let $\mathscr{F}_n(\sigma)$ denote the set of Fishburn permutations of length n avoiding the pattern σ. Fishburn-Wilf equivalence conjecture. The three pattern-avoidance c...

conjecture | open | MathDB 页面 | 原文线索

V2-043046 | MATHDB-341290

The final-matrix Sierpinski-triangle conjecture

摘录 / EXCERPT: Final-matrix Sierpinski-triangle conjecture. Suppose $n\geq0$ and $\ell\leq x*y$ in A_n. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-043047 | MATHDB-341291

The power-of-two final-matrix conjecture

摘录 / EXCERPT: Let $FM_k^{-}(2^x,2^y)$ denote the indicated final-matrix value. Power-of-two final-matrix conjecture. If $k=2^n$, then $|FM_k^{-}(2^x,2^y)|$ is a power of 2 whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-043048 | MATHDB-341292

The final-matrix exponential-pattern conjecture

摘录 / EXCERPT: Let $FM_{2^n}^{-}(2^i,2^j)$ denote the final-matrix value and let ST_n be the recursively defined Sierpinski-triangle set. Exponential-pattern conjecture (SE_n). Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-043049 | MATHDB-341293

The final-matrix periodicity conjecture

摘录 / EXCERPT: Let $FM_n^{-}(x,2^y)$ denote the negative final-matrix value. Final-matrix periodicity conjecture (EP_n). Suppose that r is a natural number and $x<2^{2^r}-1$. Then ... whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-043050 | MATHDB-341310

Universal factorial Schur P and Q duality at beta equals minus one

摘录 / EXCERPT: The conjectural combinatorial formula. One has

conjecture | open | MathDB 页面 | 原文线索

V2-043051 | MATHDB-341329

The planar discrete Brunn-Minkowski conjecture for triangulation counts

摘录 / EXCERPT: For finite two-dimensional sets $A,B one$ in the plane that are not collinear, let $[A]$ and $[B]$ denote their convex hulls, and let ${\rm tr}(A)$ and ${\rm tr}(B)$ be the numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-043052 | MATHDB-341331

The off-diagonal restricted online Ramsey number conjecture

摘录 / EXCERPT: Let $r(3,n)$ be the classical Ramsey number, and let $\tilde{r}(3,n;N)$ denote the restricted online Ramsey number on N vertices. Off-diagonal restricted online Ramsey conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-043053 | MATHDB-341349

Uniqueness conjecture for maximal-density product-free sets in free semigroups

摘录 / EXCERPT: Uniqueness conjecture. If $S\subset\mathcal{F}_{\mathscr{A}}$ is product-free and

conjecture | open | MathDB 页面 | 原文线索

V2-043054 | MATHDB-341381

Lemmens and Seidel's conjecture for equiangular lines with common angle $1/5$

摘录 / EXCERPT: Let $M_a(n)$ denote the maximum number of equiangular lines in $\boldsymbol{R}^n$ with common angle a. Lemmens and Seidel's conjecture. ... Lemmens and Seidel obtained results lead...

conjecture | open | MathDB 页面 | 原文线索

V2-043055 | MATHDB-341401

Dyson's constant-term conjecture

摘录 / EXCERPT: Dyson's conjecture. Its constant term is

conjecture | solved | MathDB 页面 | 原文线索

V2-043056 | MATHDB-341402

Andrews' q-Dyson conjecture

摘录 / EXCERPT: Let n and $a_1,a_2,\dots,a_n$ be nonnegative integers, and let q be an indeterminate. Consider the Laurent polynomial ... Define $[m]_q=1+q+\cdots+q^{m-1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043057 | MATHDB-341410

Steep-and-shallow criterion for generalized regularization and Mullineux commutation

摘录 / EXCERPT: Let a and b be positive integers with $2a<b$, and let $ u$ be a partition that is both $\operatorname{Reg}_{a,b}$-valid and $\operatorname{Colreg}_{a,b}$-valid. For each hook...

conjecture | open | MathDB 页面 | 原文线索

V2-043058 | MATHDB-341411

Vanishing and leading-term conjecture for generalized regularization q-decomposition numbers

摘录 / EXCERPT: Let a and b be positive integers. Let λ and μ be partitions of the same size, with λ $\operatorname{Reg}_{a,b}$-valid and μ (a,b)-regular. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-043059 | MATHDB-341429

Kalai-Sarkaria face-number conjecture for embeddable complexes

摘录 / EXCERPT: Kalai-Sarkaria face-number conjecture. The complex K has at most as many d-simplices as $\partial C_{r+1}(n)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-043060 | MATHDB-341430

Sarkaria's non-embeddability conjecture for 2-complexes in four-space

摘录 / EXCERPT: Sarkaria's non-embeddability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043061 | MATHDB-341439

Conjecture on log-concavity of the signed binomial-Eulerian polynomials

摘录 / EXCERPT: The signed log-concavity conjecture. The polynomial $\tilde{A}_n(t,-1)$ is log-concave for $n\geq1$.

conjecture | open | MathDB 页面 | 原文线索

V2-043062 | MATHDB-341446

Conjectured upper bound for the maximum modulus of graph independence roots

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\textnormal{maxmod}(n)$ denote the maximum modulus of an independence root among graphs on n vertices. The maximum-modulus conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-043063 | MATHDB-341447

Uniqueness conjecture for extremal graph independence roots

摘录 / EXCERPT: Let G_0, G_1, and G_2 be the graphs identified in the paper, and let $\textnormal{maxmod}(n)$ denote the maximum modulus of an independence root among graphs on n vertices....

conjecture | open | MathDB 页面 | 原文线索

V2-043064 | MATHDB-341449

Uniqueness conjecture for extremal tree independence roots

摘录 / EXCERPT: Let $T_{\frac{n-1}{2}}$ and $T'_{\frac{n-6}{2}}$ be the trees identified in the paper, and let $\textnormal{maxmod}_{T}(n)$ denote the maximum modulus of an independence root among...

conjecture | open | MathDB 页面 | 原文线索

V2-043065 | MATHDB-341486

Extremal Khovanov homology signature conjecture for Turaev genus one links

摘录 / EXCERPT: Let L be a non-split link of Turaev genus one. Write $Kh^{i,j}(L)$ for its Khovanov homology, let $j_{\min}(L)$ and $j_{\max}(L)$ denote the minimum and maximum quantum gradings...

conjecture | open | MathDB 页面 | 原文线索

V2-043066 | MATHDB-341495

One-point recursion conjecture for rational spectral curves

摘录 / EXCERPT: One-point recursion conjecture. The numbers $n_g(d)=d\,N_{g,1}(d)$ satisfy a 1-point recursion. This conjecture proposes that the one-point recursion phenomenon established for the...

conjecture | open | MathDB 页面 | 原文线索

V2-043067 | MATHDB-341496

The sharpness conjecture for strongly increasing-pattern-avoiding permutations

摘录 / EXCERPT: Sharpness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043068 | MATHDB-341525

Period-denominator equality for two-move rider chess-piece configurations

摘录 / EXCERPT: Let $u_\mathbb{P}(q;n)$ be the counting quasipolynomial for nonattacking configurations of q copies of a two-move rider $\mathbb{P}$ on the dilated unit board, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043069 | MATHDB-341526

Unbounded palindromic length in aperiodic infinite words

摘录 / EXCERPT: For an infinite word $\mathbf{w}$, let the palindromic length $|u|_{\rm pal}$ of a finite factor or prefix u be the least number Q of palindromes whose concatenation is u. Un...

conjecture | open | MathDB 页面 | 原文线索

V2-043070 | MATHDB-341534

Stevens-Meagher uniform covering array conjecture

摘录 / EXCERPT: A covering array $\operatorname{CA}(N,k,v)$ is an array with N rows, k columns, and entries from an alphabet of size v, such that every choice of two columns contains every o...

conjecture | open | MathDB 页面 | 原文线索

V2-043071 | MATHDB-341546

The Edelman-Greene coefficient bound

摘录 / EXCERPT: Let w be a permutation and let $a_{w,\lambda}$ denote the corresponding Edelman-Greene coefficient for a partition λ, with f^{λ} the number of standard Young tabl...

conjecture | open | MathDB 页面 | 原文线索

V2-043072 | MATHDB-341561

Serbanuta's conjecture on attainable M-ambiguity sequences

摘录 / EXCERPT: Let Σ be an ordered alphabet. For words $u,v\in\Sigma^*$ and a letter $a\in\Sigma$, a word is M-unambiguous if it is not M-equivalent to any distinct word, where M-equ...

conjecture | open | MathDB 页面 | 原文线索

V2-043073 | MATHDB-341578

Conjecture on the eigenvalue multiplicities of the Stern recurrence matrices

摘录 / EXCERPT: Eigenvalue multiplicity conjecture. For positive integers s, the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-043074 | MATHDB-341579

Conjecture on the exact recurrence orders for Stern's diatomic array sequences

摘录 / EXCERPT: Exact recurrence-order conjecture. The values are

conjecture | open | MathDB 页面 | 原文线索

V2-043075 | MATHDB-341584

Nonnegativity conjecture for the cd-index of regular CW spheres

摘录 / EXCERPT: A regular CW sphere is a regular CW complex whose underlying space is a sphere, and its cd-index is the polynomial $\Phi_P(c,d)$ encoding its flag f-vector, with c of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-043076 | MATHDB-341590

Ruzsa's conjecture on spanning size in abelian groups of finite torsion

摘录 / EXCERPT: Ruzsa's conjecture. If there exists a constant K such that

conjecture | open | MathDB 页面 | 原文线索

V2-043077 | MATHDB-341603

Maximum common-prefix conjecture for uniform squarefree morphisms

摘录 / EXCERPT: Let T be the alphabet, and let an n-uniform squarefree morphism be a morphism $h:T^*\to T^*$ for which every image $h(a)$ has length n and which maps squarefree words to squa...

conjecture | open | MathDB 页面 | 原文线索

V2-043078 | MATHDB-341614

Connectivity conjecture for odd balanced pairings

摘录 / EXCERPT: Connectivity conjecture. Any two balanced pairings of the same size n, where n is odd, are connected.

conjecture | open | MathDB 页面 | 原文线索

V2-043079 | MATHDB-341632

Lyngsø and Pedersen's antipalindromic subsequence conjecture

摘录 / EXCERPT: Lyngsø and Pedersen's conjecture. Every binary circular word of length n divisible by 6 with equal numbers of zeros and ones has an antipalindromic subsequence of length at lea...

conjecture | open | MathDB 页面 | 原文线索

V2-043080 | MATHDB-341633

Brevier-Preissmann-Sebő strong antipalindromic subsequence conjecture

摘录 / EXCERPT: Brevier-Preissmann-Sebő's conjecture. The word w can be partitioned into two linear words w_1,w_2 of equal length, $w=w_1w_2$, having subsequences s_1,s_2 such that s_1s_2...

conjecture | open | MathDB 页面 | 原文线索

V2-043081 | MATHDB-341634

Weak palindromic subsequence conjecture for binary circular words

摘录 / EXCERPT: Weak circular-palindrome conjecture. Every binary circular word of length n has a palindromic subsequence of length at least

conjecture | open | MathDB 页面 | 原文线索

V2-043082 | MATHDB-341635

Strong palindromic subsequence conjecture for binary circular words

摘录 / EXCERPT: Strong circular-palindrome conjecture. There is such a partition with subsequences s_1,s_2 satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-043083 | MATHDB-341636

Orthogonal-cut palindromic subsequence conjecture

摘录 / EXCERPT: Orthogonal-cut conjecture. For every such w and every such linear representation, the maximum of the lengths of $p_1p'_1$ and $p_2p'_2$ is at least

conjecture | open | MathDB 页面 | 原文线索

V2-043084 | MATHDB-341637

Three-quarter palindromic subsequence conjecture for binary words

摘录 / EXCERPT: Three-quarter palindrome conjecture. The word w has a palindromic subsequence of length at least

conjecture | open | MathDB 页面 | 原文线索

V2-043085 | MATHDB-341638

Good-relation covering conjecture for individually private information retrieval

摘录 / EXCERPT: Let $K\geq 1$, $M\geq 1$, and $D\geq 2$ satisfy $D+M\leq K$. Set $\mathcal{K}=[K]$, and let $\mathcal{I}$ and $\mathcal{J}$ be the sets of subsets $I,J\subseteq\mathcal{K}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-043086 | MATHDB-341663

Grünbaum-Kalai-Sarkaria face-number conjecture

摘录 / EXCERPT: Grünbaum-Kalai-Sarkaria conjecture. For every $d\geq 1$, there is a constant C_d such that

conjecture | open | MathDB 页面 | 原文线索

V2-043087 | MATHDB-341664

A face-number bound for embeddable simplicial complexes

摘录 / EXCERPT: Face-number bound. For every $d\geq 1$, there is a constant C_d such that

conjecture | open | MathDB 页面 | 原文线索

V2-043088 | MATHDB-341682

Nonexistence conjecture for regular 3-hypertopes with power-of-two automorphism groups

摘录 / EXCERPT: Nonexistence conjecture. There is no regular 3-hypertope of type $(2^s,2^t,2^l)$ whose automorphism group has order 2^n.

conjecture | open | MathDB 页面 | 原文线索

V2-043089 | MATHDB-341685

Ding-Li conjecture on 2-designs from ternary linear codes

摘录 / EXCERPT: Let $m\ge 3$ be an odd integer, let α be a primitive element of $\operatorname{GF}(3^m)$, and let $\mathcal C_1$ and $\mathcal C_2$ be the ternary linear codes defined by .....

conjecture | open | MathDB 页面 | 原文线索

V2-043090 | MATHDB-341686

Ding-Li conjecture on Steiner systems from dual ternary codes

摘录 / EXCERPT: Let $m\ge 3$ be an odd integer, let α be a primitive element of $\operatorname{GF}(3^m)$, and let $\mathcal C_1$ and $\mathcal C_2$ be the ternary linear codes defined above...

conjecture | open | MathDB 页面 | 原文线索

V2-043091 | MATHDB-341715

Black-Drellich-Tymoczko's range conjecture for valid plane trees

摘录 / EXCERPT: Let A be a complementary alphabet of size m, and let $R(n,m)$ be the set of positive integers k for which there is a word P of length $2n$ over A having exactly k P-v...

conjecture | open | MathDB 页面 | 原文线索

V2-043092 | MATHDB-341716

Black-Drellich-Tymoczko's universality conjecture for valid plane trees

摘录 / EXCERPT: Let $R(n,m)$ be the set of positive integers k for which some word of length $2n$ over a complementary alphabet of size m has exactly k valid rooted plane trees. Black-Drelli...

conjecture | open | MathDB 页面 | 原文线索

V2-043093 | MATHDB-341751

Balko et al.'s minimum Ramsey number conjecture for ordered paths

摘录 / EXCERPT: Let P_n be the path on n vertices, and consider all orderings of its vertices. Balko et al.'s conjecture. Among all orderings of P_n, the alternating path has minimum Ramsey...

conjecture | open | MathDB 页面 | 原文线索

V2-043094 | MATHDB-341758

The Gallai-Ramsey reduction conjecture for rainbow P_{5}

摘录 / EXCERPT: Gallai-Ramsey reduction conjecture. For any graph H with no isolated vertices, we have

conjecture | open | MathDB 页面 | 原文线索

V2-043095 | MATHDB-341763

Extremality conjecture for irreducibles of uniform matroid monoids

摘录 / EXCERPT: Let M be a matroid of rank k on n elements, and let $d4c9_p(M)$ denote its associated monoid. Write $\operatorname{Irr}(\mathcal{Q}_p(M))$ for the set of irreducibles of this...

conjecture | open | MathDB 页面 | 原文线索

V2-043096 | MATHDB-341773

Linear-length anti-power conjecture for aperiodic morphic words

摘录 / EXCERPT: Linear-length anti-power conjecture. There is a constant $C=C(W)$ such that, for all positive integers i and k, W contains a k-anti-power with blocks of length at most Ck...

conjecture | open | MathDB 页面 | 原文线索

V2-043097 | MATHDB-341798

The proportionality conjecture for the sums $P_{n,t}$ and A_n

摘录 / EXCERPT: Proportionality conjecture. For every such n, t, and $(a_1,\dots,a_n)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-043098 | MATHDB-341809

Near-linear upper bound for maximum-area triangles in line arrangements

摘录 / EXCERPT: Maximum-area triangle conjecture. The order of magnitude of $M(n)$ is $O(n^{1+\varepsilon})$ for every $\varepsilon>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-043099 | MATHDB-341823

The $8/7$-Rayleigh characterization of matroid basis sets

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\mathrm{J}\subseteq\{0,1\}^n$ be nonempty. Let f_{J} denote its generating function, and call a polynomial homogeneous $\frac{8}{7}$-Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-043100 | MATHDB-341830

The EPPA-Ramsey expansion conjecture

摘录 / EXCERPT: EPPA-Ramsey expansion conjecture. Every class with EPPA has a precompact Ramsey expansion.

conjecture | open | MathDB 页面 | 原文线索

V2-043101 | MATHDB-341850

Submodular separation conjecture for low-degree monomial families

摘录 / EXCERPT: Submodular separation conjecture. For every such r,d, there exists a submodular function f such that $f(x)$ is computable in $\mathrm{poly}(n)$ time, $f(x)\leqslant-1$ for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-043102 | MATHDB-341875

Koszulness conjecture for Chow rings of matroids

摘录 / EXCERPT: Matroid Chow-ring conjecture. The Chow ring of any matroid is Koszul.

conjecture | open | MathDB 页面 | 原文线索

V2-043103 | MATHDB-341889

Guy's conjecture on eventual periodicity of finite octal-game Grundy sequences

摘录 / EXCERPT: Guy's conjecture. Every finite octal game has an eventually periodic Grundy sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-043104 | MATHDB-341890

Undecidability of regular winning-position languages for strongly terminating games

摘录 / EXCERPT: The conjecture. Given a strongly terminating game $\mathcal{G}$, it is undecidable whether the winning positions of $\mathcal{G}$ form a regular language.

open_problem | open | MathDB 页面 | 原文线索

V2-043105 | MATHDB-341906

Balog-Shakan conjecture on sums of a set and its dilate

摘录 / EXCERPT: Balog-Shakan conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-043106 | MATHDB-341907

Bukh's linear-transformation sumset conjecture

摘录 / EXCERPT: Bukh's conjecture. For any $A\subseteq\mathbb{Z}^d$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-043107 | MATHDB-341944

The interpolation conjecture for maximal determinants of matrices with non-integer row weight

摘录 / EXCERPT: Let $T(n,\tilde{k})$ be the family of matrices with real row-weight parameter $\tilde{k}>1$, and let $M_T(n,\tilde{k})$ be its maximum determinant. Define ... and ... Let $M_R(m,n,k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-043108 | MATHDB-341975

Gallai-Ramsey number conjecture for the three-triangle fan

摘录 / EXCERPT: Let F_n be the fan graph with n triangles, defined by ... so that F_3 is the fan with three triangles, and let $gr_k(K_3;F_3)$ denote the k-color Gallai-Ramsey number of...

conjecture | open | MathDB 页面 | 原文线索

V2-043109 | MATHDB-341980

Gruenbaum-Giessen h-vector inequality for simplicial arrangements

摘录 / EXCERPT: Gruenbaum-Giessen h-vector conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-043110 | MATHDB-341981

Single-exponential growth conjecture for LUBF-gaussoids

摘录 / EXCERPT: A LUBF-gaussoid is a gaussoid in the exceptional class denoted by LUBF; let n be its dimension. Single-exponential growth conjecture. There is a single exponential upper bound fo...

conjecture | open | MathDB 页面 | 原文线索

V2-043111 | MATHDB-341987

Zhang-Ge conjecture on purely singular perfect splitter sets

摘录 / EXCERPT: Zhang-Ge conjecture. There does not exist any purely singular perfect $B-k_1,k_2$ set except when $m=1$ and possibly when $m=k_1+k_2+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-043112 | MATHDB-341988

Extremal-monomial test-configuration conjecture for isolated hypersurface singularities

摘录 / EXCERPT: Let $f(z_0,z_1,z_2,z_3)$ be an isolated hypersurface singularity. The extremal monomials are the monomials selected from the four sets I, II, III, and IV displayed in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-043113 | MATHDB-342025

Step-count conjecture for the switching algorithm

摘录 / EXCERPT: Let $(\boxtimes,T)\in \operatorname{LR}(\alpha,\scalebox{.5}{\yng(1)},\beta,\gamma)$, where $\operatorname{LR}(\alpha,\scalebox{.5}{\yng(1)},\beta,\gamma)$ is the set of type B Lit...

conjecture | open | MathDB 页面 | 原文线索

V2-043114 | MATHDB-342060

Symmetric and exterior shifting conjecture for Inc-invariant chains

摘录 / EXCERPT: Symmetric and exterior shifting conjecture. If $\mathcal{S}$ is either the symmetric or exterior shifting operation, then $(\mathcal{S}(\Delta_n))_{n\ge 1}$ is also...

conjecture | open | MathDB 页面 | 原文线索

V2-043115 | MATHDB-342072

The conjecture that the union of two arithmetic progressions is not sum-dominant

摘录 / EXCERPT: Let A and B be arithmetic progressions of integers, and consider their union $A\cup B$. A finite set S of integers is sum-dominant if $|S+S|>|S-S|$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-043116 | MATHDB-342076

Huang's maximum-diversity conjecture for intersecting families

摘录 / EXCERPT: Huang's conjecture. If $\mathcal{F}\subset 2^{[n]}$ is intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-043117 | MATHDB-342079

Unimodality and palindromicity conjecture for the polynomials $N_{n,n}(t)$

摘录 / EXCERPT: For each positive integer n, let $N_{n,n}(t)$ be the polynomial defined by the preceding generating-function construction for the expected one-dimensional Earth Mover's Distance...

conjecture | open | MathDB 页面 | 原文线索

V2-043118 | MATHDB-342094

Vertical-strip LLT polynomial identity

摘录 / EXCERPT: Let $\mathbf{a}$ be an area sequence and $\mathbf{s}$ a set of strict edges, and let $\mathrm{G}_{\mathbf{a},\mathbf{s}}(\mathbf{x};q)$ be the corresponding vertical-strip LLT poly...

conjecture | open | MathDB 页面 | 原文线索

V2-043119 | MATHDB-342128

Finiteness conjecture for minor-minimal matroids not coverable by independent sets

摘录 / EXCERPT: Let M be a matroid. A collection of independent sets covers M if every element of M belongs to at least one set in the collection. Finiteness conjecture. For every positive i...

conjecture | open | MathDB 页面 | 原文线索

V2-043120 | MATHDB-342129

Finiteness conjecture for strictly density-critical matroids

摘录 / EXCERPT: For a matroid M, let $\d(M)$ denote its density, and call M strictly t-density-critical when $d(M)>t$ while every proper minor N of M satisfies $d(N)\leq t$. Finiteness c...

conjecture | open | MathDB 页面 | 原文线索

V2-043121 | MATHDB-342130

Finiteness conjecture for density-critical matroids of prescribed density

摘录 / EXCERPT: For a matroid M, let $d(M)$ be its density, and call M density-critical when every proper minor N satisfies $d(N)<d(M)$. Finiteness conjecture. For every $t\geq 0$, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-043122 | MATHDB-342146

The simple-matrix conjecture on linear homogeneous submatrices

摘录 / EXCERPT: Simple-matrix conjecture. Let P be a simple zero-one matrix. Then every P-free $n\times n$ zero-one matrix contains a $cn\times cn$ homogeneous submatrix, for a suitable consta...

conjecture | open | MathDB 页面 | 原文线索

V2-043123 | MATHDB-342147

The acyclic-matrix conjecture for dense zero-submatrices

摘录 / EXCERPT: Acyclic-matrix conjecture. For every acyclic zero-one matrix P and every $\varepsilon>0$, there is a $\delta>0$ such that every P-free $n\times n$ zero-one matrix with at least...

conjecture | open | MathDB 页面 | 原文线索

V2-043124 | MATHDB-342148

The acyclic forbidden-matrix conjecture for homogeneous submatrices

摘录 / EXCERPT: Acyclic forbidden-matrix conjecture. Let P be an acyclic zero-one matrix. Then every $n\times n$ zero-one matrix that is both P-free and P^c-free contains a $cn\times cn$ hom...

conjecture | open | MathDB 页面 | 原文线索

V2-043125 | MATHDB-342149

The bipartite Erdős-Hajnal conjecture for separately k-intersecting curve families

摘录 / EXCERPT: Bipartite curve-family conjecture. For every k there is a constant $c_k>0$ such that there are subfamilies $\mathcal{A}_0\subseteq\mathcal{A}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043126 | MATHDB-342158

The non-translation o-polynomial block-count conjecture

摘录 / EXCERPT: Let $q=2^m$, and let f be an o-polynomial over ${\mathrm{GF}}(q)$. For $a\in {\mathrm{GF}}(q)^*$ and h satisfying $1\leq h<m$ and $\gcd(h,m)=1$, exclude the translation o-monom...

conjecture | open | MathDB 页面 | 原文线索

V2-043127 | MATHDB-342159

The non-Frobenius o-monomial stabilizer conjecture

摘录 / EXCERPT: Let $q=2^m$, let x^e be an o-monomial over ${\mathrm{GF}}(q)$, and let J_e be the associated subset used in the design construction. Write ${\mathrm{AG}}_1({\mathrm{GF}}(q))$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-043128 | MATHDB-342160

The ternary tenth-power block-size conjecture

摘录 / EXCERPT: Let $m\geq 3$ be odd, and define k_m recursively by ... For $b,c\in {\mathrm{GF}}(3^m)$, let $B_{(x^{10},b,c)}$ denote the corresponding block, and let ${\mathcal{B}}_{(x^{10},k)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-043129 | MATHDB-342161

The non-o-monomial extended block-count conjecture

摘录 / EXCERPT: Let $m\geq 5$ be odd, set $q=2^m$, and let f be an o-polynomial over ${\mathrm{GF}}(q)$. Let $\hat{\mathcal{B}}_{(f,q/2)}$ be the distinct blocks in the extended construction. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-043130 | MATHDB-342162

The extended designs from Payne, Cherowitzo and Subiaco o-polynomials

摘录 / EXCERPT: Let $m\geq 5$ be odd, set $q=2^m$, and let $\hat{\mathbb{D}}(f,k)=({\mathrm{GF}}(q),\hat{\mathcal{B}}_k)$ be the extended construction, with $k=q/2$. Define the listed o-polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-043131 | MATHDB-342195

The asymptotic cap-set conjecture for $\mathbb{Z}_4^n$

摘录 / EXCERPT: Asymptotic cap-set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043132 | MATHDB-342196

Equality of the Fibonacci gap sets

摘录 / EXCERPT: Let D_ℓ and U_ℓ be the down-gap and up-gap sets, respectively, defined by ... with U_ℓ defined analogously. Equality of the Fibonacci gap sets. For every positive inte...

conjecture | open | MathDB 页面 | 原文线索

V2-043133 | MATHDB-342197

Infinitude of Fibonacci gap fibers

摘录 / EXCERPT: For a positive integer ℓ and an integer m, let ... and define $U_\ell(m)$ analogously. Infinitude conjecture for gap fibers. For every $m\in D_\ell$, ... The conjecture expresse...

conjecture | open | MathDB 页面 | 原文线索

V2-043134 | MATHDB-342208

Existence of the logarithmic growth exponent for $\mathfrak g_3$

摘录 / EXCERPT: Let $\mathfrak g_t(m)$ denote the function defined in the paper's data section. Logarithmic growth-exponent conjecture. The limit ... exists. The supplied statement has inconsistent...

conjecture | open | MathDB 页面 | 原文线索

V2-043135 | MATHDB-342209

Positive lower logarithmic growth for $\mathfrak g_3$

摘录 / EXCERPT: Let $\mathfrak g_3(m)$ denote the function defined in the paper's data section. Positive-growth conjecture. The lower limit ... The supplied statement uses n as the limiting variab...

conjecture | open | MathDB 页面 | 原文线索

V2-043136 | MATHDB-342210

Eventual comparison conjecture for $\mathfrak g_2$ and $\mathfrak g_3$

摘录 / EXCERPT: Let $\mathfrak g_2(m)$ and $\mathfrak g_3(m)$ denote the functions defined in the paper's data section. Comparison conjecture. For every integer $m\geq 13$, we have ... This is one o...

conjecture | open | MathDB 页面 | 原文线索

V2-043137 | MATHDB-342211

Divergence conjecture for $\mathfrak g_3-\mathfrak g_2$

摘录 / EXCERPT: Let $\mathfrak g_2(m)$ and $\mathfrak g_3(m)$ denote the functions defined in the paper's data section. Divergence conjecture. Their difference diverges to infinity: ... This is pres...

conjecture | open | MathDB 页面 | 原文线索

V2-043138 | MATHDB-342250

Nonvanishing conjecture for symmetric Wigner 6j symbols

摘录 / EXCERPT: Let k,n be integers and let ... be the Wigner $6j$ symbol appearing in the description of the cubic invariant $\mathscr{P}_{n,3}(F)$. Nonvanishing conjecture. For every pair of int...

conjecture | open | MathDB 页面 | 原文线索

V2-043139 | MATHDB-342282

Mean, variance and central limit conjecture for $D_{i,K}$

摘录 / EXCERPT: For each $i\in\mathbb N$ and strip size K, let $D_{i,K}$ be the random variable defined in the paper, with expectation $E(D_{i,K})$ and standard deviation $\sigma(D_{i,K})$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043140 | MATHDB-342314

The uniqueness conjecture for cluster variables

摘录 / EXCERPT: Let $\mathcal{A}$ be an arbitrary cluster algebra with clusters $\mathbf{x}_t$ indexed by $t\in\mathbb{T}_n$, and let $F_t^{B;t_0}$ denote the associated F-polynomial data. Uniqu...

conjecture | open | MathDB 页面 | 原文线索

V2-043141 | MATHDB-342336

The optimal block-coloring conjecture for generalized Schur triples at parameter one-half

摘录 / EXCERPT: The optimal block-coloring conjecture. For $n\geq12$,

conjecture | open | MathDB 页面 | 原文线索

V2-043142 | MATHDB-342347

The self-conjugate consecutive-core and symmetric generalized Dyck path conjecture

摘录 / EXCERPT: Let s and k be positive integers. A self-conjugate $(s,s+1,\dots,s+k)$-core is a self-conjugate partition that is simultaneously an s-core, $(s+1)$-core, and so on through an...

conjecture | open | MathDB 页面 | 原文线索

V2-043143 | MATHDB-342381

APN incidence-structure conjecture for a family of 3-designs

摘录 / EXCERPT: Let $q=2^n$. Let $n\geq5$ be odd, let $s=2^{2i}-2^i+1$, and assume $\gcd(3i,n)=1$. Let B_s be the associated base block and let $\operatorname{GA}_1(q)$ denote the relevant affin...

conjecture | open | MathDB 页面 | 原文线索

V2-043144 | MATHDB-342382

APN 3-design conjecture for two quartic-dependent exponents

摘录 / EXCERPT: Let $q=2^n$, with $n\geq5$ odd, and let B_s be the associated base block. Two-exponent APN 3-design conjecture. The incidence structure ... is a 3-...

conjecture | open | MathDB 页面 | 原文线索

V2-043145 | MATHDB-342383

Pairwise non-isomorphism conjecture for Kassami APN 3-designs

摘录 / EXCERPT: Let $n\geq5$ be odd, and let ... Let $\phi(n)$ denote Euler's totient function, and let $\mathbb{KA}_{n,i}$ be the corresponding Kassami APN 3-designs. Kassami-design non-isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-043146 | MATHDB-342384

Weight-enumerator conjecture for the Kassami APN design code

摘录 / EXCERPT: Let $n\geq5$ be odd. Let $\mathsf{C}_2(\mathbb{KA}_{n,1})$ denote the binary linear code generated by the block characteristic vectors of $\mathbb{KA}_{n,1}$. Kassami-code paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-043147 | MATHDB-342385

Pairwise inequivalence conjecture for Kassami APN design codes

摘录 / EXCERPT: Let $n\geq5$ be odd, and let ... Let $\phi(n)$ denote Euler's totient function. Kassami-code inequivalence conjecture. The $\frac{\phi(n)}2$ binary linear codes...

conjecture | open | MathDB 页面 | 原文线索

V2-043148 | MATHDB-342395

Conjecture on linear type and symbolic Rees algebras of binomial edge ideals

摘录 / EXCERPT: Let G be a finite simple graph, let J_G denote its binomial edge ideal, and let C_n denote the cycle graph on n vertices. Let $\mathcal{R}(J_G)$ be the Rees algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-043149 | MATHDB-342442

Janson's conjecture on the loopy multigraph game

摘录 / EXCERPT: Janson's conjecture. In the loopy multigraph game, the Angel has a winning strategy for any starting position.

conjecture | open | MathDB 页面 | 原文线索

V2-043150 | MATHDB-342457

The sticky polymatroid conjecture

摘录 / EXCERPT: Sticky polymatroid conjecture. Every sticky polymatroid is modular.

conjecture | open | MathDB 页面 | 原文线索

V2-043151 | MATHDB-342490

Strictly balanced factors are as informative as unbalanced factors

摘录 / EXCERPT: Let $\Sigma=\{\mathtt{a},\mathtt{b}\}$, and let Σ_{sb}^{n} denote the strictly balanced words of length n. For a word w, write $\operatorname{ScatFact}_k(w)$ for its set...

conjecture | open | MathDB 页面 | 原文线索

V2-043152 | MATHDB-342491

Cardinality propagation for k-spectra of theta-palindromes

摘录 / EXCERPT: Let $\Sigma=\{\mathtt{a},\mathtt{b}\}$, let w^R denote reversal, and let $\operatorname{ScatFact}_k(w)$ be the set of length-k scattered factors of w. Consider the binary...

conjecture | open | MathDB 页面 | 原文线索

V2-043153 | MATHDB-342506

Lascoux atom formula conjecture via set-valued skyline tableaux

摘录 / EXCERPT: Let w be a permutation and λ a partition, and let $\operatorname{SLT}_{w\lambda}$ be the set of semistandard set-valued skyline tableaux of shape $w\lambda$. Define the La...

conjecture | open | MathDB 页面 | 原文线索

V2-043154 | MATHDB-342511

The square-free positivity conjecture for equivariant matroid K-classes

摘录 / EXCERPT: Square-free positivity conjecture. The Laurent polynomials $(-1)^{r(n-r)-(n-e)-|\lambda|}c_\lambda(t)\in\mathbf{Z}[t_1^{\pm1},\dots,t_n^{\pm1}]$ can be expressed, non-uniquely, as...

conjecture | open | MathDB 页面 | 原文线索

V2-043155 | MATHDB-342512

The square-free positivity conjecture for equivariant matroid Chow classes

摘录 / EXCERPT: Chow square-free positivity conjecture. The polynomial $(-1)^{r(n-r)-(n-e)-|\lambda|}d_\lambda(t)\in\mathbf{Z}[t_1,\dots,t_n]$ can be expressed, non-uniquely, as a square-free poly...

conjecture | open | MathDB 页面 | 原文线索

V2-043156 | MATHDB-342513

Nonnegative Schur coefficients for matroid Chow classes

摘录 / EXCERPT: Nonnegative Schur-coefficient conjecture. The constants d_λ are non-negative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-043157 | MATHDB-342528

The extremal configuration conjecture for d+1 unit vectors

摘录 / EXCERPT: Extremal configuration conjecture. For every such configuration, there exists $\varepsilon\in\{\pm1\}^{d+1}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043158 | MATHDB-342547

Fernandes-da Cruz conjecture on simultaneous minimum t-term ranks

摘录 / EXCERPT: Fernandes-da Cruz conjecture. If ${\cal A}(R,S)$ is nonempty, then there is a matrix $A\in {\cal A}(R,S)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043159 | MATHDB-342581

The asymptotic almost-cover conjecture for hypercubes

摘录 / EXCERPT: Almost-cover conjecture. For an arbitrary fixed integer $k 1$ and sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-043160 | MATHDB-342582

The restricted almost-cover conjecture for hypercubes

摘录 / EXCERPT: Restricted almost-cover conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-043161 | MATHDB-342583

The three-cover multiplicity conjecture for hypercubes

摘录 / EXCERPT: Three-cover multiplicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043162 | MATHDB-342600

Strict hierarchy conjecture for step-I_0 sets

摘录 / EXCERPT: A set $E\subset\mathbb{N}$ is called k-step-I_0 if it has the interpolation property for k-step nilsequences. Strict hierarchy conjecture. For every $k\in\mathbb{N}$, there e...

conjecture | open | MathDB 页面 | 原文线索

V2-043163 | MATHDB-342660

Two-line intersection conjecture for rich-line-free point sets

摘录 / EXCERPT: Two-line intersection conjecture. For all $\delta>0$, there exists $\varepsilon=\varepsilon(\delta)>0$ for which the displayed bound holds for every such P, I_1, and I_2. This conj...

conjecture | open | MathDB 页面 | 原文线索

V2-043164 | MATHDB-342675

Genericity of Assumption (J2)

摘录 / EXCERPT: Let $F(\mathbf{z})=G(\mathbf{z})/H(\mathbf{z})$ be a rational function, and let Assumption (J2) denote the Jacobian nondegeneracy condition used in the paper's critical-point syste...

conjecture | open | MathDB 页面 | 原文线索

V2-043165 | MATHDB-342688

Gyárfás-Rusza-Sárközy-Szemerédi conjecture for monochromatic paths in balanced tripartite graphs

摘录 / EXCERPT: Gyárfás-Rusza-Sárközy-Szemerédi conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-043166 | MATHDB-342694

Extendability conjecture for finite generalized Lyndon words

摘录 / EXCERPT: Let A be a finite alphabet, and let w be a finite generalized Lyndon word over A. The length of w is denoted by $|w|$. Extendability conjecture. Every finite generalized Ly...

conjecture | open | MathDB 页面 | 原文线索

V2-043167 | MATHDB-342698

Leader-Russell-Walters Hales-Jewett conjecture for finite groups

摘录 / EXCERPT: Leader-Russell-Walters Hales-Jewett conjecture. There exist positive integers d and N such that, for every r-coloring of G^N, there exist a nonempty $H\subseteq G$ and an...

conjecture | open | MathDB 页面 | 原文线索

V2-043168 | MATHDB-342705

Caterpillar and balanced species tree growth-factor conjecture

摘录 / EXCERPT: Caterpillar and balanced species tree conjecture. For a given k and sufficiently large n, the unranked species tree of size k having the largest number of...

conjecture | open | MathDB 页面 | 原文线索

V2-043169 | MATHDB-342713

Frantzikinakis' random-difference conjecture for Szemerédi's theorem

摘录 / EXCERPT: Let $S\subset\mathbb{N}$ be chosen at random with $\mathbb{P}(d\in S)=\omega(1/d)$. A set of positive upper density means a set $A\subset\mathbb{N}$ for which...

conjecture | open | MathDB 页面 | 原文线索

V2-043170 | MATHDB-342714

Nilsequence approximation conjecture for 3-term progression dual functions

摘录 / EXCERPT: For $A\subseteq[N]$, define the dual function counting normalized 3-term arithmetic progressions with common difference d by ... A 2-step nilsequence is understood in the sense of...

conjecture | open | MathDB 页面 | 原文线索

V2-043171 | MATHDB-342756

The injectivity conjecture for the classical-mechanics tree representation

摘录 / EXCERPT: Tree-representation injectivity conjecture. The map Θ is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-043172 | MATHDB-342778

The doubly nonnegative function and good graph conjecture

摘录 / EXCERPT: The doubly nonnegative function and good graph conjecture. All doubly nonnegative functions are nice, and all simple graphs are good.

conjecture | open | MathDB 页面 | 原文线索

V2-043173 | MATHDB-342779

The locally dense graph homomorphism conjecture

摘录 / EXCERPT: Locally dense graph homomorphism conjecture. For every graph G and every $\delta,d\in(0,1)$, there exists $\varepsilon=\varepsilon(\delta,d,G)$ such that every sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-043174 | MATHDB-342797

Nonvanishing conjecture for extremals of $C(k,p)$

摘录 / EXCERPT: Let $0<p<1$, and let f be an extremal for $C(k,p)$, where $C(k,p)$ denotes the coefficient-estimate constant studied in the paper. Nonvanishing conjecture. Any such extremal f...

conjecture | open | MathDB 页面 | 原文线索

V2-043175 | MATHDB-342815

Poonen's separating extremal intersection-closed sets conjecture

摘录 / EXCERPT: Poonen's conjecture. If $|S|>1$ and there is exactly one $\alpha\in X_S$ with

conjecture | open | MathDB 页面 | 原文线索

V2-043176 | MATHDB-342818

Asymptotic growth conjecture for covering systems over Z_s^q

摘录 / EXCERPT: Let $R(s,q)$ denote the minimum size of a covering system of vectors over Z_s^q. For fixed $s\geq 3$, consider the growth of $R(s,q)$ as q tends to infinity. Asymptotic growth...

conjecture | open | MathDB 页面 | 原文线索

V2-043177 | MATHDB-342819

The $s=3$ covering-system asymptotic conjecture

摘录 / EXCERPT: Let $R(s,q)$ denote the minimum size of a covering system of vectors over Z_s^q. In particular, consider the case $s=3$ as q tends to infinity. The $s=3$ asymptotic conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043178 | MATHDB-342831

Multigraph decomposition conjecture for exceptional congruence cases

摘录 / EXCERPT: Multigraph decomposition conjecture. If either

conjecture | open | MathDB 页面 | 原文线索

V2-043179 | MATHDB-342832

Neguț's positivity conjecture for generalized q,t-Catalan polynomials

摘录 / EXCERPT: Neguț's conjecture. The polynomial $f(a_1,\ldots,a_n)$ has nonnegative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-043180 | MATHDB-342833

Neguț's positivity conjecture for the tableau polynomial F

摘录 / EXCERPT: Neguț's conjecture. If $a_2\geqslant a_3\geqslant \cdots\geqslant a_n\geqslant 0$, then $F(a_2,\ldots,a_n)$ has nonnegative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-043181 | MATHDB-342855

Alon-Hefetz-Krivelevich-Tyomkyn square-root anti-concentration conjecture for graph edge statistics

摘录 / EXCERPT: Alon-Hefetz-Krivelevich-Tyomkyn's conjecture. The edge statistic satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043182 | MATHDB-342879

Conjecture on Gallai-Ramsey numbers of even wheels

摘录 / EXCERPT: Let W_{2n} denote the wheel graph with an even cycle of length $2n$, let $GR_k(H)$ denote the k-color Gallai-Ramsey number of a graph H, and let $R_2(W_{2n})$ denote the two-...

conjecture | open | MathDB 页面 | 原文线索

V2-043183 | MATHDB-342889

The spectrum conjecture for odd sun systems

摘录 / EXCERPT: Spectrum conjecture. Let $k\geq 3$ and $v>1$. There exists a k-sun system of K_v if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-043184 | MATHDB-342896

Limiting-constant conjecture for the Ramsey, Paper, Scissors game

摘录 / EXCERPT: Limiting-constant conjecture. There is some constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043185 | MATHDB-342912

Exact polynomial formulas for signaletic transition matrices

摘录 / EXCERPT: For each k, let $\mathsf{M}_k^{{||}}$ and $\mathsf{M}_k^{{\ddagger}}$ denote the parallel and series transition matrices, and let $\mathrm{Eul}_{k}(t)$ be the Eulerian polynomial...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043186 | MATHDB-342913

Eulerian polynomial conjecture for the matrices N_k

摘录 / EXCERPT: For each k, let N_k be the matrix appearing in the expression for the generating series $\mathcal{R}_k(t)$, and write $\pi_{N_k}(t)$ and $\chi_{N_k}(t)$ for its minimal and cha...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043187 | MATHDB-342926

Schelp's dense-subgraph conjecture for Ramsey paths

摘录 / EXCERPT: Let P_k denote the path with k vertices, and let G be a graph on $3n-1$ vertices. Schelp's conjecture. If n is sufficiently large and ... then G arrows P_{2n}, meaning th...

conjecture | open | MathDB 页面 | 原文线索

V2-043188 | MATHDB-342943

The exact cardinality conjecture for largest sum-dominant subsets of an interval

摘录 / EXCERPT: Fix $n\geq 16$. Let N be the cardinality of the largest sum-dominant subset(s) of $\{0,1,\ldots,n-1\}$ that contain(s) 0 and $n-1$. The exact cardinality conjecture. Then ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-043189 | MATHDB-342984

Sali's conjecture on the extremal size of Armstrong codes

摘录 / EXCERPT: Let $f(q,3)$ denote the maximum length n for which a $(q,3,n)$-Armstrong code exists. Sali's conjecture. For all $q\geq 2$, ... The construction from orthogonal double covers gives...

conjecture | open | MathDB 页面 | 原文线索

V2-043190 | MATHDB-342986

Antipalindromic fixed-point classification conjecture for primitive binary morphisms

摘录 / EXCERPT: Antipalindromic fixed-point classification conjecture. The morphism φ or its square φ^2 is conjugated to a morphism in

conjecture | open | MathDB 页面 | 原文线索

V2-043191 | MATHDB-342993

Uniqueness of extremal exchangeable pairs supported by positive dual c-vectors

摘录 / EXCERPT: Positive dual c-vector uniqueness conjecture. Any positive dual $\boldsymbol{c}$-vector supports exactly one extremal exchangeable pair.

conjecture | open | MathDB 页面 | 原文线索

V2-043192 | MATHDB-343040

Prefix palindromic length conjecture for non-ultimately periodic words

摘录 / EXCERPT: Let $u=u[1]u[2]\cdots$ be an infinite word. For $n\in\mathbb N$, let $PPL_u(n)$ denote the minimum number of concatenated palindromes needed to express the prefix of u of length...

conjecture | open | MathDB 页面 | 原文线索

V2-043193 | MATHDB-343048

The Lorentzian conjecture for normalized skew Schur polynomials

摘录 / EXCERPT: For every integer $n\ge 1$ and every pair of partitions $\mu\subseteq\lambda$, let $s_{\lambda/\mu}(x_1,\ldots,x_n)=\sum_{\alpha\in\mathbb{Z}_{\ge 0}^n}c_\alpha x^\alpha$ be the sk...

conjecture | solved | MathDB 页面 | 原文线索

V2-043194 | MATHDB-343103

Stabilization bound for standard skew tableaux

摘录 / EXCERPT: A standard skew tableau has rows whose sizes weakly decrease from top to bottom; write b for the number of rows and let $\stab$ denote its stabilization time, the least k at wh...

conjecture | open | MathDB 页面 | 原文线索

V2-043195 | MATHDB-343104

Unique permutation with maximal stabilization

摘录 / EXCERPT: Let w be a permutation of $\{1,2,\dots,n\}$, viewed as a tableau, and let $\stab(w)$ denote its stabilization time. Maximal-stabilization conjecture. The increasing permutation .....

conjecture | open | MathDB 页面 | 原文线索

V2-043196 | MATHDB-343141

Shapiro-Shapiro's multiplicity monotonicity conjecture for words

摘录 / EXCERPT: Let S_{n+1} be the symmetric group, let ${\mathbf W}_n$ be the set of words in the relevant alphabet, and let $\operatorname{mult}:S_{n+1}\to\mathbb{N}^n$ be given by ... Extend th...

conjecture | open | MathDB 页面 | 原文线索

V2-043197 | MATHDB-343167

Singer graceful-graph conjecture for finite-field graph decompositions

摘录 / EXCERPT: Let $[m]_q=(q^m-1)/(q-1)$ denote the q-analog of m. A graph is Singer-graceful when it satisfies the graceful-labeling condition associated with the Singer difference set. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043198 | MATHDB-343211

The weighted inequality for cross-intersecting pairs with profile-dependent skewness

摘录 / EXCERPT: Let $(A_1,B_1),\dots,(A_m,B_m)$ be pairs of finite non-empty subsets of $\mathbb{N}$. Write $a_i=|A_i|$ and $b_i=|B_i|$ for $1\leq i\leq m$. Assume that ... ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-043199 | MATHDB-343219

Caro-Hegarty-Morrison quadratic-order conjecture for arithmetic-progression zero sums

摘录 / EXCERPT: Let r and s be fixed positive integers. For a function $f:[n]\to\{-r,s\}$ with $f([n])=0$, let $M(r,s,k)$ denote the minimum threshold such that every sufficiently large n fo...

conjecture | open | MathDB 页面 | 原文线索

V2-043200 | MATHDB-343248

The parity-condition conjecture for large acyclic fillings

摘录 / EXCERPT: Large-filling conjecture. Over $\mathbb{F}_2$, for every $d\geq 0$ there exists a number n_d such that every non-trivial $(d-1)$-cycle Z^{d-1} on K_n^d, with $n\geq n_d$, has...

conjecture | open | MathDB 页面 | 原文线索

V2-043201 | MATHDB-343249

The bounded-deficit conjecture for the FILL procedure

摘录 / EXCERPT: Bounded-deficit conjecture. There exists a function $\alpha:\mathbb{N}\mapsto\mathbb{N}$, with $d\mapsto\alpha_d$, such that for every non-trivial $(d-1)$-cycle Z^{d-1} on...

conjecture | open | MathDB 页面 | 原文线索

V2-043202 | MATHDB-343264

Extremal critical-point conjecture for quadrature domains

摘录 / EXCERPT: Extremal critical-point conjecture. For every $n\geq 2$, and for each $k=1,2,\ldots,n+1$, such a domain exists. The bound is attained for $k=n+1$ by the extremal construction descr...

conjecture | open | MathDB 页面 | 原文线索

V2-043203 | MATHDB-343268

Primality characterization for polyomino ideals without zig-zag walks

摘录 / EXCERPT: Primality conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-043204 | MATHDB-343277

Muffin problem scaling conjecture

摘录 / EXCERPT: Scaling conjecture. For all $k,m,s\in\mathbb{N}$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043205 | MATHDB-343278

Muffin problem one-third lower-bound conjecture

摘录 / EXCERPT: One-third lower-bound conjecture. For all $m>s\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-043206 | MATHDB-343279

Balanced piece-count conjecture for supply-constrained muffin problems

摘录 / EXCERPT: Balanced piece-count conjecture. For a supply-constrained muffin problem there exists an optimal solution in which every student receives either n or $n+1$ pieces.

conjecture | open | MathDB 页面 | 原文线索

V2-043207 | MATHDB-343280

Type-1 zero-problem conjecture for 3M-DAPs

摘录 / EXCERPT: Type-1 zero-problem conjecture. If $s_t\leq(v-1)s_v$, then the optimal solution consists of (T,V)-pairs with $a/b=s_t/s_v$, with (T,V) itself being one such pair, and the optim...

conjecture | open | MathDB 页面 | 原文线索

V2-043208 | MATHDB-343281

Lower-bound conjecture for type-2 zero-problem 3M-DAPs

摘录 / EXCERPT: Type-2 zero-problem lower-bound conjecture. If $(v-1)t-v>0$ and $s_t>(v-1)s_v$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043209 | MATHDB-343282

Reduced-pair conjecture for nonintegral type-2 zero-problems

摘录 / EXCERPT: Reduced-pair conjecture. If $s_t>(v-1)s_v$ and $b^*=2s_v/r_v$ is not an integer, then there exists an optimal solution in which all maximal inseparable pairs are either $\hat b$-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-043210 | MATHDB-343283

Recursive muffin-algorithm value conjecture

摘录 / EXCERPT: Recursive value conjecture. For any 3M-DAP P that is not a 0-problem,

conjecture | open | MathDB 页面 | 原文线索

V2-043211 | MATHDB-343294

Two-line approximation conjecture for queens in exile

摘录 / EXCERPT: Let S_c be the row containing the exiled queen in column c, and let $\phi=(1+\sqrt{5})/2$ be the golden ratio. Two-line approximation conjecture. There are constants...

conjecture | open | MathDB 页面 | 原文线索

V2-043212 | MATHDB-343295

Quasi-periodic generating-function conjecture for Sprague-Grundy columns

摘录 / EXCERPT: Consider the array of Sprague-Grundy values for the single-quadrant queens-in-exile problem, and let x be the generating-function variable for a fixed column. Column generating-f...

conjecture | open | MathDB 页面 | 原文线索

V2-043213 | MATHDB-343304

Strong-avoidance Fibonacci enumeration conjecture

摘录 / EXCERPT: Strong-avoidance Fibonacci conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-043214 | MATHDB-343308

The generalized Hamming polynomial reconstruction formula for combinatroids

摘录 / EXCERPT: Generalized Hamming polynomial reconstruction conjecture. The Tutte polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043215 | MATHDB-343312

Binary necklace-splitting conjecture for an arbitrary number of thieves

摘录 / EXCERPT: Binary necklace-splitting conjecture. Given a necklace with n kinds of beads and $r\geq 4$ thieves, there exists a binary necklace splitting of size $(r-1)n$.

conjecture | open | MathDB 页面 | 原文线索

V2-043216 | MATHDB-343321

Frankl's conjecture of order $k|\ell$

摘录 / EXCERPT: Let $\mathfrak{A}_{k|\ell}$ be the class of union-closed families used in the paper, and let $\mathfrak{c}_{k|\ell}$ denote the corresponding supremal proportion of members meeting...

conjecture | open | MathDB 页面 | 原文线索

V2-043217 | MATHDB-343322

Strong Frankl's conjecture of order $k|\ell$

摘录 / EXCERPT: Let $\widetilde{\mathfrak{A}}_{k|\ell}$ be the class of weakly $k|\ell$-separated union-closed families, and let $\widetilde{\mathfrak{c}}_{k|\ell}$ be the corresponding supremal p...

conjecture | open | MathDB 页面 | 原文线索

V2-043218 | MATHDB-343333

Fibonacci prefix antipower conjecture

摘录 / EXCERPT: Let F_n be an even Fibonacci number, where the Fibonacci numbers satisfy the usual recurrence, and let $\mathbf{f}$ denote the Fibonacci word. A k-antipower is a concatenation...

conjecture | open | MathDB 页面 | 原文线索

V2-043219 | MATHDB-343345

Berger-Wang conjecture for modified EGZ constants in powers of $(\mathbb{Z}/n\mathbb{Z})$

摘录 / EXCERPT: For positive integers $n=2^k$ and $d\geq 1$, let $s'_n(G)$ denote the modified Erdős-Ginzburg-Ziv constant of a finite abelian group G. Let ℓ be the smallest integer such th...

conjecture | open | MathDB 页面 | 原文线索

V2-043220 | MATHDB-343346

Modified EGZ upper-bound conjecture for powers of $(\mathbb{Z}/n\mathbb{Z})$

摘录 / EXCERPT: Let n, t, and d be positive integers. Let $s'_{nt}(G)$ denote the modified Erdős-Ginzburg-Ziv constant associated with length nt, and let ℓ be the smallest integer suc...

conjecture | open | MathDB 页面 | 原文线索

V2-043221 | MATHDB-343356

Dreidel conjecture on the average number of spins

摘录 / EXCERPT: Dreidel conjecture. The average number of spins is approximately

conjecture | open | MathDB 页面 | 原文线索

V2-043222 | MATHDB-343386

Arithmetic progression conjecture for weakly fractal matroid classes

摘录 / EXCERPT: Let ${\mathcal M}$ be a minor-closed class of matroids, and let $\Gamma_{\mathcal M}(n)$ denote the proportion of n-element matroids in the class's boundary, namely its excluded...

conjecture | open | MathDB 页面 | 原文线索

V2-043223 | MATHDB-343387

Infinite-antichain conjecture for weakly fractal matroid classes

摘录 / EXCERPT: Let $\mathcal M$ be a class of matroids. An antichain is a collection of matroids in which no member is a minor of another. The class $\mathcal M$ is weakly fractal when the sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-043224 | MATHDB-343396

Approximate orbit structure conjecture for group actions

摘录 / EXCERPT: Let G be a group acting on a set X, let A be a subset of G, and let Y be a finite subset of X. Write $A(Y)$ for the image of Y under A, and let $K\geq 1$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-043225 | MATHDB-343399

Inverse extremal conjecture for restricted signed sumsets containing zero

摘录 / EXCERPT: Inverse conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043226 | MATHDB-343452

Maximum-row-degree exponent conjecture for zero-one matrices

摘录 / EXCERPT: Let $t\geq 2$ be a positive integer, and let A be a zero-one matrix such that every row of A contains at most t 1-entries. Maximum-row-degree exponent conjecture. ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-043227 | MATHDB-343453

Benson exponent conjecture for six-cycle matrices

摘录 / EXCERPT: Let A be a zero-one matrix corresponding to a cycle of length 6. Six-cycle exponent conjecture. ... For every such matrix, the paper gives a lower bound $\Omega(n^{4/3})$ and an...

conjecture | open | MathDB 页面 | 原文线索

V2-043228 | MATHDB-343457

The asymptotic expected homological dimension of the complex of injective words

摘录 / EXCERPT: Asymptotic homological-dimension conjecture. There exists a constant $0<\gamma<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043229 | MATHDB-343466

Kogan's five-type classification conjecture for triples of triangles

摘录 / EXCERPT: Kogan's classification conjecture. Any linking of three non-ordered triangles is combinatorially isotopic to one of the following five types: three triangles whose convex hulls are...

conjecture | open | MathDB 页面 | 原文线索

V2-043230 | MATHDB-343490

The Delta Conjecture, valley version

摘录 / EXCERPT: Let n be a positive integer and let k satisfy $k\leq n$. Let $\mathcal{M P F}_{n,k}$ be the set of valley-marked parking functions of size n with k marked valleys and label...

conjecture | open | MathDB 页面 | 原文线索

V2-043231 | MATHDB-343548

The cubic-curve conjecture for point sets with a piercing set

摘录 / EXCERPT: Let P be a set of n points in general position in the plane, meaning that no three points of P are collinear, and let R be a set of n points disjoint from P. Cubic-curv...

conjecture | open | MathDB 页面 | 原文线索

V2-043232 | MATHDB-343574

One-corner conjecture for rearrangements of 1-Naples parking functions

摘录 / EXCERPT: Let n be a positive integer, and let a parking preference be represented by its associated lattice-path diagram, with corners defined as turns of the path. Suppose that there is...

conjecture | open | MathDB 页面 | 原文线索

V2-043233 | MATHDB-343584

The flag-complex canonical-representation lower-bound conjecture

摘录 / EXCERPT: Canonical-representation lower-bound conjecture. For every $i\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-043234 | MATHDB-343597

Rainbow sunflower conjecture

摘录 / EXCERPT: Let X be a finite ground set and let $\mathcal{F}$ be a w-set system, meaning that every member of $\mathcal{F}$ has size at most w. Color each element of X independently a...

conjecture | open | MathDB 页面 | 原文线索

V2-043235 | MATHDB-343598

Monochromatic sunflower conjecture

摘录 / EXCERPT: Let $r\geq 1$ and let X be a finite ground set. Let $\mathcal{F}$ be a w-set system, meaning that every member has size at most w. A random δ-fraction coloring colors...

conjecture | open | MathDB 页面 | 原文线索

V2-043236 | MATHDB-343634

Connectivity conjecture for minors of k-polymatroids

摘录 / EXCERPT: Let M be a connected k-polymatroid and let N be a connected minor of M. A connected minor-preserving deletion or contraction is a deletion or contraction that remains conne...

conjecture | open | MathDB 页面 | 原文线索

V2-043237 | MATHDB-343655

The projective-geometry embedding conjecture for clean tangled clutters

摘录 / EXCERPT: Projective-geometry embedding conjecture. Every clean tangled clutter embeds a projective geometry over the two-element field.

conjecture | open | MathDB 页面 | 原文线索

V2-043238 | MATHDB-343696

Variance-based anti-concentration conjecture for edge counts in Ramsey graphs

摘录 / EXCERPT: Variance-based anti-concentration conjecture. For every $x\in\mathbb{Z}$,

conjecture | open | MathDB 页面 | 原文线索

V2-043239 | MATHDB-343739

Ionin-Shrikhande conjecture for Ryser design parameters

摘录 / EXCERPT: Ionin-Shrikhande conjecture. For any Ryser design on v points,

conjecture | open | MathDB 页面 | 原文线索

V2-043240 | MATHDB-343795

Regularity-independence conjecture for higher secondary polytopes

摘录 / EXCERPT: Let $\mathcal{A}$ be a point configuration, let $\mathcal{V}$ be its associated vector configuration, and let $\Upsigma_{\mathcal{A},k}$ be the higher secondary polytope defined us...

conjecture | open | MathDB 页面 | 原文线索

V2-043241 | MATHDB-343812

Finite big Ramsey degrees for Fraïssé limits with finite constraints

摘录 / EXCERPT: Let $\mathcal{K}$ be a relational Fraïssé class with finitely many relations and a finite constraint set. A constraint set is a finite set of finite structures such that membership...

conjecture | open | MathDB 页面 | 原文线索

V2-043242 | MATHDB-343813

A Borel Ramsey theorem for strong similarity types of Fraïssé limits

摘录 / EXCERPT: Let $\mathcal{K}$ be a relational Fraïssé class with finitely many relations and a finite constraint set. Suppose that $\mathcal{K}$ satisfies the Ramsey property or has finite sma...

conjecture | open | MathDB 页面 | 原文线索

V2-043243 | MATHDB-343819

The polynomial formula conjecture for six-lozenge tilings

摘录 / EXCERPT: Let $L_{n,l}$ denote the number of lozenge tilings of the equilateral triangle of side length n containing l lozenges, and let M_{n-1} be the polynomial appearing in the uppe...

conjecture | open | MathDB 页面 | 原文线索

V2-043244 | MATHDB-343832

The measure partitioning conjecture for non-atomic measures on the circle

摘录 / EXCERPT: Let $n\ge2$. Let $\mu_1,\mu_2,\dots,\mu_n$ be non-atomic probability measures on the unit circle S^1, and let $\alpha_1,\alpha_2,\dots,\alpha_n$ be non-negative reals summing to...

conjecture | open | MathDB 页面 | 原文线索

V2-043245 | MATHDB-343855

Subquadratic Boolean lattice Ramsey number conjecture

摘录 / EXCERPT: Let Q_m and Q_n denote Boolean lattices of dimensions m and n, respectively, and let $R(Q_m,Q_n)$ be the least integer N such that every red-blue coloring of the Boolean...

conjecture | open | MathDB 页面 | 原文线索

V2-043246 | MATHDB-343907

Gács-Nagy-Hegedűs-Pálvölgyi conjecture on finite-field polynomials with prescribed range

摘录 / EXCERPT: Gács-Nagy-Hegedűs-Pálvölgyi conjecture. If there is no polynomial with range M of degree less than $q-m$, then M contains an element of multiplicity at least $q-m$.

conjecture | open | MathDB 页面 | 原文线索

V2-043247 | MATHDB-343913

Lu-Milans conjecture on Lubell functions of forbidden subposets

摘录 / EXCERPT: Let P be a poset, and let $\bar{h}_n(\mathcal{F})$ denote the Lubell function of a family $\mathcal{F}$ of subsets of $[n]$. Define $\lambda_n(P)$ to be the maximum value of the...

conjecture | open | MathDB 页面 | 原文线索

V2-043248 | MATHDB-343914

Asymptotic Boolean Ramsey number conjecture for the two-element Boolean poset

摘录 / EXCERPT: For integers $k\geq 1$, let $R_k(B_2)$ be the least N such that every coloring of the Boolean lattice $\mathcal{B}_N$ with k colors contains a monochromatic copy of the Boolean...

open_problem | open | MathDB 页面 | 原文线索

V2-043249 | MATHDB-343957

Linear bound conjecture for r-wise fractional L-intersecting families

摘录 / EXCERPT: Let $cmathcal{F}$ be an r-wise fractional L-intersecting family, where $L=\{a/b\}$. The family is r-wise fractional L-intersecting if every r distinct members have pairwise f...

open_problem | open | MathDB 页面 | 原文线索

V2-043250 | MATHDB-343967

Duality conjecture for entropic complexity of access structures

摘录 / EXCERPT: Let P be a participant set, let $\mathcal A\subset 2^P$ be an access structure, and let $\mathcal A^\bot$ denote its dual access structure. Let $\sigma(\mathcal A)$ denote the en...

conjecture | open | MathDB 页面 | 原文线索

V2-043251 | MATHDB-343986

Conjecture that the Ramsey threshold exists for every pair of posets

摘录 / EXCERPT: Let P and Q be fixed non-empty posets. Define the lower and upper Ramsey exponents by ... and ... When these exponents are equal, denote their common value by...

conjecture | open | MathDB 页面 | 原文线索

V2-043252 | MATHDB-343987

Universality conjecture for Ramsey exponents of almost all posets

摘录 / EXCERPT: Let $C_3(N)$ denote the poset used in the paper's Ramsey-threshold analysis, and let $c_{\mathrm{Ram}^-}(P,P)$ and $c_{\mathrm{Ram}^+}(P,P)$ be the lower and upper Ramsey exponents...

conjecture | open | MathDB 页面 | 原文线索

V2-043253 | MATHDB-344000

The extremality criterion for $(n-1)$-element subsets of cyclic groups

摘录 / EXCERPT: Extremality criterion. An $(n-1)$-element subset of $\mathbb Z_n$ is extreme if and only if n is not congruent to 3 modulo 4.

conjecture | open | MathDB 页面 | 原文线索

V2-043254 | MATHDB-344001

The classification conjecture for six-element extreme sets in cyclic groups

摘录 / EXCERPT: Six-element classification conjecture. The only six-element extreme sets in cyclic groups are $\mathbb Z_6$, the set $\{0,1,2,3,4,7\}\subset\mathbb Z_{10}$, and their images under...

conjecture | open | MathDB 页面 | 原文线索

V2-043255 | MATHDB-344003

The differing-Sidon-constant criterion for extreme subsets

摘录 / EXCERPT: Differing-Sidon-constant conjecture. Under these assumptions, A and B are extreme.

conjecture | open | MathDB 页面 | 原文线索

V2-043256 | MATHDB-344033

Erdős's convex sumset conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}$ be a finite convex set, meaning that if $A=\{a_1<a_2<\cdots<a_k\}$, then ... for every $1<i<k$. Erdős's conjecture. For every $\epsilon>0$, ... This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-043257 | MATHDB-344070

Keller's conjecture on face-sharing pairs in cube tilings

摘录 / EXCERPT: Keller's conjecture. For all integers $d\geq 1$, there does not exist a faceshare-free tiling of $\mathbb{R}^d$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043258 | MATHDB-344084

Duality conjecture for strongly pigeonhole classes of matroids

摘录 / EXCERPT: Let $\mathcal{M}$ be a strongly pigeonhole class of matroids. Duality conjecture. The class $\{M^{*}\colon M\in\mathcal{M}\}$ is strongly pigeonhole. The preceding result establish...

conjecture | open | MathDB 页面 | 原文线索

V2-043259 | MATHDB-344085

Definability conjecture for lattice path matroids

摘录 / EXCERPT: A lattice path matroid is the matroid represented by a lattice path in the standard lattice-path construction. Let ${\mathit{MS}_{0}}$ denote monadic second-order logic over matroi...

conjecture | open | MathDB 页面 | 原文线索

V2-043260 | MATHDB-344086

CMS_0 non-definability conjecture for infinite-group gain-graphic matroids

摘录 / EXCERPT: Let H be an infinite group. Let ${\mathit{CMS}_{0}}$ denote counting monadic second-order logic over matroids without quantification over edge sets. Non-definability conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043261 | MATHDB-344087

Undecidability conjecture for rank-3 gain-graphic matroids

摘录 / EXCERPT: Let H be an infinite group, and consider the class of rank-3 H-gain-graphic matroids. Undecidability conjecture. This class has an undecidable ${\mathit{MS}_{0}}$ theory. The...

conjecture | open | MathDB 页面 | 原文线索

V2-043262 | MATHDB-344088

Efficient pigeonhole conjecture for lattice path matroids

摘录 / EXCERPT: Let $\mathcal{M}$ be the class of lattice path matroids. The class is efficiently pigeonhole when the relevant pigeonhole bounds can be obtained efficiently. Efficient pigeonhole c...

conjecture | open | MathDB 页面 | 原文线索

V2-043263 | MATHDB-344089

Efficient pigeonhole conjecture for bicircular and gain-graphic matroids

摘录 / EXCERPT: Let H be a finite group. Efficient pigeonhole conjecture. The class of bicircular matroids is efficiently pigeonhole, and the class of H-gain-graphic matroids is efficiently pi...

conjecture | open | MathDB 页面 | 原文线索

V2-043264 | MATHDB-344091

Conjecture on generalized Fibonacci growth rates below 2^n

摘录 / EXCERPT: Generalized Fibonacci growth conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043265 | MATHDB-344095

Tuza's conjecture for t-intersecting set pairs

摘录 / EXCERPT: Let $AK(n,k,t)$ denote the maximum size of a k-uniform t-intersecting family $\mathcal F\subseteq \binom{[n]}{k}$. Let $\{(A_i,B_i)\}_{i=1}^m$ be a collection of pairs of sets...

conjecture | open | MathDB 页面 | 原文线索

V2-043266 | MATHDB-344131

The nonchalant-word infinitude conjecture for three-letter alphabets

摘录 / EXCERPT: Let $\mathbb A$ be a fixed ordered alphabet, and recursively construct the sequence of nonchalant words by starting with $G_1=\mathtt a$ and, at each step, inserting the earliest p...

conjecture | open | MathDB 页面 | 原文线索

V2-043267 | MATHDB-344132

The nonexistence conjecture for extremal square-free words over four-letter alphabets

摘录 / EXCERPT: A square-free word is a word containing no factor of the form uu with u nonempty, and an extremal square-free word is a square-free word that cannot be extended by inserting a...

conjecture | open | MathDB 页面 | 原文线索

V2-043268 | MATHDB-344136

Bounded-Jacobian-exponent conjecture for connected regular matroids

摘录 / EXCERPT: Regular-matroid Jacobian exponent conjecture. For every positive integer k, there are only finitely many connected regular matroids M such that the exponent of...

conjecture | open | MathDB 页面 | 原文线索

V2-043269 | MATHDB-344167

The Genocchi cycle-counting conjecture

摘录 / EXCERPT: Genocchi cycle-counting conjecture. For all $n\geq 1$, g_n is equal to the number of cycles on $[2n]$ with only even-odd drops.

conjecture | open | MathDB 页面 | 原文线索

V2-043270 | MATHDB-344177

Experimental coincidence classification conjecture for mesh patterns

摘录 / EXCERPT: Experimental coincidence classification conjecture. The mesh patterns of size n can be coincidence classified by experimental classification with permutations of size

conjecture | open | MathDB 页面 | 原文线索

V2-043271 | MATHDB-344197

The non-concentration conjecture for random-graph Ramsey numbers

摘录 / EXCERPT: Random-graph Ramsey non-concentration conjecture. With high probability, $\log r(G_{n,p})$ is not concentrated on any interval of length $O(1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-043272 | MATHDB-344223

The asymptotic irrelevance conjecture for quasigroup identities

摘录 / EXCERPT: Asymptotic irrelevance conjecture. In the large, cancellation resulting from the quasigroup identities has a negligible effect on the asymptotic behavior of the peri-Catalan number...

conjecture | open | MathDB 页面 | 原文线索

V2-043273 | MATHDB-344224

The fitted asymptotic formula conjecture for peri-Catalan numbers

摘录 / EXCERPT: Fitted asymptotic formula conjecture. For all generator counts s, the limit exists and approximately satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043274 | MATHDB-344247

The asymptotic domination number conjecture for $G_{5,3}$

摘录 / EXCERPT: Let $G_{k,\ell}$ be the graph whose vertices are the ℓ- and k-subsets of $[n]$, with adjacency given by containment. Write $\gamma(G_{k,\ell})$ for its domination number. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-043275 | MATHDB-344248

The asymptotic tightness conjecture for domination lower bounds

摘录 / EXCERPT: Fix integers k and ℓ with ... Let $G_{k,\ell}$ be the two-level cube graph considered in the paper, let $\gamma(G_{k,\ell})$ denote its domination number, and let the lower bo...

conjecture | open | MathDB 页面 | 原文线索

V2-043276 | MATHDB-344251

A rank-three Davenport constant gap conjecture

摘录 / EXCERPT: Rank-three Davenport constant conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-043277 | MATHDB-344273

Linear Ramsey numbers and bounded co-chromatic number for finitely defined hereditary classes

摘录 / EXCERPT: Linear Ramsey-co-chromatic conjecture. A finitely defined hereditary class is of linear Ramsey numbers if and only if it has bounded co-chromatic number.

conjecture | open | MathDB 页面 | 原文线索

V2-043278 | MATHDB-344274

Forbidden-subgraph characterization of linear Ramsey numbers

摘录 / EXCERPT: Forbidden-subgraph conjecture for linear Ramsey numbers. The Ramsey numbers of X are linear if and only if the set of minimal forbidden induced subgraphs for X contains a P_3...

conjecture | open | MathDB 页面 | 原文线索

V2-043279 | MATHDB-344275

Erdős-Hajnal polynomial Ramsey-number conjecture for hereditary classes

摘录 / EXCERPT: Erdős-Hajnal conjecture. There are constants A and k such that

conjecture | open | MathDB 页面 | 原文线索

V2-043280 | MATHDB-344276

Linear bipartite Ramsey numbers and forbidden forests

摘录 / EXCERPT: Bipartite forest conjecture. Bipartite Ramsey numbers in X are linear if and only if M contains a forest and the bipartite complement of a forest.

conjecture | open | MathDB 页面 | 原文线索

V2-043281 | MATHDB-344313

Enumeration conjecture for geometric multi-chains of t-filters

摘录 / EXCERPT: Enumeration conjecture. The number of geometric multi-chains of t-filters $(V_m,V_{m-1},\ldots,V_1)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-043282 | MATHDB-344314

Conjectural combinatorial formula for the parabolic H-triangle

摘录 / EXCERPT: Combinatorial H-triangle conjecture. For positive integers m,n,t,

conjecture | open | MathDB 页面 | 原文线索

V2-043283 | MATHDB-344316

Talagrand's fractional expectation-threshold conjecture

摘录 / EXCERPT: Talagrand's fractional expectation-threshold conjecture. There is a universal constant K such that

conjecture | open | MathDB 页面 | 原文线索

V2-043284 | MATHDB-344317

Talagrand's equivalence conjecture for integral and fractional expectation-thresholds

摘录 / EXCERPT: Let X be a finite set and let ${\mathcal F}\subseteq 2^X$ be increasing. Let $q({\mathcal F})$ be the expectation-threshold and $q_f({\mathcal F})$ the fractional expectation-thre...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-043285 | MATHDB-344331

Ramsey robustness conjecture for Ramsey-infinite graphs

摘录 / EXCERPT: Let $r\geq 2$, let H be a graph, and let $\mathcal{M}_r(H)$ denote the family of r-Ramsey-minimal graphs for H. Let $\beta_r(H;G)$ be the Ramsey robustness of G with respec...

conjecture | open | MathDB 页面 | 原文线索

V2-043286 | MATHDB-344369

Graver's maximality conjecture in dimension three

摘录 / EXCERPT: Let ${\cal R}_{3,n}$ be the generic 3-dimensional rigidity matroid on $E(K_n)$, and let an abstract 3-rigidity matroid be an abstract d-rigidity matroid with $d=3$. Graver's...

conjecture | open | MathDB 页面 | 原文线索

V2-043287 | MATHDB-344370

Free-elevation rank formula for K_{d+2}-matroids

摘录 / EXCERPT: Let M_0 be the rank ${{d+2}\choose{2}}$ truncation of ${\cal R}_{d,n}$, equivalently of any K_{d+2}-matroid on $E(K_n)$, with $n\geq d+2$. A K_{d+2}-sequence is a sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-043288 | MATHDB-344384

Sharp Hilton-Milner conjecture for non-trivial d-wise intersecting families

摘录 / EXCERPT: Sharp Hilton-Milner conjecture. For $k>d\geq 2$ and $n\geq kd/(d-1)$, the unique extremal non-trivial d-wise intersecting families of k-element subsets of $[n]$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-043289 | MATHDB-344408

Latin Rectangle Conjecture

摘录 / EXCERPT: For $\alpha,\beta\in\mathbb{N}$ with $\alpha\leq\beta$ and β even, fix a set $\mathcal U$ of β vectors in $\mathbb{C}^{\alpha}$ generically. A Latin Rectangle for...

conjecture | open | MathDB 页面 | 原文线索

V2-043290 | MATHDB-344421

Uniform deficit conjecture for 1-cross intersecting set pair systems

摘录 / EXCERPT: Let $m_n(*, *, 1)$ denote the maximum size of a $(1,1,1)$-? set pair system on an n-element ground set under the notation of the paper. Uniform deficit conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-043291 | MATHDB-344422

Vanishing proportion conjecture for 1-cross intersecting set pair systems

摘录 / EXCERPT: Let $m_n(*, *, 1)$ and $m_n(*, *, *)$ be the extremal quantities for the corresponding set pair systems, with notation as in the paper. Vanishing proportion conjecture. ... The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-043292 | MATHDB-344430

The periodic perturbation conjecture for maximum-coefficient locations

摘录 / EXCERPT: Let $\mathbf{B}$ be the base vector, and let $\mathbf{u}_r$ and $\mathbf{v}_r$ denote the perturbation vectors defined by the rows of the source's tables. Periodic perturbation con...

conjecture | open | MathDB 页面 | 原文线索

V2-043293 | MATHDB-344473

Kupavskii's conjecture on disjoint cross-intersecting families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $\binom{[n]}{r}$ denote the family of all r-element subsets of $[n]$. Families $\mathcal{A},\mathcal{B}\subseteq\binom{[n]}{r}$ are cross inters...

conjecture | open | MathDB 页面 | 原文线索

V2-043294 | MATHDB-344492

Ochem's exponential growth conjecture for threshold languages

摘录 / EXCERPT: For every integer $n\geq 2$, let $A_n=\{\texttt{1},\texttt{2},\dots,\texttt{n}\}$ and let T_n be the language of all $\operatorname{RT}(n)^+$-free words over A_n, where...

conjecture | open | MathDB 页面 | 原文线索

V2-043295 | MATHDB-344493

Shur and Gorbunova's asymptotic growth-rate conjecture for threshold languages

摘录 / EXCERPT: For every $n\geq 2$, let T_n be the threshold language of order n, and define its growth rate by ... Since T_n is factorial, this limsup can equivalently be replaced by a limit...

conjecture | open | MathDB 页面 | 原文线索

V2-043296 | MATHDB-344498

The subsquare extension conjecture for Cayley tables

摘录 / EXCERPT: Subsquare extension conjecture. Then R is contained in an $m \times m$ subsquare of L.

conjecture | open | MathDB 页面 | 原文线索

V2-043297 | MATHDB-344499

Brualdi-Stein conjecture on near transversals in Latin squares

摘录 / EXCERPT: Brualdi-Stein conjecture. Every Latin square contains a near transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-043298 | MATHDB-344500

Stein's conjecture on transversals in Latin arrays

摘录 / EXCERPT: Stein's conjecture. Let R be an $(n-1)\times n$ Latin array. Then R contains a transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-043299 | MATHDB-344501

Near transversals in Latin arrays

摘录 / EXCERPT: Near-transversal conjecture for Latin arrays. Every Latin array contains a near transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-043300 | MATHDB-344509

Morales' Ehrhart positivity conjecture for Tesler polytopes

摘录 / EXCERPT: For any $n\geq 1$, let $\operatorname{Tes}_{n}(\boldsymbol{1})$ and $\operatorname{Tes}_{n}(1,0,\dots,0)$ be Tesler polytopes, where $\boldsymbol{1}=(1,\dots,1)$. A polytope is Ehr...

conjecture | open | MathDB 页面 | 原文线索

V2-043301 | MATHDB-344528

Gerbner et al.'s hemi-bundled Bollobás conjecture for intersecting families

摘录 / EXCERPT: Gerbner et al.'s conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-043302 | MATHDB-344545

Shape-independence conjecture for fillings of moon polyominoes

摘录 / EXCERPT: Shape-independence conjecture. The number $N(\mathcal{M};n;ne=u;se=v)$ depends only on the lengths of the rows of $\mathcal{M}$, not on its actual shape, and

conjecture | open | MathDB 页面 | 原文线索

V2-043303 | MATHDB-344548

Axis-parallel extremality conjecture for taxicab distance sets

摘录 / EXCERPT: Let $d,k\in\mathbb{N}$. For $P\subseteq\mathbb{R}^d$ and an ℓ^1-distance $\lambda>0$, say that λ occurs in an axis-parallel direction if there are $x\in P$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043304 | MATHDB-344568

Conjecture on the sharp three-dimensional multijoints constant

摘录 / EXCERPT: For a multijoints configuration $(\mathcal J,\mathcal L_1,\mathcal L_2,\mathcal L_3)$ in $\mathbb F^3$, let $|\mathcal L_i|$ denote the size of the ith line family. The multijoin...

conjecture | open | MathDB 页面 | 原文线索

V2-043305 | MATHDB-344569

The generalized joints bound for flats

摘录 / EXCERPT: Let $d_1,\dots,d_r$ and $m_1,\dots,m_r$ be positive integers, put $d=d_1m_1+\cdots+d_rm_r$, and let a $(d_1^{m_1},\dots,d_r^{m_r})$-joints configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-043306 | MATHDB-344582

Bolza surface conjecture for extremal-length systoles in genus 2

摘录 / EXCERPT: Let $\mathcal{M}_2$ denote the moduli space of conformal classes of closed genus-2 surfaces. For a conformal class $X\in\mathcal{M}_2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-043307 | MATHDB-344586

Griggs's dominance conjecture for chain decompositions of the Boolean lattice

摘录 / EXCERPT: Griggs's conjecture. If $s_1\geq\dots\geq s_M$ is a sequence of positive integers dominated by $\sigma_1,\dots,\sigma_M$ and

conjecture | open | MathDB 页面 | 原文线索

V2-043308 | MATHDB-344587

Upper-half dominance conjecture for chain decompositions of the Boolean lattice

摘录 / EXCERPT: Upper-half dominance conjecture. If $s_1\geq\dots\geq s_M$ is a sequence of positive integers dominated by $\sigma_1',\dots,\sigma_M'$ and

conjecture | open | MathDB 页面 | 原文线索

V2-043309 | MATHDB-344596

De Loera et al.'s Ehrhart positivity conjecture for matroid polytopes

摘录 / EXCERPT: Let M be a matroid, and let $\mathscr{P}(M)$ denote its basis polytope. Its Ehrhart polynomial is the polynomial counting lattice points in integer dilations of $\mathscr{P}(M)$....

conjecture | open | MathDB 页面 | 原文线索

V2-043310 | MATHDB-344603

The conjecture on reducing matroids to partition matroids

摘录 / EXCERPT: Let M be a k-colorable matroid, meaning that its ground set can be partitioned into k independent sets. A reduction to a partition matroid is a partition matroid on the same...

conjecture | open | MathDB 页面 | 原文线索

V2-043311 | MATHDB-344604

Aharoni-Berger's common independent-set partition conjecture

摘录 / EXCERPT: Let $M_1=(S,\mathcal{I}_1)$ and $M_2=(S,\mathcal{I}_2)$ be loopless matroids. Write $\chi(M_i)$ for the coloring number of M_i, namely the least number of independent sets into...

conjecture | open | MathDB 页面 | 原文线索

V2-043312 | MATHDB-344605

The strongly base orderable reduction conjecture for matroids

摘录 / EXCERPT: Let $M=(S,\mathcal{I})$ be a k-colorable matroid, meaning that S can be partitioned into k independent sets. A strongly base orderable matroid is a matroid for which, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-043313 | MATHDB-344621

The multiple-free subset-sum conjecture for powers of two

摘录 / EXCERPT: Let $z\in\mathbb{N}$ and let $A=\{a_1,\ldots,a_k\}\subseteq[2^z]$, where $[2^z]=\{1,2,\ldots,2^z\}$. For each non-empty subset A_i of A, write $S(A_i)$ for the sum of its eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-043314 | MATHDB-344682

Alon's universality conjecture for random permutations

摘录 / EXCERPT: Let k be a positive integer and let $\mathcal{S}_n$ be the set of permutations of $[n]$. A permutation is k-universal if it contains every pattern in $\mathcal{S}_k$. For fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-043315 | MATHDB-344706

Projective-geometry embedding conjecture for ideal tangled clutters

摘录 / EXCERPT: Projective-geometry embedding conjecture. There exists an integer $\ell\geq3$ such that every ideal tangled clutter embeds one of $PG(0,2),\ldots,PG(\ell-1,2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-043316 | MATHDB-344759

The smiling-bouquet conjecture

摘录 / EXCERPT: Smiling-bouquet conjecture. For every bouquet C, every colouring of the plane with finitely many but at least two colours contains a smiling congruent copy of C.

conjecture | open | MathDB 页面 | 原文线索

V2-043317 | MATHDB-344760

Erdős et al.'s conjecture on almost-monochromatic congruent triples

摘录 / EXCERPT: Erdős et al.'s conjecture. There is a non-monochromatic colouring of $\mathbb{R}^2$ that contains no almost-monochromatic copy congruent to (S,s_0) if and only if S is collinea...

conjecture | open | MathDB 页面 | 原文线索

V2-043318 | MATHDB-344766

Periodicity conjecture for the maximal concentration parameter

摘录 / EXCERPT: Periodicity conjecture. For sufficiently large odd n, the function

conjecture | open | MathDB 页面 | 原文线索

V2-043319 | MATHDB-344776

Brown-Erey's 2-connected graph coloring conjecture

摘录 / EXCERPT: Brown-Erey's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043320 | MATHDB-344777

Tomescu's conjectural coloring bound for ℓ-connected graphs

摘录 / EXCERPT: Tomescu's ℓ-connected generalization.

conjecture | open | MathDB 页面 | 原文线索

V2-043321 | MATHDB-344785

The cube-root fluctuation conjecture for the LCS concatenation gain

摘录 / EXCERPT: Let V and W be independent random words of length $2n$, written as concatenations $V=V_1V_2$ and $W=W_1W_2$, where each component has length n. Define ... and write...

conjecture | open | MathDB 页面 | 原文线索

V2-043322 | MATHDB-344797

Conjectured remaining 1-defective Ramsey numbers in bipartite graphs

摘录 / EXCERPT: Let $R_{1}^{\mathcal{BIP}}(4,j)$ denote the 1-defective Ramsey number for bipartite graphs. Bipartite 1-defective Ramsey conjecture. ... for $j\in\{10,11,12,18,19\}$. This fills the...

conjecture | open | MathDB 页面 | 原文线索

V2-043323 | MATHDB-344800

Kaplan's conjecture on numerical semigroup counts by multiplicity and genus

摘录 / EXCERPT: Let $N_m(g)$ denote the number of numerical semigroups S with multiplicity $\mathsf m(S)=m$ and genus $g(S)=g$. Kaplan's conjecture. For any $m\ge 2$, we have ... This refines the...

conjecture | open | MathDB 页面 | 原文线索

V2-043324 | MATHDB-344807

The square bracket polarized partition characterization of the selection property $S_1(\Omega, \Omega)$

摘录 / EXCERPT: Square bracket polarized partition conjecture. If (X,τ) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043325 | MATHDB-344810

The ball-or-sphere conjecture for polarizations of Artinian monomial ideals

摘录 / EXCERPT: Let I be an Artinian monomial ideal and let J be a polarization of I. Write $4\Delta(J)4$ for the simplicial complex associated to J. Ball-or-sphere conjecture. The simplic...

conjecture | open | MathDB 页面 | 原文线索

V2-043326 | MATHDB-344849

The extremal-function classification conjecture for binary (3, t)-formations

摘录 / EXCERPT: Extremal-function classification conjecture. Among these formations, $\operatorname{fw}(u)=2t-1$ if and only if u is one of

conjecture | open | MathDB 页面 | 原文线索

V2-043327 | MATHDB-344902

Cover-by-elementary-operation conjecture for positroids

摘录 / EXCERPT: Let σ and π be positroids on $[n]$ of ranks $k-1$ and k, respectively. For $A\subset[k]$, let $\overleftarrow{\rho_A}(\pi)$ denote the positroid obtained from π by...

conjecture | open | MathDB 页面 | 原文线索

V2-043328 | MATHDB-344962

Erdős-Frankl-Pippenger classification conjecture for intersecting permutation families

摘录 / EXCERPT: Let S_n be the symmetric group on $[n]$. A family $\mathcal{F} \subset S_n$ is t-intersecting if every two permutations in $\mathcal{F}$ agree on at least t points. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043329 | MATHDB-344963

Tightness of the compact formulation for min-stop ramping constraints

摘录 / EXCERPT: Tightness conjecture. The polytope defined for a single unit with constraints ... describes the convex hull of the feasible integer points for the min-stop ramping constraints. Equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-043330 | MATHDB-344967

Adin's nonnegativity conjecture for the cubical g-vector

摘录 / EXCERPT: Let Q be a cubical d-polytope, meaning that every proper face of Q is combinatorially a cube. Its cubical h-polynomial is written as ... and its cubical g-vector is...

conjecture | open | MathDB 页面 | 原文线索

V2-043331 | MATHDB-344998

Kommemi's sub-extremality conjecture

摘录 / EXCERPT: Let $\mathcal{Q}^+$ be the quotient spacetime, $\mathcal{I}^+$ future null infinity, and interpret “sub-extremal in the limit” as the limiting black-hole condition stated in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-043332 | MATHDB-345012

Bang-Koolen's classification conjecture for geometric distance-regular graphs

摘录 / EXCERPT: Bang-Koolen's classification conjecture. Any such graph is a Johnson graph, a Hamming graph, a Grassmann graph, a bilinear forms graph, or has a number of vertices bounded by a fun...

conjecture | open | MathDB 页面 | 原文线索

V2-043333 | MATHDB-345013

Babai's motion conjecture for primitive distance-regular graphs

摘录 / EXCERPT: Babai's motion conjecture. There exists an $\alpha>0$ such that either

conjecture | open | MathDB 页面 | 原文线索

V2-043334 | MATHDB-345015

Asymptotic sharpness conjecture for broadcast domination in the infinite grid

摘录 / EXCERPT: Let t and r be positive integers, and let $\delta_{t,r}(\mathbb{Z}^2)$ denote the minimum density of a (t,r)-broadcasting set in the infinite square grid. Asymptotic sharpnes...

conjecture | open | MathDB 页面 | 原文线索

V2-043335 | MATHDB-345016

Continuous broadcast-domination density conjecture in the plane

摘录 / EXCERPT: Let a set of towers $\mathcal{T}\subset\mathbb{R}^2$ be (t,r)-broadcasting if every point $v\in\mathbb{R}^2$ satisfies ... where d is the ℓ_1 distance. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-043336 | MATHDB-345108

Divisibility conjecture for evaluations of monomials times powers of 1+x

摘录 / EXCERPT: Let ρ be the evaluation map used in the paper, and let i,j be nonnegative integers. Divisibility conjecture. For every i,j, the number $\rho(x^i(1+x)^j)$ is divisible by .....

conjecture | open | MathDB 页面 | 原文线索

V2-043337 | MATHDB-345109

Factorial determinant conjecture for the normalized evaluation matrices

摘录 / EXCERPT: Let ρ be the evaluation map used in the paper. Assuming the divisibility conjecture, for every integer n define the square matrix $\mathbf{M}_n$ by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-043338 | MATHDB-345115

Shalaby's strong Skolem starter conjecture

摘录 / EXCERPT: Let n be an integer with $n\geq11$, and let $\mathbb{Z}_n$ be the additive cyclic group of order n. A strong Skolem starter is a starter for $\mathbb{Z}_n$ that is both Skolem...

conjecture | open | MathDB 页面 | 原文线索

V2-043339 | MATHDB-345183

Non-holonomicity conjecture for the two-player winning probability

摘录 / EXCERPT: Let $p=q=1/2$, and let $W_{n,0,0}(x)$ be the probability generating function for the two-player pile game with boundary, with n the initial pile size. Define the first player's w...

conjecture | open | MathDB 页面 | 原文线索

V2-043340 | MATHDB-345195

Gawron's conjecture on the order of longest twins in permutations

摘录 / EXCERPT: Gawron's conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-043341 | MATHDB-345196

The asymptotic restricted (k,ℓ)-sum-free set conjecture

摘录 / EXCERPT: The restricted (k,ℓ)-sum-free asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043342 | MATHDB-345207

Block diagonal form conjecture for Varchenko matrices of oriented matroids

摘录 / EXCERPT: Let an oriented matroid have its Varchenko matrix and intersection poset, with non-degenerate elements and degenerate points as specified in the conjecture. Block diagonal form con...

conjecture | open | MathDB 页面 | 原文线索

V2-043343 | MATHDB-345208

Algebraic Grünbaum-van Kampen-Flores conjecture

摘录 / EXCERPT: Let K and L be simplicial complexes, and let ϑ denote their van Kampen obstruction. Algebraic Grünbaum-van Kampen-Flores conjecture. If ... then ... The conjecture is an...

conjecture | open | MathDB 页面 | 原文线索

V2-043344 | MATHDB-345209

Algebraic Menger conjecture

摘录 / EXCERPT: Let K and L be simplicial complexes, and let ϑ denote their van Kampen obstruction. Algebraic Menger conjecture. If ... then ... This is presented as a second algebraic n...

conjecture | open | MathDB 页面 | 原文线索

V2-043345 | MATHDB-345256

The sharp Vinogradov mean value bound for integer sets

摘录 / EXCERPT: Sharp Vinogradov mean value conjecture. For every $\epsilon>0$, if $|A|$ is sufficiently large, then

conjecture | open | MathDB 页面 | 原文线索

V2-043346 | MATHDB-345262

Non-uniqueness conjecture for s-deep Zero Linear Recurrence Sequences

摘录 / EXCERPT: Non-uniqueness conjecture. Uniqueness of decomposition is lost for at least one positive integer N.

conjecture | open | MathDB 页面 | 原文线索

V2-043347 | MATHDB-345263

Uniqueness conjecture for s-deep Zero Linear Recurrence Sequences with a specified recurrence

摘录 / EXCERPT: Uniqueness conjecture. There exists a unique decomposition for each positive integer N.

conjecture | open | MathDB 页面 | 原文线索

V2-043348 | MATHDB-345264

Asymptotic constant conjecture for t-dimension of interval divisibility posets

摘录 / EXCERPT: For $0<\alpha<1$, let $\mathcal{D}_{(\alpha n,n]}$ be the divisibility poset on the integers in $(\alpha n,n]$. For an integer $t\geq2$, let $\dim_t(P)$ denote the t-dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-043349 | MATHDB-345273

Conjecture on the eternal game chromatic number of even random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the random graph on n vertices in which each edge is present independently with probability p, where n is even and p is a fixed constant in...

conjecture | open | MathDB 页面 | 原文线索

V2-043350 | MATHDB-345282

Divot classification for missing differences

摘录 / EXCERPT: Divot classification conjecture. For every $k>1$, $k(k-1)+3$ is a divot of P_n^{H} for sufficiently large n. Furthermore, these are the only divots.

conjecture | open | MathDB 页面 | 原文线索

V2-043351 | MATHDB-345283

Persistence of divots in missing-difference distributions

摘录 / EXCERPT: Divot persistence conjecture. If k is a divot of $P_{n_1}^{\mathrm{H}}$, then k is also a divot of P_n^{H} for every $n>n_1$.

conjecture | open | MathDB 页面 | 原文线索

V2-043352 | MATHDB-345284

Finite-n peak at four missing differences

摘录 / EXCERPT: Peak-at-four conjecture. For all $n\ge 15$ and every $k\ne4$,

conjecture | open | MathDB 页面 | 原文线索

V2-043353 | MATHDB-345285

No-divot conjecture for limiting missing-difference probabilities

摘录 / EXCERPT: No-divot conjecture. The limiting probabilities satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-043354 | MATHDB-345312

Bukh-Matoušek-Pálvölgyi k-fan incidence conjecture

摘录 / EXCERPT: Fix an integer $k\geq 3$. Let P be a set of n points and let $\mathcal{L}$ be a set of n lines in the plane. A k-fan is a configuration consisting of distinct points...

conjecture | open | MathDB 页面 | 原文线索

V2-043355 | MATHDB-345316

Conjecture on apwenian fixed points of non-constant-length substitutions

摘录 / EXCERPT: Let a substitution act on either the alphabet $\{0,1\}$ or the alphabet $\{-1,1\}$, and suppose its fixed point is a sequence over that alphabet. A sequence is apwenian if its Hank...

conjecture | open | MathDB 页面 | 原文线索

V2-043356 | MATHDB-345331

Conjectured identity for two-toned tiling numbers

摘录 / EXCERPT: Let $a_s(r,n)$ denote the number of two-toned tilings of a $1\times(n+r+s)$ grid with r red squares, where the last s tiles must be white. For all $s,r,n\geq 1$, the conjecture...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-043357 | MATHDB-345332

Formula for runs of a fixed part and length in compositions

摘录 / EXCERPT: Let $R(n,k,l)$ denote the number of runs of k of length l over all compositions of n, and let $a(1,m)$ be the corresponding two-toned tiling count, interpreted as zero when i...

conjecture | open | MathDB 页面 | 原文线索

V2-043358 | MATHDB-345383

Corwin-Hammond classification conjecture for extremal Brownian Gibbs line ensembles

摘录 / EXCERPT: A Brownian Gibbs $\mathbb{N}$-indexed line ensemble is a random collection of curves $\mathcal{L}=(\mathcal{L}_i)_{i\in\mathbb{N}}$ satisfying the Brownian Gibbs property. Define t...

open_problem | open | MathDB 页面 | 原文线索

V2-043359 | MATHDB-345388

Maximum monotone paths conjecture for simple 3-dimensional polytopes

摘录 / EXCERPT: Let P be a simple 3-dimensional polytope with $2n$ vertices, and let f be a generic linear functional on P. Let $\b5(P,f)$ denote the number of f-monotone paths on P, a...

conjecture | open | MathDB 页面 | 原文线索

V2-043360 | MATHDB-345389

Minimum diameter conjecture for flip graphs of 3-dimensional polytopes

摘录 / EXCERPT: Let P range over all 3-dimensional polytopes with n vertices, and let f range over all generic linear functionals on P. Write $G(P,f)$ for the flip graph associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-043361 | MATHDB-345431

The threshold conjecture for biased Maker-Breaker H-factor games

摘录 / EXCERPT: Let b_Δ denote the threshold bias for the biased Maker-Breaker H-factor game, where H has maximum degree Δ. Threshold conjecture. For all $\Delta \ge 3$, ... This p...

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V2-043362 | MATHDB-345448

The convex-polytope realization question for convex codes

摘录 / EXCERPT: Convex-polytope realization question. Can every convex code be realized with convex polytopes?

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V2-043363 | MATHDB-345451

The Genius conjecture on linearity of the functions F_i

摘录 / EXCERPT: Let p be a fixed positive integer. Let $u_1=1$, let $u_2,\dots,u_p$ and $d_1,\dots,d_p$ be variables, and let $F_1=1$ and $F_i=F_i(\{u_k\},\{d_k\})$ for $i=2,\dots,p$ be the uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-043364 | MATHDB-345452

The Genius asymptotic conjecture for the functions F_i

摘录 / EXCERPT: Let p range over the positive integers, and write the unique functions from the coefficient identity as $F_i=F_i(p,\{u_k\},\{d_k\})$ for $i=2,\dots,p$. The Genius asymptotic conj...

conjecture | open | MathDB 页面 | 原文线索

V2-043365 | MATHDB-345454

The Awesome conjecture

摘录 / EXCERPT: Let r and j be parameters, let z_i be positive integers, and let c_i be coefficients. Define ... with $a_0=1$. The Awesome conjecture. The coefficients of $\ln(F)$ satisfy ......

conjecture | open | MathDB 页面 | 原文线索

V2-043366 | MATHDB-345516

The odd-sided simple polygon intersection bound

摘录 / EXCERPT: Odd-sided intersection bound. There are at most

conjecture | open | MathDB 页面 | 原文线索

V2-043367 | MATHDB-345544

The extremal degree conjecture for eventually constant 0-Hecke operators

摘录 / EXCERPT: Extremal degree conjecture. If $T\in H_0(S_n)$ is eventually constant, then

conjecture | open | MathDB 页面 | 原文线索

V2-043368 | MATHDB-345546

The saturation conjecture for densest packings of discs with finitely many radii

摘录 / EXCERPT: Consider packings by n different given discs. A packing is saturated if no further disc can be added. Saturation conjecture. If the densest compact packing is saturated, then the...

conjecture | open | MathDB 页面 | 原文线索

V2-043369 | MATHDB-345562

Nonexistence of infinity-regular subrows in the Kolakoski sequence

摘录 / EXCERPT: Let S be the Kolakoski sequence and let $\mathcal{SR}(S)$ denote its set of subrows. An ∞-regular subrow is a subrow that is k-regular for every k. Nonexistence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-043370 | MATHDB-345563

Uniqueness of subrows from parity-history integrals

摘录 / EXCERPT: Let u and w be two subrows of the Kolakoski sequence, and let $P_n(u)$ denote the parity-history integral associated with u at index n. Parity-history uniqueness conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-043371 | MATHDB-345564

Central log-concavity conjecture for Gaussian binomial coefficients

摘录 / EXCERPT: For integers $n\geq 45$ and $13\leq k\leq n-13$, let ... be the Gaussian binomial coefficient, so that $a_r=[q^r]\left[\begin{smallmatrix}n\\k\end{smallmatrix}\right]$. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-043372 | MATHDB-345630

The biconvex-polytope realization conjecture for tropical linear spaces

摘录 / EXCERPT: Let M be a rank-k matroid on $[n]:=\{1,\dots,n\}$. A tropical linear space is a $(k-1)$-dimensional balanced polyhedral complex in tropical projective space, dual to a coherent...

conjecture | open | MathDB 页面 | 原文线索

V2-043373 | MATHDB-345632

Mandel's conjecture that every oriented matroid is Mandel

摘录 / EXCERPT: An oriented matroid (OM) is Mandel if it has an expansion in general position such that the two associated antipodal oriented matroids are Euclidean. Mandel's conjecture. Every OM...

conjecture | open | MathDB 页面 | 原文线索

V2-043374 | MATHDB-345634

Murty's Sylvester-Gallai conjecture for oriented matroids

摘录 / EXCERPT: Let G be the tope graph of an oriented matroid of rank r. Murty's conjecture. G contains a convex subgraph that is the Cartesian product of an edge and an antipodal graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-043375 | MATHDB-345635

The theta-class strengthening of Murty's conjecture for Mandel oriented matroids

摘录 / EXCERPT: Let G be a Mandel oriented matroid of rank r, and let E be any Θ-class of its tope graph. Theta-class strengthening of Murty's conjecture. The Θ-class E is in...

conjecture | open | MathDB 页面 | 原文线索

V2-043376 | MATHDB-345665

Skew shape conjecture for diagonal pattern avoidance

摘录 / EXCERPT: Let S be a skew shape, and let $k\geq 1$. For a filling of S, write $\operatorname{Tr}(S,\pi)$ for the number of transversal fillings avoiding the filling pattern π, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-043377 | MATHDB-345666

Fractional Stanley-Wilf limit conjecture for distant patterns

摘录 / EXCERPT: Let q be a classical permutation pattern, let $\rho_r(q)=\bigl| ame{Av}_n(ame{dist}_r(q))\bigr|$ denote the number of n-permutations avoiding the distant pattern...

conjecture | open | MathDB 页面 | 原文线索

V2-043378 | MATHDB-345667

Sublinear-gap Stanley-Wilf limit conjecture for distant patterns

摘录 / EXCERPT: Let q be a classical permutation pattern, and let c_2 satisfy $0<c_2<1$. Sublinear-gap Stanley-Wilf limit conjecture. There exist constants $c_q>0$ and c_2 such that, with...

conjecture | open | MathDB 页面 | 原文线索

V2-043379 | MATHDB-345669

Increasing-pattern extremal conjecture for distant-pattern avoidance

摘录 / EXCERPT: Let $m\geq3$ and $r\geq1$, and let q be any classical pattern of size m. Increasing-pattern extremal conjecture. There exists $n_0\in\mathbb{N}$ such that for every natural num...

conjecture | open | MathDB 页面 | 原文线索

V2-043380 | MATHDB-345670

Weak extremal comparison conjecture for increasing and final-descent patterns

摘录 / EXCERPT: Let $m\geq3$ and $r\geq1$. Weak extremal comparison conjecture. There exists $n_0\in\mathbb{N}$ such that for every natural number $n>n_0$, ... This is a weaker version of the two pr...

conjecture | open | MathDB 页面 | 原文线索

V2-043381 | MATHDB-345672

Extremality conjecture for boundary strata in the moduli space of stable curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable genus-g curves with n marked points. A boundary stratum is a stratum parametrizing stable curves with a fixed d...

conjecture | open | MathDB 页面 | 原文线索

V2-043382 | MATHDB-345719

The $K_{2,k}$ induced saturation conjecture

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, let $K_{2,k}$ denote the corresponding poset, and let $\operatorname{sat}^*(n,K_{2,k})$ be the smallest size of a $K_{2,k}$-saturated family of subsets...

conjecture | open | MathDB 页面 | 原文线索

V2-043383 | MATHDB-345739

De Loera's Ehrhart positivity and unimodality conjecture for matroid polytopes

摘录 / EXCERPT: Let M be a matroid, and let $\mathscr{P}(M)$ denote its matroid polytope. The h^*-polynomial of $\mathscr{P}(M)$ is the polynomial whose coefficient vector is being considered....

conjecture | open | MathDB 页面 | 原文线索

V2-043384 | MATHDB-345740

Coefficient bounds conjecture for Ehrhart polynomials of connected matroids

摘录 / EXCERPT: Let M be a connected matroid of rank k on n elements. Write $i(M,t)$ for its Ehrhart polynomial, and let $T_{k,n}$ and $U_{k,n}$ denote respectively the minimal and uniform m...

conjecture | open | MathDB 页面 | 原文线索

V2-043385 | MATHDB-345784

The Log-span conjecture

摘录 / EXCERPT: Log-span conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043386 | MATHDB-345785

The linear matroid conjecture for induced tripling

摘录 / EXCERPT: Linear matroid conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043387 | MATHDB-345786

The doubling-tripling conjecture for induced parameters

摘录 / EXCERPT: Doubling-tripling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043388 | MATHDB-345787

The independence hypothesis for induced expansion functionals

摘录 / EXCERPT: Independence hypothesis.

conjecture | open | MathDB 页面 | 原文线索

V2-043389 | MATHDB-345788

The torsion quotient conjecture for induced expansion functionals

摘录 / EXCERPT: Torsion quotient conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043390 | MATHDB-345789

The projection conjecture for induced expansion functionals

摘录 / EXCERPT: Projection conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043391 | MATHDB-345790

The multiplicativity hypothesis for induced expansion functionals

摘录 / EXCERPT: Multiplicativity hypothesis.

conjecture | open | MathDB 页面 | 原文线索

V2-043392 | MATHDB-345791

The functional isometric-tripling conjecture

摘录 / EXCERPT: Functional isometric-tripling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043393 | MATHDB-345792

The functional independence hypothesis

摘录 / EXCERPT: Functional independence hypothesis.

conjecture | open | MathDB 页面 | 原文线索

V2-043394 | MATHDB-345798

The chain-is-best conjecture for poset saturation

摘录 / EXCERPT: Chain-is-best conjecture. For any k-element poset P, we have

conjecture | open | MathDB 页面 | 原文线索

V2-043395 | MATHDB-345805

Density-one conjecture for missing tensor-product invariants

摘录 / EXCERPT: Density-one conjecture. Among the triples with $a_n=b_n=0$ and height at most X, the proportion of missing triples among those satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-043396 | MATHDB-345830

Automaton Game Correspondence

摘录 / EXCERPT: Let ${\mathrm{TB}}$ be the total budget, and let a unitary game be a game in the class defined by the paper. Associate to it the corresponding 0-bidding automaton $\mathcal{A}$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-043397 | MATHDB-345840

G-ADMS conjecture on distinct partial sums in cyclic groups

摘录 / EXCERPT: G-ADMS conjecture. There exists an ordering of the elements of A such that all partial sums are distinct.

conjecture | open | MathDB 页面 | 原文线索

V2-043398 | MATHDB-345841

CMPP conjecture on distinct partial sums for zero-sum subsets

摘录 / EXCERPT: CMPP conjecture. There exists an ordering of the elements of A such that all partial sums are distinct.

conjecture | open | MathDB 页面 | 原文线索

V2-043399 | MATHDB-345854

Unit-cube conjecture for the symmetric convex extremal

摘录 / EXCERPT: Unit-cube conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043400 | MATHDB-345862

Compact-support extremizer conjecture for autocorrelation inequalities

摘录 / EXCERPT: For $g\in L^1(\mathbb{R})$ with $g\geq 0$ and compact $I\subset\mathbb{R}$, let $\mathcal{L}_g(I)$ be the class of nonnegative L^1 functions f on $\mathbb{R}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-043401 | MATHDB-345863

Compactly supported formulation of the sharp autocorrelation constant

摘录 / EXCERPT: Let C_4 be the smallest constant such that ... for the function class specified in Theorem 4 of the source. The compactly supported formulation concerns functions...

conjecture | open | MathDB 页面 | 原文线索

V2-043402 | MATHDB-345871

The Lemmens-Seidel conjecture for equiangular lines at angle arccos(1/5)

摘录 / EXCERPT: Let $N_{\alpha}(r)$ be the maximum number of a system of equiangular lines in $\mathbb{R}^r$ with common angle $\arccos\alpha$. The Lemmens-Seidel conjecture concerns the case...

conjecture | open | MathDB 页面 | 原文线索

V2-043403 | MATHDB-345883

Planar basis evaluation conjecture for eta_J on minimal kinematics

摘录 / EXCERPT: Let $\Delta_{k,n}$ be the kth hypersimplex. For a k-element subset $J\subset\{1,\dots,n\}$, let Π_J be the associated polytope obtained by eliminating the block containing...

conjecture | open | MathDB 页面 | 原文线索

V2-043404 | MATHDB-345905

Countable-alphabet extension of Proposition valzeroun

摘录 / EXCERPT: Let X be a countable alphabet and let the weak shuffle algebra and the objects in Proposition valzeroun be defined as in the paper. Countable-alphabet extension. Proposition valz...

conjecture | open | MathDB 页面 | 原文线索

V2-043405 | MATHDB-345906

Vanishing of the kernel for the ninth weak shuffle product

摘录 / EXCERPT: Let $\underset{9}{\Box}$ be the weak shuffle product defined by C_{9}. For positive integers p and n, let $K_{(n,p)}$ denote the kernel defined in the paper. Vanishing-kernel...

conjecture | open | MathDB 页面 | 原文线索

V2-043406 | MATHDB-345907

The closure induction conjecture for union-closed families

摘录 / EXCERPT: Let $[n]$ be a finite universe, let $\mathcal F\subseteq 2^{[n]}$ be a union-closed family, and define its closure by ... Suppose that $\overline{\mathcal F}$ satisfies Frankl's unio...

conjecture | open | MathDB 页面 | 原文线索

V2-043407 | MATHDB-345943

The regularity conjecture for sign uncertainty principles

摘录 / EXCERPT: Let $s\in\{+,-\}$ and let $d\geq 1$. Write $\mathbb{A}_s(d)$ for the sign uncertainty constant in dimension d, and let $\mathbb{A}_s^{\mathcal B}(d)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043408 | MATHDB-345953

Alon-Wormald-Zhou conjecture on packing two Kneser graphs

摘录 / EXCERPT: Let $m>w\ge 1$. The Kneser graph $\operatorname{KG}(m,w)$ has as vertices the w-element subsets of an m-element set, with two vertices adjacent when the corresponding subsets a...

conjecture | open | MathDB 页面 | 原文线索

V2-043409 | MATHDB-345954

Packing conjecture for graphs with many large complete bipartite subgraphs

摘录 / EXCERPT: Let G be a graph on N vertices. Suppose there is a parameter W such that, for every induced subgraph G' and every n, all but at most ... edges of G' are contained in some...

conjecture | open | MathDB 页面 | 原文线索

V2-043410 | MATHDB-345955

Two-intersection sunflower conjecture

摘录 / EXCERPT: A w-set system is a family of sets, each of cardinality w. For a set system $\mathcal F$, sample S,S_1,S_2 independently and uniformly from $\mathcal F$. Two-intersection sun...

conjecture | open | MathDB 页面 | 原文线索

V2-043411 | MATHDB-345969

Generalised Delta conjecture, valley version

摘录 / EXCERPT: Generalised Delta conjecture, valley version.

conjecture | open | MathDB 页面 | 原文线索

V2-043412 | MATHDB-345970

Generalised Delta square conjecture, rise version

摘录 / EXCERPT: Generalised Delta square conjecture, rise version.

conjecture | open | MathDB 页面 | 原文线索

V2-043413 | MATHDB-345971

Generalised Delta square conjecture, valley version

摘录 / EXCERPT: Generalised Delta square conjecture, valley version.

conjecture | open | MathDB 页面 | 原文线索

V2-043414 | MATHDB-345972

Touching Delta conjecture, valley version

摘录 / EXCERPT: Touching Delta conjecture, valley version.

conjecture | open | MathDB 页面 | 原文线索

V2-043415 | MATHDB-345973

Modified Delta square conjecture, valley version

摘录 / EXCERPT: Modified Delta square conjecture, valley version.

conjecture | open | MathDB 页面 | 原文线索

V2-043416 | MATHDB-345977

The strengthened generating-function conjecture for $\textbackslash mathcal{P}$-Knuth graphs

摘录 / EXCERPT: Main conjecture. The claim of the stated theorem is true for all unit interval orders $\textbackslash \mathcal{P}$, except for its final assertion concerning equality of tableau-sh...

conjecture | open | MathDB 页面 | 原文线索

V2-043417 | MATHDB-345999

Cameron-Fon-Der-Flaass periodicity conjecture for plane partitions

摘录 / EXCERPT: Fix positive integers a,b,c and let $\mathsf{B}_{a,b,c}={\bf a}\times{\bf b}\times{\bf c}$ be the product of three chains. Let $J(\mathsf{B}_{a,b,c})$ denote its order ideals, an...

conjecture | open | MathDB 页面 | 原文线索

V2-043418 | MATHDB-346006

The lower-closure characterization of stably regular triangulations

摘录 / EXCERPT: Stably regular triangulation conjecture. A triangulation $T \in {\cal T}_{\mathsf{part}}(P)$ is stably regular if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-043419 | MATHDB-346018

Frankl et al.'s conjecture on intersecting subfamilies of constrained set families

摘录 / EXCERPT: Frankl et al.'s conjecture. If $n_i\geq2a_i$ for all i and $n_i>k-\sum_{j=1}^pa_j+a_i$ for all but at most one $i\in[p]$ such that $a_i>0$, then the largest intersecting subfamil...

conjecture | open | MathDB 页面 | 原文线索

V2-043420 | MATHDB-346019

DeVos-Kwon-Oum branch-depth obstruction conjecture for matroids

摘录 / EXCERPT: DeVos-Kwon-Oum's branch-depth obstruction conjecture. For every positive integer n, there is an integer d such that every matroid of branch-depth at least d contains a minor...

conjecture | open | MathDB 页面 | 原文线索

V2-043421 | MATHDB-346023

The inverse-exponent conjecture for multiplier sets

摘录 / EXCERPT: Let m, n, and k be positive integers with $m\geq 2$, and suppose ... A multiplier set M splits an abelian group G if there is a set $S\subseteq G$ such that every nonzero e...

conjecture | open | MathDB 页面 | 原文线索

V2-043422 | MATHDB-346024

The inverse-exponent conjecture for direct factors of cyclic groups

摘录 / EXCERPT: Let k, ω, and n be positive integers, and let $\mathbb{Z}_{\omega}$ be a finite cyclic group of order ω. Suppose ... A subset A is a direct factor of...

conjecture | open | MathDB 页面 | 原文线索

V2-043423 | MATHDB-346039

CJKP's list Ramsey number conjecture for triangle-free graphs

摘录 / EXCERPT: Let $R_{\chi_\ell}(3,\underline{k})$ denote the least n such that every graph on n vertices with independence number less than k has list chromatic number greater than 3. C...

conjecture | open | MathDB 页面 | 原文线索

V2-043424 | MATHDB-346042

Flag-geometric characteristic polynomial conjecture for two-step matroid flags

摘录 / EXCERPT: Flag-geometric characteristic polynomial conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-043425 | MATHDB-346043

Multidimensional branching Brownian motion extremal process conjecture

摘录 / EXCERPT: Let $d\geq 1$, let $\mathcal{N}_t$ be the particles at time t, and let $X_t(u)\in\mathbb{R}^d$ be the position of particle u. Define ... and, for $u\in\mathcal{N}_t$, define its...

conjecture | open | MathDB 页面 | 原文线索

V2-043426 | MATHDB-346063

Bounds for non-isomorphic maximal Arrow single-peaked domains

摘录 / EXCERPT: The conjectured bounds. The number $\mathcal{N}(m)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043427 | MATHDB-346066

The combinatorial-model conjecture for shifted Littlewood-Richardson coefficients

摘录 / EXCERPT: The combinatorial-model conjecture. The restriction of this map to $\overline{\mathcal{T}(\mu^t,\mu,s(\mathcal{T}_\lambda))}$ is a bijection onto...

conjecture | open | MathDB 页面 | 原文线索

V2-043428 | MATHDB-346082

The universal extremal-length constant conjecture for Liouville currents

摘录 / EXCERPT: Let Σ be a hyperbolic surface, let $\mathcal{L}_{\Sigma}$ be its Liouville current, and let $\operatorname{EL}_{\Sigma}$ denote the continuous extension of extremal length t...

conjecture | open | MathDB 页面 | 原文线索

V2-043429 | MATHDB-346096

The Borel Dilworth conjecture

摘录 / EXCERPT: Let $P=(V,\preceq)$ be a Borel poset, meaning that V is a standard Borel space and the comparability relation is a Borel subset of $V\times V$. Its width is the maximum size of a...

conjecture | open | MathDB 页面 | 原文线索

V2-043430 | MATHDB-346097

The Borel dual Dilworth conjecture

摘录 / EXCERPT: Let $P=(V,\preceq)$ be a Borel poset. Its height is the largest size of a chain in P, and an antichain is a set of pairwise incomparable elements. Borel dual Dilworth conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043431 | MATHDB-346098

The measurable Dilworth conjecture

摘录 / EXCERPT: Let $P=(V,\preceq)$ be a measurable poset, meaning an ordered probability space with probabilistic measure μ. Let G_P be its incomparability graph, and let $\chi_{\mu}(G_P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-043432 | MATHDB-346099

A strengthened Gerzon bound for spherical even-distance sets

摘录 / EXCERPT: Strengthened Gerzon bound. Then

conjecture | open | MathDB 页面 | 原文线索

V2-043433 | MATHDB-346110

Cube-section affine-cube extremality conjecture

摘录 / EXCERPT: Let $n>k\geq 1$, let $\Box^n=[-1,1]^n$, and let H be a k-dimensional linear subspace. Write $C_{\Box}(n,k)2^k$ for the maximum volume among sections $\Box^n\cap H$ that are aff...

conjecture | open | MathDB 页面 | 原文线索

V2-043434 | MATHDB-346134

Borel chromatic-number conjecture for graphs of subexponential growth

摘录 / EXCERPT: Borel chromatic-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043435 | MATHDB-346137

The extremal root path conjecture for Schur σ-groups

摘录 / EXCERPT: Extremal root path conjecture. The root path $(\pi^{i-1}(G)\to\pi^i(G))_{1\leq i\leq c-2}$ consists of edges of maximal step size, equal to the nuclear rank: for every...

conjecture | open | MathDB 页面 | 原文线索

V2-043436 | MATHDB-346139

The extremal-function formulation of Frankl's conjecture

摘录 / EXCERPT: Extremal formulation of Frankl's conjecture. For every $a,n\in\mathbb{N}^+$,

conjecture | open | MathDB 页面 | 原文线索

V2-043437 | MATHDB-346148

Shareshian's wedge-of-spheres conjecture for subgroup intervals

摘录 / EXCERPT: Shareshian's conjecture. The order complex of $\widehat{\mathcal{L}(H,G)}$ has the homotopy type of a wedge of spheres. This conjecture predicts a spherical wedge structure for the...

conjecture | open | MathDB 页面 | 原文线索

V2-043438 | MATHDB-346154

Finiteness conjecture for restricted-digit additive triples

摘录 / EXCERPT: Let $b\geq 3$ be an integer, and let B_b denote the set of positive integers whose base-b expansion contains only the digits 0 and 1. Let $a,b,c\geq 3$ be pairwise multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-043439 | MATHDB-346155

Sharp exponent conjecture for restricted-digit additive triples

摘录 / EXCERPT: Sharp-exponent conjecture. If $s>1$, then for each $\epsilon>0$ there exists a constant C such that, for every integer N, the number of solutions (x,y,z) with $x,y,z\geq 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043440 | MATHDB-346168

Kahn's coefficient conjecture for independent sets in regular graphs

摘录 / EXCERPT: Let $i_k(G)$ be the number of independent sets of size k in a graph G. For positive integers d and n with $2d$ dividing n, let $H_{d,n}$ be the union of $n/(2d)$ copies o...

conjecture | open | MathDB 页面 | 原文线索

V2-043441 | MATHDB-346169

The Upper Matching Conjecture for matchings in regular graphs

摘录 / EXCERPT: Let $m_k(G)$ be the number of matchings of size k in a graph G. For positive integers d and n with $2d$ dividing n, let $H_{d,n}$ be the union of $n/(2d)$ copies of...

conjecture | open | MathDB 页面 | 原文线索

V2-043442 | MATHDB-346253

Hajac-Tobolski nilpotent matrix extremal-product conjecture

摘录 / EXCERPT: Let N be a non-negative real and k a positive integer. For a square matrix A with non-negative real entries, define its weight by ... Assume that A is nilpotent and has weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-043443 | MATHDB-346310

Aigner-Horev and Person's asymmetric random Rado threshold conjecture

摘录 / EXCERPT: Let $A_1,\dots,A_r$ be r irredundant partition regular matrices of full rank, with $m(A_1)\geq m(A_2)\geq \dots \geq m(A_r)$. For two integer matrices A and B, define ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-043444 | MATHDB-346311

Dow-Gibson's permanent lower-bound conjecture for convex combinations of $(d-1)$-permutation matrices

摘录 / EXCERPT: Let A be a d-dimensional matrix of order n, let $\Lambda_{d-1}(d,n)$ denote the set of $(d-1)$-permutation matrices, let $\operatorname{Hull}(\Lambda_{d-1}(d,n))$ be their co...

conjecture | open | MathDB 页面 | 原文线索

V2-043445 | MATHDB-346314

Taranenko's even-dimensional decomposition conjecture for 1-polystochastic matrices

摘录 / EXCERPT: Let $\Omega_1(d,n)$ be the set of 1-polystochastic d-dimensional matrices of order n, and let a $(d-1)$-permutation matrix be an element of the corresponding set of $(d-1)$-p...

conjecture | open | MathDB 页面 | 原文线索

V2-043446 | MATHDB-346330

Bounded critical number conjecture for I_s-free triangle-free matroids

摘录 / EXCERPT: Bounded critical number conjecture. For any $s\geq 4$, the class of I_s-free and triangle-free matroids has bounded critical number.

conjecture | open | MathDB 页面 | 原文线索

V2-043447 | MATHDB-346354

Quadratic discrepancy conjecture for the Pref-opposite construction

摘录 / EXCERPT: Quadratic discrepancy conjecture. The de Bruijn sequence constructed by the Pref-opposite algorithm has discrepancy $\Theta(n^2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-043448 | MATHDB-346355

Maximum discrepancy conjecture for the de Bruijn sequence DB_max

摘录 / EXCERPT: Maximum discrepancy conjecture. The sequence $\mathcal{DB}_{max}(n)$ has discrepancy equal to

conjecture | open | MathDB 页面 | 原文线索

V2-043449 | MATHDB-346384

The coloring conjecture for the rhombus $\rho_\lozenge$

摘录 / EXCERPT: Let $\mathcal R^3$ be the unit distance graph of $\mathbb R^3$, and let $\rho_\lozenge=[uvwx]\subset\mathbb R^3$ be the unit rhombus defined in the source. A coloring is a map...

conjecture | open | MathDB 页面 | 原文线索

V2-043450 | MATHDB-346414

Conjectured recurrences for five lexicographically least pattern-avoiding words

摘录 / EXCERPT: Let $d4e6_{a/b}$ denote the lexicographically least infinite word on the nonnegative integers avoiding p/q-powers for $p/q\geq a/b$, and write $w(i)$ for the letter at position...

conjecture | open | MathDB 页面 | 原文线索

V2-043451 | MATHDB-346430

Extendable shellability of dual hypersimplices

摘录 / EXCERPT: A hypersimplex is the matroid polytope of a uniform matroid, and its dual polytope is the polytope dual to that hypersimplex. A shelling is extendable if every partial shelling can...

conjecture | open | MathDB 页面 | 原文线索

V2-043452 | MATHDB-346432

Simon's conjecture on extendable shellability of uniform matroid independence complexes

摘录 / EXCERPT: For integers $1<k\le n$, let $U_{n,k}$ be the uniform matroid and let its independence complex be the simplicial complex whose facets are the bases of $U_{n,k}$. A partial shelling...

conjecture | open | MathDB 页面 | 原文线索

V2-043453 | MATHDB-346436

The hierarchical-order conjecture for uniform polymatroid access structures

摘录 / EXCERPT: The hierarchical-order conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043454 | MATHDB-346437

Monadic characterisation conjecture for frame matroids

摘录 / EXCERPT: A matroid is a finite set system with an independence structure, and a frame matroid is a matroid arising from a biased graph; the monadic second-order language of matroids is the...

conjecture | open | MathDB 页面 | 原文线索

V2-043455 | MATHDB-346438

Finite-group gain-graphic excluded-minor conjecture

摘录 / EXCERPT: Let H be a finite group, and let $\mathcal{M}$ be any minor-closed class of H-gain-graphic matroids. A gain-graphic matroid is a frame matroid obtained from a graph whose edges...

conjecture | open | MathDB 页面 | 原文线索

V2-043456 | MATHDB-346439

Finite-group gain-graphic well-quasi-order conjecture

摘录 / EXCERPT: Let H be a finite group. A gain-graphic matroid is a frame matroid obtained from a graph with edge gains in H, and a class is well-quasi-ordered when it has neither an infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-043457 | MATHDB-346440

Non-definability conjecture for infinite-group gain-graphic matroids

摘录 / EXCERPT: Let H be an infinite group. A gain-graphic matroid is a frame matroid obtained from a graph with edge gains in H, and a class is characterised by a sentence in monadic second-o...

conjecture | open | MathDB 页面 | 原文线索

V2-043458 | MATHDB-346455

Bárány-Larman colorful topological Tverberg conjecture

摘录 / EXCERPT: Let $q \ge 2$ and $d \ge 1$ be integers. Let $C_1, \ldots, C_{d+1}$ be pairwise disjoint sets of vertices of $\Delta_{q(d+1)-1}$, each of size q, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043459 | MATHDB-346464

The 3-avoidability conjecture for nice formulas

摘录 / EXCERPT: A nice formula is a formula whose fragments satisfy the niceness condition studied in the paper. A formula is 3-avoidable if it can be avoided by an infinite word over a 3-lett...

conjecture | open | MathDB 页面 | 原文线索

V2-043460 | MATHDB-346465

The 2-avoidability conjecture for two-birds formulas

摘录 / EXCERPT: A two-birds formula is one of the formulas in the family beginning with $ABA.BAB$, $ABCBA.CBABC$, and $ABCDCBA.DCBABCD$, with the two fragments arranged symmetrically as described...

conjecture | open | MathDB 页面 | 原文线索

V2-043461 | MATHDB-346485

Capacity conjecture for nonadaptive group testing

摘录 / EXCERPT: Consider either combinatorial nonadaptive group testing, where k uniformly randomly chosen items among n are defective, or probabilistic nonadaptive group testing, where each o...

conjecture | open | MathDB 页面 | 原文线索

V2-043462 | MATHDB-346486

Ma-Ma-Yeh conjecture on partial gamma-positivity of generalized Stirling permutation polynomials

摘录 / EXCERPT: Let $n\geq 1$ and $0\leq i\leq n$. Write ... for the trivariate extension of $A_{n,i}(y)$. A trivariate polynomial $p(x,y,z)=\sum_i s_i(x,y)z^i$ is partial γ-positive when eve...

conjecture | open | MathDB 页面 | 原文线索

V2-043463 | MATHDB-346505

Martin and Thatte's maximum agreement subtree conjecture for balanced trees

摘录 / EXCERPT: Let ${\mathcal S}$ and ${\mathcal T}$ be balanced rooted binary phylogenetic X-trees, meaning that each has 2^m leaves for some non-negative integer m and height m, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043464 | MATHDB-346524

Kuhl-McGinn's completion conjecture for partial Latin squares with two rows and three columns

摘录 / EXCERPT: Kuhl-McGinn's conjecture. Every partial Latin square of order $n\geq 8$ with two completely filled rows and three completely filled columns, and with all other cells empty, is comp...

conjecture | open | MathDB 页面 | 原文线索

V2-043465 | MATHDB-346529

The TVK-shape sorting-probability conjecture

摘录 / EXCERPT: TVK-shape sorting-probability conjecture. There is a universal constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043466 | MATHDB-346568

The 1/3-2/3 Conjecture for finite Weyl groups

摘录 / EXCERPT: The 1/3-2/3 Conjecture for finite Weyl groups. If C is not a singleton, then

conjecture | open | MathDB 页面 | 原文线索

V2-043467 | MATHDB-346570

Exponential flow-count conjecture for 3-edge-connected graphs

摘录 / EXCERPT: Exponential flow-count conjecture. There exists a fixed constant $c>1$ such that, for every such G, ?, and f, there are at least c^n flows ? avoiding f.

conjecture | open | MathDB 页面 | 原文线索

V2-043468 | MATHDB-346574

Balog-Wooley decomposition conjecture

摘录 / EXCERPT: Let A be a finite subset of $\mathbb{R}$. For a finite set X, write ... where $E_k^+(X)=\sum_x r_{X-X}^k(x)$ and $E_k^\times(X)=\sum_x r_{X/X}^k(x)$. Let $\eta\leq 2/3$ be given....

conjecture | open | MathDB 页面 | 原文线索

V2-043469 | MATHDB-346587

The half-length uniqueness conjecture for bi-roots

摘录 / EXCERPT: Let w be a primitive word. A bi-root of w is a primitive pair $\{x,y\}$ whose concatenations include w. The half-length uniqueness conjecture. The word w has at most one bi...

conjecture | open | MathDB 页面 | 原文线索

V2-043470 | MATHDB-346596

Madiman-Kontoyannis conjecture for entropy of differences of log-concave random vectors

摘录 / EXCERPT: Madiman-Kontoyannis conjecture. The entropy satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043471 | MATHDB-346615

Tieredness conjecture for scalable convergence of permuton-random sequences

摘录 / EXCERPT: Tieredness conjecture. If Γ is a permuton for which every Γ-random sequence is scalably convergent, then Γ is tiered.

conjecture | open | MathDB 页面 | 原文线索

V2-043472 | MATHDB-346616

Universal scalable convergence conjecture for permuton-random sequences

摘录 / EXCERPT: Universal scalable convergence conjecture. If Γ is any permuton, then every Γ-random sequence is scalably convergent.

conjecture | open | MathDB 页面 | 原文线索

V2-043473 | MATHDB-346625

Gao-Thangadurai's zero-sum subsequence conjecture for rank-two cyclic groups

摘录 / EXCERPT: Let n be a positive integer, and let $g(G)$ denote the smallest positive integer t such that every subset X of a finite abelian group G with $|X|\geq t$ contains a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-043474 | MATHDB-346642

Conjecture on rank comparison over a field and its algebraic closure

摘录 / EXCERPT: Let K be a field, let $V=V_1\times\dots\times V_d$, and let $P:V\to K$ be a multilinear polynomial of degree d. Write $r_K(P)$ for the Schmidt rank of P, and let $\bar K$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-043475 | MATHDB-346644

Kynčl-Pach-Tóth refined necklace matching conjecture

摘录 / EXCERPT: Kynčl-Pach-Tóth refined conjecture. For every constant k, there is always a proper matching of size at least

conjecture | open | MathDB 页面 | 原文线索

V2-043476 | MATHDB-346645

Restricted heterogeneous necklace matching conjecture

摘录 / EXCERPT: Let $L\in\mathcal{N}(N)$ be a balanced necklace with $N=2n$ beads, and let $\operatorname{mono}(L)$ denote the number of maximal monochromatic arcs. In the heterogeneous model, a p...

conjecture | open | MathDB 页面 | 原文线索

V2-043477 | MATHDB-346646

Kynčl-Pach-Tóth alternating path conjecture

摘录 / EXCERPT: For every set of $N/2$ red and $N/2$ blue points on a circle, let $\ell(N)$ be the largest number such that there is a non-crossing alternating path with $\ell(N)$ vertices. Kynčl-...

conjecture | refuted | MathDB 页面 | 原文线索

V2-043478 | MATHDB-346647

Homogeneous necklace matching conjecture

摘录 / EXCERPT: Homogeneous necklace matching conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043479 | MATHDB-346676

The generalized-cone extremal conjecture for concave functions

摘录 / EXCERPT: Generalized-cone extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043480 | MATHDB-346681

Fixed-point product formula conjecture for evacuation of Kreweras words

摘录 / EXCERPT: Evacuation fixed-point conjecture. For all $n\geq 1$, the number of Kreweras words of length $3n$ satisfying $\operatorname{{Evac}}^{*}(w)=w$ is

conjecture | open | MathDB 页面 | 原文线索

V2-043481 | MATHDB-346690

The Erdős-Discrepancy majority-coloring conjecture

摘录 / EXCERPT: Majority-coloring conjecture. There exists a constant h and a red-blue coloring of $\mathbb N$ which is h-majority on every homogeneous arithmetic progression $A_{s,k}$. This i...

conjecture | open | MathDB 页面 | 原文线索

V2-043482 | MATHDB-346691

The rainbow homogeneous-progression coloring conjecture

摘录 / EXCERPT: For $s,k\in\mathbb N$, let $A_{s,k}=\{s,2s,\ldots,ks\}$. A coloring is rainbow on a set when all elements of that set receive distinct colors. Rainbow-coloring conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-043483 | MATHDB-346692

Exact rainbow-extremal sets for restricted sum-free colorings

摘录 / EXCERPT: Exact-extremal-set conjecture. Let n,r be positive integers with $r\geq4$. If n is even and $r\leq7$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043484 | MATHDB-346708

Stanley's real-rootedness conjecture for Stern-poset Eulerian row sums

摘录 / EXCERPT: Stanley's real-rootedness conjecture. For any $n\geq 1$ the polynomial $L_n(q)$ has only real zeros.

conjecture | open | MathDB 页面 | 原文线索

V2-043485 | MATHDB-346710

The asymptotic order of the smallest non-minimal complement

摘录 / EXCERPT: Asymptotic-order conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-043486 | MATHDB-346734

Lacasse's conjecture on the PAC-Bayesian combinatorial sum

摘录 / EXCERPT: Lacasse's conjecture. For every $m\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-043487 | MATHDB-346735

Finiteness conjecture for irreducible solutions of (E_N)

摘录 / EXCERPT: Let $N\in\mathbb{N}^{*}$ with $N\geq 2$. A solution of (E_N) is called irreducible when it cannot be decomposed as a sum of two nontrivial solutions, as in the preceding discussi...

conjecture | open | MathDB 页面 | 原文线索

V2-043488 | MATHDB-346736

Uniform size-bound conjecture for irreducible solutions of (E_N)

摘录 / EXCERPT: Let N be an integer greater than 2, and let the size of an irreducible solution of (E_N) mean the size notion defined earlier in the paper. Uniform size-bound conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-043489 | MATHDB-346738

The conjecture that the Schur-degree threshold equals the Schur number

摘录 / EXCERPT: For $n\ge 2$, let $S(n)$ be the nth Schur number. Let $L(n)$ be the least length such that every sequence A in $\mathbb N_+$ with $|A|=L(n)$ and average $\mu(A)\le n$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-043490 | MATHDB-346739

The conjectural recursive upper bound for Schur numbers

摘录 / EXCERPT: For $n\ge 2$, let $S(n)$ denote the nth Schur number. The recursive Schur-number upper-bound conjecture. ... This conjecture follows from the preceding conjecture on $L(n)$ togethe...

conjecture | open | MathDB 页面 | 原文线索

V2-043491 | MATHDB-346763

Generalized Sidon-set existence conjecture for perfect powers

摘录 / EXCERPT: Let $k\geq 1$, $h\geq 2$, and $\varepsilon>0$. A $B_h[g]$ set $A\subseteq(\mathbb{Z}^{+})^k$ is a set of k-th powers with counting function $A(x)$. Generalized Sidon-set conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-043492 | MATHDB-346765

Tightness conjecture for the layer number of evenly distributed point sets

摘录 / EXCERPT: Tightness conjecture. There exists an α-evenly distributed point set $X\subset\mathbb{R}^d$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043493 | MATHDB-346780

Converse CDE conjecture for ordinary Young diagrams

摘录 / EXCERPT: Converse CDE conjecture. The interval $[\emptyset,\lambda]$ has the CDE property if and only if λ is balanced.

conjecture | open | MathDB 页面 | 原文线索

V2-043494 | MATHDB-346781

Converse CDE conjecture for shifted Young diagrams

摘录 / EXCERPT: Converse CDE conjecture for shifted diagrams. The interval $[\emptyset,\lambda]$ has the CDE property if and only if λ is balanced or trapezoidal.

conjecture | open | MathDB 页面 | 原文线索

V2-043495 | MATHDB-346782

The balanced-partition conjecture for the $\mathsf a;q$-down-degree expectation

摘录 / EXCERPT: Balanced-partition product conjecture. If λ is balanced, then

conjecture | open | MathDB 页面 | 原文线索

V2-043496 | MATHDB-346809

The Lin-Ma conjecture on 0012-avoiding inversion sequences

摘录 / EXCERPT: An inversion sequence of length n is a sequence $e=e_1e_2\cdots e_n$ such that $0\le e_i\le i-1$ for $1\le i\le n$; let $\mathbf{I}_n(0012)$ denote the set of inversion sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-043497 | MATHDB-346827

Heilbronn's triangle conjecture

摘录 / EXCERPT: Let $\mathcal{D}$ be any convex shape in the plane, and let $\Delta(S)$ denote the minimal area of a triangle induced by a set S of s points in $\mathcal{D}$. Let $\Delta(s)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-043498 | MATHDB-346832

The minimal morphism conjecture for regular pairs of weighted simplicial complexes

摘录 / EXCERPT: Let $(\overline{d},\overline{a}) \in \mathcal{CS}$ be a regular pair, and let $(\mathcal{D},\mathcal{A})$ be the associated pair of weighted simplicial complexes. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-043499 | MATHDB-346882

The rook polynomial characterization of thin polyominoes

摘录 / EXCERPT: Thinness conjecture. ${\mathbb P}$ is thin if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-043500 | MATHDB-346886

The London-Paris conjecture for cocircuit paths on topes

摘录 / EXCERPT: London-Paris conjecture. There exists a tope $\operatorname{\mathcal{T}}$ in $\operatorname{\mathcal{M}}$ such that $X,Y\in\operatorname{\mathcal{T}}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-043501 | MATHDB-346887

The crabbed-path strengthening for cocircuit distances

摘录 / EXCERPT: Crabbed-path conjecture. There exists a crabbed path from X to Y whose length is no bigger than the length of any path from X to Y in $\operatorname{\mathcal{M}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-043502 | MATHDB-346895

Optimal bounds for iterated sumsets with $|5A|=100$

摘录 / EXCERPT: Optimal sumset bounds conjecture. Every such set satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043503 | MATHDB-346908

Carnevale-Voll polynomial nonvanishing conjecture

摘录 / EXCERPT: Carnevale-Voll conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043504 | MATHDB-346909

Irreducibility conjecture for the polynomials C_{X,λ_2}

摘录 / EXCERPT: Let C_{X,λ_2} denote the polynomial discussed in the concluding remarks, with coefficients in ${\mathbb Q}$. Irreducibility conjecture. For even $\lambda_2\ge2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043505 | MATHDB-346910

Irreducibility conjecture for the four polynomial families $\widetilde C_{l,i,j}(X)$

摘录 / EXCERPT: Let $l\ge3$, and let $\widetilde C_{l,0,0}(X)$, $\widetilde C_{l,0,1}(X)$, $\widetilde C_{l,1,0}(X)$ and $\widetilde C_{l,1,1}(X)$ be the four polynomial families introduced in the...

conjecture | open | MathDB 页面 | 原文线索

V2-043506 | MATHDB-346921

An upper-threshold conjecture for extremal beta-free words over the binary alphabet

摘录 / EXCERPT: Let w be a word over the binary alphabet Σ_2. For $\beta\in\mathbb{R}_{\text{ext}}$, call w extremal β-free if it is β-free and every extension of w contai...

conjecture | open | MathDB 页面 | 原文线索

V2-043507 | MATHDB-346922

The extremal repetition-threshold conjecture

摘录 / EXCERPT: Let $n\geq2$ be an integer, and let Σ_n be an alphabet with n letters. Define the repetition threshold by ... A word is extremal β-free if it is β-free and every...

conjecture | open | MathDB 页面 | 原文线索

V2-043508 | MATHDB-346923

A finite upper threshold for extremal repetition avoidance

摘录 / EXCERPT: Let $n\geq2$ be an integer, let Σ_n be an alphabet with n letters, and let $\beta\in\mathbb{R}_{\text{ext}}$. A word is extremal β-free if it is β-free and eve...

conjecture | open | MathDB 页面 | 原文线索

V2-043509 | MATHDB-346924

Kazhdan-Lusztig coefficient maximality conjecture for uniform matroids

摘录 / EXCERPT: Let M be a matroid of rank d on $m+d$ elements, and write its Kazhdan-Lusztig polynomial as ... Let $U_{m,d}$ denote the uniform matroid of rank d on $m+d$ elements, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043510 | MATHDB-346937

Hayashi's divisibility conjecture for profiles of finite connected racks

摘录 / EXCERPT: Hayashi's divisibility conjecture. Each λ_s, for $0\leq s\leq t$, divides λ_t.

conjecture | open | MathDB 页面 | 原文线索

V2-043511 | MATHDB-346957

Polyhedrality conjecture for the k-halfspace closure

摘录 / EXCERPT: Let $k\in\mathbb{N}$ be fixed. For a rational polyhedron P, define the k-halfspace closure by ... where Q_I denotes the integer hull of Q. Polyhedrality conjecture for the...

conjecture | open | MathDB 页面 | 原文线索

V2-043512 | MATHDB-346958

Inequivalence conjecture for higher-dimensional halfspace and projection closures

摘录 / EXCERPT: Let $k\geq 3$ and $n\geq k$. For a rational polyhedron $P\subseteq\mathbb{R}^n$, let $\mathcal{H}_k(P)$ be its k-halfspace closure, and let ... where $\mathcal{L}_k$ is the family...

conjecture | open | MathDB 页面 | 原文线索

V2-043513 | MATHDB-346959

Polyhedrality conjecture for the k-dimensional projection closure

摘录 / EXCERPT: Let $k\in\mathbb{N}$ be fixed. For a rational polyhedron $P\subseteq\mathbb{R}^n$, define the k-dimensional projection closure by ... where $\mathcal{L}_k$ is the family of k-dim...

conjecture | open | MathDB 页面 | 原文线索

V2-043514 | MATHDB-346974

The dual long-runs conjecture for minimal additive complements of natural-number sets

摘录 / EXCERPT: Let $C = \{c_1 < c_2 < \cdots\}$ be an infinite, bounded-below set of integers, and let ... Suppose there exists an integer $\ell > 0$ such that, for every $m \in \mathbb{N}$, the di...

conjecture | open | MathDB 页面 | 原文线索

V2-043515 | MATHDB-346985

The fair-dice decomposition conjecture

摘录 / EXCERPT: Fair-dice decomposition conjecture. If D is fair, then the length of D is a multiple of 6 and D is similar to a product of copies of $xyzzyx$.

conjecture | open | MathDB 页面 | 原文线索

V2-043516 | MATHDB-347009

Signed (p,q)-enumerator identities for type B permutations

摘录 / EXCERPT: Let B_n be the hyperoctahedral group of signed permutations, and for $sigma\in B_n$ let $\mathsf{fwex}(\sigma)$, $\mathsf{neg}(\sigma)$, $\mathsf{nest}_B(\sigma)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-043517 | MATHDB-347098

The i-Mark conjecture on positions with Sprague-Grundy value two

摘录 / EXCERPT: i-Mark conjecture. A necessary condition for $\operatorname{SG}(n)=2$ is that one of the following holds: $n=sd$, n is a term of (a_n), or n is a term of one of the sequences...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-043518 | MATHDB-347099

The central Delannoy formula for subdivisions of a 2 × n grid

摘录 / EXCERPT: Central Delannoy formula. The number of subdivisions of a $2\times n$ grid is

conjecture | open | MathDB 页面 | 原文线索

V2-043519 | MATHDB-347165

The small-dimensional subset conjecture for large spectra

摘录 / EXCERPT: Small-dimensional spectrum conjecture. For every $1\geqslant\varepsilon\gg\eta^2$, there exists $\Delta'\subset\Delta_\eta(A)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043520 | MATHDB-347166

The bounded-dimensional large-spectrum subset conjecture over finite vector spaces

摘录 / EXCERPT: Bounded-dimensional spectrum conjecture. There exists $\Delta'\subset\Delta_\eta(A)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043521 | MATHDB-347198

Combinatorial identity equivalent to the hypergeometric contribution conjecture

摘录 / EXCERPT: Combinatorial identity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-043522 | MATHDB-347219

The asymptotic extremal conjecture for forbidden subposets

摘录 / EXCERPT: Let P be a finite poset. For a family $\mathcal F\subseteq 2^{[n]}$, let $La(n,P)$ be the largest size of a P-free family, and let $La^*(n,P)$ be the largest size of an induced...

conjecture | open | MathDB 页面 | 原文线索

V2-043523 | MATHDB-347220

The supersaturation conjecture for forbidden and induced subposets

摘录 / EXCERPT: Let P be a poset and $\varepsilon>0$. Let $La(n,P)$ and $La^*(n,P)$ be the largest sizes of, respectively, P-free and induced P-free families in $2^{[n]}$. Let $e(P)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043524 | MATHDB-347221

The counting conjecture for forbidden and induced subposets

摘录 / EXCERPT: Let P be a finite poset. Let $e(P)$ and $e^*(P)$ be the consecutive-level parameters for, respectively, P-free and induced P-free families in $2^{[n]}$. The counting conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-043525 | MATHDB-347222

The random forbidden-subposet threshold conjecture

摘录 / EXCERPT: Let P be a finite connected poset. Let $e(P)$ and $e^*(P)$ be the consecutive-level parameters, and let $d(P)$ and $d^*(P)$ be the minimum ratios defined over connected subposets...

conjecture | open | MathDB 页面 | 原文线索

V2-043526 | MATHDB-347229

The extremal inequality for Kruskal-Katona functions

摘录 / EXCERPT: Let $n\ge 4$ be an even integer, and let a and k be integers satisfying ... Here, $\kappa_{n/2}$ and $\kappa^*_{n/2}$ denote the Kruskal-Katona-type functions defined in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-043527 | MATHDB-347246

Frid's conjecture on bounded palindromic length sequences

摘录 / EXCERPT: Frid's conjecture. The palindromic length sequence of an infinite word is bounded if and only if the infinite word is ultimately periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-043528 | MATHDB-347253

Bachmann's bi-bracket reduction conjecture

摘录 / EXCERPT: Bachmann's bi-bracket reduction conjecture. Every bi-bracket can be written as a $\mathbb{Q}$-linear combination of mono-brackets; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-043529 | MATHDB-347274

Mondrian partition existence conjecture for squares

摘录 / EXCERPT: Let S be a square, and let k be the number of rectangles in a partition. A proper perfect Mondrian partition is the paper's specified partition of S into k pairwise non-con...

conjecture | open | MathDB 页面 | 原文线索

V2-043530 | MATHDB-347320

Asymptotic sharpness conjecture for colored generalized Turán numbers

摘录 / EXCERPT: Colored asymptotic sharpness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043531 | MATHDB-347339

COM ample-completion conjecture

摘录 / EXCERPT: Let a complex of oriented matroids (COM) be a set family of VC-dimension d, and let an ample completion be an ample partial cube containing it. COM ample-completion conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-043532 | MATHDB-347351

Finite boundedness of Ramsey numbers for complete and anisotropic spaces

摘录 / EXCERPT: Ramsey-number conjecture. The Ramsey numbers for complete spaces and anisotropic spaces, $\mathrm{R}^{\mathrm{c}}_{\mathbb{F}}(s,t,\ell)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043533 | MATHDB-347368

Chow-Fan-Goemans-Vondrák wideness conjecture for Latin tableaux

摘录 / EXCERPT: Chow-Fan-Goemans-Vondrák conjecture. A Young diagram λ is wide if and only if there exists a Latin tableau of shape λ.

conjecture | open | MathDB 页面 | 原文线索

V2-043534 | MATHDB-347383

Conjecture on extremal real equiangular tight frames

摘录 / EXCERPT: Let $d\geq 5$. An equiangular tight frame (ETF) with parameters (d,n) is a real ETF for $\mathbb{R}^d$ of size n. ETF extension conjecture. There exists an ETF with parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-043535 | MATHDB-347401

Burr-Erdős linear Ramsey conjecture for bounded-degree graphs

摘录 / EXCERPT: Let $\Delta\in\mathbb{N}$. A graph has bounded maximum degree at most Δ when $\Delta(G)\leq\Delta$. Burr-Erdős conjecture. There exists $c=c(\Delta)>0$ such that every 2-e...

conjecture | open | MathDB 页面 | 原文线索

V2-043536 | MATHDB-347404

Positive Ramsey upper density conjecture for the graphs H_d

摘录 / EXCERPT: For each $d\in\mathbb{N}$, let H_d be the graph defined in the source's Proposition referenced by prop:constr, and let $\operatorname{\overline{Rd}}(H_d)$ denote its two-color Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-043537 | MATHDB-347405

Ramsey-cofinite versus Ramsey-lower-dense conjecture

摘录 / EXCERPT: Let G be a graph. A graph is Ramsey-cofinite when it has the cofinite monochromatic embedding property defined in the paper, and it is Ramsey-lower-dense when it has positive Ram...

conjecture | open | MathDB 页面 | 原文线索

V2-043538 | MATHDB-347432

Non-isomorphism conjecture for lattices associated with matroid codes

摘录 / EXCERPT: Non-isomorphism conjecture. The lattices $L_{C_{M_1}}$ and $L_{C_{M_2}}$ are non-isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-043539 | MATHDB-347474

Daykin-Häggkvist partial Latin square completion conjecture

摘录 / EXCERPT: Daykin-Häggkvist conjecture. Given a partial Latin square L of order n in which each row, column, and symbol is used at most $n/4$ times, it is possible to complete L into a...

conjecture | open | MathDB 页面 | 原文线索

V2-043540 | MATHDB-347506

Extended probability formula for piercing numbers in circular societies

摘录 / EXCERPT: Let S be a random fixed-length circular society with n voters, approval set length p, and piercing number $\tau(S)$. For a positive integer k, Extended probability formula....

conjecture | open | MathDB 页面 | 原文线索

V2-043541 | MATHDB-347532

The density limit conjecture for the binary Ulam set

摘录 / EXCERPT: Let $ \mathcal{W}_{2,n}$ denote the set of binary words of length n, and let $V_n=\mathcal{U}(\{0,1\})\cap\mathcal{W}_{2,n}$. The density of the binary Ulam set is measured by...

conjecture | open | MathDB 页面 | 原文线索

V2-043542 | MATHDB-347533

Regularity conjecture for even associated lattices

摘录 / EXCERPT: Let $a>2$ be even and let b be sufficiently large. Consider an Ulam set with generators v_1 and v_2 whose associated lattice is generated by $(-a,b)$. Regularity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043543 | MATHDB-347534

Infinitude conjecture for two-generator Ulam sets

摘录 / EXCERPT: Let G be a commutative group, and let an Ulam set in G be generated from two initial generators using unique sums of two earlier distinct elements. Two-generator infinitude con...

conjecture | open | MathDB 页面 | 原文线索

V2-043544 | MATHDB-347536

Existence of the exponential growth rate for max embedding dimension semigroups

摘录 / EXCERPT: For each positive integer f, let $MED(f)$ be the number of numerical semigroups with Frobenius number f and maximal embedding dimension. The notation $\log_2$ denotes the logar...

conjecture | open | MathDB 页面 | 原文线索

V2-043545 | MATHDB-347539

Coefficientwise Hankel-total positivity conjecture for the GKP recurrence

摘录 / EXCERPT: Coefficientwise Hankel-total positivity conjecture for the GKP recurrence. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-043546 | MATHDB-347540

Strong coefficientwise Hankel-total positivity conjecture for the GKP recurrence

摘录 / EXCERPT: Strong coefficientwise Hankel-total positivity conjecture for the GKP recurrence. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-043547 | MATHDB-347553

T3 facet-defining inequalities for the weak order polytope

摘录 / EXCERPT: T3 conjecture. The eight displayed inequalities are T3 valid inequalities and facet-defining inequalities of ${\bf P}^n_{WO}$ for every $n\geq5$.

conjecture | open | MathDB 页面 | 原文线索

V2-043548 | MATHDB-347554

Fixed-alternative construction conjecture for large classes of weak-order-polytope facets

摘录 / EXCERPT: Fixed-alternative construction conjecture. By fixing these $|\hat{N}|$ alternatives, new large classes of facet-defining inequalities incorporating at least one variable from each...

conjecture | open | MathDB 页面 | 原文线索

V2-043549 | MATHDB-347567

The maximum-length conjecture for circuit codes with long bit runs

摘录 / EXCERPT: Maximum-length conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-043550 | MATHDB-347579

Shur's asymptotic growth conjecture for power-free languages

摘录 / EXCERPT: For any real $b2>1$, let $b1(k,b2)$ denote the growth rate of the language of $b2$-free words over an alphabet of size k. Shur's asymptotic growth conjecture. For every fixed int...

conjecture | open | MathDB 页面 | 原文线索

V2-043551 | MATHDB-347580

Stronger tail-restricted growth conjecture for power-free languages

摘录 / EXCERPT: For a real $b2>1$, let $b1'(k,b2)$ be the growth rate of the language of words containing no $b2$-power whose tail has length at most 2. Stronger tail-restricted growth conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-043552 | MATHDB-347595

Bartley-Day conjecture on cycles in tournaments

摘录 / EXCERPT: Bartley-Day conjecture. For $\ell\ge 3$, $c(\ell)=1$ if and only if ℓ is not divisible by four. The statement is known for regular tournaments and for $\ell\le 8$. The paper p...

conjecture | open | MathDB 页面 | 原文线索

V2-043553 | MATHDB-347596

Bartley-Day conjecture for cycle lengths divisible by four

摘录 / EXCERPT: Bartley-Day conjecture. If ℓ is divisible by four, then

conjecture | open | MathDB 页面 | 原文线索

V2-043554 | MATHDB-347621

Erdős-Lovász Tihany Conjecture for graph splittability

摘录 / EXCERPT: Erdős-Lovász Tihany Conjecture. For every graph G with $\chi(G)>\omega(G)$ and any two integers $s,t\geq 2$ satisfying $s+t=\chi(G)+1$, the graph G is (s,t)-splittable.

conjecture | open | MathDB 页面 | 原文线索

V2-043555 | MATHDB-347625

Parity-dominance conjecture for numerical semigroups of fixed Frobenius number

摘录 / EXCERPT: Let $F(S)$ be the Frobenius number of a numerical semigroup S, and let $t(S)$ be its type. Frobenius-parity conjecture. For every F, ... This conjecture is based on numerical dat...

conjecture | open | MathDB 页面 | 原文线索

V2-043556 | MATHDB-347626

Equidistribution conjecture for type parity at fixed Frobenius number

摘录 / EXCERPT: Let $F(S)$ be the Frobenius number of a numerical semigroup S, let $t(S)$ be its type, and let the denominator count all numerical semigroups with Frobenius number F. Frobenius...

conjecture | open | MathDB 页面 | 原文线索

V2-043557 | MATHDB-347627

Asymptotic growth-rate conjecture for numerical semigroups of fixed type

摘录 / EXCERPT: Let $T(F,\alpha)$ denote the number of numerical semigroups with Frobenius number F and type α. For each parity $r\in\{0,1\}$, consider Frobenius numbers F satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-043558 | MATHDB-347628

Monotonicity conjecture for fixed-type Frobenius growth rates

摘录 / EXCERPT: Let a_α and b_α be the limits, when they exist, of $\log_{2}(T(F,\alpha))/F$ along even and odd Frobenius numbers, respectively, where $T(F,\alpha)$ counts numerical...

conjecture | open | MathDB 页面 | 原文线索

V2-043559 | MATHDB-347629

Odd-type dominance conjecture for numerical semigroups of fixed genus

摘录 / EXCERPT: Let $g(S)$ be the genus of a numerical semigroup S, and let $t(S)$ be its type. Odd-type genus conjecture. For every g, ... The conjecture is verified by the numerical data in th...

conjecture | open | MathDB 页面 | 原文线索

V2-043560 | MATHDB-347630

Equidistribution conjecture for type parity at fixed genus

摘录 / EXCERPT: Let $g(S)$ be the genus of a numerical semigroup S, let $t(S)$ be its type, and let the denominator count all numerical semigroups with genus g. Genus type-parity equidistribut...

conjecture | open | MathDB 页面 | 原文线索

V2-043561 | MATHDB-347631

Asymptotic growth-rate conjecture for numerical semigroups of fixed genus and type

摘录 / EXCERPT: Let $L(g,\alpha)$ denote the number of numerical semigroups of genus g and type α, and let ϕ be the golden ratio. Genus growth-rate conjecture. For any α, the...

conjecture | open | MathDB 页面 | 原文线索

V2-043562 | MATHDB-347632

Monotonicity conjecture for fixed-type genus growth rates

摘录 / EXCERPT: Let c_α be the limit of $\log_{\phi}(L(g,\alpha))/g$ as $g\to\infty$, where $L(g,\alpha)$ counts numerical semigroups of genus g and type α, and ϕ is the golde...

conjecture | open | MathDB 页面 | 原文线索

V2-043563 | MATHDB-347634

Defant-Engen-Miller's log-concavity conjecture for Lassalle's sequence

摘录 / EXCERPT: Let $A_{k+1}(ell)$ denote the terms of the sequence introduced in the paper, with k a nonnegative integer. Defant3Engen3Miller's conjecture. The sequence ... is log-concave for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-043564 | MATHDB-347636

Hou-Jagadeesan maximal partition-rank product conjecture modulo 2

摘录 / EXCERPT: Hou-Jagadeesan's maximal-value conjecture. The following are true:

conjecture | open | MathDB 页面 | 原文线索

V2-043565 | MATHDB-347644

Polynomial Density Hales-Jewett conjecture

摘录 / EXCERPT: For positive integers k and a positive real number $0<\delta<1$, define ... For $a\in K[n]$, let $\gamma\subset[N]$ and interpret γ^r as the corresponding r-fold Cartesian...

conjecture | open | MathDB 页面 | 原文线索

V2-043566 | MATHDB-347649

The 3-uniform extremal conjecture with small blocking sets

摘录 / EXCERPT: Let b be a positive integer, let $n=3s$, and let $\mathcal F$ be a family of 3-subsets of $[n]$. A blocking set is a set meeting every member of $\mathcal F$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-043567 | MATHDB-347650

The k-uniform extremal conjecture for $k\ge4$

摘录 / EXCERPT: Let k,b,s be positive integers with $k\ge4$, let $n=ks$, and let $\mathcal F$ be a family of k-subsets of $[n]$. A blocking set is a set meeting every member of $\mathcal F$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-043568 | MATHDB-347663

Bourgain-Demeter's quadratic Vinogradov mean value conjecture

摘录 / EXCERPT: Let $J_{s,2}(A)$ denote the number of solutions to the quadratic Vinogradov system for a finite set A of real numbers, with s pairs of variables. Let $s\geq 3$ be a natural num...

conjecture | open | MathDB 页面 | 原文线索

V2-043569 | MATHDB-347701

The hyperfiniteness conjecture for countable amenable Borel equivalence relations

摘录 / EXCERPT: The hyperfiniteness conjecture. Every countable amenable Borel equivalence relation is hyperfinite.

conjecture | open | MathDB 页面 | 原文线索

V2-043570 | MATHDB-347711

Concavity conjecture for the maximum m-field of the symmetric group

摘录 / EXCERPT: Concavity conjecture. For any fixed n, the function $f(m)=F(n,m)$ is concave.

conjecture | open | MathDB 页面 | 原文线索

V2-043571 | MATHDB-347712

Latin-strategy characterization conjecture for the symmetric group

摘录 / EXCERPT: Latin-strategy characterization conjecture. Optimal strategies are exactly Latin strategies.

conjecture | open | MathDB 页面 | 原文线索

V2-043572 | MATHDB-347713

Shift-strategy characterization conjecture for the symmetric group

摘录 / EXCERPT: Shift-strategy characterization conjecture. Every optimal strategy is a shift strategy.

conjecture | open | MathDB 页面 | 原文线索

V2-043573 | MATHDB-347717

Path-star extremal conjecture for random tree search

摘录 / EXCERPT: Let T be a tree on n vertices. Write P_n for the path and S_n for the star on n vertices, and let $\mathcal{P}(T)$ denote the first-player win probability when both playe...

conjecture | open | MathDB 页面 | 原文线索

V2-043574 | MATHDB-347774

Explicit formula for the shuffle-lattice interval polynomial $G_{n-1,1}(x,y)$

摘录 / EXCERPT: For integers $a,b\geq 0$, let $\mathsf{Shuf}(a,b)$ be the shuffle poset, with rank function rk, and define ... For $n>0$, consider the case $a=n-1$ and $b=1$. The interval...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043575 | MATHDB-347788

Positivity and monotonicity conjecture for the sequence f_{2^k,n}

摘录 / EXCERPT: Let $f_{2^k,n}$ denote the sequence of values defined in the paper, and let $k\geq 1$ and $\nu\geq 0$ be integers. For each interval of integers ... consider the signed sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-043576 | MATHDB-347790

Combinatoriality of lattice sheaf cohomology vanishing

摘录 / EXCERPT: Let $\mathcal{A}$ be a hyperplane arrangement in a vector space of dimension ℓ, let L_0 be its intersection lattice with the minimal element removed, and let $\mathscr{D}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-043577 | MATHDB-347800

Computer-tested inequalities for triangular ratio metric extremal points

摘录 / EXCERPT: Let $\mathbb{B}^2$ be the unit disk, and let $s_{\mathbb{B}^2}$ and $p_{\mathbb{B}^2}$ denote the triangular ratio and point-pair metrics, respectively. For $x,y\in\mathbb{B}^2$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-043578 | MATHDB-347829

Tripartite classification conjecture for connected derangement graphs

摘录 / EXCERPT: Tripartite classification conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043579 | MATHDB-347830

Conjectures for maximum intersection density by permutation-group degree

摘录 / EXCERPT: Conjectures for $I(n)$. 1. If n is even but not a power of 2, then there is a transitive group G of degree n such that Γ_G is a complete multipartite graph with...

conjecture | open | MathDB 页面 | 原文线索

V2-043580 | MATHDB-347887

Nonexistence of pairs of maximal frequency squares of half frequency

摘录 / EXCERPT: A frequency square of type $(n;\lambda)$ is an $n\times n$ array in which each symbol occurs exactly λ times in every row and every column. Two frequency squares are orthog...

conjecture | open | MathDB 页面 | 原文线索

V2-043581 | MATHDB-347910

The independence conjecture for sum-free inverse-closed subsets of finite fields

摘录 / EXCERPT: Independence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043582 | MATHDB-347914

Ma's enumeration conjecture for the {1234,3412}-equivalence

摘录 / EXCERPT: Ma's enumeration conjecture. For $n\geq7$, the number of nontrivial equivalence classes is

conjecture | open | MathDB 页面 | 原文线索

V2-043583 | MATHDB-347915

Ma's reduced enumeration conjecture for {1234,3412}-classes

摘录 / EXCERPT: Ma's reduced enumeration conjecture. For $n\geq8$, the number of nontrivial equivalence classes partitioning these permutations is

conjecture | open | MathDB 页面 | 原文线索

V2-043584 | MATHDB-347935

Wang and Zhang's refinement for neighbors of the largest letter

摘录 / EXCERPT: Wang and Zhang's refinement. For all n, d, and $2\leq j\leq n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-043585 | MATHDB-347982

The rational generating-function conjecture for stained-glass window columns

摘录 / EXCERPT: Let $t_{n,k}$, $q_{n,k}$, and $c_{n,k}$ denote the entries in the three tables of counts for the k-th square in $BC(1,n)$. For each $k\geq 3$, consider the generating function of...

conjecture | open | MathDB 页面 | 原文线索

V2-043586 | MATHDB-347983

The bounded-side conjecture for cells in $BC(2,n)$

摘录 / EXCERPT: Let $BC(2,n)$ be the stained-glass graph obtained from a $2\times n$ rectangle by the construction described in the paper, and call the bounded regions of this graph cells. Bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-043587 | MATHDB-347991

Monotonicity conjecture for the coupled Young graph dimension array

摘录 / EXCERPT: Let $K(n,k,l)$ be the recursively defined array of integers associated with the pascalized coupled Young graph, with $n\geq 3$, k,l satisfying $2k+l<n-2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043588 | MATHDB-347995

Nash-Williams's matroid intersection conjecture

摘录 / EXCERPT: Let E be a ground set, let $\mathfrak{F}(E)$ denote the relevant class of matroids on E, and for $j\in\{0,1\}$ let $\mathcal{I}_{M_j}$ be the collection of independent sets of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043589 | MATHDB-348010

Perfect-cycle density conjecture in the hypercube

摘录 / EXCERPT: Perfect-cycle density conjecture. For all $d\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-043590 | MATHDB-348011

Perfect-path density conjecture in the hypercube

摘录 / EXCERPT: Perfect-path density conjecture. For all $d\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-043591 | MATHDB-348026

Thurston-algorithm ideal-boundary conjecture

摘录 / EXCERPT: Let an admissible combinatorics be given, and consider the successive conjugacy classes generated by the Thurston algorithm. Let the shift locus denote the locus of conjugacy class...

conjecture | open | MathDB 页面 | 原文线索

V2-043592 | MATHDB-348027

Strongly obstructed combinatorics limit conjecture

摘录 / EXCERPT: Consider strongly obstructed combinatorics in the bimodal regions of topological shapes $+-+$ and $-+-$, and let (Σ,Δ) denote the moduli coordinates. In the unimodal ca...

conjecture | open | MathDB 页面 | 原文线索

V2-043593 | MATHDB-348028

Total non-hyperbolicity conjecture for minimal unimodal combinatorics

摘录 / EXCERPT: Let a minimal unimodal combinatorics be a minimal combinatorial type in the unimodal region, and call it unobstructed when it is realized by the corresponding quadratic dynamical s...

conjecture | open | MathDB 页面 | 原文线索

V2-043594 | MATHDB-348030

Grytczuk et al.'s finiteness conjecture for extremal pattern-avoiding words

摘录 / EXCERPT: Let P be an avoidable pattern, meaning that some alphabet admits arbitrarily long words avoiding P. For a fixed alphabet, an extremal P-avoiding word is a word that avoids...

conjecture | open | MathDB 页面 | 原文线索

V2-043595 | MATHDB-348057

Polytope characterization of induced subgraphs with nonzero Euler characteristic

摘录 / EXCERPT: Polytope characterization conjecture. There is a polytope P whose vertices are labelled by the edges of G and whose faces are in bijection with the induced subgraphs of G hav...

conjecture | open | MathDB 页面 | 原文线索

V2-043596 | MATHDB-348064

Existence of asymptotic limits for the number of star operations

摘录 / EXCERPT: Let $\mathbf{s}$ be a sequence of pairs of positive integers, and let $n(\mathbf{s})$ denote the associated integer appearing in the definition of $\theta(\mathbf{s},q)$. For a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-043597 | MATHDB-348065

Dependence of the asymptotic exponent only on the length parameter

摘录 / EXCERPT: Let $\mathbf{s}$ and $\mathbf{t}$ be sequences of pairs of positive integers, and let $n(\mathbf{s})$ and $n(\mathbf{t})$ be their associated integers. For each finite-field cardin...

conjecture | open | MathDB 页面 | 原文线索

V2-043598 | MATHDB-348135

Ostrovsky and Paes Leme's matroid-based valuation conjecture

摘录 / EXCERPT: Let $v:2^{\mathcal G}\to\mathbb R$ be a valuation, and define the merging operation (or convolution) of valuations v_1,v_2 by ... An $M^\natural$-concave valuation is a valuation i...

conjecture | open | MathDB 页面 | 原文线索

V2-043599 | MATHDB-348143

Equality conjecture for sumsets involving algebraic multipliers

摘录 / EXCERPT: Equality conjecture. For any real algebraic α, the inequality becomes an equality. For complex algebraic α, the analogous equality holds for $A\subset\mathbb{C}$. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-043600 | MATHDB-348144

Linear-operator sumset growth conjecture

摘录 / EXCERPT: Let $\mathcal{T}:\mathbb{R}^d\rightarrow\mathbb{R}^d$ be a linear operator with characteristic polynomial $f(\mathcal{T})$. For an irreducible polynomial $g\in\mathbb{Z}[x]$ dividi...

conjecture | open | MathDB 页面 | 原文线索

V2-043601 | MATHDB-348145

Ellis's setwise intersecting family conjecture for the symmetric group

摘录 / EXCERPT: Ellis's conjecture. If $\mathcal{F}$ is a t-setwise intersecting family of $\operatorname{sym}(n)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043602 | MATHDB-348150

Langerman's chessboard mass partition conjecture

摘录 / EXCERPT: Langerman's conjecture. For every such family of measures, there exists a set of n hyperplanes whose induced chessboard coloring splits every measure into two equal parts.

conjecture | open | MathDB 页面 | 原文线索

V2-043603 | MATHDB-348151

Holmsen-Kynčl-Vasculescu colored convex partition conjecture

摘录 / EXCERPT: Holmsen-Kynčl-Vasculescu conjecture. If the points can be partitioned into k sets of size ℓ, each containing points of at least d different colors, then they can be partit...

conjecture | open | MathDB 页面 | 原文线索

V2-043604 | MATHDB-348154

Modified Integer Round-Up Property for the cutting stock problem

摘录 / EXCERPT: Let P_{CSP} be the set of integer cutting patterns for the cutting stock problem, let a_{ip} denote the number of copies of item i in pattern p, and let d_i be the demand...

conjecture | open | MathDB 页面 | 原文线索

V2-043605 | MATHDB-348168

Katz-Tao discretized sum-product conjecture

摘录 / EXCERPT: For $0<\sigma<1$, define a $(\delta,\sigma)_1$-set to be a δ-separated subset $A\subset[1,2]$ satisfying $|A|\sim\delta^{-\sigma}$ and $|A\cap I|\lesssim |I|^{\sigma}|A|$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-043606 | MATHDB-348171

Bi-symmetric quadruple equidistribution on ascent sequences

摘录 / EXCERPT: Let $\mathcal{A}_n$ be the set of ascent sequences of length n, and let asc, rep, zero, and max denote the corresponding Euler-Stirlin...

conjecture | open | MathDB 页面 | 原文线索

V2-043607 | MATHDB-348197

Mossinghoff's symmetry conjecture for maximal-perimeter convex small polygons

摘录 / EXCERPT: Mossinghoff's symmetry conjecture. The polygon P_n^* has an axis of symmetry corresponding to one particular pendant edge in its diameter graph.

conjecture | open | MathDB 页面 | 原文线索

V2-043608 | MATHDB-348201

Existence conjecture for type 2 cyclic Legendre difference families

摘录 / EXCERPT: Type 2 existence conjecture. Legendre difference families of type 2 exist for all odd lengths $v>8$.

conjecture | open | MathDB 页面 | 原文线索

V2-043609 | MATHDB-348219

Small-doubling structural conjecture in cyclic groups

摘录 / EXCERPT: Let n be a positive integer, let $\mathbb{Z}_n$ be the cyclic group of order n, and let $A\subseteq\mathbb{Z}_n$. For a subset $S\subseteq\mathbb{Z}_n$, write $2S=S+S$. Small-d...

conjecture | open | MathDB 页面 | 原文线索

V2-043610 | MATHDB-348248

The asymptotic conjecture for the longest initial card-type subsequence

摘录 / EXCERPT: Let m and n be positive integers. For a random permutation $\boldsymbol{\pi}$ of a multiset containing n copies of each card type $1,\ldots,m$, let $L(\pi)$ be the largest in...

conjecture | open | MathDB 页面 | 原文线索

V2-043611 | MATHDB-348249

The square-root error conjecture for optimal partial-feedback card guessing

摘录 / EXCERPT: Let $\mathcal{P}_{m,n}^+$ denote the maximum expected number of correctly guessed cards in the partial-feedback card-guessing model with m card types and n copies of each type....

conjecture | open | MathDB 页面 | 原文线索

V2-043612 | MATHDB-348250

The asymptotic conjecture for minimum partial-feedback card-guessing performance

摘录 / EXCERPT: Let $\mathcal{P}_{m,n}^-$ denote the minimum expected number of correctly guessed cards in the partial-feedback card-guessing model with m card types and n copies of each type....

conjecture | open | MathDB 页面 | 原文线索

V2-043613 | MATHDB-348268

Exponential bound for near-sunflower-free uniform families

摘录 / EXCERPT: Let a near-sunflower of size r be the configuration defined in the paper, and let $\mathcal{F}$ be a family of k-element sets. Near-sunflower exponential-bound conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043614 | MATHDB-348329

The removal lemma conjecture for Latin squares

摘录 / EXCERPT: Latin-square removal lemma conjecture. For every $\varepsilon>0$ there exists $\delta>0$ such that, whenever L is a Latin square of order n with $t(A,L)<\delta$, at most...

conjecture | open | MathDB 页面 | 原文线索

V2-043615 | MATHDB-348330

The Latin-square repair conjecture

摘录 / EXCERPT: Latin-square repair conjecture. For every $\varepsilon>0$ there exists $\delta>0$ such that, if $t(A,L)<\delta$, then one can modify entries of L on at most εn^2 pos...

conjecture | open | MathDB 页面 | 原文线索

V2-043616 | MATHDB-348331

The entropy conjecture for Latin squares near a prescribed Latinon

摘录 / EXCERPT: Latin-square entropy conjecture. For every $\varepsilon>0$ there exist $\delta>0$ and n_0 such that, for every $n>n_0$, if N is the number of Latin squares K of order n sat...

conjecture | open | MathDB 页面 | 原文线索

V2-043617 | MATHDB-348334

Equivalence of the lexicographic-composition de Bruijn constructions

摘录 / EXCERPT: Let $alt(n)$ denote the alternating sequence of 0s and 1s of length n that ends with 0. Let $\d835\d835\mathcal{L}_n$ be the de Bruijn sequence obtained from the lexicograp...

conjecture | open | MathDB 页面 | 原文线索

V2-043618 | MATHDB-348354

Forbidden-minor conjecture for non-even regular oriented matroids

摘录 / EXCERPT: Forbidden-minor conjecture. A regular oriented matroid M is non-even if and only if none of its GB-minors is isomorphic to $M^*(\vec{K}_{m,n})$ for some $m,n\geq 2$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-043619 | MATHDB-348365

Existence of 1-Gray codes for Fibonacci q-decreasing words

摘录 / EXCERPT: Let $n\geq 0$ and $q\geq 1$, and let $\mathcal{W}^q_n$ denote the set of Fibonacci q-decreasing words of length n. A 1-Gray code for $\mathcal{W}^q_n$ is an ordering of its wor...

conjecture | open | MathDB 页面 | 原文线索

V2-043620 | MATHDB-348393

Linear discrepancy conjecture for non-diagonal zero-sum-square-free matrices

摘录 / EXCERPT: Linear discrepancy conjecture. For every $C>0$, there is an integer N such that, whenever $n\geq N$, every non-diagonal $n\times n$ matrix M with entries in $\{-1,1\}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-043621 | MATHDB-348405

Asymptotic conjecture for three-part Sperner partition systems

摘录 / EXCERPT: Three-part asymptotic conjecture. As $n\to\infty$ with $n\equiv 2\pmod 6$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-043622 | MATHDB-348495

Nonexistence of a quadrisection of the six-dimensional torus by subcube partitions

摘录 / EXCERPT: Let $T^6=(\mathbb{R}/4\mathbb{Z})^6=[0,4]^6/\sim$, and partition its 4^6 subcubes into four classes. Nonexistence conjecture. No partition of the subcubes of $[0,4]^6/\sim$ gives...

conjecture | open | MathDB 页面 | 原文线索

V2-043623 | MATHDB-348514

Kim's refined colorful fractional Helly conjecture

摘录 / EXCERPT: Let d be a non-negative integer. For each $i\in[d+1]$, let n_i be positive and let r_i be a non-negative integer with $n_i\ge r_i+1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-043624 | MATHDB-348515

The colorful fractional Helly conjecture for d-Leray complexes

摘录 / EXCERPT: A simplicial complex K is d-Leray if, for every induced subcomplex $\mathsf{L}\leq\mathsf{K}$ and every $i\ge d$, its reduced homology group...

conjecture | open | MathDB 页面 | 原文线索

V2-043625 | MATHDB-348541

The entropy-density conjecture for multigraph product extremal numbers

摘录 / EXCERPT: Let a,r,s,d be integers with $a,r\geq 1$, $d\in[0,a-1]$, and $s\geq (r-1)(d+1)+2$. For sufficiently large n, write $\mathrm{ex}_{\Pi}(n,s,q)$ for the maximum edge-product among...

conjecture | open | MathDB 页面 | 原文线索

V2-043626 | MATHDB-348553

Stanley's classification conjecture for indecomposable upho modular lattices

摘录 / EXCERPT: Let P be an upho modular lattice, meaning that P is of finite type and its principal order filter at every element is isomorphic to P. Suppose that P is indecomposable and...

conjecture | open | MathDB 页面 | 原文线索

V2-043627 | MATHDB-348586

Chowla's conjecture on quadratic character sums over large subsets

摘录 / EXCERPT: Let $0<\alpha\leq 1$ be a real number. Let $\mathbb{F}_p$ denote the finite field with p elements, and let χ be the quadratic character modulo p. Chowla's conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-043628 | MATHDB-348597

Boundary extremal-process conjecture between $\mathcal{C}_I$ and $\mathcal{C}_{III}$

摘录 / EXCERPT: Let $\beta>1$ and $\sigma^2=\frac{\beta}{2\beta-1}$, and let $\mathcal{N}_t^2$ be the set of type-2 particles at time t, with positions $X_u(t)$. Write $Z^{(1)}_\infty$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-043629 | MATHDB-348614

Quantum Avoid True complexity conjecture

摘录 / EXCERPT: Quantum Avoid True conjecture. Determining the outcome class of $\ruleset{Quantum Avoid True}\$ with a classical start is $\cclass{PSPACE}\$-complete. Consequently, determining the...

conjecture | open | MathDB 页面 | 原文线索

V2-043630 | MATHDB-348615

Constant-wide Demi-Quantum Nim hardness conjecture

摘录 / EXCERPT: Constant-wide Demi-Quantum Nim conjecture. Constant-wide $\ruleset{Demi-Quantum Nim}\$ is $\cclass{NP}\$-hard to play optimally.

conjecture | open | MathDB 页面 | 原文线索

V2-043631 | MATHDB-348616

Reachable poly-wide Demi-Quantum Nim hardness conjecture

摘录 / EXCERPT: Reachable poly-wide Demi-Quantum Nim conjecture. Reachable poly-wide $\ruleset{\mathrm{demi\text{-}Quantum Nim}\$ remains $\cclass{PSPACE}\$-hard to play optimally.

conjecture | open | MathDB 页面 | 原文线索

V2-043632 | MATHDB-348617

One-quantum-move Nim solvability conjecture

摘录 / EXCERPT: One-quantum-move Nim solvability conjecture. For every such $Z\$ and $\phi\$, $Z^{\mathrm{symBudgeted}[1]\mathcal{Q}(\phi,2)}\$ is polynomial-time solvable. The same is conjectured...

conjecture | open | MathDB 页面 | 原文线索

V2-043633 | MATHDB-348629

The maximal 3-term progression conjecture for the 2-dimensional sphere

摘录 / EXCERPT: Let $S^2=\{u\in\mathbb{R}^3:|u|=1\}$ be the 2-dimensional sphere, and for $n\geq 2$ let $\mu_n(S^2)$ denote the maximal number of 3-term arithmetic progressions in an n-element...

conjecture | open | MathDB 页面 | 原文线索

V2-043634 | MATHDB-348641

The q,t-symmetry conjecture for two-part partitions

摘录 / EXCERPT: q,t-symmetry conjecture. The polynomial $C_{\lambda}(q,t)$ is q,t-symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-043635 | MATHDB-348663

Row-equivalency class equality for queue inversion and HHL statistics

摘录 / EXCERPT: Row-equivalency class conjecture. For every row-equivalency class $[\tau]$,

conjecture | open | MathDB 页面 | 原文线索

V2-043636 | MATHDB-348664

Weight-preserving bijection between queue inversion and HHL statistics

摘录 / EXCERPT: Bijection conjecture. Given λ and n, there exists a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-043637 | MATHDB-348667

Bonamy-Kardoš-Kelly-Nelson-Postle conjecture on bounded critical number

摘录 / EXCERPT: Bonamy-Kardoš-Kelly-Nelson-Postle conjecture. For any $s\geq 1$, the simple I_s-free and triangle-free binary matroids have bounded critical number.

conjecture | open | MathDB 页面 | 原文线索

V2-043638 | MATHDB-348668

Generalized bounded-critical-number conjecture for $I_{s,t}$-free binary matroids

摘录 / EXCERPT: Generalized bounded-critical-number conjecture. For any $1\leq s\leq t$, the simple $I_{s,t}$-free and triangle-free binary matroids have bounded critical number.

conjecture | open | MathDB 页面 | 原文线索

V2-043639 | MATHDB-348675

Arithmetic pattern conjecture

摘录 / EXCERPT: Let k and N be positive integers. A k-term arithmetic progression has a basepoint x and a common difference. Let $G_k(N)$ be the size of the smallest set of integers contai...

conjecture | open | MathDB 页面 | 原文线索

V2-043640 | MATHDB-348676

The (U_k)-pattern conjecture

摘录 / EXCERPT: Let $(U_k)_{k\in\mathbb{N}}$ be an increasing chain of subsets of $\mathbb{Q}^+$ with $|U_k|=k$. A U_k-pattern is a set obtained from U_k by a common scaling and translation;...

conjecture | open | MathDB 页面 | 原文线索

V2-043641 | MATHDB-348677

The restricted (U_k)-pattern conjecture

摘录 / EXCERPT: Let $(U_k)_{k\in\mathbb{N}}$ be an increasing chain of subsets of $\mathbb{N}^+$ with $|U_k|=k$. Let $G_{U_k}(N)$ be the size of the smallest set of integers containing a U_k-pat...

conjecture | open | MathDB 页面 | 原文线索

V2-043642 | MATHDB-348678

Projection formulation of the arithmetic Kakeya conjecture

摘录 / EXCERPT: For $A\subset\mathbb{Z}\times\mathbb{Z}$ and $r\in\mathbb{Q}\cup\{\infty\}$, define ... with $x+\infty y=y$. Projection formulation of the arithmetic Kakeya conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-043643 | MATHDB-348679

The n-dimensional Minkowski arithmetic Kakeya conjecture

摘录 / EXCERPT: Fix a positive integer n. Let $\mathcal{f}^M_n(k)$ be the infimum of the Minkowski dimension of a set $B_k\subset\mathbb{R}^n$ containing a k-term arithmetic progression with e...

conjecture | open | MathDB 页面 | 原文线索

V2-043644 | MATHDB-348711

Heim-Neuhauser conjecture for continuously colored partition functions

摘录 / EXCERPT: Let $p_\alpha(n)$ be the continuous extension of the k-colored partition function, with $\alpha\in\mathbb{R}_{\geq 2}$. Under the assumptions $n,\ell\in\mathbb{N}$, $n>\ell$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-043645 | MATHDB-348724

The pmaj refinement of the generalized Delta conjecture

摘录 / EXCERPT: Let m,n,k be in the range for which $\mathsf{PLD}(m,n)^{\ast k}$ is defined, and let $\mathsf{pmaj}(D)$ denote the proposed partial-major-index statistic on a partially labelled...

conjecture | open | MathDB 页面 | 原文线索

V2-043646 | MATHDB-348725

The pmaj polyomino conjecture

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, let $\mathsf{LPP}(m+1,n+1)$ be the set of labelled parallelogram polyominoes of the indicated dimensions, and let $\mathsf{area}(P)$, $\mathsf{pmaj}(P)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043647 | MATHDB-348728

Categorification conjecture for higher Pascal triangles and their q-deformations

摘录 / EXCERPT: Consider higher Pascal's triangles and the Catalan-number construction by algebraic complexes. Categorification conjecture for higher Pascal triangles. There exist constructions of...

conjecture | open | MathDB 页面 | 原文线索

V2-043648 | MATHDB-348730

Bourgain-type maximal sum-free set conjecture

摘录 / EXCERPT: Let A be a set of N positive integers. A subset $\Omega\subseteq\mathbb{R}/\mathbb{Z}$ is maximal sum-free if it is sum-free and has measure $1/3$. Let $\mathbbm{1}_{\Omega}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-043649 | MATHDB-348731

Generalized unbounded improvement conjecture for (k,ℓ)-sum-free sets

摘录 / EXCERPT: Let $k<\ell$ be positive integers. For a set A of N positive integers, let $\mathscr{M}_{(k,\ell)}(A)$ be the size of its largest (k,ℓ)-sum-free subset, where a set is...

conjecture | open | MathDB 页面 | 原文线索

V2-043650 | MATHDB-348750

The upper-bound conjecture for construction lines determined by registration marks

摘录 / EXCERPT: Let $k\geq 3$ registration marks be points in the plane, and consider the construction lines determined by the arrangement, subject to the two rules in Section 1. Upper-bound conje...

conjecture | open | MathDB 页面 | 原文线索

V2-043651 | MATHDB-348755

General formula conjecture for Hankel transforms of shifted Catalan combinations

摘录 / EXCERPT: General Hankel-transform conjecture. The Hankel transform is given by

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043652 | MATHDB-348761

Three symmetry conjectures for statistics on di-sk trees

摘录 / EXCERPT: Three symmetry conjectures for di-sk trees. Over di-sk trees, the following symmetries should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-043653 | MATHDB-348766

The inverse-sumset conjecture for large subsets of the finite field

摘录 / EXCERPT: Inverse-sumset conjecture. There exists $\epsilon>0$ such that, for every $A\subseteq\mathbb{F}_p^*$ satisfying $|A|\gg p^{1-\epsilon}$,

conjecture | open | MathDB 页面 | 原文线索

V2-043654 | MATHDB-348778

The size Ramsey number conjecture for matchings versus multiple paths

摘录 / EXCERPT: Let $n\ge 5$ and $m\ge 3$. For graphs G and H, write $\hat{r}(G,H)$ for their size Ramsey number, and let K_2 be a single edge and P_m a path on m vertices. The graph...

conjecture | open | MathDB 页面 | 原文线索

V2-043655 | MATHDB-348787

Conjecture on missing numbers of reducts of ternary words

摘录 / EXCERPT: Let a ternary word be a word over a three-letter alphabet, and let $\mathcal{R}(W)$ denote the set of distinct reducts of a word W. Missing-values conjecture. There exist infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-043656 | MATHDB-348788

Existence of the asymptotic duplication-distance limit

摘录 / EXCERPT: For an alphabet with k letters, let $d(W)$ be the minimum number of single square reductions needed to turn a word W into a square-free word, and let $d_k(n)$ be the maximum va...

conjecture | open | MathDB 页面 | 原文线索

V2-043657 | MATHDB-348795

Benjamini et al.'s conjecture on consensus in sparse random-graph majority dynamics

摘录 / EXCERPT: Let $G(n,p)$ be a random graph, let P_t and N_t denote the sets of vertices playing the two strategies at time t, and let d be the parameter in the condition $p\geq d/n$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-043658 | MATHDB-348799

Balister et al.'s partition conjecture for vectors over \(\mathbb{F}_2^n\)

摘录 / EXCERPT: Balister et al.'s partition conjecture. There exists a partition of $\mathbb{F}_2^n$ into pairs of vectors $\{p_i,q_i\}$, for $i=1,\ldots,2^{n-1}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-043659 | MATHDB-348803

The product-category conjecture for C and non-crossing pair partitions

摘录 / EXCERPT: Let C be the category whose morphism sets are the finite diagram sets $C(k,l)$, and let NC_2 denote the category of non-crossing pair partitions. Let $NC_2\times NC_2$ be their...

conjecture | open | MathDB 页面 | 原文线索

V2-043660 | MATHDB-348812

The higher production matrix formula for Riordan arrays

摘录 / EXCERPT: Let $M=(g(x),f(x))$ be a Riordan array, and let $P^{(n)}_M$ denote its n-th production matrix, defined by ... Let $M^{(n)}$ be the matrix produced by $P^{(n)}_M$. Higher production...

conjecture | open | MathDB 页面 | 原文线索

V2-043661 | MATHDB-348815

Linial-Morgenstern minimum 4-cycle-density conjecture for tournaments

摘录 / EXCERPT: Let T be a tournament, and let $t(C_{\ell},T)$ denote the number of homomorphisms from the directed cycle C_{ℓ} to T divided by $|V(T)|^{\ell}$. A random blow-up of an m...

conjecture | open | MathDB 页面 | 原文线索

V2-043662 | MATHDB-348816

Minimum cycle-density conjecture for $4k+2$ cycles

摘录 / EXCERPT: Let T be a tournament, let C_{ℓ} be the directed cycle of length ℓ, and let $t(C_{\ell},T)$ denote its homomorphism density. For each ℓ, let g_{ℓ} be the func...

conjecture | open | MathDB 页面 | 原文线索

V2-043663 | MATHDB-348817

The IU-family weighted-sum conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let an IU-family be a family $\mathcal F\subset 2^{[n]}$ such that for every $F,G\in\mathcal F$, one has $F\cap G\neq\emptyset$ and $F\cup G\neq[n]$. I...

conjecture | open | MathDB 页面 | 原文线索

V2-043664 | MATHDB-348818

Gronau's complement-pair family conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\mathcal F\subset 2^{[n]}$ satisfy, for every $F,G\in\mathcal F$, either $F=[n]\setminus G$, or both $F\cap G\neq\emptyset$ and $F\cup G\neq[n]$....

conjecture | open | MathDB 页面 | 原文线索

V2-043665 | MATHDB-348823

Near-extremal intrinsic functional conjecture for hyperbolic 3-manifolds

摘录 / EXCERPT: Near-extremal intrinsic functional conjecture. There exists an $\epsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043666 | MATHDB-348828

Sticky Matroid Conjecture

摘录 / EXCERPT: Let M be a matroid. It is sticky if, for any two matroids N_1 and N_2 with $E(M)=E(N_1)\cap E(N_2)$ and $M=N_1|_{E(M)}=N_2|_{E(M)}$, there exists an amalgam of N_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-043667 | MATHDB-348829

Hypermodular Matroid Conjecture

摘录 / EXCERPT: A matroid is hypermodular if every pair of two distinct maximal proper flats is modular. A matroid is a restriction of another matroid if it is obtained by restricting that matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-043668 | MATHDB-348850

Multiplicity-freeness conjecture for antipodes of the universal Hopf class

摘录 / EXCERPT: Let the universal Hopf class be denoted by $\textbf{\textsc{pre}}$. Multiplicity-freeness conjecture. The antipode for the Hopf monoid of the universal Hopf class...

conjecture | open | MathDB 页面 | 原文线索

V2-043669 | MATHDB-348851

Inclusion conjecture for broken-circuit, Gale, and quasisymmetric complexes

摘录 / EXCERPT: Let $\textbf{\textsc{bc}}$, $\textbf{\textsc{omat}}^{\operatorname{\mathsf{gale}}}$, $\textbf{\textsc{qe}}$, and $\textbf{\textsc{qi}}$ denote the corresponding classes of prefix-p...

conjecture | open | MathDB 页面 | 原文线索

V2-043670 | MATHDB-348852

Inclusion conjecture for ordered matroid Hopf monoids and shellable complexes

摘录 / EXCERPT: Let $\textbf{\textsc{omat}}_+$ be the class of ordered matroid complexes with the indicated extension, and let $\textbf{\textsc{sls}}$ be the class of shellable prefix-pure complex...

conjecture | open | MathDB 页面 | 原文线索

V2-043671 | MATHDB-348853

Multiplicity-freeness conjecture for antipodes of Hopf-class monoids

摘录 / EXCERPT: Let $\textbf{\textsc{pre}}^\natural$ denote the Hopf monoid arising from a Hopf class; equivalently, consider every Hopf monoid arising from a Hopf class. Multiplicity-freeness con...

conjecture | open | MathDB 页面 | 原文线索

V2-043672 | MATHDB-348905

The Bruhat-compatible freeness conjecture for the polygraph module

摘录 / EXCERPT: Let W_n^+ be the set indexing the positive fixed points, let e_v be the corresponding basis elements, and let ... For $v,w\in W_n^+$, write $v\leq_{\mathrm{bru}}w$ for the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-043673 | MATHDB-348936

Hu-Hwang-Wang cutoff-point conjecture for adaptive group testing

摘录 / EXCERPT: Hu-Hwang-Wang cutoff-point conjecture. For $r\leq 3$,

conjecture | solved | MathDB 页面 | 原文线索

V2-043674 | MATHDB-348949

The group-coloring conjecture for the rational Alon-Tarsi number

摘录 / EXCERPT: Let G be a graph, let k be an integer, and let P_G denote its graph polynomial. For a field $\mathbb{F}$, write $\operatorname{A}_{\mathbb{F}}(P_G)$ for the corresponding Alo...

conjecture | open | MathDB 页面 | 原文线索

V2-043675 | MATHDB-348950

The positive average overlap density conjecture

摘录 / EXCERPT: Positive average overlap density conjecture. There exists an absolute positive constant $c>0$ such that, for every $n\in\mathbb{N}$ and every such $\mathcal{F}$, the average overla...

conjecture | open | MathDB 页面 | 原文线索

V2-043676 | MATHDB-348959

Brenti's total-positivity conjecture for the clean Eulerian triangle

摘录 / EXCERPT: Let $\bm{A}=\left(\left\langle\begin{matrix}n\\k\end{matrix}\right\rangle^{\rm clean}\right)_{n,k\geq 0}$ be the clean Eulerian triangle, where...

conjecture | open | MathDB 页面 | 原文线索

V2-043677 | MATHDB-348960

Total-positivity conjecture for the reversed Stirling subset triangle

摘录 / EXCERPT: Let $\genfrac\{\}{\}{0pt}{}{n}{k}$ denote the Stirling subset number, and define the reversed Stirling subset triangle by...

conjecture | open | MathDB 页面 | 原文线索

V2-043678 | MATHDB-348961

Coefficientwise total-positivity conjecture for the four-parameter recurrence triangle

摘录 / EXCERPT: Let a,c,d,e be algebraic indeterminates, and define $T(n,k)$ by ... for $n\geq 1$, with $T(0,k)=\delta_{k0}$. Thus $\bm{T}=(T(n,k))_{n,k\geq 0}$ is a lower-triangular matrix over...

conjecture | open | MathDB 页面 | 原文线索

V2-043679 | MATHDB-348981

Bérczi et al.'s reduction conjecture for coverable matroids

摘录 / EXCERPT: Let M be a matroid whose ground set can be covered by k independent sets; such a matroid is k-coverable. A reduction of a matroid is the reduction notion used in the source,...

conjecture | open | MathDB 页面 | 原文线索

V2-043680 | MATHDB-349028

The higher-dimensional grid Ramsey conjecture

摘录 / EXCERPT: For positive integers t, r, m, and l, let $\mathbf{m}^t$ and $\mathbf{n}^t$ denote the t-dimensional grids, ordered coordinatewise. A coloring of the $\mathbf{m}^t$ subpo...

conjecture | open | MathDB 页面 | 原文线索

V2-043681 | MATHDB-349030

Convergence conjecture for the genus-deletion ratio

摘录 / EXCERPT: Fix $\theta\in(0,1/2)$, let g_n be the genus sequence used for the high-genus quadrangulations, and let $Q(n,g)$ denote the number of quadrangulations with n faces and genus...

conjecture | open | MathDB 页面 | 原文线索

V2-043682 | MATHDB-349033

Dyson's rank conjectures for moduli 5 and 7

摘录 / EXCERPT: Dyson's rank conjectures. For all nonnegative integers n,

conjecture | solved | MathDB 页面 | 原文线索

V2-043683 | MATHDB-349035

The low-noise full-pile-removal conjecture for noisy Nim

摘录 / EXCERPT: Let $N_k(p)$ denote the probability of winning from a heap of size k when each transmitted binary digit is independently erroneous with probability p. A move transmits the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-043684 | MATHDB-349098

The extremal K-polystability conjecture for smooth Kähler test configurations

摘录 / EXCERPT: Extremal K-polystability conjecture. In the setting of Theorem, for any $\check{\mathbb T}$-equivariant smooth Kähler test configuration $({\mathscr M},{\mathscr A})$ associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-043685 | MATHDB-349115

Face-lattice conjecture for partial permutohedra

摘录 / EXCERPT: Let $\mathcal{P}(m,n)$ be the partial permutohedron and let $\mathcal{B}_m$ be the Boolean lattice of subsets of $[m]$. For a chain in $\mathcal{B}_m$, the difference between its l...

conjecture | open | MathDB 页面 | 原文线索

V2-043686 | MATHDB-349116

Normalized-volume conjecture for partial permutohedra with one parameter equal to two

摘录 / EXCERPT: Let $\mathcal{P}(m,n)$ denote the partial permutohedron. Volume conjecture. The normalized volume of $\mathcal{P}(m,2)$ is equal to 3^m-m. This conjecture gives a closed formula...

conjecture | open | MathDB 页面 | 原文线索

V2-043687 | MATHDB-349178

Both-shifts versus one-direction shift-pattern conjecture

摘录 / EXCERPT: Let $G_{m,n}$ denote the grid under discussion, and let $m,n,k,z\in\mathbb{N}^+$. A k-coloring has a shift pattern in both directions on z-subgrids, or in only one direction on...

conjecture | open | MathDB 页面 | 原文线索

V2-043688 | MATHDB-349191

Extremal configuration conjecture for $c(d,d+1,d+1)$

摘录 / EXCERPT: Let $c(d,n,k)$ denote the extremal signed-sum quantity studied in the paper, and let ω_{d+1} be the corresponding extremal vector configuration. For a linear subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-043689 | MATHDB-349196

Bean's conjecture on the maximum of the normalized area of binary forms

摘录 / EXCERPT: Bean's conjecture. The maximum value M_n of $Q(F)$ over all such forms is attained precisely when F, up to multiplication by a complex number, is equivalent under...

conjecture | open | MathDB 页面 | 原文线索

V2-043690 | MATHDB-349210

Convergence-to-random conjecture for properly bounded prisoner strategies

摘录 / EXCERPT: Let S be a properly bounded strategy, let ${\rm RS}$ denote the random strategy, and let $\epsilon_{S,\mathrm{RS}}$ be the metric distance between their P-functions for parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-043691 | MATHDB-349230

The excess bound conjecture for pairs of d-polytopes

摘录 / EXCERPT: Excess bound conjecture. If $v\notin\mathcal{E}^d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043692 | MATHDB-349290

The zero-sum 3-flow conjecture for Steiner triple systems

摘录 / EXCERPT: A Steiner triple system ${\rm STS}(v)$ is a 2-$(v,3,1)$ design. A zero-sum n-flow of a design $\mathcal{D}=(X,\mathcal{B})$ is a map ... such that the sum of the values over all...

conjecture | open | MathDB 页面 | 原文线索

V2-043693 | MATHDB-349291

The zero-sum 5-flow conjecture for non-symmetric designs

摘录 / EXCERPT: A t-(v,k,λ) design $\mathcal{D}=(X,\mathcal{B})$ consists of a v-element point set X and a collection of k-subsets of X, called blocks, such that every t-subset...

conjecture | open | MathDB 页面 | 原文线索

V2-043694 | MATHDB-349310

Longest arithmetic progressions in generalised Thue-Morse sequences

摘录 / EXCERPT: Let $p,q,n\in\mathbb{N}^{+}$, set $Q=p+q$, and let $A_{p,q}(d)$ denote the maximum length of an arithmetic progression of difference d occurring in the generalised Thue-Morse seq...

conjecture | open | MathDB 页面 | 原文线索

V2-043695 | MATHDB-349338

Large configuration-free sets over algebraic number fields

摘录 / EXCERPT: Configuration-free sets conjecture. For every $C>0$, there exist $N\in\mathbb N$ and a set $B\subseteq\{0,1,\dots,N-1\}^d$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043696 | MATHDB-349344

Asymptotically flat interval conjecture for locally sparse multigraphs

摘录 / EXCERPT: Let $r,a\in\mathbb{N}$ and let $s\geq 2r+1$. Write $\Sigma_{r,0}(a,s)$ for the maximum edge-sum in the family $\mathcal{T}_{r,0}(a,s)$ described by the partition construction in th...

conjecture | open | MathDB 页面 | 原文线索

V2-043697 | MATHDB-349383

Oliveros et al.'s transversal-number conjecture for intersecting segment systems

摘录 / EXCERPT: Let $(P,\mathcal{L})$ be an intersecting r-segment system, where $r\geq 5$ is an integer, and let τ denote its transversal number. Oliveros et al.'s conjecture. Every inters...

conjecture | open | MathDB 页面 | 原文线索

V2-043698 | MATHDB-349406

The symmetry-symbol conjecture for normalized non-exceptional PPSs

摘录 / EXCERPT: Symmetry-symbol conjecture. For each normalized and non-exceptional PPS $(v;k_1,k_2,k_3,k_4;\lambda)$ there exist PDFs with symmetry symbols $(s{**})$ and $({**}s)$.

conjecture | open | MathDB 页面 | 原文线索

V2-043699 | MATHDB-349451

The stronger s-Cohen-Macaulay M-vector conjecture

摘录 / EXCERPT: Let s be a positive integer, and let Δ be an s-Cohen-Macaulay complex of dimension $d-1$, with h-vector $(h_0,h_1,\ldots,h_d)$. The stronger s-Cohen-Macaulay M-vec...

conjecture | open | MathDB 页面 | 原文线索

V2-043700 | MATHDB-349455

Conjecture on nim-values for the divide-and-residue game

摘录 / EXCERPT: Consider the divide-and-residue game, and let ${\mathcal{SG}}(n)$ denote the nim-value of a heap of size n. Divide-and-residue conjecture. Every nim-value occurs for at least one...

conjecture | open | MathDB 页面 | 原文线索

V2-043701 | MATHDB-349483

The crystal-structure conjecture for reverse King tableaux

摘录 / EXCERPT: Crystal-structure conjecture. There exists a crystal structure on $\mathcal{T}_{\lambda}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043702 | MATHDB-349494

Lexicographic product formula for weakly-canonical poset games

摘录 / EXCERPT: Let A and B be poset games, with $B=2^n$ for some $n\in\mathbb{N}_0$, and suppose that B is weakly-canonical. Let $A\otimes B$ be their lexicographic product, whose underlyin...

conjecture | open | MathDB 页面 | 原文线索

V2-043703 | MATHDB-349497

Incidence-density conjecture for complete bipartite subgraphs

摘录 / EXCERPT: Incidence-density conjecture. For every fixed $\epsilon>0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-043704 | MATHDB-349500

Partitioned configuration conjecture for complete bipartite subgraphs

摘录 / EXCERPT: Partitioned configuration conjecture. For every fixed integer $k>1$, every such configuration satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043705 | MATHDB-349501

Almost cross-disjoint set-system conjecture

摘录 / EXCERPT: Almost cross-disjointness conjecture. For every fixed $\epsilon>0$, if $\mathcal A$ and $\mathcal B$ are ϵ-almost cross-disjoint, then there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-043706 | MATHDB-349502

Cross-intersecting set-system conjecture

摘录 / EXCERPT: Cross-intersecting set-system conjecture. For every fixed $k>0$, if $|L|=k$ and $\mathcal A,\mathcal B$ are L-cross-intersecting, then there exist $\mathcal R\subseteq\mathcal A$...

conjecture | open | MathDB 页面 | 原文线索

V2-043707 | MATHDB-349535

The small-cube conjecture for transitive sets

摘录 / EXCERPT: Small-cube conjecture. There is a unitary basis L such that

conjecture | open | MathDB 页面 | 原文线索

V2-043708 | MATHDB-349578

Subexponential NOF communication complexity conjecture for exactly-N

摘录 / EXCERPT: In the three-player number-on-forehead (NOF) model, let the exactly-N problem be the task of determining whether the sum of the players' hidden inputs equals N. Its communicati...

conjecture | open | MathDB 页面 | 原文线索

V2-043709 | MATHDB-349579

Additive-combinatorial formulation of the exactly-N conjecture

摘录 / EXCERPT: Let $\rho^{\angle}_3(N)$ denote the maximum density of a three-variable corner-free subset in the additive-combinatorial setting used by the paper. Corner-free density conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043710 | MATHDB-349640

Wilson's extremal mixing conjecture for random Cayley graphs

摘录 / EXCERPT: Let d and n diverge with $n\leq 2^d$, let G be a group of order n, and let $H=\mathbb{Z}_2^d$. For fixed $\varepsilon,\varepsilon'\in(0,1)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-043711 | MATHDB-349656

P-position semilinearity conjecture for Exact Slow k-Nim

摘录 / EXCERPT: Let ${\sc Nim}^1_{n,=k}$ be the impartial game on nonnegative integer vectors in which a move decreases exactly k coordinates by one. A position from which the previous player ca...

conjecture | open | MathDB 页面 | 原文线索

V2-043712 | MATHDB-349657

Semilinearity conjecture for nonnegative FDG games

摘录 / EXCERPT: Let d be a positive integer and let ${\cal D}\subset\mathbb{Z}^{d}$ be a finite difference set satisfying ... for every $(a_1,\dots,a_d)\in{\cal D}$. The positions are vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-043713 | MATHDB-349708

Meagher-Spiga multipartite derangement graph conjecture

摘录 / EXCERPT: Multipartite derangement graph conjecture. If n is even but not a power of 2, then there is a transitive group G of degree n such that its derangement graph is a complete m...

conjecture | open | MathDB 页面 | 原文线索

V2-043714 | MATHDB-349710

Asymptotic integrality conjecture for intersection densities

摘录 / EXCERPT: Asymptotic integrality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043715 | MATHDB-349741

Semicompleteness of the hypercube division sequences

摘录 / EXCERPT: Let C_M^N denote the maximum number of pieces into which an M-dimensional hypercube can be cut using $N-1$ hyperplanes, and let $\{C_M^N\}$ be the resulting sequence as N var...

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V2-043716 | MATHDB-349744

The case $d=3$ conjecture for weak and strong extremals of quasiconvex quadratic forms

摘录 / EXCERPT: The case $d=3$ conjecture. Any non-polyconvex weak extremal $f\in {\cal C}_3$ is an extreme ray of ${\cal C}_3$. Moreover, if $f\in {\cal C}_3$ is a non-polyconvex extreme ray of...

conjecture | open | MathDB 页面 | 原文线索

V2-043717 | MATHDB-349771

The grid correspondence conjecture for permutation patterns and fork posets

摘录 / EXCERPT: Let J_s be the permutation-pattern matrix obtained from the identity matrix by moving its last column to the beginning, and let $\bigvee_s$ be the poset with elements...

conjecture | open | MathDB 页面 | 原文线索

V2-043718 | MATHDB-349772

The monotonicity conjecture for grid forbidden-subposet densities

摘录 / EXCERPT: Let P be a poset of dimension at most d, and let $\pi_{d,P}$ and $\pi^*_{d,P}$ denote the weak and strong grid density parameters. The grid-density monotonicity conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-043719 | MATHDB-349773

The diamond-poset grid density conjecture

摘录 / EXCERPT: Let D_2 denote the diamond poset, and for $d\geq2$ let π_{d,D_2} and π^*_{d,D_2} be its weak and strong density parameters in the d-dimensional grid. The diamond-poset...

conjecture | open | MathDB 页面 | 原文线索

V2-043720 | MATHDB-349809

Submodularity criterion for a unique maximal matroid

摘录 / EXCERPT: Let $\mathcal X$ be a family of subsets of a finite set E, and let $\operatorname{val}_{\mathcal X}$ be the set function defined from proper $\mathcal X$-sequences. Suppose there...

conjecture | open | MathDB 页面 | 原文线索

V2-043721 | MATHDB-349810

Cofactor matroid maximality and the valuation rank formula

摘录 / EXCERPT: For n vertices, let K_n be the complete graph, let $\mathcal C^{d-2}_{d-1}(K_n)$ be the cofactor matroid, and let $\operatorname{val}_{K_{d+2}}$ be the valuation associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-043722 | MATHDB-349811

Valuation characterization of generic bipartite rigidity matroids

摘录 / EXCERPT: Let $K_{m,n}$ be a complete bipartite graph, and let $\mathcal R^{k,\ell}_{m,n}$ be the generic (k,ℓ)-rigidity matroid, whose rank is $\ell m+kn-k\ell$. A rooted...

conjecture | open | MathDB 页面 | 原文线索

V2-043723 | MATHDB-349812

Valuation formulas for maximal hyperconnectivity, completion and rigidity matroids

摘录 / EXCERPT: Let K_n be the complete graph. The d-hyperconnectivity matroid $\mathcal H_n^d$ is a $\{K_{d+2},K_{d+1,d+1}\}$-matroid, $\mathcal I_n^2$ is a $\{K_5,K_{3,3}\}$-matroid, and...

conjecture | open | MathDB 页面 | 原文线索

V2-043724 | MATHDB-349814

The positive-density large-triangle conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}^2$ be a set of positive upper density, and let $u,v,w\in\mathbb{R}^2$ be non-collinear points. A configuration congruent to a finite set means an image of t...

conjecture | open | MathDB 页面 | 原文线索

V2-043725 | MATHDB-349824

Nonlinear-escape conjecture for bounded locally foldable tilings

摘录 / EXCERPT: Fix the combinatorics of a periodic 2-colored tiling of a plane, defined by a bipartite graph G on the 2-torus. Suppose that a tiling with these combinatorics folds into a bo...

conjecture | open | MathDB 页面 | 原文线索

V2-043726 | MATHDB-349825

Measure-zero conjecture for nonlinear escape in locally foldable tilings

摘录 / EXCERPT: Fix the combinatorics G of a periodic 2-colored tiling of the plane, and consider the parameters of locally foldable tilings with those fixed combinatorics. Measure-zero conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-043727 | MATHDB-349831

Hirschorn's conjecture on extremal cross-intersecting families

摘录 / EXCERPT: Hirschorn's conjecture. The quantity $N_{prod}(n,a,b,t)$ is attained on some pair of families of the form

conjecture | open | MathDB 页面 | 原文线索

V2-043728 | MATHDB-349832

Extremal-parameter conjecture for the biharmonic problem P_{λ}

摘录 / EXCERPT: Extremal-parameter conjecture. The extreme value for P_{λ} is

conjecture | open | MathDB 页面 | 原文线索

V2-043729 | MATHDB-349833

Extremal-parameter conjecture for the exponential biharmonic problem Q_{λ}

摘录 / EXCERPT: Extremal-parameter conjecture. For the problem Q_{λ}, the extremal parameter is

conjecture | open | MathDB 页面 | 原文线索

V2-043730 | MATHDB-349848

Mohr et al.'s low-sum spanning forest conjecture

摘录 / EXCERPT: Let K_n be a complete graph of order n, and let $c:E(K_n)\to \{-1,1\}$ be a zero-sum labeling satisfying ... Let F be a spanning forest of K_n with maximum degree Δ. M...

conjecture | open | MathDB 页面 | 原文线索

V2-043731 | MATHDB-349850

Conorm characterization conjecture for the matroidal locus

摘录 / EXCERPT: Conorm characterization conjecture. A matrix $A\in S^n_{>0}$ lies in the matroidal locus if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-043732 | MATHDB-349881

Conjecture on lengths of extremal rational curves

摘录 / EXCERPT: Let (X,Δ) be the dlt pair and setup of Theorem 1.6(iii), with a contraction morphism $\iphi_{R_j}$ associated to a $(K_X+\Delta)$-negative extremal ray R_j, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-043733 | MATHDB-349953

Real-rootedness of the Eulerian transformation on the binomial basis cone

摘录 / EXCERPT: Let $n\geq 0$, and let f be a polynomial of the form ... Let $\operatorname*{\mathcal E}:\mathbb{R}[t]\to\mathbb{R}[t]$ be the Eulerian transformation defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-043734 | MATHDB-349968

Intersecting-family conjecture for Hamming powers of Kneser graphs

摘录 / EXCERPT: Let $B=\{0,1\}^n$, let μ be the uniform measure on B^t, and let $p_{\mathrm{intersecting}}(t)$ be the limiting maximal measure obtained when each player's winning family is a...

conjecture | open | MathDB 页面 | 原文线索

V2-043735 | MATHDB-349969

Positive random-subgraph independence-gap conjecture

摘录 / EXCERPT: Positive random-subgraph independence-gap conjecture. $\epsilon^*(\alpha)>0$ for all $\alpha\in(0,1/2)$. This conjecture is proposed as a general statement about independent sets i...

conjecture | open | MathDB 页面 | 原文线索

V2-043736 | MATHDB-349970

Binomial random-subset independence-gap conjecture

摘录 / EXCERPT: Binomial random-subset independence-gap conjecture. $\epsilon^{**}(\alpha)>0$ for all $\alpha\in(0,1/2)$. Equivalently, there is a monotone non-decreasing function...

conjecture | open | MathDB 页面 | 原文线索

V2-043737 | MATHDB-349985

Generalized Menger embedding conjecture

摘录 / EXCERPT: Generalized Menger embedding conjecture. The van Kampen obstruction of $K\times L$ is non-zero.

conjecture | open | MathDB 页面 | 原文线索

V2-043738 | MATHDB-349990

The monochromatic-solution lower-bound conjecture for linear equations

摘录 / EXCERPT: Let $[n]=\{1,2,\dots,n\}$, let $a_1,\dots,a_k\in\mathbb{Z}$, and consider the equation ... For a coloring $f:[n]\to\{-1,1\}$, call a solution $(x_1,\dots,x_k)\in[n]^k$ monochromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-043739 | MATHDB-349991

The commonness criterion for linear equations over intervals

摘录 / EXCERPT: Let $a_1,\dots,a_k\in\mathbb{Z}$ and consider the equation ... A canceling partition is a partition of the coefficients into pairs $\{a_i,a_j\}$ such that $a_i+a_j=0$. Call the equat...

conjecture | open | MathDB 页面 | 原文线索

V2-043740 | MATHDB-350024

Spiro's descent-cyclic excedance conjecture for ballot and odd-order permutations

摘录 / EXCERPT: Spiro's conjecture. For every $n\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-043741 | MATHDB-350025

Wang-Zhang's neighbor refinement of Spiro's conjecture

摘录 / EXCERPT: Let $\mathcal{B}_n$ be the set of ballot permutations of $[n]$, and let $\mathcal{O}_n$ be the set of permutations of $[n]$ whose cycles all have odd length. For distinct letters...

conjecture | open | MathDB 页面 | 原文线索

V2-043742 | MATHDB-350035

Non-collapsibility conjecture for random Vietoris-Rips complexes

摘录 / EXCERPT: Let $K\subseteq {\mathbb R}^d$ be a convex body with smooth boundary, and let the distribution be uniform on K. Write r_n for the radius parameter and ${{\mathcal R}}(n,r_n)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-043743 | MATHDB-350049

Faudree-Rousseau-Sheehan's strongly regular graph conjecture

摘录 / EXCERPT: Faudree-Rousseau-Sheehan's conjecture. There exists a constant $c\ge2$ such that for every strongly regular graph $srg(\nu,k,\lambda,\mu)$,

conjecture | open | MathDB 页面 | 原文线索

V2-043744 | MATHDB-350055

Inverse conjecture for the directional uniformity norms

摘录 / EXCERPT: Inverse conjecture for the directional uniformity norms. If $f\colon G\to\mathbb{D}$ satisfies $\|f\|_{\mathsf{U}(H_1,\dots,H_r)}\geq c$, then there exist functions...

conjecture | open | MathDB 页面 | 原文线索

V2-043745 | MATHDB-350062

Periodicity conjecture for Hankel determinants of extremal Dyck-path avoiding sets

摘录 / EXCERPT: Extremal-set periodicity conjecture. The sequence $H(D^{(m,V)})$ is periodic and has the form

conjecture | open | MathDB 页面 | 原文线索

V2-043746 | MATHDB-350112

Kalai-Meshulam's fractional Helly conjecture for polynomial homological shatter functions

摘录 / EXCERPT: Kalai-Meshulam's fractional Helly conjecture. There exists a function $\beta:(0,1)\to(0,1)$ such that, for every $\alpha>0$ and every sufficiently large set system $\mathcal{F}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-043747 | MATHDB-350118

Häggkvist's transversal cycle-factor conjecture for blow-ups of cycles

摘录 / EXCERPT: Häggkvist's conjecture. For every $k\geq 3$, if

conjecture | open | MathDB 页面 | 原文线索

V2-043748 | MATHDB-350119

Average minimum-degree conjecture for transversal cycle-factors

摘录 / EXCERPT: Average minimum-degree conjecture. For every $k\geq3$ and $\varepsilon>0$, there exists $n_0=n_0(k,\varepsilon)$ such that for every $n\geq n_0$, if there are numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-043749 | MATHDB-350141

The extremal Borsuk number conjecture for normed spaces

摘录 / EXCERPT: Let $B(n)$ be the maximum, over all norms on $\mathbb{R}^n$, of the Borsuk number of $(\mathbb{R}^n,\left\Vert\cdot\right\Vert)$, where the Borsuk number is the smallest number of...

conjecture | open | MathDB 页面 | 原文线索

V2-043750 | MATHDB-350156

Gardner's measurable equidecomposition conjecture

摘录 / EXCERPT: Let $A,B\subseteq\mathbb{R}^n$ be bounded measurable sets, and let Γ be a set of isometries of $\mathbb{R}^n$. The sets A and B are Γ-equidecomposable if they adm...

conjecture | open | MathDB 页面 | 原文线索

V2-043751 | MATHDB-350171

Cross-K_t-intersecting theorem for all large cliques

摘录 / EXCERPT: Cross-K_t-intersecting conjecture. The same theorem, including its upper bound, uniqueness, and stability conclusions, should hold for all $t\geq 5$. The source proves the cases...

conjecture | open | MathDB 页面 | 原文线索

V2-043752 | MATHDB-350179

Klurman-Pohoata conjecture on large additive or multiplicative Sidon subsets

摘录 / EXCERPT: Klurman-Pohoata conjecture. There exists a constant $c>0$ such that, for every finite $A\subset\mathbb R$,

conjecture | open | MathDB 页面 | 原文线索

V2-043753 | MATHDB-350219

Frid-Puzynina-Zamboni prefix palindromic-length conjecture

摘录 / EXCERPT: Let w be an infinite word. A prefix of arbitrarily high palindromic length means that for every integer M there is a prefix u of w with $\operatorname{PL}(u)\geq M$. Frid-P...

conjecture | open | MathDB 页面 | 原文线索

V2-043754 | MATHDB-350220

Frid's logarithmic palindromic-length conjecture for power-free words

摘录 / EXCERPT: Let w be an infinite word, and let $\operatorname{PPL}_w(n)$ denote the palindromic length of the prefix of w of length n. Frid's conjecture. If w is k-power-free for som...

conjecture | open | MathDB 页面 | 原文线索

V2-043755 | MATHDB-350224

Higher-dimensional Füredi-Hajnal conjecture

摘录 / EXCERPT: For $t\geq 2$, let P be a t-pattern, and let $f_t(n,P)$ be the maximum number of 1's in a t-dimensional $n\times\cdots\times n$ 0-1 matrix avoiding P. Higher-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-043756 | MATHDB-350225

Higher-dimensional Stanley-Wilf conjecture

摘录 / EXCERPT: For $t\geq 2$, let P be a t-pattern, and let $T_t(n,P)$ denote the set of all t-dimensional $n\times\cdots\times n$ 0-1 matrices avoiding P. Higher-dimensional Stanley-...

conjecture | open | MathDB 页面 | 原文线索

V2-043757 | MATHDB-350227

Nash-Williams' Matroid Intersection Conjecture

摘录 / EXCERPT: Nash-Williams' Matroid Intersection Conjecture. The matroids M and N admit a common independent set I for which there is a partition

conjecture | open | MathDB 页面 | 原文线索

V2-043758 | MATHDB-350244

The exponential growth conjecture for generalized numerical semigroups

摘录 / EXCERPT: Let $d\in\mathbb{N}_{>0}$, let $F\in\mathbb{N}^d$, and let $N(F)$ denote the number of generalized numerical semigroups with Frobenius element F. For $d\ne 5$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-043759 | MATHDB-350275

Frankl-Tokushige's forbidden-intersection conjecture

摘录 / EXCERPT: Let $\mathcal{F}\subseteq\binom{[n]}{k}$, and let ℓ be an integer with $2\ell<k$. Suppose that every pair $F,F'\in\mathcal{F}$ satisfies $|F\cap F'|\ne\ell$. Frankl-Tokushige'...

conjecture | open | MathDB 页面 | 原文线索

V2-043760 | MATHDB-350292

Conjecture identifying a Hankel transform with the Robbins numbers

摘录 / EXCERPT: Robbins-number conjecture. The Hankel transform of this revert transform is the Robbins number sequence

conjecture | open | MathDB 页面 | 原文线索

V2-043761 | MATHDB-350293

Conjecture on Robbins numbers from an amalgamation of Riordan arrays

摘录 / EXCERPT: Robbins-number amalgamation conjecture. The principal minor sequence of this amalgamation is the sequence of Robbins numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-043762 | MATHDB-350294

Meng-Lin-Yang's optimal-query conjecture for the Rényi-Ulam game

摘录 / EXCERPT: Meng-Lin-Yang's conjecture. With these definitions,

conjecture | open | MathDB 页面 | 原文线索

V2-043763 | MATHDB-350306

Martínez-Roldán-Rubin transversal conjecture for pairwise intersecting convex sets

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite family of convex sets in $\mathbb{R}^3$, with every two elements of $\mathcal{F}$ intersecting. A line transversal to a subfamily is a line intersecti...

conjecture | open | MathDB 页面 | 原文线索

V2-043764 | MATHDB-350308

The Kreweras interval conjecture for tree-like chord diagrams

摘录 / EXCERPT: Let $C_n(T_{\geqslant 3},B_{\geqslant 3})$ be the set of connected chord diagrams of size n avoiding all top and bottom cycles of size at least 3, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043765 | MATHDB-350309

The Catalan-square conjecture for connected chordal diagrams

摘录 / EXCERPT: Let $\mathcal{C}_{n}(T_{\geqslant 4},B_{\geqslant 4})$ be the set of connected chord diagrams of size n avoiding top and bottom cycles of size at least 4, and let C_n denote...

conjecture | open | MathDB 页面 | 原文线索

V2-043766 | MATHDB-350323

Typical triviality conjecture for intersecting uniform families

摘录 / EXCERPT: Typical triviality conjecture. For $n\geq 2k+2$ and $k\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-043767 | MATHDB-350324

Typical structure conjecture for intersecting families at the critical threshold

摘录 / EXCERPT: Typical structure conjecture. Almost all intersecting families $\mathcal{F}\subset\binom{[2k+1]}{k}$ are nice. In fact, the number of intersecting families is

conjecture | open | MathDB 页面 | 原文线索

V2-043768 | MATHDB-350348

The nonchalant-word infinitude conjecture

摘录 / EXCERPT: Nonchalant-word infinitude conjecture. The sequence of nonchalant words over $\mathbb A_n$ is infinite for every $n\geqslant 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-043769 | MATHDB-350349

The nonchalant-word convergence conjecture

摘录 / EXCERPT: Nonchalant-word convergence conjecture. The sequence of nonchalant words over $\mathbb A_n$ converges to an infinite word $\mathcal{N}_n$ for every $n\geqslant 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-043770 | MATHDB-350350

The generalized nonchalant-word infinitude conjecture

摘录 / EXCERPT: Generalized nonchalant-word infinitude conjecture. If $\ae(N_1)>2$ for the starting word N_1 of the nonchalant algorithm, then the respective sequence of nonchalant words is infi...

conjecture | open | MathDB 页面 | 原文线索

V2-043771 | MATHDB-350351

The extremal-free-word conjecture for avoidable patterns

摘录 / EXCERPT: Extremal-free-word conjecture for avoidable patterns. For every avoidable pattern P, there are no extremal P-free words over an alphabet of size $\mu(P)+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-043772 | MATHDB-350366

The even-base min-entropy formula conjecture for entropoid power shapes

摘录 / EXCERPT: Let $\mathbb{E}_{p^2}$ be an entropoid and let $g\in\mathbb{E}^*_{(p-1)^2}$ generate its maximal multiplicative subgroupoid. For every even base $\mathfrak{b}=2b_1<b_{\max}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-043773 | MATHDB-350369

Constant lower-bound conjecture for extremal rank-one POVM simulation

摘录 / EXCERPT: Let $\mathbf{M}=(M_1,\ldots,M_n)$ be an arbitrary extremal rank-one POVM on $\mathbb{C}^d$. A partition $\{X_\gamma\}_{\gamma=1}^\alpha$ of $[n]$ satisfies $|X_\gamma|\leq d-1$ whe...

conjecture | open | MathDB 页面 | 原文线索

V2-043774 | MATHDB-350372

Cyclic c-equivalence conjecture for centrally symmetric convex sets

摘录 / EXCERPT: Cyclic c-equivalence conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-043775 | MATHDB-350376

Welzl's minimum partial triangulations conjecture

摘录 / EXCERPT: Let W_n be the number of full or partial triangulations of a convex n-gon, with $W_2=1$. For a set of n points in the plane in general position, a partial triangulation is a...

conjecture | open | MathDB 页面 | 原文线索

V2-043776 | MATHDB-350406

Poisson limit conjecture for subsystems of Steiner triple systems

摘录 / EXCERPT: Let S be the set of positive integers for which Steiner triple systems of order v exist. For an Steiner triple system of order v, let the number of its subsystems of order...

conjecture | open | MathDB 页面 | 原文线索

V2-043777 | MATHDB-350417

The equivalence of Host-Kra, Weyl, true and algebraic complexity

摘录 / EXCERPT: Let $t\in\mathbb{N}_+$ and let $\vec{P}\in\mathbb{R}[x,y]^{t+1}$ be an integral polynomial progression. For each $0\leqslant i\leqslant t$, let $\mathcal{HK}_i(\vec{P})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043778 | MATHDB-350422

A conjecture on arithmetic sums of Ahlfors-regular sets

摘录 / EXCERPT: General arithmetic-sum conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043779 | MATHDB-350428

The extremal PageRank distance conjecture

摘录 / EXCERPT: Let G be any unweighted directed graph, possibly with loops and bidirected arcs, and let α_1 and α_2 be two jumping constants. For each jumping constant, write...

conjecture | open | MathDB 页面 | 原文线索

V2-043780 | MATHDB-350430

Conjecture on optimal distal cell-decomposition bounds in o-minimal structures

摘录 / EXCERPT: Conjecture on optimal o-minimal bounds. The same bounds should hold in any o-minimal structure, potentially via Davenport-Schinzel sequences. Moreover, every $\Phi(x;y)$ in an...

conjecture | open | MathDB 页面 | 原文线索

V2-043781 | MATHDB-350439

The matroid lift construction conjecture

摘录 / EXCERPT: Let M be a matroid. A lift of M is a matroid obtained from M by increasing its rank while preserving the relevant lift relation. Let $\mathcal{C}(M)$ denote the set of circui...

conjecture | open | MathDB 页面 | 原文线索

V2-043782 | MATHDB-350440

The finite-group extension conjecture for group-labeled graph lifts

摘录 / EXCERPT: Let $n\ge 3$ be an integer and let Γ be a finite group. Let K_n^{Γ} be the fully Γ-labeled complete graph, and let $\mathcal{M}_{n,\Gamma}$ denote the class of...

conjecture | open | MathDB 页面 | 原文线索

V2-043783 | MATHDB-350441

The explicit matroid construction conjecture

摘录 / EXCERPT: The explicit matroid construction conjecture. The collection $\mathcal{I}$ is the collection of independent sets of a matroid N on $\mathcal{C}(M)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-043784 | MATHDB-350482

A permutation identity supporting the Green-Liebeck code conjecture

摘录 / EXCERPT: For natural numbers k,n, define $P^a_b=b!/(b-a)!$ whenever the indices are defined. Let k,m,n be positive integers satisfying $1<2k\leq n-m$ and $1<2k\leq m$. Permutation ident...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-043785 | MATHDB-350503

Characterization of finite fields representing all graphic delta-matroids

摘录 / EXCERPT: Let Γ be a finite abelian group of order at least 2, and let $\mathbb{F}$ be a finite field. A Γ-graphic delta-matroid is a delta-matroid associated with a Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-043786 | MATHDB-350532

Extremal Betti-number conjecture for random higher-degree monomial ideals

摘录 / EXCERPT: Fix integers d,m with $d\geq m\geq 1$, and suppose that α satisfies ... Let $\Delta\sim\Delta_m(n,n^{-\alpha})$ be the corresponding multiparameter random simplicial complex...

conjecture | open | MathDB 页面 | 原文线索

V2-043787 | MATHDB-350552

Dimension-stability conjecture for exact hypercube covering numbers

摘录 / EXCERPT: For $B\subset\{0,1\}^k$ and an integer $n\geq k$, let $\operatorname{ec}(B)$ be the minimum number of hyperplanes whose union intersects the Boolean cube precisely in B. Dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-043788 | MATHDB-350553

Gedeon's real-rootedness conjecture for matroid Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let M be a matroid, and let $P_{\mathsf{M}}(t)$ denote its Kazhdan-Lusztig polynomial. Gedeon's conjecture. The polynomial $P_{\mathsf{M}}(t)$ is real-rooted. This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-043789 | MATHDB-350562

Integral surgery exact triangle for combinatorial Heegaard Floer homology

摘录 / EXCERPT: Let Y be a closed, oriented 3-manifold and let K be a framed knot in Y. Let $\mathcal H$ be a nice bordered Heegaard diagram for the infinity-framed complement...

conjecture | open | MathDB 页面 | 原文线索

V2-043790 | MATHDB-350567

Degree-two characterization conjecture for simple or cosimple connected binary matroids

摘录 / EXCERPT: Let M be a connected binary matroid that is simple or cosimple. The statements (a), (b), and (c) in the degree-two lemma are equivalent. Degree-two characterization conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-043791 | MATHDB-350568

Degree-five characterization conjecture for simple or cosimple connected binary matroids

摘录 / EXCERPT: Let M be a connected binary matroid that is simple or cosimple. The statements (a), (b), and (c) in the degree-five lemma are equivalent. Degree-five characterization conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-043792 | MATHDB-350573

The weighted Erdős matching conjecture for overlapping families

摘录 / EXCERPT: Let p be a positive integer and let $\vec{p}=(p,1,\ldots,1)$. For $n\geq (s+1)k$, define $f_{\vec{p}}(n,k,s)$ as the maximum weighted sum associated with $s+1$ overlapping families...

conjecture | open | MathDB 页面 | 原文线索

V2-043793 | MATHDB-350574

Aharoni-Howard conjecture for overlapping families

摘录 / EXCERPT: Let $\mathcal{B}_0,\ldots,\mathcal{B}_s\subset\binom{[n]}{k}$ be overlapping families, and suppose that $n\geq k(s+1)$. Aharoni-Howard conjecture. ... This is presented as a generali...

conjecture | open | MathDB 页面 | 原文线索

V2-043794 | MATHDB-350593

The multiplicative Merino-Welsh conjecture for matroids

摘录 / EXCERPT: Let M be a matroid with no loops or isthmuses, and let $T(M;x,y)$ denote its Tutte polynomial. Multiplicative Merino-Welsh conjecture. ... This is presented as a seemingly stronger...

conjecture | open | MathDB 页面 | 原文线索

V2-043795 | MATHDB-350594

The Merino-Welsh inequality for composition enumerators

摘录 / EXCERPT: Let $\bar\gamma(\underline{a};x,y)$ be the composition-indexed quantity used in the source. Let $(a_0,a_,\ldots,a_r)$ be a composition with $a_0=0$ and $a_r\geq 2$. Merino-Welsh co...

conjecture | open | MathDB 页面 | 原文线索

V2-043796 | MATHDB-350633

The extremal inversion-number conjecture for cycle transversals

摘录 / EXCERPT: Extremal inversion-number conjecture. For every positive integer n, there exists an oriented graph D such that

conjecture | open | MathDB 页面 | 原文线索

V2-043797 | MATHDB-350641

The improved uniform bound for families with bounded symmetric-difference VC-dimension

摘录 / EXCERPT: Uniform Dvir-Moran conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-043798 | MATHDB-350647

De Loera-Haws-Köppe conjecture on Ehrhart positivity of matroid basis polytopes

摘录 / EXCERPT: De Loera-Haws-Köppe conjecture. The basis polytope $\mathscr{P}(M)$ of every matroid M is Ehrhart positive.

conjecture | open | MathDB 页面 | 原文线索

V2-043799 | MATHDB-350648

Castillo-Liu conjecture on Ehrhart positivity of generalized permutohedra

摘录 / EXCERPT: Castillo-Liu conjecture. Every integral generalized permutohedron $\mathscr{P}$ is Ehrhart positive.

conjecture | open | MathDB 页面 | 原文线索

V2-043800 | MATHDB-350651

Finiteness conjecture for 2-unavoidable doubled patterns

摘录 / EXCERPT: Finiteness conjecture for 2-unavoidable doubled patterns. There exist only finitely many 2-unavoidable doubled patterns.

conjecture | open | MathDB 页面 | 原文线索

V2-043801 | MATHDB-350653

Diagonal-segment continuous Turán conjecture

摘录 / EXCERPT: Let $P\subseteq\mathbb{R}^2$ be the union of the segment with endpoints $(0,0)$ and $(1,0)$ and the segment with endpoints $(0,1)$ and $(1,2)$. Diagonal-segment conjecture. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-043802 | MATHDB-350654

Quasilinear bound for unions of horizontal segments

摘录 / EXCERPT: Let $P\subseteq\mathbb{R}^2$ be a disjoint union of horizontal segments, with no two points of P having the same x-coordinate. Quasilinear horizontal-segment conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-043803 | MATHDB-350655

Gluing inequality for continuous Turán numbers

摘录 / EXCERPT: Suppose that $P,Q\subseteq\mathbb{R}^2$ both contain (c,d). Assume that every other point of P has x- and y-coordinates at most c and that every other point of Q has...

conjecture | open | MathDB 页面 | 原文线索

V2-043804 | MATHDB-350656

Dichotomy conjecture for continuous saturation functions

摘录 / EXCERPT: For $P\subseteq\mathbb{R}^2$, call an open set $S\subseteq[0,n]^2$ saturating for P if S avoids P and every proper open superset of S in $[0,n]^2$ contains P. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-043805 | MATHDB-350657

Finite-subset saturation correspondence conjecture

摘录 / EXCERPT: For $P\subseteq\mathbb{R}^2$, let M_P denote the associated finite forbidden 0-1 matrix when P is finite, and let $\operatorname{osat}(n,P)$ be the continuous saturation fu...

conjecture | open | MathDB 页面 | 原文线索

V2-043806 | MATHDB-350676

The orientable pseudomanifold characterization via the top-dimensional Laplacian

摘录 / EXCERPT: Orientable pseudomanifold characterization. K is an orientable pseudomanifold if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-043807 | MATHDB-350677

Total positivity of the Schl"afli--Gessel--Seo polynomial matrix and sequences

摘录 / EXCERPT: For $n,k\geq 0$, let $P_{n,k}(a,b)$ be the Schl"afli--Gessel--Seo forest polynomials, let $P_n(x;a,b)=\sum_{k=0}^n P_{n,k}(a,b)x^k$, and let $P_n(a,b)=P_{n,1}(a,b)$. A polynomial m...

conjecture | open | MathDB 页面 | 原文线索

V2-043808 | MATHDB-350679

Total positivity of bivariate forest polynomials

摘录 / EXCERPT: Let $f^\sharp_{n,k}(w)$ be the entries of the exponential Riordan array with generating function $G(t)=(e^{wT(t)}-1)/w$, where $T(t)$ is the tree function, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-043809 | MATHDB-350680

Shifted Hankel total positivity for bivariate forest polynomials

摘录 / EXCERPT: Let $f^\sharp_{n,k}(w)$ and $F_n^\sharp(x,w)=\sum_{k=0}^n f^\sharp_{n,k}(w)x^k$ be as above. A polynomial sequence is coefficientwise Hankel-totally positive when all minors of its...

conjecture | open | MathDB 页面 | 原文线索

V2-043810 | MATHDB-350698

Bourgain's discrete restriction conjecture for lattice points on spheres

摘录 / EXCERPT: Let $d\geq 3$ be a natural number, let m be a natural number, let $(\mathfrak{a}_{\bm{n}})_{\bm{n}\in S_{d,m}}$ be complex numbers, let $\epsilon>0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-043811 | MATHDB-350721

Signed cycle-type restriction polynomiality conjecture

摘录 / EXCERPT: Let n be a positive integer, let k be a fixed positive integer with $1\leq k\leq n$, and let $\mathfrak{S}_n$ denote the symmetric group on n elements. For...

conjecture | open | MathDB 页面 | 原文线索

V2-043812 | MATHDB-350744

Conjecture that every extremal graph is complete multipartite

摘录 / EXCERPT: Let $\bm{k}=(k_1,\hspace{0.9pt}.\hspace{0.3pt}.\hspace{0.3pt}.\hspace{1.5pt},k_s)$ be a non-increasing sequence of natural numbers, and let a graph be $\bm{k}$-extremal if it attai...

conjecture | open | MathDB 页面 | 原文线索

V2-043813 | MATHDB-350784

The conjectured order of the clique chromatic number of random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the random graph on n vertices in which each edge is present independently with probability $p=p(n)$, and let $\chi_c(G_{n,p})$ denote its clique chromatic numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-043814 | MATHDB-350792

Finite-certificate conjecture for relaxation complexity in dimension three

摘录 / EXCERPT: Finite-certificate conjecture. There exists a finite set $Y \subseteq \mathbb{Z}^3 \setminus X$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-043815 | MATHDB-350800

Benjamini-Chan-O'Donnell-Tamuz-Tan conjecture on agreement in majority dynamics

摘录 / EXCERPT: Benjamini-Chan-O'Donnell-Tamuz-Tan conjecture. With probability $1-\varepsilon$, the vertices in $G(n,p)$ have an $(1-\varepsilon)$-proportion agreement after sufficiently many day...

conjecture | open | MathDB 页面 | 原文线索

V2-043816 | MATHDB-350803

Dowling-Wilson top-heavy conjecture for subspace numbers

摘录 / EXCERPT: Let $E=\{v_1,\ldots,v_n\}$ be a spanning subset of a d-dimensional complex vector space V, and let $w_i(E)$ denote the number of i-dimensional subspaces spanned by subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-043817 | MATHDB-350842

A coupling conjecture for the symmetry rule in random bipartite graphs

摘录 / EXCERPT: The symmetry-rule coupling conjecture. The graphs $G^L(n,n,p)$, $G(n^-,n^-,p^-)$ and $G(n^+,n^+,p^+)$ can be coupled so that, with high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-043818 | MATHDB-350848

The explicit formula conjecture for the monochromatic subset-sum constant

摘录 / EXCERPT: Explicit formula conjecture. For every integer $r\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-043819 | MATHDB-350861

The shifted generalized forest submatrix Hankel-total positivity conjecture

摘录 / EXCERPT: Let $G'(p,q,\gamma,z)=\mathsf{F}(p,q,0,0,\gamma,0,0,z)\operatorname{diag}(((pq\gamma)^{n(n-1)/2})_{n\geq0})$, and let $G'_n(x;p,q,\gamma,z)=\sum_{k=0}^n g'_{n,k}(p,q,\gamma,z)x^k$....

conjecture | open | MathDB 页面 | 原文线索

V2-043820 | MATHDB-350862

The shifted reverse Bessel polynomial Hankel-total positivity conjecture

摘录 / EXCERPT: Let $\mathsf{M}(r,s,\mu,y)$ be the multivariate matrix in the source, and define ... Let $\mathsf{M}'_0(x;r,s,\mu,y)=1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-043821 | MATHDB-350863

The crossing-nesting enumeration conjecture for the forest-matrix first column

摘录 / EXCERPT: For a perfect matching M on $[2m]$, let ${\rm cr}(M)$ and ${\rm ne}(M)$ denote its numbers of crossings and nestings, respectively, and let ${\mathcal{M}}_{2m}$ be the set of per...

conjecture | open | MathDB 页面 | 原文线索

V2-043822 | MATHDB-350920

Existence and uniqueness of homology tilings with prescribed orientations

摘录 / EXCERPT: Prescribed-orientation homology tiling conjecture. For every choice of σ, there exists a unique homology tiling of R or $-R$ such that tile w has positive area and orien...

conjecture | open | MathDB 页面 | 原文线索

V2-043823 | MATHDB-350928

A linear bound for the relative rank constant over finite fields

摘录 / EXCERPT: Let d and s be parameters, let $\mathbb F$ be a finite field with $\operatorname{char}(\mathbb F)>d$, and let $C(\mathbb F,s,d)$ denote the constant in the preceding relative-r...

conjecture | open | MathDB 页面 | 原文线索

V2-043824 | MATHDB-350935

Archer et al.'s quasi-Stirling descent conjecture

摘录 / EXCERPT: Archer et al.'s conjecture. The number of $\pi\in\overline{\mathcal{Q}}_{\mathcal{M}}$ with $des(\pi)=n$ is equal to

conjecture | open | MathDB 页面 | 原文线索

V2-043825 | MATHDB-350947

The product formula conjecture for pinnacle-set enumeration

摘录 / EXCERPT: For a finite set S of pinnacle values, let ... Write $S=\{s_1,\dots,s_p\}$ with $n\geq s_p$, define ... and set ... Define abstract expressions by $E_1=x_{1,3}+x_{1,1}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043826 | MATHDB-350973

Positivity conjecture for frieze patterns of type $\Lambda_{p_1,\ldots,p_s}$

摘录 / EXCERPT: Positivity conjecture. The only frieze patterns of Type $\Lambda_{p_1,\ldots,p_s}$ with negative entries are those which fail the realizability test. In particular, all quotient di...

conjecture | open | MathDB 页面 | 原文线索

V2-043827 | MATHDB-351002

Partial gamma-positivity conjecture for quasi-Stirling permutation polynomials

摘录 / EXCERPT: For a multiset $\mathcal{M}$, let $\overline{\mathcal{Q}}_{\mathcal{M}}$ denote the set of quasi-Stirling permutations of $\mathcal{M}$, and define ... where $asc(\pi)$, $des(\pi)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-043828 | MATHDB-351017

The freeness ordering of generic rigidity matroids

摘录 / EXCERPT: Kalai-Whiteley conjecture. For every n and d,

conjecture | open | MathDB 页面 | 原文线索

V2-043829 | MATHDB-351018

Whiteley's maximality conjecture for cofactor rigidity matroids

摘录 / EXCERPT: Whiteley's maximality conjecture. The matroid $\mathcal{C}_{d-1}^{d-2}(n)$ is the freest matroid in which every K_{d+2} is a circuit. In particular, it is the freest among all ab...

conjecture | open | MathDB 页面 | 原文线索

V2-043830 | MATHDB-351019

Kalai-Whiteley maximality conjecture for hyperconnectivity matroids

摘录 / EXCERPT: Kalai-Whiteley maximality conjecture. The matroid $\mathcal{H}_d(n)$ is the freest matroid in which every K_{d+2} and every $K_{d+1,d+1}$ are circuits.

conjecture | open | MathDB 页面 | 原文线索

V2-043831 | MATHDB-351020

Conjecture on uncommon (2 × k)-systems with maximal s(L)

摘录 / EXCERPT: Uncommon $(2\times k)$-system conjecture. For even $k\geq 6$ and sufficiently large odd q, every $(2\times k)$-system L satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-043832 | MATHDB-351028

Cylindric Naruse hook-length formula

摘录 / EXCERPT: Let $m,\ell \in \mathbb{Z}_{\geq 1}$ and let $\lambda,\mu \in \mathcal{P}_{m,\ell}$ satisfy $\lambda \supset \mu$. Put ... Here $f^{\mathring{\lambda}/\mathring{\mu}}$ denotes the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-043833 | MATHDB-351035

Ramsey-Turán tiling conjecture for cliques

摘录 / EXCERPT: Let r and k be integers satisfying $1\le r\le k$ and $k\ge 3$. The Ramsey-Turán tiling function ${\bf{RTT}}_r(K_k)$ is defined as the asymptotic minimum-degree threshold for fo...

conjecture | open | MathDB 页面 | 原文线索

V2-043834 | MATHDB-351041

Finite excluded-minor conjecture for quasi-graphic matroids

摘录 / EXCERPT: Finite excluded-minor conjecture. There are, up to isomorphism, only finitely many excluded minors for the class of quasi-graphic matroids.

conjecture | open | MathDB 页面 | 原文线索

V2-043835 | MATHDB-351063

Minimum degree conjecture for perfect clique tilings in multipartite graphs

摘录 / EXCERPT: Let $k\geq r\geq 2$ and $\gamma>0$. Let G be a k-partite graph on n vertices with parts $V_1,\ldots,V_k$ such that $|V_i|\leq n/r$ for every i. For each $i\in [k]$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-043836 | MATHDB-351064

Greedy-move conjecture for convex grabbing configurations

摘录 / EXCERPT: Let C be a cake configuration, let $\operatorname{Ex}(C)$ denote its extremal cherries, and let $\operatorname{R}(C)$ denote its red cherries. A move is optimal if it is optimal...

conjecture | open | MathDB 页面 | 原文线索

V2-043837 | MATHDB-351066

No-reveal-move conjecture for convex grabbing configurations

摘录 / EXCERPT: Let C be a cake configuration, let $\operatorname{Ex}(C)$ denote its extremal cherries, and let $\operatorname{R}(C)$ denote its red cherries. Define the set of non-revealing mov...

conjecture | open | MathDB 页面 | 原文线索

V2-043838 | MATHDB-351082

Extremal signed Wiener index conjecture for trees

摘录 / EXCERPT: Let (T,σ) be a signed tree on n vertices. For the path P_n, let $+$ denote the constant signing that assigns $+1$ to every edge, and let α denote the alternating...

conjecture | open | MathDB 页面 | 原文线索

V2-043839 | MATHDB-351089

Hurwitz stability conjecture for the alternating-run polynomial

摘录 / EXCERPT: Let $r(x)=\sqrt{\frac{1+x}{1-x}}$ and define $\mathscr{R}_n(x)=\sum_{k=0}^{2n-1}\mathscr{R}(n,k)x^k$ by ... The recurrence is ... with $\mathscr{R}_0(x)=1$ and $\mathscr{R}_1(x)=x$. Hu...

conjecture | open | MathDB 页面 | 原文线索

V2-043840 | MATHDB-351090

Hurwitz stability conjecture for the peak polynomial sequence

摘录 / EXCERPT: Hurwitz stability conjecture. The polynomial $T_{n,r}(x)$ is Hurwitz stable for all $r\ge 2$ and $n\in\mathbb{N}$. The sequence is introduced after Hurwitz stability has been estab...

conjecture | open | MathDB 页面 | 原文线索

V2-043841 | MATHDB-351093

Polynomial hitting-number conjecture for complete regular matroids

摘录 / EXCERPT: Polynomial hitting-number conjecture. The quantity $h(M)$ is bounded by a polynomial in r.

conjecture | open | MathDB 页面 | 原文线索

V2-043842 | MATHDB-351100

Polynomial-time recognition conjecture for 0/1 slack matrices

摘录 / EXCERPT: Let $S\in\{0,1\}^{m\times n}$. A matrix is a slack matrix if it is the slack matrix of a polytope. Polynomial-time recognition conjecture. There is an algorithm polynomial in m,n...

conjecture | open | MathDB 页面 | 原文线索

V2-043843 | MATHDB-351123

The one-divisor conjecture for cyclotomic divisibility in cyclic tilings

摘录 / EXCERPT: One-divisor conjecture. If $\Phi_{p_i^{n_i}}\mid A$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043844 | MATHDB-351124

The fiberedness conjecture for cyclic tilings

摘录 / EXCERPT: Fiberedness conjecture. For every i there exists $1\leq\alpha_i<n_i$ such that either A or B is $M/p_i^{\alpha_i}$-fibered in the p_i direction. In particular, if M has...

conjecture | open | MathDB 页面 | 原文线索

V2-043845 | MATHDB-351125

The line-intersection bound conjecture for cyclic tilings

摘录 / EXCERPT: Line-intersection bound conjecture. For every $x\in\mathbb{Z}_M$,

conjecture | open | MathDB 页面 | 原文线索

V2-043846 | MATHDB-351149

Conjecture on the asymptotic bound for analytic combinatorial curves

摘录 / EXCERPT: Asympture conjecture. The first terms of the sequence suggest that $a'_0=\alpha$ is sufficient; under this choice, the exponential-growth constant ρ in the bound for the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-043847 | MATHDB-351156

The q-analog of the Markoff injectivity conjecture

摘录 / EXCERPT: Let μ_q be the q-analog of the monoid homomorphism used to define q-Markoff numbers, and let $\mu_q(w)_{12}\in\mathbb{Z}[q]$ denote the corresponding entry above the diagon...

conjecture | open | MathDB 页面 | 原文线索

V2-043848 | MATHDB-351157

The q-analog of the Markoff injectivity conjecture over balanced sequences

摘录 / EXCERPT: Let $\mathcal{B}=\\{s\in\\{\mathtt{0},\mathtt{1}\\}^{\mathbb{Z}}:s\text{ is balanced}\\}$ be the set of all balanced bi-infinite binary sequences, and let ... be their language of fi...

conjecture | open | MathDB 页面 | 原文线索

V2-043849 | MATHDB-351169

Planar simple-matching domination conjecture

摘录 / EXCERPT: Let G and H be graphs with planar union, let $\mathcal{PM}(G,H)$ be the matchings pairing edges from G and H, and let s_M be the leading coefficient obtained by summing...

conjecture | open | MathDB 页面 | 原文线索

V2-043850 | MATHDB-351176

The Hankel-transform relation for Rueppel complements and first differences

摘录 / EXCERPT: Let r_n be the Rueppel sequence. Let H_n be the Hankel transform of the sequence $1-r_n$, and let h_n be the Hankel transform of the first-difference sequence $r_{n+1}-r_n$....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-043851 | MATHDB-351180

Configuration formula conjecture for the lowest skew Jack coefficient

摘录 / EXCERPT: Configuration formula conjecture. For every partition $\bbig mu$ and every $\bbig mu$-admissible configuration C, there are uniquely determined polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-043852 | MATHDB-351184

Cameron's binomial conjecture for odd rows of random Latin squares

摘录 / EXCERPT: Let a row of a Latin square be called odd if its corresponding permutation is odd, and let n be the order of the Latin square. Cameron's conjecture. The number of odd rows of a r...

conjecture | open | MathDB 页面 | 原文线索

V2-043853 | MATHDB-351273

The projective-plane completion conjecture for dense Sidon sets

摘录 / EXCERPT: Projective-plane completion conjecture. The action is faithful, $|\mathcal{P}|=(1+o(1))n$, and for some point p and line ℓ,

conjecture | open | MathDB 页面 | 原文线索

V2-043854 | MATHDB-351274

The subgroup-union conjecture for dense maximal Sidon sets

摘录 / EXCERPT: Subgroup-union conjecture. The set T is the union of $O(1)$ subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-043855 | MATHDB-351275

The elementary subgroup conjecture for Sidon sets in \(\mathbf{F}_p^2\)

摘录 / EXCERPT: Elementary subgroup conjecture. The set T is a subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-043856 | MATHDB-351286

Combinatorial positivity and monotonicity of discrete intrinsic volumes

摘录 / EXCERPT: Let $P\in\mathcal P(\mathbb{Z}^d)$ have dimension r, and let the k-th discrete intrinsic volume be represented in the binomial basis by ... A valuation is combinatorially positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-043857 | MATHDB-351293

Asymptotic conjecture for optimal Player I and random Player II in Penney-Ante

摘录 / EXCERPT: Optimal-random asymptotic conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-043858 | MATHDB-351294

Conjecture on optimal responses in the flipped Penney-Ante game

摘录 / EXCERPT: Flipped-game optimal-response conjecture. The best response strings for Player II are one or more of

conjecture | open | MathDB 页面 | 原文线索

V2-043859 | MATHDB-351295

Dudek-Grytczuk-Ruciński conjecture on permutation squares

摘录 / EXCERPT: Let π be an n-permutation. An r-square is a collection of r pairwise disjoint subsequences of π, each of length r, with the same relative order. Let $s(\pi)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-043860 | MATHDB-351304

Finitary polynomial configuration conjecture over rings of integers

摘录 / EXCERPT: Let K be a degree d number field with ring of integers $O_K\cong\mathbb{Z}^d$, and let $p(x)\in K[x]$ be an O_K-valued intersective polynomial. Finitary configuration conject...

conjecture | open | MathDB 页面 | 原文线索

V2-043861 | MATHDB-351306

Uniqueness of the holomorphic Ahlfors-Hopf extremal on the disk

摘录 / EXCERPT: Uniqueness conjecture. If $\hat{\Phi}$ is holomorphic, then $\hat{h}=\operatorname{identity}$.

conjecture | open | MathDB 页面 | 原文线索

V2-043862 | MATHDB-351334

Bollobás-Brightwell-Leader conjecture on enumerating k-SAT functions

摘录 / EXCERPT: Bollobás-Brightwell-Leader conjecture. The number of k-SAT functions on n boolean variables is

conjecture | open | MathDB 页面 | 原文线索

V2-043863 | MATHDB-351342

Permutation concentration conjecture for efficient and inefficient maximal configurations

摘录 / EXCERPT: Let maximal configurations be configurations to which the function G maps permutations, and classify them as efficient or inefficient according to the settlement-planning model.…

conjecture | open | MathDB 页面 | 原文线索

V2-043864 | MATHDB-351367

Component-wise minimality conjecture for simplicial complexes with torsion

摘录 / EXCERPT: For $p\geq 3$, consider the simplicial 2-complexes d2n83torsion and d2n84torsion, which have 3-torsion and 4-torsion, respectively. Their f-vectors record the numbers of fa...

conjecture | open | MathDB 页面 | 原文线索

V2-043865 | MATHDB-351437

Minimum-size unavoidable subsets of cyclic permutations

摘录 / EXCERPT: Minimum-cardinality conjecture. For all $k \geq 1$, the minimum cardinality of an unavoidable subset of $\overline{S_k}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-043866 | MATHDB-351449

Grytczuk-Kordulewski-Niewiadomski conjecture on extremal square-free words

摘录 / EXCERPT: Grytczuk-Kordulewski-Niewiadomski conjecture. There exists no extremal square-free word over a finite alphabet of size at least 4.

conjecture | open | MathDB 页面 | 原文线索

V2-043867 | MATHDB-351463

Global convergence conjecture for fractional combinatorial Calabi flow with surgery

摘录 / EXCERPT: Let (M,V,ε) be a marked weighted connected closed surface with $\varepsilon:V\rightarrow\{0,1\}$, and suppose there exists a PL or PH metric generated by a discrete con...

conjecture | open | MathDB 页面 | 原文线索

V2-043868 | MATHDB-351476

Non-Sidorenko conjecture for (2×5)-systems

摘录 / EXCERPT: Let L be a (2×5)-system of linear equations, and let $s(L)$ denote the parameter used in the paper. Assume that $s(L)=4$. Non-Sidorenko conjecture. Every (2×5)-sy...

conjecture | open | MathDB 页面 | 原文线索

V2-043869 | MATHDB-351501

Conjecture on threshold Ramsey multiplicity of paths

摘录 / EXCERPT: Let P_k denote the path on k vertices, and let $m(P_k)$ denote its threshold Ramsey multiplicity, namely the minimum number of monochromatic copies of P_k in a two-coloring o...

conjecture | open | MathDB 页面 | 原文线索

V2-043870 | MATHDB-351513

Hereditary chromatic recurrence inside density-recurrent sets

摘录 / EXCERPT: Let $S\subseteq\mathbb Z$ be a set of density recurrence, meaning that for every subset A of a suitable positive upper density, $(A-A)\cap S\neq\varnothing$. Hereditary recurrenc...

conjecture | open | MathDB 页面 | 原文线索

V2-043871 | MATHDB-351514

Chromatic recurrence in intersections with H-sets

摘录 / EXCERPT: Let $S\subseteq\mathbb Z$ be a set of chromatic recurrence, let $k\in\mathbb N$, let $C>0$, and let $\widetilde H(\boldsymbol\alpha;C\sqrt d,\varepsilon)$ denote the set defined ea...

conjecture | open | MathDB 页面 | 原文线索

V2-043872 | MATHDB-351525

Katznelson's recurrence conjecture for countably infinite abelian groups

摘录 / EXCERPT: Katznelson's conjecture. If $S\subseteq G$ is a set of Bohr recurrence, then S is a set of chromatic recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-043873 | MATHDB-351526

Katznelson's recurrence conjecture for iterated differences in $\mathbb Z^\omega$

摘录 / EXCERPT: Special Katznelson conjecture. If $S\subseteq\Delta_2(\mathcal E_1)$ is a set of Bohr recurrence in $\mathbb Z^\omega$, then S is a set of chromatic recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-043874 | MATHDB-351527

Katznelson's recurrence conjecture for $\mathbb F_p^\omega$

摘录 / EXCERPT: Katznelson's finite-field conjecture. Every subset of $\mathbb F_p^\omega$ which is a set of Bohr recurrence is a set of chromatic recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-043875 | MATHDB-351554

Lazar-Wachs cycle-distribution conjecture for D- and E-permutations

摘录 / EXCERPT: Lazar and Wachs' cycle-distribution conjecture. The number of D-permutations on $[2n]$ with k cycles equals the number of E-permutations on $[2n]$ with k cycles for all k. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-043876 | MATHDB-351568

Radziszowski et al.'s conjecture on the bipartite Ramsey number B(2,2,3)

摘录 / EXCERPT: Let $B(n_1,n_2,\ldots,n_t)$ be the least positive integer b such that every coloring of the edges of $K_{b,b}$ with t colors contains a monochromatic copy of K_{n_i,n_i} in t...

conjecture | open | MathDB 页面 | 原文线索

V2-043877 | MATHDB-351573

Meagher-Razafimahatratra-Spiga conjecture on derangement graphs of even-degree groups

摘录 / EXCERPT: Meagher-Razafimahatratra-Spiga's conjecture. There exists a transitive permutation group G of degree n such that Γ_G is a complete multipartite graph with $n/2$ parts.

conjecture | open | MathDB 页面 | 原文线索

V2-043878 | MATHDB-351576

The linear C^0-formulation of Strong Cosmic Censorship for asymptotically AdS black holes

摘录 / EXCERPT: Let $\boldsymbol{\psi}$ be a linear perturbation solving the wave equation on a subextremal asymptotically AdS black hole, with generic initial data and Dirichlet or more general b...

conjecture | open | MathDB 页面 | 原文线索

V2-043879 | MATHDB-351583

Conjecture on the extremal number of maximal sum-free sets in abelian groups

摘录 / EXCERPT: Extremal bounds conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043880 | MATHDB-351584

Carnevale-Voll's factorisation conjecture for multiset Euler-Mahonian polynomials

摘录 / EXCERPT: Carnevale-Voll's factorisation conjecture. The polynomial giving the joint distribution of $(\operatorname{den},\operatorname{exc})$ over S_{η} has a unitary factor if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-043881 | MATHDB-351585

Kantor's conjecture on strong embeddability of hypermodular matroids

摘录 / EXCERPT: Kantor's conjecture. Every hypermodular matroid is strongly embeddable in a modular matroid.

conjecture | open | MathDB 页面 | 原文线索

V2-043882 | MATHDB-351586

Hypermodular matroid conjecture on same-rank modular extensions

摘录 / EXCERPT: Hypermodular matroid conjecture. Every hypermodular matroid is a restriction of a modular matroid of the same rank.

conjecture | open | MathDB 页面 | 原文线索

V2-043883 | MATHDB-351587

Poljak-Turzík's sticky matroid conjecture

摘录 / EXCERPT: Sticky matroid conjecture. Stickiness of a matroid implies modularity. Thus, stickiness is equivalent to modularity.

conjecture | open | MathDB 页面 | 原文线索

V2-043884 | MATHDB-351596

Bubble-up and missing-rim conjecture for wheels of neural codes

摘录 / EXCERPT: Let $\mathcal{C}$ be a neural code, let $\Delta(\mathcal{C})$ be its simplicial complex, and let $\mathcal{W}=(\sigma_1,\sigma_2,\sigma_3,\tau)$ be a wheel of $\mathcal{C}$. A face...

conjecture | open | MathDB 页面 | 原文线索

V2-043885 | MATHDB-351630

Weak-domination conjecture for reflective tree decompositions

摘录 / EXCERPT: Let G be a connected non-trivial bigraph, and let T be a reflective tree decomposition of G with core H. For each edge $\{U_1,U_2\}\in E(T)$, assume that H weakly dominat...

open_problem | open | MathDB 页面 | 原文线索

V2-043886 | MATHDB-351641

Karasev's colorful Tverberg conjecture for hyperplanes

摘录 / EXCERPT: Let r be a positive integer. Given r blue, r red, and r green lines in the plane in general position, where a family is in general position when the normal vectors of any t...

conjecture | open | MathDB 页面 | 原文线索

V2-043887 | MATHDB-351642

Bárány-Larman colorful Tverberg conjecture

摘录 / EXCERPT: Let r,d be positive integers. Given $d+1$ sets $X_1, \ldots, X_{d+1}$, each consisting of r points of $\mathds{R}^d$, consider partitions of their union into r sets...

conjecture | open | MathDB 页面 | 原文线索

V2-043888 | MATHDB-351643

Uniqueness conjecture for maximum partially 2-intersecting uniform partition families

摘录 / EXCERPT: A (k,ℓ)-partition is a set partition of $\{1,2,\dots,k\ell\}$ with exactly ℓ blocks, each of size k. Let $S_{i,j}$ denote the canonical partially 2-intersecting family...

conjecture | open | MathDB 页面 | 原文线索

V2-043889 | MATHDB-351646

The root-of-unity derangement sum conjecture

摘录 / EXCERPT: Let $n>1$ be an integer, let ζ be a primitive n-th root of unity, and define the derangement set ... For a permutation τ, let $\operatorname{sign}(\tau)$ denote its sign...

conjecture | open | MathDB 页面 | 原文线索

V2-043890 | MATHDB-351661

Equality of weak and ordinary Boolean Ramsey numbers for balanced V-shaped posets

摘录 / EXCERPT: Let $\boldsymbol{P}_i=\mathbf{V}_{m_i,m_i}$ for $1\leq i\leq k$, where $\mathbf{V}_{m_i,m_i}$ denotes the balanced V-shaped poset with m_i elements on each branch. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-043891 | MATHDB-351723

The asymptotic density conjecture for (p,q)-sum-free subsets of the square grid

摘录 / EXCERPT: Asymptotic density conjecture. For positive integers p,q and sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-043892 | MATHDB-351732

Extremal beta-polytope intrinsic-volume limit law conjecture

摘录 / EXCERPT: Let $U_1,\ldots,U_N$ be independent random vectors in $\mathbb{R}^d$ with beta density ... where $\beta>-1$, $\mathbb B^d=\{x\in\mathbb{R}^d:\|x\|\leq 1\}$, and v_m denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-043893 | MATHDB-351775

Conjecture on the row sum polynomials of the production matrix of an almost Riordan array

摘录 / EXCERPT: Row sum polynomial conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-043894 | MATHDB-351797

The neighborhood-island conjecture for planar point sets

摘录 / EXCERPT: Neighborhood-island conjecture. The neighborhood of some point $p\in P$ contains an island of size $f(n)$, where $f(n)$ tends to infinity as n tends to infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-043895 | MATHDB-351806

Existence of arbitrarily long extremal square-free permutations

摘录 / EXCERPT: An extremal square-free permutation is a square-free permutation for which inserting any letter in any position introduces a square. The extremal-permutation conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-043896 | MATHDB-351807

Nonexistence of extremal square-free words over a four-letter alphabet

摘录 / EXCERPT: An extremal square-free word is a square-free word such that inserting any letter in any position introduces a square. The nonexistence conjecture. There are no extremal square-fre...

conjecture | open | MathDB 页面 | 原文线索

V2-043897 | MATHDB-351904

Erdős's convexity conjecture

摘录 / EXCERPT: Let $I_N=\{1,\dots,N\}$, let $\epsilon>0$, and let $f:\mathbb{R}\to\mathbb{R}$ be a strictly convex function. For a set I, write $f(I)=\{f(i)\mid i\in I\}$. Erdős's conjecture. ……

conjecture | open | MathDB 页面 | 原文线索

V2-043898 | MATHDB-351924

The asymmetric Cauchy-Schwarz complexity conjecture

摘录 / EXCERPT: Let $\Phi=(\phi_1,\dots,\phi_k)$ be a system of linear forms, with $\phi_i\colon\mathbb{F}_p^d\to\mathbb{F}_p$, and let $i_0\in[k]$. Let $s(\Phi,i_0)$ be the asymmetric true comple...

conjecture | open | MathDB 页面 | 原文线索

V2-043899 | MATHDB-351925

The bold conjecture on linear-data entailment

摘录 / EXCERPT: Let Φ and Ψ be linear data, and let $\gamma\colon I^\Psi\to I^\Phi\times\{0,1\}$ be a partial function. Assume that whenever functions...

conjecture | open | MathDB 页面 | 原文线索

V2-043900 | MATHDB-351926

The slightly less bold conjecture on elementary entailment

摘录 / EXCERPT: Let Φ and Ψ be linear data, and let γ be a partial matching map as in the paper. Write $\Phi\vdash'^k_\gamma\Psi$ for entailment using the additional elementary el...

conjecture | open | MathDB 页面 | 原文线索

V2-043901 | MATHDB-351927

Grünbaum's catalogue conjecture for simplicial line arrangements

摘录 / EXCERPT: A simplicial line arrangement is a finite arrangement of lines in $\mathbb{P}^{2}_{\mathbb{R}}$ whose complement has only open triangular connected components. Grünbaum's catalogue...

conjecture | open | MathDB 页面 | 原文线索

V2-043902 | MATHDB-351928

Cuntz-Geis bound for sporadic simplicial line arrangements

摘录 / EXCERPT: Let $\mathcal{L}$ be a sporadic simplicial line arrangement in $\mathbb{P}^{2}_{\mathbb{R}}$, meaning a simplicial line arrangement outside the known infinite families, and let d...

conjecture | open | MathDB 页面 | 原文线索

V2-043903 | MATHDB-351964

Polynomial patterns from squared fractional powers in dense sets

摘录 / EXCERPT: Multiple recurrence conjecture. The set A contains a pattern of the form

open_problem | open | MathDB 页面 | 原文线索

V2-043904 | MATHDB-351985

Configuration-count conjecture for merges of vincular patterns

摘录 / EXCERPT: Configuration-count conjecture. For every vincular pattern σ with j blocks and every $1\leq l\leq M_{\underline{\sigma}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-043905 | MATHDB-351991

Samsonov-Shur's conjecture on the Abelian repetition threshold

摘录 / EXCERPT: Samsonov-Shur's conjecture. The exact values are

conjecture | open | MathDB 页面 | 原文线索

V2-043906 | MATHDB-351992

Petrova-Shur's conjecture on the Abelian repetition threshold

摘录 / EXCERPT: Petrova-Shur's conjecture. The Abelian repetition thresholds satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-043907 | MATHDB-352002

Affine fully commutative element enumeration conjecture for $G(m,m,n)$

摘录 / EXCERPT: Let a_m denote the number of fully commutative elements in $G(m,m,n)$, for $n\geq 3$, with the affine generating set. Affine enumeration conjecture. There is a positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-043908 | MATHDB-352003

Constant-length enumeration conjecture for fully commutative elements in $G(m,m,3)$

摘录 / EXCERPT: Consider the complex reflection group $G(m,m,3)$ with its affine generating set, and let ℓ denote the length of an element. Constant-length enumeration conjecture. When...

conjecture | open | MathDB 页面 | 原文线索

V2-043909 | MATHDB-352004

Symmetry conjecture for fully commutative elements in rank-two Shephard groups

摘录 / EXCERPT: Let a and b be positive integers. In the notation $a[b]a$ for the rank-two Shephard group with endpoint parameters a,a and braid parameter b, and $b[a]b$ for the group with...

conjecture | open | MathDB 页面 | 原文线索

V2-043910 | MATHDB-352005

Supersaturation conjecture for even-town families

摘录 / EXCERPT: Supersaturation conjecture. Let $n\geq 1$ and fix $3\leq s\leq 2^{\lfloor n/2\rfloor}-2^{\lfloor n/4\rfloor}$. If $[?]\subset 2^{[n]}$ consists of even-sized subsets with...

conjecture | open | MathDB 页面 | 原文线索

V2-043911 | MATHDB-352006

Supersaturation conjecture for odd-town families

摘录 / EXCERPT: Odd-town supersaturation conjecture. Let $n\geq 1$ and fix $1\leq s\leq n$. If $[?]$ is a collection of odd-sized subsets of an n-element set with $|[?]|\geq n+s$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043912 | MATHDB-352017

Monical-Tokcan-Yong's saturated Newton polytope conjecture for double Schubert polynomials

摘录 / EXCERPT: Saturated Newton polytope conjecture. Every double Schubert polynomial ${\mathfrak{S}}_\pi({\normalfont\mathbf{t}}, {\normalfont\mathbf{s}})$ has the Saturated Newton Polytope prop...

conjecture | open | MathDB 页面 | 原文线索

V2-043913 | MATHDB-352018

The extremal sequence conjecture for short zero-sums in $C_n\oplus C_n$

摘录 / EXCERPT: Let $n\geq 2$, let $G=C_n\oplus C_n$, let $k\in [0,n-1]$, and let S be a sequence of terms from G such that ... Here, $\Sigma_{\leq r}(S)$ denotes the set of sums of nonempty sub...

conjecture | open | MathDB 页面 | 原文线索

V2-043914 | MATHDB-352019

The short zero-sum sequence structure conjecture for rank-two abelian groups

摘录 / EXCERPT: Short zero-sum sequence structure conjecture. There exists either a basis (e_1,e_2) for G with $\operatorname{ord}(e_2)=mn$ or a generating set $\{g_1,g_2\}$ for G with...

conjecture | open | MathDB 页面 | 原文线索

V2-043915 | MATHDB-352043

Luschny's permanent conjecture for median Genocchi numbers

摘录 / EXCERPT: Let n be a positive integer and define the $2n\times 2n$ matrix ... The Luschny permanent conjecture. ... Here $H_1,H_3,H_5,\ldots$ are the Genocchi numbers of the second kind, or me...

conjecture | open | MathDB 页面 | 原文线索

V2-043916 | MATHDB-352062

Typical binary words with even weight are shuffle squares

摘录 / EXCERPT: Let $n\ge 1$, and let s be chosen uniformly from the binary words $\{0,1\}^{2n}$ having an even number of ones. A typical-word shuffle-square conjecture. With high probability as...

conjecture | open | MathDB 页面 | 原文线索

V2-043917 | MATHDB-352073

The pure twin group right-angled Artin group conjecture

摘录 / EXCERPT: Pure twin group conjecture. PT_n is a right-angled Artin group for each $n\geq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-043918 | MATHDB-352083

Dual equivalence-relation conjecture for q-matroids

摘录 / EXCERPT: Dual equivalence-relation conjecture. The relation of Definition is an equivalence relation in at least one of M and M^*.

conjecture | open | MathDB 页面 | 原文线索

V2-043919 | MATHDB-352084

Circuit-cocircuit intersection conjecture for q-matroids

摘录 / EXCERPT: Circuit-cocircuit intersection conjecture. Under these hypotheses, Definition is an equivalence relation.

conjecture | open | MathDB 页面 | 原文线索

V2-043920 | MATHDB-352093

Existence of an optimal data placement for arbitrary parameters

摘录 / EXCERPT: Let V be a set of v servers, and let a data placement be a family of b blocks, each of size k, on V. For a placement $(V,\mathcal{F})$, let ... where v_j is the number of...

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V2-043921 | MATHDB-352123

The Bruhat-interval reconstruction conjecture for Weyl groups of type A

摘录 / EXCERPT: Let A_n be the Weyl group of type A_n. For $u,v,u',v'\in A_n$ with $u<v$ and $u'<v'$, let ... and ... Here $[u,v]$ denotes the Bruhat interval and $\cong$ denotes isomorphism of Br...

conjecture | open | MathDB 页面 | 原文线索

V2-043922 | MATHDB-352139

Set-wise intersection conjecture for perfect matchings

摘录 / EXCERPT: Set-wise intersection conjecture. For $k\geq 2t$, the largest set-wise t-intersecting family of perfect matchings in K_{2k} has size

conjecture | open | MathDB 页面 | 原文线索

V2-043923 | MATHDB-352149

The discretised ABC sum-product conjecture

摘录 / EXCERPT: Let $alpha,beta,gamma in (0,1)$ with $beta leq alpha$. Let $A,B,C subset [0,1]$ be $delta$-separated sets with $|A| leq delta^{-alpha}$, $|B|=delta^{-beta}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-043924 | MATHDB-352166

Brenti-Carnevale-Tenner's rank-symmetry conjecture for odd diagram classes

摘录 / EXCERPT: Brenti-Carnevale-Tenner's conjecture. Each odd diagram class is rank-symmetric in the Bruhat order.

conjecture | open | MathDB 页面 | 原文线索

V2-043925 | MATHDB-352179

Conjecture on sequencing sharply transitive group actions

摘录 / EXCERPT: Let G act sharply k-transitively on an n-set X, with $k\geq 2$. The action is sequenced if there is an enumeration of X such that the orbit representatives of its consecu...

conjecture | open | MathDB 页面 | 原文线索

V2-043926 | MATHDB-352181

Gir\~ao, Popielarz, and Snyder's oriented Ramsey conjecture for 1-subdivisions

摘录 / EXCERPT: Let T_k be the transitive tournament on k vertices, and let H_k be its 1-subdivision, obtained by subdividing every arc exactly once. For an oriented graph H, let...

conjecture | open | MathDB 页面 | 原文线索

V2-043927 | MATHDB-352200

Rademacher's conjecture on the extremal lattice for Landau's constant

摘录 / EXCERPT: Let $\Gamma \subset \mathbb{C}$ be discrete and $f_\Gamma \in \mathcal{U}$, where $\mathcal{U}$ is the class of universal covering maps of $\mathbb{D}$ onto...

conjecture | open | MathDB 页面 | 原文线索

V2-043928 | MATHDB-352205

Milićević's cubic-curve conjecture for point-line configurations

摘录 / EXCERPT: Milićević's conjecture. Then $P \cup R$ is contained in a cubic curve.

conjecture | open | MathDB 页面 | 原文线索

V2-043929 | MATHDB-352206

Karasev-Polyanskii's cubic-curve conjecture for three point sets

摘录 / EXCERPT: Karasev-Polyanskii's conjecture. Then $B \cup G \cup R$ lies on a cubic curve.

conjecture | open | MathDB 页面 | 原文线索

V2-043930 | MATHDB-352213

The generalized Borsuk covering-number conjecture

摘录 / EXCERPT: Generalized Borsuk covering-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043931 | MATHDB-352214

The cyclic-group Borsuk covering conjecture

摘录 / EXCERPT: Cyclic-group Borsuk covering conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043932 | MATHDB-352215

The strict-growth conjecture for equivariant covering numbers

摘录 / EXCERPT: Strict-growth conjecture. For all $d\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-043933 | MATHDB-352274

Seidel triangle conjecture for web permutations

摘录 / EXCERPT: Let $f(n,k)$ be the number of web permutations σ of $[n]$ such that $M(\sigma)=M^{(n)}_0$ and $\sigma_1=k$, where $M^{(n)}_0=\{\{1,2\},\dots,\{2n-1,2n\}\}$. Let $s_{i,j}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-043934 | MATHDB-352284

Stanchescu's difference-set lower-bound conjecture in Euclidean space

摘录 / EXCERPT: Let $d \geq 2$ and let $A \subset \mathbb{R}^d$ be a finite set with affine dimension $\dim(A)=d$. Stanchescu's conjecture. Then ... This conjecture proposes that Stanchescu's constr...

conjecture | open | MathDB 页面 | 原文线索

V2-043935 | MATHDB-352319

The diffsequence growth conjecture for powers of two

摘录 / EXCERPT: Let $D=\{2^i\mid i\in\mathbb{Z}_{\geq 0}\}$, and let $\Delta(D,k)$ denote the diffsequence parameter studied in the paper for the set D and positive integer k. Diffsequence gro...

conjecture | open | MathDB 页面 | 原文线索

V2-043936 | MATHDB-352341

Symmetry conjecture for optimal even small polygons

摘录 / EXCERPT: Let $n\geq 6$ be even, and let A_n^* denote the maximal area among all small n-gons. An optimal n-gon is a small n-gon whose area is A_n^*. Its diameter graph is the grap...

conjecture | open | MathDB 页面 | 原文线索

V2-043937 | MATHDB-352360

Square-root upper bound conjecture for the length of UPS-factorizations

摘录 / EXCERPT: Let A be a finite alphabet. For a finite nonempty rich word w over A, write its UPS-factorization as $w=w_pw_{p-1}\cdots w_1$, where the factors...

conjecture | open | MathDB 页面 | 原文线索

V2-043938 | MATHDB-352363

Reid-Wu-Zhou's cocircuit conjecture for minimally connected matroids

摘录 / EXCERPT: Reid-Wu-Zhou's cocircuit conjecture. If M is a minimally k-connected matroid with $|E(M)| \ge 2(k-1)$, then M has a cocircuit of size k.

conjecture | open | MathDB 页面 | 原文线索

V2-043939 | MATHDB-352364

The strong negative vertical-connectivity conjecture for minimally vertically 4-connected matroids

摘录 / EXCERPT: Strong negative vertical-connectivity conjecture. There is an infinite family of minimally vertically 4-connected matroids with no 4-cocircuits.

conjecture | open | MathDB 页面 | 原文线索

V2-043940 | MATHDB-352365

Element-specific cocircuit conjecture for minimally vertically 4-connected matroids

摘录 / EXCERPT: Element-specific cocircuit conjecture. At least one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-043941 | MATHDB-352366

Binary cocircuit conjecture for minimally vertically k-connected matroids

摘录 / EXCERPT: Binary cocircuit conjecture. For each integer $k\geq 5$, every minimally vertically k-connected binary matroid with at least $2k+2$ elements has a k-cocircuit.

conjecture | open | MathDB 页面 | 原文线索

V2-043942 | MATHDB-352367

Infinite-family conjecture for minimally vertically k-connected matroids

摘录 / EXCERPT: Infinite-family conjecture. For each integer $k\geq 4$, there is an infinite family of minimally vertically k-connected matroids with no k-cocircuits.

conjecture | open | MathDB 页面 | 原文线索

V2-043943 | MATHDB-352396

Hickman-Wright's Kakeya set conjecture over finite cyclic rings

摘录 / EXCERPT: Let $N,n\in\mathbb{N}$, and let $S\subseteq (\mathbb{Z}/N\mathbb{Z})^n$ be a Kakeya set, meaning that S contains a line in every direction in...

conjecture | open | MathDB 页面 | 原文线索

V2-043944 | MATHDB-352404

Sharp discrepancy conjecture for arithmetic progressions in grids

摘录 / EXCERPT: Let $d\geq 1$ be an integer, and let $\mathbf{N}=(N_1,N_2,\cdots,N_d)$, where $N_1,\dots,N_d$ are positive integers. Write $\mathcal{A}_{\mathbf{N}}$ for the family of arithmetic p...

conjecture | open | MathDB 页面 | 原文线索

V2-043945 | MATHDB-352419

The quadric-configuration conjecture for double $2d$ configurations

摘录 / EXCERPT: Quadric-configuration conjecture. Every double $2d$ on X consists of $4d$ lines that are the union of two quadric configurations on X.

conjecture | open | MathDB 页面 | 原文线索

V2-043946 | MATHDB-352441

Open-coloring reduction to clopen or open dense colorings

摘录 / EXCERPT: Open-coloring reduction conjecture. Every open coloring can be reduced to a clopen or to an open dense coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-043947 | MATHDB-352450

Shin-Zeng's enumeration conjecture for signed Simsun permutations

摘录 / EXCERPT: Shin-Zeng's conjecture. For $1\le k\le n$,

conjecture | open | MathDB 页面 | 原文线索

V2-043948 | MATHDB-352460

Strict increase of maximal embedding dimensions with base set size

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and for a code $\mathcal C\subseteq 2^{[n]}$ write $\operatorname{odim}(\mathcal C)$, $\operatorname{cdim}(\mathcal C)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-043949 | MATHDB-352461

Existence of codes with closed embedding dimension exceeding base set size

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $\operatorname{cdim}(\mathcal C)$ denote the closed embedding dimension of a code $\mathcal C\subseteq 2^{[n]}$. Closed-dimension existence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-043950 | MATHDB-352484

Red's one-tile advantage conjecture for Infinite Hex

摘录 / EXCERPT: Red's one-tile advantage conjecture. A game of Infinite Hex with exactly one tile of the board assigned to Red at the initial position, and none to Blue, is a win for Red.

conjecture | open | MathDB 页面 | 原文线索

V2-043951 | MATHDB-352485

Horizontal initial-position conjecture for Infinite Hex

摘录 / EXCERPT: Horizontal initial-position conjecture. A game of Infinite Hex with initial position as in figure is a win for Red.

conjecture | open | MathDB 页面 | 原文线索

V2-043952 | MATHDB-352489

NIP2 implies almost strongly quadratically atomicity

摘录 / EXCERPT: Let an elementary p-group property (epGP) be a property of elementary abelian p-groups, and let NIP_2 denote the stated second-order independence-property condition.…

conjecture | open | MathDB 页面 | 原文线索

V2-043953 | MATHDB-352490

NFOP2 properties are quadratic-atomic or contain a generalized Green-Sanders example

摘录 / EXCERPT: Let $\mathcal{P}$ be an elementary p-group property, and let NFOP_2 mean that it has no functional order property of order two. A property is quadratically atomic when...

conjecture | open | MathDB 页面 | 原文线索

V2-043954 | MATHDB-352493

Bounded VC2-dimension of quadratic Green-Sanders examples

摘录 / EXCERPT: For primes p and positive integers n, let $\mathrm{QGS}(n,p)$ be the quadratic Green-Sanders example, and let VC_2-dimension denote the paper's second-order VC-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-043955 | MATHDB-352496

The Polynexus formula for nonzero products in the rook monoid

摘录 / EXCERPT: Polynexus conjecture. For $n\in\mathbb{N}$, $\psi(n)$ equals the $(n+1)$th Polynexus number of order 7, namely

conjecture | open | MathDB 页面 | 原文线索

V2-043956 | MATHDB-352498

Multicolor Burr-Erdős-Faudree-Rousseau-Schelp conjecture for star forests

摘录 / EXCERPT: Multicolor star-forest conjecture. The multicolor size Ramsey number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-043957 | MATHDB-352539

Monotonicity and 3-power denominators of the extremal complexity sequences

摘录 / EXCERPT: Let $(a_\alpha)_{\alpha<\xi}$, $(b_\alpha)_{\alpha<\xi}$, and $(c_\alpha)_{\alpha<\xi}$ be the transfinite sequences of rational numbers from the preceding conjecture. Denominator...

conjecture | open | MathDB 页面 | 原文线索

V2-043958 | MATHDB-352545

Higher-convexity growth conjecture for difference sets

摘录 / EXCERPT: Let A be a set of reals and let f be a k-convex function. For a finite set B and positive integer r, write rB for the r-fold sumset of B and $rB-sB$ for the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-043959 | MATHDB-352567

Rainbow Erdős matching conjecture

摘录 / EXCERPT: Let $[n]=\{1,2,\dots,n\}$ and let $\binom{[n]}{k}$ denote the collection of all k-subsets of $[n]$. Let $\mathcal{F}_1,\mathcal{F}_2,\dots,\mathcal{F}_s\subseteq\binom{[n]}{k}$.…

conjecture | open | MathDB 页面 | 原文线索

V2-043960 | MATHDB-352570

Redundancy conjecture for simple odd beta-cycle inequalities

摘录 / EXCERPT: Redundancy conjecture. The simple odd β-cycle inequality corresponding to (C,σ) is redundant for $\Pcr(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-043961 | MATHDB-352590

The multicolor book graph size Ramsey number conjecture

摘录 / EXCERPT: Let $B_n^{(k)}$ denote the book graph consisting of k triangles sharing a common edge, with the shared edge contained in k pages, each having n vertices in the relevant book...

conjecture | open | MathDB 页面 | 原文线索

V2-043962 | MATHDB-352596

Strict coefficient bounds for matroid flag Hilbert-Poincaré numerators

摘录 / EXCERPT: Let M be a matroid of rank $r\geqslant 3$, let $\mathcal{N}_M(1,T)$ denote its flag Hilbert-Poincaré numerator specialized at $Y=1$, let $\pi_M(1)$ be its characteristic polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-043963 | MATHDB-352605

Compositional equivalences for pairs of length-four patterns

摘录 / EXCERPT: Let $C_n(\tau,\tau')$ be the set of all compositions of n that avoid the patterns τ and τ', and let $c_n(\tau,\tau')=|C_n(\tau,\tau')|$. Compositional equivalence conje...

conjecture | open | MathDB 页面 | 原文线索

V2-043964 | MATHDB-352620

Strong Komlós conjecture for prefix discrepancy in the Euclidean ball

摘录 / EXCERPT: Let T be a positive integer, let $v_1,\ldots,v_T\in\mathbb{B}_2^d$ be adversarial input vectors, and let $[T]=\{1,\ldots,T\}$. A signing is a vector $x\in\{\pm1\}^T$. Strong Koml...

conjecture | open | MathDB 页面 | 原文线索

V2-043965 | MATHDB-352628

3-dimensional Dinitz Conjecture

摘录 / EXCERPT: A partial Latin cube of order n is an $n\times n\times n$ array of symbols in which no symbol appears more than once in any slice. For every triple (i,j,k) with...

conjecture | open | MathDB 页面 | 原文线索

V2-043966 | MATHDB-352629

3-dimensional Rota Conjecture

摘录 / EXCERPT: Let V be a vector space over an arbitrary infinite field with $\dim V=n^2$. Suppose $B_1,\ldots,B_n$ are n sets of bases of V. For each i, choose a linear ordering of B_i...

conjecture | open | MathDB 页面 | 原文线索

V2-043967 | MATHDB-352634

Seidel-Smith conjecture on symplectic Khovanov cohomology

摘录 / EXCERPT: Let $L\subset S^3$ be a link. The singly graded symplectic Khovanov cohomology is denoted by $Kh^k_{symp}(L)$, and the bigraded combinatorial Khovanov homology by $Kh^{i,j}(L)$. Se...

conjecture | solved | MathDB 页面 | 原文线索

V2-043968 | MATHDB-352642

Pop polynomial formulas for weak, Tamari, Cambrian and root-poset lattices

摘录 / EXCERPT: Let W be a finite Coxeter group. Write $\mathrm{Weak}(W)$ for its right weak order, $\mathrm{Tamari}(W)$ for a Cambrian lattice associated to a Coxeter element obtained from a li...

conjecture | open | MathDB 页面 | 原文线索

V2-043969 | MATHDB-352652

Speyer's interior-face conjecture for matroid subdivisions

摘录 / EXCERPT: Let M be a matroid on $[n]$ of rank d, and let a matroid subdivision of M be a polyhedral subdivision of its matroid polytope in which every cell is itself a matroid polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-043970 | MATHDB-352683

The exact piercing-number conjecture for the chessboard

摘录 / EXCERPT: Let Q_n be the $n\times n$ chessboard, and let p_n denote the minimum number of lines whose union intersects every cell of Q_n. Piercing-number conjecture. For all $n \geq 3$...

conjecture | open | MathDB 页面 | 原文线索

V2-043971 | MATHDB-352710

Mayhew-Newman-Welsh-Whittle conjecture on sparse paving matroid minors

摘录 / EXCERPT: A matroid is sparse paving if its nonspanning circuits are hyperplanes; equivalently, no nonbasis of size equal to the rank can be transformed into another nonbasis by exchanging o...

conjecture | open | MathDB 页面 | 原文线索

V2-043972 | MATHDB-352732

Bérczi-Schwarcz-Yamaguchi coverability conjecture for matroids

摘录 / EXCERPT: Let ${\cal M}=(E,\mathcal{I})$ be a matroid. It is k-coverable if its ground set E can be covered by at most k independent sets from $\mathcal{I}$. A partition matroid on the...

conjecture | open | MathDB 页面 | 原文线索

V2-043973 | MATHDB-352805

Asymptotic enumeration conjecture for integer tetrahedra by perimeter

摘录 / EXCERPT: Let $\tn$ denote the number of integer tetrahedra of perimeter n, counted up to congruence. Asymptotic enumeration conjecture. The number $\tn$ satisfies ... This conjecture is mot...

conjecture | open | MathDB 页面 | 原文线索

V2-043974 | MATHDB-352806

Asymptotic enumeration conjecture for labelled tetrahedron graphs

摘录 / EXCERPT: Let T_n be the set of graphs corresponding to integer tetrahedra of perimeter n. Graph-count asymptotic conjecture. The cardinality of T_n satisfies ... This is presented as an...

conjecture | open | MathDB 页面 | 原文线索

V2-043975 | MATHDB-352807

Stable column sequence conjecture for integer tetrahedra

摘录 / EXCERPT: Let ${}^dt_m$ denote the relevant column-sequence count for integer tetrahedra, and let a_k be defined by ... where $b_0=96$ and, for $k\geq1$, b_k is given by the residue-class...

conjecture | open | MathDB 页面 | 原文线索

V2-043976 | MATHDB-352816

Asymptotic dimension conjecture for diagonal coinvariant rings of monomial groups

摘录 / EXCERPT: Let $\bell$ and m be positive integers with $\bell \ygeq 2$ and m dividing $\bell$, and let $G(\bell,m,n)$ denote the corresponding monomial group. Let $R_{G(\bell,m,n)}$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-043977 | MATHDB-352837

New conjecture on balanced-sequence critical exponents

摘录 / EXCERPT: New conjecture. For every alphabet size $d\geq 11$, the minimal critical exponent of a balanced sequence over a d-letter alphabet equals

conjecture | open | MathDB 页面 | 原文线索

V2-043978 | MATHDB-352844

Weiner's enumeration conjecture for pattern-avoiding Grassmannian permutations

摘录 / EXCERPT: Let $\mathscr{G}_m(12\cdots k)$ denote the set of Grassmannian permutations of size m that avoid the increasing pattern $12\cdots k$, and let $C_r=\frac{1}{r+1}\binom{2r}{r}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-043979 | MATHDB-352854

Lin's symmetry conjecture for ascents of restricted inversion sequences

摘录 / EXCERPT: For $n\geq 1$, let $\mathbf{I}_n(\geq,\neq,>)$ and $\mathbf{I}_n(>,\neq,\geq)$ be the two classes of inversion sequences avoiding the indicated relation triples. For an inversion s...

conjecture | open | MathDB 页面 | 原文线索

V2-043980 | MATHDB-352885

Periodicity conjecture for the k-binomial complexity of generalized Thue-Morse words

摘录 / EXCERPT: Periodicity conjecture. For every $k\geq 3$, the function $b_{\mathbf{t}_{m},k}(n)$ is ultimately periodic with period m^{k}.

conjecture | open | MathDB 页面 | 原文线索

V2-043981 | MATHDB-352896

The lower-bound conjecture for coupon collecting with friends

摘录 / EXCERPT: Lower-bound conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-043982 | MATHDB-352898

Conjecture on moduli spaces of extremal Chern character sheaves on projective three-space

摘录 / EXCERPT: Let $(r,c,d)\in\mathbb{Z}\oplus\mathbb{Z}\oplus\tfrac{1}{2}\mathbb{Z}$, and suppose that there are Gieseker-semistable sheaves E on $\mathbb{P}^3$ with ... Write $E(r,c,d)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-043983 | MATHDB-352900

Tuza's conjecture for Fano-plane-free binary matroids

摘录 / EXCERPT: Let M be a simple binary matroid. A triangle packing is a set of pairwise disjoint triangles of M, and a triangle hitting set is a set of elements meeting every triangle of M...

conjecture | open | MathDB 页面 | 原文线索

V2-043984 | MATHDB-352927

Refinement of the biased r-wise intersection threshold conjecture

摘录 / EXCERPT: Refinement of the biased r-wise intersection threshold conjecture. The condition

conjecture | open | MathDB 页面 | 原文线索

V2-043985 | MATHDB-352929

Three-fold cross-intersection product-measure conjecture

摘录 / EXCERPT: Three-fold cross-intersection product-measure conjecture. If $\mathcal A_1,\mathcal A_2,\mathcal A_3$ are 3-cross intersecting, then

open_problem | open | MathDB 页面 | 原文线索

V2-043986 | MATHDB-352941

The discretized ring conjecture of Katz and Tao

摘录 / EXCERPT: Katz-Tao's discretized ring conjecture. There exists an absolute constant c_1 such that, if $\varepsilon>0$ is sufficiently small and C is sufficiently large, then for all suff...

conjecture | open | MathDB 页面 | 原文线索

V2-043987 | MATHDB-352944

The strong stability conjecture for 3-wise t-intersecting families

摘录 / EXCERPT: Let $n\geq t\geq2$, let $[n]=\{1,2,\ldots,n\}$, and let $0<p<1$ with $q=1-p$ and $p_0=2/(\sqrt{4t+9}-1)$. For a family $\mathcal G\subset2^{[n]}$, define ... The family is 3-wise t...

conjecture | open | MathDB 页面 | 原文线索

V2-043988 | MATHDB-352964

The annular spherical independent-set conjecture

摘录 / EXCERPT: Let $\mathbb S^{d-1}$ be the unit sphere with its spherical distance, and let $\operatorname{Cap}^{d-1}(r)$ denote a spherical cap of angular radius r. Fix $\theta\in[0,\pi/2)$....

conjecture | open | MathDB 页面 | 原文线索

V2-043989 | MATHDB-352977

Lieb-Solovej generalized contractive inequality for weighted Bergman spaces

摘录 / EXCERPT: Lieb-Solovej conjecture. If $p>2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-043990 | MATHDB-352996

The free-subgroup conjecture for non-orientable cyclic presentations

摘录 / EXCERPT: Let $P_2(w)$ be a non-orientable cyclic presentation and let $G_2(w)$ be its associated group. Free-subgroup conjecture. Either $G_2(w)$ contains a non-abelian free subgroup, or...

conjecture | open | MathDB 页面 | 原文线索

V2-043991 | MATHDB-353049

Recursive 2-extension conjecture for extremal square-rich point sets

摘录 / EXCERPT: Let $S_{\square}(n)$ be the maximum number of squares determined by an n-point set in the plane, and let $\mathcal{P}_{4,1}^1$ denote the starting configuration consisting of a u...

conjecture | open | MathDB 页面 | 原文线索

V2-043992 | MATHDB-353050

Exact square count for the 13-point extension of the $(6,2)$ configuration

摘录 / EXCERPT: Let $\mathcal{P}_{6,2}^\star$ be the specified six-point, two-square configuration, and let $S_{\square}(n,\mathcal{P}_{6,2}^\star)$ denote the maximum number of squares in an n-...

conjecture | open | MathDB 页面 | 原文线索

V2-043993 | MATHDB-353051

Strict upper bound for the 16-point $(6,2)$ square configuration

摘录 / EXCERPT: Let $\mathcal{P}_{6,2}^\star$ be the specified six-point, two-square configuration, let $S_{\square}(n,\mathcal{P}_{6,2}^\star)$ denote the maximum number of squares in an n-poin...

conjecture | open | MathDB 页面 | 原文线索

V2-043994 | MATHDB-353076

The cover enumerator conjecture for type-B parabolic Tamari lattices

摘录 / EXCERPT: Cover enumerator conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-043995 | MATHDB-353079

Eventual strict increase conjectures for sixfold Thue-Morse representations

摘录 / EXCERPT: Sixfold strict increase conjectures. The sequences $r_6(n)$ and $s_6(n)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-043996 | MATHDB-353094

The inverse theorem for the U^{s+1} Gowers norm on finite abelian groups

摘录 / EXCERPT: Let G be a finite additive group, let $ta > 0$, let $s eq 1$, and let $f \colon G \to \mathbb{C}$ be a 1-bounded function with $\|f\|_{U^{s+1}(G)} \geq \eta$. A degree s filt...

conjecture | open | MathDB 页面 | 原文线索

V2-043997 | MATHDB-353099

Monotonicity conjecture for maximum inner distance Latin squares

摘录 / EXCERPT: Monotonicity conjecture. The numbers of Latin squares with each possible inner distance are nonincreasing:

conjecture | open | MathDB 页面 | 原文线索

V2-043998 | MATHDB-353112

Framed homeomorphism implies diffeomorphism

摘录 / EXCERPT: Let M and M' be smooth compact manifolds smoothly embedded in euclidean space, and let the embeddings define flat framed stratifications. Framed homeomorphism implies diffeomor...

conjecture | open | MathDB 页面 | 原文线索

V2-043999 | MATHDB-353113

Framed stratifications are dense in smooth embeddings

摘录 / EXCERPT: Let M be a smooth compact manifold smoothly embedded in euclidean space. Framed stratifications are dense in smooth embeddings. Any smooth embedding of M into euclidean space h...

conjecture | open | MathDB 页面 | 原文线索

V2-044000 | MATHDB-353119

Tidor's conjecture on approximately symmetric multilinear forms

摘录 / EXCERPT: Let G be a vector space over $\mathbb{F}_p$, and let $alpha\colon G^k\tomathbb{F}_p$ be an r-approximately symmetric multilinear form, meaning that the partition rank of...

conjecture | open | MathDB 页面 | 原文线索

V2-044001 | MATHDB-353123

The cyclic matroid wheel, whirl, spike and swirl conjecture

摘录 / EXCERPT: Cyclic matroid structure conjecture. If s is even, then M can be obtained from a spike or a swirl by a sequence of inflations. If s is odd, then M can be obtained from a wh...

conjecture | open | MathDB 页面 | 原文线索

V2-044002 | MATHDB-353152

Steady-state ratio conjecture for the doubly asymmetric simple exclusion process

摘录 / EXCERPT: Steady-state ratio conjecture. The following statements are equivalent: $t=1$; and, for all such partitions λ and all permutations $\mu,\nu\in S_n(\lambda)$,

conjecture | open | MathDB 页面 | 原文线索

V2-044003 | MATHDB-353153

Elekes-Ruzsa conjecture on product sets of small-doubling sets

摘录 / EXCERPT: Elekes-Ruzsa conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044004 | MATHDB-353171

The generalized projected hypercube conjecture for difference sets

摘录 / EXCERPT: Let b be a positive integer, and let $g(n,k,\ell)$ denote the relevant extremal function for difference sets. For the recursively constructed b-ary projected-hypercube sets, th...

conjecture | open | MathDB 页面 | 原文线索

V2-044005 | MATHDB-353173

The quadratic diameter conjecture for friends-and-strangers graphs on cycles

摘录 / EXCERPT: Let Y be a graph on n vertices, and let $\mathsf{FS}(\textsf{Cycle}_n,Y)$ denote its friends-and-strangers graph. The quadratic diameter conjecture asserts that the maximum dia...

conjecture | open | MathDB 页面 | 原文线索

V2-044006 | MATHDB-353177

Baril-Kirgizov-Vajnovszki conjecture on 1-Gray codes for restricted binary words

摘录 / EXCERPT: Let $\mathcal{W}_{q,n}$ denote the set of restricted binary words of length n associated with the positive rational parameter q. A 1-Gray code is an arrangement of all words in...

conjecture | open | MathDB 页面 | 原文线索

V2-044007 | MATHDB-353178

Existence conjecture for growth rates of real-parameter restricted words

摘录 / EXCERPT: For each positive real number r, let $\mathcal{W}_{r,n}$ be the set of restricted binary words of length n with parameter r. Existence conjecture. For given...

conjecture | open | MathDB 页面 | 原文线索

V2-044008 | MATHDB-353179

Monotonicity and discontinuity conjecture for restricted-word growth rates

摘录 / EXCERPT: Assuming the growth-rate limit exists, define ... for $r \in \mathbb{R}^+$. Monotonicity and discontinuity conjecture. The function $r \mapsto L(r)$ is increasing on $[0,+\infty)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-044009 | MATHDB-353210

The characteristic-polynomial and eigenvalue conjecture for unit-primitive matrices

摘录 / EXCERPT: Unit-primitive matrix conjecture. The characteristic polynomial of A_n is

conjecture | open | MathDB 页面 | 原文线索

V2-044010 | MATHDB-353211

The numerator-degree conjecture for row generating functions of $T(n,m)$

摘录 / EXCERPT: Numerator-degree conjecture. For $n\geq 2$, $H_n(x)$ has degree $n-2$ and is given by

conjecture | open | MathDB 页面 | 原文线索

V2-044011 | MATHDB-353212

The symmetry conjecture for the array M

摘录 / EXCERPT: Symmetry conjecture for M. For all nonnegative integers n and k,

conjecture | open | MathDB 页面 | 原文线索

V2-044012 | MATHDB-353213

The rational generating-function conjecture for rows and columns of M

摘录 / EXCERPT: Rational generating-function conjecture for M. Assuming the numerator conjecture for $H_n(x)$, the generating function for row (or column) n of M has the form

conjecture | open | MathDB 页面 | 原文线索

V2-044013 | MATHDB-353214

The column-generating-function conjecture for the first five columns of M

摘录 / EXCERPT: First-five-column conjecture. The generating functions for columns $0,1,2,3,4$ of M, in that order, are

conjecture | open | MathDB 页面 | 原文线索

V2-044014 | MATHDB-353215

The asymptotic-ratio conjecture for the array M

摘录 / EXCERPT: Asymptotic-ratio conjecture for M. Assuming the numerator conjecture for $H_n(x)$, for fixed m,

conjecture | open | MathDB 页面 | 原文线索

V2-044015 | MATHDB-353217

The extremal conjecture for triangles in non-uniform intersecting families

摘录 / EXCERPT: Let $r\ge 2$ and let $\mathcal F\subseteq 2^{[n]}$ be an r-wise intersecting family. For an $(r+1)$-set X, define ... Let $\mathcal N(\Delta_{r+1},\mathcal G)$ denote the number...

conjecture | open | MathDB 页面 | 原文线索

V2-044016 | MATHDB-353226

Generalized conjecture on extremal intersections of discriminantal arrangements

摘录 / EXCERPT: Let $\mathcal{B}(n,3,\mathcal{A}_\infty)$ be a discriminantal arrangement with the maximum number of independent intersections in...

conjecture | open | MathDB 页面 | 原文线索

V2-044017 | MATHDB-353231

Elizalde-Plante-Roby-Sagan mesicity conjecture for fence rowmotion

摘录 / EXCERPT: Let $F(\beta)$ be a fence and let $L(\beta)$ be its lattice of lower order ideals. Suppose $k\ge 2$ and $\beta=(k-1,k,k,\ldots,k,k-1)$ has an odd number of parts. Under rowmotion o...

conjecture | open | MathDB 页面 | 原文线索

V2-044018 | MATHDB-353241

The product formula conjecture for maximum (t,s)-families

摘录 / EXCERPT: Product formula conjecture. If $n\ge t+s$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044019 | MATHDB-353248

Popular-difference density conjecture for three-point matrix patterns in the integer lattice

摘录 / EXCERPT: Popular-difference density conjecture. The displayed lower bound holds for every pair of matrices satisfying the stated full-rank condition. The conjecture is a finitary form of th...

conjecture | open | MathDB 页面 | 原文线索

V2-044020 | MATHDB-353253

Combinatorial positivity conjecture for the Gaussian product inequality

摘录 / EXCERPT: Combinatorial positivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044021 | MATHDB-353277

Symmetry conjecture for an optimal pair in the minimum overlap problem

摘录 / EXCERPT: Let $f(x)$ and $M(x)$ be a pair of functions as in the source's formulation of the minimum overlap problem, let $\|M\|_{\infty}$ denote the supremum norm of M, and let μ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-044022 | MATHDB-353278

Conjectured interval for the minimum overlap constant

摘录 / EXCERPT: Let μ denote the minimum overlap constant in the Erdős minimum overlap problem. Conjectured interval for μ. The constant μ lies in the interval ... This numerical conject...

conjecture | open | MathDB 页面 | 原文线索

V2-044023 | MATHDB-353296

The finite advantage conjecture for infinite Hex

摘录 / EXCERPT: Consider infinite Hex on the triangular lattice, with Red and Blue claiming unoccupied board positions according to the usual rules, but with a board position containing finitely m...

conjecture | open | MathDB 页面 | 原文线索

V2-044024 | MATHDB-353303

Gessel and Zhuang's shuffle-compatibility conjecture for $(\mathrm{udr},\mathrm{pk},\mathrm{des})$

摘录 / EXCERPT: For a permutation τ, let $\rm udr$, $\rm pk$, and $\rm des$ denote the permutation statistics of the length of the longest up-down run, the number of peaks, and the number of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044025 | MATHDB-353321

The characterization of good ordered graphs by monotone caterpillars

摘录 / EXCERPT: Monotone-caterpillar characterization conjecture. The ordered graph $G^<$ is good if and only if it is a monotone caterpillar graph.

conjecture | open | MathDB 页面 | 原文线索

V2-044026 | MATHDB-353377

Generalized polymatroid conjecture for Grothendieck Newton polytopes

摘录 / EXCERPT: For $w\in S_n$, let $\mathfrak{G}_w$ be the Grothendieck polynomial, let $\operatorname{Newton}(\mathfrak{G}_w)$ be the convex hull of its exponent vectors, and let SNP mean that e...

conjecture | open | MathDB 页面 | 原文线索

V2-044027 | MATHDB-353401

Harrington et al.'s divisibility conjecture for modulo-dependent chip-collecting games

摘录 / EXCERPT: Let m and n be positive integers with $m\mid n$, and let $a<b<\min\{m,n\}$. Consider the random walk on $\mathbb{Z}_m\times\mathbb{Z}_n$ whose moves are $(+a,+b)$ and $(+b,+a)$...

conjecture | open | MathDB 页面 | 原文线索

V2-044028 | MATHDB-353413

Extremal great-circle distribution conjecture for six, nine, and fifteen circles

摘录 / EXCERPT: Extremal great-circle distribution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044029 | MATHDB-353449

Mészáros-Setiabrata-St. Dizier interval-support conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let S_n be the symmetric group, let $\omega\in S_n$, and let $p_1(x_1,\ldots,x_n)$ and $p_2(x_1,\ldots,x_n)$ be monomials with nonzero coefficient in the Grothendieck polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-044030 | MATHDB-353450

Up-elbow bound conjecture for bumpless pipe dreams

摘录 / EXCERPT: Let ω be a permutation, and let $P\in\operatorname{Pipes}(\omega)$. If P is non-reduced, replace the redundant crossings with bump tiles. Up-elbow bound conjecture. Pipe...

conjecture | open | MathDB 页面 | 原文线索

V2-044031 | MATHDB-353451

Stressed-hyperplane relaxation conjecture for Ehrhart positivity

摘录 / EXCERPT: Stressed-hyperplane relaxation conjecture. If M is Ehrhart positive and H is a stressed hyperplane in M, then $\operatorname{Rel}_H(M)$ is also Ehrhart positive.

conjecture | open | MathDB 页面 | 原文线索

V2-044032 | MATHDB-353474

Pascal recurrence conjecture for generalized Pascal triangles

摘录 / EXCERPT: Pascal recurrence conjecture. For $n\geq (m-1)k+1$ when $k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-044033 | MATHDB-353519

Mathon and van Trung's conjecture on perfect sequence covering arrays

摘录 / EXCERPT: For integers n and k, let $g(n,k)$ be the minimum covering multiplicity for a perfect sequence covering array with parameters (n,k). The previously known exceptional values i...

conjecture | open | MathDB 页面 | 原文线索

V2-044034 | MATHDB-353553

Zeilberger-Reed conjecture on 3-connected planar linear normal lambda-terms

摘录 / EXCERPT: A planar linear normal λ-term is a linear normal λ-term whose syntactic diagram is planar; call it 3-connected when its syntactic diagram is 3-connected. For...

conjecture | open | MathDB 页面 | 原文线索

V2-044035 | MATHDB-353579

Dartyge-Sárközy conjecture on additive decompositions of primitive elements

摘录 / EXCERPT: Dartyge-Sárközy conjecture. If p is large enough, then

conjecture | open | MathDB 页面 | 原文线索

V2-044036 | MATHDB-353582

Non-periodicity conjecture for partitions into integers congruent to one modulo three

摘录 / EXCERPT: Let ... For each $n\in\mathbb{N}$, let $f_{A,n}(t)$ be the n-th A-partition polynomial, and let $\operatorname{sign}(f_{A,n}(-1))$ be its sign. Non-periodicity conjecture. The se...

conjecture | open | MathDB 页面 | 原文线索

V2-044037 | MATHDB-353583

Factorial-set sign conjecture for weighted partition sums

摘录 / EXCERPT: For $m\in\mathbb{N}$ with $m\geq 2$, let ... For $k\in\mathbb{N}$, let $S_{U_m,k}(n)$ be the weighted alternating sum associated with the U_m-partition polynomial. Factorial-set si...

conjecture | open | MathDB 页面 | 原文线索

V2-044038 | MATHDB-353593

Asymptotic dominance conjecture for Euler characteristics of relative stable-map spaces

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_n$ denote the moduli space considered in the paper, and let $\overline{\mathcal{M}}_{0,n+1}$ denote the corresponding genus-zero moduli space. Their Eul...

conjecture | open | MathDB 页面 | 原文线索

V2-044039 | MATHDB-353628

Gedeon's equivariant Kazhdan-Lusztig coefficient bound for matroids

摘录 / EXCERPT: Gedeon's conjecture. The coefficients of

conjecture | open | MathDB 页面 | 原文线索

V2-044040 | MATHDB-353629

The perimeter bound for triangular-grid billiards

摘录 / EXCERPT: Perimeter conjecture. If P is a grid polygon, then

conjecture | open | MathDB 页面 | 原文线索

V2-044041 | MATHDB-353671

Asymptotic exponent conjecture for corner polyhedra and 3-connected Schnyder labelings

摘录 / EXCERPT: Asymptotic exponent conjecture. There exist positive constants κ and κ' such that

conjecture | open | MathDB 页面 | 原文线索

V2-044042 | MATHDB-353680

The conjecture on infinitely many grids with discrepancy 2

摘录 / EXCERPT: Let d_n denote the discrepancy associated with the grid of size n. The discrepancy-2 conjecture. There are infinitely many n such that $d_n=2$. In particular, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-044043 | MATHDB-353681

The power-index conjecture for grid discrepancies

摘录 / EXCERPT: Let d_n denote the discrepancy associated with the grid of size n. Power-index conjecture. Let a be an odd natural number. If a is not divisible by 21, then for all...

conjecture | open | MathDB 页面 | 原文线索

V2-044044 | MATHDB-353714

The connectivity dimension-dropping conjecture for rigidity matroids

摘录 / EXCERPT: Let G be a graph and let d be a positive integer. A graph is redundantly $\mathcal{R}_d$-connected when deleting any edge leaves it $\mathcal{R}_d$-connected. The connectivity...

conjecture | open | MathDB 页面 | 原文线索

V2-044045 | MATHDB-353715

The rigidity-matroid bridge conjecture

摘录 / EXCERPT: Let d be a positive integer, let G be an $\mathcal{R}_d$-connected graph, and let e be an edge of G. An edge is an $\mathcal{R}_{d+1}$-bridge when its deletion lowers the r...

conjecture | open | MathDB 页面 | 原文线索

V2-044046 | MATHDB-353719

Mcalmon-Oh-Xiang's CCW-arrow characterization of positroid quotients

摘录 / EXCERPT: Mcalmon-Oh-Xiang's conjecture. $M'\trianglelefteq_Q M$ if and only if every CCW-arrow of M is the union of CCW-arrows of M'.

conjecture | open | MathDB 页面 | 原文线索

V2-044047 | MATHDB-353800

Tightness of the function-field Sárközy bound for square differences

摘录 / EXCERPT: Function-field Sárközy tightness conjecture. Any such set satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044048 | MATHDB-353812

Gabow's sequential symmetric exchange conjecture for matroid bases

摘录 / EXCERPT: Gabow's conjecture. There are orderings $A=(a_1,\dots,a_r)$ and $B=(b_1,\dots,b_r)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044049 | MATHDB-353813

White's conjecture on equivalence of matroid basis sequences

摘录 / EXCERPT: White's conjecture. The sequences $\mathcal{X}$ and $\mathcal{Y}$ are equivalent if and only if they are compatible. Compatibility is evidently necessary, and the conjecture is con...

conjecture | open | MathDB 页面 | 原文线索

V2-044050 | MATHDB-353814

Equitability conjecture for matroids whose ground set is two bases

摘录 / EXCERPT: Equitability conjecture. If the ground set S of M can be partitioned into two bases, then M is equitable. This is a simple but surprisingly open question about equitable matr...

conjecture | open | MathDB 页面 | 原文线索

V2-044051 | MATHDB-353822

Arithmetic periodicity for Family A cut games

摘录 / EXCERPT: Family A arithmetic-periodicity conjecture. The nim-sequence for every game of textsc{cut} in this family is precisely $(0,1)^c (+2)$. This concerns one of the families whose nim-s...

conjecture | open | MathDB 页面 | 原文线索

V2-044052 | MATHDB-353823

Partial arithmetic-periodicity conjecture for Family B cut games

摘录 / EXCERPT: Family B partial conjecture. If $3x<y$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044053 | MATHDB-353824

Ultimate arithmetic periodicity for Family C cut games

摘录 / EXCERPT: Let $\mathcal{C}$ be a cut set satisfying ... and suppose that $\mathcal{C}\neq\{1,2\}$. This is Family C. Family C arithmetic-periodicity conjecture. The nim-sequence for every game...

conjecture | open | MathDB 页面 | 原文线索

V2-044054 | MATHDB-353854

Canonical form conjecture for binary Δ-matroids under handle slides

摘录 / EXCERPT: Canonical form conjecture. For each binary Δ-matroid D, there is a sequence of handle slides taking D to some $D_{i,j,k,l}$ where i is the size of the ground set minus...

conjecture | open | MathDB 页面 | 原文线索

V2-044055 | MATHDB-353862

The Helly-type compatibility conjecture for split systems on multisets

摘录 / EXCERPT: Compatibility conjecture. $\mathfrak S$ is compatible if and only if every submultiset of $\mathfrak S$ of size at most $\Delta(\mathcal M)+2$ is compatible.

conjecture | open | MathDB 页面 | 原文线索

V2-044056 | MATHDB-353863

Pilaud-Santos conjecture on multit triangulations and rigidity matroids

摘录 / EXCERPT: Let k-triangulations be the maximal graphs on $[n]$ with no $(k+1)$-crossing, and let the generic bar-and-joint rigidity matroid of dimension $2k$ be the matroid obtained from ge...

conjecture | open | MathDB 页面 | 原文线索

V2-044057 | MATHDB-353943

The graphicness conjecture for 3-connected orderable binary matroids

摘录 / EXCERPT: A matroid M is orderable if it admits an ordering of its circuits with the relevant ordering property, and it is 3-connected and binary in the usual matroid-theoretic senses. Gra...

conjecture | open | MathDB 页面 | 原文线索

V2-044058 | MATHDB-353951

Procesi's lexicographic monotonicity conjecture for the Formanek-Weingarten function

摘录 / EXCERPT: Procesi's conjecture. If $\alpha>\beta$ in lexicographic order, then

conjecture | open | MathDB 页面 | 原文线索

V2-044059 | MATHDB-353964

The improved Bukh conjecture on sums of irreducible transformations

摘录 / EXCERPT: The improved Bukh conjecture. Let $\mathcal{L}_1,\mathcal{L}_2\in\operatorname{Mat}_d(\mathbb{Z})$ be irreducible. Then, for every finite subset A of $\mathbb{Z}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-044060 | MATHDB-353979

Jamet, Popoli, and Stoll's maximum order complexity conjecture for the Fibonacci-Thue-Morse sequence

摘录 / EXCERPT: Let $\alpha=(1+\sqrt{5})/2$ be the golden ratio, let $\bf ftm$ be the Fibonacci-Thue-Morse sequence, and let $M_{\bf ftm}(n)$ denote its maximum order complexity at length n. Jam...

conjecture | open | MathDB 页面 | 原文线索

V2-044061 | MATHDB-353992

Conjecture on extremal stationary measures for half-space KPZ

摘录 / EXCERPT: Let $\mathcal{H}_{u,v}$ denote the explicitly constructed stationary processes for the half-space KPZ equation with Neumann boundary parameter u, indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-044062 | MATHDB-353993

Conjecture on extremal stationary measures for the half-space log-gamma polymer

摘录 / EXCERPT: Let $\alpha>0$, $u>-\alpha$, and let $z_{u,v}$ be the explicitly constructed stationary processes indexed by $v\in(-\alpha,\min(u,0)]$ for the half-space log-gamma polymer. For ini...

conjecture | open | MathDB 页面 | 原文线索

V2-044063 | MATHDB-353994

Strong inscribability conjecture in higher rank

摘录 / EXCERPT: Strong inscribability conjecture in higher rank. Every strongly inscribable arrangement of rank $d\geq 3$ is combinatorially isomorphic to the restriction of a reflection arrangeme...

conjecture | open | MathDB 页面 | 原文线索

V2-044064 | MATHDB-354036

Goddyn-Hochstättler-Neudauer positive-coline conjecture for gammoids

摘录 / EXCERPT: Let M be a matroid. A gammoid is a matroid arising from linkages in a directed graph; M is simple if it has no loops or parallel elements, and a coline is a rank-$r(M)-2$ flat....

conjecture | open | MathDB 页面 | 原文线索

V2-044065 | MATHDB-354038

The rainbow Sidon-coloring conjecture

摘录 / EXCERPT: Let $r\ge 2$ be an integer and let n be sufficiently large. An r-coloring of a subset of $[n]$ is rainbow $[1,-1,1,-1]$-free if it contains no rainbow 4-element set satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-044066 | MATHDB-354065

The Matching-Jack conjecture for connection coefficients

摘录 / EXCERPT: For a partition $ u$, write $p_\nu:=p_{\nu_1}p_{\nu_2}\dots$. Use the analogous notation for $\mathbf{q}$ and $\mathbf{r}$. Let $c^\lambda_{\mu,\nu}(b)$ be defined by ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-044067 | MATHDB-354066

The matching-statistic formulation of the Matching-Jack conjecture

摘录 / EXCERPT: For $n\geq1$, let $\mathcal{A}_n:=\{1,\widehat{1},...,n,\widehat{n}\}$, and let a matching on $\mathcal{A}_n$ be a partition into pairs. A matching is bipartite if every pair has t...

conjecture | open | MathDB 页面 | 原文线索

V2-044068 | MATHDB-354067

The b-conjecture for connected Jack coefficients

摘录 / EXCERPT: b-conjecture. For all partitions $\lambda,\mu,\nu\vdash n\geq 1$, the coefficient $h^\lambda_{\mu,\nu}$ is a polynomial in b with non-negative integer coefficients. The b-conje...

conjecture | open | MathDB 页面 | 原文线索

V2-044069 | MATHDB-354069

The Backman-Baker-Yuen consistency conjecture for regular matroids

摘录 / EXCERPT: Let $M=(E,\mathcal B,\chi)$ be an oriented regular matroid, let (σ,σ^*) be a pair of acyclic circuit and cocircuit signatures, and let $\alpha_{(M,\sigma,\sigma^*)}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-044070 | MATHDB-354070

The uniqueness conjecture for consistent matroidal sandpile torsor algorithms

摘录 / EXCERPT: Let $M=(E,\mathcal B,\chi)$ be an oriented regular matroid, let (σ,σ^*) be a pair of acyclic signatures, and let α and β be matroidal sandpile torsor algor...

conjecture | open | MathDB 页面 | 原文线索

V2-044071 | MATHDB-354072

Bergeron-Garsia $n!/k$ intersection conjecture

摘录 / EXCERPT: Let λ be an integer partition of $n+1$, and let $\mu^{(1)},\ldots,\mu^{(k)}$ be partitions of n, each obtained from λ by removing a removable cell from its Young...

conjecture | open | MathDB 页面 | 原文线索

V2-044072 | MATHDB-354090

Positive Schur-expansion conjecture for triangular partitions

摘录 / EXCERPT: Let τ be a triangular partition of size n, and let $\mathcal{A}_{\tau}(q,t)$ be the associated polynomial. For each partition λ of length at most 2, write...

conjecture | open | MathDB 页面 | 原文线索

V2-044073 | MATHDB-354107

Strong sequenceability conjecture for abelian groups

摘录 / EXCERPT: Let G be an abelian group. For a subset $S\subseteq G\setminus\{0\}$, a sequencing is an ordering whose partial sums are distinct, while a rotational sequencing permits the initi...

conjecture | open | MathDB 页面 | 原文线索

V2-044074 | MATHDB-354108

Costa-Morini-Pasotti-Pellegrini restricted rotational sequencing conjecture

摘录 / EXCERPT: Let G be an abelian group and let S be a finite subset of $G\setminus\{0\}$ such that ... and $|S\cap\{x,-x\}|\leq1$ for every $x\in G$. Costa-Morini-Pasotti-Pellegrini conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-044075 | MATHDB-354110

Borel CSP complexity dichotomy conjecture

摘录 / EXCERPT: Borel CSP dichotomy conjecture. For every $\mathcal{D}$, the set $\mathrm{CSP}_B(\mathcal{D})$ is either in $\mathbf{\Pi}^1_1$ or is $\mathbf{\Sigma}^1_2$-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-044076 | MATHDB-354118

O'Neill and Verstraete's uniform extremal-family conjecture

摘录 / EXCERPT: O'Neill and Verstraete's conjecture. Let $k>r\geq2$ and $n\geq kr(r-1)$. Then the unique extremal non-trivial r-wise intersecting families in $\binom{[n]}k$ are ${\rm BD}_r(n,k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-044077 | MATHDB-354150

Uniform convergence conjecture for the limiting rook-placement profile

摘录 / EXCERPT: Let $\lambda\in\mathcal{D}_n$ be a fixed partition, and let $\varepsilon>0$. For a uniformly random rook placement $\pi\sim\operatorname{Unif}(\operatorname{RP}(N\odot\lambda))$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-044078 | MATHDB-354171

Optimality of the constant-weight-code bounds for extremal designs in hypercubes

摘录 / EXCERPT: Optimality conjecture. The duals of these constant-weight codes are smallest-cardinality positively weighted extremal designs in their respective hypercubes Q_d.

conjecture | open | MathDB 页面 | 原文线索

V2-044079 | MATHDB-354194

The higher-dimensional even-incidence characterisation of colourable sphere triangulations

摘录 / EXCERPT: Let C be a triangulation of the d-sphere S^d. Its 1-skeleton is colourable with $d+1$ colours, and its $(d-2)$-faces are incident with $(d-1)$-faces. Higher-dimensional colou...

conjecture | open | MathDB 页面 | 原文线索

V2-044080 | MATHDB-354212

Bergeron's combinatorial supersymmetry conjecture

摘录 / EXCERPT: Let $\mathcal{R}(n;r,\ell)$ be the multigraded S_n-module with commuting and anticommuting variable sets, let $\operatorname{ch}\mathcal{R}(n;r,\ell)$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-044081 | MATHDB-354214

Embeddings between partial combinatory algebras at successive Turing jumps

摘录 / EXCERPT: Let $X\subseteq\omega$. Write $\mathcal{K}_1^X$, $\mathcal{K}_2^X$, and $\mathcal{B}^X$ for the partial combinatory algebras defined in the paper, and write X' for the Turing jum...

conjecture | open | MathDB 页面 | 原文线索

V2-044082 | MATHDB-354224

The crossing-degree conjecture for complete simple topological graphs

摘录 / EXCERPT: Crossing-degree conjecture. There is an absolute constant $\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044083 | MATHDB-354240

Thue-Morse arithmetic-progression bound conjecture

摘录 / EXCERPT: Thue-Morse bound conjecture. For all $d \geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044084 | MATHDB-354241

Rudin-Shapiro arithmetic-progression bound conjecture

摘录 / EXCERPT: Rudin-Shapiro bound conjecture. For all $d \geq 23$,

conjecture | open | MathDB 页面 | 原文线索

V2-044085 | MATHDB-354242

Rudin-Shapiro maximum conjecture below powers of two

摘录 / EXCERPT: Rudin-Shapiro maximum conjecture. For all integers $k \geq 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-044086 | MATHDB-354288

Merino-Welsh conjectures for matroids

摘录 / EXCERPT: Merino-Welsh conjectures. The following three inequalities should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-044087 | MATHDB-354289

Burdzy-Pitman conjecture on the maximal spread of independent coherent distributions

摘录 / EXCERPT: Burdzy-Pitman conjecture. For every $\delta\in(\frac{1}{2},1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-044088 | MATHDB-354339

Polynomial monotone diameter conjecture for $(0,k)$-lattice polytopes

摘录 / EXCERPT: Let P be a $(0,k)$-lattice polytope of dimension d, and let the monotone diameter of P be the maximum, over pairs of vertices and compatible linear objective functions, of th...

conjecture | open | MathDB 页面 | 原文线索

V2-044089 | MATHDB-354344

Cichacz's conjecture on zero-sum partitions of abelian groups

摘录 / EXCERPT: Let G be an abelian group with at least three involutions, and let $r_1,\dots,r_t\geq 3$ satisfy ... A subset is zero-sum if the sum of its elements is zero. Cichacz's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-044090 | MATHDB-354345

The classification conjecture for transversals in group subsquares

摘录 / EXCERPT: Let G be a group and let $A,B\subseteq G$ satisfy $|A|=|B|=n$. A subsquare $A\times B$ is the corresponding subarray of the multiplication table, and a transversal meets every ro...

conjecture | open | MathDB 页面 | 原文线索

V2-044091 | MATHDB-354348

Three-translate linear stability conjecture for finite subsets of the integers

摘录 / EXCERPT: Three-translate linear stability conjecture. For every $\delta>0$ there exists $\varepsilon>0$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-044092 | MATHDB-354377

The irreducible annihilation function conjecture for minimal non-basic subsets

摘录 / EXCERPT: Irreducible annihilation function conjecture. The function f satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044093 | MATHDB-354382

The monochromatic component edge-density conjecture

摘录 / EXCERPT: Let $M(n,r)$ be the largest natural number such that every r-coloring of K_n contains a monochromatic connected component with at least $M(n,r)$ edges. An affine plane of order...

conjecture | open | MathDB 页面 | 原文线索

V2-044094 | MATHDB-354402

Smallness characterisation conjecture for graph classes by twin-width

摘录 / EXCERPT: Let $\mathcal{C}$ be a class of graphs closed under induced subgraphs. Call $\mathcal{C}$ small if there is a constant c such that, for every n, it contains at most $n!\,c^n$ g...

conjecture | open | MathDB 页面 | 原文线索

V2-044095 | MATHDB-354420

The nonexistence conjecture for M-property incidence graphs of 2-quasi-symmetric designs

摘录 / EXCERPT: Nonexistence conjecture. There are no point-line incidence graphs of a 2-quasi-symmetric design with $x=0$ that have the M-property.

conjecture | open | MathDB 页面 | 原文线索

V2-044096 | MATHDB-354534

Chu-Herzog-Lu conjecture on socle ideals of polymatroidal ideals

摘录 / EXCERPT: Let K be a field, $S=K[x_1,\dots,x_n]$, and let $I\subset S$ be a polymatroidal ideal. The socle ideal $\operatorname{soc}(I)$ is the unique monomial ideal satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-044097 | MATHDB-354537

The scaled generalization of Tepper's identity

摘录 / EXCERPT: Let l be a positive integer, and let $P(x)$ be a real polynomial of degree n with leading coefficient a_n. Scaled generalized Tepper identity. One has ... The preceding theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-044098 | MATHDB-354545

The universal L^1 mixing-time conjecture for averaging on graphs

摘录 / EXCERPT: Let G be any connected graph with n nodes. Write $t_{\epsilon,1}(G)$ for the L^1 mixing time of the averaging process on G, and let $\gamma(G)$ denote the relevant graph re...

conjecture | open | MathDB 页面 | 原文线索

V2-044099 | MATHDB-354546

The L^2-to-L^1 mixing-time conjecture for averaging on graphs

摘录 / EXCERPT: Let G be any connected graph with n nodes. Write $t_{\epsilon,2\rightarrow1}(G)$ for the L^2-to-L^1 mixing time of the averaging process on G, and let $\gamma(G)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-044100 | MATHDB-354599

Quantitative generalized von Neumann conjecture for polynomial progressions

摘录 / EXCERPT: Let $P_1,\dots,P_s\in\mathbb{Z}[x]$ be linearly independent polynomials that vanish at zero. Let $f_\omega:\mathbb{Z}/N\mathbb{Z}\to\mathbb{C}$ be one-bounded functions for each...

conjecture | open | MathDB 页面 | 原文线索

V2-044101 | MATHDB-354600

Inverse theorem conjecture for polynomial progression averages

摘录 / EXCERPT: Let $P_1,\dots,P_s\in\mathbb{Z}[x]$ be linearly independent polynomials that vanish at zero. Let $f_\omega:\mathbb{Z}/N\mathbb{Z}\to\mathbb{C}$ be one-bounded functions, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-044102 | MATHDB-354601

Density bound conjecture for polynomial cube progressions

摘录 / EXCERPT: Let $P_1,\dots,P_s\in\mathbb{Z}[x]$ be linearly independent polynomials that vanish at zero. For $\omega\in\{0,1\}^s$, write $\omega=(\omega_1,\dots,\omega_s)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044103 | MATHDB-354627

Discrete covering-system convex-hull periodicity conjecture

摘录 / EXCERPT: Discrete convex-hull conjecture. There exist $j\in\{1,2,\ldots,k\}$ and $r,s\in I_j$ such that, for some $l\leq k$,

conjecture | open | MathDB 页面 | 原文线索

V2-044104 | MATHDB-354628

Characteristic-sequence conjecture for transitive permutations

摘录 / EXCERPT: Characteristic-sequence conjecture. For every transitive $f\in S_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-044105 | MATHDB-354635

Linear-deficit conjecture for antichain saturation

摘录 / EXCERPT: Linear-deficit conjecture. For each fixed positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-044106 | MATHDB-354636

Existence of a full level in antichain-saturated families

摘录 / EXCERPT: Full-level conjecture. There exists an integer i such that

open_problem | open | MathDB 页面 | 原文线索

V2-044107 | MATHDB-354653

Extremal dense hard-sequence conjecture for coTHEOREMS

摘录 / EXCERPT: Let $\mathcal{T}$ be the theory used to define H', and let H' be the collection of dense sets of true $\mathcal{T}$-unprovable sentences described in the source. Given a nondet...

conjecture | open | MathDB 页面 | 原文线索

V2-044108 | MATHDB-354691

Genericity conjecture for dope matrices

摘录 / EXCERPT: Genericity conjecture. For $m\geq 3$, $\mathcal{D}_n^{\operatorname{gen}(m)}$ consists exactly of the $m\times(n+1)$ $\{0,1\}$-matrices such that, for every $k\in[0,n]$, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-044109 | MATHDB-354692

Equivalence conjecture for triples of complex numbers

摘录 / EXCERPT: Equivalence conjecture. If $(0,1,\lambda)\sim(0,1,\lambda')$, then either λ and λ' are both transcendental, or λ and λ' are conjugate algebraic numb...

conjecture | open | MathDB 页面 | 原文线索

V2-044110 | MATHDB-354704

The d-Matoušek-to-d-Leray conjecture

摘录 / EXCERPT: Let d be a nonnegative integer, and let K be a simplicial complex. A simplicial complex is d-Matoušek and d-Leray as defined in the source paper. The d-Matoušek-to...

conjecture | open | MathDB 页面 | 原文线索

V2-044111 | MATHDB-354720

Three-layer conjecture for maximum antichains in a random subset of the Boolean lattice

摘录 / EXCERPT: Let $\mathcal{P}(n)_p$ be the random subfamily of the Boolean lattice $\mathcal{P}(n)$ obtained by including each subset independently with probability p, and let a maximum antic...

conjecture | open | MathDB 页面 | 原文线索

V2-044112 | MATHDB-354722

Homological cop number conjecture for simplicial pseudomanifolds

摘录 / EXCERPT: Suppose that X is an n-dimensional simplicial pseudomanifold, meaning that each simplex is contained in an n-simplex and each $(n-1)$-simplex is contained in at most two n-...

conjecture | open | MathDB 页面 | 原文线索

V2-044113 | MATHDB-354730

Even-parameter Nim-sum conjecture for three-dimensional chocolate-bar games

摘录 / EXCERPT: Even-parameter Nim-sum conjecture. The stated condition characterizes exactly the $\mathcal{P}$-positions in this restricted even-k family. The authors found this claim through M...

conjecture | open | MathDB 页面 | 原文线索

V2-044114 | MATHDB-354731

Shifted Nim-sum conjecture for chocolate-bar games with k congruent to 1 modulo 4

摘录 / EXCERPT: Shifted Nim-sum conjecture. The shifted nim-sum condition characterizes exactly the $\mathcal{P}$-positions for this family of chocolate-bar games. The claim was suggested by calcu...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-044115 | MATHDB-354736

Alspach-Liversidge conjecture on strong sequenceability of abelian groups

摘录 / EXCERPT: Alspach-Liversidge conjecture. Every abelian group is strongly sequenceable.

conjecture | open | MathDB 页面 | 原文线索

V2-044116 | MATHDB-354737

Conjecture on weak sequenceability of abelian groups

摘录 / EXCERPT: Weak sequenceability conjecture. The set A is t-weak sequenceable. Equivalently, every abelian group is t-weak sequenceable.

conjecture | open | MathDB 页面 | 原文线索

V2-044117 | MATHDB-354738

CMPP conjecture on sequenceability with no opposite pair

摘录 / EXCERPT: CMPP conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044118 | MATHDB-354739

Weak CMPP conjecture for abelian groups

摘录 / EXCERPT: Weak CMPP conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044119 | MATHDB-354745

Strong enumeration conjecture for q-ary codes

摘录 / EXCERPT: Fix an integer $q\ge 2$. A q-ary code of length n and distance d is a subset of $[q]^n$ whose elements are pairwise separated by Hamming distance at least d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044120 | MATHDB-354759

The principal-polynomial support-reduction conjecture

摘录 / EXCERPT: The principal-polynomial support-reduction conjecture. If E° is a principal polynomial, then either

conjecture | open | MathDB 页面 | 原文线索

V2-044121 | MATHDB-354766

Hindman's conjecture on monochromatic products and sums

摘录 / EXCERPT: Let $\mathbb{N}$ be the set of positive integers, and let a finite coloring assign one of finitely many colors to each element of $\mathbb{N}$. A set is monochromatic if all of its...

conjecture | open | MathDB 页面 | 原文线索

V2-044122 | MATHDB-354775

Lidický-Murphy conjecture on generalized Turán numbers

摘录 / EXCERPT: Lidický-Murphy conjecture. There exist such integers for which

conjecture | open | MathDB 页面 | 原文线索

V2-044123 | MATHDB-354776

Diameter-bounded blow-up conjecture for generalized Turán numbers

摘录 / EXCERPT: Let G be a graph with diameter at most $2r-2$ and chromatic number $\chi(G)<r$. Diameter-bounded blow-up conjecture. The generalized Turán number $\operatorname{ex}(n,G,K_r)$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-044124 | MATHDB-354780

The square numbers form a Lambda-p set for 2<p<4

摘录 / EXCERPT: The squares conjecture. The set A of squares is a Λ_p-set for every $2<p<4$. This conjecture is attributed in the paper to W. Rudin. Córdoba proved the corresponding ineq...

conjecture | open | MathDB 页面 | 原文线索

V2-044125 | MATHDB-354781

Cilleruelo-Córdoba's truncated square-frequency conjecture

摘录 / EXCERPT: Let $\gamma\in(0,1)$ be fixed, let N be positive, and let f be a trigonometric polynomial whose frequencies lie in $\{n^2:N\leqslant n\leqslant N+N^\gamma\}$. Cilleruelo-Córdob...

conjecture | open | MathDB 页面 | 原文线索

V2-044126 | MATHDB-354789

Generic asymptotic normality conjecture for q-Stirling coefficients

摘录 / EXCERPT: Let $S[n,k]$, $S_B[n,k]$, $c[n,k]$, and $c_B[n,k]$ denote the four families of type A and type B q-Stirling polynomials, and view each polynomial's coefficients as a probabil...

conjecture | open | MathDB 页面 | 原文线索

V2-044127 | MATHDB-354792

The projective-dimension conjecture for edge rings with 2-linear resolutions

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $n=|V|$, and let $k[G]=k[x_1,\ldots,x_n]/I(G)$ be its edge ring, where $I(G)$ is generated by $\{x_ix_j:\{i,j\}\in E\}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-044128 | MATHDB-354794

Ramsey-type characterization equivalent to the Gyárfás-Sumner conjecture

摘录 / EXCERPT: Equivalent Ramsey-type conjecture. For every finite family $\mathcal{H}$ of connected graphs, $\mathcal{H}$ satisfies $\mathrm{(P-\chi)}$ if and only if there exist an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-044129 | MATHDB-354813

The corank bound conjecture for non-negative posets of Dynkin type E

摘录 / EXCERPT: Corank bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-044130 | MATHDB-354874

Thrackle-style conjecture for almost holey convex families

摘录 / EXCERPT: Suppose that $d4c2$ is a set of n points in the plane, and let $S_1,\cdots,S_m\subseteq d4c2$ be subsets with convex hulls ... Assume that $1<|S_i|<n$ for every i, that...

conjecture | open | MathDB 页面 | 原文线索

V2-044131 | MATHDB-354875

A strengthening of the thrackle conjecture for topological trees

摘录 / EXCERPT: Let $\mathcal{P}$ be a set of n points in the plane, and let there be m topological trees such that every pair of trees intersects exactly once and every leaf of every tree bel...

conjecture | open | MathDB 页面 | 原文线索

V2-044132 | MATHDB-354878

The most likely terminal configuration conjecture for labeled chip-firing on binary trees

摘录 / EXCERPT: Most likely terminal configuration conjecture. For all n, the most likely terminal configuration is the unique binary search tree on the complete binary tree with 2^n-1 nodes.

conjecture | open | MathDB 页面 | 原文线索

V2-044133 | MATHDB-354889

Aharoni-Korman fish bone conjecture for bounded-width posets

摘录 / EXCERPT: Fish bone conjecture. There exist a chain C and a decomposition D of $V(P)$ into disjoint antichains such that every antichain in D meets C.

conjecture | open | MathDB 页面 | 原文线索

V2-044134 | MATHDB-354898

The Catalan-array conjecture for vertically-recurrent matrices

摘录 / EXCERPT: Let C_n denote the Catalan triangular array. A matrix is vertically recurrent when it satisfies the paper's vertical recurrence relation. The Catalan-array conjecture. The Catala...

conjecture | open | MathDB 页面 | 原文线索

V2-044135 | MATHDB-354899

The extremal permanent conjecture for unitarily invariant norms

摘录 / EXCERPT: Extremal permanent conjecture. For all $M\in S(n,k)$ such that $T(M)=1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044136 | MATHDB-354903

The fully reducible Venn diagram converse conjecture

摘录 / EXCERPT: Fully reducible Venn diagram converse conjecture. If $n\leq m+1$, then V is fully reducible.

conjecture | open | MathDB 页面 | 原文线索

V2-044137 | MATHDB-354933

Asymptotic growth conjecture for the line orbit in triangle solitaire

摘录 / EXCERPT: Let L_n be the line pattern on the triangular board of side length n, and let $\gamma(L_n)$ denote its orbit under the triangle-solitaire moves. For a function $f(n)$, the nota...

conjecture | open | MathDB 页面 | 原文线索

V2-044138 | MATHDB-354934

The Erdős-Tuza-Valtr cap-cup conjecture

摘录 / EXCERPT: Erdős-Tuza-Valtr conjecture. For every triplet (n,a,b),

conjecture | open | MathDB 页面 | 原文线索

V2-044139 | MATHDB-354935

The set-theoretic Erdős-Szekeres conjecture for weak polygons

摘录 / EXCERPT: Set-theoretic Erdős-Szekeres conjecture. For every $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-044140 | MATHDB-354936

The set-theoretic Erdős-Tuza-Valtr conjecture

摘录 / EXCERPT: Set-theoretic Erdős-Tuza-Valtr conjecture. For every triplet (n,a,b),

conjecture | open | MathDB 页面 | 原文线索

V2-044141 | MATHDB-354937

The mutually interweaved laced-cup conjecture

摘录 / EXCERPT: Mutually interweaved laced-cup conjecture. For every $n\geq 3$ and $1\leq k\leq n$, every 4-cap, n-cup free configuration of size

conjecture | open | MathDB 页面 | 原文线索

V2-044142 | MATHDB-354938

Kiselev-Kupavskii-Patkós minimum-degree conjecture for union antichains

摘录 / EXCERPT: Kiselev-Kupavskii-Patkós' conjecture. If $1\leq 2\ell+1<n$ and $\mathcal{F}\subseteq\mathscr{P}([n])$ is a $(2\ell+1)$-union antichain, then

conjecture | open | MathDB 页面 | 原文线索

V2-044143 | MATHDB-354942

Quadratic upper-bound conjecture for random card shuffling

摘录 / EXCERPT: Let D be a deck with n cards of each type, and let λ_D denote the expected number of steps to absorption for the random card-shuffling process started from D. As...

conjecture | open | MathDB 页面 | 原文线索

V2-044144 | MATHDB-354957

The even-like extremal binary LCD code conjecture

摘录 / EXCERPT: Let k be an even positive integer and let $n>k$ be an integer. Write $d_2^{E}(n,k)$ for the relevant extremal minimum distance of binary LCD codes, and call a binary LCD code eve...

conjecture | open | MathDB 页面 | 原文线索

V2-044145 | MATHDB-354984

Complete score conjecture for two-row grids

摘录 / EXCERPT: Let $G_{2,m}$ be the rectangular grid with two rows and m columns, and let $Ls(G_{2,m})$ and $Rs(G_{2,m})$ denote the corresponding left and right scores. Two-row grid score conj...

conjecture | open | MathDB 页面 | 原文线索

V2-044146 | MATHDB-355014

Hanging-elements conjecture for the combinatorial derived matroid

摘录 / EXCERPT: Let M be a matroid with set of circuits $\mathcal{C}$. For $S\subseteq\mathcal{C}$, let $\operatorname{supp}(S)$ be the union of the circuits in S, and let $\mathcal{A}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-044147 | MATHDB-355015

Adjoint conjecture for the combinatorial derived matroid

摘录 / EXCERPT: Let M be a matroid with ground-set size n and rank k, and let M^* denote its dual. Assume that M^* has an adjoint. Adjoint conjecture. Then δM is isomorphic to an...

conjecture | open | MathDB 页面 | 原文线索

V2-044148 | MATHDB-355016

Categorical genericity conjecture for the combinatorial derived matroid

摘录 / EXCERPT: Let M be a representable matroid with set of circuits $\mathcal{C}$. For matroids on a fixed ground set, use the weak order defined by $M_1\geq M_2$ when every dependent set of...

conjecture | open | MathDB 页面 | 原文线索

V2-044149 | MATHDB-355018

Adjoints and the combinatorial derived matroid of the dual

摘录 / EXCERPT: Let M be a matroid with circuit set $\mathcal{C}$, and suppose that an adjoint of M exists. Let δM^* denote the combinatorial derived matroid of the dual matroid. Dual-...

conjecture | open | MathDB 页面 | 原文线索

V2-044150 | MATHDB-355038

Characterization of extremal nonnegative tensors by unfolded permutation tensors

摘录 / EXCERPT: Let $\boldsymbol{\beta}$ denote the lower bound in equation (main) for the ratio of the spectral norm to the Frobenius norm. An unfolded permutation tensor is a tensor obtained by...

conjecture | open | MathDB 页面 | 原文线索

V2-044151 | MATHDB-355099

Rainbow base factorization conjecture for bounded partition classes

摘录 / EXCERPT: Let M be a k-base matroid, meaning that its ground set can be partitioned into k bases, and let $\mathcal{P}$ be a partition of $E(M)$ such that ... for every $X\in\mathcal{P}$...

conjecture | open | MathDB 页面 | 原文线索

V2-044152 | MATHDB-355100

Rainbow base factorization conjecture for 2-bounded partitions

摘录 / EXCERPT: Let M be a k-base matroid with $k\geq 3$, and let $\mathcal{P}$ be a partition of $E(M)$ satisfying ... for every $X\in\mathcal{P}$. 2-bounded rainbow factorization conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-044153 | MATHDB-355101

Aharoni-Berger covering conjecture for two matroids

摘录 / EXCERPT: Let M and N be matroids on the same ground set E. Suppose that E can be covered by k independent sets of M and by ℓ independent sets of N, where...

conjecture | open | MathDB 页面 | 原文线索

V2-044154 | MATHDB-355102

Rainbow basis covering conjecture for 2-bounded partitions

摘录 / EXCERPT: Let M be a k-base matroid, and let $\mathcal{P}$ be a 2-bounded partition of $E(M)$, meaning that every partition class has size at most 2. Rainbow basis covering conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-044155 | MATHDB-355106

The small sphere hypothesis implies the small gap property

摘录 / EXCERPT: Small sphere to small gap conjecture. Then G has the SG property. The SG property would provide the ball-intersection control needed for the SBM theory; it is known for...

conjecture | open | MathDB 页面 | 原文线索

V2-044156 | MATHDB-355115

The diagonal-monomial reachability conjecture for psd-stable polynomials

摘录 / EXCERPT: Diagonal-monomial reachability conjecture. For any monomial Z^β appearing in a psd-stable polynomial, there is a diagonal monomial Z^α appearing in f which can be re...

conjecture | open | MathDB 页面 | 原文线索

V2-044157 | MATHDB-355134

Signs-model conjecture for connectivity of the three-dimensional hypercube

摘录 / EXCERPT: Signs-model conjecture. Among models in $\mathcal{D}_{\geq p}(Q_3)$, the signs model minimises the probability that Q_3 is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-044158 | MATHDB-355135

Erdős's $B+B+t$ conjecture

摘录 / EXCERPT: Let $A e othing$ be a subset of the natural numbers with positive density. A restricted sumset consists of sums of two distinct elements. Erdős's $B+B+t$ conjecture. There exist an...

conjecture | open | MathDB 页面 | 原文线索

V2-044159 | MATHDB-355136

Higher-order shifted finite-sums conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{N}$ have positive upper Banach density, and let $k\in\mathbb{N}$. For a finite subset F of B, write its sum as $\sum_{n\in F}n$. Higher-order shifted finit...

conjecture | open | MathDB 页面 | 原文线索

V2-044160 | MATHDB-355138

The deletion conjecture for crucial permutations

摘录 / EXCERPT: Deletion conjecture. Every (k,ℓ)-crucial permutation can be obtained by deleting elements from a maximal (k,ℓ)-crucial permutation.

conjecture | open | MathDB 页面 | 原文线索

V2-044161 | MATHDB-355147

Yamanouti and nonagon extremal conjecture for the Cheeger constant at fixed minimal width and circumradius

摘录 / EXCERPT: Let $\mathcal{K}^2$ denote the class of planar convex sets, and let $\omega(\Omega)$, $h(\Omega)$, and $R(\Omega)$ denote respectively the minimal width, Cheeger constant, and circ...

conjecture | open | MathDB 页面 | 原文线索

V2-044162 | MATHDB-355148

Alvarado et al.'s conjecture on minimum dominating sets in trees

摘录 / EXCERPT: Let T be a tree with domination number $gamma$, and let the number of its minimum dominating sets be measured as a function of $gamma$. Alvarado et al.'s conjecture. A tree with...

conjecture | open | MathDB 页面 | 原文线索

V2-044163 | MATHDB-355149

Henning et al.'s conjectured upper bound for minimum total dominating sets in trees

摘录 / EXCERPT: Let T be a tree of order $n\geq 2$, with total domination number γ_t, and let $\Gamma_t(T)$ denote the number of minimum total dominating sets of T. Henning et al.'s con...

conjecture | open | MathDB 页面 | 原文线索

V2-044164 | MATHDB-355165

The conjectured enumeration of alternating involutions avoiding 1432 and 3214

摘录 / EXCERPT: Let $AI_n(\rho)$ and $RAI_n(\rho)$ denote, respectively, the sets of alternating and reverse alternating involutions of length n avoiding the pattern ρ, and let M_n be the...

conjecture | open | MathDB 页面 | 原文线索

V2-044165 | MATHDB-355166

The conjectured 123τ-321τ symmetry for alternating involutions

摘录 / EXCERPT: Let $AI_n(\rho)$ denote the set of alternating involutions of length n avoiding the pattern ρ. Let τ be any permutation of $\{4,\ldots,m\}$, where $m\geq 4$. The 123τ-3...

conjecture | open | MathDB 页面 | 原文线索

V2-044166 | MATHDB-355167

Conjecture on arbitrarily large pairwise irreducible solutions

摘录 / EXCERPT: Let a pairwise irreducible solution be a solution satisfying the pairwise irreducibility conditions defined in the paper. Conjecture. There exist pairwise irreducible solutions of...

conjecture | open | MathDB 页面 | 原文线索

V2-044167 | MATHDB-355198

Excluded-minor conjecture for 3-regular matroids

摘录 / EXCERPT: Let M be a matroid. A matroid is 3-regular if it has a representation over $\mathbb{Q}(\alpha_1,\alpha_2,\alpha_3)$ in which every non-zero subdeterminant is an integer power o...

conjecture | open | MathDB 页面 | 原文线索

V2-044168 | MATHDB-355216

Dujmović-Morin conjecture on the number of graphs with bounded obstacle number

摘录 / EXCERPT: Given a graph G, let its obstacle number be the minimum number of faces in a straight-line drawing whose union intersects every non-edge, and let $f(h,n)$ denote the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-044169 | MATHDB-355219

Kalai's reconstruction conjecture for simplicial spheres

摘录 / EXCERPT: Let P,Q) be d-dimensional simplicial spheres, and let $FR(P)$ and $FR(Q)$ denote their facet-ridge graphs. Kalai's reconstruction conjecture. Every isomorphism between their fa...

conjecture | open | MathDB 页面 | 原文线索

V2-044170 | MATHDB-355221

Stephan's conjecture relating central-binomial polynomials and poly-Bernoulli numbers

摘录 / EXCERPT: Let $(p_k(x))_{k\geq -1}$ be the sequence of polynomials defined by $p_{-1}(x)=0$ and, together with $q_{-1}(x)=1$, by ... For $n\geq 0$, let $B_m^{(-k)}$ denote the poly-Bernoulli n...

conjecture | open | MathDB 页面 | 原文线索

V2-044171 | MATHDB-355243

The correlation-to-symmetric-rank conjecture for multilinear forms

摘录 / EXCERPT: Correlation-to-symmetric-rank conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-044172 | MATHDB-355258

Hadamard matrices yielding orthogonal arrays satisfying the eight-column conditions

摘录 / EXCERPT: The conjecture. For each odd m, there exists a Hadamard matrix $4m$ which yields an orthogonal array $\operatorname{OA}(4m,8,2,2)$ satisfying the conditions of Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-044173 | MATHDB-355262

Fully symmetric Kronecker coefficient asymptotics

摘录 / EXCERPT: Fully symmetric Kronecker asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044174 | MATHDB-355263

Large fully symmetric Kronecker coefficients for staircase and square partitions

摘录 / EXCERPT: Let ρ_k denote the staircase partition and let δ_ℓ denote the square partition. Their sizes are $|\rho_k|=\binom{k}{2}$ and $|\delta_\ell|=\ell^2$. For a self-conjug…

conjecture | open | MathDB 页面 | 原文线索

V2-044175 | MATHDB-355295

Keszegh-Lemons-Martin-Pálvölgyi-Patkós extension conjecture for induced saturation

摘录 / EXCERPT: Let P be a poset on $[p]$, and let $\dot P$ be the poset on $[p+1]$ obtained by adjoining a new element that dominates every element of P. Let $\text{sat}^*(n,P)$ denote the mi…

conjecture | open | MathDB 页面 | 原文线索

V2-044176 | MATHDB-355300

Joint sumset Bohr-set conjecture for finite-index endomorphisms

摘录 / EXCERPT: Let G be a (not necessarily countable) abelian group, let $d\geq 3$, and let $\phi_1,\dots,\phi_d$ be endomorphisms of G such that $[G:\phi_j(G)]<\infty$ for each j, and such…

conjecture | open | MathDB 页面 | 原文线索

V2-044177 | MATHDB-355301

Partition sumset Bohr-set conjecture for finite-index endomorphisms

摘录 / EXCERPT: Partition sumset Bohr-set conjecture. For every finite partition $G=\bigcup_{i=1}^r A_i$, there exists $i\in [r]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044178 | MATHDB-355323

The $3/7$ conjecture for large intersecting families

摘录 / EXCERPT: The $3/7$ conjecture. Then

open_problem | open | MathDB 页面 | 原文线索

V2-044179 | MATHDB-355326

The square-partition conjecture for Downright and LCTR game trees

摘录 / EXCERPT: Let n be a positive integer, and let a partition of size n^2 be a partition whose diagram contains n^2 cells. For such a partition, consider the game trees of Downright and L…

conjecture | open | MathDB 页面 | 原文线索

V2-044180 | MATHDB-355350

Extremal bound for suboptimal $(2d+1)$-union antichains

摘录 / EXCERPT: Let V be an n-dimensional vector space over the finite field with q elements, and let $\mathcal{L}(V)$ denote its lattice of subspaces. A family…

conjecture | open | MathDB 页面 | 原文线索

V2-044181 | MATHDB-355355

Ferrara-Kay-Kramer-Martin-Reiniger-Smith-Sullivan antichain saturation asymptotic conjecture

摘录 / EXCERPT: Ferrara-Kay-Kramer-Martin-Reiniger-Smith-Sullivan conjecture. For $k\geq3$,

conjecture | open | MathDB 页面 | 原文线索

V2-044182 | MATHDB-355356

Djanković-Ivan bounded-defect antichain saturation conjecture

摘录 / EXCERPT: Djanković-Ivan bounded-defect conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044183 | MATHDB-355362

The conjecture that every sufficiently large integer is polite

摘录 / EXCERPT: Politeness conjecture. Every sufficiently large integer is polite. The paper explains that this would follow from natural, currently unproved growth assumptions for Ramsey numbers,…

conjecture | open | MathDB 页面 | 原文线索

V2-044184 | MATHDB-355374

Conjecture that the four distinguished combinatorial manifolds are the octonionic projective plane

摘录 / EXCERPT: The distinguished-manifolds conjecture. The combinatorial manifolds K_1, K_2, K_3, and K_4 are PL homeomorphic to $\mathbb{OP}^2$.

conjecture | open | MathDB 页面 | 原文线索

V2-044185 | MATHDB-355375

Conjecture that the spheres L_2 and L_3 are non-polytopal

摘录 / EXCERPT: Non-polytopality conjecture. The combinatorial spheres L_2 and L_3 are non-polytopal.

conjecture | open | MathDB 页面 | 原文线索

V2-044186 | MATHDB-355377

Higher-degree Erdős-Ginzburg-Ziv equality for equal parameters

摘录 / EXCERPT: Let q be a positive integer and let t be a parameter for which the generalized Erdős-Ginzburg-Ziv constant $4\mathsf{EGZ}({t},{q},{q})4$ and the generalized Davenport constant...

conjecture | open | MathDB 页面 | 原文线索

V2-044187 | MATHDB-355393

Makai's extremal conjecture for $d_{2,1}$

摘录 / EXCERPT: For a convex body $K\subset\mathbb R^{2}$, let $d_{2,1}(K)$ be the non-separable lattice quantity defined in the source. The Makai extremal conjecture. ... with equality possible onl...

conjecture | open | MathDB 页面 | 原文线索

V2-044188 | MATHDB-355400

Fici-Saari conjecture on the minimum number of binary abelian square factors

摘录 / EXCERPT: Fici-Saari conjecture. Every binary word of length n contains at least $\floor{n/4}$ distinct abelian square factors; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-044189 | MATHDB-355401

Mäkelä's conjecture on ternary words with only trivial abelian squares

摘录 / EXCERPT: Mäkelä's conjecture. There exists an infinite ternary word whose only abelian square factors are 00, 11, and 22.

conjecture | open | MathDB 页面 | 原文线索

V2-044190 | MATHDB-355406

Pseudoperiodic word avoidance conjecture for binary words

摘录 / EXCERPT: Let (a,b) be a pair of positive integers with $1\leq a<b$. An infinite binary word has pseudoperiod (a,b) if each position is compatible with a repetition having periods a an...

conjecture | open | MathDB 页面 | 原文线索

V2-044191 | MATHDB-355413

The Erdős-Rogers Ramsey-number separation conjecture

摘录 / EXCERPT: Let $\ell\geq 3$ be an integer. The Ramsey number $\mathbf{R}(t,m)$ is the minimum integer n such that every graph on n vertices contains either a clique K_t or an independen...

conjecture | open | MathDB 页面 | 原文线索

V2-044192 | MATHDB-355427

The asymptotic maximum-appearance conjecture for distinct-letter words in two-dimensional grids

摘录 / EXCERPT: Let w' be a word of length $k\ge 2$ whose letters are all distinct, and let $f(w',n,2)$ denote the maximum number of copies of w' appearing in an $n\times n$ grid. Distinct-let...

conjecture | open | MathDB 页面 | 原文线索

V2-044193 | MATHDB-355432

The extremal alternating-sign conjecture for transitive tournaments

摘录 / EXCERPT: Extremal alternating-sign conjecture. If n is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-044194 | MATHDB-355433

The number characterization conjecture for bidding games

摘录 / EXCERPT: Number characterization conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-044195 | MATHDB-355434

The inverse conjugation conjecture for bidding games

摘录 / EXCERPT: Inverse conjugation conjecture. If a bidding game G has an inverse, H, then

conjecture | open | MathDB 页面 | 原文线索

V2-044196 | MATHDB-355435

Number avoidance and translation conjecture for bidding games

摘录 / EXCERPT: Number avoidance/translation conjecture. For any total budget, if G is a number and H is not a number, then

conjecture | open | MathDB 页面 | 原文线索

V2-044197 | MATHDB-355436

Number simplicity conjecture for bidding games

摘录 / EXCERPT: Number simplicity conjecture. For any total budget, each number is a dyadic rational.

conjecture | open | MathDB 页面 | 原文线索

V2-044198 | MATHDB-355437

The Hackenbush characterization conjecture for bidding games

摘录 / EXCERPT: Hackenbush characterization conjecture. The numbers are exactly the Hackenbush positions:

conjecture | open | MathDB 页面 | 原文线索

V2-044199 | MATHDB-355438

Self-dual realizability conjecture for four-dimensional strongly involutive cones

摘录 / EXCERPT: Self-dual realizability conjecture. All 4-dimensional strongly involutive combinatorially self-dual polyhedral cones have self-dual realizations.

conjecture | open | MathDB 页面 | 原文线索

V2-044200 | MATHDB-355476

Liquid-regime boundary conjecture for hard-square homology

摘录 / EXCERPT: Let $C(n;p,q)$ be the hard-square complex on a $p\times q$ rectangle, let $C(n)$ be the unrestricted complex, and let ... be the natural inclusion, inducing a homology map i_*. Liq...

conjecture | open | MathDB 页面 | 原文线索

V2-044201 | MATHDB-355477

Second-order asymptotic conjecture for hard-square Betti numbers

摘录 / EXCERPT: Second-order asymptotic conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-044202 | MATHDB-355489

The iterative Glaisher-map bijection conjecture for s-regular and t-distinct partitions

摘录 / EXCERPT: Iterative Glaisher-map conjecture. There exists an integer ℓ, depending on λ, such that $({\phi_s}\phi_t)^\ell$ maps λ to a unique s-distinct, t-regular pa...

conjecture | open | MathDB 页面 | 原文线索

V2-044203 | MATHDB-355491

Higher-dimensional connectivity and chromatic number conjecture for random distance graphs on spheres

摘录 / EXCERPT: Let $d\geq1$ and let α be an angular parameter. For a random ε-distance graph on the sphere, write $\mathcal{N}(\distgraph[n,\varepsilon])$ for its neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-044204 | MATHDB-355502

Shakan's conjecture on arbitrarily large improvements for sum-free subsets

摘录 / EXCERPT: Shakan's conjecture. For every positive integer S, there exists $N_S\in\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044205 | MATHDB-355503

Shakan's stronger conjecture on the maximum of the sum-free Fourier function

摘录 / EXCERPT: Shakan's stronger conjecture. For every positive integer S, there exists $N_S\in\mathbb{Z}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044206 | MATHDB-355506

Polynomial-form volume conjecture for partial permutohedra

摘录 / EXCERPT: Let ${\mathcal{P}}(m,n)$ be the partial permutohedron, and let $v(m,n)$ denote its normalized volume. For each relevant n and i, let $p_{n,i}(m)$ be a polynomial in m. Polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-044207 | MATHDB-355514

Duchêne-Rigo conjecture on invariant games for complementary Beatty sequences

摘录 / EXCERPT: Duchêne-Rigo conjecture. There exists an invariant game having

conjecture | open | MathDB 页面 | 原文线索

V2-044208 | MATHDB-355534

Hawkes's maximal Dyck path formula for rational q,t-Catalan polynomials

摘录 / EXCERPT: Hawkes's conjecture. The rational q,t-Catalan polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044209 | MATHDB-355535

Hawkes's degreewise maximal Dyck path formula

摘录 / EXCERPT: Hawkes's degreewise conjecture. For every d,

conjecture | open | MathDB 页面 | 原文线索

V2-044210 | MATHDB-355577

Extremizer conjecture for the T-symmetric univalent polynomial problem

摘录 / EXCERPT: Let T and N be positive integers, and consider the optimization problem over real-coefficient univalent polynomials in $\mathbb D=\{z:|z|<1\}$ with T-fold symmetry and degree...

conjecture | open | MathDB 页面 | 原文线索

V2-044211 | MATHDB-355579

Kalai's conjecture on the abundance of neighborly simplicial spheres

摘录 / EXCERPT: Let $s(d,n)$ denote the number of simplicial $(d-1)$-spheres with n labeled vertices, and let $\operatorname{sn}(d,n)$ denote the number of $\lfloor d/2\rfloor$-neighborly simpli...

conjecture | open | MathDB 页面 | 原文线索

V2-044212 | MATHDB-355584

Higher-dimensional periodic Costas array conjecture

摘录 / EXCERPT: Higher-dimensional periodic Costas array conjecture. If A is periodic Costas, then $n=2^k$. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-044213 | MATHDB-355622

Mayhew-Newman-Welsh-Whittle sparse paving asymptotic predominance conjecture

摘录 / EXCERPT: For each positive integer n, let $\operatorname{mat}(n)$ and $\operatorname{mat}'(n)$ be the numbers of matroids on $[n]$, respectively counted with labels and up to isomorphism....

conjecture | open | MathDB 页面 | 原文线索

V2-044214 | MATHDB-355623

Elementary split matroid asymptotic capture conjecture

摘录 / EXCERPT: For each positive integer n, let $\operatorname{sp}(n)$ and $\operatorname{es}(n)$ denote the numbers of sparse paving and elementary split matroids on $[n]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044215 | MATHDB-355624

De Loera-Haws-Köppe positivity conjecture for matroid Ehrhart polynomials

摘录 / EXCERPT: Let M be a matroid, and let $\mathscr{P}(\mathsf{M})$ be its matroid base polytope. Denote by $\operatorname{ehr}(\mathscr{P}(\mathsf{M}),t)$ its Ehrhart polynomial. Ehr...

conjecture | open | MathDB 页面 | 原文线索

V2-044216 | MATHDB-355625

Rota-Mason conjectures on unimodality and log-concavity of Whitney numbers

摘录 / EXCERPT: Let M be a loopless matroid of rank k and cardinality n, and let W_i be the number of flats of rank i. Whitney-number coefficient conjectures. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-044217 | MATHDB-355626

Hameister-Rao-Simpson Hilbert-series identity conjecture for uniform matroids

摘录 / EXCERPT: Let $\mathsf{U}_{k,n}$ be the uniform matroid of rank k on n elements, let $\mathrm{A}(\mathsf{U}_{k,n})$ be its Chow ring, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044218 | MATHDB-355627

Speyer's positivity conjecture for matroid g-polynomials

摘录 / EXCERPT: Let M be a matroid, and let $g_{\mathsf{M}}(t)$ denote its g-polynomial. g-polynomial positivity conjecture. The polynomial $g_{\mathsf{M}}(t)$ has positive coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-044219 | MATHDB-355647

Probabilistic $(1,3/2)$-decomposability conjecture for random matroids

摘录 / EXCERPT: Probabilistic $(1,3/2)$-decomposability conjecture. With high probability as $n\to\infty$, the random matroid on ground set $[n]$ is $(1,3/2)$-decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-044220 | MATHDB-355648

Conjectured classification of Erdős-deep triples of arithmetic progressions

摘录 / EXCERPT: Let $\mathbb{Z}_n$ be the cyclic group of integers modulo n. An Erdős-deep family is a family whose distance multiset has multiplicities precisely $1,2,\dots,k-1$ for some intege...

conjecture | open | MathDB 页面 | 原文线索

V2-044221 | MATHDB-355686

Optimality conjecture for two-step lazy transposition shuffles

摘录 / EXCERPT: Optimality conjecture for $(2,n)$-shuffles.

conjecture | open | MathDB 页面 | 原文线索

V2-044222 | MATHDB-355693

The sign-vector reconstruction conjecture for affine stresses

摘录 / EXCERPT: Sign-vector reconstruction conjecture. The $(i-1)$-skeleton of ∂P together with $\mathcal V_i(P)$ determines the combinatorial type of P, equivalently the entire abstra...

conjecture | open | MathDB 页面 | 原文线索

V2-044223 | MATHDB-355696

The balanced-monotone-family success-probability conjecture

摘录 / EXCERPT: Balanced-monotone-family conjecture. The success probability $p_{\mathrm{monotone}}(t)$ tends to 0 as t grows. This is stronger than the intersecting-family conjecture because...

conjecture | open | MathDB 页面 | 原文线索

V2-044224 | MATHDB-355701

Partial converse for monochromatic triangles in symmetric v_{3} configurations

摘录 / EXCERPT: Let a symmetric v_{3} configuration be a configuration whose lines are arranged so that each line contains three points and each point lies on three lines. Suppose such a configu...

conjecture | open | MathDB 页面 | 原文线索

V2-044225 | MATHDB-355722

Iosevich-Rudnev distance-set conjecture over finite fields

摘录 / EXCERPT: Let $\mathbb F_q^d$ be the d-dimensional vector space over the finite field $\mathbb F_q$, where q is odd. For $E\subseteq\mathbb F_q^d$, define the distance set by ... where ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-044226 | MATHDB-355745

Bonichon and Morel's enumeration conjecture for 3-permutations avoiding $(12,12)$ and $(231,312)$

摘录 / EXCERPT: A 3-permutation is a permutation of three sequences, and avoidance of a tuple of patterns means that no componentwise occurrence of any listed pattern appears. Let $(12,12)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044227 | MATHDB-355747

Defant-Kravitz's bound for tangled labelings of posets

摘录 / EXCERPT: Defant-Kravitz's conjecture. P has at most $(n-1)!$ tangled labelings.

conjecture | open | MathDB 页面 | 原文线索

V2-044228 | MATHDB-355748

Maximum degree of noninvertibility for promotion on posets

摘录 / EXCERPT: Maximum noninvertibility conjecture. For any poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-044229 | MATHDB-355749

Refined bound for tangled labelings of connected posets

摘录 / EXCERPT: Refined tangled-labeling bound. P has at most

conjecture | open | MathDB 页面 | 原文线索

V2-044230 | MATHDB-355750

Coloring probability inequality for minimal elements of posets

摘录 / EXCERPT: Coloring probability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044231 | MATHDB-355751

Tangled labelings versus quasi-tangled labelings for inflated rooted trees

摘录 / EXCERPT: Tangled-quasi-tangled inequality. The number of tangled labelings of P is less than or equal to the number of quasi-tangled labelings of P.

conjecture | open | MathDB 页面 | 原文线索

V2-044232 | MATHDB-355763

Corwin-Sheffield conjecture on extremal parabolic Brownian Gibbs line ensembles

摘录 / EXCERPT: Let $\mathcal{L} = (\mathcal{L}_1, \mathcal{L}_2, \ldots)$ be a line ensemble such that $\mathcal{L}(x)+x^2$ is stationary under deterministic horizontal shifts, $\mathcal{L}$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-044233 | MATHDB-355770

Antipodal code coefficient conjecture under the Assmus-Mattson condition

摘录 / EXCERPT: Antipodal code coefficient conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-044234 | MATHDB-355806

Thomas-Williams' rowmotion order conjecture for rational Tamari lattices

摘录 / EXCERPT: Thomas-Williams' conjecture. The operator

conjecture | open | MathDB 页面 | 原文线索

V2-044235 | MATHDB-355835

Equality of maximal in-degree and out-degree subposets in ν-Tamari lattices

摘录 / EXCERPT: Maximal-degree subposet conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044236 | MATHDB-355846

Weak Grünbaum triangle conjecture for proper circle arrangements

摘录 / EXCERPT: Let ${\cal A}$ be a simple digon-free arrangement of n pairwise intersecting circles, and let $p_3({\cal A})$ denote the number of triangular cells. Weak Grünbaum triangle conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-044237 | MATHDB-355847

The triangle lower-bound conjecture for intersecting pseudocircle arrangements

摘录 / EXCERPT: Let ${\cal A}$ be a simple arrangement of $n\ge 3$ pairwise intersecting pseudocircles, and let $p_3({\cal A})$ denote the number of triangular cells. Triangle lower-bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-044238 | MATHDB-355861

Conjecture on the nonbipartite Ramsey number for C_4-free color classes

摘录 / EXCERPT: Nonbipartite Ramsey conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044239 | MATHDB-355862

Conjecture on the bipartite Ramsey number for C_4-free color classes

摘录 / EXCERPT: Bipartite Ramsey conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044240 | MATHDB-355863

Conjecture on the rarity of positive indicators in even dimensions

摘录 / EXCERPT: Even-dimensional positive-indicator conjecture. The ratio of irreducible representations of J_n of dimension t with indicator $+1$ to the total number of irreducible representa...

conjecture | open | MathDB 页面 | 原文线索

V2-044241 | MATHDB-355864

Conjecture on exceptional positive-indicator representations in even dimensions

摘录 / EXCERPT: Even-dimensional exceptional-indicator conjecture. The ratio of irreducible representations with indicator $+1$ induced from permutations that do not share their orbit with an invo...

conjecture | open | MathDB 页面 | 原文线索

V2-044242 | MATHDB-355865

Conjecture on the rarity of nonzero indicators for fixed-dimensional irreducible representations

摘录 / EXCERPT: Nonzero-indicator rarity conjecture. The ratio of irreducible representations of J_n of dimension t with nonzero indicator to the total number of irreducible representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-044243 | MATHDB-355891

The self-similarity conjecture for extremal angle configurations

摘录 / EXCERPT: A point configuration $\mathcal P$ is self-similar if there is a point $A\in\mathcal P$ such that every angle formed by three points of $\mathcal P$ can also be formed with A as...

conjecture | open | MathDB 页面 | 原文线索

V2-044244 | MATHDB-355899

The Ramsey characterisation by monochromatic line configurations

摘录 / EXCERPT: Let $X\subset\mathbb{E}^d$ be a set. Call X Ramsey if for every natural number r there exists n such that every r-colouring of $\mathbb{E}^n$ contains a monochromatic copy...

conjecture | open | MathDB 页面 | 原文线索

V2-044245 | MATHDB-355900

The non-spherical obstruction conjecture for Euclidean Ramsey sets

摘录 / EXCERPT: Let $X\subset\mathbb{E}^d$ be a set, and call it non-spherical if it is not contained in the surface of a sphere of any dimension. Write ℓ_m for the relevant m-point line co...

conjecture | open | MathDB 页面 | 原文线索

V2-044246 | MATHDB-355904

The H-to-M conjecture for shuffle lattices

摘录 / EXCERPT: Let $\mathsf{Shuf}(m,n)$ be the shuffle lattice, and let $M_{m,n}(q,t)$ denote its M-triangle. Let $H(x,y)$ denote the H-triangle associated with the bubble lattice. For intege...

conjecture | open | MathDB 页面 | 原文线索

V2-044247 | MATHDB-355905

The explicit M-triangle conjecture for shuffle lattices

摘录 / EXCERPT: Let $\mathsf{Shuf}(m,n)$ be the shuffle lattice, and let $M_{m,n}(q,t)$ denote its M-triangle. For integers $m,n\geq 0$, the explicit M-triangle conjecture. ... This formula is con...

conjecture | open | MathDB 页面 | 原文线索

V2-044248 | MATHDB-355923

Geometric orbit-growth conjecture for Bulgarian Solitaire necklaces

摘录 / EXCERPT: Let P be a primitive necklace, and let $\mathcal{O}_{P^k}$ denote the Bulgarian Solitaire orbit parametrized by P^k. For a necklace P', say that P and P' are complementar...

conjecture | open | MathDB 页面 | 原文线索

V2-044249 | MATHDB-355934

Saxton's permanent submatrix conjecture

摘录 / EXCERPT: Let $B\in\mathbb{R}^{m\times n}$ be a matrix. For each $j\in[m]$, let the support of its jth column be $\{i\in[n]:B_{ij}\neq 0\}$. Saxton's conjecture. If, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-044250 | MATHDB-355947

Staircase heavy-chain conjecture for Grothendieck polynomial weights

摘录 / EXCERPT: Fix $w\in S_n$. A heavy climbing chain of w is a climbing chain together with its marked-link data $M(C)$; write its overlined and ordinary weights as...

conjecture | open | MathDB 页面 | 原文线索

V2-044251 | MATHDB-355962

Robust Bollobás-Eldridge-Catlin universality conjecture

摘录 / EXCERPT: Robust universality conjecture. For any $k\geq 2$, there exists a constant $C>0$ such that, for all $n\in\mathbb N$ and $p\geq Cp^*_{k+1}$, every graph G with...

conjecture | open | MathDB 页面 | 原文线索

V2-044252 | MATHDB-355980

Frankl's even-parity conjecture for t-intersecting k-Sperner families

摘录 / EXCERPT: Let n and t be integers with $n+t$ even and $n>t$, and let ${\mathcal F}\subseteq 2^{[n]}$ be a t-intersecting family, meaning that $|F\cap F'|\geq t$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-044253 | MATHDB-355981

Frankl's odd-parity extremal conjecture for t-intersecting k-Sperner families

摘录 / EXCERPT: Let n,t,k be positive integers, and let ${\mathcal B}(t,k)$ denote the family constructed in the source: it contains the $(n+t-1)/2$-subsets containing $[1,t]$, all sets whose si...

conjecture | open | MathDB 页面 | 原文线索

V2-044254 | MATHDB-355998

Equality of rational blowdown depth and combinatorial depth

摘录 / EXCERPT: Let $W_{p,q}(\mathbf{n})$ denote the minimal symplectic filling of $(L(p,q),\xi_{can})$ corresponding to $\mathbf{n}=(n_1,\ldots,n_k)$ in the parameterizing set...

conjecture | open | MathDB 页面 | 原文线索

V2-044255 | MATHDB-356004

The left LBS-tree edge-activity conjecture

摘录 / EXCERPT: Left LBS-tree edge-activity conjecture. For $0\leq k\leq n-1$, the number of left LBS trees with k edges of the form $(i+1,i)$ is equal to the number of left LBS trees with k e...

conjecture | open | MathDB 页面 | 原文线索

V2-044256 | MATHDB-356059

Conjecture on highly connected graphs realizing prescribed Ramsey-number growth

摘录 / EXCERPT: Prescribed-growth connectivity conjecture. For every $k\geqslant 2$ and every admissible function f, there is a sequence of graphs $(G_n)_{n\in\mathbb N}$ with $|G_n|=n$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-044257 | MATHDB-356068

Propp's benzel tiling enumeration conjecture

摘录 / EXCERPT: Propp's conjecture. The (a,b)-benzel has

conjecture | open | MathDB 页面 | 原文线索

V2-044258 | MATHDB-356071

The strut-side perimeter conjecture for convex polygons

摘录 / EXCERPT: Strut-side perimeter conjecture. The perimeter $L(P)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044259 | MATHDB-356089

Hadwiger's conjecture for oriented matroids

摘录 / EXCERPT: Let M be a loopless oriented matroid with no $M(K_4)$ minor. A Hadwiger conjecture for oriented matroids. M has a nowhere-zero 3-coflow. This is the first non-trivial open ca...

conjecture | open | MathDB 页面 | 原文线索

V2-044260 | MATHDB-356090

Positive-coline conjecture for gammoids

摘录 / EXCERPT: A positive coline of a matroid is a coline whose copoint partition has more singular classes than multiple classes. A gammoid is a matroid arising from a directed graph by the gamm...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044261 | MATHDB-356091

Positive-double-circuit conjecture for cosimple gammoids

摘录 / EXCERPT: Let M be a cosimple gammoid of corank at least two. A positive double circuit is a double circuit whose circuit partition has more singular classes than multiple classes. Positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-044262 | MATHDB-356097

The rooted-subtree enumeration conjecture for maximally extendable transfer systems

摘录 / EXCERPT: Let $n\geqslant 0$. A transfer system in $T(n)$ is maximally extendable if it has the maximal extension property described in the surrounding definitions, and let $\mathfrak{A}_m$...

conjecture | open | MathDB 页面 | 原文线索

V2-044263 | MATHDB-356102

Finite-difference sign conjecture for the broken k-diamond partition function

摘录 / EXCERPT: Finite-difference sign conjecture. For any $k\geq 1$ and $r\geq 1$, there exists a positive integer $n_k(r)$ such that, for $n\geq n_k(r)$,

conjecture | open | MathDB 页面 | 原文线索

V2-044264 | MATHDB-356111

Equilibrium-count conjecture for symmetric finite trails

摘录 / EXCERPT: Consider the Trail of Lost Pennies on the finite trail $\llbracket -j-1,k+1\rrbracket$ in its symmetric standard form. Finite-trail equilibrium-count conjecture. The number of time...

conjecture | open | MathDB 页面 | 原文线索

V2-044265 | MATHDB-356143

List steady-word conjecture for four-letter alphabets

摘录 / EXCERPT: List steady-word conjecture. There exists a steady word $W=w_1w_2\cdots w_n$ such that $w_i\in\mathcal{A}_i$ for every $i=1,2,\ldots,n$.

conjecture | open | MathDB 页面 | 原文线索

V2-044266 | MATHDB-356144

Quaternary bifurcate-chain conjecture

摘录 / EXCERPT: Quaternary bifurcate-chain conjecture. There exists an infinite sequence of quaternary bifurcate words $W_1,W_2,\ldots$ such that W_{i+1} is a single-letter extension of W_i fo...

conjecture | open | MathDB 页面 | 原文线索

V2-044267 | MATHDB-356145

Five-letter complete bifurcate-tree conjecture

摘录 / EXCERPT: Five-letter complete bifurcate-tree conjecture. There exists a complete bifurcate tree over an alphabet of size 5.

conjecture | open | MathDB 页面 | 原文线索

V2-044268 | MATHDB-356174

The unfair 0-1-polynomial conjecture

摘录 / EXCERPT: Let X,Y be discrete independent random variables with finite support in the natural numbers, and suppose that $Z=X+Y$ is uniform on its support. Equivalently, let $R(t)=P(t)Q(t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-044269 | MATHDB-356181

The universal admissible-set conjecture for cap sets

摘录 / EXCERPT: Let $I(m,w)$ denote the admissible sets used in the extended product construction for cap sets in $\mathbb{F}_3^n$, where m and w are positive integers with $m>w$. Universal ad...

conjecture | open | MathDB 页面 | 原文线索

V2-044270 | MATHDB-356182

Quadrangulation conjecture

摘录 / EXCERPT: Quadrangulation conjecture. There is an explicit bijection preserving genus and number of edges between rooted quadrangulations and rooted maps with certain decorations that proves...

conjecture | open | MathDB 页面 | 原文线索

V2-044271 | MATHDB-356183

Cichacz-Hinc conjecture on magic rectangle sets over finite abelian groups

摘录 / EXCERPT: Let $(\Gamma,+)$ be a finite abelian group, and let $\mathcal{G}$ be the set of finite abelian groups that either have odd order or contain more than one involution. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-044272 | MATHDB-356207

The frequency-parametrization conjecture for finite-rank Gelfand-Tsetlin type graphs

摘录 / EXCERPT: Frequency-parametrization conjecture. In the finite-rank case, these limits are finite for every i, and conversely, for every $\lambda=(\lambda_1,\ldots,\lambda_d)\in\mathcal{C}$...

conjecture | open | MathDB 页面 | 原文线索

V2-044273 | MATHDB-356212

Burstein's permutation enumeration conjecture for three pattern sets

摘录 / EXCERPT: Let $a(n)$ be defined by the unique formal power series solution ... to ... For a set of patterns T, let $\mathfrak{S}_n(T)$ denote the permutations of length n avoiding every patt...

conjecture | open | MathDB 页面 | 原文线索

V2-044274 | MATHDB-356229

Inverse-peak formula for permutations avoiding $2134\cdots m$

摘录 / EXCERPT: Inverse-peak enumeration conjecture. For all $m\geq3$ and $n\geq1$, the number of permutations in $\mathfrak{S}_{n}(2134\cdots m)$ whose inverse has no peaks is

conjecture | open | MathDB 页面 | 原文线索

V2-044275 | MATHDB-356252

Positivity conjecture for copartition parity differences

摘录 / EXCERPT: Let $a,b,m\in\mathbb{N}$ and $n\in\mathbb{N}_0$. Let $\mathrm{cp}_{a,b,m}^e(n)$ and $\mathrm{cp}_{a,b,m}^o(n)$ denote the numbers of (a,b,m)-copartitions of n with an even and...

conjecture | open | MathDB 页面 | 原文线索

V2-044276 | MATHDB-356254

The asymptotic enumeration conjecture for dope matrices

摘录 / EXCERPT: Asymptotic enumeration conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-044277 | MATHDB-356257

Alspach's conjecture on distinct nonzero partial sums in cyclic groups

摘录 / EXCERPT: Let $T\subseteq \mathbb{Z}_v\setminus\{0\}$ satisfy ... An ordering of the elements of T is simple when its partial sums are all distinct and non-zero. Alspach's conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-044278 | MATHDB-356258

Graham-Archdeacon conjecture on simple orderings of cyclic groups

摘录 / EXCERPT: Let $T\subseteq \mathbb{Z}_v\setminus\{0\}$. An ordering of the elements of T is simple when its partial sums are all distinct and non-zero. Graham-Archdeacon conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-044279 | MATHDB-356259

Costa-Mella-Pasotti-Pellegrini conjecture on zero-sum simple orderings

摘录 / EXCERPT: Let $(G,+)$ be an abelian group, and let T be a finite subset of $G\setminus\{0\}$ such that no 2-subset $\{x,-x\}$ is contained in T and ... An ordering of the elements of T...

conjecture | open | MathDB 页面 | 原文线索

V2-044280 | MATHDB-356261

The smaller-game conjecture for correlated equilibrium polytopes

摘录 / EXCERPT: Smaller-game conjecture. If P_G is not of maximal dimension, then there exists a $(2\times k)$-game G' with $k<n$ such that $P_{G'}$ has maximal dimension and P_G and...

conjecture | open | MathDB 页面 | 原文线索

V2-044281 | MATHDB-356269

Brettell-Chun-Goddyn-Whittle connectivity conjecture for matroids

摘录 / EXCERPT: Brettell-Chun-Goddyn-Whittle conjecture. There exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that if M is a $(2t-1)$-connected matroid with no circuits or cocircu...

conjecture | open | MathDB 页面 | 原文线索

V2-044282 | MATHDB-356270

Generalized connectivity conjecture for matroids with circuits and cocircuits of different cardinalities

摘录 / EXCERPT: Generalized connectivity conjecture. There exists a function $f: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that if M has no circuits of size at most $2s-1$, no cocircuits of size...

conjecture | open | MathDB 页面 | 原文线索

V2-044283 | MATHDB-356297

The logarithmic-length conjecture for binary words with a prescribed number of subsequences

摘录 / EXCERPT: Logarithmic-length conjecture. For every integer $n\geqslant 1$, there exists a binary word $\mathfrak{w}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044284 | MATHDB-356302

The morphic structure conjecture for L(2)

摘录 / EXCERPT: Let $L(w)$ denote the lexicographically least infinite word on $\mathbb{N}$ beginning with w whose only square factors are contained in the prefix w. Let $L(\varepsilon)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-044285 | MATHDB-356303

The explicit limit conjecture for L(2)

摘录 / EXCERPT: The explicit limit conjecture for $L(2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-044286 | MATHDB-356304

The conjecture for extensions beginning with n1 and n2

摘录 / EXCERPT: Let $L(w)$ denote the lexicographically least infinite word on $\mathbb{N}$ beginning with w whose only square factors are contained in the prefix w. The conjecture for extensi...

conjecture | open | MathDB 页面 | 原文线索

V2-044287 | MATHDB-356307

The fixed-size MSTD-MDTS conjecture for dihedral groups

摘录 / EXCERPT: Fixed-size MSTD-MDTS conjecture. For every integer $n\geq 3$ and every $m\leq 2n$, the collection $\mathcal S_{2n,m}$ has at least as many MSTD sets as MDTS sets.

conjecture | open | MathDB 页面 | 原文线索

V2-044288 | MATHDB-356308

The generalized dihedral MSTD-MDTS conjecture

摘录 / EXCERPT: Generalized dihedral MSTD-MDTS conjecture. There are more MSTD subsets of D than MDTS subsets of D.

conjecture | open | MathDB 页面 | 原文线索

V2-044289 | MATHDB-356309

A binomial-hypergeometric identity for graph-count coefficients

摘录 / EXCERPT: For integers $g\geq 1$, $\ell\geq 0$, and $j\geq 1$, let $\binom{n}{r}$ denote the binomial coefficient, let ${}_2F_1$ denote the Gauss hypergeometric function, and let the product...

conjecture | open | MathDB 页面 | 原文线索

V2-044290 | MATHDB-356345

Myers's conjecture on a three-color Rado number

摘录 / EXCERPT: Let $R_3(\mathcal{E})$ denote the smallest n such that every 3-coloring of $[n]$ induces a monochromatic solution to the equation $\mathcal{E}$. For $m\ge 3$, consider the equati...

conjecture | open | MathDB 页面 | 原文线索

V2-044291 | MATHDB-356347

Golowich's conjecture on regularity of large homogeneous equations

摘录 / EXCERPT: Let a linear homogeneous equation have nonzero integer coefficients, not all of the same sign. Golowich's conjecture. For each positive integer k there is an integer $m(k)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-044292 | MATHDB-356351

The rainbow Turán path conjecture

摘录 / EXCERPT: Rainbow Turán path conjecture. For all $\ell \geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-044293 | MATHDB-356362

Set-weighted chromatic symmetric function coefficient-sum conjecture

摘录 / EXCERPT: Let (G,ω) be a set-weighted graph with n vertices and total weight d, and write ... Let $\mu\leq d$ be an integer, viewed as a one-part partition, that is s-allowable in...

conjecture | open | MathDB 页面 | 原文线索

V2-044294 | MATHDB-356365

Existence conjecture for D-optimal difference families over abelian groups

摘录 / EXCERPT: Existence conjecture. For each D-optimal parameter set $(v;r,s;\lambda)$ there exists a difference family over some abelian group of order v having these parameters.

conjecture | open | MathDB 页面 | 原文线索

V2-044295 | MATHDB-356368

Cassaigne's two conjectures on the Tribonacci combinatorial-game sequences

摘录 / EXCERPT: Let $(X(n))_{n\geq 0}$ and $(Y(n))_{n\geq 0}$ be the sequences defined by the recursive combinatorial-game construction in the source. Cassaigne's conjectures. For every $n\geq 1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-044296 | MATHDB-356369

Rust's conjecture on the Tribonacci combinatorial-game sequences

摘录 / EXCERPT: Let $(X(n))_{n\geq 0}$ and $(Y(n))_{n\geq 0}$ be the sequences defined by the recursive combinatorial-game construction in the source. Rust's conjecture. For all $n\geq 1$, ... The c...

conjecture | refuted | MathDB 页面 | 原文线索

V2-044297 | MATHDB-356385

The St. Ives matching conjecture for online path Ramsey numbers

摘录 / EXCERPT: St. Ives matching conjecture. For all $k\geq 1$, there is a constant c_k such that for all $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044298 | MATHDB-356482

Birch's conjecture for partitions with intersecting convex hulls

摘录 / EXCERPT: Let r and d be positive integers, and let N denote the number of points under consideration. Birch's conjecture. Any $(r-1)(d+1)+1$ points in $\mathbb{R}^{d}$ can be partitio...

conjecture | solved | MathDB 页面 | 原文线索

V2-044299 | MATHDB-356483

Bárány-Larman conjecture on the optimal colored Tverberg number

摘录 / EXCERPT: Let $n(d,r)$ be the smallest integer such that, for every map $f:\Delta_{n-1}\rightarrow\mathbb{R}^{d}$ and every coloring of the vertices of Δ_{n-1} by $d+1$ colors, with e...

conjecture | open | MathDB 页面 | 原文线索

V2-044300 | MATHDB-356492

The asymptotically better-than-random strategy conjecture

摘录 / EXCERPT: Let K be the state group of a generalized spinning switches puzzle, and measure performance by the expected number of moves when the initial hidden state is chosen uniformly. Asy...

conjecture | open | MathDB 页面 | 原文线索

V2-044301 | MATHDB-356493

The minimal surjective-strategy conjecture for wreath products

摘录 / EXCERPT: Let $G\wr H$ be a wreath product with state group K, and call a surjective strategy minimal when it has the form $\{k_i\in K\}_{i=1}^{|K|-1}$. Minimal strategy conjecture. Whenev...

conjecture | open | MathDB 页面 | 原文线索

V2-044302 | MATHDB-356522

The non-nested and non-separated matching conjectures for ordered complete graphs

摘录 / EXCERPT: Let t,n be positive integers and set ... Consider a t-coloring of the edges of the ordered complete graph K_m, whose vertices are linearly ordered. A matching is a set of pairw...

conjecture | open | MathDB 页面 | 原文线索

V2-044303 | MATHDB-356541

Fairness conjecture for random Bergman games

摘录 / EXCERPT: Fairness conjecture for random Bergman games. As $N\to\infty$, under both analogous definitions of random Bergman games, the probabilities that Player 1 and Player 2 win are each...

conjecture | open | MathDB 页面 | 原文线索

V2-044304 | MATHDB-356542

Conjecture on determinant balance for odd-order complete non-ambiguous trees

摘录 / EXCERPT: Let $T(n)$ be the number of $n\times n$ complete non-ambiguous trees (CNATs). The determinant of a CNAT is either 1 or -1. Determinant-balance conjecture. If $n>1$ is odd, then...

conjecture | open | MathDB 页面 | 原文线索

V2-044305 | MATHDB-356543

Maximum-CNAT conjecture for permutations

摘录 / EXCERPT: Let π be a permutation on n letters, and let the number of CNATs generated by π be counted as in the source. Maximum-CNAT conjecture. The maximum number of CNATs generate...

conjecture | open | MathDB 页面 | 原文线索

V2-044306 | MATHDB-356544

Conjectured relation between permutations generating two and three CNATs

摘录 / EXCERPT: Let $b(n,k)$ be the number of distinct permutations of n letters that give rise to exactly k CNATs. CNAT-count relation conjecture. For all $k\geq 1$, $b(n,2)=b(n+1,3)$. The st...

conjecture | open | MathDB 页面 | 原文线索

V2-044307 | MATHDB-356564

Sun's general conjecture on linear and circular restricted sumsets

摘录 / EXCERPT: Let G be an additive group with $|G|>1$. For finite subsets $A_1,\ldots,A_n$ of G, define ... and ... Let $n>1$ and assume that $|A_i|>1$ for every $i=1,\ldots,n$. The Sun restrict...

conjecture | open | MathDB 页面 | 原文线索

V2-044308 | MATHDB-356566

Uniqueness of optimal covers for extremal matrices

摘录 / EXCERPT: Uniqueness conjecture. Every extremal matrix has a unique optimal cover.

conjecture | open | MathDB 页面 | 原文线索

V2-044309 | MATHDB-356567

Integer-multiple deficiency conjecture for optimal covers

摘录 / EXCERPT: Integer-multiple deficiency conjecture. Every entry of Λ is an integer multiple of δ.

conjecture | open | MathDB 页面 | 原文线索

V2-044310 | MATHDB-356568

Reciprocal-integer deficiency conjecture for extremal matrices

摘录 / EXCERPT: Reciprocal-integer deficiency conjecture. There exists an $m\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044311 | MATHDB-356569

Asymptotic enumeration conjecture for extremal and threshold matrices

摘录 / EXCERPT: Asymptotic enumeration conjecture. There is a constant $c=c(n)$ such that, as $d\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-044312 | MATHDB-356582

Groenland, Johnston, Radcliffe and Scott's conjecture for two-element shuffles

摘录 / EXCERPT: For integers $k\leq n$, let $U_k(n)$ be the minimum length of a sequence of independent lazy transpositions whose product maps the elements $1,\dots,k$ uniformly to the...

conjecture | open | MathDB 页面 | 原文线索

V2-044313 | MATHDB-356601

Uniqueness conjecture for the Paley-Wiener extremal problem below $p=1$

摘录 / EXCERPT: For $0<p<\infty$, consider the extremal problem defining the Paley-Wiener extremal function, and call a function attaining the extremum a solution. Uniqueness conjecture. The extre...

conjecture | open | MathDB 页面 | 原文线索

V2-044314 | MATHDB-356602

Evenness conjecture for Paley-Wiener extremal functions

摘录 / EXCERPT: Let $0<p<1$, and let a solution mean a function solving the extremal problem defined earlier in the paper. Evenness conjecture. Any solution of the extremal problem is even. The cl...

conjecture | open | MathDB 页面 | 原文线索

V2-044315 | MATHDB-356603

Uniform-discreteness conjecture for zeros of Paley-Wiener extremal functions

摘录 / EXCERPT: Let $0<p<\infty$, and let a solution mean a function solving the extremal problem defined earlier in the paper. Uniform-discreteness conjecture. The zero set of any solution of the...

conjecture | open | MathDB 页面 | 原文线索

V2-044316 | MATHDB-356604

Asymptotic-spacing conjecture for zeros of extremal functions

摘录 / EXCERPT: Assume the extremal problem has a unique solution φ_p for each $p>0$, and let $t_n=t_n(p)$, $n\geq1$, be the positive zeros of φ_p. Zero-spacing conjecture. ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-044317 | MATHDB-356605

Asymptotic conjecture for the first positive zero

摘录 / EXCERPT: Assume the extremal problem has a unique solution φ_p, and let $t_1=t_1(p)$ be its first positive zero. First-zero asymptotic conjecture. ... and ... The source presents these...

conjecture | open | MathDB 页面 | 原文线索

V2-044318 | MATHDB-356606

Uniqueness-set conjecture for the extremal zero set

摘录 / EXCERPT: Let φ_p be the unique solution of the extremal problem, let $\mathscr{Z}(\varphi_p)$ denote its zero set, and let PW^p be the Paley-Wiener space. A set is a uniqueness se...

conjecture | open | MathDB 页面 | 原文线索

V2-044319 | MATHDB-356607

Muckenhoupt-weight conjecture for extremal functions

摘录 / EXCERPT: Let $1<p<\infty$, let φ_p be the extremal function, and let $\mathscr{Z}(\varphi_p)$ be its zero set. Write $\operatorname{dist}(x,\mathscr{Z}(\varphi_p)\cup\{0\})$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-044320 | MATHDB-356608

Integrability conjecture for the Fourier transform of extremal functions

摘录 / EXCERPT: Let φ_p be the extremal function for the Paley-Wiener extremal problem, and let $\widehat{\varphi}_p$ denote its Fourier transform. Fourier-transform regularity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-044321 | MATHDB-356609

Shape conjecture for Fourier transforms of extremal functions

摘录 / EXCERPT: Let φ be a solution of the Paley-Wiener extremal problem, and let $\widehat{\varphi}$ denote its Fourier transform. Fourier-transform shape conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-044322 | MATHDB-356611

The evolution conjecture for gluing small Kerr black holes into initial data

摘录 / EXCERPT: Let $(\hat\gamma,\hat k)$ denote boosted subextremal Kerr black hole initial data with parameters $\hat{\mathfrak{m}}$ (mass) and $\hat{\mathfrak{a}}$ (specific angular momentum)....

conjecture | open | MathDB 页面 | 原文线索

V2-044323 | MATHDB-356612

The conjectured order of twins in ordered r-matchings

摘录 / EXCERPT: For each integer $r\geqslant2$, let $t_r^{\operatorname{match}}(n)$ denote the largest size of twins that are unavoidable in every ordered r-matching of size n. Here twins are...

conjecture | open | MathDB 页面 | 原文线索

V2-044324 | MATHDB-356624

Non-word-representability conjecture for simplified de Bruijn graphs

摘录 / EXCERPT: Non-word-representability conjecture. All simplified de Bruijn graphs $S(n,k)$ are non-word-representable for $n\geq 4$ and $k\geq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-044325 | MATHDB-356649

Conjecture on monochromatic Gower sum and product subspaces

摘录 / EXCERPT: Let X_k be the underlying set of the Gower subspaces, and let $X_k=\bigcup_{i=1}^r C_i$ be a finite partition. A Gower sum subspace is generated by a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-044326 | MATHDB-356664

Golomb-Taylor conjecture on singly periodic Costas sequences

摘录 / EXCERPT: Let a Costas sequence be a sequence whose difference triangle has no repeated entry in any row. A sequence is singly periodical when every circular shift of its terms is also a Cos...

conjecture | open | MathDB 页面 | 原文线索

V2-044327 | MATHDB-356665

Golomb-Moreno conjecture on circular Costas sequences

摘录 / EXCERPT: Let $\mathbb{Z}_n$ denote the cyclic group of order n. A circular Costas sequence is equivalent to an injective map $\varphi:\mathbb{Z}_n\to\mathbb{Z}_{n+1}$ whose difference map...

conjecture | open | MathDB 页面 | 原文线索

V2-044328 | MATHDB-356666

Ortiz-Ubarri et al.'s elementary-abelian codomain conjecture for circular Costas maps

摘录 / EXCERPT: Let G_1 and G_2 be groups, and let a circular Costas map be a map $\varphi:G_1\to G_2$ that is injective and whose nonzero difference maps are injective; call it standard when...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044329 | MATHDB-356668

Homology conjecture for affine and linear fixing cycles

摘录 / EXCERPT: Let X be a simplicial manifold. A fixing cycle is a $(3n-2)$-cycle $\phi\in Z_{3n-2}(Y,\pi^*({\mathcal D}))$ satisfying ... where n is the dimension of X, ${\mathcal D}$ is its...

conjecture | open | MathDB 页面 | 原文线索

V2-044330 | MATHDB-356681

Binary-one-prefix Euler-transform recurrence conjecture for Lyndon words

摘录 / EXCERPT: A Lyndon word over $\{0,1\}$ is a word that is the unique minimum among all of its rotations. For $k\geq2$, let $\alpha=(1,0,0,\dots,0)$ be the prefix of length k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044331 | MATHDB-356718

Non-partition-regularity of the four-term pattern over bounded-degree polynomial semirings

摘录 / EXCERPT: For $d\geq 1$, let P_d be the set of polynomials in countably many variables with nonnegative integer coefficients, constant term, and degree at most d in each variable. A fini...

conjecture | open | MathDB 页面 | 原文线索

V2-044332 | MATHDB-356719

Strict hierarchy conjecture for polynomial partition regularity

摘录 / EXCERPT: For $d\geq 1$, let P_d be the set of polynomials in countably many variables with nonnegative integer coefficients, constant term, and degree at most d in each variable. A patt...

conjecture | open | MathDB 页面 | 原文线索

V2-044333 | MATHDB-356720

Equivalence of partition regularity over polynomial and integer patterns

摘录 / EXCERPT: Let T denote the polynomial setting used in the paper, and let $\mathbb{N}$ denote the positive integers. A polynomial pattern $\mathcal{F}$ is partition regular over a domain if...

conjecture | open | MathDB 页面 | 原文线索

V2-044334 | MATHDB-356725

Mauk and Stahl's logarithmic face conjecture for complete graphs

摘录 / EXCERPT: Mauk and Stahl's conjecture. The expected number of faces in a random embedding of K_n is at most

conjecture | open | MathDB 页面 | 原文线索

V2-044335 | MATHDB-356727

Boolean characterization of minimally invariant car lengths with four cars

摘录 / EXCERPT: Let $\mathbf{y}=(y_1,y_2,y_3,y_4)\in\mathbb{N}^4$. A car-length vector is minimally invariant when it has the minimal invariance property defined in the paper. Four-car minimal-inv...

conjecture | open | MathDB 页面 | 原文线索

V2-044336 | MATHDB-356741

Bajnok's limit conjecture for cyclic-group independence numbers

摘录 / EXCERPT: Bajnok's limit conjecture. The value of

conjecture | open | MathDB 页面 | 原文线索

V2-044337 | MATHDB-356742

The classification of perfect independent-spanning sets

摘录 / EXCERPT: Let G be a finite group, let $A\subset G$, and suppose that A is both t-independent and s-spanning, with t even. Write $m=t/2$ and define the Delannoy numbers by ... A set...

conjecture | open | MathDB 页面 | 原文线索

V2-044338 | MATHDB-356743

Lev's restricted sumset lower-bound conjecture

摘录 / EXCERPT: Let G be an abelian group, let m be a positive integer, and let L denote the subgroup of involutions in G. Write $\rho\hat{\;} (G,m,2)$ for the minimum size of a restricted...

conjecture | open | MathDB 页面 | 原文线索

V2-044339 | MATHDB-356748

Classification conjecture for perfect restricted s-bases

摘录 / EXCERPT: Let G be a finite abelian group, and let $m\geq2$ and $s\geq2$ be integers. A perfect restricted s-basis of size m is a subset whose restricted sums of at most s elements r...

conjecture | open | MathDB 页面 | 原文线索

V2-044340 | MATHDB-356749

Classification conjecture for perfect spanning sets

摘录 / EXCERPT: Let G be a finite abelian group, and let A be a perfect spanning set of size m and radius s, meaning that every element of G has a unique representation as a signed sum o...

conjecture | open | MathDB 页面 | 原文线索

V2-044341 | MATHDB-356783

Polynomial-move conjecture for enlarging regular polygons in the grasshopper game

摘录 / EXCERPT: Let $N>6$ and consider a configuration of grasshoppers at the vertices of a regular N-gon, with legal moves given by the grasshopper-jumping game described in the paper. Polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-044342 | MATHDB-356803

Lyubeznik-Barile-Macchia rank conjecture for monomial ideals

摘录 / EXCERPT: Let R be a polynomial ring over a field $\boldsymbol{k}$, and let $I\subseteq R$ be a monomial ideal. Let $\mathcal{F}_L$ be a Lyubeznik resolution of R/I. A Barile-Macchia res...

conjecture | open | MathDB 页面 | 原文线索

V2-044343 | MATHDB-356804

Algorithmic Lyubeznik-Barile-Macchia conjecture

摘录 / EXCERPT: Let I be a monomial ideal, let $\mathcal{F}_L$ be a Lyubeznik resolution, and let Algorithm 1 be the trimming algorithm defined in the source. The resulting complex is compared...

conjecture | open | MathDB 页面 | 原文线索

V2-044344 | MATHDB-356821

The divergence conjecture for Fibonacci and arithmetic-progression sequences

摘录 / EXCERPT: Let F denote the Fibonacci numbers, let AP_F denote the associated arithmetic-progression sequence, and let $\Delta(L,k;r)$ and $n(AP_F,k;r)$ be the parameters defined in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-044345 | MATHDB-356833

Maximal non-attacking queens on four-sided hypercubes

摘录 / EXCERPT: Four-sided hypercube queen conjecture. The maximal non-attacking queen set problem on a d-dimensional hypercube of side length 4 has solution size

conjecture | open | MathDB 页面 | 原文线索

V2-044346 | MATHDB-356852

Gao-Kitaev's enumerative conjecture for length-5 pattern-avoiding permutations

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the set of permutations of $[n]=\{1,2,\ldots,n\}$, and let $\mathfrak{S}_n(P_1,\ldots,P_r)$ denote the permutations avoiding the patterns $P_1,\ldots,P_r$....

conjecture | open | MathDB 页面 | 原文线索

V2-044347 | MATHDB-356862

The equality of combinatorial and geometric refined BPS invariants

摘录 / EXCERPT: For the local $\mathbb{P}^2$, let $\widetilde{n}_d^{i,j}$ denote the combinatorial refined BPS invariants obtained from the Nekrasov partition function, and let $n_d^{i,j}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-044348 | MATHDB-356877

The regular matroid rainbow circuit conjecture

摘录 / EXCERPT: Let M be a simple regular matroid of rank $n-1$, and let c colour $E(M)$ with n colours, each colour class having size at least 2. Regular matroid rainbow circuit conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-044349 | MATHDB-356879

Haythorpe's minimum hamiltonian-cycle conjecture for regular graphs

摘录 / EXCERPT: Let d and n be integers with $d\ge 5$ and $n\ge d+3$, and let $f(n,d):=(d-1)^2[(d-2)!]^{\frac{n}{d+1}}$. A hamiltonian d-regular graph of order n is a graph containing a ha...

conjecture | open | MathDB 页面 | 原文线索

V2-044350 | MATHDB-356969

Asymptotic online size Ramsey conjecture for even cycles versus paths

摘录 / EXCERPT: Even-cycle online Ramsey conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044351 | MATHDB-356982

Weighted exchange-distance conjecture for matroid basis pairs

摘录 / EXCERPT: Weighted exchange-distance conjecture. The weighted exchange distance of $\mathbf{P}_1$ and $\mathbf{P}_2$ is at most

conjecture | open | MathDB 页面 | 原文线索

V2-044352 | MATHDB-357000

Cilleruelo-Serra-Wójtaszczyk's structural conjecture for large Sidon systems

摘录 / EXCERPT: Cilleruelo-Serra-Wójtaszczyk's conjecture. One can remove $o(n^{k-1})$ sets from $\mathcal{F}$ to obtain a Sidon system. In particular, starting with any family satisfying the disp...

conjecture | open | MathDB 页面 | 原文线索

V2-044353 | MATHDB-357010

Whittington-Soteros asymptotic conjecture for polyominoes

摘录 / EXCERPT: Let $P(n)$ denote the number of polyominoes with n cells, and let ... be Klarner's constant. Whittington-Soteros conjecture. There exist constants A,T such that ... This is a widel...

conjecture | open | MathDB 页面 | 原文线索

V2-044354 | MATHDB-357011

Monotonicity conjecture for the inconstructible-polyomino ratio

摘录 / EXCERPT: Let $P(n)$ be the number of polyominoes with n cells. A polyomino is constructible if it is the concatenation of two polyominoes of smaller sizes, and inconstructible otherwise;...

conjecture | open | MathDB 页面 | 原文线索

V2-044355 | MATHDB-357012

Monotonicity conjecture for successive polyomino counts

摘录 / EXCERPT: Let $P(n)$ denote the number of polyominoes with n cells. Successive-ratio monotonicity conjecture. The sequence ... is increasing. The conjecture is motivated by the available val...

conjecture | open | MathDB 页面 | 原文线索

V2-044356 | MATHDB-357028

Chung et al.'s multivariate central limit conjecture for cycle counts

摘录 / EXCERPT: For fixed d, let ${\mathcal{F}}_d(n)$ be the set of permutations of $\{1,\dots,n\}$ satisfying $i-d\leq\sigma(i)\leq i+1$ for every i, and let the number of k-cycles in such...

conjecture | open | MathDB 页面 | 原文线索

V2-044357 | MATHDB-357034

The third law of black hole thermodynamics

摘录 / EXCERPT: A subextremal black hole is a black hole that is not extremal. The third law of black hole thermodynamics. A subextremal black hole cannot become extremal in finite time by any con...

conjecture | open | MathDB 页面 | 原文线索

V2-044358 | MATHDB-357043

Conrey-Gabbard-Grant-Liu-Morrison conjecture on quasirandom tournaments from random dice

摘录 / EXCERPT: Let a random n-sided die in the multiset model be a uniformly chosen multiset of n positive integers between 1 and n whose sum is $n(n+1)/2$. Let T be a tournament with v...

conjecture | open | MathDB 页面 | 原文线索

V2-044359 | MATHDB-357063

Eight-face conjecture for facets of the Newton polytope

摘录 / EXCERPT: Eight-face conjecture. The facet of $\mathbf{N}_{3,n}$ minimizing γ_{j_1,j_2,j_3} has exactly eight codimension-3 faces combinatorially isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-044360 | MATHDB-357064

General factorization conjecture for cyclic block partitions

摘录 / EXCERPT: Block-partition factorization conjecture. There is an ordering $(\mathbf{T}_1,\ldots,\mathbf{T}_{(d-1)(k-1)-1},\mathbf{S})$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-044361 | MATHDB-357094

Resolvable-derived-STS conjecture for SQS(n), n congruent to 4 modulo 6

摘录 / EXCERPT: Resolvable-derived-STS conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044362 | MATHDB-357095

Chromatic-number conjecture for derived STS of SQS(n), n congruent to 2 modulo 6

摘录 / EXCERPT: Chromatic-number conjecture for derived STS. If

conjecture | open | MathDB 页面 | 原文线索

V2-044363 | MATHDB-357132

Periodic density conjecture for directed sequences

摘录 / EXCERPT: Let T be an infinite periodic sequence over the alphabet $\{1,2\}$, and let $\mathcal{O}_T$ be its directed sequence. Denote by $d_1(T)$ and $d_1(\mathcal{O}_T)$ the densities of...

conjecture | open | MathDB 页面 | 原文线索

V2-044364 | MATHDB-357134

Lexicographic initial-segment conjecture for Frankl families

摘录 / EXCERPT: Lexicographic initial-segment conjecture. For fixed $n>k\geq3$, if $[S_m]$ is an FC-family for some positive integer m and has universe size n, then

conjecture | open | MathDB 页面 | 原文线索

V2-044365 | MATHDB-357152

Cube-cover conjecture for diagonalization games

摘录 / EXCERPT: Let n and m satisfy $2^{n-1}+1\leq m<2^n$, and set ... For a nonempty subset $J\subseteq[n]$, a J-cube is the set of binary vectors obtained by fixing all coordinates outside...

conjecture | open | MathDB 页面 | 原文线索

V2-044366 | MATHDB-357153

The cubic-scale diameter conjecture for 4-configuration graphs

摘录 / EXCERPT: Cubic-scale diameter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044367 | MATHDB-357167

The pinnacle-set enumeration conjecture for signed permutations

摘录 / EXCERPT: Let $T(n,k)$ be the sequence defined by ... and, for $0<k<n$, ... For each positive integer n, let APS^B_n denote the set of admissible pinnacle sets for signed permutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-044368 | MATHDB-357246

The antichain trace conjecture

摘录 / EXCERPT: The antichain trace conjecture. Let k be a non-negative integer and let $n\geq 2k$. If $\mathcal{F}\subset 2^{[n]}$ is an antichain satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-044369 | MATHDB-357250

The paperfolding sequence Dyck-factor length conjecture

摘录 / EXCERPT: Paperfolding sequence Dyck-factor conjecture. The paperfolding sequence has a Dyck factor of length n if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-044370 | MATHDB-357261

Maximization conjecture for nonnegative signed permutation paths

摘录 / EXCERPT: For $x\in \mathbb{R}_{>0}^n$, define ... Thus $A(x)$ consists of signed permutations whose partial sums are nonnegative. The maximization conjecture. The quantity $|A(x)|$ is maximiz...

conjecture | open | MathDB 页面 | 原文线索

V2-044371 | MATHDB-357265

The bound on lengths of perfectly clustering words in Gessel's map

摘录 / EXCERPT: Let $n\geqslant 2$ and let $(\alpha_2,\ldots,\alpha_n)$ be an $(n-1)$-tuple of nonnegative integers. Let Φ be the map from words to multisets of conjugacy classes of primitive...

conjecture | open | MathDB 页面 | 原文线索

V2-044372 | MATHDB-357268

Ruzsa's genus conjecture

摘录 / EXCERPT: Consider a linear equation ... where $a_1,\dots,a_k$ are integers. The equation has genus one if ... and $\sum_{i\in T}a_i\neq 0$ for every nonempty proper subset $T\subsetneq [k]$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-044373 | MATHDB-357279

Asymptotic unit hitting time conjecture for primitive geometric lattices

摘录 / EXCERPT: Let $L^{(n)}$ be a sequence of primitive geometric lattices, meaning geometric lattices that cannot be written as products of smaller geometric lattices, and suppose that ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044374 | MATHDB-357281

The combinatorial Jacobian conjecture

摘录 / EXCERPT: Fix an integer $d\geq 3$. For each $n\geq 1$, let $H=(H_{i,\alpha}:i\in[n],\lvert\alpha\rvert=d)$ be a collection of complex numbers, and let $\Psi_{i,j}^{\alpha}(H)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044375 | MATHDB-357282

The subtree shuffling conjecture

摘录 / EXCERPT: For integers $d\geq 2$ and $p\geq 1$, let $\mathcal{C}_k^{(d)}$ be the set of d-Catalan trees with k internal vertices, and let a length-p shuffle class be the class obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-044376 | MATHDB-357283

The shuffle-chain conjecture

摘录 / EXCERPT: Fix integers $d\geq 2$ and $p\geq 1$. Let a p-shuffle chain be the tree-valued Markov chain described in the paper, whose transitions rearrange subtrees branching from a chosen l...

conjecture | open | MathDB 页面 | 原文线索

V2-044377 | MATHDB-357284

Singer's spanning conjecture for shuffle classes

摘录 / EXCERPT: Fix integers $d\geq 2$ and $p\geq 1$. Let $\mathcal{C}_k^{(d)}$ be the set of d-Catalan trees with k internal vertices, and let the length-p shuffle classes be the classes de...

conjecture | open | MathDB 页面 | 原文线索

V2-044378 | MATHDB-357311

The conjecture on J-Artin monomials for superspace coinvariants

摘录 / EXCERPT: J-Artin monomial conjecture. For every subset $J\subseteq[n]$, the J-Artin monomials descend to a basis of

conjecture | open | MathDB 页面 | 原文线索

V2-044379 | MATHDB-357312

The Hard Lefschetz conjecture for the superspace colon quotients

摘录 / EXCERPT: Hard Lefschetz conjecture. For every $J\subseteq[n]$, the quotient ring A_J satisfies the Hard Lefschetz property.

conjecture | open | MathDB 页面 | 原文线索

V2-044380 | MATHDB-357328

Exponential induced size-Ramsey conjecture for odd cycles

摘录 / EXCERPT: Let C_n be the cycle on n vertices, and let $\hat{r}^k_{\operatorname{ind}}(C_n)$ denote the smallest number of edges in a graph whose every k-coloring contains a monochromat...

conjecture | open | MathDB 页面 | 原文线索

V2-044381 | MATHDB-357329

Exponential gadget conjecture for induced odd cycles

摘录 / EXCERPT: Let k be a positive integer. Exponential gadget conjecture. There is a graph G with $e^{O(k)}$ edges such that every k-coloring of its edges contains a monochromatic odd cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-044382 | MATHDB-357334

Day-Falgas-Ravry-Treglown multigraph extremal product conjecture

摘录 / EXCERPT: Day-Falgas-Ravry-Treglown conjecture. For all integers a,r,s,d with $a,r\geq 1$, $d\in[0,a-1]$, $s\geq(r-1)(d+1)+2$, and all sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-044383 | MATHDB-357365

Rahimpour's determinantal identity for Pascal arrays

摘录 / EXCERPT: Rahimpour's determinantal identity. The determinant of the $n\times n$ square sub-array in the Pascal array whose left edge lies in the k-th column equals the determinant of the...

conjecture | open | MathDB 页面 | 原文线索

V2-044384 | MATHDB-357379

Unbounded envelope heights for finite subsets

摘录 / EXCERPT: Let $\str H$ and $\str G$ be the structures in the source, let $\varphi\colon\str H\to\str G$ be an embedding, and let an envelope of a finite subset have the source's valuation-tr...

conjecture | open | MathDB 页面 | 原文线索

V2-044385 | MATHDB-357380

Infinite big Ramsey degrees with infinitely many relations of one arity

摘录 / EXCERPT: Let L be a relational language with finitely many unary relations and infinitely many relations of some arity $a\geq 2$. Let $\str H$ be a countable unrestricted L-structure re...

conjecture | open | MathDB 页面 | 原文线索

V2-044386 | MATHDB-357394

Reversal invariance conjecture for the Thue-Morse sequence in base $3/2$

摘录 / EXCERPT: Let $x_{3/2}$ be the binary Thue-Morse sequence in base $3/2$. For a finite word $w=w_1\dots w_m$, write $\overleftarrow{w}=w_m\dots w_1$ for its reversal. Reversal invariance conj...

conjecture | open | MathDB 页面 | 原文线索

V2-044387 | MATHDB-357395

Uniform recurrence conjecture for the Thue-Morse sequence in base $3/2$

摘录 / EXCERPT: Let $x_{3/2}$ be the binary Thue-Morse sequence in base $3/2$. A word is uniformly recurrent in $x_{3/2}$ when every occurrence of it is followed by occurrences with bounded gaps....

conjecture | open | MathDB 页面 | 原文线索

V2-044388 | MATHDB-357396

Reversal invariance conjecture for word frequencies in the Thue-Morse sequence in base $3/2$

摘录 / EXCERPT: Let $x_{3/2}$ be the binary Thue-Morse sequence in base $3/2$, and let $\mu[w]$ denote the frequency of a finite word w occurring in it. For a word $w=w_1\dots w_m$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-044389 | MATHDB-357415

Eeta-win enumeration in the Young-Fibonacci lattice

摘录 / EXCERPT: Let $f_0=f_1=1$ and $f_n=f_{n-1}+f_{n-2}$ for $n\geq 2$, where f_n is the number of elements of rank n in the Young-Fibonacci lattice. Young-Fibonacci Eeta-win conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-044390 | MATHDB-357440

Weakly connected maps conjecture for non-diagonal conformal-bootstrap spectra

摘录 / EXCERPT: Consider an n-point function of diagonal and non-diagonal fields on a Riemann surface of genus g. Let $\mathcal{M}^c_{g,n}(r_i)$ denote the set of weakly connected maps with th...

conjecture | open | MathDB 页面 | 原文线索

V2-044391 | MATHDB-357441

Map-counting conjecture with diagonal or degenerate intermediate fields

摘录 / EXCERPT: Let an n-point function of diagonal and non-diagonal fields lie on a Riemann surface of genus g. In each channel, consider conformal-bootstrap spectra containing all allowed no...

conjecture | open | MathDB 页面 | 原文线索

V2-044392 | MATHDB-357442

Critical loop-model correlation-function basis conjecture

摘录 / EXCERPT: Let M be a combinatorial map and W a choice of loop weights, and write $Z_{M,W}$ for the associated loop-model correlation function. Assume the critical limit exists. The corre...

conjecture | open | MathDB 页面 | 原文线索

V2-044393 | MATHDB-357444

Normality conjecture for the binary sequence of binomial set partition values

摘录 / EXCERPT: Let $a(n)$ be the integer sequence from Theorem, and define its binary reduction by ... A binary sequence is normal if every finite binary string of length $\ell\geq 1$ occurs in it...

conjecture | open | MathDB 页面 | 原文线索

V2-044394 | MATHDB-357466

Non-almost-realizability conjecture for the partition numbers

摘录 / EXCERPT: Partition-number conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-044395 | MATHDB-357499

The limiting function conjecture for minimal H-factors

摘录 / EXCERPT: Let H be a graph, let d^* denote its maximal density parameter, and let $F_H(alpha n,n)$ be the minimal weight of an $(alpha,H)$-factor in the random weighted complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-044396 | MATHDB-357501

The generalized approximate union-closed sets conjecture

摘录 / EXCERPT: Generalized approximate union-closed sets conjecture. There is an element contained in a $\psi_k-\delta$ fraction of the sets in $\mathcal F$. For $k=2$, this specializes to the fi...

conjecture | open | MathDB 页面 | 原文线索

V2-044397 | MATHDB-357547

Conjecture on the string attractor size of simple-Parry automatic sequences

摘录 / EXCERPT: Let c be a finite word satisfying condition (WH), and assume that one of the equivalent assertions in Proposition 4 holds. Let $\mathbf{u}$ be the associated simple-Parry automat...

conjecture | open | MathDB 页面 | 原文线索

V2-044398 | MATHDB-357559

Minkowski decomposition conjecture for flexible polytopes

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ be a flexible polytope, meaning that it admits a continuous deformation preserving its combinatorial type and edge lengths that is not a continuous isome...

conjecture | open | MathDB 页面 | 原文线索

V2-044399 | MATHDB-357638

Beck's conjecture on parts in odd and distinct partitions

摘录 / EXCERPT: Let n be a nonnegative integer. Count the total number of parts across all partitions of n whose parts are odd, and separately the total number of parts across all partitions o...

conjecture | open | MathDB 页面 | 原文线索

V2-044400 | MATHDB-357657

Non-extendability conjecture for signotopes of even rank

摘录 / EXCERPT: Let an r-signotope be a signotope of rank r, and call it 2-extendable if every pair of prescribed crossing points can be contained in an extending element of an extension to an...

conjecture | open | MathDB 页面 | 原文线索

V2-044401 | MATHDB-357658

The positroid Ehrhart-positivity conjecture

摘录 / EXCERPT: Positroid Ehrhart-positivity conjecture. Every positroid is Ehrhart positive.

conjecture | open | MathDB 页面 | 原文线索

V2-044402 | MATHDB-357664

The asymptotic poset Ramsey conjecture for fixed forbidden posets

摘录 / EXCERPT: Fixed-poset asymptotic conjecture. For every fixed poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-044403 | MATHDB-357665

The asymptotic Boolean-lattice Ramsey conjecture

摘录 / EXCERPT: Boolean-lattice asymptotic conjecture. For every fixed $m\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-044404 | MATHDB-357670

Uniqueness conjecture for the six-permutation quasirandom-forcing combination

摘录 / EXCERPT: Let ρ^* be the quasirandom-forcing linear combination of six permutations ... A linear combination is quasirandom-forcing if it has the quasirandomness-forcing property studied...

conjecture | open | MathDB 页面 | 原文线索

V2-044405 | MATHDB-357671

Lower bound for quasirandom-forcing linear combinations

摘录 / EXCERPT: A linear combination of permutation patterns is quasirandom-forcing if it has the quasirandomness-forcing property studied in the paper. The number of terms means the number of per...

conjecture | open | MathDB 页面 | 原文线索

V2-044406 | MATHDB-357735

The interval-maximization conjecture for translational tilings

摘录 / EXCERPT: Let $\alpha,n,d\in\mathbb{Z}^{+}$, let $C\subset\mathbb{Z}^{d}$ satisfy $|C|=n$, and let $\mathcal{T}(\alpha,C)$ denote the set of canonical translational tilings of C by tiles o...

conjecture | open | MathDB 页面 | 原文线索

V2-044407 | MATHDB-357762

Cooperative painting conjecture for systems of matroids

摘录 / EXCERPT: Let $M_1,M_2,\ldots,M_t$ be a system of k-colorable matroids on the same ground set E. In round i, Ben chooses a subset $B_i\subseteq E$, and Ann colors a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-044408 | MATHDB-357764

Extremal convex function conjecture for normalized Donaldson-Futaki invariant

摘录 / EXCERPT: Let P be a polytope, let u_T^{can} be the canonical family, and define the convex function ... Let $\mathrm{DF}(q)$ denote the Donaldson-Futaki invariant, let...

conjecture | open | MathDB 页面 | 原文线索

V2-044409 | MATHDB-357766

Conjecture that two paw graphs are common

摘录 / EXCERPT: Let P be the paw graph, obtained from K_3 by adding a pendant edge. A graph is common if the number of monochromatic labelled copies in a red/blue colouring of a large complete...

conjecture | open | MathDB 页面 | 原文线索

V2-044410 | MATHDB-357788

The extremal-rank mutation conjecture for nondegenerate Jacobi-finite QPs

摘录 / EXCERPT: Let $(Q,\mathcal{S})$ be a nondegenerate Jacobi-finite QP. For a pair (δ,ϵ) of δ-vectors of $(Q,\mathcal{S})$, say that the pair has completely extremal rank w...

conjecture | open | MathDB 页面 | 原文线索

V2-044411 | MATHDB-357795

Sleator-Tarjan-Thurston tet-volume conjecture for sphere triangulations

摘录 / EXCERPT: Sleator-Tarjan-Thurston tet-volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044412 | MATHDB-357814

The triangle-edge dichotomy conjecture for saturation numbers in random graphs

摘录 / EXCERPT: Triangle-edge dichotomy conjecture. Let $0<p<1$ be a constant. If every edge of F belongs to a triangle in F, then, with high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-044413 | MATHDB-357834

Braid-arrangement minimizer conjecture for matroid fans

摘录 / EXCERPT: Let M be a loopless matroid on a finite ground set E, and let $4\Sigma(M) \subseteq \mathbb{R}^E$ be its matroid fan. For a rational subspace $R \subseteq \mathbb{R}^E$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-044414 | MATHDB-357850

Roudneff's conjecture on complete cells in pseudohyperplane arrangements

摘录 / EXCERPT: Let $H(d,n)$ be an arrangement of n pseudohyperplanes in the real projective space $\mathbb{P}^d$, and call a d-cell complete if it is bounded by every pseudohyperplane. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-044415 | MATHDB-357870

Non-existence of primitive formally dual subsets of size 8 in the group \(\mathbb{Z}_8^2\)

摘录 / EXCERPT: Let $G=\mathbb{Z}_8^2$ and let a subset $S\subset G$ be primitive if it is not contained in a proper coset of a subgroup of G and generates G. A subset is formally dual if ther...

conjecture | open | MathDB 页面 | 原文线索

V2-044416 | MATHDB-357871

The online Ramsey number conjecture for cycles of fixed odd length

摘录 / EXCERPT: Online cycle Ramsey conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044417 | MATHDB-357880

Rigidity realization conjecture for reduced bipartized k-triangulations

摘录 / EXCERPT: Rigidity realization conjecture. Reduced bipartized k-triangulations of the n-gon are bases in the bar-and-joint rigidity matroid of these generic points.

conjecture | open | MathDB 页面 | 原文线索

V2-044418 | MATHDB-357881

Arbitrary-position rigidity conjecture for bipartized 2-triangulations

摘录 / EXCERPT: Arbitrary-position rigidity conjecture. Reduced bipartized 2-triangulations are bases in the bipartite bar-and-joint rigidity matroid of these points.

conjecture | open | MathDB 页面 | 原文线索

V2-044419 | MATHDB-357903

Odd-versus-distinct partition hook bias conjecture

摘录 / EXCERPT: Odd-versus-distinct hook bias conjecture. For every integer $t\geq 2$, there exists an integer N_t such that for all $n>N_t$, $a_t(n)\geq b_t(n)$. The conjectured values are

conjecture | open | MathDB 页面 | 原文线索

V2-044420 | MATHDB-357920

The greedy Tamari interval-constellation bijection conjecture

摘录 / EXCERPT: Let m be a positive integer. Consider greedy m-Tamari intervals whose maximal element has n_i ascents of length i for each $i\ge 1$, including a first ascent of length...

conjecture | open | MathDB 页面 | 原文线索

V2-044421 | MATHDB-357950

Random-walk super-coupon collection conjecture

摘录 / EXCERPT: Let the universe consist of n coupons, and let H_t be an r-subset drawn at round t. Starting from H_t, form H_{t+1} by choosing a uniformly random coupon in H_t and r...

conjecture | open | MathDB 页面 | 原文线索

V2-044422 | MATHDB-357952

A linear stability conjecture for small-doubling sets

摘录 / EXCERPT: Let $k,d\in\mathbb{N}$ and let $\epsilon>0$. Linear stability conjecture. There is $t_{k,d,\epsilon}>0$ such that, if $A\subset\mathbb{R}^k$ satisfies ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-044423 | MATHDB-357953

A convex approximation conjecture for near-minimizers of the Brunn-Minkowski inequality

摘录 / EXCERPT: Let $\Delta>0$ be an absolute constant. Convex approximation conjecture. If $\delta<\Delta$ and $A\subset\mathbb{R}^k$ satisfies ... then there is a convex set $K\subset\mathbb{R}^k$...

conjecture | open | MathDB 页面 | 原文线索

V2-044424 | MATHDB-358002

Bergeron's Gaussian-polynomial inequality conjecture

摘录 / EXCERPT: Let a,b,c,d be positive integers satisfying ... The Gaussian polynomial $\binom{n}{k}_q$ is the q-analogue of the binomial coefficient, and $f(q)\geq g(q)$ means that $f(q)-g(q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-044425 | MATHDB-358003

Zanello's unimodality strengthening of Bergeron's Gaussian inequality

摘录 / EXCERPT: Zanello's conjecture. Its coefficients are non-negative and unimodal. The source describes this as a strengthening of the preceding Gaussian inequality and notes that Zanello treat...

conjecture | open | MathDB 页面 | 原文线索

V2-044426 | MATHDB-358004

The beta-conjecture for Gaussian-polynomial inequalities

摘录 / EXCERPT: Let a and b be integers with $0<a<b$, and let $\beta\geq1$. The β-conjecture. One has ... where the inequality is coefficientwise for polynomials in q. This is the three-...

conjecture | open | MathDB 页面 | 原文线索

V2-044427 | MATHDB-358005

A coefficientwise Gaussian-polynomial product inequality

摘录 / EXCERPT: Let a and b be integers with $0\leq k\leq a<b$. The codicil conjecture. The following equivalent-looking coefficientwise inequalities hold: ... or ... Here $\binom{n}{k}_q$ denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-044428 | MATHDB-358026

The Lovász-form shadow inequality conjecture

摘录 / EXCERPT: Let k be a positive integer and let $x,y,z\real numbers$ satisfy ... with $y\ge x-1$. Lovász-form shadow inequality conjecture. One should have ... This is proposed as an analogue of...

conjecture | open | MathDB 页面 | 原文线索

V2-044429 | MATHDB-358034

The prism minimizer conjecture for the Martin invariant of 3-regular graphs

摘录 / EXCERPT: Let G be a cyclically 4-connected 3-regular graph with $2\ell+2\geq8$ vertices. Let $C_{\ell+1}$ be the cycle with $\ell+1$ sides, let $Y_{\ell+1}=K_2\times C_{\ell+1}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-044430 | MATHDB-358043

Irreducibility and normalized factorization conjecture for the coefficients of the Painlevé expansion

摘录 / EXCERPT: Let $a_0 e0$, let $ u_3(n+1)$ denote the exponent of 3 in $n+1$, and let $|n!|_3=3^{- u_3(n!)}$ be the 3-adic absolute value of $n!$. For each $n e\mathbb{N}$, write the coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-044431 | MATHDB-358045

Parity and extremal-coefficient conjecture for the asymptotic expansion

摘录 / EXCERPT: Let $a_{k,j}$ be the coefficients in the asymptotic expansion indexed by $k\in\mathbb{N}$ and $j\in\mathbb{Z}$ with $|j|\leqslant k$. Parity and extremal-coefficient conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-044432 | MATHDB-358046

Closed-form conjecture for the near-extremal asymptotic coefficients

摘录 / EXCERPT: Let $k\geqslant4$, and let $a_{k,j}$ be the coefficients in the asymptotic expansion, with parameters α, $\varkappa$, ε, b, and $a_{1,\pm1}$ as defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-044433 | MATHDB-358054

Characterization of critical positions in game NIM(n,k)

摘录 / EXCERPT: Characterization conjecture. The displayed inequalities characterize the m-critical positions of NIM(n,k).

conjecture | open | MathDB 页面 | 原文线索

V2-044434 | MATHDB-358056

Symmetry conjecture for odd-sized modular sum-difference-closed subsets

摘录 / EXCERPT: Let e be a positive integer, and let $S\subseteq\{0,1,\ldots,e-1\}$. Assume that for every $x,y\in S$, either $x+y\bmod e\in S$ or $x-y\bmod e\in S$. The odd-subset symmetry conj...

conjecture | open | MathDB 页面 | 原文线索

V2-044435 | MATHDB-358084

Schelp's minimum-degree Ramsey conjecture for paths

摘录 / EXCERPT: Let P_n denote the path on n vertices, and let $r(P_n,P_n)$ be its two-color Ramsey number. Set ... where n is sufficiently large. A graph arrows (P_n,P_n) when every 2-edge-...

conjecture | open | MathDB 页面 | 原文线索

V2-044436 | MATHDB-358093

Injectivity of the integer point transform without dilation

摘录 / EXCERPT: Integer point transform injectivity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044437 | MATHDB-358132

Mihail-Vazirani conjecture on rapid mixing of the matroid basis-exchange walk

摘录 / EXCERPT: Let $M=(X,\mathcal{I})$ be a matroid, and let its bases be its maximal independent sets. The base-exchange walk is the random walk on the bases of M that removes a uniformly chos...

conjecture | open | MathDB 页面 | 原文线索

V2-044438 | MATHDB-358147

Connectivity conjecture for local dimer dynamics in three dimensions

摘录 / EXCERPT: Let $\mathbf{n}$ be a 3-dimensional shape, and let $\mathcal{D}_3(\mathbb{Q}_{\mathbf{n}}^3)$ denote the graph of dimer configurations on $\mathbb{Q}_{\mathbf{n}}^3$ connected by...

conjecture | open | MathDB 页面 | 原文线索

V2-044439 | MATHDB-358153

The diversity bound for large intersecting families

摘录 / EXCERPT: Let $\mathcal{F}\subset\binom{[n]}{k}$ be an intersecting family, where $n>100k$, and suppose that ... Here $\varrho(\mathcal{F})$ denotes the diversity of $\mathcal{F}$. The diversi...

open_problem | open | MathDB 页面 | 原文线索

V2-044440 | MATHDB-358202

Tomon’s strong stability conjecture for 3-wise t-intersecting families

摘录 / EXCERPT: Tomon’s strong stability conjecture. There exists $\epsilon=\epsilon(p,t)>0$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-044441 | MATHDB-358203

Tomon’s stability conjecture beyond the proven range

摘录 / EXCERPT: Tomon’s conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044442 | MATHDB-358204

Tomon’s shifting conjecture for (r,t)-maximal families

摘录 / EXCERPT: Tomon’s conjecture. If $\mathcal H\subset\mathcal F_i^t(r)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044443 | MATHDB-358236

A general upper-bound conjecture for Hamming symmetric vector systems

摘录 / EXCERPT: Let $0<s\leq n$ and let $q>1$ be an integer. Let $L=\{\ell_1,\ldots,\ell_s\}\subseteq [n]$ and let $\mathcal H\subseteq \{0,1,\ldots,q-1\}^n$ be a Hamming symmetric vector system s...

conjecture | open | MathDB 页面 | 原文线索

V2-044444 | MATHDB-358286

Pak-Panova's unary Littlewood-Richardson completeness conjecture

摘录 / EXCERPT: Pak-Panova's conjecture. When the input is in unary, $\textsf{ComputeLR}$ is $\#\mathrm{P}$-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-044445 | MATHDB-358287

Panova's restricted #P conjecture for Kronecker coefficients

摘录 / EXCERPT: Panova's restricted-case conjecture. The following two statements are each equivalent to the unrestricted assertion that $\textsf{ComputeKron}$ is in $\#\mathrm{P}$:

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V2-044446 | MATHDB-358320

Existence conjecture for the polytope $\mathcal{T}_n(\epsilon_g)$

摘录 / EXCERPT: Existence conjecture. The polytope $\mathcal{T}_n(\epsilon_g)$ exists for all $n \ge 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-044447 | MATHDB-358327

Leader-Letzter-Narayanan-Walters extremal density conjecture for product-free sets

摘录 / EXCERPT: Leader-Letzter-Narayanan-Walters conjecture. If $S\subset\mathcal{F}$ is product-free and $d^{\ast}(S)=1/2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044448 | MATHDB-358328

Extremal 3-product-free set conjecture for free semigroups

摘录 / EXCERPT: Extremal 3-product-free set conjecture. One of the following holds: either the letters of $\mathcal{A}$ can be labelled by $\mathbb{Z}/3\mathbb{Z}$ so that S is contained in the...

conjecture | open | MathDB 页面 | 原文线索

V2-044449 | MATHDB-358329

Stability conjecture for near-extremal product-free sets in free semigroups

摘录 / EXCERPT: Stability conjecture. For each $\delta>0$, there exists $\varepsilon>0$ such that, whenever $S\subset\mathcal{F}$ is product-free and

conjecture | open | MathDB 页面 | 原文线索

V2-044450 | MATHDB-358334

Frankl-Kupavskii's conjecture on 3-wise and 3-intersecting families

摘录 / EXCERPT: Frankl-Kupavskii's conjecture. If $\mathcal{F} \subset \mathcal{P}(n)$ is both 3-wise intersecting and 3-intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-044451 | MATHDB-358335

Frankl-Kupavskii's incompatibility conjecture for higher intersection parameters

摘录 / EXCERPT: Frankl-Kupavskii's incompatibility conjecture. For every integer $s\geq 1$ there exists an integer $t_0(s)$ such that for all $t\geq t_0(s)$, the properties of being 3-wise s-int...

conjecture | open | MathDB 页面 | 原文线索

V2-044452 | MATHDB-358356

Maximum hyperplane intersection conjecture for odd-dimensional permutohedra

摘录 / EXCERPT: Let P_n be the permutohedron, and let H_0 be its affine hull. For an affine hyperplane $H\ne H_0$, consider the number of vertices in $H\cap P_n$. Maximum intersection conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-044453 | MATHDB-358364

The Conforti--Cornuéjols packing problem

摘录 / EXCERPT: Conforti--Cornuéjols packing problem. A clutter $\mathcal C$ has the packing property if and only if it has the max-flow min-cut property.

open_problem | open | MathDB 页面 | 原文线索

V2-044454 | MATHDB-358389

Local limit conjecture for the simplified polymer model

摘录 / EXCERPT: Simplified-model local limit conjecture. For all sufficiently large n, for the upper extremal point M_n^+ and every integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-044455 | MATHDB-358403

Conjecture on occurrence sequences in base-phi expansions

摘录 / EXCERPT: Let $\beta(N)=\beta^+(N)\cdot\beta^-(N)$ be the base-phi expansion of the number N. Let w be a word of length m, and let $R_{\cdot w}$ be the sequence of occurrences of numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-044456 | MATHDB-358443

Duncan-Steingrímsson's Bell-number conjecture for modified ascent sequences

摘录 / EXCERPT: Duncan-Steingrímsson's conjecture. On modified ascent sequences, these six patterns are all Wilf-equivalent, and the enumeration of modified ascent sequences avoiding any one of th...

conjecture | open | MathDB 页面 | 原文线索

V2-044457 | MATHDB-358479

The very colorful Helly conjecture for d-Leray complexes and matroids

摘录 / EXCERPT: Let $d\beq 1$, $r\geq d+1$, $1\leq m\leq r$ and $m\leq k\leq \min\{m+d,r\}$ be integers. Let V be a finite set with $|V|\geq \max\{m+d,r\}$. Let X be a d-Leray simplicial com...

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V2-044458 | MATHDB-358487

Aharoni-Ziv's matroidal Hall conjecture for finitary matroids

摘录 / EXCERPT: Aharoni-Ziv's matroidal Hall conjecture. There is no M-independent base of N if and only if there exists an $X\subseteq E$ such that $M\!\upharpoonright\!X$ has an $N.X$-indepe...

conjecture | open | MathDB 页面 | 原文线索

V2-044459 | MATHDB-358491

Density-maximizing conjecture for selected unequal binary disk packings

摘录 / EXCERPT: Density-maximizing conjecture. For the ratios r_b and r_c, the periodic nontriangulated binary packing depicted in Fig. 7 maximizes density among all binary packings with the s...

conjecture | open | MathDB 页面 | 原文线索

V2-044460 | MATHDB-358492

Connelly's uniformity conjecture for ternary disk packings

摘录 / EXCERPT: Connelly's uniformity conjecture. The ternary triangulated packing numbered 53 is the most uniform triangulated packing other than HCP. Its proposed deformation yields the highes...

conjecture | open | MathDB 页面 | 原文线索

V2-044461 | MATHDB-358506

Iosevich-Rudnev finite-field Falconer distance conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field of odd cardinality q, let $r\geq 2$, and let $E\subseteq\mathbb{F}_q^r$ have cardinality at least $cq^{\frac{r}{2}}$, where c is sufficient...

conjecture | open | MathDB 页面 | 原文线索

V2-044462 | MATHDB-358545

Large trees dominate even paths in the homomorphism-density order

摘录 / EXCERPT: Large-tree path conjecture. For every $k\geq 1$, there exists an integer $n_0(k)$ such that every tree H with at least $n_0(k)$ vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044463 | MATHDB-358572

Subquadratic ordered Ramsey bound for matchings of interval chromatic number two

摘录 / EXCERPT: Interval-chromatic-two conjecture. There exists an $\epsilon>0$ such that, for every ordered matching $M^<$ on n vertices with $\chi_<(M^<)=2$,

conjecture | open | MathDB 页面 | 原文线索

V2-044464 | MATHDB-358579

Motzkin-Schmidt strip-width conjecture

摘录 / EXCERPT: For every set of points $P\subset[0,1]^2$, let $\Phi(P)$ be the smallest real number $w\geqslant0$ for which some strip of width w contains at least three points of P. Define .....

conjecture | open | MathDB 页面 | 原文线索

V2-044465 | MATHDB-358589

The maximal clustering word conjecture for Arnoux-Rauzy languages

摘录 / EXCERPT: Let Λ be an Arnoux-Rauzy language satisfying condition $(Rabc)$. A clustering word is a word whose occurrences cluster in the sense defined for Λ, and a word is con...

conjecture | open | MathDB 页面 | 原文线索

V2-044466 | MATHDB-358614

Polynomial compressibility of bounded clique-number acyclic oriented graphs

摘录 / EXCERPT: Polynomial compressibility conjecture. For $k \geq 3$, the families $\mathcal A_k$ and $\mathcal R_k$ are polynomially τ-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-044467 | MATHDB-358626

McMullen's conjecture on 1-neighborly reorientations

摘录 / EXCERPT: Let $\nu(r,k)$ be the largest n such that every uniform oriented matroid of rank r on n elements has a k-neighborly reorientation. McMullen's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-044468 | MATHDB-358649

The eBugs conjecture on the existence of de Bruijn families

摘录 / EXCERPT: The eBugs conjecture. There exists a de Bruijn family in $\operatorname{deBF}(\mathcal{A},m,p,n)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-044469 | MATHDB-358684

The circumference conjecture for connected matroid pathwidth

摘录 / EXCERPT: Let M be a connected matroid with rank function r. Its circumference is the maximum size of a circuit of M, when M has at least one circuit, and its pathwidth is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-044470 | MATHDB-358689

The $r(4,t)$ asymptotic conjecture

摘录 / EXCERPT: For positive integers s and t, let $r(s,t)$ denote the smallest integer n such that every graph on n vertices contains either a clique of size s or an independent set of...

conjecture | open | MathDB 页面 | 原文线索

V2-044471 | MATHDB-358691

Roberts's lower-shadow conjecture for families of distinct k-sets

摘录 / EXCERPT: Let $[t]=\{1,\ldots,t\}$, let $[t]^{(k)}$ denote the family of k-subsets of $[t]$, and let $[t]^{(\geq k)}$ denote the family of subsets of $[t]$ of size at least k. For a fami...

conjecture | open | MathDB 页面 | 原文线索

V2-044472 | MATHDB-358692

Roberts's max-lex conjecture for minimizing unions

摘录 / EXCERPT: Let $\mathbb{N}^{(k)}$ denote the family of k-element subsets of $\mathbb{N}$. For a family $\mathcal{A}$ of sets, write $\langle\mathcal{A}\rangle$ for the union of the upward c...

conjecture | open | MathDB 页面 | 原文线索

V2-044473 | MATHDB-358753

Clifton-Huang conjecture on hyperplane covers of the hypercube

摘录 / EXCERPT: Clifton-Huang conjecture. For $k \ge 2$ and n sufficiently large,

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V2-044474 | MATHDB-358754

Conjecture on the supremum tail function for Rademacher sums on [√2,2]

摘录 / EXCERPT: Conjecture for F on $[\sqrt{2},2]$. For $x\in[\sqrt{2},2]$,

conjecture | open | MathDB 页面 | 原文线索

V2-044475 | MATHDB-358756

Strong Wilf-equivalence is characterized by trivial equivalence for binary words

摘录 / EXCERPT: Strong Wilf-equivalence conjecture. Two binary words p and q are strongly Wilf-equivalent if and only if they are trivially equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-044476 | MATHDB-358764

The vexillary Grothendieck-Schubert character conjecture

摘录 / EXCERPT: Let $w\in S_n$ be a vexillary permutation. Write $\mathfrak G_w^\mathrm{top}$ for the top-degree homogeneous component of its Grothendieck polynomial, and let $D^\mathrm{top}(w)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-044477 | MATHDB-358789

Algebraic matroid basis conjecture for Grassmannians and coherent matching fields

摘录 / EXCERPT: Let $\operatorname{Gr}(k,n)$ be the Grassmannian, and let $M_{\operatorname{Gr}(k,n)}$ be its algebraic matroid. For a coherent matching field L, let M_L be the algebraic matro...

conjecture | open | MathDB 页面 | 原文线索

V2-044478 | MATHDB-358817

Uniform boundedness of external saturation for posets

摘录 / EXCERPT: Let P be a poset, and let $\mathop{}\!\operatorname{extsat}(n,P)$ denote the minimum size of an external P-saturated family for the Boolean lattice ${\mathcal B}_n$, as define...

conjecture | open | MathDB 页面 | 原文线索

V2-044479 | MATHDB-358824

Equivalence conjecture for ternary codes and their 2-designs

摘录 / EXCERPT: Let $m\ge 2$ be a positive integer, let $p=3$, and set $n=2m+1$. Let $\mathcal C=\widetilde{\mathcal C_n}$ be the ternary code defined by the source's equation (1), with (n,d) sa...

conjecture | open | MathDB 页面 | 原文线索

V2-044480 | MATHDB-358830

Differential-operator formula for the twisted trace operator

摘录 / EXCERPT: Let ${\mathcal Z}^0_n$ be the degree-zero subspace in degree n, let $\tilde\rho_n=\rho(F(w^n))$, let $\beta(w,w^k)$ be the coefficient appearing in the bilinear form β, and...

conjecture | open | MathDB 页面 | 原文线索

V2-044481 | MATHDB-358831

Good-boundary formula for the beta and theta pairings

摘录 / EXCERPT: Let ζ and ξ be elements for which the boundary $\partial\Pi(\zeta,\xi)$ is good, in the sense defined in the paper. Let F be the map from such boundary data to the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-044482 | MATHDB-358860

Cui-Hu conjecture on two abundant elements

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite family of sets that is union-closed, meaning that the union of any two members belongs to $\mathcal{F}$. Suppose that every set in $\mathcal{F}$ has s...

conjecture | open | MathDB 页面 | 原文线索

V2-044483 | MATHDB-358863

Representability conjecture for single relaxations of K(r,t)

摘录 / EXCERPT: Representability conjecture for K(r,t). Every matroid obtained from $K(r,t)$ by relaxing a circuit-hyperplane into a basis is representable.

conjecture | open | MathDB 页面 | 原文线索

V2-044484 | MATHDB-358871

Three-run enumeration conjecture for flattened Stirling permutations

摘录 / EXCERPT: Let $\mathcal{Q}_n$ denote the set of Stirling permutations of order n, and let $\operatorname{flat}_3(\mathcal{Q}_n)$ be the set of those with exactly three runs. Three-run enum...

conjecture | open | MathDB 页面 | 原文线索

V2-044485 | MATHDB-358875

Ruzsa's discrete Brunn-Minkowski conjecture

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let ϵ be positive. For finite sets $A,B\subset\mathbb{Z}^k$, say that B is not covered by n parallel hyperplanes when no collection of n par...

conjecture | open | MathDB 页面 | 原文线索

V2-044486 | MATHDB-358890

Circuit decomposition conjecture for Eulerian binary matroids

摘录 / EXCERPT: Let M be an Eulerian binary matroid, meaning a subset of $\mathbb F_2^n\setminus\{0\}$ whose elements can be decomposed into circuits. Write $c(M)$ for the minimum number of circ...

conjecture | open | MathDB 页面 | 原文线索

V2-044487 | MATHDB-358891

Odd-cover conjecture for Eulerian binary matroids

摘录 / EXCERPT: Let M be an Eulerian binary matroid, viewed as a subset of $\mathbb F_2^n\setminus\{0\}$. Let $c_2(M)$ be the minimum size of an odd-cover of M, and let $a(M)$ be the minimum n...

conjecture | open | MathDB 页面 | 原文线索

V2-044488 | MATHDB-358893

Spectral projector kernel bounds on narrow windows for the two-dimensional torus

摘录 / EXCERPT: Let $\mathbb{T}^2=\mathbb{R}^2/\mathbb{Z}^2$, let $\lambda>2$ and $\delta<1$, and define ... Let ... so that $P_{\lambda,\delta}f=\Phi_{\lambda,\delta}*f$. Kernel conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-044489 | MATHDB-358902

Lovász-Mason flat-condition conjecture for matroid symmetric powers

摘录 / EXCERPT: Let M be a matroid with ground set E, let $(M_1,\dots,M_d)$ be a symmetric quasi power of M, and let $\sF(M)$ denote the set of flats of M. For each $F\in\sF(M)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-044490 | MATHDB-358903

Lovász-Mason contraction reduction for matroid symmetric powers

摘录 / EXCERPT: Let M be a matroid on the ground set E, let $(M_1,\dots,M_d)$ be a symmetric quasi power of M, and suppose that $F\cdot\operatorname{Sym}_{d-1}(E)$ is a flat of M_d for eve...

open_problem | open | MathDB 页面 | 原文线索

V2-044491 | MATHDB-358917

Real-representable matroid density conjecture

摘录 / EXCERPT: Let $ll,t in N$ with $ll ge 3$, and let $\falM$ be the class of R-representable matroids with no $U_{2,ll+2}$-minor and no $M(K_t)$-minor. Here $alpha$ is the constant appearing...

conjecture | open | MathDB 页面 | 原文线索

V2-044492 | MATHDB-358924

Strict positivity of the coefficients of the specialized monomial symmetric-function polynomials

摘录 / EXCERPT: Positivity conjecture. For every partition ${\Greekmath 0115}$, all coefficients a_k of $J_{{\Greekmath 0115}}(q)$ are strictly positive. This is one of three conjectures motivat...

conjecture | open | MathDB 页面 | 原文线索

V2-044493 | MATHDB-358925

Unimodality of the specialized monomial symmetric-function polynomials

摘录 / EXCERPT: Let ${\Greekmath 0115}$ be an integer partition, and let $J_{{\Greekmath 0115}}(q)$ be the polynomial associated with the specialization of the monomial symmetric function...

conjecture | open | MathDB 页面 | 原文线索

V2-044494 | MATHDB-358942

Asymptotic count conjecture for trihexes

摘录 / EXCERPT: Asymptotic count conjecture. As $v \to \infty$, the counts $\alpha(v)$ and $\beta(v)$ are respectively asymptotic to

conjecture | open | MathDB 页面 | 原文线索

V2-044495 | MATHDB-358943

Asymptotic irreducibility conjecture for maximal uniquely maximal Condorcet domains

摘录 / EXCERPT: Irreducibility conjecture. MUCDs are asymptotically almost surely irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-044496 | MATHDB-358944

Asymptotic disconnectedness conjecture for maximal uniquely maximal Condorcet domains

摘录 / EXCERPT: Disconnectedness conjecture. Asymptotically almost surely, MUCDs are not connected.

conjecture | open | MathDB 页面 | 原文线索

V2-044497 | MATHDB-358945

Asymptotic non-normality conjecture for maximal uniquely maximal Condorcet domains

摘录 / EXCERPT: Non-normality conjecture. Asymptotically almost surely, MUCDs are not normal.

conjecture | open | MathDB 页面 | 原文线索

V2-044498 | MATHDB-358952

Duality invariance of matroid p-characteristic sets

摘录 / EXCERPT: Duality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044499 | MATHDB-358982

The maximum odd-cycle and path likelihood conjecture

摘录 / EXCERPT: Maximum likelihood conjecture. For every $m\geq 3$ and every edge probability mass μ,

conjecture | open | MathDB 页面 | 原文线索

V2-044500 | MATHDB-359011

The odd-shift inclusion conjecture for symbol components

摘录 / EXCERPT: Odd-shift inclusion conjecture. If k is odd, then $X_1\subset X_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-044501 | MATHDB-359028

Determinant evaluation conjecture for the matrix T_{n,m}(x)

摘录 / EXCERPT: Determinant evaluation conjecture. The determinant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044502 | MATHDB-359045

Palindromicity and unimodality conjecture for the EMD numerator polynomials

摘录 / EXCERPT: For positive integers n, let $N_n(t)$ be the polynomial defined by the rational generating function ... where $H'_n(t)=H'_{nn}(t)$ is obtained by differentiating the bivariate gene...

conjecture | open | MathDB 页面 | 原文线索

V2-044503 | MATHDB-359046

Symmetric-difference formula conjecture for the EMD numerator

摘录 / EXCERPT: For positive integers n, let $N_n(t)$ be the numerator polynomial in ... and define ... where $2^{[m]}$ is the power set of $[m]$. Symmetric-difference formula conjecture. For all po...

conjecture | open | MathDB 页面 | 原文线索

V2-044504 | MATHDB-359052

Agoston et al.'s convex hull thrackle conjecture

摘录 / EXCERPT: Agoston et al.'s conjecture. A convex hull thrackle on n points has at most n convex hulls.

conjecture | open | MathDB 页面 | 原文线索

V2-044505 | MATHDB-359059

Ceballos-Pons realization conjecture for the s-permutahedron

摘录 / EXCERPT: Ceballos-Pons realization conjecture. The s-permutahedron $\text{Perm}_{\operatorname{s}}$ can be realized as a polyhedral subdivision of a polytope that is combin...

conjecture | open | MathDB 页面 | 原文线索

V2-044506 | MATHDB-359066

The diagonal extremality conjecture for central sections of the cube

摘录 / EXCERPT: Diagonal extremality conjecture. All locally extremal central sections of Q_n are diagonal for each $n\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-044507 | MATHDB-359086

Finitary Khintchine-type recurrence conjecture for finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group of order N, let $\textup{End}(G)$ denote its endomorphism ring, and let $A\subseteq G$. For $\alpha,\varepsilon>0$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-044508 | MATHDB-359137

Ruzsa's question on minimum Fourier-uniform 4-AP density

摘录 / EXCERPT: For $0<\alpha<1$, let $\rho_4(\alpha)$ be the largest real number such that, asymptotically in N, every sufficiently Fourier-uniform subset of $\mathbb{Z}/N\mathbb{Z}$ of density...

conjecture | open | MathDB 页面 | 原文线索

V2-044509 | MATHDB-359138

Avoiding symmetrically colored 4-APs

摘录 / EXCERPT: A 4-AP is a progression $n,n+d,n+2d,n+3d$ with $d\neq0$. It is symmetrically colored when n and $n+3d$ have the same color, and $n+d$ and $n+2d$ have the same color. Avoiding sym...

conjecture | open | MathDB 页面 | 原文线索

V2-044510 | MATHDB-359139

Ruzsa's question for minimum U^{k-2}-uniform k-AP density

摘录 / EXCERPT: For each integer $k\geq4$ and $0<\alpha<1$, let $\rho_k(\alpha)$ be the largest real number such that sufficiently U^{k-2}-uniform subsets of $\mathbb{Z}/N\mathbb{Z}$ of density...

conjecture | open | MathDB 页面 | 原文线索

V2-044511 | MATHDB-359140

Avoiding symmetrically colored even-length APs

摘录 / EXCERPT: Let $k\geq4$ be even. A symmetrically colored k-AP is a nontrivial progression $n,n+d,\ldots,n+(k-1)d$ whose terms symmetric about the center receive equal colors. Avoiding symme...

conjecture | open | MathDB 页面 | 原文线索

V2-044512 | MATHDB-359156

Jerónimo-Castro-Magazinov-Soberón piercing conjecture for translates of convex bodies

摘录 / EXCERPT: Let $n\ge 2$, and let $\mathcal{F}_1,\mathcal{F}_2,\dots,\mathcal{F}_n$ be families of translates of a convex compact set $\mathcal{K}$ in the plane. Assume that ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-044513 | MATHDB-359157

Buck et al.'s three-run enumeration conjecture for flattened 2-Stirling permutations

摘录 / EXCERPT: Buck et al.'s three-run enumeration conjecture. The number of flattened 2-Stirling permutations of order n with exactly three runs equals the displayed sum.

conjecture | open | MathDB 页面 | 原文线索

V2-044514 | MATHDB-359197

Partition regularity of the reciprocal parallelogram pattern

摘录 / EXCERPT: Reciprocal parallelogram conjecture. The pattern

conjecture | open | MathDB 页面 | 原文线索

V2-044515 | MATHDB-359198

Partition regularity of the three-term reciprocal pattern

摘录 / EXCERPT: Three-term reciprocal-pattern conjecture. The pattern

conjecture | open | MathDB 页面 | 原文线索

V2-044516 | MATHDB-359199

Alweiss's non-partition-regularity conjecture for sums and products over polynomial colorings

摘录 / EXCERPT: Alweiss's non-partition-regularity conjecture. There exists n such that, for every $d\in\mathbb{N}$, the pattern

conjecture | open | MathDB 页面 | 原文线索

V2-044517 | MATHDB-359208

Existence conjecture for symmetric $[\alpha]$-Hadamard matrices

摘录 / EXCERPT: Existence conjecture for symmetric $[\alpha]$-Hadamard matrices. For every $\alpha\geq 3$, there exist symmetric $[\alpha]$-Hadamard matrices of all orders.

conjecture | open | MathDB 页面 | 原文线索

V2-044518 | MATHDB-359209

Bipartite Ramsey number conjecture for two copies of $K_{2,n}$

摘录 / EXCERPT: Bipartite Ramsey number conjecture for two copies of $K_{2,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-044519 | MATHDB-359254

Conjecture on the Lipschitz variation of Ruzsa's number

摘录 / EXCERPT: Lipschitz conjecture for Ruzsa's number. For every positive integer m,

conjecture | open | MathDB 页面 | 原文线索

V2-044520 | MATHDB-359255

Chen's conjecture on repeated values of Ruzsa's number

摘录 / EXCERPT: Chen's conjecture. There are infinitely many positive integers m such that

conjecture | open | MathDB 页面 | 原文线索

V2-044521 | MATHDB-359256

Conjecture on the mean representation count in finite cyclic groups

摘录 / EXCERPT: Mean representation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044522 | MATHDB-359257

Conjecture on the size of bounded-representation additive bases

摘录 / EXCERPT: Size conjecture for bounded-representation bases.

conjecture | open | MathDB 页面 | 原文线索

V2-044523 | MATHDB-359258

Conjecture on attainment of the minimal mean representation count

摘录 / EXCERPT: Attainment conjecture. For all sufficiently large m,

conjecture | open | MathDB 页面 | 原文线索

V2-044524 | MATHDB-359261

Three-move outcome segmentation conjecture

摘录 / EXCERPT: Three-move segmentation conjecture. If $|S|=3$, there is an outcome segmentation.

conjecture | open | MathDB 页面 | 原文线索

V2-044525 | MATHDB-359262

Maximum segmentation conjecture for finite vector subtraction

摘录 / EXCERPT: Maximum segmentation conjecture. For each $k\in\mathbb N$, $k\ne 2$, there exists a ruleset of size k that has a $(k+1)$-segmentation. For each $k\in\mathbb N$, there is no rules...

conjecture | open | MathDB 页面 | 原文线索

V2-044526 | MATHDB-359263

Non-eventual-periodicity conjecture for finite two-dimensional subtraction

摘录 / EXCERPT: Non-eventual-periodicity conjecture. Not all finite two-dimensional rulesets are eventually periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-044527 | MATHDB-359266

Conjecture on the structure of maximum non-overlapping codes

摘录 / EXCERPT: Let $q\geq 2$ and $n\geq 3$. Let $\mathcal{M}_{q,n}$ denote the family of maximum non-overlapping codes of length n, and for a code $C\in\mathcal{M}_{q,n}$ write ... where L_i an...

conjecture | open | MathDB 页面 | 原文线索

V2-044528 | MATHDB-359276

Linear-factor conjecture for semi-external chambers of vector partition functions

摘录 / EXCERPT: Let A be a $d \times n$ integer matrix of rank d, let f be a facet of $\operatorname{pos}_{\mathbb{R}}(A)$ with inner facet normal $\bm{\iota}$, and let γ be a semi-ex...

conjecture | open | MathDB 页面 | 原文线索

V2-044529 | MATHDB-359278

Quadratic-bound reconstruction conjecture for tableau minors

摘录 / EXCERPT: Let $k\geq 2$ and let $T\in\operatorname{YT}(n)$, where $n\geq k^2+2k$. Here $\operatorname{M}_k(T)$ denotes the collection of k-minors of the standard Young tableau T. Reconst...

conjecture | open | MathDB 页面 | 原文线索

V2-044530 | MATHDB-359279

Linear-bound multiset reconstruction conjecture for tableau minors

摘录 / EXCERPT: Let $k\geq 1$ and let $T\in\operatorname{YT}(n)$, where $n\geq k+4$. Here $\operatorname{mM}_k(T)$ denotes the multiset of k-minors of the standard Young tableau T. Multiset re...

conjecture | open | MathDB 页面 | 原文线索

V2-044531 | MATHDB-359283

The two-dimensional integer covering asymptotic conjecture

摘录 / EXCERPT: Two-dimensional asymptotic conjecture. For all positive integers n and k,

conjecture | open | MathDB 页面 | 原文线索

V2-044532 | MATHDB-359284

The three-dimensional covering conjecture

摘录 / EXCERPT: Three-dimensional covering conjecture. For all $n,k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044533 | MATHDB-359285

The linear-growth conjecture for integer coverings of triangular grids

摘录 / EXCERPT: Linear-growth conjecture. For all fixed d and k, there exists a constant $C_{d,k}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044534 | MATHDB-359286

The explicit asymptotic formula conjecture for triangular-grid coverings

摘录 / EXCERPT: Explicit asymptotic formula conjecture. If d is odd and

conjecture | open | MathDB 页面 | 原文线索

V2-044535 | MATHDB-359310

Rigidity conjecture for geometric nonconvex self-intersecting polyhedra

摘录 / EXCERPT: A geometric polyhedron here may be nonconvex and self-intersecting; its faces, dihedral angles, edge lengths, and partially-flat vertices are understood in the sense of the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-044536 | MATHDB-359324

Structural conjecture for near-extremal joints configurations

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, and let $(\mathcal{J},\mathcal{L})$ be a joints configuration in $\mathbb{F}^d$, with $J=|\mathcal{J}|$ joints and $L=|\mathcal{L}|$ lines. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-044537 | MATHDB-359325

Zhao-He conjecture that joint set systems attain the optimal joints bound

摘录 / EXCERPT: For positive integers $k_1,\ldots,k_r,m_1,\ldots,m_r$, let $C_{k_1,\ldots,k_r;m_1,\ldots,m_r}$ be the optimal constant in the corresponding joints problem, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044538 | MATHDB-359331

The rational-field minor threshold conjecture for sparse random matroids

摘录 / EXCERPT: Rational-field minor threshold conjecture. Suppose that the field is $\mathbb{Q}$, the field of rational numbers, ${\mathcal P}(1,\ldots,1)=1$, and N is $\mathbb{Q}$-representabl...

conjecture | open | MathDB 页面 | 原文线索

V2-044539 | MATHDB-359332

The subfield minor threshold conjecture for sparse random matroids

摘录 / EXCERPT: Subfield minor threshold conjecture. The quantity d_k/k is the threshold, as a function of m/n, for the appearance of N as a minor of $M[A_m]$.

conjecture | open | MathDB 页面 | 原文线索

V2-044540 | MATHDB-359347

Galashin-Lam conjecture on q,t-Schur positivity for ${\mathbb Z}$-convex curves

摘录 / EXCERPT: A curve C passing through lattice points $(0,0)={\bm p}_0,{\bm p}_1,\ldots,{\bm p}_k=(m,n)$ is ${\mathbb Z}$-convex if the set of lattice points strictly above C, together with...

conjecture | open | MathDB 页面 | 原文线索

V2-044541 | MATHDB-359391

Bukh's algebraically independent distance-avoidance conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}_{>0}$ be algebraically independent, meaning that no nonzero polynomial with rational coefficients vanishes on any finite tuple of distinct elements of A....

conjecture | open | MathDB 页面 | 原文线索

V2-044542 | MATHDB-359400

Oxley's circuit-cocircuit intersection conjecture

摘录 / EXCERPT: For a matroid M, a circuit-cocircuit intersection (CCI) is a subset of its ground set that is the intersection of a circuit and a cocircuit. Oxley's circuit-cocircuit intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-044543 | MATHDB-359401

The CCI-envelope form of Oxley's circuit-cocircuit intersection conjecture

摘录 / EXCERPT: A circuit-cocircuit intersection (CCI) is a subset of a matroid's ground set that is the intersection of a circuit and a cocircuit. A CCI-envelope is a matroid that contains a CCI...

conjecture | open | MathDB 页面 | 原文线索

V2-044544 | MATHDB-359412

The conjecture on the outcome of the initial 2×3×3 Abalone constellation

摘录 / EXCERPT: Let $D0$ denote the initial constellation of $2\times 3\times 3$ Abalone, and let $o(D0)$ denote its outcome. Initial 2×3×3 Abalone conjecture. ... This is one of two experimentally...

conjecture | open | MathDB 页面 | 原文线索

V2-044545 | MATHDB-359413

The conjecture on Diamond and Trapezoid 2×3×3 Abalone constellations

摘录 / EXCERPT: Let C be a $2\times 3\times 3$ Abalone constellation, and call it a member of the “Diamond” or “Trapezoid” family according to the indicated subconfiguration. Let $o(C)$ denote i...

conjecture | open | MathDB 页面 | 原文线索

V2-044546 | MATHDB-359416

Modular Erdős distinct subset sums conjecture

摘录 / EXCERPT: Let $t\geq 0$ be fixed and let n be sufficiently large. Set $N=2^n+t$, and let A be an n-element set of positive integers whose subset sums are all distinct modulo N. Modul...

conjecture | open | MathDB 页面 | 原文线索

V2-044547 | MATHDB-359417

Kupitz-Martini-Perles conjecture on standard ball polyhedra

摘录 / EXCERPT: Let $V\subset\mathbb{R}^3$ be an extremal set for the frequent large distance problem, and let $\mathcal{B}(V)$ denote the associated ball polyhedron. The set V is strongly criti...

conjecture | open | MathDB 页面 | 原文线索

V2-044548 | MATHDB-359419

Extremal lattices as optimal unconstrained sphere packings in dimension 48

摘录 / EXCERPT: A sphere packing in dimension 48 is a set of congruent non-overlapping balls whose centers form a discrete subset of Euclidean space. An extremal lattice in dimension 48 is an...

conjecture | open | MathDB 页面 | 原文线索

V2-044549 | MATHDB-359425

Interlacing conjecture for deformed monotone Hurwitz numbers

摘录 / EXCERPT: Let $g\geqslant 0$, $n\geqslant 1$, and let $\mu_1,\ldots,\mu_n\geqslant 1$. The deformed monotone Hurwitz number $\vec{H}^t_{g,n}(\mu_1,\ldots,\mu_n)$ is a polynomial in t. Two...

conjecture | open | MathDB 页面 | 原文线索

V2-044550 | MATHDB-359426

Genus-asymptotic roots conjecture for deformed monotone Hurwitz numbers

摘录 / EXCERPT: Fix positive integers $\mu_1,\ldots,\mu_n$ whose sum is d. Let the roots of the polynomial $\vec{H}^t_{g,n}(\mu_1,\ldots,\mu_n)$ be ordered from smallest to largest. Genus-asympt...

conjecture | open | MathDB 页面 | 原文线索

V2-044551 | MATHDB-359427

A weak polynomial Freiman-Ruzsa conjecture

摘录 / EXCERPT: Weak polynomial Freiman-Ruzsa conjecture. There is an absolute constant $c>0$ such that one can find $A_1\subseteq A$ with

conjecture | open | MathDB 页面 | 原文线索

V2-044552 | MATHDB-359433

The polynomial-exponent conjecture for ordered Ramsey numbers of path powers

摘录 / EXCERPT: Let P_n^t be the tth power of the ordered path on n vertices, and let K_n be the ordered complete graph on n vertices. Write $R_{<}(G,H)$ for the ordered Ramsey number. P...

conjecture | open | MathDB 页面 | 原文线索

V2-044553 | MATHDB-359449

Pham's duality conjecture for Bulgarian Solitaire orbit-growth constants

摘录 / EXCERPT: Pham's duality conjecture. For any primitive necklace P such that c_P and c_{P^*} both exist,

conjecture | open | MathDB 页面 | 原文线索

V2-044554 | MATHDB-359450

Equal orbit-growth constants imply equal generating-function denominators

摘录 / EXCERPT: Equal-constant denominator conjecture. For any two primitive necklaces P_1 and P_2, if $c_{P_1}=c_{P_2}$, then $H_{P_1}(x)$ and $H_{P_2}(x)$ have the same denominator.

conjecture | open | MathDB 页面 | 原文线索

V2-044555 | MATHDB-359460

The hypergeometric identity conjecture for even-valence map counts

摘录 / EXCERPT: Hypergeometric identity conjecture. The following identity should hold:

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V2-044556 | MATHDB-359497

Bounded-dimensional support conjecture for minimal slice-rank decompositions

摘录 / EXCERPT: Bounded-dimensional support conjecture. There exists a positive integer $C(d,k)$ such that, for every order-d tensor T over an arbitrary field $\mathbb{F}$ with slice rank k,...

open_problem | open | MathDB 页面 | 原文线索

V2-044557 | MATHDB-359527

The structured-vector optimality conjecture for the continuous queueing problem

摘录 / EXCERPT: Structured-vector optimality conjecture. The vector $\boldsymbol{v}^*$ is an optimal solution of the paper's continuous minimization problem.

conjecture | open | MathDB 页面 | 原文线索

V2-044558 | MATHDB-359539

Salehi-Golsefidy's bounded generation conjecture for number-ring quotients

摘录 / EXCERPT: Let $0<\delta\ll 1$, let d be a positive integer, and let $N_0\gg_{d,\delta}1$ be a positive integer. For a number field K of degree at most d, write $\mathcal O$ for its rin...

conjecture | open | MathDB 页面 | 原文线索

V2-044559 | MATHDB-359575

The restricted bipartite connectivity conjecture for stable symbolic-power depth

摘录 / EXCERPT: Restricted bipartite connectivity conjecture. The stable value of the depth of the symbolic powers of $I(G)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-044560 | MATHDB-359585

The bicircular-matroid basis conciseness conjecture

摘录 / EXCERPT: Let $b(M)$ denote the number of bases of a matroid M, and restrict b to bicircular matroids. A counting function is concise if every positive integer occurs on an input of size...

conjecture | open | MathDB 页面 | 原文线索

V2-044561 | MATHDB-359586

The tournament Hamiltonian-path value conjecture

摘录 / EXCERPT: A tournament is an orientation of every edge of a complete graph. Let $h(T)$ be the number of directed Hamiltonian paths in a tournament T, and let h denote this counting funct...

conjecture | open | MathDB 页面 | 原文线索

V2-044562 | MATHDB-359588

The plane-triangulation almost-completeness conjecture

摘录 / EXCERPT: Let $X=\{p_1,\ldots,p_n\}\subset\mathbb Q^2$ be a set of points in general position, and let $t(X)$ be the number of triangulations of the convex hull of X using all points of...

conjecture | open | MathDB 页面 | 原文线索

V2-044563 | MATHDB-359590

The simply connected domino-tiling extension conjecture

摘录 / EXCERPT: A simply connected region is a domino-tileable region in the setting of the paper whose interior has no holes. Let the domino-tiling counting theorem of the paper be denoted by The...

conjecture | open | MathDB 页面 | 原文线索

V2-044564 | MATHDB-359593

Heuristic formula for generalized Stirling numbers with parameters $(-1,2;r)$

摘录 / EXCERPT: Let $r obreak\in\mathbb{Z}$, and let $\hat S_{n,k}(-1,2;r)=2^k k!\,S_{n,k}(-1,2;r)$ denote the rescaled generalized Stirling number. Heuristic formula. The identity ... should hold f...

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V2-044565 | MATHDB-359623

Alavi-Boals-Chartrand-Erdős-Oellerman ascending subgraph decomposition conjecture

摘录 / EXCERPT: Let G be a graph with $\binom{m+1}{2}$ edges. An ascending subgraph decomposition is a decomposition $H_1,\ldots,H_m$ such that H_i has i edges and is a subgraph of H_{i+1}...

conjecture | open | MathDB 页面 | 原文线索

V2-044566 | MATHDB-359643

Kajitani-Miyano-Ueno conjecture on cyclically orderable matroids

摘录 / EXCERPT: Kajitani-Miyano-Ueno conjecture. The matroid M is cyclically orderable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-044567 | MATHDB-359653

Optimality conjecture for seven points spanning four distinct triangles

摘录 / EXCERPT: A finite point set in the plane spans four distinct triangles when its triples of distinct points determine four congruence classes, up to rigid plane isometries. Four-triangle opt...

conjecture | open | MathDB 页面 | 原文线索

V2-044568 | MATHDB-359670

The linear covering-set conjecture for generalized shuffle squares

摘录 / EXCERPT: Fix an alphabet of size k. Let E_n be the set of all even k-ary words of length n, and let S_n be the set of all permutations of $[n]$. Define $m_k(n)$ to be the minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-044569 | MATHDB-359671

The shuffle anti-square existence conjecture

摘录 / EXCERPT: For a binary even word W of length $2n$, let $s(W)$ be the number of circular shifts of W that are shuffle squares. Let s_{2n} be the minimum of $s(W)$ over all binary even w...

conjecture | open | MathDB 页面 | 原文线索

V2-044570 | MATHDB-359676

Herzog-Qureshi's equality conjecture for polymatroidal ideals

摘录 / EXCERPT: Herzog-Qureshi's conjecture. Does

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V2-044571 | MATHDB-359689

Generalized Andreev conjecture for trivalent hyperbolic polyhedra

摘录 / EXCERPT: Generalized Andreev conjecture. There exists a compact convex hyperbolic polyhedron Q combinatorially equivalent to P whose dihedral angles are given by Θ, and Q is un...

conjecture | open | MathDB 页面 | 原文线索

V2-044572 | MATHDB-359698

Gerbner-Nagy-Patkós-Vizer counting conjecture for tree posets

摘录 / EXCERPT: Let P be a poset, and call a family $\mathcal F\subseteq 2^{[n]}$ P-free if it contains no copy of P. Let $La(n,P)$ denote the maximum size of a P-free family in $2^{[n]}$....

conjecture | open | MathDB 页面 | 原文线索

V2-044573 | MATHDB-359737

Asymptotic uniformity conjecture for (k,l)-uniform matroids

摘录 / EXCERPT: Let $m_n(k,l)$ be the number of all (k,l)-uniform matroids on the set $[n]$. Asymptotic uniformity conjecture. If $(k,l)\neq (1,1)$, then either ... or, dually, ... This conjecture p...

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V2-044574 | MATHDB-359741

Nonexistence conjecture for distance-regular graphs with specified classical parameters

摘录 / EXCERPT: Let q and D be integers with $q \ge 2$ and $D \ge 4$. A distance-regular graph with classical parameters ... is a graph whose classical parameters are given by this quadruple. No...

conjecture | open | MathDB 页面 | 原文线索

V2-044575 | MATHDB-359743

Weak Hurwitz stability conjecture for consecutive type A minuscule polynomials

摘录 / EXCERPT: For each positive integer n, let $N_n(x)$ be the type A minuscule polynomial ... A real polynomial is weakly Hurwitz stable if it is non-vanishing or identically zero when...

conjecture | open | MathDB 页面 | 原文线索

V2-044576 | MATHDB-359749

The higher-dimensional spacetime density conjecture

摘录 / EXCERPT: Higher-dimensional spacetime density conjecture. There exists an integer $m\geq 2$ such that, for every $\epsilon>0$, there exists a positive integer d with the property that whe...

conjecture | open | MathDB 页面 | 原文线索

V2-044577 | MATHDB-359750

The membrane extension of the sharp cycle bound

摘录 / EXCERPT: Let P be a minimal membrane of type A_2. Write $\operatorname{cyc}(P)$ for the number of cycles in its billiards permutation, $\operatorname{perim}(P)$ for its perimeter, and...

conjecture | open | MathDB 页面 | 原文线索

V2-044578 | MATHDB-359751

The perimeter conjecture for grid polygons with holes

摘录 / EXCERPT: Let P be a grid polygon with holes. Let $\operatorname{cyc}(P)$ denote the number of cycles in its billiards permutation, let $\operatorname{outerperim}(P)$ be the perimeter of t...

conjecture | open | MathDB 页面 | 原文线索

V2-044579 | MATHDB-359752

The area conjecture for grid polygons with holes

摘录 / EXCERPT: Let P be a grid polygon with holes. Let $\operatorname{cyc}(P)$ denote the number of cycles in its billiards permutation, let $\operatorname{area}(P)$ denote its area, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044580 | MATHDB-359759

Existence of the linear-threshold generalized Ramsey limit

摘录 / EXCERPT: Let $f(n,p,q)$ denote the generalized Ramsey number, and let q_{lin} denote the linear threshold. Linear-threshold limit conjecture. The limit ... exists for all $p\ge 3$. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-044581 | MATHDB-359760

Existence of the quadratic-threshold generalized Ramsey limit

摘录 / EXCERPT: Let $f(n,p,q)$ denote the generalized Ramsey number, let q_{quad} denote the quadratic threshold, and let $p\ge 4$ and $q\ge q_{quad}$. Quadratic-threshold limit conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-044582 | MATHDB-359763

Expected regular-subgraph count conjecture for random regular graphs

摘录 / EXCERPT: Let $\boldsymbol{G}_{d_1+d_2}\sim\mathcal{G}(n,d_1+d_2)$, and let $\mathcal{R}_{d_1}(G)$ be the set of d_1-regular spanning subgraphs of G. For $d_1,d_2>0$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-044583 | MATHDB-359765

Hasheminezhad-McKay conjecture on regular partitions of the complete graph

摘录 / EXCERPT: Let $d_0,d_1,\ldots,d_k\geq1$ satisfy ... and let $R(n;d_0,\ldots,d_k)$ be the number of partitions of the edges of K_n into spanning regular subgraphs of degrees $d_0,\ldots,d_k$....

conjecture | open | MathDB 页面 | 原文线索

V2-044584 | MATHDB-359779

Saxl's conjecture for staircase partitions

摘录 / EXCERPT: Let ρ_m be the staircase partition of size $n=\binom{m+1}{2}$, and let $g(\lambda,\lambda,\mu)$ denote the Kronecker coefficient. Saxl Conjecture. For every partition...

conjecture | open | MathDB 页面 | 原文线索

V2-044585 | MATHDB-359780

Bessenrodt-Panova conjecture on self-conjugate partitions

摘录 / EXCERPT: Let n be a positive integer, and let a partition be self-conjugate if it equals its conjugate. Its Durfee size is the side length of its largest square contained in its Young dia...

conjecture | open | MathDB 页面 | 原文线索

V2-044586 | MATHDB-359781

Vanishing conjecture for Kronecker coefficients of square partitions

摘录 / EXCERPT: For $m\ge 7$, let (m^m) be the square partition of m^2, let μ' denote the conjugate partition of μ, and define ... Write $g(\lambda,\lambda,\mu)$ for the Kronecker coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-044587 | MATHDB-359788

The dual first-order generalized Reed-Muller code 3-design conjecture

摘录 / EXCERPT: Dual Reed-Muller 3-design conjecture. If $q\geq 3$, then for every $\ell\in\mathbb{N}$, $(C^{\perp})_{\ell}$ is not a combinatorial 3-design.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044588 | MATHDB-359789

The density conjecture for the sets $S_{b,m}$ in SALIQUANT

摘录 / EXCERPT: For $b>0$, let $S_{b,m}$ denote the set of inputs associated with the nim-value $f(a,b,m)$ in SALIQUANT. The density conjecture for $S_{b,m}$. If $m=0$, then $S_{b,m}$ has positive...

conjecture | open | MathDB 页面 | 原文线索

V2-044589 | MATHDB-359790

Eventual vanishing conjecture for $M(a,b)$ in SALIQUANT

摘录 / EXCERPT: Eventual vanishing conjecture for $M(a,b)$. For each fixed $a>0$, there is a sufficiently large b such that

conjecture | open | MathDB 页面 | 原文线索

V2-044590 | MATHDB-359798

Matrix form of Frankl's union-closed sets conjecture

摘录 / EXCERPT: Let $M=(m_{ij})$ be a non-zero binary matrix of size $n\times m$ with distinct rows. For binary rows, write $M_{i-}\lor M_{j-}$ for their coordinatewise logical disjunction, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-044591 | MATHDB-359799

The probabilistic Graham product conjecture

摘录 / EXCERPT: For graphs G_1 and G_2, let $P_{1/2}(G)$ denote the smallest number of pebbles in a randomly generated configuration that is solvable with probability at least $1/2$. Probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-044592 | MATHDB-359816

Polynomial-growth conjecture for higher-order vector-space Ramsey numbers

摘录 / EXCERPT: For each integer $s\geqslant2$, let $R_{\mathbb F_{2}}(s,d)$ denote the corresponding vector-space Ramsey number over $\mathbb F_{2}$, with parameter s and target dimension d....

conjecture | open | MathDB 页面 | 原文线索

V2-044593 | MATHDB-359851

The fixed--modular integer programming tractability conjecture

摘录 / EXCERPT: Fixed-Δ tractability conjecture. There is a polynomial-time algorithm to solve any integer program of the form (IP) with a Δ-modular constraint matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-044594 | MATHDB-359852

The bounded-excess conjecture for generic two-row -modular matrices

摘录 / EXCERPT: Bounded-excess conjecture. There is a constant $c>0$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-044595 | MATHDB-359854

The skew corner-free set bound

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $S\subseteq[n]^2$ be skew corner-free, meaning that S contains no three points of the form … with $\delta\ne0$. The skew corner-free set conject...

conjecture | open | MathDB 页面 | 原文线索

V2-044596 | MATHDB-359855

Cohn's SDPP conjecture for abelian groups

摘录 / EXCERPT: Let G be a finite abelian group, and let $(A_i,B_i)_{i=1}^n$ be pairs of subsets satisfying the simultaneous double product property (SDPP): for each i, the equation...

conjecture | open | MathDB 页面 | 原文线索

V2-044597 | MATHDB-359889

Failure of moment matching for the multineighbor construction

摘录 / EXCERPT: Let $\Gamma_{n,m,p}$ and $Y_{n,k,q}$ be the distributions considered in the paper, let z_Kx denote the relevant count associated with a positive k-pure shape x, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-044598 | MATHDB-359900

Completeness conjecture for equality cases of Stanley's inequalities

摘录 / EXCERPT: Let $\operatorname{EqualityStanley}_k$ denote the relevant equality-verification problem for the kth Stanley inequality. EqualityStanley completeness conjecture. The problem...

conjecture | open | MathDB 页面 | 原文线索

V2-044599 | MATHDB-359902

Bóna-Dimitrov's generalized Sturm sequence conjecture for Dyck-path polynomials

摘录 / EXCERPT: Let $W_{n,k}(x)=\sum_{m=0}^{k}w_{n,k,m}x^m$, where $w_{n,k,m}$ counts Dyck paths of semilength n with k occurrences of UD and m occurrences of UUD. A sequence of real-roo...

conjecture | open | MathDB 页面 | 原文线索

V2-044600 | MATHDB-359903

Bóna-Dimitrov's Sturm-unimodality conjecture for Dyck-path polynomials

摘录 / EXCERPT: Let $W_{n,k}(x)=\sum_{m=0}^{k}w_{n,k,m}x^m$, where $w_{n,k,m}$ counts Dyck paths of semilength n with k occurrences of UD and m occurrences of UUD. For real-rooted polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-044601 | MATHDB-359904

Triangular-graph colouring enumeration conjecture

摘录 / EXCERPT: Let $\mathcal{T}_N(3)$ be the set of interlacing triangular arrays of rank 3 and height N. Let $G^{\triangle}_N$ be the triangular graph whose number of vertices along one side...

conjecture | open | MathDB 页面 | 原文线索

V2-044602 | MATHDB-359905

King-graph colouring enumeration conjecture

摘录 / EXCERPT: Let $\mathcal{T}_N(4)$ be the set of interlacing triangular arrays of rank 4 and height N. Let $G^{\varhexstar}_N$ be the square king graph with $N+1$ vertices along each side:...

conjecture | open | MathDB 页面 | 原文线索

V2-044603 | MATHDB-359968

Sárközy's conjecture for shifted multiplicative subgroups

摘录 / EXCERPT: Sárközy's conjecture. If q is sufficiently large, then for every $\lambda\in\mathbb{F}_q^*$, the shifted subgroup $(S_d-\lambda)\setminus\{0\}$ has no nontrivial multiplicative d...

conjecture | open | MathDB 页面 | 原文线索

V2-044604 | MATHDB-359969

The asymptotic advantage conjecture for distinct unit-fraction solutions

摘录 / EXCERPT: Let $f^*(k,-1)$ be the least board size for which Maker can force a solution with distinct variables in the Maker-Breaker game for the equation $1/x_1+\cdots+1/x_k=1/y$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044605 | MATHDB-360007

Extremal configuration-space homology quasi-polynomial degree conjecture

摘录 / EXCERPT: Extremal quasi-polynomial degree conjecture. If $H_{d-1}(M;\mathbb{Q})$ and $Q_M^{\nu_n-i}(n)$ are non-trivial, then the degree of the quasi-polynomial $Q_M^{\nu_n-i}(n)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-044606 | MATHDB-360018

The split-permutation conjecture for largest MVP fibre size

摘录 / EXCERPT: For $n\geq 1$, let $\mathcal{O}_{\mathrm{MVP}_{n}}$ denote the MVP outcome map on permutations of size n, and let $\mathrm{split}^{2,n-2}:=(n-1)n(n-2)(n-3)\cdots21$ for $n\geq3$....

conjecture | open | MathDB 页面 | 原文线索

V2-044607 | MATHDB-360047

Dependence of forward Zeckendorf game strategies on reverse game winners

摘录 / EXCERPT: Let the forward and reverse Zeckendorf games be played from corresponding starting positions, with each game having a winning player under optimal play. Forward-reverse Zeckendorf...

conjecture | open | MathDB 页面 | 原文线索

V2-044608 | MATHDB-360057

Parameter-family conjecture for polycyclic geometric $(7m_4)$ configurations

摘录 / EXCERPT: Let $m>3$, and consider the voltage group $\mathbb Z_m$. For parameters a,b, define the two parameter sets $F_1(m;a,b)$ and $F_2(m;a,b)$ by ... for $a\geq b$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044609 | MATHDB-360087

Young-subgroup positivity conjecture for Boolean matroid Chow rings

摘录 / EXCERPT: Young-subgroup positivity conjecture. For $i\leq j\leq k\leq\ell$ with $i+\ell=j+k$, the element

conjecture | open | MathDB 页面 | 原文线索

V2-044610 | MATHDB-360093

Linear analogue of the product-set dimension inequality

摘录 / EXCERPT: Linear product-set conjecture. Then every pair of K-subspaces $S\subset A$ and $T\subset B$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044611 | MATHDB-360094

Chowla subspace matching conjecture

摘录 / EXCERPT: Chowla matching conjecture. If B is a Chowla subspace, then A is matched to B.

conjecture | open | MathDB 页面 | 原文线索

V2-044612 | MATHDB-360136

The virtual mapping-torus conjecture for negative immersions

摘录 / EXCERPT: A compact 2-complex X has negative immersions when it satisfies the negative-immersion property defined in the source. An immersion is a map $\psi:H\rightarrow F$ between graph...

conjecture | open | MathDB 页面 | 原文线索

V2-044613 | MATHDB-360139

Conjecture on positivity of the Fibonacci characteristic function

摘录 / EXCERPT: Let $C_a(X)$ denote the characteristic function associated with the first letter a of the Fibonacci substitution. The quantities $_n$ measure the left endpoints of intervals on w...

conjecture | open | MathDB 页面 | 原文线索

V2-044614 | MATHDB-360160

Brändén and Huh's matroid characterization by homogeneous $\frac{8}{7}$-Rayleigh polynomials

摘录 / EXCERPT: Brändén-Huh's conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-044615 | MATHDB-360165

The Ramsey-type rank-pattern dichotomy for simplices

摘录 / EXCERPT: Let $\Delta^n=\{(x_1,\ldots,x_n)\in[0,1]^n: x_1\leq\cdots\leq x_n\}$, and let $F\subseteq\Delta^n$. Write $F^c=\Delta^n\setminus F$. The rank of a set is defined by the preceding r...

conjecture | open | MathDB 页面 | 原文线索

V2-044616 | MATHDB-360205

Generalized Chern number bounds for matroids

摘录 / EXCERPT: Let M be a simple matroid of rank $d+1$ on a ground set E of size n. For nonnegative integers $k_1,\ldots,k_d$ with $k_1+2k_2+\cdots+dk_d=d$, consider the Chern number...

conjecture | open | MathDB 页面 | 原文线索

V2-044617 | MATHDB-360214

Asymptotic bound for average plane-size in rank-4 complex-representable matroids

摘录 / EXCERPT: Let M be a simple n-element, rank-4, complex-representable matroid that is not the union of two lines. Its average plane-size is the average of $|P|$ over the rank-3 flats...

conjecture | open | MathDB 页面 | 原文线索

V2-044618 | MATHDB-360215

Asymptotic bound for average flat-size in complex-representable matroids

摘录 / EXCERPT: Let M be a simple n-element, rank-r complex-representable matroid, where $r\fixed$ and $r73$. The average flat-size is the average of $|F|$ over the flats F of M. Average...

conjecture | open | MathDB 页面 | 原文线索

V2-044619 | MATHDB-360216

Normalized average flat-size conjecture for complex-representable matroids

摘录 / EXCERPT: Normalized average flat-size conjecture. This average is at most $1+o_n(1)$. The source presents this as a strengthening of the preceding asymptotic flat-size conjecture; the meani...

conjecture | open | MathDB 页面 | 原文线索

V2-044620 | MATHDB-360217

Bounded average line-length from bounded average plane-size for matroids

摘录 / EXCERPT: Let M be a simple matroid of rank at least 6. The average line-length is the average size of the rank-2 flats of M, and the average plane-size is the average size of its ra...

conjecture | open | MathDB 页面 | 原文线索

V2-044621 | MATHDB-360218

Bounded average plane-size from bounded average line-length for matroids

摘录 / EXCERPT: Let M be a simple matroid of rank at least 4. The average plane-size is the average size of the rank-3 flats of M, and the average line-length is the average size of its ra...

conjecture | open | MathDB 页面 | 原文线索

V2-044622 | MATHDB-360219

Conjecture on extremal Hamiltonian paths in the unit cube

摘录 / EXCERPT: Hamiltonian-path conjecture. The equalities

conjecture | open | MathDB 页面 | 原文线索

V2-044623 | MATHDB-360227

The set-uniform Vinogradov mean value conjecture

摘录 / EXCERPT: Set-uniform Vinogradov mean value conjecture. For every real number $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-044624 | MATHDB-360260

Conjecture that cube-width is at most the poset size

摘录 / EXCERPT: Cube-width conjecture. For every finite poset $\mathcal P$,

conjecture | open | MathDB 页面 | 原文线索

V2-044625 | MATHDB-360261

Heng et al.'s conjecture on infinite families of 1-MDS codes

摘录 / EXCERPT: Heng et al.'s conjecture. For each $3\leq k\leq q-2$, the linear code $\mathcal{C}_k$ is a 1-MDS $[q-1,k,q-k-1]_q$ code, and the minimum-weight codewords of both $\mathcal{C}_k$...

conjecture | open | MathDB 页面 | 原文线索

V2-044626 | MATHDB-360273

Lexicographically minimal maximal sparse paving matroids have exponential extension complexity

摘录 / EXCERPT: For each positive integer n, let S_n be the sparse paving matroid on $[n]$ of rank $\lfloor n/2\rfloor$ with the maximal possible number of circuit-hyperplanes and lex...

conjecture | open | MathDB 页面 | 原文线索

V2-044627 | MATHDB-360284

The partition identity for multiple-cover contributions

摘录 / EXCERPT: Fix an integer $d\geq 1$. For an unordered partition $(d_1,\ldots,d_r)\vdash d$ into strictly positive parts, let $\#\operatorname{Aut}(d_1,\ldots,d_r)$ be the order of its automor...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044628 | MATHDB-360324

Existence of limiting proportions of zero types in symmetric-group character tables

摘录 / EXCERPT: For each n, let S_n be the symmetric group, and let $z_I(S_n)$, $z_{I\!I}(S_n)$, and $z(S_n)$ denote respectively the proportions of character-table zeros of type I, of type II...

conjecture | open | MathDB 页面 | 原文线索

V2-044629 | MATHDB-360326

Asymptotic predominance of type-I zeros in symmetric-group character tables

摘录 / EXCERPT: For each n, let S_n be the symmetric group, and let $z_I(S_n)$ be the proportion of character-table zeros of type I, while $z(S_n)$ is the proportion of all zeros in the charac...

conjecture | open | MathDB 页面 | 原文线索

V2-044630 | MATHDB-360327

Eventual monotonicity of type-I zero proportions for symmetric groups

摘录 / EXCERPT: For each n, let S_n be the symmetric group, and let $z_I(S_n)$ denote the proportion of character-table zeros of type I in the character table of S_n. Monotonicity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-044631 | MATHDB-360349

The bounded connectivity conjecture for breadth-critical tangles

摘录 / EXCERPT: Let $k\geq 2$ and let M be a matroid with a breadth-critical tangle $\mathcal{T}$ of order at least k. A matroid is $(s_0,s_1,\ldots,s_t)$-connected if, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-044632 | MATHDB-360350

The explicit connectivity conjecture for breadth-critical tangles

摘录 / EXCERPT: Define a sequence of non-negative integers by $t_0=0$ and $t_i=3t_{i-1}+1$ for $i\geq 1$, equivalently $t_i=(3^i-1)/2$. Let M be a matroid with a breadth-critical tangle...

conjecture | open | MathDB 页面 | 原文线索

V2-044633 | MATHDB-360351

The simultaneous breadth-preserving minor conjecture for tangle truncations

摘录 / EXCERPT: Let M be a matroid with a tangle $\mathcal{T}$ of order $k\geq 4$. For each $i\in\{4,5,\ldots,k\}$, let $T_i(\mathcal{T})$ be the truncation of $\mathcal{T}$ to order i. The si...

conjecture | open | MathDB 页面 | 原文线索

V2-044634 | MATHDB-360352

The half-breadth internally 4-connected minor conjecture

摘录 / EXCERPT: Let M be a matroid with a tangle $\mathcal{T}$ of order $k\geq 4$ and breadth m. The half-breadth internally 4-connected minor conjecture. The matroid M has an internally 4...

conjecture | open | MathDB 页面 | 原文线索

V2-044635 | MATHDB-360364

Unimodularity conjecture for super-standard inclusion matrices

摘录 / EXCERPT: Let n be a positive integer. For $0\leq i\leq \frac{n+1}{3}$ and $0\leq j\leq \frac{n+1}{3}$, let $\widetilde{\mathcal{P}}_{i,j}(n)$ be the matrix obtained by stacking the inclus...

conjecture | open | MathDB 页面 | 原文线索

V2-044636 | MATHDB-360367

Cyman-Dzido-Lapinskas-Lo exact conjecture for the online Ramsey number of P_4

摘录 / EXCERPT: Cyman-Dzido-Lapinskas-Lo conjecture. For every $n\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-044637 | MATHDB-360371

The packed-set conjecture for EvenQuads

摘录 / EXCERPT: Let an EvenQuads deck be a vector space of binary cards, and let a complete set of 2^k cards mean a k-dimensional subspace (or its affine translate). For a set of ell cards,...

conjecture | open | MathDB 页面 | 原文线索

V2-044638 | MATHDB-360427

Existence conjecture for binary upcycles of diamondicity one

摘录 / EXCERPT: Binary upcycle existence conjecture. There exist upcycles with parameters

conjecture | open | MathDB 页面 | 原文线索

V2-044639 | MATHDB-360432

Monotonicity conjecture for accurate caching-game triplets

摘录 / EXCERPT: Monotonicity conjecture. If the integer triplet (n,d,k) is accurate, then the integer triplet (n',d',k') is also accurate whenever $n'\geq n$, $d'\leq d$, and $k'\leq k$. This...

conjecture | open | MathDB 页面 | 原文线索

V2-044640 | MATHDB-360433

Ruckle's conjecture on optimal hiding in the accumulation game

摘录 / EXCERPT: Ruckle's conjecture. For all parameters, there exists an optimal hiding strategy in which the hider chooses r boxes, hides exactly d/r units of gold in each chosen box, and lea...

conjecture | open | MathDB 页面 | 原文线索

V2-044641 | MATHDB-360482

Odd-sunflower exponential bound conjecture

摘录 / EXCERPT: A k-uniform family is a family of sets in which every set has size k. An odd-sunflower is a family of at least two sets such that every element of the underlying set belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-044642 | MATHDB-360483

Sunflower-free base-size exponential bound conjecture

摘录 / EXCERPT: For a sunflower-free k-uniform family, its base elements are the elements contained in at least one set of the family. Sunflower-free base-size exponential bound conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-044643 | MATHDB-360496

Costa-Morini-Pasotti-Pellegrini simple-orderability conjecture

摘录 / EXCERPT: Let G be an abelian group and let $X\subseteq G\setminus\{0\}$. Call X simply orderable when its elements can be arranged so that every nonempty proper consecutive subsequence...

conjecture | open | MathDB 页面 | 原文线索

V2-044644 | MATHDB-360505

Conjecture on maximizers of r-star for n divisible by four

摘录 / EXCERPT: Maximizer conjecture for $r_\star$. When $n=4k$ with $k\ge 2$, the values $r_\star$ are attained by

conjecture | open | MathDB 页面 | 原文线索

V2-044645 | MATHDB-360506

Conjecture on maximizers of b-star

摘录 / EXCERPT: Maximizer conjecture for $b_\star$. For $n\ge 6$, the values $b_\star$ are attained by triples such that

conjecture | open | MathDB 页面 | 原文线索

V2-044646 | MATHDB-360507

Calbet's generalized Two Families conjecture

摘录 / EXCERPT: Let $b\geq a\geq 2$ be integers, and let $(A_i,B_i)_{i\in I}$ be a finite collection of pairs of finite sets satisfying ... and $|A_i\cap B_i|=2$ for every $i\in I$, while...

conjecture | open | MathDB 页面 | 原文线索

V2-044647 | MATHDB-360509

Biclique-intersection improvement conjecture

摘录 / EXCERPT: For a fixed graph H, let $g_H(n)$ be the maximum size of an H-intersecting family of graphs on $[n]$. Biclique-intersection conjecture. There exist $t\in\mathbb{N}$ and an abso...

conjecture | open | MathDB 页面 | 原文线索

V2-044648 | MATHDB-360510

Erdős' no-three-in-line density conjecture

摘录 / EXCERPT: Let $S\subseteq\mathbb{N}^2$ contain no three points on a common line. No-three-in-line density conjecture. ... Although finite $n\times n$ configurations have the elementary upper b...

conjecture | open | MathDB 页面 | 原文线索

V2-044649 | MATHDB-360511

Leader-Long antipodal geodesic conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, with antipodal vertices v and $\bar v$ differing in every coordinate. Leader-Long conjecture. In every red-blue colouring of the edges...

conjecture | open | MathDB 页面 | 原文线索

V2-044650 | MATHDB-360513

Modular Rado boundedness conjecture

摘录 / EXCERPT: For $k\in\mathbb{N}$, let $a_1,\ldots,a_k\in\mathbb{Z}/(2^r\mathbb{Z})$, and let $d\leq r$ be the largest non-negative integer such that 2^d divides $\sum_{i\in I}a_i$ for some n...

conjecture | open | MathDB 页面 | 原文线索

V2-044651 | MATHDB-360514

Graham's boundedness conjecture for Euclidean Ramsey sets

摘录 / EXCERPT: A finite set X in a Euclidean space is Ramsey if, for every $k\in\mathbb{N}$, some dimension contains a monochromatic congruent copy of X under every k-colouring. Graham's bo...

conjecture | open | MathDB 页面 | 原文线索

V2-044652 | MATHDB-360539

Chudnovsky-Edwards-Kim-Scott-Seymour orientation conjecture for lattice families

摘录 / EXCERPT: Chudnovsky-Edwards-Kim-Scott-Seymour conjecture. There exists an orientation D of G such that $delta_D^+(U)$ and $delta_D^-(U)$ are both nonempty for every...

conjecture | open | MathDB 页面 | 原文线索

V2-044653 | MATHDB-360554

Felsner-Weil connectivity conjecture for flip graphs of signotopes

摘录 / EXCERPT: An r-signotope is a mapping from the r-subsets of an n-element set to $\\{+,-\\}$ satisfying the relevant monotonicity condition, and a flip changes the sign of a single r-...

conjecture | open | MathDB 页面 | 原文线索

V2-044654 | MATHDB-360571

Small doubling implies small tripling for locally compact groups

摘录 / EXCERPT: Small tripling conjecture. For every $K\ge 1$ and $\varepsilon>0$, this implies

conjecture | open | MathDB 页面 | 原文线索

V2-044655 | MATHDB-360583

Locality conjecture for small-doubling sets

摘录 / EXCERPT: Let $A\subset\mathbb{R}^n$ be measurable and let $\delta<\Delta$. Locality conjecture. There is an absolute constant $\Delta>0$ such that if ... then there exists a convex set...

conjecture | open | MathDB 页面 | 原文线索

V2-044656 | MATHDB-360602

Kohayakawa-Kreuter asymmetric Ramsey threshold conjecture

摘录 / EXCERPT: Kohayakawa-Kreuter conjecture. There exist constants $c,C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044657 | MATHDB-360610

Speyer's conjecture on matroid subdivision face numbers

摘录 / EXCERPT: Let $\mathcal{S}$ be a subdivision of $\Delta_{k,n}$ into smaller matroid polytopes. For each $1\leq i\leq n$, let f_i be the number of cells of $\mathcal{S}$ of dimension $n-i$...

conjecture | open | MathDB 页面 | 原文线索

V2-044658 | MATHDB-360611

Speyer's non-negativity conjecture for matroid g-polynomials

摘录 / EXCERPT: Let M be a matroid, and let $g_{\mathsf{M}}(t)\in\mathbb{Z}[t]$ denote its g-polynomial. Speyer's conjecture. For every matroid M, the polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-044659 | MATHDB-360612

Positivity conjecture for connected matroids with series-parallel subdivisions

摘录 / EXCERPT: Connected series-parallel subdivision conjecture. The polynomial $\wwtilde{g}_{\mathsf{M}}(t-1)$ has non-negative coefficients. This is a stronger positivity statement beyond Schub...

conjecture | open | MathDB 页面 | 原文线索

V2-044660 | MATHDB-360625

The Sprout winning-strategy conjecture

摘录 / EXCERPT: Let Sprout start with n dots. In each move, a player connects a dot to itself or to another dot by a non-self-crossing curve, subdivides the curve by placing a new dot, and maint...

conjecture | open | MathDB 页面 | 原文线索

V2-044661 | MATHDB-360639

Saks's extremal gap conjecture for heights of incomparable elements

摘录 / EXCERPT: For a poset P, let $h(P,x)$ denote the average height of x, and let $\eta(P)$ be the smallest absolute difference $|h(P,x)-h(P,y)|$ over incomparable elements x,y. Set ... Saks...

conjecture | open | MathDB 页面 | 原文线索

V2-044662 | MATHDB-360647

The maximal edge-length sum conjecture for full grid lattice polygons

摘录 / EXCERPT: Let $a(n)$ be the maximum of the sum of the squares of the edge-lengths over full grid lattice polygons with n vertices, and let $\alpha(n)$ be the explicit lower-bound function...

conjecture | open | MathDB 页面 | 原文线索

V2-044663 | MATHDB-360653

The GM-rule Nash equilibrium conjecture for exact slow NIM

摘录 / EXCERPT: Consider the exact slow game NIM$(n,n-1)$ for ℓ players: a position is a non-negative n-vector, players move cyclically, and on each move a player keeps one entry unchanged...

conjecture | open | MathDB 页面 | 原文线索

V2-044664 | MATHDB-360658

Conjecture on the growth constant for harmonic measures on $\mathbb{Z}^2$

摘录 / EXCERPT: Let $\psi(\mathbb{Z}^2)$ denote the reciprocal extremal factor governing the exponential lower bound for positive harmonic measures on finite subsets of $\mathbb{Z}^2$. The propose...

conjecture | open | MathDB 页面 | 原文线索

V2-044665 | MATHDB-360659

$*$-connectivity conjecture for extremal sets in $\mathbb{Z}^d$

摘录 / EXCERPT: For $d\geq3$ and $n\geq2$, let $\mathfs{M}_n(\mathbb{Z}^d)$ be the least positive harmonic measure among subsets of $\mathbb{Z}^d$ of cardinality n. Two lattice vertices are $*$-...

conjecture | open | MathDB 页面 | 原文线索

V2-044666 | MATHDB-360664

Approximate independence for sparse partial Latin rectangles

摘录 / EXCERPT: For $k\leq n$, let $\mathcal{LR}_{k,n}$ be the set of $k\times n$ Latin rectangles with symbol set $[n]$. A partial Latin rectangle is βn-sparse if each row and column conta...

conjecture | open | MathDB 页面 | 原文线索

V2-044667 | MATHDB-360692

Perkins-Perarnau's coloring extremal conjecture for triangle-free and square-free cubic graphs

摘录 / EXCERPT: Let G be a finite graph, let K_q be the complete graph on q vertices, and let $\hom(G,K_q)$ denote the number of graph homomorphisms from G to K_q. Perkins-Perarnau's col...

conjecture | open | MathDB 页面 | 原文线索

V2-044668 | MATHDB-360706

Erdős's bounded finite-sums conjecture for positive-density sets

摘录 / EXCERPT: Erdős's conjecture. For every $k\in\mathbb{N}$, there exist an integer $t\geq0$ and an infinite set $B\subset\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044669 | MATHDB-360707

The infinite nested sumsets conjecture

摘录 / EXCERPT: Infinite nested sumsets conjecture. There is an infinite sequence of infinite sets $B_1,B_2,\ldots\subset\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044670 | MATHDB-360708

The density threshold conjecture for unrestricted sumsets

摘录 / EXCERPT: Density threshold conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-044671 | MATHDB-360709

The ordered linear sumset conjecture

摘录 / EXCERPT: Ordered linear sumset conjecture. There exist an infinite increasing set $B=\{b_1<b_2<\cdots\}\subset\mathbb{N}$ and an integer $t\geq0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044672 | MATHDB-360710

The uniform sets bounded-sums conjecture

摘录 / EXCERPT: Uniform sets bounded-sums conjecture. If A is $\mathsf{U}^k(\Phi)$-uniform for some Følner sequence Φ, then for every $\ell_1,\ldots,\ell_k\in\mathbb{N}$ there is an infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-044673 | MATHDB-360714

The logarithmic quantitative sumset conjecture

摘录 / EXCERPT: Quantitative sumset conjecture. For every $\delta>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-044674 | MATHDB-360715

The multiplicative bounded-product conjecture

摘录 / EXCERPT: Multiplicative bounded-product conjecture. There exist an infinite set $B\subset\mathbb{N}$ and $t\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044675 | MATHDB-360716

The group-theoretic two-fold sumset conjecture

摘录 / EXCERPT: Group-theoretic sumset conjecture. There exist an infinite set $B\subset G$ and $t\in G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044676 | MATHDB-360722

The augmenting-path meta-conjecture for infinite structures

摘录 / EXCERPT: Consider theorems about finite combinatorial structures that admit proofs by augmenting paths. Augmenting-path meta-conjecture. Such theorems should remain true in a structural for...

conjecture | open | MathDB 页面 | 原文线索

V2-044677 | MATHDB-360728

Gabow's exchange-distance conjecture for disjoint matroid bases

摘录 / EXCERPT: Let M be a matroid of rank r, and let X_1 and X_2 be disjoint bases. The exchange distance of two basis sequences is the minimum number of symmetric exchanges needed to tra...

conjecture | open | MathDB 页面 | 原文线索

V2-044678 | MATHDB-360774

Priestley's conjecture on approximation for weighted k-edge-connected spanning multigraphs

摘录 / EXCERPT: Priestley's conjecture. k-ECSM admits a polynomial-time $(1+O(1/k))$-approximation algorithm.

conjecture | open | MathDB 页面 | 原文线索

V2-044679 | MATHDB-360775

Improved minimal-filling bound for irreducible cycles in simplicial complexes

摘录 / EXCERPT: Let X be a d-dimensional simplicial complex in which every $(d-1)$-face is contained in at most Δ different d-faces. Let $c\in Z_{d-1}(X,\mathbb{F}_2)$ be an irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-044680 | MATHDB-360800

The feasibility conjecture for finite abelian groups

摘录 / EXCERPT: Feasibility conjecture. Every finite abelian group G is $(|G|-1)$-close.

conjecture | open | MathDB 页面 | 原文线索

V2-044681 | MATHDB-360806

Polymatroid monotonicity conjecture for matroid h*-vectors

摘录 / EXCERPT: Let E be a finite set, let P_1 and P_2 be polymatroids on E, and let $B(\mathrm{P}_1)$ and $B(\mathrm{P}_2)$ denote their base polytopes. Suppose that ... F...

conjecture | open | MathDB 页面 | 原文线索

V2-044682 | MATHDB-360814

Eberhard-Pohoata's cube-free density conjecture

摘录 / EXCERPT: Let $[N]=\{1,2,\ldots,N\}$, and let $A\subset [N]$ be 3-cube-free, meaning that it contains no projective 3-cube $\{x,y,z,x+y,y+z,x+z,x+y+z\}$. Eberhard-Pohoata's conjecture. O...

conjecture | open | MathDB 页面 | 原文线索

V2-044683 | MATHDB-360815

Cyclic cube-free density conjecture

摘录 / EXCERPT: Cyclic cube-free density conjecture. If $d\mid N$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044684 | MATHDB-360816

Cyclic progression-free density conjecture

摘录 / EXCERPT: Let $\mathbb{Z}_N$ be the cyclic group of order N, and let $A\subset\mathbb{Z}_N$ be $\{x,2x,\ldots,(d-1)x\}$-free, meaning that no x has all of $x,2x,\ldots,(d-1)x$ in A. Cy...

conjecture | open | MathDB 页面 | 原文线索

V2-044685 | MATHDB-360832

Hilton-Milner characterization conjecture for intersecting sets in 3-transitive groups

摘录 / EXCERPT: Hilton-Milner characterization conjecture. There exists an absolute constant $c_{0}<1$ such that, if S is an intersecting set in G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-044686 | MATHDB-360942

Claesson's monotonicity conjecture for 1324-avoiding permutations

摘录 / EXCERPT: For nonnegative integers n and k, let $S_n^k(1324)$ be the set of permutations of length n with k inversions that avoid the pattern 1324. Claesson's conjecture. For all n...

conjecture | open | MathDB 页面 | 原文线索

V2-044687 | MATHDB-360943

OEIS A000712 conjecture for 1324-avoiding inversion counts

摘录 / EXCERPT: For a given permutation length n, let $S_n^k(1324)$ denote the set of length-n permutations with k inversions that avoid 1324. A000712 conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-044688 | MATHDB-360944

Chow-Taylor conjecture on Latin wide Young diagrams

摘录 / EXCERPT: A Young diagram Y is wide if, for every subset S of its rows, the diagram formed by S dominates its conjugate; a filling of Y is Latin if it assigns to each row the numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-044689 | MATHDB-360947

The cycle/non-efficient contribution bound for stable primitivity rank

摘录 / EXCERPT: Let $\mathbb{\mathbf{F}}$ be a free group. Let $1\ne w\in\mathbb{\mathbf{F}}$ be a non-power, let $\sigma\in S_{d}$, and suppose that ... is an element of...

conjecture | open | MathDB 页面 | 原文线索

V2-044690 | MATHDB-360949

The stable square-length asymptotic formula for orthogonal characters

摘录 / EXCERPT: Let $1\ne w\in\mathbb{\mathbf{F}}$, let $\mu=(\mu_{1},\ldots,\mu_{k})$ be a partition, and let $e_{w,\mu}$ be the number of stable-square-length-extremal surfaces, orientable or no...

conjecture | open | MathDB 页面 | 原文线索

V2-044691 | MATHDB-360950

Modularity conjecture for uniacute spherical codes

摘录 / EXCERPT: Let $\alpha,\beta \in (0,1)$ and let $L \subseteq [-1,-\beta]\cup\{\alpha\}$ with $\alpha \in L$. An L-code is a finite set of unit vectors in Euclidean space whose pairwise inne...

conjecture | open | MathDB 页面 | 原文线索

V2-044692 | MATHDB-360955

The unimodality conjecture for plucking polynomials of hedgehog rooted trees

摘录 / EXCERPT: Let $T=T_{1,1,\ldots,1}$ be the hedgehog rooted tree (or star rooted tree) with n rays, and let f be a delay function whose values are 1 or 2. The plucking polynomial with...

conjecture | open | MathDB 页面 | 原文线索

V2-044693 | MATHDB-360957

The expansion conjecture for the generating functions $T_{a,k}(x)$

摘录 / EXCERPT: Expansion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044694 | MATHDB-360959

The compatible-uniform-coupling conjecture for coordinate-sum slices

摘录 / EXCERPT: Compatible-uniform-coupling conjecture. For every $d\in\mathbb{N}$, the triple

conjecture | open | MathDB 页面 | 原文线索

V2-044695 | MATHDB-360960

The absolutely continuous coupling conjecture for the leftover regions

摘录 / EXCERPT: Absolutely continuous coupling conjecture. For every $d\in\mathbb{N}$, there exist distributions f_1,f_2,f_3,f_4 on $\mathcal{C}$, absolutely continuous with respect to Lebesgue...

conjecture | open | MathDB 页面 | 原文线索

V2-044696 | MATHDB-360964

Conjecture that interval and integer Schur numbers coincide

摘录 / EXCERPT: Let $S(r)$ be the least positive integer N such that every r-colouring of $[N]$ contains a monochromatic solution to $x+y=z$, and let $I(r)$ be the supremum of the real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-044697 | MATHDB-360967

Sublinear graph-bootstrap running-time conjecture

摘录 / EXCERPT: Let H be a graph, and let $M_H(n)$ denote the maximum running time of the H-bootstrap process over all starting graphs on n vertices. Sublinear running-time conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-044698 | MATHDB-360995

Optimal topology conjecture for Steiner trees of the regular simplex

摘录 / EXCERPT: Let $k\geq 1$ and let d satisfy ... A good binary tree of height k is defined recursively: a height-zero tree is a single node, and for positive height at most one child-subtree...

conjecture | open | MathDB 页面 | 原文线索

V2-044699 | MATHDB-361003

Uniqueness of the IFS from the neighbor automaton

摘录 / EXCERPT: Uniqueness conjecture. For any self-similar crystallographic or self-affine tile, the combinatorial structure of the neighbor automaton uniquely determines the iterated function sy...

conjecture | open | MathDB 页面 | 原文线索

V2-044700 | MATHDB-361005

The in-degree enumeration formula for (m,n)-word lattices

摘录 / EXCERPT: The in-degree enumeration conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044701 | MATHDB-361017

Preperiod conjecture for extensions of subtraction sets of size at most three

摘录 / EXCERPT: Let A be a finite subtraction set with $|A|\leq 3$, let $S\in\{0,1\}^{\alpha}$ be a seed, and write $p=\operatorname{Per}(A,S)$. An integer of the form $kp+x$ is an extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-044702 | MATHDB-361018

Characterization of the period and preperiod for three-element subtraction sets

摘录 / EXCERPT: Let $A=\{a,b,c\}$ with $1<a<b<c$. Define integers by the division algorithm: ... ... Here $\operatorname{Per}(A)$ and $\operatorname{PrePer}(A)$ denote the eventual period and preper...

conjecture | open | MathDB 页面 | 原文线索

V2-044703 | MATHDB-361019

Longtime existence and convergence conjecture for discrete conformal curvature flows with surgery

摘录 / EXCERPT: Let (M,V,ε) be a closed connected weighted marked surface, where $\varepsilon:V\rightarrow\{0,1\}$. Consider an initial Euclidean or hyperbolic polyhedral metric induced...

conjecture | open | MathDB 页面 | 原文线索

V2-044704 | MATHDB-361039

Conjectural inversion enumerations for Magog matrices

摘录 / EXCERPT: Inversion-enumeration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044705 | MATHDB-361043

Nonreducibility conjectures for cohesive and stable Ramsey principles

摘录 / EXCERPT: Let COH denote the cohesive principle, let $\operatorname{\mathsf{Ind}}\mathscr{E}_k$ denote the indicated indivisibility problem for k-colorings, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044706 | MATHDB-361076

Diameter-boundedness conjecture for tight distance-regular graphs

摘录 / EXCERPT: Diameter-boundedness conjecture. The diameter of Γ is bounded by a function in b.

conjecture | open | MathDB 页面 | 原文线索

V2-044707 | MATHDB-361085

Sparse minimizer conjecture for induced subgraphs of the Kneser graph

摘录 / EXCERPT: Sparse minimizer conjecture. There exists a minimizer $\mathcal{F}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044708 | MATHDB-361086

Dense minimizer conjecture for induced subgraphs of the Kneser graph

摘录 / EXCERPT: Dense minimizer conjecture. There exists a family

conjecture | open | MathDB 页面 | 原文线索

V2-044709 | MATHDB-361088

Polynomiality of region counts for fixed gain-function trees

摘录 / EXCERPT: Let n be fixed, let a vary, and fix a tree of gain functions. Consider the regions of the arrangement $\mathcal{A}^{a,a}_{n-1}$ associated to that tree. Polynomiality conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-044710 | MATHDB-361117

Ramsey multiplicity conjecture for monochromatic triangles

摘录 / EXCERPT: Ramsey multiplicity conjecture. For any $c\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-044711 | MATHDB-361119

Kuca’s algebraic true complexity conjecture for polynomial progressions

摘录 / EXCERPT: Let $P_1,\dots,P_m\in\mathbf{Z}[y]$ be polynomials and let $0\leq i\leq m$. Consider the polynomial progression ... An algebraic relation among these entries is an identity ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-044712 | MATHDB-361122

Clique supersaturation conjecture for $K_{2,t}$

摘录 / EXCERPT: Let $K_{2,t}$ denote the complete bipartite graph with parts of sizes 2 and t, and let $\mathcal{N}(F,G)$ denote the number of copies of F in G. The $K_{2,t}$ clique supers...

conjecture | open | MathDB 页面 | 原文线索

V2-044713 | MATHDB-361123

Clique supersaturation conjecture for $K_{s,t}$ when $r\le s$

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with parts of sizes s and t, let K_r be the complete graph on r vertices, and let $\mathcal{N}(F,G)$ denote the number of copi...

conjecture | open | MathDB 页面 | 原文线索

V2-044714 | MATHDB-361128

Nash equilibrium conjecture for the GM-strategies in exact slow NIM

摘录 / EXCERPT: Let x be a position in the exact slow game $\operatorname{NIM}(n,n-1)$ played by ℓ players in a fixed cyclic order. A move reduces $n-1$ positive entries by 1 and leaves o...

conjecture | open | MathDB 页面 | 原文线索

V2-044715 | MATHDB-361155

The wreath conjecture for decompositions of uniform subsets

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let a wreath be the collection of intervals obtained from a cyclic permutation of $[n]$ by choosing a k-element interval and then successively choos...

conjecture | open | MathDB 页面 | 原文线索

V2-044716 | MATHDB-361172

The six-point MST-ratio conjecture for planar point sets

摘录 / EXCERPT: Let P be a set of n points in the plane. Write $w(P)$ for the length of a Euclidean minimum spanning tree of P, and let … where the maximum is over all non-trivial bipartitio...

conjecture | open | MathDB 页面 | 原文线索

V2-044717 | MATHDB-361173

Strict ML-degree monotonicity under the weak order on matroids

摘录 / EXCERPT: Let v and w be scalings for $\mathbb P^{m-1}\times\mathbb P^{n-1}$, and define … Let M_v and M_w be the linear matroids defined by the columns of $\widehat v$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044718 | MATHDB-361174

Bounded-region realization conjecture for hyperplane arrangements

摘录 / EXCERPT: Fix integers $m\geq4$ and $n\geq m$. Consider the $m-1$ coordinate hyperplanes in $\mathbb R^{m-1}$ and add n further hyperplanes. Bounded-region realization conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-044719 | MATHDB-361200

Braun-Bruegge facet bounds conjecture for symmetric edge polytopes

摘录 / EXCERPT: Let G be a connected graph with $n\geq 3$ vertices. Let ${\mathcal P}_G$ be its symmetric edge polytope, and let $N({\mathcal P}_G)$ denote the number of its facets. A 1-sum of...

conjecture | open | MathDB 页面 | 原文线索

V2-044720 | MATHDB-361201

The equality conjecture for $H(m,2)$ and $H_2(m,2)$

摘录 / EXCERPT: Let $H(m,2)$ and $H_2(m,2)$ denote the two quantities defined in the paper for integers $m\geq 3$. Equality conjecture. … Consequently, the paper's forbidden-configuration function…

conjecture | open | MathDB 页面 | 原文线索

V2-044721 | MATHDB-361202

The general equality conjecture for $H(m,\alpha)$ and $H_2(m,\alpha)$

摘录 / EXCERPT: General equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044722 | MATHDB-361212

Lattice-determined minimal Jacobian-relation degree conjecture for line arrangements

摘录 / EXCERPT: Let $\mathcal A:f=0$ be a line arrangement of degree d in the complex projective plane. Let $\operatorname{mdr}(f)$ denote the minimal degree of a Jacobian relation for f, and...

conjecture | open | MathDB 页面 | 原文线索

V2-044723 | MATHDB-361234

Bandari-Herzog's conjecture on componentwise polymatroidal ideals

摘录 / EXCERPT: Let $S=K[x_1,\dots,x_n]$ be a polynomial ring over a field K. A monomial ideal $I\subset S$ is componentwise polymatroidal when each component $I_{\langle j\rangle}$ is polymatro...

conjecture | open | MathDB 页面 | 原文线索

V2-044724 | MATHDB-361235

The power conjecture for componentwise polymatroidal ideals

摘录 / EXCERPT: Let $I\subset S$ be a componentwise polymatroidal ideal in a polynomial ring $S=K[x_1,\dots,x_n]$. Power conjecture. Every power I^k of I has linear quotients. The source gives...

conjecture | open | MathDB 页面 | 原文线索

V2-044725 | MATHDB-361280

Steiner-system extremal conjecture for constant-column-weight disjunct matrices

摘录 / EXCERPT: Let M be an $n\times m$ t-disjunct matrix with constant column weight b. Set … and assume that p is an integer. A Steiner system $\mathfrak{S}(n,b,p)$ is a family of b-su...

conjecture | open | MathDB 页面 | 原文线索

V2-044726 | MATHDB-361285

Avoidance enumeration conjecture for 123 and 121, or 132 and 212

摘录 / EXCERPT: Avoidance enumeration conjecture. The number of r-regular words over $[n]$ avoiding 123 and 121, or 132 and 212, is

conjecture | open | MathDB 页面 | 原文线索

V2-044727 | MATHDB-361310

Sergel's bi-brick-permutation C-expansion conjecture

摘录 / EXCERPT: Let μ be a partition, and let Π range over bi-brick permutations. Associate to each Π a partition statistic $\mu(\Pi)$, a composition statistic $\alpha(\Pi)$, and a non...

conjecture | open | MathDB 页面 | 原文线索

V2-044728 | MATHDB-361316

Harary and Lauri's class reconstruction conjecture for trees

摘录 / EXCERPT: Let T be a finite tree, and let $\operatorname{crn}(T)$ denote its class reconstruction number. For the class of trees, write...

conjecture | open | MathDB 页面 | 原文线索

V2-044729 | MATHDB-361335

Lazebnik-Verstraëte conjecture on k-fold Sidon sets

摘录 / EXCERPT: Lazebnik-Verstraëte conjecture. For every integer $k\geq 3$, there exists a constant $c_k>0$ such that, for every $n\geq 1$, there exists a k-fold Sidon set A in $\mathbb{Z}_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-044730 | MATHDB-361343

Nonattainment conjecture for the rectilinear crossing density of the complete graphon

摘录 / EXCERPT: Nonattainment conjecture. There exists no $f\in\mathcal F$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-044731 | MATHDB-361370

Chvátal's conjecture for hereditary set systems

摘录 / EXCERPT: Let S be a finite set. A family $\mathcal D$ of subsets of S is downward closed if it contains every subset of each of its members. Let $\mathcal F\subseteq\mathcal D$ be an in...

conjecture | open | MathDB 页面 | 原文线索

V2-044732 | MATHDB-361396

Multicolour ordered Ramsey conjecture for powers of paths

摘录 / EXCERPT: For positive integers r, t, and n, let P_n^t be the t-th power of the path on n vertices, and let $R_<^r(P_n^t)$ denote the r-colour ordered Ramsey number of P_n^t....

conjecture | open | MathDB 页面 | 原文线索

V2-044733 | MATHDB-361479

Weak minimality conjecture for the chromatic algebra generating set

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the nth-order chromatic algebra and let E_n be the generating set appearing in the paper. Weak minimality conjecture. The set E_n is a minimal generati...

conjecture | open | MathDB 页面 | 原文线索

V2-044734 | MATHDB-361491

Nasr's monotonicity conjecture for Kazhdan-Lusztig and Z-polynomials

摘录 / EXCERPT: Let M and M' be matroids, and let $\varphi:M\to M'$ be a morphism in the category $\mathcal{M}$. Let $P_M(t)$ and $P_{M'}(t)$ denote their Kazhdan-Lusztig polynomials, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044735 | MATHDB-361511

Band induction conjecture for tripod Nim bands

摘录 / EXCERPT: Let $C_c(a,b)$ denote the quantity associated with the tripod Nim array parameterized by c, and say that the numbers from 0 through m form a band when the corresponding band...

conjecture | open | MathDB 页面 | 原文线索

V2-044736 | MATHDB-361512

Period-divisor conjecture for tripod Nim orbits

摘录 / EXCERPT: Let $D(3,n)$ denote the dynamical system governing the relevant tripod Nim arrays, and consider its periodic orbits. Period-divisor conjecture. Every periodic orbit of $D(3,n)$ has...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044737 | MATHDB-361520

Parametric determinant product evaluations generalizing Di Francesco's determinant

摘录 / EXCERPT: Let n be a positive integer, let x be a parameter, and let $(\alpha)_p$ denote the Pochhammer symbol, defined by $(\alpha)_p=\alpha(\alpha+1)\cdots(\alpha+p-1)$ for $p\geq1$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-044738 | MATHDB-361524

Conjectured identities among shifted G-determinants

摘录 / EXCERPT: Shifted G-determinant identities. For all $n\geq3$, the three chains of identities displayed in the source hold:

conjecture | open | MathDB 页面 | 原文线索

V2-044739 | MATHDB-361527

Fox-Grinshpun-Pach conjecture for Gallai-Ramsey numbers of triangles and cliques

摘录 / EXCERPT: Fox-Grinshpun-Pach conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044740 | MATHDB-361543

Kravitz-Sah conjecture on the minimum size of posets with n linear extensions

摘录 / EXCERPT: Let $\mu(n)$ denote the minimum number of elements in a finite poset with exactly n linear extensions. Kravitz-Sah conjecture. ... This would improve the known bound...

conjecture | open | MathDB 页面 | 原文线索

V2-044741 | MATHDB-361599

Late-time asymptotics conjecture for the linear charged wave equation on sub-extremal Reissner-Nordström spacetime

摘录 / EXCERPT: Late-time asymptotics conjecture. If $0<|qe|<\frac{1}{2}$, then a solution ϕ with generic, regular initial data obeys the asymptotics stated in the paper's sharp asymptotic re...

conjecture | open | MathDB 页面 | 原文线索

V2-044742 | MATHDB-361608

The Breuillard-Green conjecture for minimal doubling in compact Lie groups

摘录 / EXCERPT: Breuillard-Green conjecture. The minimal value of the doubling constant for subsets of small measure should be about

conjecture | open | MathDB 页面 | 原文线索

V2-044743 | MATHDB-361609

The minimiser conjecture for doubling in compact Lie groups

摘录 / EXCERPT: Doubling minimiser conjecture. The subset H_{δ} minimises the doubling constant among measurable subsets A satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-044744 | MATHDB-361638

The fundamental-data factorization conjecture for unitary representations

摘录 / EXCERPT: Let $G=U(p,q)$, and let X be an irreducible $(\mathfrak g,K)$-module corresponding to a combinatorial θ-stable datum ... Each $\mathcal F_i$ determines a...

conjecture | open | MathDB 页面 | 原文线索

V2-044745 | MATHDB-361652

The strict first-player disadvantage conjecture for random Chomp

摘录 / EXCERPT: A Chomp board is a finite left-justified configuration of cells, and random play means that each legal move is chosen uniformly at random. Consider a starting configuration with mo...

conjecture | open | MathDB 页面 | 原文线索

V2-044746 | MATHDB-361701

Brändén-D'León relaxation conjecture for the weak half-plane property

摘录 / EXCERPT: A matroid has the weak half-plane property when its set of bases agrees with the support of a stable polynomial. A relaxation of a matroid is obtained by relaxing a circuit-hyperpl...

conjecture | open | MathDB 页面 | 原文线索

V2-044747 | MATHDB-361705

Lev-Freiman conjecture for restricted sumsets

摘录 / EXCERPT: Lev-Freiman conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-044748 | MATHDB-361706

Alon-Grynkiewicz-Kisielewicz-Przesławski Pascal-type conjecture for neighborly boxes

摘录 / EXCERPT: Let $n(0,d)=1$ for all $d\geqslant1$, and for $1\leqslant k\leqslant d$ let $n(k,d)$ be the maximum number of pairwise k-neighborly boxes in $\mathbb{R}^d$. Pascal-type conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-044749 | MATHDB-361707

Optimality conjecture for the 2-neighborly-box construction

摘录 / EXCERPT: Let $a(n)$ be the non-decreasing list of all positive integers except 1, with each number $3\cdot2^r$ repeated twice, and define ... Let $n(2,d)$ denote the maximum number of pairw...

conjecture | open | MathDB 页面 | 原文线索

V2-044750 | MATHDB-361708

Bipartite-covering formulation of the 2-neighborly-box conjecture

摘录 / EXCERPT: Let $b(d)$ be defined by ... where $a(n)$ is the sequence specified in the paper. For a graph G, let $\operatorname{bp}_2(G)$ be the least number of bipartite cliques in a 2-cove...

conjecture | open | MathDB 页面 | 原文线索

V2-044751 | MATHDB-361709

Splitting-game optimality conjecture for neighborly codes

摘录 / EXCERPT: Let S^d be the set of strings of length d over $\{0,1,*\}$. A splitting replaces one joker in a string by 0 and 1; a splitting is legal when replacing the original string p...

conjecture | open | MathDB 页面 | 原文线索

V2-044752 | MATHDB-361712

The complement-closed Lubell bound conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph that is not a matching. For a poset P, let $\mathop{}\!\mathrm{La}_{sym}(n,P)$ denote the maximum size of a P-free complement-closed family of subs...

conjecture | open | MathDB 页面 | 原文线索

V2-044753 | MATHDB-361713

The non-principal pair of posets conjecture for the incidence graph of K_4

摘录 / EXCERPT: Let $G_4=(A,B,E)$ be the bipartite incidence graph of the complete graph K_4, with $A=E(K_4)$, $B=V(K_4)$, and $(e,v)\in E$ exactly when $v\in e$. Let $P_{G_4,A}$ and $P_{G_4,B}$...

conjecture | open | MathDB 页面 | 原文线索

V2-044754 | MATHDB-361718

Kohayakawa-Kreuter conjecture on asymmetric Ramsey thresholds

摘录 / EXCERPT: Let $H_1,\dots,H_r$ be graphs. For a graph H, write $m_2(H)$ for its 2-density, and, when $m_2(H_1)\geq m_2(H_2)$, define the mixed 2-density by ... Let $G_{n,p}$ be the binomial ra...

conjecture | open | MathDB 页面 | 原文线索

V2-044755 | MATHDB-361721

Kleitman's conjecture on maximal intersecting subfamilies of the Boolean lattice

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $2^{[n]}$ denote its power set, and let $\mathcal{F}$ be a maximal intersecting subfamily of $2^{[n]}$. For each $A\subseteq[n]$, let e_A be the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-044756 | MATHDB-361746

Soft polyhedric realization conjecture for balanced normal convex tilings

摘录 / EXCERPT: Soft polyhedric realization conjecture. There exists a combinatorially equivalent, soft polyhedric tiling M'.

conjecture | open | MathDB 页面 | 原文线索

V2-044757 | MATHDB-361764

Asymptotic expected-length conjecture for grand zigzag knight's paths

摘录 / EXCERPT: Let a grand zigzag knight's path be a path whose number of steps has an expected value determined by the corresponding family of paths ending on the x-axis at size n. Asymptoti...

conjecture | open | MathDB 页面 | 原文线索

V2-044758 | MATHDB-361779

Erdős-Frankl-Füredi conjecture on the threshold for nontrivial cover-free families

摘录 / EXCERPT: Erdős-Frankl-Füredi conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044759 | MATHDB-361780

Asymptotic upper-bound conjecture for wide-sense 2-frameproof codes

摘录 / EXCERPT: Wide-sense 2-frameproof code conjecture. For large n,

conjecture | open | MathDB 页面 | 原文线索

V2-044760 | MATHDB-361786

Realizability conjecture for potential hyperbolic posets

摘录 / EXCERPT: Let S be a set of compositions of n into at most s parts, and let $\mathcal{L}(S)$ be the associated potential hyperbolic poset obtained by taking pairwise joins and then the...

conjecture | open | MathDB 页面 | 原文线索

V2-044761 | MATHDB-361798

Brändén-Huh's characterization of matroid bases by 8/7-Rayleigh polynomials

摘录 / EXCERPT: Brändén-Huh's conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-044762 | MATHDB-361799

Choe-Oxley-Sokal-Wagner's rank-three transversal matroid conjecture

摘录 / EXCERPT: Rank-three transversal matroid conjecture. Every transversal matroid of rank 3 has the half-plane property.

conjecture | open | MathDB 页面 | 原文线索

V2-044763 | MATHDB-361822

Genus bound for reconstructing non-separable matroids from Tutte polynomials

摘录 / EXCERPT: Let M be a non-separable matroid. Let C be a largest circuit of M, let D be a largest cocircuit of M, and let $\mathcal{B}(M)$ denote the set appearing in the genus bound...

conjecture | open | MathDB 页面 | 原文线索

V2-044764 | MATHDB-361823

One-step separation conjecture for genus Tutte polynomials

摘录 / EXCERPT: Let g be a positive integer, and let $\{M_i\}$ and $\{N_i\}$ be sequences of matroids. One-step separation conjecture. There exist two infinite sequences of matroids $\{M_i\}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-044765 | MATHDB-361852

Linear-order conjecture for complete bipartite poset saturation

摘录 / EXCERPT: Let $s\geq t\geq 2$ be fixed, and let $\mathcal{K}_{s,t}$ be the complete bipartite poset with s upper-layer and t lower-layer pairwise incomparable vertices, every upper-layer...

conjecture | open | MathDB 页面 | 原文线索

V2-044766 | MATHDB-361871

The extremizer conjecture for the critical Riesz projection inequality

摘录 / EXCERPT: Let $1<q<2$ or $2<q\leq\infty$, let $\mathfrak{p}(q)$ be the one-dimensional critical exponent, and let P_+ denote the Riesz projection on the unit circle. Extremizer conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-044767 | MATHDB-361881

Extremal critical collapse conjecture for vacuum gravity

摘录 / EXCERPT: Consider vacuum gravitational collapse, governed by ... The rotating extremal Kerr solution plays the role of extremal Reissner-Nordström. Vacuum extremal-collapse conjecture. Extrem...

conjecture | open | MathDB 页面 | 原文线索

V2-044768 | MATHDB-361894

The automorphism-group conjecture for extremal self-dual binary codes of length 72

摘录 / EXCERPT: Let C be a binary linear self-dual $[72,36,16]$ code, and let σ be a permutation in its permutation automorphism group $\operatorname{PAut}(C)$. Automorphism-group conject...

conjecture | open | MathDB 页面 | 原文线索

V2-044769 | MATHDB-361906

Staircase characterization conjecture for extremal star-forest decompositions

摘录 / EXCERPT: Let k be a positive integer, and let G be a complete geometric graph on $2k$ points. A k-staircase is the point-set construction defined earlier in the source. Staircase char...

conjecture | open | MathDB 页面 | 原文线索

V2-044770 | MATHDB-361958

Lu-Thompson conjecture on subquadratic diagonal poset Ramsey numbers

摘录 / EXCERPT: Let Q_n denote the Boolean lattice of subsets of an n-element set, and let $R(Q_m,Q_n)$ be the induced poset Ramsey number for the pair (Q_m,Q_n). Lu-Thompson's conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-044771 | MATHDB-361959

Subpolynomial-factor conjecture for diagonal poset Ramsey numbers

摘录 / EXCERPT: Let Q_n denote the Boolean lattice of subsets of an n-element set, let $R(Q_m,Q_n)$ be the induced poset Ramsey number, and let $\varepsilon>0$. Subpolynomial-factor conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-044772 | MATHDB-361960

Fixed-dimensional diagonal poset Ramsey asymptotic conjecture

摘录 / EXCERPT: Let Q_n denote the Boolean lattice of subsets of an n-element set, and let $R(Q_m,Q_n)$ be the induced poset Ramsey number. Fixed-m asymptotic conjecture. For fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-044773 | MATHDB-361961

Second-author conjecture on weak versus induced poset Ramsey numbers

摘录 / EXCERPT: Let Q_n denote the Boolean lattice of subsets of an n-element set, let $R(Q_n,Q_n)$ be the induced poset Ramsey number, and let $R^{\mathrm{w}}(Q_n,Q_n)$ be the weak poset Rams...

conjecture | open | MathDB 页面 | 原文线索

V2-044774 | MATHDB-361982

Sniady's positivity conjecture for hypermap coefficients

摘录 / EXCERPT: Let π, μ, and ν be partitions, and let $g^\pi_{\mu,\nu}$ denote the corresponding hypermap coefficient. Set $b:=\alpha-1$. Sniady's conjecture. For any partitions π...

conjecture | open | MathDB 页面 | 原文线索

V2-044775 | MATHDB-361983

Optimal-strategy limit distribution conjecture for semi-restricted Rock, Paper, Scissors

摘录 / EXCERPT: Consider the semi-restricted Rock, Paper, Scissors game with score S_n, and let S_n^{op} denote the score when both players use optimal strategies. Let W and...

conjecture | open | MathDB 页面 | 原文线索

V2-044776 | MATHDB-362002

The pattern-avoidance inequalities conjecture for restricted-cycle permutations

摘录 / EXCERPT: For each $n\geq 1$, let $a_n^{\{1,3\}}(\sigma)$ count length-n permutations avoiding the pattern σ whose cycle lengths belong to $\{1,3\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044777 | MATHDB-362025

Branch-depth as minor closure of contraction-deletion-depth

摘录 / EXCERPT: Branch-depth minor-closure conjecture. There exists a function f such that, if a matroid M has branch-depth d, then there exists a matroid N with contraction-deletion-depth...

open_problem | open | MathDB 页面 | 原文线索

V2-044778 | MATHDB-362031

Liu--Xu's Davenport-constant closeness conjecture

摘录 / EXCERPT: Let Γ be a finite abelian group. Its Davenport constant $D(\Gamma)$ is the least positive integer such that every sequence of $D(\Gamma)$ elements of Γ has a nonempty...

conjecture | open | MathDB 页面 | 原文线索

V2-044779 | MATHDB-362032

The weighted bounded-exchange conjecture for F-avoiding matroid bases

摘录 / EXCERPT: Let M be a matroid on a ground set E, let Γ be an abelian group, let $\psi\colon E\to\Gamma$ be a group labeling, let $F\subseteq\Gamma$ be a finite set, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044780 | MATHDB-362054

Klick-Strungaru-Tcaciuc conjecture on arithmetic progressions in cut-and-project sets

摘录 / EXCERPT: Let G be a locally compact second countable group and let $\Lambda \subset G$ be a cut-and-project set. A subset of G is an arithmetic progression of length r if it has the f...

conjecture | open | MathDB 页面 | 原文线索

V2-044781 | MATHDB-362055

Polynomial Szemerédi-type conjecture for multiplicatively closed cut-and-project sets

摘录 / EXCERPT: Polynomial Szemerédi-type conjecture. The set S_F is syndetic. This is proposed as a polynomial strengthening of the preceding corollary, motivated by polynomial extensions of Sz...

conjecture | open | MathDB 页面 | 原文线索

V2-044782 | MATHDB-362060

Conjecture on planar 2-edge-apex graphs and flexible rigidity circuits

摘录 / EXCERPT: Let G be a planar graph, and let e and f be edges added to G. A graph is $(3,6)$-tight if it has $3|V|-6$ edges and every subgraph H with at least three vertices satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-044783 | MATHDB-362063

The critical bias ratio conjecture for the square lattice

摘录 / EXCERPT: Critical bias ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044784 | MATHDB-362072

The Existence Conjecture for combinatorial designs

摘录 / EXCERPT: A design with parameters (n,q,r,λ) is a set S of q-subsets of an n-set X such that every r-subset of X belongs to exactly λ elements of S. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044785 | MATHDB-362073

Erdős' existence conjecture for high-girth Steiner triple systems

摘录 / EXCERPT: A Steiner system with parameters (n,q,r) is a design with parameters $(n,q,r,1)$. In a partial (n,q,r)-Steiner system, a (j,i)-configuration is a set of i q-element subse...

conjecture | open | MathDB 页面 | 原文线索

V2-044786 | MATHDB-362077

Absorber degeneracy conjecture for cliques

摘录 / EXCERPT: Absorber degeneracy conjecture. For each integer $q\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-044787 | MATHDB-362089

The asymptotic sum-dilate formula for fractional dilates

摘录 / EXCERPT: The asymptotic sum-dilate conjecture. $$

conjecture | open | MathDB 页面 | 原文线索

V2-044788 | MATHDB-362090

Finite-set approximation of fractional-dilate sum-dilate data

摘录 / EXCERPT: Finite-set approximation conjecture. There exists a finite subset S of the integers such that, for every integer k with $|k|\leq N$,

conjecture | open | MathDB 页面 | 原文线索

V2-044789 | MATHDB-362114

Characterization of spherical orbit spaces from affine colorings

摘录 / EXCERPT: Spherical orbit-space conjecture. The space $N(P,\lambda)$ is homeomorphic to S^n if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-044790 | MATHDB-362174

Snevily's conjecture on L-intersecting families of subsets

摘录 / EXCERPT: Let $[n]=\{1,2,\dots,n\}$. For a set of nonnegative integers $\mathcal{L}=\{l_1,l_2,\dots,l_s\}$, a family $\mathcal{A}=\{A_1,A_2,\dots,A_m\}$ of subsets of $[n]$ is $\mathcal{L}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-044791 | MATHDB-362177

Erdős-Chvátal simplex conjecture

摘录 / EXCERPT: Let $[n]=\{1,\dots,n\}$, let $\binom{[n]}{k}$ denote the family of k-subsets of $[n]$, and let a d-simplex be a collection of $d+1$ members of $\binom{[n]}{k}$ whose total inte...

conjecture | open | MathDB 页面 | 原文线索

V2-044792 | MATHDB-362191

Methuku-Tomon exponent and row-sparsity conjecture

摘录 / EXCERPT: Let $t\geq 2$, and let A be a zero-one matrix. The matrix is assumed to contain at most t 1-entries in each row. Let $\operatorname{ex}(n,A)$ denote its extremal number. Meth...

conjecture | open | MathDB 页面 | 原文线索

V2-044793 | MATHDB-362192

The diamond-free family conjecture for the subspace lattice

摘录 / EXCERPT: Let $\mathcal{L}_{n}(q)$ be the lattice of subspaces of $\mathbb{F}_{q}^{n}$, and let $\mathcal{Q}_{2}$ denote the diamond poset. Write $ex_{q}(n,\mathcal{Q}_{2})$ for the maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-044794 | MATHDB-362195

Mubayi's conjecture on d-cluster-free families of subspaces

摘录 / EXCERPT: Let V be an n-dimensional vector space over $\mathbb{F}_q$, and let $k \ge d \ge 3$ be integers satisfying ... For a family $\mathcal{F} \subset \genfrac{[}{]}{0pt}{}{V}{k}$, cal...

conjecture | open | MathDB 页面 | 原文线索

V2-044795 | MATHDB-362197

PSPACE-completeness conjecture for sums of superstars

摘录 / EXCERPT: Let a sum of superstars be a disjunctive sum of impartial combinatorial games whose components are superstars, and let ruleset{Paint Can} denote the corresponding game. PSPACE-comp...

conjecture | open | MathDB 页面 | 原文线索

V2-044796 | MATHDB-362215

Matching lower-bound conjecture for the universal exponent of sphere homeomorphs

摘录 / EXCERPT: Let d be a positive integer, and let λ_d be the universal exponent for homeomorphs of fixed d-complexes, defined by ... for every fixed d-complex S. Matching lower-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-044797 | MATHDB-362243

Existence conjecture for Johnson statistics

摘录 / EXCERPT: Let $\mathcal{L}=\mathbb{Z}^{a-1}$ be the weight lattice, let $\mathcal{R}$ be its root sublattice, and let $\mathsf{T}(x_1,\ldots,x_{a-1})=x_1+2x_2+\cdots+(a-1)x_{a-1}$. A Johnson...

conjecture | open | MathDB 页面 | 原文线索

V2-044798 | MATHDB-362278

Propp's second-quotient conjecture for $(3n,3n)$-benzels

摘录 / EXCERPT: Propp's conjecture. The numbers T_n satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-044799 | MATHDB-362279

Propp's quotient conjecture for $(3n+1,3n+1)$-benzels

摘录 / EXCERPT: Propp's conjecture. The numbers T_n satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-044800 | MATHDB-362280

Second-quotient conjecture for $(3n+1,3n+2)$-benzels

摘录 / EXCERPT: Second-quotient conjecture. The numbers T_n satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-044801 | MATHDB-362322

Unique-largest-antichain conjecture for balls in the Boolean lattice

摘录 / EXCERPT: Unique-largest-antichain conjecture. The largest layer of $B_r[p,q]$ is its unique largest antichain.

conjecture | open | MathDB 页面 | 原文线索

V2-044802 | MATHDB-362329

The cyclic sieving conjecture for decorated permutations and extended codes

摘录 / EXCERPT: Cyclic sieving conjecture. For any positive integer n and $0\leq j\leq n$, the triple

conjecture | open | MathDB 页面 | 原文线索

V2-044803 | MATHDB-362355

The perfect-game existence conjecture for PackIt! grids

摘录 / EXCERPT: Let $m \leq n$ be positive integers. Write $P(m,n)$ for the set of primes relevant to the $m \times n$ grid, let $\gamma(m,n)$ denote its gap, and let $K(m,n)$ be the total number...

conjecture | open | MathDB 页面 | 原文线索

V2-044804 | MATHDB-362371

Density of possible pentagon centers outside a disk

摘录 / EXCERPT: Let P_0 be the original pentagon in the Eden model, and let u_1,u_2,u_3,u_4,u_5 be the five possible translation vectors between centers of neighboring pentagons. The set of po...

conjecture | open | MathDB 页面 | 原文线索

V2-044805 | MATHDB-362372

Asymptotic parameter limits in the Eden model for pentagons

摘录 / EXCERPT: Let n be the number of pentagons in the model, and let $V(n)$, $E(n)$, and $H(n)$ denote the numbers of vertices, edges, and holes, respectively. The expectation is taken over th...

conjecture | open | MathDB 页面 | 原文线索

V2-044806 | MATHDB-362382

Ye's limiting efficacy distribution conjecture for numerical semigroups

摘录 / EXCERPT: A numerical semigroup Λ has efficacy $h(\Lambda)$ equal to the number of its children in the semigroup tree. Let $t(g,h)$ denote the number of numerical semigroups of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-044807 | MATHDB-362391

Nešetřil's conjecture on the chromatic bound in the GIP theorem

摘录 / EXCERPT: Let F be a graph with at least one edge. The cited theorem provides a constant $c(F)$ such that, for every natural number r, there is a graph G with $\chi(G)>r$,...

conjecture | open | MathDB 页面 | 原文线索

V2-044808 | MATHDB-362411

Ross-Yong conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let $w\in S_n$ and $\gamma\in\mathbb{Z}_{\geq 0}^n$. Write $g_{w,\gamma}=\#\{P\in {\sf Pipes}(w)\mid {\tt wt}(P)=\gamma\}$, and let ${{\sf KKoh}}(w,\gamma)$ be the set of diagrams...

conjecture | refuted | MathDB 页面 | 原文线索

V2-044809 | MATHDB-362412

Grothendieck-polynomial rule via modified Kohnert diagrams

摘录 / EXCERPT: Let $w\in S_n$ and $\gamma\in\mathbb{Z}_{\geq 0}^n$. Let $\overline{{\sf KKoh}}(w,\gamma)$ denote the modified family of diagrams associated with w and weight γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044810 | MATHDB-362439

Quasisymmetric characteristic formula for a statistic-and-descent model

摘录 / EXCERPT: Let M be an $\fS_n$-module, and let X be a set of combinatorial objects equipped with functions $\operatorname{stat}:X\to\mathbb{N}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-044811 | MATHDB-362467

Matching-size conjecture for random geometric graphs

摘录 / EXCERPT: Matching-size conjecture. Under this assumption,

conjecture | open | MathDB 页面 | 原文线索

V2-044812 | MATHDB-362473

The bounded and asymptotic quasiprimitive ratio problem

摘录 / EXCERPT: Quasiprimitive ratio conjecture. There is a constant C such that

open_problem | open | MathDB 页面 | 原文线索

V2-044813 | MATHDB-362481

The path extremal conjecture for the game cordiality number of trees

摘录 / EXCERPT: Let T be a tree of order n, and let P_n denote the path on n vertices. Path extremal conjecture. The game cordiality number satisfies ... This conjecture proposes that paths...

conjecture | open | MathDB 页面 | 原文线索

V2-044814 | MATHDB-362500

Linear asymptotics conjecture for the Waiter-Client tree degree threshold

摘录 / EXCERPT: Let $D(n)$ be the largest integer t such that for every tree T with $v(T)=n$ and $\Delta(T)\leq t$, Waiter has a winning strategy in $\operatorname{WC}(n,T)$. Linear asymptotics...

conjecture | open | MathDB 页面 | 原文线索

V2-044815 | MATHDB-362643

The atom-based parabolic combinatorial invariance conjecture

摘录 / EXCERPT: Let W_i be Coxeter groups with simple-generator subsets J_i, and let $W_i^{J_i}$ be the corresponding parabolic quotients. For $u_i,v_i\in W_i^{J_i}$, define the atoms in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-044816 | MATHDB-362667

Polynomial formulas for truncated total depth in ordered trees

摘录 / EXCERPT: Let n and ℓ be integers satisfying $n\geq \ell\geq 1$. Let $\operatorname{totalD}(n,k)$ denote the total quantity studied in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044817 | MATHDB-362678

The eventual chromatic spectrum conjecture for Kirkman triple systems

摘录 / EXCERPT: Eventual chromatic spectrum conjecture. For every integer $\delta\geqslant 3$, there exists an integer v_0 such that whenever $v\geqslant v_0$ and $v\equiv 3\pmod{6}$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-044818 | MATHDB-362689

The main conjecture on degree-\a0$k+1$ vertices in minimal k-extendable bipartite graphs

摘录 / EXCERPT: Let G be a minimal k-extendable bipartite graph, with $k\geqslant 1$, order n, and size $m=|E(G)|$. Let V_{k+1} be the set of vertices of degree precisely $k+1$. Main conje...

conjecture | open | MathDB 页面 | 原文线索

V2-044819 | MATHDB-362690

The order bound conjecture for degree-$k+1$ vertices in minimal k-extendable bipartite graphs

摘录 / EXCERPT: Let G be a minimal k-extendable bipartite graph, where $k\geqslant 1$, and let n be its order. Let V_{k+1} be the set of vertices of degree precisely $k+1$. Order bound con...

conjecture | open | MathDB 页面 | 原文线索

V2-044820 | MATHDB-362691

The eventual size bound conjecture for minimal k-extendable bipartite graphs

摘录 / EXCERPT: For each $k\geqslant 1$, let N_k be an integer, and let G be a minimal k-extendable bipartite graph of order n and size $|E|$. Eventual size bound conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-044821 | MATHDB-362692

The conjecture on P-positions for Slow SetNim with at least k playable stacks

摘录 / EXCERPT: Let n be the number of stacks, let $A=k,k+1,\ldots,n$, and consider the game $\textsc{SN}{\left(n,A\right)}$. For a position ${\mathbf p}$, let ... and let o be the number of sta...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044822 | MATHDB-362694

Structure conjecture for small quotient sets in groups

摘录 / EXCERPT: Structure conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044823 | MATHDB-362700

The Stanley-Stembridge conjecture for chromatic symmetric functions

摘录 / EXCERPT: Stanley-Stembridge conjecture. If P is a ${\bf 3}+{\bf 1}$-avoiding partial order, then X_G is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-044824 | MATHDB-362701

The generalized strand-diagram positivity conjecture

摘录 / EXCERPT: Generalized strand-diagram positivity conjecture. Let $\mathcal D$ range over all corresponding colored strand diagrams. Then

conjecture | open | MathDB 页面 | 原文线索

V2-044825 | MATHDB-362708

Interval conjecture for $(n-2,2)$ in the immersion poset

摘录 / EXCERPT: For a partition μ of n, let ... be its lower interval in the immersion poset, and let $\lessdot_I$ denote a cover relation. The $(n-2,2)$ interval conjecture states that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-044826 | MATHDB-362722

Caterpillar symbic tree basis conjecture

摘录 / EXCERPT: Let E be the ground set of matrix entries, and let $\mathcal{B}\subset E$ be a set of size $2n-1$. A symbic tree is a combinatorial object indexing a maximal cone in the tropical...

conjecture | open | MathDB 页面 | 原文线索

V2-044827 | MATHDB-362724

The even-intersection bound for strongly restricted set systems

摘录 / EXCERPT: Let t, k be integers with $t\geq 2$ and $2t-2>k$. If $(\mathcal A_1, \ldots, \mathcal A_k)$ are set families of an n element set with ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-044828 | MATHDB-362746

Marton's polynomial Freiman-Ruzsa conjecture

摘录 / EXCERPT: Let $A\subseteq \mathbb{F}_2^n$ be a set with $|A+A|\leq K|A|$. A subspace $V\subseteq \mathbb{F}_2^n$ is required to have size at most $|A|$, and translates of V are cosets of...

conjecture | open | MathDB 页面 | 原文线索

V2-044829 | MATHDB-362757

The equality-case characterization for Stanley's matroid inequality

摘录 / EXCERPT: Let M be a loopless matroid of rank n on E, let $R\subseteq E$, and put $Q=E\setminus R$. For $0\leq k\leq n$, let N_k be the number of bases having k elements in R, an...

conjecture | open | MathDB 页面 | 原文线索

V2-044830 | MATHDB-362758

The all-degree Hard Lefschetz conjecture for matroid Gorenstein rings

摘录 / EXCERPT: Let $M=(E,\mathcal I)$ be a matroid of rank d, let $\A(M)$ be its associated Gorenstein ring, and let $a\in\mathbb R^E_{>0}$. All-degree Hard Lefschetz conjecture. The ring...

conjecture | open | MathDB 页面 | 原文线索

V2-044831 | MATHDB-362767

Representation stability for uniform-matroid holonomy primitives

摘录 / EXCERPT: Uniform-matroid onset conjecture. For fixed $i geqeq 3$, the sequence $cmathcal{L}(n)_i$ is representation stable past $n=2i-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-044832 | MATHDB-362768

Onset of representation stability for braid-matroid holonomy Lie algebras

摘录 / EXCERPT: Braid-matroid primitive onset conjecture. The onset of representation stability is $n=2i$ for fixed $i geqeq 1$ when $A(n)=cmathrm{OS}(cmathrm{Br}_n)$, and is $n=2i$ for fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-044833 | MATHDB-362769

Equivariant degree-one injectivity for braid-matroid Varchenko-Gelfand duals

摘录 / EXCERPT: Braid-matroid equivariant injectivity conjecture. For all $i eggeq 1$, there exist equivariant injections

conjecture | open | MathDB 页面 | 原文线索

V2-044834 | MATHDB-362774

The consecutive Davenport constant conjecture for finite groups

摘录 / EXCERPT: Consecutive Davenport constant conjecture. For every finite group G, it holds

conjecture | open | MathDB 页面 | 原文线索

V2-044835 | MATHDB-362807

The double-caterpillar upper-boundary conjecture for tree profiles

摘录 / EXCERPT: Let $\mathrm{DCat}^{p,q}_{\infty}=(\mathrm{DCat}^{p,q}_n)_{n\geq 2}$ denote the increasing sequence of double caterpillars with ratio (p,q). Let Cat_k be the caterpill...

conjecture | open | MathDB 页面 | 原文线索

V2-044836 | MATHDB-362822

PL-structure conjecture for Alexandrov spaces without proper extremal subsets

摘录 / EXCERPT: Let X be an Alexandrov space, meaning a finite-dimensional complete length space with curvature bounded below in the Alexandrov sense, and suppose that X contains no proper ext...

conjecture | open | MathDB 页面 | 原文线索

V2-044837 | MATHDB-362857

Dominant-coefficient conjecture for the universal Drinfeld-Yetter algebra

摘录 / EXCERPT: Let $c_{n,m}$ denote the coefficient introduced in Proposition, and let r_n^id and r_m^id be the corresponding elements of the universal Drinfeld-Yetter algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-044838 | MATHDB-362858

Antidiagonal-permutation dominant-coefficient conjecture

摘录 / EXCERPT: For $n\geq 1$, define the reversal permutation ... Let $r_n^{\pi_n}$ be the corresponding element of $\mathfrak{U}_{\mathsf{DY}}^1$, and let $p_{n,m}$ denote the coefficient of...

conjecture | open | MathDB 页面 | 原文线索

V2-044839 | MATHDB-362859

Cardinality formula conjecture for looms

摘录 / EXCERPT: Let $\mathfrak{L}_{n,m}$ be the family of looms and write $H_{n,m}=\lvert\mathfrak{L}_{n,m}\rvert$. Let $T_{m,k}$ be defined by ... with $T_{0,0}=1$ and $T_{0,k}=0$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-044840 | MATHDB-362860

Pattern-avoidance image conjecture for looms

摘录 / EXCERPT: Given a set of permutations S, let $\operatorname{Av}(S)$ denote the set of permutations avoiding the patterns in S. Let $\mathfrak{L}_{n,m}$ be the family of looms and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044841 | MATHDB-362879

Critical-parameter limit for the measure of t-intersecting subspaces

摘录 / EXCERPT: Let t be a fixed positive integer. Let Ω_n be the set of all subspaces of an n-dimensional vector space over the field with q elements, let $A_n^{(t)}$ be the family o...

conjecture | open | MathDB 页面 | 原文线索

V2-044842 | MATHDB-362887

Generalized stability conjecture for binary scalar products

摘录 / EXCERPT: Generalized stability conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-044843 | MATHDB-362894

The quadratic formula conjecture for two non-bonding dominoes

摘录 / EXCERPT: Let $D(r,c,d)$ denote the number of ways to place d non-bonding dominoes on an $r\times c$ rectangular board. Quadratic formula conjecture. For integers $r,c\ge 3$, ... The formula...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-044844 | MATHDB-362895

The maximum-filling formula conjecture for mixed even and odd board dimensions

摘录 / EXCERPT: Let $\bar d(r,c)$ denote the maximum number of non-bonding dominoes that can be placed on an $r\times c$ rectangular board. Maximum-filling conjecture. If one of r and c is an...

conjecture | open | MathDB 页面 | 原文线索

V2-044845 | MATHDB-362896

Nonexistence of d-point simplex 3-designs

摘录 / EXCERPT: Let Δ_d denote the d-point probability simplex, and let a simplex t-design be a collection of points in Δ_d whose moments agree with those of the uniform simplex...

conjecture | open | MathDB 页面 | 原文线索

V2-044846 | MATHDB-362899

Lower bound for the size of simplex t-designs

摘录 / EXCERPT: Let $N_*(d,t)$ be the minimal number of elements in a t-design in the d-point simplex, and let $\binom{a}{b}$ denote a binomial coefficient. Simplex-design lower-bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-044847 | MATHDB-362901

Kamčev-Morrison conjecture on uncommon minimal rank-2 systems

摘录 / EXCERPT: Let $\boldsymbol{[?]}$ be a $(2\times k)$-system, and let $s(\Psi)$ denote the length of its shortest equation, where the length of an equation is the number of nonzero coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-044848 | MATHDB-362919

An IP-star recurrence conjecture for semigroup homomorphisms

摘录 / EXCERPT: Let $S=(S,\cdot)$ and $R=(R,\cdot)$ be semigroups, with R admitting a left invariant mean, and let $A\subseteq R$ be piecewise syndetic. For commuting homomorphisms...

conjecture | open | MathDB 页面 | 原文线索

V2-044849 | MATHDB-362920

The polynomial semiring sum-product partition conjecture

摘录 / EXCERPT: Let $\mathbb{N}[x]$ be the polynomial semiring with coefficients in $\mathbb{N}$, and let $\mathbb{N}[x]=C_0\cup C_1$ be a bipartition. The polynomial semiring conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-044850 | MATHDB-362930

The converse Bernstein-coloring conjecture for independence in \(\mathcal{B}_{m,n}(d,d)\)

摘录 / EXCERPT: Let G be a bipartite graph, let $d\geq 0$ be an integer, and let $\mathcal{B}_{m,n}(d,d)$ denote the matroid introduced in the paper. A d-coloring of the edges of G is d-Be...

conjecture | open | MathDB 页面 | 原文线索

V2-044851 | MATHDB-362958

Endpoint invariance conjecture for s-Cambrian congruences

摘录 / EXCERPT: Endpoint invariance conjecture for s-Cambrian congruences. For fixed ${\boldsymbol{s}}$, the following depend only on the endpoints of α: the cardinality of the α-Cam...

conjecture | open | MathDB 页面 | 原文线索

V2-044852 | MATHDB-362959

Orientation invariance conjecture for s-permutree lattices

摘录 / EXCERPT: Orientation invariance conjecture for s-permutree lattices. For fixed ${\boldsymbol{s}}$, changing any up decoration to a down decoration in ${\boldsymbol{\delta}}$ does not affect...

conjecture | open | MathDB 页面 | 原文线索

V2-044853 | MATHDB-362991

Bergelson-McCutcheon’s non-commuting Schur conjecture

摘录 / EXCERPT: Let G be a group whose centralizer subgroup for every element has infinite index; call such a group sufficiently non-commutative. Suppose that G is finitely colored. Bergelson-...

conjecture | open | MathDB 页面 | 原文线索

V2-044854 | MATHDB-363013

A polynomial sunflower conjecture for sparse vectors over finite cyclic groups

摘录 / EXCERPT: Polynomial sunflower conjecture. There exist constants $c_1(p),c_2(p)>0$ such that whenever

conjecture | open | MathDB 页面 | 原文线索

V2-044855 | MATHDB-363038

Huh-Matherne-Mészáros-St. Dizier's Lorentzian conjecture for normalized Schubert polynomials

摘录 / EXCERPT: Let $w\in S_n$, let $\mathfrak{S}_w(x)$ be the Schubert polynomial indexed by w, and let N be the normalization operator on monomials, defined by ... A homogeneous polynomial is...

conjecture | open | MathDB 页面 | 原文线索

V2-044856 | MATHDB-363039

Mészáros-Setiabrata-St. Dizier's unit-step support conjecture for Grothendieck polynomials

摘录 / EXCERPT: Fix $w\in S_n$ and write ... For a monomial exponent α, write $|\alpha|=\alpha_1+\cdots+\alpha_n$, and order exponents componentwise: $\alpha<\beta$ when $\alpha_i\leq\beta_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-044857 | MATHDB-363040

Mészáros-Setiabrata-St. Dizier's coefficient-sum conjecture for Grothendieck polynomials

摘录 / EXCERPT: Fix $w\in S_n$ and write ... Let $\operatorname{supp}(\mathfrak{G}_w^{\mathrm{top}}(x))$ denote the exponents of monomials of highest degree in $\mathfrak{G}_w(x)$, and order exponen...

conjecture | open | MathDB 页面 | 原文线索

V2-044858 | MATHDB-363046

Dichotomy for saturation functions of sequences

摘录 / EXCERPT: Let u be a sequence, and let $\operatorname{Sat}(u,n)$ denote its saturation function on an alphabet of size n. The saturation dichotomy. For every sequence u, either ... or .....

conjecture | open | MathDB 页面 | 原文线索

V2-044859 | MATHDB-363068

Even-dimensional monomial distance conjecture over finite p-adic rings

摘录 / EXCERPT: Even-dimensional monomial distance conjecture. If $\delta_{E_1,E_2}\gg p^{-n/2}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-044860 | MATHDB-363074

Boundary extremal-function conjecture for the bounded integral mean spectrum

摘录 / EXCERPT: Boundary extremal-function conjecture. For each $t\in\mathbb{R}$, $B_b(t)$ has at least one extremal function whose Pre-Schwarzian derivative lies in $\partial\mathcal{T}$.

conjecture | open | MathDB 页面 | 原文线索

V2-044861 | MATHDB-363099

Existence of threshold functions for monotone graph properties

摘录 / EXCERPT: Let $G_{n,p,d}$ denote the class of d-dependent random graph distributions on graphs with n vertices and marginal edge probability p. Let $d4ab$ be a monotone graph property,...

conjecture | open | MathDB 页面 | 原文线索

V2-044862 | MATHDB-363100

Bounded gap between lower and upper threshold functions

摘录 / EXCERPT: Let $G_{n,p,d}$ denote the class of d-dependent random graph distributions on graphs with n vertices and marginal edge probability p. Let $\mathcal{P}$ be a monotone graph pr...

conjecture | open | MathDB 页面 | 原文线索

V2-044863 | MATHDB-363101

Lower connectivity threshold for d-dependent random graphs

摘录 / EXCERPT: Let $G_{n,p,d}$ denote the class of d-dependent random graph distributions on graphs with n vertices and marginal edge probability p. For a graph G, connectivity means that...

conjecture | open | MathDB 页面 | 原文线索

V2-044864 | MATHDB-363105

The asymptotic extremal-size conjecture for forbidden posets

摘录 / EXCERPT: The asymptotic extremal-size conjecture. For every poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-044865 | MATHDB-363106

The counting conjecture for forbidden-poset-free set systems

摘录 / EXCERPT: Counting conjecture for forbidden posets. For every poset P, the number of P-free set systems in $2^{[n]}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-044866 | MATHDB-363118

Gowers' Reimer-conditions abundance conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\mathcal{S}\subseteq\mathcal{P}([n])$. Suppose that $\mathcal{S}$ satisfies Reimer's conditions: there is a filter...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-044867 | MATHDB-363119

The quadratic union-closure conjecture for the constructed family

摘录 / EXCERPT: Let $\mathcal{S}$ be the family and $cl(\mathcal{S})$ its union closure, defined by $A\in cl(\mathcal{S})$ if and only if A is the union of some subfamily of $\mathcal{S}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044868 | MATHDB-363120

Conjecture on bounded average degeneracy for dense random traffic-assignment instances

摘录 / EXCERPT: Bounded-degeneracy conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-044869 | MATHDB-363124

Unimodality conjecture for graph genus distributions

摘录 / EXCERPT: For a graph G and a nonnegative integer g, let $a_g(G)$ denote the number of 2-cell embeddings of G in an orientable surface of genus g, counted up to combinatorial homeo...

conjecture | open | MathDB 页面 | 原文线索

V2-044870 | MATHDB-363138

The common-bound conjecture for P-stable degree-sequence families

摘录 / EXCERPT: Let $\mathbb{D}$ be a P-stable family of degree sequences, or a P-stable simple region, and let $\partial(D)$ denote the boundary of a graphic degree sequence $D\in\mathbb{D}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044871 | MATHDB-363149

Edge irregularity strength conjecture for cycle-star graphs

摘录 / EXCERPT: Cycle-star edge irregularity strength conjecture. For $k\geq 8$ and $n-k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044872 | MATHDB-363158

Realizability conjecture for bouquets of convex geometries

摘录 / EXCERPT: A bouquet of convex geometries is the combinatorial structure considered in the paper as a generalization of ideals of convex geometries. Bouquet realizability conjecture. Every bo...

conjecture | open | MathDB 页面 | 原文线索

V2-044873 | MATHDB-363162

Asymptotic prevalence of lex-segment exterior shiftings

摘录 / EXCERPT: Lex-segment prevalence conjecture. As n tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-044874 | MATHDB-363169

The exact value of the normalized k-gon density at multiples of $k-1$

摘录 / EXCERPT: Let k be a constant, let α satisfy $\alpha\leq k-1$, and let $\Delta_k'(n)$ denote the normalized extremal area parameter for convex k-gons defined in the paper. The nor...

conjecture | open | MathDB 页面 | 原文线索

V2-044875 | MATHDB-363171

Feng-Horsley-Wang conjecture on disjoint base blocks in cyclic BIBDs

摘录 / EXCERPT: Let k and λ be fixed positive integers with $k\geqslant \lambda+1$. Let $(X,\mathcal B)$ be a cyclic (v,k,λ)-BIBD with $v\geqslant v_0$, where v_0 is an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-044876 | MATHDB-363181

The facet-halving conjecture for asymptotic contributions of minimal transverse critical points

摘录 / EXCERPT: Let $\mathbf{w}$ be a minimal transverse critical point, and let $N(\mathbf{w})$ be its normal cone. For a direction $\mathbf{r}$, the direction may lie on a facet or in the interi...

conjecture | open | MathDB 页面 | 原文线索

V2-044877 | MATHDB-363189

Structural matching conjecture for dense random graphs

摘录 / EXCERPT: Structural matching conjecture. With high probability as $n\rightarrow\infty$, the graph $G-S_r(G)$ contains a K_{r-1}-matching that covers all but $o(\sqrt{\mu_r})$ vertices of...

conjecture | open | MathDB 页面 | 原文线索

V2-044878 | MATHDB-363204

The stable defective Kreweras number formula

摘录 / EXCERPT: Let m,d,n be nonnegative integers, and let $4\lambda\vdash m$ be a partition of m of length k. For each i, let $4\mu_i(\lambda)$ be the number of parts of 4λ of siz...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-044879 | MATHDB-363290

Galvin's conjecture on two-colour homogeneous copies of the rationals

摘录 / EXCERPT: Let K be a positive natural number, and let $c:[\mathbb R]^2 \rightarrow K$ be a colouring of pairs of reals. Galvin's conjecture. There is a set of reals Y homeomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-044880 | MATHDB-363291

The unique-shape conjecture for $(\tmo,n)$-prismatic polyominoes

摘录 / EXCERPT: Let a $(\begin{tikzpicture}[scale=.25] \draw (1,0)--++(.5,0)--++(0,.5)--++(-1.5,0)--++(0,-.5)--++(1,0)--++(0,1)--++(-.5,0)--++(0,-1); \end{tikzpicture},n)$-prismatic polyomino be a...

conjecture | open | MathDB 页面 | 原文线索

V2-044881 | MATHDB-363319

Simplicity conjecture for the characteristic polynomial of a randomized subtraction game

摘录 / EXCERPT: Simplicity conjecture. The roots of $\chi_S(x)$ are all simple whenever $d_t\equiv 1\pmod{2}$ and $d_i\equiv 0\pmod{2}$ for all $1\leq i\leq t-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-044882 | MATHDB-363326

Bressoud's multi-summation conjecture for restricted partitions

摘录 / EXCERPT: Let $\alpha_1,\ldots,\alpha_\lambda$ and η be integers satisfying ... For $k\geq r\geq\lambda\geq0$, let $B(\alpha_1,\ldots,\alpha_\lambda;\eta,k,r;n)$ denote the number of part...

conjecture | open | MathDB 页面 | 原文线索

V2-044883 | MATHDB-363327

Flammenkamp's conjecture on exponential periods in subtraction games

摘录 / EXCERPT: Let a subtraction game be played on the non-negative integers using a finite ruleset $S\subset\mathbb{N}=\{1,2,\ldots\}$. The current player subtracts an element of S when possib...

conjecture | open | MathDB 页面 | 原文线索

V2-044884 | MATHDB-363346

Fu-Tang's parity-separated partition inequality

摘录 / EXCERPT: Let $\overline{\mathcal P}_{\operatorname{eu}}^{\operatorname{ou}}(n)$ be the set of partitions described in the source in which the largest even part has odd multiplicity and ever...

conjecture | open | MathDB 页面 | 原文线索

V2-044885 | MATHDB-363361

Kuhl et al.'s existence conjecture for two-repetition gerechte designs with five subsquares

摘录 / EXCERPT: Let $h_1,\dots,h_5$ be positive integers, let $n=h_1+\cdots+h_5$, and let $[5]=\{1,2,3,4,5\}$. A $\operatorname{2-RP}(h_1\dots h_5)$ is a two-repetition gerechte design whose five...

conjecture | open | MathDB 页面 | 原文线索

V2-044886 | MATHDB-363384

The corank-two building-block conjecture for matroids with vanishing T^2

摘录 / EXCERPT: Let ${\mathcal{M}}$ be a matroid, and let $T^2({\mathcal{M}})$ denote its second cotangent cohomology module. The join of matroids is their join as simplicial complexes; a matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-044887 | MATHDB-363396

The strict lower-bound conjecture for binary subword entropy

摘录 / EXCERPT: Let L_2 denote the asymptotic binary subword-entropy constant, and let $\log_2$ be the logarithm to base 2. Strict lower-bound conjecture. ... This conjecture is supported by exh...

conjecture | open | MathDB 页面 | 原文线索

V2-044888 | MATHDB-363397

The run-structure conjecture for words of minimal subword entropy

摘录 / EXCERPT: For $k \geq 2$, let w be a binary word of length $n \geq 1$ achieving the minimal subword entropy $\min S_{\mathrm{sw}}^{(k)}(n)$. A run is a maximal consecutive block of equal l...

conjecture | open | MathDB 页面 | 原文线索

V2-044889 | MATHDB-363398

The uniqueness conjecture for most frequent subwords of minimal binary words

摘录 / EXCERPT: Let w be a binary word of length n achieving the minimal subword entropy. Uniqueness conjecture. There are only finitely many values of n for which such a word w has severa...

conjecture | open | MathDB 页面 | 原文线索

V2-044890 | MATHDB-363399

The periodic-structure conjecture for most frequent subwords

摘录 / EXCERPT: Let w be a word, and write w^m for its m-fold concatenation. A subword is most frequent if it maximizes the number of occurrences in w^m. Periodic-structure conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-044891 | MATHDB-363400

The eventual monotonicity conjecture for minimal subword entropy

摘录 / EXCERPT: For $k \geq 2$, let $\min S_{\mathrm{sw}}^{(k)}(n)$ denote the minimal subword entropy among words of length n. Eventual monotonicity conjecture. There is a value N such that t...

conjecture | open | MathDB 页面 | 原文线索

V2-044892 | MATHDB-363401

The short most frequent subword conjecture

摘录 / EXCERPT: Let w be a word of length at least 2. A subword is most frequent if it maximizes the number of occurrences in w. Short-subword conjecture. At least one most frequent subword...

conjecture | open | MathDB 页面 | 原文线索

V2-044893 | MATHDB-363402

Stanton's monotonicity conjecture for t-core partitions

摘录 / EXCERPT: Let $c_t(N)$ denote the number of t-core partitions of N. Stanton's conjecture. For all integers t and N satisfying $4\leq t\ne N-1$, one has ... This conjecture strengthens...

conjecture | open | MathDB 页面 | 原文线索

V2-044894 | MATHDB-363417

Conjecture on the largest noncyclic abelian group with a 2-element spanning set

摘录 / EXCERPT: Let $s\geq 3$ be an integer. An s-spanning set of size 2 in a finite abelian group is a set of two elements whose signed sums of at most s terms cover the group. Largest nonc...

conjecture | open | MathDB 页面 | 原文线索

V2-044895 | MATHDB-363418

Conjecture on perfect spanning sets in finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group, let $A\subseteq G$ be an m-subset, and call A a perfect s-spanning set when every element of G arises as a signed sum of at most s term...

conjecture | open | MathDB 页面 | 原文线索

V2-044896 | MATHDB-363428

Multicolour induced size Ramsey conjecture for bounded-degree trees

摘录 / EXCERPT: Let $cDelta\textgreater=1$, and let T be a tree on n vertices with maximum degree $cDelta(T)\textgreater= cDelta$. For $q\textgreater=1$, write $\mathrm{ind}}^q(T)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-044897 | MATHDB-363445

The sparse paving characterization by Schubert coefficients

摘录 / EXCERPT: Let M be a connected matroid of rank r on $[n]$. Write $U_{r,n}$ for the uniform matroid of rank r on $[n]$, let $d_\lambda(M)$ denote the Schubert coefficient indexed by a p...

conjecture | open | MathDB 页面 | 原文线索

V2-044898 | MATHDB-363446

Berget-Fink non-negativity conjecture for Schubert coefficients of matroids

摘录 / EXCERPT: Let M be a matroid, and let $d_\lambda(M)$ denote its Schubert coefficient indexed by a partition λ. Berget-Fink non-negativity conjecture. For every matroid M and ever...

conjecture | open | MathDB 页面 | 原文线索

V2-044899 | MATHDB-363465

The subhypercube construction conjecture for higher-dimensional queens

摘录 / EXCERPT: Subhypercube construction conjecture. For all d and such m,n,k, if there exists a regular solution to the (n,d)-queens problem containing a subset of size k^d with p quee...

conjecture | open | MathDB 页面 | 原文线索

V2-044900 | MATHDB-363466

The asymptotic density conjecture for higher-dimensional queens

摘录 / EXCERPT: Asymptotic density conjecture. For all d,

conjecture | open | MathDB 页面 | 原文线索

V2-044901 | MATHDB-363467

The exact maximum for the four-by-four higher-dimensional queens board

摘录 / EXCERPT: Four-by-four board conjecture. For every integer $d\geq4$,

conjecture | open | MathDB 页面 | 原文线索

V2-044902 | MATHDB-363472

Asymptotic enumeration conjecture for reduced historic trees

摘录 / EXCERPT: Asymptotic enumeration conjecture. For every $m\geq 1$, the number of reduced $((2m+1))$-historic trees with n vertices is asymptotically equal to

conjecture | open | MathDB 页面 | 原文线索

V2-044903 | MATHDB-363482

The exponential multiplicity conjecture for self-overlapping polygons

摘录 / EXCERPT: Let a self-overlapping polygon have boundary length $2n$, and let its multiplicity be the number of distinct distorted disks it bounds. Exponential multiplicity conjecture. There e...

conjecture | open | MathDB 页面 | 原文线索

V2-044904 | MATHDB-363489

Even-torus conjecture for optimal plaid solutions

摘录 / EXCERPT: Let $n\in\mathbb{N}$ be even, and let (B,W) be an (a,b)-plaid on the torus $\mathbb{T}_n$ for which $\min\{|B|,|W|\}$ is maximum. Even-torus conjecture. Then ... The conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-044905 | MATHDB-363508

Kleiner's converse conjecture for combinatorially Loewner spaces

摘录 / EXCERPT: Kleiner's conjecture. Every approximately self-similar (or group boundary) combinatorially Loewner metric space is quasisymmetric to a Loewner space.

conjecture | refuted | MathDB 页面 | 原文线索

V2-044906 | MATHDB-363524

Reflection complexity periodicity conjecture

摘录 / EXCERPT: Let $\mathbf{x}$ be a sequence. The reflection complexity periodicity conjecture. ... if and only if $\mathbf{x}$ is eventually periodic. One direction is known: if $\mathbf{x}$ is e...

conjecture | open | MathDB 页面 | 原文线索

V2-044907 | MATHDB-363525

Half-or-full asymptotic reflection complexity conjecture

摘录 / EXCERPT: Let $\mathbf{x}$ be a sequence, and let $r_{\mathbf{x}}(n)$ and $\rho_{\mathbf{x}}(n)$ denote its reflection complexity and factor complexity, respectively. The half-or-full limit...

conjecture | open | MathDB 页面 | 原文线索

V2-044908 | MATHDB-363527

Conjecture on the expected number of descents in squared permutations

摘录 / EXCERPT: For $n\geq 3$, let π range uniformly over the symmetric group ${\mathcal S}_n$, and let $\operatorname{des}(\pi^2)$ denote the number of descents of the square of π. The ex...

conjecture | open | MathDB 页面 | 原文线索

V2-044909 | MATHDB-363528

Asymptotic scarcity conjecture for spine contractible fake surfaces

摘录 / EXCERPT: For each complexity n, consider the contractible fake surfaces of complexity n, partitioned into spines and non-spines. Asymptotic scarcity conjecture. The ratio of spine to no...

conjecture | open | MathDB 页面 | 原文线索

V2-044910 | MATHDB-363534

Gap conjecture for determinants of Seidel tournament matrices

摘录 / EXCERPT: For even n, let $\mathcal{D}(n)$ denote the set of square roots of determinants of $n\times n$ Seidel tournament matrices. The preceding argument shows that, assuming the existen...

conjecture | open | MathDB 页面 | 原文线索

V2-044911 | MATHDB-363570

Bukh-Griggs-Lu conjecture for the extremal number of a poset

摘录 / EXCERPT: Let P be a finite poset and let $\mathcal{B}_n=2^{[n]}$ be the Boolean lattice. Write $La(n,P)$ for the largest size of a P-free subfamily of $\mathcal{B}_n$. Define $e(P)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-044912 | MATHDB-363589

Monotonicity conjecture for metered parking functions

摘录 / EXCERPT: Let m,n be positive integers, and let $\mathrm{mpf}_{m,n}(t)$ denote the number of t-metered (m,n)-parking functions. Monotonicity conjecture. If $t_1<t_2$, then ... This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-044913 | MATHDB-363599

Critical probability conjecture for bootstrap percolation on Hamming graphs

摘录 / EXCERPT: Let $\square_{i=1}^n K_k$ be the Cartesian product of n copies of the complete graph K_k, and let $p_c(G,r)$ denote the critical probability for r-neighbor bootstrap percolat...

conjecture | open | MathDB 页面 | 原文线索

V2-044914 | MATHDB-363604

Conjecture on regularity bounds for squarefree powers of simplicial trees

摘录 / EXCERPT: Let Δ be a simplicial tree of dimension $t-1$, let $I(\Delta)$ denote its facet ideal in the polynomial ring R, and let $\nu_1(\Delta)$ and $\nu(\Delta)$ denote its induce...

conjecture | open | MathDB 页面 | 原文线索

V2-044915 | MATHDB-363605

Boros and Moll's infinite log-concavity conjecture for Boros-Moll sequences

摘录 / EXCERPT: Let $d_\ell(m)$ denote the coefficient of x^ℓ in the Boros-Moll polynomial $P_m(x)$, for $0\leq \ell\leq m$. A sequence is k-log-concave when its first k iterates under th...

conjecture | open | MathDB 页面 | 原文线索

V2-044916 | MATHDB-363606

Infinite strict log-concavity conjecture for transposed Boros-Moll sequences

摘录 / EXCERPT: Let $d_\ell(m)$ be the coefficient of x^ℓ in the Boros-Moll polynomial $P_m(x)$, and consider, for fixed ℓ, the transposed sequence $\{d_\ell(m)\}_{m\geq\ell}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-044917 | MATHDB-363607

Extended reverse ultra log-concavity conjecture for adjacent-ratio transposed Boros-Moll sequences

摘录 / EXCERPT: Let $d_\ell(m)$ be the coefficient of x^ℓ in the Boros-Moll polynomial $P_m(x)$. For each fixed $\ell\geq 0$, define the sequence $a_m=d_\ell(m-1)d_\ell(m+1)/d_\ell^2(m)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-044918 | MATHDB-363608

Half log-convexity and half log-concavity conjecture for ratios of Boros-Moll coefficients

摘录 / EXCERPT: Let $d_\ell(m)$ be the coefficient of x^ℓ in the Boros-Moll polynomial $P_m(x)$. For fixed m, consider the ratio sequence $d_\ell(m)/d_{\ell-1}(m)$. Ratio-shape conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-044919 | MATHDB-363609

Infinite log-monotonicity conjecture for transposed Boros-Moll sequences

摘录 / EXCERPT: Let $d_\ell(m)$ be the coefficient of x^ℓ in the Boros-Moll polynomial $P_m(x)$. For a sequence $\{a_i\}_{i\geq 0}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-044920 | MATHDB-363610

Asymptotic existence of symEF1 allocations

摘录 / EXCERPT: Consider instances with a fixed number n of agents and m items, with probability taken over the paper's instance model. A symEF1 allocation is an allocation satisfying symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-044921 | MATHDB-363612

Skew-circuit circumference bound conjecture

摘录 / EXCERPT: Skew-circuit circumference bound conjecture. For every nonnegative integer k, there exists an integer $alpha(k)$ depending only on k such that, whenever C_1 and C_2 are...

conjecture | open | MathDB 页面 | 原文线索

V2-044922 | MATHDB-363664

Characterization of iterated palindromes in the intersection of two clustering classes

摘录 / EXCERPT: Characterization conjecture. A word $w\in P_1\cap P_2$ if and only if $\operatorname{Pal}(u)=w$ for some $u\in\{a,c\}^*\cdot b^*$. The preceding proposition establishes this charac...

conjecture | open | MathDB 页面 | 原文线索

V2-044923 | MATHDB-363698

The even-dimensional extremal-size conjecture for intersecting temperate families

摘录 / EXCERPT: Let $n=2k$ be even, and let an intersecting temperate family on $[n]$ be a temperate family in which every two members have non-empty intersection. Even intersecting-temperate-fami...

conjecture | open | MathDB 页面 | 原文线索

V2-044924 | MATHDB-363700

Frankl and Watanabe's conjecture on the value of $m(11)$

摘录 / EXCERPT: Let $m(s)$ be the limiting maximum density associated with the arrowing relation $(n,m)\rightarrow(n-1,m-s)$: for each fixed non-negative integer s, $m(s)$ is the largest value s...

conjecture | open | MathDB 页面 | 原文线索

V2-044925 | MATHDB-363701

The anisotropy conjecture for homology spheres

摘录 / EXCERPT: Anisotropy conjecture. For an arbitrary field $\mathbb{F}$, any $\mathbb{F}$-homology sphere is generically anisotropic over $\mathbb{F}$.

conjecture | open | MathDB 页面 | 原文线索

V2-044926 | MATHDB-363723

The $(1,2)$ combinatorial Frobenius-series conjecture

摘录 / EXCERPT: Let $B_n^{(1,2)}$ be the proposed monomial basis of $R_n^{(1,2)}$. For $b\in B_n^{(1,2)}$, write $\deg_x(b)$, $\deg_\theta(b)$, and $\deg_\xi(b)$ for its degrees, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044927 | MATHDB-363735

The sharp lower-bound conjecture for multicolor Hamming graphs

摘录 / EXCERPT: Let $H(n,q)$ denote the q-ary Hamming graph, and let $b(H(n,q))$ denote its bipartite independence-related parameter as defined in the paper. Fix an integer $q\geq 3$. Sharp-boun...

conjecture | open | MathDB 页面 | 原文线索

V2-044928 | MATHDB-363762

Product-one subsequence conjecture for the ordered Davenport constant

摘录 / EXCERPT: For a finite group G, let $\mathsf{D}(G)$ be the least positive integer such that every sequence of that length over G has a nonempty product-one subsequence, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044929 | MATHDB-363774

The symmetric theta conjecture for decorated trees

摘录 / EXCERPT: Let β be a weak composition, let $\mathsf{RTT}_0(\beta)=\bigcup_{\alpha\vDash_0\lvert\beta\rvert}\mathsf{RTT}_0(\alpha,\beta)$, and let oldXi be the symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-044930 | MATHDB-363775

The symmetric theta conjecture for the nabla-star operator

摘录 / EXCERPT: Let n be a positive integer, let $\nabla_\ast$ be the operator on symmetric functions in two alphabets defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-044931 | MATHDB-363786

Shapira-Yuster conjecture for tournament forests

摘录 / EXCERPT: A tournament is a complete graph whose edges are directed, and a t-augmented tournament has t pairs of vertices carrying both directed edges. Let $t(n,H)$ be the minimum t su...

conjecture | open | MathDB 页面 | 原文线索

V2-044932 | MATHDB-363787

Weak and strong conjectures for acyclic 0-1 patterns

摘录 / EXCERPT: Let $\mathcal{P}_{\text{acyclic}}$ be the class of acyclic 0-1 patterns. Acyclic-pattern conjecture. Both of the following forms are proposed: - Weak Form: For all...

conjecture | open | MathDB 页面 | 原文线索

V2-044933 | MATHDB-363806

Non-ergodic countable-field triple-intersection conjecture

摘录 / EXCERPT: Let K be a countable field, let $(F_N)_{N\in\mathbb N}$ be a double Følner sequence in K, and let $(X,\mathcal{X},\mu)$ be a probability space on which the affine group action...

conjecture | open | MathDB 页面 | 原文线索

V2-044934 | MATHDB-363809

The upper-bound conjecture for the alphabet size of anagram-free words

摘录 / EXCERPT: Fix $k\geqslant 1$. A word is called k-anagram-free if every factor F satisfies $\alpha(F)\geqslant k+1$, where $\alpha(F)$ is the least number of cuts needed to decompose F...

conjecture | open | MathDB 页面 | 原文线索

V2-044935 | MATHDB-363812

Uniqueness conjecture for consistent sandpile torsor algorithms on oriented regular matroids

摘录 / EXCERPT: Let M be an oriented regular matroid, equipped with triangulating circuit and cocircuit signatures. A consistent sandpile torsor algorithm is a consistent algorithm assigning san...

conjecture | open | MathDB 页面 | 原文线索

V2-044936 | MATHDB-363828

Sublogarithmic bound for partially shattering permutations

摘录 / EXCERPT: Sublogarithmic-bound conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-044937 | MATHDB-363861

The minimum size conjecture for quasirandom-forcing sets of 4-point permutations

摘录 / EXCERPT: Minimum size conjecture for 4-point permutations. Every quasirandom-forcing set of 4-point permutations has cardinality at least eight.

conjecture | open | MathDB 页面 | 原文线索

V2-044938 | MATHDB-363862

The minimum size conjecture for quasirandom-forcing sets of permutations

摘录 / EXCERPT: Minimum size conjecture for permutations. Every quasirandom-forcing set of permutations has cardinality at least six.

conjecture | open | MathDB 页面 | 原文线索

V2-044939 | MATHDB-363863

Discrete Babenko-Beckner inequality for functions supported on intervals

摘录 / EXCERPT: Discrete Babenko-Beckner conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-044940 | MATHDB-363914

Finite fields distance conjecture

摘录 / EXCERPT: Finite fields distance conjecture. If $\#E\geqslant Cq^{d/2}$ with C sufficiently large, then

conjecture | open | MathDB 页面 | 原文线索

V2-044941 | MATHDB-363917

Chromatic-number bound for intersections of general matroids

摘录 / EXCERPT: Let $\mathcal{L}=\{\mathcal{M}_1,\ldots,\mathcal{M}_k\}\in\mathcal{D}^k$, where $\mathcal{D}$ is the class of matroids under consideration, and let $\bigcap\mathcal{L}$ denote thei...

conjecture | open | MathDB 页面 | 原文线索

V2-044942 | MATHDB-363918

Weighted fractional matching-cover conjecture for matroid intersections

摘录 / EXCERPT: Let $\mathcal{L}\in\mathcal{D}^k$ be a family of k matroids on V, and let $w\in\mathbb{R}_{\ge0}^V$. Weighted fractional matching-cover conjecture. One has ... This is presented...

conjecture | open | MathDB 页面 | 原文线索

V2-044943 | MATHDB-363919

k-minus-one ratio conjecture for matroid polytopes

摘录 / EXCERPT: Let $\mathcal{L}\in\mathcal{D}^k$ be a family of k matroids, and let $R(\mathcal{L})$ and $P(\bigcap\mathcal{L})$ denote the polytopes defined in the source. Polytope ratio conje...

conjecture | open | MathDB 页面 | 原文线索

V2-044944 | MATHDB-363920

Weighted fractional coloring conjecture for intersections of matroids

摘录 / EXCERPT: Let $\mathcal{L}=\{\mathcal{M}_1,\ldots,\mathcal{M}_k\}\in\mathcal{D}^k$ be matroids on V, let $\mathcal{C}=\bigcap\mathcal{L}$, and let $\vec h\in\mathbb{R}_{\ge0}^V$. Weighted...

conjecture | open | MathDB 页面 | 原文线索

V2-044945 | MATHDB-363921

Aharoni-Berger conjecture on coloring intersections of two matroids

摘录 / EXCERPT: Aharoni-Berger conjecture. For any two matroids $d4d8_1$ and $d4d8_2$ on the same ground set,

conjecture | open | MathDB 页面 | 原文线索

V2-044946 | MATHDB-363950

Higher level RSK correspondence

摘录 / EXCERPT: Let ℓ be a positive level. Denote by ... the set of bounded-partition-enhanced $\ell\times\ell$-block $\mathbb{N}$-matrices, and by ... the set of pairs of semistandard tableaux o...

conjecture | open | MathDB 页面 | 原文线索

V2-044947 | MATHDB-363965

Unimodality conjecture for Betti numbers of venustus real toric manifolds

摘录 / EXCERPT: Let $\mathcal{B}$ be a venustus building set, and let $X^\mathbb{R}_{\mathcal{B}}$ be its associated real toric manifold. Write $\beta_k(X^\mathbb{R}_{\mathcal{B}})$ for its ration...

conjecture | open | MathDB 页面 | 原文线索

V2-044948 | MATHDB-363967

The wiggly complex polytopality conjecture

摘录 / EXCERPT: Let P be any point set in the plane, and let WC_P denote its wiggly complex. Wiggly complex polytopality conjecture. For any point set P in the plane, the wiggly com...

conjecture | open | MathDB 页面 | 原文线索

V2-044949 | MATHDB-363989

The EKR conjecture for induced subgraphs of perfect matchings

摘录 / EXCERPT: EKR conjecture. $4\mathcal{H}^{(p,s)}(n)4$ is EKR: every intersecting subfamily has size at most the size of a star, with equality attained by a family consisting of all members co...

conjecture | open | MathDB 页面 | 原文线索

V2-044950 | MATHDB-363994

Subcube extremality for the Hamming cube's beta-isoperimetric profile

摘录 / EXCERPT: Subcube extremality conjecture. For all $\beta\ge\frac12$ and all $k\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-044951 | MATHDB-363999

Uniqueness conjecture for minimum universal sets of MacMahon cubes

摘录 / EXCERPT: Uniqueness conjecture. The 10 minimum universal sets are the only collections of 12 cubes that can build all 30 target cubes.

conjecture | open | MathDB 页面 | 原文线索

V2-044952 | MATHDB-364014

The minimum-degree conjecture for intersecting uniform families

摘录 / EXCERPT: Minimum-degree conjecture. For every $k>d\geq 0$ and $n\geq 2k+1$, an intersecting family $\mathcal{F}\subset\binom{[n]}{k}$ always satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-044953 | MATHDB-364029

The 3-design parameter formula for a lifted Reed-Muller code

摘录 / EXCERPT: Let $m\geq 3$. For the binary lifted Reed-Muller code ${\mathrm{RM}}_2(1,m)(2|2^2)$, let ${\mathbb{D}}_{3\times 2^{m-2}}({\mathrm{RM}}_2(1,m)(2|2^2))$ denote the design whose block...

conjecture | open | MathDB 页面 | 原文线索

V2-044954 | MATHDB-364102

Balanced triangle design existence for projective spaces of dimension $6k$

摘录 / EXCERPT: Existence conjecture. Assertion ${\rm (ii')}$ holds for each $n=6k+1$; equivalently, there is a partition of $\mathbb{Z}_{2^n-1}\setminus\{0\}$ into 18-subsets of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-044955 | MATHDB-364103

Cyclotomic-class partition conjecture for products of primes congruent to 1 modulo 6

摘录 / EXCERPT: Cyclotomic partition conjecture. For every integer $t>1$, there are $N_t/18$ pairs (a_i,b_i), $i=1,\ldots,N_t/18$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-044956 | MATHDB-364116

The Prim'-Cayley fixed-point sum conjecture

摘录 / EXCERPT: Let Prim' and Cay be the species of primitive objects and Cayley permutations, respectively. For a Cayley permutation $w=w_1\cdots w_n$, let ... be...

conjecture | open | MathDB 页面 | 原文线索

V2-044957 | MATHDB-364117

The Cayley-equivalence content conjecture

摘录 / EXCERPT: Let p and q be Cayley permutations of the same length. Write $p\sim_c q$ for Cayley-equivalence and $p\sim_{cm}q$ for Cayley-equivalence with respect to content multiplicities....

conjecture | open | MathDB 页面 | 原文线索

V2-044958 | MATHDB-364119

The Cayley-avoidance maximum inequality conjecture

摘录 / EXCERPT: Let p and q be Cayley permutations of the same length, and let $\max(p)$ denote the largest letter occurring in p. Let $\operatorname{Cay}_n(p)$ be the set of Cayley permutat...

conjecture | open | MathDB 页面 | 原文线索

V2-044959 | MATHDB-364120

The total equality conditions for all loopless matroids

摘录 / EXCERPT: Let $\mathscr{M}$ be a loopless matroid of rank $r\geq 2$ with ground set X, and let $R\subseteq X$. Suppose that $\textrm{p}(\mathscr{M},R,0)>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-044960 | MATHDB-364121

The coNP-completeness of EqualitySY for binary and bicircular matroids

摘录 / EXCERPT: For a matroid presented succinctly, let $\operatorname{EqualitySY}_0$ denote the decision problem asking whether equality holds in the $k=0$ Stanley--Yan inequality. EqualitySY coN...

conjecture | open | MathDB 页面 | 原文线索

V2-044961 | MATHDB-364122

The NP-completeness of 2-SPANNING-CIRCUIT for binary matroids

摘录 / EXCERPT: A binary matroid is a matroid representable over the two-element field. The 2-SPANNING-CIRCUIT problem asks whether a matroid has a circuit containing two specified elements. 2-SPA...

conjecture | open | MathDB 页面 | 原文线索

V2-044962 | MATHDB-364153

Tightness conjecture for the folded-form construction of minimum-density monotone 3-subwords

摘录 / EXCERPT: Folded-form tightness conjecture. The construction's upper bound is tight:

conjecture | open | MathDB 页面 | 原文线索

V2-044963 | MATHDB-364192

Equivariant gamma-positivity conjecture for matroid Chow rings

摘录 / EXCERPT: Let M be a loopless matroid of rank r, let G be a group of automorphisms of M, and let ... be its complexified matroid Chow ring, with each graded piece carrying the induced...

conjecture | open | MathDB 页面 | 原文线索

V2-044964 | MATHDB-364195

A lower bound for the average number of rounds in permutation Wordle

摘录 / EXCERPT: Average-round lower-bound conjecture. For any strategy for permutation Wordle, the average number of rounds is at least

open_problem | open | MathDB 页面 | 原文线索

V2-044965 | MATHDB-364196

CircularShift's optimality conjecture for suited permutation Wordle

摘录 / EXCERPT: Suited CircularShift optimality conjecture. For each r, the probability that CircularShift guesses the correct suited permutation within r guesses is at least as large as the c...

conjecture | open | MathDB 页面 | 原文线索

V2-044966 | MATHDB-364205

Largest rigid facet conjecture for Vietoris-Rips complexes of hypercubes

摘录 / EXCERPT: Let $Q_n=\{\pm1\}^n$ be the Hamming cube with Hamming distance. A subset is maximally diameter-r if its diameter is r and adjoining any point outside it produces diameter gre...

conjecture | open | MathDB 页面 | 原文线索

V2-044967 | MATHDB-364206

Linear lower bound for smallest facets of Vietoris-Rips complexes of hypercubes

摘录 / EXCERPT: Let $s(n;r)$ denote the size of the smallest facet in $\operatorname{\mathcal{VR}}(Q_n;r)$. The source proves $s(n;r)\leq 2r$ when a Hadamard matrix of order $2r$ exists and...

conjecture | open | MathDB 页面 | 原文线索

V2-044968 | MATHDB-364207

Row-deletion conjecture for normalized Hadamard matrices

摘录 / EXCERPT: A normalized Hadamard matrix of order n is a Hadamard matrix whose normalization is as used in the source. For a non-negative integer r and a matrix A, let A_{-r} be the su...

conjecture | open | MathDB 页面 | 原文线索

V2-044969 | MATHDB-364224

The penult token-count interval conjecture for square Tak boards

摘录 / EXCERPT: Let $n\geq 6$. Write M_n for the value given by Proposition, and let x satisfy ... A penult is a position on an $n\times n$ board with the stated token count from which the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-044970 | MATHDB-364225

The quadratic lower-bound conjecture for minimal snake diagrams

摘录 / EXCERPT: Let $L(n)$ denote the quantity measuring the size of a minimal snake diagram. Quadratic lower-bound conjecture. For all n, ... for some constants a and b. This conjecture asser...

conjecture | open | MathDB 页面 | 原文线索

V2-044971 | MATHDB-364226

The odd-board first-player conjecture for impartial Tak

摘录 / EXCERPT: Consider impartial Tak played on an $n\times n$ board, with the first player seeking a winning strategy. Odd-board first-player conjecture. If $n>5$ is odd, then the first player h...

conjecture | open | MathDB 页面 | 原文线索

V2-044972 | MATHDB-364265

McKinley-Spiro random Turán conjecture for bipartite graphs

摘录 / EXCERPT: Let F be a graph with ... for some $\alpha\in(1,2]$. Write $G_{n,p}$ for the random graph with edge probability p, and let $d_2(F)$ denote the 2-density parameter appearing in th...

conjecture | open | MathDB 页面 | 原文线索

V2-044973 | MATHDB-364281

The Crew-Narayanan-Spirkl circular disproportionate division conjecture

摘录 / EXCERPT: Crew-Narayanan-Spirkl conjecture. There is a partition $[n]=P\cup Q$ into two nonempty sets and a partition $S^1=X\cup X^c$ into two intervals such that

conjecture | open | MathDB 页面 | 原文线索

V2-044974 | MATHDB-364348

Characterization conjecture for magic rectangle sets with exceptional dimensions

摘录 / EXCERPT: A Γ-magic rectangle set $\operatorname{MRS}_{\Gamma}(a,b;c)$ on an Abelian group Γ of order abc is a collection of c arrays of size $a\times b$, whose entries are...

conjecture | open | MathDB 页面 | 原文线索

V2-044975 | MATHDB-364350

Bayer et al.'s Betti-number conjecture for total cut complexes of grid graphs

摘录 / EXCERPT: Let $\mathcal{G}_{m\times n}$ denote the $m\times n$ grid graph, and let $\Delta_k^t(G)$ be the total k-cut complex of a graph G. For a simplicial complex K, write $b_i(K)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-044976 | MATHDB-364351

Bayer et al.'s shellability conjecture for cut complexes of grid graphs

摘录 / EXCERPT: Let $\mathcal{G}_{m\times n}$ denote the $m\times n$ grid graph, and let $\Delta_k(G)$ be the k-cut complex of a graph G. A simplicial complex is shellable if its facets admit...

conjecture | open | MathDB 页面 | 原文线索

V2-044977 | MATHDB-364355

The matrix-set filling conjecture

摘录 / EXCERPT: Let $\operatorname{Mat}^{n\times n}_q$ denote the additive group of $n\times n$ matrices over $\mathbb F_q$, and let $\operatorname{GL}(n,q)$ act by left and right multiplication....

conjecture | open | MathDB 页面 | 原文线索

V2-044978 | MATHDB-364356

The many-rank matrix-sum equidistribution conjecture

摘录 / EXCERPT: Let $\operatorname{Mat}^{n\times n}_q$ be the additive group of $n\times n$ matrices over $\mathbb F_q$. For integers $r_1,\ldots,r_k,t$ with $0\le r_i,t\le n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-044979 | MATHDB-364359

Aggarwal-Borodin-Wheeler's rank-4 interlacing-array coloring conjecture

摘录 / EXCERPT: Let $\operatorname{\mathcal{T}}_{4,n}$ denote the set of interlacing triangular arrays of rank 4 and height n. Let $\square_n$ be the $n\times n$ square grid graph, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-044980 | MATHDB-364360

The rank-4 interlacing-array edge-labeling conjecture

摘录 / EXCERPT: Let $\operatorname{\mathcal{T}}_{4,n}$ be the set of interlacing triangular arrays of rank 4 and height n, and let $\square_n$ be the $n\times n$ square grid graph. An edge lab...

conjecture | open | MathDB 页面 | 原文线索

V2-044981 | MATHDB-364367

The extremal conjecture for intersecting families of binary sequences

摘录 / EXCERPT: Extremal intersection conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-044982 | MATHDB-364391

The universal exponent asymptotic conjecture

摘录 / EXCERPT: For each dimension d, let ϵ_d denote the universal exponent parameter for d-dimensional simplicial complexes. Universal exponent asymptotic conjecture. The universal e...

conjecture | open | MathDB 页面 | 原文线索

V2-044983 | MATHDB-364397

Solyom's forbidden-configuration incidence conjecture

摘录 / EXCERPT: Solyom's conjecture. For any set of points P_0 and any set of lines L_0, the maximum number of incidences between n points and n lines in the plane containing no subconfigu...

conjecture | refuted | MathDB 页面 | 原文线索

V2-044984 | MATHDB-364398

Mirzaei-Suk's power-saving forbidden-configuration incidence conjecture

摘录 / EXCERPT: Mirzaei-Suk's conjecture. For any set of points P_0 and any set of lines L_0, there is a constant $c>0$ such that the maximum number of incidences between n points and n li...

conjecture | open | MathDB 页面 | 原文线索

V2-044985 | MATHDB-364399

Asymptotic negligibility of smooth Dressian cells

摘录 / EXCERPT: Fix $r\geq 4$. Let $U(r,n)$ be the uniform matroid of rank r on an n-element ground set, and let $\mathscr{D}(U(r,n))$ be its Dressian. For a valuation...

conjecture | open | MathDB 页面 | 原文线索

V2-044986 | MATHDB-364402

Microlocal multiplicity conjecture for matroids

摘录 / EXCERPT: Microlocal multiplicity conjecture. For every loopless matroid M,

conjecture | open | MathDB 页面 | 原文线索

V2-044987 | MATHDB-364416

The valley Delta square conjecture

摘录 / EXCERPT: The valley Delta square conjecture. For all $n,k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-044988 | MATHDB-364423

Two-axis extremizer conjecture for the radius-$\sqrt{2}$ reverse Littlewood-Offord problem

摘录 / EXCERPT: For each sufficiently large integer n, let $v_1,\ldots,v_n$ be arbitrary unit vectors in $\mathbb{R}^2$, and let $\epsilon_1,\ldots,\epsilon_n$ be independent Rademacher random v...

conjecture | open | MathDB 页面 | 原文线索

V2-044989 | MATHDB-364462

Best-spread conjecture for list colorings

摘录 / EXCERPT: Let $G=(V,E)$ be a graph on n vertices. For each $v\in V$, let S_v be a list of size $d_G(v)+1$, and write $S=(S_v:v\in V)$. A distribution on S-colorings is p-spread if th...

conjecture | open | MathDB 页面 | 原文线索

V2-044990 | MATHDB-364491

Restricted van der Waerden theorem for nilprogressions

摘录 / EXCERPT: For $k\geq 2$, let $\sum_{<k}$ be the collection of words in $*_1,\ldots,*_d$ in which each letter occurs at most $k-1$ times. For elements $a,x_1,\ldots,x_d$ of an s-step nilpot...

conjecture | open | MathDB 页面 | 原文线索

V2-044991 | MATHDB-364497

Tverberg-Vrećica conjecture

摘录 / EXCERPT: Let $0 \le k < d$ be integers, and let $r_0, \dots, r_k$ be positive integers. Suppose that $P_0, \dots, P_k$ are sets of points in $\mathds{R}^d$ satisfying ... for each i. Tverbe...

conjecture | open | MathDB 页面 | 原文线索

V2-044992 | MATHDB-364510

Gowers's polynomial density Hales-Jewett conjecture

摘录 / EXCERPT: Let k,d be positive integers, let $\delta>0$, and let n be a positive integer. Write $[n]=\{1,\ldots,n\}$ and ... For a subset of a finite set, its density is its cardinality divi...

conjecture | open | MathDB 页面 | 原文线索

V2-044993 | MATHDB-364511

The powers-of-sets containment conjecture

摘录 / EXCERPT: Let d be a positive integer and let $\delta>0$. For a finite set, density means cardinality divided by the cardinality of the ambient set. The notation $[n]^d$ denotes the d-fo...

conjecture | open | MathDB 页面 | 原文线索

V2-044994 | MATHDB-364525

Asymptotic normality conjecture for coefficients of Speyer's g-polynomials

摘录 / EXCERPT: Let $d=d(n)$ be a real-valued function satisfying ... For integers n and i, let $S_i(n,d(n))$ denote the coefficient of t^i in Speyer's g-polynomial $g_{n,d(n)}(t)$. Asymptot...

conjecture | open | MathDB 页面 | 原文线索

V2-044995 | MATHDB-364527

The unrestricted classification conjecture for periodic Golay pair compressions

摘录 / EXCERPT: Let (A,B) be the $(v/2)$-decomposition of a periodic Golay pair $\operatorname{PG}(v)$ with row sums a and b. Equivalence is taken in the sense used for these decompositions....

conjecture | open | MathDB 页面 | 原文线索

V2-044996 | MATHDB-364547

Monotonicity conjecture for HB_3

摘录 / EXCERPT: Let $\mathrm{HB}_3(n)$ denote the maximum length of an unbordered partial word of length parameter n over an alphabet of size 3. Monotonicity conjecture for HB_3. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-044997 | MATHDB-364559

A coefficient pattern conjecture for the characteristic polynomial of the substitution matrix

摘录 / EXCERPT: Let n be odd with $n\geq 5$, let M_n be the substitution matrix, and write its characteristic polynomial as ... The coefficients of $q_n(x)$ in standard form are arranged accordi...

conjecture | open | MathDB 页面 | 原文线索

V2-044998 | MATHDB-364564

Kleiner-Bourdon-Kleiner conjecture on the Combinatorial Loewner Property

摘录 / EXCERPT: Kleiner-Bourdon-Kleiner conjecture. If Z satisfies the Combinatorial Loewner Property, then Z is quasisymmetric to an Ahlfors regular Loewner space.

conjecture | open | MathDB 页面 | 原文线索

V2-044999 | MATHDB-364588

Solymosi's forbidden point-line configuration conjecture

摘录 / EXCERPT: Solymosi's conjecture. For any set of points P_0 and any set of lines $\mathcal{L}_0$ in the plane, the maximum number of incidences between n points and n lines in the plane...

conjecture | open | MathDB 页面 | 原文线索

V2-045000 | MATHDB-364589

Mirzaei--Suk strengthening of Solymosi's forbidden configuration conjecture

摘录 / EXCERPT: Mirzaei--Suk strengthening. For any set of points P_0 and any set of lines $\mathcal{L}_0$ in the plane, there is a constant $\varepsilon=\varepsilon(P_0,\mathcal{L}_0)>0$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-045001 | MATHDB-364606

The non-negative coefficient conjecture for the associated Stirling series

摘录 / EXCERPT: Non-negative coefficient conjecture. For each integer $r\geq 1$, this power series has all non-negative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-045002 | MATHDB-364607

The stronger asymptotic error-bound conjecture for associated Stirling numbers

摘录 / EXCERPT: Stronger asymptotic error-bound conjecture. As $p\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-045003 | MATHDB-364636

General polynomial equation for entries of square Sinkhorn limits

摘录 / EXCERPT: For $n \geq 1$, let $D(n,n)$ be the set of minor specifications described in the paper, and let $M(S)$ be the associated product of minors of a positive $n \times n$ matrix A. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-045004 | MATHDB-364637

Rectangular Sinkhorn-limit polynomial conjecture with sign alterations

摘录 / EXCERPT: Let A be a positive $m \times n$ matrix, let $D(m,n)$ be the set of minor specifications described in the paper, and let $M(S)$ be the associated product of minors. For...

conjecture | open | MathDB 页面 | 原文线索

V2-045005 | MATHDB-364639

Concentration conjecture for the path-covering number in random trees

摘录 / EXCERPT: Let T be the random tree under consideration, let t denote its size parameter, and let $\operatorname{cov}_\ell(T)$ be the minimum number of paths of length at most ℓ need...

conjecture | open | MathDB 页面 | 原文线索

V2-045006 | MATHDB-364706

Friedgut's iterated-shadow intersection conjecture

摘录 / EXCERPT: Let n be even, let $\binom{[n]}{n/2}$ denote the middle layer of the Boolean cube, and let $\mathcal{A}\subset\binom{[n]}{n/2}$. Write $\mathcal{A}^c$ for its complement in this...

conjecture | open | MathDB 页面 | 原文线索

V2-045007 | MATHDB-364707

The extremal half-sized iterated-shadow intersection conjecture

摘录 / EXCERPT: Let n be even and let $\mathcal{A}\subset\binom{[n]}{n/2}$ satisfy $\mu(\mathcal{A})=1/2$, where μ is the uniform measure on the middle layer, $\mathcal{A}^c$ is the compleme...

conjecture | open | MathDB 页面 | 原文线索

V2-045008 | MATHDB-364735

Conjecture on products of 2-wise uniform distributions

摘录 / EXCERPT: Let H be a group, let t be a positive integer, and let X,Y be independent random variables that are 2-wise uniform over H^{t}, meaning that every pair of coordinates of eac...

conjecture | open | MathDB 页面 | 原文线索

V2-045009 | MATHDB-364748

Boundary extremality conjecture for the normalized simplex of spectra

摘录 / EXCERPT: Let $\mathbb{SL}_1^n$ denote the normalized set of spectra under consideration, let $\partial \mathbb{SL}_1^n$ be its boundary, and let $\mathbb{E}^n$ be its set of extremal points...

conjecture | open | MathDB 页面 | 原文线索

V2-045010 | MATHDB-364765

Existence of a non-modest poset

摘录 / EXCERPT: Non-modest poset conjecture. There is a fixed poset P such that

conjecture | open | MathDB 页面 | 原文线索

V2-045011 | MATHDB-364766

Lu-Thompson conjecture on off-diagonal Boolean-lattice Ramsey numbers

摘录 / EXCERPT: Lu-Thompson conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045012 | MATHDB-364767

Axenovich-author near-linear conjecture for off-diagonal Boolean lattices

摘录 / EXCERPT: Axenovich-author conjecture. For any $\varepsilon>0$, there is a large enough m_0 such that for any $m,n\in\mathbb{N}$ with $n\ge m\ge m_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-045013 | MATHDB-364768

Weak versus induced poset Ramsey numbers conjecture

摘录 / EXCERPT: Weak-versus-induced conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045014 | MATHDB-364769

Near-trivial additive term conjecture for fixed-poset Ramsey numbers

摘录 / EXCERPT: Fixed-poset asymptotic conjecture. For $n\in\mathbb{N}$ and fixed P,

conjecture | open | MathDB 页面 | 原文线索

V2-045015 | MATHDB-364770

Fixed-dimensional Boolean-lattice Ramsey conjecture

摘录 / EXCERPT: Fixed-dimensional Boolean-lattice conjecture. For every fixed $m\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045016 | MATHDB-364801

The nonexistence conjecture for extremal backbone tree colorings

摘录 / EXCERPT: Let G be a graph with maximum degree 4, and let T be a backbone tree. Write $BBC_{\lambda}(G,T)$ for the minimum number of colors in a λ-backbone coloring. Extremal t...

conjecture | open | MathDB 页面 | 原文线索

V2-045017 | MATHDB-364806

Large affine coset conjecture for sets with small approximate algebra norm

摘录 / EXCERPT: Let G be the ambient group, let $A\subseteq G$, and let $\|\mathbf{1}_A\|_{A,\epsilon}$ denote the approximate algebra norm of A at error parameter ϵ. A coset of codim...

conjecture | open | MathDB 页面 | 原文线索

V2-045018 | MATHDB-364807

Davis-Lelièvre's local-global conjecture for asymmetric pentagonal billiard periods

摘录 / EXCERPT: A combinatorial period length is the number of line segments in a periodic trajectory on the regular pentagon; a trajectory is asymmetric when it does not have 5-fold rotational sy...

conjecture | open | MathDB 页面 | 原文线索

V2-045019 | MATHDB-364815

Probabilistic upper-bound conjecture for the maximum planar EMST-ratio

摘录 / EXCERPT: Let P_n consist of n independent random points uniformly distributed in $[0,1]^2$. Probabilistic upper-bound conjecture. The maximum EMST-ratio of P_n is less than 2 with p...

conjecture | open | MathDB 页面 | 原文线索

V2-045020 | MATHDB-364816

Random planar bipartite-coloring approximation conjecture

摘录 / EXCERPT: Let P_n consist of n independent random points uniformly distributed in $[0,1]^2$, and let a bipartite coloring of P_n be any coloring induced by a Euclidean minimum spanning...

conjecture | open | MathDB 页面 | 原文线索

V2-045021 | MATHDB-364826

The general off-diagonal unordered Erdős-Rado bound

摘录 / EXCERPT: For integers $m,\ell\geq 4$, let $\operatorname{ER}(m,\ell,r)$ denote the off-diagonal unordered Erdős-Rado number for monochromatic and lexical cliques of the indicated orders aga...

conjecture | open | MathDB 页面 | 原文线索

V2-045022 | MATHDB-364827

The off-diagonal Erdős-Rado bound for monochromatic and lexical K_4

摘录 / EXCERPT: Let $\operatorname{ER}(4,4,r)$ denote the off-diagonal unordered Erdős-Rado number when monochromatic and lexical copies of K_4 are forbidden and a rainbow clique has order r....

conjecture | open | MathDB 页面 | 原文线索

V2-045023 | MATHDB-364828

The lexical-K_4 off-diagonal Erdős-Rado bound

摘录 / EXCERPT: For $m\geq4$, let $\operatorname{ER}(m,4,r)$ denote the off-diagonal unordered Erdős-Rado number in the setting where monochromatic cliques have order m, lexical cliques have ord...

conjecture | open | MathDB 页面 | 原文线索

V2-045024 | MATHDB-364837

Limit conjecture for repetition thresholds of rich words

摘录 / EXCERPT: Limit conjecture. The repetition thresholds of rich words satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-045025 | MATHDB-364839

Characterization of Castelnuovo triplets

摘录 / EXCERPT: Castelnuovo triplet conjecture. The triplet (b,c,d) is Castelnuovo if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-045026 | MATHDB-364842

The conjectured strict exclusion bounds for the cube-approximation function

摘录 / EXCERPT: Let g be the function defined in Theorem, and let k and i be the parameters occurring there. Strict exclusion conjecture. One has … and … The paper presents this as a stronge...

conjecture | open | MathDB 页面 | 原文线索

V2-045027 | MATHDB-364843

The uniform-slicing conjecture for alphabet-multiplier upcycles

摘录 / EXCERPT: Uniform-slicing conjecture. For every $0\leq i\leq n-d$, if the slicing length is $k^i|u|$, then the resulting collection is an upfamily.

conjecture | open | MathDB 页面 | 原文线索

V2-045028 | MATHDB-364881

The main conjecture on arbitrary disjoint Golomb rulers

摘录 / EXCERPT: For positive integers I and J, an (I,J,n)-DGR is a collection of I disjoint Golomb rulers, each a J-subset of $\{1,2,\ldots,n\}$. Let $H(I,J)$ be the least positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-045029 | MATHDB-364882

The incremental bound conjecture for disjoint Golomb rulers

摘录 / EXCERPT: For positive integers I and J, let $H(I,J)$ be the least n such that there are I disjoint Golomb rulers, each a J-subset of $\{1,2,\ldots,n\}$. Incremental bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-045030 | MATHDB-364883

Regularity propagation conjecture for disjoint Golomb rulers

摘录 / EXCERPT: For positive integers I and J, let $H(I,J)$ denote the least n admitting I disjoint J-mark Golomb rulers in $\{1,\ldots,n\}$. A DGR is regular when $n=IJ$. Regularity pro...

conjecture | open | MathDB 页面 | 原文线索

V2-045031 | MATHDB-364884

Extension conjecture for regular disjoint Golomb rulers

摘录 / EXCERPT: Let I and J be positive integers. A regular $(I+1,J,(I+1)J)$-DGR is a collection of $I+1$ disjoint J-mark Golomb rulers partitioning $\{1,\ldots,(I+1)J\}$. Regular extension...

conjecture | open | MathDB 页面 | 原文线索

V2-045032 | MATHDB-364886

Singer-type upper-bound conjecture for optimal Golomb rulers

摘录 / EXCERPT: Let $G(k)$ be the length of an optimal Golomb ruler with k marks. Singer-type upper-bound conjecture. For every integer $k\geq 6$, ... The source derives this as a consequence of th...

conjecture | open | MathDB 页面 | 原文线索

V2-045033 | MATHDB-364890

Conjecture on the growth and limiting behavior of hypercube domination parameters

摘录 / EXCERPT: Let χ_n denote the parameter defined in the paper for the n-dimensional hypercube. Growth and limit conjecture. The sequence χ_n satisfies both of the following propert...

conjecture | open | MathDB 页面 | 原文线索

V2-045034 | MATHDB-364944

Versteegen's even-decomposability conjecture

摘录 / EXCERPT: Versteegen's conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-045035 | MATHDB-364945

Ge-Xu-Zhang conjecture for clique odd-colourings

摘录 / EXCERPT: Ge-Xu-Zhang conjecture. For every positive integer $t\geq 4$ with $t\equiv 0,1\pmod 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-045036 | MATHDB-364946

The clique conjecture for unique-chromatic colourings

摘录 / EXCERPT: Clique uniqueness conjecture. For every positive integer $t\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-045037 | MATHDB-364947

Bachmann's conjecture on relations among Schlesinger-Zudilin q-multiple zeta values

摘录 / EXCERPT: Bachmann's conjecture. All $\mathbb{Q}$-linear relations among elements of $\mathcal{Z}_q$ are generated by the stuffle product $\ast$ and duality τ. This conjecturally gives...

conjecture | open | MathDB 页面 | 原文线索

V2-045038 | MATHDB-364956

Quadratic lower-bound conjecture for empty red-red-blue triangles

摘录 / EXCERPT: Let R and B be disjoint sets of n red and n blue points, respectively, in the plane. An empty red-red-blue triangle is a triangle whose vertices consist of two points of...

conjecture | open | MathDB 页面 | 原文线索

V2-045039 | MATHDB-364960

Archer et al.'s Pell-number conjecture for cyclic pattern-avoiding permutations

摘录 / EXCERPT: Let π be a cycle permutation in S_n, written in one-line notation as $\pi=\pi_1\pi_2\cdots\pi_n$ and in cycle form as $\pi=(c_1,c_2,\ldots,c_n)$. A cycle permutation is in...

conjecture | open | MathDB 页面 | 原文线索

V2-045040 | MATHDB-364998

A threshold conjecture for 3-wise t-intersecting uniform families

摘录 / EXCERPT: Threshold conjecture. For

conjecture | open | MathDB 页面 | 原文线索

V2-045041 | MATHDB-365000

Finite-filling shapes conjecture for the free group

摘录 / EXCERPT: Let F_2 be the free group on generators a and b, and let $S\subset F_2$ be a shape. Say that S has finite filling when the solitaire filling process generated by S can fi...

conjecture | open | MathDB 页面 | 原文线索

V2-045042 | MATHDB-365001

Contour orbit connectivity conjecture for free groups

摘录 / EXCERPT: Let F_k be the free group on k generators, let S be a shape on F_k, and let P be an S-filling closed set. An S-contour of P is a contour computed with respect to a...

conjecture | open | MathDB 页面 | 原文线索

V2-045043 | MATHDB-365002

Line-orbit cardinality conjecture for the triangle solitaire

摘录 / EXCERPT: Let L_n denote the line configuration of size n in the triangle solitaire, let $\gamma(L_n)$ denote its solitaire orbit, and let Θ denote asymptotic order up to multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-045044 | MATHDB-365006

Equidistribution of the pix and fix statistics on pattern-avoiding permutations

摘录 / EXCERPT: Equidistribution conjecture. The statistics pix and fix are equidistributed over $\mathfrak{S}_{n}(\Pi)$ for all $n\geq0$ if and only if Π is...

conjecture | open | MathDB 页面 | 原文线索

V2-045045 | MATHDB-365033

The ballot-property conjecture for stable configurations on undirected binary trees

摘录 / EXCERPT: Ballot-property conjecture. In a stable configuration, the whole tree and every subtree satisfy the ballot property.

conjecture | open | MathDB 页面 | 原文线索

V2-045046 | MATHDB-365039

The large-cardinal consistency conjecture for Kastanas Ramsey subsets

摘录 / EXCERPT: Large-cardinal consistency conjecture. It is consistent with sufficiently large cardinal assumptions that every subset of $\mathcal{R}$ is Kastanas Ramsey.

conjecture | open | MathDB 页面 | 原文线索

V2-045047 | MATHDB-365051

Conjecture on compatibility minimizers and pure connected components

摘录 / EXCERPT: Let ${\bm{\tau}}\in S_n^D$, and let $\nabla({\bm{\tau}};\eta)$ be the nonnegative quantity whose vanishing characterizes the relevant compatible configurations. For a permutation o...

conjecture | open | MathDB 页面 | 原文线索

V2-045048 | MATHDB-365061

Asymptotic density conjecture for Ulam words

摘录 / EXCERPT: Ulam-word density conjecture. The density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045049 | MATHDB-365062

Gap-distribution conjecture for Ulam words

摘录 / EXCERPT: Let $\pi:S[\{0,1\}]\to\mathbb{Z}_{\geq0}$ interpret a binary word as an integer, and let $u_1<u_2<\ldots<u_{k_n}$ be the ordered elements of $\pi(\mathscr{U}_n)$. Define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045050 | MATHDB-365063

Equidistribution conjecture for Ulam words modulo integers

摘录 / EXCERPT: Equidistribution conjecture. For every integer $N>1$ and every $a\in\mathbb{Z}/N\mathbb{Z}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045051 | MATHDB-365064

Power-of-two block conjecture for Ulam words

摘录 / EXCERPT: Let u be a binary word of length n of the form $0^a1^{2^k}0^b$, where $a,b,k\in\mathbb{Z}_{\geq1}$. Power-of-two block conjecture. The word u belongs to $\mathscr{U}$ if and...

conjecture | open | MathDB 页面 | 原文线索

V2-045052 | MATHDB-365072

Su and Liu's Ramsey-full equivalence conjecture

摘录 / EXCERPT: Su and Liu's conjecture. H is Ramsey-full if and only if H is Gallai-Ramsey-full.

conjecture | open | MathDB 页面 | 原文线索

V2-045053 | MATHDB-365073

McKay and Radziszowski's conjecture for the Ramsey number $R(5,5)$

摘录 / EXCERPT: McKay and Radziszowski's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045054 | MATHDB-365074

Fox, Grinshpun, and Pach's Gallai-Ramsey conjecture for complete graphs

摘录 / EXCERPT: Fox, Grinshpun, and Pach's conjecture. For positive integers k and p,

conjecture | open | MathDB 页面 | 原文线索

V2-045055 | MATHDB-365075

Mao, Wang, Magnant, and Schiermeyer's Gallai-Ramsey conjecture for the fan F_3

摘录 / EXCERPT: Mao, Wang, Magnant, and Schiermeyer's conjecture. For $k\ge2$,

conjecture | open | MathDB 页面 | 原文线索

V2-045056 | MATHDB-365076

Zhao and Wei's Gallai-Ramsey conjecture for kipases

摘录 / EXCERPT: Zhao and Wei's conjecture. For all $k\ge2$ and $m\ge2$,

conjecture | open | MathDB 页面 | 原文线索

V2-045057 | MATHDB-365077

Song, Wei, Zhang, and Zhao's Gallai-Ramsey conjecture for even wheels

摘录 / EXCERPT: Song, Wei, Zhang, and Zhao's conjecture. For all $k\ge2$ and even $m\ge4$,

conjecture | open | MathDB 页面 | 原文线索

V2-045058 | MATHDB-365097

The additive complexity conjecture for automatic sequences

摘录 / EXCERPT: Additive complexity conjecture. The additive complexity of a k-automatic sequence is a k-regular sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-045059 | MATHDB-365111

Gamma-positivity conjecture for dissonant canon polynomials

摘录 / EXCERPT: Let $P=[m]$, let Q be a subposet of $P\times[n]$, and let $C^{Q,\operatorname{id}}(x)$ denote the dissonant canon polynomial associated with Q. Gamma-positivity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-045060 | MATHDB-365127

Archdeacon's existence conjecture for integer Heffter arrays

摘录 / EXCERPT: Archdeacon's conjecture. There exists an integer Heffter array $\mathrm{H}(m,n;s,k)$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-045061 | MATHDB-365145

Monotonicity conjecture for the number of subset sizes in the Z-curve

摘录 / EXCERPT: Monotonicity conjecture. For given values of D and $K=n+1$, the minimum number of subsets of different sizes in $\mathbf{X}^D_K$ is at least equal to the number of subsets of dif...

conjecture | open | MathDB 页面 | 原文线索

V2-045062 | MATHDB-365146

Subset-sum conjecture for projected Z-curve subsets

摘录 / EXCERPT: Subset-sum conjecture. For any subset $\mathbb{T}\subset\mathbf{X}^D_K$ with $|\mathbb{T}|=2^D\times n$ elements,

conjecture | open | MathDB 页面 | 原文线索

V2-045063 | MATHDB-365147

Transformation sum conjecture for Z-curve subsets

摘录 / EXCERPT: Transformation sum conjecture. For any valid subset $\mathbb{T}\subset\mathbf{X}^D_S$,

conjecture | open | MathDB 页面 | 原文线索

V2-045064 | MATHDB-365148

Orthotope sum conjecture for Z-curve subsets

摘录 / EXCERPT: Orthotope sum conjecture. For every orthotope $\mathcal{O}\subset\mathbb{T}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045065 | MATHDB-365157

Stinson's strong starter conjecture for cyclic groups

摘录 / EXCERPT: Let $n\geq 5$ be an odd integer. A strong starter in the cyclic group $\mathbb{Z}_n$ is a starter satisfying the additional distinct nonzero pair-sum condition defined in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-045066 | MATHDB-365158

Horton's strong starter conjecture for finite abelian groups

摘录 / EXCERPT: Let G be an abelian group of odd order $g\geq 3$. A strong starter in G is a starter whose pair sums are distinct and nonzero. Horton's conjecture. There is a strong starter in...

conjecture | open | MathDB 页面 | 原文线索

V2-045067 | MATHDB-365223

Empirical recurrence conjecture for balanced 8 by 2n matrices

摘录 / EXCERPT: Let $b_4(n)$ denote the number of balanced $8\times 2n$ matrices. A linear recurrence of order 9 with polynomial coefficients of degree 36 should be satisfied by the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-045068 | MATHDB-365265

Wanless's conjecture on transversals in latin hypercubes

摘录 / EXCERPT: A latin hypercube of order n and dimension d is a d-dimensional array filled with n symbols so that every line contains all symbols exactly once. A transversal is a diagona...

conjecture | open | MathDB 页面 | 原文线索

V2-045069 | MATHDB-365266

Taranenko's extension to positive permanents of polystochastic matrices

摘录 / EXCERPT: A d-dimensional polystochastic matrix of order n is a nonnegative d-dimensional array whose entries on every line sum to 1. Its permanent is the sum of the products selecte...

conjecture | open | MathDB 页面 | 原文线索

V2-045070 | MATHDB-365279

Marietti's parabolic Combinatorial Invariance Conjecture

摘录 / EXCERPT: Parabolic Combinatorial Invariance Conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-045071 | MATHDB-365284

Quantum advantage threshold conjecture for grouped MAJORITY

摘录 / EXCERPT: Let n players be divided into two groups of sizes $n-k$ and k, with k fixed. Let V_Q^* and V_C^* denote the optimal quantum and classical values, respectively, for the...

conjecture | open | MathDB 页面 | 原文线索

V2-045072 | MATHDB-365339

Peczarski's Gold Partition Conjecture for finite posets

摘录 / EXCERPT: Let P be a finite poset that is not a chain. Consider two consecutive comparisons, and let $t_0=e(P)$, where $e(P)$ is the number of linearizations of P; let t_1 be the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-045073 | MATHDB-365369

The sign conjecture for Coxeter arrangements of types A and D

摘录 / EXCERPT: Let V be an n-dimensional space, let $\mathcal{H}$ be a Coxeter arrangement of type A_n or D_n in V, and let $a\in V$ satisfy $0\in\mathbb{B}(a,R)$. Assume that a lies...

conjecture | open | MathDB 页面 | 原文线索

V2-045074 | MATHDB-365370

The negativity conjecture for coefficients of the matching polynomial

摘录 / EXCERPT: Matching-coefficient negativity conjecture. The coefficient of μ in Y is negative. This coefficient-level assertion is used to establish the predicted sign in type D_n; it...

conjecture | open | MathDB 页面 | 原文线索

V2-045075 | MATHDB-365371

The nonnegative invariant-polynomial conjecture in dimension three

摘录 / EXCERPT: For $d\geq 6$, define ... Here $\mathbb{R}_{\geq0}[a_1^2+a_2^2,a_1^2a_2^2]$ denotes the polynomials in the two displayed generators with nonnegative real coefficients. Nonnegative in...

conjecture | open | MathDB 页面 | 原文线索

V2-045076 | MATHDB-365375

Odd-length linear equations are uncommon over the integers

摘录 / EXCERPT: Odd-length uncommonness conjecture. The equation is uncommon over the integers.

conjecture | open | MathDB 页面 | 原文线索

V2-045077 | MATHDB-365395

Two-point conjecture for Riesz capacity ratios

摘录 / EXCERPT: Let $n\geq 2$, let $q<p<0$, and let $K\subset\mathbb R^n$ be compact with more than one point. The quantities $\operatorname{Cap}_p(K)$ and $\operatorname{Cap}_q(K)$ are the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-045078 | MATHDB-365399

The conjecture that P-distinguishability is always trivial for M-classes

摘录 / EXCERPT: Let Σ be an ordered alphabet and let $w\in\Sigma^*$ be M-ambiguous. Write $[w]$ for the M-class of w. A class is trivially $\mathbb{P}$-distinguishable when it is a si...

conjecture | open | MathDB 页面 | 原文线索

V2-045079 | MATHDB-365423

Extension of the moduli-space conjectures to slowly rotating extremal Kerr-Newman black holes

摘录 / EXCERPT: Consider very slowly rotating extremal Kerr-Newman black holes, with parameters satisfying $M^2=e^2+a^2$ and $|a|\ll M$, in the setting without spherical symmetry. Let the regulari...

conjecture | open | MathDB 页面 | 原文线索

V2-045080 | MATHDB-365492

Intersection-support conjecture for optimal colorings

摘录 / EXCERPT: For $r\in\mathbb{Z}^{+}$, let $a,b:\mathcal{P}([r])\to[0,1]$ be an optimal solution to the coloring problem, and write $a(R)$ and $b(R)$ for the corresponding aggregate quantities...

conjecture | open | MathDB 页面 | 原文线索

V2-045081 | MATHDB-365493

Equal color-weight conjecture for optimal colorings

摘录 / EXCERPT: For $r\in\mathbb{Z}^{+}$, let $a,b:\mathcal{P}([r])\to[0,1]$ be an optimal solution to the coloring problem, and let a_i and b_i denote the quantities associated with color...

conjecture | open | MathDB 页面 | 原文线索

V2-045082 | MATHDB-365494

Second-range formula conjecture for the universal bound $g^*(r)$

摘录 / EXCERPT: Second-range formula conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-045083 | MATHDB-365498

Sorting-order characterization conjecture for Stanley coefficients

摘录 / EXCERPT: For a partition $\lambda=(\lambda_1,\ldots,\lambda_r)$, let E_λ^U be the set of λ-Escher tuples, let $\tau_\lambda:E_\lambda^U\to\mathbb{Q}^{\xi(\lambda)}$ be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-045084 | MATHDB-365505

The wasteful partial linkage hindrance conjecture for finitary matroids

摘录 / EXCERPT: Let M and N be finitary matroids on a common ground set. For a set I, write M/I and N/I for the matroid contractions, and let $r(M/I)$ and $r(N/I)$ denote their ranks. An...

conjecture | open | MathDB 页面 | 原文线索

V2-045085 | MATHDB-365513

Alishahi-Taherkhani's extremal structure conjecture for almost intersecting families

摘录 / EXCERPT: Let n, k, and s be positive integers with $k\geq 2$, and let $\mathcal{F}\subseteq \binom{[n]}{k}$ be an s-almost 1-intersecting family such that ... Assume that n is suf...

conjecture | open | MathDB 页面 | 原文线索

V2-045086 | MATHDB-365518

The strengthened row-maximum conjecture for plactic centralizers

摘录 / EXCERPT: Strengthened row-maximum conjecture. For every i with $1\leq i\leq \ell$,

conjecture | open | MathDB 页面 | 原文线索

V2-045087 | MATHDB-365567

Product of Tribes optimality conjecture for small-shadow partitions

摘录 / EXCERPT: Product of Tribes optimality conjecture. There exists a universal constant δ such that

conjecture | open | MathDB 页面 | 原文线索

V2-045088 | MATHDB-365570

Tannenbaum's zero-sum partition conjecture for Abelian groups

摘录 / EXCERPT: Tannenbaum's zero-sum partition conjecture. For every positive integer t and every integer partition $\{m_i\}_{i=1}^t$ of $m-1$, with $m_i\geq 2$ for every $i\in[1,|R|/2]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045089 | MATHDB-365571

Cichacz's 4-ZSPP conjecture for Abelian groups with more than one involution

摘录 / EXCERPT: Let Γ be a finite Abelian group, and let $I(\Gamma)$ denote its set of involutions. A finite Abelian group has 4-ZSPP if every integer partition of $|\Gamma|-1$ whose parts...

conjecture | open | MathDB 页面 | 原文线索

V2-045090 | MATHDB-365572

Cichacz's zero-sum partition conjecture for groups with one involution

摘录 / EXCERPT: Cichacz's zero-sum partition conjecture. For every positive integer t and every integer partition $\{m_i\}_{i=1}^t$ of $m-2$, with $m_i\geq 2$ for every $i\in[1,t]$, there is a s...

conjecture | open | MathDB 页面 | 原文线索

V2-045091 | MATHDB-365594

The non-real eigenvalue conjecture for RSK operators

摘录 / EXCERPT: Let (σ,π) be a pair of compositions, let $\ell(\sigma)$ and $\ell(\pi)$ denote their lengths, and let ${\sf RSK}_{\sigma,\pi}$ be the associated RSK operator. Non-real eig...

conjecture | open | MathDB 页面 | 原文线索

V2-045092 | MATHDB-365595

The non-integer eigenvalue conjecture for non-triangular RSK operators

摘录 / EXCERPT: Let (σ,π) be a reduced pair of compositions, and let ${\sf RSK}_{\sigma,\pi}$ be the associated RSK operator. A pair is triangular in the sense used in the paper. Non-inte...

conjecture | open | MathDB 页面 | 原文线索

V2-045093 | MATHDB-365596

The nonvanishing trace conjecture for permutation RSK operators

摘录 / EXCERPT: For each positive integer d, let ${\sf RSK}_{1^d,1^d}$ be the RSK operator indexed by the compositions 1^d, and let ${\sf RSK}_{m,n,d}$ be the corresponding operator in dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-045094 | MATHDB-365597

The asymptotic abundance of zero diagonal entries in RSK matrices

摘录 / EXCERPT: For each positive integer d, let C_d be the set of content arrays $\alpha\in{\sf Cont}_{1^d,1^d}$ for which the diagonal entry ${\sf RSK}_{1^d,1^d}(\alpha,\alpha)$ is zero. Asy...

conjecture | open | MathDB 页面 | 原文线索

V2-045095 | MATHDB-365641

Kajitani-Ueno-Miyano cyclic orderability conjecture

摘录 / EXCERPT: Let $M=(S,\mathcal{B})$ be a matroid with rank function r_M. It is uniformly dense if ... for every $X\subseteq S$. A matroid is cyclically orderable if its ground set has a cyclic...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045096 | MATHDB-365642

Strengthened Gabow cyclic ordering conjecture for disjoint bases

摘录 / EXCERPT: Let $M=(S,\mathcal{B})$ be a matroid and let $S=B_1\cup\dots\cup B_k$ be a partition of the ground set into k pairwise disjoint bases. Strengthened Gabow conjecture. The matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-045097 | MATHDB-365643

White's basis-sequence equivalence conjecture

摘录 / EXCERPT: Let $(B_1,\dots,B_k)$ and $(B'_1,\dots,B'_k)$ be sequences of k bases of a matroid. Two sequences are compatible if the union of the B_i as a multiset equals the union of the...

conjecture | open | MathDB 页面 | 原文线索

V2-045098 | MATHDB-365644

Bérczi-Mátravölgyi-Schwarcz weighted symmetric-exchange conjecture

摘录 / EXCERPT: Let (B_1,B_2) and (B'_1,B'_2) be compatible basis pairs of a matroid $M=(S,\mathcal{B})$, and let $w:S\to\mathbb{R}_+$. The weight of a symmetric exchange replacing B_1,B_2 b...

conjecture | open | MathDB 页面 | 原文线索

V2-045099 | MATHDB-365645

Bounded cyclic-exchange conjecture for basis sequences

摘录 / EXCERPT: Let $(B_1,\dots,B_k)$ and $(B'_1,\dots,B'_k)$ be compatible sequences of k bases of a rank-r matroid. A cyclic exchange is the simultaneous cyclic transfer of elements among di...

conjecture | open | MathDB 页面 | 原文线索

V2-045100 | MATHDB-365646

Weighted cyclic-exchange conjecture for basis sequences

摘录 / EXCERPT: Let $(B_1,\dots,B_k)$ and $(B'_1,\dots,B'_k)$ be compatible sequences of k bases of a matroid $M=(S,\mathcal{B})$, and let $w:S\to\mathbb{R}_+$. Weighted cyclic-exchange conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-045101 | MATHDB-365677

Optimal lower bounds for alpha-undominated sets

摘录 / EXCERPT: Lower-bound optimality conjecture. If $\alpha \geq \frac{2}{k+1}$, then an α-undominated set of size k always exists.

conjecture | open | MathDB 页面 | 原文线索

V2-045102 | MATHDB-365679

Sparse Sperner conjecture for three central layers

摘录 / EXCERPT: Sparse Sperner conjecture. There exists a function $\beta^*(n)=\tilde\Theta(1/n)$ such that if $\beta\geq\beta^*(n)$, then almost all antichains of size βN in B_n are cont...

conjecture | open | MathDB 页面 | 原文线索

V2-045103 | MATHDB-365705

The two-equivalence-class extremal construction conjecture for rational-value sums

摘录 / EXCERPT: Extremal construction conjecture. The maximum number of r-subsets with rational sums among an n-element set of irrational numbers is attained by this construction. The paper ha...

conjecture | open | MathDB 页面 | 原文线索

V2-045104 | MATHDB-365736

Baumgart's conjecture that every graphic matroid is SIBO

摘录 / EXCERPT: A rank-r matroid is subsequence-interchangeably base orderable (SIBO) if every pair of bases A and B admits orderings $a_1,\ldots,a_r$ of A and $b_1,\ldots,b_r$ of B such...

conjecture | open | MathDB 页面 | 原文线索

V2-045105 | MATHDB-365737

The conjecture that every sparse paving matroid is SIBO

摘录 / EXCERPT: A matroid is subsequence-interchangeably base orderable (SIBO) if every pair of bases admits the subsequence exchange orderings specified in the paper. Sparse-paving SIBO conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-045106 | MATHDB-365756

The classification conjecture for P-positions in shrinking circular Nim with six piles

摘录 / EXCERPT: Classification conjecture. The P-positions in ${\rm SCN}(6,4)$ are classified into one of the following categories:

conjecture | open | MathDB 页面 | 原文线索

V2-045107 | MATHDB-365772

Limit-point conjecture for normalized sum-product pairs

摘录 / EXCERPT: Let $\operatorname{NSPP}([3,\infty))$ denote the multiset of normalized sum-product pairs for positive-integer sets of cardinalities at least 3, with each pair lying in $[1,2]^2$...

conjecture | open | MathDB 页面 | 原文线索

V2-045108 | MATHDB-365779

Gambaudo-Tresser renormalization conjecture for Hénon maps

摘录 / EXCERPT: Let $f_{a_*} \in \mathcal Q$ be an infinitely renormalizable quadratic polynomial with bounded type combinatorics. For $a,b \in \mathbb R$, let $F_{a,b}(x,y)=(x^2+a-by,x)$ be a Hén...

conjecture | open | MathDB 页面 | 原文线索

V2-045109 | MATHDB-365783

The floor-probability bounds conjecture for convex sets

摘录 / EXCERPT: Floor-probability bounds conjecture. For all $n\geq 3$ and all $K\in\operatorname{CCSF}([0,1])$,

conjecture | open | MathDB 页面 | 原文线索

V2-045110 | MATHDB-365804

Thomason's upper-bound conjecture for book Ramsey numbers

摘录 / EXCERPT: Let $B_{t,m}$ denote the graph consisting of t triangles sharing a common edge, and let $r(B_{t,m})$ be its two-colour Ramsey number. Thomason's conjecture. For all integers...

conjecture | open | MathDB 页面 | 原文线索

V2-045111 | MATHDB-365809

Pseudoline arrangement conjecture for the Hirzebruch quadratic form

摘录 / EXCERPT: Let $\mathcal{L}\subset\mathbb{RP}^2$ be an essential pseudoline arrangement, and let M be its matroid. Let Q be the Hirzebruch quadratic form of M, and let the semistable co...

conjecture | open | MathDB 页面 | 原文线索

V2-045112 | MATHDB-365827

The optimal fifth-power bound for commuting pairs in finite real sets

摘录 / EXCERPT: Let $\mathcal{A}\subset\mathbb{R}$ be a non-empty finite set, and let $T(\mathcal{A})$ denote the quantity counting the relevant commuting pairs of 2×2 matrices associated...

conjecture | open | MathDB 页面 | 原文线索

V2-045113 | MATHDB-365840

Asymptotic equality frequencies for Hilbert depths of path and cycle ideals

摘录 / EXCERPT: Let S, I_n, and J_n be as above, and for each N let $\#\{n\leq N:\,P(n)\}$ denote the number of positive integers $n\leq N$ satisfying $P(n)$. Hilbert-depth frequency conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045114 | MATHDB-365862

Hodges's strengthened conjecture on tangled labelings

摘录 / EXCERPT: Let P be a poset with n elements and m minimal elements. A labeling of P is tangled if it requires $n-1$ applications of extended promotion to become a natural labeling. Ho...

conjecture | open | MathDB 页面 | 原文线索

V2-045115 | MATHDB-365884

Approximate Duality Conjecture for finite-field bilinear bias

摘录 / EXCERPT: Approximate Duality Conjecture. If $D_\omega(A,B)\geq\varepsilon$ for some $\varepsilon>0$, then there exist subsets $A'\subseteq A$ and $B'\subseteq B$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-045116 | MATHDB-365886

Asymptotic orders for invariants of Erdős-Rényi numerical semigroups

摘录 / EXCERPT: For $p\in(0,1)$, let ${\mathcal S}(p)$ denote the Erdős-Rényi numerical semigroup obtained by independently selecting each positive integer as a generator with probability p. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-045117 | MATHDB-365890

Andrews-Merca-Guo-Zeng conjecture on truncated Jacobi triple product coefficients

摘录 / EXCERPT: Andrews-Merca-Guo-Zeng conjecture. This coefficient is nonnegative. This conjecture is solved: it was proved analytically by Mao and combinatorially by Yee in 2015, and later repro...

conjecture | open | MathDB 页面 | 原文线索

V2-045118 | MATHDB-365902

Grundy-number and nim-sum conjecture for amalgamation Nim with restriction

摘录 / EXCERPT: A position is a triple (x,y,z) of nonnegative integers, and its Grundy number is the Sprague-Grundy value of that position; write $x\oplus y\oplus z$ for the bitwise nim-sum. For...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045119 | MATHDB-365940

M-convexity conjecture for Postnikov-Stanley polynomials

摘录 / EXCERPT: Postnikov-Stanley polynomials are a class of polynomials associated with permutations. A polynomial is M-convex when the set of exponent vectors of its monomials satisfies the disc…

conjecture | open | MathDB 页面 | 原文线索

V2-045120 | MATHDB-365953

The Rado triple conjecture for homogeneous quadratic equations

摘录 / EXCERPT: Rado triple conjecture. The equation

conjecture | open | MathDB 页面 | 原文线索

V2-045121 | MATHDB-365954

Partition regularity conjecture for generalized Pythagorean pairs

摘录 / EXCERPT: Generalized Pythagorean-pair conjecture. If at least one of ac, bc, or $(a+b)c$ is a square, then $ax^2+by^2=cz^2$ is partition regular with respect to x,y.

conjecture | open | MathDB 页面 | 原文线索

V2-045122 | MATHDB-365975

Kahn-Kalai threshold conjecture for graph appearances

摘录 / EXCERPT: Let G be a graph, and write $v(G)$ and $e(G)$ for its numbers of vertices and edges. For a subgraph $F\subseteq G$, define $p_E(G)$ to be the smallest p such that … for every…

conjecture | open | MathDB 页面 | 原文线索

V2-045123 | MATHDB-365981

The logarithmic upper-bound conjecture for planar spanning-tree numbers

摘录 / EXCERPT: Let $\alpha(t)$ be the smallest number of vertices of a planar simple graph with exactly t spanning trees. The known lower bound is $\alpha(t)=\Omega(\log t)$, and Stong proved…

conjecture | open | MathDB 页面 | 原文线索

V2-045124 | MATHDB-365982

The logarithmic edge-bound conjecture for planar spanning-tree numbers

摘录 / EXCERPT: Let $\beta(t)$ be the smallest number of edges of a planar graph, allowing multiple edges, with exactly t spanning trees. It is known that $\beta(t)=O(\log t\,\log\log t)$. Logar…

conjecture | open | MathDB 页面 | 原文线索

V2-045125 | MATHDB-365985

The exponential growth-rate limit for planar spanning-tree values

摘录 / EXCERPT: For each positive integer n, let $\mathcal T(n)$ be the set of spanning-tree numbers $\tau(G)$ attained by simple planar graphs on n vertices. The paper proves exponential lowe…

conjecture | open | MathDB 页面 | 原文线索

V2-045126 | MATHDB-366008

The f-vector conjecture for matroidal decompositions of uniform matroids

摘录 / EXCERPT: Let $\mathrm{U}_{r,n}$ be the uniform matroid of rank r on $[n]$, and let $\Delta(\mathrm{U}_{r,n})$ be its matroid base polytope. A matroidal decomposition is a decomposition of…

conjecture | open | MathDB 页面 | 原文线索

V2-045127 | MATHDB-366009

Nonnegativity conjecture for the matroid polynomial g-invariant

摘录 / EXCERPT: Let M be a matroid, and let $g_M(t)\in\mathbb{Z}[t]$ be the covaluative matroid invariant described in the paper. Nonnegativity conjecture for $g_M(t)$. For any matroid M, the…

conjecture | open | MathDB 页面 | 原文线索

V2-045128 | MATHDB-366010

Nonnegativity conjecture for the omega invariant of a matroid

摘录 / EXCERPT: Let M be a matroid of rank r, and write $g_M(t)=\sum_{i\geq 1}g_i(M)t^i$ when the ground set is nonempty. Define the omega invariant by $\omega(M)=g_r(M)$, the coefficient of…

conjecture | open | MathDB 页面 | 原文线索

V2-045129 | MATHDB-366012

Nonexistence conjecture for star-n idempotent ultrafilters

摘录 / EXCERPT: Nonexistence conjecture. For any $n\in\mathbb{N}_{>1}$, $\star_n$-idempotent ultrafilters do not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-045130 | MATHDB-366015

The two-row Schubert coefficient bound conjecture

摘录 / EXCERPT: Two-row Schubert coefficient bound conjecture. For every $vin S_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-045131 | MATHDB-366030

The asymptotic density conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let den_{n} denote the density quantity defined in the paper for Kazhdan-Lusztig polynomials indexed by permutations in S_{n}. Asymptotic density conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-045132 | MATHDB-366031

The superexponential growth and exponential decay conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: For each n, let num_{n} be the number of distinct polynomials among $\{\mathrm{P}_{w}\mid w\in\mathrm{S}_{n}\}$, and let $\mathrm{num}_{n}^{\%}$ be the ratio of this n…

conjecture | open | MathDB 页面 | 原文线索

V2-045133 | MATHDB-366032

The superexponential coefficient-growth conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: For $w\in\mathrm{S}_{n}$, let P_{w} be the first-row Kazhdan-Lusztig polynomial, and let coeff_{n} denote the maximal coefficient among these polynomials; let…

conjecture | open | MathDB 页面 | 原文线索

V2-045134 | MATHDB-366058

#P-hardness conjecture for Schubert coefficients

摘录 / EXCERPT: Let $\mathfrak{S}_w$ denote the Schubert polynomial indexed by a permutation $w\in S_\infty$, and define the Schubert coefficients by ... Writing...

conjecture | open | MathDB 页面 | 原文线索

V2-045135 | MATHDB-366059

Non-membership conjecture for Schubert coefficients in #P

摘录 / EXCERPT: Let $\mathfrak{S}_w$ denote the Schubert polynomial indexed by $w\in S_\infty$, and define $c^w_{u,v}$ by ... Writing $\operatorname{sc}_{\mathrm{Schubert}}$ for the function computi...

conjecture | open | MathDB 页面 | 原文线索

V2-045136 | MATHDB-366123

The asymptotic half-frequency conjecture for the kth-most frequent element

摘录 / EXCERPT: Let $\mathcal F$ be a union-closed set family, let $f_k(\mathcal F)$ denote its kth-highest element frequency, and fix $k\in\mathbb N$. Asymptotic half-frequency conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-045137 | MATHDB-366124

Nagel's conjecture on multiple frequent elements in union-closed families

摘录 / EXCERPT: Nagel's conjecture. There are k elements, each belonging to at least

conjecture | open | MathDB 页面 | 原文线索

V2-045138 | MATHDB-366129

Rota's excluded-minor finiteness conjecture for representable matroids

摘录 / EXCERPT: Rota's conjecture. For each finite field F, there are, up to isomorphism, only finitely many excluded minors for the class of F-representable matroids.

conjecture | open | MathDB 页面 | 原文线索

V2-045139 | MATHDB-366135

The cohomological conjecture for Dirac matrix factorizations

摘录 / EXCERPT: Cohomological conjecture. For each combinatorially spin Delzant polytope $\triangle$ such that $X_\triangle^\mathbb{R}$ is Fano, we have

conjecture | open | MathDB 页面 | 原文线索

V2-045140 | MATHDB-366140

Conjectured Vandermonde-sum identity for sixteen overlap classes

摘录 / EXCERPT: Fix nonnegative integers $x_1,\dots,x_{16}$ with $x_1+\dots+x_8=n$. Let $\mathcal{S}_{\mathbf{x}}$ be the collection of all quadruples $(\mathcal{A}_1,\dots,\mathcal{A}_4)$ whose s...

conjecture | open | MathDB 页面 | 原文线索

V2-045141 | MATHDB-366160

Structural conjecture for 3-connected binary matroids excluding M(K_4) as an induced minor

摘录 / EXCERPT: Structural conjecture. The class of 3-connected binary matroids that do not contain $M(K_4)$ as an induced minor is exactly the class of matroids that can be obtained by starting...

conjecture | open | MathDB 页面 | 原文线索

V2-045142 | MATHDB-366167

The low-power error-correcting cooling code density conjecture

摘录 / EXCERPT: Low-power error-correcting cooling code density conjecture. Under this condition,

conjecture | open | MathDB 页面 | 原文线索

V2-045143 | MATHDB-366183

Explicit Schubert-product map commuting with the word-deletion operator

摘录 / EXCERPT: Commuting-map conjecture. There exists a map ψ satisfying Proposition such that

conjecture | open | MathDB 页面 | 原文线索

V2-045144 | MATHDB-366184

Down-up Monk connectedness conjecture for back-stable Schubert products

摘录 / EXCERPT: Down-up Monk connectedness conjecture. For any $u,v\in S_{\mathbb Z}$, the set $\{w\mid\overleftarrow{c_{u,v}^w}\ne0\}$ is down-up Monk connected.

conjecture | open | MathDB 页面 | 原文线索

V2-045145 | MATHDB-366185

Action of the word-deletion operator on back-stable key polynomials

摘录 / EXCERPT: Key-polynomial operator conjecture. For any composition α,

conjecture | open | MathDB 页面 | 原文线索

V2-045146 | MATHDB-366201

Square-root deficit conjecture for transversals in equi-n-squares

摘录 / EXCERPT: An equi-n-square is an $n\times n$ array filled with n symbols, each appearing exactly n times. A transversal is a collection of cells sharing no row, column, or symbol; its...

conjecture | open | MathDB 页面 | 原文线索

V2-045147 | MATHDB-366204

The polynomial-form conjecture for lucky spots in parking functions

摘录 / EXCERPT: Let n be a positive integer and let j be a positive integer. A lucky spot is a parking spot occupied by a car whose preferred spot is that spot. Let $f_j(n)$ denote the polynom...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045148 | MATHDB-366212

Existence of the limiting slope of three-dimensional Wythoff Nim P-positions

摘录 / EXCERPT: Limiting-slope conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-045149 | MATHDB-366231

Grünbaum's digon conjecture for pairwise crossing pseudocircles

摘录 / EXCERPT: Grünbaum's digon conjecture. Every simple arrangement of $n>2$ pairwise crossing pseudocircles has at most $2n-2$ digons.

conjecture | open | MathDB 页面 | 原文线索

V2-045150 | MATHDB-366233

Trivariate symmetries on vincular patterns

摘录 / EXCERPT: For $n\geq 1$, let $\mathfrak{S}_n$ be the set of permutations of $[n]$, and define ... Here $(\textrm{13--2})$, $(\textrm{31--2})$, $(\textrm{2--13})$, and $(\textrm{2--31})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-045151 | MATHDB-366254

Speyer's tropical f-vector conjecture

摘录 / EXCERPT: Let $\Sigma(r,n)$ be the hypersimplex, and consider a subdivision of $\Sigma(r,n)$ into matroid base polytopes. An interior face is a face of the subdivision not contained in the b...

conjecture | open | MathDB 页面 | 原文线索

V2-045152 | MATHDB-366259

Generalized Lyubeznik-to-Barile-Macchia resolution conjecture

摘录 / EXCERPT: Let I be a monomial ideal. A minimal generalized Lyubeznik resolution and a minimal generalized Barile-Macchia resolution are resolutions of I with the indicated generalized co...

conjecture | open | MathDB 页面 | 原文线索

V2-045153 | MATHDB-366290

Payne-Wood's quadratic conjecture for the planar Ramsey number

摘录 / EXCERPT: Payne-Wood's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045154 | MATHDB-366291

Payne-Wood's coloring conjecture for planar point sets

摘录 / EXCERPT: Payne-Wood's coloring conjecture. Every such set P can be colored with $O(\sqrt{n})$ colors so that each color class is in general position.

conjecture | open | MathDB 页面 | 原文线索

V2-045155 | MATHDB-366299

Periodic oscillation conjecture for combinatorial classes with evenly spaced dominant singularities

摘录 / EXCERPT: Let $\mathcal{C}$ be a combinatorial class with generating function $f_\mathcal{C}(z)$ of radius of convergence $\rho<1$, and suppose ... Assume $\alpha\notin-\mathbb{N}$, and that...

conjecture | open | MathDB 页面 | 原文线索

V2-045156 | MATHDB-366309

Partition-regularity characterization by computable finite partitions

摘录 / EXCERPT: Partition-regularity characterization conjecture. There exists such a computable collection of finite partitions $\{\mathcal{C}_n\}_{n=1}^{\infty}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-045157 | MATHDB-366310

Density-regularity characterization by computable positive-density sets

摘录 / EXCERPT: Density-regularity characterization conjecture. There exists a computable collection $\{B_n\}_{n=1}^{\infty}$ of subsets of $R\setminus\{0\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-045158 | MATHDB-366320

Yokoi's Rainbow Arborescence Conjecture

摘录 / EXCERPT: Yokoi's Rainbow Arborescence Conjecture. If G is the disjoint union of $n-1$ spanning arborescences $A_1, A_2, \dots, A_{n-1}$, then G has a rainbow spanning arborescence B.

conjecture | open | MathDB 页面 | 原文线索

V2-045159 | MATHDB-366321

Equivalent Rainbow Arborescence Conjecture for at least $n-1$ colors

摘录 / EXCERPT: Equivalent Rainbow Arborescence Conjecture. If $k \ge n-1$, then the disjoint union G of k spanning arborescences has a rainbow spanning arborescence B.

conjecture | open | MathDB 页面 | 原文线索

V2-045160 | MATHDB-366369

The maximal ball-number conjecture for finite ordinal spaces

摘录 / EXCERPT: Maximal ball-number conjecture. The maximal number of balls in X is b_n, where (b_n) is sequence A263511. The paper supports the sequence by direct constructions, including o...

conjecture | open | MathDB 页面 | 原文线索

V2-045161 | MATHDB-366370

The minimal ball-number conjecture for finite ordinal spaces

摘录 / EXCERPT: Let (X,δ) be an ordinal space with $|X|=n$, and let $\mathbf B_X$ denote its set of balls. Assume that ... for every different pairs of points $\{x,y\}\neq\{z,w\}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-045162 | MATHDB-366375

Limiting-function conjecture for normalized k-mer minimizer counts

摘录 / EXCERPT: Let Σ be the underlying alphabet, and for each k,m let $f_{k,m}:[0,1]\to[0,1]$ be the piecewise linear function determined by the points $p(w)=(x,y)$ for $w\in\Sigma^m$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-045163 | MATHDB-366376

Exponential-growth conjecture for k-mer minimizer counts

摘录 / EXCERPT: Let Σ be an alphabet, let $m\geq 1$, and let $w\in\Sigma^m$ be an m-mer. Write $\pi_k(w)$ for the number of objects counted by the paper's minimizer-count function, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-045164 | MATHDB-366377

Sós's oriented Ramsey conjecture for the graphs H(n,i)

摘录 / EXCERPT: Sós's conjecture. For each n and each i satisfying $4\leq i\leq n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-045165 | MATHDB-366386

Semicontinuity conjecture for Nash equilibrium types in generic two-action games

摘录 / EXCERPT: Semicontinuity conjecture. The displayed inequality holds for every generic two-action game and every $d\in\{0,1,\ldots,m\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-045166 | MATHDB-366390

The strict-partition conjecture for Robinson-Schensted shapes

摘录 / EXCERPT: Strict-partition conjecture. begin{enumerate} item If $\nu_s>1$, then $\mathcal{S}_\nu=\mathcal{B}_\nu$. item If $\nu_s=1$ and m is odd, then $\mathcal{S}_\nu=\mathcal{B}_\nu$. i...

conjecture | open | MathDB 页面 | 原文线索

V2-045167 | MATHDB-366394

Bipartite maximizer and clique free-energy conjectures for the antiferromagnetic Ising model

摘录 / EXCERPT: Let G range over Δ-regular graphs, let $\alpha_G(B,\lambda)$ be the occupancy fraction, and let $F_G(B,\lambda)$ be the free energy density. Bipartite maximizer and clique...

conjecture | open | MathDB 页面 | 原文线索

V2-045168 | MATHDB-366397

Chen-Wen-Wu conjecture on generalized Thue-Morse k-binomial complexity

摘录 / EXCERPT: Chen-Wen-Wu conjecture. For every $k\geqslant 3$, the function $\mathsf{b}_{\mathbf{t}_m}^{(k)}$ is ultimately periodic with period m^k.

conjecture | open | MathDB 页面 | 原文线索

V2-045169 | MATHDB-366398

Kempe state-reducibility conjecture for small configurations

摘录 / EXCERPT: Kempe state-reducibility conjecture. At least one of the following configurations is Kempe state-reducible: a single pentagon, a single hexagon, two adjacent pentagons, or three pa...

conjecture | open | MathDB 页面 | 原文线索

V2-045170 | MATHDB-366414

Brenti-Carnevale unimodality conjecture for the odd-length polynomial

摘录 / EXCERPT: Brenti-Carnevale's conjecture. For any $n\geq 4$, the polynomial $B_n(q)$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-045171 | MATHDB-366415

Brenti-Carnevale unimodality conjecture for residue-class inversion polynomials

摘录 / EXCERPT: Brenti-Carnevale's conjecture. For any $n\in\mathbb{N}$ and any $k\geq 3$, the polynomial $B_{n,k,1}(q)$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-045172 | MATHDB-366422

Bóna's log-concavity conjecture for parking-function polynomials

摘录 / EXCERPT: Let PF_n be the set of parking functions of length n, and define the sum statistic by $\mathsf{sum}(\pi)=a_1+a_2+\ldots+a_n$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-045173 | MATHDB-366437

The monotonicity conjecture for normalized BGW numbers

摘录 / EXCERPT: Fix $g\ge1$, and order the partitions of $g-1$ first by increasing length and then lexicographically among partitions of equal length, with their nonzero entries arranged in increa...

conjecture | open | MathDB 页面 | 原文线索

V2-045174 | MATHDB-366447

Extremal percolation probabilities among isotropic degree-two models

摘录 / EXCERPT: Let $\mathcal{C}_p$ be the class of probability distributions on Ω that are independent and identically distributed over the vertices, isotropic, and have expected degree eq...

conjecture | open | MathDB 页面 | 原文线索

V2-045175 | MATHDB-366453

Dominance-order conjecture for the Fibonacci-word multiplicity matrix

摘录 / EXCERPT: Multiplicity-matrix conjecture. The matrix $\mathrm{\bf N}_n$ is upper triangular. Furthermore, if $N_{u,v}$ is nonzero, then $u\succeq v$ and $\mathcal{h}(u)=\mathcal{h}(v)$. This...

conjecture | open | MathDB 页面 | 原文线索

V2-045176 | MATHDB-366455

The linear subspace Ramsey conjecture over $\mathbb{F}_2$

摘录 / EXCERPT: For each positive integer n, consider the vector space $\mathbb{F}_2^n$ and a two-coloring of its nonzero vectors. A subspace $H\leq\mathbb{F}_2^n$ is monochromatic when all its...

conjecture | open | MathDB 页面 | 原文线索

V2-045177 | MATHDB-366479

Cloitre's density conjecture for the self-generating sequence

摘录 / EXCERPT: Cloitre's density conjecture. The number of 1's appearing in the prefix $a_1a_2\cdots a_n$ is $2n/3+o(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-045178 | MATHDB-366508

Conjecture on polygonal extremals in the convex (λ_1,μ_1)-diagram

摘录 / EXCERPT: Conjecture on polygonal extremals. Except for the ball, domains on the upper boundary of $\mathcal{E}^C$ are polygonal; regular polygons lie on that upper boundary; and there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-045179 | MATHDB-366544

Partition-theoretic stabilization conjecture for kth-power repulsion

摘录 / EXCERPT: Partition-theoretic stabilization conjecture. For each non-negative integer d, there is a positive integer N_d such that for $k\geq N_d$,

conjecture | open | MathDB 页面 | 原文线索

V2-045180 | MATHDB-366550

Zauner's conjecture on SICs in every dimension

摘录 / EXCERPT: Let a 1-SIC be a set of d^2 distinct rank-1 Hermitian projectors in the algebra of linear operators on ^{d} satisfying the equiangularity condition. Zauner's conjecture. 1-...

conjecture | open | MathDB 页面 | 原文线索

V2-045181 | MATHDB-366584

Maxwell's quadratic upper-bound conjecture for electrostatic equilibria

摘录 / EXCERPT: Maxwell's conjecture. The electrostatic potential defined by n point charges in $\mathbb{R}^3$ has at most $(n-1)^2$ equilibria.

conjecture | open | MathDB 页面 | 原文线索

V2-045182 | MATHDB-366604

The asymptotic bound conjecture for monochromatic progressions in the Fibonacci word

摘录 / EXCERPT: The asymptotic bound conjecture. For all $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-045183 | MATHDB-366627

Conjectured value of the feedback arc set decomposition number for maximum degree 5 and directed girth 3

摘录 / EXCERPT: Let ${\rm fasd}(\Delta,g)$ denote the supremum, over the relevant weighted orgraphs of maximum degree at most Δ and directed girth at least g, of the normalized feedback a...

conjecture | open | MathDB 页面 | 原文线索

V2-045184 | MATHDB-366638

The powers-of-two conjecture for expected coin-flip ending times

摘录 / EXCERPT: Powers-of-two conjecture. The expected value $E(S)$ equals 2^s possibly plus some lower positive powers of 2; asymptotically, $E(S)$ is about 2^s.

conjecture | open | MathDB 页面 | 原文线索

V2-045185 | MATHDB-366639

The reversal symmetry conjecture for expected coin-flip ending times

摘录 / EXCERPT: Reversal symmetry conjecture. If S' is the reversal of an ending string S, then

conjecture | open | MathDB 页面 | 原文线索

V2-045186 | MATHDB-366644

Sulanke's conjecture on 3-Narayana distributions of tableau statistics

摘录 / EXCERPT: Sulanke's conjecture. On $\operatorname{SYT}(n^3)$, each of the statistics $\st_1$ and $\st_2$ has a 3-Narayana distribution.

conjecture | open | MathDB 页面 | 原文线索

V2-045187 | MATHDB-366645

Involutions conjecture for statistics on three-row standard Young tableaux

摘录 / EXCERPT: Involutions conjecture. There exist involutions $\alpha,\delta:\operatorname{SYT}(n^3)\to\operatorname{SYT}(n^3)$ with the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-045188 | MATHDB-366670

The sumset bounds conjecture for sets with control

摘录 / EXCERPT: Let A be a finite set with control ba, meaning that it satisfies the control hypotheses introduced in the paper. For $b5>0$, write $X\gg_b5 Y$ to mean that X is bounded below...

conjecture | open | MathDB 页面 | 原文线索

V2-045189 | MATHDB-366696

The lower-bound conjecture for covering conical grids

摘录 / EXCERPT: Let Γ be a conical grid of order n. The covering number with multiplicity k is the minimum number of lines required to cover every point of Γ at least k times....

conjecture | open | MathDB 页面 | 原文线索

V2-045190 | MATHDB-366716

The real polynomial Freiman-Ruzsa inverse conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{R}$ be a finite set. If $\xi\in\mathbb{R}$ is transcendental and $K_1,K_2\geqslant 2$ satisfy ... then The real polynomial Freiman-Ruzsa conjecture. ... This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045191 | MATHDB-366750

Mandel's extension conjecture for oriented matroids

摘录 / EXCERPT: Mandel's conjecture. Every oriented matroid $\mathcal{O}$ has an extension g in general position (not a coloop) such that $(\mathcal{O}\cup g,g,f)$ is a Euclidean oriented matroi...

conjecture | open | MathDB 页面 | 原文线索

V2-045192 | MATHDB-366751

Submodular-cut extension conjecture for oriented matroids

摘录 / EXCERPT: Submodular-cut extension conjecture. There exists an extension $\mathcal{O}\cup p'$ corresponding to $\mathcal{M}'$.

conjecture | open | MathDB 页面 | 原文线索

V2-045193 | MATHDB-366769

Singh-Barman's hook bias conjecture for 2-regular partitions

摘录 / EXCERPT: Let $b_{t,i}(n)$ denote the number of hooks of length i in all t-regular partitions of n. Singh-Barman's conjecture. For every integer $k 3$, $b_{2,k}(n) b_{2,k+1}(n)$ for al...

conjecture | open | MathDB 页面 | 原文线索

V2-045194 | MATHDB-366770

Singh-Barman's hook bias conjecture for 3-regular partitions

摘录 / EXCERPT: Let $b_{t,i}(n)$ denote the number of hooks of length i in all t-regular partitions of n. Singh-Barman's conjecture. For all $n\ge 28$, $b_{3,2}(n)\ge b_{3,1}(n)$. This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045195 | MATHDB-366771

The even-parameter hook bias conjecture for 2-regular partitions

摘录 / EXCERPT: Let $b_{t,i}(n)$ denote the number of hooks of length i in all t-regular partitions of n. Even-parameter hook bias conjecture. For even $k\ge 8$, $b_{2,k}(n)\ge b_{2,k+1}(n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-045196 | MATHDB-366804

Markov polynomial Newton-polygon saturation conjecture

摘录 / EXCERPT: Saturation conjecture. The terms appearing in the numerator of M_ρ are precisely those corresponding to the integer lattice points in Δ_ρ. The conjecture is an equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-045197 | MATHDB-366805

Markov polynomial weak log-concavity conjecture

摘录 / EXCERPT: Let M_ρ be a Markov polynomial, and let its coefficients be regarded as weights on the corresponding Newton polygon. A sequence is log-concave when $x_k^2\geq x_{k-1}x_{k+1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-045198 | MATHDB-366810

Normal-form conjecture for elements of the monoid \(\mathcal{O}_n\)

摘录 / EXCERPT: Normal-form conjecture. In the second form, u and v satisfy restriction (a); in the third form, they satisfy restriction (b); and in the fourth form, they satisfy both restrict...

conjecture | open | MathDB 页面 | 原文线索

V2-045199 | MATHDB-366825

Rationality conjecture for weak saturation limits

摘录 / EXCERPT: Let F be a graph, and let w_F denote its weak saturation limit. Rationality conjecture. For any graph F, w_F is rational. This asks whether all weak saturation limits are r...

conjecture | open | MathDB 页面 | 原文线索

V2-045200 | MATHDB-366826

Density-realization conjecture for weak saturation limits

摘录 / EXCERPT: Let $\delta\geq2$ be fixed, and let $w\in\mathbb Q\cap[\delta/2,\delta-1]$. Density-realization conjecture. There exists a graph F such that ... This would extend the dense-range r...

conjecture | open | MathDB 页面 | 原文线索

V2-045201 | MATHDB-366835

Chvátal's conjecture on hereditary families

摘录 / EXCERPT: Let ${\cal F}$ be a family of sets. Call ${\cal F}$ hereditary if every subset of every member of ${\cal F}$ also belongs to ${\cal F}$. Call ${\cal F}$ EKR if some element x sat...

conjecture | open | MathDB 页面 | 原文线索

V2-045202 | MATHDB-366839

Erdős's additive irreducibility conjecture for perturbed squares

摘录 / EXCERPT: Let R be obtained from the set of positive squares by deleting some of its elements up to X and adding some positive integers, with the total number of changed elements being...

conjecture | open | MathDB 页面 | 原文线索

V2-045203 | MATHDB-366840

Hajdu-Sárközy's multiplicative irreducibility conjecture for perturbed shifted powers

摘录 / EXCERPT: Let R be obtained from the set $\{x^k+1:x\in\mathbb{N}\}$ by changing $o(X^{1/k})$ elements up to X, where $k\geq 2$. A set $S\subset\mathbb{N}$ is multiplicatively irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-045204 | MATHDB-366848

Bodirsky-Pinsker CSP dichotomy conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Bodirsky-Pinsker conjecture. If $\str M$ is a first-order reduct of a finitely bounded homogeneous structure in a finite relational language, then either...

conjecture | open | MathDB 页面 | 原文线索

V2-045205 | MATHDB-366849

Finitely bounded homogeneous structures have finitely bounded Ramsey expansions

摘录 / EXCERPT: Bodirsky's Ramsey expansion conjecture. Every finitely bounded homogeneous structure has a finitely bounded Ramsey expansion.

conjecture | open | MathDB 页面 | 原文线索

V2-045206 | MATHDB-366850

Strong amalgamation classes with EPPA have precompact Ramsey expansions

摘录 / EXCERPT: EPPA-Ramsey expansion conjecture. Every strong amalgamation class with EPPA has a precompact Ramsey expansion.

conjecture | open | MathDB 页面 | 原文线索

V2-045207 | MATHDB-366851

Polynomial-time homomorphism-embedding CSPs for homogeneous Ramsey structures

摘录 / EXCERPT: Homomorphism-embedding CSP conjecture. There is a homogeneous Ramsey structure $\str H$ in a finite relational language, first-order bi-interpretable with $\str M$, such that the h...

conjecture | open | MathDB 页面 | 原文线索

V2-045208 | MATHDB-366858

Rank-factorization conjecture for the polynomials $P_\lambda(z;q,t)$

摘录 / EXCERPT: Rank-factorization conjecture. For any partition λ of rank r, there is a polynomial $Q_\lambda(z;q,t)\in\mathbb{Z}_{\geq 0}[z,q,t]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-045209 | MATHDB-366915

Kim-Vu sandwich conjecture for random regular graphs

摘录 / EXCERPT: Let d be a degree parameter satisfying $d\gg\log n$. For sequences $p_1=p_1(n)\sim d/n$ and $p_2=p_2(n)\sim d/n$, consider a random d-regular graph $\mathbb R(n,d)$ and binomia...

conjecture | open | MathDB 页面 | 原文线索

V2-045210 | MATHDB-366921

The converse characterization of unique minimal \mathcal{X}-matroids

摘录 / EXCERPT: Let $\mathcal{X}$ be a collection of subsets defining the class of $\mathcal{X}$-matroids, and suppose that this class is nonempty. Let $v_{\mathcal{X}}$ be the function constructe...

conjecture | open | MathDB 页面 | 原文线索

V2-045211 | MATHDB-366930

Hwang-Jang-Oh web-permutation enumeration conjecture

摘录 / EXCERPT: Hwang-Jang-Oh's conjecture. For $n\geq1$, the enumerative values satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-045212 | MATHDB-366953

Rohatgi's almost-every matching conjecture for ordered Ramsey numbers

摘录 / EXCERPT: Rohatgi's conjecture. The stated subquadratic bound holds for almost every ordered matching of each fixed interval chromatic number.

conjecture | open | MathDB 页面 | 原文线索

V2-045213 | MATHDB-366954

Gir27ao-Janzer-Janzer linear-color exponent conjecture for ordered paths

摘录 / EXCERPT: Gir27ao-Janzer-Janzer conjecture. The ordered Ramsey number of $P^<_{k,n}$ in q colors admits the displayed bound with an exponent linear in q.

conjecture | open | MathDB 页面 | 原文线索

V2-045214 | MATHDB-366955

Erd6s-Szekeres conjecture for the convex-position function

摘录 / EXCERPT: A finite set of points in the plane is in general position if no three points are collinear, and it is in convex position if its points are the vertices of a convex polygon. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045215 | MATHDB-366957

Fox-Li conjecture for edge-ordered Ramsey numbers of degenerate graphs

摘录 / EXCERPT: Let $H^\prec$ be an edge-ordered d-degenerate graph on n vertices, and let $R_\prec(H^\prec)$ be its edge-ordered Ramsey number. Fox-Li conjecture. One has ... This would improve...

conjecture | open | MathDB 页面 | 原文线索

V2-045216 | MATHDB-366978

Kiselev-Kozik conjecture on satisfying sequences for rainbow partite matchings

摘录 / EXCERPT: Let s,n,k be positive integers, and let a sequence $(f_i)_{i=1}^s$ be called satisfying when it has the property that any s families in $[n]^k$ whose respective sizes exceed...

conjecture | open | MathDB 页面 | 原文线索

V2-045217 | MATHDB-366988

Finite-field extremal basis-number conjecture for rank-2 uniform minors

摘录 / EXCERPT: Bose-Burton extremal basis conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045218 | MATHDB-366989

Binary deleted-subspace basis conjecture

摘录 / EXCERPT: Binary Bose-Burton basis conjecture. Among all subsets of $\mathbb{F}_2^r$ of size 2^r-2^{r-c}, the one with the most bases is obtained by deleting a copy of the subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-045219 | MATHDB-366990

Burr-Erdős conjecture on Ramsey numbers of trees

摘录 / EXCERPT: Let T be a tree on n vertices. Burr-Erdős conjecture. ... when n is even, and ... otherwise. This is a central conjecture in the study of Ramsey numbers of trees; the source pres...

conjecture | open | MathDB 页面 | 原文线索

V2-045220 | MATHDB-366991

Caro-Roditty conjecture on zero-sum tree embeddings

摘录 / EXCERPT: Let $m \geq k \geq 2$ be integers such that $k \mid m$, and let T_{m+1} be a tree on $n=m+1$ vertices. Let G be a graph with minimum degree at least $n+k-2$. Caro-Roditty conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045221 | MATHDB-367029

Kong's modulo-8 conjecture for near-perfect matchings of odd square grids

摘录 / EXCERPT: Kong's modulo-8 conjecture. For all k, the integer c_k is congruent to 1 modulo 8.

conjecture | open | MathDB 页面 | 原文线索

V2-045222 | MATHDB-367030

Modulo-4 conjecture for near-perfect matchings of odd rectangles

摘录 / EXCERPT: Modulo-4 conjecture. Except sometimes when r and c are both congruent to 3 modulo 4, the count $a(r,c;h)$ is a multiple of 4.

conjecture | open | MathDB 页面 | 原文线索

V2-045223 | MATHDB-367050

Combinatorial Ricci flow with surgery convergence conjecture

摘录 / EXCERPT: Let M be an oriented compact 3-manifold with boundary, no component of which is a 2-sphere. Suppose that M is irreducible, atoroidal and not Seifert fibred. A combinatorial...

conjecture | open | MathDB 页面 | 原文线索

V2-045224 | MATHDB-367083

The fixed-edge extremal conjecture for t-intersecting trees

摘录 / EXCERPT: Fixed-edge extremal conjecture. If $t\le n/2$, then the largest t-intersecting family of spanning trees is

conjecture | open | MathDB 页面 | 原文线索

V2-045225 | MATHDB-367084

The complete-intersection conjecture for t-intersecting spanning trees

摘录 / EXCERPT: Complete-intersection conjecture. For any t and n, there exists a j such that $\mathcal{F}_{n,t,j}$ is the largest t-intersecting set of spanning trees.

conjecture | open | MathDB 页面 | 原文线索

V2-045226 | MATHDB-367093

Häggkvist's avoidability conjecture for (m,m,m)-arrays

摘录 / EXCERPT: An (m,m,m)-array is an $n\times n$ array of sets of size at most m, such that each number in $[n]$ occurs at most m times among the sets in any row or column. It is avoidable...

conjecture | open | MathDB 页面 | 原文线索

V2-045227 | MATHDB-367094

Campbell's tightening conjecture for minimal thick matroids

摘录 / EXCERPT: Let t be a positive integer, and let a matroid be $(t+1)$-thick when it is not the union of t hyperplanes. A tightening is the matroid operation described in the source, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-045228 | MATHDB-367097

Positivity conjecture for permanents of polystochastic matrices

摘录 / EXCERPT: A d-dimensional polystochastic matrix of order n is a nonnegative array whose sums over every line are equal to 1. Its permanent is ... where $D(A)$ is the set of diagonals of...

conjecture | open | MathDB 页面 | 原文线索

V2-045229 | MATHDB-367098

Minimal-family characterization of the ideals \mathcal{I}_{h,n}

摘录 / EXCERPT: Let $\mathcal{I}_{h,n}$ be a family of polynomial ideals in the variables $X_1,\ldots,X_n$, and let $e_i(\mathbf{X}_n)$ denote the elementary symmetric polynomials. The family is r...

conjecture | open | MathDB 页面 | 原文线索

V2-045230 | MATHDB-367108

The Scarf resolution conjecture for powers of extremal ideals

摘录 / EXCERPT: For an integer $q\geq 1$, let $\mathcal{S}_q=\mathsf{k}[x_A\colon \emptyset\neq A\subseteq [q]]$, and let $\mathcal{E}_q$ be the ideal generated by ... For $r\geq 1$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-045231 | MATHDB-367111

The polynomial formula for asymptotic ratios of Bernoulli partitions

摘录 / EXCERPT: Let $b_m(n)$ denote the Bernoulli partition entries described in the preceding discussion, and let B_{2m} be the Bernoulli numbers. For each column index $n\geq 2$, define the as...

conjecture | open | MathDB 页面 | 原文线索

V2-045232 | MATHDB-367129

Topological Erdős similarity conjecture for uncountable sets

摘录 / EXCERPT: Let P be a set in the relevant Euclidean space, and let A be a Baire set. An affine image of P is a set obtained from P by an affine transformation. Topological Erdős simil...

conjecture | open | MathDB 页面 | 原文线索

V2-045233 | MATHDB-367133

Jackson-Owen lower-bound conjecture for realisation numbers of minimally 2-rigid graphs

摘录 / EXCERPT: Jackson-Owen conjecture. Every minimally 2-rigid graph G with n vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045234 | MATHDB-367149

The high-degree double Hall cycle conjecture

摘录 / EXCERPT: High-degree double Hall cycle conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045235 | MATHDB-367166

Fixed-perimeter tipping-point conjecture for two residue classes

摘录 / EXCERPT: For positive integers d, define $h_d^{(a)}(n)$ as the number of partitions of n into parts that are d-distinct and at least a, and define $\ell_m^{(m_1,m_2)}(n)$ as the num...

conjecture | open | MathDB 页面 | 原文线索

V2-045236 | MATHDB-367167

Chen-Hattingh-Rousseau asymptotic conjecture for mixed Ramsey numbers

摘录 / EXCERPT: Let $t(m,n)$ denote the mixed Ramsey number and $r(m,n)$ the classical Ramsey number. For each fixed integer $m\geq 4$, Chen-Hattingh-Rousseau's conjecture. ... This conjecture predi...

conjecture | open | MathDB 页面 | 原文线索

V2-045237 | MATHDB-367173

Ryser's maximum permanent conjecture for regular (0,1)-matrices

摘录 / EXCERPT: Ryser's conjecture. The permanent of J^* is maximal in $\mathscr{U}(v,k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-045238 | MATHDB-367179

Infinitely many non-power lengths with universal invertibility of aperiodic necklaces

摘录 / EXCERPT: Necklace invertibility conjecture. There exist infinitely many n, different from a power of p, for which every aperiodic necklace of length n over Σ_p with non-zero we...

conjecture | open | MathDB 页面 | 原文线索

V2-045239 | MATHDB-367208

Kahn's equal-size upward-closed systems conjecture

摘录 / EXCERPT: Kahn's conjecture. If $|X|=|Y|=|Z|$, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045240 | MATHDB-367229

Wrochna's index conjecture for products of \(\mathbb{Z}_2\)-spaces

摘录 / EXCERPT: Let X and Y be $\mathbb{Z}_2$-spaces, with product equipped with the diagonal $\mathbb{Z}_2$-action, and let $\operatorname{ind}(X)$ denote the index of a $\mathbb{Z}_2$-space....

conjecture | open | MathDB 页面 | 原文线索

V2-045241 | MATHDB-367230

Wrochna's graph formulation of the index conjecture

摘录 / EXCERPT: Let G and H be graphs, let $\Omega_{2d+1}(G)$ denote the graph functor used in the source, and let n be a positive integer. Wrochna's graph-index conjecture. If ... then for so...

conjecture | open | MathDB 页面 | 原文线索

V2-045242 | MATHDB-367294

Tutte-polynomial inequality for simple co-simple matroids

摘录 / EXCERPT: Let M be a matroid. A matroid is simple if it has no loops or parallel elements, and co-simple if its dual has no loops or parallel elements. Simple co-simple matroid conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-045243 | MATHDB-367306

The direct-sum extremality conjecture for matroids

摘录 / EXCERPT: Direct-sum extremality conjecture. If $\mathsf{M}_1,\ldots,\mathsf{M}_s$ are extremal matroids, then $\mathsf{M}_1\oplus\cdots\oplus\mathsf{M}_s$ is extremal.

conjecture | open | MathDB 页面 | 原文线索

V2-045244 | MATHDB-367307

The complete-graph graphic-matroid extremality conjecture

摘录 / EXCERPT: Complete-graph graphic-matroid conjecture. For every $n\geq 1$, the graphic matroid K_n of the complete graph on n vertices is an extremal matroid.

conjecture | open | MathDB 页面 | 原文线索

V2-045245 | MATHDB-367308

The extremality conjectures for projective and Dowling geometries

摘录 / EXCERPT: Geometry extremality conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-045246 | MATHDB-367335

Maximal total persistence conjecture for extremal flag-complex filtrations

摘录 / EXCERPT: Total-persistence conjecture. The extremal filtrations described in Section 4 achieve the maximal total persistence over any edgewise filtration of flag complexes in homology degre...

conjecture | open | MathDB 页面 | 原文线索

V2-045247 | MATHDB-367339

Generic extremals avoid the strict-subgroup union in the free step-2 rank-4 group

摘录 / EXCERPT: Let $\mathbb F_4$ be the free step-2 rank-four Carnot group, and let $\mathcal H$ be the union of all strict Carnot subgroups. Equivalently, ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045248 | MATHDB-367358

Critical exponent conjecture for the generalized Thue-Morse words

摘录 / EXCERPT: For each integer $k\geq 1$, let ${\bf x}_k$ be the infinite word defined as the limit of the locally catenative sequence described above. The critical exponent conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-045249 | MATHDB-367359

Subword-complexity difference conjecture for the generalized Thue-Morse words

摘录 / EXCERPT: For each integer $k\geq 1$, let ${\bf x}_k$ be the infinite word defined as the limit of the locally catenative sequence described above, and let $\rho_{{\bf x}_k}(n)$ denote its s...

conjecture | open | MathDB 页面 | 原文线索

V2-045250 | MATHDB-367360

Caro's zero-sum Ramsey conjecture for trees modulo 3

摘录 / EXCERPT: Let T be a tree on $n \equiv 1 \pmod{3}$ vertices. Caro's conjecture. ... This conjecture extends the understanding of zero-sum Ramsey numbers from $\mathbb{Z}_2$ to the next cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-045251 | MATHDB-367387

Nonnegative SEM expansion conjecture for Schubert polynomials

摘录 / EXCERPT: Let w be a permutation, let $\mathfrak{S}_w$ denote its Schubert polynomial, and consider its expansion in the SEM basis. A polynomial is a single SEM if it is one basis element...

conjecture | open | MathDB 页面 | 原文线索

V2-045252 | MATHDB-367412

Baber-Johnson-Talbot triangle-density conjecture for tripartite graphs

摘录 / EXCERPT: Let $R=\{(\alpha,\beta,\gamma)\in[0,1]^3:\alpha\beta+\gamma>1,\alpha\gamma+\beta>1,\beta\gamma+\alpha>1\}$, and let ... Define...

conjecture | open | MathDB 页面 | 原文线索

V2-045253 | MATHDB-367430

Mean-regime conjecture for the maximum and extremal process of two-speed branching random walk

摘录 / EXCERPT: Mean-regime conjecture. There exists a constant $C_{\mathrm{m}}>0$ such that, for every $y\in\mathbb{R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045254 | MATHDB-367432

Conjecture on the balanced genus of products of spheres with the circle

摘录 / EXCERPT: Let $d\geq 3$, and let $\mathcal{G}_M$ denote the balanced genus of a PL d-manifold M. Consider the product manifold $\mathbb{S}^{d-1}\times\mathbb{S}^1$. Balanced-genus conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-045255 | MATHDB-367442

Reading's separation conjecture for cluster variables

摘录 / EXCERPT: Reading's separation conjecture. The variables u and u' are contained in the same cluster if and only if they are sign-coherent.

conjecture | open | MathDB 页面 | 原文线索

V2-045256 | MATHDB-367449

Existence conjecture for near triple arrays

摘录 / EXCERPT: Existence conjecture. $(r \times c, v)$-near triple arrays exist for any $c \geq r(r - 1)$ and $v \geq c$.

conjecture | open | MathDB 页面 | 原文线索

V2-045257 | MATHDB-367479

Fibonacci enumeration conjecture for simple permutations sortable by two parallel pop stacks

摘录 / EXCERPT: Fibonacci enumeration conjecture. The sequence (a_n) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045258 | MATHDB-367490

Lev's matching restricted-sumset conjecture

摘录 / EXCERPT: Lev's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045259 | MATHDB-367491

The integer bounded-degree restricted-sumset conjecture

摘录 / EXCERPT: Bounded-degree restricted-sumset conjecture. If the maximum degree of $\mathcal{R}$ on B is at most Δ, then

conjecture | open | MathDB 页面 | 原文线索

V2-045260 | MATHDB-367492

The small-set function restricted-sumset conjecture

摘录 / EXCERPT: Small-set restricted-sumset conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045261 | MATHDB-367510

The SPR-neighbor OLA distance bound for phylogenetic trees

摘录 / EXCERPT: Let $T\in\widehat{\mathcal{RBT}}_n$ be a fixed phylogenetic tree, and let T' be a random tree differing from T by a single subtree-prune-and-regraft (SPR) move, chosen uniforml...

conjecture | open | MathDB 页面 | 原文线索

V2-045262 | MATHDB-367522

NP-hardness of repeated-round Dealer's choice pile shuffle sort

摘录 / EXCERPT: NP-hardness conjecture. The repeated-round Dealer's choice pile shuffle sort problem is NP-hard.

conjecture | open | MathDB 页面 | 原文线索

V2-045263 | MATHDB-367544

The q-analog of Rado's theorem for independent q-transversals

摘录 / EXCERPT: Let M be a q-matroid on a vector space V, with bar nullity function $\overline{n}$, and let $\mathcal X$ be a system of subspaces of V. A q-transversal is a subspace obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-045264 | MATHDB-367545

Uniqueness of minimal presentations of q-transversal matroids

摘录 / EXCERPT: Let M be a q-transversal matroid with a presentation by a system of subspaces. A presentation is minimal if no member can be replaced by a proper subspace while preserving the...

conjecture | open | MathDB 页面 | 原文线索

V2-045265 | MATHDB-367547

The linear-growth conjecture for complete posets

摘录 / EXCERPT: Let $K_{n_1,\dots,n_k}$ denote the complete poset with consecutive layers of sizes $n_1,\dots,n_k$, where $n_1,\dots,n_k$ are positive integers and at least one n_i is not equal...

conjecture | open | MathDB 页面 | 原文线索

V2-045266 | MATHDB-367551

Kernel equality conjecture for the extremal noncommutative power-sum bound

摘录 / EXCERPT: Kernel equality conjecture. For every such tuple,

conjecture | open | MathDB 页面 | 原文线索

V2-045267 | MATHDB-367578

Unique maximality conjecture for matroids avoiding dual symmetric-tensor circuits

摘录 / EXCERPT: Unique maximality conjecture. There is a unique maximal ${\cal X}$-matroid on K_n.

conjecture | open | MathDB 页面 | 原文线索

V2-045268 | MATHDB-367580

Marković's conjecture on unique extremality for well-distributed sets

摘录 / EXCERPT: Let $E\subset\mathbb{C}$ be a discrete set. Call E well distributed if there exists $c>0$ such that E intersects every disc of Euclidean radius c in the complex plane. Markov...

conjecture | open | MathDB 页面 | 原文线索

V2-045269 | MATHDB-367599

Exponential decay of distinct quantum-invariant values

摘录 / EXCERPT: Exponential-decay conjecture. For $Q\in\{A2,A,B1,K\}$, and therefore also for $Q=J$ or $Q=KT1$,

conjecture | open | MathDB 页面 | 原文线索

V2-045270 | MATHDB-367600

Exponential growth of quantum-invariant distinguishing pairs

摘录 / EXCERPT: Distinguishing-pair growth conjecture. For $Q\in\{A2,A,B1,J,KT1\}$, and therefore also for $Q=K$,

conjecture | open | MathDB 页面 | 原文线索

V2-045271 | MATHDB-367601

Exponential growth of coefficients of quantum invariants

摘录 / EXCERPT: Exponential-growth conjecture. For $Q\in\{A2,A,B1,Kh\}$, and therefore also for $Q=J$ or $Q=KT1$,

conjecture | open | MathDB 页面 | 原文线索

V2-045272 | MATHDB-367609

Higher-degree ULC failure conjecture for the polynomials f_m and h_m

摘录 / EXCERPT: For each integer $m\geq 4$, let f_m and h_m be the polynomials defined in the paper, and let the ULC inequality refer to the ultra-log-concavity inequality for their coefficien...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045273 | MATHDB-367616

Combinatorial expansion conjecture for GS functions

摘录 / EXCERPT: GS expansion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045274 | MATHDB-367647

Sturmian maximizing measures as extremal barycentres

摘录 / EXCERPT: For integers $1\le p<q$, let $F_{0,p,q}$ be the multi-valued dynamical system on S^1 appearing in the paper, and let a Sturmian measure mean a probability measure whose support i...

conjecture | open | MathDB 页面 | 原文线索

V2-045275 | MATHDB-367672

The dismantling characterization of trivial matroid homomorphism reconfiguration

摘录 / EXCERPT: Dismantling characterization. $\operatorname{Recol}_{\mathbb{M}}(N)$ is trivial if and only if N dismantles to the loop $M^\ell(K_1)$ or the edge $M(K_2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-045276 | MATHDB-367681

Ehrhart formula conjecture for panhandle matroids

摘录 / EXCERPT: Let a,b be positive integers. Let ${\texttt{Pan}}_{b+1,a+b,a+b+2}$ be the panhandle matroid, let $i(M,t)$ denote the Ehrhart polynomial of the matroid polytope of M, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045277 | MATHDB-367700

The quadratic-logarithmic lower-bound conjecture for pebble motion on trees

摘录 / EXCERPT: Quadratic-logarithmic lower-bound conjecture. There exist instances of the textsc{Pebble Motion Problem on Trees} for which the length of the shortest solution sequences is

conjecture | open | MathDB 页面 | 原文线索

V2-045278 | MATHDB-367708

Conjecture on maximizers of the partition discrepancy

摘录 / EXCERPT: Maximizer conjecture. At most one part of λ is the sum of a part in μ and a part in ν, while the other parts of λ are distributed in μ and ν.

conjecture | open | MathDB 页面 | 原文线索

V2-045279 | MATHDB-367709

Combinatorial Foulkes conjecture for transpose-symmetric matrices

摘录 / EXCERPT: Let $M(n,m)$ be the set of $n\times m$ nonnegative integer matrices with every row and column sum equal to m and n, respectively, as in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045280 | MATHDB-367726

Bukh-Borisov deletion-distance conjecture for binary words

摘录 / EXCERPT: Bukh-Borisov deletion-distance conjecture. There is a constant c such that, for every n,

conjecture | open | MathDB 页面 | 原文线索

V2-045281 | MATHDB-367727

Two-cut conjecture for even binary words

摘录 / EXCERPT: Two-cut conjecture. Every even binary word W satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045282 | MATHDB-367728

Cutting-distance conjecture for even k-ary words

摘录 / EXCERPT: Cutting-distance conjecture. For each $k\geqslant 2$, every even k-ary word W satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045283 | MATHDB-367730

Erdős's lower-bound conjecture for off-diagonal Ramsey numbers

摘录 / EXCERPT: Let $r(s,t)$ denote the minimum integer n such that every red-blue coloring of the edges of the complete graph on n vertices contains a red copy of K_s or a blue copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-045284 | MATHDB-367731

The asymptotic order conjecture for the Ramsey number r(4,t)

摘录 / EXCERPT: Let $r(4,t)$ denote the minimum integer n such that every red-blue coloring of the edges of the complete graph on n vertices contains a red copy of K_4 or a blue copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-045285 | MATHDB-367732

Erdős's cycle-complete Ramsey conjecture for C4

摘录 / EXCERPT: Let $r(C_4,t)$ denote the minimum integer n such that every C_4-free graph on n vertices contains an independent set of size t. Erdős's C_4 Ramsey conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-045286 | MATHDB-367757

Asymptotic conjecture for equal-block mountain-valley assignments

摘录 / EXCERPT: Let $S(m,k)$ denote the MV assignment with m blocks, each of size k, and let $c(S(m,k))$ be its number of foldings. Equal-block asymptotic conjecture. For any fixed m and suf...

conjecture | open | MathDB 页面 | 原文线索

V2-045287 | MATHDB-367758

Structure conjecture for maximally foldable stamp assignments

摘录 / EXCERPT: Consider mountain-valley assignments for a $1\times n$ strip of stamps, and call an assignment maximally foldable when it has the greatest number of foldings among assignments for...

conjecture | open | MathDB 页面 | 原文线索

V2-045288 | MATHDB-367759

Asymptotic conjecture for the maximal number of stamp foldings

摘录 / EXCERPT: For a $1\times n$ strip of stamps, let $M(n)$ be the maximal number of ways to fold the strip over all mountain-valley assignments. Maximal-folding asymptotic conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-045289 | MATHDB-367770

Erdős's reverse Littlewood-Offord conjecture

摘录 / EXCERPT: Let $x_1,\dotsc,x_n$ be unit complex numbers, and let $\varepsilon_1,\dotsc,\varepsilon_n\in\{-1,+1\}$. Erdős's conjecture. The number of sums ... with ... is greater than $c2^n/n$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-045290 | MATHDB-367771

He-Juškevičius-Narayanan-Spiro conjecture for odd planar Rademacher sums

摘录 / EXCERPT: Let $v_1,\dotsc,v_n\in\mathbb{R}^2$ be unit vectors, and let $\varepsilon_1,\dotsc,\varepsilon_n$ be independent random signs, each uniformly distributed in $\{-1,1\}$. He-Juškevič...

conjecture | open | MathDB 页面 | 原文线索

V2-045291 | MATHDB-367783

Realizability of circular extensions for connected DC graphs

摘录 / EXCERPT: Fix N and let $G\in\operatorname{DC}_N$ be a connected DC graph. Let Z'_G be the partial cyclic order obtained by requiring that, for every maximal edge (i,j) of G, the tup...

conjecture | open | MathDB 页面 | 原文线索

V2-045292 | MATHDB-367787

Hypergeometric rationality conjecture for unicellular Ising maps with many monochromatic edges

摘录 / EXCERPT: Many-monochromatic-edges conjecture. For fixed $i,j\geq 0$, there exists a rational function $R_{i,j}(g)$ in g such that

conjecture | open | MathDB 页面 | 原文线索

V2-045293 | MATHDB-367788

Hypergeometric rationality conjecture for unicellular Ising maps with few monochromatic edges

摘录 / EXCERPT: Few-monochromatic-edges conjecture. For fixed $i,j\geq 0$, there exists a rational function $Q_{i,j}(g)$ in g such that

conjecture | open | MathDB 页面 | 原文线索

V2-045294 | MATHDB-367797

The asymptotic enumeration conjecture for clique factors in sparse random graphs

摘录 / EXCERPT: Let $r\geqslant 3$ be constant, let $p\leqslant n^{-2/r+o(1)}$, and let $m=\big\lfloor\binom{n}{r}p^{\binom{r}{2}}\big\rfloor$. Write $F_r(G(n,p))$ for the number of K_r-factors...

conjecture | open | MathDB 页面 | 原文线索

V2-045295 | MATHDB-367848

Pak and Robichaux's coNP-hardness conjecture for Schubert vanishing

摘录 / EXCERPT: Let $\textup{SchubertVanishing}$ denote the decision problem of determining whether a Schubert coefficient vanishes. Pak and Robichaux's conjecture. The problem...

conjecture | open | MathDB 页面 | 原文线索

V2-045296 | MATHDB-367849

The exact growth conjecture for the three-container pouring function

摘录 / EXCERPT: Exact growth conjecture. For each positive integer N, one has

conjecture | open | MathDB 页面 | 原文线索

V2-045297 | MATHDB-367907

Chvátal's conjecture on subset-closed families

摘录 / EXCERPT: Chvátal's conjecture. Every subset-closed family of sets is ${\sf EKR}$.

conjecture | open | MathDB 页面 | 原文线索

V2-045298 | MATHDB-367922

Uniformization conjecture for combinatorial Ricci flow

摘录 / EXCERPT: Uniformization of the CRF. The following statements are equivalent: the combinatorial Ricci flow on $\mathcal{T}$ converges for every initial value in hyperbolic background geometr...

conjecture | open | MathDB 页面 | 原文线索

V2-045299 | MATHDB-367923

Ideal circle-pattern uniformization conjecture for combinatorial Ricci flow

摘录 / EXCERPT: Ideal circle-pattern uniformization conjecture. The following statements are equivalent: the combinatorial Ricci flow on $\mathcal{T}$ converges for every initial value in hyperbol...

conjecture | open | MathDB 页面 | 原文线索

V2-045300 | MATHDB-367927

Classification conjecture for full-dimensional thick te-interlaces

摘录 / EXCERPT: Classification conjecture. There are only two full-dimensional thick te-interlaces, up to resigning or permuting rows and columns:

conjecture | open | MathDB 页面 | 原文线索

V2-045301 | MATHDB-367940

Second-order Eulerian formula conjecture for flattened Fubini rankings

摘录 / EXCERPT: Let FR_n denote the set of Fubini rankings on n competitors, and let $\mathrm{flat_runs}_k(\mathrm{FR}_n)$ denote the flattened Fubini rankings with k runs of ascen...

conjecture | open | MathDB 页面 | 原文线索

V2-045302 | MATHDB-367971

The positive-density k-fold sumset conjecture for abelian groups

摘录 / EXCERPT: Let $k\in\mathbb{N}$. Let Γ be a countably infinite abelian group, and suppose ... For $A\subseteq\Gamma$, write $d^*(A)$ for its upper Banach density, and for a set...

conjecture | open | MathDB 页面 | 原文线索

V2-045303 | MATHDB-367995

Bhanja-Kom-Pandey's restricted signed sumset conjecture for sets containing zero

摘录 / EXCERPT: Bhanja-Kom-Pandey's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045304 | MATHDB-367998

Bhanja-Kom-Pandey lower-bound and inverse conjecture for positive restricted signed sumsets

摘录 / EXCERPT: Let A be a set of $k\geq 4$ positive integers, and let h be an integer with $3\leq h\leq k-1$. The restricted signed h-fold sumset is denoted by $h^{\wedge}_{\pm}A$. Bhanja-K...

conjecture | open | MathDB 页面 | 原文线索

V2-045305 | MATHDB-368022

Stromquist's monochromatic powers-of-two arcs conjecture

摘录 / EXCERPT: Let $k\geq 3$. Consider 2^k-1 points evenly spaced on a circle, and colour each point arbitrarily blue or red. A set of points is said to have arcs proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-045306 | MATHDB-368023

Altman-Gaujal-Hordijk conjecture on densities of balanced sequences

摘录 / EXCERPT: Let $k\geq 3$, and let $\{r_1,\dots,r_k\}$ be a set of distinct positive reals with $r_1>\dots>r_k>0$. A sequence over a k-element set is balanced if, for every symbol, the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-045307 | MATHDB-368024

The classification conjecture for nearly-Ramsey triples

摘录 / EXCERPT: Let a triple (d_1,d_2,d_3) be nearly-Ramsey if, for every $\varepsilon>0$, every two-colouring of the circle contains a monochromatic ε-close copy of the triple. Clas...

conjecture | open | MathDB 页面 | 原文线索

V2-045308 | MATHDB-368048

Dominance threshold conjecture for random homogeneous ideals in three variables

摘录 / EXCERPT: Dominance threshold conjecture. As $d\to\infty$, for every fixed $p>0.1$, the probability that the random homogeneous ideal is dominant tends to 0.

conjecture | open | MathDB 页面 | 原文线索

V2-045309 | MATHDB-368061

Cieliebak-Mohnke extremal Lagrangian torus conjecture for the unit ball

摘录 / EXCERPT: Let an extremal Lagrangian torus in $(\bar{B}^{2n}(1),\omega_{\mathrm{std}})$ mean a Lagrangian torus whose symplectic energy attains the relevant Lagrangian capacity. Cieliebak-Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-045310 | MATHDB-368062

Extremal Lagrangian torus conjecture for convex toric domains

摘录 / EXCERPT: Let $X^{2n}_{\Omega}\subset \mathbb{C}^n$ be a convex toric domain, with standard symplectic form ω_{std}. Let $\operatorname{diagonal}(X^{2n}_{\Omega})$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-045311 | MATHDB-368063

Extremal Lagrangian torus conjecture for ellipsoids

摘录 / EXCERPT: Let $0<a_1\leq a_2\leq\dots\leq a_n<\infty$, and let $E^{2n}(a_1,a_2,\dots,a_n)$ be the corresponding ellipsoid with standard symplectic form. Ellipsoid extremal-torus conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-045312 | MATHDB-368069

Stump's interval-count conjecture for the m-Tamari and type A m-Cambrian lattices

摘录 / EXCERPT: The m-Tamari lattice and the linear type A m-Cambrian lattice are lattices associated with the same parameter m. Stump's interval-count conjecture. The m-Tamari lattice a...

conjecture | open | MathDB 页面 | 原文线索

V2-045313 | MATHDB-368085

The sublinear cop conjecture for one-dimensional Rugby

摘录 / EXCERPT: Sublinear cop conjecture. In the one-dimensional game of Rugby, for all speeds of the robber, $o(n)$ cops suffice to win.

conjecture | open | MathDB 页面 | 原文线索

V2-045314 | MATHDB-368086

The near-maximal cop-avoidance conjecture for Rugby on the d-dimensional grid

摘录 / EXCERPT: Near-maximal cop-avoidance conjecture. For all $\epsilon > 0$, there exists a speed s such that a robber of speed s can beat $n^{d-\epsilon}$ cops for sufficiently large n.

conjecture | open | MathDB 页面 | 原文线索

V2-045315 | MATHDB-368090

Asymptotic optimizer sequence for the lower bound in $g(n,k,\ell)$

摘录 / EXCERPT: For each triple (n,k,ℓ), let the right-hand side of be the lower-bound expression from Theorem, optimized over the parameter j. Asymptotic optimizer conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-045316 | MATHDB-368125

Fast convergence rate conjecture for guessing numbers

摘录 / EXCERPT: Fast convergence rate conjecture. The convergence satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045317 | MATHDB-368130

Hodge-Riemann conjecture for the intersection algebra of a line arrangement

摘录 / EXCERPT: Let V be an n-dimensional vector space, and let E be a finite set of lines in V^* that is not contained in a single hyperplane. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045318 | MATHDB-368146

Equality of degrees conjecture for Kazhdan-Lusztig and inverse Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let M be a matroid, and let $P_M(t)$ and $Q_M(t)$ denote its Kazhdan-Lusztig and inverse Kazhdan-Lusztig polynomials, respectively. Degree equality conjecture. For any matroid...

conjecture | open | MathDB 页面 | 原文线索

V2-045319 | MATHDB-368207

The density-threshold conjecture for global synchrony

摘录 / EXCERPT: A graph is globally synchronizing when its adjacency matrix has only synchronized global minima of the Kuramoto energy. Density-threshold conjecture. For every $\varepsilon>0$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-045320 | MATHDB-368210

The infinite-family lower-bound conjecture for Spencer discrepancy

摘录 / EXCERPT: Spencer discrepancy lower-bound conjecture. The infinite-family normalized discrepancy satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045321 | MATHDB-368218

Nonvanishing conjecture for order-complex characters of poset companions

摘录 / EXCERPT: Let P be a poset, let $\omega\in\mathrm{Rev}(P)$, and let $\mathrm{Comp}(P,\omega)$ denote the associated companion whose character is $\chi_{\mathrm{Comp}(P,\omega)}$. Nonvanish...

conjecture | open | MathDB 页面 | 原文线索

V2-045322 | MATHDB-368219

Automorphism count from companion multiplicities

摘录 / EXCERPT: Let P be a poset with n elements, and let $\mathcal{A}_1,\ldots,\mathcal{A}_k$ be the mutually different collections of type $\mathrm{Comp}(P,\omega)$ for $\omega\in\Sigma_P$....

conjecture | open | MathDB 页面 | 原文线索

V2-045323 | MATHDB-368224

Ardila-Billey conjecture on spread-out simplices

摘录 / EXCERPT: Let Δ_{d-1} be the standard $(d-1)$-simplex, and let $n\Delta_{d-1}$ denote its dilation by n. A unit simplex is a simplex of normalized volume one in $n\Delta_{d-1}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045324 | MATHDB-368225

Acyclic system conjecture for mixed subdivisions

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and $[d]=\{1,\ldots,d\}$. A system of permutations records the permutations obtained from the boundary data of a mixed subdivision of $n\Delta_{d-1}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045325 | MATHDB-368226

Acyclic system conjecture via matching ensembles

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and $[d]=\{1,\ldots,d\}$. A k-by-k matching is a matching between a k-subset of $[n]$ and a k-subset of $[d]$, and a matching ensemble is a collect...

conjecture | open | MathDB 页面 | 原文线索

V2-045326 | MATHDB-368227

Acyclic system conjecture on extendability of boundary subdivisions of nΔ₃

摘录 / EXCERPT: Let $n\Delta_3$ be the dilation by n of the standard 3-simplex. A fine mixed subdivision structure on the boundary facets is boundary data arising from a fine mixed subdivision...

conjecture | open | MathDB 页面 | 原文线索

V2-045327 | MATHDB-368228

Zabrocki's nonnegativity conjecture for set-partition sums

摘录 / EXCERPT: For a positive integer m, let $[m]=\{1,\dots,m\}$, and let σ and τ be set partitions of $[m]$, with $\tau\ge\sigma$ meaning that every block of σ is contained...

conjecture | open | MathDB 页面 | 原文线索

V2-045328 | MATHDB-368250

Large-girth minor conjecture for arbitrary cosimple matroids

摘录 / EXCERPT: Let $U_{2,q}$ be the q-element rank-2 uniform matroid, and more generally let $U_{t,q}$ be the q-element rank-t uniform matroid. Let $M(K_t)$ be the graphic matroid of the...

conjecture | open | MathDB 页面 | 原文线索

V2-045329 | MATHDB-368251

Finite unavoidable-minor conjecture for large-girth matroids

摘录 / EXCERPT: Let $B(G)$ denote the bicircular matroid of a graph G, let $U_{t,t+2}$ be the uniform matroid of rank t on $t+2$ elements, and let $M(K_t)$ and $M(K_t)^*$ be respectively the g...

conjecture | open | MathDB 页面 | 原文线索

V2-045330 | MATHDB-368256

Ilinca-Kahn's precise asymptotics conjecture for maximal independent sets in B(n,k)

摘录 / EXCERPT: Ilinca-Kahn's conjecture. The precise asymptotics satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-045331 | MATHDB-368277

Generic first-order rigidity conjecture for combinatorial types of convex polytopes

摘录 / EXCERPT: Let $\mathcal P$ be the combinatorial type of a convex polytope of dimension $d\geq 3$. Generic first-order rigidity conjecture. If $\mathcal P$ is the combinatorial type of a conv...

conjecture | open | MathDB 页面 | 原文线索

V2-045332 | MATHDB-368286

Archer et al.'s enumeration conjectures for cyclic permutation pattern avoidance

摘录 / EXCERPT: Archer et al.'s conjectures. For the indicated values of n, the following enumerations hold:

conjecture | open | MathDB 页面 | 原文线索

V2-045333 | MATHDB-368293

Sun's q-log-convexity conjecture for binomial sums and the polynomials S_n^{(m)}

摘录 / EXCERPT: The q-log-convexity conjecture. (i) The sequence $(\sum_{k=0}^n\binom nk^3q^k)_{n\geq8}$ is q-log-convex; for every integer $m\geq2$, the sequence...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045334 | MATHDB-368309

Conjecture on extremal sunflower-free families of vector spaces

摘录 / EXCERPT: Extremal vector-space sunflower conjecture. There exists a constant $C=C(q,s)$ such that every s-sunflower-free family ${\mathcal F}$ of k-spaces over the field with q elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-045335 | MATHDB-368315

Typical-orbit conjecture for interval-closed sets of products of two chains

摘录 / EXCERPT: Fix a positive integer m. Let $r(n)$ be the fraction of interval-closed sets of $[m]\times[n]$ that belong to rowmotion orbits of size $m+n+1$. Typical-orbit conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-045336 | MATHDB-368318

Independent mutual-visibility coloring conjecture for subdivided complete graphs

摘录 / EXCERPT: Independent mutual-visibility coloring conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-045337 | MATHDB-368344

Infinite log-convexity conjecture for the Baxter numbers

摘录 / EXCERPT: Let $\{B_n\}_{n\geq 1}$ be the sequence of Baxter numbers. The Baxter-number infinite log-convexity conjecture. The sequence ... is infinitely log-convex. The paper proves eventual...

conjecture | open | MathDB 页面 | 原文线索

V2-045338 | MATHDB-368345

Infinite log-convexity conjecture for $(1,1)$-Hoggatt sums

摘录 / EXCERPT: For a fixed integer $m\geq 2$, let $H_n^{[m]}(1,1)$ denote the $(1,1)$-Hoggatt sums. The $(1,1)$-Hoggatt-sum infinite log-convexity conjecture. For every fixed $m\geq 2$, the seque...

conjecture | open | MathDB 页面 | 原文线索

V2-045339 | MATHDB-368346

Simplex design factorial-size conjecture

摘录 / EXCERPT: Simplex design factorial-size conjecture. There exists a t-design $X\subset \mathbb{R}^d/\left(\sum_{i=1}^d x_i=-1\right)$ for σ, with $t\leq d$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-045340 | MATHDB-368376

Local asymptotic conjecture for generalized abundancy coefficients

摘录 / EXCERPT: Let $cmathcal{H}_{\ell,n}(x)=\sum_{k=0}^{n}A(\ell,n,k)x^k/n!$, and let $\lambda_{\ell}=\ell\ln(\ell-1)-\ln(\mathcal{K}_{\ell})$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-045341 | MATHDB-368381

Colbourn's conjecture on realizations with three largest equal subsquares

摘录 / EXCERPT: Let $(h_1h_2\dots h_k)$ be a partition with $h_1\geq h_2\geq\dots\geq h_k>0$, and let an RP be a realization of a partition by disjoint subsquares of a Latin square. Colbourn's con...

conjecture | open | MathDB 页面 | 原文线索

V2-045342 | MATHDB-368382

Colbourn's conjecture on realizations under a largest-part bound

摘录 / EXCERPT: Let $(h_1\dots h_k)$ be a partition with $h_1\geq\dots\geq h_k>0$, and let an RP be a realization of a partition by disjoint subsquares of a Latin square. Colbourn's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-045343 | MATHDB-368385

Freiman-Lev conjecture on restricted double sumsets

摘录 / EXCERPT: Freiman-Lev conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-045344 | MATHDB-368386

Mohan-Pandey conjecture on near-minimal restricted h-fold sumsets

摘录 / EXCERPT: Mohan-Pandey conjecture. The following implications hold:

conjecture | open | MathDB 页面 | 原文线索

V2-045345 | MATHDB-368449

The conf-connected deletion conjecture

摘录 / EXCERPT: Let $w=(\alpha_1,\dots,\alpha_{2n})$ be a conf-connected word, let a be its minimal letter, and let $I_w(a)$ denote the positions occupied by a. Set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045346 | MATHDB-368457

The intersection-polynomial determination conjecture

摘录 / EXCERPT: Let w be a word, let $p_w(x)$ be its intersection polynomial, and let $h(w)$ be the dimension of its ℓ-singular-vector space. The intersection-polynomial determination conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045347 | MATHDB-368458

The pivot containment conjecture

摘录 / EXCERPT: The pivot containment conjecture. For every $w\in W_{2n}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045348 | MATHDB-368459

The irreducible-to-steady arc-configuration conjecture

摘录 / EXCERPT: Let W_{2n} be the set of words of length $2n$, let $\mathrm{IConf}(w)$ and $\mathrm{SConf}(w)$ be the irreducible and steady arc configurations of w, and let μ_w record the...

conjecture | open | MathDB 页面 | 原文线索

V2-045349 | MATHDB-368461

Rado-number formula for ${\cal E}(3,0;a,a,b)$

摘录 / EXCERPT: Rado-number formula. Then

conjecture | open | MathDB 页面 | 原文线索

V2-045350 | MATHDB-368483

Asymptotic conjecture for the Ramsey number $R(3,k)$

摘录 / EXCERPT: Asymptotic conjecture for $R(3,k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-045351 | MATHDB-368485

Uniqueness of 2-full words modulo plane symmetries and hook-snake flips

摘录 / EXCERPT: Uniqueness conjecture for 2-full words. The 2-full words $W\in\mathcal{MW}^{\leq 2}_{h\times w}$ are unique up to symmetries of the plane and a local symmetry (a flip) on hook snak...

conjecture | open | MathDB 页面 | 原文线索

V2-045352 | MATHDB-368487

Multicoloured Erdős-Hajnal conjecture for graphs

摘录 / EXCERPT: Multicoloured Erdős-Hajnal conjecture. For all $1\le s_0\le s$ and $k\ge1$, and every s_0-colouring c_0 of K_k, there exist $\delta,C>0$ such that, for all $n\in\mathbb N$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-045353 | MATHDB-368489

Linear-dimension strengthening of the geometric Erdős-Hajnal conjectures

摘录 / EXCERPT: Linear-dimension strengthening. The multicoloured geometric Erdős-Hajnal conjecture and the geometric Erdős-Hajnal conjecture for induced restrictions should both hold with...

conjecture | open | MathDB 页面 | 原文线索

V2-045354 | MATHDB-368490

Linear sumset Ramsey bound in the binary vector space

摘录 / EXCERPT: Linear sumset bound conjecture. There is a constant $C>0$ such that, for all $n\in\mathbb N$ and every $X\subseteq\mathbb F_2^n$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-045355 | MATHDB-368499

Universality of bounded extension realization spaces

摘录 / EXCERPT: For a finite point configuration $P\subset\mathbb{R}^d$ and an integer $k\geq 0$, let $[P]_k$ be the set of point configurations $Q\in(\mathbb{R}^d)^X$ such that every chirotope of...

conjecture | open | MathDB 页面 | 原文线索

V2-045356 | MATHDB-368500

Average number of realizable one-point extensions of planar chirotopes

摘录 / EXCERPT: Let $n\geq 3$, and choose uniformly at random a planar realizable chirotope χ of size n. For each realization of χ, count the realizable one-point extensions, and avera...

conjecture | open | MathDB 页面 | 原文线索

V2-045357 | MATHDB-368501

Day, Falgas-Ravry and Treglown's generalized Turán pattern conjecture

摘录 / EXCERPT: Day, Falgas-Ravry and Treglown's conjecture. For all n sufficiently large,

conjecture | open | MathDB 页面 | 原文线索

V2-045358 | MATHDB-368502

Falgas-Ravry's flat-interval conjecture

摘录 / EXCERPT: Let $\mathrm{T}[(r);a]$ be the generalized Turán pattern with parameter r and ambient edge multiplicity a, and let $\Sigma_s(\mathrm{T}[(r);a])$ denote its associated threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-045359 | MATHDB-368503

The large ambient multiplicity conjecture for the Mubayi-Terry problem

摘录 / EXCERPT: Let $s\in\mathbb{Z}_{\geq 2}$ and let q be an integer with $0\leq q<\binom{s}{2}$. A generalized Turán pattern is a pattern $T=\mathrm{T}[\mathbf{r};a]$ whose blow-ups are...

conjecture | open | MathDB 页面 | 原文线索

V2-045360 | MATHDB-368504

The geometric stability conjecture for the Mubayi-Terry problem

摘录 / EXCERPT: Let $s\geq 2$ and let q be a non-negative integer. An (s,q)-graph is a multigraph satisfying the paper's (s,q)-constraint, and $P(G)$ denotes the product of its edge multipli...

conjecture | open | MathDB 页面 | 原文线索

V2-045361 | MATHDB-368505

The additive conjecture for star-pattern intervals

摘录 / EXCERPT: Let $s\in\mathbb{Z}_{\geq 4}$, and let $G=G^{(a)}$ be any graph pattern in the statement, with ambient edge multiplicity $a=1$. Additive conjecture. For every $k\geq 4$, if ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-045362 | MATHDB-368506

The multiplicative conjecture for star-pattern intervals

摘录 / EXCERPT: Let $s\in\mathbb{Z}_{\geq 4}$, and let $G=G^{(a)}$ be any graph pattern in the statement, with ambient edge multiplicity $a=2$. Multiplicative conjecture. If ... then ... For all...

conjecture | open | MathDB 页面 | 原文线索

V2-045363 | MATHDB-368520

The stronger q-analogue of the reverse oddtown theorem

摘录 / EXCERPT: Let n be an even positive integer and let F_q^n be an n-dimensional vector space over the finite field with q elements. Write $\mathrm{sub}(F_q^n)$ for its subspaces, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-045364 | MATHDB-368590

Uniqueness conjecture for locally minimal spherical graph configurations

摘录 / EXCERPT: For each 3-connected embedded graph $(G,\mathbf p)$ on the unit sphere with topological degree one and with every facial perimeter of length less than 2π, define a partial or...

conjecture | open | MathDB 页面 | 原文线索

V2-045365 | MATHDB-368599

Linear induced saturation conjecture for disjoint unions of chains

摘录 / EXCERPT: Let C_k denote the chain poset with k elements, and let $\bigoplus_{j=1}^m C_{i_j}$ be the disjoint union of chains with $i_1\ge i_2\ge\dots\ge i_m$. For a poset P, write...

conjecture | open | MathDB 页面 | 原文线索

V2-045366 | MATHDB-368607

Kovács-Nagy density conjecture for [k,t]-flats in binary affine spaces

摘录 / EXCERPT: Kovács-Nagy density conjecture. For every $k\ge 1$ and $0\le t\le 2^k$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-045367 | MATHDB-368627

The non-elementary abelian group obstruction to 2-designs

摘录 / EXCERPT: Non-elementary group conjecture. For any (k,x) with $1\le k\le n-1$ and $x\in G$, the incidence structure $(G,\mathcal{B}_k^x)$ is not a 2-design.

conjecture | open | MathDB 页面 | 原文线索

V2-045368 | MATHDB-368634

Alternant divisibility and Schur-positivity conjecture

摘录 / EXCERPT: Alternant divisibility and positivity conjecture. For every $\alpha\in\mathbb{N}^N$, the alternant a_α is a multiple of a_ρ in $\mathcal{P}$. Furthermore, if...

conjecture | open | MathDB 页面 | 原文线索

V2-045369 | MATHDB-368699

Shallit's factor-complexity conjecture for Thue-Morse-like sequences

摘录 / EXCERPT: Let $k\geq 1$, and let $\mathbf{x}_k$ be the Thue-Morse-like sequence obtained from the family described in the source. For a binary sequence, its factor complexity $p(n)$ is the n...

conjecture | open | MathDB 页面 | 原文线索

V2-045370 | MATHDB-368700

Shallit's critical-exponent conjecture for Thue-Morse-like sequences

摘录 / EXCERPT: Let $k\geq 1$, and let $\mathbf{x}_k$ be the Thue-Morse-like sequence described in the source. The critical exponent is the supremum of the exponents of its finite factors, while t...

conjecture | open | MathDB 页面 | 原文线索

V2-045371 | MATHDB-368709

The arithmetic-progression partition heaviness conjecture

摘录 / EXCERPT: For positive integers i,s,k, let $\llbracket i+(k-1)s,\ldots,i+s,i\rrbracket$ denote the partition whose parts form a decreasing arithmetic progression with initial part...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045372 | MATHDB-368738

Balogh-Kemkes-Lee-Young's weighted Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Let $G=(V,E,w)$ be an edge-weighted complete graph on n vertices, where $w:E\to[0,1]$. For $v\in V$, define its weighted degree by $d^w(v):=\sum_{uv\in E}w(uv)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045373 | MATHDB-368740

Even-dimensional Erdős-Falconer distance conjecture

摘录 / EXCERPT: Even-dimensional Erdős-Falconer conjecture. If $|A|\ge Cq^{d/2}$ for a sufficiently large constant C independent of q, then

conjecture | open | MathDB 页面 | 原文线索

V2-045374 | MATHDB-368741

Even-dimensional two-set Erdős-Falconer distance conjecture

摘录 / EXCERPT: Let $d\ge 2$ be an even integer, let $A,B\subseteq\mathbb F_q^d$, and define ... where $\|v\|=\sum_{i=1}^dv_i^2$. Two-set Erdős-Falconer conjecture. If ... for a sufficiently large con...

conjecture | open | MathDB 页面 | 原文线索

V2-045375 | MATHDB-368769

Contraction conjecture for the higher Bruhat to Stasheff-Tamari map

摘录 / EXCERPT: Contraction conjecture. The map f is a contraction of posets for all $n,d\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-045376 | MATHDB-368797

Bras-Amorós's ordinarization-number monotonicity conjecture

摘录 / EXCERPT: Let $n_{g,r}$ denote the number of numerical semigroups of genus g with ordinarization number r, where the ordinarization number is the length of the path from the semigroup to...

conjecture | open | MathDB 页面 | 原文线索

V2-045377 | MATHDB-368808

Buratti-Horak-Rosa conjecture on edge lengths of Hamiltonian paths

摘录 / EXCERPT: Buratti-Horak-Rosa conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045378 | MATHDB-368813

The simple-and-double-rainbow conjecture for $C_{p^nq^m}$

摘录 / EXCERPT: Let p and q be distinct primes and let $n\geqslant m\geqslant2$. Write $\mathfrak{c}(C_{p^nq^m})$ for the complexity of $C_{p^nq^m}$. A simple rainbow and a double rainbow are...

conjecture | open | MathDB 页面 | 原文线索

V2-045379 | MATHDB-368814

The explicit complexity conjecture for $C_{p^nq^m}$

摘录 / EXCERPT: Let p and q be distinct primes, let $n\geqslant m\geqslant2$, and let $\mathfrak{c}(C_{p^nq^m})$ denote the complexity. Let $\operatorname{SR}(n,m)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045380 | MATHDB-368847

Alternating Eulerian-Catalan identity for odd n

摘录 / EXCERPT: Alternating Eulerian-Catalan identity.

conjecture | open | MathDB 页面 | 原文线索

V2-045381 | MATHDB-368851

The low-degree extremal values conjecture for Sendov's refinement

摘录 / EXCERPT: For $n\ge2$, let $r_n(\beta)$ be the maximum of $d(P,\beta)$ over $P\in S(\beta)$ of degree n, and let c_n denote the corresponding sharp quadratic-refinement coefficient in th...

conjecture | open | MathDB 页面 | 原文线索

V2-045382 | MATHDB-368888

The combinatorial conjecture for the refined local Donaldson-Thomas partition function

摘录 / EXCERPT: Let λ be a Young diagram. For each box $\Box\in\mathbb{Z}^2_{\geq 0}$, let $p_{\lambda,\Box}=\prod_{\Box'\in H_\lambda(\Box)}q_{\Box'}$, where $H_\lambda(\Box)$ is the hook...

conjecture | open | MathDB 页面 | 原文线索

V2-045383 | MATHDB-368902

Orlik-Solomon Koszulness conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be a geometric lattice, and let $\mathrm{OS}(\mathcal{L})$ be its Orlik-Solomon algebra. Orlik-Solomon Koszulness conjecture. The algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-045384 | MATHDB-368903

The quasi-isomorphism refinement of the Orlik-Solomon conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be a geometric lattice. Let $\mathrm{OS}(\mathcal{L})$ be its Orlik-Solomon algebra, and let $\mathbb{I}_{\mathcal{L}}$ be the map defined in the source. Quasi-is...

conjecture | open | MathDB 页面 | 原文线索

V2-045385 | MATHDB-368931

Star decomposition conjecture for k-triangulations on the half-cylinder

摘录 / EXCERPT: Let T) be a k-triangulation on the cylinder $\mathcal{C}_n$ that lifts to $\overline{T}$ on the half-cylinder $\overline{\mathcal{C}}_n$. Let $\angle(u,v,w)$ be an angle in...

conjecture | open | MathDB 页面 | 原文线索

V2-045386 | MATHDB-368932

Bijection conjecture for k-triangulations on the half-cylinder

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the half-cylinder and let a k-triangulation be a maximal set of edges with no $(k+1)$-crossing, while an n-periodic k-triangulation on the $2nk$-gon is...

conjecture | open | MathDB 页面 | 原文线索

V2-045387 | MATHDB-368933

Enumeration conjecture for k-triangulations on the half-cylinder

摘录 / EXCERPT: Let T be a k-triangulation on the half-cylinder $\mathcal{C}_n$. Define a k-star as the corresponding local configuration of $k+1$ mutually crossing edges, and a k-relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-045388 | MATHDB-368938

Main Borsuk property conjecture for matroids

摘录 / EXCERPT: Let M be a matroid with at least two bases. It has the Borsuk property when its bases can be partitioned into finitely many parts of diameter smaller than the diameter of the set...

conjecture | open | MathDB 页面 | 原文线索

V2-045389 | MATHDB-368942

The asymptotic count of maximal sum-free subsets in elementary abelian 5-groups

摘录 / EXCERPT: Let $k\in\mathbb N$ and set $n:=5^k$. Write $f_{\max}(G)$ for the number of maximal sum-free subsets of a finite abelian group G. The elementary abelian 5-group conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-045390 | MATHDB-368943

The type I conjecture for maximal distinct sum-free subsets

摘录 / EXCERPT: Let G be a type I abelian group of order n. Write $f^\star_{\max}(G)$ for the number of maximal distinct sum-free subsets of G, and let $\mu(G)$ denote the maximum density...

conjecture | open | MathDB 页面 | 原文线索

V2-045391 | MATHDB-368999

The stabilizer conjecture for the sets W_I' in three-reflection presentations

摘录 / EXCERPT: Stabilizer conjecture. The stabilizer of W_I' is always either

conjecture | open | MathDB 页面 | 原文线索

V2-045392 | MATHDB-369018

The assignment-fixing conjecture for quadratic assignment solvers

摘录 / EXCERPT: Let a quadratic assignment problem (QAP) instance be given, and suppose that a few assignments are identified as belonging to a high-quality solution. Consider permanently fixing t...

conjecture | open | MathDB 页面 | 原文线索

V2-045393 | MATHDB-369019

The base sequence conjecture for BS(n+1,n)

摘录 / EXCERPT: Let $BS(m,n)$ denote four sequences A,B,C,D of 71 of lengths m,m,n,n with zero combined non-periodic autocorrelation, and let $BS(n+1,n)$ be the corresponding case. Base sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-045394 | MATHDB-369027

The concentration conjecture for coefficient-lattice minima of four-element sets

摘录 / EXCERPT: Concentration conjecture. There is an integer n_0 such that, for every $n\ge n_0$, at least $(1-\epsilon)\binom{n}{4}$ of the four-element subsets of $[n]$ have $h_1>h$; among th...

conjecture | open | MathDB 页面 | 原文线索

V2-045395 | MATHDB-369086

Cyclic shift's cubic-coefficient conjecture for deranged strategies

摘录 / EXCERPT: Let $f_{CS}(x)$ be the generating function for the cyclic shift strategy of length n, and let $f_S(x)$ be the generating function for a deranged strategy S, meaning a strategy...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045396 | MATHDB-369104

Hindman's generalized conjecture on finite sums and products

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let a finite coloring of $\mathbb{N}$ be given. Generalized Hindman's conjecture. There exist infinitely many $x_1<\ldots<x_k\in\mathbb{N}$ such that all f...

conjecture | open | MathDB 页面 | 原文线索

V2-045397 | MATHDB-369105

Moreira's conjecture on sums and products in positive-density sets

摘录 / EXCERPT: Let $A\subseteq\mathbb{N}$ have positive density, where its density is ... whenever this limit exists. Moreira's conjecture. There exist $x,y\in\mathbb{N}\backslash\{1\}$ such that .....

conjecture | open | MathDB 页面 | 原文线索

V2-045398 | MATHDB-369175

Sun's integrality conjecture for products of consecutive generalized polynomials

摘录 / EXCERPT: Let ${\mathbb Z}^{+}$ denote the positive integers, and for $n\in{\mathbb Z}^{+}$ define ... where ... Write $w_n(x)=w_n^{(1)}(x)$, let (a,b) denote the greatest common divisor of...

conjecture | open | MathDB 页面 | 原文线索

V2-045399 | MATHDB-369203

Troyka's q-binomial positivity conjecture

摘录 / EXCERPT: Let $a,b,c,d\in\mathbb{N}$ satisfy $ab\leqslant cd$ and $a=\min(a,b,c,d)$. Let $\genfrac{[}{]}{0pt}{}{n}{k}$ denote the Gaussian binomial coefficient and let $\mathbb{N}_0[q]$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-045400 | MATHDB-369220

Asymptotic bound for inversions of unit interval parking functions

摘录 / EXCERPT: Let $n\geq 1$ and $k\in \{0,1,\dots,n(n-1)/2\}$. The quantity $\operatorname{upf}_{n,k}^{\operatorname{inv}}$ denotes the number of unit interval parking functions of length n wi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045401 | MATHDB-369221

Foata transform conjecture for spots of unit interval parking functions

摘录 / EXCERPT: Let α be a unit interval parking function, let F denote the Foata transform, and let $\operatorname{spot}_\alpha$ denote the outcome permutation of α. Foata transfo...

conjecture | open | MathDB 页面 | 原文线索

V2-045402 | MATHDB-369253

The factor-complexity and critical-exponent conjecture for morphic images of G

摘录 / EXCERPT: Let ${\mathbf G}=\gamma^\omega(0)$, where $\gamma=[01,2,02]$, and let μ be its critical exponent. For each morphism h in Table, consider the infinite word $h({\mathbf G})$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-045403 | MATHDB-369254

The factor-complexity and critical-exponent conjecture for reversed morphic images of G

摘录 / EXCERPT: Let ${\mathbf G}=\gamma^\omega(0)$, where $\gamma=[01,2,02]$, let h be a morphism in Table, and let h^R denote the reversed morphism used in that table. Let μ be the critic...

conjecture | open | MathDB 页面 | 原文线索

V2-045404 | MATHDB-369268

Li and Rousseau's asymptotic Ramsey conjecture for complete fans

摘录 / EXCERPT: Li and Rousseau's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045405 | MATHDB-369316

Symmetric exchange conjecture for bounded powers of edge ideals

摘录 / EXCERPT: Symmetric exchange conjecture. The toric ideal

conjecture | open | MathDB 页面 | 原文线索

V2-045406 | MATHDB-369317

Phase diagram conjecture for extremal stationary measures in half-space geometric LPP

摘录 / EXCERPT: Let $\frac12\log q<0$, with $[?]$?

conjecture | open | MathDB 页面 | 原文线索

V2-045407 | MATHDB-369318

The bounded-error conjecture for equitable matroid partitions

摘录 / EXCERPT: Let $\mathcal{M}=(E,\mathcal{I})$ be a matroid whose ground set E can be partitioned into k disjoint bases, and let $S_1,S_2,\ldots,S_\ell$ be pairwise disjoint subsets of E....

conjecture | open | MathDB 页面 | 原文线索

V2-045408 | MATHDB-369319

Gabow's cyclic basis conjecture

摘录 / EXCERPT: Let $\mathcal{M}$ be a matroid and let ... be two bases. Gabow's conjecture. There exist permutations σ and σ' such that in the sequence ... every set of r cyclically c...

conjecture | open | MathDB 页面 | 原文线索

V2-045409 | MATHDB-369320

The Aharoni-Berger-Kotlar-Ziv fair representation conjecture

摘录 / EXCERPT: Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be matroids on the same ground set E, and assume that E can be partitioned into k independent sets in each matroid. For a set...

conjecture | open | MathDB 页面 | 原文线索

V2-045410 | MATHDB-369344

Partition-polytope diameter conjecture near the lower bound

摘录 / EXCERPT: Let n be a positive integer, and let $\kappa_1 \geq \cdots \geq \kappa_n$ be positive integers. The partition polytope $\mathrm{PP}(\kappa_1,\ldots,\kappa_n)$ is the polytope ass...

conjecture | open | MathDB 页面 | 原文线索

V2-045411 | MATHDB-369351

Bandari-Bayati-Herzog conjecture on homological shift ideals

摘录 / EXCERPT: Let $R=\mathbb{k}[x_1,\ldots,x_p]$ be a polynomial ring over a field $\mathbb{k}$, and let $\mathscr{P}$ be a polymatroid on $[p]$ with cage $\mathbf{m}\in\mathbb{N}^p$. Let...

conjecture | solved | MathDB 页面 | 原文线索

V2-045412 | MATHDB-369352

Castillo-Cid-Ruiz-Mohammadi-Montaño conjecture on Möbius supports

摘录 / EXCERPT: Castillo-Cid-Ruiz-Mohammadi-Montaño conjecture. The Möbius support of $\mathscr{P}$ is a generalized polymatroid, that is, a homogenization of it yields a polymatroid. The paper se...

conjecture | solved | MathDB 页面 | 原文线索

V2-045413 | MATHDB-369359

Equitable full-cookie distribution conjecture

摘录 / EXCERPT: Equitable distribution conjecture. It should be possible to cut at most $(r-1)m$ cookies and distribute them so that each child has the same amount of each kind of frosting and at...

conjecture | open | MathDB 页面 | 原文线索

V2-045414 | MATHDB-369378

Lusztig-Dyer Combinatorial Invariance Conjecture

摘录 / EXCERPT: Combinatorial Invariance Conjecture. If two Bruhat intervals, potentially from different Coxeter groups, are isomorphic as posets, then

conjecture | open | MathDB 页面 | 原文线索

V2-045415 | MATHDB-369380

Weak combinatorial invariance conjecture for dominant affine Bruhat intervals

摘录 / EXCERPT: Weak combinatorial invariance conjecture. If $x,x',y,y'\in W_+$ satisfy that $[x,y]$ and $[x',y']$ are full and poset-isomorphic, and there exists $g_0\in G$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-045416 | MATHDB-369381

The limiting codensity formula for products of chains

摘录 / EXCERPT: Let k be a positive integer, and for each $i=1,\ldots,k$ let n_i be a positive integer. Write $[n_i]$ for the corresponding finite chain, let $[n_1] \times \cdots \times [n_k]$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-045417 | MATHDB-369393

Type-monotonicity conjecture for numerical semigroups

摘录 / EXCERPT: Let $n(g,t)$ denote the number of numerical semigroups of genus g and type t. Type-monotonicity conjecture. For each pair of integers satisfying $2\leqslant t\leqslant g$, ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-045418 | MATHDB-369411

Bicategorical classifiers and truncations for partitioned groupoid assemblies

摘录 / EXCERPT: Let A be the partial combinatory algebra underlying the assembly category, and let $\mathrm{p}\mathbb{Grpd}(\operatorname{Asm}(A))$ denote the category of partitioned groupoid a...

conjecture | open | MathDB 页面 | 原文线索

V2-045419 | MATHDB-369412

Bicategorical W-types with reductions and elementary bitopos structure

摘录 / EXCERPT: Let A be the partial combinatory algebra underlying the assembly category, and let $\mathrm{p}\mathbbm{Grpd}(\operatorname{Asm}(A))$ be the category of partitioned groupoid assem...

conjecture | open | MathDB 页面 | 原文线索

V2-045420 | MATHDB-369418

The skew-Hermitian plus rank-one Hermitian principal-minor conjecture

摘录 / EXCERPT: Let K be a valued field with an automorphic involution preserving the valuation. Let $A=B+C$, where $B\in K^{n\times n}$ is skew-Hermitian and $C\in K^{n\times n}$ is Hermitian o...

conjecture | open | MathDB 页面 | 原文线索

V2-045421 | MATHDB-369419

The characteristic-two Hermitian plus rank-one conjecture

摘录 / EXCERPT: Assume $\operatorname{char} K=2$, let F be the fixed field of the involution on K with $F\ne K$, and write $K=F[\omega]$ where $\omega^2+\omega+\beta=0$ for some $\beta\in F$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-045422 | MATHDB-369446

Kronheimer-Mrowka's combinatorial dimension conjecture

摘录 / EXCERPT: Let Γ be a planar trivalent graph, let $\langle\Gamma\rangle_0$ be the state space obtained from the specialized foam evaluation over $\mathbb{F}_2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045423 | MATHDB-369447

Palindromic complexity conjecture for languages with symmetric order conditions

摘录 / EXCERPT: Let L) be a language on k letters satisfying a symmetric order condition and having no connection. Its palindromic complexity is the number of palindromic factors of each lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-045424 | MATHDB-369448

Order-condition characterization of affine interval exchange codings

摘录 / EXCERPT: Let W be a set of words, let $F(W)$ denote the associated object, and let $(<_A,<_D)$ be the pair of orders used in the order condition. Order-condition characterization conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-045425 | MATHDB-369491

Forbidden-configuration product conjecture for simple matrices

摘录 / EXCERPT: Let F be a simple $k\times \ell$ $(0,1)$-matrix. A product of matrices is obtained by stacking, in order, one column from each factor; the factors here are identity matrices I,...

conjecture | open | MathDB 页面 | 原文线索

V2-045426 | MATHDB-369498

Erdős's conjecture on equilateral triangles in six dimensions

摘录 / EXCERPT: Let $T_6(n)$ denote the maximum number of equilateral triangles determined by a set of n points in $\mathbb{R}^6$. Erdős's conjecture. Any set of n points in $\mathbb{R}^6$ can...

conjecture | open | MathDB 页面 | 原文线索

V2-045427 | MATHDB-369507

Asymptotic growth conjecture for Durfee triangle partitions

摘录 / EXCERPT: Let $R_k(n)$ denote the number of partitions of n whose Durfee triangle has size k, and let $k\in\mathbb{N}$ be fixed. Asymptotic growth conjecture. As $n\to\infty$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-045428 | MATHDB-369509

The 2-sunflowerability collapse conjecture

摘录 / EXCERPT: The 2-sunflowerability collapse conjecture. If ${\mathcal{M}}$ is a 2-sunflowerable structure, then ${\mathcal{M}}$ is sunflowerable.

conjecture | open | MathDB 页面 | 原文线索

V2-045429 | MATHDB-369511

The sunflowerability conjecture for generic K_n-free graphs

摘录 / EXCERPT: The sunflowerability conjecture for generic K_n-free graphs. For all $n\geq 3$, ${\mathcal{H}}_n$ is sunflowerable.

conjecture | open | MathDB 页面 | 原文线索

V2-045430 | MATHDB-369524

The commutativity conjecture for excedance-antiexcedance elements

摘录 / EXCERPT: Commutativity conjecture. For fixed n, the elements $\mathbf{X}_{a,b}$ commute for all $a,b\in\mathbb{N}$; equivalently,

open_problem | open | MathDB 页面 | 原文线索

V2-045431 | MATHDB-369570

Linear quantitative dependence conjecture for Ramsey-Turán density regularity

摘录 / EXCERPT: Let $\mathcal{L}: c_1x_1 + \ldots +c_k x_k = 0$ be a homogeneous linear equation with $k \ge 3$, where $c_1,\ldots,c_k \in \mathbb{Z} \setminus \{0\}$ satisfy ... but there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-045432 | MATHDB-369576

Frankl's conjecture on shattering in antichains

摘录 / EXCERPT: Frankl's conjecture. Let k be a non-negative integer and $n\geqslant 2k$. If $\mathcal{F}\subseteq 2^{[n]}$ is an antichain with $\mathcal{F}\not\rightarrow(k+1,2^{k+1})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-045433 | MATHDB-369577

Mészáros-Rónyai conjecture on deleting a member of an s-extremal family

摘录 / EXCERPT: Mészáros-Rónyai conjecture. For every nonempty s-extremal family $\mathcal{F}\subseteq 2^{[n]}$, there exists $F_0\in\mathcal{F}$ such that $\mathcal{F}\setminus\{F_0\}$ is still s...

conjecture | open | MathDB 页面 | 原文线索

V2-045434 | MATHDB-369597

Monical-Tokcan-Yong's Bruhat-order polytope conjecture

摘录 / EXCERPT: Monical-Tokcan-Yong's conjecture. If $\beta\in\widehat{\mathcal{P}}_{\alpha}$ and $\beta\in\mathbb{Z}^n_{\geq 0}$, then $\beta\leq_{S}\alpha$.

conjecture | open | MathDB 页面 | 原文线索

V2-045435 | MATHDB-369619

The original inverse Gowers conjecture

摘录 / EXCERPT: Original inverse Gowers conjecture. For every such F, a lower bound on the U^{d+1} norm implies correlation with a phase polynomial of degree d, with correlation bounded belo...

conjecture | open | MathDB 页面 | 原文线索

V2-045436 | MATHDB-369625

Standard-imaginary strategy conjecture for the two-player Levine hats game

摘录 / EXCERPT: Let $\widehat{\mathcal{S}}_1$ be the class of standard-imaginary strategies and let $\widehat{\mathcal{M}}_1$ be the class of imaginary strategies used in the continuous Levine gam...

conjecture | open | MathDB 页面 | 原文线索

V2-045437 | MATHDB-369638

Character-table characterization of totally extremal ideal Perron similarities

摘录 / EXCERPT: Character-table conjecture. If S is a normalized ideal Perron similarity and S is totally extremal, then S is the character table of a finite Abelian group.

conjecture | open | MathDB 页面 | 原文线索

V2-045438 | MATHDB-369661

Local avoidance conjecture for polynomial images of integers and primes

摘录 / EXCERPT: Let $h\in\mathbb{Z}[x]$ be a nonzero polynomial, and let $X=h(\mathbb{Z})$ or $X=h(\mathcal{P})$, where $\mathcal{P}$ denotes the set of primes. For each $m\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-045439 | MATHDB-369668

Simon's conjecture on extendable shellability of simplex skeleta

摘录 / EXCERPT: Let $\Delta_{n-1}^{(d)}$ denote the d-skeleton of the $(n-1)$-dimensional simplex. A pure d-dimensional shellable complex on n vertices is extendably shellable if every shell...

conjecture | open | MathDB 页面 | 原文线索

V2-045440 | MATHDB-369676

Bernstein's conjecture on algebraic matroids of Hadamard products

摘录 / EXCERPT: Let $L_1, \ldots, L_d \subset \mathbb{C}^m$ be linear spaces not contained in any coordinate hyperplane, and let $r_1, \ldots, r_d$ be the rank functions of the corresponding linea...

conjecture | open | MathDB 页面 | 原文线索

V2-045441 | MATHDB-369693

Additive-energy conjecture for lattice points in a three-dimensional thin annulus

摘录 / EXCERPT: Additive-energy conjecture. For $\mathcal{A}_{\lambda,\delta}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045442 | MATHDB-369767

Quadratic Rado conjecture for homogeneous three-variable equations

摘录 / EXCERPT: Quadratic Rado conjecture. The equation

conjecture | open | MathDB 页面 | 原文线索

V2-045443 | MATHDB-369793

Order-q conjecture for K-promotion on three-row rectangular increasing tableaux

摘录 / EXCERPT: An increasing tableau of shape λ with entries in $[q]\coloneqq\{1,\ldots,q\}$ is a filling of the Young diagram of λ with elements of $[q]$ whose rows and columns a...

conjecture | open | MathDB 页面 | 原文线索

V2-045444 | MATHDB-369811

Conjecture on the asymptotic injectivity of the difference-multiset map

摘录 / EXCERPT: Difference-multiset injectivity conjecture. As $n \to \infty$, the image of ϕ contains $2^{n-1}+o(2^{n})$ distinct elements. For a set A with maximum element a', the refle...

conjecture | open | MathDB 页面 | 原文线索

V2-045445 | MATHDB-369842

Upper combinatorial uniformity conjecture for geometric lattices

摘录 / EXCERPT: Upper combinatorial uniformity conjecture. Every upper combinatorially uniform geometric lattice is a TN-poset.

conjecture | open | MathDB 页面 | 原文线索

V2-045446 | MATHDB-369864

Chao-Xu-Yip-Zhang's uniform certificate conjecture

摘录 / EXCERPT: Chao-Xu-Yip-Zhang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045447 | MATHDB-369865

Frankl's chain-free conjecture for families with bounded VC-dimension

摘录 / EXCERPT: Frankl's chain-free conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045448 | MATHDB-369874

Conjecture on consistent sign patterns for the type C configuration space

摘录 / EXCERPT: The type C_n cluster configuration space $\mathcal{M}_{C_n}$ is obtained by folding from $\mathcal{M}_{0,2(n-1)-3}$. Its dihedral coordinates are indexed by pairs of chords and c...

conjecture | open | MathDB 页面 | 原文线索

V2-045449 | MATHDB-369915

Uniform asymptotic expansion for regular graph counts

摘录 / EXCERPT: Let n and d be integers with $1\leq d\leq n-2$ and dn even, and set $\lambda=d/(n-1)$ and $\varLambda=\lambda(1-\lambda)$. Let $\operatorname{RG}(n,d)$ denote the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-045450 | MATHDB-369936

Asymptotic enumeration conjecture for racks

摘录 / EXCERPT: A rack is a set equipped with a binary operation satisfying the rack axioms; two racks are identified when they are isomorphic. Let n denote the size of the rack. Rack enumeratio...

conjecture | open | MathDB 页面 | 原文线索

V2-045451 | MATHDB-369937

Exponential enumeration conjecture for set-theoretic Yang-Baxter solutions

摘录 / EXCERPT: A set-theoretic solution of the Yang-Baxter equation is a solution considered up to isomorphism, and its size is the cardinality of its underlying set. Let n denote the size of t...

conjecture | open | MathDB 页面 | 原文线索

V2-045452 | MATHDB-369947

Krachun-Petrov conjecture on sums of algebraic dilates

摘录 / EXCERPT: Krachun-Petrov conjecture. For every finite subset $A\subset\mathbb{C}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045453 | MATHDB-369952

Failure of the matroid matching property in odd-dimensional rigidity matroids

摘录 / EXCERPT: For integers $k\geq 2$ and sufficiently large n, consider the rigidity matroid ${\mathcal R}_{2k+1}(K_n)$ of the complete graph K_n in dimension $2k+1$. The matroid matching co...

conjecture | open | MathDB 页面 | 原文线索

V2-045454 | MATHDB-369971

Convergence conjecture for the Temperley-Lieb algebra rewriting system

摘录 / EXCERPT: Convergence conjecture. This rewriting system is convergent, meaning that it is terminating and confluent.

conjecture | open | MathDB 页面 | 原文线索

V2-045455 | MATHDB-369974

Conjecture on the weak saturation number of unbalanced complete bipartite graphs

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with $2<s\leq t$, and let j satisfy $2\leq j\leq t-3$. Write $\mathrm{wsat}(n,K_{s,t})$ for the weak saturation number of $K_{s,t}$....

conjecture | open | MathDB 页面 | 原文线索

V2-045456 | MATHDB-369975

Weak saturation conjecture for balanced complete bipartite graphs

摘录 / EXCERPT: Let $K_{s,s}$ be the balanced complete bipartite graph, let $\mathrm{wsat}(n,K_{s,s})$ denote its weak saturation number, and let G be an erasable graph in the corresponding eras...

conjecture | open | MathDB 页面 | 原文线索

V2-045457 | MATHDB-369988

Frankl and Wang's sturdiness conjecture for IU-families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $2^{[n]}$ denote the power set of $[n]$, and let an IU-family be a family $\mathcal{G}\subseteq 2^{[n]}$ such that ... and ... Write $\beta(\mathcal{G})$...

conjecture | open | MathDB 页面 | 原文线索

V2-045458 | MATHDB-370000

Characterization of extremal discrete Borsuk sets by cubes

摘录 / EXCERPT: Let $S \subset \mathbb{Z}^d$ be a bounded set, and let $\beta_{\mathbb{Z}}(S)$ denote its discrete Borsuk partition number. Let $\operatorname{conv}(S)$ be the convex hull of S,...

conjecture | open | MathDB 页面 | 原文线索

V2-045459 | MATHDB-370001

Frankl-Füredi forbidden delta-system conjecture

摘录 / EXCERPT: For fixed k,ℓ,s, let $f(n,k,\ell,s)$ denote the maximum size of a family of k-element sets with no $\Delta(s+1)$-system whose kernel has size ℓ. Let $\phi(\ell+1,s)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-045460 | MATHDB-370014

The asymptotic limit conjecture for neighborly boxes

摘录 / EXCERPT: A box in $\mathbb{R}^d$ is an axis-parallel d-dimensional cuboid. Two boxes are k-neighborly if their intersection has dimension strictly smaller than d but at least $d-k$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-045461 | MATHDB-370019

Rationality conjecture for submonoid generating functions

摘录 / EXCERPT: Let ... be the multivariable exponential generating function for submonoids of products of finite commutative monoids. Rationality conjecture. The generating function $G(x,y,z)$ is a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045462 | MATHDB-370020

The partite Nash-Williams conjecture

摘录 / EXCERPT: Partite Nash-Williams conjecture. For sufficiently large n, if G is a balanced partite-K_3-divisible 3-partite graph on $3n$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-045463 | MATHDB-370021

The partite folklore clique decomposition conjecture

摘录 / EXCERPT: Partite folklore conjecture. For each integer $q\geq 4$, the following holds for sufficiently large n: if G is a balanced partite-K_q-divisible q-partite graph on qn vert...

conjecture | open | MathDB 页面 | 原文线索

V2-045464 | MATHDB-370039

Monotone diameter-2 path-system conjecture

摘录 / EXCERPT: Let $f:\binom{[n]}{2}\to [n]$ be the unique résumé of a diameter-2 path system $\mathcal{P}$. The system $\mathcal{P}$ is monotone if $f(i,j)\le f(i,k)$ whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-045465 | MATHDB-370064

Erdős-Fishburn conjecture on maximum distance sets in triangular lattices

摘录 / EXCERPT: Let an m-distance set be a collection of points in the Euclidean plane such that the distances between any two points take at most m possible values. A triangular lattice is th...

conjecture | open | MathDB 页面 | 原文线索

V2-045466 | MATHDB-370065

Conjecture on the maximum 7- and 8-distance sets in triangular lattices

摘录 / EXCERPT: Let a triangular lattice be the set of points with coordinates $a(1,0)+b(\frac12,\frac{\sqrt3}{2})$, where $a,b\in\mathbb Z$. An m-distance set is a set of lattice points whose p...

conjecture | open | MathDB 页面 | 原文线索

V2-045467 | MATHDB-370066

Hexagonal construction conjecture for maximum distance sets

摘录 / EXCERPT: Let a triangular lattice be the set of points with coordinates $a(1,0)+b(\frac12,\frac{\sqrt3}{2})$, where $a,b\in\mathbb Z$. An equiangular hexagon has six sides in the lattice di...

conjecture | open | MathDB 页面 | 原文线索

V2-045468 | MATHDB-370067

Hexagon-deletion conjecture for maximum distance sets

摘录 / EXCERPT: Let a triangular lattice be the set of points with coordinates $a(1,0)+b(\frac12,\frac{\sqrt3}{2})$, where $a,b\in\mathbb Z$. Consider a regular hexagon or an equiangular hexagon w...

conjecture | open | MathDB 页面 | 原文线索

V2-045469 | MATHDB-370068

Multiplicity conjecture for maximum distance sets

摘录 / EXCERPT: Let an m-distance set be a set of points whose pairwise distances take at most m values. The multiplicity of a distance is the number of unordered pairs of points in the set re...

conjecture | open | MathDB 页面 | 原文线索

V2-045470 | MATHDB-370083

The near-quadratic energy conjecture for convex sets

摘录 / EXCERPT: Near-quadratic energy conjecture. If $A\subset\mathbb{R}$ is convex, then

conjecture | open | MathDB 页面 | 原文线索

V2-045471 | MATHDB-370089

Conlon-Luo-Tyomkyn's monochromatic component edge conjecture

摘录 / EXCERPT: Let n and r be positive integers with $r\ge 3$, and let K_n be the complete graph on n vertices whose edges are colored with r colors. A monochromatic connected component...

conjecture | open | MathDB 页面 | 原文线索

V2-045472 | MATHDB-370091

Asymptotic Fibonacci cordial labeling conjecture for circulant graphs

摘录 / EXCERPT: Let $\Gamma(n,S)$ be a circulant graph on n vertices with connection set S. Here, “small” means that S is a connection set whose size is small relative to n, although no pr...

conjecture | open | MathDB 页面 | 原文线索

V2-045473 | MATHDB-370092

Finite-line fixed-set conjecture for maximal reducible subgroups of quaternionic groups

摘录 / EXCERPT: Let G be a quaternionic reflection group, or more generally a finite irreducible group of $d\times d$ matrices over the quaternions $\mathbb{H}$. A subgroup is reducible if it pr...

conjecture | open | MathDB 页面 | 原文线索

V2-045474 | MATHDB-370104

Independence number conjecture for sparse random Cayley graphs

摘录 / EXCERPT: Let G be a group of size n, and let $G(p)$ be its random Cayley graph, formed by selecting each equivalence class $\{x,-x\}$ independently with probability p. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-045475 | MATHDB-370105

Optimal covering conjecture for small-doubling sumsets

摘录 / EXCERPT: Let A be a set of size s with doubling constant bounded by K, meaning $|A+A|\leq K|A|$. For each ell let $\mathcal{F}_\ell$ be a collection of sets, and let $\tilde{O}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045476 | MATHDB-370123

Hikita probability lower-bound conjecture

摘录 / EXCERPT: Let $\mathbf{m}\in\mathbb{E}_n$, let $\lambda\vdash n$, and let $T\in\operatorname{HikSYT}(\mathbf{m},\lambda)$. For the recursively defined Hikita weight $h_T(q)$ and every real...

conjecture | open | MathDB 页面 | 原文线索

V2-045477 | MATHDB-370133

The minimally rigidity-connected stress-independence conjecture

摘录 / EXCERPT: Minimally $\mathcal{R}_d$-connected stress-independence conjecture. Every minimally $\mathcal{R}_d$-connected graph is d-stress-independent.

conjecture | open | MathDB 页面 | 原文线索

V2-045478 | MATHDB-370175

Directed Burr-Erdős conjecture for oriented hypercubes

摘录 / EXCERPT: Let $\vv{Q_d}$ denote the oriented d-dimensional hypercube, and let $\vv{r}(\vv{Q_d})$ be its oriented Ramsey number. Directed Burr-Erdős conjecture. There is an absolute constan...

conjecture | open | MathDB 页面 | 原文线索

V2-045479 | MATHDB-370182

Conjectured special values and generating function for the polynomials \(\psi_n\)

摘录 / EXCERPT: Special-values conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-045480 | MATHDB-370217

Repetition-threshold formula for rich sequences

摘录 / EXCERPT: Let $\mathcal{C}_d$ be the class of d-ary sequences rich in palindromes. For each $n\in\mathbb{N}$, let $\mathbf{u}_n$ denote the sequence defined in the paper by the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-045481 | MATHDB-370222

Combinatorial triangulation conjecture for the Łojasiewicz exponent

摘录 / EXCERPT: Combinatorial triangulation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045482 | MATHDB-370252

Near-sharp sphere-counting conjecture

摘录 / EXCERPT: Let $S(n,d)$ denote the number of $(d+2)$-tuples of points in $[n]^d$ that lie in a $(d-1)$-dimensional sphere. Sphere-counting conjecture. For any integer $d\geqslant 2$, ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-045483 | MATHDB-370296

Equidistribution of statistics on Stoimenow matchings, Fishburn posets and Dyck paths

摘录 / EXCERPT: Let $\mathcal{M}_n(P_1)$ be the set of Stoimenow matchings avoiding P_1, let $\mathcal{P}_n{\bf (3+1)}$ be the corresponding class of $(3+1)$-avoiding Fishburn posets, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045484 | MATHDB-370332

The linear-or-bounded conjecture for induced poset saturation

摘录 / EXCERPT: The linear-or-bounded conjecture. For every finite poset $\mathcal P$, the induced saturated number $\operatorname{sat}^*(n,\mathcal P)$ is either bounded, or at least linear in...

conjecture | open | MathDB 页面 | 原文线索

V2-045485 | MATHDB-370338

Lejeune-Rigo-Rosenfeld conjecture on Arnoux-Rauzy words

摘录 / EXCERPT: Let w be an Arnoux-Rauzy infinite word. Its 2-binomial complexity and subword complexity are functions assigning to each positive integer the number of equivalence classes of f...

conjecture | open | MathDB 页面 | 原文线索

V2-045486 | MATHDB-370340

Stanley's pure O-sequence conjecture for matroid h-vectors

摘录 / EXCERPT: Let M be a matroid with h-vector $(h_0,h_1,\dots)$. A sequence is an O-sequence if it counts the monomials in each degree of an order ideal of monomials, and it is pure if all...

conjecture | open | MathDB 页面 | 原文线索

V2-045487 | MATHDB-370343

Cavenagh-Hämäläinen-Lefevre-Stones completion conjecture for partial multi-Latin squares

摘录 / EXCERPT: Cavenagh-Hämäläinen-Lefevre-Stones' conjecture. For each $r,\lambda\in\mathbb N$, there exists an integer $n(r,\lambda)$ such that, for every $n\geq n(r,\lambda)$, every simple par...

conjecture | open | MathDB 页面 | 原文线索

V2-045488 | MATHDB-370344

The joint cpen-last distribution conjecture for Jacobi permutations

摘录 / EXCERPT: The joint cpen-last distribution conjecture. For all $n\geq3$ and $i,k\geq1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-045489 | MATHDB-370345

The asc-last and des-first equidistribution conjecture for Jacobi and André permutations

摘录 / EXCERPT: The asc-last and des-first equidistribution conjecture. For all $n\geq1$, $i\geq0$, and $k\geq1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-045490 | MATHDB-370352

The 1/3-2/3 conjecture for linear extensions of posets

摘录 / EXCERPT: Let P be a finite poset that is not a chain. For distinct $x,y\in P$, write $\delta_{xy}=\min\{p(x\prec y),p(y\prec x)\}$ and $\delta(P)=\max_{x\neq y}\delta_{xy}$. The 1/3-2/3...

conjecture | open | MathDB 页面 | 原文线索

V2-045491 | MATHDB-370353

The Kahn-Saks conjecture on balance in wide posets

摘录 / EXCERPT: Let P be a finite poset, let $w(P)$ denote its width, and let $\delta(P)$ be the maximum, over distinct $x,y\in P$, of $\min\{p(x\prec y),p(y\prec x)\}$. The Kahn-Saks conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-045492 | MATHDB-370354

The conjecture that large incomparability degree forces asymptotic balance

摘录 / EXCERPT: Let P be a finite poset. For $x\in P$, let $\Pi(x)=\\{y\in P\setminus\\{x\\}:y\not\sim x\\}$, let $\pi(x)=|\Pi(x)|$, and let $\pi(P)=\max_{x\in P}\pi(x)$. Let $\delta(P)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-045493 | MATHDB-370357

The local deletion sensitivity conjecture for posets

摘录 / EXCERPT: Let P be a finite poset with n elements, let $x,y\in P$ with $y\ne x$, and let $h_P(y)$ denote the expected position of y in a uniformly random linear extension of P. The l...

conjecture | open | MathDB 页面 | 原文线索

V2-045494 | MATHDB-370358

The continuous window-variance conjecture for posets

摘录 / EXCERPT: Let P be a finite poset, let F be uniformly distributed in its order polytope, let $H(x)=\mathbb E F(x)$, and let $\operatorname{Win}(x)=\mathbb E[R(x)-Q(x)]$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-045495 | MATHDB-370389

Conjectural formula for optimal solutions with k-1 clean columns

摘录 / EXCERPT: Let $\alpha_{n,m}$ denote the number of optimal solutions for the $n\times m$ grid in the paper's notation. In the case of a grid with 2 rows and $2k+1$ columns, where $k\geq 0$,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045496 | MATHDB-370396

Bitonti-Dukes-Sagan conjecture on trivariate symmetries of vincular-pattern distributions

摘录 / EXCERPT: For $n\geq 1$, let $\mathfrak{S}_n$ be the set of permutations of $[n]$, and define ... Here $\mathrm{\underline{13}2}(\sigma)$, $\mathrm{\underline{31}2}(\sigma)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-045497 | MATHDB-370398

Cameron's conjecture on row parities in random Latin squares

摘录 / EXCERPT: Let $\mathbf{L}$ be a uniformly random $n\times n$ Latin square, and let $N_{\mathrm{row}}(\mathbf{L})$ denote the number of odd row permutations of $\mathbf{L}$. Cameron's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-045498 | MATHDB-370444

Extremal Steklov-Neumann eigenvalues on the disk

摘录 / EXCERPT: Let $\Omega=\mathbb{D}$ be the unit disk, let $\Gamma=\partial\Omega=\mathbb{S}^1$, and let $\mathcal{I}_\alpha^n\subset\mathbb{S}^1$ be the union of n evenly spaced intervals of...

conjecture | open | MathDB 页面 | 原文线索

V2-045499 | MATHDB-370457

Flip Combinatorial Invariance for Weyl-type flipclasses

摘录 / EXCERPT: Flip Combinatorial Invariance conjecture. Flip Combinatorial Invariance holds for flipclasses of Weyl type.

conjecture | open | MathDB 页面 | 原文线索

V2-045500 | MATHDB-370458

The graph form of the Combinatorial Invariance Conjecture

摘录 / EXCERPT: Graph Combinatorial Invariance conjecture. The $\widetilde{R}$-polynomial $\widetilde{R}_{u,v}(q)$ depends only on the isomorphism type of the graph $B([u,v])$.

conjecture | open | MathDB 页面 | 原文线索

V2-045501 | MATHDB-370460

Duchi-Guerrini-Rinaldi-Schaeffer's refined fighting fish-ternary tree conjecture

摘录 / EXCERPT: Duchi-Guerrini-Rinaldi-Schaeffer's conjecture. The number of fighting fish with size n, fin length k, having h tails, i ascending strips and j descending strips is equal...

conjecture | open | MathDB 页面 | 原文线索

V2-045502 | MATHDB-370466

The endpoint monotonicity conjecture for polygonal lines of combinatorial Petrie matrices

摘录 / EXCERPT: Let $A\in \mathcal{CP}(n)$ and let L be its corresponding polygonal line. If n is even, consider the endpoints of L ordered by increasing x-coordinate. Endpoint monotonicit...

conjecture | open | MathDB 页面 | 原文线索

V2-045503 | MATHDB-370467

The parity correspondence conjecture for Motzkin-path enumeration sequences

摘录 / EXCERPT: Let a_n denote the number of Motzkin paths of length n satisfying the rule specified in the source. For $n\geq 1$, let A005802 and A216947 denote the cited OEIS sequences, with...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045504 | MATHDB-370477

The generalized ballot property for labeled chips on k-ary trees

摘录 / EXCERPT: Let $N_{k,\ell}$ denote the number of labeled chips starting at the root of a k-ary tree in the setting of the paper, and let a stable configuration be one obtained from these ch...

conjecture | open | MathDB 页面 | 原文线索

V2-045505 | MATHDB-370478

Edge inducibility conjecture for odd paths

摘录 / EXCERPT: Edge inducibility conjecture for odd paths. If $k\geq 5$ is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-045506 | MATHDB-370486

Polynomial Szemerédi characterization over rings of integers of global fields

摘录 / EXCERPT: Let K be a global field with ring of integers $\mathcal{O}_K$. Let $P_1[x],\dots,P_k[x]\in\mathcal{O}_K[x]$ be nonconstant polynomials, and let $d^*(A)$ denote the upper Banach d...

conjecture | open | MathDB 页面 | 原文线索

V2-045507 | MATHDB-370487

The non-degenerate smooth Roth conjecture for thick Cantor sets

摘录 / EXCERPT: Let $K\subset\mathbb{R}$ be a Cantor set satisfying $\tau(K)>1$, and let f be a continuously differentiable function satisfying $f(0)=0$ and $f'(0)>0$. Non-degenerate smooth Roth...

conjecture | open | MathDB 页面 | 原文线索

V2-045508 | MATHDB-370495

Simplicity conjecture for extremal automata

摘录 / EXCERPT: Let an extremal automaton be a synchronizing automaton attaining equality in the Černý bound. Simplicity conjecture. Every extremal automaton is simple. The conjecture is motivated...

conjecture | open | MathDB 页面 | 原文线索

V2-045509 | MATHDB-370510

Pach-Saghafian-Schnider conjecture on decompositions of cliques into k-star-forests

摘录 / EXCERPT: Pach-Saghafian-Schnider conjecture. For any $n\ge k\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-045510 | MATHDB-370523

Terao's freeness conjecture for hyperplane arrangements

摘录 / EXCERPT: Let ${\mathcal{A}}_1$ and ${\mathcal{A}}_2$ be central hyperplane arrangements defined over the same field. Their intersection lattices $L({\mathcal{A}}_1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045511 | MATHDB-370542

The VG algebra tope-graph reconstruction conjecture

摘录 / EXCERPT: Assume $\operatorname{char} R\neq 2$. Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be real arrangements. Their filtered VG algebras are $\operatorname{\mathcal{VG}}(\mathcal{A}_i)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045512 | MATHDB-370543

The non-deterministic VG algorithm for reconstructing oriented matroids

摘录 / EXCERPT: Let $\mathcal{A}=\{H_1,\dots,H_n\}$ be a real arrangement in $\mathbb{R}^\ell$. A generalized tope graph $\operatorname{\widetilde{\mathcal{T}}}(y_1,\dots,y_n)$ is the graph constr...

conjecture | open | MathDB 页面 | 原文线索

V2-045513 | MATHDB-370547

k-Roudneff's conjecture for neighborly reorientations of oriented matroids

摘录 / EXCERPT: k-Roudneff's conjecture. For any rank $r\geq 2k+1\geq 3$ oriented matroid $\mathcal{M}$ on $n\geq r+1$ elements,

conjecture | open | MathDB 页面 | 原文线索

V2-045514 | MATHDB-370580

The linear upper-bound conjecture for point evaluation on the circle

摘录 / EXCERPT: The linear upper-bound conjecture. If $d\geq 1$ and $p\geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-045515 | MATHDB-370593

Strong Schur log-concavity conjecture for diagonals of the quantum Pascal triangle

摘录 / EXCERPT: For $n\geqslant k\geqslant0$, define the quantum binomial by ... where $[n]=(q^{-n}-q^n)/(q^{-1}-q)$ and $[0]=0$. A sequence of Laurent polynomials is strongly Schur log-concave if e...

conjecture | open | MathDB 页面 | 原文线索

V2-045516 | MATHDB-370605

Taillard's asymptotic tightness conjecture for permutation flowshop lower bounds

摘录 / EXCERPT: Taillard's conjecture. For a fixed number of machines M,

conjecture | open | MathDB 页面 | 原文线索

V2-045517 | MATHDB-370618

The extremal intersection conjecture for q-ary deletion channels

摘录 / EXCERPT: Extremal intersection conjecture. For $q\geqslant 4$ and sufficiently large n, we have

conjecture | open | MathDB 页面 | 原文线索

V2-045518 | MATHDB-370636

The periodic tiling conjecture for finite subsets of finitely generated Abelian groups

摘录 / EXCERPT: Periodic tiling conjecture. The tiling equation $F\oplus X=G$ is not aperiodic; equivalently, whenever F tiles G by translations, it admits at least one periodic tiling.

conjecture | open | MathDB 页面 | 原文线索

V2-045519 | MATHDB-370651

Uniform surface conjecture for exterior algebraic shifting

摘录 / EXCERPT: Uniform surface conjecture. For every fixed genus g, a.a.s.,

conjecture | open | MathDB 页面 | 原文线索

V2-045520 | MATHDB-370652

Barycentric subdivision conjecture for exterior algebraic shifting of surfaces

摘录 / EXCERPT: Barycentric subdivision conjecture. If K is a surface triangulation, then $\Delta^{ex}(\operatorname{sd}(K))$ is a homology lex-segment.

conjecture | open | MathDB 页面 | 原文线索

V2-045521 | MATHDB-370653

Characteristic-zero exterior shifting conjecture for surface triangulations

摘录 / EXCERPT: Characteristic-zero exterior shifting conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045522 | MATHDB-370667

The monotonicity conjecture for diagonal (m,m)-town families modulo k

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, let $\mathscr{P}(n)$ be the power set of $[n]$, and let a family be an (a,b)-town modulo k when all its sets have cardinality congruent to a modul...

conjecture | open | MathDB 页面 | 原文线索

V2-045523 | MATHDB-370685

The four matroid polynomial conjectures

摘录 / EXCERPT: The four matroid polynomial conjectures. For every matroid M, the polynomial $P_{\mathsf{M}}(x)$ is real-rooted, the polynomial $Z_{\mathsf{M}}(x)$ is real-rooted, the p...

conjecture | open | MathDB 页面 | 原文线索

V2-045524 | MATHDB-370686

Xie-Zhang's real-rootedness conjecture for normalized inverse Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Xie-Zhang's conjecture. For every matroid M, the polynomial $\mathcal{B}(Q_{\mathsf{M}})(x)$ is real-rooted.

conjecture | open | MathDB 页面 | 原文线索

V2-045525 | MATHDB-370700

Asymptotic order conjecture for higher xor-powers of Kneser graphs

摘录 / EXCERPT: Let $f_\ell(n,k)$ denote the clique number of the xor-product of ℓ copies of the Kneser graph $KG(n,k)$. Fix ℓ and suppose that k is sufficiently large. Higher-power gr...

conjecture | open | MathDB 页面 | 原文线索

V2-045526 | MATHDB-370732

The second Kahn-Kalai conjecture for graph-containment thresholds

摘录 / EXCERPT: For a graph H, let $p_c(H)=p_c(n,H)$ be the unique p such that ... Define the expectation threshold by ... The second Kahn-Kalai conjecture. There is a fixed K such that for any...

conjecture | open | MathDB 页面 | 原文线索

V2-045527 | MATHDB-370747

Lovett-Singer-Sudan conjecture on cross-intersecting set systems

摘录 / EXCERPT: Let $\mathcal{S}=\{S_1,\ldots,S_m\}$ and $\mathcal{T}=\{T_1,\ldots,T_n\}$ be families in $2^{[d]}$, and suppose that $(\mathcal{S},\mathcal{T})$ is $\{a,b\}$-cross-intersecting, me...

conjecture | open | MathDB 页面 | 原文线索

V2-045528 | MATHDB-370777

Exponential lower-bound conjecture for peeling sequences

摘录 / EXCERPT: Peeling-sequence growth conjecture. The minimal number of peeling sequences of n points in $\mathbb{R}^d$ is at least

conjecture | open | MathDB 页面 | 原文线索

V2-045529 | MATHDB-370816

Bohn-Cameron-Müller maximality conjecture for Tutte polynomial monodromy

摘录 / EXCERPT: Bohn-Cameron-Müller's conjecture. The group $G_{K,y}(T_M)$ is maximal, i.e. it is isomorphic to the symmetric group on $\operatorname{r}(M)$ letters:

conjecture | open | MathDB 页面 | 原文线索

V2-045530 | MATHDB-370817

Polynomial Gyárfás-Sumner conjecture

摘录 / EXCERPT: Let G be a finite graph, let T be a forest, and write $\chi(G)$ for the chromatic number and $\omega(G)$ for the clique number. Polynomial Gyárfás-Sumner conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-045531 | MATHDB-370821

Asymptotic growth conjecture for (P,ϕ)-Tamari lattices

摘录 / EXCERPT: Let L_n^d be the lattice of d-torsion classes associated with the (P,ϕ)-Tamari lattice, and let $J(os_n^{d+1})$ denote the set of order ideals of the poset os_n^{d+1}. F...

conjecture | open | MathDB 页面 | 原文线索

V2-045532 | MATHDB-370847

Heron-Rota-Welsh log-concavity conjecture for Whitney numbers of matroids

摘录 / EXCERPT: Let M be a matroid of rank $r(M)$, with characteristic polynomial ... The integers $w_i(M)$ are the Whitney numbers of the first kind. Heron-Rota-Welsh conjecture. The sequence ... o...

conjecture | open | MathDB 页面 | 原文线索

V2-045533 | MATHDB-370871

Refined q,t-symmetry conjecture for $\vec{k}$-Dyck paths with a- and b-tails

摘录 / EXCERPT: Let $\vec{k}$ be a vector of positive integers, let $\mathcal{K}$ denote the corresponding parameter indexing the relevant $\vec{k}$-Dyck paths, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045534 | MATHDB-370873

Beck-Brauner-? conjecture for q,t-symmetry of $(a,b,\ldots,b)$-Dyck paths

摘录 / EXCERPT: Let $\vec{k}$ be a vector of positive integers and define the area-bounce polynomial ... A polynomial is q,t-symmetric when $F(q,t)=F(t,q)$. The $(a,b,\ldots,b)$-symmetry conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-045535 | MATHDB-370883

Separation conjecture between semisaturation and packing numbers

摘录 / EXCERPT: Separation conjecture. For any $r=r(k)\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-045536 | MATHDB-370884

Packing-saturation equivalence conjecture for triangles

摘录 / EXCERPT: Packing-saturation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045537 | MATHDB-370890

The generic central slice conjecture for cubes

摘录 / EXCERPT: Generic central slice conjecture. The upper bound on the number of vertices of a k-dimensional slice of C_d can be attained by a generic central k-dimensional slice for every...

conjecture | open | MathDB 页面 | 原文线索

V2-045538 | MATHDB-370891

The central-slice extremality conjecture for centrally symmetric polytopes

摘录 / EXCERPT: Central-slice extremality conjecture. For every centrally symmetric k-dimensional polytope P, the upper bound on the number of vertices of a $(k-1)$-dimensional slice of P is...

conjecture | open | MathDB 页面 | 原文线索

V2-045539 | MATHDB-370936

The Ramsey-to-canonical Ramsey conjecture for configurations

摘录 / EXCERPT: Let $\mathcal{S}\subseteq\mathbb{E}^n$ be a configuration. It is Ramsey if for every $r\in\mathbb{N}$ there exists a dimension $n=n(\mathcal{S},r)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-045540 | MATHDB-370940

Tokushige's conjecture on non-uniform cross-t-intersecting families

摘录 / EXCERPT: Let $k\geq l\geq t\geq1$ and $n\geq(t+1)(k-t+1)$. Two families $\mathcal{A}\subseteq\binom{[n]}{k}$ and $\mathcal{B}\subseteq\binom{[n]}{l}$ are cross-t-intersecting if...

conjecture | open | MathDB 页面 | 原文线索

V2-045541 | MATHDB-370960

Conjectural enumeration of translation-equivalence classes of intervals

摘录 / EXCERPT: Let $\operatorname{WA}_{n}^{132}$ be the set of well-aligned pairs avoiding the pattern 132. Consider the equivalence relation identifying pairs whose Bruhat intervals are transl...

conjecture | open | MathDB 页面 | 原文线索

V2-045542 | MATHDB-370989

Maximality conjecture for abstract birigidity matroids

摘录 / EXCERPT: Maximality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045543 | MATHDB-371011

Vu's sharpened chromatic conjecture

摘录 / EXCERPT: Let G be a graph, with maximum degree $\operatorname{\Delta}(G)$, maximum codegree $\operatorname{\Delta_2}(G)$, and chromatic number $\chi(G)$. Vu's sharpened conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-045544 | MATHDB-371014

Folklore extremal conjecture for sum-free subsets of higher-dimensional grids

摘录 / EXCERPT: Higher-dimensional extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045545 | MATHDB-371015

Elsholtz-Rackham counting conjecture for higher-dimensional grids

摘录 / EXCERPT: Elsholtz-Rackham's higher-dimensional conjecture. The number of sum-free subsets of $[n]^d$ is

conjecture | open | MathDB 页面 | 原文线索

V2-045546 | MATHDB-371018

Leader-Russell-Walters' transitive-set conjecture for Euclidean Ramsey sets

摘录 / EXCERPT: A finite set $X\subset\mathbb{R}^d$ is Ramsey if, for every k, there exists an integer n such that every k-colouring of $\mathbb{R}^n$ contains a monochromatic isometric copy...

conjecture | open | MathDB 页面 | 原文线索

V2-045547 | MATHDB-371042

Embedding conjecture for three-dimensional symmetric designs from regular Hadamard matrices

摘录 / EXCERPT: Embedding conjecture. If H is a regular Hadamard matrix of order v^2, then such a containing cube and subcube exist.

conjecture | open | MathDB 页面 | 原文线索

V2-045548 | MATHDB-371065

Simonovits-Sós conjecture on P_4-intersecting families

摘录 / EXCERPT: Simonovits-Sós conjecture. If $\mathcal{F}$ is a P_4-intersecting family of labeled subgraphs of K_n, then

conjecture | open | MathDB 页面 | 原文线索

V2-045549 | MATHDB-371066

The Christofides bound conjecture for P_4-intersecting families

摘录 / EXCERPT: Christofides bound conjecture. For $n \ge 6$, the maximum size of a P_4-intersecting family is

conjecture | open | MathDB 页面 | 原文线索

V2-045550 | MATHDB-371067

Alon-Spencer conjecture on non-star-forest intersection families

摘录 / EXCERPT: Alon-Spencer conjecture. There exists some constant $\epsilon > 0$ such that if H is a fixed graph that is not a star-forest and $\mathcal F$ is an H-intersecting family on...

conjecture | open | MathDB 页面 | 原文线索

V2-045551 | MATHDB-371072

Adin-Alon-Roichman conjecture on circular sorting for composite sizes

摘录 / EXCERPT: Let $S(\mathbb{Z}_n)$ denote the permutations of $\mathbb{Z}_n$, let $[\pi]$ be the coset of a permutation π under cyclic shifts, and let $t([\pi])$ be the minimum number of tr...

conjecture | open | MathDB 页面 | 原文线索

V2-045552 | MATHDB-371074

Multiplicative monotonicity of the circular sorting defect

摘录 / EXCERPT: Multiplicative monotonicity conjecture. Is

open_problem | open | MathDB 页面 | 原文线索

V2-045553 | MATHDB-371086

Singh and Barman's hook length bias conjecture for t-regular partitions

摘录 / EXCERPT: Let $b_{t,k}(n)$ denote the number of hooks of length k across all t-regular partitions of n. Singh and Barman's conjecture. For every integer $t\geq 3$ and every $n\geq 0$,...

conjecture | solved | MathDB 页面 | 原文线索

V2-045554 | MATHDB-371102

Erdős's extension conjecture for finite Sidon sets

摘录 / EXCERPT: A finite Sidon set is a finite set of integers whose differences of distinct elements are all distinct. A finite perfect difference set is a finite set whose differences represent...

conjecture | open | MathDB 页面 | 原文线索

V2-045555 | MATHDB-371105

Erdős' high-girth Steiner triple system conjecture

摘录 / EXCERPT: A (j,i)-configuration in a family of sets is a collection of i members spanning at most j ground-set elements; the girth of a triangle packing is the smallest $i\geq 2$ for w...

conjecture | open | MathDB 页面 | 原文线索

V2-045556 | MATHDB-371108

Jeffs's convexity conjecture for neural codes with four maximal codewords

摘录 / EXCERPT: Jeffs's conjecture. If $\mathcal{C}$ has up to four maximal codewords, then $\mathcal{C}$ is convex if and only if $\mathcal{C}$ has no local obstructions and no wheels.

conjecture | open | MathDB 页面 | 原文线索

V2-045557 | MATHDB-371110

The asymptotic Davenport constant conjecture for Euclidean balls

摘录 / EXCERPT: Asymptotic ball Davenport conjecture. For $m\geq\sqrt d$,

conjecture | open | MathDB 页面 | 原文线索

V2-045558 | MATHDB-371128

The divisibility conjecture for the Gr_2-simplex

摘录 / EXCERPT: Arrange the entries in each layer of the Gr_2-simplex as coefficients of a polynomial in a,b,c according to the arrangement described in the paper. Divisibility conjecture. Eac...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045559 | MATHDB-371129

The reduced Gr_2-simplex recurrence conjecture

摘录 / EXCERPT: Let $xi_{\ell,r,c}$ denote the integer in the reduced Gr_2-simplex in layer ℓ, row r, and NE-SW diagonal c, and set $\xi_{\ell,r,c}=0$ when there is no entry in that pos...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045560 | MATHDB-371141

The parking-function formula for the second tree inversion enumerator

摘录 / EXCERPT: Let $I_n(q,t)$ and $\widetilde{I}_n(q,t)$ be the two t-analogues of the tree inversion enumerator, let $\mathrm{PF}(n)$ denote the set of parking functions of length n, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-045561 | MATHDB-371142

The simsun and zigzag refinements of the zigzag numbers

摘录 / EXCERPT: For $n\geq 1$, define $B(n,k)$ and $C(n,k)$ by ... and ... Here a simsun permutation is a permutation with the stated simsun property, and a big return is the stated statistic on an al...

conjecture | open | MathDB 页面 | 原文线索

V2-045562 | MATHDB-371155

Possible-size interval conjecture for sumsets

摘录 / EXCERPT: Let $\mathcal{R}(h,k)$ denote the set of possible cardinalities $|hA|$ for k-element sets A of integers, and let $\Delta_{h,k}$ be the exceptional set defined in the paper. Pos...

conjecture | open | MathDB 页面 | 原文线索

V2-045563 | MATHDB-371156

Extremal diameter conjecture for sumsets

摘录 / EXCERPT: Let $A\subseteq\mathbb{Z}$ be a finite set, let $|A|=k$, and write $\operatorname{diam}(A)=\max A-\min A=d$. For $hA=\{a_1+\cdots+a_h:a_i\in A\}$, extremal diameter conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-045564 | MATHDB-371162

Tan et al.'s labeled coupon collector difference conjecture

摘录 / EXCERPT: Let $Q_I(n,k)$ and $Q_{II}(n,k)$ denote the numbers of groups required to determine all coupon labels when the label set is known in advance and when it is not, respectively. For...

conjecture | open | MathDB 页面 | 原文线索

V2-045565 | MATHDB-371181

The first friendly-versus-antagonistic discrepancy outside the consecutive-ratio family

摘录 / EXCERPT: Let $S=\{s_2,s_1\}$ with $s_2<s_1<2s_2$, and suppose ... for every $k\in\mathbb{N}$. Let FvF and AvA denote the symmetric friendly and antagonistic self-interest games. First-discrep...

conjecture | open | MathDB 页面 | 原文线索

V2-045566 | MATHDB-371182

The inheritance of no discrepancy for dominant three-action subtraction sets

摘录 / EXCERPT: Let $S=\{s_3,s_2,s_1\}$ with $s_3<s_2<s_1$, and let a friendly/antagonistic discrepancy mean a difference between the FvF and AvA self-interest games. Dominant three-action inherit...

conjecture | open | MathDB 页面 | 原文线索

V2-045567 | MATHDB-371183

The additive three-action friendly-versus-antagonistic discrepancy criterion

摘录 / EXCERPT: Let $S=\{s_3,s_2,s_1\}$ with $s_3<s_2<s_1$, and suppose the largest subtraction is additive: ... Let a friendly/antagonistic discrepancy mean a difference between the FvF and AvA sel...

conjecture | open | MathDB 页面 | 原文线索

V2-045568 | MATHDB-371184

The three-action zero-sum discrepancy-free region

摘录 / EXCERPT: Let $S=\{s_3,s_2,s_1\}$ with $s_3<s_2<s_1$. Let o_{zs} denote the zero-sum scoring-play outcome and let the antagonistic self-interest game use AvA tie-breaking. Three-a...

conjecture | open | MathDB 页面 | 原文线索

V2-045569 | MATHDB-371197

The set-system edge-inducibility conjecture

摘录 / EXCERPT: For an integer ℓ and $p\in(0,1)$, let B_p be a random subset of a finite set B, formed by including each element independently with probability p, and define ... where the...

conjecture | open | MathDB 页面 | 原文线索

V2-045570 | MATHDB-371207

Day-Johnson nonexistence conjecture for equilibria in hypercube Voronoi games

摘录 / EXCERPT: Let Q_n be the discrete n-dimensional hypercube, and let a k-player profile be a tuple of k chosen vertices of Q_n. A profile is in equilibrium if no player can strictly...

conjecture | open | MathDB 页面 | 原文线索

V2-045571 | MATHDB-371208

Classification conjecture for bicompatible permutation statistics

摘录 / EXCERPT: Classification conjecture. Up to equivalence, the only bicompatible permutation statistics are the descent set Des, the peak set Pk, the valley se...

conjecture | open | MathDB 页面 | 原文线索

V2-045572 | MATHDB-371223

The 1/3-2/3 conjecture for linear extensions

摘录 / EXCERPT: Let P be a finite poset. For distinct elements $x,y\in P$, define ... and let ... where the probabilities are taken over a uniformly random linear extension of P. The 1/3-2/3 conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045573 | MATHDB-371224

Kahn-Saks conjecture on balancing probabilities in linear extensions

摘录 / EXCERPT: Let P be a finite poset, let $w(P)$ denote its width, and for distinct elements $x,y\in P$ define ... with ... where the probabilities are over a uniformly random linear extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-045574 | MATHDB-371225

The range conjecture for balancing probabilities in linear extensions

摘录 / EXCERPT: Let P be a finite poset. For $x\in P$, let $\Pi(x)=\{y:y\not\sim x\}$ and define the range parameter ... For distinct $x,y\in P$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-045575 | MATHDB-371226

Combinatorial dga model conjecture for surface categories

摘录 / EXCERPT: Let $(S,n,{\bf a})$ be a parametrized surface and let $R(S,n,{\bf a};k)$ be the dga generated by diagrams of k strands between the corresponding collections of arcs. Let μ_m...

conjecture | open | MathDB 页面 | 原文线索

V2-045576 | MATHDB-371232

Erdős-Falconer distance conjecture in even-dimensional finite vector spaces

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field with q elements, let $d\geq 2$ be even, and equip $\mathbb{F}_q^d$ with the quadratic form ... For $E\subseteq\mathbb{F}_q^d$, define its dista...

conjecture | open | MathDB 页面 | 原文线索

V2-045577 | MATHDB-371238

Tokushige's product-measure conjecture for cross t-intersecting families

摘录 / EXCERPT: Tokushige's conjecture. Let $p_1,p_2\in(0,\frac{1}{t+1})$. If $\mathcal F_1$ and $\mathcal F_2$ are cross t-intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-045578 | MATHDB-371239

The Gowers norm additive uncertainty conjecture

摘录 / EXCERPT: Let N,d be positive integers, let $f:\mathbb{Z}_N^d\to\mathbb{C}$ be a signal with support E and Fourier support Σ, and let $\|\cdot\|_{U^k}$ denote the Gowers U^k nor...

conjecture | open | MathDB 页面 | 原文线索

V2-045579 | MATHDB-371246

Maximum-size conjecture for strongly identity-forcing matrices

摘录 / EXCERPT: Let I_k be the $k\times k$ identity matrix. For an $n\times n$ matrix, let $\mathrm{M}(n,I_k)$ denote the maximum possible number of 1-entries in a strongly I_k-forcing matri...

conjecture | open | MathDB 页面 | 原文线索

V2-045580 | MATHDB-371252

Kalai-Meagher conjecture on intersecting families of triangulations

摘录 / EXCERPT: Kalai-Meagher conjecture. For every $n\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-045581 | MATHDB-371255

High-connectivity conjecture for completability and hyperconnectivity

摘录 / EXCERPT: Let d be a positive integer, and let G be a graph on n vertices. Write ${\mathcal{S}}_d(G)$ and ${\mathcal{H}}_d(G)$ for the completability and hyperconnectivity matroids in...

conjecture | open | MathDB 页面 | 原文线索

V2-045582 | MATHDB-371258

Blow-up conjecture for maximally imbalanced strongly playable RPS games

摘录 / EXCERPT: An m-player $(2n+1)$-RPS game is a game with $2n+1$ objects and m players interacting through the paper's RPS payoff structure. A blow-up of a game G_1 at an object repla...

conjecture | open | MathDB 页面 | 原文线索

V2-045583 | MATHDB-371261

Vanishing conjecture for the cubic generators of dual matroid toric ideals

摘录 / EXCERPT: Let L be a linear subspace of dimension r whose matroid $\mathrm{M}(L)$ is connected. Set $\mathcal E_L=\mathcal S_L\oplus\mathcal Q_L$. Vanishing conjecture for dual matroid c...

conjecture | open | MathDB 页面 | 原文线索

V2-045584 | MATHDB-371269

Multicolor Erdős-Hajnal conjecture

摘录 / EXCERPT: Multicolor Erdős-Hajnal conjecture. There exists $\varepsilon>0$ such that every coloring of the edges of K_n contains either k vertices whose edges are colored according to...

conjecture | open | MathDB 页面 | 原文线索

V2-045585 | MATHDB-371304

The population-preservation conjecture for consecutive pattern avoidance

摘录 / EXCERPT: For $m>0$, let $p_1,\ldots,p_k,p$ be consecutive patterns of length m. Define ... Suppose that $\textsc{pop}_{\mathcal{A}}(p)=0$.
Population-preservation conjecture. For every cons...

conjecture | open | MathDB 页面 | 原文线索

V2-045586 | MATHDB-371327

Berger-Aharoni conjecture on rainbow matchings

摘录 / EXCERPT: Let G be a bipartite graph, and let $M_1,\ldots,M_n$ be matchings in G, each of size n. A partial rainbow matching is a matching containing at most one edge from each M_i....

conjecture | open | MathDB 页面 | 原文线索

V2-045587 | MATHDB-371382

Stecher's cul-de-sac conjecture for rectangular-lattice HP folding

摘录 / EXCERPT: Let $J_{\mathrm{rect}}(x)$ denote the maximum number of points achievable for a binary word x in the two-dimensional rectangular lattice. For binary words x and y, Stecher's...

conjecture | open | MathDB 页面 | 原文线索

V2-045588 | MATHDB-371396

Dimension conjecture for box-product spaces of qMZV words

摘录 / EXCERPT: For positive integers z,d, let $\mathscr{S}_{z,d}$ be the $\mathbb{Q}$-span of the box products $u_{\mathbf n}\boxast u_{\boldsymbol\ell}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-045589 | MATHDB-371397

Box-kernel conjecture for qMZV box products

摘录 / EXCERPT: For positive integers z,d, let $\mathcal{K}_{z,d}$ be the explicitly defined subspace of the domain of the box map $\operatorname{Box}_{z,d}$ generated by the stuffle-box-product...

conjecture | open | MathDB 页面 | 原文线索

V2-045590 | MATHDB-371398

Basis conjecture for the box-kernel relations

摘录 / EXCERPT: For positive integers z,d with $z\leq d$, let $\mathcal{K}_{z,d}$ and $\operatorname{Box}_{z,d}$ be as defined in the source. Basis conjecture. If $z\leq d$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-045591 | MATHDB-371399

Conjectural basis for box-product spaces of qMZV words

摘录 / EXCERPT: Basis conjecture. For $1\leq z\leq d$, $\mathfrak{B}_{z,d}$ is a basis of $\mathscr{S}_{z,d}$. The set was obtained from numerical calculations, and the paper postpones a proof to...

conjecture | open | MathDB 页面 | 原文线索

V2-045592 | MATHDB-371432

The conjectured leading constant for the induced saturation number of the butterfly poset

摘录 / EXCERPT: Let $\mathcal N$ denote the poset considered in the paper, and let $\operatorname{sat}^*(n,\mathcal N)$ be its induced saturation number. Conjecture on $\mathcal N$. ... The authors...

conjecture | open | MathDB 页面 | 原文线索

V2-045593 | MATHDB-371448

Polynomial Hirsch conjecture for simplicial-complex diameter

摘录 / EXCERPT: Let $H_s(n,d)$ denote the maximum diameter of a d-dimensional simplicial complex on n vertices. The simplicial-complex diameter conjecture. For every $d\geq 3$, there is a thre...

conjecture | open | MathDB 页面 | 原文线索

V2-045594 | MATHDB-371467

The asymptotic simplicial mapping conjecture for spheres

摘录 / EXCERPT: Let T_1 be a triangulation of the n-sphere and let S_{n+2}^n be the boundary triangulation of the $(n+1)$-simplex. For a coloring $L:V(T_1)\to V(S_{n+2}^n)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-045595 | MATHDB-371476

Existence conjecture for Butson Hadamard matrices of order $2p+4$

摘录 / EXCERPT: A Butson Hadamard matrix $BH(n,q)$ is an $n\times n$ complex matrix whose entries are qth roots of unity and whose rows are pairwise orthogonal under the standard complex inner p...

conjecture | open | MathDB 页面 | 原文线索

V2-045596 | MATHDB-371484

The extremal solvabilizer conjecture for finite groups

摘录 / EXCERPT: Extremal solvabilizer conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045597 | MATHDB-371514

The generating function for words avoiding soft streaks

摘录 / EXCERPT: The generating-function conjecture. The generating function $f(z)$ is given by the displayed expression above.

conjecture | open | MathDB 页面 | 原文线索

V2-045598 | MATHDB-371515

Expected number of draws to obtain a soft streak

摘录 / EXCERPT: Expected-number conjecture. The expected number of letters drawn is given by these two equivalent expressions.

conjecture | open | MathDB 页面 | 原文线索

V2-045599 | MATHDB-371532

Ferroni-Schröter's real-rootedness conjecture for Chow polynomials

摘录 / EXCERPT: Let M be a matroid, and let its Chow polynomial be the polynomial invariant associated with its lattice of flats. Ferroni-Schröter's conjecture. The Chow polynomial of...

conjecture | open | MathDB 页面 | 原文线索

V2-045600 | MATHDB-371533

Athanasiadis-Kalampogia-Evangelinou's real-rootedness conjecture for Bergman-complex h-polynomials

摘录 / EXCERPT: Let M be a matroid, let $\mathcal{L}(\mathrm{M})$ be its lattice of flats, and let $\Delta(\widehat{\mathcal{L}}(\mathrm{M}))$ be the order complex of its proper nonempt...

conjecture | open | MathDB 页面 | 原文线索

V2-045601 | MATHDB-371534

Li's contraction interlacing conjecture for matroid Chow polynomials

摘录 / EXCERPT: Let $\mathcal{L}$ be a geometric lattice, let a be an atom of $\mathcal{L}$, and let $\underline{\mathrm{H}}_{[a,\widehat{1}]}$ and $\mathrm{H}_{[a,\widehat{1}]}$ denote the augm...

conjecture | open | MathDB 页面 | 原文线索

V2-045602 | MATHDB-371535

The TN-poset to UMEL-shellability conjecture for geometric lattices

摘录 / EXCERPT: Let M be a matroid. A TN-poset is a lower rank-uniform poset whose lower triangular matrix of lower Whitney numbers of the second kind is totally nonnegative, and UMEL-s...

conjecture | open | MathDB 页面 | 原文线索

V2-045603 | MATHDB-371548

Caoduro-Khodamoradi-Paat-Shepherd's oriented-matroid embracing-distance conjecture

摘录 / EXCERPT: Let M be an oriented matroid of rank r, let e be the distinguished element defining the embracing condition, and let A and B be bases of M. Call them e-embracing when...

conjecture | open | MathDB 页面 | 原文线索

V2-045604 | MATHDB-371549

Hamidoune's bounded symmetric-exchange conjecture

摘录 / EXCERPT: Let M be a matroid of rank r, and let (A,B) and (A',B') be compatible basis pairs, meaning that ... A symmetric exchange is an exchange of elements between the two bases that...

conjecture | open | MathDB 页面 | 原文线索

V2-045605 | MATHDB-371573

Conjecture on extremal intersections of curves in projective 4-space

摘录 / EXCERPT: Extremal-intersection conjecture. The number of intersection points of C_1 and C_2 is at most $B(d_1,d_2)$. Equality can only be achieved in one of the following situations: bo...

conjecture | open | MathDB 页面 | 原文线索

V2-045606 | MATHDB-371578

The generalized Grassmann-Plücker exchange conjecture for matroid bases

摘录 / EXCERPT: Let A and B be bases of a matroid $\mathbf{M}$ with rank function r. For any $X\subseteq A\setminus B$ and $Y\subseteq B\setminus A$ and integer p with $|Y| \le p \le |X|$,...

conjecture | open | MathDB 页面 | 原文线索

V2-045607 | MATHDB-371606

Irreducibility of extremal synchronizing automata

摘录 / EXCERPT: Let $\mathcal{A}=(Q,\Sigma,\delta)$ be a synchronizing automaton with $n=|Q|$. Its rank is the cardinality of the image of the state set under a word, and it is extremal if its sho...

conjecture | open | MathDB 页面 | 原文线索

V2-045608 | MATHDB-371608

Weak defectivity of extremal automata

摘录 / EXCERPT: Let $\mathcal{A}=(Q,\Sigma,\delta)$ be an extremal automaton, meaning an automaton whose shortest reset word reaches the relevant extremal bound. An automaton is weakly defective i...

conjecture | open | MathDB 页面 | 原文线索

V2-045609 | MATHDB-371615

The asymptotic generating-triple count conjecture for prefix reversals

摘录 / EXCERPT: For n and integers m,k with $2\leqslant k<m<n$, let $f(n)$ be the number of pairs (m,k) for which the prefix reversals r_n,r_m,r_k generate the symmetric group...

conjecture | open | MathDB 页面 | 原文线索

V2-045610 | MATHDB-371617

The maximum-row-size conjecture for prefix-reversal generating triples

摘录 / EXCERPT: For integers n and k with $n>k+1$, define ... Maximum-row-size conjecture. ... The conjecture is motivated by computational experiments and the displayed generating-triple patterns...

conjecture | open | MathDB 页面 | 原文线索

V2-045611 | MATHDB-371650

Cardinality and Ramsey-ultrafilter conjecture for Rudin-Keisler intervals

摘录 / EXCERPT: Cardinality and Ramsey-ultrafilter conjecture. For every ultrafilter $\boldsymbol u$,

open_problem | open | MathDB 页面 | 原文线索

V2-045612 | MATHDB-371659

Average-rarity conjecture for rooted-set representations

摘录 / EXCERPT: Let V be a finite ground set, and let $\mathcal C$ be an intersection-closed family on V containing $\varnothing$. Suppose that $\mathcal C$ admits a rooted-set representation...

conjecture | open | MathDB 页面 | 原文线索

V2-045613 | MATHDB-371702

Katznelson's negative answer conjecture for Bohr and topological recurrence

摘录 / EXCERPT: Katznelson's conjecture. There is a set $S\subset\mathbb{N}$ which is a set of Bohr recurrence but not a set of topological recurrence.

conjecture | open | MathDB 页面 | 原文线索

V2-045614 | MATHDB-371703

Finitary Abelian Cayley graph gap conjecture

摘录 / EXCERPT: Finitary Abelian Cayley graph conjecture. There exists an absolute constant c such that, for every constant C, it is NP-hard to distinguish whether an Abelian Cayley graph ha...

conjecture | open | MathDB 页面 | 原文线索

V2-045615 | MATHDB-371719

Hadamard-product fibre bounds for uniform matroids

摘录 / EXCERPT: For positive integers k_1,k_2, set $n=k_1+k_2-1$. Let $h_{k_1,k_2}(p)$ denote the number of pairs of matroids of ranks k_1 and k_2 on the ground set $[n]$, counted up to perm...

conjecture | open | MathDB 页面 | 原文线索

V2-045616 | MATHDB-371720

Hibi-Qureshi toric characterization of simple polyominoes

摘录 / EXCERPT: Hibi-Qureshi conjecture. The ideal $I_{\mathcal{P}}$ arises as the toric ideal of the edge ring of a finite simple graph if and only if $\mathcal{P}$ is simple.

conjecture | open | MathDB 页面 | 原文线索

V2-045617 | MATHDB-371726

Nill's facet-number conjecture for reflexive polytopes

摘录 / EXCERPT: Nill's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045618 | MATHDB-371738

Acyclicity conjecture for Gröbner-smoothable simplicial complexes

摘录 / EXCERPT: Acyclicity conjecture. If Γ is Gröbner smoothable over K, then Γ is acyclic over K.

conjecture | open | MathDB 页面 | 原文线索

V2-045619 | MATHDB-371777

Symmetric joint distribution of inversion and major-index statistics on colored derangements

摘录 / EXCERPT: Let $G_{c,n}$ be the group of colored derangements, and let $\widetilde{inv}$ and $fmaj$ denote the statistics on $G_{c,n}$. For indeterminates q and t, Symmetry conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-045620 | MATHDB-371807

The Thue list-number conjecture for the infinite path

摘录 / EXCERPT: Thue list-number conjecture. The Thue list number of the infinite path is 3.

conjecture | open | MathDB 页面 | 原文线索

V2-045621 | MATHDB-371808

The square-free word conjecture for sequences of ternary-or-larger alphabets

摘录 / EXCERPT: Square-free word conjecture. Given a sequence of alphabets $(\mathcal{A}_i)_{i\ge1}$ with $|\mathcal{A}_i|\ge3$ for all i, there exists an infinite square-free word that respects...

conjecture | open | MathDB 页面 | 原文线索

V2-045622 | MATHDB-371812

Irreducibility conjecture for extremal logarithmic 2-jet differentials

摘录 / EXCERPT: Irreducibility conjecture. For every $\epsilon>0$, there exists a nonzero negatively twisted invariant logarithmic 2-jet differential ω in this space with

conjecture | open | MathDB 页面 | 原文线索

V2-045623 | MATHDB-371845

Conjectured threshold for product-Schur subsets of random integer sets

摘录 / EXCERPT: Let $[n]_p$ be the binomial random subset of $[n]$, obtained by retaining each integer independently with probability p, and call $[n]_p$ product-Schur if every 2-colouring of...

conjecture | open | MathDB 页面 | 原文线索

V2-045624 | MATHDB-371866

A linear bound for zero-sum Ramsey numbers over composite moduli

摘录 / EXCERPT: Let k be a positive integer, let $c_k=c(k)$ be an integer, and let G be a graph on n vertices such that $k\mid E(G)$. The composite-modulus linear-bound conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-045625 | MATHDB-371876

The value of the extremal second-quotient constant for remainder-real entire functions

摘录 / EXCERPT: Extremal-constant problem. In the previous problem, $c=q_\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-045626 | MATHDB-371936

The polynomial Freiman-Ruzsa conjecture in characteristic two

摘录 / EXCERPT: Let $A\subset\mathbb{F}_{2}^{n}$ be a set with $|A+A|\leqslant K|A|$. Polynomial Freiman-Ruzsa conjecture in characteristic two. Then A is covered by at most $2K^{C}$ cosets of s...

conjecture | open | MathDB 页面 | 原文线索

V2-045627 | MATHDB-371946

Agrawal's conjecture on extremal triple arrays

摘录 / EXCERPT: Agrawal's conjecture. If there is a symmetric 2-$(r+c,r,d706_{cc})$ design with $r-d706_{cc}>2$, then there is an $(r\times c,r+c-1)$-triple array.

conjecture | open | MathDB 页面 | 原文线索

V2-045628 | MATHDB-371947

Agrawal's ordering conjecture for extremal unordered triple arrays

摘录 / EXCERPT: Agrawal's ordering conjecture. For any $(r\times c,r+c-1)$-unordered triple array U with $e>2$, there is an $(r\times c,r+c-1)$-triple array T with $U_T=U$.

conjecture | open | MathDB 页面 | 原文线索

V2-045629 | MATHDB-371950

The conjecture on energy boundedness and decay for Maxwell fields near-extremal Kerr

摘录 / EXCERPT: Fix $M>0$. Let $\eta\in(0,1)$ and $m\in\mathbb{Z}$. Let $J_2,J_3\geq J_1\geq J_{\mathrm{min}}\geq 1$, and let $C=C(m,M,J_1,J_2,J_3,\eta)>0$. For any $|a|\leq M$ and any solution...

conjecture | open | MathDB 页面 | 原文线索

V2-045630 | MATHDB-371963

Boundary Pre-Schwarzian extremal conjecture for the bounded integral means spectrum

摘录 / EXCERPT: Let $B_b(\tau)$ be the universal integral means spectrum for bounded univalent functions, let N_f denote the Pre-Schwarzian derivative of an extremal function f, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045631 | MATHDB-371964

Boundary Schwarzian extremal conjecture for the bounded integral means spectrum

摘录 / EXCERPT: Let $B_b(\tau)$ be the universal integral means spectrum for bounded univalent functions, let S_f denote the Schwarzian derivative of an extremal function f, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045632 | MATHDB-371984

Exponential decay conjecture for the crosscap-number defect

摘录 / EXCERPT: Exponential decay conjecture. Both proportions exponentially decay with respect to crossing number:

conjecture | open | MathDB 页面 | 原文线索

V2-045633 | MATHDB-372012

TAS ubiquity conjecture for oriented paths

摘录 / EXCERPT: TAS ubiquity conjecture. A $(\frac 1 2 - o_n(1))$ fraction of all oriented paths of length n are TAS.

conjecture | open | MathDB 页面 | 原文线索

V2-045634 | MATHDB-372025

The conjecture that the Stanley sequence from {0, 4} has subquadratic lower growth

摘录 / EXCERPT: Let A_4 be the Stanley sequence starting from the set $\{0,4\}$. Its growth rate is compared with the scale $n^2/\log n$. The conjecture for A_4. The sequence does not have gro...

conjecture | open | MathDB 页面 | 原文线索

V2-045635 | MATHDB-372027

Asymptotic density of binary shuffle squares

摘录 / EXCERPT: Binary shuffle-square density conjecture. As $n\to\infty$, asymptotically half of all binary words of length $2n$ are shuffle squares; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-045636 | MATHDB-372028

Asymptotic growth of shuffle squares over large alphabets

摘录 / EXCERPT: Large-alphabet shuffle-square growth conjecture. As a function of $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045637 | MATHDB-372031

Valuated bimatroid conjecture on entropy, tropical generation and probability measures

摘录 / EXCERPT: Valuated bimatroid conjecture. For every valuated bimatroid $\mathrm A$, the entropy vector is supermodular, $|\Sigma_{\mathrm A}|$ is the tropical semimodule over...

conjecture | open | MathDB 页面 | 原文线索

V2-045638 | MATHDB-372085

Maximum-size conjecture for 3-connected positroids excluding a uniform minor

摘录 / EXCERPT: Maximum-size conjecture. Every 3-connected rank-r positroid with no $U_{2,\ell+2}$-minor has at most

conjecture | open | MathDB 页面 | 原文线索

V2-045639 | MATHDB-372090

Alon's conjecture on Ramsey Cayley graphs

摘录 / EXCERPT: Let G be a finite group. A Cayley graph of G is formed from a symmetric subset $S\subset G$ by joining $x,y\in G$ when $x^{-1}y\in S$. Such a graph is C-Ramsey if it has no c...

conjecture | open | MathDB 页面 | 原文线索

V2-045640 | MATHDB-372091

The general abelian-structure conjecture for approximate groups

摘录 / EXCERPT: Let G be a group and let $A\subset G$ be a finite symmetric set satisfying ... A subset is commuting if all of its elements commute pairwise. The general abelian-structure conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-045641 | MATHDB-372117

The weak multiplicative Ramsey diagonal conjecture

摘录 / EXCERPT: Let $R(s,t)$ denote the two-colour Ramsey number. For positive integers s,t,k with $st\leqslant k^2$, the weak multiplicative Ramsey diagonal conjecture. ... This conjecture is pro...

conjecture | open | MathDB 页面 | 原文线索

V2-045642 | MATHDB-372118

The Ramsey diagonal conjecture

摘录 / EXCERPT: For positive integers s_1,s_2,t_1,t_2 satisfying ... and with $R(a,b)$ denoting the two-colour Ramsey number, the Ramsey diagonal conjecture. If ... then ... The conjecture asserts tha...

conjecture | open | MathDB 页面 | 原文线索

V2-045643 | MATHDB-372119

The multiplicative Ramsey diagonal conjecture

摘录 / EXCERPT: For positive integers s_1,s_2,t_1,t_2 satisfying ... and with $R(a,b)$ denoting the two-colour Ramsey number, the multiplicative Ramsey diagonal conjecture. If ... then ... The paper p...

conjecture | open | MathDB 页面 | 原文线索

V2-045644 | MATHDB-372160

Linear growth conjecture for the paper's sequence-production algorithm

摘录 / EXCERPT: Let u be a forbidden sequence, and let $s(n,u)$ be the sequence produced by the algorithm referenced in the source. Linear growth conjecture. The length of $s(n,u)$ grows at most...

conjecture | open | MathDB 页面 | 原文线索

V2-045645 | MATHDB-372168

Forbidden connected subsequences in bounded confidence processes

摘录 / EXCERPT: Let $(c_1,\dots,c_l)$ be a sequence of cluster sizes of a bounded confidence (BC) process. A subsequence is connected when its corresponding clusters are connected in the final con...

conjecture | open | MathDB 页面 | 原文线索

V2-045646 | MATHDB-372169

Half-density limit for the maximum number of clusters

摘录 / EXCERPT: Let $S(n)$ denote the maximum number of connected components (clusters) obtainable in a bounded confidence (BC) process with n agents. Half-density limit conjecture. ... Determinin...

conjecture | open | MathDB 页面 | 原文线索

V2-045647 | MATHDB-372172

First appearance degree conjecture for standard circulant matrices

摘录 / EXCERPT: Standard-circulant first appearance degree conjecture. For $n \geq 2$,

conjecture | solved | MathDB 页面 | 原文线索

V2-045648 | MATHDB-372173

First appearance value conjecture for standard circulant matrices

摘录 / EXCERPT: Standard-circulant first appearance value conjecture. For $n \geq 2$,

conjecture | solved | MathDB 页面 | 原文线索

V2-045649 | MATHDB-372174

First appearance degree conjecture for row-shifted second-power lattices

摘录 / EXCERPT: Row-shifted second-power lattice degree conjecture. For $n \geq 2$ and any positive integer d,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045650 | MATHDB-372175

Unified first appearance formula for row-shifted second-power lattices

摘录 / EXCERPT: Row-shifted second-power lattice value conjecture. For $n \geq 2$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045651 | MATHDB-372176

First appearance value conjecture for Hilbert matrices

摘录 / EXCERPT: Hilbert-matrix first appearance value conjecture. For $n \geq 2$,

conjecture | solved | MathDB 页面 | 原文线索

V2-045652 | MATHDB-372177

Harmonic APD relation between Hilbert and identity matrices

摘录 / EXCERPT: Harmonic APD relation. For $n \geq 1$,

conjecture | solved | MathDB 页面 | 原文线索

V2-045653 | MATHDB-372178

First appearance value conjecture for multiplication-table lattices

摘录 / EXCERPT: Multiplication-table lattice value conjecture. For $n \geq 2$,

conjecture | solved | MathDB 页面 | 原文线索

V2-045654 | MATHDB-372179

Unified APD formula for row-shifted second-power lattices

摘录 / EXCERPT: Unified row-shifted second-power lattice conjecture. For $n \geq 2$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045655 | MATHDB-372180

First appearance value conjecture for standard Vandermonde matrices

摘录 / EXCERPT: Standard-Vandermonde value conjecture. For $n \geq 2$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045656 | MATHDB-372181

Determinant formula for the first appearance value of standard Vandermonde matrices

摘录 / EXCERPT: Standard-Vandermonde determinant formula. The first appearance value is

conjecture | solved | MathDB 页面 | 原文线索

V2-045657 | MATHDB-372214

Chan-Kontorovich-Pak spanning-tree spectrum conjecture

摘录 / EXCERPT: Let $\mathcal{T}(n)$ denote the set of values of the spanning-tree count $\tau(G)$ over unrestricted graphs on n vertices. Cayley's theorem gives $\tau(K_n)=n^{n-2}$, so the poss...

conjecture | open | MathDB 页面 | 原文线索

V2-045658 | MATHDB-372236

Framing-lattice invariance of linear-interval enumerators

摘录 / EXCERPT: Let G be a graph, and let F_1 and F_2 be framings of G. For each $k=0,1,2,\ldots$, consider the linear intervals of length k in the framing lattices $\mathscr{L}_{G,F_1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-045659 | MATHDB-372242

The transposed Catalan triangle formula for partial-shuffle avoidance classes

摘录 / EXCERPT: Transposed Catalan triangle formula. The number of such permutations is

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045660 | MATHDB-372253

Stability conjecture for packed words in the plactic monoid

摘录 / EXCERPT: A word u has maximum letter m and is m-packed if it contains at least one copy of every integer in $[m]=\{1,\ldots,m\}$. A word u is m-stable when its centralizers satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-045661 | MATHDB-372280

The $3k-4$ conjecture for subsets of $\mathbb F_p$

摘录 / EXCERPT: $3k-4$ conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045662 | MATHDB-372355

Strong boundary-word existence conjecture

摘录 / EXCERPT: For each integer $n\ge 5$, let ${\bf A}_n$ be the n-letter alphabet, ${\bf A}_n^{\omega}$ the set of infinite words over it, ${\bf T}_n$ the class of boundary words, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045663 | MATHDB-372365

Conjecture on universal polar dual pairs of spherical codes

摘录 / EXCERPT: Let C be a PULB-optimal code, and let D be its set of universal minima. Assume that D is in general position, meaning that D is not contained in a hyperplane. Universal pol...

conjecture | open | MathDB 页面 | 原文线索

V2-045664 | MATHDB-372378

Conjecture for the Mattila-Sjölin problem for k-distance sets

摘录 / EXCERPT: The conjecture for the Mattila-Sjölin problem for k-distance sets. Assume that C is a sufficiently large constant independent of q. Then, for every integer $k=1,2,\ldots,d$,...

conjecture | open | MathDB 页面 | 原文线索

V2-045665 | MATHDB-372418

Xin-Zhang's tridiagonal determinant conjecture

摘录 / EXCERPT: Xin-Zhang's determinant conjecture. The determinant of C is

conjecture | open | MathDB 页面 | 原文线索

V2-045666 | MATHDB-372426

Profile conjecture for $S_{2,1}$-good graph sequences

摘录 / EXCERPT: Let $S_{2,1}$ be the relevant colored graph, and let an $S_{2,1}$-good sequence (G_n) have limiting edge density β and profile $\rho(S_{2,1},(G_n))$. Define the constructio...

conjecture | open | MathDB 页面 | 原文线索

V2-045667 | MATHDB-372428

Dowling's polynomial conjecture for independent-set polynomials of matroids

摘录 / EXCERPT: Let $M=(E,\mathcal{I})$ be a matroid with $|E|=n$ and rank r_M. For $0\leq k\leq r_M$, define the homogeneous multivariate independent-set polynomial ... For polynomials in...

conjecture | open | MathDB 页面 | 原文线索

V2-045668 | MATHDB-372429

The strengthened polynomial log-concavity conjecture for independent-set polynomials

摘录 / EXCERPT: Let $M=(E,\mathcal{I})$ be a matroid with rank r_M, and let $f_k(M)$ be its degree-k independent-set polynomial ... For $0<k<r_M$, the strengthened polynomial log-concavity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045669 | MATHDB-372437

The forbidden-configuration bound for binary matrices

摘录 / EXCERPT: Let t be given, and let ${\mathrm{forb}}(m,F(0,2^{t-2}+1,1,0))$ denote the maximum number of columns in an m-rowed binary matrix avoiding the configuration $F(0,2^{t-2}+1,1,0)$...

conjecture | open | MathDB 页面 | 原文线索

V2-045670 | MATHDB-372439

Vanishing-Λ black hole stability conjecture

摘录 / EXCERPT: Let ${\bf g}_{M,a}$ be a subextremal Kerr metric. Let (Σ_0,g_0,k_0) be a suitably small perturbation of its initial data, where Σ_0 is a 3-dimensional manifold, g_0...

conjecture | open | MathDB 页面 | 原文线索

V2-045671 | MATHDB-372442

Kirillov's saturation conjecture for Schubert coefficients

摘录 / EXCERPT: Kirillov's conjecture. For every $u,v,w\in S_n$ and every integer $N\geq 1$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045672 | MATHDB-372462

The super-exponential gap conjecture for off-diagonal Ramsey numbers

摘录 / EXCERPT: Let $R(\ell,k)$ be the two-colour Ramsey number, and suppose that k and ℓ are sufficiently large with $\log k\ll\ell\ll k$. Super-exponential gap conjecture. For every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-045673 | MATHDB-372463

The absolute-constant conjecture for multicolour Ramsey numbers

摘录 / EXCERPT: Let $R_r(k)$ denote the Ramsey number such that every r-colouring of the edges of the complete graph on $R_r(k)$ vertices contains a monochromatic copy of K_k. Absolute-constan...

conjecture | open | MathDB 页面 | 原文线索

V2-045674 | MATHDB-372467

Existence and value of the arithmetic rainbow Ramsey multiplicity limit for addition

摘录 / EXCERPT: For a positive integer n, let $[n]=\{1,2,\ldots,n\}$, and let $\operatorname{rb}_{[n]}(x+y=z)$ denote the maximum proportion of solutions to $x+y=z$ in $[n]$ that are rainbow und...

conjecture | open | MathDB 页面 | 原文线索

V2-045675 | MATHDB-372480

The sharp permutation anticoncentration conjecture for sets

摘录 / EXCERPT: Let $A=\{a_1,\dots,a_n\}$ and $B=\{b_1,\dots,b_n\}$ be sets of n distinct real numbers, and let π be uniform in S_n. For a real-valued random variable X, write...

conjecture | open | MathDB 页面 | 原文线索

V2-045676 | MATHDB-372491

Generalized sumset pattern conjecture for multiple uniform sets

摘录 / EXCERPT: Generalized sumset pattern conjecture. There exists an infinite set $B\subset\mathbb N$ such that, for every $i=1,\ldots,k$,

conjecture | open | MathDB 页面 | 原文线索

V2-045677 | MATHDB-372494

Erdős's bounded-error conjecture for dissociated sets

摘录 / EXCERPT: Erdős's conjecture. For a dissociated set $$, there is a fixed constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-045678 | MATHDB-372509

Davenport constant formula for intervals containing zero

摘录 / EXCERPT: Davenport constant conjecture for intervals.

conjecture | open | MathDB 页面 | 原文线索

V2-045679 | MATHDB-372521

Asymptotic normality conjecture for unordered rational-curve moduli spaces

摘录 / EXCERPT: Asymptotic normality conjecture. The Betti numbers of these three quotient spaces are asymptotically normally distributed, and the associated variances grow linearly at the same ra...

conjecture | open | MathDB 页面 | 原文线索

V2-045680 | MATHDB-372528

Extremal-process moderate-deviation conjecture for branching random walks

摘录 / EXCERPT: Extremal-process moderate-deviation conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-045681 | MATHDB-372536

The bottom-triangle row-count conjecture for chip-firing configurations

摘录 / EXCERPT: A configuration $F(x,y)$ consists of rows of chips, with the bottom triangle formed by the minimal rows at the bottom of the configuration. Let the longest row be a row having maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-045682 | MATHDB-372576

A bound on the clique parameters of geometric distance-regular graphs

摘录 / EXCERPT: Let Γ be a geometric distance-regular graph with diameter D and distinct eigenvalues ... Let $b = \frac{b_1}{\theta_1 + 1}$. For a Delsarte clique, let ϕ_i denote the n...

conjecture | open | MathDB 页面 | 原文线索

V2-045683 | MATHDB-372577

A bound on the intersection parameters of geometric distance-regular graphs

摘录 / EXCERPT: Let Γ be a geometric distance-regular graph with diameter $D \geq 3$ and distinct eigenvalues ... Let $b = \frac{b_1}{\theta_1 + 1}$. For vertices x,y at distance i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-045684 | MATHDB-372578

Koolen-Bang finiteness conjecture for geometric distance-regular graphs

摘录 / EXCERPT: Let $m \geq 2$ be a fixed integer. A distance-regular graph is coconnected if it is not a disjoint union of complete graphs. Let ϕ_1 and τ_2 be the clique parameters asso...

conjecture | open | MathDB 页面 | 原文线索

V2-045685 | MATHDB-372579

Koolen-Bang classification conjecture for geometric distance-regular graphs

摘录 / EXCERPT: Let $m \geq 2$ be a fixed integer. A geometric distance-regular graph has diameter $D \geq 3$ and intersection number $c_2 \geq 2$. Koolen-Bang's classification conjecture. Any geo...

conjecture | open | MathDB 页面 | 原文线索

V2-045686 | MATHDB-372589

Refined decomposition conjecture for alternating sign matrix enumerations

摘录 / EXCERPT: Let $A_n(z,\rho):=A_n(z,\rho,1)$, and for odd order define ... where $\operatorname{HSASM}(2m+1)$ is the set of horizontally symmetric ASMs of order $2m+1$, $\mu(A)$ is the statistic...

conjecture | open | MathDB 页面 | 原文线索

V2-045687 | MATHDB-372629

Asymptotic conjecture for Coxeter sorting time

摘录 / EXCERPT: Let ${\operatorname{Sort}}_n$ denote the Coxeter sorting time for permutations of size n. Asymptotic sorting-time conjecture. ... This conjecture seeks an asymptotically sharp esti...

conjecture | open | MathDB 页面 | 原文线索

V2-045688 | MATHDB-372630

Extremal Sylow 2-subgroups conjecture for even-cycle-intersecting families

摘录 / EXCERPT: Let S_n be the symmetric group on n elements. A family of permutations is even-cycle-intersecting when every pair of permutations has a common cycle of even length in the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-045689 | MATHDB-372666

The conjecture on achievable burning densities of growing grids in dimensions greater than two

摘录 / EXCERPT: Let $f: \mathbb{N} \to \mathbb{N}$ be a strictly increasing function, let ... for some $d>2$, and let $\mathcal G(f)=(G_n,n\geq 1)$. Write $P(\mathcal G(f))$ for the set of achievabl...

conjecture | open | MathDB 页面 | 原文线索

V2-045690 | MATHDB-372675

The monotone limit-object conjecture for matrix Turán density

摘录 / EXCERPT: Let H be a matrix whose entries are ordered so that $H(i,j)\le H(i',j')$ whenever $i'\ge i$ and $j'\ge j$; call such a matrix monotone. If $||H||$ is its largest symbol, define...

conjecture | open | MathDB 页面 | 原文线索

V2-045691 | MATHDB-372676

The binary minimizer conjecture for matrix Turán density

摘录 / EXCERPT: For each positive integer h, consider $h\times h$ matrices minimizing the relevant asymptotic matrix Turán density. A matrix is binary if all its entries belong to $\{0,1\}$. Bin...

conjecture | open | MathDB 页面 | 原文线索

V2-045692 | MATHDB-372741

Equivariant unimodality conjecture for inverse Z-polynomials of matroids

摘录 / EXCERPT: Equivariant unimodality conjecture. For any matroid M equipped with an action of a finite group W, the coefficients of its equivariant inverse Z-polynomial are equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-045693 | MATHDB-372749

Short rainbow circuit conjecture for regular matroids

摘录 / EXCERPT: Short rainbow circuit conjecture. If (M,c) has no colour-singular elements, then M has a short rainbow circuit. This conjecture extends the corresponding rank-two case from gra...

conjecture | open | MathDB 页面 | 原文线索

V2-045694 | MATHDB-372756

A Kneser-Pollard theorem for higher representation sums

摘录 / EXCERPT: Let t be a positive integer, let G be an abelian group, and let $A,B\subseteq G$ be finite subsets with $|A|,|B|\geq t$. Write $A+_iB$ for the set of elements of G having at...

conjecture | open | MathDB 页面 | 原文线索

V2-045695 | MATHDB-372759

Extension conjecture for type B parking-space representations

摘录 / EXCERPT: Let B_n be the signed symmetric group on n letters, with inclusion $B_n\subset B_{n+1}$ given by choosing the elements of B_{n+1} that fix the $(n+1)$st letter. For...

conjecture | open | MathDB 页面 | 原文线索

V2-045696 | MATHDB-372772

Filliman's extremal projection conjecture for the regular simplex

摘录 / EXCERPT: Filliman's conjecture. The minimum k-dimensional projection volume is attained by $T_{\ast}$:

conjecture | open | MathDB 页面 | 原文线索

V2-045697 | MATHDB-372797

Baranyai's partition characterisation of (3,6)-tight graphs

摘录 / EXCERPT: The partition characterisation conjecture. The graph G is $(3,6)$-tight if and only if, for every edge $e\in E$, there exists a partition (S_1,S_2,S_3) of E such that

conjecture | open | MathDB 页面 | 原文线索

V2-045698 | MATHDB-372801

Exact-value conjecture for Kruskal's weak tree function tree(3)

摘录 / EXCERPT: Let $\mathrm{tree}(3)$ denote the weak tree function at 3, and let ... The paper establishes $\mathrm{tree}(3)\geq N$. Exact-value conjecture. ... This conjecture asks whether the ex...

conjecture | open | MathDB 页面 | 原文线索

V2-045699 | MATHDB-372813

The high-dimensional point-line induced matching conjecture

摘录 / EXCERPT: Let $\mathrm{IM}(d,q)$ denote the maximum size of an induced matching in the point-line incidence graph of $\mathbb{F}_q^d$. High-dimensional induced matching conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-045700 | MATHDB-372827

Clique-bounded degree conjecture for derangement graphs

摘录 / EXCERPT: Let G) be a transitive permutation group of degree n, and let Γ_G denote its derangement graph. For a positive integer c, say that Γ_G contains no clique of siz...

conjecture | open | MathDB 页面 | 原文线索

V2-045701 | MATHDB-372829

Expectation conjecture for mandatory and blocking edges in Erdős-Rényi graphs

摘录 / EXCERPT: Mandatory and blocking edge expectation conjecture. The expected proportions of edges belonging to every maximum-size matching and of edges belonging to no maximum-size matching sa...

conjecture | open | MathDB 页面 | 原文线索

V2-045702 | MATHDB-372830

General expectation conjecture for mandatory and blocking edges

摘录 / EXCERPT: General mandatory and blocking edge expectation conjecture. The expected proportions of edges belonging to every maximum-size matching and of edges belonging to no maximum-size mat...

conjecture | open | MathDB 页面 | 原文线索

V2-045703 | MATHDB-372833

Björner-Las Vergnas-Sturmfels-White-Ziegler conjecture on bichromatic triangles

摘录 / EXCERPT: Björner-Las Vergnas-Sturmfels-White-Ziegler conjecture. Every bicolored arrangement has a bichromatic triangle.

conjecture | open | MathDB 页面 | 原文线索

V2-045704 | MATHDB-372834

Connectivity conjecture for the triangle-pseudoline incidence graph

摘录 / EXCERPT: Triangle-pseudoline incidence graph conjecture. The triangle-pseudoline incidence graph of a pseudoline arrangement is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-045705 | MATHDB-372835

Conjecture on bichromatic triangles or quadrangles around a blue polygon

摘录 / EXCERPT: Blue-polygon bichromatic-face conjecture. Given a blue k-gon P with at least $k-4$ red pseudolines passing through, there exists a bichromatic triangle or quadrangle on its ins...

conjecture | open | MathDB 页面 | 原文线索

V2-045706 | MATHDB-372906

The polynomial-time solvability conjecture for totally -modular integer programs

摘录 / EXCERPT: Totally Δ-modular integer-program conjecture. For any constant Δ, this integer program can be solved in polynomial time when A is totally Δ-modular.

conjecture | open | MathDB 页面 | 原文线索

V2-045707 | MATHDB-372907

The bounded-entry tree-structured integer-programming conjecture

摘录 / EXCERPT: Bounded-entry tree integer-programming conjecture. The integer program can be solved in polynomial time for constant d.

conjecture | open | MathDB 页面 | 原文线索

V2-045708 | MATHDB-372916

Transitivity conjecture for classical r-matrices with signs

摘录 / EXCERPT: Let n be a positive integer, let ${\mathfrak g}$ be a Lie algebra, and let $r\in {\mathfrak g}\otimes {\mathfrak g}$ solve the classical Yang-Baxter equation (CYBE). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045709 | MATHDB-372917

Transitivity conjecture for quantum R-matrices with signs

摘录 / EXCERPT: Transitivity conjecture. The element $\mathbf R^{(\boldsymbol{\epsilon})}$ solves the QYBE. This is expected to provide a large noncommutative class of QYBE solutions analogous to...

conjecture | open | MathDB 页面 | 原文线索

V2-045710 | MATHDB-372920

Support description conjecture for almost vexillary permutations

摘录 / EXCERPT: Almost vexillary support conjecture. For every almost vexillary permutation w, there exists a subset $A\subset D(w)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-045711 | MATHDB-372925

Three-term progression distance conjecture for fractal measures

摘录 / EXCERPT: Let $ u$ be a compactly supported probability measure on $\mathbb{R}^d$ satisfying ... and ... Assume that $\alpha>2d/3$. Three-term progression distance conjecture. Under appropriate...

conjecture | open | MathDB 页面 | 原文线索

V2-045712 | MATHDB-372994

Pak's conjecture on the size-Ramsey number of long subdivisions

摘录 / EXCERPT: Given a graph H and a function $c3:E(H)\to\mathbb{N}$, the subdivision H^{σ} is obtained by replacing each edge $e\in E(H)$ with a path of length $\sigma(e)$. For a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-045713 | MATHDB-373016

The unpinning avoidance game PSPACE-completeness conjecture

摘录 / EXCERPT: Unpinning avoidance game complexity conjecture. The problem $\mathsf{Unpin_{AVOID}}(\Sigma,s)$ is $\textsf{PSPACE}$-complete for any s and Σ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045714 | MATHDB-373017

The perfect Prishchepov group type conjecture

摘录 / EXCERPT: The perfect Prishchepov group type conjecture. If $P(r,n,k,r-1,q)$ is perfect, then it is of type $\widetilde{\mathfrak{Z}}$. This conjecture predicts that every perfect nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-045715 | MATHDB-373025

Positivity conjecture for cumulants of minimum matching costs

摘录 / EXCERPT: Let $C_{k,m,n}$ be the minimum cost of a k-matching in the random bipartite matching model, and let its cumulants κ_p be defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-045716 | MATHDB-373026

Zero-free disk conjecture for matching moment generating functions

摘录 / EXCERPT: Let $F_{k,m,n}(t)=\mathbb{E}(\exp(t\cdot C_{k,m,n}))$ be the moment generating function of the minimum cost $C_{k,m,n}$ of a k-matching in the random bipartite matching model. Ze...

conjecture | open | MathDB 页面 | 原文线索

V2-045717 | MATHDB-373047

The tropical duality conjecture for tropical linear spaces and quotients

摘录 / EXCERPT: Let p be a valuated matroid of rank d on $\mathbb{T}^n$, let L_p be its corresponding tropical linear space, and define ... Here $L_p^\vee$ denotes the tropical dual of L_p....

conjecture | open | MathDB 页面 | 原文线索

V2-045718 | MATHDB-373069

Miller-Vissuet's MSTD-MDTS conjecture for dihedral groups

摘录 / EXCERPT: Let $n\geq 3$ be an integer, and let ... be the dihedral group of order $2n$. An MSTD subset is a subset with more distinct pairwise sums than distinct pairwise differences, while an...

conjecture | open | MathDB 页面 | 原文线索

V2-045719 | MATHDB-373070

Growth conjecture for MSTD subsets of dicyclic groups

摘录 / EXCERPT: Let $m\geq 2$ be an integer. For each $n\geq 3$, let Dic_{4n} be the dicyclic group of order $4n$, and let $\mathrm{S}_{\mathrm{Dic}_{4n}}(m)$ denote the number of MSTD...

conjecture | open | MathDB 页面 | 原文线索

V2-045720 | MATHDB-373097

Strong induced log-concavity conjecture for equivariant thagomizer Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Fix $n\ge 0$, and let T_n be the thagomizer matroid with the action of the hyperoctahedral group $\mathcal{B}_n$. Write $P_{T_n}^{\mathcal{B}_n}(t)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045721 | MATHDB-373098

Conjecture on the number of classes for curve orbits on a smooth Hermitian surface

摘录 / EXCERPT: Class-number conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045722 | MATHDB-373143

Stability conjecture for consecutive permutation patterns

摘录 / EXCERPT: Stability conjecture. A consecutive pattern is stable if and only if it is monotone.

conjecture | open | MathDB 页面 | 原文线索

V2-045723 | MATHDB-373153

Lengyel's period conjecture for two transfer games

摘录 / EXCERPT: Lengyel's conjecture. (a) If b is odd and at least 3, then $L(b;b,0;1,1)$ has period $(2b(b+1),2b)$. (b) For $b\geq 2$, $L(b;1,0;1,1)$ has period $(4b,2b)$.

conjecture | open | MathDB 页面 | 原文线索

V2-045724 | MATHDB-373154

Main horizontal-period conjecture for Lengyel transfer games

摘录 / EXCERPT: Main horizontal-period conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-045725 | MATHDB-373156

Dafermos-Holzegel-Rodnianski conjecture on scattering solutions of the Einstein vacuum equations

摘录 / EXCERPT: Let $\mathcal{H}$ and $\mathcal{I}$ denote the horizon and null infinity, respectively. Consider smooth scattering data settling down to a possibly extremal Kerr solution on...

conjecture | open | MathDB 页面 | 原文线索

V2-045726 | MATHDB-373176

Block-monomial automorphism conjecture for Kimura Hadamard matrices

摘录 / EXCERPT: Let H be a Kimura Hadamard matrix (KHM) of order $8k+4$. An automorphism is a pair (R,S) acting on H by the relation $RH=HS$. The known automorphisms have block-diagonal form...

conjecture | open | MathDB 页面 | 原文线索

V2-045727 | MATHDB-373188

Unimodality conjecture for atom counts in numerical power monoids

摘录 / EXCERPT: Let H be a numerical monoid, and for $k,l\in\mathbb N$ let $\alpha_{k,l}(H)$ be the number of k-element atoms of $\mathcal P_{\mathrm{fin},0}(H)$ whose maximum is at most l....

conjecture | open | MathDB 页面 | 原文线索

V2-045728 | MATHDB-373205

The inverse Littlewood conjecture for structure of low-norm sets

摘录 / EXCERPT: Let $A\subset\mathbb{Z}$ be a finite set of size N, and let $K>0$. Write 1_A for its indicator and define ... A finite arithmetic progression is a finite subset of $\mathbb{Z}$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-045729 | MATHDB-373206

The inverse Littlewood conjecture for dissociated dimension

摘录 / EXCERPT: Let $A\subset\mathbb{Z}$ be a finite set of size N, and define $\dim A$ to be the size of the largest dissociated subset of A. Let $K>0$. The inverse Littlewood dimension conje...

conjecture | open | MathDB 页面 | 原文线索

V2-045730 | MATHDB-373207

The pivot-layer colouring conjecture for diagonal poset Ramsey numbers

摘录 / EXCERPT: Let Q_n denote the diagonal poset on n elements, and let $R(Q_n,Q_n)$ be its two-colour Ramsey number. For a positive integer L, consider the colouring construction in which...

conjecture | open | MathDB 页面 | 原文线索

V2-045731 | MATHDB-373234

Xu-Yip-Zhang conjecture on uniform witness families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $0\leqslant s\leqslant d$, and let $\mathcal{F}\subseteq\binom{[n]}{d+1}$. An s-witness family is a family such that for every $F\in\mathcal{F}$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-045732 | MATHDB-373237

Böröczky extremality conjecture for monochromatic lines

摘录 / EXCERPT: Let P be a non-collinear set of n two-colored points, and suppose that no line contains more than $(1/2-delta)n$ points, where $delta>0$. Böröczky extremality conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-045733 | MATHDB-373238

Milićević's cubic-curve conjecture for bichromatic point configurations

摘录 / EXCERPT: Let n blue points, no three on a line, and n red points disjoint from the blue points be given. Assume that every line through two blue points contains a red point. Milićević's...

conjecture | open | MathDB 页面 | 原文线索

V2-045734 | MATHDB-373249

The budget-threshold conjecture for constructing a copy of K_5

摘录 / EXCERPT: Let n be the number of vertices, let $[M]$ denote the set of admissible times, let t be a time parameter, and let b be a budget parameter. A (t,b)-strategy is a strategy in...

conjecture | open | MathDB 页面 | 原文线索

V2-045735 | MATHDB-373251

Song-Wang's cyclotomic RSK bijection conjecture

摘录 / EXCERPT: For a level ℓ and multipartition data μ,ν, let $\mathrm{ParMat}_{\nu,\mu}^{\flat}$ denote the corresponding set of enriched block matrices, let...

conjecture | open | MathDB 页面 | 原文线索

V2-045736 | MATHDB-373255

Følner-sequence density conjecture for sumsets in the integers

摘录 / EXCERPT: Let $A,B subseteq ewline \mathbb Z$ be nonempty, and let $\mathbf F=(F_n)_{n\in\mathbb N}$ be a Følner sequence of finite subsets of $\mathbb Z$, meaning that ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-045737 | MATHDB-373268

Stationary-measure and phase-diagram conjecture for the half-space log-Gamma polymer

摘录 / EXCERPT: Stationary-measure and phase-diagram conjecture. The stated family gives all extremal stationary measures, and the three parameter regimes give the corresponding weak limits of...

conjecture | open | MathDB 页面 | 原文线索

V2-045738 | MATHDB-373282

Existence conjecture for $(1,1,2\pi[D])$-extremal Poincaré-type Kähler metrics

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold and let D be a smooth divisor. Let T be a maximal compact torus in the reduced automorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-045739 | MATHDB-373314

Asymptotic multichannel conflict-avoiding code conjecture

摘录 / EXCERPT: Let M and w be positive integers with $w\geq M$, and let $K(M,L,w)$ denote the maximum number of codewords in an M-channel conflict-avoiding code of length L and weight w...

conjecture | open | MathDB 页面 | 原文线索

V2-045740 | MATHDB-373315

Enumeration conjecture for nondegenerate hypermatrices avoiding a plane partition

摘录 / EXCERPT: Enumeration conjecture. The number of nondegenerate $2\times (k+1)\times k$ hypermatrices over $\mathbb{F}_q$ which respect P is

conjecture | open | MathDB 页面 | 原文线索

V2-045741 | MATHDB-373316

The combinatorial low-degree correlation conjecture for pairwise-connected distributions

摘录 / EXCERPT: Let Σ be a finite alphabet, let $k\in\mathbb{N}$, and let μ be a pairwise-connected distribution over Σ^k in which every atom has mass at least α. A functi...

conjecture | open | MathDB 页面 | 原文线索

V2-045742 | MATHDB-373327

Switching-and-scaling equivalence conjecture for canonical gain-graph matrices

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, let G be a loopless graph with orientation D, and let ψ and ψ' be $(\mathbb{F}^+\rtimes \mathbb{F}^{\times})$-gain functions for G. Sup...

conjecture | open | MathDB 页面 | 原文线索

V2-045743 | MATHDB-373328

Representability by canonical gain-graph incidence matrices

摘录 / EXCERPT: Let $(G,\mathcal{B})$ be a biased graph, let M be an elementary lift of the frame matroid $F(G,\mathcal{B})$, and suppose that M is 3-connected. Let $\mathbb{F}$ be a field a...

conjecture | open | MathDB 页面 | 原文线索

V2-045744 | MATHDB-373329

Universal-model conjecture for Frobenius-group gain-graph matroid classes

摘录 / EXCERPT: Let Γ be a Frobenius group with Frobenius kernel Γ_1, and let $\mathcal{A}$ be a set of Frobenius complements. A sequence of universal models for a class...

conjecture | open | MathDB 页面 | 原文线索

V2-045745 | MATHDB-373330

The three-tree separation conjecture for bounded-degree trees

摘录 / EXCERPT: For integers $r\geq 2$, let $f(r,3)$ denote the limiting maximum proportion of vertices that can be guaranteed in each of two disjoint subsets separated by a cut vector for any thr...

conjecture | open | MathDB 页面 | 原文线索

V2-045746 | MATHDB-373336

Boolean Matrix Multiplication conjecture for combinatorial triangle detection

摘录 / EXCERPT: The Triangle Detection problem asks whether an n-node graph contains a triangle. A strongly subcubic combinatorial algorithm is a combinatorial algorithm whose running time is...

conjecture | refuted | MathDB 页面 | 原文线索

V2-045747 | MATHDB-373355

Strong light tails conjecture for Erdős sieves

摘录 / EXCERPT: Let K be the number field underlying the ring of integers $\mathcal{O}_K$, let R be an Erdős sieve, and let B_N be the Følner sequence used to define light tails. For a set...

conjecture | open | MathDB 页面 | 原文线索

V2-045748 | MATHDB-373403

Type B real-rootedness conjecture for the polynomials B_n

摘录 / EXCERPT: Let B_n be the type B polynomials defined in the paper, for $n\geq 1$. A polynomial is real-rooted if all of its roots are real. Type B real-rootedness conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-045749 | MATHDB-373405

The extremal ML degree bound for graphical models

摘录 / EXCERPT: Extremal ML degree bound. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-045750 | MATHDB-373406

The extremal ML threshold relation for graphical models

摘录 / EXCERPT: Extremal ML threshold relation. For any undirected graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-045751 | MATHDB-373411

Hegedüs' conjecture on skew Bollobás systems of projective subspaces

摘录 / EXCERPT: Hegedüs' conjecture. Every such system satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045752 | MATHDB-373412

Asymptotic formula for packing arithmetic progressions

摘录 / EXCERPT: Let $m(n)$ denote the maximum size of a packing of arithmetic progressions in the setting of the paper. Asymptotic packing conjecture. ... This conjecture seeks the leading constant...

conjecture | open | MathDB 页面 | 原文线索

V2-045753 | MATHDB-373413

Asymptotic formula for the maximum packing number

摘录 / EXCERPT: Let $M(n)$ denote the maximum packing number in the setting of the paper. Maximum-packing conjecture. ... The conjecture predicts the leading constant for $M(n)$; the source presents...

conjecture | open | MathDB 页面 | 原文线索

V2-045754 | MATHDB-373415

Asymptotic formula for packing progressions of intermediate length

摘录 / EXCERPT: Let $m_k(n)$ denote the packing number for progressions with parameter k. Intermediate-length packing conjecture. For every k satisfying $\Omega(\sqrt{n})=k<\sqrt{n}$, we have...

conjecture | open | MathDB 页面 | 原文线索

V2-045755 | MATHDB-373416

Piecewise asymptotic formula for packing progressions

摘录 / EXCERPT: Let $M_k(n)$ denote the packing number with parameter k. Piecewise $M_k(n)$ conjecture. If $k\ll n$, then ... while otherwise, ... The source presents these as plausible asymptotics...

conjecture | open | MathDB 页面 | 原文线索

V2-045756 | MATHDB-373417

Asymptotic sharpness of the trivial lower bound for fixed-length progression packing

摘录 / EXCERPT: Let $M_k(n)$ be the packing number for the progressions $B_d=\{d,2d,\ldots,nd\}$ with $d=1,2,\ldots,k$, with n fixed. Fixed-length progression-packing conjecture. For every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-045757 | MATHDB-373432

Kotesovic's asymptotic conjecture for the number of Matryoshkas

摘录 / EXCERPT: Let m_n be the number of maximal Matryoshkas on an $(n+2)$-gon, with $m_1=1$ and ... Kotesovic's asymptotic conjecture. The number of Matryoshkas satisfies ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-045758 | MATHDB-373438

Gallardo-Grekos-Pihko conjecture on three-fold restricted sumsets

摘录 / EXCERPT: Gallardo-Grekos-Pihko conjecture. There is a constant c such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-045759 | MATHDB-373442

Length-profile bounds and parity conjectures for hypercube drawings

摘录 / EXCERPT: Consider a length-regular drawing of the d-dimensional hypercube, with edge lengths $\ell_1\leq \ell_2\leq\cdots$. Length-profile conjectures. The following assertions are propos...

conjecture | open | MathDB 页面 | 原文线索

V2-045760 | MATHDB-373452

Hitting-angle conjecture for chromatic number and neighbourhood complexes

摘录 / EXCERPT: Fix $d\geq 2$ and $k\in\{3,\dots,d+1\}$. In the random Borsuk graph process, edges are added in order of decreasing geodesic distance as the angle parameter α increases. For...

conjecture | open | MathDB 页面 | 原文线索

V2-045761 | MATHDB-373473

Erdős-Szemerédi sum-product conjecture

摘录 / EXCERPT: Erdős-Szemerédi conjecture. For any $g\in\mathbb{N}$, any $\varepsilon>0$ and any set $A\subseteq\mathbb{C}$, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-045762 | MATHDB-373480

Odd-cycle two-line trade conjecture for Latin squares

摘录 / EXCERPT: Let $\mathcal{L}_n$ be the set of Latin squares of even order n. In each of the row, column, and symbol views, a two-line trade is obtained by swapping two lines on the support o...

conjecture | open | MathDB 页面 | 原文线索

V2-045763 | MATHDB-373482

Rank-5 toy Rota Basis Conjecture for circuit-rich pools

摘录 / EXCERPT: Let M be a rank-5 binary matroid instance from the circuit-rich pool template described in the source, including trap instances and arbitrary conjugation by...

conjecture | open | MathDB 页面 | 原文线索

V2-045764 | MATHDB-373483

Rank-7 toy Rota Basis conjecture and de-randomization

摘录 / EXCERPT: Let M be a rank-7 binary matroid instance from the circuit-rich pool template described in the source, including trap instances and conjugation by $\operatorname{GL}(7,2)$. Ran...

conjecture | open | MathDB 页面 | 原文线索

V2-045765 | MATHDB-373484

Scale-invariant toy Rota Basis conjecture on circuit-rich pools

摘录 / EXCERPT: Let n be the rank, and consider the circuit-rich pool instances generated by the source's recipe. Scale-invariant toy Rota Basis conjecture. For $n\in\{5,7,9,11,13\}$ and pool si...

conjecture | open | MathDB 页面 | 原文线索

V2-045766 | MATHDB-373497

Sharpness conjecture for k-hyperseparating set systems

摘录 / EXCERPT: Let k be a positive integer, let V be an n-element set, and let $f(n,k)$ denote the smallest size of a k-hyperseparating set system on V. A set system is k-hyperseparat...

conjecture | open | MathDB 页面 | 原文线索

V2-045767 | MATHDB-373499

The $Er{s}$-Hajnal-Szemerédi skewed coloring conjecture

摘录 / EXCERPT: Skewed coloring conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045768 | MATHDB-373540

Extremal critical collapse conjecture

摘录 / EXCERPT: Let $\widehat{\mathfrak M}$ be a moduli space of Cauchy data on $\mathbb R^3$ for a matter-model and symmetry-class combination in which an exact extremal Kerr--Newman event horizo...

conjecture | open | MathDB 页面 | 原文线索

V2-045769 | MATHDB-373541

Smooth closure of the extremal Einstein--Maxwell charged-scalar Cauchy horizon

摘录 / EXCERPT: Consider the spherically symmetric Einstein--Maxwell charged scalar field (EMCSF) model and an interpolating family of Cauchy data with critical parameter p_*, whose critical dev...

conjecture | open | MathDB 页面 | 原文线索

V2-045770 | MATHDB-373543

The mixed-alphabet language conjecture for smooth words

摘录 / EXCERPT: Let $\mathpzc{A}=\{a,b\}$ be a mixed binary alphabet, let $\mathcal{C}^\infty$ be the set of infinite smooth words over $\mathpzc{A}$, let $\mathcal{C}_f^\infty$ be the set of fini...

conjecture | open | MathDB 页面 | 原文线索

V2-045771 | MATHDB-373544

Sing's generalized power-law conjecture for f-smooth-word complexity

摘录 / EXCERPT: Sing's power-law conjecture. Over $\mathpzc{A}=\{a,b\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045772 | MATHDB-373574

Vanishing-range conjecture for Hurwitz-number coefficients with at least two fixed points

摘录 / EXCERPT: Let $d\geq10$ and let ν be a partition of d with $m_1(\nu)\geq2$. Let z_ν be the standard centralizer factor, and let $b^X_\nu(\mu^{(1)},\dots,\mu^{(s)},m)$ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-045773 | MATHDB-373581

Pach's conjecture on tangencies among precisely 1-intersecting curves

摘录 / EXCERPT: Let a family of planar curves be precisely 1-intersecting if every pair of curves has precisely one common point, either a crossing or a tangency, and assume that no three curves s...

conjecture | solved | MathDB 页面 | 原文线索

V2-045774 | MATHDB-373586

Existence of an evilly normal kernel for sets with the evil twin property

摘录 / EXCERPT: Let $\mathscr{A}$ be a closed set of games with the evil twin property. A pair $(\mathscr{A},\mathscr{K})$, where $\mathscr{K}\subseteq\mathscr{A}$ is a kernel, is evilly normal if...

conjecture | open | MathDB 页面 | 原文线索

V2-045775 | MATHDB-373610

Representability conjecture for coordinate q-matroids

摘录 / EXCERPT: Representability conjecture for coordinate q-matroids. If the corresponding matroid of a coordinate q-matroid is representable, then the q-matroid is representable.

conjecture | open | MathDB 页面 | 原文线索

V2-045776 | MATHDB-373611

Direct-sum conjecture for transversal q-matroids

摘录 / EXCERPT: Direct-sum conjecture for transversal q-matroids. The direct sum $M_1\oplus M_2$ is a transversal q-matroid with presentation $\mathcal{X}_1\oplus\mathcal{X}_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-045777 | MATHDB-373644

Large-order reverse-strategy conjecture for the permutation avoidance game

摘录 / EXCERPT: Large-order reverse-strategy conjecture. For every $k\geq 3$, there exists $n_0(k)$ such that the reverse strategy is a winning strategy for Player II on S_n for all...

conjecture | open | MathDB 页面 | 原文线索

V2-045778 | MATHDB-373645

Dyadic frequency law for the perturbed Hofstadter recursion

摘录 / EXCERPT: Dyadic frequency law. The frequencies $F(m)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-045779 | MATHDB-373646

Peak location law for the perturbed Hofstadter recursion

摘录 / EXCERPT: Let $F(m)$ be the frequency sequence and let ... For each block, let m_k be an index at which $F(m)$ attains its maximum. Peak location law. The peak locations satisfy ... Numerical...

conjecture | open | MathDB 页面 | 原文线索

V2-045780 | MATHDB-373654

The transversal ratio conjecture for 4-polytopes

摘录 / EXCERPT: For a d-polytope, let the transversal ratios be denoted by ρ_d^P and ρ_d^S, corresponding to polytopes and simplicial polytopes, respectively. A polyt...

conjecture | open | MathDB 页面 | 原文线索

V2-045781 | MATHDB-373705

Petrov's balls conjecture for biregular labeled multigraphs

摘录 / EXCERPT: Let a bipartite labeled multigraph be a tuple (L,R,E,u,v), where $u:E\to L$ and $v:E\to R$. Let $A_{n,k}$ be the set of biregular bipartite labeled multigraphs with $L=[n]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-045782 | MATHDB-373707

Matrix inequality conjecture for zero diagonal entries

摘录 / EXCERPT: Let M be a matrix with non-negative real entries, let t be an odd positive integer, and let $(x_1,y_1),\ldots,(x_t,y_t)$ be distinct entries of M satisfying $M_{x_i y_i}=0$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-045783 | MATHDB-373712

Asymptotic growth conjecture for crystallized hyperbolic polyforms of type $\{2k,3\}$

摘录 / EXCERPT: Let $k>3$ be an integer, and let $g_{2k,3}(h)$ denote the minimum number of tiles in a crystallized $\{2k,3\}$-polyform with h holes. Asymptotic growth conjecture. ... This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-045784 | MATHDB-373713

Sharpness conjecture for the crystallized $\{2k,3\}$-polyform bound

摘录 / EXCERPT: Let $k>3$ be an integer. Let A be an extremal $\{2k,k\}$-polyform with h tiles, let $\tilde{A}$ be the associated $\{2k,3\}$-polyform, and let $\mathcal{P}_{min}^{2k,k}(h)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-045785 | MATHDB-373723

The multicolored cycle matching conjecture

摘录 / EXCERPT: Let G be a cycle of length 2ℓ whose edges are colored with h colors. Let M_0 and M_1 be the perfect matchings consisting of the even and odd edges, respectively, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-045786 | MATHDB-373733

Baker-Jin conjecture on the scarcity of finite hyperfield quotients

摘录 / EXCERPT: Baker-Jin conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045787 | MATHDB-373744

The layered-permutation exponential-growth conjecture

摘录 / EXCERPT: Layered exponential-growth conjecture. The two maxima have the same exponential growth rate:

conjecture | open | MathDB 页面 | 原文线索

V2-045788 | MATHDB-373756

CMPP conjecture on sequenceability of zero-sum subsets without inverse pairs

摘录 / EXCERPT: Let G be an abelian group, and let A be a finite subset of $G\setminus\{0\}$. An ordering $a_1,\ldots,a_{|A|}$ of A is sequenceable if its partial sums ... are pairwise distinc...

conjecture | open | MathDB 页面 | 原文线索

V2-045789 | MATHDB-373758

The polynomial count conjecture for contiguous 4 by 4 super-regular matrices

摘录 / EXCERPT: Polynomial count conjecture. The count is

conjecture | open | MathDB 页面 | 原文线索

V2-045790 | MATHDB-373784

The affine type C difference-set conjecture

摘录 / EXCERPT: Let $V_C=\left(\mathbb{Z}/(2n+1)\mathbb{Z}\right)^n$ and let $\mathcal{C}_n$ be the subset defined in the paper from the affine type C residue conditions. Affine type C difference-...

conjecture | open | MathDB 页面 | 原文线索

V2-045791 | MATHDB-373809

Schefler-Zhao-Zhong full-support conjecture for extremal separating atoms

摘录 / EXCERPT: Schefler56Zhao56Zhong's conjecture. The support of A has full size:

conjecture | open | MathDB 页面 | 原文线索

V2-045792 | MATHDB-373827

Kerr stability conjecture for subextremal Kerr spacetimes

摘录 / EXCERPT: Kerr stability conjecture. The maximal Cauchy development of any initial data set for the Einstein vacuum equations, sufficiently close to a subextremal Kerr initial data set in a...

conjecture | open | MathDB 页面 | 原文线索

V2-045793 | MATHDB-373831

The projection-area conjecture for three-dimensional cube dismantling

摘录 / EXCERPT: Let $[n]^3$ be the $n\times n\times n$ cube, and let $s_3(n)$ denote the minimum possible area covered by the three orthogonal projections onto the faces of a solution to the faces...

conjecture | open | MathDB 页面 | 原文线索

V2-045794 | MATHDB-373832

Existence conjectures for solutions in small cuboids

摘录 / EXCERPT: Let $C(k,l,m)$ be the cuboid consisting of klm unit cubes, and call a configuration a solution when it has the dismantling property used throughout the paper. Small-cuboid existe...

conjecture | open | MathDB 页面 | 原文线索

V2-045795 | MATHDB-373850

The conjectured consistency probability formula for cycles

摘录 / EXCERPT: Let C_n be the cycle with n edges, and let $q(C_n)$ denote its consistency probability. Define ... The cycle consistency probability conjecture. The consistency probability of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045796 | MATHDB-373889

No interior for sums of closures of the sets of multiple zeta-star values

摘录 / EXCERPT: No-interior conjecture. For every $p\geq 2$ and every $a,b\in\mathbb{R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-045797 | MATHDB-373944

Second auto-correlation separation conjecture for binary words

摘录 / EXCERPT: Let w,w' be two words, and let $\tilde{c}(w)$ denote the second auto-correlation constant defined by the asymptotic second moment of the number of leaves associated with w. The...

conjecture | open | MathDB 页面 | 原文线索

V2-045798 | MATHDB-373984

MF-AOA asymptotic performance conjecture for the binary paint shop problem

摘录 / EXCERPT: MF-AOA asymptotic performance conjecture. In the limit $n\rightarrow\infty$, AMP algorithms such as the MF-AOA achieve a performance of approximately

conjecture | open | MathDB 页面 | 原文线索

V2-045799 | MATHDB-373993

Claesson-Jelínek-Steingrímsson inversion monotonicity conjecture

摘录 / EXCERPT: Inversion monotonicity conjecture. Every pattern, except for the identity patterns, is inversion monotone. In particular, the pattern 1324 is inversion monotone.

conjecture | open | MathDB 页面 | 原文线索

V2-045800 | MATHDB-373998

Rationality conjecture for factor-occurrence generating functions in k-Bonacci words

摘录 / EXCERPT: Factor-occurrence rationality conjecture. For every fixed $k \ge 3$ and every non-empty factor $B \sqsubseteq \mathbf{W}^{(k)}$, there exist an integer $s(B) \ge 1$ and a polynomia...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045801 | MATHDB-374004

Equality-case conjecture for cross-1-intersecting subsets of \(\mathrm{GL}_2(q)\)

摘录 / EXCERPT: Equality-case conjecture. If S_1 and S_2 are cross-1-intersecting with respect to β and

conjecture | open | MathDB 页面 | 原文线索

V2-045802 | MATHDB-374005

Asymptotic dimensionality for identical angular-momentum values

摘录 / EXCERPT: Asymptotic dimensionality conjecture. If $L\ll\sqrt{\operatorname{Var}_{\bm l}}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-045803 | MATHDB-374015

Classification conjecture for invariant vectors in CircularNim games

摘录 / EXCERPT: Classification conjecture. The only CircularNim games that admit invariant vectors are

conjecture | open | MathDB 页面 | 原文线索

V2-045804 | MATHDB-374016

Invariance Reduction Process conjecture for sub-games of SetNim

摘录 / EXCERPT: Invariance Reduction Process conjecture. Every non-trivial sub-game of G that arises through the Invariance Reduction Process also admits invariant vectors.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045805 | MATHDB-374023

The constructive generation conjecture for efficient total girth colorings

摘录 / EXCERPT: Let G be a finite connected simple cubic graph of girth 4, and let an ETGC denote an efficient total girth coloring. The six constructive tools considered in the source are spr...

conjecture | open | MathDB 页面 | 原文线索

V2-045806 | MATHDB-374053

Koszulity conjecture for Chow rings of flag or complete built matroids

摘录 / EXCERPT: Let M be a matroid and let $\mathcal{G}$ be a building set such that $(\mathrm{M},\mathcal{G})$ is a flag or complete built matroid. Koszulity conjecture. The Chow ring...

conjecture | open | MathDB 页面 | 原文线索

V2-045807 | MATHDB-374063

The computational hardness conjecture for matroid branch-depth and c*-d*-depth

摘录 / EXCERPT: A matroid M has branch-depth and c*d*-depth, two recursive depth parameters introduced in the paper. Computational hardness conjecture. Computing exactly the branch-depth a...

conjecture | open | MathDB 页面 | 原文线索

V2-045808 | MATHDB-374067

Non-empty core conjecture for correlated gift exchange coalitions

摘录 / EXCERPT: Let P be the set of players and let $\mathcal{K}$ be a coalition structure, with core stability meaning that no subset $S \subseteq P$ can deviate to form a new coalition and mak...

conjecture | open | MathDB 页面 | 原文线索

V2-045809 | MATHDB-374075

Johnson-Leader-Russell's tower-of-cubes conjecture for maximal chains

摘录 / EXCERPT: Let X be an n-element set, and let $\binom{X}{\bullet}$ denote the Boolean lattice of subsets of X. For integers $t,k\geq 1$ with $n=tk$, partition X into pairwise disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-045810 | MATHDB-374093

The structural conjecture for minimum diamond-saturated families

摘录 / EXCERPT: Structural conjecture for minimum diamond-saturated families. If

conjecture | open | MathDB 页面 | 原文线索

V2-045811 | MATHDB-374139

The dominant-zero-column singularity conjecture for random combinatorial matrices

摘录 / EXCERPT: Singularity-obstruction conjecture. If $c\in(0,1/2)$ is fixed and

conjecture | open | MathDB 页面 | 原文线索

V2-045812 | MATHDB-374151

Finite free p-Stam extremizer bifurcation conjecture

摘录 / EXCERPT: Let $n\geq 3$ and consider extremizers of the p-Stam inequality among normalized monic real-rooted polynomial pairs (f,g) of degree n; write d_{PQ} for their pair mismatch...

conjecture | open | MathDB 页面 | 原文线索

V2-045813 | MATHDB-374154

The six-times-linear wheel Ramsey conjecture

摘录 / EXCERPT: Let W_{2n} be the wheel on $2n+1$ vertices, and let $R(W_{2n})=R(W_{2n},W_{2n})$ denote the diagonal Ramsey number for two copies of this wheel. Six-times-linear wheel conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-045814 | MATHDB-374156

The signature conjecture for cones of colorings

摘录 / EXCERPT: Let $\mathcal C$ be a cone of colorings, let Q be its integral quadratic form, and let $A:\mathcal C\to\mathcal M$ be the associated locally affine map. Write $P\mathcal C$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045815 | MATHDB-374157

The global injectivity conjecture for projectivized coloring maps

摘录 / EXCERPT: Let $\mathcal C$ be a cone of colorings, let $\mathcal M$ be the relevant moduli space, and let $A:\mathcal C\to\mathcal M$ be the associated locally affine map. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-045816 | MATHDB-374165

Keszegh-Lemons-Martin-Pálvölgyi-Patkós dichotomy conjecture for induced poset saturation

摘录 / EXCERPT: Let P be a finite poset, and let $\mathrm{sat}^{\star}([2]^n,P)$ denote its induced saturation function in the Boolean lattice. Keszegh-Lemons-Martin-Pálvölgyi-Patkós' dichotomy...

conjecture | open | MathDB 页面 | 原文线索

V2-045817 | MATHDB-374166

Linear-growth dichotomy conjecture for induced saturation in the hypergrid

摘录 / EXCERPT: Let P be a poset and fix $t\in\mathbb{Z}_{\geq 2}$. Write $\mathrm{sat}^{\star}([t]^n,P)$ for the induced saturation function of P in the t-ary hypergrid. Linear-growth dicho...

conjecture | open | MathDB 页面 | 原文线索

V2-045818 | MATHDB-374167

Monotonicity conjecture for induced saturation in the hypergrid

摘录 / EXCERPT: Let P be a poset, and consider the induced saturation function $\mathrm{sat}^{\star}([t]^n,P)$ whenever it is defined. Monotonicity conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-045819 | MATHDB-374174

The asymptotic conjecture for affine-flat intersection statistics with even s

摘录 / EXCERPT: Let $d\geqslant 1$ and let $s=j\cdot 2^k$, where j is odd and $1\leqslant k\leqslant d$. The quantities $\lambda^*(d,s)$ measure the optimal proportion of affine d-flats having...

conjecture | open | MathDB 页面 | 原文线索

V2-045820 | MATHDB-374175

The asymptotic one-point conjecture for affine-flat intersection statistics

摘录 / EXCERPT: Let $d\geqslant 1$, and let $\lambda^*(d,1)$ denote the optimal proportion of affine d-flats with intersection size 1 in the paper's extremal problem. The one-point asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-045821 | MATHDB-374203

Generalized Glasby-Paseman sequence conjecture

摘录 / EXCERPT: Generalized Glasby-Paseman sequence conjecture. This sequence satisfies the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-045822 | MATHDB-374224

The exact minimum percolating-set size conjecture for 4-neighbour bootstrap percolation on hypercubes

摘录 / EXCERPT: Let Q_d be the d-dimensional hypercube, and let $m(Q_d;4)$ denote the minimum cardinality of a set that percolates under the 4-neighbour bootstrap process on Q_d. Exact-siz...

conjecture | open | MathDB 页面 | 原文线索

V2-045823 | MATHDB-374236

Cyclic Ramsey conjecture for monotone paths

摘录 / EXCERPT: Let P_a^mon and P_b^mon denote monotone paths of orders a and b, respectively, and let R_cyc be the cyclic Ramsey number. Cyclic monotone-path...

conjecture | open | MathDB 页面 | 原文线索

V2-045824 | MATHDB-374237

Mixed cyclic Ramsey conjecture for alternating and monotone paths

摘录 / EXCERPT: Let P_a^alt be an alternating path of order a and P_b^mon a monotone path of order b. Mixed path conjecture. For any $a \ge 3$ and $b \ge 4$, we have ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-045825 | MATHDB-374238

Cyclic Ramsey conjecture for alternating and reverse alternating paths

摘录 / EXCERPT: Let P_a^alt be an alternating path of order a and P_b^ralt a reverse alternating path of order b.
Alternating-reverse-alternating conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-045826 | MATHDB-374239

Ordered Ramsey conjecture for monotone cycles and alternating paths

摘录 / EXCERPT: Let C_a^mon be a monotone cycle of order a and P_b^alt an alternating path of order b. Monotone-cycle-alternating-path conjecture. For any $a \ge 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045827 | MATHDB-374240

Cyclic Ramsey conjecture for monotone cycles and alternating paths

摘录 / EXCERPT: Let C_a^mon be a monotone cycle of order a and P_b^alt an alternating path of order b. Cyclic monotone-cycle-alternating-path conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-045828 | MATHDB-374241

Cyclic Ramsey conjecture for stars and monotone paths

摘录 / EXCERPT: Let S_a be the start-central star with parameter a, and let P_b^mon be a monotone path of order b. Star-monotone-path conjecture. For any $a \ge 3$ and $b \ge 4$,...

conjecture | open | MathDB 页面 | 原文线索

V2-045829 | MATHDB-374242

Cyclic Ramsey conjecture for stars and alternating paths

摘录 / EXCERPT: Let S_a be the start-central star with parameter a, and let P_b^alt be an alternating path of order b. Star-alternating-path conjecture. For any $a \ge 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-045830 | MATHDB-374243

Ordered Ramsey conjecture for complete graphs and alternating paths

摘录 / EXCERPT: Let K_a be a complete graph of order a and P_b^alt an alternating path of order b. Complete-graph-alternating-path conjecture. For any $a \ge 2$ and $b \ge 3$, we...

conjecture | open | MathDB 页面 | 原文线索

V2-045831 | MATHDB-374244

Cyclic Ramsey conjecture for complete graphs and alternating paths

摘录 / EXCERPT: Let K_a be a complete graph of order a and P_b^alt an alternating path of order b. Complete-graph cyclic conjecture. For any $a, b \in \mathbb{N}$, we have ... The c...

conjecture | open | MathDB 页面 | 原文线索

V2-045832 | MATHDB-374245

Cyclic Ramsey conjecture for nested matchings

摘录 / EXCERPT: Let M_a^nest and M_b^nest denote nested matchings with parameters a and b. Nested-matching conjecture. Let $a, b \ge 4$ be divisible by four. Then ... A con...

conjecture | open | MathDB 页面 | 原文线索

V2-045833 | MATHDB-374246

Ordered Ramsey conjecture for nested matchings and monotone paths

摘录 / EXCERPT: Let M_a^nest be a nested matching with even parameter a, and let P_b^mon be a monotone path of order b. Nested-matching-monotone-path conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-045834 | MATHDB-374250

Completeness conjecture for regular Inverse Treblecross positions

摘录 / EXCERPT: Let R be the regular set of board states, let a state have at most one piece, and let $\mathcal{G}$ denote its Sprague-Grundy value. Completeness conjecture. The regular set R...

conjecture | open | MathDB 页面 | 原文线索

V2-045835 | MATHDB-374251

Asymptotic periodicity conjecture for single-piece Inverse Treblecross states

摘录 / EXCERPT: Let $[a X b]$ be a single-piece Inverse Treblecross state, with a and b denoting the lengths on either side of the piece, and let $\mathcal{G}$ denote its Sprague-Grundy value....

conjecture | open | MathDB 页面 | 原文线索

V2-045836 | MATHDB-374252

Generalized reflectional periodicity conjecture for odd Inverse k-cross games

摘录 / EXCERPT: Let $k=2m+1$ with $m\geq1$, and consider an Inverse k-cross game with empty-board state $[a]$. Let $\mathcal{G}([a])$ denote its Sprague-Grundy value, and let $\oplus$ denote nim...

conjecture | open | MathDB 页面 | 原文线索

V2-045837 | MATHDB-374302

The saturation conjecture for Markov polynomials

摘录 / EXCERPT: Saturation conjecture. The terms that appear in the numerator of a Markov polynomial M_{ρ} are precisely those corresponding to the integer lattice points in the Newton polygo...

conjecture | open | MathDB 页面 | 原文线索

V2-045838 | MATHDB-374329

Mihail-Vazirani conjecture on edge-expansion of 0/1 polytopes

摘录 / EXCERPT: Mihail-Vazirani conjecture. Every $0/1$ polytope P satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045839 | MATHDB-374355

The colorful-complex extremal characterization conjecture

摘录 / EXCERPT: Let G be a graph, let $\mathcal{U} = \{U_1, \ldots, U_m\}$ be a partition of $V(G)$, and let $k \ge -1$. Write $\mathbf{Col}(\mathcal{I}(G),\mathcal{U})$ for the colorful complex...

conjecture | open | MathDB 页面 | 原文线索

V2-045840 | MATHDB-374366

The combinatorial model conjecture for sutured Legendrian contact homology

摘录 / EXCERPT: Let $\Lambda \subset (\Sigma \times \mathbb{R}, \xi_\Gamma)$ be a Legendrian knot. Let $Y(\Sigma, \Gamma) \subset \Sigma \times \mathbb{R}$ be the convex sutured contact manifold a...

conjecture | open | MathDB 页面 | 原文线索

V2-045841 | MATHDB-374367

Finite-state structure conjecture for 132-avoiding permutations with bounded adjacency

摘录 / EXCERPT: Let S_n be the symmetric group, and let $A_n^{(m)}$ be the class ... Here, m is a fixed integer with $m\geq 1$. Finite-state structure conjecture. The class $A_n^{(m)}$ admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-045842 | MATHDB-374385

Fazekas-Mammoliti-Mercaş-Simpson extension of the Fici-Saarela conjecture

摘录 / EXCERPT: Fazekas-Mammoliti-Mercaş-Simpson conjecture. A binary word of length n contains at least

conjecture | open | MathDB 页面 | 原文线索

V2-045843 | MATHDB-374386

Lower bound for abelian squares from effective partitions

摘录 / EXCERPT: Effective-partition conjecture. The word w contains at least

conjecture | open | MathDB 页面 | 原文线索

V2-045844 | MATHDB-374419

Integral representation conjecture for the matroid hypergeometric system

摘录 / EXCERPT: Integral representation conjecture. For generic choices of $u\in\mathbb{C}^{n+1}$, the morphism ψ is an isomorphism of D_Z-modules.

conjecture | open | MathDB 页面 | 原文线索

V2-045845 | MATHDB-374420

Consequences conjecture for the matroid hypergeometric system

摘录 / EXCERPT: Consequences conjecture. For generic choices of $u\in\mathbb{C}^{n+1}$:

conjecture | open | MathDB 页面 | 原文线索

V2-045846 | MATHDB-374426

Doumas-Papanicolaou's finite variance-minimization conjecture for the double Dixie cup problem

摘录 / EXCERPT: Finite variance extremality. The equal-probability vector ν minimizes

conjecture | open | MathDB 页面 | 原文线索

V2-045847 | MATHDB-374433

The additive-multiplicative index dichotomy conjecture

摘录 / EXCERPT: Additive-multiplicative index dichotomy conjecture. Apart from monomials of the form $x\mapsto ax^{p^j}$, which have additive and multiplicative indices equal to 1, any function...

conjecture | open | MathDB 页面 | 原文线索

V2-045848 | MATHDB-374444

Unique Extension conjecture for 4 × n Chomp

摘录 / EXCERPT: A position in $4 \times n$ Chomp is a quadruple (a,b,c,d) of non-negative integers with $a\geq b\geq c\geq d\geq 0$; a P-position is a position from which the previous player has...

conjecture | refuted | MathDB 页面 | 原文线索

V2-045849 | MATHDB-374450

The Alon-Jaeger-Tarsi conjecture

摘录 / EXCERPT: Let $\mathbb{F}$ be a field, let M be a nonsingular matrix over $\mathbb{F}$, and let x be a vector over $\mathbb{F}$. Alon-Jaeger-Tarsi conjecture. For any field $\mathbb{F}$...

conjecture | open | MathDB 页面 | 原文线索

V2-045850 | MATHDB-374451

Conjecture on the top term of the generating function for double Hurwitz numbers

摘录 / EXCERPT: Top-term conjecture. The top term of $H_g(\mu^{(1)},\dots,\mu^{(s)};T)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-045851 | MATHDB-374457

Mathias's question on MAD families under the Ramsey property

摘录 / EXCERPT: Mathias's question. In Solovay's model, are there infinite MAD families? More generally, is it a theorem of $\mathsf{ZF}+\mathsf{DC}_{\mathbb R}$ that there are no infinite MAD fam...

open_problem | open | MathDB 页面 | 原文线索

V2-045852 | MATHDB-374464

Kirillov's eventual polynomiality conjecture for stretched Schubert coefficients

摘录 / EXCERPT: Kirillov's conjecture. $f_{u,v,w}(N)$ is eventually polynomial in N.

conjecture | open | MathDB 页面 | 原文线索

V2-045853 | MATHDB-374483

Wagner's conjecture on S- and I-Rayleigh regular matroids

摘录 / EXCERPT: Wagner's conjecture. Every regular matroid is S- and I-Rayleigh.

conjecture | open | MathDB 页面 | 原文线索

V2-045854 | MATHDB-374484

Rayleigh conjecture for spanning-forest generating functions

摘录 / EXCERPT: Spanning-forest Rayleigh conjecture. For any graph $G=(V,E)$, the SFGF $F_G(\mathbf{w})$ is Rayleigh.

conjecture | open | MathDB 页面 | 原文线索

V2-045855 | MATHDB-374488

Product-of-triangles conjecture for irreducible Ferrers-diagram polytopes

摘录 / EXCERPT: For each integer $d\geq 3$, let $\mathfrak P_d\subseteq\mathbb R^{2d-3}$ be the integral polytope whose integer points correspond to irreducible Ferrers-diagram pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-045856 | MATHDB-374505

Joint mixability of uniform cube slices

摘录 / EXCERPT: For $d\geq2$, let ... and let $U(L)$ denote the uniform distribution on a set L of nonzero finite measure. A triple of distributions is jointly mixable if it has a coupling whose c...

conjecture | open | MathDB 页面 | 原文线索

V2-045857 | MATHDB-374514

Truly linear sparsifier conjecture for set systems

摘录 / EXCERPT: Let $\mathcal S \subseteq 2^{[m]}$ be a set system, let $w:[m]\rightarrow\mathbb{R}_{\geq 0}$ be weights, and let $\epsilon>0$. Write $\operatorname{CL}(\mathcal S)$ for the chain...

conjecture | open | MathDB 页面 | 原文线索

V2-045858 | MATHDB-374516

Rauzy's conjecture on ternary words of constant abelian complexity

摘录 / EXCERPT: Rauzy's conjecture. Except for very particular vectors of letter frequencies, there do not exist any infinite ternary words with constant abelian complexity equal to 3.

conjecture | open | MathDB 页面 | 原文线索

V2-045859 | MATHDB-374528

Transversal conjecture for Latin hypercubes of odd order or dimension

摘录 / EXCERPT: Let $M(d,n)$ denote the set of Latin hypercubes of dimension d and order n. A transversal is a set of n entries containing each value in every coordinate position and each sy...

conjecture | open | MathDB 页面 | 原文线索

V2-045860 | MATHDB-374571

Erdős-Kleitman matching conjecture for set families

摘录 / EXCERPT: Erdős-Kleitman matching conjecture. Suppose that $s\ge 2$, $m\ge 1$, and $n=sm+s-\ell$ for some integer $0<\ell\leq\lceil s/2\rceil$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-045861 | MATHDB-374591

Wang-Zhang-Zhang's immanant conjectures for cyclic groups

摘录 / EXCERPT: Wang-Zhang-Zhang's immanant conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-045862 | MATHDB-374600

The growth-rate conjecture for sets with few subset sums in dimension three

摘录 / EXCERPT: Growth-rate conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-045863 | MATHDB-374607

Frankl-Kupavskii conjecture for the Erdős-Kleitman problem

摘录 / EXCERPT: For integers $n\ge s\ge2$, define ... For integers $m\ge1$ and $1\le\ell\le\lceil s/2\rceil$, set $n=(m+1)s-\ell$ and, for any $L\in\binom{[n]}{\ell-1}$, define ... and write...

conjecture | open | MathDB 页面 | 原文线索

V2-045864 | MATHDB-374612

Weighted Frankl conjecture for union-closed families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $\mathcal{F}\subseteq 2^{[n]}$ be a non-empty union-closed family, meaning that $A\cup B\in\mathcal{F}$ for all $A,B\in\mathcal{F}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045865 | MATHDB-374620

Reflection-symmetry conjecture for LEGO counting polynomials

摘录 / EXCERPT: For each positive integer n, let $p_n(w)$ be the polynomial counting the LEGO structures made from n parallel $w\times1$ tiles. Reflection-symmetry conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-045866 | MATHDB-374629

Degenerate diamonds require at least four leading terms

摘录 / EXCERPT: Let $q\geq 3$, let Π be a projective plane of order q, and consider a 4×4 determinant initial form of its combinatorial incidence matrix. A degenerate diamond is a con...

conjecture | open | MathDB 页面 | 原文线索

V2-045867 | MATHDB-374638

The Ramsey-type bound conjecture for multipartite-free chi-bounded classes

摘录 / EXCERPT: Ramsey-type bound conjecture. For every χ-bounded class $\mathcal G$ and every two integers $m,s\ge1$, there exists $C=C(\mathcal G,m,s)\ge1$ such that every $\overline{mK_s}$...

conjecture | open | MathDB 页面 | 原文线索

V2-045868 | MATHDB-374640

Four-direction LP asymptotics

摘录 / EXCERPT: Let α be the middle real root of ... For odd $n=2m+1$, let L_m^{fat} and L_m^{thin} denote the relaxation optima on the fat and thin colour classes, resp...

conjecture | open | MathDB 页面 | 原文线索

V2-045869 | MATHDB-374641

Monochromatic no-three-in-line asymptotics

摘录 / EXCERPT: Monochromatic NTIL asymptotics. The exact monochromatic no-three-in-line maxima satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-045870 | MATHDB-374642

Mixed-uniformity product Hilton-Milner conjecture

摘录 / EXCERPT: Let $n\geq 2k>2\ell\geq 4$. Fix sets $F_0\in\binom{[2,n]}{k}$ and $G_0\in\binom{[2,n]}{\ell}$ with $F_0\cap G_0\neq\emptyset$, and define ... ... Two families are cross-intersecting if...

conjecture | open | MathDB 页面 | 原文线索

V2-045871 | MATHDB-374701

Asymptotic tower formula for color-avoiding Ramsey numbers of monotone paths

摘录 / EXCERPT: Fix an integer $p\geq 1$. Let $A_k(n;q,p)$ denote the color-avoiding Ramsey number for monotone paths, and let $T_h(x)$ denote a tower function of height h. Asymptotic tower form...

conjecture | open | MathDB 页面 | 原文线索

V2-045872 | MATHDB-374706

Bergeron-Ceballos-Küstner's unimodality conjecture for q-Fibonomial coefficients

摘录 / EXCERPT: For $m,n\in\mathbb{N}$, define the q-Fibonomial coefficient by ... where $[F_n]^!_q:=\prod_{k=1}^n[F_k]_q$ and $[n]_q:=1+q+\dots+q^{n-1}$. Bergeron-Ceballos-Küstner's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-045873 | MATHDB-374707

Unimodality conjecture for products of q-analogs

摘录 / EXCERPT: Let $\pow\geq2$, $k\geq1$, and let $a_1,\ldots,a_k,b$ be positive integers. Write $[r]_q:=1+q+\dots+q^{r-1}$ and, for an integer $\pow$, write...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045874 | MATHDB-374721

The symmetric capacity conjecture for unimodular triangulations

摘录 / EXCERPT: Symmetric capacity conjecture. The symmetric capacity satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045875 | MATHDB-374728

The reverse-mex characterization for three-element subtraction Nim

摘录 / EXCERPT: The reverse-mex characterization. The subtraction Nim has the reverse-mex property if and only if, for every $\mathcal{P}$-position w with $dist(w)=2m$ for some...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045876 | MATHDB-374729

The reverse-mex condition for subtraction Nim with s_3=s_1+s_2

摘录 / EXCERPT: Consider the three-element subtraction Nim setup above, including the eventual period p and the associated distance function $dist(w)$. The reverse-mex condition. If...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045877 | MATHDB-374730

Predicted reverse-mex behavior for three families of subtraction Nim with a pass

摘录 / EXCERPT: Consider subtraction Nim with subtraction set $\{s_1,s_2,s_3\}$ and one pass. Let $\mathcal{G}(x,0)$ denote the Grundy number without a pass and $\mathcal{G}(x,1)$ the Grundy numbe...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045878 | MATHDB-374733

Bivariate combinatorial interpretation of the q,p-Catalan triangle

摘录 / EXCERPT: Bivariate combinatorial interpretation. For all $n\ge1$ and $0\le k\le n$:

conjecture | open | MathDB 页面 | 原文线索

V2-045879 | MATHDB-374734

Cyclotomic Catalan triangle conjecture

摘录 / EXCERPT: Cyclotomic Catalan conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045880 | MATHDB-374761

Hadamard's conjecture on the existence of Hadamard matrices

摘录 / EXCERPT: A Hadamard matrix is a square array with entries in $\{+1,-1\}$ such that every two distinct rows agree in exactly half of their columns. Hadamard's conjecture. A Hadamard matrix o...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045881 | MATHDB-374778

Counting-types characterization of NTP$_2$

摘录 / EXCERPT: Let $varphi(x,y)$ be a partitioned formula, and let $f_{varphi}(k,l)$ denote the number of pairwise-inconsistent partial $varphi$-types of size k over a parameter set of size l...

conjecture | open | MathDB 页面 | 原文线索

V2-045882 | MATHDB-374792

The conjectured first-difference sequence for the Fibonacci word

摘录 / EXCERPT: Let f be the infinite Fibonacci word, let Σ be its alphabet, and define ... Here $\operatorname{Cl}(w)$ is the number of distinct closed factors of w, and F_n denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-045883 | MATHDB-374793

The conjectured exact closed-rich constant of the Fibonacci word

摘录 / EXCERPT: Let f be the infinite Fibonacci word, let C_f be its closed-rich constant, and let $\phi=\frac{1+\sqrt{5}}{2}$ be the golden ratio. Fibonacci closed-rich constant conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-045884 | MATHDB-374802

Existence conjecture for symmetric 0-ULSE coloured BIBDs

摘录 / EXCERPT: Let $v,k,\ell\in\mathbb{Z}^+$ satisfy the necessary parameter equation referred to in the source as Equation, and suppose that there exists a symmetric BIBD on v points with bloc...

conjecture | open | MathDB 页面 | 原文线索

V2-045885 | MATHDB-374806

Feder-Subi conjecture on antipodal paths with few colour changes

摘录 / EXCERPT: Feder-Subi conjecture. Every 2-colouring of $E(Q_n)$ contains a path between some pair of antipodal vertices which changes colour at most once.

conjecture | open | MathDB 页面 | 原文线索

V2-045886 | MATHDB-374807

Triple-coincidence non-occurrence under single-piece extension

摘录 / EXCERPT: Triple-coincidence conjecture. For every integer $n\geq4$, no three distinct pieces in $\mathcal{P}'$ share the same strength on the $n\times n$ board.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045887 | MATHDB-374823

Carpi's conjecture on abelian power-free morphisms

摘录 / EXCERPT: Let $Sigma$ be a finite alphabet, and let a morphism on $Sigma$ be called abelian power-free when its images avoid abelian powers of the relevant exponent. Carpi's set of condition...

conjecture | open | MathDB 页面 | 原文线索

V2-045888 | MATHDB-374838

Kalai's cascade consequence for higher Tverberg partitions

摘录 / EXCERPT: Let d,r be positive integers and let k be a non-negative integer. A Tverberg $(r+1)$-partition of a finite point set is a partition into $r+1$ parts whose convex hulls have a c...

conjecture | open | MathDB 页面 | 原文线索

V2-045889 | MATHDB-374880

Ferroni-Jochemko-Schröter conjecture on Ehrhart positivity of positroids

摘录 / EXCERPT: A positroid is a matroid arising from a cell of the totally positive Grassmannian. A matroid base polytope is Ehrhart positive when all coefficients of its Ehrhart polynomial are n...

conjecture | open | MathDB 页面 | 原文线索

V2-045890 | MATHDB-374881

Ehrhart positivity conjecture for matroids with series-parallel subdivisions

摘录 / EXCERPT: Let M be a connected matroid, and let P_M be its base polytope. Say that P_M admits a series-parallel subdivision if it can be subdivided into base polytopes of series-parall...

conjecture | open | MathDB 页面 | 原文线索

V2-045891 | MATHDB-374882

Manickam-Miklós-Singhi conjecture

摘录 / EXCERPT: Manickam-Miklós-Singhi conjecture. There exist at least $\binom{n-1}{k-1}$ subsets $A\subseteq [n]$ such that $|A|=k$ and

conjecture | open | MathDB 页面 | 原文线索

V2-045892 | MATHDB-374891

The parity-pattern conjecture for generalized OOOOOOB positions

摘录 / EXCERPT: Consider a position with $k\geq 3$ piles of tokens, and let ... Assume that every pile has more than m tokens. Parity-pattern conjecture. The position is a $\mathcal{P}$-position p...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045893 | MATHDB-374892

The eventual parity conjecture for positions with many largest piles

摘录 / EXCERPT: Eventual parity conjecture. The outcome class of the position is determined solely by the parities of $a_1,a_2,\ldots,a_{n-1}$. The source reports preliminary calculations for smal...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045894 | MATHDB-374893

Version C's subtract-three conjecture

摘录 / EXCERPT: Consider a position of Version C with a_i piles of size i, for $1\leq i\leq n$, where ... Exclude the position with $1,1,2,2,3,3,3$ as its piles, equivalently...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045895 | MATHDB-374911

Conjecture on generators of the simple loopless regular matroid complex homology

摘录 / EXCERPT: Generator conjecture. The homology of the indecomposable quotient of the simple loopless regular matroid complex is generated by the classes of $U_{1,1}$ and $\mathsf{M}(W_{2k+1})$...

conjecture | open | MathDB 页面 | 原文线索

V2-045896 | MATHDB-374916

Hindman's monochromatic sum-product quadruple question

摘录 / EXCERPT: Let a finite partition of $\mathbb{N}$ be given. Hindman's question. There exist $x,y\in\mathbb{N}$ such that ... is monochromatic. This asks whether every finite coloring of the pos...

open_problem | open | MathDB 页面 | 原文线索

V2-045897 | MATHDB-374917

Monochromatic finite-sums and product-of-sums conjecture

摘录 / EXCERPT: Let $\mathbb{N}=\bigcup_{i=1}^m A_i$ be a finite partition, let $\mathcal{P}_f(\mathbb{N})$ denote the nonempty finite subsets of $\mathbb{N}$, and let $\alpha<\beta$ mean...

conjecture | open | MathDB 页面 | 原文线索

V2-045898 | MATHDB-374920

Conjecture on the center and median of a maximum-Wiener-index tree

摘录 / EXCERPT: Center-median conjecture. The center and median of T_0 have distance zero:

conjecture | open | MathDB 页面 | 原文线索

V2-045899 | MATHDB-374922

Minkowski difference conjecture for type A MV polytopes

摘录 / EXCERPT: Let Q be an MV polytope of type A, and let $Q(\theta_I)$ denote the Schubert matroid polytope associated with $I\subseteq [n]$. For collections...

conjecture | open | MathDB 页面 | 原文线索

V2-045900 | MATHDB-374950

Ahlswede-Khachatrian-type conjecture for t-intersecting families of permutations

摘录 / EXCERPT: Let n and t be positive integers, let $cSigma_n$ be the collection of permutations of $[n]$, and let cA_k be the family ... A family $\calF\subseteqcSigma_n$ is t-intersectin...

conjecture | open | MathDB 页面 | 原文线索

V2-045901 | MATHDB-374979

Complexity conjecture for unrestricted shadow inflection minimization

摘录 / EXCERPT: Complexity conjecture. For an appropriate purely combinatorial encoding of embedded shadows, the decision problem $\textnormal{\textsc{MinInflection}}$ for unrestricted shadows is...

conjecture | open | MathDB 页面 | 原文线索

V2-045902 | MATHDB-374980

Tree--necklace Gauss-load conjecture

摘录 / EXCERPT: Tree--necklace Gauss-load conjecture. For tree-like and tree--necklace shadows, the minimum Gauss load is obtained by a reduced tangent-angle realization of a coorientation minimiz...

conjecture | open | MathDB 页面 | 原文线索

V2-045903 | MATHDB-374999

Exact finite interval discrepancy conjecture

摘录 / EXCERPT: For each positive integer n, let $\operatorname{disc}(n)$ be the minimum, over all strategies of length n that repeatedly split an existing interval into two starting from...

conjecture | open | MathDB 页面 | 原文线索

V2-045904 | MATHDB-375019

Sharp consecutive gap conjecture for two-point subsets of the circle

摘录 / EXCERPT: Let $(x_n)_{n\geq 1}$ be any sequence of distinct points on the circle $\mathbb T$. For each n, let $M_n^{(2)}$ and $m_n^{(2)}$ denote the quantities defined in the paper for two...

conjecture | open | MathDB 页面 | 原文线索

V2-045905 | MATHDB-375025

Ballantine-Beck-Merca injectivity conjecture for elementary symmetric partitions

摘录 / EXCERPT: Let n be a positive integer. A partition of n is a weakly decreasing sequence of positive integers with size n and length ℓ. For $1\leq j\leq\ell$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-045906 | MATHDB-375026

Devnani-Eyyunni refined injectivity conjecture for elementary symmetric partitions

摘录 / EXCERPT: Let λ be a partition, with length denoted by ℓ, and let $\operatorname{pre}_j(\lambda)$ be the partition whose parts are all products of j distinct parts of...

conjecture | open | MathDB 页面 | 原文线索

V2-045907 | MATHDB-375027

The ordinary-partition containment conjecture

摘录 / EXCERPT: Let $\mathcal{P}$ be the set of partitions, let $\mathcal{P}_{\mathrm{2reg}}$ be the set of 2-regular partitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045908 | MATHDB-375028

The upper-ledge containment conjecture

摘录 / EXCERPT: Let Π be the set of pairs of partitions, let $i\in\mathbb{Z}_2$, and let $\mathcal{X}_i:\Pi\rightarrow\mathcal{UL}$ be the associated map to upper ledge diagrams. If...

conjecture | open | MathDB 页面 | 原文线索

V2-045909 | MATHDB-375030

The upper-ledge Specht-module conjecture

摘录 / EXCERPT: Let I be the index set of the affine type ${\tt A}_1^{(1)}$, let $\theta\in\mathbb{Z}_{\geq0}I$, and let $\mathsf{D}\in\mathcal{UL}(\theta)$ be an upper ledge diagram. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-045910 | MATHDB-375049

Nonnegativity conjecture for the auxiliary polynomial U_k

摘录 / EXCERPT: For each integer $k\geq1$, define ... Here $A(q)\geq_qB(q)$ denotes coefficientwise comparison, so $U_k(q)\geq_q0$ means that all coefficients of $U_k(q)$ are nonnegative. The U_k...

conjecture | open | MathDB 页面 | 原文线索

V2-045911 | MATHDB-375050

Conlon-Fox-Sudakov-Wei conjecture for path threshold Ramsey multiplicity

摘录 / EXCERPT: Let $M(H,n)$ be the minimum number of monochromatic copies of a graph H in a red/blue coloring of K_n, and define the threshold Ramsey multiplicity by ... where $r(H)$ is the Ram...

conjecture | open | MathDB 页面 | 原文线索

V2-045912 | MATHDB-375059

Ross-Yong's K-theoretic Kohnert rule for Grothendieck polynomials

摘录 / EXCERPT: Ross-Yong's Grothendieck-polynomial conjecture. For every permutation $w\in S_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-045913 | MATHDB-375060

Robichaux's ghost Kohnert rule for Grothendieck polynomials

摘录 / EXCERPT: Robichaux's ghost Kohnert conjecture. For every permutation $w\in S_n$ and $\alpha\in\mathbb{Z}_{\geq 0}^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-045914 | MATHDB-375072

The combinatorial criterion for kinematic support

摘录 / EXCERPT: Let $\Gamma_{L^\ast}$ be a graph in the class of loop BCFW graphs $\mathbf{\Gamma}^{\mathtt{BCFW}}_{k,n;L^\ast}$, and let ... be the corresponding branch of the loop BCFW recursion....

conjecture | open | MathDB 页面 | 原文线索

V2-045915 | MATHDB-375073

Veselinov-Marinov monotonicity conjecture for equal-parameter town families

摘录 / EXCERPT: Let $m_{k,n}(a,b)$ denote the maximum size of a family of subsets of $[n]$ whose pairwise intersections are congruent to a modulo k when the two subsets coincide and to b mod...

conjecture | open | MathDB 页面 | 原文线索

V2-045916 | MATHDB-375075

Parity conjecture for maximum equal-parameter town families

摘录 / EXCERPT: Let $m_{k,n}(a,a)$ be the maximum size of a family of subsets of $[n]$ whose intersections are all congruent to a modulo k, with $n\geq k\geq 2$. Parity conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-045917 | MATHDB-375082

Burr-Erdős-Faudree-Rousseau-Schelp conjecture for star forests

摘录 / EXCERPT: Burr-Erdős-Faudree-Rousseau-Schelp conjecture. The size Ramsey number of the two star forests satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045918 | MATHDB-375083

Davoodi-Javadi-Kamranian-Raeisi multicolor star-forest conjecture

摘录 / EXCERPT: Davoodi-Javadi-Kamranian-Raeisi conjecture. The multicolor size Ramsey number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045919 | MATHDB-375099

Exponent conjecture for k-holes avoiding a (k+1)-hole

摘录 / EXCERPT: Exponent conjecture for $h_k(n)$. For every fixed integer $k\ge 6$,

conjecture | open | MathDB 页面 | 原文线索

V2-045920 | MATHDB-375104

The Erdős-Ko-Rado conjecture for tilings with arbitrary finite tile sets

摘录 / EXCERPT: Erdős-Ko-Rado conjecture for tilings. For sufficiently large n, every intersecting family $\mathcal F\subseteq\mathcal T_n^\Lambda$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045921 | MATHDB-375112

Balanced-border characterization of half-frequency words

摘录 / EXCERPT: Let w be a binary word. A balanced border is a border of w whose numbers of 0s and 1s are equal; the statistics ρ^w and q^w are the letter-frequency quantities assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-045922 | MATHDB-375118

Exact threshold conjecture for the Semple-Steel bound with r-state characters

摘录 / EXCERPT: Let $d_r(n)$ be the maximum, over all binary phylogenetic trees with n leaves, of the minimum number of r-state characters required to define the tree. Let n_r be the least i...

conjecture | open | MathDB 页面 | 原文线索

V2-045923 | MATHDB-375126

Kra-Moreira-Richter-Robertson logarithmic growth conjecture for finite sumsets

摘录 / EXCERPT: For $cdelta cin (0,1)$ and $Ncinmathbb{N}$, let $cphi_{cdelta}(N)$ be the largest integer such that every set $Acsubseteq [N]$ with $|A|cgeq cdelta N$ contains a sumset $B+C$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-045924 | MATHDB-375137

Reversal-Wilf conjecture for contraction containment in labeled trees

摘录 / EXCERPT: Let $\mathcal T_m$ denote the set of labeled trees on $[m]=\{1,\ldots,m\}$. For $T\in\mathcal T_n$ and $U\in\mathcal T_m$, write $T\preceq U$ if U displays T, meaning that T...

conjecture | open | MathDB 页面 | 原文线索

V2-045925 | MATHDB-375140

Gutowski-Rams explicit tournament family conjecture

摘录 / EXCERPT: An explicit family of tournaments is a sequence $(T_n)_{n\geq 1}$ of tournaments, with $|V(T_n)|$ denoting the number of vertices and $\vec\omega(T_n)$ denoting the clique number....

conjecture | open | MathDB 页面 | 原文线索

V2-045926 | MATHDB-375144

The palindrome-to-factor complexity ratio conjecture for non-ultimately periodic words

摘录 / EXCERPT: Let ${\bf x}=(a_i)_{i\geq 0}$ be an infinite word over a finite alphabet Σ. Write $\rho_{\bf x}(n)$ for the number of distinct length-n factors of ${\bf x}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045927 | MATHDB-375164

Friedgut-Kahn-Kalai-Keller antipodal correlation conjecture

摘录 / EXCERPT: Friedgut-Kahn-Kalai-Keller antipodal correlation conjecture. For every increasing $f,g:\{0,1\}^n\to\{0,1\}$, if g is antipodal, then

conjecture | open | MathDB 页面 | 原文线索

V2-045928 | MATHDB-375169

Existence of a tight point in every extreme configuration

摘录 / EXCERPT: The elements of a configuration are points in the $(m-1)$-dimensional probability simplex in $\mathbb{R}^m$. A configuration is extremal if it arises as the configuration associate...

conjecture | open | MathDB 页面 | 原文线索

V2-045929 | MATHDB-375185

Cornet-Dravec-Torres conjecture for domination in odd Johnson graphs J(n, 3)

摘录 / EXCERPT: Cornet-Dravec-Torres conjecture. For every odd integer $n\ge 7$,

conjecture | open | MathDB 页面 | 原文线索

V2-045930 | MATHDB-375205

Hegedüs's balanced Sperner family conjecture

摘录 / EXCERPT: Hegedüs's conjecture. If $\mathcal{F}\subseteq 2^{[n]}$ is a Sperner family and $|\mathcal{F}|\ge n+1$, then $\mathcal{F}$ is 2-balanced.

conjecture | open | MathDB 页面 | 原文线索

V2-045931 | MATHDB-375208

Unimodular flag triangulation conjecture for polymatroid base polytopes

摘录 / EXCERPT: Let P be the base polytope of a polymatroid. A triangulation of P is unimodular if all its simplices are unimodular with respect to the relevant lattice, and it is flag if ever...

conjecture | open | MathDB 页面 | 原文线索

V2-045932 | MATHDB-375209

Ding-Zhao's adjacent Ruzsa-number conjecture

摘录 / EXCERPT: Let m be a positive integer, and let R_m denote the least integer r for which some subset of the residue class ring $4mathbb Z_m$ has between 1 and r ordered representati...

conjecture | open | MathDB 页面 | 原文线索

V2-045933 | MATHDB-375210

Ding-Zhao's revised adjacent Ruzsa-number conjecture

摘录 / EXCERPT: Let R_m be the Ruzsa number of the residue class ring $\mathbb Z_m$, namely the least integer r for which a subset has between 1 and r ordered representations of every resi...

conjecture | open | MathDB 页面 | 原文线索

V2-045934 | MATHDB-375211

The eventual value-six conjecture for Ruzsa numbers

摘录 / EXCERPT: For each positive integer m, let R_m denote the Ruzsa number of $\mathbb Z_m$. Eventual value-six conjecture. ... The conjecture is based on the table of computed values through...

conjecture | open | MathDB 页面 | 原文线索

V2-045935 | MATHDB-375222

Erdős meets Nash-Williams' conjecture for high-girth triangle decompositions

摘录 / EXCERPT: Erdős meets Nash-Williams' conjecture. For every integer g, every sufficiently large K_3-divisible graph G on n vertices satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-045936 | MATHDB-375229

The EF1 conjecture for matroids with a common valuation

摘录 / EXCERPT: Let $M=(E,\mathcal{I})$ be an independence structure, where E is a finite set and $\mathcal{I}$ is closed under taking subsets. Let $\chi(M)$ be the least number of independent s...

conjecture | open | MathDB 页面 | 原文线索

V2-045937 | MATHDB-375230

The EF1 conjecture for matroids with heterogeneous valuations

摘录 / EXCERPT: Let $M=(E,\mathcal{I})$ be a matroid on the finite set of goods E, and let $\chi(M)$ be the least number of independent sets in a partition of E. Suppose $\chi(M)\leq n$ and th...

conjecture | open | MathDB 页面 | 原文线索

V2-045938 | MATHDB-375231

Kajitani-Ueno-Miyano's Hamiltonian matroid conjecture

摘录 / EXCERPT: Let $M=(E,\mathcal{I})$ be a matroid with rank function r. It is Hamiltonian if some cyclic ordering of E has the property that every segment of $r(M)$ consecutive elements is...

conjecture | open | MathDB 页面 | 原文线索

V2-045939 | MATHDB-375232

The EF1 conjecture for individually constrained matroid bundles

摘录 / EXCERPT: Let $M_1,\ldots,M_n$ be matroids on the same finite ground set E, with $\chi(M_i)\leq n$ for every $i=1,\ldots,n$. Suppose n agents have additive valuations $v_1,\ldots,v_n$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-045940 | MATHDB-375233

The finite asymptotic separation index matching conjecture

摘录 / EXCERPT: Finite asymptotic separation index matching conjecture. If the bipartition has combinatorial expansion and $\mathsf{asi}(G)<\infty$, then G has a Borel matching covering A.

conjecture | open | MathDB 页面 | 原文线索

V2-045941 | MATHDB-375234

The type $(4,3)$ finite asymptotic separation index matching conjecture

摘录 / EXCERPT: Type $(4,3)$ finite asymptotic separation index matching conjecture. If $\mathsf{asi}(G)<\infty$, then G has a Borel matching covering A.

conjecture | open | MathDB 页面 | 原文线索

V2-045942 | MATHDB-375237

Period conjecture for three-parameter Subtraction Nim

摘录 / EXCERPT: Period conjecture. The Nim value function is periodic with a period equal to one of $a+b$, $b+c$, or $c+a$.

conjecture | open | MathDB 页面 | 原文线索

V2-045943 | MATHDB-375248

The product-beating conjecture for non-degenerate genus-one systems

摘录 / EXCERPT: Let $L=(L_1,\ldots,L_t)$ be a non-degenerate genus-one translation-invariant system of equations over $\mathbb{F}_q$, where each L_j is a translation-invariant equation in s va...

conjecture | open | MathDB 页面 | 原文线索

V2-045944 | MATHDB-375263

The strict density bound for monochromatic k-in-a-row on the integer grid

摘录 / EXCERPT: Let $D(k,\mathbb{Z}^2)$ denote the maximum density of a subset of the integer grid avoiding the relevant k-in-a-row configurations, and let $3\mathbb{N}$ denote the positive mult...

conjecture | open | MathDB 页面 | 原文线索

V2-045945 | MATHDB-375264

The strict torus-to-grid density inequality for monochromatic k-in-a-row

摘录 / EXCERPT: Let $D(k,\mathbb{Z}^2)$ and $D(k,(\mathbb{Z}/k\mathbb{Z})^2)$ denote the corresponding maximum avoiding densities on the integer grid and the discrete torus, respectively. Torus-d...

conjecture | open | MathDB 页面 | 原文线索

V2-045946 | MATHDB-375265

The hypercube maximum-density conjecture for monochromatic k-in-a-row

摘录 / EXCERPT: Let $[k]^d$ be the d-dimensional grid, with $k\in\mathbb{N}_{\geq 2}$ and $d\in\mathbb{N}$, and let $D(k,[k]^d)$ denote the maximum density of a configuration avoiding the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-045947 | MATHDB-375267

Twin-period recurrence conjecture for reversible elementary cellular automaton 115

摘录 / EXCERPT: Twin-period recurrence conjecture. Assume that p and $p+3$ are twin periods. Then the next twin pair is p' and $p'+3$, where

conjecture | open | MathDB 页面 | 原文线索

V2-045948 | MATHDB-375268

Odd-index period formula conjecture for the family $1(01)^k$

摘录 / EXCERPT: Odd-index period formula conjecture. The period function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045949 | MATHDB-375269

Even-index period formula conjecture for the family $1(01)^k$

摘录 / EXCERPT: Even-index period formula conjecture. The period function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-045950 | MATHDB-375276

The block sets conjecture

摘录 / EXCERPT: Block sets conjecture. For every pair of positive integers m and k and every template T over $[m]$, there exist positive integers n and d such that every k-colouring of...

conjecture | open | MathDB 页面 | 原文线索

V2-045951 | MATHDB-375277

The one-point extension conjecture for Euclidean Ramsey sets

摘录 / EXCERPT: One-point extension conjecture. The set $X\cup\{z\}$ is Ramsey.

conjecture | open | MathDB 页面 | 原文线索

V2-045952 | MATHDB-375281

Quadratic triangulations for dyadic matroid base polytopes

摘录 / EXCERPT: Let M be a dyadic matroid, meaning a matroid representable over the dyadic partial field. Let $P(M)$ be its matroid base polytope. Dyadic matroid triangulation conjecture. The ma...

open_problem | open | MathDB 页面 | 原文线索

V2-045953 | MATHDB-375309

Algorithmic verification criterion for the restricted asymmetric case

摘录 / EXCERPT: Algorithmic verification criterion. The following conditions are equivalent: the configuration $\lambda K+\mathbb{Z}^3$ is a lattice covering realizing the specified combinatorial...

conjecture | open | MathDB 页面 | 原文线索

V2-045954 | MATHDB-375311

Jackson-Tanigawa's hyperconnectivity maximality conjecture

摘录 / EXCERPT: Jackson-Tanigawa's hyperconnectivity conjecture. For $d\geq 1$ and $\mathcal{X}=\{K_{d+2},K_{d+1,d+1}\}$, $\mathcal{H}_n^d$ is the unique maximal $\mathcal{X}$-matroid, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045955 | MATHDB-375312

Jackson-Tanigawa's rigidity maximality conjecture

摘录 / EXCERPT: Jackson-Tanigawa's rigidity conjecture. For $d\geq 3$ and $\mathcal{X}=\{K_{d+2},K_{d+2,d+2}\}$, $\mathcal{R}_n^d$ is the unique maximal $\mathcal{X}$-matroid, and...

conjecture | open | MathDB 页面 | 原文线索

V2-045956 | MATHDB-375324

Scaffold exchange symmetry conjecture

摘录 / EXCERPT: Scaffold exchange conjecture. For every $n\ge1$, $k\ge0$, $r\ge1$, every diagonal multiset m of size n, and every scaffold Ξ,

conjecture | open | MathDB 页面 | 原文线索

V2-045957 | MATHDB-375325

Refined scaffold exchange symmetry conjecture

摘录 / EXCERPT: Fix $n\ge1$, a diagonal multiset m realizable by an area word of size n, $k\ge0$, $r\ge1$, a scaffold Ξ, and a matrix $g=(g_{t,\delta})$ recording per-gap, per-diagonal num...

conjecture | open | MathDB 页面 | 原文线索

V2-045958 | MATHDB-375347

Super-exponential growth of symmetric chain decompositions of $L(m,n)$

摘录 / EXCERPT: Let $L(m,n)$ be the minuscule lattice of partitions contained in an m by n box, and let $\#\operatorname{SCD}(L(m,n))$ denote the number of symmetric chain decompositions of th...

conjecture | open | MathDB 页面 | 原文线索

V2-045959 | MATHDB-375351

Moss-Pedersen conjecture on balanced subfamilies of maximum complement-free families

摘录 / EXCERPT: Moss-Pedersen conjecture. For all sufficiently large n, there exists a family $\mathcal F\subseteq\binom{[2n]}{n}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-045960 | MATHDB-375359

Delta-matroid invariance conjecture for ML degrees

摘录 / EXCERPT: Let $C_1,C_2\in\mathbb{C}^{d\times d}$ be scaling matrices, with associated delta-matroids, and let c_1,c_2 be the corresponding scalings. Delta-matroid invariance conjecture. If...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-045961 | MATHDB-375360

Rank-four principal-minor conjecture for scaling matrices

摘录 / EXCERPT: Let $C\in\mathbb{C}^{d\times d}$ be a scaling matrix of rank four. A principal 4-minor is non-vanishing when the corresponding principal submatrix has nonzero determinant. Rank-f...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045962 | MATHDB-375368

Near-quadratic Elekes-Rónyai expander conjecture over the reals

摘录 / EXCERPT: Let $f\in\mathbb R[x,y]$ be neither additive nor multiplicative. For a finite set $A\subset\mathbb R$, write $f(A,A)=\{f(a_1,a_2):a_1,a_2\in A\}$. Near-quadratic Elekes-Rónyai expa...

conjecture | open | MathDB 页面 | 原文线索

V2-045963 | MATHDB-375370

The even-rank conjecture for CRIM positions

摘录 / EXCERPT: A partition is a Young diagram, and its rank is the relevant rank statistic on the partition used in the CRIM analysis; a position is a $\mathcal P$-position when the next player l...

conjecture | open | MathDB 页面 | 原文线索

V2-045964 | MATHDB-375371

The Sprague-Grundy conjecture for square rectairs

摘录 / EXCERPT: A rectair is the rectair family used in CRIM, and $\mathcal G$ denotes its Sprague-Grundy value. Let r and k be integers with $0\leq k<r$. Square-rectair conjecture. ... The conj...

conjecture | open | MathDB 页面 | 原文线索

V2-045965 | MATHDB-375372

The Sprague-Grundy conjecture for near-square rectairs

摘录 / EXCERPT: A rectair is the rectair family used in CRIM, and $\mathcal G$ denotes its Sprague-Grundy value. Near-square rectair conjecture. For $r\geq7$, ... In addition, ... All other rectairs h...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045966 | MATHDB-375373

The even-by-even hook Conway-pair conjecture

摘录 / EXCERPT: A partition is a Young diagram, and its Conway pair is the pair of game-theoretic invariants used in the Conway-Gurvich-Ho classification. An even-by-even hook is a hook with an ev...

conjecture | open | MathDB 页面 | 原文线索

V2-045967 | MATHDB-375374

The thick-hook meld conjecture

摘录 / EXCERPT: A thick hook is the thick-hook family of partitions used in CRIM, and a meld is the partition operation defined in the paper. A losing position is a $\mathcal P$-position. Thick-ho...

conjecture | open | MathDB 页面 | 原文线索

V2-045968 | MATHDB-375430

Frankl-Kiselev-Kupavskii conjecture on symmetric differences of intersecting families

摘录 / EXCERPT: Frankl-Kiselev-Kupavskii conjecture. If $\mathcal{F}\subseteq\binom{[n]}{k}$ is intersecting and $n>10k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-045969 | MATHDB-375432

Cieliebak-Mohnke's monotone-extremal conjecture for projective space

摘录 / EXCERPT: Let ω_{FS} denote the Fubini-Study form on $\mathbb{CP}^n$. For a closed Lagrangian torus $L\subset(X,\omega)$, call L extremal when...

conjecture | open | MathDB 页面 | 原文线索

V2-045970 | MATHDB-375433

Fukaya-Seidel-Smith's extremal-torus conjecture for ellipsoids

摘录 / EXCERPT: Let $0<a_1\le a_2\le\cdots\le a_n<\infty$, and let $E^{2n}(a_1,\dots,a_n)$ be the corresponding ellipsoid with standard symplectic form ω_{std}. A Lagrangian torus...

conjecture | open | MathDB 页面 | 原文线索

V2-045971 | MATHDB-375441

Negative-coefficient conjecture for complementary polynomial pairs

摘录 / EXCERPT: Negative-coefficient conjecture. In the remaining 33 pairs of the polynomials A_{abcd} and $B_{2-a,2-b,2-c,2-d}$, including the case where $p\neq 2$ for $a=b=c=d=1$, one of the t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045972 | MATHDB-375445

Quadraticity conjecture for matroid base polytopes

摘录 / EXCERPT: Matroid-base quadraticity conjecture. The toric ideal $I_{P_B(\mathcal{M})}$ is quadratic.

conjecture | open | MathDB 页面 | 原文线索

V2-045973 | MATHDB-375465

Sarnak's extremal height conjecture for flat tori

摘录 / EXCERPT: Let T be an n-dimensional flat torus of volume 1, represented by a lattice in Euclidean space. Let the height of T be $\zeta'(0)$, equivalently the negative logarithm of it...

conjecture | open | MathDB 页面 | 原文线索

V2-045974 | MATHDB-375466

Bebeacua's multiplicity-two characteristic-sequence conjecture

摘录 / EXCERPT: Bebeacua's conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-045975 | MATHDB-375467

Classification conjecture for complete partitions

摘录 / EXCERPT: Complete-partition classification conjecture. A partition λ is complete if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-045976 | MATHDB-375468

Deficient-score-sequence characterization of complete partitions

摘录 / EXCERPT: Deficient-sequence characterization conjecture. A partition λ is complete if and only if, for every $\lambda'\subseteq\lambda$, the set $\mathfrak{S}^2(\lambda')$ has no de...

conjecture | open | MathDB 页面 | 原文线索

V2-045977 | MATHDB-375471

Equal-probability conjecture for the expected sibling deficit

摘录 / EXCERPT: There are N coupon types, with draws made independently according to a probability vector $\mathbf{p}=(p_1,\ldots,p_N)$ in the open simplex ... and let T_N be the first time ever...

conjecture | open | MathDB 页面 | 原文线索

V2-045978 | MATHDB-375479

Collapse conjecture for bounded-active-coordinate Hales-Jewett numbers

摘录 / EXCERPT: Collapse conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-045979 | MATHDB-375489

Slowest-chain conjecture for the Miura-ori flip graph

摘录 / EXCERPT: Slowest-chain conjecture. Among all ν-chains, the antidiagonal chain maximises

conjecture | open | MathDB 页面 | 原文线索

V2-045980 | MATHDB-375495

Kalai's Abstract Polynomial Hirsch Conjecture

摘录 / EXCERPT: Let ${\mathcal G}'(k,n)$ be the collection of graphs whose vertices are labeled by k-subsets of an n-element set, with the property that for vertices labeled by S and T, th...

conjecture | open | MathDB 页面 | 原文线索

V2-045981 | MATHDB-375496

Kalai's monomial-ideal diameter conjecture

摘录 / EXCERPT: Let $I\subset k[x_1,\ldots,x_n]$ be a square-free monomial ideal generated in degree d and having a linear presentation. Let G_I be the graph whose vertices are the minimal mono...

conjecture | open | MathDB 页面 | 原文线索

V2-045982 | MATHDB-375500

Cheong-Goaoc-Holmsen acyclicity conjecture for spaces of line transversals

摘录 / EXCERPT: Let d be a positive integer, and let ${\mathcal F}$ be a family of at least two pairwise disjoint open convex sets in ${\mathbb R}^d$. A line in ${\mathbb R}^d$ is transversal to...

conjecture | open | MathDB 页面 | 原文线索

V2-045983 | MATHDB-375506

Frankl--Wang conjecture for non-trivial cross-intersecting families of different uniformities

摘录 / EXCERPT: Let $n\ge 2a\ge 2b\ge 4$, and let $A_0\in\binom{[2,n]}{a}$ and $B_0\in\binom{[2,n]}{b}$ satisfy $A_0\cap B_0\ne\emptyset$. Define ... ... Two families are cross-intersecting if every m...

conjecture | open | MathDB 页面 | 原文线索

V2-045984 | MATHDB-375507

Balogh-Lidický-Mubayi-Pfender-Volec upper-profile conjecture for the red-blue star

摘录 / EXCERPT: Balogh-Lidický-Mubayi-Pfender-Volec's upper-profile conjecture. There exists $y\in(0,1/4)$ such that, for every $\beta\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-045985 | MATHDB-375508

Balogh-Lidický-Mubayi-Pfender-Volec lower-profile conjecture for the red-blue star

摘录 / EXCERPT: Balogh-Lidický-Mubayi-Pfender-Volec's lower-profile conjecture. For every $\beta\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-045986 | MATHDB-375509

Sharpness conjecture for the three-uniform Erdős-Frankl-Pach bound

摘录 / EXCERPT: Let $M_3(n)$ denote the maximum size of a family $\mathcal{F}\subseteq\binom{[n]}{4}$ with $\operatorname{VC}(\mathcal{F})\le 3$. The paper establishes the lower bound ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-045987 | MATHDB-375513

Maximum-width conjecture for valid-extension sets of Gilbreath sequences

摘录 / EXCERPT: For each $n\geq 2$, let $\mathcal{G}_n$ be the set of Gilbreath sequences of length n, let K_S denote the valid-extension set associated with S, and let U_n be the doubling...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045988 | MATHDB-375517

Large balanced-subset-free families of middle subsets

摘录 / EXCERPT: Balanced-subset family conjecture. For each k and all sufficiently large n, there is a family $\mathcal{F}$ of n-subsets of $[2n]$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-045989 | MATHDB-375521

Perfect-power square-maximality conjecture for induced grid subgraphs

摘录 / EXCERPT: Perfect-power square-maximality conjecture. Equality should occur only for the d-dimensional box $P_n^{\square d}$, up to lattice translation and coordinate permutation.

conjecture | open | MathDB 页面 | 原文线索

V2-045990 | MATHDB-375531

Vector-space Erdős Matching Conjecture

摘录 / EXCERPT: Let V be an n-dimensional vector space over $\mathbb{F}_q$, and let $m_q(n,k,s)$ be the maximum size of a family $\mathcal{F}\subseteq\genfrac{[}{]}{0pt}{}{V}{k}$ containing no...

conjecture | open | MathDB 页面 | 原文线索

V2-045991 | MATHDB-375578

Matsubara-Heo-Telen dual-degree conjecture for reciprocal linear spaces

摘录 / EXCERPT: Let $L\subseteq\mathbb{P}^n$ be a linear subspace of dimension d not contained in any coordinate hyperplane, let L^{-1} be its reciprocal linear space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-045992 | MATHDB-375579

Matsubara-Heo-Telen local-multiplicity conjecture for principal matroid determinants

摘录 / EXCERPT: Let $L\subseteq\mathbb{P}^n$ be a linear subspace of dimension d not contained in any coordinate hyperplane, let $M=\operatorname{M}(L)$, and let F be a flat of M. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-045993 | MATHDB-375591

Binary path configuration conjecture for avalanche homology

摘录 / EXCERPT: Let P_n be the directed path on n vertices, let β_i denote the i-th Betti number of its avalanche complex, and let a binary configuration be an initial configuration wh...

conjecture | open | MathDB 页面 | 原文线索

V2-045994 | MATHDB-375601

Sparsest-obstruction conjecture for integer pinwheel scheduling

摘录 / EXCERPT: Sparsest-obstruction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-045995 | MATHDB-375617

Grunbaum-Yaakobi conjecture on fair coupon-collecting mechanisms

摘录 / EXCERPT: Grunbaum-Yaakobi conjecture. Among all fair mechanisms on fixed parameters (n,ell), the full model maximizes the expected coverage time.

conjecture | open | MathDB 页面 | 原文线索

V2-045996 | MATHDB-375621

Monotonicity conjecture for the multifurcating-tree growth constants

摘录 / EXCERPT: For each integer $k\geqslant 2$, let γ_k be the constant in the asymptotic growth ... of the maximal ranks of at-most-k-furcating rooted trees with n leaves. Monotonicity...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-045997 | MATHDB-375625

Dombi's obstruction conjecture for strict monotonicity of higher-fold representation functions

摘录 / EXCERPT: Let $A\subseteq\mathbb N$ and let $h\geq 1$ be an integer. Define the ordered h-fold representation function by ... The complement $\mathbb N\setminus A$ is co-infinite when it is...

conjecture | open | MathDB 页面 | 原文线索

V2-045998 | MATHDB-375631

Finiteness of the single-theater misère quotient

摘录 / EXCERPT: Let the single-theater game have escalation set $S=\{1,\dots,m\}$. Its misère quotient is the commutative monoid obtained by identifying positions that are interchangeable in every...

conjecture | open | MathDB 页面 | 原文线索

V2-045999 | MATHDB-375642

Asymptotic sharpness of the layered lower bound for Boolean-lattice chain covers

摘录 / EXCERPT: Asymptotic sharpness conjecture. For every fixed $r\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-046000 | MATHDB-375653

Nonnegativity conjecture for the second gamma-number of discrete pseudomanifolds

摘录 / EXCERPT: Nonnegativity conjecture for discrete pseudomanifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-046001 | MATHDB-375663

Restricted Manickam--Miklós--Singhi conjecture near the 3k threshold

摘录 / EXCERPT: Let $\varepsilon,C>0$ be constants. For sufficiently large positive integers n,k satisfying ... let $x_1,\dots,x_n$ be real numbers with sum zero, with at most C nonnegative entr...

conjecture | open | MathDB 页面 | 原文线索

V2-046002 | MATHDB-375664

The MMS conjecture on non-negative sums in sampling without replacement

摘录 / EXCERPT: MMS conjecture. If $n\ge 4k$, then for every $P\in\mathcal{N}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-046003 | MATHDB-375665

Aydinian-Blinovsky refinement of the MMS conjecture

摘录 / EXCERPT: Aydinian-Blinovsky refinement. For every $k\in[n-1]$ and every $P\in\mathcal{N}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-046004 | MATHDB-375687

Noel's extremal complete multipartite graph conjecture

摘录 / EXCERPT: Let G be a k-chromatic graph with at most mk vertices, and let K_{m*k} denote the complete k-partite graph with m vertices in each part. Noel's conjecture. ... The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-046005 | MATHDB-375706

Ellis-Friedgut-Pilpel conjecture on extremal t-intersecting families of permutations

摘录 / EXCERPT: Ellis-Friedgut-Pilpel conjecture. For any $n\in\mathbb N$ and $1\leq t\leq n$, the maximum size of a t-intersecting family in S_n is attained by one of the families...

conjecture | open | MathDB 页面 | 原文线索

V2-046006 | MATHDB-375707

The shifted-product nonempty-interior conjecture for product sets

摘录 / EXCERPT: Let $n\ge 3$ and let $A\subseteq \mathbb R$ be a Borel set. Write $\dim_H A$ for the Hausdorff dimension of A. Shifted-product nonempty-interior conjecture. For every $n\ge 3$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-046007 | MATHDB-375720

Generating-function conjecture for greedy (m,r)-Tamari intervals

摘录 / EXCERPT: For integers $m\geq 1$ and $1\leq r\leq m-1$, let $d53c^{(m,r)}_n$ be the set of greedy (m,r)-Tamari intervals of size n, and let ... be their generating function with the weight...

conjecture | open | MathDB 页面 | 原文线索

V2-046008 | MATHDB-375723

Sharp sumset lower-bound conjecture for hypercubes

摘录 / EXCERPT: Sharp sumset lower-bound conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-046009 | MATHDB-375737

White's symmetric exchange conjecture for matroid bases

摘录 / EXCERPT: Let M be a matroid, and let $\{B_1,B_2\}$ and $\{B_1',B_2'\}$ be unordered pairs of bases such that the multiset union of B_1 and B_2 is equal to the multiset union...

conjecture | open | MathDB 页面 | 原文线索

V2-046010 | MATHDB-375739

Collapse conjecture for the Hales-Jewett bracket hierarchy

摘录 / EXCERPT: Let $[t]^n$ be the set of words of length n over an alphabet of size t, let r be the number of colors, and let $\mathrm{HJ}^{[k]}(t,r)$ denote the least dimension forcing a m...

conjecture | open | MathDB 页面 | 原文线索

V2-046011 | MATHDB-375740

Diagonal-only coloring conjecture for Hales-Jewett lines

摘录 / EXCERPT: Let $[t]^n$ be the set of words of length n over an alphabet of size t, let r be the number of colors, and let $\mathrm{HJ}(t,r)$ be the Hales-Jewett number. The diagonal is...

conjecture | open | MathDB 页面 | 原文线索

V2-046012 | MATHDB-375741

Symmetric-coloring extremality conjecture for Hales-Jewett numbers

摘录 / EXCERPT: Let $[t]^n$ be the set of words of length n over an alphabet of size t, let r be the number of colors, and let $\mathrm{HJ}(t,r)$ be the Hales-Jewett number. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-046013 | MATHDB-375751

The m-ary polynomial-form conjecture for subsequence frequencies

摘录 / EXCERPT: Let $B_p^{(m)}(n,k)$ denote the number of m-ary words of length n having exactly k occurrences of an m-ary word p of length ell with $r\geq 2$ runs. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-046014 | MATHDB-375752

Frankl-Wang's $3/7$ conjecture on maximum-degree ratios

摘录 / EXCERPT: Frankl-Wang's $3/7$ conjecture. If $\mathcal{F}\subseteq\binom{[n]}{k}$ is intersecting, $n>100k$, and $|\mathcal{F}|>\binom{n-3}{k-3}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-046015 | MATHDB-375763

Balogh-Linz-Patkós' odd-parity conjecture for intersecting k-Sperner families

摘录 / EXCERPT: Balogh-Linz-Patkós' conjecture. There exists a positive integer $n_0=n_0(k,t)$ such that if $n+t$ is odd, $n>n_0$, and $\mathcal{F}\subseteq 2^{[n]}$ is a t-intersecting k-Sper...

conjecture | open | MathDB 页面 | 原文线索

V2-046016 | MATHDB-375764

Automorphism-group conjecture for wreath-product designs

摘录 / EXCERPT: Let $\widetilde{\mathscr{D}}$ be the input design, let e and $\widetilde{k}$ be the parameters of Construction $\mathscr{D}(\widetilde{\mathscr{D}},e)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-046017 | MATHDB-375777

Ellis-Keller-Lifshitz stability conjecture for intersecting families

摘录 / EXCERPT: Let n, k, t, and d satisfy ... Let $\mathcal{F}\subseteq\binom{[n]}{k}$ be t-intersecting. Ellis-Keller-Lifshitz stability conjecture. If ... then there exists a t-subset...

conjecture | open | MathDB 页面 | 原文线索

V2-046018 | MATHDB-375792

The collapsibility conjecture for prism complexes of ample sets

摘录 / EXCERPT: Let U be a Cartesian product and let $S\subseteq U$ be an ample set. Write $\operatorname{Box}(S)$ for the boxes associated with S, and let $|\operatorname{Box}(S)|$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-046019 | MATHDB-375795

Archer-Laudone enumeration conjectures for combined classical and arrow pattern avoidance

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the set of permutations of $[n]$, and let $a_n(\pi,\alpha)$ denote the number of permutations in $\mathfrak{S}_n$ that avoid both the classical pattern...

conjecture | open | MathDB 页面 | 原文线索

V2-046020 | MATHDB-375798

Sokal's Hankel-total positivity conjecture for GKP row-generating polynomials

摘录 / EXCERPT: Let $P_n(x;\bm{\mu})$ be the row-generating polynomials of the Graham-Knuth-Patashnik recurrence, with parameters $\bm{\mu}=(\alpha,\beta,\gamma,\alpha',\beta',\gamma')$ treated as...

conjecture | open | MathDB 页面 | 原文线索

V2-046021 | MATHDB-375817

The q,t-dihedral sieving conjecture for cluster parking functions

摘录 / EXCERPT: The q,t-dihedral sieving conjecture. For every $\mathbf{F}\in\operatorname{Dih}_{n+2}$ and $\sigma\in\mathfrak{S}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-046022 | MATHDB-375819

The Euler-number specialization conjecture for Gordon's ring

摘录 / EXCERPT: Euler-number specialization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046023 | MATHDB-375830

Degree bound for remainder terms in Miura-ori flip-graph counts

摘录 / EXCERPT: Let $d\ge5$ and let (a,b) be a split with $a\le b$. Write $R_{(a,b)}$ for the remainder after removing the single-side contributions from the count associated with the split. Deg...

conjecture | open | MathDB 页面 | 原文线索

V2-046024 | MATHDB-375831

Per-split sub-top coefficient conjecture for Miura-ori flip-graph counts

摘录 / EXCERPT: Let $d\ge5$, let (a,b) satisfy $a+b=d$ and $a\le b$, and let $R_{(a,b)}(m,n)$ be the interior remainder. Denote by $[m^{d-3}]R_{(a,b)}(m,n)$ the coefficient of m^{d-3}; by symm...

conjecture | open | MathDB 页面 | 原文线索

V2-046025 | MATHDB-375832

Sub-leading Laurent coefficient conjecture for Miura-ori flip-graph counts

摘录 / EXCERPT: Sub-leading Laurent coefficient conjecture. For every $d\ge5$ and every $m\ge2$,

conjecture | solved | MathDB 页面 | 原文线索

V2-046026 | MATHDB-375833

Baxter onset defect conjecture for Miura-ori boundary corrections

摘录 / EXCERPT: Let $E_d(m,n)$ be the count and let $p_d(m,n)$ denote its closed-form polynomial in the threshold regime. For $s=\min(m,n)<d-1$, define the boundary correction, with the other vari...

conjecture | open | MathDB 页面 | 原文线索

V2-046027 | MATHDB-375834

Boundary-correction structure conjecture for Miura-ori flip-graph counts

摘录 / EXCERPT: Boundary-correction structure conjecture. For every $d\ge4$ and every m,n with exactly one of them below $d-1$, the count $E_d(m,n)$ differs from $p_d(m,n)$ by a polynomial corre...

conjecture | open | MathDB 页面 | 原文线索

V2-046028 | MATHDB-375846

Mason's rank conjecture for symmetric powers of matroids

摘录 / EXCERPT: Mason's rank conjecture. The rank of N is

conjecture | open | MathDB 页面 | 原文线索

V2-046029 | MATHDB-375847

The Bolza surface and Klein quartic extremal bass-note conjectures

摘录 / EXCERPT: Bolza-Klein bass-note conjecture. (a) The Bolza surface uniquely maximizes the bass note among closed and orientable hyperbolic surfaces. (b) The Klein quartic uniquely maximizes t...

conjecture | open | MathDB 页面 | 原文线索

V2-046030 | MATHDB-375851

Frankl-Wang product Hilton-Milner conjecture for cross-intersecting families

摘录 / EXCERPT: Frankl-Wang conjecture. Under these hypotheses,

conjecture | refuted | MathDB 页面 | 原文线索

V2-046031 | MATHDB-375882

Odd-cycle characterization of bases for analytic spread Jacobians

摘录 / EXCERPT: Let G and H be graphs on n and m vertices, respectively, with $n,m\geq 3$. Let $J_{G,H}$ be the associated binomial edge ideal, let $\ell(J_{G,H})$ denote its analytic spre...

conjecture | open | MathDB 页面 | 原文线索

V2-046032 | MATHDB-375887

Frankl's binomial norm conjecture for intersecting-union families

摘录 / EXCERPT: Frankl's conjecture. For every IU-family $\mathcal F\subseteq 2^{[n]}$,

conjecture | open | MathDB 页面 | 原文线索

V2-046033 | MATHDB-375890

The monotonicity conjecture for ternary cubic multinomial sums

摘录 / EXCERPT: Monotonicity conjecture for $S_3^{(3)}$. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-046034 | MATHDB-375899

The half-density conjecture for sum-free subsets of symmetric convex regions

摘录 / EXCERPT: Half-density conjecture. For every symmetric, convex region $R\subset\mathbb{R}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-046035 | MATHDB-375906

The universal minimum-modulus conjecture for unique multiset sums

摘录 / EXCERPT: Universal minimum-modulus conjecture. For every $n\geq 2$ and every $N<M(n)$, no set of n residues is valid mod N; equivalently, the super-increasing set...

conjecture | open | MathDB 页面 | 原文线索

V2-046036 | MATHDB-375918

Balister-Győri-Schelp conjecture on Hadamard pair solutions

摘录 / EXCERPT: Balister-Győri-Schelp conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046037 | MATHDB-375976

Two disjoint abelian sets small-doubling conjecture

摘录 / EXCERPT: Two-set small-doubling conjecture. If $|S^2|\leq 3k-3$, then $\langle S\rangle$ is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-046038 | MATHDB-375977

Three disjoint abelian sets strengthened small-doubling conjecture

摘录 / EXCERPT: Three-set small-doubling conjecture. If $|S^2|\leq 3k-4$, then $\langle S\rangle$ is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-046039 | MATHDB-375978

Torsion-free groups small-doubling abelianity conjecture

摘录 / EXCERPT: Small-doubling abelianity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046040 | MATHDB-376018

Extremal-family conjecture for non-star intersecting s-stable families

摘录 / EXCERPT: Extremal-family conjecture. If $\mathcal{F}\subseteq\binom{[n]}{k}_{s\text{-stable}}$ is a non-star intersecting family with $\tau(\mathcal{F})\geq 2$, then, for n large enough-s...

conjecture | open | MathDB 页面 | 原文线索

V2-046041 | MATHDB-376033

Belkhechine-Bouaziz-Boudabbous-Pouzet conjecture for path-reversed tournaments

摘录 / EXCERPT: Let Q_n be the tournament on $[n]=\{1,\ldots,n\}$ obtained from the natural transitive tournament by reversing precisely the consecutive pairs $12,23,\ldots,(n-1)n$. Equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-046042 | MATHDB-376070

Ostmann's inverse Goldbach conjecture

摘录 / EXCERPT: Let $\mathscr P$ denote the set of all primes. For sets S,T of positive integers, define $S\sim T$ to mean that their symmetric difference is finite. Ostmann's inverse Goldbach c...

conjecture | open | MathDB 页面 | 原文线索

V2-046043 | MATHDB-376085

Yagi-Yoshinaga's filtered VG-algebra reconstruction conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ and $\mathcal{B}$ be real central hyperplane arrangements, and let R be an integral domain with $\operatorname{char}R\neq 2$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-046044 | MATHDB-376086

The middle-regime doubling conjecture for distinct rectangle tilings

摘录 / EXCERPT: Middle-regime doubling conjecture. For every $m\ge4$ and every n in this middle regime,

conjecture | open | MathDB 页面 | 原文线索

V2-046045 | MATHDB-376109

Kaul-Mudrock conjecture for unlabeled list color functions of edgeless graphs

摘录 / EXCERPT: Let $\bar{K}_n$ be an edgeless graph on n vertices. For each $n \in \mathbb{N}$, define $P^u_{\ell}(\bar{K}_n,k)$ as the minimum, over all k-assignments L for $\bar{K}_n$, of...

conjecture | open | MathDB 页面 | 原文线索

V2-046046 | MATHDB-376110

Duality map for biperiodic Fibonacci words

摘录 / EXCERPT: Let $\mathfrak{F}^{(a,b)}$ be the infinite binary biperiodic Fibonacci word with parameters $a,b\geqslant 1$. There is an explicit, effectively computable map θ on infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-046047 | MATHDB-376131

The no-pinned-entry construction conjecture for even-order latin squares

摘录 / EXCERPT: Let $n\geqslant 28$ be an even integer. A latin square of order n is an $n\times n$ array in which each symbol occurs exactly once in each row and column, and a transversal is a...

conjecture | open | MathDB 页面 | 原文线索

V2-046048 | MATHDB-376143

The exact-value conjecture for the two-color Hales-Jewett number

摘录 / EXCERPT: Let $HJ(k,2)$ denote the least dimension such that every coloring of $[k]^{HJ(k,2)}$ with two colors contains a monochromatic combinatorial line. Exact-value conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-046049 | MATHDB-376175

Bradač-Liu-Wu-Xu conjecture on admissible colorings of ordered cliques

摘录 / EXCERPT: Bradač-Liu-Wu-Xu conjecture. For every integer $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-046050 | MATHDB-376191

O'Neill's supersaturation conjecture for eventown families

摘录 / EXCERPT: O'Neill's supersaturation conjecture. If $\mathcal{A}\subset 2^{[n]}$ consists of even-sized subsets and

conjecture | open | MathDB 页面 | 原文线索

V2-046051 | MATHDB-376196

Asymptotically almost-every-matroid conjecture for the GHV equality property

摘录 / EXCERPT: For each n, let $u(n)$ be the number of unlabeled matroids on n elements. A matroid property is a class of matroids closed under automorphism, and let $\mathcal{P}_u(n)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-046052 | MATHDB-376197

GHV upper-bound conjecture for Stanley-Reisner ideals of matroids

摘录 / EXCERPT: Let M be a matroid and let $I_{\Delta(M)}$ be its Stanley-Reisner ideal. Write $\widehat{\rho}(I_{\Delta(M)})$ for its asymptotic resurgence, $\omega(I_{\Delta(M)})$ for the larg...

conjecture | open | MathDB 页面 | 原文线索

V2-046053 | MATHDB-376201

The optimal simplex anticode conjecture in A_n

摘录 / EXCERPT: Let $A_n(D)$ be the maximum cardinality of a subset $\mathcal F\subseteq\mathbb Z^n$ whose diameter in the anticode metric $d_{\mathrm a}$ is at most D. Let $S_{n,D}$ be the simp...

conjecture | open | MathDB 页面 | 原文线索

V2-046054 | MATHDB-376207

Huang's conjecture on balanced cyclic $(132,213)$-avoiding permutations

摘录 / EXCERPT: Let $\mathcal C_n(132,213)$ be the set of cyclic permutations of size n avoiding the patterns 132 and 213, and let $\mathcal C_n^B(132,213)$ be the subset of balanced such pe...

conjecture | open | MathDB 页面 | 原文线索

V2-046055 | MATHDB-376234

Sharpening of Szabó's conjecture for arithmetic-progression-intersection families

摘录 / EXCERPT: Sharpening of Szabó's conjecture. Equality holds in this lower bound for every $N\ge 1$, that is,

conjecture | open | MathDB 页面 | 原文线索

V2-046056 | MATHDB-376236

Catalan classification conjecture for extremal even-intersecting lattice-path families

摘录 / EXCERPT: Catalan classification conjecture. There are exactly C_n families of size 2^n in which every two distinct paths have an even number of common edges.

conjecture | open | MathDB 页面 | 原文线索

V2-046057 | MATHDB-376283

Polynomial bound conjecture for the Schrijver number of the square of the Kneser graph

摘录 / EXCERPT: Let G_n be the Schrijver graph, and let $G_n^{\boxtimes 2}$ denote its strong square. The Schrijver number of a graph G, denoted by $\vartheta'(G)$, is the corresponding streng...

conjecture | open | MathDB 页面 | 原文线索

V2-046058 | MATHDB-376304

Two-barrier Ore-degree conjecture for direct-sum matchings

摘录 / EXCERPT: Two-barrier Ore-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046059 | MATHDB-376305

Frankl-Kupavskii weighted-construction conjecture for the Erdős-Kleitman problem

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, let $e(n,s)$ be the maximum of $|\mathcal F|$ over families $\mathcal F\subseteq 2^{[n]}$ with matching number $\nu(\mathcal F)<s$, and, for a weight fu...

conjecture | open | MathDB 页面 | 原文线索

V2-046060 | MATHDB-376306

Frankl-Kupavskii conjecture for the shifted family \(\mathcal P(m,s,\ell)\)

摘录 / EXCERPT: Let $n=s(m+1)-\ell$ and define ... For $s\ge 2$, $m\ge 1$, and $1\le\ell\le\lceil s/2\rceil$, this family is a weighted construction with matching number less than s. Frankl-Kupavski...

conjecture | open | MathDB 页面 | 原文线索

V2-046061 | MATHDB-376307

Completeness conjecture for weighted extremal families in the Erdős-Kleitman problem

摘录 / EXCERPT: Write $n=sm+c=s(m+1)-\ell$, and for $0\le k\le m$ define ... and let $\mathcal P(m,s,\ell)$ be the shifted family $\{F\subseteq[n]\colon |F|+|F\cap[\ell-1]|\ge m+1\}$. Completeness c...

conjecture | open | MathDB 页面 | 原文线索

V2-046062 | MATHDB-376314

Restricted-sumset analogue of the generalized Sárközy conjecture

摘录 / EXCERPT: Restricted-sumset analogue of the generalized Sárközy conjecture. Let $d\ge 2$. If $p\equiv 1\pmod d$ is sufficiently large, then there is no set $A\subset\mathbb F_p$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-046063 | MATHDB-376341

Dress's rank formula conjecture for 3-dimensional rigidity matroids

摘录 / EXCERPT: The Dress Conjecture. The rank of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-046064 | MATHDB-376342

Whiteley's conjecture equating the 3-dimensional rigidity and cofactor matroids

摘录 / EXCERPT: Whiteley's conjecture. These matroids are equal for every $n\geq 2$:

conjecture | open | MathDB 页面 | 原文线索

V2-046065 | MATHDB-376343

Kiraly-Tanigawa's body-pin rigidity characterization conjecture

摘录 / EXCERPT: Kiraly-Tanigawa's body-pin conjecture. The graph G_H is rigid in $\mathbb{R}^3$ if and only if, for every partition $\mathcal{P}=\{P_1,P_2,\dots,P_t\}$ of W,

conjecture | open | MathDB 页面 | 原文线索

V2-046066 | MATHDB-376344

Six-connectivity conjecture for rigidity of K4-covered graphs

摘录 / EXCERPT: Six-connectivity conjecture. Every 6-connected K_4-covered graph is rigid in $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-046067 | MATHDB-376345

Six-connectivity conjecture for global rigidity of 3-dimensional combinatorial zeolites

摘录 / EXCERPT: Combinatorial-zeolite conjecture. Every 6-connected 3-dimensional combinatorial zeolite is globally rigid in $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-046068 | MATHDB-376360

Suda-Tanaka-Tokushige's stability conjecture for cross-intersecting families

摘录 / EXCERPT: Suda-Tanaka-Tokushige's stability conjecture. Assume

conjecture | open | MathDB 页面 | 原文线索

V2-046069 | MATHDB-376383

The restricted density polynomial Hales-Jewett conjecture

摘录 / EXCERPT: Restricted density polynomial Hales-Jewett conjecture. There exists $N=N(d,\delta)\in\mathbb N$ such that, for every $S\subseteq\mathcal P(\{1,\ldots,N\}^d)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-046070 | MATHDB-376400

Generalized degree-power star conjecture for t-intersecting families

摘录 / EXCERPT: Let $1\leq t\leq k$, $1\leq r\leq k-1$, and let $p\geq2$ be real. Suppose that ... If $\mathcal F\subseteq\binom{[n]}k$ is t-intersecting, then, for every $T\in\binom{[n]}t$, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-046071 | MATHDB-376401

Degree-power extremal conjecture for bounded-matching families

摘录 / EXCERPT: Let $k\geq2$, $s\geq1$, $1\leq r\leq k-1$, and let $p\geq1$ be real. Suppose that ... If $\mathcal F\subseteq\binom{[n]}k$ satisfies $\nu(\mathcal F)\leq s$, let $\mathcal B_S$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-046072 | MATHDB-376512

Combinatorial semistable reduction for polyhedral maps

摘录 / EXCERPT: Polyhedral semistable reduction conjecture. There exists a projective alteration $\Delta_B^1\to\Delta_B$, with induced alteration $\Delta_X^1\to\Delta_X$, and a projective subdivis...

conjecture | open | MathDB 页面 | 原文线索

V2-046073 | MATHDB-376552

Norm formula conjecture for odd XXZ ground states

摘录 / EXCERPT: Norm formula conjecture. The norm is

conjecture | open | MathDB 页面 | 原文线索

V2-046074 | MATHDB-376555

Ferromagnetic-string probability formula conjecture

摘录 / EXCERPT: Let $N=2n+1$ be odd, and let $a_1,\ldots,a_k$ denote the local observables defining the ferromagnetic-string correlations. Ferromagnetic-string probability conjecture. The correlat...

conjecture | open | MathDB 页面 | 原文线索

V2-046075 | MATHDB-376588

The spanning-tree factorization conjecture for Sierpinski gaskets

摘录 / EXCERPT: Spanning-tree factorization conjecture. The number of spanning trees is

conjecture | open | MathDB 页面 | 原文线索

V2-046076 | MATHDB-376589

The asymptotic growth constant formula for Sierpinski gaskets

摘录 / EXCERPT: Asymptotic growth constant conjecture. The asymptotic growth constant is

conjecture | open | MathDB 页面 | 原文线索

V2-046077 | MATHDB-376624

Conjecture on the complexity of tromino tilings of extended domino-deficient rectangles

摘录 / EXCERPT: Let $N_T(m,n)$ denote the number of tromino tilings of an $m\times n$ rectangle. Consider an extended domino-deficient rectangle with minimal dimensions $m\times n'$, where $n'>n$....

conjecture | open | MathDB 页面 | 原文线索

V2-046078 | MATHDB-376638

PSPACE-completeness conjecture for unrestricted graph Ramsey games

摘录 / EXCERPT: Let G be a graph and let A be the achievement graph, with no precolored red or green edges, denoted by $(G,A,\{\},\{\})$. Unrestricted graph Ramsey-game conjecture. Graph Ramse...

conjecture | open | MathDB 页面 | 原文线索

V2-046079 | MATHDB-376639

Tractability conjecture for graph Ramsey achievement games

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let A be the achievement graph, and let E^r and E^g be the precolored red and green edge sets. Achievement-game tractability...

conjecture | open | MathDB 页面 | 原文线索

V2-046080 | MATHDB-376640

PSPACE-completeness conjecture for symmetric binary graph Ramsey avoidance games

摘录 / EXCERPT: Let K_k and K_n be complete graphs, let E^r and E^g be precolored red and green edge sets, and let $\mathrm{Ramsey}(n,n)$ denote the classic symmetric binary Ramsey number....

conjecture | open | MathDB 页面 | 原文线索

V2-046081 | MATHDB-376641

2-EXPSPACE-completeness conjecture for Ramsey-number avoidance games

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $\mathrm{Ramsey}(n,n)$ be the symmetric binary Ramsey number, and let $K_{\mathrm{Ramsey}(n,n)}$ be the complete graph on that...

conjecture | open | MathDB 页面 | 原文线索

V2-046082 | MATHDB-376688

Non-extremality of horizons in the positive-curvature cosmological case

摘录 / EXCERPT: Let k be the curvature of the symmetric surfaces, let Λ be the cosmological constant, and consider generic initial data for the surface-symmetric Einstein-Vlasov system....

conjecture | open | MathDB 页面 | 原文线索

V2-046083 | MATHDB-376753

Intertwiner-algebra determination conjecture for conformal quantum field theories

摘录 / EXCERPT: Let a conformal quantum field theory have an associated hyperfinite type $4\mathrm{II}_1$ intertwiner algebra encoding combinatorial and topological data, and let the energy-moment...

conjecture | open | MathDB 页面 | 原文线索

V2-046084 | MATHDB-376790

Asymptotic enumeration conjecture for plane curves

摘录 / EXCERPT: Let a_p and α_p denote the numbers of open and closed plane curves, respectively, with p self-intersections. Let $\gamma=-\frac{1+\sqrt{13}}6$. Asymptotic enumeration co...

conjecture | open | MathDB 页面 | 原文线索

V2-046085 | MATHDB-376791

Model I universality conjecture for colored doodles

摘录 / EXCERPT: For $0\le n<2$, let $c(n)$ denote the central charge associated with colored doodles, and write $n=-2\cos(\pi g)$ with $\frac12\le g<1$. Model I universality conjecture. Colored do...

conjecture | open | MathDB 页面 | 原文线索

V2-046086 | MATHDB-376792

Model II universality conjecture for colored doodles

摘录 / EXCERPT: For $0\le n<2$, let $c(n)$ denote the central charge associated with colored doodles. Model II universality conjecture. Colored doodles are in the universality class of models with...

conjecture | open | MathDB 页面 | 原文线索

V2-046087 | MATHDB-376809

Global validity of the regular-polygon maximizer property

摘录 / EXCERPT: Let N be the number of points and let (P_m) denote the property that the maximizer represented by regular polygons has the specified geometric extremal behavior for m points....

conjecture | open | MathDB 页面 | 原文线索

V2-046088 | MATHDB-376916

Middle-coordinate abundance conjecture for bounded ordered integer simplices

摘录 / EXCERPT: Let N be a natural number greater than 2, let d be a natural number strictly less than $N/2-1$, and let $n\geq N$ be another natural number. For the simplex $S^N(n)$ consider...

conjecture | open | MathDB 页面 | 原文线索

V2-046089 | MATHDB-376933

Asymptotic expansion conjecture for Takeuchi numbers

摘录 / EXCERPT: Let T_n denote the Takeuchi numbers, let W be the Lambert W-function, and set $w=W(n)$. Let C_T be a positive real constant. Asymptotic expansion conjecture. As n tends t...

conjecture | open | MathDB 页面 | 原文线索

V2-046090 | MATHDB-376934

Series representation conjecture for the Takeuchi asymptotic constant

摘录 / EXCERPT: Let $h(x)=\sum_{k=0}^{\infty}\lambda_kx^k$ be the function introduced in the heuristic asymptotic analysis, and let C_T be the positive constant appearing in the asymptotic formu...

conjecture | open | MathDB 页面 | 原文线索

V2-046091 | MATHDB-376960

Coppersmith-Sorkin conjecture for expected k-assignment cost

摘录 / EXCERPT: Let $F_k(m,n)$ denote the expected cost of an optimal k-assignment in an $m\times n$ matrix of independent exponentially distributed random variables with mean 1. Here a k-as...

conjecture | open | MathDB 页面 | 原文线索

V2-046092 | MATHDB-376968

Polynomiality conjecture for the singular-point contribution on ${\cal M}_N$

摘录 / EXCERPT: Let N be odd, and let ... be the contribution associated with the singular point on ${\cal M}_N$. Polynomiality conjecture. Then ... where $P_N(q,z)$ is a polynomial. This statement...

conjecture | open | MathDB 页面 | 原文线索

V2-046093 | MATHDB-376997

Connectivity conjecture for maximal weakly separated collections

摘录 / EXCERPT: Let $\operatorname{W}(k,n)$ be the set of maximal collections of pairwise weakly separated k-subsets of $[1\dots n]$, and let a $(2,4)$-move replace one of the two indicated memb...

conjecture | open | MathDB 页面 | 原文线索

V2-046094 | MATHDB-377043

The line-closed matroid-quadratic Orlik-Solomon conjecture

摘录 / EXCERPT: Line-closed matroid-quadratic Orlik-Solomon conjecture. G is line-closed if and only if $A(G)$ is quadratic.

conjecture | open | MathDB 页面 | 原文线索

V2-046095 | MATHDB-377044

The points-and-lines conjecture for free and K(pi,1) arrangements

摘录 / EXCERPT: Points-and-lines conjecture. The underlying matroid of a free or $K(\pi,1)$ arrangement is determined by its points and lines.

conjecture | open | MathDB 页面 | 原文线索

V2-046096 | MATHDB-377048

Hard Lefschetz conjecture for combinatorial intersection cohomology of arbitrary polytopes

摘录 / EXCERPT: Let $\Delta\subset\mathbb{R}^d$ be a polytope, let Φ be its dual fan, and let $\overline{M}_\Phi$ be the quotient combinatorial intersection cohomology module defined from the...

conjecture | open | MathDB 页面 | 原文线索

V2-046097 | MATHDB-377094

Theta-graph extremal conjecture for chromatic roots at fixed corank

摘录 / EXCERPT: Fixed-corank theta extremal conjecture. If $G\neq K_4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-046098 | MATHDB-377108

Krattenthaler's 1/3-phenomenon conjecture for rhombus tilings

摘录 / EXCERPT: Let a, b, c, x and y be arbitrary integers. Consider a hexagon with side lengths $2n+a$, $2n+b$, $2n+c$, $2n+a$, $2n+b$, and $2n+c$, and a horizontal rhombus whose bottom...

conjecture | open | MathDB 页面 | 原文线索

V2-046099 | MATHDB-377110

The spanning conjecture for f-vectors of ordinary polytopes

摘录 / EXCERPT: Spanning conjecture. The set of f-vectors of all ordinary d-polytopes spans the Euler hyperplane. A spanning set consists of the ordinary polytopes

conjecture | open | MathDB 页面 | 原文线索

V2-046100 | MATHDB-377120

The bilinear distance conjecture

摘录 / EXCERPT: Let Q_0,Q_1,Q_2 be three cubes in $\mathbb{B}(0,C)$ of radius ≈1 satisfying ... for all $x_0\in Q_0$, $x_1\in Q_1$, and $x_2\in Q_2$. For each $j=0,1,2$, let E_j be a...

conjecture | open | MathDB 页面 | 原文线索

V2-046101 | MATHDB-377122

The discretized Erdős ring conjecture

摘录 / EXCERPT: Let $\mathbb{A}$ be an annulus in $\mathbb{R}$, and let $0<\delta\ll1$. Let $A\subset\mathbb{A}$ be a $(\delta,1/2)_1$ set with measure $\approx\delta^{1/2}$. Discretized Erdős rin...

conjecture | open | MathDB 页面 | 原文线索

V2-046102 | MATHDB-377127

Charney-Davis flag complex conjecture

摘录 / EXCERPT: Flag complex conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046103 | MATHDB-377163

The link-pattern balance conjecture for the O(1) loop model

摘录 / EXCERPT: Link-pattern balance conjecture. For any $n=1,2,\ldots$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-046104 | MATHDB-377164

The Perron eigenvector conjecture for the O(1) loop-model Hamiltonian

摘录 / EXCERPT: Perron eigenvector conjecture. The vector Ψ is the unique eigenvector of H corresponding to its largest eigenvalue $2n$.

conjecture | open | MathDB 页面 | 原文线索

V2-046105 | MATHDB-377168

Universality conjecture for the modules $W^{(r)}_s$

摘录 / EXCERPT: Let $r\in J$ and $s\geq1$, and let $4W^{(r)}_s4$ be the module associated with the extremal weight $s\overline{\Lambda}_r$. An extremal vector is a vector satisfying the extremal-m...

conjecture | open | MathDB 页面 | 原文线索

V2-046106 | MATHDB-377169

Intrinsic-energy formula conjecture for $B^{r,s}$

摘录 / EXCERPT: Let $B^{r,s}$ be a simple affine crystal with extremal vector $u(B^{r,s})$, intrinsic energy $D_{B^{r,s}}$, level $4\operatorname{level}(B^{r,s})4$, and crystal statistic...

conjecture | open | MathDB 页面 | 原文线索

V2-046107 | MATHDB-377206

Refined recurrence conjecture for rooted trees with a specified lower critical node

摘录 / EXCERPT: Refined recurrence conjecture. For $n\geq 3$ and $1\leq i\leq n-2$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-046108 | MATHDB-377209

The algebraic eigenvalue conjecture for discrete groups

摘录 / EXCERPT: Let G be a discrete group. The algebraic eigenvalue property is the property that the eigenvalues under consideration for the combinatorial Laplacian are algebraic numbers. Algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-046109 | MATHDB-377215

Conjecture on the largest critical set in a Latin square

摘录 / EXCERPT: Let $\lcs{n}$ denote the size of the largest critical set in any Latin square of order n. Largest-critical-set bound conjecture. ... This bound is motivated by the proof of the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-046110 | MATHDB-377303

Adjacency and diameter conjecture for the ridge graph of the quasi-semimetric cone

摘录 / EXCERPT: Adjacency and diameter conjecture. (i) A triangle facet is adjacent to a facet if and only if they are non-conflicting. (ii) The non-negativity facets NN_{ij} and $NN_{i'j'}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-046111 | MATHDB-377304

Extreme rays and diameter conjecture for the oriented-multicut skeleton

摘录 / EXCERPT: Oriented-multicut skeleton conjecture. (i) All oriented multicuts are extreme rays of $QMET_n$. (ii) The oriented cuts form a dominating clique in G_{OMCUT_n}; consequently, the...

conjecture | open | MathDB 页面 | 原文线索

V2-046112 | MATHDB-377305

Facet classification conjecture for the oriented-multicut cone

摘录 / EXCERPT: Oriented-multicut facet conjecture. The following inequalities define facets of $OMCUT_n$: (i) zero-extensions of every facet of $OMCUT_{n-1}$, including oriented triangle and non-...

conjecture | open | MathDB 页面 | 原文线索

V2-046113 | MATHDB-377306

Adjacency and diameter conjecture for the oriented-multicut skeleton

摘录 / EXCERPT: Oriented-multicut adjacency conjecture. (i) An oriented multicut $\delta'(A)$ is not adjacent to every oriented multicut $\delta'(B)$ such that $B<A^T$. (ii) The orbit represented...

conjecture | open | MathDB 页面 | 原文线索

V2-046114 | MATHDB-377310

Noncanonicity conjecture for homology maps between resonance complexes

摘录 / EXCERPT: Let λ, μ, and $\tilde\mu$ be number partitions. For number partitions $\lambda>\mu$, the inclusion $i(\mu,\lambda):\delta_\mu\rightarrow\delta_\lambda$ induces a homomo...

conjecture | open | MathDB 页面 | 原文线索

V2-046115 | MATHDB-377311

Composition-dependence conjecture for maps between resonance complexes

摘录 / EXCERPT: For every $n\geq 1$, let $\operatorname{Res}\,\lambda$ denote the resonance set associated with a number partition λ of length n. If...

conjecture | open | MathDB 页面 | 原文线索

V2-046116 | MATHDB-377336

The Ramsey-ordering conjecture for permutation group orbits

摘录 / EXCERPT: Ramsey-ordering conjecture. There exists such a Ramsey ordering for which, for all $\beta,\beta'\in\mathcal{B}$ and $\gamma,\gamma'\in\mathcal{C}$,

conjecture | open | MathDB 页面 | 原文线索

V2-046117 | MATHDB-377341

Nonpolynomiality conjecture for magic-square counting functions

摘录 / EXCERPT: Let M_n, S_n, and P_n denote the counting functions for the magic squares considered in the paper, with parameter n. Nonpolynomiality conjecture. The functions M_n, S_n...

conjecture | open | MathDB 页面 | 原文线索

V2-046118 | MATHDB-377342

Nonpolynomiality conjecture for semi-magic hypercube counting functions

摘录 / EXCERPT: Let H_n^d denote the counting function for semi-magic d-dimensional hypercubes of order n. Nonpolynomiality conjecture. The function H_n^d is not a polynomial when...

open_problem | open | MathDB 页面 | 原文线索

V2-046119 | MATHDB-377351

The divided-differential-operator dimension conjecture for Vandermonde determinants

摘录 / EXCERPT: Divided-differential-operator dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-046120 | MATHDB-377361

The multivariate Brown-Colbourn conjecture for cographic matroids

摘录 / EXCERPT: Multivariate Brown-Colbourn conjecture. If $|q_e|>1$ for all e, then

conjecture | open | MathDB 页面 | 原文线索

V2-046121 | MATHDB-377362

The rank-three transversal matroid half-plane-property conjecture

摘录 / EXCERPT: Rank-three transversal matroid conjecture. Every rank-3 transversal matroid has the half-plane property.

conjecture | open | MathDB 页面 | 原文线索

V2-046122 | MATHDB-377464

Jackson-Sokal conjecture for characteristic polynomials of binary matroids

摘录 / EXCERPT: For a binary matroid M, let $\Lambda(M)$ be the minimum, over bases B of its cocycle space, of the maximum size of a cocircuit in B. Let $C(M,t)$ denote its characteristic po...

conjecture | open | MathDB 页面 | 原文线索

V2-046123 | MATHDB-377482

Andreatta-Wiśniewski equality conjecture for extremal fibers

摘录 / EXCERPT: Andreatta-Wiśniewski's equality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046124 | MATHDB-377494

Herzog's Betti-number inequality for symmetric and exterior shifting

摘录 / EXCERPT: Let Γ be a simplicial complex, let $\Delta(\Gamma)$ and $\Delta^e(\Gamma)$ denote its symmetric and exterior algebraic shifts, and let $I_{\Delta(\Gamma)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-046125 | MATHDB-377495

The iterated Betti-number inequality for symmetric and exterior shifting

摘录 / EXCERPT: Let Γ be a simplicial complex, and let $b_{i,j}(\Gamma)$ and $b^e_{i,j}(\Gamma)$ denote its symmetric and exterior iterated Betti numbers, respectively. The iterated Betti-n...

conjecture | open | MathDB 页面 | 原文线索

V2-046126 | MATHDB-377504

Equivalence and abnormal-Carathéodory-equivalence of optimal control problems

摘录 / EXCERPT: In the terminology of Constantin Carathéodory, two problems are Carathéodory-equivalent when their respective Lagrangians differ by a total derivative; the corresponding extremals...

conjecture | open | MathDB 页面 | 原文线索

V2-046127 | MATHDB-377530

Gale's subset take-away conjecture

摘录 / EXCERPT: Let A be a finite set. In subset take-away, two players alternately choose proper, non-empty subsets of A, with no chosen set containing a set chosen earlier; a player unable t...

conjecture | open | MathDB 页面 | 原文线索

V2-046128 | MATHDB-377531

Complementary-response conjecture for subset take-away

摘录 / EXCERPT: Let A be a finite set, and consider the subset take-away game in which players alternately choose proper, non-empty subsets of A, with no chosen set containing a set chosen ear...

conjecture | open | MathDB 页面 | 原文线索

V2-046129 | MATHDB-377612

The uniform distribution hypothesis for extremal B-star[g] sets

摘录 / EXCERPT: Let $R(g,n)$ denote the maximum size of a $B^{\ast}[g]$ set contained in $\{1,\dots,n\}$. A sequence $(S_n)_{n=1}^\infty$ of sets of positive integers becomes uniformly distributed...

conjecture | open | MathDB 页面 | 原文线索

V2-046130 | MATHDB-377613

The polygonal formulas for Delta three and Delta four

摘录 / EXCERPT: For $k\geq1$, let $\Delta_k(\varepsilon)$ denote the function studied in the paper, with $0\leq\varepsilon\leq1$. Delta three and Delta four polygonal conjecture. The graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-046131 | MATHDB-377660

Symmetric-function generalization of the Eremenko-Gabrielov conjecture

摘录 / EXCERPT: Let λ be a partition with ℓ parts and fixed parity for each part. Let $G_\lambda(x/x)$ be the specialized symmetric function defined by substituting $2p_{2i-1}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-046132 | MATHDB-377665

Characterization of saturated sets containing no even numbers

摘录 / EXCERPT: Let G be the abelian group under discussion, and call a subset saturated if it is a maximal avoidable set. The sets described above are those obtained as solution sets of ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-046133 | MATHDB-377666

Avoidable-set lower-density conjecture

摘录 / EXCERPT: Let U be an avoidable set of integers, and suppose that U contains infinitely many even numbers. Write $LD(U)$ for its lower logarithmic density. Lower-density conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-046134 | MATHDB-377669

Explicit generating-function form conjecture for even and odd 132-occurrence counts

摘录 / EXCERPT: Explicit generating-function form conjecture. The displayed representations hold for every $r\geq1$. The paper presents this as a stronger conjecture motivated by explicit results;...

conjecture | open | MathDB 页面 | 原文线索

V2-046135 | MATHDB-377673

Vertically symmetric nest distribution conjecture

摘录 / EXCERPT: Vertically symmetric nest distribution conjecture. The nest distribution is

conjecture | open | MathDB 页面 | 原文线索

V2-046136 | MATHDB-377676

Holroyd-Johnson conjecture for intersecting k-separated sets

摘录 / EXCERPT: Holroyd and Johnson's conjecture. If $\mathcal{A}\subseteq[n]^{(r)}_k$ is intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-046137 | MATHDB-377690

Higher Narayana number formula for local Weyl modules

摘录 / EXCERPT: Let $W_{\mathbb C^d}(\{0\}_{n\omega})$ be the indicated local Weyl module for sl_2, and let $W_{\mathbb C^d}(\{0\}_{n\omega})^{n-2i}$ denote its weight space of weight $n-2i$. Hi...

conjecture | open | MathDB 页面 | 原文线索

V2-046138 | MATHDB-377725

The layered extremal word conjecture

摘录 / EXCERPT: Layered extremal word conjecture. If Π is a set of layered patterns, then

conjecture | open | MathDB 页面 | 原文线索

V2-046139 | MATHDB-377726

The shortest common superpattern conjecture

摘录 / EXCERPT: Shortest common superpattern conjecture. For any $l\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-046140 | MATHDB-377727

The central-circuit characterization of maximal irreducible hedrites

摘录 / EXCERPT: Let i be the number of faces of an irreducible i-hedrite, and let a central circuit be a central circuit of the hedrite. An irreducible i-hedrite is maximal irreducible when...

conjecture | open | MathDB 页面 | 原文线索

V2-046141 | MATHDB-377762

Littlewood-Richardson coefficient monotonicity under the starred transformation

摘录 / EXCERPT: Given an ordered pair (λ,μ) of partitions with the same number of parts, define ... and ... For any partition ν, the starred-transformation conjecture. ... This is a combin...

conjecture | open | MathDB 页面 | 原文线索

V2-046142 | MATHDB-377813

Witten-Kontsevich conjecture on combinatorial classes and tautological classes

摘录 / EXCERPT: Let $\mathcal{M}_{g,P}$ be the moduli space of smooth genus-g curves with marked points P, and let $W_{m_*,P}$ denote the combinatorial class associated with ribbon graphs havi...

conjecture | open | MathDB 页面 | 原文线索

V2-046143 | MATHDB-377818

The partition polynomial convolution identity

摘录 / EXCERPT: Let $\Lambda=(\lambda_1,\dots,\lambda_d)$ be a finite sequence with sum ... Define the polynomial $F_\Lambda(x)$ by $F_\emptyset(x)=1$ and, for $1\leq d=\sharp(\Lambda)<\infty$, ... Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-046144 | MATHDB-377840

Cameron-Erdős conjecture on sum-free subsets

摘录 / EXCERPT: Let $[N]=\{1,\dots,N\}$. A subset A of the integers is sum-free if there do not exist $x,y,z\in A$ such that $x+y=z$. Cameron-Erdős conjecture. The number of sum-free subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-046145 | MATHDB-377852

The exact minimum transitive-triple packing conjecture for tournaments

摘录 / EXCERPT: Exact transitive-triple packing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046146 | MATHDB-377903

Nonnegativity conjecture for the invariant c of integral polytopes

摘录 / EXCERPT: Let P be an integral polytope, and let ... where the sum runs over all nonempty faces F of P. Nonnegativity conjecture. The invariant $c(P)$ is nonnegative: ... Numerical experim...

conjecture | open | MathDB 页面 | 原文线索

V2-046147 | MATHDB-377927

Prefix-exchange conjecture for increasing and decreasing patterns

摘录 / EXCERPT: Let σ and τ be patterns, and write $\sigma\sim_{\mathcal{I}}\tau$ when they are equinumerous for avoidance by involutions, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-046148 | MATHDB-377928

Symmetric-placement prefix-exchange conjecture

摘录 / EXCERPT: Let μ be a self-conjugate shape, and let $\mu_{sym}(\sigma)$ be the number of symmetric full placements on μ that avoid the pattern σ. Symmetric-placement prefix-exc...

conjecture | open | MathDB 页面 | 原文线索

V2-046149 | MATHDB-377929

The 12345-43215 involution-equivalence conjecture

摘录 / EXCERPT: For a pattern τ, let $\mathcal{I}_{n}(\tau)$ denote the number of involutions of length n avoiding τ, and write $\sigma\sim_{\mathcal{I}}\tau$ when the two patterns hav...

conjecture | open | MathDB 页面 | 原文线索

V2-046150 | MATHDB-377931

The satisfiability threshold conjecture for random k-SAT

摘录 / EXCERPT: Satisfiability threshold conjecture. For each k there exists a threshold density c_k, such that for any positive ε, for all $c<c_k-\varepsilon$, $F(n,cn)$ is almost...

conjecture | open | MathDB 页面 | 原文线索

V2-046151 | MATHDB-377932

The MAX-SAT limiting function conjecture

摘录 / EXCERPT: MAX-SAT limiting function conjecture. For every k and every constant $c>0$, as $n\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-046152 | MATHDB-377975

Conjecture on the linear coefficient of the Grassmannian Eulerian polynomial

摘录 / EXCERPT: For integers k and n in the range where the polynomial $\hat{E}_{k,n}(q)$ is defined, let $[q]\hat{E}_{k,n}(q)$ denote the coefficient of q in $\hat{E}_{k,n}(q)$. Linear-coef...

conjecture | open | MathDB 页面 | 原文线索

V2-046153 | MATHDB-378035

Stanley-Wilf conjecture for permutation pattern avoidance

摘录 / EXCERPT: Let S_n denote the set of permutations of $[n]=\{1,2,\ldots,n\}$, and let $S_n(\tau)$ be the set of permutations avoiding a pattern $\tau\in S_\ell$. Stanley-Wilf conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-046154 | MATHDB-378051

Existence conjecture for independent-set densities in sparse random graphs

摘录 / EXCERPT: Independent-set density existence conjecture. For every $c>0$ and $r\geq 3$, the limits

conjecture | open | MathDB 页面 | 原文线索

V2-046155 | MATHDB-378061

The maximum-valency connectivity conjecture for Hom complexes

摘录 / EXCERPT: Let G be a graph, let d be its maximal valency, and let K_n be the unlooped complete graph on n vertices. For an integer $k\geq -1$, write $\text{Hom}(G,K_n)$ for the Hom c...

conjecture | open | MathDB 页面 | 原文线索

V2-046156 | MATHDB-378070

Finiteness conjecture for simplicial complexes of type {3,4,5}

摘录 / EXCERPT: A simplicial complex is of type $\{3,4,5\}$ if the link of every $(n-2)$-face is one of the cycles C_3, C_4, or C_5. Finiteness conjecture. In every fixed dimension n, ther...

conjecture | open | MathDB 页面 | 原文线索

V2-046157 | MATHDB-378081

Extremal flow convergence conjecture for positive Kähler surfaces

摘录 / EXCERPT: Let (M,J) be a complex surface of positive first Chern class polarized by a Kähler class Ω. The positive first Chern class convergence conjecture. There exists an initial...

conjecture | open | MathDB 页面 | 原文线索

V2-046158 | MATHDB-378082

Extremal Kähler cone non-closedness conjecture

摘录 / EXCERPT: Let the Kähler cone be the set of Kähler classes, and let the extremal Kähler cone be the subset consisting of classes represented by extremal Kähler metrics. Suppose the flow equa...

conjecture | open | MathDB 页面 | 原文线索

V2-046159 | MATHDB-378096

The arithmetic removal conjecture for linear systems

摘录 / EXCERPT: Arithmetic removal conjecture. If $A\subseteq[N]$ and the number of vectors $x\in A^m$ satisfying $Lx=0$ is $o(N^{m-r})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-046160 | MATHDB-378110

Babson-Kozlov's contractibility conjecture for rank-selected partition-lattice quotients

摘录 / EXCERPT: Let Π_n be the partition lattice, let S be a set of ranks, and let Π_n^S denote its rank-selected subposet. Write $\Delta(\Pi_n^S)$ for the order complex of Π_n^S, le...

conjecture | open | MathDB 页面 | 原文线索

V2-046161 | MATHDB-378111

Uyemura-Reyes's integrality conjecture for random-to-random shuffling

摘录 / EXCERPT: Uyemura-Reyes's conjecture. The eigenvalues of υ_n are rational integers.

conjecture | open | MathDB 页面 | 原文线索

V2-046162 | MATHDB-378112

Edelman-Reiner conjecture on the homology of Boolean algebra h-complexes

摘录 / EXCERPT: Following the standard shelling of the order complex of the truncated Boolean algebra $B_n-\{\hat{0},\hat{1}\}$, let Δ_n be the resulting h-complex. Edelman-Reiner conject...

conjecture | open | MathDB 页面 | 原文线索

V2-046163 | MATHDB-378116

Rescaling conjecture for the MCheb polynomial

摘录 / EXCERPT: Let $P_{\rm MCheb}(z):=z^2-1.87450961730020085\ldots$ be the real quadratic polynomial constructed so that its first-return combinatorics imitate those of the Chebyshev polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-046164 | MATHDB-378126

Exact lowest degree conjecture for Schur Q-functions

摘录 / EXCERPT: Let λ be a strict partition, and let Q_{λ} be the Schur Q-function. Give each power sum symmetric function degree 1, so that $\deg(p_{\nu})=\ell(\nu)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-046165 | MATHDB-378148

Polynomial combinatorial-type conjecture for shortest paths on convex polyhedra

摘录 / EXCERPT: Polynomial combinatorial-type conjecture. The cardinality of the set of combinatorial types of shortest paths in S is polynomial in the number of facets of S when the dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-046166 | MATHDB-378198

Optimality conjecture for the Oldest random strategy in the Robin Hood game

摘录 / EXCERPT: Let $RH(r,s,A)$ be a Robin Hood game with resource functions r,s and index set A. A random strategy is independent of the index set A if its choices do not depend on A. Old...

conjecture | open | MathDB 页面 | 原文线索

V2-046167 | MATHDB-378245

The adjacent-row bigrid perfection conjecture

摘录 / EXCERPT: Let a bigrid be an array with entries $\gamma(\epsilon,i,j)$, where the indices are taken wherever the entries are defined, and let a bigrid be perfect in the sense defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-046168 | MATHDB-378246

The involutive linear-fractional bigrid perfection conjecture

摘录 / EXCERPT: Let τ be a linear fractional transformation of order 2, and let a bigrid be an array with entries $\gamma(\epsilon,i,j)$ wherever they are defined. Let a bigrid be perfect i...

conjecture | open | MathDB 页面 | 原文线索

V2-046169 | MATHDB-378271

Lovász's conjecture on the Stiefel-Whitney height of odd-cycle Hom complexes

摘录 / EXCERPT: Lovász's conjecture. The equation is true for $T=C_{2r+1}$, $r\geq 1$, and all n.

conjecture | open | MathDB 页面 | 原文线索

V2-046170 | MATHDB-378274

Small-trade conjecture for subsets of Latin squares

摘录 / EXCERPT: Let L be a Latin square of order $n\geq 3$, and let S be a subset of its entries with $|S|\leq \left\lfloor n^2/4\right\rfloor$. For a Latin trade T in L, let $R(T)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-046171 | MATHDB-378275

The minimal critical-set bounds for powers of two

摘录 / EXCERPT: Let $L(2^n)$ be the Latin square of order 2^n, with $n\geq 4$, and let C be a critical set of minimal size in $L(2^n)$. Writing S for this critical set, so that $|S|=|C|$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-046172 | MATHDB-378276

The minimal critical-set size of $L(9)$

摘录 / EXCERPT: Let $L(9)$ denote the Latin square of order 9, and let C be a critical set in $L(9)$. The minimum cardinality of a critical set in $L(9)$ is the $L(9)$ minimal critical-set con...

conjecture | open | MathDB 页面 | 原文线索

V2-046173 | MATHDB-378337

Combinatorial-dimension characterization of universal Donsker classes

摘录 / EXCERPT: Combinatorial-dimension conjecture. For every uniformly bounded class F,

conjecture | open | MathDB 页面 | 原文线索

V2-046174 | MATHDB-378379

Centrally symmetric cyclic upper bound conjecture for odd-dimensional spheres

摘录 / EXCERPT: Centrally symmetric cyclic upper bound conjecture. If d is even, then the boundary complex ∂C_d^{Δ} of the d-dimensional crosspolytope on $n=2d$ vertices is the...

conjecture | open | MathDB 页面 | 原文线索

V2-046175 | MATHDB-378395

The collapsing conjecture for facets of hierarchical model polytopes

摘录 / EXCERPT: Collapsing conjecture. Suppose that

conjecture | open | MathDB 页面 | 原文线索

V2-046176 | MATHDB-378403

Bousquet-Mélou's algebraicity conjecture for slit-plane walk generating functions

摘录 / EXCERPT: Let $\mathfrak S\subset\mathbb Z^2$ be a finite set of steps. For walks on the slit plane starting at the origin, define the generating function counted by length and final coordin...

conjecture | open | MathDB 页面 | 原文线索

V2-046177 | MATHDB-378413

Odd-dimensional complexes have zero zigzag signatures

摘录 / EXCERPT: Let ${\cal K}$ be a complex of odd dimension. A zigzag in ${\cal K}$ has a signature, recording the numbers of intersections of the zigzag with the relevant types of faces. Zero-si...

conjecture | open | MathDB 页面 | 原文线索

V2-046178 | MATHDB-378414

Zigzag structure of direct products of polygons

摘录 / EXCERPT: Let C_p and C_q be polygons, and set ... Write z for the multiset of zigzag lengths and multiplicities, and Int for intersection vectors. Product-of-polygons conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-046179 | MATHDB-378415

Zigzag structure of prisms over antiprisms

摘录 / EXCERPT: Let $APrism_m$ be the m-parameter antiprism, and let $Prism(APrism_m)$ denote the prism over it. Write z for the zigzag vector and Int for intersection vectors. Prism-over-an...

conjecture | open | MathDB 页面 | 原文线索

V2-046180 | MATHDB-378416

Zigzag structures of simplicial complexes of type {3,4}

摘录 / EXCERPT: Let ${\cal K}$ be a simplicial complex of type $\{3,4\}$, meaning that every $(d-2)$-face is contained in 3 or 4 faces of dimension $d-1$. Let $(P_1,\ldots,P_t)$ be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-046181 | MATHDB-378433

The Fomin-Zelevinsky enumerator relation for antichains

摘录 / EXCERPT: Let Φ be a finite root system of rank n. Define ... where $h_{k,\ell}$ counts antichains of cardinality k containing ℓ simple roots. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-046182 | MATHDB-378447

The triple-diagram relations-between-relations conjecture

摘录 / EXCERPT: Relations-between-relations conjecture. The 2-complex C has precisely these types of 2-cells.

conjecture | open | MathDB 页面 | 原文线索

V2-046183 | MATHDB-378451

The dominance conjecture for flag h-vector symmetry

摘录 / EXCERPT: Dominance conjecture. If $T\subseteq T\circ\beta$, then $T\circ\beta$ dominates T.

conjecture | open | MathDB 页面 | 原文线索

V2-046184 | MATHDB-378457

Neggers-Stanley conjecture for W-polynomials of labeled posets

摘录 / EXCERPT: Neggers-Stanley conjecture. For any labeled poset (P,ω), the polynomial $W(P,\omega;t)$ has only real zeros.

conjecture | open | MathDB 页面 | 原文线索

V2-046185 | MATHDB-378500

Stanley's linear-independence conjecture for tableau incidence matrices

摘录 / EXCERPT: Let λ be a partition, let K_{λ} be the matrix whose rows are indexed by horizontal tableaux of shape λ and whose columns are indexed by vertical tableaux, w...

conjecture | refuted | MathDB 页面 | 原文线索

V2-046186 | MATHDB-378501

Full-rank conjecture for tableau incidence matrices

摘录 / EXCERPT: For a partition λ, let K_{λ} be the matrix whose rows are indexed by horizontal tableaux of shape λ and whose columns are indexed by vertical tableaux, with...

conjecture | open | MathDB 页面 | 原文线索

V2-046187 | MATHDB-378558

Hatayama's structural conjecture for the crystal B^{2,s} of type D_n^{(1)}

摘录 / EXCERPT: Let $B^{2,s}$ be a crystal of type $D_n^{(1)}$, and let $B(k\varpi_2)$ denote the classical crystal of highest weight $k\varpi_2$. Let $D_{B^{2,s}}$ be an energy function on...

conjecture | open | MathDB 页面 | 原文线索

V2-046188 | MATHDB-378601

Winding Number Conjecture

摘录 / EXCERPT: Let d and q be positive integers. For a continuous map $f:S^{d-1}\to\mathbb{R}^d$ and a point $p\notin f(S^{d-1})$, define the winding number $W(f,p)\in\mathbb{Z}$ by the homol...

conjecture | open | MathDB 页面 | 原文线索

V2-046189 | MATHDB-378602

The d-skeleton conjecture for the Topological Tverberg theorem

摘录 / EXCERPT: Let d and q be positive integers, and let $\Delta_{(d+1)(q-1)}^{\le d}$ denote the d-skeleton of the simplex. A Tverberg partition is a collection of q disjoint faces whose...

conjecture | open | MathDB 页面 | 原文线索

V2-046190 | MATHDB-378610

The genus-2 extremal metric conjecture

摘录 / EXCERPT: The genus-2 extremal metric conjecture. There exists a metric on a surface of genus 2 that attains the upper bound

conjecture | open | MathDB 页面 | 原文线索

V2-046191 | MATHDB-378629

Egecioglu-Redmond-Ryavec 3-conjecture for a four-term polynomial recurrence

摘录 / EXCERPT: Let $\{p_n(z)\}_{n\in\mathbb N}$ be defined by ... where $p_{-2}(z)=p_{-1}(z)=0$, $p_0(z)=1$, and $C\in\mathbb R$. Egecioglu-Redmond-Ryavec 3-conjecture. All polynomials in the seque...

conjecture | open | MathDB 页面 | 原文线索

V2-046192 | MATHDB-378637

Green's cap-set exponent conjecture over \(\mathbb F_3^n\)

摘录 / EXCERPT: Green's conjecture. $\mu(3)<1$. Equivalently, there is an absolute constant $\delta>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-046193 | MATHDB-378640

Canonical positive monomial expression conjecture for non-commutative Schur functions

摘录 / EXCERPT: Let λ be a partition of size k, and let $s_\lambda({\mathbf u})$ and $h_\lambda({\mathbf u})$ be the non-commutative Schur and homogeneous symmetric functions in the ribb...

open_problem | open | MathDB 页面 | 原文线索

V2-046194 | MATHDB-378650

Autoconvolution norm-ratio conjecture for probability density functions

摘录 / EXCERPT: Let f be a probability density function supported on $[-\frac14,\frac14]$. The norms $\|f\ast f\|_\infty$ and $\|f\ast f\|_2$ are defined using the autoconvolution $f\ast f$. Aut...

conjecture | open | MathDB 页面 | 原文线索

V2-046195 | MATHDB-378651

Exact formula conjecture for the symmetric subset function

摘录 / EXCERPT: Let $\Delta(\varepsilon)$ denote the symmetric subset function studied in the source, with $0\leq\varepsilon\leq1$. Exact formula conjecture for Δ. ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-046196 | MATHDB-378673

Brauer-loop groundstate integrality, concatenation, and maximality conjecture

摘录 / EXCERPT: Brauer-loop groundstate conjecture. (i) For each system size, the overall normalisation can be chosen so that the smallest element of ψ_0 is 1 and all other elements are int...

conjecture | open | MathDB 页面 | 原文线索

V2-046197 | MATHDB-378682

The f-vector conjecture for tropical linear spaces

摘录 / EXCERPT: A tropical d-plane in n-space is a tropical linear space of dimension d in n-space. Let i be a face dimension, and map the space to the quotient by the diagonal line: ... A...

conjecture | open | MathDB 页面 | 原文线索

V2-046198 | MATHDB-378683

The series-parallel characterization of equality in the f-vector conjecture

摘录 / EXCERPT: Let L be a tropical linear space, and call it series-parallel when it arises from a matroidal decomposition whose facets are polytopes of series-parallel matroids. For a d-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-046199 | MATHDB-378704

The modulo-three conjecture for all-heads coin sequences

摘录 / EXCERPT: Modulo-three conjecture. The sequence 1^n is removable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-046200 | MATHDB-378714

Parisi's random assignment conjecture

摘录 / EXCERPT: For a complete bipartite graph $K_{n,n}$ with independent exponentially distributed edge appearance times of rate 1, let $L_{nn}(n)$ denote the length of a minimum n-assignment...

conjecture | open | MathDB 页面 | 原文线索

V2-046201 | MATHDB-378752

Welsh's unimodality conjecture for matroid counts

摘录 / EXCERPT: Let S_n be an n-element set. For each admissible rank r, let $\mathbbm{M}_{r}^{k}(S_n)$ denote the matroids in the relevant class indexed by k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-046202 | MATHDB-378753

Points-lines-planes conjecture for rank-4 paving matroids

摘录 / EXCERPT: Let S_n be an n-element set, and let $M\in\mathbbm{M}_{4}^{2}(S_n)$ be a rank-4 matroid in the indicated class. Write $W_i(M)$ for its number of rank-i independent sets, an...

conjecture | open | MathDB 页面 | 原文线索

V2-046203 | MATHDB-378756

The maximal-girth dimension conjecture for l-CM simplicial complexes

摘录 / EXCERPT: Let Δ be an l-CM simplicial complex of maximal girth. Let c denote the parameter used for its dimension in the paper, and suppose that Δ is not the r-skeleton o...

conjecture | open | MathDB 页面 | 原文线索

V2-046204 | MATHDB-378827

The Bernoulli matrix singularity probability conjecture

摘录 / EXCERPT: Let n be a large integer, and let M_n be a random n by n matrix whose entries are independent Bernoulli random variables, each equal to 1 or -1 with probability $1/2$....

conjecture | open | MathDB 页面 | 原文线索

V2-046205 | MATHDB-378862

Chain-independence conjecture for admissible-subset crystal graphs

摘录 / EXCERPT: Chain-independence conjecture. The colored directed graph defined by the action of root operators on the admissible subsets corresponding to any λ-chain does not depend on...

conjecture | open | MathDB 页面 | 原文线索

V2-046206 | MATHDB-378894

The Wilf-Stanley exponential-growth conjecture for permutation classes

摘录 / EXCERPT: Let X be a non-empty set of permutations, and let $s_n(X)=\left|\operatorname{Av}(X)\cap\mathcal{S}_n\right|$, where $\operatorname{Av}(X)$ is the class of permutations avoiding...

conjecture | open | MathDB 页面 | 原文线索

V2-046207 | MATHDB-378904

Classification conjecture for generalized polarized Fano manifolds with l(R_1)=n

摘录 / EXCERPT: Let M be a Fano manifold of Picard number one and let $\mathcal{E}$ be an ample vector bundle of rank r on M. Let L, the extremal ray R_1, and a minimal rational curve...

conjecture | open | MathDB 页面 | 原文线索

V2-046208 | MATHDB-378914

The partition characterization of minimal tropical halfspaces

摘录 / EXCERPT: Partition characterization. Every minimal halfspace with respect to ${\mathcal P}$ has the form

conjecture | open | MathDB 页面 | 原文线索

V2-046209 | MATHDB-378937

Xin's second determinant factorization conjecture

摘录 / EXCERPT: Xin's second determinant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046210 | MATHDB-378962

Freiman's structural conjecture for sets with doubling near three times their size

摘录 / EXCERPT: Freiman's conjecture. There exists a natural number K such that for any finite set of natural numbers A with $|A|=k>K$ and

conjecture | open | MathDB 页面 | 原文线索

V2-046211 | MATHDB-378963

Structural conjecture for sumsets bounded by a fixed multiple of the set size

摘录 / EXCERPT: Structural conjecture. There exists a $K\in\mathbb N$ such that for every finite set $A\subseteq\mathbb N$ with $|A|>K$, if

conjecture | open | MathDB 页面 | 原文线索

V2-046212 | MATHDB-378992

The generalized cluster complex shellability conjecture

摘录 / EXCERPT: Let Φ be a finite crystallographic root system of rank n, and let m be a positive integer. The generalized cluster complex $\Delta^m(\Phi)$ is the simplicial complex assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-046213 | MATHDB-379036

Csorba's Hom-complex model for Stiefel manifolds

摘录 / EXCERPT: For a graph G, let $\text{\tt Hom}(G,H)$ denote its Hom-complex. Let $V_2(\mathbb R^{n-1})$ be the Stiefel manifold of orthonormal 2-frames in $\mathbb R^{n-1}$. Csorba's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-046214 | MATHDB-379052

The two-missing-elements perfect Skolem set conjecture

摘录 / EXCERPT: Let $A\subset\{1,2,\dots,n\}$ have cardinality $n-2$, with exactly two missing elements $a_i,a_j\in\{1,2,\dots,n\}$. The two-missing-elements perfect Skolem set conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-046215 | MATHDB-379053

The perfect extremal Skolem set conjecture

摘录 / EXCERPT: Let $A=\{a_1,a_2,\dots,a_n\}$ be a set with $a_1>a_2>\dots>a_n$ and total sum $\sum_{i=1}^n a_i=n^2$. A perfect extremal Skolem set is a perfect Skolem set satisfying this extremal...

conjecture | open | MathDB 页面 | 原文线索

V2-046216 | MATHDB-379054

The $4k+1$ conjecture for perfect Skolem-type sequences

摘录 / EXCERPT: A perfect Skolem-type sequence of order n is a sequence constructed from an ordinary perfect Skolem set of cardinality n. The $4k+1$ conjecture. For every order n, the number...

conjecture | open | MathDB 页面 | 原文线索

V2-046217 | MATHDB-379055

The distinct displacement-pattern conjecture for permutations

摘录 / EXCERPT: Let S_n be the symmetric group, and let a displacement pattern be a tuple $\alpha=(a_1,a_2,\dots,a_n)$ recording the displacements of a permutation, with the entries distinct. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-046218 | MATHDB-379075

Magic permutations for covering clutters

摘录 / EXCERPT: Let $([d],\mathcal{L})$ be a covering clutter, let s be its magic subspace, and let a magic permutation be a permutation of $[d]$ realizable by a positive point in s. For a per...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046219 | MATHDB-379076

Magic permutations for linear forms

摘录 / EXCERPT: Let $f_1,\ldots,f_m$ be positive linear forms on $\mathbb{R}^d$, and let s be the subspace on which they all vanish. For a permutation σ of $[d]$, define the reverse domin...

conjecture | open | MathDB 页面 | 原文线索

V2-046220 | MATHDB-379097

Second-term oscillation conjecture for hook Young-diagram distribution functions

摘录 / EXCERPT: Let $k, ell\ge 0$ and $\beta,z>0$. Define ... and let $r_{(k,\ell),\beta}(z)$ be the corresponding limiting integral ratio. Second-term oscillation conjecture. As $n\to\infty$, the q...

conjecture | open | MathDB 页面 | 原文线索

V2-046221 | MATHDB-379098

Positive-density reciprocal conjecture for uniformly distributed sets

摘录 / EXCERPT: Positive-density reciprocal conjecture. If A contains 0, is not periodic, and is uniformly distributed modulo every power of 2, then $\overline{A}$ has positive density. The...

conjecture | open | MathDB 页面 | 原文线索

V2-046222 | MATHDB-379122

Simmons and Su's equivariant Tucker-type conjecture for $S_k^{n(k-1)}$

摘录 / EXCERPT: Simmons and Su's conjecture. There must exist k adjacent vertices in the triangulation whose labels are $\{\omega^j m:1\leq j\leq k\}$ for some fixed m.

conjecture | open | MathDB 页面 | 原文线索

V2-046223 | MATHDB-379147

Sundaram-Welker vanishing conjecture for polynomial root-multiplicity strata

摘录 / EXCERPT: Sundaram-Welker's conjecture. For every number partition λ,

conjecture | open | MathDB 页面 | 原文线索

V2-046224 | MATHDB-379148

Sundaram-Welker acyclicity conjecture for partition-poset complexes

摘录 / EXCERPT: Sundaram-Welker's acyclicity conjecture. For $\lambda\vdash\mu$ with $\lambda\neq\mu$, the space $X_{\lambda,\mu}$ is ${\mathbb Q}$-acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-046225 | MATHDB-379167

Folkman's conjecture on dense sequences and subcompleteness

摘录 / EXCERPT: Folkman's conjecture. There is a constant C such that, if $A(n)\geq Cn$ for all sufficiently large n, then A is subcomplete.

conjecture | open | MathDB 页面 | 原文线索

V2-046226 | MATHDB-379168

Folkman's conjecture on square-root-density sequences being subcomplete

摘录 / EXCERPT: Folkman's conjecture. There is a constant c such that every increasing sequence satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-046227 | MATHDB-379174

The shifting conjecture for non-reverse-lexicographic term orders

摘录 / EXCERPT: Let S be a polynomial ring, let $\succ$ be a term order on S other than the graded reverse lexicographic order, and let $k\geq 2$ be the smallest degree on which $\succ$ and th...

conjecture | open | MathDB 页面 | 原文线索

V2-046228 | MATHDB-379213

The Lex Plus Powers conjecture on extremal graded Betti numbers

摘录 / EXCERPT: The Lex Plus Powers conjecture. If $\mathcal H$ is $\mathbb A$-lpp valid and $L_{\mathcal H,\mathbb A}$ is the $\mathbb A$-lex plus powers ideal attaining $\mathcal H$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-046229 | MATHDB-379231

The finite-field Erdős distance conjecture

摘录 / EXCERPT: Finite-field Erdős distance conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046230 | MATHDB-379276

The fish-scale conjecture for posets

摘录 / EXCERPT: Let P be a poset containing no infinite antichain. A chain is a pairwise comparable subset of P, and an antichain is a pairwise incomparable subset. Fish-scale conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-046231 | MATHDB-379281

Lower bound for the number of distinct multinomial coefficients

摘录 / EXCERPT: Let $M(n)$ be the number of distinct multinomial coefficients arising from partitions of n, and let $p^*(n)$ be the total number of partitions of n into parts that are either a...

conjecture | open | MathDB 页面 | 原文线索

V2-046232 | MATHDB-379282

Asymptotic growth conjecture for the number of distinct multinomial coefficients

摘录 / EXCERPT: Let $M(n)$ denote the number of distinct multinomial coefficients arising from partitions of n. Asymptotic-growth conjecture. There exists a positive constant C such that ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-046233 | MATHDB-379283

Exact asymptotic constant for the number of distinct multinomial coefficients

摘录 / EXCERPT: Let $M(n)$ denote the number of distinct multinomial coefficients arising from partitions of n. Exact-asymptotic-constant conjecture. ... The paper explains that this would follow...

conjecture | open | MathDB 页面 | 原文线索

V2-046234 | MATHDB-379318

Orientability conjecture for graph coloring manifolds

摘录 / EXCERPT: Let G be the complement of the 1-skeleton of a flag simplicial PL sphere, and let s be the maximal valency of G. The associated graph coloring manifolds are the Hom complex...

conjecture | open | MathDB 页面 | 原文线索

V2-046235 | MATHDB-379373

Neil White's symmetric-exchange conjecture for matroid toric ideals

摘录 / EXCERPT: Let M be a matroid on the ground set $\{1,2,\ldots,n\}$. Fix a field k, let $S_M=k[y_B:B\text{ a base of }M]$, and let I_M be the kernel of the homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-046236 | MATHDB-379375

The matroid toric-ideal quadratic Gröbner-basis conjecture

摘录 / EXCERPT: Let M be a matroid and let I_M be its toric ideal, the kernel of the map sending each base variable y_B to the corresponding squarefree monomial. The matroid quadratic Gröbne...

conjecture | open | MathDB 页面 | 原文线索

V2-046237 | MATHDB-379376

The quadratic-generation conjecture for matroid toric ideals

摘录 / EXCERPT: Let M be a matroid and let I_M be its toric ideal. A quadratic binomial in I_M is a binomial homogeneous of degree two in the base variables. The quadratic-generation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-046238 | MATHDB-379377

The quadratic symmetric-exchange generation conjecture for matroid toric ideals

摘录 / EXCERPT: Let M be a matroid, with toric ideal I_M. For bases B_1,B_2,D_1,D_2, call $y_{B_1}y_{B_2}-y_{D_1}y_{D_2}$ a double-swap binomial when the pair D_1,D_2 is obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-046239 | MATHDB-379381

Villarreal's torsion-free quotient conjecture for uniform clutters

摘录 / EXCERPT: Let A be a $\{0,1\}$-matrix whose columns each contain the same number $d\geq 2$ of 1's, with column vectors $v_1,\ldots,v_q$. Let $\mathcal{C}$ be the clutter associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-046240 | MATHDB-379385

Strong USP capacity conjecture

摘录 / EXCERPT: Strong USP capacity conjecture. The strong USP capacity equals

conjecture | open | MathDB 页面 | 原文线索

V2-046241 | MATHDB-379386

Simultaneous double product property conjecture for abelian groups

摘录 / EXCERPT: Simultaneous double product property conjecture. For arbitrarily large n, there exists an abelian group H with n pairs of subsets A_i,B_i satisfying the simultaneous double...

conjecture | open | MathDB 页面 | 原文线索

V2-046242 | MATHDB-379453

No-cut set conjecture for the metric polytope

摘录 / EXCERPT: Let met_n be the metric polytope, and let its fractional vertices be the vertices that are not integral. The restriction of met_n to its fractional vertices i...

conjecture | open | MathDB 页面 | 原文线索

V2-046243 | MATHDB-379521

Phelps-Rodriguez conjecture on extremal Sendov polynomials

摘录 / EXCERPT: Let S_n be the set of monic complex polynomials of degree $n\ge 2$ whose zeros lie in the closed unit disk. For $p\in S_n$, define ... Call p extremal for Sendov's conjecture if...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046244 | MATHDB-379541

Sturmfels-Ziegler extension-space conjecture for realizable oriented matroids

摘录 / EXCERPT: Let $\mathcal{M}$ be a realizable oriented matroid of rank r. Its extension space is the simplicial complex associated to the natural poset of one-point extensions of...

conjecture | open | MathDB 页面 | 原文线索

V2-046245 | MATHDB-379565

The combinatorial principal-coefficients exchange-graph conjecture

摘录 / EXCERPT: Let $\mathbb{T}_n$ be the n-regular tree, and let $(\tilde B_t)_{t\in\mathbb{T}_n}$ be the family of $2n\times n$ extended exchange matrices for the principal-coefficients cluste...

conjecture | open | MathDB 页面 | 原文线索

V2-046246 | MATHDB-379618

The symmetrization conjecture for Rayleigh partition functions

摘录 / EXCERPT: Symmetrization conjecture. If $Z(\omega;\mathbf y)$ is Rayleigh, then $\widetilde Z(\omega;\mathbf y)$ is Rayleigh. This is stated as an equivalent form of the Big Conjecture and r...

conjecture | open | MathDB 页面 | 原文线索

V2-046247 | MATHDB-379620

The averaged symmetrization inequality for Rayleigh partition functions

摘录 / EXCERPT: Let $Z(\omega;\mathbf y)$ be Rayleigh, and let $\widetilde Z$ be its symmetrization. For each $e\in E$ and each pair $\{e,f\}\subseteq E$, use the auxiliary quantities $z(e)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-046248 | MATHDB-379621

The pairwise symmetrization inequality for Rayleigh partition functions

摘录 / EXCERPT: Let $Z(\omega;\mathbf y)$ be Rayleigh, let $\widetilde Z$ be its symmetrization, and let $z(e)$ and $z(ef)$ be the auxiliary quantities defined in the source. Pairwise symmetrizati...

conjecture | open | MathDB 页面 | 原文线索

V2-046249 | MATHDB-379625

Containment conjecture for l-Rayleigh matroids

摘录 / EXCERPT: Let a matroid belong to one of the listed classes: planar-graph matroids, graph matroids, regular matroids, sixth-root-of-unity matroids, half-plane-property matroids, or B-Rayleig...

conjecture | open | MathDB 页面 | 原文线索

V2-046250 | MATHDB-379626

Narayana polynomials form a q-log-convex sequence

摘录 / EXCERPT: The Narayana number $N(n,k)$ counts Dyck paths of length $2n$ with exactly k peaks, and the Narayana polynomials are $N_n(q)=\sum_{k=0}^{n}N(n,k)q^k$. A sequence of polynomials i...

conjecture | open | MathDB 页面 | 原文线索

V2-046251 | MATHDB-379627

Narayana transformation preserves log-convexity

摘录 / EXCERPT: Narayana transformation conjecture. The Narayana transformation preserves log-convexity. The source presents this as a problem following known log-convexity results for specializat...

conjecture | open | MathDB 页面 | 原文线索

V2-046252 | MATHDB-379628

Squared-binomial triangle satisfies conditions (C1) and (C2)

摘录 / EXCERPT: Consider the triangular array with entries $\binom{n}{k}^2$ for $0\leq k\leq n$. Conditions (C1) and (C2) are the two conditions imposed on a triangular array in the source's theor...

conjecture | open | MathDB 页面 | 原文线索

V2-046253 | MATHDB-379629

Eulerian transformation preserves log-convexity

摘录 / EXCERPT: Eulerian transformation conjecture. The Eulerian transformation preserves log-convexity. The source presents this as a related problem after discussing q-log-convexity of Eulerian...

conjecture | open | MathDB 页面 | 原文线索

V2-046254 | MATHDB-379654

The Upper Matching Conjecture

摘录 / EXCERPT: The Upper Matching Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-046255 | MATHDB-379662

The K-theoretic matroid invariant conjecture

摘录 / EXCERPT: K-theoretic matroid invariant conjecture. For any matroids M, M_1, and M_2,

conjecture | open | MathDB 页面 | 原文线索

V2-046256 | MATHDB-379664

The combinatorial characterization conjecture for the matroid invariant

摘录 / EXCERPT: Combinatorial characterization conjecture. If the assignment $M\mapsto G_M$ satisfies 1. for every such decomposition,

conjecture | open | MathDB 页面 | 原文线索

V2-046257 | MATHDB-379696

Grove formula conjecture for the bounded cube recurrence

摘录 / EXCERPT: Let S be a section, let f be a state, and let (x_0,y_0,t_0) be a point in $\overset{\circ}{Y}\cap\mathcal{L}$ in the future of S. Assume that the cone $\mathcal{C}$ meets...

conjecture | open | MathDB 页面 | 原文线索

V2-046258 | MATHDB-379698

The Stability Conjecture for generic relator tuples in free groups

摘录 / EXCERPT: Fix $k\ge 2$ and $m\ge 1$, and let $F=F(a_1,\dots,a_k)$ be the free group of rank k. For an m-tuple σ of elements of F, write $R(\sigma)$ for the corresponding set of...

conjecture | open | MathDB 页面 | 原文线索

V2-046259 | MATHDB-379755

Steingrímsson's Euler-Mahonian conjecture for ordered partitions

摘录 / EXCERPT: Steingrímsson's conjecture. The statistics

conjecture | open | MathDB 页面 | 原文线索

V2-046260 | MATHDB-379757

The tropical halfspace representation conjecture for pure tropical polytopes

摘录 / EXCERPT: Tropical halfspace representation conjecture. If P is pure, then the halfspaces from a generic lift $\overline{P}$ of P map to tropical halfspaces whose intersection is P its...

conjecture | open | MathDB 页面 | 原文线索

V2-046261 | MATHDB-379758

The tropical halfspace characterization conjecture for pure full-dimensional polytopes

摘录 / EXCERPT: Tropical halfspace characterization conjecture. A tropical polytope P is pure and full dimensional if and only if it has a halfspace description whose apices are in general posit...

conjecture | open | MathDB 页面 | 原文线索

V2-046262 | MATHDB-379759

The face-structure conjectures for tropical polytopes

摘录 / EXCERPT: Face-structure conjectures for tropical polytopes. The following assertions hold: (1) k-faces of tropical polytopes are extreme sets; (2) the topological boundary of a k-face i...

conjecture | open | MathDB 页面 | 原文线索

V2-046263 | MATHDB-379762

The subspace approximation conjecture for functions on \(\mathbb{F}_2^n\)

摘录 / EXCERPT: Let $G=\mathbb{F}_2^n$, let $f\in A(G)$, and let $\epsilon\in(0,1]$. Write ... Here $\operatorname{codim}V$ denotes the codimension of a subspace $V\leq G$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-046264 | MATHDB-379765

The fine mixed subdivision basis conjecture for the matroid \(\mathcal{T}_{n,d}\)

摘录 / EXCERPT: Let $\mathcal{T}_{n,d}$ be the matroid introduced from the combinatorial encoding of arrangements of d generic complete flags in $\mathbb{C}^n$. Let $n\Delta_{d-1}$ be the Minkow...

conjecture | open | MathDB 页面 | 原文线索

V2-046265 | MATHDB-379770

Euler-Mahonian conjecture for the LSB statistics on ordered partitions

摘录 / EXCERPT: LSB Euler-Mahonian conjecture. The statistics $\mathop{\text{\text{\rm cmaj}\text{LSB}}}$ and $\mathop{\text{\text{\rm cinv}\text{LSB}}}$ are Euler-Mahonian on ordered partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-046266 | MATHDB-379831

The colored-permutation formula conjecture for normalized character polynomials

摘录 / EXCERPT: Let λ be the partition of n whose diagram is the union of m rectangles described above. Let $\mu\vdash k$ and fix a permutation $w_\mu\in\mathfrak S_k$ of cycle type...

conjecture | open | MathDB 页面 | 原文线索

V2-046267 | MATHDB-379909

Collinear-triple conjecture for permutation graphs

摘录 / EXCERPT: Let $\sigma:{\mathbb{F}_q}\rightarrow{\mathbb{F}_q}$ be a permutation, and let $\Gamma_\sigma=\{(x,\sigma(x)):x\in{\mathbb{F}_q}\}\subset{\mathbb{F}_q}^2$ be its graph. A collinear...

conjecture | open | MathDB 页面 | 原文线索

V2-046268 | MATHDB-379921

Ruzsa's sumset conjecture for finite sets of squares

摘录 / EXCERPT: For a finite set E of squares, write $E+E=\{x+y:x,y\in E\}$. Ruzsa's conjecture. For every $\epsilon>0$, ... The source states that Chang's conjecture implies this one, while the B...

conjecture | open | MathDB 页面 | 原文线索

V2-046269 | MATHDB-379926

The logarithmic additive-energy conjecture for subsets of squares

摘录 / EXCERPT: For a set $E\subset\{1^2,\ldots,N^2\}$ of squares, let $r_{E+E}(m)$ count representations of m as a sum of two elements of E. The logarithmic energy conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-046270 | MATHDB-379928

The local square-energy conjecture in logarithmic intervals

摘录 / EXCERPT: For a finite set E of squares in the short interval ... let $r_{E+E}(m)$ count representations of m as a sum of two elements of E. The local energy conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-046271 | MATHDB-379929

The short square-set energy conjecture

摘录 / EXCERPT: The short square-set energy conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-046272 | MATHDB-379935

Limiting proportions of sum-dominant, difference-dominant, and balanced sets

摘录 / EXCERPT: Limiting-proportion conjecture. The three limiting proportions all exist and are positive.

conjecture | open | MathDB 页面 | 原文线索

V2-046273 | MATHDB-379936

Positive limiting proportions for prescribed missing sums and differences

摘录 / EXCERPT: For $n\ge 1$, let S range over subsets of $\{0,1,\dots,n-1\}$. For nonnegative integers j and k, define ... assuming the limit exists. Since $S-S$ is symmetric about 0, only...

conjecture | open | MathDB 页面 | 原文线索

V2-046274 | MATHDB-379937

Existence of sets with prescribed numbers of missing sums and differences

摘录 / EXCERPT: Let j and k be nonnegative integers with k even. Prescribed-defect existence conjecture. There exists a positive integer n and a set $S\subset\{0,1,\dots,n-1\}$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-046275 | MATHDB-379961

The seed-by-cluster conjecture for cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}=\mathcal{A}(\mathcal{S})$ be a normalized skew-symmetrizable cluster algebra, where $\mathcal{S}$ is a mutation equivalence class of seeds, and let a cluster mean...

conjecture | open | MathDB 页面 | 原文线索

V2-046276 | MATHDB-379962

The flag-complex conjecture for cluster complexes

摘录 / EXCERPT: Let $\mathcal{A}=\mathcal{A}(\mathcal{S})$ be a cluster algebra, let $\mathcal{X}$ be its set of cluster variables, and let the cluster complex be the simplicial complex on...

conjecture | open | MathDB 页面 | 原文线索

V2-046277 | MATHDB-379976

Avoidability threshold for powers in odd-difference arithmetic progressions

摘录 / EXCERPT: Let an infinite word be a sequence over a 4-letter alphabet. An r-power is a word consisting of repetitions of a nonempty block with exponent r, and a word avoids r-powers in...

conjecture | open | MathDB 页面 | 原文线索

V2-046278 | MATHDB-379977

Orbit-closure classification of square-free words in odd-difference progressions

摘录 / EXCERPT: Let ${\bf f}=f_0f_1f_2\cdots$ be a paperfolding word over $\{1,4\}$, and define ${\bf v}=v_0v_1v_2\cdots$ by ... for all $n\geq 0$, with the even-position symbols of ${\bf f}$ recode...

conjecture | open | MathDB 页面 | 原文线索

V2-046279 | MATHDB-379994

The conjecture on symmetric six-tuples solving Problem 1

摘录 / EXCERPT: Symmetric six-tuple conjecture. All symmetric six-tuples (a,b,c,a,b,c) are solutions of Problem 1.

conjecture | open | MathDB 页面 | 原文线索

V2-046280 | MATHDB-380014

Minimality conjecture for non-reconstructible pedigrees

摘录 / EXCERPT: Let $\mathcal{T}(X_0)$ be a pedigree of order n. A pedigree is $(n-1)$-reconstructible if it is determined up to congruence by the collection of its restrictions to subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-046281 | MATHDB-380031

The (n,k)-set dimension conjecture

摘录 / EXCERPT: (n,k)-set dimension conjecture. If $E\subset\mathbb{R}^n$ is an (n,k) set, then

conjecture | open | MathDB 页面 | 原文线索

V2-046282 | MATHDB-380046

Generalized skeletal expansion conjecture for colored regular graphs

摘录 / EXCERPT: Let Γ be a regular $(n+1)$-valent graph with a “good” $(\mathbb Z_2)^{n+1}$-coloring α. A generalized $(n+1)$-skeletal expansion is an expansion denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-046283 | MATHDB-380100

The maximal-imaginary-part conjecture for roots of SNN polynomials

摘录 / EXCERPT: Let $M_d(t)$ be the polynomial associated with degree d, and let γ_d be a root of $M_d(t)$ with largest norm. An SNN polynomial is a degree-d polynomial satisfying Stanl...

conjecture | open | MathDB 页面 | 原文线索

V2-046284 | MATHDB-380101

The Beck-de Loera-Pfeifle-Stanley vertical strip conjecture for Ehrhart polynomial roots

摘录 / EXCERPT: Let P be a lattice polytope and let $L_P(t)$ be its Ehrhart polynomial of degree d, with roots α_i. Vertical strip conjecture. Every root satisfies ... for all i. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-046285 | MATHDB-380116

Positivity conjecture for normalized characters in twisted Boolean cumulants

摘录 / EXCERPT: For positive integers $k_1,\ldots,k_l$, let $\Sigma_{k_1,\ldots,k_l}$ denote the normalized character and let $\hat{B}_i$ denote the twisted Boolean cumulants. Positivity conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-046286 | MATHDB-380117

Positivity conjecture for the two-variable Kerov polynomial correction

摘录 / EXCERPT: Let r,s be positive integers, let $\Phi(x,j)$ be the generating-function expression used in the preceding propositions, and write $[x^{r+1}]$ and $[y^{s+1}]$ for coefficient extr...

conjecture | open | MathDB 页面 | 原文线索

V2-046287 | MATHDB-380119

Uniform-contraction conjecture for bounded complexes of affine oriented matroids

摘录 / EXCERPT: Let $(E,\mathcal{L},g)$ be an affine oriented matroid, where g is the distinguished element and $\mathcal{L}/g$ denotes contraction by g. The bounded complex is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-046288 | MATHDB-380120

Simplicial affine oriented matroid bounded-complex conjecture

摘录 / EXCERPT: Let $(E,\mathcal{L},g)$ be a simplicial affine oriented matroid, where g is the distinguished element and $\mathcal{L}^{++}$ denotes its bounded complex. Simplicial bounded-compl...

conjecture | open | MathDB 页面 | 原文线索

V2-046289 | MATHDB-380128

Schilling-Shimozono conjecture for the branching-component automorphism

摘录 / EXCERPT: Let D be a $\pm$ diagram of shape Λ/λ whose ambient $r\times s$ rectangle has complement of Λ tiled by vertical dominos. Let m_{i}°, m_{i}^{+},...

conjecture | open | MathDB 页面 | 原文线索

V2-046290 | MATHDB-380137

Unimodality conjecture for the two-variable Shi distance enumerator

摘录 / EXCERPT: Unimodality conjecture. For fixed n,k, the numbers $\mathsf{Dist}_n(k,\ell)$ as ℓ increases are unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-046291 | MATHDB-380154

Conjecture on the order-complex structure of pure diagrams

摘录 / EXCERPT: Let $\varPi^M_m$ be the finite poset of pure diagrams between $\pi(m)$ and $\pi(M)$, and let $\Delta(\varPi^M_m)$ be its order complex, whose faces are chains in $\varPi^M_m$. Rega...

conjecture | open | MathDB 页面 | 原文线索

V2-046292 | MATHDB-380179

Extension of the crystal commutor realization to quasi-minuscule embeddings

摘录 / EXCERPT: Let $B_{\boldsymbol{\pi}'}$ and $B_{\boldsymbol{\pi}}$ be the crystals embedded as in equation, and let Theorem denote the stated agreement between the combinatorial realization an...

conjecture | open | MathDB 页面 | 原文线索

V2-046293 | MATHDB-380186

Converse to the tropical-rank criterion for tropical bases

摘录 / EXCERPT: Let L be an $(n-k)$-dimensional linear subspace of K^n whose tropical Plücker coordinates are all finite, and let $M\in K^{m\times n}$, with $m\geq k$, have nonzero rows in the...

conjecture | open | MathDB 页面 | 原文线索

V2-046294 | MATHDB-380251

Cameron-Ku stability conjecture for intersecting families of permutations

摘录 / EXCERPT: Let G be the graph whose vertices are the elements of the symmetric group S_n, with two vertices π and π' adjacent when $\pi(i)\neq\pi'(i)$ for every $1\leq i\leq n$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-046295 | MATHDB-380326

Conjecture on the structure of a minimal configuration in Moser's worm problem

摘录 / EXCERPT: Let $\mathcal{L}$ be the fixed line segment, $\mathcal{S}$ the square, and let $\mathcal{T}$ denote the triangular objects in the configuration. Consider configurations with...

conjecture | open | MathDB 页面 | 原文线索

V2-046296 | MATHDB-380337

The existence conjecture for universal cycles of multisets

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let a Universal Cycle for t-multisets of $[n]$ be a cyclic sequence of $\binom{n+t-1}{t}$ integers from $[n]$ in which every t-multiset of $[n]$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-046297 | MATHDB-380344

The maximum-independent-set structure conjecture for qualitative independence graphs

摘录 / EXCERPT: Maximum-independent-set structure conjecture. For all positive integers k, every maximum independent set in $QI(k^2,k)$ is of the form $S_{\{a,b\}}$ for distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-046298 | MATHDB-380348

The covering-array nonexistence conjecture for parameters $(12,8,3)$

摘录 / EXCERPT: Covering-array nonexistence conjecture. A $CA(12,8,3)$ does not exist; consequently, ...

conjecture | open | MathDB 页面 | 原文线索

V2-046299 | MATHDB-380350

The almost-uniform extremal conjecture for Sperner partition systems

摘录 / EXCERPT: Almost-uniform extremal conjecture. For positive integers n,k, a largest Sperner partition system in $\mathcal{P}^n_k$ is an almost-uniform partition system.

conjecture | open | MathDB 页面 | 原文线索

V2-046300 | MATHDB-380351

The complete-intersection conjecture for uniform partition systems

摘录 / EXCERPT: Complete-intersection conjecture. For $n\geq k\geq t\geq1$, if $\mathcal{P}\subseteq\mathcal{U}^n_k$ is t-intersecting, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-046301 | MATHDB-380352

Czabarka's conjecture for partially intersecting partition systems

摘录 / EXCERPT: Czabarka's conjecture. If $n\leq2k-1$ and $\mathcal{P}\subseteq\mathcal{P}^n_k$ is a partially 2-intersecting partition system, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-046302 | MATHDB-380353

The uniform partially-intersecting partition-system conjecture

摘录 / EXCERPT: Uniform partial-intersection conjecture. Let k,c,t be positive integers with $t\leq c$ and $n=ck$. If $\mathcal{P}\subseteq\mathcal{U}^n_k$ is partially t-intersecting, then ……

conjecture | open | MathDB 页面 | 原文线索

V2-046303 | MATHDB-380376

Erdős et al.'s conjecture on avoiding triangles in two-colored planes

摘录 / EXCERPT: Let T be a triangle in the Euclidean plane, and let a coloring be a partition of $\mathbb{R}^2$ into two color classes. A coloring contains T if it has a monochromatic copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-046304 | MATHDB-380377

Erdős et al.'s conjecture on monochromatic copies of nonequilateral triangles

摘录 / EXCERPT: Let a coloring χ be a partition of $\mathbb{R}^2$ into two color classes. A coloring contains a triangle T if there is a monochromatic copy of T, where copies are obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-046305 | MATHDB-380406

The universal-cover conjecture for extremally Ricci-pinched closed G_2 structures

摘录 / EXCERPT: Universal-cover conjecture. The universal covering space $\widetilde{M}$ is isometric to $G/\operatorname{\textsl{SU}}(2)$ equipped with its unique G_2 structure with parallel to...

conjecture | open | MathDB 页面 | 原文线索

V2-046306 | MATHDB-380408

The corrected Welsh conjecture on the possible numbers of matroid bases

摘录 / EXCERPT: Corrected Welsh conjecture. For every triple (n,r,b) such that $0 \leq r \leq n$ and $1 \leq b \leq \binom{n}{r}$, there is a matroid of rank r on n elements with exactly b...

conjecture | open | MathDB 页面 | 原文线索

V2-046307 | MATHDB-380409

Morris's conjecture on the threshold for 3-uniform FC-families

摘录 / EXCERPT: For integers $n\geqslant 4$, let $FC(3,n)$ denote the threshold quantity for 3-uniform families on an n-element ground set as defined in the paper. Morris's conjecture. ... The con...

conjecture | open | MathDB 页面 | 原文线索

V2-046308 | MATHDB-380410

Morris's asymptotic conjecture for FC-family thresholds

摘录 / EXCERPT: For integers $k\geqslant 2$ and n, let $FC(k,n)$ denote the threshold quantity for k-uniform families on an n-element ground set as defined in the paper, and let Θ hav...

conjecture | open | MathDB 页面 | 原文线索

V2-046309 | MATHDB-380420

The Stanley-Wilf conjecture for hereditary properties of tournaments

摘录 / EXCERPT: Let $\mathcal{P}$ be a hereditary property of tournaments, and let $\mathcal{P}_n$ denote the tournaments in $\mathcal{P}$ on n vertices. There exists a constant $\alpha>0$ and a...

conjecture | open | MathDB 页面 | 原文线索

V2-046310 | MATHDB-380427

The exact value conjecture for the combinatorial exceptional-point constant

摘录 / EXCERPT: Exact value conjecture. The universal constant $\delta_{\mathcal K}$ is the only real root of

conjecture | open | MathDB 页面 | 原文线索

V2-046311 | MATHDB-380431

Pommerenke's integral conjecture for univalent meromorphic functions

摘录 / EXCERPT: Let Σ_0 be the class of univalent meromorphic functions F in the exterior of the unit disk, normalized by $F(z)=z+O(z^{-1})$ as $z\to\infty$. Let $K=F(\mathbb{B}^c)^c$ con...

conjecture | open | MathDB 页面 | 原文线索

V2-046312 | MATHDB-380444

Recursive structure conjecture for extremal polyominoes

摘录 / EXCERPT: Let P be a d-dimensional polyomino with parameters $l_1,\dots,l_d$ and $v_1,\dots,v_d$, and suppose that the convex hull of P has maximum volume. Let P' be a subpolyomino o...

conjecture | open | MathDB 页面 | 原文线索

V2-046313 | MATHDB-380446

Stanley's equidistribution conjecture for alternating permutations and derangements

摘录 / EXCERPT: Stanley's conjecture. For $n\geq4$ and $n\geq5$, respectively,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046314 | MATHDB-380491

A uniform bound for the correction term in integer matrix enumeration

摘录 / EXCERPT: Bounded-correction conjecture. For every such 4-tuple,

conjecture | open | MathDB 页面 | 原文线索

V2-046315 | MATHDB-380583

Fermionic formula conjecture for generalized Kostka polynomials

摘录 / EXCERPT: Let $N\geq 1,n\geq 2$ be integers, λ a partition with $\operatorname{height}(\lambda)\leq n$, and $\boldsymbol L$ an $n\times N$ matrix with nonnegative integer entries. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-046316 | MATHDB-380622

The homotopy-type conjecture for connected central arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ be a central hyperplane arrangement with connected underlying matroid, and let $C({\mathcal A})$ denote its complement. Homotopy-type conjecture. The homotopy ty...

conjecture | open | MathDB 页面 | 原文线索

V2-046317 | MATHDB-380636

The disjoint-transversal conjecture for row-Latin rectangles

摘录 / EXCERPT: Disjoint-transversal conjecture. If $m\ge2n-1$, then A has $m-(n-1)$ pairwise disjoint transversals.

conjecture | open | MathDB 页面 | 原文线索

V2-046318 | MATHDB-380638

The disjoint-independent-transversal conjecture for matroid-valued matrices

摘录 / EXCERPT: Disjoint-independent-transversal conjecture. If $m\ge2n-1$, then A has $m-(n-1)$ pairwise disjoint ITs.

conjecture | open | MathDB 页面 | 原文线索

V2-046319 | MATHDB-380675

Narayana refinement conjecture for the sets ${\cal A}_n$

摘录 / EXCERPT: Narayana refinement conjecture. The number D_{nk} is divisible by

conjecture | open | MathDB 页面 | 原文线索

V2-046320 | MATHDB-380676

Enumeration conjecture for the generalized sets ${\cal A}_n^j$

摘录 / EXCERPT: Enumeration conjecture for ${\cal A}_n^j$. The number of elements of ${\cal A}_n^j$ is

conjecture | open | MathDB 页面 | 原文线索

V2-046321 | MATHDB-380677

Recurrence conjecture for the first two diagonal entries of the volume triangle

摘录 / EXCERPT: Recurrence conjecture for a_n and b_n. The first diagonal entry satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-046322 | MATHDB-380704

A three-block symmetry conjecture for multiple zeta values

摘录 / EXCERPT: Three-block symmetry conjecture. For any non-negative integers a_1,a_2,a_3,b_1,b_2,

conjecture | open | MathDB 页面 | 原文线索

V2-046323 | MATHDB-380719

A partition identity involving selected Ferrers-diagram points

摘录 / EXCERPT: The partition identity conjecture. For all integers $n,r,s e 0$, the following equivalent identities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-046324 | MATHDB-380755

The symmetric-polynomial form conjecture for transitive factorisation generating series

摘录 / EXCERPT: Let $u,z,p_1,p_2,\ldots$ be indeterminates, let $p_\alpha=p_{\alpha_1}p_{\alpha_2}\cdots$, and define ... For a partition α with m parts, define...

conjecture | open | MathDB 页面 | 原文线索

V2-046325 | MATHDB-380853

Reconstruction conjecture for cubical polytopes

摘录 / EXCERPT: Cubical-polytope reconstruction conjecture. Every cubical polytope can be reconstructed from its dual graph.

conjecture | open | MathDB 页面 | 原文线索

V2-046326 | MATHDB-380896

The three-halves-unit shifted triangular-hole tiling enumeration conjecture

摘录 / EXCERPT: Three-halves-unit shifted-hole tiling conjecture. The number of lozenge tilings of this hexagon with the triangular hole equals

conjecture | open | MathDB 页面 | 原文线索

V2-046327 | MATHDB-380912

Fomin-Kirillov additive-generation conjecture for Dunkl evaluations

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathcal{E}_n^+$ be the positive cone generated by noncommutative monomials in the Bruhat generators $[ij]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-046328 | MATHDB-380931

The extremal-eigenstate convergence conjecture for the symmetric open baker

摘录 / EXCERPT: Extremal-eigenstate convergence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-046329 | MATHDB-380942

Macdonald-Stanley conjecture on Jack polynomial coefficients

摘录 / EXCERPT: Let λ and μ be partitions, and let $m_i(\mu)$ denote the multiplicity of the part i in μ. Define ... where $v_{\lambda\mu}(\alpha)$ are the coefficients defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-046330 | MATHDB-380972

Dimension formula conjecture for ${\cal E}_{n,k}^0$

摘录 / EXCERPT: Let ${\cal E}_{n,k}^0$ denote the homogeneous degree-k component of the quadratic algebra ${\cal E}_n^0$, and let n and k be positive integers. Dimension formula conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-046331 | MATHDB-380973

Factorization conjecture for the Hilbert polynomial of ${\cal E}_n^0$

摘录 / EXCERPT: Let ${\cal E}_n^0$ be the quadratic algebra, let $Z_{n,k}$ be the sets of normal monomials defined in the paper, and write $H(A;t)$ for the Hilbert polynomial of a graded object...

conjecture | open | MathDB 页面 | 原文线索

V2-046332 | MATHDB-381007

Stability conjecture for the combinatorial R-matrix on the limiting crystal

摘录 / EXCERPT: Let $B_\natural$ be the limiting crystal obtained from the crystals B_l, and let $g_l:B_l\to B_\natural$ be its set-theoretic embedding. For fixed $u\in B_\natural$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-046333 | MATHDB-381021

The level-count conjecture for forbidden subposets

摘录 / EXCERPT: Let Psubseteq be a finite poset. Let $\operatorname{Mon}(\mathbb{Z})$ be the set of functions $f\colon\mathbb{Z}\to\{0,1\}$ such that $f(n)=1$ and $f(-n)=0$ for all sufficiently...

conjecture | open | MathDB 页面 | 原文线索

V2-046334 | MATHDB-381022

Folkman's conjecture on subset sums and infinite arithmetic progressions

摘录 / EXCERPT: Let A be a strictly increasing sequence of positive integers, and write $A(n):=|A\cap[n]|$, where $[n]:=\{1,2,\dots,n\}$. Thus A has asymptotic density at least $C\sqrt n$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-046335 | MATHDB-381023

Folkman's second conjecture for increasing sequences and infinite arithmetic progressions

摘录 / EXCERPT: Let A be an increasing sequence of positive integers, allowing repetitions, and let $A(n)$ denote the number of elements of A, counted with multiplicity, that are at most n....

conjecture | open | MathDB 页面 | 原文线索

V2-046336 | MATHDB-381026

The equality characterization for the boundary h_2-bound

摘录 / EXCERPT: Let Δ be a connected $(d-1)$-dimensional homology manifold with nonempty orientable boundary, where $d\geq 4$. For each face F of Δ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-046337 | MATHDB-381028

The inner-product formula for the patterns $J_{k,n}$

摘录 / EXCERPT: The $J_{k,n}$ inner-product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046338 | MATHDB-381044

The Polynomial inverse conjecture for the U^3 norm over $\mathbb{F}_2^n$

摘录 / EXCERPT: Let $f:\mathbb{F}_2^n\to\mathbb{C}$ be a K-approximate quadratic, meaning that f has large U^3 norm in the sense used in the paper. A quadratic phase is a function of the for...

conjecture | open | MathDB 页面 | 原文线索

V2-046339 | MATHDB-381061

The little Schröder polynomial analogue of the q-Catalan identity

摘录 / EXCERPT: Let $R_{n,k}(q,t)$ be the (q,t)-Schröder polynomial obtained by weighting little Schröder paths with the statistics defined above, and let $[m]_q$ and $[m]!_q$ denote the q-int...

conjecture | open | MathDB 页面 | 原文线索

V2-046340 | MATHDB-381071

The nimber conjecture for Sprouts starting positions

摘录 / EXCERPT: Let p be the number of spots in the starting position of a game of Sprouts, and let its nimber be the associated Sprague-Grundy value. Nimber conjecture. The nimber for the start...

conjecture | refuted | MathDB 页面 | 原文线索

V2-046341 | MATHDB-381106

Property H conjecture for Artin-Tits presentations

摘录 / EXCERPT: Property H conjecture. Every Artin-Tits presentation satisfies Property H. The conjecture asks whether transformations of type ∞ are unnecessary for reducing every word re...

conjecture | open | MathDB 页面 | 原文线索

V2-046342 | MATHDB-381109

The bigraded Hilbert-series conjecture for alternating polynomials

摘录 / EXCERPT: Let I_n be the ideal generated by alternating polynomials in $R=\mathbb{C}[x_1,y_1,\ldots,x_n,y_n]$, and let q and t record the two gradings. Write s_1^n for the n-fold p...

conjecture | open | MathDB 页面 | 原文线索

V2-046343 | MATHDB-381110

Polynomial-size uniform subset conjecture for transitive group actions

摘录 / EXCERPT: Polynomial-size uniform subset conjecture. There exists an X-uniform subset $T\subset G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-046344 | MATHDB-381117

The extremely amenable subgroup conjecture for closed oligomorphic permutation groups

摘录 / EXCERPT: Let G be a closed oligomorphic permutation group, meaning a closed permutation group with finitely many orbits on n-tuples for every $n\geq 1$. An expansion of an ω-cate...

conjecture | open | MathDB 页面 | 原文线索

V2-046345 | MATHDB-381119

Eventual period-two conjecture for impartial scoring subtraction games

摘录 / EXCERPT: Eventual period-two conjecture. There exists an integer N such that

conjecture | open | MathDB 页面 | 原文线索

V2-046346 | MATHDB-381127

Optimality of the Theorem uj1 algorithm for one defective search

摘录 / EXCERPT: Optimality conjecture. Under these assumptions, the algorithm used in the proof of Theorem uj1 is optimal for $m=1$. The preceding discussion indicates that the algorithm seems opt...

conjecture | open | MathDB 页面 | 原文线索

V2-046347 | MATHDB-381143

Chung-Goldwasser conjecture on maximal measurable 3-sum-free subsets of the unit interval

摘录 / EXCERPT: Chung-Goldwasser conjecture. Every measurable 3-sum-free subset A of $[0,1]$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-046348 | MATHDB-381168

Quadratic bound conjecture for products of primitive matrices

摘录 / EXCERPT: Quadratic bound conjecture. There is a constant K such that every such set has a product of length smaller than Kn^2 with positive entries.

conjecture | open | MathDB 页面 | 原文线索

V2-046349 | MATHDB-381180

Polynomial-growth conjecture for permutations avoiding a fixed pattern

摘录 / EXCERPT: Polynomial-growth conjecture. If π avoids some t-permutation, then

conjecture | open | MathDB 页面 | 原文线索

V2-046350 | MATHDB-381184

Partition-dependence conjecture for divisor and orbit polynomial leading coefficients

摘录 / EXCERPT: Let X be a Z_n curve described by the family in the source. For each positive integer c, let x_c be the number of indices i with $c_i=c$, and for each positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-046351 | MATHDB-381244

Polynomial-growth conjecture for the number of curve diagrams

摘录 / EXCERPT: Let $g_{n,k}$ denote the number of curve diagrams of genus n and complexity k, as in the paper. Let $n\geq 2$ be an integer. Polynomial-growth conjecture. There exist positive...

conjecture | open | MathDB 页面 | 原文线索

V2-046352 | MATHDB-381274

Erdős's conjecture on Sidon bases of order 3

摘录 / EXCERPT: A sequence of positive integers is a Sidon sequence if all sums $a+a'$ with a,a' in the sequence and $a\leq a'$ are distinct; it is an asymptotic basis of order g if every suff...

conjecture | open | MathDB 页面 | 原文线索

V2-046353 | MATHDB-381292

A stronger rainbow degree-sequence conjecture for bipartite graphs

摘录 / EXCERPT: Let F_i, $i\leq k$, be subgraphs of $K_{n,n}$, and let $A=(a_{ij})$ be the $k\times n$ matrix whose entry a_{ij} is the degree of v_j in F_i. Let $r_i(A)$ be the sum of the...

conjecture | open | MathDB 页面 | 原文线索

V2-046354 | MATHDB-381341

The permutation-degree conjecture for friendly triangulations

摘录 / EXCERPT: Permutation-degree conjecture. The labeling L induces a single-valued labeling $\ell:\operatorname{Vert}(T)\to [n]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-046355 | MATHDB-381349

Shao's polynomial bound conjecture for the primitive degree of tensors

摘录 / EXCERPT: Let $\mathbb{A}$ be a nonnegative primitive tensor of fixed order m and dimension n. Its primitive degree $\gamma(\mathbb{A})$ is the smallest positive integer r such that...

conjecture | open | MathDB 页面 | 原文线索

V2-046356 | MATHDB-381364

Bonin's hyperplane bound for matroids with no uniform-minor

摘录 / EXCERPT: Bonin's conjecture. The matroid M has at most

conjecture | open | MathDB 页面 | 原文线索

V2-046357 | MATHDB-381366

Function-field Freiman conjecture for spaces of small combinatorial genus

摘录 / EXCERPT: Let K be an algebraically closed field and let F be an extension field of K. Let S be a finite-dimensional K-subspace of F such that $K\subset S$. Define the combinator...

conjecture | open | MathDB 页面 | 原文线索

V2-046358 | MATHDB-381380

The USPN conjecture for non-adaptive one-sided permutation-freeness testing

摘录 / EXCERPT: USPN conjecture. For any permutation π of any length, π-freeness has a non-adaptive one-sided ε-test making

conjecture | open | MathDB 页面 | 原文线索

V2-046359 | MATHDB-381395

The asymptotic bicommutative-system bound conjecture

摘录 / EXCERPT: For integers $k\geq 2$, let $n=4k+7$ and $m=2k+3$. Let $\mathrm{Spec}(k,2)$ be the minimum cardinality of a $(2k+3)$-bicommutative system in $\mathcal{P}_{n}(m)$. Asymptotic bicomm...

conjecture | open | MathDB 页面 | 原文线索

V2-046360 | MATHDB-381415

Extremal localization conjecture for the Gaussian STFT

摘录 / EXCERPT: Let $\Delta\subset\mathbb{R}^{2}$ be a measurable set of finite measure, let $A=|\Delta|$, let $\varphi=h_{0}$ be the Gaussian, and let $V_{\varphi}f$ denote the short-time Fourier...

conjecture | open | MathDB 页面 | 原文线索

V2-046361 | MATHDB-381419

Conjecture on equiangular lines saturating the relative and incoherence bounds

摘录 / EXCERPT: Classification conjecture. Then $d=2,3,6,7,23$, and Ω is isometric to the corresponding example found in the examples section.

conjecture | open | MathDB 页面 | 原文线索

V2-046362 | MATHDB-381422

Existence of 3-inflatable permutations in every admissible residue class modulo 144

摘录 / EXCERPT: Existence conjecture. For every positive integer x whose length belongs to an admissible residue class modulo 144, there exists a 3-inflatable permutation of length x.

conjecture | open | MathDB 页面 | 原文线索

V2-046363 | MATHDB-381450

Symmetry conjecture for crossing distributions of restricted permutations

摘录 / EXCERPT: Symmetry conjecture. For all integers n and k satisfying $1\leq k\leq n$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-046364 | MATHDB-381453

Extension of the main theorem without condition (nu)

摘录 / EXCERPT: The Bachet's game with lottery moves model is governed by the assumptions preceding the main theorem, including condition ... is violated. The condition is used technically in the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-046365 | MATHDB-381465

The generalized Erdős-Ginzburg-Ziv conjecture for finite groups

摘录 / EXCERPT: Generalized Erdős-Ginzburg-Ziv conjecture. For every finite group G,

conjecture | open | MathDB 页面 | 原文线索

V2-046366 | MATHDB-381507

Erdős-Fishburn triangular-lattice conjecture for few-distance configurations

摘录 / EXCERPT: Let $L_{\Delta}=\left\{\left(a+\frac{1}{2}b,\frac{\sqrt{3}}{2}b\right):a,b\in\mathbb{Z}\right\}$ be the triangular lattice. A configuration is k-optimal if it has the maximum pos...

conjecture | open | MathDB 页面 | 原文线索

V2-046367 | MATHDB-381524

Global chain decomposition conjecture for q,t-Catalan numbers

摘录 / EXCERPT: Global chain decomposition conjecture. There exist collections of partitions $\mathcal{C}_{\mu}$, indexed by deficit partitions μ, and a size-preserving involution...

conjecture | open | MathDB 页面 | 原文线索

V2-046368 | MATHDB-381550

Polynomial dimension conjecture for multiplicatively invariant sets

摘录 / EXCERPT: Fix $n,m\in\mathbb{N}$ and let ... for $A,B\subseteq\mathbb{N}_0$. Let $r,s\in\mathbb{N}$ be multiplicatively independent, and let $A,B\subseteq\mathbb{N}_0$ be $\times r$- and...

conjecture | open | MathDB 页面 | 原文线索

V2-046369 | MATHDB-381594

The unrestricted cylindrical width conjecture for finite transitive subsets of complex spheres

摘录 / EXCERPT: Let $1\le k\le d$. Let X be a finite transitive subset of the unit sphere in $\mathbb{C}^d$. Unrestricted cylindrical width conjecture. There is a complex k-dimensional subspac...

conjecture | open | MathDB 页面 | 原文线索

V2-046370 | MATHDB-381601

Khabibullin's conjecture for nonnegative logarithmically convex functions

摘录 / EXCERPT: Khabibullin's conjecture. The stated integral implication holds for every such S, every $\rho>1$, and every integer $n\geq 2$. This is an extremal integral inequality arising in...

conjecture | open | MathDB 页面 | 原文线索

V2-046371 | MATHDB-381628

Linear upper bound conjecture for the longest solvable board

摘录 / EXCERPT: Linear upper bound conjecture. As $n\to\infty$, $ML(n)=O(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-046372 | MATHDB-381649

Pinheiro-Machado-Firer extension conjecture for partition-induced combinatorial metrics

摘录 / EXCERPT: Let $\mathbb{F}_2$ be the binary field and let $\mathbf{H}=\mathbb{F}_2^\Omega$. For $k\in[1,|\Omega|]$, write $\mathcal{P}(k,\Omega)$ for the covering of Ω by all k-eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-046373 | MATHDB-381659

Bijection between constrained Delannoy paths and Catalan paths avoiding symmetric peaks

摘录 / EXCERPT: Delannoy-Catalan bijection conjecture. For every positive integer n, there exists a bijection between $E\mathcal{P}^{(1)}_{n,n}(NE,EN)$ and the set of Catalan paths of size $n+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-046374 | MATHDB-381757

Gedeon's dominance conjecture for matroid Kazhdan--Lusztig and Z-polynomials

摘录 / EXCERPT: Let M be a loopless matroid of rank k on a ground set of size n. Let $\mathsf{U}_{k,n}$ be the uniform matroid of rank k on n elements, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-046375 | MATHDB-381760

The ordinary-flat conjecture for complex-representable matroids

摘录 / EXCERPT: Let M be a matroid, and call a rank-k flat ordinary if it is the direct sum of a point and a rank-$(k-1)$ flat. Ordinary-flat conjecture. For each integer $k\ge 2$, every compl...

conjecture | open | MathDB 页面 | 原文线索

V2-046376 | MATHDB-381788

The strong Fox conjecture for alternating knots

摘录 / EXCERPT: Let $K \subset S^3$ be an alternating knot. Its Alexander polynomial is ... Here a polynomial is log-concave when its coefficient sequence satisfies $a_{i-1}a_{i+1}\leq a_i^2$ for ev...

conjecture | open | MathDB 页面 | 原文线索

V2-046377 | MATHDB-381793

The Möbius-function conjecture for signed Wachs permutations

摘录 / EXCERPT: Let $n\in\mathbb{P}$, and let $\mu(u,v)$ denote the Möbius function of the Bruhat-order interval $[u,v]$ in the poset of signed Wachs permutations $\operatorname{\mathcal{W}}(B_n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-046378 | MATHDB-381801

Total-lamination conjecture for neighborly box families

摘录 / EXCERPT: Let $S=\{0,1,*\}$, and let S^d be the set of length-d strings over S. A family $F\subseteq S^d$ is a k-neighborly total lamination if it is k-neighborly and is a total la...

conjecture | open | MathDB 页面 | 原文线索

V2-046379 | MATHDB-381814

The conjectural formula for the one-wrapping count gamma

摘录 / EXCERPT: Let $b3^\ell(s,n)$ denote the number of continuous multiline queues with the specified one-wrapping configuration, and let $f_{(n-1,s+1)}$ denote the relevant hook-length count. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-046380 | MATHDB-381815

The conjectural formula for rho sub 2,3

摘录 / EXCERPT: Let $c1_{2,3}(s,n)$ denote the quantity counting the relevant continuous multiline-queue configurations, let N be the total size parameter used in the paper, and let $f_{(s,s)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-046381 | MATHDB-381816

The recurrence equivalent to the conjectural formula for rho sub 2,3

摘录 / EXCERPT: Let $c1_{2,3}(s,n)$ denote the quantity appearing in the preceding conjecture, with s fixed and n varying. The recurrence conjecture. ... The source explicitly says that, using t...

conjecture | open | MathDB 页面 | 原文线索

V2-046382 | MATHDB-381842

Frankl-Győri-He-Lv-Salia-Tompkins-Varga-Zhu conjecture for Gallai 3-colouring templates

摘录 / EXCERPT: Frankl-Győri-He-Lv-Salia-Tompkins-Varga-Zhu conjecture. As n tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-046383 | MATHDB-381851

The simply connected level-set conjecture for checkerboard switching

摘录 / EXCERPT: Let $R\in\mathbb{N}^p$, $C\in\mathbb{N}^q$, and let $\mathbf{M}\in\mathcal{M}_{p,q}(\{-1,0,1\})$. For each integer i, let $\mathcal{P}_i(\mathbf{M})$ denote the set of cells at l...

conjecture | open | MathDB 页面 | 原文线索

V2-046384 | MATHDB-381853

Snevily's conjecture for non-uniform L-intersecting families

摘录 / EXCERPT: Let $L=\{\ell_{1},\ell_{2},\ldots,\ell_{s}\}$ be a set of s non-negative integers and $K=\{k_{1},k_{2},\ldots,k_{r}\}$ be a set of r positive integers with ... Let an L-interse...

conjecture | open | MathDB 页面 | 原文线索

V2-046385 | MATHDB-381854

The Sperner-system strengthening of Snevily's conjecture

摘录 / EXCERPT: Let n,s be integers with $n\geq 2s-1$, let $L=\{\ell_{1},\ell_{2},\ldots,\ell_{s}\}$ be a set of s non-negative integers, and let an L-intersecting Sperner system be a Sperne...

conjecture | open | MathDB 页面 | 原文线索

V2-046386 | MATHDB-381865

The binary-form area lower-bound conjecture

摘录 / EXCERPT: Binary-form area lower-bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-046387 | MATHDB-381866

Bean's discriminant-area maximality conjecture for binary forms

摘录 / EXCERPT: Bean's conjecture. The maximum is attained precisely when F, up to multiplication by a complex number, is equivalent under $\operatorname{GL}_2(\mathbb R)$ to $F_{n,n}$. In parti...

conjecture | open | MathDB 页面 | 原文线索

V2-046388 | MATHDB-381877

Poonen's twin-free union-closed family conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite union-closed family of sets with $\emptyset \notin\mathcal{F}$. Elements a and b are twins if every set $A\in\mathcal{F}$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-046389 | MATHDB-381878

Poonen's twin-free structural formulation

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite union-closed family of sets, and let M be its largest set. Assume that for every pair of elements $a,b\in M$, the family contains a set A with...

conjecture | open | MathDB 页面 | 原文线索

V2-046390 | MATHDB-381896

Ranked cluster-variable counting formulas for Grassmannian cluster algebras

摘录 / EXCERPT: For positive integers k,n,r, let $N_{k,n,r}$ denote the number of cluster variables of rank r in $\mathbb{C}[\operatorname{Gr}(k,n)]$. Cluster-variable counting conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-046391 | MATHDB-381909

Markoff injectivity conjecture for Christoffel words

摘录 / EXCERPT: A Markoff triple is a positive solution of $x^2+y^2+z^2=3xyz$. A Markoff number is an element of a Markoff triple. Let μ be the monoid homomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-046392 | MATHDB-381915

Sturm-unimodality conjecture for refined Narayana polynomials

摘录 / EXCERPT: Let $W_{n,k}(t)$ be the polynomials considered in the paper. A finite sequence of real-rooted polynomials is Sturm-unimodal if it increases by interlacing up to one index and then...

conjecture | open | MathDB 页面 | 原文线索

V2-046393 | MATHDB-381916

Explicit formulas conjecture for special symmetric-decomposition polynomials

摘录 / EXCERPT: Let C_k denote the kth Catalan number, let $\operatorname{Nar}_{k}(t)$ denote the Narayana polynomial, and define ... Let $W_{n,k}^{+}(t)$ and $W_{n,k}^{-}(t)$ be the symmetric-d...

conjecture | open | MathDB 页面 | 原文线索

V2-046394 | MATHDB-381930

The compatibility-count conjecture for $(3,n)$ generalized Feynman diagrams

摘录 / EXCERPT: Let a $(3,n)$ generalized Feynman diagram (GFD) be a combinatorial object compatible with color orderings in the sense of the paper. Compatibility-count conjecture. Every $(3,n)$ G...

conjecture | open | MathDB 页面 | 原文线索

V2-046395 | MATHDB-381931

The chirotopal tropical Grassmannian conjecture for rank three

摘录 / EXCERPT: Fix a realizable chirotope $\chi\in\{-1,1\}^{\binom{n}{3}}$. Let $\operatorname{Dr}^\chi_{3,n}$ be the chirotopal Dressian, consisting of the χ-tropical Plücker vectors, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-046396 | MATHDB-381951

Dijoin conjecture for joins of 2-invertible oriented graphs

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $D_1,\dots,D_k$ be oriented graphs satisfying ... Write $[D_1,\dots,D_k]$ for their join, and let inv denote inversion number. Dijoin co...

conjecture | refuted | MathDB 页面 | 原文线索

V2-046397 | MATHDB-381980

Conjectured initial values of the ML-degree polynomial

摘录 / EXCERPT: Let $\phi(n,d)$ denote the polynomial that gives the dimension of the space of quadrics in n variables passing through d general points and tangent to $\binom{n+1}{2}-d-1$ gene...

conjecture | open | MathDB 页面 | 原文线索

V2-046398 | MATHDB-381991

Möbius support conjecture for base polymatroids

摘录 / EXCERPT: Möbius support conjecture. For any base polymatroid $\mathscr{P}$, the Möbius support $\mu\text{-supp}(\mathscr{P})$ is a generalized polymatroid. The theorem preceding this conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-046399 | MATHDB-382024

Fingerhut's max-sum matching conjecture

摘录 / EXCERPT: Let S be an even set of points in the plane, and let $\{a_ib_i:i=1,\dots,n\}$ be a max-sum matching of S, where $|S|=2n$. A point o is required to satisfy, for every matched...

conjecture | open | MathDB 页面 | 原文线索

V2-046400 | MATHDB-382025

Suri's minimum Steiner star conjecture for max-sum matchings

摘录 / EXCERPT: Let S be an even set of points in the plane, and let $\mathcal M$ be a max-sum matching of S. Define $t(S)$ as the minimum, over points y in the plane, of...

conjecture | open | MathDB 页面 | 原文线索

V2-046401 | MATHDB-382027

The arbitrary-position rigidity conjecture for 2-triangulations

摘录 / EXCERPT: Let P be a convex n-gon, and place n distinct points on the moment curve in $\mathbb{R}^{4}$. The associated bar-and-joint rigidity matroid is represented by the rigidity mat...

conjecture | open | MathDB 页面 | 原文线索

V2-046402 | MATHDB-382034

Minimal-weight-vector conjecture for tight irreducible subcube partitions

摘录 / EXCERPT: Minimal-weight-vector conjecture. For every $n\geq 3$, the minimal weight vectors, under majorization, of tight irreducible subcube partitions of length n are

conjecture | open | MathDB 页面 | 原文线索

V2-046403 | MATHDB-382035

Maximum-minimum-dimension conjecture for tight irreducible subcube partitions

摘录 / EXCERPT: Maximum-minimum-dimension conjecture. Every tight irreducible subcube partition F of length n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-046404 | MATHDB-382036

Minimal-size conjecture for tight irreducible affine vector space partitions

摘录 / EXCERPT: Affine-vector-space minimal-size conjecture. The minimal size of a tight irreducible affine vector space partition of length n is

conjecture | open | MathDB 页面 | 原文线索

V2-046405 | MATHDB-382090

Conjecture on matroidal 3-colorability of paths

摘录 / EXCERPT: Path 3-colorability conjecture. Every path is matroidally 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-046406 | MATHDB-382099

Susskind-Uglum conjecture on bulk entropy and Newton's constant

摘录 / EXCERPT: In the exterior region of a black hole, the bulk entropy $S_{bulk}(\rho)$ is regulated by a bulk UV cutoff, while the area term in the quantum extremal surface formula depends on N...

conjecture | open | MathDB 页面 | 原文线索

V2-046407 | MATHDB-382107

Asymptotic conjecture for products of cross-Sperner families

摘录 / EXCERPT: Asymptotic product conjecture. For fixed $k\geq 2$ and sufficiently large n with respect to k,

conjecture | open | MathDB 页面 | 原文线索

V2-046408 | MATHDB-382181

Conjecture on totally chain-intersecting families for p at least q

摘录 / EXCERPT: Conjecture for totally chain-intersecting families. If $\mathcal F$ is a totally (p,q)-chain intersecting family and $p\geq q$, then

conjecture | open | MathDB 页面 | 原文线索

V2-046409 | MATHDB-382182

Optimal Power of Few conjecture for majority dynamics

摘录 / EXCERPT: Optimal Power of Few conjecture. There is a function $f(x)$ such that $f(x)>1/2$ for $x>0$, $f(x)$ tends to 1 as x tends to infinity, and, in majority dynamics on $G(n,p)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-046410 | MATHDB-382187

Conjecture on the power-series coefficients of the expected largest BP-factorization width

摘录 / EXCERPT: Let E_k denote the limit of the expected width of the largest block-palindrome factorization of a length-n word over an alphabet of size k, whose existence is established for...

conjecture | open | MathDB 页面 | 原文线索

V2-046411 | MATHDB-382189

Polynomial product formula conjecture for the generalized Aztec-diamond matrix

摘录 / EXCERPT: Generalized product formula conjecture. There exists a sequence of polynomials $o_n(k,t)\in\mathbb{Z}[k,t]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-046412 | MATHDB-382194

Lusztig-Dyer combinatorial invariance conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let (W_1,S_1) and (W_2,S_2) be Coxeter systems, and let $\sigma_1,\omega_1 easymp W_1$ and $\sigma_2,\omega_2 easymp W_2$. For each pair, let $G[\sigma_i,\omega_i]$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-046413 | MATHDB-382195

Relative combinatorial invariance conjecture for relative R-polynomials

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system. For a subset $J\subseteq S$, write W_J for the corresponding standard parabolic subgroup. For pairs in finite Coxeter systems, let...

conjecture | open | MathDB 页面 | 原文线索

V2-046414 | MATHDB-382196

Filtered combinatorial invariance conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system equipped with a canonical filtration ... This filtration gives a reflection-coloring map $\alpha:T\to\{1,\ldots,r\}$ by assigning to each refle...

conjecture | open | MathDB 页面 | 原文线索

V2-046415 | MATHDB-382199

The ABK Conjecture on countable unions of bqos

摘录 / EXCERPT: A well-quasi-order (wqo) is a quasi-order with no infinite descending sequence and no infinite antichain; a better-quasi-order (bqo) is a wqo satisfying the stronger Ramsey-theoret...

conjecture | open | MathDB 页面 | 原文线索

V2-046416 | MATHDB-382200

The asymptotic scarcity conjecture for blockwise simple permutations

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let W_n be the set of permutations of size n, let Simp_n be the set of blockwise simple permutations, and define ... where S_n is th...

conjecture | open | MathDB 页面 | 原文线索

V2-046417 | MATHDB-382207

The genus-three formula conjecture for the coefficients $S_{n,k}^{(3)}$

摘录 / EXCERPT: Let $S_{n,k}^{(g)}$ denote the coefficients indexed by n, k, and genus g, and let the expression referred to as ... , with $g=3$, gives $S_{n,k}^{(3)}$ for all n and k. The...

conjecture | open | MathDB 页面 | 原文线索

V2-046418 | MATHDB-382208

The all-genus formula conjecture for the coefficients $S_{n,k}^{(g)}$

摘录 / EXCERPT: For each positive genus g, let $S_{n,k}^{(g)}$ be the coefficients indexed by positive integers n and k, and let $\chi^{(g)}$ be a triangular array of constants. Let the expr...

conjecture | open | MathDB 页面 | 原文线索

V2-046419 | MATHDB-382231

Density conjecture for integral points at spacetime distance

摘录 / EXCERPT: Density conjecture. For every $\epsilon>0$, there exists some positive integer d such that if $I\subseteq\mathbb{Z}^2$ has upper density greater than ϵ, then there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-046420 | MATHDB-382232

The converse characterization of matroidal cycles under integer linear equivalence

摘录 / EXCERPT: Let M and M' be non-isomorphic matroids, and let Z_M and $Z_{M'}$ denote their matroidal cycles in their respective ambient vector spaces. Suppose there exists an invertible...

conjecture | open | MathDB 页面 | 原文线索

V2-046421 | MATHDB-382246

Weaker variant of Woodall's dijoin decomposition conjecture

摘录 / EXCERPT: Weaker dijoin decomposition conjecture. The arc set A can be decomposed into a k-dijoin and a $(\tau-k)$-dijoin, for every $k\in\{1,\ldots,\tau-1\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-046422 | MATHDB-382250

Asymptotic conjectures for the crossroads sequences L_n and M_n

摘录 / EXCERPT: Conjectures for L_n and M_n. The following five propositions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-046423 | MATHDB-382312

Asymptotic inversion constant for the non-reduced random permutation

摘录 / EXCERPT: Let $p\in[0,1)$ and let $\boldsymbol{w}=\boldsymbol{w}(n)\in S_n$ be the random permutation in the non-reduced model. Write $\operatorname{inv}(\boldsymbol{w})$ for its number of i...

conjecture | open | MathDB 页面 | 原文线索

V2-046424 | MATHDB-382322

Minimum-volume asymptotics for Boltzmann random triangulations of the d-sphere

摘录 / EXCERPT: Let $\mathcal{S}^{(d)}_n$ be the collection of triangulations of the d-sphere, and let $\mu^{(d)}_{n,\beta}$ denote the median of the minimum volume of the Boltzmann random spher...

conjecture | open | MathDB 页面 | 原文线索

V2-046425 | MATHDB-382335

Ehrenborg-Morel-Readdy's conjecture on pizza quantities for Coxeter arrangements

摘录 / EXCERPT: Let $\mathcal{H}$ be a Coxeter arrangement on a finite-dimensional inner product space V, let $a\in V$, and let R satisfy $\|a\|\leq R$. Assume that the number of hyperplanes o...

conjecture | open | MathDB 页面 | 原文线索

V2-046426 | MATHDB-382344

The entropic Marton-type conjecture for random variables over \(\mathbb{F}_2^m\)

摘录 / EXCERPT: Entropic Marton-type conjecture. There exists $c_2=\exp(O(1/c_1))$ such that, for every $m\in\mathbb{Z}_+$ and every such X, there is a subspace G with

conjecture | open | MathDB 页面 | 原文线索

V2-046427 | MATHDB-382363

Bergeron's Fibonacci conjecture for the sign character of the $(1,3)$-bosonic-fermionic coinvariant ring

摘录 / EXCERPT: Let $R_n^{(1,3)}$ be the $(1,3)$-bosonic-fermionic coinvariant ring, let $\operatorname{Frob}(R_n^{(1,3)};1;1,1,1)$ denote its Frobenius series with all grading variables specializ...

conjecture | open | MathDB 页面 | 原文线索

V2-046428 | MATHDB-382375

Cyclic difference matrix conjecture for the Fibonacci-like partition construction

摘录 / EXCERPT: Let $q_{i,j}=q_{i,j}^{[m,n]}$ be the non-negative integers obtained in Algorithm; these integers are independent of m and n. For each $i\geq 1$, consider the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-046429 | MATHDB-382378

More precise asymptotics of quasinormal modes near extremality

摘录 / EXCERPT: Let σ_0 be a simple quasinormal mode of the Klein--Gordon equation on the near-horizon geometry with parameters $r_{\rm e}$ and $r_{\rm c}$. Let $\epsilon>0$ be small and se...

open_problem | open | MathDB 页面 | 原文线索

V2-046430 | MATHDB-382385

Growth conjecture for simple permutations in the avoidance class

摘录 / EXCERPT: Growth conjecture. For $n \geq 12$,

conjecture | open | MathDB 页面 | 原文线索

V2-046431 | MATHDB-382389

Tightness of extremal points of discrete bridge arches to the Brownian limit

摘录 / EXCERPT: Tightness conjecture. The highest points of sufficiently long positive arches, and the lowest points of sufficiently long negative arches, are tight to the Brownian limit.

conjecture | open | MathDB 页面 | 原文线索

V2-046432 | MATHDB-382390

Overflow conjecture for families of bounded diameter

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $2^{[n]}$ denote its power set. For a family $\mathcal{F}\subset 2^{[n]}$, write ... and define its diametral overflow by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-046433 | MATHDB-382399

Superlinear monotone-path covering conjecture for dense point sets

摘录 / EXCERPT: Let A be a dense n-element point set in general position in the plane, and let $K_n[A]$ denote the complete geometric graph induced by A. A path is monotone if its edge set i...

conjecture | open | MathDB 页面 | 原文线索

V2-046434 | MATHDB-382408

Basis conjecture for the alternating component of generalized diagonal coinvariants

摘录 / EXCERPT: Let m,n be positive integers, let $L_{mn,n}^+$ denote the set of rational m-Dyck paths, and let $L_{n,n}^+$ denote the set of Dyck paths of size n. For any rational Dyck path...

conjecture | open | MathDB 页面 | 原文线索

V2-046435 | MATHDB-382416

Eigenvalue bound conjecture for covariance matrices of volume power functionals

摘录 / EXCERPT: Eigenvalue bound conjecture. These eigenvalues satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-046436 | MATHDB-382419

Feasible indexing conjecture for successor and ancestor sets of weakly connected DAGs

摘录 / EXCERPT: Feasible indexing conjecture. There exists an indexing satisfying this successor-set ordering and, whenever two consecutive nodes have equal successor-set order,

conjecture | open | MathDB 页面 | 原文线索

V2-046437 | MATHDB-382424

Functional equation conjecture for well-aligned pairs

摘录 / EXCERPT: Functional equation conjecture. The exponential generating function satisfies

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046438 | MATHDB-382442

The recursive upper-bound conjecture for multidimensional egg-drop problems

摘录 / EXCERPT: Recursive upper-bound conjecture. In a d-dimensional setting,

conjecture | open | MathDB 页面 | 原文线索

V2-046439 | MATHDB-382454

The integer-numerator conjecture for the stationary distribution of the finite k-chain

摘录 / EXCERPT: Let $k\geq 2$, let $\mathcal{R}_k$ be the state space of the finite k-chain, let π be its stationary distribution, and set ... For each $\lambda\in\mathcal{R}_k$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-046440 | MATHDB-382464

The transitive-array conjecture for solutions of the classical Yang-Baxter equation

摘录 / EXCERPT: Let C be a set, let $\boldsymbol{a}$ be a transitive $n\times n$ matrix with entries in C, and let $\mathbf r=\{r^{(c)}\}_{c\in C}\subset \mathfrak{g}\otimes\mathfrak{g}$ be a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046441 | MATHDB-382465

Linear cumulant conjecture for colored interlacing triangles

摘录 / EXCERPT: Linear cumulant conjecture. Every cumulant is a linear function of n for all $k\geq 1$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046442 | MATHDB-382477

Linear tangency conjecture for pairwise intersecting bi-infinite x-monotone curves

摘录 / EXCERPT: Let k be a fixed positive integer, and let a family consist of n pairwise intersecting bi-infinite x-monotone curves such that any two curves intersect at most k times. Lin...

conjecture | open | MathDB 页面 | 原文线索

V2-046443 | MATHDB-382478

Baker-Chen-Li-Qian refinement for cyclic V-shaped permutations

摘录 / EXCERPT: Let n be a positive integer. A cyclic V-shaped permutation of $[n]$ is a permutation with a unique local minimum that is also a cycle, and let the minimum occur at position k....

conjecture | open | MathDB 页面 | 原文线索

V2-046444 | MATHDB-382484

Aluffi-Chen-Marcolli real-rootedness conjecture for the moduli space of stable pointed curves

摘录 / EXCERPT: Aluffi-Chen-Marcolli conjecture. The polynomial $P_{\overline{\mathcal{M}}_{0,n+1}}(t)$ is real-rooted.

conjecture | open | MathDB 页面 | 原文线索

V2-046445 | MATHDB-382489

Non-rank-symmetry conjecture for hull configurations without blank sides

摘录 / EXCERPT: A hull configuration is a point configuration lying either on a line segment or on the boundary of a convex polygon; a blank side is an edge of the convex hull containing only two...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046446 | MATHDB-382507

Bérczi-Nádor conjecture on e-embracing exchange distance

摘录 / EXCERPT: Let $\mathcal{M}=(E,\mathcal{C})$ be an oriented matroid of rank r, let $e\in E$, and let $\mathcal{C}^+$ denote its positive circuits. An e-positive circuit is a positive circ...

conjecture | open | MathDB 页面 | 原文线索

V2-046447 | MATHDB-382508

Extremality conjecture for symmetric Hales-Jewett thresholds

摘录 / EXCERPT: Extremality conjecture. For all $t,r\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-046448 | MATHDB-382513

Frankl's conjecture for even $n+t$ and t-intersecting k-Sperner families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $\mathcal{F}\subseteq 2^{[n]}$. The family $\mathcal{F}$ is t-intersecting if $|A\cap B|\geq t$ for all $A,B\in\mathcal{F}$, and it is k-Spern...

conjecture | open | MathDB 页面 | 原文线索

V2-046449 | MATHDB-382514

Automorphism-group conjecture for rectangular-combination designs

摘录 / EXCERPT: Let $\widetilde{\mathscr{D}}$ be a design and let ${\mathscr{D}}(\widetilde{\mathscr{D}},(e_1,e_2))$ denote the design obtained from it by the rectangular combination construction,...

conjecture | open | MathDB 页面 | 原文线索

V2-046450 | MATHDB-382522

Polynomial-time minor containment conjecture for fixed codes

摘录 / EXCERPT: Fix a code ${\mathcal M}$. Given a code ${\mathcal C}$ of length n, polynomial-time minor-containment conjecture. It is decidable in time polynomial in n whether ${\mathcal C}$...

conjecture | open | MathDB 页面 | 原文线索

V2-046451 | MATHDB-382537

Strict inequality conjecture for permutation pattern variance

摘录 / EXCERPT: Strict inequality conjecture. The inequality in is strict for all $m>0$. This would imply that $\operatorname{Var}(X_\tau)$ has order $2m-1$ in n, so the standard deviation of...

conjecture | open | MathDB 页面 | 原文线索

V2-046452 | MATHDB-382538

Asymptotic extremal covariance conjecture for permutation patterns

摘录 / EXCERPT: Asymptotic extremal covariance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-046453 | MATHDB-382953

Brenti’s log-concavity conjecture for \widetilde{R}-polynomials

摘录 / EXCERPT: Conjecture 2 (Brenti). Let W be a finite Coxeter group. For $u,v\in W$ with $u\leq v$ in Bruhat order, define $\widetilde{R}_{u,v}$ by

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046454 | MATHDB-382985

Reading-Segovia question on order dimension of posets of regions

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite simplicial hyperplane arrangement of rank r, and let $W(\mathcal{A})$ be its poset of regions with the standard region order. Is...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046455 | MATHDB-383003

Nim with pass, problem A13

摘录 / EXCERPT: Given a Nim position $\mathbf{x}=(x_1,\ldots,x_n)\in\mathbb{Z}_{\ge 0}^n$ together with a pass-status $\varepsilon\in\{0,1\}$, where $\varepsilon=1$ means that one pass remains ava...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046456 | MATHDB-383028

Kelly-Trotter product conjecture for posets

摘录 / EXCERPT: For all finite posets P and Q, with $P\times Q$ ordered coordinatewise, $\dim(P\times Q)\ge \dim(P)+\dim(Q)-2$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046457 | MATHDB-383095

O’Neill’s eventown supersaturation conjecture

摘录 / EXCERPT: For every integer $n\ge 1$ and every integer s satisfying $1\le s\le 2^{\lfloor n/2\rfloor}-2^{\lfloor n/4\rfloor}$, let $\mathcal F$ be a family of even-sized subsets of $[n]$ w...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046458 | MATHDB-383134

Theta conjecture

摘录 / EXCERPT: Conjecture 9.1 proposes the identity ... where $\mathsf{LD}(0,n)^{\ast k,\circ r}$ is the set of positively labelled Dyck paths of size n with k decorated rises and r decorated...

conjecture | open | MathDB 页面 | 原文线索

V2-046459 | MATHDB-383200

Athanasiadis-Douvropoulos-Kalampogia-Evangelinou conjecture on noncrossing-partition h-polynomials

摘录 / EXCERPT: For every finite Coxeter group W of rank r, the h-polynomial of the order complex of its noncrossing-partition lattice admits a nonnegative real-rooted symmetric decompositio...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046460 | MATHDB-383210

Hadwiger conjecture (combinatorial geometry)

摘录 / EXCERPT: Let $n \ge 1$ and let $K \subset \mathbb{R}^n$ be a convex body, i.e. a bounded closed convex set with nonempty interior. Call a set of the form $sK + v = \{sx + v : x \in K\}$, wi...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046461 | MATHDB-383239

Han-Mao-Zeng and Baril-Kirgizov descent conjectures

摘录 / EXCERPT: The conjectures concern permutation statistics on $\mathfrak{S}_n$. First, the generating function formed from...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046462 | MATHDB-383263

Nonabelian Erdős-Ginzburg-Ziv conjecture

摘录 / EXCERPT: For every finite nonabelian group G, let $\exp(G)$ be the exponent of G. Define $s(G)$ to be the least integer ℓ such that every sequence over G of length at least...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046463 | MATHDB-383305

Mahmoodian-Mirzakhani conjecture

摘录 / EXCERPT: For positive integers $r\le s\le t$, the complete tripartite graph $K_{r,s,t}$ admits a decomposition into 5-cycles if and only if $r\equiv s\equiv t\pmod{2}$,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046464 | MATHDB-383336

Erdős unit-distance problem and square-grid optimality conjecture

摘录 / EXCERPT: For each positive integer n, let $U(n)$ be the maximum, over all sets $P\subseteq\mathbb{R}^2$ with $|P|=n$, of the number of unordered pairs $\{p,q\}\subseteq P$ satisfying...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046465 | MATHDB-383345

Chapoton-Athanasiadis conjecture on h-polynomials of preorders

摘录 / EXCERPT: For every finite preorder τ, the h-polynomial of τ equals that of its dual preorder: $h_{\tau}(t)=h_{\tau^{*}}(t)$, where τ^{*} denotes the dual preorder.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046466 | MATHDB-383437

Richards-Sahi conjecture

摘录 / EXCERPT: Let X, Y, and Z be upward closed subsets of Q_n. It is conjectured that ... Notation: the poset Q_n is the power set of a set with n elements, ordered by subset relation....

open_problem | open | Kada Williams | MathDB 页面 | 原文线索

V2-046467 | MATHDB-383493

Erdős Matching Conjecture

摘录 / EXCERPT: For integers $k\ge 2$, $s\ge 1$, and $n\ge (s+1)k$, every family $\mathcal{F}\subseteq\binom{[n]}{k}$ with matching number $\nu(\mathcal{F})\le s$ satisfies ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046468 | MATHDB-383496

Norin-Turcotte exact random-cover conjecture

摘录 / EXCERPT: Let T be a finite metric tree of total length L, and let $r>0$ satisfy $L\ge 2r$. Then there exists a probability distribution on 0-good ball covers of T, with all ball rad...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046469 | MATHDB-383499

Schiffler Problem 6.3

摘录 / EXCERPT: For admissible parameters a,b and any two lattice paths $P,Q\in\mathcal{D}(a,b)$, does equality of their Lagrange numbers imply that their associated band graphs are isomorphic?...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046470 | MATHDB-383540

Samuels' Conjecture

摘录 / EXCERPT: For $0\leq\mu_1\leq\cdots\leq\mu_n$ and $\lambda>\sum_{i=1}^n\mu_i$, let $X_1,\ldots,X_n$ range over independent nonnegative random variables satisfying $\mathbb{E}X_i=\mu_i$. Defi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046471 | MATHDB-383581

Qiu-Remmel conjecture on quadrant marked mesh patterns

摘录 / EXCERPT: ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046472 | MATHDB-383686

Greedy superstring conjecture

摘录 / EXCERPT: Let S be a finite set of finite strings. For strings x and y, let $\operatorname{ov}(x,y)$ be the maximum length of a string that is simultaneously a suffix of x and a pref...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046473 | MATHDB-383689

Bergelson's partition-regularity question for nice recurrence

摘录 / EXCERPT: A set $R\subseteq\mathbb{N}_{>0}$ is called a set of nice recurrence if, for every measure-preserving system (X,μ,T), every measurable set $A\subseteq X$, and every...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046474 | MATHDB-383717

Polynomial Freiman-Ruzsa conjecture

摘录 / EXCERPT: For every integer $n \ge 0$, every real number $K \ge 1$, and every nonempty set $A \subseteq \mathbb{F}_2^n$ satisfying $\lvert A+A\rvert \le K\lvert A\rvert$, there exist a subsp...

conjecture | partial | MathDB 页面 | 原文线索

V2-046475 | MATHDB-383761

Non-Cancelling-Intersections conjecture

摘录 / EXCERPT: For every finite set X and every finite family of subsets $A_1,\ldots,A_n\subseteq X$, let $\mathcal{I}$ be the set of distinct nonempty intersections $\bigcap_{i\in S}A_i$ for...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046476 | MATHDB-384485

Understanding properties of extremal Blaschke products (problem 2)

摘录 / EXCERPT: What are geometric properties of the sub-classes $A\in\mathbb{C}^{d\times d}$ with $\mathrm{deg}\,B_A=k$, $k=0,\ldots,d$?

open_problem | open | MathDB 页面 | 原文线索

V2-046477 | MATHDB-384486

Understanding properties of extremal Blaschke products (problem 3)

摘录 / EXCERPT: Are there relations between the Blaschke product b generating a compression S_b of the shift (Pamela Gorkin's talk) and its (extremal) Crouzeix Blaschke product B?

open_problem | open | MathDB 页面 | 原文线索

V2-046478 | MATHDB-384605

Poincaré-Wirtinger extremal domain

摘录 / EXCERPT: Prove that the equality in the Poincaré-Wirtinger inequality is attained only for infinite strips.

open_problem | open | MathDB 页面 | 原文线索

V2-046479 | MATHDB-384699

Uniformly prove ... by interpreting the right-hand side as fibers of the Lyashko-Looijenga map.

摘录 / EXCERPT: Uniformly prove ... by interpreting the right-hand side as fibers of the Lyashko-Looijenga map.

open_problem | open | MathDB 页面 | 原文线索

V2-046480 | MATHDB-384702

Why measure it? (What are the real-world applications?)

摘录 / EXCERPT: Why measure it? (What are the real-world applications?)

open_problem | open | MathDB 页面 | 原文线索

V2-046481 | MATHDB-384707

What precise notions of negative curvature imply congestion?

摘录 / EXCERPT: What precise notions of negative curvature imply congestion?

open_problem | open | MathDB 页面 | 原文线索

V2-046482 | MATHDB-384708

What are the data required to define congestion?

摘录 / EXCERPT: What are the data required to define congestion?

open_problem | open | MathDB 页面 | 原文线索

V2-046483 | MATHDB-384713

Turán functions for books (problem 1)

摘录 / EXCERPT: Determine $\operatorname{ex}(n,B_{5,2})$ and $\operatorname{ex}(n,B_{6,2})$ exactly for all large n.

open_problem | open | MathDB 页面 | 原文线索

V2-046484 | MATHDB-384714

Turán functions for books (problem 2)

摘录 / EXCERPT: ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-046485 | MATHDB-384715

Ruzsa-Szemer\'edi Theorem and Relatives (problem 1)

摘录 / EXCERPT: For any $r\ge 3$ and $s\ge 4$ we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-046486 | MATHDB-384717

Tight paths versus cliques

摘录 / EXCERPT: Improve the bounds. Is it $o(t^2)$? Perhaps the lower bound is the truth. emph{Related results:} (Phelps--R"odl) $r_3(P_2, K_t) = \Theta(t^2/\log t)$. This is the same as giving th...

open_problem | open | MathDB 页面 | 原文线索

V2-046487 | MATHDB-384720

Online Ramsey numbers

摘录 / EXCERPT: Show that $\tilde r(n,n) \ge (\sqrt 2 + \epsilon)^n$ for some $\epsilon > 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-046488 | MATHDB-384722

Local trees of triangles

摘录 / EXCERPT: For every ℓ, does there exist a graph $G \xrightarrow{2} K_3$ which on every ℓ vertices induces a subgraph of a tree of triangles. The property in the conclusion of the q...

open_problem | open | MathDB 页面 | 原文线索

V2-046489 | MATHDB-384723

Ramsey minimal graphs

摘录 / EXCERPT: textbf{Conjecture:} (Fox--Grinshpun--Liebenau--Person--Szab'o) $s_r(k) \le s_r(k+1)$ for all k. emph{Results:} $s_2(k) = (k-1)^2$ (ErdH os--Burr--Lov'asz 1976)...

open_problem | open | MathDB 页面 | 原文线索

V2-046490 | MATHDB-384724

Semi-algebraic Ramsey numbers

摘录 / EXCERPT: What is the size of the largest homogeneous set (i.e., clique or independent set) that is guaranteed to exist in every such H_n? Let...

open_problem | open | MathDB 页面 | 原文线索

V2-046491 | MATHDB-384726

Chromatic number of graphs with everywhere high independence number

摘录 / EXCERPT: Is there an $\epsilon > 0$ so that $\chi(G) \le d^{1-\epsilon}$ for every graph G of maximum degree d such that $\alpha(H) \geq \frac13 |V(H)|$ for every subgraph...

open_problem | open | MathDB 页面 | 原文线索

V2-046492 | MATHDB-384728

Local conditions for distinct distances

摘录 / EXCERPT: textbf{Question:} (ErdH{o}s 1986) What is the least possible number of distinct distances among n points in the plane if every 5 of them span at least 9 distinct distances?

open_problem | open | MathDB 页面 | 原文线索

V2-046493 | MATHDB-384729

Ramsey number of simplices

摘录 / EXCERPT: Estimate $r_k(k+1):= r_k(k+1,k+1)$. emph{Bounds:} $2^{\Omega(k)} \le r_k(k+1) \le 2^{2^{.^{.^{.^{2^C}}}}}$, where the tower of 2's has height $\approx k$. C.f. (Duffus--Lefmann--...

open_problem | open | MathDB 页面 | 原文线索

V2-046494 | MATHDB-384731

Increasing paths in edge labelings

摘录 / EXCERPT: Is it true that $\max_v t(v) \ge cn$ for some $c > 0$? Furthermore, is it true that for any graph G, edges labeled $1, 2, \dots, |E(G)|$, we always have...

open_problem | open | MathDB 页面 | 原文线索

V2-046495 | MATHDB-384733

Ordered Ramsey numbers

摘录 / EXCERPT: Does there exist $\epsilon > 0$ such that $r_<(K_3,M) = O(n^{2-\epsilon})$ for every ordered matching M on n vertices. emph{Known:}...

open_problem | open | MathDB 页面 | 原文线索

V2-046496 | MATHDB-384734

Euclidean Ramsey sets

摘录 / EXCERPT: What is the minimum number of colors needed above?

open_problem | open | MathDB 页面 | 原文线索

V2-046497 | MATHDB-384736

Koml\'{o}s conjecture

摘录 / EXCERPT: There is a universal constant $c>0$ s.t. for every $v_1,v_2,...,v_n \in \mathbb{B_2}^d$, there exist signs $\epsilon_1,...,\epsilon_n\in \{-1,1\}$ s.t....

open_problem | open | MathDB 页面 | 原文线索

V2-046498 | MATHDB-384737

Constructive Koml\'{o}s

摘录 / EXCERPT: Find a polynomial time algorithm to find the signs above.

open_problem | open | MathDB 页面 | 原文线索

V2-046499 | MATHDB-384738

Beck-Fiala conjecture and making it constructive

摘录 / EXCERPT: Given a set system $(V,\mathcal{S})$ with $V=[n]$, $\mathcal{S}=S_1,...,S_m$. Each $i\in V$ occurs in at most t sets from $\mathcal{S}$. Then there is a coloring to the elements...

open_problem | open | MathDB 页面 | 原文线索

V2-046500 | MATHDB-384739

Beck-Fiala for random set system

摘录 / EXCERPT: Each element $i\in [n]$ lies in t randomly selected sets from $\mathcal{S}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046501 | MATHDB-384742

Matrix Spencer

摘录 / EXCERPT: Given matrices $A_1,...,A_n \in\mathbb{R}^{n\times n}$ satisfying $||A_i||_{op} \le 1$ where $||.||_{op}$ is the operator norm of matrices. Do there exists signs...

open_problem | open | MathDB 页面 | 原文线索

V2-046502 | MATHDB-384743

Steinitz conjecture

摘录 / EXCERPT: Given vectors $v_1,...,v_n\in\mathbb{R}^d$ and a norm X satisfying $\sum_{i=1}^n v_i=0$ and $||v_i||_X\le 1$ for all i. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-046503 | MATHDB-384744

Vector Balancing

摘录 / EXCERPT: Is this lower bound also an upper bound for $\alpha(X,Y,d)$ upto $\log d$ factors?

open_problem | open | MathDB 页面 | 原文线索

V2-046504 | MATHDB-384749

Polysize list of colorings

摘录 / EXCERPT: Is there a list of colorings of polynomial size such that for any subset of the columns, one of these colorings gives it a discrepancy of O(h), where h is the hereditary discrepa...

open_problem | open | MathDB 页面 | 原文线索

V2-046505 | MATHDB-384750

Determine the precise locations of the roots of Eulerian Polynomials.

摘录 / EXCERPT: Determine the precise locations of the roots of Eulerian Polynomials. In particular, prove an outstanding conjecture of the asymptotic behavior of the largest root (and other roots...

open_problem | open | MathDB 页面 | 原文线索

V2-046506 | MATHDB-384751

Polynomials Arising From Combinatorics (problem 2)

摘录 / EXCERPT: Let $D(\lambda)$ be the size of the Durfee square of λ. Define $D_n(x):=\sum_{\lambda\vdash n} x^{D(\lambda)}$. Prove or disprove that $D_n(x)$ has all real roots.

open_problem | open | MathDB 页面 | 原文线索

V2-046507 | MATHDB-384752

Study unimodality of families of palindromic polynomials via alternating ... -vectors.

摘录 / EXCERPT: Study unimodality of families of palindromic polynomials via alternating γ-vectors. Some polynomials to consider are sums over partitions with distinct parts or...

open_problem | open | MathDB 页面 | 原文线索

V2-046508 | MATHDB-384753

Polynomials Arising From Combinatorics (problem 4)

摘录 / EXCERPT: Define $\displaystyle\mathcal{A}_n(t) =\sum_{\lambda\vdash n}\prod_{\Box\in\lambda}\frac{h_{\Box}^2+t}{h_{\Box}}$ where $h_\Box$ is the hook length of $\Box$. Consider the problems...

open_problem | open | MathDB 页面 | 原文线索

V2-046509 | MATHDB-384754

Polynomials Arising From Combinatorics (problem 5)

摘录 / EXCERPT: Let $C(n,a,b,m)= {\rm ConstantTerm}_{x_1,\cdots,x_n}\left(\prod_{i=1}^{n}(1-x_i)^bx_i^{-a}\prod_{{1\leq i,j\leq n} \\ }(x_i-x_j)^{-m}\right)$. Morris's identity is...

open_problem | open | MathDB 页面 | 原文线索

V2-046510 | MATHDB-384755

General Polynomials and Real Rootedness (problem 1)

摘录 / EXCERPT: Given a palindromic polynomial $\sum_{i=0}^Nc_ix^i$, consider the following conjecture: There exists a cone C such that $(c_0,c_1,\cdots,c_N)\in C$ if and only if the roots of th...

open_problem | open | MathDB 页面 | 原文线索

V2-046511 | MATHDB-384756

General Polynomials and Real Rootedness (problem 2)

摘录 / EXCERPT: Given a polynomial $ \sum_{n=0}^N a_nx^n$ where a_n is hypergeometric (i.e $\frac{a_{n+1}}{a_n}$ is a rational function in n), determine real rootedness algorithmically.

open_problem | open | MathDB 页面 | 原文线索

V2-046512 | MATHDB-384757

General Polynomials and Real Rootedness (problem 3)

摘录 / EXCERPT: Let $\sum_{i=0}^n b_ix^i$ be a real rooted polynomial with $b_i\geq 0$ for all i and let $A_i(x)$ denote the ith Eulerian polynomial. Is the polynomial $\sum_{i=0}^nb_iA_i(x)$...

open_problem | open | MathDB 页面 | 原文线索

V2-046513 | MATHDB-384758

General Polynomials and Real Rootedness (problem 4)

摘录 / EXCERPT: Let $s(q):=\prod_{j=1}^\infty\frac{(1-q^{2j})^2}{1-q^j}$. Suppose that $s(q)=\sum_{n\geq 0}s_n q^n$, is $s_n\geq 0$ for all n?

open_problem | open | MathDB 页面 | 原文线索

V2-046514 | MATHDB-384759

General Polynomials and Real Rootedness (problem 5)

摘录 / EXCERPT: Let $p(n)$ be a polynomial with positive coefficients. $p(n)=a(n)+nb(n)$ uniquely where $a(n)$ and $b(n)$ are palidromic. Do $a(n)$ and $b(n)$ have nonnegative coefficients?

open_problem | open | MathDB 页面 | 原文线索

V2-046515 | MATHDB-384760

Characterize simultaneous core partitions in terms of lecture hall partitions.

摘录 / EXCERPT: Characterize simultaneous core partitions in terms of lecture hall partitions.

open_problem | open | MathDB 页面 | 原文线索

V2-046516 | MATHDB-384761

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 2)

摘录 / EXCERPT: Consider deformations of the real braid arrangement by s-lecture hall partitions $\left(\frac{x_i}{s_i}=\frac{x_j}{s_j}\right)$. Starting with the case of $s=(1,2,\cdots,n)$, stu...

open_problem | open | MathDB 页面 | 原文线索

V2-046517 | MATHDB-384762

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 3)

摘录 / EXCERPT: Consider plane lecture hall partitions on the poset $[0,n]\times[0,m]\subset{\\mathbb{Z}}^2$ which are defined to be functions f such that ... when $x\leq y$. Are there nice combin...

open_problem | open | MathDB 页面 | 原文线索

V2-046518 | MATHDB-384763

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 4)

摘录 / EXCERPT: Consider valuations of s-lecture hall polytopes different than the usual Ehrhart polynomial.

open_problem | open | MathDB 页面 | 原文线索

V2-046519 | MATHDB-384764

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 5)

摘录 / EXCERPT: Find algebraic interpretation (e.g. Coxeter etc.) of s-lecture hall partitions.

open_problem | open | MathDB 页面 | 原文线索

V2-046520 | MATHDB-384765

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 6)

摘录 / EXCERPT: Consider principle evaluations of symmetric functions in the context of s-lecture hall partitions.

open_problem | open | MathDB 页面 | 原文线索

V2-046521 | MATHDB-384766

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 7)

摘录 / EXCERPT: What is the Hilbert basis for $L_n^{(s)}$ for general s?

open_problem | open | MathDB 页面 | 原文线索

V2-046522 | MATHDB-384767

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 8)

摘录 / EXCERPT: What are the multigraded Betti numbers $\beta_{i,m}$ of $L_n^{(s)}\cap \mathbb{Z}^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-046523 | MATHDB-384768

Lecture Hall Combinatorics: Partitions, Polytopes, and Valuations (problem 9)

摘录 / EXCERPT: The ℓ-sequences are defined by the recurrence $\{0\}_{\ell,-1}=0, \{1\}_{\ell,-1}=1$ and for $n\geq2$: ... Write $\{n\}$ short for $\{n\}_{\ell,-1}$. The Generalized Lecture Hal...

open_problem | open | MathDB 页面 | 原文线索

V2-046524 | MATHDB-384769

Investigate the volume of flow polytopes by way of geometry.

摘录 / EXCERPT: Investigate the volume of flow polytopes by way of geometry. There are many interesting known results such as ${\rm Vol}(K_{n+1}(1,0,0,\cdots,0,-1))=\prod C_i$, a product of Catala...

open_problem | open | MathDB 页面 | 原文线索

V2-046525 | MATHDB-384770

Polytopes and Lattice Point Enumeration (problem 2)

摘录 / EXCERPT: Consider the following exponential generating function identity ... where h_n denotes the complete homogeneous symmetric functions, and for a parition $\mu=(\mu_1,\cdots, \mu_j)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-046526 | MATHDB-384771

Study the quasipolynomial leading coefficients for ... -row vector partition functions.

摘录 / EXCERPT: Study the quasipolynomial leading coefficients for 2-row vector partition functions. Find efficient computational methods for finding these coefficients.

open_problem | open | MathDB 页面 | 原文线索

V2-046527 | MATHDB-384772

Vector Partition Functions and Hyperplanes (problem 2)

摘录 / EXCERPT: Let A be a finite, real, linear hyperplane arrangement with isometric chambers. Is A a Coxeter arrangment?

open_problem | open | MathDB 页面 | 原文线索

V2-046528 | MATHDB-384773

What is the easiest way to compute the pieces of a vector partition function?

摘录 / EXCERPT: What is the easiest way to compute the pieces of a vector partition function?

open_problem | open | MathDB 页面 | 原文线索

V2-046529 | MATHDB-384774

Vector Partition Functions and Hyperplanes (problem 4)

摘录 / EXCERPT: onsider the A_{n-1} vector partition function. begin{itemize} item How many pieces are there? This is known for $n\leq 7$, and there is known lower bound of...

open_problem | open | MathDB 页面 | 原文线索

V2-046530 | MATHDB-384775

Formal Products, Power Series, and Super Catalan Numbers (problem 1)

摘录 / EXCERPT: Consider the formal product ... (a) If $z=2$ then on the one hand we get Euler's ... on the other we get Pólya's formula (here the "cycle index decomposition", see Stanley's EC2, p.19)...

open_problem | open | MathDB 页面 | 原文线索

V2-046531 | MATHDB-384776

Formal Products, Power Series, and Super Catalan Numbers (problem 2)

摘录 / EXCERPT: Let $(q;q)_n=(1-q)(1-q^2)\cdots(1-q^n)$ with $(q;q)_0:=1$. Define a q-exponential by ... There is a notion of q-Eulerian polynomials. I like to introduce it q-Eulerian polynomi...

open_problem | open | MathDB 页面 | 原文线索

V2-046532 | MATHDB-384777

Formal Products, Power Series, and Super Catalan Numbers (problem 3)

摘录 / EXCERPT: Let S_n be the permutation group and fix $k\in\mathbb{N}$. Then, ...

open_problem | open | MathDB 页面 | 原文线索

V2-046533 | MATHDB-384778

Formal Products, Power Series, and Super Catalan Numbers (problem 4)

摘录 / EXCERPT: Ira Gessel "dubbed" the name mathit{ super Catalan} for the sequence ... and offers an algebraic proof. Not combinatorial! textbf{ Note.} The numbers $\frac{1}{2}S(m,n)$ are also int...

open_problem | open | MathDB 页面 | 原文线索

V2-046534 | MATHDB-384779

Problem List (problem 1)

摘录 / EXCERPT: If G is a graph with n vertices, m edges and diameter d are there functions $f(n,m,d)$ and $g(n,m,d)$ that can serve as lower and upper bounds, respectively, for $Z(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046535 | MATHDB-384780

Can we find asymptotic results based on density, ... ?

摘录 / EXCERPT: Can we find asymptotic results based on density, $\delta = \dfrac{|E|}{ {{n}\choose{2}} }$? See Butler and Young paper (https://ajc.maths.uq.edu.au/pdf/57/ajcv57p065.pdf). Note:...

open_problem | open | MathDB 页面 | 原文线索

V2-046536 | MATHDB-384781

Does ... have a better' combinatorial version, maybe based on inertia?

摘录 / EXCERPT: Does Z_q have a better' combinatorial version, maybe based on inertia? C.C.R. for Z_q: This is a two player game and player 1 is trying to minimize how much money is spent an...

open_problem | open | MathDB 页面 | 原文线索

V2-046537 | MATHDB-384782

Problem List (problem 4)

摘录 / EXCERPT: For $Z_{\infty}(G) = Z(G)$ we know path cover is an bound, for $Z_0(G) = Z_+(G)$ we know tree cover number is a bound for PSD zero forcing, are there related bounds for $Z_1(G)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-046538 | MATHDB-384783

Problem List (problem 5)

摘录 / EXCERPT: Can we characterize graphs for which $Z_1(G) = 0,1,2$ and/or $Z_1(G) = |G|, |G| -1, |G| - 2$.

open_problem | open | MathDB 页面 | 原文线索

V2-046539 | MATHDB-384784

Can we design a zero forcing number/color change rule for colin de Verdiere type parameter ... , i.e.

摘录 / EXCERPT: Can we design a zero forcing number/color change rule for colin de Verdiere type parameter $\mu(G)$, i.e. can we design $Z_{\mu}(G) \ge \mu(G)$.

open_problem | open | MathDB 页面 | 原文线索

V2-046540 | MATHDB-384785

Is there a practical reduction of zero forcing to some known, well studied, NP-problem?

摘录 / EXCERPT: Is there a practical reduction of zero forcing to some known, well studied, NP-problem?

open_problem | open | MathDB 页面 | 原文线索

V2-046541 | MATHDB-384786

Problem List (problem 8)

摘录 / EXCERPT: Is there a good way to compute $Z(G)$ for G planar or outer planar? (See related results for power domination: Jiong Guo, Rolf Biedermeier, 2007, Improved Algorithms and complexi...

open_problem | open | MathDB 页面 | 原文线索

V2-046542 | MATHDB-384787

Is the minor-monotone floor version of zero forcing easy (polynomial time) to compute for planar graphs?

摘录 / EXCERPT: Is the minor-monotone floor version of zero forcing easy (polynomial time) to compute for planar graphs? Probably NP-hard by tree-width relationship.

open_problem | open | MathDB 页面 | 原文线索

V2-046543 | MATHDB-384788

Problem List (problem 10)

摘录 / EXCERPT: Given $\text{Aut}(G)$, what are the number of non-isomorphic zero forcing sets. In particular, what about circulates? This can have implications for propagation time because it is...

open_problem | open | MathDB 页面 | 原文线索

V2-046544 | MATHDB-384789

Problem List (problem 11)

摘录 / EXCERPT: If G is bipartite, $V = X \cup Y$, $|X| \ge |Y|$ and for all $v\in X$ $\text{deg}(v) = 3$ is $\text{M}(G) = \text{Z}(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046545 | MATHDB-384790

Problem List (problem 12)

摘录 / EXCERPT: What about approximate zero forcing? This is a question about quantum mechanical systems where typically neighboring quantum particles interact strongly with each other and quantum...

open_problem | open | MathDB 页面 | 原文线索

V2-046546 | MATHDB-384791

What can we say about probabalistic zero forcing, i.e.

摘录 / EXCERPT: What can we say about probabalistic zero forcing, i.e. edges have weights and if standard zero forcing says we can force we do so with probability of the edge weight.

open_problem | open | MathDB 页面 | 原文线索

V2-046547 | MATHDB-384792

Problem List (problem 14)

摘录 / EXCERPT: For positive semidefinite (PSD) zero forcing consider the minimum PSD propagation time $\text{pt}_+(G):= \min\{\text{propagation times given a minimum PSD zero forcing set}\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-046548 | MATHDB-384793

Investigate the convexity of zero forcing sets by creating a simplicial complex based on the complements of minimal zero forcing sets.

摘录 / EXCERPT: Investigate the convexity of zero forcing sets by creating a simplicial complex based on the complements of minimal zero forcing sets. Can the geometry of the complex tell us anyth...

open_problem | open | MathDB 页面 | 原文线索

V2-046549 | MATHDB-384794

Problem List (problem 16)

摘录 / EXCERPT: Can we find $G_1, G_2, G_3,\dots $ where $\displaystyle\lim_{i\to \infty} \dfrac{|Z(G_1)|}{|G_i|} = 0$ and $Z(G_i) \ge \dfrac{n}{\log(n)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046550 | MATHDB-384795

Throttling is concerned about the sum of the size of a zero forcing set and the associated propagation time, i.e.

摘录 / EXCERPT: Throttling is concerned about the sum of the size of a zero forcing set and the associated propagation time, i.e. we are willing to not use the minimum zero forcing number if it de...

open_problem | open | MathDB 页面 | 原文线索

V2-046551 | MATHDB-384796

Problem List (problem 18)

摘录 / EXCERPT: What about variations of throttling, like the product of $|B|$ and $\text{pt}(G,B)$? Note that ℓ-round power domination seems like it is similar to throttling.

open_problem | open | MathDB 页面 | 原文线索

V2-046552 | MATHDB-384797

Problem List (problem 19)

摘录 / EXCERPT: Can we beat the wavefront algorithm for computing $\text{Z}(G)$, power domination is around $(1.8)^n$, which is better than brute force.

open_problem | open | MathDB 页面 | 原文线索

V2-046553 | MATHDB-384799

Develop theory of partial zero forcing/unique paths that Bryan Shader talked about.

摘录 / EXCERPT: Develop theory of partial zero forcing/unique paths that Bryan Shader talked about.

open_problem | open | MathDB 页面 | 原文线索

V2-046554 | MATHDB-384800

Problem List (problem 22)

摘录 / EXCERPT: Given a set S, what is the smallest zero forcing set that contains S (restrained zero forcing). Similarly, given a set S what is the smallest zero forcing set that is contain...

open_problem | open | MathDB 页面 | 原文线索

V2-046555 | MATHDB-384802

Find all graphs where all minimal zero forcing sets are also minimum zero forcing sets.

摘录 / EXCERPT: Find all graphs where all minimal zero forcing sets are also minimum zero forcing sets. (Greedoids?)

open_problem | open | MathDB 页面 | 原文线索

V2-046556 | MATHDB-384803

Can we use the geometry of the simplicial complex generated by complements of zero forcing sets to answer questions about zero forcing or propagation time?

摘录 / EXCERPT: Can we use the geometry of the simplicial complex generated by complements of zero forcing sets to answer questions about zero forcing or propagation time?

open_problem | open | MathDB 页面 | 原文线索

V2-046557 | MATHDB-384804

Establish relationship between zero forcing sets and matroids.

摘录 / EXCERPT: Establish relationship between zero forcing sets and matroids.

open_problem | open | MathDB 页面 | 原文线索

V2-046558 | MATHDB-384805

Establish relationship between zero forcing sets and Hodge theory.

摘录 / EXCERPT: Establish relationship between zero forcing sets and Hodge theory.

open_problem | open | MathDB 页面 | 原文线索

V2-046559 | MATHDB-384807

What is the generalization to zero forcing with multiple colors?

摘录 / EXCERPT: What is the generalization to zero forcing with multiple colors? Maybe the colors are linearly ordered, i.e. red can force purple, purple can force blue, etc. Maybe the graph has a...

open_problem | open | MathDB 页面 | 原文线索

V2-046560 | MATHDB-384809

Problem List (problem 31)

摘录 / EXCERPT: k-forcing edge weighted: let $w(u,v)$ be the weight of edge uv. let u be filled, if $\displaystyle \sum_{v \text{unfilled neighbor of u}} w(u,v) \le k$ then u forces all un...

open_problem | open | MathDB 页面 | 原文线索

V2-046561 | MATHDB-384810

Can we use art gallery problem results/techniques to help answer zero forcing questions and power domination questions.

摘录 / EXCERPT: Can we use art gallery problem results/techniques to help answer zero forcing questions and power domination questions. In particular, what about near triangulation graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-046562 | MATHDB-384811

Is there a natural relaxation for fort and forcing chain IPs?

摘录 / EXCERPT: Is there a natural relaxation for fort and forcing chain IPs?

open_problem | open | MathDB 页面 | 原文线索

V2-046563 | MATHDB-384812

Can Daniel Burgarth's ideas relating the zero forcing number and controllability be extended to PSD matrices?

摘录 / EXCERPT: Can Daniel Burgarth's ideas relating the zero forcing number and controllability be extended to PSD matrices?

open_problem | open | MathDB 页面 | 原文线索

V2-046564 | MATHDB-384813

We can use zero forcing to create and' and or' gates, can we extend this to model cellular automata evolution?

摘录 / EXCERPT: We can use zero forcing to create and' and or' gates, can we extend this to model cellular automata evolution?

open_problem | open | MathDB 页面 | 原文线索

V2-046565 | MATHDB-384814

Problem List (problem 36)

摘录 / EXCERPT: Does Z_q equal M_q for caterpillar graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-046566 | MATHDB-384815

Problem List (problem 37)

摘录 / EXCERPT: If $a_0,a_1,a_2,\dots$ is a sequence of integers and $a_0 = 1 \le a_1 \le a_2 \le \dots$ can we find a caterpillar graph C such that $Z_0(C) = a_0$, $Z_1(C) = a_1$,...?

open_problem | open | MathDB 页面 | 原文线索

V2-046567 | MATHDB-384817

Triangle closure

摘录 / EXCERPT: Let $G \sim G_{n,p}$ with $p = \Theta(1)/n$. For each pair of vertices at distance 2, add an edge between them with probability q. Call the resulting graph H. Is it possible to...

open_problem | open | MathDB 页面 | 原文线索

V2-046568 | MATHDB-384818

Graph localization

摘录 / EXCERPT: The following graph localization problem is a non-mean field model, which means that many standard tools from statistical physics do not apply. Let $R \subseteq \mathbb{R}^d$ and l...

open_problem | open | MathDB 页面 | 原文线索

V2-046569 | MATHDB-384821

Node vs.\ edge information

摘录 / EXCERPT: Some real-world graphs (e.g. facebook) have information both on the nodes and on the edges. Can you quantify how much information is contained in the nodes versus in the edges? An...

open_problem | open | MathDB 页面 | 原文线索

V2-046570 | MATHDB-384822

Can local changes shift the threshold?

摘录 / EXCERPT: Let $G \sim \mathrm{SBM}$. Now let G be changed by local rules. For what types of rules does this change the threshold?

open_problem | open | MathDB 页面 | 原文线索

V2-046571 | MATHDB-384823

ML vs.\ Bayes for spiked tensors

摘录 / EXCERPT: Let x_0 be a uniformly random unit vector and observe $Y = \lambda x_0^{\otimes d} + W$ where W is a symmetric tensor with $W_{i_1,\ldots,i_d} \sim \mathcal{N}(0,1/n)$ independ...

open_problem | open | MathDB 页面 | 原文线索

V2-046572 | MATHDB-384824

ML vs.\ Bayes for SBM

摘录 / EXCERPT: In the SBM and related models, are there regimes in which maximum-likelihood reconstruction does not achieve the optimal threshold?

open_problem | open | MathDB 页面 | 原文线索

V2-046573 | MATHDB-384825

Separation in trees and graphs

摘录 / EXCERPT: There are many connections between the SBM and the broadcast model in trees (which describes local neighborhoods). Are there questions (e.g. threshold for ML) for which the answer...

open_problem | open | MathDB 页面 | 原文线索

V2-046574 | MATHDB-384826

Suboptimality of SDP

摘录 / EXCERPT: Are there regimes (even in the 2-block symmetric SBM) in which the SDP does not achieve the information-theoretic threshold?

open_problem | open | MathDB 页面 | 原文线索

V2-046575 | MATHDB-384827

In what sense is the ``hard'' regime hard?

摘录 / EXCERPT: Can we formalize in what sense we believe that conjectured “hard” regions are hard? There are some caveats here because, for instance, statistical physics and the sum-of-squares hi...

open_problem | open | MathDB 页面 | 原文线索

V2-046576 | MATHDB-384828

Refuting colorability with SoS

摘录 / EXCERPT: Can constant-degree sum-of-squares refute k-colorability for random graphs of average degree less than k^2 (i.e. below the Kesten-Stigum bound)?

open_problem | open | MathDB 页面 | 原文线索

V2-046577 | MATHDB-384829

Bounding independent set size with SoS

摘录 / EXCERPT: For a random graph of average degree d, what upper bound can constant-degree sum-of-squares prove on the size of an independent set. It is conjectured that the answer is...

open_problem | open | MathDB 页面 | 原文线索

V2-046578 | MATHDB-384830

Failure of BP

摘录 / EXCERPT: Show that BP fails on the SBM below the KS bound.

open_problem | open | MathDB 页面 | 原文线索

V2-046579 | MATHDB-384831

Failure of MC

摘录 / EXCERPT: Show that Monte-Carlo fails on the SBM below the KS bound. This is related to the quantity $Z(\alpha) = \sum_{\tau: \alpha(\sigma,\tau) \approx \alpha} P(G | \tau)$. We want to und...

open_problem | open | MathDB 页面 | 原文线索

V2-046580 | MATHDB-384832

Analyze BP

摘录 / EXCERPT: For the symmetric 2-groups SBM, show that BP (with random initialization) achieves ε correlation with the truth, above the KS bound.

open_problem | open | MathDB 页面 | 原文线索

V2-046581 | MATHDB-384833

BP with good initialization

摘录 / EXCERPT: Run BP with a good' initialization (e.g. using a spectral method based on the non-backtracking walk matrix). Show that it achieves the optimal overlap with the truth.

open_problem | open | MathDB 页面 | 原文线索

V2-046582 | MATHDB-384834

Analyze Gibbs sampling

摘录 / EXCERPT: For the symmetric 2-groups SBM, show that Gibbs sampling achieves ε correlation with the truth, above the KS bound.

open_problem | open | MathDB 页面 | 原文线索

V2-046583 | MATHDB-384835

Ramanujan graphs

摘录 / EXCERPT: Let G be Chung-Lu with power law degree distribution. Is G “Ramanujan”? More precisely, we expect that the non-backtracking operator B has one eigenvalue $\tilde d$ and all o...

open_problem | open | MathDB 页面 | 原文线索

V2-046584 | MATHDB-384836

Non-backtracking operator

摘录 / EXCERPT: Take the SBM with k communities of equal size. Take a random second-largest eigenvector $e_2 \in \mathbb{R}^{2E}$ of the non-backtracking operator B. Pass to a vector...

open_problem | open | MathDB 页面 | 原文线索

V2-046585 | MATHDB-384837

2-lift of Ramanujan graph

摘录 / EXCERPT: Is a random 2-lift of a Ramanujan graph also Ramanujan (or nearly) with constant probability? More precisely, let G be a d-regular Ramanujan graph with adjacency matrix A. Co...

open_problem | open | MathDB 页面 | 原文线索

V2-046586 | MATHDB-384838

Alon-Boppana for non-regular graphs

摘录 / EXCERPT: A proposed definition of Ramanujan for non-regular graphs is that the non-backtracking matrix B has spectrum satisfying ... For this definition, prove the result analagous to Alon-...

open_problem | open | MathDB 页面 | 原文线索

V2-046587 | MATHDB-384839

Construction of non-regular Ramanujan graphs

摘录 / EXCERPT: What are some constructions of non-regular Ramanujan graphs (using the definition of Ramanujan from the previous problem)?

open_problem | open | MathDB 页面 | 原文线索

V2-046588 | MATHDB-384840

Rudin's problem for $\mathbb{F}_p[x], p>2$

摘录 / EXCERPT: Suppose $A\subset \mathbb{F}_p[x]$ is of finite size, and every element of A is a square. Show that $|A+A|\geq |A|^{1+\epsilon}$ for some constant ϵ.

open_problem | open | MathDB 页面 | 原文线索

V2-046589 | MATHDB-384841

3APs in sets of small growth

摘录 / EXCERPT: Obtain a proof of the following theorem using the polynomial method. Theorem. There is an absolute constant $c>0$ that the following holds. If $A\subset \mathbb{F}_p^n$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-046590 | MATHDB-384842

Polynomial method via tensorizing

摘录 / EXCERPT: Let $\mathbf{r}$ be either $arank$ or $prank$. Let A be a tensor, and Id is the identity tensor on the same space as A. Suppose $\mathbf{r}(A)< \mathbf{r}(Id)$. Is it true th...

open_problem | open | MathDB 页面 | 原文线索

V2-046591 | MATHDB-384843

Arithmetic removal lemma for cyclic groups

摘录 / EXCERPT: Obtain improved bound for arithmetic regularity lemma for cyclic groups.

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V2-046592 | MATHDB-384844

A polynomial method proof of Frankl-Rodl theorem

摘录 / EXCERPT: Find a proof of Frankl-Rodl theorem using the polynomial method, in particular Croot-Lev-Pach method.

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V2-046593 | MATHDB-384845

Doubling of generic set of points

摘录 / EXCERPT: Let $A\subset \mathbb{R}^3$ be a finite set with no five points on a plane. What is the best inequality ... that we can have?

open_problem | open | MathDB 页面 | 原文线索

V2-046594 | MATHDB-384846

A stronger BSG theorem

摘录 / EXCERPT: Let $A\subset G$, and suppose $E(A)\geq \frac{|A|^3}{K}$. Then there is a set $A'\subset A$ so that ... ... Can we moreover obtain the following? ...

open_problem | open | MathDB 页面 | 原文线索

V2-046595 | MATHDB-384847

Finding examples.

摘录 / EXCERPT: Find counter examples to the polynomial Bogolyubov Ruzsa theorem. Find interesting examples of subset $A\subset \mathbb{F}_2^n$ so that $A+A$ is far from all of $\mathbb{F}_2^n$. F...

open_problem | open | MathDB 页面 | 原文线索

V2-046596 | MATHDB-384848

Approximate Caratheodory theorem

摘录 / EXCERPT: Q. First, assuming some assumptions on the point set A, find a better dependence (sub-linear) on p. Q. Characterize configurations for which the bound $O(p\varepsilon^{-2})$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-046597 | MATHDB-384849

Finding good modeling

摘录 / EXCERPT: Suppose $A\subset G$, and $|A+A|\leq K|A|$. Can we find a subset $A'\subset A$ with $|A'|\geq K^{-O(1)}|A|$ and a group H so that A' is Freiman isomorphic to a subset of H a...

open_problem | open | MathDB 页面 | 原文线索

V2-046598 | MATHDB-384850

Product sets in iterated sumsets

摘录 / EXCERPT: There exists n,ε,δ for which the following holds. Let $A\subset \mathbb{F}_p$ and suppose $|A+A|\leq |A|^{1+\varepsilon}$. Then $nA-nA$ contains a product set XY...

open_problem | open | MathDB 页面 | 原文线索

V2-046599 | MATHDB-384851

Entropy of sum of random variables.

摘录 / EXCERPT: Show ...

open_problem | open | MathDB 页面 | 原文线索

V2-046600 | MATHDB-384854

Finding structure inside $A+A$

摘录 / EXCERPT: Let $A\subset \mathbb{F}_2^n$ be a subset of density 0.01. Then there is a subset $B\subset \mathbb{F}_2^n$ with density absolute constant, so that ... and ... for some sets S_i's...

open_problem | open | MathDB 页面 | 原文线索

V2-046601 | MATHDB-384855

A conjecture about generalized additive energy

摘录 / EXCERPT: Suppose ... . Then there is $H\subset (n-1)A, |H|\geq \frac{\rho_{2n}^{\frac{1}{n-1}}}{K^c}$ so that ...

open_problem | open | MathDB 页面 | 原文线索

V2-046602 | MATHDB-384858

Nonbacktracking Matrix

摘录 / EXCERPT: Let G be any graph on n vertices and m edges, and let B_G denote the emph{nonbacktracking matrix} of G, which is a $2m\times 2m$ matrix whose rows and columns are indexed...

open_problem | open | MathDB 页面 | 原文线索

V2-046603 | MATHDB-384859

Positive and negative ℓ_2-mass on spectrum

摘录 / EXCERPT: Let G be a connected graph and let $\lambda_1\ge\dots\ge\lambda_s\ge 0>\lambda_{s+1}\ge\dots\ge\lambda_n$ be the eigenvalues of the adjacency matrix of G. Defining: ... Some conj...

open_problem | open | MathDB 页面 | 原文线索

V2-046604 | MATHDB-384860

Ehrhart Positivity (problem 1)

摘录 / EXCERPT: Let $[n]:=\{1,\ldots,n\}$ be a groundset, and let $i(P; t)$ denote the Ehrhart polynomial of a lattice polytope P. Then coefficient-wise, ... where $T_{r,n}$ is the rank r minima...

open_problem | open | MathDB 页面 | 原文线索

V2-046605 | MATHDB-384861

What is known (or can be shown) about the Ehrhart positivity of polytopes related to Birkhoff polytopes (e.g., transportation polytopes)?

摘录 / EXCERPT: What is known (or can be shown) about the Ehrhart positivity of polytopes related to Birkhoff polytopes (e.g., transportation polytopes)?

open_problem | open | MathDB 页面 | 原文线索

V2-046606 | MATHDB-384862

Ehrhart Positivity (problem 3)

摘录 / EXCERPT: Are d-dimensional smooth polytopes with $\leq d+3$ facets Ehrhart positive or satisfy the higher integrality condition?

open_problem | open | MathDB 页面 | 原文线索

V2-046607 | MATHDB-384863

Ehrhart Positivity (problem 4)

摘录 / EXCERPT: begin{enumerate} item[(a)]Is there a closed formula for the Ehrhart polynomial of order polytopes of fence posets? Further, are these Ehrhart polynomials Ehrhart positive? item[(b)...

open_problem | open | MathDB 页面 | 原文线索

V2-046608 | MATHDB-384864

Does a rational polygon ... with quasi-period 1 satisfy Scott's inequality?

摘录 / EXCERPT: Does a rational polygon $P\subseteq {\\mathbb{R}}^2$ with quasi-period 1 satisfy Scott's inequality? That is, assuming $i\geq 1$, is the inequality $b\leq 2i+7$ satisfied, where...

open_problem | open | MathDB 页面 | 原文线索

V2-046609 | MATHDB-384865

Ehrhart Positivity (problem 6)

摘录 / EXCERPT: Let ASM_n denote the nth alternating sign matrix polytope (defined as the convex hull in $\mathbb{R}^{n^2}$ of the $n\times n$ alternating sign matrices). It is known that...

open_problem | open | MathDB 页面 | 原文线索

V2-046610 | MATHDB-384867

In types B, C, and D, the stretched Clebsch-Gordon coefficients are period-2 quasi-polynomials.

摘录 / EXCERPT: In types B, C, and D, the stretched Clebsch-Gordon coefficients are period-2 quasi-polynomials. Are the coefficients of the constituent polynomials always positive?

open_problem | open | MathDB 页面 | 原文线索

V2-046611 | MATHDB-384868

Ehrhart Positivity (problem 9)

摘录 / EXCERPT: Consider rational polygons of denominator 2. The Ehrhart quasi-polynomial has two constituent polynomials, one of which is an Ehrhart polynomial. Classify the other constituent pol...

open_problem | open | MathDB 页面 | 原文线索

V2-046612 | MATHDB-384869

Ehrhart Positivity (problem 10)

摘录 / EXCERPT: Characterize Ehrhart polynomials of tropical polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-046613 | MATHDB-384870

Study the Ehrhart polynomial of Harmonic polytopes and/or bipermutahedra.

摘录 / EXCERPT: Study the Ehrhart polynomial of Harmonic polytopes and/or bipermutahedra.

open_problem | open | MathDB 页面 | 原文线索

V2-046614 | MATHDB-384871

Ehrhart Positivity (problem 12)

摘录 / EXCERPT: begin{enumerate} item[(a)]Consider the Ehrhart polynomial of a lattice polytope P. Below are two well-known representations of $i(P;t)$: ... Fix $k\in {\\mathbb{Z}}_{>0}$ and write...

open_problem | open | MathDB 页面 | 原文线索

V2-046615 | MATHDB-384872

Does the Pitman-Stanley polytope have real-rooted ... -polynomial?

摘录 / EXCERPT: Does the Pitman-Stanley polytope have real-rooted h^*-polynomial?

open_problem | open | MathDB 页面 | 原文线索

V2-046616 | MATHDB-384873

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 2)

摘录 / EXCERPT: Study the h^*-polynomials of (not necessarily convex) 3-polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-046617 | MATHDB-384874

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 3)

摘录 / EXCERPT: For a lattice polytope P, let P^n denote the Cartesian product of P with itself n times ($n\in {\\mathbb{N}}$). Investigate what happens to the h^*-polynomial of P^n. D...

open_problem | open | MathDB 页面 | 原文线索

V2-046618 | MATHDB-384875

For a symmetric edge polytope, can we obtain regular unimodular triangulations using liftings with low heights?

摘录 / EXCERPT: For a symmetric edge polytope, can we obtain regular unimodular triangulations using liftings with low heights? If so, how many such triangulations can be obtained?

open_problem | open | MathDB 页面 | 原文线索

V2-046619 | MATHDB-384876

What are the ways of “measuring” root polytopes, i.e., edge polytopes?

摘录 / EXCERPT: What are the ways of “measuring” root polytopes, i.e., edge polytopes? Furthermore, how should Grothendieck polynomials be specialized to obtain Ehrhart h^*-polynomials?

open_problem | open | MathDB 页面 | 原文线索

V2-046620 | MATHDB-384877

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 6)

摘录 / EXCERPT: Consider the order polynomial, denoted $\Omega(P,t)$, of a finite poset P. Let n be the number of elements in P. textbf{Conjecture} [Kahn, Saks] $\frac{\Omega(P,t)}{t^n}$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-046621 | MATHDB-384878

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 7)

摘录 / EXCERPT: Note that $i(P;1;n) = i(P;n)$. It is known that $i(P;\omega;n)$ is a polynomial with leading coefficient given by $\int_P \omega$. Investigate ω-Ehrhart function for well-kn...

open_problem | open | MathDB 页面 | 原文线索

V2-046622 | MATHDB-384879

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 8)

摘录 / EXCERPT: Identify polytopes with extremal h^*-vectors. That is, for a family of polytopes $\mathcal{P}$ and index j, find $P\in \mathcal{P}$ that minimize or maximize h_j^*. Potential...

open_problem | open | MathDB 页面 | 原文线索

V2-046623 | MATHDB-384880

What relationship, if any, is there between heights of Hilbert basis elements and differences in ... -coefficients?

摘录 / EXCERPT: What relationship, if any, is there between heights of Hilbert basis elements and differences in h^*-coefficients? For example, consider $h^*_{j+1} - h^*_j$ or more generally...

open_problem | open | MathDB 页面 | 原文线索

V2-046624 | MATHDB-384881

Take the nonnegative hull of ... -vectors of polytopes in a fixed dimension.

摘录 / EXCERPT: Take the nonnegative hull of h^*-vectors of polytopes in a fixed dimension. Does this form a polyhedral cone?

open_problem | open | MathDB 页面 | 原文线索

V2-046625 | MATHDB-384882

Are there “holes” in the set of ... -vectors of polytopes in a fixed dimension which are not a result of failing known linear inequalities?

摘录 / EXCERPT: Are there “holes” in the set of h^*-vectors of polytopes in a fixed dimension which are not a result of failing known linear inequalities?

open_problem | open | MathDB 页面 | 原文线索

V2-046626 | MATHDB-384883

How many facets do symmetric edge polytopes have?

摘录 / EXCERPT: How many facets do symmetric edge polytopes have?

open_problem | open | MathDB 页面 | 原文线索

V2-046627 | MATHDB-384884

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 13)

摘录 / EXCERPT: Consider the convex hull of parking functions. Does this admit a unimodular triangulation? Furthermore, can the Ehrhart polynomial be obtained from this?

open_problem | open | MathDB 页面 | 原文线索

V2-046628 | MATHDB-384885

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 14)

摘录 / EXCERPT: begin{enumerate} item[(a)] Find $0/1$-polytopes with non-unimodal, non-log-concave, etc. h^*-vectors. item[(b)] Find $0/1$-polytopes with non-unimodal, non-log-concave, etc. f-...

open_problem | open | MathDB 页面 | 原文线索

V2-046629 | MATHDB-384886

Determine a Veronese construction for rational polytopes.

摘录 / EXCERPT: Determine a Veronese construction for rational polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-046630 | MATHDB-384887

How many chambers are there for transportation polytopes for ... matrices?

摘录 / EXCERPT: How many chambers are there for transportation polytopes for $n\times n$ matrices? For example, how many Ehrhart polynomials are needed?

open_problem | open | MathDB 页面 | 原文线索

V2-046631 | MATHDB-384888

Identify classes of simplices with unimodal box polynomials.

摘录 / EXCERPT: Identify classes of simplices with unimodal box polynomials.

open_problem | open | MathDB 页面 | 原文线索

V2-046632 | MATHDB-384889

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 18)

摘录 / EXCERPT: Investigate a,b-decompositions for “nice” h^*-polynomials.

open_problem | open | MathDB 页面 | 原文线索

V2-046633 | MATHDB-384890

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 19)

摘录 / EXCERPT: Investigate Stapledon's Effectiveness Conjecture. textbf{Conjecture.} [Stapledon] Let P be a lattice polytope invariant under the action of a group G. Then the characters of th...

open_problem | open | MathDB 页面 | 原文线索

V2-046634 | MATHDB-384892

Equivariant Log-Concavity of Stirling Numbers of the First Kind

摘录 / EXCERPT: The sequence of Stirling numbers of the first kind count the number of permutations in S_n decomposing into k disjoint cycles. Precisely, ... We say a sequence $\{a_i\}$ of nonne...

open_problem | open | MathDB 页面 | 原文线索

V2-046635 | MATHDB-384893

Ehrhardt Volume of the Normal Complex

摘录 / EXCERPT: Given a matroid M, one can associate to it a certain polytopal complex called the normal complex of M. One can ask whether the normal complex of M admits an analogue of the E...

open_problem | open | MathDB 页面 | 原文线索

V2-046636 | MATHDB-384894

Distribution of Intersection Numbers of Undirected Graphs

摘录 / EXCERPT: Given an undirected graph G, a clique of G is an induced subgraph of G that is complete (that is, any two distinct vetices are adjacent). An edge clique cover is a set of cli...

open_problem | open | MathDB 页面 | 原文线索

V2-046637 | MATHDB-384895

Ehrhart Theory of Lattice Path and Panhandle Matroids

摘录 / EXCERPT: Given a panhandle matroid that can be realized as a lattice path matroid on a $n \times k$ grid, is it true that the coeffcients of the Ehrhart polynomial are bounded above by the...

open_problem | open | MathDB 页面 | 原文线索

V2-046638 | MATHDB-384898

Taking Intent to Action

摘录 / EXCERPT: How can we go from ideas/intention to action with our DEI work? In particular, can we create a list of ways to take action and/or find literature on best practices for taking inten...

open_problem | open | MathDB 页面 | 原文线索

V2-046639 | MATHDB-384899

The History of Women in Combinatorics

摘录 / EXCERPT: What is the history of women in combinatorics? Answering this question might involve researchers from many different areas, including historians, sociologists, mathematicians, and...

open_problem | open | MathDB 页面 | 原文线索

V2-046640 | MATHDB-384900

Starting DEI Conversations

摘录 / EXCERPT: How can we start DEI conversations in departments outside the US?

open_problem | open | MathDB 页面 | 原文线索

V2-046641 | MATHDB-384901

Improving Conference Organization/Resources for Organizers

摘录 / EXCERPT: What might it look like to make our events (conferences, workshops, etc.) accessible to all participants? How can we make it accessible to parent participants, trans participants,...

open_problem | open | MathDB 页面 | 原文线索

V2-046642 | MATHDB-384902

Less Actionable But Nonetheless Important Issues

摘录 / EXCERPT: How do we deal with the structural problem of expecting mathematicians to move every 2 years for post-docs? How can we work to change the gender pay gap? How can we financially sup...

open_problem | open | MathDB 页面 | 原文线索

V2-046643 | MATHDB-384903

Gender Diversity Beyond the Gender Binary

摘录 / EXCERPT: How can we address and work towards gender diversity outside of the gender binary? Can "Women in Math" organizations clarify what they mean by "women?" And can we push for greater...

open_problem | open | MathDB 页面 | 原文线索

V2-046644 | MATHDB-384904

How Does the Gender Binary Influence the Math we Create?

摘录 / EXCERPT: How does the gender binary (and more generally how we perceive the world) influence the math we create? (not just how we teach and share it)

open_problem | open | MathDB 页面 | 原文线索

V2-046645 | MATHDB-384905

Entropy Freiman-Ruzsa lower bounds

摘录 / EXCERPT: Can we give lower bounds for $d[X,U_H]$?

open_problem | open | MathDB 页面 | 原文线索

V2-046646 | MATHDB-384906

Entropy proof of Sidorenko

摘录 / EXCERPT: Define a proof system for graph homomorphism inequalities using entropy.

open_problem | open | MathDB 页面 | 原文线索

V2-046647 | MATHDB-384907

Inverse theorems with polynomial bounds

摘录 / EXCERPT: Is there a direct entropic proof of the above results with polynomial bounds?

open_problem | open | MathDB 页面 | 原文线索

V2-046648 | MATHDB-384910

Roth with constrained differences

摘录 / EXCERPT: For which $0\in D\subseteq \mathbb{F}_p$ is it true that any $A\subset \mathbb{F}_p^n$ avoiding ... has size $\leq C^n$ for some $C < p$?

open_problem | open | MathDB 页面 | 原文线索

V2-046649 | MATHDB-384911

Inverse theorem for equilateral triangles on the sphere

摘录 / EXCERPT: If f is 1-bounded and $|Tf|>\delta$, does f correlate with a function that depends on $O_{\delta}(1)$ coordinates (with respect to some basis)?

open_problem | open | MathDB 页面 | 原文线索

V2-046650 | MATHDB-384912

Easier arithmetic Kakeya

摘录 / EXCERPT: Let $S\subset\mathbb{Z}^n$ be such that for all $(d_1,\ldots,d_r)\in ([N]^n)^r$, S contains the GAP (for some translate a_0) ... Then $|S|\geq N^{c_kn+o_{k\to\infty}(1)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046651 | MATHDB-384913

Popular Furstenberg-Sárközy

摘录 / EXCERPT: (1) Prove lower bounds of the form $N_0(\epsilon)>superpoly(1/\epsilon)$, or even just $N_0(\epsilon)>1/\epsilon^{10}$. (2) What happens in the function field case...

open_problem | open | MathDB 页面 | 原文线索

V2-046652 | MATHDB-384915

Expansion of non-Sidorenko graphs

摘录 / EXCERPT: Does there exist a non-Sidorenko (hyper)graph H whose r-expansion is Sidorenko?

open_problem | open | MathDB 页面 | 原文线索

V2-046653 | MATHDB-384916

Induced packing and covering

摘录 / EXCERPT: Can we get better bounds, even for the case $H=C_5$?

open_problem | open | MathDB 页面 | 原文线索

V2-046654 | MATHDB-384918

Equi-isoclinic subspaces

摘录 / EXCERPT: Let $N_\alpha^k(d)$ be the maximum number of k-dimensional subspaces of $\mathbb{R}^d$, every pair of which is equi-isoclinic with angle α. Then (1)...

open_problem | open | MathDB 页面 | 原文线索

V2-046655 | MATHDB-384920

Kupitz conjecture in three dimensions

摘录 / EXCERPT: Given n points in $\mathbb{R}^3$, is there always a plane through at least two points, such that the number of points on each side differ by $O(1)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046656 | MATHDB-384921

EMT labeling of disjoint union of cycles

摘录 / EXCERPT: For even n, determine if the disjoint union $C_n \ \dot{\cup} \ C_n$ is edge-magic total. For $m \ge n$, determine if the disjoint union $C_m \ \dot{\cup} \ C_n$ is edge-magic to...

open_problem | open | MathDB 页面 | 原文线索

V2-046657 | MATHDB-384923

Radio graceful Cartesian graph products

摘录 / EXCERPT: Determine conditions on graphs G and H that guarantee $G \ \square \ H$ is radio graceful. In the other direction, given $G \ \square \ H$ is radio graceful, determine if G a...

open_problem | open | MathDB 页面 | 原文线索

V2-046658 | MATHDB-384924

EMT labeling for grid graphs

摘录 / EXCERPT: Let $m \ge n$. Determine the validity of the existence of edge total labelings for $P_m \ \square \ P_n$. Determine the existence of edge total labelings for $C_m \ \square \ P_n$.

open_problem | open | MathDB 页面 | 原文线索

V2-046659 | MATHDB-384925

A linear algebra interpretation of labelings

摘录 / EXCERPT: Determine linear algebra properties that correspond to the existence of particular labelings.

open_problem | open | MathDB 页面 | 原文线索

V2-046660 | MATHDB-384926

Variations on total labelings

摘录 / EXCERPT: Determine which graphs admit such a graph labeling. For such graphs, determine the minimum number of labels required.

open_problem | open | MathDB 页面 | 原文线索

V2-046661 | MATHDB-384927

Fractional labeling

摘录 / EXCERPT: Determine graphs that admit such a fractional labeling. In particular, determine graphs that admit a fractional labeling but not an integer labeling.

open_problem | open | MathDB 页面 | 原文线索

V2-046662 | MATHDB-384929

Distance antimagic labeling

摘录 / EXCERPT: A graph is distance antimagic if and only if all the vertices have a distinct neighborhood.

open_problem | open | MathDB 页面 | 原文线索

V2-046663 | MATHDB-384930

Distance magic labeling

摘录 / EXCERPT: Let r be even and n be odd. For a r-regular graph of order n, determine if such a labeling exists.

open_problem | open | MathDB 页面 | 原文线索

V2-046664 | MATHDB-384931

Neighborhood balanced coloring

摘录 / EXCERPT: Determine graphs that admit such a coloring.

open_problem | open | MathDB 页面 | 原文线索

V2-046665 | MATHDB-384932

Power domination with zero forcing variants

摘录 / EXCERPT: Determine the minimum number of elements in a set S so that such a propagation process covers the vertices of G after a time step $t \in \mathbb{N}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046666 | MATHDB-384933

Reversible zero forcing

摘录 / EXCERPT: For $t=3$ and a graph G, determine the minimum number of elements in a set $S \subseteq V(G)$ so that such a propagation process covers the vertices of G.

open_problem | open | MathDB 页面 | 原文线索

V2-046667 | MATHDB-384934

Zero forcing and planarity

摘录 / EXCERPT: Let G be a planar graph. Determine conditions for which a zero forcing set S exists. If such a set S exists, determine bounds on the size of the zero forcing set S of G.

open_problem | open | MathDB 页面 | 原文线索

V2-046668 | MATHDB-384935

Power domination for d-regular random graphs

摘录 / EXCERPT: Determine if the power domination result for 3-regular random graphs can be generalized to d-regular random graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-046669 | MATHDB-384936

Random forcing set

摘录 / EXCERPT: Given a graph G with a minimum zero forcing set of size s_0, determine the probability that a randomly chosen set S of size $s \ge s_0$ is a forcing set.

open_problem | open | MathDB 页面 | 原文线索

V2-046670 | MATHDB-384937

Redefining as poset problems

摘录 / EXCERPT: Determine conditions for a poset $(\mathcal{G}, \preceq)$ where $\mathcal{G}$ is the set of graphs with a minimum size power dominating (zero forcing) set.

open_problem | open | MathDB 页面 | 原文线索

V2-046671 | MATHDB-384938

Proportional zero forcing

摘录 / EXCERPT: Determine the minimum number of vertices needed to guarantee a graph to be fully forced.

open_problem | open | MathDB 页面 | 原文线索

V2-046672 | MATHDB-384939

Upper bound on the maximum nullity of a graph

摘录 / EXCERPT: Determine if the enhanced (odd cycle) forcing number bounds the maximum nullity of any families of graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-046673 | MATHDB-384940

Maximal zero forcing sets

摘录 / EXCERPT: Determine if the zero forcing number is equivalent to the maximal number of zero forcing sets.

open_problem | open | MathDB 页面 | 原文线索

V2-046674 | MATHDB-384941

Thickness of a graph

摘录 / EXCERPT: Characterize the nonplanar graphs that are thickness-2.

open_problem | open | MathDB 页面 | 原文线索

V2-046675 | MATHDB-384942

Minimum rank clique covers

摘录 / EXCERPT: Given a graph G, determine if there exists a clique cover whose minimum rank matrices admit a minimum rank matrix of G.

open_problem | open | MathDB 页面 | 原文线索

V2-046676 | MATHDB-384943

Forbidden separation

摘录 / EXCERPT: Given a forbidden set of nonnegative integers S, find an infinite sequence such that for any two numbers in the sequence, their difference is not in S. Furthermore, determine t...

open_problem | open | MathDB 页面 | 原文线索

V2-046677 | MATHDB-384944

Permutation assignment on lift graphs

摘录 / EXCERPT: Given a base graph G with property P, determine if there exists a permutation assignment to the edge of the base graph such that the lift of the graph inherits property P.

open_problem | open | MathDB 页面 | 原文线索

V2-046678 | MATHDB-384945

Induced subgraphs

摘录 / EXCERPT: Determine the possible induced subgraphs of G.

open_problem | open | MathDB 页面 | 原文线索

V2-046679 | MATHDB-384946

Graph Homomorphisms

摘录 / EXCERPT: Determine a family of trees $\mathscr{T}$ that admits the minimum cardinality of S.

open_problem | open | MathDB 页面 | 原文线索

V2-046680 | MATHDB-384947

Crossing number and edge addition

摘录 / EXCERPT: Is there a constant $c < 1$ such that any graph G with m edges and any two non-adjacent vertices u,v in G satisfy $\text{cr}(G + uv) \leq \text{cr}(G) + c \cdot m$?

open_problem | open | MathDB 页面 | 原文线索

V2-046681 | MATHDB-384948

Crossing number and edge deletion

摘录 / EXCERPT: In any graph G, can we always find an edge e such that $\text{cr}(G - e) \geq \text{cr}(G) - o\left(\text{cr}(G)\right)$.

open_problem | open | MathDB 页面 | 原文线索

V2-046682 | MATHDB-384949

Crossing number and edge partition

摘录 / EXCERPT: Is there a constant $c < 3/8$ such that any graph G can be decomposed as a union of two subgraphs $G = G_1 \cup G_2$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-046683 | MATHDB-384950

Crossing number (and its variations) in random graphs

摘录 / EXCERPT: Let $G \sim G(n,m)$ with $m \geq \omega(n)$, is it true with high probability that the crossing number $\text{cr}(G) \geq \Omega(m^2)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046684 | MATHDB-384951

Crossing number on surfaces

摘录 / EXCERPT: Study $\text{cr}_g(K_n)$ and obtain better bounds.

open_problem | open | MathDB 页面 | 原文线索

V2-046685 | MATHDB-384952

Chromatic number of generalized Kneser graphs

摘录 / EXCERPT: Is it true that for $n > sk$, the chromatic number $\chi(KG_s(n,k)) = n - sk +s$?

open_problem | open | MathDB 页面 | 原文线索

V2-046686 | MATHDB-384953

Coloring distance on planar graphs

摘录 / EXCERPT: Fix a planar graph G and $k=8$, is it true that the distance between any two proper k-colorings of G is at most $O(n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046687 | MATHDB-384954

Coloring distance and matching number

摘录 / EXCERPT: For a fixed graph G with n vertices and $k = \Delta(G) + 2$, is it true that the distance between any two proper k-colorings of G is at most $n + \mu(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046688 | MATHDB-384955

Planar-critical graphs

摘录 / EXCERPT: Is there a constant k such that every 1-planar-critical graph is k-planar?

open_problem | open | MathDB 页面 | 原文线索

V2-046689 | MATHDB-384956

Quasi-planar graphs

摘录 / EXCERPT: Is it true that for every fixed k the genus $g(G)$ (Euler or orientable) of all k-quasi-planar graphs G is in $O(n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046690 | MATHDB-384957

Collinear triples and self-intersection paths

摘录 / EXCERPT: What is the minimal number of collinear triples in any set P of n points in order to guarantee a self-intersection 3-path in G_P, i.e. a path $abcd$ with ab and cd cros...

open_problem | open | MathDB 页面 | 原文线索

V2-046691 | MATHDB-384958

Plane paths in simple topological graphs

摘录 / EXCERPT: Is there a constant $c > 0$ such that $f(n) \geq \Omega(n^c)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046692 | MATHDB-384959

Projections and Lorentzian Polynomials

摘录 / EXCERPT: If $X \subseteq \mathbb{R}^n$ is an M-convex set, and we take a linear projection $L: R^n → R$ that preserves integer points (i.e. $L(x) = \langle v,x \rangle$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-046693 | MATHDB-384960

Preserving real roots on a subspace

摘录 / EXCERPT: Let $S \subseteq {\\mathbb{R}}[x]_d$ be a subspace of univariate polynomials, characterize linear maps sending real rooted polynomials in S to real rooted polynomials in S. As...

open_problem | open | MathDB 页面 | 原文线索

V2-046694 | MATHDB-384961

Preservers of Real Rootedness with Nonnegative Roots

摘录 / EXCERPT: What are linear maps $T:\mathbb{R}[t] \rightarrow \mathbb{R}[t]$ so that for any real rooted polynomial f with nonnegative real coefficients, $T(f)$ is real rooted.

open_problem | open | MathDB 页面 | 原文线索

V2-046695 | MATHDB-384962

Thin Tree Conjecture

摘录 / EXCERPT: There is a k_0 (independent of $|V|$) so that every k_0-edge connected graph has a $\frac{1}{2}$-thin spanning tree.

open_problem | open | MathDB 页面 | 原文线索

V2-046696 | MATHDB-384963

Negative correlation for random forests

摘录 / EXCERPT: For what ϵ is it true that $\Pr(e \in F \text{ and }f \in F) \le (1+\epsilon)\Pr(e \in F)\Pr(f \in F)$ for F a uniformly random spanning forest?

open_problem | open | MathDB 页面 | 原文线索

V2-046697 | MATHDB-384965

Even Delta Matroids, Jump Systems and Fractional Log-Concavity (problem 1)

摘录 / EXCERPT: Let $J \subseteq \mathbb{Z}^n$ be a jump system so that either every element of $\{\sum_{i=0}^n a_i: a \in J\}$ is even, or every element of it is odd. For every $k \in \mathbb{N}$...

open_problem | open | MathDB 页面 | 原文线索

V2-046698 | MATHDB-384966

Even Delta Matroids, Jump Systems and Fractional Log-Concavity (problem 2)

摘录 / EXCERPT: Suppose that we know that some posinomial $f(x)$ does not vanish whenever it is evaluated when x is in the right half-plane, does that imply that $f(x)$ is log concave on the pos...

open_problem | open | MathDB 页面 | 原文线索

V2-046699 | MATHDB-384967

New Polynomials Arising from Matroids (problem 1)

摘录 / EXCERPT: Let M be a matroid (say graphic) of rank d with ground set E with m elements. Let $S_1, \dots, S_n \subseteq E$ (not necessarily disjoint). Let...

open_problem | open | MathDB 页面 | 原文线索

V2-046700 | MATHDB-384968

New Polynomials Arising from Matroids (problem 2)

摘录 / EXCERPT: Here, M_F is the result of contracting F in M, M^F is the restriction of M to F and $\overline{\chi_{M^F}}(t)$ is the reduced characteristic polynomial of M. Is this...

open_problem | open | MathDB 页面 | 原文线索

V2-046701 | MATHDB-384969

Misc (problem 1)

摘录 / EXCERPT: Let P be a convex body with baricenter at 0. Suppose that P contains no integer points in its interior besides 0. The volume of P is at most $\frac{(n+1)^n}{n!}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046702 | MATHDB-385138

Find a criterion that implies a metric space has a quasi-symmetric metric that is a PI space.

摘录 / EXCERPT: Find a criterion that implies a metric space has a quasi-symmetric metric that is a PI space. (Is it the combinatorial Loewner property?)

open_problem | open | MathDB 页面 | 原文线索

V2-046703 | MATHDB-385377

Extra combinatorial data

摘录 / EXCERPT: How can we describe such a trivialization in terms of the skeleton of W? More basically, how can we describe the first Chern class of W in terms of its skeleton?

open_problem | open | MathDB 页面 | 原文线索

V2-046704 | MATHDB-385378

Combinatorial moduli of skeleta

摘录 / EXCERPT: Is there an explicit list of moves on a Lagrangian skeleton that preserve the Weinstein structure?

open_problem | open | MathDB 页面 | 原文线索

V2-046705 | MATHDB-385473

For geodesic flows on non-compact metric spaces, can the variational principle have an extremal measure and metric?

摘录 / EXCERPT: For geodesic flows on non-compact metric spaces, can the variational principle have an extremal measure and metric? That is, if $\mathcal M_f$ is the set of f-invariant probabili...

open_problem | open | MathDB 页面 | 原文线索

V2-046706 | MATHDB-385512

Rigidity (problem 6)

摘录 / EXCERPT: In L. Lovasz’s paper ["Flats in Matroids and geometric graphs”], he considers the dimension of $span(F\cap h)$, where F is a family of linear subspaces of R^d, h is a subspac...

open_problem | open | MathDB 页面 | 原文线索

V2-046707 | MATHDB-385513

Determine which ... -regular graphs are independent in the generic 3-dimensional rigidity matroid.

摘录 / EXCERPT: Determine which k-regular graphs are independent in the generic 3-dimensional rigidity matroid.

open_problem | open | MathDB 页面 | 原文线索

V2-046708 | MATHDB-385612

Is the converse also true (Is ... for the big positroid cell exactly the points with disconnected matroid)?

摘录 / EXCERPT: Is the converse also true (Is P for the big positroid cell exactly the points with disconnected matroid)?

open_problem | open | MathDB 页面 | 原文线索

V2-046709 | MATHDB-385615

How does this relate to the combinatorics of the subword complex?

摘录 / EXCERPT: How does this relate to the combinatorics of the subword complex?

open_problem | open | MathDB 页面 | 原文线索

V2-046710 | MATHDB-385619

What is the relationship between the piecewise linear combinatorics in the Casals-Gorsky-Gorsky-Le-Shen-Simental cluster algebra construction and the PL-combinatorics of MV-polytopes?

摘录 / EXCERPT: What is the relationship between the piecewise linear combinatorics in the Casals-Gorsky-Gorsky-Le-Shen-Simental cluster algebra construction and the PL-combinatorics of MV-polytop...

open_problem | open | MathDB 页面 | 原文线索

V2-046711 | MATHDB-385626

Is there a local operation of plabic graphs which induces Lagrangian cobordisms between the associated Legendrian knots?

摘录 / EXCERPT: Is there a local operation of plabic graphs which induces Lagrangian cobordisms between the associated Legendrian knots? In general, what can be done on the combinatorial side to g...

open_problem | open | MathDB 页面 | 原文线索

V2-046712 | MATHDB-385679

Can we give topological/combinatorial criteria for when the minimal resolution of a toric ideal is cellular?

摘录 / EXCERPT: Can we give topological/combinatorial criteria for when the minimal resolution of a toric ideal is cellular?

open_problem | open | MathDB 页面 | 原文线索

V2-046713 | MATHDB-385687

Can M2 encode and output the combinatorial description of certain Fukaya categories?

摘录 / EXCERPT: Can M2 encode and output the combinatorial description of certain Fukaya categories?

open_problem | open | MathDB 页面 | 原文线索

V2-046714 | MATHDB-385702

Complexity of combinatorial questions and dichotomy theorems (problem 1)

摘录 / EXCERPT: Are the following decidable? begin{enumerate}[a.] item Given a finite set of tiles in ${\\mathbb{Z}}^n$, is there a continuous tiling of $F(2^{{\\mathbb{Z}}^n})$ with those tiles?...

open_problem | open | MathDB 页面 | 原文线索

V2-046715 | MATHDB-385705

Complexity of combinatorial questions and dichotomy theorems (problem 4)

摘录 / EXCERPT: Let G be an acyclic analytic Borel graph with uncountable Borel chromatic number. Is there an injective continuous reduction f from $\mathbb{G}_0$ to G ($x \mathbb{G}_0 y$ if...

open_problem | open | MathDB 页面 | 原文线索

V2-046716 | MATHDB-385706

Dynamics and measurable combinatorics of group actions (problem 1)

摘录 / EXCERPT: Let $d < n \in \omega$. Is there a compact subflow of the space of all bounded displacement actions of ${\\mathbb{Z}}^d$ on ${\\mathbb{Z}}^n$ with finitely many orbits?

open_problem | open | MathDB 页面 | 原文线索

V2-046717 | MATHDB-385707

Dynamics and measurable combinatorics of group actions (problem 2)

摘录 / EXCERPT: Let G be a Schreier graph of a free pmp (probability measure preserving) action of a finitely generated group Γ. begin{enumerate}[a.] item If Γ is non-amenable, doe...

open_problem | open | MathDB 页面 | 原文线索

V2-046718 | MATHDB-385708

Dynamics and measurable combinatorics of group actions (problem 3)

摘录 / EXCERPT: Let Γ be an infinite property (T) group. Let G be one of the following: begin{enumerate}[a.] item A Borel graphing of the orbit equivalence relation of a free pmp action o...

open_problem | open | MathDB 页面 | 原文线索

V2-046719 | MATHDB-385709

Dynamics and measurable combinatorics of group actions (problem 4)

摘录 / EXCERPT: Let G be a Polish group and N a locally compactable normal subgroup of G. Is the translation action of N on G hypersmooth? (It is if N is countable.)

open_problem | open | MathDB 页面 | 原文线索

V2-046720 | MATHDB-386425

#3 Enumerating partitions with ZDD

摘录 / EXCERPT: Is ZDD the best algorithm to enumerate all balanced partitions that are ε-balanced? Or is there some other method? (a) Is there a combinatorial/geometric way to underst...

open_problem | open | MathDB 页面 | 原文线索

V2-046721 | MATHDB-386514

Give a classification of big mapping classes by understanding their dynamics on some combinatorial complex associated with the surface.

摘录 / EXCERPT: Give a classification of big mapping classes by understanding their dynamics on some combinatorial complex associated with the surface.

open_problem | open | MathDB 页面 | 原文线索

V2-046722 | MATHDB-386515

What combinatorial objects are "good" analogues of the curve complex, either uniformly for all big surfaces or for some class of big surfaces?

摘录 / EXCERPT: What combinatorial objects are "good" analogues of the curve complex, either uniformly for all big surfaces or for some class of big surfaces? Here "good" means that there exist re...

open_problem | open | MathDB 页面 | 原文线索

V2-046723 | MATHDB-386569

Trisections of 4-manifolds (problem 15)

摘录 / EXCERPT: What invariants of 4-manifolds can be computed using the trisection diagrams (i.e. use combinatorial information of curve intersections) What invariants are not computable? Give...

open_problem | open | MathDB 页面 | 原文线索

V2-046724 | MATHDB-386665

Underlying matroid of the graph.

摘录 / EXCERPT: What properties of the configuration spaces on a graph depend only on the underlying matroid of the graph?

open_problem | open | MathDB 页面 | 原文线索

V2-046725 | MATHDB-388107

Smooth well-balanced orientations with prescribed in-degrees

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected graph, and $T \subseteq V$ a set of nodes of odd degree. When does an orientation D of G exist which is i) smooth (the in-degree and out-degree o...

open_problem | open | MathDB 页面 | 原文线索

V2-046726 | MATHDB-388685

Slicing Axes-Parallel Rectangles

摘录 / EXCERPT: Let us say that two rectangles in the place are independent if both their x- and y-axis projections are disjoint. A set of rectangles is then independent if the rectangles are...

open_problem | open | MathDB 页面 | 原文线索

V2-046727 | MATHDB-390517

Facet-gap problem for order and chain polytopes

摘录 / EXCERPT: For every finite poset P with $n=|P|$, determine the facet-count gap $\operatorname{gap}(P):=f_{n-1}(C(P))-f_{n-1}(O(P))$, where $O(P)$ and $C(P)$ are respectively the order and...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046728 | MATHDB-390531

Growth rate of Av(1324)

摘录 / EXCERPT: Determine the exact value of the Stanley-Wilf limit $\operatorname{gr}(\operatorname{Av}(1324)):=\lim_{n\to\infty}\lvert\operatorname{Av}_n(1324)\rvert^{1/n}$, where...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046729 | MATHDB-390599

Frankston-Kahn-Narayanan regular 3-wise intersection problem

摘录 / EXCERPT: For every sequence of families $\mathcal{A}_n\subseteq 2^{[n]}$ such that each $\mathcal{A}_n$ is increasing, regular, and 3-wise intersecting, prove that...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046730 | MATHDB-390600

Erdős-Ko-Rado problem for permutations with a fixed number of cycles

摘录 / EXCERPT: Let $\mathrm{Sym}(n,k)$ be the set of permutations of $[n]=\{1,\ldots,n\}$ having exactly k cycles. A family $\mathcal{F}\subseteq\mathrm{Sym}(n,k)$ is intersecting if, for every...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046731 | MATHDB-391897

Elsholtz-Rackham conjecture on sum-free subsets of lattice cubes

摘录 / EXCERPT: For every fixed integer $d\geq 1$, let $[n]^d=\{1,\ldots,n\}^d$, let $M([n]^d)$ denote the maximum cardinality of a sum-free subset of $[n]^d$, and let $N_d(n)$ be the number of su...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046732 | MATHDB-391898

Dennin’s Conjecture 8.9 on dual RSK correspondence

摘录 / EXCERPT: For every binary matrix A, let $A^\dagger$ be its transpose-complement, and let $\operatorname{ins}(A)$ and $\operatorname{rec}(A)$ denote the insertion and recording tableaux in...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046733 | MATHDB-391907

Zeeman’s collapsibility conjecture

摘录 / EXCERPT: For every finite contractible 2-dimensional polyhedron P, the product $P\times I$, where $I=[0,1]$, is collapsible to a point.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046734 | MATHDB-391910

Zhuang-Gao conjecture

摘录 / EXCERPT: For every finite nonabelian group G, the Gao constant satisfies $E(G)=\mathsf d(G)+|G|$, where $\mathsf d(G)$ is the largest integer ℓ for which there exists a product-one-f...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046735 | MATHDB-391926

Open group-specific existence problems for signed difference sets

摘录 / EXCERPT: Given a finite abelian group G of order v and integers k and λ, determine whether there exists a function $x:G\to\{-1,0,1\}$ such that $\sum_{g\in G}x(g)^2=k$ and, fo...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046736 | MATHDB-391935

Existence of nonconstant continuous or smooth finitely forcible kernels

摘录 / EXCERPT: Does there exist a nonconstant continuous, or even smooth, symmetric kernel $W\colon[0,1]^2\to\mathbb{R}$ that is finitely forcible? More precisely, do there exist such a kernel...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046737 | MATHDB-392092

Generalized covering-code rate problem

摘录 / EXCERPT: Let G_q be an alphabet of size $q\ge 2$, let $t\ge 1$ be fixed, and let $C\subseteq G_q^n$. For $X,Y\in G_q^{t\times n}$, define the column Hamming distance by...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046738 | MATHDB-392103

Conjectures of Colomo-Pronko and Lukas Riegler on antisymmetrizer determinant formulas

摘录 / EXCERPT: The conjecture concerns a determinantal expression for a related antisymmetrizer arising in the study of symmetric functions and alternating-sign-matrix enumeration. The supplied s...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046739 | MATHDB-392105

Weak generalized topological Hedetniemi conjecture

摘录 / EXCERPT: For a finite group G and finite free G-spaces (or finite free G-simplicial complexes) X and Y, the generalized topological Hedetniemi conjecture asserts that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046740 | MATHDB-392277

Dimension of the best-worst choice polytope

摘录 / EXCERPT: For $n\ge 2$, let $[n]=\{1,\ldots,n\}$ be a set of alternatives. For each linear ranking $\pi=(\pi(1),\ldots,\pi(n))$ of $[n]$, define its deterministic best-worst choice vector...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046741 | MATHDB-392459

Reiner-Rhoades q-Ehrhart conjecture

摘录 / EXCERPT: For every lattice polytope P, the associated q-Ehrhart series $E_P(q,t)$ is a rational function and satisfies the corresponding q-reciprocity relation; moreover, the associat...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046742 | MATHDB-392472

Fukuda-Ziegler question on g(n)

摘录 / EXCERPT: For each positive integer n, define $g(n)$ to be the maximum number of facets of a full-dimensional polytope $P\subseteq\mathbb{R}^n$ whose vertices belong to $\{0,1\}^n$. Determ...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046743 | MATHDB-392475

Erez Lapid’s conjecture on socles of induced representations

摘录 / EXCERPT: Let F be a local non-Archimedean field and let $\mathfrak m$ be any multisegment. If $\Lambda(\mathfrak m)$ denotes the associated RSK-standard representation of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046744 | MATHDB-392477

Cigler and Cigler-Krattenthaler Hankel determinant conjectures

摘录 / EXCERPT: Prove the collection of explicit conjectural evaluations, due to Cigler and to Cigler-Krattenthaler, for shifted Hankel determinants of Catalan-like sequences arising from weighted...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046745 | MATHDB-392623

Extremal linear extensions of bipartite posets

摘录 / EXCERPT: Let P be a bipartite poset on n elements, with two ranks A and B, such that every element of A is below exactly u elements of B, and every element of B is above exa...

open_problem | open | MathDB 页面 | 原文线索

V2-046746 | MATHDB-392636

Consistency of simple Ramsey systems

摘录 / EXCERPT: Are any two simple Ramsey systems consistent? Equivalently, for fixed positive integers $a_1,\ldots,a_m$ and $b_1,\ldots,b_n$, does every finite colouring of $\mathbb{N}$ admit seq...

open_problem | open | MathDB 页面 | 原文线索

V2-046747 | MATHDB-392641

Ramsey-Turán non-principality for graphs

摘录 / EXCERPT: Do there exist 2-graphs G_1,G_2 for which ... What about if we require $\rho(\{G_1,G_2\})>0$ as well?

open_problem | open | MathDB 页面 | 原文线索

V2-046748 | MATHDB-392649

Erdős-Simonovits conjecture for hexagons

摘录 / EXCERPT: Is the extremal number for the 6-cycle asymptotic to $\frac{1}{2}n^{4/3}$ as $n\to\infty$? The report answers this negatively by asserting that, for infinitely many positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-046749 | MATHDB-392745

Extremal additive intersective sets: negative solution to Erdős’s problem

摘录 / EXCERPT: Let $\omega:\mathbb N\to\mathbb R$ be any increasing function with $\omega(n)\to\infty$ as $n\to\infty$. Then there exists a set $S=\{s_1,s_2,\ldots\}\subseteq\mathbb N$, with coun...

open_problem | open | MathDB 页面 | 原文线索

V2-046750 | MATHDB-392796

Yau’s conjecture on the Hitchin-Kobayashi correspondence for polarized manifolds

摘录 / EXCERPT: For a polarized algebraic manifold (M,L), the polarization class admits a Kähler metric of constant scalar curvature (or, more generally, an extremal Kähler metric) if and only i...

conjecture | open | MathDB 页面 | 原文线索

V2-046751 | MATHDB-392800

Upper-edge Lifshits exponent for the bond-percolation Laplacian

摘录 / EXCERPT: In the non-percolating phase for bond-percolation graphs in $\mathbb{Z}^d$, the integrated density of states N of the combinatorial Laplacian is conjectured to have Lifshits expo...

open_problem | open | MathDB 页面 | 原文线索

V2-046752 | MATHDB-392827

Positivity of interpolatory cubature weights for extremal fundamental systems on S^2

摘录 / EXCERPT: Let $P_n(S^2)$ be the space of spherical polynomials of degree at most n, and let $d_n=\\dim P_n(S^2)=(n+1)^2$. For an extremal fundamental system...

open_problem | open | MathDB 页面 | 原文线索

V2-046753 | MATHDB-392898

General interpretation of Kazhdan-Lusztig polynomial coefficients

摘录 / EXCERPT: Find, in general, a combinatorial or commutative-algebra interpretation of the coefficients of the Kazhdan-Lusztig polynomials $P_{u,v}(q)$ for $u,v\in S_n$.

open_problem | open | MathDB 页面 | 原文线索

V2-046754 | MATHDB-393059

Extremal gcd products for integers and shifted primes

摘录 / EXCERPT: Define ... and ... With $\vartheta(n)=(\log n)^2/\log\log n$, prove that ...

open_problem | open | MathDB 页面 | 原文线索

V2-046755 | MATHDB-393195

Kahn’s conjecture on inequivalent matroid representations

摘录 / EXCERPT: For every finite field F, there exists an integer $\\mu(F)$ such that every 3-connected matroid has at most $\\mu(F)$ inequivalent representations over F.

conjecture | open | MathDB 页面 | 原文线索

V2-046756 | MATHDB-393309

Combinatorial interpretation of coefficients in the inverse-dimension expansion of Weingarten functions

摘录 / EXCERPT: Interpret combinatorially the coefficients in the development in d^{-1} of the functions Wg, as numbers enumerating a combinatorial structure related to the endpoints of the pa...

open_problem | open | MathDB 页面 | 原文线索

V2-046757 | MATHDB-393423

Radial symmetry of extremals in the remaining Caffarelli-Kohn-Nirenberg parameter region

摘录 / EXCERPT: The conjecture is that all extremal functions for the Caffarelli-Kohn-Nirenberg inequalities, for parameters $a\le 0$ and $h(a)\le b<a+1$, are radially symmetric.

open_problem | open | MathDB 页面 | 原文线索

V2-046758 | MATHDB-393728

Countable quotients of combinatorial structures

摘录 / EXCERPT: Let L be a finite relational language and $[U+007F]\mathcal{C}$ a class of L-structures. Given a structure $X\in\mathcal{C}$, when is there a countable quotient of X which is still...

open_problem | open | MathDB 页面 | 原文线索

V2-046759 | MATHDB-393750

Bipartite distance of K_4-free graphs

摘录 / EXCERPT: Every K_4-free graph G with n vertices can be made bipartite by deleting at most $n^2/9$ edges. Moreover, the only extremal graph which requires deletion of that many edges i...

open_problem | open | MathDB 页面 | 原文线索

V2-046760 | MATHDB-393755

Extremal triangle-free versus bipartite subgraphs of random graphs

摘录 / EXCERPT: Let $G=G(n,m)=(V,E)$ be a uniformly random graph with $|V|=n$ and $|E|=m$. Let $T(G)$ be a largest subset of E containing no triangle, and let $B(G)$ be a largest subset of E s...

open_problem | open | MathDB 页面 | 原文线索

V2-046761 | MATHDB-393915

Combinatorial description of isolated blocks in finite classical groups

摘录 / EXCERPT: Let $G=\operatorname{Sp}(2n,q)$ or $\operatorname{O}^{\pm}(2n,q)$, where q is odd and $q>2n$. Let B be an isolated ℓ-block of G, where ℓ is odd, and let e be the...

open_problem | open | MathDB 页面 | 原文线索

V2-046762 | MATHDB-394161

Combinatorial interpretation of the IIC approximation error in planar percolation

摘录 / EXCERPT: Give a combinatorial interpretation of why the error in approximating the four-armed incipient infinite-cluster measure by a finitely conditioned measure is governed by the probabi...

open_problem | open | MathDB 页面 | 原文线索

V2-046763 | MATHDB-394207

Erdős-Faudree-Rousseau-Schelp question on the size-Ramsey number of paths

摘录 / EXCERPT: Is it true that ... Here P_n is the path on n vertices and $\widehat r(P_n)=\min\{|E(\Gamma)|:\Gamma\to(P_n)^2\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046764 | MATHDB-394208

Size-Ramsey numbers of long subdivisions

摘录 / EXCERPT: There is an absolute constant c such that the following holds: for every integer D, there is a constant C_D such that, if H is a graph with $ ...

open_problem | open | MathDB 页面 | 原文线索

V2-046765 | MATHDB-394216

Asymmetric Ramsey threshold conjecture for two graphs

摘录 / EXCERPT: Let F_1,F_2 be graphs with $1<m_2(F_1)\le m_2(F_2)$. Then there exists a constant $b>0$ such that, for every $\varepsilon>0$, ... where ...

conjecture | open | MathDB 页面 | 原文线索

V2-046766 | MATHDB-394240

Rainbow Ramsey versus Ramsey for higher arities

摘录 / EXCERPT: Is it the case that RRT^{n}_{2} implies RT^{n}_{2} for any $n\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-046767 | MATHDB-394270

Mean-width extremality of the inscribed regular simplex

摘录 / EXCERPT: Is the mean width of a polytope with $d+1$ vertices inscribed in the unit ball B^d maximal when the polytope is a regular simplex?

open_problem | open | MathDB 页面 | 原文线索

V2-046768 | MATHDB-394340

Combinatorial invariance of complete cd-coefficients

摘录 / EXCERPT: Let b$[u,v]\d$ be a Bruhat interval and let b$[w]_{u,v}\d$ denote the coefficient of the cd-word b$w\d$ in its complete cd-index b$\widetilde{\Psi}_{u,v}\d$. Do the cd-coefficients...

open_problem | open | MathDB 页面 | 原文线索

V2-046769 | MATHDB-394354

Asymptotic extremality of bicyclic polytopes

摘录 / EXCERPT: The bicyclic polytopes provide asymptotically the largest number of faces in all dimensions among all centrally symmetric polytopes with n vertices.

open_problem | open | MathDB 页面 | 原文线索

V2-046770 | MATHDB-394471

Kurek-Ruciński conjecture on the on-line size Ramsey number

摘录 / EXCERPT: Conjecture 1. For all positive integers c and t, ...

conjecture | open | MathDB 页面 | 原文线索

V2-046771 | MATHDB-394603

Relative Stability Theorem for extremal subsets of Alexandrov spaces

摘录 / EXCERPT: Let $X_i^n \xrightarrow{d_{GH}} X^n$ be a noncollapsing sequence of compact Alexandrov spaces with curvature at least κ and diameter at most D. Let $E_i\subset X_i$ be a s...

open_problem | open | MathDB 页面 | 原文线索

V2-046772 | MATHDB-394883

Combinatorial characterization of the fundamental group of a locally finite graph with ends

摘录 / EXCERPT: Let G be a locally finite connected graph, let T be a topological spanning tree of G, and let $e_0,e_1,\\ldots$ be its chords, arbitrarily oriented. Then the map...

open_problem | open | MathDB 页面 | 原文线索

V2-046773 | MATHDB-394890

Leader-Russell-Walters transitivity conjecture in Euclidean Ramsey theory

摘录 / EXCERPT: A finite set $S\subseteq\mathbb{R}^d$ is Ramsey if and only if it embeds, possibly in a higher-dimensional Euclidean space, in a finite transitive set; here transitive means that t...

conjecture | open | MathDB 页面 | 原文线索

V2-046774 | MATHDB-394897

Burr-Erdős bounded-degree graph Ramsey conjecture

摘录 / EXCERPT: For each positive integer Δ, there is a constant $c(\Delta)$ such that every graph H with n vertices and maximum degree Δ satisfies $r(H)\le c(\Delta)n$.

conjecture | open | MathDB 页面 | 原文线索

V2-046775 | MATHDB-394898

Burr-Erdős conjecture for Ramsey numbers of d-degenerate graphs

摘录 / EXCERPT: For each positive integer d, there exists a constant $c(d)$ such that every d-degenerate graph H on n vertices satisfies $r(H)\le c(d)n$, where $r(H)$ is the least positive...

conjecture | open | MathDB 页面 | 原文线索

V2-046776 | MATHDB-394899

Linear Ramsey numbers for hypercubes

摘录 / EXCERPT: Burr and Erdős conjectured that $r(Q_d)$ is linear in the number of vertices of the d-cube.

open_problem | open | MathDB 页面 | 原文线索

V2-046777 | MATHDB-394900

Exponential upper bound for Ramsey numbers of bounded-degree graphs

摘录 / EXCERPT: It seems plausible that $r(H)\le 2^{c\Delta}n$ holds in general for every graph H with n vertices and maximum degree Δ.

open_problem | open | MathDB 页面 | 原文线索

V2-046778 | MATHDB-394913

Semi-Combinatorial Multiplicity Conjecture

摘录 / EXCERPT: Let Δ be a simplicial complex on the ground set $[n]$, of dimension $d-1$, and let k be a coefficient field. For $1\le i\le n-d$, set ... where Δ_W is the induced sub...

conjecture | open | MathDB 页面 | 原文线索

V2-046779 | MATHDB-394914

Combinatorial Multiplicity Conjecture

摘录 / EXCERPT: Let Δ be a simplicial complex on ground set $[n]$ of dimension $d-1$. Let $\mathcal{MN}(\Delta)$ denote the set of minimal non-faces of Δ, and set ... Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-046780 | MATHDB-395061

The conformally invariant embedding of a planar lattice

摘录 / EXCERPT: Given a lattice L, considered as a combinatorial object without an embedding into the plane, which periodic embedding of L corresponds to a conformally invariant scaling limit...

open_problem | open | MathDB 页面 | 原文线索

V2-046781 | MATHDB-395224

Equidistribution of Fekete configurations on the sphere

摘录 / EXCERPT: Let $E\subset S^n$ be a compact subset of the round n-sphere. For each k, let $P_k\in E^{N_k}$ be a Fekete configuration of degree k for E (also called an extremal fundamen...

open_problem | open | MathDB 页面 | 原文线索

V2-046782 | MATHDB-395293

Extremal density of sets avoiding a fixed number of distances

摘录 / EXCERPT: For fixed positive integers n and N, what is the value of ... Here $m_{d_1,\ldots,d_N}(\mathbb{R}^n)$ denotes the supremum of the upper densities of measurable subsets of...

open_problem | open | MathDB 页面 | 原文线索

V2-046783 | MATHDB-395312

Exponential upper bound for simplicial 3-spheres by number of facets

摘录 / EXCERPT: Is the number of combinatorial types of simplicial 3-spheres on N facets bounded by an exponential function C^N?

open_problem | open | MathDB 页面 | 原文线索

V2-046784 | MATHDB-395322

Connectivity of weighted rank 3 matroids

摘录 / EXCERPT: Let M be a finite simple rank-3 matroid on n points, which have real weights a_i satisfying $\sum_{i=1}^{n} a_i\ge 0$. Is it true that the points of non-negative weight are...

open_problem | open | MathDB 页面 | 原文线索

V2-046785 | MATHDB-395388

Weighted matroid matching for linearly represented matroids

摘录 / EXCERPT: Given a matroid $M=(S,r)$, with ground set S partitioned into subsets of size 2 called lines E, and nonnegative line weights $w\in\mathbb{R}_+^E$, find a maximum-weight matro...

open_problem | open | MathDB 页面 | 原文线索

V2-046786 | MATHDB-395389

Weighted fractional matroid matching

摘录 / EXCERPT: Given a matroid $M=(S,r)$, a set E of subsets of S (called lines), each of rank 1 or 2, and nonnegative weights $w\in\mathbb{R}_+^E$, maximize $w\cdot x$ over vectors...

open_problem | open | MathDB 页面 | 原文线索

V2-046787 | MATHDB-395432

Uniqueness of the extremal gradient Gibbs measure under small deformations

摘录 / EXCERPT: Is there, for sufficiently small temperature and small affine deformations at the boundary, a unique extremal gradient Gibbs measure?

open_problem | open | MathDB 页面 | 原文线索

V2-046788 | MATHDB-395476

The Reach Problem for 3D revolute-jointed robots

摘录 / EXCERPT: Given a 3D revolute-jointed robot, compute the extremal values of the endpoint distance function and the corresponding extremal configurations.

open_problem | open | MathDB 页面 | 原文线索

V2-046789 | MATHDB-395487

Geometric realizations of toroidal triangulations in prescribed isotopy classes

摘录 / EXCERPT: For a given combinatorial triangulation of a two-torus, in which isotopy classes can it be realized geometrically in $\mathbb{R}^3$?

open_problem | open | MathDB 页面 | 原文线索

V2-046790 | MATHDB-395568

Extremal constants and domains for four inequalities on simply connected planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be an arbitrary simply connected domain. Define ... where $d_\Omega(z)$ is the distance from z to the boundary of Ω. Let $\sigma_\Omega(z)$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-046791 | MATHDB-395707

Extremals of commuting-contraction families for more than two variables

摘录 / EXCERPT: Identify the extremals for the family of commuting contractions consisting of commuting d-tuples when $d>2$. Here an extremal tuple T is one for which every extension R in th...

open_problem | open | MathDB 页面 | 原文线索

V2-046792 | MATHDB-395765

Doubly exponential lower bound for the two-color 3-uniform Ramsey number

摘录 / EXCERPT: Determine whether there exists a constant $c>0$ such that $r_3(n,n)>2^{2^{cn}}$ for all sufficiently large n.

open_problem | open | MathDB 页面 | 原文线索

V2-046793 | MATHDB-395899

On the extremality of an 80-dimensional lattice

摘录 / EXCERPT: The specific even unimodular lattice of dimension 80 described by Watkins is extremal: the minimum length of a nonzero vector is 8. Its automorphism group contains a subgroup i...

open_problem | open | MathDB 页面 | 原文线索

V2-046794 | MATHDB-395900

Extremality of a specific 80-dimensional even unimodular lattice

摘录 / EXCERPT: The specific even unimodular lattice of dimension 80 described in the report is extremal: its minimum length is 8, equivalently it has no nonzero vectors of norm 2, 4, or...

open_problem | open | MathDB 页面 | 原文线索

V2-046795 | MATHDB-395995

Extremal-graph reduction for the Nash mapping

摘录 / EXCERPT: If the Nash mapping is bijective for singularities whose minimal good resolution graph is extremal, then it is bijective in general. Equivalently, if the Nash mapping is bijective...

open_problem | open | MathDB 页面 | 原文线索

V2-046796 | MATHDB-396055

Non-existence of equilibrium between two aligned sub-extremal rotating black holes

摘录 / EXCERPT: There is no stationary equilibrium configuration consisting of two aligned, rotating, axisymmetric sub-extremal black holes, where sub-extremality is defined by the existence of tr...

open_problem | open | MathDB 页面 | 原文线索

V2-046797 | MATHDB-396131

Extremality of Hausdorff measure for a hyperbolic exponential map

摘录 / EXCERPT: Let $f(z)=\lambda e^z$ be a hyperbolic exponential function, and let h be the Hausdorff dimension of the radial Julia set. It was proved that the Hausdorff measure $\mathcal H^h$...

open_problem | open | MathDB 页面 | 原文线索

V2-046798 | MATHDB-396227

Rota’s Conjecture for finite-field-representable matroids

摘录 / EXCERPT: For a finite field F, the class of matroids representable over F can be characterised by a finite number of forbidden minors.

conjecture | open | MathDB 页面 | 原文线索

V2-046799 | MATHDB-396314

Boundary amenability for groups acting on buildings

摘录 / EXCERPT: Let G be a group acting properly on a locally finite building X. Then the action of G on the combinatorial compactification $\mathscr C_{\mathrm{ch}}(X)$ is topologically ame...

open_problem | open | MathDB 页面 | 原文线索

V2-046800 | MATHDB-396806

Finite forcibility of extremal graphons

摘录 / EXCERPT: Lovász and Szegedy conjecture that every extremal problem-such as maximizing $\sum_{i=1}^{n} c_i t(F_i,G)$ over simple graphs G subject to $\sum_{i=1}^{n} a_i t(F_i,G)\ge 0$, wit...

open_problem | open | MathDB 页面 | 原文线索

V2-046801 | MATHDB-396839

Euclidean Ramsey conjecture for spherical finite configurations

摘录 / EXCERPT: For every finite point set K that can be embedded in a sphere and every positive integer r, does there exist an integer $d_0(K,r)$ such that, for every integer $d\ge d_0(K,r)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-046802 | MATHDB-396853

Whether the Ramsey property alone is equivalent to universality

摘录 / EXCERPT: Is the Ramsey property alone equivalent to universality? The report answers: no, in general.

open_problem | open | MathDB 页面 | 原文线索

V2-046803 | MATHDB-396859

Combinatorial characterization of Boolean algebras supporting non-atomic separable measures

摘录 / EXCERPT: Find a combinatorial characterization of Boolean algebras that support a non-atomic separable strictly positive finitely additive measure.

open_problem | open | MathDB 页面 | 原文线索

V2-046804 | MATHDB-396910

Whether the complexes $B(i,d)$ are products with a ball

摘录 / EXCERPT: Is $B(i,d)$ a combinatorial triangulation of $S^i \times B^{d-i-1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-046805 | MATHDB-396955

Combinatorial positivity conjecture

摘录 / EXCERPT: Let C be an $r\times r$ matrix with non-negative integer entries, not necessarily symmetric. Define T on formal power series in $x_1,\ldots,x_r$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-046806 | MATHDB-397056

Coalescents representable by cutting down a combinatorial tree

摘录 / EXCERPT: Find other coalescents that can be represented by cutting down a combinatorial tree.

open_problem | open | MathDB 页面 | 原文线索

V2-046807 | MATHDB-397141

Extremal process of branching Brownian motion

摘录 / EXCERPT: For the branching Brownian motion defined on pages 13-14, let ... and let Z be the almost-sure limit of the derivative martingale $Z(t)$. Conditionally on Z, let ... be a Poisson p...

open_problem | open | MathDB 页面 | 原文线索

V2-046808 | MATHDB-397254

Existence of an extremal binary code of length 72

摘录 / EXCERPT: Does there exist an extremal doubly-even self-dual binary code of length 72?

open_problem | open | MathDB 页面 | 原文线索

V2-046809 | MATHDB-397323

Füredi-Hajnal conjecture

摘录 / EXCERPT: For every permutation matrix P, the extremal function $\operatorname{ex}_P(n)$-the maximum number of 1-entries in an $n\times n$ $(0,1)$-matrix avoiding P-is linear in n.

conjecture | open | MathDB 页面 | 原文线索

V2-046810 | MATHDB-397325

Bounds for the modified Füredi-Hajnal extremal function $p_3(n)$

摘录 / EXCERPT: Let $p_k(n)$ be the maximum number of 1-entries in an $n\times n$ $(0,1)$-matrix such that no $(k+1)$-tuple of columns contains all $(k+1)$-permutation matrices. If $\alpha(n)$ d...

open_problem | open | MathDB 页面 | 原文线索

V2-046811 | MATHDB-397330

Finiteness of combinatorial types up to projective transformations

摘录 / EXCERPT: For a fixed dimension d, is it true that, up to projective transformations, there are only finitely many combinatorial types of d-dimensional convex polytopes?

open_problem | open | MathDB 页面 | 原文线索

V2-046812 | MATHDB-397336

Exceptional set in extremal favourite-distance configurations in three-space

摘录 / EXCERPT: For all sufficiently large n, if $S\subset\mathbb{R}^{3}$ with $|S|=n$ and $r:S\to(0,\infty)$ satisfies $e_r(S)=f_3(n)$, then the exceptional set T in Theorem B can be taken to...

open_problem | open | MathDB 页面 | 原文线索

V2-046813 | MATHDB-397338

Dimension of the space of convex partitions of 3-space

摘录 / EXCERPT: For an integer $n>3$, what is the dimension of the space of all partitions of $\\mathbb{R}^3$ into n convex pieces? Equivalently, since there are only finitely many combinatorial...

open_problem | open | MathDB 页面 | 原文线索

V2-046814 | MATHDB-397415

Maximum feasible vector in a matroid independence polytope

摘录 / EXCERPT: Let $M=(S,r)$ be a matroid and let $g:S\to\mathbb{R}_{+}$ be a function. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-046815 | MATHDB-397416

Bound of n bases for representing an integer combination of matroid bases

摘录 / EXCERPT: Any nonnegative integer combination of incidence vectors of bases of a matroid can be written as such a combination using at most n bases, where n is the size of the ground set...

open_problem | open | MathDB 页面 | 原文线索

V2-046816 | MATHDB-397419

Seymour’s flowing conjecture for signed matroids

摘录 / EXCERPT: If a signed matroid (M,Σ) with edge weights w satisfies the cut-condition and does not contain any of three special signed minors, then there exists a fractional...

conjecture | open | MathDB 页面 | 原文线索

V2-046817 | MATHDB-397422

Minimum-cost integer multiflow problem for tree metrics

摘录 / EXCERPT: The minimum-cost version of $\bc$-IMP with tree-metric $\bc$ has a combinatorial minmax relation extending Mader’s edge-disjoint S-path theorem and Karzanov’s minimum-cost genera...

open_problem | open | MathDB 页面 | 原文线索

V2-046818 | MATHDB-397424

Weighted Linear Matroid Parity

摘录 / EXCERPT: Let A be a row-full-rank matrix with row set U and column set V, where $r=|U|$ and $n=|V|$ are even. Partition V into pairs, called lines, and let $w_\ell\in\mathbb{R}$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-046819 | MATHDB-397431

Weighted linear matroid matching

摘录 / EXCERPT: Given a vector space V, a set E of lines in V (with a basis-two vectors in V-specified for each line), and a weight function $w:E\to\mathbb{R}_+$, a subset $M\subseteq E$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-046820 | MATHDB-397432

Running time for weighted linear matroid matching

摘录 / EXCERPT: The author conjectures that the running time for the weighted linear matroid matching problem could eventually be reduced to $O(r(V)|E|^3)$, matching the running time of the unweig...

open_problem | open | MathDB 页面 | 原文线索

V2-046821 | MATHDB-397523

ω-model separation of Stable Ramsey’s Theorem for Pairs from Ramsey’s Theorem for Pairs

摘录 / EXCERPT: Is every $\c9$-model of $\\mathrm{RCA}_0 + \mathrm{SRT}^2_2$ also a model of RT^2_2?

open_problem | open | MathDB 页面 | 原文线索

V2-046822 | MATHDB-397725

Black hole stability conjecture for sub-extremal Kerr solutions

摘录 / EXCERPT: The black hole stability conjecture asserts that small perturbations of Kerr initial data with sub-extremal parameters $|a_0|<M_0$ asymptote in time, under evolution by the Einstei...

conjecture | open | MathDB 页面 | 原文线索

V2-046823 | MATHDB-397853

Rigid transmitter polytopes

摘录 / EXCERPT: There is an infinite family $(\mathrm{RTP}_{81}[n])$, $n\in\mathbb{N}$, of combinatorially distinct 81-dimensional polytopes, all of which are projectively unique.

open_problem | open | MathDB 页面 | 原文线索

V2-046824 | MATHDB-397919

Removing axisymmetry from the extremal Kerr stability result

摘录 / EXCERPT: Removing the axisymmetry assumption for the stability result remains an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-046825 | MATHDB-398068

Invariants on generic subsets of matroid realization spaces

摘录 / EXCERPT: Which invariants of hyperplane arrangement complements hold on an open set or a countable intersection of open sets of the realization space of the associated matroid?

open_problem | open | MathDB 页面 | 原文线索

V2-046826 | MATHDB-398079

Classification of Ramsey configurations in Euclidean Ramsey theory

摘录 / EXCERPT: Conjecture: A configuration is Ramsey if and only if it is spherical. Here, for a k-point configuration $K=\{x_1,\ldots,x_k\}\subset\mathbb{R}^n$ and a natural number r, the Ra...

open_problem | open | MathDB 页面 | 原文线索

V2-046827 | MATHDB-398130

The Polynomial Freiman-Ruzsa Conjecture in Additive Combinatorics

摘录 / EXCERPT: For a subset $A\subseteq \mathbb{F}_2^n$, if $|A+A|\le K|A|$, then there exists a subset $A'\subseteq A$ such that $|A'|\ge \frac{1}{\operatorname{poly}(K)}|A|$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-046828 | MATHDB-398198

Packing-covering conjecture for infinite matroids

摘录 / EXCERPT: For any family of matroids on the same ground set, there is a partition of the ground set into two parts, the packing side and the covering side, such that the family obtained by c...

conjecture | open | MathDB 页面 | 原文线索

V2-046829 | MATHDB-398247

Equality of algebraic and combinatorial ranks of divisors on graphs

摘录 / EXCERPT: Let G be a graph and let $ ...

open_problem | open | MathDB 页面 | 原文线索

V2-046830 | MATHDB-398342

Finiteness of the extremal agenda-size function G(n)

摘录 / EXCERPT: For an $n\times m$ matrix $A=(a_{ij})$ with entries in $\{-1,1\}$, let $Q\subseteq[m]$ be a winning agenda if more than half of the n rows satisfy $\sum_{j\in Q}a_{ij}>0$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-046831 | MATHDB-398352

Small complete minors above the extremal edge density

摘录 / EXCERPT: Fiorini et al. [3] conjectured that any graph G with $d(G) \ge c(t) + \varepsilon$ contains a K_t-minor of order $C(\varepsilon)\log n$.

open_problem | open | MathDB 页面 | 原文线索

V2-046832 | MATHDB-398370

Extremal edge bound for $K_{t,t}$-free string graphs

摘录 / EXCERPT: Is there an absolute constant c such that every $K_{t,t}$-free string graph with n vertices has at most $c\,t\log t\,n$ edges?

open_problem | open | MathDB 页面 | 原文线索

V2-046833 | MATHDB-398448

Existence of contractions of negative extremal rays

摘录 / EXCERPT: Let X be a normal $\factorial compact Kähler threefold with at most terminal singularities such that KX is pseudoeffective. Let $mathbb{R}+[Gammai]$ be a $KX...

open_problem | open | MathDB 页面 | 原文线索

V2-046834 | MATHDB-398629

Character rigidity of \mathrm{PSL}_2(R)

摘录 / EXCERPT: Let R be either an infinite field or a ring of algebraic integers with infinitely many units. Then $\mathrm{PSL}_2(R)$ is character rigid; that is, its only extremal characters c...

open_problem | open | MathDB 页面 | 原文线索

V2-046835 | MATHDB-398773

The tropical Laplacian of a matroid has exactly one negative eigenvalue

摘录 / EXCERPT: Let M be a matroid of rank $r+1$ on the ground set $[U+007F]\{0,1,\ldots,n\u007f\}$. For each subset $F\subseteq\u007f\{0,1,\ldots,n\u007f\}$, set $e_F:=\sum_{i\in F}e_i$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-046836 | MATHDB-398783

Ramsey numbers of bounded-degree graphs

摘录 / EXCERPT: Conjecture 7. We have $c(\Delta)=2^{\Theta(\Delta)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-046837 | MATHDB-398794

Order of magnitude of the Ramsey number $r(P_3,t)$

摘录 / EXCERPT: The order of magnitude of $r(P_3,t)$ remains open, where P_3 is the tight 3-uniform path with three edges.

open_problem | open | MathDB 页面 | 原文线索

V2-046838 | MATHDB-398921

Finiteness of extremal rays for klt pairs with nonnegative Kodaira dimension

摘录 / EXCERPT: Suppose that (X,Δ) is a klt pair with $\kappa(X,\Delta)\ge 0$. Then the set of curves C_j in Theorem 1 may be taken to be finite.

open_problem | open | MathDB 页面 | 原文线索

V2-046839 | MATHDB-398990

Extremal effective divisors on the moduli space of genus-one curves

摘录 / EXCERPT: Let $\mathbf a=(a_1,\ldots,a_n)\u0000$ be integers with $\sum_{i=1}^n a_i=0$, not all zero. Let $Q(\mathbf a)$ be the stratum of quadratic differentials on smooth genus-one curves...

open_problem | open | MathDB 页面 | 原文线索

V2-046840 | MATHDB-399001

Terao’s conjecture for hyperplane arrangements

摘录 / EXCERPT: For hyperplane arrangements A and A' in $\d53c^n$ having the same combinatorial type, if the logarithmic tangent bundle $T_{\d53c^n}(-\log D_A)$ splits as a direct sum of line...

conjecture | open | MathDB 页面 | 原文线索

V2-046841 | MATHDB-399384

Polynomial-time solvability of weighted linear matroid matching

摘录 / EXCERPT: Determine whether weighted linear matroid matching is solvable in polynomial time.

open_problem | open | MathDB 页面 | 原文线索

V2-046842 | MATHDB-399395

Maximum-weight linear matroid matching

摘录 / EXCERPT: Let V be a vector space over a given field $\[3m]\Gamma\u001b[0m$, let E be a set of lines (linear subspaces of rank 2), and let $w:E\to\mathbb{R}_+$. A subset $M\subseteq E$...

open_problem | open | MathDB 页面 | 原文线索

V2-046843 | MATHDB-399415

No quasicrystals in two dimensions at positive temperature

摘录 / EXCERPT: Conjecture 2. In two dimensions at positive temperature, all extremal Gibbs measures for finite-range interactions are periodic (that is, quasicrystals do not exist for short-range...

open_problem | open | MathDB 页面 | 原文线索

V2-046844 | MATHDB-399482

Cauchy problem for 2D supercyclidic nets

摘录 / EXCERPT: For an entire two-dimensional Q-net with regular combinatorics: (1) given the supporting Q-net $x:\mathbb{Z}^{2}\to\mathbb{R}\mathrm{P}^{3}$; (2) given two tangent congruences alon...

open_problem | open | MathDB 页面 | 原文线索

V2-046845 | MATHDB-399497

Approximately inscribed polytopes

摘录 / EXCERPT: Does every combinatorial type of a d-polytope have a realization with all vertices outside a $(d-1)$-sphere and all facets cutting into the same $(d-1)$-sphere?

open_problem | open | MathDB 页面 | 原文线索

V2-046846 | MATHDB-399859

Zero Hausdorff dimension of the DGFF extremal measure

摘录 / EXCERPT: For the random measure Z_D introduced above, Z_D is a.s. supported on a set of zero Hausdorff dimension.

open_problem | open | MathDB 页面 | 原文线索

V2-046847 | MATHDB-399887

Shortest non-contractible or non-separating closed curve

摘录 / EXCERPT: Given an edge-weighted graph G embedded on a surface S, with positive edge weights and the combinatorial surface model in which the length of a curve drawn on G is the sum of...

open_problem | open | MathDB 页面 | 原文线索

V2-046848 | MATHDB-399888

Shortest homotopic walks in the combinatorial surface model

摘录 / EXCERPT: Given a weighted embedded graph G and a path (or walk) p on G, compute, in the combinatorial surface model, a shortest walk with the same endpoints as p that is homotopic t...

open_problem | open | MathDB 页面 | 原文线索

V2-046849 | MATHDB-399893

Explicit local combinatorial formulas for rational Pontryagin classes

摘录 / EXCERPT: Find explicit expressions for local combinatorial cocycles representing the rational Pontryagin classes $p_k(M)$ of a triangulated manifold M, beyond the known pure existence the...

open_problem | open | MathDB 页面 | 原文线索

V2-046850 | MATHDB-399983

Explicit constructions of optimal Ramsey graphs

摘录 / EXCERPT: Construct an explicit family of $2\log n$-Ramsey graphs on n vertices, matching Erdős's 1947 existence result.

open_problem | open | MathDB 页面 | 原文线索

V2-046851 | MATHDB-400125

SLP invariance under combinatorics of line arrangements

摘录 / EXCERPT: Let $I=(\ell_1^b,\ldots,\ell_{2b+1}^b)$ be a minimally generated ideal that has the SLP at range 2 in degree $b-2$, and let $J=(h_1^b,\ldots,h_{2b+1}^b)$ be a minimally generated...

open_problem | open | MathDB 页面 | 原文线索

V2-046852 | MATHDB-400164

Rota’s characteristic-polynomial log-concavity conjecture

摘录 / EXCERPT: Let M be a matroid with characteristic polynomial ... Then the coefficient sequence is log-concave in absolute value: ... for every index $1\leq i\leq r$. Equivalently, the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-046853 | MATHDB-400171

White’s conjecture on toric ideals of matroids

摘录 / EXCERPT: For every matroid M on a ground set E, with set of bases $\mathcal B$, let K be a field and define $\varphi_M:K[y_B:B\in\mathcal B]\to K[x_e:e\in E]$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-046854 | MATHDB-400239

Allen-Brightwell-Skokan conjecture on Ramsey goodness of paths

摘录 / EXCERPT: Let H be a fixed graph with chromatic number k and let $n \ge k|H|$. Then $R(P_n,H)=(n-1)(k-1)+\sigma(H)$, where $R(P_n,H)$ is the two-colour Ramsey number and $\sigma(H)$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-046855 | MATHDB-400399

Symmetry theorem for extremal functions in Caffarelli-Kohn-Nirenberg inequalities

摘录 / EXCERPT: Under the conditions ... let $H^p_{\beta,\gamma}(\mathbb{R}^d)$ be the functional space specified for inequality (1), and define ... If either $\gamma\ge 0$, or $\gamma<0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-046856 | MATHDB-400586

Additional hypotheses ensuring extremality in positive residue characteristic

摘录 / EXCERPT: Let (K,v) be a valued field of positive residue characteristic. Assume that either vK is a $\mathbb{Z}$-group, or that vK is divisible and Kv is an imperfect large field....

open_problem | open | MathDB 页面 | 原文线索

V2-046857 | MATHDB-400696

Injectivity conjecture for the graded Möbius algebra of a matroid

摘录 / EXCERPT: Let M be a rank-r simple matroid on E, let L be the set of all flats of M, and let L_p be the set of rank-p flats. Define $B_p(M)=\bigoplus_{F\in L_p}\mathbb{Q}y_F$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-046858 | MATHDB-400820

Ramsey property and the non-existence of mad families

摘录 / EXCERPT: Does $\mathrm{ZF}+\mathrm{DC}+\text{“every set of reals has the Ramsey property”}$ imply that there are no mad families?

open_problem | open | MathDB 页面 | 原文线索

V2-046859 | MATHDB-400821

Connections between the Ramsey property and the non-existence of mad families

摘录 / EXCERPT: What are the connections/similarities/differences between the Ramsey property and the non-existence of mad families?

open_problem | open | MathDB 页面 | 原文线索

V2-046860 | MATHDB-400921

Bárány-Kalai-Meshulam connectivity conjecture for deleted joins of matroids

摘录 / EXCERPT: Conjecture 1 (Bárány, Kalai, and Meshulam, 2016). For any integer $k \ge 1$ there exists an integer $n_k \ge 1$ depending only on k such that for any matroid M of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-046861 | MATHDB-400927

Rota’s unimodality conjecture for matroids

摘录 / EXCERPT: For every matroid, the sequence of its numbers f_k is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-046862 | MATHDB-400970

Characterization of combinatorial core structures of supermodular functions

摘录 / EXCERPT: Characterize those collections of classes of subsets of a finite non-empty set N which are core structures for supermodular functions.

open_problem | open | MathDB 页面 | 原文线索

V2-046863 | MATHDB-401049

Extremal compact Riemann surfaces for the determinant of the Laplacian

摘录 / EXCERPT: For a compact Riemann surface X of genus $g\ge 2$, find the surfaces that are extremal points of $\det(\Delta_0)$ on Teichmüller space. The report states that locally extremal po...

open_problem | open | MathDB 页面 | 原文线索

V2-046864 | MATHDB-401122

Wiener’s lemma and extremality along subsequences

摘录 / EXCERPT: For which subsequences $(k_n)\subset\mathbb{N}$ do the assertions of Wiener’s lemma and its corollary hold along (k_n)? Here Wiener’s lemma is the assertion that every complex Bo...

open_problem | open | MathDB 页面 | 原文线索

V2-046865 | MATHDB-401343

Combinatorial determination of Alexander polynomials for line arrangements

摘录 / EXCERPT: Is the Alexander polynomial of an arrangement of lines in $\mathbb{P}^2$ determined by the combinatorics of the arrangement, without restricting the intersection points to be only...

open_problem | open | MathDB 页面 | 原文线索

V2-046866 | MATHDB-401528

Non-negativity of matroid Kazhdan-Lusztig polynomial coefficients

摘录 / EXCERPT: For every matroid M, all coefficients of its matroid Kazhdan-Lusztig polynomial $P(M,t)$ are non-negative.

open_problem | open | MathDB 页面 | 原文线索

V2-046867 | MATHDB-401529

Computing the top coefficient of the braid-matroid Kazhdan-Lusztig polynomial

摘录 / EXCERPT: Could Theorems 2, 3, and 4 help in computing the top coefficient of $P(B_n,t)$, conjectured in [2]?

open_problem | open | MathDB 页面 | 原文线索

V2-046868 | MATHDB-401599

Combinatorial proof of Fischer’s operator formula for monotone triangles

摘录 / EXCERPT: Find, by combinatorial means, a Laurent polynomial P such that ... for $k_1<\cdots<k_n$, and prove that ... lies in the ideal generated by ... where e_r denotes the rth elementary...

open_problem | open | MathDB 页面 | 原文线索

V2-046869 | MATHDB-401601

Combinatorics of the Ehrhart h^*-polynomial for sign-sequence adjacent-sum polytopes

摘录 / EXCERPT: For a sign sequence varepsilon:{1,dots,n-1}to{pm}, consider the polytope ... The open problem is to understand the combinatorics of the Ehrhart h^*-polynomial of these polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-046870 | MATHDB-401602

Dyson’s rank conjecture for the mod 5 and mod 7 partition congruences

摘录 / EXCERPT: For a partition, Dyson defined the rank as the largest part minus the number of parts, and conjectured that this rank statistic would account combinatorially for the Ramanujan part...

conjecture | open | MathDB 页面 | 原文线索

V2-046871 | MATHDB-401606

Combinatorial interpretation of the substitution formula for bicolored planar maps

摘录 / EXCERPT: For the generating series of bounded-degree bicolored Eulerian rooted planar maps defined in the report, where ... and the rational parametrization $x=x(z)$, $y=y(z)$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-046872 | MATHDB-401608

Explicit cyclic descents and actions for SYT of arbitrary non-ribbon shape

摘录 / EXCERPT: Find an explicit combinatorial description of the cyclic descent set of a standard Young tableau of arbitrary non-ribbon shape, as well as an explicit cyclic action on standard You...

open_problem | open | MathDB 页面 | 原文线索

V2-046873 | MATHDB-401609

Explicit cyclic descents for arbitrary non-ribbon Young tableaux

摘录 / EXCERPT: Find an explicit combinatorial description of the cyclic descent set of a standard Young tableau of arbitrary non-ribbon shape.

open_problem | open | MathDB 页面 | 原文线索

V2-046874 | MATHDB-401611

A combinatorial proof of the identity $P_n(i)=a_n$

摘录 / EXCERPT: For the graph G_n and matrix $\bar\Delta^{(n)}$ defined in the report, let $P_n(\alpha)=\det\bar\Delta^{(n)}$. The report conjectures, and then notes that the identity is analyti...

open_problem | open | MathDB 页面 | 原文线索

V2-046875 | MATHDB-401758

The top-heavy conjecture for matroids

摘录 / EXCERPT: Let M be a rank-d matroid, and let W_k be the number of rank-k flats of M. Then ... for every $k\le \frac d2$.

conjecture | open | MathDB 页面 | 原文线索

V2-046876 | MATHDB-401759

Kähler package for combinatorially defined intersection cohomology groups

摘录 / EXCERPT: The key problem is to prove the Kähler package-Poincaré duality, the hard Lefschetz property, and the Hodge-Riemann relation-of the combinatorially defined intersection cohomology...

open_problem | open | MathDB 页面 | 原文线索

V2-046877 | MATHDB-401798

Scattering map for linear waves in a subextremal Reissner-Nordström interior

摘录 / EXCERPT: Let $g_{M,Q}=-(1-2M/r+Q^2/r^2)\,dt^2+(1-2M/r+Q^2/r^2)^{-1}\,dr^2+r^2g_{\mathbb S^2}$ be a subextremal Reissner-Nordström metric with $0<|Q|<M$. For solutions of $\Box_g\psi=0$ in t...

open_problem | open | MathDB 页面 | 原文线索

V2-046878 | MATHDB-401835

Extremal decomposition of the free Ising state on a Cayley tree

摘录 / EXCERPT: For the ferromagnetic nearest-neighbor Ising model on the Cayley tree τ_k with branching ratio k, and in particular for temperatures below the spin-glass temperature...

open_problem | open | MathDB 页面 | 原文线索

V2-046879 | MATHDB-401891

Infinite-board strong Ramsey-game separation problem

摘录 / EXCERPT: Does there exist a graph H such that $R(K_\omega,H)$ is a first-player win, but $R(K_\omega\sqcup K_\omega,H)$ is a draw?

open_problem | open | MathDB 页面 | 原文线索

V2-046880 | MATHDB-401894

Ramsey equivalence of a clique and a clique plus a star

摘录 / EXCERPT: Determine the largest $t=t(k)$ such that $K_k\\sim_2 K_k+K_{1,t}$; here $\\sim_2$ denotes equality of the sets of 2-Ramsey-minimal graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-046881 | MATHDB-401935

Matroid Max-Min Allocation

摘录 / EXCERPT: Given a matroid $M=(X,\mathcal{I})$, a bipartite graph $G=(X\mathbin{\dot\cup}W,E)$, and nonnegative resource sizes $p_j\ge 0$ for $j\in W$, find a basis $S\in\mathcal{B}(M)$ and a...

open_problem | open | MathDB 页面 | 原文线索

V2-046882 | MATHDB-401943

Matroids with cardinality constraints on the intersection

摘录 / EXCERPT: Given two matroids $M_1=(E,\mathcal B_1)$ and $M_2=(E,\mathcal B_2)$ on a common ground set E, with base sets $\mathcal B_1$ and $\mathcal B_2$, two cost functions...

open_problem | open | MathDB 页面 | 原文线索

V2-046883 | MATHDB-401946

Equality of the longest-\a15-sequence matroid and generic 3-rigidity

摘录 / EXCERPT: Our ambitious conjecture is that eqa0 displaystyle (E(Kn),r5)nn) is the generic 3-rigidity matroid.

open_problem | open | MathDB 页面 | 原文线索

V2-046884 | MATHDB-401948

A special case of submodular function minimization

摘录 / EXCERPT: Let r be the rank function on a ground set V of a linear matroid, provided via an explicit representation over a field, and let $z:V\to\mathbb{R}$. Solving ... is a special case...

open_problem | open | MathDB 页面 | 原文线索

V2-046885 | MATHDB-401949

Linearly representable functions

摘录 / EXCERPT: Let f be a polymatroid function on a ground set V, meaning that f is integer-valued, monotone increasing, and submodular. The function f is called linearly representable if...

open_problem | open | MathDB 页面 | 原文线索

V2-046886 | MATHDB-402023

Largest connected graph Ramsey-equivalent to a clique

摘录 / EXCERPT: Determine the largest value of $t=t(k)$ such that there is a connected graph H on t vertices for which K_k and K_k+H are Ramsey equivalent.

open_problem | open | MathDB 页面 | 原文线索

V2-046887 | MATHDB-402113

Conlon-Lee conjecture on extremal numbers of bipartite graphs

摘录 / EXCERPT: Let H be a bipartite graph such that in one of the parts all the degrees are at most r and H does not contain $K_{r,r}$ as a subgraph. Then there exists some $\elta>0$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-046888 | MATHDB-402116

Conlon-Lee subdivision extremal-number conjecture

摘录 / EXCERPT: Let $k\ge 2$ be an even integer and let H be a graph. Then there exists some $\delta>0$ such that $\operatorname{ex}(n,H^{k-1})=O\!\left(n^{1+1/k-\delta}\right)$, where H^{k-1}...

conjecture | open | MathDB 页面 | 原文线索

V2-046889 | MATHDB-402117

Conjecture 7 on online random Ramsey numbers

摘录 / EXCERPT: The diagonal online random Ramsey numbers satisfy ... The off-diagonal online random Ramsey numbers, for fixed $m\ge 3$ and $n\to\infty$, satisfy ...

conjecture | open | MathDB 页面 | 原文线索

V2-046890 | MATHDB-402144

Exponential upper bound for edge-ordered Ramsey numbers

摘录 / EXCERPT: Does every edge-ordered graph H on n vertices satisfy $r_{\mathrm{edge}}(H)=2^{O(n)}$?

open_problem | open | MathDB 页面 | 原文线索

V2-046891 | MATHDB-402302

Log-concavity and unimodality of the matroid independence-set sequence

摘录 / EXCERPT: For a matroid M, let $f(M)=(f_0,\dots,f_r)$, where f_d is the number of independent sets of size d. Then $f(M)$ is unimodal and log-concave.

open_problem | open | MathDB 页面 | 原文线索

V2-046892 | MATHDB-402303

Finite-ground-set Matroid-Based Valuation Conjecture

摘录 / EXCERPT: For each n, does there exist a finite integer $m\ge n$ such that the matroid-based valuations on $[m]$ generate all gross-substitute functions on $[n]$?

conjecture | open | MathDB 页面 | 原文线索

V2-046893 | MATHDB-402321

Positivity of Matroids

摘录 / EXCERPT: Let $P(M)$ be the matroid polytope of M. Conjecture 1. - The coefficients of the Ehrhart polynomial of $P(M)$ are positive. - The h^*-vector of $P(M)$ is unimodal. The conjectu...

open_problem | open | MathDB 页面 | 原文线索

V2-046894 | MATHDB-402615

Frankl-Füredi extremal problem for t-intersecting codes

摘录 / EXCERPT: Given a code $\mathcal F\subseteq [m]^n$ such that every $x,y\in\mathcal F$ satisfy ... determine the maximum possible value of $|\mathcal F|$. The report states that Ahlswede-Khacha...

open_problem | open | MathDB 页面 | 原文线索

V2-046895 | MATHDB-402625

Girth Ramsey theorem (Theorem 3)

摘录 / EXCERPT: Given an integer $g\ge 2$, a graph F with $\operatorname{girth}(F)>g$, and a natural number r, there exists a graph H with $\operatorname{girth}(H)>g$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-046896 | MATHDB-402626

Girth Ramsey theorem for induced cycles

摘录 / EXCERPT: Given a graph F and positive integers r and n, there exists a graph H with $H\longrightarrow(F)_r$ such that, for every $m\in[2,n]$, every induced m-cycle of H is conta...

open_problem | open | MathDB 页面 | 原文线索

V2-046897 | MATHDB-402822

Extremal number of fixed-length discrete chains

摘录 / EXCERPT: Let $\delta=(\delta_1,\ldots,\delta_k)$ be a fixed sequence of k positive real numbers. A $(k+1)$-tuple $(p_1,\ldots,p_{k+1})$ of distinct points in $\mathbb{R}^d$ is called a...

open_problem | open | MathDB 页面 | 原文线索

V2-046898 | MATHDB-403015

Positive combinatorial interpretation for Kronecker and plethysm decomposition multiplicities

摘录 / EXCERPT: Find a positive combinatorial interpretation for the decomposition multiplicities arising in the Kronecker and plethysm products of Schur functions.

open_problem | open | MathDB 页面 | 原文线索

V2-046899 | MATHDB-403123

Combinatorial equivalence of a computability-theoretic question

摘录 / EXCERPT: Does every coloring c on the d-ary string space admit a c-computable variable-word infinite solution?

open_problem | open | MathDB 页面 | 原文线索

V2-046900 | MATHDB-403124

Does Ramsey’s theorem for pairs imply the tree-coloring principle?

摘录 / EXCERPT: Does $\mathsf{RCA}_0 + \mathsf{RT}^2_2$ imply TT_1?

open_problem | open | MathDB 页面 | 原文线索

V2-046901 | MATHDB-403129

Uniform reduction of closed choice to open Ramsey’s theorem

摘录 / EXCERPT: Does the Weihrauch reduction $\mathrm{C}_{\mathbb N}\leq_W\Sigma^0_1\text{-RT}$ hold, where $\Sigma^0_1\text{-RT}$ is the problem of finding a homogeneous solution for an open set...

open_problem | open | MathDB 页面 | 原文线索

V2-046902 | MATHDB-403169

Vanishing of higher cohomology of the determinant line bundle on a matroid wonderful compactification

摘录 / EXCERPT: Let $L\subset K^E$ be a realization of a matroid M over an algebraically closed field K of arbitrary characteristic, and let W_L be the associated wonderful compactification....

open_problem | open | MathDB 页面 | 原文线索

V2-046903 | MATHDB-403435

Baez-Dolan Stabilization Theorem

摘录 / EXCERPT: Let $0 \le n$ and let M be an n-truncated monoidal combinatorial model category with cofibrant unit. Then, for $k \ge n+2$, the comparison functors...

open_problem | open | MathDB 页面 | 原文线索

V2-046904 | MATHDB-403463

Terao’s freeness conjecture

摘录 / EXCERPT: Terao’s conjecture asserts that the algebraic notion of freeness of a hyperplane arrangement depends only on the combinatorics of the underlying arrangement over a fixed field.

conjecture | open | MathDB 页面 | 原文线索

V2-046905 | MATHDB-403500

Combinatorial interpretations of Kostka-Foulkes coefficients beyond type A

摘录 / EXCERPT: Beyond type A, find combinatorial interpretations for the coefficients of the Kostka-Foulkes polynomials $K_{\\lambda,\\mu}(q)$.

open_problem | open | MathDB 页面 | 原文线索

V2-046906 | MATHDB-403677

Enumeration of rank-3 matroids

摘录 / EXCERPT: Let $m(n,r)$ denote the number of rank-r matroids on a ground set of size n. As $n\to\infty$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-046907 | MATHDB-403685

Erdős’s limit problem for Ramsey numbers

摘录 / EXCERPT: Prove that the quantity $R(n)^{1/n}$ tends to a limit.

open_problem | open | MathDB 页面 | 原文线索

V2-046908 | MATHDB-403833

Combinatorial interpretation of restriction coefficients

摘录 / EXCERPT: Let V^λ be a polynomial representation of $\mathrm{GL}_n(\mathbb{C})$ indexed by a partition λ with at most n parts. Under restriction to the subgroup S_n of pe...

open_problem | open | MathDB 页面 | 原文线索

V2-046909 | MATHDB-403923

Matroid invariance of the geproci property

摘录 / EXCERPT: If Z and Z' realize the same matroid, is it true that Z is geproci if and only if Z' is?

open_problem | open | MathDB 页面 | 原文线索

V2-046910 | MATHDB-403927

Geproci realizations of uniform matroids

摘录 / EXCERPT: “Realizations of a (nontrivial) uniform matroid are never geproci. These correspond to point configurations in linear general position. Is it true that such point configurations ar...

open_problem | open | MathDB 页面 | 原文线索

V2-046911 | MATHDB-403944

Refined combinatorial models and rational generalization of Theorem 1

摘录 / EXCERPT: Refine the combinatorial models for rational Catalan objects and generalize Theorem 1 in that setting.

open_problem | open | MathDB 页面 | 原文线索

V2-046912 | MATHDB-403968

Combinatorial interpretations for regions of generalized Catalan and Shi arrangements

摘录 / EXCERPT: For positive integers (mathbf{k}=(k1,ldots,kn)), define the arrangements displayed in the report by ... and ... Find a combinatorial interpretation for the regions of...

open_problem | open | MathDB 页面 | 原文线索

V2-046913 | MATHDB-404007

Erdős-McKay conjecture for induced subgraph edge counts in C-Ramsey graphs

摘录 / EXCERPT: There is a constant $\[?δ_C>0$ such that, for any C-Ramsey graph G on n vertices and any integer x with $0\,\le x\le δ_C n^2$, there is a subset $U\subseteq V(G)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-046914 | MATHDB-404012

Exact combinatorial characterization of the rank of sparse random graphs

摘录 / EXCERPT: For a fixed constant $c\ne e$, let $G\sim G(n,c/n)$ be an Erdős-Rényi random graph, with adjacency matrix $A(G)$, and let $G\sim G(n,n,c/n)$ be a bipartite Erdős-Rényi random graph...

open_problem | open | MathDB 页面 | 原文线索

V2-046915 | MATHDB-404057

Odd powers of extremal positive semidefinite forms

摘录 / EXCERPT: For every number of variables n, every even degree m, every extremal form $p\in P_{n,m}\setminus\Sigma_{n,m}$, and every integer $k\ge 0$, is it true that...

open_problem | open | MathDB 页面 | 原文线索

V2-046916 | MATHDB-404071

A matroid whose beta-invariant equals the degree of the inverse linear space

摘录 / EXCERPT: For a linear space $L\subseteq\mathbb{C}^m$ viewed as a space of diagonal $m\times m$ matrices, let L^{-1} be the Zariski closure of the inverse points and let $d=\dim(L)$. Is th...

open_problem | open | MathDB 页面 | 原文线索

V2-046917 | MATHDB-404083

A matroidal formula for amoeba dimension

摘录 / EXCERPT: Let M be an arbitrary loopless matroid on $[n]$, and let $F\subseteq\mathbb{R}^n$ be the support of the matroid fan of M. Is ... equal to ...

open_problem | open | MathDB 页面 | 原文线索

V2-046918 | MATHDB-404350

Koszulness of Chow rings of matroids

摘录 / EXCERPT: Dotsenko’s conjecture: the Chow rings of matroids are Koszul.

open_problem | open | MathDB 页面 | 原文线索

V2-046919 | MATHDB-404407

Extremal elastic materials that remain functional under finite deformation

摘录 / EXCERPT: Can one obtain all possible materials that, in addition to being extremal, still function under finite deformation?

open_problem | open | MathDB 页面 | 原文线索

V2-046920 | MATHDB-404659

Combinatorial interpretation of analytically extended k-arrangement moments

摘录 / EXCERPT: Extend the definition of $A(n,k)$ by ... to $n\in\mathbb{N}$ and $k\in\mathbb{R}$. Does $A(n,k)$ have a combinatorial interpretation when k is an arbitrary negative integer?

open_problem | open | MathDB 页面 | 原文线索

V2-046921 | MATHDB-404727

Combinatorial structure of KRW and Lipschitz polytopes

摘录 / EXCERPT: Understand the combinatorial structure (e.g., computing face numbers) of Kantorovich-Rubinstein-Wasserstein (KRW) and Lipschitz polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-046922 | MATHDB-404944

Hypercube statistics: determine or estimate the extremal densities

摘录 / EXCERPT: For integers $d\ge 1$ and $0\le s\le 2^d$, let Q_n be the n-dimensional hypercube, with $n\ge d$. For a subset A of the vertices of Q_n, let $\lambda(n,d,s,A)$ be the fract...

open_problem | open | MathDB 页面 | 原文线索

V2-046923 | MATHDB-404953

Fox-Grinshpun-Liebenau-Person-Szabó conjecture for the minimum degree of minimal Ramsey graphs

摘录 / EXCERPT: For every fixed integer $k\ge 3$, $P_r(k)=\Theta\!\left(r\,\bigl(f_{k,k+1}(r)\bigr)^2\right)$, where $P_r(k)$ is the minimum order of a K_{k+1}-free r-color pattern forcing a s...

conjecture | open | MathDB 页面 | 原文线索

V2-046924 | MATHDB-404959

Asymptotic value of the off-diagonal Ramsey number $R(3,k)$

摘录 / EXCERPT: Conjecture 2. $R(3,k)=\left(\frac12+o(1)\right)\frac{k^2}{\log k}$ as $k\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-046925 | MATHDB-404992

Rayleigh inequality for matroid basis-generating polynomials

摘录 / EXCERPT: Conjecture (Huh-Schröter-Wang). If M is the set of bases of a matroid on ground set E, then $R(8/7,\{\mathbf{1}\})$ holds for ... where $x^B=\prod_{e\in B}x_e$. The same statemen...

open_problem | open | MathDB 页面 | 原文线索

V2-046926 | MATHDB-405044

Exact value of the Tverberg-type extremal function for unions of convex sets

摘录 / EXCERPT: What is the least integer $f=f_r(d,s_1,\ldots,s_r)$ such that for any set P of f points in $\mathbb{R}^d$ there is a partition into r pairwise disjoint sets...

open_problem | open | MathDB 页面 | 原文线索

V2-046927 | MATHDB-405051

Conjectured exponent for odd cycle-complete Ramsey numbers

摘录 / EXCERPT: For each fixed odd integer $\ell>1$, as $k\to\infty$, the cycle-complete Ramsey number satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-046928 | MATHDB-405052

Fox's conjecture on multicolour Ramsey numbers of odd cycles

摘录 / EXCERPT: For every $\varepsilon>0$, there exists $\ell"); such that, for all sufficiently large k, ...

conjecture | open | MathDB 页面 | 原文线索

V2-046929 | MATHDB-405092

Combinatorial explanation of a three-quarter-plane walk identity

摘录 / EXCERPT: Fix a step set $S\subseteq\{-1,0,1\}^2$ and let $\mathcal C=\{(i,j):i\ge 0\text{ or }j\ge 0\}$, with boundary $\partial\mathcal C=\{(0,j):j\le 0\}\cup\{(i,0):i\le 0\}$. For positiv...

open_problem | open | MathDB 页面 | 原文线索

V2-046930 | MATHDB-405094

Combinatorial interpretation of coefficients related to the inverse Bessel function

摘录 / EXCERPT: Let ... Does the sequence $(k_n)_{n\ge 0}$, or the integer sequence $((n+1)!(n+2)!k_n)_{n\ge 0}$, have a simple combinatorial interpretation?

open_problem | open | MathDB 页面 | 原文线索

V2-046931 | MATHDB-405145

Existence of weighted envy-freeness up to one item (WEF1)

摘录 / EXCERPT: Given a finite set of indivisible items M, agents $N=\{1,\ldots,n\}$ with positive entitlements $w_i>0$, and additive valuations $v_i:2^M\to\mathbb{R}$, does there always exist a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046932 | MATHDB-405155

Quadratic Easy Coefficients Conjecture

摘录 / EXCERPT: Let f be a finite sum of quadratic monomial rotation-symmetric Boolean functions, and let $W_f(n)$ denote the Hamming weight of its induced Boolean function on n variables. Let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046933 | MATHDB-405166

Real-rootedness conjecture for toric g-contribution polynomials

摘录 / EXCERPT: For every toric g-contribution polynomial $p(t)$ in the family defined by Ehrenborg, Hetyei, and Readdy, all zeros of $p(t)$ are real; equivalently, $p(t)$ is real-rooted.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046934 | MATHDB-405178

Brenti's problem on palindromic real-rooted polynomials and h-vectors of polytopes

摘录 / EXCERPT: For every polynomial $f(t)\in\mathbb{Z}[t]$ that is monic, has nonnegative coefficients, is real-rooted, and is palindromic in the sense that $f(t)=t^d f(t^{-1})$ for $d=\deg f$, t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046935 | MATHDB-405184

Fey-Meester-Redig divisible-sandpile percolation question

摘录 / EXCERPT: Fix $d\ge 2$. Let $\eta=(\eta(x))_{x\in\mathbb{Z}^d}$ be an i.i.d. initial mass configuration for the divisible sandpile, with mean density ρ, and let...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046936 | MATHDB-405190

Pairwise independent correlation gap conjecture

摘录 / EXCERPT: For every integer $n\ge 1$, every monotone submodular function $f:2^{[n]}\to\mathbb{R}_{\ge 0}$, and every marginal vector $x\in[0,1]^n$ with $f^{++}(x)>0$, the correlation gap und...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046937 | MATHDB-405195

Conjecture on facet ideals of chessboard complexes

摘录 / EXCERPT: Let $\Delta_{m,n}$ be the chessboard complex, whose faces are the matchings in the complete bipartite graph $K_{m,n}$, with $n\geq m\geq 1$. Let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046938 | MATHDB-405203

Green’s Problem 24 on affine copies of {0,1,3}

摘录 / EXCERPT: For a finite set $A\subset\mathbb Z$, let $M_{\{0,1,3\}}(A)$ be the number of pairs $(x,d)\in\mathbb Z\times(\mathbb Z\setminus\{0\})$ such that $\{x,x+d,x+3d\}\subseteq A$. Determ...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046939 | MATHDB-405206

Kauers-Koutschan 2023 conjecture on long increasing subsequences

摘录 / EXCERPT: Let $b_{2n,n}$ denote the number of permutations of length $2n$ that contain an increasing subsequence of length n, and define $B(z)=\sum_{n\geq 0} b_{2n,n}z^n$. The conjecture i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046940 | MATHDB-405209

Minimum forward and integral cycle basis problems

摘录 / EXCERPT: Given a weighted graph $G=(V,E,w)$, where $w:E\to\mathbb{R}_{\ge 0}$, find an integral cycle basis $\mathcal{B}$ minimizing its total weight...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-046941 | MATHDB-405211

Kiss-Sándor-Yang conjecture on sparse k-AP covering sets

摘录 / EXCERPT: ...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046942 | MATHDB-405217

Optimal alphabet size for synchronization strings

摘录 / EXCERPT: Let an ε-synchronization string of length n over an alphabet Σ be a word $S\in\Sigma^n$ such that, for every $0\le i<j<k\le n$, the insertion-deletion edit dis...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046943 | MATHDB-405230

Huang-Huh-Soskin-Wang conjecture on bounded ratios of Lorentzian polynomials

摘录 / EXCERPT: For every number of variables n and every normalized bounded ratio R on Lorentzian polynomials in n variables, the optimal bounding constant of R is at most 2:...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046944 | MATHDB-405245

Complexity of recognizing SDP exactness for Max-Cut on simple unweighted graphs

摘录 / EXCERPT: Given a finite simple unweighted graph $G=(V,E)$, decide whether the standard Goemans-Williamson semidefinite relaxation of Max-Cut has the same optimal value as the integer Max-Cu...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046945 | MATHDB-405250

Cohen-Macaulayness classification for Inc(N)-invariant chains of edge ideals

摘录 / EXCERPT: Let k be a field, let $R_m=k[x_1,\ldots,x_m]$, and let $I(C_n)\subseteq R_n$ be the edge ideal of the cycle graph C_n. For $n\ge 4$ and $r\ge 0$, set...

conjecture | claimed_progress | MathDB 页面 | 原文线索

B11 图论 / Graph Theory | 7,316 条

V2-046946 | MATHDB-315496

Four-particle monochromatic quantum graphs for all $D\geq4$

摘录 / EXCERPT: For which values of n and d do there exist bi-colored weighted graphs on n vertices and with d different colors such that all d monochromatic inherited vertex colorings h...

open_problem | open | MathDB 页面 | 原文线索

V2-046947 | MATHDB-315505

Chemical reaction networks and CRN-computable reals

摘录 / EXCERPT: Can the central equivalence and universality results connecting polynomial ODEs, chemical reaction networks, linear production protocols, and stochastic mean-field limits be assemb...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046948 | MATHDB-315506

Multivariate independence-polynomial lower bound

摘录 / EXCERPT: Can the known degree-sensitive lower bound for the hard-core partition function be extended from one common activity to arbitrary nonnegative vertex fugacities?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046949 | MATHDB-315528

Monochromatic quantum graphs in the diagonal family

摘录 / EXCERPT: Can a complete edge-coloured, complex-weighted graph be chosen so that perfect-matching amplitudes equal one for every monochromatic inherited vertex colouring and vanish for every...

open_problem | open | MathDB 页面 | 原文线索

V2-046950 | MATHDB-315534

Weak bipartite reconstruction with distinct vertex types

摘录 / EXCERPT: If a suitably 2-connected bipartite graph has distinct vertex types $\tau(v)=(\deg v,\{\!\{\deg u:u\sim v\}\!\})$, is it determined up to bipartite isomorphism by the multiset of i...

open_problem | open | MathDB 页面 | 原文线索

V2-046951 | MATHDB-315595

Irreducible vertices in definite tree lattices

摘录 / EXCERPT: Let T be a positive-definite weighted tree with exactly one vertex v satisfying $w(v)<d(v)$. Must T contain a vertex that is irreducible in its lattice $L(T)$?

open_problem | open | MathDB 页面 | 原文线索

V2-046952 | MATHDB-315667

Fajtlowicz’s graph-energy conjecture

摘录 / EXCERPT: Let G be a graph on n vertices with independence number $4\alpha(G)$. Let A be the adjacency matrix of G with eigenvalues $\lambda_1\geq\cdots\geq\lambda_n$. Fajtlowicz's g...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046953 | MATHDB-315678

Feige’s hypergraph Moore-bound conjecture

摘录 / EXCERPT: Feige's hypergraph Moore bound conjecture. For every $k\ge 3$, there exist constants $c_k,C_k>0$ such that, whenever

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046954 | MATHDB-315743

Stanley’s claw-free Schur-positivity conjecture

摘录 / EXCERPT: Stanley--Gasharov conjecture (1998). Every claw-free graph is Schur-positive.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046955 | MATHDB-315770

Ramsey-style hypergraph construction

摘录 / EXCERPT: Let $H(n)$ be the largest integer k such that there exists a hypergraph $(V,\mathcal H)$ with $|V|=k$, no isolated vertices, and no partition of size greater than n, where a hy...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046956 | MATHDB-315988

Shephard's problem (a.k.a. Dürer's conjecture)

摘录 / EXCERPT: Let $P \subset \mathbb{R}^3$ be a convex polytope of dimension 3, with vertex set V, edge set E, and set of (planar, convex) faces $\mathcal{F}$; let $G = (V, E)$ be its edge...

conjecture | open | MathDB 页面 | 原文线索

V2-046957 | MATHDB-315992

Brouwer's conjecture

摘录 / EXCERPT: Let G be a simple undirected graph on n vertices with $m(G)$ edges, let $A(G)$ be its adjacency matrix, let $D(G)$ be the diagonal matrix of its vertex degrees, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-046958 | MATHDB-315993

Open questions

摘录 / EXCERPT: Let $G=(V,E)$ be a finite simple graph. A set $D\subseteq V$ is a dominating set of G if every vertex in $V\setminus D$ has a neighbour in D, and $\gamma(G)$ denotes the minimu...

open_problem | open | MathDB 页面 | 原文线索

V2-046959 | MATHDB-315995

1-factorization conjecture

摘录 / EXCERPT: Let G be a finite simple graph. A 1-factor of G is a spanning 1-regular subgraph of G, i.e. a perfect matching. A 1-factorization of G is a partition of the edge set...

conjecture | open | MathDB 页面 | 原文线索

V2-046960 | MATHDB-315996

Cereceda's conjecture

摘录 / EXCERPT: For a finite simple undirected graph G, the degeneracy of G is the least integer d such that every non-empty subgraph of G has a vertex of degree at most d. For a positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-046961 | MATHDB-316001

Hadwiger conjecture (graph theory)

摘录 / EXCERPT: Let G be a finite loopless undirected graph. A graph H is a minor of G if H can be obtained from a subgraph of G by a sequence of edge contractions; equivalently, K_t i...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046962 | MATHDB-316004

List edge coloring conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite loopless graph (multiple edges permitted). Two edges are adjacent if they share an endpoint. A proper edge coloring of G with color set C is a map...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046963 | MATHDB-316005

Overfull conjecture

摘录 / EXCERPT: All graphs here are finite, simple and undirected. For a graph G write $V(G)$ for its vertex set, $E(G)$ for its edge set, $n=|V(G)|$, and $\Delta(G)$ for its maximum degree. Cal...

conjecture | open | MathDB 页面 | 原文线索

V2-046964 | MATHDB-316011

Negami's conjecture

摘录 / EXCERPT: All graphs are finite and undirected. For graphs C and H, a covering map from C to H is a surjection f from the vertex set of C onto the vertex set of H such that, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-046965 | MATHDB-316016

Barnette's conjecture

摘录 / EXCERPT: Let G be a finite undirected graph with the following four properties: 1. G is planar: it can be embedded in the Euclidean plane with no two edges crossing. 2. G is 3-verte...

conjecture | open | MathDB 页面 | 原文线索

V2-046966 | MATHDB-316017

Gilbert-Pollak conjecture

摘录 / EXCERPT: For a finite set $P\subset\mathbb{R}^2$ with $|P|\ge 2$, and for any finite set $Q\subset\mathbb{R}^2$, let ... the minimum being taken over all spanning trees T of the complete gr...

conjecture | open | MathDB 页面 | 原文线索

V2-046967 | MATHDB-316022

Grünbaum-Nash-Williams conjecture

摘录 / EXCERPT: A graph G is toroidal if it admits an embedding in the torus, i.e. there is a drawing of G on the closed orientable surface of genus 1 in which vertices are distinct points,...

conjecture | open | MathDB 页面 | 原文线索

V2-046968 | MATHDB-316023

Lovász conjecture

摘录 / EXCERPT: Call a finite simple graph Γ vertex-transitive if its automorphism group acts transitively on $V(\Gamma)$, and call a cycle in Γ Hamiltonian if it passes through ever...

conjecture | open | MathDB 页面 | 原文线索

V2-046969 | MATHDB-316024

Oberwolfach problem

摘录 / EXCERPT: For an integer $t\ge 1$ and integers $m_1,m_2,\dots,m_t\ge 3$, let ... denote the vertex-disjoint union of cycles of lengths $m_1,\dots,m_t$, a 2-regular graph on...

open_problem | open | MathDB 页面 | 原文线索

V2-046970 | MATHDB-316025

reconstruction conjecture

摘录 / EXCERPT: (1) Reconstruction. For every pair of graphs G and H with $|V(G)| = |V(H)| \ge 3$, if $D(G) = D(H)$ then $G \cong H$.

conjecture | open | MathDB 页面 | 原文线索

V2-046971 | MATHDB-316026

Sumner's conjecture

摘录 / EXCERPT: Call a digraph T a polytree if T is obtained from a (finite, undirected) tree by assigning to each edge one of its two possible directions, and call a digraph D a tournament...

conjecture | open | MathDB 页面 | 原文线索

V2-046972 | MATHDB-316027

Szymanski's conjecture

摘录 / EXCERPT: For $n\ge 1$ let $V_n=\{0,1\}^n$ and let H_n be the doubly directed n-dimensional hypercube: the directed graph on vertex set V_n whose arc set is ... so that each edge of the...

conjecture | open | MathDB 页面 | 原文线索

V2-046973 | MATHDB-316028

Tuza's conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite simple undirected graph. Call a set of three edges of G a triangle if these edges form a cycle of length 3, i.e. if they are $\{u,v\},\{v,w\},\{u,w\}$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046974 | MATHDB-316029

Vizing's conjecture

摘录 / EXCERPT: All graphs considered are finite, simple, and undirected. For a graph G and a vertex $v \in V(G)$, let $N[v] = \{v\} \cup \{u \in V(G) : uv \in E(G)\}$ denote the closed neighbou...

conjecture | open | MathDB 页面 | 原文线索

V2-046975 | MATHDB-316030

Walecki's theorem for hypergraphs

摘录 / EXCERPT: In graph theory, a branch of mathematics, a Hamiltonian decomposition of a given graph is a partition of the edges of the graph into Hamiltonian cycles. Hamiltonian decompositions...

open_problem | open | MathDB 页面 | 原文线索

V2-046976 | MATHDB-316031

Zarankiewicz problem

摘录 / EXCERPT: For integers $m,n\ge 1$ and $s,t\ge 1$, let $G=(U\cup V,E)$ be a bipartite graph with disjoint vertex parts U,V satisfying $|U|=m$, $|V|=n$, and with edge set...

open_problem | open | MathDB 页面 | 原文线索

V2-046977 | MATHDB-316032

imbalance conjecture

摘录 / EXCERPT: Let G be a finite simple undirected graph with vertex set $V(G)$ and edge set $E(G)$, and for $u\in V(G)$ let $\deg(u)$ denote the number of neighbours of u in G. For an edge...

conjecture | solved | MathDB 页面 | 原文线索

V2-046978 | MATHDB-316034

Ryser's conjecture

摘录 / EXCERPT: Let $r \ge 2$ be an integer and let $H = (V, E)$ be a hypergraph, i.e. V is a finite set and E is a finite collection of subsets of V. Say that H is r-uniform if...

conjecture | open | MathDB 页面 | 原文线索

V2-046979 | MATHDB-316036

Sidorenko's conjecture

摘录 / EXCERPT: For a finite simple graph H with vertex set $V(H)=\{1,\dots,k\}$ and edge set $E(H)$, and for a graphon W (that is, a symmetric measurable function $W\colon [0,1]^2 \to [0,1]$)...

conjecture | open | MathDB 页面 | 原文线索

V2-046980 | MATHDB-316037

Woodall's conjecture

摘录 / EXCERPT: Let $D=(V,E)$ be a finite directed graph. For a set $S\subseteq V$ with $S\neq\emptyset$ and $S\neq V$, write ... Call $\partial^+(S)$ a $\emph{dicut}$ of D if...

conjecture | open | MathDB 页面 | 原文线索

V2-046981 | MATHDB-316187

Dinitz-Garg-Goemans conjecture

摘录 / EXCERPT: Let $G=(V,A)$ be a directed graph with a distinguished source $s\in V$, a set of terminals $T\subseteq V$, and a demand $d_t\geq 0$ for each $t\in T$. A flow $x=(x_a)_{a\in A}$ wit...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046982 | MATHDB-316188

Kahn-Kalai conjecture

摘录 / EXCERPT: Let X be a finite set and let 2^X denote the family of all subsets of X. For $p \in [0,1]$ let μ_p be the product probability measure on 2^X given by ... and for...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046983 | MATHDB-316189

Blankenship-Oporowski conjecture

摘录 / EXCERPT: For an integer $k\ge 1$, a k-page book consists of a line ℓ (the spine) together with k half-planes (the pages), each having ℓ as its boundary. A book embedding of a...

conjecture | open | MathDB 页面 | 原文线索

V2-046984 | MATHDB-316190

Hedetniemi's conjecture

摘录 / EXCERPT: In graph theory, Hedetniemi's conjecture, formulated by Stephen T. Hedetniemi in 1966, concerns the connection between graph coloring and the tensor product of graphs. This conject...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-046985 | MATHDB-316191

Kelmans-Seymour conjecture

摘录 / EXCERPT: Let G be a finite simple graph. Say that G is 5-connected if $|V(G)| \ge 6$ and, for every set $S \subseteq V(G)$ with $|S| \le 4$, the graph $G - S$ is connected. Say that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046986 | MATHDB-316192

Goldberg-Seymour conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite loopless multigraph, i.e. parallel edges are permitted but no edge joins a vertex to itself. For $v \in V$ let $d_G(v)$ be the number of edges of E inci...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046987 | MATHDB-316193

Alspach's conjecture

摘录 / EXCERPT: For a graph G, a decomposition of G into cycles of lengths $m_1,\dots,m_t$ is a set of t simple cycles $C_1,\dots,C_t$ in G, with C_i having length m_i, such that no tw...

conjecture | open | MathDB 页面 | 原文线索

V2-046988 | MATHDB-316194

Alon-Saks-Seymour conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite simple graph. For disjoint nonempty sets $L,R\subseteq V$ such that $\{u,v\}\in E$ for all $u\in L$ and $v\in R$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-046989 | MATHDB-316195

Read-Hoggar conjecture

摘录 / EXCERPT: Read's conjecture is a conjecture, first made by Ronald Read, about the unimodality of the coefficients of chromatic polynomials in the context of graph theory. In 1974, S. G. Hogg...

conjecture | open | MathDB 页面 | 原文线索

V2-046990 | MATHDB-316196

Scheinerman's conjecture

摘录 / EXCERPT: In mathematics, Scheinerman's conjecture, now a theorem, states that every planar graph is the intersection graph of a set of line segments in the plane. This conjecture was formul...

conjecture | open | MathDB 页面 | 原文线索

V2-046991 | MATHDB-316197

Erdős-Menger conjecture

摘录 / EXCERPT: In the mathematical discipline of graph theory, Menger's theorem says that in a finite graph, the size of a minimum cut set is equal to the maximum number of disjoint paths that ca...

conjecture | open | MathDB 页面 | 原文线索

V2-046992 | MATHDB-316198

Road coloring conjecture

摘录 / EXCERPT: In graph theory the road coloring theorem, known previously as the road coloring conjecture, deals with synchronized instructions. The issue involves whether by using such instruct...

conjecture | open | MathDB 页面 | 原文线索

V2-046993 | MATHDB-316199

Robertson-Seymour theorem

摘录 / EXCERPT: In graph theory, the Robertson-Seymour theorem (also called the graph minors theorem) states that the undirected graphs, partially ordered by the graph minor relationship, form a w...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-046994 | MATHDB-316200

Strong perfect graph conjecture

摘录 / EXCERPT: All graphs are finite, simple, and undirected. For a graph G, let $\omega(G)$ denote the number of vertices in a largest clique of G, and let $\chi(G)$ denote the least number...

conjecture | open | MathDB 页面 | 原文线索

V2-046995 | MATHDB-316201

Toida's conjecture

摘录 / EXCERPT: Let n be a positive integer, let $\mathbb{Z}_n=\mathbb{Z}/n\mathbb{Z}$, and for a subset $S\subseteq\mathbb{Z}_n\setminus\{0\}$ let $\mathrm{Circ}(n,S)$ denote the circulant digr...

conjecture | open | MathDB 页面 | 原文线索

V2-046996 | MATHDB-316215

Burr-Erdős conjecture

摘录 / EXCERPT: For an undirected graph G, the degeneracy of G is the minimum integer p such that every subgraph of G contains a vertex of degree at most p; a graph of degeneracy at most...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-046997 | MATHDB-316243

Seymour's image-size conjecture for finite reflexive relations

摘录 / EXCERPT: Let $\Gamma=(V,E)$ be a finite reflexive relation, and let j be an integer. Write $\Gamma^j(x)$ for the j-fold image of x and $\Gamma^{-1}(x)$ for its inverse image. Seymour'...

conjecture | open | MathDB 页面 | 原文线索

V2-046998 | MATHDB-316252

Normality and Cohen-Macaulayness conjecture for Gaussian Bayesian network ideals

摘录 / EXCERPT: Normality and Cohen-Macaulayness conjecture. The quotient ring

conjecture | open | MathDB 页面 | 原文线索

V2-046999 | MATHDB-316253

Gorenstein conjecture for directed tree Gaussian models

摘录 / EXCERPT: Gorenstein tree conjecture. The quotient ring $\mathbb{C}[\Sigma]/I_T$ is Gorenstein if and only if every vertex of T has degree at most three.

conjecture | open | MathDB 页面 | 原文线索

V2-047000 | MATHDB-316269

Jackson's conjecture on chromatic roots of 3-connected graphs

摘录 / EXCERPT: Let G be a 3-connected graph that is not bipartite of odd order. A chromatic root is a real or complex zero of the chromatic polynomial $P(G,x)$. Jackson's conjecture. The graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047001 | MATHDB-316270

Dong-Koh conjecture on independent separators of graphs with chromatic roots

摘录 / EXCERPT: Let G be a graph, let $c(G-S)$ denote the number of connected components of $G-S$, and let $(1,2)$ be the interval of real numbers between 1 and 2. Dong-Koh's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-047002 | MATHDB-316287

The unsolvability conjecture for twisted graphs

摘录 / EXCERPT: Unsolvability conjecture. For every twisted graph T, there is some $\boldsymbol{\theta}$ respecting $\prec$ for which the system

conjecture | open | MathDB 页面 | 原文线索

V2-047003 | MATHDB-316288

The constraining conjecture for irredundant minimal twisted graphs

摘录 / EXCERPT: Constraining conjecture. Every irredundant $\prec$-minimal twisted graph $T=(\vec{G},G,\prec,A)$ is constraining.

conjecture | open | MathDB 页面 | 原文线索

V2-047004 | MATHDB-316289

The constraining conjecture for minimal simply three-edge-connected graphs

摘录 / EXCERPT: Constraining conjecture. For every minimal simply three edge connected, totally cyclic directed graph $\vec{G}$, there is a total order $<$ and a constraining twisted graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047005 | MATHDB-316290

The constraining conjecture for minimal three-edge-connected simple graphs

摘录 / EXCERPT: Constraining conjecture. For every minimal three edge connected simple graph G, there is a directed graph $\vec{G}$, a total order $<$, and a constraining twisted graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047006 | MATHDB-316304

Existence of bounded local encoding delays for cyclic multicast networks

摘录 / EXCERPT: Bounded-delay existence conjecture. There exists a finite value I, depending on the graph and in particular on $T(e)$, such that one can choose delays satisfying $i_e(p)<I$ for e...

conjecture | open | MathDB 页面 | 原文线索

V2-047007 | MATHDB-316314

Linear upper-bound conjecture for random-walk speed-up

摘录 / EXCERPT: Let G be a graph, let $k\geq 1$ be the number of random walks, and write $S^k(G)$ for their speed-up in covering the graph. Linear speed-up conjecture. For any graph G and any...

conjecture | open | MathDB 页面 | 原文线索

V2-047008 | MATHDB-316315

Logarithmic lower-bound conjecture for random-walk speed-up

摘录 / EXCERPT: Let G be a graph on n vertices, let k satisfy $n\geq k\geq 1$, and write $S^k(G)$ for the speed-up in covering the graph. Logarithmic speed-up conjecture. For any graph G a...

conjecture | open | MathDB 页面 | 原文线索

V2-047009 | MATHDB-316327

The geometric realization conjecture for triangulated closed orientable surfaces of genus at most four

摘录 / EXCERPT: Geometric realization conjecture. Every triangulation of a closed orientable surface of genus at most 4 is realizable in $\mathbb{R}^3$ by straight edges, flat triangles, and wit...

conjecture | open | MathDB 页面 | 原文线索

V2-047010 | MATHDB-316383

Oriented-cycle conjecture for knot cycle presentations

摘录 / EXCERPT: Let K be a non-trivial knot and let D be an n-cycle presentation of K. Let $G(D)$ be the directed graph whose vertices correspond to the edges of D, with an oriented edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047011 | MATHDB-316460

Vertex-transitive balanced graph cycle conjecture

摘录 / EXCERPT: Let X be a connected vertex-transitive balanced graph. An (l,t)-cycle is the graph described by the paper's preceding definitions. Vertex-transitive balanced graph cycle conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-047012 | MATHDB-316461

Cubic balanced graph twin conjecture

摘录 / EXCERPT: Let X be a cubic balanced graph. Two vertices are non-trivial twins when they form a non-trivial twin class, namely distinct vertices with the same relevant neighbourhood as defi...

conjecture | open | MathDB 页面 | 原文线索

V2-047013 | MATHDB-316497

The two-dimensionality conjecture for ideals in the triplet vertex algebra

摘录 / EXCERPT: Let $p\geq 2$, let $A(\mathcal{W}(p))$ be the Zhu algebra of the triplet vertex algebra, and let $h_{i,1}$ denote the relevant conformal weights. For each i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-047014 | MATHDB-316506

Bollobás-Komlós conjecture on embedding bounded-bandwidth graphs

摘录 / EXCERPT: Let H and G be graphs with $|V(H)|=|V(G)|=n$, and write $\bw(H)$ for the band-width of H, $\chi(H)$ for its chromatic number, $\Delta(H)$ for its maximum degree, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047015 | MATHDB-316548

Uniqueness conjecture for non-cyclic graphs of finite non-abelian simple groups

摘录 / EXCERPT: For a group G, let Γ_G be its non-cyclic graph, whose vertices are the elements outside … and whose edges join pairs of vertices that do not generate a cyclic subgroup. Un...

conjecture | open | MathDB 页面 | 原文线索

V2-047016 | MATHDB-316559

The unmixed admissible clutter conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be an admissible clutter, and let $I(\mathcal{C})$ denote its edge ideal. The clutter is unmixed when all its minimal vertex covers have the same cardinality, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047017 | MATHDB-316663

Linear-edge construction conjecture for sparse intersecting hypergraphs

摘录 / EXCERPT: Linear-edge construction conjecture. The function $f(r)$ grows linearly:

conjecture | open | MathDB 页面 | 原文线索

V2-047018 | MATHDB-316665

The nice perfect-elimination characterization of graph polynomials

摘录 / EXCERPT: Nice perfect-elimination conjecture. The polynomial $R_G(q)$ can be factorized as a product of q-numbers if and only if the graph G has a nice perfect elimination order.

conjecture | open | MathDB 页面 | 原文线索

V2-047019 | MATHDB-316668

The Colin de Verdière stress-freeness conjecture

摘录 / EXCERPT: Colin de Verdière stress-freeness conjecture. If $\mu(G)\leq k$ for a positive integer k, then G is generically k-stress free.

conjecture | open | MathDB 页面 | 原文线索

V2-047020 | MATHDB-316686

The vertex-barycentre conjecture for permutahedra and associahedra

摘录 / EXCERPT: Vertex-barycentre conjecture. The vertex barycentres of ${\mathsf{Perm}}^{\pmb a}(W)$ and ${\mathsf{Asso}}^{\pmb a}_c(W)$ coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-047021 | MATHDB-316692

The maximum G-different permutation conjecture

摘录 / EXCERPT: The maximum G-different permutation conjecture. For every $\ell\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-047022 | MATHDB-316730

The VPN Tree Routing Conjecture

摘录 / EXCERPT: The VPN Tree Routing Conjecture. For each sVPND instance (G,c,W,b), there exists an optimal virtual private network which is a tree solution.

conjecture | open | MathDB 页面 | 原文线索

V2-047023 | MATHDB-316731

The Pyramidal Routing Conjecture

摘录 / EXCERPT: The Pyramidal Routing Conjecture. For each Pyramidal Routing instance (G,c,W,i), there exists an optimal solution which is a tree solution.

conjecture | open | MathDB 页面 | 原文线索

V2-047024 | MATHDB-316755

Hedetniemi's multiplicativity conjecture for non-k-colourable graphs

摘录 / EXCERPT: Hedetniemi's multiplicativity conjecture. For every positive integer k, the class $\bar{\mathcal K}_k$ is multiplicative.

conjecture | open | MathDB 页面 | 原文线索

V2-047025 | MATHDB-316756

Hedetniemi's irreducibility conjecture for complete graphs

摘录 / EXCERPT: Hedetniemi's irreducibility conjecture. The complete graph K_k is irreducible for every positive integer k.

conjecture | open | MathDB 页面 | 原文线索

V2-047026 | MATHDB-316765

The adjacency-component conjecture for perfect Delaunay polytopes

摘录 / EXCERPT: Let P and P_1 be perfect Delaunay polytopes for $\mathbb{Z}^n$ and call them adjacent when the quadratic rank of their common vertex set is 2. An adjacency component is a col...

conjecture | open | MathDB 页面 | 原文线索

V2-047027 | MATHDB-316820

The exterior face algebra characterization for tree braid groups

摘录 / EXCERPT: Exterior face algebra conjecture. $H^*(B_nT)$ is an exterior face algebra if and only if $n<6$ or T is linear.

conjecture | open | MathDB 页面 | 原文线索

V2-047028 | MATHDB-316862

Energy lower bound for large subcubic trees

摘录 / EXCERPT: Let c be the constant defined by ... where $\mathcal{B}_{n}$ is the tree obtained from three disjoint copies of the balanced binary tree of order $2^{n+1}-1$ by joining an addition...

conjecture | open | MathDB 页面 | 原文线索

V2-047029 | MATHDB-316883

Adamus's degree-sum conjecture for long cycles

摘录 / EXCERPT: Let G be a finite, undirected graph with no loops or multiple edges. Write $d(v)$ for the degree of a vertex v. Suppose that G is 2-connected, has order $n\geq3$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-047030 | MATHDB-316884

The classification conjecture for moonshine-type vertex operator algebras with central charge one

摘录 / EXCERPT: Let V be a simple, rational, C_2-cofinite vertex operator algebra of moonshine type, with central charge $c=\tilde{c}=1$ and $\dim V_2>1$. Classification conjecture. Then ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-047031 | MATHDB-316893

The small quantum group equivalence conjecture for the supertriplet algebra

摘录 / EXCERPT: Let $m\geq 1$, let $\mathcal{SW}(m)$ be the $N=1$ supertriplet vertex operator superalgebra, and set ... Let $U_q^{\mathrm{small}}(sl_2)$ be the small quantum group at this root of u...

conjecture | open | MathDB 页面 | 原文线索

V2-047032 | MATHDB-316894

The polynomial formula conjecture for the supertriplet residue polynomial

摘录 / EXCERPT: Let $m\geq 1$ and let $G_m(t)$ be the polynomial obtained from the action of $U^{F,E}(0)$ on the highest-weight vector v_λ of an $SM(1)$-module, with...

conjecture | open | MathDB 页面 | 原文线索

V2-047033 | MATHDB-316931

The bipartite graph Betti-number lower-bound conjecture

摘录 / EXCERPT: Let G be a bipartite graph on vertex set $X\sqcup Y$, and let $I(G)$ be its edge ideal. For each $x\in X$, write $\deg_G(x)$ for its degree. Let R_G be the unique, up to isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-047034 | MATHDB-316976

The additive height conjecture for two pegs in peg solitaire

摘录 / EXCERPT: Let A and A' be points of a peg-solitaire board, and let I be an initial position. Define $\operatorname{Height}(A,I)$ as the minimum number of legal moves needed to put a pe...

conjecture | open | MathDB 页面 | 原文线索

V2-047035 | MATHDB-317074

Aldous's spectral-gap conjecture for the interchange process

摘录 / EXCERPT: Let $X_N=\{0,1\}^N$ be the vertex set of the N-dimensional hypercube, let $X_{N,2}=\{\{i,j\}:1\leq i<j\leq N\}$ be the set of coordinate pairs, and let $q:X_{N,2}\to[0,\infty)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-047036 | MATHDB-317111

Quantum maxflow/mincut conjecture for tensor-network maps

摘录 / EXCERPT: Let N be a v-valent graph with k input cut edges and l output cut edges, with the incoming edges at each vertex locally ordered. For a v-index tensor W on a vector spac...

conjecture | open | MathDB 页面 | 原文线索

V2-047037 | MATHDB-317148

Conjecture on the cop number of graphs embedded in non-orientable surfaces

摘录 / EXCERPT: Let $c(g)$ denote the maximum cop number of a graph embedded in an orientable surface of genus g, and let $\tilde c(g)$ denote the corresponding maximum for graphs embedded in a...

conjecture | open | MathDB 页面 | 原文线索

V2-047038 | MATHDB-317174

Logarithmic concavity conjecture for degree-constrained spanning subgraph coefficients

摘录 / EXCERPT: Let $G=(V,E)$ be a finite graph, and let $f,g:V\rightarrow\mathbb{N}$ be any two functions. Fix the vertex activities $\mathbf{u}$ as in (4.1). Let N_j denote the coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-047039 | MATHDB-317192

The bounded-degree-3 four-edge decomposition conjecture

摘录 / EXCERPT: Four-edge decomposition conjecture. The edges of G can be partitioned into subgraphs with at most 4 edges, in such a way that each vertex belongs to at most 2 subgraphs.

conjecture | open | MathDB 页面 | 原文线索

V2-047040 | MATHDB-317193

Lexicographic eigenvalue-ordering conjecture for the derangement graph

摘录 / EXCERPT: Lexicographic eigenvalue-ordering conjecture. For every partition $\lambda=(\lambda_1,\ldots,\lambda_s)\vdash n$,

conjecture | open | MathDB 页面 | 原文线索

V2-047041 | MATHDB-317220

Grimmett-Winkler correlation conjecture for spanning forests

摘录 / EXCERPT: Let G be a (weighted) graph. Let $\mathcal{F}$ be the set of all spanning forests of G, and let $\mathcal{B}$ be the probability distribution on spanning forests in which each...

conjecture | open | MathDB 页面 | 原文线索

V2-047042 | MATHDB-317259

The minimum oriented diameter bound in terms of vertex-cover number

摘录 / EXCERPT: Minimum oriented diameter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047043 | MATHDB-317278

LC-orbit minimum-degree conjecture for the interlace polynomial Q

摘录 / EXCERPT: LC-orbit minimum-degree conjecture. Then there is no graph on n vertices whose minimum vertex degree among all graphs in its LC orbit is greater than δ. The statement prop...

conjecture | open | MathDB 页面 | 原文线索

V2-047044 | MATHDB-317311

The subdivided complete-graph conjecture for the cop number

摘录 / EXCERPT: Subdivided complete-graph conjecture. For every such graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047045 | MATHDB-317327

The conjecture that optimal rectilinear drawings of K_n are 3-decomposable

摘录 / EXCERPT: 3-decomposability conjecture. For each positive integer n divisible by 3, all optimal rectilinear drawings of K_n are 3-decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-047046 | MATHDB-317328

The conjecture that every divisible-order complete graph has a 3-symmetric optimal drawing

摘录 / EXCERPT: 3-symmetric optimal-drawing conjecture. For each positive integer n divisible by 3, there is an optimal geometric drawing of K_n that is 3-symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-047047 | MATHDB-317333

The nucleus characterization conjecture for graph braid groups

摘录 / EXCERPT: For a graph Γ and a braid index $n\geq 2$, let $B_n\Gamma$ denote its graph braid group. An n-nucleus is a minimal graph whose n-braid group is not a right-angled Artin...

conjecture | open | MathDB 页面 | 原文线索

V2-047048 | MATHDB-317334

The four- and three-nuclei classification conjecture

摘录 / EXCERPT: An n-nucleus is a minimal graph whose n-braid group is not a right-angled Artin group, whereas the n-braid group of every graph contained in it is a right-angled Artin group.…

conjecture | open | MathDB 页面 | 原文线索

V2-047049 | MATHDB-317335

The planarity characterization by torsion conjecture for graph braid groups

摘录 / EXCERPT: For a graph Γ and an integer $n\geq 2$, let $B_n\Gamma$ be its graph braid group and let $H_1(B_n\Gamma)$ be its first homology group. Planarity-torsion conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-047050 | MATHDB-317351

The 18-vertex lower bound for simplicial triangulations of hyperbolic 3-manifolds

摘录 / EXCERPT: A hyperbolic 3-manifold is a 3-manifold admitting a complete hyperbolic structure. 18-vertex lower-bound conjecture. At least 18 vertices are needed to triangulate a hyperbol...

conjecture | open | MathDB 页面 | 原文线索

V2-047051 | MATHDB-317419

Sturmfels-Sullivant conjecture on normality and Cohen-Macaulayness of cut ideals

摘录 / EXCERPT: Let G be a finite simple graph with vertex set $V(G)$ and edge set $E(G)$. For each unordered partition $A|B$ of $V(G)$, let $q_{A|B}$ be a variable, and let I_G be the kernel...

conjecture | open | MathDB 页面 | 原文线索

V2-047052 | MATHDB-317422

The minimum-semidegree conjecture for cycles of given length in oriented graphs

摘录 / EXCERPT: Minimum-semidegree conjecture for ell-cycles. There exists an integer $n_0=n_0(ell)$ such that every oriented graph G on $n g eq n_0$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-047053 | MATHDB-317423

The cycle-type conjecture for arbitrarily oriented cycles

摘录 / EXCERPT: Cycle-type conjecture. There exists an integer $n_0=n_0(ell,k)$ such that every oriented graph G on $n g eq n_0$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-047054 | MATHDB-317431

The algebraic Conforti-Cornuéjols conjecture on normality of the Rees algebra

摘录 / EXCERPT: Let $\cal C$ be a clutter, let $\tau({\cal C}')$ and $\nu({\cal C}')$ denote the minimum vertex-cover and maximum matching sizes of a minor $\cal C'$, and let $R[It]$ be the Rees a...

conjecture | open | MathDB 页面 | 原文线索

V2-047055 | MATHDB-317441

The phi-beta conjecture for word maps

摘录 / EXCERPT: Let $\mathbf{F}_k$ be the free group on generators $g_1,\ldots,g_k$. For a word $w\in\mathbf{F}_k$, write ... let $\phi(w)$ be the least index i for which $a_i(w)\neq 0$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-047056 | MATHDB-317451

Non-recursive enumerability of the minimum equational graph theory

摘录 / EXCERPT: Minimum-theory conjecture. The minimum equational graph theory is non-r.e.

conjecture | open | MathDB 页面 | 原文线索

V2-047057 | MATHDB-317455

Asymptotic n/6 domination conjecture for degree-six triangulations

摘录 / EXCERPT: A triangulation is a plane graph in which every face, including the outer face, is bounded by a triangle. A dominating set of a graph is a set containing every vertex or a neighbor...

conjecture | open | MathDB 页面 | 原文线索

V2-047058 | MATHDB-317456

Domination conjecture for triangulations with finitely many non-six-valent vertices

摘录 / EXCERPT: A triangulation is a plane graph in which every face, including the outer face, is bounded by a triangle. The degree of a vertex is its number of incident edges. The bounded-except...

conjecture | open | MathDB 页面 | 原文线索

V2-047059 | MATHDB-317460

Dymacek's conjecture on regular Steinhaus graphs of even degree

摘录 / EXCERPT: Dymacek's conjecture. The regular Steinhaus graphs of even degree are the zero-edge graph on n vertices, for all positive integers n, and the Steinhaus graph $G(s)$ on $n=3m+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-047060 | MATHDB-317474

Conjecture on zero-free intervals for multivariate Tutte polynomials of 2-connected graphs

摘录 / EXCERPT: Zero-free interval conjecture. Suppose $0<q<1$. Then there exists a strictly increasing sequence of self-dual intervals ${\mathcal{V}}_m(q)$, $m\geq 2$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-047061 | MATHDB-317475

Conjecture on sign variation for multivariate Tutte polynomials in uncovered regions

摘录 / EXCERPT: Sign-variation conjecture. Fix real q and v in one of these regions. Then, for all sufficiently large n (depending on q and v) and all sufficiently large m (depending o...

conjecture | open | MathDB 页面 | 原文线索

V2-047062 | MATHDB-317476

Conjecture on density of bivariate Tutte-polynomial zeros in uncovered regions

摘录 / EXCERPT: Zero-density conjecture. As G ranges over all graphs, the zeros of $Z_G(q,v)$ are dense in regions (a)--(e). More precisely, for each fixed q the zeros are dense in v, or for...

conjecture | open | MathDB 页面 | 原文线索

V2-047063 | MATHDB-317483

The center-and-trace dimension conjecture for twisted and untwisted Zhu algebras

摘录 / EXCERPT: Center-and-trace dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047064 | MATHDB-317484

The dimension conjecture for the twisted Zhu algebra of the supertriplet algebra

摘录 / EXCERPT: Twisted Zhu-algebra dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047065 | MATHDB-317496

Probabilistic formula for the energy-kernel representative on infinite resistance networks

摘录 / EXCERPT: Let G be an infinite resistance network with origin o, and let v_x be the representative of an element of the energy kernel normalized by $v_x(o)=0$. Write $R^F(o,x)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-047066 | MATHDB-317546

Nash-Williams' weak Hamilton degree-sequence conjecture for digraphs

摘录 / EXCERPT: Let G be a digraph on $n\geq 3$ vertices, with nondecreasing outdegree and indegree sequences $d^+_1\leq\dots\leq d^+_n$ and $d^-_1\leq\dots\leq d^-_n$. Nash-Williams' conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-047067 | MATHDB-317573

Nonvanishing conjecture for eigenfunctions of random graphs

摘录 / EXCERPT: Nonvanishing conjecture. Almost surely, every eigenfunction f of G has empty zero set:

conjecture | open | MathDB 页面 | 原文线索

V2-047068 | MATHDB-317580

Multipartite Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Let G be a balanced q-partite graph on qn vertices, with vertex classes of size n. A clique factor is a collection of n vertex-disjoint copies of K_q covering all verti...

conjecture | open | MathDB 页面 | 原文线索

V2-047069 | MATHDB-317602

Non-wedge homotopy conjecture for theta complexes of cube hypergraphs

摘录 / EXCERPT: Let $\operatorname{Cube}(n,k)$ be the hypergraph whose vertices are the vertices of an n-cube and whose hyperedges come from the k-dimensional faces of the n-cube, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047070 | MATHDB-317607

Generalized theorem for simplicial maps between arbitrary graphs

摘录 / EXCERPT: Generalized graph-map conjecture. Theorem1.5 is true for φ.

conjecture | open | MathDB 页面 | 原文线索

V2-047071 | MATHDB-317608

Disjoint-arc criterion for approximability by embeddings

摘录 / EXCERPT: Disjoint-arc criterion. The path φ is approximable by embeddings if and only if, for every pair of disjoint arcs $I_1,I_2\subset I$, the restrictions...

conjecture | open | MathDB 页面 | 原文线索

V2-047072 | MATHDB-317624

The 3-connectivity conjecture for cubic graphs of maximal complexity

摘录 / EXCERPT: 3-connectivity conjecture. A cubic graph with maximal complexity is 3-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-047073 | MATHDB-317625

The large-girth conjecture for cubic graphs of maximal complexity

摘录 / EXCERPT: Large-girth conjecture. A sequence $\{\Gamma_i\}_{i\in\mathbb N}$ of cubic graphs with maximal complexity has large girth.

conjecture | open | MathDB 页面 | 原文线索

V2-047074 | MATHDB-317654

The top homology conjecture for hypergraph matching complexes

摘录 / EXCERPT: Let $p>1$ and $k\geq 1$ be integers, and set $n=kp+1$. Let $\mathsf M_p(n)$ denote the p-uniform hypergraph matching complex on n vertices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047075 | MATHDB-317659

Universal-cycle conjecture for isomorphism classes of graphs

摘录 / EXCERPT: Let k be a positive integer. A U-cycle of isomorphism classes of graphs on k nodes is a cyclic sequence in which every k-window represents a distinct isomorphism class of gra...

conjecture | open | MathDB 页面 | 原文线索

V2-047076 | MATHDB-317662

Erdős-Hajnal conjecture for the extremal function $h_1^{(3)}$

摘录 / EXCERPT: Let $\mathcal{F}$ be the minimal family of 3-uniform hypergraphs generated from the empty hypergraphs on 1 and 2 vertices and an edge by the operation $H(G,v)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047077 | MATHDB-317663

Erdős-Hajnal equality conjecture for the functions F_1 and $g_1^{(3)}$

摘录 / EXCERPT: For a positive integer s, let $F_1(s)$ be the maximum number of triples (a,b,c) with $1\leq a<b<c\leq s$ satisfying a prescribed three-coloring pattern on the edges (a,b),...

conjecture | open | MathDB 页面 | 原文线索

V2-047078 | MATHDB-317671

The rationality conjecture for brane-tiling Donaldson-Thomas generating functions

摘录 / EXCERPT: Let G be a consistent brane tiling, let (Q,W) be the corresponding quiver with potential, and let $A=\mathbb{C}Q/(\partial W)$ be its quiver-potential algebra. Fix a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-047079 | MATHDB-317672

The optimality conjecture for distance-based tree reconstruction

摘录 / EXCERPT: Optimality conjecture. No distance-based algorithm has l_∞-radius greater than $\frac{1}{4}$ for general trees. Consequently, if this assertion holds, the RNJ algorithm achi...

conjecture | open | MathDB 页面 | 原文线索

V2-047080 | MATHDB-317705

The extremal graph conjecture for the first eigenvalue of bipartite graphs

摘录 / EXCERPT: Extremal graph conjecture. Under these assumptions, an extremal graph solving this maximal problem is obtained from a complete bipartite graph by adding one vertex and a correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-047081 | MATHDB-317716

Babson-Reiner connectivity conjecture for Coxeter-like complexes

摘录 / EXCERPT: Let T be a tree with n nodes and b leaves, and let $cDelta_T$ be its Coxeter-like complex. Babson-Reiner conjecture. The complex $cDelta_T$ is at least $(n-1-b)$-connected. T...

conjecture | open | MathDB 页面 | 原文线索

V2-047082 | MATHDB-317721

The power-of-two factor degree conjecture for spanning-tree transfer matrices

摘录 / EXCERPT: Let G be a graph with k vertices, let $G_n=G\times P_n$, and let M be the integer $B_k\times B_k$ matrix from the theorem such that $T_n=M^nv[1]$, where T_n is the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-047083 | MATHDB-317722

The alternating-sign recurrence conjecture for spanning trees of graph products

摘录 / EXCERPT: Let G be any graph, and write $G_n=G\times P_n$. Let T_n denote the number of spanning trees of G_n, which satisfies a linear recurrence by the preceding result. Alternating-...

conjecture | open | MathDB 页面 | 原文线索

V2-047084 | MATHDB-317723

The order conjecture for spanning-tree recurrences of grid graphs

摘录 / EXCERPT: Let $G_k(n)$ denote the grid graph considered in the paper, with the recurrence for its spanning-tree counts taken with respect to n. Grid recurrence order conjecture. The recurr...

conjecture | open | MathDB 页面 | 原文线索

V2-047085 | MATHDB-317724

The complete-graph product recurrence order conjecture

摘录 / EXCERPT: Let K_k be the complete graph on k vertices, let P_n be the path graph on n vertices, and consider the spanning-tree recurrence for the graph product $K_k\times P_n$. Compl...

conjecture | open | MathDB 页面 | 原文线索

V2-047086 | MATHDB-317733

F-free hypergraph coloring conjecture

摘录 / EXCERPT: Let F be a k-graph. A k-graph is F-free if it contains no copy of F. Let Δ denote the maximum degree. F-free coloring conjecture. There is a constant c_F dependi...

conjecture | open | MathDB 页面 | 原文线索

V2-047087 | MATHDB-317747

Conjecture on the maximal ideal and automorphism of the parafermion algebra

摘录 / EXCERPT: Let $V(k,0)$ be the universal affine vertex operator algebra for $ obreak\mathfrak{sl}_2$ at positive integer level k, let N_0 be its Heisenberg commutant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047088 | MATHDB-317748

Conjecture on the modules and rationality of the parafermion algebra

摘录 / EXCERPT: Let k be a positive integer, let $M^{i,j}$ be the modules indexed by $0\leq i\leq k$ and $0\leq j\leq i$, and let $M^{0,0}$ be the parafermion vertex operator algebra, denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-047089 | MATHDB-317749

The B-factorizability conjecture for complete graphs

摘录 / EXCERPT: Complete-graph B-factorizability conjecture. The complete graph $K_{|V|}$ is B-factorizable.

conjecture | open | MathDB 页面 | 原文线索

V2-047090 | MATHDB-317750

The prism B-factorizability conjecture

摘录 / EXCERPT: Prism B-factorizability conjecture. The graph of every k-gonal prism is B-factorizable.

conjecture | open | MathDB 页面 | 原文线索

V2-047091 | MATHDB-317751

The complete-bipartite B-factorizability conjecture

摘录 / EXCERPT: Complete-bipartite B-factorizability conjecture. Every complete bipartite graph $K_{k,k}$ is B-factorizable.

conjecture | open | MathDB 页面 | 原文线索

V2-047092 | MATHDB-317752

The Petersen-minor Hilbert-basis conjecture for perfect matchings

摘录 / EXCERPT: Petersen-minor Hilbert-basis conjecture. For any graph without the Petersen graph as a graph minor, the characteristic vectors χ_q of perfect matchings q form the Hilbert ba...

conjecture | open | MathDB 页面 | 原文线索

V2-047093 | MATHDB-317755

Polynomiality conjecture for the capped Donaldson-Thomas vertex

摘录 / EXCERPT: Capped vertex polynomiality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047094 | MATHDB-317807

Alev's uniqueness conjecture for simple graphs solving the B_n equation

摘录 / EXCERPT: Simple-graph uniqueness conjecture. For every integer $n\geq 2$, there exists a unique polynomial of degree $4(n-1)$ in $S^W_{\mathfrak{sl}_2}$ solving the equation. It is represen...

conjecture | open | MathDB 页面 | 原文线索

V2-047095 | MATHDB-317808

Alev's completeness conjecture for the constructed B_n graph solutions

摘录 / EXCERPT: Alev's completeness conjecture. For every integer $n\geq 2$, these $\pi(n)$ constructed graphs are the only solutions of the equation.

conjecture | open | MathDB 页面 | 原文线索

V2-047096 | MATHDB-317817

DeVos-Mohar-Samal conjecture on crossing-free drawings of disconnected graphs

摘录 / EXCERPT: DeVos-Mohar-Samal conjecture. For every optimal drawing of G on Σ, the restrictions to H and K do not intersect.

conjecture | open | MathDB 页面 | 原文线索

V2-047097 | MATHDB-317831

Nonexistence of one-vertex JSJ quotients of a free product

摘录 / EXCERPT: Let F be the free group appearing in the free product $F*\langle x,y\rangle$. A quotient is fully residually F if it has the residual property relative to F, and its JSJ deco...

conjecture | open | MathDB 页面 | 原文线索

V2-047098 | MATHDB-317838

The enhanced conflict graph extreme-point conjecture for 2×N switches

摘录 / EXCERPT: Consider a $2\times N$ switch with traffic consisting of unicasts and broadcasts, and let G be its enhanced conflict graph. Let $QSTAB(G)$ denote the fractional stable set polyto...

conjecture | open | MathDB 页面 | 原文线索

V2-047099 | MATHDB-317839

The 1.25 speedup conjecture for unicasts and broadcasts

摘录 / EXCERPT: Consider a $K\times N$ switch with traffic patterns consisting only of unicasts and broadcasts. The 1.25 speedup conjecture. The minimum speedup required to achieve 100% through...

conjecture | open | MathDB 页面 | 原文线索

V2-047100 | MATHDB-317859

The switching-equivalent reconstruction conjecture

摘录 / EXCERPT: Let G and H be graphs on the same number of vertices. Their decks are switching equivalent, written $\mathcal{D}(G) \sim \mathcal{D}(H)$, when there is a bijection between thei...

conjecture | open | MathDB 页面 | 原文线索

V2-047101 | MATHDB-317860

The deck characterization of switching equivalence

摘录 / EXCERPT: Let X_1 and X_2 be graphs on n vertices, and let $\mathcal{D}(X_i)$ denote the deck of vertex-deleted subgraphs of X_i. The decks are switching equivalent when there is a b...

conjecture | open | MathDB 页面 | 原文线索

V2-047102 | MATHDB-317861

The one-norm characterization of switching equivalence

摘录 / EXCERPT: Let X be a graph on n vertices with Seidel adjacency matrix S. For $1\leq m\leq n$, let $\mathcal{D}_m$ be the set of diagonal matrices having exactly m diagonal entries eq...

conjecture | open | MathDB 页面 | 原文线索

V2-047103 | MATHDB-317864

Positivity conjecture for coefficients of the graph series R_g

摘录 / EXCERPT: Positivity conjecture. All the coefficients C_G of the elements $R_g\in\mathcal{W}$ are positive.

conjecture | open | MathDB 页面 | 原文线索

V2-047104 | MATHDB-317867

Erdős's conjecture on a quasi-isometric graph of the Euclidean plane

摘录 / EXCERPT: The integer lattice points in the Euclidean plane form the vertex set of a graph, with edges chosen so that the resulting graph is $(1,k)$-quasi-isometric to the Euclidean plane. E...

conjecture | open | MathDB 页面 | 原文线索

V2-047105 | MATHDB-317900

The edge-density conjecture for intrinsic knotting of bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph with at least five vertices in each part. Write $v(G)=|V(G)|$ for its number of vertices and $e(G)=|E(G)|$ for its number of edges. Intrinsic-knotting...

conjecture | open | MathDB 页面 | 原文线索

V2-047106 | MATHDB-317923

The generalized Bunkbed conjecture for conditioned edge percolation

摘录 / EXCERPT: Generalized Bunkbed conjecture. For every $u,v\in V(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-047107 | MATHDB-317924

The generalized Bunkbed conjecture for two-color edge percolation

摘录 / EXCERPT: Two-color Bunkbed conjecture. For every $u,v\in V(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-047108 | MATHDB-317925

The hypergraph Bunkbed conjecture

摘录 / EXCERPT: Hypergraph Bunkbed conjecture. For every $u,v\in V(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-047109 | MATHDB-317973

Cutoff conjecture for simple random walk on vertex-transitive expander graphs

摘录 / EXCERPT: Let (G_n) be any family of finite vertex-transitive expander graphs, and let the simple random walk (SRW) be the walk that at each step moves uniformly to a neighboring vertex. V...

conjecture | open | MathDB 页面 | 原文线索

V2-047110 | MATHDB-317996

Bonato-Tardif rooted tree alternative conjecture

摘录 / EXCERPT: A rooted tree (T,r) is a tree T with a distinguished vertex r. Two rooted trees (T,r) and (T',r') are twins, or mutually embeddable, when there are injective graph homomo...

conjecture | solved | MathDB 页面 | 原文线索

V2-047111 | MATHDB-317998

Vanishing matching-sum conjecture for odd square-grid cylinders

摘录 / EXCERPT: Let $C_{m,n}$ be the square-grid cylinder, let Q_2 be the class of configurations specified in the surrounding construction, and let π be the associated projection map. For...

conjecture | open | MathDB 页面 | 原文线索

V2-047112 | MATHDB-318027

The regular Ramsey number comparison conjecture

摘录 / EXCERPT: Let $R(k,\ell)$ be the Ramsey number, and let $R^{\mathrm{reg}}(k,\ell)$ denote the corresponding Ramsey number restricted to regular graphs. Regular Ramsey comparison conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047113 | MATHDB-318058

Kitakubo's planar emulator conjecture

摘录 / EXCERPT: All graphs considered are finite and simple. A finite planar emulator of a graph is a finite emulator whose source graph is planar, and a graph embeds in the projective plane when...

conjecture | open | MathDB 页面 | 原文线索

V2-047114 | MATHDB-318059

Fellows' conjecture on finite planar emulators and covers

摘录 / EXCERPT: All graphs considered are finite and simple. A finite planar emulator (respectively, finite planar cover) of a graph is a finite emulator (respectively, cover) whose source graph i...

conjecture | refuted | MathDB 页面 | 原文线索

V2-047115 | MATHDB-318127

Simple 3-graph multiplicity conjecture

摘录 / EXCERPT: Simple 3-graph conjecture. There exists a simple 3-graph H for which

conjecture | open | MathDB 页面 | 原文线索

V2-047116 | MATHDB-318130

The relative category equality for distinguished conjugacy classes in symplectic groups

摘录 / EXCERPT: Let $\operatorname{Sp}(n)$ be the compact symplectic group, let $\mathcal O_k$ be the distinguished conjugacy class associated with the k-th vertex of the fundamental alcove, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047117 | MATHDB-318156

Nash-Williams's Pósa-type degree-sequence conjecture for Hamilton cycles in digraphs

摘录 / EXCERPT: Let G be a digraph on $n\geq 3$ vertices, with nondecreasing outdegree and indegree sequences $d^+_1\leq\cdots\leq d^+_n$ and $d^-_1\leq\cdots\leq d^-_n$. Nash-Williams's Pósa-ty...

conjecture | open | MathDB 页面 | 原文线索

V2-047118 | MATHDB-318163

Lovász-Plummer exponential perfect-matching conjecture for cubic bridgeless graphs

摘录 / EXCERPT: Let G be a cubic bridgeless graph with n vertices; parallel edges are allowed. A perfect matching is a set of edges meeting every vertex exactly once. Lovász-Plummer conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-047119 | MATHDB-318177

Feigin-Gainutdinov-Semikhatov-Tipunin equivalence conjecture for triplet W-algebras

摘录 / EXCERPT: Let $p\geq 2$, let $W(p)$ be the triplet vertex operator algebra, and let $\overline{U}_q(\mathfrak{sl}_2)$ be the restricted quantum enveloping algebra associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-047120 | MATHDB-318257

The logarithmic module-count conjecture for the triplet vertex algebra

摘录 / EXCERPT: Let $p,p'\in\mathbb{Z}_{>0}$ with $p\ne p'$ and $p\geq 2$. Let $L=\mathbb{Z}\alpha$ satisfy $\langle\alpha,\alpha\rangle=2pp'$, and let $\mathcal{W}(p,p')$ be the vertex operator a...

conjecture | open | MathDB 页面 | 原文线索

V2-047121 | MATHDB-318272

The polygonal spectral enclosure conjecture for standardized Laplacian matrices

摘录 / EXCERPT: The polygonal spectral enclosure conjecture. All eigenvalues of every standardized Laplacian matrix $\widetilde L\in\widetilde\Lambda_n$ belong to $S(n)$. The conjecture would iden...

conjecture | open | MathDB 页面 | 原文线索

V2-047122 | MATHDB-318283

Balanced blow-up conjecture for triangle densities

摘录 / EXCERPT: Balanced blow-up conjecture. For every $t\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-047123 | MATHDB-318302

MacLane-type planarity criterion for compact one-dimensional continua

摘录 / EXCERPT: MacLane-type planarity conjecture. The space X is embeddable in S^2 if and only if there are a simple set S of circles in X and a metric d inducing the topology of X su...

conjecture | open | MathDB 页面 | 原文线索

V2-047124 | MATHDB-318305

Euler-tour characterization for metric graph realizations

摘录 / EXCERPT: Metric Euler-tour conjecture. The metric realization $|G|_{\ell}$ has a topological Euler tour if and only if $|G|_{\ell}$ is compact and the topological realization $|G|$ has a to...

conjecture | open | MathDB 页面 | 原文线索

V2-047125 | MATHDB-318334

Alon-Saks-Seymour conjecture on biclique partitions

摘录 / EXCERPT: Alon-Saks-Seymour conjecture. The minimum number of bicliques in such a partition is $k-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-047126 | MATHDB-318335

Katona-Szemerédi-type conjecture for biclique covers

摘录 / EXCERPT: Katona-Szemerédi-type conjecture. The sum of the orders of the bicliques in any biclique cover of $E(G)$ is at least

conjecture | open | MathDB 页面 | 原文线索

V2-047127 | MATHDB-318338

Hanner inequality conjecture for Type I semi-norming hypergraph pairs

摘录 / EXCERPT: Hanner inequality conjecture. Every L_H space satisfies the $|H|$-Hanner inequality. The inequality is known in the paper for non-factorizable semi-norming hypergraph pairs of Ty...

conjecture | open | MathDB 页面 | 原文线索

V2-047128 | MATHDB-318380

The vertex bound for weakly neighbourly antipodal convex polytopes

摘录 / EXCERPT: Let ${\bf P}\subset {\bf E}^d$ be a d-dimensional convex polytope. It is weakly neighbourly if any two vertices lie on a face, and antipodal if any two vertices lie on parallel s...

conjecture | open | MathDB 页面 | 原文线索

V2-047129 | MATHDB-318385

Classification conjecture for regular graphs of complexity less than one

摘录 / EXCERPT: Regular-graph complexity conjecture. If $c(G)<1$, then G is either isomorphic to C_7, or there is an integer $n\geq2$ such that G is isomorphic to either $K_{n,n}$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-047130 | MATHDB-318419

F-polynomial constant-term conjecture

摘录 / EXCERPT: Fix a positive integer n, an n-regular tree $\mathbb{T}_n$, a vertex $t_0\mathbin{\in}\mathbb{T}_n$, and a skew-symmetrizable integer matrix $B=(b_{i,j})$. For each...

conjecture | solved | MathDB 页面 | 原文线索

V2-047131 | MATHDB-318420

F-polynomial maximal-monomial conjecture

摘录 / EXCERPT: Fix a positive integer n, an n-regular tree $\mathbb{T}_n$, a vertex $t_0\in\mathbb{T}_n$, and a skew-symmetrizable integer matrix $B=(b_{i,j})$. For each $t\in\mathbb{T}_n$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-047132 | MATHDB-318424

G-vector mutation conjecture

摘录 / EXCERPT: Fix a positive integer n, an n-regular tree $\mathbb{T}_n$, a vertex $t_0\in\mathbb{T}_n$, and a skew-symmetrizable integer matrix $B=(b_{i,j})$. Let...

conjecture | solved | MathDB 页面 | 原文线索

V2-047133 | MATHDB-318448

Linear decoding and delivery delay for the new three-receiver module

摘录 / EXCERPT: Let $\rho=\lambda/\mu<1$ be the load factor in the three-receiver system, and consider the newly proposed coding module. Define decoding delay as the time from packet arrival at th...

conjecture | open | MathDB 页面 | 原文线索

V2-047134 | MATHDB-318450

The Sands-Sauer-Woodrow conjecture for 3-coloured tournaments

摘录 / EXCERPT: Let T be a finite 3-coloured tournament, meaning a tournament whose edges are coloured with at most three colours. A tournament is a T_3 if three of its vertices span a non-tra...

conjecture | open | MathDB 页面 | 原文线索

V2-047135 | MATHDB-318453

Eigenvalue conjecture for the degenerate elliptic operator

摘录 / EXCERPT: Eigenvalue conjecture. The eigenvalues of $\big[\eta(z;p q^{-1}t)\big]_1$ on $\mathcal{F}$ are given by

conjecture | open | MathDB 页面 | 原文线索

V2-047136 | MATHDB-318460

The minimum-light-number gap conjecture for trees with loops

摘录 / EXCERPT: Tree gap conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047137 | MATHDB-318461

The maximum-degree bound for the configuration-wise minimum-light-number gap

摘录 / EXCERPT: Maximum-degree gap conjecture. For every graph G and every configuration x,

conjecture | open | MathDB 页面 | 原文线索

V2-047138 | MATHDB-318472

Linear Euler-characteristic bound for nonnegative-curvature vertices in polyhedral maps

摘录 / EXCERPT: The conjecture. For each surface $\mathbb{M}$, there exists a constant $c_{\mathbb{M}}$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-047139 | MATHDB-318480

The realization conjecture for independence polynomials at -1

摘录 / EXCERPT: Realization conjecture. For every positive integer ν and each integer q such that $|q|\leq 2^{\nu}$, there is a connected graph G with $\nu(G)=\nu$ and $I(G;-1)=q$.

conjecture | open | MathDB 页面 | 原文线索

V2-047140 | MATHDB-318482

Convergence of iterative maximum-likelihood estimators

摘录 / EXCERPT: Let V be the finite set of agents, let $X_v(t)$ denote the estimator held by agent $v\in V$ at iteration t, and let $X(\infty)$ be a common limiting estimator. Convergence conj...

conjecture | open | MathDB 页面 | 原文线索

V2-047141 | MATHDB-318484

Finite-time consensus for agents with memory

摘录 / EXCERPT: Let V be the finite set of agents, let $X_v(t)$ be the estimator of agent v at time t in the model where agents retain their previous values, and let $X(\infty)$ denote the l...

conjecture | open | MathDB 页面 | 原文线索

V2-047142 | MATHDB-318509

Two-partition conjecture for high-chromatic digraphs and path families

摘录 / EXCERPT: Let $\mathcal{P}_{3k,k-1}$ be the family of paths used in the source, and let a partition into two sets be written as ... A digraph G admits a homomorphism from a path P when the...

conjecture | open | MathDB 页面 | 原文线索

V2-047143 | MATHDB-318517

The counting conjecture for stable non-3-partite triple systems

摘录 / EXCERPT: Say that a 3-graph F is stable if, for sufficiently large n, the extremal number $\operatorname{ex}(n,F)$ is attained uniquely by an n-vertex 3-graph ${\cal H}(n)$, and every...

conjecture | open | MathDB 页面 | 原文线索

V2-047144 | MATHDB-318536

Odd-parameter vertex-target pebbling conjecture

摘录 / EXCERPT: Let G and H be graphs, let s and t be positive odd integers, and let $x\in V(G)$ and $y\in V(H)$. Odd-parameter vertex-target pebbling conjecture. For all such graphs, inte...

conjecture | open | MathDB 页面 | 原文线索

V2-047145 | MATHDB-318543

Realizability conjecture for alternating independent-set numbers

摘录 / EXCERPT: Realizability conjecture. For every positive integer k and each integer q such that $|q|\leq 2^k$, there is a graph G with $\varphi(G)=k$ and $I(G;-1)=q$.

conjecture | open | MathDB 页面 | 原文线索

V2-047146 | MATHDB-318552

Postle-Streib-Yerger's general diameter conjecture for graph pebbling

摘录 / EXCERPT: Postle-Streib-Yerger's general diameter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047147 | MATHDB-318553

The extremal graph conjecture for pebbling numbers at fixed order and diameter

摘录 / EXCERPT: Extremal graph conjecture. The pebbling number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047148 | MATHDB-318565

The successor conjecture for $(3m-1,1^3)$ in the 3-cohomology graph

摘录 / EXCERPT: Let $m\in\mathbb Z_{>0}$, and let $\mathcal C_3^2$ denote the graph whose vertices are the relevant partitions for second cohomology at $p=3$; a successor is a vertex joined by the...

conjecture | open | MathDB 页面 | 原文线索

V2-047149 | MATHDB-318582

The degree-two vertex conjecture for random tournaments

摘录 / EXCERPT: Degree-two vertex conjecture. If s has degree at most two, then the events $\{a\to s\}$ and $\{s\to b\}$ are independent or negatively correlated.

conjecture | open | MathDB 页面 | 原文线索

V2-047150 | MATHDB-318603

Nikiforov's limiting energy conjecture for ternary complete-tree extensions

摘录 / EXCERPT: Let C_n be the complete 2-ary tree of height $n-1$, and let $\mathcal{B}_n$ be the tree formed from three disjoint copies of C_n by adjoining one vertex to their roots. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047151 | MATHDB-318604

Nikiforov's lower-bound conjecture for the energy of subcubic trees

摘录 / EXCERPT: Let T be a tree, let $\Delta(T)$ denote its maximum degree, let $|T|$ denote its number of vertices, and let c be the constant from Nikiforov's limiting energy conjecture. Niki...

conjecture | open | MathDB 页面 | 原文线索

V2-047152 | MATHDB-318615

Global resilience conjecture for odd cycles in pseudorandom graphs

摘录 / EXCERPT: Global resilience conjecture. Then G has global resilience $(1/4+o(1))nd$ with respect to being C_k-free.

conjecture | open | MathDB 页面 | 原文线索

V2-047153 | MATHDB-318666

Rectified Füredi-Mubayi signed domination conjecture

摘录 / EXCERPT: Let G be a graph with n vertices, minimum degree δ, and signed domination number $\gamma_s(G)$. Rectified signed domination conjecture. There are constants C and...

conjecture | open | MathDB 页面 | 原文线索

V2-047154 | MATHDB-318668

Uniqueness of the maximum-excess irreducible triangulation

摘录 / EXCERPT: Uniqueness conjecture. Suppose that K_ω triangulates Σ. Then K_ω is the only irreducible triangulation of Σ with maximum excess.

conjecture | open | MathDB 页面 | 原文线索

V2-047155 | MATHDB-318669

General exact upper-bound conjecture for cliques on surfaces

摘录 / EXCERPT: General exact upper-bound conjecture. Every graph G embeddable in Σ has at most

conjecture | open | MathDB 页面 | 原文线索

V2-047156 | MATHDB-318670

Complete-graph containment in maximum-excess irreducible triangulations

摘录 / EXCERPT: Complete-subgraph conjecture. Every irreducible triangulation of Σ with maximum excess contains K_ω as a subgraph.

conjecture | open | MathDB 页面 | 原文线索

V2-047157 | MATHDB-318671

Vertex-minimal maximum-excess triangulation conjecture

摘录 / EXCERPT: Vertex-minimality conjecture. For every surface Σ, the maximum excess is attained by some vertex-minimal triangulation of Σ that contains K_ω as a subgraph. Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-047158 | MATHDB-318727

S. B. Rao's well-quasi-ordering conjecture for degree sequences

摘录 / EXCERPT: Let a graphic degree sequence be a multiset of non-negative integers realized as the degree sequence of a simple graph. For a degree sequence D, let ... be its set of realizations....

conjecture | open | MathDB 页面 | 原文线索

V2-047159 | MATHDB-318730

Richter and Salazar's bounded-bandwidth conjecture for crossing-critical graphs

摘录 / EXCERPT: A graph is k-crossing-critical if its crossing number is at least k, but every proper subgraph has crossing number smaller than k. Richter and Salazar's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-047160 | MATHDB-318731

Bound on the number of high-degree vertices in crossing-critical graphs

摘录 / EXCERPT: A graph is k-crossing-critical if its crossing number is at least k, but every proper subgraph has crossing number smaller than k. High-degree vertex conjecture. For every po...

conjecture | open | MathDB 页面 | 原文线索

V2-047161 | MATHDB-318743

The capacity trichotomy conjecture for three-source three-terminal sum-networks

摘录 / EXCERPT: Capacity trichotomy conjecture. The capacity of the sum-network is either 0, $2/3$, or at least 1.

conjecture | open | MathDB 页面 | 原文线索

V2-047162 | MATHDB-318748

The Duchet-Maffray-Woodall product conjecture for independent sets and complete minors

摘录 / EXCERPT: Let G be a graph. Write $\alpha(G)$ for its independence number, the maximum size of an independent set, and let $h(G)$ be the size of a largest complete minor of G. Duchet-Maf...

conjecture | open | MathDB 页面 | 原文线索

V2-047163 | MATHDB-318769

Erdős bipartification conjecture

摘录 / EXCERPT: Let G be a triangle-free graph. A bipartification is a set of edges whose deletion makes G bipartite. Erdős bipartification conjecture. Every triangle-free graph G can be mad...

conjecture | open | MathDB 页面 | 原文线索

V2-047164 | MATHDB-318770

Minimum bipartification conjecture for Andrásfai graphs

摘录 / EXCERPT: Let $\textnormal{And}_k$ be the Andrásfai graph, and let F_k be the bipartification defined in Theorem; thus, deleting F_k from $\textnormal{And}_k$ makes it bipartite. Minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-047165 | MATHDB-318771

Bipartification conjecture for homomorphic preimages of Andrásfai graphs

摘录 / EXCERPT: Let $\textnormal{And}_k$ be the Andrásfai graph, and let H be an arbitrary homomorphic preimage of $\textnormal{And}_k$. Let F_k be the bipartification defined in Theorem. Homo...

conjecture | open | MathDB 页面 | 原文线索

V2-047166 | MATHDB-318792

Twisted chiral three-dimensional quantum gravity conjecture for Monster elements

摘录 / EXCERPT: Let $\mathbb M$ be the Monster, let $g\in\mathbb M$, let Γ_g be the associated group, and let Γ_{∞} denote the cusp subgroup. Let $T_{\Gamma}^{(1)}$ be the norma...

conjecture | open | MathDB 页面 | 原文线索

V2-047167 | MATHDB-318837

The spanning conjecture for short paths in polytopes

摘录 / EXCERPT: Let P be a $(d,2d-1)$-polytope with graph G_P, and let x,y be vertices of P. A collection of (x,y) paths spans the graph if removing one vertex from every path disconnect...

conjecture | open | MathDB 页面 | 原文线索

V2-047168 | MATHDB-318841

The polynomial expansion conjecture for 0/1 polytopes

摘录 / EXCERPT: A 0/1 d-polytope is a d-dimensional polytope whose vertices lie in $\{0,1\}^d$. Let edge expansion mean the minimum ratio of boundary edges to the size of a vertex subset of si...

conjecture | open | MathDB 页面 | 原文线索

V2-047169 | MATHDB-318842

Tutte's strong embedding conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A strong embedding of G is an embedding on a surface in which the perimeter of every region is a circuit. Tutte's conjecture. Every bridgeles...

conjecture | open | MathDB 页面 | 原文线索

V2-047170 | MATHDB-318846

Finiteness conjecture for coconnected geometric distance-regular graphs

摘录 / EXCERPT: Let $m\geq 2$ be fixed. A distance-regular graph is geometric if it has a family of Delsarte cliques whose incidence structure gives the graph, and it is coconnected if its complem...

conjecture | open | MathDB 页面 | 原文线索

V2-047171 | MATHDB-318874

The regular oriented graph Hamilton decomposition conjecture above the $3/8$ threshold

摘录 / EXCERPT: For $\gamma>0$, let $n_0=n_0(\gamma)$ be a threshold. An r-regular oriented graph on n vertices is an oriented graph in which every vertex has indegree and outdegree r. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-047172 | MATHDB-318881

Kohayakawa-Łuczak-Rödl conjecture on typical sparse regular graph embeddings

摘录 / EXCERPT: Kohayakawa-Łuczak-Rödl conjecture. For any $\beta>0$, there exist constants $\varepsilon_0>0$, $C>0$, and $n_0>0$ such that, for all $m\geq Cn^{2-1/d_2(H)}$, $n\geq n_0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047173 | MATHDB-318895

Isometry conjecture for Eisenstein flow lattices of sixth-root-of-unity matroids

摘录 / EXCERPT: Isometry conjecture. If M and M' represent the same matroid $\mathcal{M}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047174 | MATHDB-318896

Eisenstein flow-lattice characterization conjecture for sixth-root-of-unity matroids

摘录 / EXCERPT: Characterization conjecture. The lattices $\Lambda_{\mathbb{E}}(\mathcal{M})$ and $\Lambda_{\mathbb{E}}(\mathcal{N})$ are isometric if and only if $\mathcal{M}_{\bullet}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-047175 | MATHDB-318909

Nash-Williams' Hamilton decomposition conjecture for regular graphs

摘录 / EXCERPT: Nash-Williams' conjecture. G contains $\lfloor d/2 \rfloor$ edge-disjoint Hamilton cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-047176 | MATHDB-318924

The near-linear-order Hajós conjecture for critical graphs

摘录 / EXCERPT: Let G be an r-critical graph on $r+o(r)$ vertices, where an r-critical graph has chromatic number r and every proper subgraph has smaller chromatic number. A graph satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-047177 | MATHDB-318953

Strong embedding conjecture for 2-connected non-planar graphs

摘录 / EXCERPT: Let G be a 2-connected non-planar graph contained in a closed surface F. For a pair (F,G), define its representativity by ... where $\mathcal{C}_F$ is the set of all essentia...

conjecture | open | MathDB 页面 | 原文线索

V2-047178 | MATHDB-318957

PL equivalence of the 17-vertex and 16-vertex K3 triangulations

摘录 / EXCERPT: Let $(K3)_{16}$ and $(K3)_{17}$ denote the simplicial complexes with 16 and 17 vertices, respectively, constructed in the dilatation process described in the paper. The PL-equi...

conjecture | open | MathDB 页面 | 原文线索

V2-047179 | MATHDB-318982

Strong meets empirical coordination conjecture

摘录 / EXCERPT: Let ω denote common randomness, with enough common randomness available-for example, $\omega \sim \operatorname{Unif}\{[0,1]\}$. For a fixed network setting, let...

conjecture | open | MathDB 页面 | 原文线索

V2-047180 | MATHDB-318986

Chudnovsky-Seymour-Sullivan conjecture on cycles in triangle-free digraphs

摘录 / EXCERPT: Chudnovsky-Seymour-Sullivan conjecture. If G is a 3-free digraph, then

conjecture | open | MathDB 页面 | 原文线索

V2-047181 | MATHDB-319001

Limiting domination number conjecture for proximity catch digraphs on simplices

摘录 / EXCERPT: Limiting domination number conjecture. As $n\to\infty$, the domination number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047182 | MATHDB-319084

The H9 non-extremality conjecture in the second parameter region

摘录 / EXCERPT: Let $\alpha,\beta,\gamma\in[0,1]$, let $R_2\subseteq[0,1]^3$ be the second region defined in the paper, and let H_9 be the specified tripartite graph. For a weighting w,...

conjecture | open | MathDB 页面 | 原文线索

V2-047183 | MATHDB-319105

The extremal 2-fold sumgraph conjecture for regular graphs

摘录 / EXCERPT: Let d,n be positive integers, and let $\mathscr{G}$ be a d-regular graph on n vertices, with edge set $\mathscr{E}$. The graph $2\mathscr{G}$ denotes the 2-fold sumgraph, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047184 | MATHDB-319110

Generalized Hovey conjecture for p-uniform hypertrees

摘录 / EXCERPT: Generalized Hovey conjecture. All p-uniform hypertrees are k-cordial for all k.

conjecture | open | MathDB 页面 | 原文线索

V2-047185 | MATHDB-319114

Vertex bound for regular minimal graphs

摘录 / EXCERPT: Vertex-bound conjecture. For all $d \geq 3$, the number of vertices of a d-regular minimal graph attached to n points is

conjecture | open | MathDB 页面 | 原文线索

V2-047186 | MATHDB-319132

The forbidden-isometric-subgraph characterization of half-hyperbolic 6-chordal graphs

摘录 / EXCERPT: Forbidden-subgraph conjecture. A 6-chordal graph is $\frac{1}{2}$-hyperbolic if and only if it does not contain any of the ten special graphs G_1,G_2,G_3,C_4,C_6,H_i,...

conjecture | open | MathDB 页面 | 原文线索

V2-047187 | MATHDB-319200

Archdeacon, Bonnington, and Širáň's crossing sequence conjecture

摘录 / EXCERPT: Archdeacon, Bonnington, and Širáň's conjecture. If $(a_1,a_2,\ldots,0)$ is a sequence of integers which strictly decreases until 0, then there is a graph whose crossing sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-047188 | MATHDB-319226

The nonorientable-to-orientable closed 2-cell embedding conjecture

摘录 / EXCERPT: Let G be a 2-connected graph. A closed 2-cell embedding is an embedding in a connected compact surface without boundary in which every face is an open disk bounded by a cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-047189 | MATHDB-319231

Dimension equality for Zhu and C_2-algebras of classical Lie algebras

摘录 / EXCERPT: Let $\g$ be a classical Lie algebra, let $k\ge 0$, and let $A(\g;k)$ and $A_{[2]}(\g;k)$ denote respectively the Zhu algebra and the C_2-algebra of the level-k WZW model associ...

conjecture | open | MathDB 页面 | 原文线索

V2-047190 | MATHDB-319241

Conjecture on bias forcing oriented six-cycles

摘录 / EXCERPT: Bias-forcing six-cycle conjecture. There exists a constant $\varepsilon>0$ such that every oriented graph D satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-047191 | MATHDB-319242

General bias-forcing conjecture for oriented even cycles

摘录 / EXCERPT: Even-cycle bias conjecture. For each integer $k\geq 2$, there exists a constant $\varepsilon>0$ such that every oriented graph D satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-047192 | MATHDB-319243

The orthogonal web-flow conjecture for weighted webs

摘录 / EXCERPT: Orthogonal web-flow conjecture. In every weighted web there exists a web-flow f and an A-B separating set orthogonal to f.

conjecture | open | MathDB 页面 | 原文线索

V2-047193 | MATHDB-319244

The loose weighted-web linkability conjecture

摘录 / EXCERPT: Loose weighted-web conjecture. A loose weighted web is linkable.

conjecture | open | MathDB 页面 | 原文线索

V2-047194 | MATHDB-319245

Half-degree conjecture for complementary tuples in balanced multipartite hypergraphs

摘录 / EXCERPT: Let H) be an n-balanced r-partite r-graph, and let I be a subset of $[r]:=\{1,2,\ldots,r\}$. An I-tuple is an element of $\times_{i\in I}V_i$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-047195 | MATHDB-319246

Equal complementary-tuple degrees conjecture for balanced multipartite hypergraphs

摘录 / EXCERPT: Let H be an n-balanced r-partite r-graph, and let I be a subset of $[r]:=\{1,2,\ldots,r\}$. An I-tuple is an element of $\times_{i\in I}V_i$, and an I^c-tuple is defi...

conjecture | open | MathDB 页面 | 原文线索

V2-047196 | MATHDB-319247

Strongly minimal covers by independent sets

摘录 / EXCERPT: Let G be a graph. An independent set is a set of vertices containing no edge of G, and a cover of the vertex set is a family of such sets whose union is the vertex set. Strong...

conjecture | open | MathDB 页面 | 原文线索

V2-047197 | MATHDB-319248

Chain and independent-set partition conjecture for posets of bounded width

摘录 / EXCERPT: Let P be a poset of bounded width, viewed through its comparability structure. A chain is a subset of pairwise comparable elements, and an independent set is a subset containing...

conjecture | open | MathDB 页面 | 原文线索

V2-047198 | MATHDB-319249

Conjecture on the limiting probability that a random lift is Ramanujan

摘录 / EXCERPT: Ramanujan-lift probability conjecture. There exists a constant c with $0<c<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047199 | MATHDB-319250

Generalized Montgomery conjecture for dynamic chromatic number

摘录 / EXCERPT: Let G be a nontrivial connected graph. Write $\Delta(G)$ and $\delta(G)$ for its maximum and minimum degrees, respectively, and let $\chi(G)$ and $\chi_2(G)$ denote its chromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-047200 | MATHDB-319275

Tight upper-bound conjecture for admissible vertex counts of normal surfaces

摘录 / EXCERPT: Tight upper-bound conjecture. For every positive $n\neq 1,2,3,5$, equation gives a tight upper bound on the admissible vertex count σ. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-047201 | MATHDB-319276

Fibonacci-growth conjecture for mean admissible vertex counts

摘录 / EXCERPT: Fibonacci-growth conjecture. For every $n\geq 3$, the mean admissible vertex count satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047202 | MATHDB-319308

The blocker number of simple drawings of complete graphs

摘录 / EXCERPT: A drawing of a graph represents vertices by distinct plane points and edges by simple curves, with a vertex meeting an edge only at an endpoint. A drawing is simple if any two edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047203 | MATHDB-319311

The quadratic blocker conjecture for point sets in convex position

摘录 / EXCERPT: A finite point set is in convex position if every point is a vertex of its convex hull. A blocker for such a set is a point, outside the set, lying on an edge segment determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-047204 | MATHDB-319346

Thiéry's maximal-rank conjecture for the forest-tree incidence matrix

摘录 / EXCERPT: Thiéry's conjecture. The labeled matrix M_n has maximal rank, with linearly independent rows. The same conclusion holds in the unlabeled case.

conjecture | open | MathDB 页面 | 原文线索

V2-047205 | MATHDB-319360

Relation between forcibly k-tough and forcibly k-factor graphical degree conditions

摘录 / EXCERPT: Toughness-to-factor conjecture. If pi is forcibly k-tough graphical by Theorem, then pi is forcibly k-factor graphical by Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-047206 | MATHDB-319426

The monochromatic cycle interval conjecture for dense 2-colored graphs

摘录 / EXCERPT: Monochromatic cycle interval conjecture. If $n\geq4$ and $\delta(G)>3n/4$, then every 2-coloring of $E(G)$ has, for every $k\in[4,\lceil n/2\rceil]$, either a red k-cycle or a...

conjecture | open | MathDB 页面 | 原文线索

V2-047207 | MATHDB-319427

The linear-length monochromatic cycle conjecture for dense graphs

摘录 / EXCERPT: Linear-length monochromatic cycle conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-047208 | MATHDB-319458

The linear edge bound conjecture for graphs with at most two bends

摘录 / EXCERPT: Linear edge bound conjecture. A graph G on n vertices belonging to the class R_2 can have at most $O(n)$ edges.

conjecture | open | MathDB 页面 | 原文线索

V2-047209 | MATHDB-319496

Conjecture that the path-poset complex computes digraph homology

摘录 / EXCERPT: Let Γ be a directed graph and let $\mathcal{F}$ be a functor on its path poset. Let $\mathbb{H}_*(\Gamma;\mathcal{F})$ denote the homology of the chain complex...

conjecture | open | MathDB 页面 | 原文线索

V2-047210 | MATHDB-319518

Multipartite universal-cover ball spectral-radius convergence conjecture

摘录 / EXCERPT: Let D be a multipartite degree matrix, let T_D be its universal cover, and let $T_{D,r}(s)$ denote the radius-r ball around $s\in V(T_D)$; write $\rho(\mathord\cdot)$ for spe...

conjecture | open | MathDB 页面 | 原文线索

V2-047211 | MATHDB-319519

Infinitely many D-Ramanujan graphs under the second-eigenvalue condition

摘录 / EXCERPT: Let D be a degree matrix of order $t\geq 2$, let T_D be its universal cover, and let $\lambda_2(D)$ denote the second eigenvalue of D. A finite graph with degree matrix D i...

conjecture | open | MathDB 页面 | 原文线索

V2-047212 | MATHDB-319525

Equality between resistor count and sum-of-squares length for rational circuit resistances

摘录 / EXCERPT: Let $q(z)\in \mathbb{Q}(z)$ be the resistance of a circuit formed by resistors of resistance 1 and z. Define the resistor count of such a circuit to be the number of resistors...

conjecture | open | MathDB 页面 | 原文线索

V2-047213 | MATHDB-319557

Strong-recurrence threshold conjecture for branching random walks on bounded-degree graphs

摘录 / EXCERPT: Strong-recurrence threshold conjecture. Let G be a graph with bounded degrees. Then the BRW is strongly recurrent if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-047214 | MATHDB-319558

Transience conjecture for simple random walk on branching-random-walk traces

摘录 / EXCERPT: Trace-transience conjecture. The trace of a transient BRW is almost surely transient for SRW.

conjecture | open | MathDB 页面 | 原文线索

V2-047215 | MATHDB-319582

Density conjecture for chromatic roots of the graphs $S_{m,n}$

摘录 / EXCERPT: Let $S_{m,n}$ be the graph family considered in the paper, and let a chromatic root mean a zero of its chromatic polynomial in the complex q-plane. There exists a finite constant...

conjecture | open | MathDB 页面 | 原文线索

V2-047216 | MATHDB-319597

Existence of difficult subgraph-counting identities

摘录 / EXCERPT: Existence conjecture. For every $K\geq 4$, there is such an identity that holds for every graph G with n vertices, has one graph $J_{a,i}$ or $J'_{a,j}$ with K vertices, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047217 | MATHDB-319598

Uniqueness of difficult subgraph-counting identities

摘录 / EXCERPT: Uniqueness conjecture. For every $K\geq4$, there is only one subgraph-counting identity modulo the easily described families and the difficult identities for lower K.

conjecture | open | MathDB 页面 | 原文线索

V2-047218 | MATHDB-319622

Dense fixed-subgraph conjecture for the final H-free process

摘录 / EXCERPT: Dense fixed-subgraph conjecture. With high probability, the final graph of the H-free process contains no copy of F. This strengthens the proved result for graphs whose number...

conjecture | open | MathDB 页面 | 原文线索

V2-047219 | MATHDB-319677

Extremal-value conjecture for irredundant families when n equals 2k minus 1

摘录 / EXCERPT: Let $M(n,k)$ be the maximum size of an irredundant family of k-dimensional subcubes of $Q_n=\{0,1\}^n$. Let O be the centre vertex used in the source, and let $\mathcal F_O$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-047220 | MATHDB-319776

The asymptotic conjecture for the partitioning function $f(A,1)$

摘录 / EXCERPT: Let $f(A,B)$ be the partitioning-function quantity discussed in the paper, with A and B positive integers. The asymptotic conjecture for $f(A,1)$. ... The source notes that only...

conjecture | open | MathDB 页面 | 原文线索

V2-047221 | MATHDB-319782

Keslassy-McKeown conjecture on average delay under MW-α

摘录 / EXCERPT: An input-queued switch has queues $Q_{i,j}$, and the MW-α policy is the scheduling policy that selects a schedule using queue lengths raised componentwise to the power...

conjecture | open | MathDB 页面 | 原文线索

V2-047222 | MATHDB-319787

Algebraic form of the Kelly-Ulam vertex-reconstruction conjecture

摘录 / EXCERPT: Let $V=\{1,\ldots,f\}$, let E be the set of two-element subsets of V, and let $\mathcal{G}_f$ be the quotient of the real polynomial algebra in variables $x_{\{i,j\}}$ by the r...

conjecture | open | MathDB 页面 | 原文线索

V2-047223 | MATHDB-319796

The two-bases conjecture for the Tutte polynomial

摘录 / EXCERPT: Let M be a matroid, and let $T_M(x,y)$ denote its Tutte polynomial. Say that M contains two disjoint bases if it has bases B_1 and B_2 with $B_1\cap B_2=\varnothing$; alter...

conjecture | open | MathDB 页面 | 原文线索

V2-047224 | MATHDB-319804

Injectivity conjecture for souped-up Lie algebras

摘录 / EXCERPT: Let $d4e2$ be a souped-up Lie algebra, let $F(d4e2)$ be the infinite free algebra generated by $d4e2$, and let $I(d4e2)$ be the defining ideal used to construct...

conjecture | open | MathDB 页面 | 原文线索

V2-047225 | MATHDB-319831

The forcing conjecture for graphs

摘录 / EXCERPT: A graph is called forcing if it is p-forcing for every $0<p<1$, where a graph H is p-forcing when, for every graph sequence of edge density tending to p, the condition...

conjecture | open | MathDB 页面 | 原文线索

V2-047226 | MATHDB-319847

Level classification conjecture for scalar Poisson lambda-brackets

摘录 / EXCERPT: Let a non-quasiconstant coefficient Poisson λ-bracket on $\widetilde{\mathcal{R}}_1$ have odd order N and level m, where the level is defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-047227 | MATHDB-319848

CFT-type Poisson vertex algebra classification conjecture

摘录 / EXCERPT: Let $A=\mathbb{C}[D]L\oplus\mathbb{C}[D]W_1\oplus\cdots\oplus\mathbb{C}[D]W_{\ell-1}\oplus\mathbb{C}C$ be a $\mathbb{C}[D]$-module with $DC=0$. Endow $S(A)$ with a Poisson...

conjecture | open | MathDB 页面 | 原文线索

V2-047228 | MATHDB-319863

NP-completeness conjecture for generalized k_1-connectivity

摘录 / EXCERPT: The generalized-connectivity NP-completeness conjecture. Given G and k, deciding whether

conjecture | open | MathDB 页面 | 原文线索

V2-047229 | MATHDB-319870

Small K_5-models at average degree greater than six

摘录 / EXCERPT: Let G be a graph, let $|G|$ denote its number of vertices, and let a K_5-model be a model of the complete graph K_5 in G. For a real number $\varepsilon>0$, suppose that th...

conjecture | open | MathDB 页面 | 原文线索

V2-047230 | MATHDB-319972

Perfect-matching conjecture for regularly colored cycles

摘录 / EXCERPT: Let G be a $2d$-regular graph whose edges are colored so that the edges of each color form a cycle of length $2d$. Perfect-matching conjecture. The graph G has a perfect matchi...

conjecture | open | MathDB 页面 | 原文线索

V2-047231 | MATHDB-319991

The strongly 2-connected regular digraph Hamiltonicity conjecture

摘录 / EXCERPT: Hamiltonicity conjecture. If $d\ge n/3$, then G contains a Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-047232 | MATHDB-319993

Jackson and Ordaz's connectivity and independence conjecture for digraphs

摘录 / EXCERPT: Jackson and Ordaz's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-047233 | MATHDB-320026

King and Pelsmajer's bounded-exception domination conjecture for plane triangulations

摘录 / EXCERPT: King and Pelsmajer's conjecture. For any constant t, there exists n_t such that every n-vertex plane triangulation with $n>n_t$ and at most t vertices of degree other than...

conjecture | open | MathDB 页面 | 原文线索

V2-047234 | MATHDB-320048

The fractional Young convolution inequality for regular hypergraphs

摘录 / EXCERPT: Let $M\geq 2$ and write $[M]=\{1,2,\ldots,M\}$. Let $\mathcal{G}$ be a d-regular hypergraph on $[M]$, let r and $p_{\mathbf{s}}$ for $\mathbf{s}\in\mathcal{G}$ be real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-047235 | MATHDB-320049

The fractional Brunn-Minkowski inequality for hypergraph fractional partitions

摘录 / EXCERPT: The fractional Brunn-Minkowski conjecture. For every such fractional partition,

conjecture | open | MathDB 页面 | 原文线索

V2-047236 | MATHDB-320054

The discrete-torus path-length conjecture

摘录 / EXCERPT: Let C_3^n be the n-dimensional discrete torus, and let G be a subgraph of C_3^n with minimum degree at least d. The discrete-torus path-length conjecture. G must contai...

conjecture | open | MathDB 页面 | 原文线索

V2-047237 | MATHDB-320055

The hypercube average-degree path conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, and let G be a subgraph of Q_n with average degree at least d. The hypercube average-degree path conjecture. G contains a path o...

conjecture | open | MathDB 页面 | 原文线索

V2-047238 | MATHDB-320062

Autocatalytic networks and critical siphons

摘录 / EXCERPT: Let G be a weakly-reversible chemical reaction network (CRN). A critical siphon is a siphon that is stoichiometrically compatible with a positive point, and G is autocatalytic...

conjecture | open | MathDB 页面 | 原文线索

V2-047239 | MATHDB-320071

Completion-class invariance of c_2 modulo the Lefschetz motive

摘录 / EXCERPT: Let G_1 and G_2 be primitive-divergent graphs in ϕ^4 theory. For a graph G, let $\widehat G$ be its completion, obtained by adjoining a vertex to the four three-valent v...

conjecture | open | MathDB 页面 | 原文线索

V2-047240 | MATHDB-320075

Ivančo-Jendrol' conjecture on total edge irregularity strength

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with $m:=|E|$ edges, maximum degree Δ, and let $tes(G)$ denote the least positive integer s for which there is a weighting...

conjecture | open | MathDB 页面 | 原文线索

V2-047241 | MATHDB-320077

Borel summability of the loop vertex expansion in non-integer dimension

摘录 / EXCERPT: Let D be real with $0\leq D<2$, and let $A_{D,\bar{\mathcal{T}}}$ denote the D-dimensional tree amplitude obtained by summing the Feynman amplitudes over graphs containing the...

conjecture | open | MathDB 页面 | 原文线索

V2-047242 | MATHDB-320105

Nonvanishing remainder conjecture for orthogonal Heisenberg invariants

摘录 / EXCERPT: For each $n\geq 1$, let R_0 denote the remainder of the element D_0 in the construction of the minimal relation for ${\mathcal H}(n)^{O(n)}$. Nonvanishing remainder conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047243 | MATHDB-320136

Connelly-Whiteley vertex-splitting conjecture for 3-circuits and global rigidity

摘录 / EXCERPT: Consider a vertex splitting in a three-dimensional framework, with both the added vertex and the split vertex at least 4-valent. Connelly-Whiteley vertex-splitting conjecture. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-047244 | MATHDB-320139

Vertex-splitting conjecture for circuits and global rigidity in 3-space

摘录 / EXCERPT: Consider vertex splitting in a three-dimensional framework, with both the added vertex and the split vertex at least 4-valent. Vertex-splitting circuit conjecture. The operation...

conjecture | open | MathDB 页面 | 原文线索

V2-047245 | MATHDB-320153

The near-acyclic characterization of chromatic threshold zero

摘录 / EXCERPT: A graph G is near acyclic if it has a stable set S such that $G-S$ is a forest and every odd cycle has at least two vertices in S. Near-acyclic chromatic-threshold conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-047246 | MATHDB-320154

The locally bipartite chromatic-threshold conjecture

摘录 / EXCERPT: A graph is locally bipartite if the neighborhood of every vertex induces a bipartite graph. Locally bipartite chromatic-threshold conjecture. The class of locally bipartite graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-047247 | MATHDB-320173

Small-diameter obstruction for four pairs of points

摘录 / EXCERPT: Let $A_i=\{p_i,q_i\}$, $i=1,2,3,4$, be four sets of two points each in $\mathbf{R}^3$. For a set of points, call it convexly independent when every point is a vertex of its convex...

conjecture | open | MathDB 页面 | 原文线索

V2-047248 | MATHDB-320191

Asymptotic clique partition conjecture for uniform hypergraphs

摘录 / EXCERPT: Asymptotic clique partition conjecture. For every fixed $r\ge2$,

conjecture | open | MathDB 页面 | 原文线索

V2-047249 | MATHDB-320192

Bounded-endvertex path-packing conjecture for degenerate graphs

摘录 / EXCERPT: Let k be a positive integer, let G be a graph of degeneracy d, and let c_k denote a constant depending on k. A collection of paths is edge-disjoint if no two paths share...

open_problem | open | MathDB 页面 | 原文线索

V2-047250 | MATHDB-320200

The good-vertex conjecture for the independent-set bound

摘录 / EXCERPT: For a graph G, let ${\rm ind}(G)$ be its number of independent sets, let $d(u)$ be the degree of u, let ${\rm iso}(G)$ be the number of isolated vertices, and define ... A vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-047251 | MATHDB-320203

Alon-Kahn conjecture on independent sets in regular graphs

摘录 / EXCERPT: Let G be a simple, finite, undirected graph with N vertices that is d-regular, and let $i(G)$ denote its number of independent sets. The graph $\frac{N}{2d}K_{d,d}$ is the di...

conjecture | open | MathDB 页面 | 原文线索

V2-047252 | MATHDB-320274

The pointedness conjecture for highway-path homology classes

摘录 / EXCERPT: Let S be an orientable surface and let u,v be boundary vertices. Fix a basepath p from u to v, and let $H(u,v)$ be the set of homology classes...

conjecture | open | MathDB 页面 | 原文线索

V2-047253 | MATHDB-320275

The rationality conjecture for generating functions of surface-network measurements

摘录 / EXCERPT: Let S be an orientable surface, let u,v be boundary vertices, and let $M_{u,v}$ denote the generating function associated with the highway paths from u to v, expressed usin...

conjecture | open | MathDB 页面 | 原文线索

V2-047254 | MATHDB-320276

The discreteness conjecture for monodromy of reduced oriented networks

摘录 / EXCERPT: Let N be an oriented network, and call it reduced when it satisfies the paper's reducedness condition. Let the monodromy action on N be the action induced by the network's mono...

conjecture | open | MathDB 页面 | 原文线索

V2-047255 | MATHDB-320288

Behsaz-Hatami-Mahmoodian's minimum vertex-cover conjecture for generalized Petersen graphs

摘录 / EXCERPT: Let $P(n,k)$ be the generalized Petersen graph, and let $\beta(P(n,k))$ denote the size of a minimum vertex cover. Behsaz-Hatami-Mahmoodian's conjecture. For all n and k, ... Sin...

conjecture | open | MathDB 页面 | 原文线索

V2-047256 | MATHDB-320300

The exponential properly coloured path or cycle conjecture

摘录 / EXCERPT: Let G be an edge-coloured graph, and let $delta^c(G)$ denote its minimum colour degree. A properly coloured (p.c.) path or cycle is one whose consecutive edges have distinct colo...

conjecture | open | MathDB 页面 | 原文线索

V2-047257 | MATHDB-320312

The implication from Delta-SOS to Phi-SOS for graphs

摘录 / EXCERPT: Let $\mathcal G$ be a graph. Say that $\mathcal G$ is Δ-SOS if every relevant Rayleigh difference $\Delta G\{e,f\}$ has the required sum-of-squares representation, and say t...

conjecture | open | MathDB 页面 | 原文线索

V2-047258 | MATHDB-320326

Fon-der-Flaass conjecture for the Turán (3,4)-problem

摘录 / EXCERPT: Fon-der-Flaass conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047259 | MATHDB-320327

Read's log-concavity conjecture for chromatic polynomial coefficients

摘录 / EXCERPT: Let G be a finite graph, and write its chromatic polynomial as ... A sequence $a_0,a_1,\ldots,a_n$ is log-concave if $a_{i-1}a_{i+1}\leq a_i^2$ for every $0<i<n$. Read's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-047260 | MATHDB-320351

The directed graph square edge-growth conjecture

摘录 / EXCERPT: Directed square edge-growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047261 | MATHDB-320363

Lovász's mobility-chromatic number conjecture for finite loopless graphs

摘录 / EXCERPT: Let H be a finite loopless graph. Its chromatic number is denoted by $\chi(H)$, and its mobility $m(H)$ is the largest integer d for which H is d-mobile, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-047262 | MATHDB-320373

The minimum vertex-degree conjecture for perfect matchings in uniform hypergraphs

摘录 / EXCERPT: Minimum vertex-degree conjecture. For each integer $r\geq 3$ there exists an integer $n_0=n_0(r)$ such that the following holds. Suppose that H is an r-uniform hypergraph whose...

conjecture | open | MathDB 页面 | 原文线索

V2-047263 | MATHDB-320378

Depth-seven conjecture for edge rings of finite graphs

摘录 / EXCERPT: Let G be a finite graph on the vertex set $[d]=\{1,\ldots,d\}$ with $d\geq 7$, and let $K[G]$ be its edge ring. Depth-seven conjecture. ... The claim arises from computations showi...

conjecture | open | MathDB 页面 | 原文线索

V2-047264 | MATHDB-320382

Zhang et al.'s conjecture on adjacent vertex distinguishing total colourings

摘录 / EXCERPT: Let G be a graph, and let $\chi_{at}(G)$ denote its adjacent vertex distinguishing total chromatic number, the least number of colours in a proper total colouring such that every...

conjecture | open | MathDB 页面 | 原文线索

V2-047265 | MATHDB-320386

Wojciechowski-Keller volume-boundary conjecture for stochastic completeness

摘录 / EXCERPT: Let (V,E) be a graph, choose a fixed root $x^*\in V$, and let B_r be the ball of radius r about x^*. Let ∂B_r denote its outer vertex boundary, and write $\#A$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-047266 | MATHDB-320391

The Global Attractor Conjecture for detailed balanced reaction systems

摘录 / EXCERPT: Consider a reversible reaction system and a detailed balanced equilibrium, meaning an equilibrium satisfying detailed balance for every reversible reaction. A trajectory is a solut...

conjecture | open | MathDB 页面 | 原文线索

V2-047267 | MATHDB-320403

Combinatorial tiling conjecture for paired non-acyclic graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a non-acyclic graph, with a fixed-point-free involution $\mu:V\to V$ and, for every vertex v, a bijection $\sigma_v:\delta(v)\to\delta(\mu(v))$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-047268 | MATHDB-320405

Welsh's conjecture on real flow roots

摘录 / EXCERPT: Let G be a bridgeless graph, and let $\Phi_G(Q)$ denote its flow polynomial. Welsh's conjecture. For any bridgeless graph G, ... The conjecture parallels the Birkhoff-Lewis resul...

conjecture | open | MathDB 页面 | 原文线索

V2-047269 | MATHDB-320406

Haggard-Pearce-Royle conjecture on flow roots above five

摘录 / EXCERPT: Let G be a bridgeless graph, and let $\Phi_G(Q)$ denote its flow polynomial. Haggard-Pearce-Royle conjecture. For any bridgeless graph G, ... This is a proposed weakening of Wels...

conjecture | open | MathDB 页面 | 原文线索

V2-047270 | MATHDB-320408

Dominant-eigenvalue pattern conjecture for generalized Petersen limiting curves

摘录 / EXCERPT: For a family of generalized Petersen graphs $G(nk,k)$, let $Q_c(k)$ be the real crossing value and let $\mu_{k+1,\ell,(\ell),\star}$ denote the indicated dominant eigenvalues. Domi...

conjecture | open | MathDB 页面 | 原文线索

V2-047271 | MATHDB-320409

Eigenvalue-crossing conjecture for non-planar transfer matrices

摘录 / EXCERPT: Consider a family $G_{n,L}$ of non-planar connected graphs as above. Let Q_ℓ be the smallest $Q>0$ for which the transfer matrix $\widetilde{\sf T}_{L,\ell,(\ell)}$ for the fu...

conjecture | open | MathDB 页面 | 原文线索

V2-047272 | MATHDB-320434

Free-generation conjecture for the closed-diagram endomorphism algebra

摘录 / EXCERPT: Let I be the vertex set of the Dynkin diagram. For $i\in I$ and $k\in\mathbb{N}$, let c_k^i be a clockwise circle labeled i containing exactly k hollow dots, and let c_k...

conjecture | open | MathDB 页面 | 原文线索

V2-047273 | MATHDB-320441

The extremal-family conjecture for cliques in graphs with bounded minimum degree

摘录 / EXCERPT: Let n and β be such that the family of graphs $\mathcal{G}(n,\beta)$ is defined, and let (n,β) be feasible when $G[V_0]$ is triangle-free for...

conjecture | open | MathDB 页面 | 原文线索

V2-047274 | MATHDB-320466

Conjecture on the facets of the network model cone

摘录 / EXCERPT: Let B denote the model matrix for the directed random graph network model with reciprocation, and let $\mathrm{cone}(B)$ be its cone. For a network on n nodes, the facets conje...

conjecture | open | MathDB 页面 | 原文线索

V2-047275 | MATHDB-320492

Subpolynomial vertex-stabiliser growth for almost simple tetravalent arc-transitive graphs

摘录 / EXCERPT: Let $c>0$. Let ${\mathcal F}_c$ be a finite family of graphs, let G be an almost simple group, and let Γ be a connected tetravalent G-arc-transitive graph with vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-047276 | MATHDB-320513

The conjecture that infinite Ramanujan graphs are not Liouville

摘录 / EXCERPT: Let G be an infinite d-regular graph. Call G Ramanujan if the spectral radius of its Markov operator on l^2 of the vertices equals ... the spectral radius of the d-regular...

conjecture | open | MathDB 页面 | 原文线索

V2-047277 | MATHDB-320514

The conjecture that quasi-isometry to a nonamenable Cayley graph prevents the Liouville property

摘录 / EXCERPT: Let G be a graph that is quasi-isometric to a nonamenable Cayley graph. The quasi-isometric non-Liouville conjecture. The graph G is not Liouville. The source presents this as...

conjecture | open | MathDB 页面 | 原文线索

V2-047278 | MATHDB-320546

The kernel-vertex K_4-minor conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a 3-connected graph with chromatic number 4 and no K_5 minor. A kernel vertex is a vertex in the kernel-vertex set used by the source, and let $Ex(U)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-047279 | MATHDB-320562

Cross-triangle-intersecting family product conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ and $\mathcal{G}$ be families of graphs on the same set of n vertices, and call them cross-triangle-intersecting if $F\cap G$ contains a triangle for every...

conjecture | open | MathDB 页面 | 原文线索

V2-047280 | MATHDB-320564

Asymptotic structure of the minimizer near a singular point

摘录 / EXCERPT: Let $H:\mathbb{R}^2\to\mathbb{R}$ be the scaling-limit function associated with the minimizer near a singular point of the minimal height function. The asymptotic structure conject...

conjecture | open | MathDB 页面 | 原文线索

V2-047281 | MATHDB-320565

Bethe-root condensation on a symmetric curve

摘录 / EXCERPT: Let $N\to\infty$ and $n\to\infty$ with $\alpha=n/N$ fixed. Consider the Bethe equations for the six-vertex transfer matrix and the maximal eigenvalue in the subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-047282 | MATHDB-320566

Intersection alternatives for six-vertex Bethe contours

摘录 / EXCERPT: Let C be the smooth Bethe-root curve and define ... Assume $\Delta\neq0$, and let $\alpha=n/N$ be fixed. Bethe-contour intersection conjecture. Exactly one of the following occurs:...

conjecture | open | MathDB 页面 | 原文线索

V2-047283 | MATHDB-320567

Convexity and concavity of six-vertex phase interfaces

摘录 / EXCERPT: Let $q=1$, and impose stabilizing fixed boundary conditions producing an outer interface between the ferroelectric and disordered phases and an inner interface between the disorder...

conjecture | open | MathDB 页面 | 原文线索

V2-047284 | MATHDB-320574

Degeneracy-bandwidth always-goodness conjecture

摘录 / EXCERPT: For fixed d, let $\beta(n)=o(n)$ be a function, and let $\mathcal G'_{d,\beta}$ be the class of graphs G whose degeneracy is at most d and whose bandwidth is at most...

conjecture | open | MathDB 页面 | 原文线索

V2-047285 | MATHDB-320587

Extremal family conjecture for nonuniform simplex-free set systems

摘录 / EXCERPT: Let X be an n-element set, and write $X^{(\le m)}=\bigcup_{0\le i\le m}X^{(i)}$ and $X_x^{(\le m)}=\{A\in X^{(\le m)}:x\in A\}$. A collection of $d+1$ sets is a d-simplex if...

conjecture | open | MathDB 页面 | 原文线索

V2-047286 | MATHDB-320598

Holonomy operator ordering conjecture for U(N) spin-network states

摘录 / EXCERPT: Holonomy operator ordering conjecture. The holonomy operator around $\mathcal L$ acts on spin-network states as

conjecture | open | MathDB 页面 | 原文线索

V2-047287 | MATHDB-320606

Nonexistence conjecture for bipartite graphs of defect four

摘录 / EXCERPT: Nonexistence conjecture. There is no bipartite $(D835,D,-4)$-graph with $D835 e 3$ and $D\ge 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-047288 | MATHDB-320612

Uniqueness conjecture for graphs with cyclic defect or excess

摘录 / EXCERPT: A graph with defect or excess 2 is said to have cyclic defect or excess when its defect or excess matrix is the adjacency matrix of a cycle. The Möbius ladder on 8 vertices is...

conjecture | open | MathDB 页面 | 原文线索

V2-047289 | MATHDB-320650

The asymptotic t-broom conjecture for distance graphs

摘录 / EXCERPT: The asymptotic t-broom conjecture. For every integer $k\geq 3$, there exists a function $h=h(k)$ such that, whenever $n\geq h(k)$, the maximum of $e(G_k)$ over all graphs G with...

conjecture | open | MathDB 页面 | 原文线索

V2-047290 | MATHDB-320696

Relative Turán threshold conjecture for sparse random graphs

摘录 / EXCERPT: Relative Turán threshold conjecture. For every $\epsilon>0$ and every graph H, there exist positive constants c and C such that

conjecture | open | MathDB 页面 | 原文线索

V2-047291 | MATHDB-320698

Random hypergraph removal conjecture

摘录 / EXCERPT: Random hypergraph removal conjecture. For every $\delta>0$ and every integer $k\geq 3$, there exist constants $\epsilon>0$ and C such that, if $p\geq Cn^{-1/k}$, then with probab...

conjecture | open | MathDB 页面 | 原文线索

V2-047292 | MATHDB-320721

Caporossi-Cvetković-Gutman-Hansen conjecture on maximal energy of unicyclic graphs

摘录 / EXCERPT: Caporossi-Cvetković-Gutman-Hansen conjecture. Among all unicyclic graphs on n vertices, the cycle C_n has maximal energy if $n\leq 7$ or $n\in\{9,10,11,13,15\}$. For all other...

conjecture | open | MathDB 页面 | 原文线索

V2-047293 | MATHDB-320743

Vertexwise admissibility conjecture for minuscule cocharacters

摘录 / EXCERPT: Vertexwise admissibility conjecture. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-047294 | MATHDB-320777

The semicircle law conjecture for sparse random regular graphs

摘录 / EXCERPT: The semicircle law conjecture. If $d\rightarrow\infty$, then the empirical spectral distribution of $\frac{1}{\sqrt n}M'_n$ converges to the standard semicircle distribution. This...

conjecture | open | MathDB 页面 | 原文线索

V2-047295 | MATHDB-320844

Lovász-Plummer conjecture for regular matching-covered graphs

摘录 / EXCERPT: Let G be a k-regular matching-covered graph, meaning that every edge of G belongs to a perfect matching. Lovász-Plummer regular-graph conjecture. For every $k\geq 3$, there e...

conjecture | open | MathDB 页面 | 原文线索

V2-047296 | MATHDB-320847

The edge-connectivity conjecture for UBB cardinality

摘录 / EXCERPT: Edge-connectivity UBB conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-047297 | MATHDB-320863

Structural conjecture for planar graph complexes

摘录 / EXCERPT: Let G be a connected planar graph. A cube complex is a chain complex ... indexed by the vertices of an n-dimensional hypercube. For a vertex v, write $|v|=\sum_i v_i$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047298 | MATHDB-320916

Conjectured skeletons and non-adjacencies for partial metric and weighted quasi-metric cones

摘录 / EXCERPT: Skeleton conjecture. (i) ${{\operatorname{Sk}}}(\text{\sl OCUT}_n)=K_{2S(n,2)}$ and is a subgraph of ${{\operatorname{Sk}}}(\text{\sl WQMET}_n)$. (ii) The complement of...

conjecture | open | MathDB 页面 | 原文线索

V2-047299 | MATHDB-320917

Conjectured ridge graphs of partial metric and weighted quasi-metric cones

摘录 / EXCERPT: Ridge-graph conjecture. (i) ${{\operatorname{Ri}}}(\text{\sl PMET}_n)$ has diameter 2, with all non-adjacencies given by $L_{ii}\nsim M_{ik}$;...

conjecture | open | MathDB 页面 | 原文线索

V2-047300 | MATHDB-320922

The asymptotic cluster-weight conjecture for H-coloring tori

摘录 / EXCERPT: Let H be a graph, let $Lambda$ be a vector of vertex weights, and let $mgeq 2$ be even. Write $Hom({\mathbb Z}^d_m,H)$ for the set of H-colorings of the m-ary d-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-047301 | MATHDB-320924

The polynomial bottleneck conjecture for local Glauber dynamics on H-coloring tori

摘录 / EXCERPT: Fix a graph H, vertex-weight vector Λ, and even $m\geq 2$. Let $Z_\Lambda({\mathbb Z}^d_m,H)$ be the partition function, and let $C(0)$ denote the central part of a parti...

conjecture | open | MathDB 页面 | 原文线索

V2-047302 | MATHDB-320955

Local induced-cycle conjecture for vertex-prescribed deletion

摘录 / EXCERPT: Vertex-prescribed induced-cycle conjecture. For every k, there exists an $h(k)$ such that for every vertex v of every $h(k)$-connected graph G, there exists an induced cycle...

open_problem | open | MathDB 页面 | 原文线索

V2-047303 | MATHDB-320960

Perfect tunneling conjecture for asymmetric equilateral triple wells

摘录 / EXCERPT: Let $x,y,z\in X$ be three wells in a graph with equal pairwise distances $a=b=c$, and let c_{uv} denote the sum over shortest paths from u to v of the products of inverse squ...

conjecture | open | MathDB 页面 | 原文线索

V2-047304 | MATHDB-320973

Burton's worst-case vertex-normal-surface growth conjecture

摘录 / EXCERPT: Let n be a positive integer other than $1,2,3,5$, and let k be the integer determined by the congruence class of n modulo 4. A vertex normal surface is a vertex solution in...

conjecture | open | MathDB 页面 | 原文线索

V2-047305 | MATHDB-320974

Burton's average vertex-normal-surface growth conjecture

摘录 / EXCERPT: For each n, let $\overline{\sigma}_n$ denote the average number of vertex normal surfaces over all closed 3-manifold triangulations of size n, counted up to isomorphism. A vert...

conjecture | open | MathDB 页面 | 原文线索

V2-047306 | MATHDB-320997

Uniform lower-bound conjecture for critical-graph density constants

摘录 / EXCERPT: For each integer $\ell\geq 3$, let $c_{\ell,k}$ be the quadratic-density constant for k-critical graphs of odd-girth at least ℓ, with $k\geq 4$. Uniform lower-bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-047307 | MATHDB-321016

Gromov-Hausdorff convergence conjecture for random quadrangulations

摘录 / EXCERPT: Let Q_n be the random quadrangulation with vertex set $V(Q_n)$ and graph distance d_{Q_n}. Consider the rescaled pointed metric spaces $(V(Q_n),n^{-1/4}d_{Q_n})$. Gromov-Hausdo...

conjecture | open | MathDB 页面 | 原文线索

V2-047308 | MATHDB-321022

Degree condition for maximum-length cycles in bipartite digraphs

摘录 / EXCERPT: Maximum-cycle degree conjecture. If this inequality holds whenever u and v lie in opposite colour classes and $uv\notin A(D)$, then D contains an oriented cycle of length...

conjecture | open | MathDB 页面 | 原文线索

V2-047309 | MATHDB-321024

The semiprimitivity characterization of graph-restrictive permutation groups

摘录 / EXCERPT: Let L be a transitive permutation group. It is semiprimitive if each of its normal subgroups is either transitive or semiregular, and it is graph-restrictive if there exists a co...

conjecture | open | MathDB 页面 | 原文线索

V2-047310 | MATHDB-321038

Caro's three-quarters conjecture for rainbow connection

摘录 / EXCERPT: Let G be a connected graph with n vertices, and write $\delta(G)$ for its minimum degree and $rc(G)$ for its rainbow connection number. Caro's conjecture. If $\delta(G)\geq 3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-047311 | MATHDB-321042

Hàn-Person-Schacht minimum degree conjecture for perfect matchings in hypergraphs

摘录 / EXCERPT: Hàn-Person-Schacht conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047312 | MATHDB-321049

The diameter-three conjecture for small Weyl graphs

摘录 / EXCERPT: Let Φ be an irreducible classical root system of rank at least 2, and let $\Gamma_s(\Phi)$ be its small Weyl graph. The diameter of a graph is the maximum distance between t...

conjecture | open | MathDB 页面 | 原文线索

V2-047313 | MATHDB-321066

Bruhn's Hamilton-circle conjecture for locally finite planar graphs

摘录 / EXCERPT: Bruhn's conjecture. Every 4-connected locally finite planar graph has a Hamilton circle.

conjecture | solved | MathDB 页面 | 原文线索

V2-047314 | MATHDB-321071

Generalized lower bound conjecture for vertex-minimal triangulated manifolds

摘录 / EXCERPT: Let M be an m-vertex connected closed triangulated d-manifold, and let β_i be its Betti numbers over some field. Let ${\cal K}_l^{\ast}(d)$ denote the class used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-047315 | MATHDB-321074

Counting word-representable digraphs with a prescribed number of strong components

摘录 / EXCERPT: Let an ℓ-word be a word of length ℓ over an n-letter alphabet, and suppose it has a unique factorisation into factors corresponding to alphabet-disjoint partitions. A w...

conjecture | open | MathDB 页面 | 原文线索

V2-047316 | MATHDB-321110

The minimum critical independent set count conjecture for graphs

摘录 / EXCERPT: Minimum critical independent set count conjecture. The number of inclusion minimal independent sets S such that $d(S)>0$ is greater than or equal to $d_c(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-047317 | MATHDB-321114

Bonnington-Watkins conjecture on infinite face-boundaries

摘录 / EXCERPT: Let G be a locally finite vertex-transitive graph of connectivity 3, and consider a planar embedding of G. Bonnington-Watkins conjecture. No such embedding has a vertex incid...

conjecture | open | MathDB 页面 | 原文线索

V2-047318 | MATHDB-321138

Markov width conjecture for cycles

摘录 / EXCERPT: Let C be a cycle of length n, with edges $12, 23, \ldots, n1$. For a vertex i, let d_i denote its associated model parameter, and let $\mu(C,d)$ be the Markov width of the...

conjecture | open | MathDB 页面 | 原文线索

V2-047319 | MATHDB-321194

Transience of the degree-three, two-parameter fractal mother graph

摘录 / EXCERPT: Transience conjecture. The graph $\mathcal G(3,2,\infty)$ is transient.

conjecture | open | MathDB 页面 | 原文线索

V2-047320 | MATHDB-321195

Subpolynomial maximal resistance in degree-two fractal mother graphs

摘录 / EXCERPT: Subpolynomial resistance conjecture. The maximal resistance in $\mathcal G(2,m,n)$ grows slower than any power of n:

conjecture | open | MathDB 页面 | 原文线索

V2-047321 | MATHDB-321196

Root-antiroot resistance maximization conjecture for fractal mother graphs

摘录 / EXCERPT: Root-antiroot conjecture. The root and the antiroot maximize the resistance in $\mathcal G(d,m,n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-047322 | MATHDB-321197

Equality of group and group-list edge chromatic numbers

摘录 / EXCERPT: Let G be a simple graph. Denote its group edge chromatic number by $\chi'_g(G)$ and its group list edge chromatic number by $\chi'_{gl}(G)$. Equality conjecture. For every simple...

conjecture | open | MathDB 页面 | 原文线索

V2-047323 | MATHDB-321216

The extremal cover-size conjecture for unstable hypergraph Kneser colorings

摘录 / EXCERPT: Let $k\geq 5$, and let r and ℓ be positive integers satisfying ... For a cover C with elements t_i, let $H_{C,r}(n)$ denote the associated hypergraph, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047324 | MATHDB-321229

Asymptotic conjecture for the minimum size of semisaturated cycle graphs

摘录 / EXCERPT: Let C_k be the cycle on k vertices, and let ${\rm ssat}(n,C_k)$ denote the minimum number of edges in an n-vertex graph that is C_k-semisaturated, meaning that adding any m...

conjecture | open | MathDB 页面 | 原文线索

V2-047325 | MATHDB-321230

Emms, Hancock, Severini and Wilson's spectral characterization conjecture for strongly regular graphs

摘录 / EXCERPT: Let G and H be strongly regular graphs with the same set of parameters. For a graph F, let ${\bf U}(F)$ denote its transition matrix, let ${\bf U}(F)^3$ be its third power, l...

conjecture | open | MathDB 页面 | 原文线索

V2-047326 | MATHDB-321247

Abstract duality invariance conjecture for the interior polynomial

摘录 / EXCERPT: Let $G=(V_0,V_1,E)$ be a connected bipartite graph inducing hypergraphs $d4a2_0$ and $d4a2_1$ by taking the neighborhoods of the vertices in V_0 and V_1, respectively. The inte...

conjecture | open | MathDB 页面 | 原文线索

V2-047327 | MATHDB-321248

Cayley polytope volume conjecture

摘录 / EXCERPT: Let C_j be the Cayley polytope associated with Cayley compositions, and let its normalized volume be ... where $\mathrm{vol}(C_j)$ is the Euclidean volume of C_j. Cayley polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-047328 | MATHDB-321265

Gyárfás's conjecture on few-coloured matchings in uniform hypergraphs

摘录 / EXCERPT: Let r, s, t, k be positive integers. A t-colouring assigns one of t colours to every edge of a complete r-uniform hypergraph, and an s-coloured matching of size k...

conjecture | open | MathDB 页面 | 原文线索

V2-047329 | MATHDB-321277

Universal bounded-tail conjecture for Biggs potentials of distance-regular graphs

摘录 / EXCERPT: Let G be a distance-regular graph of diameter D, and let $\phi_0,\ldots,\phi_{D-1}$ denote its Biggs potentials. For an integer m with $0\leq m\leq D-2$, the universal bounde...

conjecture | open | MathDB 页面 | 原文线索

V2-047330 | MATHDB-321279

Finite-time constant-topology conjecture for SBC and SBI opinion dynamics

摘录 / EXCERPT: Let n interacting agents have opinion vector $x(t)\in\mathbb{R}^n$ and bounds vector $r\in\mathbb{R}^n_{>0}$. In the synchronized bounded confidence (SBC) and synchronized bounde...

conjecture | open | MathDB 页面 | 原文线索

V2-047331 | MATHDB-321299

Kolmogorov-Barzdin embedding conjecture for higher-dimensional simplicial complexes

摘录 / EXCERPT: Let X be a k-dimensional simplicial complex with N simplices, and suppose that each vertex lies in at most L simplices. Let $n\ge 2k+1$. An embedding is 1-thick if the dist...

conjecture | open | MathDB 页面 | 原文线索

V2-047332 | MATHDB-321315

Generalised squashed cube conjecture for exact clause-sets

摘录 / EXCERPT: Generalised squashed cube conjecture. Then F is matching lean.

conjecture | open | MathDB 页面 | 原文线索

V2-047333 | MATHDB-321316

Alon-Saks-Seymour conjecture for clause-sets

摘录 / EXCERPT: Alon-Saks-Seymour conjecture. For every such clause-set F,

conjecture | open | MathDB 页面 | 原文线索

V2-047334 | MATHDB-321321

The two-part decomposition conjecture for q(k)

摘录 / EXCERPT: For a positive integer k, define ... where the minimum is over positive integers $k_1,\ldots,k_s$ with $k_1+\cdots+k_s=k$ and $s\geq 1$, and set $q(0):=0$. Here $\omega(m)$ denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-047335 | MATHDB-321365

Generation conjecture for Vogel's algebra

摘录 / EXCERPT: Let Λ be Vogel's algebra of three-legged Jacobi diagrams that are antisymmetric under permutations of their legs, with product defined by insertion at a vertex. Generation...

conjecture | open | MathDB 页面 | 原文线索

V2-047336 | MATHDB-321378

The even-order non-separable subgraph conjecture

摘录 / EXCERPT: Let G be a non-separable graph of even order, meaning that no subset W of its vertices with $2\leq |W|\leq |V(G)|-2$ becomes disconnected from its complement after any switchin...

conjecture | open | MathDB 页面 | 原文线索

V2-047337 | MATHDB-321380

Degree-$n-1$ quasipolynomiality conjecture for reachable chip-firing configurations

摘录 / EXCERPT: Let G be an arbitrary graph with n vertices, and consider the configuration $(c,0,0,\dots,0)$, with c chips on one vertex and zero chips on the remaining vertices. Reachable-...

conjecture | open | MathDB 页面 | 原文线索

V2-047338 | MATHDB-321405

Brenti-Welsh log-concavity conjecture for chromatic polynomials

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $P_G(q)$ be its chromatic polynomial, and write $P(q)=P_G(q)$. Let $\chi(G)$ denote the chromatic number of G. Brenti-Welsh conjecture. The chromati...

conjecture | open | MathDB 页面 | 原文线索

V2-047339 | MATHDB-321411

Transitive-graph nonpercolation conjecture for Brownian-moving nodes

摘录 / EXCERPT: Let G be an infinite vertex-transitive graph of bounded degree, with every edge assigned length 1 and the resulting graph metric. Let $\Pi_0(G)$ place one node at each vertex,...

conjecture | open | MathDB 页面 | 原文线索

V2-047340 | MATHDB-321436

Hamiltonian path conjecture for Cayley graphs of finite groups

摘录 / EXCERPT: Hamiltonian path conjecture. There exists a simple path in the Cayley graph of G with respect to A that passes through every vertex. Equivalently, there is a word $w\in A^+$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-047341 | MATHDB-321481

The Zagreb index inequality conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a simple connected graph with $n=|V|$ vertices and $m=|E|$ edges. Its first and second Zagreb indices are ... where d_i is the degree of vertex i. Zagreb index i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-047342 | MATHDB-321501

Regularity conjecture for uniquely K_r-saturated graphs

摘录 / EXCERPT: Regularity conjecture. For each r, a uniquely K_r-saturated graph with no dominating vertex is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-047343 | MATHDB-321507

Erdős-Hajnal-type conjecture for multipartite subgraphs in hypergraphs

摘录 / EXCERPT: Erdős-Hajnal-type hypergraph conjecture. Any $\mathcal{H}$-free k-uniform hypergraph on n vertices contains a complete or empty k-partite subgraph each part of which has orde...

conjecture | open | MathDB 页面 | 原文线索

V2-047344 | MATHDB-321530

Grytczuk's conjecture on bounded nonrepetitive chromatic number of planar graphs

摘录 / EXCERPT: Let G be a graph. A vertex coloring is nonrepetitive if no path has a color sequence of the form $s_1,s_2,\dots,s_t,s_1,s_2,\dots,s_t$. Let $\pi(G)$ denote the minimum number of...

conjecture | open | MathDB 页面 | 原文线索

V2-047345 | MATHDB-321534

The daisy density conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$. For a set X, write $X^{(r)}$ for its r-subsets. An r-graph is a subset of $X^{(r)}$. Given an $(r-2)$-set P and a disjoint 4-set Q, the r-da...

conjecture | open | MathDB 页面 | 原文线索

V2-047346 | MATHDB-321535

The 3-daisy density conjecture

摘录 / EXCERPT: Let $\mathcal D_3$ be the daisy formed from a fixed 1-set and a disjoint 4-set, and let $\pi(\mathcal D_3)$ be its limiting Turán density. The 3-daisy density conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-047347 | MATHDB-321536

The exact power-of-seven bound for daisy-free triple systems

摘录 / EXCERPT: Let $\mathcal A$ be a family of 3-sets of $[n]$, where $n=7^k$, and suppose that $\mathcal A$ contains no daisy. The power-of-seven daisy bound. ... This is proposed as the exact u...

conjecture | open | MathDB 页面 | 原文线索

V2-047348 | MATHDB-321537

The generalized daisy density conjecture

摘录 / EXCERPT: For fixed integers s and t, let $\mathcal D_r(s,t)$ consist of all r-sets A containing a fixed $(r-t)$-set P and contained in $P\cup Q$, where Q is a fixed disjoint s...

conjecture | open | MathDB 页面 | 原文线索

V2-047349 | MATHDB-321549

The Hartke-Stolee-West-Yancey upper-bound conjecture for p-extremal excess

摘录 / EXCERPT: Let p, k, and t be integers such that $k\in\{1,\dots,2t\}$ and ... Set $C_p=t^2-t+k-1$. Hartke-Stolee-West-Yancey conjecture. For these parameters, the maximum excess c_p of...

conjecture | open | MathDB 页面 | 原文线索

V2-047350 | MATHDB-321550

The logarithmic-size conjecture for p-extremal elementary graphs

摘录 / EXCERPT: Let $p\geq1$ be an integer. A p-extremal elementary graph is an elementary graph with p perfect matchings and maximum possible excess c_p. Elementary-graph size conjecture. E...

conjecture | open | MathDB 页面 | 原文线索

V2-047351 | MATHDB-321551

The existence conjecture for p-extremal elementary graphs

摘录 / EXCERPT: Let $p\geq1$ be an integer. A p-extremal elementary graph is an elementary graph with p perfect matchings and maximum possible excess c_p. p-extremal elementary-graph existen...

conjecture | open | MathDB 页面 | 原文线索

V2-047352 | MATHDB-321552

The superadditivity conjecture for excess under factorization

摘录 / EXCERPT: Let $k\geq1$ and let $p,p_1,\dots,p_k$ be integers satisfying ... Here c_q denotes the maximum excess among graphs with q perfect matchings. Excess superadditivity conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-047353 | MATHDB-321622

Weakly--Chordal Conjecture for right-angled Artin groups

摘录 / EXCERPT: Weakly--Chordal Conjecture. If $Gamma$ is weakly chordal, then

conjecture | open | MathDB 页面 | 原文线索

V2-047354 | MATHDB-321623

Häggkvist-Markström Kotzig spanning-minor conjecture

摘录 / EXCERPT: Häggkvist-Markström conjecture. Every 3-connected cubic graph contains a Kotzig graph as a spanning minor.

conjecture | open | MathDB 页面 | 原文线索

V2-047355 | MATHDB-321624

Semi-Kotzig-frame double cover conjecture

摘录 / EXCERPT: Semi-Kotzig-frame conjecture. Every bridgeless cubic graph with a semi-Kotzig frame has a 6-even-subgraph double cover.

conjecture | open | MathDB 页面 | 原文线索

V2-047356 | MATHDB-321625

Connected 3-edge-colorable spanning-minor conjecture

摘录 / EXCERPT: Connected 3-edge-colorable spanning-minor conjecture. If a cubic bridgeless graph contains a connected 3-edge-colorable cubic graph as a spanning minor, then G has a 6-even-sub...

conjecture | open | MathDB 页面 | 原文线索

V2-047357 | MATHDB-321633

Woo-Neumaier conjecture on regular graphs with smallest eigenvalue at least $-1-\sqrt 2$

摘录 / EXCERPT: Let G be a regular graph with smallest eigenvalue at least $-1-\sqrt 2$, and suppose its valency is sufficiently large. Woo-Neumaier conjecture. Then G is a line graph or a coc...

conjecture | open | MathDB 页面 | 原文线索

V2-047358 | MATHDB-321649

Full-rank conjecture for regular and irregular banded random binary matrices

摘录 / EXCERPT: Let $n,k,alpha$ and $gamma$ be integers with $k\leq n$ and $\gamma<\alpha$. Let M be a (γ,α) regular or irregular symmetric or asymmetric banded random matrix of size...

conjecture | open | MathDB 页面 | 原文线索

V2-047359 | MATHDB-321658

Došlić-Vukićević explicit conjecture for bipartizing fullerenes

摘录 / EXCERPT: Let H be a fullerene graph with n vertices, and let $b(H)$ be the minimum number of edges whose removal makes H bipartite. Došlić-Vukićević explicit conjecture. ... This is pr...

conjecture | open | MathDB 页面 | 原文线索

V2-047360 | MATHDB-321711

Feit's finiteness conjecture for sources of simple modules

摘录 / EXCERPT: Feit's conjecture. There are only finitely many isomorphism classes of indecomposable FQ-modules occurring as sources of simple FG-modules with vertex Q; here G varies over...

conjecture | open | MathDB 页面 | 原文线索

V2-047361 | MATHDB-321713

Feit's conjecture for vertex-source pairs

摘录 / EXCERPT: Feit's conjecture. Then $\mathcal{V}_{\mathcal{G}}(Q)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-047362 | MATHDB-321714

Feit's finiteness conjecture for interior algebras of vertex-source pairs

摘录 / EXCERPT: Feit's conjecture. There are, up to isomorphism in $\mathcal{A}$, only finitely many such interior algebras (Q,α,E_L).

conjecture | open | MathDB 页面 | 原文线索

V2-047363 | MATHDB-321755

The factorized vertex-operator matrix-element conjecture

摘录 / EXCERPT: Let $\mathbf{u}=(u_1,\ldots,u_m)$ and $\mathbf{v}=(v_1,\ldots,v_m)$, and let $\Phi(w)=\Phi_{\mathbf{u}}^{\mathbf{v}}(w):\mathcal{F}_{\mathbf{u}}\to\mathcal{F}_{\mathbf{v}}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-047364 | MATHDB-321767

Resource-cost lower bound for finite-time average-consensus algorithms

摘录 / EXCERPT: Let $\mathcal{A}$ be a distributed algorithm, let $C_{[0,t_1]}$ be a communication sequence, and let $\mathrm{S\mathcal{V}SC}$ denote the relevant sufficient vertex-strongly-connec...

conjecture | open | MathDB 页面 | 原文线索

V2-047365 | MATHDB-321768

Randomized DDA average-consensus conjecture under condition (C)

摘录 / EXCERPT: Let $C_{[0,t_1]}$ be a communication sequence, and let the $\mathrm{S\mathcal{V}CC}$ condition be defined by condition (C) of the cited theorem. Upon reception of each signal, choo...

conjecture | open | MathDB 页面 | 原文线索

V2-047366 | MATHDB-321769

RIS and ARIS asymptotic finite-time average-consensus conjecture

摘录 / EXCERPT: Let $C_{[0,t_1]}$ be a communication sequence satisfying the $\mathrm{S\mathcal{V}SC}$ condition. Let the RIS algorithm be the algorithm proposed by Shah and Hassibi (2008), let th...

conjecture | open | MathDB 页面 | 原文线索

V2-047367 | MATHDB-321770

ARIS characterization by infinite-vertex-strong connectivity

摘录 / EXCERPT: Let r be a positive integer, let $C_{[0,t_1]}$ be a communication sequence, and let the $\mathrm{I\mathcal{V}SC}$ condition be the condition specified by the source. Let ARIS den...

conjecture | open | MathDB 页面 | 原文线索

V2-047368 | MATHDB-321780

Strong monotonicity conjecture for flag vertex-induced homology subdivisions

摘录 / EXCERPT: Let Δ be a flag homology sphere, and let Δ' be a flag vertex-induced homology subdivision of Δ. Let $\gamma(\Delta)$ and $\gamma(\Delta')$ denote their...

conjecture | open | MathDB 页面 | 原文线索

V2-047369 | MATHDB-321781

Nonnegativity conjecture for local gamma-vectors of flag vertex-induced subdivisions

摘录 / EXCERPT: Let V be the vertex set of a simplex 2^V, and let Γ be a flag vertex-induced homology subdivision of 2^V. Let $\xi_V(\Gamma)$ denote the local γ-vector associat...

conjecture | open | MathDB 页面 | 原文线索

V2-047370 | MATHDB-321784

The infinite motion conjecture for locally finite denumerable graphs

摘录 / EXCERPT: Let Γ be a connected, locally finite, denumerable graph. Its automorphism group has infinite motion if it contains no automorphism with finite support, where the support of...

conjecture | open | MathDB 页面 | 原文线索

V2-047371 | MATHDB-321800

Successive interference alignment for grouped 3K-user channels

摘录 / EXCERPT: Consider a $3K$-user interference channel with $K otin\mathbb{Z}^+$? No: let $K\in\mathbb{Z}^+$, and partition the transmitter-receiver pairs into groups $G_1,G_2,\ldots,G_K$, each...

conjecture | open | MathDB 页面 | 原文线索

V2-047372 | MATHDB-321816

Maximum-energy conjecture for integral weighted trees

摘录 / EXCERPT: Let $n\ge 5$ and $m\ge n$. Write ${\mathcal T}(n,m)$ for the class of weighted trees on n vertices with total edge weight m, and let the energy of a weighted tree be the sum of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-047373 | MATHDB-321835

The FKKR upper-bound conjecture for identifying codes

摘录 / EXCERPT: Let G be a twin-free graph, let $|V(G)|$ denote its number of vertices, and let $\gamma^{\text{\tiny ID}}(G)$ be its identifying code number. Write $\Delta(G)$ for the maximum deg...

conjecture | open | MathDB 页面 | 原文线索

V2-047374 | MATHDB-321856

Retract-smoothness conjecture for Sidorenko graphs

摘录 / EXCERPT: Retract-smoothness conjecture. If H is a Sidorenko graph and T is a retract of H, then T is smooth in H.

conjecture | open | MathDB 页面 | 原文线索

V2-047375 | MATHDB-321857

The asymptotic minimum d-degree matching conjecture for uniform hypergraphs

摘录 / EXCERPT: Let $m_d(k,n)$ denote the minimum d-degree forcing a matching of the relevant size in a k-uniform hypergraph, with $1\le d\le k-1$. Asymptotic minimum d-degree matching conje...

conjecture | open | MathDB 页面 | 原文线索

V2-047376 | MATHDB-321858

The asymptotic fractional matching threshold conjecture for uniform hypergraphs

摘录 / EXCERPT: Let $f_d(k,n)$ denote the minimum d-degree forcing a fractional matching of the relevant size in a k-uniform hypergraph, with $1\le d\le k-1$. Fractional matching threshold con...

conjecture | open | MathDB 页面 | 原文线索

V2-047377 | MATHDB-321875

The bounded-boxicity conjecture for large-edgewidth graphs on a fixed surface

摘录 / EXCERPT: Large-edgewidth boxicity conjecture. There exists an integer e_Σ such that every graph G embeddable on Σ with edgewidth at least e_Σ has boxicity at most thr...

conjecture | open | MathDB 页面 | 原文线索

V2-047378 | MATHDB-321881

Asymptotic formula for the random hypergraph 2-colorability threshold

摘录 / EXCERPT: Threshold-asymptotic conjecture. There is a sequence $\varepsilon_k\to0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047379 | MATHDB-321893

The clique-number conjecture for graphs with chromatic number near the order

摘录 / EXCERPT: Let $Q(n,r)$ denote the minimum clique number among graphs on n vertices with chromatic number r. The clique-number conjecture. Let k be a positive integer. If n is suffici...

conjecture | open | MathDB 页面 | 原文线索

V2-047380 | MATHDB-321922

The categorical Cardy condition for full conformal field theories

摘录 / EXCERPT: Let V^L and V^R be rational vertex operator algebras, with modular tensor categories of representations $4\mathcal{C}_{V^L}4$ and $4\mathcal{C}_{V^R}4$. Let U_i, $i\in I$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-047381 | MATHDB-321923

Sufficiency of the modular invariance and Cardy conditions for open-closed CFT

摘录 / EXCERPT: Let V be a rational vertex operator algebra, and let an open-closed field algebra over V consist of the genus-zero open-closed operations. The modular invariance condition is t...

conjecture | open | MathDB 页面 | 原文线索

V2-047382 | MATHDB-321949

Scott-White multicolour pancyclicity conjecture

摘录 / EXCERPT: Let $n\geq 3$, let k be an integer, and let G be a graph of order n with minimum degree ... A k-edge colouring is a decomposition ... where each G_i is a spanning colour clas...

conjecture | open | MathDB 页面 | 原文线索

V2-047383 | MATHDB-321950

Exact monochromatic circumference conjecture at minimum degree three quarters

摘录 / EXCERPT: Let G be a graph of order n with minimum degree ... Write $n=3t+r$ with $r\in\{0,1,2\}$, and let ... be a 2-edge colouring. The exact monochromatic circumference conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-047384 | MATHDB-321955

Multicolored Erdős matching conjecture

摘录 / EXCERPT: Let $\binom{[n]}{k}$ be the family of k-element subsets of $[n]$, and let $\mathcal{F}_1,\ldots,\mathcal{F}_t$ be families of such subsets. A rainbow matching of size t is a co...

conjecture | open | MathDB 页面 | 原文线索

V2-047385 | MATHDB-321973

Dress's tight-span realisation conjecture

摘录 / EXCERPT: Let (X,d) be a finite metric space. Its tight-span $T(d)$ is the polytopal complex of minimal elements, under the product order, of ... Let G_d be the weighted graph consisting o...

conjecture | open | MathDB 页面 | 原文线索

V2-047386 | MATHDB-322019

Diameter growth conjecture for bounded-degree graph families

摘录 / EXCERPT: Let a family of graphs have a common largest degree, and let the vertex number denote the number of vertices in a graph in the family. Diameter growth conjecture. The graph diamete...

conjecture | open | MathDB 页面 | 原文线索

V2-047387 | MATHDB-322021

Łuczak-Thomassé conjecture on chromatic thresholds and near-acyclic graphs

摘录 / EXCERPT: Łuczak-Thomassé conjecture. The chromatic threshold of H satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047388 | MATHDB-322022

Conjecture on bounded chromatic number at the chromatic threshold

摘录 / EXCERPT: Boundedness-at-threshold conjecture. For every graph H with $\delta_\chi(H)=\lambda(H)$, there exists a constant $C(H)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047389 | MATHDB-322062

The triangle-free conjecture for gap-extremal graphs

摘录 / EXCERPT: Let a gap-extremal graph be a graph attaining the maximum possible value of the gap among graphs of its order, where the gap is the difference between its chromatic number and inde...

conjecture | open | MathDB 页面 | 原文线索

V2-047390 | MATHDB-322065

The equality of the unrestricted and triangle-free gap sequences

摘录 / EXCERPT: Let ${\rm gap}(n)$ be the maximum gap among graphs on n vertices, let ${\rm gap}_2(n)$ be the corresponding maximum restricted to triangle-free graphs, and let $s(t)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-047391 | MATHDB-322073

Fischer's multipartite Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Fischer's conjecture. There exists an integer $a_{n,t}$ such that if

conjecture | open | MathDB 页面 | 原文线索

V2-047392 | MATHDB-322147

Conjecture on oriented diameter of bridgeless graphs of diameter at most 3

摘录 / EXCERPT: Diameter-three oriented-diameter conjecture. Every bridgeless graph with diameter at most 3 has oriented diameter at most 9.

conjecture | open | MathDB 页面 | 原文线索

V2-047393 | MATHDB-322148

The linear-order conjecture for the maximum degree ratio in maximal intersecting families

摘录 / EXCERPT: Let $M(n,r)$ denote the minimum, over maximal intersecting families of r-subsets of an n-element set, of the ratio between the maximum and minimum vertex degrees. If $r=o(n)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-047394 | MATHDB-322153

Modified Araújo-Cohen-Giroire-Havet critical-graph finiteness conjecture

摘录 / EXCERPT: Modified Araújo-Cohen-Giroire-Havet conjecture. For every $c<4$, the number of pairwise non-isomorphic critical graphs with girth at least five and average degree at most c is fi...

conjecture | open | MathDB 页面 | 原文线索

V2-047395 | MATHDB-322154

Modified Araújo-Cohen-Giroire-Havet average-degree conjecture for critical graphs

摘录 / EXCERPT: Modified Araújo-Cohen-Giroire-Havet conjecture. There is no critical graph with average degree less than 3, with the exception of C_3, $K_{2,3}$, and H.

conjecture | open | MathDB 页面 | 原文线索

V2-047396 | MATHDB-322155

Araújo-Cohen-Giroire-Havet average-degree conjecture for critical graphs

摘录 / EXCERPT: Araújo-Cohen-Giroire-Havet conjecture. There is no critical graph with average degree less than 3, with the exception of C_3 and $K_{2,3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-047397 | MATHDB-322183

The independent-set polynomial cover inequality for bipartite graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a finite bipartite graph, let $\tilde{G}$ be its M-cover, and let $p(G)=\sum_{I\text{ independent}}\prod_{v\in I}x_v$ be its multivariate independent-set polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-047398 | MATHDB-322210

Perfect distance-dominating set conjecture for integer grids

摘录 / EXCERPT: Perfect distance-dominating set conjecture. For $k\geq 1$, there is no r-PDDS$[P_k]$ in Λ_n for $n\geq 3$ and $r>1$, with the exception of a 2-PDDS$[P_2]$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-047399 | MATHDB-322223

Minimum outdegree forcing immersions of transitive tournaments

摘录 / EXCERPT: Immersion conjecture for transitive tournaments. There exists a function f so that every simple digraph of minimum outdegree $f(t)$ contains an immersion of the transitive tourna...

open_problem | open | MathDB 页面 | 原文线索

V2-047400 | MATHDB-322228

Bondage-number bound for mesh networks with at least five rows

摘录 / EXCERPT: Mesh-network bondage conjecture. For $m\geqslant 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-047401 | MATHDB-322229

Vertex-connectivity analogue for the Roman bondage number

摘录 / EXCERPT: Vertex-connectivity analogue. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-047402 | MATHDB-322245

The Weinstein skeleton conjecture for toric hypersurface mirrors

摘录 / EXCERPT: Suppose $\triangle$ is vertex-simplicial and $\triangle^\vee$ is facet-simplicial. Let $Z^\vee_\infty$ be the large volume limit and let Λ^∞ be the associated Legendri...

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V2-047403 | MATHDB-322250

Bollobás and Scott's logarithmic heavy cycle conjecture

摘录 / EXCERPT: Let D be a weighted loopless digraph with $n\geq 2$ vertices. For each vertex $v\in V(D)$, write $d^{w+}(v)$ for its weighted outdegree, and write $w(C)$ for the total weight of...

conjecture | open | MathDB 页面 | 原文线索

V2-047404 | MATHDB-322271

The cubic-graph cycle-length conjecture for powers of two

摘录 / EXCERPT: Let G be a cubic graph, and let l be the length of a cycle in G. Cubic-graph cycle-length conjecture. There exist a cycle length l in G and a positive integer k such th...

conjecture | open | MathDB 页面 | 原文线索

V2-047405 | MATHDB-322280

Ineffective-leg orbifold vertex GW/DT correspondence

摘录 / EXCERPT: Ineffective-leg orbifold GW/DT conjecture. After the identifications $q_0=e^{\sqrt{-1}(\lambda-x)}$ and $q_1=-e^{\sqrt{-1}x}$,

conjecture | open | MathDB 页面 | 原文线索

V2-047406 | MATHDB-322296

Georgakopoulos's Hamiltonian-circle conjecture for line graphs

摘录 / EXCERPT: Let G be a locally finite graph, and let $L(G)$ denote its line graph. A Hamiltonian circle is a circle in the Freudenthal compactification of $L(G)$ that contains every vertex o...

conjecture | open | MathDB 页面 | 原文线索

V2-047407 | MATHDB-322298

Strong connectivity comparison conjecture for bridge-addable graph sets

摘录 / EXCERPT: For a positive integer n, let $\mathcal{A}$ be a non-empty bridge-addable set of graphs on $\{1,\ldots,n\}$: whenever a graph belongs to $\mathcal{A}$, adding an edge between ver...

conjecture | open | MathDB 页面 | 原文线索

V2-047408 | MATHDB-322345

Erdős-Sós bipartite-links conjecture for 3-graphs

摘录 / EXCERPT: A 3-graph is a 3-uniform hypergraph, and the link graph of a vertex is the graph whose vertices are the remaining vertices, with an edge for each 3-edge containing the given vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-047409 | MATHDB-322346

Independent-neighbourhoods conjecture for 3-graph Turán density

摘录 / EXCERPT: Let $\pi(\operatorname{odd\ cycle\ in\ link\ graph},F_{3,2})$ denote the maximum asymptotic edge density of a 3-graph containing no copy of $F_{3,2}$ and having no odd cycle in any...

conjecture | open | MathDB 页面 | 原文线索

V2-047410 | MATHDB-322347

Stability conjecture for 3-graphs with independent neighbourhoods

摘录 / EXCERPT: A 3-graph has independent neighbourhoods when the neighbourhood of every pair of vertices is an independent set. Stability conjecture. Turán problems for 3-graphs with independent...

conjecture | open | MathDB 页面 | 原文线索

V2-047411 | MATHDB-322351

Base-cobase graph diameter conjecture for block matroids

摘录 / EXCERPT: Base-cobase graph diameter conjecture. The diameter of G is equal to n.

conjecture | open | MathDB 页面 | 原文线索

V2-047412 | MATHDB-322375

Critical-threshold conjecture for recurrence in the symmetric flat-network model

摘录 / EXCERPT: Recurrence-threshold conjecture. For some $a_K>0$, the process is positive recurrent if $0<\mu/\lambda<a_K$ and null recurrent if $\mu/\lambda>a_K$.

conjecture | open | MathDB 页面 | 原文线索

V2-047413 | MATHDB-322376

Cameron's reachability graph conjecture for highly arc-transitive digraphs

摘录 / EXCERPT: Let D be a connected highly arc transitive digraph. Suppose there is a homomorphism ... such that $f^{-1}(0)$ is finite, and let $R(D)$ be the reachability graph of D. Cameron's...

conjecture | open | MathDB 页面 | 原文线索

V2-047414 | MATHDB-322383

The degree-sequence conjecture for perfect clique packings

摘录 / EXCERPT: Degree-sequence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-047415 | MATHDB-322384

The degree-sequence conjecture for perfect graph packings

摘录 / EXCERPT: Perfect-packing conjecture. There is an integer $n_0=n_0(\gamma,H)$ such that, whenever $n\geq n_0$, $|H|$ divides n, and

conjecture | open | MathDB 页面 | 原文线索

V2-047416 | MATHDB-322399

The small-order extremal conjecture for subcube intersection graphs

摘录 / EXCERPT: Small-order extremal conjecture. For any $n\leq 2\times2^{d/2}+(r-2)$, the maximum number of edges in a K_{r+1}-free graph in $\mathcal{I}(n,d)$ is attained by a graph G which...

conjecture | open | MathDB 页面 | 原文线索

V2-047417 | MATHDB-322400

The large-order extremal construction conjecture for subcube intersection graphs

摘录 / EXCERPT: Large-order extremal construction conjecture. For any $n\geq 2\times2^{d/2}+(r-2)$, the maximum number of edges in a K_{r+1}-free graph in $\mathcal{I}(n,d)$ is attained by a gra...

conjecture | open | MathDB 页面 | 原文线索

V2-047418 | MATHDB-322403

Conjecture on the three-dimensional vertex index

摘录 / EXCERPT: Let ${\bf K}$ be an arbitrary 0-symmetric convex body in ${\mathbb R}^3$. A truncated octahedron of the form ${\bf T}-{\bf T}$ is obtained from an arbitrary tetrahedron ${\bf T}$...

conjecture | open | MathDB 页面 | 原文线索

V2-047419 | MATHDB-322411

Depth-bound conjecture for vertex-transitive graphs

摘录 / EXCERPT: Let G be a connected vertex-transitive graph, let $A\subseteq V(G)$ be finite with $0<|A|\leq \tfrac12|V(G)|$, let ∂A denote the vertex boundary of A, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047420 | MATHDB-322412

Regular digraph analogue of Reed's chromatic bound

摘录 / EXCERPT: Regular digraph analogue of Reed's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047421 | MATHDB-322420

Montgomery's dynamic chromatic number conjecture for regular graphs

摘录 / EXCERPT: Let G be a regular graph. Its chromatic number is denoted by $\chi(G)$, and its dynamic chromatic number, the smallest number of colors in a dynamic proper vertex coloring, is de...

conjecture | open | MathDB 页面 | 原文线索

V2-047422 | MATHDB-322439

Bounded excess-height conjecture for 3-sphere triangulations

摘录 / EXCERPT: Let $\mathscr{P}_1(S^3)$ be the graph whose nodes are one-vertex triangulations of the 3-sphere, arranged by level n, and let a simplification path be a path reducing the level t...

conjecture | open | MathDB 页面 | 原文线索

V2-047423 | MATHDB-322469

Vempala's bipartite partition conjecture

摘录 / EXCERPT: Let $G=(U,V,E)$ be an $N\times k$ complete bipartite graph with $N\geq k$. Let $\mathcal{P}$ be a partition of E. For each part $p\in\mathcal{P}$ and vertex i, let p_i be the...

conjecture | open | MathDB 页面 | 原文线索

V2-047424 | MATHDB-322472

Huang-Sudakov's biclique partition-chromatic number conjecture

摘录 / EXCERPT: Huang-Sudakov's conjecture. For each integer $k>0$, there exists a graph G with biclique partition number k and chromatic number at least

conjecture | open | MathDB 页面 | 原文线索

V2-047425 | MATHDB-322478

Flipping or non-flipping behavior on regions II and IV

摘录 / EXCERPT: Let $a,b\in(0,1)$, let n be the graph parameter, and assume ... Let Regions emph{II} and emph{IV} denote the regions of the state space described in the paper, and let iterates mea...

conjecture | open | MathDB 页面 | 原文线索

V2-047426 | MATHDB-322479

Unique non-trivial fixed point on generalized spoke and star graphs

摘录 / EXCERPT: Consider a generalized spoke and star graph with ℓ levels. Level one consists of one node, the hub; level two consists of n_1 spokes connected to the hub; and each node at l...

conjecture | open | MathDB 页面 | 原文线索

V2-047427 | MATHDB-322522

Cluster $\mathbf g$-vector fan conjecture

摘录 / EXCERPT: For each vertex v of the exchange graph, let $\operatorname{Cone}(v)$ be the cone in V^* spanned by the weights $\mathbf g(x^v_e)$ for $e\in I(v)$. A fan is a collection of con...

conjecture | open | MathDB 页面 | 原文线索

V2-047428 | MATHDB-322525

Sign-coherence conjecture for the rows of H^v

摘录 / EXCERPT: For each vertex $v\in\mathrm{Ex}_0(B)$, let H^v be the associated matrix. A collection of entries is sign-coherent when, in each coordinate, the relevant entries are all nonnegat...

conjecture | open | MathDB 页面 | 原文线索

V2-047429 | MATHDB-322534

Guo's lower-bound conjecture for Laplacian eigenvalues and degrees

摘录 / EXCERPT: Let G be a simple connected graph of order n. Let $\rho_1(G)\ge\rho_2(G)\ge\cdots\ge\rho_n(G)=0$ be its Laplacian eigenvalues, and let $d_1\ge d_2\ge\cdots\ge d_n$ be its verte...

conjecture | open | MathDB 页面 | 原文线索

V2-047430 | MATHDB-322535

Grone-Merris majorization conjecture for Laplacian eigenvalues

摘录 / EXCERPT: Let G be a connected graph of order $n\ge2$, with nonincreasing degree sequence $d_1\ge d_2\ge\cdots\ge d_n$. For nonincreasing sequences, write...

conjecture | open | MathDB 页面 | 原文线索

V2-047431 | MATHDB-322552

The spanning-tree upper-bound conjecture for partition dimension of unicyclic graphs

摘录 / EXCERPT: Let G be a unicyclic graph and let T be a spanning tree of G. The spanning-tree upper-bound conjecture. If T is a spanning tree of a unicyclic graph G, then ... This conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-047432 | MATHDB-322558

Conjecture on the number of independent sets in triangle-free graphs

摘录 / EXCERPT: Let G be an n-vertex triangle-free graph, and let $i(G)$ denote its number of independent sets. Independent-set counting conjecture. There is an absolute positive constant c...

conjecture | open | MathDB 页面 | 原文线索

V2-047433 | MATHDB-322562

The generic full-rank vertex-map conjecture for hom-polytopes

摘录 / EXCERPT: Let $P(M)$ and $Q(M')$ be polytopes of dimensions d and e, respectively, with $M\in R_{d,m}$ and $M'\in R'_{e,n}$, where $m\geq d+1$ and $n\geq e+1$. A vertex map...

conjecture | open | MathDB 页面 | 原文线索

V2-047434 | MATHDB-322568

Asymptotic conjecture for the maximal number of independent sets

摘录 / EXCERPT: Let $L_{ciota_m,\nu}^{\mathcal{B}}(n)$ denote the quantity defined in the paper for graphs on n vertices. Asymptotic conjecture. As n tends to infinity, ... The authors state tha...

conjecture | open | MathDB 页面 | 原文线索

V2-047435 | MATHDB-322599

Online Ohba conjecture

摘录 / EXCERPT: Let G be a graph, with vertex number $|V(G)|$, chromatic number $c7(G)$, and on-line choice number ${\rm ch}^{\rm OL}(G)$. A graph is on-line chromatic-choosable when...

conjecture | open | MathDB 页面 | 原文线索

V2-047436 | MATHDB-322607

Exact codegree threshold conjecture for (K_4^3-e)-factors

摘录 / EXCERPT: Let $t_2^3(n,K_4^3-e)$ be the smallest integer d such that every 3-uniform hypergraph on n vertices with minimum 2-degree at least d contains a spanning collection of vertex-...

conjecture | open | MathDB 页面 | 原文线索

V2-047437 | MATHDB-322611

Erdős's asymptotic conjecture on two complete 3-uniform subhypergraphs

摘录 / EXCERPT: Let s be a positive integer, and let $t(n,s,3)$ denote the maximum number of edges in a 3-uniform hypergraph on n vertices containing no copy of K_s^3. Erdős's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-047438 | MATHDB-322612

Conjectured separation of Turán densities for 3-uniform hypergraphs

摘录 / EXCERPT: For $s=5,6$, let K_s^3 denote the complete 3-uniform hypergraph on s vertices, and let K_{s+1}^{3-} denote the 3-uniform hypergraph obtained from K_{s+1}^3 by deleting one...

conjecture | open | MathDB 页面 | 原文线索

V2-047439 | MATHDB-322619

Lovász-Saks graph-rank formulation of the log-rank conjecture

摘录 / EXCERPT: Lovász-Saks graph-rank conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047440 | MATHDB-322646

Extremal hypergraph conjecture for bounded-degree spanning structures

摘录 / EXCERPT: Let $k\in\mathbb N$. A k-graph is a hypergraph whose edges have size k. Let F be a forbidden hypergraph of bounded maximum vertex degree, and for each $v\in V$ let $F(v)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-047441 | MATHDB-322661

Vu's nonsingularity conjecture for random regular graph adjacency matrices

摘录 / EXCERPT: Let $d\geq 3$ be a number that may depend on n, and let $R_{n,d}$ be the adjacency matrix of a d-regular graph. Vu's nonsingularity conjecture. The matrix $R_{n,d}$ is nonsingu...

conjecture | open | MathDB 页面 | 原文线索

V2-047442 | MATHDB-322665

Conjecture on embeddings of graphs in \(\mathcal{F}_{xy}^k\)

摘录 / EXCERPT: Embedding conjecture. Every graph $G\in\mathcal{F}_{xy}^k$ embeds in $\mathbb{S}_k$.

conjecture | open | MathDB 页面 | 原文线索

V2-047443 | MATHDB-322680

DeVos-Mohar depth expansion conjecture for vertex-transitive graphs

摘录 / EXCERPT: DeVos-Mohar conjecture. There exists a fixed constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047444 | MATHDB-322700

The HOMFLY polynomial conjecture for hypergraph interior polynomials

摘录 / EXCERPT: Let G be the connected bipartite graph from the construction above, with bipartition denoted by V and E, and let L_G be its associated link. Write $P_{L_G}(v,z)$ for the HO...

conjecture | open | MathDB 页面 | 原文线索

V2-047445 | MATHDB-322710

The Kuratowski-edge conjecture for crossing-critical graphs

摘录 / EXCERPT: Let G be a graph. An edge of G is a Kuratowski edge if it belongs to a subgraph homeomorphic to $K_{3,3}$ or K_5. Kuratowski-edge conjecture. If every edge of G is a Kurato...

conjecture | open | MathDB 页面 | 原文线索

V2-047446 | MATHDB-322711

The two-Kuratowski-edge conjecture for simultaneous crossing-criticality

摘录 / EXCERPT: Let G be a graph, and let e and f be Kuratowski edges of G, meaning that each belongs to a subgraph homeomorphic to $K_{3,3}$ or K_5. A graph obtained by multiplying edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047447 | MATHDB-322713

Four-bend necessity conjecture for planar graph EPG representations

摘录 / EXCERPT: An EPG representation of a graph represents each vertex by a path in the square grid, with adjacency corresponding to sharing a grid edge; a path has a bend at each change of grid...

conjecture | open | MathDB 页面 | 原文线索

V2-047448 | MATHDB-322733

Michels-Knauer eventual polynomiality conjecture for path congruence classes

摘录 / EXCERPT: Michels-Knauer's conjecture. There exists a polynomial $f_k\in\mathbb{Q}[x]$ with $\deg(f_k)=\lceil(k-2)/2\rceil$ such that, for a fixed n_k (most probably $n_k=2k$),

conjecture | open | MathDB 页面 | 原文线索

V2-047449 | MATHDB-322752

Cioabă-van Dam-Koolen-Lee minimizer conjecture for graphs of prescribed diameter

摘录 / EXCERPT: Cioabă-van Dam-Koolen-Lee conjecture. For any $e\geq 1$ and sufficiently large n with $n+e$ even,...

conjecture | open | MathDB 页面 | 原文线索

V2-047450 | MATHDB-322757

Hegarty's edge-growth conjecture for graph squares

摘录 / EXCERPT: Let G be a connected d-regular graph with n vertices, and suppose that G^2 is not a complete graph. Hegarty's conjecture. The number of edges added by squaring the graph sa...

conjecture | open | MathDB 页面 | 原文线索

V2-047451 | MATHDB-322781

The sandpile Betti-number conjecture via connected partitions

摘录 / EXCERPT: Let $G=(V,E,s)$ be an undirected sandpile graph. For $U\subseteq V$, let $G|_U$ be the subgraph induced by U. A connected k-partition of G is a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-047452 | MATHDB-322809

The sharp finite hexagonal-grid isoperimetric inequality

摘录 / EXCERPT: Let G_r be the finite hexagonal grid graph of radius r, let $V(G_r)$ be its vertex set, and let $N(W)$ and ${\mathscr E}(W)$ denote the relevant vertex and edge neighborhoods o...

conjecture | open | MathDB 页面 | 原文线索

V2-047453 | MATHDB-322835

Flow scalability characterizes weighted α-fairness for arbitrary network topologies

摘录 / EXCERPT: Flow-scalability conjecture. A network is flow scalable if and only if it maximizes a weighted α-fair utility function.

conjecture | open | MathDB 页面 | 原文线索

V2-047454 | MATHDB-322842

The quasi-transitive digraph colored-kernel conjecture

摘录 / EXCERPT: Quasi-transitive colored-kernel conjecture. D has a k-colored kernel for every $k\geq l+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-047455 | MATHDB-322843

The minimum-cycle-color conjecture for colored kernels

摘录 / EXCERPT: Minimum-cycle-color conjecture. Then D has a k-colored kernel.

conjecture | open | MathDB 页面 | 原文线索

V2-047456 | MATHDB-322845

Quantum-dimension conjecture for compact-group decompositions of vertex operator algebras

摘录 / EXCERPT: Quantum-dimension conjecture. For every $\chi\in\operatorname{Irr}(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-047457 | MATHDB-322876

The Keevash-Mubayi extremal uniqueness conjecture for K_t^-

摘录 / EXCERPT: Keevash-Mubayi conjecture. $G_t(n)$ is the unique, up to isomorphism, 3-graph with $\operatorname{ex}(n,K_t)$ edges and $\operatorname{ex}_{K_t^-}(n,K_t)$ induced copies of...

conjecture | open | MathDB 页面 | 原文线索

V2-047458 | MATHDB-322877

The K_4^--density conjecture in C_5-free 3-graphs

摘录 / EXCERPT: The K_4^--density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047459 | MATHDB-322878

Turán's conjecture for the Turán density of K_4^-

摘录 / EXCERPT: Frankl-Füredi-Razborov conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047460 | MATHDB-322879

The 4.2-density conjecture in K_4^--free 3-graphs

摘录 / EXCERPT: The 4.2-density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047461 | MATHDB-322880

The K_4^--density conjecture for (K_4^-,C_5)-free 3-graphs

摘录 / EXCERPT: The K_4^--density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047462 | MATHDB-322881

The 4.2-density conjecture in (K_4^-,C_5)-free 3-graphs

摘录 / EXCERPT: The 4.2-density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047463 | MATHDB-322903

Minimal-cycle Betti-number conjecture for graph curves

摘录 / EXCERPT: Let G be a graph on d vertices, embedded as in Theorem 1.3. Let n denote the girth of G, and suppose that $d=2g+1+p$ and $n-2\leq p$. Minimal-cycle Betti-number conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047464 | MATHDB-322922

The odd-order exact formula for replication numbers of paths

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $\rho_R(P_n)$ denote its replication number. For odd $n\geq 1$, write the upper-bound formula according to the residue class of n m...

conjecture | open | MathDB 页面 | 原文线索

V2-047465 | MATHDB-322923

The quadratic asymptotic conjecture for replication numbers of paths

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $\rho_R(P_n)$ denote its replication number. The notation $O(n)$ means a quantity bounded in absolute value by a constant multiple of...

conjecture | open | MathDB 页面 | 原文线索

V2-047466 | MATHDB-322924

The equality of replication numbers for cycles and paths

摘录 / EXCERPT: Let P_n and C_n denote, respectively, the path and cycle on n vertices, and let $\rho_R(G)$ denote the replication number of a graph G. The cycle-path replication conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-047467 | MATHDB-322997

Spectral outlier conjecture for biased random regular graphs

摘录 / EXCERPT: Consider either of the regular planted-partition graph constructions described in the source: a d-regular graph with two planted classes, where ϵ is the cross-class edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047468 | MATHDB-322999

The rational generating-function denominator conjecture for hard squares on cylinders

摘录 / EXCERPT: Generating-function denominator conjecture. For every $n\geq 0$, there are polynomials $h_n(t)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047469 | MATHDB-323000

The reduced-denominator multiplicity conjecture for hard squares on cylinders

摘录 / EXCERPT: Let P_m be the path graph on m vertices and C_n the cycle graph on n vertices. Define ... where $Z(G)=1-\chi(\operatorname{Ind}(G))$ is the Witten index associated with the i...

conjecture | open | MathDB 页面 | 原文线索

V2-047470 | MATHDB-323001

The cycle-length conjecture for necklace dynamics

摘录 / EXCERPT: For integers k and n, let $\operatorname{Neck}(k,n)$ be the directed graph whose vertices are (k,n)-necklaces and whose transition map is T. A cycle in this graph is a peri...

conjecture | open | MathDB 页面 | 原文线索

V2-047471 | MATHDB-323043

The monochromatic triangle-and-quadrilateral conjecture for semicomplete multipartite digraphs

摘录 / EXCERPT: Let D be an m-colored semicomplete r-partite digraph with $r\geq 2$. A directed cycle is monochromatic if all of its arcs have the same color, and a 1-colored kernel is a k...

conjecture | open | MathDB 页面 | 原文线索

V2-047472 | MATHDB-323057

Pseudo-periodicity conjecture for the inhomogeneous eight-vertex ground-state eigenvector

摘录 / EXCERPT: Let $L=2n+1$, let $T_L(u|x_1,\ldots,x_L)$ be the inhomogeneous eight-vertex transfer matrix, and let $\Psi_L(x_1,\ldots,x_L)$ satisfy the eigenvector equations with eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-047473 | MATHDB-323059

Extremal conjecture for maximal independent sets in regular graphs

摘录 / EXCERPT: Regular-graph maximal-independent-set conjecture. Every d-regular n-vertex graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047474 | MATHDB-323075

Existence conjecture for large two-dimensional (α,β)-tours

摘录 / EXCERPT: Existence conjecture. There exists an integer M such that an (α,β)-tour exists on every $n_1\times n_2$ chessboard for which n_1 is even and

conjecture | open | MathDB 页面 | 原文线索

V2-047475 | MATHDB-323109

Acyclic Edge Coloring Conjecture

摘录 / EXCERPT: In a finite, simple, undirected graph G, let $\Delta(G)$ denote its maximum degree. The acyclic chromatic index $\chi'_a(G)$ is the least number of colors in a proper edge colori...

conjecture | open | MathDB 页面 | 原文线索

V2-047476 | MATHDB-323133

Černý-Kynčl-Tóth dense stability conjecture

摘录 / EXCERPT: Černý-Kynčl-Tóth dense stability conjecture. There exists an $\overline{\epsilon}\in(1/3,1)$ such that, for each $\epsilon\in(\overline{\epsilon},1]$, the family of graphs with...

conjecture | open | MathDB 页面 | 原文线索

V2-047477 | MATHDB-323149

Finiteness conjecture for r-primitive graphs

摘录 / EXCERPT: Let $r\geq 3$. An r-primitive graph is a uniquely K_r-saturated graph with no dominating vertex. Finiteness conjecture. For each $r\geq 3$, there are a finite number of r-pri...

conjecture | open | MathDB 页面 | 原文线索

V2-047478 | MATHDB-323163

The refined flow-series recursion conjecture

摘录 / EXCERPT: Let T be a rooted tree, let v be a vertex of T, let u be its parent, let w be a rooted tree grafted at v, and let S and S' be the rooted trees defined by removing e...

conjecture | open | MathDB 页面 | 原文线索

V2-047479 | MATHDB-323202

The component and fragment conjecture for bridge-addable graph classes

摘录 / EXCERPT: Let ${\mathcal A}$ be a set of graphs, let $R_n \in_u {\mathcal A}$ be uniformly distributed over its graphs on vertex set $[n]$, let $\kappa(R_n)$ denote the number of components...

conjecture | open | MathDB 页面 | 原文线索

V2-047480 | MATHDB-323205

Non-existence conjecture for bipartite diameter-3 defect-4 graphs

摘录 / EXCERPT: Non-existence conjecture. There is no bipartite $(d,3,-4)$-graph with $d\ge 6$.

conjecture | open | MathDB 页面 | 原文线索

V2-047481 | MATHDB-323218

The conjectured value of the minimum 4-clique density with bounded independence number

摘录 / EXCERPT: Conjectured value of $c_{4,4}$.

conjecture | open | MathDB 页面 | 原文线索

V2-047482 | MATHDB-323228

First-author-Schmitt asymptotic upper-bound conjecture for the potential function

摘录 / EXCERPT: First-author-Schmitt conjecture. For every graph H and every $\epsilon>0$, there exists $n_0=n_0(\epsilon,H)$ such that, for every $n>n_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-047483 | MATHDB-323249

The scaling-limit conjecture for vertex-transitive graphs with polynomial ball growth

摘录 / EXCERPT: Scaling-limit conjecture. Then $(X_n,d/\Delta_n)$ has a subsequence converging for the pointed GH-topology to a connected nilpotent Lie group equipped with a Carnot-Carathéodory me...

conjecture | open | MathDB 页面 | 原文线索

V2-047484 | MATHDB-323322

Three-path conjecture for the maximum gap between average distance and proximity

摘录 / EXCERPT: Three-path conjecture. Among all such graphs, the difference $\bar{l}-\pi$ is maximum for a graph composed of three paths of almost equal lengths with a common end point.

conjecture | open | MathDB 页面 | 原文线索

V2-047485 | MATHDB-323323

Cycle and crossed-edge extremal conjecture for remoteness minus radius

摘录 / EXCERPT: Remoteness-radius conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-047486 | MATHDB-323325

Specialisation conjecture for equivalent realisations

摘录 / EXCERPT: Specialisation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-047487 | MATHDB-323370

Galvin's fixed-size independent-set conjecture for minimum-degree graphs

摘录 / EXCERPT: Galvin's conjecture. Except perhaps for some small values of t, for every fixed δ and sufficiently large n, the graph $K_{\delta,n-\delta}$ maximizes $i_t(G)$ among n-...

conjecture | open | MathDB 页面 | 原文线索

V2-047488 | MATHDB-323371

Strong level-set conjecture for independent sets in minimum-degree graphs

摘录 / EXCERPT: Let $\mathcal{G}(n,\delta)$ be the family of n-vertex graphs with minimum degree at least δ, and let $i_t(G)$ denote the number of independent sets of size t in G. Str...

conjecture | open | MathDB 页面 | 原文线索

V2-047489 | MATHDB-323396

Conjecture on bondage number and minimum edge-degree

摘录 / EXCERPT: Let G be a connected graph other than $K_{1,n}$ and K_3. Write $b(G)$ for its bondage number, $\delta(G)$ for its minimum degree, and $\lambda'(G)$ for its restricted edge-conn...

conjecture | open | MathDB 页面 | 原文线索

V2-047490 | MATHDB-323398

Teschner's universal three-halves conjecture

摘录 / EXCERPT: Let G be any graph, with $b(G)$ its bondage number and $\Delta(G)$ its maximum degree. Teschner's conjecture. ... The source notes that equality is attained by Cartesian products...

conjecture | open | MathDB 页面 | 原文线索

V2-047491 | MATHDB-323399

Conjecture on bondage numbers of Cartesian products of paths

摘录 / EXCERPT: Let P_n and P_m be paths, and let $P_n\times P_m$ denote their Cartesian product. Write $b(\,\cdot\,)$ for the bondage number. Path-product bondage conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-047492 | MATHDB-323401

Carlson-Develin conjecture for digraph bondage number

摘录 / EXCERPT: Let G be a digraph, and let $b(G)$ denote its bondage number and $\Delta(G)$ its maximum degree. Carlson-Develin conjecture. ... The source presents this as the directed analogue o...

conjecture | open | MathDB 页面 | 原文线索

V2-047493 | MATHDB-323464

Invariant clique conjecture for dismantlable graphs

摘录 / EXCERPT: Let H be a finite group of automorphisms of an infinite dismantlable graph Γ. Here, dismantlable means that every finite set of vertices of Γ is contained in a fini...

conjecture | open | MathDB 页面 | 原文线索

V2-047494 | MATHDB-323476

Chartrand et al.'s realizability conjecture for metric and upper dimensions

摘录 / EXCERPT: Chartrand et al.'s realizability conjecture. There exists a connected graph G such that

conjecture | open | MathDB 页面 | 原文线索

V2-047495 | MATHDB-323481

The simple multiplicity conjecture for fusion rules of the extended algebra

摘录 / EXCERPT: Simple multiplicity conjecture. The fusion rules of $\mathfrak W_{n,\ell}$ are

conjecture | open | MathDB 页面 | 原文线索

V2-047496 | MATHDB-323483

The local track-number conjecture for planar graphs

摘录 / EXCERPT: Local track-number conjecture. The local track-number of G is at most 3.

conjecture | open | MathDB 页面 | 原文线索

V2-047497 | MATHDB-323504

Recognition of edge-clique graphs

摘录 / EXCERPT: An edge-clique graph is a graph obtained as the edge-clique graph $K_e(G)$ of some graph G, where vertices of $K_e(G)$ represent edges of G and adjacency represents incident ed...

open_problem | open | MathDB 页面 | 原文线索

V2-047498 | MATHDB-323515

The 1-2-3-conjecture for vertex-coloring edge weightings

摘录 / EXCERPT: 1-2-3-conjecture. For every connected graph G with at least three vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-047499 | MATHDB-323535

Ilinca-Kahn conjectured bipartite matching entropy bound

摘录 / EXCERPT: Ilinca-Kahn conjectured matching bound. If, for each $x\in X$, $t_x=\ell d_x/M$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047500 | MATHDB-323537

Murty-Simon conjecture on diameter-two edge-critical graphs

摘录 / EXCERPT: Let G be a simple graph on n vertices. Write $E(G)$ for its edge set, and call G diameter-2 edge-critical when $\operatorname{diam}(G)=2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-047501 | MATHDB-323539

The decoupling-relation conjecture for the Kac-Wang-Yan algebra at negative half-integer charge

摘录 / EXCERPT: Let $n\geq1$, and let w^{2m+1} denote the standard fields in ${\mathcal V}_{-n+1/2}$. The minimal-weight invariant-theoretic relation has weight $4n^2+8n+4$, with remainder R_0...

conjecture | open | MathDB 页面 | 原文线索

V2-047502 | MATHDB-323555

Approximate Erdős-Gyárfás-Pyber cycle covering conjecture

摘录 / EXCERPT: Let K_n be a complete graph whose edges are coloured with r colours. A vertex-disjoint monochromatic cycle packing is a collection of r vertex-disjoint monochromatic cycles....

conjecture | open | MathDB 页面 | 原文线索

V2-047503 | MATHDB-323556

Non-disjoint monochromatic cycle covering conjecture

摘录 / EXCERPT: Let K_n be a complete graph whose edges are coloured with r colours. A cycle covering is a collection of r monochromatic cycles whose union of vertex sets contains all vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-047504 | MATHDB-323557

Bipartite monochromatic path partition conjecture

摘录 / EXCERPT: Let $K_{n,n}$ be a balanced complete bipartite graph whose edges are coloured with r colours. A vertex-partition into monochromatic paths is a partition of all vertices into path...

conjecture | open | MathDB 页面 | 原文线索

V2-047505 | MATHDB-323558

Liu et al.'s domination formula for Cartesian products of directed cycles

摘录 / EXCERPT: Let $k\geq2$ and let n be a positive integer. For directed cycles $\overrightarrow{C_{3k}}$ and $\overrightarrow{C_n}$, write $\gamma(G)$ for the domination number of a directed...

conjecture | open | MathDB 页面 | 原文线索

V2-047506 | MATHDB-323573

The characterization of positive graphs

摘录 / EXCERPT: Positive-graph characterization conjecture. A graph G is positive if and only if it is symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-047507 | MATHDB-323625

The $(k+1)$-kernel conjecture for k-quasi-transitive digraphs

摘录 / EXCERPT: $(k+1)$-kernel conjecture. Every k-quasi-transitive digraph has a $(k+1)$-kernel.

conjecture | open | MathDB 页面 | 原文线索

V2-047508 | MATHDB-323639

Avoidance conjecture for fractional colorings of 4-cliques

摘录 / EXCERPT: Let G be a graph with maximum degree 5 and clique number 4, such that no two 4-cliques intersect and no vertex outside any maximum clique C has more than one neighbour in...

conjecture | open | MathDB 页面 | 原文线索

V2-047509 | MATHDB-323662

The weighted bipartite-graph product conjecture

摘录 / EXCERPT: Weighted bipartite-graph product conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-047510 | MATHDB-323665

Independent-set extremal conjecture for graphs of prescribed minimum degree

摘录 / EXCERPT: Minimum-degree extremal conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-047511 | MATHDB-323674

The in-degree bound for arc-coloured digraph constructions

摘录 / EXCERPT: Let $D=(N,A_0\cup A_1)$ be an arc-coloured digraph. Suppose every triangle has the form … or … and every node has exactly d incoming arcs in each of A_0 and A_1. The in-degre...

conjecture | open | MathDB 页面 | 原文线索

V2-047512 | MATHDB-323692

NP-completeness conjecture for edge-clique cover on cographs

摘录 / EXCERPT: A cograph is a graph obtained from a single vertex by repeatedly applying disjoint union and complementation. For a graph G, the edge-clique cover problem asks whether the edges...

conjecture | open | MathDB 页面 | 原文线索

V2-047513 | MATHDB-323710

The endpoint-rooted path conjecture for cover cost-to-time ratio

摘录 / EXCERPT: Let G be a rooted graph on n vertices, with root r, and let $cc_r(G)$ and $CT_r(G)$ denote its cover cost and cover time from r. The path on n vertices rooted at an endpo...

conjecture | open | MathDB 页面 | 原文线索

V2-047514 | MATHDB-323732

Golumbic-Monma-Trotter conjecture on tolerance and cocomparability graphs

摘录 / EXCERPT: Golumbic-Monma-Trotter conjecture. The intersection of cocomparability graphs with tolerance graphs is exactly the class of bounded tolerance graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-047515 | MATHDB-323739

Independence of multispins in rooted forests of Cartesian products of complete graphs

摘录 / EXCERPT: Let $G=(U,A)$ be a weighted digraph, let K_p be the complete graph with vertex set $[p]$, and let $H=G\times K_p$ be their Cartesian product. A rooted forest of H has a G-pro...

conjecture | open | MathDB 页面 | 原文线索

V2-047516 | MATHDB-323748

The authors' characterization of radically solvable rigid graphs

摘录 / EXCERPT: The authors' rigid-graph characterization conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-047517 | MATHDB-323775

The local rigidity conjecture for graphs with infinite neighbourhood differences

摘录 / EXCERPT: Let Γ be an infinite graph, and write $\Gamma(x)$ for the neighbourhood of a vertex x. Assume that for every pair of distinct vertices x,y, ... is infinite. A graph is loc...

conjecture | open | MathDB 页面 | 原文线索

V2-047518 | MATHDB-323786

The AWGN multiway relay coding-strategy dominance conjecture

摘录 / EXCERPT: Consider an additive white Gaussian noise multiway relay channel (AWGN MWRC) with relay power $P_0=LP$, where L is the number of users and P is the user power. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047519 | MATHDB-323819

Hajós' cycle decomposition conjecture

摘录 / EXCERPT: A simple even graph is a simple graph in which every vertex has even degree. A decomposition into cycles is a partition of its edge set into cycles. Hajós' conjecture. If G is a...

conjecture | open | MathDB 页面 | 原文线索

V2-047520 | MATHDB-323827

Generalised lower bound conjecture for neighbourly triangulated manifolds

摘录 / EXCERPT: Let M be an m-vertex connected triangulated closed d-manifold with Betti numbers β_i, and let ${\cal K}_l^{\ast}(d)$ denote the class of $(l+1)$-neighbourly members of...

conjecture | open | MathDB 页面 | 原文线索

V2-047521 | MATHDB-323839

Faudree-Schelp conjecture for three-color Ramsey numbers of paths

摘录 / EXCERPT: For a graph G, let $R_3(G)=R(G,G,G)$ denote its three-color diagonal Ramsey number, and let P_n be the path on n vertices. Faudree-Schelp conjecture. ... holds for all...

conjecture | open | MathDB 页面 | 原文线索

V2-047522 | MATHDB-323840

Conjectured three-color Ramsey numbers for cycles

摘录 / EXCERPT: For a graph G, let $R_3(G)=R(G,G,G)$ denote its three-color diagonal Ramsey number, and let C_n be the cycle on n vertices. Conjectured cycle formula. ... The source presents t...

conjecture | open | MathDB 页面 | 原文线索

V2-047523 | MATHDB-323849

Nonexistence conjecture for hoppers of levels 1 and 2

摘录 / EXCERPT: A hopper of level 1 or 2 is the corresponding hopper object defined in the paper's graph-embedding framework. The no-hoppers conjecture. There are no hoppers of level 1 or 2. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-047524 | MATHDB-323882

Type D Chan-Robbins-Yuen volume conjecture

摘录 / EXCERPT: Let K_n^D be the complete signed graph with vertex set $[n]$, containing all edges $(i,j,\pm)$ with $1\leq i<j\leq n$, and let ... be its flow polytope. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047525 | MATHDB-323895

Extended permanence conjecture for confined endotactic reaction systems

摘录 / EXCERPT: Let N be a confined endotactic reaction system, and let $\mathfrak M^\ell(N)$ denote its associated differential inclusion. A differential inclusion is permanent when its traject...

conjecture | open | MathDB 页面 | 原文线索

V2-047526 | MATHDB-323896

Weakly reversible reaction-system boundedness and repulsion conjecture

摘录 / EXCERPT: Let N be a confined, weakly reversible reaction system, and let $\mathfrak M(N)$ denote its associated mass-action differential inclusion. Say that $\mathfrak M(N)$ is repelled b...

conjecture | open | MathDB 页面 | 原文线索

V2-047527 | MATHDB-323935

Zhang-Liu-Wang conjecture on the adjacent vertex distinguishing chromatic index

摘录 / EXCERPT: Let G be a finite simple graph with no isolated vertices. Write $V(G)$ and $E(G)$ for its vertex and edge sets, let $\Delta(G)$ be its maximum degree, and let $\chi'_{a}(G)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-047528 | MATHDB-323940

Cubic or 3-connected hardness conjecture for Pirates and Treasure

摘录 / EXCERPT: Let G be the graph input to the Pirates and Treasure game. A graph is cubic if every vertex has degree 3, and 3-connected if deleting fewer than three vertices does not disconn...

conjecture | open | MathDB 页面 | 原文线索

V2-047529 | MATHDB-323968

Zhang's conjecture on cyclically 5-edge-connected permutation snarks

摘录 / EXCERPT: Zhang's conjecture. If G is a cyclically 5-edge-connected cycle permutation graph and G is a snark, then G must be the Petersen graph.

conjecture | open | MathDB 页面 | 原文线索

V2-047530 | MATHDB-323969

Fleischner's conjecture on compatible cycle decompositions

摘录 / EXCERPT: Fleischner's conjecture. The pair (G,T) has no compatible cycle decomposition if and only if (G,T) is the bad loop or the bad K_5.

conjecture | open | MathDB 页面 | 原文线索

V2-047531 | MATHDB-323970

Berge-Jackson-Zhang conjecture on circuit double covers

摘录 / EXCERPT: Berge-Jackson-Zhang conjecture. If G is a cyclically 5-edge-connected cubic graph and D is a set of pairwise disjoint circuits of G, then D is a subset of a CDC, unless...

conjecture | open | MathDB 页面 | 原文线索

V2-047532 | MATHDB-323971

Zhang's conjecture on circuit double covers of cycle permutation graphs

摘录 / EXCERPT: Zhang's conjecture. If G is cyclically 5-edge-connected and there is no CDC that includes both C_1 and C_2, then G must be the Petersen graph.

conjecture | open | MathDB 页面 | 原文线索

V2-047533 | MATHDB-323999

The weak virtual periodicity conjecture for maximal coclass trees

摘录 / EXCERPT: Let $r\in\mathbb N_0$ and let K be an arbitrary field. For an algebra A in $\mathcal G_{r,K}$, write $\mathcal T(A)$ for the rooted subgraph consisting of all paths starting at...

conjecture | open | MathDB 页面 | 原文线索

V2-047534 | MATHDB-324000

The strong virtual periodicity conjecture for maximal coclass trees

摘录 / EXCERPT: Let $r\in\mathbb N_0$ and let K be an arbitrary field. Let $\mathcal T$ be a maximal coclass tree in $\mathcal G_{r,K}$ with maximal infinite path $A_1\to A_2\to\cdots$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-047535 | MATHDB-324076

Socle lower bound conjecture for homology manifolds

摘录 / EXCERPT: Let Δ be a connected orientable homology $(d-1)$-manifold without boundary whose vertex links have the weak Lefschetz property. Let Θ and w be as in Theorem 5.4, se...

open_problem | open | MathDB 页面 | 原文线索

V2-047536 | MATHDB-324087

The topological-surface asymptotic bondage-number conjecture

摘录 / EXCERPT: Let G be a connected graph of orientable genus h and non-orientable genus k. Let $b(G)$ be its bondage number and $\Delta(G)$ its maximum vertex degree. Let c_h and c'_k...

conjecture | open | MathDB 页面 | 原文线索

V2-047537 | MATHDB-324097

Gyárfás-Sárközy-Selkow conjecture on multicolored matchings in uniform hypergraphs

摘录 / EXCERPT: Gyárfás-Sárközy-Selkow conjecture. Every t-coloring of the edges of K_n^r contains an s-colored matching of size k provided that

conjecture | open | MathDB 页面 | 原文线索

V2-047538 | MATHDB-324119

The high-vertex decomposition conjecture for smooth Fano polytopes

摘录 / EXCERPT: Let P be a d-dimensional smooth Fano polytope with n vertices, where $n\ge 3d-k$ and $k\le\tfrac{d}{3}$. Let P_6 denote the six-dimensional smooth Fano polytope used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-047539 | MATHDB-324120

The inertia lower-bound conjecture for the chromatic number

摘录 / EXCERPT: The inertia lower-bound conjecture. The chromatic number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047540 | MATHDB-324121

Hansen-Lucas conjectures on the signless Laplacian spectral radius

摘录 / EXCERPT: Hansen-Lucas conjectures. The following inequalities and equality characterizations hold:

conjecture | open | MathDB 页面 | 原文线索

V2-047541 | MATHDB-324126

Orthogonal-complement spanning conjecture for simplicial rook graphs

摘录 / EXCERPT: Let $N=|V(d,n)|$, let $\mathcal{H}_{d,n}\subseteq\mathbb{R}^N$ be the linear span of the constructed permutohedron eigenvectors, and let ${\mathbf{X}}^{(i,j)}_\alpha$ denote the as...

conjecture | open | MathDB 页面 | 原文线索

V2-047542 | MATHDB-324127

Small-dilate least-eigenvalue and multiplicity conjecture for simplicial rook graphs

摘录 / EXCERPT: Let $SR(d,n)$ be the simplicial rook graph, let $\tau(SR(d,n))$ denote its least adjacency eigenvalue, and let $M(d,n)$ be the Mahonian number. Small-dilate eigenvalue conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047543 | MATHDB-324128

The strong saturation conjecture for bipartite graphs

摘录 / EXCERPT: Let $S_n(p,q)$ be the minimum number of edges in an $n\times n$ bipartite graph such that adding any new edge between its two classes creates a copy of $K_{p,q}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047544 | MATHDB-324129

The global attractor conjecture for weakly reversible reaction networks

摘录 / EXCERPT: Global attractor conjecture. The rate equation has exactly one equilibrium solution in each positive stoichiometric compatibility class, and any positive solution that starts in th...

conjecture | open | MathDB 页面 | 原文线索

V2-047545 | MATHDB-324130

Regularity conjecture for critical graphs at maximum degree eight

摘录 / EXCERPT: Let G be a critical graph, meaning that deleting any vertex lowers its chromatic number, with chromatic number χ and maximum degree Δ. Regularity conjecture. Every cr...

conjecture | open | MathDB 页面 | 原文线索

V2-047546 | MATHDB-324139

Integer-basis conjecture for g-vectors

摘录 / EXCERPT: Let B be an exchange matrix, let t_0 be a fixed initial vertex of the n-regular tree T_n, and let t be any vertex. The vectors $\boldsymbol{g}_{i;t}^{B;t_0}$ are t...

conjecture | open | MathDB 页面 | 原文线索

V2-047547 | MATHDB-324140

g-vector compatibility criterion for common clusters

摘录 / EXCERPT: Let B be an exchange matrix with principal coefficients and initial vertex t_0. Let $x_{i;t'}^{B;t_0}$ and $x_{j;t”}^{B;t_0}$ be cluster variables. For any vertex t of the...

conjecture | open | MathDB 页面 | 原文线索

V2-047548 | MATHDB-324150

The weak-closedness characterization of F-purity for binomial edge ideals in characteristic 2

摘录 / EXCERPT: Weak-closedness characterization. The quotient S/J_G is F-pure if and only if G is weakly closed.

conjecture | open | MathDB 页面 | 原文线索

V2-047549 | MATHDB-324213

The tree domination-game pair conjecture

摘录 / EXCERPT: Tree domination-game pair conjecture. No pair of the form $(k,k-1)$ can be realized by a tree.

conjecture | open | MathDB 页面 | 原文线索

V2-047550 | MATHDB-324251

Cao's strengthened Randić index-radius conjecture

摘录 / EXCERPT: Cao's strengthened conjecture. For any connected graph G except even paths,

conjecture | open | MathDB 页面 | 原文线索

V2-047551 | MATHDB-324252

The Harmonic index-radius conjecture

摘录 / EXCERPT: Harmonic index-radius conjecture. For all connected graphs G except even paths,

conjecture | open | MathDB 页面 | 原文线索

V2-047552 | MATHDB-324271

The degree-ordering conjecture for subtree-maximizing trees

摘录 / EXCERPT: Degree-ordering conjecture. The degrees along the odd-length path must satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-047553 | MATHDB-324305

Norine and Thomas's linear abundance conjecture for degree-3 vertices in minimal bricks

摘录 / EXCERPT: A minimal brick is a brick G such that $G-e$ is not a brick for every edge $e\in E(G)$. A brick is a bicritical and 3-connected graph, and a graph is bicritical if $G-\{u,v\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-047554 | MATHDB-324307

Torpid mixing conjecture for q-coloring Glauber dynamics on high-dimensional tori

摘录 / EXCERPT: Let ${\mathbb Z}^d_n$ be the even discrete torus with vertex set $[n]^d$, where n is even, and nearest-neighbor edges taken modulo n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047555 | MATHDB-324344

Turán codegree conjecture for complete hypergraphs

摘录 / EXCERPT: For integers $k\geq 2$, let K_{k+1}^k be the complete k-graph on $k+1$ vertices. For a k-graph H, let $\delta_{k-1}(H)$ be its minimum $(k-1)$-degree, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047556 | MATHDB-324345

Asymptotic enumeration conjecture for graphical regular representations

摘录 / EXCERPT: Let $d\text{-valent}$ vertex-transitive graphs, Cayley graphs, and graphical regular representations (GRRs) be counted up to order at most n by $VT_d(n)$, $CAY_d(n)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047557 | MATHDB-324346

McKay-Praeger conjecture for fixed-valency vertex-transitive graphs

摘录 / EXCERPT: For fixed $d\geq 3$, let $VT_d(n)$ be the number of isomorphism classes of d-valent vertex-transitive graphs of order at most n, and let $CAY_d(n)$ count the corresponding Cayl...

conjecture | open | MathDB 页面 | 原文线索

V2-047558 | MATHDB-324356

Existence conjecture for perfect distance-dominating sets with path components

摘录 / EXCERPT: Existence conjecture. Let H be a finite path or a Cartesian product of two finite paths. Then a t-PDDS$[H]$ in Λ_n exists if and only if at least one of the following h...

conjecture | open | MathDB 页面 | 原文线索

V2-047559 | MATHDB-324369

Hansen's Szeged-Wiener index conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a finite, simple, connected bipartite graph with $n \geq 4$ vertices and $m \geq n$ edges. Its Wiener index is ... and its Szeged index is ... where $n_u(e)$ and $n_v(e)$ co...

conjecture | open | MathDB 页面 | 原文线索

V2-047560 | MATHDB-324384

The 1-2 Conjecture for total weightings

摘录 / EXCERPT: The 1-2 Conjecture. Every graph has a total 2-weighting whose resulting vertex sums form a proper colouring; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-047561 | MATHDB-324390

Exact minimum-degree threshold conjecture for perfect matchings in uniform hypergraphs

摘录 / EXCERPT: Let $k,ll in \mathbb N$ satisfy ... Let H be a k-uniform hypergraph on $n\geq n_0$ vertices, where $n_0\in\mathbb N$ and k divides n. Write $\delta_\ell(H)$ for the minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-047562 | MATHDB-324408

Cyclically 5-edge-connected odd 2-factor classification conjecture

摘录 / EXCERPT: Let G be a cyclically 5-edge-connected odd 2-factored snark, meaning that G is a snark, every cycle in every 2-factor of G is odd, and no edge cut of size at most four...

conjecture | open | MathDB 页面 | 原文线索

V2-047563 | MATHDB-324409

Wilmes' multiplicity conjecture for boundary divisors

摘录 / EXCERPT: Wilmes' multiplicity conjecture. If there are exactly l distinct partitions $\Pi_j\in\operatorname{\mathcal{P}}_{k+1}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047564 | MATHDB-324423

Total sequence list 1-2-3 Conjecture

摘录 / EXCERPT: Let G be any graph, and fix an ordering of the edges and vertices used in a total weighting. A total sequence k-list-weighting assigns to each edge and vertex a weight from its...

conjecture | open | MathDB 页面 | 原文线索

V2-047565 | MATHDB-324431

Linear growth conjecture for rainbow matching thresholds

摘录 / EXCERPT: Let $f(r,t)$ be the largest number of colors for which there exists an (f,t)-colored r-partite r-graph without a rainbow t-matching. Linear growth conjecture. For every r...

conjecture | open | MathDB 页面 | 原文线索

V2-047566 | MATHDB-324435

The mirror extension conjecture for vertex operator algebras

摘录 / EXCERPT: Mirror extension conjecture. Under these hypotheses, rationality of the extension U^e implies rationality of the mirror extension $(U^c)^e$.

conjecture | open | MathDB 页面 | 原文线索

V2-047567 | MATHDB-324440

Generalized hypertree width is W[1]-hard parameterized by incidence treewidth

摘录 / EXCERPT: Let H be a hypergraph, and let the incidence treewidth of H be the treewidth of its incidence graph. Generalized-hypertree-width conjecture. Generalized Hypertree Width is W[1]...

conjecture | open | MathDB 页面 | 原文线索

V2-047568 | MATHDB-324442

Wilfong-Haxell-Winkler delay-colouring conjecture for bipartite multigraphs

摘录 / EXCERPT: Let G be a bipartite multigraph with partition classes A,B and maximum degree d. Each edge e has an associated integer delay $r(e)$. An edge-colouring is a map...

conjecture | open | MathDB 页面 | 原文线索

V2-047569 | MATHDB-324443

Aharoni-Berger-Raisman's hypergraph matching conjecture

摘录 / EXCERPT: Let H be a 3-partite 3-uniform hypergraph with partition classes V_1,V_2,V_3 satisfying ... For every $x\in V_1$, the hyperedges containing x induce a perfect matching of...

conjecture | open | MathDB 页面 | 原文线索

V2-047570 | MATHDB-324455

The even-cycle decomposition conjecture for odd-K_5-minor-free signed graphs

摘录 / EXCERPT: A signed graph is a graph whose edges are designated even or odd; a cycle is even when it contains an even number of odd edges. A graph is even cycle decomposable if its edge set c...

conjecture | open | MathDB 页面 | 原文线索

V2-047571 | MATHDB-324470

The cycles-bridge-number conjecture for knot diagrams

摘录 / EXCERPT: Let a knot diagram determine its universe, and let the associated balanced overlaid Tait graph have a number of cycles. Let the bridge number be the minimum number of bridges among...

conjecture | open | MathDB 页面 | 原文线索

V2-047572 | MATHDB-324487

Nash-Williams's regular-subgraph conjecture for Hamilton-cycle packings

摘录 / EXCERPT: Nash-Williams's conjecture. If G is a graph on n vertices with minimum degree $\delta\ge n/2$, then G contains at least

conjecture | open | MathDB 页面 | 原文线索

V2-047573 | MATHDB-324488

The regular-subgraph conjecture for Hamilton-cycle packings

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\operatorname{reg}_{\operatorname{even}}(G)$ denote the largest degree of an even-regular spanning subgraph of G. The regular-subgrap...

conjecture | open | MathDB 页面 | 原文线索

V2-047574 | MATHDB-324500

Maximum-degree bound for equitable vertex arboricity

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ be its maximum degree, and let $va^{\equiv}_{\infty,\infty}(G)$ denote the strong equitable (∞,∞)-vertex-arboricity. Maximum-degree...

conjecture | open | MathDB 页面 | 原文线索

V2-047575 | MATHDB-324501

Meyer equitable chromatic number conjecture

摘录 / EXCERPT: Let G be a connected graph. A proper equitable t-coloring is a proper vertex coloring whose color classes differ in size by at most one, and $\chi^{=}(G)$ is the least t for...

conjecture | open | MathDB 页面 | 原文线索

V2-047576 | MATHDB-324505

The crossing-number conjecture for the crown graph

摘录 / EXCERPT: Crossing-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047577 | MATHDB-324517

Characterization of assembly graphs with genus range {0,1}

摘录 / EXCERPT: Genus-range characterization. Any DOW whose corresponding graph has genus range $\{0,1\}$ is obtained from the word

conjecture | open | MathDB 页面 | 原文线索

V2-047578 | MATHDB-324518

Maximum genus range of assembly graphs with an even number of vertices

摘录 / EXCERPT: Maximum-genus-range conjecture. The maximum genus range of assembly graphs with $2n$ vertices is

conjecture | open | MathDB 页面 | 原文线索

V2-047579 | MATHDB-324531

Liu-Wang-Zhang's conjecture on the neighbourhood distinguishing index

摘录 / EXCERPT: Liu-Wang-Zhang's conjecture. If $G\notin\{K_2,C_5\}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047580 | MATHDB-324532

Baril-Togni's multigraph neighbourhood distinguishing conjecture

摘录 / EXCERPT: Baril-Togni's conjecture. If $G\neq C_5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047581 | MATHDB-324535

Zhang-Chen-Li-Yao-Lu-Wang's adjacent vertex distinguishing total-colouring conjecture

摘录 / EXCERPT: Zhang-Chen-Li-Yao-Lu-Wang's conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047582 | MATHDB-324536

Czerwiński-Grytczuk-Żelazny's additive colouring conjecture

摘录 / EXCERPT: Czerwiński-Grytczuk-Żelazny's conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047583 | MATHDB-324587

Linear covering-number conjecture for game-saturated intersecting families

摘录 / EXCERPT: Let $k\text{-uniform}$ intersecting families be played in the game-saturation game on the complete k-uniform hypergraph $4\mathbb{I}_{n,k}4$, with $gsat_F(\mathbb{I}_{n,k})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-047584 | MATHDB-324593

Cameron-Sheehan-Spiga conjecture on semiregular automorphisms of cubic vertex-transitive graphs

摘录 / EXCERPT: Cameron-Sheehan-Spiga conjecture. There exists a function $f:\mathbb{N}\to\mathbb{N}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-047585 | MATHDB-324594

Connected tree-width duality conjecture

摘录 / EXCERPT: Let G be a graph. A bramble in G is a set of pairwise touching connected subsets of $V(G)$, where two subsets touch if they have a vertex in common or G contains an edge betw...

conjecture | open | MathDB 页面 | 原文线索

V2-047586 | MATHDB-324627

The converse augmenting-cycle conjecture for higher-dimensional maxflow

摘录 / EXCERPT: Let (X,T,c) be a d-dimensional network, let f be a feasible flow on X, and let an augmenting cycle be a d-cycle in the residual complex X_f that contains the target fac...

conjecture | open | MathDB 页面 | 原文线索

V2-047587 | MATHDB-324630

The weighted expanding formulation of Seymour's conjecture

摘录 / EXCERPT: Let D be a digraph without loops or two-cycles. Give each vertex a nonnegative real weight, and call a vertex weakly expanding when the weight of its second out-neighborhood is a...

conjecture | open | MathDB 页面 | 原文线索

V2-047588 | MATHDB-324631

The arc-weighted second neighborhood conjecture

摘录 / EXCERPT: Let D be an arc-weighted digraph without loops or two-cycles. For each vertex v, let δ_v be the second-neighborhood weight minus the first-neighborhood weight, as define...

conjecture | open | MathDB 页面 | 原文线索

V2-047589 | MATHDB-324641

Fujita-Nakamigawa conjecture on the balanced decomposition number

摘录 / EXCERPT: Fujita-Nakamigawa's conjecture. If G is 2-connected, then its balanced decomposition number is at most

conjecture | open | MathDB 页面 | 原文线索

V2-047590 | MATHDB-324659

King's local Reed conjecture for graph coloring

摘录 / EXCERPT: Let G be a finite simple graph. For a vertex v, let $\widetilde N(v)=\{v\}\cup N(v)$ be its closed neighbourhood, let $G[\widetilde N(v)]$ be the induced subgraph, and define .....

conjecture | open | MathDB 页面 | 原文线索

V2-047591 | MATHDB-324670

Beck's Chooser-Picker conjecture

摘录 / EXCERPT: Let G be a hypergraph. Suppose that Breaker wins the Maker-Breaker game on G, with Maker moving first. Beck's Chooser-Picker conjecture. Picker wins the Chooser-Picker game on...

conjecture | open | MathDB 页面 | 原文线索

V2-047592 | MATHDB-324706

Graph formulation of the union-closed sets conjecture

摘录 / EXCERPT: A stable set in a graph is a set of vertices no two of which are adjacent; it is maximal if it is maximal under inclusion. Let G be a finite graph with at least one edge. Graph f...

conjecture | open | MathDB 页面 | 原文线索

V2-047593 | MATHDB-324720

Global persistency conjecture for compartmental reaction-diffusion networks

摘录 / EXCERPT: Let $X_0\in\mathbb{R}_+^{mN}$ be an initial condition for the closed compartmental reaction-diffusion system, with trajectories denoted by $t\mapsto X(t)$. Global persistency conje...

conjecture | open | MathDB 页面 | 原文线索

V2-047594 | MATHDB-324721

Friedman's adjacency and Markov operator conjecture for random graph coverings

摘录 / EXCERPT: Let Ω be a finite connected graph, let Γ be a random n-covering of Ω, and let $\lambda_A(\Gamma)$ and $\lambda_M(\Gamma)$ denote the largest absolute values...

conjecture | open | MathDB 页面 | 原文线索

V2-047595 | MATHDB-324723

ADE orbifold C2-cofiniteness conjecture for the triplet vertex algebra

摘录 / EXCERPT: Let $\mathcal{W}(p)$ be the triplet vertex algebra, and let Γ be a finite subgroup of $PSL(2,\mathbb{C})$ in the ADE classification. ADE orbifold C2-cofiniteness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047596 | MATHDB-324734

Li et al.'s conjecture on rainbow connection and vertex connectivity

摘录 / EXCERPT: Li et al.'s conjecture. There exists a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-047597 | MATHDB-324737

Chang-Narayanan conjecture on strong edge-colorings of degenerate graphs

摘录 / EXCERPT: Chang-Narayanan conjecture. There exists an absolute constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-047598 | MATHDB-324761

Bollobás-Erdős conjecture on properly coloured Hamiltonian cycles

摘录 / EXCERPT: Let K_n^c be an edge-coloured complete graph on n vertices. Define $\operatorname*{\Delta_{{\rm mon}}}(K_n^c)$ to be the maximum number of edges of one colour incident with any...

conjecture | open | MathDB 页面 | 原文线索

V2-047599 | MATHDB-324764

Long-screening kernel conjecture for principal W-algebras

摘录 / EXCERPT: Let $L=\sqrt{p}Q$ with $p\geq 2$, let $\mathfrak{g}$ be simply laced of rank l, and let $\mathcal{W}_k(\mathfrak{g})$ be the associated principal W-algebra with $p=k+h^\vee$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-047600 | MATHDB-324765

Maximally extended W-algebra conjecture for lattice W-algebras

摘录 / EXCERPT: Let $L=\sqrt{p}Q$, let V_L be the associated lattice vertex algebra, and define ... Write $\mathcal{W}_k(\mathfrak{g})$ for the principal W-algebra and...

conjecture | open | MathDB 页面 | 原文线索

V2-047601 | MATHDB-324769

Ryser's biclique covering conjecture

摘录 / EXCERPT: Ryser's biclique covering conjecture. In every r-coloring of the edges of a biclique, the vertex set can be covered by the vertices of at most $2r-2$ monochromatic components.

conjecture | open | MathDB 页面 | 原文线索

V2-047602 | MATHDB-324777

The short-orientation conjecture for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph on n vertices, where $d\ge3$. Write $l_1(G)$ for the relevant shortest-orientation parameter and let $\chi(G)$ denote the chromatic number of G....

conjecture | open | MathDB 页面 | 原文线索

V2-047603 | MATHDB-324778

Bollobás and Erdős's conjecture on properly colored Hamiltonian cycles

摘录 / EXCERPT: Let K_n^c be an edge-colored complete graph on n vertices, and let $\Delta^{\mathrm{mon}}(K_n^c)$ be the maximum number of edges of the same color incident with a vertex of...

conjecture | open | MathDB 页面 | 原文线索

V2-047604 | MATHDB-324797

The splitting-set formula for the excessive 4-index of graphs

摘录 / EXCERPT: Let G be an $[4]$-coverable graph. For each t, let $s^t(G)$ denote the cardinality of a largest t-splitting set of G, where a t-splitting set is a subgraph whose $[t+1]$-...

conjecture | open | MathDB 页面 | 原文线索

V2-047605 | MATHDB-324808

Conjecture on Turán densities of iterated suspensions

摘录 / EXCERPT: Iterated-suspension conjecture. For any $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-047606 | MATHDB-324809

Conjecture on limiting Turán densities of iterated suspensions

摘录 / EXCERPT: Iterated-suspension limit conjecture. For all hypergraphs H,

conjecture | open | MathDB 页面 | 原文线索

V2-047607 | MATHDB-324812

Conjecture on sparse graphs with average degree and strong chromatic index of the same order

摘录 / EXCERPT: Let $d(n)$ be a function tending to infinity. Average-degree strong-index conjecture. There exists such a function $d(n)$ that, for every constant c, there is a graph on n vert...

conjecture | open | MathDB 页面 | 原文线索

V2-047608 | MATHDB-324819

Clique-Stable Set separation conjecture for graphs

摘录 / EXCERPT: Let G be a graph on n vertices. A clique-stable set separator is a family of cuts such that every disjoint clique and stable set of G are separated by one of the cuts. Clique...

conjecture | open | MathDB 页面 | 原文线索

V2-047609 | MATHDB-324848

Dankelmann-Laskar's simultaneous domination conjecture for two graphs

摘录 / EXCERPT: Let $\gamma_{\rm sd}(2,2,n)$ denote the maximum, over pairs of graphs of order n, of the minimum size of a set that simultaneously dominates both graphs. Dankelmann-Laskar's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-047610 | MATHDB-324884

Li's directed C_4 conjecture for oriented bipartite graphs

摘录 / EXCERPT: Let D be an oriented bipartite graph with bipartition (A,B), and let $d^+(x)$ denote the out-degree of a vertex x. A directed C_4 is a directed cycle of length four in D....

conjecture | open | MathDB 页面 | 原文线索

V2-047611 | MATHDB-324954

Conjecture that partitionable convergence is equivalent to left-convergence

摘录 / EXCERPT: Let $(\mathscr{P}_n)$ be a sequence of branching partitions and let $\widetilde{\mathscr{P}}$ be a branching partition. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-047612 | MATHDB-324957

Asymptotic equivalence of even-diameter surface and projective-plane graphs

摘录 / EXCERPT: Let Σ be a surface, let $k\geq 2$ be an even diameter, and let $N(\Delta,k,\Sigma)$ denote the maximum order of a graph of maximum degree Δ and diameter k embeddabl...

conjecture | open | MathDB 页面 | 原文线索

V2-047613 | MATHDB-324961

Convexity and reciprocal concavity of the coupon-collection objective function

摘录 / EXCERPT: Convexity and reciprocal-concavity conjecture. For each $N\in\mathbb{N}$, the function p_N is convex; indeed, $1/p_N$ is concave. This claim concerns the network objective functi...

conjecture | open | MathDB 页面 | 原文线索

V2-047614 | MATHDB-324969

DeMarco-Kahn conjecture for sparse random Turán identities

摘录 / EXCERPT: Let $t\geq2$ be an integer, let $G(n,p)$ be the binomial random graph, let $\operatorname{ex}_G(K_{t+1})$ denote the maximum number of edges in a K_{t+1}-free subgraph of G, an...

conjecture | open | MathDB 页面 | 原文线索

V2-047615 | MATHDB-324970

Kohayakawa-Łuczak-Rödl sparse embedding conjecture

摘录 / EXCERPT: Let $\alpha>0$, $d>0$, and let F be a graph with vertex set $V(F)=[\ell]$. For $p\in(0,1]$ and integers m,M, let $\mathcal G(F,m,M,\varepsilon,p,d)$ be the family of balanced...

conjecture | open | MathDB 页面 | 原文线索

V2-047616 | MATHDB-324980

Mubayi-Verstraëte conjecture on Turán numbers of linear cycles

摘录 / EXCERPT: Mubayi-Verstraëte conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-047617 | MATHDB-324992

One-versus-t hardness conjecture for acquaintance time

摘录 / EXCERPT: Let $t\in\mathbb{N}$ be a constant and let G be a graph. Write $\mathcal{AC}(G)$ for the acquaintance time of G. One-versus-t hardness conjecture. It is $\mathcal{NP}$-hard t...

conjecture | open | MathDB 页面 | 原文线索

V2-047618 | MATHDB-324993

Hardness of distinguishing low equitable chromatic number from intersecting graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a graph on n vertices. For $t\in\mathbb{N}$ and $\beta>0$, call G (β,t)-intersecting if, for every t subsets $S_1,\dots,S_t\subseteq V$ of size...

conjecture | open | MathDB 页面 | 原文线索

V2-047619 | MATHDB-325042

Non-scaling conjecture for multi-hop interference networks with delayed CSI

摘录 / EXCERPT: Let K users communicate over a layered M-hop interference network with delayed channel-state information (CSI), and let its network sum-DoF be the supremum of achievable sum-Do...

conjecture | open | MathDB 页面 | 原文线索

V2-047620 | MATHDB-325097

Fox-Pach sharp edge bound conjecture for K_{t,t}-free string graphs

摘录 / EXCERPT: For an integer $t\geq 2$, a graph is $K_{t,t}$-free if it contains no complete bipartite subgraph with t vertices in each part. Fox-Pach's sharp edge bound conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-047621 | MATHDB-325098

Crossing-pairs conjecture for topological graphs

摘录 / EXCERPT: A topological graph is a graph drawn in the plane with vertices represented by points and edges by curves connecting their endpoints; two edges cross when their curves intersect aw...

conjecture | open | MathDB 页面 | 原文线索

V2-047622 | MATHDB-325100

The complete-partite graph cycle-space conjecture for K^{s,s-1}

摘录 / EXCERPT: The complete-partite graph cycle-space conjecture. For every $s\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-047623 | MATHDB-325110

Multiple-copy Ramsey numbers for loose and tight hypergraph paths and cycles

摘录 / EXCERPT: Multiple-copy path and cycle conjecture. For every $k\geq 3$, $m\geq n$ and $r\geq s\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-047624 | MATHDB-325119

Bounds on the critical interaction strength for graphs with one negative edge

摘录 / EXCERPT: Let Γ be a graph with N vertices, let Γ_+ denote its positive-edge subgraph, and let $t^\star(\Gamma)$ be the critical interaction strength at which stability is lo...

conjecture | open | MathDB 页面 | 原文线索

V2-047625 | MATHDB-325152

Nordhaus et al.'s conjecture on maximum and minimum genus

摘录 / EXCERPT: Let G be a graph, with minimum genus $\gamma(G)$ and maximum genus $\Gamma(G)$. A graph is upper-embeddable if it has a one- or two-face embedding and hence satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-047626 | MATHDB-325159

Symmetry characterization for independence polynomials of lexicographic products

摘录 / EXCERPT: Symmetry characterization. $I(G\circ H;x)$ is symmetric for every graph G if and only if $H=K_r-e$ for some $r\geq 2$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-047627 | MATHDB-325163

The connected domination bound for connected (P_9,C_9,H)-free graphs

摘录 / EXCERPT: Let P_9 denote the path on 9 vertices, let C_9 denote the cycle on 9 vertices, and let H be the graph shown in Figure H. A graph is (P_9,C_9,H)-free if it contains none...

conjecture | open | MathDB 页面 | 原文线索

V2-047628 | MATHDB-325227

Aalipour-Hafshejani et al.'s domination-polynomial uniqueness conjecture for complete bipartite graphs

摘录 / EXCERPT: Aalipour-Hafshejani et al.'s conjecture. For all $a,b\in\mathbb{N}$, if $|a-b|\geq 2$, then ${\cal K}(a,b)$ is ${\cal D}$-unique; that is, every graph with the same domination poly...

conjecture | open | MathDB 页面 | 原文线索

V2-047629 | MATHDB-325238

The bounded-degree modeling FO-limit characterization conjecture

摘录 / EXCERPT: Let a modeling be a measurable graph equipped with a probability measure for which first-order definable sets are measurable. A graph has bounded degree if its vertex degrees are u...

conjecture | open | MathDB 页面 | 原文线索

V2-047630 | MATHDB-325239

Francisco-Hà-Van Tuyl's expansion conjecture for critically chromatic graphs

摘录 / EXCERPT: Let G be a graph that is critically s-chromatic, meaning that its chromatic number is s and deleting any vertex lowers the chromatic number. For a subset W of the vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-047631 | MATHDB-325240

Algebraic form of the Conforti-Cornuéjols conjecture for edge ideals

摘录 / EXCERPT: Let G be a simple hypergraph with edge ideal $I=I(G)$, and let $\operatorname{gr}_I=\bigoplus_{q\geq 0}I^q/I^{q+1}$ be the associated graded ring. A hypergraph is König when its...

conjecture | open | MathDB 页面 | 原文线索

V2-047632 | MATHDB-325241

Ideal-theoretic Conforti-Cornuéjols conjecture for squarefree monomial ideals

摘录 / EXCERPT: Let I be a squarefree monomial ideal. A minor of I is obtained by a sequence of quotients and localizations at variables, and the monomial grade of I is the maximum length of...

conjecture | open | MathDB 页面 | 原文线索

V2-047633 | MATHDB-325247

Two disjoint shortest paths in the equal-hop path construction

摘录 / EXCERPT: Two-path conjecture. Node k always has two disjoint shortest paths p_{0} and p_{1} satisfying one of the following alternatives: (1) p_{0} runs from X_{0} through the col...

conjecture | open | MathDB 页面 | 原文线索

V2-047634 | MATHDB-325260

Conjecture on a braided tensor category of strongly graded generalized modules

摘录 / EXCERPT: Let V be a vertex algebra and let $\mathcal{C}$ be a category of strongly graded generalized V-modules. Assume that $\mathcal{C}$ is closed under $P(z)$-tensor products for som...

conjecture | open | MathDB 页面 | 原文线索

V2-047635 | MATHDB-325277

The monochromatic cycle-cover conjecture for t-colored complete graphs

摘录 / EXCERPT: Let K be a complete graph whose edges are colored with t colors. A monochromatic cycle is a cycle all of whose edges have the same color; the empty set, one vertex, and one edg...

conjecture | open | MathDB 页面 | 原文线索

V2-047636 | MATHDB-325279

Ryser's transversal conjecture for t-partite hypergraphs

摘录 / EXCERPT: Let H be an intersecting t-partite hypergraph, meaning that every two hyperedges intersect. A transversal is a set of vertices meeting every hyperedge. Ryser's conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-047637 | MATHDB-325281

The two-colored matching conjecture for 4-colored complete graphs

摘录 / EXCERPT: Let $k\ge1$. A 2-colored matching is a matching whose edges use at most two colors. The 4-color matching conjecture. If ... then every edge-coloring of K_n with four colors contain...

conjecture | open | MathDB 页面 | 原文线索

V2-047638 | MATHDB-325282

Sills-Wang characteristic-polynomial conjecture for the Wiener-index matrix

摘录 / EXCERPT: Let $A_k=(a_{ij})$ be the $k\times k$ matrix with $a_{ij}=\frac{1}{2}|i-j|$, and let I_k be the $k\times k$ identity matrix. Define $C_k(\lambda)=\det(A_k-\lambda I_k)$ to be the...

conjecture | open | MathDB 页面 | 原文线索

V2-047639 | MATHDB-325283

Sills-Wang balanced-ordering conjecture for extremal caterpillars

摘录 / EXCERPT: Sills-Wang's balanced-ordering conjecture. The quadratic form $q(x)$ is maximized by this arrangement, where $|l_i-r_i|\le 1$ and $l_i+r_i=m_i$ for $i=2,\dots,s$.

conjecture | open | MathDB 页面 | 原文线索

V2-047640 | MATHDB-325291

Non-negativity conjecture for the cd-index of balanced labeled acyclic digraphs

摘录 / EXCERPT: Let G be a bounded labeled acyclic digraph with a balanced linear edge labeling. Non-negativity conjecture. The $\mathbf{c}\mathbf{d}$-index of G is non-negative. This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-047641 | MATHDB-325293

The largest Laplacian H-eigenvalue characterization of hyperstars

摘录 / EXCERPT: Hyperstar characterization conjecture. The largest Laplacian H-eigenvalue $\lambda(\mathcal L)$ is equal to d if and only if G is a hyperstar.

conjecture | open | MathDB 页面 | 原文线索

V2-047642 | MATHDB-325295

Regular hypergraph loose Hamilton cycle conjecture

摘录 / EXCERPT: Let $\mathbb H^{(k)}(n,d)$ denote the random d-regular k-uniform hypergraph on n vertices, and let a loose Hamilton cycle be a loose cycle containing all vertices. Loose Hami...

conjecture | open | MathDB 页面 | 原文线索

V2-047643 | MATHDB-325296

Regular hypergraph overlapping Hamilton cycle conjecture

摘录 / EXCERPT: Let $\mathbb H^{(k)}(n,d)$ denote the random d-regular k-uniform hypergraph on n vertices. For integers $k>\ell\geq 2$, an ℓ-overlapping cycle is a k-uniform hypergrap...

conjecture | open | MathDB 页面 | 原文线索

V2-047644 | MATHDB-325330

The conjecture on the complementary bounding function of the almost identity function

摘录 / EXCERPT: Let g be the almost identity function ... For a graph G, let $\alpha(G)$ be its independence number and let $\theta(G)$ be its minimum clique-cover number. The complementary boun...

conjecture | open | MathDB 页面 | 原文线索

V2-047645 | MATHDB-325340

Rational superpolynomial Dyck-path formula

摘录 / EXCERPT: Let $Y_{m/n}$ be the set of m/n Dyck paths, let $v(D)$ be the set of internal vertices of D, and let $\beta(P)$ be the stated vertex statistic. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047646 | MATHDB-325362

Maker's bounded-degree tree-building conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let ${\mathcal T}_n$ denote the family of spanning trees on these vertices. In the Maker-Breaker game $(E(K_n),{\mathcal T}_n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-047647 | MATHDB-325363

Finiteness conjecture for forbidden graphs of bounded algebraic co-rank

摘录 / EXCERPT: Let $\Gamma_{\leq k}$ denote the class of simple graphs with algebraic co-rank at most k, and let ${\bf Forb}(\Gamma_{\leq k})$ be the set of graphs that do not contain a graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-047648 | MATHDB-325390

Spectral robustness conjecture for fully connected modular networks

摘录 / EXCERPT: Let T be the matrix associated with two fully connected modules, with $\gamma=1$ (equivalently, $S=R=M$), and let α and β be the inter-modular densities of the block...

conjecture | open | MathDB 页面 | 原文线索

V2-047649 | MATHDB-325393

The main conjecture on exponential growth of product replacement graphs

摘录 / EXCERPT: Let G be an infinite group generated by d elements. The main conjecture asserts that the product replacement graphs $\Gamma_k(G)$ have exponential growth for all $k\ge d+1$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-047650 | MATHDB-325395

Non-amenability conjecture for product replacement graphs of the Grigorchuk group

摘录 / EXCERPT: Let $\mathbb G$ be the Grigorchuk group. The Grigorchuk non-amenability conjecture asserts that the product replacement graphs $\Gamma_k(\mathbb G)$ are non-amenable for all...

conjecture | open | MathDB 页面 | 原文线索

V2-047651 | MATHDB-325396

Exponential Nielsen growth conjecture for infinite finitely generated groups

摘录 / EXCERPT: Let G be an infinite finitely generated group. The exponential Nielsen growth conjecture asserts that there are an integer n and a generating n-tuple $S\in\Gamma_n(G)$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-047652 | MATHDB-325400

Characterization of block graphs by connected fair sets

摘录 / EXCERPT: Block-graph characterization conjecture. The graph G is a block graph if and only if the induced subgraph of every fair set of G is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-047653 | MATHDB-325403

Extension conjecture for dense-triangle K_9-minor-free graphs

摘录 / EXCERPT: A triangle is a cycle of length 3, and a graph is K_9-minor free if it has no K_9 minor. Dense-triangle K_9 extension conjecture. If every edge of a graph G belongs to at...

conjecture | open | MathDB 页面 | 原文线索

V2-047654 | MATHDB-325420

Li's conjecture on the complexity of generalized connectivity

摘录 / EXCERPT: Li's conjecture. Given G, S, and an integer $cell$ with $2\leq cell\leq n-2$, deciding whether

conjecture | open | MathDB 页面 | 原文线索

V2-047655 | MATHDB-325428

Universality of normalized genus-two partition functions under sewing

摘录 / EXCERPT: Let V be a vertex operator algebra with positive integer central charge C. Define the normalized genus-two partition functions ... Universality conjecture. The normalized partiti...

conjecture | open | MathDB 页面 | 原文线索

V2-047656 | MATHDB-325432

The infinite motion conjecture for graphs

摘录 / EXCERPT: Infinite motion conjecture for graphs. If $\operatorname{Aut}(G)$ has infinite motion, then

conjecture | open | MathDB 页面 | 原文线索

V2-047657 | MATHDB-325433

The motion conjecture for graphs

摘录 / EXCERPT: Motion conjecture for graphs. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-047658 | MATHDB-325452

Conjecture on Laplacian and signless Laplacian H-eigenvalues of even-uniform power hypergraphs

摘录 / EXCERPT: Let $G=(V,E)$ be an ordinary graph, let $k=2r$ be even, and let $G^k=(V^k,E^k)$ be the k-power hypergraph of G. Let $\mathcal L^k$ and $\mathcal Q^k$ be the Laplacian and signl...

conjecture | open | MathDB 页面 | 原文线索

V2-047659 | MATHDB-325453

Perfect matching minimum degree conjecture for uniform hypergraphs

摘录 / EXCERPT: Let G be a k-uniform hypergraph on n vertices. For integers n, d, and k with $1\leq d\leq k-1$ and $n/k\in\mathbb{N}$, let $m_d^{n/k}(k,n)$ denote the minimum integer...

conjecture | open | MathDB 页面 | 原文线索

V2-047660 | MATHDB-325454

Large matching minimum degree conjecture for uniform hypergraphs

摘录 / EXCERPT: Let G be a k-uniform hypergraph on n vertices. For integers n, d, k, and s with $1\leq d\leq k-1$ and $0\leq s\leq (1-\varepsilon)n/k$, let $m_d^s(k,n)$ denote the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-047661 | MATHDB-325496

The Henneberg 2-extension conjecture in three-dimensional rigidity

摘录 / EXCERPT: Let $G=(V,E)$ be a generically 3-isostatic graph, and let $X\subseteq V$ satisfy ... Let e and f be distinct edges whose vertices lie in X. An implied K_4 is a complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047662 | MATHDB-325502

Non-realizability of K_{3k+4} as a semi-arc k-visibility graph

摘录 / EXCERPT: Let K_{3k+4} denote the complete graph on $3k+4$ vertices, and let a semi-arc k-visibility graph be a graph represented by semi-arcs with visibility determined by at most k b...

conjecture | open | MathDB 页面 | 原文线索

V2-047663 | MATHDB-325520

Gutman-Furtula revised conjecture on trees with minimal ABC index

摘录 / EXCERPT: Let G be a tree with minimal atom-bond connectivity (ABC) index among all trees of size n. The trees $T_0,T_1,T_2,T_3,T_4,T_5,T_6$ are the structures depicted in Figure. Gutman...

conjecture | open | MathDB 页面 | 原文线索

V2-047664 | MATHDB-325548

Digraph form of the CSP dichotomy conjecture

摘录 / EXCERPT: Let $\mathbb{H}$ be a finite digraph, viewed as a relational structure, and let $\operatorname{CSP}(\mathbb{H})$ ask whether a finite input digraph admits a homomorphism to...

conjecture | open | MathDB 页面 | 原文线索

V2-047665 | MATHDB-325555

Non-amenability conjecture for product replacement graphs

摘录 / EXCERPT: Let G be an infinite finitely generated group, and let $\Gamma_n(G)$ denote its product replacement graph, whose vertices are generating n-tuples of G joined by Nielsen moves...

conjecture | open | MathDB 页面 | 原文线索

V2-047666 | MATHDB-325559

The KŁR conjecture on sparse regular graphs

摘录 / EXCERPT: Let H) be a fixed graph with vertex set $\{1,2,\dots,k\}$. For positive integers n,m and positive reals p,ϵ, let $\mathcal{G}(H,n,m,p,\epsilon)$ be the collection of g...

conjecture | open | MathDB 页面 | 原文线索

V2-047667 | MATHDB-325562

Hamiltonian cycle interval conjecture for generalized polygons

摘录 / EXCERPT: Let $s\ge 2$ and $n\ge 3$, and let $\mathcal{G}=({{\mathcal P}},{{\mathcal L}};{\mathcal I})$ be a generalized n-gon of order s, so its Levi graph is $(s+1)$-regular, has diame...

conjecture | open | MathDB 页面 | 原文线索

V2-047668 | MATHDB-325563

The quantum Hamiltonian reduction conjecture for \(W^{(2)}_n\)

摘录 / EXCERPT: Let $n>1$, and let $\varphi_n:\mathfrak{sl}(2)\rightarrow\mathfrak{sl}(n)$ be an embedding such that, as $\mathfrak{sl}(2)$-modules, $\mathfrak{sl}(n)$ decomposes as ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-047669 | MATHDB-325564

The surjective homomorphism conjecture for \(W^{(2)}_n\) and \(\mathcal B_{n+1}\)

摘录 / EXCERPT: For general n, let $W^{(2)}_n$ be the Feigin-Semikhatov algebra at level $k=-n^2/(n+1)$, let $\mathcal B_{n+1}$ be the subalgebra constructed in the $(1,n+1)$-theory, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047670 | MATHDB-325565

The fusion-spectral-flow compatibility conjecture for affine \(\widehat{\mathfrak{sl}}(2)\)-modules

摘录 / EXCERPT: Let V and W be admissible level k modules of $\widehat{\mathfrak{sl}}(2)$, let $s,t\in\mathbb Z$, and let $k\in\{-1/2,-4/3\}$. Write V^s for the module obtained from V by...

conjecture | open | MathDB 页面 | 原文线索

V2-047671 | MATHDB-325570

The asymptotic judicious partition conjecture for directed graphs

摘录 / EXCERPT: Let d be an integer satisfying $d\ge4$. For a directed graph, let m denote its number of edges, and for disjoint vertex sets V_1,V_2, let $e(V_1,V_2)$ denote the number of di...

conjecture | open | MathDB 页面 | 原文线索

V2-047672 | MATHDB-325572

Cui-Hou switching-equivalence conjecture for skew spectra of bipartite graphs

摘录 / EXCERPT: Cui-Hou conjecture. For every orientation σ of G,

conjecture | open | MathDB 页面 | 原文线索

V2-047673 | MATHDB-325576

Cranston-Rabern conjecture on maximum-degree cliques

摘录 / EXCERPT: Let G be a graph. Write Δ for its maximum degree, ω for its clique number, and χ for its chromatic number. Let $\mathcal{H}(G)$ be the subgraph induced by the...

conjecture | open | MathDB 页面 | 原文线索

V2-047674 | MATHDB-325583

Dominating-cycle conjecture for cubic graphs with 4-connected matching contractions

摘录 / EXCERPT: Let G be a cubic graph with a perfect matching $\mathcal M$ such that $G / \mathcal M$ is 4-connected. A cycle dominates $\mathcal M$ if it contains at least one endpoint of every...

conjecture | open | MathDB 页面 | 原文线索

V2-047675 | MATHDB-325603

Strong Hanani-Tutte conjecture for clustered graphs whose cluster graph is a path

摘录 / EXCERPT: Let (G,T) be a flat clustered graph, and let G_T be the simple graph obtained by contracting the clusters and deleting loops and multiple edges. An independently even clustered...

conjecture | open | MathDB 页面 | 原文线索

V2-047676 | MATHDB-325626

Feinberg's persistence conjecture for weakly reversible mass-action systems

摘录 / EXCERPT: A weakly reversible mass-action kinetics ODE system is a system whose reaction network is weakly reversible, with dynamics governed by mass-action kinetics. A system is persistent...

conjecture | open | MathDB 页面 | 原文线索

V2-047677 | MATHDB-325627

Craciun-Nazarov-Pantea extended permanence conjecture for endotactic networks

摘录 / EXCERPT: An endotactic reaction network is a reaction network satisfying the endotactic geometric condition, and a network is permanent when its mass-action differential inclusion is perman...

conjecture | open | MathDB 页面 | 原文线索

V2-047678 | MATHDB-325628

Anderson's boundedness conjecture for weakly reversible mass-action systems

摘录 / EXCERPT: A weakly reversible mass-action kinetics ODE system is a system whose reaction network is weakly reversible and whose dynamics follow mass-action kinetics. Boundedness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047679 | MATHDB-325633

The resoluble-gem invariance conjecture under blob and valid flip moves

摘录 / EXCERPT: Resoluble-gem invariance conjecture. A resoluble gem remains resoluble under a blob move, a valid flip move, and straightforward simplifications using $TS\rho$-moves.

conjecture | open | MathDB 页面 | 原文线索

V2-047680 | MATHDB-325636

The collapse conjecture for expected dimension with sparse collapsed graph

摘录 / EXCERPT: Let G be a graph with n vertices and at most $2n-2$ edges with an exchange, and let G' be the resulting collapsed graph. The collapse conjecture. If G' has $n-1$ edges, the...

conjecture | open | MathDB 页面 | 原文线索

V2-047681 | MATHDB-325638

Linear-query priority-queue algorithm for maximum st-flow in directed planar graphs

摘录 / EXCERPT: Priority-queue implementation conjecture. Such an algorithm can be implemented with $O(n)$ queries to a priority queue.

conjecture | open | MathDB 页面 | 原文线索

V2-047682 | MATHDB-325645

Edge-colouring conjecture for bipartite hypergraphs

摘录 / EXCERPT: A bipartite r-uniform hypergraph with special side S is one in which every edge meets S in exactly one vertex. Assume ... for every $u\in S$ and $v\in V\setminus S$. Edge-colou...

conjecture | open | MathDB 页面 | 原文线索

V2-047683 | MATHDB-325648

Generalized Ryser-Brualdi-Stein matching conjecture

摘录 / EXCERPT: Let H be a simple d-regular 3-partite 3-uniform hypergraph whose three sides each have size n, so that H is an $n\times n\times n$ 3-partite hypergraph. A matching is...

conjecture | open | MathDB 页面 | 原文线索

V2-047684 | MATHDB-325649

Conjecture on collecting all but one twelfth of a planar graph

摘录 / EXCERPT: Let G be a planar graph. To delete a vertex means to remove it and its incident edges. To collect a vertex means to remove it when its current degree is at most 4; a set is col...

conjecture | open | MathDB 页面 | 原文线索

V2-047685 | MATHDB-325658

Horak-Rosa conjecture on edge-length lists of Hamiltonian paths

摘录 / EXCERPT: Let K_v be the complete graph on $\\{0,1,\ldots,v-1\\}$. For an edge $[x,y]$ of K_v, define its length by ... and let $\ell(H)$ be the multiset of edge-lengths of a Hamiltonian...

conjecture | open | MathDB 页面 | 原文线索

V2-047686 | MATHDB-325670

Strict inequality for infinite Turán numbers of bipartite graphs

摘录 / EXCERPT: Infinite Turán strict-inequality conjecture. The infinite Turán number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047687 | MATHDB-325672

The surviving-rate conjecture for square grids

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $P_n\square P_n$ be the Cartesian product of two such paths. Write $\rho(G)$ for the expected percentage of vertices saved when a fir...

conjecture | open | MathDB 页面 | 原文线索

V2-047688 | MATHDB-325681

The nested chromatic Nordhaus-Gaddum conjecture

摘录 / EXCERPT: Nested chromatic Nordhaus-Gaddum conjecture. If G is a finite simple graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-047689 | MATHDB-325688

Vertex-contraction ideal conjecture for the Tutte polynomial

摘录 / EXCERPT: Let G be a connected undirected graph, let s be a vertex of G, and let N be the set of neighbors of s, excluding s. For each nonempty subset $W\subseteq N$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-047690 | MATHDB-325702

Xu-Liu conjecture on degree sequences of chromatically equivalent complements

摘录 / EXCERPT: Xu-Liu conjecture. Then G and $\overline{G}$ have the same degree sequence. This conjecture concerns graphs whose chromatic polynomial agrees with that of their complement. The p...

conjecture | open | MathDB 页面 | 原文线索

V2-047691 | MATHDB-325706

Signless Laplacian power-sum conjecture for bipartite graphs

摘录 / EXCERPT: Bipartite signless Laplacian power-sum conjecture. If $\alpha>1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047692 | MATHDB-325707

Signless Laplacian power-sum conjecture for graphs with bounded vertex connectivity

摘录 / EXCERPT: Let $\mathcal{F}_n$ be the family of graphs on n vertices, let $\mathcal{V}_n^k$ denote the graphs in $\mathcal{F}_n$ with vertex connectivity at most k, and let n,k be posit...

conjecture | open | MathDB 页面 | 原文线索

V2-047693 | MATHDB-325721

Galvin's minimum-degree conjecture for independent sets

摘录 / EXCERPT: Let G be a graph on n vertices, let $\delta(G)$ denote its minimum degree, and let $i(G)$ denote its number of independent sets. Assume that $n\geq 2d$ and $\delta(G)\geq d$. G...

conjecture | open | MathDB 页面 | 原文线索

V2-047694 | MATHDB-325733

Patel's covering-fraction conjecture for three perfect matchings

摘录 / EXCERPT: Patel's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047695 | MATHDB-325734

Wilmes' conjecture on Betti numbers of graph parking function ideals

摘录 / EXCERPT: Wilmes' conjecture. For all $i\geq 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-047696 | MATHDB-325740

The snark shortest cycle cover conjecture

摘录 / EXCERPT: Let G be a snark, meaning a cyclically 4-edge-connected cubic graph that is not 3-edge-colourable, and let m be the number of edges of G. Write $scc(G)$ for the length of a s...

conjecture | open | MathDB 页面 | 原文线索

V2-047697 | MATHDB-325802

Erdős-Faudree-Gould-Gyárfás-Rousseau-Schelp 2/3 conjecture

摘录 / EXCERPT: Let G be a complete graph whose edges are coloured with three colours. A set of vertices c-dominates a vertex if the vertex is joined to at least one member of the set by an ed...

conjecture | open | MathDB 页面 | 原文线索

V2-047698 | MATHDB-325809

Laison's eventual even-universality conjecture for checkerboard graphs

摘录 / EXCERPT: Let $R_{m,n}$ denote the checkerboard graph whose vertices are the squares of an $m\times n$ checkerboard, with adjacency determined by the graph construction above. A graph derang...

conjecture | open | MathDB 页面 | 原文线索

V2-047699 | MATHDB-325817

Bohman-Frieze-Mubayi-Pikhurko's codegree density conjecture for $F_{3,2}$

摘录 / EXCERPT: Bohman-Frieze-Mubayi-Pikhurko's conjecture. The balanced construction $T_{A,B,C}$ is asymptotically optimal for the codegree problem for $F_{3,2}$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-047700 | MATHDB-325828

Faben and Jerrum's homomorphism-counting dichotomy conjecture for graphs

摘录 / EXCERPT: Faben and Jerrum's conjecture. The problem $\oplus \operatorname{HomsTo}(H)$ is in FP if the involution-free reduction of H is the empty graph, a singleton vertex with...

conjecture | open | MathDB 页面 | 原文线索

V2-047701 | MATHDB-325832

The alternating grid system periodicity and continued-fraction tiling conjecture

摘录 / EXCERPT: Let R_+ denote the positive real numbers, and let $A_s=(R_{1,0}R_{s,0})^2$ be the alternating grid system. Let Q_s be the square of $G_{s,0}$ whose bottom-left vertex is the or...

conjecture | open | MathDB 页面 | 原文线索

V2-047702 | MATHDB-325858

Chromatic-polynomial coefficient unimodality conjecture

摘录 / EXCERPT: Let P be the chromatic polynomial of a graph. Unimodality conjecture. The sequence of coefficients of P is unimodal. The source reports this as a long-standing conjecture from...

conjecture | open | MathDB 页面 | 原文线索

V2-047703 | MATHDB-325864

Jackson-Sokal maxmaxflow bound conjecture for chromatic roots

摘录 / EXCERPT: For a loopless graph G, let $\lambda_G(x,y)$ be the maximum number of edge-disjoint paths between distinct vertices x and y, and let the maxmaxflow be ... A chromatic root is a...

conjecture | open | MathDB 页面 | 原文线索

V2-047704 | MATHDB-325865

Erdős-Faudree-Gould-Scheinerman conjecture on cycles in sparse graphs

摘录 / EXCERPT: Erdős-Faudree-Gould-Scheinerman conjecture. Every $G\in G(n)$ contains cycles of length i for every integer $3\le i\le k$, where k tends to infinity with n.

conjecture | open | MathDB 页面 | 原文线索

V2-047705 | MATHDB-325866

Erdős-Gould-Yuster-Oz conjecture on dense chordal subgraphs

摘录 / EXCERPT: Erdős-Gould-Yuster-Oz conjecture. Any graph with n vertices and more than $n^2/3$ edges contains a chordal subgraph with at least $8n/3-4$ edges. The complete tripartite graph wi...

conjecture | open | MathDB 页面 | 原文线索

V2-047706 | MATHDB-325867

Erdős-Gould-Yuster-P conjecture on monochromatic cycle partitions

摘录 / EXCERPT: Erdős-Gould-Yuster-P conjecture. The cycle partition number of any r-colored complete graph is at most r.

conjecture | open | MathDB 页面 | 原文线索

V2-047707 | MATHDB-325868

The path-spanning 3-colorability conjecture

摘录 / EXCERPT: Path-spanning 3-colorability conjecture. If each path of a graph spans a subgraph with chromatic number at most 3, then the graph is c-colorable, perhaps with $c=4$.

conjecture | open | MathDB 页面 | 原文线索

V2-047708 | MATHDB-325869

The nearly bipartite deletion conjecture

摘录 / EXCERPT: Nearly bipartite deletion conjecture. A nearly bipartite graph can be made bipartite by deleting at most 5 vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-047709 | MATHDB-325875

The 1-2-3 Conjecture on vertex-coloring edge-weightings

摘录 / EXCERPT: Let G be a finite, undirected, simple connected graph with no isolated edge. A vertex-coloring 3-edge-weighting assigns to each edge of G a weight from $\{1,2,3\}$ so that, f...

conjecture | open | MathDB 页面 | 原文线索

V2-047710 | MATHDB-325877

Uniqueness conjecture for monochromatic 3-connected blinks

摘录 / EXCERPT: Blink uniqueness conjecture. If B and B' are monochromatic 3-connected blinks inducing a 3-manifold, then

conjecture | open | MathDB 页面 | 原文线索

V2-047711 | MATHDB-325878

Sārā's maximal-clique bound for binomial edge ideals

摘录 / EXCERPT: Let G be a graph, let $S=K[x_1,\ldots,x_n,y_1,\ldots,y_n]$, and let J_G be its binomial edge ideal. Denote by r the number of maximal cliques of G. Sārā's conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-047712 | MATHDB-325888

The k(k-3)-certificate conjecture for perfect matchings

摘录 / EXCERPT: Under the paper's setup, H is a k-graph satisfying the stated setup hypotheses, and a C-certificate is the obstruction defined there. The $k(k-3)$-certificate conjecture. H...

conjecture | open | MathDB 页面 | 原文线索

V2-047713 | MATHDB-325889

The polynomial-time threshold conjecture for minimum ℓ-degree perfect matching

摘录 / EXCERPT: Let $b4\in[0,1]$, let $k\geq 2$, and let $\ell\in[k-1]$. Define $\mathbf{PM}(k,\ell,\delta)$ to be the problem of deciding whether a k-graph on n vertices with minimum ℓ-d...

conjecture | open | MathDB 页面 | 原文线索

V2-047714 | MATHDB-325899

Asymptotic convergence to the zero set in nonlinear preferential attachment models

摘录 / EXCERPT: Let N be the number of types, let m be the number of sampled vertices added at each step, and let $p_{\mathbf{u}}^i$ be the probability that a new vertex is assigned type i w...

conjecture | open | MathDB 页面 | 原文线索

V2-047715 | MATHDB-325908

Asymptotic Turán-density upper-bound conjecture for suspensions

摘录 / EXCERPT: For $s\geq2$, let $K_s^{(2)}$ be the complete 2-graph on s vertices, let $S(K_s^{(2)})$ be its suspension, and let $4\pi(S(K_s^{(2)}))4$ denote its Turán density. Asymptotic su...

conjecture | open | MathDB 页面 | 原文线索

V2-047716 | MATHDB-325922

Information retention conjecture for bounded-degree majority dynamics

摘录 / EXCERPT: Information retention conjecture. For every such sequence and every $\frac12<p<1$,

conjecture | open | MathDB 页面 | 原文线索

V2-047717 | MATHDB-325935

Full-clique tiling conjecture for standard multigraphs

摘录 / EXCERPT: Let M be a standard multigraph on sk vertices. The full s-clique associated with an acyclic standard multigraph on s vertices is its complement in the complete standard mul...

conjecture | open | MathDB 页面 | 原文线索

V2-047718 | MATHDB-325937

Universal-clique tiling conjecture for standard multigraphs

摘录 / EXCERPT: Let M be a standard multigraph on sk vertices. An universal s-clique is an s-vertex clique whose every orientation contains every tournament on s vertices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047719 | MATHDB-325953

The asymptotic q-internal-partition conjecture

摘录 / EXCERPT: Let d be a positive integer and let $0<q<1$. A q-internal partition of a d-regular graph is a non-trivial partition in which every vertex has at least a proportion q of its...

conjecture | open | MathDB 页面 | 原文线索

V2-047720 | MATHDB-325976

Asymptotic multipartite Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Fix an integer $k\geq 3$. Let $G=(V_1,\ldots,V_k;E)$ be a balanced k-partite graph on kn vertices, and let ... be its minimum bipartite degree. Asymptotic multipartite Hajnal-Sze...

conjecture | open | MathDB 页面 | 原文线索

V2-047721 | MATHDB-325978

Extremal homomorphism bound for graphs with fixed minimum degree

摘录 / EXCERPT: Fix an integer $\delta\geq 1$ and a graph H. Let $\mathcal{G}(n,\delta)$ be the family of all n-vertex graphs with minimum degree δ, and let $c(\delta,H)$ be a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-047722 | MATHDB-325979

Lin et al.'s distance spectral characterization conjecture for complete multipartite graphs

摘录 / EXCERPT: Let $G=K_{n_1,n_2,...,n_k}$ be a complete k-partite graph. For a graph H, let $Sp(D(H))$ denote the multiset of eigenvalues of its distance matrix, and say that G is determin...

conjecture | open | MathDB 页面 | 原文线索

V2-047723 | MATHDB-325981

The typical-partite structure conjecture for sparse strictly 2-balanced edge-color-critical graphs

摘录 / EXCERPT: Let H be a strictly 2-balanced, edge-color-critical graph. Write $e(H)$ for its number of edges, $\chi(H)$ for its chromatic number, and ... where $v(H)$ is the number of vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-047724 | MATHDB-325997

The almost-commuting Laplacians conjecture

摘录 / EXCERPT: Almost-commuting Laplacians conjecture. There exists a function $\delta(\epsilon)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-047725 | MATHDB-326005

The planar-graph decontamination-time conjecture

摘录 / EXCERPT: Planar-graph decontamination-time conjecture. Any planar graph G on n vertices can be decontaminated with

conjecture | open | MathDB 页面 | 原文线索

V2-047726 | MATHDB-326013

Tree-degree conjecture for S-Roth joins

摘录 / EXCERPT: Tree-degree conjecture. Let $t>s \geq 6$ and let $H=\overline{K}_{s} \vee G$. If G is a tree and if $\Delta(G) \leq s$, then H is S-Roth.

conjecture | open | MathDB 页面 | 原文线索

V2-047727 | MATHDB-326014

Strict maximum-degree conjecture for S-Roth joins

摘录 / EXCERPT: Strict maximum-degree conjecture. Let $t>s \geq 6$ and let $H=\overline{K}_{s} \vee G$. If $\Delta(G)<s$, then H is S-Roth.

conjecture | open | MathDB 页面 | 原文线索

V2-047728 | MATHDB-326021

Generalization of the alternating-word lemmas

摘录 / EXCERPT: Let w_0 be a word in the free group on two generators, and let $\phi_1,\phi_2,\phi_3,\phi_4$ be the automorphisms and $[\,\cdot\,]$ the graph-vertex notation defined in the surro...

conjecture | open | MathDB 页面 | 原文线索

V2-047729 | MATHDB-326023

Weight-four P1 class enumeration conjecture

摘录 / EXCERPT: Let F_2 be the free group on two generators. An automorphic conjugacy class is of type (P1) as defined in the paper; its size is the cardinality of its vertex set, and the weight...

conjecture | open | MathDB 页面 | 原文线索

V2-047730 | MATHDB-326024

Eventual constancy for P3 class counts

摘录 / EXCERPT: Let F_2 be the free group on two generators. For an automorphic conjugacy class, its size is the cardinality of its vertex set, and its minimal words are those of least length in...

conjecture | open | MathDB 页面 | 原文线索

V2-047731 | MATHDB-326032

The Turán and stability conjecture for \(\mathcal{K}_{r,r}^r\)

摘录 / EXCERPT: The Turán and stability conjecture. For $r\ge4$ and sufficiently large $n>n_0(r)$,

conjecture | open | MathDB 页面 | 原文线索

V2-047732 | MATHDB-326037

Exponential localization conjecture for pointwise sampling functions on the Sierpinski gasket

摘录 / EXCERPT: Let β_m be the level-m vertex set of the Sierpinski gasket, let y be a non-boundary vertex in β_m, and let $\phi_y^{(m)}$ be the pointwise sampling function satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-047733 | MATHDB-326042

Conjecture on anti-directed ladders and universal anti-directed 2-factors

摘录 / EXCERPT: Let D be a directed graph on $2n$ vertices. Let L_n be the bipartite ladder on vertex set $\{u_1,\dots,u_n,v_1,\dots,v_n\}$, with $\{u_i,v_j\}$ an edge exactly when...

conjecture | open | MathDB 页面 | 原文线索

V2-047734 | MATHDB-326049

The hypergraph 1-2-3 Conjecture

摘录 / EXCERPT: Hypergraph 1-2-3 Conjecture. There is such a weighting for which the induced vertex weights properly color $V(H)$.

conjecture | open | MathDB 页面 | 原文线索

V2-047735 | MATHDB-326080

Reconstruction conjecture for quasi Cartesian products

摘录 / EXCERPT: A quasi Cartesian product is a graph with the local product-like structure described above. Reconstruction conjecture. Quasi Cartesian products can be reconstructed in essentially...

conjecture | open | MathDB 页面 | 原文线索

V2-047736 | MATHDB-326120

Nullity conjecture for even hyperbanana rigidity matrices

摘录 / EXCERPT: An even hyperbanana is the graph $H^+_{d,b}$ embedded in $\mathbb{R}^d$ with $d=2b$ and $b\geq 2$. Its rigidity matrix records the infinitesimal edge-length constraints of the embe...

conjecture | open | MathDB 页面 | 原文线索

V2-047737 | MATHDB-326129

The bipartite-complement conjecture for extremal edge polytopes

摘录 / EXCERPT: Bipartite-complement conjecture. The complement of G is a bipartite graph.

conjecture | open | MathDB 页面 | 原文线索

V2-047738 | MATHDB-326152

The obstacle-number graph-counting conjecture

摘录 / EXCERPT: Obstacle-number graph-counting conjecture. The number of such graphs satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047739 | MATHDB-326154

Exact-spectrum conjecture for interval colorings of K_{1,m,n}

摘录 / EXCERPT: Let $K_{1,m,n}$ be the complete tripartite graph with part sizes $1,m,n$, and let an interval t-coloring mean an interval edge-coloring using the colors $1,\ldots,t$. Exact-spect...

conjecture | open | MathDB 页面 | 原文线索

V2-047740 | MATHDB-326157

Edge-density criterion for spanning subtree probabilities

摘录 / EXCERPT: Let (G_n) be a graph family with n vertices, and let E denote its edge set. Consider selecting a spanning tree of G_n under either the uniform or weighted probability. Edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047741 | MATHDB-326158

Non-isomorphic graphs with the same subtree polynomial

摘录 / EXCERPT: Let n be a positive integer, and let the subtree polynomial of a graph G be ... where a_k counts the subtrees with k edges. Subtree-polynomial multiplicity conjecture. For an...

open_problem | open | MathDB 页面 | 原文线索

V2-047742 | MATHDB-326159

Density monotonicity conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph, let e join two distinct vertices of G, and let $G+e$ be the graph obtained by adding that edge. Write $\mu(G)$ for the subtree density of G. Den...

conjecture | open | MathDB 页面 | 原文线索

V2-047743 | MATHDB-326177

The random-eigenvector probability conjecture for transitive graphs

摘录 / EXCERPT: Random-eigenvector probability conjecture. For any transitive graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047744 | MATHDB-326212

The one-parameter primitive-solution conjecture for non-perturbative spectral p-q duality

摘录 / EXCERPT: In the (p,q)-system, let $\mathfrak g$ be defined by $2\mathfrak g=(p-1)(q-1)$, and let $\{\theta_a\}_{a=1}^{2\mathfrak g}$ denote the D-instanton and ghost D-instanton fugacitie...

conjecture | open | MathDB 页面 | 原文线索

V2-047745 | MATHDB-326219

Do-Galvin conjecture on asymptotic normality of graph Stirling sequences for acyclic graphs

摘录 / EXCERPT: For each $n\geq 0$, let G_n be an acyclic graph on n vertices, and write ${G_n \brace k}$ for the number of partitions of the vertex set of G_n into k non-empty independent...

conjecture | open | MathDB 页面 | 原文线索

V2-047746 | MATHDB-326232

The bipartisan graph quasi-parity conjecture

摘录 / EXCERPT: Maffray-Thomas conjecture. Every bipartisan graph is a quasi-parity graph.

conjecture | open | MathDB 页面 | 原文线索

V2-047747 | MATHDB-326233

The triangle-probability phase-transition conjecture at b6/b1=3/2

摘录 / EXCERPT: Triangle phase-transition conjecture. The sharp phase transition in the probability of triangles occurs at

conjecture | open | MathDB 页面 | 原文线索

V2-047748 | MATHDB-326249

Generalized entropy-ratio conjecture for the delayed-CSIT X-channel

摘录 / EXCERPT: Consider the 2-transmitter 2-receiver network setting of the paper's linear-subspace-ratio lemma. Let $n\in\mathbb N$, let $\{f_1^{(n)},f_2^{(n)}\}$ be arbitrary encoding functions...

conjecture | open | MathDB 页面 | 原文线索

V2-047749 | MATHDB-326268

The induced-subdivision dichotomy conjecture for digraphs

摘录 / EXCERPT: Induced-subdivision dichotomy conjecture. i_D is NP-complete unless D is the disjoint union of spiders and at most one 2-cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-047750 | MATHDB-326269

Monotonicity conjecture for induced-subdivision detection

摘录 / EXCERPT: Monotonicity conjecture. If i_D (respectively, i'_D) is NP-complete, then for every digraph D' containing D as an induced subdigraph, $i_{D'}$ (respectively, $i'_{D'}$) is...

conjecture | open | MathDB 页面 | 原文线索

V2-047751 | MATHDB-326283

Conforti and Rao's conjecture on balanced bipartite graphs

摘录 / EXCERPT: A $0,1$ matrix is balanced if every square submatrix with two ones per row and column has a number of ones divisible by four. A bipartite graph is balanced when it is the bipartite...

conjecture | open | MathDB 页面 | 原文线索

V2-047752 | MATHDB-326284

Conforti, Cornuéjols and Vušković's conjecture on balanceable bipartite graphs

摘录 / EXCERPT: A bipartite graph is balanceable if its edges can be assigned weights $+1$ or -1 so that the resulting signed bipartite graph has every hole's edge-weight sum divisible by four....

conjecture | open | MathDB 页面 | 原文线索

V2-047753 | MATHDB-326285

Bounded-loss conjecture for single-letter solutions in multi-terminal AWGN networks

摘录 / EXCERPT: Bounded-loss conjecture. For any multi-terminal AWGN information network, an achievable single-letter solution is within a bounded distance from the optimal solution, regardless of...

conjecture | open | MathDB 页面 | 原文线索

V2-047754 | MATHDB-326287

The inverse-dimension conjecture for average resistance of toroidal grids

摘录 / EXCERPT: Let T_{M^d} be the d-dimensional toroidal grid with all side lengths equal to M, and let $R_{\rm ave}(T_{M^d})$ denote its average effective resistance. More generally, let...

conjecture | open | MathDB 页面 | 原文线索

V2-047755 | MATHDB-326292

Manickam-Miklós-Singhi conjecture for sets

摘录 / EXCERPT: Let n real numbers $w_1,\u2026,w_n$ satisfy ... A k-subset A is nonnegative when $sum_{w_i\in A}w_i\geq 0$. Manickam-Miklf3sSinghi conjecture. If $n\geq 4k$, then at least ... of...

conjecture | open | MathDB 页面 | 原文线索

V2-047756 | MATHDB-326295

The lexicographic shellability conjecture for the electrical poset

摘录 / EXCERPT: Let EP_n be the graded poset of electrical networks on n boundary vertices, with rank given by the number of edges in a critical representative. Lexicographic shellability conj...

conjecture | open | MathDB 页面 | 原文线索

V2-047757 | MATHDB-326297

The rank-unimodality conjecture for the electrical poset

摘录 / EXCERPT: Rank-unimodality conjecture. EP_n is rank-unimodal; equivalently, the sequence $(|EP_{n,r}|)_r$ is unimodal. The conjecture concerns the distribution of electrical networks by ra...

conjecture | open | MathDB 页面 | 原文线索

V2-047758 | MATHDB-326304

Erd s-Simonovits even-cycle extremal-number conjecture

摘录 / EXCERPT: Let $\ell\geqslant2$, and write ... for the maximum number of edges in an n-vertex graph containing none of the cycles $C_3,C_4,\ldots,C_{2\ell}$. Erd s-Simonovits conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-047759 | MATHDB-326319

Frankl's bipartite graph conjecture

摘录 / EXCERPT: Frankl's graph conjecture. Each bipartition class of G contains a low-frequency vertex.

conjecture | open | MathDB 页面 | 原文线索

V2-047760 | MATHDB-326347

PageRank singular-gap conjecture

摘录 / EXCERPT: Let G be a directed graph with $d_{min}^{out}>0$. For a parameter α and associated PageRank random walk, let $\mathbf{R}$ be the probability transition matrix, given by .....

conjecture | open | MathDB 页面 | 原文线索

V2-047761 | MATHDB-326363

Conjecture on the asymptotic quotient of hypergraph ordering complexity

摘录 / EXCERPT: Let $f_{3}(n,n)$ be the maximum, over families of n sets of an n-element ground set with each set of size at most 3, of the minimum ordering excess defined by the maximum of...

conjecture | open | MathDB 页面 | 原文线索

V2-047762 | MATHDB-326366

Bauer-Fan-Veldman circumference conjecture for 1-tough graphs

摘录 / EXCERPT: Let G be a graph of order n, let $\tau(G)$ denote its toughness, let $\rho(G)$ denote the independence number, and let $\sigma_3(G)$ be its smallest union degree of order 3:...

conjecture | open | MathDB 页面 | 原文线索

V2-047763 | MATHDB-326386

The Verlinde relations for the Grothendieck ring of the singlet algebra

摘录 / EXCERPT: The singlet vertex algebra $ \mathcal{W}(2,2p-1)$ is considered together with its module category, which is expected to be a braided tensor category, or at least to contain a suita...

conjecture | open | MathDB 页面 | 原文线索

V2-047764 | MATHDB-326436

Ahlswede-Khachatrian conjecture on hypergraphs without perfect fractional matchings

摘录 / EXCERPT: Let $H(n,k)=([n],E)$ be a k-uniform hypergraph, where $[n]=\{1,\ldots,n\}$, and let a perfect fractional matching be a collection of nonnegative real numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-047765 | MATHDB-326437

Ma et al.'s signature bounds conjecture for simple graphs

摘录 / EXCERPT: Let G be a simple graph with adjacency matrix $A(G)$. Its signature is $s(G)=p(G)-n(G)$, where $p(G)$ and $n(G)$ are respectively the numbers of positive and negative eigenvalues...

conjecture | open | MathDB 页面 | 原文线索

V2-047766 | MATHDB-326439

The 7/8 circumference-ratio conjecture for cyclically 4-edge-connected cubic graphs

摘录 / EXCERPT: A cubic graph is a graph in which every vertex has degree 3, and it is cyclically 4-edge-connected if at least four edges must be removed to disconnect it into components conta...

conjecture | open | MathDB 页面 | 原文线索

V2-047767 | MATHDB-326440

The order bound conjecture for critical graphs

摘录 / EXCERPT: The order bound conjecture. Every critical graph with tree-depth k has at most

conjecture | open | MathDB 页面 | 原文线索

V2-047768 | MATHDB-326441

The maximum-degree conjecture for critical graphs

摘录 / EXCERPT: The maximum-degree conjecture. Every critical graph with tree-depth k has maximum degree at most

conjecture | open | MathDB 页面 | 原文线索

V2-047769 | MATHDB-326449

The symmetric doubly stochastic matrix spectral determination conjecture

摘录 / EXCERPT: Spectral determination conjecture. For $0<a\leq n$, the only elements of $\Delta_n^s(a)$ that are DS in Δ_n^s are points on the line segments $[I_n,C_n]$, $[I_n,P]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-047770 | MATHDB-326457

The Q-index extremal conjecture for forbidden odd and even cycles

摘录 / EXCERPT: Let G be a graph of order n. For $k\geq 2$, let $S_{n,k}=K_k\vee\overline{K}_{n-k}$, the graph obtained by joining every vertex of a complete graph of order k to every vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-047771 | MATHDB-326473

Even cyclic permutation conjecture for twisted modules of vertex operator superalgebras

摘录 / EXCERPT: Let V) be a vertex operator superalgebra and let k be an even positive integer. Consider weak, weak admissible, ordinary, parity-stable, and parity-unstable modules for the cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-047772 | MATHDB-326474

Lattice-realization conjecture for even cyclic permutations of free fermions

摘录 / EXCERPT: Let V_{fer} denote the free fermion vertex operator superalgebra, let k be an even positive integer, and let $V_{\mathbb{Z}\alpha}$ be the rank-one lattice vertex operator supe...

conjecture | open | MathDB 页面 | 原文线索

V2-047773 | MATHDB-326524

Six-vertex criterion for infinite-dimensional Fomin-Kirillov algebras

摘录 / EXCERPT: Six-vertex infinite-dimensionality conjecture. The algebra $\mathcal E_G$ is infinite-dimensional if and only if G contains a subgraph isomorphic to one of the graphs shown in Fi...

conjecture | open | MathDB 页面 | 原文线索

V2-047774 | MATHDB-326535

Minimum-energy active-constraint conjecture for EAST+ optimization

摘录 / EXCERPT: Minimum-energy active-constraint conjecture. Under this assumption, the optimal solution requires only the constraint corresponding to the node with minimum energy, E_1, to be ac...

conjecture | open | MathDB 页面 | 原文线索

V2-047775 | MATHDB-326539

Conjectured crossing numbers for $K_{1,1,3,n}$ and $K_{2,4,n}$

摘录 / EXCERPT: Let $K_{a_1,a_2,a_3,n}$ denote the complete multipartite graph whose parts have sizes a_1,a_2,a_3,n, let $K_{2,4,n}$ denote the complete multipartite graph with parts of sizes...

conjecture | open | MathDB 页面 | 原文线索

V2-047776 | MATHDB-326541

Beck's conjecture for annihilating ideal-graphs of commutative rings

摘录 / EXCERPT: Beck's conjecture. For every commutative ring R with unity,

conjecture | open | MathDB 页面 | 原文线索

V2-047777 | MATHDB-326542

Behboodi-Rakeei conjecture on the girth of annihilating ideal-graphs

摘录 / EXCERPT: Behboodi-Rakeei conjecture. Every reduced ring with more than two minimal primes has annihilating ideal-graph of girth 3. The paper states that this conjecture is settled by prov...

conjecture | open | MathDB 页面 | 原文线索

V2-047778 | MATHDB-326550

The bipartite Turán graph conjecture for cycle-maximal triangle-free graphs

摘录 / EXCERPT: Let a cycle-maximal triangle-free graph be a triangle-free graph to which no edge can be added without creating a triangle. For a graph on n vertices, let...

conjecture | open | MathDB 页面 | 原文线索

V2-047779 | MATHDB-326561

The treewidth formula for Kneser graphs

摘录 / EXCERPT: Treewidth conjecture. If $n \geq 3k$ and $k \geq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047780 | MATHDB-326576

Strengthening of the Schrijver graph quadrangulation theorem

摘录 / EXCERPT: Let n and k be integers with $n>2k\geq 2$, and let $SG(n,k)$ denote the Schrijver graph. A quadrangulation of P^{n-2k} is a graph embedded in the projective space P^{n-2k}...

conjecture | open | MathDB 页面 | 原文线索

V2-047781 | MATHDB-326599

Conjecture on decomposing even-dimensional hypercubes into paths

摘录 / EXCERPT: Let n be an even positive integer, and let k be a positive integer. The n-dimensional hypercube $\mathcal{Q}_n$ has vertex set $\{0,1\}^n$ and edges joining vertices at...

conjecture | open | MathDB 页面 | 原文线索

V2-047782 | MATHDB-326625

Conjectured circular-flow bound for odd-regular graphs

摘录 / EXCERPT: Let $t>1$ be an integer and let G be a $(2t+1)$-graph, meaning a graph in which every vertex has degree $2t+1$. The odd-regular flow conjecture. Every such graph satisfies ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-047783 | MATHDB-326639

The complete-bipartite majority out-domination conjecture

摘录 / EXCERPT: Complete-bipartite majority out-domination conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047784 | MATHDB-326640

The majority out-dominating number conjecture for complete bipartite graphs

摘录 / EXCERPT: Majority out-dominating number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047785 | MATHDB-326670

Ilić and Ilić's minimum Laplacian-coefficient conjecture for unicyclic graphs

摘录 / EXCERPT: Ilić and Ilić's conjecture. Among all n-vertex unicyclic graphs, the graph $U_{n,l}^{3,0}$ has the minimum Laplacian coefficients c_k, $k=0,\dots,n$; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-047786 | MATHDB-326679

Cyclic Hamiltonian-path decomposition conjecture for Cayley multigraphs

摘录 / EXCERPT: Let $\mathbb{Z}_v$ be the cyclic group of order v, let L be a list of edge-lengths, and let $Cay[\mathbb{Z}_v:\Lambda]$ denote the corresponding Cayley multigraph. Cyclic decom...

conjecture | open | MathDB 页面 | 原文线索

V2-047787 | MATHDB-326680

Kierstead-Szemerédi-Trotter sharpness question for locally colorable graphs

摘录 / EXCERPT: Kierstead-Szemerédi-Trotter sharpness conjecture. There exists a graph G on $O(r^{n-1})$ vertices such that

open_problem | solved | MathDB 页面 | 原文线索

V2-047788 | MATHDB-326684

Lima et al.'s lower-bound conjecture for the least signless Laplacian eigenvalue

摘录 / EXCERPT: Lima et al.'s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047789 | MATHDB-326698

Strong-geometry conjecture for geometric preferential attachment

摘录 / EXCERPT: For $\gamma>1$, define ... where $\log x=\max\{0,\log x\}$. Let G_n be the geometric preferential attachment graph under the paper's standing assumptions, let v_n be the vertex s...

conjecture | open | MathDB 页面 | 原文线索

V2-047790 | MATHDB-326701

Erde, Golé nia and Golé nia's generalized knight's tour conjecture

摘录 / EXCERPT: Let a and b be positive integers satisfying ... with a and b not both odd. An (a,b) knight's tour is a cycle visiting every vertex of the corresponding knight's graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-047791 | MATHDB-326702

Engbers-Galvin conjecture on independent sets of fixed size

摘录 / EXCERPT: Engbers-Galvin conjecture. The complete bipartite graph $K_{\delta,n-\delta}$ maximizes $i_t(G)$ over all n-vertex graphs with minimum degree at least δ.

conjecture | open | MathDB 页面 | 原文线索

V2-047792 | MATHDB-326720

Strengthened Erdős-Gallai-Tuza conjecture for bipartite edge covers

摘录 / EXCERPT: Strengthened Erdős-Gallai-Tuza conjecture. For every n-vertex graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047793 | MATHDB-326738

Brualdi-Quinn Massey's $\Delta+2$ conjecture for 4-cycle-free bipartite graphs

摘录 / EXCERPT: Brualdi-Quinn Massey's conjecture. The strong chromatic index satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047794 | MATHDB-326751

Spectral-network singular-locus conjecture for universal buildings

摘录 / EXCERPT: Let X be a Riemann surface and let ϕ be a spectral cover. Let $\mathcal W$ be the spectral network associated to ϕ, and let $\widetilde{\mathcal W}$ be its lift to...

conjecture | open | MathDB 页面 | 原文线索

V2-047795 | MATHDB-326767

The USCS edge-removal conjecture for asymptotic non-linear index coding

摘录 / EXCERPT: Let a USCS graph be a graph whose edges all lie on directed cycles. Suppose that the graph supports a given set of rates for asymptotic non-linear index coding, and let e be an e...

conjecture | open | MathDB 页面 | 原文线索

V2-047796 | MATHDB-326769

Caporossi's maximal-energy conjecture for unicyclic graphs

摘录 / EXCERPT: Let P_n be a path, C_n a cycle, and let P_n^6 be the graph obtained by connecting a vertex of C_6 to a terminal vertex of P_{n-6}. Among all unicyclic graphs on n verti...

conjecture | open | MathDB 页面 | 原文线索

V2-047797 | MATHDB-326770

Gutman-Vidović's maximal-energy conjecture for bicyclic molecular graphs

摘录 / EXCERPT: Let $P_n^{6,6}$ denote the bicyclic molecular graph of order n corresponding to the alpha,beta diphenyl-polyene $C_6H_5(CH)_{n-12}C_6H_5$. Gutman-Vidović's conjecture. For $n=14$...

conjecture | open | MathDB 页面 | 原文线索

V2-047798 | MATHDB-326771

Aouchiche-Hansen-Cioabă's maximal-energy conjecture for tricyclic graphs

摘录 / EXCERPT: Let $P_n^{6,6,6}$, for $n\geq 20$, be the graph obtained from three copies of C_6 and a path P_{n-18} by adding a single edge between each of two copies of C_6 and one endpoi...

conjecture | open | MathDB 页面 | 原文线索

V2-047799 | MATHDB-326775

The bipartite linkless-embeddability conjecture

摘录 / EXCERPT: Let $G=(A\uplus B,E)$ be a bipartite linklessly embeddable graph, and let $<$ be a $(3,3)$-admissible order. Write G^b for the balanced shift of G. Bipartite linkless-embeddabi...

conjecture | open | MathDB 页面 | 原文线索

V2-047800 | MATHDB-326780

Strict monotonicity conjecture for the minimum size of pancyclic graphs

摘录 / EXCERPT: Let $m(n)$ be the minimum number of edges in a pancyclic graph on n vertices, where a pancyclic graph contains a cycle of every length k with $3\leq k\leq n$. Strict monotonici...

conjecture | open | MathDB 页面 | 原文线索

V2-047801 | MATHDB-326811

Asymptotic upper-bound conjecture for the maximum order of graphs embedded in surfaces

摘录 / EXCERPT: Let Σ be a surface of genus g, let $\mathcal{G}_\Sigma$ denote the class of graphs embeddable in Σ, and let $N(\Delta,k,\mathcal{G}_\Sigma)$ be the maximum order of...

conjecture | open | MathDB 页面 | 原文线索

V2-047802 | MATHDB-326815

Eigenvalue criterion for many edge-disjoint spanning trees

摘录 / EXCERPT: Let $d\geq 8$ and $4\leq k\leq \left\lfloor\frac{d}{2}\right\rfloor$ be two integers. Let G be a d-regular graph, and write $\lambda_2(G)$ for its second largest eigenvalue. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-047803 | MATHDB-326841

Chvátal's line-count conjecture for graph metrics

摘录 / EXCERPT: Let G be a connected graph on n vertices, with $n\ge 2$, and consider its induced metric space using graph distance. A line is defined from metric betweenness, and a line is un...

conjecture | open | MathDB 页面 | 原文线索

V2-047804 | MATHDB-326857

The generalized Nase conjecture on triple cumulative edge bounds

摘录 / EXCERPT: Let D be a simple drawing of a graph, and let $E_{\le}{\le}{\le}k}(D)$ denote the number of its $\le!\le!\le k$-edges, with ... Here $E_i(D)$ denotes the number of i-edges in...

conjecture | open | MathDB 页面 | 原文线索

V2-047805 | MATHDB-326882

Monotonicity conjecture for simple random walk on growing graphs

摘录 / EXCERPT: Let $\mathbb{G}_t$ and $\mathbb{G}'_t$ be graphs non-decreasing in t, of uniformly bounded degrees, with $\mathbb{G}_t\subseteq\mathbb{G}'_t$ for every t. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047806 | MATHDB-326883

Fixed-graph recurrence conjecture for simple random walk on growing subgraphs

摘录 / EXCERPT: Let $\mathbb{G}_\infty$ be a fixed graph of uniformly bounded degrees on which simple random walk is recurrent, and let $\mathbb{G}_t$ be non-decreasing subgraphs with...

conjecture | open | MathDB 页面 | 原文线索

V2-047807 | MATHDB-326893

Erdős-Tuza conjecture on K_4-saturating edges

摘录 / EXCERPT: Erdős-Tuza conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047808 | MATHDB-326915

The 1-2-3-Conjecture for graphs

摘录 / EXCERPT: The 1-2-3-Conjecture. For every graph G without isolated edges, there is a weighting $\omega:E(G)\to\{1,2,3\}$ such that the induced vertex weights properly color $V(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-047809 | MATHDB-326916

The 1-2-Conjecture for total graph weightings

摘录 / EXCERPT: The 1-2-Conjecture. For every graph G, there is a weighting $\omega:E(G)\cup V(G)\to\{1,2\}$ such that the induced total vertex weights properly color $V(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-047810 | MATHDB-326917

The list 1-2-3 conjecture for graphs

摘录 / EXCERPT: The list 1-2-3 conjecture. For every graph G without isolated edges and every assignment of lists of size 3 to the edges of G, there exists a weighting...

conjecture | open | MathDB 页面 | 原文线索

V2-047811 | MATHDB-326956

Peng-Zhao Motzkin-Straus-type conjecture for hypergraphs with a large clique

摘录 / EXCERPT: Let t, m, and $r\ge 3$ be positive integers satisfying ... Let G be an r-graph with m edges that contains a clique of order $t-1$. Peng-Zhao's Motzkin-Straus-type conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-047812 | MATHDB-326959

The square-root bound for the cop number of diameter-two graphs

摘录 / EXCERPT: Square-root bound conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-047813 | MATHDB-326960

The trap bound for restricted cop number in diameter-two graphs

摘录 / EXCERPT: Restricted cop-number trap conjecture. If H is an s-trap in G, then

conjecture | open | MathDB 页面 | 原文线索

V2-047814 | MATHDB-326985

Jackson's conjecture on Hamilton cycles in 3-connected regular graphs

摘录 / EXCERPT: Jackson's conjecture. The graph G contains a Hamilton cycle. This is the $t=3$ case of the Bollobás-Häggkvist conjecture. The source reports that the conjecture was believed to r...

conjecture | open | MathDB 页面 | 原文线索

V2-047815 | MATHDB-327022

The planar subcubic spectral-radius ratio conjecture

摘录 / EXCERPT: Let G be a planar subcubic graph, and let $R(G)$ denote the ratio defined earlier in the paper. Planar subcubic conjecture. If G is a planar subcubic graph, then $R(G)\le 1$. N...

conjecture | open | MathDB 页面 | 原文线索

V2-047816 | MATHDB-327031

The domination-number conjecture for total graphs of finite rings

摘录 / EXCERPT: Let R be an arbitrary ring with Jacobson radical J, and suppose that ... where each F_i is a finite field and $n_i\geq 2$ for all i. For a finite ring A, let $\tau(A)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-047817 | MATHDB-327033

The gap-cycle conjecture for 2-coloring simple planar digraphs

摘录 / EXCERPT: Let D be a simple planar digraph. Say that D has no cycles of lengths $4,\dots,k$ when it contains no directed cycles whose lengths are among those integers. Gap-cycle conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-047818 | MATHDB-327034

Hefetz's acyclic-set conjecture for planar digraphs

摘录 / EXCERPT: Let D be a simple planar digraph on n vertices. An acyclic set is a vertex set inducing no directed cycle in D. Hefetz's conjecture. Every simple n-vertex planar digraph ha...

conjecture | open | MathDB 页面 | 原文线索

V2-047819 | MATHDB-327035

The framed vertex correspondence for transverse A_n singularities

摘录 / EXCERPT: Let q, q_k, ξ_n, ξ_{2n}, u, and x_i be the variables and roots of unity appearing in the framed vertex theories, and let τ^+, τ^-, and μ be partitions...

conjecture | open | MathDB 页面 | 原文线索

V2-047820 | MATHDB-327039

Sublinear polynomial upper bound for the on-line ranking number of trees

摘录 / EXCERPT: On-line ranking conjecture. There exist universal constants a and b satisfying $0<a<1<b$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-047821 | MATHDB-327063

List rankability of trees with more leaves than internal vertices

摘录 / EXCERPT: Let T be a tree with p internal vertices and q leaves, and let $\rho _{\ell}(T)$ denote its list-ranking number. Tree list-ranking conjecture. If $p<q$, then ... This would giv...

conjecture | open | MathDB 页面 | 原文线索

V2-047822 | MATHDB-327064

Signed list-ranking of trees with many leaves

摘录 / EXCERPT: Let T be a tree, let p be its number of internal vertices, and let $\rho^{\pm}_{\ell}(T)$ denote its signed list-ranking number. Signed tree list-ranking conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-047823 | MATHDB-327065

A threshold for Taxer's win on paths in the signed list-ranking game

摘录 / EXCERPT: For a positive integer n, let P_n be the path on n vertices, let f be a nonnegative integer-valued function on $V(P_n)$, and define ... Let $R^{\pm}(P_n,f)$ be the signed on-...

conjecture | open | MathDB 页面 | 原文线索

V2-047824 | MATHDB-327066

The path-linear-forest quasar Ramsey number conjecture

摘录 / EXCERPT: Let P_n be the path on n vertices, let F be a non-empty linear forest on m vertices, let $K_1\vee F$ denote the join of F with a single vertex, and let $o(F)$ be the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-047825 | MATHDB-327083

Cilleruelo's minimum-degree conjecture for infinite C_4-free graphs

摘录 / EXCERPT: Let $\mathcal{G}$ be a countable infinite graph with no cycle of length four, and let $\mathcal{G}_n$ be the subgraph induced by its first n vertices. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-047826 | MATHDB-327084

Cilleruelo's lower-bound conjecture for infinite C_4-free graphs

摘录 / EXCERPT: Let $\epsilon>0$. A countable infinite graph with no cycle of length four is called C_4-free, and let $\mathcal{G}_n$ denote the subgraph induced by the first n vertices. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-047827 | MATHDB-327095

The minimum-degree lower-bound conjecture for H-free subgraphs

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of graphs, and let $h(\delta,\Delta,\mathcal{F})$ be the minimum, over graphs G with minimum degree δ and maximum degree Δ, of the lar...

conjecture | open | MathDB 页面 | 原文线索

V2-047828 | MATHDB-327120

Existence of non-isomorphic graphs with equal homomorphism polynomial

摘录 / EXCERPT: Let q be fixed, and let $\mathcal{P}_q(G)$ denote the homomorphism polynomial of a graph G. Two graphs are non-isomorphic if there is no graph isomorphism between them. Homomor...

conjecture | open | MathDB 页面 | 原文线索

V2-047829 | MATHDB-327121

Liu-Sousa conjecture on monochromatic triangle decompositions

摘录 / EXCERPT: Let $k\geq 2$, and let $\phi_k(n,K_3)$ denote the maximum, over k-edge-coloured graphs on n vertices, of the minimum number of parts in a monochromatic K_3-decomposition. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047830 | MATHDB-327142

Bounded diameter conjecture for connected proper power graphs

摘录 / EXCERPT: For a finite group G, let $bdom?\mathcal{P}^*(G)$ denote its proper power graph, and suppose that this graph is connected. Bounded diameter conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-047831 | MATHDB-327159

Independence ratio conjecture for the distance graphs $\{1,6,k\}$

摘录 / EXCERPT: Let $G(S)$ be the distance graph on the integers with generating set S, and let $\overline{\alpha}(S)$ denote its maximum density of an independent set. Let $k>6$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-047832 | MATHDB-327160

Independence ratio conjecture for the distance graphs $\{1,k,k+7\}$

摘录 / EXCERPT: Let $G(S)$ be the distance graph on the integers with generating set S, and let $\overline{\alpha}(S)$ denote its maximum density of an independent set. Let $k\geq8$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-047833 | MATHDB-327172

Non-equality conjecture for the geometric-arithmetic and atom-bond connectivity indices

摘录 / EXCERPT: Non-equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047834 | MATHDB-327184

Path-pairability conjecture for odd-dimensional hypercubes

摘录 / EXCERPT: Let Q_n denote the n-dimensional hypercube. A graph is path-pairable if every pairing of its vertices can be linked by pairwise edge-disjoint paths. Odd-dimensional hypercube c...

conjecture | open | MathDB 页面 | 原文线索

V2-047835 | MATHDB-327189

The harmonic-measure support conjecture for graphs with the doubling property

摘录 / EXCERPT: Let G be an infinite graph with the doubling property: there exists a universal constant $C>0$ such that for every $r>0$ and every vertex x, ... where $B(x,r)$ is the ball of rad...

conjecture | open | MathDB 页面 | 原文线索

V2-047836 | MATHDB-327190

Gracefulness of lobsters with almost perfect matchings

摘录 / EXCERPT: Let T be a lobster, and let an almost perfect matching mean a matching that leaves at most one vertex unmatched. Gracefulness conjecture for lobsters with almost perfect matching...

conjecture | open | MathDB 页面 | 原文线索

V2-047837 | MATHDB-327206

The p-spectral Turán conjecture for k-chromatic uniform hypergraphs

摘录 / EXCERPT: Let G be a k-chromatic r-graph of order n, where $k\geq 2$ and $n>(r-1)k$. For $p\geq 1$, write $\lambda^{(p)}(G)$ for the p-spectral radius of G, and let $Q_k^r(n)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-047838 | MATHDB-327207

The explicit p-spectral Turán conjecture for k-chromatic uniform hypergraphs

摘录 / EXCERPT: Let G be a k-chromatic r-graph of order n, where $k\geq 2$ and $n>(r-1)k$. For $p\geq 1$, let $\lambda^{(p)}(G)$ be the p-spectral radius of G, and let $Q_k^r(n)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-047839 | MATHDB-327211

Erdős-Sós and Kalai's conjecture for Turán numbers of r-trees

摘录 / EXCERPT: Let $r\geq 2$, and let T be an r-tree on v vertices. An r-tree is an r-uniform hypergraph with edges $e_1,\ldots,e_q$ such that, for each i, there is some $k<i$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-047840 | MATHDB-327212

Characterization of class III graphs for the adjacent vertex distinguishing index by sum

摘录 / EXCERPT: For a graph Γ, let $tndi_{\sum}(\Gamma)$ be the minimum number of colors in a proper total coloring distinguishing adjacent vertices by sums. The graph is called...

conjecture | open | MathDB 页面 | 原文线索

V2-047841 | MATHDB-327213

Class II characterization for regular graphs under adjacent vertex distinguishing total coloring by sum

摘录 / EXCERPT: Let Γ be a k-regular graph. The graph is called $tndi_{\sum}$ class II when its adjacent vertex distinguishing index by sum attains the class-II value. Class II characteri...

conjecture | open | MathDB 页面 | 原文线索

V2-047842 | MATHDB-327217

NP-completeness of PL realizability for 2-hypergraphs in three dimensions

摘录 / EXCERPT: A 2-hypergraph is a hypergraph whose hyperedges have size two, and PL realizability in $\mathbb R^d$ means realizability by a piecewise-linear representation in $\mathbb R^d$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-047843 | MATHDB-327220

Conjecture on powers of Hamilton cycles in pseudorandom graphs

摘录 / EXCERPT: For $k\geq 2$, let an $(\varepsilon,p,k-1,k)$-pseudorandom graph mean a graph satisfying the stated pseudorandomness condition with parameters $(\varepsilon,p,k-1,k)$. The kth po...

conjecture | open | MathDB 页面 | 原文线索

V2-047844 | MATHDB-327261

Regularity conjecture for weakly closed graphs

摘录 / EXCERPT: Let G be a connected weakly closed graph, let $S=K[x_1,\ldots,x_n,y_1,\ldots,y_n]$, and let J_G be its binomial edge ideal. Let ℓ denote the length of the longest induced...

conjecture | open | MathDB 页面 | 原文线索

V2-047845 | MATHDB-327263

Fractional covering conjecture for separated d-intervals

摘录 / EXCERPT: Let H be a hypergraph of separated d-intervals. Write $ u(H)$ for its matching number and $\tau^*(H)$ for its fractional covering number. Fractional covering conjecture. One sh...

conjecture | open | MathDB 页面 | 原文线索

V2-047846 | MATHDB-327264

Weighted covering conjecture for d-interval hypergraphs

摘录 / EXCERPT: Let H be a hypergraph of d-intervals, let $\tau_w(H)$ be its weighted covering number, and let $\nu_w(H)$ be its weighted matching number. Weighted covering conjecture. One sho...

conjecture | open | MathDB 页面 | 原文线索

V2-047847 | MATHDB-327265

Edge-colouring conjecture for d-interval hypergraphs

摘录 / EXCERPT: Let H be a hypergraph of d-intervals. Let $\chi_e(H)$ denote its edge chromatic number, and let Δ be the maximum degree of a point on any line. Edge-colouring conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-047848 | MATHDB-327267

The connected 3-colouring conjecture for tricoloured 4-sets

摘录 / EXCERPT: Let $K_n^{(3)}$ be the complete 3-uniform hypergraph on n vertices, and call a colouring connected when each colour class spans a connected subhypergraph. A 4-set is tricoloured...

conjecture | open | MathDB 页面 | 原文线索

V2-047849 | MATHDB-327268

The connected 3-colouring conjecture for general uniform hypergraphs

摘录 / EXCERPT: Let $K_n^{(r)}$ be the complete r-uniform hypergraph on n vertices. Call an r-uniform hypergraph connected when there is a strong path between every pair of $(r-1)$-sets, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-047850 | MATHDB-327285

Laplacian spread equality characterization conjecture

摘录 / EXCERPT: Let G be a graph with n vertices, let $\overline G$ be its complement, and let K_1 denote the one-vertex complete graph. The join of graphs is denoted by $\vee$, and a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047851 | MATHDB-327286

Laplacian Nordhaus-Gaddum product equality characterization conjecture

摘录 / EXCERPT: Let G be a graph with n vertices, let $\overline G$ be its complement, and let K_1 denote the one-vertex complete graph. The join of graphs is denoted by $\vee$. Laplacian pr...

conjecture | open | MathDB 页面 | 原文线索

V2-047852 | MATHDB-327287

Aouchiche-Hansen signless Laplacian Nordhaus-Gaddum product conjecture

摘录 / EXCERPT: Let G be a simple graph on $n\geq 2$ vertices, let $\overline G$ be its complement, and let $q_1(H)$ denote the largest signless Laplacian eigenvalue of a graph H. Aouchiche-Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-047853 | MATHDB-327304

Bermond's lobster conjecture

摘录 / EXCERPT: Let T be a tree, and let P be a longest path in T. A tree is 2-distant, or a lobster, if every vertex of T has distance at most 2 from P. A tree is graceful if its vert...

conjecture | open | MathDB 页面 | 原文线索

V2-047854 | MATHDB-327306

Conjectured fixation-time bound for the iPod model

摘录 / EXCERPT: Let $\mathfrak{G}$ be a graph on N vertices with spectral gap λ, let η be the interaction parameter, let σ be the number of songs, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-047855 | MATHDB-327307

Cayley graph Hamilton-cycle conjecture

摘录 / EXCERPT: Let H be a finite group and let $S\subseteq H$ satisfy $S=S^{-1}$ and exclude the identity. The Cayley graph $G(H;S)$ has vertex set H, with $g,h\in H$ adjacent when $g=sh$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-047856 | MATHDB-327313

Rödl-Ruciński tight Hamilton-cycle conjecture for hypergraphs

摘录 / EXCERPT: Let G be a k-uniform hypergraph on n vertices, and let $\delta_1(G)$ denote its minimum vertex degree. Rödl-Ruciński conjecture. For every integer $k\ge3$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-047857 | MATHDB-327344

The Chen-Fu-Kim-Stehr-Watts extremal spectral-radius conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a simple bipartite graph with bipartition orders p and q, and let $\mathcal{K}(p,q,e)$ denote the family of e-edge subgraphs of the complete bipartite graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047858 | MATHDB-327345

The Chen-Fu-Kim-Stehr-Watts refined conjecture for dense bipartite graphs

摘录 / EXCERPT: Let G be a simple bipartite graph in $\mathcal{K}(p,q,e)$, the family of e-edge subgraphs of $K_{p,q}$. Let $K_{p,q}^{\{e\}}$ be obtained from $K_{p,q}$ by deleting $pq-e$ edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047859 | MATHDB-327346

The general refined extremal spectral-radius conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a simple bipartite graph in $\mathcal{K}(p,q,e)$, the family of e-edge subgraphs of $K_{p,q}$. For positive integers $s\leq p$ and $t\leq q$, let $K_{s,t}^{\{e\}}$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-047860 | MATHDB-327349

Faudree et al.'s subcubic strong edge coloring conjectures

摘录 / EXCERPT: Faudree et al.'s subcubic conjectures. begin{enumerate} item The strong chromatic index is at most 10. item If G is bipartite, then $\chiup_{s}'(G) \leq 9$. item If G is plan...

conjecture | open | MathDB 页面 | 原文线索

V2-047861 | MATHDB-327362

Level-matching conjecture for similarities between graph-directed IFS attractors

摘录 / EXCERPT: Let $\bigl(V, E^*, i, t, r, ((\mathbb{R},\left| \ \ \right|))_{v \in V}, (S_e)_{e \in E^1} \bigr)$ be an n-vertex (CSSC) IFS defined on the unit interval, with attractor...

conjecture | open | MathDB 页面 | 原文线索

V2-047862 | MATHDB-327370

The corrected bipartite decomposition conjecture for random graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, and let $\tau(G)$ denote the minimum number of pairwise edge-disjoint complete bipartite subgraphs of G whose edges partition E. Let $\beta(G)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-047863 | MATHDB-327433

The structural conjecture for nowhere dense classes without bounded expansion

摘录 / EXCERPT: Structural conjecture for nowhere dense classes. If $\mathcal C$ does not have bounded expansion, then there exists an integer p such that $\mathcal C$ includes p-subdivisions...

conjecture | open | MathDB 页面 | 原文线索

V2-047864 | MATHDB-327440

Variance bound for the degree deviation of connected graphs

摘录 / EXCERPT: Let G be a connected graph, and let $\rho(G)$ denote its spectral radius, d its degree vector, and $\operatorname{var}(G)$ its degree variance. Define the degree deviation by...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-047865 | MATHDB-327441

Extremal pineapple graph conjecture for degree deviation

摘录 / EXCERPT: Let $PA(n,q)$ be the pineapple graph consisting of a clique on q vertices together with a stable set on $n-q$ vertices, with one vertex of the clique adjacent to every vertex of...

conjecture | open | MathDB 页面 | 原文线索

V2-047866 | MATHDB-327445

The diffeomorphism conjecture for the 16- and 17-vertex triangulations of the K3 surface

摘录 / EXCERPT: Let $(K3)_{16}$ and $(K3)_{17}$ denote the 16-vertex and 17-vertex triangulations, respectively, of the K3 surface. K3 triangulation diffeomorphism conjecture. ... The smooth type of...

conjecture | open | MathDB 页面 | 原文线索

V2-047867 | MATHDB-327446

Strong equitable vertex arboricity bounded by maximum degree

摘录 / EXCERPT: Let G be a graph, let $Delta(G)$ denote its maximum degree, and let $a^*_{eq}(G)$ be the smallest integer t such that G has an equitable t'-tree-coloring for every...

conjecture | open | MathDB 页面 | 原文线索

V2-047868 | MATHDB-327447

Gyárfás-Lehel-Sárközy-Szemerédi conjecture on monochromatic Hamiltonian Berge-cycles

摘录 / EXCERPT: Gyárfás-Lehel-Sárközy-Szemerédi conjecture. For sufficiently large n, every $(r-1)$-edge coloring of K_n^r contains a monochromatic Hamiltonian Berge-cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-047869 | MATHDB-327448

Dorbec-Gyárfás-Sárközy conjecture on monochromatic Hamiltonian tight Berge-cycles

摘录 / EXCERPT: Dorbec-Gyárfás-Sárközy conjecture. Assume that $c\geq 2$, $2\leq t\leq r$, $c+t\leq r+1$, and n is sufficiently large. Then every c-edge coloring of K_n^r contains a monochro...

conjecture | open | MathDB 页面 | 原文线索

V2-047870 | MATHDB-327450

Typical outer-string graph edge-density and degree-distribution conjecture

摘录 / EXCERPT: For a graph G on n vertices, let X_n be the degree of a uniformly random vertex of G. Let $\mathcal{OSG}_n$ and $\mathcal{OSG}^L_n$ denote the classes of unlabeled and labe...

conjecture | open | MathDB 页面 | 原文线索

V2-047871 | MATHDB-327458

The relativized Alon conjecture for random graph covers

摘录 / EXCERPT: Let B be a fixed graph, let $\widehat B$ be its universal cover, and let $A_{\widehat B}$ be the adjacency operator on $\widehat B$. Let $\mathcal{C}_n(B)$ denote the model of de...

conjecture | open | MathDB 页面 | 原文线索

V2-047872 | MATHDB-327550

The shrub-depth characterization by vertex-minors

摘录 / EXCERPT: Shrub-depth characterization conjecture. The class $\mathcal{C}$ is of bounded shrub-depth if, and only if, there exists an integer t such that no graph $G\in\mathcal{C}$ contain...

conjecture | open | MathDB 页面 | 原文线索

V2-047873 | MATHDB-327553

Athanasia's local gamma-vector nonnegativity conjecture for flag vertex-induced subdivisions

摘录 / EXCERPT: Let V be a d-element set, and let Γ be a flag vertex-induced homology subdivision of the simplex 2^V. The local h-polynomial has a unique expansion ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-047874 | MATHDB-327560

Characterization of Kyle graphs among symmetric directed graphs

摘录 / EXCERPT: Kyle graph characterization conjecture. The graph D is a Kyle graph if and only if all of the following conditions hold: (i) m is odd and $m=2a_l+1$, where a_l is determined...

conjecture | open | MathDB 页面 | 原文线索

V2-047875 | MATHDB-327602

The second-minimal total irregularity conjecture for connected graphs

摘录 / EXCERPT: Let G be a simple connected graph with n vertices, and let $\operatorname{irr}_{t}(G)$ denote its total irregularity. A graph is regular if all its vertices have the same degre...

conjecture | open | MathDB 页面 | 原文线索

V2-047876 | MATHDB-327612

Catalan-path expansion conjecture for standard Hermitian networks

摘录 / EXCERPT: A standard Hermitian network has face variables and interior vertex variables, and its matrix entries are Laurent polynomials in these variables. A Catalan-path expansion conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-047877 | MATHDB-327613

Kinnersley-West-Zamani 3/5-conjecture for isolate-free forests

摘录 / EXCERPT: Kinnersley-West-Zamani 3/5-conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047878 | MATHDB-327622

The Clifford-Heisenberg structure conjecture for full COHA of one-vertex quivers

摘录 / EXCERPT: Let Q be a quiver with one vertex and m arrows. For even m, let Cl_c denote the infinite Clifford algebra with generators ξ_n^{+} and ξ_n^{-} for...

conjecture | open | MathDB 页面 | 原文线索

V2-047879 | MATHDB-327634

Optimality of the square lattice among Steiner networks

摘录 / EXCERPT: Let the cities be the points $\{(i,j): i,j\in\mathbb{Z}\}$ in the plane, and consider a Steiner network on this square-grid configuration. Its normalized length is its total networ...

conjecture | open | MathDB 页面 | 原文线索

V2-047880 | MATHDB-327667

Gyárfás-Lehel-Sárközy-Schelp conjecture on monochromatic Hamiltonian Berge-cycles

摘录 / EXCERPT: For fixed $r2$, let K_n^r be the complete r-uniform hypergraph on n vertices, and let a Berge-cycle of length n be an r-uniform hypergraph with core vertices $v_1,,v_n$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-047881 | MATHDB-327678

Polyhedral eigenvalue conjecture for regular matroid representations

摘录 / EXCERPT: Let $\mathcal{M}$ be a regular matroid and let M be a unimodular $m \times n$ representation of $\mathcal{M}$ with corank greater than 1. Let L denote the associated matrix w...

conjecture | open | MathDB 页面 | 原文线索

V2-047882 | MATHDB-327679

Horak-Rosa Hamiltonian path conjecture

摘录 / EXCERPT: Let K_v denote the complete graph on $\{0,1,\dots,v-1\}$, and let $\ell(H)$ be the multiset of edge-lengths of a graph H, with edge lengths in K_v defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-047883 | MATHDB-327680

The exact online Ramsey number conjecture for paths

摘录 / EXCERPT: Exact path Ramsey conjecture. For all integers $\ell\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-047884 | MATHDB-327681

The asymptotic and exact online Ramsey conjecture for paths

摘录 / EXCERPT: Path Ramsey conjecture. For $k\ge 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-047885 | MATHDB-327694

Erdős's conjecture on odd cycle lengths in triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph with chromatic number k, and let $\frac{}{ }$ denote a positive real parameter. Erdős's conjecture. For every $\frac{}{ } > 0$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-047886 | MATHDB-327695

Near-Hamiltonian cycle conjecture for minimal triangle-free k-chromatic graphs

摘录 / EXCERPT: Let G be a k-chromatic triangle-free graph, and let n_k be the minimum number of vertices in a k-chromatic triangle-free graph. Near-Hamiltonian cycle conjecture. The graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047887 | MATHDB-327712

Manoussakis's degree-sum conjecture for Hamilton cycles in digraphs

摘录 / EXCERPT: Manoussakis's conjecture. If x,y and w,z are distinct pairs of nonadjacent vertices and

conjecture | open | MathDB 页面 | 原文线索

V2-047888 | MATHDB-327713

Manoussakis's pancyclicity conjecture for digraphs

摘录 / EXCERPT: Manoussakis's conjecture. If, for every triple $x,y,z\in V(D)$ with x nonadjacent to y, the inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-047889 | MATHDB-327727

The linear-backwards-edge conjecture for forbidden-cycle oriented graphs and digraphs

摘录 / EXCERPT: Linear-backwards-edge conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-047890 | MATHDB-327733

Tree reconstruction conjecture for the two-state broadcast process with revealed nodes

摘录 / EXCERPT: Let T be an infinite tree with root ρ. In the two-state broadcast process, each vertex label $\tau_v\in\{\pm 1\}$ is generated by assigning the root a uniformly random label...

conjecture | open | MathDB 页面 | 原文线索

V2-047891 | MATHDB-327738

Cranston-Rabern's chromatic bound conjecture for vertex-transitive graphs

摘录 / EXCERPT: Let G be a vertex-transitive graph, with chromatic number $\chi(G)$, clique number $\omega(G)$, and maximum degree $\Delta(G)$. Cranston-Rabern's conjecture. Every vertex-transit...

conjecture | open | MathDB 页面 | 原文线索

V2-047892 | MATHDB-327739

Cranston-Rabern's strong-coloring conjecture for vertex-transitive graphs

摘录 / EXCERPT: Let G be a vertex-transitive graph with maximum degree Δ, and let $s\chi(G)$ denote its strong chromatic number, the smallest r such that every partition of $V(G)$ into...

conjecture | open | MathDB 页面 | 原文线索

V2-047893 | MATHDB-327797

Foucaud-Krivelevich-Perarnau spanning subgraph conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of fixed graphs and let d be sufficiently large. For a d-regular graph G, a subgraph $H\subseteq G$ is spanning if it has the same vertex set as...

conjecture | open | MathDB 页面 | 原文线索

V2-047894 | MATHDB-327816

Fiamčík's acyclic edge coloring conjecture

摘录 / EXCERPT: Acyclic edge coloring conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047895 | MATHDB-327827

Subleading-order conjecture for cover times of the two-dimensional discrete torus

摘录 / EXCERPT: Let C_N be the time taken by discrete- or continuous-time random walk to visit every vertex of the two-dimensional discrete torus graph $(\mathbb{Z}/N\mathbb{Z})^2$. Subleading-o...

conjecture | open | MathDB 页面 | 原文线索

V2-047896 | MATHDB-327841

Minimum-dimension conjecture for collapse-free Hecke algebras

摘录 / EXCERPT: Let (W,S) be a Coxeter system whose Coxeter diagram is the simply laced bipartite graph G. Let $\mathcal{H}(\mathbf{q})$ be the Hecke algebra with independent parameters, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-047897 | MATHDB-327856

Conjectured upper bound for the saturation number of fullerene graphs

摘录 / EXCERPT: Fullerene saturation-number conjecture. There is a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-047898 | MATHDB-327871

Pikhurko-Zhao Motzkin-Straus conjecture for hypergraphs without a large clique

摘录 / EXCERPT: Pikhurko-Zhao conjecture. The Lagrangian of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047899 | MATHDB-327891

Containability bound by cop number and maximum degree

摘录 / EXCERPT: Let G be a graph. Write $\xi(G)$ for its containability number, the minimum number of cops needed to contain the robber, $c(G)$ for its cop number in the original Cops and Robber...

conjecture | open | MathDB 页面 | 原文线索

V2-047900 | MATHDB-327935

Non-degeneracy conjecture for the simple nineteen-vertex-model eigenvalue

摘录 / EXCERPT: Let $T^{(2)}(z)$ be the transfer matrix of the nineteen-vertex model with twist angle $\phi=\pi$, and let ... be its eigenvalue for inhomogeneity parameters $w_1,\dots,w_N$. Non-dege...

conjecture | open | MathDB 页面 | 原文线索

V2-047901 | MATHDB-327944

Rationality conjecture for crossing limits of periodic tiles

摘录 / EXCERPT: Let T be a tile, and let $c_n(T)=\operatorname{cr}(\bigcirc(T^n))$. The limit ... exists. Rationality conjecture for crossing limits. There exists a computable function f that as...

conjecture | open | MathDB 页面 | 原文线索

V2-047902 | MATHDB-327965

The cycle extension distinguishing number conjecture

摘录 / EXCERPT: Cycle extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047903 | MATHDB-327985

Best monotone condition for 2-factor graphs to be Hamiltonian

摘录 / EXCERPT: 2-factor Hamiltonicity conjecture. This degree condition is a best monotone degree condition for 2-factor $\Rightarrow$ hamiltonian. The source states that a best monotone condit...

conjecture | open | MathDB 页面 | 原文线索

V2-047904 | MATHDB-328006

The expansion conjecture for critically chromatic graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a simple graph, where V is its vertex set. A graph is critically s-chromatic if it has chromatic number s and deleting any vertex lowers its chromatic number...

conjecture | open | MathDB 页面 | 原文线索

V2-047905 | MATHDB-328009

The cut avoidance conjecture for robust ear decompositions

摘录 / EXCERPT: Let H be a trigraph, let H_0 be an induced cycle of H, and let S be a set of disjoint paths on 3 vertices. Let $\mathcal{H}$ be a robust ear decomposition of H that admit...

conjecture | open | MathDB 页面 | 原文线索

V2-047906 | MATHDB-328015

Invariant junction models and well-defined traffic statics for LTM networks

摘录 / EXCERPT: A junction model is invariant when its resulting junction behavior is unchanged under the relevant interior-state representation. A traffic statics problem for a junction network i...

conjecture | open | MathDB 页面 | 原文线索

V2-047907 | MATHDB-328045

The girth-minimum-degree lower-bound conjecture for zero forcing number

摘录 / EXCERPT: The girth-minimum-degree conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047908 | MATHDB-328046

The triangle-free lower-bound conjecture for zero forcing number

摘录 / EXCERPT: Triangle-free zero-forcing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047909 | MATHDB-328048

NP-completeness of induced cycle for cubic graphs

摘录 / EXCERPT: Cubic induced-cycle NP-completeness conjecture. INDUCED CYCLE is NP-complete for cubic graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-047910 | MATHDB-328050

The common connectivity-threshold conjecture for directed and strict undirected nearest-neighbour graphs

摘录 / EXCERPT: Let the directed model and the strict undirected model be the nearest-neighbour graph models considered in the paper, with n points and parameter k. An isolated vertex is a ver...

conjecture | open | MathDB 页面 | 原文线索

V2-047911 | MATHDB-328087

Quadratic saturation score conjecture for Hamiltonicity

摘录 / EXCERPT: Let ${\mathcal H}$ be the graph property of admitting a Hamilton cycle, and let $s(n, {\mathcal H})$ denote the score of the corresponding saturation game on graphs with n vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-047912 | MATHDB-328101

The induced matching bound for graphs of maximum degree at least three

摘录 / EXCERPT: The induced matching bound. If $G\notin\{C_5^2,K_{3,3}^+\}$ and $\Delta\geq 3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047913 | MATHDB-328105

Exact upper bound for edges of edge-minimal hypertrees

摘录 / EXCERPT: Let k be a positive integer, and let $\mathcal{F}=(V,\mathcal{E})$ be a k-uniform edge-minimal hypertree on $n=|V|$ vertices. Edge-minimal hypertree upper-bound conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-047914 | MATHDB-328106

Asymptotic lower bound for edges of 3-uniform edge-maximal hypertrees

摘录 / EXCERPT: Let n be the number of vertices, and let a 3-uniform edge-maximal hypertree be a 3-uniform hypertree to which no further edge can be added without destroying the hypertree proper...

conjecture | open | MathDB 页面 | 原文线索

V2-047915 | MATHDB-328145

Expected positive and negative Hashimoto eigenvalue conjecture

摘录 / EXCERPT: For fixed even $d\ge4$, let $\mathcal{G}_{n,d}$ be the random model of d-regular graphs, and let Hashimoto eigenvalues mean the eigenvalues of the non-backtracking matrix H_G....

conjecture | open | MathDB 页面 | 原文线索

V2-047916 | MATHDB-328186

Recursive construction conjecture for redundantly rigid graphs on concentric cylinders

摘录 / EXCERPT: Let G be a graph, and let K_5-e, H_1, and H_2 be the three specified base graphs. The operations under consideration are edge addition, 1-extension, and 1-, 2- and...

conjecture | open | MathDB 页面 | 原文线索

V2-047917 | MATHDB-328204

Nash-Williams's Hamilton cycle conjecture for digraphs

摘录 / EXCERPT: Let G be a strongly connected digraph on $n\geq 3$ vertices. Write its nondecreasing outdegree sequence as $d^+_1\leq d^+_2\leq\cdots\leq d^+_n$ and its nondecreasing indegree se...

conjecture | open | MathDB 页面 | 原文线索

V2-047918 | MATHDB-328210

Ore-Plesník-Murty-Simon conjecture on extremal diameter-2-critical graphs

摘录 / EXCERPT: A graph G is diameter-critical if deleting every edge increases its diameter, and it is diameter-k-critical if it is diameter-critical with diameter k. For a graph on n ver...

conjecture | open | MathDB 页面 | 原文线索

V2-047919 | MATHDB-328211

The connected triangle-free Ramsey simplicity conjecture

摘录 / EXCERPT: Connected triangle-free Ramsey simplicity conjecture. For every connected triangle-free graph H on at least 2 vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-047920 | MATHDB-328212

The BEL-gadget conjecture for 2-connected graphs

摘录 / EXCERPT: BEL-gadget conjecture. Every 2-connected graph has BEL gadgets.

conjecture | open | MathDB 页面 | 原文线索

V2-047921 | MATHDB-328213

The linear upper-bound conjecture for minimum degrees of minimal Ramsey graphs

摘录 / EXCERPT: Linear upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047922 | MATHDB-328245

McKay-Praeger Cayley graph proportion conjecture

摘录 / EXCERPT: For each positive integer n, let b_n be the ratio of the number of Cayley graphs on at most n vertices to the number of vertex-transitive graphs on at most n vertices: ... Mc...

conjecture | open | MathDB 页面 | 原文线索

V2-047923 | MATHDB-328248

The chromatic extremal conjecture for regular graphs

摘录 / EXCERPT: Let G be an n-vertex, d-regular graph, let K_q be the complete graph on q vertices, and let ${\rm hom}(G,K_q)$ denote the number of graph homomorphisms from G to K_q....

conjecture | open | MathDB 页面 | 原文线索

V2-047924 | MATHDB-328279

Zhu-You-Yang's lower-bound conjecture for non-regular graphs

摘录 / EXCERPT: Zhu-You-Yang's conjecture. If G is non-regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-047925 | MATHDB-328291

Finiteness and non-semisimplicity conjecture for modules of ${\mathcal V}^{(p)}$

摘录 / EXCERPT: For $p\geq 3$, let ${\mathcal V}^{(p)}$ be the generalized vertex algebra introduced by ... Consider its modules in the category $\mathcal{O}$. Module-category conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-047926 | MATHDB-328292

Identification conjecture for ${\mathcal R}^{(3)}$ with a simple affine W-algebra

摘录 / EXCERPT: Let ${\mathcal R}^{(p)}$ be the subalgebra generated inside ${\mathcal V}^{(p)}\otimes F_{-p/2}$ by the affine A_1 generators, the Heisenberg field, and the four fields displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-047927 | MATHDB-328306

The k-optimal set edge-colouring conjecture

摘录 / EXCERPT: Let G be a graph, let $k\in\mathbb{Z}^+$, and let D be a k-optimal set, meaning a k-dependent set maximizing $\phi_k(D)=k|D|-|E(G[D])|$. A subgraph is k-edge-chromatic if...

conjecture | open | MathDB 页面 | 原文线索

V2-047928 | MATHDB-328307

The sequential decomposition conjecture for graph orientations

摘录 / EXCERPT: Let G be a graph. A sequential decomposition of a digraph J is an ordered sequence of spanning subdigraphs $H_1,H_2,\ldots$ such that every arc of J belongs to exactly one...

conjecture | open | MathDB 页面 | 原文线索

V2-047929 | MATHDB-328340

The diameter conjecture for divisibility graphs of symmetric and alternating groups

摘录 / EXCERPT: For $n\geq 1$, let $D(S_n)$ and $D(A_n)$ be the divisibility graphs whose vertices correspond to conjugacy classes of S_n and A_n, respectively, with adjacency determined by di...

conjecture | open | MathDB 页面 | 原文线索

V2-047930 | MATHDB-328341

The forbidden-matching avoidance conjecture

摘录 / EXCERPT: Let G be a multigraph with maximum degree $\Delta(G)$ and maximum multiplicity $\mu(G)$. A forbidden matching is a matching whose edges have been assigned arbitrary, not necessar...

conjecture | open | MathDB 页面 | 原文线索

V2-047931 | MATHDB-328354

The finite-hat upper-bound conjecture for Levine's hat puzzle

摘录 / EXCERPT: Let $p=\frac{1}{2}$ be the probability that each hat is white, and consider strategies for the hat puzzle using finitely many hats. Finite-hat upper-bound conjecture. There is no s...

conjecture | open | MathDB 页面 | 原文线索

V2-047932 | MATHDB-328367

The additive lower bound for m-eternal vertex cover

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph, let H be a subgraph of G, and let $G[V\setminus V(H)]$ be the subgraph induced by the vertices outside H. Write $\delta(\cdot)$ for minimu...

conjecture | open | MathDB 页面 | 原文线索

V2-047933 | MATHDB-328398

Log-concavity conjecture for partial genus distributions

摘录 / EXCERPT: For a singly or doubly vertex-rooted or edge-rooted graph, the associated partial genus distributions are the sequences counting embeddings in the corresponding root-incidence clas...

conjecture | open | MathDB 页面 | 原文线索

V2-047934 | MATHDB-328406

Asymptotic bound for the triangle-saturated minimum-degree constant

摘录 / EXCERPT: For fixed t, define $c(t,3)$ by ... for all sufficiently large n. The preceding construction gives a lower bound of order $2^{t}t^{3/2}$ for $c(t,3)$. The asymptotic bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-047935 | MATHDB-328414

Bi-stability conjecture for the fluid model

摘录 / EXCERPT: Let x be a fluid solution with initial condition $x(0)$, let ${\cal O}$ be the set of initial conditions whose trajectories oscillate indefinitely, and let u^* denote a periodi...

conjecture | open | MathDB 页面 | 原文线索

V2-047936 | MATHDB-328423

Elphick's graph energy lower-bound conjecture in terms of irregularity

摘录 / EXCERPT: Elphick's conjecture. The graph energy satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047937 | MATHDB-328424

Elphick's graph energy upper-bound conjecture via the spectral radius

摘录 / EXCERPT: Let G be a connected graph with n vertices, m edges, adjacency eigenvalues including the largest eigenvalue λ_1, and graph energy $E=\sum_i|\lambda_i|$. Elphick's con...

conjecture | open | MathDB 页面 | 原文线索

V2-047938 | MATHDB-328435

General Star Decomposition Conjecture

摘录 / EXCERPT: Let H be a graph with edge densities $b3_e$. For each proper labeling f, consider the weighted monotone-path tree $T_f(H)$, with the edge weights interpreted as densities, and...

conjecture | open | MathDB 页面 | 原文线索

V2-047939 | MATHDB-328436

Uniform Star Decomposition Conjecture

摘录 / EXCERPT: Let H be a graph, let $S(H)$ be the set of proper labelings of H, and let $T_f(H)$ denote the weighted monotone-path tree associated with f. Write d_{crit} for the critical...

conjecture | open | MathDB 页面 | 原文线索

V2-047940 | MATHDB-328437

Critical density formula for complete bipartite graphs

摘录 / EXCERPT: Let $K_{n,m}$ be the complete bipartite graph. Denote its critical edge density by $d_{crit}(K_{n,m})=d(n,m)$ and the best edge density obtained from star decomposition by...

conjecture | open | MathDB 页面 | 原文线索

V2-047941 | MATHDB-328439

Škrekovski's acyclic 2-coloring conjecture for oriented planar graphs

摘录 / EXCERPT: An oriented planar graph is a planar digraph with no directed 2-cycles. An acyclic coloring is a vertex coloring in which every color class induces an acyclic subgraph; being acycl...

conjecture | open | MathDB 页面 | 原文线索

V2-047942 | MATHDB-328455

The $m(G)/17$ bound for induced matchings without C_5^2 components

摘录 / EXCERPT: Let G be a graph, let $m(G)$ denote its number of edges, and let C_5^2 denote the graph obtained by replacing each vertex of a 5-cycle with an independent set of order 2. A...

conjecture | open | MathDB 页面 | 原文线索

V2-047943 | MATHDB-328456

The induced-matching bound for connected graphs beyond C_5^2 and $K_{3,3}^+$

摘录 / EXCERPT: Let G be a connected graph with maximum degree $ \Delta\geq 3$. Let C_5^2 be the graph obtained from a 5-cycle by replacing its vertices with independent sets of order 2, a...

conjecture | open | MathDB 页面 | 原文线索

V2-047944 | MATHDB-328459

The acyclic Client graph conjecture for the Waiter-Client game

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. In a $(q:1)$ Waiter-Client game on $E(K_n)$, Waiter offers $q+1$ unclaimed edges in each round, and Client claims one offered edge...

conjecture | open | MathDB 页面 | 原文线索

V2-047945 | MATHDB-328475

Enomoto and Ota's path-partition conjecture

摘录 / EXCERPT: Let G be a graph of order n, let $k\geq 3$ be an integer, and let $n_{1},\dots,n_{k}$ be positive integers satisfying ... Write $\sigma_2(G)$ for the minimum degree sum of two no...

conjecture | open | MathDB 页面 | 原文线索

V2-047946 | MATHDB-328498

You and Yang's signless Laplacian power-sum conjecture for bipartite graphs

摘录 / EXCERPT: You and Yang's conjecture. If $\alpha>1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047947 | MATHDB-328499

You and Yang's connectivity-constrained signless Laplacian power-sum conjecture

摘录 / EXCERPT: You and Yang's conjecture. (i) If $0<\alpha<1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047948 | MATHDB-328505

Ore's recurrence conjecture for the minimum size of critical graphs

摘录 / EXCERPT: Let $f_k(n)$ be the minimum number of edges in an n-vertex k-critical graph, where a graph is k-critical if it is not $(k-1)$-colorable but every proper subgraph is $(k-1)$-c...

conjecture | open | MathDB 页面 | 原文线索

V2-047949 | MATHDB-328526

Caporossi-Cioabă-Gutman-Hansen conjecture on minimum-energy connected graphs

摘录 / EXCERPT: Let G be a connected graph with n vertices and e edges. For a graph with adjacency eigenvalues $\lambda_1,\ldots,\lambda_n$, its energy is $E(G)=\sum_{i=1}^n|\lambda_i|$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-047950 | MATHDB-328543

The pre-colored leaf extension conjecture

摘录 / EXCERPT: Let $f(k)$ be the smallest integer f such that for every $d\in[k]$ the following holds. Let G be an arbitrary finite graph such that every degree is at most d, except at most...

conjecture | open | MathDB 页面 | 原文线索

V2-047951 | MATHDB-328558

Homotopy triviality of digital images without simple loops of length at least five

摘录 / EXCERPT: Let X be a finite digital image, viewed as a graph, and let a simple m-loop mean a simple loop in X with m vertices. Homotopy triviality conjecture. If X has no simple...

conjecture | open | MathDB 页面 | 原文线索

V2-047952 | MATHDB-328573

Adjacency perfect state transfer conjecture for trees

摘录 / EXCERPT: Let T be a tree, let P_2 and P_3 denote the paths on two and three vertices, and say that T has adjacency perfect state transfer if its adjacency-matrix continuous-time qua...

conjecture | open | MathDB 页面 | 原文线索

V2-047953 | MATHDB-328576

Zhang-Liu-Wang's neighbour set distinguishing index conjecture

摘录 / EXCERPT: Zhang-Liu-Wang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-047954 | MATHDB-328613

Grothendieck-Verlinde fusion ring isomorphism conjecture for bosonic ghosts

摘录 / EXCERPT: Grothendieck-Verlinde fusion ring isomorphism conjecture. The product on the resulting Grothendieck ring may be identified with the Verlinde product under the group isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-047955 | MATHDB-328614

Projective-cover conjecture for typical and atypical ghost modules

摘录 / EXCERPT: Let $\mathscr{C}$ be the abelian category of ghost vertex operator algebra modules generated by imposing closure under extensions from the typical modules...

conjecture | open | MathDB 页面 | 原文线索

V2-047956 | MATHDB-328628

Lehel's strengthened Erdős-Gallai-Tuza conjecture

摘录 / EXCERPT: Lehel's strengthened Erdős-Gallai-Tuza conjecture. For every n-vertex graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047957 | MATHDB-328630

Cycle-count conjecture for degree 3-critical graphs

摘录 / EXCERPT: A degree 3-critical graph is a graph on n vertices with $2n-2$ edges and no proper induced subgraph of minimum degree 3. Cycle-count conjecture. Every degree 3-critical graph...

conjecture | open | MathDB 页面 | 原文线索

V2-047958 | MATHDB-328631

Many short leaf-leaf path lengths in 1-3 trees

摘录 / EXCERPT: A 1-3 tree is a tree whose vertices have degrees 1 or 3, and a leaf-leaf path is a path with leaf endpoints. Many-short-paths conjecture. There is a constant $\alpha>0$ and a f...

conjecture | open | MathDB 页面 | 原文线索

V2-047959 | MATHDB-328638

Threshold conjecture for the index of squarefree powers of edge ideals

摘录 / EXCERPT: Let G be a simple graph, let I be its edge ideal, and let $I^{[k]}$ denote its kth squarefree power. Let $\nu_0(G)$ be the maximum size of a restricted matching in G, with...

conjecture | open | MathDB 页面 | 原文线索

V2-047960 | MATHDB-328645

The cycle rank conjecture for metric dimension and zero forcing number

摘录 / EXCERPT: Let G be a graph. Write $\dim(G)$ for its metric dimension, $Z(G)$ for its zero forcing number, and $r(G)$ for its cycle rank. Cycle Rank Conjecture. ... The conjecture proposes th...

conjecture | open | MathDB 页面 | 原文线索

V2-047961 | MATHDB-328717

Threshold conjecture for the local pooling factor in random graphs

摘录 / EXCERPT: Let $G(n,r)$ be the random graph model under consideration, with edge-probability parameter r^2, and let LoP denote the graph property that the local pooling fac...

conjecture | open | MathDB 页面 | 原文线索

V2-047962 | MATHDB-328728

Albertson and Berman's induced forest conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph of order n. An induced forest of G is a vertex-induced subgraph that is acyclic. Albertson and Berman's conjecture. Every planar graph of order n ad...

conjecture | open | MathDB 页面 | 原文线索

V2-047963 | MATHDB-328729

Akiyama-Watanabe and Albertson-Rhaas conjecture for bipartite planar graphs

摘录 / EXCERPT: Let G be a bipartite planar graph of order n. An induced forest of G is a vertex-induced subgraph that is acyclic. Akiyama-Watanabe and Albertson-Rhaas conjecture. Every bipa...

conjecture | open | MathDB 页面 | 原文线索

V2-047964 | MATHDB-328752

The componentwise inertia sum-of-squares conjecture

摘录 / EXCERPT: Componentwise inertia sum-of-squares conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-047965 | MATHDB-328764

Five-list capital-colouring conjecture for plane graphs

摘录 / EXCERPT: Let G be a plane graph, and assign to each vertex v a list of five integers. A capital colouring is a colouring in which every face contains a unique vertex coloured with the m...

conjecture | open | MathDB 页面 | 原文线索

V2-047966 | MATHDB-328770

The n-free generators conjecture for vertex initial ideals of I_{S_n}

摘录 / EXCERPT: Let S_n be the polytope of symmetric doubly-stochastic matrices, and consider the toric ideal I_{S_n} using only the vertices of S_n. For $n eq 2$, let an n-free monomial m...

conjecture | open | MathDB 页面 | 原文线索

V2-047967 | MATHDB-328775

Pyber's conjecture on Hamiltonian cycles in distance-regular graphs

摘录 / EXCERPT: A distance-regular graph is a connected regular graph whose distance partition around every vertex has the same intersection numbers; connected strongly regular graphs are exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-047968 | MATHDB-328806

Hong-Lai conjecture on trees preserving graph connectivity

摘录 / EXCERPT: Hong-Lai conjecture. For all positive integers k and r, there exists a positive integer $h(k,r)$ such that, for every $h(k,r)$-connected graph G and subset X with r verti...

conjecture | open | MathDB 页面 | 原文线索

V2-047969 | MATHDB-328810

The polygonization maximum-angle conjecture

摘录 / EXCERPT: Let S be a set of n points in the plane, and let a polygonization of S be a simple polygon whose vertex set is S. Natural conjecture. There exists a polygonization of S w...

conjecture | open | MathDB 页面 | 原文线索

V2-047970 | MATHDB-328820

Broersma's degree condition conjecture for hamiltonian claw-free graphs

摘录 / EXCERPT: Broersma's conjecture. If every end-vertex of an induced copy of N in G has degree at least $(n-2)/3$, then G is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-047971 | MATHDB-328828

Structured-cover conjecture for intersecting multipartite hypergraphs

摘录 / EXCERPT: Let H be an intersecting r-partite hypergraph with sides $V_1,\ldots,V_r$. A cover is a set of vertices meeting every edge. The structured-cover conjecture. There exists either...

conjecture | open | MathDB 页面 | 原文线索

V2-047972 | MATHDB-328829

Generalized structured-cover conjecture for multipartite hypergraphs

摘录 / EXCERPT: Let H be an r-partite hypergraph with matching number $ u(H)=k$. A set is contained in a side or in an edge if it is a subset of one of the sides or of one edge, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-047973 | MATHDB-328832

Mansour-Song-Yuster conjecture on the size of extremal Ryser hypergraphs

摘录 / EXCERPT: Mansour-Song-Yuster conjecture. The minimum number of edges grows linearly with r:

conjecture | open | MathDB 页面 | 原文线索

V2-047974 | MATHDB-328851

The planar dual conjecture for Laplacian quadratic forms

摘录 / EXCERPT: Planar dual conjecture. The quadratic forms over $\mathbb{Z}$ represented by the combinatorial Laplacians of G_1 and G_2 are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-047975 | MATHDB-328852

The higher-genus dual conjecture for Laplacian quadratic forms

摘录 / EXCERPT: Higher-genus dual conjecture. The quadratic forms over $\mathbb{Z}$ represented by the combinatorial Laplacians of G_1 and G_2 are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-047976 | MATHDB-328856

O's spectral conjecture for edge-connectivity of regular graphs

摘录 / EXCERPT: O's conjecture. For $t\geq 3$, if

conjecture | open | MathDB 页面 | 原文线索

V2-047977 | MATHDB-328861

Uniqueness conjecture for the 16-vertex triangulation of the real projective 4-space

摘录 / EXCERPT: Let $\mathbb{R}P^4_{16}$ denote the constructed 16-vertex triangulation of the real projective 4-space $\mathbb{R}P^4$. Uniqueness conjecture. $\mathbb{R}P^4_{16}$ is the uniqu...

conjecture | open | MathDB 页面 | 原文线索

V2-047978 | MATHDB-328876

The treewidth upper-bound conjecture for line graphs

摘录 / EXCERPT: Let G be a graph, let $L(G)$ be its line graph, let $\Delta(G)$ be its maximum degree, and let $\textsf{tw}(G)$ denote its treewidth. The treewidth upper-bound conjecture. One ha...

conjecture | open | MathDB 页面 | 原文线索

V2-047979 | MATHDB-328899

McDiarmid-Mohar conjecture on acyclic chromatic number of oriented graphs

摘录 / EXCERPT: McDiarmid-Mohar conjecture. Every oriented graph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047980 | MATHDB-328900

Extension of the McDiarmid-Mohar bound to m-degenerate chromatic number

摘录 / EXCERPT: m-degenerate extension conjecture. Every oriented graph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-047981 | MATHDB-328902

Benjamini-Georgakopoulos conjecture on local indistinguishability of Cayley graphs

摘录 / EXCERPT: Benjamini-Georgakopoulos conjecture. There exists $r\in\mathbb{N}$ such that, whenever a graph G is r-locally F, F covers G.

conjecture | open | MathDB 页面 | 原文线索

V2-047982 | MATHDB-328907

Karp's evasiveness conjecture for monotone graph and digraph properties

摘录 / EXCERPT: Let a graph or digraph property be monotone if it is preserved under deletion of edges or arcs, and let a property be non-trivial if it is neither always true nor always false. A p...

conjecture | open | MathDB 页面 | 原文线索

V2-047983 | MATHDB-328910

The generalized-star path-packing conjecture in hypercubes

摘录 / EXCERPT: The generalized-star path-packing conjecture. Let

conjecture | open | MathDB 页面 | 原文线索

V2-047984 | MATHDB-328911

The minimum-degree and central-path conjecture for tree saturation in hypercubes

摘录 / EXCERPT: Let T be a tree, let $\operatorname{emin}(T)=\delta$ denote the minimum degree parameter used in the paper, and let C be the maximum distance from any given vertex of T to th...

conjecture | open | MathDB 页面 | 原文线索

V2-047985 | MATHDB-328947

Xu's polynomial-time conjecture for the maximum forcing number of elementary polyominoes

摘录 / EXCERPT: Let G be an elementary polyomino, meaning a polyomino whose associated plane bipartite graph is elementary. The maximum forcing number conjecture. The maximum forcing number of...

conjecture | open | MathDB 页面 | 原文线索

V2-047986 | MATHDB-328950

Wu, Zhang, and Li's equitable induced-forest partition conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph. An equitable partition of G is a partition of its vertex set into parts whose sizes differ by at most one, and an induced forest is a vertex-induced su...

conjecture | open | MathDB 页面 | 原文线索

V2-047987 | MATHDB-328951

Equitable induced-forest partition conjecture for bounded-degree graphs

摘录 / EXCERPT: Let G be a graph with maximum degree Δ. An equitable partition is a partition of the vertex set into parts whose sizes differ by at most one, and an induced forest is a ve...

conjecture | open | MathDB 页面 | 原文线索

V2-047988 | MATHDB-328958

Häggkvist's conjecture on compatible Hamilton cycles in Dirac graphs

摘录 / EXCERPT: Häggkvist's conjecture. For every 1-bounded incompatibility system $\mathcal{F}$ over a Dirac graph G, there exists a Hamilton cycle compatible with $\mathcal{F}$.

conjecture | open | MathDB 页面 | 原文线索

V2-047989 | MATHDB-328966

The acyclic chromatic index conjecture

摘录 / EXCERPT: Acyclic chromatic index conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-047990 | MATHDB-328977

de Freitas-Nikiforov-Patuzzi conjecture on the Q-index of graphs forbidding even cycles

摘录 / EXCERPT: de Freitas-Nikiforov-Patuzzi conjecture. For $k\geq 2$ and a graph G of sufficiently large order n, if G has no cycle of length $2k+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-047991 | MATHDB-328978

Wang's conjecture that graphs in \(\mathcal{F}_n\) are DGS

摘录 / EXCERPT: Let $\mathcal{F}_n$ be the family of graphs whose walk-matrix determinant satisfies that $\det(W)/2^{\lfloor n/2\rfloor}$ is an odd square-free integer, where W is the walk-matri...

conjecture | open | MathDB 页面 | 原文线索

V2-047992 | MATHDB-328989

The fractional clique-cover conjecture for triangle-free graphs

摘录 / EXCERPT: Let G be an undirected triangle-free graph. The fractional clique-cover conjecture. ... The fractional clique cover gives a general lower bound on the guessing number, and equality...

conjecture | open | MathDB 页面 | 原文线索

V2-047993 | MATHDB-328993

Continuity of the connective constant under local convergence

摘录 / EXCERPT: Continuity conjecture. The connective constant is continuous with respect to local convergence:

conjecture | open | MathDB 页面 | 原文线索

V2-047994 | MATHDB-329006

Zhu's conjecture for categorical products of hypergraphs

摘录 / EXCERPT: Let $\mathcal{G}=(V,E)$ and $\mathcal{G}'=(V',E')$ be hypergraphs. Their categorical product $\mathcal{G}\times\mathcal{G}'$ has vertex set $V\times V'$ and edges whose projections...

conjecture | open | MathDB 页面 | 原文线索

V2-047995 | MATHDB-329015

Finiteness conjecture for game distinguishing numbers

摘录 / EXCERPT: Let G be a graph, and let $D_\mathcal{G}(G)$ and $D_\mathcal{R}(G)$ denote its Gentle-game and Rascal-game distinguishing numbers. An automorphism of order 2 is an automorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-047996 | MATHDB-329044

Bollobás-Scott spanning balanced bipartite subgraph conjecture

摘录 / EXCERPT: Let G be a graph of even order. A spanning balanced bipartite subgraph B is a spanning bipartite subgraph whose two vertex classes have equal size. Bollobás-Scott's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-047997 | MATHDB-329045

Multipartite tiling threshold conjecture

摘录 / EXCERPT: Let H be a graph with order h and chromatic number r. Let $\chi_{cr}(H)$ denote its critical chromatic number and let $hcf(H)$ denote its highest common factor. For a balance...

conjecture | open | MathDB 页面 | 原文线索

V2-047998 | MATHDB-329058

Bounded multiplicity conjecture for Jacobians of biconnected graphs

摘录 / EXCERPT: Let n be a positive integer, and let k_n be an integer depending on n. A graph is biconnected if it has the connectivity property intended in the paper. Bounded multiplicity...

conjecture | open | MathDB 页面 | 原文线索

V2-047999 | MATHDB-329069

The square-root singular-value conjecture for random regular digraphs

摘录 / EXCERPT: Square-root singular-value conjecture. Assume $1\leq d\leq n$. Then asymptotically almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-048000 | MATHDB-329072

Folklore clique-decomposition conjecture for complete graphs

摘录 / EXCERPT: For a graph G, being K_{r+1}-divisible means that the necessary divisibility conditions for a K_{r+1}-decomposition hold. Folklore clique-decomposition conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-048001 | MATHDB-329078

The two-value genus-range conjecture for chord diagrams

摘录 / EXCERPT: Two-value genus-range conjecture. If the genus range of D consists of two numbers, then it is either

conjecture | open | MathDB 页面 | 原文线索

V2-048002 | MATHDB-329079

The uniqueness conjecture for chord diagrams with genus range {0,1}

摘录 / EXCERPT: Uniqueness conjecture for genus range $\{0,1\}$. For any $n\neq 2$, there is a unique, up to equivalence, double-occurrence word

conjecture | open | MathDB 页面 | 原文线索

V2-048003 | MATHDB-329080

The uniqueness conjecture for chord diagrams with genus range {1,2}

摘录 / EXCERPT: Uniqueness conjecture for genus range $\{1,2\}$. For any $n\neq 4$, there is a unique, up to equivalence, chord diagram with genus range

conjecture | open | MathDB 页面 | 原文线索

V2-048004 | MATHDB-329136

Christofides-Markström conjecture on the fractional clique-cover bound

摘录 / EXCERPT: Let G be an undirected graph. Write $\operatorname{gn}(G)$ for its guessing number, let $V(G)$ be its vertex set, and let $\kappa_f(G)$ denote the fractional clique-cover number...

conjecture | open | MathDB 页面 | 原文线索

V2-048005 | MATHDB-329141

Intersection conjecture for the difference set S

摘录 / EXCERPT: Intersection conjecture for the difference set S. The intersection $S\cap(-S)$ is exactly as follows:

conjecture | open | MathDB 页面 | 原文线索

V2-048006 | MATHDB-329173

Polynomial singularity bound conjecture for random regular digraphs

摘录 / EXCERPT: Let M be a uniform random element of the set of $n\times n$ zero-one matrices with every row and column sum equal to d. Polynomial singularity conjecture. There are absolute co...

conjecture | open | MathDB 页面 | 原文线索

V2-048007 | MATHDB-329174

General upper-bound conjecture for interval cyclic colorings

摘录 / EXCERPT: Let G be a connected graph with at least two vertices, let $V(G)$ be its vertex set, and let $\mathfrak{N}_{c}$ denote the class of interval cyclically colorable graphs. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-048008 | MATHDB-329182

Erdős's conjecture on extremal odd-cycle edges

摘录 / EXCERPT: Let G be an n-vertex graph with e edges, and let a C_{2k+1}-edge mean an edge contained in a cycle of length $2k+1$. Assume that $e>n^2/4$, $k\geq 3$, and $n>n_k$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-048009 | MATHDB-329186

Square-root upper-bound conjecture for the synchronous-to-asynchronous guaranteed spreading ratio

摘录 / EXCERPT: Synchronous-to-asynchronous ratio conjecture. For any n-vertex graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-048010 | MATHDB-329202

Generalisation of the frustrated-triangle extremal conjecture to frustrated cycles

摘录 / EXCERPT: Let G be a graph, let $\overline{G}$ denote its complement, and for $k\geq 3$ let $f_k(G)$ be the number of frustrated k-cycles, where a cyclic ordering $v_1v_2\dots v_k$ is fr...

conjecture | open | MathDB 页面 | 原文线索

V2-048011 | MATHDB-329209

The near-bipartite characterization conjecture for ${\operatorname{LS}}_+$-perfect graphs

摘录 / EXCERPT: For a graph G, let $\operatorname{STAB}(G)$ be its stable set polytope, $\operatorname{NB}(G)$ the near-bipartite polyhedral relaxation, and ${\operatorname{LS}}_+(G)$ the Lovász...

conjecture | open | MathDB 页面 | 原文线索

V2-048012 | MATHDB-329224

The Janson-Sós conjecture on quasi-random properties from graph averages

摘录 / EXCERPT: Let $F \neq K_2$ be a non-empty graph with vertex set $\{1,\ldots,m\}$, and let $\alpha_1,\ldots,\alpha_m \in (0,1)$ satisfy ... Write $\mathcal{P}(F,\alpha_1,\ldots,\alpha_m)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-048013 | MATHDB-329231

Generic optimality of arbitrary sampling sets for connected graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph and let ${\mathcal S}\subset V$ be an arbitrary set with $|{\mathcal S}|=m$. Let $0=\lambda_1\leq\cdots\leq\lambda_n$ be the eigenvalues of the n...

conjecture | open | MathDB 页面 | 原文线索

V2-048014 | MATHDB-329237

Grothendieck-ring Verlinde relations conjecture for singlet modules

摘录 / EXCERPT: Let $\mathcal{M}(p_+,p_-)$ be the singlet vertex operator algebra, and let a suitable quotient category of its singlet modules have Grothendieck ring K_0. The relations obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-048015 | MATHDB-329252

Primality and complete-intersection conjecture for graph ideals

摘录 / EXCERPT: Let $d\geq 1$, let K be an arbitrary field, and let G be a graph on vertex set $[n]$. Primality and complete-intersection conjecture. If G is $(n-d)$-connected, then L_G is...

conjecture | open | MathDB 页面 | 原文线索

V2-048016 | MATHDB-329275

The exact count of subdivisions of K_4

摘录 / EXCERPT: Let $\phi(n)$ denote the maximum number of K_4-subdivisions in a graph with n vertices. Exact-count conjecture. For every n, ... The theorem preceding this conjecture gives the...

conjecture | open | MathDB 页面 | 原文线索

V2-048017 | MATHDB-329279

The extremal conjecture for tight paths and cycles in uniform hypergraphs

摘录 / EXCERPT: Extremal conjecture for tight paths and cycles. For any ℓ, every n-vertex k-graph with more than

conjecture | open | MathDB 页面 | 原文线索

V2-048018 | MATHDB-329281

Proper Hamiltonian cycles in 2-connected edge-colored multigraphs

摘录 / EXCERPT: Let G^c be a 2-connected c-edge-colored multigraph on n vertices, where $n\geq 10$ and $c\geq 3$. Let $rd(G^c)$ denote its rainbow degree and let m be its number of edges...

conjecture | open | MathDB 页面 | 原文线索

V2-048019 | MATHDB-329286

Reed-Johnson-Robertson-Seymour-Thomas directed grid conjecture

摘录 / EXCERPT: Let $f\colon\mathbb{N}\to\mathbb{N}$ be a function, let $k\in\mathbb{N}$, and let a digraph have directed tree width at least $f(k)$. A cylindrical grid of order k is the directe...

conjecture | open | MathDB 页面 | 原文线索

V2-048020 | MATHDB-329338

Positivity testing conjecture for minimal circular planar networks

摘录 / EXCERPT: Let L be the response matrix of a minimal circular planar network with m edges, and let a noninterlaced minor of L mean a minor whose row and column index sets are noninterla...

conjecture | open | MathDB 页面 | 原文线索

V2-048021 | MATHDB-329339

Base-set Laurent phenomenon conjecture for minimal circular planar networks

摘录 / EXCERPT: Let L be the response matrix of a minimal circular planar network with m edges, and call a minor of L noninterlaced when its row and column index sets are noninterlaced. Base...

conjecture | open | MathDB 页面 | 原文线索

V2-048022 | MATHDB-329340

Domino-tiling conjecture for semicontiguous minors

摘录 / EXCERPT: A semicontiguous minor is a minor M_A^B in which only one of A or B is contiguous. Domino-tiling conjecture. Every semicontiguous minor has an associated domino-tiling region...

conjecture | open | MathDB 页面 | 原文线索

V2-048023 | MATHDB-329360

Kenyon's spanning-tree entropy conjecture for finite planar graphs

摘录 / EXCERPT: Let G be a finite planar graph, with $e(G)$ edges and $\tau(G)$ spanning trees. Let ${\rm C}\approx0.9160$ be Catalan's constant. Kenyon's conjecture. ... This is equivalent in the...

conjecture | open | MathDB 页面 | 原文线索

V2-048024 | MATHDB-329404

Balogh, Kostochka and Treglown's degree-sequence Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Let $n,r\in\mathbb N$ with r dividing n. Let G be a graph on n vertices with degree sequence $d_1\leq\dots\leq d_n$. Balogh, Kostochka and Treglown's conjecture. If ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-048025 | MATHDB-329405

Chepoi-Estellon-Vaxès linear Erdős-Pósa conjecture for planar ball hypergraphs

摘录 / EXCERPT: Let G be a planar graph and let $ u_{\ell}(G)$ and $\tau_{\ell}(G)$ denote, respectively, the packing number and transversality of its hypergraph of balls of radius ℓ. Chepo...

conjecture | open | MathDB 页面 | 原文线索

V2-048026 | MATHDB-329406

Distance VC-dimension versus clique number conjecture for chi-bounded graph classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a class of graphs. Write $\chi(G)$ for the chromatic number, $\omega(G)$ for the clique number, and say that $\mathcal{G}$ is χ-bounded when there is a fu...

conjecture | open | MathDB 页面 | 原文线索

V2-048027 | MATHDB-329409

Existence of an edge inactive in every spanning tree for strongly Tutte-descriptive activities

摘录 / EXCERPT: Inactivity conjecture. There exists an edge e of G such that, for every spanning tree T, e does not belong to $\psi(T)$; equivalently, e is active in no spanning tree.

conjecture | open | MathDB 页面 | 原文线索

V2-048028 | MATHDB-329418

Garijo et al.'s locating-dominating set conjecture for twin-free graphs

摘录 / EXCERPT: Garijo et al.'s conjecture. Every twin-free graph G of order n without isolated vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048029 | MATHDB-329426

Garijo-González-Márquez conjecture on location-domination in twin-free connected graphs

摘录 / EXCERPT: Let $\lambda_{|\mathcal C^*}(n)$ denote the maximum location-domination number of a twin-free, connected graph on n vertices. Garijo-González-Márquez conjecture. There exists a p...

conjecture | open | MathDB 页面 | 原文线索

V2-048030 | MATHDB-329427

Foucaud-Henning conjecture on location-domination in twin-free graphs

摘录 / EXCERPT: Let G be a graph of order n with no isolated vertices. A graph is twin-free if no two distinct vertices have the same open or closed neighborhood, and let $\gamma_L(G)$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-048031 | MATHDB-329428

Erdős-Pósa conjecture for long cycles through prescribed vertices

摘录 / EXCERPT: Long S-cycle Erdős-Pósa conjecture. For every graph G, every subset of vertices S, and every pair of positive integers k,ℓ, there is either a set of k disjoint S-cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-048032 | MATHDB-329431

Cycle goodness conjecture

摘录 / EXCERPT: For a connected graph H, define its goodness to be the largest integer p for which H is p-good, where H is p-good if $R(H,K_p)=(|H|-1)(p-1)+1$. Cycle goodness conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048033 | MATHDB-329442

Degree-sequence conjecture for spanning triangulations

摘录 / EXCERPT: Degree-sequence conjecture for spanning triangulations. For every $\eta>0$, there exists $n_0\in\mathbb N$ such that, whenever $n\geq n_0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-048034 | MATHDB-329445

The fCyclabilityf completeness conjecture

摘录 / EXCERPT: Cyclability completeness conjecture. The problem textsc{Cyclability} is $\mathsf{\Pi}_{2}^{\mathrm{P}}$-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-048035 | MATHDB-329448

The worst-case value conjecture for unweighted trees in Tron

摘录 / EXCERPT: Let an unweighted instance of texttt{Tron} on a tree be an instance in which all vertex weights equal $1/n$, where n is the number of vertices. The unweighted-tree value conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-048036 | MATHDB-329449

The worst-case value conjecture for Tron on cycles

摘录 / EXCERPT: Let an instance of texttt{Tron} on a cycle be a game instance whose underlying graph is a cycle, with arbitrary vertex weights. The cycle value conjecture. Any instance of texttt{T...

conjecture | open | MathDB 页面 | 原文线索

V2-048037 | MATHDB-329457

Michael-Traves conjecture: the Roller Coaster Conjecture

摘录 / EXCERPT: Let q be a positive integer, let π be a permutation of $\{\lceil q/2\rceil,\lceil q/2\rceil+1,\ldots,q\}$, and let $i_k(G)$ denote the number of independent sets of cardinali...

conjecture | open | MathDB 页面 | 原文线索

V2-048038 | MATHDB-329458

Alavi-Erdős-Malde-Schwenk conjecture on unimodality for trees and forests

摘录 / EXCERPT: A tree is a connected acyclic graph, and a forest is an acyclic graph. For a graph G, its independence sequence $(i_0(G),i_1(G),\ldots,i_{\alpha(G)}(G))$ records the numbers of i...

conjecture | open | MathDB 页面 | 原文线索

V2-048039 | MATHDB-329459

The Global Attractor Conjecture for complex-balanced mass action systems

摘录 / EXCERPT: Global Attractor Conjecture. The unique complex-balanced steady state in each positive stoichiometric compatibility class of a complex-balanced mass action system is a global attra...

conjecture | open | MathDB 页面 | 原文线索

V2-048040 | MATHDB-329460

The Persistence Conjecture for weakly reversible mass action systems

摘录 / EXCERPT: Persistence Conjecture. Every weakly reversible mass action system is persistent.

conjecture | open | MathDB 页面 | 原文线索

V2-048041 | MATHDB-329466

Perrot-Pham sink-independence conjecture for the Biggs-Merino polynomial

摘录 / EXCERPT: Perrot-Pham's conjecture. The polynomial $\mathcal B(G,s;y)$ is independent of the choice of the sink vertex s.

conjecture | open | MathDB 页面 | 原文线索

V2-048042 | MATHDB-329469

The generic-and-Hilton-Milner obstruction conjecture for EKR in random hypergraphs

摘录 / EXCERPT: Let ${\cal H}$ be the random k-uniform hypergraph with edge-probability φ, let Δ denote its maximum degree, and let $\Lambda'(\Delta)$ be the parameter from Theore...

conjecture | open | MathDB 页面 | 原文线索

V2-048043 | MATHDB-329474

Nondegenerate multistationarity conjecture for the sequestration networks

摘录 / EXCERPT: Let $K_{m,n}$ be the sequestration network with n reactions and n species, and let $\widetilde K_{m,n}$ denote its fully open extension. A steady state is nondegenerate when it...

conjecture | open | MathDB 页面 | 原文线索

V2-048044 | MATHDB-329486

Cavers et al.'s maximum skew-spectral radius conjecture for odd-cycle graphs

摘录 / EXCERPT: Cavers et al.'s conjecture. If G is an odd-cycle graph of order n, then

conjecture | open | MathDB 页面 | 原文线索

V2-048045 | MATHDB-329493

The complete-bipartite extremal conjecture for algebraic connectivity

摘录 / EXCERPT: Let G be a graph with exactly n vertices and $m=2(n-2)$ edges. The complete-bipartite extremal conjecture. Among all such graphs, a graph maximizing the algebraic connectivity...

conjecture | open | MathDB 页面 | 原文线索

V2-048046 | MATHDB-329494

The asymptotically optimal algebraic-connectivity bound for bounded-degree trees

摘录 / EXCERPT: Let T be a tree with n vertices and maximum degree d. The asymptotically optimal tree bound conjecture. As $n\rightarrow\infty$ for fixed d, ... Here $\lambda_{2}(T)$ denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-048047 | MATHDB-329495

The well-balanced Bethe tree extremal conjecture

摘录 / EXCERPT: Let T be a tree with ... vertices and maximum degree d. The well-balanced Bethe tree extremal conjecture. Its algebraic connectivity is less than the algebraic connectivity of th...

conjecture | open | MathDB 页面 | 原文线索

V2-048048 | MATHDB-329496

The improved diameter and order bounds for cubic graphs

摘录 / EXCERPT: Let G be a cubic graph, and let D be its diameter. The cubic-graph algebraic-connectivity bound conjecture. Its algebraic connectivity satisfies ... Moreover, if G has order...

conjecture | open | MathDB 页面 | 原文线索

V2-048049 | MATHDB-329509

Gutman-Wagner matching energy conjecture for random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the random graph on n vertices in which each edge is present independently with fixed probability $p\in(0,1)$. For a graph G, let $ME(G)$ denote its matching e...

conjecture | open | MathDB 页面 | 原文线索

V2-048050 | MATHDB-329517

Gyárfás-Györi-Simonovits partition conjecture for 3-uniform hypergraphs

摘录 / EXCERPT: Let H be a 3-uniform hypergraph, and let $b1(H)$ denote the cardinality of a largest independent set of H. A linear cycle is a sequence of at least three edges in which only...

conjecture | open | MathDB 页面 | 原文线索

V2-048051 | MATHDB-329525

The extremal average communicability angle conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph with n nodes. Write P_n for the path graph on n nodes, K_n for the complete graph on n nodes, and let the average communicability angle be th...

conjecture | open | MathDB 页面 | 原文线索

V2-048052 | MATHDB-329560

Faudree-Schelp-Gyárfás-Tuza conjectures for subcubic strong edge coloring

摘录 / EXCERPT: Faudree-Schelp-Gyárfás-Tuza conjectures. The following assertions hold:

conjecture | solved | MathDB 页面 | 原文线索

V2-048053 | MATHDB-329571

Extremal edge conjecture for unbalanced bipartite 1-planar graphs

摘录 / EXCERPT: Let x and y be integers with $x\geq 3$ and $y\geq 6x-12$, and let G be a bipartite 1-planar graph whose partite sets have sizes x and y. Extremal edge conjecture. The gra...

conjecture | open | MathDB 页面 | 原文线索

V2-048054 | MATHDB-329573

The chromatic upper-bound conjecture for Euclidean dimension of graphs

摘录 / EXCERPT: Let G be a graph. Its Euclidean dimension, denoted $dim(G)$, is the minimum dimension of a Euclidean space in which G can be represented with adjacent vertices at unit distance...

conjecture | open | MathDB 页面 | 原文线索

V2-048055 | MATHDB-329594

Durocher-Gunderson-Li-Skala conjecture on cycles in triangle-free graphs

摘录 / EXCERPT: Let $n\geq 4$. A triangle-free graph is a graph containing no triangle, and $K_{a,b}$ denotes the complete bipartite graph with parts of sizes a and b. Durocher-Gunderson-Li-Sk...

conjecture | open | MathDB 页面 | 原文线索

V2-048056 | MATHDB-329595

Conjecture on cycles in odd-cycle-free graphs

摘录 / EXCERPT: Let $k>1$, and let C_{2k+1} denote the cycle of length $2k+1$. A graph is C_{2k+1}-free if it contains no subgraph isomorphic to C_{2k+1}; $K_{a,b}$ denotes the complete bipa...

conjecture | open | MathDB 页面 | 原文线索

V2-048057 | MATHDB-329599

The threshold conjecture for random graphs and critical subgraphs

摘录 / EXCERPT: Let H be a graph with a color-critical edge e, meaning that $\chi(H-e)<\chi(H)$. For a graph G, let $t_H(G)$ and $b_H(G)$ be the maximum values of $|K|$ over, respectively,...

conjecture | open | MathDB 页面 | 原文线索

V2-048058 | MATHDB-329600

The max-cut local concentration conjecture for random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the binomial random graph. An ordinary max cut is a partition of the vertex set into two parts attaining the maximum possible number of crossing edges. Max-cut loc...

conjecture | open | MathDB 页面 | 原文线索

V2-048059 | MATHDB-329608

Concavity of coverage probability in tier association probabilities

摘录 / EXCERPT: Let K) be the number of tiers, and let A_i denote the tier-association probability for tier i, with $A_i\geq 0$ and $\sum_{i=1}^{K}A_i=1$. Let P_c^* be the maximum overall...

conjecture | open | MathDB 页面 | 原文线索

V2-048060 | MATHDB-329657

Unbounded normalized induced decomposition extremal function

摘录 / EXCERPT: Let F be a graph, and let $\overline{\operatorname{ex}^*}(n,F)$ denote the relevant induced decomposition extremal function for F. Unboundedness conjecture. There exist graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-048061 | MATHDB-329658

Power-law exponent conjecture for induced decomposition extremal functions

摘录 / EXCERPT: Let F be a non-complete graph without isolated vertices, and let $\overline{\operatorname{ex}^*}(n,F)$ denote the relevant induced decomposition extremal function for F. Power-...

conjecture | open | MathDB 页面 | 原文线索

V2-048062 | MATHDB-329659

Rationality conjecture for induced decomposition growth exponents

摘录 / EXCERPT: Let F be a graph for which the power-law exponent $c(F)$ is defined by the preceding conjecture. Rationality conjecture. The constant $c=c(F)$ is rational for every graph F. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048063 | MATHDB-329660

Nagys conjecture for induced decomposition extremal functions

摘录 / EXCERPT: Let F be a non-complete graph without isolated vertices, and let $\overline{\operatorname{ex}^*}(n,F)$ denote the relevant induced decomposition extremal function. Nagys conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-048064 | MATHDB-329661

Limit conjecture for induced decomposition extremal functions

摘录 / EXCERPT: Let F be a non-complete graph without isolated vertices, and let $\overline{\operatorname{ex}^*}(n,F)$ denote the relevant induced decomposition extremal function. Positive-limit...

conjecture | open | MathDB 页面 | 原文线索

V2-048065 | MATHDB-329705

The adjacency dimension bound for the join of a graph with a universal vertex

摘录 / EXCERPT: Let H be a graph of order $n\ge 2$, let K_1+H denote the graph obtained by adding a universal vertex to H, and let $\mathcal{C}(K_1+H)$ be its adjacency threshold. For...

conjecture | open | MathDB 页面 | 原文线索

V2-048066 | MATHDB-329726

The 3-extendability conjecture for bow-tie graphs

摘录 / EXCERPT: Let C_m be a cycle of even length $m\ge6$, let P_n be a path of odd order $n\ge5$, and let $C_m\bowtie P_n$ denote their bow-tie graph. 3-extendability conjecture. For any even...

conjecture | open | MathDB 页面 | 原文线索

V2-048067 | MATHDB-329729

The angle-labeling characterization of essentially 3-connected maps

摘录 / EXCERPT: Angle-labeling characterization of essentially 3-connected maps. The map M admits an edge, $\mathbb{N}^*$-vertex, $\mathbb{N}^*$-face angle labeling if and only if it is essentia...

conjecture | open | MathDB 页面 | 原文线索

V2-048068 | MATHDB-329730

The three-\B_1-branch conjecture for minimal-ABC trees

摘录 / EXCERPT: Three-B_1-branch conjecture. A minimal-ABC tree can contain at most three B_1-branches.

conjecture | open | MathDB 页面 | 原文线索

V2-048069 | MATHDB-329731

The nine-\B_2-branch conjecture for minimal-ABC trees

摘录 / EXCERPT: Nine-B_2-branch conjecture. A minimal-ABC tree can contain at most nine B_2-branches.

conjecture | open | MathDB 页面 | 原文线索

V2-048070 | MATHDB-329734

The odd-order nonzero domination-root conjecture for generalized friendship graphs

摘录 / EXCERPT: Let $F_{4,n}$ be the generalized friendship graph with parameter n, and let a domination root mean a real zero of its domination polynomial $D(F_{4,n},x)$. Odd-order nonzero-root...

conjecture | open | MathDB 页面 | 原文线索

V2-048071 | MATHDB-329747

A strengthened extremal conjecture for trees

摘录 / EXCERPT: Let T be a tree on k vertices, and let ${\rm ex}(n,T)$ denote the largest number of edges in an n-vertex graph with no copy of T. The strengthened tree extremal conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-048072 | MATHDB-329781

Faben-Jerrum dichotomy conjecture for homomorphisms to graphs modulo 2

摘录 / EXCERPT: Faben-Jerrum's conjecture. Let H be a graph. If its involution-free reduction H^* has at most one vertex, then $\oplus\operatorname{HomsTo}(H)$ is in P; otherwise,...

conjecture | open | MathDB 页面 | 原文线索

V2-048073 | MATHDB-329788

Zhang et al.'s Ramsey-number conjecture for odd cycles and even wheels

摘录 / EXCERPT: Zhang et al.'s conjecture. For integers $n<m$, with n odd and m even,

conjecture | open | MathDB 页面 | 原文线索

V2-048074 | MATHDB-329804

A strict asymptotic bound conjecture for Ky Fan graph eigenvalue sums

摘录 / EXCERPT: Strict asymptotic bound conjecture. There exist infinitely many integers k such that

conjecture | open | MathDB 页面 | 原文线索

V2-048075 | MATHDB-329812

Walk-power chromatic-number conjecture for projective cubes

摘录 / EXCERPT: Let r and k be integers with $r\geq k$. For a graph G and a positive integer l, let $G^{(l)}$ be the graph on the same vertex set in which two vertices are adjacent when th...

conjecture | open | MathDB 页面 | 原文线索

V2-048076 | MATHDB-329816

No phase transition for the lower-tail variational problem of bipartite graphs

摘录 / EXCERPT: For $0\leq q\leq p\leq 1$, let $\mathsf{LT}_p(H,q)$ be the problem of minimizing $\mathbb{E}[I_p(W)]$ over graphons W subject to $t(H,W)\leq q^{e(H)}$, where I_p is the rate fu...

conjecture | open | MathDB 页面 | 原文线索

V2-048077 | MATHDB-329817

The bipartite lower-tail conjecture in the sparse limit

摘录 / EXCERPT: Let H be a graph and $0\leq r\leq 1$. Define $h(x)=x\log x-x+1$ and let $\mathsf{LT}(H,r)$ be the problem of minimizing $\mathbb{E}[h(W)]$ over graphons W subject to...

conjecture | open | MathDB 页面 | 原文线索

V2-048078 | MATHDB-329822

Optimality of extremely branched trees for the Basic regression

摘录 / EXCERPT: Let $\mathcal{R}(n)$ be the class of trees on n vertices, and let BP^0 denote the Basic regression with minimum value $BP^{0*}_\mathcal{R}(n)$ over $\mathcal{R}(n)$. A tree is...

conjecture | open | MathDB 页面 | 原文线索

V2-048079 | MATHDB-329830

Termination-time conjecture for random greedy independent sets in hypergraphs

摘录 / EXCERPT: Consider the random greedy independent set algorithm on a sufficiently nice hypergraph, meaning one that is almost uniform, almost regular, and not too sparse. Let Q denote the n...

conjecture | open | MathDB 页面 | 原文线索

V2-048080 | MATHDB-329836

Half-order immersion conjecture for graphs with independence number at most two

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\alpha(G)$ denote its independence number. An immersion of a graph H in G is a subgraph obtainable from G by lifting edges and de...

conjecture | open | MathDB 页面 | 原文线索

V2-048081 | MATHDB-329840

The escaping-trajectory conjecture for isosceles triangle tilings

摘录 / EXCERPT: Isosceles escaping-trajectory conjecture. An isosceles triangle tiling has an escaping trajectory if and only if its vertex angle is not of the form

conjecture | open | MathDB 页面 | 原文线索

V2-048082 | MATHDB-329849

Large-uniformity conjecture for hypergraphs without regular subgraphs

摘录 / EXCERPT: For a positive integer r, let k_r and n_k be thresholds, and let H be an n-vertex k-uniform hypergraph. A full k-star is the family of all k-edges containing a fixe...

conjecture | open | MathDB 页面 | 原文线索

V2-048083 | MATHDB-329850

Mubayi-Verstraëte conjecture for odd-uniform hypergraphs without 2-regular subgraphs

摘录 / EXCERPT: Let k be an integer with $2\nmid k$, and let H be an n-vertex k-uniform hypergraph. A full k-star is the family of all k-edges containing a fixed vertex, and a matching...

conjecture | open | MathDB 页面 | 原文线索

V2-048084 | MATHDB-329854

Conjectured free-boundary solution for the EWF in Case IID

摘录 / EXCERPT: Let $B=(B_1,B_2)$ be the two-dimensional Brownian motion on a filtered probability space $(\mathnormal{\Omega},\mathcal{F},\{\mathcal{F}_t\}_{t\ge 0},\mathbb{P})$, and consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-048085 | MATHDB-329857

Tournament-forest conjecture for transitive tournaments

摘录 / EXCERPT: Let H be a tournament whose vertex set can be partitioned into two transitive sets L and R such that the bipartite graph on L and R, joining $\ell\in L$ to $r\in R$ exact...

conjecture | open | MathDB 页面 | 原文线索

V2-048086 | MATHDB-329866

Monotonicity of the minimum degree of minimal Ramsey graphs in the clique size

摘录 / EXCERPT: Let $s_r(K_k)$ denote the minimum degree of an r-edge-coloured graph that is minimal Ramsey for K_k, with $r\geqslant 3$ and $k\geqslant 3$. Monotonicity conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-048087 | MATHDB-329867

Asymptotic formula for the minimum degree of minimal Ramsey graphs

摘录 / EXCERPT: Let $s_r(K_k)$ be the minimum degree of an r-edge-coloured graph that is minimal Ramsey for K_k, and let $f_{k-1,k}(r)$ denote the Erdős-Rogers function. Asymptotic conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-048088 | MATHDB-329868

Sharp order of growth for the minimum degree of minimal Ramsey graphs for triangles

摘录 / EXCERPT: Let $s_r(K_3)$ denote the minimum degree of an r-edge-coloured graph that is minimal Ramsey for the triangle K_3. Triangle-case conjecture. ... The conjecture asserts that the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-048089 | MATHDB-329885

The Reed-Wood conjecture on minors of 2-regular graphs

摘录 / EXCERPT: Reed-Wood conjecture. Every graph with average degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048090 | MATHDB-329886

The minimum-degree conjecture for disjoint long cycles

摘录 / EXCERPT: Long-cycle packing conjecture. Every graph with minimum degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048091 | MATHDB-329887

Hwang's odd-cycle packing conjecture

摘录 / EXCERPT: Hwang's odd-cycle packing conjecture. If G has at least rk vertices and minimum degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048092 | MATHDB-329888

Hwang's even-cycle packing conjecture

摘录 / EXCERPT: Hwang's even-cycle packing conjecture. If G has at least rk vertices and minimum degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048093 | MATHDB-329889

The average-degree conjecture for even long-cycle packings

摘录 / EXCERPT: Even average-degree cycle-packing conjecture. If G has average degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048094 | MATHDB-329890

The average-degree conjecture for minors of 2-regular graphs

摘录 / EXCERPT: General 2-regular minor conjecture. Every graph with average degree at least

conjecture | open | MathDB 页面 | 原文线索

V2-048095 | MATHDB-329896

Erdős-Rubin-Taylor conjecture on scaling choosability parameters

摘录 / EXCERPT: Let G be a graph, and let (a,b)-choosability mean that every assignment of lists of size a to the vertices of G admits a coloring in which each vertex receives a color from...

conjecture | open | MathDB 页面 | 原文线索

V2-048096 | MATHDB-329912

Sparse graph decomposition conjecture with no overfull set

摘录 / EXCERPT: Let G be a graph and let $k,d\in\mathbb{N}$. A vertex subset $A\subseteq V(G)$ is overfull if ... A graph is (k,d)-sparse when it satisfies the corresponding (k,d)-sparseness i...

conjecture | open | MathDB 页面 | 原文线索

V2-048097 | MATHDB-329924

The maximum-degree conjecture for critical graphs of tree-depth

摘录 / EXCERPT: Let G be a critical graph with tree-depth k, meaning that G has tree-depth k and every proper minor of G has smaller tree-depth. The maximum-degree conjecture. The maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-048098 | MATHDB-329925

The 1-uniqueness conjecture for critical graphs

摘录 / EXCERPT: Let G be a critical graph, meaning that it has tree-depth k for some k and every proper minor of G has smaller tree-depth. A graph is 1-unique if, for every vertex v of...

conjecture | open | MathDB 页面 | 原文线索

V2-048099 | MATHDB-329948

The planarity conjecture for non-rigid irreducible digital images

摘录 / EXCERPT: Planarity conjecture. If X is a non-rigid irreducible digital image that is not isomorphic to C_n, then the adjacency graph of X is not planar.

conjecture | open | MathDB 页面 | 原文线索

V2-048100 | MATHDB-329950

Teschner's conjecture on the upper bondage number of critical graphs

摘录 / EXCERPT: Let $\mathcal{P} \subseteq \mathcal{G}$ be additive and hereditary. A graph G is vertex-$\gamma_{\mathcal{P}}$-critical if its vertex domination number with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-048101 | MATHDB-329980

The Pressing Game Conjecture on successful pressing sequences

摘录 / EXCERPT: Let (G,c) be a bicolored graph, where a press may be performed only at a black vertex, and call a sequence of presses successful if it transforms (G,c) into an all-white, empty...

conjecture | open | MathDB 页面 | 原文线索

V2-048102 | MATHDB-330004

The center conjecture for modular affine vertex algebras

摘录 / EXCERPT: Center conjecture. For $\kappa\neq\kappa_c$, the center of $V^{\kappa}(\mathfrak g)$ coincides with $\mathfrak z_0(V^{\kappa}(\mathfrak g))$. At the critical level, the center of...

conjecture | open | MathDB 页面 | 原文线索

V2-048103 | MATHDB-330008

Non-uniqueness conjecture for percolating clusters below the spectral threshold

摘录 / EXCERPT: Let H be the non-backtracking matrix of the directed graph under consideration, let $\rho(H)$ denote its spectral radius, let p be the bond-occupation probability, and let n...

conjecture | open | MathDB 页面 | 原文线索

V2-048104 | MATHDB-330010

The odd-order dense quasirandom edge-colouring conjecture

摘录 / EXCERPT: For $0<p<1$, suppose there exist $\varepsilon,\eta>0$ and $n_0\in\mathbb{N}$ such that G is a lower-(p,ε)-regular graph on $n\geq n_0$ vertices, n is odd, and...

conjecture | open | MathDB 页面 | 原文线索

V2-048105 | MATHDB-330018

Cycle Ramsey-number conjecture for loose cycles in uniform hypergraphs

摘录 / EXCERPT: Let $\mathcal{C}^k_n$ denote the k-uniform loose cycle with n edges, and let $R(\mathcal{H},\mathcal{G})$ be the Ramsey number for hypergraphs $\mathcal{H}$ and $\mathcal{G}$....

conjecture | open | MathDB 页面 | 原文线索

V2-048106 | MATHDB-330026

The no-BPS finite stabilization conjecture

摘录 / EXCERPT: No-BPS finite stabilization conjecture. The sequence Z_i is well-defined until it reaches a construction with no more fourfold points or sixfold points with triple fold lines; th...

conjecture | open | MathDB 页面 | 原文线索

V2-048107 | MATHDB-330027

The bounded-region finiteness conjecture

摘录 / EXCERPT: Bounded-region finiteness conjecture. On bounded regions, the process stabilizes in finitely many steps.

conjecture | open | MathDB 页面 | 原文线索

V2-048108 | MATHDB-330028

The pre-building Finsler isometric-embedding conjecture

摘录 / EXCERPT: Pre-building Finsler isometric-embedding conjecture. The factorization $f:B^{\rm pre}_{\phi}\rightarrow B$ is an isometric embedding for the Finsler distance.

conjecture | open | MathDB 页面 | 原文线索

V2-048109 | MATHDB-330029

The spectral-network determination conjecture for WKB exponents

摘录 / EXCERPT: WKB spectral-network determination conjecture. The WKB exponents determined by the process coincide with the exponents constructed by the spectral network procedure of Gaiotto, Moo...

conjecture | open | MathDB 页面 | 原文线索

V2-048110 | MATHDB-330038

Conjecture on the distance parameter for three longest paths with two transition vertices

摘录 / EXCERPT: Let G be a connected graph, let $\mathcal{L}(G)$ be the set of longest paths of G, and let $f(G,\mathcal{P})$ be the minimum, over vertices $v\in V(G)$, of...

conjecture | open | MathDB 页面 | 原文线索

V2-048111 | MATHDB-330039

Sublinear bound conjecture for three longest paths

摘录 / EXCERPT: Sublinear bound conjecture. There exists a sublinear non-decreasing function g such that, for every connected graph G of order n and every subset...

conjecture | open | MathDB 页面 | 原文线索

V2-048112 | MATHDB-330040

Lattice VOA strong locality and net-identification conjecture

摘录 / EXCERPT: Lattice VOA conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048113 | MATHDB-330043

Relative-primality conjecture for the D_m auxiliary polynomials

摘录 / EXCERPT: Let $h_{p,m}(t)$ and $\tilde{h}_{p,m}(t)$ be the polynomials defined by the factorizations of the residue polynomials $H_{p,m}(t)$ and $\tilde{H}_{p,m}(t)$ for the D_m orbifold v...

conjecture | open | MathDB 页面 | 原文线索

V2-048114 | MATHDB-330044

Constant term identity of type A_m, I

摘录 / EXCERPT: Let m and p be the parameters of the A_m constant-term identity, let t be an indeterminate, and let $\operatorname{Res}_{x_0,x_1,\ldots,x_{m+1}}$ denote the iterated residu...

conjecture | open | MathDB 页面 | 原文线索

V2-048115 | MATHDB-330045

Constant term identity of type D_m, $m>2$

摘录 / EXCERPT: Let $m>2$, let p be the parameter of the D_m identity, let t be an indeterminate, and let $\operatorname{Res}_{x_0,x_1,\ldots,x_{m+2}}$ denote the iterated residue in the dis...

conjecture | open | MathDB 页面 | 原文线索

V2-048116 | MATHDB-330046

Constant term identity of type D_2, I

摘录 / EXCERPT: Let p be the parameter of the D_2 identity, let t be an indeterminate, and let $\operatorname{Res}_{x_0,x_1,x_2,x_3,x_4}$ denote the iterated residue in the displayed variabl...

conjecture | open | MathDB 页面 | 原文线索

V2-048117 | MATHDB-330048

MSOL recognizability conjecture for 3-connected and general k-outerplanar graphs

摘录 / EXCERPT: Fix k and consider k-outerplanar graphs, with the subclass of 3-connected k-outerplanar graphs. A graph property is recognizable if it is recognized by a finite-state tree au...

conjecture | open | MathDB 页面 | 原文线索

V2-048118 | MATHDB-330049

MSOL recognizability conjecture for (3-connected) Hamiltonian partial k-trees

摘录 / EXCERPT: Fix k and consider partial k-trees, namely graphs of treewidth at most k, together with the subclass of 3-connected partial k-trees. A graph property is recognizable if it...

conjecture | open | MathDB 页面 | 原文线索

V2-048119 | MATHDB-330067

Sigma clique covering growth conjecture for complete multipartite graphs

摘录 / EXCERPT: Sigma clique covering growth conjecture. For every positive integers t and d, if $t\geq f(d)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-048120 | MATHDB-330075

The locating-number conjecture for twin-free graphs

摘录 / EXCERPT: Locating-number conjecture. Every twin-free graph G of order n with no isolated vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048121 | MATHDB-330079

Gould, Horn and Magnant's chorded-cycle packing conjecture

摘录 / EXCERPT: Let c,k be integers with $c\geq 2$ and $k\geq 1$. For an integer $r\geq 0$, an r-chorded cycle is a cycle with r chords. Gould, Horn and Magnant's conjecture. Every graph G...

conjecture | open | MathDB 页面 | 原文线索

V2-048122 | MATHDB-330080

General minimum codegree conjecture for perfect matchings in uniform hypergraphs

摘录 / EXCERPT: Suppose that $k,\ell\in\mathbb N$ with $\ell\leq k-1$. For a k-uniform hypergraph H, let $\delta_\ell(H)$ denote its minimum ℓ-degree, and let $m_\ell(k,n)$ be the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-048123 | MATHDB-330081

Multipartite book-size conjecture

摘录 / EXCERPT: A balanced ℓ-partite graph is an ℓ-partite graph $G=(V_1\cup\cdots\cup V_{\ell},E)$ whose vertex classes have equal size. For distinct parts, let $d(V_i,V_j)$ denote thei...

conjecture | open | MathDB 页面 | 原文线索

V2-048124 | MATHDB-330094

Perfect network coding solution conjecture for receiver packet-intersection instances

摘录 / EXCERPT: Let $\boldsymbol A$ be an instance of the paper's network-coding problem with r receivers, where each receiver wants r data packets and every pair of receivers wants at most on...

conjecture | open | MathDB 页面 | 原文线索

V2-048125 | MATHDB-330096

The secure domination conjecture for digraphs with minimum semidegree at least one

摘录 / EXCERPT: Let D be a digraph of order n. Write δ^{0} for its minimum semidegree, and let $\gamma_{oso}(D)$ and $\gamma_{iso}(D)$ denote its open secure out-domination and secure i...

conjecture | open | MathDB 页面 | 原文线索

V2-048126 | MATHDB-330128

Characterization of competition graphs induced by 123-avoiding permutations

摘录 / EXCERPT: 123-avoidance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048127 | MATHDB-330129

Vertex algebra-conformal net correspondence conjecture

摘录 / EXCERPT: Vertex algebra-conformal net correspondence conjecture. There is a bijective correspondence between completely rational local conformal nets and unitary C_2-cofinite vertex opera...

conjecture | open | MathDB 页面 | 原文线索

V2-048128 | MATHDB-330138

Subparking-function and acyclic-cut orientation conjecture for graph Tutte polynomials

摘录 / EXCERPT: Subparking-function and acyclic-cut orientation conjecture. For any graph G and choice of sink $q\in V(G)$, there exists an ordered, q-rooted spanning tree T of G such that

conjecture | open | MathDB 页面 | 原文线索

V2-048129 | MATHDB-330146

Equivalence conjecture for connected-group loop-group representation categories

摘录 / EXCERPT: Let G be a connected Lie group and let $k\in H^4(BG,\mathbb{Z})$ be a positive level. Consider the categories $\operatorname{Rep}^k(LG)$ defined via vertex operator algebras and...

conjecture | open | MathDB 页面 | 原文线索

V2-048130 | MATHDB-330171

The largest-Laplacian-eigenvalue conjecture for Hamming graphs

摘录 / EXCERPT: Largest-Laplacian-eigenvalue conjecture. If $q\geq 2$ and $j\geq d-\frac{d-1}{q}$, with j even if $q=2$, then λ is the largest Laplacian eigenvalue of $H(d,q,j)$. This co...

conjecture | open | MathDB 页面 | 原文线索

V2-048131 | MATHDB-330178

The 0-sum 3-flow conjecture for 2-edge-connected 5-regular graphs

摘录 / EXCERPT: 0-sum 3-flow conjecture. Every 2-edge connected 5-regular graph admits a 0-sum 3-flow.

conjecture | open | MathDB 页面 | 原文线索

V2-048132 | MATHDB-330190

The permanent bound for 1-factors of simple 3-uniform hypergraphs

摘录 / EXCERPT: Let G be a simple 3-uniform hypergraph, let $M(G)$ be its adjacency matrix, let $\operatorname{per} M(G)$ denote the permanent of that matrix, and let $\varphi(G)$ denote the n...

conjecture | open | MathDB 页面 | 原文线索

V2-048133 | MATHDB-330192

Nešetřil-Ossona de Mendez conjecture on bounded tree-depth components in 3-colorings of planar graphs

摘录 / EXCERPT: Nešetřil-Ossona de Mendez conjecture. There exists a constant t such that every planar graph has a vertex coloring with 3 colors in which every monochromatic component has tree-d...

conjecture | open | MathDB 页面 | 原文线索

V2-048134 | MATHDB-330193

The vertex Folkman number conjecture for maximum parameter 5

摘录 / EXCERPT: The vertex Folkman number conjecture. If $p\geq 4$, then $r_0(p)=2$ and, for every $r\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-048135 | MATHDB-330215

The Kneser graph Hamiltonicity conjecture

摘录 / EXCERPT: For integers n and k with $k\geq 1$ and $n\geq 2k+1$, let the Kneser graph $K(n,k)$ be the graph whose vertices are the k-element subsets of $[n]:=\{1,2,\ldots,n\}$, with an...

conjecture | open | MathDB 页面 | 原文线索

V2-048136 | MATHDB-330216

The bipartite Kneser graph Hamiltonicity conjecture

摘录 / EXCERPT: For integers n and k with $k\geq 1$ and $n\geq 2k+1$, let the bipartite Kneser graph $H(n,k)$ have as vertices all k-element and all $(n-k)$-element subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-048137 | MATHDB-330217

The odd graph Hamiltonicity conjecture

摘录 / EXCERPT: For $k\geq 1$, let the odd graph $K(2k+1,k)$ be the Kneser graph whose vertices are the k-element subsets of $[2k+1]$, with edges between disjoint sets. Odd graph Hamiltonicity c...

conjecture | open | MathDB 页面 | 原文线索

V2-048138 | MATHDB-330221

The asymptotic exponent conjecture for hypergraphs avoiding complete multipartite hypergraphs

摘录 / EXCERPT: Let $r\ge 2$ and let $2\le \ell_1\le \cdots \le \ell_r$ be integers. Let $K^{(r)}_{\ell_1,\dots,\ell_r}$ be the r-uniform hypergraph whose vertex set is a disjoint union...

conjecture | open | MathDB 页面 | 原文线索

V2-048139 | MATHDB-330229

Nonvanishing eigenvector conjecture for random graphs

摘录 / EXCERPT: Let $G(n,p)$ be a random graph with constant p, and let an eigenvector mean an eigenvector of its adjacency matrix. A strong nodal domain is a maximal connected component on whic...

conjecture | open | MathDB 页面 | 原文线索

V2-048140 | MATHDB-330252

The feedback vertex set bound for planar graphs of prescribed girth

摘录 / EXCERPT: Feedback vertex set conjecture for prescribed girth. There exists a feedback vertex set S of G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-048141 | MATHDB-330259

The list colouring conjecture for chromatic index

摘录 / EXCERPT: An instance of list edge-colouring consists of a graph G and an assignment of lists $L:E(G)\rightarrow\mathcal{P}(\mathbb{N})$, where each edge e receives a list $L(e)$ of colo...

conjecture | open | MathDB 页面 | 原文线索

V2-048142 | MATHDB-330262

The bounded-degree transversal conjecture for uniform hypergraphs

摘录 / EXCERPT: Let $k\ge 2$, and let H be a k-uniform hypergraph on n vertices with m edges. Write $\tau(H)$ for its transversal number and $\Delta(H)$ for its maximum degree. The bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-048143 | MATHDB-330263

The linear 4-uniform hypergraph transversal conjecture

摘录 / EXCERPT: Let H be a 4-uniform hypergraph on n vertices with m edges. Recall that H is linear if every two edges intersect in at most one vertex, and let $\tau(H)$ denote its trans...

conjecture | open | MathDB 页面 | 原文线索

V2-048144 | MATHDB-330264

The spectral characterization of geodesic growth for right-angled Coxeter groups on trees

摘录 / EXCERPT: Let T and T' be trees. Assume that they are simultaneously co-spectral, their complements are co-spectral, and their line graphs are co-spectral. Geodesic-growth conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048145 | MATHDB-330266

Amicability-anti-amicability permutation conjecture for Clifford basis matrices

摘录 / EXCERPT: Let $m\geqslant 0$. Consider the 4^m canonical basis matrices of the real representation of the Clifford algebra $\mathbb{R}_{m,m}$; a pair has disjoint support when the supports...

conjecture | open | MathDB 页面 | 原文线索

V2-048146 | MATHDB-330267

Pairing conjecture for non-self-complementary transversal graphs

摘录 / EXCERPT: Let $m\geqslant 0$ and let $\mathbb{R}_{m,m}$ be the real representation of the Clifford algebra. A transversal graph is an induced complete subgraph on 2^m vertices, and its edg...

conjecture | open | MathDB 页面 | 原文线索

V2-048147 | MATHDB-330268

Existence conjecture for edge-colour complements of transversal graphs

摘录 / EXCERPT: Let $m\geqslant 0$ and let $\mathbb{R}_{m,m}$ be the real representation of the Clifford algebra. A transversal graph is an induced complete subgraph on 2^m vertices whose edges...

conjecture | open | MathDB 页面 | 原文线索

V2-048148 | MATHDB-330270

The eigenvalue bound for interpolating graphs

摘录 / EXCERPT: Eigenvalue bound conjecture. For all graphs $G_{n,a}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-048149 | MATHDB-330273

The square-root-of-six ninth-root conjecture for 2-dominating independent sets

摘录 / EXCERPT: The $\sqrt[9]{6}$-conjecture. The maximal number of 2-dominating independent sets in n-vertex graphs is $\Theta(\sqrt[9]{6}^{\,n})$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-048150 | MATHDB-330274

Existence of a graph attaining the asymptotic growth rate of k-dominating independent sets

摘录 / EXCERPT: Attainment conjecture. For every k, there exists a graph G such that

conjecture | open | MathDB 页面 | 原文线索

V2-048151 | MATHDB-330275

Extremality of disjoint unions of complete bipartite graphs for k-dominating independent sets

摘录 / EXCERPT: Complete-bipartite-union conjecture. The maximum number of k-dominating independent sets in an n-vertex graph is attained by the disjoint union of $n/(2k)$ copies of $K_{k,k}$.

conjecture | open | MathDB 页面 | 原文线索

V2-048152 | MATHDB-330276

Non-polynomial growth for k-dominating independent sets in incidence graphs of projective planes

摘录 / EXCERPT: Projective-plane incidence-graph conjecture. The function $\text{mi}_k(n,\mathcal{F})$ is not bounded above by any polynomial in n.

conjecture | open | MathDB 页面 | 原文线索

V2-048153 | MATHDB-330281

Daskalakis-Mehta-Papadimitriou's bounded-cycle-or-undominated-set conjecture

摘录 / EXCERPT: Daskalakis-Mehta-Papadimitriou's conjecture. There are integers k and l such that every digraph either has a cycle of length at most k or an undominated set of l vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-048154 | MATHDB-330282

Myers's bounded-cycle-or-pair-of-undominated-vertices conjecture

摘录 / EXCERPT: Myers's conjecture. There is an integer k such that every digraph either has a cycle of length at most k or an undominated set of two vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-048155 | MATHDB-330284

Doubling conjecture for dimension-banned graphs

摘录 / EXCERPT: Let a graph be banned in dimension k when it has no orthogonal representation in dimension k. If a vertex of valence k in a graph banned in dimension k is doubled, then the...

conjecture | open | MathDB 页面 | 原文线索

V2-048156 | MATHDB-330285

Isomorphic-vertex conjecture for critical banned graphs

摘录 / EXCERPT: Let a graph be critical banned in dimension d when it is banned in dimension d but deleting any vertex lowers the required dimension. Two vertices are isomorphic when deleting...

conjecture | open | MathDB 页面 | 原文线索

V2-048157 | MATHDB-330286

Matching-deletion conjecture for critical graph dimension

摘录 / EXCERPT: Let G be a critical graph (not banned), let M be a maximal matching, and suppose that its orthogonal representation dimension satisfies $\rho ^\text{\perp}(G)=d$. The matching-...

conjecture | open | MathDB 页面 | 原文线索

V2-048158 | MATHDB-330287

Self-complementary vertex-transitive graph conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a self-complementary, vertex-transitive graph with $|V|=n$. The self-complementary vertex-transitive conjecture. Its orthogonal representation dimension is ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-048159 | MATHDB-330288

Complementary-dimension bounds conjecture

摘录 / EXCERPT: Let G be a graph, let $\overline G$ be its complement, and let $\rho ^\perp$ denote orthogonal representation dimension. The complementary-dimension bounds conjecture. The two di...

conjecture | open | MathDB 页面 | 原文线索

V2-048160 | MATHDB-330299

The nonexistence conjecture for nontrivial uniquely cycle-saturated graphs

摘录 / EXCERPT: Nonexistence conjecture. For $t\ge 6$, there are no nontrivial uniquely C_t-saturated graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-048161 | MATHDB-330331

Shearer's conjecture on properly colored copies of graphs with few cherries

摘录 / EXCERPT: Shearer's conjecture. For every two integers s and k, there exists an integer n_0 such that, if $n\ge n_0$ and G is an n-vertex graph with at most sn cherries, then any...

conjecture | open | MathDB 页面 | 原文线索

V2-048162 | MATHDB-330334

Finite classification conjecture for graphs of bounded algebraic co-rank

摘录 / EXCERPT: For $k\geq1$, let ... and let $\mathcal{T}_{\delta}^*(G)$ denote the set of induced subgraphs of one graph in $\mathcal{T}_{\delta}(G)$. Finite classification conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-048163 | MATHDB-330335

Eventual family conjecture for nested graphs of bounded algebraic co-rank

摘录 / EXCERPT: Let $\mathcal{G}=\{G_i\}_{i=1}^{\infty}$ be an infinite family of simple graphs such that G_i is a proper induced subgraph of G_j for all $i<j$. Eventual family conjecture. Eit...

conjecture | open | MathDB 页面 | 原文线索

V2-048164 | MATHDB-330336

Cograph cotree lower-bound conjecture

摘录 / EXCERPT: Let C be a cograph, and let $\widetilde{T}$ be the tree obtained from its cotree by erasing the leaves. Cograph cotree lower-bound conjecture. ... The proposed bound would sharpen...

conjecture | open | MathDB 页面 | 原文线索

V2-048165 | MATHDB-330337

Twin-pair conjecture for vertex-invariant algebraic co-rank

摘录 / EXCERPT: Let G be a simple graph, let $v\in V(G)$, and let $G-v$ denote the graph obtained by deleting v. Twin-pair conjecture. If ... then G has at least one pair of twin vertices. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-048166 | MATHDB-330354

Harutyunyan's acyclic-set conjecture for planar oriented graphs

摘录 / EXCERPT: Let an oriented graph be a digraph without loops and multiple arcs, and let an acyclic set be a set of vertices inducing a subgraph with no directed cycles. Harutyunyan's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048167 | MATHDB-330367

Bollobás-Leader directed vertex-disjoint paths conjecture

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, identified with the power set $\mathcal{P}[n]$. For disjoint subsets $A,B\subseteq Q_n$, let $\overrightarrow{p}_v(A,B)$ denote the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-048168 | MATHDB-330404

Gross-Ruan topological conjecture for special Lagrangian torus fibrations

摘录 / EXCERPT: Gross-Ruan topological conjecture. The discriminant and monodromy data of the fibration satisfy properties (i)--(iv) above, including the specified positive and negative vertex mod...

conjecture | open | MathDB 页面 | 原文线索

V2-048169 | MATHDB-330411

Chen-Han-O-Shan-Tsuchiya conjecture on spanning Halin subgraphs

摘录 / EXCERPT: Let $\mathcal{H}$ be a forbidden pair, meaning a family of two forbidden subgraphs. For two families $\mathcal{H}_1$ and $\mathcal{H}_2$ of forbidden subgraphs, write...

conjecture | open | MathDB 页面 | 原文线索

V2-048170 | MATHDB-330472

Kalai's Turán conjecture for tight uniform hypergraph trees

摘录 / EXCERPT: Let $r\geq 2$ and let ${\cal H}$ be a tight r-tree on v vertices. The Turán number ${\rm\bf ex}(n,{\cal H})$ is the largest size of an r-uniform hypergraph on n vertices th...

conjecture | open | MathDB 页面 | 原文线索

V2-048171 | MATHDB-330473

Finiteness conjecture for DHS complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be the complete bipartite graph with positive integers m and n. A mixed graph is DHS if it is determined by its Hermitian adjacency spectrum among mixed graphs. F...

conjecture | open | MathDB 页面 | 原文线索

V2-048172 | MATHDB-330488

Liang's matching and node-disjoint link conjecture for subcubic bipartite graphs

摘录 / EXCERPT: Liang's conjecture. If every node in S has degree at most 4 and every node in T has degree at most 3, then G has a matching M and a family $\mathcal{F}$ of node-disjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-048173 | MATHDB-330491

The N_+-Perfect Graph Conjecture

摘录 / EXCERPT: N_+-Perfect Graph Conjecture. The stable set polytope of every N_+-perfect graph can be described by facet-defining inequalities with near-bipartite support.

conjecture | open | MathDB 页面 | 原文线索

V2-048174 | MATHDB-330505

Cao-Dehmer-Schaumann's extremal degree-based entropy conjecture for trees

摘录 / EXCERPT: Cao-Dehmer-Schaumann's conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-048175 | MATHDB-330506

Cao and Dehmer's monotonicity conjecture for degree-based graph entropy

摘录 / EXCERPT: Cao and Dehmer's conjecture. For $I>0$, $I_{f^I_d}(G)$ is a monotonically increasing function of I for connected graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-048176 | MATHDB-330507

Chen-Dehmer-Shi's extremal entropy conjecture for trees

摘录 / EXCERPT: Chen-Dehmer-Shi's conjecture. The balanced double star and the comet $CS(n,t_0)$ attain the maximum and minimum values of $I_{f^n_2}(T)$, respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-048177 | MATHDB-330508

Dehmer-Kraus's extremal dendrimer entropy conjecture

摘录 / EXCERPT: Dehmer-Kraus's conjecture. For every sequence

conjecture | open | MathDB 页面 | 原文线索

V2-048178 | MATHDB-330510

Dehmer-Kraus's two-tailed-comet conjecture for tree entropies

摘录 / EXCERPT: Dehmer-Kraus's conjecture. Among all trees $T_{n,d}$ with $d\ll n$, the two-tailed comet

conjecture | open | MathDB 页面 | 原文线索

V2-048179 | MATHDB-330511

Kraus-Dehmer-Schaumann's minimal exponential-entropy graph conjecture

摘录 / EXCERPT: Kraus-Dehmer-Schaumann's conjecture. A graph minimizing $I_{f^2}(G)$ for the exponential sequence is a tree. Moreover, it is a generalized star of diameter approximately...

conjecture | open | MathDB 页面 | 原文线索

V2-048180 | MATHDB-330512

Kraus-Dehmer-Schaumann's highly connected minimal-entropy conjecture

摘录 / EXCERPT: Kraus-Dehmer-Schaumann's conjecture. A minimal graph for $I_{f^\sigma}(G)$ is highly connected. In particular, a minimal graph on n vertices has at least $m\geq n/2$ vertices of...

conjecture | open | MathDB 页面 | 原文线索

V2-048181 | MATHDB-330515

Original Burr-Erdős conjecture for pairs of bounded-arboricity graphs

摘录 / EXCERPT: For graphs H_1 and H_2, let $r(H_1,H_2)$ be the least integer n such that every red-blue edge-coloring of K_n contains a red copy of H_1 or a blue copy of H_2. The arbo...

conjecture | open | MathDB 页面 | 原文线索

V2-048182 | MATHDB-330517

Brualdi and Massey's incidence coloring conjecture

摘录 / EXCERPT: An incidence in a graph G is a pair (v,e) where v is a vertex of G and e is an edge of G incident to v; two incidences are adjacent if they share a vertex, share an e...

conjecture | open | MathDB 页面 | 原文线索

V2-048183 | MATHDB-330518

Pai et al.'s hypercube incidence coloring conjecture

摘录 / EXCERPT: For an integer $n\geq 1$, let Q_n be the n-dimensional hypercube, with vertex set $\{0,1\}^n$ and edges between vertices differing in exactly one coordinate. Let $\chi_i(Q_n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-048184 | MATHDB-330521

Borsuk's transversality conjecture

摘录 / EXCERPT: Let $K\subseteq\mathbb R^d$ be a set of points, let D be the set of pairs of points in K at the diameter distance, and call (K,D) general if every continuous deformation of t...

conjecture | open | MathDB 页面 | 原文线索

V2-048185 | MATHDB-330522

Stress-free geometric graph conjecture

摘录 / EXCERPT: Let G be a geometric graph embedded in $\mathbb R^d$, with edges joining pairs at the diameter distance. Call G stress-free if there are no not-all-zero edge weights whose weig...

conjecture | open | MathDB 页面 | 原文线索

V2-048186 | MATHDB-330525

Pinchasi-Sharir conjecture on planar tangent graphs

摘录 / EXCERPT: A tangent graph has one vertex for each disc and an edge when the corresponding discs are tangent. Pinchasi-Sharir tangent-incidence conjecture. (i) Planar tangent graphs with n...

conjecture | open | MathDB 页面 | 原文线索

V2-048187 | MATHDB-330526

Ringle circle problem

摘录 / EXCERPT: A tangent graph of a finite collection of circles in the plane has one vertex for each circle and an edge when two circles are tangent. Assume that no more than two circles pass th...

conjecture | open | MathDB 页面 | 原文线索

V2-048188 | MATHDB-330545

Transitivity conjecture for the Galois action on enharmonic functions

摘录 / EXCERPT: Let G be a graph with boundary, and suppose the energies and boundary values are rational, with positive energies. Let Σ_u denote the set of compatible orientations, equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-048189 | MATHDB-330624

Diadem-nucleus characterization of König-Egerváry graphs

摘录 / EXCERPT: Diadem-nucleus conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048190 | MATHDB-330625

The ker-diadem inequality for graphs

摘录 / EXCERPT: Ker-diadem inequality. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-048191 | MATHDB-330641

Simplicity conjecture for the intermediate affine vertex algebra quotients

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type D_l with $l\geqslant 4$, or of type E_6, E_7, or E_8. For $n\in\mathbb{Z}_{\geqslant0}$, let k_n be defined by ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048192 | MATHDB-330647

Path formation conjecture for vertex-reinforced non-backtracking random walks

摘录 / EXCERPT: Let X be a vertex-reinforced non-backtracking random walk on a connected finite undirected graph, with reinforcement function $W(k)=(1+k)^\alpha$. Let $C=\{i_1,\dots,i_\ell\}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-048193 | MATHDB-330652

The spectral determination conjecture for Kite graphs

摘录 / EXCERPT: For integers p and q with $q>2$, let $Kite_{p,q}$ denote the graph obtained by appending a complete graph K_p to a pendant vertex of a path P_q. A graph is determined by ad...

conjecture | open | MathDB 页面 | 原文线索

V2-048194 | MATHDB-330655

Exoo's conjecture on the chromatic number of epsilon-unit distance graphs

摘录 / EXCERPT: Let G_ε be the graph with vertex set $\mathbb{R}^2$ in which two points are adjacent when their distance d satisfies $1-\varepsilon\leq d\leq 1+\varepsilon$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048195 | MATHDB-330673

Garijo, González and Márquez's location-domination conjecture for twin-free graphs

摘录 / EXCERPT: Garijo, González and Márquez's conjecture. There exists an integer n_1 such that for any $n\geq n_1$, the maximum value of the location-domination number of a connected twin-free...

conjecture | open | MathDB 页面 | 原文线索

V2-048196 | MATHDB-330687

Edge-disjoint triangle conjecture for K4-free graphs

摘录 / EXCERPT: Let G be a K_4-free graph on n vertices with $t_2(n)+m$ edges, where $t_2(n)=\lfloor n^2/4\rfloor$ is the number of edges in the 2-partite Turán graph on n vertices. Edge...

conjecture | open | MathDB 页面 | 原文线索

V2-048197 | MATHDB-330688

Greedy-partition bound for K4-free graphs

摘录 / EXCERPT: Let G be a K_4-free graph with n vertices, e edges, and an arbitrary greedy partition P of size $r=r(P)$. Let t and t_e denote the quantities used in the paper's func...

conjecture | open | MathDB 页面 | 原文线索

V2-048198 | MATHDB-330690

Hu, Qi and Shao's monotonicity conjecture for Laplacian eigenvalues of power hypergraphs

摘录 / EXCERPT: Hu, Qi and Shao's conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-048199 | MATHDB-330699

Conditional Turán number conjecture for linear paths and triangles

摘录 / EXCERPT: Let P^3_3 be the 3-uniform linear path with three edges, C^3_3 the 3-uniform triangle, and M^3_2 the matching of two 3-edges. For an F-free 3-uniform hypergraph that is not...

conjecture | open | MathDB 页面 | 原文线索

V2-048200 | MATHDB-330715

Balogh-Bollobás asymptotic conjecture for bootstrap percolation in the hypercube

摘录 / EXCERPT: Let $r\geq 3$ be fixed, and let Q_d be the graph with vertex set $\{0,1\}^d$ in which two vertices are adjacent when they differ in exactly one coordinate. For the r-neighbour...

conjecture | open | MathDB 页面 | 原文线索

V2-048201 | MATHDB-330764

The extremal number conjecture for powers of balanced rooted trees

摘录 / EXCERPT: Let (T,R) be a balanced rooted tree, let T^p be the graph formed from the union of p distinct labelled copies of T that agree on the roots R and are otherwise disjoint, a...

conjecture | open | MathDB 页面 | 原文线索

V2-048202 | MATHDB-330804

EKR conjecture for depth-two claws

摘录 / EXCERPT: Depth-two-claw EKR conjecture. Let n and r be positive integers, with $n > r$, and let G be a depth-two claw with n leaves. Then G is r-EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-048203 | MATHDB-330821

Bang-Jensen-Gutin-Li hamiltonicity conjecture for dominating pairs

摘录 / EXCERPT: Bang-Jensen-Gutin-Li conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048204 | MATHDB-330830

Schur-positivity conjecture for linear interval hypergraphs

摘录 / EXCERPT: Let H be a hypergraph with chromatic symmetric function X_H. An interval hypergraph is a hypergraph isomorphic to one with vertex set $[n]$ in which every edge has the form...

conjecture | open | MathDB 页面 | 原文线索

V2-048205 | MATHDB-330841

Ulam reconstruction conjecture for infinite graphs with non-isomorphic vertex-deleted subgraphs

摘录 / EXCERPT: Let G) be an infinite graph with vertex set E. For $x\in E$, write $G|E\setminus\{x\}$ for the subgraph induced by deleting x, and write $\sim$ for graph isomorphism. Ulam re...

conjecture | open | MathDB 页面 | 原文线索

V2-048206 | MATHDB-330842

Ulam reconstruction conjecture from pairwise isomorphic vertex-deleted subgraphs

摘录 / EXCERPT: Let G be an infinite graph with vertex set E. For distinct vertices $x,y\in E$, write $G|E\setminus\{x\}$ for the subgraph induced by deleting x, and write $\sim$ for graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-048207 | MATHDB-330843

Ulam reconstruction conjecture for infinite graphs of finite vertex degrees

摘录 / EXCERPT: Let G be an infinite graph with vertex set E, and let $\deg(x)$ denote the degree of x. Ulam reconstruction conjecture. If ... then G is Ulam-reconstructible. The source give...

conjecture | open | MathDB 页面 | 原文线索

V2-048208 | MATHDB-330858

The Q-index extremal conjecture for $K_{t,s+1}$-free graphs

摘录 / EXCERPT: Let G be a graph of order n, with adjacency matrix A and diagonal degree matrix D. Its signless Laplacian is $Q(G)=A+D$, and its Q-index $q(G)$ is the largest eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-048209 | MATHDB-330873

The digital capacity conjecture for the analogue communication network

摘录 / EXCERPT: Let the communication network be the one depicted in figure labelled analogue, and let its digital capacity denote the largest rate achievable using digital communication. Digital...

conjecture | open | MathDB 页面 | 原文线索

V2-048210 | MATHDB-330874

The analogue-over-digital capacity ratio conjecture

摘录 / EXCERPT: Analogue-over-digital ratio conjecture. The maximal ratio in the class of problems where digital communication outperforms analogue communication is strictly larger than 1. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-048211 | MATHDB-330879

Goubko's BFS-tree conjecture for distance spectral radius

摘录 / EXCERPT: Let a tree degree sequence be fixed, and let $BFS(\mathbf{d})$ denote the breadth-first-search (greedy) tree generated by it. The BFS-tree conjecture for distance spectral radius....

conjecture | open | MathDB 页面 | 原文线索

V2-048212 | MATHDB-330880

Goubko's BFS-tree conjecture for terminal distance spectral radius

摘录 / EXCERPT: Let $\mathbf{d}$ be a tree degree sequence, and let $BFS(\mathbf{d})$ denote its breadth-first-search (greedy) tree. For a graph G, let $RD(G)$ be its terminal distance matrix, a...

conjecture | open | MathDB 页面 | 原文线索

V2-048213 | MATHDB-330881

Goubko's extremal degree-sequence conjecture for terminal distance error

摘录 / EXCERPT: For a tree degree sequence $\mathbf{d}$, write $BFS(\mathbf{d})$ for the associated breadth-first-search tree and define ... Let $TErr$ denote the relative error between the upper an...

conjecture | open | MathDB 页面 | 原文线索

V2-048214 | MATHDB-330883

Alon's acyclic edge-colouring conjecture for graphs

摘录 / EXCERPT: Let G be a graph, and let $a'(G)$ denote its acyclic chromatic index and $\Delta(G)$ its maximum degree. Alon's conjecture. ... This is the foundational conjecture in acyclic edge...

conjecture | open | MathDB 页面 | 原文线索

V2-048215 | MATHDB-330884

Vertex-chromatic-index conjecture for subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning that every vertex of G has degree at most 3. Let $\chi_{\nu}(G)$ denote the new vertex chromatic number introduced in the paper. Subcubic v...

conjecture | open | MathDB 页面 | 原文线索

V2-048216 | MATHDB-330886

Non-embeddability conjecture for De Bruijn and Kautz graphs

摘录 / EXCERPT: De Bruijn-Kautz non-embeddability conjecture. (i) All $Br(m,n)$ with $(m,n)\neq(2,2)$ are not 2-tr.embeddable. (ii) All $Ka(m,n)$ with $m\geq 3$ are not 2-tr.embeddable.

conjecture | open | MathDB 页面 | 原文线索

V2-048217 | MATHDB-330887

Non-embeddability conjecture for cube-connected cycles

摘录 / EXCERPT: Cube-connected-cycles non-embeddability conjecture. For the tested values $n=3,4,5$, CCC_n is not $(n+1)$-tr.embeddable.

conjecture | open | MathDB 页面 | 原文线索

V2-048218 | MATHDB-330888

Uniform height-tail estimate for Boltzmann triangulations

摘录 / EXCERPT: Let $\mathbf{t}$ be a Boltzmann triangulation with a boundary of perimeter $p\geq 1$, let $\mathsf{Height}(\mathbf{t})$ denote the maximal height of a vertex, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048219 | MATHDB-330889

The near-perfect matching threshold conjecture with divisibility barriers

摘录 / EXCERPT: Let $k\ge2$, let $1\le d\le k-1$, and for $0\le\ell\le k-1$ define ... where $H_\ell^j(x)$ is the divisibility-barrier k-graph introduced in the paper. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-048220 | MATHDB-330908

The minimum-degree conjecture for distance-preserving graphs

摘录 / EXCERPT: Minimum-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048221 | MATHDB-330909

The induced-cycle conjecture for distance-preserving graphs

摘录 / EXCERPT: Induced-cycle conjecture. If G does not contain an induced cycle of length 5 or greater, then G is a distance-preserving graph.

conjecture | open | MathDB 页面 | 原文线索

V2-048222 | MATHDB-330915

Conjecture on the maximizing configuration for path graphs

摘录 / EXCERPT: Let G be a path with $k-1$ edges, where $k\geq 3$. For a configuration $\mathbf{n}^{\max}=(n^{\max}_j:j=1,\dots,k-1)$ in $n\,\mathcal{S}_G$ maximizing $p_G(\mathbf{n})$, define .....

conjecture | open | MathDB 页面 | 原文线索

V2-048223 | MATHDB-330958

Orientation conjecture for triangulations on orientable surfaces

摘录 / EXCERPT: Orientation conjecture. T admits an orientation of its edges such that every vertex has outdegree at least 3, divisible by 3, and such that there is no oriented non-contracti...

conjecture | open | MathDB 页面 | 原文线索

V2-048224 | MATHDB-330961

One-high-vertex conjecture for fixable trees

摘录 / EXCERPT: One-high-vertex conjecture. The tree T is f-fixable if $f(v)=d_T(v)$ for at most one non-leaf vertex v of T.

conjecture | open | MathDB 页面 | 原文线索

V2-048225 | MATHDB-330962

Moonshine conjecture on strict degree surplus and fixability

摘录 / EXCERPT: Moonshine conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048226 | MATHDB-330963

Star-with-one-edge-subdivided fixability conjecture

摘录 / EXCERPT: Star-with-one-edge-subdivided conjecture. The graph G is L-fixable if at least one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-048227 | MATHDB-330972

The monochromatic path partition conjecture for infinite complete bipartite graphs

摘录 / EXCERPT: Infinite bipartite monochromatic path conjecture. The vertices can be partitioned into $2r-1$ disjoint monochromatic paths.

conjecture | open | MathDB 页面 | 原文线索

V2-048228 | MATHDB-330982

Extremal sum-color cost conjecture for k-trees

摘录 / EXCERPT: A k-tree is a graph obtained from K_k by iteratively adding a vertex whose neighborhood is a k-clique in the existing graph. The join $G\vee H$ is obtained from the disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-048229 | MATHDB-330993

Weyl-like perturbation conjecture for hypermatrix eigenvalues

摘录 / EXCERPT: Let A and B be hypermatrices such that ... for some norm $\lVert\cdot\rVert$ (or spectral radius), where $\epsilon>0$. Weyl-like perturbation conjecture. There is a bijection...

conjecture | open | MathDB 页面 | 原文线索

V2-048230 | MATHDB-330999

Ryjáček's conjecture on locally connected graphs

摘录 / EXCERPT: Ryjáček's conjecture. Every locally connected graph is weakly pancyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-048231 | MATHDB-331004

Yan's spanning-tree formula for line graphs of subdivisions

摘录 / EXCERPT: Let G be a connected graph of order $n+s$ and size $m+s$, in which s vertices have degree 1 and all other vertices have degree k. Let $S(G)$ be the graph obtained by insert...

conjecture | open | MathDB 页面 | 原文线索

V2-048232 | MATHDB-331023

Chromatic number conjecture for matching graphs

摘录 / EXCERPT: Let G be a connected graph, let r be a positive integer, and let ${\rm ex}(G,rK_2)$ denote the maximum number of edges in an rK_2-free subgraph of G. The matching graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048233 | MATHDB-331026

Busch-Ferrara-Hartke-Jacobson-Kaul-West conjecture on edge-disjoint perfect matchings

摘录 / EXCERPT: Busch-Ferrara-Hartke-Jacobson-Kaul-West conjecture. The sequences π and $\pi-k$ are graphic and of even length if and only if π potentially has k edge-disjoint 1-factor...

conjecture | open | MathDB 页面 | 原文线索

V2-048234 | MATHDB-331030

Symmetric-matrix extension of the quadratic theorem for totally real algebraic integers

摘录 / EXCERPT: Work over $\mathbb{C}$. For a totally real algebraic integer λ of degree $d>1$, let $E_s(n,\lambda)$ be the maximum multiplicity of λ as an eigenvalue of an n-ver...

conjecture | open | MathDB 页面 | 原文线索

V2-048235 | MATHDB-331034

Bordenave-Chafaï conjecture for random regular digraph matrices

摘录 / EXCERPT: Let A_n be a random regular digraph matrix, meaning a uniformly random element of $\mathop{\mathrm{M}}_n(d)$, where $\mathop{\mathrm{M}}_n(d)$ is the set of $n\times n$ zero-one...

conjecture | open | MathDB 页面 | 原文线索

V2-048236 | MATHDB-331056

The 1-2-3 Conjecture on neighbour sum distinguishing edge colourings

摘录 / EXCERPT: 1-2-3 Conjecture. Every graph G containing no isolated edges admits a neighbour sum distinguishing 3-edge colouring.

conjecture | open | MathDB 页面 | 原文线索

V2-048237 | MATHDB-331065

The Kempe-coloring clique-minor conjecture

摘录 / EXCERPT: Let G be a graph with a Kempe-coloring ${\frak C}$ of size k, and let a transversal T of ${\frak C}$ contain one vertex from each color class. A clique minor traversed by T...

conjecture | open | MathDB 页面 | 原文线索

V2-048238 | MATHDB-331074

Asymptotic smallest Q-eigenvalue conjecture for K_{r+1}-free graphs

摘录 / EXCERPT: Let c_{r} be the asymptotic extremal constant associated with the smallest signless Laplacian eigenvalue of K_{r+1}-free graphs. Asymptotic smallest Q-eigenvalue conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-048239 | MATHDB-331104

Asymptotic unit bar visibility number conjecture for complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be the complete bipartite graph with $m\ge n\ge 2$, and let $ub(G)$ denote the unit bar visibility number of a graph G. Asymptotic unit bar visibility conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-048240 | MATHDB-331133

Żak's graph-packing conjecture

摘录 / EXCERPT: Let G_1 and G_2 be graphs on n vertices, and let $E(G_i)$ and $\Delta(G_i)$ denote the edge set and maximum degree of G_i, respectively. Two graphs pack if there is a bijec...

conjecture | open | MathDB 页面 | 原文线索

V2-048241 | MATHDB-331142

Infinitely many Ramsey-close but non-equivalent pairs

摘录 / EXCERPT: Ramsey-closeness conjecture. There are infinitely many pairs of Ramsey-infinite graphs which are Ramsey close but not Ramsey equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-048242 | MATHDB-331144

The chromatic-threshold conjecture for graphs with chromatic number four

摘录 / EXCERPT: Let H be a graph, let $\chi(H)$ denote its chromatic number, let $m_2(H)$ denote its 2-density, and let $\delta_\chi(H,p)$ be the chromatic threshold of H at edge-probability par...

conjecture | open | MathDB 页面 | 原文线索

V2-048243 | MATHDB-331147

Kupek, Petrovic, and Yero's torus EOD-graph conjecture

摘录 / EXCERPT: Kupek, Petrovic, and Yero's torus EOD-graph conjecture. Let r and t be integers such that $r\geq 3$ and $t\geq 3$. The torus $C_r\,\Box\,C_t$ is an EOD-graph if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-048244 | MATHDB-331181

All-time proportionality conjecture for the wireless-network size estimator

摘录 / EXCERPT: All-time proportionality conjecture. The estimator is proportional to the BLUE and can be written as

conjecture | open | MathDB 页面 | 原文线索

V2-048245 | MATHDB-331182

The r-adjacent strong chromatic index conjecture

摘录 / EXCERPT: The r-adjacent strong chromatic index conjecture. For each positive integer r there exist constants δ_0 and C such that

conjecture | open | MathDB 页面 | 原文线索

V2-048246 | MATHDB-331186

Minimum codegree conjecture for loose cycle factors

摘录 / EXCERPT: Loose-cycle factor conjecture. If n is sufficiently large and

conjecture | open | MathDB 页面 | 原文线索

V2-048247 | MATHDB-331201

The Normal Graph Conjecture

摘录 / EXCERPT: Let G be a graph. A graph is normal if its vertex set has coverings by cliques and by independent sets such that every clique in the first covering intersects every independent s...

conjecture | open | MathDB 页面 | 原文线索

V2-048248 | MATHDB-331202

The strong normal graph characterization conjecture

摘录 / EXCERPT: Let G be a graph. A graph is strongly normal if every induced subgraph of G is normal; let $\overline{G}$ denote the complement of G, and let C_k denote the cycle on k ve...

conjecture | open | MathDB 页面 | 原文线索

V2-048249 | MATHDB-331214

Rödl-Ruciński conjecture relating Hamilton cycles and perfect matchings

摘录 / EXCERPT: Let $k\ge3$ and $1\le d\le k-2$. For an n-vertex k-uniform hypergraph, let $h_d(k,n)=h_d^{k-1}(k,n)$ be the smallest integer such that minimum d-degree at least $h_d(k,n)$ gu...

conjecture | open | MathDB 页面 | 原文线索

V2-048250 | MATHDB-331225

Flandrin et al.'s neighbour sum distinguishing index conjecture

摘录 / EXCERPT: Let G be a connected graph of order at least three, different from the cycle C_5. Write $\chi'_{\sum}(G)$ for its neighbour sum distinguishing index and $\Delta(G)$ for its max...

conjecture | open | MathDB 页面 | 原文线索

V2-048251 | MATHDB-331226

The smaller matching threshold conjecture

摘录 / EXCERPT: Smaller matching threshold conjecture. Given $1\leq d\leq k-2$, there exist n_0 and C such that

conjecture | open | MathDB 页面 | 原文线索

V2-048252 | MATHDB-331277

Bordeaux Conjecture on 3-coloring planar graphs

摘录 / EXCERPT: Let G be a planar graph. Two cycles are intersecting if they have at least one common vertex. Bordeaux Conjecture. Every planar graph without intersecting 3-cycles and without...

conjecture | open | MathDB 页面 | 原文线索

V2-048253 | MATHDB-331278

Conjecture on consecutive odd cycle lengths in 2-connected non-bipartite graphs

摘录 / EXCERPT: Conjecture on consecutive odd cycle lengths. If G is a 2-connected non-bipartite graph with minimum degree at least $k+1$, then G contains $\lceil k/2\rceil$ cycles with consec...

conjecture | open | MathDB 页面 | 原文线索

V2-048254 | MATHDB-331305

The 1-factorization conjecture for regular graphs

摘录 / EXCERPT: Let G be a regular graph of even order. The 1-factorization conjecture. Every regular graph of even order with sufficiently high degree is 1-factorizable. Here, 1-factorizabl...

conjecture | open | MathDB 页面 | 原文线索

V2-048255 | MATHDB-331312

Qian's conjecture on extremal functions of disjoint graph minors

摘录 / EXCERPT: Qian's conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-048256 | MATHDB-331313

Conjecture on partitioning a dense graph into two dense spanning subgraphs

摘录 / EXCERPT: Partition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048257 | MATHDB-331314

Conjecture on the extremal function of 2-regular graphs

摘录 / EXCERPT: Extremal-function conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048258 | MATHDB-331327

Mazzuoccolo's excessive-index conjecture for r-graphs

摘录 / EXCERPT: Let r be a positive integer, let G be an r-graph, and let $\chi'_e(G)$ denote the excessive index of G, namely the minimum number of 1-factors needed to cover $E(G)$. Mazzu...

conjecture | open | MathDB 页面 | 原文线索

V2-048259 | MATHDB-331331

2-degeneracy conjecture for strictly convex vertex and edge-midpoint drawings

摘录 / EXCERPT: 2-degeneracy conjecture. If $G\in\mathcal{G}_s^s$, then G is 2-degenerate; that is, every non-empty induced subgraph of G has a vertex of degree at most 2.

conjecture | open | MathDB 页面 | 原文线索

V2-048260 | MATHDB-331332

Subdivision conjecture for strictly convex vertex and edge-midpoint drawings

摘录 / EXCERPT: Subdivision conjecture. Every graph has a subdivision obtained by multiplying subdividing its edges such that the resulting graph belongs to $\mathcal{G}_s^s$.

conjecture | open | MathDB 页面 | 原文线索

V2-048261 | MATHDB-331333

Lavrov-Loh conjecture on Hamiltonian monotone paths in random edge-orderings

摘录 / EXCERPT: Lavrov-Loh conjecture. With probability tending to 1 as $n\to\infty$, a random edge-ordering of K_n contains a Hamiltonian monotone path.

conjecture | open | MathDB 页面 | 原文线索

V2-048262 | MATHDB-331341

Meshkat-Sullivant's collapsed-graph conjecture for expected dimension

摘录 / EXCERPT: Let G be a graph with n vertices, $2n-2$ edges, and an exchange with i. The collapsed graph G' is obtained by identifying vertices 1 and i. Assume that G' has $2n-4$...

conjecture | open | MathDB 页面 | 原文线索

V2-048263 | MATHDB-331342

Expected-dimension conjecture for subdividing an edge in a graph

摘录 / EXCERPT: Let G be a graph on $n-1$ vertices, and let $k\to l$ be an edge in G. Construct a graph G' on n vertices by subdividing the edge $k\to l$: add vertex n and replace the ed...

conjecture | open | MathDB 页面 | 原文线索

V2-048264 | MATHDB-331377

The middle-level graph conjecture for vertex-transitive partial cubes of girth six

摘录 / EXCERPT: Middle-level graph conjecture. The graph G is a middle-level graph.

conjecture | open | MathDB 页面 | 原文线索

V2-048265 | MATHDB-331389

Erdős-Gyárfás monochromatic-same-color path cover conjecture

摘录 / EXCERPT: Erdős-Gyárfás same-color path cover conjecture. The vertex set of every 2-colored K_n can be covered by at most

conjecture | open | MathDB 页面 | 原文线索

V2-048266 | MATHDB-331390

Grinshpun-Sárközy linear bound for powers-of-cycles covers

摘录 / EXCERPT: Grinshpun-Sárközy powers-of-cycles conjecture. The vertex set of every 2-colored complete graph can be covered by at most

conjecture | open | MathDB 页面 | 原文线索

V2-048267 | MATHDB-331391

Sárközy's cycle partition conjecture for edge-colored graphs

摘录 / EXCERPT: Let G be a t-colored graph, and let $\alpha(G)$ denote its independence number. Let the cycle partition number be the minimum number of vertex-disjoint monochromatic cycles cov...

conjecture | open | MathDB 页面 | 原文线索

V2-048268 | MATHDB-331392

Asymptotic Sárközy cycle-cover conjecture for edge-colored graphs

摘录 / EXCERPT: Let G be a t-colored graph with independence number $\alpha(G)=\alpha$, and let $n=|V(G)|$. Asymptotic Sárközy cycle-cover conjecture. There exists a constant $c=c(\alpha,t)$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-048269 | MATHDB-331393

H-free-complement monochromatic path-cover conjecture

摘录 / EXCERPT: Let H be a graph with chromatic number $k+1$, and let G be an t-edge-colored graph on n vertices such that H is not a subgraph of the complement $\overline{G}$. H-free-co...

conjecture | open | MathDB 页面 | 原文线索

V2-048270 | MATHDB-331435

Low-degree spanning-tree conjecture for consecutive Boolean-lattice levels

摘录 / EXCERPT: Low-degree spanning-tree conjecture. There exists some real $t>0$ such that, taking

conjecture | open | MathDB 页面 | 原文线索

V2-048271 | MATHDB-331437

The Short Cycle Cover Conjecture for bridgeless graphs

摘录 / EXCERPT: Let G be a bridgeless graph with m edges, and let the length of a cycle cover be the sum of the lengths of all cycles in the cover. Short Cycle Cover Conjecture. Every bridgele...

conjecture | open | MathDB 页面 | 原文线索

V2-048272 | MATHDB-331459

The converse conjecture for Postnikov-Shapiro algebras and bridge-free matroids

摘录 / EXCERPT: Converse conjecture. The algebras ${\mathcal B}_{G_1}^T$ and ${\mathcal B}_{G_2}^T$ are isomorphic if and only if the bridge-free matroids of G_1 and G_2 are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-048273 | MATHDB-331490

The graphic Merino-Welsh conjectures

摘录 / EXCERPT: Let G be a graph with no bridges and no loops. Let $\tau(G)$ be its number of spanning trees, let (G) be its number of acyclic orientations, and let $^*(G)$ be its number of to...

conjecture | open | MathDB 页面 | 原文线索

V2-048274 | MATHDB-331503

Maximum-order conjecture for extendable actions on minimal graphs

摘录 / EXCERPT: Maximum-order conjecture. Under these hypotheses, M_g has the values specified above. The claim concerns the still-unknown maximum orders of extendable finite group actions on mi...

conjecture | open | MathDB 页面 | 原文线索

V2-048275 | MATHDB-331532

Faber-Krahn conjecture for unicyclic degree sequences

摘录 / EXCERPT: Let $\pi=(d_0,d_1,\ldots,d_{k-1},1,\ldots,1)$ be a graphic unicyclic degree sequence with $2\le d_0\le d_1\le\ldots\le d_{k-1}$ and $d_k=\ldots=d_{n-1}=1$. Let $\mathcal{U}_{\pi}$...

conjecture | open | MathDB 页面 | 原文线索

V2-048276 | MATHDB-331534

Goddyn's conjecture on cycle double covers

摘录 / EXCERPT: Goddyn's conjecture. There exists a cycle double cover of G containing C.

conjecture | open | MathDB 页面 | 原文线索

V2-048277 | MATHDB-331535

The regular edge-connected cycle cover conjecture

摘录 / EXCERPT: Regular edge-connected cycle cover conjecture. There exists a cycle $(k-1)$-cover of G.

conjecture | open | MathDB 页面 | 原文线索

V2-048278 | MATHDB-331537

The strict inequality between the largest Laplacian H-eigenvalue and spectral radius

摘录 / EXCERPT: Strict inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048279 | MATHDB-331553

The strict upper-bound conjecture for graph Ky Fan norms

摘录 / EXCERPT: Strict upper-bound conjecture. There exist infinitely many positive integers k such that

conjecture | open | MathDB 页面 | 原文线索

V2-048280 | MATHDB-331554

The sharp Schatten norm conjecture for graphs with fixed size

摘录 / EXCERPT: Sharp Schatten norm conjecture. If $p>2$ and G is a graph of size m, then

conjecture | open | MathDB 页面 | 原文线索

V2-048281 | MATHDB-331555

The strict Schatten norm conjecture for graphs of fixed order

摘录 / EXCERPT: Strict Schatten norm conjecture. If $p>2$ and G is a graph of order n, then

conjecture | open | MathDB 页面 | 原文线索

V2-048282 | MATHDB-331556

The Schatten norm conjecture for r-partite graphs

摘录 / EXCERPT: Schatten norm conjecture for r-partite graphs. If G is an r-partite graph of order n, then

conjecture | open | MathDB 页面 | 原文线索

V2-048283 | MATHDB-331559

Shi's conjecture on UPC graphs with at least four extra edges

摘录 / EXCERPT: Let a UPC graph be a graph that is (r)-pancyclic for every positive integer r in the relevant range, and let v be its number of vertices and $v+m$ its number of edges. Shi's...

conjecture | open | MathDB 页面 | 原文线索

V2-048284 | MATHDB-331577

Erdős's conjecture on the dichromatic number of digon-free digraphs

摘录 / EXCERPT: Let D be a digon-free digraph, meaning that no pair of opposite directed edges joins the same two vertices, and let Δ be its maximum total degree. Erdős's conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-048285 | MATHDB-331599

Site recurrence for non-backtracking coalescing random walk on bounded degree trees

摘录 / EXCERPT: A non-backtracking coalescing random walk is a coalescing random walk in which each particle moves to a neighboring vertex other than the vertex from which it most recently arrived...

conjecture | open | MathDB 页面 | 原文线索

V2-048286 | MATHDB-331605

Hartsfield-Ringle antimagic labeling conjecture for connected graphs

摘录 / EXCERPT: Let G be a simple connected graph, and let an antimagic labeling be a bijection from $E(G)$ to $\{1,\dotsc,|E(G)|\}$ for which the vertex sums-the sums of labels on edges inciden...

conjecture | open | MathDB 页面 | 原文线索

V2-048287 | MATHDB-331606

Saito-Woei's plateaux conjecture for graph Laplacian eigenvalues

摘录 / EXCERPT: Saito-Woei's plateaux conjecture. For any positive integer k and any graph G, some Laplacian eigenvalue of G has multiplicity at least

conjecture | open | MathDB 页面 | 原文线索

V2-048288 | MATHDB-331612

Existence of spanning trees with floor((n+2)/6) leaves in large cubic graphs

摘录 / EXCERPT: Spanning-tree conjecture. There exists $n\in\mathbb{N}$ such that every 3-regular graph with at least n vertices has a spanning tree with...

conjecture | open | MathDB 页面 | 原文线索

V2-048289 | MATHDB-331644

The k-optimal set covering conjecture

摘录 / EXCERPT: The k-optimal set covering conjecture. If D is a k-optimal set in G, then G has a k-edge-colorable subgraph M such that

conjecture | open | MathDB 页面 | 原文线索

V2-048290 | MATHDB-331645

Subquadratic maximum running time conjecture for graph bootstrap percolation

摘录 / EXCERPT: For $r\geqslant 5$, let $M_r(n)$ denote the maximum number of time steps for which the K_r-bootstrap process on n vertices can run before stabilizing. Subquadratic running-time...

conjecture | open | MathDB 页面 | 原文线索

V2-048291 | MATHDB-331649

Bollobás's consecutive-order tree packing conjecture

摘录 / EXCERPT: Let $k\geq 1$ and let $T_n,T_{n-1},\dots,T_{n-k+1}$ be trees, where T_{n-j} has $n-j$ vertices. The trees pack into K_n when they can be found as pairwise edge-disjoint subgrap...

conjecture | open | MathDB 页面 | 原文线索

V2-048292 | MATHDB-331675

Weighted-network formula for tropical determinantal valuations

摘录 / EXCERPT: Let x_1,x_2,x_3 be points in the affine Grassmannian, and let $\omega_i,\omega_j,\omega_k$ be fundamental weights of SL_n with $i+j+k=n$. For points p,q in the affine Grassma...

conjecture | open | MathDB 页面 | 原文线索

V2-048293 | MATHDB-331676

Binary-network formula for ternary determinantal valuations

摘录 / EXCERPT: Let L,M,N be lattices in $\mathbb{C}((t))^n$, and let i,j,k be nonnegative integers with $i+j+k=n$. For lattices $L_1,\ldots,L_r$, define the tropical determinantal valuation...

conjecture | open | MathDB 页面 | 原文线索

V2-048294 | MATHDB-331683

Daniel-Jolliffe conjecture on the 2-elementary divisors of the hypercube adjacency matrix

摘录 / EXCERPT: Daniel-Jolliffe conjecture. For every i, the multiplicity of 2^i as a 2-elementary divisor of A equals the number of eigenvalues of A with exact 2-power divisor...

conjecture | open | MathDB 页面 | 原文线索

V2-048295 | MATHDB-331690

The achromatic number bound by the harmonic index

摘录 / EXCERPT: Let G be a graph. The achromatic number $psi(G)$ is the largest number of colors in a complete proper vertex coloring, and $R'(G)$ denotes the inverse degree-based index used in...

conjecture | open | MathDB 页面 | 原文线索

V2-048296 | MATHDB-331692

Extremal maximal-matchings conjecture for benzenoid chains

摘录 / EXCERPT: Extremal maximal-matchings conjecture. For every benzenoid chain B_n of length n,

conjecture | open | MathDB 页面 | 原文线索

V2-048297 | MATHDB-331712

Extremal-tree characterization for exponential domination in subcubic trees

摘录 / EXCERPT: Let T be a tree of maximum degree at most 3, let $n(T)$ denote its number of vertices, and let $\gamma_e(T)$ denote its exponential domination number. For two trees T and...

conjecture | open | MathDB 页面 | 原文线索

V2-048298 | MATHDB-331726

The (5, 2)-cycle-cover conjecture

摘录 / EXCERPT: The (5, 2)-cycle-cover conjecture. The graph G contains five even subgraphs such that every edge of G belongs to exactly two of them.

conjecture | open | MathDB 页面 | 原文线索

V2-048299 | MATHDB-331768

The directed Antimagic Graph Conjecture

摘录 / EXCERPT: Let G be a connected directed graph. An antimagic labeling is an edge labeling in which each label is a distinct element of $\{1,2,\dots,|E|\}$ and the oriented sums at the verti...

conjecture | open | MathDB 页面 | 原文线索

V2-048300 | MATHDB-331778

The separability characterization of matching-stable graphs

摘录 / EXCERPT: Separability characterization conjecture. The only connected and non-bipartite graphs G for which the matching queue $(G,\lambda,\Phi)_{\scriptsize{\textsc{c}}}$ is stable for an...

conjecture | open | MathDB 页面 | 原文线索

V2-048301 | MATHDB-331785

Finite obstruction conjecture for semiacyclic tournaments

摘录 / EXCERPT: Finite obstruction conjecture. There is a finite list of tournaments that are minimally not semiacyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-048302 | MATHDB-331786

Finite obstruction conjecture for dominance graphs of coin systems

摘录 / EXCERPT: Finite obstruction conjecture. There is a finite list of tournaments that are minimally not dominance graphs of systems of coins.

conjecture | open | MathDB 页面 | 原文线索

V2-048303 | MATHDB-331788

Strong weighting conjecture for nice uniform hypergraphs

摘录 / EXCERPT: Strong weighting conjecture. For every $r\geq 3$, there is a constant $w=w(r)$ such that each nice r-uniform hypergraph is strongly w-weighted.

conjecture | open | MathDB 页面 | 原文线索

V2-048304 | MATHDB-331789

The weak 3-weighting conjecture for uniform hypergraphs

摘录 / EXCERPT: Weak 3-weighting conjecture. For each $r\geq 3$, every r-uniform hypergraph without isolated edges is weakly 3-weighted.

conjecture | open | MathDB 页面 | 原文线索

V2-048305 | MATHDB-331790

The clique-count conjecture for complete minor-free graphs

摘录 / EXCERPT: Let $\textup{cliques}(n,t)$ denote the maximum number of cliques in an n-vertex graph with no K_t-minor. Clique-count conjecture. ... The bound is motivated by complete multipart...

conjecture | open | MathDB 页面 | 原文线索

V2-048306 | MATHDB-331791

The large-clique conjecture for graphs excluding a complete minor

摘录 / EXCERPT: Let $\textup{cliques}(n,t,k)$ denote the maximum number of k-cliques in an n-vertex graph with no K_t-minor. Large-clique conjecture. There are constants $c_1,c_2>0$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-048307 | MATHDB-331792

The very-large-clique conjecture for graphs excluding a complete minor

摘录 / EXCERPT: Let $\textup{cliques}(n,t,k)$ denote the maximum number of k-cliques in an n-vertex graph with no K_t-minor. Very-large-clique conjecture. There is some...

conjecture | open | MathDB 页面 | 原文线索

V2-048308 | MATHDB-331800

Vertex-algebra cohomology conjecture for the smooth cone

摘录 / EXCERPT: Let $\mathcal{C}$ be the cone in the source, let A_f denote the corresponding finite or smooth-part algebra, and let $H_{\mathfrak{a}}^{i+\frac{\infty}{2}}(A_f)$ be the semi-infi...

conjecture | open | MathDB 页面 | 原文线索

V2-048309 | MATHDB-331819

Hamilton path conjecture for connected graphs with exactly two spanning paths after vertex deletion

摘录 / EXCERPT: Hamilton path conjecture. The following implication holds

conjecture | open | MathDB 页面 | 原文线索

V2-048310 | MATHDB-331833

Polynomial dependence of sparse connected graph invariants on smaller graphs

摘录 / EXCERPT: Let $n\geqslant 3$ and let G be a connected graph on n nodes with at most ... edges. Write $\mathcal{P}_{n,G}$ for the corresponding graph invariant, and let H range over graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048311 | MATHDB-331841

Bonato, Janssen, and Roshanbin's burning number conjecture

摘录 / EXCERPT: Let G be a connected graph of order n, and let $b(G)$ denote its burning number. Bonato, Janssen, and Roshanbin's conjecture. ... This conjecture would improve the general bound...

conjecture | open | MathDB 页面 | 原文线索

V2-048312 | MATHDB-331842

Bosák's classification conjecture for mixed Moore graphs

摘录 / EXCERPT: Bosák's conjecture. A finite graph G is a mixed Moore graph of degree d and diameter k if and only if either $d=1$ and G is $\vec{C}_{k+1}$, or $d=2$ and G is C_{k+1}.

conjecture | open | MathDB 页面 | 原文线索

V2-048313 | MATHDB-331859

The decomposed-spider conjecture for graph burning

摘录 / EXCERPT: Decomposed-spider conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048314 | MATHDB-331864

Li and Zhang's signless Laplacian conjecture for cacti with given matching number

摘录 / EXCERPT: Li and Zhang's conjecture. If $n=2m+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-048315 | MATHDB-331873

Bean-Hansen-Swart conjecture on the (k,r)-domination number

摘录 / EXCERPT: Bean-Hansen-Swart conjecture. For positive integers k and r,

conjecture | open | MathDB 页面 | 原文线索

V2-048316 | MATHDB-331880

Zhang-Broersma-Chen conjecture on Ramsey numbers of trees versus fans

摘录 / EXCERPT: Let T_n be a tree on n vertices and let F_m be the fan consisting of m triangles sharing a common vertex. For integers $m\geq 6$ and $n\geq m^2-m+1$, Zhang-Broersma-Chen co...

conjecture | open | MathDB 页面 | 原文线索

V2-048317 | MATHDB-331886

Conjectures on higher-genus sewing matrices and differentials

摘录 / EXCERPT: Let N be an integer, let $\mathcal{D}_{\mathrm{sew}}$ be the sewing domain, and let $\widetilde{\Lambda}_a$ and $\widetilde{\Lambda}_{\bar a}$ be the matrices appearing in the se...

conjecture | open | MathDB 页面 | 原文线索

V2-048318 | MATHDB-331888

Bushaw and Kettle's piecewise Turán-number conjecture for disjoint copies of P_3

摘录 / EXCERPT: Let P_3 be the path on three vertices, let k be a positive integer, and let $ex(n,H)$ denote the maximum number of edges in an n-vertex graph containing no subgraph isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-048319 | MATHDB-331893

Warmth-index conjecture for graph homomorphism complexes

摘录 / EXCERPT: Let G be a finite graph. Write $\text{Hom}(K_2,G)$ for the homomorphism complex from the edge K_2 to G, and let $\operatorname{ind}(X)$ denote the least integer m for which...

conjecture | open | MathDB 页面 | 原文线索

V2-048320 | MATHDB-331922

Rationality conjecture for affine osp(1|2) super VOAs

摘录 / EXCERPT: Let k be a positive integer. The affine algebra $L_k({\mathfrak{sl}}_2)$ is expected to be a sub-VOA of $L_k({\mathfrak{osp}}(1|2))$. Affine osp(1|2) rationality conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-048321 | MATHDB-331923

The $2t+2$-cycle conjecture for chromatic numbers of graph powers

摘录 / EXCERPT: Let G be a graph of maximum degree at most d, and let G^t denote its tth power. Suppose that G contains no cycle of length $2t+2$ as a subgraph. The $2t+2$-cycle conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-048322 | MATHDB-331967

Jonckheere-Lou-Bonahon-Baryshnikov conjecture on highest demand vertices

摘录 / EXCERPT: Let G be a large but finite graph with negative curvature. Let the demand of a vertex mean the quantity measuring how many shortest paths pass through it, as defined in the paper...

conjecture | open | MathDB 页面 | 原文线索

V2-048323 | MATHDB-331968

Jonckheere-Lou-Bonahon-Baryshnikov conjecture on demand and inertia

摘录 / EXCERPT: Let G be a large but finite graph with negative curvature. Let the demand and inertia of a vertex be the graph quantities defined in the paper. Jonckheere-Lou-Bonahon-Baryshnikov...

conjecture | open | MathDB 页面 | 原文线索

V2-048324 | MATHDB-331969

Jonckheere-Lou-Bonahon-Baryshnikov conjecture on curvature and balance

摘录 / EXCERPT: Let G be a large but finite graph with positive curvature. Let the demand and inertia of a vertex be the graph quantities defined in the paper. Jonckheere-Lou-Bonahon-Baryshnikov...

conjecture | open | MathDB 页面 | 原文线索

V2-048325 | MATHDB-331970

Jonckheere-Lou-Bonahon-Baryshnikov vertex-transitivity conjecture

摘录 / EXCERPT: Let G be a large but finite graph with positive curvature. A graph is vertex-transitive when for every pair of vertices there is a graph isomorphism taking one to the other. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-048326 | MATHDB-331971

Conjecture on the unique extremal vertex in negatively curved symmetric graphs

摘录 / EXCERPT: Let G be a large but finite graph with negative curvature. Suppose its symmetric group fixes a unique point v_0. Let the inertia and demand of a vertex be the quantities define...

conjecture | open | MathDB 页面 | 原文线索

V2-048327 | MATHDB-331973

Moment-curve extremality conjecture for crossing pairs in convex drawings

摘录 / EXCERPT: Moment-curve extremality conjecture. Placing the n vertices on the d-dimensional moment curve maximizes the number of crossing pairs of hyperedges among all d-dimensional con...

conjecture | open | MathDB 页面 | 原文线索

V2-048328 | MATHDB-331982

Kernel-diadem inequality for graphs

摘录 / EXCERPT: Let G be a finite graph. Write $\ker(G)$ for its kernel, $\diadem(G)$ for its diadem, and $\alpha(G)$ for its independence number. Kernel-diadem conjecture. For every graph G,...

conjecture | open | MathDB 页面 | 原文线索

V2-048329 | MATHDB-331987

The linear formula for vertex Folkman numbers with parameters 3 and p

摘录 / EXCERPT: Let p be an integer and, for $r\geq1$, let $F_v(2_r,3,p;r+p+1)$ denote the vertex Folkman number with r entries equal to 2, together with parameters 3 and p. The...

conjecture | open | MathDB 页面 | 原文线索

V2-048330 | MATHDB-331992

Linear vertex deletion conjecture for bounded boxicity of surface graphs

摘录 / EXCERPT: Linear vertex deletion conjecture. There is a constant $c>0$ such that, for every graph G embedded on a surface of Euler genus g, at most cg vertices can be removed so that t...

conjecture | open | MathDB 页面 | 原文线索

V2-048331 | MATHDB-331993

Canonical sequence conjecture for tangle-distinguishing tree-decompositions

摘录 / EXCERPT: Canonical sequence conjecture. For every graph G there exists a canonical sequence of tree-decompositions $(T_i,\mathcal{V}_i)$ for $1 \leq i \leq n$ of G such that:

conjecture | open | MathDB 页面 | 原文线索

V2-048332 | MATHDB-331994

Multi-separation star refinement conjecture

摘录 / EXCERPT: Multi-separation star refinement conjecture. There exists a set of separations $\{\overleftarrow{u}_i:i\in[n]\}$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-048333 | MATHDB-331996

Convexity conjecture for matching numbers of nearly bipartite graphs

摘录 / EXCERPT: Nearly bipartite matching conjecture. For any $k\geq 2$ and any nearly bipartite graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-048334 | MATHDB-331997

The strong geometric description conjecture for non-parallelizable tensegrities

摘录 / EXCERPT: Let G be a graph and let $G(P)$ be a framework whose vertex configuration P is in general position. A system of strong geometric conditions is a system of algebraic geometric c...

conjecture | open | MathDB 页面 | 原文线索

V2-048335 | MATHDB-332026

The conjecture that planar graphs are (4,2)-choosable

摘录 / EXCERPT: A (k,s)-list assignment L assigns a list $L(v)$ of k colors to each vertex of a graph, with $|L(u)\cap L(v)|\leq s$ whenever uv is an edge. A graph is (k,s)-choosable if...

conjecture | open | MathDB 页面 | 原文线索

V2-048336 | MATHDB-332038

High-dimensional permutation discrepancy conjecture

摘录 / EXCERPT: High-dimensional permutation discrepancy conjecture. For every $d\geq2$ there exist arbitrarily large d-dimensional permutations A such that, for every box $\mathcal T$,

conjecture | open | MathDB 页面 | 原文线索

V2-048337 | MATHDB-332100

Lehel-Puleo conjecture on triangle-independent sets and bipartite deletion

摘录 / EXCERPT: Let G be a simple undirected graph on n vertices. A triangle-independent set is an edge set containing at most one edge from each triangle of G. Let $\tau_{B}(G)$ be the mini...

conjecture | open | MathDB 页面 | 原文线索

V2-048338 | MATHDB-332101

Sudakov's K_r-free bipartization conjecture

摘录 / EXCERPT: Let G be an n-vertex K_r-free graph, where $r\geq 5$, and let $\tau_{B}(G)$ denote the minimum number of edges whose deletion makes G bipartite. Sudakov's conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-048339 | MATHDB-332143

Matrix-element conjecture for the crystal-limit vertex operator

摘录 / EXCERPT: Let $\widetilde\Phi(z):\mathcal F_{\vec u}\to\mathcal F_{\vec v}$ be the crystal-limit vertex operator, with $\vec u=(u_1,u_2)$ and $\vec v=(v_1,v_2)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048340 | MATHDB-332146

Wiener-Randić distance conjecture for trees

摘录 / EXCERPT: Let T and T' be any two trees with n vertices. Wiener-Randić distance conjecture. It holds ... The conjecture compares graph distance measures induced by the Wiener and Randić...

conjecture | open | MathDB 页面 | 原文线索

V2-048341 | MATHDB-332147

Energy-graph-entropy distance conjecture for trees

摘录 / EXCERPT: Let T and T' be any two trees with n vertices. Energy-graph-entropy distance conjecture. It holds ... Here E is the graph energy and Ig is the entropy based on the absolute...

conjecture | open | MathDB 页面 | 原文线索

V2-048342 | MATHDB-332148

Randić-degree-power entropy distance conjecture for trees

摘录 / EXCERPT: Let T and T' be any two trees with n vertices. Randić-degree-power entropy distance conjecture. It holds ... Here R is the Randić index and If_1 is the degree-power entropy...

conjecture | open | MathDB 页面 | 原文线索

V2-048343 | MATHDB-332173

Decelle et al.'s achievability conjecture for stochastic block model detection

摘录 / EXCERPT: Let (X,G) be drawn from the stochastic block model $\operatorname{SBM}(n,k,a,b)$: X is uniformly drawn among partitions of $[n]$ into k balanced clusters, and G has vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-048344 | MATHDB-332206

Jewel characterization conjecture for regularity of binomial edge ideals

摘录 / EXCERPT: Let T be a tree, and let J_T be its binomial edge ideal in $S=K[x_1,\ldots,x_n,y_1,\ldots,y_n]$. Denote by $\#\{\text{internal vertices of }T\}$ the number of internal vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-048345 | MATHDB-332242

Optimal covering ratio conjecture for the infinite square grid

摘录 / EXCERPT: Let G be an infinite square grid, and let D be a pebble distribution on G. Say that a vertex is reachable from D if pebbles can be moved to it under the pebbling rules, and...

conjecture | open | MathDB 页面 | 原文线索

V2-048346 | MATHDB-332255

Generalised Thue chromatic index bound for Jaco graphs

摘录 / EXCERPT: Generalised Jaco graph conjecture. For every such Jaco graph,

conjecture | open | MathDB 页面 | 原文线索

V2-048347 | MATHDB-332259

The total monochromatic connection conjecture

摘录 / EXCERPT: Total monochromatic connection conjecture. For a connected graph G, it always holds that

conjecture | open | MathDB 页面 | 原文线索

V2-048348 | MATHDB-332288

The at-most-n-minus-one generators conjecture for hulls

摘录 / EXCERPT: Let a hull be a graph arising as the hull of a set of transformations under the paper's construction, and let n be its number of vertices. At-most-$n-1$ generators conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-048349 | MATHDB-332289

The special-vertex shortest-path conjecture for Hanoi graphs

摘录 / EXCERPT: Let $p\geq 3$, let ${\cal A}_p$ be an alphabet of p letters, and let $A,B\in{\cal A}_p$. For $n\geq 2$, consider vertices labeled A^n and B^n. A vertex on a path is special i...

conjecture | open | MathDB 页面 | 原文线索

V2-048350 | MATHDB-332290

The eventual special-vertex conjecture for Hanoi border paths

摘录 / EXCERPT: Let $p\geq 3$, let $G^{(p)}_{n+1}$ contain the copy $AG^{(p)}_n$, and let S_{n_0} be the recursively defined border set. Let a be the vertex labeled A^{n_0+1}, and call a ver...

conjecture | open | MathDB 页面 | 原文线索

V2-048351 | MATHDB-332315

Tightness of the communication-rate upper bounds for hypergraphical sources

摘录 / EXCERPT: Let $X_{\mathcal{M}}$ be a hypergraphical source, and let R_{SK} denote the minimum communication rate required to achieve secret-key capacity. Theorem gives an upper bo...

conjecture | open | MathDB 页面 | 原文线索

V2-048352 | MATHDB-332317

Fiamčík-Alon-Sudakov-Zaks conjecture on the acyclic chromatic index

摘录 / EXCERPT: Fiamčík-Alon-Sudakov-Zaks conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-048353 | MATHDB-332325

Five-line Fano-flow conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A 5-line Fano-flow is a Fano-flow in which at most five lines of the Fano plane occur as flow values at vertices. Five-line Fano-flow conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048354 | MATHDB-332326

Bipartite core conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let a bipartite core mean a core whose underlying graph is bipartite. Bipartite core conjecture. Every bridgeless cubic graph has a biparti...

conjecture | solved | MathDB 页面 | 原文线索

V2-048355 | MATHDB-332327

Cyclic 1-weak core conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A 1-weak core is a weak core arising from three joins, with precisely one of the joins not a 1-factor; it is cyclic when it is a cycle. Cyclic...

conjecture | open | MathDB 页面 | 原文线索

V2-048356 | MATHDB-332328

Acyclic complement of two 1-factors conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let M_1,M_2 be 1-factors of G. Acyclic two-matchings conjecture. There exist two 1-factors such that the complement of their union is a...

conjecture | open | MathDB 页面 | 原文线索

V2-048357 | MATHDB-332341

The braided equivalence conjecture for symplectic fermion representation categories

摘录 / EXCERPT: Braided equivalence conjecture. For $\beta=e^{-\pi i d/8}$, the functor F is a $\mathbb{C}$-linear braided monoidal equivalence. This conjecture identifies the representation cat...

conjecture | open | MathDB 页面 | 原文线索

V2-048358 | MATHDB-332343

Pemantle-Stacey conjecture on the invariance of the contact-process critical rate under edge addition

摘录 / EXCERPT: Let $G=(V,E)$ and $G'=(V,E')$ be connected graphs with the same vertex set V, where ... for some $x,y\in V$. Write $\lambda_c(G)$ for the critical infection rate of the contact pro...

conjecture | open | MathDB 页面 | 原文线索

V2-048359 | MATHDB-332362

The zero-sum 6-flow conjecture

摘录 / EXCERPT: Let G be a finite undirected graph. A zero-sum flow assigns a nonzero integer to each edge so that the sum of the assignments on all edges incident with every vertex is zero. A z...

conjecture | open | MathDB 页面 | 原文线索

V2-048360 | MATHDB-332372

Log-convexity conjecture for bipartite graphical degree-sequence counts

摘录 / EXCERPT: For each positive integer n, let a_n denote the number of bipartite graphical degree sequences on two parts of size n, written as $n+n$ vertices. Log-convexity conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048361 | MATHDB-332393

The bounded-colour-diameter covering conjecture for countable graphs

摘录 / EXCERPT: For each integer $k\geq2$, there is a positive constant C_k. Given a k-colouring of a countably infinite graph G, let $\operatorname{diam}_{c}G[A]$ denote the diameter of the...

conjecture | open | MathDB 页面 | 原文线索

V2-048362 | MATHDB-332421

Gyárfás's conjecture on domination numbers of 3-tournaments

摘录 / EXCERPT: Gyárfás's conjecture. There are 3-tournaments with arbitrarily large domination number, and the domination number of a 3-tournament such that any four of its vertices induce at lea...

conjecture | refuted | MathDB 页面 | 原文线索

V2-048363 | MATHDB-332433

Block decomposition conjecture for permutation matrices

摘录 / EXCERPT: Block decomposition conjecture. There should be a continuous function u from Π_n into $U(n)$ such that each vertex is mapped to the corresponding permutation matrix and the b...

conjecture | open | MathDB 页面 | 原文线索

V2-048364 | MATHDB-332434

Unitarity conjecture for simple current extensions of unitary vertex operator algebras

摘录 / EXCERPT: Let W be a unitary vertex operator algebra, let $\pi\subset\mathrm{Rep}^\times(W)$ be a subgroup of its invertible unitary modules, and suppose that the simple current extension...

conjecture | open | MathDB 页面 | 原文线索

V2-048365 | MATHDB-332441

The edge-density conjecture for K_{k-2}-free k-critical graphs

摘录 / EXCERPT: Edge-density conjecture. For every $k\ge 4$, there exists $\epsilon_k>0$ such that if G is a k-critical K_{k-2}-free graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-048366 | MATHDB-332453

Maximum vertex-periodicity conjecture for fractal graphs

摘录 / EXCERPT: Maximum periodicity conjecture. The maximum periodicity of any vertex v in each listed fractal graph is given by the corresponding value above.

conjecture | open | MathDB 页面 | 原文线索

V2-048367 | MATHDB-332463

Identical lists maximize the probability of a non-distinguishing coloring

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{L}=\mathcal{L}$ be a collection of equal-sized lists, each of size k, indexed by the vertices $v eq 0$ of G. Choose a color from each vert...

conjecture | open | MathDB 页面 | 原文线索

V2-048368 | MATHDB-332470

The 3-dimensional shadow inclusion connectivity conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^3$ be finite and let $v\in X$. Write X_v for the subset associated with the vertex v, X^v for the corresponding closed neighborhood, and $X-v$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-048369 | MATHDB-332481

The pseudoloop lemma conjecture for smooth digraphs

摘录 / EXCERPT: Let $\mathcal{G}$ be an oligomorphic permutation group on a set G, and let $\mathbb{G}$ be a countable smooth digraph of algebraic length 1 on G that is invariant under...

conjecture | open | MathDB 页面 | 原文线索

V2-048370 | MATHDB-332485

Erdős-Gallai-Tuza conjecture on triangle-independent and triangle-covering sets

摘录 / EXCERPT: Erdős-Gallai-Tuza conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048371 | MATHDB-332503

Monotonicity conjecture for the computational capabilities of oblivious robot systems

摘录 / EXCERPT: Let G be a network and let $k\geq 1$. Write $\mathcal F(G,k)$ for the system of k oblivious mobile robots on G, and write $\preceq$ for the computational-capability preorder....

conjecture | open | MathDB 页面 | 原文线索

V2-048372 | MATHDB-332504

Characterization conjecture for universal classes of unlabeled robot systems

摘录 / EXCERPT: Let G_i be unlabeled networks, let $k\geq 3$, and let $\mathcal F(G_i,k)$ denote the corresponding system of k oblivious mobile robots. Let $G_i^\ast$ be the quotient graph ass...

conjecture | open | MathDB 页面 | 原文线索

V2-048373 | MATHDB-332514

The strengthened Szeged-Wiener lower bound for 2-connected graphs

摘录 / EXCERPT: Let G be a 2-connected graph of order $n\ge 10$. Let K_n denote the complete graph, and let K_n^2 and K_n^{n-2} be the graphs obtained from K_{n-1} by adding one vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-048374 | MATHDB-332520

Ilinca-Kahn conjecture for maximal independent sets in the Boolean-layer graph

摘录 / EXCERPT: For positive integers n and k, let $\mathcal{B}_{n,k}$ be the graph with vertex set $\binom{[n]}{k}\cup\binom{[n]}{k+1}$ and edges given by inclusion, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048375 | MATHDB-332540

Jackson's hamiltonicity conjecture for diregular oriented graphs

摘录 / EXCERPT: Jackson's conjecture. The graph G contains a directed Hamiltonian circuit.

conjecture | open | MathDB 页面 | 原文线索

V2-048376 | MATHDB-332556

Li and Magnant's diameter-three proper connection conjecture

摘录 / EXCERPT: Let G be a 2-connected noncomplete graph, and let $diam(G)$ denote its diameter. Let $pc(G)$ denote the minimum number of colors in an edge-coloring such that every pair of disti...

conjecture | open | MathDB 页面 | 原文线索

V2-048377 | MATHDB-332573

Closure of co-bipartite unit disk graphs under bipartite complementation

摘录 / EXCERPT: Let G be a co-bipartite unit disk graph, meaning that its vertex set can be partitioned into two cliques, and let the bipartite complement of G be the graph obtained by complem...

conjecture | open | MathDB 页面 | 原文线索

V2-048378 | MATHDB-332601

ICC optimality conjecture for K-IC structures

摘录 / EXCERPT: Let D_K be a K-IC structure, and let $t\geq 1$ be the number of bits in each message. Write $\ell_{\mathsf{ICC}}(D_K)$ for the scalar linear index-code length of the...

conjecture | open | MathDB 页面 | 原文线索

V2-048379 | MATHDB-332602

Equal-savings conjecture for minimal partial cliques

摘录 / EXCERPT: Let D' be a minimal partial clique, meaning that its savings cannot be matched or exceeded by the sum of the savings of any partition into smaller subdigraphs. Let $\delta^+(D')$...

conjecture | open | MathDB 页面 | 原文线索

V2-048380 | MATHDB-332604

Uniqueness of the local maximum of the aerial-base-station coverage function

摘录 / EXCERPT: Let γ be a QoS requirement and let f_c be an operating frequency. For the propagation environments whose parameters are listed in Table 1, define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-048381 | MATHDB-332607

The anchor-cell excess conjecture for canonical systems on metric graphs with skeleton K_4

摘录 / EXCERPT: Let Γ be a metric graph whose skeleton is the complete graph K_4, and let $D=K$ be its canonical divisor. Write $|D|$ for the associated linear system, and call the cells...

conjecture | open | MathDB 页面 | 原文线索

V2-048382 | MATHDB-332637

The converse characterization for 1-dimensional bipartite hypergraphs

摘录 / EXCERPT: Converse characterization conjecture. Then $\operatorname{pd}(\mathcal{H})=|V(\mathcal{H})|=2$ if and only if $\mathcal{H}$ satisfies $(\sharp)$. The authors are asking whether thi...

conjecture | open | MathDB 页面 | 原文线索

V2-048383 | MATHDB-332662

Quadratic additive bound conjecture for sparse spanning strongly connected subgraphs of tournaments

摘录 / EXCERPT: Let k be a positive integer, and let T be a strongly k-connected tournament on n vertices. A strongly k-connected spanning subgraph is a spanning subgraph with strong ver...

conjecture | open | MathDB 页面 | 原文线索

V2-048384 | MATHDB-332667

Pippinger's inducibility conjecture for cycles

摘录 / EXCERPT: Let C_k denote the cycle on k vertices. For a graph H, its induced density is the number of induced copies of H divided by $\binom{|G|}{|H|}$; write $I(H;n)$ for the maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-048385 | MATHDB-332668

Conjecture on extremal graphs for linearly long induced cycles

摘录 / EXCERPT: Let $c\in(0,1)$, let $l(n)=\lceil cn\rceil$, and let $c(n,l)$ be the maximum number of induced cycles of length l in a graph on n vertices. Let $C(n,l)$ be the set of graphs at...

conjecture | open | MathDB 页面 | 原文线索

V2-048386 | MATHDB-332677

The threshold conjecture for arbitrary oriented Hamilton cycles in random digraphs

摘录 / EXCERPT: Let $D\sim\mathcal D(n,p)$ be a random directed graph on vertex set $[n]$, where each possible arc is present independently with probability p, and let C be a Hamilton cycle wi...

conjecture | open | MathDB 页面 | 原文线索

V2-048387 | MATHDB-332682

Frieze-Pegden homomorphism conjecture for sparse random graphs

摘录 / EXCERPT: Let $c>1$ be fixed, let $G\sim G(n,c/n)$, and let $C_{2\ell+1}$ be the cycle of length $2\ell+1$. A graph homomorphism from G to $C_{2\ell+1}$ is a vertex map preserving adjacenc...

conjecture | open | MathDB 页面 | 原文线索

V2-048388 | MATHDB-332708

Kowalik-Lużar-Skrekovski conjecture for planar graphs of girth at least five

摘录 / EXCERPT: A feedback vertex set of a graph G is a set $S\subseteq V(G)$ such that $G-S$ is a forest. Let $\phi(G)$ denote the minimum size of a feedback vertex set of G. The girth of a g...

conjecture | open | MathDB 页面 | 原文线索

V2-048389 | MATHDB-332711

The bridge-free matroid criterion for isomorphism of graph algebras

摘录 / EXCERPT: Bridge-free matroid criterion. The algebras $\mathcal C_{G_1}^T$ and $\mathcal C_{G_2}^T$ are isomorphic if and only if their bridge-free matroids are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-048390 | MATHDB-332713

Gustavsson-Nash-Williams conjecture on clique decompositions

摘录 / EXCERPT: Gustavsson-Nash-Williams conjecture. For every $r\ge 3$, there exists an $n_0=n_0(r)$ such that every K_r-divisible graph G on $n\ge n_0$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-048391 | MATHDB-332724

Treewidth-maximum degree conjecture for the chromatic index

摘录 / EXCERPT: Let G be a graph with treewidth k and maximum degree $Delta(G)$. The treewidth-maximum degree conjecture asserts that if ... then ... The conjecture would improve the known suffici...

conjecture | open | MathDB 页面 | 原文线索

V2-048392 | MATHDB-332766

The nondegeneracy conjecture for reaction-network steady states

摘录 / EXCERPT: Nondegeneracy conjecture. Every network with finitely many positive steady states has the same capacity when capacity is counted using all positive steady states or only nondegener...

conjecture | open | MathDB 页面 | 原文线索

V2-048393 | MATHDB-332775

Benjamini-Schramm conjecture on hyperbolic transient spaces and bounded harmonic functions

摘录 / EXCERPT: Benjamini-Schramm conjecture. The manifold M admits nonconstant bounded harmonic functions; similarly, the graph admits nonconstant bounded harmonic functions.

open_problem | open | MathDB 页面 | 原文线索

V2-048394 | MATHDB-332776

Plummer-Toft conjecture on cyclic coloring of 3-connected plane graphs

摘录 / EXCERPT: Let G be a 3-connected plane graph, and let $Delta^\star$ be its maximum face size. A cyclic coloring of G is a vertex coloring in which any two vertices incident with the sa...

conjecture | open | MathDB 页面 | 原文线索

V2-048395 | MATHDB-332842

Akbari-Kano conjecture on two-valued factors of odd-regular graphs

摘录 / EXCERPT: Akbari-Kano conjecture. Every r-regular graph has an $\{r-t,t\}$-factor.

conjecture | open | MathDB 页面 | 原文线索

V2-048396 | MATHDB-332843

The 5-regular graph {4,1}-factor conjecture

摘录 / EXCERPT: 5-regular $\{4,1\}$-factor conjecture. Every 5-regular graph has an $\{4,1\}$-factor.

conjecture | open | MathDB 页面 | 原文线索

V2-048397 | MATHDB-332865

A tight chromatic bound in terms of triangle count and local triangle bound

摘录 / EXCERPT: Let G be a graph with n vertices, t triangles, and local triangle bound y. Here, the local triangle bound y is an upper bound on the number of triangles containing any gi...

conjecture | open | MathDB 页面 | 原文线索

V2-048398 | MATHDB-332867

The fractional chromatic bound for triangle-free degenerate graphs

摘录 / EXCERPT: Let G be a triangle-free graph that is d-degenerate, meaning every subgraph of G has a vertex of degree at most d. The fractional chromatic bound. Then ... The claim follows...

conjecture | open | MathDB 页面 | 原文线索

V2-048399 | MATHDB-332885

Edge-criticality conjecture for the quadrangulation graph $QG(n,k)$

摘录 / EXCERPT: For integers $k\geq 1$ and $n\geq 2k+1$, let $QG(n,k)$ be the spanning subgraph of the Schrijver graph $SG(n,k)$ constructed in Theorem 1.1, which is a quadrangulation of...

conjecture | open | MathDB 页面 | 原文线索

V2-048400 | MATHDB-332895

The subdivision conjecture for K_4

摘录 / EXCERPT: Subdivision conjecture for K_4. Every subdivision of K_4 in which all six edges were subdivided, each at least once, is SPN.

conjecture | open | MathDB 页面 | 原文线索

V2-048401 | MATHDB-332907

The spanning-coloring covering conjecture for complete partite hypergraphs

摘录 / EXCERPT: The complete-partite covering conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048402 | MATHDB-332908

Generalized covering conjecture for complete (r,ℓ)-partite hypergraphs

摘录 / EXCERPT: Generalized covering conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048403 | MATHDB-332935

Conjecture on automorphisms of graph squares

摘录 / EXCERPT: Let G be a graph, and write G^2 for its square, with $Aut(G)$ and $Aut(G^2)$ denoting their respective automorphism groups. For a connected graph, let d be its diameter and...

conjecture | open | MathDB 页面 | 原文线索

V2-048404 | MATHDB-332937

The metric-locating-dominating bound for doubly resolving sets

摘录 / EXCERPT: Let G be a graph. Write ${\rm dim}(G)$ for its metric dimension, $\gamma(G)$ for its domination number, and $\psi(G)$ for its doubly resolving number. Metric-locating-dominating...

conjecture | open | MathDB 页面 | 原文线索

V2-048405 | MATHDB-332945

Asymptotically extremal energy of sparse regular graphs

摘录 / EXCERPT: Let G be a k-regular graph of order n, where n and k are positive integers and $\kn$ is even. Write $\lVert G\rVert _{\ast}$ for the graph energy in the notation of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-048406 | MATHDB-332946

Asymptotically extremal energy for graphs of bounded density

摘录 / EXCERPT: Let G be a graph of order n, and let $\lVert G\rVert _{\ast}$ denote the graph energy in the notation of the source. Bounded-density energy conjecture. Let $0<c\leq1/2$. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-048407 | MATHDB-332960

Connectedness conjecture for uni-Julia sets of coupled logistic-map networks

摘录 / EXCERPT: Connectedness conjecture. For any of the three models described, and for any equi-parameter $c\in\mathbb{C}$, the uni-J set is connected only if c is in the equi-M set of the net...

conjecture | open | MathDB 页面 | 原文线索

V2-048408 | MATHDB-333017

Singleton-hypergraph extremal conjecture for the isolation lemma

摘录 / EXCERPT: Singleton-hypergraph extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048409 | MATHDB-333018

Layer-one extremal conjecture for the isolation lemma

摘录 / EXCERPT: Layer-one extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048410 | MATHDB-333020

Füredi's extremal conjecture for generalized 4-cycle-free uniform hypergraphs

摘录 / EXCERPT: Let $f_k(n)$ denote the maximum number of edges in an n-vertex k-uniform hypergraph containing no generalized 4-cycle. For $k\geq 4$ and $n\in\mathbb{N}$, Füredi's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-048411 | MATHDB-333021

Conjecture on 2k-vertex-2-regular-free uniform hypergraphs

摘录 / EXCERPT: Let $k\geq 3$ be an integer. An n-vertex k-uniform hypergraph is said to have no 2-regular subgraphs on $2k$ vertices if it contains no subhypergraph on exactly $2k$ vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-048412 | MATHDB-333024

Impossibility of distributed algorithms for TP under limited communication constraints

摘录 / EXCERPT: Impossibility claim. Given these communication constraints, there exists no distributed algorithm that can solve TP without further conditions on the sequence of communication grap...

conjecture | open | MathDB 页面 | 原文线索

V2-048413 | MATHDB-333029

Conjecture on Laplace transforms of interference in stationary birth-death networks

摘录 / EXCERPT: Let ϕ_0 be the point process on $\mathbf{S}$ corresponding to the stationary distribution of ϕ_t, with intensity β. Let ψ be an independently marked Poisson Po...

conjecture | open | MathDB 页面 | 原文线索

V2-048414 | MATHDB-333031

Broere-Pilśniak's infinite edge motion conjecture

摘录 / EXCERPT: Let G be a countable, connected graph. An automorphism moves infinitely many edges if it does not fix infinitely many edges of G. A 2-edge colouring is distinguishing if the...

conjecture | open | MathDB 页面 | 原文线索

V2-048415 | MATHDB-333036

Nordhaus-Gaddum product lower-bound conjecture for the Hadwiger number

摘录 / EXCERPT: Let $G_1,\ldots,G_r$ be graphs on a common vertex set of size n, whose edge sets partition the edges of K_n, and let $\ngpl{\eta}(r;n)$ denote the minimum of...

conjecture | open | MathDB 页面 | 原文线索

V2-048416 | MATHDB-333037

Akiyama-Watanabe conjecture on induced forests in bipartite planar graphs

摘录 / EXCERPT: Let G be a simple bipartite planar graph. Write $|G|=|V(G)|$, and let $a(G)$ denote the maximum number of vertices in an induced forest of G. Akiyama-Watanabe conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-048417 | MATHDB-333055

Kuhlemann-Vassilevski conjecture on Laplacian eigenvalue approximation under graph disaggregation

摘录 / EXCERPT: Let G be a graph and let G_D be the graph obtained by disaggregating a vertex of G, with weights on the internal edges of the disaggregation. Kuhle...

conjecture | open | MathDB 页面 | 原文线索

V2-048418 | MATHDB-333056

The asymptotic conjecture for the tight Hamiltonicity threshold of triple systems

摘录 / EXCERPT: Asymptotic threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048419 | MATHDB-333065

The Arctic Curve Conjecture for the six-vertex model on the square domain

摘录 / EXCERPT: Let $r(z)$ be the scaled logarithmic derivative of the boundary correlation function $h_N(z)$, namely ... For the six-vertex model with domain wall boundary conditions, let t and...

conjecture | open | MathDB 页面 | 原文线索

V2-048420 | MATHDB-333066

The external Arctic curve envelope conjecture for generic six-vertex domains

摘录 / EXCERPT: Let Λ be a generic domain for the six-vertex model, and consider a south side ending at a corner, with the relevant external portion of the Arctic curve lying above that si...

conjecture | open | MathDB 页面 | 原文线索

V2-048421 | MATHDB-333073

Conjecture on the spectral radii of three linear bicyclic hypergraphs

摘录 / EXCERPT: Spectral-radius ordering conjecture. For $m\geqslant5$,

conjecture | open | MathDB 页面 | 原文线索

V2-048422 | MATHDB-333078

The augmented Zagreb index-atom-bond connectivity index extremal tree conjecture

摘录 / EXCERPT: Extremal tree conjecture. The n-vertex tree T has the maximum AZI value if and only if T has the minimum ABC index.

conjecture | open | MathDB 页面 | 原文线索

V2-048423 | MATHDB-333085

Connected forcing number lower bound conjecture

摘录 / EXCERPT: Connected forcing number lower bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048424 | MATHDB-333101

Alon's independence-number conjecture for Cayley graphs

摘录 / EXCERPT: Let Γ be an abelian group of odd order n, and let $G(\Gamma,T)$ denote the graph associated with a subset $T\subseteq \Gamma$. For a graph G, write $\alpha(G)$ for its i...

conjecture | open | MathDB 页面 | 原文线索

V2-048425 | MATHDB-333112

Equality in the generalised Ramsey number lower bound

摘录 / EXCERPT: Let $\mathfrak{C}$ be the set of all pairs of non-empty sets of cycles, and let $R(\Ga_1,\Ga_2)$ be the corresponding generalised Ramsey number. For $(\Ga_1,\Ga_2)\in\mathfrak{C}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-048426 | MATHDB-333117

Zhu's planar graph multiple list-colouring conjecture

摘录 / EXCERPT: A graph is (k,m)-choosable if, for every assignment of lists of k colours to its vertices, it has a colouring in which each vertex receives m colours from its list and adjace...

conjecture | open | MathDB 页面 | 原文线索

V2-048427 | MATHDB-333118

The $(9,2)$-paintability conjecture for planar graphs

摘录 / EXCERPT: A graph is (k,m)-paintable if, in the multiple painting game, Painter can ensure that every vertex is assigned m colours from the colours offered to it, with adjacent vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-048428 | MATHDB-333119

Rigidity, pivotality and factorisability conjecture for the representation category of a vertex operator algebra

摘录 / EXCERPT: Let $\operatorname{Rep}(\mathcal{V})$ be the representation category of a vertex operator algebra $\mathcal{V}$. Rigidity, pivotality and factorisability conjecture. The category...

conjecture | open | MathDB 页面 | 原文线索

V2-048429 | MATHDB-333123

The exponential domination bound for subcubic trees

摘录 / EXCERPT: The exponential domination bound.

conjecture | open | MathDB 页面 | 原文线索

V2-048430 | MATHDB-333124

The characterization of equality between porous and fractional porous exponential domination

摘录 / EXCERPT: The equality characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048431 | MATHDB-333126

Gaberdiel-Goddard's semisimplicity criterion for strongly-finite VOAs

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly-finite VOA, and let $A_0(\mathcal{V})$ be Zhu's algebra. Gaberdiel-Goddard's conjecture. The VOA $\mathcal{V}$ is strongly-rational if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-048432 | MATHDB-333127

The log-modularity conjecture for strongly-finite VOAs

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly-finite VOA, and let $\mathrm{Mod}^{g.r}(\mathcal{V})$ denote its category of grading-restricted generalized modules. A log-modular tensor category i...

conjecture | open | MathDB 页面 | 原文线索

V2-048433 | MATHDB-333128

The semi-simplification conjecture for strongly-finite VOAs

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly-finite VOA, and let $\mathrm{Mod}^{g.r}(\mathcal{V})$ be its category of grading-restricted generalized modules. Its semi-simplification is obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-048434 | MATHDB-333129

The unimodularity conjecture for strongly-finite VOAs

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly-finite VOA, let $P_{\mathcal{V}}$ be the projective cover of the $\mathcal{V}$-module $\mathcal{V}$, and interpret containing $\mathcal{V}$ as a sub...

conjecture | open | MathDB 页面 | 原文线索

V2-048435 | MATHDB-333130

The open Hopf link representation conjecture

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly-finite VOA. Let $\mathcal{N}^{gr}_M$ denote the stated Grothendieck-ring representation, B^h_M an indecomposable subrepresentation of it, P_j th...

conjecture | open | MathDB 页面 | 原文线索

V2-048436 | MATHDB-333133

Multiple list-colouring conjecture for planar graphs

摘录 / EXCERPT: Let m be a positive integer. A graph is (k,m)-choosable if, whenever each vertex is assigned a list of at least k colours, there is an m-fold colouring assigning each verte...

conjecture | open | MathDB 页面 | 原文线索

V2-048437 | MATHDB-333134

Multiple list-colouring conjecture for planar graphs at ratio $(9,2)$

摘录 / EXCERPT: A graph is (k,m)-choosable if, for every assignment of lists of at least k colours to its vertices, it has an m-fold colouring in which each vertex receives m colours from...

conjecture | open | MathDB 页面 | 原文线索

V2-048438 | MATHDB-333144

Finite forbidden-subgraph characterization of hereditary equality of domination numbers

摘录 / EXCERPT: Let G be a graph, and let $\gamma(G)$ and $\gamma_e(G)$ denote its domination number and exponential domination number, respectively. A graph is hereditarily equal when...

conjecture | open | MathDB 页面 | 原文线索

V2-048439 | MATHDB-333145

Forbidden graphs for hereditary equality of domination numbers have values 3 and 2

摘录 / EXCERPT: Let ${\cal F}$ be a finite forbidden set as in the finite forbidden-subgraph conjecture, and let $\gamma(F)$ and $\gamma_e(F)$ denote the domination number and exponential dominati...

conjecture | open | MathDB 页面 | 原文线索

V2-048440 | MATHDB-333146

Hereditary equality of domination and porous exponential domination

摘录 / EXCERPT: Let G be a graph. For a vertex set D of G, define its porous exponential domination weight at u by ... The porous exponential domination number $\gamma_e^*(G)$ is the minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-048441 | MATHDB-333158

Simple-current fusion rules for singlet-algebra modules

摘录 / EXCERPT: Let $p\in\mathbb{Z}_{\ge 2}$. For $j\in\mathbb{Z}$, let π_j be the irreducible $\overline{M(1)}$-module defined in the surrounding construction, and let M_t denote the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-048442 | MATHDB-333159

Brualdi-Stein hypergraph matching conjecture

摘录 / EXCERPT: Let H be an n-regular 3-partite hypergraph with sides A, B, and C, each of size n. A pair of sides is simple if no pair of vertices from those sides belongs to more t...

conjecture | open | MathDB 页面 | 原文线索

V2-048443 | MATHDB-333160

Symmetric Stein conjecture for simple regular 3-partite hypergraphs

摘录 / EXCERPT: Let H be a simple 3-partite hypergraph, meaning that no edge is repeated, and suppose that H is n-regular with each side of size n. Let $\nu(H)$ denote the maximum size o...

conjecture | open | MathDB 页面 | 原文线索

V2-048444 | MATHDB-333161

Asymmetric degree conjecture for 3-partite hypergraph matchings

摘录 / EXCERPT: Let H be a simple 3-partite hypergraph with sides A, B, and C. Write $\delta(A)$ for the minimum degree of a vertex in A, $\Delta(B\cup C)$ for the maximum degree of a...

conjecture | open | MathDB 页面 | 原文线索

V2-048445 | MATHDB-333162

Simple 3-partite hypergraph generalization of Drisko's theorem

摘录 / EXCERPT: Let H be a simple 3-partite hypergraph with sides A, B, and C, where simplicity means that no edge is repeated. Let $\nu(H)$ denote the maximum matching size. Generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-048446 | MATHDB-333168

The rainbow matching conjecture for hypergraph families

摘录 / EXCERPT: Rainbow matching conjecture. (i) If $F_1,\ldots,F_k$ are sub-hypergraphs of $[n]^r$, each with more than $(k-1)n^{r-1}$ edges, then there exist pairwise disjoint edges $e_i\in F_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-048447 | MATHDB-333173

Gyárfás-Sumner conjecture on induced trees in bounded-clique graphs

摘录 / EXCERPT: Let k be an integer, let T be a tree, and let G be a graph. Write $ ... $ for the clique number and chromatic number conditions. **Gyárfás-Sumner conjecture.** For every intege...

conjecture | open | MathDB 页面 | 原文线索

V2-048448 | MATHDB-333187

The multiple-constraint Erdős-Pósa conjecture

摘录 / EXCERPT: Multiple-constraint Erdős-Pósa conjecture. The set of all $\bigoplus_{i=1}^t\Gamma_i$-labeled graphs has the half-integral Erdős-Pósa property for $(\Gamma_1,\dots,\Gamma_t)$-non-z...

conjecture | open | MathDB 页面 | 原文线索

V2-048449 | MATHDB-333190

Independence of twisted representations and Zhu's algebras under inner automorphisms

摘录 / EXCERPT: Inner-automorphism invariance conjecture. There is a one-to-one correspondence between the simple objects in the category of lower bounded logarithmic g-twisted modules and the s...

conjecture | open | MathDB 页面 | 原文线索

V2-048450 | MATHDB-333198

Degree version of the Erdős hypergraph matching conjecture

摘录 / EXCERPT: Let H be a k-uniform hypergraph on n vertices. Write $\delta_1(H)$ for its minimum vertex degree, and say that a matching has size s when it consists of s vertex-disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-048451 | MATHDB-333201

Strong spanning subdigraphs in highly arc-strong and arc-antistrong digraphs

摘录 / EXCERPT: Special-case conjecture. There exists a natural number k such that every digraph D which is both k-arc-strong and k-arc-antistrong has arc-disjoint strong spanning subdigra...

conjecture | open | MathDB 页面 | 原文线索

V2-048452 | MATHDB-333208

Conjecture that the complete graph is the unique extremal graph for $\widehat{F}_3$

摘录 / EXCERPT: Let G be a graph, let $c_{3,\widehat{F}_3}(G)$ denote the number of feasible three-edge-colorings of G avoiding the prescribed colored pattern $\widehat{F}_3$, and let K_n be...

conjecture | open | MathDB 页面 | 原文线索

V2-048453 | MATHDB-333238

Erdős-Faudree-Rousseau conjecture on edges occurring in odd cycles

摘录 / EXCERPT: Let G be an n-vertex graph with ... Let C_{2k+1} denote the cycle of length $2k+1$, and say that an edge occurs in C_{2k+1} if it belongs to a copy of that cycle. Fix an inte...

conjecture | open | MathDB 页面 | 原文线索

V2-048454 | MATHDB-333239

Universality conjecture for local topological convergence of network flows

摘录 / EXCERPT: Let $\mathcal{G}^n$ denote the class of embedded graphs in $\mathbb{R}^n$, and let a network flow $G(t)$ have homogeneous initial condition $G(0)\in\mathcal{G}^n$. Let σ_n b...

conjecture | open | MathDB 页面 | 原文线索

V2-048455 | MATHDB-333240

Classification of orbits under curvature flow for homogeneous topological cloths

摘录 / EXCERPT: Let $\mathcal{G}^n$ be the class of embedded graphs in $\mathbb{R}^n$. Suppose $G\in\mathcal{G}^n$ is an embedded graph with a homogeneous topological cloth and a well-defined time...

conjecture | open | MathDB 页面 | 原文线索

V2-048456 | MATHDB-333265

Completeness conjecture for Legendrian graph stabilizations and vertex twists

摘录 / EXCERPT: Let G be a graph and let $g:G\rightarrow S^3$ be a smooth embedding. A Legendrian realization of g is a Legendrian embedding of G in (S^3,ξ_{std}) representing the smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-048457 | MATHDB-333294

Bishop comparison conjecture for graphs with nonnegative Bakry-Émery curvature

摘录 / EXCERPT: Let $d_{\max}\in{\mathbb N}$. A graph $G=(V,E)$ has vertex degrees $d_x\leq d_{\max}$ for all $x\in V$ and satisfies $CD(0,\infty)$. Bishop comparison conjecture. There are constan...

conjecture | open | MathDB 页面 | 原文线索

V2-048458 | MATHDB-333307

Aouchiche et al. pineapple graph conjecture for spectral radius minus average degree

摘录 / EXCERPT: Let G be a connected graph on n vertices. Its spectral radius is the largest eigenvalue of its adjacency matrix, and its average degree is $2|E(G)|/n$. A pineapple graph is the...

conjecture | open | MathDB 页面 | 原文线索

V2-048459 | MATHDB-333321

The tope-graph conjecture for finite graphs in $\mathcal{S}_4$

摘录 / EXCERPT: The tope-graph conjecture for finite graphs in $\mathcal{S}_4$. Every finite graph in $\mathcal{S}_4$ is the tope graph of a COM.

conjecture | open | MathDB 页面 | 原文线索

V2-048460 | MATHDB-333345

Vanishing-boundary conjecture for configuration spaces of one-connected graphs

摘录 / EXCERPT: Vanishing-boundary conjecture. For indistinguishable particles on one-connected graphs from this class, the map δ always has trivial image:

conjecture | open | MathDB 页面 | 原文线索

V2-048461 | MATHDB-333356

The cycle-with-loops homomorphism-density conjecture

摘录 / EXCERPT: Cycle-with-loops conjecture. For every $ne4$,

conjecture | open | MathDB 页面 | 原文线索

V2-048462 | MATHDB-333357

The star-with-loops homomorphism-density conjecture

摘录 / EXCERPT: Star-with-loops conjecture. For $d\geq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-048463 | MATHDB-333359

Obstacle number two conjecture for gyroelongated bipyramids

摘录 / EXCERPT: Let X_n denote the gyroelongated n-bipyramid, and let ${\operatorname{{obs}}}_\mathrm{out}(X_n)$ be its outside obstacle number. Obstacle-number conjecture. If $n\geq 4$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-048464 | MATHDB-333368

Compatible-completion conjecture for genus-zero full conformal field theories

摘录 / EXCERPT: Compatible-completion conjecture. If the graded state space has a compatible inner product, then these two completions are isomorphic as locally convex topological vector spaces wi...

conjecture | open | MathDB 页面 | 原文线索

V2-048465 | MATHDB-333369

FLM uniqueness conjecture for the moonshine module vertex operator algebra

摘录 / EXCERPT: Moonshine uniqueness conjecture. Then V is isomorphic to the moonshine module vertex operator algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-048466 | MATHDB-333383

Upper-cluster-algebra conjecture for quiver T-system invariants

摘录 / EXCERPT: Let Q be a quiver in the classification, and let J_k be the conserved quantities appearing in the minimal linear recurrence for a vertex sequence $T_v(t)$. The upper-cluster-al...

conjecture | open | MathDB 页面 | 原文线索

V2-048467 | MATHDB-333384

Positivity conjecture for quiver T-system invariants

摘录 / EXCERPT: Let Q be a quiver in the classification, and let J_k be the conserved quantities in the minimal recurrence for a vertex sequence. A positive Laurent expression is a Laurent pol...

conjecture | open | MathDB 页面 | 原文线索

V2-048468 | MATHDB-333387

Palindromicity conjecture for recurrence polynomials

摘录 / EXCERPT: Let v be a vertex in a classified T-system outside the tensor-product case with finite factor of type A_m, D_{2m+1}, or E_6, and let $J_v(z)$ be its recurrence polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-048469 | MATHDB-333400

Seymour-Thomas extremal conjecture for complete graph minors

摘录 / EXCERPT: Seymour-Thomas conjecture. For every $p\ge1$, there exists a constant $N=N(p)$ such that every $(p-2)$-connected graph on $n\ge N$ vertices and at least

conjecture | open | MathDB 页面 | 原文线索

V2-048470 | MATHDB-333401

Mader's bound conjecture for complete graph minors

摘录 / EXCERPT: Mader's bound conjecture. For every $p\ge1$, every graph G on n vertices with at least

conjecture | open | MathDB 页面 | 原文线索

V2-048471 | MATHDB-333406

The vertex-degree conjecture for tight Hamilton cycles in 3-uniform hypergraphs

摘录 / EXCERPT: The vertex-degree conjecture. For every $\eta > 0$, there exists n_0 such that if H is a 3-graph on $n \geq n_0$ vertices with...

conjecture | open | MathDB 页面 | 原文线索

V2-048472 | MATHDB-333415

Neumann-Lara's 2-colourability conjecture for planar digraphs

摘录 / EXCERPT: Let D be an oriented planar graph, meaning a planar digraph without directed cycles of length at most 2. A 2-colouring of D is a function $f:V(D)\to\{1,2\}$ such that the s...

conjecture | open | MathDB 页面 | 原文线索

V2-048473 | MATHDB-333459

The maximum-degree bound for lazy cop number

摘录 / EXCERPT: Maximum-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048474 | MATHDB-333460

The classification conjecture for ten-vertex graphs with lazy cop number three

摘录 / EXCERPT: Ten-vertex classification conjecture. The only graphs G on 10 vertices with $c_L(G)=3$ are either the Petersen graph, or a copy of R_3 with an additional vertex v whose degre...

conjecture | open | MathDB 页面 | 原文线索

V2-048475 | MATHDB-333471

Asymptotic conjecture for the extremal function of cycles modulo k

摘录 / EXCERPT: For positive integers k and even integers $\ell\geq2$, let $c_k(\ell)$ be the least average-degree threshold such that every graph of average degree at least $c_k(\ell)$ contains...

conjecture | open | MathDB 页面 | 原文线索

V2-048476 | MATHDB-333472

Chromatic extremal-function conjecture for cycles modulo k

摘录 / EXCERPT: For positive integers k and $\ell\geq3$, let $\chi_k(\ell)$ be the largest possible chromatic number of a graph with no cycle of length congruent to ℓ modulo k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048477 | MATHDB-333473

Block structure conjecture for graphs avoiding consecutive even cycle lengths

摘录 / EXCERPT: Let $t\geq1$ and let G be a graph with a maximum number of edges among graphs that do not contain cycles of t consecutive even lengths. A block is a maximal connected subgraph...

conjecture | open | MathDB 页面 | 原文线索

V2-048478 | MATHDB-333480

The perfect-induced-subgraph formulation of the Erdős-Hajnal conjecture

摘录 / EXCERPT: Perfect-subgraph formulation. For every graph H, there exists a constant $\psi(H) > 0$, such that every H-free graph G has a perfect induced subgraph with at least...

conjecture | open | MathDB 页面 | 原文线索

V2-048479 | MATHDB-333481

The narrowness conjecture for H-free graphs

摘录 / EXCERPT: Narrowness conjecture. For every graph H, there exists a constant $\alpha(H)\geq 1$ such that every H-free graph G is $\alpha(H)$-narrow.

conjecture | open | MathDB 页面 | 原文线索

V2-048480 | MATHDB-333482

The self-complementary Erdős-Hajnal conjecture

摘录 / EXCERPT: Self-complementary Erdős-Hajnal conjecture. For every graph H, there exists a constant $\epsilon(H)>0$ such that every $\{H,H^c\}$-free graph G has either a clique or a stable...

conjecture | open | MathDB 页面 | 原文线索

V2-048481 | MATHDB-333484

The heroic-family conjecture

摘录 / EXCERPT: Heroic-family conjecture. A finite family of graphs is heroic if and only if it contains a complete multipartite graph, the complement of a complete multipartite graph, a forest, a...

conjecture | open | MathDB 页面 | 原文线索

V2-048482 | MATHDB-333486

Gyárfás's infinite heroic-family conjecture

摘录 / EXCERPT: Gyárfás's infinite-family conjecture. For every complete graph K, and every integer $t>0$, the family consisting of K and all cycles of length at least t is heroic.

conjecture | open | MathDB 页面 | 原文线索

V2-048483 | MATHDB-333516

Eventual crossing-free-edge conjecture for extremal subgraphs of D_n

摘录 / EXCERPT: Let D_n be the graph under consideration, let $e_\ell(n)$ denote the maximum number of edges in a subgraph of D_n with local crossing number ℓ, and call an edge crossing f...

conjecture | open | MathDB 页面 | 原文线索

V2-048484 | MATHDB-333517

Asymptotic linearity conjecture for the extremal edge function $e_\ell(n)$

摘录 / EXCERPT: Fix an integer $\ell\geq 0$. Let $e_\ell(n)$ denote the maximum number of edges in a subgraph of D_n with local crossing number ℓ, and let C_ℓ be the coefficient define...

conjecture | open | MathDB 页面 | 原文线索

V2-048485 | MATHDB-333521

Minimum distance signless Laplacian spread among bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph with n vertices. The minimum distance signless Laplacian spread conjecture. ... Equality holds if and only if ... This proposes that the balanced complet...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-048486 | MATHDB-333532

Affine version of the generalized Kneser-hypergraph lower bound

摘录 / EXCERPT: Affine-map conjecture. The lower bound of Corollary remains valid for every integer $r\geq2$ when f is required to be affine.

conjecture | open | MathDB 页面 | 原文线索

V2-048487 | MATHDB-333533

Ziegler-Alon-Drewnowski-Łuczak conjecture on stable Kneser hypergraphs

摘录 / EXCERPT: Ziegler-Alon-Drewnowski-Łuczak conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048488 | MATHDB-333548

Sugawara nature of the modes of the pfaffian fields

摘录 / EXCERPT: Sugawara question. Are the modes of the elements $C_n(u)$ Sugawara elements? If not, how should they be changed to obtain Sugawara elements? The question concerns whether these Cap...

open_problem | open | MathDB 页面 | 原文线索

V2-048489 | MATHDB-333554

Monotonicity conjecture for Eulerian percolation measures

摘录 / EXCERPT: Let $G=(V,E)$ be an Eulerian graph, and let $(\mu_p)_{p\in[0,1]}$ be the corresponding Eulerian percolation measures on $\{0,1\}^E$. Monotonicity conjecture for Eulerian percolatio...

conjecture | open | MathDB 页面 | 原文线索

V2-048490 | MATHDB-333561

Probabilistic threshold conjecture for non-2-colourability games

摘录 / EXCERPT: Non-2-colourability threshold conjecture. The threshold biases satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-048491 | MATHDB-333562

Probabilistic threshold conjecture for k-SAT games

摘录 / EXCERPT: k-SAT threshold conjecture. The threshold biases satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-048492 | MATHDB-333584

Lehel's cycle partition conjecture

摘录 / EXCERPT: Let G be a graph, and let $\overline G$ denote its complement. Lehel's conjecture. The vertices of G can be covered by a cycle C_1 in G and a vertex-disjoint cycle C_2 in...

conjecture | open | MathDB 页面 | 原文线索

V2-048493 | MATHDB-333585

Erdős-Gyárfás-Pyber monochromatic cycle partition conjecture

摘录 / EXCERPT: Let r be a positive integer, and let an r-edge-coloured complete graph be a complete graph whose edges receive one of r colours.
Erdős-Gyárfás-Pyber conjecture. The vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-048494 | MATHDB-333594

Potočnik-Spiga-Verret's semiprimitive graph-restrictiveness conjecture

摘录 / EXCERPT: Let G be a finite permutation group. Call G semiprimitive if every normal subgroup of G is transitive or semiregular, and call G graph-restrictive if there is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-048495 | MATHDB-333611

The jump-separation conjecture for generalized directed hypergraph theories

摘录 / EXCERPT: Let T' and T be r-ary generalized directed hypergraph theories for $r \geq 3$. Write J_T for the set of jumps of T and m_T for its multiplicity parameter. Jump-separati...

conjecture | open | MathDB 页面 | 原文线索

V2-048496 | MATHDB-333636

Extremal construction conjecture for generalized Ramsey-Turán numbers

摘录 / EXCERPT: Let $t>s\ge 3$. For $t\le 2s-1$, let $\mathcal{H}(n,s,t)$ be the family of graphs from the stated construction: partition the vertex set into $V_1\cup\cdots\cup V_s$, use an extrem...

conjecture | open | MathDB 页面 | 原文线索

V2-048497 | MATHDB-333649

The 2-ranking conjecture for subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning that every vertex of G has degree at most 3, and let $\chi_{2}(G)$ denote its 2-ranking number. The 2-ranking conjecture for subcubic graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048498 | MATHDB-333664

The minimum-degree threshold for perfect triangle-tilings in K4-free graphs

摘录 / EXCERPT: Minimum-degree conjecture. For every $\omega>0$ there exist $\gamma,n_0>0$ such that every K_4-free graph on $n\geq n_0$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-048499 | MATHDB-333675

Conjectured equality of spectral and positive-eigenvalue Nordhaus-Gaddum maxima

摘录 / EXCERPT: Let G range over graphs on n vertices, let $\overline G$ be its complement, let $\mu(G)$ be the spectral radius, and let $s^+(G)$ be the sum of the squares of the positive adja...

conjecture | open | MathDB 页面 | 原文线索

V2-048500 | MATHDB-333676

Conference-graph lower-bound conjecture for complementary positive eigenvalue sums

摘录 / EXCERPT: Let G be a graph on n vertices, let $\overline G$ be its complement, and let $s^+(G)$ denote the sum of the squares of the positive adjacency eigenvalues. A conference graph is...

conjecture | open | MathDB 页面 | 原文线索

V2-048501 | MATHDB-333677

Randić-index inequality for triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph, let $R(G)$ be its Randić index, and let $s^+(G)$ be the sum of the squares of the positive adjacency eigenvalues. Triangle-free Randić conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-048502 | MATHDB-333687

Zamfirescu's three-longest-path conjecture

摘录 / EXCERPT: Let a longest path of a connected graph be a path of maximum length. Zamfirescu's conjecture. For every connected graph, any 3 of its longest paths have a common vertex. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-048503 | MATHDB-333689

Sublinear-distance conjecture for three to six longest paths

摘录 / EXCERPT: Let G be a connected graph of order n, let $\mathcal{L}(G)$ denote the set of longest paths of G, and let $f(G,\mathcal{P})$ denote the distance parameter for a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-048504 | MATHDB-333715

The maximum-degree-three zero forcing conjecture

摘录 / EXCERPT: Maximum-degree-three zero forcing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048505 | MATHDB-333716

Horadam in-degree subsequence conjecture for Jaco-type graphs

摘录 / EXCERPT: Let $H_0=p\in\mathbb{N}_0$, $H_1=q\in\mathbb{N}_0$, and ... where $r,s\in\mathbb{N}_0$. Let $J_\infty(\{H_n\})$ be the Horadam Jaco-type graph, with vertices v_i and in-degree subs...

conjecture | open | MathDB 页面 | 原文线索

V2-048506 | MATHDB-333724

Packing conjecture for type-l Hamilton cycles

摘录 / EXCERPT: Packing conjecture. There exists a constant C such that, if $(k-l)\mid n$ and

conjecture | open | MathDB 页面 | 原文线索

V2-048507 | MATHDB-333729

Graham--Sloane harmonious tree conjecture

摘录 / EXCERPT: Let T be a tree with n vertices. A harmonious labelling assigns distinct vertex labels and gives an edge uv the label $(\psi(u)+\psi(v))\bmod(n-1)$. Harmonious Tree Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048508 | MATHDB-333735

The logarithmic diameter conjecture for the SPA model giant component

摘录 / EXCERPT: Let $m=2$ and let $p,A_1 in(0,1)$ be parameters of the SPA model, with A_2 chosen sufficiently large. The giant component is the largest connected component of the graph produced...

conjecture | open | MathDB 页面 | 原文线索

V2-048509 | MATHDB-333738

A majorization-product conjecture for bipartite graph degree and Laplacian data

摘录 / EXCERPT: Let $n=p+q>1$ be an integer. Let $a\vdash p$ and $b\vdash q$ be integer partitions, and let $d=d_1\geq d_2\geq\dots\geq d_n$ be their union. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-048510 | MATHDB-333739

Godsil's controllability conjecture for random graphs

摘录 / EXCERPT: Let $G(n,1/2)$ be the Erdős-Rényi random graph on n vertices, and let G be controllable and minimally controllable when it has the corresponding controllability properties for...

conjecture | solved | MathDB 页面 | 原文线索

V2-048511 | MATHDB-333766

The weak Hedetniemi conjecture

摘录 / EXCERPT: Let $\chi(G)$ denote the chromatic number of a graph G. The weak Hedetniemi conjecture. For every integer n, there exists an integer $f(n)$ such that whenever G and H are g...

conjecture | open | MathDB 页面 | 原文线索

V2-048512 | MATHDB-333772

Győri-Katona-Lemons conjecture for Berge paths in uniform hypergraphs

摘录 / EXCERPT: Let r and k satisfy $k=r+1>2$, and let $\mathcal{H}$ be an r-uniform hypergraph, meaning that every hyperedge has size r. A Berge path of length k consists of k distinc...

conjecture | open | MathDB 页面 | 原文线索

V2-048513 | MATHDB-333774

Douglas's asymptotic Kac-polynomial conjecture

摘录 / EXCERPT: Let Γ be a quiver with vertex set Γ_0, let $\R \Gamma_0$ be the associated real vector space, let $\N \Gamma_0$ be the monoid of dimension vectors, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048514 | MATHDB-333775

The forbidden Berge hypergraph conjecture for the 4-cycle

摘录 / EXCERPT: Forbidden Berge hypergraph conjecture for the 4-cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-048515 | MATHDB-333783

Seven-vertex conjecture for overtwisted contact structures on the 3-sphere

摘录 / EXCERPT: Let xi be an overtwisted contact structure on the 3-sphere $\mathbb{S}^3$, and let a contact triangulation of $(\mathbb{S}^3,xi)$ be a triangulation compatible with the contact s...

conjecture | open | MathDB 页面 | 原文线索

V2-048516 | MATHDB-333808

Asymptotic optimality conjecture for the construction H_{r,s,t}(n)

摘录 / EXCERPT: Let $g_r(n,m)$ denote the maximum number of vertices that one is required to remove from an n-vertex, $(r+1)$-saturated graph with at least $t_r(n)-m$ edges so that the remaining...

conjecture | open | MathDB 页面 | 原文线索

V2-048517 | MATHDB-333822

Odd-cycle-different Hamiltonian paths conjecture

摘录 / EXCERPT: Odd-cycle-different Hamiltonian paths conjecture. If $k>1$ and n is large enough, the maximal number of pairwise C_{2k+1}-different Hamiltonian paths on n vertices is equal t...

conjecture | open | MathDB 页面 | 原文线索

V2-048518 | MATHDB-333823

Asymptotic odd-cycle-different Hamiltonian paths conjecture

摘录 / EXCERPT: Asymptotic odd-cycle-different Hamiltonian paths conjecture. If $k>1$, then the maximal number of pairwise C_{2k+1}-different Hamiltonian paths on n vertices is at least

conjecture | open | MathDB 页面 | 原文线索

V2-048519 | MATHDB-333829

The 2-dimensional Dowker persistence conjecture for cycle networks

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geq 3$, and let G_n be a cycle network. Write $\operatorname{Dgm}_2^{\mathfrak{D}}(G_n)$ for its 2-dimensional Dowker persistence diagram. Cycle-netw...

conjecture | open | MathDB 页面 | 原文线索

V2-048520 | MATHDB-333834

Bondy-Erdős conjecture for multicolour Ramsey numbers of odd cycles

摘录 / EXCERPT: Let $k\geq 2$ and let $n>3$ be odd. Write $R_k(C_n)$ for the least integer N such that every colouring of the edges of the complete graph K_N with k colours contains a monoch...

conjecture | open | MathDB 页面 | 原文线索

V2-048521 | MATHDB-333845

The circular altitude conjecture for the third Mycielskian of K_6

摘录 / EXCERPT: Circular altitude conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048522 | MATHDB-333862

The regular-graph zero-sum 5-flow conjecture

摘录 / EXCERPT: Let G be an r-regular graph, meaning every vertex of G has degree r, with $r\geq 3$. A zero-sum 5-flow is a zero-sum flow whose integer edge values satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-048523 | MATHDB-333867

Maximum atom-bond connectivity index for graphs with given chromatic number

摘录 / EXCERPT: Let G be an n-vertex connected graph with chromatic number $\chi\geq 3$. For $t\geq 1$, let $T_{n,t}$ denote the complete t-partite graph of order n whose partition sizes d...

conjecture | open | MathDB 页面 | 原文线索

V2-048524 | MATHDB-333875

Steffen's conjecture on hypohamiltonian snarks

摘录 / EXCERPT: Steffen's conjecture. If G is a hypohamiltonian snark, then

conjecture | open | MathDB 页面 | 原文线索

V2-048525 | MATHDB-333882

The tight-bound conjecture for TSP walks in 2-connected subcubic graphs

摘录 / EXCERPT: Let G be a 2-connected subcubic graph with n vertices, and let n_2 denote the number of vertices of degree two in G. A TSP walk is a walk that visits every vertex of G....

conjecture | open | MathDB 页面 | 原文线索

V2-048526 | MATHDB-333885

General graph conjecture for social connectivity time

摘录 / EXCERPT: Let G be a connected graph with n vertices and average degree d. Let $\mathrm{SC}(G)$ denote its social connectivity time. General graph conjecture. There exists a universal...

conjecture | open | MathDB 页面 | 原文线索

V2-048527 | MATHDB-333886

Monotonicity in the number of walkers

摘录 / EXCERPT: Let G be a vertex-transitive graph with stationary distribution π. For $k\;\geqslant\;2$, let $\mathrm{SC}(G,k)$ be the social connectivity time when k walkers independentl...

conjecture | open | MathDB 页面 | 原文线索

V2-048528 | MATHDB-333887

Infinite-class conjecture for social connectivity on infinite graphs

摘录 / EXCERPT: Let G be a connected infinite bounded-degree graph whose critical density for independent percolation is strictly less than 1. At each vertex v, independently place a Poisson...

conjecture | open | MathDB 页面 | 原文线索

V2-048529 | MATHDB-333891

Javadi-Maleki-Omoomi conjecture on local clique cover numbers

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\operatorname{lcc}(G)$ denote its local clique cover number. Let $\overline{G}$ be the complement of G. Javadi-Maleki-Omoomi conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-048530 | MATHDB-333893

The clique cover bound for triad-free graphs

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\alpha(G)$ denote its independence number, the maximum size of a set of pairwise nonadjacent vertices. The graph is triad-free when...

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V2-048531 | MATHDB-333944

The star conjecture for minimum von Neumann entropy of connected graphs

摘录 / EXCERPT: Let G be a connected graph on n vertices. Its scaled Laplacian is $\rho(G):=L(G)/\operatorname{tr}L(G)$, where $L(G)=D(G)-A(G)$, with $D(G)$ the diagonal degree matrix and...

conjecture | open | MathDB 页面 | 原文线索

V2-048532 | MATHDB-333945

The path conjecture for maximum von Neumann entropy of trees

摘录 / EXCERPT: The path conjecture. For all trees on n vertices, the maximum von Neumann entropy is attained by the path P_n.

conjecture | open | MathDB 页面 | 原文线索

V2-048533 | MATHDB-333946

The star conjecture for minimum Rényi entropy of connected graphs

摘录 / EXCERPT: Let G be a graph with at least one edge, let $L(G)=D(G)-A(G)$ be its combinatorial Laplacian, and define the scaled Laplacian by $\rho(G):=L(G)/\operatorname{tr}L(G)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-048534 | MATHDB-333950

Simple-trajectory conjecture for denominators of two-move riders

摘录 / EXCERPT: Let $\mathbf z$ be a vertex of an inside-out polytope for q two-move riders, with components $z_i\in\mathbb R^2$. A simple trajectory is a primitive trajectory with no equal cons...

conjecture | open | MathDB 页面 | 原文线索

V2-048535 | MATHDB-333951

Maximal-denominator conjecture for semiqueens

摘录 / EXCERPT: Let $\mathbb Q^{21}$ be the semiqueen, and let F_m denote the mth Fibonacci number. Semiqueen maximal-denominator conjecture. The largest denominator of a vertex configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-048536 | MATHDB-333952

Maximal-denominator conjecture for frontal queens

摘录 / EXCERPT: Let $\mathbb Q^{12}$ be the frontal queen, and let F_m denote the mth Fibonacci number. Frontal-queen maximal-denominator conjecture. The largest denominator of a vertex config...

conjecture | open | MathDB 页面 | 原文线索

V2-048537 | MATHDB-333953

Maximal-denominator conjecture for queens

摘录 / EXCERPT: Let F_q denote the qth Fibonacci number, and consider vertex configurations of the inside-out polytope for q queens. Queen maximal-denominator conjecture. The largest denomin...

conjecture | open | MathDB 页面 | 原文线索

V2-048538 | MATHDB-333954

Conjecture on the cop number of fast robbers on grids

摘录 / EXCERPT: Fast-robber grid conjecture. For all sufficiently large $R\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-048539 | MATHDB-333966

The asymptotic frequency conjecture for torus graphs with one, two, or three diagonals

摘录 / EXCERPT: Asymptotic frequency conjecture. The limiting frequencies are

conjecture | open | MathDB 页面 | 原文线索

V2-048540 | MATHDB-333968

The Graovac-Ghorbani index conjecture for almost dendrimers

摘录 / EXCERPT: Let G be a tree on n vertices with maximum degree $\Delta\leq n-1$. An almost dendrimer $T_{n,d}$ is a rooted tree with n vertices in which every non-pendant vertex, except pe...

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V2-048541 | MATHDB-333976

The extremal K_4-free graph conjecture for fractional separation dimension

摘录 / EXCERPT: The extremal K_4-free graph conjecture. For $n\ge 10$, the n-vertex graph not containing K_4 that maximizes π_f is $K_{1,\lfloor (n-1)/2\rfloor,\lceil (n-1)/2\rceil}$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048542 | MATHDB-333977

Distinct-degree strengthening for graphs with small homogeneous sets

摘录 / EXCERPT: Let $k\ge 2$ be a positive integer, let n be sufficiently large, and let G be an n-vertex graph. Write $f(G)$ for the maximum number of distinct degrees in an induced subgrap...

conjecture | open | MathDB 页面 | 原文线索

V2-048543 | MATHDB-333978

Many distinct degrees from large homogeneous sets

摘录 / EXCERPT: Let G be an n-vertex graph. Write $f(G)$ for the maximum number of distinct degrees in an induced subgraph of G, and let $\hom(G)$ denote the maximum size of a homogeneous se...

conjecture | open | MathDB 页面 | 原文线索

V2-048544 | MATHDB-333979

Erdős-Faudree-Sós conjecture on size pairs of induced subgraphs of Ramsey graphs

摘录 / EXCERPT: Erdős-Faudree-Sós conjecture. For any $C>0$, there exists a $\delta=\delta(C)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048545 | MATHDB-333980

Uniform-density conjecture for edge counts of induced subgraphs

摘录 / EXCERPT: Uniform-density conjecture. For any fixed $\varepsilon>0$, if G is a uniformly ε-dense graph on n vertices, then

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V2-048546 | MATHDB-333995

The conjecture that graph conditioning reduces average resistance

摘录 / EXCERPT: Graph-conditioning resistance conjecture. The process of conditioning a graph reduces the average resistance between any two vertices on the graph.

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V2-048547 | MATHDB-334015

Pivotto's finite excluded-minor conjecture for graphic or bicircular lift matroids

摘录 / EXCERPT: Let BL be the class of bicircular lift matroids, and let $\overline{BL}$ be the union of BL with the class of graphic matroids. The class $\overline{BL}$ is minor-closed. Pivot...

conjecture | open | MathDB 页面 | 原文线索

V2-048548 | MATHDB-334037

Alon and Tarsi's shortest circuit cover conjecture

摘录 / EXCERPT: Let G be a 2-edge-connected cubic graph. A shortest circuit cover of G is a circuit cover having minimum total length, denoted by $\operatorname{scc}(G)$. Alon and Tarsi's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-048549 | MATHDB-334038

The Circuit Double Cover Conjecture of Szekeres and Seymour

摘录 / EXCERPT: Let G be a 2-edge-connected graph, and let a circuit double cover be a family of circuits in which every edge of G is covered twice. Circuit Double Cover Conjecture. Every 2-ed...

conjecture | open | MathDB 页面 | 原文线索

V2-048550 | MATHDB-334043

Permutation-process convergence to the Archimedean process

摘录 / EXCERPT: Archimedean process convergence conjecture. $\mathbf{RSN}^n$ converges in probability, as a permutation process, to the Archimedean process.

conjecture | open | MathDB 页面 | 原文线索

V2-048551 | MATHDB-334080

The 2-Decomposition Conjecture for graphs with separating cycles

摘录 / EXCERPT: 2-Decomposition Conjecture. The graph G can be decomposed into a spanning tree and a nonempty matching.

conjecture | open | MathDB 页面 | 原文线索

V2-048552 | MATHDB-334086

Turan's conjecture on graph-property sensitivity

摘录 / EXCERPT: Let f be a non-trivial graph property for graphs with n vertices, so that f is a Boolean function of the ${n \choose 2}$ possible edges. Turan's conjecture. The sensitivity s...

conjecture | open | MathDB 页面 | 原文线索

V2-048553 | MATHDB-334087

The min-term graph-property sensitivity conjecture

摘录 / EXCERPT: Let f be a Boolean graph property on the ${n \choose 2}$ possible edges. A min-term graph property is a graph property of the type defined in Chakraborty's work, and f is non-t...

conjecture | open | MathDB 页面 | 原文线索

V2-048554 | MATHDB-334103

The tightness conjecture for diffusion games

摘录 / EXCERPT: Tightness conjecture. Every finite graph and initial configuration is tight.

conjecture | open | MathDB 页面 | 原文线索

V2-048555 | MATHDB-334114

Keevash-Knox-Mycroft complexity conjecture for dense hypergraph perfect matchings

摘录 / EXCERPT: Keevash-Knox-Mycroft conjecture. The decision problem

conjecture | open | MathDB 页面 | 原文线索

V2-048556 | MATHDB-334115

Hypergeometric formula for matching polynomials of complete multipartite graphs

摘录 / EXCERPT: Let $K_{(n)}$ be the complete k-partite graph with $N=n_1+\dots+n_k$ vertices, whose k color classes have sizes $n_1,\dots,n_k$. For odd r, let $M_r(K_{(n)})$ denote its matchi...

conjecture | open | MathDB 页面 | 原文线索

V2-048557 | MATHDB-334130

Thomassen's finiteness conjecture for 5-list-critical graphs on surfaces

摘录 / EXCERPT: Let $d4a2$ be a surface, and let G be a graph embedded in $d4a2$. A list assignment L assigns a set of colors to each vertex; G is L-critical if it has no L-coloring but...

conjecture | open | MathDB 页面 | 原文线索

V2-048558 | MATHDB-334135

Zaare-Nahandi's semi-perfectness conjecture for multipartite well-covered graphs

摘录 / EXCERPT: Zaare-Nahandi's conjecture. G is semi-perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-048559 | MATHDB-334146

The distinguishing stability bound for simple connected graphs

摘录 / EXCERPT: Distinguishing stability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048560 | MATHDB-334151

Signed Talaska formula for perfectly oriented networks on surfaces

摘录 / EXCERPT: Signed Talaska formula. There exists a map

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V2-048561 | MATHDB-334155

The ordered Erdős-Hajnal tower-growth conjecture for tight paths

摘录 / EXCERPT: For integers $2\le k < s < n$ and $2\le t\le {s\choose k}$, let $r_k(s,t;P_n)$ be the minimum N such that every red/blue coloring of the k-sets of $[N]$ contains a monochromati...

conjecture | open | MathDB 页面 | 原文线索

V2-048562 | MATHDB-334188

The power-bound conjecture for $(u,u-4)$-free hypergraphs

摘录 / EXCERPT: $(u,u-4)$-free hypergraph power-bound conjecture. For every u there exists $\epsilon>0$ such that G has at most

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V2-048563 | MATHDB-334195

Mou and Pasechnik's Hamiltonicity conjecture for tough $2K_2$-free graphs

摘录 / EXCERPT: Mou and Pasechnik's conjecture. Every 2-tough $2K_2$-free graph with at least three vertices is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-048564 | MATHDB-334196

The $\frac{5}{4}$-tough spanning 2-trail conjecture for $2K_2$-free graphs

摘录 / EXCERPT: The $\frac{5}{4}$-tough spanning 2-trail conjecture. Any $\frac{5}{4}$-tough $2K_2$-free graph with at least three vertices has a spanning 2-trail.

conjecture | open | MathDB 页面 | 原文线索

V2-048565 | MATHDB-334216

Layer-width conjecture for ideality in multilayer random networks

摘录 / EXCERPT: Multilayer random-network conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-048566 | MATHDB-334223

Kühn-Lapinskas-Osthus conjecture on Hamilton decompositions of Dirac graphs

摘录 / EXCERPT: Let G be a Dirac graph, meaning an n-vertex graph with minimum degree at least $n/2$, and let $\operatorname{reg}(G)$ be the largest even integer r for which G contains an...

conjecture | open | MathDB 页面 | 原文线索

V2-048567 | MATHDB-334229

Edge-size conjecture for diameter-2, 3_t-critical graphs

摘录 / EXCERPT: Let G be a simple graph of order n. A graph is 3_t-critical if its total domination number is 3 and adding an edge between any nonadjacent vertices lowers its total dominat...

conjecture | open | MathDB 页面 | 原文线索

V2-048568 | MATHDB-334238

Mader's conjecture for oriented trees

摘录 / EXCERPT: Let a δ^+-maderian digraph be a digraph contained as a subdivision in every digraph whose minimum out-degree is sufficiently large. An oriented tree is an orientation of an...

conjecture | open | MathDB 页面 | 原文线索

V2-048569 | MATHDB-334239

Alon's partition conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph, let $D\langle V_i\rangle$ denote the subdigraph induced by V_i, and let k be a positive integer. Alon's partition conjecture. There exists a function h...

conjecture | open | MathDB 页面 | 原文线索

V2-048570 | MATHDB-334240

Lovász's conjecture on even directed cycles

摘录 / EXCERPT: A digraph is p-strongly connected when its strong connectivity is at least p. Lovász's conjecture. There exists an integer p such that every p-strongly connected digraph ha...

conjecture | open | MathDB 页面 | 原文线索

V2-048571 | MATHDB-334241

The oriented-cycle minimum-out-degree conjecture

摘录 / EXCERPT: Let C be an oriented cycle, and let $|V(C)|$ denote its order. Oriented-cycle conjecture. Every digraph D with $\delta^+(D)\geq 2k-1$ contains a subdivision of any oriented cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-048572 | MATHDB-334242

The dichromatic Mader bound for oriented cycles

摘录 / EXCERPT: Let C be an oriented cycle, and let $\vec\chi(D)$ be the dichromatic number of a digraph D. Dichromatic-cycle conjecture. Every oriented cycle C satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-048573 | MATHDB-334257

Lexicographic edge-isoperimetric conjecture for generalized expanded Sierpinski graphs

摘录 / EXCERPT: Let $S(n,m)$ be the generalized and expanded Sierpinski graph, with vertex set $V_{S(n,m)}=\left\{0,1,\ldots,m-1\right\}^{n}$. Let Lex denote the lexicographic order on this vert...

conjecture | open | MathDB 页面 | 原文线索

V2-048574 | MATHDB-334258

Generalized edge-isoperimetric conjecture for Sierpinski graphs with exterior boundary conditions

摘录 / EXCERPT: Let $I=\left\{0,1,\ldots,s-1\right\}$, $J=\left\{s,s+1,\ldots,s+t-1\right\}$, and $K=\left\{s+t,s+t+1,\ldots,m-1\right\}$, where $s,t\geq0$ and $s+t\leq m$. Let $S_{s,t}(n,m)$ be t...

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V2-048575 | MATHDB-334260

The complete multipartite extremal conjecture for the maximum ABC index

摘录 / EXCERPT: The complete multipartite extremal conjecture.

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V2-048576 | MATHDB-334269

Nordhaus-Gaddum conjecture for power domination

摘录 / EXCERPT: Let G be a graph of order n, and let $\overline{G}$ denote its complement. Assume that every component of both G and $\overline{G}$ has order at least 3. Write...

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V2-048577 | MATHDB-334290

Quasi-line graph 5/6 conjecture

摘录 / EXCERPT: A quasi-line graph is a graph in which the neighborhood of every vertex is covered by two cliques. For a graph Q, let $\chi(Q)$, $\omega(Q)$, and $\Delta(Q)$ denote its chromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-048578 | MATHDB-334310

The extremal susceptibility conjecture for regular graphs

摘录 / EXCERPT: Let $d\geq2$ and $\lambda>0$. Let $\mathfrak{G}(n,d)$ be the collection of all connected n-vertex d-regular graphs, and let $H_{d,n}$ be the graph defined in the source by arra...

conjecture | open | MathDB 页面 | 原文线索

V2-048579 | MATHDB-334312

Chromatic-number criterion for zero-forcing graphs

摘录 / EXCERPT: Let $mathcal G$ be a finite abelian group, let $D(mathcal G)$ denote its Davenport constant, and let G be a graph. A graph is zero-forcing for $mathcal G$ if every $mathcal G$-la...

conjecture | open | MathDB 页面 | 原文线索

V2-048580 | MATHDB-334318

Jaeger-Linial-Payan-Tarsi conjecture on _3-connected graphs

摘录 / EXCERPT: Let G be a finite graph without loops, possibly with multiple edges. A graph is $\mathbb{Z}_3$-connected if, for every zero-sum function $\beta:V(G)\to\mathbb{Z}_3$, there is an...

conjecture | open | MathDB 页面 | 原文线索

V2-048581 | MATHDB-334319

Extendability conjecture for 5-edge-connected essentially 6-edge-connected graphs

摘录 / EXCERPT: Let G be a graph and let $z_0\in V(G)$. A graph is $\langle\mathbb{Z}_3\rangle$-extendable at z_0 if every pre-orientation of the edges incident with z_0 whose out-degree min...

conjecture | open | MathDB 页面 | 原文线索

V2-048582 | MATHDB-334332

The homomorphism threshold conjecture for odd-cycle-free graphs

摘录 / EXCERPT: Homomorphism threshold conjecture.

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V2-048583 | MATHDB-334396

The triangle-freeness conjecture for Cheeger graphs

摘录 / EXCERPT: Let G be a Cheeger graph, meaning a graph attaining the first Cheeger constant of the relevant simplex. Triangle-freeness conjecture. All Cheeger graphs are triangle-free. This i...

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V2-048584 | MATHDB-334397

The bipartiteness conjecture for Cheeger graphs

摘录 / EXCERPT: Let G be a Cheeger graph, meaning a graph attaining the first Cheeger constant of the relevant simplex. Bipartiteness conjecture. All Cheeger graphs are bipartite. This is propos...

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V2-048585 | MATHDB-334398

The staircase representation conjecture for Cheeger graphs

摘录 / EXCERPT: Let $G_2(1)$ be the indicated two-vertex graph, let $\sqcup$ denote disjoint union, and let a staircase graph mean a graph of the staircase type defined in the paper. Staircase rep...

conjecture | open | MathDB 页面 | 原文线索

V2-048586 | MATHDB-334418

Connectivity threshold conjecture for heterogeneous random key graphs

摘录 / EXCERPT: Connectivity threshold conjecture. Possibly under some additional conditions, the heterogeneous random graph $\mathbb{H}(n;\pmb{\mu},\pmb{\Theta}_n)$ should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-048587 | MATHDB-334437

Davila-Kenter forcing-number conjecture for graphs with given girth

摘录 / EXCERPT: Davila-Kenter conjecture.

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V2-048588 | MATHDB-334438

Potts internal-energy bounds for regular graphs

摘录 / EXCERPT: Regular-graph Potts bounds. The conjectured inequalities are

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V2-048589 | MATHDB-334451

The fixed-size independent-set extremal conjecture

摘录 / EXCERPT: Let G be a d-regular graph with $2ad$ vertices, and let $i_t(G)$ denote the number of independent sets of size t in G. Recall that $aK_{d,d}$ is the disjoint union of a c...

conjecture | open | MathDB 页面 | 原文线索

V2-048590 | MATHDB-334454

The fixed-size matching extremal conjecture

摘录 / EXCERPT: Let G be a d-regular graph on $2ad$ vertices, and let $m_t(G)$ denote the number of matchings of G with t edges. Fixed-size matching conjecture. For every integer $t\ge 0$,...

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V2-048591 | MATHDB-334457

Type C coset conjecture for principal minimal W-algebras

摘录 / EXCERPT: Type C coset conjecture. The vertex algebra ${\mathcal C}_k(n)$ is isomorphic to the C_2-cofinite, rational principal ${\mathcal W}$-algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-048592 | MATHDB-334458

Type A coset conjecture for subregular W-algebras

摘录 / EXCERPT: Type A coset conjecture. There is an isomorphism

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V2-048593 | MATHDB-334507

The disjoint-paths Turán formula conjecture

摘录 / EXCERPT: Let P_k denote the path with k vertices. For integers $k_1\geq k_2\geq\cdots\geq k_m\geq 3$ with $k_1>3$, define ... Write $e(n,F_m)$ for the maximum number of edges in an n-ve...

conjecture | open | MathDB 页面 | 原文线索

V2-048594 | MATHDB-334534

Vertex-transitive graph power decomposition conjecture

摘录 / EXCERPT: For a finite graph G, let G^n be the graph with vertex set $V(G)^n$, in which two vertices are adjacent exactly when they differ in one coordinate and the entries in that coord...

conjecture | open | MathDB 页面 | 原文线索

V2-048595 | MATHDB-334536

Quantitative forcing conjecture for bipartite graphs

摘录 / EXCERPT: Let H be a fixed bipartite graph of girth g, and let p be a fixed constant with $0<p<1$. For each $\epsilon>0$, there is a parameter $\delta=\Omega(\epsilon^g)$ such that any...

conjecture | open | MathDB 页面 | 原文线索

V2-048596 | MATHDB-334540

The three-color conjecture for locally irregular chromatic index

摘录 / EXCERPT: Let G be a finite simple decomposable graph, meaning that it admits a locally irregular edge-coloring. Its locally irregular chromatic index, denoted by $\chi_{\mathrm{irr}}'(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-048597 | MATHDB-334547

Heavy-traffic queue-length distribution conjecture for roving server networks

摘录 / EXCERPT: Let L_i be the queue length at queue i, let $\mathcal{L}^{\textit{fluid}}_{i,k}$ be the standardised number of type-i particles in the fluid model during visit period V_k,...

conjecture | open | MathDB 页面 | 原文线索

V2-048598 | MATHDB-334548

Heavy-traffic waiting-time distribution conjecture for roving server networks

摘录 / EXCERPT: Let W_i be the waiting time of a type-i customer, and let $\mathcal{W}^{\textit{fluid}}_i$ be the standardised waiting time of a type-i particle in the fluid model. Let...

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V2-048599 | MATHDB-334549

Heavy-traffic path-time distribution conjecture for roving server networks

摘录 / EXCERPT: Let $W_{i_1,i_2,\dots,i_M}$ be the path time along the customer route $i_1,i_2,\dots,i_M$, and let $\mathcal{W}^{\textit{fluid}}_{i_1,i_2,\dots,i_M}$ be the corresponding standardi...

conjecture | open | MathDB 页面 | 原文线索

V2-048600 | MATHDB-334553

The bounded-degree tree Ramsey goodness conjecture

摘录 / EXCERPT: Let T be a tree, let Δ and k be positive integers, and let H be a graph with chromatic number $\chi(H)=k$. A graph T is H-good if its Ramsey number with H satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-048601 | MATHDB-334554

The one-law for k-connectivity in inhomogeneous random key graphs with unreliable links

摘录 / EXCERPT: Let $\mathbb{K \cap G}(n;\pmb{\Theta}_n)$ denote the intersection of an inhomogeneous random key graph and an Erdős-Rényi graph, with class probability distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-048602 | MATHDB-334558

Engbers's asymptotic extremal conjecture for graph homomorphisms

摘录 / EXCERPT: Let $\delta\geq1$ and let H be a graph. Let G be a graph on n vertices with minimum degree at least δ. Engbers's conjecture. There exists a constant $c(\delta,H)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-048603 | MATHDB-334561

The equirepresentation matching conjecture for K_{n,n}

摘录 / EXCERPT: Let the edge set of $K_{n,n}$ be partitioned into sets $E_1,\ldots,E_n$, each of size n. A matching is a set of pairwise vertex-disjoint edges. Equirepresentation matching conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-048604 | MATHDB-334562

Awanless-Barát rainbow matching conjecture

摘录 / EXCERPT: Let G be a bipartite graph and let $E_1,\ldots,E_m$ be sets of edges. Write $\Delta(H)$ for the maximum degree of a graph H. A rainbow matching is a matching containing an edge...

conjecture | open | MathDB 页面 | 原文线索

V2-048605 | MATHDB-334563

Fair matching conjecture for graph edge partitions

摘录 / EXCERPT: Let H be a graph with maximum degree $\Delta(H)$, and partition its edge set into $E_1,\ldots,E_m$. A matching is a set of pairwise vertex-disjoint edges. Fair matching conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048606 | MATHDB-334565

Chromatic number conjecture for powers of cycle Kneser hypergraphs

摘录 / EXCERPT: For integers $n\geqslant s\geqslant2$ with $n>ks$, let C_n^{s-1} be the $(s-1)$-st power of the cycle of length n, and let $K(n,k,s)$ be the s-uniform hypergraph whose vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-048607 | MATHDB-334568

Gyárfás-Sárközy's monochromatic loose path partition conjecture

摘录 / EXCERPT: Let K_n^k be the complete k-uniform hypergraph on n vertices, with its edges colored using two colors. A loose path is a hypergraph path in which consecutive edges intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-048608 | MATHDB-334574

Aharoni-Berger-Chudnovsky-Howard-Seymour rainbow matching conjecture

摘录 / EXCERPT: Let n be a positive integer. A rainbow matching for a family of matchings is a matching whose edges are chosen from distinct members of the family. Rainbow matching conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-048609 | MATHDB-334584

Captain inclusion conjecture for the tournament equilibrium set

摘录 / EXCERPT: Let T be a tournament. A captain vertex is a vertex with the captain property used in the paper, and let A be the set of all captain vertices of T. Captain inclusion conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-048610 | MATHDB-334585

The genus existence and finiteness conjecture for unitary modular tensor categories

摘录 / EXCERPT: Genus existence and finiteness conjecture. 1. There exists a central charge c such that the admissible genus $(\mathcal{C},c)$ is realizable. 2. Each realizable genus...

conjecture | open | MathDB 页面 | 原文线索

V2-048611 | MATHDB-334587

The finiteness conjecture for extremal vertex operator algebras

摘录 / EXCERPT: Extremal VOA finiteness conjecture. 1. There are finitely many isomorphism classes of extremal VOAs realizing $(\mathcal{C},c)$. 2. There are finitely many character vectors realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-048612 | MATHDB-334595

Lower heavy-tail conjecture for the nearest-interferer path-loss ratio

摘录 / EXCERPT: Lower heavy-tail conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048613 | MATHDB-334596

Heavy-tail conjecture for unbounded signal power in SIR models

摘录 / EXCERPT: Heavy-tail conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048614 | MATHDB-334597

Exponential-tail conjecture for bounded signal power with Rayleigh fading

摘录 / EXCERPT: Exponential-tail conjecture. For Rayleigh fading,

conjecture | open | MathDB 页面 | 原文线索

V2-048615 | MATHDB-334605

The 8/9-density conjecture for monochromatic paths

摘录 / EXCERPT: Let $K_\NN$ be the complete graph on the positive integers, and define the upper density of a vertex set $A\subseteq\NN$ by $\bar d(A)=\limsup_{n\to\infty}|A\cap[n]|/n$. The upper...

conjecture | open | MathDB 页面 | 原文线索

V2-048616 | MATHDB-334606

The connected-subgraph density conjecture for r-colorings

摘录 / EXCERPT: Let $K_\NN$ be the complete graph on the positive integers. For a subgraph T of $K_\NN$, write $\bar d(T)$ for the upper density of its vertex set and $\bar d_S(T)$ for its stron...

conjecture | open | MathDB 页面 | 原文线索

V2-048617 | MATHDB-334607

The bounded-degree monochromatic density conjecture

摘录 / EXCERPT: Let $\bar d(G)$ denote the upper density of the vertex set of a graph G embedded in $K_\NN$. Bounded-degree density conjecture. For every $\Delta\in\NN$, there exists $c>0$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-048618 | MATHDB-334618

The edge-transitivity characterization of weakly norming graphs

摘录 / EXCERPT: Let H be a bipartite graph. Its cut involution group is the group of graph symmetries generated by cut involutions, and H is weakly norming when its associated homomorphism-den...

conjecture | open | MathDB 页面 | 原文线索

V2-048619 | MATHDB-334619

The norming graph characterization by stable involutions

摘录 / EXCERPT: Let H be a bipartite graph, let $t_H(f)$ denote its homomorphism density functional, and let the stable involution group be the corresponding group of stable involutions of H....

conjecture | open | MathDB 页面 | 原文线索

V2-048620 | MATHDB-334632

Lovász's Hamiltonicity conjecture for vertex-transitive graphs

摘录 / EXCERPT: A graph is vertex transitive if, for every pair of vertices v,w, there is a graph automorphism mapping v to w. A graph is Hamiltonian if it admits a Hamiltonian circuit, name...

conjecture | open | MathDB 页面 | 原文线索

V2-048621 | MATHDB-334645

Davila-Kenter's lower-bound conjecture for the zero forcing number

摘录 / EXCERPT: Davila-Kenter's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048622 | MATHDB-334674

Häggkvist-Kriesell conjecture on locally self-avoiding Eulerian tours

摘录 / EXCERPT: Häggkvist-Kriesell conjecture. For every positive integer ℓ, there is an integer d_ℓ such that every Eulerian graph G with minimum degree at least d_ℓ admits an...

conjecture | open | MathDB 页面 | 原文线索

V2-048623 | MATHDB-334675

Aharoni-Zerbib's generalization of Tuza's conjecture

摘录 / EXCERPT: Aharoni-Zerbib's generalization of Tuza's conjecture. Every 3-uniform hypergraph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048624 | MATHDB-334676

Aharoni-Zerbib's extremal ratio conjecture for generalized matchings

摘录 / EXCERPT: Aharoni-Zerbib's extremal ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048625 | MATHDB-334677

Aharoni-Zerbib's conjecture for generalized $(k-1)$-covers

摘录 / EXCERPT: Aharoni-Zerbib's conjecture for generalized $(k-1)$-covers.

conjecture | open | MathDB 页面 | 原文线索

V2-048626 | MATHDB-334693

The road closure conjecture for primitive basic permutation groups

摘录 / EXCERPT: Road closure conjecture. If G does not satisfy condition (c) or (d) of Theorem 4, then G has the road closure property.

conjecture | open | MathDB 页面 | 原文线索

V2-048627 | MATHDB-334694

Generic shrinker conjecture for network singularities

摘录 / EXCERPT: Consider the curvature flow of a network and the listed dynamically stable shrinkers: the line, unit circle, standard triod, standard cross, Brakke spoon, lens, and three-ray star....

conjecture | open | MathDB 页面 | 原文线索

V2-048628 | MATHDB-334697

The general DoF upper-bound conjecture for cached two-way 2×2×2 relay networks

摘录 / EXCERPT: General DoF upper-bound conjecture. The total degrees of freedom satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048629 | MATHDB-334705

He and Ye's free-vertex conjecture for 4-regular 4-uniform hypergraphs

摘录 / EXCERPT: He and Ye's free-vertex conjecture. Every 4-regular 4-uniform hypergraph contains a free vertex.

conjecture | open | MathDB 页面 | 原文线索

V2-048630 | MATHDB-334722

The 2-connected chromatic polynomial ear conjecture

摘录 / EXCERPT: Let G be a 2-connected k-chromatic graph of order $n>k\geq 4$. An ear is a new path or cycle that overlaps an existing graph only in its two endpoints. The 2-connected ear conj...

conjecture | open | MathDB 页面 | 原文线索

V2-048631 | MATHDB-334737

The classification conjecture for nice holomorphic vertex operator algebras of central charge 24

摘录 / EXCERPT: Let V be a nice, holomorphic vertex operator algebra of central charge 24, and let V_1 denote its weight-one subspace with its induced Lie algebra structure. Classification c...

conjecture | open | MathDB 页面 | 原文线索

V2-048632 | MATHDB-334740

The tree-width bound for real parts of chromatic roots

摘录 / EXCERPT: Let G be a graph with tree-width k, and let $\z$ be a root of its chromatic polynomial $\pi(G,x)$. Tree-width conjecture. The real part of z satisfies ... This conjecture propo...

conjecture | open | MathDB 页面 | 原文线索

V2-048633 | MATHDB-334759

Tightened voltage stability index approximation bound

摘录 / EXCERPT: Let a power distribution network have one slack bus and n PQ loads, and let it be described by the relaxed branch flow model. In the voltage stability region, set ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-048634 | MATHDB-334765

Conjecture on highly valenced graphs with two main and two plain eigenvalues

摘录 / EXCERPT: Valency conjecture. There exists a positive integer C such that if $t\geq C$, then Γ is a strong graph.

conjecture | open | MathDB 页面 | 原文线索

V2-048635 | MATHDB-334766

Largest-eigenvalue conjecture for equitable quotient matrices

摘录 / EXCERPT: Largest-eigenvalue conjecture. The largest eigenvalue of $B(M)$ is the largest eigenvalue of M.

conjecture | open | MathDB 页面 | 原文线索

V2-048636 | MATHDB-334795

The three-color homothetic-copy conjecture for convex sets

摘录 / EXCERPT: Let D be a convex set in the plane, and call a scaled and translated copy of D without rotation a homothetic copy of D. Let S be a finite set of points. The three-color hom...

conjecture | open | MathDB 页面 | 原文线索

V2-048637 | MATHDB-334796

The three-color pseudo-disk arrangement conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a pseudo-disk arrangement, meaning a family of planar bodies whose boundaries are Jordan curves and such that the boundary of any member intersects the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-048638 | MATHDB-334797

The dual three-color pseudo-disk arrangement conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a pseudo-disk arrangement, and let its members be the objects to be colored. The dual pseudo-disk arrangement conjecture. The members of any pseudo-disk arrange...

conjecture | open | MathDB 页面 | 原文线索

V2-048639 | MATHDB-334806

Monotonicity conjecture for computing the Laplacian p-spectral radius

摘录 / EXCERPT: Let G be a graph, and for $p\geq 1$ let $\mu^{(p)}(G)$ denote its Laplacian p-spectral radius. Consider the computational problem of finding $\mu^{(p)}(G)$ from the input graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048640 | MATHDB-334807

The LS+-perfect graph conjecture

摘录 / EXCERPT: ${\rm LS}_+$-perfect graph conjecture. A graph G is ${\rm LS}_+$-perfect if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-048641 | MATHDB-334808

Non-realizability conjecture for complete bipartite strong resolving graphs

摘录 / EXCERPT: Complete bipartite strong resolving graph conjecture. The graph equation $G_{SR}\cong K_{r,s}$ has no solution for any $r,s\ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-048642 | MATHDB-334831

The overhauled Roller-Coaster Conjecture for well-covered graphs of fixed order

摘录 / EXCERPT: Let $\alpha\geq2$ and $n\geq4$ be integers satisfying ... For a graph G, let $\alpha(G)$ denote its independence number, let $|V(G)|$ denote its order, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-048643 | MATHDB-334832

The Roller-Coaster Conjecture for 1-well-covered graphs

摘录 / EXCERPT: Let G be a 1-well-covered graph, let $\alpha(G)$ be its independence number, let $n=|V(G)|$, and write $I(G;x)=\sum_k s_kx^k$ for its independence polynomial. Roller-Coaster Con...

conjecture | open | MathDB 页面 | 原文线索

V2-048644 | MATHDB-334834

Sign conjecture for the abstract matrix-tree Laplacian

摘录 / EXCERPT: Sign conjecture. All coefficients x_H are integers of the same sign.

conjecture | open | MathDB 页面 | 原文线索

V2-048645 | MATHDB-334855

Generalized Shareshian-Wachs conjecture for proper circular arc digraphs

摘录 / EXCERPT: Let $\overrightarrow{G}$ be a proper circular arc digraph, meaning a digraph with no induced subdigraph isomorphic to either $\overrightarrow{K_{12}}$ or $\overrightarrow{K_{21}}$....

conjecture | open | MathDB 页面 | 原文线索

V2-048646 | MATHDB-334897

The looped-path sorting conjecture

摘录 / EXCERPT: Looped-path sorting conjecture. Then $\mathbb{Z}\langle S\rangle$ sorts Δ^n. Here, sorting means that every labeled stabilization of the initial configuration Δ^n is...

conjecture | open | MathDB 页面 | 原文线索

V2-048647 | MATHDB-334898

The parallel-edge path sorting conjecture

摘录 / EXCERPT: For $r\geq1$, let $r\mathbb{Z}$ be the graph obtained from the infinite path $\mathbb{Z}$ by replacing each edge with r parallel edges. Parallel-edge path sorting conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-048648 | MATHDB-334899

The generalized parallel-loop sorting conjecture

摘录 / EXCERPT: Generalized parallel-loop sorting conjecture. Then $r(\mathbb{Z}\langle S\rangle)$ sorts Δ^{rn} for each $r\geq1$. This simultaneously generalizes the looped-path and parall...

conjecture | open | MathDB 页面 | 原文线索

V2-048649 | MATHDB-334901

The random labeled chip-firing probability conjecture

摘录 / EXCERPT: For each of the three random labeled chip-firing protocols described in the source-uniformly choosing a legal firing move, uniformly choosing an unstable vertex and then a pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-048650 | MATHDB-334926

Simultaneous deficit conjecture for independent sets of paths

摘录 / EXCERPT: Let P be a path with vertex set V, and let $(A_1,\ldots,A_m)$ be a partition of V. Simultaneous deficit conjecture. There exists a subset $S\subseteq V$ independent in P su...

conjecture | open | MathDB 页面 | 原文线索

V2-048651 | MATHDB-334930

Conjecture on the eigenvalues of the Laplacian on $C^i(X^n_\varnothing)$

摘录 / EXCERPT: Let q be the parameter in the construction of $X^n_\varnothing$, let $0\leq i\leq n-1$, and let Δ act on $C^i(X^n_\varnothing)$. The positive eigenvalues of Δ are a...

conjecture | open | MathDB 页面 | 原文线索

V2-048652 | MATHDB-334944

Milans-Stolee-West conjecture on unbounded track number of line graphs

摘录 / EXCERPT: Let $(G_n)_{n=1}^{}$ be a sequence of graphs. Write $\chi(G)$ for the chromatic number of G, $L(G)$ for its line graph, and $\tau(G)$ for the minimum number of interval graphs wh...

conjecture | open | MathDB 页面 | 原文线索

V2-048653 | MATHDB-334960

Partition conjecture for graphs with no holes of length at least five

摘录 / EXCERPT: Let G be a graph with no holes of length at least 5. A partition conjecture asserts that $V(G)$ can be partitioned into two sets, neither of which contains a maximum clique of...

conjecture | open | MathDB 页面 | 原文线索

V2-048654 | MATHDB-334975

Three-node equivalence conjecture for persistent path and Dowker homology

摘录 / EXCERPT: Three-node equivalence conjecture. On three-node networks, 1-dimensional PPH is isomorphic to 1-dimensional Dowker persistent homology.

conjecture | open | MathDB 页面 | 原文线索

V2-048655 | MATHDB-334976

Cycle-network characterization conjecture for persistent path homology

摘录 / EXCERPT: Cycle-network characterization conjecture. On cycle networks having any number of nodes, 1-dimensional PPH is isomorphic to 1-dimensional Dowker persistent homology. More specifica...

conjecture | open | MathDB 页面 | 原文线索

V2-048656 | MATHDB-335032

Balanced dumbbell conjecture for the minimum sum-Balaban index

摘录 / EXCERPT: Balanced dumbbell conjecture. Among all dumbbell graphs $D_{a,b,a'}$ on at least 14 vertices, the minimum value of sum-Balaban index is achieved for one with $a'=a$ or $a'=a+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-048657 | MATHDB-335033

Asymptotic dumbbell conjecture for the minimum sum-Balaban index

摘录 / EXCERPT: Asymptotic dumbbell conjecture. Among all dumbbell graphs $D_{a,b,a'}$ on n vertices, the minimum is achieved for one with

conjecture | open | MathDB 页面 | 原文线索

V2-048658 | MATHDB-335034

Sum-Balaban extremal dumbbell-like graph conjecture

摘录 / EXCERPT: Dumbbell-like graph conjecture. Dumbbell-like graphs $D_{a,b,a'}^{\ell}$ attain the minimum value of sum-Balaban index.

conjecture | open | MathDB 页面 | 原文线索

V2-048659 | MATHDB-335040

Feasible-partition conjecture for edge-colored graphs

摘录 / EXCERPT: Let G be a finite simple edge-colored graph. For a vertex v, let $d^c_G(v)$ be the number of colors on edges adjacent to v, and let ... For integers $a\geq b\geq 1$, an (a,b)...

conjecture | open | MathDB 页面 | 原文线索

V2-048660 | MATHDB-335042

Quantum Drinfeld-Sokolov reduction equivalence for generic parameters

摘录 / EXCERPT: Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, let ${}^L\mathfrak{g}$ be its Langlands dual, and let $\widehat{{}^L\mathfrak{g}}_{{}^Lk}\text{-mod}^0$ be the semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-048661 | MATHDB-335044

Conformal-block correspondence under quantum Drinfeld-Sokolov reduction

摘录 / EXCERPT: Let ${\mc C}$ be a smooth projective algebraic curve, let p_i be marked points, let $L_{\lambda_i,{}^Lk}$ be the relevant modules for the affine algebra at level ${}^Lk$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048662 | MATHDB-335055

Determinant inequality for Gaussian Markov random fields

摘录 / EXCERPT: Let $G=(V,E)$ be a finite graph. For $x otin(-1,1)$? [sic] the paper defines $\tau(G,x)=\det(\Sigma(G,x))$, where $\Sigma(G,x)$ is the unique positive-definite matrix with diagonal...

conjecture | open | MathDB 页面 | 原文线索

V2-048663 | MATHDB-335059

Kreutzer-Oum-Seymour-van der Zypen-Wood conjecture on majority 3-colourings of digraphs

摘录 / EXCERPT: Let D be a digraph. A colouring of D is $\frac{1}{2}$-majority if every vertex has the same colour as at most half of its out-neighbours. Kreutzer-Oum-Seymour-van der Zypen-Woo...

conjecture | open | MathDB 页面 | 原文线索

V2-048664 | MATHDB-335074

Kohayakawa-Kreuter asymmetric Ramsey conjecture for random graphs

摘录 / EXCERPT: Let $H_1,\ldots,H_r$ be graphs with $m_2(H_1) \geq \ldots \geq m_2(H_r)>1$, and let $m_2(H_1,H_2)$ denote their asymmetric density as defined by ... For a graph G, being...

conjecture | open | MathDB 页面 | 原文线索

V2-048665 | MATHDB-335076

Colour-critical graph Ramsey extremal conjecture

摘录 / EXCERPT: Let r be a positive integer. A graph H is colour-critical if there exists an edge $e\in E(H)$ such that $\chi(H-e)<\chi(H)$. Let $\mathrm{ex}^r(n;H,\ldots,H)$ be the maximum nu...

conjecture | open | MathDB 页面 | 原文线索

V2-048666 | MATHDB-335078

Distinct leverage centralities in Cartesian powers of path powers

摘录 / EXCERPT: Let P_n^k denote the kth power of the path graph on n vertices, and let ... be the Cartesian product of m copies of P_n^k, with $n\geq 4k+1$. The leverage-centrality count...

conjecture | open | MathDB 页面 | 原文线索

V2-048667 | MATHDB-335093

The extremal middle-parts conjecture for binary trees

摘录 / EXCERPT: Extremal middle-parts conjecture. Among binary trees of order n, the tree T that maximizes $d(CT(T), C(T))$ is formed by identifying the root of an rgood binary tree with a ver...

conjecture | open | MathDB 页面 | 原文线索

V2-048668 | MATHDB-335099

The existence of Kotzig-frames in cyclically 4-edge-connected cubic graphs

摘录 / EXCERPT: Kotzig-frame existence conjecture. Every cyclically 4-edge connected cubic graph has a Kotzig-frame.

conjecture | open | MathDB 页面 | 原文线索

V2-048669 | MATHDB-335100

Amiable colorings of 3-row graphs

摘录 / EXCERPT: Amiable-coloring conjecture. If R is a 3-row graph such that R_C is eulerian, then R admits an amiable coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-048670 | MATHDB-335102

Conjecture on conformal embeddings in the family of W-algebras

摘录 / EXCERPT: Let n,m be positive integers with $n\geq 2$. Define $\mathcal{W}_{n,m}$ as the quantum Hamiltonian reduction of $V_k(\mathfrak{sl}_{n+m})$ at the level satisfying ... Here the redu...

conjecture | open | MathDB 页面 | 原文线索

V2-048671 | MATHDB-335104

The degree bound for the regular number of connected graphs

摘录 / EXCERPT: The degree-bound conjecture. For every connected graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-048672 | MATHDB-335105

Ahadi et al.'s three-subgraph conjecture

摘录 / EXCERPT: Ahadi et al.'s conjecture. Every graph can be partitioned into three subgraphs, each of which is locally irregular or regular.

conjecture | open | MathDB 页面 | 原文线索

V2-048673 | MATHDB-335106

The three-subgraph locally irregular or weakly semiregular conjecture

摘录 / EXCERPT: Three-subgraph conjecture. Every graph can be partitioned into three subgraphs, such that each subgraph is locally irregular or weakly semiregular.

conjecture | open | MathDB 页面 | 原文线索

V2-048674 | MATHDB-335117

Hefetz-Keevash's Turán conjecture for extensions of two-edge matchings

摘录 / EXCERPT: Let $r\geq 4$ and let $H_{2r}^{M_2^r}$ be the extension of the r-uniform matching M_2^r consisting of two pairwise disjoint edges. Let G be an r-graph on $[n]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048675 | MATHDB-335118

Peng-Zhao's clique conjecture for dense uniform hypergraphs

摘录 / EXCERPT: Peng-Zhao's conjecture. The Lagrangian of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048676 | MATHDB-335129

Aharoni-Kotlar-Ziv matching conjecture for regular simple tripartite 3-graphs

摘录 / EXCERPT: Let $\mathcal{H}$ be an r-regular simple 3-partite 3-graph with n vertices in each class. Aharoni-Kotlar-Ziv conjecture. ... This conjecture concerns the matching number of s...

conjecture | open | MathDB 页面 | 原文线索

V2-048677 | MATHDB-335130

Aharoni-Berger-Kotlar-Ziv degree-constrained matching conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a simple 3-partite 3-graph with vertex classes A, B and C. Suppose each vertex in A has degree at least r, and each vertex in $B \cup C$ has degr...

conjecture | open | MathDB 页面 | 原文线索

V2-048678 | MATHDB-335160

The large-support 3-flow conjecture for 2-edge-connected graphs

摘录 / EXCERPT: Let G be a graph, and let $E(G)$ denote its edge set. A 3-flow on G is a flow whose values lie in a group of order three, and its support is the set of edges on which the flow...

conjecture | open | MathDB 页面 | 原文线索

V2-048679 | MATHDB-335177

The 3-uniform tight-cycle Turán density conjecture

摘录 / EXCERPT: Let C_s^3 denote the 3-uniform tight cycle of length s, and let $\operatorname{ex}_2(n,C_s^3)$ be the maximum number of edges in a C_s^3-free 3-uniform hypergraph on n...

conjecture | open | MathDB 页面 | 原文线索

V2-048680 | MATHDB-335178

The odd-uniform tight-cycle tiling threshold formula

摘录 / EXCERPT: Let (k,s) be an admissible pair with $k\ge 3$ odd and $s\ge 5k^2$, and let $t_{k-1}(n,C_s)$ denote the minimum $(k-1)$-degree forcing a perfect C_s-tiling in a k-uniform hype...

conjecture | open | MathDB 页面 | 原文线索

V2-048681 | MATHDB-335181

The conjecture on decomposing colored complete bipartite graphs into transversals

摘录 / EXCERPT: Let $l^{*}(n)$ be the least integer such that every proper edge-coloring of $K_{n,n}$ with at least $l^{*}(n)$ colors can be decomposed into n disjoint multicolored perfect match...

conjecture | open | MathDB 页面 | 原文线索

V2-048682 | MATHDB-335182

The rainbow 2-factor conjecture for properly colored complete graphs

摘录 / EXCERPT: Let K_{2n} be a complete graph with a proper edge-coloring using exactly $2n-1$ colors. A multicolored 2-factor is a 2-factor whose edges have pairwise distinct colors. The r...

open_problem | open | MathDB 页面 | 原文线索

V2-048683 | MATHDB-335186

Asymptotic conjecture that the normalized constants α_r vanish

摘录 / EXCERPT: Vanishing-α_r conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-048684 | MATHDB-335229

The outerplanar graph fractional-defect conjecture

摘录 / EXCERPT: A 2-coloring assigns each vertex a red usage in $[0,1]$, with blue usage $1-r(v)$; the defect of a vertex is the sum, over its neighbors, of the overlap in their color usages, an...

conjecture | open | MathDB 页面 | 原文线索

V2-048685 | MATHDB-335231

The eventual fractional-defect conjecture for connected 4-regular graphs

摘录 / EXCERPT: A graph G is 4-regular if every vertex has degree 4. For a graph G, let $D(G,2)$ denote the minimum defect in a fractional 2-coloring. The exceptional graphs are the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-048686 | MATHDB-335233

Scott-Seymour local-neighborhood chromatic conjecture

摘录 / EXCERPT: Scott-Seymour conjecture. For every k, there exists K such that, whenever G has chromatic number at least K, there is a vertex v such that the subgraph of G induced by...

conjecture | open | MathDB 页面 | 原文线索

V2-048687 | MATHDB-335246

The Spin(7) global-section conjecture for the chiral de Rham complex

摘录 / EXCERPT: Let M be an 8-dimensional Riemannian manifold with holonomy $Spin_7$, and let Ω^{ch}_M denote its chiral de Rham complex. Write $C^\infty(M,\Omega^{ch}_M)$ for the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-048688 | MATHDB-335260

Dense-graph upper-bound conjecture for the number of cycles

摘录 / EXCERPT: Let $C(G)$ denote the number of cycles in a graph G. For $\alpha\in(0,1]$, let n be sufficiently large, let G be a graph on n vertices, and let $d=\alpha n$ be its average d...

conjecture | open | MathDB 页面 | 原文线索

V2-048689 | MATHDB-335261

Cubic-root-of-three lower-bound conjecture for cycles in graphs

摘录 / EXCERPT: Let m be a positive integer, and let $C(G)$ denote the number of cycles in a graph G. Cubic-root-of-three lower-bound conjecture. For all sufficiently large m, there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-048690 | MATHDB-335270

Extremal conjecture for trees with bounded leaf count in bipartite graphs

摘录 / EXCERPT: Extremal conjecture for $\mathbb{T}_{k,l}$. The following formulas hold:

conjecture | open | MathDB 页面 | 原文线索

V2-048691 | MATHDB-335273

Vertex/facet trade-off conjecture for 2-level polytopes

摘录 / EXCERPT: Vertex/facet trade-off conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048692 | MATHDB-335279

Piśniak's distinguishing-index conjecture for 2-connected graphs

摘录 / EXCERPT: Piśniak's conjecture. If G is a 2-connected graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-048693 | MATHDB-335280

A distinguishing-index bound for graphs of minimum degree at least three

摘录 / EXCERPT: Proposed distinguishing-index conjecture. (i) If G is a δ-minimally graph with $\delta(G)\geq 3$ and G is neither a complete bipartite graph nor a δ-regular graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048694 | MATHDB-335309

The eventual non-ℓ-distance-balance conjecture for generalized Petersen graphs

摘录 / EXCERPT: The eventual non-ℓ-distance-balance conjecture. For any $n>n_k$, the graph $GP(n,k)$ is not ℓ-distance-balanced for any integer ℓ with $1\leq\ell<D$. Moreover, n_k...

conjecture | open | MathDB 页面 | 原文线索

V2-048695 | MATHDB-335311

Deterministic equivalent conjecture for test error in random-feature neural networks

摘录 / EXCERPT: Let $X\in\mathbb{R}^{p\times T}$ and $Y\in\mathbb{R}^{d\times T}$ be the training data matrices, let $\hat X\in\mathbb{R}^{p\times\hat T}$ and $\hat Y\in\mathbb{R}^{d\times\hat T}$...

conjecture | open | MathDB 页面 | 原文线索

V2-048696 | MATHDB-335316

Double-exponential lower bound conjecture for r_4(5,n)

摘录 / EXCERPT: Let $r_4(5,n)$ be the off-diagonal Ramsey number for 4-uniform hypergraphs, the least N such that every red-blue coloring of the 4-edges of an N-vertex complete hypergraph cont...

conjecture | open | MathDB 页面 | 原文线索

V2-048697 | MATHDB-335330

Twisted intertwining operators for finite automorphism groups

摘录 / EXCERPT: Let V) be a simple vertex operator algebra satisfying the following conditions: $V_{(0)}=\mathbb{C}\mathbf{1}$, $V_{(n)}=0$ for $n<0$, and the contragredient V' is equivalent t...

conjecture | open | MathDB 页面 | 原文线索

V2-048698 | MATHDB-335346

Conjecture that long screenings represent the corresponding Lie algebra

摘录 / EXCERPT: Let $zem_{\alpha_i^{long}}$ be the long screening operators associated with the long basis of the integral lattice, and let $\mathfrak{g}$ denote the corresponding Lie algebra (or...

conjecture | open | MathDB 页面 | 原文线索

V2-048699 | MATHDB-335347

Free-field realization conjecture for W-algebras

摘录 / EXCERPT: Free-field realization conjecture. In the Lie-algebra case, W should be isomorphic to the corresponding W-algebra, namely the Hamiltonian or quantum Drinfeld-Sokolov red...

conjecture | open | MathDB 页面 | 原文线索

V2-048700 | MATHDB-335348

Conjecture on the logarithmic extension generated by highest-root exponentials

摘录 / EXCERPT: Let W be the kernel intersection of the short and long screenings inside $\mathcal{V}_{\Lambda^{long}}$, and let $\mathcal{W}^{log}$ be the larger subspace obtained by i...

conjecture | open | MathDB 页面 | 原文线索

V2-048701 | MATHDB-335350

Intermediate-step conjectures for the logarithmic Kazhdan-Lusztig correspondence

摘录 / EXCERPT: Let $\mathcal{W}^{log}$ be the proposed logarithmic conformal field theory obtained from the screening-kernel construction, and let lattice-VOA modules be restricted to it. Interme...

open_problem | open | MathDB 页面 | 原文线索

V2-048702 | MATHDB-335368

Petersen graph vertex-irreducibility conjecture

摘录 / EXCERPT: Petersen graph vertex-irreducibility conjecture. The Petersen graph is the only vertex-irreducible bridgeless cubic class 2 graph.

conjecture | open | MathDB 页面 | 原文线索

V2-048703 | MATHDB-335369

Jaeger-Swart cyclic-connectivity conjecture for snarks

摘录 / EXCERPT: Jaeger-Swart cyclic-connectivity conjecture. If G is a snark, then its cyclic connectivity is at most 6.

conjecture | open | MathDB 页面 | 原文线索

V2-048704 | MATHDB-335370

Hypohamiltonian oddness-resistance conjecture

摘录 / EXCERPT: Hypohamiltonian oddness-resistance conjecture. If G is a cubic class 2 graph and G is hypohamiltonian, then

conjecture | open | MathDB 页面 | 原文线索

V2-048705 | MATHDB-335371

Flow resistance versus resistance conjecture

摘录 / EXCERPT: Flow resistance conjecture. If G is a bridgeless cubic graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-048706 | MATHDB-335372

Oddness-four order conjecture for snarks

摘录 / EXCERPT: Oddness-four order conjecture. If $\omega(G)=4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-048707 | MATHDB-335373

Snark order-to-oddness ratio conjecture

摘录 / EXCERPT: Snark order-to-oddness ratio conjecture. The order of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048708 | MATHDB-335374

Weak oddness versus resistance conjecture

摘录 / EXCERPT: Weak oddness versus resistance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048709 | MATHDB-335375

Oddness versus resistance conjecture for snarks

摘录 / EXCERPT: Oddness versus resistance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048710 | MATHDB-335382

Existence of angle labelings for essentially 3-connected maps

摘录 / EXCERPT: Let G be a map on a genus $g\geq 1$ orientable surface. An edge, positive-vertex, positive-face angle labeling means a labeling with the edge condition and positive vertex and fa...

conjecture | open | MathDB 页面 | 原文线索

V2-048711 | MATHDB-335383

Dual-acyclic orientation conjecture for positive-genus triangulations

摘录 / EXCERPT: Let G be a triangulation on a genus $g\geq 1$ orientable surface. An orientation of G has vertex outdegree divisible by 3 and at least 3 when every vertex has outdegree div...

conjecture | open | MathDB 页面 | 原文线索

V2-048712 | MATHDB-335389

Geelen's chi-boundedness conjecture for graphs excluding a vertex-minor

摘录 / EXCERPT: Let H be a graph. A class of graphs is χ-bounded if there exists a function $f:\mathbb N\rightarrow\mathbb N$ such that for every graph G in the class and every induced su...

conjecture | open | MathDB 页面 | 原文线索

V2-048713 | MATHDB-335435

Erdős's smaller subgraphs of minimum degree conjecture

摘录 / EXCERPT: Erdős's conjecture. For every $k\geq 2$ there exists an $\epsilon_k>0$ such that every graph on $n\geq k+1$ vertices and $t_k(n)+1$ edges contains a subgraph of minimum degree k...

conjecture | open | MathDB 页面 | 原文线索

V2-048714 | MATHDB-335438

Flandrin-Michałek-Przybyło-Sudakov-Wozniak neighbour sum distinguishing conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph. A neighbour sum distinguishing edge colouring is a proper edge colouring $c:E\to\{1,2,\ldots,k\}$ such that, writing ... we have $d_c(u)\neq d_c(v)$ for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-048715 | MATHDB-335482

The vertex-facet product conjecture for 2-level polytopes

摘录 / EXCERPT: Let P be a 2-level d-polytope, and let $f_0(P)$ and $f_{d-1}(P)$ denote its numbers of vertices and facets, respectively. Vertex-facet product conjecture. For every 2-level...

conjecture | open | MathDB 页面 | 原文线索

V2-048716 | MATHDB-335494

Cohen-Havet-López-Neumann's oriented-cycle subdivision conjecture

摘录 / EXCERPT: Let C be an oriented cycle, and let a subdivision of C be obtained by replacing each arc of C by a directed path. A digraph is strong if every ordered pair of vertices is joi...

conjecture | open | MathDB 页面 | 原文线索

V2-048717 | MATHDB-335495

The bispindle subdivision conjecture

摘录 / EXCERPT: For positive integers k_1,k_2,k_3, let $B(k_1,k_2;k_3)$ be the digraph formed by three internally disjoint paths between vertices x and y: two directed paths from x to y,...

conjecture | open | MathDB 页面 | 原文线索

V2-048718 | MATHDB-335496

Kostochka-Nakprasit equitable coloring conjecture for d-degenerate graphs

摘录 / EXCERPT: Let Δ be an integer, let $d\in\{2,\dotsc,\Delta\}$, and let G be a d-degenerate graph with maximum degree at most Δ. An equitable coloring is a proper vertex colo...

conjecture | open | MathDB 页面 | 原文线索

V2-048719 | MATHDB-335509

Vertex characterization conjecture for key-polynomial Newton polytopes

摘录 / EXCERPT: For compositions γ and α, write $\gamma\preceq\alpha$ when $\lambda(\gamma)=\lambda(\alpha)$ and $w(\gamma)\leq w(\alpha)$ in Bruhat order, where $\lambda(\gamma)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-048720 | MATHDB-335528

Typical structure conjecture for blow-up transitive tournaments

摘录 / EXCERPT: Let $r,t\functionality e \mathbb{N}$ with $r\ge 2$ and $t\ge 1$. Write T^t_{r+1} for the t-blow-up of the transitive tournament T_{r+1}. An oriented graph is r-partite if i...

conjecture | open | MathDB 页面 | 原文线索

V2-048721 | MATHDB-335537

Jha's conjecture on tensor products of Hamilton-decomposable graphs

摘录 / EXCERPT: Let G and H be Hamilton cycle decomposable graphs, and suppose that their tensor product $G\times H$ is connected. Jha's conjecture. The graph $G\times H$ is Hamilton cycle dec...

conjecture | open | MathDB 页面 | 原文线索

V2-048722 | MATHDB-335540

Weighted independence-domination inequality

摘录 / EXCERPT: Weighted independence-domination inequality. For every weight function $w:V\to\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-048723 | MATHDB-335544

Alazemi-Andelić-Simić conjecture on eigenvalues of chain graphs

摘录 / EXCERPT: Alazemi-Andelić-Simić conjecture. In any chain graph, every vertex is downer with respect to every non-zero eigenvalue. Equivalently, for every chain graph G, every $v\in V(G)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-048724 | MATHDB-335554

Ma's Nordhaus-Gaddum conjecture for total-rainbow connection number

摘录 / EXCERPT: Let G and $\overline{G}$ be complementary connected graphs with n vertices. Ma's conjecture. There exist constants C_1 and C_2 such that ... and this upper bound is tight. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048725 | MATHDB-335573

The distant sum distinguishing index conjecture

摘录 / EXCERPT: Let r be a positive integer and let $G=(V,E)$ be a graph without isolated edges. A proper edge colouring $c:E\to\{1,2,\ldots,k\}$ sum-distinguishes r-neighbours when...

conjecture | open | MathDB 页面 | 原文线索

V2-048726 | MATHDB-335589

Oriented Kesten-McKay law conjecture for random regular digraphs

摘录 / EXCERPT: Fix an integer $d\ge 3$, and let $A^{(d)}_n$ be uniformly drawn from the set $\mathcal{A}_{n,d}$ of adjacency matrices of d-regular directed graphs on n vertices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-048727 | MATHDB-335651

Normality conjecture for Cayley graphs generated by transpositions

摘录 / EXCERPT: Let S be a set of transpositions generating S_n, with $n\ge 3$, and let $T(S)$ be the transposition graph of S. Write $\Cay(S_n,S)$ for the Cayley graph generated by S,...

conjecture | open | MathDB 页面 | 原文线索

V2-048728 | MATHDB-335652

Erdős-Sands-Sauer-Woodrow conjecture for quasi-order multidigraphs

摘录 / EXCERPT: Let T be a complete multidigraph whose arcs are the union of k quasi-orders, and let $\gamma(T)$ denote the minimum size of a dominating set, where a set S is dominating when...

conjecture | open | MathDB 页面 | 原文线索

V2-048729 | MATHDB-335653

Sands-Sauer-Woodrow stable-set domination problem for quasi-order multidigraphs

摘录 / EXCERPT: Let D be a multidigraph whose arcs are the union of k quasi-orders. A stable set is a set of vertices containing no arc between distinct vertices, and a dominating set S sati...

open_problem | open | MathDB 页面 | 原文线索

V2-048730 | MATHDB-335666

The necessity of having no isolatable vertex for well-covered prisms

摘录 / EXCERPT: Prism necessity conjecture. If $G \Box K_2$ is well-covered, then G has no isolatable vertex.

conjecture | open | MathDB 页面 | 原文线索

V2-048731 | MATHDB-335668

Adjacency-spectral determination conjecture for complements of multicone cycle graphs

摘录 / EXCERPT: Let K_w be the complete graph on w vertices, let C_n be the cycle graph on n vertices, and let mC_n denote the disjoint union of m copies of C_n. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-048732 | MATHDB-335670

Diregularity conjecture for geodetic cages

摘录 / EXCERPT: Diregularity conjecture. All geodetic cages are diregular.

conjecture | open | MathDB 页面 | 原文线索

V2-048733 | MATHDB-335671

Vertex-splitting conjecture for smallest non-diregular geodetic digraphs

摘录 / EXCERPT: Vertex-splitting conjecture. All smallest possible non-diregular $(d,k,+\epsilon)$-digraphs can be derived from a diregular (d,k)-geodetic cage by the vertex splitting constructi...

conjecture | open | MathDB 页面 | 原文线索

V2-048734 | MATHDB-335678

Trofimov's conjecture on vertex-transitive finite-valency graphs

摘录 / EXCERPT: Trofimov's conjecture. Either the action is block-discrete, or X essentially includes a tree.

conjecture | open | MathDB 页面 | 原文线索

V2-048735 | MATHDB-335681

Thomas's local negative-circuit conjecture for sustained oscillations

摘录 / EXCERPT: Let f be a Boolean evolution function with asynchronous state transition graph $\Gamma(f)$ and local, or type-1, functional circuits in its interaction graph. A type-1 functional...

conjecture | open | MathDB 页面 | 原文线索

V2-048736 | MATHDB-335700

Extremal spectral-radius conjecture for graphs excluding a complete bipartite minor

摘录 / EXCERPT: Let $2\leq s\leq t$, let $0\leq p<t$, and write ... Here, K_r denotes the complete graph on r vertices, kK_t denotes the disjoint union of k copies of K_t, and $G\vee H$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-048737 | MATHDB-335703

The bounded geodecity conjecture for subdivisions of graphs

摘录 / EXCERPT: Bounded geodecity conjecture. For every graph G there exists an integer m such that every subdivision of G lies in $\H_m$.

conjecture | open | MathDB 页面 | 原文线索

V2-048738 | MATHDB-335726

Brown-Colbourn conjecture on all-terminal reliability roots

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph, and let $\mathrm{Rel}(G;q)$ denote its all-terminal reliability polynomial in the edge-failure probability q. A complex number z is an all-t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-048739 | MATHDB-335740

Aldous's continuum random tree conjecture for drainage network clusters

摘录 / EXCERPT: Aldous's continuum random tree conjecture. The conditional distribution of $T_n(0,0)$ given that $L(0,0)=n$ converges in distribution, as $n\to\infty$, to a continuum random tree i...

conjecture | open | MathDB 页面 | 原文线索

V2-048740 | MATHDB-335741

The uniqueness conjecture for strongly rational holomorphic vertex operator algebras of central charge 24

摘录 / EXCERPT: Let V be a strongly rational, holomorphic vertex operator algebra of central charge 24, and let V_1 denote its weight-one Lie algebra. Uniqueness conjecture. The Lie algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-048741 | MATHDB-335750

Graph reconstruction conjecture for polytopes with few nonsimple vertices

摘录 / EXCERPT: Let P be a d-polytope. A vertex of P is nonsimple if its degree is greater than d. Graph reconstruction conjecture. If P has at most $d-2$ nonsimple vertices, then the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-048742 | MATHDB-335751

The asymptotic cardinality conjecture for Wiener indices of trees

摘录 / EXCERPT: Let $\mathcal{T}_{n}$ denote the class of trees on n vertices, and let $W[\mathcal{T}_{n}]$ be the set of their Wiener indices. Cardinality conjecture. The cardinality of...

conjecture | open | MathDB 页面 | 原文线索

V2-048743 | MATHDB-335752

The Wiener inverse interval conjecture for trees

摘录 / EXCERPT: Let $\mathcal{T}_{n}$ denote the class of trees on n vertices, and let $W[\mathcal{T}_{n}]$ be the set of their Wiener indices. Define $W^{int}[\mathcal{T}_{n}]$ to be the larges...

conjecture | open | MathDB 页面 | 原文线索

V2-048744 | MATHDB-335753

Pikhurko-Sousa conjecture on asymptotic extremal decompositions

摘录 / EXCERPT: Pikhurko-Sousa conjecture. There is an integer $n_0=n_0(H)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048745 | MATHDB-335757

The anchor-orbit conjecture for graph reconstruction

摘录 / EXCERPT: The anchor-orbit conjecture. Every graph has either an $(n-2)$-vertex (connectional) anchor in which the relative position of shadows on the anchor is reconstructible, or an orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-048746 | MATHDB-335758

Fixed-point computation conjecture for harmonic-map polygons

摘录 / EXCERPT: Let P be a polynomial with only simple zeroes, and let $Y(P)$ be the convex polygon associated to the minimal harmonic map determined by P. Let ζ be the variable on which...

conjecture | open | MathDB 页面 | 原文线索

V2-048747 | MATHDB-335792

Fair splitting into arbitrarily many stable sets for colored paths

摘录 / EXCERPT: Fair stable-set splitting conjecture. There always exist pairwise disjoint q-stable sets $S_1,\ldots,S_q$ covering all vertices but $q-1$ in each V_j, with sizes differing by a...

conjecture | open | MathDB 页面 | 原文线索

V2-048748 | MATHDB-335797

The tree degree sequence packing conjecture

摘录 / EXCERPT: A tree degree sequence is a list of positive integers whose sum is $2n-2$; a realization is a graph with exactly those vertex degrees. Let $D_1,D_2,\ldots,D_k$ be tree degree seque...

conjecture | open | MathDB 页面 | 原文线索

V2-048749 | MATHDB-335803

Quadratic upper-bound conjecture for pair-digraph diameters

摘录 / EXCERPT: Let S_n be the symmetric group, and let $A\subseteq S_n$ be a subset whose pair digraph has diameter under consideration. Pair-digraph diameter conjecture. The diameter of the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-048750 | MATHDB-335804

The asymptotic metric-dimension conjecture for maximal planar graphs

摘录 / EXCERPT: Let G be a maximal planar graph with n vertices, and let β denote its metric dimension. The asymptotic metric-dimension conjecture. For maximal planar graphs, ... The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-048751 | MATHDB-335806

Two-label conjecture for distinguishing indices of regular graphs

摘录 / EXCERPT: Let G be a k-regular graph, meaning that every vertex of G has degree k. The distinguishing index of G, denoted $D'(G)$, is the least number of edge labels needed so that...

conjecture | open | MathDB 页面 | 原文线索

V2-048752 | MATHDB-335826

Devron-property conjecture for affine-affine ADE T-systems

摘录 / EXCERPT: Let G be an affine-affine ADE bigraph in the classification. A backward Devron singularity is an initial state with $T_v(-1)=0$ for every vertex of color $\epsilon_v=1$, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-048753 | MATHDB-335830

Linear recurrence conjecture for arbitrary cylindrical networks

摘录 / EXCERPT: Let $\widetilde{N}$ be a cylindrical network and let $\widetilde{\mathbf{u}},\widetilde{\mathbf{v}}$ be two r-vertices. Let $f(\ell)$ be the sequence from Theorem. Cylindrical-ne...

conjecture | open | MathDB 页面 | 原文线索

V2-048754 | MATHDB-335831

Pólya frequency conjecture for cycle-cover polynomials

摘录 / EXCERPT: Let $\widetilde{N}$ be a planar cylindrical network, and define ... where $\mathcal{C}^r(N)$ denotes the r-cycle covers of N, with $H_r=0$ for $r>d$. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-048755 | MATHDB-335832

Positive-roots conjecture for cylindrical-network characteristic polynomials

摘录 / EXCERPT: Let N be the finite quotient network associated with a planar cylindrical network, and let $Q_N^{[h_r]}$ and $Q_N^{[e_r]}$ denote the characteristic polynomials associated with t...

conjecture | open | MathDB 页面 | 原文线索

V2-048756 | MATHDB-335833

Totally positive lift conjecture for cylindrical networks

摘录 / EXCERPT: Let $\widetilde{N}$ be the relevant planar cylindrical network, substitute positive real numbers for the variables in $\mathbf{x}$, and let L be a local lift. Let S be the matr...

conjecture | open | MathDB 页面 | 原文线索

V2-048757 | MATHDB-335835

Catabolizable-multitableau formula for quiver Kostka-Shoji polynomials

摘录 / EXCERPT: Let Q be a quiver whose vertex set Q_0 has at most one incoming and one outgoing arrow at each vertex. Let $(\mathbf{i},\mathbf{a},\mu(\bullet))$ be a triple such that...

conjecture | open | MathDB 页面 | 原文线索

V2-048758 | MATHDB-335848

Hamid's independent transversal domination bound for connected graphs

摘录 / EXCERPT: Let G be a non-complete connected graph on n vertices. For a graph G, an independent transversal dominating set is a dominating set $S\subseteq V(G)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-048759 | MATHDB-335849

Hamid's independent transversal domination conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a connected bipartite graph. Write $\gamma(G)$ for its domination number and $\gamma_{it}(G)$ for its independent transversal domination number, the minimum cardinality...

conjecture | open | MathDB 页面 | 原文线索

V2-048760 | MATHDB-335855

Bounded-path partition conjecture for $K^*_{s,t}$-minor-free graphs

摘录 / EXCERPT: Bounded-path partition conjecture. For all $t\geqslant s\geqslant1$, there exists an integer p such that every $K^*_{s,t}$-minor-free graph G has a connected partition...

conjecture | open | MathDB 页面 | 原文线索

V2-048761 | MATHDB-335867

The conjecture that every critical graph is 1-unique

摘录 / EXCERPT: Every-critical-graph conjecture. Every critical graph is 1-unique.

conjecture | open | MathDB 页面 | 原文线索

V2-048762 | MATHDB-335873

Davies et al.'s free-volume conjecture for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph on n vertices. For an independent set or matching of size k, let $f_k^{\mathrm{ind}}(G)$ and $f_k^{\mathrm{match}}(G)$ denote the expected number...

conjecture | open | MathDB 页面 | 原文线索

V2-048763 | MATHDB-335875

Athanasiadis's nonnegativity conjecture for local gamma-vectors

摘录 / EXCERPT: Athanasiadis's local gamma-vector conjecture. The local γ-vector of a flag vertex-induced homology subdivision of a simplex is nonnegative.

conjecture | open | MathDB 页面 | 原文线索

V2-048764 | MATHDB-335879

Forest maximum-degree repetition conjecture

摘录 / EXCERPT: Let F be a forest on n vertices, and let $f(F)=f_2(F)$ denote the minimum number of vertices that must be deleted to obtain an induced subgraph with at least two vertices attai...

conjecture | open | MathDB 页面 | 原文线索

V2-048765 | MATHDB-335880

Maximum-degree boundedness conjecture for equating degrees

摘录 / EXCERPT: Bounded-degree conjecture. For every $k\geq 2$ there is a constant $g(k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048766 | MATHDB-335886

The spectral index conjecture for tensor powers

摘录 / EXCERPT: Let G be a connected hypergraph, let $G^{m,s}$ denote its generalized power hypergraph with parameters m and s, and let $c(G)$ denote the cyclic index of G. Spectral index...

conjecture | open | MathDB 页面 | 原文线索

V2-048767 | MATHDB-335899

Extension conjecture for edge-disjoint branchings in digraphs

摘录 / EXCERPT: Let $D=(V,A)$ be a digraph, let κ be an infinite cardinal, and let $\mathcal{B}_i=(V_i,A_i)$ for $i<\kappa$ be edge-disjoint branchings in D. Define ... Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-048768 | MATHDB-335906

Erdős's bounded monochromatic reachability conjecture

摘录 / EXCERPT: Let k be a positive integer. A k-edge-coloured finite tournament is a finite tournament whose edges are assigned one of k colours, and a set $S\subseteq V(T)$ is reachable fr...

conjecture | open | MathDB 页面 | 原文线索

V2-048769 | MATHDB-335908

The minimum codegree conjecture for perfect matchings in K_4^3-free 3-graphs

摘录 / EXCERPT: K_4^3-free matching conjecture. For any $\gamma>0$, for sufficiently large $n\in 3\mathbb{N}$, if H contains no copy of K_4^3 and

conjecture | open | MathDB 页面 | 原文线索

V2-048770 | MATHDB-335910

The complementarity conditions conjecture for infinite matroid-rooted digraphs

摘录 / EXCERPT: Complementarity conditions conjecture. There always exist an (I,t)-linkage $\mathcal{P}$ and a vertex set $X\ni t$ such that $\mathcal{P}$ and X satisfy the complementarity con...

conjecture | open | MathDB 页面 | 原文线索

V2-048771 | MATHDB-335924

Dvořák-Norin polynomial expansion exponent conjecture

摘录 / EXCERPT: Dvořák-Norin exponent conjecture. There exists a real $c>0$ such that, under these hypotheses, the expansion of $\mathcal{C}$ is bounded by some function

conjecture | open | MathDB 页面 | 原文线索

V2-048772 | MATHDB-335940

Class 1 conjecture for relative non-commuting graphs of proper non-commutative subrings

摘录 / EXCERPT: Class 1 conjecture. The relative non-commuting graph $\Gamma_{S,R}$ is of class 1.

conjecture | open | MathDB 页面 | 原文线索

V2-048773 | MATHDB-336004

Circles in the Morse boundaries of random right-angled Coxeter groups

摘录 / EXCERPT: Let n-vertex graphs be sampled from a random graph model with density greater than $1/n$ and bounded away from 1, and let the associated right-angled Coxeter group be the group...

conjecture | open | MathDB 页面 | 原文线索

V2-048774 | MATHDB-336012

Asymptotic conjecture for the Turán number of Berge-$K_{2,t}$

摘录 / EXCERPT: Let $t\in\{3,4,5,6\}$, and let $ex_3(n,K_{2,t})$ denote the maximum number of edges in a 3-uniform hypergraph on n vertices containing no Berge-$K_{2,t}$. Asymptotic conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-048775 | MATHDB-336031

Topological Tree Alternative Conjecture for locally finite trees

摘录 / EXCERPT: Topological Tree Alternative Conjecture. For a given locally finite tree T, the number of isomorphism classes of trees that are mutually topological minors with T is either 1...

conjecture | open | MathDB 页面 | 原文线索

V2-048776 | MATHDB-336032

Fiamčík's acyclic chromatic index conjecture

摘录 / EXCERPT: Fiamčík's conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048777 | MATHDB-336049

Pilsniak's distinguishing index conjecture for 2-connected graphs

摘录 / EXCERPT: Pilsniak's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048778 | MATHDB-336051

The NP-completeness conjecture for recognizing 2-divisible graphs

摘录 / EXCERPT: The NP-completeness conjecture. It is NP-complete to determine whether a graph is 2-divisible.

conjecture | open | MathDB 页面 | 原文线索

V2-048779 | MATHDB-336052

The NP-completeness conjecture for detecting odd holes

摘录 / EXCERPT: The odd-hole detection conjecture. It is NP-complete to determine whether a graph contains an odd hole.

conjecture | open | MathDB 页面 | 原文线索

V2-048780 | MATHDB-336068

The admissible-genus realization conjecture for unitary modular categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a unitary modular category, and let $(\mathcal{C},c)$ denote an admissible genus consisting of $\mathcal{C}$ together with a compatible central charge c. Adm...

conjecture | open | MathDB 页面 | 原文线索

V2-048781 | MATHDB-336081

Equivalence of hypergraph index, thickness and divergence for right-angled Coxeter groups

摘录 / EXCERPT: Let Γ be a simplicial graph and let W_{Γ} be the corresponding right-angled Coxeter group. Fix an order n. Equivalence conjecture. The following are equivalent:...

conjecture | open | MathDB 页面 | 原文线索

V2-048782 | MATHDB-336094

Restriction to the toric boundary under mirror symmetry

摘录 / EXCERPT: Let D_0 be the union of the irreducible toric divisors of Z_0, and let $i_{D_0}:D_0\hookrightarrow Z_0$ be the inclusion. Assume that $0\in A$ is a vertex of every maximal cell...

conjecture | open | MathDB 页面 | 原文线索

V2-048783 | MATHDB-336102

Ando's colouring conjecture for cubic graphs

摘录 / EXCERPT: Ando's colouring conjecture. A cubic graph G admits a 2-vertex colouring c_V such that the monochromatic induced subgraphs are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-048784 | MATHDB-336111

The path extremal conjecture for conflict-free vertex-connection number

摘录 / EXCERPT: Let G be a connected graph of order n. The path extremal conjecture. ... Here $vcfc(G)$ denotes the conflict-free vertex-connection number of G, and P_n is the path on n ve...

conjecture | open | MathDB 页面 | 原文线索

V2-048785 | MATHDB-336112

Caro and Yuster's maximum-degree repetition conjecture

摘录 / EXCERPT: Caro and Yuster's conjecture. For every integer k at least 2, there is a constant c_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-048786 | MATHDB-336113

Caro, Lauri, and Zarb's forest bound conjecture for f_2

摘录 / EXCERPT: Caro, Lauri, and Zarb's conjecture. If F has order at most

conjecture | open | MathDB 页面 | 原文线索

V2-048787 | MATHDB-336130

The uniform central graph appendage conjecture for complete central graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let P be a graph, and write $\operatorname{diam}(P)$ for its diameter, $r(P)$ for its radius, and $A_{ucg}(K_n,P)$ for the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-048788 | MATHDB-336131

The uniform central graph appendage conjecture for central graphs of diameter greater than one

摘录 / EXCERPT: Let C and P be graphs, let $\operatorname{diam}(C)$ and $\operatorname{diam}(P)$ denote their diameters, let $r(P)$ denote the radius of P, and let $A_{ucg}(C,P)$ be the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-048789 | MATHDB-336136

Li et al.'s generalized 3-connectivity conjecture for Cartesian product graphs

摘录 / EXCERPT: Let G be a nontrivial connected graph, and let H be a 3-connected graph. The generalized 3-connectivity of a graph G, denoted by $\kappa_{3}(G)$, and its vertex-connectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-048790 | MATHDB-336138

The permanent-non-singular $(0,2)$-matrix conjecture

摘录 / EXCERPT: Let G be a graph, and define the graph matrix A_G and the matrices $A_G(\eta)$ as above. An (a,b)-matrix is a square matrix $A_G(\eta)$ with $\eta(v)\le a$ for every vertex a...

conjecture | open | MathDB 页面 | 原文线索

V2-048791 | MATHDB-336139

The permanent-non-singular $(0,k)$-matrix conjecture

摘录 / EXCERPT: Let G be a graph, and let an (a,b)-matrix mean a square matrix $A_G(\eta)$ in which each vertex column occurs at most a times and each edge column occurs at most b times. S...

conjecture | open | MathDB 页面 | 原文线索

V2-048792 | MATHDB-336150

Egawa-Furuya conjecture on path-factors and odd components

摘录 / EXCERPT: Let $k\geq 3$ be an integer, let G be a graph, and for a vertex set $X\subseteq V(G)$ let $c_{2j+1}(G-X)$ denote the number of components of $G-X$ having order $2j+1$. A {...

conjecture | open | MathDB 页面 | 原文线索

V2-048793 | MATHDB-336162

Caro-Lauri-Zarb conjecture on induced subgraphs with repeated maximum degree

摘录 / EXCERPT: Caro-Lauri-Zarb conjecture. For every $k\geq 2$ there is a constant $g(k)$ such that one can remove at most $g(k)\sqrt{\Delta}$ vertices from G so that the remaining subgraph...

conjecture | open | MathDB 页面 | 原文线索

V2-048794 | MATHDB-336172

Gyárfás's monochromatic component covering conjecture

摘录 / EXCERPT: Gyárfás's conjecture. For fixed k, there exist sets $A_1,A_2,\dots,A_{k-1}$ whose union is $[n]$, and colours $c_1,c_2,\dots,c_{k-1}$, such that $K_n[A_i,c_i]$ is connected for e...

conjecture | open | MathDB 页面 | 原文线索

V2-048795 | MATHDB-336173

The finite-subset covering conjecture for direct products of complete graphs

摘录 / EXCERPT: Finite-subset covering conjecture. There exist sets $X_1,\dots,X_l\subseteq X$ whose union is X, such that for every i, either X_i is contained in a hyperplane of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-048796 | MATHDB-336175

The cubic solid-brick odd-intercyclic conjecture

摘录 / EXCERPT: A solid brick is a brick with no nontrivial separating cut, and a graph is odd-intercyclic if every two odd cycles have a common vertex. A graph is cubic if every vertex has degree...

conjecture | open | MathDB 页面 | 原文线索

V2-048797 | MATHDB-336182

Variance-optimal functions determine isoperimetric extremizers

摘录 / EXCERPT: Isoperimetric extremizer conjecture. For n sufficiently large and d,r in appropriate ranges,

open_problem | open | MathDB 页面 | 原文线索

V2-048798 | MATHDB-336183

Variance-optimal functions on trees are centered at a vertex

摘录 / EXCERPT: Tree characterization conjecture. There exists a vertex r such that, for every vertex $u\in V(T)$,

conjecture | open | MathDB 页面 | 原文线索

V2-048799 | MATHDB-336184

Bobkov-Houdré-Tetali's optimal functions conjecture for odd cycles

摘录 / EXCERPT: Bobkov-Houdré-Tetali conjecture. There exists a vertex $x_0\in V(C_n)$ such that, for every $v\in V(C_n)$,

conjecture | open | MathDB 页面 | 原文线索

V2-048800 | MATHDB-336185

The impossibility of big-picture curvature in power-law graphs

摘录 / EXCERPT: Power-law curvature conjecture. It is impossible for a power-law graph to have “big picture” curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-048801 | MATHDB-336189

Gentner et al.'s zero forcing bound for connected subcubic graphs

摘录 / EXCERPT: Gentner et al.'s conjecture.

conjecture | refuted | MathDB 页面 | 原文线索

V2-048802 | MATHDB-336197

Distance-two antimagic labelling conjecture

摘录 / EXCERPT: Let G be a graph with m edges and no isolated edge. For each vertex v, let S_v be the sum of the labels on edges incident with v. A distance-two antimagic labelling uses...

conjecture | open | MathDB 页面 | 原文线索

V2-048803 | MATHDB-336198

Erdős-Faudree-Rousseau-Schelp subgraph conjecture for graphs of minimum degree k

摘录 / EXCERPT: Let $k\geq 2$ and let G be a graph on $n\geq k-1$ vertices with ... edges. A subgraph has minimum degree at least k if every vertex in it has degree at least k within the subgr...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-048804 | MATHDB-336207

The TDC-stability conjecture for graphs with a vertex of degree one or two

摘录 / EXCERPT: Let G be a graph with a vertex of degree one or two. TDC-stability conjecture. Then $St_d^t(G)=1$ or $St_d^t(G)=2$. This conjecture concerns Nordhaus-Gaddum-type questions for th...

conjecture | open | MathDB 页面 | 原文线索

V2-048805 | MATHDB-336212

Strict-corner LLT e-positivity difference conjecture

摘录 / EXCERPT: Let Γ be a circular unit arc digraph, and let H be a graph obtained from Γ by marking k corner edges strict. Strict-corner difference conjecture. Then ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-048806 | MATHDB-336213

Circular LLT power-sum positivity conjecture

摘录 / EXCERPT: Let ${\boldsymbol\nu}$ determine a circular vertical strip digraph. Circular LLT power-sum positivity conjecture. Then $\omega \mathrm{G}_{\boldsymbol\nu}(\mathbf{x};q+1)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-048807 | MATHDB-336220

DP-completeness of recognizing minimally t-tough graphs

摘录 / EXCERPT: Min-t-Tough complexity conjecture. ${\scshape Min-}t$-Tough is DP-complete for any positive rational number t.

conjecture | open | MathDB 页面 | 原文线索

V2-048808 | MATHDB-336227

Spectral-radius asymptotic conjecture for directed Chung-Lu random graphs

摘录 / EXCERPT: Let A be the adjacency matrix of a realization from the directed Chung-Lu model with expected in-degree and out-degree vectors $\mathbf{a}$ and $\mathbf{b}$, respectively, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-048809 | MATHDB-336247

The Jacobian nonresonance conjecture for oscillation in chemical reaction networks

摘录 / EXCERPT: Jacobian nonresonance conjecture. If, for all $x\in\mathbb{R}^n_{\gg 0}$ and all $v\in\mathcal{K}$, the Jacobian matrix $\Gamma Dv(x)$ has no purely imaginary eigenvalues, then...

conjecture | open | MathDB 页面 | 原文线索

V2-048810 | MATHDB-336296

W-algebra conjecture for the VOAs of $(J^b[k],Y)$ Argyres-Douglas theories

摘录 / EXCERPT: Let $(J^b[k],Y)$ be an indecomposable Argyres-Douglas theory, where $J^b[k]$ denotes an irregular singularity and Y is a regular singularity. Let h be the dual Coxeter number o...

conjecture | open | MathDB 页面 | 原文线索

V2-048811 | MATHDB-336318

Conjecture on algebraic connectivity and Laplacian integrality of power graphs of cyclic groups

摘录 / EXCERPT: Let $n\geq 2$ be an integer, let $\mathbb{Z}_n$ be the cyclic group of order n, and let $\mathcal{G}(\mathbb{Z}_n)$ denote its power graph. The cyclic power-graph conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-048812 | MATHDB-336330

Regular-or-biregular conjecture for powers of the adjacency matrix and normalized Laplacian

摘录 / EXCERPT: Let G be a connected graph, let A be its adjacency matrix, and let $\mathcal{L}$ be its normalized Laplacian. A graph is regular if all its vertices have the same degree, and b...

conjecture | open | MathDB 页面 | 原文线索

V2-048813 | MATHDB-336331

Regular-or-biregular eigenvector conjecture for the square of the adjacency matrix

摘录 / EXCERPT: Let G be a connected graph, let A be its adjacency matrix, let d_v denote the degree of vertex v, and let D be the diagonal degree matrix. Thus $D^{1/2}\mathbf{1}$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-048814 | MATHDB-336359

Jiang, Li and Zhang's total monochromatic connection conjecture

摘录 / EXCERPT: Let G be a connected graph. Jiang, Li and Zhang's conjecture. It always holds that ... The conjecture compares the total monochromatic connection number with the monochromatic conn...

conjecture | open | MathDB 页面 | 原文线索

V2-048815 | MATHDB-336369

Conditional Greedy conjecture for dense multigraphs

摘录 / EXCERPT: Let G be a multigraph with chromatic index $\chi'(G)$, maximum degree $\Delta(G)$, and density $\omega(G)$. Let ConditionalGreedy be the algorithm that processes the edges in the...

conjecture | open | MathDB 页面 | 原文线索

V2-048816 | MATHDB-336381

Conjecture on Verlinde coefficients and fusion rules for simple VOSA modules

摘录 / EXCERPT: Verlinde-coefficient conjecture. The Verlinde coefficient $N^{\mu_2}_{\lambda,\mu_1}$ can be expressed as a certain delta function with a non-negative integer coefficient, and this...

conjecture | open | MathDB 页面 | 原文线索

V2-048817 | MATHDB-336401

Feghali-Johnson-Thomas conjecture on independent sets of depth-two claws

摘录 / EXCERPT: Let T_n be the depth-two claw consisting of pairwise disjoint edges $x_1y_1,\dots,x_ny_n$ and a vertex x_0 adjacent to $y_1,\dots,y_n$. For $r\geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-048818 | MATHDB-336404

Every Hist-snark has a cycle double cover containing all outer cycles

摘录 / EXCERPT: Let a Hist-snark be a snark with a Hist, where a Hist is a spanning tree having only vertices of degree three and one, and the outer cycles are the vertex-disjoint cycles induced b...

open_problem | open | MathDB 页面 | 原文线索

V2-048819 | MATHDB-336445

The Euclidean ball minimizer conjecture for the fractional vertex index

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^{n}$ be a centered convex body, and let $\fvein K$ denote its fractional vertex index. Let $B_{2}^{n}=\{x\in\mathbb{R}^{n}:|x|^{2}\leq 1\}$ be the Euclide...

conjecture | open | MathDB 页面 | 原文线索

V2-048820 | MATHDB-336461

The modular tensor category conjecture for even symplectic fermions

摘录 / EXCERPT: Let $\mathcal{V}_{\mathcal{SF}_n^{\mathrm{even}}}$ be the even part of the vertex superalgebra of n pairs of symplectic fermions, and let $\mathcal{C}_0\oplus\mathcal{C}_1$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-048821 | MATHDB-336462

Restricted size Ramsey number conjecture for paths versus cycles

摘录 / EXCERPT: Let P_3 be the path on three vertices and let C_n be the cycle on n vertices. Write $r^{*}(P_3,C_n)$ for the restricted size Ramsey number of P_3 versus C_n. Restricted s...

conjecture | open | MathDB 页面 | 原文线索

V2-048822 | MATHDB-336463

The pseudo-trace isomorphism conjecture for the even symplectic fermion theory

摘录 / EXCERPT: Pseudo-trace functions define an injective linear map ... Here $\mathcal{E}$ is the relevant endomorphism algebra, $C(\mathcal{E})$ denotes its space of symmetric linear forms, and...

conjecture | open | MathDB 页面 | 原文线索

V2-048823 | MATHDB-336467

The Deligne-series module-classification conjecture for negative-level affine vertex algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be one of D_4, E_6, E_7, or E_8, and set $b=2,3,4,6$, respectively. Let $k\in\{-1,-2,\ldots,-b\}$, and consider the vertex algebra $L(k\Lambda_0)$ and it...

conjecture | open | MathDB 页面 | 原文线索

V2-048824 | MATHDB-336474

The strong scaling-limit conjecture for unitary minimal-model VOAs

摘录 / EXCERPT: Let $\mathcal{V}=\mathcal{V}_{c,0}$ be a unitary minimal-model (UMM) vertex operator algebra and let $\mathcal{V}_{c,h}$ be a chiral representation. A quantum theory has a strong s...

conjecture | open | MathDB 页面 | 原文线索

V2-048825 | MATHDB-336476

Space-locality conjecture for a spanning set of VOA fields

摘录 / EXCERPT: Let $\mathcal{V}$ be a vertex operator algebra, let $\mathcal{S}={a}_{a\in\mathcal{V}}$ be a spanning set of $\mathcal{V}$, and let $Y(a,f)$ denote the smeared field. A smeared fie...

conjecture | open | MathDB 页面 | 原文线索

V2-048826 | MATHDB-336478

Strong SL-algebra conjecture for point-like field operators

摘录 / EXCERPT: Let $\mathcal{V}$ be the VOA and let $Y(a,z)$ be its point-like field operators. Their product is defined on $\mathcal{V}\times\mathcal{V}$ when the relevant field-operator product...

conjecture | open | MathDB 页面 | 原文线索

V2-048827 | MATHDB-336492

Seyed Fakhari-Kiani clique-bound conjecture for binomial edge ideals

摘录 / EXCERPT: Let G be a graph, let J_G be its binomial edge ideal, and let $c(G)$ be the number of maximal cliques of G. Seyed Fakhari-Kiani's conjecture. ... This conjecture is proposed as...

conjecture | open | MathDB 页面 | 原文线索

V2-048828 | MATHDB-336520

The CMSO transduction characterization of bounded shrub-depth

摘录 / EXCERPT: Let $\cf G$ be a class of graphs, let ${\mathrm{CMSO}_1}\xspace$ denote counting monadic second-order logic with one free set-variable type, let τ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-048829 | MATHDB-336525

Erdős's high-girth conjecture for Steiner triple systems

摘录 / EXCERPT: A Steiner Triple System is a combinatorial design consisting of triples on a vertex set such that every pair of vertices lies in exactly one triple. Its girth is the smallest integ...

conjecture | open | MathDB 页面 | 原文线索

V2-048830 | MATHDB-336555

A weighted Complete Intersection conjecture for Lagrangians

摘录 / EXCERPT: Let G be a t-intersecting r-graph, meaning that any two edges of G have intersection of size at least t. For integers $t\geq 1$, $i\geq 0$, and $r\geq t+i$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-048831 | MATHDB-336575

Gould-Hirohata-Keller conjecture on degree sums for disjoint cycles

摘录 / EXCERPT: Let G be a graph of sufficiently large order. For an integer $t\geq 1$, let $\sigma_t(G)$ denote the minimum degree sum over all sets of t pairwise nonadjacent vertices of G....

conjecture | open | MathDB 页面 | 原文线索

V2-048832 | MATHDB-336579

Aharoni-Howard-Huang-Loh-Sudakov rainbow matching conjecture

摘录 / EXCERPT: A t-matching is a k-uniform hypergraph consisting of t pairwise disjoint edges. Families $\mathcal{F}_1,\ldots,\mathcal{F}_t\subseteq\binom{[n]}{k}$ are cross free of an orde...

conjecture | open | MathDB 页面 | 原文线索

V2-048833 | MATHDB-336583

Sufficiency of the correspondence-graph connectivity condition for rigidity

摘录 / EXCERPT: Let a configuration consist of sensor and anchor points in $\mathbb{R}^d$, with correspondence graph Γ. Assume the paper's standing Assumption, and suppose that any...

conjecture | open | MathDB 页面 | 原文线索

V2-048834 | MATHDB-336591

Odd-valence subsequence conjecture for face-homogeneous tessellations

摘录 / EXCERPT: Let σ be the valence sequence of a face-homogeneous tessellation. Suppose that σ contains $[p,q,r]$ as a subsequence, where q is odd and $p\neq r$. Odd-valence subs...

conjecture | open | MathDB 页面 | 原文线索

V2-048835 | MATHDB-336592

Extremal growth-rate conjecture for polymorphic valence sequences

摘录 / EXCERPT: Extremal growth-rate conjecture. The sets $\mathscr{L}_\sigma$ and $\mathscr{H}_\sigma$ are nonempty. Thus the lower and upper growth-rate bounds are realized by tessellations with...

conjecture | open | MathDB 页面 | 原文线索

V2-048836 | MATHDB-336593

Strong growth-order conjecture for valence sequences

摘录 / EXCERPT: Let σ_1 and σ_2 be valence sequences with $\sigma_1<\sigma_2$, and let $\underline{\lambda}_\sigma$ and $\overline{\lambda}_\sigma$ denote the lower and upper bounds...

conjecture | open | MathDB 页面 | 原文线索

V2-048837 | MATHDB-336594

Monotonicity conjecture for lower growth bounds of valence sequences

摘录 / EXCERPT: Let σ_1 and σ_2 be valence sequences with $\sigma_1<\sigma_2$, and let $\underline{\lambda}_\sigma$ denote the lower growth-rate bound among face-homogeneous tessella...

conjecture | open | MathDB 页面 | 原文线索

V2-048838 | MATHDB-336598

Conjecture that every graph admits an antimagic orientation

摘录 / EXCERPT: Let G be a finite simple graph. An antimagic orientation of G is an orientation D whose arcs have a bijective labeling by $\{1,\ldots,|A(D)|\}$ with pairwise distinct vertex-...

conjecture | open | MathDB 页面 | 原文线索

V2-048839 | MATHDB-336615

Erdős-Hajnal off-diagonal hypergraph Ramsey conjecture

摘录 / EXCERPT: For integers $4\leq k<s$ and n, let $r_k(s,n)$ be the least N such that every red/blue coloring of the edges of the complete k-uniform hypergraph on N vertices contains a r...

conjecture | open | MathDB 页面 | 原文线索

V2-048840 | MATHDB-336616

Erdős-Hajnal tower-growth conjecture for $r_k(k+1,t;n)$

摘录 / EXCERPT: For integers $k\geq3$ and $2\leq t\leq k$, let $r_k(k+1,t;n)$ be the Erdős-Hajnal Ramsey parameter. Define $\operatorname{twr}_1(x)=x$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-048841 | MATHDB-336617

Kostochka-Mubayi-Verstraëte loose-triangle Ramsey conjecture

摘录 / EXCERPT: For fixed $k\geq3$, let $LC_3^{(k)}$ be the k-uniform loose 3-cycle, whose consecutive edges intersect in exactly one vertex and whose nonconsecutive edges are disjoint. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-048842 | MATHDB-336618

Kostochka-Mubayi-Verstraëte loose 5-cycle Ramsey conjecture

摘录 / EXCERPT: For each $k\geq3$, let $LC_5^{(k)}$ be the k-uniform loose 5-cycle, and let $r_k(LC_5,n)$ denote its Ramsey number against a complete k-uniform hypergraph on n vertices. Kost...

conjecture | open | MathDB 页面 | 原文线索

V2-048843 | MATHDB-336619

Mubayi-Stein strengthening for tight-path Ramsey numbers

摘录 / EXCERPT: For fixed $k\geq4$, let P_{k+1} denote the relevant k-uniform tight path, and let $\overline r_k(P_{k+1},n)$ be its Ramsey number against a complete k-uniform hypergraph on...

conjecture | open | MathDB 页面 | 原文线索

V2-048844 | MATHDB-336620

Mubayi-Stein threshold conjecture for tight-path Ramsey parameters

摘录 / EXCERPT: For integers $3\leq t\leq k$, let $\overline r_k(k+1,t;P_n)$ be the least N such that every red/blue coloring of the k-sets of an N-vertex set contains a blue tight path...

conjecture | open | MathDB 页面 | 原文线索

V2-048845 | MATHDB-336640

The odd induced subgraph conjecture

摘录 / EXCERPT: Odd induced subgraph conjecture. There exists a constant $c>0$ such that for every graph G without isolated vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-048846 | MATHDB-336647

The classification conjecture for groups admitting oriented regular representations

摘录 / EXCERPT: The ORR classification conjecture. Every finite group G admits an ORR, unless one of the following occurs:

conjecture | open | MathDB 页面 | 原文线索

V2-048847 | MATHDB-336654

Colorability conjecture for $G_\alpha(m;1,1,c)$ with nontrivial gcd

摘录 / EXCERPT: Let $G_\alpha(m;1,1,c)$ be the alpha family, and suppose $\gcd(c,m)>1$. Nontrivial-gcd colorability conjecture. The graph $G_\alpha(m;1,1,c)$ is 3-edge-colorable. The paper notes t...

conjecture | open | MathDB 页面 | 原文线索

V2-048848 | MATHDB-336656

Linear equivalence conjecture for hereditary and rooted quasirandomness

摘录 / EXCERPT: Let H be a graph, let $p\in[0,1]$, and let G be a graph. The property $\mathcal{P}_{H,p}^{*}(\delta)$ means that every $U\subseteq V(G)$ satisfies ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-048849 | MATHDB-336662

Conjecture on connectedness of Morse boundaries of right-angled Coxeter groups

摘录 / EXCERPT: Let Γ be the defining graph of a right-angled Coxeter group G_Γ. An induced loop σ in Γ has length greater than 4 and its vertex set contains no pair...

conjecture | open | MathDB 页面 | 原文线索

V2-048850 | MATHDB-336723

Chung's conjecture on Braess's paradox for Erdős-Rényi graphs

摘录 / EXCERPT: Chung's conjecture. For fixed $p\in(0,1)$, there exists a constant $c(p)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048851 | MATHDB-336735

The optical-index ratio bound for trees

摘录 / EXCERPT: Let T be a tree, and let $w(T)$ and $\pi(T)$ denote its undirected optical index and path-packing number, respectively. The optical-index ratio conjecture. For any tree T, ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-048852 | MATHDB-336736

The feasibility interval for the undirected optical-index ratio

摘录 / EXCERPT: The feasibility-interval conjecture. Any rational number δ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-048853 | MATHDB-336740

Fabrici et al.'s four-color conjecture for facial unique-maximum edge-coloring

摘录 / EXCERPT: Fabrici et al.'s conjecture. If G is a 2-edge-connected plane graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-048854 | MATHDB-336741

Wendland's five-list conjecture for facial unique-maximum coloring

摘录 / EXCERPT: Wendland's conjecture. If each vertex of a plane graph is assigned a list of 5 integers, then there exists a FUM-coloring assigning each vertex a color from its list.

conjecture | open | MathDB 页面 | 原文线索

V2-048855 | MATHDB-336742

The four-list conjecture for facial unique-maximum edge-coloring

摘录 / EXCERPT: Four-list conjecture. If each edge of a plane graph is assigned a list of 4 integers, then there exists a FUM-edge-coloring assigning each edge a color from its list.

conjecture | open | MathDB 页面 | 原文线索

V2-048856 | MATHDB-336745

Faudree-Li conjecture on prescribed distances between two vertices

摘录 / EXCERPT: Let G be a graph of order $n\geq 3$, with minimum degree $\delta(G)\geq \frac{n}{2}+1$. For vertices x,y in G, write $dist_C(x,y)$ for their distance on a Hamiltonian cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-048857 | MATHDB-336747

Polynomial domination conjecture for directed-triangle-free digraphs

摘录 / EXCERPT: Polynomial domination conjecture. There is an integer ℓ such that if G is a directed-triangle-free digraph with $\alpha(D)=\alpha$, then

conjecture | open | MathDB 页面 | 原文线索

V2-048858 | MATHDB-336756

The infinite-coupling affine vertex operator superalgebra extension conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive simply-laced Lie algebra, let n be a positive integer, and let P^+_n be the set of dominant integral weights of $\mathfrak{g}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-048859 | MATHDB-336757

The W-algebra chain extension conjecture

摘录 / EXCERPT: W-algebra chain extension conjecture. The module

conjecture | open | MathDB 页面 | 原文线索

V2-048860 | MATHDB-336771

The extremal representation conjecture for graphs

摘录 / EXCERPT: Let G be a connected graph of order n, and let a representation of G mean a graph representation in the sense of the paper. The extremal representation conjecture. The number...

conjecture | open | MathDB 页面 | 原文线索

V2-048861 | MATHDB-336772

Upper domination conjecture for direct products of complete multipartite graphs

摘录 / EXCERPT: Let $K[a_i,b_i]$ denote the complete b_i-partite graph with a_i vertices in each part, and let G be its direct product ... Assume $2\leq b_1\leq b_2\leq\cdots\leq b_t$. The upp...

conjecture | open | MathDB 页面 | 原文线索

V2-048862 | MATHDB-336782

The conjecture that the upper bound for beta_2 is sharp

摘录 / EXCERPT: For integers k and r satisfying ... let $\beta_2(k,r)$ be the parameter defined in the paper. The sharpness conjecture for β_2. One has ... This conjecture would make the upp...

conjecture | open | MathDB 页面 | 原文线索

V2-048863 | MATHDB-336783

The conjecture for alpha in the intermediate partite range

摘录 / EXCERPT: Let $\alpha(k,r)$ be the parameter defined as the minimum number of edges between a specified independent transversal and its complement over all K_r-partite-saturated k-partit...

conjecture | open | MathDB 页面 | 原文线索

V2-048864 | MATHDB-336785

Bang-Jensen et al.'s subdivision-problem dichotomy conjecture

摘录 / EXCERPT: Let F be a fixed strict digraph. An F-subdivision is a digraph obtained from F by replacing each arc ab with a directed (a,b)-path; the associated problem asks whether an...

conjecture | open | MathDB 页面 | 原文线索

V2-048865 | MATHDB-336801

Bipartite-subgraph Ore-degree clique conjecture

摘录 / EXCERPT: Let G be a graph and H a subgraph of G. Define the Ore-degree of H in G by ... Let $L(G)^2$ be the square of the line graph of G, and suppose that H is bipartite and...

conjecture | open | MathDB 页面 | 原文线索

V2-048866 | MATHDB-336802

Bukh's odd-reciprocal conjecture for equiangular lines

摘录 / EXCERPT: Let $N_\alpha(n)$ denote the maximum number of equiangular lines in $\mathbb{R}^n$ with common angle $\arccos\alpha$. Bukh's conjecture. If $1/\alpha$ is an odd natural number, the...

conjecture | open | MathDB 页面 | 原文线索

V2-048867 | MATHDB-336803

The spectral-radius-order conjecture for equiangular lines

摘录 / EXCERPT: Let $N_\alpha(n)$ denote the maximum number of equiangular lines in $\mathbb{R}^n$ with angle $\arccos\alpha$. For $\alpha\in(0,1)$, define ... and let $k(\lambda)$ be the smallest o...

conjecture | open | MathDB 页面 | 原文线索

V2-048868 | MATHDB-336804

Multiplicity conjecture for the second-largest eigenvalue

摘录 / EXCERPT: For $\lambda>0$, let $k(\lambda)$ be the smallest order of a graph with spectral radius λ, or ∞ if no such graph exists, and let $v(G)$ denote the order of a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-048869 | MATHDB-336807

Thomassé's two-acyclic-subgraphs conjecture for cycle reversions

摘录 / EXCERPT: Let D be a finite or infinite digraph, and let $\b1\mathcal{C}\in \mathbb{RS}(D)$ be a reversal sequence, with $D\hspace{-0.08\,\mathrm{cm}}\rcirclearrowright\mathcal{C}$ denotin...

conjecture | open | MathDB 页面 | 原文线索

V2-048870 | MATHDB-336813

Conditional connectivity conjecture for transposition networks generated by connected graphs

摘录 / EXCERPT: Let $G(\mathcal{T})$ be a connected graph, and let G_n be the Cayley graph generated by $G(\mathcal{T})$. Write $\kappa^2(G_n)$ for its conditional connectivity parameter and...

open_problem | open | MathDB 页面 | 原文线索

V2-048871 | MATHDB-336831

Rankin-type conjecture for Hamilton circles in two-connected Cayley graphs

摘录 / EXCERPT: Let $G=\langle S\rangle$ be a group with $S=\{a^{\pm},b^{\pm}\}$, satisfying $(ab)^2=1$, and suppose that the vertex-connectivity of $\Gamma(G,S)$ satisfies ... A Hamilton circle is...

conjecture | open | MathDB 页面 | 原文线索

V2-048872 | MATHDB-336839

Čada-Kaneko-Ryjáček conjecture on minimum color degree and rainbow cycles

摘录 / EXCERPT: Čada-Kaneko-Ryjáček conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048873 | MATHDB-336854

The tight-star conjecture for betweenness structures

摘录 / EXCERPT: The tight-star conjecture. For all integers $k\geq 0$ and c, there exists a threshold $N_c^k>0$ such that for all $n\geq N_c^k$ and every betweenness structure...

conjecture | open | MathDB 页面 | 原文线索

V2-048874 | MATHDB-336861

Conjecture on flexible labelings of non-triangle-connected Laman graphs

摘录 / EXCERPT: Let G be a Laman graph. A graph is triangle-connected if any two edges can be joined by a sequence of edges in which consecutive edges lie in a common triangle. A NAC-coloring is...

conjecture | open | MathDB 页面 | 原文线索

V2-048875 | MATHDB-336865

Meng-Puleo-Zhu's characterization conjecture for $(4,2)$-choosable graphs

摘录 / EXCERPT: Meng-Puleo-Zhu's conjecture. The graph G is $(4,2)$-choosable if and only if its core is one of the following: K_1; C_{2s}; $\theta_{2,2s,2t}$; $\theta_{1,2s+1,2t+1}$;...

conjecture | open | MathDB 页面 | 原文线索

V2-048876 | MATHDB-336875

Dankelmann-Swart-Oellermann conjecture on Steiner diameter of highly connected graphs

摘录 / EXCERPT: Let C_n^{2k} denote the $2k$-th power of the cycle C_n. Let G be a $2k$-connected graph of order n. Dankelmann-Swart-Oellermann conjecture. ... This conjectures that the $2k$...

conjecture | open | MathDB 页面 | 原文线索

V2-048877 | MATHDB-336876

Dankelmann-Oellermann-Swart conjecture on average Steiner distance

摘录 / EXCERPT: Let G be a connected graph of order $n\geq k$, and let $\mu(G)$ denote its average distance and $\mu_k(G)$ its average Steiner k-distance. Dankelmann-Oellermann-Swart conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-048878 | MATHDB-336880

The orbit-lattice fixed-point category conjecture

摘录 / EXCERPT: Let $g\in O(\Lambda)$ be an element whose frame shape equals the type of an orbit lattice $N(Z)$, and let $\Lambda_g=(\Lambda^g)^\perp$. Choose a lift $\hat g$ to an automorphism o...

conjecture | open | MathDB 页面 | 原文线索

V2-048879 | MATHDB-336888

Aharoni-Hartman-Hoffman dual conjecture for orthogonal colorings and optimal k-packs

摘录 / EXCERPT: Aharoni-Hartman-Hoffman's conjecture. If $\mathcal{P}$ is an optimal k-pack of D, then there exists a coloring of D orthogonal to $\mathcal{P}$.

conjecture | open | MathDB 页面 | 原文线索

V2-048880 | MATHDB-336889

Linial's dual inequality for colorings and k-packs

摘录 / EXCERPT: Linial's dual conjecture. For every digraph D and every positive integer k, we have

conjecture | open | MathDB 页面 | 原文线索

V2-048881 | MATHDB-336899

Conjectured upper bound for terminal-pairable complete bipartite graphs

摘录 / EXCERPT: Let $K_{n,n}^q$ consist of every demand multigraph on the complete balanced bipartite graph $K_{n,n}$ with maximum degree at most q; the demand graph need not be bipartite. A dem...

conjecture | open | MathDB 页面 | 原文线索

V2-048882 | MATHDB-336919

Genus and planarity structure across the second phase transition

摘录 / EXCERPT: Let $S_g(n,m)$ be the random graph with n vertices and m edges embedded on the orientable surface $\mathbb{S}_g$, and let the giant component mean the component containing a li...

conjecture | open | MathDB 页面 | 原文线索

V2-048883 | MATHDB-336920

Minimum semidefinite rank conjecture for graphs of large girth

摘录 / EXCERPT: Large-girth minimum semidefinite rank conjecture. If G has girth at least k, then

conjecture | open | MathDB 页面 | 原文线索

V2-048884 | MATHDB-336930

Seymour-Wood connectivity conjecture for clique-minor-free graphs

摘录 / EXCERPT: Let $p\ge2$ be an integer. A graph is $(p-2)$-connected if it remains connected after deletion of any set of fewer than $p-2$ vertices; a minor of a graph is obtained by vertex del...

conjecture | open | MathDB 页面 | 原文线索

V2-048885 | MATHDB-336931

McCarty's spectral-radius conjecture for clique-minor-free graphs

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\mu(G)$ denote its spectral radius. A graph has no K_p minor when no graph obtained from it by vertex deletions, edge deletions, and...

conjecture | open | MathDB 页面 | 原文线索

V2-048886 | MATHDB-336935

Heterochromatic number conjecture for circuits in projective geometries

摘录 / EXCERPT: Let $PG(r-1,q)$ be the projective geometry of rank r over the finite field with q elements, and let $C_p(PG(r-1,q))$ be the hypergraph whose hyperedges are the p-circuits of...

conjecture | open | MathDB 页面 | 原文线索

V2-048887 | MATHDB-336938

Conjecture on disjoint properly colored cycles in edge-colored complete graphs

摘录 / EXCERPT: Let G be an edge-colored complete graph on n vertices. Write $cDelta^{mon}(G)$ for its maximum monochromatic degree, and call a cycle properly colored if consecutive edges have...

conjecture | open | MathDB 页面 | 原文线索

V2-048888 | MATHDB-336957

Benzi's single-temperature walk-regularity conjecture

摘录 / EXCERPT: Let G be a simple graph, let n_G be its number of vertices, and let $S^V(G,\beta)$ denote its walk entropy at temperature $\beta\geq0$. The graph G is walk-regular when every...

conjecture | open | MathDB 页面 | 原文线索

V2-048889 | MATHDB-336960

Bound on maximum-entropy temperatures for non-walk-regular graphs

摘录 / EXCERPT: Let G be a simple graph, let n_G be its number of vertices, and let $S^V(G,\beta)$ denote its walk entropy at temperature $\beta>0$. Temperature-count conjecture. If G is not...

conjecture | open | MathDB 页面 | 原文线索

V2-048890 | MATHDB-336964

The disjoint domination number with a perfect matching is bounded by independence number

摘录 / EXCERPT: Let G be a connected graph containing a swap set. Let $DD_{\operatorname{m}}(G)$ denote the minimum size of a disjoint dominating set whose two parts are joined by a perfect matc...

conjecture | open | MathDB 页面 | 原文线索

V2-048891 | MATHDB-336971

de Lima-Oliveira-de Abreu-Nikiforov lower-bound conjecture for the least signless Laplacian eigenvalue

摘录 / EXCERPT: Let G be a graph with order n and size m. de Lima-Oliveira-de Abreu-Nikiforov conjecture. ... The source presents this as a conjectured lower bound for the least signless Lapla...

conjecture | open | MathDB 页面 | 原文线索

V2-048892 | MATHDB-336972

Star-minimization conjecture for the least A_α-eigenvalue

摘录 / EXCERPT: Let G be a connected graph on n vertices with $\frac{1}{2}<\alpha<1$. Star-minimization conjecture. ... with equality if and only if $G\cong K_{1,n-1}$. This conjecture asks whet...

conjecture | open | MathDB 页面 | 原文线索

V2-048893 | MATHDB-336974

e-positivity and e-unimodality conjecture for circular indifference digraphs

摘录 / EXCERPT: Circular indifference digraph conjecture. The palindromic polynomial $X_{\overrightarrow{G}}({\bf x},t)$ is e-positive and e-unimodal: $a_j({\bf x})$ is e-positive for every...

conjecture | open | MathDB 页面 | 原文线索

V2-048894 | MATHDB-336976

Schaefer's bounded-crossing Hanani-Tutte conjecture

摘录 / EXCERPT: Schaefer's conjecture. If a graph G has an independently even drawing on a surface S such that every pair of edges crosses at most t times, then G has an embedding on S.

conjecture | open | MathDB 页面 | 原文线索

V2-048895 | MATHDB-336981

Strong tau-sequence conjecture for antimorphic-digraph R-systems

摘录 / EXCERPT: Antimorphic-digraph tau-sequence conjecture. The R-systems associated with antimorphic digraphs possess strong τ-sequences $Y(t)$ with the substitution

conjecture | open | MathDB 页面 | 原文线索

V2-048896 | MATHDB-336987

Win's conjecture on disjoint perfect matchings under an Ore-type condition

摘录 / EXCERPT: Win's conjecture. If G is of Ore-type-$(k-2)$, then G contains k disjoint perfect matchings.

conjecture | open | MathDB 页面 | 原文线索

V2-048897 | MATHDB-336995

Moment computation conjecture for the optimal estimator

摘录 / EXCERPT: Let $\mathcal{P}_{\bm{t}}$ and $\bm{V}_s$ be as in Proposition. For a coordinate k, define the moment ... Let Δ, $\boldsymbol{\lambda}$, and the inverse Laplace operators...

conjecture | open | MathDB 页面 | 原文线索

V2-048898 | MATHDB-336999

Li et al.'s maximum conflict-free vertex-connection number conjecture

摘录 / EXCERPT: Let G be a connected graph of order n. The conflict-free vertex-connection number $vcfc(G)$ is the smallest number of colors required for a vertex-coloring in which every two v...

conjecture | open | MathDB 页面 | 原文线索

V2-048899 | MATHDB-337000

The minimum conflict-free connection number conjecture for trees

摘录 / EXCERPT: Let T be a tree of order n, and let $cfc(T)$ denote its conflict-free connection number, the smallest number of edge colors needed so that every two vertices are joined by a pa...

conjecture | open | MathDB 页面 | 原文线索

V2-048900 | MATHDB-337001

Even fans with handles and two spokes removed conjecture

摘录 / EXCERPT: Let F^*_n be the fan with a handle whose spokes are all edges connected to the hub except the edge connected to the degree 1 vertex. For all even $n\geqslant 4$, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-048901 | MATHDB-337025

The maximum-degree conjecture for rainbow matchings in bipartite multigraphs

摘录 / EXCERPT: Let G be a bipartite multigraph with maximum degree $\Delta(G)$, and allow its edges to be coloured not necessarily properly. A full rainbow matching is a matching containing one...

conjecture | open | MathDB 页面 | 原文线索

V2-048902 | MATHDB-337037

Dvořák-Mohar-Šámal's six-color conjecture for subcubic graphs

摘录 / EXCERPT: Dvořák-Mohar-Šámal's conjecture. Every subcubic graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048903 | MATHDB-337042

The large-grid hypothesis for forthcoming translation results

摘录 / EXCERPT: Large-grid conjecture. The forthcoming results apply for grid graphs such that

conjecture | open | MathDB 页面 | 原文线索

V2-048904 | MATHDB-337044

The seven-eighths conjecture for induced 2-degenerate subgraphs of triangle-free planar graphs

摘录 / EXCERPT: Seven-eighths conjecture. Every triangle-free planar graph contains an induced 2-degenerate subgraph on at least

conjecture | open | MathDB 页面 | 原文线索

V2-048905 | MATHDB-337060

The edge-transitive 3-circulant modular-product conjecture

摘录 / EXCERPT: Let $C_n(1,a,b)$ be an edge-transitive circulant graph, where the connection parameters are taken modulo n. Edge-transitive 3-circulant conjecture. At least one of the following...

conjecture | open | MathDB 页面 | 原文线索

V2-048906 | MATHDB-337061

The edge-transitive circulant modular-product conjecture

摘录 / EXCERPT: Let $C_n(1,a_2,\dots,a_m)$ be an edge-transitive circulant graph. The parameters $a_2,\dots,a_m$ are connection parameters modulo n. General edge-transitive circulant conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-048907 | MATHDB-337062

Uniform edge betweenness centrality in seven circulant graph classes

摘录 / EXCERPT: Uniform edge betweenness centrality conjecture. Each of these seven classes of circulant graphs has uniform edge betweenness centrality but is not edge-transitive.

conjecture | open | MathDB 页面 | 原文线索

V2-048908 | MATHDB-337063

The bounded-defect cycle conjecture for Hamiltonian graphs

摘录 / EXCERPT: Let G be an n-vertex graph, and write $\delta(G)$ for its minimum degree. A Hamiltonian cycle is a cycle spanning all vertices of G. The bounded-defect cycle conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-048909 | MATHDB-337092

Bonato-Tardif's tree alternative conjecture

摘录 / EXCERPT: Let G and H be non-isomorphic graphs. We call H a twin of G if there are embeddings $G\to H$ and $H\to G$, that is, injective maps $V(G)\to V(H)$ and $V(H)\to V(G)$ preserv...

conjecture | open | MathDB 页面 | 原文线索

V2-048910 | MATHDB-337102

Jaeger's strong embedding conjecture for 2-connected graphs

摘录 / EXCERPT: Let Γ be a 2-connected graph. A strong embedding is an embedding of a graph in a surface in which every facial walk is simple. Jaeger's strong embedding conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-048911 | MATHDB-337103

The dihedral 2-flow conjecture for 3-edge-colorable cubic graphs

摘录 / EXCERPT: Let Γ be a 3-edge-colorable cubic graph. An embedding of Γ in a surface is an embedding whose genus is the genus of Γ, and a nowhere-identity dihedral 2-fl...

conjecture | open | MathDB 页面 | 原文线索

V2-048912 | MATHDB-337157

Generic complete-graph rigidity conjecture for admissible matrix norms

摘录 / EXCERPT: Let X be either ${\mathcal{M}}_n({\mathbb{F}})$ or ${\mathcal{H}}_n({\mathbb{F}})$, let $\|\cdot\|$ be an admissible norm on X, and write $k=\dim X$. For a placement $p\in X^V$...

conjecture | open | MathDB 页面 | 原文线索

V2-048913 | MATHDB-337162

The lower-threshold conjecture for Avoider-Enforcer games

摘录 / EXCERPT: For a non-empty graph H, define ... where $v(F)$ and $e(F)$ are the numbers of vertices and edges of F, respectively. Let $f^-_H(n)$ be the lower threshold of the strict...

conjecture | open | MathDB 页面 | 原文线索

V2-048914 | MATHDB-337163

The upper-threshold conjecture for unicyclic graphs

摘录 / EXCERPT: Let H be a unicyclic graph, meaning a connected graph containing exactly one cycle, and let $v(H)$ denote its number of vertices. Let $f^+_H(n)$ be the upper threshold of the str...

conjecture | open | MathDB 页面 | 原文线索

V2-048915 | MATHDB-337164

The edge-deletion conjecture for upper thresholds

摘录 / EXCERPT: Let H be a graph with at least three edges. For an edge e of H, let $H-e$ be the graph obtained by deleting e, together with a vertex of degree one if that vertex is an end...

conjecture | open | MathDB 页面 | 原文线索

V2-048916 | MATHDB-337188

Alspach's conjecture on Hamilton decompositions of abelian Cayley graphs

摘录 / EXCERPT: Let $(\Gamma,+)$ be an abelian group, let S be a generating set, and let $G(\Gamma,S)$ be the Cayley multigraph with vertex set Γ. Its simplification is the simple graph o...

conjecture | open | MathDB 页面 | 原文线索

V2-048917 | MATHDB-337213

Korpelainen-Lozin-Razgon conjecture on labelled well-quasi-ordering of graph classes

摘录 / EXCERPT: Korpelainen-Lozin-Razgon conjecture. A class of graphs which is wqo by the induced subgraph relation is lwqo if and only if it is defined by finitely many minimal forbidden induced...

conjecture | open | MathDB 页面 | 原文线索

V2-048918 | MATHDB-337228

The graph positivity conjecture for random regular bipartite graphs

摘录 / EXCERPT: Let G be an r-regular bipartite graph with $v=2n$ vertices, let m_i denote its number of i-matchings, and define ... where $\overline{m}_i$ is the number of i-matchings in...

conjecture | open | MathDB 页面 | 原文线索

V2-048919 | MATHDB-337232

Ghebleh-Niepel conjecture on locating-dominating codes in circulant graphs

摘录 / EXCERPT: Let $C_n(1,3)$ be the circulant graph on vertex set $\mathbb{Z}_n$ in which vertices at cyclic distances 1 or 3 are adjacent. A locating-dominating code is a dominating vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-048920 | MATHDB-337233

Ghebleh-Niepel conjecture on identifying codes in circulant graphs

摘录 / EXCERPT: Let $C_n(1,3)$ be the circulant graph on vertex set $\mathbb{Z}_n$ in which vertices at cyclic distances 1 or 3 are adjacent. An identifying code is a dominating vertex set who...

conjecture | open | MathDB 页面 | 原文线索

V2-048921 | MATHDB-337250

The star characterization conjecture for graphs with one large Laplacian eigenvalue

摘录 / EXCERPT: Let G be a finite simple graph with Laplacian eigenvalues $\mu_1\geq\cdots\geq\mu_n$ and average degree $\overline{d}(G)=2m/n$, where m is the number of edges and n the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-048922 | MATHDB-337257

Mader's nonseparating dipath conjecture

摘录 / EXCERPT: Mader's nonseparating dipath conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048923 | MATHDB-337261

Exact feedback vertex number of generalized Sierpiński triangle graphs

摘录 / EXCERPT: Let $\hat{S_p^n}$ be the generalized Sierpiński triangle graph with integers $n\geq 3$ and $p\geq 4$, and let $f(G)$ denote the feedback vertex number of a graph G, namely the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-048924 | MATHDB-337277

Kawasetsu's coset coincidence conjecture

摘录 / EXCERPT: For integers $n\geq4$ and $k\geq0$, let $\mathcal C_k(n)$ be the simple quotient of the coset of $L_{k+1}(\mathfrak{gl}_{n-2})$ in $\mathcal W_k(\mathfrak{sl}_n,f_{\mathrm{min}})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-048925 | MATHDB-337278

Kawasetsu's explicit ideal conjecture for minimal W-algebra cosets

摘录 / EXCERPT: For $n\geq4$, let K_n be the ideal in $\mathbb C[c,\lambda]$ defining the two-parameter quotient associated with the minimal $\mathcal W$-algebra coset. Kawasetsu's explicit idea...

conjecture | open | MathDB 页面 | 原文线索

V2-048926 | MATHDB-337279

Subregular W-algebra coset truncation conjecture

摘录 / EXCERPT: For $n\geq4$, let $\mathcal C^k(n)$ be the coset of the Heisenberg algebra in $\mathcal W^k(\mathfrak{sl}_n,f_{\mathrm{subreg}})$, and let $\mathcal W^{K_n}_{R_n}(c,\lambda)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-048927 | MATHDB-337280

Subregular coset coincidence conjecture

摘录 / EXCERPT: For $n\geq4$ and $m\geq3$, let $\mathcal C_k(n)$ be the simple quotient of the Heisenberg coset in the subregular $\mathcal W$-algebra of $\mathfrak{sl}_n$. Subregular coset coinci...

conjecture | open | MathDB 页面 | 原文线索

V2-048928 | MATHDB-337289

Deterministic approximation conjecture for Nakagami fading channels

摘录 / EXCERPT: Deterministic approximation conjecture. If h has a Nakagami-m distribution with $m \geq 3$, then the fading channel can be well approximated by a deterministic channel gain wit...

conjecture | open | MathDB 页面 | 原文线索

V2-048929 | MATHDB-337292

Characterization of clustered coloring number in minor-closed classes

摘录 / EXCERPT: Let $t\geq 1$ be an integer. For a graph class $\mathcal{G}$, let $\operatorname{col}_\star(\mathcal{G})$ be the least integer t for which there is a constant $C\geq 1$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-048930 | MATHDB-337297

Resistance exponent conjecture for stable causal carpets

摘录 / EXCERPT: Resistance exponent conjecture. The exponent $\mathfrak{r}$ exists and satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-048931 | MATHDB-337325

Exact Berge-path saturation conjecture

摘录 / EXCERPT: Let $\operatorname{sat}_k(n,\text{Berge-}P_m)$ denote the minimum number of edges in a k-uniform hypergraph on n vertices that is Berge-P_m-saturated, and let $a^{(k)}_m$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-048932 | MATHDB-337326

Linear growth conjecture for Berge saturation numbers

摘录 / EXCERPT: Let k be fixed, and let $\mathcal{F}$ be any fixed finite family of graphs. Write $\operatorname{sat}_k(n,\text{Berge-}\mathcal{F})$ for the minimum number of edges in a k-unif...

conjecture | open | MathDB 页面 | 原文线索

V2-048933 | MATHDB-337328

Countability-free vertex-flame conjecture

摘录 / EXCERPT: Countability-free vertex-flame conjecture. We may omit the countability of D in Theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-048934 | MATHDB-337345

Bipartite extremal-number growth conjecture

摘录 / EXCERPT: Bipartite extremal-number growth conjecture. For every nonempty bipartite graph H, there exist $\alpha\in[1,2]$ and constants $c_2>c_1>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048935 | MATHDB-337352

The diameter conjecture for powers-of-two Bell-type Riordan graphs

摘录 / EXCERPT: Let CG_{2^k} be the relevant io-decomposable Riordan graph of the Bell type, and let $\operatorname{diam}(G)$ denote graph diameter. Diameter conjecture. ... and there are no io-de...

conjecture | open | MathDB 页面 | 原文线索

V2-048936 | MATHDB-337354

Generalized Erdős' matching conjecture for k-matchings

摘录 / EXCERPT: Let $\mathcal{H}=(V,\mathcal{E})$ be an r-uniform hypergraph on n vertices. For a positive integer $k<r$, let $\nu_k(\mathcal{H})$ denote its k-matching number, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048937 | MATHDB-337360

Degree-sum conjecture for matchings in 3-uniform hypergraphs with isolated vertices allowed

摘录 / EXCERPT: Let H be a 3-graph of order n, let s be an integer with $2\le s\le n/3$, and let $H_{n,3,s}^2$ and $H_{n,3,s}^3$ be the two extremal 3-graphs used in the degree-sum compa...

conjecture | open | MathDB 页面 | 原文线索

V2-048938 | MATHDB-337386

Infinitude conjecture for rotation snarks

摘录 / EXCERPT: A rotation snark is a snark constructed by the rotation method described in the paper. Infinitude conjecture for rotation snarks. There are infinitely many rotation snarks. This wo...

conjecture | open | MathDB 页面 | 原文线索

V2-048939 | MATHDB-337390

The Ramsey equivalence conjecture for uniform trees

摘录 / EXCERPT: Let $r\ge 2$, let T_1 and T_2 be r-uniform trees of order m, and let $K_n^{(r)}$ be the complete r-uniform hypergraph on n vertices. The Ramsey equivalence conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-048940 | MATHDB-337391

The conjecture that every uniform tree is good

摘录 / EXCERPT: Let $r\ge 2$, let T be an r-uniform tree, and let n be a positive integer. An r-uniform tree T is n-good when its Ramsey number against the complete r-uniform hypergr...

conjecture | open | MathDB 页面 | 原文线索

V2-048941 | MATHDB-337406

Grötschel-Padberg diameter conjecture for the symmetric traveling salesperson polytope

摘录 / EXCERPT: Grötschel-Padberg conjecture. For every integer $n\geq 5$, the diameter of the skeleton of the symmetric traveling salesperson polytope is

conjecture | open | MathDB 页面 | 原文线索

V2-048942 | MATHDB-337427

Gruslys, Leader and Tan's hypercube edge-packing conjecture

摘录 / EXCERPT: Gruslys, Leader and Tan's conjecture. For $kge 1$, there exists a positive integer n such that the edges of Q_n can be covered by edge-disjoint copies of H; the copies of H...

conjecture | open | MathDB 页面 | 原文线索

V2-048943 | MATHDB-337434

The -free graph sum conjecture

摘录 / EXCERPT: Let $\mathcal{B}:=\mathcal{F}\setminus\{H_1,H_{26},H_{27}\}=\{H_2,\ldots,H_{25}\}$, where $\mathcal{F}$ is the family of 27 forbidden induced subgraphs characterizing sum-perfect...

conjecture | open | MathDB 页面 | 原文线索

V2-048944 | MATHDB-337435

The bounded-deficiency conjecture for graphs excluding six-vertex bipartite graphs

摘录 / EXCERPT: Let $\alpha(G)$ denote the stability number of a graph G, let $\omega(G)$ denote its clique number, and call G free of a family of graphs if it contains none of them as an indu...

conjecture | open | MathDB 页面 | 原文线索

V2-048945 | MATHDB-337440

Conjecture on Turán numbers of ordered 6-cycles

摘录 / EXCERPT: Let C be an ordered 6-cycle with interval chromatic number 2, and let $\operatorname{ex}_{<}(n,C)$ denote the maximum number of edges in an n-vertex ordered graph containing...

conjecture | open | MathDB 页面 | 原文线索

V2-048946 | MATHDB-337442

Hereditary characterization conjecture for ${\cal G}_k$

摘录 / EXCERPT: Hereditary characterization conjecture. For every positive integer k, the set ${\cal G}_k$ equals the set of all graphs G such that

conjecture | open | MathDB 页面 | 原文线索

V2-048947 | MATHDB-337449

Rainbow clique saturation conjecture

摘录 / EXCERPT: Let K_r be the complete graph on r vertices, and let $\operatorname{sat}_{t}\left(n,\mathfrak{R}(K_r)\right)$ denote the minimum number of edges in an n-vertex graph whose ed...

conjecture | open | MathDB 页面 | 原文线索

V2-048948 | MATHDB-337463

Győri-Tuza conjecture on triangle-and-edge decompositions

摘录 / EXCERPT: For a graph G, let $\pi_k(G)$ be the minimum integer m such that the edges of G can be decomposed into complete graphs $C_1,\ldots,C_\ell$ of order at most k with ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048949 | MATHDB-337485

The second-moment finiteness conjecture for minimal stationary queue lengths

摘录 / EXCERPT: Let $x_{0;\infty}(0)$ denote the queue length at the origin at time 0 in the minimal stationary solution of the interference queueing network on the grid, with arrival rate...

conjecture | open | MathDB 页面 | 原文线索

V2-048950 | MATHDB-337486

The instability conjecture for interference queueing networks on grids

摘录 / EXCERPT: Consider an interference queueing network on the d-dimensional grid with arrival rate λ and interference sequence $(a_i)_{i\in\mathbb{Z}}$. Instability conjecture. For al...

conjecture | open | MathDB 页面 | 原文线索

V2-048951 | MATHDB-337492

Acyclic matching bound for connected subcubic graphs

摘录 / EXCERPT: Let G be a connected subcubic graph, let $n(G)$ denote its number of vertices, and let $ u_{ac}(G)$ denote its acyclic matching number. The acyclic matching conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-048952 | MATHDB-337493

Acyclic matching lower bound for graphs of bounded maximum degree

摘录 / EXCERPT: Let G be a graph with no isolated vertices, let Δ be its maximum degree, let $n(G)$ denote its number of vertices, and let $\nu_{ac}(G)$ denote its acyclic matching number...

conjecture | open | MathDB 页面 | 原文线索

V2-048953 | MATHDB-337510

Breuer-Guralnick-Kantor Hamiltonian generating-graph conjecture

摘录 / EXCERPT: Let G be a finite group with $|G|\geq 4$, and let $\Gamma(G)$ be its generating graph: the vertices are the non-identity elements, and x and y are adjacent exactly when...

conjecture | open | MathDB 页面 | 原文线索

V2-048954 | MATHDB-337513

Multicolor Ramsey conjecture for disjoint unions of three-vertex paths

摘录 / EXCERPT: Let P_3 be the path on three vertices, and let $gr_k(K_3:n_1P_3,\ldots,n_kP_3)$ denote the corresponding Gallai-Ramsey number. For positive integers $k,n_1,n_2,\ldots,n_k$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-048955 | MATHDB-337516

Li-Feng pendent-path transfer conjecture for the alpha-index

摘录 / EXCERPT: Let G be a connected graph, let u be a vertex of G, and let $G_{p,q}(u)$ denote the graph obtained by attaching pendent paths of lengths p and q at u. For...

conjecture | open | MathDB 页面 | 原文线索

V2-048956 | MATHDB-337517

Pendent-path transfer conjecture for adjacent attachment vertices

摘录 / EXCERPT: Let G be a connected graph, and let u and v be adjacent vertices of G with degree at least 2. Let $G_{p,q}(u,v)$ denote the graph obtained by attaching pendent paths of l...

conjecture | open | MathDB 页面 | 原文线索

V2-048957 | MATHDB-337518

The characterization of graph random-walk evolution algebras

摘录 / EXCERPT: Graph evolution-algebra isomorphism conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048958 | MATHDB-337528

Blue path cover conjecture for the complete symmetric infinite digraph

摘录 / EXCERPT: Let $\vec{K}_{\mathbb{N}}$ be the complete symmetric digraph on the natural numbers, with each directed edge coloured red or blue. A red directed path of length r is a directed p...

conjecture | open | MathDB 页面 | 原文线索

V2-048959 | MATHDB-337543

Mader's subdivision conjecture for transitive tournaments

摘录 / EXCERPT: Let $k\geq 1$. A simple digraph has no loops and at most one arc from x to y for any two vertices x,y; opposite arcs are allowed. The transitive tournament TT_k is the tour...

conjecture | open | MathDB 页面 | 原文线索

V2-048960 | MATHDB-337544

Devos-McDonald-Mohar-Scheide immersion conjecture for transitive tournaments

摘录 / EXCERPT: Let $k\geq 1$. A simple digraph has no loops and at most one arc from x to y for any two vertices x,y; opposite arcs are allowed. An immersion of a digraph H maps the verti...

conjecture | open | MathDB 页面 | 原文线索

V2-048961 | MATHDB-337563

Generic-level simplicity conjecture for modular affine vertex algebras

摘录 / EXCERPT: Let $\mathbb{F}$ be an algebraically closed field of characteristic p, let $\mathfrak{g}$ be the finite-dimensional classical simple Lie algebra fixed in the section, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-048962 | MATHDB-337567

The average Second Neighborhood Conjecture for Eulerian digraphs

摘录 / EXCERPT: Let D be an Eulerian digraph with no loops or digons. For each vertex $v\in V(D)$, let $N^{+}(v)$ and $N^{+2}(v)$ denote its first and second out-neighborhoods, respectively. Ave...

conjecture | open | MathDB 页面 | 原文线索

V2-048963 | MATHDB-337568

The biplanar crossing number conjecture for the 8-dimensional hypercube

摘录 / EXCERPT: Let Q_8 be the 8-dimensional hypercube, and let $cr_2(Q_8)$ denote its biplanar crossing number, the minimum total number of crossings in a drawing of Q_8 on two planes. Biplan...

conjecture | open | MathDB 页面 | 原文线索

V2-048964 | MATHDB-337577

Divisibility conjecture for matching and characteristic polynomials of hypertrees

摘录 / EXCERPT: Divisibility conjecture. If $H\sqsubseteq\mathcal{H}$ are k-trees for $k\geq 3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-048965 | MATHDB-337595

The threshold conjecture for Steiner systems in random hypergraphs

摘录 / EXCERPT: Let K_k^n be the complete k-uniform hypergraph on n vertices, let $\mathcal{H}_k(n;p)$ be the random k-uniform hypergraph obtained by retaining each edge independently with...

conjecture | open | MathDB 页面 | 原文线索

V2-048966 | MATHDB-337596

The hitting-time conjecture for Steiner systems in random hypergraph processes

摘录 / EXCERPT: Let K_k^n be the complete k-uniform hypergraph on n vertices, and expose its edges one by one in a uniformly random order to obtain the random hypergraph process. A facet is...

conjecture | open | MathDB 页面 | 原文线索

V2-048967 | MATHDB-337597

The threshold conjecture for fractional decompositions of random hypergraphs

摘录 / EXCERPT: Fractional-decomposition threshold conjecture. The threshold for the appearance of fractional K_k-decompositions in $\mathcal{H}_k(n;p)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-048968 | MATHDB-337603

The complete-uniform-hypergraph host conjecture

摘录 / EXCERPT: Complete-host conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-048969 | MATHDB-337638

Positive fading number conjecture for odd non-cycle graphs

摘录 / EXCERPT: Positive fading number conjecture. For every such graph,

conjecture | open | MathDB 页面 | 原文线索

V2-048970 | MATHDB-337641

Rob Pratt's generating-function conjecture for the packing number of the double vertex graph of a path

摘录 / EXCERPT: Let $A085680(n)$ denote the packing number of the double vertex graph of the path graph P_n, equivalently $\rho(F_2(P_n))$, where $F_2(P_n)$ is the 2-token graph of P_n. Rob Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-048971 | MATHDB-337702

Harutyunyan's half-degree conjecture for regular digraphs

摘录 / EXCERPT: Harutyunyan's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048972 | MATHDB-337703

Cichacz-Görlich-Tuza conjecture on cordial hypertrees

摘录 / EXCERPT: Let $H=(V,E)$ be a hypertree, meaning a connected hypergraph without cycles. A vertex labeling $f:V\to\mathbb{Z}_2$ induces an edge labeling by $f(e)=\sum_{v\in e}f(v)$ modulo 2....

conjecture | open | MathDB 页面 | 原文线索

V2-048973 | MATHDB-337704

Cordial labeling conjecture for hypertrees

摘录 / EXCERPT: Let $H=(V,E)$ be a hypertree, meaning a connected hypergraph without cycles. For an integer $k\ge 2$, a vertex labeling $f:V\to\mathbb{Z}_k$ induces an edge labeling by...

conjecture | open | MathDB 页面 | 原文线索

V2-048974 | MATHDB-337705

2-cordiality conjecture for 1-degenerate connected hypergraphs

摘录 / EXCERPT: Let H be a connected hypergraph. It is 1-degenerated if every subhypergraph of H has a vertex of degree at most 1. A hypergraph is 2-cordial when a labeling of its vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-048975 | MATHDB-337719

Alon-Drewnowski-Łuczak conjecture for stable Kneser hypergraphs

摘录 / EXCERPT: For positive integers n,k,r with $k\geq 2$, $r\geq 2$, and $n\geq rk$, let $KG^{r}(n,k)_{r\text{-}stab}$ be the r-stable r-uniform Kneser hypergraph, and let $KG^{r}(n,k)$ be...

conjecture | solved | MathDB 页面 | 原文线索

V2-048976 | MATHDB-337761

Guggiari's path-cover conjecture for 2-edge-coloured complete symmetric digraphs

摘录 / EXCERPT: Let $\vec{K}_{\mathbb{N}}$ be the complete symmetric digraph on the positive integers, and let a 2-edge-colouring assign one of two colours, red or blue, to every directed edge. A...

conjecture | open | MathDB 页面 | 原文线索

V2-048977 | MATHDB-337765

Daligault-Rao-Thomassé conjecture on well-quasi-ordering and clique-width

摘录 / EXCERPT: Daligault-Rao-Thomassé conjecture. If a finitely defined hereditary class of graphs $\mathcal{G}$ is well-quasi-ordered by the induced subgraph relation, then $\mathcal{G}$ has bou...

conjecture | open | MathDB 页面 | 原文线索

V2-048978 | MATHDB-337823

Tree-cycle decomposition cycle double cover conjecture

摘录 / EXCERPT: Let G be a 2-edge connected cubic graph with a decomposition into edge-disjoint subgraphs covering $E(G)$, consisting of a tree T and a cycle C. Tree-cycle decomposition cy...

conjecture | open | MathDB 页面 | 原文线索

V2-048979 | MATHDB-337824

Non-separating cycle with a nowhere-zero 4-flow conjecture

摘录 / EXCERPT: Let G be a cyclically 4-edge connected cubic graph. For a cycle C of G, let $G/E(C)$ denote the graph obtained by contracting the edges of C, and let a nowhere-zero 4-f...

conjecture | open | MathDB 页面 | 原文线索

V2-048980 | MATHDB-337830

The $(1,1,2,2,2,2)$ and $(1,1,2,2,2)$ coloring conjecture for cubic graphs

摘录 / EXCERPT: $(1,1,2,2,2,2)$ and $(1,1,2,2,2)$ coloring conjecture. Every cubic graph is $(1,1,2,2,2,2)$-colorable, and every 3-edge-colorable cubic graph is $(1,1,2,2,2)$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-048981 | MATHDB-337835

The subregular W-algebra extension conjecture for A(n,r)

摘录 / EXCERPT: Subregular extension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-048982 | MATHDB-337836

Revised conjecture for $(t,3)$ and $(t-1,1)$ broadcast densities

摘录 / EXCERPT: Let G_∞ be the infinite grid, and let $\delta_{t,r}(G_\infty)$ denote the minimum density of a (t,r) broadcast. Revised broadcast-density conjecture. For all $t>2$, ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-048983 | MATHDB-337837

Blessing et al.'s broadcast-density isomorphism conjecture

摘录 / EXCERPT: Let G_∞ be the infinite grid, and for positive integers t and r let $\delta_{t,r}(G_\infty)$ denote the optimal density of a (t,r) broadcast. Blessing et al.'s conject...

conjecture | open | MathDB 页面 | 原文线索

V2-048984 | MATHDB-337843

Conjecture on the correspondence between rational conformal nets and vertex operator algebras

摘录 / EXCERPT: A completely rational local conformal net is a local conformal net with finite representation theory in the sense used in the source, and a simple unitary C_2-cofinite vertex ope...

conjecture | open | MathDB 页面 | 原文线索

V2-048985 | MATHDB-337847

VC-density conjecture for subgraphs of Cartesian products

摘录 / EXCERPT: Let G be a subgraph of the Cartesian product ... Here, $\operatorname{VC-dens}^*(G)$ denotes the VC-density of G. VC-density conjecture. ... The conjecture strengthens the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-048986 | MATHDB-337848

Barnette's conjecture in the language of legal systems

摘录 / EXCERPT: Let Q be a 3-connected planar quadrangulation, and let (A,B) be the bipartition of the vertices of Q. Suppose that every vertex of A has valence three, and that A admit...

conjecture | open | MathDB 页面 | 原文线索

V2-048987 | MATHDB-337853

Barrus-Ferrara-Vandenbussche-Wenger conjecture on rainbow clique saturation

摘录 / EXCERPT: Let K_k be the complete graph on k vertices, let $\mathcal{R}(K_k)$ be the family of rainbow edge-colorings of K_k, and let $\operatorname{sat}_t(n,\mathcal{R}(K_k))$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-048988 | MATHDB-337857

Lovász-Simonovits conjecture on minimum clique counts

摘录 / EXCERPT: Lovász-Simonovits conjecture. For every integer $r\geq 3$, there exists $n_0=n_0(r)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-048989 | MATHDB-337858

Exact extremal-graph conjecture for the Lovász-Simonovits triangle problem

摘录 / EXCERPT: Let $g_3(n,e)$ be the minimum number of triangles in an (n,e)-graph. Let $\mathcal{H}_0^*(n,e)$ and $\mathcal{H}_2^*(n,e)$ denote the subclasses of $\mathcal{H}_0(n,e)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-048990 | MATHDB-337861

Conjecture on regularity and component structure of distinguishing critical graphs

摘录 / EXCERPT: Regularity and component conjecture. (i) If G is a d-distinguishing critical graph, then G is a k-regular graph for some $k\leq d$. (ii) If G is a disconnected d-distin...

conjecture | open | MathDB 页面 | 原文线索

V2-048991 | MATHDB-337879

The vertex-maximum-clique criterion for weakly perfect graphs

摘录 / EXCERPT: Vertex-maximum-clique criterion. Any weakly perfect graph G is a perfect graph if each vertex $v\in V(G)$ is in some maximum clique of G.

conjecture | open | MathDB 页面 | 原文线索

V2-048992 | MATHDB-337884

Benjamini-Gurel-Gurevich-Schramm conjecture on cutpoints of random walk paths

摘录 / EXCERPT: Benjamini-Gurel-Gurevich-Schramm conjecture. The path graph PATH has almost surely infinitely many cutpoints.

conjecture | open | MathDB 页面 | 原文线索

V2-048993 | MATHDB-337900

Partitioned Kneser hypergraph chromatic-number conjecture

摘录 / EXCERPT: Let $r\ge 2$, $k\ge 1$, and $n\ge rk$ be integers, and let $\mathcal P=\{P_1,\dots,P_\ell\}$ be a partition of $[n]$ such that $|P_i|\le r$ for every i. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-048994 | MATHDB-337901

Abyazi Sani-Alishahi conjecture for deleted Kneser hypergraphs

摘录 / EXCERPT: Let $A\subset[n]$, and let $\mathrm{KG}^r_A(n,k)$ be the subhypergraph of the r-uniform Kneser hypergraph induced by the k-subsets $\sigma\subset[n]$ with $\sigma\not\subset A$...

conjecture | open | MathDB 页面 | 原文线索

V2-048995 | MATHDB-337903

The hypersurface conjecture for edge rings with q-linear resolutions

摘录 / EXCERPT: Let G be a finite connected simple graph, and let $K[G]$ denote its edge ring. A graded ring has a q-linear resolution when its minimal graded free resolution is q-linear, an...

conjecture | open | MathDB 页面 | 原文线索

V2-048996 | MATHDB-337912

The Saxl graph diameter-two conjecture for primitive base-two groups

摘录 / EXCERPT: Let G be a finite primitive permutation group with base size $b(G)=2$, and let $\alpha0\Sigma(G)$ be its Saxl graph, whose vertices are the points of the permutation domain and w...

conjecture | open | MathDB 页面 | 原文线索

V2-048997 | MATHDB-337930

Colored Erdős-Sós conjecture for trees with leaves of different colors

摘录 / EXCERPT: Colored Erdős-Sós conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-048998 | MATHDB-337962

Resistance increment conjecture for straight linear k-trees

摘录 / EXCERPT: Let G be the straight linear k-tree with $k\geq 1$ and n vertices, and let H be the straight linear k-tree with $n+1$ vertices. Let r_G and r_H denote resistance dist...

conjecture | open | MathDB 页面 | 原文线索

V2-048999 | MATHDB-337963

Unbounded resistance conjecture for linear 2-trees of growing diameter

摘录 / EXCERPT: Let G be a linear 2-tree, and let its diameter tend to infinity. Let the maximal resistance distance of G mean the maximum of $r_G(u,v)$ over pairs of vertices. Unbounded resis...

conjecture | open | MathDB 页面 | 原文线索

V2-049000 | MATHDB-337965

Invariance of network uni-asymptotic classes under equi-parameter changes

摘录 / EXCERPT: Consider networks ${\cal N}_1$ and ${\cal N}_2$ of coupled quadratic nodes. They are in the same uni-asymptotic class when, for every uni-initial condition $z_0\in\mathbb{C}$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-049001 | MATHDB-337973

Hochstättler's even-subgraph contraction conjecture for 3-edge-connected digraphs

摘录 / EXCERPT: Let $D=(V,A)$ be a 3-edge-connected digraph. An even subgraph $E\subseteq A$ is an edge-disjoint union of cycles of D, where the cycles need not be directed, and D/E is the gra...

conjecture | open | MathDB 页面 | 原文线索

V2-049002 | MATHDB-337974

Hochstättler's perfect-matching formulation for cubic digraphs

摘录 / EXCERPT: Let G be a 3-edge-connected, cubic digraph. A perfect matching without a directed cut is a perfect matching that contains no directed edge-cut. Hochstättler's equivalent conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-049003 | MATHDB-337975

The bipartite directed perfect-matching conjecture

摘录 / EXCERPT: Let G be a 3-connected, cubic, bipartite digraph. A perfect matching without a directed cut is a perfect matching that contains no directed cut. Bipartite directed perfect-matchi...

open_problem | open | MathDB 页面 | 原文线索

V2-049004 | MATHDB-337979

The local characterization of f-convex drawings

摘录 / EXCERPT: Let D be an h-convex drawing of K_n, meaning that every induced subdrawing on K_5 is convex, and let $D[J]$ denote the subdrawing induced by an isomorph J of K_8. The loc...

conjecture | open | MathDB 页面 | 原文线索

V2-049005 | MATHDB-337980

The natural deficiency conjecture for convex drawings of complete graphs

摘录 / EXCERPT: For a drawing D of K_n, define its deficiency by ... The drawing D has the natural deficiency property if, for every vertex v of K_n, ... Here a convex drawing is a drawing i...

conjecture | open | MathDB 页面 | 原文线索

V2-049006 | MATHDB-337994

Strong form of Erdős's girth conjecture

摘录 / EXCERPT: Strong girth conjecture. For every positive integer k, there exists a family of almost-regular graphs $\{G_n\}$ such that $|V(G_n)|=n$, $|E(G_n)|\ge \frac{n^{1+1/k}}{2}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049007 | MATHDB-337995

Subquadratic conjecture for odd cycles with an additional forbidden odd cycle

摘录 / EXCERPT: Subquadratic odd-cycle conjecture. There exists an $\epsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-049008 | MATHDB-337997

The quasi-Hopf algebra correspondence for the triplet vertex operator algebra

摘录 / EXCERPT: Let $p\geq 2$, let $q=e^{\pi i/p}$, let $\overline{U}^{(\Phi)}_{q} s\ell(2)$ be the factorisable ribbon quasi-Hopf algebra described above, and let $\mathcal{W}(p)$ be the triplet...

conjecture | open | MathDB 页面 | 原文线索

V2-049009 | MATHDB-337998

The singlet quantum-group correspondence

摘录 / EXCERPT: Let $\overline{U}_{q}^{H}s\ell(2)$ be the unrolled restricted quantum group, and let $\mathrm{\bf Rep}_{\!\mathrm{wt}\,}\overline{U}_{q}^{H}s\ell(2)$ denote its finite-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-049010 | MATHDB-337999

The singlet VOA rigidity conjecture

摘录 / EXCERPT: Let $\mathcal{M}(p)$ be the singlet vertex operator algebra, and let $\mathrm{\bf Rep}\,\mathcal{M}(p)$ be its representation category equipped with the $P(1)$-tensor product. The...

conjecture | open | MathDB 页面 | 原文线索

V2-049011 | MATHDB-338000

Beem-Rastelli Higgs branch conjecture for four-dimensional N=2 SCFTs

摘录 / EXCERPT: Let $\mathcal{T}$ be a four-dimensional $N=2$ superconformal field theory, let $\Phi(\mathcal{T})$ be its associated vertex algebra, and let $Higgs_{\mathcal{T}}$ denote its Higgs...

conjecture | open | MathDB 页面 | 原文线索

V2-049012 | MATHDB-338002

The affine DT4 vertex conjecture for solid partitions

摘录 / EXCERPT: The affine DT4 vertex conjecture. There exists a way of choosing the signs of the weights w_π such that

conjecture | open | MathDB 页面 | 原文线索

V2-049013 | MATHDB-338006

Weakly convex domination can decrease under permutation in diameter-two graphs

摘录 / EXCERPT: Let G be a connected graph with $\operatorname{diam}(G)=2$, and let π be a permutation of its vertex set V_G. Write $\gamma_{wcon}(G)$ for the weakly convex domination numb...

conjecture | open | MathDB 页面 | 原文线索

V2-049014 | MATHDB-338010

Triangle extremality conjecture for the graph \(\operatorname{BT}(3)\)

摘录 / EXCERPT: For $k\geq 2$, let $\widetilde{\operatorname{BT}}(k)$ be obtained from $T_2(2k)$ by deleting an edge and adjoining a new vertex to the two endpoints of that edge, and let ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049015 | MATHDB-338011

Existence of extremal graphs for clique density

摘录 / EXCERPT: Let $\mathcal{G}(\Delta,\omega)$ be the class of graphs with maximum degree at most Δ and clique number at most ω, and let $f_t(\Delta,\omega)$ be the supremum of...

conjecture | open | MathDB 页面 | 原文线索

V2-049016 | MATHDB-338014

Fleischner's dominating circuit conjecture for cyclically 4-edge-connected snarks

摘录 / EXCERPT: Fleischner's dominating circuit conjecture. Every cyclically 4-edge-connected snark has a dominating circuit.

conjecture | open | MathDB 页面 | 原文线索

V2-049017 | MATHDB-338028

Asymptotic online coloring of the graph C_5^+

摘录 / EXCERPT: Let C_5^+ be the graph obtained from the 5-cycle C_5 by adding the edge indicated in the paper, and let $f(w,G)$ denote the maximum number of colors that an adversary can force...

conjecture | open | MathDB 页面 | 原文线索

V2-049018 | MATHDB-338039

Asymmetric density criterion for Ramsey-infinite ordered graph pairs

摘录 / EXCERPT: Let H and H' be ordered graphs with asymmetric 2-density parameters satisfying $m_2(H)\geq m_2(H')$. Let $m(F)$ denote the relevant density of a graph F, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049019 | MATHDB-338040

Asymmetric Ramsey-density conjecture for graphs

摘录 / EXCERPT: Let H and H' be graphs with $m_2(H)\geq m_2(H')$. Asymmetric Ramsey-density conjecture. If neither H nor H' is a matching and at least one of H or H' contains a cycle,...

conjecture | open | MathDB 页面 | 原文线索

V2-049020 | MATHDB-338069

Moonshine-like function characterization conjecture

摘录 / EXCERPT: Let f be a moonshine-like function, meaning a completely replicable function given by the graded trace of a finite-order automorphism of a C_2-cofinite, holomorphic vertex oper...

conjecture | open | MathDB 页面 | 原文线索

V2-049021 | MATHDB-338090

Elimination of null edge vectors on reducible PBDTP networks

摘录 / EXCERPT: Let $({\mathcal N},\mathcal O,\mathfrak l)$ be a reducible PBDTP network representing a point...

conjecture | open | MathDB 页面 | 原文线索

V2-049022 | MATHDB-338098

Anti-commonality monotonicity conjecture

摘录 / EXCERPT: Let H be a graph and let r be a positive integer. Say that H is r-anti-common when its rainbow-copy behavior satisfies the paper's definition for r-edge-colorings. Anti-c...

conjecture | open | MathDB 页面 | 原文线索

V2-049023 | MATHDB-338109

The asymptotic conjecture for odd cycle-creating paths

摘录 / EXCERPT: Let $H(n,k)$ denote the maximum size of a family of paths in the complete graph K_n such that the union of every pair contains a cycle of length k. The established cases includ...

conjecture | open | MathDB 页面 | 原文线索

V2-049024 | MATHDB-338113

Conjecture that the exact induced density formula holds for all graph sizes

摘录 / EXCERPT: Let H be a strongly asymmetric graph on h vertices. Let $i_H(n)$ be the maximum number of induced copies of H in an n-vertex graph, and let $g(n,h)$ be the recursively defi...

conjecture | open | MathDB 页面 | 原文线索

V2-049025 | MATHDB-338118

Homological span conjecture for chromatic homology over \mathcal{A}_m

摘录 / EXCERPT: Let G be a graph with v vertices and b blocks, and let $\mathcal{A}_m=\mathbb{Z}[x]/(x^m=0)$. Write $H_{\mathcal{A}_m}(G)$ for its chromatic homology and...

conjecture | open | MathDB 页面 | 原文线索

V2-049026 | MATHDB-338119

Girth-determined torsion-width conjecture over \mathcal{A}_3

摘录 / EXCERPT: Let G be a simple, connected graph with v vertices and girth ℓ, where $\ell=2k$ or $\ell=2k-1$ according to its parity. Let P_ℓ denote the cycle graph of length...

conjecture | open | MathDB 页面 | 原文线索

V2-049027 | MATHDB-338135

Sparse strongly connected spanning subgraph conjecture with optimal degree-dependent bound

摘录 / EXCERPT: Let $k,n\geq 1$ and $\overline{\Delta}\geq 0$. Sparse spanning subgraph conjecture. There is $C>0$ such that every strongly k-connected n-vertex digraph D with...

conjecture | open | MathDB 页面 | 原文线索

V2-049028 | MATHDB-338136

Almost-regular spanning subgraph conjecture for semicomplete digraphs

摘录 / EXCERPT: Let $k,n\geq 1$. In the first setting, let D be a strongly k-connected semicomplete digraph; in the second, let D be a strongly k-arc-connected semicomplete directed multig...

conjecture | open | MathDB 页面 | 原文线索

V2-049029 | MATHDB-338146

Conjecture on the extremal degree power of broom-free graphs

摘录 / EXCERPT: Let $B_{\ell,s}$ be a broom graph, let $H(n,\ell)$ denote the graph used in the paper, and let $\operatorname{ex}_p(n,B_{\ell,s})$ be the maximum degree power among $B_{\ell,s}$-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-049030 | MATHDB-338154

The logarithmic localization conjecture for invisible agents on graphs

摘录 / EXCERPT: Logarithmic localization conjecture. The cat can localize the mouse up to distance $O(\log n)$ on G.

conjecture | open | MathDB 页面 | 原文线索

V2-049031 | MATHDB-338159

The asymptotic scarcity conjecture for collapsible hypertrees

摘录 / EXCERPT: Let ${\mathcal T_{n,d}}$ denote the family of n-vertex d-hypertrees, and let ${\mathcal C_{n,d}}$ denote the set of n-vertex d-collapsible d-hypertrees. Here, a d-colla...

conjecture | open | MathDB 页面 | 原文线索

V2-049032 | MATHDB-338193

Schellekens' conjecture on holomorphic tame vertex algebras of central charge 24

摘录 / EXCERPT: Schellekens' conjecture. There are exactly 71 holomorphic tame vertex algebras of central charge 24.

conjecture | open | MathDB 页面 | 原文线索

V2-049033 | MATHDB-338199

The generalized Fibonacci cube perfect-code conjecture

摘录 / EXCERPT: Generalized Fibonacci cube perfect-code conjecture. If C is a perfect code in $\Gamma_n(1^s)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049034 | MATHDB-338204

Akbari et al.'s zero-sum 6-flow conjecture

摘录 / EXCERPT: Let G be a graph. A zero-sum k-flow of G is an edge labeling with labels in $\{\pm1,\ldots,\pm(k-1)\}$ such that the sum of the labels on all edges incident with every vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-049035 | MATHDB-338234

Kanté-Kwon conjecture on vertex-minors of forests in graphs of large linear rank-width

摘录 / EXCERPT: Kanté-Kwon conjecture. For every fixed forest T, there is an integer $f(T)$ such that every graph of linear rank-width at least $f(T)$ contains a vertex-minor isomorphic to T.

conjecture | open | MathDB 页面 | 原文线索

V2-049036 | MATHDB-338254

The K10-minor extremal graph conjecture

摘录 / EXCERPT: K10-minor extremal graph conjecture. A graph on $V\geq 9$ vertices with no K_{10}-minor has at most $8V-36$ edges, unless it is isomorphic to one of the following graphs: a...

conjecture | open | MathDB 页面 | 原文线索

V2-049037 | MATHDB-338255

The bipartite excluded-minor extremal conjecture

摘录 / EXCERPT: Bipartite excluded-minor extremal conjecture. A bipartite graph on $V\geq 2p-5$ vertices with no K_p-minor has at most

conjecture | open | MathDB 页面 | 原文线索

V2-049038 | MATHDB-338258

Odd-coordinate kernel conjecture for singular io-decomposable Riordan graphs

摘录 / EXCERPT: Odd-coordinate kernel conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-049039 | MATHDB-338259

Nullity equality conjecture for singular io-decomposable Riordan graphs

摘录 / EXCERPT: Let ${\cal A}_n$ be the adjacency matrix of a singular io-decomposable Riordan graph $G_n(g,zg)$ of the form specified in the preceding structural theorem, and let B be the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-049040 | MATHDB-338260

Catalan graph determinant conjecture

摘录 / EXCERPT: Catalan graph determinant conjecture. The determinant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049041 | MATHDB-338303

Mycielskian symmetry-breaking conjecture

摘录 / EXCERPT: Let G be a connected graph of order $n\geq 3$, and let $\mu(G)$ denote its Mycielskian. Write $D(G)$ for the distinguishing number of G and $D'(G)$ for its distinguishing index...

conjecture | open | MathDB 页面 | 原文线索

V2-049042 | MATHDB-338334

The nondegeneracy conjecture for positive steady states

摘录 / EXCERPT: Let G be a reaction network, and suppose that G does not admit infinitely many positive steady states in any stoichiometric compatibility class. Recall that G admits k posi...

conjecture | open | MathDB 页面 | 原文线索

V2-049043 | MATHDB-338335

The three-species reversible-irreversible network conjecture

摘录 / EXCERPT: Let G be a reaction network consisting of one reversible-reaction pair $y\leftrightarrows y'$ and one irreversible reaction $\widetilde y\to\widetilde y'$. The reaction vectors a...

conjecture | open | MathDB 页面 | 原文线索

V2-049044 | MATHDB-338351

Dominating-vertex conjecture for anti-van der Waerden numbers

摘录 / EXCERPT: Let G be a graph with vertex set $V(G)$, and let $v\in V(G)$ be dominating if its closed neighborhood satisfies ... For an integer $k\geq 3$, let $\operatorname{aw}(G,k)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-049045 | MATHDB-338360

Henning-Yeo transversal bound for uniform linear hypergraphs

摘录 / EXCERPT: Let H be a k-uniform linear hypergraph on n vertices with m edges, where $\tau(H)$ denotes the minimum number of vertices intersecting every edge. Henning-Yeo conjecture. F...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049046 | MATHDB-338361

Henning-Yeo four-uniform transversal bound

摘录 / EXCERPT: Let H be a 4-uniform linear hypergraph on n vertices with m edges, and let $\tau(H)$ be its transversal number. Henning-Yeo four-uniform conjecture. ... The paper proves this...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049047 | MATHDB-338362

Henning-Yeo bound for four-uniform linear hypergraphs

摘录 / EXCERPT: Let H be a 4-uniform linear hypergraph on n vertices with m edges, and let $\tau(H)$ be its transversal number. Henning-Yeo conjecture. ... The source presents this as a conj...

conjecture | open | MathDB 页面 | 原文线索

V2-049048 | MATHDB-338363

Conjecture that the transversal-bound threshold exceeds five

摘录 / EXCERPT: Let $k_{{\rm min}}$ be the smallest integer k for which there exists a k-uniform linear hypergraph violating $\tau(H)\le(n+m)/(k+1)$. Threshold conjecture. ... The paper proves...

conjecture | open | MathDB 页面 | 原文线索

V2-049049 | MATHDB-338366

Maximum volume growth conjecture for products of regular trees

摘录 / EXCERPT: Let T_p be the infinite p-regular tree, and let T_p^q be its q-fold Cartesian product, which is pq-regular. For a graph G and vertex x, write...

conjecture | open | MathDB 页面 | 原文线索

V2-049050 | MATHDB-338374

Topological determination by the log fiber for smooth proper exact morphisms

摘录 / EXCERPT: Let P) be a sharp toric monoid, let V_P be a neighborhood of the vertex v of the associated toric space, and let $f\colon X\to Y=V_P$ be a smooth proper and exact morphism of...

conjecture | open | MathDB 页面 | 原文线索

V2-049051 | MATHDB-338387

The conjecture on the image of the conformal embedding map

摘录 / EXCERPT: The conjecture on the image of the conformal embedding map. The image of the homomorphism Φ is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-049052 | MATHDB-338398

Domination conjecture for Turán hypergraphs

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $C(n)$ be the family ... For a 3-uniform hypergraph H, say that H dominates $C(n)$ when the corresponding rows and columns of the third compou...

conjecture | open | MathDB 页面 | 原文线索

V2-049053 | MATHDB-338404

Separation choosability grows with minimum degree

摘录 / EXCERPT: Let G be a graph with minimum degree d, and let $\operatorname{ch}_{\operatorname{sep}}(G)$ denote its separation choosability. Separation-choosability conjecture. There is a f...

conjecture | open | MathDB 页面 | 原文线索

V2-049054 | MATHDB-338405

Logarithmic bipartite induced subgraphs in triangle-free graphs

摘录 / EXCERPT: Let d be a positive integer. A triangle-free graph is a graph containing no triangle, and a bipartite induced subgraph is an induced subgraph that is bipartite. Triangle-free bip...

conjecture | open | MathDB 页面 | 原文线索

V2-049055 | MATHDB-338406

Girth-forces-dense-bipartite-induced-subgraph conjecture

摘录 / EXCERPT: Let d_0 and g_0 be constants. A bipartite induced subgraph is an induced subgraph that is bipartite. Girth-induced-subgraph conjecture. There exist d_0 and g_0 such that an...

conjecture | open | MathDB 页面 | 原文线索

V2-049056 | MATHDB-338408

Matching-threshold conjecture for partial immunization of graphs

摘录 / EXCERPT: Let G be a graph with $m(G)$ edges, let d_G denote its degree function, and let b be an integer satisfying $0\leq b\leq 2m(G)$. For a matching M of G, define the threshol...

conjecture | open | MathDB 页面 | 原文线索

V2-049057 | MATHDB-338422

Klee-Wolfe nonrevisiting path conjecture for polytopes

摘录 / EXCERPT: Let P be a d-dimensional polytope, and let u and v be any two of its vertices. A path is nonrevisiting if, after leaving a facet, it does not later return to that facet. No...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049058 | MATHDB-338423

Strict monotone Hirsch conjecture for polytope orientations

摘录 / EXCERPT: Let P be a d-dimensional polytope with n facets, and let ${\bf c}$ be a generic linear functional. Orient the graph of P according to increasing values of ${\bf c}$, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-049059 | MATHDB-338437

The bunkbed conjecture for independent bond percolation

摘录 / EXCERPT: The bunkbed graph $G=(V,E)$ of a graph $\widetilde{G}=(\widetilde{V},\widetilde{E})$ is formed by superposing two identical copies of $\widetilde{G}$ and adding edges between corre...

conjecture | open | MathDB 页面 | 原文线索

V2-049060 | MATHDB-338441

Wilson's classification conjecture for nontrivially unstable circulants

摘录 / EXCERPT: Let G be a group and let S be a nonempty inverse-closed subset of $G\setminus\{1\}$, where 1 is the identity. The Cayley graph $\operatorname{Cay}(G,S)$ has vertex set G, w...

conjecture | open | MathDB 页面 | 原文线索

V2-049061 | MATHDB-338455

Conjecture on spectra and positive semistability of forest chromatic operators

摘录 / EXCERPT: Let F be a forest on n vertices, let π be an ordering of its edges, and let $\alpha_{F,\pi}\in\mathbb{C}[S_n]$ be its chromatic operator. A chromatic operator is positive s...

conjecture | open | MathDB 页面 | 原文线索

V2-049062 | MATHDB-338478

Bean-Tannock-Ulfarsson purity conjecture for boundary-grid downcore graphs

摘录 / EXCERPT: Let π be a 123-avoiding permutation, and let its boundary grid be the associated skew Young diagram. The downcore graph has the boxes of the boundary grid as vertices, with a...

conjecture | open | MathDB 页面 | 原文线索

V2-049063 | MATHDB-338499

Forward-closure conjecture for linkage-class-one weakly reversible systems

摘录 / EXCERPT: Let a weakly reversible chemical reaction network have linkage class one, meaning that its underlying graph has only one strongly connected component. Consider a trajectory that re...

conjecture | open | MathDB 页面 | 原文线索

V2-049064 | MATHDB-338511

Extremal hyperedge conjecture for FF4-hypergraphs

摘录 / EXCERPT: Extremal hyperedge conjecture. If $n\equiv2\pmod{4}$, then $\mathcal{H}$ has at most

conjecture | open | MathDB 页面 | 原文线索

V2-049065 | MATHDB-338535

Jahanbekam-West anti-Ramsey conjecture for edge-disjoint rainbow spanning trees

摘录 / EXCERPT: For positive integers n and t, let $r(n,t)$ be the maximum number of colors in an edge-coloring of K_n that has no t edge-disjoint rainbow spanning trees. Jahanbekam-West c...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-049066 | MATHDB-338536

Archimedean limit conjecture for sorting-network permutation matrices

摘录 / EXCERPT: Angel-Holroyd-Romik-Virág Archimedean-limit conjecture. For every $t\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-049067 | MATHDB-338537

Fluctuation-variance conjecture for halfway permutation paths

摘录 / EXCERPT: Fluctuation-variance conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-049068 | MATHDB-338538

Maximum fluctuation conjecture for halfway permutation paths

摘录 / EXCERPT: Maximum fluctuation conjecture. For every $epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-049069 | MATHDB-338539

Local-limit conjecture for halfway permutation point processes

摘录 / EXCERPT: Local-limit conjecture. There exists a rotationally symmetric, translation-invariant point process Pi on $\mathbb{R}^2$ such that, for every (x,y) in the open unit disk,

conjecture | open | MathDB 页面 | 原文线索

V2-049070 | MATHDB-338540

Exponential concentration conjecture for sorting-network permutation measures

摘录 / EXCERPT: Exponential concentration conjecture. There exist constants $c_1,c_2>0$ such that, for every n,

conjecture | open | MathDB 页面 | 原文线索

V2-049071 | MATHDB-338541

Local trajectory reconstruction conjecture for sorting networks

摘录 / EXCERPT: Local trajectory reconstruction conjecture. For every $epsilon>0$, there exists $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-049072 | MATHDB-338557

The 3-core conjecture for maximal typical rank

摘录 / EXCERPT: 3-core conjecture. The maximal typical rank of G is at least 3.

conjecture | open | MathDB 页面 | 原文线索

V2-049073 | MATHDB-338558

The generic-rank conjecture for bipartite circulant graphs

摘录 / EXCERPT: Bipartite circulant generic-rank conjecture. Every graph $G(n,l)$ has generic completion rank predicted by the dimension count; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-049074 | MATHDB-338568

A closed-form conjecture for the MS-index of uniform hypergraphs

摘录 / EXCERPT: MS-index conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-049075 | MATHDB-338569

The vanishing proportion conjecture for magic n_3-configurations

摘录 / EXCERPT: An n_3-configuration is a 3-uniform configuration with n points and n lines, in which every point lies on three lines and every line contains three points. Call such a conf...

conjecture | open | MathDB 页面 | 原文线索

V2-049076 | MATHDB-338576

The downer-vertex conjecture for chain graphs

摘录 / EXCERPT: Downer-vertex conjecture. In any chain graph, every vertex is downer with respect to every non-zero eigenvalue.

conjecture | open | MathDB 页面 | 原文线索

V2-049077 | MATHDB-338579

Diameter bounds conjecture for cyclic shift components in patience sorting monoids

摘录 / EXCERPT: Let $\smash{\mathrm{K}}(\smash{\mathrm{rps}})$ be the cyclic shift graph of the right patience sorting monoid, and consider an element with $n\geq 3$ symbols, allowing symbols to o...

conjecture | open | MathDB 页面 | 原文线索

V2-049078 | MATHDB-338601

Positivity conjecture for simple VOAs generated by sigma-type Ising vectors

摘录 / EXCERPT: Let V be a simple VOA satisfying the paper's Conditions 1, let E_V be the set of Ising vectors of V of σ-type, and let $G_V=\langle \sigma_e\mid e\in E_V\rangle$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-049079 | MATHDB-338603

Conjecture on chromatic core subgraphs of strong graph products

摘录 / EXCERPT: Let G and H be graphs, and let G' and H' be chromatic core subgraphs of G and H, respectively. The strong product $G\boxtimes H$ has vertex set $V(G)\times V(H)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-049080 | MATHDB-338604

Conjecture on chromatic core subgraphs of general graph products

摘录 / EXCERPT: Let G and H be graphs with chromatic core subgraphs G' and H', respectively, and let $G\circledast H$ denote the general graph product used in the source. General-product c...

conjecture | open | MathDB 页面 | 原文线索

V2-049081 | MATHDB-338609

Anderson et al.'s asymptotic lower-bound conjecture for exponential domination of Cartesian products of cycles

摘录 / EXCERPT: Anderson et al.'s conjecture. For all m and n,

conjecture | open | MathDB 页面 | 原文线索

V2-049082 | MATHDB-338624

The broadcast domination-multipacking bound conjecture

摘录 / EXCERPT: Broadcast domination-multipacking conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-049083 | MATHDB-338630

The critical-window conjecture for giant j-components in random hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}^k(n,p)$ be the random k-uniform hypergraph with edge probability p, and let j-components denote components under the adjacency notion used in the paper. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-049084 | MATHDB-338639

Constant-error rainbow tree embedding conjecture

摘录 / EXCERPT: Constant-error rainbow tree conjecture. There is a constant C such that every properly coloured K_n has a rainbow copy of every tree on $n-C$ vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-049085 | MATHDB-338647

The sharp snowflake exponent conjecture for modulus metrics on networks

摘录 / EXCERPT: Let G be a connected network, let d_p denote its modulus metric for $p\in[1,\infty]$, and let $s(p)$ be the supremum of the exponents t such that d_p^t is a metric on every...

conjecture | open | MathDB 页面 | 原文线索

V2-049086 | MATHDB-338648

Alon-Tarsi shortest cycle cover conjecture for bridgeless graphs

摘录 / EXCERPT: Let G be a bridgeless graph, and let a circuit cover be a family of circuits covering every edge of G. Its length is the sum of the numbers of edges in its circuits. Alon-Tarsi...

conjecture | open | MathDB 页面 | 原文线索

V2-049087 | MATHDB-338650

Positivity conjecture for conformal weights of twisted modules

摘录 / EXCERPT: Let V be a strongly rational, holomorphic vertex operator algebra and let $g\neq\operatorname{id}$ be an automorphism of V of finite order. Write $V(g)$ for the unique irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-049088 | MATHDB-338651

Strict positivity conjecture for twisted conformal weights

摘录 / EXCERPT: Let V be a holomorphic, C_2-cofinite vertex operator algebra and let $g\neq\operatorname{id}$ be an automorphism of V of finite order. Write $V(g)$ for the irreducible g-tw...

conjecture | open | MathDB 页面 | 原文线索

V2-049089 | MATHDB-338652

Gromov's bounded-imbalance conjecture for planar geodesic nets

摘录 / EXCERPT: Let $G=(B,U,E)$ be a geodesic net in the plane, with balanced vertices B and unbalanced vertices U. For a vertex v, let $\operatorname{imb}(v)$ denote the norm of the sum of...

conjecture | open | MathDB 页面 | 原文线索

V2-049090 | MATHDB-338655

The nullity conjecture for connected block graphs with blocks of order at least 3

摘录 / EXCERPT: Nullity conjecture. If each block of G has order at least 3, then the nullity of G is at most 1.

conjecture | open | MathDB 页面 | 原文线索

V2-049091 | MATHDB-338656

The nonsingular edge-extension conjecture for connected block graphs

摘录 / EXCERPT: Nonsingular edge-extension conjecture. If an edge is added between vertices $v_1\rtimes V(G_1)$ and $v_2\rtimes V(G_2)$, then the resulting block graph is nonsingular.

conjecture | open | MathDB 页面 | 原文线索

V2-049092 | MATHDB-338681

Espenschied's conjecture on complete multipartite graphs of polytopes

摘录 / EXCERPT: Espenschied's conjecture. If $K_{n_1,n_2,\dots, n_t}$ is the graph of a polytope, then $\\{n_1, n_2, \dots n_t\\} \subset \\{1, 2\\}$ as sets. The classification of polytopes and c...

conjecture | open | MathDB 页面 | 原文线索

V2-049093 | MATHDB-338692

Pikhurko's saturation limit conjecture for hypergraphs

摘录 / EXCERPT: Pikhurko's conjecture. For every r-graph F, the limit

conjecture | open | MathDB 页面 | 原文线索

V2-049094 | MATHDB-338716

The classification conjecture for unsplittable cyclic (n_3) configurations

摘录 / EXCERPT: Classification conjecture. A cyclic (n_3) configuration is unsplittable if and only if its Levi graph belongs to one of the following three infinite families:

conjecture | open | MathDB 页面 | 原文线索

V2-049095 | MATHDB-338738

AMRZ finite-exception conjecture for graph lines and bridges

摘录 / EXCERPT: Let G be a connected graph, let $\ell(G)$ be the number of distinct graph-metric lines, and let $\textsc{br}(G)$ denote the number of bridges of G. A pendant edge is an edge in...

conjecture | open | MathDB 页面 | 原文线索

V2-049096 | MATHDB-338739

AMRZ conjecture on graph lines and universal pairs

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph with at least two vertices. Let $\ell(G)$ be the number of distinct graph-metric lines, and let a universal pair be a pair of vertices generating...

conjecture | open | MathDB 页面 | 原文线索

V2-049097 | MATHDB-338749

Vanishing of cubical homology for graphs without triangles or quadrilaterals

摘录 / EXCERPT: Let G be an undirected graph, and let $\mathcal{H}_n^{\normalfont {\textrm{Cube}}}(G)$ denote its n-th cubical homology group. Suppose that G contains no 3-cycles and no...

conjecture | open | MathDB 页面 | 原文线索

V2-049098 | MATHDB-338762

The two-thirds excess conjecture for hypergraph r-cuts

摘录 / EXCERPT: Two-thirds excess conjecture. Every k-graph has an r-cut of size

conjecture | open | MathDB 页面 | 原文线索

V2-049099 | MATHDB-338770

Higuchi's conjecture on positive combinatorial curvature

摘录 / EXCERPT: Let a tessellation be a graph embedded in the plane so that every face is a polygon and the embedding satisfies the usual tessellation conditions. At each vertex v, let its combi...

conjecture | open | MathDB 页面 | 原文线索

V2-049100 | MATHDB-338779

Broersma's induced-net Hamiltonicity conjecture

摘录 / EXCERPT: Let G be a graph, and call the connected graph with degree sequence $3,3,3,1,1,1$ a net, with its vertices of degree 1 called endvertices. Call G claw-free if it has no induc...

conjecture | open | MathDB 页面 | 原文线索

V2-049101 | MATHDB-338792

Bernhart-Kainen conjecture on dispersable regular bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph that is k-regular. Its dispersable book thickness $dbt(G)$ is the minimum number of pages in a dispersable book embedding, and a graph is dispersable...

conjecture | open | MathDB 页面 | 原文线索

V2-049102 | MATHDB-338799

Average-degree gap conjecture for simple 3-connected crossing-critical graphs

摘录 / EXCERPT: A graph is k-crossing-critical if its crossing number is at least k while deleting any edge reduces the crossing number below k. Let $g:\mathbb N\to\mathbb R^+$ be a function...

conjecture | open | MathDB 页面 | 原文线索

V2-049103 | MATHDB-338806

High-girth approximation conjecture for uniquely restricted matching number

摘录 / EXCERPT: For a graph G, let $\nu(G)$ denote its matching number and $\nu_{ur}(G)$ its uniquely restricted matching number. For every positive real number ϵ and every positive int...

conjecture | open | MathDB 页面 | 原文线索

V2-049104 | MATHDB-338811

Monotonicity of strong geodetic number under the Cartesian product with K_2

摘录 / EXCERPT: Let G be a graph, let $n(G)$ denote its order, and let $G \,\square\, K_2$ be the Cartesian product of G with the complete graph on two vertices. Write $\operatorname{sg}(G)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-049105 | MATHDB-338813

The conjectured upper bound for minimum dominating sets in trees

摘录 / EXCERPT: Upper-bound conjecture. A tree with domination number γ has at most $O\left(\frac{\gamma 2^{\gamma}}{\ln \gamma}\right)$ minimum dominating sets. The conjecture refines the...

conjecture | open | MathDB 页面 | 原文线索

V2-049106 | MATHDB-338817

Uniform integrability conjecture for graphs with smallest eigenvalue at least -3

摘录 / EXCERPT: Let G be a connected graph, and let its smallest eigenvalue be denoted by $4\lambda_{\min}(G)4$. A graph is σ-integrable if its associated integral lattice admits an embed...

conjecture | open | MathDB 页面 | 原文线索

V2-049107 | MATHDB-338818

Uniform integrability conjecture for irreducible lattices generated by norm-three vectors

摘录 / EXCERPT: Let Λ be an irreducible integral lattice generated by vectors of norm 3. A lattice is σ-integrable if its bilinear form, after multiplication by σ, admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-049108 | MATHDB-338831

The composite-graph permitted motif conjecture

摘录 / EXCERPT: Composite-graph permitted motif conjecture. G is a permitted motif if and only if every component G_i is permitted.

conjecture | open | MathDB 页面 | 原文线索

V2-049109 | MATHDB-338836

Keevash-Mubayi simplex-cluster strengthening conjecture

摘录 / EXCERPT: Keevash-Mubayi strengthening conjecture. For $d<k\le\frac{d}{d+1}n$, if $\mathcal{F}$ contains no d-simplex-cluster, then

conjecture | open | MathDB 页面 | 原文线索

V2-049110 | MATHDB-338837

The orbifold quantum-dimension conjecture for the Heisenberg vertex algebra

摘录 / EXCERPT: Orbifold quantum-dimension conjecture. Every irreducible ordinary $\mathcal{H}(3)^{S_3}$-module M appears in the decomposition of a g-twisted $\mathcal{H}(3)$-module for some...

conjecture | open | MathDB 页面 | 原文线索

V2-049111 | MATHDB-338848

Proper-source death conjecture for network attractors

摘录 / EXCERPT: Proper-source death conjecture. Proper sources always die and are never active in an attractor of a network.

conjecture | open | MathDB 页面 | 原文线索

V2-049112 | MATHDB-338854

Adjacency distribution conjecture for random sorting networks

摘录 / EXCERPT: Adjacency distribution conjecture. The random sorting-network statistic satisfies $Y_n(\alpha)/\mathbb{E}[Y_n(1)]\to c\alpha$ in probability for some constant c, and the random...

conjecture | open | MathDB 页面 | 原文线索

V2-049113 | MATHDB-338877

Kozma-Puder conjecture on spectral gaps of normal elements

摘录 / EXCERPT: For each n, let S_n be the symmetric group and let $\Sigma\in\mathbb{R}[S_n]$ be a symmetric non-negative element. Let $\rho^{(1)},\ldots,\rho^{(m)}$ be finitely many families...

conjecture | open | MathDB 页面 | 原文线索

V2-049114 | MATHDB-338887

Conjecture on recurrent component cardinalities in sinkless rotor networks

摘录 / EXCERPT: Recurrent-component cardinality conjecture. For $n\geq 3$, $m\geq 1$, and $i,j\in\mathbb{Z}_n$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-049115 | MATHDB-338910

The self-exponential formulation of Generalized Moonshine

摘录 / EXCERPT: Let $V^\natural$ be the Moonshine vertex operator algebra, let $\mathbb{M}$ be the Monster group, and let $\alpha^\natural\in H^3(\mathbb{M},U(1))$ be the Moonshine anomaly. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049116 | MATHDB-338911

Borcherds's unified Moonshine conjecture

摘录 / EXCERPT: Let $\mathbb{M}$ be the Monster group. For each $g\in\mathbb{M}$, seek a $\frac{1}{|g|}\mathbb{Z}$-graded super-module ${}^g\hat V$ over $\mathbb{Z}[e^{2\pi i/|g|}]$, equipped with...

conjecture | open | MathDB 页面 | 原文线索

V2-049117 | MATHDB-338912

The non-Fricke vertex superalgebra conjecture

摘录 / EXCERPT: Let $\mathbb{M}$ be the Monster group, let $g\in\mathbb{M}$ be non-Fricke, and let ${}^g\hat V$ be the module assigned to g in the proposed unified Moonshine construction. Non-Fr...

conjecture | open | MathDB 页面 | 原文线索

V2-049118 | MATHDB-338932

Existence and polynomial dependence of ramified irregular vertex operators

摘录 / EXCERPT: Let r be a positive integer and let $M^{[r]}_{\Lambda}$ be an irregular Verma module with $\Lambda_{2r-1}\neq 0$ and $\Lambda_{2r}=0$. A ramified irregular vertex operator has an...

conjecture | open | MathDB 页面 | 原文线索

V2-049119 | MATHDB-338933

Singular ramified irregular vertex operator conjecture

摘录 / EXCERPT: Let p,q be positive integers, let ... and let $\chi_{p,q}$ be the singular vector of level pq in $M_{\Delta_{p,q}}$. A ramified irregular vertex operator is called singular when...

conjecture | open | MathDB 页面 | 原文线索

V2-049120 | MATHDB-338934

Unique construction of general ramified irregular vertex operators

摘录 / EXCERPT: Let $\Phi^{\Delta}_{\Lambda,\Lambda}(\alpha,\beta,c_{\emptyset};z)$ be a ramified irregular vertex operator with expansion coefficients $c_{\lambda}^{(m)}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049121 | MATHDB-338947

The Leader-Spink maze solvability conjecture with one-column vertical gaps

摘录 / EXCERPT: Let HNEs and VNEs denote horizontal and vertical edges removed from the square lattice maze, respectively. Consider mazes with arbitrarily many HNEs and arbitrarily many VNEs in on...

conjecture | open | MathDB 页面 | 原文线索

V2-049122 | MATHDB-338948

Solvability conjecture for mazes whose origin component is a path

摘录 / EXCERPT: Let $\mathcal{N}$ be the subset of square-lattice mazes in which the connected component of the origin is a simple, possibly infinite, path. Path-component maze solvability conject...

conjecture | open | MathDB 页面 | 原文线索

V2-049123 | MATHDB-338951

Nonsolvability conjecture for the three-column restricted maze class

摘录 / EXCERPT: Let $\mathcal{M}$ be the class of all mazes, and let $\mathcal{N}_3\subset\mathcal{M}$ be the set of mazes for which the only horizontal edges removed between the pairs of columns...

conjecture | open | MathDB 页面 | 原文线索

V2-049124 | MATHDB-338958

Topological association of Powell classes along the disk-pair complex

摘录 / EXCERPT: Disk-pair association conjecture. There is a topological method associating to every such vertex (a,b) a Powell equivalence class of homeomorphisms...

conjecture | open | MathDB 页面 | 原文线索

V2-049125 | MATHDB-338978

The detachable-pair classification conjecture for simple 3-connected graphs

摘录 / EXCERPT: Detachable-pair classification conjecture. Either G has a pair of edges $\{x,y\}$ such that either $G\backslash x\backslash y$ or $G/x/y$ is a simple 3-connected graph; G is...

conjecture | open | MathDB 页面 | 原文线索

V2-049126 | MATHDB-338998

Cohomological characterization of complete reducibility for meromorphic open-string vertex algebras

摘录 / EXCERPT: Cohomological complete-reducibility conjecture. If every left V-module of finite length is completely reducible, then

conjecture | open | MathDB 页面 | 原文线索

V2-049127 | MATHDB-339009

Bounded-degree hypertree universality conjecture for Steiner triple systems

摘录 / EXCERPT: A Steiner triple system on n vertices is a 3-uniform hypergraph in which every pair of vertices is contained in exactly one edge. A hypertree is a connected, simple 3-uniform hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-049128 | MATHDB-339048

AHRV frozen-region limit-shape conjecture for random sorting networks

摘录 / EXCERPT: AHRV frozen-region conjecture. As $n\to\infty$, the boundary of the limit shape of F_t is

conjecture | open | MathDB 页面 | 原文线索

V2-049129 | MATHDB-339058

Fuller-Gould conjecture on dense K_4^--saturated graphs

摘录 / EXCERPT: Fuller-Gould conjecture. The K_4^--saturated graphs with sizes in the interval

conjecture | open | MathDB 页面 | 原文线索

V2-049130 | MATHDB-339072

Yadin's SIT-EIT equivalence conjecture for Cayley graphs

摘录 / EXCERPT: Let G be a Cayley graph of a finitely generated group. A probability measure on the set of infinite self-avoiding paths from a fixed vertex is SIT if its two independent samples...

conjecture | open | MathDB 页面 | 原文线索

V2-049131 | MATHDB-339074

Strong bunk bed connectivity criterion

摘录 / EXCERPT: Let G be a graph, let X be a specified collection of vertices with outside edges, and let s and f be the vertices considered in the bunk bed conjecture. Let $G-X$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-049132 | MATHDB-339135

The singularity conjecture for biased random walks

摘录 / EXCERPT: Let G be an infinite graph with critical parameter $\lambda_c(G)$ and growth rate $\operatorname{gr}(G)$ satisfying ... For a starting vertex o and parameter λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049133 | MATHDB-339150

Falgas-Ravry and Vaughan's extremal hypergraph conjecture for K_4^- and C_5

摘录 / EXCERPT: Let K_4^- denote the 3-uniform hypergraph obtained from the complete 3-uniform hypergraph on four vertices by deleting one edge, and let C_5 denote the 3-uniform 5-cycle. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-049134 | MATHDB-339153

Mohar's conjecture on edge-critical graphs and strong chromatic-choosability

摘录 / EXCERPT: Mohar's conjecture. If G is a $(\Delta(G)+1)$-edge-critical graph, then $L(G)$ is strong $(\Delta(G)+1)$-chromatic-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-049135 | MATHDB-339154

Asymptotic upper bound for the strong chromatic index of planar multigraphs

摘录 / EXCERPT: Let G be a planar multigraph of maximum degree d, and let $\chi'_2(G)$ denote its strong chromatic index, the minimum number of colours in a distance-2 edge-colouring of G....

conjecture | open | MathDB 页面 | 原文线索

V2-049136 | MATHDB-339173

Bang-Jensen-Bessy-Thomassé conjecture on cycles and girth

摘录 / EXCERPT: Let k and g be positive integers, and let $f(k,g)$ be the minimum integer such that every finite simple digraph of girth g and minimum outdegree at least $f(k,g)$ contains...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-049137 | MATHDB-339174

Thomassé's path-length conjecture for oriented graphs

摘录 / EXCERPT: An oriented graph is a digraph without 2-cycles. Let h and g be positive integers, and let $l(h,g)$ be the minimum integer such that every oriented graph of girth g and minim...

conjecture | open | MathDB 页面 | 原文线索

V2-049138 | MATHDB-339208

Galvin-Tetali conjecture on colorings of regular graphs

摘录 / EXCERPT: Let $c_q(G)$ denote the number of proper q-colorings of a graph G. Let G be an n-vertex d-regular graph, with $q\ge 3$, and let $K_{d,d}$ be the complete bipartite graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049139 | MATHDB-339209

The path-graph invariant-subspace completeness conjecture

摘录 / EXCERPT: Let $G=P_n$ be the path graph on n cells, and let $\mathcal D_{G,\ell}$ denote the collection of transformations whose invariant subspaces are being classified by the algorithm i...

conjecture | open | MathDB 页面 | 原文线索

V2-049140 | MATHDB-339211

The conjectured validity of the main Berge-cycle theorem for $k=r+2$

摘录 / EXCERPT: Extension of the main Berge-cycle theorem. If $\mathcal H$ has no Berge cycle of length k or longer, then

conjecture | open | MathDB 页面 | 原文线索

V2-049141 | MATHDB-339220

The logarithmic-distance conjecture for localising a mouse on graphs

摘录 / EXCERPT: Let G be a connected graph of order n. In the Cat and Mouse localisation game, the cat receives only the information specified by the game rules and seeks to determine a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-049142 | MATHDB-339245

Kamat's cross-intersection conjecture for independent sets of graphs

摘录 / EXCERPT: Kamat's conjecture. If $\mu(\mathcal{I}_G)\geq 2r$ and $\mathcal{A},\mathcal{B}\subseteq\mathcal{I}_G^{(r)}$ are cross-intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-049143 | MATHDB-339249

Conjectures on complete hyperenergeticity of random graphs

摘录 / EXCERPT: Let a graph be completely hyperenergetic when every vertex has energy exceeding the corresponding hyperenergetic threshold, and let a uniform random graph on n points and a rando...

conjecture | open | MathDB 页面 | 原文线索

V2-049144 | MATHDB-339252

Lee, Loh, and Sudakov's conjecture on judicious bipartitions of digraphs

摘录 / EXCERPT: Let D be a digraph with m arcs and minimum outdegree at least an integer $d\geq 2$. For a bipartition $V(D)=V_1\cup V_2$, write $e(V_i,V_j)$ for the number of arcs directed fro...

conjecture | open | MathDB 页面 | 原文线索

V2-049145 | MATHDB-339255

Mycroft's constant-error conjecture for perfect tilings of partite hypergraphs

摘录 / EXCERPT: Mycroft's constant-error conjecture. There exists a constant $C=C(F)$ such that the error terms in $$ can be replaced by C.

conjecture | open | MathDB 页面 | 原文线索

V2-049146 | MATHDB-339256

Theorem on Laplacian H-eigenvalues of a k-uniform loose path of length three

摘录 / EXCERPT: Let k be a positive integer, and let the k-uniform loose path of length 3 be the hypergraph considered in Theorem. The conjecture. Theorem is true for the k-uniform loose p...

conjecture | open | MathDB 页面 | 原文线索

V2-049147 | MATHDB-339257

Nikiforov-Rojo conjectures on the A_{b1}-spectral radius of graphs with pendent paths

摘录 / EXCERPT: Let G be a connected graph and let $u,v\tnobreak\text{ be vertices with }d(u),d(v)geq 2$. Suppose that u and v are joined by a path ... where $d(w_i)=2$ for $1\leq i\leq s-1$....

conjecture | open | MathDB 页面 | 原文线索

V2-049148 | MATHDB-339259

Asymptotic independence-number conjecture for edge-chromatic critical graphs

摘录 / EXCERPT: Let G be a simple finite graph with maximum degree $\Delta(G)$ and minimum degree $\delta(G)$. The graph G is Δ-critical when it is connected, $\chi'(G)=\Delta(G)+1$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-049149 | MATHDB-339261

Bollobás-Eldridge small-size graph packing conjecture

摘录 / EXCERPT: Let $H_1,H_2,\ldots,H_k$ be graphs and let K_n be the complete graph on n vertices. A packing of $(H_1,H_2,\ldots,H_k)$ in K_n is a collection of injections of the graphs int...

conjecture | open | MathDB 页面 | 原文线索

V2-049150 | MATHDB-339262

Bollobás-Eldridge degree-product packing conjecture

摘录 / EXCERPT: Let H_1 and H_2 be graphs, let K_n be the complete graph on n vertices, and write $\Delta(H_i)$ for the maximum degree of H_i. A packing of (H_1,H_2) in K_n is a pair...

conjecture | open | MathDB 页面 | 原文线索

V2-049151 | MATHDB-339265

The universal deviation inequality for good drawings of complete graphs

摘录 / EXCERPT: Universal deviation inequality. For every face $F\in\mathcal{F}(D)$,

conjecture | open | MathDB 页面 | 原文线索

V2-049152 | MATHDB-339268

Large Inducibility Conjecture for nontrivial graphs

摘录 / EXCERPT: For a finite graph H, let $\operatorname{ind}(H)$ denote its inducibility, and let $K_{|H|}$ and $\overline{K}_{|H|}$ be respectively the complete and edgeless graphs on $|H|$ ve...

conjecture | open | MathDB 页面 | 原文线索

V2-049153 | MATHDB-339269

Super-linear edge-statistics conjecture

摘录 / EXCERPT: Let $\operatorname{ind}(k,\ell)$ denote the limiting maximum proportion of k-vertex subsets inducing exactly ℓ edges. Super-linear edge-statistics conjecture. For all pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-049154 | MATHDB-339270

Quadratic edge-statistics conjecture

摘录 / EXCERPT: Let $\operatorname{ind}(k,\ell)$ denote the limiting maximum proportion of k-vertex subsets inducing exactly ℓ edges. Quadratic edge-statistics conjecture. For all pairs...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-049155 | MATHDB-339281

Gu et al.'s vertex-transitivity conjecture for Dcell networks

摘录 / EXCERPT: Let $D_{k,n}$ be the data center network Dcell, with $k\geq0$ and $n\geq2$. Gu et al.'s vertex-transitivity conjecture. $D_{k,n}$ is vertex-transitive for all $k\geq0$ and $n\geq2$...

conjecture | open | MathDB 页面 | 原文线索

V2-049156 | MATHDB-339283

The constant-degree-tree version of the $2k$-$\frac{k}{2}$ conjecture

摘录 / EXCERPT: Let $k,\Delta\in\mathbb{N}$, and let G be a graph. Write $\delta(G)$ and $\Delta(G)$ for the minimum and maximum degrees of G, respectively. Constant-degree-tree conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-049157 | MATHDB-339284

The interpolating degree-condition conjecture for tree embeddings

摘录 / EXCERPT: Let $k\in\mathbb{N}$, let $0\leq\alpha\leq \frac{1}{2}$, and let G be a graph. Write $\delta(G)$ and $\Delta(G)$ for the minimum and maximum degrees of G, respectively. Interpo...

conjecture | open | MathDB 页面 | 原文线索

V2-049158 | MATHDB-339285

The large first- and second-neighbourhood conjecture for tree embeddings

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let G be a graph. For a vertex $x\in V(G)$, let $N(x)$ denote its neighbourhood and let $N_2(x)$ denote its second neighbourhood. Large first- and secon...

conjecture | open | MathDB 页面 | 原文线索

V2-049159 | MATHDB-339296

The sharper bounds conjecture for the Neumann surplus on quantum graphs

摘录 / EXCERPT: Let Γ be a compact metric graph, let $\partial\Gamma$ denote its boundary, let β be its first Betti number, and let ω_n denote the Neumann surplus associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-049160 | MATHDB-339302

AMKL conjecture for feature-code performance

摘录 / EXCERPT: AMKL conjecture. A feature code with bigger AMKL has better performance.

conjecture | open | MathDB 页面 | 原文线索

V2-049161 | MATHDB-339303

The large-convex-combination conjecture for non-vertex integer partitions

摘录 / EXCERPT: Let P_n be the convex hull of the integer partitions of n. For a non-vertex partition $x\in P(n)$, let $\xi(x)$ be the minimum number of partitions of n needed to express x...

conjecture | open | MathDB 页面 | 原文线索

V2-049162 | MATHDB-339339

Quadratic-size K_3-intersecting family conjecture

摘录 / EXCERPT: Let K_n be a complete geometric graph on n points in general position in the plane. A family of subgraphs is K_3-intersecting if every two members intersect in a triangle. Qu...

conjecture | open | MathDB 页面 | 原文线索

V2-049163 | MATHDB-339355

Existence of constant-curvature ball packings under bounded vertex-degree variation

摘录 / EXCERPT: Constant-curvature packing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049164 | MATHDB-339358

Höhn's coinvariant Leech lattice conjecture for semisimple cases

摘录 / EXCERPT: Let Λ be the Leech lattice, let $O(\Lambda)$ be its isometry group, and let Λ_g denote the sublattice orthogonal to the fixed-point sublattice of $g\in O(\Lambda)$....

conjecture | open | MathDB 页面 | 原文线索

V2-049165 | MATHDB-339360

ACKL's truncation conjecture for the symplectic minimal coset

摘录 / EXCERPT: ACKL's conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-049166 | MATHDB-339361

Completeness conjecture for orthogonal orbifold coincidences

摘录 / EXCERPT: For $n\geq2$ and $m\geq3$, let $\mathcal F_k(n)$ denote the simple quotient of the symplectic minimal coset family, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049167 | MATHDB-339402

The approximation conjecture for the strong geodetic number of complete bipartite graphs

摘录 / EXCERPT: Approximation conjecture. If $n \leq m \leq \binom{n}{2}$, then $e(m,n)<2$.

conjecture | open | MathDB 页面 | 原文线索

V2-049168 | MATHDB-339410

Farber's topological complexity conjecture for graph configuration spaces

摘录 / EXCERPT: Let G be a connected graph, and let $V_{\ge3}$ denote the set of vertices of G of degree at least 3. Assume that $|V_{\ge3}|\ge2$ and let $n\ge2|V_{\ge3}|$. Farber's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-049169 | MATHDB-339417

The odd-order obstruction conjecture for transitive n-partition orientations

摘录 / EXCERPT: Odd-order obstruction conjecture. No σ_n-partition with n odd has a transitive σ_n-orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-049170 | MATHDB-339418

The odd-order obstruction conjecture for multi-part transitive n-partition orientations

摘录 / EXCERPT: Odd-order obstruction conjecture. No σ_n-k-partition with $k>1$ and n odd has a transitive σ_n-orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-049171 | MATHDB-339419

Matsuda-Ozeki-Yamashita's branch-vertex conjecture for claw-free graphs

摘录 / EXCERPT: Let k be a non-negative integer and let G be a connected claw-free graph of order n. For an integer $r\geq 2$, let $\sigma_r(G)$ be the minimum degree sum of an independent s...

conjecture | open | MathDB 页面 | 原文线索

V2-049172 | MATHDB-339444

Unimodality conjecture for intervals in graph containment posets

摘录 / EXCERPT: Let $\mathcal{G}$ be the poset of graphs ordered by induced containment, and let $\mathcal{G}^c$ be its induced subposet of connected graphs. For an interval in either poset, its r...

conjecture | open | MathDB 页面 | 原文线索

V2-049173 | MATHDB-339455

Intermediate matching number conjecture for tree degree sequences

摘录 / EXCERPT: Intermediate matching number conjecture. For every positive integer k satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-049174 | MATHDB-339456

The rectilinear crossing-number conjecture for complete uniform hypergraphs

摘录 / EXCERPT: Let K_{2d}^d be the complete d-uniform hypergraph on $2d$ vertices, and let $\overline{cr}_d(K_{2d}^d)$ denote its rectilinear crossing number in d dimensions. Rectilinear c...

conjecture | open | MathDB 页面 | 原文线索

V2-049175 | MATHDB-339457

The signless Laplacian determination conjecture for multicone graphs

摘录 / EXCERPT: Let K_r be the complete graph on r vertices, let sK_t be the disjoint union of s copies of K_t, and let $K_r\triangledown sK_t$ denote their join. A graph is determined b...

conjecture | open | MathDB 页面 | 原文线索

V2-049176 | MATHDB-339478

The large bond-dimension conjecture for maximal flattening rank on 2 by N grids

摘录 / EXCERPT: Large bond-dimension conjecture. For $d=2$ and k sufficiently large, there exists a tensor $T \in \operatorname{TNS}(G_{2 \times N},d,k)$ such that all flattenings of T have ma...

conjecture | open | MathDB 页面 | 原文线索

V2-049177 | MATHDB-339479

Conjecture on large co-edge-regular graphs with four eigenvalues

摘录 / EXCERPT: Let Γ be a connected co-edge-regular graph with four distinct eigenvalues. Let $t\geq 2$ be an integer and write $|V(\Gamma)|=n(\Gamma)$. An s-clique extension is a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049178 | MATHDB-339492

The closedness and value conjecture for the degree-three density constant

摘录 / EXCERPT: Let d denote the degree of a Cayley digraph, and let $Delta_3$ and $Delta_3'$ be the associated density constants introduced in the paper. Degree-three density conjecture. The de...

conjecture | open | MathDB 页面 | 原文线索

V2-049179 | MATHDB-339503

Surface-inversion correspondence for spherical parking functions and birooted trees

摘录 / EXCERPT: Let n be a positive integer. A spherical parking function of length n has a degree, and a birooted tree on vertex set $[n]$ has a number of surface inversions, where a surface...

conjecture | open | MathDB 页面 | 原文线索

V2-049180 | MATHDB-339516

The minimum-leaf conjecture for 3-connected cubic graphs

摘录 / EXCERPT: Let G be a 3-connected cubic graph of order n, and let $\mathrm{ml}(G)$ denote its minimum leaf number. Minimum-leaf conjecture for 3-connected cubic graphs. ... The conjecture i...

conjecture | open | MathDB 页面 | 原文线索

V2-049181 | MATHDB-339523

Regular spanning 2-factor decomposition conjecture

摘录 / EXCERPT: Let G be an n-vertex r-regular graph with even $r\geq \frac{3}{4}n+o(n)$, and let F be a 2-regular graph on n vertices. Regular spanning 2-factor decomposition conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-049182 | MATHDB-339561

Laplacian spread conjecture for graphs

摘录 / EXCERPT: Let G be a simple graph of order $n\ge 2$, with Laplacian eigenvalues ... The Laplacian spread conjecture. The inequality ... holds, or equivalently ... Moreover, equality holds if and...

conjecture | open | MathDB 页面 | 原文线索

V2-049183 | MATHDB-339594

Ramsey numbers of uniform loose paths and cycles

摘录 / EXCERPT: Let $\mathcal{P}^k_n$ and $\mathcal{C}^k_n$ denote the k-uniform loose path and loose cycle with n edges, respectively, and let $R(H_1,H_2)$ be the two-colour Ramsey number for...

conjecture | open | MathDB 页面 | 原文线索

V2-049184 | MATHDB-339616

Asymptotic Erdős Matching Conjecture

摘录 / EXCERPT: Fix an integer $k\geq2$ and a positive real number $x\leq1/k$. Let $m(n,k,xn)$ denote the maximum size of a k-uniform family on $[n]$ whose matching number is less than $xn+1$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-049185 | MATHDB-339617

Diversity bound for intersecting-free hypergraph families

摘录 / EXCERPT: Let $\mathcal F\subset {[n]\choose k}$ satisfy $\nu(\mathcal F)\leq s$ and $n\geq k(s+1)$. Define the s-diversity by ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-049186 | MATHDB-339626

Minimum edge condition for proper connection number two

摘录 / EXCERPT: The minimum edge condition conjecture. If $\delta\geq 3$ and

conjecture | open | MathDB 页面 | 原文线索

V2-049187 | MATHDB-339631

The chromatic-number conjecture for extremal vertex Folkman graphs

摘录 / EXCERPT: Let $a_1,\ldots,a_s$ be positive integers, let m and p be defined by the paper's preceding notation, and let $\mathcal{H}_{\mathrm{extr}}(a_1,\ldots,a_s;m-1)$ denote the extrem...

conjecture | open | MathDB 页面 | 原文线索

V2-049188 | MATHDB-339632

The conjecture that the minimum is attained at two copies of 2

摘录 / EXCERPT: For a fixed integer $p\geq4$, let $F_v(a_1,\ldots,a_s;q)$ denote the vertex Folkman number, and let 2_r denote r occurrences of 2. The minimum-attainment conjecture. ... that i...

conjecture | open | MathDB 页面 | 原文线索

V2-049189 | MATHDB-339633

Faudree-Gould-Jacobson-Magnant cycle spectra conjecture

摘录 / EXCERPT: Faudree-Gould-Jacobson-Magnant conjecture. If G is a 2-connected graph on $n\geq 2d$ vertices with minimum degree $\delta(G)=d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049190 | MATHDB-339638

Induced-cycle conjecture for triangle-free graphs

摘录 / EXCERPT: A graph is triangle-free if it contains no cycle of length 3, and an induced cycle is a cycle whose vertices span no additional edges. For each integer $k\geq 5$, let the graph h...

conjecture | open | MathDB 页面 | 原文线索

V2-049191 | MATHDB-339639

Odd-cycle generalized Turán conjecture for forbidden smaller odd cycles

摘录 / EXCERPT: For integers k and ℓ, write $\operatorname{ex}(n,C_k,C_\ell)$ for the maximum number of copies of the cycle C_k in an n-vertex graph containing no copy of C_ℓ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049192 | MATHDB-339649

The vertex-distinguishing total coloring bound

摘录 / EXCERPT: Vertex-distinguishing total coloring conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049193 | MATHDB-339650

The linear bound for 1-strongly vertex-distinguishing total coloring

摘录 / EXCERPT: Linear vertex-distinguishing total coloring conjecture. For some positive constant c,

conjecture | open | MathDB 页面 | 原文线索

V2-049194 | MATHDB-339718

Frankl-Füredi conjecture on hypergraph Lagrangians

摘录 / EXCERPT: Frankl-Füredi conjecture. If G is an r-graph with m edges, then

conjecture | open | MathDB 页面 | 原文线索

V2-049195 | MATHDB-339731

The complete multipartite super-graceful classification conjecture

摘录 / EXCERPT: Let G be a complete t-partite graph, meaning that its vertex set is partitioned into t independent parts and every two vertices in different parts are adjacent. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049196 | MATHDB-339747

Uniqueness of the predominant orientation type for accordion graphs

摘录 / EXCERPT: Predominant orientation-type conjecture. The partition $\Lambda(n,v,d)$ is predominant among all non-vanishing orientation types of G.

conjecture | open | MathDB 页面 | 原文线索

V2-049197 | MATHDB-339754

The Bakry-Émery curvature Bonnet-Myers inequality conjecture for regular graphs

摘录 / EXCERPT: Bakry-Émery curvature Bonnet-Myers conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-049198 | MATHDB-339763

Common-neighborhood and common-antineighborhood conjecture for dense graphs

摘录 / EXCERPT: Let G be a graph on n vertices with edge density ε, meaning that G has $\varepsilon\binom{n}{2}$ edges, up to the interpretation of the source's phrase “...

conjecture | open | MathDB 页面 | 原文线索

V2-049199 | MATHDB-339798

The stronger Ban-Linial conjecture for orientable bisections

摘录 / EXCERPT: Let G be a cubic graph that admits a perfect matching. An orientable 4-weak bisection is a 4-weak bisection with the orientability property used in the source; a...

conjecture | open | MathDB 页面 | 原文线索

V2-049200 | MATHDB-339799

The $(4\frac14,\frac14)$-flow conjecture for simple cubic graphs

摘录 / EXCERPT: Let G be a simple cubic graph, and let $bed(G)$ denote the set of flow parameters admitted by G in the source. $(4\frac14,\frac14)$-flow conjecture. For every simple cubic grap...

conjecture | open | MathDB 页面 | 原文线索

V2-049201 | MATHDB-339808

The extremal bound for hypergraphs with circumference less than the uniformity

摘录 / EXCERPT: Let r be a positive integer and let $\mathcal H$ be an r-uniform hypergraph on n vertices with no cycle of length r or longer. The extremal bound conjecture. Then ... This co...

conjecture | open | MathDB 页面 | 原文线索

V2-049202 | MATHDB-339810

Golden-parallelogram maximal-denominator conjecture for three-move pieces

摘录 / EXCERPT: Let a piece have exactly three moves, and consider the vertex configurations arising from its golden parallelogram constructions. Three-move golden-parallelogram conjecture. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-049203 | MATHDB-339812

Largest-denominator conjecture for queens

摘录 / EXCERPT: Let F_q be the qth Fibonacci number, and consider vertex configurations of q queens in the square-board inside-out polytope. Queen maximal-denominator conjecture. The largest...

conjecture | open | MathDB 页面 | 原文线索

V2-049204 | MATHDB-339815

Twisted-Fibonacci-spiral maximality conjecture for rider pieces

摘录 / EXCERPT: Let $\mathbb P$ be a piece, and let a twisted Fibonacci spiral be the vertex configuration defined by four selected moves of $\mathbb P$ in the Fibonacci-spiral pattern, together w...

conjecture | open | MathDB 页面 | 原文线索

V2-049205 | MATHDB-339817

Refined covering-number conjecture for intersecting segment hypergraphs

摘录 / EXCERPT: Refined intersecting-segment conjecture. If H is an intersecting r-segment hypergraph and $r\geq 5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049206 | MATHDB-339827

Alon-Hefetz-Krivelevich-Tyomkyn 1/e inducibility conjecture for graphs

摘录 / EXCERPT: Alon-Hefetz-Krivelevich-Tyomkyn's 1/e conjecture. For all $0<\ell<\binom{k}{2}$ we have

conjecture | open | MathDB 页面 | 原文线索

V2-049207 | MATHDB-339828

Alon-Hefetz-Krivelevich-Tyomkyn superlinear sparsity conjecture

摘录 / EXCERPT: Alon-Hefetz-Krivelevich-Tyomkyn's superlinear sparsity conjecture. For all k,ℓ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-049208 | MATHDB-339829

Logarithm-free inducibility bound conjecture

摘录 / EXCERPT: Logarithm-free inducibility bound conjecture. For all k and all $0\le\ell\le\binom{k}{2}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049209 | MATHDB-339830

Hypergraph logarithm-free inducibility bound conjecture

摘录 / EXCERPT: Hypergraph logarithm-free inducibility bound conjecture. For any r,k and any $0\le\ell\le\binom{k}{r}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049210 | MATHDB-339848

Second-largest signless Laplacian spectral radius conjecture for uniform supertrees

摘录 / EXCERPT: Let $S_{2}(m,d,k)$ be the k-uniform supertree obtained from a loose path of diameter d by attaching $m-d$ edges at the vertex $v_{\lfloor d/2\rfloor}$. Let $\mathbb{S}(m,d,k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-049211 | MATHDB-339854

Finite-family conjecture for $(4,2)$ pillars

摘录 / EXCERPT: Let an equiangular set have angle $\alpha=\frac{1}{5}$ and base size $K=4$, and consider a $(4,2)$ pillar and its Seidel graph. Finite-family conjecture for $(4,2)$ pillars. In the...

conjecture | open | MathDB 页面 | 原文线索

V2-049212 | MATHDB-339860

Lianna's nearly bipartite graph convexity conjecture

摘录 / EXCERPT: Lianna's conjecture. For each $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-049213 | MATHDB-339861

Nearly bipartite defect convexity conjecture for maximum edge-colorable subgraphs

摘录 / EXCERPT: Defect convexity conjecture. For each $k\geq 0$ and each $i\in\{0,1,\ldots,k\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049214 | MATHDB-339869

The edge b1-critical graph conjecture on disjoint maximum independent sets

摘录 / EXCERPT: Let G be a graph. It is edge α-critical if $\alpha(G-e)>\alpha(G)$ for every $e\in E(G)$; an independent set is maximum if it has cardinality $\alpha(G)$, and G has two...

conjecture | open | MathDB 页面 | 原文线索

V2-049215 | MATHDB-339874

Higher-uniformity Berge saturation conjecture

摘录 / EXCERPT: Higher-uniformity Berge saturation conjecture. For every $3\leq r<k$ and every r-uniform hypergraph $F^{(r)}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049216 | MATHDB-339891

The (5,2)-even-subgraph-cover conjecture

摘录 / EXCERPT: The $(5,2)$-even-subgraph-cover conjecture. There exist five even subgraphs of G such that every edge of G belongs to exactly two of them.

conjecture | open | MathDB 页面 | 原文线索

V2-049217 | MATHDB-339901

Multipartite form of the generalized Ohba conjecture

摘录 / EXCERPT: Let $r\ge 2$, and let $p_1,p_2,\ldots,p_k$ be positive integers. For each i, let V_i be a part of size p_i, and let $K_{p_1,p_2,\ldots,p_k}^r$ be the r-complete k-partite...

conjecture | open | MathDB 页面 | 原文线索

V2-049218 | MATHDB-339908

The square-root logarithmic fluctuation conjecture for IDLA on the Sierpinski gasket

摘录 / EXCERPT: Let SG be the graphical Sierpinski gasket, let o be its source vertex, and let $B_o(n)$ denote the graph-metric ball of radius n centered at o. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049219 | MATHDB-339909

Limit shape universality for nested fractal graphs

摘录 / EXCERPT: Let G be a nested fractal graph, and let o be a starting vertex such that every graph-metric ball $B_o(n)$ has spatial symmetry. Consider the four single-source Laplacian growt...

conjecture | open | MathDB 页面 | 原文线索

V2-049220 | MATHDB-339911

Levcovitz's equivalence conjecture for hypergraph index and divergence

摘录 / EXCERPT: Let W_Γ be a right-angled Coxeter group, and let n denote its hypergraph index. Let thickness of order n and divergence x^{n+1} have their standard meanings for the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-049221 | MATHDB-339933

The graphsums generation conjecture for invariants of sums of fundamental representations

摘录 / EXCERPT: Graphsums generation conjecture. The graphsums from Theorem generate the ideal of relations of $\mathbb{C}[W]^{{\rm SL}_4}$.

conjecture | open | MathDB 页面 | 原文线索

V2-049222 | MATHDB-339934

Square Hamilton cycle conjecture for graphs with sublinear independence number

摘录 / EXCERPT: Let $\varepsilon>0$. A graph G has independence number $\alpha(G)$ equal to the maximum size of a set of pairwise nonadjacent vertices, and the square of a Hamilton cycle is the...

conjecture | open | MathDB 页面 | 原文线索

V2-049223 | MATHDB-339945

Reed's path partition conjecture for 2-connected cubic graphs

摘录 / EXCERPT: Reed's conjecture. The path partition number satisfies

conjecture | solved | MathDB 页面 | 原文线索

V2-049224 | MATHDB-339953

Conjecture on the tree with largest imaginary-part Wiener root

摘录 / EXCERPT: Let T be a tree of order n, and let a Wiener root mean a zero of its Wiener polynomial. For each $n\geq 9$, let $\widetilde T_n$ be the tree shown in the source figure: a path...

conjecture | open | MathDB 页面 | 原文线索

V2-049225 | MATHDB-339962

The completely reversible edge-coloring form of the linear arboricity conjecture

摘录 / EXCERPT: Let Σ be a simple signed graph, let $\Delta(\Sigma)$ be its maximum degree, and let $\chi'_R(\Sigma)$ be the minimum number of colors in a completely reversible zero-free pr...

conjecture | open | MathDB 页面 | 原文线索

V2-049226 | MATHDB-339971

Reconstructibility of degree sequences from incomplete decks

摘录 / EXCERPT: Degree-sequence reconstructibility conjecture. For any graph G on n vertices, the degree sequence of G is reconstructible from any $n-k$ cards.

conjecture | open | MathDB 页面 | 原文线索

V2-049227 | MATHDB-339975

The spider lower-bound equality conjecture for maximum rectilinear crossing numbers

摘录 / EXCERPT: Spider lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049228 | MATHDB-339989

Bollobás-Erdős-type conjecture for the high-girth triple-process

摘录 / EXCERPT: Let $\ell \ge 4$. Let m be the total number of steps in the high-girth triple-process that produces a partial Steiner system with girth greater than ℓ. High-girth process co...

conjecture | open | MathDB 页面 | 原文线索

V2-049229 | MATHDB-339994

Graph-realizable consumption-order conjecture

摘录 / EXCERPT: Let λ and μ be partitions of n, and restrict the consumption ordering to subgroups of S_n that are realizable as automorphism groups of graphs on n vertices. The...

conjecture | open | MathDB 页面 | 原文线索

V2-049230 | MATHDB-340001

Second-eigenvalue conjecture for the auxiliary graph $FJ_1(n,2)$

摘录 / EXCERPT: For $n\geq 4$, let $FJ_1(n,2)=\mathrm{Cay}(S_n,RP_1^{(2)})$ be the connected auxiliary Cayley graph defined in the source, and let $B_n^{(1)}$ be its displayed quotient matrix. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-049231 | MATHDB-340004

The apex crown graph extremal representation-number conjecture

摘录 / EXCERPT: Let G_n be the graph obtained from the crown graph $H_{n,n}$ by adding an apex, that is, a vertex adjacent to every vertex of $H_{n,n}$. The apex crown graph extremal conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-049232 | MATHDB-340024

Erdős-Hajnal-Thomassen conjecture on high-chromatic subgraphs of prescribed girth and average degree

摘录 / EXCERPT: Let k be a positive integer and let $g\ge3$ be an integer. For a graph, its chromatic number is denoted by $\chi(G)$, its average degree is the average of its vertex degrees, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049233 | MATHDB-340030

The Δ-Equitable Coloring Conjecture for connected graphs

摘录 / EXCERPT: Let G) be a connected graph, and let $\Delta(G)$ denote its maximum vertex degree. An equitable k-coloring is a proper coloring whose color classes differ in size by at most...

conjecture | open | MathDB 页面 | 原文线索

V2-049234 | MATHDB-340031

Nešetřil-Šámal conjecture on cores of cubelike graphs

摘录 / EXCERPT: Let a cubelike graph be a Cayley graph for an elementary abelian group $\mathbb{Z}_2^n$, and let the core of a graph be its minimal homomorphically equivalent graph. Nešetřil-Šámal...

conjecture | open | MathDB 页面 | 原文线索

V2-049235 | MATHDB-340037

Two-vertex second neighborhood conjecture for oriented graphs without sinks

摘录 / EXCERPT: Let G be an oriented graph, and call a vertex a sink if it has out-degree zero. A vertex v has a large second neighborhood when its second out-neighborhood $N^{++}(v)$ satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-049236 | MATHDB-340038

Unique large second neighborhood implies a sink

摘录 / EXCERPT: Let G be an oriented graph. A vertex v has a large second neighborhood when $|N^{++}(v)|\geq |N^+(v)|$, and a sink is a vertex with no out-neighbors. Unique-large-second-neighb...

conjecture | open | MathDB 页面 | 原文线索

V2-049237 | MATHDB-340053

Flandrin-Fouquet-Li conjecture on Hamilton cycles in $S_{3,3}$-free bipartite graphs

摘录 / EXCERPT: Let G be a connected balanced $S_{3,3}$-free bipartite graph, where $S_{3,3}$ denotes the corresponding forbidden subgraph. Flandrin-Fouquet-Li conjecture. There exists a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-049238 | MATHDB-340057

Sun-Gutin-Yeo-Zhang's computational conjecture for strong subgraph k-connectivity

摘录 / EXCERPT: Let D be a digraph, and let $k\geq 2$ and $\ell\geq 2$ be fixed integers. The parameter $\kappa_k(D)$ is the maximum, over all k-vertex subsets of D, of the number of interna...

conjecture | open | MathDB 页面 | 原文线索

V2-049239 | MATHDB-340073

The strongly e-positive graph characterization conjecture

摘录 / EXCERPT: Strongly e-positive graph characterization conjecture. A graph G is strongly e-positive if and only if it is (claw,net)-free.

conjecture | open | MathDB 页面 | 原文线索

V2-049240 | MATHDB-340074

The induced claw-or-net conjecture for non-e-positive graphs

摘录 / EXCERPT: Induced claw-or-net conjecture. If G is not e-positive, then G contains an induced claw or an induced net.

conjecture | open | MathDB 页面 | 原文线索

V2-049241 | MATHDB-340080

Odd-cycle saturation conjecture

摘录 / EXCERPT: Let $\mathcal{C}_{2k+1}$ denote the family consisting of the cycle C_{2k+1}, and let $\mathrm{sat}(\mathcal{C}_{2k+1};n)$ be its saturation number on n vertices. Odd-cycle satu...

conjecture | open | MathDB 页面 | 原文线索

V2-049242 | MATHDB-340081

Consecutive odd-cycle saturation-game monotonicity conjecture

摘录 / EXCERPT: Let $\mathcal{C}_{2k+1}$ denote the family consisting of the cycle C_{2k+1}, and let $\mathrm{sat}_g(\mathcal{C}_{2k+1};n)$ be the corresponding saturation-game value on n vert...

conjecture | open | MathDB 页面 | 原文线索

V2-049243 | MATHDB-340089

Li's Hamilton cycle or long path conjecture for colour degree

摘录 / EXCERPT: Let G be an edge-coloured connected graph, and let $d^c_G(v)$ denote the number of distinct colours on edges incident to a vertex v. Write ... for the minimum colour degree. A su...

conjecture | open | MathDB 页面 | 原文线索

V2-049244 | MATHDB-340093

Entropy lower-bound conjecture for circular hypergraphs

摘录 / EXCERPT: Fix a positive integer $r\ge 3$. For each $n>r$, let $\mathcal{C}_{n,r}$ be the circular r-uniform hypergraph on n vertices whose edges are all paths of size r in a cycle, an...

conjecture | open | MathDB 页面 | 原文线索

V2-049245 | MATHDB-340103

The Seifert-cycle tree characterization of Gauss codes

摘录 / EXCERPT: Seifert-cycle tree conjecture. The graph $\overline{G}(\omega)$ is a tree if and only if ω is the Gauss code of a closed curve in the plane.

conjecture | open | MathDB 页面 | 原文线索

V2-049246 | MATHDB-340117

Four-partite network decomposition conjecture

摘录 / EXCERPT: Let Σ be a network with four boundary vertices $A_1,\ldots,A_4$, and let $I(A_i:A_j)$ denote the mutual information between A_i and A_j on Σ. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049247 | MATHDB-340118

The arbitrary-boundary network decomposition conjecture

摘录 / EXCERPT: Let Σ be a network with $n\geq2$ boundary vertices. An extremal-ray realization is a subnetwork whose entropy vector realizes an extremal ray of the n-boundary-region holo...

conjecture | open | MathDB 页面 | 原文线索

V2-049248 | MATHDB-340120

The limiting proportions of graph subclasses

摘录 / EXCERPT: Let $\mathcal{G}_n$ be the class of connected simple graphs of order n, and let $\mathcal{G}_n^1,\mathcal{G}_n^2,\mathcal{G}_n^3,$ and $\mathcal{G}_n^4$ be the four subclasses co...

conjecture | open | MathDB 页面 | 原文线索

V2-049249 | MATHDB-340131

Conjecture on the CSM classes of Dynkin quiver orbits

摘录 / EXCERPT: Let Q be a Dynkin quiver with vertex set Q_0, and let β be a positive root of Q. For each vertex $i\in Q_0$, write $\alpha_{i,1},\ldots,\alpha_{i,\beta(i)}$ for the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-049250 | MATHDB-340135

Ryser's symbolic-power containment conjecture

摘录 / EXCERPT: Ryser's containment conjecture. For all $k \geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-049251 | MATHDB-340136

Conforti-Cornuéjols integer round-down conjecture

摘录 / EXCERPT: Conforti-Cornuéjols integer round-down conjecture. If ${\mathcal H}$ has the packing property, then it has the integer round-down property.

conjecture | open | MathDB 页面 | 原文线索

V2-049252 | MATHDB-340140

Cioabă-Wong spanning tree packing conjecture

摘录 / EXCERPT: Let G be a graph with minimum degree ... . For an integer $k\geq 2$, let ... be the maximum number of edge-disjoint spanning trees in G. Cioabă-Wong conjecture. If ... then ... This sp...

conjecture | open | MathDB 页面 | 原文线索

V2-049253 | MATHDB-340146

Subset version of Seymour's Second Neighborhood Conjecture

摘录 / EXCERPT: Let D be an oriented simple graph. For a nonempty proper subset $S\subset V(D)$, let $N_k^+(S)$ be the set of vertices u such that $\min_{s\in S}d(s,u)=k$, where $d(s,u)$ is di...

conjecture | open | MathDB 页面 | 原文线索

V2-049254 | MATHDB-340152

The minimum codegree conjecture for spanning spheres in 3-graphs

摘录 / EXCERPT: Spanning-sphere conjecture. Every such $\mathcal{H}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-049255 | MATHDB-340153

The codegree conjecture for spanning spheres in uniform hypergraphs

摘录 / EXCERPT: Higher-dimensional spanning-sphere conjecture. If $\mathcal{H}$ comprises a single tight component and every set Z of $r-1$ vertices contained in an edge of $\mathcal{H}$ is cont...

conjecture | open | MathDB 页面 | 原文线索

V2-049256 | MATHDB-340156

Regularity bound by the h-polynomial degree for binomial edge ideals

摘录 / EXCERPT: Let G be a finite simple graph on the vertex set $[n]=\{1,\ldots,n\}$, let $K[\boldsymbol{x},\boldsymbol{y}]=K[x_1,\ldots,x_n,y_1,\ldots,y_n]$ be the standard graded polynomial r...

conjecture | open | MathDB 页面 | 原文线索

V2-049257 | MATHDB-340172

The equivalence of $athfrak{I}$-contractibility and homotopical collapsibility

摘录 / EXCERPT: Let an $athfrak{I}$-contractible graph be a graph reducible to a single vertex by $athfrak{I}$-contractible transformations. An expansion is the inverse of a collapse, and a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049258 | MATHDB-340184

Chung-Diaconis-Graham conjecture on universal cycles

摘录 / EXCERPT: Chung-Diaconis-Graham conjecture. For every $k\in\mathbb{N}$, there exists $n_0\in\mathbb{N}$ such that for all $n\ge n_0$, there exists a universal cycle for $\binom{[n]}{k}$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-049259 | MATHDB-340212

Parallel dispersion time bound by sequential dispersion and cover times

摘录 / EXCERPT: Dispersion-time conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049260 | MATHDB-340235

Well-definedness of Kronheimer-Mrowka's foam evaluation algorithm

摘录 / EXCERPT: Kronheimer-Mrowka's well-definedness conjecture. The quantity $J^\flat(F)$ is well-defined: it does not depend on the choices made in the seam-vertex cancellation step.

conjecture | open | MathDB 页面 | 原文线索

V2-049261 | MATHDB-340262

Generalized 2-colour conjecture for planar digraphs of prescribed digirth

摘录 / EXCERPT: Let D be a planar digraph with digirth at least g, where $g\geq 3$, and let $\vec{\chi}^\ast(D)$ denote its star dichromatic number. Generalized 2-colour conjecture. ... Equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-049262 | MATHDB-340268

Perles' conjecture for subgraphs of simple polytopes

摘录 / EXCERPT: Let P be a simple d-polytope, and consider its vertex-edge graph. A subgraph is non-separating if its deletion does not separate the relevant graph, as in the source's formulat...

conjecture | open | MathDB 页面 | 原文线索

V2-049263 | MATHDB-340284

The Edge-statistics Conjecture for edge-inducibility

摘录 / EXCERPT: The Edge-statistics Conjecture. For all integers k and ℓ with $0<\ell<{k\choose 2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049264 | MATHDB-340294

Sufficiency of total-Hurwitzness for global exponential stability

摘录 / EXCERPT: Consider the network dynamics defined by equation (dyn-d), with weight matrix $\mathbf{W}$ satisfying Assumption 1. Let $\mathcal{H}$ denote the class of totally Hurwitz matrices,...

conjecture | open | MathDB 页面 | 原文线索

V2-049265 | MATHDB-340299

The asymptotic excess conjecture for regular graphs with fixed second largest eigenvalue

摘录 / EXCERPT: Conjecture on $T(\lambda)$. Let λ be a positive integer. Then

conjecture | open | MathDB 页面 | 原文线索

V2-049266 | MATHDB-340306

Braided equivalence conjecture for $\bar{u}_q(\mathfrak{g})$ and $\mathcal{W}(\mathfrak{g},p)$

摘录 / EXCERPT: Let $(\mathfrak{g},p)$ be as above, let $\bar{u}_q(\mathfrak{g})$ be the small quasi-quantum group in the source, and set ... Let $\mathcal{W}(\mathfrak{g},p)$ be the corresponding V...

conjecture | open | MathDB 页面 | 原文线索

V2-049267 | MATHDB-340307

Braided equivalence conjecture for $\bar{u}_q(\mathfrak{g})$ and $\mathcal{W}(\mathfrak{g},p)$ via the alternate quasi-quantum group

摘录 / EXCERPT: Let $(\mathfrak{g},p)$ be as above, let $\bar{u}_q(\mathfrak{g})$ denote the alternate small quasi-quantum group specified in the source, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049268 | MATHDB-340328

Alon-Hefetz-Krivelevich-Tyomkyn edge-statistics conjecture

摘录 / EXCERPT: Edge-statistics conjecture. For all $k,\ell\in\mathbb{N}$ with $0<\ell<\binom{k}{2}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049269 | MATHDB-340329

Alon-Hefetz-Krivelevich-Tyomkyn sparse-edge decay conjecture

摘录 / EXCERPT: Sparse-edge decay conjecture. For all $k,\ell\in\mathbb{N}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-049270 | MATHDB-340330

Alon-Hefetz-Krivelevich-Tyomkyn quadratic-range edge-statistics conjecture

摘录 / EXCERPT: Quadratic-range edge-statistics conjecture. For all $k,\ell\in\mathbb{N}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-049271 | MATHDB-340339

Conjecture that most rooted digraphs are not determined by their greedoid polynomials

摘录 / EXCERPT: Let a rooted digraph be determined by its greedoid polynomial if no other non-isomorphic rooted digraph has the same greedoid polynomial. Greedoid-polynomial non-determination conj...

conjecture | open | MathDB 页面 | 原文线索

V2-049272 | MATHDB-340346

Hansen's maximum eccentric connectivity index conjecture for graphs with fixed order and size

摘录 / EXCERPT: Hansen's conjecture. The eccentric connectivity index of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049273 | MATHDB-340348

Alishahi-Meunier conjecture on stable almost fair splittings of paths

摘录 / EXCERPT: Let q be a positive integer, and let P be a path whose vertex set is partitioned into m subsets $V_1,\dots,V_m$, each of size at least $q-1$. A set of vertices is q-stable...

conjecture | open | MathDB 页面 | 原文线索

V2-049274 | MATHDB-340364

Brualdi-Hoffman-type conjecture for sums of adjacency eigenvalues

摘录 / EXCERPT: Let G be a graph of order n with e edges, let $g(t,e')$ denote the graph formed from K_t by adding one vertex with e' incident edges and isolated vertices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049275 | MATHDB-340378

Asymptotic induced path factor conjecture for regular graphs

摘录 / EXCERPT: Asymptotic conjecture for c_k. We have

conjecture | open | MathDB 页面 | 原文线索

V2-049276 | MATHDB-340389

Strengthened Defant-Iyer conjecture for upper irredundance in multipartite graph products

摘录 / EXCERPT: Let $G=\prod_{i=1}^t K[a_i,b_i]$ be a direct product of balanced complete multipartite graphs, where $2\leq b_1\leq\cdots\leq b_t$ and $K[a_i,b_i]$ denotes the complete multipartit...

conjecture | open | MathDB 页面 | 原文线索

V2-049277 | MATHDB-340431

Wu-Gu-Sun conjecture on flat PL metrics of the hexagonal triangulation

摘录 / EXCERPT: Let l be a PL metric on $\mathbb{C}$ with standard hexagonal triangulation, PL conformal to $l_0\equiv 1$. Let V be the vertex set of the triangulation, identified with points...

conjecture | open | MathDB 页面 | 原文线索

V2-049278 | MATHDB-340436

Rolek-Song's coloring conjecture for graphs with no K_t minor

摘录 / EXCERPT: Let $t\geq 1$, and let G be a graph on n vertices. A graph is $(t-1)$-colorable if its vertices can be properly colored with at most $t-1$ colors. Rolek-Song's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-049279 | MATHDB-340449

Hahn's rainbow Hamilton path conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. An edge-colouring is k-bounded if no colour appears on more than k edges. A subgraph is rainbow if no two of its edges have the...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049280 | MATHDB-340468

Lovász's stronger Ryser conjecture

摘录 / EXCERPT: Lovász's stronger Ryser conjecture. There exists a set S of vertices with $|S|\leq r-1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-049281 | MATHDB-340489

The Case I achievable-rate conjecture for two-sender unicast index coding

摘录 / EXCERPT: Let $\mathcal{D}^k$, $k \in \{13,14,\cdots,25\}$, be a side-information digraph whose sub-digraphs $\mathcal{D}_{i}^{k,\mathcal{P}}$, $i \in \{1,2,3\}$, have any type of interactio...

conjecture | open | MathDB 页面 | 原文线索

V2-049282 | MATHDB-340501

Wang-Hou's conjecture on sums of signed-graph Laplacian eigenvalues

摘录 / EXCERPT: Let G_σ be a signed graph obtained from a simple graph G, and let $L_{G_\sigma}=D(G)-A(G_\sigma)$ be its signed-graph Laplacian matrix. For $k,n\in\mathbb{N}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-049283 | MATHDB-340514

Reed's domination conjecture for connected cubic graphs

摘录 / EXCERPT: Let G be a connected cubic graph, and let $V(G)$ denote its vertex set. A dominating set is a subset of vertices such that every vertex outside the subset has a neighbor in it. R...

conjecture | open | MathDB 页面 | 原文线索

V2-049284 | MATHDB-340527

Conjecture on decompositions of complete uniform hypergraphs into regular Hamiltonian cycles

摘录 / EXCERPT: Let K_n^h denote the complete h-uniform hypergraph on n vertices, and let $\mathbb{K}_n^h$ be the hypergraph with $\mathbb{$ copies of every edge. A regular Hamiltonian cycle is a Hamilt...

conjecture | open | MathDB 页面 | 原文线索

V2-049285 | MATHDB-340551

So's Cayley-DS conjecture for integral circulant graphs

摘录 / EXCERPT: Two graphs are cospectral when their adjacency matrices have the same eigenvalue multiset, and a circulant graph is Cayley-DS when every cospectral Cayley graph on the same group i...

conjecture | open | MathDB 页面 | 原文线索

V2-049286 | MATHDB-340558

The many-colour edge-density conjecture for non-monochromatic triangles

摘录 / EXCERPT: Many-colour edge-density conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049287 | MATHDB-340559

The many-colour minimum colour-class conjecture for non-monochromatic triangles

摘录 / EXCERPT: Many-colour minimum colour-class conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049288 | MATHDB-340560

Closed-form conjecture for the precoloring polynomial of the hexwheel network

摘录 / EXCERPT: Let N be the network from the hexwheel example, let $m\geq 3$, and write $\operatorname{cp}_m(\lambda)=\operatorname{rcp}(M(N))$. Let $\omega_\pm=\lambda-2\pm\sqrt{\lambda^2+4}$....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-049289 | MATHDB-340561

The non-annihilating graph conjecture for Monster-type axial algebras

摘录 / EXCERPT: Let A be an axial algebra of Monster-type with generating axes X, and let $\Delta(X)$ be the graph whose vertices are the axes in X, with distinct vertices a,b adjacent exa...

conjecture | open | MathDB 页面 | 原文线索

V2-049290 | MATHDB-340563

The invariant global sections conjecture for chiral de Rham complexes

摘录 / EXCERPT: Let V be the vertex algebra associated with the geometric data above, let ${\mathcal W}(V)$ denote the corresponding global-section algebra, let $\mathcal Vect^A(V)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-049291 | MATHDB-340565

Amitava's extremal spectral-radius conjecture for bipartite graphs

摘录 / EXCERPT: Let $2 \le p \le q$ and $1<e<pq$ be integers, and let ${\cal K}(p,q,e)$ be the family of noncomplete bipartite subgraphs of $K_{p,q}$ with e edges and no isolated vertices. The p...

conjecture | open | MathDB 页面 | 原文线索

V2-049292 | MATHDB-340566

Ehrenborg's Ferrers bound conjecture for bipartite graphs

摘录 / EXCERPT: Ferrers bound conjecture. Every such graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049293 | MATHDB-340567

Normalized-Laplacian form of the Ferrers bound conjecture

摘录 / EXCERPT: Let G be a bipartite graph on $n\geq 3$ vertices with parts X and Y, and let $\rho(G)=|E(G)|/(|X||Y|)$ be its bipartite density. If G is connected, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049294 | MATHDB-340586

Large planar graph expansions lack the edge-Erdős-Pósa property

摘录 / EXCERPT: Large-expansion conjecture. There is an integer c such that for every planar graph H of treewidth (or even pathwidth) at least c, the family of H-expansions does not have t...

conjecture | open | MathDB 页面 | 原文线索

V2-049295 | MATHDB-340587

Condensed-wall criterion for the edge-Erdős-Pósa property

摘录 / EXCERPT: Condensed-wall conjecture. If there is an integer r such that the condensed wall of size r contains an H-expansion, then the family of H-expansions has the edge-Erdős-Pósa...

conjecture | open | MathDB 页面 | 原文线索

V2-049296 | MATHDB-340606

Cut-at-vertex Bernardi conjecture for the exterior polynomial

摘录 / EXCERPT: Cut-at-vertex exterior-polynomial conjecture. For every such $\mathcal H$, $\tilde{X}_{\mathcal H}=X_{\mathcal H}$, $\tilde{X}_{\mathcal H}$ is independent of the ribbon structure,...

conjecture | open | MathDB 页面 | 原文线索

V2-049297 | MATHDB-340637

Bipartite structure conjecture for edge-optimal minimally k-edge-connected graphs

摘录 / EXCERPT: Let $k\geq 3$, and let G be an edge-optimal minimally k-edge-connected graph of order n, meaning a minimally k-edge-connected graph with largest average edge-connectivity a...

conjecture | open | MathDB 页面 | 原文线索

V2-049298 | MATHDB-340666

Saeedi and Kiani's regularity bound by maximal cliques

摘录 / EXCERPT: Let G be a graph, let $S={\mathbb K}[x_1,\ldots,x_n,y_1,\ldots,y_n]$, and let J_G be its binomial edge ideal. Denote by $c(G)$ the number of maximal cliques of G. Saeedi and...

conjecture | solved | MathDB 页面 | 原文线索

V2-049299 | MATHDB-340667

Seacrest-Seacrest conjecture that every bipartite graph belongs to $\mathcal{H}$

摘录 / EXCERPT: Seacrest-Seacrest's conjecture. Every bipartite graph belongs to $\mathcal{H}$.

conjecture | open | MathDB 页面 | 原文线索

V2-049300 | MATHDB-340699

Sharp logarithmic blowup-size conjecture for Erdős-Stone type results

摘录 / EXCERPT: Fix integers $r\geq s\geq 3$ and $\epsilon>0$. In the Erdős-Stone type result under discussion, m denotes the number of edges and t denotes the size of the blowup. Sharp logari...

conjecture | open | MathDB 页面 | 原文线索

V2-049301 | MATHDB-340700

Supersaturation conjecture for clique counts in K_{r+1}-free graphs

摘录 / EXCERPT: Let $r\geq s\geq 3$, let $\epsilon>0$, and let G be a graph with $m=e(G)$ edges. Write $\operatorname{k}_s(G)$ for the number of copies of K_s in G. Supersaturation conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-049302 | MATHDB-340704

Path energy monotonicity conjecture for unicyclic graphs

摘录 / EXCERPT: Let G be a unicyclic graph of order n whose cycle has size k. The path energy $PE(G)$ is defined for such graphs. Path energy conjecture for unicyclic graphs. $PE(G)$ depends...

conjecture | open | MathDB 页面 | 原文线索

V2-049303 | MATHDB-340718

The cycle independence-equivalence conjecture

摘录 / EXCERPT: For a positive integer n, let C_n be the cycle graph on n vertices, let D_n be the graph obtained from C_n by adding a loop structure as defined in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049304 | MATHDB-340719

The total color degree conjecture for rainbow matchings

摘录 / EXCERPT: Let G be an edge-colored graph on n vertices. The total color degree of G is ... where $\hat d(v)$ is the number of distinct colors on edges incident with v. A rainbow matchi...

conjecture | open | MathDB 页面 | 原文线索

V2-049305 | MATHDB-340728

Hall's matching-polynomial conjecture for cycle covers

摘录 / EXCERPT: Let C_n be the cycle graph on n vertices, let P_m be the path graph on m vertices, and let $d4d0_{G,d}(x)$ denote the d-matching polynomial of a graph G, defined as the...

conjecture | open | MathDB 页面 | 原文线索

V2-049306 | MATHDB-340730

The extremal characterization of complete bidirected graphs without rainbow triangles

摘录 / EXCERPT: Extremal characterization conjecture. If D contains no rainbow triangles and

conjecture | open | MathDB 页面 | 原文线索

V2-049307 | MATHDB-340747

Cioabă-Guo-Haemers conjecture on the chromatic index of connected strongly regular graphs

摘录 / EXCERPT: A strongly regular graph (SRG) G with parameters (n,k,λ,μ) is a k-regular graph on n vertices such that every adjacent pair has exactly λ common neighbors a...

conjecture | open | MathDB 页面 | 原文线索

V2-049308 | MATHDB-340761

Diestel's Hamilton-circle conjecture for locally finite graphs

摘录 / EXCERPT: Diestel's conjecture. The graph G has a Hamilton circle if

conjecture | open | MathDB 页面 | 原文线索

V2-049309 | MATHDB-340762

Hamilton-circle conjecture from 2-connected radius-two balls

摘录 / EXCERPT: Hamilton-circle conjecture. The graph G has a Hamilton circle if every ball of radius 2 in G is 2-connected and

conjecture | open | MathDB 页面 | 原文线索

V2-049310 | MATHDB-340768

The #P-hardness conjecture for counting pressing sequences

摘录 / EXCERPT: A graph G has a set of pressing sequences, namely vertex orderings satisfying the pressing condition defined in the paper. The P-hardness conjecture. Exactly counting the pressin...

conjecture | open | MathDB 页面 | 原文线索

V2-049311 | MATHDB-340769

The 2-factor polynomial counting conjecture

摘录 / EXCERPT: Let G be a connected planar trivalent graph with perfect matching M. Let $[G:M]_2$ denote the set of 2-factors of G that factor through M, and let $|G:M|_2$ be its cardin...

conjecture | solved | MathDB 页面 | 原文线索

V2-049312 | MATHDB-340770

The bridgeless planar trivalent graph Tait polynomial positivity conjecture

摘录 / EXCERPT: Let G be a connected planar trivalent graph, and let $T_G(z)$ denote its Tait polynomial. Tait polynomial positivity conjecture. If G is bridgeless, then G has a perfect matc...

conjecture | solved | MathDB 页面 | 原文线索

V2-049313 | MATHDB-340771

The spatial trivalent graph polynomial and cohomology invariance conjecture

摘录 / EXCERPT: Let G be a spatial trivalent graph with a perfect matching M. Let $\langle G:M\rangle_2$ be its 2-factor polynomial and let the cohomology theory constructed in the paper be...

conjecture | open | MathDB 页面 | 原文线索

V2-049314 | MATHDB-340772

The dumbbell graph Tait polynomial characterization conjecture

摘录 / EXCERPT: Let G be a connected planar trivalent graph, and let $T_G(z)$ denote its Tait polynomial. Dumbbell graph characterization conjecture. If ... then G is the dumbbell graph. The sou...

conjecture | open | MathDB 页面 | 原文线索

V2-049315 | MATHDB-340773

The even perfect matching positivity conjecture

摘录 / EXCERPT: Let G be a planar trivalent graph with perfect matching M. Call M even when every cycle of $G\setminus M$ has an even number of edges, and let $\langle G:M\rangle_2$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-049316 | MATHDB-340774

The Tait polynomial characterization of four-colorability

摘录 / EXCERPT: Let G be a planar trivalent graph, and let T_G denote its Tait polynomial. Tait polynomial four-colorability conjecture. The graph G is 4-colorable if and only if ... The sou...

conjecture | open | MathDB 页面 | 原文线索

V2-049317 | MATHDB-340796

The n=8 maximal-vertex neighborhood conjecture

摘录 / EXCERPT: Consider the vertex graph for the blind-decoding objective at matrix size $n=8$, and call a vertex maximal when it corresponds to a globally maximal objective value. Each vertex ha...

conjecture | open | MathDB 页面 | 原文线索

V2-049318 | MATHDB-340797

The n=10 maximal-vertex neighborhood conjecture

摘录 / EXCERPT: Consider the vertex graph for the blind-decoding objective at matrix size $n=10$, and call a vertex maximal when it corresponds to a globally maximal objective value. Each vertex h...

conjecture | open | MathDB 页面 | 原文线索

V2-049319 | MATHDB-340809

The disk conjecture for subtree roots of trees

摘录 / EXCERPT: Disk conjecture. The subtree roots of T lie in D_n.

conjecture | open | MathDB 页面 | 原文线索

V2-049320 | MATHDB-340810

The annulus conjecture for subtree roots of trees

摘录 / EXCERPT: Annulus conjecture. The subtree roots of T are contained in this annulus.

conjecture | open | MathDB 页面 | 原文线索

V2-049321 | MATHDB-340826

Lovász-Woodall conjecture on cycles through prescribed independent edges

摘录 / EXCERPT: Lovász-Woodall conjecture. If k is even or $G-S$ is connected, then G contains a cycle containing every edge in S.

conjecture | open | MathDB 页面 | 原文线索

V2-049322 | MATHDB-340831

The tree cover bound for connected triangle-free graphs

摘录 / EXCERPT: Triangle-free tree cover conjecture. For all connected triangle-free graphs,

conjecture | open | MathDB 页面 | 原文线索

V2-049323 | MATHDB-340832

Chartrand et al.'s odd-order maximum-size conjecture for rainbow disconnection number

摘录 / EXCERPT: Chartrand et al.'s conjecture. The maximum size of a connected graph G of order n with $rd(G)=k$ is

conjecture | open | MathDB 页面 | 原文线索

V2-049324 | MATHDB-340837

Alon-Bollobás-Krivelevich-Sudakov conjecture on Max-Cut in H-free graphs

摘录 / EXCERPT: Alon-Bollobás-Krivelevich-Sudakov conjecture. There exist constants $\varepsilon=\varepsilon(H)>0$ and $c=c(H)>0$ such that, for all H-free graphs G,

conjecture | open | MathDB 页面 | 原文线索

V2-049325 | MATHDB-340841

The conjectural vertex algebras for ind-constructible Calabi-Yau categories

摘录 / EXCERPT: The vertex-algebra existence conjecture. One can associate to $\mathcal{C}$ a class of vertex algebras parametrized by rays and by a choice of framing object; for almost all rays,...

conjecture | open | MathDB 页面 | 原文线索

V2-049326 | MATHDB-340858

The n^{-1} convergence-rate conjecture for deep-limit minimisers

摘录 / EXCERPT: The n^{-1} convergence-rate conjecture. Under these assumptions, the distance between the convergent subsequence of finite-depth minimisers and its limit is $O(n_k^{-1})$. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-049327 | MATHDB-340876

Characterisation conjecture for $g_{B,A}$-perfect graphs

摘录 / EXCERPT: Characterisation conjecture. A graph is $g_{B,A}$-perfect if and only if it contains no odd antihole of order $k\geq 7$ or any of the 14 configurations listed above as an induced s...

conjecture | open | MathDB 页面 | 原文线索

V2-049328 | MATHDB-340900

Conjecture on perfect linear hypergraphs and disjoint unions

摘录 / EXCERPT: Perfectness conjecture. (1) For $r\ge 3$, there exists n such that every linear r-uniform hypergraph with at least n vertices is perfect. (2) For $r\ge 3$, there exists n s...

conjecture | open | MathDB 页面 | 原文线索

V2-049329 | MATHDB-340901

Conjecture on perfect linear cycles and extremal Lagrangians

摘录 / EXCERPT: Perfect linear-cycle conjecture. C_t^3 is perfect for every $t\ge 3$. Moreover, a C_t^3-free 3-graph achieves the maximum Lagrangian density only if it contains K_{2t-1}^3.

conjecture | open | MathDB 页面 | 原文线索

V2-049330 | MATHDB-340929

Total regularity conjecture for mixed graphs with defect or excess one

摘录 / EXCERPT: Total regularity conjecture. All $(r,z,k;-1)$- and $(r,z,k;+1)$-graphs are totally regular.

conjecture | open | MathDB 页面 | 原文线索

V2-049331 | MATHDB-340937

Exponential lower-bound conjecture for high-covering-number intersecting hypergraphs

摘录 / EXCERPT: For integers $0\le a<b\le k$, let ... where the maximum is over intersecting families ${\mathcal F}\subset {[n]\choose k}$ with covering number $\tau({\mathcal F})=b$. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-049332 | MATHDB-340981

Ore's asymptotic density conjecture for critical graphs without large cliques

摘录 / EXCERPT: Let k be an integer with $k\geq 4$, and let G be a k-critical graph, meaning that $\chi(G)=k$ and every proper subgraph of G has chromatic number less than k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049333 | MATHDB-340982

The potential-function conjecture for critical graphs

摘录 / EXCERPT: Potential-function conjecture. For every $k\geq 6$, there exist $\varepsilon_k,\delta_k,P_k>0$ such that the (ε_k,δ_k)-potential satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049334 | MATHDB-340993

Generalized graph realization conjecture for heteroclinic networks

摘录 / EXCERPT: Generalized realization conjecture. The conclusion of Theorem holds even for directed graphs G that may contain 2-cycles and Δ-cliques.

conjecture | open | MathDB 页面 | 原文线索

V2-049335 | MATHDB-341020

Irreducibility conjecture for Whittaker modules of the symmetric-group orbifold

摘录 / EXCERPT: Let $M(1)$ be the rank-ℓ Heisenberg vertex operator algebra, let S_{ℓ} act on it by permuting a basis $h_1,\dots,h_{\ell}$ of the underlying space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049336 | MATHDB-341031

Creutzig-Huang-Yang rigidity conjecture for admissible affine categories

摘录 / EXCERPT: Creutzig-Huang-Yang conjecture. The braided tensor category structure on $\mathcal{O}_{\ell,\,\operatorname{ord}}$ is rigid and thus is a ribbon tensor category.

conjecture | open | MathDB 页面 | 原文线索

V2-049337 | MATHDB-341036

Caterpillar conjecture for block graphs maximizing mean CIS order

摘录 / EXCERPT: Let $n\geq 5$, and let G be a block graph of order n. Suppose that G has maximum mean CIS order among all block graphs of order n. Block-graph caterpillar conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-049338 | MATHDB-341046

Subquadratic growth conjecture for acyclic subgraphs of tournaments

摘录 / EXCERPT: Subquadratic growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049339 | MATHDB-341060

Bollobás-Gyárfás conjecture on highly connected monochromatic subgraphs

摘录 / EXCERPT: Bollobás-Gyárfás conjecture. If $n>4(k-1)$, then every 2-coloring of K_n contains a monochromatic k-connected subgraph of order at least

conjecture | open | MathDB 页面 | 原文线索

V2-049340 | MATHDB-341061

Conjecture on large two-colored highly connected subgraphs in Gallai 3-colorings

摘录 / EXCERPT: Gallai-coloring connectivity conjecture. For integers $k\geq 1$ and sufficiently large n, if $k\neq 4t$ for every integer $t\geq 1$, then every Gallai 3-coloring of K_n contain...

conjecture | open | MathDB 页面 | 原文线索

V2-049341 | MATHDB-341063

Hahn's nonincreasing-distance conjecture for cop-win graphs

摘录 / EXCERPT: Let G be an undirected cop-win graph, and consider optimal play, meaning that the cop minimizes capture time while the robber maximizes it. Hahn's conjecture. The distance betwee...

conjecture | open | MathDB 页面 | 原文线索

V2-049342 | MATHDB-341072

MLLS-reconstructibility of binary valid networks

摘录 / EXCERPT: MLLS-reconstructibility conjecture. The class of binary valid networks is MLLS-reconstructible.

conjecture | open | MathDB 页面 | 原文线索

V2-049343 | MATHDB-341078

Subdivision conjecture for bipartite graphs

摘录 / EXCERPT: For a graph F, let $\operatorname{sub}(F)$ be its 1-subdivision, obtained by replacing every edge of F with a pairwise internally vertex-disjoint path of length two. Subdivis...

conjecture | open | MathDB 页面 | 原文线索

V2-049344 | MATHDB-341108

The generalized distance spectral radius conjecture for digraphs

摘录 / EXCERPT: Let n,k be positive integers, let $0<\alpha<1$, and let $G\in\mathcal{G}_{n,k}$. Here, $\mu_\alpha(G)$ denotes the generalized distance spectral radius of G, and $K(n,k,1)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-049345 | MATHDB-341110

Mkrtchyan-Petrosyan-Zerafa frequency formulation of the Fan-Raspaud conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, let $a\in E(G)$, and let $i\in\{0,1,2\}$. For a list of perfect matchings, define $\nu_G[a:M_1,M_2,M_3]$ to be the number of those matchings co...

conjecture | open | MathDB 页面 | 原文线索

V2-049346 | MATHDB-341111

Mkrtchyan et al.'s vertex-frequency strengthening of the Fan-Raspaud conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, let w be a vertex of G, and let $i,j,k\in\{0,1,2\}$ satisfy $i+j+k=3$. For a list of perfect matchings, the frequency of an edge is the num...

conjecture | open | MathDB 页面 | 原文线索

V2-049347 | MATHDB-341112

Edge-frequency conjecture for S_4-pairs

摘录 / EXCERPT: Let G be a bridgeless cubic graph, let $a\in E(G)$, and let $i\in\{0,1,2\}$. An S_4-pair is the pair of perfect matchings used in the paper's S_4-formulation, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049348 | MATHDB-341117

Bounded-part List Colouring Conjecture for edge-colourable graphs

摘录 / EXCERPT: Let s be an integer. An edge λ-choosable graph is a graph whose line graph is colourable from every list assignment whose list-size pattern is the partition λ. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-049349 | MATHDB-341122

Gutman-Furtula-Bozkurt conjecture on maximal Randić energy

摘录 / EXCERPT: Let G be a connected graph on n vertices. Let Sun denote the sun graph and let $Balanced\ double\ sun$ denote the balanced double sun. Gutman-Furtula-Bozkurt conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-049350 | MATHDB-341123

Closed-form maximal Randić energy conjecture for selected orders

摘录 / EXCERPT: Let G be a connected graph on n vertices, and let S^p and $DS^{p,q}$ denote the sun and double-sun graphs used in the source. Let $E_{\mathcal{L}}$ denote the normalized Lapl...

conjecture | open | MathDB 页面 | 原文线索

V2-049351 | MATHDB-341133

Ergodicity equivalence for delayed unimolecular stochastic reaction networks

摘录 / EXCERPT: Ergodicity conjecture. Any delayed unimolecular stochastic reaction network with constant deterministic or stochastic delays, or with time-varying stochastic delays, is ergodic if...

conjecture | open | MathDB 页面 | 原文线索

V2-049352 | MATHDB-341142

The six-fifths conjecture for 2-edge-connected multigraphs

摘录 / EXCERPT: Six-fifths conjecture. If $x\in\mathrm{LP}(G)$, then $\frac{6}{5}x$ dominates a convex combination of 2-edge-connected multigraphs of G. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-049353 | MATHDB-341145

Uniqueness conjecture for optimal hyperplane covers of extremal matrices

摘录 / EXCERPT: Uniqueness conjecture. Every extremal matrix has the unique optimal hyperplane cover.

conjecture | open | MathDB 页面 | 原文线索

V2-049354 | MATHDB-341147

Diagonal-extremality conjecture for multidimensional extremal matrices

摘录 / EXCERPT: Diagonal-extremality conjecture. Every multidimensional extremal matrix is diagonally extremal.

conjecture | open | MathDB 页面 | 原文线索

V2-049355 | MATHDB-341148

Polydiagonal-to-diagonal conjecture for hyperplane-cover matrices

摘录 / EXCERPT: Polydiagonal-to-diagonal conjecture. Then A contains a diagonal.

conjecture | open | MathDB 页面 | 原文线索

V2-049356 | MATHDB-341187

Harris's conjecture on the Hall ratio and fractional chromatic number

摘录 / EXCERPT: Let G be a graph, let $\chi_f(G)$ denote its fractional chromatic number, and let $\rho(G)$ denote its Hall ratio, the maximum of $v(H)/\alpha(H)$ over all non-null subgraphs...

conjecture | solved | MathDB 页面 | 原文线索

V2-049357 | MATHDB-341232

The complete bipartite rook equivalence graph conjecture for Ferrers boards

摘录 / EXCERPT: Let $K_{a,b}$ be a complete bipartite graph, and let B be a Ferrers board whose rook equivalence graph is $K_{a,b}$. Complete bipartite rook equivalence conjecture. Then $a=b=1$....

conjecture | open | MathDB 页面 | 原文线索

V2-049358 | MATHDB-341233

The explicit bijection conjecture for edges of rook equivalence graphs

摘录 / EXCERPT: Let B be a board, let $G(B)$ be its rook equivalence graph, and suppose that $\{B_1,B_2\}$ is an edge of $G(B)$. For a nonnegative integer k, consider the placements of k roo...

conjecture | open | MathDB 页面 | 原文线索

V2-049359 | MATHDB-341250

Wang's Turán-number conjecture for spanning linear forests

摘录 / EXCERPT: Let r-linear forests be r-uniform hypergraphs consisting of vertex-disjoint tight paths or isolated vertices, and let $\mathcal{L}^r_{n,k}$ be the set of all r-linear forests...

conjecture | open | MathDB 页面 | 原文线索

V2-049360 | MATHDB-341255

Uniqueness of local moves preserving electrical equivalence in punctured-disk networks

摘录 / EXCERPT: Local-move completeness conjecture. Any two electrically equivalent graphs associated to rnpds are connected by the local moves stated in this section.

conjecture | open | MathDB 页面 | 原文线索

V2-049361 | MATHDB-341256

Hexagon conditional local-move conjecture for punctured-disk resistor networks

摘录 / EXCERPT: Hexagon conditional local-move conjecture. There is a hexagon conditional local move of the displayed form, with the conjectural conductance formulas described in the paper, that p...

conjecture | open | MathDB 页面 | 原文线索

V2-049362 | MATHDB-341257

Conditional local moves on polygonal networks for every n

摘录 / EXCERPT: Conditional local-move existence conjecture. For every integer $n\geq 3$, there exists a conditional local move on Π_n such that:

conjecture | open | MathDB 页面 | 原文线索

V2-049363 | MATHDB-341267

Sullivan's compromise conjectures for the second neighbourhood

摘录 / EXCERPT: Sullivan's compromise conjectures. Every oriented graph D has a vertex x satisfying at least one of the following two assertions:

conjecture | open | MathDB 页面 | 原文线索

V2-049364 | MATHDB-341276

The tight linear forest Turán conjecture for k congruent to 1 modulo r

摘录 / EXCERPT: Let $r\geq 2$, and let $\mathcal{L}_{n,k}^{(r)}$ be the family of all tight linear forests of order n with k edges in r-graphs. Here a tight linear forest is an r-graph who...

conjecture | open | MathDB 页面 | 原文线索

V2-049365 | MATHDB-341280

Bushlanov et al.'s tensor equivalence conjecture for the $(1,p)$-logarithmic minimal model

摘录 / EXCERPT: Let p be such that q has order $2p$. Let $4\mathscr{C}_p^+4$ be the positive part of the $4\mathbb{Z}/2\mathbb{Z}4$-graded category of representations of the divided-power quan...

conjecture | open | MathDB 页面 | 原文线索

V2-049366 | MATHDB-341281

The invariant-functor conjecture for the $(1,p)$-logarithmic minimal model

摘录 / EXCERPT: Fix q of order $2p$. Let $\mathcal{W}_p$ be the triplet vertex operator algebra, with a PSL_2-action, and let $T^\vee$ be its one-dimensional torus. The invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-049367 | MATHDB-341282

The $4\operatorname{PSL}_24$-equivariance conjecture for the triplet quantum-group equivalence

摘录 / EXCERPT: Fix q of order $2p$, and let $u_q^{\operatorname{M}}(\mathfrak{sl}_2)$ be the simply connected log-modular quantum group and $\mathcal{W}_p$ the triplet vertex operator algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-049368 | MATHDB-341305

DNA graph conjecture for ladder digraphs

摘录 / EXCERPT: Let P_n denote the path digraph on n vertices, and let $\Box$ denote the Cartesian product of digraphs. A digraph is a DNA graph if it admits a DNA labeling, namely a vertex la...

conjecture | open | MathDB 页面 | 原文线索

V2-049369 | MATHDB-341320

Thomassé-Yeo conjecture on total domination in graphs of minimum degree five

摘录 / EXCERPT: Thomassé-Yeo conjecture. Every n-vertex graph G with minimum degree

conjecture | open | MathDB 页面 | 原文线索

V2-049370 | MATHDB-341330

The mixed-subdivision strengthening of the planar discrete Brunn-Minkowski conjecture

摘录 / EXCERPT: Let $A,B\subset\mathbb R^2$ be finite two-dimensional sets. A triangulation T_A of $[A]$ uses A as its vertex set, and a triangulation T_B of $[B]$ uses B as its vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-049371 | MATHDB-341358

Havet-Thomassé leaf-sensitive oriented-tree unavoidability conjecture

摘录 / EXCERPT: An oriented tree is an orientation of an undirected tree, a leaf is a vertex of degree one in its underlying tree, and a digraph is m-unavoidable if it is contained as a subdigra...

conjecture | open | MathDB 页面 | 原文线索

V2-049372 | MATHDB-341360

Leaf-sensitive oriented-tree universality conjecture

摘录 / EXCERPT: A digraph is k-universal if it is contained in every digraph with chromatic number k. An oriented tree is an orientation of an undirected tree, and a leaf is a vertex of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-049373 | MATHDB-341373

Campos et al.'s Hilton condition conjecture for 3-clique graphs

摘录 / EXCERPT: Let G be a 3-clique graph, and let Hilton's condition mean the condition for a simple graph with an even number of vertices and a universal vertex that ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-049374 | MATHDB-341384

The De Bruijn-Erdős conjecture for finite metric spaces

摘录 / EXCERPT: Let (V,d) be a finite metric space with $n=|V|$ and $n\geq 2$. For distinct points $x,y\in V$, let the line determined by x and y be the set consisting of x, y, and every...

conjecture | open | MathDB 页面 | 原文线索

V2-049375 | MATHDB-341385

The De Bruijn-Erdős conjecture for finite connected graphs

摘录 / EXCERPT: Let G be a finite connected graph with $n=|V(G)|$ and $n\geq 2$, equipped with its graph metric. For distinct vertices u,v, let the line $L_G(u,v)$ be the metric line determine...

conjecture | open | MathDB 页面 | 原文线索

V2-049376 | MATHDB-341386

The $\Omega(n^{4/3})$ line conjecture for graphs

摘录 / EXCERPT: Let G be a finite graph with $n=|V(G)|$, and call a line universal if it contains all vertices of G. Line-count conjecture. Every graph G without a universal line has...

conjecture | open | MathDB 页面 | 原文线索

V2-049377 | MATHDB-341387

The complete multipartite reduction conjecture for graph lines

摘录 / EXCERPT: Let G be a finite graph with no universal line, and let a line be a metric line in the graph metric. Complete multipartite reduction conjecture. There is a complete multipartite...

conjecture | open | MathDB 页面 | 原文线索

V2-049378 | MATHDB-341388

The AMRZ line-and-bridge conjecture for graphs without pendant edges

摘录 / EXCERPT: Let G be a finite graph with no pendant edges. Let a line be a metric line in G, and let a bridge be an edge whose deletion disconnects G. AMRZ line-and-bridge conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-049379 | MATHDB-341389

Finite termination of splitting processes for accessible quasi-transitive graphs

摘录 / EXCERPT: Finite termination conjecture. Every process of splittings must end after finitely many steps.

conjecture | open | MathDB 页面 | 原文线索

V2-049380 | MATHDB-341412

Benevides-Hoppen-Sampaio's conjecture on Gallai 3-extremal graphs

摘录 / EXCERPT: Benevides-Hoppen-Sampaio's conjecture. The only Gallai 3-extremal graph of order n is the complete graph K_n.

conjecture | open | MathDB 页面 | 原文线索

V2-049381 | MATHDB-341413

Hoppen-Lefmann-Odermann's conjecture on Gallai 4-extremal graphs

摘录 / EXCERPT: Hoppen-Lefmann-Odermann's conjecture. The only Gallai 4-extremal graph of order n is the complete bipartite graph

conjecture | open | MathDB 页面 | 原文线索

V2-049382 | MATHDB-341414

JSJ characterization conjecture for commensurably coHopfian hyperbolic groups

摘录 / EXCERPT: Let Γ be a one-ended hyperbolic group that is not Fuchsian. Its JSJ decomposition is the canonical graph-of-groups decomposition over two-ended subgroups, with vertex groups...

conjecture | open | MathDB 页面 | 原文线索

V2-049383 | MATHDB-341420

Universal TCS characterization of containment for non-binary tree-child networks

摘录 / EXCERPT: Let N and N' be non-binary tree-child networks on the same leaf set. A tree-child sequence (TCS) of N is a sequence of reductions associated with the tree-child network. Univ...

conjecture | open | MathDB 页面 | 原文线索

V2-049384 | MATHDB-341426

Balbuena-Hansberg-Haynes-Henning conjecture for non-bipartite diameter-2-critical graphs

摘录 / EXCERPT: Let G be a non-bipartite diameter-2-critical graph without a dominating edge, and let n be its order. Let $\mathcal{C}_5^+$ denote the family of expanded 5-cycles described...

conjecture | open | MathDB 页面 | 原文线索

V2-049385 | MATHDB-341427

Strengthened conjecture for non-bipartite diameter-2-critical graphs

摘录 / EXCERPT: Let G be a non-bipartite diameter-2-critical graph of order n, and let H_5 be the exceptional graph and $\mathcal{C}_5^+$ the family of expanded 5-cycles described in the...

conjecture | open | MathDB 页面 | 原文线索

V2-049386 | MATHDB-341444

Entropy conjecture for tournament digraphs

摘录 / EXCERPT: Entropy conjecture. For α sufficiently large,

conjecture | open | MathDB 页面 | 原文线索

V2-049387 | MATHDB-341457

The cubic-order conjecture for bridge-burning capture time

摘录 / EXCERPT: Let G be an n-vertex graph. Write $c_b(G)$ for its bridge-burning cop number and $\mathrm{capt}_b(G)$ for its bridge-burning capture time. Cubic-order conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-049388 | MATHDB-341477

List-colouring bound for graphs with sparse neighbourhoods

摘录 / EXCERPT: Let G be a graph of maximum degree Δ such that the neighbourhood of every vertex spans at most Δ^2/f edges. List-colouring conjecture. For every $\varepsilon>0$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-049389 | MATHDB-341478

Monotonicity of worst-case bottleneck values for degree-bounded spanning trees

摘录 / EXCERPT: Let v be a root vertex with children $\{v_1, \dots, v_k\}$, where $k \geq 2$, the angles between children with respect to v are at least 60°, and all children have rad...

conjecture | open | MathDB 页面 | 原文线索

V2-049390 | MATHDB-341479

Monotonicity of worst-case bottleneck values for the (δ,k)-PKRY algorithm

摘录 / EXCERPT: For each m, let S_m be a star with m children rooted at its centre vertex, chosen to maximise the length of the longest edge in the tree produced by the (δ,k)-PKRY alg...

conjecture | open | MathDB 页面 | 原文线索

V2-049391 | MATHDB-341485

The maximum-size matching conjecture for odd generalised Petersen graphs

摘录 / EXCERPT: Let $P(2n+1,1)$ be the generalised Petersen graph, with $n\geq 4$. A matching is a set of pairwise vertex-disjoint edges of the graph. Maximum-size matching conjecture. Any matchin...

conjecture | open | MathDB 页面 | 原文线索

V2-049392 | MATHDB-341491

Connected {2,4}-factor conjecture for 2-tough graphs

摘录 / EXCERPT: Let G be a graph. A connected $\{2,4\}$-factor is a connected spanning subgraph of G in which every vertex has degree either 2 or 4. Connected $\{2,4\}$-factor conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-049393 | MATHDB-341494

Aharoni-Howard rainbow matching conjecture

摘录 / EXCERPT: Aharoni-Howard's conjecture. If each H_i has $f(n,r,k)$ hyperedges, then there is a rainbow matching

conjecture | open | MathDB 页面 | 原文线索

V2-049394 | MATHDB-341523

Neumann-Lara's conjecture on NL-2-coflows of planar digraphs

摘录 / EXCERPT: Neumann-Lara's conjecture. Every loopless planar digraph admits an NL-2-coflow.

conjecture | open | MathDB 页面 | 原文线索

V2-049395 | MATHDB-341527

Kaiser et al.'s prism-toughness conjecture

摘录 / EXCERPT: Kaiser et al.'s prism-toughness conjecture. There exists a constant t_1 such that the prism over any t_1-tough graph is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-049396 | MATHDB-341533

Schur-positive trees with a degree-floor vertex

摘录 / EXCERPT: For each integer $n\geq 2$, consider n-vertex trees and vertices whose degree is measured in the usual graph-theoretic sense. The degree-floor tree conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-049397 | MATHDB-341542

Hagborg-Oblomkov capped-vertex DT/PT correspondence conjecture

摘录 / EXCERPT: Let $X=\mathbb{P}^1\times\mathbb{P}^1\times\mathbb{P}^1$ and let $D=D_1\cup D_2\cup D_3$ be the union of the three coordinate divisors, with...

conjecture | open | MathDB 页面 | 原文线索

V2-049398 | MATHDB-341580

The extremal weighted Szeged index conjecture for connected graphs

摘录 / EXCERPT: Extremal weighted Szeged index conjecture. The maximum weighted Szeged index among n-vertex graphs is attained by the balanced complete bipartite graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049399 | MATHDB-341595

Erdős-Hajnal three-color triangle conjecture

摘录 / EXCERPT: Let the edges of the complete graph on vertex set $[n]$ be colored with three colors I, II, and III. For a triangle with vertices $a<b<c$, consider the pattern in which $\{a,b\}$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-049400 | MATHDB-341596

Berge cycle conjecture for uniform covering hypergraphs

摘录 / EXCERPT: Let a k-uniform covering hypergraph be a hypergraph in which every hyperedge has size k and every pair of vertices is contained in a hyperedge. Let n be the number of vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-049401 | MATHDB-341598

Győri's conjecture on bipartite Turán numbers of even cycles

摘录 / EXCERPT: Győri's conjecture. Suppose that m,n,t are integers satisfying $n\geq m^2$ and $m\geq t\geq 3$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-049402 | MATHDB-341599

A sharp bipartite Turán formula for large even cycles

摘录 / EXCERPT: Sharp bipartite Turán formula. Suppose that m,n,t are positive integers satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-049403 | MATHDB-341639

D-graph external mother vertex-set conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a directed graph without parallel edges on K nodes. For a node $v\in V$, write $d_{\mathrm{in}}(v)$ and $d_{\mathrm{out}}(v)$ for its in-degree and out-degree. A...

conjecture | open | MathDB 页面 | 原文线索

V2-049404 | MATHDB-341692

Generalized union property conjecture

摘录 / EXCERPT: Generalized union property conjecture. For all choices of n,k,s,q, one has $m(n,k,s,q)=|{\mathcal A}_{p,r}|$ for an appropriate choice of $p,r>0$ such that ${\mathcal A}_{p,r}$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-049405 | MATHDB-341693

Three-candidate conjecture for 3-graphs with bounded union

摘录 / EXCERPT: Three-candidate conjecture. If ${\mathcal F}$ has property $U(s,2s+1)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049406 | MATHDB-341706

Erdős-Faudree-Győri minimum-book-number conjecture

摘录 / EXCERPT: Let $x\in\mathbb{Q}$ satisfy $\frac12<x<1$, and let ... be its unique greedy representation, where $3\leq r_1$ and $(r_{i-1}-1)^2<r_i$ for $2\leq i\leq k$. Set ... For a graph, its boo...

conjecture | open | MathDB 页面 | 原文线索

V2-049407 | MATHDB-341707

Erdős-Faudree-Győri size-condition conjecture for book numbers

摘录 / EXCERPT: Erdős-Faudree-Győri size-condition conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049408 | MATHDB-341711

The potential crossing pair characterization of graphs with crossing number at least 2

摘录 / EXCERPT: Potential crossing pair conjecture. The graph G has crossing number at least 2 if and only if it does not have a potential crossing pair.

conjecture | open | MathDB 页面 | 原文线索

V2-049409 | MATHDB-341717

The general sparse sum-of-squares certificate conjecture for Vizing's conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ and $\mathcal{H}$ be the graph objects used in the paper's polynomial model, with vertex sets $V(\mathcal{G})$ and $V(\mathcal{H})$, parameters $n_\mathcal{G}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-049410 | MATHDB-341718

Gupta's 1967 conjecture on multigraph edge-coloring

摘录 / EXCERPT: Gupta's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-049411 | MATHDB-341719

Jakobsen's bounded-order conjecture for critical multigraphs

摘录 / EXCERPT: Jakobsen's bounded-order conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049412 | MATHDB-341720

The elementary multigraph reformulation of the Goldberg-Seymour conjecture

摘录 / EXCERPT: Elementary-multigraph conjecture. Every multigraph G with

conjecture | open | MathDB 页面 | 原文线索

V2-049413 | MATHDB-341726

Alon-Tarsi-Jaeger short cycle cover conjecture

摘录 / EXCERPT: Short cycle cover conjecture. Every bridgeless graph with m edges has a cycle cover of length at most $1.4m$.

conjecture | open | MathDB 页面 | 原文线索

V2-049414 | MATHDB-341735

Diameter extremality conjecture for Bell-type io-decomposable Riordan graphs

摘录 / EXCERPT: Let G_n be an io-decomposable Riordan graph of the Bell type. The Pascal and Catalan graphs are ... Here ${\rm diam}(G)$ denotes the graph diameter. Diameter extremality conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-049415 | MATHDB-341737

Thomassen's red-blue decomposition conjecture for 3-connected cubic graphs

摘录 / EXCERPT: Let G be a 3-connected cubic graph on at least 8 vertices. A red-blue vertex coloring partitions its vertices into red and blue classes; the blue subgraph is the subgraph ind...

conjecture | open | MathDB 页面 | 原文线索

V2-049416 | MATHDB-341738

Subcubic red-blue decomposition conjecture

摘录 / EXCERPT: Let G be a subcubic graph, meaning that its maximum degree is at most 3, and suppose that G has at least 7 vertices. In a red-blue vertex coloring, the induced blue compone...

conjecture | open | MathDB 页面 | 原文线索

V2-049417 | MATHDB-341774

The constructive lower-bound conjecture for Gallai-Ramsey numbers of hypergraphs

摘录 / EXCERPT: The constructive lower-bound conjecture. The displayed inequality holds for all such r, t, and hypergraphs $H_1,\dots,H_t$. The preceding constructive theorems provide evidence...

conjecture | open | MathDB 页面 | 原文线索

V2-049418 | MATHDB-341796

The eccentricity spectral bound conjecture for trees

摘录 / EXCERPT: Let T be a tree on n vertices, with $n\geq 3$, and let $\epsilon_n(T)$ denote the least eigenvalue of its eccentricity matrix. Eccentricity spectral bound conjecture. One has ……

conjecture | open | MathDB 页面 | 原文线索

V2-049419 | MATHDB-341825

Benjamini-Yadin cylinder growth conjecture for external DLA

摘录 / EXCERPT: Let $(\mathcal{G}^n)_{n\geq 0}$ be a family of d-regular graphs with $|\mathcal{G}^n|\to\infty$. For each n, form the cylinder graph $\mathcal{G}^n\times\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049420 | MATHDB-341846

The Evasiveness conjecture for vertex-transitive graphs

摘录 / EXCERPT: Evasiveness conjecture for graphs. If $X\in D_{\infty}$ and X is vertex-transitive, then X is a complete graph.

conjecture | open | MathDB 页面 | 原文线索

V2-049421 | MATHDB-341862

Asymptotically sharp lower bound for oriented $L(2,1)$-labelings

摘录 / EXCERPT: Asymptotic lower-bound conjecture. There is an oriented graph G for which every vertex satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049422 | MATHDB-341872

The sparse hypergraph extremal-function conjecture

摘录 / EXCERPT: Sparse hypergraph extremal-function conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-049423 | MATHDB-341908

The Brown-Erdős-Sós conjecture for groups

摘录 / EXCERPT: Let $t\in\mathbb{Z}^+$ and let $\epsilon>0$. Let G be a group, and let A be a subset of the multiplication table of G with density at least ϵ. A $(t+3,t)$-configurat...

conjecture | open | MathDB 页面 | 原文线索

V2-049424 | MATHDB-341916

Sharp distinction between expansive and abelian expansive networks

摘录 / EXCERPT: Let q be an alphabet size, and let $E[D,q]$ denote the set of expansive networks on a digraph D over an alphabet of size q, while $EA[D,q]$ denotes the set of abelian expansi...

conjecture | open | MathDB 页面 | 原文线索

V2-049425 | MATHDB-341918

Arman-Gunderson-Tsaturian odd-cycle-free extremal conjecture

摘录 / EXCERPT: Let $k>1$ and let $T_2(n)$ be the Turán graph, the complete bipartite graph with its two vertex classes as equal in size as possible. For a graph G, write $m(n;C_{2k+1})$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-049426 | MATHDB-341926

The minimum-degree-three conjecture for strong parity property

摘录 / EXCERPT: Minimum-degree-three conjecture. Every 2-edge-connected graph of minimum degree at least three has the strong parity property.

conjecture | open | MathDB 页面 | 原文线索

V2-049427 | MATHDB-341963

The polynomial-size DFJ relaxation conjecture for cubic graphs

摘录 / EXCERPT: Let $amma$ be a cubic graph. Let $ Mathcal{P}_b$ be the feasible region defined by constraints ... , and let $ Mathcal{P}_c$ be the feasible region defined by the degree, subtour, sy...

conjecture | open | MathDB 页面 | 原文线索

V2-049428 | MATHDB-341964

The crossing-number conjecture for Sunlet-Star Cartesian products

摘录 / EXCERPT: Let $\mathcal{S}_n$ be the Sunlet graph on $2n$ vertices, obtained by attaching n pendant edges to the cycle C_n, and let $K_{1,m}$ be the star graph on $m+1$ vertices. For a g...

conjecture | open | MathDB 页面 | 原文线索

V2-049429 | MATHDB-341971

The non-negative curvature obstruction for increasing regular expander families

摘录 / EXCERPT: Let $D\in\mathbb{N}$ be fixed. An increasing D-regular expander graph family is a sequence of D-regular expander graphs $\{G_k\}_{k\in\mathbb{N}}$ whose sizes increase with k...

conjecture | open | MathDB 页面 | 原文线索

V2-049430 | MATHDB-341976

Alspach-Mason-Pullman conjecture on path decompositions of even tournaments

摘录 / EXCERPT: Let D be a directed graph with vertex set $V(D)$ and edge set $E(D)$. A path decomposition of D is a collection of directed paths whose edge sets partition $E(D)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049431 | MATHDB-341985

Long edges in the scenario-B potential infection graph

摘录 / EXCERPT: Long-edge conjecture for scenario B. There exists $\phi>0$ such that, if $\lambda=n^{-\phi}$, a.a.s. I_n contains an edge of length greater than λ.

conjecture | open | MathDB 页面 | 原文线索

V2-049432 | MATHDB-342013

The cop-number conjecture for connected P_t-free graphs

摘录 / EXCERPT: Cop-number conjecture. The robber can be captured by $t-3$ cops.

conjecture | open | MathDB 页面 | 原文线索

V2-049433 | MATHDB-342014

Digraph conjecture on minimum total distance at fixed outradius

摘录 / EXCERPT: Let n and r be two positive integers with $n \ge 2r$ and $r \ge 3$. For a digraph D, let $W(D)$ denote the sum of distances over all ordered pairs of vertices, let the outrad...

conjecture | open | MathDB 页面 | 原文线索

V2-049434 | MATHDB-342015

Digraph analogue of Vizing's extremal-size theorem

摘录 / EXCERPT: Let $r \ge 3$ and $n \ge 2r$. A biconnected digraph is a digraph with no cutvertex, and $D_{n,r,s}$ denotes the family defined in the paper. Digraph Vizing conjecture. The maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-049435 | MATHDB-342017

Conjecture on smooth arithmetical structures on paths with a doubled edge

摘录 / EXCERPT: Smooth-structure growth conjecture. The number of smooth arithmetical structures on $\mathcal{P}_{m,n}$ grows at the same rate as

conjecture | open | MathDB 页面 | 原文线索

V2-049436 | MATHDB-342042

Maximum positive and stable steady states in the full ERK network

摘录 / EXCERPT: Consider the full ERK network and the same network with $k_{\mathrm{on}}=0$, and let a positive steady state mean a steady state with all relevant concentrations positive; call a s...

conjecture | open | MathDB 页面 | 原文线索

V2-049437 | MATHDB-342047

Discrete surface-area conjecture for simplices

摘录 / EXCERPT: Let S be a d-simplex with the origin in its interior and with rational vertex directions. Let $\operatorname{Vol}_{\mathbb{Z}^d}(S)$ be normalized volume and let ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-049438 | MATHDB-342049

Boundary discrete surface-area conjecture for simplices

摘录 / EXCERPT: Let $S=\operatorname{conv}(\{v_0,\ldots,v_d\})$ be a d-simplex with $\mathbf{0}\in S\setminus\{v_0,\ldots,v_d\}$ and with rational vertex directions. For each i, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049439 | MATHDB-342050

The Two-Colour-Conjecture for oriented planar digraphs

摘录 / EXCERPT: Two-Colour-Conjecture. Every oriented planar digraph D is 2-colourable.

conjecture | open | MathDB 页面 | 原文线索

V2-049440 | MATHDB-342051

The higher-girth packing conjecture for strongly planar digraphs

摘录 / EXCERPT: Higher-girth packing conjecture. There exists a packing of g pairwise disjoint feedback vertex sets in D. Equivalently, D can be vertex g-coloured such that every directed...

conjecture | open | MathDB 页面 | 原文线索

V2-049441 | MATHDB-342062

Uniqueness conjecture for connected well-indumatched graphs of girth 11

摘录 / EXCERPT: Let G be a connected well-indumatched graph of girth 11, and let C_{11} denote the cycle on 11 vertices. The girth-11 uniqueness conjecture. The cycle C_{11} is the only...

conjecture | open | MathDB 页面 | 原文线索

V2-049442 | MATHDB-342075

Packing-property conjecture for symbolic and ordinary powers of edge ideals

摘录 / EXCERPT: Packing-property conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-049443 | MATHDB-342096

Brualdi-Hollingsworth conjecture on rainbow spanning-tree decompositions

摘录 / EXCERPT: Let K_n be a complete graph and let a 1-factorization be an edge-colouring whose colour classes form a decomposition of K_n into perfect matchings. A subgraph is rainbow if a...

conjecture | open | MathDB 页面 | 原文线索

V2-049444 | MATHDB-342098

Kaneko-Kano-Suzuki conjecture on rainbow spanning trees in proper colourings

摘录 / EXCERPT: Let K_n be a complete graph with a proper edge-colouring, meaning that any two incident edges have different colours. A subgraph is rainbow if all its edges have distinct colours...

conjecture | open | MathDB 页面 | 原文线索

V2-049445 | MATHDB-342103

Pokrovskiy's bounded-defect monochromatic cycle conjecture

摘录 / EXCERPT: Let r be a positive integer, and let a complete graph have its edges coloured with r colours. A collection of vertex-disjoint monochromatic cycles consists of cycles with pairw...

conjecture | open | MathDB 页面 | 原文线索

V2-049446 | MATHDB-342104

Gyárfás's monochromatic tight-cycle partition conjecture

摘录 / EXCERPT: Let k and r be positive integers. A k-uniform hypergraph is a hypergraph whose edges are k-element subsets of its vertex set, and a tight cycle is a cyclically ordered sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-049447 | MATHDB-342106

Uniqueness of the minimum normalized distance Laplacian spectral radius

摘录 / EXCERPT: Let G be a graph on $n\geq 2$ vertices. Write $\partial^{\mathcal L}_n$ for the largest eigenvalue of its normalized distance Laplacian and $\rho_{\mathcal D^{\mathcal L}}$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-049448 | MATHDB-342110

Weighted Graham-type conjecture for cover pebbling numbers

摘录 / EXCERPT: Let G and H be graphs. A nonnegative function on a graph assigns a nonnegative real number to each vertex. Let ω_1 be a nonnegative function on G and ω_2 a nonn...

conjecture | open | MathDB 页面 | 原文线索

V2-049449 | MATHDB-342134

Normal 6-edge-coloring conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let $\chi'_N(G)$ denote the least number of colors in a normal edge-coloring of G, where every edge is poor or rich. Normal 6-edge-colori...

conjecture | open | MathDB 页面 | 原文线索

V2-049450 | MATHDB-342135

The flow conjecture for 3-edge-connected cubic graphs excluding the Petersen graph

摘录 / EXCERPT: Let G be a 3-edge-connected cubic graph different from the Petersen graph P_{10}. A nowhere-zero $\mathbb{Z}_2^3$-flow is a flow $f:E(G)\to\mathbb{Z}_2^3\setminus\{0\}$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-049451 | MATHDB-342139

The reconstruction conjecture for a class of hypergraphs

摘录 / EXCERPT: Let V be a finite vertex set, let D be the presheaf of hypergraphs, and let $C\subset D$ be a subfunctor. For each vertex $x\in V$, write $\jmath_x:V-x\hookrightarrow V$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-049452 | MATHDB-342140

Disjoint reconstruction conjecture for proper hypergraph presheaves

摘录 / EXCERPT: Let D be the presheaf of hypergraphs and let $C\subset D$ be a proper presheaf, meaning that disjoint union preserves membership in C. For each finite vertex set V, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049453 | MATHDB-342141

The deleting-process reconstruction conjecture

摘录 / EXCERPT: A hypergraph reconstruction context consists of a category $\mathfrak{C}$ of applicable prestacks, a deleting process...

conjecture | open | MathDB 页面 | 原文线索

V2-049454 | MATHDB-342191

Asymptotic lex-order conjecture for maximizing independent sets in uniform hypergraphs

摘录 / EXCERPT: Fix integers r and s with $1\le s\le r$, and let $\eta>0$. Let $\mathcal{H}$ be an r-uniform hypergraph on n vertices with e edges, where ... and suppose that $\mathcal{H}$...

conjecture | open | MathDB 页面 | 原文线索

V2-049455 | MATHDB-342192

Strong worst-case hardness conjecture for counting k-cliques

摘录 / EXCERPT: Strong worst-case clique-counting hardness conjecture. Any randomized algorithm A for $\textsc{\#$(k,s)$-clique}$ with error probability less than $1/3$ takes time...

conjecture | open | MathDB 页面 | 原文线索

V2-049456 | MATHDB-342198

Degeneracy conjecture for the equitable vertex arborable threshold

摘录 / EXCERPT: Let k be a nonnegative integer, and let G be a k-degenerate graph, meaning that every subgraph $H\subseteq G$ has a vertex of degree at most k. Let $va_{eq}^*(G)$ be the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-049457 | MATHDB-342200

Characterization conjecture for forcing number and parallel paths

摘录 / EXCERPT: Let G be a graph with maximum degree at most 3, let n be the number of vertices of G, and let $F(G)$ denote its forcing number. A k-parallel path graph is a graph of k-...

conjecture | open | MathDB 页面 | 原文线索

V2-049458 | MATHDB-342220

Mohar's finiteness conjecture for complete bipartite WHDS graphs

摘录 / EXCERPT: Let $K_{m,n}$ be the complete bipartite graph on parts of sizes m and n, and let WHDS denote the property of being determined by the Hermitian spectrum up to the relevant equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-049459 | MATHDB-342221

Asymptotic scarcity conjecture for self-converse digraphs

摘录 / EXCERPT: Let $\mathcal{SC}_n$ be the number of self-converse digraphs on n vertices, and let $\mathcal{D}_n$ be the number of digraphs on n vertices. Asymptotic scarcity conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-049460 | MATHDB-342230

The transitive digraph dimension realization conjecture

摘录 / EXCERPT: Let d be a nonnegative integer. A transitive digraph is a digraph D in which, whenever $u\to v$ and $v\to w$ are arcs, $u\to w$ is also an arc. Let $\dim(D)$ denote the weak ma...

conjecture | open | MathDB 页面 | 原文线索

V2-049461 | MATHDB-342235

Unbounded ratio of maximum throttling number to maximum cop number

摘录 / EXCERPT: For each positive integer n, let $mc(n)$ and $mt(n)$ denote, respectively, the maximum cop number and maximum cop throttling number over all connected graphs of order n. The un...

conjecture | open | MathDB 页面 | 原文线索

V2-049462 | MATHDB-342236

Polynomial-time recognition conjecture for good orderings of 2T-graphs

摘录 / EXCERPT: A 2T-graph is a graph consisting of two edge-disjoint spanning trees. A good ordering is an ordering of the vertices whose associated orientation has both an in-branching and an ou...

open_problem | open | MathDB 页面 | 原文线索

V2-049463 | MATHDB-342242

Extremal bound conjecture for subdivisions of uniform hypergraphs

摘录 / EXCERPT: Uniform hypergraph subdivision conjecture. For any r-uniform hypergraph $\mathcal{H}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049464 | MATHDB-342245

The conjecture that almost every graph is determined by its spectrum

摘录 / EXCERPT: Spectral determination conjecture. Almost every graph is determined by its spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-049465 | MATHDB-342269

Nekrasov's sign-uniqueness conjecture for the zero-leg DT vertex

摘录 / EXCERPT: Let $\lambda_1,\lambda_2,\lambda_3$ be the equivariant parameters for the torus action on $\mathbb C^4$, with the Calabi-Yau relation determining the fourth parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049466 | MATHDB-342272

Fractional Erdős matching conjecture

摘录 / EXCERPT: Fractional Erdős matching conjecture. For integers $\ell,d\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-049467 | MATHDB-342273

The inverse-reward optimal strategy conjecture for booby trap search games

摘录 / EXCERPT: Consider a booby trap search game played on a hypergraph whose hyperedges S have positive rewards $r(S)$. A Searcher strategy assigns probabilities $p(S)$ to choosing hyperedges....

conjecture | open | MathDB 页面 | 原文线索

V2-049468 | MATHDB-342295

Unbounded balanced vertices for planar geodesic nets with four unbalanced vertices

摘录 / EXCERPT: Let a geodesic net in the Euclidean plane have four unbalanced vertices and some number of balanced vertices. Four-vertex unboundedness conjecture. There exist geodesic nets in the...

conjecture | open | MathDB 页面 | 原文线索

V2-049469 | MATHDB-342323

Balogh-Kostochka-Treglown degree-sequence conjecture for perfect clique packings

摘录 / EXCERPT: Let $n,r\in\mathbb{N}$ with r dividing n, and let G be a graph on n vertices whose degree sequence is $d_1\leq\cdots\leq d_n$.
Balogh-Kostochka-Treglown conjecture. If ... fo...

conjecture | open | MathDB 页面 | 原文线索

V2-049470 | MATHDB-342331

Babu-Diwan-type subdivision conjecture for vertex-disjoint even cycles

摘录 / EXCERPT: Subdivision conjecture. If the minimum degree of G is at least $n/2-k+1$, then G contains a subdivision of H.

conjecture | open | MathDB 页面 | 原文线索

V2-049471 | MATHDB-342337

Minimum independence number conjecture for $(3,3)$-Ramsey graphs with clique number four

摘录 / EXCERPT: Let $\mathcal{H}_e(3,3;4)$ denote the class of graphs G such that $G\overset{e}{\rightarrow}(3,3)$ and $\omega(G)=3$. For a graph G, let $\alpha(G)$ be its independence number....

conjecture | open | MathDB 页面 | 原文线索

V2-049472 | MATHDB-342350

Burcroff's irredundance conjecture for direct products of balanced complete multipartite graphs

摘录 / EXCERPT: Let $G=\prod_{i=1}^n K[u_i,t_i]$ be a direct product of balanced complete multipartite graphs. The independence number $\alpha(G)$ is the largest size of an independent set, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049473 | MATHDB-342351

The irredundance conjecture for direct products of complete multipartite graphs

摘录 / EXCERPT: Let $H_1,\ldots,H_n$ be complete multipartite graphs, and let $G=\prod_{i=1}^n H_i$. The independence number $\alpha(G)$ is the largest size of an independent set, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049474 | MATHDB-342352

The Eulerianity conjecture for Peano continua

摘录 / EXCERPT: The Eulerianity conjecture. A Peano continuum X is Eulerian if and only if every edge cut of X is even.

conjecture | open | MathDB 页面 | 原文线索

V2-049475 | MATHDB-342353

The open Eulerianity conjecture for Peano continua

摘录 / EXCERPT: The open Eulerianity conjecture. A Peano continuum X is open Eulerian if and only if all but two vertices of $X_\sim$ have even degree.

conjecture | open | MathDB 页面 | 原文线索

V2-049476 | MATHDB-342354

The normal 5-edge-coloring conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a cubic graph. A normal edge-coloring is a proper edge-coloring in which every edge uv satisfies $|S_c(u)\cup S_c(v)|\in\{3,5\}$, where $S_c(v)$ is the set of colors o...

conjecture | open | MathDB 页面 | 原文线索

V2-049477 | MATHDB-342355

Characterization of BE-diperfect digraphs by blocking odd cycles

摘录 / EXCERPT: Proposed characterization. A digraph D is BE-diperfect if, and only if, D has no blocking odd cycle as an induced subdigraph.

conjecture | open | MathDB 页面 | 原文线索

V2-049478 | MATHDB-342372

Clark's infinite-lift conjecture for finite graphs

摘录 / EXCERPT: Let G be a finite graph, and let H be a lift of G. Write $\mathsf{spec}(G)$ for the spectrum of G, and let $\rho(\mathsf{UCT}(G))$ denote the relevant universal-cover spect...

conjecture | open | MathDB 页面 | 原文线索

V2-049479 | MATHDB-342373

Bilu-Linial's 2-lift conjecture

摘录 / EXCERPT: Let G be a finite regular graph, and let H be a 2-lift of G. An eigenvalue of H not inherited from G is called new, and let $\rho(\mathsf{UCT}(G))$ denote the relevant un...

conjecture | open | MathDB 页面 | 原文线索

V2-049480 | MATHDB-342374

Mohar's quasi-Ramanujan conjecture for multipartite trees

摘录 / EXCERPT: Mohar's conjecture. For every degree matrix D, if $\mathbb{T}_D$ is weakly Ramanujan, then it is k_D-quasi-Ramanujan; moreover, if $k_D=1$, then $\mathbb{T}_D$ is Ramanujan. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-049481 | MATHDB-342386

The Sierpiński product automorphism conjecture for 2-connected graphs

摘录 / EXCERPT: Let G and H be 2-connected graphs, and let $f: V(G) \to V(H)$ be any mapping. The group $\tilde A(G,H,f)$ and the automorphism group of the Sierpiński product $G \otimes_f H$...

conjecture | open | MathDB 页面 | 原文线索

V2-049482 | MATHDB-342387

The semidirect-product conjecture for Sierpiński product automorphisms

摘录 / EXCERPT: Let G and H be connected graphs on the same vertex set, and let $f: V(G) \to V(H)$ be a bijection. The groups $\tilde A(G,H,f)$, $\bar A(G,H,f)$, and $\hat B(G,H,f)$ are the au...

conjecture | open | MathDB 页面 | 原文线索

V2-049483 | MATHDB-342396

Körner's reversing permutations conjecture

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$. Two permutations of $[n]$ are reversing if, viewed as vectors of length n, two coordinates contain the same two elements $\{a,b\}\subset[n]$ in reversed...

conjecture | open | MathDB 页面 | 原文线索

V2-049484 | MATHDB-342414

The edge-pattern convex-combination conjecture

摘录 / EXCERPT: Let P be a d-regular edge-rooted pattern of depth r and girth $2r+2$. For each edge-rooted pattern, let $\mathbf{e}(P)$ denote its constraint vector, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049485 | MATHDB-342424

The undecidability conjecture for infinite stature in graphs of free groups

摘录 / EXCERPT: Let G be a group that splits as a finite graph of finitely generated free groups. An algorithm is said to certify that G has infinite stature with respect to its vertex groups...

conjecture | open | MathDB 页面 | 原文线索

V2-049486 | MATHDB-342439

The multi-hypergraph Berge-tree Turán conjecture

摘录 / EXCERPT: Multi-hypergraph Berge-tree Turán conjecture. For all $r\geq k+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-049487 | MATHDB-342461

Hanson and Toft's minimum-edge conjecture for Ramsey co-critical graphs

摘录 / EXCERPT: Let $K_{t_1},\ldots,K_{t_r}$ be complete graphs, let $R=R(K_{t_1},\ldots,K_{t_r})$ be their classical Ramsey number, and let G be a $(K_{t_1},\ldots,K_{t_r})$-co-critical graph o...

conjecture | open | MathDB 页面 | 原文线索

V2-049488 | MATHDB-342465

Andreae-Schroeder conjecture for toroidal graphs

摘录 / EXCERPT: Andreae-Schroeder conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-049489 | MATHDB-342467

Sudakov-Verstraëte chromatic-number conjecture for consecutive cycle lengths

摘录 / EXCERPT: For $k\geq 2$, let χ_k be the largest chromatic number of a graph that does not contain k cycles of consecutive lengths. Sudakov-Verstraëte's conjecture. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-049490 | MATHDB-342534

Quasi-isometric rigidity conjecture for graphs of rigid free groups

摘录 / EXCERPT: Quasi-isometric rigidity conjecture. If Γ is the fundamental group of a graph of groups with a JSJ decomposition in which all non-cyclic vertex groups are rigid free groups,...

conjecture | open | MathDB 页面 | 原文线索

V2-049491 | MATHDB-342559

Product-form Poisson conjecture for binary reaction networks

摘录 / EXCERPT: Product-form Poisson conjecture. If $B(u(t))=0$, then the distribution of the process X_t is a product of Poisson distributions for all time.

conjecture | open | MathDB 页面 | 原文线索

V2-049492 | MATHDB-342566

Kelly's conjecture on reconstructibility from vertex-deleted decks

摘录 / EXCERPT: Kelly's conjecture. For $l\in\mathbb{N}$, there is an integer M_l such that any graph with at least M_l vertices is reconstructible from its deck of cards obtained by deleting...

conjecture | open | MathDB 页面 | 原文线索

V2-049493 | MATHDB-342584

Brondani-de Lima-Oliveira conjecture on the second signless Laplacian eigenvalues

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\lambda_2(G)$ denote its second signless Laplacian eigenvalue. Brondani-de Lima-Oliveira conjecture. ... This conjecture slightly improve...

conjecture | open | MathDB 页面 | 原文线索

V2-049494 | MATHDB-342585

Zhai-Shu-Hong conjecture on the Laplacian spread

摘录 / EXCERPT: Let G be a graph of order $n\geq 2$, and let $\mu_1(G)\geq\cdots\geq\mu_{n-1}(G)$ be its Laplacian eigenvalues. The Laplacian spread is $\mu_1(G)-\mu_{n-1}(G)$. Zhai-Shu-Hong con...

conjecture | open | MathDB 页面 | 原文线索

V2-049495 | MATHDB-342603

The critical-window product conjecture for random digraph components

摘录 / EXCERPT: Let $D(n,p)$ be the random directed graph and $G(n,p)$ the random graph, with $p=n^{-1}+\lambda n^{-4/3}$. Let $X^\lambda=n^{-2/3}|\mathcal{C}_1(G(n,p))|$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-049496 | MATHDB-342630

Maximal Wiener index conjecture for fullerene graphs

摘录 / EXCERPT: Maximal Wiener index conjecture. If a fullerene graph of an arbitrary order has the maximal Wiener index, then it is a nanotubical fullerene graph with caps of types a--d, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049497 | MATHDB-342631

Wiener complexity and diameter conjecture for maximal-Wiener fullerene graphs

摘录 / EXCERPT: Wiener complexity and diameter conjecture. The Wiener complexity and the diameter of fullerene graphs of an arbitrary order having the maximal Wiener index are given in Proposition...

conjecture | open | MathDB 页面 | 原文线索

V2-049498 | MATHDB-342665

The extremal inverse sum indegree energy conjecture for trees

摘录 / EXCERPT: Let G be a tree on $n\geq 2$ vertices. Write $E_{\mathrm{ISI}}(G)$ for its inverse sum indegree energy, and let S_n and P_n denote the star and path on n vertices, respecti...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-049499 | MATHDB-342666

Matching sequencibility conjecture for complete multi-partite graphs

摘录 / EXCERPT: Matching sequencibility conjecture. For any integers $n\geq 2$ and $s\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-049500 | MATHDB-342681

Dong-Wang's perfect matching conjecture for folded hypercubes

摘录 / EXCERPT: Dong-Wang's conjecture. E^m is a perfect matching if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-049501 | MATHDB-342689

The exact 3-partite path Ramsey conjecture

摘录 / EXCERPT: Let $K_{n,n,n}$ be the complete tripartite graph with three parts of size n, and let P_m denote the path on m vertices. For graphs G_0,G_1,G_2, write $G_0\mapsto(G_1,G_2)$...

conjecture | open | MathDB 页面 | 原文线索

V2-049502 | MATHDB-342701

Witt-realization conjecture for UMTCs at prescribed central charge

摘录 / EXCERPT: Witt-realization conjecture. There is at least one unitary vertex operator algebra (VOA) V of central charge c such that Mod_V is a UMTC Witt equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-049503 | MATHDB-342717

Pokrovskiy-type conjecture on covering finite complete hypergraphs by monochromatic Berge-cycles

摘录 / EXCERPT: For $s,k\in\mathbb{N}$ with $k\geq 2$, let $c=c(s,k)\in\mathbb{N}$ be a constant independent of n, and let $\mathcal{K}_{n}^{(k)}$ denote the complete k-uniform hypergraph on...

conjecture | open | MathDB 页面 | 原文线索

V2-049504 | MATHDB-342719

Extremal triangle-count conjecture for bounded book number

摘录 / EXCERPT: Let G be a graph on n vertices, and let $b(G)$ be its book number. For nonnegative integers b and n with $b\leq n/4$, let $S_{b,n}$ be the graph obtained by blowing up the...

conjecture | open | MathDB 页面 | 原文线索

V2-049505 | MATHDB-342730

The five-site localization conjecture for vertex-reinforced random walks

摘录 / EXCERPT: Let w be a nondecreasing weight sequence satisfying ... Extend w to the positive reals by $w(t)=w(\lfloor t\rfloor)$, define ... and let W^{-1} be its inverse. Set ... ... and, with...

conjecture | open | MathDB 页面 | 原文线索

V2-049506 | MATHDB-342732

Edge-disjoint caterpillar realization conjecture for tree degree matrices

摘录 / EXCERPT: Caterpillar realization conjecture. Every tree degree matrix without common leaves has a caterpillar realization.

conjecture | open | MathDB 页面 | 原文线索

V2-049507 | MATHDB-342733

Hadwiger-based strong chromatic index bound for multigraphs

摘录 / EXCERPT: Let $k\ge 4$ and let $G=(V,E)$ be a K_k-minor-free multigraph of maximum degree Δ. Multigraph strong chromatic index conjecture. The strong chromatic index of G satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-049508 | MATHDB-342736

Eulerian conjecture for covering hypergraphs

摘录 / EXCERPT: Let H be an ℓ-covering k-hypergraph with $2\leq\ell<k$ and at least two edges. Eulerian conjecture. H is eulerian. The preceding result establishes that such hypergraphs...

conjecture | open | MathDB 页面 | 原文线索

V2-049509 | MATHDB-342739

The cover inequality for Eulerian orientations

摘录 / EXCERPT: Cover inequality. Then

open_problem | open | MathDB 页面 | 原文线索

V2-049510 | MATHDB-342751

The b-invariant edge conjecture for essentially 4-edge-connected near-bipartite cubic bricks

摘录 / EXCERPT: The b-invariant edge conjecture. Every essentially 4-edge-connected cubic near-bipartite brick G, distinct from K_4, has at least

conjecture | open | MathDB 页面 | 原文线索

V2-049511 | MATHDB-342764

Nikiforov-Rojo edge-grafting conjecture for the α-spectral radius

摘录 / EXCERPT: Nikiforov-Rojo's edge-grafting conjecture. If $q\geq1$ and $p\geq q+2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049512 | MATHDB-342765

Conjecture on the α-spectral radius of hypertrees in NC(m,d)

摘录 / EXCERPT: The $NC(m,d)$ extremal conjecture. For $G\in NC(m,d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-049513 | MATHDB-342791

Conjecture on optimal gaps in cycle spectra of 3-connected cubic planar graphs

摘录 / EXCERPT: Let $\mathcal{C}(G)$ denote the cycle spectrum of a graph G, and let the circumference of G be the length of its longest cycle. Conjecture on optimal gaps. If $k\in\mathbb{N}$...

conjecture | open | MathDB 页面 | 原文线索

V2-049514 | MATHDB-342803

Faudree-Gould-Jacobson-Lesniak-Saito conjecture on 2-factors in Hamiltonian graphs

摘录 / EXCERPT: Faudree-Gould-Jacobson-Lesniak-Saito conjecture. For any $k\in\mathbb{N}$ there are constants $c_k<1/2$, n_k and a_k such that any Hamiltonian graph G of order $n\ge n_k$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-049515 | MATHDB-342826

Erdős-Simonovits-Sós rainbow cycle conjecture

摘录 / EXCERPT: Let G be an edge-colored graph of order n, and let $c(G)$ denote the number of colors appearing on its edges. Erdős-Simonovits-Sós conjecture. For all $n\geq k\geq 3$, if ... the...

conjecture | open | MathDB 页面 | 原文线索

V2-049516 | MATHDB-342830

Packing conjecture for triple systems

摘录 / EXCERPT: Packing conjecture. If $p\neq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049517 | MATHDB-342834

The spanning eulerian subdigraph avoidance conjecture for semicomplete digraphs

摘录 / EXCERPT: Spanning eulerian subdigraph avoidance conjecture. Every $(k+1)$-arc-strong semicomplete digraph D has a spanning eulerian subdigraph that avoids any prescribed set of k arcs.

conjecture | open | MathDB 页面 | 原文线索

V2-049518 | MATHDB-342835

The fixed-arc deletion algorithm conjecture for semicomplete digraphs

摘录 / EXCERPT: Fixed-arc deletion algorithm conjecture. For each fixed positive integer k, there exists a polynomial-time algorithm which, given a semicomplete digraph $D=(V,A)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-049519 | MATHDB-342837

Homology concentration conjecture for nucleus complexes

摘录 / EXCERPT: Homology concentration conjecture. The group $\widetilde H_k(\Delta^G_U;\mathbb{R})$ is nonzero only in the following cases:

conjecture | open | MathDB 页面 | 原文线索

V2-049520 | MATHDB-342838

Fusion-rule conjecture for a family of rational lisse vertex algebras

摘录 / EXCERPT: Fusion-rule conjecture. There exists such a rational lisse vertex algebra whose irreducible modules satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-049521 | MATHDB-342839

Unitarity conjecture for the exceptional vertex operator algebras U_6 and U_7

摘录 / EXCERPT: Let ... The source observes that both vertex algebras are positive, meaning that every irreducible module other than the algebra itself has positive conformal dimension. Unitarity co...

conjecture | open | MathDB 页面 | 原文线索

V2-049522 | MATHDB-342841

Dorbec et al.'s power domination bound for connected regular graphs

摘录 / EXCERPT: Let G be a connected r-regular graph of order n, with $r\ge 3$, and let $k\ge 1$. Write $\gamma_{p,k}(G)$ for the minimum cardinality of a k-power dominating set of G. As...

conjecture | open | MathDB 页面 | 原文线索

V2-049523 | MATHDB-342864

The DWBC3 domino-tiling conjecture for the 20V model

摘录 / EXCERPT: Let B_n be the sequence associated with 20V configurations with DWBC3 boundary conditions, and let $b_n=T({\mathcal T}_n)$ denote the number of domino tilings of the triangle...

conjecture | open | MathDB 页面 | 原文线索

V2-049524 | MATHDB-342865

The extended DWBC3 conjecture for 20V configurations and domino tilings

摘录 / EXCERPT: Let $P_{n,k}$ be the pentagonal domain carrying the extended DWBC3 boundary conditions, and let ${\mathcal T}_{n,k}$ be the corresponding extended triangle whose domino tilings are...

conjecture | open | MathDB 页面 | 原文线索

V2-049525 | MATHDB-342866

The DWBC4 Schröder-path enumeration conjecture

摘录 / EXCERPT: For nonnegative integers a,b,c, let $N_{a,b,c}$ denote the number of 20V configurations on the rectangular grid of size $(a+b+1)\times(b+c+1)$ with DWBC4-type boundary conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-049526 | MATHDB-342870

Universal-enveloping-algebra conjecture for the local line Quot CoHA module

摘录 / EXCERPT: Let Q be the three-loop quiver obtained from Q_r by removing the framing vertex and all arrows containing it. For framing vector $\mathbf f=1$ and stability parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-049527 | MATHDB-342880

Benjamini's conjecture on expanders at all scales

摘录 / EXCERPT: An infinite, connected, bounded-degree graph $\mathcal G$ is an expander at all scales if there exists $\varepsilon>0$ such that every ball $B\subset \mathcal G$ and subset...

conjecture | open | MathDB 页面 | 原文线索

V2-049528 | MATHDB-342881

Gaiotto-Creutzig conjecture on vertex operator superalgebras for S-duality

摘录 / EXCERPT: Gaiotto-Creutzig conjecture. The object $A^n[\mathfrak g,\psi]$ can be given the structure of a simple vertex operator superalgebra.

conjecture | open | MathDB 页面 | 原文线索

V2-049529 | MATHDB-342882

The c-unbounded lower-bound conjecture for hypergraph Ramsey numbers

摘录 / EXCERPT: Let $k\geq 10$ be an integer, and let p and q be integers satisfying $p\geq k+2$ and $q\geq k+1$. Write $r_k(p,q)$ for the smallest integer n such that every red-blue colorin...

conjecture | open | MathDB 页面 | 原文线索

V2-049530 | MATHDB-342909

The general Turán number conjecture for disjoint cliques in multipartite graphs

摘录 / EXCERPT: The general multipartite Turán conjecture. The Turán number of the disjoint union kK_t in the complete r-partite graph is

conjecture | open | MathDB 页面 | 原文线索

V2-049531 | MATHDB-342911

The eventual extremal conjecture for triangles in K_4-saturated graphs

摘录 / EXCERPT: Let t be an integer with $t\geq 4$. For an n-vertex K_4-saturated graph, let $\operatorname{sat}_t(n,K_3,K_4)$ denote the minimum number of triangles among graphs with minimu...

conjecture | open | MathDB 页面 | 原文线索

V2-049532 | MATHDB-342925

Li-Nikiforov-Schelp cycle conjecture for dense graphs

摘录 / EXCERPT: Let G be a graph of order $n\geq 4$, let $\delta(G)$ denote its minimum degree, and write a 2-edge-coloring as $E(G)=E(R)\cup E(B)$. Let C_ℓ denote the cycle with ℓ v...

conjecture | open | MathDB 页面 | 原文线索

V2-049533 | MATHDB-342927

Benevides-Łuczak-Scott-Skokan-White monochromatic circumference conjecture

摘录 / EXCERPT: Let G be a graph of order n, let $n=3t+r$ with $r\in\{0,1,2\}$, and write a 2-edge-coloring as $E(G)=E(R_G)\cup E(B_G)$. The monochromatic circumference is the largest length...

conjecture | open | MathDB 页面 | 原文线索

V2-049534 | MATHDB-342942

Stahl's multichromatic-number conjecture for Kneser graphs

摘录 / EXCERPT: Let $K(m,n)$ be the Kneser graph whose vertices are the n-subsets of $\{1,\ldots,m\}$, with two vertices adjacent when the corresponding subsets are disjoint. For an integer r,...

conjecture | open | MathDB 页面 | 原文线索

V2-049535 | MATHDB-342950

Hefetz, Krivelevich and Tan's asymptotic Client-Waiter odd cycle game conjecture

摘录 / EXCERPT: Let $\mathcal{OC}_n$ denote the family of odd cycles on the board of the Client-Waiter game, and let $b_{cw}(\mathcal{OC}_n)$ be its threshold bias. Hefetz, Krivelevich and Tan's c...

conjecture | open | MathDB 页面 | 原文线索

V2-049536 | MATHDB-342955

The gated-hull characterization conjecture for two-dimensional partial cubes

摘录 / EXCERPT: Gated-hull characterization conjecture. If the gated hull of each isometric cycle of G is a disk or a full subdivision, then G is two-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-049537 | MATHDB-342956

Pach-Tardos conjecture for ordered uniform forests

摘录 / EXCERPT: Pach-Tardos conjecture for ordered forests. Let $r\geq 2$. Then for any ordered r-uniform forest F with interval chromatic number r,

conjecture | open | MathDB 页面 | 原文线索

V2-049538 | MATHDB-342957

The maximum-diamond conjecture for tournaments of order congruent to 1 modulo 4

摘录 / EXCERPT: Maximum-diamond conjecture. The number of diamonds in T is at most

conjecture | open | MathDB 页面 | 原文线索

V2-049539 | MATHDB-342964

Expectation-threshold conjecture for nonlinear Hamiltonian cycles

摘录 / EXCERPT: Let $G^{(r)}(n,p)$ be the binomial random r-uniform hypergraph, and let X_ℓ count its Hamiltonian ℓ-cycles. Expectation-threshold conjecture. For all integers...

conjecture | open | MathDB 页面 | 原文线索

V2-049540 | MATHDB-342965

Degree-threshold conjecture for linear Hamiltonian cycles

摘录 / EXCERPT: Let $G^{(r)}(n,p)$ be the binomial random r-uniform hypergraph. For $r\geq 3$, define ... This is the sharp threshold for the disappearance of isolated vertices. Degree-threshold c...

conjecture | open | MathDB 页面 | 原文线索

V2-049541 | MATHDB-342980

Lovász-Simonovits conjecture on the Erdős-Rademacher problem

摘录 / EXCERPT: Let K_r denote the complete graph on r vertices. For integers n and m with $0\leq m\leq {n\choose 2}$, let $G_r(n,m)$ be the minimum number of copies of K_r in an (n,m)...

conjecture | open | MathDB 页面 | 原文线索

V2-049542 | MATHDB-342985

The edge-addition conjecture for rooted phylogenetic networks

摘录 / EXCERPT: Let N be a rooted phylogenetic network on leaf set X. The quantities $\ell(N)$ and $e(N)$ measure, respectively, the tree-based proximity of N and the minimum number of direc...

conjecture | open | MathDB 页面 | 原文线索

V2-049543 | MATHDB-342989

Gupta's co-density conjecture for multigraph cover index

摘录 / EXCERPT: Let G be a multigraph. Its cover index $\xi(G)$ is the maximum number of pairwise disjoint edge covers of G, and its minimum degree is $\delta(G)$. Define the co-density by ... w...

conjecture | open | MathDB 页面 | 原文线索

V2-049544 | MATHDB-342990

Gupta's characterization conjecture for the cover-index lower bound

摘录 / EXCERPT: Let G be a multigraph, with minimum degree $\delta(G)$ and maximum edge multiplicity $\mu(G)$. Suppose that $\delta(G)$ cannot be expressed in the form ... for any integers p and...

conjecture | open | MathDB 页面 | 原文线索

V2-049545 | MATHDB-342998

The unrolled quantum group-singlet correspondence conjecture

摘录 / EXCERPT: Quantum group-singlet correspondence conjecture. The category of weight modules of $\overline{U}_q^H(\mathfrak{sl}_{2})$ is braided equivalent to that category of $\mathcal{M}(p)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-049546 | MATHDB-343012

The maximal 3-gamma_c-vertex critical graph Hamiltonian connectivity conjecture

摘录 / EXCERPT: Hamiltonian connectivity conjecture. Then G is either the 5-cycle C_{5} or hamiltonian connected.

conjecture | open | MathDB 页面 | 原文线索

V2-049547 | MATHDB-343017

Nekrasov's conjecture for the empty fourfold DT vertex

摘录 / EXCERPT: Let t_1,t_2,t_3,t_4,y,q be formal variables satisfying $t_1t_2t_3t_4=1$. For any formal variable x, write $[x]=x^{\frac12}-x^{-\frac12}$, and let Exp denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-049548 | MATHDB-343019

Cao-Kool's cohomological DT/PT vertex correspondence

摘录 / EXCERPT: Let λ,μ,ν,ρ be finite plane partitions, with at most two of them non-empty in the stable-pairs case. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049549 | MATHDB-343027

Polynomial bound conjecture for edge-ordered Ramsey numbers of degenerate graphs

摘录 / EXCERPT: An edge-ordered graph is a graph whose edges are equipped with a linear ordering. A graph is d-degenerate if every induced subgraph has a vertex of degree at most d. For an edg...

conjecture | open | MathDB 页面 | 原文线索

V2-049550 | MATHDB-343068

Domination versus edge domination in regular graphs

摘录 / EXCERPT: Let G be a finite, simple, undirected Δ-regular graph with $\Delta\geq 1$. The domination number $\gamma(G)$ is the minimum cardinality of a dominating vertex set, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-049551 | MATHDB-343074

Partite-set comparison conjecture for expected propagation times of complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ have partite vertex sets R and R' of orders m and n respectively, and let $u\in R$ and $v\in R'$. Partite-set comparison conjecture. If $n>3$ and $m<n$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-049552 | MATHDB-343075

Limiting increment conjecture for expected propagation times of n-Suns

摘录 / EXCERPT: For the $n\operatorname{-Sun}$ graph, let $\operatorname{ept}(n\operatorname{-Sun})$ denote its expected propagation time. Limiting increment conjecture. ... The conjecture is motiva...

conjecture | open | MathDB 页面 | 原文线索

V2-049553 | MATHDB-343113

Linear degree-three-vertex conjecture for minimal bricks

摘录 / EXCERPT: Linear degree-three-vertex conjecture. There exists $\alpha > 0$ such that every minimal brick G has at least

conjecture | open | MathDB 页面 | 原文线索

V2-049554 | MATHDB-343114

Neumann-Lara-Škrekovski conjecture on the dichromatic number of planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let $\vec{\chi}(G)$ denote the maximum, over all orientations of G, of the minimum number of acyclic colour classes in a vertex colouring. Neumann-...

conjecture | open | MathDB 页面 | 原文线索

V2-049555 | MATHDB-343124

Bang-Jensen-Thomassé's arc-connectivity conjecture for supereulerian digraphs

摘录 / EXCERPT: Bang-Jensen-Thomassé's conjecture. Every digraph D with $\lambda(D)\geq\alpha(D)$ is supereulerian.

conjecture | open | MathDB 页面 | 原文线索

V2-049556 | MATHDB-343125

Arc-avoidance conjecture for supereulerian semicomplete digraphs

摘录 / EXCERPT: Arc-avoidance conjecture. The digraph $D\setminus A'$ is supereulerian.

conjecture | open | MathDB 页面 | 原文线索

V2-049557 | MATHDB-343145

The independence-number bound for clique immersions

摘录 / EXCERPT: Let G be an n-vertex graph, and let $\alpha(G)$ denote its independence number. Independence-number clique immersion conjecture. The graph G contains ... as an immersion. This...

conjecture | open | MathDB 页面 | 原文线索

V2-049558 | MATHDB-343164

Jacobian-dual formulation of the Caccetta-Häggkvist conjecture

摘录 / EXCERPT: Let D be a digraph on n vertices, let $G(D)$ be its associated bipartite graph, let M_D be the perfect matching associated with D, and let ϕ be the Taylor presentation...

conjecture | open | MathDB 页面 | 原文线索

V2-049559 | MATHDB-343165

Rees-algebra formulation of the Caccetta-Häggkvist conjecture

摘录 / EXCERPT: Let D be a digraph on n vertices, let $G(D)$ be its associated bipartite graph, and let M_D be the perfect matching associated with D. If J is the defining ideal of the R...

conjecture | open | MathDB 页面 | 原文线索

V2-049560 | MATHDB-343173

Aldous-Fill spectral-gap conjecture for regular graphs

摘录 / EXCERPT: Aldous-Fill spectral-gap conjecture. The spectral gap of a connected k-regular graph on n vertices is at least

conjecture | open | MathDB 页面 | 原文线索

V2-049561 | MATHDB-343174

Conjectured structure of quartic graphs with minimum spectral gap

摘录 / EXCERPT: Quartic minimum-gap conjecture. The connected quartic graph on $n\geq 11$ vertices with minimum spectral gap is the unique graph G described as follows: G consists of q middl...

conjecture | open | MathDB 页面 | 原文线索

V2-049562 | MATHDB-343177

The maximum-independent-sets conjecture for k-chromatic ℓ-connected graphs with k at most ℓ

摘录 / EXCERPT: Let G^* denote the graph characterized in the paper as the extremal n-vertex k-chromatic ... -connected graph, then ... Theorem characterizes G^* for large n, and this conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-049563 | MATHDB-343178

The maximum-independent-sets conjecture for k-chromatic ℓ-connected graphs with k greater than ℓ

摘录 / EXCERPT: Let G^* denote the graph characterized in the paper as the extremal n-vertex k-chromatic ... -connected graph, then ... The conjecture is known for $k=3$ and $=2$ when $n5$, whil...

conjecture | open | MathDB 页面 | 原文线索

V2-049564 | MATHDB-343183

Beck-Fiala conjecture for regular hypergraphs

摘录 / EXCERPT: Beck-Fiala conjecture. For a t-regular hypergraph $\mathcal{H}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049565 | MATHDB-343184

Constant discrepancy conjecture for random regular hypergraphs

摘录 / EXCERPT: Constant discrepancy conjecture. There is an absolute constant $K>0$ such that, for every integer $t>0$, a random t-regular hypergraph $\mathcal{H}$ on n vertices and...

conjecture | open | MathDB 页面 | 原文线索

V2-049566 | MATHDB-343185

Koolen-Yang integrability conjecture for graphs with smallest eigenvalue at least -3

摘录 / EXCERPT: Koolen-Yang conjecture. There exists a constant c such that every unsigned graph G with smallest eigenvalue at least -3 is c-integrable.

conjecture | open | MathDB 页面 | 原文线索

V2-049567 | MATHDB-343186

Gregory's signing conjecture for irregular graphs

摘录 / EXCERPT: Gregory's conjecture. There exists a signature σ such that

conjecture | open | MathDB 页面 | 原文线索

V2-049568 | MATHDB-343187

Bukh-Balla-Dräxler-Keevash-Sudakov conjecture on equiangular-line bounds

摘录 / EXCERPT: Bukh-Balla-Dräxler-Keevash-Sudakov conjecture. For sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-049569 | MATHDB-343188

The p-centered coloring conjecture for planar graphs

摘录 / EXCERPT: A p-centered coloring of a graph is a vertex coloring such that every connected subgraph either receives more than p colors or has a color appearing exactly once. Planar graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049570 | MATHDB-343192

Non-biregular construction conjecture for critical bipartite graphs

摘录 / EXCERPT: Non-biregular construction conjecture. The graph $(U,V;E)$ is k-critical bipartite.

conjecture | open | MathDB 页面 | 原文线索

V2-049571 | MATHDB-343193

Brown-Erdős-Sós and Alon-Shapira conjecture on sparse hypergraph exponents

摘录 / EXCERPT: Let $f_r(n,v,e)$ denote the maximum number of edges in an r-graph on n vertices such that every set of v vertices spans at most $e-1$ edges. For fixed integers $r>k\ge 2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-049572 | MATHDB-343209

The positive-density patterns conjecture for affine buildings

摘录 / EXCERPT: Positive-density patterns conjecture. Every positive-density subset of the vertex set of an irreducible affine building contains these arbitrarily large prescribed configurations w...

conjecture | open | MathDB 页面 | 原文线索

V2-049573 | MATHDB-343220

The toroidal graph without configurations in E DP-4-degeneracy conjecture

摘录 / EXCERPT: Let G be a toroidal graph with no subgraph isomorphic to any configuration in the family denoted by E. Let (H,f) be a cover of G, where f assigns to each vertex of H a...

conjecture | open | MathDB 页面 | 原文线索

V2-049574 | MATHDB-343221

The planar and toroidal graphs without intersecting triangles DP-4-degeneracy conjecture

摘录 / EXCERPT: Let G be a planar or toroidal graph without intersecting triangles. Let (H,f) be a cover of G, where f assigns to each vertex of H a value in $\{0,1,2\}$. A strictly f-...

conjecture | open | MathDB 页面 | 原文线索

V2-049575 | MATHDB-343223

Condensed Ricci curvature conjecture for strongly regular conference graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a strongly regular conference graph with parameters ... where $\beta\geq 2$. The condensed Ricci curvature conjecture. The condensed Ricci curvature satisfies ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-049576 | MATHDB-343225

The independence number conjecture for even Johnson-type graphs $J_{\pm}(n,4,1)$

摘录 / EXCERPT: Let $J_{\pm}(n,4,1)$ be the Johnson-type graph with parameters n, 4, and 1, and let $\alpha[G]$ denote the independence number of a graph G. Independence number conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-049577 | MATHDB-343238

Conjectural Brouwer inequality for Laplacian spectra of simplicial complexes

摘录 / EXCERPT: Generalized Brouwer conjecture. For every $t \in \mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049578 | MATHDB-343240

Henning-Oellermann-Swart conjecture on the Steiner diameter-to-radius ratio

摘录 / EXCERPT: Let G be a connected graph of order at least k, and let $\operatorname{srad}_k(G)$ and $\operatorname{sdiam}_k(G)$ denote its Steiner k-radius and Steiner k-diameter, respe...

conjecture | open | MathDB 页面 | 原文线索

V2-049579 | MATHDB-343251

Jackson's Tutte-cycle conjecture for 2-connected claw-free graphs

摘录 / EXCERPT: Jackson's conjecture. Every 2-connected claw-free graph has a Tutte cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-049580 | MATHDB-343252

Li et al.'s Tutte-path conjecture for rank-three hypergraph line graphs

摘录 / EXCERPT: Li et al.'s conjecture. For every pair of vertices a,b of a connected line graph of a hypergraph of rank at most 3, there is a maximal (a,b)-path which is a Tutte path.

conjecture | open | MathDB 页面 | 原文线索

V2-049581 | MATHDB-343253

Tutte-path conjecture for connected claw-free graphs

摘录 / EXCERPT: Claw-free Tutte-path conjecture. For every pair of vertices a,b of a connected claw-free graph, there is a maximal (a,b)-path which is a Tutte path.

conjecture | open | MathDB 页面 | 原文线索

V2-049582 | MATHDB-343254

Conjectured analogue of the minimum-degree zero-one law for k-connectivity

摘录 / EXCERPT: Zero-one law conjecture for k-connectivity. Under this scaling,

conjecture | open | MathDB 页面 | 原文线索

V2-049583 | MATHDB-343270

Approximate Jackson conjecture for regular bipartite digraphs

摘录 / EXCERPT: Let G be a cn-regular bipartite digraph on $2n$ vertices, meaning that every vertex has indegree and outdegree cn. A Hamilton cycle decomposition is a partition of the edge s...

conjecture | open | MathDB 页面 | 原文线索

V2-049584 | MATHDB-343271

Approximate Hamilton packing conjecture for diregular bipartite tournaments

摘录 / EXCERPT: Let G be a diregular bipartite tournament on $2n$ vertices, so G is a complete bipartite orientation with equal indegree and outdegree at every vertex. Edge-disjoint Hamilton c...

conjecture | open | MathDB 页面 | 原文线索

V2-049585 | MATHDB-343272

Approximate Hamilton packing conjecture for regular tripartite digraphs

摘录 / EXCERPT: Let G be a cn-regular tripartite digraph with three vertex classes, each of size n, meaning that every vertex has indegree and outdegree cn. A Hamilton cycle is a directed...

conjecture | open | MathDB 页面 | 原文线索

V2-049586 | MATHDB-343315

Nikiforov's conjecture on the Lagrangian of colex hypergraphs

摘录 / EXCERPT: Let r-graphs be finite uniform hypergraphs, let $\lambda(m,r)$ denote the maximum Lagrangian of an r-graph with m edges, and let $\binom{x}{r}$ be the generalized binomial co...

conjecture | open | MathDB 页面 | 原文线索

V2-049587 | MATHDB-343319

Conjecture on the injective chromatic index of subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning a graph with maximum degree at most 3. The injective chromatic index, denoted by $\chi'_{\operatorname{inj}}(G)$, is the minimum number of co...

conjecture | open | MathDB 页面 | 原文线索

V2-049588 | MATHDB-343320

Conjecture on the injective chromatic index of subcubic bipartite graphs

摘录 / EXCERPT: Let G be a subcubic bipartite graph, meaning a bipartite graph with maximum degree at most 3. Write $\chi'_{\operatorname{inj}}(G)$ for its injective chromatic index. Subcubic...

conjecture | open | MathDB 页面 | 原文线索

V2-049589 | MATHDB-343348

Type-one tree complement conjecture for multiplicity bipartition

摘录 / EXCERPT: Let $T=S^k_{m,n}$ be the tree formed by taking a path on k vertices and adding m leaves to one endpoint and n leaves to the other, where $k\in\{2,3,4\}$ and $m,n>1$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049590 | MATHDB-343349

Join conjecture with maximum multiplicity parameter

摘录 / EXCERPT: Let G and H be graphs such that $q(G)=q(H)=2$, with $MB(G)=k_1$ and $MB(H)=k_2$, and suppose that the quantities denoted in the source by $\mrr+(G)$ and $\mrr+(H)$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-049591 | MATHDB-343360

Common minimal-tree conjecture for totally real algebraic integers

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $\lambda_1,\ldots,\lambda_k\in\mathtt{TRAI}\setminus\{0\}$. For each λ_i, let $\mathcal{T}_{\mathrm{min},\lambda_i}$ denote the set of trees ha...

conjecture | open | MathDB 页面 | 原文线索

V2-049592 | MATHDB-343361

Infinite minimal Laplacian-tree conjecture

摘录 / EXCERPT: Let μ be a Laplacian eigenvalue of some tree T, with $\mu\neq 1$. Let $\mathcal{LT}_{\mathrm{min},\mu}^{0}$ and $\mathcal{LT}_{\mathrm{min},\mu}^{1}$ denote the two classes o...

conjecture | open | MathDB 页面 | 原文线索

V2-049593 | MATHDB-343362

Totally positive algebraic integer Laplacian-eigenvalue conjecture

摘录 / EXCERPT: A totally positive algebraic integer is an algebraic integer all of whose algebraic conjugates are positive real numbers. Totally positive algebraic integer Laplacian-eigenvalue co...

conjecture | open | MathDB 页面 | 原文线索

V2-049594 | MATHDB-343364

The avoidable induced path conjecture

摘录 / EXCERPT: Let G be a graph, and let P_k denote an induced path on k vertices. An induced path P is avoidable if every extension of P is contained in an induced cycle, where an exte...

conjecture | open | MathDB 页面 | 原文线索

V2-049595 | MATHDB-343365

The Arman-Tsaturian extremal cycle-count conjecture

摘录 / EXCERPT: Arman-Tsaturian conjecture. Every simple graph with m edges has at most

conjecture | open | MathDB 页面 | 原文线索

V2-049596 | MATHDB-343369

Jiang-Polyanskii conjecture for infinite spectral radius order

摘录 / EXCERPT: Jiang-Polyanskii conjecture. If $k(\lambda)=\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049597 | MATHDB-343370

The feasible ideal-point conjecture for the MIN NODES-MIN DOUBLE problem

摘录 / EXCERPT: Let $\mathcal{V}$ be the vertex set of a discretizable distance geometry problem, let $rank$ assign vertices to ranks, and let $nodes(rank(v))$ denote the number of nodes at the ra...

conjecture | open | MathDB 页面 | 原文线索

V2-049598 | MATHDB-343382

Independent-set bound for graphs excluding a clique minor

摘录 / EXCERPT: Let t be a positive integer, let G be an n-vertex graph, and let $\alpha(G)$ denote the size of its largest independent set. Independent-set bound for clique-minor-free graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049599 | MATHDB-343415

Gyárfás's extremal conjecture for monochromatic components in Steiner triple systems

摘录 / EXCERPT: Let $\mathcal{S}_n$ be the family of Steiner triple systems on n points. For a Steiner triple system S, let $\alpha^*_3(S)$ denote the maximum size of a 3-independent set, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049600 | MATHDB-343418

Vertex-function formula for tangent characters under 3d-mirror symmetry

摘录 / EXCERPT: Vertex-function character formula. The vertex functions of X with vanishing equivariant parameters are given by the Taylor series expansions of

conjecture | open | MathDB 页面 | 原文线索

V2-049601 | MATHDB-343424

Leaf-reflection conjecture for the uniformed patroller game

摘录 / EXCERPT: Leaf-reflection conjecture. For any network and any attack duration, it is optimal to always reflect at leaf nodes.

conjecture | open | MathDB 页面 | 原文线索

V2-049602 | MATHDB-343428

Extremal number conjecture for matching blow-ups of complete bipartite graphs

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with $s\leq t$, let $p\geq 3$, and let $\mathcal{M}(K^{p+1}_{s,t})$ denote the matching blow-up family used in the paper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049603 | MATHDB-343429

Revised conjecture for the non-monochromatic extremal function

摘录 / EXCERPT: Let G be a given graph, and let $g(n,G)$ denote the maximum number of edges not contained in any monochromatic copy of G in a 2-edge-coloring of K_n. Let $\emph{ex}(n,G)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-049604 | MATHDB-343435

Janzer's conjecture on blowups of spiders

摘录 / EXCERPT: Let $s\geq2$, $k,b,t\geq1$ be integers. Let S be an s-legged spider whose longest leg has length k, and suppose that $e(S)=(s-1)k+b$, where $0\leq b\leq k$. Let $t*S$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-049605 | MATHDB-343451

The regularity conjecture for graphs with bounded edge-ideal regularity

摘录 / EXCERPT: Let G be a finite simple graph, let $I(G)$ be its edge ideal, and let G^c denote its complement. Regularity conjecture for graphs with bounded edge-ideal regularity. If ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-049606 | MATHDB-343459

The von Neumann neighborhood conjecture for graph codes

摘录 / EXCERPT: Let A be the adjacency matrix of a graph Γ on n vertices with vertex set V. Let C be the binary linear code generated by $[I_n|A]$, and let $d(C)$ denote its minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-049607 | MATHDB-343473

Conjectured Laplacian spectrum of Sierpiński graphs

摘录 / EXCERPT: Laplacian spectrum conjecture. The Laplacian spectrum of $S(n,k)$ consists of the following eigenvalues:

conjecture | open | MathDB 页面 | 原文线索

V2-049608 | MATHDB-343478

Signless Laplacian extremal conjecture for large $K_{s,t}$-minor free graphs

摘录 / EXCERPT: Signless Laplacian extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049609 | MATHDB-343481

Democracy coefficient boundedness and balance conjecture

摘录 / EXCERPT: Let G be a weakly connected directed graph, and let $\eta_f(G)$ and $\eta_b(G)$ denote its forward and backward democracy coefficients. Democracy coefficient boundedness and bala...

conjecture | open | MathDB 页面 | 原文线索

V2-049610 | MATHDB-343497

Equitable Vertex Arboricity Conjecture for bounded-degree graphs

摘录 / EXCERPT: Equitable Vertex Arboricity Conjecture. Every graph with maximum degree at most Δ is equitably tree-k-colorable for every integer $k\geq (\Delta+1)/2$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-049611 | MATHDB-343498

Planar equitable vertex arboricity conjecture

摘录 / EXCERPT: Planar equitable vertex arboricity conjecture. There exists a constant k such that every planar graph is equitably tree-k'-colorable for every integer $k'\geq k$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-049612 | MATHDB-343499

Planar equitable tree-3-coloring conjecture

摘录 / EXCERPT: Planar equitable tree-3-coloring conjecture. Every planar graph is equitably tree-3-colorable; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-049613 | MATHDB-343500

Bounded equitable vertex arborable threshold for planar graphs

摘录 / EXCERPT: Let G be a simple finite planar graph, and let $va^{\equiv}(G)$ denote its equitable vertex arborable threshold: the minimum integer k such that G admits an equitable tree-...

conjecture | open | MathDB 页面 | 原文线索

V2-049614 | MATHDB-343507

The Kővári-Sós-Turán conjecture for complete bipartite graphs

摘录 / EXCERPT: Kővári-Sós-Turán conjecture. For every $t,s\in\mathbb{N}$ with $t\leq s$,

conjecture | open | MathDB 页面 | 原文线索

V2-049615 | MATHDB-343509

The ESN embedding conjecture for generic reservoir matrices

摘录 / EXCERPT: Let M) be a compact m-manifold and let $n\in\mathbb{N}$ satisfy $n>2m$. Let A be an $n\times n$ matrix with $\lVert A\rVert_2<\min(1/\lVert D\phi^{-1}\rVert_{\infty},1)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049616 | MATHDB-343526

The linkedness characterization of the two-agent price of connectivity

摘录 / EXCERPT: Let G be a biconnected graph. For positive integers a,b, say that G is (a,b)-linked if, for every pair of disjoint vertex sets M_1,M_2 with $|M_1|=a$ and $|M_2|=b$, there...

conjecture | open | MathDB 页面 | 原文线索

V2-049617 | MATHDB-343527

Generalized weak-subdivision conjecture for complete graphs

摘录 / EXCERPT: Generalized weak-subdivision conjecture. For every $\epsilon>0$ and integer $t\geq 3$, there exists $\delta>0$ such that, if G has at most

conjecture | open | MathDB 页面 | 原文线索

V2-049618 | MATHDB-343528

Admissible-subgraph density conjecture for partial subdivisions of complete graphs

摘录 / EXCERPT: Admissible-subgraph density conjecture. If (Q,ϕ) has no H-admissible subgraph for any $H\in\mathcal{H}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049619 | MATHDB-343530

The $K_{4,6}$ counting conjecture for projective norm graphs

摘录 / EXCERPT: Let $\operatorname{NG}(q,4)$ be the projective norm graph over the parameter q, and let $K_{4,6}$ be the complete bipartite graph with parts of sizes 4 and 6. $K_{4,6}$ count...

conjecture | open | MathDB 页面 | 原文线索

V2-049620 | MATHDB-343531

The $K_{4,6}$ extremal-number conjecture

摘录 / EXCERPT: Let $\operatorname{ex}(n,K_{4,6})$ be the maximum number of edges in an n-vertex graph containing no copy of $K_{4,6}$. $K_{4,6}$ extremal-number conjecture. ... The conjecture ref...

conjecture | open | MathDB 页面 | 原文线索

V2-049621 | MATHDB-343533

Hibi-Matsuda-Tsuchiya conjecture on linear resolutions of edge rings

摘录 / EXCERPT: Let G be a finite connected simple graph, let $K[G]$ denote its edge ring over a field K, and let $q\geq 3$. A graded ring has a q-linear resolution when its minimal graded f...

conjecture | open | MathDB 页面 | 原文线索

V2-049622 | MATHDB-343536

Equitable partitions of rectangular grid graphs

摘录 / EXCERPT: Grid-graph equitable-partition conjecture. The equitable partitions of $P_m\oblong P_n$ are the orbit partitions of natural actions of the subgroups of the automorphism group of th...

conjecture | open | MathDB 页面 | 原文线索

V2-049623 | MATHDB-343537

The multiset-dimension upper-bound conjecture for graphs

摘录 / EXCERPT: Multiset-dimension upper-bound conjecture. If G has finite multiset dimension, then

conjecture | open | MathDB 页面 | 原文线索

V2-049624 | MATHDB-343547

Minimum-cost homologous circulation algorithm for embedded directed graphs

摘录 / EXCERPT: Let D be a directed graph cellularly embedded on a possibly non-orientable surface $\mathbb{S}$ of Euler genus g, let $c\colon A(D)\to\mathbb{R}_{\geq 0}$ be a cost function, a...

open_problem | open | MathDB 页面 | 原文线索

V2-049625 | MATHDB-343554

Polynomial-time safe-set algorithm for clique-acyclic digraphs

摘录 / EXCERPT: Safe-set algorithm conjecture. There exists a polynomial-time algorithm to find a minimum safe set in a clique acyclic digraph with a constant independence number α.

conjecture | open | MathDB 页面 | 原文线索

V2-049626 | MATHDB-343555

Safe set is NP-complete for strong tournaments

摘录 / EXCERPT: Strong-tournament safe-set conjecture. The safe-set problem is NP-complete for strong tournaments.

conjecture | open | MathDB 页面 | 原文线索

V2-049627 | MATHDB-343595

The degree bound conjecture for Snort on boards

摘录 / EXCERPT: Let B be a board for Snort, and let the degree of B denote its maximum vertex degree. The temperature of a Snort position on B measures the incentive to move. The degree boun...

conjecture | open | MathDB 页面 | 原文线索

V2-049628 | MATHDB-343596

Kneser graph packing conjecture

摘录 / EXCERPT: For integers $n>k\geq 1$, let $\operatorname{KG}(n,k)$ be the graph whose vertices are the k-subsets of $[n]$, with two vertices adjacent exactly when the corresponding sets are...

conjecture | open | MathDB 页面 | 原文线索

V2-049629 | MATHDB-343607

The cherry bound for minimal support networks of HGT networks

摘录 / EXCERPT: Let N be a level-k HGT network with a biconnected component B. Let $N(B)$ denote the minimal support network of B, let $T_{N(B)}$ denote its underlying tree, and call a pai...

conjecture | open | MathDB 页面 | 原文线索

V2-049630 | MATHDB-343608

The reticulation lower-bound conjecture for tree-metrizable minimal support networks

摘录 / EXCERPT: Let $N(B)$ be a tree-metrizable minimal support network of a biconnected component B, and suppose that $N(B)$ has n leaves. A reticulation arc is an arc entering a reticulation...

conjecture | open | MathDB 页面 | 原文线索

V2-049631 | MATHDB-343612

Link Conjecture for flag homology spheres

摘录 / EXCERPT: Let Δ be a flag homology sphere and let v be a vertex. Its vertex link is $\operatorname{lk}_v(\Delta)$. Link Conjecture. The link's γ-vector is coefficientwise at...

conjecture | open | MathDB 页面 | 原文线索

V2-049632 | MATHDB-343628

Conjecture on minimally _g-imperfect graphs

摘录 / EXCERPT: Let G be a graph. It is minimally γ_g-imperfect if G is not γ_g-perfect, while every proper induced subgraph of G is γ_g-perfect. The anti-holes...

conjecture | open | MathDB 页面 | 原文线索

V2-049633 | MATHDB-343642

Existence of a topological spanning tree-like subcontinuum in the tangle compactification

摘录 / EXCERPT: Let G be a graph, and let ΛG be its tangle compactification. A TST of ΛG is a spanning standard subspace of ΛG that is also a TST-like subcontinuum. Ex...

conjecture | open | MathDB 页面 | 原文线索

V2-049634 | MATHDB-343643

Characterisation of wild auxiliary arcs between vertices

摘录 / EXCERPT: Let G be a graph and let x and y be distinct vertices of G. Write $\overset{{}_{\;\rightarrow}}{S}(G-xy)$ for the partially ordered set of oriented vertex separators of...

conjecture | open | MathDB 页面 | 原文线索

V2-049635 | MATHDB-343666

The cop-number conjecture for $2P_2$-free graphs

摘录 / EXCERPT: Let G be a graph, and write $c(G)$ for its cop number. The graph G is $2P_2$-free if it has no induced subgraph consisting of two vertex-disjoint copies of the path on two vert...

conjecture | open | MathDB 页面 | 原文线索

V2-049636 | MATHDB-343674

The inclusion chromatic index bound for connected graphs

摘录 / EXCERPT: Let G be a connected graph with minimum degree $\delta\geq 2$ and maximum degree Δ, and let $\chi'_{\subset}(G)$ denote its inclusion chromatic index, the least number of c...

conjecture | open | MathDB 页面 | 原文线索

V2-049637 | MATHDB-343681

Thomassé's cycle-reversion conjecture for dichromatic number

摘录 / EXCERPT: Let D be a potentially infinite digraph. A cycle reversion reverses the edges of a directed cycle; iteratively perform such reversions, requiring that each edge is reversed only...

conjecture | open | MathDB 页面 | 原文线索

V2-049638 | MATHDB-343702

Degree and transmission values in quasi-(λ,n)-distance-balanced graphs

摘录 / EXCERPT: Let G be a quasi-λ-distance-balanced graph. For a vertex u of G, write $\deg(u)$ for its degree and $D(u)=\sum_{v\in V(G)}d(u,v)$ for its total distance (transmission...

conjecture | open | MathDB 页面 | 原文线索

V2-049639 | MATHDB-343720

Eventual vanishing conjecture for discrete singular cubical homology of graphs

摘录 / EXCERPT: Let G be a graph. Write $4\mathcal{H}_d^{\normalfont {\textrm{Cube}}}(G)4$ for its discrete singular cubical homology group in dimension d. Eventual vanishing conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-049640 | MATHDB-343721

Covering conjecture for discrete singular cubical homology of graphs

摘录 / EXCERPT: Let G be a graph, and let ${\mathcal K}=\{K_i\}$ be a covering of G by subgraphs such that every edge, 3-cycle, and 4-cycle of G is contained in some K_i. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-049641 | MATHDB-343777

Linear upper bound conjecture for expected propagation time

摘录 / EXCERPT: Let G be a connected graph with n vertices. The expected propagation time is denoted by $4\operatorname{ept}(G)4$. Expected propagation time conjecture. ... Paths are suggested a...

conjecture | open | MathDB 页面 | 原文线索

V2-049642 | MATHDB-343778

Square-root throttling conjecture for trees

摘录 / EXCERPT: Let G be a connected tree of order n, and let $\operatorname{th_{pzf}}(G)$ denote its probabilistic zero forcing throttling number. Tree throttling conjecture. The maximum poss...

open_problem | open | MathDB 页面 | 原文线索

V2-049643 | MATHDB-343779

Double-logarithmic conjecture for random-graph propagation time

摘录 / EXCERPT: Let $G(n,p)$ be the random graph of order n in which each edge is present independently with probability p. Let $\operatorname{ept}(G(n,p))$ denote its expected propagation tim...

conjecture | open | MathDB 页面 | 原文线索

V2-049644 | MATHDB-343814

Density separation conjecture for polynomial Ramsey classes

摘录 / EXCERPT: For a family $\mathcal{F}$ of 3-graphs, define ... Let $\mathcal{F}_{\mathrm{poly}}$, $\mathcal{F}_{\mathrm{part\text{-}poly}}$, $\mathcal{F}_{\mathrm{polyfact}}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049645 | MATHDB-343827

The inertia-bicoloring conjecture for full-rank typical graphs

摘录 / EXCERPT: Let G be a full-rank typical graph. A G-partial matrix is a partial symmetric matrix whose specified off-diagonal entries correspond to the edges of G; it is minimally comple...

conjecture | open | MathDB 页面 | 原文线索

V2-049646 | MATHDB-343831

McDonald-Puleo-Tennenhouse conjecture for directed triangle packing and covering

摘录 / EXCERPT: Let D be a directed multigraph. Write $\nu_t(D)$ for the maximum number of arc-disjoint directed triangles in D, and write $\tau_t(D)$ for the minimum number of arcs whose remo...

conjecture | open | MathDB 页面 | 原文线索

V2-049647 | MATHDB-343850

Nested Infinite Lucchesi-Younger Conjecture

摘录 / EXCERPT: Let D be a weakly connected digraph, and let $(F,\mathcal{B})$ be an optimal pair, where F is a finitary dijoin and $\mathcal{B}$ is a set of disjoint finite dicuts. The pair i...

conjecture | open | MathDB 页面 | 原文线索

V2-049648 | MATHDB-343851

Finite-character König property conjecture for infinite hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}=(\mathcal{V},\mathcal{E})$ be a hypergraph. It has the König property if there are a matching $\mathcal{M}$ and a cover A such that...

conjecture | open | MathDB 页面 | 原文线索

V2-049649 | MATHDB-343852

Finite-extension König property conjecture for infinite hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}=(\mathcal{V},\mathcal{E})$ be a hypergraph of finite character. It has the König property if there are a matching $\mathcal{M}$ and a cover A such that...

conjecture | open | MathDB 页面 | 原文线索

V2-049650 | MATHDB-343875

Gyárfás and Sárközy's minimum-degree conjecture for monochromatic components

摘录 / EXCERPT: Fix $k\geq 3$. Let G be a graph with n vertices and minimum degree satisfying ... If the edges of G are k-coloured, meaning that each edge receives one of k colours, then G...

conjecture | open | MathDB 页面 | 原文线索

V2-049651 | MATHDB-343876

Limiting characteristic function of the total solvency shock

摘录 / EXCERPT: Let $S^{(0)}_{1}=\sum_{w\ne 1}S^{(0)}_{w1}$ be the total solvency shock transmitted to bank 1 in step 0, conditioned on the type $T_1=T$. Let $\hat f^{(0)}_\Delta(k'\mid T')$,...

conjecture | open | MathDB 页面 | 原文线索

V2-049652 | MATHDB-343878

Surface bound conjecture for signed graph genus

摘录 / EXCERPT: Let S be a surface, let n_S be the largest order of a complete graph that embeds into S, and let $\gamma(G,\sigma)$ and $\gamma^*(G,\sigma)$ denote the relevant orientable an...

conjecture | open | MathDB 页面 | 原文线索

V2-049653 | MATHDB-343883

The conjecture on exactly six exceptional autonomous components

摘录 / EXCERPT: Let a Goldbach factorization graph be a graph associated with an even integer and let an exceptional autonomous component (EAC) denote the special source strongly connected compone...

conjecture | open | MathDB 页面 | 原文线索

V2-049654 | MATHDB-343890

The neighbour-distinguishing index conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph. Write $\operatorname{ndi}(D)$ for its neighbour-distinguishing index and let $\Delta^*(D)$ denote the maximum of the indegree and outdegree of D. Neighbour-...

conjecture | open | MathDB 页面 | 原文线索

V2-049655 | MATHDB-343900

The maximum-average-degree partition conjecture

摘录 / EXCERPT: Maximum-average-degree partition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049656 | MATHDB-343937

Balogh, Kostochka and Treglown's conjecture on degree sequences forcing perfect clique tilings

摘录 / EXCERPT: Balogh, Kostochka and Treglown's conjecture. Then G contains a perfect K_r-tiling, that is, a collection of vertex-disjoint copies of K_r covering all vertices of G.

conjecture | open | MathDB 页面 | 原文线索

V2-049657 | MATHDB-343946

The general-graph rainbow matching conjecture

摘录 / EXCERPT: Let $f_{\mathcal G}(n,m)$ denote the least number of n-element matchings in an arbitrary graph that guarantees a rainbow matching of size m. Rainbow matching conjecture. ... and,...

conjecture | open | MathDB 页面 | 原文线索

V2-049658 | MATHDB-343952

The bounded-degree rainbow independent-set conjecture

摘录 / EXCERPT: Let $\mathcal D(k)$ be the class of graphs whose vertex degrees are at most k. Bounded-degree conjecture. ... The source gives a general upper bound from graph colouring and says t...

conjecture | open | MathDB 页面 | 原文线索

V2-049659 | MATHDB-343972

Badakhshian-Katona-Tuza asymptotic conjecture for the domination number of $G_{k,2}$

摘录 / EXCERPT: Let $G_{k,2}$ be the bipartite graph with vertex classes $\binom{[n]}{k}$ and $\binom{[n]}{2}$, where a k-element set S is adjacent to a 2-element set T exactly when...

conjecture | open | MathDB 页面 | 原文线索

V2-049660 | MATHDB-343973

Structural conjecture for near-extremal K_{r+1}-free graphs

摘录 / EXCERPT: Structural conjecture. There exists $\alpha_0>0$ such that, for every $0\leq\alpha\leq\alpha_0$ and every K_{r+1}-free graph G on n vertices, there is an unbalanced blow-up...

conjecture | open | MathDB 页面 | 原文线索

V2-049661 | MATHDB-343974

Coproduct conjecture for \(\mathcal{W}_{m|n\times\infty}\)

摘录 / EXCERPT: Let N_1,N_2 be the orders of two pseudo-differential Miura factors, let $\mathcal{L}^{(1)}$ and $\mathcal{L}^{(2)}$ be their product as displayed in the source, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-049662 | MATHDB-343991

Spectral characterization conjecture for fullerene isomers

摘录 / EXCERPT: Spectral characterization conjecture. (a) For all feasible $n\ne44$, two C_n-isomers are isomorphic if and only if they are cospectral with respect to T_n.

conjecture | open | MathDB 页面 | 原文线索

V2-049663 | MATHDB-344015

Strong rationality conjecture for diagonal cosets

摘录 / EXCERPT: Let $C_{k,n}$ be the simple graded quotient of C^{k_1,k_2} at $(k_1,k_2)=(k,n)$, where n is a positive integer and k is a positive integer or an admissible level for...

conjecture | open | MathDB 页面 | 原文线索

V2-049664 | MATHDB-344028

The critical-probability conjecture for linear graphs

摘录 / EXCERPT: Let p_c be the infimum of the probabilities p such that the vertex set is saturated with probability at least $1/2$ for every $p_{\text{open}}\ge p$, and assume that G_0 is t...

conjecture | open | MathDB 页面 | 原文线索

V2-049665 | MATHDB-344052

The clique-and-hub conjecture for the hypergraph variational problem

摘录 / EXCERPT: Clique-and-hub conjecture. The variational problem is asymptotically optimized by planting a clique and a hub.

conjecture | refuted | MathDB 页面 | 原文线索

V2-049666 | MATHDB-344053

The stable-labeling variational asymptotic conjecture

摘录 / EXCERPT: Stable-labeling variational asymptotic conjecture. If $n\to\infty$ and $p=p(n)$ satisfies $n^{-1/\Delta}\ll p\ll 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-049667 | MATHDB-344055

The quarter-order bound for power domination in connected hypergraphs

摘录 / EXCERPT: Quarter-order bound.

conjecture | open | MathDB 页面 | 原文线索

V2-049668 | MATHDB-344110

Independent domination conjecture for Cartesian products

摘录 / EXCERPT: Let G and H be undirected graphs. Write $\gamma(G)$ for the domination number of a graph, and let $i(G)$ denote its independent domination number, the smallest size of an indep...

conjecture | open | MathDB 页面 | 原文线索

V2-049669 | MATHDB-344112

Convergence conjectures for deep linear neural network gradient flows

摘录 / EXCERPT: Assume that $X X^T$ has full rank. Let N be the depth, let $W(t)=W_N(t)\cdots W_1(t)$ denote the product along the gradient flow, and let $\mathcal{M}_k$ denote the rank-constrai...

conjecture | open | MathDB 页面 | 原文线索

V2-049670 | MATHDB-344114

Bang-Jensen et al.'s Meyniel-type conjecture for strong digraphs

摘录 / EXCERPT: Bang-Jensen et al.'s conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049671 | MATHDB-344115

Connectivity-minimum-degree conjecture for Fibonacci (p,r)-cubes

摘录 / EXCERPT: Let $I\Gamma_{n}^{(p,r)}$ and $O\Gamma_{n}^{(p,r)}$ be the Fibonacci-like cubes considered above, with $p\geq1$, $r\geq1$, and $n\geq1$. Connectivity-minimum-degree conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-049672 | MATHDB-344119

Benjamini-Chan-O'Donnell-Tamuz conjecture on limiting majority-dynamics cycles

摘录 / EXCERPT: Let $G(n,p)$ be a random graph with vertex set V_n, let $d=np$, and let $S_t(v)\in\{-1,+1\}$ be the state of vertex v after t rounds of synchronous majority dynamics. Benjami...

conjecture | open | MathDB 页面 | 原文线索

V2-049673 | MATHDB-344135

Baker-Shokrieh's bounded-Jacobian-exponent conjecture for biconnected graphs

摘录 / EXCERPT: Baker-Shokrieh's conjecture. For every positive integer k, there are only finitely many biconnected graphs G such that the exponent of...

conjecture | open | MathDB 页面 | 原文线索

V2-049674 | MATHDB-344153

Chin et al.'s edge-transitive dense graph conjecture for spanning subtrees

摘录 / EXCERPT: Let G_n be a sequence of connected graphs, each with n vertices and $Omega(n^2)$ edges, and suppose that each G_n is edge-transitive. Chin et al.'s conjecture. Then ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-049675 | MATHDB-344154

Chin et al.'s sparse graph conjecture for spanning-subtree probability

摘录 / EXCERPT: Let G_n be a sequence of connected graphs, each with n vertices and $O(n^{\alpha})$ edges, where $\alpha<2$ is fixed. Here $P(G)$ denotes the probability that a uniformly chose...

conjecture | open | MathDB 页面 | 原文线索

V2-049676 | MATHDB-344196

The bounded vertex-deletion growth conjecture for Ramsey numbers

摘录 / EXCERPT: Vertex-deletion Ramsey conjecture. There is an absolute constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-049677 | MATHDB-344198

Thomassé's connected-matching conjecture for graphs with independence number two

摘录 / EXCERPT: Thomassé's conjecture. There is a constant C such that every n-vertex graph G with $\alpha(G)=2$ contains a connected matching of size at least

conjecture | open | MathDB 页面 | 原文线索

V2-049678 | MATHDB-344204

The EKR conjecture for unions of length-2 paths

摘录 / EXCERPT: Let $r \leq n$, and let G be the vertex-disjoint union of n paths each of length 2. A family of independent r-sets of G is intersecting if any two of its members have a c...

conjecture | open | MathDB 页面 | 原文线索

V2-049679 | MATHDB-344227

The 6-cover conjecture for flow-admissible signed graphs

摘录 / EXCERPT: 6-cover conjecture. Every flow-admissible signed graph has a family of signed circuits such that every edge belongs to exactly 6 members of the family.

conjecture | open | MathDB 页面 | 原文线索

V2-049680 | MATHDB-344253

Conlon-Janzer-Lee conjecture for K_{2,t}-free graphs

摘录 / EXCERPT: Let $t\geq 2$ be an integer, and let H be a $K_{2,t}$-free bipartite graph such that every vertex in one of the parts of H has degree at most t. Conlon-Janzer-Lee conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-049681 | MATHDB-344268

The diameter conjecture for one-vertex random Belyi surfaces

摘录 / EXCERPT: Let S_n^O be the random surface in the BM model, and let $\widetilde{S}_n^O$ denote this surface conditioned on having a single vertex. Let $\widetilde{S}_n^C$ be the compactifie...

conjecture | open | MathDB 页面 | 原文线索

V2-049682 | MATHDB-344295

Vega's wedge-of-spheres conjecture for matching complexes of caterpillar graphs

摘录 / EXCERPT: Let G be a caterpillar graph, and let $M_k(G)$ denote its k-matching complex. Vega's conjecture. The k-matching complex $M_k(G)$ is either contractible or homotopy equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-049683 | MATHDB-344319

Identifiability conjecture for Fin and Wing submodels of cycle graphs

摘录 / EXCERPT: Fin-and-Wing submodel conjecture. The linear compartmental model $(G,In,Out,Leak)$ is generically locally identifiable.

conjecture | open | MathDB 页面 | 原文线索

V2-049684 | MATHDB-344324

Classification conjecture for irreducible ordinary \mathcal W_k-modules

摘录 / EXCERPT: Let $k\in\mathbb Z$ with $k\geq -1$, define ... and let $L(x,y)$ denote the irreducible highest-weight $\mathcal W_k$-module with highest weight (x,y). Classification conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-049685 | MATHDB-344350

Bistability at all processivity levels in the minimally bistable ERK subnetwork

摘录 / EXCERPT: Bistability at all processivity levels. For every choice of $p_k,p_{\ell}\in(0,1)$, there is such a rate-constant vector for which the resulting system admits multiple exponentiall...

conjecture | open | MathDB 页面 | 原文线索

V2-049686 | MATHDB-344351

Three-positive-steady-state bound for the minimally bistable ERK subnetwork

摘录 / EXCERPT: Consider the minimally bistable ERK subnetwork, and let a compatibility class be determined by the conserved quantities of the reaction system. Three-steady-state conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-049687 | MATHDB-344364

Almost-all-graphs irreducible characteristic-polynomial conjecture

摘录 / EXCERPT: Irreducible characteristic-polynomial conjecture. For almost all graphs, $\operatorname{char}_G(x)$ is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-049688 | MATHDB-344371

K6 and K6,6 sequence rank formula for four-dimensional rigidity

摘录 / EXCERPT: Let K_n be the complete graph and let $F\subseteq E(K_n)$. A proper $\{K_6,K_{6,6}\}$-sequence is the sequence construction defined in the source using copies of K_6 and...

conjecture | open | MathDB 页面 | 原文线索

V2-049689 | MATHDB-344375

The forbidden-subgraph characterization of 12-representable line grid graphs

摘录 / EXCERPT: A line grid graph is a graph obtained as the line graph of a grid graph. A graph is 12-representable if there is a word over its vertices in which every pair of distinct vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-049690 | MATHDB-344396

Schröder's cop-number conjecture for graphs of bounded genus

摘录 / EXCERPT: Schröder's conjecture. For every $g\in\mathbb{N}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049691 | MATHDB-344401

Gir3o et al.'s generalized majority coloring conjecture

摘录 / EXCERPT: For $3b1\in[0,1]$, an $3b1$-majority coloring of a digraph is a vertex-coloring in which, for every vertex v, at most $3b1\cdot d^+(v)$ out-neighbors have the same color as v....

conjecture | open | MathDB 页面 | 原文线索

V2-049692 | MATHDB-344412

Reed's maximum-average-degree strengthening of the chromatic bound

摘录 / EXCERPT: Let G be a graph. Write $\chi(G)$ for its chromatic number, $\omega(G)$ for the size of a largest clique, and $\operatorname{mad}(G)$ for its maximum average degree: ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-049693 | MATHDB-344413

Conjecture on Euler impure graphs on Dyck's surface

摘录 / EXCERPT: Let $\mathbb{N}_3$ denote Dyck's surface, and let an Euler impure graph be an edge-maximal graph embeddable on a surface that does not triangulate that surface. For a graph G and...

conjecture | open | MathDB 页面 | 原文线索

V2-049694 | MATHDB-344415

Local version of Reed's list-coloring conjecture

摘录 / EXCERPT: Let G be a graph and let $L=(L(v):v\in V(G))$ be a list-assignment. For each vertex v, let $d(v)$ denote its degree and let $\omega(v)$ denote the size of the largest clique co...

conjecture | open | MathDB 页面 | 原文线索

V2-049695 | MATHDB-344416

King's strengthening of Reed's conjecture

摘录 / EXCERPT: Let G be a graph. For each vertex $v\in V(G)$, let $d(v)$ be its degree and let $\omega(v)$ be the size of the largest clique containing v. Write $\chi(G)$ for the chromatic nu...

conjecture | open | MathDB 页面 | 原文线索

V2-049696 | MATHDB-344427

Inhomogeneous transfer-matrix eigenvalue conjecture for the open eight-vertex model

摘录 / EXCERPT: Let $L\geqslant 1$, let $u_1,\dots,u_L$ be inhomogeneity parameters, and define the inhomogeneous open-boundary transfer matrix ... where $K^-(u)=K(u)$, $K^+(u)=K(u+2\eta)$, ... Set...

conjecture | open | MathDB 页面 | 原文线索

V2-049697 | MATHDB-344440

Pandharipande-Thomas topological vertex conjecture

摘录 / EXCERPT: Let $V_{,,}$ be the DT topological vertex and let $W_{,,}$ be the weighted labelled-box vertex. Let ... be the plane-partition generating function. Pandharipande-Thomas topological v...

conjecture | open | MathDB 页面 | 原文线索

V2-049698 | MATHDB-344448

The converse to the $(1,1)$-degeneracy characterization of subgraph query complexity

摘录 / EXCERPT: Converse to the $(1,1)$-degeneracy characterization. If H is a 2-degenerate graph that is not $(1,1)$-degenerate, then

conjecture | open | MathDB 页面 | 原文线索

V2-049699 | MATHDB-344449

Existence of extremal graphs for every degeneracy

摘录 / EXCERPT: Extremal graph existence conjecture. For every integer $d\geq 2$, there exists a d-degenerate graph H such that

conjecture | open | MathDB 页面 | 原文线索

V2-049700 | MATHDB-344454

The matching bound conjecture for 1-planar graphs of minimum degree 6

摘录 / EXCERPT: Let N be a fixed positive integer. A 1-planar graph is a graph that admits a drawing in which each edge is crossed at most once. Consider a 1-planar graph with minimum degree at...

conjecture | open | MathDB 页面 | 原文线索

V2-049701 | MATHDB-344482

Bienia-Meyniel conjecture for Eulerian digraphs

摘录 / EXCERPT: Let D be an Eulerian digraph on n vertices. Bienia-Meyniel conjecture. There exists $\alpha\in\mathbb{R}$ such that D can be decomposed into at most ... dicycles. The source in...

conjecture | open | MathDB 页面 | 原文线索

V2-049702 | MATHDB-344494

The weaker mean subtree order conjecture for connected noncomplete graphs

摘录 / EXCERPT: Let a graph be finite, loopless, and without multiple edges, and let $\mu(G)$ denote the mean subtree order of a multigraph G, namely the average order of its subtrees. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-049703 | MATHDB-344495

Compact DV representations for GIRS matrices

摘录 / EXCERPT: Let $(A,\mathbb{G})$ be a GIRS-c pair, and let $\mathbb{V}_\mathbb{G}$ denote the vertex set of $\mathbb{G}$. A DV representation is a representation with rank parameters r_i i...

conjecture | open | MathDB 页面 | 原文线索

V2-049704 | MATHDB-344496

Compact graph-induced semiseparable representations for two-dimensional mesh graphs

摘录 / EXCERPT: Let $M\times M$ 2D mesh graphs be graphs with the corresponding vertex sets $\mathbb{V}_\mathbb{G}$, and let a Hamiltonian path be a path $\mathbb{P}$ visiting every vertex exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-049705 | MATHDB-344520

The successive algebraic-power conjectures for basic models over regular graphs

摘录 / EXCERPT: Let B be a d-regular graph and let $\{\mathcal{C}_n(B)\}_{n\in N}$ be one of the basic models over B, with tangle power $\tau_{\rm tang}$ and algebraic power $\tau_{\rm alg}$...

conjecture | open | MathDB 页面 | 原文线索

V2-049706 | MATHDB-344550

Deleting from step-diagonals

摘录 / EXCERPT: Let $\mathcal R_{2r\times 2r}$ be the $2r\times 2r$ grid graph shown in Figure 1. If G is a subgraph with the same vertex set, constructed by deleting any k of the highlighted...

conjecture | open | MathDB 页面 | 原文线索

V2-049707 | MATHDB-344559

The relaxed partition number conjecture

摘录 / EXCERPT: The relaxed partition number conjecture. For any d-regular graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-049708 | MATHDB-344560

Critical-regime clique-count conjecture for bounded-degree graphs

摘录 / EXCERPT: Let $n=a(r+1)+b$, and let m satisfy ... For fixed t, or for $k=\sum_{t \ge 2} k_t$, consider graphs of order n, size m, and maximum degree at most r. Critical-regime clique...

conjecture | open | MathDB 页面 | 原文线索

V2-049709 | MATHDB-344577

A large independent set of low-degree vertices in non-chorded connected graphs

摘录 / EXCERPT: Low-degree independent set conjecture. H contains an independent set I of order at least $n/6$ such that every vertex of I has degree at most 2 in H.

conjecture | open | MathDB 页面 | 原文线索

V2-049710 | MATHDB-344580

The sharp lower-bound conjecture for tight Hamilton cycles in Dirac hypergraphs

摘录 / EXCERPT: Fix an integer $k\geq 2$ and a constant $\gamma>0$. Let G be an n-vertex k-graph with minimum codegree ... A tight Hamilton cycle is a cyclic ordering of the vertices of G su...

conjecture | open | MathDB 页面 | 原文线索

V2-049711 | MATHDB-344584

Xu's asymptotic conjecture for normal Cayley digraphs

摘录 / EXCERPT: Xu's conjecture. The minimum, over all groups R of order r, of the proportion of inverse-closed subsets S of R such that $\mathop{\Gamma}(R,S)$ is a normal Cayley digraph t...

conjecture | open | MathDB 页面 | 原文线索

V2-049712 | MATHDB-344623

The infinite-cardinal distinguishing-index conjecture

摘录 / EXCERPT: Let G be a connected $d4f$-regular graph for an infinite cardinal $d4f$, and call an edge-colouring distinguishing if the only colour-preserving automorphism of G is the identi...

conjecture | open | MathDB 页面 | 原文线索

V2-049713 | MATHDB-344630

Brill-Noether existence conjecture for real divisors with equal degree

摘录 / EXCERPT: Equal-degree Brill-Noether existence conjecture. If $\rho(g,r,d)\geq 0$, then there exists an $\mathbb{R}$-divisor of degree at most d and rank equal to r; moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-049714 | MATHDB-344631

Brill-Noether non-existence conjecture for real divisors

摘录 / EXCERPT: Let $g\geq 1$. Brill-Noether non-existence conjecture for real divisors. There exists a family of undirected, connected multigraphs, one for each genus, such that for every pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-049715 | MATHDB-344632

Covering radius conjecture for graph critical sets

摘录 / EXCERPT: Let G be an undirected, connected multigraph with genus $g\geq 1$ and $n\geq 2$ vertices. Let $\mathcal{N}_G$, ${\rm Crit}_{\triangle}(L_G)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-049716 | MATHDB-344636

Point-line incidence conjecture for even-cycle-free graphs

摘录 / EXCERPT: Point-line incidence conjecture. If G is C_{2k}-free, then

conjecture | open | MathDB 页面 | 原文线索

V2-049717 | MATHDB-344687

The walk-matrix criterion for generalized spectral determination

摘录 / EXCERPT: Let G be a self-converse mixed graph of order n, with walk-matrix ... Here $\mathcal{G}_n^{sc}$ denotes the self-converse mixed graphs of order n, and G is DGS when it is det...

conjecture | open | MathDB 页面 | 原文线索

V2-049718 | MATHDB-344688

The Gaussian square-free walk-matrix conjecture

摘录 / EXCERPT: Let $G\in\mathcal{G}_n^{sc}$ be a self-converse mixed graph, let $W=W(G)$ be its walk-matrix, let $\mathscr{U}_G$ be the associated set of Gaussian rational unitary matrices, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-049719 | MATHDB-344697

The q-degree zero homology conjecture for cycles

摘录 / EXCERPT: Let C_n be the cycle graph with n vertices. Cycle homology conjecture. For $0\leq i\leq n-1$, $H_i(C_n;\mathbb{Z})$ is a free $\mathbb{Z}$-module of rank ... The source supplies...

conjecture | open | MathDB 页面 | 原文线索

V2-049720 | MATHDB-344698

The block bound conjecture for chromatic symmetric homology span

摘录 / EXCERPT: Let G be a graph with n vertices and m edges, and let b be the number of blocks of G. Let $\operatorname{span}_0(G)$ denote the homological span of the degree-zero chroma...

conjecture | open | MathDB 页面 | 原文线索

V2-049721 | MATHDB-344718

McCuaig's polarity-graph conjecture for extremal C_4-free graphs

摘录 / EXCERPT: Let C_4 be the 4-cycle, and let $\operatorname{ex}(n,C_4)$ be the maximum number of edges in an n-vertex graph containing no copy of C_4. A polarity graph is the graph asso...

conjecture | open | MathDB 页面 | 原文线索

V2-049722 | MATHDB-344719

Galashin-Pylyavskyy positivity conjecture for arborescence ratios

摘录 / EXCERPT: Let Γ be a directed graph, let $\tilde{\Gamma}$ be a k-cover of Γ, let v be a vertex of Γ, and let $\tilde{v}$ be a lift of v in $\tilde{\Gamma}$. If the...

conjecture | open | MathDB 页面 | 原文线索

V2-049723 | MATHDB-344720

Vector-field expansion conjecture for arborescence ratios

摘录 / EXCERPT: Let Γ be a graph and let $\tilde{\Gamma}$ be a k-fold cover of Γ. Let $\mathcal V$ be the set of vector fields of Γ, let $\operatorname{wt}(\gamma)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-049724 | MATHDB-344721

Random-cover expectation conjecture for arborescence ratios

摘录 / EXCERPT: Let $\Gamma=(E,V)$ be a graph, fix a vertex v with non-trivial arborescence, and let $\mathcal V$ be the set of vector fields of Γ. Let Γ' be a uniformly random k...

conjecture | open | MathDB 页面 | 原文线索

V2-049725 | MATHDB-344724

The h-vector form conjecture for odd cycle toric rings

摘录 / EXCERPT: Let C_{2s+1} be the cycle graph on $2s+1$ vertices, and let $K[C_{2s+1}]$ denote its toric ring. Write its h-vector as a sequence with entries $h_1,h_2,\ldots,h_{s-1}$ appearin...

conjecture | open | MathDB 页面 | 原文线索

V2-049726 | MATHDB-344728

Jones' Conjecture for planar graphs

摘录 / EXCERPT: Jones' Conjecture. Every planar graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049727 | MATHDB-344734

Gan-Loh-Sudakov triangle extremal conjecture

摘录 / EXCERPT: Let G be a graph on n vertices with maximum degree d. Write ... Here q and r are the quotient and remainder in the division of n by $d+1$. Gan-Loh-Sudakov's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-049728 | MATHDB-344751

Polynomial Erdős-Hajnal bound for graphs of bounded VC-dimension

摘录 / EXCERPT: Let $d\geq 2$. A graph has VC-dimension at most d when its VC-dimension is bounded by d. Bounded-VC-dimension Erdős-Hajnal conjecture. There exists a constant $\varepsilon(d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-049729 | MATHDB-344754

The subcubic star chromatic index conjecture

摘录 / EXCERPT: Let G be a graph, and let $\chi'_{st}(G)$ denote its star chromatic index, the minimum number of colors in a proper edge coloring with no two-colored path or cycle of length four...

conjecture | open | MathDB 页面 | 原文线索

V2-049730 | MATHDB-344758

Tucker's two-colour conjecture for connected locally finite graphs

摘录 / EXCERPT: Let G be a connected, locally finite graph. The graph G has infinite motion if every non-identity automorphism moves infinitely many vertices. A vertex colouring is asymmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-049731 | MATHDB-344763

The asymptotic Turán conjecture for the square of a path

摘录 / EXCERPT: Let P_k be the path on k vertices, let P_k^2 denote its square, and let ${\rm ex}(n,P_k^2)$ be the maximum number of edges in an n-vertex graph containing no copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-049732 | MATHDB-344764

The Wiener index conjecture for 5-connected maximal planar graphs

摘录 / EXCERPT: Let G be a maximal 5-connected planar graph with $n\geq 12$ vertices. The 5-connected Wiener index conjecture. ... This is presented as a conjectured sharp bound for 5-connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-049733 | MATHDB-344813

The sharp Loebl-Komlós-Sós conjecture

摘录 / EXCERPT: Let G be an n-vertex graph, and let k be a positive integer. Sharp Loebl-Komlós-Sós conjecture. If more than...

conjecture | open | MathDB 页面 | 原文线索

V2-049734 | MATHDB-344814

The skew Loebl-Komlós-Sós conjecture

摘录 / EXCERPT: Let a k-edge tree be r-skew if one of its colour classes has size at most $r(k+1)$. Skew Loebl-Komlós-Sós conjecture. If a graph G on n vertices has more than rn vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-049735 | MATHDB-344815

The two-thirds minimum/maximum degree conjecture

摘录 / EXCERPT: Let G be a graph, and let k be a positive integer. Two-thirds conjecture. If $\delta(G)\ge\left\lfloor\frac{2k}{3}\right\rfloor$ and $\Delta(G)\ge k$, then G contains every t...

conjecture | open | MathDB 页面 | 原文线索

V2-049736 | MATHDB-344816

The $2k$-$k/2$ degree conjecture

摘录 / EXCERPT: Let G be a graph, and let k be a positive integer. $2k$-$k/2$ conjecture. If $\delta(G)\ge\frac{k}{2}$ and $\Delta(G)\ge2k$, then G contains every tree with k edges. This i...

conjecture | open | MathDB 页面 | 原文线索

V2-049737 | MATHDB-344817

The intermediate-range minimum/maximum degree conjecture

摘录 / EXCERPT: Let k be a positive integer and let $\alpha\in[0,\frac13)$. Intermediate-range conjecture. Every graph with minimum degree at least $(1+\alpha)\frac k2$ and maximum degree at lea...

conjecture | open | MathDB 页面 | 原文线索

V2-049738 | MATHDB-344818

The sparse minimum/maximum degree conjecture for tree containment

摘录 / EXCERPT: Let G be an n-vertex graph, and let k be a positive integer. Sparse degree conjecture. If $\delta(G)\ge\frac k2$ and at least $\frac{n}{2\sqrt{k}}$ vertices of G have degre...

conjecture | open | MathDB 页面 | 原文线索

V2-049739 | MATHDB-344819

The minimum-semidegree conjecture for oriented paths

摘录 / EXCERPT: For a digraph D, define the semidegree of a vertex by $d^0(v):=\min\{d^-(v),d^+(v)\}$ and the minimum semidegree by $\delta^0(D):=\min\{d^0(v):v\in V(D)\}$. Minimum-semidegree pa...

open_problem | open | MathDB 页面 | 原文线索

V2-049740 | MATHDB-344854

The adjacency-matrix supremum conjecture for nilpotent nonnegative matrices

摘录 / EXCERPT: Adjacency-matrix supremum conjecture. For every $k\in\{1,\ldots,[N]\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-049741 | MATHDB-344874

Small-maximum-degree conjecture for tree versus wheel Ramsey numbers

摘录 / EXCERPT: Small-maximum-degree conjecture. For even m and $n>m\geq4$,

conjecture | open | MathDB 页面 | 原文线索

V2-049742 | MATHDB-344905

Parity conjecture for the number of Sylow-2 cyclic factors

摘录 / EXCERPT: Let M define a connected Cayley graph on $\mathbb{F}_2^r$, let $d(M)$ be the number of Sylow-2 cyclic factors in its sandpile group, and let the graph's eigenvalues be those as...

conjecture | open | MathDB 页面 | 原文线索

V2-049743 | MATHDB-344924

Maximum-nullity and zero-forcing equality for circulant graphs

摘录 / EXCERPT: Let k be a positive integer and let $[...ELLIPSIZATION...]rname{Circ}[(\ell^2-1)k,\{1,\ell\}])=2\ell.\]$ and field independent minimum rank with universally optimal matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-049744 | MATHDB-344925

Field-independent minimum rank for extended cube graphs

摘录 / EXCERPT: Let t be an integer satisfying $t\equiv 0,1,2\pmod{6}$, and let r be an integer with $r>\lfloor t/6\rfloor$. The extended cube graphs are denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-049745 | MATHDB-344926

Pasotti-Pellegrini's odd-order near-factor conjecture

摘录 / EXCERPT: Pasotti-Pellegrini conjecture. There exists a near 1-factor F of K_v with $\ell(F)=L$ if and only if L is admissible.

conjecture | open | MathDB 页面 | 原文线索

V2-049746 | MATHDB-344930

Conjecture on regular graphs with a clique joined to an independent set as a star complement

摘录 / EXCERPT: Let G be an r-regular graph, let $ abla$ denote the graph join, and let K_s be the complete graph on s vertices and tK_1 the disjoint union of t isolated vertices, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-049747 | MATHDB-344932

Three-quarters exponent conjecture for acyclic subgraphs of tournaments

摘录 / EXCERPT: Let n be a positive integer and let G be an n-vertex tournament. An acyclic subgraph is a subgraph with no directed cycle, and its chromatic number is the chromatic number of...

conjecture | open | MathDB 页面 | 原文线索

V2-049748 | MATHDB-344940

The conjecture that blowup Ramsey numbers must depend on the base graph

摘录 / EXCERPT: Let $G \overset{r}{\longrightarrow} H$ mean that every r-edge-coloring of G contains a monochromatic copy of H, and let $\mathrm{B}(G \overset{r}{\longrightarrow} H;t)$ be th...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049749 | MATHDB-344941

Dehmer-Pickl-Shi-Yu conjecture on tree Laplacian and adjacency spectral distances

摘录 / EXCERPT: Dehmer-Pickl-Shi-Yu conjecture. For every such pair of trees,

conjecture | refuted | MathDB 页面 | 原文线索

V2-049750 | MATHDB-344942

Dehmer-Pickl-Shi-Yu conjecture comparing degree-power and Laplacian spectral distances

摘录 / EXCERPT: Dehmer-Pickl-Shi-Yu conjecture. For every such pair of trees,

conjecture | refuted | MathDB 页面 | 原文线索

V2-049751 | MATHDB-344972

Clique-counting conjecture for bounded-degree graphs with a fixed number of edges

摘录 / EXCERPT: Let $t\geq 3$, let G be a graph with m edges and maximum degree at most r, and write ... Here $k_t(G)$ denotes the number of copies of K_t in G, K_{r+1} is the complete g...

conjecture | open | MathDB 页面 | 原文线索

V2-049752 | MATHDB-344975

Algebraicity and absence of singular continuous spectrum for amalgamated free-product graphs

摘录 / EXCERPT: Consider a graph constructed via the amalgamated free product of graphs, and a Jacobi operator on it whose Jacobi coefficients are lifted from the factor graphs in a sensible way....

conjecture | open | MathDB 页面 | 原文线索

V2-049753 | MATHDB-344976

The regular Turán bound for 3-chromatic graphs built from bipartite graphs

摘录 / EXCERPT: Regular Turán bound. For such a graph F,

conjecture | open | MathDB 页面 | 原文线索

V2-049754 | MATHDB-345018

Boots-Royle-Cao-Vince conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph on $n\geq 9$ vertices, and let P_k denote the path on k vertices. The join $P_2\vee P_{n-2}$ is also a planar graph. Boots-Royle-Cao-Vince conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-049755 | MATHDB-345028

Upper bound conjecture for total vertex product irregularity strength

摘录 / EXCERPT: Let G be a d-regular graph of order n, and let $tvps(G)$ denote its total vertex product irregularity strength, namely the least integer k for which G has a product-irreg...

conjecture | open | MathDB 页面 | 原文线索

V2-049756 | MATHDB-345041

Collapsibility bound for independence complexes of bounded-degree graphs

摘录 / EXCERPT: Collapsibility conjecture. If G has maximum degree at most Δ, then

conjecture | open | MathDB 页面 | 原文线索

V2-049757 | MATHDB-345042

Link collapsibility conjecture for independence complexes of bounded-degree graphs

摘录 / EXCERPT: Link collapsibility conjecture. If A is an independent set of size $n-1$ in G, then

conjecture | open | MathDB 页面 | 原文线索

V2-049758 | MATHDB-345058

Conjecture on starting-vertex variability of the nearest unvisited vertex walk

摘录 / EXCERPT: Let G be a finite graph, let V be a uniformly random initial vertex, and let $L_{NUV}(G,V)$ denote the length of the nearest unvisited vertex walk started at V. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049759 | MATHDB-345071

The torsion-freeness and rank conjecture for square polyominoes

摘录 / EXCERPT: Let C_4 denote a square cycle, and let G^4_S be a square polyomino formed from S copies of C_4. Write $V(G^4_S)$ and $E(G^4_S)$ for its vertex and edge sets, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049760 | MATHDB-345072

The diagonal magnitude homology conjecture for triangles attached to a square

摘录 / EXCERPT: Let C_3 and C_4 denote cycle graphs of lengths three and four, respectively. Form a graph by gluing two copies of C_3 along single edges to a single copy of C_4, and suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-049761 | MATHDB-345073

The diagonal magnitude homology conjecture for triangle attachments to a square

摘录 / EXCERPT: Let C_3 and C_4 denote cycle graphs of lengths three and four, respectively. Form a graph by gluing any number of copies of C_3 along single edges to a single copy of C_4,...

conjecture | open | MathDB 页面 | 原文线索

V2-049762 | MATHDB-345074

The threshold conjecture for rainbow even cliques in randomly perturbed dense graphs

摘录 / EXCERPT: Even-clique threshold conjecture. For every real $d>0$ and every integer $r\geq 5$, the threshold for the property

conjecture | open | MathDB 页面 | 原文线索

V2-049763 | MATHDB-345075

Basis conjecture for curvature-zero functions on positroid subdivisions

摘录 / EXCERPT: Let $\Delta_{k,n}$ be the hypersimplex, and let $e_J\in\Delta_{k,n}$ range over its $\binom{n}{k}-n$ nonfrozen vertices. For each such vertex, let ρ_J be the corresponding fun...

conjecture | open | MathDB 页面 | 原文线索

V2-049764 | MATHDB-345086

Strong Fractional Tuza's conjecture

摘录 / EXCERPT: Strong Fractional Tuza's conjecture. For every $k\in\mathbb{N}$ with $k\geq 2$ and every graph G, ...

conjecture | open | MathDB 页面 | 原文线索

V2-049765 | MATHDB-345090

The 16-fold way conjecture for super-modular categories

摘录 / EXCERPT: The 16-fold way conjecture. Every super-modular category $\mathcal{B}$ admits a minimal modular extension, and it has exactly 16 minimal modular extensions up to braided monoidal e...

conjecture | open | MathDB 页面 | 原文线索

V2-049766 | MATHDB-345091

The cop-number conjecture for graphs without long holes

摘录 / EXCERPT: Cop-number conjecture. The graph G can be won by $t-4$ cops.

conjecture | open | MathDB 页面 | 原文线索

V2-049767 | MATHDB-345095

The near-bipartite stability conjecture for low fractional triangle-packing density

摘录 / EXCERPT: Let G be a graph on n vertices, let $\overline{G}$ denote its complement, and define ... where $\nu^*(G)$ is the fractional triangle-packing number of G. Near-bipartite stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-049768 | MATHDB-345113

Linear routing strengthening of the Langberg-Médard conjecture

摘录 / EXCERPT: Let a k-pair network have sources $s_1,\dots,s_k$ and sinks $t_1,\dots,t_k$, and suppose it is strongly reachable, meaning that for every sink t_j there are edge-disjoint direc...

conjecture | open | MathDB 页面 | 原文线索

V2-049769 | MATHDB-345114

Satisfying sequences for rainbow matchings in multipartite hypergraphs

摘录 / EXCERPT: Let $n\geq s$ and let k be a positive integer. A nondecreasing sequence $f_1\leq\cdots\leq f_s$ is called satisfying if every collection of families...

conjecture | open | MathDB 页面 | 原文线索

V2-049770 | MATHDB-345120

Stefán's circular-flow infimum conjecture for odd-regular class 2 graphs

摘录 / EXCERPT: Stefán's circular-flow infimum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049771 | MATHDB-345121

Stefán's circular-flow bound conjecture for class 1 odd-regular graphs

摘录 / EXCERPT: Stefán's class 1 circular-flow bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049772 | MATHDB-345127

Huang-Zhao matching conjecture for minimum vertex degree

摘录 / EXCERPT: Huang-Zhao conjecture. For positive integers m,k,n satisfying $m<n/k$, if

conjecture | open | MathDB 页面 | 原文线索

V2-049773 | MATHDB-345131

Sivaraman-Testa conjecture on the cop number of induced-path-free graphs

摘录 / EXCERPT: Let G be a connected graph, and let P_t denote the path on t vertices. A graph is P_t-free if it has no induced subgraph isomorphic to P_t, and $c(G)$ denotes its cop num...

conjecture | open | MathDB 页面 | 原文线索

V2-049774 | MATHDB-345158

Exact-value conjecture for Ryser's function

摘录 / EXCERPT: For integers r and t with $2\leq t\leq r$, let an (r,t)-graph be an r-uniform, r-partite, t-intersecting hypergraph, and define ... where $\tau(\mathcal H)$ is the cover...

conjecture | open | MathDB 页面 | 原文线索

V2-049775 | MATHDB-345179

MIMO SINR and area spectral efficiency scaling conjecture

摘录 / EXCERPT: Let $N_t(\lambda)$ and $N_r(\lambda)$ be the numbers of transmit and receive antennas, respectively, with ... Consider eigenbeamforming with a single data stream and a physically fea...

conjecture | open | MathDB 页面 | 原文线索

V2-049776 | MATHDB-345184

The pendant-vertex conjecture for local antimagic chromatic number of trees

摘录 / EXCERPT: Let T_k be a tree with $k\ge 2$ pendant vertices. Caterpillars, spiders, and lobsters are standard classes of trees, and $\chi_{la}(T_k)$ denotes the local antimagic chromatic nu...

conjecture | open | MathDB 页面 | 原文线索

V2-049777 | MATHDB-345185

The exceptional-graph conjecture for local antimagic chromatic number

摘录 / EXCERPT: Let G be a graph with $k\ge 1$ pendant vertices and chromatic number $\chi(G)$ satisfying $k\ge \chi(G)-1$. The graphs $K(2;a,b)$, caterpillar graphs in the cited theorem,...

conjecture | open | MathDB 页面 | 原文线索

V2-049778 | MATHDB-345188

The unlabelled bridge-addable connectivity conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a class of graphs, let $\tilde{\mathcal A}_n$ be the set of unlabelled graphs in $\mathcal A$ on n vertices, and let $\tilde{R}_n \in_u \tilde{\mathcal A}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-049779 | MATHDB-345189

The unlabelled bridge-addable connectivity lower-bound conjecture

摘录 / EXCERPT: Let $\mathcal A$ be a class of graphs that is bridge-addable, meaning that adding an edge between vertices in distinct components preserves membership, and decomposable, meaning th...

conjecture | open | MathDB 页面 | 原文线索

V2-049780 | MATHDB-345194

Homotopy type conjecture for higher independence complexes of rectangular grid graphs

摘录 / EXCERPT: For $m,n\geq 2$, let $G_{m,n}$ be the rectangular grid graph with vertex set ... and edges joining (i,j) to $(i+1,j)$ or $(i,j+1)$ whenever those vertices exist. For an integer r...

conjecture | open | MathDB 页面 | 原文线索

V2-049781 | MATHDB-345199

The Rail Conjecture for knight's tours

摘录 / EXCERPT: Let an $n \times n$ knight's tour be a closed knight's tour on an $n \times n$ board, and let a rail be the structure defined by the preceding discussion: a pair of edges that can...

conjecture | open | MathDB 页面 | 原文线索

V2-049782 | MATHDB-345229

The weaker generalised second neighbourhood conjecture

摘录 / EXCERPT: Weaker generalised second neighbourhood conjecture. There is a vertex v such that

conjecture | open | MathDB 页面 | 原文线索

V2-049783 | MATHDB-345230

Derived Reid's recipe conjecture for consistent dimer models

摘录 / EXCERPT: Let Q_0 be the vertex set of the dual quiver, let θ be a generic stability parameter, and let $theta$ denote the corresponding moduli space with fan $Sigma_theta$. For a n...

conjecture | open | MathDB 页面 | 原文线索

V2-049784 | MATHDB-345311

Hasse-diagram chromatic number gap conjecture

摘录 / EXCERPT: Let G be a Hasse diagram on n vertices, and let $\chi(G)$ denote its chromatic number. Hasse-diagram chromatic number gap conjecture. There exists $\epsilon>0$ such that ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-049785 | MATHDB-345325

Benjamini's transient triangulation conjecture

摘录 / EXCERPT: Let G be a bounded-degree plane triangulation. Write $G^{1/2}$ for site percolation on G in which each vertex is retained independently with probability $1/2$. The graph G is...

conjecture | open | MathDB 页面 | 原文线索

V2-049786 | MATHDB-345343

The Andr\e1sfai blow-up conjecture for triangle-free graphs

摘录 / EXCERPT: Let n and s be integers with $n s\e 0$. For $k\e 2$, define ... Define $g(n,s)$ by ... Here $\operatorname{ex}(n,s)$ denotes the maximum number of edges in a triangle-free n-vert...

conjecture | open | MathDB 页面 | 原文线索

V2-049787 | MATHDB-345344

The predicted piecewise-quadratic formula for the Ramsey-Tur\e1n density

摘录 / EXCERPT: Ramseye1Ture1n density formula. The function f is given by

conjecture | open | MathDB 页面 | 原文线索

V2-049788 | MATHDB-345349

Wang's arbitrary cycle-factor conjecture for graphs

摘录 / EXCERPT: Wang's conjecture. If $|W|\geq 3k$ and $\delta(W)\geq 2n/3$, then G contains an arbitrary W-cycle-factor.

conjecture | open | MathDB 页面 | 原文线索

V2-049789 | MATHDB-345350

Wang's minimum semi-degree conjecture for disjoint directed cycles

摘录 / EXCERPT: Wang's conjecture. If $\delta^0(D)\geq (9k-3)/4$, then D contains k vertex-disjoint directed cycles, each of order at least 3.

conjecture | open | MathDB 页面 | 原文线索

V2-049790 | MATHDB-345351

Wang's arbitrary directed W-cycle-factor conjecture

摘录 / EXCERPT: Wang's conjecture. If $\delta(W)\geq (3n-3)/2$, then D contains an arbitrary W-cycle-factor.

conjecture | open | MathDB 页面 | 原文线索

V2-049791 | MATHDB-345352

The 2n/3 minimum semi-degree conjecture for directed W-cycle-factors

摘录 / EXCERPT: The $2n/3$ minimum semi-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049792 | MATHDB-345363

The 3/5-conjecture for the game domination number

摘录 / EXCERPT: Consider a simple undirected graph G with vertex set $V(G)$ and order $n=|V(G)|$. The game domination number $\gamma_g(G)$ is the value of the domination game on G when Dominat...

conjecture | open | MathDB 页面 | 原文线索

V2-049793 | MATHDB-345398

The total Roman domination function support conjecture

摘录 / EXCERPT: Support conjecture. If G is a γ_{tR}-edge-supercritical graph and $v\in V(G)$, then there exists a $\gamma_{tR}(G)$-function f such that $v\in V_f^+$.

conjecture | open | MathDB 页面 | 原文线索

V2-049794 | MATHDB-345399

The total Roman domination join conjecture

摘录 / EXCERPT: Join conjecture. If G is a k-γ_{tR}-edge-supercritical graph, then $G\cup K_n$ is $(k+3)$-γ_{tR}-edge-critical, for $n\geq3$.

conjecture | open | MathDB 页面 | 原文线索

V2-049795 | MATHDB-345403

Reiher-LPR extremal function conjecture for triangle-free graphs

摘录 / EXCERPT: Reiher-LPR conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-049796 | MATHDB-345404

Extremal-graph classification conjecture for triangle-free graphs

摘录 / EXCERPT: Let n and s be integers with $n/3<s\le n/2$, and let $\mathrm{ex}(n,s)$ denote the maximum number of edges in a triangle-free graph on n vertices with independence number at...

conjecture | open | MathDB 页面 | 原文线索

V2-049797 | MATHDB-345430

Monotonicity conjecture for the projective-dimension-regularity spectrum

摘录 / EXCERPT: Monotonicity conjecture. Let $r,n\geq 2$ be arbitrary integers. If $(p,r)\in\operatorname{pdreg}(n)$, then $(p,r-1)\in\operatorname{pdreg}(n)$. The preceding results show that the...

conjecture | open | MathDB 页面 | 原文线索

V2-049798 | MATHDB-345453

The graph positivity conjecture

摘录 / EXCERPT: Consider r-regular bipartite graphs with $v=2n$ vertices. Let m_i be the number of i-matchings, and let $\overline{m}_i$ be the number of i-matchings in the complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-049799 | MATHDB-345463

Coordinated beamforming scaling conjecture for MISO cellular networks

摘录 / EXCERPT: Let $N_t(\lambda)$ be the number of transmit antennas in a MISO cellular network with a physically feasible path loss model. Suppose coordinated beamforming perfectly cancels the i...

conjecture | open | MathDB 页面 | 原文线索

V2-049800 | MATHDB-345465

Woodall's circumference conjecture

摘录 / EXCERPT: Let G) be a 2-connected graph on n vertices, and let $c(G)$ denote the length of a longest cycle in G. Woodall's conjecture. If there are at least $\frac{n}{2}+k$ vertices of...

conjecture | open | MathDB 页面 | 原文线索

V2-049801 | MATHDB-345466

Häggkvist-Jackson strengthening of Woodall's conjecture

摘录 / EXCERPT: Let G be a 2-connected graph on n vertices. Häggkvist-Jackson conjecture. If G contains at least ... vertices of degree at least k, then G has a cycle of length at least .....

conjecture | open | MathDB 页面 | 原文线索

V2-049802 | MATHDB-345467

The alpha-generalized Erdős-Gallai path conjecture

摘录 / EXCERPT: Let G be a 2-connected graph on n vertices, and let $x,y\in V(G)$. Let $0<\alpha\leq\frac{1}{2}$. Alpha-generalized path conjecture. If $G-\{x,y\}$ contains more than...

conjecture | open | MathDB 页面 | 原文线索

V2-049803 | MATHDB-345480

The finite-grid lower-bound conjecture for coded-packet forwarding

摘录 / EXCERPT: Let $p_{k,n,\delta}(\Gamma_m)$ denote the forwarding probability for a grid network of size m, and let Y be a binomial random variable with parameters n and $(\theta^+(p))^2$...

conjecture | open | MathDB 页面 | 原文线索

V2-049804 | MATHDB-345481

Eppstein's integer-power conjecture for planar subgraph counts

摘录 / EXCERPT: Eppstein's conjecture. For every graph H, there exists a non-negative integer k such that

conjecture | open | MathDB 页面 | 原文线索

V2-049805 | MATHDB-345482

Integer-power conjecture for forbidden-subgraph planar counts

摘录 / EXCERPT: Integer-power conjecture. For all finite sets of graphs $\mathcal{F}$ and all graphs H, there is an integer k such that

conjecture | open | MathDB 页面 | 原文线索

V2-049806 | MATHDB-345483

Generalized planar Turán conjecture for cycles with forbidden even cycles

摘录 / EXCERPT: Generalized cycle-count conjecture. For sufficiently large k,

conjecture | open | MathDB 页面 | 原文线索

V2-049807 | MATHDB-345505

The quantum-state realization conjecture for hypergraph entropy rays

摘录 / EXCERPT: Let G_K be a hypergraph defining a K-party state construction, and let $\Ket{G_K}$ denote the resulting K-party quantum state. The construction uses AME tensors at bulk verti...

conjecture | open | MathDB 页面 | 原文线索

V2-049808 | MATHDB-345506

The hypergraph-QLR cone equality conjecture

摘录 / EXCERPT: For each n, let the n-party hypergraph cone be the cone of entropy vectors obtained from n-party hypergraphs, and let the n-party QLR cone be the cone defined by the quantu...

conjecture | open | MathDB 页面 | 原文线索

V2-049809 | MATHDB-345507

The hypergraph-stabilizer cone equality conjecture

摘录 / EXCERPT: For each n, let the n-party hypergraph cone be the cone of entropy vectors obtained from n-party hypergraphs, and let the n-party stabilizer cone be the cone of entropy vec...

conjecture | open | MathDB 页面 | 原文线索

V2-049810 | MATHDB-345518

C1-cofiniteness and finite-length conjecture for vertex operator algebras

摘录 / EXCERPT: Let V be a vertex operator algebra such that all irreducible ordinary V-modules are C_1-cofinite, and let W be a generalized V-module. C1-cofiniteness and finite-length c...

conjecture | open | MathDB 页面 | 原文线索

V2-049811 | MATHDB-345519

Commutant and fixed-point-free extension conjecture beyond collapsing levels

摘录 / EXCERPT: Let $n\geq 4$ and $r\geq 0$, set $\mathfrak{g}=\mathfrak{sl}(n+r+2|r)$, and let Π be a rank-one Heisenberg vertex operator algebra. Define ... and ... Let $h^\vee$ be the dual Coxe...

conjecture | open | MathDB 页面 | 原文线索

V2-049812 | MATHDB-345521

Minimal-diagram conjecture for reducible handlebody links

摘录 / EXCERPT: A handlebody link HL is a union of finitely many handlebodies of positive genus embedded in the 3-sphere $\mathbb S^3$. It is reducible if there is a cutting 2...

conjecture | open | MathDB 页面 | 原文线索

V2-049813 | MATHDB-345552

The list chromatic number versus list vertex arboricity conjecture

摘录 / EXCERPT: List colouring conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049814 | MATHDB-345553

The bounded-order list vertex arboricity conjecture

摘录 / EXCERPT: Bounded-order list vertex arboricity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049815 | MATHDB-345566

The Odd Alternating Cycle Conjecture

摘录 / EXCERPT: Odd Alternating Cycle Conjecture. For every field $\mathbb{F}$ there exist $\varepsilon>0$ and an odd integer ℓ such that every n-vertex directed graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-049816 | MATHDB-345574

Weaker Borcherds conjecture on vanishing of Tate cohomology for Fricke elements

摘录 / EXCERPT: Let V be a self-dual integral form of the monster vertex algebra, let g be a Fricke element, and suppose that g^j is a Fricke element for every integer $j>0$. Weaker Borcherd...

conjecture | open | MathDB 页面 | 原文线索

V2-049817 | MATHDB-345583

de Oliveira et al.'s degree deviation conjecture for connected graphs

摘录 / EXCERPT: Let G be a graph with vertex set $V(G)$, edge set $E(G)$, $n=|V(G)|$ vertices, and $m=|E(G)|$ edges. Its degree deviation measure is ... For integers n and k, let $CS(n,k)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-049818 | MATHDB-345589

Kalai's tight-tree conjecture for tight paths

摘录 / EXCERPT: For $n \geq r \geq 2$ and $k \geq 1$, let P_k^r be the tight k-path in an r-uniform hypergraph, and let ${\rm ex}(n,P_k^r)$ denote the maximum number of edges in an n-verte...

conjecture | open | MathDB 页面 | 原文线索

V2-049819 | MATHDB-345595

Alice's winning conjecture for triangular grid graphs of order 2^m-3

摘录 / EXCERPT: Let T_n be the triangular grid graph with starting vertex v^0_0, and consider the feedback game on T_n in which Alice and Bob play according to the game's rules. Alice's winn...

conjecture | open | MathDB 页面 | 原文线索

V2-049820 | MATHDB-345596

Alice's coprimality conjecture for rectangular grid graphs

摘录 / EXCERPT: Let $Q(m,n)$ be the rectangular grid graph with side parameters m,n, and let $\gcd(m,n)$ denote their greatest common divisor. Consider the feedback game on $Q(m,n)$. Alice's cop...

conjecture | open | MathDB 页面 | 原文线索

V2-049821 | MATHDB-345607

The hypergraph independent-set extremal conjecture

摘录 / EXCERPT: Hypergraph independent-set extremal conjecture. If $\mathcal{G}$ is a d-regular r-graph on n vertices, then

conjecture | open | MathDB 页面 | 原文线索

V2-049822 | MATHDB-345612

Class 1 conjecture for K_5-minor free graphs of maximum degree six

摘录 / EXCERPT: Edge-coloring conjecture. Then G is class 1.

conjecture | open | MathDB 页面 | 原文线索

V2-049823 | MATHDB-345631

Las Vergnas's simplicial vertex conjecture for oriented matroids

摘录 / EXCERPT: Let G be the tope graph of an oriented matroid (OM), and let $r(G)$ be its rank. A vertex v is simplicial if $\deg(v)=r(G)$. Las Vergnas's conjecture. Every OM has a simplicial...

conjecture | open | MathDB 页面 | 原文线索

V2-049824 | MATHDB-345633

Cordovil-Las Vergnas connectivity conjecture for oriented matroid mutation graphs

摘录 / EXCERPT: For positive integers r and n, let $\mathcal{G}^{n,r}$ be the mutation graph of uniform oriented matroids of rank r on n elements. Cordovil-Las Vergnas conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-049825 | MATHDB-345691

Asymptotic extremal conjecture for path blowups

摘录 / EXCERPT: Path-blowup conjecture. The family $\Psi_{k-1}(n,r)$ gives the correct asymptotic of the Turán number in all the above cases.

conjecture | open | MathDB 页面 | 原文线索

V2-049826 | MATHDB-345692

General asymptotic and stability conjecture for tree blowups

摘录 / EXCERPT: Let $\mathcal{T}\in\mathcal{T}_{s,t}(a,b)$, where $b\leq a<r-1$, and set $\sigma=\sigma(\mathcal{T})=\min\{s,t\}$. Let H be a $\mathcal{T}$-free n-vertex r-graph. General tre...

conjecture | open | MathDB 页面 | 原文线索

V2-049827 | MATHDB-345693

Füredi's conjecture for blowups of the 4-cycle

摘录 / EXCERPT: Füredi's conjecture. The extremal number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-049828 | MATHDB-345705

Piecewise edge bound for graphs without large highly connected subgraphs

摘录 / EXCERPT: Piecewise extremal edge-bound conjecture. If G has no $(k+1)$-connected subgraph, then

conjecture | open | MathDB 页面 | 原文线索

V2-049829 | MATHDB-345720

Yuster's hypergraph independent transversal asymptotic conjecture

摘录 / EXCERPT: For integers n,k,r,s, let $f(k,r,s)$ be the extremal function for independent transversals in (n,k,r,s)-graphs. Fix $r\ge 3$. Yuster's conjecture. For all $1\le s\le k$, ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-049830 | MATHDB-345723

The characterization of strongly perfect graphs by forbidden induced subgraphs

摘录 / EXCERPT: Let G be a graph. A graph is strongly perfect if every induced subgraph has a stable set meeting every nonempty maximal clique. An odd hole is an induced cycle of odd length, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049831 | MATHDB-345726

The independence-number-two conjecture for prescribed roots

摘录 / EXCERPT: Prescribed-roots conjecture. Every 3-arc-strong digraph $D=(V,A)$ with $\alpha(D)=2$ has a pair of arc-disjoint branchings B_s^+,B_t^- for every choice of $s,t\in V$.

conjecture | open | MathDB 页面 | 原文线索

V2-049832 | MATHDB-345733

Discrete Hurewicz conjecture for higher homotopy groups of graphs

摘录 / EXCERPT: Let G be a graph, let $A_k(G)$ denote its kth discrete homotopy group, and let $\hc_k(G)$ denote its kth discrete singular cubical homology group. Suppose $n\geq 2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-049833 | MATHDB-345734

Injectivity conjecture for discrete homotopy and homology of the graphs G_n

摘录 / EXCERPT: Let G_1 be a cycle graph of length 5, and let G_n be the graphs defined recursively in the source via the discrete suspension construction. For each n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-049834 | MATHDB-345735

Discrete connectivity conjecture for the graphs G_n

摘录 / EXCERPT: Let G_1 be a cycle graph of length 5, and let G_n be the graphs defined recursively in the source via the discrete suspension construction. Let $A_i(G_n)$ denote the ith di...

conjecture | open | MathDB 页面 | 原文线索

V2-049835 | MATHDB-345742

The list Unfriendly Partition Conjecture for countable graphs

摘录 / EXCERPT: Let G be a countable graph, and assign to each vertex v a list $L(v)$ of available colors. The graph is majority k-choosable if, for every such assignment with $|L(v)|=k$, it...

conjecture | open | MathDB 页面 | 原文线索

V2-049836 | MATHDB-345743

The 2-coloring conjecture for countable acyclic digraphs

摘录 / EXCERPT: Let D be a countable acyclic digraph. A vertex coloring of D is a majority coloring if, at every vertex, at most half of its outgoing edges are bad, meaning that their endpoint...

conjecture | open | MathDB 页面 | 原文线索

V2-049837 | MATHDB-345744

The list 2-coloring conjecture for countable acyclic digraphs

摘录 / EXCERPT: Let D be a countable acyclic digraph, and assign to each vertex v a list $L(v)$ of available colors. The digraph is majority k-choosable if, for every such assignment with...

conjecture | open | MathDB 页面 | 原文线索

V2-049838 | MATHDB-345746

The additive 1-2 Conjecture

摘录 / EXCERPT: Let G be a graph, and let $\chi(G)$ denote its chromatic number. For a vertex decoration f, define $S(v)=\sum_{x\in N(v)}f(x)$, and call f cool if $S(u)\ne S(v)$ for every ad...

conjecture | open | MathDB 页面 | 原文线索

V2-049839 | MATHDB-345749

The ironic decoration conjecture

摘录 / EXCERPT: Let G be a graph, let d_v be the degree of v, and assign a number $f(v)$ to each vertex. Define $M(v)=f(v)d_v$. An assignment is ironic if $M(u)\ne M(v)$ for every adjacent p...

conjecture | open | MathDB 页面 | 原文线索

V2-049840 | MATHDB-345756

The bounded-decoration conjecture for hypergraphs

摘录 / EXCERPT: Let $\mathcal F$ be a hypergraph, meaning a family of subsets of a finite set V. Assign a real number $f(v)$ to each $v\in V$, and write ... for $A\in\mathcal F$. The decoration is...

conjecture | open | MathDB 页面 | 原文线索

V2-049841 | MATHDB-345761

The P_{4k-4} Cartesian-product non-Hamiltonicity conjecture

摘录 / EXCERPT: For $k\geq 3$, let P_{4k-4} be the path on $4k-4$ vertices, and let G be a connected graph with a path factor and maximum degree $\Delta(G)=k$. The Cartesian product...

conjecture | open | MathDB 页面 | 原文线索

V2-049842 | MATHDB-345762

The balanced-bipartite Cartesian-product Hamiltonicity conjecture

摘录 / EXCERPT: Let G be a graph with a path factor, let $\Delta(G)$ be its maximum degree, and let P_n be the path on n vertices. The Cartesian product $P_n\square G$ is balanced bipartite...

conjecture | open | MathDB 页面 | 原文线索

V2-049843 | MATHDB-345766

The Up-Down last-passage percolation distributional identity

摘录 / EXCERPT: Let $(X_{i,j})_{i,j\ge 1}$ be an array of independent, identically distributed exponential random variables of rate 1. For each $n\ge 2$, let $\bm{V}_n$ denote the vector of last...

conjecture | open | MathDB 页面 | 原文线索

V2-049844 | MATHDB-345767

The staircase Young tableaux-sorting networks generating-function identity

摘录 / EXCERPT: For $n\ge 2$, let δ_n be the staircase Young diagram, let $\operatorname{SYT}(\delta_n)$ be its standard Young tableaux, and let SN_n be the set of sorting...

conjecture | open | MathDB 页面 | 原文线索

V2-049845 | MATHDB-345796

Primary stranded-graph bound for traceless Hermitian matrix models

摘录 / EXCERPT: Let $\mathcal G$ be a connected vacuum Feynman graph, let $\alpha(\mathcal G)$ denote its large-D exponent, and let $P_{\mathcal G}$ be the associated polynomial. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049846 | MATHDB-345802

Empty-intersection conjecture for copies of complete graphs above the Turán threshold

摘录 / EXCERPT: Let $V_1,V_2,\ldots,V_{k-1}$ be the vertex classes of the balanced Turán graph $T_{k-1}(n)$, with sizes satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-049847 | MATHDB-345803

Triangle-covering conjecture for graphs above the Mantel threshold

摘录 / EXCERPT: Let G be an n-vertex graph, let $T(G)$ denote its number of triangles, and let $\tau_{\triangle}(G)$ be the minimum size of a vertex set meeting every triangle. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-049848 | MATHDB-345810

NP-completeness conjecture for graph balanceability

摘录 / EXCERPT: Let G be a graph. A graph is balanceable if it admits a balanced signing, namely an assignment of signs to its edges such that every cycle has equally many positive and negative...

conjecture | open | MathDB 页面 | 原文线索

V2-049849 | MATHDB-345811

The regularity formula for neighborhood ideals of forests

摘录 / EXCERPT: Regularity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049850 | MATHDB-345825

Conjecture on invariant extensions and regularity of edge-ideal powers

摘录 / EXCERPT: Invariant-extension conjecture. If $\reg(I(G)^k)=2k$ for all $k\geq 2$, then $\reg(I(G')^k)=2k$ for all $k\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-049851 | MATHDB-345826

Generalized Nevo-Peeva conjecture on powers of edge ideals

摘录 / EXCERPT: Generalized Nevo-Peeva conjecture. If $\reg(I(G))=n$, then $I(G)^s$ has a linear resolution for all $s\geq n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-049852 | MATHDB-345827

Induced-subgraph family conjecture for gap-free graphs

摘录 / EXCERPT: Induced-subgraph family conjecture. For all integers $n\geq2$, there exists a family of graphs $\{G_i^{(n)}\}$ such that $I(G_i^{(n)})^k$ has a linear resolution for all...

conjecture | open | MathDB 页面 | 原文线索

V2-049853 | MATHDB-345836

Gutman et al.'s E-LE conjecture for graph energy

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph. Its energy is $E(\mathcal{G})$, and its Laplacian energy is $LE(\mathcal{G})$. E-LE conjecture. For any graph $\mathcal{G}$, ... Gutman et al. posed thi...

conjecture | open | MathDB 页面 | 原文线索

V2-049854 | MATHDB-345852

Generalised t-sum conjecture for the d-dimensional rigidity matroid

摘录 / EXCERPT: Let $G=(V,E)$, $G_1=(V_1,E_1)$ and $G_2=(V_2,E_2)$ be graphs. Say that G is a t-sum of G_1,G_2 along an edge e when ... Here ${\mathcal R}_d$ denotes the d-dimensional rigi...

conjecture | open | MathDB 页面 | 原文线索

V2-049855 | MATHDB-345884

Kuperberg's sharpness conjecture for triangulation enumeration

摘录 / EXCERPT: Let $a_0=1$, and for each positive integer n, let a_n be the number of triangulations of a regular n-gon such that every internal vertex has degree at least 6. Define the g...

conjecture | open | MathDB 页面 | 原文线索

V2-049856 | MATHDB-345900

Conjecture on AKIA and KIA approximation performance in packet-switched networks

摘录 / EXCERPT: Let PSNs denote packet-switched networks, and let AKIA and KIA be the two information-assumption regimes used for the delay approximations. Let R_{hi} and R_{lo} denote the hig...

conjecture | open | MathDB 页面 | 原文线索

V2-049857 | MATHDB-345908

The degree-sensitive density conjecture for 3-flow-critical graphs

摘录 / EXCERPT: Degree-sensitive density conjecture. For every 3-flow-critical graph G on n vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-049858 | MATHDB-345919

The natural orientation conjecture for finite graphs

摘录 / EXCERPT: A finite graph Γ is called naturally oriented if $H^1(\Gamma)$ has a basis whose supports are isomorphic, in the orientation-preserving sense, to cycles whose edges all poin...

conjecture | open | MathDB 页面 | 原文线索

V2-049859 | MATHDB-345928

Faudree et al.'s bipartite strong chromatic index conjecture

摘录 / EXCERPT: Faudree et al.'s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049860 | MATHDB-345931

Universality conjecture for heptagon relations over fields

摘录 / EXCERPT: A heptagon relation is an algebraic relation involving the 6×6 matrices associated with the colorings of a 6-vertex simplex; here the colors are elements of a field, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-049861 | MATHDB-345961

The minimum codegree threshold conjecture for two tight paths in 3-graphs

摘录 / EXCERPT: Let H be an n-vertex 3-graph, and let $[?]$ denote its minimum codegree, the minimum number of common neighbours over all pairs of vertices. A collection of tight paths is ve...

conjecture | open | MathDB 页面 | 原文线索

V2-049862 | MATHDB-345980

Pitones-Reyes-Toledo conjecture on Cohen-Macaulay vertex-weighted edge ideals

摘录 / EXCERPT: Let $\mathcal{D}=(V(\mathcal{D}),E(\mathcal{D}))$ be a vertex-weighted oriented graph, let G be its underlying graph, and let $S=\mathbb{K}[x_1,\ldots,x_n]$ be the polynomial rin...

conjecture | open | MathDB 页面 | 原文线索

V2-049863 | MATHDB-345986

Xiao-Katona-Xiao-Zamora conjecture on the Turán number of squared paths

摘录 / EXCERPT: Xiao-Katona-Xiao-Zamora conjecture. For the square of the path P_k, one has

conjecture | open | MathDB 页面 | 原文线索

V2-049864 | MATHDB-345989

The pagenumber conjecture for complete expansion graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $\mathcal{E}_{c}(K_n)$ denote its complete expansion graph, and let $pn(G)$ be the pagenumber of a graph G. For a positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-049865 | MATHDB-345990

Pivot-minor Erdős-Hajnal conjecture

摘录 / EXCERPT: Let H be a graph. For a graph G, let $\boldsymbol{\alpha}(G)$ be its maximum independent-set size and $\boldsymbol{\omega}(G)$ its maximum clique size. Pivot-minor Erdős-Hajnal...

conjecture | open | MathDB 页面 | 原文线索

V2-049866 | MATHDB-345991

Strong pivot-minor Erdős-Hajnal conjecture

摘录 / EXCERPT: Let H be a graph. A set of vertices A is complete to a disjoint set B if every vertex of A is adjacent to every vertex of B, and anti-complete if no such edges exist. Str...

conjecture | open | MathDB 页面 | 原文线索

V2-049867 | MATHDB-345998

The 1-2-3 Conjecture for vertex-colouring edge weightings

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph with at least two edges. A 3-weighting of G is a mapping $\omega:E\to\{1,2,3\}$, and it is vertex-colouring if the weighted degree...

conjecture | open | MathDB 页面 | 原文线索

V2-049868 | MATHDB-346001

Rainbow saturation conjecture for uniform hypergraphs

摘录 / EXCERPT: Rainbow hypergraph saturation conjecture. For every k-uniform hypergraph H, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049869 | MATHDB-346002

Upper edge-connectivity bound for the rainbow disconnection number

摘录 / EXCERPT: Rainbow disconnection bound. The conjecture states

conjecture | open | MathDB 页面 | 原文线索

V2-049870 | MATHDB-346015

The simultaneous clique-maximisation conjecture for graphs on surfaces

摘录 / EXCERPT: Simultaneous clique-maximisation conjecture. If $C(G)=C(\Sigma,n)$, then for every $s\geqslant 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-049871 | MATHDB-346016

The triangle-maximisation conjecture for total clique counts on surfaces

摘录 / EXCERPT: Triangle-maximisation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049872 | MATHDB-346040

Thomassen's conjecture on dense bipartite subgraphs of large girth

摘录 / EXCERPT: Thomassen's conjecture. There is a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that, for all $t,k\in\mathbb{N}$, every bipartite graph of average degree at least $f(t,k)$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-049873 | MATHDB-346041

Dense induced subgraphs of large girth in bipartite graphs

摘录 / EXCERPT: The induced-subgraph conjecture. There is a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that, for all $t,k\in\mathbb{N}$, every bipartite graph of average degree at least...

conjecture | open | MathDB 页面 | 原文线索

V2-049874 | MATHDB-346057

The covariance-matrix conjecture for planar leaf-rooted trees

摘录 / EXCERPT: Let G and G' be planar leaf-rooted trees with no vertices of degree 2. For each tree, let $\mathbf{\Sigma}_G$ and $\mathbf{\Sigma}_{G'}$ be the covariance matrices associated...

conjecture | open | MathDB 页面 | 原文线索

V2-049875 | MATHDB-346058

The weak covariance-matrix conjecture for planar leaf-rooted trees

摘录 / EXCERPT: Let G and G' be planar leaf-rooted trees with no vertices of degree 2. For each tree, let $\mathbf{\Sigma}_G$ and $\mathbf{\Sigma}_{G'}$ be the covariance matrices determined...

conjecture | open | MathDB 页面 | 原文线索

V2-049876 | MATHDB-346060

Seymour's Exact Conjecture for planar multigraphs

摘录 / EXCERPT: Let G be a planar multigraph, and let $\chi'(G)$ be its chromatic index and $\chi'_f(G)$ its fractional chromatic index. Seymour's Exact Conjecture. Every planar multigraph G s...

conjecture | open | MathDB 页面 | 原文线索

V2-049877 | MATHDB-346061

Hilton's Overfull Conjecture for dense graphs

摘录 / EXCERPT: Let G be a graph with vertex set $V(G)$, maximum degree Δ, chromatic index $\chi'(G)$, and fractional chromatic index $\chi'_f(G)$. Hilton's Overfull Conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-049878 | MATHDB-346062

HZ-graph classification equivalent to the Core Conjecture

摘录 / EXCERPT: Let G be an HZ-graph, meaning a connected Class 2 graph with $\Delta(G_\Delta)\le2$, and let P^* be the Petersen graph with one vertex removed. Let $\mathcal{O}_\Delta$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-049879 | MATHDB-346068

Orbifold conjecture for twisted intertwining operators

摘录 / EXCERPT: Twisted intertwining-operator conjecture. The twisted intertwining operators among the g-twisted V-modules, for all $g\in G$, satisfy associativity, commutativity and modular i...

conjecture | open | MathDB 页面 | 原文线索

V2-049880 | MATHDB-346069

G-crossed tensor-category conjecture for twisted modules

摘录 / EXCERPT: G-crossed tensor-category conjecture. This category has a natural structure of a G-crossed tensor category satisfying additional properties.

conjecture | open | MathDB 页面 | 原文线索

V2-049881 | MATHDB-346084

Conjecture on the growth of two-colorings avoiding monochromatic even cycles

摘录 / EXCERPT: Let $f_k(n,H)$ denote the maximum number of edges in a k-edge-coloring of a graph on n vertices that contains no monochromatic copy of H. For an integer $ \ell\geqslant 3$, c...

conjecture | open | MathDB 页面 | 原文线索

V2-049882 | MATHDB-346086

Polynomial-time recognition of supereulerian 2-edge-coloured complete multipartite graphs

摘录 / EXCERPT: A 2-edge-coloured complete multipartite graph is a complete multipartite graph whose edges are assigned one of two colours, and a graph is supereulerian if it contains a spanning e...

conjecture | open | MathDB 页面 | 原文线索

V2-049883 | MATHDB-346093

Alon-Tarsi bound for vertex-face graphs of plane graphs

摘录 / EXCERPT: Let G be a plane graph, and let G_{vf} denote its vertex-face graph. The Alon-Tarsi conjecture. Every plane graph G satisfies ... This would extend the corresponding list-colou...

conjecture | open | MathDB 页面 | 原文线索

V2-049884 | MATHDB-346094

Alon-Tarsi bound for combined vertex-edge-face graphs

摘录 / EXCERPT: Let G be a plane graph, and let $\overline{G}_{vef}$ denote its combined vertex-edge-face graph. The Alon-Tarsi conjecture. Every plane graph G satisfies ... If true, this would...

conjecture | open | MathDB 页面 | 原文线索

V2-049885 | MATHDB-346108

Global stability conjecture for the CRC-CRN

摘录 / EXCERPT: The CRC-CRN is the colorectal-cancer-cell chemical reaction network described in the paper. The global stability condition is the condition introduced in Definition. Global stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-049886 | MATHDB-346109

The universal exponent conjecture for homeomorphs of 3-graphs

摘录 / EXCERPT: Universal exponent conjecture. For each 3-graph H, there exists $C_H>0$ such that every 3-graph G on n vertices with at least

conjecture | open | MathDB 页面 | 原文线索

V2-049887 | MATHDB-346143

Closure of (strongly) sat-sharp graphs under disjoint union

摘录 / EXCERPT: Let H_1 and H_2 be graphs that are (strongly) sat-sharp, and let H_1+H_2 denote their disjoint union. Disjoint-union conjecture. If H_1 and H_2 are (strongly) sat-sharp g...

conjecture | open | MathDB 页面 | 原文线索

V2-049888 | MATHDB-346147

Kelly's $[?]$-reconstruction conjecture

摘录 / EXCERPT: For a graph G, write $\mathcal{D}_{|V(G)|-\ell}(G)$ for the multiset of induced subgraphs obtained by deleting ℓ vertices, and call G ℓ-reconstructible if this deck d...

conjecture | open | MathDB 页面 | 原文线索

V2-049889 | MATHDB-346149

Feigin-Tipunin conjecture for irreducible and complete modules

摘录 / EXCERPT: Feigin-Tipunin conjecture. 1. The $W(p)_Q$-module $H^0(\xi_\lambda)$ is irreducible. 2. For every $\alpha\in P_+\cap Q$, $\mathcal{W}_{-\sqrt{p}\alpha+\lambda}$ is an irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-049890 | MATHDB-346150

Conjecture on minimal-expression independence of narrow screening operators

摘录 / EXCERPT: Narrow-screening conjecture. For every $\lambda\in\Lambda$ and $\sigma\in W$, $F_{\sigma,\lambda}$ is independent of the choice of a minimal expression of σ. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-049891 | MATHDB-346167

The rainbow vertex-disconnection Nordhaus-Gaddum conjecture

摘录 / EXCERPT: Let G and $\overline{G}$ be nontrivial connected graphs of order $n\geq 8$. Write $rvd(G)$ for the rainbow vertex-disconnection number of G. Rainbow vertex-disconnection Nordha...

conjecture | open | MathDB 页面 | 原文线索

V2-049892 | MATHDB-346193

The rainbow matching conjecture for general graphs

摘录 / EXCERPT: Let $(m,n)\to k$ mean that every m matchings of size n in any graph have a rainbow matching of size k. The notation $(m,n)\to_{\mathcal{B}} k$ has the same meaning for bipart...

conjecture | open | MathDB 页面 | 原文线索

V2-049893 | MATHDB-346194

The rainbow matching conjecture $(n,n)\to n-1$

摘录 / EXCERPT: Let $(m,n)\to k$ mean that every m matchings of size n in any graph have a rainbow matching of size k. Rainbow matching conjecture. ... The conjecture is stated as a general-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-049894 | MATHDB-346206

Chang-Roussel's upper-bound conjecture for power domination in uniform hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}$ be a connected r-uniform hypergraph on n vertices, and let $\gamma_p^k(\mathcal{H})$ denote its k-power domination number. Assume $k+r\leq n$. Chang-Roussel...

conjecture | open | MathDB 页面 | 原文线索

V2-049895 | MATHDB-346209

Jackson's Hamiltonian cycle conjecture for 2-connected bipartite graphs

摘录 / EXCERPT: Let n, m, and $b4$ be integers with $n > b4$. Let $\mathcal G(n,m,b4)$ denote the set of bipartite graphs with parts of sizes n and m in which every vertex in the part of s...

conjecture | open | MathDB 页面 | 原文线索

V2-049896 | MATHDB-346212

Planar maximal 3-degenerate graphs are good

摘录 / EXCERPT: Planar maximal 3-degenerate graph conjecture. Every planar maximal 3-degenerate graph is good.

open_problem | open | MathDB 页面 | 原文线索

V2-049897 | MATHDB-346217

Turán's conjectures for 3-uniform hypergraphs

摘录 / EXCERPT: Let $t_r(q,p)=\lim_{n\to\infty}T_r(n,q,p)/\binom{n}{r}$, where $T_r(n,q,p)$ is the minimum number of edges in an r-uniform hypergraph on n vertices having the property that eve...

conjecture | open | MathDB 页面 | 原文线索

V2-049898 | MATHDB-346218

Large-p conjecture for Turán densities of uniform hypergraphs

摘录 / EXCERPT: Let X be a finite set, and let $\mathcal{H}\subseteq\binom{X}{r}$ be an r-uniform hypergraph. For integers $q\geq p\geq r\geq2$, say that $\mathcal{H}$ has the (q,p)-property...

conjecture | open | MathDB 页面 | 原文线索

V2-049899 | MATHDB-346219

Asymptotic two-block conjecture for forbidden hypergraph pairs

摘录 / EXCERPT: Let $t(n,\mathcal{R}_1,\mathcal{R}_2)$ denote the extremal quantity for the pair of 3-uniform hypergraphs $\mathcal{R}_1$ and $\mathcal{R}_2$. Two-block conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-049900 | MATHDB-346225

Bethe quantum-number conjecture for the maximal six-vertex-model eigenvalue

摘录 / EXCERPT: Let n be the fixed number of Bethe roots, let H be the parameter in the transfer matrix, and let I_j denote the corresponding Bethe quantum numbers. Bethe quantum-number conj...

conjecture | open | MathDB 页面 | 原文线索

V2-049901 | MATHDB-346226

Thermodynamic Bethe-root contour conjecture for the homogeneous six-vertex model

摘录 / EXCERPT: Fix the ratio n/N, where n is the number of Bethe roots and N is the system size, and let C denote the contour described in the paper. Thermodynamic Bethe-root contour conj...

conjecture | open | MathDB 页面 | 原文线索

V2-049902 | MATHDB-346227

Asymptotic conjecture for the number of even-length paths in planar graphs

摘录 / EXCERPT: Let P_k be a path of length k, and let $f(n,P_k)$ denote the maximum number of copies of P_k in a planar graph on n vertices. Even-length path conjecture. For every positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-049903 | MATHDB-346234

The classification conjecture for winning hat games on cycles with a vertex of hatness 2

摘录 / EXCERPT: Let G be a cycle C_n and let h be a hat function such that $2 \leq h(v) \leq 4$ for every vertex v. Let $A\in V(G)$ be such that $h(A)=2$. The game...

conjecture | open | MathDB 页面 | 原文线索

V2-049904 | MATHDB-346238

Inequivalence conjecture for rooted spanning trees and initial states in de Bruijn sequence generation

摘录 / EXCERPT: Let F_h be the feedback function defined from a nonsingular function h in Equation, with $n\geq m+2$. An input pair consists of a rooted spanning tree and an initial state, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049905 | MATHDB-346254

Conjecture on sparse triple systems with few triangle-free subgraphs

摘录 / EXCERPT: Sparse triple-system conjecture. For $\triangle\geq 1$, there exists a triple system G with maximum degree $\triangle$ such that, as $\triangle\to\infty$, every T^3-free subgra...

conjecture | open | MathDB 页面 | 原文线索

V2-049906 | MATHDB-346255

Conjectured interpolation formula for triangle-free subgraphs of random hypergraphs

摘录 / EXCERPT: Interpolation conjecture. For $\frac{3}{2}<x\leq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-049907 | MATHDB-346257

The transition conjecture for unicyclic graphs

摘录 / EXCERPT: Let U and U' be unicyclic graphs with the same degree sequence. A 2-switch is an edge-switching operation on a graph. Transition conjecture. There is a sequence of 2-switches t...

conjecture | open | MathDB 页面 | 原文线索

V2-049908 | MATHDB-346267

Ziegler's vertex-subset conjecture for duals of simple polytopes

摘录 / EXCERPT: Let P be a simple polytope of dimension at least 3, and let P^* denote its dual polytope. For a subset S of the vertices of P, write $\operatorname{conv}(S)$ for its conv...

conjecture | open | MathDB 页面 | 原文线索

V2-049909 | MATHDB-346270

The equivariant reconstruction conjecture for VOA representation categories

摘录 / EXCERPT: Let $\mathcal{V}$ be a vertex operator algebra with symmetry group G, and let G_0 be the identity component of G. Let $\operatorname{Rep}(\mathcal{V})$ denote the representat...

conjecture | open | MathDB 页面 | 原文线索

V2-049910 | MATHDB-346271

The non-lattice realization conjecture for abelian anyon models

摘录 / EXCERPT: Let $\mathcal{B}$ be an abelian anyon model. A vertex operator algebra is extremal in the sense used by the source, and a realization is non-lattice when it is not a lattice realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-049911 | MATHDB-346284

Aharoni-Howard conjecture on rainbow matchings

摘录 / EXCERPT: For a positive integer n, write $[n]=\{1,\ldots,n\}$, and let $\binom{[n]}{k}$ be the family of k-subsets of $[n]$. Let $\mathcal F=\{F_1,\ldots,F_t\}$ be a family of subsets o...

conjecture | open | MathDB 页面 | 原文线索

V2-049912 | MATHDB-346303

Star-graph averaging conjecture

摘录 / EXCERPT: Let S_n be the star graph on $n+1$ vertices, and let x be the weight vector on S_n that gives weight $1-\frac{1}{n+1}$ to the central vertex and weight $-\frac{1}{n+1}$ to ev...

conjecture | open | MathDB 页面 | 原文线索

V2-049913 | MATHDB-346304

Path-graph averaging conjecture without a logarithmic factor

摘录 / EXCERPT: Let P_n be the path graph on n vertices. For a weight vector x on P_n, define ... where $\overline{x}$ denotes the average-weight vector. Path-graph averaging conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-049914 | MATHDB-346307

Class II augmentation conjecture for a perfect matching

摘录 / EXCERPT: Let G be a graph admitting a perfect matching M, and let $G+tM$ denote the multigraph obtained by adding t copies of M to G. Class II augmentation conjecture. If $G+M$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-049915 | MATHDB-346308

Perfect-matching index four characterization of shortest cycle covers

摘录 / EXCERPT: Let G be a bridgeless cubic graph. Write $scc(G)$ for the minimum total length of a cycle cover, and let $\chi'_{e}(G)$ denote the edge-chromatic number of G. Perfect-matching...

conjecture | open | MathDB 页面 | 原文线索

V2-049916 | MATHDB-346329

Dolbeault homology vertex-algebra conjecture for affine Grassmannian bundles

摘录 / EXCERPT: Let G be a group, with formal disc and loop groups $G(\mathcal{O})$ and $G(\mathcal{K})$, and let Gr_G be the affine Grassmannian. Let $\mathcal{V}$ be any verte...

conjecture | open | MathDB 页面 | 原文线索

V2-049917 | MATHDB-346365

The vector-valued generating-function identity for standard Young tableaux and sorting networks

摘录 / EXCERPT: For $n\geq 2$, let $\operatorname{SYT}(\delta_n)$ be the set of standard Young tableaux of staircase shape δ_n, let SN_n be the set of sorting networks on...

conjecture | open | MathDB 页面 | 原文线索

V2-049918 | MATHDB-346368

The Graovac-Ghorbani index upper-bound conjecture for bicyclic graphs

摘录 / EXCERPT: Let $\mathcal{B}'_n$ be the family of all bicyclic graphs on n vertices, and let $G\in\mathcal{B}'_n$ have order $n\geq 8$. The Graovac-Ghorbani atom-bond connectivity index is d...

conjecture | open | MathDB 页面 | 原文线索

V2-049919 | MATHDB-346369

The minimum Graovac-Ghorbani index conjecture for the B2 family

摘录 / EXCERPT: Let $B_2(n)$ denote the family of bicyclic graphs specified in the paper, and let $G\in B_2(n)$ have order $n\geq 11$. The Graovac-Ghorbani atom-bond connectivity index is denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-049920 | MATHDB-346378

De Caen-Gregory-Pritikin biclique-cover conjecture for complete graphs

摘录 / EXCERPT: De Caen-Gregory-Pritikin conjecture. For every positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-049921 | MATHDB-346379

De Caen-Erdős-Pullman-Wormald clique-partition conjecture

摘录 / EXCERPT: De Caen-Erdős-Pullman-Wormald conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-049922 | MATHDB-346391

The nonsingularity conjecture for random regular graph adjacency matrices

摘录 / EXCERPT: Random regular graph singularity conjecture. For every $3\leq d\leq n-1$, $A_{n,d}$ is nonsingular with probability $1-o(1)$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-049923 | MATHDB-346405

The fair cake-cutting conjecture for uniformly most reliable hamiltonian graphs

摘录 / EXCERPT: Let $FCG_{n,c}$ denote the fair cake-cutting graph with parameters n and c. A graph is uniformly most reliable for fixed order n and size m if its reliability polynomial is...

conjecture | open | MathDB 页面 | 原文线索

V2-049924 | MATHDB-346406

Asymptotic six-color conjecture for cycle-path Cartesian products

摘录 / EXCERPT: Asymptotic six-color conjecture. There exist constants K_1 and K_2 such that for every pair of integers m and n, where $m \ge K_1$ and $n \ge K_2$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049925 | MATHDB-346407

Asymptotic six-color conjecture for Cartesian products of cycles

摘录 / EXCERPT: Asymptotic cycle-product six-color conjecture. There exists a constant L such that for every pair of integers m and n, where $m,n \ge L$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-049926 | MATHDB-346431

Pinned broken-line shelling conjecture for matroid h-vectors

摘录 / EXCERPT: Let M be a matroid. A pinned broken line shelling order is a shelling order obtained from a broken-line shelling whose normal vectors lie in the normal cone of one fixed vertex o...

conjecture | open | MathDB 页面 | 原文线索

V2-049927 | MATHDB-346435

Bounded-degree Cayley graphs without a bounded eigenbasis

摘录 / EXCERPT: Bounded-degree eigenbasis conjecture. There exists some d such that for every C there exists a d-regular Cayley graph without an orthonormal C-bounded eigenbasis.

conjecture | open | MathDB 页面 | 原文线索

V2-049928 | MATHDB-346450

The theta-pyramid-prism-turtle minimal separator conjecture

摘录 / EXCERPT: A graph is a finite undirected graph, and a minimal separator is a vertex set that minimally separates some pair of vertices. Theta-pyramid-prism-turtle conjecture. There is a poly...

conjecture | open | MathDB 页面 | 原文线索

V2-049929 | MATHDB-346451

The k-creature minimal separator conjecture

摘录 / EXCERPT: For an integer $k \geq 3$, a graph G is a k-creature if ... such that $G[A]$ and $G[B]$ are connected, A is anticomplete to B, for $i=1,\dots,k$, $x_i y_i\in E(G)$, x_i has...

conjecture | open | MathDB 页面 | 原文线索

V2-049930 | MATHDB-346457

Odd-uniformity Erdős-Gallai threshold conjecture for link graphs

摘录 / EXCERPT: Odd-uniformity link-graph conjecture. When $k-d$ is odd, $\mathrm{eg}(k-d)$ is attained by the $(k-d)$-uniform link graphs of the Han-Zhao construction.

conjecture | open | MathDB 页面 | 原文线索

V2-049931 | MATHDB-346493

MDS conjecture for general codes

摘录 / EXCERPT: Let $G=([n],E)$ be a k-uniform hypergraph, and let $q(G)$ be the smallest alphabet size for which there are encoding and decoding functions recovering every message from the symb...

conjecture | solved | MathDB 页面 | 原文线索

V2-049932 | MATHDB-346503

Xiao and Katona's clique-covering conjecture

摘录 / EXCERPT: Let $t_{k-1}(n)$ denote the maximum number of edges in a K_{k-1}-free graph on n vertices, and let $\tau_k(G)$ be the minimum size of a vertex set meeting every copy of K_k i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-049933 | MATHDB-346530

The permutation-orbifold isomorphism for the principal mathfrak{f}_4 affine W-algebra

摘录 / EXCERPT: Let $\mathcal{L}_{-22/5}$ be the simple Virasoro vertex algebra of central charge $-22/5$, and let S_4 act on its fourth tensor power by permuting the tensor factors. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-049934 | MATHDB-346551

The conjecture on the extremal function for three forbidden triangle colorings

摘录 / EXCERPT: Let $\mathcal H_1$ be the family of three edge-colored triangles consisting of a monochromatic red triangle, a triangle with two red edges and one blue edge, and a triangle with tw...

conjecture | open | MathDB 页面 | 原文线索

V2-049935 | MATHDB-346566

Tucker's infinite motion conjecture for locally finite graphs

摘录 / EXCERPT: Infinite motion conjecture. If X is a connected, locally finite graph with infinite motion, then

conjecture | open | MathDB 页面 | 原文线索

V2-049936 | MATHDB-346578

The simple vertex-superalgebra structure conjecture for A[sl_N,f,ψ]

摘录 / EXCERPT: Let $N=n+m$, let f be any nilpotent element of $\mathfrak{sl}_N$, and let ... Here $s(\lambda)$ is the lattice-label map defined in the source, and k and ℓ are related to...

conjecture | open | MathDB 页面 | 原文线索

V2-049937 | MATHDB-346579

The semi-infinite-cohomology simple-superalgebra conjecture

摘录 / EXCERPT: Let m,n and the levels k,ℓ be as in the source, and let $\pi^{k-\ell}$ be the rank-one Heisenberg vertex algebra. Semi-infinite-cohomology conjecture. The relative semi-infi...

conjecture | open | MathDB 页面 | 原文线索

V2-049938 | MATHDB-346580

The braid-reversed vertex-tensor-category conjecture for D^ψ(n,m)

摘录 / EXCERPT: Let $m,n\geq2$, let ψ be generic, and set $k=\psi-m+1$ and $\ell=\phi-n+1$, where $\phi^{-1}+\psi^{-1}=1$. Let $\mathcal H$ be a rank-one Heisenberg algebra. The D-algebra cat...

conjecture | open | MathDB 页面 | 原文线索

V2-049939 | MATHDB-346581

The orthogonal-even invariant strong-generation conjecture

摘录 / EXCERPT: Let $n\geq1$ and even $k\geq2$. Let $\mathcal O_{\mathrm{ev}}(n,k)$ be the indicated free-field vertex algebra with an O_n-action, and write $\mathcal O_{\mathrm{ev}}(n,k)^{O_n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-049940 | MATHDB-346589

HPC detection computational hardness conjecture

摘录 / EXCERPT: Let $\mathcal{G}_d(N,1/2,\kappa)$ be the hypergraphic planted clique model on N vertices with fixed integer $d\geq2$, and let $\mathcal{E}_{{\rm HPC}_D}$ denote the sum of Type-I...

conjecture | open | MathDB 页面 | 原文线索

V2-049941 | MATHDB-346590

HPDS recovery computational hardness conjecture

摘录 / EXCERPT: Let $G\sim\mathcal{G}_d(N,\kappa,q_1,q_2)$ be a hypergraphic planted dense subgraph with $q_1>q_2$, adjacency tensor ${\mathbfcal A}$, and recovery error...

conjecture | open | MathDB 页面 | 原文线索

V2-049942 | MATHDB-346593

Folklore conjecture on the order of regular graphs with prescribed girth

摘录 / EXCERPT: Let g and d be positive integers. A graph is d-regular if every vertex has degree d, its girth is the length of its shortest cycle, and its order is its number of vertices....

conjecture | open | MathDB 页面 | 原文线索

V2-049943 | MATHDB-346597

Cruickshank-Garamvölgyi-Jordán-Nixon conjecture on rigidity with added loops

摘录 / EXCERPT: Let $G=(V,E,L)$ be a looped simple graph. For a positive integer k, let $G^{[k]}$ be the graph obtained by adding k new loops at every vertex, and call G k-tight if every v...

conjecture | open | MathDB 页面 | 原文线索

V2-049944 | MATHDB-346639

Conjectures on ribbon categories of modules for W_Q(r) and B_Q(r)

摘录 / EXCERPT: The conjectures. $\operatorname{Rep}_{\langle s\rangle}\mathcal{W}_Q(r)$ is a finite non-degenerate ribbon category with...

conjecture | open | MathDB 页面 | 原文线索

V2-049945 | MATHDB-346651

Alon-Yuster conjecture on orientations avoiding directed triangles

摘录 / EXCERPT: Given a graph G and an oriented graph $\vec H$, let $D(G,\vec H)$ be the number of orientations of G containing no copy of $\vec H$, and define ... Let $K^\circlearrowright_3$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-049946 | MATHDB-346664

Rank formula for the first homology of token graphs of stars

摘录 / EXCERPT: Let S_m be the star graph on $m+1$ vertices, and let $\tok{S_m}{n}$ denote its token graph. Write $A_1(-)$ for the first homology group. The graph $\tok{S_m}{n}$ is known to have...

conjecture | open | MathDB 页面 | 原文线索

V2-049947 | MATHDB-346665

The affine character conjecture for the $F_k(q)$ series

摘录 / EXCERPT: Let $F_k(q)$ be defined by ... Here $k\geq 1$, and $L_{\mathfrak{sp}(2k)}(\Lambda_0)$ denotes the suitably normalized level-one affine vertex algebra of type $C^{(1)}_k$, with charac...

conjecture | open | MathDB 页面 | 原文线索

V2-049948 | MATHDB-346703

Robertson's conjecture on bounded Robertson chains

摘录 / EXCERPT: Let a Robertson chain of length k be the graph obtained from a path of length k by duplicating each edge. A graph contains another graph as a topological minor if the latter ca...

conjecture | open | MathDB 页面 | 原文线索

V2-049949 | MATHDB-346704

Robertson's labelled conjecture for topological minors

摘录 / EXCERPT: Let k be a positive integer. Let $G_1,G_2,\ldots$ be graphs that do not contain a Robertson chain of length at least k as a topological minor. Let Q be a set equipped with a...

conjecture | open | MathDB 页面 | 原文线索

V2-049950 | MATHDB-346722

Asymptotic planar Turán conjectures for the two six-vertex theta graphs

摘录 / EXCERPT: Asymptotic planar Turán conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-049951 | MATHDB-346725

Keevash-Mubayi-Sudakov-Verstraëte rainbow Turán conjecture for even cycles

摘录 / EXCERPT: Let C_{2k} denote the cycle of length $2k$, and let $\mathrm{ex}^*(n,H)$ be the maximum number of edges in a properly edge-coloured n-vertex graph containing no rainbow copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-049952 | MATHDB-346726

Xu-Zhang-Jing-Ge conjecture on colour-isomorphic even cycles

摘录 / EXCERPT: For an integer $r\geq 2$ and a graph H, let $f_r(n,H)$ be the smallest number C such that some proper edge-colouring of K_n with C colours contains no r vertex-disjoint c...

conjecture | refuted | MathDB 页面 | 原文线索

V2-049953 | MATHDB-346729

Lee-Loh-Sudakov conjecture on judicious bipartitions of digraphs

摘录 / EXCERPT: Lee-Loh-Sudakov conjecture. Every digraph D with m arcs and minimum outdegree at least d admits a bipartition $V(D)=V_1\cup V_2$ with

conjecture | open | MathDB 页面 | 原文线索

V2-049954 | MATHDB-346730

The diameter-two conjecture for extremal connected induced subgraph counts

摘录 / EXCERPT: Let $n\geq 5$ be an integer, and let G be an n-vertex graph. Write $\overline{G}$ for its complement and let $\eta(G)$ denote the number of connected induced subgraphs of G....

conjecture | open | MathDB 页面 | 原文线索

V2-049955 | MATHDB-346753

Azar et al.'s stationary-probability boosting conjecture for biased random walks

摘录 / EXCERPT: Let G be a graph, let p be the stationary probability of a vertex under simple random walk, and let $ obreak\varepsilon$ be the controller's bias parameter. Azar et al.'s conje...

conjecture | open | MathDB 页面 | 原文线索

V2-049956 | MATHDB-346754

The undirected BestStep PSPACE-hardness conjecture

摘录 / EXCERPT: Let $\mathtt{BestStep}$ denote the problem of computing an optimal next step for the online covering problem for the time-dependent biased random walk on a graph. BestStep hardness...

conjecture | open | MathDB 页面 | 原文线索

V2-049957 | MATHDB-346758

Gallai-Ramsey conjecture for blow-ups of complete graphs

摘录 / EXCERPT: Let $k\geq 1$ and $m\geq 2$. For a graph H, let $GR_k(H)$ denote the least order of a complete graph such that every exact k-edge-coloring contains a monochromatic copy of H,...

conjecture | open | MathDB 页面 | 原文线索

V2-049958 | MATHDB-346762

Rall's 1/2-conjecture on the domination game

摘录 / EXCERPT: Rall's 1/2-conjecture. If G is traceable, then

conjecture | open | MathDB 页面 | 原文线索

V2-049959 | MATHDB-346778

The anti-regular graph conjecture on extremal eigenvalues of threshold graphs

摘录 / EXCERPT: A threshold graph is a graph obtained from an isolated vertex by repeatedly adding either an isolated vertex or a dominating vertex. Let A_n denote the anti-regular graph on n...

conjecture | open | MathDB 页面 | 原文线索

V2-049960 | MATHDB-346797

Eventual Turán-goodness conjecture for every graph

摘录 / EXCERPT: Let H be any graph. For a positive integer k, say that H is k-Turán-good if the relevant Turán graph $T_{k-1}(n)$ is extremal for copies of H in every sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-049961 | MATHDB-346798

Turán-goodness of paths and even cycles for odd cycles

摘录 / EXCERPT: Let k and l be positive integers. Write P_k for the path with index k, C_{2k} for the even cycle with index $2k$, and C_{2l+1} for the odd cycle with index $2l+1$. A gr...

conjecture | open | MathDB 页面 | 原文线索

V2-049962 | MATHDB-346810

Dirac's conjecture for hypergraph perfect matching thresholds

摘录 / EXCERPT: Let Hleft be an n-vertex k-graph, and for a set S of vertices with $|S|=d$, let $\delta_d(H)$ denote the minimum number of edges containing any such S. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-049963 | MATHDB-346811

Random hypergraph transference conjecture for perfect matchings

摘录 / EXCERPT: Random transference conjecture. Asymptotically almost surely, every spanning subgraph $G'\subseteq G$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-049964 | MATHDB-346824

Esperet-Kang-Thomassé conjecture on dense induced bipartite subgraphs

摘录 / EXCERPT: Let d be a positive real number, and let G be a triangle-free graph with minimum degree at least d. An induced bipartite subgraph of G is a bipartite subgraph induced by a...

conjecture | open | MathDB 页面 | 原文线索

V2-049965 | MATHDB-346828

Levit-Mandrescu conjecture on independence-equivalent well-covered trees

摘录 / EXCERPT: Let G be a connected graph, let T be a well-covered tree, and let $I(G,x)$ denote the independence polynomial of G. Levit-Mandrescu conjecture. If ... then G is a well-covere...

conjecture | open | MathDB 页面 | 原文线索

V2-049966 | MATHDB-346829

Oboudi's comparability conjecture for trees under independence-polynomial order

摘录 / EXCERPT: Let T_1 and T_2 be trees of order n. Write $T_1\triangleleft T_2$ when the independence polynomial order satisfies the strict relation, and $T_2\trianglelefteq T_1$ for the c...

open_problem | open | MathDB 页面 | 原文线索

V2-049967 | MATHDB-346837

Grzesik-Janzer-Nagy conjecture on Turán numbers of graph blow-ups

摘录 / EXCERPT: Grzesik-Janzer-Nagy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-049968 | MATHDB-346838

Grzesik-Janzer-Nagy conjecture for blow-ups of even cycles

摘录 / EXCERPT: Grzesik-Janzer-Nagy conjecture for even cycles. For every pair of integers $r,k\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-049969 | MATHDB-346849

Rainbow connection number bound by forest number

摘录 / EXCERPT: Rainbow connection conjecture. A connected graph G has

conjecture | open | MathDB 页面 | 原文线索

V2-049970 | MATHDB-346850

Strong rainbow connection number bound by forest number

摘录 / EXCERPT: Strong rainbow connection conjecture. A connected graph G has

conjecture | open | MathDB 页面 | 原文线索

V2-049971 | MATHDB-346854

Finite-forbidden-subgraph conjecture for signed-graph eigenvalue multiplicities

摘录 / EXCERPT: For a signed graph $G^\pm$, let $\chi(G^\pm)$ be its signed chromatic number, let $\lambda_i(G^\pm)$ denote its ordered signed adjacency eigenvalues, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-049972 | MATHDB-346855

Maximum-action conjecture for triangle and square lattice graphs

摘录 / EXCERPT: Let a triangle or square lattice graph be equipped with edge lengths, and consider the action obtained by summing the Ollivier-Ricci curvatures over its edges. The maximum-action c...

conjecture | open | MathDB 页面 | 原文线索

V2-049973 | MATHDB-346856

Perfect-matching minimum-action conjecture for complete graphs

摘录 / EXCERPT: Let K_n be a complete graph with an even number of vertices, and let the action be the sum of the edge curvatures. A perfect-matching minimum-action conjecture states that the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-049974 | MATHDB-346857

Constant-solution conjecture for positive-edge-length tEoM settings on T_q

摘录 / EXCERPT: Let T_q be the graph under consideration, let M_q be a setting of its edge lengths, and let the tEoM denote the discrete time equations of motion. Assume that all edge lengths...

conjecture | open | MathDB 页面 | 原文线索

V2-049975 | MATHDB-346858

Non-constant-solution nonexistence conjecture for the tEoM on T_2

摘录 / EXCERPT: Let T_2 be the graph appearing in the paper, and let a setting be an assignment of edge lengths satisfying the tEoM. The non-constant-solution nonexistence conjecture states that...

conjecture | open | MathDB 页面 | 原文线索

V2-049976 | MATHDB-346863

Saeedi Madani-Kiani regularity bound conjecture for binomial edge ideals

摘录 / EXCERPT: Let G be a graph on $[n]$, let $S=K[x_1,\ldots,x_n,y_1,\ldots,y_n]$, and let J_G be its binomial edge ideal. Let $c(G)$ denote the number of maximal cliques in G. Saeedi Mada...

conjecture | open | MathDB 页面 | 原文线索

V2-049977 | MATHDB-346949

The equivalence of maximal matching and maximal independent matching

摘录 / EXCERPT: The computational-strength conjecture. $athsf{MM}$ and $athsf{MIM}$ are equivalent over $peratorname{\mathsf{RCA}}_0$, $i^1_1\text{-}\mathsf{TR}}_0$ proves $athsf{PM}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049978 | MATHDB-346950

The equivalence of finite-path perfect matching and \(\mathsf{ATR}_0\)

摘录 / EXCERPT: Finite-path perfect matching conjecture. $ext{Finite Path }\mathsf{PM}$ is equivalent to $peratorname{\mathsf{ATR}}_0$ over $peratorname{\mathsf{RCA}}_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-049979 | MATHDB-346984

Qin-Xu-Zhou's conjecture on nontrivially unstable circulants of odd order

摘录 / EXCERPT: Let X be a circulant, that is, a Cayley graph on a cyclic group. Its canonical double cover $B(X)$ is the bipartite graph with vertex set $V(X)\times\mathbb{Z}_2$ in which...

conjecture | open | MathDB 页面 | 原文线索

V2-049980 | MATHDB-347023

The logarithm-free extremal-number bound for split graphs

摘录 / EXCERPT: Let H be a graph, let ${\rm ex}(nk,H)$ denote the maximum number of edges in an H-free graph on nk vertices, and let $f(n,H)$ denote the smallest integer k for which there...

conjecture | open | MathDB 页面 | 原文线索

V2-049981 | MATHDB-347048

Jackson's degree conjecture for cycles in 2-connected bigraphs

摘录 / EXCERPT: Let $m,n,b4$ be integers, and let ${\mathcal G}(n,m,b4)$ be the set of bipartite graphs with bipartition (X,Y) such that $|X|=n\geq 2$, $|Y|=m$, and every $x\in X$ has degree at...

conjecture | open | MathDB 页面 | 原文线索

V2-049982 | MATHDB-347049

The necessary-conditions conjecture for super-cyclic bigraphs

摘录 / EXCERPT: Let G be an (X,Y)-bigraph, meaning a bipartite graph with specified ordered bipartition (X,Y). For $A\subseteq X$, define its super-neighborhood by ... The bigraph G is super...

conjecture | open | MathDB 页面 | 原文线索

V2-049983 | MATHDB-347066

Rainbow tight Hamilton cycle conjecture for Dirac hypergraphs

摘录 / EXCERPT: Rainbow tight Hamilton cycle conjecture. For every $k\geq 3$ and $\gamma>0$ there exist $c>0$ and $n_0>0$ such that if $(\mathcal{H},\phi)$ is an n-vertex colored k-graph with...

conjecture | open | MathDB 页面 | 原文线索

V2-049984 | MATHDB-347074

The conjecture on minimizing the matching partition function for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph. For a graph H, let $M_H(\lambda)=\sum_k m_k(H)\lambda^k$ be its matching partition function, where $m_k(H)$ is the number of matchings of size k...

conjecture | open | MathDB 页面 | 原文线索

V2-049985 | MATHDB-347080

Dichotomy conjecture for finite families of vertex-critical H-free graphs

摘录 / EXCERPT: Dichotomy conjecture. There is a finite number of k-vertex-critical H-free graphs if and only if H is an induced subgraph of P_4+sP_1 for some $s\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-049986 | MATHDB-347087

The min-max eigenvalue conjecture for the p-Laplacian

摘录 / EXCERPT: Let $1<p<\infty$, and let Δ_p denote the p-Laplacian of the oriented hypergraph under consideration. An eigenvalue is called a min-max eigenvalue if it is expressible in t...

conjecture | open | MathDB 页面 | 原文线索

V2-049987 | MATHDB-347097

Nonzero-coefficient conjecture for the Combinatorial Nullstellensatz framework

摘录 / EXCERPT: Nonzero-coefficient conjecture. One can choose the identifier spanning trees in a way that there is a Vandermonde-completable monomial with nonzero coefficient in

conjecture | open | MathDB 页面 | 原文线索

V2-049988 | MATHDB-347109

Conjecture on gradient-descent performance in overparameterized neural networks

摘录 / EXCERPT: Gradient-descent performance conjecture. For a proper constant η, and for a random initial point drawn from a certain distribution, such as Xavier initialization,

conjecture | open | MathDB 页面 | 原文线索

V2-049989 | MATHDB-347110

Conjecture on two-dimensional loss landscapes of wide neural networks

摘录 / EXCERPT: Let F be the loss of a neural network and let θ^* be a global minimum. Choose two vectors v_1,v_2 according to a specified rule, for example a Gaussian distribution, and...

conjecture | open | MathDB 页面 | 原文线索

V2-049990 | MATHDB-347135

Arc-disjoint branchings in 2-arc-strong digraphs of independence number 2

摘录 / EXCERPT: Let $D=(V,A)$ be a digraph. Write $\alpha(D)$ for its independence number, let $\lambda(D)$ denote its arc-connectivity, and let B_s^+ and B_s^- denote, respectively, an out-br...

conjecture | open | MathDB 页面 | 原文线索

V2-049991 | MATHDB-347136

Non-separating spanning-tree conjecture for independence number 2

摘录 / EXCERPT: Let D be a digraph on n vertices. Write $\lambda(D)$ for its arc-connectivity and $\alpha(D)$ for its independence number. A non-separating spanning tree is a spanning tree who...

conjecture | open | MathDB 页面 | 原文线索

V2-049992 | MATHDB-347137

Large-order non-separating out-branching conjecture

摘录 / EXCERPT: Let D be a digraph on at least K vertices. Write $\lambda(D)$ for its arc-connectivity and $\alpha(D)$ for its independence number. A non-separating out-branching is an out-bra...

conjecture | open | MathDB 页面 | 原文线索

V2-049993 | MATHDB-347138

Rooted non-separating out-branching conjecture

摘录 / EXCERPT: Let D be a digraph on at least L vertices. Write $\lambda(D)$ for its arc-connectivity and $\alpha(D)$ for its independence number, and let B_s^+ denote an out-branching root...

conjecture | open | MathDB 页面 | 原文线索

V2-049994 | MATHDB-347142

Realizability of rational exponents by a single bipartite graph

摘录 / EXCERPT: Let $r\in(1,2)$ be rational. For a graph F, write $\operatorname{ex}(n,F)$ for the maximum number of edges in an n-vertex graph containing no copy of F. Realizability of rati...

conjecture | open | MathDB 页面 | 原文线索

V2-049995 | MATHDB-347143

Existence of Bukh-Conlon densities in every fractional class

摘录 / EXCERPT: Existence of Bukh-Conlon densities in every fractional class. For every $s,a\in\mathbb{N}$ with $s<a$, there exists $m\in\mathbb{N}^+$ such that $m+s/a$ is a Bukh-Conlon density.

conjecture | open | MathDB 页面 | 原文线索

V2-049996 | MATHDB-347215

Lovász's perfect-matching conjecture for cubic vertex transitive graphs

摘录 / EXCERPT: Let Γ be a finite cubic vertex transitive graph, and let M be a perfect matching in Γ. Write $\Gamma \backslash M$ for the graph obtained by deleting the edges of M. Lo...

conjecture | open | MathDB 页面 | 原文线索

V2-049997 | MATHDB-347216

Equivariant partite-presentation refinement for vertex transitive graphs

摘录 / EXCERPT: Let Γ be a vertex transitive graph with a partite presentation P such that $\Gamma=\operatorname{PCay}(P)$, and let $\Gamma\leq\operatorname{Aut}_{c\text{-}loc}(\Gamma)$ be a tran...

open_problem | open | MathDB 页面 | 原文线索

V2-049998 | MATHDB-347227

Even-path inverse Turán conjecture

摘录 / EXCERPT: Let t be a positive integer, let P_{2t} denote the path of length $2t$, and let $\operatorname{ex}^{-1}(k,H)$ denote the inverse Turán number of H, namely the maximum number...

conjecture | open | MathDB 页面 | 原文线索

V2-049999 | MATHDB-347228

Inverse Turán conjecture for the four-cycle

摘录 / EXCERPT: Let C_4 denote the cycle of length four, and let $\operatorname{ex}^{-1}(k,H)$ denote the inverse Turán number of H, namely the maximum number of edges in a graph whose every...

conjecture | open | MathDB 页面 | 原文线索

V2-050000 | MATHDB-347235

Hendry's cycle-extendibility conjecture for Hamiltonian chordal graphs

摘录 / EXCERPT: Hendry's conjecture. Every Hamiltonian chordal graph is fully cycle extendible.

conjecture | refuted | MathDB 页面 | 原文线索

V2-050001 | MATHDB-347241

The strict spectral-radius minimization conjecture for block graphs

摘录 / EXCERPT: Let $\mathcal{B}(n,q)$ be the class of block graphs with parameters n and q, let P_b^q denote the corresponding q-clique path with b blocks, and let $\prec$ be the graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050002 | MATHDB-347254

Estrada-Rodríguez-Velázquez conjecture on subgraph centrality and walk-regularity

摘录 / EXCERPT: Let $\mathcal{G}=(V,E)$ be a simple undirected graph with adjacency matrix A. For $\beta>0$, the β-subgraph centrality of vertex i is $[e^{\beta A}]_{ii}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-050003 | MATHDB-347255

Estrada-Rodríguez-Velázquez conjecture on centralities of subgraph-regular graphs

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph whose vertices all have the same 1-subgraph centrality; equivalently, $[e^A]_{ii}$ is independent of the vertex i. The degree, closeness, eigenvect...

conjecture | open | MathDB 页面 | 原文线索

V2-050004 | MATHDB-347256

Kloster-Král-Sullivan rational subgraph-regularity conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph. It is walk-regular if $[A^r]_{ii}=[A^r]_{jj}$ for all vertices i,j and every positive integer r, and it is β-subgraph regular if...

conjecture | open | MathDB 页面 | 原文线索

V2-050005 | MATHDB-347257

Conjecture on exponential diagonal equality for non-diagonalizable directed graphs

摘录 / EXCERPT: Let A be the adjacency matrix of a directed, unweighted graph, and assume that A is non-diagonalizable. Let i and j be vertices, and let $\beta\in\overline{\mathbb{Q}}$. Ex...

conjecture | open | MathDB 页面 | 原文线索

V2-050006 | MATHDB-347263

The equivalence conjecture for Hadamard diagonalizable graphs

摘录 / EXCERPT: The equivalence conjecture for Hadamard diagonalizable graphs. If H_1 and H_2 are equivalent normalized Hadamard matrices, then G is Hadamard diagonalizable by H_1 if and o...

conjecture | open | MathDB 页面 | 原文线索

V2-050007 | MATHDB-347264

The bounded-family conjecture for Hadamard diagonalizable graphs

摘录 / EXCERPT: The bounded-family conjecture for Hadamard diagonalizable graphs. For $n=16k+8$, $k=1,2,\ldots$, there are at most 26 distinct graphs which are Hadamard diagonalizable for some H...

conjecture | open | MathDB 页面 | 原文线索

V2-050008 | MATHDB-347275

Scaling-limit conjecture for ancestral lineages in ranked tree-child networks

摘录 / EXCERPT: Let $X^{(\ell)}_k$ denote the number of lineages in the ancestry of a leaf after moving k steps away from that leaf in a uniformly sampled ranked tree-child network with ℓ l...

conjecture | open | MathDB 页面 | 原文线索

V2-050009 | MATHDB-347287

Piecewise growth-rate conjecture for $\operatorname{ex}(G_{n,p}^{(3)},C_4^{(3)})$

摘录 / EXCERPT: Let $G_{n,p}^{(3)}$ be the random 3-uniform hypergraph on n vertices, let $C_4^{(3)}$ be the 3-uniform expansion of the 4-cycle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050010 | MATHDB-347288

Minimum weighted Szeged index graphs are attained by trees

摘录 / EXCERPT: Let G be an n-vertex graph, and let $wSz(G)$ denote its weighted Szeged index. Tree attainment conjecture. The minimum weighted Szeged index among n-vertex graphs is attained...

conjecture | open | MathDB 页面 | 原文线索

V2-050011 | MATHDB-347289

Regular main branches in minimum weighted Szeged index trees

摘录 / EXCERPT: Let a regular ending branch be a minimal weighted-Szeged-index ending branch whose children have the same order, and let a main ending branch be an ending branch directly connected...

conjecture | open | MathDB 页面 | 原文线索

V2-050012 | MATHDB-347290

Non-increasing root-to-leaf degrees in minimum weighted Szeged index trees

摘录 / EXCERPT: Let T be a minimum weighted Szeged index tree, rooted at its designated root. Root-to-leaf degree conjecture. Along every path from the root to a leaf, the sequence of vertex deg...

conjecture | open | MathDB 页面 | 原文线索

V2-050013 | MATHDB-347291

Degree bound for vertices in minimum weighted Szeged index trees

摘录 / EXCERPT: Let T be a minimum weighted Szeged index tree. Maximum-degree conjecture. No vertex of T has degree greater than 6. This is an experimentally motivated structural conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050014 | MATHDB-347301

The chromatic symmetric function conjecture for triangle-free graphs

摘录 / EXCERPT: Triangle-free graph conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050015 | MATHDB-347302

The outerplane adjacency-polytope volume formula

摘录 / EXCERPT: Let G be an outerplane graph. For each connected component G_i of G, let $B_{i,1},\dots,B_{i,b_i}$ be its blocks; let $\mathscr{F}(B_{i,j})$ denote the bounded faces of...

conjecture | open | MathDB 页面 | 原文线索

V2-050016 | MATHDB-347303

The wheel-graph adjacency-polytope volume conjecture

摘录 / EXCERPT: Let $W_N=K_1\vee C_N$ be the wheel graph obtained by joining a single vertex to the cycle C_N, and let $\operatorname{NVol}(\nabla^{\operatorname{PQ}}_{W_N})$ denote the normaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-050017 | MATHDB-347304

Mubayi-Verstraëte's generalized intersecting-subgraph conjecture

摘录 / EXCERPT: Let $d\ge k\ge 4$ and let n be sufficiently large. Suppose that $\mathcal{H}\subset\binom{[n]}{k}$ contains no non-trivial intersecting subgraph of size $d+1$. A hypergraph is a...

conjecture | open | MathDB 页面 | 原文线索

V2-050018 | MATHDB-347305

Mubayi-Verstraëte's Steiner-system extremal conjecture for 3-graphs

摘录 / EXCERPT: Let $m\ge4$ and let n be sufficiently large. Suppose that $\mathcal{H}\subset\binom{[n]}{3}$ contains no non-trivial intersecting family of size $3m+1$. A Steiner $(n,3,m-1)$-sys...

conjecture | open | MathDB 页面 | 原文线索

V2-050019 | MATHDB-347324

Cameron-Lucchini-Roney-Dougal's conjecture on the generating graph and replacement number

摘录 / EXCERPT: Let G be a finite group. The generating graph $\Gamma(G)$ has vertex set G, with two distinct vertices adjacent when they generate G. A vertex is isolated if it has no adjace...

conjecture | open | MathDB 页面 | 原文线索

V2-050020 | MATHDB-347328

Strong Taylor's conjecture on finite subgraphs of shift graphs

摘录 / EXCERPT: For a cardinal κ, the shift graph $\operatorname{Sh}_n(\kappa)$ has as vertices the increasing n-tuples of ordinals less than κ, with adjacency defined by the shift...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050021 | MATHDB-347337

Boros, Caro, Füredi and Yuster's asymptotic conjecture for non-repeated cycle lengths

摘录 / EXCERPT: Let $f_2(n)$ denote the maximum, over all n-vertex 2-connected graphs, of the number of cycle lengths that occur exactly once. The authors' conjecture is ... This conjecture assert...

conjecture | open | MathDB 页面 | 原文线索

V2-050022 | MATHDB-347353

The Strong Erdős-Szemerédi conjecture for sums and products along graph edges

摘录 / EXCERPT: Strong Erdős-Szemerédi conjecture. For every $c>0$ and $\varepsilon>0$, there is a threshold n_0 such that, if $n\geq n_0$, then for every n-element set of reals $\mathcal{A}$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050023 | MATHDB-347356

Italian domination conjecture for the Cartesian product of a 4-cycle and an odd cycle

摘录 / EXCERPT: Let C_n be the cycle on an odd integer n, and let $C_4\square C_n$ denote their Cartesian product. For an Italian dominating function on a graph, assign each vertex a value in...

conjecture | open | MathDB 页面 | 原文线索

V2-050024 | MATHDB-347363

Diagonalizability conjecture for non-backtracking matrices and the 1-shell

摘录 / EXCERPT: Let G be a graph, let its non-backtracking matrix be the matrix whose entries record directed-edge transitions that do not immediately reverse direction, and let the 1-shell of...

conjecture | open | MathDB 页面 | 原文线索

V2-050025 | MATHDB-347384

Fujita-Magnant conjecture on properly colored Hamiltonian cycles

摘录 / EXCERPT: Let (G,c) be an edge-colored graph on $n\geq 3$ vertices. For each vertex v, let $d^c_G(v)$ be the number of colors on edges incident with v, and let $\delta^c(G)$ be the min...

conjecture | open | MathDB 页面 | 原文线索

V2-050026 | MATHDB-347385

Fujita-Magnant conjecture on proper vertex-pancyclicity

摘录 / EXCERPT: Let (G,c) be an edge-colored complete graph on $n\geq 3$ vertices. For each vertex v, let $d^c_G(v)$ be the number of colors on edges incident with v, and let $\delta^c(G)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-050027 | MATHDB-347402

Burr-Erdős conjecture for degenerate graphs

摘录 / EXCERPT: Let $d\in\mathbb{N}$. A graph is d-degenerate when its vertices admit an ordering in which every vertex has at most d neighbors earlier in the ordering. Burr-Erdős conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-050028 | MATHDB-347403

The finite-exception partition conjecture for multicolored infinite complete bipartite graphs

摘录 / EXCERPT: Let $r\in\mathbb{N}$. An r-colored $K_{\mathbb{N},\mathbb{N}}$ is an assignment of one of r edge colors to every edge of the infinite complete bipartite graph. Partition conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050029 | MATHDB-347417

Chung-Graham conjecture for sparse halves in K_4-free graphs

摘录 / EXCERPT: Let G be a K_4-free graph on n vertices. Chung-Graham conjecture. G contains a vertex set of size ... that spans at most $n^2/18$ edges. The bound is best possible, as shown...

conjecture | open | MathDB 页面 | 原文线索

V2-050030 | MATHDB-347428

Local Vizing's theorem

摘录 / EXCERPT: Local Vizing's theorem. There is an L-edge-colouring of G.

conjecture | open | MathDB 页面 | 原文线索

V2-050031 | MATHDB-347439

Postle's density conjecture for clique-free critical graphs

摘录 / EXCERPT: Postle's density conjecture. For every $k\geq 4$, there exists $\varepsilon_k>0$ such that if G does not contain a K_{k-2} subgraph, then

conjecture | open | MathDB 页面 | 原文线索

V2-050032 | MATHDB-347457

Protection number conjecture for Cartesian products with K_2

摘录 / EXCERPT: Protection number conjecture. For any integer $t\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050033 | MATHDB-347464

Kanté and Kwon's tree vertex-minor conjecture for linear rank-width

摘录 / EXCERPT: Kanté and Kwon's conjecture. For every tree T, the class of T-vertex-minor-free graphs has bounded linear rank-width.

conjecture | open | MathDB 页面 | 原文线索

V2-050034 | MATHDB-347465

Non-orientable realizability conjecture for degree sequences

摘录 / EXCERPT: Non-orientable realizability conjecture. Every feasible pair $(\boldsymbol{d};\boldsymbol{t})$ with $\chi(\boldsymbol{d};\boldsymbol{t})\leq 0$ can be realized on a non-orientable...

conjecture | open | MathDB 页面 | 原文线索

V2-050035 | MATHDB-347470

Reed's constrained colouring conjecture

摘录 / EXCERPT: Let G be a graph, and let L be a $(k+1)$-list assignment of G, meaning that each vertex v has a list $L(v)$ of $k+1$ available colours. Suppose that for every vertex v of...

conjecture | open | MathDB 页面 | 原文线索

V2-050036 | MATHDB-347475

Fractional and exact hypergraph clique-decomposition threshold conjecture

摘录 / EXCERPT: Threshold equality conjecture. For all $r>k\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050037 | MATHDB-347477

Minimum-degree Baranyai factorization conjecture

摘录 / EXCERPT: Minimum-degree Baranyai conjecture. For all sufficiently large n, an n-vertex k-graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-050038 | MATHDB-347489

The finite-colour Seurat game conjecture for digraphs

摘录 / EXCERPT: Let ${\bf G}^n(G,H)$ be the Seurat game with n colours played on digraphs G and H, with players $\forall$ and $\exists$, where $\forall$ seeks to distinguish non-isomorphic d...

conjecture | open | MathDB 页面 | 原文线索

V2-050039 | MATHDB-347502

The truly subquadratic map-isomorphism conjecture for unrestricted genus

摘录 / EXCERPT: Map-isomorphism conjecture. There is no $\varepsilon > 0$ for which there is an algorithm for testing isomorphism of maps of unrestricted genus in time

conjecture | open | MathDB 页面 | 原文线索

V2-050040 | MATHDB-347513

Baste-Fürst-Henning-Mohr-Rautenbach bound for minimum maximal matchings

摘录 / EXCERPT: Baste-Fürst-Henning-Mohr-Rautenbach conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050041 | MATHDB-347514

The 7/18 bound conjecture for minimum maximal matchings in cubic graphs

摘录 / EXCERPT: The 7/18 bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050042 | MATHDB-347515

Exceptional-case strengthening of the subcubic minimum maximal matching bound

摘录 / EXCERPT: Subcubic bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050043 | MATHDB-347518

Weak Hippchen conjecture for highly connected graphs

摘录 / EXCERPT: Let G be a k-connected graph on n vertices. Weak Hippchen conjecture. If $k \geq n/4$, then every pair of longest paths intersect in at least k vertices. This weakens Hippc...

conjecture | open | MathDB 页面 | 原文线索

V2-050044 | MATHDB-347519

Hippchen's conjecture for 5-connected graphs

摘录 / EXCERPT: Let G be a 5-connected graph. 5-connected Hippchen conjecture. Every pair of longest paths intersect each other in at least five vertices. The paper proves the analogous statemen...

conjecture | open | MathDB 页面 | 原文线索

V2-050045 | MATHDB-347524

A quadratic-degree bound for minimal Ramsey graphs for cliques

摘录 / EXCERPT: Let $s_r(K_k)$ denote the minimum number of edges in an r-Ramsey-minimal graph for the clique K_k. Quadratic-degree bound conjecture. For all $r\geqslant 2$ and $k\geqslant 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-050046 | MATHDB-347541

Aldous-Fill conjecture on maximum relaxation time

摘录 / EXCERPT: Let G be a connected regular graph on n vertices, and let τ denote the relaxation time of its random walk, defined by ... where η_2 is the second-largest eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-050047 | MATHDB-347542

Uniqueness conjecture for minimum-spectral-gap quartic graphs

摘录 / EXCERPT: For each $n\ge 11$, let $\mathcal{G}_n$ be the quartic graph of order n constructed from the blocks displayed in the paper, and call a quartic graph minimal when it has minimum s...

conjecture | open | MathDB 页面 | 原文线索

V2-050048 | MATHDB-347561

List Colouring Conjecture

摘录 / EXCERPT: List Colouring Conjecture. If G is a graph of maximum degree Δ and L is a list assignment to $E(G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-050049 | MATHDB-347582

Kreutzer et al.'s majority 3-colouring conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph. A majority colouring of D is a vertex colouring c such that at least half of the out-neighbours of every vertex v have a colour different from $c(v)$. K...

conjecture | open | MathDB 页面 | 原文线索

V2-050050 | MATHDB-347583

Erdős's monochromatic triangle packing conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices whose edges are coloured with two colours. A collection of triangles is pairwise edge-disjoint if no edge belongs to more than one t...

conjecture | open | MathDB 页面 | 原文线索

V2-050051 | MATHDB-347584

Jacobson's same-colour monochromatic triangle packing conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices with its edges coloured red and blue. A collection of triangles is pairwise edge-disjoint if no edge occurs in more than one triangl...

conjecture | open | MathDB 页面 | 原文线索

V2-050052 | MATHDB-347585

Akaria-Yuster's directed triangle packing conjecture for regular tournaments

摘录 / EXCERPT: Let T be a regular tournament on n vertices, where n is odd. A collection of directed triangles is pairwise edge-disjoint if no directed edge belongs to more than one triangl...

conjecture | open | MathDB 页面 | 原文线索

V2-050053 | MATHDB-347589

Alon-Cambie-Kang asymmetric choosability conjecture for bipartite graphs

摘录 / EXCERPT: Alon-Cambie-Kang conjecture. Suppose that $\Delta_A,\Delta_B\geq 2$ and that k_A,k_B are positive integers. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-050054 | MATHDB-347594

The monotonicity sandwich conjecture for average dominating-set order

摘录 / EXCERPT: Let G be a nonempty graph. For a vertex v and an edge e, write $G-v$ and $G-e$ for the graphs obtained by deleting v and e, respectively. The monotonicity sandwich conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050055 | MATHDB-347599

Tait's conjecture on Hamiltonian planar cubic graphs

摘录 / EXCERPT: Let G be a 3-regular, 3-connected planar graph. Tait's conjecture. Every such graph is Hamiltonian. The conjecture was disproved by W. T. Tutte, who constructed a counterexam...

conjecture | open | MathDB 页面 | 原文线索

V2-050056 | MATHDB-347600

Tutte's conjecture on Hamiltonian cubic bipartite graphs

摘录 / EXCERPT: Let G be a 3-regular, 3-connected bipartite graph. Tutte's conjecture. Every such graph is Hamiltonian. The supplied text introduces this conjecture after noting that the kno...

conjecture | open | MathDB 页面 | 原文线索

V2-050057 | MATHDB-347608

The regularity-threshold conjecture for random Steiner complexes

摘录 / EXCERPT: Regularity-threshold conjecture. The condition on k in Theorem 1 can be improved to

conjecture | open | MathDB 页面 | 原文线索

V2-050058 | MATHDB-347633

The toughness conjecture for Kneser graphs

摘录 / EXCERPT: The toughness conjecture for Kneser graphs. If $k\geq 5$ and $n\geq 2k+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050059 | MATHDB-347648

Small-k extension of the graph proposition

摘录 / EXCERPT: Let k be the uniformity parameter in the graph proposition referred to in the source. Small-k graph conjecture. The statement of that proposition holds whenever $k\ge4$. The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-050060 | MATHDB-347653

Wu-Zhang-Li equitable vertex arboricity conjecture

摘录 / EXCERPT: Wu-Zhang-Li conjecture. Graph G is equitably vertex k-arborable whenever

conjecture | open | MathDB 页面 | 原文线索

V2-050061 | MATHDB-347656

Conjecture on the threshold for large monochromatic k-connected subgraphs

摘录 / EXCERPT: Threshold conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050062 | MATHDB-347659

Sullivan's conjecture on feedback arc sets in m-free digraphs

摘录 / EXCERPT: Let $m\ge 3$, and let G be an m-free digraph, meaning that G has no directed cycle of length at most m. Let $\beta(G)$ be the minimum number of edges whose removal makes...

conjecture | open | MathDB 页面 | 原文线索

V2-050063 | MATHDB-347665

Fujita and Magnant's vertex-pancyclic conjecture for properly colored cycles

摘录 / EXCERPT: Fujita and Magnant's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050064 | MATHDB-347666

Bollobás and Erdős's properly colored Hamilton cycle conjecture

摘录 / EXCERPT: Bollobás and Erdős's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050065 | MATHDB-347669

The subcontraction conjecture for facial 3-colorability and 3-flows

摘录 / EXCERPT: The subcontraction conjecture. If G is a facially 3-colorable graph which does not have a subcontraction isomorphic to $K_{3,n}^+$ for some $n\geq 4$, then G is 3-flowable.

conjecture | open | MathDB 页面 | 原文线索

V2-050066 | MATHDB-347675

Aboulker et al.'s directed cycle subdivision conjecture

摘录 / EXCERPT: Let C_ℓ be the undirected cycle of length ℓ, and let C be an orientation of C_ℓ. For a digraph F, let $\operatorname{mader}_{\vec{\chi}}(F)$ be the smallest inte...

conjecture | open | MathDB 页面 | 原文线索

V2-050067 | MATHDB-347676

The linear Mader-number conjecture for Mader-perfect digraphs

摘录 / EXCERPT: For a positive integer k, let $m(k)$ denote the maximum number of arcs in a Mader-perfect digraph on k vertices. A digraph is Mader-perfect when its dichromatic Mader number eq...

conjecture | open | MathDB 页面 | 原文线索

V2-050068 | MATHDB-347677

The quadratic Mader-number conjecture for bioriented cliques

摘录 / EXCERPT: Let $\accentset{\leftrightarrow}{K}_n$ denote the bioriented complete digraph on n vertices, and let $\operatorname{mader}_{\vec{\chi}}(F)$ be the smallest integer $k\ge1$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-050069 | MATHDB-347678

The bioriented-cycle subdivision conjecture

摘录 / EXCERPT: Let D be a digraph, let $\vec{\chi}(D)$ denote its dichromatic number, and let $\accentset{\leftrightarrow}{C}_\ell$ denote the bioriented cycle of length ℓ. Bioriented-cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-050070 | MATHDB-347689

Bipartiteness conjecture for graphs with the complete identity property after attaching trees

摘录 / EXCERPT: Bipartiteness conjecture. G is bipartite.

conjecture | open | MathDB 页面 | 原文线索

V2-050071 | MATHDB-347690

Complete maximal identity conjecture for Cartesian products of paths

摘录 / EXCERPT: Path-product conjecture. The only graphs $P_i\square P_j$ with the complete maximal identity property are $P_2\square P_j$ when $j\equiv1\pmod{3}$, together with $P_2\square P_2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-050072 | MATHDB-347697

The asymptotic Zarankiewicz conjecture for crossing numbers

摘录 / EXCERPT: Let $cr(K_{n,n})$ be the crossing number of the complete bipartite graph $K_{n,n}$, and let $cr_0(K_{n,n})$ be its geodesic crossing number. Define ... and ... The limits exist, and sa...

conjecture | open | MathDB 页面 | 原文线索

V2-050073 | MATHDB-347730

Quasi-star or quasi-clique conjecture for path supersaturation

摘录 / EXCERPT: Let k and ℓ be positive integers with $k>\ell$. For an n-vertex graph containing at least m copies of the path P_k, let $\operatorname{satex}(n,P_k:m,P_{\ell})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-050074 | MATHDB-347754

Liu-Shu-Xue conjecture on distance spectral radius and maximum transmission

摘录 / EXCERPT: Let G be a connected non-transmission-regular graph with n vertices. Its distance spectral radius $\lambda_1(G)$ is the largest eigenvalue of its distance matrix, and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050075 | MATHDB-347762

Family-size stabilization conjecture for complete multipartite graphs

摘录 / EXCERPT: Let $n \geq 3$, let $1 \leq a_1 \leq \cdots \leq a_n$, and define ... If $a_1 + \cdots + a_{n-1} > 6$ and $a_n \geq e$, then the family-size stabilization conjecture asserts that ... H...

conjecture | open | MathDB 页面 | 原文线索

V2-050076 | MATHDB-347765

Mader's forest-orientation conjecture

摘录 / EXCERPT: Mader's forest-orientation conjecture. Every orientation of a forest is δ^+-maderian.

conjecture | open | MathDB 页面 | 原文线索

V2-050077 | MATHDB-347766

Aboulker et al.'s cycle-orientation conjecture

摘录 / EXCERPT: Aboulker et al.'s cycle-orientation conjecture. Every orientation of a cycle is δ^+-maderian.

conjecture | open | MathDB 页面 | 原文线索

V2-050078 | MATHDB-347767

Disjoint-union conjecture for oriented cycles

摘录 / EXCERPT: Disjoint-union conjecture for oriented cycles. Any disjoint union of orientations of cycles is δ^+-maderian.

conjecture | open | MathDB 页面 | 原文线索

V2-050079 | MATHDB-347768

Subdivision-preservation conjecture for maderian digraphs

摘录 / EXCERPT: Subdivision-preservation conjecture. If a digraph F is δ^+-maderian, all subdivisions of F are δ^+-maderian as well.

conjecture | open | MathDB 页面 | 原文线索

V2-050080 | MATHDB-347769

Long-subdivision extraction conjecture for digraphs

摘录 / EXCERPT: Long-subdivision extraction conjecture. There is a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that for every $k\in\mathbb{N}$ and every digraph D with

conjecture | open | MathDB 页面 | 原文线索

V2-050081 | MATHDB-347770

Out-leaf extension conjecture for maderian digraphs

摘录 / EXCERPT: Out-leaf extension conjecture. The digraph $F^\ast$ is δ^+-maderian as well.

conjecture | open | MathDB 页面 | 原文线索

V2-050082 | MATHDB-347771

In-dominating-set conjecture for digraphs of large outdegree

摘录 / EXCERPT: In-dominating-set conjecture. There exists a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that, for every $k\geq1$, if

conjecture | open | MathDB 页面 | 原文线索

V2-050083 | MATHDB-347798

The connectivity bound for monochromatic disconnection numbers

摘录 / EXCERPT: Let G be a graph of order n that is k-connected, meaning that deleting fewer than k vertices does not disconnect G or reduce it to a single vertex. Let $md(G)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-050084 | MATHDB-347807

Conlon's conjecture on the extremal number of fixed-length tight cycles

摘录 / EXCERPT: Let $r\geq 2$ and let $C^{(r)}_{\ell}$ denote the r-uniform tight cycle of length ℓ. Conlon's conjecture. There exists $c=c(r)>0$ such that, for every $\ell\geq r+1$ divisib...

conjecture | open | MathDB 页面 | 原文线索

V2-050085 | MATHDB-347815

Brubach-Sankararaman-Srinivasan-Xu sampling conjecture for hypergraph matchings

摘录 / EXCERPT: Let $H=(V,E)$ be a hypergraph, let $w\in\mathbb{R}_{\geq 0}^E$ be edge weights, and let $x\in[0,1]^E$ be a fractional matching, meaning that ... where $\delta(v)=\\{e\in E:v\in e\\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-050086 | MATHDB-347816

Erdős's girth conjecture

摘录 / EXCERPT: Let l,n be positive integers. A graph with girth $2l+1$ and n vertices is said to have many edges if its number of edges is bounded below by a positive constant times...

conjecture | open | MathDB 页面 | 原文线索

V2-050087 | MATHDB-347828

Rational exponent conjecture for the bipartite subdivision function

摘录 / EXCERPT: Let $r\in(1,2)$ be rational. For a bipartite graph H, let $f_2(n,H)$ denote the corresponding bipartite subdivision function. Rational exponent conjecture. For every rational num...

conjecture | open | MathDB 页面 | 原文线索

V2-050088 | MATHDB-347841

The surface Hanani-Tutte conjecture for independent crossings

摘录 / EXCERPT: Let G be a graph drawn on a surface, and call two edges independent if they have no common endpoint. Surface Hanani-Tutte conjecture. If G has a drawing on a surface in which n...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050089 | MATHDB-347842

Quadratic bound between surface crossing numbers

摘录 / EXCERPT: Let G be a graph, and for a surface $[200~\Sigma$ let $\mathop{\rm iocr}\nolimits_{\Sigma}(G)$ denote the fewest number of pairs of independent edges that cross oddly in a drawin...

conjecture | open | MathDB 页面 | 原文线索

V2-050090 | MATHDB-347862

Relative Turán lower-bound conjecture for graph hosts

摘录 / EXCERPT: Let F and H be graphs, let H have maximum degree Δ, and let $\operatorname{ex}(H,F)$ denote the maximum number of edges in an F-free subgraph of H. Relative Turán...

conjecture | open | MathDB 页面 | 原文线索

V2-050091 | MATHDB-347863

Injective coloring conjecture for regular uniform hypergraphs

摘录 / EXCERPT: Let $r\geq 2$. An injective q-coloring of an r-uniform hypergraph is a coloring of its vertices with q colors such that any two edges sharing $r-1$ vertices have distinct col...

conjecture | open | MathDB 页面 | 原文线索

V2-050092 | MATHDB-347866

Optimality conjecture for the general construction of message-passing structures

摘录 / EXCERPT: Let $n\geq 2$ and $\tau\geq \lceil\log(n-1)\rceil$, and let $S_{n,\tau}$ be the structure returned by Construction, where $S_{n,\tau}\in\mathcal{S}_{n,\tau}$, $c(S)$ denotes its co...

conjecture | open | MathDB 页面 | 原文线索

V2-050093 | MATHDB-347867

Hypergraph planted clique conjecture

摘录 / EXCERPT: Hypergraph planted clique conjecture. Suppose that $p=1/2$ and that $D\geq 2$ is a fixed integer. If

conjecture | open | MathDB 页面 | 原文线索

V2-050094 | MATHDB-347868

Modified clique-free diameter conjecture

摘录 / EXCERPT: Modified diameter conjecture. Under either the K_{k+1}-free hypothesis, or the weaker k-colorable hypothesis, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-050095 | MATHDB-347888

Vertex distortion characterizes the unknot

摘录 / EXCERPT: Let $[K]$ be the lattice-knot equivalence class of a knot K, let $[U]$ be the equivalence class of the unknot, and let $delta_V$ denote vertex distortion. Unknot characterization...

conjecture | open | MathDB 页面 | 原文线索

V2-050096 | MATHDB-347901

Grinshpun's conjecture on Ramsey-simple triangle-free graphs

摘录 / EXCERPT: Let H be a connected triangle-free graph. A graph H is 2-Ramsey-simple when $s_2(H)=2(\delta(H)-1)+1$, where $s_2(H)$ is the minimum degree that can occur in a minimal 2-Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-050097 | MATHDB-347902

Conjecture that every 3-connected graph is s_q-abundant

摘录 / EXCERPT: Let $q\geq 2$ be an integer, and let H be a graph. A graph H is s_q-abundant if there are minimal q-Ramsey graphs for H containing arbitrarily many vertices of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-050098 | MATHDB-347913

Convexity conjecture for the hypergraph-based polynomial f_d

摘录 / EXCERPT: Let $n,L,d\geq 2$ be integers, let f_d be the polynomial ... and let Δ_m be the simplex on which it is defined. Convexity conjecture. The polynomial f_d is convex over the...

conjecture | open | MathDB 页面 | 原文线索

V2-050099 | MATHDB-347943

The two-colour proper-walk connection conjecture for Hamiltonian digraphs

摘录 / EXCERPT: Let D be a Hamiltonian digraph, and let $d_D^+(v)$ and $d_D^-(v)$ denote the out-degree and in-degree of a vertex $v\in V(D)$, respectively. A vertex-colouring of D is properly...

conjecture | open | MathDB 页面 | 原文线索

V2-050100 | MATHDB-347959

The S versus complement conjecture for monochromatic covers

摘录 / EXCERPT: S versus complement conjecture. There exists a monochromatic cover of order at most $r-1$ whose subgraphs all have colors in S, or a monochromatic cover of order at most $r-1$ wh...

conjecture | open | MathDB 页面 | 原文线索

V2-050101 | MATHDB-347962

The tightly connected monochromatic subgraph conjecture for complete uniform hypergraphs

摘录 / EXCERPT: Tightly connected hypergraph conjecture. For every $r\geq 3$, every r-coloring of K_n^r contains a monochromatic tightly connected subgraph covering all vertices of K_n^r.

conjecture | open | MathDB 页面 | 原文线索

V2-050102 | MATHDB-347977

The conjectured asymptotic value of the minimum degree threshold d_n

摘录 / EXCERPT: Let d_n denote the parameter defined in the paper’s minimum-degree theorem for friends-and-strangers graphs. Minimum-degree threshold conjecture. We have ... This conjecture predic...

conjecture | open | MathDB 页面 | 原文线索

V2-050103 | MATHDB-347978

The conjectured bipartite minimum-degree threshold $d_{r,r}$

摘录 / EXCERPT: Let $d_{r,r}$ denote the parameter defined in the paper’s bipartite minimum-degree theorem, and let r be a positive integer. Bipartite minimum-degree threshold conjecture. We hav...

conjecture | open | MathDB 页面 | 原文线索

V2-050104 | MATHDB-347985

Holub-Lužar-Mihaliková-Mockovčiaková-Soták's path-cycle product conjecture

摘录 / EXCERPT: Holub-Lužar-Mihaliková-Mockovčiaková-Soták's conjecture. There exist constants c_1 and c_2 such that for all integers $m\geq c_1$ and $n\geq c_2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050105 | MATHDB-347986

Holub-Lužar-Mihaliková-Mockovčiaková-Soták's cycle product existence conjecture

摘录 / EXCERPT: Holub-Lužar-Mihaliková-Mockovčiaková-Soták's conjecture. There exists a constant c such that for every integer $m\geq c$, there is an integer n satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050106 | MATHDB-347987

Holub-Lužar-Mihaliková-Mockovčiaková-Soták's large cycle product conjecture

摘录 / EXCERPT: Holub-Lužar-Mihaliková-Mockovčiaková-Soták's stronger conjecture. There exists a constant c such that for all integers $m,n\geq c$,

conjecture | open | MathDB 页面 | 原文线索

V2-050107 | MATHDB-348006

The monotonicity conjecture for polychromatic 2-regular subgraphs

摘录 / EXCERPT: Monotonicity conjecture. Let $n\geq 6$ and let j be an integer such that $3\leq j<n$; if $j=5$, assume $n\geq 9$. If an edge-coloring φ of K_n has the property that eve...

conjecture | open | MathDB 页面 | 原文线索

V2-050108 | MATHDB-348042

Chromatic lower-bound conjecture for almost stable general Kneser hypergraphs

摘录 / EXCERPT: Let $n\ge 1$, $r\ge 2$, and $s\ge r$, and let $\cal F$ be a family of subsets of $[n]$. For the almost s-stable subfamily $\cal F_s$, let $\operatorname{KG}^r(\cal F_s)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-050109 | MATHDB-348043

Polynomially bounded nonlinearities in the NetsorT mean-convergence theorem

摘录 / EXCERPT: Let ψ be the nonlinearity appearing in the NetsorT mean-convergence theorem, and let polynomially bounded mean that ψ has polynomial growth in its inputs. The theorem den...

conjecture | open | MathDB 页面 | 原文线索

V2-050110 | MATHDB-348053

Kalai-Meshulam chromatic conjecture on induced cycles

摘录 / EXCERPT: Kalai-Meshulam chromatic conjecture. A graph with high enough chromatic number has an induced cycle of length divisible by three.

conjecture | open | MathDB 页面 | 原文线索

V2-050111 | MATHDB-348054

Kalai-Meshulam Euler characteristic conjecture for independence complexes

摘录 / EXCERPT: Kalai-Meshulam Euler characteristic conjecture. For any graph G, we have

conjecture | open | MathDB 页面 | 原文线索

V2-050112 | MATHDB-348055

Kalai-Meshulam total Betti number conjecture for independence complexes

摘录 / EXCERPT: Kalai-Meshulam total Betti number conjecture. For any graph G, we have

conjecture | open | MathDB 页面 | 原文线索

V2-050113 | MATHDB-348056

Engström's sphere-or-contractible conjecture for independence complexes

摘录 / EXCERPT: Engström's sphere-or-contractible conjecture. For any graph G, we have that $\mathrm{Ind}(H)$ is contractible or homotopy equivalent to a sphere for every induced subgraph H of...

conjecture | open | MathDB 页面 | 原文线索

V2-050114 | MATHDB-348074

Fractional Albertson-Berman conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let $\fva(G)$ denote its fractional vertex-arboricity. Fractional Albertson-Berman conjecture. Every planar graph has fractional vertex-arboricity at...

conjecture | open | MathDB 页面 | 原文线索

V2-050115 | MATHDB-348075

Fractional Akiyama-Watanabe conjecture for triangle-free planar graphs

摘录 / EXCERPT: Let G be a triangle-free planar graph, and let $\fva(G)$ denote its fractional vertex-arboricity. Fractional Akiyama-Watanabe conjecture. Every triangle-free planar graph has fra...

conjecture | open | MathDB 页面 | 原文线索

V2-050116 | MATHDB-348076

Fractional Kowalik-Luzar-Škrekovski conjecture for high-girth planar graphs

摘录 / EXCERPT: Let G be a planar graph of girth at least five, and let $\fva(G)$ denote its fractional vertex-arboricity. Fractional Kowalik-Luv{z}ar-Škrekovski conjecture. Every planar graph o...

conjecture | open | MathDB 页面 | 原文线索

V2-050117 | MATHDB-348092

Degenerate-graph density conjecture

摘录 / EXCERPT: Degenerate-graph density conjecture. For every fixed k-degenerate graph H,

conjecture | open | MathDB 页面 | 原文线索

V2-050118 | MATHDB-348100

Heroic-set conjecture for transitive tournaments and oriented forests

摘录 / EXCERPT: Let F be an oriented forest, and let K_k denote the complete symmetric digraph on k vertices. Heroic-set conjecture. For every integer k, the set ... is heroic. This is prese...

conjecture | open | MathDB 页面 | 原文线索

V2-050119 | MATHDB-348101

Dichromatic-number conjecture for triangle-free digraphs with an oriented 2-star

摘录 / EXCERPT: Let $\overleftrightarrow K_2$ be the complete symmetric digraph on two vertices, let $\overrightarrow C_3$ be the directed 3-cycle, and let S^+_2 be the oriented 2-star. For a fa...

conjecture | open | MathDB 页面 | 原文线索

V2-050120 | MATHDB-348117

Zinn-Justin's eigenvector conjecture for the supersymmetric eight-vertex model

摘录 / EXCERPT: Let $L=2n+1$ with $n\geqslant 0$, let $\eta=\pi/3$, and let V be the two-dimensional spin space. For inhomogeneity parameters $u_1,\dots,u_{2n+1}$, consider the eigenvalue probl...

conjecture | open | MathDB 页面 | 原文线索

V2-050121 | MATHDB-348148

The injective chromatic index conjecture for subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning that its maximum degree is at most 3. The subcubic injective edge-coloring conjecture. ... The injective chromatic index is the minimum number...

conjecture | open | MathDB 页面 | 原文线索

V2-050122 | MATHDB-348174

Critical recurrence and transience conjecture for biased random networks on lattices

摘录 / EXCERPT: Let p_c denote the critical probability for Bernoulli bond percolation on ${\mathbb{Z}}^d$, and let $({\mathbb{Z}}^d,\, \mathbf{C}_{\lambda_1},\, \mathbf{C}_{\lambda_2},\,p)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-050123 | MATHDB-348175

Current uniqueness conjecture for biased random networks on lattices

摘录 / EXCERPT: Let $({\mathbb{Z}}^d,\, \mathbf{C}_{\lambda_1},\, \mathbf{C}_{\lambda_2},\,p)$ be a biased disordered random network, with $d\geq 3$ and $0<\lambda_1\leq 1<\lambda_2$. A network is...

conjecture | open | MathDB 页面 | 原文线索

V2-050124 | MATHDB-348196

Rigidity conjecture for modules over self-dual quasi-lisse vertex operator algebras

摘录 / EXCERPT: Let V be a finitely strongly generated, self-dual, quasi-lisse vertex operator algebra of CFT type, and let $\mathcal{C}_V^{\mathrm{ord}}$ denote its category of ordinary modules...

conjecture | open | MathDB 页面 | 原文线索

V2-050125 | MATHDB-348240

NP-hardness conjecture for promise CSPs between $\mathbf{LO}_k$ and $\mathbf{LO}_l$

摘录 / EXCERPT: For integers $2\leq k<l$, let $\mathbf{LO}_k$ and $\mathbf{LO}_l$ denote the corresponding symmetric relational templates, and let $\mathrm{PCSP}(\mathbf{LO}_k,\mathbf{LO}_l)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-050126 | MATHDB-348242

Planar 3-uniform hypergraph analogue of the Boots-Royle-Cao-Vince conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a planar 3-uniform hypergraph on n vertices, and let its shadow be the graph obtained by replacing each hyperedge by the edges joining its pairs of vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-050127 | MATHDB-348243

Dixon-Pressman conjecture on the nullity of an alternating matrix operator

摘录 / EXCERPT: Let F be a field and let $A_1,\dots,A_k\in\operatorname{Mat}_n(F)$. Define the linear operator ... where ... The matrices are in general position when they satisfy the genericity con...

conjecture | open | MathDB 页面 | 原文线索

V2-050128 | MATHDB-348244

Conjecture on limiting pheromone weights and geodesics

摘录 / EXCERPT: Limiting-weight conjecture. Almost surely, as $n\to+\infty$, for every $e\in E$,

conjecture | open | MathDB 页面 | 原文线索

V2-050129 | MATHDB-348251

Brouwer's toughness conjecture for regular graphs

摘录 / EXCERPT: Let G be a connected d-regular graph, let $t(G)$ denote its toughness, and let λ be the second largest absolute eigenvalue of the adjacency matrix of G. Brouwer's tou...

conjecture | open | MathDB 页面 | 原文线索

V2-050130 | MATHDB-348265

Smallest Fibonacci-run graph containing a hypercube

摘录 / EXCERPT: Let Q_n denote the n-dimensional hypercube graph, and let $\mathcal{R}_m$ denote the Fibonacci-run graph of parameter m. Hypercube embedding conjecture. The smallest integer...

conjecture | open | MathDB 页面 | 原文线索

V2-050131 | MATHDB-348299

The real-projective-plane extremal-number conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a 3-uniform hypergraph on n vertices, and let a triangulation of the real projective plane mean a 3-uniform hypergraph whose associated simplicial comple...

conjecture | open | MathDB 页面 | 原文线索

V2-050132 | MATHDB-348300

Linial's universal surface extremal-number conjecture

摘录 / EXCERPT: Let $\mathcal{S}$ be a 3-uniform hypergraph, and let $\operatorname{ex}_{hom}(n,\mathcal{S})$ denote the maximum number of edges in a 3-uniform hypergraph on n vertices conta...

conjecture | open | MathDB 页面 | 原文线索

V2-050133 | MATHDB-348303

Edge-size conjecture for graceful zillion graphs

摘录 / EXCERPT: Let Γ be a zillion graph with ω odd cycles and μ even cycles, and let $|E(\Gamma)|$ denote its number of edges. Edge-size conjecture. There exist a quadratic poly...

conjecture | open | MathDB 页面 | 原文线索

V2-050134 | MATHDB-348342

One-extra-vertex conjecture for majority dynamics

摘录 / EXCERPT: Let $G_{n,p}$ be a random graph on n vertices, and consider majority dynamics from an initial red-blue coloring with more red vertices than blue. For fixed $p\in(0,1)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-050135 | MATHDB-348389

Equality of pressure on a CMS and its completion

摘录 / EXCERPT: Let (Σ,σ) be a finite entropy countable Markov shift, let ρ be a totally bounded metric on its vertex set, and let d be the corresponding metric on Σ. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-050136 | MATHDB-348394

The cyclomatic-number bound for the difference between metric dimensions

摘录 / EXCERPT: Cyclomatic-number bound. For every such graph,

conjecture | open | MathDB 页面 | 原文线索

V2-050137 | MATHDB-348396

Halin's end degree conjecture

摘录 / EXCERPT: Halin's conjecture. Every graph contains ray graphs for all its ends.

conjecture | open | MathDB 页面 | 原文线索

V2-050138 | MATHDB-348400

The mixed metric dimension bound for graphs

摘录 / EXCERPT: Let $G\ne C_n$ be a graph, and let $c(G)$ denote its cyclomatic number. The quantity $L_1(G)$ is the parameter used in the mixed metric dimension bound, and $\mathrm{mdim}(G)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-050139 | MATHDB-348408

Cada et al.'s rainbow cycle length conjecture

摘录 / EXCERPT: Let G be an edge-colored graph of order n, and let k be a positive integer. For a vertex v of G, write $d^c(v)$ for its color degree, the number of distinct edge colors i...

conjecture | open | MathDB 页面 | 原文线索

V2-050140 | MATHDB-348409

Cohen et al.'s $(2+1)$-bispindle conjecture

摘录 / EXCERPT: A $(2+1)$-bispindle $B(k_1,k_2;k_3)$ is the union of two internally disjoint xy-dipaths of lengths k_1 and k_2, and one internally disjoint yx-dipath of length k_3. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-050141 | MATHDB-348417

The extremal conjecture for S_1-free cyclic geometric hypergraphs

摘录 / EXCERPT: The S_1 extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050142 | MATHDB-348418

Howroyd's periodicity conjecture for maximum independent sets

摘录 / EXCERPT: Let $G_3(d)$ be the graph on the monomials of degree d in three variables, with two monomials adjacent exactly when their least common multiple has degree $d+1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-050143 | MATHDB-348419

Eventual uniqueness of maximum independent sets in monomial graphs

摘录 / EXCERPT: For positive integers n and nonnegative integers d, let $G_n(d)$ be the graph whose vertices are the monomials of degree d in n variables, with two monomials adjacent exact...

conjecture | open | MathDB 页面 | 原文线索

V2-050144 | MATHDB-348441

Polynomial bound conjecture for excluding grasped clique minors

摘录 / EXCERPT: Let f and g be polynomials. For integers $r,p\geq 0$, set ... Let G be a graph and let W be an R-wall in G. A flat wall is a wall that is flat in the relevant surface dec...

conjecture | open | MathDB 页面 | 原文线索

V2-050145 | MATHDB-348453

Hat-guessing number bound for complete bipartite graphs

摘录 / EXCERPT: Complete bipartite hat-guessing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050146 | MATHDB-348454

Maximum-degree bound for the hat-guessing number

摘录 / EXCERPT: Maximum-degree hat-guessing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050147 | MATHDB-348457

Monotonicity conjecture for the number of galled networks

摘录 / EXCERPT: Let $\mathrm{GN}_{n,k}$ denote the number of galled networks with parameters n and k. Monotonicity conjecture. The sequence $\mathrm{GN}_{n,k}$ is increasing for...

conjecture | open | MathDB 页面 | 原文线索

V2-050148 | MATHDB-348462

The edge-containing long-cycle clique conjecture

摘录 / EXCERPT: Let G be a 2-connected graph on n vertices, and let ab be an edge of G. Let $r\geq 4$ and $s\geq 2$ be integers, and write ... for some $0\leq t\leq r-4$. Here $N_s(G)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-050149 | MATHDB-348476

Dvořák-Mohar-Šámal's subcubic star chromatic index conjecture

摘录 / EXCERPT: Let G be a subcubic graph. The Dvořák-Mohar-Šámal conjecture. ... The star chromatic index is the minimum number of colors in a star edge-coloring, in which every bichromatic subgr...

conjecture | open | MathDB 页面 | 原文线索

V2-050150 | MATHDB-348477

Best-bound conjecture for the star chromatic index of outerplanar graphs

摘录 / EXCERPT: Let G be an outerplanar graph with maximum degree Δ. The outerplanar star chromatic index conjecture. ... This would improve the known bound...

conjecture | open | MathDB 页面 | 原文线索

V2-050151 | MATHDB-348481

Quadratic universal graph conjecture for bipartite permutation graphs

摘录 / EXCERPT: Quadratic universal graph conjecture. The minimum number of vertices in a bipartite permutation graph containing all n-vertex bipartite permutation graphs is

conjecture | open | MathDB 页面 | 原文线索

V2-050152 | MATHDB-348494

The path-extension lengthening conjecture for permutations

摘录 / EXCERPT: Let $p=(a_1,\ldots,a_{n-1})\in\mathbf{P}_n$, and let $i\geq 2$ satisfy $a_{i-1}=a_i$. Here, $L(p)$ denotes the set of objects associated with p in the paper, and Φ_i is the...

conjecture | open | MathDB 页面 | 原文线索

V2-050153 | MATHDB-348536

Extension conjecture for fractionally balanced multipartite hypergraphs

摘录 / EXCERPT: Let $(a_1;\ldots,a_d)\xrightarrow{~\tiny \textsc{ad}~}m$ denote the property that the corresponding multipartite parameter admits the stated extension conclusion. Extension conject...

open_problem | open | MathDB 页面 | 原文线索

V2-050154 | MATHDB-348537

Tightness conjecture for the $(n,n,\binom{n}{2})$ balanced matching bound

摘录 / EXCERPT: Let $\textsc{bm}(a_1,a_2,t)$ denote the balanced matching parameter for the indicated complete bipartite setting. Tightness conjecture. For every integer $n\geq4$, ... This would mak...

conjecture | open | MathDB 页面 | 原文线索

V2-050155 | MATHDB-348538

Conjecture that the balanced matching parameter equals two

摘录 / EXCERPT: Let $\textsc{bm}(n,n,\ldots,n)$ denote the balanced matching parameter with n entries, each equal to n. Balanced matching conjecture. ... The source says that a theorem gives the...

conjecture | open | MathDB 页面 | 原文线索

V2-050156 | MATHDB-348542

The merged-star conjecture for second-smallest distance-unbalancedness of trees

摘录 / EXCERPT: Let $n\geq 5$. For even n, let $S_{(n-2)/2}\mathbin{\cdot}S_{(n-2)/2}$ denote the merged star formed from two stars of parameter $(n-2)/2$; for odd n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-050157 | MATHDB-348550

Generalized extremal permanent conjecture for C4-free bipartite graphs

摘录 / EXCERPT: Let $A_{k,6}$ denote the smallest k-regular bipartite graph with girth 6, and let $2v_k$ be the number of vertices in such a graph. Let $A\in\{0,1\}^{n\times n}$ be C_4-free...

conjecture | open | MathDB 页面 | 原文线索

V2-050158 | MATHDB-348564

Strong edge-coloring conjecture for bridgeless subcubic graphs

摘录 / EXCERPT: Strong edge-coloring conjecture. Every bridgeless subcubic graph G, not isomorphic to the Wagner graph or the complete bipartite graph $K_{3,3}$ with one edge subdivided, admits...

conjecture | open | MathDB 页面 | 原文线索

V2-050159 | MATHDB-348565

Albertson-Haas conjecture on planar subcubic edge-coloring

摘录 / EXCERPT: Albertson-Haas conjecture. Every bridgeless subcubic planar graph with at least two vertices of degree 2 admits a $(1,1,1)$-packing edge-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-050160 | MATHDB-348620

Fujita's connectivity-preserving connected subgraph conjecture

摘录 / EXCERPT: Let k and m be positive integers. A graph G is k-connected when $ba(G)\geq k$, where $ba(G)$ is the minimum size of a vertex set whose deletion disconnects G or leaves on...

conjecture | open | MathDB 页面 | 原文线索

V2-050161 | MATHDB-348635

Barát-Gyárfás-Sárközy conjecture for rainbow matchings in multigraphs

摘录 / EXCERPT: Barát-Gyárfás-Sárközy conjecture. Then G has a rainbow matching using n colors.

conjecture | open | MathDB 页面 | 原文线索

V2-050162 | MATHDB-348649

Hypergraph Ringel conjecture for k-trees

摘录 / EXCERPT: Let $k,n\in\mathbb{N}\setminus\{1\}$. A k-tree is defined recursively: a single edge is a k-tree, and a k-tree with ℓ edges is obtained from one with $\ell-1$ edges by a...

conjecture | open | MathDB 页面 | 原文线索

V2-050163 | MATHDB-348650

Balanced multipartite conjecture for decompositions of k-trees

摘录 / EXCERPT: Let $k,n\in\mathbb{N}\setminus\{1\}$. Let T be a k-tree, meaning a k-uniform tree defined recursively by starting with one edge and successively adding a vertex together with...

conjecture | open | MathDB 页面 | 原文线索

V2-050164 | MATHDB-348651

Hypergraph tree-packing conjecture

摘录 / EXCERPT: Let $k,n\in\mathbb{N}\setminus\{1\}$. Let $\mathcal{T}$ be a family of k-trees, where a k-tree is the recursively defined k-uniform tree obtained from one edge by repeatedly...

conjecture | open | MathDB 页面 | 原文线索

V2-050165 | MATHDB-348657

Jerline-Michaelraj harmonic-index conjecture for non-trees

摘录 / EXCERPT: Let G be a simple connected graph that is not a tree, of order $n\geq 7$. Let $H(G)$ be its harmonic index and $D(G)$ its diameter. Let $U_{n,4}^{1,n-5}$ denote the graph obtaine...

conjecture | open | MathDB 页面 | 原文线索

V2-050166 | MATHDB-348659

Petrovic's 4-king conjecture for multipartite hypertournaments

摘录 / EXCERPT: Let H be a multipartite k-hypertournament, where $k\ge 2$. A vertex is a transmitter if it has no incoming arc with any vertex from a different partite set, and a 4-king is a v...

conjecture | open | MathDB 页面 | 原文线索

V2-050167 | MATHDB-348660

Petrovic's two 4-kings conjecture for bipartite hypertournaments

摘录 / EXCERPT: Let B be a bipartite k-hypertournament, where $k\ge 2$. A vertex is a transmitter if it has no incoming arc with any vertex from a different partite set, and a 4-king is a vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050168 | MATHDB-348661

Self-distinguishability conjecture for trees

摘录 / EXCERPT: Self-distinguishability conjecture. $X_{T_1}^{T_1} \neq X_{T_2}^{T_2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-050169 | MATHDB-348662

Star-target p-monotonicity conjecture for H-chromatic symmetric functions

摘录 / EXCERPT: Star-target p-monotonicity conjecture. The symmetric function $\omega(X_G^H)$ is p-monotone for every graph G.

conjecture | open | MathDB 页面 | 原文线索

V2-050170 | MATHDB-348681

The good-function conjecture for hypergraph matching

摘录 / EXCERPT: Good-function conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050171 | MATHDB-348689

Uniqueness conjecture for optimal surveillance strategies on directed line graphs

摘录 / EXCERPT: Let $\mathcal{G}=(V,\mathcal{E})$ be a directed line graph with $n\geq 3$ nodes, and consider Problem, which seeks an optimal surveillance strategy for the graph. Uniqueness of the...

conjecture | open | MathDB 页面 | 原文线索

V2-050172 | MATHDB-348697

Mohar's APX-hardness conjecture for the genus of cubic graphs

摘录 / EXCERPT: A cubic graph is a graph in which every vertex has degree three. Mohar's conjecture. Approximating the genus of cubic graphs is APX-hard. This conjecture asserts computational hard...

conjecture | open | MathDB 页面 | 原文线索

V2-050173 | MATHDB-348713

Hippchen's conjecture on intersections of longest paths

摘录 / EXCERPT: Hippchen's conjecture. Any two longest paths in G share at least k vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-050174 | MATHDB-348717

Large-t oddtown conjecture for set-family tuples

摘录 / EXCERPT: Let t,k be integers with $t\geq 2$, and let $(\mathcal{A}_1,\mathcal{A}_2,\dots,\mathcal{A}_k)$ be set families of an n-element set, where ... Suppose that $2t-2>k$ and that...

conjecture | open | MathDB 页面 | 原文线索

V2-050175 | MATHDB-348718

Minimum-size multipartite covering conjecture

摘录 / EXCERPT: Let $k\geq 2$ and $\ell\geq 1$. A complete k-partite k-graph is a complete k-partite k-uniform hypergraph, and $H_1,\ldots,H_m$ form a k-partite covering of order ℓ...

conjecture | open | MathDB 页面 | 原文线索

V2-050176 | MATHDB-348734

Boyer et al.'s sphere-of-influence dimension conjecture

摘录 / EXCERPT: Boyer et al.'s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050177 | MATHDB-348747

Herscovici-Hester-Hurlbert Target Conjecture for graph pebbling

摘录 / EXCERPT: Let G be a graph and let D be a target distribution on G. Write $|D|$ for the total demand, $\pi(G,D)$ for the smallest integer m such that every configuration of m pebbl...

conjecture | open | MathDB 页面 | 原文线索

V2-050178 | MATHDB-348748

The UMR conjecture for the graph family $H_{n,m}$

摘录 / EXCERPT: The UMR conjecture for $H_{n,m}$. Among all m-cyclic graphs, $H_{n,m}$ is UMR with respect to node cop-win reliability for all n sufficiently large.

conjecture | open | MathDB 页面 | 原文线索

V2-050179 | MATHDB-348760

The leaf-end conjecture for alternating paths in bipartite graphs

摘录 / EXCERPT: Let G be a connected graph with matching number $ u(G)$ and saturation number $\mu(G)$ satisfying ... Let M,H,H' satisfy the conditions that M is a maximum matching and that ar...

conjecture | open | MathDB 页面 | 原文线索

V2-050180 | MATHDB-348764

Korándi, Roberts and Scott's blowup conjecture for odd-cycle-free graphs

摘录 / EXCERPT: Let $k\geq 2$, and let G be a C_{2k+1}-free graph with n vertices and at least $\frac{n^2}{4}-\varepsilon n^2$ edges. For a graph H, write $e(H)$ for its number of edges an...

conjecture | open | MathDB 页面 | 原文线索

V2-050181 | MATHDB-348776

Vertex-preservation conjecture for the type D string-to-Gelfand-Tsetlin map

摘录 / EXCERPT: Vertex-preservation conjecture. The map ϕ_λ induces a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-050182 | MATHDB-348789

Finite-component conjecture for square-reduction graphs

摘录 / EXCERPT: For each $k\geqslant 2$, let $\mathcal{G}_k$ be the graph whose vertices are square-free words over a k-letter alphabet, with edges arising from the square-reduction relation. Fi...

conjecture | open | MathDB 页面 | 原文线索

V2-050183 | MATHDB-348793

Higher-uniformity unavoidable hypergraph conjecture

摘录 / EXCERPT: For $r\ge 2$, let $\mathrm{un}_r(n,e)$ denote the maximum number of edges in an r-uniform hypergraph contained in every r-uniform hypergraph on n vertices with e edges. For...

conjecture | open | MathDB 页面 | 原文线索

V2-050184 | MATHDB-348794

Higher-uniformity sunflower Turán-number conjecture

摘录 / EXCERPT: For integers $r\ge 5$ and $t<r$, let $\mathit{Sf}_r(t,k)$ denote the r-uniform sunflower with kernel size t and k petals, and let $\mathrm{ex}(n,\mathit{Sf}_r(t,k))$ be its T...

conjecture | open | MathDB 页面 | 原文线索

V2-050185 | MATHDB-348800

Random-walk betweenness conjecture for asymptotically pairwise stable networks

摘录 / EXCERPT: Let an asymptotically pairwise stable network (APSN) be a network satisfying asymptotic pairwise stability for the relevant centrality game, and let $\emptyset_n and K_n denote...

conjecture | open | MathDB 页面 | 原文线索

V2-050186 | MATHDB-348801

Eigenvector centrality conjecture for asymptotically pairwise stable networks

摘录 / EXCERPT: Let K_n denote the complete graph on n vertices, and call a network asymptotically pairwise stable if it satisfies asymptotic pairwise stability for the eigenvector-centrality...

conjecture | open | MathDB 页面 | 原文线索

V2-050187 | MATHDB-348809

The unlabelled weighted graph conjecture for horizontal-strip LLT polynomials

摘录 / EXCERPT: Let $\bm\lambda$ and $\bm\mu$ be horizontal strips, and let $\Pi(\bm\lambda)$ and $\Pi(\bm\mu)$ be their weighted graphs. If ... as weighted graphs, then the associated LLT polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-050188 | MATHDB-348810

Fang's multipartite cycle conjecture

摘录 / EXCERPT: Let $r\ge 3$ and let $n_{1}\ge n_{2}\ge \cdots\ge n_{r}\ge 1$. Let $K_{n_{1},n_{2},\ldots,n_{r}}$ be the complete r-partite graph with parts of these sizes, let G be a subgraph...

conjecture | open | MathDB 页面 | 原文线索

V2-050189 | MATHDB-348872

Zhang's metric dimension conjecture for folded hypercubes

摘录 / EXCERPT: Let F_n be the folded n-cube obtained from the hypercube Q_n by identifying each vertex u with its antipode; write $\beta(F_n)$ for its metric dimension. Zhang's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050190 | MATHDB-348873

The folded-hypercube doubly resolving set conjecture

摘录 / EXCERPT: Let F_n be the folded n-cube, and let $\beta(F_n)$ and $\Psi(F_n)$ denote its metric dimension and minimal doubly resolving set number, respectively. Folded-hypercube conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-050191 | MATHDB-348875

First law of cubology for Rubik groups of oriented maps

摘录 / EXCERPT: Let M be an oriented map with vertex set $V(M)$ and edge set $E(M)$, and let $H_1(M)$, $H_2(M)$, and $H_3(M)$ be the kernels of the successive homomorphisms in the decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-050192 | MATHDB-348889

Maximum total eccentricity conjecture for graphs with fixed cut vertices

摘录 / EXCERPT: Maximum total eccentricity conjecture. For $n\geq 5$ and $2\leq s\leq n-4$, if $G\in\mathfrak{C}_{n,s}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050193 | MATHDB-348915

The circulant orbit-eigenvalue conjecture

摘录 / EXCERPT: Circulant orbit-eigenvalue conjecture. The number of orbits of the action of $\operatorname{Aut}(G)$ on $V*V$ coincides with the number of distinct eigenvalues of its adjacency mat...

conjecture | open | MathDB 页面 | 原文线索

V2-050194 | MATHDB-348916

Weakley's domination-number equality conjecture for orders congruent to 1 modulo 4

摘录 / EXCERPT: Let $\mathcal{Q}(n)$ be the n-Queens' graph, and let $\gamma(\mathcal{Q}(n))$ denote its domination number. For $k \in \mathbb{N} \cup \{0\}$, consider the order $n=4k+1$. Weakle...

conjecture | open | MathDB 页面 | 原文线索

V2-050195 | MATHDB-348917

Cockayne's monotonicity conjecture for domination numbers of Queens' graphs

摘录 / EXCERPT: Let $\mathcal{Q}(n)$ be the n-Queens' graph, and let $\gamma(\mathcal{Q}(n))$ denote its domination number. Cockayne's conjecture. For every $n \in \mathbb{N}$, ... This asks wheth...

conjecture | open | MathDB 页面 | 原文线索

V2-050196 | MATHDB-348918

Integer-spectrum conjecture for the n-Queens' graph

摘录 / EXCERPT: Let $\mathcal{Q}(n)$ be the n-Queens' graph, let $\sigma(\mathcal{Q}(n))$ denote its spectrum, and let $\mathbb{Z}$ denote the integers. Integer-spectrum conjecture. For $n\geq 4$...

conjecture | open | MathDB 页面 | 原文线索

V2-050197 | MATHDB-348919

Apex conjecture for connected bipartite graphs satisfying Sidorenko's conjecture

摘录 / EXCERPT: Apex conjecture. For every positive integer a, the graph H^{+a} is common. In particular, every complete tripartite graph $K_{r,s,t}$ is common.

conjecture | open | MathDB 页面 | 原文线索

V2-050198 | MATHDB-348938

The strong persistent Cheeger inequality for graph pairs

摘录 / EXCERPT: Let $K=(V_K,E_K)$ and $L=(V_L,E_L)$ be graphs with $K\hookrightarrow L$. The strong persistent Cheeger inequality conjectures that ... where $h^{K,L}_s$ is the strong persistent Chee...

conjecture | open | MathDB 页面 | 原文线索

V2-050199 | MATHDB-348947

Product-form characterization for conservative weakly reversible reaction networks

摘录 / EXCERPT: Product-form characterization conjecture. The network $(\mathcal{G},\kappa)$ has a product-form stationary distribution if and only if it is complex balanced.

conjecture | open | MathDB 页面 | 原文线索

V2-050200 | MATHDB-348948

Non-product-form conjecture for conservative non-weakly reversible reaction networks

摘录 / EXCERPT: Non-product-form conjecture. Then $\mathcal{G}$ has a non-product-form stationary distribution independently of the reaction rate; equivalently, it belongs to class N.

conjecture | open | MathDB 页面 | 原文线索

V2-050201 | MATHDB-348951

The conjecture that every 3-edge-connected graph has Frank number at most 3

摘录 / EXCERPT: Let G be a 3-edge-connected graph. Its Frank number $f(G)$ is the minimum number of orientations needed so that every edge is deletable in at least one of them. Frank-number co...

conjecture | open | MathDB 页面 | 原文线索

V2-050202 | MATHDB-348952

The matching-deletability conjecture for 3-edge-connected graphs

摘录 / EXCERPT: Let G be a 3-edge-connected graph. A matching is a set of pairwise vertex-disjoint edges, and a 3-edge-cut is an edge cut of size three. A matching is deletable when all its ed...

conjecture | open | MathDB 页面 | 原文线索

V2-050203 | MATHDB-348962

Polynomial degeneracy conjecture for graphs excluding subdivisions

摘录 / EXCERPT: Polynomial degeneracy conjecture for graphs excluding subdivisions. For every graph H, every H-subdivision-free graph G that does not contain $K_{\ell,\ell}$ as a subgraph ha...

conjecture | open | MathDB 页面 | 原文线索

V2-050204 | MATHDB-348969

Almost-complete realisation conjecture for excitable CTRNN networks

摘录 / EXCERPT: Let G be a directed graph with N vertices satisfying the hypotheses in Theorem 1. Assume that $\epsilon>0$ is small and $\theta>0$. Let the input-free CTRNN have N cells, pie...

conjecture | open | MathDB 页面 | 原文线索

V2-050205 | MATHDB-348979

Fujita-Kawarabayashi connected-subgraph removal conjecture

摘录 / EXCERPT: Fujita-Kawarabayashi conjecture. There is a least non-negative integer $f_k(m)$ such that every k-connected graph G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050206 | MATHDB-348995

The linkedness conjecture for simple polytopes

摘录 / EXCERPT: Let P be a simple d-polytope, meaning a d-dimensional polytope in which every vertex is incident to exactly d edges. A polytope is k-linked when its graph is k-linked,...

conjecture | open | MathDB 页面 | 原文线索

V2-050207 | MATHDB-349032

Logarithmic lower bound conjecture for separating non-contractible cycles

摘录 / EXCERPT: For any map m, let $\operatorname{SNC}(m)$ be the size of the shortest separating non-contractible cycle of m. Let $q^{(n)}$ be the random uniform high-genus quadrangulation, a...

conjecture | open | MathDB 页面 | 原文线索

V2-050208 | MATHDB-349034

The boundary-isoperimetric conjecture for triangular grids

摘录 / EXCERPT: Let P_n be the triangular grid graph with n vertices on each side, and let ∂C denote the boundary vertices of a vertex set C. A row packing fills each row completel...

conjecture | open | MathDB 页面 | 原文线索

V2-050209 | MATHDB-349070

Conjecture on the adjacency-matrix inequality for graph normalization

摘录 / EXCERPT: Graph-normalization adjacency inequality conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-050210 | MATHDB-349082

The normality conjecture for cut polytopes

摘录 / EXCERPT: Let $G(V,E)$ be a graph, and let $\operatorname{Cut}^\square(G)$ be the convex hull of the cut points $\delta_{A\vert B}\in\mathbb{R}^{\lvert E\rvert}$ over all unordered vertex pa...

conjecture | open | MathDB 页面 | 原文线索

V2-050211 | MATHDB-349096

Characterization of equality in the mixed metric dimension bound

摘录 / EXCERPT: Let G be a graph, let $c(G)$ be its cyclomatic number, let $L_1(G)$ be its number of leaves, and let $\mathrm{mdim}(G)$ be its mixed metric dimension. A cactus graph is a graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-050212 | MATHDB-349120

The conjecture on monophonic position numbers of cages

摘录 / EXCERPT: A (d,g)-cage is a graph of minimum possible order among graphs with degree d and girth g, and ${\rm mp}(G)$ denotes the monophonic position number of G. The cage monophonic...

conjecture | open | MathDB 页面 | 原文线索

V2-050213 | MATHDB-349127

Perfect stability conjecture for inducibility of complete partite graphs

摘录 / EXCERPT: Let F be a complete partite graph. The inducibility problem for F asks for the maximum possible induced F-density in graphs of increasing order, and is perfectly stable when...

conjecture | open | MathDB 页面 | 原文线索

V2-050214 | MATHDB-349128

Singleton-optimizer stability conjecture

摘录 / EXCERPT: Let OPT be the set of optimal vectors for the graph-density optimization problem under consideration, and let a problem be stable when every near-optimal graph is close to an opt...

conjecture | open | MathDB 页面 | 原文线索

V2-050215 | MATHDB-349129

Finite-support conjecture for clique-density optimizers

摘录 / EXCERPT: For $2\leq i\leq k$, let a_i be coefficients and define a graph parameter by ... where $p(K_i,G)$ is the induced density of the clique K_i. Let OPT denote the set of optimal ve...

conjecture | open | MathDB 页面 | 原文线索

V2-050216 | MATHDB-349130

Positive-isolated-part conjecture for clique-density optimizers

摘录 / EXCERPT: For $2\leq i\leq k$, let a_i be coefficients and define ... where $p(K_i,G)$ is the induced density of K_i. Let OPT denote the set of optimal vectors, and write x_0 for the c...

conjecture | open | MathDB 页面 | 原文线索

V2-050217 | MATHDB-349135

Recurrence conjecture for IID p-rotor walks

摘录 / EXCERPT: p-rotor recurrence conjecture. For almost every such ρ, the corresponding p-rotor walk on $\mathbb{Z}^2$ visits every vertex infinitely many times almost surely.

conjecture | open | MathDB 页面 | 原文线索

V2-050218 | MATHDB-349136

The graph positivity conjecture for regular bipartite graphs

摘录 / EXCERPT: The graph positivity conjecture. As n goes to infinity, the fraction of r-regular bipartite graphs with $2n$ vertices that are graph positive approaches one.

conjecture | open | MathDB 页面 | 原文线索

V2-050219 | MATHDB-349137

The awesome conjecture for coefficients of the logarithmic expansion

摘录 / EXCERPT: The awesome conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050220 | MATHDB-349147

Extremal Ky Fan norms of the LI-matrix for trees

摘录 / EXCERPT: Tree extremal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050221 | MATHDB-349148

Guan et al.'s extremal graph conjecture for Ky Fan norms of LI-matrices

摘录 / EXCERPT: Extremal graph conjecture. Among all connected graphs with n vertices and m edges satisfying $n\leq m\leq 2n-3$, the graph $G_{m,n}$ is the unique graph attaining the maximum o...

conjecture | open | MathDB 页面 | 原文线索

V2-050222 | MATHDB-349149

Ashraf's signless Laplacian eigenvalue-sum conjecture

摘录 / EXCERPT: Let G be a graph with n vertices and $e(G)$ edges, let $Q(G)$ be its signless Laplacian matrix, and let $S_k(Q(G))$ denote the sum of the k largest eigenvalues of $Q(G)$. Ash...

conjecture | open | MathDB 页面 | 原文线索

V2-050223 | MATHDB-349150

Amaro et al.'s strengthened signless Laplacian eigenvalue-sum conjecture

摘录 / EXCERPT: Let G be a graph with $n\geq5$ vertices and $3\leq k\leq n-2$, let $Q(G)$ be its signless Laplacian matrix, and let $S_k(Q(G))$ denote the sum of its k largest eigenvalues. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-050224 | MATHDB-349151

The A-alpha signless Laplacian eigenvalue-sum conjecture

摘录 / EXCERPT: Let G be a graph with n vertices and $e(G)$ edges. For a real parameter $\alpha\in[0,1]$, define ... where $D(G)$ is the diagonal degree matrix and $A(G)$ is the adjacency matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-050225 | MATHDB-349177

Nonexistence conjecture for dendrimer nanostars of at least four stages

摘录 / EXCERPT: A dendrimer nanostar is the nanostar structure obtained by growing a dendrimer through a specified number of stages; a graph is a G_ϕ-graph if it is realized as $G_\phi(A)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050226 | MATHDB-349189

Conjecture on limit points of the Laplacian spectral radius

摘录 / EXCERPT: Let $\omega = \frac{1}{3}\left((19 + 3\sqrt{33})^{\frac{1}{3}} + (19 - 3\sqrt{33})^{\frac{1}{3}}+1 \right)$, and let $\rho_L(G)$ denote the Laplacian spectral radius of a graph G...

conjecture | open | MathDB 页面 | 原文线索

V2-050227 | MATHDB-349208

Cohen-Perkins-Sarantis-Tetali conjecture for independent sets in linear hypergraphs

摘录 / EXCERPT: Cohen-Perkins-Sarantis-Tetali conjecture. For every such hypergraph,

conjecture | open | MathDB 页面 | 原文线索

V2-050228 | MATHDB-349224

The Shapley value is in the core of the energy game

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, with the energy game assigning each coalition its graph energy and with $\boldsymbol{\varphi}$ denoting the Shapley value. Energy-game core conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050229 | MATHDB-349225

Attractor-invariant formula under quiver mutation

摘录 / EXCERPT: Let (Q,W) and (Q',W') be quivers with potential related by mutation at a vertex k, and let d,d' and stability weights θ,θ' be related by the mutation formulas i...

conjecture | open | MathDB 页面 | 原文线索

V2-050230 | MATHDB-349226

Attractor invariants for the local Hirzebruch-surface quiver

摘录 / EXCERPT: Let Q be the quiver for the local Hirzebruch surface $K_{\mathbb{F}_0}$, let e_i be its vertex dimension vectors, and let $\mathrm{d}=(1,1,1,1)$ be the dimension vector of a si...

conjecture | solved | MathDB 页面 | 原文线索

V2-050231 | MATHDB-349227

Attractor invariants for the local second del Pezzo phase

摘录 / EXCERPT: Let Q be the quiver for the local phase labelled PdP_2, let e_i be its vertex dimension vectors, and let d be the dimension vector of a single D0-brane. L...

conjecture | solved | MathDB 页面 | 原文线索

V2-050232 | MATHDB-349258

Nondegeneracy conjecture for positive steady states in reaction networks

摘录 / EXCERPT: Nondegeneracy conjecture. Given a network G, if

conjecture | open | MathDB 页面 | 原文线索

V2-050233 | MATHDB-349259

The expander spanning-subgraph conjecture with large girth

摘录 / EXCERPT: Let G be a finite graph with vertex set $V(G)$. For a subset $S\subset V(G)$ with $0<|S|\leq |V(G)|/2$, let $\partial_G S$ denote its edge boundary, and define the Cheeger consta...

conjecture | open | MathDB 页面 | 原文线索

V2-050234 | MATHDB-349271

The cardinal-inequality realization conjecture for $cotimes$ graphs

摘录 / EXCERPT: Cardinal-inequality realization conjecture. For every algorithm $\mathcal{A}$ and input x, there exists a $\otimes$ graph $\mathcal{G}$ with cardinal inequalities such that

conjecture | open | MathDB 页面 | 原文线索

V2-050235 | MATHDB-349298

Koledin-Stanić conjecture on maximal index of complete signed graphs

摘录 / EXCERPT: Koledin-Stanić conjecture. The complete signed graph with n vertices and m negative edges that maximizes the index has the following structure: if $m<n-1$, the negative edges i...

conjecture | open | MathDB 页面 | 原文线索

V2-050236 | MATHDB-349325

Grünbaum's Hamiltonicity conjecture for 4-connected maps on the torus

摘录 / EXCERPT: Grünbaum-Nash-Williams conjecture. Every 4-connected map on the torus is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-050237 | MATHDB-349348

The Nagy conjecture on stability for saturated graphs

摘录 / EXCERPT: Nagy's conjecture. Every F-saturated graph G on n vertices with $t_r(n)-o(n^{\frac{r+1}{r}})$ edges contains a complete r-partite subgraph on $(1-o(1))n$ vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-050238 | MATHDB-349349

The color-critical-edge stability conjecture for saturated graphs

摘录 / EXCERPT: Color-critical-edge stability conjecture. Every F-saturated graph G on n vertices with $t_r(n)-o(n^{\frac{r+1}{r}})$ edges contains a complete r-partite subgraph G' on...

conjecture | open | MathDB 页面 | 原文线索

V2-050239 | MATHDB-349357

The circumference conjecture for essentially 4-connected planar graphs

摘录 / EXCERPT: Let G be an essentially 4-connected planar graph on n vertices, and let $\operatorname{circ}(G)$ denote the length of a longest cycle in G. Circumference conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-050240 | MATHDB-349361

The packing problem for hypergraphs

摘录 / EXCERPT: Let $H=(V,E)$ be a hypergraph, and let $I(H)$ be its edge ideal. A minor of H is obtained by deleting and contracting vertices. The hypergraph H has the packing property if eve...

conjecture | open | MathDB 页面 | 原文线索

V2-050241 | MATHDB-349362

The partite-structure conjecture for packed hypergraphs

摘录 / EXCERPT: Let H be a hypergraph with edge ideal $I=I(H)$. Write $\alpha(I)$ for the least degree of a nonzero homogeneous element of I, and let $\operatorname{ht}(I)$ denote the height o...

conjecture | open | MathDB 页面 | 原文线索

V2-050242 | MATHDB-349370

The polylogarithmic improvement conjecture for independent sets in omitting systems

摘录 / EXCERPT: Let $k\geq 3$ and $2\leq \ell<k$ satisfy $k\leq 2\ell+1$. An (n,k,ℓ)-omitting system is an n-vertex k-uniform hypergraph with no two edges whose intersection has size...

conjecture | open | MathDB 页面 | 原文线索

V2-050243 | MATHDB-349382

Berge-Füredi-Meyniel conjecture on the chromatic index of linear hypergraphs

摘录 / EXCERPT: Berge-Füredi-Meyniel conjecture. For any linear hypergraph H, its chromatic index is at most $\Delta^*(H)+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-050244 | MATHDB-349385

Hocquard-Lajou-Lužar conjecture on nine-coloring bridgeless subcubic graphs

摘录 / EXCERPT: Hocquard-Lajou-Lužar conjecture. If G is not isomorphic to the Wagner graph or to $K_{3,3}$ with one edge subdivided, then G admits a strong 9-edge-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-050245 | MATHDB-349391

Elphick-Wocjan eigenvalue-sum conjecture for clique number

摘录 / EXCERPT: Let G be a non-empty graph with m edges, adjacency eigenvalues $mu_1\geq\cdots\geqmu_n$, clique number $omega$, and n^+ positive eigenvalues. Set $ell=\min(n^+,omega)$. Elphi...

conjecture | open | MathDB 页面 | 原文线索

V2-050246 | MATHDB-349403

Arithmetical structures conjecture on extremal graph orders

摘录 / EXCERPT: Arithmetical structures conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050247 | MATHDB-349405

The maximal-vertex realization conjecture for Newton-Okounkov bodies on surfaces

摘录 / EXCERPT: Let S be a smooth projective algebraic surface, let D be a big divisor on S, and write $\rho=\rho(S)$. A smooth projective surface $\tilde S$ with a proper projective morphis...

conjecture | open | MathDB 页面 | 原文线索

V2-050248 | MATHDB-349418

Uniform-mass conjecture for the path optimization parameter

摘录 / EXCERPT: For an integer $m\geq 2$, let C_m be the cycle on m edges, let $E(C_m)$ be its edge set, and let $\rho(m)$ be the path optimization parameter defined in the paper. Uniform-mass...

conjecture | open | MathDB 页面 | 原文线索

V2-050249 | MATHDB-349419

Uniform-mass conjecture for cycles

摘录 / EXCERPT: Let C_m be the cycle on m edges, let $E(C_m)$ be its edge set, and let $\beta(C_m,1)$ be the optimization parameter defined in the paper. Uniform-mass conjecture for cycles. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050250 | MATHDB-349420

Characterization of graphs for which the blow-up optimization is unattained

摘录 / EXCERPT: Let H be a graph with no isolated vertices, let k be a positive integer, and let $\beta(H,k)$ be the optimization parameter defined in the paper. Attainment conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-050251 | MATHDB-349421

Asymptotic reduction conjecture for graph blow-ups

摘录 / EXCERPT: Let H be a graph on m edges, let k be a positive integer, let $\delta(H)$ be the minimum degree of H, and let $\mathcal{G}_C$ be the graph class defined in the paper. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-050252 | MATHDB-349425

The vertex bound conjecture for minimal forbidden induced subgraphs of subgraph complementation number

摘录 / EXCERPT: Let $c_2(G)$ denote the minimum cardinality of a subgraph complementation system for a graph G, and fix a natural number k. A minimal forbidden induced subgraph for the propert...

conjecture | open | MathDB 页面 | 原文线索

V2-050253 | MATHDB-349435

The girth conjecture for the hypergraphs D'(k,q)

摘录 / EXCERPT: Let $D'(k,q)$ be the hypergraph defined in the paper, and let its girth be the length of its shortest cycle. Girth conjecture for $D'(k,q)$. The hypergraph $D'(k,q)$ has girth at l...

conjecture | open | MathDB 页面 | 原文线索

V2-050254 | MATHDB-349441

Estrada-Mugnolo's Kirchhoff-index ordering conjecture for hubs-biased resistance distances

摘录 / EXCERPT: Kirchhoff-index ordering conjecture. For $\alpha\in\{-1,1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-050255 | MATHDB-349468

The macaron criterion for reconstructibility of level-2 networks

摘录 / EXCERPT: Let a level-2 network be a network whose blobs have level at most 2, let a macaron be a level-2 blob with two cut-edges, and let an alt-path structure be the structure descri...

conjecture | open | MathDB 页面 | 原文线索

V2-050256 | MATHDB-349481

NP-completeness conjecture for finding (\alpha,\beta)-modular decompositions

摘录 / EXCERPT: NP-completeness conjecture. For every (α,β) with $\max\{\alpha,\beta\}>1$, finding an (α,β)-modular decomposition of G is an NP-complete problem.

conjecture | open | MathDB 页面 | 原文线索

V2-050257 | MATHDB-349520

The K_4^- and L_2-free 3-graph Turán conjecture

摘录 / EXCERPT: Let K_4^- be the 3-uniform hypergraph obtained from the complete 3-uniform hypergraph on four vertices by deleting one edge, and let ... For a family of forbidden 3-uniform hypergr...

conjecture | open | MathDB 页面 | 原文线索

V2-050258 | MATHDB-349527

The nebula conjecture

摘录 / EXCERPT: Let T be a tournament. A nebula is a tournament admitting an ordering whose vertices are partitioned into the vertex sets of stars and singleton components, with no restriction o...

conjecture | open | MathDB 页面 | 原文线索

V2-050259 | MATHDB-349537

The zero-sum tree-factor conjecture

摘录 / EXCERPT: Let k and n be positive integers such that ... are both even integers. Let T be a tree of order k, and let $c:E(K_n)\to\{-1,1\}$ be a zero-sum labeling of the complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050260 | MATHDB-349545

Bollobás-Scott conjecture on long cycles in Eulerian digraphs

摘录 / EXCERPT: Let G be a finite loopless directed graph, allowing both orientations of an edge, and write $d^+(v)$ and $d^-(v)$ for the outdegree and indegree of a vertex v. The graph is Eul...

conjecture | open | MathDB 页面 | 原文线索

V2-050261 | MATHDB-349546

Bollobás-Scott partition conjecture for Eulerian digraphs

摘录 / EXCERPT: Let G be an Eulerian directed graph on n vertices, meaning that its indegree equals its outdegree at every vertex. Bollobás-Scott partition conjecture. The edge set of G can...

conjecture | open | MathDB 页面 | 原文线索

V2-050262 | MATHDB-349557

Conlon's conjecture on norming graphs and stable reflection graphs

摘录 / EXCERPT: Conlon's conjecture. Every norming graph is a stable reflection graph. Equivalently, a graph is norming if and only if it is isomorphic to a special type of $(G_1,G_2;G)$-graph wit...

conjecture | open | MathDB 页面 | 原文线索

V2-050263 | MATHDB-349568

Broadcasting information conjecture for regular and Poisson Galton-Watson trees

摘录 / EXCERPT: Let T be a regular tree or a Galton-Watson tree with Poisson offspring distribution, with root vertex ρ. Let (T,ρ,θ,W) denote the corresponding broadcast model wit...

conjecture | open | MathDB 页面 | 原文线索

V2-050264 | MATHDB-349573

Thin Q-polynomial distance-regular graphs are pseudo-vertex-transitive

摘录 / EXCERPT: Let Γ be a thin Q-polynomial distance-regular graph with diameter $D\geq 3$. Thinness conjecture. The graph Γ is pseudo-vertex-transitive. This conjecture is motiva...

conjecture | open | MathDB 页面 | 原文线索

V2-050265 | MATHDB-349574

Positivity and definiteness of dominant transfer-matrix eigenvectors for real iTRs

摘录 / EXCERPT: Positivity and definiteness conjecture. The dominant eigenvalue η is real and positive, and

conjecture | open | MathDB 页面 | 原文线索

V2-050266 | MATHDB-349575

Almost-all-graphs unimodality conjecture for upward monotone augmented graph properties

摘录 / EXCERPT: Almost-all-graphs unimodality conjecture. For almost all graphs G, the sequence $c^{\mathcal{P}}_i(G)$, $0\leq i\leq n(G)$, is unimodal.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050267 | MATHDB-349584

The near-trivial-bound conjecture for domination in Knödel graphs

摘录 / EXCERPT: Let KG_n be the Knödel graph on even n vertices, let $m=\lfloor\log n\rfloor$, and let $\gamma(KG_n)$ denote its domination number. The near-trivial-bound conjecture. ... The low...

conjecture | open | MathDB 页面 | 原文线索

V2-050268 | MATHDB-349585

Sharpness of the edge-distinguishing bound for infinite locally finite star-free graphs

摘录 / EXCERPT: Sharpness conjecture. For every natural number n greater than three there exists a connected infinite locally finite graph G such that

conjecture | open | MathDB 页面 | 原文线索

V2-050269 | MATHDB-349589

Neumann Lara's conjecture on oriented planar graph dichromatic number

摘录 / EXCERPT: Neumann Lara's conjecture. Every oriented planar graph is 2-dicolourable.

conjecture | open | MathDB 页面 | 原文线索

V2-050270 | MATHDB-349594

Balogh-Keevash-Sudakov conjecture on triangle-free and bipartite subgraphs

摘录 / EXCERPT: Let G be a graph on n vertices. A triangle-free subgraph contains no triangle, and a bipartite subgraph has a bipartition of its vertex set with no edges within either part. Ba...

conjecture | open | MathDB 页面 | 原文线索

V2-050271 | MATHDB-349607

Equality of dynamic user-equilibrium delays and DSO multipliers in evening commute

摘录 / EXCERPT: Let $\mathcal{T}$ be the set of departure times and $\mathcal{N}$ the set of bottlenecks in a corridor network for the evening commute. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-050272 | MATHDB-349612

Hamiltonicity conjecture for randomly perturbed graphs with geometric randomness

摘录 / EXCERPT: Let $dgeq1$ be an integer, let $\alpha=\alpha(n)\in(0,1/2)$, and let H be an n-vertex graph with minimum degree at least αn. Write $G^d(n,r)$ for the random geometric g...

conjecture | open | MathDB 页面 | 原文线索

V2-050273 | MATHDB-349627

Spectral condition for consecutive cycle lengths

摘录 / EXCERPT: Let k be a fixed positive integer and let G be a graph of sufficiently large size m without isolated vertices. Write $\rho(G)$ for the spectral radius of G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050274 | MATHDB-349643

Factorization conjecture for complete multipartite 3-uniform hypergraphs with unequal part sizes

摘录 / EXCERPT: Let $\mathscr K_{m_1,\ldots,m_n}^{3}$ be the complete multipartite 3-uniform hypergraph whose parts have sizes $m_1,\ldots,m_n$, and let $\lambda \mathscr K_{m_1,\ldots,m_n}^{3}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-050275 | MATHDB-349647

Conjectured enumeration formula for 20-vertex configurations and Aztec-triangle domino tilings

摘录 / EXCERPT: For each $n\geq 1$, let Z^{20V}_n denote the total number of 20-vertex configurations with DWBC3 boundary conditions on the quadrangle ${\mathcal Q}_n$, and let Z^{DT}_n denote...

conjecture | open | MathDB 页面 | 原文线索

V2-050276 | MATHDB-349650

Maximum-index cactus conjecture for signed complete graphs

摘录 / EXCERPT: Let G_t be the cactus graph with k edges and t cycles, with $G_0=K_{1,k}$. Let Γ be a signed complete graph of order $n>5$ whose negative edges induce a cactus graph w...

conjecture | open | MathDB 页面 | 原文线索

V2-050277 | MATHDB-349652

Dimension and primality conjecture for CFN sunlet network ideals

摘录 / EXCERPT: Let J_n be the ideal of invariants of the CFN model on the n-sunlet network, and let I_n be the ideal generated by all quadratic invariants in J_n. Dimension and primality...

conjecture | open | MathDB 页面 | 原文线索

V2-050278 | MATHDB-349655

Lando's degree conjecture for the \mathfrak{sl}_2 weight system on complete bipartite graphs

摘录 / EXCERPT: Lando's complete-bipartite reformulation. The value of the $\mathfrak{sl}_2$ weight system at $K_{l,n}$ is a polynomial of degree $\min(l,n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-050279 | MATHDB-349668

Kostochka-Mubayi-Verstraëte conjecture for loose-cycle Ramsey numbers

摘录 / EXCERPT: For integers $r,k\geq 3$, let C_k^r be the loose k-cycle in an r-uniform hypergraph: its edges $e_1,\dots,e_k$ satisfy $|e_i\cap e_{i+1}|=1$ cyclically, and every other pair...

conjecture | open | MathDB 页面 | 原文线索

V2-050280 | MATHDB-349676

Lovász-Cherkassky theorem without the countability hypothesis

摘录 / EXCERPT: Lovász-Cherkassky conjecture. Under these hypotheses, there exists a system $\mathcal{P}$ of edge-disjoint T-paths having the required cut property for every $t\in T$.

conjecture | open | MathDB 页面 | 原文线索

V2-050281 | MATHDB-349677

Infinite Gallai theorem for vertex-disjoint T-paths

摘录 / EXCERPT: Gallai's conjecture. There exists such a system $\mathcal{P}$ and set X.

conjecture | open | MathDB 页面 | 原文线索

V2-050282 | MATHDB-349678

Infinite Mader theorem for internally vertex-disjoint T-paths

摘录 / EXCERPT: Mader's internally-disjoint-path conjecture. Such a system $\mathcal{P}$, set X, and partition $\mathcal{Y}$ exist.

conjecture | open | MathDB 页面 | 原文线索

V2-050283 | MATHDB-349680

Erdős-Faudree-Schelp-Simonovits conjecture on high-degree vertices and long paths

摘录 / EXCERPT: For positive integers $n>d\geq k$, let $\phi(n,d,k)$ be the smallest integer such that every n-vertex graph with at least $\phi(n,d,k)$ vertices of degree at least d contains a...

conjecture | open | MathDB 页面 | 原文线索

V2-050284 | MATHDB-349689

Dividing-edge conjecture for linear compartmental models

摘录 / EXCERPT: Let $\mathcal{M}$ be a strongly connected linear compartmental model that is generically locally identifiable, with singular-locus equation f. An edge $j\to i$ is dividing if its...

conjecture | open | MathDB 页面 | 原文线索

V2-050285 | MATHDB-349692

The existence conjecture for 12-regular circulant nut graphs

摘录 / EXCERPT: A circulant nut graph $C(n,S)$ is a circulant graph on n vertices whose adjacency matrix has nullity one and whose nullvector has no zero entries; here $S=\{a_1,a_2,\dots,a_6\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-050286 | MATHDB-349694

Zhang's prescribed-vertex complementary cycle conjecture for regular bipartite tournaments

摘录 / EXCERPT: Let D be a k-regular bipartite tournament, where k is an integer greater than 2, and let u and v be two specified vertices of D. Let F_{4k} and the other specified...

conjecture | open | MathDB 页面 | 原文线索

V2-050287 | MATHDB-349696

Complementary cycle-factor conjecture for fully regular multipartite tournaments

摘录 / EXCERPT: Let D be a k-fully regular c-partite tournament with $c\geq5$, meaning that for every pair of distinct partite sets, their induced bipartite tournament is k-regular. A...

conjecture | open | MathDB 页面 | 原文线索

V2-050288 | MATHDB-349703

Extremal size and uniqueness conjecture for strongly-connected k-geodetic digraphs

摘录 / EXCERPT: Let $ex^*(n;k)$ denote the largest number of arcs in a strongly-connected k-geodetic digraph of order n, and let $G(kr,k)$ be the construction described in the paper. The extre...

conjecture | open | MathDB 页面 | 原文线索

V2-050289 | MATHDB-349704

Asymptotic cycle-count conjecture for k-geodetic digraphs

摘录 / EXCERPT: Let $ex(n;C_{k+1};k)$ denote the maximum number of directed copies of the cycle C_{k+1} in a k-geodetic digraph of order n. The asymptotic cycle-count conjecture asserts that...

conjecture | open | MathDB 页面 | 原文线索

V2-050290 | MATHDB-349705

Asymptotic formula for generalized Catalan numbers from hypergraphs

摘录 / EXCERPT: Generalized Catalan asymptotic conjecture. As $n\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-050291 | MATHDB-349735

The directed-cycle fractalizer conjecture

摘录 / EXCERPT: For a directed cycle $\vec{C}_k$, call a directed graph a fractalizer if its extremal induced-copy density is attained by iterated balanced blow-ups of $\vec{C}_k$. Directed-cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-050292 | MATHDB-349765

The 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a graph, and call it nice if it has no connected component isomorphic to K_2. For an edge labelling ℓ, let $\sigma_\ell(v)$ be the sum of the labels on the edges...

conjecture | open | MathDB 页面 | 原文线索

V2-050293 | MATHDB-349785

Generalized gap threshold conjecture for tangled cakes

摘录 / EXCERPT: Let $\mathcal{T}$ be a tangle, and let n agents have monotone continuous valuations. The Generalized gap threshold conjecture. A tangle $\mathcal{T}$ guarantees connected envy-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-050294 | MATHDB-349788

Bowler, Brown and Fenner's extremal graph reconstruction conjecture

摘录 / EXCERPT: For graphs G and H on n vertices, let $cc(G,H)$ be the number of cards common to their decks, and define ... A vertex-deleted subgraph is a card. Bowler, Brown and Fenner's con...

conjecture | open | MathDB 页面 | 原文线索

V2-050295 | MATHDB-349789

Groenland, Guggiari and Scott's constant-missing-card conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let its degree sequence be the multiset of vertex degrees. A deck with k missing cards consists of any $n-k$ cards from the vertex-deleted...

conjecture | open | MathDB 页面 | 原文线索

V2-050296 | MATHDB-349807

Elliot-Rödl hypertree embedding conjecture for Steiner triple systems

摘录 / EXCERPT: A hypertree is a connected, simple 3-uniform hypergraph in which every two vertices are joined by a unique path. A Steiner triple system is a 3-uniform hypergraph in which ever...

conjecture | open | MathDB 页面 | 原文线索

V2-050297 | MATHDB-349828

Admissible-level diagonal coset rationality conjecture

摘录 / EXCERPT: Diagonal coset rationality conjecture. The coset $\mathcal C$ is lisse and rational. This generalizes the known simply-laced diagonal-coset cases and the lattice-subalgebra situati...

conjecture | open | MathDB 页面 | 原文线索

V2-050298 | MATHDB-349829

Lin, Liu and Lu's maximum distance energy conjecture for clique trees

摘录 / EXCERPT: Let G be a clique tree of order n whose cliques are $K_{n_1},\ldots,K_{n_k}$. For a clique path $\mathbb{P}_{m_1,\ldots,m_k}$, let $\mathbb{P}_{m_1,\ldots,m_k}$ denote the grap...

conjecture | open | MathDB 页面 | 原文线索

V2-050299 | MATHDB-349834

Diameter formula for generalized Petersen graphs

摘录 / EXCERPT: Let $GPG(n,s)$ be the generalized Petersen graph and let $C_n(1,s)$ be the circulant graph with step sizes 1 and s. Let $n\geq 5$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050300 | MATHDB-349840

The extremal digraph conjecture for directed paths

摘录 / EXCERPT: The extremal digraph conjecture for directed paths. For every $k,n\in \mathbb{N}^*$ with $n\ge 3k$,

conjecture | open | MathDB 页面 | 原文线索

V2-050301 | MATHDB-349841

The anti-directed cycle Turán conjecture

摘录 / EXCERPT: The anti-directed cycle Turán conjecture. For every $k\in \mathbb{N}^*$, there exists a constant n_0 such that for $n\ge n_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-050302 | MATHDB-349858

The even-stretch lower-bound conjecture for fault-tolerant spanners

摘录 / EXCERPT: Let n and f be positive integers, and let k be an even integer. An f-EFT $(2k-1)$-spanner of a graph is an edge fault-tolerant spanner resilient to any set of at most f f...

conjecture | open | MathDB 页面 | 原文线索

V2-050303 | MATHDB-349863

Generalized niceness conjecture for subdivided complete bipartite graphs

摘录 / EXCERPT: Let a,b,ℓ,r be positive integers with $a\leq b$ and ... Let $K_{a,b}^{\ell}$ denote the graph considered in the paper. Generalized niceness conjecture. Under these conditions,...

conjecture | open | MathDB 页面 | 原文线索

V2-050304 | MATHDB-349908

Square DLA exponent conjecture

摘录 / EXCERPT: Let ${\sf TDLA}_0(n,[0,+\infty)^2)$ be the square DLA tree with n vertices, rooted at the corner ${\bf c}$, and let ${\bf v}$ be a uniformly sampled vertex of the tree. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-050305 | MATHDB-349917

The star spanning tree minimum intersection conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph that admits a star spanning tree T_s. Let $\mathscr{T}_G$ denote the set of spanning trees of G, and let $\cap(T)$ be the intersection number of a span...

conjecture | open | MathDB 页面 | 原文线索

V2-050306 | MATHDB-349926

Rationality conjecture for admissible W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, f a nilpotent element, and k an admissible level. Let $\mathbb{O}_k$ denote the corresponding nilpotent orbit. Rationality conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-050307 | MATHDB-349945

The Graham-Lovász conjecture on unimodality of normalized tree coefficients

摘录 / EXCERPT: Let T be a tree of order $n\geq 3$. For a graph G of order n, write $\delta_k(G)$ for the coefficients of its distance characteristic polynomial and define the normalized coe...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-050308 | MATHDB-349946

Shor's conjecture on the peak of normalized tree coefficients

摘录 / EXCERPT: Let T be a tree with $n\geq 3$ vertices. Let $d_k(T)$ denote the normalized coefficients of the distance characteristic polynomial, defined from the coefficients $\delta_k(T)$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-050309 | MATHDB-349947

The conjecture on extremal normalized distance-Laplacian spectral radius

摘录 / EXCERPT: For integers $n_1,n_3\geq 1$ and $n_2\geq 2$, let KPK_{n_1,n_2,n_3} be formed by taking the vertex sum of a vertex in K_{n_1} with one end of P_{n_2} and the vertex sum of a...

conjecture | open | MathDB 页面 | 原文线索

V2-050310 | MATHDB-349948

The conjecture on the unique minimizer of normalized distance-Laplacian spectral radius

摘录 / EXCERPT: Let G be a connected graph on $n\geq 2$ vertices, let K_n be the complete graph, and let $\rho(\mathcal D^{\mathcal L}(G))$ denote the spectral radius of the normalized distanc...

conjecture | open | MathDB 页面 | 原文线索

V2-050311 | MATHDB-349959

Borradaile et al.'s lexicographic optimality conjecture for SC-Path-Reversal

摘录 / EXCERPT: Borradaile et al.'s conjecture. The algorithm SC-Path-Reversal finds a strongly connected orientation that minimizes the lexicographic order of indegrees.

conjecture | open | MathDB 页面 | 原文线索

V2-050312 | MATHDB-349971

Blocker-family conjecture for maximal independent sets

摘录 / EXCERPT: Let $\alpha\in(0,1/2)$. Blocker-family conjecture. There exist k and $\tau>0$ such that, for every graph G on n vertices with a maximal independent set of size αn, th...

conjecture | open | MathDB 页面 | 原文线索

V2-050313 | MATHDB-349972

The chromatic symmetric homology planarity conjecture

摘录 / EXCERPT: Chromatic symmetric homology planarity conjecture. The graph G is non-planar if and only if its chromatic symmetric homology in bidegree $(1,0)$ contains $\mathbb{Z}_2$-torsion.

conjecture | open | MathDB 页面 | 原文线索

V2-050314 | MATHDB-349974

Finite-convergence conjecture for the theta hierarchy of a graph

摘录 / EXCERPT: Finite-convergence conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-050315 | MATHDB-349978

Matsumoto-Ohno interpolation conjecture for face hypergraphs of planar triangulations

摘录 / EXCERPT: Matsumoto-Ohno conjecture. If H is 3-colorable, then it admits a complete t-coloring for every integer t satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050316 | MATHDB-349981

Tan-Koolen-Xia conjecture on co-edge-regular graphs with four eigenvalues

摘录 / EXCERPT: All graphs considered are finite, undirected and simple. A graph is co-edge-regular with parameters (n,k,c) if it is k-regular on n vertices and every two distinct non-adjace...

conjecture | open | MathDB 页面 | 原文线索

V2-050317 | MATHDB-350006

Dominance-order conjecture for permutahedron Laplacian eigenvalues

摘录 / EXCERPT: Dominance-order conjecture. If $\nu \vartriangleright \gamma$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050318 | MATHDB-350007

Aboulker et al.'s conjecture on heroic triples of oriented star forests

摘录 / EXCERPT: An oriented graph is a digraph with no digons, and its dichromatic number $\vec{\chi}(D)$ is the least number of acyclic sets partitioning its vertex set. A digraph is a heroic set...

conjecture | open | MathDB 页面 | 原文线索

V2-050319 | MATHDB-350008

Aboulker et al.'s conjecture for out-stars and the directed triangle

摘录 / EXCERPT: Let $\accentset{\leftrightarrow}{K}_2$ be the complete digraph on two vertices, let S_2^+ be the orientation of a two-leaf star with all arcs directed outwards, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050320 | MATHDB-350009

The strong four-vertex tournament conjecture

摘录 / EXCERPT: Let $\accentset{\leftrightarrow}{K}_2$ be the complete digraph on two vertices, let S_2^+ be the two-out-star, and let $\vec{K}_4^s$ be the unique strong tournament on four verti...

conjecture | open | MathDB 页面 | 原文线索

V2-050321 | MATHDB-350012

Conjecture that LPS graphs are lossless expanders

摘录 / EXCERPT: Let q be fixed and large, and let X_n be the family of $(q+1)$-regular Ramanujan graphs constructed by Lubotzky, Phillips, and Sarnak. For a vertex set $Y\subset X_n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-050322 | MATHDB-350033

The linear expected-face conjecture for simple graphs

摘录 / EXCERPT: Let G be a simple graph on n vertices, and choose an orientable embedding of G uniformly at random. Write F_G for the number of faces of the resulting embedding. Linear exp...

conjecture | open | MathDB 页面 | 原文线索

V2-050323 | MATHDB-350034

The linear expected-face conjecture for multigraphs

摘录 / EXCERPT: Let G be a multigraph on n vertices, and let μ be its maximum edge-multiplicity. Choose an orientable embedding of G uniformly at random. Linear multigraph expected-face...

conjecture | open | MathDB 页面 | 原文线索

V2-050324 | MATHDB-350037

Ore-type toughness conjecture for hamiltonicity

摘录 / EXCERPT: Ore-type toughness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050325 | MATHDB-350041

The two-vertex transversal conjecture for longest cycles in partial 4-trees

摘录 / EXCERPT: Let G be a 2-connected partial 4-tree, and let a longest cycle of G mean a cycle with maximum length among the cycles of G. Two-vertex transversal conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-050326 | MATHDB-350044

Lower-bound conjecture for spanning trees containing a forest in complete tripartite graphs

摘录 / EXCERPT: Let X_1,X_2 and X_3 be the partite sets of the complete tripartite graph K_{n_1,n_2,n_3}, where $|X_i|=n_i$ for $i\in \llbracket 3\rrbracket$. Let F be a spanning forest in...

conjecture | open | MathDB 页面 | 原文线索

V2-050327 | MATHDB-350045

Alikhani-Soltani distinguishing-index conjecture for Mycielskians

摘录 / EXCERPT: Alikhani-Soltani's conjecture. For all but a finite number of connected graphs G with at least three vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-050328 | MATHDB-350050

Friedgut-Kalai-Kindler packing conjecture for maximum independent sets

摘录 / EXCERPT: Friedgut-Kalai-Kindler conjecture. For every $\alpha\in(0,1/2)$ there exist k and $\tau>0$ such that, if G has a maximum independent set of size αn, then there are pair...

conjecture | open | MathDB 页面 | 原文线索

V2-050329 | MATHDB-350054

Existence of a small asymmetric fractalizer

摘录 / EXCERPT: Asymmetric fractalizer conjecture. There exists an asymmetric fractalizer on at most 9 vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-050330 | MATHDB-350069

Carousel extremal tournament conjecture for inducibility of C_k

摘录 / EXCERPT: Let C_k be the k-vertex carousel tournament, and let C_n be the carousel tournament on n vertices. For a tournament T, write $I(C_k,T)$ for the number of induced copies o...

conjecture | open | MathDB 页面 | 原文线索

V2-050331 | MATHDB-350083

Polynomial -boundedness conjecture for capped graphs

摘录 / EXCERPT: An ordered graph is a graph equipped with a vertex order. An ordered graph $(G,\prec)$ is capped if, for every four vertices $a\prec b\prec c\prec d$, the edges ac and bd imply...

conjecture | open | MathDB 页面 | 原文线索

V2-050332 | MATHDB-350085

The heroic forbidden-set conjecture for oriented forests and heroes

摘录 / EXCERPT: Let H be a hero, meaning a tournament such that every tournament not containing H has bounded dichromatic number, and let F be an oriented forest. A set of digraphs is heroic...

conjecture | open | MathDB 页面 | 原文线索

V2-050333 | MATHDB-350111

The conjecture on factors of independent transversals in $[k,n,1]$-graphs

摘录 / EXCERPT: The conjecture on $f(k)$. For every $k\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050334 | MATHDB-350126

Boruzanli and Gauci's super-connectivity conjecture for Kneser graphs

摘录 / EXCERPT: Let $KG(n,k)$ be the graph whose vertices are the k-subsets of $[n]=\{1,\ldots,n\}$, with two vertices adjacent exactly when the corresponding subsets are disjoint. A vertex cut...

conjecture | open | MathDB 页面 | 原文线索

V2-050335 | MATHDB-350130

Baker and Wang's conjecture on Bernardi and rotor-routing actions

摘录 / EXCERPT: Baker and Wang's conjecture. There exists a vertex q of G such that

conjecture | open | MathDB 页面 | 原文线索

V2-050336 | MATHDB-350132

Erdős-Hajnal-Simonovits-Sós-Szemerédi periodicity conjecture for Ramsey-Turán extremal graphs

摘录 / EXCERPT: Erdős-Hajnal-Simonovits-Sós-Szemerédi conjecture. The asymptotic extremal graphs G for $\varrho_p(q)$ have a partition

conjecture | open | MathDB 页面 | 原文线索

V2-050337 | MATHDB-350143

Las Vergnas-Meyniel quasi-minor conjecture

摘录 / EXCERPT: Let K_t denote the complete graph on t vertices. A quasi-K_t-minor in a graph G consists of t nonempty, pairwise disjoint vertex sets $B_1,\ldots,B_t$ such that each unio...

conjecture | open | MathDB 页面 | 原文线索

V2-050338 | MATHDB-350147

Laborde's path partition conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph, and let $\ell(D)$ denote the order of a longest directed path in D. For positive integers ℓ_1 and ℓ_2 satisfying ... a partition of D into two sub...

conjecture | open | MathDB 页面 | 原文线索

V2-050339 | MATHDB-350149

Arroyo-Galeana-Sánchez's equivalent path partition conjecture

摘录 / EXCERPT: Let D be a digraph, and let $\ell(D)$ denote the order of a longest directed path in D. For a positive integer $q<\ell(D)$, a partition (A,B) of $V(D)$ determines the induced...

conjecture | open | MathDB 页面 | 原文线索

V2-050340 | MATHDB-350158

The cubic edge diameter conjecture for maximum degree

摘录 / EXCERPT: Cubic edge diameter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050341 | MATHDB-350159

The asymptotic upper-bound conjecture for edge diameter

摘录 / EXCERPT: Asymptotic upper-bound conjecture. For $t\ne 2$ and any $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-050342 | MATHDB-350166

Generic existence conjecture for basic WKB cameral networks

摘录 / EXCERPT: Let $\mathcal A$ be the Hitchin base. A basic WKB cameral network associated to $a\in\mathcal A$ consists of a suitable homotopy-equivalent open subset of the curve together with f...

conjecture | open | MathDB 页面 | 原文线索

V2-050343 | MATHDB-350189

Spinoza-West sharp threshold conjecture for connectedness recognition

摘录 / EXCERPT: Let G be an n-vertex graph, and let $\mathcal{D}_\ell(G)$ be the multiset of all induced subgraphs of G on ℓ vertices. The connectedness of G is determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-050344 | MATHDB-350197

Taylor's conjecture on finite subgraphs of shift graphs

摘录 / EXCERPT: Let G be a graph, and let $\chi(G)$ denote its chromatic number. For $n\in\mathbb{N}$, write $\operatorname{Sh}_n(\omega)$ for the corresponding shift graph on ω. Taylor's...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050345 | MATHDB-350231

Frankl and Kupavskii's matching-cover conjecture for uniform hypergraphs

摘录 / EXCERPT: Let n,s,k be positive integers with $k\geq 2$ and $n\geq ks+k-1$. Let H be a k-graph on vertex set $[n]$, and let $\nu(H)$ and $\tau(H)$ denote its matching number and vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-050346 | MATHDB-350253

Optimal energy bounds for primary vertex operators

摘录 / EXCERPT: Let V be a simple unitary VOA, and let $Y(a,z)$ be a primary vertex operator with $a\in V\setminus V_1$. Optimal energy bounds conjecture. Every such operator $Y(a,z)$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-050347 | MATHDB-350266

Réti et al.'s maximum Sombor index conjecture for connected cyclic graphs

摘录 / EXCERPT: Réti et al.'s conjecture. Among all connected ν-cyclic graphs of order n, only the graph $H_{n,\nu}$ has the maximum Sombor index.

conjecture | open | MathDB 页面 | 原文线索

V2-050348 | MATHDB-350270

Universal logarithmic-to-linear speed-up conjecture for covering graphs with independent walks

摘录 / EXCERPT: Universal speed-up conjecture. There exist universal constants C and C' such that, for every graph G and every k,

conjecture | open | MathDB 页面 | 原文线索

V2-050349 | MATHDB-350273

Avoidance conjecture for partial edge colorings of hypercubes

摘录 / EXCERPT: Let Q_d be the d-dimensional hypercube, and let φ be a partial d-edge coloring of Q_d, meaning that some edges of Q_d receive colors, without requiring the colorin...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050350 | MATHDB-350274

Induced-matching avoidance conjecture for partial edge colorings of hypercubes

摘录 / EXCERPT: Let Q_d be the d-dimensional hypercube, and let φ be a partial d-edge coloring of Q_d. A color class is the set of edges receiving one fixed color, and an induced ma...

conjecture | open | MathDB 页面 | 原文线索

V2-050351 | MATHDB-350283

Isomorphism of minimum-eigenvector-induced Gilbert graphs

摘录 / EXCERPT: Let G_t be the graph obtained at step t of the construction, with vertex set V_t. For each vector v, let $a^{(t)}_v$ be the eigenvector of G_t with eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-050352 | MATHDB-350295

Cioabă and Wong's spectral spanning-tree packing conjecture

摘录 / EXCERPT: Let $m\geq 1$ be an integer, and let G be a d-regular graph with $d\geq 2m+2$. Let $\lambda_2(G)$ be the second-largest adjacency eigenvalue of G, and let $\sigma(G)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-050353 | MATHDB-350310

The cubic-map enumeration conjecture for bipartite chord diagrams

摘录 / EXCERPT: Let $\mathcal{C}_{n}(\{T_{2k+1},B_{2k+1}\}_{k\geqslant 1})$ be the set of connected chord diagrams of size n avoiding all odd top and bottom cycles. The cubic-map enumeration c...

conjecture | open | MathDB 页面 | 原文线索

V2-050354 | MATHDB-350311

The forbidden-subdiagram characterization conjecture for k-terminal-minimality

摘录 / EXCERPT: A chord diagram is k-terminal-minimal if it is k-terminal and minimal with respect to that property under the reduction considered in the paper. The forbidden-subdiagram charac...

conjecture | open | MathDB 页面 | 原文线索

V2-050355 | MATHDB-350318

Wiener-eccentricity monotonicity conjecture under edge contraction

摘录 / EXCERPT: Wiener-eccentricity monotonicity conjecture. For every such edge e,

conjecture | open | MathDB 页面 | 原文线索

V2-050356 | MATHDB-350331

Haemers's Laplacian eigenvalue conjecture for graph toughness

摘录 / EXCERPT: Let G be a simple graph with minimum degree δ. Let μ_2 and μ_n denote the second-smallest and largest eigenvalues of the Laplacian matrix of G, respectively, and...

conjecture | open | MathDB 页面 | 原文线索

V2-050357 | MATHDB-350337

Generic simplicity conjecture for principal and nonprincipal W-algebra Verma modules

摘录 / EXCERPT: Let (f,h,e) be an $\mathfrak{sl}_2$-triple in a Lie algebra $\mathfrak{g}$, let k be a level, and let $M^{\mathsf{k}}(\lambda,f)$ denote the Verma module for the W-alg...

conjecture | open | MathDB 页面 | 原文线索

V2-050358 | MATHDB-350364

The exact-one-edge transversal conjecture for 5-cycles in triangle-free subcubic graphs

摘录 / EXCERPT: Let G be a triangle-free subcubic graph. Exact-one-edge 5-cycle conjecture. There exists a set E' of edges such that every 5-cycle in G contains exactly one edge from E'....

conjecture | open | MathDB 页面 | 原文线索

V2-050359 | MATHDB-350377

Conjecture on chi-boundedness of graphs excluding induced k-fans

摘录 / EXCERPT: For a positive integer k, a k-fan is a graph consisting of an induced path and one additional vertex with at least k neighbours on the path; a 3-fan is called a fan. The k-...

conjecture | open | MathDB 页面 | 原文线索

V2-050360 | MATHDB-350380

Subcubic multigraph star chromatic index conjecture

摘录 / EXCERPT: Subcubic multigraph star chromatic index conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050361 | MATHDB-350381

Outerplanar graph star chromatic index conjecture

摘录 / EXCERPT: Outerplanar graph star chromatic index conjecture. Every such graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050362 | MATHDB-350410

Harrington-Henninger-Voss-Karhadkar-Robinson-Wong sum-difference index inequality conjecture

摘录 / EXCERPT: Let G be a finite simple graph, with sum index $\operatorname{sum}(G)$ and difference index $\operatorname{dif}(G)$ defined using injective integer vertex labellings and, respect...

conjecture | open | MathDB 页面 | 原文线索

V2-050363 | MATHDB-350413

Carlson et al.'s Shearer-type surplus conjecture for clique-free graphs

摘录 / EXCERPT: Carlson et al.'s conjecture. Every K_r-free d-degenerate graph with m edges has surplus

conjecture | open | MathDB 页面 | 原文线索

V2-050364 | MATHDB-350426

Verstraëte's minimum-degree conjecture on cycle lengths in Hamiltonian graphs

摘录 / EXCERPT: Let G be an n-vertex Hamiltonian graph, and let $\delta(G)$ denote its minimum degree. Verstraëte's conjecture. If ... then G has $\Omega(n)$ different cycle lengths. This stre...

conjecture | open | MathDB 页面 | 原文线索

V2-050365 | MATHDB-350427

Concentration and separation conjecture for cycle lengths in Hamiltonian graphs

摘录 / EXCERPT: Let G be an n-vertex Hamiltonian graph with minimum degree at least 3. Concentration and separation conjecture. Both of the following hold: 1. G has $\Omega(\sqrt n)$ cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-050366 | MATHDB-350429

Spectral extremal graphs conjecture for Turán graphs plus $O(1)$ edges

摘录 / EXCERPT: General spectral extremal conjecture. For sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-050367 | MATHDB-350432

Uniqueness conjecture for cograph minimal (∞,k)-polar obstructions

摘录 / EXCERPT: Uniqueness conjecture. There exists exactly one cograph minimal (∞,k)-polar obstruction of type (c,i). This extends the cases for which uniqueness is established and leave...

conjecture | open | MathDB 页面 | 原文线索

V2-050368 | MATHDB-350444

Conjectured Laplacian coefficient bounds for graphs via closed walks

摘录 / EXCERPT: Let G be a graph with adjacency matrix $A=A(G)$, and let $A_1=A(L_1(G))$ be the adjacency matrix of its line graph. Write $c_{n-k}(G)$ for the corresponding Laplacian coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-050369 | MATHDB-350449

The asymptotic existence conjecture for irreducible non-metrizable path systems

摘录 / EXCERPT: Asymptotic existence conjecture. Asymptotically almost every graph has an irreducible non-metrizable path system; specifically, this holds with probability

conjecture | open | MathDB 页面 | 原文线索

V2-050370 | MATHDB-350451

Jet graphs of very well covered graphs are very well covered

摘录 / EXCERPT: Let G be a very well covered graph, and for $s\in\mathbb{N}$ let $\mathcal{J}_s(G)$ denote its s-th jet graph. A graph is very well covered if all its minimal vertex covers hav...

conjecture | open | MathDB 页面 | 原文线索

V2-050371 | MATHDB-350452

Non-IR conjecture for paths and cycles

摘录 / EXCERPT: Non-IR conjecture for paths and cycles. The path P_n is not an IR-graph for each $n\geq 3$, and the cycle C_n is not an IR-graph for each $n\geq 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-050372 | MATHDB-350476

Beaton-Brown-Cameron conjecture on independence equivalence of odd cycles

摘录 / EXCERPT: Beaton-Brown-Cameron conjecture. If $3\not\vert n$, then a graph G is independence equivalent to C_n if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-050373 | MATHDB-350483

Large-system response-time conjecture for speculative queueing networks

摘录 / EXCERPT: Large-system response-time conjecture. Provided that $\rho(\tau)<1$,

conjecture | open | MathDB 页面 | 原文线索

V2-050374 | MATHDB-350489

Universal generator-set conjecture for circulant nut graphs

摘录 / EXCERPT: Universal generator-set conjecture. For each odd $t\geq 3$, there exists p_t such that $\operatorname{Circ}(n,\{1,\dots,2t+1\}\setminus\{p_t\})$ is a nut graph for every even...

conjecture | open | MathDB 页面 | 原文线索

V2-050375 | MATHDB-350496

W[1]-hardness of M-EPVCB parameterized by the smaller bipartition

摘录 / EXCERPT: Let $G=(X,Y,E)$ be the edge-weighted bipartite graph in an instance of M-EPVCB, and let $|X|$ and $|Y|$ denote the sizes of its two bipartition classes. The minimum-side hardness c...

conjecture | open | MathDB 页面 | 原文线索

V2-050376 | MATHDB-350497

W[1]-hardness of M-EPVCB parameterized by the degree deficit

摘录 / EXCERPT: Let $G=(V,E)$ be the graph in an instance of M-EPVCB, and let $\Delta(G)$ denote its maximum degree. The degree-deficit hardness conjecture. M-EPVCB is W[1]-hard with respect to th...

conjecture | open | MathDB 页面 | 原文线索

V2-050377 | MATHDB-350519

Zhai-Lin-Shu booksize conjecture for graphs satisfying Nosal's inequality

摘录 / EXCERPT: Zhai-Lin-Shu's booksize conjecture. For every natural number k, there exists $m(k)$ such that if $m>m(k)$ and $\lambda_{1}^{2}\geq m$, then $bk(G)>k$, unless G is a complete bi...

conjecture | open | MathDB 页面 | 原文线索

V2-050378 | MATHDB-350520

Minimal-embedding conjecture for three-dimensional grid graphs with an even parameter

摘录 / EXCERPT: Let $G(\alpha_1,\alpha_2,\alpha_3)$ be a three-dimensional grid graph with at least one even parameter. Let the embedding constructed in Proposition 3.3 be the 2-cell embedding d...

conjecture | open | MathDB 页面 | 原文线索

V2-050379 | MATHDB-350528

Rainbow directed cycle-or-path conjecture for proper arc-colourings

摘录 / EXCERPT: Let $\overleftrightarrow{K_n}$ be the digraph obtained from the complete graph K_n by replacing each edge with two oppositely directed arcs and adding a directed loop at each ver...

conjecture | open | MathDB 页面 | 原文线索

V2-050380 | MATHDB-350542

Goulden-Rattan square-coefficient inequalities for expanders

摘录 / EXCERPT: Let $Y_k(u)$ be the set of expanders with k edges, $u-1$ white vertices and one black vertex. Let $X_k(i,j)$ be the set of expanders with k edges, $i+j-2$ white vertices and tw...

conjecture | open | MathDB 页面 | 原文线索

V2-050381 | MATHDB-350548

Auchiche-Hansen's proximity and distance eigenvalue conjecture

摘录 / EXCERPT: Let G be a connected graph on $n\geq 4$ vertices with diameter D, proximity π and distance spectrum $\partial_1\geq \ldots \geq \partial_n$. Auchiche-Hansen's conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-050382 | MATHDB-350549

Collins's matching peak conjecture for adjacency and distance polynomials of trees

摘录 / EXCERPT: Let T be a tree. Let $CPA(T)$ and $CPD(T)$ denote the characteristic polynomials of its adjacency and distance matrices, respectively. Let $p_A(T)$ be the position of the peak of...

conjecture | open | MathDB 页面 | 原文线索

V2-050383 | MATHDB-350550

Shor's bounds for the distance-polynomial peak of trees

摘录 / EXCERPT: Let T be a tree on n vertices, and let $p_D(T)$ denote the index of the peak of the normalized coefficients of the characteristic polynomial of the distance matrix of T. Shor...

conjecture | open | MathDB 页面 | 原文线索

V2-050384 | MATHDB-350580

The growable-realizations conjecture for admissible multisets

摘录 / EXCERPT: Growable-realizations conjecture. For any fixed set U, there is a finite set of growable realizations with underlying set U that implies the existence of realizations for all b...

conjecture | open | MathDB 页面 | 原文线索

V2-050385 | MATHDB-350587

The positive-integer edge-Kempe class conjecture for bipartite cubic graphs

摘录 / EXCERPT: Positive-integer edge-Kempe class conjecture. For any positive integer m, there is a bipartite cubic graph with exactly m edge-Kempe equivalence classes.

conjecture | open | MathDB 页面 | 原文线索

V2-050386 | MATHDB-350588

Bessy-Havet extremal 3-edge-coloring conjecture for connected cubic graphs

摘录 / EXCERPT: Bessy-Havet extremal coloring conjecture. The connected cubic graph of order n with the largest number of 3-edge-colorings has

conjecture | open | MathDB 页面 | 原文线索

V2-050387 | MATHDB-350595

The planar 3-path vertex cover conjecture

摘录 / EXCERPT: Planar 3-path vertex cover conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050388 | MATHDB-350596

The chordal graph k-path vertex cover conjecture

摘录 / EXCERPT: Chordal graph k-path vertex cover conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050389 | MATHDB-350600

Kim-Kim-Kostochka-O spectral-radius conjecture for saturated graphs

摘录 / EXCERPT: Kim-Kim-Kostochka-O conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050390 | MATHDB-350601

The odd-even cycle ratio conjecture for 2-connected claw-free graphs

摘录 / EXCERPT: Odd-even cycle ratio conjecture. Both ratios

conjecture | open | MathDB 页面 | 原文线索

V2-050391 | MATHDB-350602

Super-polynomial one-color oriented Ramsey growth for bounded-degree acyclic digraphs

摘录 / EXCERPT: For $k,\Delta\geq 1$ and $n>\Delta$, let $H_{k,n,\Delta}$ be an acyclic digraph with n vertices and maximum degree Δ maximizing $\ovar{r_k}(H)$. Super-polynomial growth co...

conjecture | open | MathDB 页面 | 原文线索

V2-050392 | MATHDB-350603

Quantitative growth conjecture for multicolor oriented Ramsey numbers

摘录 / EXCERPT: For $k,\Delta\geq 1$ and $n>\Delta$, let $H_{k,n,\Delta}$ be an acyclic digraph with n vertices and maximum degree Δ maximizing $\ovar{r_k}(H)$. Quantitative growth conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050393 | MATHDB-350604

Linear oriented Ramsey number of sparse random digraphs

摘录 / EXCERPT: Fix $d\geq 2$, and let $H=\ovar{G}(n,d)$ be the sparse random digraph used in the paper. Write $\ovar{r_1}(H)$ for its one-color oriented Ramsey number. Linear random-digraph conje...

conjecture | open | MathDB 页面 | 原文线索

V2-050394 | MATHDB-350605

Super-polynomial directed Ramsey numbers with three colors

摘录 / EXCERPT: Let $\overleftrightarrow{r_3}(H)$ be the three-color directed Ramsey number, and let an acyclic digraph have maximum degree at most Δ. Three-color directed-Ramsey conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050395 | MATHDB-350613

The alpha-spectral radius conjecture for theta digraphs

摘录 / EXCERPT: Let $alpha$ satisfy $\frac{1}{2}<alpha<1$, and let $\widetilde{Theta}(3,1;n-3)$ and $\widehat{Theta}(n)$ be the theta digraphs defined in the paper. Write $\rho_{alpha}(G)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-050396 | MATHDB-350616

A logarithmic upper bound for the moment-to-Perron-value ratio of rooted trees

摘录 / EXCERPT: Let T be a rooted tree of order n, and let $\rho(T)$ denote its Perron value and $\mu(T)$ its moment. Logarithmic bound conjecture. There exists $\alpha_0\in\mathbb{R}$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-050397 | MATHDB-350623

The integer-move conjecture for the fractional transversal game

摘录 / EXCERPT: Let $H=(V,\mathcal E)$ be a finite hypergraph. An integer move in the fractional transversal game is a move of the form $(v_{i_1},1)$. Integer-move conjecture. If each of the first...

conjecture | open | MathDB 页面 | 原文线索

V2-050398 | MATHDB-350624

The fractional transversal game inequality conjecture

摘录 / EXCERPT: Let H be a finite hypergraph, and let $\tau_g(H)$ denote its ordinary transversal game number and $\tau_g^*(H)$ its fractional transversal game number. Fractional transversal gam...

conjecture | open | MathDB 页面 | 原文线索

V2-050399 | MATHDB-350625

The largest-legal-weight strategy conjecture for Staller

摘录 / EXCERPT: Let a partial covering function be a current position in the fractional transversal game from which legal submoves are defined. Largest-legal-weight strategy conjecture. Starting f...

conjecture | open | MathDB 页面 | 原文线索

V2-050400 | MATHDB-350630

The uberhomology conjecture for trees, complete bipartite graphs and paths

摘录 / EXCERPT: Uberhomology conjecture for trees, complete bipartite graphs and paths. The following assertions hold in general:

conjecture | open | MathDB 页面 | 原文线索

V2-050401 | MATHDB-350631

The vanishing conjecture for second uberhomology

摘录 / EXCERPT: Vanishing conjecture for second uberhomology.

conjecture | open | MathDB 页面 | 原文线索

V2-050402 | MATHDB-350678

Total positivity of bivariate functional-digraph polynomials

摘录 / EXCERPT: Let $\Psi_n(x,y)$ be the generating polynomial for functional digraphs on $[n]$, with weight x for each cyclic vertex and weight y for each component. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-050403 | MATHDB-350682

Almost color-balanced perfect matching conjecture

摘录 / EXCERPT: Almost color-balanced perfect matching conjecture. There is a perfect matching M of K_{2kn} with

conjecture | open | MathDB 页面 | 原文线索

V2-050404 | MATHDB-350692

Piecewise convexity conjecture for hypergraph Ollivier curvature

摘录 / EXCERPT: Let $H=(V,E)$ be a hypergraph, let x,y be vertices of H, and let h be the concave cost function used to define the transport distance W_h and the associated curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-050405 | MATHDB-350693

Coroničová Hurajová-Gago-Madaras blow-up conjecture for betweenness-uniform graphs

摘录 / EXCERPT: Coroničová Hurajová-Gago-Madaras conjecture. For any graph G on a vertex set $\{v_1,\dots,v_n\}$ there exist graphs $H_1,\dots,H_n$ such that $G[H_1,\dots,H_n]$ is betweenness-un...

conjecture | open | MathDB 页面 | 原文线索

V2-050406 | MATHDB-350694

Vertex-cut obstruction for betweenness-uniform blow-ups

摘录 / EXCERPT: Vertex-cut obstruction. There are no graphs $H_1,\dots,H_n$ such that $G[H_1,\dots,H_n]$ is betweenness-uniform.

conjecture | open | MathDB 页面 | 原文线索

V2-050407 | MATHDB-350695

Bencs-Csikvári conjecture on spanning forests of regular graphs

摘录 / EXCERPT: Let G be a d--regular graph on n vertices, and let $F(G)$ denote its number of spanning forests. Bencs-Csikvári conjecture. ... This is the proposed sharp analogue for spanning...

conjecture | open | MathDB 页面 | 原文线索

V2-050408 | MATHDB-350696

Negative correlation conjecture for uniformly random spanning forests

摘录 / EXCERPT: Let G be a graph, and let $\mathbf{F}$ be a uniformly random forest chosen from all forests of G. For edges $e,f\in E(G)$, negative correlation conjecture. ... This conjecture as...

conjecture | open | MathDB 页面 | 原文线索

V2-050409 | MATHDB-350724

The 3412-pattern upper-bound conjecture for reduced-word graphs

摘录 / EXCERPT: Let S_n be the symmetric group, let $w\in S_n$, let $G(w)$ be the graph of reduced words of w, let $L_2(w)$ denote the associated set of rank-two root subsystems, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050410 | MATHDB-350736

Polynomial removal threshold exceeds the homomorphism threshold for odd cycles

摘录 / EXCERPT: For a graph H, let $\delta_\chi(H)$ and $\delta_{\hom}(H)$ denote its chromatic and homomorphism thresholds, and let $\delta_{\mathrm{poly\mhyphen rem}}(H)$ denote its polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-050411 | MATHDB-350737

Realizability of all orders among graph thresholds

摘录 / EXCERPT: For a graph H, let $\delta_\chi(H)$ and $\delta_{\hom}(H)$ be its chromatic and homomorphism thresholds, and let $\delta_{\mathrm{lin\mhyphen rem}}(H)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-050412 | MATHDB-350742

Rall's complete-factor conjecture for well-dominated Cartesian products

摘录 / EXCERPT: Rall's conjecture. If $G\Box H$ is well-dominated, then at least one of G or H is a complete graph.

conjecture | open | MathDB 页面 | 原文线索

V2-050413 | MATHDB-350747

Rainbow Hajnal--Szemerédi conjecture for graph systems

摘录 / EXCERPT: Let t and n be positive integers, and let $\textbf{G}=\{G_1,G_2,\ldots,G_{\frac{n}{t}\binom{t}{2}}\}$ be an n-vertex graph system. Rainbow Hajnal--Szemerédi conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-050414 | MATHDB-350762

Vertex conjecture for subdifferentials of bivariate rational functions

摘录 / EXCERPT: Let r be the rational function defined in the paper, let P be a polytope, let $f(x)=r(x)+I_P(x)$, and let v be a vertex of P satisfying $\xi_1(v)=\xi_2(v)=0$. Vertex conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050415 | MATHDB-350763

The exceptional-graph conjecture for average color degree and rainbow matchings

摘录 / EXCERPT: Let G be an edge-colored graph, let $\hat d(G)$ be its average color degree, and let $\widetilde{K_4}$ denote the exceptional edge-colored graph referred to in the source. A rain...

conjecture | open | MathDB 页面 | 原文线索

V2-050416 | MATHDB-350767

Second-, third- and fourth-minimum conjecture for the A_α spectral radius of strongly connected digraphs

摘录 / EXCERPT: Second-, third- and fourth-minimum conjecture. Among all the strongly connected digraphs with order $n\geq 5$ and $0\leq \alpha<1$, $\theta(0,1,n-3)$, $\theta(1,1,n-4)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050417 | MATHDB-350796

Vince-Wang's minimum-density conjecture for graphs of minimum degree three

摘录 / EXCERPT: Vince-Wang's conjecture. If G is a connected graph all of whose vertices have degree at least 3, then

conjecture | open | MathDB 页面 | 原文线索

V2-050418 | MATHDB-350798

Distinctness conjecture for graph invariants of finite polar spaces

摘录 / EXCERPT: Let Δ_i denote the graph associated with the i-dimensional subspaces of the finite polar space, and let κ_i, χ_i, and ξ_i be its corresponding graph parame...

conjecture | open | MathDB 页面 | 原文线索

V2-050419 | MATHDB-350802

The DCI and CI classification conjecture for dihedral groups

摘录 / EXCERPT: Let n be a positive integer and let D_{2n} be the dihedral group of order $2n$. A finite group is a DCI-group if every Cayley digraph on it is a CI-digraph, and a CI-g...

conjecture | open | MathDB 页面 | 原文线索

V2-050420 | MATHDB-350808

Tran-Vu conjecture on one-voter bias in majority dynamics

摘录 / EXCERPT: Let $\mathbb{G}(2n+\Delta,1/2)$ be the random graph with vertex sets initially colored red and blue as ... A color wins when all vertices eventually have that color. Tran-Vu's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050421 | MATHDB-350837

The strict depth hierarchy conjecture for ReLU networks

摘录 / EXCERPT: For $n\boxin\mathbb{N}$, let ... Here, $\operatorname{ReLU}_n(k)$ denotes the class of functions representable by ReLU neural networks with n input variables and k hidden layers,...

conjecture | open | MathDB 页面 | 原文线索

V2-050422 | MATHDB-350840

Facial edge-coloring conjecture for plane graphs

摘录 / EXCERPT: Let G be a plane graph, and let $ u$ be a positive integer. An ν-facial edge-coloring of G is an edge-coloring in which all edges on every facial trail of length at most...

conjecture | open | MathDB 页面 | 原文线索

V2-050423 | MATHDB-350853

The extremal digraph blow-up conjecture for walks with common endpoints

摘录 / EXCERPT: Extremal digraph blow-up conjecture. Let $k\ge z(t)$ and let n be sufficiently large. Then $D\in EX(n,\mathscr{F}_{k,t+1})$ if and only if D is a balanced blow-up of the transi...

conjecture | open | MathDB 页面 | 原文线索

V2-050424 | MATHDB-350880

Kamatchi-Arumugam tree characterization conjecture for distance-antimagic trees

摘录 / EXCERPT: Kamatchi-Arumugam tree conjecture. T is distance antimagic if and only if every support vertex v has precisely one leaf adjacent to v.

conjecture | open | MathDB 页面 | 原文线索

V2-050425 | MATHDB-350919

The Stanley-Schwartz-Zaks conjecture on distinguishing trees by their chromatic symmetric functions

摘录 / EXCERPT: Let T and T' be trees, and call them non-isomorphic when they do not become the same graph after removing vertex labels. The chromatic symmetric function of a graph G is deno...

conjecture | open | MathDB 页面 | 原文线索

V2-050426 | MATHDB-350924

Elementary equivalence from isomorphic cores for graph products

摘录 / EXCERPT: Let $\mathcal G=\mathcal G(\Gamma,\{G_v\}_{v\in\Gamma})$ and $\mathcal G'=\mathcal G(\Delta,\{G'_v\}_{v\in\Delta})$ be graph products with finite graphs, where each vertex group sa...

conjecture | open | MathDB 页面 | 原文线索

V2-050427 | MATHDB-350925

The core classification conjecture for right-angled Artin groups

摘录 / EXCERPT: Let $\mathbb G(\Gamma)$ and $\mathbb G(\Delta)$ be right-angled Artin groups, and let $\operatorname{Core}(\Gamma)$ denote the core graph. RAAG core classification conjecture. If .....

conjecture | open | MathDB 页面 | 原文线索

V2-050428 | MATHDB-350932

Degree-partition conjecture for optimal minimally connected graphs

摘录 / EXCERPT: Degree-partition conjecture. If G is an optimal minimally k-(edge-)connected graph of order $n\geq 2k+1$, then G is degree-partitioned.

conjecture | open | MathDB 页面 | 原文线索

V2-050429 | MATHDB-350974

The regular-orbit conjecture for cubic vertex-transitive graphs

摘录 / EXCERPT: Let Γ be a cubic vertex-transitive graph of order n that is not isomorphic to $K_{3,3}$ or a split Praeger-Xu graph, and let $g\in\operatorname{Aut}(\Gamma)$. A regular or...

conjecture | open | MathDB 页面 | 原文线索

V2-050430 | MATHDB-350979

Circulant graph conjecture for monophonic position number two

摘录 / EXCERPT: Let n be an integer, and let a circulant graph mean a graph whose vertices are arranged cyclically with adjacency determined by a fixed set of cyclic differences. The monophonic...

conjecture | open | MathDB 页面 | 原文线索

V2-050431 | MATHDB-350980

Stochastic dominance of nearest-neighbor distances in PLM-PPP and PSP-PPP

摘录 / EXCERPT: Let $n>1$ and consider the distance from a typical general vehicle to its nth-nearest neighbor in the PLM-PPP and in the PSP-PPP, with Rayleigh-distributed half-lengths. Nearest-...

conjecture | open | MathDB 页面 | 原文线索

V2-050432 | MATHDB-350999

Akbari et al.'s switching conjecture for main eigenvalues of signed graphs

摘录 / EXCERPT: Akbari et al.'s conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050433 | MATHDB-351003

Kamčev-Liebenau-Wormald conjecture on typical degree sequences of uniform hypergraphs

摘录 / EXCERPT: Kamčev-Liebenau-Wormald conjecture. There exists a set $\mathfrak{W}$ having probability $1-O(n^{-\omega(1)})$ in both $\mathcal{D}_r(n,m)$ and $\mathcal{T}_r(n,m)$ such that, unif...

conjecture | open | MathDB 页面 | 原文线索

V2-050434 | MATHDB-351021

Holroyd-Talbot conjecture for pendant path graphs

摘录 / EXCERPT: Pendant-path conjecture. The pendant path graph P_n^* is r-EKR whenever

conjecture | open | MathDB 页面 | 原文线索

V2-050435 | MATHDB-351026

A sharp edge bound in terms of the forcing number

摘录 / EXCERPT: Let G be a graph of order $2n$ with a perfect matching. Write $e(G)$ for the number of edges of G, and let $F(G)$ denote its maximum forcing number. The edge-forcing number con...

conjecture | open | MathDB 页面 | 原文线索

V2-050436 | MATHDB-351027

The maximal b1-spectral-radius conjecture for bicyclic graphs

摘录 / EXCERPT: Maximal α-spectral-radius conjecture. If G has the maximal α-spectral radius among graphs in $\mathscr{B}(n,d)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050437 | MATHDB-351059

Vertex decomposability of higher independence complexes of trees

摘录 / EXCERPT: Let T be a tree and let $r\geq 1$. The complex $\operatorname{Ind}_r(T)$ is the r-independence complex of T. Tree higher-independence conjecture. For any tree T and...

conjecture | open | MathDB 页面 | 原文线索

V2-050438 | MATHDB-351060

Cioabă-Gregory's maximum principal ratio conjecture for connected graphs

摘录 / EXCERPT: Let G be a simple, undirected connected graph on n vertices. Let ${\bf x}$ be the positive unit eigenvector of its adjacency matrix corresponding to the largest eigenvalue, and...

conjecture | open | MathDB 页面 | 原文线索

V2-050439 | MATHDB-351068

Conjecture on extinction and non-halting in multitype branching processes in a Markovian environment

摘录 / EXCERPT: Let $G=(V,E)$ be a finite directed graph. For each vertex $v\in V$, let S_v be a finite state space, let P_v be an irreducible stochastic transition matrix, and let R_v be a...

conjecture | open | MathDB 页面 | 原文线索

V2-050440 | MATHDB-351079

Glock-Joos-Kim-Kühn-Osthus conjecture on packing bounded-regular graphs

摘录 / EXCERPT: Fix an integer r. Bounded-regular graph packing conjecture. There exists $n_0\in\mathbb{N}$ such that, for every $n\ge n_0$, any family of n-vertex graphs $(G_s)_{s\in[N]}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-050441 | MATHDB-351083

Fiorini-Huynh-Joret-Pashkovich conjecture on spanning tree polytopes in minor-closed classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a proper minor-closed graph class, and let G be a connected n-vertex graph in $\mathcal{G}$. The spanning tree polytope of G is the convex hull of the in...

conjecture | open | MathDB 页面 | 原文线索

V2-050442 | MATHDB-351097

The Brill-Noether gonality conjecture for finite graphs

摘录 / EXCERPT: Let G be a connected loopless multigraph, and let $g:= |E(G)| - |V(G)| + 1$ denote its cyclomatic number. Brill-Noether gonality conjecture. Then ... This is the $r=1$ case of the...

conjecture | open | MathDB 页面 | 原文线索

V2-050443 | MATHDB-351098

Baker's subdivision and metric-graph gonality conjecture

摘录 / EXCERPT: Let G be a connected loopless multigraph, let $\Gamma(G)$ be the corresponding metric graph with unit edge lengths, and let $r \geq 1$. For each integer $k \geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-050444 | MATHDB-351104

Abdi-Ghorbani's uniqueness conjecture for minimum-gap quartic graphs

摘录 / EXCERPT: Abdi-Ghorbani's uniqueness conjecture. For every $n\geq11$, the n-vertex graph of ${\cal F}_{\rm qua}$ is the unique graph with minimum spectral gap among connected quartic graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050445 | MATHDB-351105

Maximal-gap conjecture for the spectra of the cubic graphs Δ_n

摘录 / EXCERPT: Maximal-gap conjecture for Δ_n. The interval

conjecture | open | MathDB 页面 | 原文线索

V2-050446 | MATHDB-351108

Conjecture on non-Hamiltonian graphs with an arbitrarily high fraction of Hamiltonian vertex pairs

摘录 / EXCERPT: Let G be a graph, and let r_G denote the fraction of pairs of vertices of G that are connected by a Hamiltonian path. Equivalently, if G has order n, define ... where G i...

conjecture | open | MathDB 页面 | 原文线索

V2-050447 | MATHDB-351109

Power domination number of the middle graph

摘录 / EXCERPT: Let G be any graph, and let $M(G)$ denote its middle graph. Write $\gamma_p(M(G))$ for the power domination number of $M(G)$ and $\gamma^{'}(G)$ for the edge domination number of...

conjecture | open | MathDB 页面 | 原文线索

V2-050448 | MATHDB-351150

The non-expanding links conjecture for bounded-degree graph sequences

摘录 / EXCERPT: Non-expanding links conjecture. There is no sequence of bounded-degree finite graphs, with size growing to infinity, such that all links in all the graphs form an expander family.

conjecture | open | MathDB 页面 | 原文线索

V2-050449 | MATHDB-351160

The clique-density construction conjecture for induced 4-cycles

摘录 / EXCERPT: Induced 4-cycle construction conjecture. For every real number $x\in[1/2,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-050450 | MATHDB-351162

Equal-component failure-state conjecture for proportional edge connectivity

摘录 / EXCERPT: Equal-component failure-state conjecture. There exists $G\in G(n,m)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-050451 | MATHDB-351163

Fajtlowicz's Randić index-radius conjecture

摘录 / EXCERPT: Let G) be a graph. Write $R=R(G)$ for its Randić index and r for its radius. An even path is a path with an even number of vertices. Fajtlowicz's conjecture. If G is an even...

conjecture | open | MathDB 页面 | 原文线索

V2-050452 | MATHDB-351164

Aouchiche's Randić index-diameter bounds conjecture

摘录 / EXCERPT: Let G be a graph with n vertices. Write $R=R(G)$ for its Randić index and d for its diameter. Aouchiche's conjecture. The two bounds ... and ... should hold. These inequalities c...

conjecture | open | MathDB 页面 | 原文线索

V2-050453 | MATHDB-351166

Tight noise thresholds for root confidence sets in preferential and uniform attachment

摘录 / EXCERPT: Noise-threshold conjecture. For linear preferential attachment (LPA), corresponding to $\alpha=0$ and $\beta=1$, if $\theta=o(n^{-1/2})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050454 | MATHDB-351185

Győri-Paulos-Salia-Tompkins-Zamora planar forbidden-subgraph counting conjecture

摘录 / EXCERPT: Let $\mathcal F$ be a finite set of graphs, let H be a graph, and let $\mathcal G$ be the class of all planar graphs with no subgraph isomorphic to any member of $\mathcal F$. As...

conjecture | open | MathDB 页面 | 原文线索

V2-050455 | MATHDB-351186

Huynh-Wood conjecture on subgraph counts in degenerate graphs

摘录 / EXCERPT: Let d be a nonnegative integer, let $\mathcal D_d$ be the class of d-degenerate graphs, and let H be a d-degenerate graph. For a graph H, let $\operatorname{flap}_d(H)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-050456 | MATHDB-351187

Huynh-Wood complete-bipartite-minor-free counting conjecture

摘录 / EXCERPT: Let s and t be positive integers with $s\leq t$, and let $\mathcal G$ be the class of graphs containing no $K_{s,t}$ minor. Let H be a graph containing no $K_{s,t}$ minor. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050457 | MATHDB-351188

Győri-Paulos-Salia-Tompkins-Zamora even-cycle-free planar counting conjecture

摘录 / EXCERPT: Let $\ell\geq 2$ be an integer, let p be a sufficiently large integer, let H be the cycle of length p, and let $\mathcal G$ be the class of all planar graphs with no even cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-050458 | MATHDB-351189

Conjecture on dispersability of bipartite cubic planar graphs

摘录 / EXCERPT: Let G be a bipartite cubic planar graph, meaning that G is bipartite, planar, and every vertex has degree three. A graph is dispersable if it has a proper $\Delta(G)$-edge-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-050459 | MATHDB-351190

Ore's periodicity conjecture for the minimum size of k-critical graphs

摘录 / EXCERPT: Ore's conjecture. If $k\geq4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050460 | MATHDB-351193

Cycle-rank conjecture for vertex metric dimension

摘录 / EXCERPT: Let G be a connected graph. Let $L(G)$ denote the number of leaves, and let $c(G)=\left\vert E(G)\right\vert-\left\vert V(G)\right\vert+1$ be its cyclomatic number. The cycle-ran...

conjecture | open | MathDB 页面 | 原文线索

V2-050461 | MATHDB-351194

Zero-forcing cycle-rank conjecture for edge metric dimension

摘录 / EXCERPT: Let G be a connected graph. Let $Z(G)$ denote its zero forcing number, let $c(G)=\left\vert E(G)\right\vert-\left\vert V(G)\right\vert+1$ be its cyclomatic number, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050462 | MATHDB-351225

The vertex algebra Hilbert conjecture for reductive automorphism groups

摘录 / EXCERPT: Let ${\mathcal V}$ be a simple, strongly finitely generated VOA that is $\mathbb N$-graded by conformal weight, with finite-dimensional graded pieces. Let G be a reductive group...

conjecture | open | MathDB 页面 | 原文线索

V2-050463 | MATHDB-351235

Characterization of divergence by order via non-cyclic subgraphs

摘录 / EXCERPT: Divergence characterization conjecture. The graph G has a sequence that diverges by order if and only if there exists a k such that $HL^k(G)$ has a connected subgraph G' wher...

conjecture | open | MathDB 页面 | 原文线索

V2-050464 | MATHDB-351236

Divergence from two non-isomorphic minimal convergent subgraphs

摘录 / EXCERPT: Non-isomorphic subgraph conjecture. If $G_1\not\cong G_2$ and $G_1,G_2\in\Lambda_n$, then G has a sequence that diverges by order.

conjecture | open | MathDB 页面 | 原文线索

V2-050465 | MATHDB-351237

Unicyclic-components conjecture for minimally n-convergent graphs

摘录 / EXCERPT: Unicyclic-components conjecture. If $G\in\lambda_n$, then G has unicyclic components.

conjecture | open | MathDB 页面 | 原文线索

V2-050466 | MATHDB-351238

Unique minimal convergent subgraph conjecture

摘录 / EXCERPT: Unique minimal subgraph conjecture. If $G\in\Lambda_n$ and G is not the disconnected union of two graphs in Λ_n, then there exists a unique graph $G'\in\lambda_n$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-050467 | MATHDB-351252

Alishahi-Hajiabolhassan conjecture on matching Kneser graph chromatic numbers

摘录 / EXCERPT: Let G be a connected graph and let $r\geq 2$ be a positive integer. The matching Kneser graph $\textsf{KG}(G,rK_2)$ has as vertices the r-matchings of G, with two vertices ad...

conjecture | open | MathDB 页面 | 原文线索

V2-050468 | MATHDB-351254

Weak virial positivity conjecture for random regular bipartite graphs

摘录 / EXCERPT: Let G be an r-regular bipartite graph with $v=2n$ vertices, let m_i denote the number of i-matchings, and define ... For the finite-difference operator defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-050469 | MATHDB-351278

Tong-Viet's regular character-graph conjecture

摘录 / EXCERPT: Let G be a finite group, and let $\Delta(G)$ be its character graph: its vertices are the primes dividing some irreducible character degree of G, with two distinct primes adjac...

conjecture | open | MathDB 页面 | 原文线索

V2-050470 | MATHDB-351288

Martin's finite CRG conjecture for edit distance functions

摘录 / EXCERPT: Let $d4d8$ be a non-trivial hereditary property, and let $d4d2(d4d8)$ be the set of CRGs avoiding its forbidden graphs. For each $p a[0,1]$, let $g_K(p)$ denote the quadratic minim...

conjecture | open | MathDB 页面 | 原文线索

V2-050471 | MATHDB-351289

The bipartite-forbidden-graph conjecture for accumulation points

摘录 / EXCERPT: For positive integers s and t, let $K_{s,t}$ be the complete bipartite graph with parts of sizes s and t, and let $\operatorname{Forb}(K_{s,t})$ denote the hereditary prope...

conjecture | open | MathDB 页面 | 原文线索

V2-050472 | MATHDB-351297

The fully cycle extendable conjecture for locally connected regular graphs with bounded second neighborhoods

摘录 / EXCERPT: Let G be a graph in $\mathcal{G}(k)$, where $\mathcal{G}(k)$ denotes the class of connected k-regular graphs whose vertices have at most k vertices at distance two, and suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-050473 | MATHDB-351300

A join upper-bound conjecture for the hyperopic cop number

摘录 / EXCERPT: Let G and J be connected graphs. Write $G\vee J$ for their graph join, $c_H(X)$ for the hyperopic cop number of a graph X, and let $\Upsilon(X)$ denote the maximum size of a...

conjecture | open | MathDB 页面 | 原文线索

V2-050474 | MATHDB-351310

Mohar's Kempe-equivalence conjecture for edge-colourings

摘录 / EXCERPT: Let G be a simple graph. Two k-edge-colourings are Kempe-equivalent if one can be reached from the other through a series of Kempe changes using colours from $\{1,\ldots,k\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-050475 | MATHDB-351316

Planarity conjecture for alternate-generator origami orbit graphs

摘录 / EXCERPT: Let n be a positive integer. Using the generating set $\{P,R\}$ for $\operatorname{SL}(2,\mathbb{Z})$, where ... let $\overline{\mathcal{G}}_{n}$, $\overline{\mathcal{G}}_{n}^{A}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-050476 | MATHDB-351328

Laplacian energy conjecture for trees

摘录 / EXCERPT: Laplacian energy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050477 | MATHDB-351329

Radenković-Gutman conjecture on the Laplacian energy of trees

摘录 / EXCERPT: Let G be a simple graph with n vertices, m edges, and Laplacian eigenvalues $\mu_1,\mu_2,\ldots,\mu_{n-1},\mu_n=0$. Its Laplacian energy is ... where $\overline{d}=\frac{2m}{n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-050478 | MATHDB-351335

Turán-density conjecture for partially directed k-graphs

摘录 / EXCERPT: Partially directed triangle conjecture. For all sufficiently large n, every n-vertex partially directed k-graph with $\alpha\binom{n}{k}$ undirected edges and...

conjecture | open | MathDB 页面 | 原文线索

V2-050479 | MATHDB-351336

Optimal construction conjecture for the partially directed k-triangle Turán problem

摘录 / EXCERPT: Optimal construction conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050480 | MATHDB-351380

The Turán number conjecture for the triangular pyramid of 3 layers

摘录 / EXCERPT: Let TP_3 denote the triangular pyramid of 3 layers, and let $\operatorname{ex}(n,TP_3)$ be the maximum number of edges in an n-vertex graph containing no copy of TP_3. Turá...

conjecture | open | MathDB 页面 | 原文线索

V2-050481 | MATHDB-351381

The asymptotic Turán number conjecture for the triangular pyramid of 4 layers

摘录 / EXCERPT: Let TP_4 denote the triangular pyramid of 4 layers, and let $\operatorname{ex}(n,TP_4)$ be the maximum number of edges in an n-vertex graph containing no copy of TP_4. Asym...

conjecture | open | MathDB 页面 | 原文线索

V2-050482 | MATHDB-351395

Tree equidistant dimension conjecture

摘录 / EXCERPT: Tree equidistant dimension conjecture.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-050483 | MATHDB-351415

The maximal Randić energy conjecture for connected graphs

摘录 / EXCERPT: Maximal Randić energy conjecture. The connected graph with maximal Randić energy is a tree.

conjecture | open | MathDB 页面 | 原文线索

V2-050484 | MATHDB-351416

The sun and balanced double sun conjecture for maximal Randić energy

摘录 / EXCERPT: Sun and balanced double sun conjecture. The connected graph of odd order $n\geq 1$ having maximal Randić energy is the sun. The connected graph of even order $n\geq 2$ having maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-050485 | MATHDB-351420

Bilu-Linial conjecture for arbitrary graphs

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta(G)>1$. An edge-signing of G is a map from $E(G)$ to $\{-1,1\}$; write G^σ for the resulting edge-signed graph and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050486 | MATHDB-351456

Existence of isolated n-PTMCs in Γn

摘录 / EXCERPT: Let Γ_n denote the graph considered in the paper, and let an isolated n-PTMC mean an isolated perfect total-multipacking code with parameter n. Existence conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-050487 | MATHDB-351478

The critical-configuration conjecture for crown-free linear 3-graphs

摘录 / EXCERPT: Critical-configuration conjecture. Every minimal counterexample to

conjecture | open | MathDB 页面 | 原文线索

V2-050488 | MATHDB-351481

Conjectural formulas for cluster variables of star graphs

摘录 / EXCERPT: Let S_n be the star graph on n vertices, with central vertex labeled by 1, and let ... For a vertex subset S, write Y_S for the associated cluster-algebra element and...

conjecture | open | MathDB 页面 | 原文线索

V2-050489 | MATHDB-351485

The distance-seven containment conjecture for the Pyro game

摘录 / EXCERPT: Consider the Pyro game on the infinite Cartesian grid, with the original burned vertex at $(0,0)$ and one firefighter protecting vertices over successive steps. Here, a vertex's di...

conjecture | open | MathDB 页面 | 原文线索

V2-050490 | MATHDB-351486

The one-firefighter containment conjecture for the strong grid

摘录 / EXCERPT: The strong grid is the graph obtained from the square grid by joining vertices that are at king-move distance one, so each vertex is adjacent also to its diagonal neighbours. In th...

conjecture | open | MathDB 页面 | 原文线索

V2-050491 | MATHDB-351491

Conjecture on extremal induced cycles in planar graphs

摘录 / EXCERPT: Let $k\geq 6$. Consider a k-cycle and choose $\left\lfloor\frac{k}{2}\right\rfloor$ pairwise non-adjacent vertices on it. Blowing up a vertex means replacing it by an independent...

conjecture | open | MathDB 页面 | 原文线索

V2-050492 | MATHDB-351500

Threshold Ramsey multiplicity conjecture for odd cycles

摘录 / EXCERPT: For a positive integer k, let C_k denote the cycle on k vertices, and let $m(C_k)$ be its threshold Ramsey multiplicity. Threshold Ramsey multiplicity conjecture. For any suf...

conjecture | open | MathDB 页面 | 原文线索

V2-050493 | MATHDB-351502

Ryser's conjecture in the remaining 2-packing cases

摘录 / EXCERPT: Ryser's conjecture in the remaining 2-packing cases. Every such system satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050494 | MATHDB-351504

The kernel existence conjecture for colored quasi-transitive digraphs

摘录 / EXCERPT: Kernel existence conjecture. For every $k\geq r+2$, the digraph D has a (k,H)-kernel.

conjecture | open | MathDB 页面 | 原文线索

V2-050495 | MATHDB-351520

Primitive-matrix generation conjecture for integral vectors

摘录 / EXCERPT: Let v be an n-dimensional integral vector whose entries are all nonzero modulo p. Suppose that ... Here e denotes the all-one vector, and a vector generates a primitive matri...

conjecture | open | MathDB 页面 | 原文线索

V2-050496 | MATHDB-351528

Hypergraph-to-Cayley chromatic conjecture over $\mathbb F_p^\omega$

摘录 / EXCERPT: Hypergraph-to-Cayley conjecture. If the hypergraph with edge set $\mathcal F$ has infinite chromatic number, then

conjecture | open | MathDB 页面 | 原文线索

V2-050497 | MATHDB-351540

Aharoni-Briggs-Kim-Kim conjecture for graphs of maximum degree two

摘录 / EXCERPT: Let G be a graph, let n and m be integers with $m\le n$, and let $f_G(n,m)$ denote the minimum number of independent n-sets whose collection has a rainbow independent m-s...

conjecture | open | MathDB 页面 | 原文线索

V2-050498 | MATHDB-351541

Aharoni-Briggs-Kim-Kim conjecture for cycles

摘录 / EXCERPT: Let C_t be a cycle with t vertices, and let $f_{C_t}(n,n)$ denote the minimum number of independent n-sets in C_t whose collection has a rainbow independent n-set. Aharon...

conjecture | open | MathDB 页面 | 原文线索

V2-050499 | MATHDB-351544

Pokrovskiy's strong tournament linking conjecture

摘录 / EXCERPT: A tournament is r-strong if deleting fewer than r vertices leaves a strongly connected tournament, and its minimum in-degree and minimum out-degree are the least in-degree and...

conjecture | open | MathDB 页面 | 原文线索

V2-050500 | MATHDB-351545

Alon and Wei's balanced degree-count conjecture for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph on n vertices. For a graph F and a nonnegative integer k, let $m(F,k)$ denote the number of vertices of degree k in F. Balanced degree-coun...

conjecture | open | MathDB 页面 | 原文线索

V2-050501 | MATHDB-351546

Alon and Wei's bounded repeated-degree conjecture

摘录 / EXCERPT: Let G be a graph on n vertices with minimum degree δ. For a graph F, let $m(F)$ denote the maximum number of vertices having the same degree in F. Bounded repeated-d...

conjecture | open | MathDB 页面 | 原文线索

V2-050502 | MATHDB-351575

Hriňáková-Knor-Škrekovski strong conjecture on variable Wiener and Szeged indices

摘录 / EXCERPT: For a connected graph G, let ... where $d(u,v)$ is the distance between vertices u and v, and $n_u(v)$ and $n_v(u)$ count vertices closer to u and v, respectively, among th...

conjecture | open | MathDB 页面 | 原文线索

V2-050503 | MATHDB-351595

Abrishami's common Laplacian eigenvalues conjecture for equivalent cographs

摘录 / EXCERPT: Abrishami's conjecture. G and H have at least

conjecture | open | MathDB 页面 | 原文线索

V2-050504 | MATHDB-351602

Realization conjecture for the reverse-braided module category of a vertex operator algebra

摘录 / EXCERPT: Let V be a rational, C_2-cofinite vertex operator algebra, and let $\coverline{\mathcal{C}}_V}$ denote its modular tensor category equipped with the reverse braiding. Realization...

conjecture | open | MathDB 页面 | 原文线索

V2-050505 | MATHDB-351620

Vizing's conjecture for subcubic planar graphs of girth 6

摘录 / EXCERPT: A subcubic planar graph is a planar graph in which every vertex has degree at most 3, and its girth is the length of its shortest cycle. Such a graph is class I if its edge-chrom...

conjecture | open | MathDB 页面 | 原文线索

V2-050506 | MATHDB-351622

Bound for maximal independent sets in triangle-free multipartite graphs

摘录 / EXCERPT: Let G be an n-vertex k-partite graph with vertex partition $V_1\cup\cdots\cup V_k$. Assume there are no indices a,b,c such that there is an edge with both endpoints in...

conjecture | open | MathDB 页面 | 原文线索

V2-050507 | MATHDB-351637

Harborth's edge-count conjecture for matchstick graphs

摘录 / EXCERPT: A matchstick graph is a graph drawn in the plane with each edge a straight-line segment of unit length, such that no two edges have a point in common unless that point is an endpoi...

conjecture | open | MathDB 页面 | 原文线索

V2-050508 | MATHDB-351640

The edge-count conjecture for random graphs embeddable in a surface

摘录 / EXCERPT: Let S be a surface, let $n\in\mathbb{N}$, and let R^S_n be chosen uniformly at random from the graphs on n vertices embeddable in S. Write $e(G)$ for the number of edges of...

conjecture | open | MathDB 页面 | 原文线索

V2-050509 | MATHDB-351649

Gao et al.'s conjecture on rainbow matchings in multigraphs

摘录 / EXCERPT: Let G be an edge-coloured multigraph with n colours such that each colour class is a matching of size $n+2$. A rainbow matching is a matching containing at most one edge of eac...

conjecture | open | MathDB 页面 | 原文线索

V2-050510 | MATHDB-351650

Grinblat's conjecture on rainbow matchings in (n,v)-multigraphs

摘录 / EXCERPT: An (n,v)-multigraph is an n-edge-coloured multigraph in which the edges of each colour span a disjoint union of non-trivial cliques whose total number of vertices is at least...

conjecture | open | MathDB 页面 | 原文线索

V2-050511 | MATHDB-351662

Thomassen's crumby coloring conjecture for 3-connected cubic graphs

摘录 / EXCERPT: A crumby coloring of a graph is a red-blue vertex coloring in which the blue subgraph has maximum degree at most 1 and the red subgraph has minimum degree at least 1 and contai...

conjecture | open | MathDB 页面 | 原文线索

V2-050512 | MATHDB-351663

The crumby coloring conjecture for subcubic bipartite graphs

摘录 / EXCERPT: A crumby coloring is a red-blue vertex coloring in which the blue subgraph has maximum degree at most 1 and the red subgraph has minimum degree at least 1 and contains no path...

conjecture | open | MathDB 页面 | 原文线索

V2-050513 | MATHDB-351674

The conjecture that no graph has integer-valued Sombor energy

摘录 / EXCERPT: Let G be a graph, and let $En_{SO}(G)$ denote its Sombor energy, the sum of the absolute values of the eigenvalues of its Sombor matrix. Sombor energy integrality conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-050514 | MATHDB-351696

The leafless-graph upper-bound conjecture for edge metric dimension

摘录 / EXCERPT: Leafless-graph upper-bound conjecture. Every leafless graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050515 | MATHDB-351715

The codegree-squared extremal conjecture for the Fano plane

摘录 / EXCERPT: Let $\mathbb{F}$ be the Fano plane, and let B_n be the complete balanced bipartite 3-graph on n vertices. For a 3-graph H, write $\operatorname{co}_2(H)$ for its codegree...

conjecture | open | MathDB 页面 | 原文线索

V2-050516 | MATHDB-351716

The iterated-blow-up conjecture for $\{K_4^{3-},C_5\}$

摘录 / EXCERPT: Let G be the iterated blow-up of an edge, obtained from a complete balanced 3-partite 3-graph by inserting a complete balanced 3-partite 3-graph recursively in each part....

conjecture | open | MathDB 页面 | 原文线索

V2-050517 | MATHDB-351717

The complement-of-Fano blow-up conjecture for J_4 in codegree squared density

摘录 / EXCERPT: Let G be the iterated blow-up of the complement of the Fano plane. It is J_4-free and has codegree squared density at least 0.28. The complement-of-Fano blow-up conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050518 | MATHDB-351718

The lower-bound sharpness conjecture for $K_5^<$

摘录 / EXCERPT: The lower-bound sharpness conjecture for $K_5^<$. All four lower bounds are sharp:

conjecture | open | MathDB 页面 | 原文线索

V2-050519 | MATHDB-351719

The codegree-squared density conjecture for K_5^4

摘录 / EXCERPT: The codegree-squared density conjecture for K_5^4. The lower bound is tight:

conjecture | open | MathDB 页面 | 原文线索

V2-050520 | MATHDB-351728

The characterization conjecture for F-factors in k-uniform hypergraphs

摘录 / EXCERPT: The characterization conjecture. The k-graph F belongs to $\textbf{FACTOR}_k$ if and only if it satisfies:

conjecture | open | MathDB 页面 | 原文线索

V2-050521 | MATHDB-351745

The bar visibility number conjecture for the transitive tournament on 11 vertices

摘录 / EXCERPT: Let T_n denote the transitive tournament on n vertices, and let $b(G)$ be the bar visibility number of a digraph G. The conjecture. ... This was one of two conjectures posed by...

conjecture | open | MathDB 页面 | 原文线索

V2-050522 | MATHDB-351761

Thomassen's coloring-growth conjecture for triangle-free planar graphs

摘录 / EXCERPT: Let G be an n-vertex triangle-free planar graph. A proper 3-coloring of G is a vertex coloring with three colors in which adjacent vertices receive different colors. Thomasse...

conjecture | open | MathDB 页面 | 原文线索

V2-050523 | MATHDB-351763

The $T_{16}(n)$ conjecture for $(5;4)$- and $(5;5)$-extremal graphs

摘录 / EXCERPT: Let $T_{16}(n)$ be the balanced complete 16-partite graph on n vertices. For a parameter tuple $(k;s)$, an $(k;s)$-extremal graph is an n-vertex graph attaining the maximum i...

conjecture | open | MathDB 页面 | 原文线索

V2-050524 | MATHDB-351778

The second-eigenvalue conjecture for Cayley graphs of the symmetric group

摘录 / EXCERPT: Let $\Gamma_{n,k}$ be the Cayley graph under consideration on the symmetric group, where $n\in\mathbb{N}$ and $2\leq k\leq n-2$. Let $[n-1,1]$ denote the corresponding partition of...

conjecture | open | MathDB 页面 | 原文线索

V2-050525 | MATHDB-351804

The extremal A_α spectral radius conjecture for non-strongly connected digraphs

摘录 / EXCERPT: Let $\mathcal{G}_n^m$ be the class of digraphs under consideration, let $G”$ denote the specific extremal digraph constructed in the preceding results, and let A_α be the...

conjecture | open | MathDB 页面 | 原文线索

V2-050526 | MATHDB-351811

Conjectured minimum degree threshold for Hamilton ell-cycles

摘录 / EXCERPT: For $k\ge3$, $1\le\ell<k/2$, and $k-\ell\le d\le k-1$, let $h_d^\ell(k)$ be the limiting normalized minimum d-degree threshold for Hamilton ℓ-cycles. Let $t(k,d,\ell)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-050527 | MATHDB-351812

Hypergraph independence number conjecture for inhomogeneous random graphs

摘录 / EXCERPT: Let $\boldsymbol{G}=\boldsymbol{G}(n)$ be a random graph on vertex set $[n]$, with edges ij appearing independently with probabilities $p_{ij}=p_{ij}(n)\in(0,1)$. Suppose there a...

conjecture | open | MathDB 页面 | 原文线索

V2-050528 | MATHDB-351817

The cycle-vertex conjecture for the Loosened Collatz Graph

摘录 / EXCERPT: Let $\mathbb{N}_1$ denote the positive integers, and let the Loosened Collatz Graph be the graph whose vertices and cycles are defined by the Loosened Collatz Function. Cycle-verte...

conjecture | open | MathDB 页面 | 原文线索

V2-050529 | MATHDB-351819

The zero-axis conjecture for dual objects of Loosened Collatz cycles

摘录 / EXCERPT: Let a satisfying cycle determine its dual object in Euclidean space by interpreting its associated tuple and its bitwise rotations as vertices in $\mathbb{R}^n$. A vertex is said t...

conjecture | open | MathDB 页面 | 原文线索

V2-050530 | MATHDB-351844

Edge-label ratio conjecture for triangular grids

摘录 / EXCERPT: Edge-label ratio conjecture. For the $T_1(k,c)$ grid,

conjecture | open | MathDB 页面 | 原文线索

V2-050531 | MATHDB-351851

Conjecture on rank-one solutions for edge sums in the max-cut SDP

摘录 / EXCERPT: Let G_1 and G_2 be two graphs with vertex sets ... Let (X_1,S_1) and (X_2,S_2) be the primal-dual solution pairs to the max-cut SDPs on G_1 and G_2, respectively. Let G...

conjecture | open | MathDB 页面 | 原文线索

V2-050532 | MATHDB-351856

Maximum 3-separator conjecture for planar quadrangulations

摘录 / EXCERPT: Let $n\geq 14$. A 3-connected planar quadrangulation is a 3-connected planar graph in which every face is bounded by a cycle of length four. Let $\mathcal{G}^\ast$ be the class con...

conjecture | open | MathDB 页面 | 原文线索

V2-050533 | MATHDB-351859

Girth-at-least-six formulation of Funk et al.'s classification conjecture

摘录 / EXCERPT: Girth-at-least-six formulation of Funk et al.'s conjecture. Every bipartite cyclically 4-edge-connected cubic graph with the PMH-property and girth at least 6, except the Heawood...

conjecture | open | MathDB 页面 | 原文线索

V2-050534 | MATHDB-351862

Gregory-Hershkowitz-Kirkland maximum-spread conjecture for graphs

摘录 / EXCERPT: Let G be a simple undirected graph of order n, with adjacency eigenvalues $\lambda_1(G)\geq\lambda_2(G)\geq\cdots\geq\lambda_n(G)$ and spread $s(G):=\lambda_1(G)-\lambda_n(G)$....

conjecture | open | MathDB 页面 | 原文线索

V2-050535 | MATHDB-351885

Koolen-Gebremichel conjecture on primitive strongly regular graphs with smallest eigenvalue -3

摘录 / EXCERPT: Koolen-Gebremichel conjecture. Either $\mu\in\{6,9\}$ or $n\leqslant 276$.

conjecture | open | MathDB 页面 | 原文线索

V2-050536 | MATHDB-351887

The edge-deletion conjecture for graphs of odd chromatic index four

摘录 / EXCERPT: Edge-deletion conjecture. If $4_o'(G)=4$, then there is an edge e such that $G-e$ admits an odd 3-edge-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-050537 | MATHDB-351888

The three-color conjecture for S-edge-colorings

摘录 / EXCERPT: Three-color conjecture for $\mathcal{S}$-edge-colorings. If G is a connected loopless graph that is not a Shannon triangle of type $(2,2,1)$ with multiplicity at least 3, then

conjecture | open | MathDB 页面 | 原文线索

V2-050538 | MATHDB-351890

Szabó-Zumstein-Zürcher's conjecture on Ramsey simplicity of bipartite graphs

摘录 / EXCERPT: Let H be a bipartite graph. A graph is 2-Ramsey simple when its minimum-degree parameter satisfies $s_2(H)=2(\delta(H)-1)+1$. Szabó-Zumstein-Zürcher's conjecture. Every biparti...

conjecture | open | MathDB 页面 | 原文线索

V2-050539 | MATHDB-351897

Spectral odd-cycle containment conjecture for the graph construction R_k

摘录 / EXCERPT: Spectral odd-cycle containment conjecture. Under this condition, G contains at least one cycle from

conjecture | open | MathDB 页面 | 原文线索

V2-050540 | MATHDB-351948

The minimal Dunce Hat graph conjecture

摘录 / EXCERPT: Let G be a graph, and let $\Delta(G)$ denote its clique complex. Say that G is strong $\mathfrak{I}$-contractible when it can be reduced by the permitted strong $\mathfrak{I}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-050541 | MATHDB-351949

The minimal edge-deletion example for strong I-contractibility

摘录 / EXCERPT: Let G be a graph, let $N(\{v,w\})$ denote the common neighborhood of an edge $\{v,w\}$, and let $\mathfrak{I}_S$ denote the class of $\mathfrak{I}_S$-contractible graphs. The min...

conjecture | open | MathDB 页面 | 原文线索

V2-050542 | MATHDB-351977

Extremal complete-bipartite multiplication-table conjecture

摘录 / EXCERPT: Extremal complete-bipartite multiplication-table conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-050543 | MATHDB-351979

Two-size undecidability conjecture for partially fixed-size network coding

摘录 / EXCERPT: A partially fixed-size network has vertex set V, edge set E, source-message sets $\\{A_v\\}$, demanded-message sets $\\{B_v\\}$, and size specifications s_M and s_E. The sp...

conjecture | open | MathDB 页面 | 原文线索

V2-050544 | MATHDB-351980

Undecidability of the asymptotic capacity threshold for partially fixed-size networks

摘录 / EXCERPT: Asymptotic-capacity undecidability conjecture. The following problem is undecidable: given a partially fixed-size network $(V,E,\{A_v\},\{B_v\},s_M,s_E)$, decide whether its asympt...

conjecture | open | MathDB 页面 | 原文线索

V2-050545 | MATHDB-351981

Coset conjecture for symplectic bosons and a subregular W-algebra

摘录 / EXCERPT: Coset conjecture. For all $n\geq 2$, there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-050546 | MATHDB-351987

Near-goodness conjecture for multipartite Ramsey numbers and books

摘录 / EXCERPT: Let $k,p,t\geq 2$ be integers. For some $\delta>0$, let $n\geq 1$, and let $a_1\leq\dotsb\leq a_{p-1}\leq t$ be positive integers with $a_p\leq\delta n$. For a graph H, define...

conjecture | open | MathDB 页面 | 原文线索

V2-050547 | MATHDB-351988

Sidorenko order reverses the average hitting-time index for trees

摘录 / EXCERPT: Let T and T' be trees with the same number of vertices, and let $T\succcurlyeq T'$ mean that $\operatorname{hom}(T,G)\geq \operatorname{hom}(T',G)$ for every simple graph G,...

conjecture | open | MathDB 页面 | 原文线索

V2-050548 | MATHDB-352001

Endpoint minimum conjecture for cosine sequences of distance-regular graphs

摘录 / EXCERPT: Let G be a distance-regular graph of diameter d with eigenvalues $\theta_0>\theta_1>\cdots>\theta_d$. For the cosine sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-050549 | MATHDB-352007

Saturation conjecture for the convex geometric hypergraph $M_1^{(r)}$

摘录 / EXCERPT: Let $r\ge 3$ and let $H_n^{(r)}$ be the convex geometric hypergraph defined in the source. Write $\operatorname{sat}_\circlearrowright(n,M_1^{(r)})$ for the minimum number of edges...

conjecture | open | MathDB 页面 | 原文线索

V2-050550 | MATHDB-352008

Asymptotic saturation conjecture for the convex geometric hypergraph S_3

摘录 / EXCERPT: Let S_3 be the two-edge convex geometric 3-uniform hypergraph denoted by S_3 in the source, and let $\operatorname{sat}_\circlearrowright(n,S_3)$ denote its saturation number...

conjecture | open | MathDB 页面 | 原文线索

V2-050551 | MATHDB-352009

Quadratic saturation conjecture for the convex geometric matching M_3

摘录 / EXCERPT: Let M_3 be the convex geometric matching denoted by M_3 in the source, and let $\operatorname{sat}_\circlearrowright(n,M_3)$ denote its saturation number on n vertices. Quadr...

conjecture | open | MathDB 页面 | 原文线索

V2-050552 | MATHDB-352031

Hurajová-Coroničová-Madaras diameter conjecture for betweenness-uniform graphs

摘录 / EXCERPT: Hurajová-Coroničová-Madaras conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-050553 | MATHDB-352059

Component structure and vertex cardinalities of the digraph ${\cal D}^{*}_m$

摘录 / EXCERPT: Let $m\geq 2$. The digraph ${\cal D}^{*}_m$ has components ${\cal A}^{*}_m$ and ${\cal B}^{*}_m$, where ${\cal B}^{*}_m$ consists of components...

conjecture | open | MathDB 页面 | 原文线索

V2-050554 | MATHDB-352060

Vertex cardinalities of the reduced digraphs ${\cal R}^{*}_m$ and ${\cal R}^{**}_m$

摘录 / EXCERPT: Let m be an integer, and let ${\cal R}^{*}_m$ and ${\cal R}^{**}_m$ be the reduced digraphs associated with the rectangular-grid construction. Vertex-cardinality conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-050555 | MATHDB-352061

Perron eigenvalue conjecture for the component ${\cal A}^{*}_m$

摘录 / EXCERPT: Let $m\geq 2$, let ${\cal D}^{*}_m$ be the relevant digraph, and let ${\cal A}^{*}_m$ be its distinguished component. The maximum eigenvalue of a digraph means the largest eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-050556 | MATHDB-352071

Finite-convergence conjecture for the copositive stability-number hierarchy

摘录 / EXCERPT: Let G be a graph, let M_G be its graph matrix, let $F_G(x)=(x^{\circ 2})^\top Gx^{\circ 2}$, and let ... Define $\vartheta^{(r)}(G)=\min\{t:t(A_G+I)-J\in\mathcal K_n^{(r)}\}$. The fi...

conjecture | open | MathDB 页面 | 原文线索

V2-050557 | MATHDB-352072

Gvozdenović-Laurent bound for the copositive stability-number hierarchy

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $\vartheta^{(r)}(G)$ be its order-r copositive hierarchy bound, and let $S^\perp$ denote the vertices adjacent to or equal to a stable set S. For...

conjecture | open | MathDB 页面 | 原文线索

V2-050558 | MATHDB-352082

Vlasev-Yeats conjecture on free variables in quadratic spanning forest identities

摘录 / EXCERPT: Let m be the number of marked vertices in a graph. A quadratic spanning forest identity of the indicated type has coefficients as free variables, where the left-hand side consist...

conjecture | open | MathDB 页面 | 原文线索

V2-050559 | MATHDB-352088

Czabarka-Dankelmann-Székely conjecture on diameters of clique-free graphs

摘录 / EXCERPT: Czabarka-Dankelmann-Székely conjecture. For every such G,

conjecture | refuted | MathDB 页面 | 原文线索

V2-050560 | MATHDB-352121

Spider leaf-deletion conjecture for domination-polynomial modes

摘录 / EXCERPT: Let S be a spider, meaning a tree with one vertex of degree greater than 2, whose legs all have length at most 3. Let S' be obtained from S by deleting a leaf, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050561 | MATHDB-352122

The subgraph core conjecture for connected graphs

摘录 / EXCERPT: Subgraph core conjecture. The subgraph core of G is contained in a block of G.

conjecture | open | MathDB 页面 | 原文线索

V2-050562 | MATHDB-352130

The asymptotic B_k-free hypergraph bound

摘录 / EXCERPT: The B_k-free hypergraph conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-050563 | MATHDB-352134

Ghosh-Győri-Martin-Paulos-Xiao conjecture on planar Turán numbers of cycles

摘录 / EXCERPT: Let C_{ℓ} denote the cycle of length ℓ, and let $\operatorname{ex}_{\mathcal{P}}(n,C_{\ell})$ be the maximum number of edges in an n-vertex planar graph containing no c...

conjecture | open | MathDB 页面 | 原文线索

V2-050564 | MATHDB-352135

Extremality conjecture for substituted planar triangulations

摘录 / EXCERPT: Let $\ell\ge 7$. Let G be a dense graph of girth $\ell+1$, and let B be an n-vertex planar triangulation with no ℓ-cycle, chosen to maximize n. Form G' by substituti...

conjecture | open | MathDB 页面 | 原文线索

V2-050565 | MATHDB-352136

Logarithmic-exponent upper-bound conjecture for planar Turán numbers of cycles

摘录 / EXCERPT: For a positive integer ℓ, let C_{ℓ} be the cycle of length ℓ, and let $\operatorname{ex}_{\mathcal{P}}(n,C_{\ell})$ denote the maximum number of edges in an n-vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050566 | MATHDB-352180

Asymptotic prevalence of constructible cospectral cubic graphs

摘录 / EXCERPT: Let C_N be the set of all cubic graphs of order N that are cospectral with at least one other cubic graph, and let $C^*_N\subseteq C_N$ consist of the graphs produced by one of...

conjecture | open | MathDB 页面 | 原文线索

V2-050567 | MATHDB-352188

Principal representation conjecture for normal homogeneous subrings of graphs

摘录 / EXCERPT: Let G be an unmixed graph, let S be its homogeneous subring, and let B be the configuration defining S. Write $(\mathbb{R}_+B)^\circ$ for the relative interior of the cone...

conjecture | open | MathDB 页面 | 原文线索

V2-050568 | MATHDB-352189

Gorenstein characterization conjecture for normal homogeneous subrings of graphs

摘录 / EXCERPT: Let G be a graph with homogeneous subring S, and assume that S is normal. Say that G is unmixed when all its relevant minimal vertex covers have the same cardinality, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-050569 | MATHDB-352190

The positive hypergraph stable-involution conjecture

摘录 / EXCERPT: Let H be an r-uniform hypergraph, or r-graph. Call H positive when its corresponding hypergraph homomorphism-density functional is nonnegative for every bounded measurable...

conjecture | open | MathDB 页面 | 原文线索

V2-050570 | MATHDB-352192

Subquadratic upper bound conjecture for the longest solvable board

摘录 / EXCERPT: Let $n\geq 3$ be odd, and let $ML(n)$ denote the maximal length of a solvable board of size n. The paper establishes the lower bound $ML(n)\geq 3n$ for all $n\geq 11$ and the tri...

conjecture | open | MathDB 页面 | 原文线索

V2-050571 | MATHDB-352193

Symmetry classification conjecture for board isomorphisms

摘录 / EXCERPT: Let $\mathcal{D}$ be the set of eight arrow directions, let $\operatorname{Mat}_{n\times n}(\mathcal{D})$ be the set of boards, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050572 | MATHDB-352229

The generalized outerplanar Turán conjecture for P_6

摘录 / EXCERPT: Let $f_{\mathcal{OP}}(n,P_6)$ denote the generalized outerplanar Turán number of the path P_6 on 6 vertices, namely the maximum number of copies of P_6 in an outerplanar grap...

conjecture | open | MathDB 页面 | 原文线索

V2-050573 | MATHDB-352243

Savery's extremal conjecture for induced cycles in planar graphs

摘录 / EXCERPT: Let $f_I(n,C_k)$ denote the maximum number of induced copies of the cycle C_k in an n-vertex planar graph. Let $F_{n,k}$ be the planar graph obtained by blowing up three pairwi...

conjecture | open | MathDB 页面 | 原文线索

V2-050574 | MATHDB-352251

Bermudo's differential bound for graphs with forbidden induced cycles

摘录 / EXCERPT: Let G be a graph of order $n\ge 6k+9$ with minimum degree $\delta\ge 2$, and suppose that G contains no induced cycles of lengths in $\{5,8,\ldots,3k+2\}$. The differential of...

conjecture | open | MathDB 页面 | 原文线索

V2-050575 | MATHDB-352253

Subgraph criterion for minimal doubled 3-regular graphs

摘录 / EXCERPT: Minimality criterion conjecture. The graph G' is a minimal graph if and only if it does not contain such a chain of 3-cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-050576 | MATHDB-352257

Large-deviation conjecture for the typical Poisson zero-cell vertex number

摘录 / EXCERPT: Let $\mathcal Z_d^{\text{typ}}$ denote the typical Poisson zero cell, and let $f_0(\mathcal Z_d^{\text{typ}})$ be its number of vertices. Large-deviation conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-050577 | MATHDB-352259

The NFM positive-semidefinite subgraph conjecture under Dirichlet noise

摘录 / EXCERPT: NFM positive-semidefinite subgraph conjecture. There exists a scalar $t(k,\bm{\alpha})\in(0.5,1/\sqrt{2})$ such that, for every $j\in[k]$, $G[V_j']$ satisfies the hypothesis of The...

conjecture | open | MathDB 页面 | 原文线索

V2-050578 | MATHDB-352260

The NFM separated-eigenvector conjecture under Dirichlet noise

摘录 / EXCERPT: NFM separated-eigenvector conjecture. There exists a scalar $t(k,\bm{\alpha})\in(0.5,1/\sqrt{2})$ such that, for every $j\in[k]$, with probability not converging to 0 as...

conjecture | open | MathDB 页面 | 原文线索

V2-050579 | MATHDB-352266

Exponential graph-size conjecture for alternating-group quotients

摘录 / EXCERPT: Exponential graph-size conjecture. There exists a constant $C>1$ such that, if $k\geq3$ and $\operatorname{Aut}(X)$ has an epimorphism onto A_k, then

conjecture | open | MathDB 页面 | 原文线索

V2-050580 | MATHDB-352285

The lower-bound conjecture for secure domination in subdivisions

摘录 / EXCERPT: Secure-domination lower-bound conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-050581 | MATHDB-352293

Mean normalized delay profile convergence for Oldest-Useful and Random-Useful

摘录 / EXCERPT: Delay-profile convergence conjecture. For $D=\mathrm{OU}$ or $D=\mathrm{RU}$, and for a given load $\lambda<1$, as $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-050582 | MATHDB-352294

Poisson and delay-profile conjecture for the large reshuffling system

摘录 / EXCERPT: Large reshuffling-system conjecture. As $N\to\infty$: (i) the number of useful packets on a link at a communication epoch converges in distribution to...

conjecture | open | MathDB 页面 | 原文线索

V2-050583 | MATHDB-352295

Hydrodynamic-limit conjecture for the reshuffling stage profile

摘录 / EXCERPT: Reshuffling hydrodynamic-limit conjecture. Under these assumptions, the process $\phi^N_{\mathrm{RS}}(\cdot,\cdot)$ converges, in an appropriate sense, to the hydrodynamic model...

conjecture | open | MathDB 页面 | 原文线索

V2-050584 | MATHDB-352298

Separation conjecture for adjacent spine vertices

摘录 / EXCERPT: Let G be an augmented caterpillar drawn as a great-circle thrackle, and let v and v' be adjacent vertices in the spine of G. A separating 2-path at a vertex is a 2-path...

conjecture | open | MathDB 页面 | 原文线索

V2-050585 | MATHDB-352300

The γ-θ conjecture for eternal domination and clique covering

摘录 / EXCERPT: γ-θ conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-050586 | MATHDB-352303

The extremal biharmonic index conjecture for trees

摘录 / EXCERPT: Let T_n be a tree on $n\geq 5$ vertices. Write $BH(G)$ for the biharmonic index of a graph G, and let $K_{1,n-1}$ and P_n denote the star and path on n vertices, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-050587 | MATHDB-352312

Exact planar Turán number conjecture for $S_{2,4}$

摘录 / EXCERPT: Let $S_{2,4}$ be the double star with two centers having 2 and 4 leaves, and let $\operatorname{ex}_{\mathcal{P}}(n,S_{2,4})$ denote the maximum number of edges in an $S_{2,4}$...

conjecture | open | MathDB 页面 | 原文线索

V2-050588 | MATHDB-352313

Exact planar Turán number conjecture for $S_{3,3}$

摘录 / EXCERPT: Let $S_{3,3}$ be the double star with two centers having three leaves each, and let $\operatorname{ex}_{\mathcal{P}}(n,S_{3,3})$ denote the maximum number of edges in an $S_{3,3}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-050589 | MATHDB-352314

Asymptotic planar Turán number conjecture for $S_{3,5}$

摘录 / EXCERPT: Let $S_{3,5}$ be the double star with two centers having three and five leaves, and let $\operatorname{ex}_{\mathcal{P}}(n,S_{3,5})$ denote the maximum number of edges in an...

conjecture | open | MathDB 页面 | 原文线索

V2-050590 | MATHDB-352315

The Burning Graph Conjecture for connected graphs

摘录 / EXCERPT: Burning Graph Conjecture. Every connected graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050591 | MATHDB-352316

The Burning Graph Conjecture for trees

摘录 / EXCERPT: Burning Graph Conjecture. Every tree T on n vertices is $\{0,\ldots,\lceil\sqrt{n}\rceil-1\}$-burnable.

conjecture | open | MathDB 页面 | 原文线索

V2-050592 | MATHDB-352326

Balogh-Morris-Samotij-Warnke threshold conjecture for strictly 2-balanced edge-critical graphs

摘录 / EXCERPT: A graph H is strictly 2-balanced if, with ... we have $m_2(K)<m_2(H)$ for every proper subgraph $K\subsetneq H$. Let $\mathcal{F}_{n,m}(H)$ be the family of H-free graphs with...

conjecture | open | MathDB 页面 | 原文线索

V2-050593 | MATHDB-352331

ZFC realization conjecture for Ext-graphs of abelian groups

摘录 / EXCERPT: Let (S,R) be a graph, meaning that $R\subseteq S\times S$. An Ext-graph of a family of abelian groups $(\mathbb G_s:s\in S)$ is the graph on vertex set S whose relation is dete...

conjecture | open | MathDB 页面 | 原文线索

V2-050594 | MATHDB-352334

Rainbow Caccetta-Häggkvist conjecture for edge-set families

摘录 / EXCERPT: Let G be an undirected graph, and let $F_1,F_2,\ldots,F_n$ be sets of edges in G. For a family of edge sets, let $rg(F_1,\ldots,F_n)$ denote the length of its shortest rainbow...

conjecture | open | MathDB 页面 | 原文线索

V2-050595 | MATHDB-352343

Distance split tree as the tree of blobs of an equivalent network

摘录 / EXCERPT: Let N be a metric semidirected network on taxon set X, let d_N denote its average distances on X, and let T be the distance split tree reconstructed from d_N. Distance-...

conjecture | open | MathDB 页面 | 原文线索

V2-050596 | MATHDB-352345

Minimality conjecture for initial threshold graphs generated by RFI

摘录 / EXCERPT: Let $N>0$ and $r\geq 2$ be fixed, and let $I=(0,N]$. Let $T_G(a_1,\ldots,a_r)$ be the initial threshold graph obtained from RFI(N,r). Minimality conjecture. The graph G is mini...

conjecture | open | MathDB 页面 | 原文线索

V2-050597 | MATHDB-352349

The positive matching decomposition formula for hypercubes

摘录 / EXCERPT: For $n\geq1$, let Q_n be the n-dimensional hypercube and let $\mathrm{pmd}(Q_n)$ denote its positive matching decomposition number. Hypercube formula. ... for all $n\geq1$. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-050598 | MATHDB-352370

Uniform bound conjecture for balanced spanning tree factorizations

摘录 / EXCERPT: Let k be a positive integer, and let G be a k-multiple tree, meaning that the edge set of G is the disjoint union of the edge sets of spanning trees $(T_1,\ldots,T_k)$. Suc...

conjecture | open | MathDB 页面 | 原文线索

V2-050599 | MATHDB-352374

Time for IDLA to fill a transitive graph

摘录 / EXCERPT: Let V be a transitive graph with one vertex designated as a sink. Time for IDLA to fill a transitive graph. The IDLA filling time satisfies ... in probability as $\#V \to \infty$....

conjecture | open | MathDB 页面 | 原文线索

V2-050600 | MATHDB-352382

The off-diagonal book Ramsey random-bound conjecture for two books

摘录 / EXCERPT: Let $B_m^{(2)}$ denote the book graph consisting of m triangles sharing a common edge, and let $r(B_{cn}^{(2)},B_n^{(2)})$ be the smallest integer N such that every red-blue co...

conjecture | open | MathDB 页面 | 原文线索

V2-050601 | MATHDB-352383

Rainbow extension of the main bandwidth theorem

摘录 / EXCERPT: Rainbow bandwidth conjecture. There are $\nu,\beta,z>0$ and $n_0\in\mathbb{N}$ such that, whenever G additionally has an edge-colouring in which every colour appears on at most...

conjecture | open | MathDB 页面 | 原文线索

V2-050602 | MATHDB-352390

The star conjecture for big vertices in minimal-ABC trees

摘录 / EXCERPT: Star conjecture. The subgraph of a minimal-ABC tree induced by its big vertices is a star.

conjecture | open | MathDB 页面 | 原文线索

V2-050603 | MATHDB-352391

The branch-type conjecture for sufficiently large minimal-ABC trees

摘录 / EXCERPT: Branch-type conjecture. For sufficiently large n, minimal-ABC trees have only B_3-branches besides the big vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-050604 | MATHDB-352402

Source-vertex broken hexagon conjecture for modified crystal operator components

摘录 / EXCERPT: Let C be a connected component of the graph given by the modified operators ϕ_i, and call a hexagon broken by ϕ_i after ϕ_j when it has the broken-hexagon property...

conjecture | open | MathDB 页面 | 原文线索

V2-050605 | MATHDB-352405

The tight Hamiltonian cycle counting conjecture for Dirac hypergraphs

摘录 / EXCERPT: Let H be a $b4$-Dirac k-graph on n vertices, meaning that every $(k-1)$-set of vertices has degree at least $b4 n$. A tight Hamiltonian cycle is a cyclic ordering of the vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050606 | MATHDB-352408

The cluster-concentrated eigenvector basis conjecture for graph Paley-Wiener spaces

摘录 / EXCERPT: Cluster-concentrated basis conjecture. There are dim_K-1 eigenvalues of PQ larger than $1/2$, including $\operatorname{dim}_K-(K+1)$ eigenvalues equal to one and...

conjecture | open | MathDB 页面 | 原文线索

V2-050607 | MATHDB-352413

Minimal-clique-cover conjecture for realizable graphs

摘录 / EXCERPT: Let H be a realizable graph. A clique cover of H is a collection of cliques whose union covers all edges of H, and let n be the smallest number of cliques in such a cover....

conjecture | open | MathDB 页面 | 原文线索

V2-050608 | MATHDB-352439

The three-color conjecture for weak conflict-free vertex-to-vertex guarding

摘录 / EXCERPT: Three-color guarding conjecture. Every simple polygon admits a weak conflict-free vertex-to-vertex chromatic guarding with at most 3 colors.

conjecture | open | MathDB 页面 | 原文线索

V2-050609 | MATHDB-352443

Beaton-Brown conjecture on the modes of domination polynomials of trees

摘录 / EXCERPT: Let T be a tree and let $v\in V(T)$ be a leaf. Let $D(T)$ and $D(T\backslash v)$ denote the domination polynomials of T and the tree obtained by deleting v, respectively. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-050610 | MATHDB-352491

An unknown-order-property characterization of quadratic atomicity and binary error

摘录 / EXCERPT: Let XOP_2 be a prospective, as-yet-undefined generalization of the order property, and write NXOP_2 for its negation. Let a property be quadratically atomic i...

conjecture | open | MathDB 页面 | 原文线索

V2-050611 | MATHDB-352509

Conjecture on the polygon protocol's optimal verification value

摘录 / EXCERPT: Let $M\geq3$ be an integer and let $0\leq C\leq1$. For the polygon protocol, $g(C,M)$ denotes its optimal verification value. Polygon protocol conjecture. ... Numerical calculation i...

conjecture | open | MathDB 页面 | 原文线索

V2-050612 | MATHDB-352510

Conjecture for the dodecahedron protocol

摘录 / EXCERPT: Let μ be the dodecahedron protocol, and let $g(C,\mu)$ and $\gamma_2(C,\mu)$ denote its verification quantities. An intelligent direction is a direction specified by the vector...

conjecture | open | MathDB 页面 | 原文线索

V2-050613 | MATHDB-352530

Bollobás-Nikiforov signless Laplacian conjecture for clique-free graphs

摘录 / EXCERPT: Let G be a graph containing no copy of K_{r+1}, and let m be its number of edges. Write $q(G)$ for the signless Laplacian spectral radius. Bollobás-Nikiforov conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-050614 | MATHDB-352531

Generalized-book spectral extremal conjecture

摘录 / EXCERPT: Let $B_{r,k}=K_r\vee I_k$ be the generalized book formed by joining every vertex of a clique K_r to every vertex of an independent set I_k. Let G be a graph with m edges, a...

conjecture | open | MathDB 页面 | 原文线索

V2-050615 | MATHDB-352532

Spectral Erdős-Stone-Simonovits conjecture for forbidden graphs

摘录 / EXCERPT: Let F be a graph, let $\chi(F)$ be its chromatic number, and let G be an F-free graph with m edges. Let $\lambda(G)$ denote the adjacency spectral radius. Spectral Erdős-St...

conjecture | open | MathDB 页面 | 原文线索

V2-050616 | MATHDB-352533

Spectral cycle-length constant conjecture

摘录 / EXCERPT: Spectral cycle-length conjecture. The largest such constant C is characterized by the property that every such graph contains a cycle of every length $t\le (C-\varepsilon)n$. The...

open_problem | open | MathDB 页面 | 原文线索

V2-050617 | MATHDB-352534

The C_8 extremal-number conjecture

摘录 / EXCERPT: Let C_8 be the cycle on eight vertices, and let $\operatorname{ex}(n,H)$ denote the maximum number of edges in an H-free graph on n vertices. C_8 extremal-number conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050618 | MATHDB-352535

Chen-Liu-Zhang signless Laplacian conjecture for intersecting cliques

摘录 / EXCERPT: For integers $k\ge 1$ and $r\ge 3$, let $F_{k,r}$ be the forbidden graph from the source, let $S_{n,a}=K_a\vee I_{n-a}$, and let $q(G)$ denote the signless Laplacian spectral radiu...

conjecture | open | MathDB 页面 | 原文线索

V2-050619 | MATHDB-352551

High-ordered spectral characterization conjecture for trees

摘录 / EXCERPT: High-ordered spectral characterization conjecture. Two trees are isomorphic if and only if they have the same high-ordered spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-050620 | MATHDB-352554

Geometric-packability conjecture for the plane graphs \Theta_2, \Theta_3, and \Theta_4

摘录 / EXCERPT: Let $\mathcal{P}(G)$ denote the set of plane drawings of a graph G, and let Θ_2, Θ_3, and Θ_4 be the three plane triangulated cycles shown in the source. A s...

conjecture | open | MathDB 页面 | 原文线索

V2-050621 | MATHDB-352555

Asymptotic extremal-edge conjecture for convex-packable plane paths

摘录 / EXCERPT: Let $f(k)$ be the maximum number of extremal edges of a convex-packable plane path with k edges. The source establishes the upper bound $f(k)\leq 2\sqrt{k}$ for every positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-050622 | MATHDB-352589

The complete bipartite graph size Ramsey number conjecture

摘录 / EXCERPT: Let $K_{s,t}$ denote the complete bipartite graph with parts of sizes s and t, and let $\hat r(H)$ denote the size Ramsey number of a graph H. For functions of the parameters,...

conjecture | open | MathDB 页面 | 原文线索

V2-050623 | MATHDB-352591

The disjoint-edge bound for geometric graphs

摘录 / EXCERPT: Let m be a non-negative integer and G be a geometric graph such that $|DJ(uv)|\leq m$ for any edge $uv\in E(G)$. The disjoint-edge bound. Then ... The conjecture proposes a sharp...

conjecture | open | MathDB 页面 | 原文线索

V2-050624 | MATHDB-352592

The disjoint-edge lower bound for dense geometric graphs

摘录 / EXCERPT: Let G be a geometric graph with $2|E(G)|\geq |V(G)|$. The disjoint-edge lower-bound conjecture. Then ... This conjecture gives a lower bound for the number of disjoint edge pairs i...

conjecture | open | MathDB 页面 | 原文线索

V2-050625 | MATHDB-352598

C2-cofiniteness characterization of Neveu-Schwarz minimal series representations

摘录 / EXCERPT: Neveu-Schwarz minimal-series conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-050626 | MATHDB-352603

Weak degeneracy lower-bound conjecture for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph, and let $\mathsf{wd}(G)$ denote its weak degeneracy. Regular-graph weak degeneracy conjecture. Every d-regular graph G satisfies ... The co...

open_problem | open | MathDB 页面 | 原文线索

V2-050627 | MATHDB-352616

Degree-sequence conditions for r-Hamiltonian hypergraphs

摘录 / EXCERPT: Let $(d_1,d_2,\ldots,d_n)$ be an integer sequence with ... where $n>2r$ and $r\geq 3$. A hypergraph is r-Hamiltonian if it remains Hamiltonian after the deletion of any set of fewe...

conjecture | open | MathDB 页面 | 原文线索

V2-050628 | MATHDB-352630

Converse Lyapunov conjecture for robust delay stability

摘录 / EXCERPT: Fix a network with queue m, tail-exponent set Γ, and the notions of RDS and special ε-Lyapunov function as in the preceding corollary. Converse Lyapunov conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-050629 | MATHDB-352631

Generic equivalence conjecture for 0-RJF and G-RDS

摘录 / EXCERPT: Fix a network and a tail-exponent vector ${\boldsymbol{\gamma}}$. Let ${\boldsymbol{\lambda}}^*$ range over nonnegative nominal arrival vectors, and let 0-RJF and G-RDS denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-050630 | MATHDB-352636

Eroh-Kang-Yi cycle-rank conjecture for metric dimension and zero forcing

摘录 / EXCERPT: Let G be a connected graph. Write $Z(G)$ for its zero forcing number, let $\dim(G)$ denote its metric dimension, and let $c(G)$ be the number of edges that must be removed from...

conjecture | open | MathDB 页面 | 原文线索

V2-050631 | MATHDB-352639

Karskiński-Łuczak-Thomason 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a graph without isolated edges. An edge-coloring is neighbor-sum-distinguishing if any two neighboring vertices have different sums of the colors on their incident edges...

conjecture | open | MathDB 页面 | 原文线索

V2-050632 | MATHDB-352647

Minimum domination conjecture for the augmented Cartesian product

摘录 / EXCERPT: Let G and H be graphs, let D_G and D_H be minimum dominating sets of G and H, respectively, and let $A_{G,H}[F_G,F_H]$ denote the graph obtained by the paper's edge-adj...

conjecture | open | MathDB 页面 | 原文线索

V2-050633 | MATHDB-352677

The recursive profile conjecture for distance games on complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be the complete bipartite graph with part sizes m and n, and let $P_{\textsc{Col},K_{m,n}}(1)$ denote the number of positions in Col (equivalently, by the biparti...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-050634 | MATHDB-352679

Seneta-Heyde scaling conjecture for the critical local-time measure on a regular tree

摘录 / EXCERPT: Let $\mathbb L_n$ be the set of vertices at level n of the rooted regular tree with branching factor b, let $L_t(x)$ denote the local time at vertex x up to time t for simp...

conjecture | open | MathDB 页面 | 原文线索

V2-050635 | MATHDB-352681

BNY property conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph. A k-path subdigraph of D is a collection of k vertex-disjoint paths, and let $\lambda_k(D)$ denote the maximum order of a k-path subdigraph; in partic...

conjecture | open | MathDB 页面 | 原文线索

V2-050636 | MATHDB-352682

The maximal extended spectral radius conjecture for bipartite connected graphs

摘录 / EXCERPT: Let G be a bipartite connected graph of order $n\geq 5$, and let S_n denote the star on n vertices. The extended spectral radius of G is denoted by $\eta_1(G)$. Maximal ext...

conjecture | open | MathDB 页面 | 原文线索

V2-050637 | MATHDB-352694

The probabilistic universal graph conjecture for stable matrix families

摘录 / EXCERPT: Let $\mathcal M$ be a sequence of $n \times n$ Boolean matrices. For a sequence of matrices, let $\operatorname{cl}(\mathcal M)$ be the set of all square submatrices of matrices in...

conjecture | open | MathDB 页面 | 原文线索

V2-050638 | MATHDB-352695

Odd-homology vanishing conjecture for moduli spaces of planar trivalent graphs

摘录 / EXCERPT: Odd-homology vanishing conjecture. For any planar trivalent graph G and for i odd, one has

conjecture | open | MathDB 页面 | 原文线索

V2-050639 | MATHDB-352696

Łuczak-Peled conjecture on torsion in random Linial-Meshulam complexes

摘录 / EXCERPT: Łuczak-Peled conjecture. For every $d \geq 2$ and $p=p(n)$ such that $|np-c_d|$ is bounded away from 0, $H_{d-1}(Y_d(n,p))$ is torsion-free asymptotically almost surely.

conjecture | open | MathDB 页面 | 原文线索

V2-050640 | MATHDB-352705

Exact Berge-book Turán conjecture in the large-book regime

摘录 / EXCERPT: Let r and t be integers, and let ${\mathrm{ex}}_r(n,\operatorname{Berge-}B_2)$ denote the maximum number of hyperedges in an r-uniform n-vertex hypergraph containing no Ber...

conjecture | open | MathDB 页面 | 原文线索

V2-050641 | MATHDB-352708

Odd-level graded twisting equivalence conjecture for quantum group representation categories

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let P be its weight lattice and Q its root lattice, and let $Q_{\mathfrak{g}}^N:P/Q\to\mathbb{C}^\times$ be the quadratic form defin...

conjecture | open | MathDB 页面 | 原文线索

V2-050642 | MATHDB-352711

The maximum-spread outerplanar graph conjecture

摘录 / EXCERPT: Maximum-spread outerplanar graph conjecture. For n sufficiently large, the unique n-vertex outerplanar graph of maximum spread is

conjecture | open | MathDB 页面 | 原文线索

V2-050643 | MATHDB-352721

Sambinelli-Silva-Lee's conjecture on BE-diperfect digraphs

摘录 / EXCERPT: Sambinelli-Silva-Lee's conjecture. A digraph D is BE-diperfect if and only if D does not contain a blocking odd cycle as an induced subdigraph.

conjecture | open | MathDB 页面 | 原文线索

V2-050644 | MATHDB-352740

Aravind's induced rainbow path conjecture

摘录 / EXCERPT: A graph is properly colored if adjacent vertices receive distinct colors, and a vertex-colored graph is rainbow if no two vertices have the same color. A graph is triangle-free if...

conjecture | open | MathDB 页面 | 原文线索

V2-050645 | MATHDB-352741

The interior-vertex lower-bound conjecture for grid pebbling distributions

摘录 / EXCERPT: Interior-vertex lower-bound conjecture. For every interior vertex X,

conjecture | open | MathDB 页面 | 原文线索

V2-050646 | MATHDB-352752

Kurek-Ruciński conjecture on the online-to-size Ramsey ratio for cliques

摘录 / EXCERPT: Kurek-Ruciński conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050647 | MATHDB-352753

Exact online size Ramsey number of a triangle versus a path

摘录 / EXCERPT: Conjecture on the triangle-path online size Ramsey number.

conjecture | open | MathDB 页面 | 原文线索

V2-050648 | MATHDB-352768

Improved upper bound for feedback vertex sets in graphs of even degeneracy

摘录 / EXCERPT: Let k be an even degeneracy bound, and let G be an n-vertex graph of degeneracy k. Write f for the size of a minimum feedback vertex set of G. There is an...

conjecture | open | MathDB 页面 | 原文线索

V2-050649 | MATHDB-352769

Lower bound for feedback vertex sets in digraphs of bounded degeneracy

摘录 / EXCERPT: Let k be the degeneracy of a digraph, let n be a positive integer, and write f for the size of a minimum feedback vertex set. The paper conjectures that there is an...

conjecture | open | MathDB 页面 | 原文线索

V2-050650 | MATHDB-352779

Akbari-Alazemi-Andjelić energy bound conjecture for graphs

摘录 / EXCERPT: Akbari-Alazemi-Andjelić conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-050651 | MATHDB-352782

Alon et al.'s conjecture on iterativity and strong adversaries

摘录 / EXCERPT: Let a graph be evaluated under an opinion-forming model with initially selected experts, binary labels, and an adversary that chooses the experts' labels. In the iterative propagat...

conjecture | open | MathDB 页面 | 原文线索

V2-050652 | MATHDB-352787

Ma and Yuan's clique-cycle conjecture

摘录 / EXCERPT: Let G be a 2-connected n-vertex graph with $n\ge 3$, and let uv be an edge of G. Let $k\ge 4$ and $s\ge 2$ be integers, and write ... where $0\le t\le k-4$. Here $N_s(G)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-050653 | MATHDB-352795

The sparsity conjecture for minimally AP-irreducible sign pattern matrices

摘录 / EXCERPT: Let A be a minimally AP-irreducible sign pattern matrix of order n whose diagonal entries are all zero. A sign pattern matrix A is minimally AP-irreducible if it is AP-irredu...

conjecture | open | MathDB 页面 | 原文线索

V2-050654 | MATHDB-352801

Dvořák-Feghali linear diameter conjecture for list recolouring of planar graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a planar graph on n vertices. A list assignment L assigns a set $L(v)$ of colours to each vertex $v\in V$, and let $\mathcal{G}(G,L)$ be the graph whose vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-050655 | MATHDB-352809

Facet-replacement conjecture for the multicut dominant

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $S\subseteq\binom{V}{2}$, and let $\widehat{G}$ be obtained from G by replacing an edge by a connected graph. A facet-replacement conjecture asserts...

conjecture | open | MathDB 页面 | 原文线索

V2-050656 | MATHDB-352810

Facet description conjecture for the multicut dominant of an even cycle

摘录 / EXCERPT: Even-cycle multicut conjecture. The multicut dominant $\textsc{MultC}(C_{2n},S)$ is completely defined by the edge inequalities, the path inequalities, and the inequalities from th...

conjecture | open | MathDB 页面 | 原文线索

V2-050657 | MATHDB-352811

Two-terminal-pair multicut dominant description conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph and let S be a set of terminal pairs with $|S|=2$. The multicut dominant $\textsc{MultC}(G,S)$ is the dominant associated with multicut incidence vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-050658 | MATHDB-352812

Chen et al.'s refined BFP conjecture for near-complete bipartite graphs

摘录 / EXCERPT: Let p,q,e be positive integers with $p\leq q$ and $pq-p<e<pq$. Let $K_{p,q}$ be the complete bipartite graph with partite-set orders p and q, and let $\mathcal{K}(p,q,e)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-050659 | MATHDB-352818

Generalized large-girth conjecture for strong edge-coloring of regular bipartite graphs

摘录 / EXCERPT: For each integer $k\ge 3$, let g_k be a proposed girth threshold. Let G be a bipartite graph of girth at least g_k, maximum degree k, and with m edges. A generalized larg...

conjecture | open | MathDB 页面 | 原文线索

V2-050660 | MATHDB-352821

Logarithmic level-rank duality for boundary VOAs

摘录 / EXCERPT: Let $\widetilde{\mathcal N}_{n,k}$ be the slight modification of the boundary VOA ${\mathcal N}_{n,k}$ obtained by successive extension and orbifold, let...

conjecture | open | MathDB 页面 | 原文线索

V2-050661 | MATHDB-352826

Monotonicity conjecture for the diameter of Schrijver graphs

摘录 / EXCERPT: Let k and r be integers with $r\geq 1$, and let $D(G)$ denote the diameter of a graph G. For $n=2k+r$, write $\operatorname{SG}(n,k)$ for the Schrijver graph. Monotonicity co...

conjecture | open | MathDB 页面 | 原文线索

V2-050662 | MATHDB-352827

Unit-gap conjecture for diameters of Schrijver graphs

摘录 / EXCERPT: Let k and r be integers with $2\leq r\leq k-2$, and let $D(G)$ denote the diameter of a graph G. For $n=2k+r$, write $\operatorname{SG}(n,k)$ for the Schrijver graph. Unit-ga...

conjecture | open | MathDB 页面 | 原文线索

V2-050663 | MATHDB-352831

The linear induced-subgraph conjecture for circle graphs

摘录 / EXCERPT: A circle graph is the intersection graph of chords of a circle. Let r and n be positive integers, and let the circle graph have n vertices. The linear induced-subgraph conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050664 | MATHDB-352839

The bounded leftover conjecture for ordered graph tilings

摘录 / EXCERPT: Let H be an ordered graph. An H-tiling in an ordered graph G is a collection of vertex-disjoint copies of H in G, and let $\delta(G)$ denote the minimum degree of G. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-050665 | MATHDB-352846

No perfect state transfer across the longer-path bridge

摘录 / EXCERPT: Let Z be the graph shown in Figure 6: a longer path X joins the relevant subgraphs, with vertices a,b,c,d labelled as in that figure. No-perfect-state-transfer conjecture. Pe...

conjecture | open | MathDB 页面 | 原文线索

V2-050666 | MATHDB-352853

The edge clique cover conjecture for graphs with independence number two

摘录 / EXCERPT: Let G be a simple graph with n vertices. A clique is a set of vertices inducing a complete graph, and the edge clique cover number $ecc(G)$ is the minimum number of cliques suc...

conjecture | open | MathDB 页面 | 原文线索

V2-050667 | MATHDB-352855

Hartsfield's antimagic conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph. An antimagic labeling of G is a bijection $f:E(G)\to\{1,\ldots,|E(G)|\}$ such that the vertex weights ... are distinct for every pair of distinct vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050668 | MATHDB-352856

Hartsfield's antimagic conjecture for trees

摘录 / EXCERPT: Let T be a tree. An antimagic labeling of T is a bijection $f:E(T)\to\{1,\ldots,|E(T)|\}$ such that the vertex weights ... are distinct for every pair of distinct vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-050669 | MATHDB-352865

Koh-Rogers-Lee-Toh conjecture on graceful variable windmills

摘录 / EXCERPT: Let C_n be a cycle of length $n\geq 3$, and let C_n^t be the graph obtained from the union of t copies of C_n with one vertex in common, called the central vertex. A graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050670 | MATHDB-352904

Dahlberg-Shepherd conjecture on the maximum degree of e-positive trees

摘录 / EXCERPT: Dahlberg-Shepherd conjecture. Any tree with a vertex of degree at least 4 is not e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-050671 | MATHDB-352911

Toric promotion order conjecture for path graphs

摘录 / EXCERPT: Toric promotion order conjecture. If G is a path graph with n vertices and $1\leq h\leq \left\lfloor n/2\right\rfloor$, then the operator...

conjecture | open | MathDB 页面 | 原文线索

V2-050672 | MATHDB-352912

Polynomial-time reconstruction for a non-series base edge

摘录 / EXCERPT: Polynomial-time lifting conjecture. When the chosen base edge e is not a series edge, the above process can be carried out in at worst polynomial time in the number of vertices a...

conjecture | open | MathDB 页面 | 原文线索

V2-050673 | MATHDB-352919

Frank's vertex-connected orientation conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph and let k be a positive integer. A k-vertex-connected orientation is an orientation of G that is k-vertex-connected. Frank's conjecture. G has a...

conjecture | open | MathDB 页面 | 原文线索

V2-050674 | MATHDB-352946

The exact $(k+1)$-metric dimension of three-dimensional grids

摘录 / EXCERPT: The conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050675 | MATHDB-352960

Nagle's codegree density conjecture for K_4^-

摘录 / EXCERPT: Nagle's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050676 | MATHDB-352971

Bosek et al.'s chromatic bound for hat guessing numbers

摘录 / EXCERPT: Let G be a graph, let $\chi(G)$ denote its chromatic number, and let $\operatorname{HG}(G)$ denote its hat guessing number. Bosek et al.'s conjecture. Every graph G satisfies .....

conjecture | open | MathDB 页面 | 原文线索

V2-050677 | MATHDB-352972

Balanced-graph conjecture for flat graphs

摘录 / EXCERPT: Balanced-graph conjecture. Any flat graph is a balanced graph.

conjecture | open | MathDB 页面 | 原文线索

V2-050678 | MATHDB-352973

Clique-root conjecture for l-connected chordal K(l+3)-free graphs

摘录 / EXCERPT: Clique-root conjecture. If G is K_{l+3}-free, then G has only clique roots.

conjecture | open | MathDB 页面 | 原文线索

V2-050679 | MATHDB-352987

Erdős-Purdy-Agarwal-Sharir conjecture on congruent simplices

摘录 / EXCERPT: Let $d\geq 4$, let $k\leq d$, and let $\mathrm{C}(d,k,n)$ denote the maximum, over all k-vertex simplices S and all n-element sets $P\subseteq\mathbb{R}^d$, of the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-050680 | MATHDB-353008

Caporossi-Dobrynin-Gutman-Hansen conjecture on odd-diameter H-palindromic trees

摘录 / EXCERPT: Let G be a graph of diameter D, and write $\alpha(G,k)$ for the number of unordered pairs of vertices at distance k. The graph is H-palindromic when ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-050681 | MATHDB-353016

Verstraëte's girth-six conjecture for independent domination in cubic graphs

摘录 / EXCERPT: Verstraëte's conjecture. If G is a cubic graph with girth at least 6, then

conjecture | open | MathDB 页面 | 原文线索

V2-050682 | MATHDB-353064

Vertex tensor category and Grothendieck fusion conjecture for $L_{-\frac{3}{2}}(\mathfrak{sl}_3)$

摘录 / EXCERPT: Let $\mathscr{D}$ be the category of weight $L_{-\frac{3}{2}}(\mathfrak{sl}_3)$-modules with finite-dimensional weight spaces and real weights. Let $\mathbin{\boxtimes}$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-050683 | MATHDB-353065

Partial Kazhdan-Lusztig correspondence for $W^0_{\!A_2}(2)$

摘录 / EXCERPT: Let $\mathscr{E}$ be the category formed by the $W^0_{\!A_2}(2)$-modules obtained by decomposing the $L_{-\frac{3}{2}}(\mathfrak{sl}_3)$-modules in $\mathscr{D}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-050684 | MATHDB-353066

Full Kazhdan-Lusztig correspondence for $W^0_{\!A_2}(2)$

摘录 / EXCERPT: Let $\mathscr{E}$ be the vertex tensor category of $W^0_{\!A_2}(2)$-modules obtained by decomposing the relevant $L_{-\frac{3}{2}}(\mathfrak{sl}_3)$-modules, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050685 | MATHDB-353078

The spectral-network stability conjecture for Fukaya categories with coefficients

摘录 / EXCERPT: Spectral-network stability conjecture. There is a stability condition on the homotopy category of $\mathcal F(S,M;\mathcal E)_{\mathbf K}$ with central charge Z, whose semistable...

conjecture | solved | MathDB 页面 | 原文线索

V2-050686 | MATHDB-353124

Conjecture relating normalized single-hypergraph Turán densities across uniformities

摘录 / EXCERPT: For an integer $r\geq 2$, let $\Pi_{1}^{(r)}$ denote the set of Turán densities of single r-graphs. The normalized Turán-density conjecture. If ... is in $\Pi_{1}^{(r)}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-050687 | MATHDB-353126

Zhang-Sun-Wu and Gholami-Rowshan conjecture on bipartite Ramsey numbers of even cycles

摘录 / EXCERPT: Zhang-Sun-Wu-Gholami-Rowshan conjecture. For positive integers n and m,

conjecture | open | MathDB 页面 | 原文线索

V2-050688 | MATHDB-353133

Zhang-Liu-Shen cycle-length conjecture for large graphs

摘录 / EXCERPT: Let $k\geq 2$ be a fixed positive integer, and let G be a graph with sufficiently large edge count m and no isolated vertices. Write $\lambda(G)$ for the adjacency spectral rad...

conjecture | open | MathDB 页面 | 原文线索

V2-050689 | MATHDB-353154

Chen-Wu-An conjecture on minimum total distance of graphs with given radius

摘录 / EXCERPT: Let G be a graph of order n, and let $W(G)$ denote the sum of distances between all unordered pairs of vertices. For positive integers n and r satisfying $n\ge 2r$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050690 | MATHDB-353155

Digraph analogue of the Chen-Wu-An conjecture on minimum total distance

摘录 / EXCERPT: Let D be a digraph of order n, and let $W(D)$ denote the sum of distances over all ordered pairs of vertices. The outradius of D is the least integer r for which some verte...

conjecture | open | MathDB 页面 | 原文线索

V2-050691 | MATHDB-353156

Maximum-size biconnected digraphs of given radius

摘录 / EXCERPT: Let $r\geq \frac52$ and $n\geq 2r+1$. Let $\overline{\Gamma}_{2r+1}$ be the digraph specified in the paper, and let $\overline{\Gamma}_{n,2r,i,s}$ denote the family obtained by the...

conjecture | open | MathDB 页面 | 原文线索

V2-050692 | MATHDB-353172

Cone conjecture for characteristic-dependent Betti numbers of graph ideals

摘录 / EXCERPT: Cone conjecture. If the Betti numbers of $I(G)$ depend on the field, then the Betti numbers of J^2 also depend on the field.

conjecture | open | MathDB 页面 | 原文线索

V2-050693 | MATHDB-353174

The quadratic double-flip distance conjecture for acyclic orientations

摘录 / EXCERPT: Let G be an n-vertex graph, and let $\alpha,\alpha”\in\operatorname{Acyc}(G)$ be double-flip equivalent, meaning that they can be connected by a sequence of allowed double-flip...

conjecture | open | MathDB 页面 | 原文线索

V2-050694 | MATHDB-353175

The biconnected friends-and-strangers connectivity conjecture

摘录 / EXCERPT: Let Y be a graph on $n\geq 3$ vertices such that its complement is a forest consisting of trees $\mathcal T_1,\dots,\mathcal T_r$ with ... If X is a biconnected graph on n vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050695 | MATHDB-353176

The cycle-barbell-theta girth conjecture for star friends-and-strangers graphs

摘录 / EXCERPT: Let X be a connected graph with finite girth, let $\mathcal C$ be the set of cycle subgraphs of $\mathsf{FS}(X,\textsf{Star}_n)$ that achieve its girth, and for each...

conjecture | open | MathDB 页面 | 原文线索

V2-050696 | MATHDB-353183

Flandrin et al.'s spanning-tree branch-vertex conjecture

摘录 / EXCERPT: Let k be a positive integer and let G be a connected graph of order n. Flandrin et al.'s conjecture. If ... then G has a spanning tree with at most k branch vertices. This...

conjecture | open | MathDB 页面 | 原文线索

V2-050697 | MATHDB-353184

França-Brondani conjecture on the main spectrum of generalized Bethe trees

摘录 / EXCERPT: Let $k\ge 2$ and $d_1,d_2,\ldots,d_{k-1}$ be integers of at least 2. A generalized Bethe tree $\mathcal{B}(d_1,d_2,\ldots,d_{k-1})$ with k levels is a rooted tree in which all...

conjecture | open | MathDB 页面 | 原文线索

V2-050698 | MATHDB-353192

Fei's vertex-coefficient and saturated-support conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be a TSSS cluster algebra with principal coefficients. For a coefficient-free cluster monomial, let its Newton polytope be the convex polytope determined by its...

conjecture | open | MathDB 页面 | 原文线索

V2-050699 | MATHDB-353203

Uniqueness conjecture for power domination reconfiguration graphs of complete bipartite graphs

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with part sizes s and t, and let $\mathscr{PD}^{TAR}(K_{s,t})$ denote its power domination reconfiguration graph under token addit...

conjecture | open | MathDB 页面 | 原文线索

V2-050700 | MATHDB-353216

Balakrishnan et al.'s characterization conjecture for strongly distance-balanced graphs

摘录 / EXCERPT: Let Γ be a graph. For vertices $u,v\in V(\Gamma)$, let $W_{u,v}$ denote the set of vertices closer to u than to v, and define $d(u,S)=\sum_{x\in S}d(u,x)$ for a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-050701 | MATHDB-353236

Exact anti-Ramsey number of vertex-disjoint triangles

摘录 / EXCERPT: Let kC_3 denote the union of k vertex-disjoint copies of the triangle C_3. For a positive integer n, let $ar(n,kC_3)$ be the maximum number of colors in an edge-coloring of...

conjecture | open | MathDB 页面 | 原文线索

V2-050702 | MATHDB-353242

Esperet's polylogarithmic induced-path conjecture for k-degenerate graphs

摘录 / EXCERPT: Esperet's conjecture. There is a constant d such that every k-degenerate graph that has a path of order n also has an induced path of order at least

conjecture | open | MathDB 页面 | 原文线索

V2-050703 | MATHDB-353250

Algebraic lower-bound conjecture for the complement connectivity pair

摘录 / EXCERPT: Algebraic lower-bound conjecture. The pair (x,y) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050704 | MATHDB-353251

Weighted Laplacian Spread Conjecture

摘录 / EXCERPT: Weighted Laplacian Spread Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-050705 | MATHDB-353252

SE(s) extremal conjecture for weighted Laplacian spread

摘录 / EXCERPT: SE(s) extremal conjecture. The quantities x and y satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-050706 | MATHDB-353317

Hypergraphic planted clique detection conjecture

摘录 / EXCERPT: Let H be an order-K hypergraph in the HPC detection problem, with null distribution $H_0:H\sim\mathcal{H}_K(p,1/2)$ and planted-clique alternative...

conjecture | open | MathDB 页面 | 原文线索

V2-050707 | MATHDB-353336

Esperet-Galvin-Rödl conjecture on high-chromatic triangle-free induced subgraphs

摘录 / EXCERPT: Esperet-Galvin-Rödl conjecture. For all $k,r\in\mathbb{N}$ there is an $n\in\mathbb{N}$ such that every graph G with $\chi(G)\geq n$ and $\omega(G)\leq k$ has an induced subgraph...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050708 | MATHDB-353345

The upper-density conjecture for monochromatic paths in infinite complete graphs

摘录 / EXCERPT: Upper-density conjecture. For every integer $k\geq 3$, every k-edge-colouring of $K_{\mathbb{N}}$ contains a monochromatic path P such that

conjecture | open | MathDB 页面 | 原文线索

V2-050709 | MATHDB-353356

Woodall's list-colouring conjecture for complete bipartite minor-free graphs

摘录 / EXCERPT: Woodall's list-colouring conjecture. For every $s,t \in \mathbb{N}$, if G is a graph with $G \not\succeq K_{s,t}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050710 | MATHDB-353357

Woodall's chromatic conjecture for complete bipartite minor-free graphs

摘录 / EXCERPT: Woodall's chromatic conjecture. For every $s,t \in \mathbb{N}$, if G is a graph with $G \not\succeq K_{s,t}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-050711 | MATHDB-353367

Lower-bound conjecture for deterministic locating-dominating sets on the infinite king grid

摘录 / EXCERPT: Let K denote the infinite king grid, and let $\operatorname{DET:LD}(K)$ denote its deterministic locating-dominating density. Deterministic locating-dominating density conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050712 | MATHDB-353369

The analytic torsion formula for three-dimensional spheres

摘录 / EXCERPT: Let G be a 3-sphere, with $f_2(S_i)$ denoting the number of 2-faces in the unit sphere S_i of a vertex and $f_0(G)$ the number of vertices of G. The analytic torsion form...

conjecture | open | MathDB 页面 | 原文线索

V2-050713 | MATHDB-353384

Gamma-zero characterization conjecture for k-connected graph symmetric edge polytopes

摘录 / EXCERPT: Gamma-zero characterization conjecture. If $k\in\mathbb{N}$ and G is a k-connected graph on n vertices, then

conjecture | open | MathDB 页面 | 原文线索

V2-050714 | MATHDB-353406

The directed Nine Dragon Tree Conjecture for branchings

摘录 / EXCERPT: Let D be a digraph. Write $\gamma(D)$ for the fractional arboricity of its underlying graph, and let $\Delta^-(D)$ and $\Delta^+(D)$ denote its maximum in-degree and maximum out-...

conjecture | open | MathDB 页面 | 原文线索

V2-050715 | MATHDB-353417

Resistance-distance bound for connected balanced digraphs

摘录 / EXCERPT: Resistance-distance conjecture. The resistance distance is bounded above by the directed distance:

conjecture | solved | MathDB 页面 | 原文线索

V2-050716 | MATHDB-353420

Odd-defect list edge-colouring bound

摘录 / EXCERPT: Odd-defect list edge-colouring conjecture. For every odd integer d and for every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-050717 | MATHDB-353421

Defective list edge-colouring conjecture

摘录 / EXCERPT: Defective list edge-colouring conjecture. For every graph G and every integer d,

conjecture | open | MathDB 页面 | 原文线索

V2-050718 | MATHDB-353422

Defective Goldberg-Seymour conjecture

摘录 / EXCERPT: Defective Goldberg-Seymour conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050719 | MATHDB-353428

The chromatic quasisymmetric function conjecture for directed trees

摘录 / EXCERPT: Let $\overrightarrow{G}$ and $\overrightarrow{H}$ be directed trees, and let $X_{\overrightarrow{G}}(\mathbf{x},t)$ denote the chromatic quasisymmetric function of a directed graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050720 | MATHDB-353430

The principal specialization conjecture of Loehr and Warrington for trees

摘录 / EXCERPT: For a graph G, let $X_G(\mathbf{x})$ denote its chromatic symmetric function, and for a positive integer n specialize the variables by $x_i=q^{i-1}$ for $1\leq i\leq n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050721 | MATHDB-353435

Maximum Wiener index conjecture for directed grids

摘录 / EXCERPT: Let $G_{m,n}=P_m\Box P_n$ be the Cartesian product of paths on $m,n\geq 2$ vertices. Let ${D}_{m,n}$ be the orientation of $G_{m,n}$ with all P_m-layers oriented up except the la...

conjecture | open | MathDB 页面 | 原文线索

V2-050722 | MATHDB-353438

The pleated bowtie spectral-rank conjecture

摘录 / EXCERPT: Let B_t denote the t-pleated bowtie graph, and let $\mathfrak{u}(B_t)$ be its spectral rank. Pleated bowtie conjecture. We have ... for $t\geq 8$. This claim extends the proved c...

conjecture | open | MathDB 页面 | 原文线索

V2-050723 | MATHDB-353439

The Chebyshev-distance asymptotic conjecture for hypergraph clique-shadows

摘录 / EXCERPT: For each $k\geq 3$, let Δ_k be the maximum Chebyshev distance between the coordinatewise powers of the principal eigenvectors of a connected k-graph and its clique-shadow:...

conjecture | open | MathDB 页面 | 原文线索

V2-050724 | MATHDB-353461

The nonnegative excess conjecture for braid edges

摘录 / EXCERPT: The nonnegative excess conjecture. For any $d70e i d53e$,

conjecture | open | MathDB 页面 | 原文线索

V2-050725 | MATHDB-353462

The maximal braid-edge density conjecture for layered permutations

摘录 / EXCERPT: Let $d53e$ be the symmetric group on n letters, and let $w_0^{(k,i)}$ denote the family of permutations specified in the paper. For $d70e i d53e$, let $G(d70e)$ be the graph of r...

conjecture | open | MathDB 页面 | 原文线索

V2-050726 | MATHDB-353470

Connectivity conjecture for Hamilton cycles in hypergraph line graphs

摘录 / EXCERPT: Hypergraph line-graph connectivity conjecture. For any $r\geq 2$, there is an integer $\phi(r)$ such that every $\phi(r)$-connected line graph of a rank r hypergraph is Hamiltoni...

conjecture | open | MathDB 页面 | 原文线索

V2-050727 | MATHDB-353476

The proper-subset reconstruction conjecture for graphs

摘录 / EXCERPT: Let G and H be graphs with n vertices. A proper subset means a subset S_G or S_H of the corresponding vertex and edge sets that is not the whole set. Proper-subset recons...

conjecture | open | MathDB 页面 | 原文线索

V2-050728 | MATHDB-353477

Invariance of semi-infinite cohomology under the higher-genus ab-move

摘录 / EXCERPT: Let $\mathbf{V}_{G_u,3}$ be the trinion vertex algebra with maximal untwisted punctures and let $\mathbf{V}_{1,1}$ be the mixed trinion vertex algebra. Let i_1,i_2,i_3 denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-050729 | MATHDB-353484

Four's a Crowd conjecture for eigenvector dispersion

摘录 / EXCERPT: Let G be a connected graph on $n\geq 6$ vertices. Write $P_{n-3}K_4$ for the kite graph consisting of a path attached to a complete graph K_4, and let $c_e(G)$ denote the eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-050730 | MATHDB-353486

Planar Barnette-type conjecture for sink-regular bipartite digraphs

摘录 / EXCERPT: Barnette-type conjecture. There exists a rounded 1-factor J such that

conjecture | open | MathDB 页面 | 原文线索

V2-050731 | MATHDB-353487

Dijoin-clutter tau equals two conjecture

摘录 / EXCERPT: Dijoin-clutter $\tau=2$ conjecture. The weighted digraph (D,w) has a dicut of weight two.

conjecture | open | MathDB 页面 | 原文线索

V2-050732 | MATHDB-353488

Dijoin weight decomposition conjecture

摘录 / EXCERPT: Dijoin weight decomposition conjecture. There exist weighted digraphs (D,c) and (D,c') such that

conjecture | open | MathDB 页面 | 原文线索

V2-050733 | MATHDB-353489

Pippenger-Golumbic's inducibility conjecture for cycles

摘录 / EXCERPT: Let C_k denote the cycle graph on k vertices, and let $\mathcal{I}(C_k)$ be its inducibility, the limiting maximum proportion of induced copies of C_k in graphs with an incre...

conjecture | open | MathDB 页面 | 原文线索

V2-050734 | MATHDB-353499

Aboulker-Charbit-Naserasr conjecture on oriented-tree-free digraphs

摘录 / EXCERPT: An oriented tree is an orientation of a tree. For a digraph G, let $[?\vec{\chi}(G)$ denote its dichromatic number, and say that a class of digraphs is $\vec{\chi}$-bounded if th...

conjecture | open | MathDB 页面 | 原文线索

V2-050735 | MATHDB-353540

Brualdi-Quinn Massey's strong chromatic index conjecture for bipartite graphs

摘录 / EXCERPT: Let G be an (a,b)-bipartite graph, meaning that the two vertex classes of G have maximum degrees at most a and b, respectively. Brualdi-Quinn Massey's conjecture. The str...

conjecture | open | MathDB 页面 | 原文线索

V2-050736 | MATHDB-353558

Sokal's conjecture on functional-digraph component polynomials

摘录 / EXCERPT: Sokal's functional-digraph conjecture. The following assertions hold: (i) $[\psi_{n,k}]_{n,k}$ is totally positive; (ii) $(\bm{\psi}_n(y))_{n\geq0}$ is coefficientwise Hankel-total...

conjecture | open | MathDB 页面 | 原文线索

V2-050737 | MATHDB-353570

The Alon-Frankl-Huang-Rödl-Ruciński-Sudakov conjecture on fractional matching thresholds

摘录 / EXCERPT: For $k,\ell\in\mathbb N$ with $\ell\leq k-1$, let $c^*_{k,\ell}$ be the smallest number c such that every k-graph H on n vertices with ... contains a perfect fractional match...

conjecture | open | MathDB 页面 | 原文线索

V2-050738 | MATHDB-353571

The Hàn-Person-Schacht perfect matching threshold conjecture

摘录 / EXCERPT: Let k and ℓ be integers with $1\leq\ell<k$, and let H be a k-graph on n vertices. Write $\delta_\ell(H)$ for the minimum ℓ-degree and let $c^*_{k,\ell}$ be the fr...

conjecture | open | MathDB 页面 | 原文线索

V2-050739 | MATHDB-353590

Krenn-Gu conjecture on the matching index

摘录 / EXCERPT: Krenn-Gu conjecture. If G is non-isomorphic to K_4, then

conjecture | open | MathDB 页面 | 原文线索

V2-050740 | MATHDB-353608

Shi-Shan conjecture for $(P_2\cup kP_1)$-free graphs

摘录 / EXCERPT: Shi-Shan conjecture. Let $k\geq 4$ be an integer and let G be a 1-tough and $2k$-connected $(P_2\cup kP_1)$-free graph. Then G is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-050741 | MATHDB-353637

Larson-Pepper's characterisation of graphs with equal independence and annihilation number

摘录 / EXCERPT: Larson-Pepper's conjecture. The equality $\alpha=a$ holds if and only if G is a König-Egerváry graph and every maximum independent set of G is a maximal annihilating set.

conjecture | refuted | MathDB 页面 | 原文线索

V2-050742 | MATHDB-353642

The linear expected-face conjecture for random orientable embeddings

摘录 / EXCERPT: Let G be a simple graph of order n. Select an orientable embedding of G uniformly at random, and let F denote its number of faces. Linear expected-face conjecture. The expe...

conjecture | open | MathDB 页面 | 原文线索

V2-050743 | MATHDB-353643

The multigraph linear-logarithmic expected-face conjecture

摘录 / EXCERPT: Let G be an n-vertex multigraph with maximum edge-multiplicity μ. Select an orientable embedding of G uniformly at random, and let F denote its number of faces. Multigr...

conjecture | open | MathDB 页面 | 原文线索

V2-050744 | MATHDB-353644

The chain-of-triangles bound for expected faces

摘录 / EXCERPT: Let G be a simple graph of order n, and let F denote the number of faces of a uniformly random orientable embedding of G. Chain-of-triangles conjecture. For every such grap...

conjecture | open | MathDB 页面 | 原文线索

V2-050745 | MATHDB-353672

Thomassen's bipartite high-girth subgraph conjecture

摘录 / EXCERPT: Thomassen's conjecture. For every integer k, every graph of sufficiently large minimum degree contains a bipartite subgraph of girth at least k and large minimum degree.

conjecture | open | MathDB 页面 | 原文线索

V2-050746 | MATHDB-353676

The stability conjecture for infinite set-homogeneous hypergraphs

摘录 / EXCERPT: The stability conjecture. Any infinite set-homogeneous k-hypergraph with stable theory is complete or has complete complement.

open_problem | open | MathDB 页面 | 原文线索

V2-050747 | MATHDB-353677

The odd-clique b2 conjecture for odd covers

摘录 / EXCERPT: Let $b_2(G)$ denote the minimum dimension of a binary vector space in which the vertices of a graph G can be represented as an odd cover, and let K_m be the complete graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-050748 | MATHDB-353695

Quadratic bound conjecture for the pair-matching decomposition number

摘录 / EXCERPT: Let $H=(V,E)$ be a 3-uniform hypergraph with n vertices. Quadratic bound conjecture. Its pair-matching decomposition number satisfies ... This bound would follow from the precedi...

conjecture | open | MathDB 页面 | 原文线索

V2-050749 | MATHDB-353697

Robust flow model gap conjecture

摘录 / EXCERPT: Robust flow gap conjecture. For any $\Gamma\in\mathbb{N}$, the inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-050750 | MATHDB-353711

The extremal sparsity conjecture for minimally globally rigid graphs

摘录 / EXCERPT: Let d be a positive integer and let $G=(V,E)$ be a graph on at least $d+2$ vertices that is minimally globally rigid in $\mathbb{R}^d$. The extremal sparsity conjecture. ... and th...

conjecture | open | MathDB 页面 | 原文线索

V2-050751 | MATHDB-353712

The linked-pairs dimension-dropping conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph. A pair of vertices is linked in G in $\mathbb{R}^k$ when its distance is fixed across all generic equivalent realizations, and globally linked when that...

conjecture | open | MathDB 页面 | 原文线索

V2-050752 | MATHDB-353713

The globally rigid graph bridge conjecture

摘录 / EXCERPT: Let G be a globally rigid graph in $\mathbb{R}^d$, and let e be an edge such that $G-e$ is not globally rigid. An edge is an $\mathcal{R}_{d+1}$-bridge when its deletion lowers...

conjecture | open | MathDB 页面 | 原文线索

V2-050753 | MATHDB-353716

The Lovász-Yemini connectivity conjecture for generic rigidity

摘录 / EXCERPT: Let d be a positive integer. A graph is $d(d+1)$-connected when deleting fewer than $d(d+1)$ vertices leaves it connected. The Lovász-Yemini conjecture. If a graph is $d(d+1)$-co...

conjecture | open | MathDB 页面 | 原文线索

V2-050754 | MATHDB-353745

The $(d-2)$-leaky forcing number conjecture for hypercubes

摘录 / EXCERPT: Hypercube leaky forcing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050755 | MATHDB-353750

The determinant formula for the reduced walk matrix of the Dynkin graph D_n

摘录 / EXCERPT: Let D_n ($n\ge 4$) be the Dynkin graph, let $W(D_n)$ be its walk matrix, and let $\hat{W}(D_n)$ be the $(n-1)\times(n-1)$ matrix obtained from $W(D_n)$ by removing the first row...

conjecture | open | MathDB 页面 | 原文线索

V2-050756 | MATHDB-353771

Tower lower-bound conjecture for Ramsey numbers of hedgehogs

摘录 / EXCERPT: Let $H_t^{(k)}$ denote the k-uniform hedgehog with parameter t, let $r_k(H_t^{(k)})$ be its two-colour Ramsey number, and let T_m denote the tower function of height m. Hed...

conjecture | open | MathDB 页面 | 原文线索

V2-050757 | MATHDB-353833

A weaker stable-set colorability conjecture for set systems

摘录 / EXCERPT: Let $r\geq 2$ and let $\mathcal{F}$ be a set system. Write $\mathcal{F}_{\widetilde{r-\operatorname{stab}}}$ for the family obtained by retaining the source's weakened notion of...

conjecture | open | MathDB 页面 | 原文线索

V2-050758 | MATHDB-353848

Bernstein-generality conjecture for conservative reaction networks

摘录 / EXCERPT: A conservative reaction network is a chemical reaction network in which each species occurs in at least one nonnegative conservation law. Let the mixed-volume bound be the upper bo...

conjecture | open | MathDB 页面 | 原文线索

V2-050759 | MATHDB-353865

Berge's conjecture on alpha-diperfect digraphs

摘录 / EXCERPT: Let D be a digraph. A stable set is a set of pairwise non-adjacent vertices, and a path partition is a collection of disjoint paths containing every vertex of D exactly once. A...

conjecture | open | MathDB 页面 | 原文线索

V2-050760 | MATHDB-353871

Turán number conjecture for Cartesian products of trees

摘录 / EXCERPT: Let T and S be trees, each with at least one edge, and let $T\Box S$ denote their Cartesian product. For a positive integer n, let $\mathrm{ex}(n,T\Box S)$ be the maximum num...

conjecture | open | MathDB 页面 | 原文线索

V2-050761 | MATHDB-353872

Turán bound conjecture for higher-dimensional grids

摘录 / EXCERPT: For positive integers d and t, let $F_t^{(d)}$ be the d-dimensional grid with vertex set $[t]^d$, where two vertices are adjacent when they differ by exactly one in exactly o...

conjecture | open | MathDB 页面 | 原文线索

V2-050762 | MATHDB-353873

The binary polyhedral groups conjecture for the orbifold VOA category

摘录 / EXCERPT: Let L_2 be the A_1 root lattice and let $V:=V_{L_2}$ be the corresponding lattice vertex operator algebra. Let $SU(2)$ be the universal central extension of $SO_3(\mathbb{R})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-050763 | MATHDB-353901

Negami's joint crossing number conjecture

摘录 / EXCERPT: Let G_1 and G_2 be graphs embedded on a closed surface S, and let the joint crossing number be the minimum number of crossing points between $h(G_1)$ and G_2 over all homeo...

conjecture | open | MathDB 页面 | 原文线索

V2-050764 | MATHDB-353922

The vertex metric dimension bound for connected graphs

摘录 / EXCERPT: Let G be a connected graph. The invariant $L(G)$ is the graph parameter measuring the contribution of leaves, and $c(G)$ is the cyclomatic number of G. The vertex metric dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-050765 | MATHDB-353923

The leafless vertex metric dimension bound conjecture

摘录 / EXCERPT: Let G be a graph distinct from the cycle C_n, with minimum degree $\delta(G)\geq2$. Here $\mathrm{dim}(G)$ denotes the vertex metric dimension and $c(G)$ the cyclomatic number.…

conjecture | open | MathDB 页面 | 原文线索

V2-050766 | MATHDB-353924

The equality characterization for leafless vertex metric dimension

摘录 / EXCERPT: Let G be a connected graph with minimum degree $\delta(G)\geq2$. A daisy graph is a graph consisting of at least two cycles sharing one vertex, and a petal is one of its cycles;...

conjecture | open | MathDB 页面 | 原文线索

V2-050767 | MATHDB-353929

Limit law conjecture for reticulation nodes in d-combining tree-child networks

摘录 / EXCERPT: Limit law conjecture. The following limit behavior holds:

conjecture | open | MathDB 页面 | 原文线索

V2-050768 | MATHDB-353948

Non-collapsing conjecture for the first hook level

摘录 / EXCERPT: Let $W_k(sl(n+m),x,f_{m,n})$ be the hook W--algebra at level k, and let $k^{(1)}_{m,n}$ denote the first distinguished level. Non-collapsing conjecture. The level … is always n...

conjecture | open | MathDB 页面 | 原文线索

V2-050769 | MATHDB-353958

Dunbar et al.'s bondage-number conjecture for planar graphs

摘录 / EXCERPT: Dunbar et al.'s conjecture. If G is a planar graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-050770 | MATHDB-353969

Kühn-Osthus conjecture for strongly 2-connected regular digraphs

摘录 / EXCERPT: Kühn-Osthus conjecture. The digraph G contains a Hamilton cycle. The source introduces this as the digraph analogue of the $t=2$ case of the Bollobás-Häggkvist conjecture; no res...

conjecture | open | MathDB 页面 | 原文线索

V2-050771 | MATHDB-353973

The spanning 2-tree conjecture for maximal planar graphs

摘录 / EXCERPT: Let a maximal planar graph be a planar graph to which no edge can be added while preserving planarity, and let a spanning 2-tree be a spanning subgraph that is a 2-tree. Spanning 2...

conjecture | open | MathDB 页面 | 原文线索

V2-050772 | MATHDB-353974

The linear Hamiltonian cycle conjecture for 4-connected maximal planar graphs

摘录 / EXCERPT: Let a 4MP be a 4-connected maximal planar graph, and let a linear Hamiltonian cycle be a Hamiltonian cycle such that the regions on one side of the cycle share edges with at most t...

conjecture | open | MathDB 页面 | 原文线索

V2-050773 | MATHDB-353976

Cycle, barbell, or theta structure conjecture for girth-achieving cycles

摘录 / EXCERPT: Let X be an arbitrary connected graph with finite girth, and let $\mathcal C$ be the set of cycle subgraphs in $\mathsf{FS}(X,\operatorname{Star}_n)$ that achieve its girth. For...

conjecture | open | MathDB 页面 | 原文线索

V2-050774 | MATHDB-353987

The non-backtracking Laplacian spectrum determination conjecture

摘录 / EXCERPT: Non-backtracking Laplacian spectrum determination conjecture. Almost all graphs with minimum degree at least 2 are determined by the spectrum of their non-backtracking Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-050775 | MATHDB-353988

The non-backtracking mate exclusion conjecture for the non-backtracking Laplacian

摘录 / EXCERPT: Non-backtracking mate exclusion conjecture. Almost all graphs with minimum degree at least 2 that have an $8\mathcal{A}$-mate have no $8\mathcal{L}$-mate.

conjecture | open | MathDB 页面 | 原文线索

V2-050776 | MATHDB-353990

Arc-disjoint hamiltonian paths in 2-generated Cayley digraphs of finite abelian groups

摘录 / EXCERPT: Finite abelian Cayley-digraph conjecture. The digraph $\mathop{\mathsf{C}\mkern-1mu\overrightarrow{\mathsf{ay}}}(G;a,b)$ has two arc-disjoint hamiltonian paths.

conjecture | open | MathDB 页面 | 原文线索

V2-050777 | MATHDB-354031

Nowakowski-Rall's independent domination conjecture for direct products

摘录 / EXCERPT: Let G and H be graphs, let $G\times H$ denote their direct product, and write $i(G)$ for the independent domination number of G. Nowakowski-Rall's conjecture. For all graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-050778 | MATHDB-354035

Dorpalen-Barry et al.'s homology conjecture for U-nucleus complexes

摘录 / EXCERPT: Let G be a connected undirected simple graph with at least 3 vertices, and let $U\subseteq V(G)$. Define the U-nucleus complex by ... where $\mathcal{N}(G)$ is the set of nucle...

conjecture | open | MathDB 页面 | 原文线索

V2-050779 | MATHDB-354086

Erdős-Faudree-Rousseau-Schelp unbalanced sparse-subset conjecture

摘录 / EXCERPT: Let G be an n-vertex graph and let α be an arbitrary constant with $53/120\leq \alpha\leq 1$. Let β satisfy ... If n is sufficiently large and every...

conjecture | open | MathDB 页面 | 原文线索

V2-050780 | MATHDB-354087

Sudakov's bipartisation conjecture for K_{r+1}-free graphs

摘录 / EXCERPT: For a graph G, let $D_2(G)$ be the minimum number of edges that must be deleted to make G bipartite. Fix $r\geq3$, and let G be an n-vertex K_{r+1}-free graph. Sudakov's...

conjecture | open | MathDB 页面 | 原文线索

V2-050781 | MATHDB-354103

Kündgen-Ramamurthi's weak 4-choosability conjecture for planar graphs

摘录 / EXCERPT: A list assignment L of a graph is symmetric if its colours are integers and, for every vertex v and integer i, $i\in L(v)$ implies that $-i\in L(v)$. A graph is weakly k-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-050782 | MATHDB-354131

Brandt's adjacency-spectrum conjecture for regular triangle-free graphs

摘录 / EXCERPT: Let G be a regular triangle-free graph on n vertices. Let $\lambda_n(G)\leq\cdots\leq\lambda_1(G)$ be the eigenvalues of the adjacency matrix of G. Brandt's conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-050783 | MATHDB-354147

Pósa-type degree-sequence conjecture for monochromatic cycle partitions

摘录 / EXCERPT: Let $d_1\leq\dots\leq d_n$ be the degree sequence of a graph G. The degree-sequence conjecture. There is a function f such that, for every $\beta>0$, every integer $r\geq 2$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-050784 | MATHDB-354152

The monadic dependence characterization of fixed-parameter tractable FO model checking

摘录 / EXCERPT: The monadic dependence conjecture. For every hereditary class $d4a2$ of structures, FO model checking is FPT on $d4a2$ if and only if $d4a2$ is monadically dependent.

conjecture | open | MathDB 页面 | 原文线索

V2-050785 | MATHDB-354163

Existence of double scaling limit for random neural networks

摘录 / EXCERPT: Consider a random depth L neural network with input dimension n_0, hidden layer widths ... output dimension n_{L+1} and non-linearity σ. Suppose that the network is tune...

conjecture | open | MathDB 页面 | 原文线索

V2-050786 | MATHDB-354173

The large-r feedback arc set conjecture for r-free digraphs

摘录 / EXCERPT: Let $G=(V,E)$ be a digraph on n vertices, let $\beta(G)$ denote the minimum size of a feedback arc set, and let an $(r-1)$-free digraph be one containing no directed cycle of len...

conjecture | open | MathDB 页面 | 原文线索

V2-050787 | MATHDB-354182

Vertex-splitting conjecture for global rigidity

摘录 / EXCERPT: Let G be a graph, let d be a positive integer, and let $uv\in E(G)$. Suppose that G is $(d+1)$-connected, $|N_G(u)\cap N_G(v)|=d-1$, and G/uv is globally rigid in...

conjecture | open | MathDB 页面 | 原文线索

V2-050788 | MATHDB-354195

Szemerédi-Petruska conjecture on 3-uniform witness hypergraphs

摘录 / EXCERPT: Szemerédi-Petruska conjecture. The maximum order of a 3-uniform (n,k)-witness hypergraph is

conjecture | open | MathDB 页面 | 原文线索

V2-050789 | MATHDB-354205

Mutation invariance of unrestricted red size

摘录 / EXCERPT: Let Q be a quiver and let $\mu_k(Q)$ denote the quiver obtained by mutation at vertex k. The unrestricted red size $\operatorname{uRed}(Q)$ is the minimum red size over general...

conjecture | open | MathDB 页面 | 原文线索

V2-050790 | MATHDB-354206

Red-size conjectures for connected quivers without reddening sequences

摘录 / EXCERPT: Let Q be a connected quiver, let $V(Q)$ be its vertex set, and suppose that Q does not admit a reddening sequence. A general maximal green sequence is a mutation sequence resul...

conjecture | open | MathDB 页面 | 原文线索

V2-050791 | MATHDB-354213

Fox, He and Wigderson's Ramsey goodness conjecture for complete multipartite graphs and books

摘录 / EXCERPT: Let $k,p,t\geq 2$ be integers. For a complete p-partite graph $H=K_p(a_1,\dots,a_p)$ with $1\leq a_1\leq\cdots\leq a_{p-1}\leq t$ and $a_p\leq\delta n$, let $B_{k,n}$ be the book...

conjecture | open | MathDB 页面 | 原文线索

V2-050792 | MATHDB-354217

Steiner triple system hitting-time conjecture

摘录 / EXCERPT: Let $n\equiv 1,3 \pmod 6$, and let $T_1,T_2,\ldots$ be a uniformly random ordering of the triples in $\binom{[n]}{3}$. Steiner triple system hitting-time conjecture. With high prob...

conjecture | open | MathDB 页面 | 原文线索

V2-050793 | MATHDB-354223

The plane-path conjecture for complete simple topological graphs

摘录 / EXCERPT: Plane-path conjecture. There is an absolute constant $\varepsilon>0$ such that every complete n-vertex simple topological graph contains a plane path of length n^{ε}.

conjecture | open | MathDB 页面 | 原文线索

V2-050794 | MATHDB-354231

Aharoni-Zerbib generalized Tuza conjecture for uniform hypergraphs

摘录 / EXCERPT: Let G be an r-graph. An $(r-1)$-matching is a set of edges such that any two intersect in fewer than $r-1$ vertices, and an $(r-1)$-cover is a collection of $(r-1)$-sets of ver...

conjecture | open | MathDB 页面 | 原文线索

V2-050795 | MATHDB-354260

Shortest Circuit Cover Conjecture

摘录 / EXCERPT: Let G be a 2-edge-connected graph, and let $scc(G)$ denote the minimum length of a circuit cover of G, where the length of a circuit cover is the sum of the lengths of its ci...

conjecture | open | MathDB 页面 | 原文线索

V2-050796 | MATHDB-354267

The coarse bounded H-complexity conjecture

摘录 / EXCERPT: Coarse bounded H-complexity conjecture. The coarse bounded sublevel H-complexity (respectively, the coarse bounded superlevel H-complexity) of F is a lower bound for the glob...

conjecture | open | MathDB 页面 | 原文线索

V2-050797 | MATHDB-354268

The generic total H-complexity conjecture for ReLU neural network maps

摘录 / EXCERPT: Generic total H-complexity conjecture. With probability 1, the total H-complexity of F is less than or equal to the number of 0-cells of $\mathcal{C}(F)$.

conjecture | open | MathDB 页面 | 原文线索

V2-050798 | MATHDB-354273

Structural reformulation of the 3-colorability conjecture for ${\cal G}_2$

摘录 / EXCERPT: Structural reformulation. Graphs in ${\cal G}_0$ induce neither θ nor $\theta^- $.

conjecture | open | MathDB 页面 | 原文线索

V2-050799 | MATHDB-354287

Merino-Welsh conjecture for graphs

摘录 / EXCERPT: Merino-Welsh conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-050800 | MATHDB-354292

Frankl's product conjecture for rainbow-triangle-free graph triples

摘录 / EXCERPT: Let G_1,G_2,G_3 be graphs with a common vertex set on n vertices. A rainbow triangle is a triangle whose three edges belong to three distinct graphs among G_1,G_2,G_3. Frankl...

conjecture | open | MathDB 页面 | 原文线索

V2-050801 | MATHDB-354293

Asymptotic product conjecture for rainbow-triangle-free graph triples

摘录 / EXCERPT: Let γ be the maximum on $[0,1]$ of ... and suppose that the maximum is attained at $x=x_0$. Consider three graphs G_1,G_2,G_3 on a common vertex set of size n, with no rai...

conjecture | open | MathDB 页面 | 原文线索

V2-050802 | MATHDB-354294

Gerbner-Palmer's conjecture that paths are Turán-good

摘录 / EXCERPT: Gerbner-Palmer's conjecture. Every path is K_{k+1}-Turán-good for every k.

conjecture | open | MathDB 页面 | 原文线索

V2-050803 | MATHDB-354295

Recursive limit-law conjecture for fringe patterns in ranked tree-child networks

摘录 / EXCERPT: Recursive limit-law conjecture. The limit law of F is determined recursively as follows. If P is a normal pattern, then F is a Poisson pattern; in all other cases for P,...

conjecture | open | MathDB 页面 | 原文线索

V2-050804 | MATHDB-354327

Bollobás-Leader-Malvenuto conjecture on Turán densities of daisies

摘录 / EXCERPT: Bollobás-Leader-Malvenuto conjecture. For all $t\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050805 | MATHDB-354329

Erdős's high-girth Steiner system conjecture

摘录 / EXCERPT: Given $n\ge q>r\ge2$, an (n,q,r)-Steiner system is a collection of q-element subsets of an n-element set in which every r-element subset lies in exactly one block. A...

conjecture | open | MathDB 页面 | 原文线索

V2-050806 | MATHDB-354330

Glock-Kühn-Lo-Osthus high-girth Steiner system conjecture

摘录 / EXCERPT: For integers $q>r\ge2$, an (n,q,r)-Steiner system is a collection of q-element subsets of an n-element set in which every r-element subset lies in exactly one block. A...

conjecture | open | MathDB 页面 | 原文线索

V2-050807 | MATHDB-354331

Alspach's rainbow matching conjecture for 2-factorizations

摘录 / EXCERPT: A 2-factor is a spanning 2-regular subgraph, and a full rainbow matching is a matching containing one edge from every color class. Alspach's conjecture. If G is a simple $2d$...

conjecture | open | MathDB 页面 | 原文线索

V2-050808 | MATHDB-354332

Grinblat's rainbow matching conjecture

摘录 / EXCERPT: A rainbow matching is a matching whose edges have pairwise distinct colors. Grinblat's conjecture. If G is a multigraph that is not necessarily properly edge colored with n col...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050809 | MATHDB-354346

The perfect-matching conjecture for near-complete group hypergraphs

摘录 / EXCERPT: Let G be a group of order n, and let G_A,G_B,G_C denote the three vertex classes used to define the hypergraph $H_G[X,Y,Z]$. Let $X\subseteq G_A$, $Y\subseteq G_B$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-050810 | MATHDB-354350

Maximum spectral radius conjecture for odd-cycle-free non-bipartite graphs

摘录 / EXCERPT: Maximum spectral radius conjecture. If G contains no member of $\{C_3,C_5,\ldots,C_{2k+1}\}$ and is non-bipartite, then

conjecture | open | MathDB 页面 | 原文线索

V2-050811 | MATHDB-354351

Booksize lower-bound conjecture for graphs with large spectral radius

摘录 / EXCERPT: Booksize lower-bound conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050812 | MATHDB-354371

Nonexistence of uniformly bounded edge-length regular geometric triangulations

摘录 / EXCERPT: Let T be a geometric triangulation of $\mathbb{E}^2$ or $\mathbb{H}^2$, and let k denote its vertex degree. Say that T has uniformly bounded edge lengths if there are constan...

conjecture | open | MathDB 页面 | 原文线索

V2-050813 | MATHDB-354389

Grötzsch's conjecture on edge-coloring subcubic planar graphs

摘录 / EXCERPT: Let G be a 2-connected planar graph with maximum degree $\Delta(G)\leq 3$. Grötzsch's conjecture. The graph G is 3-edge-colorable unless it has exactly one vertex of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-050814 | MATHDB-354391

Thomason's balanced clique-subdivision conjecture

摘录 / EXCERPT: A balanced subdivision of a graph H is obtained by replacing every edge of H with an internally vertex-disjoint path of the same length; write this as $TH^{(\ell)}$ when that l...

conjecture | open | MathDB 页面 | 原文线索

V2-050815 | MATHDB-354392

Thomason's conjecture on balanced subdivisions

摘录 / EXCERPT: Given a graph H and an integer ℓ, a balanced H-subdivision is obtained by replacing every edge of H with an internally vertex-disjoint path of length ℓ, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-050816 | MATHDB-354410

Intermediate fractional colorability conjecture for subcubic triangle-free graphs

摘录 / EXCERPT: Intermediate fractional colorability conjecture. Every such graph G is fractionally $19/7$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-050817 | MATHDB-354413

Gyárfás-Sárközy conjecture on long monochromatic Berge cycles

摘录 / EXCERPT: Let $\eta>0$ and let r be a positive integer. For each r, an r-hyperedge coloring of the complete r-uniform hypergraph $\mathcal{K}_n^r$ assigns one of r colors to every...

conjecture | open | MathDB 页面 | 原文线索

V2-050818 | MATHDB-354414

The bunkbed conjecture for finite graphs

摘录 / EXCERPT: The bunkbed conjecture. For every such G, the following three inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-050819 | MATHDB-354431

The 1-2-3 conjecture for neighbor sum distinguishing edge weightings

摘录 / EXCERPT: Let G be a nice graph, meaning a simple undirected graph with no component isomorphic to K_2. For an integer edge weighting $w:E(G)\rightarrow\{1,2,\ldots,k\}$, define ... The we...

conjecture | open | MathDB 页面 | 原文线索

V2-050820 | MATHDB-354432

Baudon's neighbor full sum distinguishing total coloring conjecture

摘录 / EXCERPT: Let G be a nice graph and let $w:V(G)\cup E(G)\rightarrow\{1,2,\ldots,k\}$ be a total integer weighting. For each vertex v, define ... where $N(v)$ is the open neighborhood of...

conjecture | open | MathDB 页面 | 原文线索

V2-050821 | MATHDB-354433

The 1-2-3 conjecture for 3-uniform hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}=(V,E)$ be a hypergraph, and call it 3-uniform when every edge has size 3. A hypergraph vertex coloring is proper when every edge contains at least two vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-050822 | MATHDB-354434

The 1-2-3 conjecture for uniform hypergraphs

摘录 / EXCERPT: Let $r\geq 3$ and let $\mathcal{H}$ be an r-uniform hypergraph, meaning that every edge has size r. A hypergraph vertex coloring is proper when every edge contains at least two...

conjecture | open | MathDB 页面 | 原文线索

V2-050823 | MATHDB-354435

Kalkowski-Karoński-Pfender's twin-free hypergraph conjecture

摘录 / EXCERPT: Let $r\geq 3$. Define ... Two vertices are twins when the sets of edges containing them are equal, and a hypergraph is twin-free when it contains no twins. Kalkowski-Karoński-Pfender...

conjecture | open | MathDB 页面 | 原文线索

V2-050824 | MATHDB-354440

The Seidel energy minimization conjecture for chain graphs

摘录 / EXCERPT: Let G be a chain graph of order n with binary string ... where ... and $s_i\geq 1$ and $t_i\geq 1$ for $1\leq i\leq k$. Let $SE(G)$ denote the Seidel energy, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050825 | MATHDB-354471

Gukov-Pei-Putrov-Vafa conjecture on homological blocks and WRT invariants

摘录 / EXCERPT: Let $\Gamma=(V,E,(w_v)_{v\in V})$ be a finite tree with integral vertex weights, and let W be its negative-definite linking matrix. Let $M(\Gamma)$ be the integral homology spher...

conjecture | open | MathDB 页面 | 原文线索

V2-050826 | MATHDB-354475

Brualdi-Busch conjecture on edge-disjoint 1-factors in realizations of degree sequences

摘录 / EXCERPT: Brualdi-Busch conjecture. Some realization of $d70$ has k edge-disjoint 1-factors if and only if $\mathcal{D}_{k}(\pi)$ is graphic.

conjecture | open | MathDB 页面 | 原文线索

V2-050827 | MATHDB-354495

Structural conjecture for maximal independent sets in generalized Kneser graphs

摘录 / EXCERPT: Let $d\ge 2$ be an integer, and let $q\Gamma_{2d+1,\{d,d+1\}}$ be the Kneser graph whose vertices are the relevant subspaces and whose adjacency is determined by dimensions d and...

conjecture | open | MathDB 页面 | 原文线索

V2-050828 | MATHDB-354506

The independence-dissociation bound for connected triangle-free subcubic graphs

摘录 / EXCERPT: Let G be a connected triangle-free subcubic graph. Write $\alpha(G)$ for its independence number and ${\rm diss}(G)$ for its dissociation number, the maximum cardinality of a ver...

conjecture | open | MathDB 页面 | 原文线索

V2-050829 | MATHDB-354531

Connected size Ramsey number conjecture for matchings versus C_4

摘录 / EXCERPT: Connected size Ramsey conjecture. For all positive integers n,

conjecture | open | MathDB 页面 | 原文线索

V2-050830 | MATHDB-354539

Kropholler's conjecture on almost invariant sets and tree actions

摘录 / EXCERPT: Kropholler's conjecture. If there is a proper H-almost invariant set A such that $A=AH$, then G has a non-trivial action on a tree in which H fixes a vertex v and every e...

conjecture | open | MathDB 页面 | 原文线索

V2-050831 | MATHDB-354571

The 2-balanced double tree decomposition conjecture

摘录 / EXCERPT: 2-balanced double tree decomposition conjecture. Any double tree has a 2-balanced double tree decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-050832 | MATHDB-354574

Chvátal's spanning closed 1-trail conjecture

摘录 / EXCERPT: Let a t-tough graph G satisfy $\omega(G\setminus S)\leq\max\\{1,\frac{1}{t}|S|\\}$ for every $S\subseteq V(G)$. A spanning closed 1-trail is a spanning closed trail meeting e...

conjecture | open | MathDB 页面 | 原文线索

V2-050833 | MATHDB-354575

The improved degree bound for spanning closed trails in edge-connected graphs

摘录 / EXCERPT: Let G be a k-edge-connected graph, meaning every edge cut of G has size at least k, with $k\geq 4$. A spanning closed trail is a closed trail containing every vertex of G...

conjecture | open | MathDB 页面 | 原文线索

V2-050834 | MATHDB-354576

The toughness condition for connected {2,4}-factors

摘录 / EXCERPT: Let G be a graph of order at least three, and write $\omega(G\setminus S)$ for the number of components of $G\setminus S$. A connected $\\{2,4\\}$-factor is a connected spanning...

conjecture | open | MathDB 页面 | 原文线索

V2-050835 | MATHDB-354577

The path-matrix conjecture for 2-connected graphs

摘录 / EXCERPT: The path-matrix conjecture. If M is 2-connected, then its path matrix has exactly one positive eigenvalue.

conjecture | open | MathDB 页面 | 原文线索

V2-050836 | MATHDB-354578

The path-energy conjecture for graphs with 2-connected components

摘录 / EXCERPT: The path-energy conjecture. If the graph M can be decomposed into k 2-connected components, then its path matrix has exactly k positive eigenvalues.

conjecture | open | MathDB 页面 | 原文线索

V2-050837 | MATHDB-354590

Membership conjecture for graph matrices in the copositive hierarchy

摘录 / EXCERPT: Let G be a graph with stability number $\alpha(G)$, and let A_G, I, and J denote its adjacency, identity, and all-ones matrices, respectively. Define ... By the Motzkin-Strau...

conjecture | open | MathDB 页面 | 原文线索

V2-050838 | MATHDB-354594

Second-largest eigenvalue conjecture for balanced complete-graph configurations

摘录 / EXCERPT: Let $d\geq 3$, and let $p:[n]\to{\mathbb R}^d$ satisfy $\|p(i)\|=1$ for every $i\in[n]$ and ... Assume that the image of p has size at least 3. Second-largest eigenvalue conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-050839 | MATHDB-354595

Large-eigenvalue sum conjecture for complete-graph Laplacians

摘录 / EXCERPT: Let $p:[n]\to{\mathbb R}^d$ be injective. Large-eigenvalue sum conjecture. The sum of the n largest eigenvalues of $L(K_n,p)$ is at least ... The conjecture improves the previously...

conjecture | open | MathDB 页面 | 原文线索

V2-050840 | MATHDB-354596

Essential tightness conjecture for the complete-graph rigidity parameter

摘录 / EXCERPT: Let $a_d(K_n)$ denote the parameter associated with d-dimensional algebraic connectivity for the complete graph K_n. Complete-graph tightness conjecture. If $n\geq 2d$, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-050841 | MATHDB-354597

Vanishing algebraic connectivity for regular graphs

摘录 / EXCERPT: For $d\geq 1$, let the maximum range over all $2d$-regular graphs G on n vertices be taken. Regular-graph vanishing conjecture. ... This is presented as a possible generalization...

conjecture | open | MathDB 页面 | 原文线索

V2-050842 | MATHDB-354598

Repeated-point scaling conjecture for the spectral gap

摘录 / EXCERPT: Let $p:[n]\to{\mathbb R}^d$ be injective, let G be a graph with vertex set $[n]$, and write $\lambda(G,p)=\lambda_{\binom{d+1}{2}+1}(L(G,p))$. For $k\geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-050843 | MATHDB-354609

Horak-Rosa conjecture on realizable and admissible chord multisets

摘录 / EXCERPT: Consider n equally spaced points on a circle and write $m=\lfloor n/2\rfloor$. A multiset is represented as $[\ell_1,\ldots,\ell_m]$, where ℓ_j counts chords of type j. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-050844 | MATHDB-354612

The four-color neighborhood conjecture for near-triangulations

摘录 / EXCERPT: Let G be a near-triangulation, meaning a plane graph whose bounded faces are triangles, and let $N(v)$ denote the neighborhood of a vertex v. Four-color neighborhood conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-050845 | MATHDB-354613

The planar total-domination conjecture for minimum degree three

摘录 / EXCERPT: Let G be a planar graph, let $d_t(G)$ denote its total domatic number, and let the minimum degree of G be the least degree of any vertex. Planar total-domination conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-050846 | MATHDB-354632

Edge-negative association conjecture for uniform random forests

摘录 / EXCERPT: The edge-negative association conjecture. For every such graph G, the random forest Φ is edge-negatively associated. This is a longstanding problem concerning negative depen...

conjecture | open | MathDB 页面 | 原文线索

V2-050847 | MATHDB-354640

The density conjecture for \mathbb{Z}_3-flow-critical graphs

摘录 / EXCERPT: A graph G is $\mathbb{Z}_3$-flow-critical if it is flow-critical with respect to nowhere-zero $\mathbb{Z}_3$-flows, as defined in the source. Density conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-050848 | MATHDB-354641

Jaeger et al.'s group-connectivity analogue of the 3-flow conjecture

摘录 / EXCERPT: For an abelian group Γ, a graph G is Γ-connected if (G,β) has a nowhere-zero flow for every Γ-boundary β. Jaeger et al.'s group-connectivity conj...

conjecture | open | MathDB 页面 | 原文线索

V2-050849 | MATHDB-354659

Canonical Jordan recoverability criterion for subcategories of representations of A_n-type quivers

摘录 / EXCERPT: Let $d4ac$ be an A_n-type quiver, and let $d524$ be a subcategory of $d53d\operatorname{rep}(d4ac)$. For strings, write $\mathsf{Supp}_0(\rho)$ for the relevant vertex support an...

conjecture | open | MathDB 页面 | 原文线索

V2-050850 | MATHDB-354687

Alon-Friedland-Kalai conjecture on modulo k-regular subgraphs

摘录 / EXCERPT: Alon-Friedland-Kalai conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-050851 | MATHDB-354706

Egerváry decomposition characterization and independence-number conjecture

摘录 / EXCERPT: Let G be a graph with a perfect matching and a Deming decomposition ... A Deming-K_4 subgraph is denoted by K_j, the residual subgraph by R, and M_R is the matching induced...

conjecture | open | MathDB 页面 | 原文线索

V2-050852 | MATHDB-354717

The almost-peripheral-vertex conjecture for graph products

摘录 / EXCERPT: Let G and H be graphs, and let $G\square H$ denote their Cartesian product. For a vertex u of a graph, let $\epsilon(u)$ be its eccentricity, let $\operatorname{d}(u,v)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-050853 | MATHDB-354719

The cone-type minimality conjecture for spherical triangulations

摘录 / EXCERPT: Cone-type minimality conjecture. If τ has no bad cycle, then the cone-type triangulation at a degree-6 vertex is minimal, and

conjecture | open | MathDB 页面 | 原文线索

V2-050854 | MATHDB-354721

Schroeder's genus conjecture for geodesic surfaces

摘录 / EXCERPT: Let X be a geodesic surface of genus g, and let $c(X)$ denote its cop number. Schroeder's genus conjecture. ... The paper compares this conjecture with a linear upper bound and a...

conjecture | open | MathDB 页面 | 原文线索

V2-050855 | MATHDB-354746

Kleitman-Winston conjecture on enumerating H-free graphs

摘录 / EXCERPT: Let H be a fixed bipartite graph that contains a cycle, and let $\operatorname{ex}(n,H)$ denote the maximum number of edges in an n-vertex graph containing no copy of H. Klei...

conjecture | open | MathDB 页面 | 原文线索

V2-050856 | MATHDB-354761

Fox-Luo-Wigderson's dependence-on-the-ground-graph conjecture for blowup Ramsey numbers

摘录 / EXCERPT: Fox-Luo-Wigderson's conjecture. There exists a graph H and integers $r,t\geq 2$ such that there are graphs $G_1,G_2,\dots$ with $G_i\stackrel{r}{\to}H$ for every i and

conjecture | open | MathDB 页面 | 原文线索

V2-050857 | MATHDB-354762

Cycle obstruction conjecture for ground-graph-independent blowup Ramsey numbers

摘录 / EXCERPT: Cycle obstruction conjecture. There exists an integer t such that, for every s, there is a graph G satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050858 | MATHDB-354807

Conjecture on residues of perfect-matching counts in random regular bipartite multigraphs

摘录 / EXCERPT: Fix an integer $q\ge 3$. For an even sufficiently large integer n, let G be a random q-regular bipartite multigraph on n vertices, and let the number of its perfect matchin...

conjecture | open | MathDB 页面 | 原文线索

V2-050859 | MATHDB-354817

Gu's Laplacian spectral Hamiltonicity conjecture

摘录 / EXCERPT: Let G be the graph under consideration, with Laplacian eigenvalues μ_2 and μ_n, and say that G is Hamiltonian when it contains a Hamilton cycle. Gu's conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-050860 | MATHDB-354818

Hamiltonicity conjecture for weakly pseudorandom graphs

摘录 / EXCERPT: Let a weakly (n,β)-graph be a graph in the sense defined in the source, and let G be Hamiltonian when it contains a Hamilton cycle. Hamiltonicity conjecture for weakly...

conjecture | open | MathDB 页面 | 原文线索

V2-050861 | MATHDB-354823

Conjecture on edge coloring signed complete bipartite graphs with equal parts

摘录 / EXCERPT: Let $r \in \mathbf{N}$ and let S be a signed complete bipartite graph $K_{r,r}$. Write $\chi'(S)$ for its signed edge-chromatic number and $\Delta(S)$ for its maximum degree. Sig...

conjecture | open | MathDB 页面 | 原文线索

V2-050862 | MATHDB-354848

The 3-connected subgraph conjecture for chromatic-critical vertex pairs

摘录 / EXCERPT: Let $m\ge4$, and let G be a graph with chromatic number at least m. An optimal coloring of G is a coloring using exactly $\chi(G)$ colors. The chromatic-critical-pair conject...

conjecture | open | MathDB 页面 | 原文线索

V2-050863 | MATHDB-354851

Le's polynomial bound on induced paths in $\mathcal{O}_2$-free graphs

摘录 / EXCERPT: Le's induced-path conjecture. There is a constant c such that every $\mathcal{O}_2$-free n-vertex graph has at most n^c distinct induced paths.

conjecture | open | MathDB 页面 | 原文线索

V2-050864 | MATHDB-354853

The almost-stable Kneser hypergraph colorability bound

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathcal G$ be a hypergraph over the ground set $[n]$, and let $r\geq s\geq 2$. Write $\mathcal G_{\widetilde{\{s\text{-stable}\}}}$ for the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-050865 | MATHDB-354859

Elekes' tic-tac-toe conjecture for linear 3-uniform hypergraphs

摘录 / EXCERPT: Elekes' conjecture. If H contains no tic-tac-toe, then it has $o(n^2)$ edges. Equivalently, for every $\varepsilon>0$ there is an $n_0=n_0(\varepsilon)$ such that, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-050866 | MATHDB-354861

The classification conjecture for Laplacian-realizable sets $S_{\{i,n\}_n^m}$

摘录 / EXCERPT: For integers n, m, and i with $1\leqslant i<n$, let $S_{\{i,n\}_n^m}$ denote the multiset obtained from $\{0,1,\ldots,n\}$ by deleting i and n, and repeating m once, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-050867 | MATHDB-354872

Iterated quantum Hamiltonian reduction stable-equivalence conjecture

摘录 / EXCERPT: Let ${\mathfrak g}$ be a simple Lie algebra, let $({\mathfrak a}_1,\ldots,{\mathfrak a}_L)$ be ordered subalgebras whose direct sum is a subalgebra of ${\mathfrak g}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050868 | MATHDB-354873

Iterated W-algebra isomorphism conjecture for ordered partitions

摘录 / EXCERPT: Let $N=m_1s_1+\cdots+m_Ls_L$ with $0<s_1<\cdots<s_L$ and $m_i>0$, set $N_i=m_is_i$, and let f_i be principal nilpotent elements of ${\mathfrak {gl}}_{N_i}$. Let ... be the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-050869 | MATHDB-354888

Strong maximality and minimality conjecture for bounded-edge hypergraphs

摘录 / EXCERPT: Strong maximality and minimality conjecture. Every such hypergraph has a strongly maximal matching, a strongly minimal vertex cover, and a strongly minimal edge cover.

conjecture | open | MathDB 页面 | 原文线索

V2-050870 | MATHDB-354890

Aharoni-Lovász strong-perfectness conjecture for perfect graphs with finite independent sets

摘录 / EXCERPT: Aharoni-Lovász conjecture. Every perfect graph in which all independent sets are finite is strongly perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-050871 | MATHDB-354900

The cubic hypergraph bootstrap quadratic running-time conjecture

摘录 / EXCERPT: Let $M_4^3(n)$ be the maximum running time of the K_4^3-bootstrap process over all initial 3-uniform hypergraphs on n vertices. The paper gives a quadratic lower bound for th...

conjecture | open | MathDB 页面 | 原文线索

V2-050872 | MATHDB-354919

Cvetković-Rowlinson-Simić conjecture on the signless Laplacian spread

摘录 / EXCERPT: Cvetković-Rowlinson-Simić conjecture. Over all connected graphs on $n\geq6$ vertices, $s_Q(G)$ is maximized by K_{n-1}^+ and minimized by P_n and, when n is odd, by C_n.

conjecture | open | MathDB 页面 | 原文线索

V2-050873 | MATHDB-354922

Vertex attainment conjecture for the distance-to-conformal-radius ratio

摘录 / EXCERPT: Vertex attainment conjecture. The maximum

conjecture | open | MathDB 页面 | 原文线索

V2-050874 | MATHDB-354927

Exponential decay conjecture for the Hamming-cube theta number

摘录 / EXCERPT: Hamming-cube decay conjecture. For every fixed $c>2$, the quantity

conjecture | open | MathDB 页面 | 原文线索

V2-050875 | MATHDB-354928

The graph complement conjecture for minimum rank and maximum nullity

摘录 / EXCERPT: For a graph G on n vertices, let $S(G)$ be the set of real symmetric matrices whose off-diagonal zero pattern is prescribed by adjacency in G. Define $mr(G)$ as the minimum r...

conjecture | open | MathDB 页面 | 原文线索

V2-050876 | MATHDB-354951

Inherited correspondence between disjoint cycles and non-zero eigenvalues

摘录 / EXCERPT: Inherited correspondence conjecture. The maximum number of edges in a subgraph H of G in which every collection of disjoint cycles covers at most k vertices equals the larges...

conjecture | open | MathDB 页面 | 原文线索

V2-050877 | MATHDB-354975

Conjecture on vertex-gluing closed nonseparating planar graphs

摘录 / EXCERPT: Vertex-gluing conjecture. The resulting graph is IPPI3L.

conjecture | open | MathDB 页面 | 原文线索

V2-050878 | MATHDB-354976

Conjecture on minor-minimal IPPI3L graphs from vertex-gluing

摘录 / EXCERPT: Minor-minimal vertex-gluing conjecture. The resulting graph is minor-minimal IPPI3L.

conjecture | open | MathDB 页面 | 原文线索

V2-050879 | MATHDB-354982

Duchêne et al.'s ultimate periodicity conjecture for segment scores

摘录 / EXCERPT: A segment of n vertices starting with a black vertex is denoted by S_n, and a segment starting with a white vertex by S_{-n}. Let $Ls(S_n)$ and $Rs(S_n)$ denote the left and...

conjecture | open | MathDB 页面 | 原文线索

V2-050880 | MATHDB-354983

First-player conjecture for rectangular grid instances of INFLUENCE

摘录 / EXCERPT: Let $G_{n,m}$ be the rectangular grid with n rows and m columns, with alternated black and white vertices and a black vertex in the top-left corner. Let $Ls(G_{n,m})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050881 | MATHDB-355025

Volkman's cactus domination-irredundance conjecture

摘录 / EXCERPT: Let G be a cactus, that is, a graph in which every edge belongs to at most one cycle. Write $\gamma(G)$ for its domination number and $ir(G)$ for its irredundance number. Volkman...

conjecture | solved | MathDB 页面 | 原文线索

V2-050882 | MATHDB-355026

The degree-dependent partial-isolation bound conjecture

摘录 / EXCERPT: Let G be a graph with maximum degree Δ. For a positive integer k, let $\iota_k(G)$ denote the minimum size of a vertex set whose closed neighborhood leaves no induced co...

conjecture | open | MathDB 页面 | 原文线索

V2-050883 | MATHDB-355027

The asymptotic domination-irredundance bound conjecture

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta\geq 2$. Write $\gamma(G)$ for its domination number and $ir(G)$ for its irredundance number. The asymptotic domination-irredundance c...

conjecture | open | MathDB 页面 | 原文线索

V2-050884 | MATHDB-355044

The convex lower-bound conjecture for optimal resistor networks

摘录 / EXCERPT: Convex lower-bound conjecture. For all x,

conjecture | open | MathDB 页面 | 原文线索

V2-050885 | MATHDB-355045

The leaf-density phase-transition conjecture for optimal resistor networks

摘录 / EXCERPT: Leaf-density phase-transition conjecture. There is a threshold α_0 on the average degree such that below it all optimal graphs have a positive proportion of leaves, whereas...

conjecture | open | MathDB 页面 | 原文线索

V2-050886 | MATHDB-355046

The maximum-degree phase-transition conjecture for optimal resistor networks

摘录 / EXCERPT: Maximum-degree phase-transition conjecture. There is a threshold α_1 on the average degree such that below it an optimal graph has a vertex of degree $\Omega(n)$, whereas ab...

conjecture | open | MathDB 页面 | 原文线索

V2-050887 | MATHDB-355131

Kelly's conjecture on Hamiltonian decompositions of regular tournaments

摘录 / EXCERPT: A regular tournament is a tournament in which every vertex has the same indegree and outdegree. Kelly's conjecture. Every regular tournament on $n=2k+1$ vertices has a decompositio...

conjecture | open | MathDB 页面 | 原文线索

V2-050888 | MATHDB-355140

Improved upper bounds for Staller-Maker-Breaker domination numbers

摘录 / EXCERPT: Improved SMBD-number bounds.

conjecture | open | MathDB 页面 | 原文线索

V2-050889 | MATHDB-355143

The polylogarithmic conjecture for edge-ordered forests

摘录 / EXCERPT: An edge-ordered graph is a finite simple graph equipped with a linear order on its edges; its extremal function ${\mathrm{ex}}_<(n,H)$ is the maximum number of edges in an n-vert...

conjecture | open | MathDB 页面 | 原文线索

V2-050890 | MATHDB-355144

The general structure conjecture for higher level Zhu algebras of the Heisenberg vertex operator algebra

摘录 / EXCERPT: Higher level Zhu algebra conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-050891 | MATHDB-355158

Product formula for successive vertex orderings of the line graph of a bipartite 3-uniform hypergraph

摘录 / EXCERPT: Product-formula conjecture. The number of successive vertex orderings satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050892 | MATHDB-355160

Hypergraph Hadwiger conjecture for the chromatic number

摘录 / EXCERPT: For an integer $t\geq 2$, let $h(t)$ be the largest chromatic number of a hypergraph with no K_t-minor; the paper establishes that this quantity exists. A hypergraph is K_t-min...

conjecture | open | MathDB 页面 | 原文线索

V2-050893 | MATHDB-355184

Random threshold conjecture for Steiner designs

摘录 / EXCERPT: Let $t,k,\lambda\in\mathbb N$ with $k>t$, let V be an n-element set satisfying the divisibility conditions $\binom{k-i}{t-i}\mid\lambda\binom{n-i}{t-i}$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-050894 | MATHDB-355189

The cubic graph girth bound conjecture

摘录 / EXCERPT: Let $g(n)$ denote the largest girth among all cubic graphs on n vertices. Girth bound conjecture. There is a constant $\epsilon_0>0$ such that ... The Moore bound gives only...

conjecture | open | MathDB 页面 | 原文线索

V2-050895 | MATHDB-355212

The balanced-digraph Laplacian spread gap conjecture

摘录 / EXCERPT: Let Γ be an unweighted balanced digraph, and let $\operatorname{sp}(\Gamma)$ denote its Laplacian spread. The balanced-digraph gap conjecture. There is no balanced digraph s...

conjecture | open | MathDB 页面 | 原文线索

V2-050896 | MATHDB-355213

The balanced-digraph algebraic-connectivity inequality conjecture

摘录 / EXCERPT: Let Γ be an unweighted balanced digraph of order $n\geq 2$. Let $\alpha(\Gamma)$ denote its algebraic connectivity, and set $x=\alpha(\Gamma)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-050897 | MATHDB-355214

The polygonal-digraph endpoint characterization conjecture

摘录 / EXCERPT: Let Γ be an unweighted polygonal digraph of order $n\geq 2$, and let $\operatorname{sp}(\Gamma)$ denote its Laplacian spread. A k-imploding star is the digraph family name...

conjecture | open | MathDB 页面 | 原文线索

V2-050898 | MATHDB-355215

The linear-slope-one bound conjecture for Laplacian spread

摘录 / EXCERPT: Let the Laplacian spread of a digraph be bounded by the bound referred to in the source as equation (unwght-spread-bound). The linear-slope-one bound conjecture. That bound can be...

conjecture | open | MathDB 页面 | 原文线索

V2-050899 | MATHDB-355217

Strict middle-graph increase conjecture for outer-connected domination

摘录 / EXCERPT: Let G be a graph of order $n\ge 2$. The strict middle-graph increase conjecture asserts that ... The paper notes that this inequality holds for trees and for several standard graph...

conjecture | open | MathDB 页面 | 原文线索

V2-050900 | MATHDB-355218

Line-middle graph inequality conjecture for outer-connected domination

摘录 / EXCERPT: Let G be a graph of order $n\ge 2$, and let $L(G)$ and $M(G)$ denote its line graph and middle graph, respectively. The line-middle graph inequality conjecture asserts that ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-050901 | MATHDB-355220

Strong vertex decomposability of infinite-type spherical subword complexes

摘录 / EXCERPT: Let $\mathcal{S}(Q,\pi)$ be a spherical subword complex of infinite type. Strong vertex-decomposability conjecture. The complex $\mathcal{S}(Q,\pi)$ is strongly vertex decomposable...

conjecture | open | MathDB 页面 | 原文线索

V2-050902 | MATHDB-355222

The non-uniform 1-factorability congruence conjecture

摘录 / EXCERPT: Let $cmathcal{L}$ be a set of distinct positive integers whose largest element is k. The non-uniform set system $cbinom{[n]}{cmathcal{L}}$ consists of the subsets of $[n]$ whose...

conjecture | open | MathDB 页面 | 原文线索

V2-050903 | MATHDB-355252

Conjecture on small regular subgraphs with prescribed average degree

摘录 / EXCERPT: Let $s\geq 3$ be an integer. For sufficiently large n, let G be an n-vertex graph with average degree at least d, where $C\log\log n\leq d\leq n^{\frac{s-2}{s}}$. Regular-s...

conjecture | open | MathDB 页面 | 原文线索

V2-050904 | MATHDB-355253

Conjecture on bounded-size regular subgraphs

摘录 / EXCERPT: Let $s\geq 3$ be an integer and let $\varepsilon>0$. Bounded-size regular-subgraph conjecture. There is a positive integer t such that, for all sufficiently large n, every n-...

conjecture | open | MathDB 页面 | 原文线索

V2-050905 | MATHDB-355255

Ahanjideh-Ekim-Yıldız conjecture for intermediate triangle-free extremal values

摘录 / EXCERPT: Intermediate-value conjecture. For $7\leq d<i<Z(d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-050906 | MATHDB-355256

Ahanjideh-Ekim-Yıldız general formula for triangle-free extremal graphs in terms of $Z(d)$

摘录 / EXCERPT: General formula in terms of $Z(d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-050907 | MATHDB-355257

Ahanjideh-Ekim-Yıldız conjecture for $Z(d)$

摘录 / EXCERPT: $Z(d)$ conjecture. For odd $d\geq 21$,

conjecture | open | MathDB 页面 | 原文线索

V2-050908 | MATHDB-355268

The asymptotic minimum-degree characterization for H-factors with acyclic-partition hcf 1

摘录 / EXCERPT: The asymptotic H-factor conjecture. Given $\mu>0$, $h\in\mathbb{N}$, and an h-vertex graph H with $\mathrm{hcf}(H,\mathrm{AP})=1$, there exists $\alpha>0$ such that, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-050909 | MATHDB-355274

The four-color bound for locally irregular edge colorings of colorable graphs

摘录 / EXCERPT: Four-color bound. Every colorable connected graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050910 | MATHDB-355289

Cheng-Kano-Wang conjecture on PC Hamilton paths in balanced bipartite graphs

摘录 / EXCERPT: Cheng-Kano-Wang conjecture. If $K_{n,n}$ contains a PC 1-path-cycle factor, then $K_{n,n}$ contains a PC Hamilton path.

conjecture | open | MathDB 页面 | 原文线索

V2-050911 | MATHDB-355290

Gutin-Kim conjecture on recognizing PC Hamilton cycles

摘录 / EXCERPT: Gutin-Kim conjecture. It is polynomial-time checkable whether an edge-colored complete graph contains a PC Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-050912 | MATHDB-355293

The four-color bound for locally irregular edge colorings

摘录 / EXCERPT: A graph is locally irregular if the degrees of the end-vertices of every edge are distinct. An edge coloring is locally irregular if every color induces a locally irregular subgrap...

conjecture | open | MathDB 页面 | 原文线索

V2-050913 | MATHDB-355307

The average-degree energy lower-bound conjecture for non-singular graphs

摘录 / EXCERPT: Average-degree energy conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-050914 | MATHDB-355310

Ning-Lu-Wang spectral-radius conjecture for edge connectivity

摘录 / EXCERPT: Let $\mathcal{A}_{n}^{\kappa',\delta}$ be the set of graphs of order n with minimum degree δ and edge connectivity κ'. Let $B_{n,\delta+1}^{\kappa'}$ be the graph o...

conjecture | open | MathDB 页面 | 原文线索

V2-050915 | MATHDB-355363

The conjecture on mixed blowup colorings

摘录 / EXCERPT: Let $r(k)$ denote the diagonal two-color Ramsey number, and let $K_{r(k)-1}[2]$ be the blowup of $K_{r(k)-1}$ in which each vertex is replaced by two vertices. A mixed blowup color...

conjecture | open | MathDB 页面 | 原文线索

V2-050916 | MATHDB-355364

The generalized-lollipop conjecture for Ramsey multiplicity

摘录 / EXCERPT: A graph H is a (k,t)-generalized lollipop if it has t vertices and contains a K_k whose deletion yields a forest; equivalently, it is obtained from K_k by attaching trees...

conjecture | open | MathDB 页面 | 原文线索

V2-050917 | MATHDB-355365

The lollipop conjecture for Ramsey multiplicity

摘录 / EXCERPT: For integers $k\geq4$ and sufficiently large t, let $L_{k,t}$ be the graph obtained from K_k by appending to one vertex a path with $t-k$ edges. A graph is a two-color bonbon w...

conjecture | open | MathDB 页面 | 原文线索

V2-050918 | MATHDB-355390

The forest characterization conjecture for the strong Erdős-Hajnal property

摘录 / EXCERPT: Forest characterization conjecture. If a tournament H has an ordering of its vertices for which the backward arc digraph is a forest, then H has the strong Erdős-Hajnal propert...

conjecture | open | MathDB 页面 | 原文线索

V2-050919 | MATHDB-355409

Logarithmic improvement conjecture for defective colouring of linear hypergraphs

摘录 / EXCERPT: Defective-colouring conjecture. Every $(r+1)$-uniform linear hypergraph with maximum degree at most Δ has a d-defective colouring with

conjecture | open | MathDB 页面 | 原文线索

V2-050920 | MATHDB-355412

Conjecture on controllable graphs with complete auxiliary graphs

摘录 / EXCERPT: For a graph G, let $\Omega(G)$ denote the auxiliary graph used in the paper's generalized spectral characterization, and call G controllable when it satisfies the controllabili...

conjecture | open | MathDB 页面 | 原文线索

V2-050921 | MATHDB-355421

The classification conjecture for essentially 4-edge-connected pseudo 2-factor isomorphic cubic bipartite graphs

摘录 / EXCERPT: A pseudo 2-factor isomorphic graph is a graph admitting a 2-factor such that the parity of the number of cycles is the same for all its 2-factors. A graph is cubic if every v...

conjecture | open | MathDB 页面 | 原文线索

V2-050922 | MATHDB-355425

Vanishing Ones Conjecture for transformed triangular grids

摘录 / EXCERPT: Let $T_{n,k}$ denote the triangular grid obtained after k transformations from the n-grid, and let $T_{n,n-s}^s$ denote an s-subgrid of $T_{n,n-s}$. For an edge, its edge val...

conjecture | open | MathDB 页面 | 原文线索

V2-050923 | MATHDB-355431

Keevash-Sudakov conjecture on monochromatic copies and Turán numbers

摘录 / EXCERPT: Let H be a graph, and let $f(n,H)$ denote the maximum number of edges not contained in any monochromatic copy of H in a 2-edge-coloring of the complete graph K_n. Keevash-S...

conjecture | open | MathDB 页面 | 原文线索

V2-050924 | MATHDB-355443

Conjecture that color-critical forbidden graphs inherit weak Turán-goodness

摘录 / EXCERPT: Let F be a graph with chromatic number $r+1$ and a color-critical edge, meaning an edge whose removal decreases the chromatic number. Let H be weakly F-Turán-good if, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-050925 | MATHDB-355444

Conjecture on separating asymptotic weak Turán-goodness by blow-up containment

摘录 / EXCERPT: Let F and F' be graphs with chromatic number $r+1$. A graph H is asymptotically weakly F-Turán-good if ${\mathrm{ex}}(n,H,F)=(1+o(1))\mathcal N(H,T)$ for some complete r-...

conjecture | open | MathDB 页面 | 原文线索

V2-050926 | MATHDB-355449

Nonfull-queue routing property for minimum loss policies

摘录 / EXCERPT: Consider the static-routing optimization problem ... where $\widetilde{\boldsymbol{\Pi}}^{\textup{SR}}$ is the class of stationary randomized policies that route every arrival to som...

conjecture | open | MathDB 页面 | 原文线索

V2-050927 | MATHDB-355479

The bunkbed conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite connected graph and let $T\subset V$. The bunkbed graph G_T has vertex set $V\times\{0,1\}$, two horizontal copies of every edge of G, and a vertical...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050928 | MATHDB-355539

Erdős's jump conjecture for hypergraph Turán densities

摘录 / EXCERPT: For an integer $r\ge 2$, an r-graph is an r-uniform hypergraph. A real number $\alpha\in[0,1]$ is a jump for r if there exists a constant $c>0$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-050929 | MATHDB-355540

Diagonalizability conjecture for vertex-primitive arc-transitive digraphs

摘录 / EXCERPT: Diagonalizability conjecture. Every vertex-primitive arc-transitive digraph is diagonalizable.

conjecture | open | MathDB 页面 | 原文线索

V2-050930 | MATHDB-355542

Quasi-polynomial upper-bound conjecture for asymmetric semi-algebraic Ramsey numbers

摘录 / EXCERPT: Quasi-polynomial upper-bound conjecture. There exists a constant $c=c(s,d,D,m)>0$ such that, for every sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-050931 | MATHDB-355543

Dimension-degree tower-height conjecture for semi-algebraic Ramsey numbers

摘录 / EXCERPT: Dimension-degree tower-height conjecture. There exists a function $k:\mathbb{N}\to\mathbb{N}$ such that, for integers r,d,D,m, there is a constant $c=c(r,d,D,m)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050932 | MATHDB-355576

The common blow-up conjecture for K4-free graphs

摘录 / EXCERPT: A blow-up of a graph F is obtained by replacing each vertex of F by a non-empty independent set, possibly of a different size, and joining two such sets exactly when their corr...

conjecture | open | MathDB 页面 | 原文线索

V2-050933 | MATHDB-355592

Characteristic-polynomial conjecture for parameter matrices of hypergraph transversals

摘录 / EXCERPT: Let $\mathbb{S}$ be the parameter matrix of a k-transversal in a d-uniform r-regular hypergraph $\mathcal{G}$. Let ξ be a d-th primitive root of unity. Characteristic-p...

conjecture | open | MathDB 页面 | 原文线索

V2-050934 | MATHDB-355617

The maximal Euler characteristic conjecture for multipath complexes

摘录 / EXCERPT: Maximal Euler characteristic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-050935 | MATHDB-355637

The parity conjecture for Game of Cycles gameboards without degree-one vertices

摘录 / EXCERPT: Parity conjecture. Player 1 has a winning strategy when the gameboard has an odd number of total edges, and Player 2 has a winning strategy when it has an even number of total edge...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050936 | MATHDB-355661

The lower-bound conjecture for DET:IC density in cubic graphs

摘录 / EXCERPT: Let G be a cubic graph, and let $\textrm{DET:IC\%}(G)$ denote the minimum density of a deterministic error-detecting identifying code in G. Lower-bound conjecture. ... The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-050937 | MATHDB-355673

Joshi-Shiu's multistationarity conjecture for one-dimensional reaction networks

摘录 / EXCERPT: Joshi-Shiu's conjecture. A reaction network with one-dimensional stoichiometric subspace and more than one source complex has the capacity for multistationarity if and only if it h...

conjecture | open | MathDB 页面 | 原文线索

V2-050938 | MATHDB-355687

Minimum length of 2-uniformity networks

摘录 / EXCERPT: A 2-uniformity network is a network of lazy transpositions whose output permutation is uniformly distributed on every subset of two positions; let $U_2(n)$ denote the minimum lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-050939 | MATHDB-355695

The intersecting-family success-probability conjecture

摘录 / EXCERPT: Intersecting-family conjecture. The success probability $p_{\mathrm{intersecting}}(t)$ tends to 0 as t grows. This is stronger than the dictatorship version because every dicta...

conjecture | open | MathDB 页面 | 原文线索

V2-050940 | MATHDB-355697

The positively correlated random-subgraph independence conjecture

摘录 / EXCERPT: Random-subgraph independence conjecture. $\epsilon^*(\alpha)>0$ for all $\alpha\in(0,1/2)$. This conjecture would provide a uniform drop in the largest independent-set density afte...

conjecture | open | MathDB 页面 | 原文线索

V2-050941 | MATHDB-355698

The binomial random-subset independence conjecture

摘录 / EXCERPT: Binomial random-subset independence conjecture. There exists a monotone non-decreasing function $\epsilon^{**}:(0,1/2)\to(0,1/2)$, positive throughout its domain, such that if G...

conjecture | open | MathDB 页面 | 原文线索

V2-050942 | MATHDB-355699

The Bollobás-Erdős-Tuza blocker conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, let $\alpha(G)$ be the size of a maximum independent set, and let $h(G)$ be the minimum cardinality of a vertex set intersecting every maximum i...

conjecture | open | MathDB 页面 | 原文线索

V2-050943 | MATHDB-355700

The partitioned blocker conjecture for almost-maximum independent sets

摘录 / EXCERPT: Let $\alpha>0$. There exist a positive integer B, numbers $\tau>0$, $\delta>0$, and $\epsilon>0$ with $\tau4^{-B}>\delta$, such that for all sufficiently large n, if G is a g...

conjecture | open | MathDB 页面 | 原文线索

V2-050944 | MATHDB-355702

Minimum-order conjecture for odd and even D(k) graphs

摘录 / EXCERPT: Minimum-order conjecture. For all $k\geq 3$, there exists no $D^{odd}_{k,n}$ for all $n < 4k-3$ and $D^{even}_{k,n}$ for all $n < 3k$.

conjecture | open | MathDB 页面 | 原文线索

V2-050945 | MATHDB-355710

Mao-Wang's determinant conjecture for walk matrices of rooted products with paths

摘录 / EXCERPT: Mao-Wang's conjecture. For any positive integer $m\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-050946 | MATHDB-355717

Quantitative restricted partition conjecture for graphs

摘录 / EXCERPT: For $\varepsilon>0$, a graph G is (N,ε)-restricted if its vertex set can be partitioned into at most N subsets that are ε-restricted in G, where a sub...

conjecture | open | MathDB 页面 | 原文线索

V2-050947 | MATHDB-355718

The asymptotic formula for cliques in a random uniform hypergraph

摘录 / EXCERPT: Asymptotic clique-count conjecture. The expected number of cliques satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-050948 | MATHDB-355736

Odd-cycle Maker-Breaker metric resolving game conjecture

摘录 / EXCERPT: Odd-cycle conjecture. For every odd integer $n\ge5$,

conjecture | open | MathDB 页面 | 原文线索

V2-050949 | MATHDB-355756

The 1-2-3 Conjecture on neighbor-sum-distinguishing edge colorings

摘录 / EXCERPT: Let G be a graph with no isolated edges. An edge coloring $f:E(G)\to\{1,2,\dots,k\}$ assigns a sum $\sigma(x)=\sum_{x\in e}f(e)$ to each vertex x; adjacent vertices are disting...

conjecture | open | MathDB 页面 | 原文线索

V2-050950 | MATHDB-355759

The (2,2) Conjecture for graph edge colorings

摘录 / EXCERPT: A (p,q)-coloring of a graph G is a decomposition into at most p subgraphs such that, in each subgraph, neighboring vertices can be distinguished by sums using at most q col...

conjecture | open | MathDB 页面 | 原文线索

V2-050951 | MATHDB-355760

The multigraph formulation of the 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{M}^{[3]}(G)$ be the family of multigraphs obtained from G by edge multiplication with edge multiplicities at most 3. A multigraph is local...

conjecture | open | MathDB 页面 | 原文线索

V2-050952 | MATHDB-355761

The multigraph formulation of the (2,2) Conjecture

摘录 / EXCERPT: Let G be a connected graph of order $n\geq 4$. Let $\mathcal{M}^{[2]}(H)$ denote the family of multigraphs obtained from a graph H by edge multiplication with edge multipliciti...

conjecture | open | MathDB 页面 | 原文线索

V2-050953 | MATHDB-355772

The integral-roots characterization of free connected subgraph arrangements

摘录 / EXCERPT: Let G be a connected graph, let $\mathcal{A}_G$ be its connected subgraph arrangement, and let $\chi(\mathcal{A}_G)$ denote the characteristic polynomial of this arrangement. Int...

conjecture | open | MathDB 页面 | 原文线索

V2-050954 | MATHDB-355799

Elliott-Rödl's hypertree embedding conjecture for Steiner triple systems

摘录 / EXCERPT: A hypertree is a 3-uniform hypergraph in which every two vertices are connected by a unique path. A Steiner triple system is a 3-uniform hypergraph in which every pair of verti...

conjecture | open | MathDB 页面 | 原文线索

V2-050955 | MATHDB-355810

Beaton-Brown average dominating-set order conjecture

摘录 / EXCERPT: Let G be a graph with $n\geq 2$ vertices, and let $\operatorname{avd}(G)$ denote its average order of dominating sets. If G has no isolated vertices, then Beaton-Brown's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-050956 | MATHDB-355812

Morris-Saxton's balanced supersaturation conjecture for bipartite graphs

摘录 / EXCERPT: Morris-Saxton's conjecture. The stated balanced supersaturation bound should hold for every bipartite graph containing a cycle. Its validity would yield enumeration results for H...

conjecture | open | MathDB 页面 | 原文线索

V2-050957 | MATHDB-355813

Ferber-McKinley-Samotij's conjecture on nontrivial extremal growth

摘录 / EXCERPT: Ferber-McKinley-Samotij's conjecture. There exists an $\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-050958 | MATHDB-355828

Conlon-Fox-Sudakov vertex-deletion conjecture for Ramsey numbers

摘录 / EXCERPT: Let $r(G)$ denote the two-color Ramsey number of a graph G. Let G be a graph, and let H be obtained from G by deleting a single vertex. Conlon-Fox-Sudakov conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-050959 | MATHDB-355829

Polynomial vertex-deletion sensitivity conjecture for Ramsey numbers

摘录 / EXCERPT: Polynomial sensitivity conjecture. There is some $c>0$ such that $f(n)\geq n^c$ for all sufficiently large n.

conjecture | open | MathDB 页面 | 原文线索

V2-050960 | MATHDB-355830

Average-case deck conjecture for vertex-deleted Ramsey numbers

摘录 / EXCERPT: For a graph G, let $D(G)$ be the multiset of all vertex-deleted induced subgraphs of G. Average-case deck conjecture. There exists an absolute constant $c>0$ such that, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-050961 | MATHDB-355837

Szemerédi-Petruska's sharp bound for (n,m)-structures

摘录 / EXCERPT: Let ${\cal N} = \{N_1,\ldots,N_\ell\}$ be a collection of k-subsets of a finite set V with $|V|=n$, where $\ell\geq 2$, $k\geq 3$, and $m=n-k$. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-050962 | MATHDB-355849

The star conjecture for maximum Wiener-entropy among trees

摘录 / EXCERPT: Let T be a tree of order $n\geq 5$, and let S_n denote the star on n vertices. The Wiener-entropy $I_w(T)$ is defined from the vertex transmissions of T. The star conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-050963 | MATHDB-355850

The minimum eccentricity-entropy conjecture for graphs

摘录 / EXCERPT: Let G be a graph of order n, and let $I_{ecc}(G)$ denote its eccentricity-entropy. An extremal graph is one attaining the minimum value of this entropy. The minimum eccentricit...

conjecture | open | MathDB 页面 | 原文线索

V2-050964 | MATHDB-355851

The fixed-diameter extremal tree conjecture for distance entropies

摘录 / EXCERPT: Let T be a tree of order n and diameter $d\ll n$. Let $I_{f_s}(T)$ and $I_{f_e}(T)$ be the distance-based entropies associated with the functionals f_s and f_e. The fixed-d...

conjecture | open | MathDB 页面 | 原文线索

V2-050965 | MATHDB-355852

The asymptotic broom and $G_{n,k,1}$ conjecture for minimum Wiener-entropy

摘录 / EXCERPT: For integers n,k,j, let $G_{n,k,j}$ be the graph formed from a path P_k and a clique K_{n-k} by joining one end vertex of the path to j vertices of the clique. Let a broom...

conjecture | open | MathDB 页面 | 原文线索

V2-050966 | MATHDB-355860

Smith normal form conjecture for the walk matrix of the extended Dynkin graph

摘录 / EXCERPT: Smith normal form conjecture. The Smith normal form of $W(\tilde{D}_n)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-050967 | MATHDB-355869

Electrical canonical basis conjecture

摘录 / EXCERPT: Let $G_{d,n}$ be the degree-d homogeneous piece of the electrical network coordinate ring, let $\mathcal{C}^{(d)}_n$ index the proposed basis, let $E_{\mathbf{P}}$ denote its ele...

conjecture | open | MathDB 页面 | 原文线索

V2-050968 | MATHDB-355871

The two-edge rooted-path inherence conjecture

摘录 / EXCERPT: Two-edge rooted-path conjecture. The two-rooted path (P,s,t) is inherent.

conjecture | open | MathDB 页面 | 原文线索

V2-050969 | MATHDB-355872

The component-deletion conjecture for inherent two-rooted graphs

摘录 / EXCERPT: Component-deletion conjecture. If $(H+H',s,t)$ is inherent and s,t are in H, then (H,s,t) is also inherent.

conjecture | open | MathDB 页面 | 原文线索

V2-050970 | MATHDB-355880

Path extremal conjecture for random set zero forcing

摘录 / EXCERPT: Let G be an n-vertex graph, let P_n denote the path on n vertices, and let $B_p(G)$ be the random subset of vertices obtained by including each vertex independently with pr...

conjecture | open | MathDB 页面 | 原文线索

V2-050971 | MATHDB-355881

Path extremal conjecture for zero forcing sets of fixed size

摘录 / EXCERPT: Let G be an n-vertex graph, and let $z(G;k)$ denote the number of zero forcing sets of G having size k. Let P_n be the path on n vertices. Fixed-size zero forcing conje...

conjecture | open | MathDB 页面 | 原文线索

V2-050972 | MATHDB-355888

Maximum-degree forcing conjecture for graphs with large spectral radius

摘录 / EXCERPT: Let $s\geq 1$ be an integer, and let $f(s)$ be a function of s. Let G be a graph of order n with $n\geq f(s)$, and let $\rho(G)$ denote its spectral radius. Maximum-degree fo...

conjecture | open | MathDB 页面 | 原文线索

V2-050973 | MATHDB-355892

Semistrong chromatic index bound beyond complete bipartite graphs

摘录 / EXCERPT: Let G be a connected graph, let $K_{n,n}$ denote a complete bipartite graph, and let $\Delta(G)$ and $\chi_{\mathrm{ss}}'(G)$ denote the maximum degree and semistrong chromatic i...

conjecture | open | MathDB 页面 | 原文线索

V2-050974 | MATHDB-355893

Subcubic semistrong chromatic index conjecture

摘录 / EXCERPT: Let G be a connected graph with maximum degree 3, let $K_{3,3}$ be the complete bipartite graph, let the 5-prism be the graph named in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-050975 | MATHDB-355906

The regularity conjecture for powers of edge ideals of gap-free graphs

摘录 / EXCERPT: Let G be a gap-free graph and let $I=I(G)$ be its edge ideal. For each integer $s\ge 2$, write $I^{(s)}$ for the s-th symbolic power of I. Regularity conjecture. One has ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-050976 | MATHDB-355914

Hartnell's characterization of bipartite graphs partitionable into two-colored domination sets

摘录 / EXCERPT: Hartnell's characterization. If $V(G)$ can be partitioned into two-colored γ-sets, then G is the 4-cycle or G can be obtained from $K_{2t,2t}$ by removing the edges of...

conjecture | open | MathDB 页面 | 原文线索

V2-050977 | MATHDB-355915

Han-Zhao conjecture on vertex-disjoint cliques in multipartite graphs

摘录 / EXCERPT: Let $r\geq t\geq 3$ and $k\geq 2$, and let K_{t} denote the complete graph on t vertices. For positive integers $n_1,\ldots,n_r$, write $K_{n_1,\ldots,n_r}$ for the complete...

conjecture | open | MathDB 页面 | 原文线索

V2-050978 | MATHDB-355916

The simplified exponents-region conjecture for K-hop networks

摘录 / EXCERPT: Simplified exponents-region conjecture. The region $\mathcal{E}^*(R_1,\ldots,R_K,\epsilon_1,\ldots,\epsilon_K)$ is the set of all exponent tuples $(\theta_1,\ldots,\theta_K)$ for w...

conjecture | open | MathDB 页面 | 原文线索

V2-050979 | MATHDB-355921

Divisibility conjecture for finite components of triangular hitomezashi patterns

摘录 / EXCERPT: Divisibility conjecture. The number of vertices in a finite connected component of a triangular hitomezashi pattern must be divisible by 16.

conjecture | open | MathDB 页面 | 原文线索

V2-050980 | MATHDB-355929

The finite-alternet conjecture for distance-transitive digraphs

摘录 / EXCERPT: Let D be a distance-transitive, weakly descendant-homogeneous digraph of finite out-valency $m>1$ with property Z. An alternet is the subdigraph induced by the out-neighbours a...

conjecture | open | MathDB 页面 | 原文线索

V2-050981 | MATHDB-355932

The Davies conjecture on chi-boundedness of graphs containing a k-fan

摘录 / EXCERPT: For an integer $k\geq 1$, a k-fan is a graph formed from a path P by adding one vertex with exactly k neighbours on P. A family of graphs is chi-bounded if there is a funct...

conjecture | open | MathDB 页面 | 原文线索

V2-050982 | MATHDB-355939

Conjecture on super-maximized genera of graphs in surfaces

摘录 / EXCERPT: Super-maximization conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-050983 | MATHDB-355954

Liu-You-Tang-Liu conjecture on the exponential reduced Sombor index

摘录 / EXCERPT: Liu-You-Tang-Liu's conjecture. For every $T\in {\cal CT}_n$ with $n\geq 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-050984 | MATHDB-355963

Robust Hamilton-power conjecture at the minimum-degree threshold

摘录 / EXCERPT: Robust Hamilton-power conjecture. For every k there is a constant C such that, if $p\geq Cn^{-1/k}$, then every n-vertex graph G with $\delta(G)\geq \tfrac{k}{k+1}n$ has th...

conjecture | open | MathDB 页面 | 原文线索

V2-050985 | MATHDB-355964

Madani-Kiani's maximal-clique regularity conjecture for binomial edge ideals

摘录 / EXCERPT: Let G be a simple graph, let S be the polynomial ring used to define its binomial edge ideal J_G, and let $c(G)$ be the number of maximal cliques of G. Madani-Kiani's conje...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-050986 | MATHDB-355966

Linear-type conjecture for trees and unicyclic graphs

摘录 / EXCERPT: Let G be a graph, let J_G be its binomial edge ideal, and say that J_G is of linear type when its Rees algebra is generated by the degree-one part in the standard sense. Line...

conjecture | open | MathDB 页面 | 原文线索

V2-050987 | MATHDB-355968

Madani-Kiani's generalized regularity bound for connected graphs

摘录 / EXCERPT: Let G be a connected graph on vertex set $[n]$, let K_m be the complete graph on $[m]$, let $J_{K_m,G}$ be the generalized binomial edge ideal, let $c(G)$ be the number of maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-050988 | MATHDB-355971

Cranston-Lidický-Liu-Shantanam conjecture on planar Turán numbers of cycles

摘录 / EXCERPT: Let C_k denote the cycle of length k, and let $ex_{\mathcal{P}}(n,C_k)$ be the maximum number of edges in an n-vertex planar graph containing no copy of C_k. Cranston-Lidic...

conjecture | refuted | MathDB 页面 | 原文线索

V2-050989 | MATHDB-355995

Jackson's Hamilton decomposition conjecture for regular bipartite tournaments

摘录 / EXCERPT: A regular bipartite tournament is an orientation of the complete balanced bipartite graph $K_{2n,2n}$ in which every vertex has indegree and outdegree n. A Hamilton decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-050990 | MATHDB-356007

Distance-matrix complement conjecture for coalescing cospectral sets

摘录 / EXCERPT: Let H_1 and H_2 be graphs, and let $B_1\subseteq V(H_1)$ and $B_2\subseteq V(H_2)$ be vertex sets. Let $\mathcal{D}$ denote the distance matrix, whose (u,v)-entry records the...

conjecture | open | MathDB 页面 | 原文线索

V2-050991 | MATHDB-356077

The $O(G)$ conjecture for strongly connected graphs

摘录 / EXCERPT: $O(G)$ conjecture. For every strongly connected graph G, there is a unique $\leq_S$-minimal graph $O(G)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-050992 | MATHDB-356078

The bunchy factor conjecture

摘录 / EXCERPT: Bunchy factor conjecture. Every strongly connected graph has a bunchy synchronizing factor.

conjecture | open | MathDB 页面 | 原文线索

V2-050993 | MATHDB-356079

The bunchy factor conjecture in terms of the maximal bunchy factor

摘录 / EXCERPT: Bunchy factor conjecture. For every strongly connected graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-050994 | MATHDB-356083

The 106-color obstruction conjecture for partial 6-factorizations

摘录 / EXCERPT: A partial 6-factorization is a collection of edge-disjoint 6-factors, called colors, in a sub-hypergraph of the complete 3-uniform hypergraph K_{10}^3; an extension is a...

conjecture | open | MathDB 页面 | 原文线索

V2-050995 | MATHDB-356128

Mubayi-Rödl conjecture for the Turán density of the tight 5-cycle

摘录 / EXCERPT: Let $\mathcal{C}_5^3$ be the 3-uniform tight cycle of length 5, with vertex set $\{1,\dots,5\}$ and edges of the form $\{x,x+1,x+2\pmod 5\}$. Let $\pi(\mathcal{C}_5^3)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-050996 | MATHDB-356129

Conjecture for the Turán density of the tight 5-cycle with one edge removed

摘录 / EXCERPT: Let $\mathcal{C}_5^3$ be the 3-uniform tight cycle on 5 vertices, and let $\mathcal{C}_5^{-}$ be the 3-uniform hypergraph obtained from $\mathcal{C}_5^3$ by removing one edge...

conjecture | open | MathDB 页面 | 原文线索

V2-050997 | MATHDB-356155

The two-orientation deletability conjecture for cubic 3-edge-connected graphs

摘录 / EXCERPT: Two-orientation deletability conjecture. For every cubic 3-edge-connected graph G, there exist two strongly connected orientations D_1 and D_2 of G such that for every ve...

conjecture | open | MathDB 页面 | 原文线索

V2-050998 | MATHDB-356157

Knor-Škrekovski-Tepeh conjecture on maximum-Wiener-index orientations of trees

摘录 / EXCERPT: Knor-Škrekovski-Tepeh conjecture. If D maximises $W(D)$ among all orientations of T, then there exists a vertex v in D such that for every vertex u there exists either a...

conjecture | open | MathDB 页面 | 原文线索

V2-050999 | MATHDB-356161

Akbari-Dalirrooyfard-Ehsani-Ozeki-Sherkati list-avoiding orientation conjecture

摘录 / EXCERPT: Let G be a graph and let $F:V(G)\to 2^{\mathbb{N}}$. An orientation D of G is F-avoiding if $\deg_D^+(v)\notin F(v)$ for every $v\in V(G)$. The list-avoiding orientation co...

conjecture | open | MathDB 页面 | 原文线索

V2-051000 | MATHDB-356169

The half-integral reduction conjecture for labelled cycles

摘录 / EXCERPT: For an abelian group Γ, a subset $A\subseteq\Gamma$, and a Γ-labelled graph (G,γ), let $\mathcal O^A_{G,\gamma}$ denote the set of cycles whose γ-values...

conjecture | open | MathDB 页面 | 原文线索

V2-051001 | MATHDB-356170

The independent spanning trees conjecture

摘录 / EXCERPT: Let G be a graph and let v be any specified vertex of G. A collection of spanning trees of G is independent spanning trees rooted at v if, for every vertex u, the paths...

conjecture | open | MathDB 页面 | 原文线索

V2-051002 | MATHDB-356180

The idempotent criterion conjecture for Mathieu-Zhao subspaces of vertex operator algebras

摘录 / EXCERPT: The idempotent criterion conjecture. A subspace $M\subseteq V$ is a Mathieu-Zhao subspace of V if and only if, whenever $e_i\in M$, one has $V^i\subseteq M$. This is proposed as...

conjecture | open | MathDB 页面 | 原文线索

V2-051003 | MATHDB-356201

The linking conjecture for deranged matchings in balanced complete multipartite graphs

摘录 / EXCERPT: For a positive integer r, let $K_{r\times {2n}/{r}}$ be the $2n$-vertex balanced complete r-partite graph, and let M be a perfect matching in it. Write ${\mathrm{pm}}(G)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-051004 | MATHDB-356209

Zhai-Lin-Shu's spectral-radius conjecture for forbidden odd cycles

摘录 / EXCERPT: Let $k\ge2$, and let G be a graph with sufficiently large size m and no isolated vertices. Write $\lambda(G)$ for its spectral radius, let C_r denote the cycle of length r,...

conjecture | open | MathDB 页面 | 原文线索

V2-051005 | MATHDB-356210

Li-Lin-Huang's conjecture for forbidden triangle-extended cycles

摘录 / EXCERPT: Let $k\ge2$, and let G be a graph with sufficiently large size m and no isolated vertices. Write $\lambda(G)$ for its spectral radius, let $C_t^{\triangle}$ be the graph obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-051006 | MATHDB-356214

Jayanthan-Raghavan conjecture on linear type binomial edge ideals

摘录 / EXCERPT: Let G be a graph, and let its binomial edge ideal be the ideal generated by the binomials associated to the edges of G. A graph is unicyclic if it has exactly one cycle. Jayant...

conjecture | open | MathDB 页面 | 原文线索

V2-051007 | MATHDB-356242

Yan-Peng conjecture on large linear hypergraphs being lambda-perfect

摘录 / EXCERPT: Let $r\geq 3$. An r-graph is linear if any two edges have at most one vertex in common, and it is lambda-perfect when its Lagrangian density equals the natural complete-hypergrap...

conjecture | open | MathDB 页面 | 原文线索

V2-051008 | MATHDB-356243

Large sparse hypergraphs are lambda-perfect conjecture

摘录 / EXCERPT: Let $r\geq 3$. An r-graph G has m edges and is lambda-perfect when its Lagrangian density equals the natural complete-hypergraph lower bound determined by its number of verti...

conjecture | open | MathDB 页面 | 原文线索

V2-051009 | MATHDB-356244

The $E_8(a_7)$ permutation-orbifold conjecture for the subregular W-algebra

摘录 / EXCERPT: $E_8(a_7)$ permutation-orbifold conjecture. We have an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-051010 | MATHDB-356255

The clique counterexample conjecture for balanced edge-colorings

摘录 / EXCERPT: Let $S(N)$ be the set of natural numbers q such that $4\leq q\leq N$ and, for every n_0, there is some $n\geq n_0$ with $n\equiv 1\pmod{\ell}$ and a balanced coloring of K_n...

conjecture | open | MathDB 页面 | 原文线索

V2-051011 | MATHDB-356264

Brockman-Kay-Snively's gucycle conjecture for unlabeled graphs

摘录 / EXCERPT: Let a gucycle be a cyclic ordering whose windows represent each isomorphism class of graphs on n vertices exactly once. Brockman-Kay-Snively's conjecture. For each $n\neq 2$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-051012 | MATHDB-356265

Corrected gucycle conjecture for unlabeled graphs

摘录 / EXCERPT: Let a gucycle be a cyclic ordering whose windows represent each isomorphism class of graphs on n vertices exactly once. Corrected gucycle conjecture. For each $n\geq 4$, there ex...

conjecture | open | MathDB 页面 | 原文线索

V2-051013 | MATHDB-356266

Hurlbert's ordering conjecture for universal cycles of permutations

摘录 / EXCERPT: Let S_n be the set of permutations of $[n]$, and let $\mathcal{O}(n)$ be the clustered graph of overlapping n-permutations: its edges are the permutations in S_n, and consecu...

conjecture | open | MathDB 页面 | 原文线索

V2-051014 | MATHDB-356275

Parity conjecture for trees

摘录 / EXCERPT: Parity conjecture for trees. The Grundy value of a tree is zero if and only if its size is even.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-051015 | MATHDB-356277

Hypergraph-index conjecture for thickness and divergence of Coxeter groups

摘录 / EXCERPT: Hypergraph-index conjecture. These upper bounds are equalities: the following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-051016 | MATHDB-356278

The KSTH conjecture on spanning-tree decompositions of complete graphs

摘录 / EXCERPT: Let K_{2m} be the complete graph on $2m$ vertices. A vertex-split is a decomposition of its edge set into mutually edge-disjoint spanning trees; write the resulting group as...

conjecture | open | MathDB 页面 | 原文线索

V2-051017 | MATHDB-356296

Harary's Edge Reconstruction Conjecture for graphs

摘录 / EXCERPT: Let G be a graph. An edge-deck is the multiset ... A graph is edge-reconstructible if it is uniquely determined, up to isomorphism, by its edge-deck. Harary's Edge Reconstruction C...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-051018 | MATHDB-356298

Nongsiang-Saikia order conjecture for non-nilpotent graphs

摘录 / EXCERPT: Let G and H be finite non-nilpotent groups. For a group K, let $\mathfrak{R}_K$ denote the graph obtained from the graph on K in which x and y are adjacent exactly when...

conjecture | open | MathDB 页面 | 原文线索

V2-051019 | MATHDB-356366

The exact constant conjecture for cycles of length two modulo $2k$

摘录 / EXCERPT: Exact constant conjecture. For every integer $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-051020 | MATHDB-356373

The local antimagic chromatic number conjecture for s-bridge graphs

摘录 / EXCERPT: Let θ_s be an s-bridge graph, and let $\chi_{la}(\theta_s)$ denote its local antimagic chromatic number. The graphs covered by Theorem are precisely the characterized s-...

conjecture | open | MathDB 页面 | 原文线索

V2-051021 | MATHDB-356375

Sedlar-Škrekovski conjecture on the locally irregular chromatic index of decomposable graphs

摘录 / EXCERPT: A graph is decomposable if it admits a locally irregular edge-coloring (LIEC), and let $\chi_{\mathrm{irr}}'(G)$ be the smallest number of colors in such a coloring. Let H_0 be t...

conjecture | open | MathDB 页面 | 原文线索

V2-051022 | MATHDB-356376

Subcubic decomposable graph conjecture on the locally irregular chromatic index

摘录 / EXCERPT: A graph is decomposable if it admits a locally irregular edge-coloring (LIEC), and let $\chi_{\mathrm{irr}}'(G)$ denote the smallest number of colors in a LIEC of a decomposable gr...

conjecture | open | MathDB 页面 | 原文线索

V2-051023 | MATHDB-356399

Rainbow Hamiltonian cycle conjecture for strongly edge-colored graphs

摘录 / EXCERPT: Let G^c be a strongly edge-colored graph with n vertices, where strongly edge-colored means that any two adjacent edges and any two edges joined by an edge have distinct colors...

conjecture | open | MathDB 页面 | 原文线索

V2-051024 | MATHDB-356404

Brualdi-Quinn bipartite strong edge-coloring conjecture

摘录 / EXCERPT: Let G be a bipartite graph with parts A and B, and let Δ_A and Δ_B be the maximum degrees of vertices in A and B, respectively. Brualdi-Quinn conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-051025 | MATHDB-356405

Stronger subcubic strong edge-coloring conjectures

摘录 / EXCERPT: Let G be a subcubic graph. Stronger subcubic conjectures. (a) If G is bridgeless and is neither the Wagner graph nor the graph formed from $K_{3,3}$ by subdividing one edge, th...

conjecture | open | MathDB 页面 | 原文线索

V2-051026 | MATHDB-356406

Hudák-Lužar-Soták-Škrekovski planar strong edge-coloring conjecture

摘录 / EXCERPT: Let G be a planar graph of girth $g\ge5$ and maximum degree Δ.
Hudák-Lužar-Soták-Škrekovski conjecture. There exists a constant C such that ... The conjecture proposes tha...

conjecture | open | MathDB 页面 | 原文线索

V2-051027 | MATHDB-356462

Bašić et al.'s divisible-by-four degree conjecture for circulant nut graphs

摘录 / EXCERPT: Let n be the order of a circulant graph, and let $\mathrm{Circ}(n,S)$ denote the circulant graph with generator set S. A nut graph is a non-trivial graph whose adjacency matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-051028 | MATHDB-356463

The even-parameter order-four circulant nut graph conjecture

摘录 / EXCERPT: Let t be an integer and let n be the order of a circulant graph. A circulant nut graph is a circulant graph whose adjacency matrix has nullity one and whose non-zero null-space...

conjecture | open | MathDB 页面 | 原文线索

V2-051029 | MATHDB-356464

The Weak (n,k,t)-Conjecture on minimum (n,k,t)-graphs

摘录 / EXCERPT: Let n,k,t be positive integers with $n\geq k\geq t$. A graph G is an (n,k,t)-graph if $|V(G)|=n$ and every induced subgraph on k vertices contains a clique on t vertices;...

conjecture | open | MathDB 页面 | 原文线索

V2-051030 | MATHDB-356537

The cograph formulation of the Erdős-Hajnal conjecture

摘录 / EXCERPT: Cograph formulation of the Erdős-Hajnal conjecture. For every graph H, there exists an $\varepsilon>0$ such that every n-vertex H-free graph G contains an induced cograph o...

conjecture | open | MathDB 页面 | 原文线索

V2-051031 | MATHDB-356538

The ternary-tree minimization conjecture for connected partitions

摘录 / EXCERPT: Ternary-tree minimization conjecture. $P(G,2)$ is minimized by $T(n,3)$ over all connected graphs on n vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-051032 | MATHDB-356539

Yuan's additive-constant conjecture for NIM-H edges

摘录 / EXCERPT: Let c be a 2-coloring of the edges of K_n, let $E(c,H)$ be the set of edges contained in no monochromatic copy of H, and define ... Let ${\rm ex}(n,H)$ denote the Turán numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-051033 | MATHDB-356555

Manvel's Kelly conjecture on large-graph \ell-reconstructibility

摘录 / EXCERPT: For a graph G, its k-deck is the multiset of induced subgraphs with k vertices, and G is ℓ-reconstructible when it is determined by the deck obtained by deleting...

conjecture | open | MathDB 页面 | 原文线索

V2-051034 | MATHDB-356556

Hussain-Niepel-Kinawi's optimal LPDS density conjecture for king grids

摘录 / EXCERPT: Let $G=(V,E)$) be the infinite king grid with vertex set $V=\mathbb{Z}\times\mathbb{Z}$, in which two vertices are adjacent exactly when their Euclidean distance is at most...

conjecture | open | MathDB 页面 | 原文线索

V2-051035 | MATHDB-356560

Generalized Turán extremal graph conjecture for edge blow-ups of cycles and paths

摘录 / EXCERPT: Let C_k^3 and P_k^3 denote the edge blow-ups of the cycle C_k and path P_k, respectively. For integers p,t, let $T_p(n-t+1)$ be the balanced p-partite complete graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-051036 | MATHDB-356576

Quantum causal correlations conjecture for chordless siblings-on-cycles graphs

摘录 / EXCERPT: Let W be a quantum process with causal structure $G=(V,E)$. A process produces causal correlations when, for every collection of local experiments μ, the resulting correlatio...

conjecture | open | MathDB 页面 | 原文线索

V2-051037 | MATHDB-356593

The permanence conjecture for weakly reversible mass-action systems

摘录 / EXCERPT: A mass-action system $(G,\boldsymbol{\kappa})$ has associated ODEs and is permanent if, on every compatibility class, all solutions eventually enter a compact set that remains attr...

conjecture | open | MathDB 页面 | 原文线索

V2-051038 | MATHDB-356595

Characterisation of graphs of linear growth by tree blow-ups

摘录 / EXCERPT: Linear-growth product-structure conjecture. There exist functions $g\colon\mathbb{R}\to\mathbb{N}$ and $h\colon\mathbb{R}\to\mathbb{R}$ such that for any $c\geqslant 1$, every grap...

conjecture | open | MathDB 页面 | 原文线索

V2-051039 | MATHDB-356596

Characterisation of polynomial-growth graphs by products of trees

摘录 / EXCERPT: Polynomial-growth product-structure conjecture. There exist functions $g\colon\mathbb{R}\times\mathbb{N}\to\mathbb{N}$ and $h\colon\mathbb{R}\times\mathbb{N}\to\mathbb{R}$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-051040 | MATHDB-356632

Tree patrolling conjecture for the E-patrolling strategy

摘录 / EXCERPT: Let Q be a tree network, let $\alpha>0$ be the attack duration, let E denote the set used to define the E-patrolling strategy, let μ denote the measure of Q, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051041 | MATHDB-356645

The distinguished-orbit collapsing conjecture for W-algebras

摘录 / EXCERPT: Distinguished-orbit collapsing conjecture. If f is distinguished, then

conjecture | open | MathDB 页面 | 原文线索

V2-051042 | MATHDB-356651

Chvátal's Erdős-Chvátal simplex conjecture

摘录 / EXCERPT: Erdős-Chvátal simplex conjecture. Every family $\mathcal{F}\subseteq\binom{[n]}{k}$ without an r-simplex contains at most

conjecture | open | MathDB 页面 | 原文线索

V2-051043 | MATHDB-356686

The nearly connected partition conjecture

摘录 / EXCERPT: Let G be a 2-connected graph of order n, and let $n=n_1+n_2+\cdots+n_k$ be a partition of n. A subset $A\subseteq V(G)$ is nearly connected if it is contained in a subtree...

conjecture | open | MathDB 页面 | 原文线索

V2-051044 | MATHDB-356691

The pairwise-adjacent vertices-or-edges conjecture for triangle-free graphs

摘录 / EXCERPT: Pairwise-adjacent vertices-or-edges conjecture. Every $coverline{K_3}$-free graph on n vertices has a set of

conjecture | open | MathDB 页面 | 原文线索

V2-051045 | MATHDB-356702

Equality of hypergraph and graph zero-free loci

摘录 / EXCERPT: Zero-free locus conjecture. The zero-free locus of hypergraphs of maximum degree Δ is identical to the zero-free locus of graphs of maximum degree Δ:

conjecture | open | MathDB 页面 | 原文线索

V2-051046 | MATHDB-356712

5-regular planar graph 4-choosability conjecture

摘录 / EXCERPT: Let G be a planar graph that is 5-regular, meaning every vertex has degree 5. Write $\chi_\ell(G)$ for its list chromatic number. 5-regular planar graph conjecture. Every 5...

conjecture | open | MathDB 页面 | 原文线索

V2-051047 | MATHDB-356716

Chromatic symmetric homology interval conjecture

摘录 / EXCERPT: Chromatic symmetric homology interval conjecture. The groups $H_{i,0}(G;\mathbb{C})$ are non-trivial for every integer i satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-051048 | MATHDB-356717

Bounds for the degree-zero chromatic symmetric homology span

摘录 / EXCERPT: Span bounds conjecture. The homological span satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051049 | MATHDB-356731

The maximum edge derivative conjecture for normalized Laplacian eigenvalues

摘录 / EXCERPT: Let G be a graph, let x,y be vertices, and let λ be an eigenvalue of the normalized Laplacian whose edge derivative is being considered. The edge derivative with respec...

conjecture | open | MathDB 页面 | 原文线索

V2-051050 | MATHDB-356732

The lollipop extremal conjecture for the edge derivative of Kemeny's constant

摘录 / EXCERPT: Let $L_{r,s}$ denote a lollipop graph: a complete graph K_r connected to a path graph on s vertices by a bridge. For each n, consider simple graphs of order n and the deriv...

conjecture | open | MathDB 页面 | 原文线索

V2-051051 | MATHDB-356733

The path extremal conjecture for the minimum edge derivative of Kemeny's constant

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $\{n,1\}$ be the non-edge joining its two ends. For simple graphs of order n, consider the derivative of Kemeny's constant...

conjecture | open | MathDB 页面 | 原文线索

V2-051052 | MATHDB-356764

The extremal function for circumference in k-cyclable graphs

摘录 / EXCERPT: Let G be a graph, let $c(G)$ denote its circumference, and call G k-cyclable if every set of k vertices is contained in a cycle. For a given n, let $f(n)$ be the largest...

open_problem | open | MathDB 页面 | 原文线索

V2-051053 | MATHDB-356765

The growth rate of the circumference threshold in k-cyclable graphs

摘录 / EXCERPT: Let G be a graph, let $c(G)$ denote its circumference, and call G k-cyclable if every set of k vertices is contained in a cycle. For a given n, let $g(n)$ be the largest...

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V2-051054 | MATHDB-356774

Glock, Kühn and Osthus's cycle-decomposition threshold conjecture

摘录 / EXCERPT: Let k be an integer with $k\geq 3$. For a k-graph H, a cycle-decomposition is an edge partition of H into tight cycles, and the cycle-decomposition threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-051055 | MATHDB-356775

Glock, Kühn and Osthus's Euler-tour threshold conjecture

摘录 / EXCERPT: Let k be an integer with $k\geq 3$. An Euler tour in a k-graph is a cyclic sequence of vertices in which every k cyclically consecutive vertices form an edge and every edge a...

conjecture | open | MathDB 页面 | 原文线索

V2-051056 | MATHDB-356778

Equivalence of FO- and FOM-transductions

摘录 / EXCERPT: Let ... scr D$ be infinite classes of finite graphs. An FO-transduction interprets graphs using first-order formulas, while an FOM-transduction allows first-order formulas with modul...

conjecture | open | MathDB 页面 | 原文线索

V2-051057 | MATHDB-356792

Conjecture that every nontrivial tree is harmonious

摘录 / EXCERPT: A graph G is harmonious when it admits a harmonious labeling, namely an injective vertex labeling whose induced edge labels are also injective modulo $|E(G)|$; for trees, the def...

conjecture | open | MathDB 页面 | 原文线索

V2-051058 | MATHDB-356796

The weighted clique-packing conjecture for uniform weights on a clique independent set

摘录 / EXCERPT: Let G be a graph, let $m\geq 1$ and $k\geq 2$, and let w be an m-weighting of G. A k-clique independent set is a set I of vertices such that every k-clique of G con...

conjecture | open | MathDB 页面 | 原文线索

V2-051059 | MATHDB-356799

The minimum maximal-independent-set conjecture for connected twin-free triangle-free graphs

摘录 / EXCERPT: Let G be a connected, twin-free and triangle-free graph of order n. Write $i_{\text{max}}(G)$ for the number of maximal independent sets of G. Minimum maximal-independent-set...

conjecture | open | MathDB 页面 | 原文线索

V2-051060 | MATHDB-356825

Local vertex-star basis conjecture for the Stokes potential space on Freudenthal meshes

摘录 / EXCERPT: Let $V_h\subset[H^1(\Omega)]^3$ be constructed with continuous Lagrange elements of degree k. Let $\Sigma_h\subset H^1(\operatorname{curl},\Omega)$ be the space preceding V_h i...

conjecture | open | MathDB 页面 | 原文线索

V2-051061 | MATHDB-356836

Mkrtchyan-Steffen decomposition conjecture for multigraphs

摘录 / EXCERPT: Let G be a graph with maximum degree $4\Delta(G)=\Delta$4 and chromatic index $4\chi'(G)=\Delta+k$4$, where $kgeq 1$. A **maximum $4Delta$4-edge-colorable subgraph** is a $4Delta...

conjecture | open | MathDB 页面 | 原文线索

V2-051062 | MATHDB-356838

Asymptotic sharpness of the lower bound for edge intersection cycles

摘录 / EXCERPT: Asymptotic sharpness conjecture. For any $k ge 3$ there exists an $n_k in \mathbb{N}$ such that for every $n ge n_k$,

conjecture | open | MathDB 页面 | 原文线索

V2-051063 | MATHDB-356874

Linear-size separating path systems conjecture

摘录 / EXCERPT: Let G be a graph on n vertices. A separating path system is a collection of paths in G that separates the edges of G, either weakly or strongly. Linear-size separating path...

conjecture | open | MathDB 页面 | 原文线索

V2-051064 | MATHDB-356876

Seymour's non-uniform Caccetta-Häggkvist conjecture

摘录 / EXCERPT: Seymour's non-uniform Caccetta-Häggkvist conjecture. D contains a directed cycle of length at most

conjecture | open | MathDB 页面 | 原文线索

V2-051065 | MATHDB-356912

Kajitani, Ueno and Miyano's conjecture on cyclically orderable graphs

摘录 / EXCERPT: Let G be a connected graph. Its density is ... The graph G is uniformly dense if $d(H)\leq d(G)$ for every connected subgraph H of G. A graph is cyclically orderable if it ha...

conjecture | open | MathDB 页面 | 原文线索

V2-051066 | MATHDB-356917

The density domination exponent conjecture for complete bipartite graphs

摘录 / EXCERPT: The density domination exponent conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051067 | MATHDB-356933

Caro's linear odd induced subgraph conjecture

摘录 / EXCERPT: Caro's conjecture. There exists a positive constant c such that every n-vertex graph without isolated vertices contains an odd induced subgraph with at least cn vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-051068 | MATHDB-356934

Berman-Wang-Wargo conjecture on odd induced subgraphs

摘录 / EXCERPT: Berman-Wang-Wargo conjecture. For any connected simple graph G of order n,

conjecture | open | MathDB 页面 | 原文线索

V2-051069 | MATHDB-356940

Chen-Zhang's signless Laplacian extremal conjecture for $K_{s,t}$-minor-free graphs

摘录 / EXCERPT: Chen-Zhang's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051070 | MATHDB-356958

Lincoln-Vassilevska-Williams-Williams hyperclique conjecture

摘录 / EXCERPT: Let $k>r\geq 3$ and let an r-uniform hypergraph be a hypergraph whose edges have size r. A k-hyperclique is a set of k vertices such that every r-subset is an edge. The L...

conjecture | open | MathDB 页面 | 原文线索

V2-051071 | MATHDB-356960

The Erdős-Goodman-Pósa conjecture for graphs with independence number at most two

摘录 / EXCERPT: Erdős-Goodman-Pósa conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051072 | MATHDB-356961

Hajebi's conjecture on subquadratic clique covers of K_{t,t}-free graphs

摘录 / EXCERPT: Hajebi's conjecture. For every integer $t\ge 1$, there exists $\varepsilon>0$ such that every $K_{t,t}$-free graph G has a clique cover of size

conjecture | open | MathDB 页面 | 原文线索

V2-051073 | MATHDB-356977

Kanade-Russell conjecture modulo 16 for restricted partition sets

摘录 / EXCERPT: Let F be the set of the following 13 partitions: ... Define ... ... For $1\leq a\leq3$, let $\mathsf L_a$ be the set of partitions that do not begin with any $\boldsymbol c\in I_a$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-051074 | MATHDB-356981

Mubayi-Pikhurko-Sudakov conjecture on Turán densities of tight cycles minus one edge

摘录 / EXCERPT: Mubayi-Pikhurko-Sudakov conjecture. If $\ell\geq 5$ is not divisible by three, then

conjecture | open | MathDB 页面 | 原文线索

V2-051075 | MATHDB-356988

The μ-PageRank monotonicity and range conjecture

摘录 / EXCERPT: μ-PageRank monotonicity and range conjecture. The function $([?])_i$ is monotonic for all i and

conjecture | open | MathDB 页面 | 原文线索

V2-051076 | MATHDB-357001

The two-thirds conjecture for locating-total dominating sets

摘录 / EXCERPT: Two-thirds conjecture. Every twin-free isolate-free graph G of order n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051077 | MATHDB-357009

Bucko-Czap conjecture on 1-planarity of lexicographic products

摘录 / EXCERPT: Let G be a graph, and let K_2 denote the complete graph with two vertices. A graph is 1-planar if it has a drawing in the plane in which every edge is crossed by at most one ot...

open_problem | open | MathDB 页面 | 原文线索

V2-051078 | MATHDB-357036

Weak polymatroidality of Alexander duals of vertex decomposable complexes

摘录 / EXCERPT: Let Δ be a pure simplicial complex on vertex set $[n]$, and let $I_{\Delta}^{\vee}$ denote the Alexander dual of its Stanley-Reisner ideal. Recall that a pure simplicial com...

conjecture | open | MathDB 页面 | 原文线索

V2-051079 | MATHDB-357037

Weak polymatroidality of vertex cover ideals of unmixed vertex decomposable graphs

摘录 / EXCERPT: Let G be an unmixed vertex decomposable graph, and let $J(G)$ be its vertex cover ideal. Weak graph conjecture. Then ... This is presented as a weak form of the conjecture for pure...

conjecture | open | MathDB 页面 | 原文线索

V2-051080 | MATHDB-357038

Fox-Grinshpun-Liebenau-Person-Szabó conjecture on the monotonicity of the minimum degree parameter

摘录 / EXCERPT: Let $s_q(K_t)$ denote the minimum degree parameter for q-color Ramsey problems with clique target K_t. Fox-Grinshpun-Liebenau-Person-Szabó conjecture. For all $q\geqslant 3$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-051081 | MATHDB-357050

Henning-Klavžar-Rall 3/4-conjecture for the total domination game

摘录 / EXCERPT: Let G be a graph without isolated vertices or edges, let $n=|V(G)|$, and let $\gamma_{tg}(G)$ denote the game total domination number of G, the number of selected vertices when...

conjecture | open | MathDB 页面 | 原文线索

V2-051082 | MATHDB-357051

Kinnersley-West-Zamani 3/5-conjecture for the domination game

摘录 / EXCERPT: Let G be an isolate-free graph of order n, and let $\gamma_g(G)$ denote the game domination number, the number of vertices selected when Dominator starts and both players play...

conjecture | open | MathDB 页面 | 原文线索

V2-051083 | MATHDB-357052

Bondy's reverse degree-sum conjecture for long cycles

摘录 / EXCERPT: Bondy's reverse degree-sum conjecture. If $\overline{p}\geq\Greekmath 0114 -1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051084 | MATHDB-357053

Jung's reverse minimum-degree conjecture for long cycles

摘录 / EXCERPT: Jung's reverse minimum-degree conjecture. If $\overline{p}\geq\Greekmath 0114 -1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051085 | MATHDB-357072

The eight-vertex bound for forbidden GaTEx subgraphs

摘录 / EXCERPT: Eight-vertex bound. For all $F\in\mathfrak{F}_{\mathrm{GT}}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-051086 | MATHDB-357092

Brill-Noether existence conjecture for once-marked graphs

摘录 / EXCERPT: Brill-Noether existence conjecture for once-marked graphs. For any once-marked graph (G,u) of genus g, every partition λ with $|\lambda|\leq g$ belongs to the divisor c...

conjecture | solved | MathDB 页面 | 原文线索

V2-051087 | MATHDB-357110

Minimum spectral radius conjecture for graphs with odd order and prescribed domination number

摘录 / EXCERPT: Minimum spectral radius conjecture. For every graph $G\in\mathbb{G}_{n,\lfloor\frac{n}{2}\rfloor}$,

conjecture | open | MathDB 页面 | 原文线索

V2-051088 | MATHDB-357118

Jackson and Ringel's bichromatic number conjecture

摘录 / EXCERPT: Jackson and Ringel's conjecture. For all $g\geq 1$, the bound in this inequality is tight:

conjecture | open | MathDB 页面 | 原文线索

V2-051089 | MATHDB-357119

Bosek's asymptotic harmonious coloring conjecture for hypergraphs

摘录 / EXCERPT: Let $k,\Delta\geq 2$. A harmonious coloring of a k-uniform hypergraph H is a vertex coloring that is rainbow on every edge and assigns distinct color sets to distinct edges; le...

conjecture | open | MathDB 页面 | 原文线索

V2-051090 | MATHDB-357155

Largest-order critical group conjecture for arithmetical structures on star and complete graphs

摘录 / EXCERPT: Let $a_1=1$ and $a_n=a_{n-1}^2+a_{n-1}$ for $n\geq 2$. For an arithmetical structure on the star graph S_n or the complete graph K_n, let $\mathcal{K}$ denote its critical grou...

conjecture | open | MathDB 页面 | 原文线索

V2-051091 | MATHDB-357156

Largest cyclic critical group conjecture for arithmetical structures on star and complete graphs

摘录 / EXCERPT: Let $a_1=1$ and $a_n=a_{n-1}^2+a_{n-1}$ for $n\geq 2$. For an arithmetical structure on the star graph S_n or the complete graph K_n, its critical group is a finite abelian gro...

conjecture | open | MathDB 页面 | 原文线索

V2-051092 | MATHDB-357182

Digraph analogue of Cereceda's diameter conjecture

摘录 / EXCERPT: Let D be a digraph on n vertices, and let $\delta^*_{\min}(D)$ denote its min-degeneracy. Let ${\cal D}_k(D)$ be the graph whose vertices are the k-dicolourings of D, with...

conjecture | open | MathDB 页面 | 原文线索

V2-051093 | MATHDB-357183

Maximum-average-degree conjecture for non-2-mixing oriented graphs

摘录 / EXCERPT: Let an oriented graph be a digraph with no pair of oppositely directed parallel arcs, and let its maximum average degree be the maximum of $2|A(H)|/|V(H)|$ over all non-empty subdi...

conjecture | open | MathDB 页面 | 原文线索

V2-051094 | MATHDB-357195

Extended Knuth conjecture for joint arc-count distributions in random digraph depth-first search

摘录 / EXCERPT: Consider a random multidigraph on n vertices in which the outdegree of each vertex is independently distributed as $\operatorname{Ge}(1-p)$, with $p\in(0,1)$, and endpoints of ou...

conjecture | open | MathDB 页面 | 原文线索

V2-051095 | MATHDB-357221

Quadratic-diameter conjecture for digraph redicolouring

摘录 / EXCERPT: Quadratic-diameter redicolouring conjecture. There is an absolute constant d such that for every integer k and every digraph D on n vertices satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-051096 | MATHDB-357226

Bujtás-Iršič-Klavžar's strict 3/5 improvement conjecture for the domination game

摘录 / EXCERPT: Let G be a graph on $n \geq 6$ vertices, with minimum degree at least 2. The game domination number $\gamma_g(G)$ is the number of moves in the domination game when Dominator s...

conjecture | open | MathDB 页面 | 原文线索

V2-051097 | MATHDB-357227

The 1/2-conjecture for the domination game

摘录 / EXCERPT: Let G be a graph with n vertices and minimum degree at least 2. The game domination number $\gamma_g(G)$ is the number of moves in the domination game when Dominator starts a...

conjecture | open | MathDB 页面 | 原文线索

V2-051098 | MATHDB-357228

Rall's Hamiltonian-path conjecture for the domination game

摘录 / EXCERPT: Let G be a graph on n vertices. A Hamiltonian path is a path containing every vertex of G exactly once. The game domination number $\gamma_g(G)$ is the number of moves in the...

conjecture | open | MathDB 页面 | 原文线索

V2-051099 | MATHDB-357229

The additive 1/2-conjecture for the domination game

摘录 / EXCERPT: Let G be a graph on n vertices with minimum degree at least 2. The game domination number $\gamma_g(G)$ is the number of moves in the domination game when Dominator starts an...

conjecture | open | MathDB 页面 | 原文线索

V2-051100 | MATHDB-357236

Frohmader's localized clique-weight conjecture

摘录 / EXCERPT: Frohmader's localized clique-weight conjecture. For every m-edge graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-051101 | MATHDB-357237

Chakraborti-Chen localized cycle-weight conjecture

摘录 / EXCERPT: Chakraborti-Chen localized cycle-weight conjecture. The quantity $c'_G(T)$ is well-defined and decreasing in $\gamma_G(T)$, and

conjecture | open | MathDB 页面 | 原文线索

V2-051102 | MATHDB-357238

Localized hypergraph clique-weight conjecture

摘录 / EXCERPT: Localized hypergraph clique-weight conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-051103 | MATHDB-357240

Knuth's conjecture on forward and back arcs in geometric-outdegree digraphs

摘录 / EXCERPT: For a random multidigraph on n vertices with geometric outdegree distribution, independently assign each vertex v a geometrically distributed outdegree $d^+(v)$ with mode 0 a...

conjecture | open | MathDB 页面 | 原文线索

V2-051104 | MATHDB-357241

Janson's joint distribution conjecture for DFS arc counts

摘录 / EXCERPT: Consider a random multidigraph on n vertices with geometric outdegree distribution: independently for each vertex v, let $d^+(v)$ be geometrically distributed with mode 0 and...

conjecture | open | MathDB 页面 | 原文线索

V2-051105 | MATHDB-357245

The general arrow-relation conjecture for finite-set traces

摘录 / EXCERPT: The general arrow-relation conjecture. For all integers $n>\ell>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-051106 | MATHDB-357269

Constant deficiency Brown-Erdős-Sós conjecture

摘录 / EXCERPT: Let e be a positive integer, and let a (v,e)-configuration be a set of e edges in a 3-uniform hypergraph spanned by at most v vertices. Constant deficiency Brown-Erdős-Só...

conjecture | open | MathDB 页面 | 原文线索

V2-051107 | MATHDB-357270

Weak Conlon conjecture for sparse degenerate graphs

摘录 / EXCERPT: For integers k and t, let $\mathcal{H}_{k,t}$ be the family of 2-degenerate graphs on k vertices and $2k-t$ edges. Weak Conlon conjecture. There exist absolute constants...

conjecture | open | MathDB 页面 | 原文线索

V2-051108 | MATHDB-357273

Conjecture on the integer eigenvalues of the n-Queens graph

摘录 / EXCERPT: Integer-eigenvalue conjecture. For $n \geq 4$, these are the unique integer eigenvalues of $\mathcal{Q}(n)$. If n is even, $n-4$ has multiplicity $\frac{n-2}{2}$; if n is odd,...

conjecture | open | MathDB 页面 | 原文线索

V2-051109 | MATHDB-357277

The infinite Toeplitz trace conjecture for return probabilities

摘录 / EXCERPT: Consider the Markov chain on $\mathbb{N}$ in which every vertex i has an edge of weight r to $i-1$ and edges of weight c_l to $i+l$ for $0\leq l\leq k$, with ... Let p_l be t...

conjecture | open | MathDB 页面 | 原文线索

V2-051110 | MATHDB-357298

Lattice-cohomology morphism conjecture for deformations of isolated curve singularities

摘录 / EXCERPT: Let $\{(C_t,o)\}_{t\in(\mathbb C,0)}$ be a flat deformation of isolated curve singularities. For each level n, let ℓ^* denote a vertex of the level set of the generic fibre...

conjecture | open | MathDB 页面 | 原文线索

V2-051111 | MATHDB-357318

The half-minimum-degree conjecture for restricted optimal pebbling

摘录 / EXCERPT: Half-minimum-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051112 | MATHDB-357319

The dominating-set reformulation of the half-minimum-degree conjecture

摘录 / EXCERPT: Dominating-set reformulation. If

conjecture | open | MathDB 页面 | 原文线索

V2-051113 | MATHDB-357323

Polynomial chromatic-number conjecture for high-girth intersection graphs of lines

摘录 / EXCERPT: Let $g\in\mathbb{N}$. An intersection graph of lines in $\mathbb{R}^3$ has one vertex for each line and edges joining intersecting lines; its girth is the length of its shortest cy…

conjecture | open | MathDB 页面 | 原文线索

V2-051114 | MATHDB-357324

Polynomial chromatic-number conjecture for triangle-free intersection graphs of projective lines

摘录 / EXCERPT: An intersection graph of lines in $\mathbb{PR}^3$ has one vertex for each projective line and edges joining intersecting lines; it is triangle-free when it has no 3-cycle. Polyno…

conjecture | open | MathDB 页面 | 原文线索

V2-051115 | MATHDB-357325

Small-independent-set conjecture for triangle-free intersection graphs of boxes

摘录 / EXCERPT: For $\alpha>0$, an intersection graph of n boxes in $\mathbb{R}^3$ has one vertex for each box and edges joining intersecting boxes; it is triangle-free when it has no 3-cycle,…

conjecture | open | MathDB 页面 | 原文线索

V2-051116 | MATHDB-357333

Johnston, Palmer and Sarkar's rainbow Turán conjecture for paths

摘录 / EXCERPT: Let P_k denote a path on k vertices, and let $\operatorname{ex}^{*}(n,P_k)$ be the maximum number of edges in a properly edge-colored n-vertex graph containing no rainbow cop…

conjecture | open | MathDB 页面 | 原文线索

V2-051117 | MATHDB-357342

Splitting conjecture for unordered multiplicity lists of trees

摘录 / EXCERPT: Splitting conjecture. For any $j\in [k]$ with $m_j\geq 2$, the unordered multiplicity list

conjecture | open | MathDB 页面 | 原文线索

V2-051118 | MATHDB-357358

Marc's conjecture on vertex-transitive partial cubes of girth 6

摘录 / EXCERPT: Let G be a vertex-transitive partial cube with girth 6. Marc's conjecture. Doubled Odd graphs are the only vertex-transitive partial cubes with girth 6. This conjecture propo…

conjecture | open | MathDB 页面 | 原文线索

V2-051119 | MATHDB-357372

The three strongly connected components conjecture for two-generated clone-minimal dispersive algebras

摘录 / EXCERPT: Three strongly connected components conjecture. The digraph $D_{\mathbb{A}}$ has at least three strongly connected components.

conjecture | open | MathDB 页面 | 原文线索

V2-051120 | MATHDB-357373

Conjecture on extension of zero-pattern results to arbitrary sample size

摘录 / EXCERPT: Consider linear neural networks trained on m data points. The results referred to as Proposition … , Corollary … are observed computationally to hold when $m=2$. Extension conjec…

conjecture | open | MathDB 页面 | 原文线索

V2-051121 | MATHDB-357401

Frankl-Kupavskii stability conjecture for bounded matching number

摘录 / EXCERPT: Frankl-Kupavskii stability conjecture. If $\nu(F)\le t$, then $\tau(F)\le t$ or

conjecture | open | MathDB 页面 | 原文线索

V2-051122 | MATHDB-357402

Bang-Jensen-Yeo arc-partition conjecture for highly arc-connected digraphs

摘录 / EXCERPT: Bang-Jensen-Yeo's conjecture. There exists an integer K such that every K-arc-strong digraph $D=(V,A)$ has an arc-partition

conjecture | open | MathDB 页面 | 原文线索

V2-051123 | MATHDB-357403

Wrochna's topological Hedetniemi conjecture for free involutive complexes

摘录 / EXCERPT: Let $\mathcal{K}$ and $\mathcal{L}$ be finite free $\mathbb Z/2$-simplicial complexes, with diagonal action on their Cartesian product, and let $\operatorname{ind} X$ denote the le…

conjecture | open | MathDB 页面 | 原文线索

V2-051124 | MATHDB-357416

Large-degree conjecture for multipartite hypergraphs with a heavy edge

摘录 / EXCERPT: Large-degree conjecture. There exists $d_r>0$ such that if H has an edge of multiplicity at least $\nu+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051125 | MATHDB-357420

Gallai vertex conjecture for connected graphs with independence number at most five

摘录 / EXCERPT: Let G be a connected graph, and let $\alpha(G)$ denote its independence number. A vertex of G is a Gallai vertex if it belongs to every longest path in G. Gallai vertex conje…

conjecture | open | MathDB 页面 | 原文线索

V2-051126 | MATHDB-357427

Conjecture on fixed-descent subgraphs of the skeleton of a crystal

摘录 / EXCERPT: Let λ be a partition, and let $Skeleton(\lambda)$ be the skeleton whose vertices are standard tableaux of shape λ. For each positive integer s, let H_s be the i…

conjecture | open | MathDB 页面 | 原文线索

V2-051127 | MATHDB-357431

The 27-29 geodesic networks conjecture for Liouville quantum gravity

摘录 / EXCERPT: Fix $\gamma \in (0,2)$ and consider γ-Liouville quantum gravity (γ-LQG). Let Star_k denote the set of points joined to the origin by at least k geo...

conjecture | open | MathDB 页面 | 原文线索

V2-051128 | MATHDB-357432

Erdős-Hajnal-Simonovits-Sós-Szemerédi periodic structure conjecture for Ramsey-Turán extremal graphs

摘录 / EXCERPT: Let G be an asymptotically extremal graph for the Ramsey-Turán density $\varrho_p(q)$, where the p-independence number is the largest size of a vertex set inducing a K_p-free...

conjecture | open | MathDB 页面 | 原文线索

V2-051129 | MATHDB-357439

Four-point map-counting lower-bound conjecture

摘录 / EXCERPT: Let $\mathcal{M}_{0,4}(r_1,r_2,r_3,r_4)$ be the set of maps with the specified vertex data, and order signatures componentwise: $\sigma ot greater than or equal to\sigma'$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-051130 | MATHDB-357445

Bang-Jensen and Gutin's algorithmic conjecture for good pairs in quasi-transitive digraphs

摘录 / EXCERPT: Bang-Jensen and Gutin's conjecture. There exists a polynomial algorithm for deciding, given a quasi-transitive digraph $D=(V,A)$ and two vertices u,v of V, whether D has a go...

conjecture | open | MathDB 页面 | 原文线索

V2-051131 | MATHDB-357453

Asymptotic oriented discrepancy conjecture for directed rooted trees

摘录 / EXCERPT: Let T be a tree with ℓ leaves, and let $\mathcal{DT}$ be the set of all directed rooted trees, namely trees with a distinguished root and all edges oriented away from it. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-051132 | MATHDB-357454

Complex discrepancy conjecture for trees

摘录 / EXCERPT: Let T be a tree with ℓ leaves, let $\mathcal{T}$ denote the family of trees used in the discrepancy definition, and let $\mathcal{D}^1(T,\mathcal{T})$ denote the one-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-051133 | MATHDB-357457

Equality conjecture for balanced bipartite biindependence in hypercubes

摘录 / EXCERPT: Let Q_r be the r-dimensional hypercube, let $\alpha_{\text{bal}}(Q_r)$ denote the maximum total size of a balanced bipartite biindependent pair in Q_r, and let $a(r)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-051134 | MATHDB-357482

You and Liu's Laplacian spectral ratio conjecture for trees

摘录 / EXCERPT: Let T be a tree with $n\geq 3$ vertices. The Laplacian spectral ratio is ... where $\mu_1(G)$ and $\mu_{n-1}(G)$ are the largest and second smallest Laplacian eigenvalues of a conn...

conjecture | open | MathDB 页面 | 原文线索

V2-051135 | MATHDB-357488

Reinhart's normalized distance Laplacian spectral-radius characterization

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\partial_n^\mathcal{L}$ denote the largest eigenvalue of its normalized distance Laplacian matrix $\mathcal{D}^\mathcal{L}(G)$. Reinhar...

conjecture | open | MathDB 页面 | 原文线索

V2-051136 | MATHDB-357489

Reinhart's extremal normalized distance Laplacian conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\mathcal{D}^\mathcal{L}(G)$ be its normalized distance Laplacian matrix. Write KPK_{n_1,n_2,n_3} for the graph obtained by connecting...

conjecture | open | MathDB 页面 | 原文线索

V2-051137 | MATHDB-357500

Jiang-Polyanskii conjecture on the bounded secondary term for equiangular lines

摘录 / EXCERPT: For $\alpha\in(0,1)$, let $\lambda=(1-\alpha)/(2\alpha)$, and let $k(\lambda)$ denote the minimum number of vertices of a graph whose largest eigenvalue is exactly λ, or...

conjecture | open | MathDB 页面 | 原文线索

V2-051138 | MATHDB-357513

Hertrich-Lindner-Raginsky-Bach depth-separation conjecture for ReLU networks

摘录 / EXCERPT: Let $\operatorname{ReLU}_n(k)$ denote the set of continuous piecewise linear functions defined on $\mathbb{R}^n$ and computable with k hidden layers. Hertrich-Lindner-Raginsky-Ba...

conjecture | open | MathDB 页面 | 原文线索

V2-051139 | MATHDB-357514

Max-function formulation of the ReLU depth-separation conjecture

摘录 / EXCERPT: Let $n=2^k$, and let $\operatorname{ReLU}_n(k)$ be the set of continuous piecewise linear functions on $\mathbb{R}^n$ computable with k hidden layers. Max-function formulation of...

conjecture | open | MathDB 页面 | 原文线索

V2-051140 | MATHDB-357521

Vizing's edge-recoloring conjecture

摘录 / EXCERPT: Vizing's conjecture. Every k-edge-coloring of a graph G is equivalent to a $\Delta(G)$-edge-coloring of G if there is any.

conjecture | open | MathDB 页面 | 原文线索

V2-051141 | MATHDB-357522

The $(\Delta(G)+2)$-edge-coloring equivalence conjecture

摘录 / EXCERPT: The $(\Delta(G)+2)$-edge-coloring equivalence conjecture. All $(\Delta(G)+2)$-edge-colorings of a graph G are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-051142 | MATHDB-357524

The non-optimal edge-coloring equivalence conjecture

摘录 / EXCERPT: The non-optimal edge-coloring equivalence conjecture. Any two non-optimal colorings are equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-051143 | MATHDB-357541

Day's conical-vertex conjecture for extremal K_r-saturated graphs

摘录 / EXCERPT: Let $r\geq 4$ and $t\geq r-2$ be integers. Let $C(r,t)$ denote the family of graphs defined by $C(r,t)=C(r,t,-c(r,t))$, where $c(r,t)$ is the corresponding extremal constant for...

conjecture | open | MathDB 页面 | 原文线索

V2-051144 | MATHDB-357543

The even-degree intersection conjecture for partition path polytopes

摘录 / EXCERPT: Let G be a graph with terminal set T, and let $Q(G,\mathcal{T})$ denote the polytope associated with a partition $\mathcal{T}$ of T. A vector is $\mathcal{T}_1-\mathcal{T}_2$...

conjecture | open | MathDB 页面 | 原文线索

V2-051145 | MATHDB-357555

Monotone edge-shrinking and vertex-expansion conjecture for polytopes

摘录 / EXCERPT: Let $P,Q\subset\mathbb{R}^d$ be polytopes with isomorphic edge-graphs. Assume that Q contains the origin in its interior, edges in Q are not longer than the corresponding edges...

conjecture | open | MathDB 页面 | 原文线索

V2-051146 | MATHDB-357557

Doolittle's local-cone conjecture for graph embeddings

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ be a polytope and let $q\colon V(G_P)\to\mathbb{R}^e$ be a graph embedding of its edge-graph G_P. For each vertex $i\in V(G_P)$, define ... Assume that e...

conjecture | open | MathDB 页面 | 原文线索

V2-051147 | MATHDB-357558

Central symmetry reconstruction conjecture for polytopes

摘录 / EXCERPT: A centrally symmetric polytope is uniquely determined, up to isometry, by its edge-graph, edge lengths, and vertex-origin distances. Central symmetry reconstruction conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-051148 | MATHDB-357560

Extension of the main inequality to arbitrary total lifetimes

摘录 / EXCERPT: Let $(X_i(t))_{t\in[t_i]}$, for $i=1,\ldots,k$, be the discrete Markov chains considered in the paper, with total lifetime ... The total lifetime need not be even. Lifespan-parity co...

conjecture | open | MathDB 页面 | 原文线索

V2-051149 | MATHDB-357577

The non-cycle bound for the non-backtracking Laplacian parameter

摘录 / EXCERPT: Let G be a graph with minimum degree ... denote the parameter introduced above. Assume that G is not the cycle graph. Non-cycle bound. ... This conjecture refines the preceding gen...

conjecture | open | MathDB 页面 | 原文线索

V2-051150 | MATHDB-357578

Frankl-Huang-Rödl conjecture on local Turán densities

摘录 / EXCERPT: For integers $r\ge 2$ and sufficiently large integers p, let $t_r(2p+1,p+1)$ denote the local Turán density of the complete r-uniform hypergraph on $p+1$ vertices in a hypergra...

conjecture | open | MathDB 页面 | 原文线索

V2-051151 | MATHDB-357600

The Aharoni-Korman conjecture for ordered sets without infinite antichains

摘录 / EXCERPT: Let P be an ordered set with no infinite antichains, and let k be a positive integer. Aharoni and Korman's conjecture asserts that there are k chains $C_1,\ldots,C_k$ and a p...

conjecture | open | MathDB 页面 | 原文线索

V2-051152 | MATHDB-357605

The vertex-deletion conjecture for correspondence packing number

摘录 / EXCERPT: Let G be a graph and let $v\in V(G)$. Vertex-deletion conjecture. Then ... Equivalently, adding a new vertex adjacent to all existing vertices increases the correspondence packing...

conjecture | open | MathDB 页面 | 原文线索

V2-051153 | MATHDB-357611

Domination conjecture for neighborhoods in vertex-transitive reflection graphs

摘录 / EXCERPT: Neighborhood domination conjecture. If H is a vertex-transitive reflection graph with diameter d, then $H(\ell)$ dominates $H(k)$ for all $1\leq k\leq \ell\leq d$.

conjecture | open | MathDB 页面 | 原文线索

V2-051154 | MATHDB-357629

Optimal union vertex-distinguishing edge colorings of forests

摘录 / EXCERPT: Let F be a forest on n vertices, and let $\chi'_{\cup}(F)$ denote its union vertex-distinguishing edge chromatic number, where edge colors are sets and the union of the colors...

conjecture | open | MathDB 页面 | 原文线索

V2-051155 | MATHDB-357655

Multi-layer analogue of Meyniel's conjecture

摘录 / EXCERPT: Multi-layer analogue of Meyniel's conjecture. For every fixed $\tau\geqslant 1$ and every connected n-vertex graph G, one has

conjecture | open | MathDB 页面 | 原文线索

V2-051156 | MATHDB-357663

Smallness of the class $\mathcal{Q}$

摘录 / EXCERPT: Let $\mathcal{Q}$ be the graph class defined in the preceding construction. A graph class is small if there is a constant c such that the number of its labelled n-vertex graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-051157 | MATHDB-357673

Hansen's biregular interval-colourability conjecture

摘录 / EXCERPT: A graph is biregular if it is bipartite and all vertices in the same part have the same degree. A graph is interval colourable if it has a proper edge-colouring in which the colour...

conjecture | open | MathDB 页面 | 原文线索

V2-051158 | MATHDB-357694

Akbari-Etesami-Mahini-Mahmoody conjecture on rainbow Hamilton cycles

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $\chi'(K_n)$ denote its edge-chromatic number. A properly edge-coloured graph is one in which adjacent edges receive distin...

conjecture | open | MathDB 页面 | 原文线索

V2-051159 | MATHDB-357697

Evans-type conjecture for cartesian products with K_2

摘录 / EXCERPT: Let G be a graph, let k be an integer, and write $G\square K_2$ for the cartesian product of G and the complete graph on two vertices. Write $\chi'(G)$ for the edge-chromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-051160 | MATHDB-357700

Distance-three matching conjecture for Cartesian powers of even cycles

摘录 / EXCERPT: Let C_{2k} be an even cycle and let C^d_{2k} denote the d-fold Cartesian product of C_{2k} with itself. A distance-three matching is a matching whose distinct edges are at...

conjecture | open | MathDB 页面 | 原文线索

V2-051161 | MATHDB-357702

Zhao's nonnegative-real-part conjecture for directed bearing Laplacians

摘录 / EXCERPT: Let L_{B} be the bearing Laplacian of a directed formation. Zhao's conjecture. Every eigenvalue of L_{B} has nonnegative real part. This conjecture concerns the s...

conjecture | open | MathDB 页面 | 原文线索

V2-051162 | MATHDB-357705

The graph-monoid equality conjecture for two-vertex directed graphs

摘录 / EXCERPT: Let G be a graph on the vertex set x,y with adjacency matrix ... Assume $A\geq D$, $A,B,C,D>0$, and $(A-C)(B-D)<0$, and let $\mathcal{M}(G)$ denote its graph monoid. Graph-monoid...

conjecture | open | MathDB 页面 | 原文线索

V2-051163 | MATHDB-357706

Hilton's conjecture on subgraph-overfull graphs

摘录 / EXCERPT: Hilton's conjecture. If $\Delta(G)>\frac{|V|}{3}$, then G is Class 1 if and only if it is not subgraph-overfull. The supplied paper states that this conjecture holds for the clas...

conjecture | open | MathDB 页面 | 原文线索

V2-051164 | MATHDB-357707

Figueiredo et al.'s conjecture on chordal graph edge-coloring

摘录 / EXCERPT: Figueiredo et al.'s conjecture. G is Class 2 if and only if it is subgraph-overfull. This conjecture proposes a characterization of edge-chromatic Class 2 chordal graphs by the s...

conjecture | open | MathDB 页面 | 原文线索

V2-051165 | MATHDB-357719

Core-edge conjecture for one-vertex triangulations of lens spaces

摘录 / EXCERPT: Core-edge conjecture. Every one-vertex triangulation of a lens space has a core edge.

conjecture | open | MathDB 页面 | 原文线索

V2-051166 | MATHDB-357720

Tunnel-edge conjecture for one-vertex ideal triangulations of tunnel-number-one knots

摘录 / EXCERPT: Tunnel-edge conjecture. Every one-vertex ideal triangulation of the complement of K has a tunnel edge.

conjecture | open | MathDB 页面 | 原文线索

V2-051167 | MATHDB-357721

Seifert-fibre-edge conjecture for one-vertex triangulations of small Seifert fibre spaces

摘录 / EXCERPT: Seifert-fibre-edge conjecture. Every one-vertex triangulation of a small Seifert fibre space has an edge isotopic to a Seifert fibre.

conjecture | open | MathDB 页面 | 原文线索

V2-051168 | MATHDB-357722

Infinite counterexample conjecture for core edges in lens spaces

摘录 / EXCERPT: Infinite core-edge counterexample conjecture. Every lens space admits infinitely many one-vertex triangulations with no core edges.

conjecture | open | MathDB 页面 | 原文线索

V2-051169 | MATHDB-357724

Infinite counterexample conjecture for Seifert-fibre edges

摘录 / EXCERPT: Infinite Seifert-fibre counterexample conjecture. For every small Seifert fibre space that is not a lens space or prism manifold, there exist infinitely many one-vertex triangulati...

conjecture | open | MathDB 页面 | 原文线索

V2-051170 | MATHDB-357777

Shi and Shan's conjecture for tough connected $(P_2\cup kP_1)$-free graphs

摘录 / EXCERPT: Let k be an integer with $k\ge 4$, and let G be a graph. Here P_n denotes the path on n vertices, $P_2\cup kP_1$ is the disjoint union of a two-vertex path and k isolated...

conjecture | open | MathDB 页面 | 原文线索

V2-051171 | MATHDB-357778

Bose et al.'s maximum distance spectral radius conjecture for cacti

摘录 / EXCERPT: Bose et al.'s conjecture. The graph

conjecture | open | MathDB 页面 | 原文线索

V2-051172 | MATHDB-357780

Henning's Sweller-start competition-independence conjecture for trees

摘录 / EXCERPT: Henning's conjecture. For every tree T of order n, $I_s(T) \geq \frac{3}{7}n$. The paper disproves this conjecture by exhibiting arbitrarily large trees for which the Sweller-s...

conjecture | open | MathDB 页面 | 原文线索

V2-051173 | MATHDB-357787

Eventual stabilization of eigenvalues below -2 in graph subdivisions

摘录 / EXCERPT: Stabilization conjecture. There exists $t_0\in \mathbb{N}$ such that $m_{G_t}(-\infty,-2)$ is constant for all $t\ge t_0$.

conjecture | open | MathDB 页面 | 原文线索

V2-051174 | MATHDB-357791

The counting circuit double cover conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph with n vertices. A circuit double cover is a family of circuits covering every edge of G exactly twice. The counting circuit double cover co...

conjecture | open | MathDB 页面 | 原文线索

V2-051175 | MATHDB-357792

Quantitative max-sum matching conjecture for diametral balls

摘录 / EXCERPT: Let d be a positive integer. Let X be an even set of distinct points in $\mathbb R^d$ with minimum distance $r>0$, and let $\mathcal M$ be a max-sum matching of X. The closed...

conjecture | open | MathDB 页面 | 原文线索

V2-051176 | MATHDB-357800

Ahanjideh-Akbari-Fakharan-Trevisan conjecture on Laplacian eigenvalues and diameter

摘录 / EXCERPT: Let G be a connected graph of order n, let d be its diameter with $d\ge 2$, and let m_GI denote the number of Laplacian eigenvalues of G in an interval $I\subseteq[0,n]$....

conjecture | open | MathDB 页面 | 原文线索

V2-051177 | MATHDB-357802

Nonexistence of constant-factor approximation algorithms for directed strengthening inversions

摘录 / EXCERPT: Approximation-hardness conjecture. Unless P=NP, there is no α-approximation algorithm for computing $\operatorname{sinv}_k(D)$ or $\operatorname{sinv}'_k(D)$.

conjecture | open | MathDB 页面 | 原文线索

V2-051178 | MATHDB-357804

The extremal graph conjecture for unions of cliques

摘录 / EXCERPT: Extremal graph conjecture for unions of cliques. Suppose $2\leqslant p\leqslant q$, $n>(t-1)(q+1)^2+3(p-2)$ and $t\geqslant 4$. Then $H(n;p,q,\ldots,q)$ is the unique extremal grap...

conjecture | open | MathDB 页面 | 原文线索

V2-051179 | MATHDB-357807

Regularity conjecture for Šoltés graphs

摘录 / EXCERPT: A Šoltés graph is a graph G for which deleting any vertex leaves the Wiener index unchanged. Regularity conjecture. If G is a Šoltés graph, then G is regular. This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-051180 | MATHDB-357808

Vertex-transitivity conjecture for Šoltés graphs

摘录 / EXCERPT: Let G be a graph, and let $W(G)$ denote its Wiener index. A Šoltés graph is a graph for which deleting any vertex leaves the Wiener index unchanged. A graph is vertex transitive...

conjecture | open | MathDB 页面 | 原文线索

V2-051181 | MATHDB-357809

Šoltés graph uniqueness conjecture

摘录 / EXCERPT: Let C_{11} denote the cycle on eleven vertices. A Šoltés graph is a graph for which deleting any vertex leaves the Wiener index unchanged. Šoltés graph uniqueness conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-051182 | MATHDB-357816

Hong, Kang and Yu's induced-cycle connectivity conjecture

摘录 / EXCERPT: Hong-Kang-Yu conjecture. There exists a smallest positive integer $g(k,l)$ such that, for every $g(k,l)$-connected graph G, every edge $e otin E(G[X])$ with $X e V(G)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-051183 | MATHDB-357829

Authors-Shibata quadratic stable-set-ring conjecture for perfectly contractile graphs

摘录 / EXCERPT: Authors-Shibata conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-051184 | MATHDB-357830

Kempe-equivalence characterization of perfectly contractile graphs

摘录 / EXCERPT: Kempe-equivalence conjecture. The graph G is perfectly contractile if and only if, for every replication graph H of an arbitrary induced subgraph of G and every...

conjecture | open | MathDB 页面 | 原文线索

V2-051185 | MATHDB-357837

He-Li-Feng's even-prism extremal-number conjecture

摘录 / EXCERPT: Let $C_{2\ell}^{\square}=C_{2\ell}\square K_2$ be the 2ℓ-prism, formed from two vertex-disjoint 2ℓ-cycles with a matching joining corresponding vertices. For a graph H,...

conjecture | open | MathDB 页面 | 原文线索

V2-051186 | MATHDB-357838

The hexagonal-prism extremal-number conjecture

摘录 / EXCERPT: Let $C_6^{\square}=C_6\square K_2$ be the hexagonal prism, and let $\mathrm{ex}(n,H)$ denote the maximum number of edges in an n-vertex H-free graph. Hexagonal-prism conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-051187 | MATHDB-357842

Chappell-Pelsmajer's induced subgraph conjecture for bounded-treewidth graphs

摘录 / EXCERPT: Let G be a graph of order n and treewidth at most k, where $k\geq 0$. An induced subgraph $G[S]$ has maximum degree at most 1 if every vertex in S is adjacent to at most...

conjecture | open | MathDB 页面 | 原文线索

V2-051188 | MATHDB-357843

One-third conjecture for clustered sets in planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let $S\subseteq V(G)$. Write $\alpha_{\mathcal{P},2}(G)$ for the maximum size of a set S whose induced subgraph has maximum degree at most 1, equ...

conjecture | open | MathDB 页面 | 原文线索

V2-051189 | MATHDB-357849

Super vertex total local antimagic labeling existence conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite simple undirected graph without isolated vertices. A super vertex total local antimagic labeling is a bijection $f:V\cup E\to\{1,2,\dots,|V|+|E|\}$ satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-051190 | MATHDB-357873

Lexicographic homological linear quotients conjecture for powers of cover ideals

摘录 / EXCERPT: Lexicographic homological quotients conjecture. For all $k\geq 0$ and all $\ell\geq 1$, the ideal $\operatorname{HS}_k((I(G)^\vee)^\ell)$ has linear quotients with respect to the l...

conjecture | open | MathDB 页面 | 原文线索

V2-051191 | MATHDB-357883

The neighborhood expansion conjecture for positive entropic curvature on graph spaces

摘录 / EXCERPT: Let $G=({\mathcal X},E)$ be a graph endowed with the counting measure. For $z\in {\mathcal X}$, let $S_2(z)$ denote the vertices at graph distance two from z, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-051192 | MATHDB-357889

Ryser's conjecture on rainbow cycle packings

摘录 / EXCERPT: Ryser's conjecture. If n is odd, then $\vec{K}_n$ contains a rainbow spanning subgraph in which every vertex has in-degree and out-degree equal to one. Equivalently, $\vec{K}_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-051193 | MATHDB-357910

Order bounds for 2K2-split minimal (∞, k)-polar obstructions

摘录 / EXCERPT: Order-bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-051194 | MATHDB-357916

Triangular realisation conjecture for graphical degree sequences

摘录 / EXCERPT: Triangular realisation conjecture. Any graphical sequence all of whose terms are greater than or equal to 4 admits a triangular realisation.

conjecture | open | MathDB 页面 | 原文线索

V2-051195 | MATHDB-357942

The exact formula conjecture for the minimum arc count of oriented cliques

摘录 / EXCERPT: Let $f_2(n)$ denote the minimum number of arcs in an oriented graph with weak diameter 2 on n vertices, and let x_n be the upper bound constructed in the paper. An oriented c...

conjecture | open | MathDB 页面 | 原文线索

V2-051196 | MATHDB-357943

The cut-vertex extremal oriented clique conjecture

摘录 / EXCERPT: Let $f_2(n)$ be the minimum number of arcs in an oriented graph with weak diameter 2 on n vertices. An oriented clique is an oriented graph in which every pair of distinct vert...

conjecture | open | MathDB 页面 | 原文线索

V2-051197 | MATHDB-357951

Convergence-radius conjecture for STARLINK temporal visibility graphs

摘录 / EXCERPT: Let n be the number of nodes, and let a sub-TVG be a randomly sampled sub-temporal visibility graph of the STARLINK satellite system. Let r denote the convergence radius of its...

conjecture | open | MathDB 页面 | 原文线索

V2-051198 | MATHDB-357956

Limit-point conjecture for the top eigenvalue vectors of regular graphs

摘录 / EXCERPT: Top-eigenvalue vector limit-point conjecture. For any $d\geq 3$ and any fixed k, the set of all limit points of the vectors

conjecture | open | MathDB 页面 | 原文线索

V2-051199 | MATHDB-357957

Extremality conjecture for generalized path graphs

摘录 / EXCERPT: Let $d\geq 1$ and let $P_{n,d}$ be the generalized path graph, an n-vertex minimally d-rigid graph. For a graph G, let $a_d(G)$ denote its d-dimensional algebraic connectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-051200 | MATHDB-357970

Ahanjideh-Ekim-Yıldız formula for maximum edges in triangle-free graphs

摘录 / EXCERPT: Ahanjideh-Ekim-Yıldız conjecture. For all natural numbers $d\geq 2$ and m, we have

conjecture | open | MathDB 页面 | 原文线索

V2-051201 | MATHDB-357975

The normalized Laplacian maximum-connectivity conjecture

摘录 / EXCERPT: Normalized Laplacian maximum-connectivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051202 | MATHDB-357976

The equality characterization for the normalized Laplacian maximum bound

摘录 / EXCERPT: Let G be a graph on $n\geq5$ vertices, with normalized Laplacian eigenvalues $0=\lambda_1(G)\leq\lambda_2(G)\leq\cdots\leq\lambda_n(G)$. Normalized Laplacian maximum-bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-051203 | MATHDB-357977

The connected-complement normalized Laplacian sum conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\lambda_2(G)$ denote the second-smallest eigenvalue of its normalized Laplacian. Connected-complement normalized Laplacian conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051204 | MATHDB-357980

Ustimenko's eigenvalue conjecture for Lazebnik-Ustimenko-Woldar graphs

摘录 / EXCERPT: Ustimenko's conjecture. For all (k,q), every nontrivial eigenvalue of $D(k,q)$, and therefore of $CD(k,q)$, is at most $2\sqrt q$.

conjecture | open | MathDB 页面 | 原文线索

V2-051205 | MATHDB-357981

Harcos-Soltész conjecture on $K_{2,4}$-creating Hamilton paths

摘录 / EXCERPT: Harcos-Soltész conjecture. There is a constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051206 | MATHDB-357982

Hansen-Vukicevic's conjecture for commuting and non-commuting graphs of finite groups

摘录 / EXCERPT: Hansen-Vukiv{c}evi'{c}'s conjecture for group graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-051207 | MATHDB-357985

Gan-Han-Sun-Wang conjecture on large $Y_{k,b}$-tilings

摘录 / EXCERPT: Gan-Han-Sun-Wang conjecture. The displayed asymptotic upper bound should hold. This conjecture generalizes the matching problem to tilings by two-edge hypergraphs with prescribed i...

conjecture | open | MathDB 页面 | 原文线索

V2-051208 | MATHDB-358017

The characterization conjecture for 2-extremal digraphs

摘录 / EXCERPT: Let $\mathcal H_2$ be the smallest class of digraphs containing symmetric odd cycles and closed under taking directed Hajós joins and 2-Hajós tree joins. A digraph is 2-extrema...

conjecture | open | MathDB 页面 | 原文线索

V2-051209 | MATHDB-358020

Asymptotic formula for linear-system chromatic number

摘录 / EXCERPT: For integers $k\ge 3$, let $f_1(e,q,k)$ denote the maximum chromatic number of a 1-(q,k)-system with e q-cliques. The asymptotic chromatic-number conjecture. Fix $k\ge 3$....

conjecture | open | MathDB 页面 | 原文线索

V2-051210 | MATHDB-358028

The looped-subgraph energy conjecture for simple graphs

摘录 / EXCERPT: Let G be a simple graph, let $V(G)$ be its vertex set, and for $S\subseteq V(G)$ let G_S denote the graph obtained by adding loops at the vertices in S. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-051211 | MATHDB-358042

Cranston et al.'s revised planar Turán conjecture for cycles

摘录 / EXCERPT: Let C_k be the cycle on k vertices, and let $\operatorname{ex}_{\mathcal{P}}(n,C_k)$ denote the maximum number of edges in an n-vertex planar graph containing no copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-051212 | MATHDB-358060

Li's planar Turán conjecture for two 4-cycles

摘录 / EXCERPT: Let ${\rm ex}_{\mathcal{P}}(n,2C_4)$ denote the maximum number of edges in an n-vertex planar graph containing no two vertex-disjoint copies of C_4. Li's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-051213 | MATHDB-358082

Li, Nikiforov and Schelp's monochromatic cycle conjecture

摘录 / EXCERPT: Let $n\geq4$ and let G be an n-vertex graph with minimum degree $\delta(G)>\frac{3}{4}n$. A 2-edge-coloring of G is a partition $E(G)=E(B)\uplus E(R)$ into blue and red edges...

conjecture | open | MathDB 页面 | 原文线索

V2-051214 | MATHDB-358083

Benevides, Łuczak, Scott, Skokan and White's long monochromatic cycle conjecture

摘录 / EXCERPT: Let G be an n-vertex graph with minimum degree $\delta(G)\geq\frac{3}{4}n$, and write ... A 2-edge-coloring assigns one of two colors to every edge of G. Benevides, Łuczak, Sco...

conjecture | open | MathDB 页面 | 原文线索

V2-051215 | MATHDB-358120

Esperet-Lemoine-Maffray conjecture on induced paths in degenerate graphs

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let G be a k-degenerate graph with a path of order $n\geq 1$. An induced path is a path whose vertices induce exactly the edges of the path in G. Es...

conjecture | open | MathDB 页面 | 原文线索

V2-051216 | MATHDB-358166

An upper-bound conjecture for the essential spectral radius index

摘录 / EXCERPT: Upper-bound conjecture for $j^{\star}$. There exists $\overline{j}\in\{2,\ldots,n\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051217 | MATHDB-358173

Characteristic-polynomial equality for type B Shi and Ish arrangements

摘录 / EXCERPT: Characteristic-polynomial equality conjecture. The arrangements $\mathcal{S}(G)$ and $\mathcal{I}(G)$ have the same characteristic polynomial for any digraph G. This conjecture e...

conjecture | open | MathDB 页面 | 原文线索

V2-051218 | MATHDB-358179

The common-zero graph component conjecture

摘录 / EXCERPT: Common-zero graph component conjecture. The number of connected components of $\Gamma_v(G)$ is at most 3.

conjecture | open | MathDB 页面 | 原文线索

V2-051219 | MATHDB-358180

Conlon, Fox, Lee and Sudakov's Erdős-Gyárfás function conjecture

摘录 / EXCERPT: Let $f_r(n,p,q)$ denote the least number of colours in an r-uniform colouring of the complete hypergraph on n vertices with no induced copy of a p-vertex hypergraph using at...

conjecture | open | MathDB 页面 | 原文线索

V2-051220 | MATHDB-358191

Siggers-behavior characterization of k-consistency for network satisfaction problems

摘录 / EXCERPT: Siggers-behavior conjecture. Dropping the requirement that ${\bf A}$ has all 1-cycles should yield a necessary and sufficient condition for solvability of...

conjecture | open | MathDB 页面 | 原文线索

V2-051221 | MATHDB-358193

The weak K-two-threes conjecture for metric matrices

摘录 / EXCERPT: Let M be an n-by-n metric matrix, meaning a hollow, symmetric, nonnegative matrix satisfying $m_{jk}\leq m_{ij}+m_{jk}$ for all $1\leq i,j,k\leq n$. Equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-051222 | MATHDB-358194

The strong K-two-threes conjecture for weighted graph distance matrices

摘录 / EXCERPT: Let G be a graph, let ϕ be an edge-weighting, and let $D_{\phi G}$ denote the weighted distance matrix of G. For a positive integer n, let $K_{2,3,\ldots,3}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-051223 | MATHDB-358205

Homotopy type conjecture for cut complexes of squared cycles

摘录 / EXCERPT: Let W_n be the squared cycle graph, with vertex set $[n]$ and edges $(i,i+1\bmod n)$ and $(i,i+2\bmod n)$. For $k\ge3$, consider its 4-dimensional cut complex $\Delta_k(W_{k+5})$...

conjecture | open | MathDB 页面 | 原文线索

V2-051224 | MATHDB-358215

Exact 2-dimensional flow number conjecture for the Petersen graph

摘录 / EXCERPT: Petersen graph flow-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051225 | MATHDB-358217

The converse conjecture for irreducible graded-local conformal nets

摘录 / EXCERPT: Let $\mathcal{A}$ be an irreducible graded-local conformal net, meaning a graded-local conformal net satisfying the irreducibility condition. Converse conformal-net conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051226 | MATHDB-358225

Mubayi-Yepremyan conjecture for random 3-graphs and linear 4-cycles

摘录 / EXCERPT: Let $G^3_{n,p}$ be the random 3-uniform hypergraph on n vertices in which each 3-edge is present independently with probability p, and let C^3_4 be the 3-uniform linear...

conjecture | open | MathDB 页面 | 原文线索

V2-051227 | MATHDB-358233

Piecewise asymptotic conjecture for random 3-uniform linear 4-cycles

摘录 / EXCERPT: Piecewise asymptotic conjecture for random 3-uniform linear 4-cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-051228 | MATHDB-358241

Everett-Reed's conjecture on even-contractile Berge graphs

摘录 / EXCERPT: Let G be a Berge graph, meaning a graph with no induced odd hole and no induced odd antihole. An antihole is the complement of a hole, and an odd prism is a prism whose three run...

conjecture | open | MathDB 页面 | 原文线索

V2-051229 | MATHDB-358243

The total sign-skew-symmetric principal-coefficients conjecture

摘录 / EXCERPT: Let B be a totally sign-skew-symmetric matrix, and let $\bar{\mathcal{A}}$ be the cluster algebra with principal coefficients. For each vertex t of the exchange tree, write the...

conjecture | open | MathDB 页面 | 原文线索

V2-051230 | MATHDB-358262

The average-degree generalization of the independent-set domination inequality

摘录 / EXCERPT: Let G be a graph with average vertex degree at most k, where $k\geq 1$. Let $i(G)$ denote the independent domination number of G, and let $\partial_k(G)$ denote the minimum n...

conjecture | open | MathDB 页面 | 原文线索

V2-051231 | MATHDB-358270

The strict lower-bound conjecture for the robust clique covering number

摘录 / EXCERPT: Let G be an isolate-free graph. Write $\theta(G)$ for its clique covering number and $\theta_1(G)$ for the corresponding robustness parameter, so that the general lower bound is...

conjecture | open | MathDB 页面 | 原文线索

V2-051232 | MATHDB-358275

Frank number conjecture for cyclically 4-edge-connected graphs

摘录 / EXCERPT: Let G be a cyclically 4-edge-connected graph, meaning that deleting fewer than 4 edges cannot separate G into two components both containing a cycle. Let $fn(G)$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-051233 | MATHDB-358311

The self-duality conjecture for finite LR-structures

摘录 / EXCERPT: Let $(\Gamma,\mathcal{C})$ be a finite LR-structure, and let $\operatorname{Aut}^+(\Gamma,\mathcal{C})$ denote its orientation-preserving automorphism group. The LR-structure is se...

conjecture | open | MathDB 页面 | 原文线索

V2-051234 | MATHDB-358313

A Menger-type conjecture for many induced paths

摘录 / EXCERPT: Many-path Menger-type conjecture. There exists a constant c such that, for all such G, X, Y, d, and k, either there exist k disjoint X-Y paths $P_1,\dots,P_k$ satis...

conjecture | open | MathDB 页面 | 原文线索

V2-051235 | MATHDB-358324

Asymptotic success probability conjecture for the edge-addition process

摘录 / EXCERPT: Edge-addition success-probability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051236 | MATHDB-358325

Quadratic edge-count conjecture for the (1,1) edge-addition process

摘录 / EXCERPT: Quadratic edge-count conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051237 | MATHDB-358326

Logarithmic path-length conjecture for the (1,1) edge-removal process

摘录 / EXCERPT: Logarithmic path-length conjecture. The expected maximum directed path length of the resulting task-dependency graph is

conjecture | open | MathDB 页面 | 原文线索

V2-051238 | MATHDB-358340

Connected transversal threshold conjecture for complete multipartite graphs

摘录 / EXCERPT: Connected transversal conjecture. The lower bound in Theorem is tight. In particular, for any $r\geq 4$, every r-partite graph with r-partite density at least...

conjecture | open | MathDB 页面 | 原文线索

V2-051239 | MATHDB-358341

Multipartite Dirac conjecture for Hamiltonicity

摘录 / EXCERPT: Multipartite Dirac conjecture. The Hamiltonicity threshold should converge to $\frac{1}{2}$ as $r\rightarrow\infty$:

conjecture | open | MathDB 页面 | 原文线索

V2-051240 | MATHDB-358344

Conjecture that fewer graphs lie in A than in B

摘录 / EXCERPT: The A-B cardinality conjecture. $|A|<|B|$.

conjecture | open | MathDB 页面 | 原文线索

V2-051241 | MATHDB-358355

Hyperplane covering conjecture for the permutohedron

摘录 / EXCERPT: Let P_n be the permutohedron with vertex set contained in the hyperplane ... Let $H_1,\ldots,H_m$ be affine hyperplanes different from H_0 whose union contains the vertex set of...

conjecture | open | MathDB 页面 | 原文线索

V2-051242 | MATHDB-358359

Linear minimum-out-degree conjecture for transitive tournament immersions

摘录 / EXCERPT: Let D be a digraph, let k be a positive integer, and write $\delta^{+}(D)$ for its minimum out-degree. A transitive tournament on k vertices is the tournament whose vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-051243 | MATHDB-358384

Coarse Thomassen theorem for vertex-transitive graphs

摘录 / EXCERPT: Coarse Thomassen theorem. Either G is quasi-isometric to a planar graph, or G contains every finite graph as an asymptotic minor.

conjecture | open | MathDB 页面 | 原文线索

V2-051244 | MATHDB-358392

The conjecture on the number of 5-cycles in integer distance graphs

摘录 / EXCERPT: Let T be the set of positive integers under consideration in the paper, let $T_1=\{22,58,70,82,142,190,298,330,358,382,478,658,742,862,1222,1978\}$, and let $C_3(t)$ denote the n...

conjecture | open | MathDB 页面 | 原文线索

V2-051245 | MATHDB-358393

The conjecture on odd girth of rational three-dimensional distance graphs

摘录 / EXCERPT: Let T be the set of positive integers under consideration, and let $G(\mathbb{Q}^3,\sqrt{t})$ denote the graph whose vertices are points of $\mathbb{Q}^3$ with adjacency determin...

conjecture | open | MathDB 页面 | 原文线索

V2-051246 | MATHDB-358396

SyCo-AE conjecture on approximating full-space constraints

摘录 / EXCERPT: SyCo-AE conjecture. The full-space constraint can be approximated by

conjecture | open | MathDB 页面 | 原文线索

V2-051247 | MATHDB-358402

The sharp upper-bound conjecture for the Halin Turán number of the 6-cycle

摘录 / EXCERPT: Let $\operatorname{ex}_{\mathcal{H}}(n,C_6)$ denote the maximum number of edges in an n-vertex Halin graph containing no cycle of length 6. The sharp upper-bound conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-051248 | MATHDB-358411

Strong orientation conjecture for maximum total arc-connectivity

摘录 / EXCERPT: Let G be a 2-edge-connected graph, let $\vec{G}$ be an orientation of G, and let $tac(\vec{G})$ denote its total arc-connectivity. An orientation maximizes total arc-connectivi...

conjecture | open | MathDB 页面 | 原文线索

V2-051249 | MATHDB-358412

Well-balanced orientation conjecture for maximum total arc-connectivity

摘录 / EXCERPT: Let G be a graph, let $\vec{G}$ be an orientation of G, and let $tac(\vec{G})$ denote its total arc-connectivity. An orientation maximizes total arc-connectivity when it maximi...

conjecture | open | MathDB 页面 | 原文线索

V2-051250 | MATHDB-358422

The convex-drawing extension conjecture for plane Hamiltonian cycles

摘录 / EXCERPT: Convex-drawing extension conjecture. Every plane Hamiltonian cycle C in D can be extended to a plane Hamiltonian subdrawing on $2n-3$ edges.

conjecture | open | MathDB 页面 | 原文线索

V2-051251 | MATHDB-358423

The h-convex drawings conjecture on crossed edges

摘录 / EXCERPT: h-convex crossed-edges conjecture. There exists an h-convex drawing of K_n in which every edge is crossed.

conjecture | open | MathDB 页面 | 原文线索

V2-051252 | MATHDB-358473

Logarithmic clique-width conjecture for classes of bounded stretch-width

摘录 / EXCERPT: Let $\mathcal C$ be a class of graphs of bounded stretch-width. For an n-vertex graph $G\in\mathcal C$, let the clique-width of G be the minimum number of labels needed to cons...

conjecture | open | MathDB 页面 | 原文线索

V2-051253 | MATHDB-358482

Upper decay-bound conjecture for NIMFA SIS prevalence below threshold

摘录 / EXCERPT: Let G be a graph, let K_N be the complete graph on N vertices, let u denote the all-infected initial infection-probability vector, and let $V(0)$ be an initial infection-pr...

conjecture | open | MathDB 页面 | 原文线索

V2-051254 | MATHDB-358483

Above-threshold prevalence-distance decay conjecture for NIMFA SIS epidemics

摘录 / EXCERPT: Let G be a graph, let K_N be the complete graph on N vertices, let u be the all-infected initial infection-probability vector, and let V_∞ be the endemic steady-stat...

conjecture | open | MathDB 页面 | 原文线索

V2-051255 | MATHDB-358501

Tournament strengthening of the Lefmann-Rödl-Thomas conjecture

摘录 / EXCERPT: A tournament is an orientation of a complete graph. For non-negative integers l and k, an l-flash is a directed walk of length l whose edge colours are all the same, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-051256 | MATHDB-358502

Tournament path strengthening of the flash-and-rainbow conjecture

摘录 / EXCERPT: Let l and k be positive integers, and let T be a tournament with $l^{k-1}+1$ vertices. A directed monochromatic path has all edges of one colour, while a directed rainbow pat...

open_problem | open | MathDB 页面 | 原文线索

V2-051257 | MATHDB-358510

Workshop characterization conjecture for generically universally rigid graphs on the line

摘录 / EXCERPT: Workshop characterization conjecture. The graph G is generically universally rigid in $\mathbb{R}^1$ if and only if it can be obtained from a set of triangles by edge reduced att...

conjecture | open | MathDB 页面 | 原文线索

V2-051258 | MATHDB-358525

Weakening conjecture for compatible-inclusion problems

摘录 / EXCERPT: Let $c\geq 1$ be an odd integer, and let (P,Q) be a compatible-inclusion problem, with its c-weakening defined by retaining directed edges represented by paths of length at mos...

conjecture | open | MathDB 页面 | 原文线索

V2-051259 | MATHDB-358527

The cycle in an out-neighbourhood conjecture

摘录 / EXCERPT: Cycle in an out-neighbourhood conjecture. For every graph G with sufficiently large chromatic number and every tournament T on the same vertex set, there exists a vertex v su...

conjecture | open | MathDB 页面 | 原文线索

V2-051260 | MATHDB-358546

The random Turán conjecture for bipartite graphs

摘录 / EXCERPT: The random Turán conjecture. With high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-051261 | MATHDB-358562

Staden-Treglown conjecture for squares of Hamilton cycles

摘录 / EXCERPT: Staden-Treglown conjecture. For every $\mu>0$, there exist $\alpha>0$ and $ _0\in \mathbb{N}$ such that the following holds. For every n-vertex graph G with $ \geq n_0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-051262 | MATHDB-358563

Conjectured threshold for powers of Hamilton cycles

摘录 / EXCERPT: Connecting-barrier conjecture. Given $\mu>0$ and $r\geq 4$, there exists $\alpha>0$ such that the following holds for sufficiently large n. If

conjecture | open | MathDB 页面 | 原文线索

V2-051263 | MATHDB-358568

Galvin-McKinley-Perkins-Sarantis-Tetali zero-free region conjecture for linear hypergraphs

摘录 / EXCERPT: Galvin-McKinley-Perkins-Sarantis-Tetali conjecture. For each $k\geq 2$, there exists a constant $C_k>0$ such that, if G is a k-uniform linear hypergraph of maximum degree...

conjecture | open | MathDB 页面 | 原文线索

V2-051264 | MATHDB-358587

Bašić et al.'s existence conjecture for circulant nut graphs of degree divisible by four

摘录 / EXCERPT: A circulant nut graph is a circulant graph whose adjacency matrix has a one-dimensional null space spanned by a full vector. Let d be a degree satisfying $d \equiv_4 0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051265 | MATHDB-358590

Genericity conjecture for g-rigidity matroids and sparsity matroids

摘录 / EXCERPT: Let $g:(\mathbb{F}^d)^k\to\mathbb{F}$ be a polynomial map, let ${\cal M}_{g,n}$ be its g-rigidity matroid on the k-uniform complete hypergraph, and let ${\cal S}_n^{d,0}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-051266 | MATHDB-358610

Kostochka-Stocker equality conjecture for connected cubic graphs

摘录 / EXCERPT: Kostochka-Stocker equality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051267 | MATHDB-358618

Steiner distance hyperdeterminant conjecture for trees

摘录 / EXCERPT: Let T be a tree on $n eq 3$ vertices, let k be a positive integer, and let $d4ae_T$ be the order-k Steiner distance hypermatrix of T, whose $(v_1,d7e2,d7e3,d7e4,d7e5,d7e6)$...

conjecture | open | MathDB 页面 | 原文线索

V2-051268 | MATHDB-358641

The zero forcing conjecture for Johnson graphs

摘录 / EXCERPT: Let $G_n=K(n,2)$ be the Johnson graph whose vertices are the 2-subsets of an n-element set, with two vertices adjacent when the corresponding subsets intersect in one element....

conjecture | open | MathDB 页面 | 原文线索

V2-051269 | MATHDB-358648

Fan's weighted refinement of the 4-flow conjecture

摘录 / EXCERPT: Fan's weighted refinement. If there is a $\mathbb{Z}_2\times\mathbb{Z}_2$-flow f such that

conjecture | open | MathDB 页面 | 原文线索

V2-051270 | MATHDB-358651

Wilson's conjecture on non-trivially unstable Rose Window graphs

摘录 / EXCERPT: Wilson's conjecture. Every non-trivially unstable Rose Window graph is isomorphic to a graph in one of the families W1--W9.

conjecture | open | MathDB 页面 | 原文线索

V2-051271 | MATHDB-358657

MAX-VP completeness conjecture

摘录 / EXCERPT: Let G be a graph, let k be an integer, and let a numbering assign the integers $\{1,\dots,n\}$ distinctly to the vertices of G. A valid path is a 2-path whose middle vertex h...

conjecture | open | MathDB 页面 | 原文线索

V2-051272 | MATHDB-358662

Conjecture relating function-space and finite-set closedness for sparse ReLU networks

摘录 / EXCERPT: Closedness-equivalence conjecture. An architecture $\boldsymbol{I}$ has a closed function space on the cube if and only if its realization set is closed on arbitrary finite domains...

conjecture | open | MathDB 页面 | 原文线索

V2-051273 | MATHDB-358663

Necessary-and-sufficient closedness conjecture for shallow sparse ReLU networks

摘录 / EXCERPT: Shallow-network closedness conjecture. The set $\mathcal{F}_{\boldsymbol{I}}(\Omega)$ is closed in $(C^0(\Omega),\|\cdot\|_\infty)$ if and only if $\boldsymbol{I}$ satisfies the se...

conjecture | open | MathDB 页面 | 原文线索

V2-051274 | MATHDB-358667

Liu-Shan-Xu conjecture on the maximum edge Mostar index of bicyclic graphs

摘录 / EXCERPT: Let m be the size of a bicyclic graph, and let B_m and B^5_m denote the bicyclic graphs shown in the source's Figures X and Y, respectively. Liu-Shan-Xu's conjecture. If m...

conjecture | open | MathDB 页面 | 原文线索

V2-051275 | MATHDB-358688

Spanning subdivision conjecture for dense graphs

摘录 / EXCERPT: Spanning subdivision conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051276 | MATHDB-358705

Conlon-Fox-Wigderson's book Ramsey-number conjecture

摘录 / EXCERPT: For graphs G and H, the Ramsey number $r(G,H)$ is the smallest positive integer N such that every red/blue edge-coloring of the complete graph K_N contains either a red G...

conjecture | open | MathDB 页面 | 原文线索

V2-051277 | MATHDB-358735

The stronger-than relation conjecture for graphs without semi-edges

摘录 / EXCERPT: The stronger-than relation conjecture. If A has no semi-edges, then

conjecture | open | MathDB 页面 | 原文线索

V2-051278 | MATHDB-358841

Conjecture on Nichols algebras of screening operators and logarithmic Kazhdan-Lusztig correspondences

摘录 / EXCERPT: Nichols-algebra and screening-operator conjecture. The screening operators generate a corresponding Nichols algebra $\mathfrak{N}$ in the braided tensor category $\mathcal{C}$;...

conjecture | open | MathDB 页面 | 原文线索

V2-051279 | MATHDB-358857

Keevash-Saks-Sudakov-Verstraëte rainbow Turán conjecture for critical graphs

摘录 / EXCERPT: Keevash-Saks-Sudakov-Verstraëte conjecture. Suppose $k\ge h$ and n is sufficiently large. Then

conjecture | open | MathDB 页面 | 原文线索

V2-051280 | MATHDB-358858

Weighted rainbow clique conjecture

摘录 / EXCERPT: Weighted rainbow clique conjecture. If $n\ge r$, G_0 contains no K_r with weight sequence bound $(1,2,\ldots,\binom{r}{2})$, then: (i) if $k=k_1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-051281 | MATHDB-358865

The density conjecture for distance-restricted firefighting on square grids

摘录 / EXCERPT: Let P_n be the path graph on n vertices, let $P_n\mathbin{\square}P_n$ be the Cartesian-product square grid, and let $\operatorname{DRMVS}(G,s;d,b)$ denote the maximum number o...

conjecture | open | MathDB 页面 | 原文线索

V2-051282 | MATHDB-358866

The DET:OLD density conjecture for C_4-free cubic graphs

摘录 / EXCERPT: Let G be a C_4-free cubic graph, and let $\textrm{DET:OLD}\%(G)$ denote the minimum density of an error-detecting open-locating-dominating set in G. DET:OLD density conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-051283 | MATHDB-358867

Davila-Henning zero-forcing conjecture for cubic graphs and total domination

摘录 / EXCERPT: Davila-Henning zero-forcing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051284 | MATHDB-358883

Cofiniteness conjecture for relative semi-infinite cohomology of admissible affine algebras

摘录 / EXCERPT: Let $L_k(\mathfrak{sl}_n)$ be the simple affine vertex algebra at the admissible level $k=-n+\frac{p}{q}$. Cofiniteness conjecture. The following vertex algebras are C_2-cofinite...

conjecture | open | MathDB 页面 | 原文线索

V2-051285 | MATHDB-358884

Semikhatov-Tipunin's logarithmic Kazhdan-Lusztig correspondence conjecture

摘录 / EXCERPT: Let V be one of the vertex algebras specified by the paper's extension pairs, let A be the corresponding free field algebra, and let $\mathfrak{N}$ be the Nichols algebra gener...

conjecture | open | MathDB 页面 | 原文线索

V2-051286 | MATHDB-358888

Ikeda-Kitabeppu-Takai-Uehara's curvature formula conjecture

摘录 / EXCERPT: Ikeda-Kitabeppu-Takai-Uehara's conjecture. For any $x,y\in V$,

conjecture | open | MathDB 页面 | 原文线索

V2-051287 | MATHDB-358889

Interchange-of-limit-and-infimum conjecture for weak Kantorovich differences

摘录 / EXCERPT: Interchange-of-limit-and-infimum conjecture. For any $x,y\in V$,

conjecture | open | MathDB 页面 | 原文线索

V2-051288 | MATHDB-358900

Subpolynomial growth conjecture for even-edge cliques

摘录 / EXCERPT: Let K_p be the complete graph on p vertices, and let $f(n,H)$ be the smallest number of colors in an edge coloring of K_n in which every copy of H intersects at least one c...

conjecture | open | MathDB 页面 | 原文线索

V2-051289 | MATHDB-358934

Asymptotic Euler characteristic conjecture for moduli spaces of graphs

摘录 / EXCERPT: Let $\mathcal{MG}_{g,n}$ be the moduli space of graphs of rank g with n legs, let $\chi(\Gamma_{g,n})$ denote the virtual Euler characteristic of the corresponding group, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-051290 | MATHDB-358935

The bipartite extremal bound for the eccentricity strength measure

摘录 / EXCERPT: Bipartite extremal bound conjecture. The maximum value of $\mathrm{espr}(G)$ over all bipartite n-vertex graphs G is

conjecture | open | MathDB 页面 | 原文线索

V2-051291 | MATHDB-358936

Trinajstić's uniqueness conjecture for the extremal graph

摘录 / EXCERPT: Trinajstić extremal uniqueness conjecture. The graph with the maximal value of the Trinajstić index is unique and is the graph consisting of a complete subgraph...

conjecture | open | MathDB 页面 | 原文线索

V2-051292 | MATHDB-358937

Trinajstić's regularity conjecture for zero index

摘录 / EXCERPT: Trinajstić regularity conjecture. The minimum value of the Trinajstić index is 0, and being regular is a necessary but not sufficient condition for a graph to attain this minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-051293 | MATHDB-358938

The equivalence of NT-balanced and ultra NT-balanced graphs

摘录 / EXCERPT: NT-balance equivalence conjecture. Every NT-balanced graph is also ultra NT-balanced.

conjecture | open | MathDB 页面 | 原文线索

V2-051294 | MATHDB-358948

Cycle-length conjecture for toppled digraphs of uniform trees

摘录 / EXCERPT: Let $\mathcal{T}$ be an r-tree with maximum degree Δ, where $r\geq 3$. A toppled digraph is a digraph associated with $\mathcal{T}$ through a choice of ordering and toppli...

conjecture | open | MathDB 页面 | 原文线索

V2-051295 | MATHDB-358949

Clark-Cooper divisibility conjecture for matching polynomials of uniform trees

摘录 / EXCERPT: Let $\mathcal{H}$ be an r-tree with $r\geq 3$, and let H be a subgraph induced by some vertex subset. Let φ_H denote the matching polynomial of H, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051296 | MATHDB-358953

The accessible-graph characterization of Cohen-Macaulay binomial edge ideals

摘录 / EXCERPT: Let G be a finite simple graph on vertex set $[n]$. For $S\subseteq[n]$, let $c_G(S)$ be the number of connected components of $G-S$, and call S a cut set if...

conjecture | open | MathDB 页面 | 原文线索

V2-051297 | MATHDB-358956

The characterization of $\mathbf{W_2}$ graphs by neighborhood independence

摘录 / EXCERPT: $\mathbf{W_2}$ characterization conjecture. For every graph G, the following assertions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-051298 | MATHDB-358958

Algebraicity conjecture for Ramsey multiplicity constants

摘录 / EXCERPT: Algebraicity conjecture. There do not exist graphs H_1 and H_2 such that $c(H_1,H_2)$ is transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-051299 | MATHDB-358987

Krivelevich-Sudakov Hamiltonicity conjecture for pseudorandom graphs

摘录 / EXCERPT: Let G be an (n,d,λ)-graph, meaning an n-vertex d-regular graph whose non-trivial eigenvalues have absolute value at most λ. A Hamilton cycle is a cycle contai...

conjecture | open | MathDB 页面 | 原文线索

V2-051300 | MATHDB-358996

The conjectured convergence-rate bound for the disagreement component

摘录 / EXCERPT: Let $\vec{z}_t$ denote the disagreement component of the opinion vector at time t, and let λ be the parameter appearing in the model. Here $\mathbb{E}$ denotes expectatio...

conjecture | open | MathDB 页面 | 原文线索

V2-051301 | MATHDB-359026

Nonexistence of odd-dimensional weakly Hadamard diagonalizable graphs with diagonalizable Laplacians

摘录 / EXCERPT: Odd-dimensional nonexistence conjecture. For any odd dimension, there is no weakly Hadamard diagonalizable graph whose Laplacian eigenspaces have equal algebraic and geometric mult...

conjecture | open | MathDB 页面 | 原文线索

V2-051302 | MATHDB-359029

Neural-collapse characterization for the graph unconstrained feature model

摘录 / EXCERPT: Consider the graph unconstrained feature model (gUFM) in the objective defined by equation (... (s{c1,1},ldots,s{cC,1})=cdots=(s{c1,n},ldots,s{cC,n}),qquad forall cin[C]. ...), if an...

conjecture | open | MathDB 页面 | 原文线索

V2-051303 | MATHDB-359053

Graph-theoretic partition conjecture for quantum-code connectivity graphs

摘录 / EXCERPT: Graph theory conjecture. There exists a partition

conjecture | open | MathDB 页面 | 原文线索

V2-051304 | MATHDB-359064

The conjecture that all odd cycles are strongly common

摘录 / EXCERPT: Let W be a graphon, meaning a symmetric measurable function $W:[0,1]^2\to[0,1]$, and let $t(H,W)$ denote the homomorphism density of a graph H in W. A graph H is strongly c...

conjecture | open | MathDB 页面 | 原文线索

V2-051305 | MATHDB-359065

The conjecture that an off-diagonal common pair has a common component

摘录 / EXCERPT: Let H_1 and H_2 be graphs, and let $p\in(0,1)$. A pair (H_1,H_2) is $(p,1-p)$-common when it satisfies the off-diagonal common-pair inequality for every complementary pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-051306 | MATHDB-359070

Erdős-Hajnal-Soukup partition conjecture for infinite digraphs

摘录 / EXCERPT: Erdős-Hajnal-Soukup partition conjecture. There is a partition

conjecture | open | MathDB 页面 | 原文线索

V2-051307 | MATHDB-359096

Regularity conjecture for graphs with lexicographically optimal Cartesian squares

摘录 / EXCERPT: Let G be a graph, and consider the Cartesian square $G\times G$. An ordering of the vertices is optimal for the edge-isoperimetric problem (EIP) when it minimizes the number of b...

conjecture | open | MathDB 页面 | 原文线索

V2-051308 | MATHDB-359120

The expansion characterization conjecture for graph independence ratios

摘录 / EXCERPT: For graphs G and H, define $\operatorname{Expand}(H)$ as the set of graphs obtainable from H by iterating vertex deletion, edge addition, and replacement of a vertex by a cli...

conjecture | open | MathDB 页面 | 原文线索

V2-051309 | MATHDB-359136

Ghosh et al.'s planar Turán bound for cycles

摘录 / EXCERPT: Ghosh et al.'s conjecture. For each $k\geq 7$ and sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-051310 | MATHDB-359145

Authors' conjecture on the non-realizability of $S_{\{i,n\}_{n}^{1}}$

摘录 / EXCERPT: The $m=1$ non-realizability conjecture. For $n\geqslant3$, the set $S_{\{i,n\}_{n}^{1}}$ is not Laplacian realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-051311 | MATHDB-359147

Maximum algebraic connectivity implies maximum girth

摘录 / EXCERPT: Maximum-connectivity-girth conjecture. For fixed degree d and order n, a graph maximizing $AC(G)$ also has maximum possible girth.

conjecture | open | MathDB 页面 | 原文线索

V2-051312 | MATHDB-359148

Monotonicity of maximum algebraic connectivity with girth

摘录 / EXCERPT: Algebraic-connectivity monotonicity conjecture. The function $f(g;d,n)$ is increasing as a function of g.

conjecture | open | MathDB 页面 | 原文线索

V2-051313 | MATHDB-359149

Girth constraints for diameter-maximal regular graphs

摘录 / EXCERPT: Diameter-maximal girth conjecture. If D is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-051314 | MATHDB-359150

Cichacz-Suchan conjecture on irregular minimum k-critical-bipartite graphs

摘录 / EXCERPT: Cichacz-Suchan's conjecture. The graph $\widehat{G}_{n,m}^{\lceil a\rceil}$ is k-critical-bipartite. This means that after the removal of any k vertices from U, every vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-051315 | MATHDB-359173

The short-edge crossing-number conjecture

摘录 / EXCERPT: Let $h(n)$ denote the maximum, over complete n-vertex simple topological graphs, of the minimum number of crossings of an edge in a suitable noncrossing matching construction. Sh...

conjecture | open | MathDB 页面 | 原文线索

V2-051316 | MATHDB-359174

The polynomial noncrossing-path conjecture

摘录 / EXCERPT: A complete n-vertex simple topological graph is a drawing of the complete graph on n vertices in which vertices are points and edges are simple curves, with no edge passing thr...

conjecture | open | MathDB 页面 | 原文线索

V2-051317 | MATHDB-359184

Discrete positive mass conjecture for asymptotically flat graphs

摘录 / EXCERPT: Discrete positive mass conjecture. For an asymptotically flat graph G with non-negative scalar curvature, one has $E(G)\geq 0$. Moreover, $E(G)=0$ if and only if G is a standar...

conjecture | open | MathDB 页面 | 原文线索

V2-051318 | MATHDB-359186

The radial tree-width conjecture for graphs excluding quasi-geodesic subdivisions of the triangle

摘录 / EXCERPT: Radial tree-width conjecture. There is a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that, if a connected graph G does not contain a c-quasi-geodesic $(\geqslant ck)$-sub...

conjecture | open | MathDB 页面 | 原文线索

V2-051319 | MATHDB-359190

Existence of high-girth bipartite graphs with prescribed logarithmic degree ratio

摘录 / EXCERPT: High-girth graph existence conjecture. For every $\ell \in \mathbb{N}$ and $\alpha \geq 1$, there exists an infinite sequence of (m,n,a,b)-graphs of girth larger than $4\ell + 2$...

conjecture | open | MathDB 页面 | 原文线索

V2-051320 | MATHDB-359193

Universality conjecture for overparameterized neural networks as neural channels

摘录 / EXCERPT: Neural-channel universality conjecture. Every overparameterized neural network is a Neural Channel.

conjecture | open | MathDB 页面 | 原文线索

V2-051321 | MATHDB-359194

Neural Channel capacity conjecture for spare subnetworks

摘录 / EXCERPT: Neural Channel capacity conjecture. The capacity of a Neural Channel increases with the size of the spare subnetwork.

conjecture | open | MathDB 页面 | 原文线索

V2-051322 | MATHDB-359200

Alon-Bollobás-Krivelevich-Sudakov surplus conjecture for H-free graphs

摘录 / EXCERPT: Let G be a graph with m edges, and let $\mathrm{sp}(G)=\mathrm{mc}(G)-m/2$ denote its surplus, where $\mathrm{mc}(G)$ is the number of edges in a largest bipartite subgraph of...

conjecture | open | MathDB 页面 | 原文线索

V2-051323 | MATHDB-359204

The bounded enumeration conjecture for induced cuts

摘录 / EXCERPT: For a graph with the parameters and family of cuts defined in Lemma, let $\mathcal{C}^{\mathrm{ind}}$ denote the corresponding family of induced cuts. Bounded enumeration conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-051324 | MATHDB-359206

The edge-deletion conjecture for two-eigenvalue realizations of complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let H be a subgraph of K_n with $|E(H)|\leq n-3$. Write $K_n\backslash H$ for the graph obtained by removing the edges of...

conjecture | open | MathDB 页面 | 原文线索

V2-051325 | MATHDB-359207

The conjectured threshold for graphs with two distinct eigenvalues

摘录 / EXCERPT: For each integer $n\geq 2$, let $m(n)$ denote the least number of edges such that every graph on n vertices with at least $m(n)$ edges has a matrix realization with two distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-051326 | MATHDB-359218

Manvel's eventual reconstructibility conjecture for graphs

摘录 / EXCERPT: For $l\in\mathbb{N}$, let an l-deck be the multiset of l-vertex induced subgraphs of a graph, and call a graph ℓ-reconstructible if it is determined by its $(n-\ell)$-deck...

conjecture | open | MathDB 页面 | 原文线索

V2-051327 | MATHDB-359220

Levin-Peres conjecture on cutoff for vertex-transitive expanders

摘录 / EXCERPT: Let $(G_n)_{n\ge 1}$ be a sequence of finite vertex-transitive expander graphs, and consider simple random walk on each G_n. Levin-Peres conjecture. The sequence $(G_n)_{n\ge 1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-051328 | MATHDB-359247

Choudum-Karthick-Shalu conjecture on quadratic coloring of P_5-free graphs

摘录 / EXCERPT: For a graph G, let $\chi(G)$ denote its chromatic number and $\omega(G)$ its clique number. A graph is P_5-free if it has no induced subgraph isomorphic to the five-vertex path...

conjecture | open | MathDB 页面 | 原文线索

V2-051329 | MATHDB-359248

Karthick-Kishore-Sahu conjecture on perfect divisibility of fork-free graphs

摘录 / EXCERPT: A graph is perfectly divisible if, for every induced subgraph H, its vertex set can be partitioned into A and B such that $H[A]$ is perfect and $\omega(H[B])<\omega(H)$. A fo...

conjecture | open | MathDB 页面 | 原文线索

V2-051330 | MATHDB-359253

The linear polychromatic coloring conjecture for hypergraph families

摘录 / EXCERPT: Let $\mathcal{H}$ be a hypergraph family, and let $m_\mathcal{H}(k)$ denote the minimum number of vertices needed to guarantee a polychromatic k-coloring in the relevant hypergra...

conjecture | open | MathDB 页面 | 原文线索

V2-051331 | MATHDB-359271

The 3-edge-colouring conjecture for cubic Cayley graphs

摘录 / EXCERPT: 3-edge-colouring conjecture. Every cubic Cayley graph admits a 3-edge colouring.

conjecture | open | MathDB 页面 | 原文线索

V2-051332 | MATHDB-359290

Sintiari and Trotignon's bounded-treewidth conjecture for even-hole-free graphs

摘录 / EXCERPT: For a graph $G=(V(G),E(G))$, a tree decomposition (T,χ) consists of a tree T and a map $\chi:V(T)\to 2^{V(G)}$ satisfying the usual vertex coverage, edge coverage, and conne...

conjecture | open | MathDB 页面 | 原文线索

V2-051333 | MATHDB-359299

The strong parity property at the critical number of unoccupied vertices

摘录 / EXCERPT: Let (d,k) be an admissible tuple of integers, and let $S(d,k)$ be the minimum integer such that $puz[d,k,S(d,k)]$ has the even solvability property. A configuration is k-mobile...

conjecture | open | MathDB 页面 | 原文线索

V2-051334 | MATHDB-359300

The diameter formula for one-rule higher-dimensional cubical sliding puzzles

摘录 / EXCERPT: Let $puz[d,1,l]$ denote the puzzle graph for the d-dimensional cube with one movable token and l unoccupied vertices, and let $diam$ denote its diameter. Diameter conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-051335 | MATHDB-359302

Ongie et al.'s finite-representation-cost conjecture for continuous piecewise linear functions

摘录 / EXCERPT: Let f be a continuous piecewise linear function, and consider shallow ReLU neural networks, allowing either finite width or infinite width with a representation cost. Ongie et al...

conjecture | open | MathDB 页面 | 原文线索

V2-051336 | MATHDB-359303

Exceptional-level identification of the even-spin W-algebra

摘录 / EXCERPT: Let $n\geq 2$, and write $h_m^\vee=2m-1$. Let $\mathcal{W}_{-h_n^\vee+1}^{D^+}(n,1)$ denote the indicated even-spin vertex algebra, and let $\mathcal{A}(n-1)^{\mathrm{Sp}_{2n-2}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-051337 | MATHDB-359304

Spectral-flow and quantum-Hamiltonian-reduction conjecture for admissible modules

摘录 / EXCERPT: Spectral-flow conjecture. For every $\lambda\in\mathrm{Pr}_{\mathbb{Z}}^k$ and $0\leq\theta\leq\bar q$, there exists $\mu\in\mathfrak{h}^*$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051338 | MATHDB-359305

Grothendieck-ring conjecture for principal exceptional W-algebras

摘录 / EXCERPT: Let $n\geq 2$ and let $k=-h^\vee+\frac{p}{2n-1}$ be an exceptional level. Let $\mathcal{K}(L_{p-h^\vee}(\mathfrak{so}_{2n+1}))$ be the Grothendieck ring of the corresponding affine...

conjecture | open | MathDB 页面 | 原文线索

V2-051339 | MATHDB-359306

Braided tensor equivalence conjecture for the quantum Hamiltonian reduction functor

摘录 / EXCERPT: Braided tensor equivalence conjecture. The functor H_f^0 is an equivalence of braided tensor categories. The source already states that this functor is an equivalence of abelian...

conjecture | open | MathDB 页面 | 原文线索

V2-051340 | MATHDB-359308

Cioabă, Desai and Tait's spectral Turán containment conjecture

摘录 / EXCERPT: Let F be a graph. Write $\operatorname{EX}(n,F)$ for the family of F-free graphs attaining the extremal number, and let $\operatorname{SPEX}(n,F)$ denote the family of F-free...

conjecture | open | MathDB 页面 | 原文线索

V2-051341 | MATHDB-359327

Levit-Tankus characterization conjecture for \mathbf{W_2} graphs

摘录 / EXCERPT: Levit-Tankus conjecture. The graph G is $\mathbf{W_2}$ if and only if, for every vertex v in G and every maximal independent set S in $G-N[v]$, the largest independent set...

conjecture | open | MathDB 页面 | 原文线索

V2-051342 | MATHDB-359329

Liu-Li's minimum spectral radius conjecture for odd-order graphs

摘录 / EXCERPT: Let $n\geq 3$ be odd, and let G be a graph in $\mathbb{G}_{n,\lfloor n/2\rfloor}$. Write $\rho(G)$ for its spectral radius and $T^{(n+3)/2}_{(n-3)/2,0}$ for the specified graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-051343 | MATHDB-359333

Conjecture that nonlinear neural-network loss functions are not rotationally invariant

摘录 / EXCERPT: Rotational-invariance conjecture. If the activation function $csigma$ is a smooth function that is not linear and the data set contains $cgeq 2$ data points, then

conjecture | open | MathDB 页面 | 原文线索

V2-051344 | MATHDB-359343

Forest deletion conjecture for maximum 2-density

摘录 / EXCERPT: For a graph G, let $m(G)=\max_{G'\subseteq G,\,v(G')\geq 1}e(G')/v(G')$ denote its maximum average density, and let $m_2(G)$ denote its maximum 2-density. Forest deletion conje...

conjecture | open | MathDB 页面 | 原文线索

V2-051345 | MATHDB-359346

Hyde's conjecture on valid edge-colourings of bounded-density graphs

摘录 / EXCERPT: Let H_1 and H_2 be non-empty graphs with $H_1\neq H_2$ and $m_2(H_1)\geq m_2(H_2)>1$. Assume that H_2 is strictly 2-balanced. If $m_2(H_1)>m_2(H_2)$, assume that H_1 is s...

conjecture | open | MathDB 页面 | 原文线索

V2-051346 | MATHDB-359351

Eternal distance-k domination versus ordinary domination in trees

摘录 / EXCERPT: Let T be a tree and let $k>2$. Write $\gamma_k(T)$ for the distance-k domination number, $\gamma_{all,k}^{\infty}(T)$ for the eternal distance-k domination number, and...

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V2-051347 | MATHDB-359353

Equivalence of left, partitionable and tensor-action convergence for uniform hypergraphs

摘录 / EXCERPT: Convergence-equivalence conjecture. Left-convergence, partitionable convergence, and action convergence of the $(r-1)$-action of the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-051348 | MATHDB-359361

Hanson-Toft conjecture on the minimum size of clique co-critical graphs

摘录 / EXCERPT: Let $k\geq 1$, let $t_1,\ldots,t_k$ be positive integers, and let ... A non-complete graph G is $(K_{t_1},\ldots,K_{t_k})$-co-critical if $G\not\rightarrow(K_{t_1},\ldots,K_{t_k})$...

conjecture | open | MathDB 页面 | 原文线索

V2-051349 | MATHDB-359376

Infinite family attaining the bipartite annihilation bound

摘录 / EXCERPT: Infinite-family conjecture. There exists an infinite family of connected bipartite graphs that illustrates the tight bound in the bipartite-graph inequality

conjecture | open | MathDB 页面 | 原文线索

V2-051350 | MATHDB-359377

Minimum order of an extremal bipartite graph

摘录 / EXCERPT: Minimum-order conjecture. The minimum number of vertices of an extremal bipartite graph is 16.

conjecture | open | MathDB 页面 | 原文线索

V2-051351 | MATHDB-359379

Generic identifiability conjecture for all non-complete graph families

摘录 / EXCERPT: Let $\mathcal{G}$ be any family of non-complete graphs, and let M_G denote the homoscedastic linear structural equation model associated with $G\in\mathcal{G}$. Generic identifia...

conjecture | open | MathDB 页面 | 原文线索

V2-051352 | MATHDB-359380

Complete-digraph edge-reversal conjecture for Jacobian matroids

摘录 / EXCERPT: Let G_1 and G_2 be complete digraphs on p nodes, with all edge directions agreeing except for the edge between nodes $p-1$ and p. Complete-digraph edge-reversal conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051353 | MATHDB-359398

Goddard-Henning independent domination conjecture for planar triangulations

摘录 / EXCERPT: Let G be a planar triangulation on n vertices, meaning a plane graph in which every face is bounded by a triangle, and let $\iota(G)$ denote its independent domination number,...

conjecture | open | MathDB 页面 | 原文线索

V2-051354 | MATHDB-359423

Crew's generalized degree sequence conjecture for trees

摘录 / EXCERPT: Crew's conjecture. For any tree T, the generalized degree sequence of T is determined by X_T.

conjecture | open | MathDB 页面 | 原文线索

V2-051355 | MATHDB-359435

Modified reconstruction conjecture for trees beyond the known exceptions

摘录 / EXCERPT: Modified tree reconstruction conjecture. For $n\ge 2\ell+1$, except when $(n,\ell)\in\{(5,2),(13,6)\}$, every n-vertex tree is ℓ-reconstructible. The threshold on n is kno...

conjecture | open | MathDB 页面 | 原文线索

V2-051356 | MATHDB-359441

Conjectural duality for intertwining operators of quasi-lisse vertex superalgebras

摘录 / EXCERPT: Duality conjecture. There exists a maximally extended multivalued analytic function $f(z_1,z_2;w_1,w_2,w_3,w_4)$ on

conjecture | open | MathDB 页面 | 原文线索

V2-051357 | MATHDB-359474

Traceability conjecture for graphs with degree sequence $\{m^{m-1},n^{n-1}\}$

摘录 / EXCERPT: Let $G_{m,n}$ denote the class of graphs with degree sequence $\{m^{m-1},n^{n-1}\}$. Traceability conjecture. Every graph $G\in G_{m,n}$ is traceable. The preceding proposition sho...

conjecture | open | MathDB 页面 | 原文线索

V2-051358 | MATHDB-359475

Graffiti's average-distance lower bound for the independence number

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ be its independence number, and let $\bar d$ be the average distance between distinct vertices of G. Graffiti's average-distance conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-051359 | MATHDB-359476

Graffiti's radius lower bound for the independence number

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ be its independence number, and let $r(G)$ be its graph radius. Graffiti's radius conjecture. ... The source verifies the intended bound for the l...

conjecture | open | MathDB 页面 | 原文线索

V2-051360 | MATHDB-359477

Graffiti's cut-vertex lower bound for the independence number

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ be its independence number, and let $c(G)$ be the number of cut-vertices of G. Graffiti's cut-vertex conjecture. ... The paper verifies this ine...

conjecture | open | MathDB 页面 | 原文线索

V2-051361 | MATHDB-359478

AGX's square-root lower bound for the independence number

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ be its independence number, let v be the number of vertices of G, and let Δ be its maximum degree. AGX's square-root conjecture. ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-051362 | MATHDB-359479

AGX's order-to-degree lower bound for the independence number

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ be its independence number, let v be the number of vertices of G, and let Δ be its maximum degree. AGX's order-to-degree conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051363 | MATHDB-359499

Budden and DeJonge's star-critical Ramsey conjecture for stars

摘录 / EXCERPT: Let $t\geq 2$ be an integer, and let $m_{1},\dots,m_{t}$ be integers larger than one, exactly k of which are even. For graphs $H_{1},\dots,H_{t}$, the star-critical Ramsey number...

conjecture | open | MathDB 页面 | 原文线索

V2-051364 | MATHDB-359504

Premalatha et al.'s local antimagic conjecture for connected graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a finite connected graph with no isolated vertices, and let $f:E\to\{1,2,\ldots,|E|\}$ be a bijection. For each vertex $u\in V$, define its weight by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-051365 | MATHDB-359532

Papikian's spectral conjecture for flag complex Laplacians

摘录 / EXCERPT: Papikian's conjecture. The number of distinct eigenvalues of $\Delta_k^+(\text{Fl}_{n,q})$ does not depend on q. Moreover, for every $\epsilon>0$ there exists an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-051366 | MATHDB-359533

Conjecture beyond Hamilton connectivity for hypergraph cycle embeddings

摘录 / EXCERPT: For a positive real number α and $k\in\mathbb N$, let $\delta^{\mathrm{CON}}_{k,\ell,d}$ be the threshold appearing in the main vertex-spread theorem, let...

conjecture | open | MathDB 页面 | 原文线索

V2-051367 | MATHDB-359534

Robust exact-threshold conjecture for Hamilton cycle embeddings

摘录 / EXCERPT: For $k\in\mathbb N$, $d\in[k-1]$, $\ell\in\{0,\dots,k-1\}$, and n divisible by $k-\ell$, let $h_{k,\ell,d}(n)$ be the least integer $D\geq0$ such that every n-vertex k-unifor...

conjecture | open | MathDB 页面 | 原文线索

V2-051368 | MATHDB-359535

Robust Katona-Kierstead conjecture for 3-uniform tight cycles

摘录 / EXCERPT: Let H be an n-vertex 3-uniform hypergraph, and let $C_{n,3,2}$ denote its tight Hamilton cycle. A distribution on embeddings is vertex-spread if it has the vertex-spread prop...

conjecture | open | MathDB 页面 | 原文线索

V2-051369 | MATHDB-359536

Robust Han-Zhao conjecture for loose Hamilton cycles

摘录 / EXCERPT: Let $C_{n,k,1}$ denote a loose Hamilton cycle in an n-vertex k-uniform hypergraph, where $k-1$ divides n. A distribution on embeddings is vertex-spread if it has the vertex-s...

conjecture | open | MathDB 页面 | 原文线索

V2-051370 | MATHDB-359598

Mohar's perfect cross-cap drawing conjecture

摘录 / EXCERPT: Let G be a simple graph. A perfect cross-cap drawing of G is a drawing with $g(G)$ cross-caps in which every edge intersects each cross-cap at most once, where $g(G)$ is the no...

conjecture | open | MathDB 页面 | 原文线索

V2-051371 | MATHDB-359599

Mohar's pseudo-triangulation cross-cap drawing conjecture

摘录 / EXCERPT: Let g be a positive integer, and let N_g be the non-orientable surface of genus g. A loopless pseudo-triangulation of N_g is a cellularly embedded loopless multigraph in wh...

conjecture | open | MathDB 页面 | 原文线索

V2-051372 | MATHDB-359644

Lopsided fractional Erdős matching conjecture for hypergraph tilings

摘录 / EXCERPT: Lopsided fractional Erdős matching conjecture. For every k-graph F and $0\leq\beta\leq 1/v(F)$,

conjecture | open | MathDB 页面 | 原文线索

V2-051373 | MATHDB-359648

Signless Brouwer conjecture for graphs

摘录 / EXCERPT: Signless Brouwer conjecture. For an integer k with $1\leq k\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-051374 | MATHDB-359649

Cameron-Mol conjecture on decreasing mean subtree order after adding k edges

摘录 / EXCERPT: Cameron-Mol's conjecture. For every positive integer k, there exist such graphs satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-051375 | MATHDB-359650

Cameron-Mol conjecture comparing complete-join and complete-bipartite graphs

摘录 / EXCERPT: Let m and n be positive integers. Write K_m+nK_1 for the disjoint union of a complete graph on m vertices and n isolated vertices, and let $K_{m,n}$ be the complete bipar...

conjecture | open | MathDB 页面 | 原文线索

V2-051376 | MATHDB-359657

The critical-snark defect-3 conjecture

摘录 / EXCERPT: Let a critical snark be a snark in which deleting any edge yields a 3-edge-colourable graph, and let the colouring defect of a cubic graph be the minimum number of uncovered edge...

conjecture | open | MathDB 页面 | 原文线索

V2-051377 | MATHDB-359658

The critical-snark Fulkerson-cover conjecture

摘录 / EXCERPT: Let G be a critical snark, let a hexagon be a 6-cycle, and let a hexagon be double-core when it is a double-core hexagon for a 3-array of perfect matchings. An optimal 3-ar...

conjecture | open | MathDB 页面 | 原文线索

V2-051378 | MATHDB-359659

The irreducible-snark defect-3 conjecture

摘录 / EXCERPT: Let G be an irreducible snark, meaning a snark for which deleting any pair of distinct vertices produces a 3-edge-colourable graph, and let defect denote colouring defect. Irre...

conjecture | open | MathDB 页面 | 原文线索

V2-051379 | MATHDB-359660

The irreducible-snark girth conjecture

摘录 / EXCERPT: Let an irreducible snark be a snark for which deleting any pair of distinct vertices produces a 3-edge-colourable graph, and let the girth of a graph be the length of its shortes...

conjecture | open | MathDB 页面 | 原文线索

V2-051380 | MATHDB-359661

Cohen et al.'s general oriented-cycle subdivision conjecture

摘录 / EXCERPT: For positive integers $k_1,k_2,\ldots,k_n$, let $C(k_1,k_2,\ldots,k_n)$ denote the oriented cycle formed by blocks of those lengths. A digraph is strongly connected if every ordere...

conjecture | open | MathDB 页面 | 原文线索

V2-051381 | MATHDB-359672

The non-occurrence conjecture for PGST in double blow-ups of paths

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $\overset{2}{\uplus}~P_n$ denote its double blow-up, with corresponding vertices $(0,u)$ and $(1,u)$ for each vertex u of P_n. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-051382 | MATHDB-359678

The ordinary diversity bound for intersecting uniform hypergraphs

摘录 / EXCERPT: Let $\mathcal{F}\subset \binom{[n]}{k}$ be an intersecting family, and let $\gamma(\mathcal{F})$ denote its ordinary diversity. Suppose that $n>4k$. Ordinary diversity conjecture....

open_problem | open | MathDB 页面 | 原文线索

V2-051383 | MATHDB-359713

Liu-Morris-Prince logarithmic-factor conjecture for multicolour connectivity

摘录 / EXCERPT: Logarithmic-factor conjecture. For any

conjecture | open | MathDB 页面 | 原文线索

V2-051384 | MATHDB-359719

The Hochbaum-Nishizeki-Shmoys conjecture on polynomial-time edge coloring

摘录 / EXCERPT: Hochbaum-Nishizeki-Shmoys conjecture. There exists a polynomial-time algorithm that finds a proper edge-coloring of every multigraph G using $\max\{\Delta(G)+1,\Gamma(G)\}$ col...

conjecture | open | MathDB 页面 | 原文线索

V2-051385 | MATHDB-359722

Aravind-Subramanian conjecture for the oriented chromatic number of bounded-genus graphs

摘录 / EXCERPT: Aravind-Subramanian conjecture. There is an absolute constant implicit in the O-notation such that

conjecture | open | MathDB 页面 | 原文线索

V2-051386 | MATHDB-359738

Bacher's invertible-distance conjecture for the seating couple problem

摘录 / EXCERPT: Let K_v be the complete graph on v circularly ordered vertices, with edge-length $ll(u,w)=\min(|u-w|,v-|u-w|)$. A solution to the king's problem is a near 1-factor of K_v w...

conjecture | open | MathDB 页面 | 原文线索

V2-051387 | MATHDB-359739

The odd-order near-factor characterization conjecture

摘录 / EXCERPT: Let $v=2n+1$ be an odd integer, let K_v be the complete graph on $\{0,1,\ldots,v-1\}$, and let L be a list of n positive integers not exceeding n. A near 1-factor F is...

conjecture | open | MathDB 页面 | 原文线索

V2-051388 | MATHDB-359742

Axenovich-Martin conjecture on complete partitions of bipartite-free graphs

摘录 / EXCERPT: Let H be a bipartite graph that is not a forest. Write $f(r,H)$ for the least integer k such that every H-free (r,k)-graph does not exist, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-051389 | MATHDB-359746

Jordán and Garamvölgyi's separator conjecture for globally linked pairs

摘录 / EXCERPT: Let $G=G_1\cup G_2$, where $|V(G_1)\cap V(G_2)|\leq d+1$, and let $u,v\in V(G_1)\cap V(G_2)$. A pair is $\mathcal{R}_d$-linked when it is contained in an $\mathcal{R}_d$-circuit. J...

conjecture | open | MathDB 页面 | 原文线索

V2-051390 | MATHDB-359747

The uniform affine measurement matroid conjecture

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{A}_d(G)$ denote its d-dimensional generic affine edge measurement matroid. Uniform affine measurement matroid conjecture. If G has at leas...

conjecture | open | MathDB 页面 | 原文线索

V2-051391 | MATHDB-359776

The bridgeless-intersection extremal conjecture

摘录 / EXCERPT: Bridgeless-intersection conjecture. If the intersection of any two graphs in $\mathcal{F}$ is 2-edge-connected, then

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V2-051392 | MATHDB-359777

The two-component-intersection extremal conjecture

摘录 / EXCERPT: Two-component-intersection conjecture. If the intersection of any two graphs in $\mathcal{F}$ has at most two components, then

conjecture | open | MathDB 页面 | 原文线索

V2-051393 | MATHDB-359778

The three-component-intersection extremal conjecture for odd n

摘录 / EXCERPT: Three-component-intersection conjecture. If the intersection of any two graphs in $\mathcal{F}$ has at most three components, then

conjecture | open | MathDB 页面 | 原文线索

V2-051394 | MATHDB-359786

Strengthened Chen-Lih-Wu and Kostochka-Pelsmajer-West conjecture

摘录 / EXCERPT: Let G be an r-colorable graph, with maximum degree $\Delta(G)\leq r$. Let an r-list assignment L assign r available colors to each vertex, and call G equitably r-choo...

conjecture | open | MathDB 页面 | 原文线索

V2-051395 | MATHDB-359803

Miklavič-Šparl conjecture on non--distance-balanced generalized Petersen graphs

摘录 / EXCERPT: Let $k\ge 2$ be an integer, and define ... For a graph G, write D for its diameter. Miklavič-Šparl conjecture. For every $n>n_k$, the generalized Petersen graph $GP(n,k)$ is not...

conjecture | open | MathDB 页面 | 原文线索

V2-051396 | MATHDB-359829

Cambie-de Verclos-Kang conjecture on triangles in regular graphs

摘录 / EXCERPT: Cambie-de Verclos-Kang conjecture. For every odd integer n and even integer k with

conjecture | open | MathDB 页面 | 原文线索

V2-051397 | MATHDB-359838

Gyárfás-Lehel covering conjecture for cross-intersecting partite hypergraphs

摘录 / EXCERPT: Gyárfás-Lehel conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-051398 | MATHDB-359839

Projective-plane extremal conjecture for cross-intersecting hypergraphs

摘录 / EXCERPT: Projective-plane extremal conjecture. For $m\geqslant r+2$,

conjecture | open | MathDB 页面 | 原文线索

V2-051399 | MATHDB-359840

Fractional covering conjecture for looms

摘录 / EXCERPT: Loom fractional covering conjecture. If $\mathbb{L}=(A,B)$ is an (r,s)-loom, then

conjecture | open | MathDB 页面 | 原文线索

V2-051400 | MATHDB-359841

Individual fractional covering conjecture for looms

摘录 / EXCERPT: Loom component conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-051401 | MATHDB-359842

Pinnable cross-intersection covering conjecture

摘录 / EXCERPT: Pinnable covering conjecture. If $A\cup B$ is pinnable, then

conjecture | open | MathDB 页面 | 原文线索

V2-051402 | MATHDB-359843

Pinnable covering conjecture for hypergraphs of different uniformities

摘录 / EXCERPT: General pinnable covering conjecture. If $A\cup B$ is pinnable, then

conjecture | open | MathDB 页面 | 原文线索

V2-051403 | MATHDB-359844

Pinnable rainbow matching conjecture

摘录 / EXCERPT: Pinnable rainbow matching conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051404 | MATHDB-359866

Polychromatic finiteness conjecture for hereditary hypergraph families

摘录 / EXCERPT: Polychromatic finiteness conjecture. If $m_2(\mathcal F)<\infty$, then $m_k(\mathcal F)<\infty$ for every k for any hereditary family $\mathcal F$.

conjecture | open | MathDB 页面 | 原文线索

V2-051405 | MATHDB-359867

Keszegh-Pálvölgyi linear bound conjecture for polychromatic coloring

摘录 / EXCERPT: Keszegh-Pálvölgyi linear bound conjecture. For every k and every hereditary family $\mathcal F$,

conjecture | open | MathDB 页面 | 原文线索

V2-051406 | MATHDB-359890

Linear bound for Kemeny's constant of diameter-two graphs

摘录 / EXCERPT: Diameter-two Kemeny bound. The Kemeny constant of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051407 | MATHDB-359912

The Abdollahi-Akbari-Maimani non-commuting graph conjecture

摘录 / EXCERPT: Abdollahi-Akbari-Maimani conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051408 | MATHDB-359919

Polynomial Hamiltonian-cycle conjecture for split digraphs with bounded independent sets

摘录 / EXCERPT: Bounded-independent-set conjecture. For every fixed integer k, the Hamiltonian-cycle problem is polynomial-time solvable for the class of split digraphs $D=(V_1,V_2;A)$ in which...

conjecture | open | MathDB 页面 | 原文线索

V2-051409 | MATHDB-359926

The Eulerian graph recurrence conjecture for bd(G)

摘录 / EXCERPT: Let G be a graph, and define $\nu(G)$ as the absolute value of $A_D(-1)$ for any orientation D of G. An Eulerian graph is a graph in which every vertex has even degree. Euler...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-051410 | MATHDB-359927

The multipartite orientation conjecture for the multiplicity of -1

摘录 / EXCERPT: Let D be an orientation of a complete multipartite graph, and let r be the number of its parts having odd size. Write $A_D(t)$ for the Eulerian polynomial of D, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-051411 | MATHDB-359937

The bounded-degree induced Menger conjecture

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ denote its maximum degree, and let $X,Y\subseteq V(G)$. A set of vertices separates X and Y if it intersects every X-Y-path; paths are p...

conjecture | open | MathDB 页面 | 原文线索

V2-051412 | MATHDB-359959

Extremal kth-eigenvalue conjecture for connected outerplanar graphs

摘录 / EXCERPT: Extremal kth-eigenvalue conjecture. If $n=kq+1$, then for fixed k and sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-051413 | MATHDB-359985

Nonadjacency conjecture for degree-5 vertices in edge-to-edge plane graphs

摘录 / EXCERPT: Nonadjacency conjecture. Are there any two degree-5 vertices adjacent in EP? No way! Equivalently, no two degree-5 vertices in EP are adjacent.

conjecture | open | MathDB 页面 | 原文线索

V2-051414 | MATHDB-360002

Zhu et al.'s conjecture on acyclic colouring of cubic planar 3-connected graphs

摘录 / EXCERPT: A cubic planar 3-connected graph is a planar graph in which every vertex has degree three and which remains connected after deletion of any two vertices. Let K_4 and Q_3 denote...

conjecture | open | MathDB 页面 | 原文线索

V2-051415 | MATHDB-360010

The coherent decycling partition conjecture for 3-connected cubic graphs

摘录 / EXCERPT: Let G be a 3-connected cubic graph. A coherent decycling partition is a partition of the edges of G with the coherent decycling property, and G is upper-embeddable when it...

conjecture | open | MathDB 页面 | 原文线索

V2-051416 | MATHDB-360011

The almost-all cubic graphs induced-tree conjecture

摘录 / EXCERPT: Let G be a cubic graph. An induced tree is a tree that is an induced subgraph of G. The almost-all cubic graphs induced-tree conjecture. Almost all cubic graphs contain an indu...

conjecture | open | MathDB 页面 | 原文线索

V2-051417 | MATHDB-360012

Aouchiche-Hansen Randić index diameter bound

摘录 / EXCERPT: Let G be a graph, with Randić index $\operatorname{R}(G)$, diameter $\operatorname{d}(G)$, and order $\#V(G)$. Aouchiche-Hansen's conjectured bound. ... This is a proposed lower bo...

conjecture | open | MathDB 页面 | 原文线索

V2-051418 | MATHDB-360014

Magnant-Martin path-cover conjecture for regular digraphs and oriented graphs

摘录 / EXCERPT: Magnant-Martin's strengthened conjecture. If G is a d-regular digraph on n vertices, then

conjecture | open | MathDB 页面 | 原文线索

V2-051419 | MATHDB-360015

Magnant-Wang-Yuan path-cover conjecture for graphs with prescribed degree bounds

摘录 / EXCERPT: Magnant-Wang-Yuan's conjecture. The path-cover number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051420 | MATHDB-360016

Enomoto-Kaneko-Tuza conjecture for edge-disjoint cycle covers in digraphs

摘录 / EXCERPT: Directed edge-disjoint cycle-cover conjecture. For every $k\in\mathbb{N}$ with $k\geq 2$, if the minimum semi-degree of G is at least n/k, then $V(G)$ can be covered by $k-1$ e...

conjecture | open | MathDB 页面 | 原文线索

V2-051421 | MATHDB-360042

Universality of six-vertex edge fluctuations for \Delta<1

摘录 / EXCERPT: Let a_1,a_2,b_1,b_2,c_1,c_2 be the six-vertex weights, let Δ be the associated interaction parameter, and consider the six-vertex model with domain wall boundary condition...

conjecture | open | MathDB 页面 | 原文线索

V2-051422 | MATHDB-360045

Bernshteyn-Lee's lower-bound conjecture for weak degeneracy of regular graphs

摘录 / EXCERPT: A graph G is weakly d-degenerate if it is weakly f-degenerate for the constant function with value d, and its weak degeneracy $\operatorname{wd}(G)$ is the minimum such int...

conjecture | open | MathDB 页面 | 原文线索

V2-051423 | MATHDB-360050

Weak Sidorenko expansion conjecture

摘录 / EXCERPT: Let F be a bipartite graph, and let $\mathrm{E}^r(F)$ denote its r-uniform expansion: each edge of F is extended to an r-edge by adding new vertices. An r-graph is Sidore...

conjecture | open | MathDB 页面 | 原文线索

V2-051424 | MATHDB-360051

Sidorenko conjecture for linear hypergraphs containing an expanded clique

摘录 / EXCERPT: Let F be an r-uniform hypergraph and let k be the uniformity of the clique K_{k+1}^k. Distinct edges of F satisfy the k-linear condition when $|e\cap f|<k$ for every pa...

conjecture | open | MathDB 页面 | 原文线索

V2-051425 | MATHDB-360062

The perfect diameter-two graph conjecture for 2-bootstrap percolation

摘录 / EXCERPT: Let $\mathcal{G}$ be the collection of all connected graphs of order at least 3 with at most two blocks. A graph is 2-bootstrap good if it contains at least one pair of vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-051426 | MATHDB-360101

Haemers' minimum energy conjecture for Seidel matrices

摘录 / EXCERPT: Let A be a symmetric matrix of order n whose diagonal entries are zero and whose off-diagonal entries belong to $\{\pm 1\}$. Let J_n be the all-ones matrix and I_n the id...

conjecture | open | MathDB 页面 | 原文线索

V2-051427 | MATHDB-360114

Mutual-visibility number conjecture for Cartesian products of paths and cycles

摘录 / EXCERPT: Let P_s be the path on s vertices and C_t the cycle on t vertices, and let $cmu(G)$ denote the mutual-visibility number of a graph G. Mutual-visibility number conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051428 | MATHDB-360119

Gharakhloo-Welker positive matching conjecture for complete 3-uniform hypergraphs

摘录 / EXCERPT: Gharakhloo-Welker's conjecture. For every $3\leq l_1\leq 2n-3$ and $5\leq l_2\leq 2n-1$, E_{l_1,l_2} is a positive matching.

conjecture | open | MathDB 页面 | 原文线索

V2-051429 | MATHDB-360120

Gharakhloo-Welker upper-bound conjecture for positive matching decomposition numbers

摘录 / EXCERPT: Gharakhloo-Welker's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051430 | MATHDB-360146

Directed Hedetniemi conjecture for dichromatic number

摘录 / EXCERPT: Directed Hedetniemi conjecture. For any two digraphs G and H,

conjecture | refuted | MathDB 页面 | 原文线索

V2-051431 | MATHDB-360150

Equivalence of componentwise linearity, linear quotients, and vertex splittability for weighted oriented graph edge ideals

摘录 / EXCERPT: Let D be a weighted oriented graph, and let $I(D)$ denote its edge ideal. The properties in question are componentwise linearity of $I(D)$, having linear quotient property, and b...

conjecture | open | MathDB 页面 | 原文线索

V2-051432 | MATHDB-360151

Equality of (3,3)- and (3,4)-spreading numbers on grid graphs

摘录 / EXCERPT: Let P_m and P_n be paths on m and n vertices, respectively, and let $P_m\square P_n$ denote their Cartesian product. For a graph G, write $\sigma_{(p,q)}(G)$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-051433 | MATHDB-360168

Hyperbolicity theorem conjecture for local girth correspondence colouring

摘录 / EXCERPT: Local girth correspondence-colouring hyperbolicity conjecture. There exists a theorem analogous to the paper's main hyperbolicity theorem for local girth correspondence colouring.

conjecture | open | MathDB 页面 | 原文线索

V2-051434 | MATHDB-360169

Spectral gap, Poisson fluctuations, and eigenvector localization for the Erdős-Rényi graph Laplacian

摘录 / EXCERPT: Spectral-gap, fluctuation, and localization conjecture. With high probability, $\lambda_2\in[a,b]$; the point measure

conjecture | open | MathDB 页面 | 原文线索

V2-051435 | MATHDB-360179

Zheng's zero Laplacian eigenvalue multiplicity conjecture for uniform hypertrees

摘录 / EXCERPT: Zheng's conjecture. For every k-uniform hypertree T with $k\geq 3$, the multiplicity of the zero Laplacian eigenvalue is $k^{|E(T)|(k-2)}$. This conjecture concerns the Laplaci...

conjecture | open | MathDB 页面 | 原文线索

V2-051436 | MATHDB-360199

Asymptotic order conjecture for twins in ordered r-matchings

摘录 / EXCERPT: An ordered r-matching is an ordered matching in which every vertex belongs to one of r edge labels, and let $t^{(r)}(n)$ be the minimum, over ordered r-matchings of size n,...

conjecture | open | MathDB 页面 | 原文线索

V2-051437 | MATHDB-360213

Powerlessness conjecture for late changepoint detection below the square-root scale

摘录 / EXCERPT: Powerlessness conjecture. When $\gamma<\tfrac{1}{2}$, all tests based on the vertex degrees $\{D_v(n)\}_{v\in[n]}$ are powerless, and all tests based on G_n are powerless.

conjecture | open | MathDB 页面 | 原文线索

V2-051438 | MATHDB-360222

Finiteness and convergence conjecture for tree network flow

摘录 / EXCERPT: Consider the evolution of a tree under network flow. A singular time is a time at which the flow develops a singularity, and a boundary curve is a curve of the network ending at th...

conjecture | open | MathDB 页面 | 原文线索

V2-051439 | MATHDB-360235

ReLU network approximation error conjecture

摘录 / EXCERPT: ReLU approximation error conjecture. Under these assumptions, the displayed bound holds.

conjecture | open | MathDB 页面 | 原文线索

V2-051440 | MATHDB-360241

Conjecture P(k,r) for intersections of longest paths

摘录 / EXCERPT: Let $k\geq 1$ and $r\leq k$. A graph is k-connected if, for any two distinct vertices, it has k internally disjoint paths between them. A longest path is a path of maximum leng...

conjecture | open | MathDB 页面 | 原文线索

V2-051441 | MATHDB-360249

The ordered matching conjecture

摘录 / EXCERPT: An ordered graph is a graph together with a total vertex ordering. The ordered matching conjecture. Let $(M,\prec)$ be an ordered graph with maximum degree 1. Then the class of...

conjecture | open | MathDB 页面 | 原文线索

V2-051442 | MATHDB-360255

The substitution conjecture for dichromatically bounded digraph classes

摘录 / EXCERPT: A class of digraphs $\mathcal C$ is dichromatically bounded if its dichromatic number is bounded as a function of its clique number. The substitution conjecture. If a class of digr...

conjecture | open | MathDB 页面 | 原文线索

V2-051443 | MATHDB-360262

Fujita-Lo-Zhang conjecture on properly colored cycles in edge-colored complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be a complete bipartite graph with parts of sizes m and n, equipped with an edge-coloring, and let $\delta^c(K_{m,n})$ be its minimum color degree, where the colo...

conjecture | open | MathDB 页面 | 原文线索

V2-051444 | MATHDB-360288

The conjecture that all cubes are stackable

摘录 / EXCERPT: Let Q^d denote the d-dimensional hypercube, and call a graph stackable if, starting with one cup at each vertex, all cups can be moved onto any prescribed target vertex accordi...

conjecture | open | MathDB 页面 | 原文线索

V2-051445 | MATHDB-360289

The conjecture that all higher-dimensional grids are stackable

摘录 / EXCERPT: Let P_k be the path graph on k vertices, let d be a positive integer, and let $k_1,\ldots,k_d$ be positive integers. Write the Cartesian product grid as ... Call a graph stacka...

conjecture | open | MathDB 页面 | 原文线索

V2-051446 | MATHDB-360291

Existence of minimal uncolorable uniform bi-hypergraphs

摘录 / EXCERPT: Existence conjecture. For any $n,r\in \mathbb{N}$ with $r\ge 3$ and $n\ge (r-1)^2+1$, the set $\mathcal{MUC}(n,r)$ is not empty.

conjecture | open | MathDB 页面 | 原文线索

V2-051447 | MATHDB-360292

The value of the minimum size for minimal uncolorable triple bi-hypergraphs of order seven

摘录 / EXCERPT: Conjecture on $\ell(7,3)$.

conjecture | open | MathDB 页面 | 原文线索

V2-051448 | MATHDB-360293

The minimum-size conjecture for minimal uncolorable uniform bi-hypergraphs

摘录 / EXCERPT: Minimum-size conjecture. For any $r\in \mathbb{N}$ with $r\ge 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-051449 | MATHDB-360294

The colorability-extension conjecture for uniform bi-hypergraphs

摘录 / EXCERPT: Colorability-extension conjecture. For any $r,m,n\in \mathbb{N}$ with $r\ge 3$ and $n\ge (r-1)^2+1$, if all bi-hypergraphs in $Bi\mathcal{H}yp(n,r,m)$ are colorable, then all bi-hy...

open_problem | open | MathDB 页面 | 原文线索

V2-051450 | MATHDB-360297

Ghosh-Győri-Martin-Paulos-Xiao planar Turán conjecture for cycles

摘录 / EXCERPT: Let C_k denote the cycle of length k, and let $\operatorname{ex}_{\mathcal P}(n,C_k)$ be the maximum number of edges in an n-vertex planar graph containing no subgraph isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-051451 | MATHDB-360333

Ricci-vertex proximity conjecture for Kähler-Ricci flow fibres

摘录 / EXCERPT: Let $g(t)$ be the maximal Kähler-Ricci flow on $X\times[0,1)$, let $\Phi:X\to Y$ be the limiting fibration, and let $\mathcal{F}_q=\Phi^{-1}(q)$. A Ricci vertex associated to a smo...

conjecture | open | MathDB 页面 | 原文线索

V2-051452 | MATHDB-360334

Fibre diameter and scalar curvature conjecture near Ricci vertices

摘录 / EXCERPT: Let $g(t)$ be the maximal Kähler-Ricci flow on $X\times[0,1)$, let $\theta_Y\in\vartheta$ be a smooth closed $(1,1)$-form on Y, and let p_t be a Ricci vertex associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-051453 | MATHDB-360342

Conjecture on the construction code maximizing Kemeny's constant for threshold graphs

摘录 / EXCERPT: Let G be a threshold graph of order n, represented by a construction code, and let $\mathcal{K}(G)$ denote its Kemeny constant. Maximizing construction-code conjecture. The thr...

conjecture | open | MathDB 页面 | 原文线索

V2-051454 | MATHDB-360343

Asymptotic maximum conjecture for Kemeny's constant of threshold graphs

摘录 / EXCERPT: Let $n\geq 3$. For a threshold graph G of order n, let $\mathcal{K}(G)$ denote its Kemeny constant, and let r be one of the integers specified in the preceding proposition:...

conjecture | open | MathDB 页面 | 原文线索

V2-051455 | MATHDB-360344

Optimal minimum-degree conjecture for Tverberg partition graphs

摘录 / EXCERPT: Let S be a set of n points in $\mathbb{R}^d$ in strong general position, and let $G_T[S,r]$ be its Tverberg r-partition graph. Write $\delta(G_T[S,r])$ for the minimum degree...

conjecture | open | MathDB 页面 | 原文线索

V2-051456 | MATHDB-360345

Bounded-excess connectivity conjecture for Tverberg partition graphs

摘录 / EXCERPT: The bounded-excess connectivity conjecture. There exists a constant $k(d)$ depending only on d such that $G_T[S,r]$ is connected whenever

conjecture | open | MathDB 页面 | 原文线索

V2-051457 | MATHDB-360353

Oliveira et al.'s extremal conjecture for the signless Laplacian spectrum

摘录 / EXCERPT: Let G be a connected simple graph on n vertices, let $S_2(G)$ denote the sum of its two largest signless Laplacian eigenvalues, and define ... Let $K^+_{1,n-1}$ denote the star g...

conjecture | open | MathDB 页面 | 原文线索

V2-051458 | MATHDB-360354

Ashraf et al.'s signless Laplacian eigenvalue-sum conjecture

摘录 / EXCERPT: Let G be a connected simple graph on n vertices, and let $S_k(G)$ denote the sum of the first k largest signless Laplacian eigenvalues of G. Ashraf et al.'s conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051459 | MATHDB-360358

The one-fifth conjecture for isolated cycles in connected graphs

摘录 / EXCERPT: Let $\mathcal{C}_4=\{C_k\colon k\geq 4\}$, and let $\iota(G,\mathcal{C}_4)$ denote the maximum number of vertex-disjoint members of $\mathcal{C}_4$ in a graph G. Problem 3 asks f...

conjecture | open | MathDB 页面 | 原文线索

V2-051460 | MATHDB-360368

Exact codegree-squared-sum conjecture for t-intersecting families

摘录 / EXCERPT: Exact codegree-squared-sum conjecture. If $t\ge1$ and $n\ge(t+1)(k-t+1)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051461 | MATHDB-360369

Hilton-Milner codegree-squared-sum conjecture for intersecting families

摘录 / EXCERPT: Hilton-Milner codegree-squared-sum conjecture. For $k\ge3$ and $n>2k$,

conjecture | open | MathDB 页面 | 原文线索

V2-051462 | MATHDB-360391

Universal inverse-degree upper bound for critical percolation

摘录 / EXCERPT: Let G be an infinite, connected, transitive, simple graph of vertex degree d that is not one-dimensional, and let $p_c(G)$ be its Bernoulli bond percolation critical probabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-051463 | MATHDB-360397

Alon-Shapira-Sudakov conjecture on the complexity of graph removal

摘录 / EXCERPT: Let G and H be graphs, and let $\mathrm{Rem}_H(G)$ denote the maximum number of edges in an H-free spanning subgraph of G. A graph H is a star forest if every connected c...

conjecture | open | MathDB 页面 | 原文线索

V2-051464 | MATHDB-360400

Rigidity conjecture for generic symmetric hypergraphs

摘录 / EXCERPT: Let $n\geq 2$. Let $\mathscr{L}_n=\{R\}$ be a relational language with R an n-ary relation, and let $\operatorname{AgeK}_{\mathrm{sym},n}$ be the class of all finite...

conjecture | open | MathDB 页面 | 原文线索

V2-051465 | MATHDB-360421

Jia-Song's second distance signless Laplacian conjecture

摘录 / EXCERPT: Let G be a connected graph of order $n\geq 4$, with remoteness ρ, and suppose that $G\ncong K_n$ and $G\ncong K_n-e$. Here K_n-e is the complete graph with one edge remove...

conjecture | open | MathDB 页面 | 原文线索

V2-051466 | MATHDB-360423

De Klerk-Pasechnik conjecture on the copositive hierarchy for the stability number

摘录 / EXCERPT: De Klerk-Pasechnik conjecture. For every non-empty graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-051467 | MATHDB-360455

The vanishing minimal deviation angle conjecture for networks

摘录 / EXCERPT: Let N be the dimension of the network, and let the minimal deviation angle be the minimum deviation angle arising from the network's admissible vectors as defined in the surround...

conjecture | open | MathDB 页面 | 原文线索

V2-051468 | MATHDB-360464

Accessibility conjecture for quasi-transitive graphs quasi-isometric to minor-excluded graphs

摘录 / EXCERPT: A graph Γ is minor-excluded if some finite graph is not a minor of Γ. A graph is accessible if it admits a finite decomposition over finite separators with pieces hav...

conjecture | open | MathDB 页面 | 原文线索

V2-051469 | MATHDB-360465

Planarity conjecture for quasi-transitive graphs quasi-isometric to minor-excluded graphs

摘录 / EXCERPT: A graph Γ is minor-excluded if some finite graph is not a minor of Γ. Two graphs are quasi-isometric when their large-scale metric spaces are equivalent up to multipl...

conjecture | open | MathDB 页面 | 原文线索

V2-051470 | MATHDB-360473

Akbari et al.'s conjecture on avoiding forbidden out-degree lists

摘录 / EXCERPT: Let G be a graph, and let $F:V(G)\to 2^N$ assign a set of forbidden out-degrees to each vertex. An orientation O of G is F-avoiding if $d^+_O(v)\notin F(v)$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-051471 | MATHDB-360474

The minimum codegree threshold for resolvable Steiner triple systems

摘录 / EXCERPT: Let n be sufficiently large with $n\equiv 3\pmod{6}$, and let H be a hypergraph on n vertices. A resolvable Steiner triple system is a Steiner triple system whose triples can...

conjecture | open | MathDB 页面 | 原文线索

V2-051472 | MATHDB-360475

The minimum codegree threshold for transversal resolvable Steiner triple systems

摘录 / EXCERPT: Let n be sufficiently large with $n\equiv 3\pmod{6}$, and let V be a common n-vertex set. Let $\mathcal{H}=\{H_1,\ldots,H_{n(n-1)/6}\}$ be a collection of 3-uniform hypergr...

conjecture | open | MathDB 页面 | 原文线索

V2-051473 | MATHDB-360486

Monotonicity conjecture for spectral radii of token graphs

摘录 / EXCERPT: Let G be a graph on n vertices. For each k with $1\le k\le \lfloor n/2\rfloor$, let $\rho_k(G)$ denote the spectral radius of its k-token graph. Monotonicity conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-051474 | MATHDB-360497

Haythorpe's lower-bound conjecture for Hamiltonian cycles in regular graphs

摘录 / EXCERPT: Let r and n be integers with $r\geq 5$ and $n\geq r+3$, and let G be a Hamiltonian r-regular graph on n vertices. Haythorpe's conjecture. The graph G has at least ... Ham...

conjecture | open | MathDB 页面 | 原文线索

V2-051475 | MATHDB-360504

The zero forcing lower bound for Cartesian products

摘录 / EXCERPT: Let G and G' be graphs, each containing an edge. The Cartesian-product zero forcing conjecture. ... The conjecture would extend the sharp lower bound known when the maximum nulli...

conjecture | open | MathDB 页面 | 原文线索

V2-051476 | MATHDB-360508

Quadratic weakening of the chromatic-intersection conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of labelled graphs on the vertex-set $[n]$, and let $\chi(G)$ denote chromatic number. Quadratic chromatic-intersection conjecture. There is an absolu...

conjecture | open | MathDB 页面 | 原文线索

V2-051477 | MATHDB-360517

The equivalence of abundance for graphs and coloured graphs

摘录 / EXCERPT: Abundance equivalence conjecture. A graph H is F-abundant if and only if there is an F-colouring σ of H such that (H, σ) is F-abundant.

conjecture | open | MathDB 页面 | 原文线索

V2-051478 | MATHDB-360520

Bipartite-graph density dichotomy conjecture

摘录 / EXCERPT: Let H be a bipartite graph. For a graph G, write $d(G)$ for its average degree. Bipartite-graph density dichotomy conjecture. For every bipartite graph H, there is a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-051479 | MATHDB-360522

Havet's chromatic conjecture for trees with few leaves

摘录 / EXCERPT: Let $k\in\mathbb N$, let $\ell\in\mathbb N$, let D be a digraph, and let T be an oriented tree with k edges and ℓ leaves. Havet's conjecture. Every $(k+\ell)$-chromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-051480 | MATHDB-360523

Addario-Berry-Havet-Linhares Sales-Reed-Thomassé density conjecture

摘录 / EXCERPT: Let $k\in\mathbb N$, let D be a digraph, and let T be an antidirected tree with k edges. Addario-Berry et al.'s conjecture. Every digraph D with more than $(k-1)|V(D)|$ edg...

conjecture | open | MathDB 页面 | 原文线索

V2-051481 | MATHDB-360526

Minimum-semidegree conjecture for bounded-degree spanning oriented trees

摘录 / EXCERPT: Let $\Delta\in\mathbb N$, and let c be a constant. The bounded-degree spanning-tree conjecture. For every Δ there is a c such that every sufficiently large n-vertex di...

conjecture | open | MathDB 页面 | 原文线索

V2-051482 | MATHDB-360532

Kohayakawa-Nagle-Parczyk conjecture on squares of Hamilton cycles in random graphs

摘录 / EXCERPT: Let $G(n,p)$ be the binomial random graph on vertex set $[n]$, with each possible edge included independently with probability $p=p(n)$. A square of a Hamilton cycle is the second...

conjecture | open | MathDB 页面 | 原文线索

V2-051483 | MATHDB-360559

Exponential concentration of graphs with even decompositions

摘录 / EXCERPT: Let n be a positive integer. An even decomposition of a graph is the decomposition referred to in the paper, and consider graphs on n vertices having an even number of edges. E...

conjecture | open | MathDB 页面 | 原文线索

V2-051484 | MATHDB-360560

Subpolynomial growth for graphs without even decompositions

摘录 / EXCERPT: Let H be a graph without an even decomposition, and let $r_H(n)$ denote the graph parameter used in the paper. Subpolynomial-growth conjecture. For all graphs H without an even...

conjecture | open | MathDB 页面 | 原文线索

V2-051485 | MATHDB-360565

PSPACE-completeness conjecture for the B- and B'-games

摘录 / EXCERPT: The Builder-Blocker general position games are played on a graph, with Builder and Blocker alternately selecting vertices according to the rules defining the games; the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-051486 | MATHDB-360569

Anđelić-da Fonseca-Simić-Du conjecture on cospectral connected chain graphs

摘录 / EXCERPT: Anđelić-da Fonseca-Simić-Du conjecture. There do not exist non-isomorphic cospectral connected chain graphs with respect to the adjacency spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-051487 | MATHDB-360587

The extremal complement conjecture for independent and connected sets

摘录 / EXCERPT: Let d be a positive integer and let n satisfy ... For a graph G, write $i(G)$ for its number of independent sets, $N(G)$ for its number of connected sets, and $N_{dom}(G)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-051488 | MATHDB-360588

Moore graph extremality conjecture for the connected-set growth constant

摘录 / EXCERPT: Let $c_d(n)$ denote the exponential growth constant for the maximum number of connected vertex subsets among d-regular graphs of order n. A Moore graph is a d-regular graph o...

conjecture | open | MathDB 页面 | 原文线索

V2-051489 | MATHDB-360589

Egan and Nikolayevsky's conjecture on degree sequences of triangular simple graphs

摘录 / EXCERPT: Let $n\geq 3$ and let $(d_1,\ldots,d_n)$ be a sequence of integers satisfying ... ... and ... Here a triangular simple graph is a simple graph in which every edge is contained in a trian...

conjecture | open | MathDB 页面 | 原文线索

V2-051490 | MATHDB-360614

Regular Turán extremal conjecture for graphs with chromatic number at most k

摘录 / EXCERPT: Let H be a graph with chromatic number $\chi(H)\le k$ and let $k\ge 3$. Write $T(n,k)$ for the n-vertex Turán graph with k parts, and let $\mathcal N(H,G)$ denote the number...

conjecture | open | MathDB 页面 | 原文线索

V2-051491 | MATHDB-360616

Nonexistence of the specified planar semi-cover of $K_{1,2,2,2}$

摘录 / EXCERPT: Nonexistence conjecture. There is no planar semi-cover G' of $K_{1,2, 2, 2}$ having the properties listed in Lemma $$.

conjecture | open | MathDB 页面 | 原文线索

V2-051492 | MATHDB-360617

Mohar's HL-index conjecture for planar subcubic graphs

摘录 / EXCERPT: Let G be a simple graph of order n, and let its adjacency eigenvalues be ... A graph is subcubic if its maximum degree is at most 3, and it is planar if it can be drawn in the...

conjecture | open | MathDB 页面 | 原文线索

V2-051493 | MATHDB-360618

Pavez-Signé-Piga-Sanhueza-Matamala conjecture on exact Ramsey growth approximation

摘录 / EXCERPT: Let $r(G)$ denote the two-colour Ramsey number of a graph G, and let $f\colon\mathbb{N}\to\mathbb{N}$ satisfy ... for all n. For $\epsilon>0$, seek graphs G_n with $|V(G_n)|=n$...

conjecture | open | MathDB 页面 | 原文线索

V2-051494 | MATHDB-360619

Verstraete's positive discrepancy conjecture for moderately dense graphs

摘录 / EXCERPT: Verstraete's conjecture.

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V2-051495 | MATHDB-360620

The far-from-Turán positive discrepancy conjecture

摘录 / EXCERPT: Far-from-Turán discrepancy conjecture. For every $\varepsilon>0$, there exists $c>0$ such that, if G is an n-vertex graph that is ε-far from every Turán graph, incl...

conjecture | open | MathDB 页面 | 原文线索

V2-051496 | MATHDB-360621

The far-from-Turán second-eigenvalue conjecture

摘录 / EXCERPT: Far-from-Turán eigenvalue conjecture. For every $\varepsilon>0$, there exists $c>0$ such that, if G is a regular n-vertex graph that is ε-far from every Turán graph...

conjecture | open | MathDB 页面 | 原文线索

V2-051497 | MATHDB-360626

Cochordality conjecture for higher independence hypergraphs

摘录 / EXCERPT: Let G be a graph, and for each integer $r\geq 1$ let $\mathrm{Con}_r(G)$ denote the hypergraph defined by the r-independence construction. A graph is cochordal if its complemen...

conjecture | open | MathDB 页面 | 原文线索

V2-051498 | MATHDB-360627

Caputo's shuffle conjecture for weighted hypergraphs

摘录 / EXCERPT: Let G be a finite weighted hypergraph with non-negative weights. At each step, choose a hyper-edge with probability proportional to its weight and then choose a uniformly random...

conjecture | open | MathDB 页面 | 原文线索

V2-051499 | MATHDB-360651

List chromatic number conjecture for multipartite uniform hypergraphs

摘录 / EXCERPT: Let $k\geqslant 2$, let Δ be sufficiently large, and let H be a k-partite k-graph of maximum degree at most Δ. Write $\chi_\ell(H)$ for the list chromatic numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-051500 | MATHDB-360652

The palette-size Erdős-Hajnal conjecture for edge-colourings

摘录 / EXCERPT: Size-version of the Erdős-Hajnal conjecture. There is a positive constant ϵ such that every colouring c of K_n in colours from $[s]$ avoiding T satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051501 | MATHDB-360672

Edmonds-Giles weighted dijoin packing conjecture

摘录 / EXCERPT: Edmonds-Giles conjecture. If the minimum weight of a dicut is τ, then D can pack τ dijoins.

conjecture | open | MathDB 页面 | 原文线索

V2-051502 | MATHDB-360673

Mixed-graph pseudo-dicut orientation conjecture

摘录 / EXCERPT: Mixed-graph orientation conjecture. There is a constant $k>2$ such that for every mixed graph $M=(V,E\cup A)$, there is an orientation E^+ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051503 | MATHDB-360674

Mixed-graph pseudo-dicut orientation threshold conjecture

摘录 / EXCERPT: Mixed-graph threshold orientation conjecture. There is a constant $k>2$ such that every mixed graph with minimum pseudo dicut size at least k has an orientation E^+ satisfying

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V2-051504 | MATHDB-360687

Universality conjecture for graphs with bounded density

摘录 / EXCERPT: For $d \in \mathbb{Q}$ and $n \in \mathbb{N}$, let $\mathcal{H}_d(n)$ be the family of all graphs with n vertices and density at most d, where the density of a graph H is ... A...

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V2-051505 | MATHDB-360688

The 3-connected case of Häggkvist's Hamiltonicity conjecture

摘录 / EXCERPT: Häggkvist's conjecture in the 3-connected case. Every such graph on at most $4k$ vertices is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-051506 | MATHDB-360696

Cleanliness conjecture for t-clock-free graphs

摘录 / EXCERPT: A hole is an induced cycle on four or more vertices. A t-clock is a clock consisting of a hole C and a vertex v such that there are two neighbours $x,y\in V(C)$ of v where th...

conjecture | open | MathDB 页面 | 原文线索

V2-051507 | MATHDB-360723

Infinite Gallai-Milgram conjecture

摘录 / EXCERPT: Let D be a digraph without infinite directed paths. Infinite Gallai-Milgram conjecture. There is a vertex-partition $d4ab$ of D into directed paths and an independent set U o...

conjecture | open | MathDB 页面 | 原文线索

V2-051508 | MATHDB-360738

Conjecture on eigenvector coordinate averages for large S-regular matrices

摘录 / EXCERPT: Eigenvector coordinate conjecture. With high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-051509 | MATHDB-360739

Conjecture on zero-dimensionality of the walk-generating-function polynomial system

摘录 / EXCERPT: Zero-dimensionality conjecture. The ideal generated by P is zero dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-051510 | MATHDB-360747

EPPA number of cycles

摘录 / EXCERPT: Cycle EPPA conjecture. If $n\geq 7$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051511 | MATHDB-360758

Chudnovsky-Nevo conjecture on independent sets in flag spheres

摘录 / EXCERPT: Chudnovsky-Nevo conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051512 | MATHDB-360781

The affine vertex algebra-small quantum group correspondence conjecture

摘录 / EXCERPT: Affine vertex algebra-small quantum group correspondence conjecture. The category ... associated with $$.

conjecture | open | MathDB 页面 | 原文线索

V2-051513 | MATHDB-360782

Vertex tensor and ribbon category conjecture for admissible affine Lie algebra weight modules

摘录 / EXCERPT: Let ell be an admissible level for the simple Lie algebra $\bf g$. Vertex tensor and ribbon category conjecture. 1. $L_{ell}(\bf g)\operatorname{-wtmod}$ is a vertex tensor categ...

conjecture | open | MathDB 页面 | 原文线索

V2-051514 | MATHDB-360791

The extremal homology-manifold conjecture in dimensions 8 and 16

摘录 / EXCERPT: Extremal homology-manifold conjecture. 1. Suppose that K is a 15-vertex Z-homology 8-manifold such that

conjecture | open | MathDB 页面 | 原文线索

V2-051515 | MATHDB-360798

Elphick-Liu-Ning conjecture on the square energies of connected graphs

摘录 / EXCERPT: Elphick-Liu-Ning conjecture. For any connected graph,

conjecture | open | MathDB 页面 | 原文线索

V2-051516 | MATHDB-360853

The crown-subgraph conjecture for representation numbers of bipartite graphs

摘录 / EXCERPT: Crown-subgraph conjecture. The representation number of G equals either

conjecture | open | MathDB 页面 | 原文线索

V2-051517 | MATHDB-360878

Li-Li-Wang constructibility conjecture for independent-set numbers of trees

摘录 / EXCERPT: A positive integer n is constructible on trees if there exists a tree T such that $i(T)=n$, where $i(T)$ denotes the number of independent sets in T. Li-Li-Wang's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051518 | MATHDB-360879

Ratio conjecture for independent sets in linear chains of hexagons

摘录 / EXCERPT: Let f_n be the number of $2\times n$ matrices containing a 1 in the top left entry, with integer entries such that adjacent entries differ by at most 1. Let a_n be the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-051519 | MATHDB-360893

Norine-Thomas linear-density conjecture for degree-three vertices in minimal bricks

摘录 / EXCERPT: Norine-Thomas conjecture. There exists a constant $\alpha>0$ such that every minimal brick G has at least $\alpha|V(G)|$ vertices of degree three.

conjecture | open | MathDB 页面 | 原文线索

V2-051520 | MATHDB-360904

Vertex-deletion conjecture for the upper orientable total domination number

摘录 / EXCERPT: Let G be a graph in $\mathcal{C}$ and let $v\in V(G)$ be such that $G-v\in\mathcal{C}$. Vertex-deletion conjecture. ... The conjecture proposes the missing lower bound from the cit...

conjecture | open | MathDB 页面 | 原文线索

V2-051521 | MATHDB-360905

Intermediate-value conjecture for orientable total domination numbers

摘录 / EXCERPT: Let $G\in\mathcal{C}$ be a graph, and let ${\rm dom}_t(G)$ and ${\rm DOM}_t(G)$ denote its lower and upper orientable total domination numbers. For an orientation f of G, write...

conjecture | open | MathDB 页面 | 原文线索

V2-051522 | MATHDB-360913

The relative fractional packing characterization by graph expansion

摘录 / EXCERPT: Let G and H be graphs. For graphs G and H, write $[?]$ for the relative fractional packing number of G with respect to H, and let $Expand(H)$ be the family of graphs ob...

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V2-051523 | MATHDB-360916

Bowler-Brown-Fenner common-card bound for trees and connected non-trees

摘录 / EXCERPT: Let G and H be graphs on n vertices, where one is a tree and the other is a connected non-tree. A common card is an unlabeled graph occurring as a card in the decks of both...

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V2-051524 | MATHDB-360917

Bowler-Brown-Fenner tree-recognition conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let a card be an unlabeled vertex-deleted subgraph $G-v$. Bowler-Brown-Fenner conjecture. For $n\geq 44$, it can be determined whether G i...

conjecture | open | MathDB 页面 | 原文线索

V2-051525 | MATHDB-360918

Common-card bound for bipartite and non-bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph and H a non-bipartite graph, both on n vertices. A common card is an unlabeled vertex-deleted subgraph occurring in the decks of both graphs. Bipar...

conjecture | open | MathDB 页面 | 原文线索

V2-051526 | MATHDB-360924

Saturation spectrum conjecture for three unions of cycles

摘录 / EXCERPT: Saturation spectrum conjecture.

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V2-051527 | MATHDB-360945

Wide Young diagram matching conjecture

摘录 / EXCERPT: For a hypergraph H, a 2-matching is a set of edges such that every two edges share fewer than two vertices, and let $\nu^{(2)}(H)$ denote its maximum size. For a Young diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-051528 | MATHDB-360951

Conjecture on the number of ideals in C-vines

摘录 / EXCERPT: C-vine ideal-count conjecture. The number of ideals in the C-vine of dimension $\ell\ge1$ is given by the OEIS sequence A047970:

conjecture | open | MathDB 页面 | 原文线索

V2-051529 | MATHDB-360965

Amended Pikhurko-Razborov conjecture on extremal clique-minimizing graphs

摘录 / EXCERPT: Fix $r\ge4$. For positive integers n,e with $e\le\binom{n}{2}$, let $t_{r-1}(n)$ be the number of edges in the balanced complete $(r-1)$-partite graph, let $g_r(n,e)$ be the mini...

conjecture | open | MathDB 页面 | 原文线索

V2-051530 | MATHDB-360981

The tripartite hypergraph degree-sequence threshold conjecture

摘录 / EXCERPT: Let c be the positive real number specified by ... A tripartite hypergraph degree sequence on $n+n+n$ vertices consists of degree data for the three vertex classes, and it is graph...

conjecture | open | MathDB 页面 | 原文线索

V2-051531 | MATHDB-360982

The quarter-range tripartite hypergraph degree-sequence conjecture

摘录 / EXCERPT: A tripartite hypergraph degree sequence on $n+n+n$ vertices assigns degrees to the three vertex classes; it is graphic if it has a tripartite hypergraph realization. The quarter-ra...

conjecture | open | MathDB 页面 | 原文线索

V2-051532 | MATHDB-360983

The near-quarter hypergraph degree-sequence conjecture

摘录 / EXCERPT: A hypergraph degree sequence on n vertices is a sequence of vertex degrees, and it is graphic if it is realized by a 3-uniform hypergraph; its degree sum is the sum of all vert...

conjecture | open | MathDB 页面 | 原文线索

V2-051533 | MATHDB-360985

Unfriendly Partition Conjecture for countably infinite graphs

摘录 / EXCERPT: A majority 2-vertex-coloring of a graph is a coloring of its vertices with two colors such that, at every vertex, the number of neighbors having the same color is at most the num...

conjecture | open | MathDB 页面 | 原文线索

V2-051534 | MATHDB-360986

Majority 3-edge-coloring conjecture for graphs without pendant edges

摘录 / EXCERPT: A majority 3-edge-coloring of a graph is an edge-coloring with three colors in which, at every vertex, the number of incident edges of its most frequent color is at most the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-051535 | MATHDB-360994

Star embedding conjecture for Euclidean Steiner trees

摘录 / EXCERPT: Let G be a graph with m edges, and embed each edge by its characteristic vector, as in the Vertex Cover reduction discussed in the source. Consider the Euclidean Steiner tree o...

conjecture | open | MathDB 页面 | 原文线索

V2-051536 | MATHDB-361001

Eventual equality of tournament counting exponents

摘录 / EXCERPT: For a positive integer k, let $c(k)$ be the exponent for counting copies of an arbitrary k-vertex tournament, and let $c^*(k)$ be the exponent for counting copies of the transi...

conjecture | open | MathDB 页面 | 原文线索

V2-051537 | MATHDB-361002

The recursive upper-bound conjecture for tournament counting

摘录 / EXCERPT: For each positive integer k, let $c(k)$ denote the exponent for counting copies of an arbitrary k-vertex tournament. Recursive counting conjecture. For all $k \ge 3$ it holds t...

conjecture | open | MathDB 页面 | 原文线索

V2-051538 | MATHDB-361013

The asymptotic $k/2$ conjecture for directed cycle packing and covering

摘录 / EXCERPT: Asymptotic directed-cycle packing and covering conjecture. For all sufficiently large n, every n-vertex directed graph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051539 | MATHDB-361014

Yuster's conjecture on inversion number for 3-decycling sets

摘录 / EXCERPT: Let T be an n-vertex tournament. An inversion reverses all edges whose endpoints lie in a specified vertex set, and ${\rm inv}_k(T)$ is the minimum length of a sequence of inve...

conjecture | open | MathDB 页面 | 原文线索

V2-051540 | MATHDB-361027

Edge-coloring conjecture for two cliques joined by an edge

摘录 / EXCERPT: Two-cliques edge-coloring conjecture. The signed edge-chromatic number satisfies

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-051541 | MATHDB-361033

Existence of nut graphs with two vertex orbits and three edge orbits

摘录 / EXCERPT: Let $n\geq 9$ be a composite integer. A nut graph is a graph whose adjacency matrix has nullity one and whose nullspace is spanned by a vector with no zero entries; write $o_v(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-051542 | MATHDB-361036

Zhu and Wu's domination conjecture for 2-connected cubic planar graphs

摘录 / EXCERPT: Let G be a 2-connected cubic planar graph. Write $V(G)$ for its vertex set and let $\gamma(G)$ be the domination number of G, the minimum size of a dominating set. Zhu and Wu...

conjecture | open | MathDB 页面 | 原文线索

V2-051543 | MATHDB-361067

Random-graph inertia conjecture

摘录 / EXCERPT: Let G be sampled from the binomial random graph $\mathbb G(n,\frac 12)$, and let A be a weighted adjacency matrix of G. Write $n_{\geq 0}(A)$ for the number of non-negative e...

conjecture | open | MathDB 页面 | 原文线索

V2-051544 | MATHDB-361072

Complete-network maximizer conjecture for homogeneous finite networks

摘录 / EXCERPT: Fix parameters p,q and a positive integer M, and let … For ${\cal N}\in{\cal G}(p,q,M)$, write $f(t;{\cal N})$ for its expected adoption level, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051545 | MATHDB-361075

Five-color conjecture for majority distinguishing edge colorings

摘录 / EXCERPT: Five-color conjecture. Every such graph G has a majority distinguishing edge 5-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-051546 | MATHDB-361080

Brewster-McGuinness-Moore-Noel conjecture on cycles in non-fcolorable graphs

摘录 / EXCERPT: Let $k\ge 3$ and let G be a graph, with chromatic number denoted by $\chi(G)$. A cycle has length $0\bmod k$ when its length is divisible by k. BrewsterfMcGuinnessfMoorefNoel c...

conjecture | open | MathDB 页面 | 原文线索

V2-051547 | MATHDB-361096

Spiro's clique Ramsey-number conjecture

摘录 / EXCERPT: Spiro's conjecture. For all $s\leq t$,

conjecture | open | MathDB 页面 | 原文线索

V2-051548 | MATHDB-361097

The sparse r-chromatic graph triangle conjecture

摘录 / EXCERPT: Sparse r-chromatic graph triangle conjecture. There exists $c>0$ such that every r-chromatic graph with at most

conjecture | open | MathDB 页面 | 原文线索

V2-051549 | MATHDB-361098

Odd-order graphs do not exhibit Laplacian perfect state transfer

摘录 / EXCERPT: Let X be a simple connected unweighted graph on an odd number of vertices. Odd-order Laplacian PST conjecture. Laplacian perfect state transfer does not occur in X. This extend...

conjecture | open | MathDB 页面 | 原文线索

V2-051550 | MATHDB-361120

Caterpillar decomposition conjecture for fully leafed Penrose subtrees

摘录 / EXCERPT: Let T be a fully leafed Penrose subtree. A Penrose tree is saturated if $n_{1}(T)=\overline{L_{P2}}(n(T))$, where $\overline{L_{P2}}(n)=8n/17+b$ is the unique linear upper bound...

conjecture | open | MathDB 页面 | 原文线索

V2-051551 | MATHDB-361121

Caterpillar classification conjecture for Penrose caterpillars

摘录 / EXCERPT: Let C be a Penrose caterpillar, let S be the graded poset of caterpillars used in the source, and let f_{boat} transform a Penrose caterpillar by replacing each relevant subp...

conjecture | open | MathDB 页面 | 原文线索

V2-051552 | MATHDB-361133

Matrix product factorization conjecture for complete graphs of order $4n+1$

摘录 / EXCERPT: Let K_{4n+1} be a complete graph on $4n+1$ vertices, where ... and both m_1 and m_2 are even. A graphical pair $((G,g),(H,h))$ without loops consists here of graphs G and H...

conjecture | open | MathDB 页面 | 原文线索

V2-051553 | MATHDB-361135

The poor-edge conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Poor-edge conjecture. If $G\not=P_{10}$, then G has a normal 5-edge-coloring with at least one poor edge. Moreover, if additionally $G\not=P_{10}^{\Delta}$, then G has a norm...

conjecture | open | MathDB 页面 | 原文线索

V2-051554 | MATHDB-361151

Sharp component-count bound for digraphs

摘录 / EXCERPT: First digraph component conjecture. The number of connected components satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051555 | MATHDB-361177

The hypergraph generalization conjecture for cycle-flow entropy vectors

摘录 / EXCERPT: Hypergraph generalization conjecture. Every entropy vector representable by a hypergraph can be represented using the cycle-flow prescription on a nesting balanced network.

conjecture | open | MathDB 页面 | 原文线索

V2-051556 | MATHDB-361178

Upper bound on the change in positive semidefinite leaky forcing under edge deletion

摘录 / EXCERPT: Let G be a graph and let $e\in E(G)$ be an edge. Write $G-e$ for the graph obtained by deleting e, and let $Z^+_{(1)}(G)$ denote the positive semidefinite 1-leaky forcing num...

conjecture | open | MathDB 页面 | 原文线索

V2-051557 | MATHDB-361236

Polynomial-time precoloring extension for bounded outer faces

摘录 / EXCERPT: Polynomial-time precoloring-extension conjecture. For every positive integer ℓ, there is a polynomial-time algorithm that, given a planar near-Eulerian-triangulation G with...

conjecture | open | MathDB 页面 | 原文线索

V2-051558 | MATHDB-361247

Miura-quiver free-field compatibility conjecture

摘录 / EXCERPT: Let $\sigma=[l^{w_l},\ldots,1^{w_1}]$ be a partition, let $q_i=\sum_{j=i}^l w_j$, and set $n=q_1+\cdots+q_l$. Consider the Miura map and Wakimoto realization ... and, assuming the Ka...

conjecture | open | MathDB 页面 | 原文线索

V2-051559 | MATHDB-361258

NP-completeness of recognizing graphs with minimum separating number

摘录 / EXCERPT: Let G be a graph, let $V(G)$ be its vertex set, and let $f(G)$ denote the exact minimum size of a vertex-separating tree system of G. Threshold-complexity conjecture. Determini...

conjecture | open | MathDB 页面 | 原文线索

V2-051560 | MATHDB-361260

Orthogonality conjecture for half-space six-vertex symmetric functions

摘录 / EXCERPT: Orthogonality conjecture. In the limit $c\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-051561 | MATHDB-361269

Impropriety bound for 2-trees

摘录 / EXCERPT: Let G be a 2-tree, that is, a graph obtained from a triangle by repeatedly adding a new vertex adjacent to both endpoints of an existing edge. Let $\mu_{\operatorname{int}}(G)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-051562 | MATHDB-361275

Fox-Sudakov conjecture on the polynomial Rödl property

摘录 / EXCERPT: Fox-Sudakov conjecture. Every graph H has the polynomial Rödl property.

conjecture | open | MathDB 页面 | 原文线索

V2-051563 | MATHDB-361278

Kostochka-Luo-Shan conjecture for quasi-kernels and sinks

摘录 / EXCERPT: Kostochka-Luo-Shan conjecture. Every digraph D has a quasi-kernel of size at most

conjecture | open | MathDB 页面 | 原文线索

V2-051564 | MATHDB-361283

The minimum-degree spectral conjecture for spanning-tree packing

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let G be a graph, and let δ be its minimum degree. Write $\lambda_2(G)$ for the second-largest adjacency eigenvalue and $\tau(G)$ for the span...

conjecture | open | MathDB 页面 | 原文线索

V2-051565 | MATHDB-361289

Li, Feng, and Liu's signless Laplacian spectral Turán conjecture

摘录 / EXCERPT: Let G be a graph with m edges. A graph is K_{r+1}-free if it contains no complete subgraph on $r+1$ vertices. Let $q(G)$ denote the largest eigenvalue of the signless Laplaci...

conjecture | open | MathDB 页面 | 原文线索

V2-051566 | MATHDB-361290

Gregory's Perron-vector conjecture for independent sets

摘录 / EXCERPT: Let G be a connected graph with chromatic number $k\geq 2$, let n be its order, and let S be an independent set. Let $\bm{x}$ be the Perron vector of G, with component...

conjecture | open | MathDB 页面 | 原文线索

V2-051567 | MATHDB-361291

The modified Gregory conjecture on Perron-vector mass of independent sets

摘录 / EXCERPT: Let G be a connected graph of order n with chromatic number $k\geq 2$. Let S be an independent set, and let $\bm{x}$ be the Perron vector of G. Modified Gregory conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-051568 | MATHDB-361293

Automorphism invariance of the Lorentzian lattice sublattices

摘录 / EXCERPT: Automorphism invariance conjecture. Any automorphism $f\in\operatorname{Aut}(\Lambda)$ preserves Λ_i^0 for $i=1,2$.

conjecture | open | MathDB 页面 | 原文线索

V2-051569 | MATHDB-361303

The C3-free graph bound for positive Lin-Lu-Yau Ricci curvature

摘录 / EXCERPT: The C3-free curvature bound. Then

conjecture | open | MathDB 页面 | 原文线索

V2-051570 | MATHDB-361304

The exponential order bound for graphs with positive Lin-Lu-Yau Ricci curvature

摘录 / EXCERPT: The exponential curvature bound. There exists a positive constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-051571 | MATHDB-361305

Cyman-Dzido-Lapinskas-Lo asymptotic conjecture for online Ramsey numbers of paths

摘录 / EXCERPT: Cyman-Dzido-Lapinskas-Lo conjecture. For every fixed $k \geq 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-051572 | MATHDB-361332

Gago-Coroničová-Hurajová-Madaras conjecture on betweenness-uniform graphs

摘录 / EXCERPT: Gago-Coroničová-Hurajová-Madaras conjecture. For every rational α in the interval

conjecture | open | MathDB 页面 | 原文线索

V2-051573 | MATHDB-361333

Dense hypergraphs contain large linear subhypergraphs conjecture

摘录 / EXCERPT: A 3-uniform hypergraph is linear if any two of its edges intersect in at most one vertex. Linear-subhypergraph conjecture. There is a constant $c>0$ such that for every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-051574 | MATHDB-361362

The complete, complete-bipartite, or 2-degenerate induced-subgraph conjecture

摘录 / EXCERPT: Let $\operatorname{tw}(G)$ denote the treewidth of G. A graph is 2-degenerate if every induced subgraph has a vertex of degree at most 2. The complete, complete-bipartite, or...

conjecture | refuted | MathDB 页面 | 原文线索

V2-051575 | MATHDB-361366

Equality of adoption levels in one-sided and two-sided Cartesian Bass models

摘录 / EXCERPT: Equality conjecture. The expected adoption levels are identical:

conjecture | open | MathDB 页面 | 原文线索

V2-051576 | MATHDB-361379

Ruskey-Savage conjecture on extending hypercube matchings to Hamilton cycles

摘录 / EXCERPT: The d-dimensional hypercube Q_d has vertex set consisting of all subsets of $[d]:=\{1,\ldots,d\}$, with edges between sets that differ in a single element. A matching is a set...

conjecture | open | MathDB 页面 | 原文线索

V2-051577 | MATHDB-361397

Mutual-visibility conjecture for distance-hereditary graphs

摘录 / EXCERPT: Let G be a distance-hereditary graph. A big-μ cograph without universal vertices is a cograph of the form $G=(K_1\cup K_t)+H$ without universal vertices, as described in the...

conjecture | open | MathDB 页面 | 原文线索

V2-051578 | MATHDB-361402

Cereceda-van den Heuvel-Johnson conjecture for graph recoloring

摘录 / EXCERPT: Let G be a graph and let k be an integer with $k\geq 4$. A k-coloring of G is a proper vertex coloring using colors from a set of k colors, and k-Mixing asks whether an...

conjecture | open | MathDB 页面 | 原文线索

V2-051579 | MATHDB-361403

PSPACE-completeness conjecture for 4-To-3 recoloring

摘录 / EXCERPT: Let G be a graph. A 4-To-3 instance asks whether a proper 4-coloring of G can be transformed into a proper 3-coloring by repeatedly recoloring one vertex while maintain...

conjecture | open | MathDB 页面 | 原文线索

V2-051580 | MATHDB-361422

Zhang-Lu matching extremal conjecture for 3-uniform hypergraphs

摘录 / EXCERPT: Let H be a 3-graph of order n, let $sigma_2(H)$ be the minimum of $deg(u)+deg(v)$ over adjacent vertices u,v, and let $H_{n,s}^2$ be the corresponding extremal 3-graph. Zhang...

conjecture | open | MathDB 页面 | 原文线索

V2-051581 | MATHDB-361423

Conjecture on the extremal obstruction to perfect matchings

摘录 / EXCERPT: Let H be a 3-graph of order n, let $sigma_2(H)$ denote the minimum $deg(u)+deg(v)$ over adjacent vertices u,v, and let $H^{1,2}_{n,x,y}$ be the construction defined in the so...

conjecture | open | MathDB 页面 | 原文线索

V2-051582 | MATHDB-361424

Conjecture on the Laplacian eigenvalue distribution of graphs with given diameter

摘录 / EXCERPT: Laplacian eigenvalue distribution conjecture. If $c=0,\dots,d-2$ and

conjecture | open | MathDB 页面 | 原文线索

V2-051583 | MATHDB-361456

Addario-Berry-Havet-Linhares Sales-Reed-Thomassé anti-directed-tree conjecture

摘录 / EXCERPT: Anti-directed-tree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051584 | MATHDB-361462

Malnegro-Ozeki's $\hat{A}$-coloring conjecture for line graphs of cubic graphs

摘录 / EXCERPT: Let G be a 2-edge-connected simple cubic graph with an even number of edges. Its line graph $L(G)$ has one vertex for each edge of G, with two vertices adjacent when the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-051585 | MATHDB-361468

Nikiforov's spectral Erdős-Sós conjecture for trees

摘录 / EXCERPT: Let F be a tree of order $\ell\geq 4$. For integers n,k,p with $n>k>0$ and $p\in\{0,\dots,\lfloor(n-k)/2\rfloor\}$, let $S_{n,k}^{p}$ be the graph obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-051586 | MATHDB-361470

Gishboliner-Steiner-Szabó conjecture on subdivisions of the bidirected triangle

摘录 / EXCERPT: Let F be a digraph, and let $\mathrm{mader}_{\vec{\chi}}(F)$ be the least integer k such that every digraph D with dichromatic number $\vec{\chi}(D)\geq k$ contains a subdivi...

conjecture | open | MathDB 页面 | 原文线索

V2-051587 | MATHDB-361471

Large-digirth dichromatic subdivision bound by maximum degree

摘录 / EXCERPT: Let F be a digraph with maximum degree Δ. For an integer g, let $\mathrm{mader}_{\vec{\chi}}^{(g)}(F)$ be the least integer k, if it exists, such that every digraph...

conjecture | open | MathDB 页面 | 原文线索

V2-051588 | MATHDB-361476

Delcourt-Postle conjecture on full rainbow matchings in multigraphs

摘录 / EXCERPT: Let G be a multigraph with maximum degree Δ, and let $E_1,\ldots,E_n$ be color classes of a proper edge-coloring of G. A full rainbow matching is a matching containing e...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-051589 | MATHDB-361477

Gao-Ramadurai-Wanless-Wormald conjecture on rainbow matchings in multigraphs

摘录 / EXCERPT: Let G be a multigraph, and let $E_1,\ldots,E_n$ be color classes of a proper edge-coloring of G. A full rainbow matching is a matching containing exactly one edge from each col...

conjecture | open | MathDB 页面 | 原文线索

V2-051590 | MATHDB-361486

Kac-Sridharan conjecture on integral irreducible objects of the subregular vertex-algebra category

摘录 / EXCERPT: For a complex simple Lie algebra $\mathfrak{g}$, let $V_{-b}(\mathfrak{g})$ be the simple affine vertex algebra at level $-b$, where b is the maximal coefficient in the highest c...

conjecture | open | MathDB 页面 | 原文线索

V2-051591 | MATHDB-361496

Equivalence conjecture for amoeba immortality notions

摘录 / EXCERPT: Immortality-equivalence conjecture. The following properties of an amoeba A are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-051592 | MATHDB-361497

Undecidability conjecture for amoeba-colony growth

摘录 / EXCERPT: Amoeba-colony undecidability conjecture. The following algorithmic problem is undecidable:

conjecture | open | MathDB 页面 | 原文线索

V2-051593 | MATHDB-361517

Araujo-Pardo-Leemans family conjecture for edge-girth-regular graphs

摘录 / EXCERPT: Let $egr(v,k,g,\lambda)$ denote an edge-girth-regular graph with v vertices, valency k, girth g, and each edge contained in exactly λ cycles of length g. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-051594 | MATHDB-361519

Di Francesco's determinant evaluation for twenty-vertex configurations

摘录 / EXCERPT: Di Francesco's conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-051595 | MATHDB-361526

Expressiveness hierarchy conjecture for Local k-FGNN

摘录 / EXCERPT: For an integer $k\ge 2$, let Local k-FGNN, Local k-GNN, and k-FGNN denote the corresponding higher-order graph neural network architectures, ordered by their ability to disti...

conjecture | open | MathDB 页面 | 原文线索

V2-051596 | MATHDB-361549

Maximum-weight bisection bound for graphs of bounded maximum degree

摘录 / EXCERPT: Bounded-degree bisection conjecture. If k is odd, then G has a bisection of weight at least

conjecture | open | MathDB 页面 | 原文线索

V2-051597 | MATHDB-361554

Asymptotic order conjecture for the clique ratio of regular graphs with bounded smallest eigenvalue

摘录 / EXCERPT: Asymptotic clique-ratio conjecture. There exist positive constants ε_1 and ε_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-051598 | MATHDB-361568

Spreer-Tobin's sharp vertex bound conjecture in arbitrary dimensions

摘录 / EXCERPT: Let $\mathcal{T}$ be a triangulation of a closed and connected d-dimensional manifold, with $d>0$, f_0 vertices, and f_d facets. Spreer-Tobin's sharp vertex bound conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051599 | MATHDB-361583

Characterization of maximum-chromatic integer distance graphs

摘录 / EXCERPT: Let $D\subseteq\mathbb N$ be a distance set, let $G(D)$ be its integer distance graph, let $\chi(D)$ be its chromatic number, and let $\omega(G(D))$ be the clique number of $G(D)$....

conjecture | open | MathDB 页面 | 原文线索

V2-051600 | MATHDB-361584

Hriňáková-Knor-Škrekovski conjecture on extremal iterated-line-graph Wiener ratios

摘录 / EXCERPT: Hriňáková-Knor-Škrekovski conjecture. For large n and $k\geq 2$, among all graphs G on n vertices, $R_k(G)$ attains its maximum at $G=K_n$ and its minimum at $G=P_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-051601 | MATHDB-361592

The ADKMV conjecture for the topological vertex

摘录 / EXCERPT: Let μ^1,μ^2,μ^3 be partitions, and let $W_{\mu^1,\mu^2,\mu^3}(q)$ denote the topological vertex. For $i,j\in\{1,2,3\}$ and $m,n\geq 0$, let $A_{mn}^{ij}(q)$ be the explicit...

conjecture | open | MathDB 页面 | 原文线索

V2-051602 | MATHDB-361593

The framed ADKMV conjecture for the framed topological vertex

摘录 / EXCERPT: Framed ADKMV conjecture. The framed topological vertex is given by

conjecture | open | MathDB 页面 | 原文线索

V2-051603 | MATHDB-361596

The restrained-polynomial deck reconstructs the chromatic polynomial

摘录 / EXCERPT: Let G_1 and G_2 be graphs. For each vertex v, let $\rho_{r_v^{(2)}}(k)$ denote the corresponding restrained chromatic polynomial, and consider the multiset of these polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-051604 | MATHDB-361623

Conditional lower-bound conjecture for mixed cages

摘录 / EXCERPT: Let $n[z,r;g]$ be the minimum order of a $[z,r;g]$-mixed cage, and let $n_{\operatorname{AHM}}[1,r;g]$ denote the AHM bound appearing in the source. Conditional lower-bound conject...

conjecture | open | MathDB 页面 | 原文线索

V2-051605 | MATHDB-361653

Independence-number rationality conjecture for FI-graphs

摘录 / EXCERPT: Independence-number rationality conjecture. The generating function is rational. In particular, the function $n \mapsto \alpha(G_n)$ agrees with a quasi-polynomial for all...

conjecture | open | MathDB 页面 | 原文线索

V2-051606 | MATHDB-361656

The center-realization conjecture for 2-connected plane graphs

摘录 / EXCERPT: Let H be a 2-connected plane graph. A vertex v of H is quasi-eccentric to $H[f]$ for a face f of H when the relevant quasi-eccentricity condition holds for the subgraph i...

conjecture | open | MathDB 页面 | 原文线索

V2-051607 | MATHDB-361662

The one-branch conjecture for real Collatz graphs

摘录 / EXCERPT: For each $x\in(0,\infty)$, let G_x be the directed graph constructed from the balanced ternary data associated with x. A branch is a directed path in G_x of the type used in...

conjecture | open | MathDB 页面 | 原文线索

V2-051608 | MATHDB-361717

Multipartite Hajnal-Szemerédi perfect packing conjecture

摘录 / EXCERPT: Let $k\ge 2$ and let G be a k-partite graph with parts $V_1,\ldots,V_k$ of the same size n. Define the partite minimum degree of G to be the largest integer d such that e...

conjecture | open | MathDB 页面 | 原文线索

V2-051609 | MATHDB-361719

Kuperwasser-Samotij-Wigderson forest-removal conjecture

摘录 / EXCERPT: Let $m>0$ be a real number. For a graph G, write $m(G)$ for its maximum average density and $m_2(G)$ for its 2-density. Kuperwasser-Samotij-Wigderson conjecture. Every graph G...

conjecture | open | MathDB 页面 | 原文线索

V2-051610 | MATHDB-361730

An FPRAS for counting list packings

摘录 / EXCERPT: Let Δ be a maximum-degree bound, let $q\ge 2\Delta$, and consider graphs of maximum degree at most Δ. A q-list packing assigns colors from each vertex's list so tha...

conjecture | open | MathDB 页面 | 原文线索

V2-051611 | MATHDB-361741

Quantum version of Deligne's conjecture for the exceptional series

摘录 / EXCERPT: Let $\mathsf{QExc}_{R,v,w}$ be the quantum exceptional diagram category. A trivalent ribbon category is a ribbon category generated by a trivalent vertex, and a ribbon functor pres...

conjecture | open | MathDB 页面 | 原文线索

V2-051612 | MATHDB-361742

Finite-dimensionality conjecture for the classical F4(d) family

摘录 / EXCERPT: Let $F4(d)$ be the symmetric ribbon category generated by a trivalent vertex subject to the defining $F4(d)$ relations, and let Hom denote a morphism space in this...

conjecture | open | MathDB 页面 | 原文线索

V2-051613 | MATHDB-361744

Häggkvist's Hamiltonicity conjecture for oriented graphs

摘录 / EXCERPT: Häggkvist's conjecture. Every oriented graph D with

conjecture | open | MathDB 页面 | 原文线索

V2-051614 | MATHDB-361769

Domination-packing conjecture for connected graphs

摘录 / EXCERPT: Domination-packing conjecture. For every such graph,

conjecture | open | MathDB 页面 | 原文线索

V2-051615 | MATHDB-361770

Domination-packing conjecture for maximal outerplanar graphs

摘录 / EXCERPT: Maximal outerplanar domination-packing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051616 | MATHDB-361774

Matroid-rooted bounded regular packing conjecture for mixed hyperarborescences

摘录 / EXCERPT: Matroid-rooted bounded regular packing conjecture. There exists an M-rooted (f,g)-bounded k-regular packing of mixed hyperarborescences in $\mathcal{F}$ if and only if the co...

conjecture | open | MathDB 页面 | 原文线索

V2-051617 | MATHDB-361775

Orientation conjecture for mixed hypergraphs with supermodular and submodular bounds

摘录 / EXCERPT: Mixed-hypergraph orientation conjecture. There exists an orientation $\overrightarrow{\mathcal{E}}$ of $\mathcal{E}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051618 | MATHDB-361783

The acyclic digraph cellular-singular cubical homology conjecture

摘录 / EXCERPT: Let G be a digraph. Write $H_*^{\operatorname{cell}}(G;\mathbb R)$ for its cellular homology and $H_*^c(G;\mathbb R)$ for its singular cubical homology. The cellular-singular cub...

conjecture | open | MathDB 页面 | 原文线索

V2-051619 | MATHDB-361788

Mohar's cop-number conjecture for geodesic surfaces

摘录 / EXCERPT: Let S be a geodesic surface of genus g, and let $c(S)$ denote its cop number. Mohar's surface conjecture. ... The paper describes this as a tough conjecture because it implies th...

conjecture | open | MathDB 页面 | 原文线索

V2-051620 | MATHDB-361791

The proper abelian coloring conjecture for exceptional groups

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A proper A-coloring is a proper edge-coloring of G by the non-zero elements of an abelian group A such that the sum of the colors on the...

conjecture | open | MathDB 页面 | 原文线索

V2-051621 | MATHDB-361793

The resistance and edge-reduction conjecture for snarks

摘录 / EXCERPT: Let G be a snark. Define the resistance $r(G)$ to be the minimum number of edges that can be removed from G so that the resulting graph is 3-edge-colorable, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-051622 | MATHDB-361805

The DT-PT₀ vertex correspondence without PT₀ moduli

摘录 / EXCERPT: Let $\boldsymbol\mu=\{\mu_a\}_{a=1}^{4}$ be plane partitions with no PT$_0$ moduli, and let $\mathsf V^{\mathrm{DT}}_{\mu_1\mu_2\mu_3\mu_4}(q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-051623 | MATHDB-361806

Uniqueness of triangle colours within components of the 2-coloured flip graph

摘录 / EXCERPT: Let P_{n+2} be a convex polygon, and consider its 2-coloured flip graph, whose vertices are triangulations with triangles coloured using two colours. A triangle is understood b...

conjecture | open | MathDB 页面 | 原文线索

V2-051624 | MATHDB-361809

Krivelevich-Sudakov conjecture on Hamiltonicity of pseudorandom graphs

摘录 / EXCERPT: Krivelevich-Sudakov conjecture. There exists $C>0$ such that if

conjecture | open | MathDB 页面 | 原文线索

V2-051625 | MATHDB-361851

Ivan's complete bipartite poset saturation conjecture

摘录 / EXCERPT: For $s\geq 2$, let $\mathcal{K}_{s,t}$ be the complete bipartite poset with s upper-layer and t lower-layer pairwise incomparable vertices, every upper-layer vertex larger than...

conjecture | open | MathDB 页面 | 原文线索

V2-051626 | MATHDB-361860

Li-Li edge-connection conjecture

摘录 / EXCERPT: Let G be a k-edge-connected graph, where $k\ge 2$, and let $H\subset G$ be a minimum-edge spanning k-edge-connected subgraph, meaning a spanning k-edge-connected subgraph w...

conjecture | open | MathDB 页面 | 原文线索

V2-051627 | MATHDB-361875

Casselgren's random-list colouring conjecture for graphs of unbounded degree

摘录 / EXCERPT: Given positive integers $k \leq m$, a random (k,m)-list-assignment assigns independently to each vertex a uniformly random k-element subset of $\{1,\ldots,m\}$. An n-vertex g...

conjecture | open | MathDB 页面 | 原文线索

V2-051628 | MATHDB-361876

Tensor-stabilizer characterization for forests

摘录 / EXCERPT: Let G be a forest on vertex set $V=\\{1,\ldots,d\\}$, with each connected component a tree having at least three nodes, and let A be its adjacency matrix. Let $\mathcal{V}_G$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-051629 | MATHDB-361888

Hefetz-Mütze-Schwartz conjecture on antimagic orientations

摘录 / EXCERPT: Let G be a connected undirected graph. An antimagic orientation consists of an orientation D of G and a bijection from $E(G)$ to $\{1,2,\ldots,|E(G)|\}$ such that the vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-051630 | MATHDB-361897

Paired coalition number conjecture for complete binary trees

摘录 / EXCERPT: Complete binary tree conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051631 | MATHDB-361907

Minimum-order conjecture for Maker-win hypergraphs with Ophelia draws

摘录 / EXCERPT: A hypergraph is a pair consisting of a vertex set and a collection of subsets of that vertex set, called hyperedges. In the Tic-Tac-Toe game on a hypergraph, Maker wins if Maker ca...

conjecture | open | MathDB 页面 | 原文线索

V2-051632 | MATHDB-361910

The forbidden biclique and path conjecture for tree-independence number

摘录 / EXCERPT: For positive integers d and s, a graph is $\{K_{d,d},P_s\}$-free if it has no induced subgraph isomorphic to $K_{d,d}$ or the s-vertex path P_s. The biclique-and-path conje...

conjecture | open | MathDB 页面 | 原文线索

V2-051633 | MATHDB-361913

The $K_{2,d}$-free path conjecture for tree-independence number

摘录 / EXCERPT: For positive integers d and s, a graph is $\{K_{2,d},P_s\}$-free if it has no induced subgraph isomorphic to $K_{2,d}$ or the s-vertex path P_s. The $K_{2,d}$-and-path conj...

conjecture | open | MathDB 页面 | 原文线索

V2-051634 | MATHDB-361920

Kostochka-Stiebitz gap conjecture for oriented dicritical digraphs

摘录 / EXCERPT: Let $d_k(n)$ be the minimum number of arcs in a k-dicritical digraph of order n, and let $o_k(n)$ be the minimum number of arcs in a k-dicritical oriented graph of order n,...

conjecture | open | MathDB 页面 | 原文线索

V2-051635 | MATHDB-361921

Hoshino-Kawarabayashi maximum-density conjecture for oriented dicritical digraphs

摘录 / EXCERPT: Let D be a k-dicritical oriented graph on n vertices, and let $m(D)$ denote its number of arcs. Hoshino-Kawarabayashi conjecture. For every integer $k\geq3$, ... Hoshino and Ka...

conjecture | open | MathDB 页面 | 原文线索

V2-051636 | MATHDB-361922

Conjecture on the finiteness of dense 3-dicritical digraphs

摘录 / EXCERPT: Let D be a 3-dicritical digraph on n vertices, and let $m(D)$ denote its number of arcs. Dense 3-dicritical finiteness conjecture. There are only finitely many 3-dicritical dig...

conjecture | open | MathDB 页面 | 原文线索

V2-051637 | MATHDB-361923

Finiteness conjecture for k-dicritical semi-complete digraphs

摘录 / EXCERPT: A semi-complete digraph is a digraph in which every pair of distinct vertices is joined by at least one directed arc. Semi-complete-digraph finiteness conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-051638 | MATHDB-361933

Linear obstruction-size conjecture for graphs of bounded lettericity

摘录 / EXCERPT: Let k be a positive integer, and call a graph an obstruction for the class of k-letter graphs if it is not a k-letter graph while all of its proper induced subgraphs are k-...

conjecture | open | MathDB 页面 | 原文线索

V2-051639 | MATHDB-361937

Triangle-free extremizers for warm-started QAOA lower bounds

摘录 / EXCERPT: Let $\textbf{LB}_{\kappa}(\theta)$ and $\textbf{LB}'_{\kappa}(\theta)$ be the stated inner-minimization problems, with variables $\mathbf r$, $\mathbf s$, r_1, r_6, s, t,...

conjecture | open | MathDB 页面 | 原文线索

V2-051640 | MATHDB-361941

A $k^{2/3}$ lower bound for the Randić index in terms of matching number

摘录 / EXCERPT: Randić-index matching conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051641 | MATHDB-361990

The min-plus dot product dimension is bounded by the max-plus dot product dimension

摘录 / EXCERPT: Dimension inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-051642 | MATHDB-361991

The generalized edge reconstruction conjecture

摘录 / EXCERPT: Let $n\geq 4$, and let $G_1,\dots,G_k$ be a collection of n-edged graphs for some $k\geq 1$. Write $\coprod_{i=1}^k\operatorname{ED}(G_i)$ for the disjoint union of their edge de...

conjecture | open | MathDB 页面 | 原文线索

V2-051643 | MATHDB-361995

Equivalent maximal-ReLU-function minimal-depth conjecture

摘录 / EXCERPT: Hertrich et al.'s equivalent conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-051644 | MATHDB-361996

Geometric Newton-polytope formulation of the maximal-ReLU-function conjecture

摘录 / EXCERPT: Hertrich et al.'s geometric conjecture. There do not exist polytopes $P,Q\in\Delta(m)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051645 | MATHDB-362008

Tree-independence conjecture for biclique-free graphs of bounded induced matching treewidth

摘录 / EXCERPT: Tree-independence conjecture. For any two integers $k,t\in\mathbb{N}$ there exists an integer r such that every graph with induced matching treewidth at most k and no induced s...

conjecture | open | MathDB 页面 | 原文线索

V2-051646 | MATHDB-362022

The lower bound for negative oriented distinguishing index

摘录 / EXCERPT: Lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051647 | MATHDB-362023

The conjecture on negative oriented distinguishing index when the distinguishing index is two

摘录 / EXCERPT: Conjecture for distinguishing index two. If

conjecture | open | MathDB 页面 | 原文线索

V2-051648 | MATHDB-362038

Linear-connectivity partition conjecture for highly connected digraphs

摘录 / EXCERPT: Linear-connectivity partition conjecture. There exists a constant $c>0$ such that the vertices of every strongly $c\cdot ktl$-connected digraph D with $\delta(D)\geq n-l$ can be...

conjecture | open | MathDB 页面 | 原文线索

V2-051649 | MATHDB-362039

Thomassé's conjecture for oriented graphs

摘录 / EXCERPT: Let D be an oriented graph, and let δ denote its minimum out-degree. Thomassé's conjecture. The graph D contains a directed path of length 2δ. This is the $g=3$ ca...

conjecture | open | MathDB 页面 | 原文线索

V2-051650 | MATHDB-362040

A linear lower bound for directed-path length in terms of girth and minimum out-degree

摘录 / EXCERPT: For a digraph D, let $\ell(D)$ be the length of a longest directed path, let $g(D)$ be its girth, and let $\delta^+(D)$ be its minimum out-degree. Proposed weaker Thomassé conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-051651 | MATHDB-362064

The critical bias ratio conjecture for vertex-transitive graphs

摘录 / EXCERPT: Critical bias ratio conjecture. If Λ is a connected vertex-transitive graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-051652 | MATHDB-362074

The high-girth existence conjecture for Steiner systems

摘录 / EXCERPT: A Steiner system with parameters (n,q,r) is a design in which every r-subset belongs to exactly one q-subset. For a partial (n,q,r)-Steiner system, the girth is the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-051653 | MATHDB-362078

Odd-vertex perfect matching conjecture in random graphs

摘录 / EXCERPT: Odd-vertex matching conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051654 | MATHDB-362080

Treewidth order conjecture for token graphs of trees

摘录 / EXCERPT: Let G) be a tree on n vertices, and let $F_k(G)$ denote its k-token graph, whose vertices are the k-subsets of $V(G)$. For a fixed integer k satisfying $1\le k\le n-1$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-051655 | MATHDB-362085

The proper rainbow triangle-packing conjecture for edge-colored graphs

摘录 / EXCERPT: Let G be an edge-colored graph on n vertices, let $e(G)$ denote its number of edges, and let $c(G)$ denote its number of colors. A proper mK_3 is a collection of m vertex-d...

conjecture | open | MathDB 页面 | 原文线索

V2-051656 | MATHDB-362123

Jin-Tan conjecture on connected two-distance transitive dihedrants

摘录 / EXCERPT: Jin-Tan's conjecture. Every connected 2-distance transitive dihedrant either is a known 2-arc transitive dihedrant, is isomorphic to $\mathsf{K}_{m[b]}$ for some $m\geq 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-051657 | MATHDB-362128

Exterior Stanley-Reisner cohomology conjecture for oriented pro-p RAAGs

摘录 / EXCERPT: Let $\Gamma=(\mathcal{V},\mathcal{E})$ be a special digraph, let G be the oriented pro-p right-angled Artin group associated to Γ, and let q be a p-power with...

conjecture | open | MathDB 页面 | 原文线索

V2-051658 | MATHDB-362129

Wong and Zhu's $(1,3)$-choosability conjecture for graphs without isolated edges

摘录 / EXCERPT: Wong and Zhu's conjecture. Every graph with no isolated edges is $(1,3)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-051659 | MATHDB-362130

Wong and Zhu's permanent-index conjecture for the total-weighting matrix

摘录 / EXCERPT: Wong and Zhu's conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-051660 | MATHDB-362138

Steiner distance Graham-Pollak determinant conjecture for trees

摘录 / EXCERPT: Let $D_k(T)$ denote the Steiner distance hypermatrix of order k associated with a tree T on n vertices. Steiner distance Graham-Pollak determinant conjecture. The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-051661 | MATHDB-362139

Harutyunyan-McDiarmid conjecture on acyclic sets in H-free oriented graphs

摘录 / EXCERPT: Let H be an oriented graph, and let an H-free oriented graph be one that does not contain H as a not necessarily induced subdigraph. For an oriented graph D, write...

conjecture | open | MathDB 页面 | 原文线索

V2-051662 | MATHDB-362140

Alon-Pachs-Solymosi conjecture on acyclic sets in H-free tournaments

摘录 / EXCERPT: Let H be a tournament. An H-free tournament is one that does not contain H as a not necessarily induced subdigraph. For a tournament T, write $\vec\alpha(T)$ for its maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-051663 | MATHDB-362143

Nichols-algebra conjecture for screening operators

摘录 / EXCERPT: Let W be a vertex operator algebra containing Heisenberg fields α_i, and let $S_i=\operatorname{Res}(e^{\alpha_i})$. Assume that W decomposes into integer eigenvalues un...

conjecture | open | MathDB 页面 | 原文线索

V2-051664 | MATHDB-362163

Botler, Colucci and Kohayakawa's additive bound conjecture for the mod k chromatic index

摘录 / EXCERPT: Let G be a simple graph and let $k\geq 2$ be an integer. A χ'_{k}-coloring of G is an edge coloring such that the subgraph induced by the edges of each color has all degre...

conjecture | open | MathDB 页面 | 原文线索

V2-051665 | MATHDB-362165

Weak finiteness conjecture for prescribed numbers of different cycle lengths

摘录 / EXCERPT: Let $h(k,\ell)$ be the source's minimum-outdegree threshold for forcing k vertex-disjoint directed cycles with ℓ different lengths. Weak finiteness conjecture. For every pos...

conjecture | open | MathDB 页面 | 原文线索

V2-051666 | MATHDB-362185

Adamović's orbifold quasi-lisse conjecture

摘录 / EXCERPT: Let ... be a simple positively graded quasi-lisse vertex algebra with $V_0\cong\mathbb{C}$, and let G be a finite solvable automorphism group of V. Adamović's conjecture. The fix...

conjecture | open | MathDB 页面 | 原文线索

V2-051667 | MATHDB-362187

Kac-Roan-Wakimoto's Drinfeld-Sokolov reduction conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let f be a nilpotent element of $\mathfrak{g}$, and let $L_k(\mathfrak{g})$ be the simple affine vertex algebra at level k. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-051668 | MATHDB-362204

The sparse Strong Nine Dragon Tree Conjecture

摘录 / EXCERPT: Let k and d be positive integers. For a graph G, call it (k,d)-sparse if every subgraph H satisfies ... Call a subgraph H $(k+1)$-overfull if $e(H)>(k+1)(v(H)-1)$. Sparse...

conjecture | open | MathDB 页面 | 原文线索

V2-051669 | MATHDB-362224

Thomassen's high-girth subgraph conjecture

摘录 / EXCERPT: For a graph G, let $\delta(G)$ be its minimum degree and let $\operatorname{girth}(G)$ be the length of its shortest cycle, with value ∞ when G is acyclic. Thomassen's c...

conjecture | open | MathDB 页面 | 原文线索

V2-051670 | MATHDB-362225

The induced Thomassen conjecture

摘录 / EXCERPT: For a graph G, write $\delta(G)$ for its minimum degree and $\operatorname{girth}(G)$ for its girth. A graph is induced-subgraph-free of the two basic obstructions when it contai...

conjecture | open | MathDB 页面 | 原文线索

V2-051671 | MATHDB-362226

Hunter-Milojević-Sudakov-Tomon hereditary Kővári-Sós-Turán conjecture

摘录 / EXCERPT: For a graph H and an integer n, let $\operatorname{ex}(n,H)$ be the maximum number of edges in an n-vertex graph with no subgraph isomorphic to H. Hereditary Kővári-Sós-Tur...

conjecture | open | MathDB 页面 | 原文线索

V2-051672 | MATHDB-362277

Bang-Jensen and Gutin's longest path algorithm conjecture for semicomplete digraphs

摘录 / EXCERPT: Let D be a semicomplete digraph, and let x,y be distinct vertices of D. A longest (x,y)-path is an (x,y)-path containing the maximum possible number of arcs. Longest path...

conjecture | open | MathDB 页面 | 原文线索

V2-051673 | MATHDB-362291

Successive quantum Hamiltonian reduction conjecture for type A W-algebras

摘录 / EXCERPT: Let $\lambda=(\lambda_1\leq\lambda_2\leq\dots\leq\lambda_n)\vdash N$, and set $\widehat{\lambda}_i=(1,\dots,1,\lambda_i)\vdash N_i$ with $N_i=N-\sum_{j>i}\lambda_j$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-051674 | MATHDB-362292

Conformal coset embedding conjecture for type A W-algebras

摘录 / EXCERPT: Let $\lambda=(\lambda_1\leq\dots\leq\lambda_n)\vdash N$, let $N_i=N-\sum_{j>i}\lambda_j$, and let $C^{k_i^\sharp}(\mathfrak{sl}_{N_i},f_{\widehat{\lambda}_i})$ be the affine coset...

conjecture | open | MathDB 页面 | 原文线索

V2-051675 | MATHDB-362293

Inverse Hamiltonian reduction embedding conjecture for type A W-algebras

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\dots,\lambda_n)\vdash N$, let a be the unique integer with $\lambda_{a-1}=1$ and $\lambda_a>1$, and define ... Set $m=\lambda_1+\dots+\lambda_a$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-051676 | MATHDB-362294

Collapsing-level singlet decomposition conjecture for W-algebras

摘录 / EXCERPT: Let p be a parameter for which the displayed level and W-algebra are defined, let $S_{2;n}$ and $S_{p;n}$ denote the indicated singlet-algebra modules, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051677 | MATHDB-362295

Feigin-Frenkel duality conjecture for regular W-superalgebras

摘录 / EXCERPT: Let $n,r\geq0$, let $\mathcal{W}^k(\mathfrak{sl}_{n+r|n},f_{n+r|n})$ be the universal regular W-superalgebra, let $\mathrm{SB}^{p|q}=\beta\gamma^{\otimes p}\otimes bc^{\otimes q}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-051678 | MATHDB-362297

Exceptional-level conformal embedding conjecture for type A W-algebras

摘录 / EXCERPT: Let $N\in\mathbb{Z}_{>0}$ and let $k=-N+p/n$ be admissible. Write π for a rank-one Heisenberg vertex algebra, and set $L_\bullet(\mathfrak{sl}_i)=\mathbb{C}$ for $i=0,1$. Excep...

conjecture | open | MathDB 页面 | 原文线索

V2-051679 | MATHDB-362298

Fox-Sudakov polynomial Rödl conjecture

摘录 / EXCERPT: Fox-Sudakov's conjecture. For any graph H there exists a $C_H>0$ such that for any $0<\varepsilon \le 1/2$ any n-vertex H-free graph contains an ε-homogeneous set...

conjecture | open | MathDB 页面 | 原文线索

V2-051680 | MATHDB-362299

Polynomial Rödl property for graphs close to a proper hereditary family

摘录 / EXCERPT: Conjecture on hereditary families close to $\mathcal{F}$. For each proper hereditary family $\mathcal{F}$ of graphs there are $C_{\mathcal{F}},d_{\mathcal{F}}>0$ such that if...

conjecture | open | MathDB 页面 | 原文线索

V2-051681 | MATHDB-362302

Borozan et al.'s conjecture on k-proper partitions

摘录 / EXCERPT: Let G be a graph of order n, and let k be an integer at least 2. A partition $\mathcal{P}$ of $V(G)$ is k-proper if every part $P\in\mathcal{P}$ induces a k-connected s...

conjecture | open | MathDB 页面 | 原文线索

V2-051682 | MATHDB-362310

The graft transformation conjecture for the distance spectral radius

摘录 / EXCERPT: Graft transformation conjecture. For $s\geq t\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-051683 | MATHDB-362311

Gutman's non-hyperenergetic graph conjecture

摘录 / EXCERPT: Let Γ be a finite graph, and let $K_{|v(\mathcal{G})|}$ denote the complete graph on $|v(\mathcal{G})|$ vertices. A graph is hyperenergetic if $E(\Gamma)>E(K_{|V(\Gamma)|})$...

conjecture | open | MathDB 页面 | 原文线索

V2-051684 | MATHDB-362319

Transversal Hajnal-Szemerédi conjecture

摘录 / EXCERPT: Let $k\geq 2$ be an integer, and let n be a sufficiently large multiple of k. Let ... be a collection of graphs on a common vertex set of size n. Write $\delta(\bm{G})$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-051685 | MATHDB-362336

Apex-outerplanar minor conjecture

摘录 / EXCERPT: Let H be an outerplanar graph, and let H^+ denote the graph obtained from H by adding an apex vertex adjacent to every vertex of H. Apex-outerplanar minor conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-051686 | MATHDB-362337

Diwan's subdivision conjecture for planar maximal 3-degenerate graphs

摘录 / EXCERPT: Let H be a planar maximal 3-degenerate graph. A subdivision of H is a graph obtained by replacing edges by internally vertex-disjoint paths. Diwan's conjecture. Every graph wit...

conjecture | open | MathDB 页面 | 原文线索

V2-051687 | MATHDB-362338

Common minimum in- and out-degree conjecture for cycle orientations

摘录 / EXCERPT: Let t be a positive integer. An orientation of the cycle on t vertices is obtained by assigning a direction to each edge of that cycle. Common degree conjecture. Every digraph...

conjecture | open | MathDB 页面 | 原文线索

V2-051688 | MATHDB-362345

Tie-breaking independence conjecture for scale-free cascading failures

摘录 / EXCERPT: Let G be a simple, connected graph, and let an exceedance tie-breaking rule T be used in the cascading-failure process described by Theorem. Tie-breaking independence conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-051689 | MATHDB-362370

The universal-cycle conjecture for pairwise balanced designs

摘录 / EXCERPT: Let a (v,K,λ)-pairwise balanced design be a v-vertex hypergraph whose 2-degree is λ and whose edge cardinalities belong to a set of integers K. A base block i...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-051690 | MATHDB-362378

Aichholzer-Orthaber-Vogtenhuber's plane Hamiltonian-connectedness conjecture

摘录 / EXCERPT: Aichholzer-Orthaber-Vogtenhuber's conjecture. For each pair of vertices s,t in a simple drawing of K_n, there exists a plane Hamiltonian path from s to t.

conjecture | open | MathDB 页面 | 原文线索

V2-051691 | MATHDB-362379

The $2n-3$-edge plane Hamiltonian subdrawing conjecture

摘录 / EXCERPT: $2n-3$-edge conjecture. Every simple drawing of K_n with $n\geq 3$ contains a plane Hamiltonian subdrawing on $2n-3$ edges.

conjecture | open | MathDB 页面 | 原文线索

V2-051692 | MATHDB-362380

Extremal partition conjecture for generalized Ramsey-Turán clique density

摘录 / EXCERPT: Let s and t be integers with $3\leq s\leq t-2$. An extremal graph for the generalized Ramsey-Turán density $\varrho_s(K_t)$ is a graph admitting either an $(s,t-1-s)$-partition...

conjecture | open | MathDB 页面 | 原文线索

V2-051693 | MATHDB-362381

Weighted-graph reformulation of the extremal partition conjecture

摘录 / EXCERPT: Fix integers s and t with $3\leq s\leq t-2$. A weighted graph admitting a $(b,t-1-b)$-partition has b vertices and $t-1-b$ parts satisfying the defining partition conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-051694 | MATHDB-362384

The identification number bound for connected identifiable graphs

摘录 / EXCERPT: Let G be a connected identifiable graph of order $n\ge 2$ and maximum degree $\Delta\ge 2$. Write $\gamma^{{\rm ID}}(G)$ for the minimum size of an identifying code of G. Ident...

conjecture | open | MathDB 页面 | 原文线索

V2-051695 | MATHDB-362386

Extremal degree-based weighted adjacency conjecture for functions with property P^{**}

摘录 / EXCERPT: Let $f(x,y)>0$ be a symmetric real function that is increasing and convex in x, and suppose that whenever $x_1+y_1=x_2+y_2$ and $|x_1-y_1|>|x_2-y_2|$, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-051696 | MATHDB-362427

Grzesik-Lee-Lidický-Volec conjecture on apices of Sidorenko graphs

摘录 / EXCERPT: Let G be a connected graph with the Sidorenko property. For an integer k, the k-apex of G is obtained by adding k independent vertices, each adjacent to every vertex of...

conjecture | open | MathDB 页面 | 原文线索

V2-051697 | MATHDB-362430

Kromatic symmetric function distinguishes all graphs

摘录 / EXCERPT: Kromatic distinguishing conjecture. There do not exist nonisomorphic graphs G and H with

conjecture | open | MathDB 页面 | 原文线索

V2-051698 | MATHDB-362434

Transversal-matroid characterization of minimally angle-rigid colored graphs

摘录 / EXCERPT: Let (G,c) be a colored graph, let $T_{c,j}(G)$ denote the family of transversal edge sets associated with color j, and let H be a minimally rigid graph. Transversal character...

conjecture | open | MathDB 页面 | 原文线索

V2-051699 | MATHDB-362435

Equality of unlabelled point-determining and endpoint-free graphs

摘录 / EXCERPT: Point-determining graph conjecture. The number of unlabelled point-determining graphs of a given order equals the number of unlabelled endpoint-free graphs of that order.

conjecture | open | MathDB 页面 | 原文线索

V2-051700 | MATHDB-362436

Maximum diameter of redundant non-surrounded graphs

摘录 / EXCERPT: Maximum-diameter conjecture. The maximum diameter of such a graph is

conjecture | open | MathDB 页面 | 原文线索

V2-051701 | MATHDB-362437

Partial-short reduction conjecture for weak graphs

摘录 / EXCERPT: Partial-short reduction conjecture. Starting with a weak graph, identify vertices having the least number of degree-2 neighbours, then among those select vertices of least degree...

conjecture | open | MathDB 页面 | 原文线索

V2-051702 | MATHDB-362440

Brinkmann, Preissmann and Sasaki's large-girth Type 2 conjecture

摘录 / EXCERPT: Let G be a cubic graph, meaning that every vertex has degree 3, and let its girth be the length of its shortest cycle. A cubic graph is Type 2 when its total chromatic number i...

conjecture | open | MathDB 页面 | 原文线索

V2-051703 | MATHDB-362442

The local antimagic chromatic number of joins of disjoint paths with a universal vertex

摘录 / EXCERPT: Let P_n be the path on n vertices, let mP_n denote the disjoint union of m copies of P_n, let K_1 be a single vertex, and let $\chi_{la}(G)$ denote the local antimagic...

conjecture | open | MathDB 页面 | 原文线索

V2-051704 | MATHDB-362468

Akbari et al.'s Caro-Wei bound for induced linear forests

摘录 / EXCERPT: All graphs are finite, simple, and undirected; for a graph G, let $V(G)$ be its vertex set and let $d(v)$ denote the degree of v. A linear forest is a forest in which every com...

conjecture | open | MathDB 页面 | 原文线索

V2-051705 | MATHDB-362516

The traceability conjecture for k-traceable oriented graphs

摘录 / EXCERPT: Traceability conjecture. For every integer $k\geq 2$, every k-traceable oriented graph of order at least $2k-1$ is traceable.

conjecture | open | MathDB 页面 | 原文线索

V2-051706 | MATHDB-362520

Leading-coefficient conjecture for the q-chromatic polynomial of trees

摘录 / EXCERPT: Let G be a finite tree, and let $\widetilde{\chi}_G^{\mathbf{1}}(q,x)$ denote the q-chromatic polynomial of G with respect to the all-ones weight vector $\mathbf{1}$. Leading...

conjecture | open | MathDB 页面 | 原文线索

V2-051707 | MATHDB-362525

Hajebi-Li-Spirkl conjecture for graphs without large induced matchings

摘录 / EXCERPT: For a graph G, let $h(G)$ be the smallest size of a subset of $V(G)$ intersecting every maximum independent set, and let $\omega(G)$ denote its clique number. An induced matching...

conjecture | open | MathDB 页面 | 原文线索

V2-051708 | MATHDB-362569

Butler et al.'s complement coalescing conjecture for distance matrices

摘录 / EXCERPT: Let G_1 and G_2 be graphs with $B_1\subseteq V(G_1)$ and $B_2\subseteq V(G_2)$. Coalescing conjecture. If coalescing the same connected rooted graph onto every vertex of B_1...

conjecture | open | MathDB 页面 | 原文线索

V2-051709 | MATHDB-362575

Low-complexity encoding conjecture for hypergraph Ramsey colorings

摘录 / EXCERPT: A pair construction of a 3-graph coloring is a coloring of $K_N^{(3)}$ obtained from functions $f:\binom{[N]}{2}\to[p]$ and $g:[p]^3\to\{\operatorname{red},\operatorname{blue}\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-051710 | MATHDB-362576

Linear-exponential growth conjecture for off-diagonal hypergraph Ramsey numbers

摘录 / EXCERPT: For a fixed 3-graph H, write $r(H,K_n^{(3)})$ for the least N such that every red-blue coloring of the edges of $K_N^{(3)}$ contains a red copy of H or a blue copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-051711 | MATHDB-362577

Polynomial-exponent upper-bound conjecture for linear hypergraph Ramsey numbers

摘录 / EXCERPT: A 3-graph F is linear if any two of its edges share at most one vertex. For fixed F, let $r(F,K_n^{(3)})$ be the least N such that every red-blue coloring of $K_N^{(3)}$ co...

conjecture | open | MathDB 页面 | 原文线索

V2-051712 | MATHDB-362578

Tower-growth conjecture for hereditary 3-uniform hypergraph properties

摘录 / EXCERPT: Let $\mathcal{H}$ be a hereditary property of 3-uniform hypergraphs with infinite VC_2-dimension. Let $L_{\mathcal{H}}(\epsilon_1,\epsilon_2)$ denote the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-051713 | MATHDB-362603

Linear-threshold conjecture for connected size Ramsey numbers of star matchings

摘录 / EXCERPT: Let $nK_{1,p}$ be the disjoint union of n copies of the star $K_{1,p}$, and let $K_{1,m}$ be the star with m edges. The connected size Ramsey number $\hat r_c(G_1,G_2)$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-051714 | MATHDB-362607

Li-Lu-Peng conjecture for spectral extremal F_k-free graphs

摘录 / EXCERPT: Li-Lu-Peng conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051715 | MATHDB-362608

Fan-free spectral extremal conjecture for odd and even fans

摘录 / EXCERPT: Fan-free spectral extremal conjecture. If G is a V_{2k+1}-free or V_{2k+2}-free graph with m edges, then

conjecture | open | MathDB 页面 | 原文线索

V2-051716 | MATHDB-362612

Exact depth-three bound for two-colored random tournaments

摘录 / EXCERPT: Let $f^*(2,3,n)$ denote the random variable measuring the largest number of vertices that can be reached from a vertex by monochromatic directed paths of depth at most 3 in a two...

conjecture | open | MathDB 页面 | 原文线索

V2-051717 | MATHDB-362702

The Small Quasikernel Conjecture

摘录 / EXCERPT: Small Quasikernel Conjecture. If D is a source-free digraph, then D contains a quasikernel Q with

conjecture | open | MathDB 页面 | 原文线索

V2-051718 | MATHDB-362703

The Large Quasikernel Conjecture

摘录 / EXCERPT: Large Quasikernel Conjecture. Every digraph D contains a quasikernel Q with

conjecture | open | MathDB 页面 | 原文线索

V2-051719 | MATHDB-362704

The subexponential bound conjecture for disjoint q-kernels

摘录 / EXCERPT: Subexponential disjoint q-kernel conjecture. There exist constants $\varepsilon>0$ and r_0 such that, whenever $r\ge r_0$ and D contains no $(r-1)$-source sets, D contains...

conjecture | open | MathDB 页面 | 原文线索

V2-051720 | MATHDB-362705

The minimum in-degree conjecture for small quasikernels

摘录 / EXCERPT: Minimum in-degree quasikernel conjecture. There exists an integer $\delta\ge 2$ such that every digraph D with $\delta^-(D)\ge\delta$ contains a quasikernel Q with

conjecture | open | MathDB 页面 | 原文线索

V2-051721 | MATHDB-362706

The large-quasikernel structural conjecture

摘录 / EXCERPT: Large-quasikernel structural conjecture. Every digraph D either contains a subset $A\subseteq V(D)$ such that $D[A]$ is acyclic and

conjecture | open | MathDB 页面 | 原文线索

V2-051722 | MATHDB-362763

Alon et al.'s additivity conjecture for joins of low-inversion graphs

摘录 / EXCERPT: Let $k\in\mathbb{N}$, and let $D_1,\ldots,D_k$ be oriented graphs. The k-join $[D_1,\ldots,D_k]$ is formed from vertex-disjoint copies of these graphs by adding every edge from...

conjecture | open | MathDB 页面 | 原文线索

V2-051723 | MATHDB-362766

Addario-Berry-Havet-Linhares Sales-Reed-Thomassé conjecture for antidirected trees

摘录 / EXCERPT: Let D be a digraph, let k be a positive integer, and let T be an antidirected tree with k arcs. Addario-Berry-Havet-Linhares Sales-Reed-Thomassé conjecture. If D has more...

conjecture | open | MathDB 页面 | 原文线索

V2-051724 | MATHDB-362810

Harary-Hill conjecture for the crossing number of complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and define ... Here $\operatorname{cr}(G)$ denotes the minimum number of crossings in a plane drawing of a graph G. Harary-Hill con...

conjecture | open | MathDB 页面 | 原文线索

V2-051725 | MATHDB-362811

Zarankiewicz's conjecture for complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be the complete bipartite graph with part sizes m and n, and define ... Here $\operatorname{cr}(G)$ denotes the minimum number of crossings in a plane drawing of a...

conjecture | open | MathDB 页面 | 原文线索

V2-051726 | MATHDB-362816

Derangement representation bound for finite graphs

摘录 / EXCERPT: Derangement representation conjecture. For every finite graph G except the graphs in $\mathcal{F}$,

conjecture | open | MathDB 页面 | 原文线索

V2-051727 | MATHDB-362829

Bang-Jensen et al.'s dijoin conjecture for inversion number

摘录 / EXCERPT: Bang-Jensen et al.'s dijoin conjecture. For oriented graphs L and R,

conjecture | open | MathDB 页面 | 原文线索

V2-051728 | MATHDB-362830

Aubian et al.'s universal counterexample conjecture for dijoin inversion

摘录 / EXCERPT: Aubian et al.'s conjecture. For every integer $k\ge 3$, there exists a tournament T_k with

conjecture | open | MathDB 页面 | 原文线索

V2-051729 | MATHDB-362831

The direction-independence conjecture for dijoin inversion number

摘录 / EXCERPT: Direction-independence conjecture. For every oriented graphs L and R,

conjecture | open | MathDB 页面 | 原文线索

V2-051730 | MATHDB-362836

Faught-Nordhaus-Gaddum spectral-gap sum conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\overline{G}$ denote its complement. Assume that both G and $\overline{G}$ are connected; $\operatorname{Gap}(G)$ denotes the differe...

conjecture | open | MathDB 页面 | 原文线索

V2-051731 | MATHDB-362865

The W-algebra conformal-block conjecture for multi-leg PT vertices

摘录 / EXCERPT: Let $W_{q_{1,2,3,4}}(\widehat{A}_{0})$ be the W-algebra associated with the quantum parameters $q_{1,2,3,4}$, and consider conformal blocks of multi-screened vertex operators. The...

conjecture | open | MathDB 页面 | 原文线索

V2-051732 | MATHDB-362873

The edge-isoperimetric conjecture for the permutahedron at size $6^{(n+1)/3}$

摘录 / EXCERPT: Edge-isoperimetric conjecture for the permutahedron.

conjecture | open | MathDB 页面 | 原文线索

V2-051733 | MATHDB-362874

The inverse-polynomial Cheeger conjecture for zonotopes

摘录 / EXCERPT: Let Z be a zonotope of the form ... and let G be its 1-skeleton. Let $i(G)$ denote the Cheeger constant, or edge-expansion, of G. Inverse-polynomial Cheeger conjecture for zo...

conjecture | open | MathDB 页面 | 原文线索

V2-051734 | MATHDB-362884

The positive graph conjecture

摘录 / EXCERPT: The positive graph conjecture. A graph H is positive if and only if it has a stable involution.

conjecture | open | MathDB 页面 | 原文线索

V2-051735 | MATHDB-362891

Nonintegrality conjecture for elliptic Sombor energy

摘录 / EXCERPT: For a graph G, let its elliptic Sombor energy be the sum of the absolute values of the eigenvalues of its elliptic Sombor matrix. Nonintegrality conjecture. There is no graph wit...

conjecture | open | MathDB 页面 | 原文线索

V2-051736 | MATHDB-362912

The lower general position product bound

摘录 / EXCERPT: The lower general position product bound.

conjecture | open | MathDB 页面 | 原文线索

V2-051737 | MATHDB-362913

The terminal set existence conjecture

摘录 / EXCERPT: Terminal set existence conjecture. Every graph has a terminal set.

conjecture | open | MathDB 页面 | 原文线索

V2-051738 | MATHDB-362937

The e-positivity conjecture for theta graphs

摘录 / EXCERPT: For any integers $a\ge b\ge c\ge 1$, let θ_{abc} be the theta graph formed from two distinct vertices joined by three pairwise disjoint paths of lengths a, b, and c. A...

conjecture | open | MathDB 页面 | 原文线索

V2-051739 | MATHDB-362939

Boyer et al.'s zero forcing set domination conjecture for paths

摘录 / EXCERPT: Boyer et al.'s conjecture. For every graph G on n vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-051740 | MATHDB-362946

Balogh-et al. extremal-construction conjecture for generalized Ramsey-Turán numbers

摘录 / EXCERPT: Let $p>q\geq 3$ be integers, and let $\mathcal{G}(n;s,t)$ denote the class of constructions with parameters s and t used for the generalized Ramsey-Turán problem. Balogh-et al....

conjecture | refuted | MathDB 页面 | 原文线索

V2-051741 | MATHDB-362951

The odd signable graph conjecture for induced $K_{3,3}$ minors

摘录 / EXCERPT: Odd signable graph conjecture. If G is an odd signable graph (in particular, if G is an even-hole-free graph), then G does not contain $K_{3,3}$ as an induced minor.

conjecture | open | MathDB 页面 | 原文线索

V2-051742 | MATHDB-362953

Dong's vertex-isolation conjecture for mean color numbers

摘录 / EXCERPT: Let G be a graph and let w be a vertex of G. Write $G-w$ for the graph obtained by deleting w, and let K_1 be the one-vertex graph. Dong's vertex-isolation conjecture. On...

conjecture | open | MathDB 页面 | 原文线索

V2-051743 | MATHDB-362954

Dong's one-edge-at-a-vertex conjecture for mean color numbers

摘录 / EXCERPT: Let G be a graph, let w be a vertex of G with $d(w)\geq 1$, and let H be obtained from G by deleting all but one of the edges incident to w. Dong's one-edge-at-a-vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-051744 | MATHDB-362965

Uniqueness conjecture for (t,λ)-liking digraphs

摘录 / EXCERPT: Let $t\geq 3$ and let a (t,λ)-liking digraph be a digraph in which every set of t vertices has exactly λ common out-neighbors. A complete digraph is a digraph hav...

conjecture | open | MathDB 页面 | 原文线索

V2-051745 | MATHDB-363019

Conjecture that regular Bonnet-Myers sharp graphs are self-centered

摘录 / EXCERPT: Let G be a regular Bonnet-Myers sharp graph, and suppose the classification theorem for Bonnet-Myers sharp graphs assumes that G is self-centered. Regular-graph conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051746 | MATHDB-363022

Aliniaeifard, van Willigenburg and Wang's spider e-positivity conjecture

摘录 / EXCERPT: For a partition $ambda=ambda_1\dotsmambda_d$ of $n-1$ with $d\ge 3$, the spider $S(\lambda)$ is the n-vertex tree consisting of paths of lengths $\lambda_1,\dots,\lambda_d$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-051747 | MATHDB-363027

The odd balanced complete bipartite graph conjecture for Trail Trap

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with equal parts of size n, and let ${\rm P}{2}$ denote the second player in Trail Trap. A graph is ${\rm P}{2}$-win if the second p...

conjecture | open | MathDB 页面 | 原文线索

V2-051748 | MATHDB-363028

The larger grid graph conjecture for Trail Trap

摘录 / EXCERPT: Let P_m and P_n be paths on m and n vertices, respectively, and let $P_m\square P_n$ be their Cartesian product, the (m,n) grid graph. Let ${\rm P}{2}$ denote the second...

conjecture | open | MathDB 页面 | 原文线索

V2-051749 | MATHDB-363029

The PSPACE-completeness conjecture for Trail Trap

摘录 / EXCERPT: Consider the decision problem ${\sc Trail Trap}\ {\rm P}{2}\text{-win}$: given a graph, determine whether the second player has a winning strategy in Trail Trap. Trail Trap PSPACE-...

conjecture | open | MathDB 页面 | 原文线索

V2-051750 | MATHDB-363030

The asymptotic rarity conjecture for first-player wins in Trail Trap

摘录 / EXCERPT: For each n, consider the collection of graphs on n vertices and the subset consisting of graphs on which the first player has a winning strategy in Trail Trap. Asymptotic rarit...

conjecture | open | MathDB 页面 | 原文线索

V2-051751 | MATHDB-363036

Kneser's conjecture on the chromatic number of the Kneser graph

摘录 / EXCERPT: Let $[n]$ be a set with n elements, let k satisfy $n\geq 2k-1$, and let $\binom{[n]}{k}$ denote the family of all k-element subsets of $[n]$. A proper coloring assigns colors...

conjecture | open | MathDB 页面 | 原文线索

V2-051752 | MATHDB-363043

Sparse-host induced subdivision extremal conjecture

摘录 / EXCERPT: Let H be a graph, and let $\mathcal{D}_H$ be the property of not containing a subdivision of H as an induced subgraph. For a graph Γ, write...

conjecture | open | MathDB 页面 | 原文线索

V2-051753 | MATHDB-363044

Regularity conjecture for finite-state string machines without a meta-vertex

摘录 / EXCERPT: Let a string machine be composed only of finite-state transducers, with the input and output categories of each transducer being tape categories. Suppose it has one free input...

conjecture | open | MathDB 页面 | 原文线索

V2-051754 | MATHDB-363045

Kainen's conjecture for complete graphs in arbitrary surfaces

摘录 / EXCERPT: For each nonnegative integer $g\neq 2$, let ... and let S_g denote the orientable surface of genus g. A Kainen drawing is a drawing attaining Kainen's lower bound for the surface...

conjecture | open | MathDB 页面 | 原文线索

V2-051755 | MATHDB-363083

Extremal construction conjecture for short cycles in triangle-free planar graphs

摘录 / EXCERPT: Let n be sufficiently large and $k\geq 2$. For even cycles, let ${\mathcal G}_{\mathrm{even}}$ be the balanced blow-up construction described in the paper. For odd cycles, let...

conjecture | open | MathDB 页面 | 原文线索

V2-051756 | MATHDB-363129

The quasi-center conjecture for zigzag 6-cycles in type D_n Artin complexes

摘录 / EXCERPT: Let Λ be the Dynkin diagram of type D_n, and let Δ_Λ be its associated Artin complex. Order the vertex types by...

conjecture | open | MathDB 页面 | 原文线索

V2-051757 | MATHDB-363187

The Local Irregularity Conjecture excluding the bow-tie graph

摘录 / EXCERPT: Local Irregularity Conjecture. If $G\neq B$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051758 | MATHDB-363188

The (2,2)-Conjecture for graph decompositions

摘录 / EXCERPT: (2,2)-Conjecture. The graph G can be decomposed into two subgraphs G_1 and G_2 such that there exist locally irregular multigraphs M_1 and M_2 satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-051759 | MATHDB-363190

Conjecture on chromatic-number fluctuations from isolated vertices

摘录 / EXCERPT: Isolated-vertex fluctuation conjecture. The typical deviation of $\chi(G(n,p))$ from its expectation is of order

conjecture | open | MathDB 页面 | 原文线索

V2-051760 | MATHDB-363215

Ansensio-García-Marco-Knauer conjecture on induced degree-one subsets of the ternary Hamming graph

摘录 / EXCERPT: Ansensio-García-Marco-Knauer conjecture. All such subsets satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-051761 | MATHDB-363221

The v-number conjecture for cycles and binary trees

摘录 / EXCERPT: Conjecture for cycles and binary trees. The following equalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-051762 | MATHDB-363222

Liu-Montgomery's crux conjecture for clique subdivisions

摘录 / EXCERPT: Liu-Montgomery's crux conjecture. There exists some constant $c>0$ such that every graph G contains a subdivision of a clique with at least

conjecture | open | MathDB 页面 | 原文线索

V2-051763 | MATHDB-363223

Akbari et al.'s energy-increasing self-loop conjecture

摘录 / EXCERPT: Akbari et al.'s conjecture. There exists a subset $S\subseteq V(G)$ such that

conjecture | refuted | MathDB 页面 | 原文线索

V2-051764 | MATHDB-363235

Regularity conjecture for t-connected ideals of trees

摘录 / EXCERPT: Let T be a tree, let t be a positive integer, and let $J_t(T)$ denote the t-connected ideal of T. Let $\mathcal H$ be the hypergraph corresponding to $J_t(T)$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-051765 | MATHDB-363236

Projective-dimension conjecture for t-connected ideals of trees

摘录 / EXCERPT: Let T be a tree, let t be a positive integer, and let $J_t(T)$ denote the t-connected ideal of T. Let $\operatorname{bight}(J_t(T))$ denote the big height of this ideal. Pr...

conjecture | open | MathDB 页面 | 原文线索

V2-051766 | MATHDB-363238

Hopf-bifurcation invariance under removing a reverse reaction

摘录 / EXCERPT: Let G be a reaction network with n species containing a motif ... such that the species Y does not appear in any other complex of the network. Let G' be obtained from G by...

conjecture | open | MathDB 页面 | 原文线索

V2-051767 | MATHDB-363239

Purely imaginary roots under reverse-reaction removal

摘录 / EXCERPT: Let G be a reaction network containing the motif ... where Y appears in no other complex. Let $p_{(h,\ell)}$ and $p'_{(h,\ell')}$ be the reduced characteristic polynomials of G...

conjecture | open | MathDB 页面 | 原文线索

V2-051768 | MATHDB-363244

Grünbaum's torus conjecture for cubic graph colorings

摘录 / EXCERPT: Let G be a 2-connected cubic graph embedded in the torus. The graph G is 3-edge-colorable if its edges can be colored with three colors so that adjacent edges receive differe...

conjecture | open | MathDB 页面 | 原文线索

V2-051769 | MATHDB-363259

The second-largest 1-nearly independent edge-subset count

摘录 / EXCERPT: Let F be a forest of order $n\geq 12$, let P_n denote the path on n vertices, and let $[P_3,P_3,P_{n-7}]$ denote the tree formed by joining one endpoint of each of three path...

conjecture | open | MathDB 页面 | 原文线索

V2-051770 | MATHDB-363284

The location-domination conjecture for twin-free graphs

摘录 / EXCERPT: Location-domination conjecture. If G is a twin-free graph on n vertices, then

conjecture | open | MathDB 页面 | 原文线索

V2-051771 | MATHDB-363286

The induced even-cycle-free Bollobás-Erdős-Tuza conjecture

摘录 / EXCERPT: Induced even-cycle-free conjecture. If $\alpha(G)=\Omega(n)$, then $h(G)=o(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-051772 | MATHDB-363294

Kahn's independent-set bound for regular graphs

摘录 / EXCERPT: Let G be a d-regular graph with n vertices, and let $i(G)$ denote its number of independent sets. Alon and Kahn's conjecture. One has ... This formalized the proposed extremal...

conjecture | open | MathDB 页面 | 原文线索

V2-051773 | MATHDB-363306

Chromatic invariance conjecture for generalized Sierpiński gasket graphs

摘录 / EXCERPT: Let G be a simple graph, let t be a positive integer, and let $S[G,t]$ denote the generalized Sierpiński gasket graph based on G at level t. Write $\chi(G)$ for the chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-051774 | MATHDB-363317

Bounded chromatic number for bounded-degree color classes with no lonely colors

摘录 / EXCERPT: No-lonely-color conjecture. There is a function f such that, whenever no color is lonely on any cycle of G, one has

conjecture | open | MathDB 页面 | 原文线索

V2-051775 | MATHDB-363337

Divisor class group conjecture for Lovász-Saks-Schrijver rings of forests

摘录 / EXCERPT: Divisor class group conjecture. The divisor class group of $R_G(d)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-051776 | MATHDB-363351

Federici's conjecture on finite geodetic Cayley graphs

摘录 / EXCERPT: Let G be a finite group and let Σ be a generating set. The undirected Cayley graph $\operatorname{Cay}(G,\Sigma)$ has vertex set G and edges joining g to ga for...

conjecture | open | MathDB 页面 | 原文线索

V2-051777 | MATHDB-363366

The e-positivity conjecture for hat-chains

摘录 / EXCERPT: Hat-chain e-positivity conjecture. Every hat-chain is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-051778 | MATHDB-363373

Linusson's bunkbed conjecture for hypergraph percolation

摘录 / EXCERPT: Let $\mathcal{H}$ be a hypergraph and let $T\subseteq V(\mathcal{H})$ be a set of posts. In the conditioned model $E_4^T(\widetilde{\mathcal{H}})$, precisely one of the two copies...

conjecture | open | MathDB 页面 | 原文线索

V2-051779 | MATHDB-363374

The simple-digraph counterexample conjecture for the fully directed bunkbed model

摘录 / EXCERPT: Let D be a simple digraph. Construct $\widetilde{D}$ by adding both directed vertical edges $v^{(0)}v^{(1)}$ and $v^{(1)}v^{(0)}$ for every $v\in V(D)$, and let E_8 select a su...

conjecture | open | MathDB 页面 | 原文线索

V2-051780 | MATHDB-363379

The first- and second-degree bound for Snort temperature

摘录 / EXCERPT: Let G be a graph. In the impartial graph game Snort, let $t(G)$ denote the temperature of the position on G, let $\deg(G)$ be the maximum degree of G, and let $\deg_2(G)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-051781 | MATHDB-363391

Plummer-Stiebitz-Toft reformulation for independence number two

摘录 / EXCERPT: Plummer-Stiebitz-Toft conjecture. Every n-vertex graph G with $\alpha(G)\leq2$ contains $K_{\lceil n/2\rceil}$ as a minor.

conjecture | open | MathDB 页面 | 原文线索

V2-051782 | MATHDB-363392

Botler et al.'s bipartite minor conjecture for independence number two

摘录 / EXCERPT: Botler et al.'s conjecture. Let G be an n-vertex graph with $\alpha(G)\leq2$. For any positive integer ℓ with $\ell<\lceil n/2\rceil$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-051783 | MATHDB-363404

The consensus-time conjecture for evolving scale-free networks

摘录 / EXCERPT: Consider the evolving scale-free network and the voter model in the same setup as Theorem, with arbitrary update rate κ. Let T_{cons} denote the consensus time and...

conjecture | open | MathDB 页面 | 原文线索

V2-051784 | MATHDB-363414

Cycle-spectrum conjecture for the associahedron

摘录 / EXCERPT: Let $\cG_N$ be the associahedron graph, let $S(\cG_N)$ be the set of lengths of cycles in $\cG_N$, and let $\cD_{N,3}$ denote the comparison set used in the source. Cycle-spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-051785 | MATHDB-363426

Asymptotic Erdős-Hajnal conjecture for forbidden order-size pairs

摘录 / EXCERPT: Let G be an n-vertex r-graph, let $s(G;m)$ denote the number of values of f for which G contains m vertices spanning exactly f edges, and let $g_r(m)$ be the threshol...

conjecture | open | MathDB 页面 | 原文线索

V2-051786 | MATHDB-363438

Generalized forbidden out-degree conjecture

摘录 / EXCERPT: Let G be a loopless graph, and let $d(v)$ denote the degree of each vertex $v\in V(G)$. Given a function $F:V(G)\to 2^{\mathbb N}$, an orientation is F-avoiding if its out-degr...

conjecture | open | MathDB 页面 | 原文线索

V2-051787 | MATHDB-363447

Insensitivity of the minimum eigenvalue under edge addition at the optimal path-graph port

摘录 / EXCERPT: Let n be odd, and let P_n be a path graph with Laplacian matrix L_n. For 1-port selection, perturb L_n to ... where $p^*=(n+1)/2$, and denote the resulting optimal perturbed...

conjecture | open | MathDB 页面 | 原文线索

V2-051788 | MATHDB-363450

The spectral Mubayi conjecture for color-critical graphs

摘录 / EXCERPT: Let F be a color-critical graph with chromatic number $\chi(F)=r+1\ge 3$. Let $T_{n,r}$ be Turán's graph, let $c(n,F)$ be the minimum number of copies of F obtained by adding o...

conjecture | open | MathDB 页面 | 原文线索

V2-051789 | MATHDB-363455

Ghosh-Győri-Paulos-Xiao conjecture on the planar Turán number of the balanced double star S_{3,3}

摘录 / EXCERPT: A balanced double star $S_{m,m}$ is the graph obtained from an edge whose endpoints are joined respectively to m and m additional distinct vertices. For a planar graph family,...

conjecture | open | MathDB 页面 | 原文线索

V2-051790 | MATHDB-363458

Jayanthan-A R conjecture on linear type binomial edge ideals

摘录 / EXCERPT: Let G be a graph, let $S=\mathbb{K}[x_1,\dots,x_n,y_1,\dots,y_n]$, and let ... be its binomial edge ideal. An ideal $I\subset A$ is of linear type when its symmetric algebra is nat...

conjecture | open | MathDB 页面 | 原文线索

V2-051791 | MATHDB-363469

Conjecture on the rarity of graphs close to graphs with nontrivial two-fold automorphisms

摘录 / EXCERPT: Let n be a positive integer, let $\mathcal{P}_n(\varepsilon)$ be the probability that a random labeled graph Γ on n vertices can be transformed into a graph Γ' wi...

conjecture | open | MathDB 页面 | 原文线索

V2-051792 | MATHDB-363470

Conjecture on the expected size of the two-fold automorphism group of a random graph

摘录 / EXCERPT: Let Γ be a random labeled graph on n vertices, let $\operatorname{Aut}^{\pi}(\Gamma)$ denote its two-fold automorphism group, and let $\mathbb{E}$ denote expectation over...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-051793 | MATHDB-363473

Kahn's matching-uncovered probability conjecture for regular linear hypergraphs

摘录 / EXCERPT: Let H be a d-regular linear k-graph, and let M be a matching chosen uniformly from the set of all matchings of H. For a vertex v, write $\mathcal{P}_H(\overline{v})$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-051794 | MATHDB-363474

Kahn-Kim variance conjecture for random matchings in regular linear hypergraphs

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let H be an n-vertex d-regular linear k-graph, and let M be a matching chosen uniformly from the set of all matchings of H. For a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-051795 | MATHDB-363475

Asymptotic uncovered-vertex probability for dense almost-regular hypergraphs

摘录 / EXCERPT: Let $k\geq 3$ be a positive integer and let $0<\delta,\varepsilon<1$ be positive real numbers. There is an n_0 such that, for every $n\geq n_0$, if H is an n-vertex k-graph...

conjecture | open | MathDB 页面 | 原文线索

V2-051796 | MATHDB-363504

The coarse Menger-type conjecture for asymptotic minors

摘录 / EXCERPT: Coarse Menger-type conjecture. If a graph is not coarsely bottlenecked, it must contain D_n as an asymptotic minor for every $n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-051797 | MATHDB-363505

Quasi n-bottlenecking conjecture

摘录 / EXCERPT: Quasi n-bottlenecking conjecture. If G is coarsely n-bottlenecked, then it is quasi-isometric to an n-edge bottlenecked graph.

conjecture | open | MathDB 页面 | 原文线索

V2-051798 | MATHDB-363507

Thomassen's chord conjecture for longest cycles in 3-connected graphs

摘录 / EXCERPT: A graph is 3-connected if deleting fewer than three vertices leaves it connected. A chord of a cycle is an edge joining two nonconsecutive vertices of the cycle. Thomassen's chord...

conjecture | open | MathDB 页面 | 原文线索

V2-051799 | MATHDB-363531

Faudree-Lehel conjecture on the irregularity strength of regular graphs

摘录 / EXCERPT: Let G be a d-regular graph with n vertices, where $d\geq 2$. Its irregularity strength $s(G)$ is the least positive integer k for which there is a weighting...

conjecture | open | MathDB 页面 | 原文线索

V2-051800 | MATHDB-363540

Petruševski-Škrekovski edge-deletion conjecture for odd edge-colorings

摘录 / EXCERPT: Let G be a connected graph that is not a Shannon triangle of type $(2,2,2)$ or type $(2,2,1)$. Write $\chi_{\text{odd}}'(G)$ for the odd chromatic index, and call an edge special...

conjecture | open | MathDB 页面 | 原文线索

V2-051801 | MATHDB-363549

Finite forbidden-subgraph characterization of the class $\mathfrak{D}_4$

摘录 / EXCERPT: Finite forbidden-subgraph conjecture for $\mathfrak{D}_4$. There exists a finite family $\mathscr{F}$ of graphs such that $\mathfrak{D}_4$ is the class of maximal triangle-free gra...

conjecture | open | MathDB 页面 | 原文线索

V2-051802 | MATHDB-363572

Jamison and Sprague's threshold-number conjecture for graphs and their complements

摘录 / EXCERPT: Jamison and Sprague's conjecture. For every integer $k\ge 1$, there is a graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-051803 | MATHDB-363577

Kohayakawa-Nagle-Rödl-Schacht forcing conjecture

摘录 / EXCERPT: Kohayakawa-Nagle-Rödl-Schacht forcing conjecture. If H is not a forest, then H is KNRS-forcing.

conjecture | open | MathDB 页面 | 原文线索

V2-051804 | MATHDB-363594

Füredi-Maleki structural conjecture for minimizing triangular edges

摘录 / EXCERPT: Let $m>n^2/4$ and let G be an n-vertex graph with m edges that minimizes the number of triangular edges. For integers a,b,c, let $G(a,b,c)$ be the graph consisting of a cli...

conjecture | open | MathDB 页面 | 原文线索

V2-051805 | MATHDB-363595

Spectral supersaturation-stability conjecture for bipartiteness

摘录 / EXCERPT: Let G be a graph on n vertices, let $t>0$, and let $t(G)$ denote the number of triangles. A graph is t-far from being bipartite if at least t edges must be removed to make...

conjecture | open | MathDB 页面 | 原文线索

V2-051806 | MATHDB-363596

Spectral triangular-edge counting conjecture

摘录 / EXCERPT: Let G be a graph with m edges, let $\lambda(G)$ denote its spectral radius, and let $t(G)$ denote the number of triangular edges. Spectral triangular-edge counting conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051807 | MATHDB-363597

Spectral extremal conjecture for the graph family G(a,b,c)

摘录 / EXCERPT: For integers a,b,c, let $G(a,b,c)$ be the graph on $a+b+c$ vertices consisting of a clique of size a and independent sets of sizes b and c, with all edges between the set o...

conjecture | open | MathDB 页面 | 原文线索

V2-051808 | MATHDB-363613

Aharoni's transversal Dirac conjecture

摘录 / EXCERPT: Let $\mathcal{G} = \{G_1, \ldots, G_n\}$ be a collection of graphs with common vertex set V of size n. Define $\delta(\mathcal{G})=\min\{\delta(G_i):1\leq i\leq n\}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-051809 | MATHDB-363618

The treewidth formula for the fall-off value of graph polytopes

摘录 / EXCERPT: Treewidth fall-off conjecture. The fall-off value is

conjecture | open | MathDB 页面 | 原文线索

V2-051810 | MATHDB-363626

Haxell-Łuczak-Peng-Rödl-Ruciński-Skokan conjecture for tight-cycle Ramsey numbers

摘录 / EXCERPT: Haxell-Łuczak-Peng-Rödl-Ruciński-Skokan conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-051811 | MATHDB-363627

Lehel's approximate conjecture for uniform tight cycles

摘录 / EXCERPT: Approximate Lehel conjecture. There exist constants $c_k\geq1$ and $n_0(k)\in\mathbb N$ such that, for every $n\geq n_0(k)$, the coloured complete k-graph contains vertex-disjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-051812 | MATHDB-363641

K_l-critical graph conjecture

摘录 / EXCERPT: For an integer $\ell\geq 2$, call a graph G K_{ℓ}-critical if it contains a copy of K_{ℓ}, is vertex-critical, and removing the vertex set of any copy of K_{ℓ} red...

conjecture | open | MathDB 页面 | 原文线索

V2-051813 | MATHDB-363648

The dominated-or-dominating pair degree-sum conjecture for supereulerian digraphs

摘录 / EXCERPT: Let D be a strong digraph with n vertices. A pair is dominated or dominating when it is a pair of dominated or dominating nonadjacent vertices of D as specified in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-051814 | MATHDB-363649

Elliptic corner VOA correlation conjecture for elliptic Macdonald polynomials

摘录 / EXCERPT: Let $m,n\in\mathbb{Z}^{\geq 1}$ with $m\leq n$, and let $\lambda\in\operatorname{Par}(m)$. Let $\widetilde{\psi}^{(q,\xi)}_{\lambda}$ be the specialization map that sends a symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-051815 | MATHDB-363650

Partial symmetry conjecture for elliptic corner VOA correlation functions

摘录 / EXCERPT: Let $\vec{c}=(c_1,\dots,c_n)=(1^L2^M3^N)$ and suppose that $q_{\vec{c}}\neq1$. Let $f^{\vec{c}}_{11}$ be the corresponding elliptic corner-VOA structure function and...

conjecture | open | MathDB 页面 | 原文线索

V2-051816 | MATHDB-363656

STS extremal construction conjecture for the $(2,p)$-norm Turán problem of F_5

摘录 / EXCERPT: Let F_5 be the 3-uniform hypergraph considered in the $(2,p)$-norm Turán problem, and let $\rho_{2,p}(F_5)$ denote its asymptotic extremal value. For a Steiner triple system (STS...

conjecture | open | MathDB 页面 | 原文线索

V2-051817 | MATHDB-363666

The non-generation conjecture for off-diagonal cycles in deleted graph products

摘录 / EXCERPT: Off-diagonal cycle conjecture. In $K^{\square\underline 2}$, no off-diagonal cycle is a sum of boundaries and symmetrized cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-051818 | MATHDB-363667

The generation conjecture for cycles in deleted graph products

摘录 / EXCERPT: Cycle-generation conjecture. In $K^{\square\underline2}$:

conjecture | open | MathDB 页面 | 原文线索

V2-051819 | MATHDB-363683

Adamus-Adamus conjecture on long cycles in balanced bipartite graphs

摘录 / EXCERPT: Let G be a balanced bipartite graph of order $2n$, with minimum degree $\delta(G)\ge r\ge 1$, where $n\ge 2k+2r$ and $k\in\mathbb{Z}$. Adamus-Adamus conjecture. If ... then G cont...

conjecture | open | MathDB 页面 | 原文线索

V2-051820 | MATHDB-363684

Conjecture on clique counts forcing long cycles in bipartite graphs

摘录 / EXCERPT: Let G be a connected bipartite graph with bipartition (X,Y). Suppose ... let $h=\left\lfloor\frac{n-k}{2}\right\rfloor$, and assume that $\delta(G)\ge r\ge 1$, where $n\ge 2k+2r$...

conjecture | open | MathDB 页面 | 原文线索

V2-051821 | MATHDB-363690

The path-linked triangles 3-colourability conjecture

摘录 / EXCERPT: Let H be a graph which is a disjoint union of triangles and paths of length at most 12, and let G be obtained from H by gluing on vertex-disjoint triangles. For two triangl...

conjecture | open | MathDB 页面 | 原文线索

V2-051822 | MATHDB-363691

The cycles plus K_4's conjecture

摘录 / EXCERPT: Let H be a graph with maximum degree at most 2, and let G be obtained from H by gluing on vertex-disjoint copies of K_4. Cycles plus K_4's conjecture. Then ... The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-051823 | MATHDB-363705

The half-order conjecture for locating-dominating sets in twin-free graphs

摘录 / EXCERPT: Let G be a finite graph. A set $C\subseteq V(G)$ is a locating-dominating set if it dominates every vertex outside C and, for every two distinct vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-051824 | MATHDB-363722

Győri et al.'s planar Turán conjectures for the two particular Θ_6-graphs

摘录 / EXCERPT: Let $ex_{\mathcal{P}}(n,H)$ denote the maximum number of edges in a planar graph with n vertices that does not contain H as a subgraph. For $k\geq 4$, let Θ_k denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-051825 | MATHDB-363730

Brouwer's matching conjecture for Steiner triple systems

摘录 / EXCERPT: A Steiner triple system of order n is a 3-uniform hypergraph on n vertices in which every pair of vertices lies in exactly one edge. A matching is a set of pairwise vertex-di...

conjecture | open | MathDB 页面 | 原文线索

V2-051826 | MATHDB-363731

Ferber-Kwan resolvability conjecture for random Steiner triple systems

摘录 / EXCERPT: Let a Steiner triple system of order n be a 3-uniform hypergraph on n vertices in which every pair lies in exactly one edge, and let a perfect matching be a set of $n/3$ pair...

conjecture | open | MathDB 页面 | 原文线索

V2-051827 | MATHDB-363732

Andersen's rainbow path conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let a proper edge-colouring be an edge-colouring in which edges of the same colour do not meet. A rainbow path is a path whose...

conjecture | open | MathDB 页面 | 原文线索

V2-051828 | MATHDB-363761

Characterization of factors guaranteeing well-dominated strong products

摘录 / EXCERPT: Characterization conjecture. If G is a well-dominated graph that is not trivially well-dominated, then there exists a well-dominated graph H such that $G\boxtimes H$ is not wel...

conjecture | open | MathDB 页面 | 原文线索

V2-051829 | MATHDB-363770

Balogh et al.'s linear path-separating-system conjecture

摘录 / EXCERPT: Let G be a graph on n vertices. A path separating system of G is a collection of paths in G such that, for every ordered pair of distinct edges (e,f), some path contains...

conjecture | open | MathDB 页面 | 原文线索

V2-051830 | MATHDB-363782

Jerebic's eventual non-distance-balancedness conjecture for generalized Petersen graphs

摘录 / EXCERPT: Jerebic's conjecture. For any $k\ge 2$, there exists a positive integer n_0 such that $GP(n,k)$ is not distance-balanced for every $n\ge n_0$. The conjecture was positively resol...

conjecture | open | MathDB 页面 | 原文线索

V2-051831 | MATHDB-363783

Füredi-Hajnal conjecture for arbitrary forbidden patterns

摘录 / EXCERPT: Let $P\in\{0,1\}^{k\times l}$ be any forbidden pattern, and let $G(P)$ be its unordered bipartite graph. Let $\operatorname{Ex}(P,n)$ denote the maximum number of ones in an...

conjecture | open | MathDB 页面 | 原文线索

V2-051832 | MATHDB-363784

Füredi-Hajnal conjecture for acyclic patterns

摘录 / EXCERPT: Let $P\in\{0,1\}^{k\times l}$ be an acyclic forbidden pattern, and let $G(P)$ be its unordered bipartite graph. Füredi-Hajnal acyclic-pattern conjecture. One should have ... The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-051833 | MATHDB-363785

Füredi-Jiang-Kostochka-Mubayi-Verstraëte conjecture for ordered hypergraph forests

摘录 / EXCERPT: Let F be an r-uniform forest with interval chromatic number r. Füredi-Jiang-Kostochka-Mubayi-Verstraëte conjecture. The maximum number of edges in a vertex-ordered r-unifor...

conjecture | open | MathDB 页面 | 原文线索

V2-051834 | MATHDB-363790

TxGraffiti's independence-matching conjecture for regular graphs

摘录 / EXCERPT: Let G be a connected r-regular graph with $r>0$. Let $\alpha(G)$ denote the independence number of G, and let $\mu(G)$ denote its matching number. TxGraffiti's independence-m...

conjecture | open | MathDB 页面 | 原文线索

V2-051835 | MATHDB-363816

The supersaturation conjecture for independent transversals

摘录 / EXCERPT: Let G be a graph with maximum degree Δ and let $\mathcal{U}$ be a (2Δ)-thick partition of $V(G)$. An independent transversal is a set containing exactly one vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-051836 | MATHDB-363823

The triangle-count conjecture under a spectral-radius condition

摘录 / EXCERPT: Triangle-count conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051837 | MATHDB-363824

The spectral extremal conjecture for bowties with q added edges

摘录 / EXCERPT: Spectral bowtie conjecture. For $q\ge 3$, if G is a graph of sufficiently large order n satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-051838 | MATHDB-363831

Harutyunyan-Mohar conjecture for oriented graphs

摘录 / EXCERPT: Harutyunyan-Mohar conjecture. Every oriented graph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051839 | MATHDB-363832

The directed-clique conjecture for the minimum degree of digraphs

摘录 / EXCERPT: Directed-clique conjecture. There exists $\varepsilon>0$ such that every digraph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051840 | MATHDB-363833

The biclique corollary for the minimum degree of digraphs

摘录 / EXCERPT: Biclique corollary conjecture. There exists $\varepsilon>0$ such that every digraph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-051841 | MATHDB-363837

The distance-packing Erdős-Pósa conjecture for cycles

摘录 / EXCERPT: For a positive integer d, a distance-d packing of cycles in a graph G is a set of cycles such that no path of length at most d joins two distinct cycles. For a vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-051842 | MATHDB-363838

The long-cycle induced Erdős-Pósa conjecture

摘录 / EXCERPT: For a graph G, an induced packing of cycles is a collection of cycles with no edge between distinct cycles. For a vertex set X, let $B_G(X,1)$ be its closed distance-one neighb...

conjecture | open | MathDB 页面 | 原文线索

V2-051843 | MATHDB-363839

The induced Erdős-Pósa conjecture for S-cycles

摘录 / EXCERPT: Let G be a graph and $S\subseteq V(G)$. An S-cycle is a cycle containing a vertex of S, and an induced packing is a collection of cycles with no edge between distinct cycles....

conjecture | open | MathDB 页面 | 原文线索

V2-051844 | MATHDB-363841

ELW's Hamiltonicity conjecture for subgraphs of powers of cycles

摘录 / EXCERPT: Let C_n^k be the k-th power of the cycle on n vertices, and let $\delta(G)$ denote the minimum degree of a graph G. ELW's Hamiltonicity conjecture. For all integers...

conjecture | open | MathDB 页面 | 原文线索

V2-051845 | MATHDB-363842

Hamiltonicity conjecture for subgraphs of powers of paths

摘录 / EXCERPT: Let P_n^k be the k-th power of the path on n vertices, and let $\deg_G(v)$ denote the degree of v in G. Hamiltonicity conjecture for path powers. Let $n\geq4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-051846 | MATHDB-363857

Kirillov-Bazhanov-Reshetikhin dilogarithm identity for constant Y-systems

摘录 / EXCERPT: Let X be a simply-laced connected Dynkin diagram of finite type with rank r, Coxeter number $h=h(X)$, and vertex set $I=1,\dots,r1$. Let $\ell\geq 2$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051847 | MATHDB-363885

The planar graph equilibrium conjecture

摘录 / EXCERPT: Let G be a strongly connected planar graph, and let g denote its undirected girth. A static equilibrium is an equilibrium of the static type, while a 2-chase equilibrium and...

conjecture | open | MathDB 页面 | 原文线索

V2-051848 | MATHDB-363906

The diagonal recurrence conjecture for Betti numbers of squared-path cut complexes

摘录 / EXCERPT: Diagonal recurrence conjecture. For each fixed $r\ge 3$ and $k\ge r+3$, these numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-051849 | MATHDB-363909

The balanced-tree conjecture for the maximum second Steklov eigenvalue

摘录 / EXCERPT: Balanced-tree conjecture. For sufficiently large ℓ, the tree $T_b^*(\ell,D)$ attains the maximum λ_2 among all trees in $\mathcal{T}_S(\ell,D)$.

conjecture | open | MathDB 页面 | 原文线索

V2-051850 | MATHDB-363916

Layered 3-graph characterization of vanishing codegree Turán density

摘录 / EXCERPT: A 3-graph F is layered if there is a function $f:V(F)\to\mathbb N$ such that: every edge has exactly one vertex whose label is strictly greater than the other two; edges with t...

conjecture | open | MathDB 页面 | 原文线索

V2-051851 | MATHDB-363923

Extremal-size conjecture for regular graphs with positive Lin-Lu-Yau curvature

摘录 / EXCERPT: Let $\mathcal{M}_{d}$ denote the maximum number of vertices of a connected d-regular graph with positive Lin-Lu-Yau curvature. For even degree, write $d\in2\mathbb{N}$. Extremal-...

conjecture | open | MathDB 页面 | 原文线索

V2-051852 | MATHDB-363934

Conjecture on the k-coalition number of complete bipartite graphs

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with part sizes s and t, where $s\leq t$, and let $C_k(G)$ denote the k-coalition number of a graph G. Complete bipartite grap...

conjecture | open | MathDB 页面 | 原文线索

V2-051853 | MATHDB-363952

Lew's extremal characterization conjecture for simplicial complexes

摘录 / EXCERPT: Let X be a simplicial complex on vertex set V of size n, with $h(X)=d$. Write $\mu_k(X)$ for the parameter appearing in the equality below, and let Δ_m denote the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-051854 | MATHDB-363953

Equatorial graph existence conjecture

摘录 / EXCERPT: Equatorial graph existence conjecture. There exists an equatorial graph with degree δ and girth g if and only if there exists a Moore graph with degree δ and girth...

conjecture | open | MathDB 页面 | 原文线索

V2-051855 | MATHDB-363954

Drier-Linial conjecture on the Hajós number of random lifts

摘录 / EXCERPT: An ℓ-lift of K_n is obtained by replacing every vertex of K_n with an independent set of size ℓ and every edge with a matching of size ℓ between the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-051856 | MATHDB-363959

Kohayakawa-Nagle-Rödl-Schacht conjecture for locally dense graphs

摘录 / EXCERPT: Let H and G be graphs, let $t(H,G)$ denote the homomorphism density of H in G, and call an n-vertex graph G (ρ,d)-dense when every subset X of at least ρn...

conjecture | open | MathDB 页面 | 原文线索

V2-051857 | MATHDB-363960

Bradač-Sudakov-Wigderson conjecture for nearly regular locally dense graphs

摘录 / EXCERPT: Let H and G be graphs, let $t(H,G)$ denote the homomorphism density of H in G, and call an n-vertex graph G (ρ,d)-nearly-regular when all but at most ρn vert...

conjecture | open | MathDB 页面 | 原文线索

V2-051858 | MATHDB-363966

Houdré-Tetali conjecture on isoperimetric constants of reversible Markov chains

摘录 / EXCERPT: Let (V,P,pi) be an irreducible and reversible Markov chain. For $p\in[0,1]$, let $\varphi_p(P)$ denote its isoperimetric constant, and let $\lambda_2(I-P)$ be the second-smallest...

conjecture | open | MathDB 页面 | 原文线索

V2-051859 | MATHDB-363968

The Hamiltonicity conjecture for the wiggly flip graph

摘录 / EXCERPT: Let WFG_n be the wiggly flip graph. Hamiltonicity conjecture. The wiggly flip graph WFG_n admits a Hamiltonian path, or even a Hamiltonian cycle. The authors...

conjecture | open | MathDB 页面 | 原文线索

V2-051860 | MATHDB-363987

Asymptotic sparsity conjecture for digraph placement game values

摘录 / EXCERPT: Let $F(b)$ denote the largest number of vertices needed to construct a game value born by day b, and let $g(b)$ be the comparison function defined in the paper. Asymptotic sparsi...

conjecture | open | MathDB 页面 | 原文线索

V2-051861 | MATHDB-363988

Nimber construction conjecture for digraph placement games

摘录 / EXCERPT: For a game value X, let $f(X)$ be the minimum number of vertices in a digraph placement game having value X. For an integer $n\geq1$, let ${\ast n}$ denote the nimber of order...

conjecture | open | MathDB 页面 | 原文线索

V2-051862 | MATHDB-364010

Involution-equivalence conjecture for cycle-free d-partitions of K_{2d}

摘录 / EXCERPT: Let K_{2d} be the complete graph on $2d$ vertices. A cycle-free d-partition is a cycle-free edge d-partition of K_{2d}, and two such partitions are involution equivalent if...

conjecture | open | MathDB 页面 | 原文线索

V2-051863 | MATHDB-364055

Converse-invariant orientations of trees with maximum degree at least three

摘录 / EXCERPT: Conjecture on converse-invariant tree orientations. D is converse invariant if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-051864 | MATHDB-364060

The weighted Caro-Wei obstruction conjecture for odd-cycle blowups

摘录 / EXCERPT: Weighted Caro-Wei obstruction conjecture. Unless G contains either a clique $K\subseteq V(G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-051865 | MATHDB-364062

Fuentes-Kamat conjecture on Erdős-Ko-Rado subfamilies of perfect matchings

摘录 / EXCERPT: Let M_n be the graph with vertex set ... and edge set ... Let $\mathcal{H}^{(p,s)}(n)$ be the family of subsets of $V(M_n)$ that contain exactly $2p+s$ vertices and span exactly p...

conjecture | open | MathDB 页面 | 原文线索

V2-051866 | MATHDB-364080

The cycle-factor conjecture for dense digraphs

摘录 / EXCERPT: Let D be a digraph of order n, with minimum semi-degree $\delta^0(D)$. Let $n=n_1+\cdots+n_k$ be a positive integer partition with $n_i\geq 3$ for each i. A cycle factor of t...

conjecture | open | MathDB 页面 | 原文线索

V2-051867 | MATHDB-364094

The general-coloring palette conjecture for complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $\tau_m(n)$ denote the minimum number of colors in an edge-coloring of K_n such that all triangles have distinct color pa...

conjecture | open | MathDB 页面 | 原文线索

V2-051868 | MATHDB-364095

The set-palette conjecture for complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. In a general edge-coloring of K_n, let $\tau_s(n)$ denote the minimum number of colors needed so that all triangles have distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-051869 | MATHDB-364104

Elphick-Linz-Wocjan square-sum generalization of the Bollobás-Nikiforov conjecture

摘录 / EXCERPT: Elphick-Linz-Wocjan conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-051870 | MATHDB-364187

Monotonicity conjecture for the signature of the interaction matrix

摘录 / EXCERPT: Let $M(\epsilon)$ be the matrix governing the stability of the graph-based gradient dynamics, with ϵ the interaction-strength parameter introduced above. The signature mon...

conjecture | open | MathDB 页面 | 原文线索

V2-051871 | MATHDB-364203

The NE/EN vertex conjecture for inner polynomials

摘录 / EXCERPT: Let Z be the innermost polynomial and $\zeta(F)$ the inner polynomial arising from the paper's construction. For a convex polygon, the NE vertex is the uppermost vertex with larg...

conjecture | open | MathDB 页面 | 原文线索

V2-051872 | MATHDB-364212

Chandler-Sazdanović-Stella-Yip multiplicity conjecture for the $(2^2,1^{n-4})$ Specht module of a star

摘录 / EXCERPT: Star Specht-multiplicity conjecture. The multiplicity of $\mathcal{S}_{2^2 1^{n-4}}$ in $H_{1,0}(G)$ is

conjecture | solved | MathDB 页面 | 原文线索

V2-051873 | MATHDB-364213

Vanishing multiplicities conjecture for degree-zero chromatic symmetric homology of star graphs

摘录 / EXCERPT: Star-homology vanishing conjecture. Whenever $i\ge 2$ or $\lambda_2\ge 3$, the multiplicity of $\mathcal{S}_\lambda$ in $H_{i,0}(G)$ is 0.

conjecture | open | MathDB 页面 | 原文线索

V2-051874 | MATHDB-364217

Brown-Erdős-Sós limit conjecture for $(\ell,\ell-2)$-free 3-uniform hypergraphs

摘录 / EXCERPT: Let $F(n;s,k)$ be the maximum number of edges in an (s,k)-free 3-uniform hypergraph on n vertices, where an (s,k)-configuration is a set of s vertices inducing at least...

conjecture | open | MathDB 页面 | 原文线索

V2-051875 | MATHDB-364233

The planar attacking cop number conjecture

摘录 / EXCERPT: Planar attacking cop number conjecture. For every planar graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-051876 | MATHDB-364234

The unbounded attacking-cop-number gap conjecture

摘录 / EXCERPT: Unbounded attacking-cop-number gap conjecture. For every non-negative integer k, there exists a graph H such that

conjecture | open | MathDB 页面 | 原文线索

V2-051877 | MATHDB-364235

Robber strategies and non-principal single ultrafilters

摘录 / EXCERPT: Let G be a graph or digraph inducing a connectivity system, and consider an elimination game with an invisible robber in which cop positions satisfy the relevant connectivity bou...

conjecture | open | MathDB 页面 | 原文线索

V2-051878 | MATHDB-364237

Characterization of connected graphs whose geodesic metric satisfies the Menger condition

摘录 / EXCERPT: Let G be a nonempty connected graph. Write $V(G)$ for its vertex set and d_G for its geodesic distance. Let $\mathfrak{MB}$ denote the class of metric spaces (X,d) such that...

conjecture | open | MathDB 页面 | 原文线索

V2-051879 | MATHDB-364239

Erdős, Gimbel and Straight's conjecture for graphs with clique number less than five

摘录 / EXCERPT: Let G be a graph, and write $\omega(G)$ for its clique number, $\zeta(G)$ for its cochromatic number, and $\chi(G)$ for its chromatic number. Erdős, Gimbel and Straight's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-051880 | MATHDB-364242

Norine's conjecture on monochromatic antipodal paths in cube graphs

摘录 / EXCERPT: Norine's conjecture. For $n\ge 2$, any antipodal edge-coloring of Q_n contains antipodal vertices u and $\bar u$ such that u and $\bar u$ are joined by a monochromatic path.

conjecture | open | MathDB 页面 | 原文线索

V2-051881 | MATHDB-364253

Odd-multiplicity determinant conjecture for Hodge-Riemann forms on pseudomanifolds

摘录 / EXCERPT: Let k be a field of arbitrary characteristic, and let Δ be a connected oriented simplicial pseudomanifold over k of dimension $d-1$ with vertex set V. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051882 | MATHDB-364254

Primitive determinant reformulation of the pseudomanifold Hodge-Riemann conjecture

摘录 / EXCERPT: Let k be a field of arbitrary characteristic and let Δ be a connected oriented simplicial pseudomanifold over k of dimension $d-1$ with vertex set V. Let ℓ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-051883 | MATHDB-364256

Exceptional-parameter conjecture for biased edge coloring games

摘录 / EXCERPT: Let m_1 and m_2 be positive integers satisfying $m_1>m_2$, and suppose that there is no positive integer k such that $m_1=km_2+k-1$. Let $\chi'_g(G;m,1)$ denote the game chro...

conjecture | open | MathDB 页面 | 原文线索

V2-051884 | MATHDB-364257

Conjecture on univariate distance ideals of directed circuits

摘录 / EXCERPT: Univariate distance-ideal conjecture. Based on computational experiments, for $4\leq k\leq n-2$,

conjecture | open | MathDB 页面 | 原文线索

V2-051885 | MATHDB-364270

KNRS conjecture on locally dense graphs

摘录 / EXCERPT: KNRS conjecture. Let H be a graph. For every $d,\eta\in(0,1)$, there exists $\varepsilon=\varepsilon(d,\eta,H)$ such that if G is (ε,d)-dense, then

conjecture | open | MathDB 页面 | 原文线索

V2-051886 | MATHDB-364295

Rationality conjecture for non-admissible minimal W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra of type d_4, e_6, e_7, or e_8, and let $m\geq -1$ be an integer. Set ... where $h^{\vee}$ is the dual Coxeter number of...

conjecture | open | MathDB 页面 | 原文线索

V2-051887 | MATHDB-364296

Coset rationality and lisse-ness conjecture

摘录 / EXCERPT: Coset conjecture. The commutant $\operatorname{Com}(W,V)$ is also rational and lisse. This is presented as a widely believed assertion and is used in the paper to deduce rationalit...

conjecture | open | MathDB 页面 | 原文线索

V2-051888 | MATHDB-364306

The finite-list conjecture for 5-vertex-critical co-gem-free graphs

摘录 / EXCERPT: Finite-list conjecture. There are only finitely many 5-vertex-critical co-gem-free graphs, and the list of 327 is complete.

conjecture | open | MathDB 页面 | 原文线索

V2-051889 | MATHDB-364307

Hameed's conjecture on semi-transitivity of Mycielski graphs

摘录 / EXCERPT: Let G be a graph, and let $\mu(G)$ denote its Mycielski graph. A graph is semi-transitive if it admits an acyclic orientation with no shortcut. Hameed's conjecture. For every gra...

conjecture | open | MathDB 页面 | 原文线索

V2-051890 | MATHDB-364311

Conjecture on the adjacency spectrum of burnt pancake graphs

摘录 / EXCERPT: Let $\mathbb{BP}_n$ be the burnt pancake graph, and let $\operatorname{sp}(\mathbb{BP}_n)$ denote its adjacency spectrum. Write $[\overline{(n-1)},n]$ for the interval from $n-1$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-051891 | MATHDB-364312

Conjecture on the spectral gap of burnt pancake graphs

摘录 / EXCERPT: Let $\mathbb{BP}_n$ be the burnt pancake graph, and let $\operatorname{sg}(\mathbb{BP}_n)$ denote its spectral gap. Spectral-gap conjecture. ... and ... This conjecture is supported by...

conjecture | open | MathDB 页面 | 原文线索

V2-051892 | MATHDB-364315

The exact WKB analytic continuation conjecture for strongly GMN spectral curves

摘录 / EXCERPT: Let ∇ be a WKB-regular $\hbar$-flat connection with strongly GMN spectral curve, let Ψ be a WKB formal solution, and let $\cL\Psi$ be its Borel-Laplace dual. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-051893 | MATHDB-364321

A stable shift-graph embedding conjecture

摘录 / EXCERPT: Stable shift-graph embedding conjecture. If, for every cardinal μ,

conjecture | open | MathDB 页面 | 原文线索

V2-051894 | MATHDB-364336

Kao's sharpness conjecture for Hamiltonicity of graph-path products

摘录 / EXCERPT: A path factor of a graph is a factor whose components are paths on at least two vertices. Let P_m be the path on m vertices, and let $\Delta(G)$ denote the maximum degree of a...

conjecture | open | MathDB 页面 | 原文线索

V2-051895 | MATHDB-364337

Gowers' clique-difference conjecture for graphs

摘录 / EXCERPT: Let $\delta>0$. Consider the collection of non-oriented graphs on the vertex set $[n]$, with density measured relative to the number of such graphs. Gowers' clique-difference conje...

conjecture | open | MathDB 页面 | 原文线索

V2-051896 | MATHDB-364342

Weak reconstruction conjecture for vertex operator algebras

摘录 / EXCERPT: Weak reconstruction conjecture. If there is a positive strongly rational vertex operator algebra V with

conjecture | open | MathDB 页面 | 原文线索

V2-051897 | MATHDB-364343

Finiteness conjecture for bulk genera of vertex operator algebras

摘录 / EXCERPT: Bulk genera finiteness conjecture. Every bulk genus contains only finitely many isomorphism classes of vertex operator algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-051898 | MATHDB-364344

Conjugacy conjecture for Cartan subalgebras of tensor products

摘录 / EXCERPT: Cartan-subalgebra conjugacy conjecture. All Cartan subalgebras of the conformal vertex algebra $V\otimes V_L$ are conjugate.

conjecture | open | MathDB 页面 | 原文线索

V2-051899 | MATHDB-364376

Buchanan-odd-cover conjecture for complete graphs

摘录 / EXCERPT: For a graph G, let $b_2(G)$ denote its biclique partition number over the field of two elements, and let K_n denote the complete graph on n vertices. Let k be a positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-051900 | MATHDB-364405

The induced-cycle conjecture for triangle-free graphs with nonnegative Bakry-Émery curvature

摘录 / EXCERPT: Let G be a triangle-free graph satisfying $CD(0,\infty)$ with both the unweighted normalized and non-normalized Laplacians. An induced k-cycle conjecture. G contains no induc...

conjecture | open | MathDB 页面 | 原文线索

V2-051901 | MATHDB-364417

Hu, Li and Yang's conjecture on rainbow triangle tilings

摘录 / EXCERPT: Hu, Li and Yang's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051902 | MATHDB-364421

The colour-bias threshold conjecture for tight Hamilton cycles in 2-coloured 4-graphs

摘录 / EXCERPT: Let H be a 2-edge-coloured 4-uniform hypergraph on n vertices, and let $\delta_1(H)$ denote its minimum vertex degree. A tight Hamilton cycle is a cyclic ordering of all ve...

conjecture | open | MathDB 页面 | 原文线索

V2-051903 | MATHDB-364446

Wood's conjecture on cliques in graphs with a forbidden clique minor

摘录 / EXCERPT: Wood's conjecture. The number of k-cliques in a K_t-minor free graph on n vertices is at most

conjecture | open | MathDB 页面 | 原文线索

V2-051904 | MATHDB-364464

Exponential multicolor Ramsey bound for balanced complete tripartite graphs

摘录 / EXCERPT: Let $K_{k,k,k}$ denote the complete tripartite graph with three parts of size k, and let $r(F;q)$ be the least n such that every q-edge-coloring of the complete graph on n...

conjecture | open | MathDB 页面 | 原文线索

V2-051905 | MATHDB-364478

Density upper-bound conjecture for the truncated square tiling graph

摘录 / EXCERPT: Let $H_{\infty,\infty}$ be the infinite truncated square tiling graph, and let $\delta_{t,r}(H_{\infty,\infty})$ denote its optimal (t,r) broadcast density. Density conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-051906 | MATHDB-364479

Conjecture on single-vertex broadcasts in square truncated tiling graphs

摘录 / EXCERPT: Let $H_{m,m}$ be the finite truncated square tiling graph with $m\geq 1$, let $\gamma_{t,r}(H_{m,m})$ denote its (t,r) broadcast domination number, and let m and r be positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-051907 | MATHDB-364500

Uniform positivity and boundedness conjecture for the entropy-maximizing permuton density

摘录 / EXCERPT: Uniform bounds conjecture. There exist constants $0<c<C$, depending on all the data, such that $$

conjecture | open | MathDB 页面 | 原文线索

V2-051908 | MATHDB-364523

Extremal blowup conjecture for the maximum spectral gap $s_{1,j}$

摘录 / EXCERPT: Extremal blowup conjecture. For any $j\geq 1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-051909 | MATHDB-364524

Korándi-Roberts-Scott conjecture on blowups of odd-cycle-free graphs

摘录 / EXCERPT: Fix $k\ge 2$. Let $\delta>\delta_0>0$ and let n be sufficiently large. For a graph G, write $D_2(G)$ for the maximum number of edges in a bipartite subgraph of G, and call a...

conjecture | open | MathDB 页面 | 原文线索

V2-051910 | MATHDB-364538

Asymptotically optimal degree balance in regular graphs

摘录 / EXCERPT: Let G be a d-regular graph of order n, and let $m(H,k)$ denote the number of vertices of degree k in a spanning subgraph H. Asymptotic degree-balance conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-051911 | MATHDB-364561

Linear almost-spanning monochromatic tight-cycle conjecture

摘录 / EXCERPT: Let $K_n^{(k)}$ be the complete k-uniform hypergraph, with its edges coloured using r colours. A collection of cycles covers $(1-o(1))n$ vertices if the number of uncovered ver...

conjecture | open | MathDB 页面 | 原文线索

V2-051912 | MATHDB-364563

Generic identifiability of normalized deep self-attention parametrizations

摘录 / EXCERPT: Let normalized deep self-attention networks be parametrized via $(\mathbf{M},L)$, where these parameters encode the network as defined in the paper. The normalization removes layer...

conjecture | open | MathDB 页面 | 原文线索

V2-051913 | MATHDB-364571

Minimum-volume scaling conjecture for random triangulations of the d-sphere

摘录 / EXCERPT: Let $\mathcal{S}^{(d)}_n$ be the set of n-vertex triangulations of the d-sphere, let $M(\mathcal{S}^{(d)}_n)$ denote their minimum volume, and let $\mu^{(d)}_n$ be the median o...

conjecture | open | MathDB 页面 | 原文线索

V2-051914 | MATHDB-364574

Even-diameter order conjecture for chordal ring graphs

摘录 / EXCERPT: Let $CR(N,c)$ be a chordal ring graph with diameter k, and let $N(k)$ denote the maximum number of vertices among chordal ring graphs of diameter k. For even $k>2$, the conject...

conjecture | open | MathDB 页面 | 原文线索

V2-051915 | MATHDB-364575

Even-diameter order conjecture for chordal ring mixed graphs

摘录 / EXCERPT: Let $CRM(N,c)$ be a chordal ring mixed graph with vertex set $\mathbb Z_N$, directed-cycle arcs $i\mathbin{\rightarrow}i+1$, and chords determined by the odd parameter c as in th...

conjecture | open | MathDB 页面 | 原文线索

V2-051916 | MATHDB-364576

Even-diameter order conjecture for chordal multi-ring mixed graphs

摘录 / EXCERPT: Let $CMRM(m,n,c)$ be a chordal m-ring mixed graph on vertex set $\mathbb Z_m\times\mathbb Z_n$, with the arcs and chords defined by the source, and let k be an even diameter sa...

conjecture | open | MathDB 页面 | 原文线索

V2-051917 | MATHDB-364578

Constant-additive bound for the open conflict-free chromatic index

摘录 / EXCERPT: Let G be a graph of maximum degree Δ without isolated vertices and edges. The constant-additive open conflict-free edge-coloring conjecture. There is a constant C such t...

conjecture | open | MathDB 页面 | 原文线索

V2-051918 | MATHDB-364581

The characterization of 2-quasi-regularizable connected W_2 graphs

摘录 / EXCERPT: The 2-quasi-regularizability characterization. G is 2-quasi-regularizable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-051919 | MATHDB-364586

Facet- and vertex-minimality conjecture for the degree construction

摘录 / EXCERPT: Minimality conjecture. The triangulated n-sphere constructed in Theorem is facet-minimal, and it is vertex-minimal as well.

conjecture | open | MathDB 页面 | 原文线索

V2-051920 | MATHDB-364604

Conjecture on the structure of the extremal tree for odd diameter

摘录 / EXCERPT: Structural conjecture. If $d\ge 5$ is odd, then the following hold for $T^\#$:

conjecture | open | MathDB 页面 | 原文线索

V2-051921 | MATHDB-364608

The Permanence Conjecture for endotactic mass-action systems

摘录 / EXCERPT: An endotactic mass-action system is a mass-action system whose reaction network is endotactic, and a system is permanent when its trajectories remain uniformly separated from the b...

conjecture | open | MathDB 页面 | 原文线索

V2-051922 | MATHDB-364612

Bergold-Fabila-Monroy-Roldán-Pensado-Schaefer conjecture on empty cycles in simple drawings

摘录 / EXCERPT: Bergold-Fabila-Monroy-Roldán-Pensado-Schaefer conjecture. Every simple drawing of K_n contains an empty k-cycle for each $3 leq k leq n$. The supplied paper proves the case...

conjecture | open | MathDB 页面 | 原文线索

V2-051923 | MATHDB-364622

Generalized Lehman-Ron conjecture for directed hypercube layers

摘录 / EXCERPT: Let L_i) and L_j be layers of the directed hypercube, with $i<j$ and $r:=j-i$. Let $(S,T;\phi)$ be a matched pair with $S\subseteq L_i$ and $T\subseteq L_j$. Generalized Lehman...

conjecture | open | MathDB 页面 | 原文线索

V2-051924 | MATHDB-364623

Flow-capacity conjecture for directed hypercube separation distance

摘录 / EXCERPT: Let S be a source set in the directed hypercube with separation distance r. Consider a flow network with unit edge capacities and vertex capacities r^2. Flow-capacity conject...

conjecture | open | MathDB 页面 | 原文线索

V2-051925 | MATHDB-364624

Frankl's conjecture for bipartite graphs with exactly one pendant vertex

摘录 / EXCERPT: Let G be a bipartite graph, and call a vertex pendant when it has degree one. Frankl's one-pendant-vertex conjecture. Every bipartite graph with exactly one pendant vertex satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-051926 | MATHDB-364625

Frankl's conjecture for bipartite graphs with pendant vertices

摘录 / EXCERPT: Let G be a bipartite graph, and call a vertex pendant when it has degree one. Frankl's pendant-vertex conjecture. Every bipartite graph with at least one pendant vertex satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-051927 | MATHDB-364626

Finiteness conjecture for connected graphs with odd metamour period

摘录 / EXCERPT: Finiteness conjecture. For each odd $k\in \mathbb{Z}^+$, there exist only finitely many connected graphs with metamour period k.

conjecture | open | MathDB 页面 | 原文线索

V2-051928 | MATHDB-364654

Exact connected Turán number conjecture for Berge paths

摘录 / EXCERPT: Let r and k be integers satisfying ... For an r-uniform hypergraph on n vertices, let $\operatorname{ex}^{\mathrm{conn}}_r(n,\mathcal{B}P_k)$ denote the maximum number of edg...

conjecture | open | MathDB 页面 | 原文线索

V2-051929 | MATHDB-364657

Linear growth conjecture for maximal facet dimensions in Eulerian magnitude homology

摘录 / EXCERPT: Linear growth conjecture. There exists a linear function φ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-051930 | MATHDB-364662

Conjecture on irreducible Zhu algebra modules for admissible affine sl_3 levels

摘录 / EXCERPT: Let $k=-3+\frac{2}{2m+1}$ with $m\geqslant 0$. Write $\bar{\Lambda}_1$ and $\bar{\Lambda}_2$ for the fundamental weights of sl_3, $A(L_k(sl_3))$ for the Zhu algebra of...

conjecture | open | MathDB 页面 | 原文线索

V2-051931 | MATHDB-364683

Kim, Kim and Liu's conjecture for the Ramsey-Turán function \(\rho(3,6,\delta)\)

摘录 / EXCERPT: Let $\rho(3,6,\delta)$ denote the asymptotic maximum edge density of an n-vertex graph with independence number at most δn admitting a 2-edge-coloring with no monochromat...

conjecture | open | MathDB 页面 | 原文线索

V2-051932 | MATHDB-364684

The general Ramsey-Turán formula conjecture for \(\rho(3,q)\)

摘录 / EXCERPT: For each integer $q\ge 5$, let $\rho(3,m)$ denote the Ramsey-Turán density for graphs admitting a 2-edge-coloring with no monochromatic blue K_3 and no monochromatic red K_m, a...

conjecture | open | MathDB 页面 | 原文线索

V2-051933 | MATHDB-364696

Linear neighborhood-complexity conjecture for hereditary small classes

摘录 / EXCERPT: Linear neighborhood-complexity conjecture. Every hereditary small class of graphs has neighborhood complexity $O(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-051934 | MATHDB-364699

Asymptotic Pauling estimate for residual entropy of simple graphs

摘录 / EXCERPT: Let $G=G(n)$ be a sequence of simple graphs with even degree sequence $\boldsymbol{d}=\boldsymbol{d}(n)$. Write d_{hm} for the harmonic mean degree, and let $\rho(G)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-051935 | MATHDB-364700

Schrijver's upper-bound conjecture for Eulerian orientations

摘录 / EXCERPT: Let G be a simple graph with even degrees $d_1,\ldots,d_n$. Let $\operatorname{EO}(G)$ denote the number of Eulerian orientations of G, and let $\operatorname{RT}(d+1)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-051936 | MATHDB-364714

Conjecture on the k-general position number of infinite grids

摘录 / EXCERPT: Let $P^{\Box,n}_\infty$ be the n-dimensional infinite grid, and let $\mathrm{gp}^k(G)$ denote its k-general position number. Infinite-grid conjecture. ... This conjecture is moti...

conjecture | open | MathDB 页面 | 原文线索

V2-051937 | MATHDB-364715

Conjecture on the complexity of the k-general d-position problem

摘录 / EXCERPT: The k-general d-position problem takes as input a graph G, integers $k\geq 2$ and $d\geq 1$, and a positive integer r, and asks whether $\mathrm{gp}^k_d(G)$ is greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-051938 | MATHDB-364720

Füredi-Gyárfás-Simonyi conjecture on connected matchings

摘录 / EXCERPT: Füredi-Gyárfás-Simonyi conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-051939 | MATHDB-364723

Whiskered-cycle depth formula conjecture for square-free powers

摘录 / EXCERPT: Let $W(C_m)$ be the graph obtained by attaching a whisker to each vertex of the cycle C_m, let $I(W(C_m))$ be its edge ideal in the corresponding polynomial ring...

conjecture | open | MathDB 页面 | 原文线索

V2-051940 | MATHDB-364731

Bau-Dankelmann oriented diameter conjecture for bridgeless graphs

摘录 / EXCERPT: Let G be a bridgeless graph of order n and minimum degree $\delta\geq 2$, and let $\overrightarrow{\operatorname{diam}}(G)$ denote the minimum diameter over all strongly connec...

conjecture | open | MathDB 页面 | 原文线索

V2-051941 | MATHDB-364733

Subtraction-free characterization conjecture for bounded determinantal ratios

摘录 / EXCERPT: Let A be a totally positive element of GL_n, and let R be a ratio of products of minors as above. In a weighted planar-network parametrization, call $R=p/q$ subtraction free...

conjecture | open | MathDB 页面 | 原文线索

V2-051942 | MATHDB-364737

Exactness conjecture for the plus Drinfeld-Sokolov reduction

摘录 / EXCERPT: Let $\check{k}=m-r-1$, let $\lambda\in\Lambda$ satisfy $(p\lambda_\bullet+\rho,{}^L\theta)\leq p$, and let $\mathcal{O}_{\check{k}}^{[\lambda]}$ be the indicated subcategory of mod...

conjecture | open | MathDB 页面 | 原文线索

V2-051943 | MATHDB-364738

Collapsing-level conjecture for boundary VOAs of 3d mirror theories

摘录 / EXCERPT: Let a non-rank-zero 3d $\mathcal{N}=4$ SQFT be a three-dimensional supersymmetric quantum field theory with nonzero rank, and let its boundary VOA be obtained after a topological t...

conjecture | open | MathDB 页面 | 原文线索

V2-051944 | MATHDB-364757

Uniqueness conjecture for the 12-vertex flag triangulation of the 3-sphere

摘录 / EXCERPT: Uniqueness conjecture. T_{12} is, up to isomorphism, the only flag triangulation of S^3 with 12 vertices in which every edge is contained in a square.

conjecture | open | MathDB 页面 | 原文线索

V2-051945 | MATHDB-364778

The tree antimagic conjecture for finite Abelian groups

摘录 / EXCERPT: Let A be a finite Abelian group, and let T be a tree. An A-antimagic labeling is an edge labeling whose induced vertex sums are pairwise distinct. Tree antimagic conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-051946 | MATHDB-364797

Katona-Xiao conjecture for the path-and-clique Turán problem

摘录 / EXCERPT: Katona-Xiao conjecture. If k is odd and $(k-1)\mid n$, then the disjoint union of $n/(k-1)$ copies of $T(k-1,m-1)$ gives the maximum number of edges in a graph containing neither...

conjecture | open | MathDB 页面 | 原文线索

V2-051947 | MATHDB-364810

Kang-Kim-Liu conjecture on single-graph extremal exponents

摘录 / EXCERPT: Kang-Kim-Liu conjecture. For every rational number $r\in[1,2]$, there exists a graph H such that

conjecture | open | MathDB 页面 | 原文线索

V2-051948 | MATHDB-364817

Versteegen's random-graph conjecture for even-non-decomposable graphs

摘录 / EXCERPT: Let $H\sim G(n,\frac{1}{2})$, where $G(n,p)$ is the binomial random graph on n vertices with edge probability p. An even-non-decomposable graph is a graph with an even number o...

conjecture | open | MathDB 页面 | 原文线索

V2-051949 | MATHDB-364834

Ju's palindromicity conjecture for graph polytope Ehrhart numerators

摘录 / EXCERPT: Ju's conjecture. The polynomial in the numerator of this Ehrhart series is symmetric, also known as palindromic.

conjecture | open | MathDB 页面 | 原文线索

V2-051950 | MATHDB-364835

Bóna-Ju-Ann palindromicity conjecture for circular graph polytopes

摘录 / EXCERPT: Bóna-Ju-Ann conjecture. If $k=2\ell$, then

conjecture | open | MathDB 页面 | 原文线索

V2-051951 | MATHDB-364856

Beem-Ballin conjecture for SQED[n]

摘录 / EXCERPT: Let V^A be the boundary VOA associated with SQED[n], let $L_1(\mathfrak{psl}_{n|n})$ be the level-one simple vertex algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051952 | MATHDB-364857

Maximal-submodule conjecture for \(V^1(\mathfrak{psl}_{n|n})\)

摘录 / EXCERPT: Let $V^1(\mathfrak{psl}_{n|n})$ be the universal level-one vertex algebra of $\mathfrak{psl}_{n|n}$, let χ be the singular vector constructed in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-051953 | MATHDB-364870

Li-Lu-Peng conjecture for the spectral radius of F_k-free graphs

摘录 / EXCERPT: Let F_k be the friendship graph consisting of k triangles sharing a common vertex. Let G be an F_k-free graph with m edges, and let $\rho(G)$ denote its spectral radius....

conjecture | open | MathDB 页面 | 原文线索

V2-051954 | MATHDB-364871

Li-Lu-Peng conjecture for fan-free graphs

摘录 / EXCERPT: Let $V_{k+1}=K_1\vee P_k$ be the fan graph on $k+1$ vertices, and let G be a graph with m edges and spectral radius $\rho(G)$. Li-Lu-Peng fan-graph conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051955 | MATHDB-364872

The three-sevenths upper-bound conjecture for the isolation game

摘录 / EXCERPT: Let G be a graph, let $n(G)=|V(G)|$ denote its order, and let $\iota_{\rm g}(G)$ and $\iota_{\rm g}'(G)$ denote the game isolation numbers when Dominator and Staller, respectivel...

conjecture | open | MathDB 页面 | 原文线索

V2-051956 | MATHDB-364874

The all-dimensional SIP characterization

摘录 / EXCERPT: Let G be a graph, let f be a nonedge of G, and let $G \cup f$ denote the graph obtained by adding f. An atom of $G \cup f$ is a graph-theoretic atom containing f, and an...

conjecture | open | MathDB 页面 | 原文线索

V2-051957 | MATHDB-364875

The edge-deletion characterization of forbidden flattenability minors

摘录 / EXCERPT: Let $d \geq 2$. A graph is a forbidden minor for d-flattenability when it is a minimal obstruction to d-flattenability. For an edge e of a graph G, $(G \setminus e,e)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-051958 | MATHDB-364895

The common-zero independence-number conjecture for finite groups

摘录 / EXCERPT: Let G be a finite group. The common-zero graph $\Gamma_v(G)$ has the non-linear irreducible characters of G as vertices, with two characters adjacent when they have a common ze...

conjecture | open | MathDB 页面 | 原文线索

V2-051959 | MATHDB-364899

Loebl-Komlós-Sós conjecture for trees

摘录 / EXCERPT: Loebl-Komlós-Sós conjecture. If at least $n/2$ vertices of G have degree at least d, then G contains a copy of T.

conjecture | open | MathDB 页面 | 原文线索

V2-051960 | MATHDB-364900

Klimošová-Piguet-Rohzoň skew-tree conjecture

摘录 / EXCERPT: Klimošová-Piguet-Rohzoň conjecture. If at least rn vertices of G have degree at least d, then G contains a copy of every r-skew tree with d edges.

conjecture | open | MathDB 页面 | 原文线索

V2-051961 | MATHDB-364901

Klimošová-Piguet-Rozhoň local degree conjecture

摘录 / EXCERPT: Klimošová-Piguet-Rozhoň local degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051962 | MATHDB-364903

Addario-Berry-Havet-Linhares Sales-Thomassé-Reed antidirected tree conjecture

摘录 / EXCERPT: Addario-Berry-Havet-Linhares Sales-Thomassé-Reed conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051963 | MATHDB-364913

The $6/7$ positive square-energy exponent conjecture

摘录 / EXCERPT: Let G be a graph with m edges, and let $s^+(G)$ denote its positive square energy. Positive square-energy exponent conjecture. ... The paper proves the same lower bound with a do...

conjecture | open | MathDB 页面 | 原文线索

V2-051964 | MATHDB-364914

Negative square energy versus graph surplus

摘录 / EXCERPT: For a graph G, let $\operatorname{surp}(G)$ denote its surplus, and let $s^-(G)$ denote its negative square energy. Negative surplus exponent conjecture. ... The paper constructs e...

conjecture | open | MathDB 页面 | 原文线索

V2-051965 | MATHDB-364915

Jackson-Yoshimoto's spanning even tree conjecture for regular nonbipartite graphs

摘录 / EXCERPT: Let G be a finite connected graph, possibly with multiple edges but no loops. A spanning tree T of G is even if all leaves of T belong to the same part of the bipartition o...

conjecture | open | MathDB 页面 | 原文线索

V2-051966 | MATHDB-364916

Jackson-Yoshimoto's spanning weakly even tree conjecture

摘录 / EXCERPT: Jackson-Yoshimoto's conjecture. Every connected graph that is not a regular bipartite graph has a spanning weakly even tree.

conjecture | open | MathDB 页面 | 原文线索

V2-051967 | MATHDB-364930

Brualdi-Hoffman conjecture on maximal spectral radius for fixed-size graphs

摘录 / EXCERPT: Let $G=(V(G),E(G))$ be a simple finite graph of size m without isolated vertices. For graphs G and H, write $G\vee H$ for their join, and let K_r denote the complete graph...

conjecture | open | MathDB 页面 | 原文线索

V2-051968 | MATHDB-364931

Bhattacharya-Friedland-Peled conjecture on maximal spectral radius of bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph of size m with bipartition sets S and T, satisfying $2\leqslant |S|\leqslant |T|$ and $0<m<|S||T|$. A graph achieves the maximum spectral radius...

conjecture | open | MathDB 页面 | 原文线索

V2-051969 | MATHDB-364932

Yu-Li-Peng conjecture on spectral radius of fan-free graphs

摘录 / EXCERPT: Let $F_k=K_1\vee P_{k-1}$ be the fan graph on k vertices, with K_1 the one-vertex graph, P_{k-1} the path on $k-1$ vertices, and $\vee$ the graph join. Let $\mathcal{G}(m,F)$...

conjecture | open | MathDB 页面 | 原文线索

V2-051970 | MATHDB-364937

Strong m-ary sensitivity conjecture

摘录 / EXCERPT: Let m be a positive integer, let ε be an m-th primitive root of unity, and let $H(n,m)$ be the Hamming graph. Write $H_1,\dots,H_m$ for possibly empty induced subgr...

conjecture | open | MathDB 页面 | 原文线索

V2-051971 | MATHDB-364966

Quadratic embeddability conjecture for theta graphs

摘录 / EXCERPT: Quadratic embeddability conjecture. Assume $1\leq\alpha\leq\beta\leq\gamma$ and $\beta\geq2$. Then $\Theta(\alpha,\beta,\gamma)$ is of QE class if and only if either

conjecture | open | MathDB 页面 | 原文线索

V2-051972 | MATHDB-364967

Forest-cut conjecture for sparse graphs

摘录 / EXCERPT: Let G be a finite, simple, undirected graph of order n, and call a vertex set a forest cut if it is a vertex cut whose induced subgraph is a forest. Forest-cut conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-051973 | MATHDB-364968

Minimum-size conjecture for 3-connected graphs with cyclic neighborhoods

摘录 / EXCERPT: Let G be a graph of order n and size m. Assume that G is 3-connected and that every neighborhood of a vertex contains a cycle. Minimum-size conjecture. Then ... The question...

open_problem | open | MathDB 页面 | 原文线索

V2-051974 | MATHDB-364971

Alternating orientations minimize arborescences in Eulerian planar graphs

摘录 / EXCERPT: Let G be an Eulerian planar graph. An orientation is alternatingly in- and outward oriented at each vertex when, in the cyclic order around every vertex, its incident edges alter...

conjecture | open | MathDB 页面 | 原文线索

V2-051975 | MATHDB-364985

Alikhani-Soltani conjecture on distinguishing parameters of Mycielskian graphs

摘录 / EXCERPT: Let G be a connected graph of order $n\geq 3$, and let $\mu(G)$ denote its Mycielskian graph. Write $\operatorname{Dist}(G)$ for the distinguishing number and...

conjecture | open | MathDB 页面 | 原文线索

V2-051976 | MATHDB-364988

Extension of the critical-point asymptotic formula to inhomogeneous potentials

摘录 / EXCERPT: Inhomogeneous extension conjecture. The asymptotic formula

conjecture | open | MathDB 页面 | 原文线索

V2-051977 | MATHDB-364993

Four-colorability of Eulerian torus triangulations of representativity at least 8

摘录 / EXCERPT: Let G be an Eulerian triangulation of the torus, and let its representativity be at least 8. Four-colorability conjecture. The graph G is 4-colorable. The paper presents th...

conjecture | open | MathDB 页面 | 原文线索

V2-051978 | MATHDB-364999

Contour orbit diameter conjecture for S-polygons

摘录 / EXCERPT: Let an S-polygon be a polygonal configuration equipped with a contour, and let the orbit of a contour be the set of contours reachable from it by solitaire moves, with diameter m...

conjecture | open | MathDB 页面 | 原文线索

V2-051979 | MATHDB-365009

Independent-set conjecture for linear uniform hypergraphs

摘录 / EXCERPT: Let $r\geq 2$, let $a_r>0$ be a constant satisfying $a_r=1-o_r(1)$ as $r\to\infty$, and let H be an $(r+1)$-uniform n-vertex linear hypergraph with maximum degree $d\geq 1$. In...

conjecture | open | MathDB 页面 | 原文线索

V2-051980 | MATHDB-365010

Independence-number conjecture for linear triple systems

摘录 / EXCERPT: Let H be an n-vertex linear triple system, meaning a linear 3-uniform hypergraph. Triple-system independence conjecture. Its independence number should satisfy ... The source d...

conjecture | open | MathDB 页面 | 原文线索

V2-051981 | MATHDB-365011

Boundary-size conjecture for random independent sets in linear hypergraphs

摘录 / EXCERPT: Let $r\geq 2$, let H be an n-vertex linear $(r+1)$-uniform hypergraph of maximum degree $d\geq 1$, and let $\mathcal{I}(H)$ denote its independent sets. For...

conjecture | open | MathDB 页面 | 原文线索

V2-051982 | MATHDB-365032

Conjecture on the codegree threshold for spanning loose-cycle factors

摘录 / EXCERPT: There exists an integer n_0 such that for all $n\geq n_0$, the following holds. Let $\mathcal{C}$ be a 3-graph consisting of vertex-disjoint loose cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-051983 | MATHDB-365041

Cvetković-Rowlinson outerplanar spectral-radius conjecture

摘录 / EXCERPT: Let n be the number of vertices, let P_{n-1} be the path on $n-1$ vertices, and let $\vee$ denote the join of graphs. Cvetković-Rowlinson's conjecture. The outerplanar graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-051984 | MATHDB-365042

Piecewise planar spectral-extremal graph conjecture for fixed edge count

摘录 / EXCERPT: Let G_m denote the graph defined earlier in the paper as the large-edge spectral-extremal planar graph, and let H_m be defined by ... Here $\cup$ denotes disjoint union and...

conjecture | open | MathDB 页面 | 原文线索

V2-051985 | MATHDB-365045

Liu-Ma conjecture on consecutive odd cycles

摘录 / EXCERPT: Liu-Ma conjecture. G contains $eil k/2$ cycles with consecutive odd lengths.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-051986 | MATHDB-365046

Conjecture on consecutive cycle lengths in 3-connected graphs

摘录 / EXCERPT: Conjecture on consecutive cycle lengths. Except when G is K_{k+1}, the graph G contains k cycles of consecutive lengths.

conjecture | open | MathDB 页面 | 原文线索

V2-051987 | MATHDB-365050

Strong rationality conjecture for finite orbifolds of vertex operator algebras

摘录 / EXCERPT: Let $\mathcal{V}$ be a strongly rational vertex operator algebra, and let G be a finite group of automorphisms of $\mathcal{V}$. The fixed-point vertex operator algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-051988 | MATHDB-365053

The cubic graph list strong chromatic index conjecture at index five

摘录 / EXCERPT: Cubic index-five list-coloring conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-051989 | MATHDB-365056

The edge-coloring conjecture for dual graphs of alcoved triangulations

摘录 / EXCERPT: Edge-coloring conjecture. The edge-coloring of G determined by the hyperplane types prescribes a choice of connecting sets such that

conjecture | open | MathDB 页面 | 原文线索

V2-051990 | MATHDB-365079

The maximum-degree surplus conjecture for 3-multigraph cuts

摘录 / EXCERPT: Maximum-degree surplus conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-051991 | MATHDB-365080

Conlon-Fox-Kwan-Sudakov conjecture on surplus in hypergraph cuts

摘录 / EXCERPT: Conlon-Fox-Kwan-Sudakov conjecture. The surplus should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-051992 | MATHDB-365093

Optimal dual-separable embeddings of complete graphs

摘录 / EXCERPT: Let $\delta_1(c)$ denote the minimum genus of a dual-separable embedding of a c-connected graph, and let $\gamma(K_{c+1})$ denote the genus of the complete graph K_{c+1}. Optim...

conjecture | open | MathDB 页面 | 原文线索

V2-051993 | MATHDB-365096

Petruševski-Škrekovski conjecture on odd colorings of planar graphs

摘录 / EXCERPT: Let G be a planar graph. An odd coloring of G is a proper vertex coloring such that every non-isolated vertex has a color appearing an odd number of times in its neighborhood;...

conjecture | open | MathDB 页面 | 原文线索

V2-051994 | MATHDB-365130

The normal chromatic-index formulation of Petersen coloring

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let $\chi'_{N}(G)$ denote its normal chromatic index, the least number of colors in a normal edge-coloring of G. The normal chromatic-ind...

conjecture | open | MathDB 页面 | 原文线索

V2-051995 | MATHDB-365132

The non-conflicting flow conjecture for 3-edge-connected cubic graphs

摘录 / EXCERPT: Let G be a 3-edge-connected cubic graph different from the Petersen graph. A nowhere-zero $Z_2\times Z_2\times Z_2$-flow is a flow f whose edge values are nonzero elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-051996 | MATHDB-365141

Spanning triangulated subsurface conjecture for triangulations of surfaces

摘录 / EXCERPT: Spanning triangulated subsurface conjecture. For all $g \geq 0$, every triangulation of a surface of Euler genus g contains a spanning subgraph which is a triangulated...

conjecture | open | MathDB 页面 | 原文线索

V2-051997 | MATHDB-365143

Improved matching-number bound for Laplacian eigenvalue sums

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with n non-isolated vertices, and let $\nu(G)$ denote the maximum size of a matching in G. Let $L(G)$ be the Laplacian matrix of G, with eigenvalues...

conjecture | open | MathDB 页面 | 原文线索

V2-051998 | MATHDB-365144

Covering-number bound for Laplacian eigenvalue sums

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $\tau(G)$ denote its covering number, and let $L(G)$ be the Laplacian matrix of G. For $1\leq i\leq |V|$, write $\lambda_i(L(G))$ for the i-th lar...

conjecture | open | MathDB 页面 | 原文线索

V2-051999 | MATHDB-365149

Cumulative-XOR conjecture for dictionaries and subset generators

摘录 / EXCERPT: Cumulative-XOR conjecture. After substitution, the cumulative XOR yields a string of ones across all paths or dimensional components, regardless of the subset size or dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-052000 | MATHDB-365162

The optimal-girth conjecture for Wiener-minimizing unicyclic graphs

摘录 / EXCERPT: Let $D=(d_1,d_2,\dots,d_{t-1},d_t,1,\dots,1)$ be a degree sequence with $d_t\geq 2$ and $d_2\neq 2$, and let G be a unicyclic graph with degree sequence D. Define ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052001 | MATHDB-365170

The bounded multiplicity extension for properly edge-coloured graphs

摘录 / EXCERPT: Let m and k be positive integers. The bounded multiplicity extension. There exists a graph with chromatic number at least k that has a proper edge-colouring such that each cy...

conjecture | open | MathDB 页面 | 原文线索

V2-052002 | MATHDB-365207

The hitting-time conjecture for random graph properties associated with a strictly 2-balanced graph

摘录 / EXCERPT: Hitting-time conjecture. With high probability, every index at which G_i belongs to $\mathcal Q_H$ is an index at which G_i belongs to $\mathcal T_H$.

conjecture | open | MathDB 页面 | 原文线索

V2-052003 | MATHDB-365216

Tomon's incomparability-graph blowup conjecture

摘录 / EXCERPT: Let $k\ge 3$, let $\varepsilon>0$, and let G be an n-vertex incomparability graph. Here $\varrho_2(G)$ denotes the edge density of G, and $K_k[t]$ denotes the complete k-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-052004 | MATHDB-365217

The conjectured exact value of the Ramsey parameter f(k)

摘录 / EXCERPT: Let $f(k)$ be the Ramsey parameter defined in the source: the minimum parameter governing the relevant dependency-digraph Ramsey problem, and let $g(k)$ be the corresponding ordere...

conjecture | open | MathDB 页面 | 原文线索

V2-052005 | MATHDB-365218

The clique-density conjecture for complements of string graphs

摘录 / EXCERPT: Let $s\ge 1$, let $\varepsilon>0$, and let G be the complement of a string graph. Here $\varrho_{4s+1}(G)$ denotes the density of copies of K_{4s+1} in G, and $K_{s+1}[t]$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-052006 | MATHDB-365224

Edge bound conjecture for minimally ${\cal R}_d$-connected graphs

摘录 / EXCERPT: Edge-bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052007 | MATHDB-365225

Asymptotic extremal conjecture for 3-uniform hypergraphs with bounded matching number

摘录 / EXCERPT: Let F be a graph, let $\rchi(F)$ denote its chromatic number, let $q(F)$ be the minimum number of red vertices in a strong red-blue coloring of F, and let H_i be the 3-unifor...

conjecture | open | MathDB 页面 | 原文线索

V2-052008 | MATHDB-365226

Asymptotic extremal conjecture for powers of graphs with bounded matching number

摘录 / EXCERPT: Let G be a graph with chromatic number k, let G^r denote its r-uniform power, and let $m'(G)$ be the number of edges between a specified pair of color classes of a $(k-1)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-052009 | MATHDB-365229

Wang's conjecture on e-positivity of theta graphs

摘录 / EXCERPT: Wang's conjecture. All theta graphs θ_{abc} are e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-052010 | MATHDB-365230

Submodularity characterization of resistance nonnegativity

摘录 / EXCERPT: Let G be a graph with vertex set V, let c be normalized edge weights, and let $\mathcal{S}(G)\subseteq[0,1]^{2^V}$ be the space of resistance capacities. A set function on...

conjecture | open | MathDB 页面 | 原文线索

V2-052011 | MATHDB-365232

Pardey-Rautenbach conjecture on almost colour-balanced perfect matchings

摘录 / EXCERPT: Pardey-Rautenbach conjecture. If $c\colon E(K_{2kn})\to [k]$ is colour-balanced, then there is a perfect matching M of K_{2kn} such that

conjecture | open | MathDB 页面 | 原文线索

V2-052012 | MATHDB-365238

Nevo-Petersen-Tenner balanced-complex conjecture for gamma vectors of flag spheres

摘录 / EXCERPT: Let K be a flag simplicial sphere, and let $\gamma(K)$ denote its gamma vector. A simplicial complex is balanced if, for its vertex set V, there is a map $\kappa:V\to[d]$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-052013 | MATHDB-365244

Metric dimension formula for Villarceau grids

摘录 / EXCERPT: Metric dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052014 | MATHDB-365256

The bunkbed conjecture for graph percolation

摘录 / EXCERPT: Let G be a graph, and let $BB(G)$ be its bunkbed graph with two layers, whose corresponding vertices are denoted x^- and x^+ for $x obreak\text{ in } obreak V(G)$. Let ... be a...

conjecture | open | MathDB 页面 | 原文线索

V2-052015 | MATHDB-365275

Guo's monotonicity conjecture for positive square energy under edge addition

摘录 / EXCERPT: Guo's monotonicity conjecture. Adding an edge should not decrease positive square energy:

conjecture | open | MathDB 页面 | 原文线索

V2-052016 | MATHDB-365276

Nikiforov's extremal p-energy conjecture for trees

摘录 / EXCERPT: Nikiforov's extremal p-energy conjecture. For $1\leq p\leq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-052017 | MATHDB-365277

Positive p-energy path-minimization conjecture for connected graphs

摘录 / EXCERPT: Positive p-energy path-minimization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052018 | MATHDB-365278

The realization conjecture for winding-number invariants of almost embeddings of K_4

摘录 / EXCERPT: Realization conjecture. For any integers n_1,n_2,n_3,n_4 whose sum is odd, there exists an almost embedding $f:K_4\to\mathbb R^2$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052019 | MATHDB-365285

Rafla's conjecture on crossing-free Hamiltonian cycles in simple drawings

摘录 / EXCERPT: Let K_n be the complete graph, and let a simple drawing be a drawing in which any two edges have at most one common point and no two edges incident to the same vertex cross. A cr...

conjecture | open | MathDB 页面 | 原文线索

V2-052020 | MATHDB-365286

The Hamiltonian-path strengthening of Rafla's conjecture

摘录 / EXCERPT: Let K_n be the complete graph, let $\mathcal{D}$ be a simple drawing of it on $n\ge 2$ vertices, and let v and w be distinct vertices of $\mathcal{D}$. A crossing-free Hamilt...

conjecture | open | MathDB 页面 | 原文线索

V2-052021 | MATHDB-365298

Schmeichel's basis-number conjecture for toroidal graphs

摘录 / EXCERPT: A graph G is toroidal if it can be embedded on the torus, and its basis number $\operatorname{bn}(G)$ is the smallest integer k for which G has a cycle-space basis in which e...

conjecture | open | MathDB 页面 | 原文线索

V2-052022 | MATHDB-365299

Uniqueness conjecture for the toroidal penny graph embedding of $K_{3,3}$

摘录 / EXCERPT: Uniqueness conjecture for the $K_{3,3}$ embedding. The proposed embedding of $K_{3,3}$, with coordinates as described in the table, is unique up to an isometry.

conjecture | open | MathDB 页面 | 原文线索

V2-052023 | MATHDB-365300

Bickel-Hong's contact-order conjecture for colored graph representations

摘录 / EXCERPT: Bickel-Hong's contact-order conjecture. The order of contact of f is $2n$. This conjecture concerns the behavior of level curves near boundary singularities; it has been proved i...

conjecture | open | MathDB 页面 | 原文线索

V2-052024 | MATHDB-365301

Contact-order distance conjecture for colored graph representations

摘录 / EXCERPT: Let G be a graph whose vertices are colored with z and w, and let i be a vertex from which the representation is computed. The contact order is the order describing how the...

conjecture | open | MathDB 页面 | 原文线索

V2-052025 | MATHDB-365311

Partite saturation limit conjecture

摘录 / EXCERPT: Fix $k\geq 2$, let K_k^n be the complete k-partite graph with n vertices in each part, and let F be a subgraph of K_k^n. Partite saturation limit conjecture. The limit .....

conjecture | open | MathDB 页面 | 原文线索

V2-052026 | MATHDB-365312

Complete bipartite even-cycle partite saturation conjecture

摘录 / EXCERPT: Let K_{n_1,n_2} be a complete bipartite graph, and let $C_{2\ell}$ denote the cycle of length 2ℓ. Even-cycle bipartite saturation conjecture. For $\ell\geq 4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-052027 | MATHDB-365314

Extremal structure for saturation by cycles of length at least ℓ

摘录 / EXCERPT: Let K_k^n be the complete k-partite graph with n vertices in each part, let $C_{\geq\ell}$ be the family of cycles of length at least ℓ, and let $\Omega^{(\ell,k,n)}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-052028 | MATHDB-365341

Thomassen's conjecture on good vertex pairs in arc-strong digraphs

摘录 / EXCERPT: Thomassen's conjecture. There exists an integer K such that every K-arc-strong digraph $D=(V,A)$ has a good (u,v)-pair for every choice of $u,v\in V$.

conjecture | open | MathDB 页面 | 原文线索

V2-052029 | MATHDB-365342

Bang-Jensen and Yeo's strong arc decomposition conjecture

摘录 / EXCERPT: Bang-Jensen and Yeo's conjecture. There exists an integer K such that every K-arc-strong digraph has a strong arc decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-052030 | MATHDB-365343

Bang-Jensen and Wang's conjecture on branchings in semicomplete split digraphs

摘录 / EXCERPT: Bang-Jensen and Wang's conjecture. Every 2-arc-strong semicomplete split digraph D contains a good (u,v)-pair for every choice of vertices u,v in D.

conjecture | open | MathDB 页面 | 原文线索

V2-052031 | MATHDB-365353

The Hamiltonian decomposition conjecture for regular graphs

摘录 / EXCERPT: Let G be an N-regular graph on n vertices. A Hamiltonian cycle is a cycle that meets every vertex, and a perfect matching is a matching covering every vertex. The Hamiltonian...

conjecture | open | MathDB 页面 | 原文线索

V2-052032 | MATHDB-365354

The perfect 1-factorization conjecture for complete graphs

摘录 / EXCERPT: Let K_{2n} be the complete graph on $2n$ vertices. A 1-factorization is a decomposition of its edges into perfect matchings, and it is perfect when the union of any two distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-052033 | MATHDB-365362

Exact bipartite Komlós-Sárközy-Szemerédi conjecture

摘录 / EXCERPT: Exact bipartite Komlós-Sárközy-Szemerédi conjecture. For each $\gamma>0$, there are $c,n_0>0$ such that for every $n\ge n_0$, if $G=(A,B,E)$ is a balanced bipartite graph on $2n$ v...

conjecture | open | MathDB 页面 | 原文线索

V2-052034 | MATHDB-365381

Conjecture on regular nut graphs from circulant and Cartesian graph constructions

摘录 / EXCERPT: Let $\mathcal{D}(n,t)$ denote the graph used in the construction, let F_3 and F_5 be the graphs shown in the paper, and let $\operatorname{\Box}$ denote the Cartesian product o...

conjecture | open | MathDB 页面 | 原文线索

V2-052035 | MATHDB-365387

Polarisation conjecture for the co-evolutionary opinion model

摘录 / EXCERPT: Let $u\in[0,1]$, let $\vec{v}(t,u)=(f_+(t,u),f_-(t,u))^\top$ be the vector of opinion densities, and let $\pi^g_+,\pi^g_-,\pi^r_+,\pi^r_-\in[0,1]$ and $q\in\mathbb N$ be the model...

conjecture | open | MathDB 页面 | 原文线索

V2-052036 | MATHDB-365391

Partial-fraction decomposition conjecture for flat-space wavefunctions

摘录 / EXCERPT: Flat-space partial-fraction decomposition conjecture. The flat-space wavefunction can be decomposed into partial fractions as

conjecture | open | MathDB 页面 | 原文线索

V2-052037 | MATHDB-365411

The fixed-manifold correspondence for unpunctured class-S theories

摘录 / EXCERPT: Let $V_{g,0}$ be the vertex operator algebra associated with the unpunctured genus-g theory, with simple modules $\{L_a\}$ and conformal dimensions $h(L_a)$. Let ${\cal M}_{g,0}$...

conjecture | open | MathDB 页面 | 原文线索

V2-052038 | MATHDB-365412

The fixed-manifold correspondence for punctured class-S theories

摘录 / EXCERPT: Let $V_{g,n}$ be the vertex operator algebra associated with the genus-g theory with n regular punctures, and let $\{L_a\}$ be its simple modules. Let ${\cal M}_{g,n}$ be the m...

conjecture | open | MathDB 页面 | 原文线索

V2-052039 | MATHDB-365466

Moonshine uniqueness conjecture for the monster vertex operator algebra

摘录 / EXCERPT: A vertex operator algebra is strong-rational if it satisfies the paper's strong rationality conditions, and it is holomorphic if its representation category is trivial. The moonshi...

conjecture | open | MathDB 页面 | 原文线索

V2-052040 | MATHDB-365467

Primitive holomorphic extension conjecture for positive vertex operator algebras

摘录 / EXCERPT: Let V be a positive, strong-rational vertex operator algebra. A primitive embedding is an embedding into a holomorphic vertex operator algebra W whose commutant...

conjecture | open | MathDB 页面 | 原文线索

V2-052041 | MATHDB-365468

Countability conjecture for Witt classes of strong-rational vertex operator algebras

摘录 / EXCERPT: A Witt class is the collection of strong-rational vertex operator algebras having the same Witt class data. Countability conjecture. Every Witt class of strong-rational vertex oper...

conjecture | open | MathDB 页面 | 原文线索

V2-052042 | MATHDB-365470

The bulk-genus orbifold conjecture for anisotropic representations

摘录 / EXCERPT: Let V and V' be positive, strong-rational vertex operator algebras in the same bulk genus, and suppose their representation categories are anisotropic. Bulk-genus orbifold conj...

conjecture | open | MathDB 页面 | 原文线索

V2-052043 | MATHDB-365491

Wheel characterization of the circumference gap

摘录 / EXCERPT: Wheel characterization conjecture. Let G be a 2-connected graph with

conjecture | open | MathDB 页面 | 原文线索

V2-052044 | MATHDB-365502

The character conjecture for distinguishing dual fullerene graphs

摘录 / EXCERPT: Character conjecture. For $lpha,\beta\rtf]\in(0,1)$ and feasible n_1,n_2, if T_{n_1} and T_{n_2} are dual fullerenes, then

conjecture | open | MathDB 页面 | 原文线索

V2-052045 | MATHDB-365508

Cambie-Goedgebeur-Jooken conjecture on independent sets in dense regular graphs

摘录 / EXCERPT: Let E_{n-d} denote the edgeless graph on $n-d$ vertices, and let the mutual join of graphs mean the graph obtained by joining every vertex in one graph to every vertex in each of...

conjecture | open | MathDB 页面 | 原文线索

V2-052046 | MATHDB-365521

The asymptotic minimum-size conjecture for highly connected chordal bipartite graphs

摘录 / EXCERPT: Asymptotic minimum-size conjecture. There exists an integer N such that, for every $n\ge N$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-052047 | MATHDB-365524

Laplacian zero-eigenvalue multiplicity conjecture for connected uniform hypergraphs

摘录 / EXCERPT: Let H be a connected k-uniform hypergraph, let $\mathcal{L}(H)$ be its Laplacian tensor, let $\mathcal{A}(H)$ be its adjacency tensor, and let rho be the spectral radius of...

conjecture | open | MathDB 页面 | 原文线索

V2-052048 | MATHDB-365525

The image conjecture for the Dehn-Nielsen-Baer map of a graph-like surface

摘录 / EXCERPT: Let S be an infinite-type surface with compact boundary, properly homotopy equivalent to a locally finite infinite graph $\\G$, and let Θ be the induced homomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-052049 | MATHDB-365533

Terwilliger's finiteness conjecture for amply regular graphs

摘录 / EXCERPT: Let G be an amply regular graph with parameters (n,d,α,β), where n is possibly infinite. Terwilliger's conjecture. If $\beta\geq 2$, then G is finite. Terwilliger'...

conjecture | open | MathDB 页面 | 原文线索

V2-052050 | MATHDB-365534

Qiao, Park and Koolen's bounded-diameter conjecture for amply regular graphs

摘录 / EXCERPT: Let G be a connected amply regular graph with parameters (n,d,α,β), and let $0<\varepsilon<1$ be a real number. Qiao, Park and Koolen's conjecture. There exists a cons...

conjecture | open | MathDB 页面 | 原文线索

V2-052051 | MATHDB-365573

Cichacz-Suchan irregular-labeling conjecture for digraphs

摘录 / EXCERPT: Let $\overrightarrow{G}$ be a digraph of order n whose weakly connected components all have order at least 3, and let Γ be an Abelian group. A Γ-irregular labelin...

conjecture | open | MathDB 页面 | 原文线索

V2-052052 | MATHDB-365610

The constant-factor vertex-deletion conjecture for acyclic digraphs

摘录 / EXCERPT: Vertex-deletion conjecture. There exists a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-052053 | MATHDB-365611

The extension conjecture for oriented trees

摘录 / EXCERPT: Extension conjecture. There is an absolute constant C such that, for every integer k, every k-extension of an oriented tree of order n is $C^k(2n-2)$-unavoidable; that is,

conjecture | open | MathDB 页面 | 原文线索

V2-052054 | MATHDB-365612

The extension-preservation conjecture for linearly unavoidable families

摘录 / EXCERPT: Extension-preservation conjecture. If $\mathcal F$ is linearly unavoidable, then the family of k-extensions of digraphs in $\mathcal F$ is also linearly unavoidable.

conjecture | open | MathDB 页面 | 原文线索

V2-052055 | MATHDB-365614

The non-natural tree conjecture for matroidal derived families

摘录 / EXCERPT: A tree is a hypergraph with the relevant tree property, and a hypergraph is natural when it belongs to a family containing a natural tree. A matroidal family is a family arising fr...

conjecture | open | MathDB 页面 | 原文线索

V2-052056 | MATHDB-365628

The nonzero-solution conjecture for distance-d polynomials of $H(d,3)$

摘录 / EXCERPT: Let $H(d,3)$ be the Hamming graph on length-d words over an alphabet of size 3, and let $\Phi_d(H(d,3))$ be the associated system of polynomials. A solution is all-nonzero if e...

conjecture | open | MathDB 页面 | 原文线索

V2-052057 | MATHDB-365651

The generalized Berge-Fulkerson conjecture for r-graphs

摘录 / EXCERPT: Let $r\geq 1$. An r-graph is the graph class defined in the source, and a perfect matching is a set of edges meeting every vertex exactly once. The generalized Berge-Fulkerson co...

conjecture | open | MathDB 页面 | 原文线索

V2-052058 | MATHDB-365652

The per-degree fractional perfect-matching conjecture for r-graphs

摘录 / EXCERPT: Let G be an r-graph. A (t,r)-PM is a multiset of $t\cdot r$ perfect matchings of G such that every edge of G is contained in exactly t of them. The per-degree fractiona...

conjecture | open | MathDB 页面 | 原文线索

V2-052059 | MATHDB-365660

The characterization of graphs with tree reconfiguration graphs

摘录 / EXCERPT: Tree reconfiguration graph conjecture. $\mathcal{R}(G)$ is a tree if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-052060 | MATHDB-365685

The partition conjecture for k-clique isolating sets

摘录 / EXCERPT: Let $k \geq 3$ be an integer. A k-clique isolating set of a graph is a vertex set whose closed neighborhood leaves no copy of K_k. The partition conjecture. Every connected gra...

conjecture | open | MathDB 页面 | 原文线索

V2-052061 | MATHDB-365686

The planar cycle-isolation partition conjecture

摘录 / EXCERPT: A cycle isolating set of a graph is a vertex set whose closed neighborhood leaves no cycle. The planar cycle-isolation partition conjecture. Every connected planar graph, except...

conjecture | open | MathDB 页面 | 原文线索

V2-052062 | MATHDB-365687

The isolation-number lower-bound conjectures

摘录 / EXCERPT: For a graph G, let ${\rm iso}(G,k)$ be the maximum number of pairwise disjoint k-clique isolating sets in a partition of $V(G)$, and let ${\rm iso}_c(G)$ be the analogous numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-052063 | MATHDB-365699

Füredi-Kahn-Seymour conjecture on chromatic indices of uniform hypergraphs

摘录 / EXCERPT: Füredi-Kahn-Seymour conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052064 | MATHDB-365700

Cambie's bounded-overlap conjecture for simultaneous edge-colourings

摘录 / EXCERPT: Cambie's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052065 | MATHDB-365707

Homology conjecture for flag complexes of complete digraphs

摘录 / EXCERPT: Let G_n be a complete digraph on n nodes, and let $\dFl(G_n)$ denote its flag complex. Write $H_k(\dFl(G_n))$ for the k-th homology group, and let $!n$ be the number of deran...

conjecture | open | MathDB 页面 | 原文线索

V2-052066 | MATHDB-365710

Maximum sigma_t-irregularity conjecture for connected triangle-free graphs

摘录 / EXCERPT: Triangle-free extremal conjecture. Among connected triangle-free graphs on n vertices, the extremal complete bipartite graphs have the maximum $sigma_t(G)$-irregularity.

conjecture | open | MathDB 页面 | 原文线索

V2-052067 | MATHDB-365711

Tree sigma_t-irregularity inequality conjecture

摘录 / EXCERPT: Tree irregularity inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052068 | MATHDB-365728

Hocquard-Lajou-Luar conjecture for subcubic planar graphs

摘录 / EXCERPT: For a sequence $S=(s_1,dots,s_k)$ of non-decreasing positive integers, an S-packing edge-coloring of a graph G is a partition of $E(G)$ into $E_1,dots,E_k$ such that the distan...

conjecture | open | MathDB 页面 | 原文线索

V2-052069 | MATHDB-365733

Kaneko's spanning-tree leaf-distance conjecture

摘录 / EXCERPT: Let $d\geq 3$ be an integer and let G be a connected graph of order $n\geq d+1$. For $S\subseteq V(G)$, let $i(G-S)$ denote the number of isolated vertices in $G-S$. Kaneko's con...

conjecture | open | MathDB 页面 | 原文线索

V2-052070 | MATHDB-365761

Pavez-Signé's spanning subdivision conjecture

摘录 / EXCERPT: Pavez-Signé's conjecture. For all $C\geq C_0$ and any integer m, if G is a graph on $n\geq Cm$ vertices with minimum degree at least $(1/2+\varepsilon)n$, then G contains a s...

conjecture | open | MathDB 页面 | 原文线索

V2-052071 | MATHDB-365762

The spanning subdivision conjecture for digraphs under non-edge degree-sum conditions

摘录 / EXCERPT: Spanning subdivision conjecture. There is a constant n_0 and a real number C_H depending only on H such that, whenever D has order n and the displayed condition holds for...

open_problem | open | MathDB 页面 | 原文线索

V2-052072 | MATHDB-365766

The conjectured fractional codegree threshold for generalized Steiner systems

摘录 / EXCERPT: For integers $n\geq r\geq 2$, a fractional $(r-1,r,n)$-Steiner system in an r-uniform hypergraph H is a function $\varphi:E(H)\to\mathbb{R}_{\geq0}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-052073 | MATHDB-365767

Largest-component conjecture for Boolean sublattices

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $2^{[n]}$ be its power set, and let $G_{\mathcal{F}}$ be the graph whose vertex set is a family $\mathcal{F}\subseteq 2^{[n]}$, with two vertices adja...

conjecture | open | MathDB 页面 | 原文线索

V2-052074 | MATHDB-365774

Weighted Turán conjecture for two-dimensional vector weights

摘录 / EXCERPT: Let ω be sufficiently large and let $G=(V,E)$ be a graph with clique number ω. Assign a vector $\vv_i\in\mathbb R^2$ to each vertex i, and define ... for each pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-052075 | MATHDB-365775

Generic circumsphere rigidity of complete uniform hypergraphs

摘录 / EXCERPT: Let d and n be positive integers such that ... The complete $(d+1)$-uniform hypergraph is generically circumsphere rigid in dimension d if its circumsphere framework is rigid f...

conjecture | open | MathDB 页面 | 原文线索

V2-052076 | MATHDB-365781

Li-Zhai-Shu cycle spectral extremal conjecture

摘录 / EXCERPT: Li-Zhai-Shu's cycle spectral extremal conjecture. For $k\geqslant 2$ and m sufficiently large, if G is C_{2k+1}-free or C_{2k+2}-free with given size m, then

conjecture | open | MathDB 页面 | 原文线索

V2-052077 | MATHDB-365782

Kazama-Suzuki duality conjecture for subregular and principal W-algebras

摘录 / EXCERPT: Let n be the parameter in the families of vertex algebras, and set ... Here $f_{\rm pr}$ and $f_{\rm sub}$ denote respectively principal and subregular nilpotent elements. Kazama-S...

conjecture | open | MathDB 页面 | 原文线索

V2-052078 | MATHDB-365808

The forbidden-octahedral-graph characterization of gentle graphic arrangements

摘录 / EXCERPT: Gentle graphic-arrangement conjecture. A graphic arrangement $\mathcal{A}(G)$ is gentle if and only if G_{oct} cannot be obtained from G by a series of edge contractio...

conjecture | open | MathDB 页面 | 原文线索

V2-052079 | MATHDB-365811

Máčajová-Škoviera odd-cut conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A perfect matching is a set of edges meeting every vertex exactly once, and an edge-cut is the set of edges joining a vertex subset to its comp...

conjecture | open | MathDB 页面 | 原文线索

V2-052080 | MATHDB-365835

The two-step hook-reduction conjecture for type A W-algebras

摘录 / EXCERPT: Let $\lambda=(\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_\ell)$ be a partition of N, and set ... Let $\mathrm{H}_{\mathbb{O}_{\widehat{\lambda}_i}}^0$ denote the BRST reduction as...

conjecture | open | MathDB 页面 | 原文线索

V2-052081 | MATHDB-365836

The successive thick-hook reduction conjecture for classical W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be $\mathfrak{so}_N$ or $\mathfrak{sp}_N$, and let ... be a partition parametrizing a nilpotent orbit, with $m_i\in\{1,2\}$ subject to the parity conditions for th...

conjecture | open | MathDB 页面 | 原文线索

V2-052082 | MATHDB-365838

The even-spin W-infinity extension conjecture for the symplectic universal algebra

摘录 / EXCERPT: Let $\mathcal{W}^{\mathfrak{sp}}_{\infty}(c,\mathsf{k})$ be the universal two-parameter vertex algebra, let $\mathrm{H}^0_{\ydiagram{2}}$ denote its Virasoro-type reduction, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-052083 | MATHDB-365842

Gray-code canonical coloring conjecture for powers of paths

摘录 / EXCERPT: Gray-code canonical coloring conjecture. The canonical edge-coloring ϕ is an optimal strong parity edge-coloring of P_n^ℓ.

conjecture | open | MathDB 页面 | 原文线索

V2-052084 | MATHDB-365857

The quadratic-logarithmic conjecture for regular subgraphs

摘录 / EXCERPT: Let $d(r,n)$ be the smallest average degree such that every n-vertex graph with average degree at least $d(r,n)$ contains an r-regular subgraph. Quadratic-logarithmic conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-052085 | MATHDB-365858

Regular-subhypergraph conjecture for linear hypergraphs

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let $\lambda\geq 1$, and let a k-uniform linear hypergraph be a hypergraph in which every edge has k vertices and any two distinct edges intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-052086 | MATHDB-365865

Lescure-Meyniel-Abu-Khzam-Langston conjecture on chromatic number and clique immersions

摘录 / EXCERPT: Let G be a graph, let $\chi(G)$ denote its chromatic number, and let K_t be the complete graph on t vertices. An H-immersion is an injective mapping of $V(H)$ to $V(G)$ tog...

conjecture | open | MathDB 页面 | 原文线索

V2-052087 | MATHDB-365866

Thomassen's conjecture on balanced clique subdivisions

摘录 / EXCERPT: Let G be a graph, let $d(G)$ denote its average degree, and let a balanced K_t-subdivision be a subdivision of K_t in which all subdivided edge-paths have the same length. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-052088 | MATHDB-365880

Tower-bound problem for induced Ramsey numbers of cliques

摘录 / EXCERPT: For graphs G and F, let $G\mathrel{\overset{\makebox[0pt]{\text{\normalfont\tiny\sffamily ind}}}{\rightarrow}} (F)^{K_r}$ mean that every 2-coloring of the complete subgraphs...

open_problem | open | MathDB 页面 | 原文线索

V2-052089 | MATHDB-365881

Narayanan-Schacht conjecture on the threshold for non-linear Hamilton cycles

摘录 / EXCERPT: Let $G_r(n,p)$ be the random r-uniform hypergraph, and let $C^{(r)}_{n,\ell}$ be the r-uniform ℓ-cycle on n vertices. Write $Z(C^{(r)}_{n,\ell})$ for the number of copie...

conjecture | open | MathDB 页面 | 原文线索

V2-052090 | MATHDB-365885

Existence of the holomorphic E(7)-orbifold of the Barnes-Wall lattice VOA

摘录 / EXCERPT: Let V_L be the lattice vertex operator algebra associated with the 128-dimensional Barnes-Wall lattice, and let $E(7)$ act on V_L. A holomorphic conformal field theory has cent...

conjecture | open | MathDB 页面 | 原文线索

V2-052091 | MATHDB-365887

The iterated-tripartite classification conjecture for off-diagonal 3-graph Ramsey numbers

摘录 / EXCERPT: The iterated-tripartite classification conjecture. For a 3-graph H, there exists a constant c depending only on H such that

conjecture | open | MathDB 页面 | 原文线索

V2-052092 | MATHDB-365888

The tower lower-bound conjecture for tightly connected hypergraph Ramsey numbers

摘录 / EXCERPT: The tower lower-bound conjecture. If $k>s$ and H is an s-tightly connected k-graph that is not k-partite, then there exists a positive constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-052093 | MATHDB-365889

The family-level tower lower-bound conjecture for tightly connected hypergraphs

摘录 / EXCERPT: The family-level tower lower-bound conjecture. If $k>s$, then there exists a positive constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-052094 | MATHDB-365918

Rational exponent conjecture for degenerate hypergraph Turán numbers

摘录 / EXCERPT: Let $r\ge 3$, and let $\mathcal{F}$ be a degenerate finite family of r-graphs satisfying ... for some constant $\alpha>r-2$. Then there exist constants $\beta>0$, $c>0$, and $C>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-052095 | MATHDB-365930

The spectral extremal containment conjecture for graphs with prescribed Turán deficit

摘录 / EXCERPT: Spectral extremal containment conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052096 | MATHDB-365943

Nonexistence conjecture for cubic 8-circulant nut graphs

摘录 / EXCERPT: A cubic 8-circulant nut graph is a cubic nut graph admitting an 8-circulant structure. Nonexistence conjecture for cubic 8-circulant nut graphs. There exists no cubic 8-cir...

conjecture | open | MathDB 页面 | 原文线索

V2-052097 | MATHDB-365948

Liu's modified degree-sequence conjecture for extremal nonregular graphs

摘录 / EXCERPT: Let $\mathcal{G}(n,\Delta)$ be the set of connected nonregular graphs of order n with maximum degree Δ that attain the maximum spectral radius. Suppose $\Delta\geq 3$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-052098 | MATHDB-365964

The 4-connected 2-cyclic graph edge conjecture

摘录 / EXCERPT: 4-connected 2-cyclic edge conjecture. If G is 4-connected and 2-cyclic, then

conjecture | open | MathDB 页面 | 原文线索

V2-052099 | MATHDB-365965

The alpha forest-cut conjecture

摘录 / EXCERPT: α-FC conjecture. If G has no forest cut, then

conjecture | open | MathDB 页面 | 原文线索

V2-052100 | MATHDB-365968

Nonisomorphic trees are not LV-equivalent

摘录 / EXCERPT: A tree is a hypergraph whose hyperedges all have size two, and an LV-system is a Lotka-Volterra system associated with such a hypergraph. Two associated systems are LV-equivalent w...

conjecture | open | MathDB 页面 | 原文线索

V2-052101 | MATHDB-365976

The spectral extremal conjecture for 3-chromatic graph families

摘录 / EXCERPT: Let $\mathcal F$ be a finite graph family, let $\chi(F)$ denote the chromatic number of a graph F, let $\mathrm{ex}(n,\mathcal F)$ be the maximum number of edges in an...

conjecture | open | MathDB 页面 | 原文线索

V2-052102 | MATHDB-365979

Rainbow Hamilton cycle conjecture for bounded edge-colourings of Dirac graphs

摘录 / EXCERPT: Let G be a Dirac graph on n vertices, meaning a graph with minimum degree at least $n/2$, and let its edge-colouring be proper if every pair of incident edges receives differen...

conjecture | open | MathDB 页面 | 原文线索

V2-052103 | MATHDB-365986

The superexponential number-of-values conjecture for general graphs

摘录 / EXCERPT: Let $\mathcal T'(n)$ be the set of spanning-tree numbers $\tau(G)$ as G ranges over all simple graphs on n vertices, without a planarity assumption. Superexponential growth con...

conjecture | open | MathDB 页面 | 原文线索

V2-052104 | MATHDB-365990

Yu, Li and Peng's spectral extremal conjecture for fan-free graphs

摘录 / EXCERPT: Let $k\geq 2$ and let m be sufficiently large. For a graph G, write $\rho(G)$ for its spectral radius, let H_{r} denote the fan graph with r vertices, let K_k be the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-052105 | MATHDB-365999

Asymptotic equivalence of the edge functions for uniform systems

摘录 / EXCERPT: Let r and t be integers with $r\geq 3$ and $t\geq r$. For an (r,t)-system $(H,\mathcal{F})$, let $e_{r,t}(s)$ be the minimum of $e(H)$ over systems with $|\mathcal{F}|=s$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-052106 | MATHDB-366000

Conical-vertex conjecture for minimum-degree twin-free saturated graphs

摘录 / EXCERPT: Let $\operatorname{tsat}(n,K_r,t)$ denote the minimum number of edges in a twin-free K_r-saturated graph on n vertices with minimum degree at least t. A vertex is conical if...

conjecture | open | MathDB 页面 | 原文线索

V2-052107 | MATHDB-366001

Conical-vertex conjecture for extremal uniform set systems

摘录 / EXCERPT: For integers $r\geq 3$, $t\geq r-2$, and $m\geq 0$, let $s_{r,t}(m)$ be the maximum size of the family in an (r,t)-system $(H,\mathcal{F})$ with $|H|=m$. An extremal system is on...

conjecture | open | MathDB 页面 | 原文线索

V2-052108 | MATHDB-366002

Conical-vertex conjecture for extremal edge-minimizing systems

摘录 / EXCERPT: For integers $r\geq 3$ and $t\geq r$, let $e_{r,t}(s)$ and $e_{r,t}'(s)$ be the minimum number of edges in the underlying graph of an (r,t)-system and a maximal (r,t)-system, r...

conjecture | open | MathDB 页面 | 原文线索

V2-052109 | MATHDB-366007

Optimal universal hypergraph conjecture for partition-injective tree homomorphisms

摘录 / EXCERPT: Let $D\in\mathbb{N}$. An (n,d,λ)-graph is a graph on n vertices with degree d and second eigenvalue bounded by λ. Let T be a tree of maximum degree at most...

conjecture | open | MathDB 页面 | 原文线索

V2-052110 | MATHDB-366011

Maximal-status conjecture for radial Moore graphs

摘录 / EXCERPT: Let G_d be the radial Moore graph of degree d, and let the status of a vertex be the sum of its distances to all vertices of G_d. Maximal-status conjecture. The radial Moore...

conjecture | open | MathDB 页面 | 原文线索

V2-052111 | MATHDB-366017

Exponential upper-bound conjecture for the three-uniform color-avoiding Ramsey function

摘录 / EXCERPT: Let $R_3(n;3,2)$ be the least integer N such that every three-coloring of the edges of the complete three-uniform hypergraph $K_N^{(3)}$ contains a copy of $K_n^{(3)}$ using at m...

conjecture | open | MathDB 页面 | 原文线索

V2-052112 | MATHDB-366098

Mirror-symmetry correspondence for punctured class-S theories

摘录 / EXCERPT: Mirror-symmetry conjecture. When $n>0$, there is a bijection between the simple modules $\{L_a\}$ of $V_{g,n}$ and the fixed manifolds $\mathcal{M}_{g,n}^T=\{M_a\}$ such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-052113 | MATHDB-366099

Mixed-Hodge-polynomial conjecture for modular data

摘录 / EXCERPT: Mixed-Hodge-polynomial conjecture. The total number of fixed manifolds of $\mathcal{M}_{g,n}$, equivalently the number of simple modules of $V_{g,n}$, is

conjecture | open | MathDB 页面 | 原文线索

V2-052114 | MATHDB-366131

Hliněný's higher-genus non-orientable cover conjecture

摘录 / EXCERPT: Let Σ be a connected compact non-orientable surface without boundary. A finite Σ-cover is a finite cover of a connected graph that embeds in Σ. Hliněný's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-052115 | MATHDB-366132

The orientable higher-genus extension of Negami's conjecture

摘录 / EXCERPT: Let Σ be an orientable surface of Euler genus g, and let a finite Σ-cover mean a finite cover of a connected graph that embeds in Σ. Orientable higher-genus...

conjecture | open | MathDB 页面 | 原文线索

V2-052116 | MATHDB-366133

The high-genus surface-cover extension of Negami's conjecture

摘录 / EXCERPT: Let Σ be a connected compact surface without boundary and let g be its Euler genus. A finite Σ-cover is a finite cover of a connected graph that embeds in Σ....

conjecture | open | MathDB 页面 | 原文线索

V2-052117 | MATHDB-366137

Cutset finiteness conjecture for graphs of dimension greater than one

摘录 / EXCERPT: Let G be a graph, and let $\operatorname{Dim}(G)$ denote its dimension parameter. Cutset finiteness conjecture. Every graph satisfying ... has ... The conjecture proposes that finite...

conjecture | open | MathDB 页面 | 原文线索

V2-052118 | MATHDB-366145

Morell and Skutella's two-sided convex-combination conjecture for acyclic single-source flows

摘录 / EXCERPT: Morell and Skutella's conjecture. Any fractional flow $(x_e)_{e\in E}$ can be expressed as a convex combination of unsplittable flows $(y_e)_{e\in E}$ satisfying both bounds. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-052119 | MATHDB-366150

Burris-Schelp conjecture on the vertex-distinguishing chromatic index

摘录 / EXCERPT: Burris-Schelp conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052120 | MATHDB-366151

The two-value criterion for generalized distance-matrix cospectrality

摘录 / EXCERPT: Let G and H be graphs, and suppose that a single similarity matrix establishes cospectrality of their generalized distance matrices for two distinct values of q, neither of w...

conjecture | open | MathDB 页面 | 原文线索

V2-052121 | MATHDB-366155

Conjecture on the additive gap between integral and fractional t-clique covers

摘录 / EXCERPT: For a graph G, let $\mathcal{C}\mathcal{C}_t(G)$ and $\mathcal{C}\mathcal{C}^*_t(G)$ denote its integral and fractional t-clique cover numbers, respectively. Additive-gap conje...

conjecture | open | MathDB 页面 | 原文线索

V2-052122 | MATHDB-366168

Tait's extremal A_0-spectral radius conjecture for $K_{a,b}$-minor-free graphs

摘录 / EXCERPT: Let a and b be integers with $b\geqslant a\geqslant 2$, and write ... For graphs, let $A_0(G)=A(G)$, let $K_{a-1}\vee (kK_b\cup K_t)$ denote the join of K_{a-1} with the disjoi...

conjecture | open | MathDB 页面 | 原文线索

V2-052123 | MATHDB-366169

Nevo-Peeva conjecture on eventual linear resolutions of gapfree graph edge ideals

摘录 / EXCERPT: Let G be a finite simple graph with vertex set $V(G)=\{1,\ldots,n\}$, let $S=K[x_1,\ldots,x_n]$ be a polynomial ring over a field K, and let $I(G)\subseteq S$ be the edge ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-052124 | MATHDB-366255

Dual Barnette conjecture for Eulerian planar triangulations

摘录 / EXCERPT: Let G be an Eulerian planar triangulation, meaning a planar triangulation whose vertices all have even degree. A vertex set is acyclic when it induces a forest. Dual Barnette con...

conjecture | open | MathDB 页面 | 原文线索

V2-052125 | MATHDB-366256

Neumann-Lara's dicoloring conjecture for oriented planar triangulations

摘录 / EXCERPT: Let G be an oriented planar triangulation, meaning a directed planar triangulation with no directed cycles of length 1 or 2. A vertex set is acyclic when it induces a subdigr...

conjecture | open | MathDB 页面 | 原文线索

V2-052126 | MATHDB-366257

Eulerian Neumann-Lara conjecture for oriented planar graphs

摘录 / EXCERPT: Let G be an Eulerian oriented planar graph, meaning a connected oriented planar graph in which every vertex has even degree. A vertex set is acyclic when it induces a subdigraph...

conjecture | open | MathDB 页面 | 原文线索

V2-052127 | MATHDB-366258

Treewidth-connectivity partition conjecture for graphs

摘录 / EXCERPT: Let G be a graph, and let k be an integer such that G has treewidth at most k and connectivity at least k. A vertex set inducing a graph of treewidth at most $k-1$ and co...

conjecture | open | MathDB 页面 | 原文线索

V2-052128 | MATHDB-366272

The odd-diameter almost seesaw conjecture for the maximum second Steklov eigenvalue

摘录 / EXCERPT: Odd-diameter almost seesaw conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052129 | MATHDB-366279

Alspach et al.'s wreath-product conjecture for hamiltonian decomposable digraphs

摘录 / EXCERPT: Alspach et al.'s wreath-product conjecture. If G and H are hamiltonian decomposable directed graphs, then $G\wr H$ is also hamiltonian decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-052130 | MATHDB-366283

Feige's even-cover girth conjecture for uniform hypergraphs

摘录 / EXCERPT: Feige's conjecture. There exists a sufficiently large constant $C>0$ such that, for sufficiently large n and every ℓ, every such H with

conjecture | open | MathDB 页面 | 原文线索

V2-052131 | MATHDB-366284

Keevash-Mubayi-Sudakov-Verstraëte rainbow-cycle conjecture

摘录 / EXCERPT: Keevash-Mubayi-Sudakov-Verstraëte conjecture. There is a constant C such that every properly edge-colored n-vertex graph with at least $Cn\log n$ edges contains a rainbow cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-052132 | MATHDB-366300

Lin-Miao-Guo conjecture on the maximum A_{α}-spread

摘录 / EXCERPT: Lin-Miao-Guo conjecture. If $1/2\leq\alpha<1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052133 | MATHDB-366303

Dense odd-uniform hypergraphs have large tight components

摘录 / EXCERPT: Odd-uniformity component conjecture. The hypergraph G contains a tight component C with

conjecture | open | MathDB 页面 | 原文线索

V2-052134 | MATHDB-366305

Pósa's conjecture for squares of Hamilton cycles in triple systems

摘录 / EXCERPT: Pósa-type conjecture for triple systems. For every $\varepsilon>0$, there is n_0 such that every 3-uniform hypergraph G of order $n\geq n_0$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-052135 | MATHDB-366306

Tight connectivity of the 4-clique graph in dense triple systems

摘录 / EXCERPT: 4-clique connectivity conjecture. For every $\varepsilon>0$, there is n_0 such that every 3-uniform hypergraph G of order $n\geq n_0$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-052136 | MATHDB-366307

Polylogarithmic cluster-size extension of the hypergraph bandwidth theorem

摘录 / EXCERPT: Cluster-size conjecture. Theorem remains valid for H with clusters of size at most

conjecture | open | MathDB 页面 | 原文线索

V2-052137 | MATHDB-366314

Bennett-Bohman folklore conjecture for complete-hypergraph removal

摘录 / EXCERPT: Let $K^{(k)}_\ell$ be the k-uniform complete hypergraph on ℓ vertices, and let $R_n(K^{(k)}_\ell)$ denote the final number of edges in its removal process. Bennett-Bohman fo...

conjecture | open | MathDB 页面 | 原文线索

V2-052138 | MATHDB-366315

Heuristic asymptotic conjecture for the free process

摘录 / EXCERPT: Let $k\geq2$, and let $\mathcal F$ be a strictly k-balanced k-uniform hypergraph with k-density ρ. Write $v(\mathcal F)$ and $e(\mathcal F)$ for its numbers of vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-052139 | MATHDB-366336

Unique NAC-colouring conjecture for minimally rigid graphs

摘录 / EXCERPT: Let G be a minimally rigid graph. Unique NAC-colouring conjecture. We have ... if and only if either $G\in\mathcal{G}_{sc}$ and it has exactly one 3-prism subgraph, or G is a...

conjecture | open | MathDB 页面 | 原文线索

V2-052140 | MATHDB-366357

Ghalavand-Klavžar-Li local metric dimension bound by clique number

摘录 / EXCERPT: Ghalavand's conjecture. The local metric dimension of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052141 | MATHDB-366388

The characterization conjecture for uniquely C_{4}^{+}-saturated graphs

摘录 / EXCERPT: Uniquely C_{4}^{+}-saturation conjecture. A graph G is nontrivial uniquely C_{4}^{+}-saturated if and only if G is a strongly regular graph with parameters $(n,k,0,2)$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-052142 | MATHDB-366395

Pósa-Seymour conjecture for powers of Hamilton cycles

摘录 / EXCERPT: Let G be a graph on $n\geq k\geq 2$ vertices, and let $\delta(G)$ denote its minimum degree. The Pósa-Seymour conjecture. If ... then G contains the kth power of a Hamilton cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-052143 | MATHDB-366407

The off-diagonal F-Ramsey conjecture

摘录 / EXCERPT: Let G be a graph, and let $r_{K_s}(t_1,t_2)$ denote the minimum number of copies of K_s in a graph that is (t_1,t_2)-Ramsey. Let $r(t_1,t_2)$ be the Ramsey number, the smalle...

conjecture | open | MathDB 页面 | 原文线索

V2-052144 | MATHDB-366410

Odd-region obstruction conjecture for optimal 1-embedded graphs

摘录 / EXCERPT: Let G be an optimal 1-embedded graph on the torus or the Klein bottle. Let $Q(G)$ denote its quadrangulation, and call a region odd weighted when its associated weight is odd....

conjecture | open | MathDB 页面 | 原文线索

V2-052145 | MATHDB-366412

The uniquely restricted edge coloring bound for planar graphs

摘录 / EXCERPT: Uniquely restricted edge coloring conjecture. Every planar graph G with maximum degree Δ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052146 | MATHDB-366413

The tight tree Ramsey growth conjecture for non-trivial tight hypergraph trees

摘录 / EXCERPT: Tight tree Ramsey growth conjecture. For $r\geq 5$, if T is a non-trivial tight r-tree, then there exist constants $c_1,c_2>0$ such that, for every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-052147 | MATHDB-366426

Conjecture on truncated generalized block Laplacian spectra

摘录 / EXCERPT: Let G be a graph on n vertices, with distinct degrees $\delta_1 > \delta_2 > \cdots > \delta_p$. Set $r = \lceil\log_2 n\rceil + 1$, partition $V = \bigsqcup_{i=1}^r V_i$ by ta...

conjecture | open | MathDB 页面 | 原文线索

V2-052148 | MATHDB-366434

Behzad-Chartrand-Wall conjecture for oriented digraphs

摘录 / EXCERPT: Behzad-Chartrand-Wall conjecture. Every n-vertex oriented digraph with minimum out-degree and minimum in-degree at least $n/3$ contains a directed triangle.

conjecture | open | MathDB 页面 | 原文线索

V2-052149 | MATHDB-366438

The metric dimension of the supertoken graph of the complete graph

摘录 / EXCERPT: Metric-dimension conjecture. The metric dimension of the supertoken graph ${\cal F}_k(K_n)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-052150 | MATHDB-366443

Ath-Sobel conjecture on sparse uniformly most reliable graphs

摘录 / EXCERPT: Let $\mathcal C_{n,m}$ be the class of connected simple graphs on n vertices and m edges, and define its corank by $c=m-n+1$. A uniformly most reliable graph (UMRG) is a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-052151 | MATHDB-366444

Boesch's conjecture on uniformly most reliable graphs

摘录 / EXCERPT: Boesch's conjecture. Each uniformly most reliable graph in $\mathcal{C}_{n,m}$ is a 0-element in $\mathcal{C}_{n,m}$.

conjecture | open | MathDB 页面 | 原文线索

V2-052152 | MATHDB-366469

Conlon-Lee conjecture on improved extremal bounds for bipartite graphs

摘录 / EXCERPT: Conlon-Lee conjecture. There exists $c=c_H>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052153 | MATHDB-366471

Kelly-Postle local fractional Shearer conjecture

摘录 / EXCERPT: Let G be a triangle-free graph, and let $d_G(v)$ denote the degree of a vertex $v\in V(G)$. Consider a probability distribution on the independent sets of G. Kelly-Postle's loc...

conjecture | solved | MathDB 页面 | 原文线索

V2-052154 | MATHDB-366473

Spectral-radius chromatic bound for triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph, let $\chi(G)$ denote its chromatic number, and let $\rho(G)$ denote its spectral radius. Spectral-radius chromatic bound. Every triangle-free grap...

open_problem | open | MathDB 页面 | 原文线索

V2-052155 | MATHDB-366489

Furtula-Oz conjecture on the extremal complementary second Zagreb index graph

摘录 / EXCERPT: Let G be a connected graph of order n, and let ... where $d_u(G)$ is the degree of vertex u. Let G^* be a graph attaining the maximum value of cM_2 among all connected grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052156 | MATHDB-366513

Bounded-diameter conjecture for finite Goldbach graphs

摘录 / EXCERPT: Let a finite Goldbach graph be a finite graph associated with Goldbach representations, and let a finite near Goldbach graph be a finite graph obtained from the corresponding near...

conjecture | open | MathDB 页面 | 原文线索

V2-052157 | MATHDB-366514

Classification conjecture for extremal third-eigenvalue graphs

摘录 / EXCERPT: Let G be a graph and let $H_{a,b}$ denote the family of graphs introduced in the paper, where a,b are non-negative integers. Classification conjecture for extremal graphs. If ……

conjecture | open | MathDB 页面 | 原文线索

V2-052158 | MATHDB-366515

Weighted-graph third-eigenvalue conjecture

摘录 / EXCERPT: For $n\geq 2$, let $\mathcal{S}_n$ be the set of $n\times n$ real symmetric matrices with entries in $[0,1]$, and let $\lambda_{n-1}(M)$ denote the penultimate eigenvalue of M. W...

conjecture | open | MathDB 页面 | 原文线索

V2-052159 | MATHDB-366516

Vertex-multiplication conjecture for the matrix T

摘录 / EXCERPT: Let T be the 11×11 matrix displayed in the paper, and let $\tilde{T}$ be any vertex multiplication of T. Vertex-multiplication conjecture. One should have ... This conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-052160 | MATHDB-366521

3-extendability conjecture for optimal 1-embedded graphs on the Klein bottle

摘录 / EXCERPT: 3-extendability conjecture. Every 8-regular optimal 1-embedded graph G on the Klein bottle is 3-extendable.

conjecture | open | MathDB 页面 | 原文线索

V2-052161 | MATHDB-366528

Aharoni-Berger conjecture for graphs

摘录 / EXCERPT: Aharoni-Berger conjecture. Every collection of n matchings of size n in a graph admits a rainbow matching of size $n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-052162 | MATHDB-366530

Aharoni-Berger non-bipartite strong conjecture

摘录 / EXCERPT: Aharoni-Berger non-bipartite strong conjecture. Every collection of n matchings of size $n+2$ in a graph admits a rainbow matching of size n.

conjecture | open | MathDB 页面 | 原文线索

V2-052163 | MATHDB-366531

Asymptotic rainbow matching conjecture for hypergraphs

摘录 / EXCERPT: Asymptotic rainbow matching conjecture. Fix $r\geq2$. For every $x\in\{g(r,n),g'(r,n),h(r,n),h'(r,n)\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-052164 | MATHDB-366560

Gajarsky-Pilipczuk-Toruńczyk conjecture on cliquewidth obstruction classes via walls

摘录 / EXCERPT: Gajarsky-Pilipczuk-Toruńczyk's cliquewidth obstruction conjecture. A class of graphs $\mathscr{C}$ has unbounded cliquewidth if and only if $\mathscr{C}$ transduces a class...

conjecture | open | MathDB 页面 | 原文线索

V2-052165 | MATHDB-366561

Transduction order of graph classes embeddable in surfaces

摘录 / EXCERPT: Surface transduction-order conjecture. Let Σ and Γ be surfaces such that $\Sigma\not\preccurlyeq\Gamma$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052166 | MATHDB-366577

The sublinear upper-bound conjecture for the crossing profile of complete graphs

摘录 / EXCERPT: For a rectilinear drawing of the complete graph K_n, let $e_k(K_n)$ denote the number of edges crossed exactly k times, and let ... denote the maximum of $e_k(K_n)$ over all rect...

conjecture | open | MathDB 页面 | 原文线索

V2-052167 | MATHDB-366578

Kamatchi-Arumugam's distance antimagic conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected simple graph. A bijection $h:V(G)\rightarrow\{1,2,\dots,|V(G)|\}$ is a distance antimagic labeling when the vertex weights ... are distinct for all ver...

conjecture | open | MathDB 页面 | 原文线索

V2-052168 | MATHDB-366579

The D-antimagic neighborhood conjecture

摘录 / EXCERPT: Let G be a graph, let $D\subseteq\{0,1,2,\dots,\operatorname{diam}(G)\}$ be a non-empty distance set, and define the D-neighborhood of a vertex u by ... A bijection...

conjecture | open | MathDB 页面 | 原文线索

V2-052169 | MATHDB-366623

Eventual exact rainbow saturation conjecture for C_5

摘录 / EXCERPT: Let C_5 denote the cycle on five vertices, and let $rsat(n,C_5)$ be its rainbow saturation number. Eventual exact rainbow saturation conjecture for C_5. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-052170 | MATHDB-366624

Beck's bounded-move conjecture for strong Ramsey games

摘录 / EXCERPT: Beck's conjecture. For sufficiently large n and every integer $t\geq 3$, there is a constant c_t, depending only on t, such that

conjecture | open | MathDB 页面 | 原文线索

V2-052171 | MATHDB-366625

Wang-Wu's forest-number conjecture for Cartesian products of trees

摘录 / EXCERPT: Wang-Wu's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052172 | MATHDB-366628

Negative type conjecture for 3-subdivisions of graphs with a $K_{2,3}$ minor

摘录 / EXCERPT: Negative type conjecture. There is no graph G with a $K_{2,3}$ minor where the shortest distance metric of the 3-subdivision of G has negative type.

conjecture | open | MathDB 页面 | 原文线索

V2-052173 | MATHDB-366634

Baranyai-Katona wreath conjecture

摘录 / EXCERPT: The wreath conjecture. For any positive integers $k\leq n$, there is a decomposition of $\mathbb{Z}_{n}^{(k)}$ into disjoint wreaths.

conjecture | open | MathDB 页面 | 原文线索

V2-052174 | MATHDB-366635

Dyck-path strengthening of the wreath conjecture

摘录 / EXCERPT: Dyck-path strengthening. There exists a set $\Pi=\{\pi_1,\pi_2,\ldots,\pi_{C_k}\}$ of C_k permutations of $\mathbb{Z}_{2k+1}$, each fixing 0, and a bijection...

conjecture | open | MathDB 页面 | 原文线索

V2-052175 | MATHDB-366636

Reflection-symmetric Dyck-path wreath conjecture

摘录 / EXCERPT: Reflection-symmetric strengthening. There exists a set $\Pi=\{\pi_1,\pi_2,\ldots,\pi_{C_k}\}$ of C_k permutations of $\mathbb{Z}_{2k+1}$, each fixing 0, and a bijection...

conjecture | open | MathDB 页面 | 原文线索

V2-052176 | MATHDB-366641

Strengthened lower bound for the sum of the two smallest graph eigenvalues

摘录 / EXCERPT: Let G be a graph of order n, and let λ_{n-1} and λ_n denote its two smallest eigenvalues. Strengthened eigenvalue conjecture. One has ... This strengthens the ori...

conjecture | open | MathDB 页面 | 原文线索

V2-052177 | MATHDB-366661

The forbidden-minor characterization of strictly metrizable graphs

摘录 / EXCERPT: A graph is strictly metrizable if every pair of vertices has a unique shortest path under some positive edge weighting. The forbidden-minor conjecture. A graph is strictly metrizab...

conjecture | open | MathDB 页面 | 原文线索

V2-052178 | MATHDB-366693

Logarithmic Kazhdan-Lusztig correspondence for Feigin-Tipunin vertex algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple finite-dimensional complex Lie algebra with simple roots $\alpha_1,\ldots,\alpha_n$ and Killing form $(-,-)$. For an integer $p\geq 2$ divisible...

conjecture | open | MathDB 页面 | 原文线索

V2-052179 | MATHDB-366705

Fiamčík-Alon conjecture on the acyclic chromatic index

摘录 / EXCERPT: Fiamčík-Alon conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052180 | MATHDB-366707

Lang's asymptotic tiling conjecture for spanning subgraphs of complete multipartite hypergraphs

摘录 / EXCERPT: Let $r\ge 2$ and $1 \le s_1 \le \cdots \le s_r$ be integers, and let $m \coloneqq s_1 + \cdots + s_r$. Suppose that F is a spanning subgraph of $K_{s_1, \ldots, s_r}^{r}$. For in...

conjecture | open | MathDB 页面 | 原文线索

V2-052181 | MATHDB-366708

Revised asymptotic tiling conjecture for rigid spanning hypergraphs

摘录 / EXCERPT: Let $r\ge 3$ and $1 \le s_1 \le \cdots \le s_r$ be integers, and let $m\coloneqq s_1+\cdots+s_r$. A spanning subgraph F of $K_{s_1,\ldots,s_r}^{r}$ is rigid if, for some such int...

conjecture | open | MathDB 页面 | 原文线索

V2-052182 | MATHDB-366723

Havet, Reed, Wood and Stein's conjecture on embedding trees in graphs

摘录 / EXCERPT: Let k be a positive integer, and let G be a graph with minimum degree $\delta(G)\ge \lfloor 2k/3\rfloor$ and maximum degree $\Delta(G)\ge k$. A k-edge tree is a tree with exact...

conjecture | open | MathDB 页面 | 原文线索

V2-052183 | MATHDB-366726

Balanced-regime confinement and collapse conjecture for strongly coupled networks

摘录 / EXCERPT: Balanced-regime confinement and collapse conjecture. If the set $\mathcal{B}$ is nonempty, there exist asymptotic solutions confined to the balanced regime for all times. Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-052184 | MATHDB-366733

Triangular Gröbner-basis conjecture for chemical reaction networks

摘录 / EXCERPT: Consider a chemical reaction network (CRN) whose steady-state ideal satisfies Assumptions 1-3: mass-action kinetics, inclusion of conservation laws, and absence of independent subn...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-052185 | MATHDB-366746

Positivity conjecture for colored unknot invariants in real projective three-space

摘录 / EXCERPT: Let U be an unknot in $\mathbb{R}P^3$ colored by an n-dimensional skew-symmetric or symmetric representation R of $su(2)$. Let $Z_{R\emptyset}(U;Q,q)$ be its invariant, and I...

conjecture | open | MathDB 页面 | 原文线索

V2-052186 | MATHDB-366752

The bipartite graph gluing conjecture for extremal numbers

摘录 / EXCERPT: The bipartite graph gluing conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052187 | MATHDB-366802

Polylogarithmic tree-independence conjecture for induced-minor-free graphs

摘录 / EXCERPT: Polylogarithmic tree-independence conjecture. For every positive integer t, there is an integer $d=d(t)$ such that for every $n\ge2$, every n-vertex graph with no induced minor...

conjecture | open | MathDB 页面 | 原文线索

V2-052188 | MATHDB-366811

Orbifold conjecture for twisted modules of vertex operator algebras

摘录 / EXCERPT: Let V be a vertex operator algebra satisfying the following conditions: $V_{(n)}=0$ for $n<0$, $V_{(0)}=\mathbb{C}\mathbf{1}$, and V is equivalent to its contragredient module;...

conjecture | open | MathDB 页面 | 原文线索

V2-052189 | MATHDB-366834

The caterpillar maximum-center neighbor conjecture

摘录 / EXCERPT: Let T be a caterpillar, let $r\geq 1$, and let a maximum r-center be a vertex z maximizing the number of independent r-sets containing it. Caterpillar center-neighbor conje...

conjecture | open | MathDB 页面 | 原文线索

V2-052190 | MATHDB-366852

Locally series-parallel graph characterization conjecture

摘录 / EXCERPT: Locally series-parallel graph characterization conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-052191 | MATHDB-366864

Harris's fractional chromatic number conjecture for triangle-free degenerate graphs

摘录 / EXCERPT: Let G be a graph, and let d be a nonnegative integer such that G is d-degenerate, meaning that every subgraph of G has a vertex of degree at most d. Let $\triangle$-fre...

conjecture | open | MathDB 页面 | 原文线索

V2-052192 | MATHDB-366883

Hamilton-cycle reconstruction conjecture from induced subgraphs

摘录 / EXCERPT: Hamilton-cycle reconstruction conjecture. There are constants $0\leq c<1$ and $n_0\geq 0$ such that, for all integers $n\geq n_0$, the number of Hamilton cycles of an n-vertex gr...

conjecture | open | MathDB 页面 | 原文线索

V2-052193 | MATHDB-366885

The non-bipartite matching-polytope toric ideal degree conjecture

摘录 / EXCERPT: Matching-polytope degree conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052194 | MATHDB-366886

The perfect-matching-polytope toric ideal degree conjecture

摘录 / EXCERPT: Perfect-matching-polytope degree conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052195 | MATHDB-366888

The wheel-graph matching-polytope degree conjecture

摘录 / EXCERPT: Wheel-graph degree conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052196 | MATHDB-366896

Thomas's cubic-graph monophonic position conjecture

摘录 / EXCERPT: Let G be a cubic graph of order n, and let $\operatorname{mp}(G)$ denote its monophonic position number. Thomas's conjecture. The largest possible monophonic position number of...

conjecture | open | MathDB 页面 | 原文线索

V2-052197 | MATHDB-366897

Thomas's cage monophonic position conjecture

摘录 / EXCERPT: Let a (d,g)-cage G be a graph of degree d and girth g, and let $\operatorname{mp}(G)$ denote its monophonic position number. Thomas's conjecture. For sufficiently large g...

conjecture | open | MathDB 页面 | 原文线索

V2-052198 | MATHDB-366898

Tuite-Thomas-Chartrand circulant-graph conjecture

摘录 / EXCERPT: Let n be the order of a graph, and let a circulant graph be a graph whose vertices and adjacency relation are invariant under cyclic translation. Let $\operatorname{mp}(G)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-052199 | MATHDB-366899

Kruft's terminal-set conjecture

摘录 / EXCERPT: Let G be a graph. A set $S\subseteq V(G)$ is a terminal set if S is a general position set of G and adding any vertex $u\in V(G)-S$ to S creates three-in-a-line with u as...

conjecture | open | MathDB 页面 | 原文线索

V2-052200 | MATHDB-366900

Kruft's lower general position number conjecture for Cartesian products

摘录 / EXCERPT: Let G and H be graphs, let $G\mathbin{\square}H$ denote their Cartesian product, and let $\operatorname{gp}^-(G)$ denote the lower general position number of G. Kruft's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-052201 | MATHDB-366901

Nordhaus-Gaddum conjecture for independent position chromatic numbers

摘录 / EXCERPT: Let G be a graph of order n, let $\overline{G}$ be its complement, and choose $\pi\in\{\operatorname{gp},\operatorname{mp}\}$, where gp and mp...

conjecture | open | MathDB 页面 | 原文线索

V2-052202 | MATHDB-366902

Fitzpatrick's fractional general position conjecture for square grids

摘录 / EXCERPT: Let P_n be the path on n vertices, let $P_n\mathbin{\square}P_n$ be the $n\times n$ Cartesian grid, and let gp_f denote the fractional general position number....

conjecture | open | MathDB 页面 | 原文线索

V2-052203 | MATHDB-366903

Klavžar-Tian-Tuite complexity conjecture for Builder/Blocker games

摘录 / EXCERPT: Consider the Builder/Blocker games denoted by the B-game and B'-game, together with their decision versions. Klavžar-Tian-Tuite's conjecture. The decision versions of the B- and B'...

conjecture | open | MathDB 页面 | 原文线索

V2-052204 | MATHDB-366906

The integer 2-domination formula for three-row grids

摘录 / EXCERPT: Let $G_{m,n}$ denote the m-by-n grid graph, and let $\gamma_{\{2\}}(G_{m,n})$ be its integer $\{2\}$-domination number. The three-row grid conjecture. For every positive intege...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052205 | MATHDB-366907

The integer 2-domination formula for four-row grids

摘录 / EXCERPT: Let $G_{m,n}$ denote the m-by-n grid graph, and let $\gamma_{\{2\}}(G_{m,n})$ be its integer $\{2\}$-domination number. The four-row grid conjecture. For every integer...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052206 | MATHDB-366908

Rainbow Erdős-Sós conjecture for hypercubes

摘录 / EXCERPT: For a graph G and a graph F, let $\mathrm{ex^*}(G,F)$ be the maximum number of edges in a subgraph of G that has a proper edge-coloring with no rainbow copy of F. The n-d...

conjecture | open | MathDB 页面 | 原文线索

V2-052207 | MATHDB-366909

Rainbow minimum-degree Erdős-Sós conjecture for hypercubes

摘录 / EXCERPT: For a graph G and a graph F, let $\delta^*(G,F)$ be the maximum minimum degree of a subgraph $H\subset G$ that has a proper edge-coloring with no rainbow copy of F. Let Q_n...

conjecture | open | MathDB 页面 | 原文线索

V2-052208 | MATHDB-366914

Linear realizations for small k

摘录 / EXCERPT: Let L be a multiset of linear lengths, where $\{1^a,(y-k)^b,y^c\}$ denotes the multiset containing a, b, and c copies of 1, $y-k$, and y, respectively. A multiset is li...

conjecture | open | MathDB 页面 | 原文线索

V2-052209 | MATHDB-366917

Hamiltonicity of a regular graph plus a random 2-regular graph

摘录 / EXCERPT: Let G be an n-vertex regular graph, and let $F\sim G_{n,2}$ be a random 2-regular graph on the same vertex set as G. Here, $G\cup F$ denotes the graph formed by adding the...

open_problem | open | MathDB 页面 | 原文线索

V2-052210 | MATHDB-366918

Bet et al.'s changepoint detection threshold conjecture for preferential attachment models

摘录 / EXCERPT: Let G_n be the final snapshot of a preferential attachment network on n vertices, and suppose the changepoint is in the late-change regime $\tau_n=n-\Delta$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-052211 | MATHDB-366932

The four-point obstruction conjecture for labeled star graphs

摘录 / EXCERPT: Four-point obstruction conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-052212 | MATHDB-366933

The compactness characterization conjecture for infinite labeled star ultrametric spaces

摘录 / EXCERPT: Compactness characterization conjecture. The following statements are equivalent: (X,d) is compact; and there is a labeled ray $R(l)$ generating (X_0,d_0) such that

conjecture | open | MathDB 页面 | 原文线索

V2-052213 | MATHDB-366938

Li et al.'s density conjecture for connected-flow-critical graphs

摘录 / EXCERPT: Let G be a connected-flow-critical graph, meaning that G does not admit a nowhere-zero 3-flow and, for every non-trivial partition $\mathcal{P}$ of $V(G)$ whose parts each in...

conjecture | open | MathDB 页面 | 原文线索

V2-052214 | MATHDB-366939

The degree-sequence conjecture for non-trivial flow-critical tame canvases

摘录 / EXCERPT: Let $(\mathfrak{G},z)$ be a non-trivial flow-critical tame canvas, and let $C(\mathfrak{G},z)$ denote its associated degree sequence. If ... and $\mathfrak{G}$ has a vertex of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-052215 | MATHDB-366956

Mubayi-Suk tower-growth conjecture for off-diagonal hypergraph Ramsey numbers

摘录 / EXCERPT: Let $K^{<(k)}_n$ denote the ordered complete k-uniform hypergraph, let $MP^{<(k)}_{k+1}$ denote the monotone $(k+1)$-vertex path in the k-uniform setting, and let t_{k-1} be...

conjecture | open | MathDB 页面 | 原文线索

V2-052216 | MATHDB-366968

Goddyn's thin tree conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be an unweighted undirected graph. A set $T\subseteq E$ is α-thin with respect to G if, for every nonempty set $S\subsetneq V$, ... A graph is d-edge-connect...

conjecture | open | MathDB 页面 | 原文线索

V2-052217 | MATHDB-366973

Additive lower-bound conjecture for positive matching decomposition numbers

摘录 / EXCERPT: Let Γ_1 and Γ_2 be graphs, let $\Gamma_1\square\Gamma_2$ denote their Cartesian product, and let $\mathrm{pmd}(\Gamma)$ denote the positive matching decomposition num...

conjecture | open | MathDB 页面 | 原文线索

V2-052218 | MATHDB-366974

Positive matching decomposition conjecture for Cartesian products of cycles

摘录 / EXCERPT: Let C_m and C_n be cycles with $m,n\geq3$, and let $\mathrm{pmd}(G)$ denote the positive matching decomposition number. Positive matching decomposition conjecture for Cartesian...

conjecture | open | MathDB 页面 | 原文线索

V2-052219 | MATHDB-367016

Monotonicity conjecture for the regularity of graph-power edge ideals

摘录 / EXCERPT: Let G be a simple graph on vertex set $V(G)=[n]$, let $R=\Bbbk[x_1,\ldots,x_n]$, and let $I(G^d)$ be the edge ideal of the d-th power of G, where d is a positive integer. T...

conjecture | open | MathDB 页面 | 原文线索

V2-052220 | MATHDB-367018

A diameter-bound conjecture for distance-regular graphs with classical parameters

摘录 / EXCERPT: Diameter-bound conjecture. There is a constant ℓ such that if $D\geq\ell$, then one of the following statements holds:

conjecture | open | MathDB 页面 | 原文线索

V2-052221 | MATHDB-367033

The parity conjecture for ensnaring graph complements

摘录 / EXCERPT: Let G be an m-vertex graph. For an integer $n\ge m$, define the n-vertex complement of G, denoted $\operatorname{compl}_n(G)$, to be the n-vertex graph whose complement c...

conjecture | open | MathDB 页面 | 原文线索

V2-052222 | MATHDB-367034

The tree-complement ensnaring conjecture

摘录 / EXCERPT: Let T be an m-vertex tree, and let $n>m$ be an even integer. Using the n-vertex complement notation $\operatorname{compl}_n(T)$, tree-complement ensnaring conjecture. The gra...

conjecture | open | MathDB 页面 | 原文线索

V2-052223 | MATHDB-367035

The orbit-size conjecture for complements of odd paths

摘录 / EXCERPT: Let P_m be a path graph with m vertices, and let $G=\operatorname{compl}_n(P_m)$ be the n-vertex complement of P_m. Let Θ_G denote the associated dynamical system, a...

conjecture | open | MathDB 页面 | 原文线索

V2-052224 | MATHDB-367036

Hypergraph matching conjecture for edge-modification complexity

摘录 / EXCERPT: Let $k\geq 3$, let F be a k-uniform hypergraph, and let G be an input k-uniform hypergraph. Write $\mathrm{rem}_F(G)$ for the minimum number of edges that must be removed f...

conjecture | open | MathDB 页面 | 原文线索

V2-052225 | MATHDB-367040

The self-avoiding-walk bunkbed question

摘录 / EXCERPT: Suppose $G=(V,E)$ is a finite, connected, simple graph. For $x,y\in V$, let $\mathscr{S}(x,y)$ be the set of self-avoiding walks from x to y, and let $\mathscr{S}(u_i,v_j)$ den...

open_problem | open | MathDB 页面 | 原文线索

V2-052226 | MATHDB-367063

The unrestricted upper-bound conjecture for the total 2-coalition number

摘录 / EXCERPT: Let G be a graph with minimum degree $\delta=\delta(G)$ and maximum degree $\Delta=\Delta(G)$, where $\delta\neq 2$. The unrestricted upper-bound conjecture states that ... The authors r...

conjecture | open | MathDB 页面 | 原文线索

V2-052227 | MATHDB-367072

The corrected Local Irregularity Conjecture excluding the bow-tie graph

摘录 / EXCERPT: Let G be a connected locally irregular colorable graph, let B be the bow-tie graph, and let $\mathrm{lir}(G)$ denote the minimum number of locally irregular subgraphs whose edg...

conjecture | refuted | MathDB 页面 | 原文线索

V2-052228 | MATHDB-367073

Bensmail-Renault conjecture on locally irregular colorings of oriented graphs

摘录 / EXCERPT: For $a,b\in\{+,-\}$, a digraph is (a,b)-locally irregular if, for every arc uv, $d^+(u)\neq d^-(v)$; let $\mathrm{lir}^{(+,+)}(D)$ be the minimum number of colors in an arc col...

conjecture | open | MathDB 页面 | 原文线索

V2-052229 | MATHDB-367074

Bensmail-Filasto-Hocquard-Marcille conjecture on mixed local irregularity

摘录 / EXCERPT: A sink-source path is an oriented path containing an arc directed into a vertex and an arc directed out of that vertex, as used in the source. For $a,b\in\{+,-\}$, define (a,b)-l...

conjecture | open | MathDB 页面 | 原文线索

V2-052230 | MATHDB-367075

Two-color conjecture for weak local irregularity of digraphs

摘录 / EXCERPT: For a digraph D, two adjacent vertices are weakly distinguished when their outdegree-indegree pairs differ. Let $\mathrm{lir}(D)$ be the minimum number of colors in an arc colori...

conjecture | open | MathDB 页面 | 原文线索

V2-052231 | MATHDB-367076

Two-color conjecture for strong local irregularity of digraphs

摘录 / EXCERPT: For a digraph D, define the balanced degree of a vertex x by ... A digraph is strongly locally irregular if $\sigma(x)\neq\sigma(y)$ for every arc xy, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052232 | MATHDB-367091

Czabarka-Singgih-Székely diameter conjecture for graphs with bounded clique number

摘录 / EXCERPT: Czabarka-Singgih-Székely conjecture. If $\omega(G)\leq k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052233 | MATHDB-367092

Conjecture that the two repetition ratios agree

摘录 / EXCERPT: Repetition-ratio conjecture. For every $\delta\geq4$,

conjecture | open | MathDB 页面 | 原文线索

V2-052234 | MATHDB-367106

Homomorphism and blowup threshold conjecture

摘录 / EXCERPT: Homomorphism-blowup threshold conjecture. For any graph H,

conjecture | open | MathDB 页面 | 原文线索

V2-052235 | MATHDB-367107

VC-dimension and blowup threshold conjecture

摘录 / EXCERPT: VC-dimension-blowup threshold conjecture. For any graph H,

conjecture | open | MathDB 页面 | 原文线索

V2-052236 | MATHDB-367137

Kadrawi-Levit conjecture on the locations of log-concavity breakdowns in tree independence sequences

摘录 / EXCERPT: Let T be a tree, let $\alpha(T)$ denote its independence number, and let $(a_i)_{i=0}^{\alpha(T)}$ be its independent set sequence. Log-concavity is broken at k when...

conjecture | open | MathDB 页面 | 原文线索

V2-052237 | MATHDB-367148

Kostochka et al.'s super-neighborhood cycle conjecture for bigraphs

摘录 / EXCERPT: Kostochka et al.'s conjecture. If a bigraph $G=(X,Y)$ is snp, then there is a cycle containing all vertices of X.

conjecture | open | MathDB 页面 | 原文线索

V2-052238 | MATHDB-367159

Alon-Shapira conjecture for generalized hypergraph Turán numbers

摘录 / EXCERPT: Let $f_r(n,v,e)$ be the maximum number of edges in an n-vertex r-uniform hypergraph such that every set of v vertices spans fewer than e edges. For fixed integers...

conjecture | open | MathDB 页面 | 原文线索

V2-052239 | MATHDB-367168

The minimum-degree conjecture for majority edge colourings

摘录 / EXCERPT: Majority edge-colouring conjecture. For every integer $k\geq 2$, if a graph G has minimum degree $\delta\geq k^2$, then G is $1/k$-majority edge $(k+1)$-colourable.

conjecture | open | MathDB 页面 | 原文线索

V2-052240 | MATHDB-367169

The list strengthening of the minimum-degree conjecture

摘录 / EXCERPT: List strengthening conjecture. For every integer $k\geq 2$, if a graph G has minimum degree $\delta(G)\geq k^2$, then G has a $1/k$-majority edge colouring from any lists of si...

conjecture | open | MathDB 页面 | 原文线索

V2-052241 | MATHDB-367171

The concentric-group conjecture for 4-HAT-stabilizers

摘录 / EXCERPT: A finite group H is called concentric if it satisfies the concentricity condition used in the classification of 4-HAT-stabilizers, and a 4-HAT-stabilizer is a group occurring...

conjecture | open | MathDB 页面 | 原文线索

V2-052242 | MATHDB-367191

Expansion conjecture for joint-ancestor-free subsets

摘录 / EXCERPT: Let $G=(V,E)$ be a formal Markov chain, and let K_1 and K_2 be connected components of the cut graph $C_1(G)$. Let $I\subseteq K_1$ and $J\subseteq K_2$ be joint-ancestor-free...

conjecture | open | MathDB 页面 | 原文线索

V2-052243 | MATHDB-367192

Rücker, Rücker and Gutman's minimum principal-eigenvector concentration conjecture

摘录 / EXCERPT: Let G be a finite simple connected graph with vertex set $V(G)$, adjacency matrix A_G, and principal eigenvector $\underline{x}=(x_v)_{v\in V(G)}$ corresponding to the largest...

conjecture | open | MathDB 页面 | 原文线索

V2-052244 | MATHDB-367197

Linear domination conjecture for graph covers

摘录 / EXCERPT: Linear domination conjecture. There exists a constant $c>0$ such that for every k-fold cover G of a graph F,

conjecture | open | MathDB 页面 | 原文线索

V2-052245 | MATHDB-367200

Bírógyiköğlü-Hordijk-Leydold-Pisanski-Stadler conjecture on strong nodal domains of hypercubes

摘录 / EXCERPT: Bírógyiköğlü-Hordijk-Leydold-Pisanski-Stadler conjecture. For every $1\leq i\leq n-2$, there is an eigenfunction f of $H(n,2)$ with eigenvalue $2i$ such that $\mathrm{SND}(f)=2$.

conjecture | open | MathDB 页面 | 原文线索

V2-052246 | MATHDB-367201

Conjecture on strong nodal domains of ternary Hamming graphs

摘录 / EXCERPT: Conjecture on ternary strong nodal domains. For any eigenfunction f of $H(n,3)$, $n\geq 1$, with eigenvalue $3n$ we have $\mathrm{SND}(f)\geq n+1$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052247 | MATHDB-367205

The bounded-twin-width subclass conjecture for hereditary classes

摘录 / EXCERPT: A hereditary class is a graph class closed under induced subgraphs, and twin-width is the graph parameter measuring the minimum contraction complexity under sequences of vertex ide...

conjecture | open | MathDB 页面 | 原文线索

V2-052248 | MATHDB-367206

Cocks's essential-family conjecture for unbounded treewidth

摘录 / EXCERPT: A family of unbounded treewidth is a family of graphs whose treewidth is unbounded, and it is essential when its hereditary closure is a minimal hereditary class of unbounded treew...

conjecture | open | MathDB 页面 | 原文线索

V2-052249 | MATHDB-367213

Lexicode conjecture for clique covering numbers of Johnson graphs

摘录 / EXCERPT: Lexicode conjecture. Assume that $k>1$ is a power of 2. The lexicode $$ has cardinality

conjecture | open | MathDB 页面 | 原文线索

V2-052250 | MATHDB-367225

Georgakopoulos-Papasoglu's quasi-isometric bounded-treewidth conjecture

摘录 / EXCERPT: Let G be a graph, and say that G excludes a graph as an asymptotic minor if there is an integer k such that the graph is not an asymptotic minor of G. Georgakopoulos-Papaso...

conjecture | refuted | MathDB 页面 | 原文线索

V2-052251 | MATHDB-367231

Burr-Erdős-Lovász conjecture for chromatic Ramsey numbers

摘录 / EXCERPT: For a graph G, let $R_\chi(G)$ be its chromatic Ramsey number, and define ... Burr-Erdős-Lovász conjecture. For every positive integer n, ... The source states this conjecture as a...

conjecture | open | MathDB 页面 | 原文线索

V2-052252 | MATHDB-367238

Győri's asymptotic clique-packing conjecture

摘录 / EXCERPT: Győri's conjecture. For every fixed $r\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-052253 | MATHDB-367239

Correctness of the proposed algorithm for solving the black-box problem

摘录 / EXCERPT: Let $\mathcal{E}$ be a well-connected network with an odd number of boundary nodes and response matrix $M_R(\mathcal{E})$. Let $A(\mathcal{E})$ be the matrix constructed from this...

conjecture | open | MathDB 页面 | 原文线索

V2-052254 | MATHDB-367275

The almost bounded merge-width characterization conjecture

摘录 / EXCERPT: Let $\mathcal C$ be a hereditary graph class. It has almost bounded merge-width if, for each fixed $r\in\mathbb N$, every n-vertex graph in $\mathcal C$ has radius-r merge-widt...

conjecture | open | MathDB 页面 | 原文线索

V2-052255 | MATHDB-367277

The 6-edge-connectivity conjecture for strongly connected modulo 3-orientations

摘录 / EXCERPT: The 6-edge-connectivity conjecture. Every 6-edge-connected graph has a strongly connected modulo 3-orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-052256 | MATHDB-367286

The large empty bipartite pair conjecture under induced four-cycle-free restrictions

摘录 / EXCERPT: Let G be a bipartite graph with vertex classes A and B, each of size n. The maximum degree of G is at most εn, and an induced four cycle-free subgraph is an...

conjecture | open | MathDB 页面 | 原文线索

V2-052257 | MATHDB-367313

Forest sufficient condition for polynomial approximation homomorphism complexity

摘录 / EXCERPT: Let F and H be graphs. Let an F-forest T be the forest construction used in the source, and let $T^{\star}$ denote its associated graph. Forest sufficient-condition conject...

conjecture | open | MathDB 页面 | 原文线索

V2-052258 | MATHDB-367332

The bowtie extremal-orientation conjecture

摘录 / EXCERPT: Let $B=F_{2,3}$ be the bowtie graph, with a distinguished center vertex. Let G be a complete tripartite graph with part sizes 1, $\left\lfloor\frac{n-1}{2}\right\rfloor$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052259 | MATHDB-367333

The bowtie three-in-or-three-out conjecture

摘录 / EXCERPT: Let $B=F_{2,3}$ be the bowtie graph, with a distinguished center vertex, and let $\vec{B}$ be an orientation of B having exactly three edges directed toward the center vertex or...

conjecture | open | MathDB 页面 | 原文线索

V2-052260 | MATHDB-367334

The matching-orientation conjecture for directed graphs

摘录 / EXCERPT: Let H be a graph and let $\vec{H}$ be a directed graph with underlying graph H. For a graph or directed graph, let $\mathcal{M}$ denote the family of maximal matchings, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052261 | MATHDB-367336

Complete-bipartite spanning-subgraph conjecture for extremal flag Betti numbers

摘录 / EXCERPT: Complete-bipartite spanning-subgraph conjecture. If $G\in\mathbf{G}(n,m)$ and

conjecture | open | MathDB 页面 | 原文线索

V2-052262 | MATHDB-367343

Keszegh-Pálvölgyi conjecture on 2-coloring one-headed directed hypergraphs

摘录 / EXCERPT: Keszegh-Pálvölgyi conjecture. If every hyperedge has more tail-vertices than head-vertices, and for every $e_{1},e_{2}\in E(H)$ with $|e_{1}\cap e_{2}|=1$ the common vertex is a he...

conjecture | open | MathDB 页面 | 原文线索

V2-052263 | MATHDB-367347

Extremal cubic graph conjecture for the subpath number

摘录 / EXCERPT: Assume that $n\geq 10$ is even. If $n=4q-2$ for some integer q, let L_n be obtained from $(n-10)/4$ copies of K_4-e by adding $(n-10)/4-1$ edges so that the resulting graph i...

conjecture | open | MathDB 页面 | 原文线索

V2-052264 | MATHDB-367356

Nguyen-Scott-Seymour conjecture on additive quasi-isometries to bounded-tree-width graphs

摘录 / EXCERPT: Nguyen-Scott-Seymour conjecture. There is a constant k such that if a graph G admits a quasi-isometry to a graph of tree-width at most two, then G admits a quasi-isometry wit...

conjecture | open | MathDB 页面 | 原文线索

V2-052265 | MATHDB-367363

Furtula-Oz conjecture on extremal complementary second Zagreb graphs

摘录 / EXCERPT: Let G be a graph with n vertices, and let $cM_{2}(G)$ denote its complementary second Zagreb index, defined by ... Here $K_m\vee\overline{K_{n-m}}$ is the join of the complete gr...

conjecture | open | MathDB 页面 | 原文线索

V2-052266 | MATHDB-367365

Rank characterization conjecture for the (k,d)-volume rigidity matroid

摘录 / EXCERPT: Let X be a k-dimensional simplicial complex on vertex set X_0, with X_1 its set of edges and X_k its set of k-faces. For disjoint families of sets ${\cal A},{\cal B}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-052267 | MATHDB-367371

Robust Hajnal-Szemerédi conjecture for induced K_r-factors

摘录 / EXCERPT: Let $r\geq 2$, let n be divisible by r, and let G be an $(\tfrac{(r-1)n}{r}+1)$-regular graph on n vertices. A subset of $V(G)$ induces a K_r-factor if its induced subgra...

conjecture | open | MathDB 页面 | 原文线索

V2-052268 | MATHDB-367383

Jain's connectivity and diameter conjecture for unit graphs over \(\mathbb{Z}_n\)

摘录 / EXCERPT: Let n be a natural number, and let $G(\mathbb{Z}_n)$ be the unit graph. Write $U(\mathbb{Z}_n)$ for its set of units, $N_U(\mathbb{Z}_n)$ for its set of non-units, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052269 | MATHDB-367410

Turán bound for suspensions of arbitrary trees

摘录 / EXCERPT: Let T be a tree whose smaller color class has size k, and let $\widehat{T}$ denote the suspension of T. Let $f(n,k)$ be the bound defined earlier in the paper. Conjecture on...

conjecture | open | MathDB 页面 | 原文线索

V2-052270 | MATHDB-367411

The improved ratio conjecture for vertex-disjoint triangles in tripartite graphs

摘录 / EXCERPT: Let $k\geq 2$ and let n be sufficiently large. For $1\geq\alpha,\beta,\gamma\geq 0$, an (k,n)-cyclic triple is a triple satisfying the six cyclic density inequalities stated in...

conjecture | open | MathDB 页面 | 原文线索

V2-052271 | MATHDB-367414

Complete characterization of the tropical multiple Horn solution set for large networks

摘录 / EXCERPT: Tropical multiple Horn conjecture. For sufficiently large networks Π_1,Π_2,Π_3, for example standard networks, these conditions are necessary and sufficient. Equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-052272 | MATHDB-367420

Minimum-edge bound for k-edge-proper graphs

摘录 / EXCERPT: Let G be a graph of order n, where $k\geq 4$ and n is sufficiently large. A graph is k-edge-proper if it contains cycles of every length from k through n. Minimum-edge...

conjecture | open | MathDB 页面 | 原文线索

V2-052273 | MATHDB-367421

Asymptotic edge lower bound for edge-pancyclic graphs

摘录 / EXCERPT: Let G be an edge-pancyclic graph with n vertices; that is, G contains a cycle containing each edge for every possible cycle length. Assume that n is sufficiently large. Asy...

conjecture | open | MathDB 页面 | 原文线索

V2-052274 | MATHDB-367427

The nilpotency-index bound for four-dimensional ${\cal N}=2$ SCFTs

摘录 / EXCERPT: Let $\mathfrak{rank}$ be the complex dimension of the Coulomb branch of a four-dimensional ${\cal N}=2$ SCFT. Let $\mathfrak{n}$ be the smallest positive integer such that the VOA...

conjecture | open | MathDB 页面 | 原文线索

V2-052275 | MATHDB-367428

Monotonicity of the nilpotency index under DS reduction

摘录 / EXCERPT: Let ${\cal V}$ be a VOA corresponding to a four-dimensional ${\cal N}=2$ SCFT, and let DS reduction produce another VOA corresponding to a four-dimensional ${\cal N}=2$ SCFT. Denot...

conjecture | open | MathDB 页面 | 原文线索

V2-052276 | MATHDB-367448

Chen-Yang-Yuan-Zhang conjecture on cliques in graphs avoiding disjoint copies of P_3

摘录 / EXCERPT: Let n, k, and s be integers, and let P_3 denote the path on three vertices. Write kP_3 for the disjoint union of k copies of P_3, and let M_n be the graph consistin...

conjecture | open | MathDB 页面 | 原文线索

V2-052277 | MATHDB-367450

The asymptotic Sudoku number conjecture for random cubic graphs

摘录 / EXCERPT: Let $\mathcal{G}_{n,3}$ be the random 3-regular graph on n vertices, and let $s(G)$ denote the smallest size of a Sudoku set for a graph G. Here, a Sudoku set is a vertex sub...

conjecture | open | MathDB 页面 | 原文线索

V2-052278 | MATHDB-367452

Cyclic edge-connectivity conjecture for cages

摘录 / EXCERPT: Cyclic edge-connectivity conjecture. Every (k,g)-cage is cyclically $(k-2)g$-edge-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-052279 | MATHDB-367453

Cycle-separating cut conjecture for cages

摘录 / EXCERPT: Cycle-separating cut conjecture. For each (k,g)-cage G, every cycle-separating edge-cut of size $(k-2)g$ in G separates a g-cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-052280 | MATHDB-367457

Nguyen-Scott-Seymour edge-weighting conjecture for graph quasi-isometries

摘录 / EXCERPT: Nguyen-Scott-Seymour conjecture. For all $L,C\in\mathbb{N}$, there exists $C'\in\mathbb{N}$ such that if φ is an (L,C)-quasi-isometry from a graph G to a graph H, the...

conjecture | open | MathDB 页面 | 原文线索

V2-052281 | MATHDB-367458

Edge-contraction and subdivision conjecture for graph quasi-isometries

摘录 / EXCERPT: Edge-contraction and subdivision conjecture. For all $L,C\in\mathbb{N}$, there exists $C'\in\mathbb{N}$ such that if a graph G is (L,C)-quasi-isometric to a graph in...

conjecture | open | MathDB 页面 | 原文线索

V2-052282 | MATHDB-367463

The marginal-edge probability conjecture for ERGMs

摘录 / EXCERPT: Let X be sampled from the ERGM measure μ on graphs with vertex set $[n]$, let e be any edge in $\binom{[n]}{2}$, and let p^* denote the corresponding Erdős-Rényi edge pro...

conjecture | open | MathDB 页面 | 原文线索

V2-052283 | MATHDB-367465

Triangle moment conjecture for ERGMs in the phase-coexistence regime

摘录 / EXCERPT: Let μ be an ERGM measure on graphs with vertex set $[n]$, and let e_1,e_2,e_3 be edges spanning a triangle in K_n. Let e be an arbitrary edge. Triangle moment conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-052284 | MATHDB-367476

Ustimenko's spectral bound conjecture for components of $D(k,q)$

摘录 / EXCERPT: Let $C\!D(k,q)$ be a connected component of $D(k,q)$, and let λ_2 denote its second largest eigenvalue. Ustimenko's conjecture. For all k and q, ... This is a universal s...

conjecture | open | MathDB 页面 | 原文线索

V2-052285 | MATHDB-367480

The extremal size conjecture for resilient 3-graphs

摘录 / EXCERPT: A 2-resilient 3-graph is a 3-uniform hypergraph $\mathcal{T}$ for which deleting any set of at most two vertices does not decrease its matching number. Let $\nu(\mathcal{T})$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-052286 | MATHDB-367481

The asymptotic extremal-size conjecture for resilient hypergraphs

摘录 / EXCERPT: Let $m(k,s)$ denote the maximum number of edges in a $(k-1)$-resilient k-uniform hypergraph with matching number s. Resilient hypergraph extremal conjecture. For fixed $k\geq4$...

conjecture | open | MathDB 页面 | 原文线索

V2-052287 | MATHDB-367482

The covering-number weakening for resilient hypergraph extremal size

摘录 / EXCERPT: Let $\mathcal{F}$ be a k-uniform hypergraph, let $\nu(\mathcal{F})$ be its matching number, and let $\tau(\mathcal{F})$ be its covering number, the minimum size of a vertex set m...

conjecture | open | MathDB 页面 | 原文线索

V2-052288 | MATHDB-367484

Persistence and lack of persistence in the PAVD model without higher-order assumptions

摘录 / EXCERPT: Let the PAVD model be the preferential attachment model with vertex death from Definition. Suppose that the attachment and death-rate sequences b and d satisfy Assumptions and....

conjecture | open | MathDB 页面 | 原文线索

V2-052289 | MATHDB-367495

Dahlberg-She-van Willigenburg's non-e-positivity conjecture for trees

摘录 / EXCERPT: Let T be a tree, let $\Delta(T)$ denote its maximum degree, and say that T is e-positive when its chromatic symmetric function X_T has a nonnegative expansion in the elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-052290 | MATHDB-367498

The Modified Bellows Conjecture for flexible spherical polyhedra

摘录 / EXCERPT: Let $P(u)$ be a flexible polyhedron in the sphere $\mathbb{S}^n$, where $n\ge 3$. Replacing a vertex by its antipode means replacing that vertex of $P(u)$ with its antipodal point...

conjecture | open | MathDB 页面 | 原文线索

V2-052291 | MATHDB-367508

Occupancy-fraction conjecture for independent sets in triangle-free graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a triangle-free graph of average degree d, let $Z_G(\lambda)$ denote its independence polynomial, and write $Z_G'(1)$ for its derivative at 1. Occupancy-fracti...

conjecture | open | MathDB 页面 | 原文线索

V2-052292 | MATHDB-367509

Parameterized occupancy-fraction conjecture for triangle-free graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a triangle-free graph of average degree d, let $Z_G(\lambda)$ denote its independence polynomial, and let W be the Lambert W-function, defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-052293 | MATHDB-367511

The H-union-K_2 perfect-divisibility conjecture

摘录 / EXCERPT: Let H be a graph, let K_2 be the complete graph on two vertices, and let $k\in\mathbb{N}_{>2}$. Assume that every H-free graph is perfectly $(k-2)$-divisible. The H-union-K2...

conjecture | open | MathDB 页面 | 原文线索

V2-052294 | MATHDB-367512

The eventual wheel saturation recurrence

摘录 / EXCERPT: Wheel saturation recurrence. For every integer $k\geq 4$, there exists a constant N_k such that, for all $n>N_k$,

conjecture | open | MathDB 页面 | 原文线索

V2-052295 | MATHDB-367516

The $2/3$ lower-bound conjecture for normalized distance Laplacian spectral gaps

摘录 / EXCERPT: Spectral-gap conjecture. The spectral gap satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052296 | MATHDB-367518

The full Brouwer's Laplacian spectrum conjecture

摘录 / EXCERPT: Let G be a graph on n vertices with m edges and Laplacian eigenvalues $\mu_1\geq\mu_2\geq\cdots\geq\mu_n$, and define ... For integers $k\in\{1,2,\cdots,n-1\}$, let $G_{k,r,s}$...

conjecture | open | MathDB 页面 | 原文线索

V2-052297 | MATHDB-367519

The Nordhaus-Gaddum full Brouwer conjecture for Laplacian spectra

摘录 / EXCERPT: Let G be a graph with n vertices, let $\overline{G}$ be its complement, and let $s_k(G)$ denote the sum of the k largest Laplacian eigenvalues of G. Nordhaus-Gaddum full Br...

conjecture | open | MathDB 页面 | 原文线索

V2-052298 | MATHDB-367533

Aanderaa-Karp-Rosenberg conjecture for finite graph properties

摘录 / EXCERPT: Aanderaa-Karp-Rosenberg conjecture. Every nontrivial monotone graph property is elusive.

conjecture | open | MathDB 页面 | 原文线索

V2-052299 | MATHDB-367536

Discriminant magnitude homology conjecture for trees

摘录 / EXCERPT: Tree discriminant magnitude homology conjecture. For $k=3,4$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052300 | MATHDB-367537

Recurrence conjecture for Eulerian magnitude homology of path trees

摘录 / EXCERPT: Path-tree recurrence conjecture. For $0\leq k\leq n$ and all $\ell\geq 0$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052301 | MATHDB-367538

Tan et al.'s conjecture on co-edge-regular graphs with four eigenvalues

摘录 / EXCERPT: Let G be a connected k-regular graph with n vertices and co-edge-regular with parameter μ, having four distinct eigenvalues. Let $t\geq 2$ be an integer. Tan et al.'s con...

conjecture | open | MathDB 页面 | 原文线索

V2-052302 | MATHDB-367539

Goldberg's eigenvalue inequality conjecture for edge-regular graphs

摘录 / EXCERPT: Let G be a k-regular and edge-regular graph with parameter λ, and let θ,θ' be two distinct eigenvalues of G, different from k. Goldberg's conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-052303 | MATHDB-367540

Bolognini-Macchia-Strazzanti's Cohen-Macaulayness conjecture for binomial edge ideals

摘录 / EXCERPT: Bolognini-Macchia-Strazzanti's conjecture. J_G is Cohen-Macaulay if and only if G is accessible.

conjecture | open | MathDB 页面 | 原文线索

V2-052304 | MATHDB-367560

Shi-Shan's toughness conjecture for forbidden linear forests

摘录 / EXCERPT: Let $k\ge 4$ be an integer. A graph is t-tough if its toughness satisfies $\tau(G)\ge t$, a graph is r-connected if deleting fewer than r vertices leaves it connected, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052305 | MATHDB-367576

Hoàng's degree-sequence conjecture for tough graphs

摘录 / EXCERPT: Hoàng's conjecture. Let $n\ge 3$ and $t\ge 1$ be integers. If G is t-tough and, for every $i<\frac n2$, $d_i\le i$ implies $d_{n-i+t}\ge n-i$, then G is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-052306 | MATHDB-367590

The Pell-equation existence conjecture for $(1,1)$-antimagic graph unions

摘录 / EXCERPT: Let P_3 be the path on three vertices. For a graph H, a $(1,1)$-antimagic labeling is an antimagic labeling whose vertex sums are exactly $1,2,\ldots,|V(H)|$. Pell-equation exi...

conjecture | open | MathDB 页面 | 原文线索

V2-052307 | MATHDB-367593

Holroyd-Talbot-Borg conjecture for intersecting faces of simplicial complexes

摘录 / EXCERPT: Holroyd-Talbot-Borg conjecture. If $r\leq d/2$, then there is some vertex v of Δ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052308 | MATHDB-367605

Heinig's Hamilton-generated conjecture for dense balanced bipartite graphs

摘录 / EXCERPT: Let G be a balanced bipartite graph with $2n$ vertices, and write $\delta(G)$ for its minimum degree. Heinig's conjecture. If n is sufficiently large and ... then G is Hamilton...

conjecture | open | MathDB 页面 | 原文线索

V2-052309 | MATHDB-367620

Eventual saturation number for cycles of lengths from 4 to r

摘录 / EXCERPT: For a positive integer set I, let $\mathcal{C}_I$ denote the family of cycles whose lengths belong to I, and let $\operatorname{sat}(n,\mathcal{C}_I)$ be the minimum number of...

conjecture | open | MathDB 页面 | 原文线索

V2-052310 | MATHDB-367621

Eventual stabilization of saturation numbers for finite cycle intervals

摘录 / EXCERPT: Let $\mathcal{C}_I$ denote the family of cycles whose lengths belong to an integer set I, and let $\operatorname{sat}(n,\mathcal{C}_I)$ be the minimum number of edges in an n-v...

conjecture | open | MathDB 页面 | 原文线索

V2-052311 | MATHDB-367622

Saturation number for cycles of lengths congruent to 1 modulo 3

摘录 / EXCERPT: For an integer set I, let $\mathcal{C}_I$ denote the family of cycles whose lengths belong to I, and let $\operatorname{sat}(n,\mathcal{C}_I)$ be the minimum number of edges in...

conjecture | open | MathDB 页面 | 原文线索

V2-052312 | MATHDB-367643

Ulam-Kelly reconstruction conjecture for graph homomorphism counts

摘录 / EXCERPT: Let $Hom(F,G)$ denote the number of graph homomorphisms from F to G, and let $\mathcal{G}$ be the family of all graphs. For a family of graphs $\mathcal{F}$, say that...

conjecture | open | MathDB 页面 | 原文线索

V2-052313 | MATHDB-367653

The c3_{k+1} degree-sum conjecture for 2-factors with at most k cycles

摘录 / EXCERPT: Let G be a graph of order n, let k be a positive integer, and let $c3_{k+1}(G)$ denote the minimum degree sum over every set of $k+1$ pairwise nonadjacent vertices of G. De...

conjecture | open | MathDB 页面 | 原文线索

V2-052314 | MATHDB-367654

Levit-Mandrescu unimodality conjecture for pure flag complexes

摘录 / EXCERPT: Let $ Delta$ be a $(d-1)$-dimensional pure flag complex on $2d$ vertices, and let its f-vector be the vector counting faces by dimension. Levit-Mandrescu conjecture. The f-vect...

conjecture | open | MathDB 页面 | 原文线索

V2-052315 | MATHDB-367655

Pruesse-Ruskey Hamiltonicity conjecture for Venn-diagram graphs

摘录 / EXCERPT: For an n-Venn diagram, consider the graph whose vertices are the crossings of the curves and whose edges are the curve segments between consecutive crossings; this graph may have...

conjecture | open | MathDB 页面 | 原文线索

V2-052316 | MATHDB-367656

Polynomial-expansion path-degeneracy lower-bound conjecture

摘录 / EXCERPT: Polynomial-expansion lower-bound conjecture. For every real $b>0$, there exists a graph class with expansion $O(r^b)$ such that for infinitely many $p\in\mathbb N$ the class contai...

conjecture | open | MathDB 页面 | 原文线索

V2-052317 | MATHDB-367657

Cubic-cage order conjecture

摘录 / EXCERPT: Cubic-cage conjecture. For infinitely many integers g, there exists a cubic graph G such that

conjecture | open | MathDB 页面 | 原文线索

V2-052318 | MATHDB-367674

Ellzey's e-positivity conjecture for proper circular arc digraphs

摘录 / EXCERPT: Ellzey's conjecture. If $\vec G$ is a proper circular arc digraph, then $X_{\vec G}(\bm x;q)$ is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-052319 | MATHDB-367680

Extremal conjecture for equal-degree endpoint paths of odd length

摘录 / EXCERPT: Odd-length extremal conjecture. For any odd integer $\ell\geq 3$ and sufficiently large n, it holds that

conjecture | open | MathDB 页面 | 原文线索

V2-052320 | MATHDB-367685

Graph isomorphism conjectures for edge-, spanning-tree-, and subset-deleted graphs

摘录 / EXCERPT: Let G,H,G_a,G_b be graphs, let $E_G\subset E(G)$ and $E_H\subset E(H)$ be edge subsets, let T_a,T_b be spanning trees, and let $S_G\in V(G)\cup E(G)$ and $S_H\in V(H)\cup E(H)$...

conjecture | open | MathDB 页面 | 原文线索

V2-052321 | MATHDB-367686

Hypergraph reconstruction conjecture from hyperedge-deleted subhypergraphs

摘录 / EXCERPT: Let $\mathcal{H}_{\operatorname{yper}}=(\Lambda,\mathcal{E})$ and $\mathcal{H}^*_{\operatorname{yper}}=(\Lambda^*,\mathcal{E}^*)$ be hypergraphs based on 3l-hyperedge sets...

conjecture | open | MathDB 页面 | 原文线索

V2-052322 | MATHDB-367688

Gyárfás-Lehel-Bollobás K-T conjecture

摘录 / EXCERPT: Let $n\geq 3$, and for each $1\leq k\leq n$ let T_k be a tree on k vertices, with the trees mutually vertex disjoint. Let K_n be the complete graph on n vertices. K-T c...

conjecture | open | MathDB 页面 | 原文线索

V2-052323 | MATHDB-367689

The maximum partial-dual genus bound for planar graphs

摘录 / EXCERPT: Maximum partial-dual genus conjecture. If $G\not\cong C_3$ is a λ-edge-connected planar graph with $\lambda\in\{1,2\}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052324 | MATHDB-367724

Existence threshold conjecture for regular link-irregular graphs

摘录 / EXCERPT: Let G be a graph on n vertices. It is link-irregular if $G[N(u)]$ and $G[N(v)]$ are non-isomorphic for every pair of distinct vertices u,v, where $N(u)$ denotes the neighborh...

conjecture | open | MathDB 页面 | 原文线索

V2-052325 | MATHDB-367733

The Erdős-Rogers function asymptotic conjecture

摘录 / EXCERPT: For integers $t>s\geq2$, let $f_{s,t}(n)$ be the largest integer m such that every n-vertex K_t-free graph has an induced K_s-free subgraph with at least m vertices. The...

conjecture | open | MathDB 页面 | 原文线索

V2-052326 | MATHDB-367773

Conjecture on endomorphisms of antipodal non-bipartite distance-regular graphs

摘录 / EXCERPT: Let X be an antipodal, non-bipartite distance-regular graph of diameter 3. An endomorphism of X is a graph homomorphism from X to itself, and a subgraph has diameter 2 wh...

conjecture | open | MathDB 页面 | 原文线索

V2-052327 | MATHDB-367774

The linear-forest conjecture for intersectionwise self-χ-guarding graph classes

摘录 / EXCERPT: Linear-forest conjecture. If H is a linear forest, then the class of H-free graphs is intersectionwise self-χ-guarding.

conjecture | open | MathDB 页面 | 原文线索

V2-052328 | MATHDB-367780

Geelen's weak vertex-minor structure conjecture

摘录 / EXCERPT: Let $d53d$ be a proper vertex-minor-closed class of graphs. For $k,p displaymath \in \mathbb{N}$, a graph is k-rank-connected if it has at least $2k$ vertices and satisfies ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-052329 | MATHDB-367781

Superlinear CZ-distance for circle graphs

摘录 / EXCERPT: Let G be an n-vertex circle graph, and let $\operatorname{CZ}(G)$ denote its CZ-distance. Circle-graph lower-bound conjecture. There exist n-vertex circle graphs G with ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-052330 | MATHDB-367789

Helme-Guizon-Przytycki-Rong torsion conjecture for chromatic homology

摘录 / EXCERPT: Helme-Guizon-Przytycki-Rong torsion conjecture. The cohomology ${{H_{*,*}}^{\mathcal{A}_m}(G)}$ contains a torsion part if and only if G has no loops and contains a cycle of orde...

conjecture | open | MathDB 页面 | 原文线索

V2-052331 | MATHDB-367794

The balanced separator conjecture for sphere intersection graphs

摘录 / EXCERPT: Let $t,d greater than or equal to 1$ be integers, and let G be an n-vertex graph in the class $\SphereInt{d}_t$ of sphere intersection graphs. A balanced separator is a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-052332 | MATHDB-367799

Algebraic connectivity bound for complete multipartite graphs

摘录 / EXCERPT: Let $d\geq 1$ and $k\geq 2$. For positive integers $n_1,\ldots,n_k$, write $n=n_1+\cdots+n_k$, and let $K_{n_1,\ldots,n_k}$ be the complete k-partite graph whose sides have sizes...

conjecture | open | MathDB 页面 | 原文线索

V2-052333 | MATHDB-367800

Elphick-Wocjan conjecture on positive and negative p-energies

摘录 / EXCERPT: Let G be a connected graph with n vertices, and let $\mathcal{E}_p^+(G)$ and $\mathcal{E}_p^-(G)$ denote its positive and negative p-energies. Elphick-Wocjan conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-052334 | MATHDB-367801

Vector-chromatic p-energy lower-bound conjecture

摘录 / EXCERPT: Let G be a graph, let $p\geq 2$, and let $\chi_v(G)$ denote its vector chromatic number. Let $\mathcal{E}_p^+(G)$ and $\mathcal{E}_p^-(G)$ denote its positive and negative p-en...

conjecture | open | MathDB 页面 | 原文线索

V2-052335 | MATHDB-367804

The lower-threshold conjecture for star counts in extremal graphs

摘录 / EXCERPT: Let $G_n^{\ell}(1)$ and $G_n^{\ell}(i_0)$ be the two graph constructions appearing in Corollary, and let $s_t(G)$ denote the number of stars of size t in a graph G. Assume the...

conjecture | open | MathDB 页面 | 原文线索

V2-052336 | MATHDB-367816

Laborde-Payan-Xuong conjecture on independent sets meeting longest paths

摘录 / EXCERPT: Let D be a digraph, and call a set of vertices independent if no two of its vertices are joined by an arc. A path in D is longest if it has maximum length among all paths in...

conjecture | open | MathDB 页面 | 原文线索

V2-052337 | MATHDB-367817

Generation conjecture for local face modules of vertex-induced triangulations

摘录 / EXCERPT: Let $\sigma \\: \Gamma \to 2^V$ be a vertex-induced triangulation of a $(d-1)$-dimensional simplex, and let k be a field. Generation conjecture. The local face module $L(\Gamma)$...

conjecture | open | MathDB 页面 | 原文线索

V2-052338 | MATHDB-367818

Anisotropy conjecture for the middle socle of local face modules

摘录 / EXCERPT: Let $\sigma \\: \Gamma \to 2^V$ be a vertex-induced triangulation of a $(d-1)$-dimensional simplex, and let k be a field. Let B be the symmetric bilinear form on the local face...

conjecture | open | MathDB 页面 | 原文线索

V2-052339 | MATHDB-367822

Global rigidity conjecture for complete graphs in the infinity norm

摘录 / EXCERPT: Let $d\geq 1$ and let K_{2d+1} be the complete graph on $2d+1$ vertices. A realisation of K_{2d+1} in ℓ_∞^d assigns a point of $\mathbb{R}^d$ to each vertex, and it i...

conjecture | open | MathDB 页面 | 原文线索

V2-052340 | MATHDB-367829

The twin-width gap conjecture for random Latin square graphs

摘录 / EXCERPT: Twin-width gap conjecture. The twin-width of G is asymptotically larger than $\operatorname{lb}_1(G)$, that is,

conjecture | open | MathDB 页面 | 原文线索

V2-052341 | MATHDB-367830

The reconstruction conjecture for finite simple graphs

摘录 / EXCERPT: Let G be a finite simple undirected graph. For each vertex $v\in V(G)$, let $G_v=[G-v]_{\cong}$ be its card, and let ... be its deck. The graph G is reconstructible if every grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052342 | MATHDB-367835

All-pairs plane Hamiltonian path conjecture for simple drawings

摘录 / EXCERPT: Let K_n be drawn simply in the plane, and let a,b be two vertices. A path is plane if no two of its edges cross. All-pairs Hamiltonian-path conjecture. For every simple drawing...

conjecture | open | MathDB 页面 | 原文线索

V2-052343 | MATHDB-367836

Bergold et al.'s plane subdrawing conjecture for simple drawings

摘录 / EXCERPT: Let K_n be drawn simply in the plane. A plane subdrawing is a subdrawing whose edges do not cross, and a Hamiltonian cycle is a cycle through all n vertices. Bergold et al.'s c...

conjecture | open | MathDB 页面 | 原文线索

V2-052344 | MATHDB-367837

Hoffmann-Tóth convex matching conjecture for simple drawings

摘录 / EXCERPT: Let K_n have a convex drawing, and let M be a plane matching, meaning a set of pairwise vertex-disjoint edges no two of which cross. Hoffmann-Tóth convex-matching conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-052345 | MATHDB-367838

Bergold et al.'s empty-cycle conjecture for simple drawings

摘录 / EXCERPT: An empty k-cycle in a simple drawing of K_n is a plane cycle through k vertices such that all remaining vertices lie on one common side. Bergold et al.'s conjecture. Every si...

conjecture | open | MathDB 页面 | 原文线索

V2-052346 | MATHDB-367839

Harborth-Mengersen's uncrossed-edge conjecture for crossing-maximal drawings

摘录 / EXCERPT: A simple drawing of K_n is crossing-maximal if it has $\binom{n}{4}$ crossings; an uncrossed edge is an edge crossed by no other edge. Harborth-Mengersen's conjecture. Every cros...

conjecture | open | MathDB 页面 | 原文线索

V2-052347 | MATHDB-367840

H-convex all-edges-crossed conjecture

摘录 / EXCERPT: An h-convex drawing of K_n is a drawing in which the vertices are in convex position with respect to the relevant hull structure. H-convex all-edges-crossed conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-052348 | MATHDB-367841

Crossing-maximal large crossing-family conjecture

摘录 / EXCERPT: A k-crossing family is a set of k independent edges that cross pairwise. A simple drawing of K_n is crossing-maximal when it has the maximum possible number $\binom{n}{4}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-052349 | MATHDB-367842

Harborth's empty-triangle conjecture for simple drawings

摘录 / EXCERPT: For a simple drawing of K_n, let $\triangle(n)$ denote the minimum number of empty triangles among all such drawings. An empty triangle is a triangle induced by three vertices su...

conjecture | open | MathDB 页面 | 原文线索

V2-052350 | MATHDB-367847

Salia's cycle-cover conjecture for infinite bipartite graphs

摘录 / EXCERPT: Salia's conjecture. If G has the double Hall property, then for every $X\subseteq A$ with $|X|\geq 2$ there is a cycle C_X in G such that

conjecture | open | MathDB 页面 | 原文线索

V2-052351 | MATHDB-367857

Conlon-Fox-Gunby-He-Mubayi-Suk-Verstraëte-Yu conjecture for generalized Erdős-Rogers numbers

摘录 / EXCERPT: Conlon-Fox-Gunby-He-Mubayi-Suk-Verstraëte-Yu conjecture. There exists a constant $c=c(G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052352 | MATHDB-367858

Generalized Erdős-Rogers polynomial-growth conjecture for hypergraphs

摘录 / EXCERPT: Generalized Erdős-Rogers polynomial-growth conjecture. There exists a constant $c=c(F,G)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052353 | MATHDB-367859

Conjecture that the logarithmic upper bound holds without the extra restriction

摘录 / EXCERPT: Logarithmic upper-bound conjecture. If G is tightly connected and is not homomorphic to F, then there exists a constant c depending only on F such that, for large enough...

conjecture | open | MathDB 页面 | 原文线索

V2-052354 | MATHDB-367868

The domination game conjecture for Cartesian products of paths

摘录 / EXCERPT: Let P_k and P_m be paths on k and m vertices, respectively, and let $G=P_k\mathbin{\square}P_m$ denote their Cartesian product. For a graph G, write ${\mathbb D}(G)$ for...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052355 | MATHDB-367872

The odd-partite independent-transversal blowup conjecture

摘录 / EXCERPT: Let $\mathcal{G}_r(n)$ be the class of r-partite graphs with parts of size n. An independent transversal with exactly s vertices in each part is denoted by $IT(s)$. Odd-parti...

conjecture | open | MathDB 页面 | 原文线索

V2-052356 | MATHDB-367881

Uniqueness conjecture for extremal connected graphs maximizing the two largest signless Laplacian eigenvalues

摘录 / EXCERPT: Let G be a connected graph with n vertices, let $c(G)=e(G)-n(G)+\omega(G)$ be its cycle-space dimension, and let $S_2(G)$ denote the sum of the two largest signless Laplacian e...

conjecture | open | MathDB 页面 | 原文线索

V2-052357 | MATHDB-367887

Elphick-Wocjan conjecture on the spectral lower bound for clique number

摘录 / EXCERPT: Let G be a graph with n vertices, let $ \lambda_1,\dots,\lambda_n$ be the eigenvalues of its adjacency matrix, and let $ s^+$ be the sum of the squares of its positive eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-052358 | MATHDB-367891

Converse characterization conjecture for mixed memories

摘录 / EXCERPT: Let F be a smooth function whose derivative satisfies $F'(x)>0$ for all $x>0$. Let $n\in\boldsymbol{N}$ be odd, let m be a configuration in the model, and let $\xi^{(N)}(m)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-052359 | MATHDB-367892

NP-hardness of Minimum Eternal Vertex Cover on series-parallel graphs

摘录 / EXCERPT: NP-hardness conjecture. Minimum Eternal Vertex Cover is NP-hard on series-parallel graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-052360 | MATHDB-367898

Simonovits's supersaturation conjecture for bipartite graphs

摘录 / EXCERPT: Let F be a bipartite graph such that ... for some $0<\alpha<1$. Let G be an n-vertex graph, and write $v(F)$ and $e(F)$ for the numbers of vertices and edges of F. Simonovits...

conjecture | open | MathDB 页面 | 原文线索

V2-052361 | MATHDB-367899

Supersaturation conjecture for r-partite hypergraphs with Sidorenko gap

摘录 / EXCERPT: For an r-partite r-graph F, let $s(F)$ be its Sidorenko gap, defined by ... Assume $r\geq 3$ and ... for some $0<\alpha<r-1$. The hypergraph supersaturation conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-052362 | MATHDB-367900

Sidorenko-gap conjecture for odd linear cycles

摘录 / EXCERPT: For an r-uniform linear cycle $C^r_{2\ell+1}$ of odd length $2\ell+1$, let $s(C^r_{2\ell+1})$ denote its Sidorenko gap. Odd linear-cycle Sidorenko-gap conjecture. For $r\geq 3$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-052363 | MATHDB-367909

Completeness conjecture for deep NN-QM representations

摘录 / EXCERPT: Completeness conjecture. Every reflection-positive process admits a representation as a deep NN-QM whose inputs $y_t^{(i)}$ are symmetric Markov processes.

conjecture | open | MathDB 页面 | 原文线索

V2-052364 | MATHDB-367920

Type II classification conjecture for circulant graphs of order twice an odd integer

摘录 / EXCERPT: Type II classification conjecture. If Γ is nontrivially unstable and of Type II, then

conjecture | open | MathDB 页面 | 原文线索

V2-052365 | MATHDB-367921

Type I conjecture for nontrivially unstable circulant graphs of order divisible by eight

摘录 / EXCERPT: Type I conjecture. Every such graph is of Type I.

conjecture | open | MathDB 页面 | 原文线索

V2-052366 | MATHDB-367954

Positive recurrence of the fork-join LLM engine network

摘录 / EXCERPT: Consider a fork-join network of four LLM agents indexed by $j\in\{0,1,2,3\}$. Requests arrive at agent 0 with rate λ, fork into tasks processed by agents 1 and 2, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052367 | MATHDB-367965

Rooted-minor obstruction conjecture for colorings

摘录 / EXCERPT: For a graph H in a graph G, an X-rooted minor is a model of H in G whose branch subgraph for every vertex of H intersects a specified vertex set X. A graph G is X...

conjecture | open | MathDB 页面 | 原文线索

V2-052368 | MATHDB-367966

Hill's crossing-number conjecture for complete graphs

摘录 / EXCERPT: Hill's conjecture. The crossing number satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052369 | MATHDB-367969

Signless Laplacian extremal conjecture for forbidding $K_{s,t}^+$

摘录 / EXCERPT: Let $2le sle t$ be positive integers. For n-vertex graphs, let $K_{s,t}^+$ denote the graph obtained from $K_{s,t}$ by adding an edge within the part of size s, let...

conjecture | open | MathDB 页面 | 原文线索

V2-052370 | MATHDB-367970

Signless Laplacian Simonovits-type conjecture for color-k-critical graphs

摘录 / EXCERPT: Let $k\ge 2$, and let F be a color-k-critical graph with chromatic number $\chi(F)=r+1\ge4$. Define $H_{n,r,k}:=K_{k-1}\vee T_{n-k+1,r}$, where $T_{n-k+1,r}$ is the r-partite...

conjecture | open | MathDB 页面 | 原文线索

V2-052371 | MATHDB-367983

The strong 4-coloring conjecture for graphs with one crossing

摘录 / EXCERPT: Strong one-crossing 4-coloring conjecture. If G has no separating triangles and every vertex not incident with the crossed edges has degree at least five, then G is 4-colorable...

conjecture | open | MathDB 页面 | 原文线索

V2-052372 | MATHDB-367984

The internal-degree conjecture for one-crossing triangulation graphs

摘录 / EXCERPT: Internal-degree conjecture. Every non-4-colorable graph in ${\cal C}_0$ has an internal vertex of degree four.

conjecture | open | MathDB 页面 | 原文线索

V2-052373 | MATHDB-367985

The generation conjecture for bichromatic-forbidding 4-candidates

摘录 / EXCERPT: 4-candidate generation conjecture. Every bichromatic-forbidding 4-candidate can be obtained from the diamond by a sequence of the operations described in Lemmas 10ext, extproj, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052374 | MATHDB-368009

Calabi-Yau realization conjecture for unitary full vertex operator algebras

摘录 / EXCERPT: Let X be a compact Calabi-Yau manifold of complex dimension d. Let F_X be a unitary $N=(2,2)$ full vertex operator algebra, and let $h_{p,q}^A(F_X)$ denote its A-Hodge number...

conjecture | open | MathDB 页面 | 原文线索

V2-052375 | MATHDB-368016

The freeness conjecture for GPTW-generated Lie subalgebras

摘录 / EXCERPT: Let $\mathcal{K}$ be a simplicial complex on $[m]$, let $\mathbf{k}$ be a commutative ring with unit, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-052376 | MATHDB-368018

Weak Target Conjecture

摘录 / EXCERPT: Weak Target Conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052377 | MATHDB-368019

Strong Target Conjecture

摘录 / EXCERPT: Strong Target Conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052378 | MATHDB-368034

Williams' shift-equivalence conjecture for essential graphs

摘录 / EXCERPT: Let E and F be essential graphs, and let A_E and A_F be their adjacency matrices. Write $A_E\sim_{SE}A_F$ for shift equivalence and $A_E\sim_{SSE}A_F$ for strong shift equi...

conjecture | open | MathDB 页面 | 原文线索

V2-052379 | MATHDB-368036

Narins-Pokrovskiy-Szabó conjecture on cycle-length diversity in degree 3-critical graphs

摘录 / EXCERPT: A graph is degree 3-critical if it has n vertices, $2n-2$ edges, and no proper induced subgraph with minimum degree at least 3. Narins-Pokrovskiy-Szabó conjecture. Every degree...

conjecture | open | MathDB 页面 | 原文线索

V2-052380 | MATHDB-368037

Narins-Pokrovskiy-Szabó conjecture on leaf-to-leaf path lengths in 1-3 trees

摘录 / EXCERPT: A 1-3 tree is a tree in which every vertex has degree either 1 or 3. Narins-Pokrovskiy-Szabó conjecture. Every 1-3 tree T of order n has leaf-to-leaf paths of at least ... di...

conjecture | open | MathDB 页面 | 原文线索

V2-052381 | MATHDB-368038

Narins-Pokrovskiy-Szabó conjecture on short leaf-to-leaf path lengths

摘录 / EXCERPT: A 1-3 tree is a tree in which every vertex has degree either 1 or 3. For a tree, a leaf-to-leaf path is a path whose endpoints are leaves. Narins-Pokrovskiy-Szabó conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-052382 | MATHDB-368044

Fleischner-Jackson-Litsyn-Swart's classification conjecture for 2-factor Hamiltonian regular bipartite graphs

摘录 / EXCERPT: Fleischner-Jackson-Litsyn-Swart's conjecture. Either $k=2$ and G is a cycle, or $k=3$ and G can be obtained from $K_{3,3}$ and the Heawood graph H_0 by repeated star products...

conjecture | open | MathDB 页面 | 原文线索

V2-052383 | MATHDB-368045

Heawood graph uniqueness conjecture for cyclically 4-edge-connected cubic bipartite graphs

摘录 / EXCERPT: Heawood graph uniqueness conjecture. The Heawood graph is the only 2-factor Hamiltonian cyclically 4-edge-connected cubic bipartite graph of girth at least 6.

conjecture | open | MathDB 页面 | 原文线索

V2-052384 | MATHDB-368046

Abreu-Diwan-Jackson-Labbate-Schwenk's classification conjecture for 3-edge-connected pseudo 2-factor isomorphic graphs

摘录 / EXCERPT: Abreu-Diwan-Jackson-Labbate-Schwenk's conjecture. G is pseudo 2-factor isomorphic if and only if G can be obtained from $K_{3,3}$, the Heawood graph or the Pappus graph by repe...

conjecture | open | MathDB 页面 | 原文线索

V2-052385 | MATHDB-368047

Abreu-Diwan-Jackson-Labbate-Schwenk's constituent classification conjecture for pseudo 2-factor isomorphic graphs

摘录 / EXCERPT: Abreu-Diwan-Jackson-Labbate-Schwenk's conjecture. G must be $K_{3,3}$, the Heawood graph or the Pappus graph.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052386 | MATHDB-368079

Quasi-lisse, unique-module, and tensor-category conjecture for $L_{k_m}(D_4)$

摘录 / EXCERPT: Conjecture on $L_{k_m}(D_4)$. For each $m\geq1$, $L_{k_m}(D_4)$ is a quasi-lisse vertex algebra, it has a unique irreducible ordinary module, and $KL_{k_m}(D_4)$ is a semisimple br...

conjecture | open | MathDB 页面 | 原文线索

V2-052387 | MATHDB-368101

Erdős's clique-partition conjecture

摘录 / EXCERPT: Let H be a graph on $n=km$ vertices. A partition of $V(H)$ into k sets each inducing an m-clique is called a k-partition into m-cliques. Erdős's conjecture. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-052388 | MATHDB-368104

Strong Chen-Lih-Wu decomposition conjecture

摘录 / EXCERPT: Let $k\geq3$, and let a k-decomposition of a graph be a partition of its vertex set into induced subgraphs that are k-basic. Strong Chen-Lih-Wu conjecture. If G is a k-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-052389 | MATHDB-368113

Multiplicity Ryser-Brualdi-Stein conjecture

摘录 / EXCERPT: Let G be a complete bipartite graph on $2n$ vertices whose edge set is decomposed into perfect matchings M_i, for $i=1,\ldots,n$. Let $a_1,\ldots,a_n\in\mathbb{N}_0$ be nonnega...

conjecture | open | MathDB 页面 | 原文线索

V2-052390 | MATHDB-368114

Anastos-Fabian-Müyesser-Szabó conjecture for three disjoint perfect matchings

摘录 / EXCERPT: Let G be a graph on $2n$ vertices that is the union of three disjoint perfect matchings. An (a_1,a_2,a_3)-matching is a matching containing exactly a_i edges from the ith p...

conjecture | open | MathDB 页面 | 原文线索

V2-052391 | MATHDB-368119

Conjectured exact value of the Ramsey number of a book graph versus an odd cycle

摘录 / EXCERPT: The authors' conjecture. For each integer $n\geq 3493$, if $n\geq 2m+499$ and $m\geq 7$ is an odd integer, then

conjecture | open | MathDB 页面 | 原文线索

V2-052392 | MATHDB-368124

The bicirculant Hamiltonicity conjecture

摘录 / EXCERPT: Bicirculant Hamiltonicity conjecture. Every connected bicirculant, except for K_2 and the generalized Petersen graphs $G(m,2)$ with $m\equiv 5\pmod 6$, is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-052393 | MATHDB-368163

The Shortest Cycle Cover Conjecture

摘录 / EXCERPT: For a bridgeless graph G, let $scc(G)$ denote the length of a shortest cycle cover of G. Shortest Cycle Cover Conjecture. If G is a bridgeless graph, then ... This conjecture g...

conjecture | open | MathDB 页面 | 原文线索

V2-052394 | MATHDB-368165

The equality conjecture for shortest cycle covers and triangle expansions

摘录 / EXCERPT: Let G be a cubic graph. For $U\subseteq V(G)$, let G_U be the cubic graph obtained by expanding each vertex of U into a triangle. For a bridgeless cubic graph G, let $T(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-052395 | MATHDB-368166

Zhang's depth-two shortest cycle cover conjecture

摘录 / EXCERPT: A cycle cover of a graph is a collection of cycles covering every edge. The depth of an edge in a cycle cover is the number of cycles containing it, and the depth of the cover is t...

conjecture | open | MathDB 页面 | 原文线索

V2-052396 | MATHDB-368167

The 1/10 upper-bound conjecture for triangle expansions

摘录 / EXCERPT: Let G be a bridgeless cubic graph. For $U\subseteq V(G)$, let G_U be obtained by expanding each vertex of U into a triangle, and let $T(G)$ be the minimum size of a set U f...

conjecture | open | MathDB 页面 | 原文线索

V2-052397 | MATHDB-368169

Equivalence of the 3/5 and 2/5 5-cycle double cover bounds

摘录 / EXCERPT: Let G be a bridgeless graph, and let a 5-cycle double cover be a collection ${\cal C}=(E_0,\ldots,E_4)$ of five cycles in which every edge belongs to exactly two cycles. Equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-052398 | MATHDB-368202

Devriendt-Ottolini-Steinerberger minimum constant resistance curvature conjecture

摘录 / EXCERPT: Let G be a connected graph on $n\geq 3$ vertices. For each vertex v_i, let κ_i be its resistance curvature, defined by the unique solution κ of ... where R_G is t...

conjecture | open | MathDB 页面 | 原文线索

V2-052399 | MATHDB-368214

The odd chromatic number conjecture for k-trees

摘录 / EXCERPT: Let a k-tree be a graph obtained from a complete graph on $k+1$ vertices by repeatedly adding a vertex adjacent to all vertices of an existing k-clique. An odd coloring of a grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052400 | MATHDB-368221

The complete-pairs formulation of the Gyárfás-Sumner conjecture

摘录 / EXCERPT: Let G be a finite simple graph. For disjoint vertex sets $A,B\subseteq V(G)$, call (A,B) a complete pair if every possible edge between A and B is present. For a forest T...

conjecture | open | MathDB 页面 | 原文线索

V2-052401 | MATHDB-368222

The colourful induced-subgraph conjecture for forests

摘录 / EXCERPT: Let G be a finite simple graph. For a vertex $v\in V(G)$, write $N_G[v]=N_G(v)\cup\{v\}$, and call G ε-colourful if ... for every $v\in V(G)$. For an induced subgraph...

conjecture | open | MathDB 页面 | 原文线索

V2-052402 | MATHDB-368229

Rigidity conjecture for special mixed graphs without satellites

摘录 / EXCERPT: Rigidity conjecture. If Γ has no satellites, then it is rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-052403 | MATHDB-368236

The 1-2-3 Conjecture for edge sum-labellings

摘录 / EXCERPT: Let G be a simple graph. For a positive integer k, an edge k-labelling is a map $\ell:E(G)\to\{1,\dots,k\}$. For each vertex u, define its incident sum by ... The labelling i...

conjecture | open | MathDB 页面 | 原文线索

V2-052404 | MATHDB-368237

The 1-2 Conjecture for total product-labellings

摘录 / EXCERPT: Let G be a graph. A total k-labelling is a map $\ell:V(G)\cup E(G)\to\{1,\dots,k\}$. For each vertex u, let $\pi_\ell(u)$ be the product of the labels incident to u, includ...

conjecture | open | MathDB 页面 | 原文线索

V2-052405 | MATHDB-368238

The 1-2 Conjecture for total multiset-labellings

摘录 / EXCERPT: Let G be a graph. A total k-labelling is a map $\ell:V(G)\cup E(G)\to\{1,\dots,k\}$. For each vertex u, let $\mu_\ell(u)$ be the multiset of labels incident to u, including...

conjecture | open | MathDB 页面 | 原文线索

V2-052406 | MATHDB-368242

Hudák-Madaras-Suzuki conjecture on 3-connected maximal 1-planar graph size

摘录 / EXCERPT: Let $\overline{\mathcal{P}^\star}$ be the family of 3-connected maximal 1-planar graphs, and let $m(\overline{\mathcal{P}^\star},n)$ denote the minimum number of edges among graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-052407 | MATHDB-368243

Maximality conjecture for the graph families XH^k and YH^k

摘录 / EXCERPT: For each integer $k\geq 1$, let XH^k and YH^k denote the graph families constructed in the paper. Maximality conjecture for XH^k and YH^k. The graphs XH^k and YH^k are...

conjecture | open | MathDB 页面 | 原文线索

V2-052408 | MATHDB-368246

Pancyclicity conjecture for Bang-Jensen-Gutin-Li type conditions

摘录 / EXCERPT: A digraph D has order $n\ge 3$. A pair of nonadjacent vertices is dominated if it has a common in-neighbor, and dominating if it has a common out-neighbor. Let $d(u)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-052409 | MATHDB-368260

Perfect-or-bounded-clique-width conjecture for claw-, $4K_1$-, and bridge-free graphs

摘录 / EXCERPT: Perfect-or-bounded-clique-width conjecture. One of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-052410 | MATHDB-368265

Nonexistence conjecture for face-transitive Klein bottles

摘录 / EXCERPT: A face-transitive Klein bottle is a simplicial Klein bottle whose face-transitive symmetry property is understood in the sense of the paper's census. Face-transitive Klein bottle c...

conjecture | open | MathDB 页面 | 原文线索

V2-052411 | MATHDB-368266

Caro-Yuster conjecture on equating maximum degrees

摘录 / EXCERPT: Caro-Yuster conjecture. For any fixed integer $k\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-052412 | MATHDB-368298

Jiang et al.'s asymptotic conjecture for spherical two-distance sets

摘录 / EXCERPT: Let $-1 \leq \beta < 0 \leq \alpha < 1$. A spherical two-distance set in $\mathbb{R}^d$ is a collection of unit vectors whose pairwise inner products lie in $\{\alpha,\beta\}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-052413 | MATHDB-368299

Jiang et al.'s forbidden-subgraph multiplicity conjecture

摘录 / EXCERPT: Let $\lambda>0$ and $p\in\mathbb{N}^+$. For a finite family $\mathcal{F}$ of signed graphs, let $M_{p,\mathcal{F}}(\lambda,N)$ be the maximum of $\operatorname{mult}(-\lambda,G)$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-052414 | MATHDB-368305

Bensmail-Mc Inerney's vertex-capturing game conjecture

摘录 / EXCERPT: Bensmail-Mc Inerney's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052415 | MATHDB-368306

Second-player conjecture for the clique game on edge-transitive graphs

摘录 / EXCERPT: Edge-transitive clique-game conjecture. Except for K_2, every edge-transitive graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052416 | MATHDB-368307

Strict monotonicity conjecture for Maker-Breaker clique games

摘录 / EXCERPT: Strict monotonicity conjecture. For fixed b,m, for all sufficiently large N,

conjecture | open | MathDB 页面 | 原文线索

V2-052417 | MATHDB-368308

Sharp biased clique-game conjecture

摘录 / EXCERPT: Sharp biased clique-game conjecture. For every fixed p, if n is sufficiently large, then $\operatorname{Clique}_{(p,p+1)}(n)$ is won by player 2.

conjecture | open | MathDB 页面 | 原文线索

V2-052418 | MATHDB-368337

Kirwan surjectivity for negatively paired indivisible dimension vectors

摘录 / EXCERPT: Let Q be a quiver, let d be an indivisible dimension vector such that ... where (d,e_i) is the relevant bilinear form and Q_0 is the vertex set. Let $\mathcal M_d^\chi(Q)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-052419 | MATHDB-368348

Self-negative games attain the maximum birthday game size

摘录 / EXCERPT: Let $F(b)$ be the maximum value of $f(X)$ among games X with birthday b, and let $-X$ denote the negation of the game X. Self-negation conjecture. For every integer $b\geq 0$...

conjecture | open | MathDB 页面 | 原文线索

V2-052420 | MATHDB-368352

Degree-sequence and structural invariants conjecture for chromatic symmetric functions of unicyclic graphs

摘录 / EXCERPT: Let G_1 and G_2 be connected unicyclic graphs, where either each has a 3-cycle and an odd number of vertices, or each has a c-cycle for some $c\geq4$. Let $\mathbf{X}_G$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-052421 | MATHDB-368355

Georgakopoulos coarse grid minor conjecture

摘录 / EXCERPT: Georgakopoulos's coarse grid minor conjecture. For every planar graph H, there exist $k,q\in\mathbb{N}$ such that every H-induced-minor-free graph is q-quasi-isometric to a g...

conjecture | open | MathDB 页面 | 原文线索

V2-052422 | MATHDB-368358

Graphical distinct-realisation conjecture from the weak pinned distance conjecture

摘录 / EXCERPT: Graphical distinct-realisation conjecture. For every $\varepsilon>0$, there exists a positive constant C_ε, independent of P, such that

conjecture | open | MathDB 页面 | 原文线索

V2-052423 | MATHDB-368360

Dot-product realisation conjecture for locally completable graphs

摘录 / EXCERPT: Dot-product realisation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052424 | MATHDB-368373

Grünbaum's toroidal conjecture on non-3-edge-colorable cubic graphs

摘录 / EXCERPT: Let G be a 2-connected cubic graph embedded in the torus. Grünbaum's conjecture. If G is not 3-edge-colorable, then G contains two edges on the same face of the embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-052425 | MATHDB-368374

Robertson's representativity conjecture for toroidal graphs

摘录 / EXCERPT: Let G be a 2-edge-connected graph embedded in the torus. Robertson's conjecture. If the representativity of the embedding is at least 3, then G has a nowhere-zero 4-flow....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052426 | MATHDB-368375

Robertson's surface representativity conjecture for nowhere-zero 4-flows

摘录 / EXCERPT: Let S be a surface. Robertson's surface conjecture. There is a constant r such that every 2-edge-connected graph embedded in S with representativity at least r has a nowh...

conjecture | open | MathDB 页面 | 原文线索

V2-052427 | MATHDB-368383

Kwan-Wigderson's unbounded inertia conjecture for graphs with independence number two

摘录 / EXCERPT: Kwan-Wigderson's conjecture. For every integer k, there exists a graph G with $\alpha(G)=2$ and

conjecture | open | MathDB 页面 | 原文线索

V2-052428 | MATHDB-368384

Unweighted Nordhaus-Gaddum conjecture for graph inertia

摘录 / EXCERPT: Unweighted Nordhaus-Gaddum conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-052429 | MATHDB-368396

Katona-Varga's generalized Kriesell conjecture

摘录 / EXCERPT: Let $t>0$, and let a minimally t-tough graph be a t-tough graph whose toughness decreases after deleting any edge. Katona-Varga's generalized conjecture. Every minimally t-to...

conjecture | open | MathDB 页面 | 原文线索

V2-052430 | MATHDB-368397

Zheng-Sun's revised generalized Kriesell conjecture for non-regular graphs

摘录 / EXCERPT: Let $t>0$. A graph is non-regular if its vertex degrees are not all equal, and a minimally t-tough graph is a t-tough graph whose toughness decreases after deleting any edge. Z...

conjecture | open | MathDB 页面 | 原文线索

V2-052431 | MATHDB-368406

Admissibility conjecture for wiring diagrams without oriented faces

摘录 / EXCERPT: Admissibility conjecture. If G includes no oriented faces, then it is admissible.

conjecture | open | MathDB 页面 | 原文线索

V2-052432 | MATHDB-368434

Sign conjecture for Steiner distance hyperdeterminants

摘录 / EXCERPT: Sign conjecture. For any tree on n vertices,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052433 | MATHDB-368448

The degeneracy-graph connectivity conjecture

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{>0}$, and let $DG(2n)$ be the degeneracy graph. The degeneracy-graph connectivity conjecture. 1. The graph $DG(2n)$ is connected when regarded as an undirected...

conjecture | open | MathDB 页面 | 原文线索

V2-052434 | MATHDB-368450

The conf-configuration invariance conjecture for $h(w)$

摘录 / EXCERPT: Let A be a vertex and let v,w be A-generic words, meaning that their arc-configuration sets equal the set defining A. The conf-configuration invariance conjecture. ... Equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-052435 | MATHDB-368455

The tensor-product admissible-edge conjecture

摘录 / EXCERPT: Let V and R be modules satisfying the hypotheses under which the vertex bijection ι sends (R_i,h_i) to...

conjecture | open | MathDB 页面 | 原文线索

V2-052436 | MATHDB-368492

Generalised Common Neighbour Conjecture for Saxl hypergraphs

摘录 / EXCERPT: Let $G\leq \operatorname{Sym}(\Omega)$ be a finite permutation group with base size $b(G)\geq 2$, and let $\mathcal{H}(G)$ be its Saxl hypergraph; in the original graph formulation...

conjecture | open | MathDB 页面 | 原文线索

V2-052437 | MATHDB-368493

Exceptional-groups conjecture for intersecting Saxl-hypergraph edges

摘录 / EXCERPT: Let $G\leq \operatorname{Sym}(\Omega)$ be a finite primitive permutation group with base size $b(G)\geq 2$, and let $\mathcal{H}(G)$ be its Saxl hypergraph. Exceptional-groups conj...

conjecture | open | MathDB 页面 | 原文线索

V2-052438 | MATHDB-368515

The asymptotic spectral extremal conjecture for trees outside the embedding condition

摘录 / EXCERPT: Let $\mathcal{T}_{m,l+1}^{\delta}$ denote the class of trees used in the paper, and let $\overline{K}_l$ be the edgeless graph on l vertices and S_{δ} the star used in the...

conjecture | open | MathDB 页面 | 原文线索

V2-052439 | MATHDB-368527

Chernyshev-Rauch-Rautenbach conjecture on sparse graphs and forest cuts

摘录 / EXCERPT: Chernyshev-Rauch-Rautenbach conjecture. If G has fewer than $3n-6$ edges, then G has a forest cut.

conjecture | open | MathDB 页面 | 原文线索

V2-052440 | MATHDB-368539

Uniform fixed-index Gaussian nodal surplus conjecture

摘录 / EXCERPT: Uniform Gaussian nodal surplus conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-052441 | MATHDB-368568

Weak k-metric dimension formula for Hamming graphs $K_n\square K_m$

摘录 / EXCERPT: Let $K_n\square K_m$ be the Cartesian product of complete graphs, with $n\ge 3$, $m\ge n+1$, and $3\le k\le 2n$. The weak k-metric dimension formula asserts that ... This conjectur...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052442 | MATHDB-368569

Bounded treedepth for even-cycle-subgraph-free graphs of diameter three

摘录 / EXCERPT: Let $r\geq 4$. A graph is C_{2r}-subgraph-free if it contains no subgraph isomorphic to the cycle C_{2r}. The diameter of a graph is the maximum distance between two vertices....

conjecture | open | MathDB 页面 | 原文线索

V2-052443 | MATHDB-368570

Chen et al.'s conjecture on the strong chromatic index and maximum edge weight

摘录 / EXCERPT: Chen et al.'s conjecture. If $W(G)\ge 5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052444 | MATHDB-368574

Covering graph conjecture for token graphs of odd stars

摘录 / EXCERPT: Covering conjecture for odd-star token graphs. The token graph $F_k(K_{1,n})$ is isomorphic to a covering graph $X^{(\alpha,\omega)}$ of a base graph X on

conjecture | open | MathDB 页面 | 原文线索

V2-052445 | MATHDB-368575

Covering graph conjecture for 2-token graphs of stars

摘录 / EXCERPT: Covering conjecture for 2-token star graphs. The token graph $F_2(K_{1,n})$ is isomorphic to a covering graph $X^{(\alpha,\omega)}$ of a base graph X on $n-k$ vertices, with comb...

conjecture | open | MathDB 页面 | 原文线索

V2-052446 | MATHDB-368584

KAMAK tree conjecture for grounded forests

摘录 / EXCERPT: KAMAK tree conjecture. Every grounded forest is δ^+-enforcible.

conjecture | open | MathDB 页面 | 原文线索

V2-052447 | MATHDB-368585

Talagrand's bounded-uniformity fractional expectation threshold conjecture

摘录 / EXCERPT: Let X be a finite set, let $k\in\mathbb{N}$ satisfy $k\leq\log_L|X|$, and let $g\colon {\mleft(\kern-.1em\genfrac{}{}{0pt}{}{X}{k}\kern-.1em\mright)}\to[0,1]$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-052448 | MATHDB-368589

Uniqueness conjecture for stressed geodesic graphs on the sphere

摘录 / EXCERPT: Let G be a vertex 3-connected planar graph embedded in the unit-radius sphere. Assume that each edge is a geodesic of length less than pi and each face has total perimeter le...

conjecture | open | MathDB 页面 | 原文线索

V2-052449 | MATHDB-368614

The Graph Complement Conjecture for minimum rank

摘录 / EXCERPT: Graph Complement Conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052450 | MATHDB-368616

The Graph Complement Conjecture for the Colin de Verdière parameter ν

摘录 / EXCERPT: Graph Complement Conjecture for ν. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052451 | MATHDB-368618

The strong δ-conjecture for the Colin de Verdière parameter

摘录 / EXCERPT: Strong δ-conjecture for ν. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052452 | MATHDB-368622

The enumeration conjecture for 3-minimal graphs

摘录 / EXCERPT: Enumeration conjecture. There are exactly 49 non-isomorphic 3-minimal graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-052453 | MATHDB-368623

The sparsest minimal-graph conjecture

摘录 / EXCERPT: Sparsest minimal-graph conjecture. For every positive integer ℓ, a sparsest ℓ-minimal graph is a sparse stretched clique in $\hat{\mathcal{K}}_{\ell+2,\ell-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-052454 | MATHDB-368624

The maximal-gamma conjecture for minimal graphs

摘录 / EXCERPT: Maximal-gamma conjecture. For every positive integer ℓ, the maximum value of $\gamma_{\ell-1}(G,a)$ among ℓ-minimal graphs is attained by a sparse stretched clique...

conjecture | open | MathDB 页面 | 原文线索

V2-052455 | MATHDB-368625

The clique-rank bound conjecture for stretched-clique classes

摘录 / EXCERPT: Clique-rank bound conjecture. For every positive integer ℓ, if $G \in \mathcal{K}_{\ell+2,\ell-1}$ and $\omega(G) \geq 4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052456 | MATHDB-368638

The extremal edge bound for 4-connected maximal bipartite IC-plane graphs

摘录 / EXCERPT: Extremal edge-bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052457 | MATHDB-368680

Gerbner's color-critical-edge conjecture for weak Turán-goodness

摘录 / EXCERPT: A graph H is weakly K_k-Turán-good when, for all sufficiently large n, some complete $(k-1)$-partite graph attains ${\mathrm{ex}}(n,H,K_k)$. An edge of a graph F is color-c...

conjecture | open | MathDB 页面 | 原文线索

V2-052458 | MATHDB-368681

Győri-Salia-Tompkins-Zamora path extremal graph conjecture

摘录 / EXCERPT: For odd $k\geq5$, let $H_{n,k}$ be the graph obtained from a clique $K_{\lfloor k/2\rfloor}$ by choosing a vertex v, adding m independent vertices adjacent to v, and adding...

conjecture | open | MathDB 页面 | 原文线索

V2-052459 | MATHDB-368705

Zhai-Lin-Shu spectral Turán conjecture for books

摘录 / EXCERPT: Let B_{r+1} denote the graph consisting of $r+1$ triangles sharing a common edge, and let $\rho(G)$ be the spectral radius of a graph G. A graph is B_{r+1}-free if it contain...

conjecture | open | MathDB 页面 | 原文线索

V2-052460 | MATHDB-368710

Corona characterization conjecture for well-hued graphs

摘录 / EXCERPT: Let G be a connected well-hued graph, with independence number $\alpha(G)$, chromatic number k, and colorability sequence recording the maximum orders of induced i-colorable...

conjecture | open | MathDB 页面 | 原文线索

V2-052461 | MATHDB-368711

Characterization conjecture for well-hued graphs with linear colorability sequence

摘录 / EXCERPT: Let G be a connected well-hued graph, let $\alpha(G)$ be its independence number, and let $k=\chi(G)$ be its chromatic number. Its colorability sequence is the sequence of maximu...

conjecture | open | MathDB 页面 | 原文线索

V2-052462 | MATHDB-368742

Strong positive square-energy conjecture for graphs with many edges

摘录 / EXCERPT: Let G be a connected graph of order n and size m. Strong positive square-energy conjecture. If $m\ge n+1$, then ... This conjecture strengthens the positive-energy side of the...

conjecture | open | MathDB 页面 | 原文线索

V2-052463 | MATHDB-368743

Unicyclic graph square-energy conjecture

摘录 / EXCERPT: Let G be a unicyclic graph of order n, and let k be the odd length of its unique cycle. Unicyclic graph square-energy conjecture. ... The conjecture is motivated by an exact ca...

conjecture | open | MathDB 页面 | 原文线索

V2-052464 | MATHDB-368744

Clique-number refinement of the positive square-energy conjecture

摘录 / EXCERPT: Let G be a connected graph of order n, and let $\omega(G)$ denote its clique number. Clique-number refinement. If ... then ... This conjecture is motivated by the expectation that...

conjecture | open | MathDB 页面 | 原文线索

V2-052465 | MATHDB-368745

Positive square-energy bounds for maximal planar and outerplanar graphs

摘录 / EXCERPT: A graph is maximal planar if it is planar and no edge can be added while preserving planarity, and maximal outerplanar if it is outerplanar and no edge can be added while preservin...

conjecture | open | MathDB 页面 | 原文线索

V2-052466 | MATHDB-368752

Expander-family conjecture for undirected prefix-reversal graphs

摘录 / EXCERPT: Let $\mathbb{P}_m(n)$ be the undirected prefix-reversal graph, which is $2n$-regular for $m>1$. A sequence of regular graphs is an expander family if there exists $\varepsilon>0$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-052467 | MATHDB-368759

Verstraëte's conjecture on consecutive even cycles

摘录 / EXCERPT: Let G be an n-vertex graph, and let $e(G)$ denote its number of edges. Say that G contains k cycles of consecutive even lengths when it contains such a collection of cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-052468 | MATHDB-368760

Sudakov-Verstraëte's extremal block conjecture for consecutive even cycles

摘录 / EXCERPT: Let G be a graph with the maximum possible number of edges among graphs that do not contain k cycles of consecutive even lengths. A block is a maximal connected subgraph with n...

conjecture | open | MathDB 页面 | 原文线索

V2-052469 | MATHDB-368766

Locally finite perturbation conjecture for edge-coloured complete graphs

摘录 / EXCERPT: Let $k\ge2$ be an integer, let K be a finite clique, and let c be a k-colouring of the edges of K such that there is an edge of every colour. A locally finite perturbation...

conjecture | open | MathDB 页面 | 原文线索

V2-052470 | MATHDB-368790

Faudree-Schelp conjecture on path lengths in hamiltonian-connected graphs

摘录 / EXCERPT: Faudree-Schelp conjecture. Every such pair u,v has a path of every length k in this range. The conjecture was disproved by Thomassen, who constructed hamiltonian-connected grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052471 | MATHDB-368793

Füredi-Kim conjecture on saturation numbers of long cycles

摘录 / EXCERPT: Let C_k denote the cycle graph on k vertices, and let $\mathrm{sat}(n,C_k)$ be its saturation number. Füredi-Kim conjecture. There exists a constant k_0 such that ... holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-052472 | MATHDB-368807

The neighborhood-tour conjecture for ASEP vertices

摘录 / EXCERPT: Let P_{ASEP}^{n} be the asymmetric subtour elimination polytope, let $\boldsymbol{x}$ be a vertex of this polytope, and let $\mathcal{N}(\boldsymbol{x})$ denote its neighb...

conjecture | open | MathDB 页面 | 原文线索

V2-052473 | MATHDB-368820

Stevanović's Nordhaus-Gaddum extremal conjecture for graph spectral radius

摘录 / EXCERPT: Stevanović's conjecture. The maximum value of

conjecture | open | MathDB 页面 | 原文线索

V2-052474 | MATHDB-368826

English-McCourt-Mattes-Phillips cocktail-party graph conjecture

摘录 / EXCERPT: English-McCourt-Mattes-Phillips conjecture. In every 2-coloring of the edges of G^c, there exist $A,B\subseteq V$ and colors $i,j\in[2]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052475 | MATHDB-368827

Cairns-Nikolayevsky thrackle conjecture for orientable surfaces

摘录 / EXCERPT: Let S_g be a compact orientable surface of genus $g>0$, and let G be a graph with n vertices and m edges. Cairns-Nikolayevsky's conjecture. If G can be thrackled on S_g...

conjecture | open | MathDB 页面 | 原文线索

V2-052476 | MATHDB-368828

Linear thrackle-genus conjecture for complete bipartite graphs $K_{2,n}$

摘录 / EXCERPT: Let $\operatorname{tg}(G)$ denote the minimum genus of an orientable surface on which the graph G can be thrackled. Linear thrackle-genus conjecture. ... The source has already est...

conjecture | open | MathDB 页面 | 原文线索

V2-052477 | MATHDB-368829

Addario-Berry et al.'s antidirected-tree conjecture

摘录 / EXCERPT: An oriented graph G on n vertices has an edge for each oriented edge of its underlying graph. An antidirected tree is an orientation of a tree in which every vertex has either...

conjecture | open | MathDB 页面 | 原文线索

V2-052478 | MATHDB-368830

Subdivision conjecture for percolated vertex expanders

摘录 / EXCERPT: Fix $d\in\mathbb{N}$ and let H be a graph of maximum degree 3. Let $k\in\mathbb{N}$, and let G be an $(=\hspace{-0.2em}k,d)$-vertex expander, meaning that every set...

conjecture | open | MathDB 页面 | 原文线索

V2-052479 | MATHDB-368836

The clique-cutset conjecture for minimally non-perfectly divisible graphs

摘录 / EXCERPT: Clique-cutset conjecture. No MNPD graph contains a clique cutset.

conjecture | open | MathDB 页面 | 原文线索

V2-052480 | MATHDB-368837

The P5-free perfect divisibility conjecture

摘录 / EXCERPT: P5-free conjecture. Every P_5-free graph is perfectly divisible.

conjecture | open | MathDB 页面 | 原文线索

V2-052481 | MATHDB-368838

The even-hole-free perfect divisibility conjecture

摘录 / EXCERPT: Even-hole-free conjecture. Every even-hole-free graph is perfectly divisible.

conjecture | open | MathDB 页面 | 原文线索

V2-052482 | MATHDB-368839

The bisimplicial-vertex conjecture for minimally non-perfectly divisible graphs

摘录 / EXCERPT: Bisimplicial-vertex conjecture. No MNPD graph contains a bisimplicial vertex.

conjecture | open | MathDB 页面 | 原文线索

V2-052483 | MATHDB-368840

The even-hole-free 3-divisibility conjecture

摘录 / EXCERPT: Even-hole-free 3-divisibility conjecture. Every even-hole-free graph is 3-divisible.

conjecture | open | MathDB 页面 | 原文线索

V2-052484 | MATHDB-368869

Zhao's bipartite swapping conjecture

摘录 / EXCERPT: Let H be a graph and let $q\geq 2$. Write K_q for the complete graph on q vertices, and let $H\times K_2$ denote the categorical product with the two-vertex complete graph. Z...

conjecture | open | MathDB 页面 | 原文线索

V2-052485 | MATHDB-368870

Sah-Sawhney-Stoner-Zhao conjecture on antiferromagnetic homomorphism maxima

摘录 / EXCERPT: Let H be a d-regular graph, and let G be a graph possibly with loops. Regard G as its symmetric adjacency matrix, and suppose that it has at most one positive eigenvalue. S...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-052486 | MATHDB-368871

Universal bipartite-swapping conjecture for antiferromagnetic graphs

摘录 / EXCERPT: Let G be an antiferromagnetic graph, and let H be any graph. Universal bipartite-swapping conjecture. Then ... Here $H\times K_2$ is the categorical product with the two-vertex c...

conjecture | open | MathDB 页面 | 原文线索

V2-052487 | MATHDB-368874

The distinguishing proper arc-colouring conjecture for complete symmetric digraphs

摘录 / EXCERPT: Let $\overleftrightarrow{K_n}$ be the complete symmetric digraph on n vertices. Write $\chi'_{D_{2}}(\overleftrightarrow{K_n})$ for its distinguishing proper arc-colouring number...

conjecture | open | MathDB 页面 | 原文线索

V2-052488 | MATHDB-368875

Botler-Naia conjecture on linear-size subdivision separating systems

摘录 / EXCERPT: Let H be a graph with at least one edge. A subdivision of H is obtained by replacing edges of H with pairwise internally vertex-disjoint paths; let $\operatorname{Sub}(H)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-052489 | MATHDB-368877

Existence of cubic graphs outside minimal Cayley graphs

摘录 / EXCERPT: Cubic graph noncontainment conjecture. For every integer $g\geq 3$, there is a connected cubic graph of girth g that is not a subgraph of a minimal Cayley graph.

conjecture | open | MathDB 页面 | 原文线索

V2-052490 | MATHDB-368878

Generalized Petersen graph induced-subgraph conjecture

摘录 / EXCERPT: Generalized Petersen graph conjecture. Every generalized Petersen graph is an induced subgraph of a minimal Cayley graph.

conjecture | open | MathDB 页面 | 原文线索

V2-052491 | MATHDB-368885

Wong-Zhu's $(1,3)$-choosability conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a nice graph, meaning a graph without isolated edges. A graph is (k,k')-choosable if every assignment of a k-element list of real numbers to each vertex and a...

conjecture | open | MathDB 页面 | 原文线索

V2-052492 | MATHDB-368886

Three-weight edge-weighting problem for nice graphs

摘录 / EXCERPT: Let G be a nice graph, meaning a graph without isolated edges, and let a,b,c be pairwise distinct real numbers. A proper edge weighting is an edge weighting for which adjacent...

open_problem | open | MathDB 页面 | 原文线索

V2-052493 | MATHDB-368889

Characterization of 2-distinguishable trees by branch orbit counts

摘录 / EXCERPT: Let T be a tree. For a vertex w and a neighbor x of w, let T^x denote the component containing x after deleting w, let $\mu(x)$ denote the number of isomorphism types...

conjecture | open | MathDB 页面 | 原文线索

V2-052494 | MATHDB-368890

Conjecture on the eigenvalue interval of the transposition graph

摘录 / EXCERPT: Let S_n be the symmetric group on $[n]:=\{1,\dots,n\}$, let T_n be the set of transpositions in S_n, and let $Cay(S_n,T_n)$ be the Cayley graph of S_n generated by T_n, w...

conjecture | open | MathDB 页面 | 原文线索

V2-052495 | MATHDB-368891

The minimum-edge non-TRVG conjecture for $K_{3,5}$

摘录 / EXCERPT: Minimum-edge non-TRVG conjecture. $K_{3,5}$ is a non-TRVG with the least number of edges.

conjecture | open | MathDB 页面 | 原文线索

V2-052496 | MATHDB-368892

The non-TRVG conjecture for D^a_n in the remaining range

摘录 / EXCERPT: The D^a_n non-TRVG conjecture. D^a_n is a non-TRVG when $2a+5\le n\le 2a+7$ and $a\ge 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-052497 | MATHDB-368893

The D^2_n non-TRVG conjecture

摘录 / EXCERPT: The D^2_n non-TRVG conjecture. D^2_n is a non-TRVG when $n\ge10$.

conjecture | open | MathDB 页面 | 原文线索

V2-052498 | MATHDB-368908

Polynomial dependence in the weak regularity-to-homogeneity proposition

摘录 / EXCERPT: For every $D\geq 1$ and $\varepsilon>0$, there exists $\delta>0$ such that every 3-partite 3-graph H of slicewise VC-dimension at most D that is weakly δ-regular sat...

conjecture | open | MathDB 页面 | 原文线索

V2-052499 | MATHDB-368909

Polynomial Erdős-Hajnal-type dependence for tripartite 3-graphs

摘录 / EXCERPT: Let H_0 be a tripartite 3-graph one of whose parts is a singleton. For every $\varepsilon>0$, there exists $\delta>0$ such that, whenever H is a 3-partite 3-graph on part...

conjecture | open | MathDB 页面 | 原文线索

V2-052500 | MATHDB-368916

Kurz-Lätsch conjecture on oriented diameter and domination number

摘录 / EXCERPT: Kurz-Lätsch conjecture. For every bridgeless graph G with $\gamma(G)=\gamma$,

conjecture | open | MathDB 页面 | 原文线索

V2-052501 | MATHDB-368935

Bender et al.'s reduced-graph conjecture for complemented zero-divisor graphs

摘录 / EXCERPT: Let S be a finite commutative semigroup with zero, and let $G(S)$ be its zero-divisor graph. A graph is complemented if every vertex is orthogonal to some vertex, where adjacent...

conjecture | open | MathDB 页面 | 原文线索

V2-052502 | MATHDB-368944

The maximal-independent-set conjecture for graphs with perfect matchings

摘录 / EXCERPT: Let Γ be a graph on n vertices containing a perfect matching. A maximal independent set is an independent vertex set that is not properly contained in any larger independe...

conjecture | open | MathDB 页面 | 原文线索

V2-052503 | MATHDB-368949

Greedy gossiping conjecture for temporal k-uniform hypergraphs

摘录 / EXCERPT: Greedy gossiping conjecture. The total number of gossip items known by the dons is maximal when every call maximises the amount of new gossip overheard.

conjecture | open | MathDB 页面 | 原文线索

V2-052504 | MATHDB-368967

The 3-sum formula for Speyer's invariant

摘录 / EXCERPT: 3-sum conjecture. The invariant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052505 | MATHDB-368968

The twist invariance conjecture for Speyer's invariant

摘录 / EXCERPT: Twist invariance conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052506 | MATHDB-368969

The 3-edge-cut congruence for Speyer's polynomial

摘录 / EXCERPT: 3-edge-cut conjecture. The Speyer polynomials satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-052507 | MATHDB-368970

The 4-edge-twist congruence for Speyer's polynomial

摘录 / EXCERPT: 4-edge-twist conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052508 | MATHDB-368971

The cubic graph relation between Speyer and Tutte polynomials

摘录 / EXCERPT: Cubic Tutte relation conjecture. The second derivative of Speyer's polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052509 | MATHDB-368972

The prism and Möbius ladder formulas for Speyer's polynomial

摘录 / EXCERPT: Prism and Möbius ladder conjecture. Their Speyer polynomials are

conjecture | open | MathDB 页面 | 原文线索

V2-052510 | MATHDB-368973

The complete bipartite and 4-partite Speyer polynomial formulas

摘录 / EXCERPT: Complete multipartite formula conjecture. Their Speyer polynomials are

conjecture | open | MathDB 页面 | 原文线索

V2-052511 | MATHDB-368974

The planar 4-connected conjecture for Speyer's third invariant

摘录 / EXCERPT: Planar 4-connectivity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052512 | MATHDB-368975

The join zero conjecture for Speyer's polynomial

摘录 / EXCERPT: Join zero conjecture. The polynomial $g_{K_{2+r}\vee\overline{K}_{s+r}}(t)$ has a zero of order s at

conjecture | open | MathDB 页面 | 原文线索

V2-052513 | MATHDB-368976

The Feynman-period characterization of Speyer's invariant

摘录 / EXCERPT: Period--Speyer conjecture. If two cyclically 6-connected 4-regular graphs, or two cyclically 4-connected 3-regular graphs, satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-052514 | MATHDB-368977

The planar-duality-after-vertex-deletion conjecture

摘录 / EXCERPT: Planar deletion-duality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-052515 | MATHDB-368978

The Martin-sequence determination conjecture for Speyer's invariant

摘录 / EXCERPT: Martin-sequence conjecture. The Martin sequence $\operatorname{\mathsf{M}}(G^\bullet)$ determines the value of $\mathsf{N}_2(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-052516 | MATHDB-368982

Non-trivial percolation transition conjecture for supercritical random planar maps

摘录 / EXCERPT: Fix $Q\in(0,2)$ and consider site percolation on $\mathcal M_k$ with open boundary condition: boundary vertices are open, while each interior vertex is independently open with prob...

conjecture | open | MathDB 页面 | 原文线索

V2-052517 | MATHDB-368991

Near-maximum-degree realizable exponents conjecture for graphs

摘录 / EXCERPT: Let H be a graph, with $v(H)$ vertices, $e(H)$ edges, and maximum degree $\Delta(H)$. A rational number is realizable for H if it occurs as an exponent in the generalized Turán...

conjecture | open | MathDB 页面 | 原文线索

V2-052518 | MATHDB-369010

Ding and Marshall's minor conjecture for 3-connected non-Hamiltonian graphs

摘录 / EXCERPT: Let G) be a 3-connected non-Hamiltonian graph. Let $K_{3,4}$ be the complete bipartite graph, let $\mathfrak{Q}^+$ be obtained from the cube by adding a new vertex adjacent to th...

conjecture | open | MathDB 页面 | 原文线索

V2-052519 | MATHDB-369037

The 3m conjecture for irregularising walks

摘录 / EXCERPT: Let G be a nice graph of size m, and let ${\rm ML}^{\rm W}(G)$ denote the minimum size of an irregularising walk of G. The 3m conjecture for irregularising walks states that...

conjecture | open | MathDB 页面 | 原文线索

V2-052520 | MATHDB-369038

Lovász's matching-reduction conjecture for r-partite hypergraphs

摘录 / EXCERPT: Let H be an r-partite hypergraph containing at least one edge, and let $\nu(H)$ denote its matching number, the maximum number of pairwise disjoint edges. For a set S of vert...

conjecture | open | MathDB 页面 | 原文线索

V2-052521 | MATHDB-369051

Draganić-Keevash conjecture on Hamilton cycles after adding a random 2-factor

摘录 / EXCERPT: Let G be an n-vertex d-regular graph, and let $F \sim G_{n,2}$ be a uniformly random 2-factor, that is, a spanning 2-regular graph, on the same vertex set. Draganić-Keevash c...

conjecture | open | MathDB 页面 | 原文线索

V2-052522 | MATHDB-369063

Kasteleyn's Bunkbed Conjecture for Bernoulli bond percolation

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected graph, and let (V^-,E^-) and (V^+,E^+) be two copies of G. For each $v\in V$, write v^- and v^+ for its copies, and form the bunkbed graph...

conjecture | open | MathDB 页面 | 原文线索

V2-052523 | MATHDB-369073

Restricted antidirected subdivision conjecture for oriented graphs

摘录 / EXCERPT: Restricted antidirected subdivision conjecture. Every oriented graph D with

conjecture | open | MathDB 页面 | 原文线索

V2-052524 | MATHDB-369088

Classification conjecture for equal determinants of colored paths

摘录 / EXCERPT: Let P) and Q be two colored paths on m vertices with concentration matrices K_P and K_Q satisfying ... A reflection of a colored path is the path obtained by reversing its...

conjecture | open | MathDB 页面 | 原文线索

V2-052525 | MATHDB-369089

Itai-Zehavi conjecture on independent spanning trees

摘录 / EXCERPT: Let $d\geq 1$, let G be a d-vertex-connected graph, and let r be a vertex of G. A family of spanning trees $T_1,\ldots,T_d$ is said to provide independent paths from r if...

conjecture | open | MathDB 页面 | 原文线索

V2-052526 | MATHDB-369091

Conjecture on optimal independent spanning trees in random regular graphs

摘录 / EXCERPT: Let $d=d(n)\in[3,n-1]$, and let $G\sim G(n,d)$ be a random d-regular graph. An independent spanning tree (IST) family is a collection of spanning trees whose root-to-vertex paths...

conjecture | open | MathDB 页面 | 原文线索

V2-052527 | MATHDB-369109

NP-completeness conjecture for regular completions of trees

摘录 / EXCERPT: Let T be a tree of maximum degree Δ. For $d\geq\Delta$, let $\overline{T_d}$ be a d-regular graph obtained from T by adding semi-edges or loops so that every vertex of...

open_problem | open | MathDB 页面 | 原文线索

V2-052528 | MATHDB-369116

The two 1-visibility cops cleaning conjecture

摘录 / EXCERPT: Two 1-visibility cops cleaning conjecture. Two 1-visibility cops can clean at least 10 vertices of G, or they can clean the whole graph.

conjecture | open | MathDB 页面 | 原文线索

V2-052529 | MATHDB-369134

Rödl-Szemerédi conjecture on superlinear size-Ramsey numbers

摘录 / EXCERPT: Rödl-Szemerédi conjecture. For every $d\geq 3$ there exist $\varepsilon>0$ and a sequence of graphs (G_n) on n vertices and maximum degree at most d such that

conjecture | open | MathDB 页面 | 原文线索

V2-052530 | MATHDB-369136

The broadcast domination-multipacking inequality

摘录 / EXCERPT: For a graph $G=(V,E)$, a multipacking is a set $M\subseteq V$ such that, for every vertex $v\in V$ and every integer $k\geq 1$, $|N_k[v]\cap M|\leq k$, where $N_k[v]$ is the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-052531 | MATHDB-369139

Marcello's completion conjecture for connected graphs with at most one pendant vertex

摘录 / EXCERPT: Marcello's completion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052532 | MATHDB-369142

Signless Laplacian supersaturation conjecture for cliques

摘录 / EXCERPT: Let G be an n-vertex graph, and let $q(G)$ denote its signless Laplacian spectral radius. For fixed $k\geq 3$ and sufficiently large n, consider graphs whose signless Laplaci...

conjecture | open | MathDB 页面 | 原文线索

V2-052533 | MATHDB-369143

Signless Laplacian spectral extremal graph conjecture for graph families

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite family of graphs, let $\chi(\mathcal{F})$ denote its chromatic number, and let $\operatorname{ex}(n,\mathcal{F})$ be the maximum number of edges in an...

conjecture | open | MathDB 页面 | 原文线索

V2-052534 | MATHDB-369154

The $K_7^{=}$ extremal minor conjecture for 5-connected graphs

摘录 / EXCERPT: $K_7^{=}$ extremal minor conjecture. Every 5-connected graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-052535 | MATHDB-369159

Bollobás--Eldridge--Catlin conjecture for bounded-degree spanning subgraphs

摘录 / EXCERPT: Bollobás--Eldridge--Catlin conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-052536 | MATHDB-369161

Equality of the B-Z constants for powers of cycles and paths

摘录 / EXCERPT: Let $\mathcal{C}_k$ and $\mathcal{P}_k$ denote the graph families of k-th powers of cycles and paths, respectively, and let their B-Z constants be defined. B-Z constant equality...

conjecture | open | MathDB 页面 | 原文线索

V2-052537 | MATHDB-369162

Bounded-edge-deletion invariance of the B-Z constant

摘录 / EXCERPT: Bounded-edge-deletion invariance conjecture. Under these assumptions, $\mathcal{G}$ and $\mathcal{H}$ have the same B-Z constant. The source explicitly marks this proposed statemen...

conjecture | open | MathDB 页面 | 原文线索

V2-052538 | MATHDB-369166

The 0_k-graph modular chromatic-index conjecture

摘录 / EXCERPT: Let k be a positive integer. A 0_k-graph is the graph class defined in the source, and $\chi'_k(G)$ is the mod k chromatic index. 0_k-graph conjecture. For every 0_k-grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052539 | MATHDB-369167

The $0_{q(k)}$-graph modular chromatic-index conjecture

摘录 / EXCERPT: Let k be a positive integer. A $0_{q(k)}$-graph is the graph class defined in the source, and $\chi'_k(G)$ is the mod k chromatic index. $0_{q(k)}$-graph conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-052540 | MATHDB-369168

Nash-Williams' Chvátal-type conjecture for Hamilton cycles in digraphs

摘录 / EXCERPT: Let G be a strongly connected digraph on $n\geq 3$ vertices, and let $d^+_1\leq\cdots\leq d^+_n$ and $d^-_1\leq\cdots\leq d^-_n$ be its ordered outdegree and indegree sequences....

conjecture | open | MathDB 页面 | 原文线索

V2-052541 | MATHDB-369174

Aharoni's rainbow Caccetta-Häggkvist conjecture

摘录 / EXCERPT: Let G be an n-vertex graph whose edges are colored with n colors, and let the rainbow girth be the minimum length of a rainbow cycle, with value ∞ if no rainbow cycle e...

conjecture | open | MathDB 页面 | 原文线索

V2-052542 | MATHDB-369195

Matthews-Sumner conjecture on Hamilton cycles in 4-connected claw-free graphs

摘录 / EXCERPT: Let G be a graph. It is claw-free if it has no induced subgraph isomorphic to a claw, and G is 4-connected if deleting fewer than four vertices does not disconnect it. Matthews...

conjecture | open | MathDB 页面 | 原文线索

V2-052543 | MATHDB-369196

Milićević's bounded-diameter conjecture for monochromatic component covers

摘录 / EXCERPT: Milićević's conjecture. For every r, there is a constant $D(r)$ such that every r-edge-coloured complete graph can be covered by $r-1$ monochromatic components of diameter at m...

conjecture | open | MathDB 页面 | 原文线索

V2-052544 | MATHDB-369198

DeBiasio-Kamel-McCourt-Sheats bounded-diameter extension of Ryser's conjecture

摘录 / EXCERPT: DeBiasio-Kamel-McCourt-Sheats conjecture. There exists a constant $d=d(\alpha)$, depending only on α, such that every r-colouring of G has $(r-1)\alpha$ monochromatic co...

conjecture | open | MathDB 页面 | 原文线索

V2-052545 | MATHDB-369199

Equivalence conjecture for the vertex-operator-algebra tensor products

摘录 / EXCERPT: Tensor-product equivalence conjecture. The map above induces an equivalence of monoidal categories

conjecture | open | MathDB 页面 | 原文线索

V2-052546 | MATHDB-369204

Song's refined-character formula for the Macdonald index

摘录 / EXCERPT: Let ${\cal V}$ be the strongly finitely generated vertex operator algebra associated with a four-dimensional ${\cal N}=2$ superconformal field theory, and let ... be its refined char...

conjecture | open | MathDB 页面 | 原文线索

V2-052547 | MATHDB-369233

The three-sevenths upper-bound conjecture for the game isolation number

摘录 / EXCERPT: Three-sevenths upper-bound conjecture. For any graph G with no K_2-components,

conjecture | open | MathDB 页面 | 原文线索

V2-052548 | MATHDB-369255

Transience conjecture for the two-environment reaction-network example

摘录 / EXCERPT: Example transience conjecture. There exists $\alpha>0$ such that the mixed process from Example is transient for all sufficiently small values of κ.

conjecture | open | MathDB 页面 | 原文线索

V2-052549 | MATHDB-369256

Erdős-Hajnal conjecture for the iterated-blowup threshold

摘录 / EXCERPT: Let $g_k(n)$ be the maximum number of edges in an n-vertex iterated blowup of an edge. Thus $g_k(n)=0$ for $n<k$, and for $n\geq k$, ... where the maximum is over compositions of...

conjecture | open | MathDB 页面 | 原文线索

V2-052550 | MATHDB-369257

Linear-exponent conjecture for 3-uniform Ramsey numbers

摘录 / EXCERPT: Let $g_3(s)$ denote the maximum number of edges in an s-vertex iterated blowup of a 3-uniform edge, and let $r_3(s,e;t)$ be the corresponding hypergraph Ramsey number. Conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-052551 | MATHDB-369258

CFG+Y characterization of polynomial 3-uniform Ramsey growth

摘录 / EXCERPT: Let H be a 3-uniform hypergraph, let K_n^3 be the complete 3-uniform hypergraph on n vertices, and let $r(H,K_n^3)$ be the corresponding Ramsey number. Call H iterated...

conjecture | open | MathDB 页面 | 原文线索

V2-052552 | MATHDB-369259

Higher-uniformity monochromatic clique bound for k-partite colorings

摘录 / EXCERPT: Let $g_k(n)$ be the maximum number of edges in an n-vertex iterated blowup of a k-uniform edge. A k-partite coloring of the complete graph K_n is a coloring arising from a...

conjecture | open | MathDB 页面 | 原文线索

V2-052553 | MATHDB-369260

Jiang-Tidor-Yao-Zhang-Zhao bounded-error conjecture for equiangular lines

摘录 / EXCERPT: Let $\alpha\in(0,1)$, define ... and let $\kappa=\kappa(\lambda)$ be the spectral radius order of λ, namely the least number of vertices of a graph whose largest adjacency-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-052554 | MATHDB-369270

Räty-Sudakov-Tomon conjecture on the second eigenvalue of graphs far from Turán graphs

摘录 / EXCERPT: Räty-Sudakov-Tomon conjecture. For any $epsilon>0$, there exists a $c>0$ such that, if G is $epsilon$-far from every Turán graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-052555 | MATHDB-369271

Räty-Sudakov-Tomon surplus conjecture for graphs far from clique unions

摘录 / EXCERPT: Räty-Sudakov-Tomon conjecture. For any $epsilon>0$, there exists a $c>0$ such that, if G is $epsilon$-far from every disjoint union of cliques, then

conjecture | open | MathDB 页面 | 原文线索

V2-052556 | MATHDB-369273

Lo's conjecture on distinguishing connected graphs by subgraph sums

摘录 / EXCERPT: For a graph G and a positive integer N, let ... Here $S(G,c)$ is the set of sums $c(H)=\sum_{v\in V(H)}c(v)$ over connected subgraphs H of G. Lo's conjecture. For any two non...

conjecture | open | MathDB 页面 | 原文线索

V2-052557 | MATHDB-369274

Kileel's degree-independent bound conjecture for powers of polynomials

摘录 / EXCERPT: Let d, k, and s be positive integers, and let $p_1,\ldots,p_k\in\mathbb{R}[x_1,\ldots,x_d]$ be homogeneous polynomials of the same degree s, no two of which are linearly de...

conjecture | open | MathDB 页面 | 原文线索

V2-052558 | MATHDB-369276

Kawarabayashi-Picasarri-Arrieta Reed-type conjecture for digraphs

摘录 / EXCERPT: Let D be a digraph. Define its maximum geometric-mean degree by ... let $\overset{\leftrightarrow}{\omega}(D)$ be its biclique number, and let $\vec{\chi}(D)$ be its dichromatic nu...

conjecture | open | MathDB 页面 | 原文线索

V2-052559 | MATHDB-369277

Directed Borodin-Kostochka conjecture with a biclique-directed-cycle obstruction

摘录 / EXCERPT: Let D be a digraph, let ... let $\overset{\leftrightarrow}{\omega}(D)$ be the biclique number, and let $\vec{\chi}(D)$ be the dichromatic number. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-052560 | MATHDB-369284

Leaf-to-leaf path length conjecture for trees with a given degree sequence

摘录 / EXCERPT: Leaf-to-leaf path length conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052561 | MATHDB-369293

The CVR dimension conjecture for hierarchical T-mesh spline spaces

摘录 / EXCERPT: Let $\mathscr{T}$ be a hierarchical T-mesh, and let $\mathscr{C}$ be its CVR (cross-vertex relationship) graph. For a spline space with the highest order of smoothness, write...

conjecture | open | MathDB 页面 | 原文线索

V2-052562 | MATHDB-369310

The six-vertex Mahler-type conjecture for centrally symmetric planar convex bodies

摘录 / EXCERPT: Let $B\in\mathcal{K}^2_o$ be an o-symmetric convex body in $\mathbb{R}^2$, let $Q_6(B)$ denote the associated six-vertex convex polygon, and let B° be the polar body of...

conjecture | open | MathDB 页面 | 原文线索

V2-052563 | MATHDB-369343

Kanté and Kwon's linear rank-width conjecture for vertex-minor-closed classes

摘录 / EXCERPT: Kanté and Kwon's conjecture. A vertex-minor-closed class of graphs has bounded linear rank-width if and only if it does not contain some tree.

conjecture | open | MathDB 页面 | 原文线索

V2-052564 | MATHDB-369347

Hung's Hamiltonian decomposition conjecture for augmented cubes

摘录 / EXCERPT: Hung's conjecture. An n-dimensional augmented cube admits a Hamiltonian decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-052565 | MATHDB-369349

The ultra-fat half-grid conjecture for quasi-transitive locally finite graphs

摘录 / EXCERPT: Ultra-fat half-grid conjecture. Every connected, quasi-transitive, locally finite graph with a thick end contains an ultra-fat model of the half-grid.

conjecture | open | MathDB 页面 | 原文线索

V2-052566 | MATHDB-369358

The tight upper-bound conjecture for generalized Ramsey numbers of even cycles

摘录 / EXCERPT: Tight upper-bound conjecture. For all $k\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-052567 | MATHDB-369368

Hu-Ye multiplicity conjecture for hypermatrices

摘录 / EXCERPT: Let $\mathcal{M}$ be an order h hypermatrix of dimension n, and let μ be an eigenvalue of $\mathcal{M}$. Write $am_{\mathcal{M}}(\mu)$ for its algebraic multiplicity and...

conjecture | open | MathDB 页面 | 原文线索

V2-052568 | MATHDB-369388

Conjecture on the cop number of graphs forbidding an induced cycle

摘录 / EXCERPT: For an integer $\ell\geq 3$, let C_ℓ be the cycle with ℓ vertices. A graph G is C_ℓ-free if it contains no induced subgraph isomorphic to C_ℓ; write $c(G)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-052569 | MATHDB-369390

Degree-bounded expected value polynomial conjecture for fragile power domination

摘录 / EXCERPT: Degree-bounded expected value polynomial conjecture. The polynomial $\mathcal{E}(G,S,q)$ has degree at most ℓ if and only if, for every $1\leq k\leq s$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052570 | MATHDB-369409

Asymptotic normality criterion for H-linear and H-circular genus distributions

摘录 / EXCERPT: Let H be a graph with a self-gluing $d711: U_1 \to U_2$, and let K be a valid minimal cut for this self-gluing. Construct the swapping $(H',\psi)=\operatorname{Sw}(H,d711,K)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-052571 | MATHDB-369421

The representation-number conjecture for larger toroidal grid graphs

摘录 / EXCERPT: Let $\text{TGr}_{m,n}$ be the toroidal grid graph formed from the Cartesian product of the cycle graphs C_m and C_n, with $m,n\geq 3$, and let $\mathcal{R}(G)$ denote the repre...

conjecture | open | MathDB 页面 | 原文线索

V2-052572 | MATHDB-369423

Maximum-degree formula conjecture for finite linear Jaco graphs

摘录 / EXCERPT: Let $J_n(x)$ be the finite linear Jaco graph of order n, and let $\Delta(J_n(x))$ denote its maximum degree. A sequence is p-graphical for a graph family $\mathcal{F}$ when it...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052573 | MATHDB-369424

Delta-set cardinality conjecture for finite linear Jaco graphs

摘录 / EXCERPT: Let $J_n(x)$ be a finite linear Jaco graph of order $n\geq3$, and let X be its Δ-set, equivalently its Jaconian set, consisting of vertices attaining the maximum degree. A...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052574 | MATHDB-369425

Vertices that never form singleton Delta-sets in finite linear Jaco graphs

摘录 / EXCERPT: Let $J_n(x)$ be a finite linear Jaco graph and let X be its Δ-set. Non-singleton-Delta-vertex conjecture. The vertex subscripts i for which v_i does not yield a single...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052575 | MATHDB-369426

Primary minimal dom-path length conjecture for linear Jaco graphs

摘录 / EXCERPT: Let $J_n(x)$ be a linear Jaco graph of order $n\geq1$. A diam-path is a path of diameter length, and let P_d be a primary minimal dom-path, meaning a minimal path from v_1 to...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052576 | MATHDB-369427

Euler-exponential gamma-set conjecture for the infinite linear Jaco graph

摘录 / EXCERPT: Let $J_\infty(x)$ be the infinite linear Jaco graph, and let X be a γ-set, that is, a minimum dominating set. The vertex subscripts in the displayed set are...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052577 | MATHDB-369445

Bickle-White conjecture on attaining the genus-sum lower bound

摘录 / EXCERPT: Let G be a graph of order n, let $\overline{G}$ be its complement, and let $\gamma(G)$ denote the genus of G. For $n\geq 11$, consider the lower bound ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-052578 | MATHDB-369454

Local-resilience conjecture for structurally stable color-biased Hamilton cycles

摘录 / EXCERPT: Local-resilience conjecture. There exists a sufficiently large constant $C>0$ such that, whenever G satisfies this local resilience condition and every Hamilton cycle in G has...

conjecture | open | MathDB 页面 | 原文线索

V2-052579 | MATHDB-369455

TxGraffiti's harmonic-index bound for minimal maximal matchings

摘录 / EXCERPT: Let G be a nontrivial connected graph. Let $\mu^*(G)$ denote the minimum cardinality of a maximal matching, and let $H(G)$ denote the harmonic index of G. TxGraffiti's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-052580 | MATHDB-369464

Integrality conjecture for the linear program on directed graphs

摘录 / EXCERPT: Integrality conjecture. If the optimal value of $\operatorname{Lp}(G,G_1,Z,P,Y_a)$ is $|Z|+|P|$, then there is an integer solution.

conjecture | open | MathDB 页面 | 原文线索

V2-052581 | MATHDB-369466

Upper-bound conjecture for the d-distance p-packing domination number of trees

摘录 / EXCERPT: Tree upper-bound conjecture. If $3 \leq p \leq d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052582 | MATHDB-369478

The integer-multiple criterion for early state exclusion in 7-qubit spin chains

摘录 / EXCERPT: Let J be a 7×7 Jacobi matrix with symmetric spectrum that realizes perfect state transfer (PST). Its early state exclusion (ESE) property means that no state is excluded...

conjecture | open | MathDB 页面 | 原文线索

V2-052583 | MATHDB-369497

Czap, Przybyło and Škrabul'áková's bipartite 1-planar graph size conjecture

摘录 / EXCERPT: Czap, Przybyło and Škrabul'áková's conjecture. The size of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052584 | MATHDB-369527

Unbounded gap conjecture between outerthickness and uncrossed number

摘录 / EXCERPT: Let $\theta_o(G)$ denote the outerthickness of a graph G, and let $\operatorname{unc}(G)$ denote its uncrossed number. Unbounded-gap conjecture. For every positive integer k, t...

conjecture | open | MathDB 页面 | 原文线索

V2-052585 | MATHDB-369559

Tightness of the mindegree-1 Maker-PhantomBreaker lower bound

摘录 / EXCERPT: Mindegree-1 tightness conjecture. Maker has a randomized strategy to win with probability at least

conjecture | open | MathDB 页面 | 原文线索

V2-052586 | MATHDB-369567

Path-count recurrence for generalized action graphs of the super Catalan numbers

摘录 / EXCERPT: Let G_n be the graph associated with $S(0,n)$, and let $K_{\ell,v,n}$ denote the number of paths of length ℓ in G_n that start at a vertex labeled v and end at a vertex...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052587 | MATHDB-369568

Super Catalan number recurrence from generalized action graph path counts

摘录 / EXCERPT: Let G_n be the action graph associated with $S(0,n)$, and let $K_{\ell,v,n}$ be the number of paths of length ℓ in G_n that start at a vertex labeled v and end at a vert...

conjecture | open | MathDB 页面 | 原文线索

V2-052588 | MATHDB-369571

Cochran's multiplier conjecture for generalized action graphs

摘录 / EXCERPT: Let (s_n) be an appropriate sequence of positive integers, and let G_n be the action graph associated with s_n. For $n\geq 1$, write z_n for the number of new vertices labe...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052589 | MATHDB-369572

The strong conjecture on subpolynomial additive error in weighted grid graphs

摘录 / EXCERPT: Let Grid be the integer grid graph with edge set $E(\mathsf{Grid})$, and let $w:E(\mathsf{Grid})\to\mathbb{R}_{\geq 0}$ be an edge weighting. For vertices u,v, write...

conjecture | open | MathDB 页面 | 原文线索

V2-052590 | MATHDB-369573

Odd-subdivision degree-boundedness conjecture

摘录 / EXCERPT: An odd subdivision of a graph G is a graph obtained from G by replacing each edge by a path of odd length. For a positive integer s, a graph is $K_{s,s}$-free if it contains...

conjecture | open | MathDB 页面 | 原文线索

V2-052591 | MATHDB-369575

Turán's conjecture for complete 3-uniform hypergraphs

摘录 / EXCERPT: Turán's conjecture. For any integer $k\geqslant 4$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-052592 | MATHDB-369578

Mubayi-Zhao conjecture for forbidden traces of complete uniform hypergraphs

摘录 / EXCERPT: Mubayi-Zhao conjecture. Given positive integers n,r,s,t with $2\leqslant t<\min\{r,s\}$, there exists $n_0>0$ such that for $n>n_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-052593 | MATHDB-369584

High-girth Erdős-Nash-Williams conjecture

摘录 / EXCERPT: Let G be a sufficiently large K_3-divisible graph on n vertices, meaning $3\mid e(G)$ and $2\mid d_G(v)$ for every vertex v. The girth of a triangle decomposition is the sm...

conjecture | open | MathDB 页面 | 原文线索

V2-052594 | MATHDB-369585

Generic horofunction boundary conjecture for infinite Sierpinski polygon graphs

摘录 / EXCERPT: Generic horofunction boundary conjecture. If S is empty, the horofunction boundary is as in the stated theorem: it has exactly two Busemann points and countably many non-Busemann...

conjecture | open | MathDB 页面 | 原文线索

V2-052595 | MATHDB-369594

The algebraic-connectivity conjecture for complete graphs in ℓ_∞^d

摘录 / EXCERPT: Let $a(G,X)$ denote the algebraic connectivity of a graph G in a normed space X, let K_{2d} be the complete graph on $2d$ vertices, and let T_d be the tree defined in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-052596 | MATHDB-369595

The edge-redundant rigidity conjecture for complete graphs in ℓ_∞^d

摘录 / EXCERPT: Let $d\geq 2$, let $n\geq 2d+2$, and let K_n denote the complete graph on n vertices. A graph is edge-redundantly rigid in a normed space if it has a framework that is infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-052597 | MATHDB-369610

Integer eigenvalue conjecture for the n-Queens' graph

摘录 / EXCERPT: Let $n\geq 4$, and let $\mathcal{Q}(n)$ denote the n-Queens' graph. The integers already identified as eigenvalues are $n-4$ together with, when n is odd,...

conjecture | open | MathDB 页面 | 原文线索

V2-052598 | MATHDB-369613

The reduced-graph order conjecture of Akbari, Cameron, and Khosrovshahi

摘录 / EXCERPT: Let $r\geq 2$, and let a reduced graph mean a graph with no pair of vertices that can be removed without changing the relevant adjacency-matrix rank. Write $m(r)$ for ... Akbari-Came...

conjecture | open | MathDB 页面 | 原文线索

V2-052599 | MATHDB-369614

Mohammadian's reduced-graph inertia conjecture

摘录 / EXCERPT: Let k be a non-negative integer, and let a reduced graph be a graph with no redundant vertices in the sense used for reduced graphs in the paper. Mohammadian's conjecture. The or...

conjecture | open | MathDB 页面 | 原文线索

V2-052600 | MATHDB-369615

Ma, Yang, and Li's signature cycle-count conjecture

摘录 / EXCERPT: Let G be a graph with signature $s(G)$, where the signature is the difference between the numbers of positive and negative eigenvalues of its adjacency matrix. Let $c_3(G)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-052601 | MATHDB-369616

The line-graph inertia conjecture

摘录 / EXCERPT: Let G be a connected graph, and let $L(G)$ denote its line graph. Let $n^+(L(G))$ and $n^-(L(G))$ be the numbers of positive and negative eigenvalues of the adjacency matrix of...

conjecture | open | MathDB 页面 | 原文线索

V2-052602 | MATHDB-369617

The weakened graph inertia absolute-bound conjecture

摘录 / EXCERPT: Let G be a graph of order n, with $n^+(G)$ denoting the number of positive eigenvalues of its adjacency matrix. Weakened inertia absolute-bound conjecture. ... The paper proposes...

conjecture | open | MathDB 页面 | 原文线索

V2-052603 | MATHDB-369627

The Fibonacci upper-bound conjecture for source-to-sink paths

摘录 / EXCERPT: Let $F_1=F_2=1$ and $F_n=F_{n-1}+F_{n-2}$ for $n 3$. Consider a 3-regular, 3-edge connected, acyclic directed graph with a unique source and a unique sink on $2n$ vertices. Fibonac...

conjecture | open | MathDB 页面 | 原文线索

V2-052604 | MATHDB-369628

The simple 2-connected cubic graph source-to-sink path-count conjecture

摘录 / EXCERPT: Let D be a simple, 2-connected, 3-regular acyclic directed graph on $2n$ vertices with a unique source and a unique sink. Simple 2-connected path-count conjecture. The graph D...

conjecture | open | MathDB 页面 | 原文线索

V2-052605 | MATHDB-369645

The cycle-free chromatic discrepancy conjecture

摘录 / EXCERPT: Cycle-free chromatic discrepancy conjecture. For every integer $\ell\ge 2$, every $C_{\ell+1}$-free graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052606 | MATHDB-369646

The strengthened cycle-free chromatic discrepancy conjecture

摘录 / EXCERPT: Strengthened cycle-free chromatic discrepancy conjecture. For every integer $\ell\ge 3$, every $C_{\ell+1}$-free graph $G\ne K_{\ell}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052607 | MATHDB-369660

Zhang et al.'s AVD total coloring conjecture

摘录 / EXCERPT: Let G be a finite, simple, undirected graph. For each vertex u, let $N_G(u)$ be its set of neighbours, let $\partial_G(u)$ be the set of edges incident with u, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052608 | MATHDB-369666

The exact growth conjecture for treedepth in P_t-free graphs

摘录 / EXCERPT: For integers $t\geqslant 2$ and $k\geqslant 2$, define ... Here, $\operatorname{td}(G)$ is the treedepth of G, $\operatorname{tdtwo}(G)$ is its 2-treedepth, and P_t denotes the p...

conjecture | open | MathDB 页面 | 原文线索

V2-052609 | MATHDB-369670

Aichholzer's compatible triangulation conjecture with prescribed convex-hull bijection

摘录 / EXCERPT: Let P and Q be two arbitrary point sets in general position with $|P|=|Q|$. Write $\operatorname{CH}(P)$ and $\operatorname{CH}(Q)$ for their convex hull vertex sets, and let a...

conjecture | open | MathDB 页面 | 原文线索

V2-052610 | MATHDB-369703

Jackson-Wormald conjecture on walks in highly connected star-free graphs

摘录 / EXCERPT: Let $j\ge 1$ and $k\ge 2$ be integers, and let G be a j-connected $K_{1,j(k-1)+1}$-free graph. Jackson-Wormald's conjecture. Then G has a $(k-1)$-walk. Jin and Li provided co...

conjecture | open | MathDB 页面 | 原文线索

V2-052611 | MATHDB-369710

The nonintegrality conjecture for diminished Sombor energy

摘录 / EXCERPT: Let G be a graph, and let its diminished Sombor energy be the sum of the absolute values of the eigenvalues of its diminished Sombor matrix. Nonintegrality conjecture. There does...

conjecture | open | MathDB 页面 | 原文线索

V2-052612 | MATHDB-369714

Neveu-Schwarz unitarity conjecture for minimal W-algebra highest weight modules

摘录 / EXCERPT: Let $W^k_{\min}(\mathfrak g)$ be the minimal W-algebra, let P^+_k denote the relevant set of dominant integral weights, and let $A(k,\nu)$ be the lower bound for the conformal...

conjecture | open | MathDB 页面 | 原文线索

V2-052613 | MATHDB-369715

Ramond unitarity conjecture for minimal W-algebra highest weight modules

摘录 / EXCERPT: Ramond unitarity conjecture. The set of unitary highest weight modules over $W^k_{\min}(\mathfrak g)$ is the union of

conjecture | open | MathDB 页面 | 原文线索

V2-052614 | MATHDB-369771

Brown-Erdős-Sós conjecture for dense linear triple systems

摘录 / EXCERPT: Let $k\geq 4$ and $0<\delta\leq 1$. A linear 3-graph is a 3-uniform hypergraph in which any two edges share at most one vertex; for a linear 3-graph $\mathcal{H}$ with n ve...

conjecture | open | MathDB 页面 | 原文线索

V2-052615 | MATHDB-369774

Double-tower upper-bound conjecture for regularity of bounded-VC$_2$ hypergraphs

摘录 / EXCERPT: Let $\mathcal{H}$ be a 3-graph with bounded VC$_2$ dimension, and let $\mathcal{Q}_V$ denote the vertex partition in the upper-bound theorem. Double-tower upper-bound conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-052616 | MATHDB-369775

Polynomial graph-partition conjecture for bounded-VC$_2$ hypergraphs

摘录 / EXCERPT: Polynomial graph-partition conjecture. If $\mathcal{H}$ has bounded VC$_2$ dimension, then it has an ε-graph partition with $M=\operatorname{poly}(1/\varepsilon)$ many...

conjecture | open | MathDB 页面 | 原文线索

V2-052617 | MATHDB-369787

Localization capture time conjecture for connected graphs

摘录 / EXCERPT: Localization Capture Time Conjecture (LCTC). The family of all connected graphs is well-localizable; equivalently, there exists a constant D such that every connected graph G s...

conjecture | open | MathDB 页面 | 原文线索

V2-052618 | MATHDB-369805

Cioabă-Desai-Tait conjecture on spectral-consistent graphs

摘录 / EXCERPT: Let H be a graph, and let $T_p(n)$ denote the p-partite Turán graph on n vertices. A graph family is spectral-consistent if, for all sufficiently large n, every graph maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-052619 | MATHDB-369807

Higher-rank DT/PT correspondence for the four-fold vertex with leg boundary conditions

摘录 / EXCERPT: Let $\vec{\pi}_a$ be n-tuples of plane partitions for the four legs, and let $\vec{v}=(v_1,\ldots,v_n)$ and $\vec{w}=(w_1,\ldots,w_n)$ be the flavor fugacities of the n D8 and...

conjecture | open | MathDB 页面 | 原文线索

V2-052620 | MATHDB-369817

Matsuda-Murai's characterization of maximal regularity for binomial edge ideals

摘录 / EXCERPT: Let G be a finite simple graph, with $n(G)$ denoting its number of vertices and J_G its binomial edge ideal in $S=K[x_i,y_i:i\in V(G)]$. Matsuda-Murai's conjecture. ... if and on...

conjecture | open | MathDB 页面 | 原文线索

V2-052621 | MATHDB-369860

Schrijver's rainbow path conjecture

摘录 / EXCERPT: Schrijver's conjecture. If G is a properly edge-colored d-regular graph, then G contains a rainbow path of length $d-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-052622 | MATHDB-369861

The extremal upper-bound conjecture for the exponential hyper-Zagreb index

摘录 / EXCERPT: Let $G\in\mathcal{G}_{n,m}$ be a connected graph with $n\geq 6$ and ... where $1\leq k\leq n-1$ and $0\leq a<n-k-1$. Let $F(G)$ denote the function in the statement. The extremal upp...

conjecture | open | MathDB 页面 | 原文线索

V2-052623 | MATHDB-369862

The redundant-tree conjecture for highly connected graphs

摘录 / EXCERPT: Let T be a tree of order m, and let G be a k-connected graph. Write $\delta(G)$ for the minimum degree of G. A tree subgraph is k-redundant if it has the redundancy pro...

conjecture | open | MathDB 页面 | 原文线索

V2-052624 | MATHDB-369867

Li-Liu-Feng conjecture for generalized books

摘录 / EXCERPT: Li-Liu-Feng conjecture. For fixed integers $r\ge 2$ and $k\ge 1$, and for m sufficiently large, every $B_{r,k}$-free graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052625 | MATHDB-369868

Li-Feng-Peng conjecture on triangular edges in Nosal graphs

摘录 / EXCERPT: Li-Feng-Peng conjecture. Every m-edge Nosal graph has at least $\sqrt{m}$ triangular edges.

conjecture | open | MathDB 页面 | 原文线索

V2-052626 | MATHDB-369881

Necessity and sufficiency of D-unstable cores for network instability

摘录 / EXCERPT: A D-unstable core is a D-unstable Child-Selection matrix $\mathbb{S}[\pmb{\kappa}]$ such that none of its principal submatrices are D-unstable. A network is said to admit ins...

conjecture | open | MathDB 页面 | 原文线索

V2-052627 | MATHDB-369883

The Coarse Grid Theorem

摘录 / EXCERPT: Let $K,n \in \mathbb{N}$. A graph has a K-fat $(n\times n)$-grid minor if it contains the corresponding fat minor model, and graphs are (M,A)-quasi-isometric when they satisfy...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052628 | MATHDB-369884

The induced-minor coarse grid conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$. An induced $(n\times n)$-grid minor is the indicated induced-minor model, and (M,A)-quasi-isometry and tree-width have their usual meanings. Then there exis...

conjecture | open | MathDB 页面 | 原文线索

V2-052629 | MATHDB-369885

The bounded-degree fat-grid tree-width conjecture

摘录 / EXCERPT: Let $K,d,n\in\mathbb{N}$. A graph has a K-fat $(n\times n)$-grid minor when it contains the corresponding fat minor model. Then there exists some $g\in\mathbb{N}$ such that every...

conjecture | open | MathDB 页面 | 原文线索

V2-052630 | MATHDB-369886

The coarse connectivity witness conjecture

摘录 / EXCERPT: Let $K,n,m,r\in\mathbb{N}$. A graph is (M,A)-quasi-isometric to a graph of tree-width at most g in the usual coarse sense, and connected sets are at least K apart when every...

conjecture | open | MathDB 页面 | 原文线索

V2-052631 | MATHDB-369902

Brandt's weak pancyclicity conjecture

摘录 / EXCERPT: A graph is weakly pancyclic if the set of lengths of its cycles forms an interval. Brandt's conjecture. Every non-bipartite graph on n vertices with more than ... edges is weakly p...

conjecture | open | MathDB 页面 | 原文线索

V2-052632 | MATHDB-369903

The $T^{**}(r,n)$ extremal stability conjecture

摘录 / EXCERPT: Let k,r,b be integers with $k\geq 2$, $r=2k+b$, and $3\leq b\leq 2k$. Let $T^{**}(r,n)$ be the three-block graph consisting of a complete bipartite block with parts of sizes...

conjecture | open | MathDB 页面 | 原文线索

V2-052633 | MATHDB-369904

The blowup monotonicity conjecture of Korándi, Roberts and Scott

摘录 / EXCERPT: Fix $k\geq 2$, and let δ be small enough. For any $\delta>\delta_0>0$ and sufficiently large n, consider every C_{2k-1}-free graph G on n vertices satisfying ... Korán...

conjecture | open | MathDB 页面 | 原文线索

V2-052634 | MATHDB-369905

Isolation-number analogue of Vizing's conjecture

摘录 / EXCERPT: Let G and H be graphs, let $G\,\Box\,H$ denote their Cartesian product, and let $\iota(G)$ denote the isolation number of G. Isolation-number product conjecture. For any two...

conjecture | open | MathDB 页面 | 原文线索

V2-052635 | MATHDB-369910

The induced C_4-freeness polynomial testability conjecture

摘录 / EXCERPT: Let C_4 denote the cycle with four vertices, and let $q_{\text{ind-}C_4\text{-free}}(\varepsilon)$ be the query complexity for testing induced C_4-freeness. Induced C_4-freen...

conjecture | open | MathDB 页面 | 原文线索

V2-052636 | MATHDB-369920

Conjectured edge threshold for bounded-diameter orientations of bridgeless graphs

摘录 / EXCERPT: Conjecture on $m(n,d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-052637 | MATHDB-369926

Pierson's conjecture that the Kromatic symmetric function distinguishes all graphs

摘录 / EXCERPT: Let G be a graph, and let $fbar{X}_G$ denote its Kromatic symmetric function. Two graphs are isomorphic when they differ only by a relabeling of their vertices. Pierson's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-052638 | MATHDB-369933

BFP conjecture on nut graphs with two vertex and three edge orbits

摘录 / EXCERPT: Let n be the order of a graph, and let a nut graph be a graph whose adjacency matrix is singular with one-dimensional null space spanned by an eigenvector having no zero entries....

conjecture | open | MathDB 页面 | 原文线索

V2-052639 | MATHDB-369934

Dewdney's circuit-and-enumeration conjecture for (m,n)-trees

摘录 / EXCERPT: Let $n\geq 1$, let $0\leq m\leq n-1$, and let K be a pure n-simplicial complex. Write $\alpha_k(K)$ for the number of k-simplices of K, and call a sequence an (m,n)-circu...

conjecture | open | MathDB 页面 | 原文线索

V2-052640 | MATHDB-369935

Simplicial-tree characterisation from an intermediate face number

摘录 / EXCERPT: Let $n\geq 1$, let p be the number of vertices, and let K be a pure n-simplicial complex. For some $0\leq m\leq n-1$, an (m,n)-simplicial cycle sequence is the cycle-sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-052641 | MATHDB-369953

Double-circuit formulation of the X-replacement conjecture

摘录 / EXCERPT: Let D be a double ${\mathcal R}_3$-circuit, and let w be a technicolour vertex of degree 5 in D. The principal partition of D is the partition associated with its double-...

conjecture | open | MathDB 页面 | 原文线索

V2-052642 | MATHDB-369958

Griess-algebra conjecture for commutants in affine W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple Lie algebra, let f be a nilpotent element, and set ... Let V_2 be the weight-two Griess algebra of V. Griess-algebra conjecture. The algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-052643 | MATHDB-369959

The three-color majority-coloring conjecture for digraphs

摘录 / EXCERPT: A majority k-coloring of a digraph G is a vertex coloring $c:V(G)\rightarrow\{1,\ldots,k\}$ such that every vertex of color i has at least as many out-neighbours of colors di...

conjecture | open | MathDB 页面 | 原文线索

V2-052644 | MATHDB-369970

Kelly's reconstruction conjecture for graphs

摘录 / EXCERPT: Let G be a graph on n vertices. For each vertex $v\in V(G)$, let G_v be the subgraph obtained by deleting v from G, and let ... be the multiset of all vertex-deleted subgra...

conjecture | open | MathDB 页面 | 原文线索

V2-052645 | MATHDB-370009

The path and clique extremality conjecture for mean subtree order

摘录 / EXCERPT: Mean subtree order extremality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052646 | MATHDB-370010

Connected-graph contraction conjecture for mean subtree order

摘录 / EXCERPT: Connected-graph contraction conjecture. Contracting any edge should reduce the mean subtree order by at least $\frac13$, with equality only for a path:

conjecture | open | MathDB 页面 | 原文线索

V2-052647 | MATHDB-370011

Almost-spanning subtree ratio conjecture for the complete graph

摘录 / EXCERPT: Almost-spanning subtree ratio conjecture. For every graph G of order n,

conjecture | open | MathDB 页面 | 原文线索

V2-052648 | MATHDB-370012

Spanning-subtree probability conjecture for the complete graph

摘录 / EXCERPT: Spanning-subtree probability conjecture. For every graph G with n vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-052649 | MATHDB-370013

Complete-graph extremality conjecture for mean subtree order

摘录 / EXCERPT: Complete-graph extremality conjecture. For every graph G with n vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-052650 | MATHDB-370016

Characterization of zero uniform Turán density for r-graphs

摘录 / EXCERPT: Let F be an r-graph. An ordering $\preceq$ on $V(F)$ and a function ... assign to each pair of vertices the role it plays in an edge containing that pair. The zero-density charac...

conjecture | open | MathDB 页面 | 原文线索

V2-052651 | MATHDB-370022

Erdős's conjecture on 5-cycles in triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph on n vertices. Erdős's conjecture. The graph G contains at most $(n/5)^5$ cycles of length 5. The balanced blow-up of the 5-cycle C_5 att...

conjecture | open | MathDB 页面 | 原文线索

V2-052652 | MATHDB-370023

Erdős's balanced blow-up conjecture for 5-cycles

摘录 / EXCERPT: For $n\geq5$, consider the balanced blow-up of the 5-cycle C_5, obtained by replacing each vertex of C_5 by an independent set of size $\lfloor n/5\rfloor$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-052653 | MATHDB-370024

Erdős-Pach-Pollack-Tuza conjecture on diameters of clique-free graphs

摘录 / EXCERPT: Let $r,\delta\geq2$ be fixed integers, and let G be a connected graph of order n and minimum degree δ. Erdős-Pach-Pollack-Tuza conjecture. (i) If G is K_{2r}-free an...

conjecture | open | MathDB 页面 | 原文线索

V2-052654 | MATHDB-370060

The distance-edge-monitoring conjecture

摘录 / EXCERPT: Distance-edge-monitoring conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052655 | MATHDB-370094

The linear cycle-and-edge cover conjecture for 2-regular graphs

摘录 / EXCERPT: Let G be an n-vertex graph. A cycle-and-edge cover is a cover of the edge set of G by subgraphs that are 2-regular graphs or single edges. The linear cycle-and-edge cover c...

conjecture | open | MathDB 页面 | 原文线索

V2-052656 | MATHDB-370095

The regular linear cycle-and-edge cover conjecture

摘录 / EXCERPT: Let G be an n-vertex $(2k+1)$-regular graph, where k is a nonnegative integer. Let $\operatorname{f_{re}}(G)$ denote the minimum number of 2-regular graphs and edges in a c...

conjecture | open | MathDB 页面 | 原文线索

V2-052657 | MATHDB-370106

Quantitative dynamical exclusion-sensitivity threshold conjecture for the voter model

摘录 / EXCERPT: Let $(G_n)_{n\in\mathbb N}$ be a sequence of connected graphs, with n vertices, edge set E_n, and maximum degree Δ_n. Let t_n be a non-negative sequence of times, let...

conjecture | open | MathDB 页面 | 原文线索

V2-052658 | MATHDB-370125

Conjecture on the limiting cut-off for Katz similarity and resistance rankings

摘录 / EXCERPT: For each n, let α_n be the cut-off value at which the Katz similarity index and effective resistance distance rankings for the path graph P_n change from agreeing to dis...

conjecture | open | MathDB 页面 | 原文线索

V2-052659 | MATHDB-370153

Mubayi-Mukherjee conjecture on triangles avoiding a suspended path

摘录 / EXCERPT: Let $n\geq k\geq 4$. For a graph F, let $\widehat{F}$ denote the graph obtained by adding a vertex adjacent to every vertex of F, and let $\operatorname{ex}(n,H,F)$ be the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-052660 | MATHDB-370179

Asymptotic tightness conjecture for limited-visibility cops on Hamming graphs

摘录 / EXCERPT: Asymptotic tightness conjecture. For any fixed positive integer d, we have

conjecture | open | MathDB 页面 | 原文线索

V2-052661 | MATHDB-370190

The Chang-Hsu-Rogers rainbow embedding conjecture

摘录 / EXCERPT: Chang-Hsu-Rogers conjecture. For every n-vertex tree T, the edge-coloured graph $K_{\mathbb{Z}_n}$ contains a rainbow copy of T.

conjecture | open | MathDB 页面 | 原文线索

V2-052662 | MATHDB-370191

The rainbow tree conjecture for properly coloured regular graphs

摘录 / EXCERPT: Rainbow tree conjecture. For every tree T on d vertices, every properly coloured d-regular graph G contains a rainbow copy of T.

conjecture | open | MathDB 页面 | 原文线索

V2-052663 | MATHDB-370192

Gronau-Mullin-Rosa orthogonal double cover conjecture for trees

摘录 / EXCERPT: Gronau-Mullin-Rosa conjecture. For every n-vertex tree T other than the path on 4 vertices, K_n has an orthogonal double cover by copies of T.

conjecture | open | MathDB 页面 | 原文线索

V2-052664 | MATHDB-370194

The cyclic difference rainbow embedding conjecture

摘录 / EXCERPT: Cyclic difference rainbow embedding conjecture. The resulting edge-coloured bipartite graph contains a rainbow embedding of T.

conjecture | open | MathDB 页面 | 原文线索

V2-052665 | MATHDB-370195

The oriented rainbow tree conjecture

摘录 / EXCERPT: Oriented rainbow tree conjecture. Every d-regular properly coloured digraph D contains a rainbow copy of every oriented tree T on $d-1$ edges.

conjecture | open | MathDB 页面 | 原文线索

V2-052666 | MATHDB-370208

Haemers' conjecture on cospectral mates of graphs

摘录 / EXCERPT: Haemers' conjecture. Almost all graphs are determined by their spectra; equivalently, almost all graphs have no cospectral mate.

conjecture | open | MathDB 页面 | 原文线索

V2-052667 | MATHDB-370211

The projective-rank lower bound for graph energy

摘录 / EXCERPT: Let G be an n-vertex graph, and let $\xi_f(G)$ denote the projective rank of G. Projective-rank energy conjecture. ... This is proposed as a weakening of Fajtlowicz's graph ene...

conjecture | open | MathDB 页面 | 原文线索

V2-052668 | MATHDB-370230

Constant-offset conjecture for proper conflict-free degree-choosability

摘录 / EXCERPT: Let G be a graph. A list assignment L assigns a set of positive integers to each vertex, and G is proper conflict-free $({\rm degree}+k)$-choosable if it has a proper conflic...

conjecture | open | MathDB 页面 | 原文线索

V2-052669 | MATHDB-370232

Voss's bridge-length conjecture for longest cycles

摘录 / EXCERPT: Voss's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052670 | MATHDB-370233

Scott Smith's longest-cycle intersection conjecture

摘录 / EXCERPT: Scott Smith's conjecture. The cycles C_1 and C_2 intersect in at least k vertices:

conjecture | open | MathDB 页面 | 原文线索

V2-052671 | MATHDB-370247

Chen-Huang-Yu-Zhou conjecture on strong edge-coloring and Ore-degree seven

摘录 / EXCERPT: Chen-Huang-Yu-Zhou conjecture. If $\theta(G)=7$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052672 | MATHDB-370255

The first-second Zagreb index comparison conjecture

摘录 / EXCERPT: Zagreb index comparison conjecture. If G is a connected graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-052673 | MATHDB-370257

Conjecture on the unique signed graph realizing $n_s(4,3)$

摘录 / EXCERPT: Let $n_s(\lambda,p)$ be the minimum number of vertices of a signed graph with balanced chromatic number at least p and negative girth at least λ. Signed-graph extremal co...

conjecture | open | MathDB 页面 | 原文线索

V2-052674 | MATHDB-370263

The common-difference set conjecture for cyclic arithmetic progressions

摘录 / EXCERPT: Common-difference set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052675 | MATHDB-370271

Zdeborová-Boettcher conjecture on max-cut and bisection width

摘录 / EXCERPT: Let G be a random regular graph, and let the max-cut be a partition of its vertex set maximizing the number of crossing edges. Let the bisection width be the minimum number of cr...

conjecture | open | MathDB 页面 | 原文线索

V2-052676 | MATHDB-370277

Near-Sidorenko conjecture for 3-uniform tight cycles

摘录 / EXCERPT: Let $C^{(3)}_{3\ell}$ denote the 3-uniform tight cycle with 3ℓ edges, and let $s(F)$ denote the Sidorenko exponent of a hypergraph F. Near-Sidorenko conjecture for 3-unifo...

conjecture | open | MathDB 页面 | 原文线索

V2-052677 | MATHDB-370278

Bounded Sidorenko exponents for higher-uniformity tight cycles

摘录 / EXCERPT: For each integer $r\geq 4$, let $C^{(r)}_{r\ell}$ denote the r-uniform tight cycle with $r\ell$ edges, and let $s(F)$ denote the Sidorenko exponent of a hypergraph F. Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-052678 | MATHDB-370279

KST-type lower-bound conjecture from Sidorenko exponents

摘录 / EXCERPT: Let $r\geq 2$ be an integer and let F be an $(r-1)$-graph. Let $F(t)$ denote the r-uniform hypergraph obtained by adjoining a new vertex to every edge of F, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052679 | MATHDB-370280

Extremal lower bound for the suspended 6-cycle

摘录 / EXCERPT: Let C_6 be the 6-cycle, let $C_6(t)$ denote the associated 3-uniform hypergraph obtained by adjoining a new vertex to every edge of C_6, and let $\operatorname{ex}(n,H)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-052680 | MATHDB-370303

Ferdjallah et al.'s subcubic injective edge-coloring conjecture

摘录 / EXCERPT: Ferdjallah et al.'s conjecture. For every subcubic graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052681 | MATHDB-370304

Blanco-Buehrle spectral gap conjecture for generalised pancake graphs

摘录 / EXCERPT: Let $\mathcal P_m(n)$ denote the generalised pancake graph, and let $\lambda_1(\mathcal P_m(n))$ and $\lambda_2(\mathcal P_m(n))$ be its largest and second-largest signless Laplaci...

conjecture | open | MathDB 页面 | 原文线索

V2-052682 | MATHDB-370305

Spectral-gap equality conjecture for generalised pancake graphs

摘录 / EXCERPT: Let $\mathcal P_m(n)$ be the generalised pancake graph and let $Q(\mathfrak P_{m,n})$ be the associated matrix used in the paper. For a symmetric matrix or graph, write λ_1...

conjecture | open | MathDB 页面 | 原文线索

V2-052683 | MATHDB-370316

The generalized k-independence lower-bound conjecture

摘录 / EXCERPT: Generalized k-independence conjecture. For every n-vertex simple graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052684 | MATHDB-370334

k-regular bipartite domination number bound

摘录 / EXCERPT: Let G be a k-regular bipartite graph with bipartition $V(G)=A\cup B$ and $|A|=|B|=n$. Let $\gamma(G)$ be the domination number of G, and let ρ_G denote its normalised do...

conjecture | open | MathDB 页面 | 原文线索

V2-052685 | MATHDB-370335

Sivaraman's cop-number conjecture for path-free graphs

摘录 / EXCERPT: Let P_k denote the path on k vertices, let G be a graph, and let $c(G)$ denote its cop number. A graph is P_k-free if it contains no induced subgraph isomorphic to P_k. S...

conjecture | open | MathDB 页面 | 原文线索

V2-052686 | MATHDB-370341

Binary matroid conjecture for matroids with cycle systems

摘录 / EXCERPT: Let M be a matroid admitting a cycle system, meaning a cycle-system structure as defined in the source. Binary matroid conjecture. Every matroid with a cycle system is binary. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-052687 | MATHDB-370342

Cycle-system extremal conjecture for glued-triangle graphs

摘录 / EXCERPT: For n vertices, let G_n be the connected graph obtained by gluing $n-2$ triangles along one common edge. Cycle-system extremal conjecture. The graph G_n has exactly ... cycle s...

conjecture | open | MathDB 页面 | 原文线索

V2-052688 | MATHDB-370375

Győri-Keszegh's greedy-partition triangle conjecture for K_4-free graphs

摘录 / EXCERPT: Győri-Keszegh's conjecture. For every such graph G and every greedy partition P,

conjecture | open | MathDB 页面 | 原文线索

V2-052689 | MATHDB-370414

Holroyd-Talbot conjecture on the EKR property of graphs

摘录 / EXCERPT: Let G be a graph, let $\cal I^{(r)}(G)$ denote the family of independent r-subsets of $V(G)$, and let $\cal I^{(r)}_v(G)$ denote the star centered at $v\in V(G)$. Write $\mu(G)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-052690 | MATHDB-370420

Determination of corona cycles by Laplacian permanental polynomials

摘录 / EXCERPT: Let C_n be the cycle graph on n vertices, let K_1 be the one-vertex graph, and let $C_n\odot K_1$ denote their corona product. A graph is determined by its (signless) Laplaci...

conjecture | open | MathDB 页面 | 原文线索

V2-052691 | MATHDB-370430

The oriented strong embedding conjecture

摘录 / EXCERPT: Oriented strong embedding conjecture. Every 2-connected graph has a strong embedding on some orientable surface.

conjecture | open | MathDB 页面 | 原文线索

V2-052692 | MATHDB-370434

Cereceda's quadratic recoloring diameter conjecture for degenerate graphs

摘录 / EXCERPT: Let G be a d-degenerate graph, and let $R_t(G)$ be the graph whose vertices are the proper t-colorings of G, with two colorings adjacent when they differ on one vertex. The...

conjecture | open | MathDB 页面 | 原文线索

V2-052693 | MATHDB-370435

Flawless 1-factorization conjecture for even orders

摘录 / EXCERPT: For an even integer n, let $f(n)$ be the maximum integer k such that some k-regular graph G on n vertices admits a flawless 1-factorization, meaning a perfect 1-factoriza...

conjecture | open | MathDB 页面 | 原文线索

V2-052694 | MATHDB-370439

Precoloring extension conjecture for Cartesian products with complete bipartite graphs

摘录 / EXCERPT: Cartesian-product precoloring extension conjecture. This precoloring is extendable to a proper edge coloring.

open_problem | open | MathDB 页面 | 原文线索

V2-052695 | MATHDB-370515

Merris's algebraic-connectivity conjecture for graph accessibility

摘录 / EXCERPT: Let G be a graph on n vertices. Let $a(G)$ denote its algebraic connectivity, and let $\omega(G)$ denote the minimum entry of its matrix of relative forest accessibilities. Mer...

conjecture | open | MathDB 页面 | 原文线索

V2-052696 | MATHDB-370516

Merris's anti-regular graph accessibility conjecture

摘录 / EXCERPT: Let E_n be the degree anti-regular graph, meaning the unique connected graph whose vertex degrees attain every value from 1 through $n-1$. Let $\omega(E_n)$ denote its minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-052697 | MATHDB-370517

The arboreal-gas bunkbed conjecture

摘录 / EXCERPT: Let G be a graph, let $\widetilde{G}$ be its associated bunkbed graph, and let $\mu^{\text{F}}_{\widetilde{G},\lambda}$ denote the arboreal gas measure with parameter λ....

conjecture | open | MathDB 页面 | 原文线索

V2-052698 | MATHDB-370518

The alternate arboreal-gas bunkbed conjecture

摘录 / EXCERPT: Let G be a graph, let $T\subset V(G)$ be a set of posts, and let $\nu^{\text{F}}_{G,T}$ denote the alternate arboreal gas measure on red-blue edge-colourings, restricted so that...

conjecture | refuted | MathDB 页面 | 原文线索

V2-052699 | MATHDB-370519

The arboreal-gas three-point correlation conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph and let $u,v,w\in V$. Let $\mu^{\text{F}}_{G,\lambda}$ denote the arboreal gas measure with parameter λ, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-052700 | MATHDB-370522

Edwards et al.'s distance-2 matching edge-coloring conjecture

摘录 / EXCERPT: Edwards et al.'s conjecture. If H is a distance-2 matching, then every $(\Delta+1)$-edge-coloring of H extends to a $(\Delta+1)$-edge-coloring of G.

conjecture | open | MathDB 页面 | 原文线索

V2-052701 | MATHDB-370527

Hassler-Treglown conjecture on maximal independent sets in graphs with a perfect matching

摘录 / EXCERPT: Let Γ be an n-vertex graph, and let $\mathrm{mis}(\Gamma)$ denote its number of maximal independent sets. Hassler-Treglown conjecture. If Γ contains a perfect match...

conjecture | open | MathDB 页面 | 原文线索

V2-052702 | MATHDB-370553

Pauling's residual entropy conjecture for regular graphs of growing degree

摘录 / EXCERPT: Let $G=G(n)$ be a sequence of d-regular graphs of order n, with even degree $d=d(n)$. Let $\rho(G)$ denote the residual entropy, ... where $\operatorname{EO}(G)$ is the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-052703 | MATHDB-370568

Conjecture on maximal vertex numbers of maxout polytopes

摘录 / EXCERPT: A (d,n,m)-maxout polytope is a polytope specified by the parameters d, n, and m, and $f_0(P)$ denotes the number of vertices of a polytope P. Maxout polytopes vertex-numb...

conjecture | open | MathDB 页面 | 原文线索

V2-052704 | MATHDB-370578

Meunier's stability conjecture for stable Kneser hypergraphs

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, and let $\mathrm{KG}^r(n,k)_{s\textup{-stab}}$ be the induced r-uniform Kneser hypergraph whose vertices are the s-stable k-subsets of $[n]$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-052705 | MATHDB-370584

Constant-curvature convergence conjecture for normalized Ricci flow on trees

摘录 / EXCERPT: Let $T=(V,E)$ be a tree with an initial metric $w\in\mathbb{R}_{>0}^{E}$, and let the metric $w(t)$ evolve under the normalized Ricci flow. The flow converges to a metric with cons...

conjecture | open | MathDB 页面 | 原文线索

V2-052706 | MATHDB-370616

Path P_6 edge-inducibility conjecture

摘录 / EXCERPT: Path P_6 edge-inducibility conjecture. The lower bound is exact:

conjecture | open | MathDB 页面 | 原文线索

V2-052707 | MATHDB-370645

Quantum upper cluster algebra generation conjecture for projected stated ${\rm SL}_n$-skein algebras

摘录 / EXCERPT: Let $\mathfrak{S}$ be a triangulable pb surface, let $\mathcal V$ be the vertex set of the associated quantum seed, and let $\mathcal V_{\mathrm{mut}}$ be its mutable vertices. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-052708 | MATHDB-370647

Verstraëte's conjecture on packing graph subdivisions

摘录 / EXCERPT: Given graphs F and G, a subdivision packing of F in G is a collection of pairwise vertex-disjoint copies of subdivisions of F. For a real number $\eta > 0$ and a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-052709 | MATHDB-370676

Boundary-level rationality and C_2-cofiniteness conjecture for affine $\mathfrak{g}$

摘录 / EXCERPT: Boundary-level rationality conjecture. The vertex operator superalgebra $L_{\widehat{\mathfrak{g}}}(\mathcal{k},0)$ is rational in the category $\mathcal{O}$, and its irreducible w...

conjecture | open | MathDB 页面 | 原文线索

V2-052710 | MATHDB-370678

The finite-orbit bound conjecture for the curve graph of spherical Artin-Tits groups

摘录 / EXCERPT: Let G be an Artin-Tits group of spherical type with rank n, let $\mathcal C(G)$ be the curve graph, and let $\alpha\in G$. Consider a vertex of $\mathcal C(G)$ with a finite or...

conjecture | open | MathDB 页面 | 原文线索

V2-052711 | MATHDB-370679

The periodic-order bound conjecture for finite orbits in spherical Artin-Tits groups

摘录 / EXCERPT: Let G be an Artin-Tits group of spherical type, let $\mathcal C(G)$ be its curve graph, and let $\alpha\in G$. Suppose a vertex of $\mathcal C(G)$ has a finite orbit under...

conjecture | open | MathDB 页面 | 原文线索

V2-052712 | MATHDB-370691

Upper bound for outer general position number after vertex removal

摘录 / EXCERPT: Let G be a graph and let x be a vertex that is not a cut vertex of G. Write ${\rm gp}_{\rm o}(G)$ for the outer general position number of G and $\deg_G(x)$ for the degree...

conjecture | open | MathDB 页面 | 原文线索

V2-052713 | MATHDB-370712

The $K_{1,3}$ characterization conjecture for ultrametric spaces

摘录 / EXCERPT: $K_{1,3}$ characterization conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-052714 | MATHDB-370718

Bogdanov-Neustroeva-Sokolov-Volostnov-Russkin-Voronov conjecture for bipartite cuts

摘录 / EXCERPT: Let G be an n-vertex graph. A bipartite cut is a cut whose induced subgraph on one side is bipartite. Bogdanov-Neustroeva-Sokolov-Volostnov-Russkin-Voronov conjecture. Any n-...

conjecture | open | MathDB 页面 | 原文线索

V2-052715 | MATHDB-370719

General chromatic cut conjecture

摘录 / EXCERPT: Let k be an integer, let G be a graph with at least k vertices, and let X be one side of a cut of G. Write $V(G)$ and $E(G)$ for the vertex and edge sets, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052716 | MATHDB-370726

Modified anti-Ramsey formula conjecture for vertex-disjoint triangles

摘录 / EXCERPT: Let n and k be integers with $n\geq 3k$, let kC_3 be the vertex-disjoint union of k triangles, and let $G_1(n,k),\ldots,G_4(n,k)$ be the four explicitly defined edge-colore...

conjecture | open | MathDB 页面 | 原文线索

V2-052717 | MATHDB-370733

Gullerud-Johnson-Mbirika conjectures on roots of Fibonacci-shifted path polynomials

摘录 / EXCERPT: Let P_n be the path graph on n vertices, let ... be its characteristic polynomial, and let $F_0=0$, $F_1=1$, with F_n denoting the Fibonacci numbers. For each n, let R_n be...

conjecture | open | MathDB 页面 | 原文线索

V2-052718 | MATHDB-370754

Spectral stability conjecture for minimum Q-degree hypergraphs

摘录 / EXCERPT: The spectral stability conjecture. There exists n_{0} such that for any $\mathcal{F}$-free r-graph H on $n\geq n_{0}$ vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-052719 | MATHDB-370769

Minimum-degree conjecture for minimal extendable claw-free graphs

摘录 / EXCERPT: Let G be a minimal k-extendable claw-free graph, meaning that G is k-extendable and deleting any edge produces a graph that is not k-extendable. Let $\delta(G)$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-052720 | MATHDB-370779

Rainbow anti-Ramsey conjecture for unions of triangles

摘录 / EXCERPT: Rainbow anti-Ramsey conjecture. The rainbow anti-Ramsey number for $(t+2)$ vertex-disjoint triangles satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052721 | MATHDB-370825

Linear degree-count separation conjecture for super-minimally and uniformly connected graphs

摘录 / EXCERPT: Linear degree-count separation conjecture. There are constants a_k and b_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-052722 | MATHDB-370830

Lorentzianity conjecture for graph-coloring generating polynomials

摘录 / EXCERPT: Let G be a graph with vertex set V, and for each $v\in V$ let $S(v)$ be its list of available colors. Let $p_\varnothing$ be the polynomial recursively defined from the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-052723 | MATHDB-370838

The extremal bound conjecture for the Sigma index of trees

摘录 / EXCERPT: Extremal bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052724 | MATHDB-370849

Glock-Kühn-Osthus high-degree conjecture for hypergraph decompositions

摘录 / EXCERPT: Glock-Kühn-Osthus conjecture. For each integer $r\ge 3$, there exists a constant $C>0$ such that, for all integers $q>r$ and all sufficiently large n, every K_q^r-divisible r...

conjecture | open | MathDB 页面 | 原文线索

V2-052725 | MATHDB-370857

Berge-Füredi conjecture for linear hypergraphs

摘录 / EXCERPT: Berge-Füredi conjecture. A linear (loopless) hypergraph (V,E) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052726 | MATHDB-370870

Lee's lower-bound conjecture for apex partite hypergraphs

摘录 / EXCERPT: Let $d\ge 2$ and let $\mathcal{H}$ be a $(d-1)$-partite $(d-1)$-uniform hypergraph. For $k\in\mathbb{N}$, let $\mathcal{H}(k)$ be the d-partite d-uniform hypergraph whose dth...

conjecture | open | MathDB 页面 | 原文线索

V2-052727 | MATHDB-370893

Isospectrality conjecture for generalized projective graphs over finite local rings

摘录 / EXCERPT: Let p, r, and d be as above, and let $2\leq m<n\leq d$. Consider the graphs $\mathbb{P}_{m,n}^{d-1}\left(\mathbb{Z}/(p^{r})\right)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-052728 | MATHDB-370899

Fixed-parameter tractability of Map-ε-PDD for $k+\delta+h$

摘录 / EXCERPT: Let k, δ, and h be the parameters of the Map-ε-PDD problem, where h is the value of H in a phylogenetic tree. Map-ε-PDD conjecture. Map-...

conjecture | open | MathDB 页面 | 原文线索

V2-052729 | MATHDB-370928

The conjecture that all orientations of odd cycles are avoidable

摘录 / EXCERPT: Odd-cycle avoidability conjecture. All orientations of odd cycles are avoidable.

conjecture | open | MathDB 页面 | 原文线索

V2-052730 | MATHDB-370951

Finiteness conjecture for resistance-positive simple 3-polytopes

摘录 / EXCERPT: Let a polytope be resistance-positive when its resistance curvature is positive at every vertex. Finiteness conjecture for resistance-positive simple 3-polytopes. There are finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-052731 | MATHDB-370963

Conjectured formula for the uniformly dense Turán density of stars

摘录 / EXCERPT: For each integer k with $4\le k\le 10$, let S_k denote the star 3-graph considered in the paper, and let $\pi_{\mathrm{U}}(S_k)$ be its uniformly dense Turán density. Conject...

conjecture | open | MathDB 页面 | 原文线索

V2-052732 | MATHDB-370964

Conjectured value of the uniformly dense Turán density of the four-vertex star

摘录 / EXCERPT: Let S_4 be the four-vertex star 3-graph, and let $\pi_{\mathrm{V}}(S_4)$ denote the corresponding Turán-density parameter. Conjecture for the four-vertex star. ... This is the se...

conjecture | open | MathDB 页面 | 原文线索

V2-052733 | MATHDB-370991

Net graph asymptotic regularity conjecture

摘录 / EXCERPT: Let N be the net graph. Its binomial edge ideal is denoted by J_N, and write $\widehat{\operatorname{reg}}(J_N)$ for its asymptotic regularity. Net graph asymptotic regularity...

conjecture | open | MathDB 页面 | 原文线索

V2-052734 | MATHDB-370992

Asymptotic regularity conjecture for initial ideals of binomial edge ideals

摘录 / EXCERPT: Let G be an enumerated graph. Let ℓ be the length of the longest induced path in G, and let $\ell_{<}$ be the length of the longest admissible path in G. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-052735 | MATHDB-371075

The dim-boundedness conjectures for planar posets

摘录 / EXCERPT: A poset has a planar diagram if its cover graph can be drawn in the plane with every cover relation represented by a curve directed upwards. Its cover graph is the graph whose vert...

conjecture | open | MathDB 页面 | 原文线索

V2-052736 | MATHDB-371077

The Tree Unimodality Conjecture

摘录 / EXCERPT: Let T be a tree, let $\alpha=\alpha(T)$ be its independence number, and let a_i denote the number of independent sets of cardinality i, so that $a_0=1$. The Tree Unimodality...

conjecture | open | MathDB 页面 | 原文线索

V2-052737 | MATHDB-371078

The independence-number lower-bound conjecture for log-concavity failures

摘录 / EXCERPT: Let T be a tree on N vertices, with independence sequence (a_i) and independence number $\alpha(T)$. Independence-number lower-bound conjecture. If the independence sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-052738 | MATHDB-371081

Hajebi's treewidth-clique boundedness conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary graph class. A graph is 2-degenerate if every induced subgraph has a vertex of degree at most 2, and $\mathcal{G}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-052739 | MATHDB-371098

The wheel-free Burling-control conjecture

摘录 / EXCERPT: A wheel is a graph consisting of an induced cycle of length at least 4 and one additional vertex adjacent to at least three vertices on the cycle. A graph class is Burling-contro...

conjecture | open | MathDB 页面 | 原文线索

V2-052740 | MATHDB-371101

Campos-Jenssen-Michelen-Sahasrabudhe conjecture for the asymptotic constant in $R(3,k)$

摘录 / EXCERPT: Campos-Jenssen-Michelen-Sahasrabudhe conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052741 | MATHDB-371107

Kang-Kelly-Kühn-Methuku-Osthus random design existence conjecture

摘录 / EXCERPT: Let $\mathcal{G}^{(q)}(n,p)$ be the random q-uniform hypergraph on n vertices, and let an (n,q,r)-Steiner system be a collection of q-sets containing every r-set exactly...

conjecture | open | MathDB 页面 | 原文线索

V2-052742 | MATHDB-371138

The two-dimensional Chebyshev flow-number conjecture

摘录 / EXCERPT: Two-dimensional Chebyshev flow-number conjecture. For every bridgeless graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-052743 | MATHDB-371139

The half-flow-pair conjecture

摘录 / EXCERPT: Half-flow-pair conjecture. Every bridgeless graph admits a $1/2$-flow-pair.

conjecture | open | MathDB 页面 | 原文线索

V2-052744 | MATHDB-371140

The unimodality conjecture for multidimensional flow numbers

摘录 / EXCERPT: Unimodality conjecture for flow numbers. As a function of p, $\Phi_d^p(G)$ is unimodal: there exists $p^*\in[1,\infty]$ such that it is non-decreasing for $p\leq p^*$ and non-inc...

conjecture | open | MathDB 页面 | 原文线索

V2-052745 | MATHDB-371148

Bayer et al.'s homotopy conjecture for total cut complexes of squared cycles

摘录 / EXCERPT: Bayer et al.'s conjecture. The $(n-k-1)$-dimensional complex $$ is the void complex if $n e 3k+i$ for the relevant ranges below; otherwise, it is homotopy equivalent to a single sp...

conjecture | open | MathDB 页面 | 原文线索

V2-052746 | MATHDB-371149

Homotopy conjecture for 2-total cut complexes of powers of cycles

摘录 / EXCERPT: The 2-total cut complex conjecture. For $p\ge 3$ and $n\ge 3p+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-052747 | MATHDB-371150

Homotopy conjecture for total cut complexes of powers of cycles

摘录 / EXCERPT: The total cut complex conjecture. For $p\ge 3$ and $n\ge (2k-1)p+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-052748 | MATHDB-371170

The extremal conjecture for average size of 1-nearly independent sets in trees

摘录 / EXCERPT: Let R_n be the tree obtained, for $n\geq 4$, by subdividing once an edge of the star S_{n-1}, and let $av_1(T)$ denote the average size of the 1-nearly independent vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-052749 | MATHDB-371172

Linear-graph connection optimality conjecture for approximate unitary designs

摘录 / EXCERPT: Linear-graph connection optimality conjecture. No other graph on n qudits requires more connections to form an ϵ-approximate t-design than the linear graph, which requ...

conjecture | open | MathDB 页面 | 原文线索

V2-052750 | MATHDB-371187

The maximal semi-inducibility conjecture for the 4-vertex graph H_3

摘录 / EXCERPT: Semi-inducibility conjecture for H_3. The semi-induced density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-052751 | MATHDB-371211

Krivelevich-Lew-Michaeli edge-counting conjecture for graph rigidity

摘录 / EXCERPT: Let $f(n,d)$ be the smallest integer such that every n-vertex graph with minimum degree at least $f(n,d)$ is d-rigid. A graph is d-rigid when every generic framework in...

conjecture | open | MathDB 页面 | 原文线索

V2-052752 | MATHDB-371212

Degree-sum conjecture for graph rigidity

摘录 / EXCERPT: For a graph G, define … and let $g(n,d)$ be the smallest integer such that every n-vertex graph G with $\eta(G)\geq g(n,d)$ is d-rigid. Degree-sum conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-052753 | MATHDB-371234

Bollobás-Scott weighted long-cycle conjecture for Eulerian digraphs

摘录 / EXCERPT: A digraph is a directed graph; it is Eulerian when it is strongly connected and every vertex has equal in-degree and out-degree. For a digraph G, let $A(G)$ denote its arc set, l...

conjecture | open | MathDB 页面 | 原文线索

V2-052754 | MATHDB-371235

Bollobás-Scott linear long-cycle conjecture for Eulerian digraphs

摘录 / EXCERPT: A digraph is Eulerian when it is strongly connected and every vertex has equal in-degree and out-degree. Its average out-degree is the average of the out-degrees of its vertices; d...

conjecture | open | MathDB 页面 | 原文线索

V2-052755 | MATHDB-371247

The totally sorted configuration and SYT probability conjecture for labeled chip-firing on star graphs

摘录 / EXCERPT: Let $\Delta^{k,m}$ be the star graph with k branches, each receiving m labeled chips, and consider the random labeled chip-firing process in which every firable vertex is equal...

conjecture | open | MathDB 页面 | 原文线索

V2-052756 | MATHDB-371253

Asymptotic chromatic-number conjecture for Kneser hypergraphs of triangulations

摘录 / EXCERPT: Asymptotic chromatic-number conjecture. For any $r\geq 2$, there exists an integer $n_0(r)$ such that, for all $n\geq n_0(r)$,

conjecture | refuted | MathDB 页面 | 原文线索

V2-052757 | MATHDB-371282

Yuan-Zhang conjecture on exact Turán numbers of vertex-disjoint paths

摘录 / EXCERPT: Let $k_1\geq k_2\geq\ldots\geq k_m\geq 3$ with $k_1>3$, and let … be the disjoint union of paths. For integers $n\geq m\geq 3$, write $c(n,a,b)$ and $c(n,a)$ as in the preceding de...

conjecture | open | MathDB 页面 | 原文线索

V2-052758 | MATHDB-371286

The bipartite unique domination bound

摘录 / EXCERPT: Bipartite unique domination bound. If G has a unique minimum dominating set, $\gamma\geq 2$, and $n\geq 3\gamma$, then its size $s(G)$ is bounded above by

conjecture | open | MathDB 页面 | 原文线索

V2-052759 | MATHDB-371289

Flandrin et al.'s neighbor sum distinguishing proper edge-coloring conjecture

摘录 / EXCERPT: Let G be a nice graph, meaning that it has no K_2 component. A proper edge-coloring is an edge-coloring in which adjacent edges receive different colors, and $\chi'_{\sum}(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-052760 | MATHDB-371290

Quasi-majority neighbor sum distinguishing conjecture

摘录 / EXCERPT: Let G be a nice graph, meaning that it has no K_2 component, and let $\chi_{\sum}^{QM}(G)$ denote its quasi-majority neighbor sum distinguishing index. Quasi-majority neighbor...

conjecture | open | MathDB 页面 | 原文线索

V2-052761 | MATHDB-371291

Majority neighbor sum distinguishing conjecture

摘录 / EXCERPT: Let G be a graph with minimum degree $\delta(G)\geq 2$, and let $\chi_{\sum}^{M}(G)$ denote its majority neighbor sum distinguishing index. Majority neighbor sum distinguishing c...

conjecture | open | MathDB 页面 | 原文线索

V2-052762 | MATHDB-371321

Generalized asymmetric three-colour Ramsey conjecture

摘录 / EXCERPT: Generalized asymmetric three-colour Ramsey conjecture. For any integers $k \geq \ell \geq 2$, there exists a graph G such that $G \not\rightarrow (K_k,K_k,K_{\ell})$, but...

conjecture | open | MathDB 页面 | 原文线索

V2-052763 | MATHDB-371323

Stabilization conjecture for chromatic numbers of rank-r Abelian Cayley graphs

摘录 / EXCERPT: Let r be a positive integer. Let M_X be an $m\times r$ integer matrix and let $X=M_X^{SACG}$ denote the Abelian Cayley graph associated with M_X. Assume that M_X has no zer...

conjecture | open | MathDB 页面 | 原文线索

V2-052764 | MATHDB-371333

Bounded neighborhoods in K4-free, induced-path-free signed graphs

摘录 / EXCERPT: For a signed graph $\widehat{G}$ and a vertex u, let $N[u]$ denote the closed neighborhood of u, and let χ_b denote balanced chromatic number. Bounded-neighborhood conject...

conjecture | open | MathDB 页面 | 原文线索

V2-052765 | MATHDB-371337

Harvey's minimum-degree conjecture on chords of longest cycles

摘录 / EXCERPT: Let G be a graph on n vertices, let $\delta(G)$ denote its minimum degree, and let a cycle of maximum order mean a cycle containing the maximum possible number of vertices in...

conjecture | open | MathDB 页面 | 原文线索

V2-052766 | MATHDB-371338

Chord conjecture for longest cycles containing a linear forest

摘录 / EXCERPT: Let G be a k-connected graph with $k\geq 2$, and let F be a linear forest, meaning a graph whose components are paths, that is a subgraph of G with l edges and t isolat...

conjecture | open | MathDB 页面 | 原文线索

V2-052767 | MATHDB-371348

Kierstead-Kostochka Ore-type analogue of the Chen-Lih-Wu Conjecture

摘录 / EXCERPT: Let an equitable k-coloring be a proper k-coloring whose color classes have sizes differing by at most one, and let $d(x)$ denote the degree of a vertex x. Kierstead-Kostochk...

conjecture | open | MathDB 页面 | 原文线索

V2-052768 | MATHDB-371350

Balogh-Kostochka-Treglown degree-sequence conjecture for equitable colorings

摘录 / EXCERPT: Let $n,k\in\mathbb{N}$ with $k\leq n$. Let G be an n-vertex graph whose degree sequence is $d_1\geq\cdots\geq d_n$. Let an equitable k-coloring be a proper k-coloring whose color classes hav...

conjecture | open | MathDB 页面 | 原文线索

V2-052769 | MATHDB-371352

A reduction conjecture for the complexes $\Gamma_n^{\alpha,r}$

摘录 / EXCERPT: For fixed $m>0$, let $\mathcal{G}_m^n$, $\Delta_m^{n,r}$, and $\Gamma_n^{\alpha,r}$ be the complexes defined in the paper. Let $V(\Delta)$ denote the vertex set of an induced subco...

conjecture | open | MathDB 页面 | 原文线索

V2-052770 | MATHDB-371360

Reconfiguration conjecture for multipartite hypergraphs

摘录 / EXCERPT: Let H be an r-partite r-graph, let A be one of its r partition classes, and let $\mathbf{RG}_{\mathrm{Mat}}(H,A;k)$ denote the matching reconfiguration graph on matchings...

conjecture | open | MathDB 页面 | 原文线索

V2-052771 | MATHDB-371376

The target-free clique conjecture for combinatorial threshold-linear networks

摘录 / EXCERPT: The target-free clique conjecture. A subset σ is the support of a stable fixed point if and only if σ is a target-free clique. Target-free cliques are known to suppor...

conjecture | open | MathDB 页面 | 原文线索

V2-052772 | MATHDB-371383

Mader's connectivity-keeping tree conjecture

摘录 / EXCERPT: Mader's conjecture. For any tree $\T$ of order $\m$, every $\k$-connected graph $\G$ with minimum degree

conjecture | open | MathDB 页面 | 原文线索

V2-052773 | MATHDB-371384

Luo-Tian-Wu's bipartite connectivity-keeping tree conjecture

摘录 / EXCERPT: Luo-Tian-Wu's conjecture. Every $\k$-connected bipartite graph $\G$ with

conjecture | open | MathDB 页面 | 原文线索

V2-052774 | MATHDB-371394

EISPN conjectures for Cartesian products of paths

摘录 / EXCERPT: EISPN path-product conjectures. If $k \ge 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052775 | MATHDB-371401

Edge-count conjecture for minimal kinematics graphs

摘录 / EXCERPT: Let $[n]=\{1,2,\dots,n\}$, and call a graph on $[n]$ a minimal kinematics graph when its edge set is inclusion-maximal among the graphs realizing minimal kinematics. Minimal kinema...

conjecture | open | MathDB 页面 | 原文线索

V2-052776 | MATHDB-371402

Copiousness conjecture for irreducible hypertrees

摘录 / EXCERPT: Let T be an irreducible hypertree, and identify it with its graph G, whose edges are the pairs occurring in the triples of T. Irreducible hypertree copiousness conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-052777 | MATHDB-371410

A spectral bound for the chromatic number involving the number of edges

摘录 / EXCERPT: Let G be a non-empty graph with m edges, chromatic number χ, and least adjacency eigenvalue λ_n. The spectral edge-count conjecture. ... This conjecture is motivated...

conjecture | open | MathDB 页面 | 原文线索

V2-052778 | MATHDB-371415

Nabutovsky's unbounded balanced-vertex conjecture for geodesic nets

摘录 / EXCERPT: Nabutovsky's conjecture. There exist geodesic nets in the Euclidean plane with 4 unbalanced vertices and an arbitrarily large number of balanced vertices. Moreover, this may alre...

conjecture | open | MathDB 页面 | 原文线索

V2-052779 | MATHDB-371425

The self-CSF conjecture for sufficiently large spiders

摘录 / EXCERPT: Let a spider be a tree with at most one vertex of degree greater than two, and let X_T^T denote the self-chromatic symmetric function of a spider T. Spider self-CSF conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-052780 | MATHDB-371449

Chemistry nonexplosivity implies coarse-grained nonexplosivity

摘录 / EXCERPT: Let N be the coarse-grained model associated to the general content-dependent fragmentation reaction network, and let $\mathcal I_\mathcal K$ denote its chemistry. Nonexplosivity...

conjecture | open | MathDB 页面 | 原文线索

V2-052781 | MATHDB-371450

Boundary-case nonexplosivity conjecture for the coarse-grained model

摘录 / EXCERPT: Let N be the model described in the preceding proposition, with parameters α and p satisfying $\alpha=\log(1/p)$. Boundary-case nonexplosivity conjecture. The model N...

conjecture | open | MathDB 页面 | 原文线索

V2-052782 | MATHDB-371458

Second covering allocation conjecture for Young diagrams

摘录 / EXCERPT: Let Y be a Young diagram and let $H(Y)$ be the associated 3-partite 3-uniform hypergraph. Write $|Y|$ for the number of cells of Y, let $\tau^{(2)}(H(Y))$ denote its second cov...

conjecture | open | MathDB 页面 | 原文线索

V2-052783 | MATHDB-371468

Maximum independent edge set conjecture for 3-token graphs of paths

摘录 / EXCERPT: Maximum independent edge set conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052784 | MATHDB-371478

Ore-type conjecture for all orientations of Hamilton cycles

摘录 / EXCERPT: An oriented graph is a directed graph with at most one directed edge between any pair of vertices. For an oriented graph G, let $V(G)$ be its vertex set, let $n=|V(G)|$, and writ...

conjecture | open | MathDB 页面 | 原文线索

V2-052785 | MATHDB-371510

Dvořák et al.'s star chromatic index conjecture for subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning that every vertex of G has degree at most 3, and let $\chi'_{st}(G)$ denote the minimum number of colors in a proper edge-coloring of G w...

conjecture | open | MathDB 页面 | 原文线索

V2-052786 | MATHDB-371547

The edge-irregularity deletion conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph, and let $m=|E|$ be its number of edges. An edge-irregulator of G is a set $S\subseteq E$ such that $G-S$ is locally irregular, meaning that ev...

conjecture | open | MathDB 页面 | 原文线索

V2-052787 | MATHDB-371561

Localised representation conjecture for string graphs on surfaces

摘录 / EXCERPT: Let Σ be a surface with Euler genus g, and let G be a string graph in Σ. A string representation $\mathcal{C}$ of G in Σ assigns a non-self-intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-052788 | MATHDB-371564

The girth-five frustration index conjecture for signed subcubic graphs

摘录 / EXCERPT: Let (G,σ) be a signed connected simple subcubic graph, where $v(G)$ denotes the number of vertices of G, $F(G,\sigma)$ denotes its frustration index, and the girth is the...

conjecture | open | MathDB 页面 | 原文线索

V2-052789 | MATHDB-371569

Spectral Turán conjecture for color-critical graphs

摘录 / EXCERPT: Let F be a color-critical graph with chromatic number $\chi(F)=r+1\ge 4$. For sufficiently large m, let G be an F-free graph with m edges, and write $\lambda_1(G)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-052790 | MATHDB-371570

Full-range low-degree vertex conjecture for intersecting hypergraphs

摘录 / EXCERPT: Full-range low-degree vertex conjecture. If $n>2k$, then at least $n-2k$ vertices have degree at most

conjecture | open | MathDB 页面 | 原文线索

V2-052791 | MATHDB-371571

Linear-range low-degree conjecture for t-intersecting hypergraphs

摘录 / EXCERPT: Linear-range low-degree conjecture. There is an absolute constant c such that, whenever $n\geq ckt$,

conjecture | open | MathDB 页面 | 原文线索

V2-052792 | MATHDB-371577

Path-graph incompatibility robustness conjecture

摘录 / EXCERPT: Path-graph robustness conjecture. For all $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-052793 | MATHDB-371594

The conjecture characterizing graphs with fixed PSD propagation time one

摘录 / EXCERPT: Let G be a connected graph. A PSD fast join is a graph of order $n\geq 2$ such that $G\cong K_n$ or ... for $t\geq 2$ and positive integers n_i,m_i with $n=\sum_{i=1}^t n_i+m_i$....

conjecture | open | MathDB 页面 | 原文线索

V2-052794 | MATHDB-371618

Gerbner's asymptotic conjecture for forbidden bipartite 3-graphs with bounded matching number

摘录 / EXCERPT: Let F be a 3-graph, let $\chi(F)$ denote its chromatic number, and let $q(F)$ be the parameter defined by the ordered link-graph chromatic numbers in the source. For...

conjecture | open | MathDB 页面 | 原文线索

V2-052795 | MATHDB-371619

Exact star-colored Turán bound conjecture

摘录 / EXCERPT: For integers r and $s\geq r-1$, let $G_1,\ldots,G_s\subseteq\binom{[n]}{2}$ be graphs whose edges are $(r-1)$-star edge-colored and whose underlying graphs are K_r-free. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-052796 | MATHDB-371634

Hahn-Hell-Poljak conjecture for odd wheels

摘录 / EXCERPT: Let W_t be the join of the cycle C_t with K_1, and let $d4b0(G)$ denote the ultimate independence ratio of a graph G. For odd wheels, $d4b0(W_{2t+1})$ is bounded below by...

conjecture | open | MathDB 页面 | 原文线索

V2-052797 | MATHDB-371635

Conjecture on the largest 3-colourable subgraph of odd-wheel squares

摘录 / EXCERPT: Let W_t be the join of the cycle C_t with K_1, let $\alpha(G)$ be the independence number of G, and let $G\Box H$ denote the Cartesian product of graphs. For $t\geq 2$, con...

conjecture | open | MathDB 页面 | 原文线索

V2-052798 | MATHDB-371636

Conjecture on the independence number of the fourth power of the 5-wheel

摘录 / EXCERPT: Let W_5 be the join of the cycle C_5 with K_1, let $\alpha(G)$ denote the independence number of G, and let $W_5^{\Box 4}$ be the fourth Cartesian power of W_5. Fourth-po...

conjecture | open | MathDB 页面 | 原文线索

V2-052799 | MATHDB-371637

Conjecture on the fractional chromatic number of odd-wheel squares

摘录 / EXCERPT: Let W_t be the join of the cycle C_t with K_1, let $\chi_f(G)$ denote the fractional chromatic number of a graph G, and let $W_{2t+1}^{\Box 2}$ be the Cartesian square of t...

conjecture | open | MathDB 页面 | 原文线索

V2-052800 | MATHDB-371638

Conjecture on the triangle bound for odd-wheel Cartesian powers

摘录 / EXCERPT: Let W_t be the join of the cycle C_t with K_1, let $\alpha(G)$ be the independence number, let $\chi_f(G)$ be the fractional chromatic number, and let $\Box$ denote the Carte...

conjecture | open | MathDB 页面 | 原文线索

V2-052801 | MATHDB-371639

Conjectural fractional-chromatic bound from a largest low-chromatic induced subgraph

摘录 / EXCERPT: Let G be a graph, let $\chi(G)$ and $\omega(G)$ denote its chromatic and clique numbers, let $\chi_f(G)$ denote its fractional chromatic number, and let H be a largest induced...

conjecture | open | MathDB 页面 | 原文线索

V2-052802 | MATHDB-371641

Gao-Yang's directed Nine Dragon Tree conjecture

摘录 / EXCERPT: Let D be a digraph. A branching is a digraph whose components are arborescences, and write $\gamma(D)$ for the directed fractional packing parameter and Δ^- and Δ^+...

conjecture | open | MathDB 页面 | 原文线索

V2-052803 | MATHDB-371647

The gonality-tight graph conjecture

摘录 / EXCERPT: Gonality-tight graph conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-052804 | MATHDB-371665

Adamović-Wang duality conjecture for affine and Virasoro vertex algebras

摘录 / EXCERPT: Let $q\geq 2$ with $3\nmid q$, and define the generalized vertex operator algebras ... ... Duality means that each simple vertex (super)algebra can be equipped with a module structure...

conjecture | open | MathDB 页面 | 原文线索

V2-052805 | MATHDB-371679

Saturation-number formula for joins of cliques and paths

摘录 / EXCERPT: Let P_k be the path on k vertices, let K_s be the complete graph on s vertices, and let $sat(n,H)$ denote the minimum number of edges in an H-saturated graph on n verti...

conjecture | open | MathDB 页面 | 原文线索

V2-052806 | MATHDB-371697

Conjecture on the matching-restricted Turán number of Berge cliques

摘录 / EXCERPT: The matching-restricted Berge-clique conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052807 | MATHDB-371698

Strict realizability conjecture for $K_{2,3}^{'}$-free graphs

摘录 / EXCERPT: Strict realizability conjecture. If G is a $K_{2,3}^{'}$-free graph which is independent in $\mathcal{R}_\ell$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052808 | MATHDB-371699

The extremal algebraic-connectivity conjecture for k-path graphs

摘录 / EXCERPT: Extremal algebraic-connectivity conjecture. Given fixed $n\geq k+1$ and $k\geq 2$, the unique k-path graph that maximizes the algebraic connectivity is $K_{k-1}\vee P_{n-k+1}$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-052809 | MATHDB-371700

The extremal alpha-index and algebraic-connectivity conjecture for k-path graphs

摘录 / EXCERPT: Extremal alpha-index and algebraic-connectivity conjecture. Given fixed $n\geq k+1$ and $k\geq 2$, the unique k-path graph that maximizes the α-index for $\alpha\in[0,1]$...

conjecture | open | MathDB 页面 | 原文线索

V2-052810 | MATHDB-371701

Frankl-Kleitman-type conjecture for the non-uniform Erdős matching problem

摘录 / EXCERPT: Frankl-Kleitman conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-052811 | MATHDB-371710

The large flame extension conjecture for rooted digraphs

摘录 / EXCERPT: Large flame extension conjecture. In every r-rooted digraph D, every flame extends to a large flame. In particular, every r-rooted digraph D admits a large flame.

conjecture | open | MathDB 页面 | 原文线索

V2-052812 | MATHDB-371727

Bounded clique-width from bounded chordless-cycle subgraphs

摘录 / EXCERPT: Let G be a graph, and let ${\cal P}=(A_1,\ldots,A_k)$ be a monotone partition of $V(G)$ into cliques. The box graph $b(G;{\cal P})$ has a chordless cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-052813 | MATHDB-371728

Logarithmic clique-width for cyclic box graphs

摘录 / EXCERPT: Let G be a graph and let ${\cal P}$ be a monotone partition of $V(G)$ into cliques. Assume that the box graph $b(G;{\cal P})$ is a chordless cycle. The logarithmic cyclic-box con...

conjecture | open | MathDB 页面 | 原文线索

V2-052814 | MATHDB-371731

Nýdl's reconstructibility conjecture for trees

摘录 / EXCERPT: For an n-vertex graph, the m-deck is the multiset of its unlabeled induced m-vertex subgraphs, and a family of graphs is weakly $\celltext{ell}$-reconstructible if no two grap...

conjecture | open | MathDB 页面 | 原文线索

V2-052815 | MATHDB-371744

Orbit length conjecture for iterated bridge sums of trees and complete graphs

摘录 / EXCERPT: Let $G_1=(V_1,E_1),\ldots,G_q=(V_q,E_q)$ be graphs on $n_1,\ldots,n_q$ vertices, respectively, where each G_i is either a complete graph or a tree. Form G by bridge summing...

conjecture | open | MathDB 页面 | 原文线索

V2-052816 | MATHDB-371745

Complete-graph-cycle bridge-sum orbit-length conjecture

摘录 / EXCERPT: Let...

conjecture | open | MathDB 页面 | 原文线索

V2-052817 | MATHDB-371755

Elphick-Linz-Wocjan conjecture on the positive spectral sum

摘录 / EXCERPT: Let G be a graph of order n and clique number ω. Let $s^+(G)$ denote the sum of the squares of the positive eigenvalues of the adjacency matrix of G. Elphick-Linz-Wocja...

conjecture | open | MathDB 页面 | 原文线索

V2-052818 | MATHDB-371756

Vertex-localized Elphick-Linz-Wocjan conjecture

摘录 / EXCERPT: Let G be a graph with vertex set V. For each vertex v, let $c(v)$ denote the relevant local clique parameter, and let $s^+(G)$ denote the sum of the squares of the positive e...

conjecture | open | MathDB 页面 | 原文线索

V2-052819 | MATHDB-371757

Li-Liu-Zhang color-critical strengthening of the Bollobás-Nikiforov conjecture

摘录 / EXCERPT: Let F be a color-critical graph with chromatic number $\chi(F)\geq 4$, and let G be an F-free graph with sufficiently large size m. Let $\lambda_1(G)$ and $\lambda_2(G)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-052820 | MATHDB-371758

Asymptotic feedback vertex set conjecture for bounded-degree digraphs

摘录 / EXCERPT: For any integer $k\ge 2$, let $f(k)$ be the supremum of ${\rm fvs}(D)/|V(D)|$ over all orgraphs D with maximum degree at most k, where ${\rm fvs}(D)$ denotes the minimum feedba...

conjecture | open | MathDB 页面 | 原文线索

V2-052821 | MATHDB-371761

Generalized Kriesell's conjecture for minimally tough graphs

摘录 / EXCERPT: Let t be a real number, and let G be a graph. A graph is minimally t-tough when $\tau(G)=t$ and $\tau(G-e)<t$ for every edge $e\in E(G)$, where $\tau(G)$ denotes its toughnes...

conjecture | open | MathDB 页面 | 原文线索

V2-052822 | MATHDB-371782

Irreducibility conjecture for Wakimoto modules with Whittaker Heisenberg factors

摘录 / EXCERPT: Let $\frak{g}$ be the Lie algebra under consideration, let $\frak{h}$ be its Cartan subalgebra, and let $ u$ be a non-critical invariant bilinear form with corresponding affine lev...

conjecture | open | MathDB 页面 | 原文线索

V2-052823 | MATHDB-371783

Irreducibility conjecture for generalized Whittaker modules of \widehat{\mathfrak{sl}}_2

摘录 / EXCERPT: Let $M_1(\bm{\lambda},\bm{\mu})$ and $N_1(\bm{\eta})$ be the Weyl and Heisenberg modules described in the source, with $\mu_M\ne0$, $\lambda_N\ne0$, and $\eta\ne0$. Let ψ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-052824 | MATHDB-371790

The conjecture on excess two digraphs

摘录 / EXCERPT: Let a $(d,k;+\epsilon)$-digraph be a k-geodetic digraph with minimum out-degree d and order ... where ... Thus a $(d,k;+2)$-digraph has excess two. The conjecture on excess two dig...

conjecture | open | MathDB 页面 | 原文线索

V2-052825 | MATHDB-371816

Existence conjecture for Steiner systems

摘录 / EXCERPT: Let $n>q>r\geq 1$ be integers. An (n,q,r)-Steiner system is a collection of q-subsets of an n-element set such that every r-subset is contained in exactly one member. Equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-052826 | MATHDB-371817

Hypergraph Nash-Williams conjecture for clique decompositions

摘录 / EXCERPT: Let $q>r\geq 3$ be integers. An r-graph is an r-uniform hypergraph, and it is K_q^r-divisible when it satisfies the divisibility conditions necessary for a decomposition into...

conjecture | open | MathDB 页面 | 原文线索

V2-052827 | MATHDB-371824

Aragão-Marciano-Mendonça's minimum-degree conjecture for path-clique Ramsey goodness

摘录 / EXCERPT: Aragão-Marciano-Mendonça's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-052828 | MATHDB-371828

The homotopy conjecture for total 2-cut complexes of cycle powers in the intermediate range

摘录 / EXCERPT: The intermediate-range homotopy conjecture. If $n=2p+\frac{p}{r}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052829 | MATHDB-371829

The homology conjecture for total 3-cut complexes of complete graph-cycle products

摘录 / EXCERPT: Let K_m be the complete graph on m vertices, let C_n be the cycle graph on n vertices, and let $\Delta_{3}^t(K_m\square C_n)$ denote their total 3-cut complex. The total...

conjecture | open | MathDB 页面 | 原文线索

V2-052830 | MATHDB-371851

Barnette's conjecture on simple separating non-contractible cycles

摘录 / EXCERPT: A simple separating non-contractible cycle (SNCC) is a simple cycle in a triangulation that is both separating and non-contractible. Let t be an arbitrary triangulation without l...

conjecture | open | MathDB 页面 | 原文线索

V2-052831 | MATHDB-371852

Existence of short simple separating cycles in random high-genus triangulations

摘录 / EXCERPT: Fix $\theta\in(0,1/2)$ and a sequence (g_n) such that $g_n/n\rightarrow\theta$. Let $\mathbf{T}_{2n,g_n}$ be a uniform triangulation of genus g_n with $2n$ faces. A simple sepa...

conjecture | open | MathDB 页面 | 原文线索

V2-052832 | MATHDB-371854

Reduced form conjecture for shortest separating non-contractible cycles

摘录 / EXCERPT: A separating non-contractible cycle (SNCC) is a cycle in a triangulation that is separating and non-contractible. Let t be a triangulation and let C be an SNCC of t of shorte...

conjecture | open | MathDB 页面 | 原文线索

V2-052833 | MATHDB-371857

Word-representability conjecture for 5-regular circulant graphs

摘录 / EXCERPT: Word-representability conjecture. Every 5-regular circulant graph is word-representable.

conjecture | open | MathDB 页面 | 原文线索

V2-052834 | MATHDB-371858

Existence conjecture for non-trivial F-irregular graphs

摘录 / EXCERPT: Existence conjecture. For every connected graph F of order 3 or more, there exists a non-trivial F-irregular graph.

conjecture | open | MathDB 页面 | 原文线索

V2-052835 | MATHDB-371911

Hilton-Zhao's vertex-splitting conjecture

摘录 / EXCERPT: Let G^* be an n-vertex connected class 1 Δ-regular graph with $\Delta>\frac{n}{3}$. A vertex-splitting replaces a vertex by two adjacent vertices whose neighborhoods par...

conjecture | open | MathDB 页面 | 原文线索

V2-052836 | MATHDB-371918

Marginal b-conjecture for map enumeration

摘录 / EXCERPT: For all $n,f\geq 1$ and partitions $\nu\vdash 2n$, let $c_{\mu,\nu,n}(b)$ denote the coefficient associated with maps having n edges and vertex and face profiles μ and ν....

conjecture | open | MathDB 页面 | 原文线索

V2-052837 | MATHDB-371970

The hereditary-family polychromatic coloring conjecture

摘录 / EXCERPT: Let $\mathcal F$ be a hereditary family of hypergraphs. For $k\ge1$, let $m_k(\mathcal F)$ be the smallest integer m such that every m-heavy hypergraph in $\mathcal F$ has a po...

conjecture | open | MathDB 页面 | 原文线索

V2-052838 | MATHDB-371971

The linear-growth conjecture for hereditary-family polychromatic thresholds

摘录 / EXCERPT: Let $\mathcal F$ be a hereditary family of hypergraphs, and let $m_k(\mathcal F)$ be the least threshold guaranteeing a polychromatic k-coloring for every m-heavy member of...

conjecture | open | MathDB 页面 | 原文线索

V2-052839 | MATHDB-371972

The bottomless-rectangle linear decomposition conjecture

摘录 / EXCERPT: Let $\mathcal F$ be the family of bottomless rectangles in the plane, and let $m_k^*(\mathcal F)$ denote the corresponding threshold for decomposing coverings into k coverings. B...

conjecture | open | MathDB 页面 | 原文线索

V2-052840 | MATHDB-371975

Ziegler's conjecture for r-stable Kneser hypergraphs

摘录 / EXCERPT: Let $KG^r(n,k)$ be the Kneser hypergraph whose vertices are the k-element subsets of $[n]$, and call a subset $S\subseteq[n]$ r-stable when any two elements $x,y\in S$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-052841 | MATHDB-371996

Geometric threshold of endemicity in curvature-weighted contact networks

摘录 / EXCERPT: Let $\overline{\kappa}$ denote the mean curvature of the network, let $\kappa^\star>0$ be a constant, and let R_0 denote the basic reproduction number of the unweighted system. T...

conjecture | open | MathDB 页面 | 原文线索

V2-052842 | MATHDB-372026

Chen-Chvátal conjecture for finite metric spaces

摘录 / EXCERPT: Let (V,d) be a finite metric space with n points. For distinct points $x,y,z\in V$, say that z lies between x and y when ... For distinct $x,y\in V$, define the line genera...

conjecture | open | MathDB 页面 | 原文线索

V2-052843 | MATHDB-372036

Gartland-Lokshtanov dominated balanced separator conjecture

摘录 / EXCERPT: Gartland-Lokshtanov's conjecture. For every planar graph H there exists an integer k_H such that every H-induced-minor-free graph G has a balanced separator dominated by at...

conjecture | open | MathDB 页面 | 原文线索

V2-052844 | MATHDB-372043

The zero-forcing consequence of the minimal-forts bound

摘录 / EXCERPT: Let G be a graph, let n be its order, and let $\mathcal{F}_G$ denote the family of minimal forts of G. Let $\operatorname{Z}(G)$ denote the zero forcing number of G. Zero-f...

conjecture | open | MathDB 页面 | 原文线索

V2-052845 | MATHDB-372046

Rooted-triple reconstruction conjecture for binary normal networks

摘录 / EXCERPT: Let X be the leaf set, and let N_1 and N_2 be binary normal networks on X with no near-sibling reticulations. Write $R(N)$ for the set of rooted triples displayed by a netw...

conjecture | open | MathDB 页面 | 原文线索

V2-052846 | MATHDB-372089

Beck-Fiala discrepancy conjecture

摘录 / EXCERPT: Let $\mathcal H$ be a hypergraph on a finite ground set X, where every vertex has degree at most d. Beck-Fiala's conjecture. There exists a function $f:X\rightarrow\{-1,1\}$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-052847 | MATHDB-372105

The panchromatic-bipanchromatic number equality conjecture

摘录 / EXCERPT: Let H be a hypergraph. Write $\chi_p(H)$ for its panchromatic number, $\chi^2_p(H)$ for its bipanchromatic number, and let $\alpha_{\chi_p(H)}(H)$ be the minimum number of unique...

conjecture | open | MathDB 页面 | 原文线索

V2-052848 | MATHDB-372121

Valence-four vertex conjecture for Dirichlet domains of m125 and m129

摘录 / EXCERPT: Let M be either of the census manifolds texttt{m125} or texttt{m129}, and let D be a Dirichlet domain for M. A finite vertex of valence four is a finite vertex of D inciden...

conjecture | open | MathDB 页面 | 原文线索

V2-052849 | MATHDB-372127

Edge-deletion gamma-nonnegativity conjecture for symmetric edge polytopes

摘录 / EXCERPT: Let G be a 2-connected graph with edge set E, and let P_G denote its symmetric edge polytope. For an edge $e\in E$, let $P_{G\setminus e}$ be the symmetric edge polytope of t...

conjecture | open | MathDB 页面 | 原文线索

V2-052850 | MATHDB-372128

Nonnegativity conjecture for the summed edge-deletion gamma-polynomial

摘录 / EXCERPT: Summed edge-deletion nonnegativity conjecture. The polynomial $Z_G(t)$ has nonnegative coefficients. This is proposed as a stronger statement whose gamma-nonnegativity would imply...

conjecture | open | MathDB 页面 | 原文线索

V2-052851 | MATHDB-372137

Besomi-Pavez-Signé-Stein second-neighbourhood conjecture for tree embeddings

摘录 / EXCERPT: Let k be a positive integer. For a graph G and a vertex $x\in V(G)$, let $N_G(x)$ be the neighbourhood of x, and let $N_G^2(x)$ be the set of vertices other than x sharing...

conjecture | open | MathDB 页面 | 原文线索

V2-052852 | MATHDB-372140

Quantum vertex algebra structure conjecture for the induced extension of cmathfrak{d}

摘录 / EXCERPT: Let $cmathfrak{d}$ be the cotangent Lie algebra under consideration, let $cwidehat{DY}_{chbar,cell}(cmathfrak{d})$ be its completed double Yangian, and define ... with $cmathcal M_V$...

conjecture | open | MathDB 页面 | 原文线索

V2-052853 | MATHDB-372141

Quantum extension conjecture for the beta-gamma and bc vertex algebras

摘录 / EXCERPT: Let $cmathfrak{gl}(1|1)$ be the smallest type-A Lie superalgebra, let $cmathcal V_{chbar,k}(cmathfrak{gl}(1|1))$ be its quantum vertex algebra, and set ... Here $csigma^{n,0}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-052854 | MATHDB-372142

Seymour's epsilon conjecture for graphs with independence number two

摘录 / EXCERPT: Let G be a finite simple graph and let $\operatorname{had}(G)$ be its Hadwiger number. Seymour's epsilon conjecture. There is a constant $\varepsilon>0$ such that every graph G...

conjecture | open | MathDB 页面 | 原文线索

V2-052855 | MATHDB-372144

Füredi's 4-CM conjecture

摘录 / EXCERPT: Let G be a finite simple graph, and let a connected matching be a matching whose distinct edges have adjacent endpoint sets. Füredi's 4-CM conjecture. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-052856 | MATHDB-372145

Seymour's half-order matching conjecture

摘录 / EXCERPT: Let G be a finite simple graph, and let a model consist of pairwise vertex-disjoint connected branch sets, with adjacent branch sets joined by an edge. Seymour's half-order match...

conjecture | open | MathDB 页面 | 原文线索

V2-052857 | MATHDB-372146

Connected dominating matching conjecture for graphs with independence number two

摘录 / EXCERPT: Let G be a finite simple graph. A connected dominating matching is a connected matching M such that every vertex outside its endpoints is adjacent to every edge of M. Connect...

conjecture | open | MathDB 页面 | 原文线索

V2-052858 | MATHDB-372147

Sparse triangle-free diameter-two graph conjecture

摘录 / EXCERPT: Let G be a finite simple graph. A graph is triangle-free if it contains no triangle, and has diameter 2 if every two vertices are at distance at most 2. Sparse triangle-free...

conjecture | open | MathDB 页面 | 原文线索

V2-052859 | MATHDB-372150

The Induced Menger Conjecture

摘录 / EXCERPT: Let G be a graph and let $X,Y\subseteq V(G)$. An (X,Y)-path is a path with one endpoint in X and the other in Y, and two paths are anti-complete if they are vertex-disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-052860 | MATHDB-372162

Conjecture on bounded-conflict antimagic labellings

摘录 / EXCERPT: Let G be a finite, simple, undirected graph. A graph is C-antimagic if it admits an edge labelling whose vertex sums have at most C conflicts, as defined in the paper. C-an...

conjecture | open | MathDB 页面 | 原文线索

V2-052861 | MATHDB-372165

Linear zero-sum Ramsey bound for graphs of bounded degeneracy

摘录 / EXCERPT: Let $d\geq 2$ be an integer. For a graph G, write $R(G,\Gamma_0)$ for its zero-sum Ramsey number with respect to a finite abelian group Γ_0, and let $n=e(G)$ be the number...

conjecture | open | MathDB 页面 | 原文线索

V2-052862 | MATHDB-372167

Polynomial-time solvability of Directed Plane Strong Connectivity Augmentation

摘录 / EXCERPT: Let D be a connected plane digraph and let k be an integer. The Directed Plane Strong Connectivity Augmentation problem asks whether there is a set $X\subseteq V(D)^2$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-052863 | MATHDB-372171

Biro, Collado and Zamora's indeque ratio conjecture for series-parallel graphs

摘录 / EXCERPT: Let $\mathcal S$ be the class of series-parallel graphs. Biro, Collado and Zamora's conjecture. ... The indeque ratio measures the asymptotic minimum proportion of vertices that can...

conjecture | open | MathDB 页面 | 原文线索

V2-052864 | MATHDB-372195

The Ban-Linial conjecture for cubic graphs

摘录 / EXCERPT: Ban-Linial conjecture. Every cubic graph has an external split (X,Y) satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-052865 | MATHDB-372202

The characterization of locally interval graphs as circular-arc graphs

摘录 / EXCERPT: Rzk{a}.{z}ewski's conjecture. Locally interval graphs are exactly circular-arc graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-052866 | MATHDB-372203

Subquadratic maximal-clique conjecture for locally chordal graphs

摘录 / EXCERPT: Let G be a finite graph, and call it r-locally chordal if every ball of radius r in G is chordal. Let $|V(G)|$ denote the number of vertices of G and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052867 | MATHDB-372204

Minimal vertex counts for constant-curvature polyhedra

摘录 / EXCERPT: Minimal vertex-count conjecture. (i) For a CCP of a topological torus without self-intersection, the minimal number of vertices is 9. (ii) For an orientable CCP of genus g, if...

conjecture | open | MathDB 页面 | 原文线索

V2-052868 | MATHDB-372205

Existence of regular open and closed XOR-magic graphs of power-of-two order

摘录 / EXCERPT: Existence conjecture. For every permitted degree d in the two displayed sets, the corresponding regular open or closed XOR-magic graph exists. This would complete the existence p...

conjecture | open | MathDB 页面 | 原文线索

V2-052869 | MATHDB-372208

Vector-valued Sprecher representation conjecture

摘录 / EXCERPT: Let $n,m\in\mathbb{N}$ with $m>1$, and let $f:[0,1]^n\to\mathbb{R}^m$ be continuous. A Sprecher Network has architecture $n\to[d_1]\to m$ with one hidden block of width...

conjecture | open | MathDB 页面 | 原文线索

V2-052870 | MATHDB-372209

Deep Sprecher Network universality conjecture

摘录 / EXCERPT: Let $n\geq 1$, $L\geq 1$, and let $f:[0,1]^n\to\mathbb{R}$ or $f:[0,1]^n\to\mathbb{R}^m$ be continuous. Consider a Sprecher Network with architecture $n\to[d_1,\dots,d_L]\to 1$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-052871 | MATHDB-372225

Erdős-Faudree conjecture on Hamiltonian subsets of regular Dirac graphs

摘录 / EXCERPT: Let G be an $(n+1)$-regular graph on $2n$ vertices. A subset $S\subseteq V(G)$ is called Hamiltonian when the induced subgraph $G[S]$ contains a Hamilton cycle. Erdős-Faudree con...

conjecture | open | MathDB 页面 | 原文线索

V2-052872 | MATHDB-372226

Draganić-Keevash-Müyesser conjecture on induced K_r-factors

摘录 / EXCERPT: Let $r\geq 2$, and let G be an $((r-1)n+1)$-regular graph on rn vertices. A subset $S\subseteq V(G)$ is said to induce a K_r-factor when $G[S]$ contains a collection of verte...

conjecture | open | MathDB 页面 | 原文线索

V2-052873 | MATHDB-372229

Gerards-Seymour odd Hadwiger conjecture

摘录 / EXCERPT: Odd Hadwiger conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-052874 | MATHDB-372238

Cho et al.'s spectral conjecture for [a,b]-factors

摘录 / EXCERPT: Let G be an n-vertex graph, and let $b\ge a$ be two positive integers such that $na\equiv 0\pmod{2}$ and $n\ge a+1$. Here, $\lambda(G)$ denotes the spectral radius of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-052875 | MATHDB-372239

Spectral conjecture for (r,k)-critical graphs

摘录 / EXCERPT: Let $r\ge 2$ and $k\ge 0$, and let G be a connected graph of order $n\ge 2(2r+k+2)(r+k+2)$ with minimum degree $\delta(G)\ge r+k$. Let $F_n^{r,k}$ be the extremal graph appearing...

conjecture | open | MathDB 页面 | 原文线索

V2-052876 | MATHDB-372243

The conjectured upper bound for the largest Neumann eigenvalue of a path graph

摘录 / EXCERPT: Let P be a path graph with n vertices, and let $\lambda_{n-2}(P)$ denote its largest Neumann eigenvalue. Path-graph Neumann eigenvalue conjecture. The largest Neumann eigenvalu...

conjecture | open | MathDB 页面 | 原文线索

V2-052877 | MATHDB-372246

Dhawan's multipartite hypergraph choosability conjecture

摘录 / EXCERPT: Dhawan's conjecture. For every $k\geqslant 2$, there is a constant $c=c(k)>0$ such that, for all sufficiently large Δ,

conjecture | open | MathDB 页面 | 原文线索

V2-052878 | MATHDB-372256

Shellability conjecture for k-cut complexes of hexagonal line tilings

摘录 / EXCERPT: Hexagonal line tiling shellability conjecture. For $k=4,5,$ and 6, the k-cut complex of the hexagonal line tiling is shellable.

conjecture | open | MathDB 页面 | 原文线索

V2-052879 | MATHDB-372272

Liu-Zhang's edge bound in terms of maximum forcing number

摘录 / EXCERPT: Let G be a finite simple graph of order $2n$ with a perfect matching. Write $e(G)$ for the number of edges of G, and let $F(G)$ denote its maximum forcing number. Liu-Zhang's c...

conjecture | open | MathDB 页面 | 原文线索

V2-052880 | MATHDB-372279

Brešar-Togni conjecture on packing coloring of subdivided subcubic graphs

摘录 / EXCERPT: Let G be a graph in which every vertex has degree at most 3, and let $S(G)$ be the graph obtained by subdividing every edge of G, replacing each edge uv by a path u-w-...

conjecture | open | MathDB 页面 | 原文线索

V2-052881 | MATHDB-372295

Extremal formula for Berge-disjoint paths

摘录 / EXCERPT: Let ℓ,r be integers and let $k\geq 2$. Write ... For an r-uniform hypergraph, let $\mathrm{ex}_r(n,\text{Berge-} kP_\ell)$ denote the maximum number of hyperedges in an n-ve...

conjecture | open | MathDB 页面 | 原文线索

V2-052882 | MATHDB-372296

Extremal formula for Berge linear forests

摘录 / EXCERPT: Let $\mathcal{P}=P_{\ell_1}\cup\dots\cup P_{\ell_k}$ be a linear forest with $k\geq 2$ and $\ell_k\geq\dots\geq\ell_1\geq 3$. Write ... For an r-uniform hypergraph, let...

conjecture | open | MathDB 页面 | 原文线索

V2-052883 | MATHDB-372314

The beta-catastrophe conjecture for emergence of a giant wandering strongly connected component

摘录 / EXCERPT: Let $G_{\beta,n}$ be the support digraph for a discrete family, let V_n be its vertex set, and let S_n be the set of solution states. Write ... A phase transition at...

conjecture | open | MathDB 页面 | 原文线索

V2-052884 | MATHDB-372315

Asymptotic formula for the Dirac over-threshold

摘录 / EXCERPT: For integers $k\geq2$ and m, let $\bar\eta_{k,m}$ denote the (k,m)-Dirac over-exponent, and let $f(\ell_{k,m})$ be the quantity defined in the paper. The asymptotic over-thresh...

conjecture | open | MathDB 页面 | 原文线索

V2-052885 | MATHDB-372316

Partition concentration conjecture for powers of paths

摘录 / EXCERPT: Fix integers $k\geq2$ and $m\geq k+1$. Let P be an m-path whose vertex set is partitioned as ... For a vertex subset V_j, write $P[V_j]$ for the induced subgraph, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-052886 | MATHDB-372320

Conjectured dominion formula for cycles

摘录 / EXCERPT: Let G be a cycle on $n\ge 3$ vertices. Its dominion is the invariant denoted by $\zeta(G)$. Cycle dominion conjecture. If G is a cycle on $n\ge 3$ vertices, then ... The cases...

conjecture | open | MathDB 页面 | 原文线索

V2-052887 | MATHDB-372329

The minimum vertex-degree conjecture for spanning components in hypergraphs

摘录 / EXCERPT: Let G be a k-graph on n vertices. Write $\delta_1(G)$ for its minimum vertex degree, the minimum number of edges containing any one vertex. Spanning-component conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-052888 | MATHDB-372330

The tetrahedron-component conjecture for dense 3-graphs

摘录 / EXCERPT: Let G be a 3-graph on n vertices, and let $\delta_2(G)$ denote its minimum codegree. Define the 4-graph H on $V(G)$ by making a 4-set an edge whenever it spans a tetrah...

conjecture | open | MathDB 页面 | 原文线索

V2-052889 | MATHDB-372331

The minimum codegree conjecture for spanning components in hypergraphs

摘录 / EXCERPT: Let G be a k-graph on n vertices, and let $\delta_{k-1}(G)$ denote its minimum $(k-1)$-degree. Minimum-codegree spanning-component conjecture. If ... then G contains a spanni...

conjecture | open | MathDB 页面 | 原文线索

V2-052890 | MATHDB-372358

The exact edge-count conjecture for order-7 triangular tiling spirals

摘录 / EXCERPT: Let $e_{d(7)}(n)$ denote the maximum number of edges in a penny graph with n vertices and distance $d(7)$. Exact edge-count conjecture. For all $n obreak\geq 1$, $e_{d(7)}(n)$ eq...

conjecture | open | MathDB 页面 | 原文线索

V2-052891 | MATHDB-372359

Order-7 spiral optimality conjecture for fixed vertex number

摘录 / EXCERPT: For a given positive integer n, let $e_d(n)$ be the maximum number of edges among penny graphs with n vertices and distance d, and let $d(7)$ be the distance associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-052892 | MATHDB-372360

Strict asymptotic optimality of the order-7 triangular tiling

摘录 / EXCERPT: Let $c(d)$ denote the asymptotic edge-density constant for distance d, and let $d(7)$ be the distance associated with the order-7 triangular tiling. Strict order-7 optimality c...

conjecture | open | MathDB 页面 | 原文线索

V2-052893 | MATHDB-372361

Rounded upper-bound conjecture for distances from triangular tilings

摘录 / EXCERPT: Let $d(k)$ be the distance associated with the order-k triangular tiling, and let α be the corresponding angle parameter. For $n\geq 1$ and $k\geq 7$, define the displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-052894 | MATHDB-372373

The $(3,4)$-tournament spanning-path conjecture

摘录 / EXCERPT: $(3,4)$-tournament spanning-path conjecture. Every $(3,4)$-tournament has a spanning path. That is, $f(n,3,4)=n$.

conjecture | open | MathDB 页面 | 原文线索

V2-052895 | MATHDB-372374

The acyclic $(3,4)$-tournament spanning-path conjecture

摘录 / EXCERPT: Acyclic $(3,4)$-tournament spanning-path conjecture. If G is a $(3,4)$-tournament with no closed walk, then G has a spanning path.

conjecture | open | MathDB 页面 | 原文线索

V2-052896 | MATHDB-372375

Anderson-Kim's ergodicity conjecture for weakly reversible stochastic mass-action reaction systems

摘录 / EXCERPT: A stochastic mass-action reaction system (SMART) is a continuous-time Markov chain describing molecule counts in a mass-action reaction network, and it is weakly reversible when ev...

conjecture | open | MathDB 页面 | 原文线索

V2-052897 | MATHDB-372379

The induced rainbow path conjecture for triangle-free graphs

摘录 / EXCERPT: Let G be a finite simple triangle-free graph, let $\chi(G)$ be its chromatic number, and let $\phi:V(G)\to[k]$ be a proper vertex coloring, where $k=\chi(G)$. A path in G is ra...

conjecture | open | MathDB 页面 | 原文线索

V2-052898 | MATHDB-372390

Mubayi-Zhao's rational codegree Turán density conjecture for single hypergraphs

摘录 / EXCERPT: Let k-graphs be k-uniform hypergraphs, let $\beta(F)$ denote the codegree Turán density of a k-graph F, and let $\beta(\rho)$ denote the corresponding density for a family...

conjecture | open | MathDB 页面 | 原文线索

V2-052899 | MATHDB-372391

Linear upper bound for set mappings of uniform hypergraphs

摘录 / EXCERPT: Let k be a positive integer and let G be a k-uniform hypergraph with m edges and no isolated vertices. The parameter $w(G)$ is the minimum N such that every map ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-052900 | MATHDB-372392

Conjectured upper bound for the free-set parameter g_0

摘录 / EXCERPT: Let G be a graph with n vertices, m edges and no isolated vertices. The parameter $g_0(G)$ is the corresponding set-mapping parameter defined in the paper. Conjecture on...

conjecture | open | MathDB 页面 | 原文线索

V2-052901 | MATHDB-372394

Turcotte's cop-number conjecture for graphs with bounded independence number

摘录 / EXCERPT: Turcotte's conjecture. For any positive integer p and any graph G such that $\alpha(G)<p$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-052902 | MATHDB-372399

The Delta Conjecture for maximum nullity of graphs

摘录 / EXCERPT: Let G be a simple graph, and let $\delta(G)$ denote its minimum degree. The maximum nullity $\mathrm{M}(G)$ is the greatest nullity among matrices associated with G in the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-052903 | MATHDB-372400

Strong Delta Conjecture for the SAP maximum nullity parameter

摘录 / EXCERPT: Let G be a simple graph, and let $\delta(G)$ denote its minimum degree. Write $\nu(G)$ for the greatest codimension of a faithful orthogonal representation whose Gram matrix has...

conjecture | open | MathDB 页面 | 原文线索

V2-052904 | MATHDB-372406

Universal determinant formula conjecture for adjacency walk matrices of rooted-product preservers

摘录 / EXCERPT: Let $H^{(v)}$ be an $\mathcal{F}$-preserver of order m, and let G be any graph. Write $A(G)$ for the adjacency matrix of G, $W_A(G)$ for its adjacency walk matrix, and...

conjecture | open | MathDB 页面 | 原文线索

V2-052905 | MATHDB-372424

The two-thirds conjecture for the total isolation number

摘录 / EXCERPT: Let G be a graph of order n, and suppose that every component of G has order at least 3. Two-thirds conjecture. The total isolation number of G satisfies ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-052906 | MATHDB-372425

Semi-inducibility conjecture for symmetric paths

摘录 / EXCERPT: Let $P_{E^\ell N^\ell}$ be a path of length 2ℓ whose first ℓ edges are red and whose remaining ℓ edges are blue. Let $K(a,n)$ be the graph formed by a clique on a...

conjecture | open | MathDB 页面 | 原文线索

V2-052907 | MATHDB-372430

Characterization of the stability number of connected graphs

摘录 / EXCERPT: Stability characterization conjecture. For every integer $p\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-052908 | MATHDB-372435

Spread versus width conjecture for grid tree decompositions

摘录 / EXCERPT: Let $m\geq n\geq 1$ be integers, and let a denote the additional width parameter in a tree-decomposition of the $(n\times m)$-grid whose width is $n+a$. The spread versus width c...

conjecture | open | MathDB 页面 | 原文线索

V2-052909 | MATHDB-372450

Fan's Laplacian multiplicity conjecture for connected uniform hypergraphs

摘录 / EXCERPT: Let H be a connected k-uniform hypergraph, and let ρ be the spectral radius of its adjacency tensor $\mathcal{A}(H)$. Let $\mathcal{L}(H)$ denote its Laplacian tensor. The...

conjecture | open | MathDB 页面 | 原文线索

V2-052910 | MATHDB-372451

Li-Feng-Peng's spectral supersaturation conjecture for the bowtie

摘录 / EXCERPT: Let F_2 be the bowtie graph, let $\tau(G)$ denote the number of copies of F_2 in a graph G, and let $K_{s,t}^{q}$ be obtained from $K_{s,t}$ by adding q pairwise disjoint e...

conjecture | open | MathDB 页面 | 原文线索

V2-052911 | MATHDB-372452

General spectral supersaturation conjecture for friendship graphs

摘录 / EXCERPT: For an integer $k\geq 1$, define ... Let F_k be the friendship graph and let $\tau_k(G)$ denote the number of copies of F_k in G. Let $K_{s,t}^{q}$ be obtained from $K_{s,t}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-052912 | MATHDB-372461

Erdős's exponential conjecture for induced Ramsey numbers

摘录 / EXCERPT: Let H be a graph with k vertices, and let $R^{\mathrm{ind}}(H)$ denote the minimum number of vertices of a graph G such that every red-blue edge-colouring of G contains an...

conjecture | open | MathDB 页面 | 原文线索

V2-052913 | MATHDB-372471

Homology conjecture for a bidirectional polygonal graph with one edge removed

摘录 / EXCERPT: Let ${\tt BP}_n$ be the bidirectional polygonal graph, and let ${\tt BP^{-1}_n}$ be the graph obtained by removing a single edge from it. Single-edge-removal homology conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-052914 | MATHDB-372479

Delocalization conjecture for eigenvectors of general binary networks

摘录 / EXCERPT: Let $\mathbf{A}=\mathbf{P}+\mathbf{E}$ be the observed adjacency matrix, with rank parameter r, and suppose that Assumptions (deloc)-(moments) hold, except that the moment assump...

conjecture | open | MathDB 页面 | 原文线索

V2-052915 | MATHDB-372498

Exponential non-isomorphism conjecture for spanning trees of regular graphs

摘录 / EXCERPT: Let $n,d\geq 3$ be integers, and let G be a connected n-vertex d-regular graph. Write $T_{\mathrm{unlabeled}}(G)$ for the set of isomorphism classes of spanning trees of G....

conjecture | open | MathDB 页面 | 原文线索

V2-052916 | MATHDB-372499

Exponential anticoncentration conjecture for random spanning trees

摘录 / EXCERPT: Let $\varepsilon>0$ be real. There is a constant $d_{\varepsilon}>0$ such that for every integer $d\geq d_{\varepsilon}$, every connected n-vertex d-regular graph G, and a un...

conjecture | open | MathDB 页面 | 原文线索

V2-052917 | MATHDB-372500

Polynomial anticoncentration conjecture under a minimum-degree condition

摘录 / EXCERPT: Let $d_0,c>0$ be universal constants. For every integer $d\geq d_0$, let G be a connected n-vertex graph with minimum degree at least d, and let $\mathcal{T}$ be a uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-052918 | MATHDB-372552

Ghosh-Győri-Paulos-Xiao-Zamora's extremal conjecture for the triangular pyramid TP_4

摘录 / EXCERPT: Let TP_4 denote the triangular pyramid graph with four levels, and let $\operatorname{ex}(n,TP_4)$ be the maximum number of edges in an n-vertex graph containing no copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-052919 | MATHDB-372555

Pons and Batle's enumeration conjecture for tree-child networks

摘录 / EXCERPT: Let $\mathcal{TC}_{n,k}$ be the set of all tree-child networks with n leaves and k reticulation nodes. The numbers $a_{n-1,k}$ are the coefficients appearing in the paper's pre...

conjecture | open | MathDB 页面 | 原文线索

V2-052920 | MATHDB-372569

Jovović-Koledin-Stanić spectral-gap conjecture for trees

摘录 / EXCERPT: Let $\mathcal{T}(n)$ be the family of trees on n vertices. For a tree, write λ_1 and λ_2 for its two largest adjacency eigenvalues, and let $DC(k,k,\ell)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-052921 | MATHDB-372570

Ebrahimi-Mohar-Nikiforov-Ahmady conjecture on the spectral sum of graphs

摘录 / EXCERPT: Let G be any graph of order n, and let $\lambda_1(G)$ and $\lambda_2(G)$ denote its two largest adjacency eigenvalues. Ebrahimi-Mohar-Nikiforov-Ahmady conjecture. The spectral...

conjecture | open | MathDB 页面 | 原文线索

V2-052922 | MATHDB-372572

Liu-Miao's spectral extremal conjecture for non-bipartite book-free graphs

摘录 / EXCERPT: Let G be a non-bipartite graph with m edges, and let B_{r+1} denote the graph formed by $r+1$ triangles sharing a common edge. Let $S^+_{m,1}$ be the graph obtained by adding...

conjecture | open | MathDB 页面 | 原文线索

V2-052923 | MATHDB-372581

Gaussian-universality conjecture for quadratic neural-network sensing

摘录 / EXCERPT: Let $X_1,\ldots,X_n$ be independent sensing matrices of the form ... Consider the quadratic-neural-network sensing model with these matrices and fixed output weights. Gaussian-univer...

conjecture | open | MathDB 页面 | 原文线索

V2-052924 | MATHDB-372596

Dewar's degree-threshold conjecture for globally rigid vertex-transitive graphs

摘录 / EXCERPT: Dewar's conjecture. There exists a function $f \colon \mathbb{N}\rightarrow \mathbb{N}$ with the following property. For $d\in \mathbb{N}$, if G is a vertex-transitive graph of d...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052925 | MATHDB-372608

The Small Quasi-Kernel Conjecture

摘录 / EXCERPT: Small Quasi-Kernel Conjecture. The digraph D contains a quasi-kernel of order at most $\frac{n}{2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-052926 | MATHDB-372609

The out-degree-three expansion conjecture for quasi-kernels

摘录 / EXCERPT: Out-degree-three expansion conjecture. There exists a vertex $v\in V(D)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052927 | MATHDB-372610

The worst-case output-size conjecture for the quasi-kernel algorithm

摘录 / EXCERPT: Worst-case output-size conjecture. For every $\varepsilon>0$, there exists a sourceless digraph D such that, for every vertex v of D, every quasi-kernel returned by the algor...

conjecture | open | MathDB 页面 | 原文线索

V2-052928 | MATHDB-372614

Jain's S^2-map conjecture

摘录 / EXCERPT: Let S^2 be the unit sphere in $\mathbb{R}^3$. Jain's S^2-map conjecture. There exists a map ... such that antipodal points of S^2 receive opposite values and the three vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-052929 | MATHDB-372615

Monotonicity conjecture for the p-normed flow index

摘录 / EXCERPT: Let G be a bridgeless graph and let $d\ge 2$. The quantity $\phi_{d,p}(G)$ is the d-dimensional p-normed flow index. Monotonicity conjecture. As a function of p,...

conjecture | open | MathDB 页面 | 原文线索

V2-052930 | MATHDB-372616

Duality conjecture for normed flow indices

摘录 / EXCERPT: Let G be a bridgeless graph and let $d\ge 2$ be an integer. For $p\in[1,2]$ and $q\in[2,\infty]$, write $\phi_{d,p}(G)$ and $\phi_{d,q}(G)$ for the corresponding flow indices. Du...

conjecture | open | MathDB 页面 | 原文线索

V2-052931 | MATHDB-372617

Circular-flow threshold conjecture for two-dimensional normed flows

摘录 / EXCERPT: Let G be a graph and let a circular $(3+\epsilon)$-NZF mean a circular nowhere-zero flow with parameter $3+\epsilon$. Circular-flow threshold conjecture. For $\epsilon\in[0,1]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-052932 | MATHDB-372633

Sublinear improvement conjecture for the three-page index

摘录 / EXCERPT: Let L be a non-split link, let $c(L)$ denote its crossing number, and let $\alpha_3(L)$ denote its three-page index. Sublinear improvement conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-052933 | MATHDB-372634

Linear non-separating independent-set conjecture for link-diagram dual graphs

摘录 / EXCERPT: Let D be a reduced link diagram with n crossings, and let $\Gamma(D)$ be its dual graph. An independent set in $\Gamma(D)$ is non-separating if deleting its vertices does not d...

conjecture | open | MathDB 页面 | 原文线索

V2-052934 | MATHDB-372636

Wojda's conjecture on packing digraphs

摘录 / EXCERPT: Let D and D' be digraphs, and let $|A(D)|$ denote the size of D, meaning its number of arcs. For every integer n and every m satisfying ... write $\mu(n,k)$ for the smalles...

conjecture | open | MathDB 页面 | 原文线索

V2-052935 | MATHDB-372662

The treewidth-linear basis number conjecture

摘录 / EXCERPT: Let G be a graph, let k be a nonnegative integer, and let $\operatorname{bn}(G)$ denote the basis number of G. Treewidth-linear basis number conjecture. Every graph G of tr...

conjecture | open | MathDB 页面 | 原文线索

V2-052936 | MATHDB-372663

Biedl's logarithmic basis number conjecture

摘录 / EXCERPT: Let G be an n-vertex graph, and let $\operatorname{bn}(G)$ denote the basis number of G. Biedl's logarithmic basis number conjecture. Every n-vertex graph G has ... The con...

conjecture | open | MathDB 页面 | 原文线索

V2-052937 | MATHDB-372669

Alikhani-Peng unimodality conjecture for domination sequences

摘录 / EXCERPT: Let G be a finite simple undirected graph with vertex set $V(G)$, let $n=|V(G)|$, and let $d_k(G)$ denote the number of dominating sets of G of size k. The polynomial ... recor...

conjecture | open | MathDB 页面 | 原文线索

V2-052938 | MATHDB-372685

Matsuda's weakly closed graph characterization of F-purity

摘录 / EXCERPT: Let G be a simple graph on n vertices, let $\Bbbk$ be a field, and set $R=\Bbbk[x_{1},\ldots,x_{n},y_{1},\ldots,y_{n}]$. For $\{i,j\}\in E(G)$, write...

conjecture | refuted | MathDB 页面 | 原文线索

V2-052939 | MATHDB-372686

Matsuda's eventual F-purity conjecture for binomial edge ideals

摘录 / EXCERPT: Let G be a graph, let $\Bbbk$ be a field, and let R_G denote the quotient by the binomial edge ideal associated to G. Write p for the characteristic of the base field. Mats...

conjecture | refuted | MathDB 页面 | 原文线索

V2-052940 | MATHDB-372692

Degree-plus-one conjecture for graphs of minimum degree three

摘录 / EXCERPT: For a graph G, write $d_G(v)$ for the degree of a vertex v. A $({\rm degree}+1)$-list assignment assigns each vertex v a list $L(v)$ with $|L(v)|\geq d_G(v)+1$. A graph is pr...

conjecture | open | MathDB 页面 | 原文线索

V2-052941 | MATHDB-372693

Degeneracy conjecture for proper conflict-free degree-plus list coloring

摘录 / EXCERPT: A graph G is d-degenerate if every subgraph of G has a vertex of degree at most d. For a graph G, a $({\rm degree}+d)$-list assignment assigns each vertex v a list...

conjecture | open | MathDB 页面 | 原文线索

V2-052942 | MATHDB-372694

Strengthened list-size conjecture for proper conflict-free coloring

摘录 / EXCERPT: Let G be a connected graph distinct from C_5, and let L be a list assignment of G. For each vertex v, assume that ... Here a coloring is proper conflict-free when it is bot...

conjecture | open | MathDB 页面 | 原文线索

V2-052943 | MATHDB-372699

Equality of the hybrid Floer-Morse and holomorphic curve-counting maps

摘录 / EXCERPT: Let C be the underlying curve, let Σ be a real exact Lagrangian in T^*C, and let $q_1,\dots,q_\kappa$ be points of C. The maps ... and ... are respectively defined by holo...

conjecture | open | MathDB 页面 | 原文线索

V2-052944 | MATHDB-372706

The sparsifying-transduction characterization of dense class properties

摘录 / EXCERPT: Let $\mathscr C$ be a class of TOWS graphs, and let $\mathsf{Sp}(\mathscr C)$ be the class obtained by the paper's sparsifying transduction. Let Π_S be a downset of weakly spar...

conjecture | open | MathDB 页面 | 原文线索

V2-052945 | MATHDB-372709

Cheng, Sun, Tan and Wang's rainbow Hamilton cycle conjecture

摘录 / EXCERPT: Let G be a strongly edge-colored graph on n vertices, and let $\delta(G)$ denote its minimum degree. Cheng, Sun, Tan and Wang's conjecture. If ... then G has a rainbow Hamilton...

conjecture | open | MathDB 页面 | 原文线索

V2-052946 | MATHDB-372710

Rich flow number conjecture

摘录 / EXCERPT: Rich flow number conjecture. If $\Delta\geq 5$, then G admits a rich $\lfloor 1.5\Delta+1\rfloor$-flow.

conjecture | open | MathDB 页面 | 原文线索

V2-052947 | MATHDB-372711

Rich flows for 3-edge-connected graphs

摘录 / EXCERPT: Rich-flow conjecture for 3-edge-connected graphs. Every 3-edge-connected graph with maximum degree Δ admits a rich $(\Delta+3)$-flow.

conjecture | open | MathDB 页面 | 原文线索

V2-052948 | MATHDB-372712

Logarithmic bound for confluent-edge-free flows

摘录 / EXCERPT: Confluent-edge-free flow conjecture. Every rich flow admissible graph with maximum degree Δ has a nowhere-zero $O(\ln(\Delta))$-flow containing no pair of confluent edges.

conjecture | open | MathDB 页面 | 原文线索

V2-052949 | MATHDB-372739

Hurlbert-Kamat leaf-centred maximum-star conjecture for trees

摘录 / EXCERPT: Let T be a tree, let $k\geq 1$, and let $\mathcal{I}_{T}^{k}(v)$ denote the family of independent sets of size k in T containing v. A leaf-centred maximum-star conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-052950 | MATHDB-372742

The sharp linear Turán bound conjecture for 4-edge linear paths

摘录 / EXCERPT: Let P_4^r be the r-uniform linear path with four edges, and let $ex_r^{\mathrm{lin}}(n,P_4^r)$ denote the maximum number of edges in an n-vertex linear r-uniform hypergraph...

conjecture | open | MathDB 页面 | 原文线索

V2-052951 | MATHDB-372746

Apte-Parekh-Sud token-graph Laplacian conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph and let $F_k(G)$ be its k-th token graph, whose vertices are the k-element subsets of V. For $1\le k\le |V|/2$, let $L(F_k(G))$ denote the Laplacian...

conjecture | solved | MathDB 页面 | 原文线索

V2-052952 | MATHDB-372747

H-free extension of Brouwer's Laplacian eigenvalue-sum conjecture

摘录 / EXCERPT: H-free Brouwer conjecture. For every H-free graph $G=(V,E)$ with $|V|\ge k$,

conjecture | solved | MathDB 页面 | 原文线索

V2-052953 | MATHDB-372748

Apte-Parekh-Sud token-excess matching conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $F_k(G)$ be its k-th token graph, and define ... Let $\nu(G)$ denote the matching number of G. Apte-Parekh-Sud's token-excess conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-052954 | MATHDB-372751

Conjecture on the proper open conflict-free chromatic index

摘录 / EXCERPT: Proper open conflict-free chromatic-index conjecture. For every graph G with maximum degree Δ,

conjecture | open | MathDB 页面 | 原文线索

V2-052955 | MATHDB-372758

Weak Loewner spectral dominance of the connection matrix over the Dirac matrix

摘录 / EXCERPT: Weak Loewner spectral-dominance conjecture. The matrices L and D satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-052956 | MATHDB-372769

The comparison conjecture for A_α-eigenvalue bounds

摘录 / EXCERPT: Let G be a simple graph, with maximum degree Δ, minimum degree δ, and let $\alpha\in[0,1]$. Comparison conjecture. One has ... The inequality asserts that the lower b...

conjecture | open | MathDB 页面 | 原文线索

V2-052957 | MATHDB-372773

Coarse Gallai conjecture for A-paths

摘录 / EXCERPT: Let G be a graph and let A be a subset of the vertices of G. An A-path is a path in G whose two endpoints belong to A. For subsets X of vertices of G and integers...

conjecture | open | MathDB 页面 | 原文线索

V2-052958 | MATHDB-372789

Győri et al.'s linear Turán conjecture for linear paths in 3-graphs

摘录 / EXCERPT: Győri et al.'s linear Turán conjecture. If G contains no linear path of length $k\ge 5$, then the number of edges in G is at most

conjecture | open | MathDB 页面 | 原文线索

V2-052959 | MATHDB-372803

The minimum-fort-size conjecture for marginal observances

摘录 / EXCERPT: Let G be a graph. Write $\underline{f}(G)$ for the minimum fort size of G, let $\mathrm{MObs}(G;i)$ denote the marginal observance at step i, and let $\gamma_P(G)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-052960 | MATHDB-372850

Weak identifiability of MLP parameters under arbitrary activation functions

摘录 / EXCERPT: Weak identifiability conjecture. With mild assumptions on the activation function σ, for every parameter $\theta \in \Theta$ of an MLP, there exists a parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-052961 | MATHDB-372851

Weak identifiability of multi-head attention parameters

摘录 / EXCERPT: MHA weak identifiability conjecture. For every parameter $\theta\in\Theta$ of an MHA model, the parameter θ is weakly identifiable with respect to the symmetry group

conjecture | open | MathDB 页面 | 原文线索

V2-052962 | MATHDB-372853

Ito's duality conjecture for regular \mathcal{W}-superalgebras

摘录 / EXCERPT: Ito's conjecture. There is an isomorphism of vertex superalgebras

conjecture | open | MathDB 页面 | 原文线索

V2-052963 | MATHDB-372854

Critical-level duality conjecture for regular \mathcal{W}-superalgebras

摘录 / EXCERPT: Assume $n\geqslant m$, and let κ_c denote the critical level. Let $\mathcal{A}^m_{\mathrm{deg}}$ be the degenerate $\beta\gamma$-system associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-052964 | MATHDB-372855

Center conjecture for critical-level regular \mathcal{W}-superalgebras

摘录 / EXCERPT: Assume $n\geqslant m$, with κ_c the critical level, and let $\mathfrak{z}$ denote the commutative center. Let $\mathcal{A}^m_{\mathrm{deg}}$ be the degenerate $\beta\gamma$-...

conjecture | open | MathDB 页面 | 原文线索

V2-052965 | MATHDB-372869

Bal-DeBiasio conjecture on monochromatic tree covers

摘录 / EXCERPT: Let G be an n-vertex r-edge-coloured graph, and let $tc(G)$ be the smallest number of not necessarily vertex-disjoint monochromatic trees whose vertices cover $V(G)$. Bal-DeB...

conjecture | open | MathDB 页面 | 原文线索

V2-052966 | MATHDB-372870

Tight cycle-partition conjecture for dense edge-coloured graphs

摘录 / EXCERPT: For $r\geq2$ and $\delta\in(0,1/2)$, let ... where $\mathcal{G}(n,r,\delta)$ consists of the n-vertex r-edge-coloured graphs with $\delta(G)\geq(1-\delta)n$, and $cp(G)$ is the s...

conjecture | open | MathDB 页面 | 原文线索

V2-052967 | MATHDB-372871

Tight tree-cover conjecture for dense edge-coloured graphs

摘录 / EXCERPT: For $r\geq2$ and $\delta\in(0,1)$, let ... where $\mathcal{G}(n,r,\delta)$ consists of the n-vertex r-edge-coloured graphs with $\delta(G)\geq(1-\delta)n$, and $tc(G)$ is the sma...

conjecture | open | MathDB 页面 | 原文线索

V2-052968 | MATHDB-372887

Symmetry conjecture for graphs with edge-disjoint cycles

摘录 / EXCERPT: Let G be a graph, and let $T\chi_G^o(q)$ denote its total oriented chromatic quasisymmetric function. Say that cycles of G are edge-disjoint when no two cycles share an edge. S...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052969 | MATHDB-372894

Han-Zhao's exact minimum co-degree conjecture for Hamilton ll-cycles

摘录 / EXCERPT: Han-Zhao's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-052970 | MATHDB-372897

Logarithmic Kazhdan-Lusztig conjecture for kernels of screening operators

摘录 / EXCERPT: Let $\mathcal{V}$ be a vertex algebra, let $\mathfrak{Z}_{\mathcal{M}_1},\ldots,\mathfrak{Z}_{\mathcal{M}_n}$ be screening operators, and let $\mathcal{W}\subset\mathcal{V}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-052971 | MATHDB-372912

Unimodality conjecture for graphical Stirling numbers of planar graphs

摘录 / EXCERPT: Planar-graph unimodality conjecture. For any planar graph G of order n, the sequence $\{\st{G}{k}\}_{k=1}^n$ is unimodal. This property is known for paths, cycles, trees, lolli...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052972 | MATHDB-372913

Monotonicity conjecture for restricted graphical Stirling numbers

摘录 / EXCERPT: Let G be a graph, let k be fixed, and let $\st{G}{k}_r$ denote the r-restricted graphical Stirling number, where the restricted-set size is indexed by r. Restricted-number...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-052973 | MATHDB-372918

Catalan-type formula for transitive partitions of the complete directed graph

摘录 / EXCERPT: For each integer $n\ge 3$, let $\vec K_n$ denote the complete directed graph on n vertices, and let $\genfrac{\{} {\}}{0pt}1{\vec K_n}{n-2}$ denote the number of admissible parti...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052974 | MATHDB-372921

Antiferromagnetic models as clique-minimisers

摘录 / EXCERPT: Let G be a graph, let $d_v\coloneqq\deg_G(v)$, and let H be an edge-weighted graph, possibly with loops. Define the weighted homomorphism count by ... where $H(a,b)$ is the edge...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-052975 | MATHDB-372943

Chen et al.'s Ore-degree conjecture for the strong chromatic index

摘录 / EXCERPT: Chen et al.'s conjecture. If $\theta(G)\ge 5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-052976 | MATHDB-372950

Vanishing conjecture for higher path homology of circulant digraphs

摘录 / EXCERPT: Higher path-homology vanishing conjecture. The path homology of $\vec{C}_n^S$ vanishes in every degree at least three:

conjecture | open | MathDB 页面 | 原文线索

V2-052977 | MATHDB-372962

Canonical vertex-Ramsey characterization of the finite sunflower property

摘录 / EXCERPT: Let $\mathcal{K}$ be a relational Fraïssé class with strong amalgamation. The canonical vertex-Ramsey characterization. ... The source also conjectures that having the finite 2-sun...

conjecture | open | MathDB 页面 | 原文线索

V2-052978 | MATHDB-372995

The graphical biclosed ornamentation lattice conjecture

摘录 / EXCERPT: For a finite undirected graph G, let B be the graphical pointed building set of G, and let $\operatorname{Bicl}(\mathsf{B})$ denote the subposet of biclosed ornament...

conjecture | open | MathDB 页面 | 原文线索

V2-052979 | MATHDB-373012

Regular cycle-conformal braces are complete bipartite graphs

摘录 / EXCERPT: A brace is a bipartite matching covered graph, and a graph is cycle-conformal if every even cycle C has the property that deleting its vertices leaves a graph with a perfect matc...

conjecture | open | MathDB 页面 | 原文线索

V2-052980 | MATHDB-373023

Ghosh et al.'s Turán-number conjecture for the triangular pyramid of 4-layers

摘录 / EXCERPT: Let TP_4 be the triangular pyramid with 4 layers, and let $\textup{ex}(n,TP_4)$ denote the maximum number of edges in an n-vertex graph containing no copy of TP_4. Ghosh et...

conjecture | open | MathDB 页面 | 原文线索

V2-052981 | MATHDB-373027

Hansen-Vukićević conjecture on Zagreb indices

摘录 / EXCERPT: Hansen-Vukićević conjecture. For any simple finite graph $\mathcal{G}$,

conjecture | refuted | MathDB 页面 | 原文线索

V2-052982 | MATHDB-373067

Polynomial pivot-minor Ramsey number conjecture

摘录 / EXCERPT: Let $R_{\mathrm{piv}}(k)$ be the minimum n such that every n-vertex graph contains an independent set or clique of size k as a pivot-minor. Pivot-minor Ramsey conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-052983 | MATHDB-373068

Sparse random graph vertex-minor universality conjecture

摘录 / EXCERPT: Let $p=\omega(1/\sqrt n)$ with $p\leq 1/2$, and let G be sampled from either $\mathbb{G}(n,p)$ or $\mathbb{G}(n,1-p)$. A graph is k-vertex-minor universal if every graph on any...

conjecture | open | MathDB 页面 | 原文线索

V2-052984 | MATHDB-373084

Giant Spider Conjecture for directed and oriented graphs

摘录 / EXCERPT: For integers $k\geq 2$ and $\ell\geq 1$, a (k,ℓ)-spider is the $(k-1)$-subdivision of an in-star with ℓ leaves. A directed graph is considered with its minimum out-degree...

conjecture | open | MathDB 页面 | 原文线索

V2-052985 | MATHDB-373087

The regularity-h-degree band conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph on n vertices. Write R for the polynomial ring associated with G, $J(G)$ for its cover ideal, $\operatorname{reg}(R/J(G))$ for the Castelnuovo-Mu...

conjecture | open | MathDB 页面 | 原文线索

V2-052986 | MATHDB-373088

Linear obstruction conjecture for regularity-h-degree pairs of connected graphs

摘录 / EXCERPT: Let $r,d\in\mathbb{Z}_{\ge 0}$, and let G be a connected graph. Write R for the polynomial ring associated with G, $J(G)$ for its cover ideal, $\operatorname{reg}(R/J(G))$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-052987 | MATHDB-373103

Vertex-nonamenability implies linear range

摘录 / EXCERPT: Vertex-nonamenability implies linear range. If G is vertex-nonamenable, then there exists $c=c(G)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-052988 | MATHDB-373107

Wedge-of-spheres conjecture for robust clique complexes of square-sequence graphs

摘录 / EXCERPT: Wedge-of-spheres conjecture. For all $k\geq 2$, the complex $\mathsf{Cliq}_{k}(G)$ is homotopy equivalent to a wedge sum of $(2k-3)$-dimensional spheres. Moreover, the number of sp...

conjecture | open | MathDB 页面 | 原文线索

V2-052989 | MATHDB-373109

Explicit polynomial directed-clique construction conjecture for tournaments

摘录 / EXCERPT: Let $T_1,T_2,\ldots$ be tournaments, and write $V(T_n)$ for the vertex set of T_n. Explicit tournament construction conjecture. There is an explicit construction of tournaments...

conjecture | open | MathDB 页面 | 原文线索

V2-052990 | MATHDB-373111

Existence of the asymptotic limit for universal tree graphs

摘录 / EXCERPT: Let $s(n)$ denote the minimum number of vertices in a universal graph for all trees with n vertices. Asymptotic-limit conjecture. The limit ... exists. Determining this limit would...

conjecture | open | MathDB 页面 | 原文线索

V2-052991 | MATHDB-373112

Finiteness conjecture for symmetric abstract 2-rigidity matroids

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many values of n for which there exists a symmetric abstract 2-rigidity matroid $\mathcal{M}_n\neq\mathcal{R}_n$ in which...

conjecture | open | MathDB 页面 | 原文线索

V2-052992 | MATHDB-373126

Bollobás-Przykucki-Riordan-Sahasrabudhe conjecture for the K_5 running time

摘录 / EXCERPT: Let $M_{K_5}(n)$ denote the maximum running time of the K_5 infection process over starting graphs on n vertices. Bollobás-Przykucki-Riordan-Sahasrabudhe conjecture. ... The conj...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-052993 | MATHDB-373133

The bipartite extremal-separation conjecture

摘录 / EXCERPT: Let H be a bipartite graph, let $M_H(n)$ denote the maximum running time of the H-process, and let $\operatorname{ex}(n,H)$ be the Turán extremal number of H. Bipartite extre...

conjecture | open | MathDB 页面 | 原文线索

V2-052994 | MATHDB-373134

The edge-deletion extremal upper-bound conjecture

摘录 / EXCERPT: Let H be an infection rule with at least three edges, let $M_H(n)$ denote the maximum running time of the H-process, and let $\operatorname{ex}(n,H-e)$ be the Turán extremal nu...

conjecture | open | MathDB 页面 | 原文线索

V2-052995 | MATHDB-373142

ChatGPT 5.2's local smoothability conjecture for valence-4 polyhedral Lagrangian vertices

摘录 / EXCERPT: Let K be an orientable polyhedral Lagrangian in $(\mathbb{R}^4,\omega)$, let v be a valence-4 vertex, and let $\Gamma_{v,\delta}\subset (S^3,\xi_{\mathrm{std}})$ be the Legen...

conjecture | open | MathDB 页面 | 原文线索

V2-052996 | MATHDB-373146

The C_4-factor conjecture for digraphs

摘录 / EXCERPT: The C_4-factor conjecture. There exists $n_0=n_0(\eta)\in\mathbb N$ such that, for every $n\geq n_0$ divisible by 4, if

conjecture | open | MathDB 页面 | 原文线索

V2-052997 | MATHDB-373167

The spectral conjecture for detection in random geometric graphs

摘录 / EXCERPT: Spectral detection conjecture. The sharp threshold for distinguishing the RGG from its Erdős-Rényi counterpart is determined by

conjecture | open | MathDB 页面 | 原文线索

V2-052998 | MATHDB-373173

Polynomial-time conjecture for subgraphs with boundedly many components

摘录 / EXCERPT: Let G be a finite graph and let H be a subgraph of G with at most k components, where k is a natural number. The notions S-Eulerian and semi S-Eulerian mean, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-052999 | MATHDB-373193

The initial-vertex conjecture for randomized zero forcing on complete bipartite graphs

摘录 / EXCERPT: Let $K_{a,b}$ be a bidirectional complete bipartite graph with all edges of equal weight, where $a>b$. Let A and B be its partite sets, with $|A|=a$ and $|B|=b$. For vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-053000 | MATHDB-373195

Linear perimeter-gap conjecture for vertex-transitive digraphs

摘录 / EXCERPT: For a directed graph, its circumference is the maximum length of a directed cycle, and its perimeter gap is the difference between its order and its circumference. Linear perimeter...

conjecture | open | MathDB 页面 | 原文线索

V2-053001 | MATHDB-373196

Asymptotic equivalence of graph and digraph circumferences

摘录 / EXCERPT: Let $c(n)$ be the minimum circumference of a connected vertex-transitive graph on n vertices, and let $d(n)$ be the minimum circumference of a connected vertex-transitive digraph...

conjecture | open | MathDB 页面 | 原文线索

V2-053002 | MATHDB-373229

Akbari-Alazemi-Anđelić's energy-matching number conjecture

摘录 / EXCERPT: Akbari-Alazemi-Anđelić's conjecture. For any connected graph $G \not\cong C_3, C_5, C_7$ with $\Delta(G) \in \{ 2, 3, 4, 5 \}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-053003 | MATHDB-373230

Mostar index extremal graph conjecture

摘录 / EXCERPT: Mostar index extremal graph conjecture. For any $n \ge 3$, the graph

conjecture | open | MathDB 页面 | 原文线索

V2-053004 | MATHDB-373233

The extension of Theorem on the forbidden graphs $\overline{K}_3$ and C_5

摘录 / EXCERPT: Let $\mathrm{er}_{s}(\{\overline{K}_3,C_5\})$ denote the minimum asymptotic density of s-cliques among graphs avoiding both $\overline{K}_3$ and C_5. Theorem's extension conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-053005 | MATHDB-373245

The special 4-cycle center conjecture for Artin complexes

摘录 / EXCERPT: Let A_S be an Artin group whose Coxeter diagram is a tree. A vertex of the Artin complex Δ_S is of type $\hat s=S\setminus\{s\}$ if it corresponds to a coset of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-053006 | MATHDB-373247

Chromatic-index conjecture for signed complete graphs of even order

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let σ be a signature on K_n, and let $\chi' (K_n, \sigma)$ denote its signed chromatic index. Chromatic-index conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-053007 | MATHDB-373260

Kühn-Osthus-Townsend conjecture on minimum d-degree thresholds for matchings

摘录 / EXCERPT: Let H be a k-graph, let d be an integer with $1\leq d\leq k-1$, and let $m_d^s(k,n)$ denote the minimum integer m such that every n-vertex k-graph H with minimum d-...

conjecture | open | MathDB 页面 | 原文线索

V2-053008 | MATHDB-373261

Discrete homotopy hypothesis

摘录 / EXCERPT: Let Graph be the category of graphs and cSet the category of cubical sets. Let $\mathrm{N}_{\infty}\colon \mathsf{Graph}\to\mathsf{cSet}$ be the nerve functor...

conjecture | open | MathDB 页面 | 原文线索

V2-053009 | MATHDB-373305

Bobby's spherical flow conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. An $\mathbb{S}^2$-flow is a unit vector flow with values on the unit sphere $\mathbb{S}^2$. Bobby's spherical flow conjecture. Every bridgeless...

conjecture | open | MathDB 页面 | 原文线索

V2-053010 | MATHDB-373306

The spherical coloring conjecture for the 2-sphere

摘录 / EXCERPT: Let $\mathbb{S}^2$ be the unit 2-sphere. Spherical coloring conjecture. There exists a map $q:\mathbb{S}^2 \rightarrow \{\pm 1, \pm 2, \pm 3, \pm 4\}$ such that antipodal points...

conjecture | open | MathDB 页面 | 原文线索

V2-053011 | MATHDB-373307

Cho-Hyun-O-Park spectral-radius conjecture for [a,b]-factors

摘录 / EXCERPT: Let a and b be positive integers with $a\le b$, and let G be a finite undirected simple connected graph of order n satisfying ... For a graph H, write $\rho(H)$ for its spe...

conjecture | open | MathDB 页面 | 原文线索

V2-053012 | MATHDB-373325

Spectral characterization conjecture for the 2-core of a sparse random graph

摘录 / EXCERPT: Let $G(n,p)$ be an Erdős-Rényi graph with $p=\lambda/n$, let $C_{\textnormal{giant}}$ denote its largest component, and let $\operatorname{Core}_2(C_{\textnormal{giant}})$ be the g...

conjecture | open | MathDB 页面 | 原文线索

V2-053013 | MATHDB-373337

The conjecture on existence of F-irregular graphs

摘录 / EXCERPT: Existence conjecture. For every connected graph F on three or more vertices, there exists an F-irregular graph.

conjecture | open | MathDB 页面 | 原文线索

V2-053014 | MATHDB-373338

The super-strong conjecture on F-irregular graphs

摘录 / EXCERPT: Super-strong conjecture on F-irregular graphs. For each integer $n\geqslant 3$, there exist infinitely many n-hyper-irregular graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-053015 | MATHDB-373343

Illingworth-Lang-Müyesser-Parczyk-Sgueglia's spanning tight component conjecture

摘录 / EXCERPT: Illingworth-Lang-Müyesser-Parczyk-Sgueglia's conjecture. If H has minimum codegree at least n/k, then H has a spanning tight component.

conjecture | open | MathDB 页面 | 原文线索

V2-053016 | MATHDB-373350

Colourful Hamilton cycle packing conjecture for Dirac graphs

摘录 / EXCERPT: Let c satisfy $1/2<c\leqslant1$, let G be an n-vertex graph with minimum degree $\delta(G)\geqslant cn$, and suppose the edges of G are properly edge-coloured. Colourful Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-053017 | MATHDB-373351

Colourful almost-spanning tree embedding conjecture for Dirac graphs

摘录 / EXCERPT: Let c satisfy $1/2<c\leqslant1$, and let G be an n-vertex graph with minimum degree $\delta(G)\geqslant cn$. Colourful tree embedding conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-053018 | MATHDB-373359

Havet et al.'s inversion diameter conjecture

摘录 / EXCERPT: Let G be a graph, let $\operatorname{diam}(G)$ denote its inversion diameter, and let $\Delta(G)$ denote its maximum degree. Havet et al.'s inversion diameter conjecture. Every g...

conjecture | open | MathDB 页面 | 原文线索

V2-053019 | MATHDB-373364

Degenerate tiling incidence conjecture over division rings

摘录 / EXCERPT: Let $\mathcal T$ be a simple tiling of a surface of positive genus, meaning that its underlying graph is $(n+1)$-degenerate, and let ${\mathrm D}$ be a division ring. The incidence...

conjecture | open | MathDB 页面 | 原文线索

V2-053020 | MATHDB-373365

Uniform dimension threshold conjecture for positive-genus tilings

摘录 / EXCERPT: Let $g>0$. A simple tiling $\mathcal T$ of a genus-g surface has an associated incidence theorem over a division ring ${\mathrm D}$, and n denotes the dimension of the ambient...

conjecture | open | MathDB 页面 | 原文线索

V2-053021 | MATHDB-373368

Characterization of separating cuts preserving the pfaffian number

摘录 / EXCERPT: Let a separating cut be a cut in a matchable graph whose contraction produces a graph, and let $\operatorname{pf}(G)$ denote the pfaffian number of a graph G. Separating-cut pres...

open_problem | open | MathDB 页面 | 原文线索

V2-053022 | MATHDB-373369

Caputo's hypergraph conjecture for symmetric-group representations

摘录 / EXCERPT: Let $\Gamma=([n],w)$ be a weighted hypergraph on $[n]$, with non-negative weights, and let ρ be a non-trivial irreducible representation of $\mathrm{Sym}(n)$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-053023 | MATHDB-373370

Aldous-Caputo-type conjecture for the spectral gap of $\mathrm{U}(n)$

摘录 / EXCERPT: Let $\Gamma=([n],w)$ be an arbitrary hypergraph with non-negative weights. For each irreducible representation ρ of $\mathrm{U}(n)$, let $\lambda_{\min}(\Gamma,\rho)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-053024 | MATHDB-373371

KMP$_2$ conjecture for arbitrary weighted hypergraphs

摘录 / EXCERPT: Let $\Gamma=([n],w)$ be an arbitrary hypergraph with non-negative weights. Let $\mathcal{KMP}_k$ be the discrete KMP process with k indistinguishable particles on Γ, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-053025 | MATHDB-373372

Monotonicity conjecture for new KMP eigenvalues

摘录 / EXCERPT: Let Γ be a hypergraph with non-negative weights. For every $k\in\mathbb{Z}_{\geq 1}$, let $\omega_k=\omega_k(\Gamma)$ denote the smallest new eigenvalue of the Laplacian...

conjecture | open | MathDB 页面 | 原文线索

V2-053026 | MATHDB-373385

Erdős-Neumann-Lara conjecture on chromatic and dichromatic numbers

摘录 / EXCERPT: Erdős-Neumann-Lara conjecture. For every integer k there is an integer $f(k)$ such that, for every graph G, $\chi(G)\geq f(k)$ implies $\vec\chi(G)\geq k$.

conjecture | open | MathDB 页面 | 原文线索

V2-053027 | MATHDB-373386

List Erdős-Neumann-Lara conjecture

摘录 / EXCERPT: List Erdős-Neumann-Lara conjecture. For every integer k there is an integer $f(k)$ such that, for every graph G, $\chi_{\ell}(G)\geq f(k)$ implies $\vec\chi_{\ell}(G)\geq k$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053028 | MATHDB-373421

Strong Erdős-Hajnal conjecture for simple digraphs

摘录 / EXCERPT: Let T be a tournament, let D be a simple digraph on n vertices with no subdigraph isomorphic to T, and let $\vec{\alpha}(D)$ denote its largest acyclic set. Strong Erdős-Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-053029 | MATHDB-373422

Strong Erdős-Hajnal property for the directed triangle

摘录 / EXCERPT: Let the directed triangle mean the cyclic tournament on three vertices, and say that a tournament has the Strong Erdős-Hajnal property when it satisfies the strong Erdős-Hajnal con...

conjecture | open | MathDB 页面 | 原文线索

V2-053030 | MATHDB-373423

Hartsfield-Ringel antimagic graph conjecture

摘录 / EXCERPT: A finite, simple, undirected graph $G=(V,E)$ has an antimagic labelling if a bijection $f:E\to\{1,2,\ldots,|E|\}$ gives each vertex u the sum of labels on its incident edges, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-053031 | MATHDB-373424

Alon-Bertay-Noga-? large-maximum-degree antimagic conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph with $|V|=n$, and let x be a vertex such that $d(x)=\Delta(G)=n-k$ for some $k\geq4$. Alon et al.'s large-maximum-degree conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-053032 | MATHDB-373425

Density-dependent antimagic conjecture for large-maximum-degree graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph with $|V|=n$, $|E|=m$, and let k be a positive integer. Density-dependent antimagic conjecture. If $\Delta(G)=n-k$ and $m\geq f(k)n$ for some p...

conjecture | open | MathDB 页面 | 原文线索

V2-053033 | MATHDB-373430

The hypergraph connector extension of the Cockayne-Lorimer theorem

摘录 / EXCERPT: Let $q,s\ge 1$ and $r\ge 2$ be integers. An s-connector is an r-uniform hypergraph such that every collection of pairwise disjoint vertex sets $A_1,\dots,A_r$ with $|A_i|\ge s$...

conjecture | open | MathDB 页面 | 原文线索

V2-053034 | MATHDB-373434

The inverse-linear bound for the exponent of random k-majority tournaments

摘录 / EXCERPT: Let $\mathcal{G}(n,k)$ be the probability space of k-majority tournaments on vertex set $[n]$ obtained by uniformly choosing, with replacement, $2k-1$ linear orders of $[n]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-053035 | MATHDB-373444

Gyárfás-Lehel biclique monochromatic-component conjecture

摘录 / EXCERPT: Gyárfás-Lehel conjecture. In every r-coloring of the edges of G, the vertex set can be covered by the vertices of at most $2r-2$ monochromatic components.

conjecture | open | MathDB 页面 | 原文线索

V2-053036 | MATHDB-373449

Szegö's construction conjecture for tight multigraphs

摘录 / EXCERPT: Let $\Gamma=(V,E)$ be a (d,ℓ)-tight multigraph, with $\ell<\frac{4d+2}{3}$. Let $K_2^{2k-\ell}$ denote the graph with two vertices and $2k-\ell$ parallel edges, and let a d-...

conjecture | open | MathDB 页面 | 原文线索

V2-053037 | MATHDB-373461

Higher-degree vanishing-order conjecture

摘录 / EXCERPT: For integers $k\geq4$ and $\ell\geq3$ with $\ell<k$, let F be a k-graph. An ℓ-vanishing order of F is an ordering of $V(F)$ admitting an ℓ-vanishing coloring with t...

conjecture | open | MathDB 页面 | 原文线索

V2-053038 | MATHDB-373462

Apex extension conjecture for degree Turán densities

摘录 / EXCERPT: Let F be a $(k-1)$-graph, and let $\mathcal{S}_F$ be the k-graph obtained by adjoining an apex vertex v and including $v\cup e$ as an edge for every $e\in E(F)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053039 | MATHDB-373463

Reiher-Rödl-Schacht conjecture on uniform Turán density and vanishing orders

摘录 / EXCERPT: Let F be a k-graph. For integers $1\leq\ell\leq k-3$, let $\pi_{\rm u}^{(\ell)}(F)$ denote its ℓ-uniform Turán density. An $(\ell+1)$-vanishing order is an ordering of...

conjecture | open | MathDB 页面 | 原文线索

V2-053040 | MATHDB-373471

Ore-degree Erdős Matching Conjecture for uniform hypergraphs

摘录 / EXCERPT: Ore-degree Erdős Matching Conjecture. If $n>rs$ and

conjecture | open | MathDB 页面 | 原文线索

V2-053041 | MATHDB-373477

Generic bipartite reconstruction by low-order type constraints

摘录 / EXCERPT: Let $G=(U,V,E)$ be a bipartite graph with $|U|=u$ and $|V|=v$, and let $\mathcal{D}(G)$ denote its vertex-deletion deck. Assume that G is 2-connected, not regular, and has minimu...

conjecture | open | MathDB 页面 | 原文线索

V2-053042 | MATHDB-373478

Biconnected planar non-maximal reconstruction conjecture

摘录 / EXCERPT: Let G be a planar biconnected graph with minimum degree $\delta(G)\geq 3$, and suppose that G is not maximal planar. Let $\mathcal{D}(G)$ be its vertex-deletion deck. Biconnect...

conjecture | open | MathDB 页面 | 原文线索

V2-053043 | MATHDB-373488

Conjectural description of maximizers for the maximum product of distances

摘录 / EXCERPT: Maximizer structure conjecture. The following properties hold:

conjecture | open | MathDB 页面 | 原文线索

V2-053044 | MATHDB-373490

The tangled-cord extremal conjecture for Hamiltonian sets in simple assembly graphs

摘录 / EXCERPT: Let G) be a simple assembly graph with n rigid vertices, each of degree 4, and let F_k denote the kth Fibonacci number. A Hamiltonian set of polygonal paths is a Hamilton...

conjecture | open | MathDB 页面 | 原文线索

V2-053045 | MATHDB-373496

Vertex-minimality conjecture for the triangulation of real projective 5-space from \mathcal{P}_{6,48}

摘录 / EXCERPT: Let $\mathcal{P}_{6,48}$ be the centrally symmetric simplicial 6-polytope whose antipodal quotient of the boundary gives a triangulation of $\mathbb{R}P^5$. A triangulation is ve...

conjecture | open | MathDB 页面 | 原文线索

V2-053046 | MATHDB-373510

Rappaport-Strasser conjecture for Hamiltonian Cayley graphs

摘录 / EXCERPT: Given a finite group G and a symmetric subset $S\subset G$, the Cayley graph $\operatorname{Cay}_G(S)$ has vertex set G and edges $\{g,gs\}$ for $g\in G$ and $s\in S$. A graph...

conjecture | open | MathDB 页面 | 原文线索

V2-053047 | MATHDB-373512

Duality conjecture for preclusters and maximal invariant polyhedral supports

摘录 / EXCERPT: Precluster-support duality conjecture. For conservative mass-action networks that do not have two species with exactly the same rates and have at least one positive steady state, t...

conjecture | open | MathDB 页面 | 原文线索

V2-053048 | MATHDB-373513

DP-completeness of exact circular chromatic index

摘录 / EXCERPT: DP-completeness conjecture. The problem $\mathrm{CCHI}^{=_r}$ is DP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-053049 | MATHDB-373514

Geelen's simulation conjecture for vertex-minor-closed graph classes

摘录 / EXCERPT: A graph class is vertex-minor-closed if it contains every vertex-minor of each of its graphs. Geelen's simulation conjecture. Measurement-based quantum computation (MBQC) is effici...

conjecture | open | MathDB 页面 | 原文线索

V2-053050 | MATHDB-373515

Oum's rank-width conjecture for bipartite circle graphs

摘录 / EXCERPT: A bipartite circle graph is a circle graph whose vertices can be divided into two parts such that edges join only vertices in different parts. A graph H is a pivot-minor of G i...

conjecture | open | MathDB 页面 | 原文线索

V2-053051 | MATHDB-373516

The edge-deletion energy conjecture for degree-based weighted adjacency matrices

摘录 / EXCERPT: Let K_n be the complete graph, let e be an edge of K_n, and let $\boldsymbol{\dot{A}}(K_n)$ and $\boldsymbol{\dot{A}}(K_n-e)$ denote the degree-based weighted adjacency matri...

conjecture | open | MathDB 页面 | 原文线索

V2-053052 | MATHDB-373524

Perfect divisibility conjecture for odd hole-free graphs

摘录 / EXCERPT: Let G be a graph with no induced odd cycle of length at least five. A graph is perfectly divisible if, for every induced subgraph H with at least one edge, its vertex set can b...

conjecture | open | MathDB 页面 | 原文线索

V2-053053 | MATHDB-373532

Levit's matching-cycle conjecture for almost bipartite non-König-Egerváry graphs

摘录 / EXCERPT: Let G be an almost bipartite non-König-Egerváry graph, and let C be its unique odd cycle. A maximum matching of G is a matching with the largest possible number of edges. Lev...

conjecture | open | MathDB 页面 | 原文线索

V2-053054 | MATHDB-373533

Determinant factorization conjecture for R-disjoint graphs

摘录 / EXCERPT: Let G be an R-disjoint graph with flower decomposition $B(G),R_{1},\dots,R_{k}$, and let $A(G)$ denote its adjacency matrix. Determinant factorization conjecture. The determina...

conjecture | open | MathDB 页面 | 原文线索

V2-053055 | MATHDB-373535

Rodríguez's half-integer curvature conjecture for binary Bayesian networks

摘录 / EXCERPT: Let a bitnet be a binary Bayesian network equipped with its Fisher information metric, and let $\langle R\rangle$ denote its volume-averaged Ricci scalar. Rodríguez's conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-053056 | MATHDB-373538

The corona-kernel bound for graphs with odd cycles

摘录 / EXCERPT: Corona-kernel conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-053057 | MATHDB-373539

Completeness conjecture at the extremal parameters for planar factor-critical graphs

摘录 / EXCERPT: Completeness conjecture. If G is a minimal t-$\{1,2\}$-factor-critical graph, then G is a complete graph.

conjecture | open | MathDB 页面 | 原文线索

V2-053058 | MATHDB-373545

Pairwise negative correlation conjecture for random-cluster measures

摘录 / EXCERPT: Random-cluster p-NC conjecture. The measures $\phi_{p,q}$ satisfy the p-NC property.

conjecture | open | MathDB 页面 | 原文线索

V2-053059 | MATHDB-373546

Pairwise negative correlation conjecture for the uniform forest measure

摘录 / EXCERPT: Uniform forest p-NC conjecture. The measure $\mathbb{P}_{\mathrm{UF}}$ satisfies the p-NC property.

conjecture | open | MathDB 页面 | 原文线索

V2-053060 | MATHDB-373564

The augmented acyclicity conjecture for the Collatz return graph

摘录 / EXCERPT: Let G_M^+ denote the augmented return graph at modular depth M, incorporating the mixed-state excursion, exit, and return dynamics, and let $K\ge1$ be the parameter indexing th...

conjecture | open | MathDB 页面 | 原文线索

V2-053061 | MATHDB-373575

The four-cycle conjecture for genus-one configuration curves

摘录 / EXCERPT: Let G be a 1-dof graph, meaning a graph whose generic configuration fiber has one degree of freedom, and let its generic fiber have genus one. A graph is said to be equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-053062 | MATHDB-373577

The arbor-polytope word-enumerator conjecture

摘录 / EXCERPT: Let τ be an arbor of size n, let $\mathcal Q_\tau$ be its associated arbor polytope, and let $\mathcal D(v)$ denote the union of the blocks at the descendants of a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-053063 | MATHDB-373578

Yin and Wu's size interpolation conjecture for 2-connected subgraphs

摘录 / EXCERPT: Yin and Wu's size interpolation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053064 | MATHDB-373579

Asymptotic minimum-degree interpolation conjecture for 2-connected subgraphs

摘录 / EXCERPT: Asymptotic interpolation conjecture. If $k\geqslant3$ and

conjecture | open | MathDB 页面 | 原文线索

V2-053065 | MATHDB-373590

The path-graph variational constant conjecture

摘录 / EXCERPT: Path-graph conjecture. For every integer $n\geqslant3$,

conjecture | open | MathDB 页面 | 原文线索

V2-053066 | MATHDB-373604

Karoński-Łuczak-Thomason 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a graph without isolated edges. An edge coloring $f:E\to\{1,2,\dots,k\}$ assigns a weight $\sigma(v)=\sum_{v\in e}f(e)$ to each vertex; adjacent vertices are distinguish...

conjecture | open | MathDB 页面 | 原文线索

V2-053067 | MATHDB-373605

Przybyło-Woźniak total 1-2 Conjecture

摘录 / EXCERPT: Let G be a graph. A total coloring is a function $g:V\cup E\to\{1,2,\dots,k\}$, with total vertex weight $\sigma^t(v)=g(v)+\sum_{v\in e}g(e)$. Adjacent vertices are distinguished...

conjecture | open | MathDB 页面 | 原文线索

V2-053068 | MATHDB-373609

The domatic game first-player difference conjecture

摘录 / EXCERPT: First-player difference conjecture. For all graphs G,

conjecture | open | MathDB 页面 | 原文线索

V2-053069 | MATHDB-373613

Ascoli-Frederickson-Frederickson-McFarland-Post vertex-minor universality conjecture

摘录 / EXCERPT: Vertex-minor universality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053070 | MATHDB-373627

Favaron's forbidden-subgraph conjecture for irredundance perfect graphs

摘录 / EXCERPT: Let G be a graph, and let P_6, G_1, and G_2 be the graphs described in Figure 5 of the source. Favaron's conjecture. If G does not contain P_6, G_1, or G_2 as induc...

conjecture | open | MathDB 页面 | 原文线索

V2-053071 | MATHDB-373628

Faudree-Favaron-Li conjecture on P_5-free irredundance perfect graphs

摘录 / EXCERPT: A graph is P_5-free if it has no induced subgraph isomorphic to the path P_5. Faudree-Favaron-Li conjecture. Any P_5-free graph is irredundance perfect. Puech proved this con...

conjecture | open | MathDB 页面 | 原文线索

V2-053072 | MATHDB-373629

Puech's forbidden-subgraph conjecture for irredundance perfect graphs

摘录 / EXCERPT: Let G be a graph, and let P_6, G_4, and G_5 be the graphs described in Figure 5 of the source. Puech's conjecture. If G does not contain P_6, G_4, or G_5 as induced...

conjecture | open | MathDB 页面 | 原文线索

V2-053073 | MATHDB-373634

Mahdian's conjecture for strong edge coloring of K_{t,t}-free graphs

摘录 / EXCERPT: Let $t\geq 2$ be fixed, and let G be a $K_{t,t}$-free graph of maximum degree d. Let $\chi'_s(G)$ denote its strong chromatic index. Mahdian's conjecture. ... The paper proves a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053074 | MATHDB-373655

Monotonicity of the average density for stochastic sandpiles

摘录 / EXCERPT: Let $G=(V\cup\{s\},E)$ be a finite graph with designated sink vertex s. Under the p-toppling rule, each edge incident to a toppled vertex independently sends a particle with pr...

conjecture | open | MathDB 页面 | 原文线索

V2-053075 | MATHDB-373656

The abelian sandpile minimizes average density

摘录 / EXCERPT: Let $G=(V\cup\{s\},E)$ be a finite graph with designated sink vertex s, and let $\rho_G(p)$ denote the stationary average density for the p-toppling rule, where $p\in[0,1]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053076 | MATHDB-373672

Linial's path-partition conjecture for digraphs

摘录 / EXCERPT: Linial's conjecture. For every digraph D and every positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-053077 | MATHDB-373677

Schrijver's partition formulation of Woodall's conjecture

摘录 / EXCERPT: Let $tau\textgreater=2$ be an integer, and let $D=(V,A)$ be a digraph whose minimum dicut size is tau. A strengthening is an arc set whose reversal makes the digraph strongly con...

conjecture | open | MathDB 页面 | 原文线索

V2-053078 | MATHDB-373684

Lee's polynomial anticoncentration conjecture for random spanning trees

摘录 / EXCERPT: Lee's conjecture. For every tree T,

conjecture | open | MathDB 页面 | 原文线索

V2-053079 | MATHDB-373685

The sharp anticoncentration conjecture for random spanning trees

摘录 / EXCERPT: Sharp anticoncentration conjecture. For every tree T,

conjecture | open | MathDB 页面 | 原文线索

V2-053080 | MATHDB-373686

The polynomial lower-bound conjecture for non-isomorphic spanning trees

摘录 / EXCERPT: Non-isomorphic spanning-tree lower-bound conjecture. The number of non-isomorphic spanning trees of G is at least

conjecture | open | MathDB 页面 | 原文线索

V2-053081 | MATHDB-373688

Fractional coloring conjecture for degenerate triangle-free hypergraphs

摘录 / EXCERPT: Let H be an r-uniform d-degenerate triangle-free hypergraph, with $r\geq 3$. Here, $\chi_f(H)$ denotes the fractional chromatic number, and $c_r>0$ is a constant depending on...

conjecture | open | MathDB 页面 | 原文线索

V2-053082 | MATHDB-373689

Besomi-Pavez-Signé-Stein conjecture on minimum and maximum degree

摘录 / EXCERPT: Besomi-Pavez-Signé-Stein conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053083 | MATHDB-373690

Havet-Reed-Stein-Wood conjecture on tree containment

摘录 / EXCERPT: Havet-Reed-Stein-Wood conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053084 | MATHDB-373703

Higgins's perfect matching conjecture for bipartite tensor products

摘录 / EXCERPT: Let M be a binary $n\times k$ matrix. Define $M\otimes M^T$ to be the symmetric $nk\times nk$ matrix indexed by pairs in $[n]\times[k]$, with entries ... A perfect matching is an i...

conjecture | open | MathDB 页面 | 原文线索

V2-053085 | MATHDB-373704

Weak balls conjecture for biregular bipartite multigraphs

摘录 / EXCERPT: Let M be a binary $n\times k$ matrix with non-negative integer entries, row sums k, and column sums n. Let $M\otimes M^T$ be the matrix associated with the corresponding tens...

conjecture | open | MathDB 页面 | 原文线索

V2-053086 | MATHDB-373706

Four-parts conjecture for biregular bipartite labeled multigraphs

摘录 / EXCERPT: Let L_1,R_1,L_2,R_2 be vertex sets of sizes n_1,k_1,n_2,k_2, respectively, satisfying ... Let (L_1,R_1,E_1,u_1,v_1) and (L_2,R_2,E_2,u_2,v_2) be biregular bipartite labeled m...

conjecture | open | MathDB 页面 | 原文线索

V2-053087 | MATHDB-373727

Burr-Erdős-Graham-Sós maximal anti-Ramsey conjecture for odd cycles

摘录 / EXCERPT: Burr-Erdős-Graham-Sós conjecture. For every integer $k\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-053088 | MATHDB-373737

Kohayakawa-Prömel-Rödl conjecture for induced Ramsey numbers

摘录 / EXCERPT: Let G be a fixed graph, and let H be a graph on n vertices. The quantity $r_{\mathrm{ind}}(G,H)$ is the minimum order of a graph whose every red-blue edge-coloring contains e...

conjecture | open | MathDB 页面 | 原文线索

V2-053089 | MATHDB-373761

Leonida-Li's weighted third-eigenvalue conjecture

摘录 / EXCERPT: Let $\mathcal{S}_n$ denote the set of real symmetric $n\times n$ matrices with entries in $[0,1]$, and let $\lambda_1(M)\geq\cdots\geq\lambda_n(M)$ be the eigenvalues of M. Leoni...

conjecture | open | MathDB 页面 | 原文线索

V2-053090 | MATHDB-373763

Core-structure conjecture for conditionally non-redundant hypergraphs

摘录 / EXCERPT: Core-structure conjecture. Under these hypotheses, every vertex incident with more than linearly many hyperedges shares a second common vertex with all those hyperedges.

conjecture | open | MathDB 页面 | 原文线索

V2-053091 | MATHDB-373768

Karthick-Kumar-Sivaraman conjecture for fork-free graphs

摘录 / EXCERPT: Let G be a graph. A graph is perfectly divisible if, for every induced subgraph H, the vertex set $V(H)$ can be partitioned into sets A and B such that $H[A]$ is perfect an...

conjecture | open | MathDB 页面 | 原文线索

V2-053092 | MATHDB-373773

Bollobás-Meir conjecture for the power-weighted Euclidean traveling-salesman problem

摘录 / EXCERPT: Bollobás-Meir conjecture. For any finite set of points $X\subseteq[0,1]^k$, there exists a Hamiltonian cycle H on X with $S_k(H)\leq 2\cdot k^{k/2}$ if $k\neq3$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-053093 | MATHDB-373791

Bounded spinal concentration for local invariants of the partition graph

摘录 / EXCERPT: Let G_n be the partition graph, let Ax_n and Sp_n denote its axial and spinal vertex sets, and let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053094 | MATHDB-373805

Casselgren, Petros and Fufa's Cartesian-product edge-precoloring conjecture

摘录 / EXCERPT: Casselgren, Petros and Fufa's conjecture. If every precoloring of at most $k < \chi'(G)$ edges of G can be extended to a proper $\chi'(G)$-edge-coloring, then every precoloring o...

conjecture | open | MathDB 页面 | 原文线索

V2-053095 | MATHDB-373806

Cartesian-product edge-precoloring conjecture for two graphs

摘录 / EXCERPT: General Cartesian-product conjecture. Every precoloring of at most $k+l+1$ edges of $G \square H$ is extendable to a proper $(\chi'(G)+\chi'(H))$-edge-coloring of $G \square H$.

conjecture | open | MathDB 页面 | 原文线索

V2-053096 | MATHDB-373807

Edge-precoloring conjecture for Cartesian products with balanced complete bipartite graphs

摘录 / EXCERPT: Balanced complete-bipartite Cartesian-product conjecture. If any precoloring of at most $k < r$ edges of G can be extended to a proper $\chi'(G)$-edge-coloring of G, then any p...

conjecture | open | MathDB 页面 | 原文线索

V2-053097 | MATHDB-373808

Casselgren, Markström and Pham's hypercube precoloring-extension conjecture

摘录 / EXCERPT: Casselgren, Markström and Pham's conjecture. If n and d are positive integers, and φ is a proper edge-precoloring of $K_{n,n}^d$ with at most $nd-1$ precolored edges, t...

conjecture | open | MathDB 页面 | 原文线索

V2-053098 | MATHDB-373810

Triangular threshold conjecture for simplex layers in the partition graph

摘录 / EXCERPT: Let G_n be the partition graph, let K_n be its clique complex, let $L_r(n)$ denote the vertices of local simplex dimension r, let $\Delta(n)$ be the maximal local simplex dim...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053099 | MATHDB-373812

Low-dimensional framework conjecture for the partition graph

摘录 / EXCERPT: Let Fw_n denote the boundary framework, let $L_r(n)$ be the vertices of local simplex dimension r, and let (n) and (1^n) be the two antenna vertices. Low-dimension...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053100 | MATHDB-373825

Liu-Zhang-Zhang conjecture on packing colorings of subcubic graphs

摘录 / EXCERPT: A packing $(1,1,2,2)$-coloring of a graph is a partition of its vertex set into four classes whose pairwise vertex distances are respectively at least 2, 2, 3, and 3. Liu-Z...

conjecture | open | MathDB 页面 | 原文线索

V2-053101 | MATHDB-373826

Gastineau-Togni conjecture on packing colorings of subcubic graphs

摘录 / EXCERPT: A packing $(1,1,2,3)$-coloring is a partition of the vertex set into four classes whose pairwise vertex distances are respectively at least 2, 2, 3, and 4. Gastineau-Togni...

conjecture | open | MathDB 页面 | 原文线索

V2-053102 | MATHDB-373843

Verstraëte's even-cycle subgraph conjecture

摘录 / EXCERPT: Verstraëte's conjecture. There exists a positive constant c such that every C_{2k}-free bipartite graph G has a $C_{2\ell}$-free subgraph F satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-053103 | MATHDB-373862

The common-zero independence-number conjecture

摘录 / EXCERPT: Let G be a finite group, and let $\Gamma_v(G)$ be the graph whose vertices are the nonlinear irreducible characters of G, with two vertices adjacent when they have a common zer...

conjecture | open | MathDB 页面 | 原文线索

V2-053104 | MATHDB-373886

The Bollobás-Nikiforov conjecture for K_4-free graphs

摘录 / EXCERPT: Let G be a K_4-free graph with m edges, and let ... be the adjacency eigenvalues of G. Bollobás-Nikiforov conjecture for K_4-free graphs. If $G\neq K_3$, then ... This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-053105 | MATHDB-373904

Zhang, Liu and Zhou's extremal Eccentric Connectivity Index conjecture

摘录 / EXCERPT: For integers n and m, define ... and let $E_{n,m}$ be the graph obtained from a clique K_{n-d-1} and a path ... by joining each clique vertex to both v_d and v_{d-1}, and by...

conjecture | open | MathDB 页面 | 原文线索

V2-053106 | MATHDB-373907

Post-LayerNorm smuggled-bias degeneracy conjecture

摘录 / EXCERPT: Because the outputs are constrained by $\mathbf{1}^T x=1$, let $x'=x-\frac{1}{d}\mathbf{1}$, so that $\mathbf{1}^T x'=0$, and let $W\in\mathbb{R}^{m\times d}$ act on the input. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-053107 | MATHDB-373908

Degree-four median-eigenvalue conjecture excluding projective-plane incidence graphs

摘录 / EXCERPT: Let G be a simple graph of order n, with adjacency eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, and let λ_h and λ_ℓ be its median eigenvalues, where ……

conjecture | open | MathDB 页面 | 原文线索

V2-053108 | MATHDB-373919

Kawarabayashi-Maharry conjecture for 5-connected non-planar graphs

摘录 / EXCERPT: Let G be a 5-connected non-planar graph with at least seven vertices. Kawarabayashi-Maharry conjecture. The graph G contains both a $K_{3,4}$ minor and a K_6^- minor. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-053109 | MATHDB-373920

Liu and Ma's admissible-cycle conjecture

摘录 / EXCERPT: Liu and Ma's conjecture. Every graph with minimum degree at least $k+1$ contains k admissible cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-053110 | MATHDB-373925

Rear-central nucleation conjecture for higher simplicial behaviour in the partition graph

摘录 / EXCERPT: Let G_n be the partition graph on the partitions associated with the parameter n, and consider its simplicial thick zones, including the tetrahedral and higher-order regions. T...

conjecture | open | MathDB 页面 | 原文线索

V2-053111 | MATHDB-373927

The minimum-codegree conjecture for squares of tight Hamilton cycles in 3-graphs

摘录 / EXCERPT: Let H be a 3-graph on n vertices, and let $\delta_2(H)$ denote its minimum codegree, the minimum number of vertices completing a pair to an edge of H. The square of a tight...

conjecture | open | MathDB 页面 | 原文线索

V2-053112 | MATHDB-373928

The v-number bound for closed neighborhood ideals

摘录 / EXCERPT: The v-number bound. For every simple connected graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-053113 | MATHDB-373932

The odd-graph Thurston-Bennequin symmetry conjecture

摘录 / EXCERPT: Let O_n denote the odd graph with parameter $n\geq 2$. A graph is TB-symmetrical when its total Thurston-Bennequin number admits the symmetry described in the paper. Odd-graph TB...

conjecture | open | MathDB 页面 | 原文线索

V2-053114 | MATHDB-373949

Critical-level conjecture for superelliptic affine Lie algebras

摘录 / EXCERPT: Let r and m be the parameters of a superelliptic algebra, and set ... Let $\mathfrak{g}$ be the underlying Lie algebra, let $h^{\vee}$ be its dual Coxeter number, and let f be...

conjecture | open | MathDB 页面 | 原文线索

V2-053115 | MATHDB-373950

The exact extremal number of C_4-free subgraphs of Q_8

摘录 / EXCERPT: Let Q_8 be the 8-dimensional hypercube and let C_4 denote the cycle of length four. Write $\mathrm{ex}(Q_8,C_4)$ for the maximum number of edges in a subgraph of Q_8 containi...

conjecture | open | MathDB 页面 | 原文线索

V2-053116 | MATHDB-373951

The exact extremal number of C_4-free subgraphs of Q_7

摘录 / EXCERPT: Let Q_7 be the 7-dimensional hypercube and let C_4 denote the cycle of length four. Write $\mathrm{ex}(Q_7,C_4)$ for the maximum number of edges in a subgraph of Q_7 containi...

conjecture | open | MathDB 页面 | 原文线索

V2-053117 | MATHDB-373966

Distance-transitive Seymour-tight orientation conjecture

摘录 / EXCERPT: Distance-transitive Seymour-tightness conjecture. Every distance-transitive digraph is a Seymour-tight orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-053118 | MATHDB-373969

Gerke-Marciniszyn-Steger sparse counting lemma

摘录 / EXCERPT: Let H be a fixed graph. For each vertex $x\in V(H)$, let V_x be a pairwise disjoint set of n vertices, and let $\mathcal G(H,n,m)$ be the family of graphs with vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-053119 | MATHDB-373972

Neural-network complexity conjecture for C-GNP latent domains

摘录 / EXCERPT: Neural-network complexity conjecture. The complexity of the architecture, measured by the number of layers and its width, is controlled by $4\|\tau_{\Omega}\|_{\infty}4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053120 | MATHDB-373981

The 3-line conjecture for minimal braces

摘录 / EXCERPT: A brace is a bipartite matching covered graph with no nontrivial barrier. A 3-line is an edge whose two endpoints have degree 3. Let G be a minimal brace with at least 6 vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-053121 | MATHDB-373982

The topological determination of convergence and limit compositions for infinity-foods

摘录 / EXCERPT: Food-quiver topology conjecture. The topology of the food quiver determines the convergence properties and limit compositions of its infinity-foods. In particular, the presence of...

conjecture | open | MathDB 页面 | 原文线索

V2-053122 | MATHDB-373983

The path as the unique minimum spectral-sum graph

摘录 / EXCERPT: Let n denote the order of a connected graph, and let the spectral sum be $\lambda_1(G)+\lambda_2(G)$, where $\lambda_1(G)$ and $\lambda_2(G)$ are the largest and second-largest a...

conjecture | open | MathDB 页面 | 原文线索

V2-053123 | MATHDB-374000

Davies et al.'s variance lower-bound conjecture for the hard-core model

摘录 / EXCERPT: Davies et al.'s variance conjecture. For any $\lambda>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-053124 | MATHDB-374003

Bounded cliquewidth under proper separator decompositions

摘录 / EXCERPT: Cliquewidth preservation conjecture. There is a function f such that every graph in $\mathcal C$ has cliquewidth at most $f(a)$.

conjecture | open | MathDB 页面 | 原文线索

V2-053125 | MATHDB-374011

Conjecture on the roots of the parity-specific functions g_0 and g_1

摘录 / EXCERPT: Root-separation conjecture. For any $\alpha\the 2$, $g_0(n,\alpha)=0$ has exactly one real root for $x\the(3,+\infty)$. For any $\alpha>2$, $g_1(n,\alpha)=0$ has at most one real r...

conjecture | open | MathDB 页面 | 原文线索

V2-053126 | MATHDB-374012

Extension of the random Turán upper-bound theorems to (c,r)-bounded graphs

摘录 / EXCERPT: Extension conjecture. Theorems ... should hold for (c,r)-bounded graphs with $1\le c\le r$.

conjecture | open | MathDB 页面 | 原文线索

V2-053127 | MATHDB-374022

The periodic efficient total girth coloring conjecture for cubic maps

摘录 / EXCERPT: Let $M(G)$ be a connected simple cubic map of girth 4 embedded in a genus-realizing orientable surface, and suppose all its belts have lengths ℓ with $\ell\equiv 0\pmod 4$....

conjecture | open | MathDB 页面 | 原文线索

V2-053128 | MATHDB-374036

Refined non-Abelian dominance hypothesis for graph automorphism groups

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph on n vertices with non-abelian automorphism group $G=\operatorname{Aut}(\mathcal{G})$ of order n, and let $\mathbf{x}$ be a single observation of a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053129 | MATHDB-374056

Connectivity conjecture for the friendship graph of knots

摘录 / EXCERPT: Let Γ_{fr} be the graph whose vertices are knot types and whose edges represent friendship, meaning that each knot is represented by a simple closed curve on a mini...

conjecture | open | MathDB 页面 | 原文线索

V2-053130 | MATHDB-374069

Hunter-Milojević-Sudakov-Tomon conjecture on induced Turán numbers

摘录 / EXCERPT: For positive integers n,s, let $K_{s,s}$ be the complete bipartite graph with s vertices in each part. For a graph H, let $\mathrm{ex}(n,H)$ be the maximum number of edges in...

conjecture | open | MathDB 页面 | 原文线索

V2-053131 | MATHDB-374070

Dong-Gao-Li-Liu induced rational exponents conjecture

摘录 / EXCERPT: For positive integers n,s, let $K_{s,s}$) be the complete bipartite graph with s vertices in each part. For a bipartite graph H, let $\mathrm{ex}^*(n,H,s)$ be the maximum num...

conjecture | open | MathDB 页面 | 原文线索

V2-053132 | MATHDB-374071

Erdős-Simonovits rational exponents conjecture

摘录 / EXCERPT: For a graph H, let $\mathrm{ex}(n,H)$ denote the maximum number of edges in an n-vertex graph containing no copy of H. Erdős-Simonovits rational exponents conjecture. For eve...

conjecture | open | MathDB 页面 | 原文线索

V2-053133 | MATHDB-374083

The supertoken graph independence-number conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph with independent sets C_1 and C_2, with $|C_1|=c_1\leq c_2=|C_2|$, as in Theorem. Let ${\mathcal F}_k(G)$ be the k-supertoken graph of G, and s...

conjecture | open | MathDB 页面 | 原文线索

V2-053134 | MATHDB-374103

The gamma lower-bound conjecture for weak saturation of hypergraphs

摘录 / EXCERPT: Let $r\ge 2$, $s\ge 2$, and $k\ge r+1$ be integers. Let ... Let H be an r-uniform hypergraph with $s(H)=s$ and $\delta_{s-1}(H)=\delta$. The gamma lower-bound conjecture. Then .....

conjecture | open | MathDB 页面 | 原文线索

V2-053135 | MATHDB-374104

Asymptotic equivalence of weak saturation and rho-saturation

摘录 / EXCERPT: Let $r\ge 1$ and $s\ge 1$ be integers, and let H be an r-uniform hypergraph with $s(H)=s$. Asymptotic equivalence conjecture. Then ... The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-053136 | MATHDB-374123

Sharp Heisenberg eigenvalue and minimal profile

摘录 / EXCERPT: Sharp Heisenberg eigenvalue and minimal profile. The explicit family satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053137 | MATHDB-374128

Pardey and Rautenbach's bounded-deviation conjecture for colour-balanced matchings

摘录 / EXCERPT: Pardey and Rautenbach's conjecture. For all integers $k\geq 2$ and $t\geq 1$, every colour-balanced k-edge-coloured K_{2kt} admits a perfect matching M satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-053138 | MATHDB-374129

The square-root discrepancy conjecture for representative perfect matchings in bipartite graphs

摘录 / EXCERPT: Square-root discrepancy conjecture. There is a perfect matching M of $K_{n,n}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-053139 | MATHDB-374130

The constant-deviation conjecture for colour-balanced perfect matchings

摘录 / EXCERPT: Constant-deviation conjecture. There exists an absolute constant C such that for all positive integers k and t, every colour-balanced k-edge-colouring of K_{2kt} admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-053140 | MATHDB-374142

Lu-Ma's non-trivial matching conjecture for partite hypergraphs

摘录 / EXCERPT: For integers $r\ge 3$ and $n_1\ge\dots\ge n_r\ge 2$, let $X_1,\dots,X_r$ be disjoint sets with $|X_i|=n_i$. Define $m_0(s,n_1,\dots,n_r)$ as the maximum size of a family...

conjecture | open | MathDB 页面 | 原文线索

V2-053141 | MATHDB-374152

Li-Schiermeyer’s star-wheel Ramsey conjecture

摘录 / EXCERPT: Let $R(G,H)$ denote the Ramsey number of graphs G and H, let $K_{1,m}$ be the star with m leaves, and let W_{2n} be the wheel on $2n+1$ vertices. Li-Schiermeyer’s conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-053142 | MATHDB-374153

The minimum-degree wheel conjecture

摘录 / EXCERPT: Let W_{2k} be the wheel on $2k+1$ vertices, let G be a graph on n vertices, and let $\delta(G)$ denote its minimum degree. Minimum-degree wheel conjecture. For all integers...

conjecture | open | MathDB 页面 | 原文线索

V2-053143 | MATHDB-374171

Dong-Ge-Gong-Ning-Ouyang-Tay higher-derivative conjecture for chromatic polynomials

摘录 / EXCERPT: Let G be a graph of order n, and let $P(G,x)$ denote its chromatic polynomial. For $x<0$, the quantity $(-1)^nP(G,x)$ is positive, so its logarithm is defined. Dong-Ge-Gong-Nin...

conjecture | open | MathDB 页面 | 原文线索

V2-053144 | MATHDB-374179

The vertex-minor Ramsey number conjecture for five vertices

摘录 / EXCERPT: Vertex-minor Ramsey number conjecture for $k=5$.

conjecture | open | MathDB 页面 | 原文线索

V2-053145 | MATHDB-374187

Hoang-Reed conjecture on cycles with small overlaps

摘录 / EXCERPT: Let $r\in\mathbb{N}$, and let $\delta^+(D)$ denote the minimum out-degree of a digraph D. A sequence $C_1,\ldots,C_r$ of directed cycles has at most one overlap per cycle if, for...

conjecture | open | MathDB 页面 | 原文线索

V2-053146 | MATHDB-374188

Mader's conjecture on openly disjoint cycles in regular digraphs

摘录 / EXCERPT: For a digraph D, let $c(D)$ be the largest integer k for which there are directed cycles $C_1,\ldots,C_k$ through a common vertex v such that $C_1-v,\ldots,C_k-v$ are pairwis...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053147 | MATHDB-374189

Mader's subdivision conjecture for acyclic digraphs

摘录 / EXCERPT: Let F be an acyclic digraph, and let $\delta^+(D)$ denote the minimum out-degree of a digraph D. A subdivision of F is obtained by replacing the arcs of F by directed paths...

conjecture | open | MathDB 页面 | 原文线索

V2-053148 | MATHDB-374190

Local spectral matching conjecture for 3-graphs

摘录 / EXCERPT: Local spectral matching conjecture. If, for every $v\in V(H)$,

conjecture | open | MathDB 页面 | 原文线索

V2-053149 | MATHDB-374191

The wheel-clique Ramsey number conjecture

摘录 / EXCERPT: Wheel-clique Ramsey conjecture. For all positive integers n, we have

conjecture | open | MathDB 页面 | 原文线索

V2-053150 | MATHDB-374204

Dai-Zhang-Broerama-Zhang conjecture on Hamiltonicity of connected K_{1,r}-free split graphs

摘录 / EXCERPT: Dai-Zhang-Broerama-Zhang conjecture. Every $(r-1)$-connected $K_{1,r}$-free split graph is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-053151 | MATHDB-374225

Bender et al.'s isomorphism conjecture for uniquely complemented zero-divisor graphs

摘录 / EXCERPT: Bender et al.'s conjecture. If $G(S)$ is uniquely complemented with clique number 3 or greater and every vertex has a unique complement, then $G(S)$ is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-053152 | MATHDB-374228

Elphick-Tang-Zhang square-energy conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph on n vertices, and let $\mathcal{E}_2^+(G)$ and $\mathcal{E}_2^-(G)$ denote its positive and negative 2-energies, respectively. Elphick-Tang-Zhang'...

conjecture | open | MathDB 页面 | 原文线索

V2-053153 | MATHDB-374229

Q. Tang, Y. Liu, and W. Wang's positive p-energy conjecture

摘录 / EXCERPT: Let n and p be positive integers with $p\geq 2$, let G be a connected graph on n vertices, and let P_n denote the path on n vertices. Tang-Liu-Wang's positive p-energ...

conjecture | open | MathDB 页面 | 原文线索

V2-053154 | MATHDB-374230

S. Akbari, H. Kumar, and B. Mohar's negative p-energy conjecture

摘录 / EXCERPT: Let n and p be positive integers with $p\geq 2$, let G be a connected graph on n vertices, and let K_n denote the complete graph on n vertices. Akbari-Kumar-Mohar's neg...

conjecture | open | MathDB 页面 | 原文线索

V2-053155 | MATHDB-374231

3-connected minimum-degree conjecture for odd edge-colorings

摘录 / EXCERPT: 3-connected minimum-degree conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053156 | MATHDB-374232

Yokoi's minimum-degree conjecture for odd edge-colorings

摘录 / EXCERPT: Yokoi's conjecture. There exists a constant d satisfying the preceding condition: for every 2-connected graph G of odd order with minimum degree at least d, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-053157 | MATHDB-374249

Lyapunov diagonal stability conjecture for global stability of linear-threshold networks

摘录 / EXCERPT: LDS implies GAS of LTN equilibria. If $W-D\in\mathcal{LDS}$, then for every $u\in\mathbb{R}^n$, the linear-threshold network admits a unique globally asymptotically stable equilibr...

conjecture | open | MathDB 页面 | 原文线索

V2-053158 | MATHDB-374269

Burr-Erdős-Faudree-Rousseau-Schelp conjecture on adding matching components

摘录 / EXCERPT: Let G and H be graphs, and let K_2 denote the complete graph on two vertices. A pair (G,H) is Ramsey-finite if it has only finitely many Ramsey-minimal graphs. Burr-Erdős-F...

conjecture | open | MathDB 页面 | 原文线索

V2-053159 | MATHDB-374270

Burr-Erdős-Faudree-Rousseau-Schelp one-sided Ramsey-infiniteness conjecture

摘录 / EXCERPT: Let G and H be graphs. A pair (G,H) is Ramsey-infinite if it has infinitely many Ramsey-minimal graphs. An odd star is a star with an odd number of leaves, and a K_2-compon...

conjecture | open | MathDB 页面 | 原文线索

V2-053160 | MATHDB-374271

Burr's stronger finite-characterization conjecture for Ramsey-minimal graph families

摘录 / EXCERPT: Let G and H be graphs, and let $\mathcal{R}(G,H)$ denote the family of Ramsey-minimal graphs for the pair (G,H). A matching is a graph whose connected components are all isom...

conjecture | open | MathDB 页面 | 原文线索

V2-053161 | MATHDB-374282

Distance Laplacian integrality conjecture for cyclic-group power graphs

摘录 / EXCERPT: Let $n\geq 2$ be an integer, and let $\mathcal{P}(\mathbb{Z}_{n})$ denote the power graph of the cyclic group $\mathbb{Z}_{n}$. Write ∂_{1}^{L} for the relevant distance L...

conjecture | open | MathDB 页面 | 原文线索

V2-053162 | MATHDB-374297

General-graph isolated-matching threshold conjecture

摘录 / EXCERPT: Let $\delta^{\mathbf{no\ iso}}_k(n)$ denote the minimum-degree threshold such that the k-switch graph of an n-vertex graph, when nonempty, is guaranteed to have positive minimu...

conjecture | open | MathDB 页面 | 原文线索

V2-053163 | MATHDB-374303

Distance-set rigidity conjecture for finite GH-spaces generated by labeled trees

摘录 / EXCERPT: Let $i=1,2$ and let $T_i=T_i(l_i)$ be finite labeled trees with non-degenerate labelings $l_i:V(T_i)\to\mathbb{R}^+$. Suppose that $(V(T_1),d_{l_1})$ and $(V(T_2),d_{l_2})$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-053164 | MATHDB-374312

The effective Linek conjecture for independent sets of trees

摘录 / EXCERPT: Let $i(T)$ denote the number of independent sets of a tree T. Effective Linek's Problem. Every integer greater than 88013 appears as $i(T)$ for some tree T. This is an effect...

conjecture | open | MathDB 页面 | 原文线索

V2-053165 | MATHDB-374313

The planar Linek conjecture for independent sets

摘录 / EXCERPT: Let $i(P)$ denote the number of independent sets of a connected planar graph P. Planar Linek's Problem. Every positive integer is equal to $i(P)$ for some connected planar graph...

conjecture | open | MathDB 页面 | 原文线索

V2-053166 | MATHDB-374321

Scott's odd induced subgraph conjecture

摘录 / EXCERPT: Let G be a finite simple graph, let $f_o(G)$ be the maximum order of an induced subgraph of G whose every vertex has odd degree, and let $\chi(G)$ be the chromatic number of...

conjecture | open | MathDB 页面 | 原文线索

V2-053167 | MATHDB-374338

Minimal strong generating type conjecture for the N=2 coset ${\mathcal C}^{\psi}_{{\mathcal N}=2}(n,r|s)$

摘录 / EXCERPT: Let n,r,s be non-negative integers, and define ... Let ${\mathcal C}^{\psi}_{{\mathcal N}=2}(n,r|s)$ be the corresponding coset vertex algebra. Minimal strong generating type conje...

conjecture | open | MathDB 页面 | 原文线索

V2-053168 | MATHDB-374339

Procházka-Rapčák Feigin-Frenkel duality conjecture for N=2 cosets

摘录 / EXCERPT: Let n,r,s be non-negative integers, and let ${\mathcal C}^{\psi}_{{\mathcal N}=2}(n,r|s)$ denote the corresponding N=2 coset vertex algebra. Procházka-Rapčák duality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-053169 | MATHDB-374341

Feige-Pauzner conjecture for enabling graphs

摘录 / EXCERPT: Feige-Pauzner conjecture. For all $n>0$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053170 | MATHDB-374344

Existence conjecture for doubly saturated Ramsey-good graphs

摘录 / EXCERPT: Let $s,t \ge 3$ with $s \le t$ and $(s,t) \notin \{(3,4),(3,6)\}$. A graph is doubly saturated $R(s,t)$-good if it is $R(s,t)$-good and remains so under the relevant double-saturat...

conjecture | open | MathDB 页面 | 原文线索

V2-053171 | MATHDB-374345

Circulant construction conjecture for doubly saturated $R(3,t)$-good graphs

摘录 / EXCERPT: For odd $t \ge 17$, let G_t be the circulant graph on $5t-10$ vertices whose distance set is ... A graph is doubly saturated $R(3,t)$-good if it is $R(3,t)$-good and satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-053172 | MATHDB-374353

The spectral-radius conjecture for extremal graphs with edge-disjoint spanning trees

摘录 / EXCERPT: Let k and c be integers with $0\le c\le k-1$. For $0\le c\le k-2$, let $\mathcal{H}_1(n,k,c)$ be the class of all $(k+c)$-edge-connected graphs H of order n for which there...

conjecture | open | MathDB 页面 | 原文线索

V2-053173 | MATHDB-374354

Conjecture on the strong chromatic extremal function of Berge paths

摘录 / EXCERPT: Let $r\geq 3$ be a fixed integer. For a path P_k with k edges, let $\mathcal{B}(P_k)$ denote its Berge hypergraph, and let $\mathrm{schex}_r(n,\mathcal{B}(P_k))$ be the strong...

conjecture | open | MathDB 页面 | 原文线索

V2-053174 | MATHDB-374356

Jung-Keszegh-Pálvölgyi-Yuditsky conjecture on piercing maximum hypergraph cliques

摘录 / EXCERPT: Jung-Keszegh-Pálvölgyi-Yuditsky conjecture. For all $k\geq2$ there exists a constant $c_k<1$ such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-053175 | MATHDB-374357

The extremal edge-count conjecture for odd-order triangle-free 1-planar graphs

摘录 / EXCERPT: Extremal edge-count conjecture. For any odd $n\geq 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-053176 | MATHDB-374372

Jaeger orientations in random regular graphs

摘录 / EXCERPT: Let $\mathcal{G}_{n,d}$ be the uniform model of random d-regular graphs on the vertex set $[n]=\{1,2,\ldots,n\}$. A p-orientation is an orientation of a d-regular graph in wh...

conjecture | open | MathDB 页面 | 原文线索

V2-053177 | MATHDB-374380

Hardness conjecture for bounded-size inversion approximation

摘录 / EXCERPT: Bounded-size inversion approximation hardness conjecture. There exists $\varepsilon>0$ such that, unless $P=NP$, for every $p\geq 3$ and every $k\geq 1$, no polynomial-time...

conjecture | open | MathDB 页面 | 原文线索

V2-053178 | MATHDB-374382

Burning Number Conjecture for connected graphs

摘录 / EXCERPT: Let G be a connected graph on n vertices, and let $b(G)$ denote the minimum number of rounds needed to burn all vertices of G. Burning Number Conjecture. ... This conjecture pr...

conjecture | open | MathDB 页面 | 原文线索

V2-053179 | MATHDB-374429

Conjecture on near-perfect triangle packings in randomly perturbed regular graphs

摘录 / EXCERPT: Near-perfect packing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053180 | MATHDB-374443

The induced P_6 planar Turán number conjecture

摘录 / EXCERPT: Let P_6 be the path on six vertices, and let ${\mathrm{ex}}_{\mathcal P}(n,P_6^{\mathrm{ind}})$ be the maximum number of edges in an n-vertex planar graph containing no induced...

conjecture | open | MathDB 页面 | 原文线索

V2-053181 | MATHDB-374452

Lin-Zhou's distance spectral extremal conjecture

摘录 / EXCERPT: Lin-Zhou's conjecture. If $1\leqslant s\leqslant \frac{n-7}{2}$ and $G\in\mathcal G(m)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-053182 | MATHDB-374476

Nash-Williams' directed degree-sequence conjecture

摘录 / EXCERPT: Let D be a strongly connected digraph on n vertices, with nondecreasing out-degree sequence $d_1^+\leq\cdots\leq d_n^+$ and in-degree sequence $d_1^-\leq\cdots\leq d_n^-$. Nash...

conjecture | open | MathDB 页面 | 原文线索

V2-053183 | MATHDB-374477

Nash-Williams' directed Pósa-type conjecture

摘录 / EXCERPT: Let D be a digraph on $n\geq 3$ vertices, with nondecreasing out-degree sequence $d_1^+\leq\cdots\leq d_n^+$ and in-degree sequence $d_1^-\leq\cdots\leq d_n^-$. Nash-Williams' di...

conjecture | open | MathDB 页面 | 原文线索

V2-053184 | MATHDB-374478

Rainbow Hamiltonian cycle conjecture for graph families satisfying an Ore-type condition

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n\geq 3$, and let $\mathcal{G}=\{G_i:i\in[n]\}$ be a family of n-vertex graphs with the same vertex set; the graphs in the family may be identical. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053185 | MATHDB-374492

Engbers-Erey's extremal conjecture for average maximal matching size in trees

摘录 / EXCERPT: Engbers-Erey's conjecture. The average size of maximal matchings, $avm(T)$, is uniquely minimized by

conjecture | open | MathDB 页面 | 原文线索

V2-053186 | MATHDB-374493

Tyomkyn-Uzzell's balanced double-broom conjecture for triangle-free distance graphs

摘录 / EXCERPT: Tyomkyn-Uzzell's conjecture. For $k\geq 3$ and $n\geq k+1$, except when $k=3$ and $n=8$, every n-vertex graph G such that G_k is triangle-free satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053187 | MATHDB-374512

Dallard-Krnc-Kwon-Milanič-Munaro-Štorgel-Wiederrecht forbidden-subgraph characterization

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite class of graphs, and let $\mathcal{F}$-free mean having no induced subgraph isomorphic to a member of $\mathcal{F}$. A multiclaw is a graph each compo...

conjecture | open | MathDB 页面 | 原文线索

V2-053188 | MATHDB-374513

The random embedding edge-type conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Random embedding edge-type conjecture. The expected numbers of bad singular edges, good singular edges, and regular edges are respectively

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053189 | MATHDB-374520

The two-place Laplacian matching root integral variation conjecture

摘录 / EXCERPT: Let G be a connected graph and let e be a non-edge of G. Write the nonnegative real roots of the Laplacian matching polynomial $4\mathscr{LM}_G(x)\4$ in non-increasing order...

conjecture | open | MathDB 页面 | 原文线索

V2-053190 | MATHDB-374521

The multi-edge Laplacian matching root integral variation conjecture

摘录 / EXCERPT: Let G be a connected graph, let $F\subseteq E(G^c)$ be nonempty, and set $H=G+F$. Let $\lambda_1(G),\dots,\lambda_n(G)$ denote the roots of the Laplacian matching polynomial of...

conjecture | open | MathDB 页面 | 原文线索

V2-053191 | MATHDB-374522

Gerbner-Tompkins-Zhou's asymptotic conjecture for expanded graphs with bounded matching number

摘录 / EXCERPT: Let G be a graph with chromatic number $\chi(G)=\ell>r$, and let U be an independent set of G such that deleting U results in a graph G_0 of chromatic number $\ell-1$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-053192 | MATHDB-374523

Yang-Zeng-Zhang's conjecture for expanded cliques with bounded matching number

摘录 / EXCERPT: Let $K^{(r)+}_{\ell+1}$ be the r-uniform expansion of the complete graph on $\ell+1$ vertices, let $M^{(r)+}_{s+1}$ be an r-uniform matching with $s+1$ edges, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-053193 | MATHDB-374524

The rainbow Turán threshold conjecture for expanded cliques

摘录 / EXCERPT: Let ${\mathrm{ex}}^{\sum}_r(n,k,K_\ell^{(r)+})$ denote the rainbow hyper-Turán number for the r-uniform expansion $K_\ell^{(r)+}$, and let $t_r(n,\ell-1)$ be the number of edges...

conjecture | open | MathDB 页面 | 原文线索

V2-053194 | MATHDB-374525

The refined threshold conjecture for expanded cliques with bounded matching number

摘录 / EXCERPT: Let $K_{\ell+1}^{(r)+}$ be the r-uniform expansion of the complete graph on $\ell+1$ vertices, let $M_{s+1}^{(r)+}$ be an r-uniform matching with $s+1$ edges, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-053195 | MATHDB-374532

Unimodality conjecture for dominating-set sequences of finite graphs

摘录 / EXCERPT: Let G be a finite undirected graph on n vertices. A dominating set is a set of vertices such that every vertex is either in the set or adjacent to a vertex in the set. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053196 | MATHDB-374538

Linear vertex threshold conjecture for isolated cliques in the minimum-degree Kruskal-Katona problem

摘录 / EXCERPT: Linear threshold conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053197 | MATHDB-374540

Conlon-Fox-Sudakov conjecture on books versus triangles

摘录 / EXCERPT: Conlon-Fox-Sudakov conjecture. If $b(G)\le b$ and G is not the balanced complete bipartite graph, then G has at least

conjecture | open | MathDB 页面 | 原文线索

V2-053198 | MATHDB-374541

Dean's conjecture on cycles divisible by the minimum-degree bound

摘录 / EXCERPT: All graphs under consideration are finite and simple. For a graph G and a vertex v, let $d_G(v)$ denote the degree of v. Dean's conjecture. For every integer $k \ge 3$, every...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053199 | MATHDB-374544

Wu-Magnant-Nowbandegani-Xia conjecture for multicolor Ramsey numbers of cherries

摘录 / EXCERPT: Wu-Magnant-Nowbandegani-Xia conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053200 | MATHDB-374545

The k-planar conjecture for quasi-transitive graphs

摘录 / EXCERPT: The k-planar conjecture. The graph Γ is almost planar if and only if it is quasi-isometric to some planar graph.

conjecture | open | MathDB 页面 | 原文线索

V2-053201 | MATHDB-374548

Phase transition conjecture for the lcm-density of edge ideals

摘录 / EXCERPT: Phase transition conjecture. The lcm-density of edge ideals of graphs exhibits a phase transition phenomenon when considering the set of all graphs on n vertices as...

conjecture | open | MathDB 页面 | 原文线索

V2-053202 | MATHDB-374552

Conjecture on identifiable parameters in ReLU architectures with width-one layers

摘录 / EXCERPT: Width-one identifiability conjecture. 1. There exist architectures $\mathcal{A}=(n_0,\ldots,n_{L+1})$ with $n_\ell=1$ for some $\ell\leq L$ for which no identifiable parameters exi...

conjecture | open | MathDB 页面 | 原文线索

V2-053203 | MATHDB-374553

Chen and Ma's conjecture on equal-degree endpoints joined by odd paths

摘录 / EXCERPT: Let n be a positive integer, and let $\ell\geq 2$ be fixed. Consider graphs on $2n+1$ vertices, and say that two vertices are joined by a path of length $2\ell+1$ when such a pat...

conjecture | open | MathDB 页面 | 原文线索

V2-053204 | MATHDB-374558

The path-biclique conjecture for bounded alpha-degeneracy

摘录 / EXCERPT: Let P_t denote the t-vertex path and $K_{,}$ the complete bipartite graph with ... . A graph is ... or $K_{,}$. Path-biclique alpha-degeneracy conjecture. For every two positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-053205 | MATHDB-374559

The strong odd Hadwiger-type conjecture for clique immersions

摘录 / EXCERPT: Let $t\in\mathbb{Z}^+$ and let G be a graph. An immersion of K_t is strong odd if its paths are pairwise edge-disjoint, have odd length, and no terminal is an interior vertex o...

conjecture | open | MathDB 页面 | 原文线索

V2-053206 | MATHDB-374561

Effective Lipschitz concentration for transformer language models

摘录 / EXCERPT: Effective Lipschitz concentration. The effective Lipschitz constant satisfies $\Lambda_{\mathrm{eff}}\ll\Lambda(W)$, specifically $\Lambda_{\mathrm{eff}}=O(\mathrm{poly}(H))$, whil...

conjecture | open | MathDB 页面 | 原文线索

V2-053207 | MATHDB-374567

Mubayi-Suk double-exponential lower-bound conjecture for $r_4(5,n)$

摘录 / EXCERPT: Let $r_4(5,n)$ be the smallest integer N such that every N-vertex 4-uniform hypergraph contains either a copy of $K_5^{(4)}$ or an independent set of size n. Mubayi-Suk con...

conjecture | open | MathDB 页面 | 原文线索

V2-053208 | MATHDB-374589

Guiduli's spectral dense-neighborhood conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, let $T_r(n)$ denote the Turán graph, let $N(v)$ denote the neighborhood of a vertex v, let $d(v)$ denote its degree, and let $bb(G)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-053209 | MATHDB-374596

Balogh-Clemen-Lidický conjecture on the balanced bipartite distance of K_4-free graphs

摘录 / EXCERPT: Let G be a K_4-free graph on n vertices. The Balogh-Clemen-Lidický conjecture. G can be made balanced bipartite by removing at most ... edges. This conjecture asks whether th...

conjecture | open | MathDB 页面 | 原文线索

V2-053210 | MATHDB-374606

Allen-Brightwell-Skokan cycle goodness conjecture

摘录 / EXCERPT: For a graph H, let $\chi(H)$ denote its chromatic number and $|H|$ its number of vertices. A graph G is H-good if...

conjecture | open | MathDB 页面 | 原文线索

V2-053211 | MATHDB-374617

Linear asymptotic formula for K^s_t-saturation numbers

摘录 / EXCERPT: Let K^s_t be the complete s-uniform hypergraph on t vertices, and let $\mathrm{sat}(n,K^s_t)$ denote its saturation number on n vertices. Saturation-number conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053212 | MATHDB-374618

Akbari-Elphick-Kumar-Pragada-Tang order-inertia conjecture

摘录 / EXCERPT: Let G be a graph of order n, and let $n^+(G)$ denote the number of positive eigenvalues of its adjacency matrix, counted with multiplicity. Akbari-Elphick-Kumar-Pragada-Tang's...

conjecture | open | MathDB 页面 | 原文线索

V2-053213 | MATHDB-374619

Characterization of trees attaining the independent locating-dominating bound

摘录 / EXCERPT: Characterization problem. Characterize those trees T for which

open_problem | open | MathDB 页面 | 原文线索

V2-053214 | MATHDB-374621

Johnson's bounded-degree Eulerian digraph immersion conjecture

摘录 / EXCERPT: Johnson's conjecture. For every $d\geq 1$, the class of Eulerian digraphs of maximum degree $2d$ is well-quasi-ordered by strong immersion.

conjecture | open | MathDB 页面 | 原文线索

V2-053215 | MATHDB-374622

Weak immersion conjecture for Eulerian digraphs

摘录 / EXCERPT: Weak immersion conjecture. The class of Eulerian digraphs is well-quasi-ordered by weak immersion.

conjecture | open | MathDB 页面 | 原文线索

V2-053216 | MATHDB-374623

Treewidth-bounded Eulerian digraph immersion conjecture

摘录 / EXCERPT: Treewidth-bounded immersion conjecture. The class of Eulerian digraphs of treewidth at most k is well-quasi-ordered by immersion.

conjecture | open | MathDB 页面 | 原文线索

V2-053217 | MATHDB-374643

Hoàng's 3-divisibility conjecture for even-hole-free graphs

摘录 / EXCERPT: Let a hole be a chordless cycle of length at least four; it is even if its length is even. For an integer $k\geqslant 2$, a graph G with at least one edge is k-divisible if, fo...

conjecture | open | MathDB 页面 | 原文线索

V2-053218 | MATHDB-374646

The N-regularity conjecture for mutable lattices

摘录 / EXCERPT: N-regularity conjecture. Every vertex of the underlying undirected graph of $\operatorname{Hasse}(L)$ has degree N; equivalently, the graph is N-regular.

conjecture | open | MathDB 页面 | 原文线索

V2-053219 | MATHDB-374652

The strong coarse Menger conjecture

摘录 / EXCERPT: Let G be a finite or infinite graph, and let $X,Y\subseteq V(G)$. An X-Y path is a path with at least one end in X and at least one end in Y. For sets $S,T\subseteq V(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-053220 | MATHDB-374653

Nguyen-Scott-Seymour's weak coarse Menger conjecture

摘录 / EXCERPT: Let G be a finite or infinite graph, and let $X,Y\subseteq V(G)$. An X-Y path is a path with at least one end in X and at least one end in Y. A set $Z\subseteq V(G)$ is...

conjecture | refuted | MathDB 页面 | 原文线索

V2-053221 | MATHDB-374654

Geelen's coarse Gallai conjecture for A-paths

摘录 / EXCERPT: Let G be a finite or infinite graph and let $A\subseteq V(G)$. An A-path is a path in G between two distinct vertices of A. Two subgraphs are at distance at least r when...

conjecture | open | MathDB 页面 | 原文线索

V2-053222 | MATHDB-374656

The half-integral coarse Menger conjecture

摘录 / EXCERPT: Let G be a graph and let $X,Y\subseteq V(G)$. An X-Y path is a path with at least one end in X and at least one end in Y. A set is (a,b)-centered if it is contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-053223 | MATHDB-374679

Conjecture on the possible chromatic thresholds of pairs of graphs

摘录 / EXCERPT: Let $r,s\ge 3$, and for graphs H_1,H_2 let $\delta_\chi(H_1,H_2)$ denote the minimum-degree density above which every graph admitting a red-blue edge-coloring with no red copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-053224 | MATHDB-374680

Pons-Batle's word-encoding conjecture for tree-child networks

摘录 / EXCERPT: Let $\mathcal{TC}_{n,k}$ be the class of tree-child networks with n leaves and k reticulation nodes, and let $\mathcal{C}_{n-1,k}$ be the class of words over the alphabet...

conjecture | open | MathDB 页面 | 原文线索

V2-053225 | MATHDB-374686

Conjectured Betti-diagram shape for edge ideals of generalized Andrásfai graph complements

摘录 / EXCERPT: Let $I({\rm GA}(t,k)')$ be the edge ideal of the complement of the generalized Andrásfai graph ${\rm GA}(t,k)$, and let $\beta_{i,i+j}(I({\rm GA}(t,k)'))$ denote its graded Betti n...

conjecture | open | MathDB 页面 | 原文线索

V2-053226 | MATHDB-374697

The minimum-energy conjecture for normal cycle sets

摘录 / EXCERPT: The minimum-energy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053227 | MATHDB-374698

The low-density normal-set conjecture

摘录 / EXCERPT: Low-density normal-set conjecture. Every plane digraph G admits a normal set of cycles with low density and of size $fvs(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-053228 | MATHDB-374699

Albertson-Berman's planar graph feedback vertex set conjecture

摘录 / EXCERPT: Albertson-Berman's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053229 | MATHDB-374700

Neumann-Lara's acyclic 2-coloring conjecture for planar digraphs

摘录 / EXCERPT: Neumann-Lara's conjecture. Every planar digraph of digirth at least 3 can be vertex-partitioned into two acyclic sets.

conjecture | open | MathDB 页面 | 原文线索

V2-053230 | MATHDB-374708

Eventual sign coherence for quivers with frozen vertices

摘录 / EXCERPT: Let Q be a connected quiver with at least one frozen vertex, and let $\mathbf M$ be a reduced (weakly) balanced and monotone mutation sequence. Write $Q^{(j)}_{\mathbf M}$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-053231 | MATHDB-374717

Circuit characterization of distance reduction for toric ideals of graphs

摘录 / EXCERPT: Let G be a simple graph, let A_G be its incidence matrix, let I_G be the corresponding toric ideal, and let M be a minimal Markov basis for I_G. A Markov basis is distanc...

conjecture | open | MathDB 页面 | 原文线索

V2-053232 | MATHDB-374723

Wu-Dong-Lai conjecture on Laplacian ratios of trees

摘录 / EXCERPT: Let T be a tree on n vertices. For a graph G with no isolated vertices, define its Laplacian ratio by ... where $L(G)$ is the Laplacian matrix, $d(v)$ is the degree of v, and...

conjecture | open | MathDB 页面 | 原文线索

V2-053233 | MATHDB-374735

Kang-Nikiforov-Yuan conjecture on the spectral radius of chromatic hypergraphs

摘录 / EXCERPT: Let G be an r-uniform hypergraph, and let $\binom{n}{r}$ denote the number of r-subsets of an n-vertex set. For $p\geq 1$, let $\lambda^{(p)}(G)$ be the maximum of the poly...

conjecture | open | MathDB 页面 | 原文线索

V2-053234 | MATHDB-374736

Kang-Nikiforov-Yuan's explicit spectral-radius conjecture for chromatic hypergraphs

摘录 / EXCERPT: Let G be a k-chromatic r-graph of order $n>(r-1)k$, where $k\geq 2$ and $r\geq 4$. For $p\geq 1$, let $\lambda^{(p)}(G)$ denote its p-spectral radius. Kang-Nikiforov-Yuan's...

conjecture | open | MathDB 页面 | 原文线索

V2-053235 | MATHDB-374743

The conjecture on the local maxima of the inducibility profile of $K_{2,2,1}$

摘录 / EXCERPT: The conjectured profile of $K_{2,2,1}$. The function is conjectured to satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-053236 | MATHDB-374744

The conjecture on unboundedly many local maxima of inducibility profiles

摘录 / EXCERPT: Let $\mathcal{M}$ be the family of complete multipartite graphs, and let $I_H(x)$ denote the inducibility profile of a graph H. Unbounded-local-maxima conjecture. For every $k>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-053237 | MATHDB-374765

Lv et al.'s generalized Turán conjecture for triangles and expanded paths and cycles

摘录 / EXCERPT: Lv et al.'s conjecture. For $k\ge 4$ and sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-053238 | MATHDB-374766

Spectral support-forest-profile conjecture for trees

摘录 / EXCERPT: Spectral support-forest-profile conjecture. If T_1 and T_2 have the same Laplacian spectrum and the same $(n-2,2)$ support-forest profile, then T_1 and T_2 are isomorphic....

conjecture | open | MathDB 页面 | 原文线索

V2-053239 | MATHDB-374767

Subtree-separation conjecture for the $(n-2,2)$-spectrum of trees

摘录 / EXCERPT: Subtree separation conjecture. The universal support-forest-counting combinations arising from $M_r^{(2)}(T)$, together with the Laplacian spectrum, determine all embedded forest c...

conjecture | open | MathDB 页面 | 原文线索

V2-053240 | MATHDB-374782

Cox-McGinnis-Marchant conjecture on finiteness of accumulation points

摘录 / EXCERPT: Let ${\mathcal H}$ be a non-trivial hereditary property of graphs, and call $p\in[0,1]$ an accumulation point when the edit-distance function of ${\mathcal H}$ is not locally deter...

conjecture | open | MathDB 页面 | 原文线索

V2-053241 | MATHDB-374869

Hypergraph Nash-Williams-Tutte conjecture in literal form

摘录 / EXCERPT: Hypergraph Nash-Williams-Tutte conjecture. For positive integers t and k, every k-weakly-partition-connected hypergraph H on t vertices has a k-distinguishable tree ass...

conjecture | open | MathDB 页面 | 原文线索

V2-053242 | MATHDB-374870

Critical-excess conjecture for distinguishable tree assignments

摘录 / EXCERPT: Critical-excess conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053243 | MATHDB-374875

MSN-integrality conjecture for finite non-commutative CC-rings

摘录 / EXCERPT: Let R be a finite non-commutative CC-ring, meaning that all centralizers of non-central elements of R are commutative, and let Γ_R be its commuting graph. Let MSN-integr...

conjecture | open | MathDB 页面 | 原文线索

V2-053244 | MATHDB-374894

Edge-localized Turán inequality for the signless Laplacian spectral radius

摘录 / EXCERPT: Let G) be a connected graph, let $E(G)$ be its edge set, let $\deg(u)$ denote the degree of a vertex u, let $q(G)$ denote the spectral radius of the signless Laplacian of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-053245 | MATHDB-374904

The half-degree matching-removability conjecture

摘录 / EXCERPT: For $k\ge 1$, a vertex set S in a k-connected graph G is k-removable when $G-S$ remains k-connected; a matching is k-removable when its edge deletion leaves a k-conne...

conjecture | open | MathDB 页面 | 原文线索

V2-053246 | MATHDB-374909

The 2-tone edge coloring conjecture for cubic graphs excluding K_4

摘录 / EXCERPT: Let G be a cubic graph that does not contain K_4, and let $\tau'_2(G)$ denote its 2-tone edge chromatic number, namely the minimum number of colors needed for a 2-tone edge col...

conjecture | open | MathDB 页面 | 原文线索

V2-053247 | MATHDB-374919

DeLaViña-Waller conjecture on the Wiener index of graphs of order $2d+1$

摘录 / EXCERPT: Let G be a finite connected graph, let d be its diameter, let $W(G)$ denote its Wiener index, and let C_{2d+1} denote the cycle on $2d+1$ vertices. DeLaViña-Waller conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-053248 | MATHDB-374968

Friedan's reconstruction conjecture for self-dual vertex operator algebras

摘录 / EXCERPT: Reconstruction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053249 | MATHDB-374971

Compact orbifold uniqueness conjecture for unitary vertex operator algebras

摘录 / EXCERPT: Compact orbifold uniqueness conjecture. The unitary V^G-module V has a unique unitary VOA structure up to isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-053250 | MATHDB-374972

Holomorphic vertex operator algebra unitarity conjecture

摘录 / EXCERPT: Holomorphic VOA unitarity conjecture. Every holomorphic VOA is unitary.

conjecture | open | MathDB 页面 | 原文线索

V2-053251 | MATHDB-374976

Properness of the hierarchy of graph classes $\mathcal{G}_k$

摘录 / EXCERPT: Let $\mathcal{G}_k$ be the graph class defined in the paper for each $k\in\mathbb{N}$. Hierarchy properness conjecture. For each $k\in\mathbb{N}$, the inclusion ... is proper. The co...

conjecture | open | MathDB 页面 | 原文线索

V2-053252 | MATHDB-374981

Cactus-shadow extension conjecture for the necklace formula

摘录 / EXCERPT: Cactus-shadow extension conjecture. For any embedded shadow whose block-adjacency graph is a cactus and whose cyclic blocks are annular in the sense of the necklace definition, the...

conjecture | open | MathDB 页面 | 原文线索

V2-053253 | MATHDB-375018

NF-number formula for dumbbell graphs

摘录 / EXCERPT: Let $n,m\ge 2$ and let $G=B_{n,m}$ be the dumbbell graph. The NF-number conjecture. ... The preceding results determine the first two iterates of the NF-operator for $B_{n,m}$ and mo...

conjecture | open | MathDB 页面 | 原文线索

V2-053254 | MATHDB-375046

The diagonal maximal-covering-number conjecture for cross-intersecting families

摘录 / EXCERPT: Let $m(k,k)$ denote the maximum product of the sizes of two cross-intersecting families with maximal covering number k in the setting of the paper. The diagonal conjecture. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-053255 | MATHDB-375051

Espuny Díaz and Girão's Hamiltonicity threshold conjecture for random clique-factors

摘录 / EXCERPT: For a fixed integer $r\ge 2$, let K_r denote the complete graph on r vertices, and let $\alpha^*(K_r)$ be the minimum-degree threshold of Hamiltonicity for perturbation by a un...

conjecture | open | MathDB 页面 | 原文线索

V2-053256 | MATHDB-375054

Balogh-Clemen-Lidický's exact $cell_2$-norm Turán conjecture for K_5^3

摘录 / EXCERPT: Let $\mathcal{H}$ be an r-graph, and let $d_{\mathcal{H}}(e)$ denote the codegree of an $(r-1)$-set e. Define ... For an r-graph $\mathcal{H}$, let ... Given a balanced partition...

conjecture | open | MathDB 页面 | 原文线索

V2-053257 | MATHDB-375055

Frankl-Gryaznov-Talebanfard's clique-counting conjecture for hypergraphs

摘录 / EXCERPT: Frankl-Gryaznov-Talebanfard's conjecture. If $\mathcal{H}$ contains no clique of size $ell+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-053258 | MATHDB-375061

Conjecture on unique Hamiltonian cycles in P_4+sP_1-free graphs

摘录 / EXCERPT: Conjecture. There exists a constant n_s depending on s such that G is not uniquely Hamiltonian whenever $n\geq n_s$.

conjecture | open | MathDB 页面 | 原文线索

V2-053259 | MATHDB-375062

Extremal number of 2-connected graphs avoiding cycles of length 2 modulo 2k

摘录 / EXCERPT: Conjecture on the 2-connected extremal number.

conjecture | open | MathDB 页面 | 原文线索

V2-053260 | MATHDB-375067

The asymptotic extremal-number conjecture for $K_{t,t}$ in $K_{2,t+1}$-free graphs

摘录 / EXCERPT: Asymptotic extremal-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053261 | MATHDB-375084

The reflection-permutation conjecture for the free energy of random tensor networks

摘录 / EXCERPT: Reflection-permutation conjecture. The reflection permutation π gives the true minimum for all (q,n), namely

conjecture | open | MathDB 页面 | 原文线索

V2-053262 | MATHDB-375095

Orbifold rationality conjecture for strongly rational vertex operator algebras

摘录 / EXCERPT: Orbifold rationality conjecture. The conformal subalgebra V^G is also strongly rational.

conjecture | open | MathDB 页面 | 原文线索

V2-053263 | MATHDB-375102

Cycle-space spanning conjecture for highly connected graphs

摘录 / EXCERPT: Let G be an n-vertex graph, where n is odd. Write $\kappa(G)$ for its vertex-connectivity, $\alpha(G)$ for its independence number, $\mathcal{C}(G)$ for its cycle space, and...

conjecture | open | MathDB 页面 | 原文线索

V2-053264 | MATHDB-375116

Erdős-Nešetřil conjecture on the strong chromatic index

摘录 / EXCERPT: Erdős-Nešetřil conjecture. The strong chromatic index of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053265 | MATHDB-375117

Planar D-chromatic index conjecture

摘录 / EXCERPT: Planar D-chromatic index conjecture. For every such graph,

conjecture | open | MathDB 页面 | 原文线索

V2-053266 | MATHDB-375129

Dyadic gap-label conjecture for the KNS measure

摘录 / EXCERPT: Let the KNS measure be the spectral measure considered in the paper, and let a gap label be the value of its spectral counting function on a gap interval. Dyadic gap-label conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-053267 | MATHDB-375177

Generalized Turán number exponent conjecture

摘录 / EXCERPT: Generalized Turán number exponent conjecture. There exists t_0 such that, for all $t\geq t_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-053268 | MATHDB-375182

Partition graphs maximize the remaining perfect matchings after one-user de-anonymization

摘录 / EXCERPT: Partition-graph de-anonymization conjecture. If a single user is de-anonymized by an optimizing attacker from G, the number of remaining perfect matchings is at most the number r...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053269 | MATHDB-375212

Non-negative discrete Bakry-Émery curvature implies polynomial growth

摘录 / EXCERPT: Non-negative curvature implies polynomial growth. If G has bounded degrees and satisfies $\mathrm{CD}(0,\infty)$, then G has polynomial growth. More precisely,

conjecture | open | MathDB 页面 | 原文线索

V2-053270 | MATHDB-375213

Regular-graph self-improvement conjecture for the Bakry-Émery condition

摘录 / EXCERPT: Regular-graph self-improvement conjecture. Any d-regular graph satisfying $\mathrm{CD}(0,\infty)$ satisfies $\mathrm{CD}(0,d)$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053271 | MATHDB-375220

Nash-Williams' conjecture for triangle decompositions

摘录 / EXCERPT: Nash-Williams' conjecture. If G is K_3-divisible and

conjecture | open | MathDB 页面 | 原文线索

V2-053272 | MATHDB-375221

Fractional Nash-Williams' conjecture

摘录 / EXCERPT: Fractional Nash-Williams' conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053273 | MATHDB-375223

Approximate Oberwolfach Nash-Williams' conjecture

摘录 / EXCERPT: Approximate Oberwolfach Nash-Williams' conjecture. For every $\varepsilon>0$, for all sufficiently large n, if

conjecture | open | MathDB 页面 | 原文线索

V2-053274 | MATHDB-375235

The unbalanced bipartite measurable edge-coloring conjecture

摘录 / EXCERPT: Unbalanced bipartite measurable edge-coloring conjecture. For every probability measure μ on V,

conjecture | open | MathDB 页面 | 原文线索

V2-053275 | MATHDB-375236

The nonadjacent maximum-degree measurable edge-coloring conjecture

摘录 / EXCERPT: Nonadjacent maximum-degree measurable edge-coloring conjecture. For every probability measure μ on V,

conjecture | open | MathDB 页面 | 原文线索

V2-053276 | MATHDB-375238

Aouchiche-Hansen conjecture on the second adjacency eigenvalue

摘录 / EXCERPT: Aouchiche-Hansen conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-053277 | MATHDB-375251

Average degree conjecture for critical multigraphs

摘录 / EXCERPT: Let G be a finite, undirected, loopless multigraph. Write $\Delta=\Delta(G)$ for its maximum degree, $\chi'(G)$ for its chromatic index, and $\overline{d}(G)$ for its average deg...

conjecture | open | MathDB 页面 | 原文线索

V2-053278 | MATHDB-375252

Goldberg's density conjecture

摘录 / EXCERPT: Goldberg's density conjecture. For any graph G, if $\Gamma(G)\le\Delta(G)-1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-053279 | MATHDB-375260

The Nordhaus-Gaddum upper-bound conjecture for the Cheeger constant

摘录 / EXCERPT: Cheeger-constant Nordhaus-Gaddum conjecture. If $G\neq S_n$, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053280 | MATHDB-375261

The normalized algebraic-connectivity Nordhaus-Gaddum conjecture

摘录 / EXCERPT: Normalized algebraic-connectivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053281 | MATHDB-375262

The higher-outerplanarity obstruction conjecture for completely independent spanning trees

摘录 / EXCERPT: Higher-outerplanarity obstruction conjecture. For any $k\geq 4$, there exists a 3-connected k-outerplanar triangulated disc with no completely independent spanning trees.

conjecture | open | MathDB 页面 | 原文线索

V2-053282 | MATHDB-375299

Zhu's rationality conjecture on C_2-cofiniteness

摘录 / EXCERPT: Zhu's conjecture. If V is rational, then R_V is finite-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-053283 | MATHDB-375306

Boundary-vertex conjecture for minimal covering bodies

摘录 / EXCERPT: Boundary-vertex conjecture. The vertices of Q lie on the boundary of $K-\mathbf{x}$.

conjecture | open | MathDB 页面 | 原文线索

V2-053284 | MATHDB-375345

Subquartic saturation-number conjecture

摘录 / EXCERPT: Let G be a connected simple graph, and let $\mu^{*}(G)$ denote its saturation number, the minimum cardinality of a maximal matching. Let ... be its harmonic index. A graph is subqu...

conjecture | open | MathDB 页面 | 原文线索

V2-053285 | MATHDB-375346

Exact spectral threshold conjecture for perfect matchings in balanced 3-partite 3-graphs

摘录 / EXCERPT: Exact spectral threshold conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053286 | MATHDB-375350

Hilton's cycle multiplicity conjecture for dense hamiltonian graphs

摘录 / EXCERPT: Hilton's conjecture. For every integer ℓ with $3\le \ell\le n$,

conjecture | open | MathDB 页面 | 原文线索

V2-053287 | MATHDB-375377

The exact asymptotic bound conjecture for small q-kernels

摘录 / EXCERPT: Exact asymptotic bound conjecture. For all $\delta\geq 1$ and $q\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-053288 | MATHDB-375383

Fu-Peng-Zhang's coloring conjecture for finitary upho posets

摘录 / EXCERPT: Fu-Peng-Zhang's coloring conjecture. Every finitary upho poset, in particular every finite-type $\mathbb{N}$-graded upho poset, has a coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-053289 | MATHDB-375384

Fu-Peng-Zhang's multiplicability conjecture for finitary upho posets

摘录 / EXCERPT: Fu-Peng-Zhang's multiplicability conjecture. Every finitary upho poset, in particular every finite-type $\mathbb{N}$-graded upho poset, is multiplicable.

conjecture | open | MathDB 页面 | 原文线索

V2-053290 | MATHDB-375387

The coarse Menger conjecture for graphs and geodesic metric spaces

摘录 / EXCERPT: Coarse Menger conjecture. There is a function $g:\mathbb{N}^2\to\mathbb{N}$ such that for every $k,d\in\mathbb{N}$, every graph or geodesic metric space G, and every two sets...

conjecture | open | MathDB 页面 | 原文线索

V2-053291 | MATHDB-375388

Georgakopoulos-Papasoglu fat minor conjecture

摘录 / EXCERPT: Fat minor conjecture. For every graph X there exists a function $f:\mathbb{N}\to\mathbb{N}^2$ such that, for every graph G and $K\in\mathbb{N}$, if G does not contain X as...

conjecture | open | MathDB 页面 | 原文线索

V2-053292 | MATHDB-375389

The bounded-cycle-space fat minor conjecture

摘录 / EXCERPT: Bounded-cycle-space fat minor conjecture. For every graph X there exists a function $f:\mathbb{N}^2\to\mathbb{N}^2$ such that, for every graph G whose cycle space is generated...

conjecture | open | MathDB 页面 | 原文线索

V2-053293 | MATHDB-375392

Havet-Horsch-Rambaud inversion-diameter conjecture

摘录 / EXCERPT: Havet-Horsch-Rambaud's conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-053294 | MATHDB-375393

Havet-Horsch-Rambaud lexicographic-product conjecture for inversion diameter

摘录 / EXCERPT: Havet-Horsch-Rambaud's lexicographic-product conjecture. For every graph G and every positive integer t,

conjecture | open | MathDB 页面 | 原文线索

V2-053295 | MATHDB-375395

Folklore conjecture on pseudorandom hypergraphs and the shattering threshold

摘录 / EXCERPT: A pseudorandom class of hypergraphs is a hereditary class whose sparse random model contains, with high probability, an induced subhypergraph on at least $(1-o_d(1))n$ vertices bel...

open_problem | open | MathDB 页面 | 原文线索

V2-053296 | MATHDB-375396

Verstraëte-Wilson conjecture on independent sets in linear hypergraphs

摘录 / EXCERPT: Let $k\geq 3$, and let $a_k>0$ satisfy $a_k=1-o_k(1)$ as $k\to\infty$. A k-uniform hypergraph is linear if it contains no 2-cycle. Verstraëte-Wilson conjecture. Every n-verte...

conjecture | open | MathDB 页面 | 原文线索

V2-053297 | MATHDB-375397

Shattering-threshold conjecture for independent sets in linear hypergraphs

摘录 / EXCERPT: Let $k\geq 3$ be an integer. A k-uniform hypergraph is linear if it contains no 2-cycle, and let $\alpha({\mathcal H})$ denote its independence number. Shattering-threshold con...

conjecture | open | MathDB 页面 | 原文线索

V2-053298 | MATHDB-375401

The domination-isolation ratio conjecture for cubic graphs

摘录 / EXCERPT: Let G be a cubic graph, with domination number $\gamma(G)$ and isolation number $\iota(G)$. Domination-isolation ratio conjecture. For every cubic graph G, ... The conjecture wou...

conjecture | open | MathDB 页面 | 原文线索

V2-053299 | MATHDB-375407

Alon's asymptotic subgraph-count conjecture

摘录 / EXCERPT: Alon's asymptotic subgraph-count conjecture. There is a positive constant $b(H)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-053300 | MATHDB-375408

Alon's star-forest extremal-graph conjecture

摘录 / EXCERPT: Alon's star-forest extremal-graph conjecture. For every integer $m>0$, or at least for all sufficiently large m, there exists a star forest G_m with $e(G_m)=m$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-053301 | MATHDB-375413

Winkler-Zuckerman blanket-time conjecture for finite graphs

摘录 / EXCERPT: Let G be a finite graph with vertex set V. For a starting vertex $v\in V$, write $\mathbb{E}_v$ for expectation for the simple random walk started at v, and let the...

conjecture | solved | MathDB 页面 | 原文线索

V2-053302 | MATHDB-375415

Spanning-tree formula conjecture for the Fitting ideal of a graph

摘录 / EXCERPT: Let $\Gamma=(\mathsf{V},\mathsf{E})$ be a connected graph, let $\overline{\mathsf{E}}$ denote the relevant complementary edge set, and let n be the number of vertices. Define ... w...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053303 | MATHDB-375419

Hamiltonian path pattern conjecture for labelled graph notations

摘录 / EXCERPT: Hamiltonian path pattern conjecture. In order for a Hamiltonian path to exist on $I[G]$, it is necessary and sufficient that $I[G]$ satisfy one of the four possible patterns:

conjecture | open | MathDB 页面 | 原文线索

V2-053304 | MATHDB-375422

Caoduro-Khodamoradi-Paat-Shepherd simplex exchange conjecture

摘录 / EXCERPT: Let S and T be d-dimensional simplices whose interiors contain the origin. A vertex exchange replaces one vertex while producing another d-dimensional simplex whose interio...

conjecture | open | MathDB 页面 | 原文线索

V2-053305 | MATHDB-375423

PageRank fairness conjecture for homophilic and heterophilic two-colored PAMs

摘录 / EXCERPT: Consider the two-colored PAM with $m_1=m_2=m$, attachment kernel ... and minority-vertex probability μ_1. PageRank fairness conjecture. For every sufficiently small $c>0$, there...

conjecture | open | MathDB 页面 | 原文线索

V2-053306 | MATHDB-375428

Monotonicity conjecture for the largest upper-Laplacian eigenvalue

摘录 / EXCERPT: Let X be a finite simplicial complex, let $csigma_{\max}(\partial_k)$ denote the largest singular value of its k-th boundary operator, and let L_k^{up} denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-053307 | MATHDB-375477

The full-degree vertex necessity conjecture for saturation of joins

摘录 / EXCERPT: Full-degree vertex necessity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053308 | MATHDB-375480

Zaks's conjecture on perfect matchings in connected graphs

摘录 / EXCERPT: Let $k \ge 3$ and $n \ge 2k$ be integers, and let G be a matchable k-connected graph with n vertices. Zaks's conjecture. Every such graph satisfies ... moreover, infinitely man...

conjecture | refuted | MathDB 页面 | 原文线索

V2-053309 | MATHDB-375481

Minimum perfect matchings conjecture for planar 4-connected triangulations

摘录 / EXCERPT: A double wheel is the planar triangulation obtained by joining two vertices to every vertex of a cycle. A planar triangulation is 4-connected if it has no separating set of at most...

conjecture | open | MathDB 页面 | 原文线索

V2-053310 | MATHDB-375514

Day and Sarkar's sparse threshold graphon conjecture

摘录 / EXCERPT: Let H be a fixed graph without isolated vertices. For $0<\beta<1$, let $\mathcal M_H(\beta)$ be the supremum of $t(H,W)$ over graphons W with $t(K_2,W)\le\beta$. For...

conjecture | refuted | MathDB 页面 | 原文线索

V2-053311 | MATHDB-375515

Gerbner-Nagy-Patkós-Vizer quasi-complete bipartite extremal conjecture

摘录 / EXCERPT: Let H be a fixed bipartite graph. For positive integers n,m, let $N_{\mathrm{bip}}(n,m,H)$ be the maximum number of unlabelled copies of H over bipartite graphs G with...

conjecture | refuted | MathDB 页面 | 原文线索

V2-053312 | MATHDB-375530

Cashman-Kelley’s odd-cycle Cayley-Turán conjecture

摘录 / EXCERPT: Cashman-Kelley’s conjecture. For $p>2\ell+1$, the Cayley-Turán number of the odd cycle $C_{2\ell+1}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053313 | MATHDB-375536

Tutte's three nowhere-zero flow conjectures

摘录 / EXCERPT: Tutte's three flow conjectures. The following three statements all hold:

conjecture | open | MathDB 页面 | 原文线索

V2-053314 | MATHDB-375537

Esperet et al.'s flow reconfiguration equivalence conjecture

摘录 / EXCERPT: Esperet et al.'s flow reconfiguration conjecture. For each graph G and each positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-053315 | MATHDB-375542

The deletion-normality conjecture for non-complete graphs

摘录 / EXCERPT: For a graph H, an H-deletion-saturated graph is an H-free graph G with $E(G)\neq\varnothing$ such that, for every $e\in E(G)$, the graph $G-e$ is not H-free. A graph H...

conjecture | open | MathDB 页面 | 原文线索

V2-053316 | MATHDB-375561

The upper bound on the strong periodicity of connected graphs under the 2-distance operator

摘录 / EXCERPT: Upper-bound conjecture. Any connected graph $G=(V,E)$ has

conjecture | open | MathDB 页面 | 原文线索

V2-053317 | MATHDB-375581

Duval-Reiner conjecture for simplicial-complex Laplacians

摘录 / EXCERPT: Duval-Reiner conjecture. The eigenvalue sequence is majorized by the conjugate vertex degree sequence:

conjecture | open | MathDB 页面 | 原文线索

V2-053318 | MATHDB-375582

Mubayi's supersaturation conjecture for stable non-r-partite hypergraphs

摘录 / EXCERPT: Mubayi's conjecture. If $\mathcal F$ is a stable non-r-partite r-graph, then, for every positive integer q and all sufficiently large n, every n-vertex r-graph with...

conjecture | open | MathDB 页面 | 原文线索

V2-053319 | MATHDB-375599

Erde-Kang-Lehner-Mohar-Schmid conjecture for cop numbers of uniform hypergraphs

摘录 / EXCERPT: Erde-Kang-Lehner-Mohar-Schmid conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053320 | MATHDB-375611

Negami's Planar Cover Conjecture

摘录 / EXCERPT: Let G be a connected finite simple graph. A finite planar cover of G is a finite graph that covers G in the graph-theoretic sense, with the covering graph planar. Negami's Pl...

conjecture | open | MathDB 页面 | 原文线索

V2-053321 | MATHDB-375616

Asymptotic independence-ratio conjecture for finite unit-distance graphs

摘录 / EXCERPT: Let $f(n)$ denote the minimum independence number among n-vertex unit-distance graphs in the plane, and let $m_1(\mathbb R^2)$ be the supremum of the upper densities of measurabl...

conjecture | open | MathDB 页面 | 原文线索

V2-053322 | MATHDB-375644

Balla's conjecture on equiangular lines

摘录 / EXCERPT: Balla's conjecture. For any $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-053323 | MATHDB-375645

Balla's conjecture for reciprocal odd integers

摘录 / EXCERPT: Balla's conjecture for $1/\alpha$ is odd. For every integer $k\geq 1$ and every $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-053324 | MATHDB-375646

TxGraffiti's annihilation-number bound for connected graphs

摘录 / EXCERPT: TxGraffiti's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-053325 | MATHDB-375670

Alon-Wei asymptotic nearly-uniform degree-distribution conjecture

摘录 / EXCERPT: Let G be a d-regular graph on n vertices, let H be a spanning subgraph of G, and suppose that $d=o(n)$. For each $k\in\{0,1,\dots,d\}$, write $\#\{v\in V(H):d_H(v)=k\}$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-053326 | MATHDB-375673

Tang-Liu-Wang positive p-energy conjecture

摘录 / EXCERPT: For a graph G, define its positive adjacency p-energy by ... Let $p\ge2$ be a real number, and let G be a connected graph with n vertices. Tang-Liu-Wang's positive p-energy...

conjecture | open | MathDB 页面 | 原文线索

V2-053327 | MATHDB-375674

Jia-Fan-Liu's spectral-radius conjecture for fractional ID-[a,b]-factor-critical graphs

摘录 / EXCERPT: Jia-Fan-Liu's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053328 | MATHDB-375692

Häggkvist-Chetwynd line graph list-coloring conjecture

摘录 / EXCERPT: For a graph G, its line graph $L(G)$ has one vertex for each edge of G, with two vertices adjacent exactly when the corresponding edges share an endpoint. Let χ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053329 | MATHDB-375693

List Total Coloring Conjecture

摘录 / EXCERPT: For a multigraph G, a total coloring colors vertices and edges so that adjacent vertices, incident edges, and incident vertex-edge pairs receive distinct colors. Let $\chi”(G)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-053330 | MATHDB-375696

Borowiecki-Jozef Cartesian product list-coloring conjecture

摘录 / EXCERPT: For graphs G_1 and G_2, their Cartesian product $G_1\Box G_2$ has vertex set $V(G_1)\times V(G_2)$, with (u,v) adjacent to (u',v') when either $u=u'$ and $vv'\in E(G_2)$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-053331 | MATHDB-375699

Prowse-Woodall $(a:b)$-choosability equivalence conjecture

摘录 / EXCERPT: Let L be an a-list assignment on a graph G. The graph is $(a:b)$-choosable when every such assignment admits a choice of a b-element sublist at each vertex whose selected l...

conjecture | open | MathDB 页面 | 原文线索

V2-053332 | MATHDB-375704

Injective edge-function conjecture for length-area transferal

摘录 / EXCERPT: Let $G=(\mathcal{V},\mathcal{E})$ be a simplicial complex. A triangle edge choice function assigns to each triangle of G one of its edges, and it is injective when distinct trian...

conjecture | open | MathDB 页面 | 原文线索

V2-053333 | MATHDB-375710

Berikkyzy-Hogenson-Kirsch-McDonald conjecture on extremal star counts

摘录 / EXCERPT: Berikkyzy-Hogenson-Kirsch-McDonald conjecture. If $t\leq \left\lfloor\frac{n+\ell}{2}\right\rfloor$, then, for sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-053334 | MATHDB-375716

Fang-Lin-Zhai spectral supersaturation conjecture for color-critical graphs

摘录 / EXCERPT: Let F be a color-critical graph with order $f=|F|$ and chromatic number $\chi(F)=r+1\geqslant4$. Write $N_F(G)$ for the number of copies of F in an m-edge graph G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-053335 | MATHDB-375722

Van der Holst's classification of the obstructions to Colin de Verdière invariant at most 5

摘录 / EXCERPT: Van der Holst's classification conjecture. The graphs in the K_7 and $K_{3,3,1,1}$ families are the obstructions to $\mu\leq 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-053336 | MATHDB-375727

Girao, Kittipassorn, and Narayanan's long-cycle conjecture

摘录 / EXCERPT: Let G be a simple graph with n vertices and minimum degree $\delta(G)\geq 3$, and suppose that G contains a Hamiltonian cycle. Girao, Kittipassorn, and Narayanan's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-053337 | MATHDB-375745

Dirac's 1-Factorization Conjecture

摘录 / EXCERPT: Let G be a graph of even order, and let k be an integer. A graph is 1-factorable if its edges can be partitioned into 1-factors, equivalently if its chromatic index equals its...

conjecture | open | MathDB 页面 | 原文线索

V2-053338 | MATHDB-375755

Cambie-Cames van Batenburg-Joannis de Verclos-Kang conjecture for $h_3(\Delta)$

摘录 / EXCERPT: For positive integers Δ and t, let $h_t(\Delta)$ be the smallest integer such that every graph G with at least $h_t(\Delta)$ edges and maximum degree...

conjecture | open | MathDB 页面 | 原文线索

V2-053339 | MATHDB-375781

Strong Roberson Conjecture on homomorphism distinguishing closed graph classes

摘录 / EXCERPT: Let $\mathcal{F}$ be a class of graphs. It is homomorphism distinguishing closed if it is maximal among the classes defining its homomorphism indistinguishability relation. A graph...

conjecture | open | MathDB 页面 | 原文线索

V2-053340 | MATHDB-375782

Weak Roberson Conjecture on non-isomorphic homomorphism indistinguishability

摘录 / EXCERPT: Let $\mathcal{F}$ be a class of graphs, and let $\equiv_{\mathcal{F}}$ denote the relation on graphs defined by equality of homomorphism counts from every graph in $\mathcal{F}$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-053341 | MATHDB-375784

Alon-Kim conjecture on the chromatic index of simple hypergraphs

摘录 / EXCERPT: Alon-Kim conjecture. For every $k\ge t\ge 1$ and $\varepsilon>0$, there exists D_0 such that, for every $D\ge D_0$, every k-uniform, t-simple hypergraph with maximum degree a...

conjecture | open | MathDB 页面 | 原文线索

V2-053342 | MATHDB-375789

The cycle extremal conjecture for perfect edge dominating sets

摘录 / EXCERPT: Cycle extremal conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-053343 | MATHDB-375790

The extremal graph classification conjecture for perfect edge dominating sets

摘录 / EXCERPT: Extremal graph classification conjecture. Every such extremal graph satisfies

open_problem | open | MathDB 页面 | 原文线索

V2-053344 | MATHDB-375796

The affine upper-bound conjecture for sequential cable width

摘录 / EXCERPT: Let G be a graph, let $\mathrm{lrw}(G)$ denote its linear rank-width, and let $\mathrm{lsfw}(G)$ denote its sequential cable width. Affine upper-bound conjecture. For every graph...

conjecture | open | MathDB 页面 | 原文线索

V2-053345 | MATHDB-375797

Mullin's root-of-unity conjecture for uniform mixing

摘录 / EXCERPT: Let X be a graph, and suppose that X admits uniform mixing at time t. Mullin's conjecture. Then e^{it} is a root of unity. This conjecture concerns an arithmetic constraint...

conjecture | open | MathDB 页面 | 原文线索

V2-053346 | MATHDB-375815

Depth formula conjecture for squarefree powers of wheel edge ideals

摘录 / EXCERPT: Let $\mathcal W_n$ be the wheel graph on $n+1$ vertices, let $I(\mathcal W_n)$ be its edge ideal in $R=\mathbbk[x_1,\ldots,x_n,v]$, and let $\nu(\mathcal W_n)$ denote its matching...

conjecture | open | MathDB 页面 | 原文线索

V2-053347 | MATHDB-375820

The $(2,0)$-tight spanning subgraph characterization of rigidity in S_g

摘录 / EXCERPT: Let $g\geq 2$, let S_g be the closed orientable surface of genus g, and let a simple graph be called rigid in S_g when it has a well-positioned realisation whose associated g...

conjecture | open | MathDB 页面 | 原文线索

V2-053348 | MATHDB-375825

The wedge-of-spheres conjecture for path-free complexes of complete multipartite graphs

摘录 / EXCERPT: Wedge-of-spheres conjecture. The complex $\operatorname{PF}_t(G)$ is homotopy equivalent to a wedge of spheres.

conjecture | open | MathDB 页面 | 原文线索

V2-053349 | MATHDB-375829

Lew-Noy-Plaza-Eberhardt conjecture on the second largest eigenvalue of normalized complete frameworks

摘录 / EXCERPT: Let $d\geq 3$, and let $p:[n]\to\mathbb{R}^d$ satisfy ... for all $i\in[n]$ and ... Assume that the image of p has size at least 3. Lew-Noy-Plaza-Eberhardt's conjecture. The second...

conjecture | open | MathDB 页面 | 原文线索

V2-053350 | MATHDB-375837

Rationality conjecture for simple Virasoro VOAs in positive characteristic

摘录 / EXCERPT: Let $\mathbb{F}$ be a field of characteristic $p>2$, let $c\in\mathbb{F}$, and let $L_{\operatorname{Vir}}(c,0)_{\mathbb{F}}$ denote the simple quotient Virasoro vertex operator al...

conjecture | open | MathDB 页面 | 原文线索

V2-053351 | MATHDB-375842

Bucić-Hendrey-Mohar-Steiner-Yepremyan linear perimeter-gap conjecture

摘录 / EXCERPT: Let the perimeter gap of a digraph be the difference between its number of vertices and the length of its longest directed cycle. Bucić-Hendrey-Mohar-Steiner-Yepremyan's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-053352 | MATHDB-375850

The $7/5$ cycle-cover conjecture for bridgeless graphs

摘录 / EXCERPT: Let G be a bridgeless graph, not necessarily cubic, and let E denote its edge set. A cycle cover here is a list of cycles in which every edge of G lies on at least one cycle....

conjecture | open | MathDB 页面 | 原文线索

V2-053353 | MATHDB-375853

Optimality conjecture for tree-suspension transfer functions

摘录 / EXCERPT: Tree-suspension optimality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-053354 | MATHDB-375860

Dihedral Ramsey conjecture for alternating paths and start-central stars

摘录 / EXCERPT: Let P^alt_a be an alternating path, let S^sc_b be a start-central star, and let R_dih denote the corresponding dihedral Ramsey number. Dihedral Ram...

conjecture | open | MathDB 页面 | 原文线索

V2-053355 | MATHDB-375861

Dihedral Ramsey conjecture for alternating and monotone paths

摘录 / EXCERPT: Let P^alt_a be an alternating path, let P^mon_b be a monotone path, and let R_dih denote the corresponding dihedral Ramsey number. Dihedral Ramsey...

conjecture | open | MathDB 页面 | 原文线索

V2-053356 | MATHDB-375863

Erdős-Pósa-type conjecture for cycles of distinct lengths

摘录 / EXCERPT: For a graph G, let $\Lambda(G)=\{\ell\in\mathbb{N}\mid G\text{ contains a cycle of length }\ell\}$, and let $G-X$ denote the graph obtained by deleting the vertices in X. Disti...

conjecture | open | MathDB 页面 | 原文线索

V2-053357 | MATHDB-375864

Half-integral Erdős-Pósa conjecture for directed cycles of distinct lengths

摘录 / EXCERPT: For a digraph D, a set of directed cycles has distinct lengths when no two of its cycles have the same length. Half-integral distinct-length directed-cycle conjecture. For every...

open_problem | open | MathDB 页面 | 原文线索

V2-053358 | MATHDB-375865

Lichiardopol's minimum out-degree conjecture for directed cycles of distinct lengths

摘录 / EXCERPT: For a digraph D, its minimum out-degree is the minimum number of outgoing edges over all vertices of D. Lichiardopol's conjecture. For every $k\geq1$, there exists an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-053359 | MATHDB-375867

Kawarabayashi-Picasarri-Arrieta dichromatic bound conjecture

摘录 / EXCERPT: Let D be a digraph. Define ... and let $\overset{\text{tiny}}{\leftrightarrow}{\omega}(D)$ be the largest size of a biclique in D. Let $\vec{\chi}(D)$ denote the dichromatic numb...

conjecture | open | MathDB 页面 | 原文线索

V2-053360 | MATHDB-375877

The Feedback-Exponent Conjecture for bounded minimum feedback vertex number

摘录 / EXCERPT: Let $k_0\geq 1$ be an integer. For a graph H, write $\operatorname{ex}(n,H)$ for its Turán number, and call a rational number $r\in[1,2]$ realizable as a Turán exponent if there...

conjecture | open | MathDB 页面 | 原文线索

V2-053361 | MATHDB-375879

Erdős Matching Conjecture for uniform hypergraphs

摘录 / EXCERPT: Let M_s^r be a matching of size s in an r-uniform hypergraph, and let $\operatorname{ex}_r(n,M_s^r)$ denote the maximum number of edges in an n-vertex r-uniform hypergrap...

conjecture | open | MathDB 页面 | 原文线索

V2-053362 | MATHDB-375881

Buchanan et al.'s even-order conjecture for odd covers

摘录 / EXCERPT: Buchanan et al.'s conjecture. For such k,

conjecture | solved | MathDB 页面 | 原文线索

V2-053363 | MATHDB-375883

Path-maximizes analytic spread among trees

摘录 / EXCERPT: Let T_1 and T_2 be trees on n and m vertices, respectively, and fix $i\in V(T_1)$ and $j\in V(T_2)$. For a positive integer l, define G_1 as the tree on $n+m+l$ vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-053364 | MATHDB-375894

Bubeck-Li-Nagaraj robustness law for arbitrary-weight two-layer networks

摘录 / EXCERPT: Bubeck-Li-Nagaraj conjecture. The stated lower bound should hold for every Lipschitz activation and arbitrary weights, with no restriction on their magnitudes. The conjecture quant...

conjecture | open | MathDB 页面 | 原文线索

V2-053365 | MATHDB-375900

The exponential convergence conjecture for triangle-free unit-distance digraphs

摘录 / EXCERPT: Exponential convergence conjecture. There exist absolute constants $c>0$ and $d_0\in\mathbb{N}$ such that, for all $d\geqslant d_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-053366 | MATHDB-375902

Hafidh's sharp tree multiset dimension bound

摘录 / EXCERPT: Let T be a tree with order n and diameter $\operatorname{diam}(T)$, and let $md(T)$ denote its multiset dimension. Hafidh's conjecture. If $md(T)<\infty$, then ... and this bound...

conjecture | open | MathDB 页面 | 原文线索

V2-053367 | MATHDB-375907

Block Khovanskii bound for multi-layer Pfaffian neural networks

摘录 / EXCERPT: Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be the output of a fully connected neural network with ℓ hidden layers of widths $n_1,\ldots,n_\ell$ and a Pfaffian activation functi...

conjecture | open | MathDB 页面 | 原文线索

V2-053368 | MATHDB-375909

Lew's matching-number conjecture for Laplacian eigenvalue sums

摘录 / EXCERPT: Let G be a finite simple graph with n non-isolated vertices. Let $L(G)$ be its Laplacian matrix, let $\lambda_1(G)\ge\cdots\ge\lambda_n(G)$ be the Laplacian eigenvalues, and wr...

conjecture | solved | MathDB 页面 | 原文线索

V2-053369 | MATHDB-375921

Vertex bound conjecture for primitive polytopes

摘录 / EXCERPT: Let P be a primitive polytope in $\mathbb{E}^d$, where $d\geq 4$. Primitive-polytope vertex conjecture. The polytope P has at most 2^d vertices, and it has fewer than 2^d v...

conjecture | open | MathDB 页面 | 原文线索

V2-053370 | MATHDB-375922

Spiro's bipartite quasikernel improvement conjecture

摘录 / EXCERPT: Let D be a source-free bipartite digraph, meaning that every vertex has a nonempty set of external in-neighbors. A quasikernel of D is an independent set $Q\subseteq V(D)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-053371 | MATHDB-375923

Spiro's disjoint generalized quasikernel conjecture

摘录 / EXCERPT: Let D be a digraph. For an integer $s\ge 1$, call a set $S\subseteq V(D)$ an s-source set if ... where $N^-(S)$ is the set of external in-neighbors of S. For an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-053372 | MATHDB-375924

Erdős-Hajnal conjecture for hereditary graph classes

摘录 / EXCERPT: Erdős-Hajnal conjecture. Every proper hereditary class of graphs has the Erdős-Hajnal property.

conjecture | open | MathDB 页面 | 原文线索

V2-053373 | MATHDB-375932

Kao's conjecture on Hamiltonicity of Cartesian products with odd paths

摘录 / EXCERPT: Kao's conjecture. If $n\geq 4\Delta(G)-2$ and $G\Box P_n$ is balanced bipartite, then $G\Box P_n$ is hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-053374 | MATHDB-375944

Kenyon-Wilson's response-matrix characterization conjecture for lower-dimensional electrical-network cells

摘录 / EXCERPT: Kenyon-Wilson's conjecture. The matrix M is the response matrix of a circular electrical network belonging to a cell of co-dimension r in the poset EP_n if and only if the fi...

conjecture | open | MathDB 页面 | 原文线索

V2-053375 | MATHDB-375945

Bhargava-de Pascale-Koenig conjecture on Sylow subgroups of random bipartite sandpile groups

摘录 / EXCERPT: Let $n_1\geq n_2$ be positive integers, let $0<u<1$, and let $G(n_1,n_2,u)$ be the Erdős-Rényi random bipartite graph whose bipartition has vertex sets of sizes n_1 and n_2, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-053376 | MATHDB-375960

Exact dimension formula for vertex operator algebra bundles of general signature

摘录 / EXCERPT: Exact dimension formula. The dimension is

conjecture | open | MathDB 页面 | 原文线索

V2-053377 | MATHDB-375966

Zhu et al.'s generalized Turán conjecture for cycles with bounded circumference

摘录 / EXCERPT: For a graph G, let $N(C_q,G)$ denote the number of copies of the cycle C_q in G, and let $\operatorname{ex}(n,C_q,\mathcal C_{\ge L+1})$ be the maximum possible value of...

conjecture | open | MathDB 页面 | 原文线索

V2-053378 | MATHDB-375971

The monadic dependence characterization of fixed-parameter tractability

摘录 / EXCERPT: Let ${\mathscr C}$ be a hereditary class of graphs. A class is monadically dependent if one cannot interpret all graphs in vertex-colored graphs from the class using a fixed first-...

conjecture | open | MathDB 页面 | 原文线索

V2-053379 | MATHDB-375993

Chartrand-Holbert-Oellermann-Swart conjecture on F-irregular graphs

摘录 / EXCERPT: Chartrand-Holbert-Oellermann-Swart conjecture. For every connected graph F of order at least 3, there exists a nontrivial F-irregular graph.

conjecture | open | MathDB 页面 | 原文线索

V2-053380 | MATHDB-375994

Strong conjecture on rooted F-irregular graphs

摘录 / EXCERPT: Strong conjecture on rooted F-irregular graphs. For every connected graph F of order $|F| \ge 3$ and each root r of F, unless $F=K_{1,n-1}$ and r is the center of...

conjecture | open | MathDB 页面 | 原文线索

V2-053381 | MATHDB-376002

Cayley-graph diameter and minimal-generator conjecture

摘录 / EXCERPT: Let G be a finite group, let S be a generating set, let n denote the group order, let g_{min} be the minimal number of generators of G, and let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053382 | MATHDB-376003

Cayley-graph disorder-diameter asymptotic conjecture

摘录 / EXCERPT: Let G be a finite group, let S be a generating set, let $n=|G|$, and let $\mathfrak{D}(\operatorname{Cay}(G,S))$ and $\mathcal{A}(\operatorname{Cay}(G,S))$ denote the graph dia...

conjecture | open | MathDB 页面 | 原文线索

V2-053383 | MATHDB-376004

Central-quotient eigengap conjecture for nilpotent Cayley graphs

摘录 / EXCERPT: Central-quotient eigengap conjecture. Either $k_{>1}=|G|-1$, or

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053384 | MATHDB-376007

Coarse Erdős-Pósa conjecture for fat triangle minor-models

摘录 / EXCERPT: Coarse Erdős-Pósa conjecture. There exist functions

conjecture | open | MathDB 页面 | 原文线索

V2-053385 | MATHDB-376030

Finite exact-density tile families for multiset profile graphs

摘录 / EXCERPT: Finite exact-density tile-family conjecture. For every fixed q, there is a finite coordinate-symmetric family of induced templates of independence density $1/q$ such that, in eve...

conjecture | open | MathDB 页面 | 原文线索

V2-053386 | MATHDB-376035

DeBiasio's conjecture on the minimum total degree for powers of Hamilton cycles

摘录 / EXCERPT: DeBiasio's conjecture. Every n-vertex digraph D satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-053387 | MATHDB-376039

Gao-Lam-Xu's electrical canonical basis conjecture

摘录 / EXCERPT: Let $G_{m,n}$ be the degree-m part of the grove algebra, and let c generate the cyclic action ... The basis elements are to be indexed by height m staircase plane partitions of...

conjecture | open | MathDB 页面 | 原文线索

V2-053388 | MATHDB-376054

The universal golden-ratio threshold for decorated unicyclic graphs

摘录 / EXCERPT: Let G be a decorated unicyclic graph with cycle length parameter g, let R_G be its curvature operator, and let $w^\ast$ denote the Perron metric. Write ... and let φ be...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053389 | MATHDB-376061

Li-Peng square-root booksize conjecture for Nosal graphs

摘录 / EXCERPT: Li-Peng conjecture. For every m-edge Nosal graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-053390 | MATHDB-376066

Baker's Brill-Noether existence conjecture for finite graphs

摘录 / EXCERPT: Let G be a finite, undirected, connected, loop-free multigraph of genus g. Define the Brill-Noether number by ... A divisor on G has a degree and a rank in the usual graph-theo...

conjecture | open | MathDB 页面 | 原文线索

V2-053391 | MATHDB-376073

Bollobás-Erdős-Tuza conjecture on transversals of maximum independent sets

摘录 / EXCERPT: Let $0<c<1$ be a constant. For a graph G, let $\alpha(G)$ be the size of a largest independent set, and let $h(G)$ be the size of a smallest set meeting every maximum independent...

conjecture | open | MathDB 页面 | 原文线索

V2-053392 | MATHDB-376074

The beta conjecture for maximal independent sets

摘录 / EXCERPT: Beta conjecture. For all graphs G on n vertices,

conjecture | open | MathDB 页面 | 原文线索

V2-053393 | MATHDB-376075

Strong conjecture on transversals of large maximal independent sets

摘录 / EXCERPT: Let $0<c<1$ be a constant. For a graph G, let $\operatorname{MIS}(G)$ denote its family of maximal independent sets, and define the family of large maximal independent sets by ……

conjecture | open | MathDB 页面 | 原文线索

V2-053394 | MATHDB-376076

Voss's conjecture on chords in odd cycles of critical graphs

摘录 / EXCERPT: All graphs are finite and simple. A graph is k-critical if its chromatic number is k and every proper subgraph is $(k-1)$-colorable. Let $g_k(n)$ be the largest integer...

conjecture | open | MathDB 页面 | 原文线索

V2-053395 | MATHDB-376077

Kára and Král's exact minimum degree for 31-vertex cycle chords

摘录 / EXCERPT: For integers n and c, let $f(n,c)$ be the least integer d such that every n-vertex graph with minimum degree at least d contains a cycle with at least c chords. Kára an...

conjecture | open | MathDB 页面 | 原文线索

V2-053396 | MATHDB-376087

Extremal construction conjecture for $\alpha_{1,k}$

摘录 / EXCERPT: For an integer $k\geq 3$, let $\alpha_{1,k}$ be the asymptotic extremal constant studied in the paper, and let J_r and K_r denote the indicated graph constructions. Extremal co...

conjecture | open | MathDB 页面 | 原文线索

V2-053397 | MATHDB-376099

Theo-Conjecture for connected cographs

摘录 / EXCERPT: Let G) be a connected cograph with no universal vertex. For the co-components $H_{(1)}$ and $H_{(2)}$ defined as in Proposition, Theo-Conjecture. ... The cograph condition is essen...

conjecture | open | MathDB 页面 | 原文线索

V2-053398 | MATHDB-376100

Theo-Conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Theo-Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053399 | MATHDB-376107

Li--Feng--Peng's conjecture on triangular edges in spectral graphs

摘录 / EXCERPT: Let G be a graph with m edges, and let $\lambda(G)$ denote its spectral radius. An edge is triangular if it lies in a triangle. Li--Feng--Peng's conjecture. If ... then G conta...

conjecture | open | MathDB 页面 | 原文线索

V2-053400 | MATHDB-376111

Matching upper-bound conjecture for the linear Turán number of the four-edge hypertree

摘录 / EXCERPT: The P_4^r upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053401 | MATHDB-376117

Jackson-Thomassen conjecture on strong orientations of digraphs

摘录 / EXCERPT: Jackson-Thomassen conjecture. Every $2k$-strong digraph has a spanning k-strong oriented subdigraph.

conjecture | open | MathDB 页面 | 原文线索

V2-053402 | MATHDB-376118

Bang-Jensen-DeVos-Mütze conjecture on strong tournaments in semicomplete digraphs

摘录 / EXCERPT: Bang-Jensen-DeVos-Mütze conjecture. Every $(2k-1)$-strong semicomplete digraph on at least $2k+1$ vertices contains a spanning k-strong tournament.

conjecture | open | MathDB 页面 | 原文线索

V2-053403 | MATHDB-376119

Thomassen's orientation conjecture for highly connected graphs

摘录 / EXCERPT: A graph $G=(V,E)$ is k-connected if it remains connected after the deletion of any set of at most $k-1$ vertices. An orientation of G is k-strong if its corresponding digraph...

conjecture | open | MathDB 页面 | 原文线索

V2-053404 | MATHDB-376120

Weak Jackson-Thomassen orientation conjecture

摘录 / EXCERPT: A digraph D is k-strong if it has at least $k+1$ vertices and remains strongly connected after the deletion of any set of at most $k-1$ vertices. An orientation of a digraph is...

conjecture | open | MathDB 页面 | 原文线索

V2-053405 | MATHDB-376121

Bang-Jensen-Jordán conjecture for semicomplete digraphs

摘录 / EXCERPT: A digraph is semicomplete if it has no pair of non-adjacent vertices. A tournament is an orientation of a complete graph, hence a semicomplete digraph with no directed 2-cycles. A...

conjecture | open | MathDB 页面 | 原文线索

V2-053406 | MATHDB-376123

The full Brouwer conjecture on equality cases

摘录 / EXCERPT: Let G be a finite simple graph with n vertices and m edges. Let $\mu_1(G)\geq\mu_2(G)\geq\cdots\geq\mu_n(G)=0$ be its Laplacian eigenvalues, and define ... A threshold graph is...

conjecture | solved | MathDB 页面 | 原文线索

V2-053407 | MATHDB-376132

Huang-Zhang-Chen conjecture on tree-cycle Ramsey numbers

摘录 / EXCERPT: Let T_n be an arbitrary tree on n vertices, let C_m be a cycle on m vertices, and let $R(T_n,C_m)$ denote the Ramsey number of T_n versus C_m. Define $f(m)$ to be the l...

conjecture | open | MathDB 页面 | 原文线索

V2-053408 | MATHDB-376134

Akiyama-Watanabe-Albertson-Hass conjecture for bipartite planar graphs

摘录 / EXCERPT: Akiyama-Watanabe-Albertson-Hass conjecture. Every bipartite planar graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053409 | MATHDB-376135

Dross-Montassier-Pinlou conjecture on feedback vertex sets in large-girth planar graphs

摘录 / EXCERPT: Dross-Montassier-Pinlou conjecture. Every planar graph G of girth at least g satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053410 | MATHDB-376136

Elphick-Farber-Goldberg-Wocjan positive and negative square-energy conjecture

摘录 / EXCERPT: Elphick-Farber-Goldberg-Wocjan conjecture. Every connected graph G on n vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053411 | MATHDB-376137

Elphick-Wocjan positive square-energy strengthening of Wilf's inequality

摘录 / EXCERPT: Elphick-Wocjan conjecture. For every graph G on $n\ge 1$ vertices,

conjecture | solved | MathDB 页面 | 原文线索

V2-053412 | MATHDB-376138

The strong Seymour vertex conjecture for oriented graphs

摘录 / EXCERPT: Let D be an oriented graph. For a vertex x, let $N_D^+(x)$ and $N_D^{++}(x)$ denote its out-neighborhood and second out-neighborhood, respectively. A complete matching from X...

conjecture | open | MathDB 页面 | 原文线索

V2-053413 | MATHDB-376139

The strong Seymour vertex conjecture for tournaments

摘录 / EXCERPT: Let T be a tournament, namely an oriented graph in which every pair of distinct vertices is joined by exactly one directed arc. For a vertex x, let $N_T^+(x)$ and $N_T^{++}(x)$...

conjecture | open | MathDB 页面 | 原文线索

V2-053414 | MATHDB-376144

Chen-Yang-Yuan-Zhang conjecture for generalized Turán numbers of linear paths

摘录 / EXCERPT: Chen-Yang-Yuan-Zhang conjecture. For $n\geq 3k$ and $s\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-053415 | MATHDB-376147

Global spectral minimizer conjecture for signed circulants $C_n(1,2)$

摘录 / EXCERPT: Let $C_n(1,2)$ be the circulant graph on even n vertices, let A_σ range over its signed adjacency matrices, and define ... The switching classes are coordinatized by triangl...

conjecture | open | MathDB 页面 | 原文线索

V2-053416 | MATHDB-376155

Liu and Ning's interpolation conjecture for 2-connected subgraphs

摘录 / EXCERPT: For a fixed integer $q\ge 3$, let $n_0(q)$ be an integer threshold. Let G be a 2-connected graph of order $n\ge n_0(q)$, and write $\delta(G)$ for its minimum degree. Liu and Nin...

conjecture | open | MathDB 页面 | 原文线索

V2-053417 | MATHDB-376174

Bukh-Conlon conjecture for powers of balanced rooted trees

摘录 / EXCERPT: Bukh-Conlon conjecture. For any balanced rooted tree (T,R) and any natural number ℓ, we have

conjecture | open | MathDB 页面 | 原文线索

V2-053418 | MATHDB-376182

Lovász's Hamilton-path conjecture for connected vertex-transitive graphs

摘录 / EXCERPT: Lovász's conjecture. Every connected vertex-transitive graph has a Hamilton path.

conjecture | open | MathDB 页面 | 原文线索

V2-053419 | MATHDB-376183

Rapaport-Strasser conjecture for Hamilton cycles in connected Cayley graphs

摘录 / EXCERPT: Rapaport-Strasser conjecture. Every connected Cayley graph on a finite group with at least three elements has a Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-053420 | MATHDB-376189

Strong lottery ticket conjecture at virtually all sparsity levels

摘录 / EXCERPT: Strong lottery ticket conjecture at virtually all sparsity levels. There exists N such that, with probability at least δ, the network contains a mask H with sparsity s...

conjecture | open | MathDB 页面 | 原文线索

V2-053421 | MATHDB-376192

The tiling conjecture for syzygy supports in strongly connected quiver algebras

摘录 / EXCERPT: Let $\Lambda=kQ/I$ be a finite-dimensional algebra whose quiver Q is strongly connected, and let S_x be the simple module associated with a vertex x. Write $\Omega^n(S_x)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-053422 | MATHDB-376198

Duval-Reiner majorization conjecture for simplicial-complex Laplacians

摘录 / EXCERPT: Let K be an r-dimensional simplicial complex. Using the spectrum $\bm{\lambda}_{r-1}(K)$ of the up-Laplacian and the conjugate vertex-degree sequence $ \mathbf d_0^\top(K)$, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-053423 | MATHDB-376199

Lew's higher-dimensional Brouwer Laplacian conjecture

摘录 / EXCERPT: Let K be an r-dimensional simplicial complex, let $f_r(K)$ denote the number of r-faces, and let $\lambda_{r-1,i}(K)$ be the eigenvalues of the up-Laplacian on $(r-1)$-faces,...

conjecture | open | MathDB 页面 | 原文线索

V2-053424 | MATHDB-376209

Stanley's claw-free graph conjecture for chromatic symmetric functions

摘录 / EXCERPT: Let G be a simple graph, and let $X_G({\bf x})$ denote its chromatic symmetric function. A graph is claw-free if it has no induced subgraph isomorphic to the claw graph $K_{1,3}$...

conjecture | open | MathDB 页面 | 原文线索

V2-053425 | MATHDB-376213

Girão, Hurley, Illingworth, and Michel's colouring characterization of abundance

摘录 / EXCERPT: Girão, Hurley, Illingworth, and Michel's conjecture. The graph H is F-abundant if and only if there is an F-colouring $sigma$ of H such that $(H,sigma)$ is F-abundant.

conjecture | open | MathDB 页面 | 原文线索

V2-053426 | MATHDB-376232

Exponential lower-bound conjecture for the oriented chromatic number of degenerate graphs

摘录 / EXCERPT: Let $d\text{-degenerate}$ graphs be graphs in which every subgraph has a vertex of degree at most d, and let $\chi_o(G)$ denote the oriented chromatic number of a graph G. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053427 | MATHDB-376252

Spencer's Bus Stop Problem for integral stationary geodesic networks

摘录 / EXCERPT: Fix a Riemannian n-manifold (M,g), $n \geq 2$, and let $U \subset\subset V \subset\subset M$ be open. An integral stationary geodesic network is a geodesic network with integer...

conjecture | open | MathDB 页面 | 原文线索

V2-053428 | MATHDB-376256

Aouchiche-Hansen-Zheng conjecture on the Randić index and matching number

摘录 / EXCERPT: Aouchiche-Hansen-Zheng conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-053429 | MATHDB-376257

Gerbner-Palmer's generalized Turán conjecture for trees

摘录 / EXCERPT: Let T be a tree on k vertices and let $r\geq3$. Write ... For a graph G, let ${\mathcal N}(K_r,G)$ denote the number of copies of K_r in G, and let ${\mathrm{ex}}(n,K_r,T)$...

conjecture | open | MathDB 页面 | 原文线索

V2-053430 | MATHDB-376263

A recurrent relation for the oscillator specialization of the c(2)-weight system

摘录 / EXCERPT: Suppose that $\xi(G)$ is a 4-invariant of graphs extending the specialization of the $\mathfrak{sl}(2)$-weight system at $c=-\frac{3}{32}$. Let G be a graph, let x be a vertex...

conjecture | open | MathDB 页面 | 原文线索

V2-053431 | MATHDB-376276

Hušek-Šámal's exponential circuit double cover conjecture

摘录 / EXCERPT: Hušek-Šámal's conjecture. The graph G has at least

conjecture | open | MathDB 页面 | 原文线索

V2-053432 | MATHDB-376294

The proper hat-guessing number conjecture for two-spine book graphs

摘录 / EXCERPT: Let $B_{2,n}$ denote the two-spine book graph with n pages, and let $\operatorname{HGP}(G)$ be the proper hat-guessing number of a graph G. Two-spine book graph conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-053433 | MATHDB-376299

Hoàng's conjecture on bisimplicial vertices in minimally nonperfectly divisible graphs

摘录 / EXCERPT: All graphs considered are finite and simple. For a graph G, write $\rho(G)$ for its chromatic number and $\tau(G)$ for its clique number. A graph is minimally nonperfectly divisi...

conjecture | open | MathDB 页面 | 原文线索

V2-053434 | MATHDB-376322

The dual-cube lifting conjecture for completely independent spanning trees

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube and let F_{n+1} be the dual-cube of dimension $n+1$. Let k be a positive integer. Dual-cube lifting conjecture. If Q_n has k comp...

conjecture | open | MathDB 页面 | 原文线索

V2-053435 | MATHDB-376327

Kühn-Osthus lower-bound conjecture for complete-graph subdivisions

摘录 / EXCERPT: Let $d(t)$ be the smallest real number such that every graph with average degree more than $d(t)$ contains a subdivision of K_t. Kühn-Osthus lower-bound conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-053436 | MATHDB-376329

Erdős-Gallai linear cycle-and-edge decomposition conjecture

摘录 / EXCERPT: A decomposition of a graph is a partition of its edge set into subgraphs of the indicated types. Erdős-Gallai conjecture. Every n-vertex graph has a decomposition into $O(n)$ cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-053437 | MATHDB-376330

Nested cycles with no geometric crossings conjecture

摘录 / EXCERPT: Two edge-disjoint cycles are nested with no geometric crossings when the vertex set of one is contained in that of the other and the two cycles can be drawn together in the plane w...

conjecture | open | MathDB 页面 | 原文线索

V2-053438 | MATHDB-376331

Rainbow-cycle Turán conjecture

摘录 / EXCERPT: A proper edge-colouring is one in which incident edges receive distinct colours, and a rainbow cycle has pairwise distinct edge colours. Rainbow-cycle conjecture. Every properly-co...

conjecture | open | MathDB 页面 | 原文线索

V2-053439 | MATHDB-376332

Rainbow complete-graph subdivision conjecture

摘录 / EXCERPT: A proper edge-colouring is one in which incident edges receive distinct colours, and a rainbow K_k-subdivision is a subdivision of K_k whose edges have pairwise distinct colour...

conjecture | open | MathDB 页面 | 原文线索

V2-053440 | MATHDB-376333

Erdős-Faudree-Rousseau-Schelp cycle-versus-clique Ramsey conjecture

摘录 / EXCERPT: For graphs G and H, let $R(G,H)$ denote the least integer N such that every red-blue colouring of the edges of K_N contains a red copy of G or a blue copy of H. Erdős-F...

conjecture | open | MathDB 页面 | 原文线索

V2-053441 | MATHDB-376334

Pokrovskiy-Sudakov cycle Ramsey goodness conjecture

摘录 / EXCERPT: Let $R(G,H)$ denote the two-colour Ramsey number, let C_n be the cycle on n vertices, and let $\chi(H)$ and $\sigma(H)$ denote the chromatic number and minimum colour-class size...

conjecture | open | MathDB 页面 | 原文线索

V2-053442 | MATHDB-376357

The extremal arc-count conjecture for strongly connected digraphs without acyclic separators

摘录 / EXCERPT: Let D be a strongly connected digraph with $n\geq 3$ vertices and m arcs. An acyclic separator is a separator whose induced subdigraph is acyclic. Extremal arc-count conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-053443 | MATHDB-376366

The Ramsey-Sidorenko conjecture for homomorphism-multiplicity bounds

摘录 / EXCERPT: Let H be a graph, and let G be a graph of order n. Write $v(H)$ and $e(H)$ for the numbers of vertices and edges of H, $\hom(H,G)$ for the number of homomorphisms from H...

conjecture | open | MathDB 页面 | 原文线索

V2-053444 | MATHDB-376368

The injection-multiplicity Ramsey conjecture

摘录 / EXCERPT: Let H be a graph, and let G be a graph of order n with average degree $\overline{d}(G)$. Write $v(H)$ and $e(H)$ for the numbers of vertices and edges of H, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-053445 | MATHDB-376372

Hadwiger's conjecture for graphs

摘录 / EXCERPT: Let G be a finite graph and let $t\in\mathbb{N}$. A minor of G is a graph obtainable from G by a sequence of vertex deletions, edge deletions, and edge contractions; write...

conjecture | open | MathDB 页面 | 原文线索

V2-053446 | MATHDB-376374

Ghosh et al.'s conjecture on even-length paths in planar graphs

摘录 / EXCERPT: For a finite simple graph G, let $N(G,H)$ be the number of unlabeled, not necessarily induced copies of H in G. Define ... Let P_k denote the path on k vertices, so...

conjecture | open | MathDB 页面 | 原文线索

V2-053447 | MATHDB-376375

Cox and Martin's weighted optimization conjecture for even-length paths

摘录 / EXCERPT: Let U be a finite set, and let μ be a probability measure on the edges of the complete graph on U, so that $\mu(uv)\ge 0$ and ... For $u\in U$, define the weighted degree by...

conjecture | open | MathDB 页面 | 原文线索

V2-053448 | MATHDB-376377

Fajtlowicz's conjecture on average neighbour degree and distance spectra

摘录 / EXCERPT: Let G be a finite simple connected graph of order $n=|V(G)|\ge2$. For each vertex v, let $d(v)$ be its degree and $N(v)$ its open neighbourhood, and define ... Let $D(G)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-053449 | MATHDB-376380

Sedlar-Škrekovski's mixed metric dimension conjecture

摘录 / EXCERPT: Let G be a connected graph that is not a cycle. Write $O(G)=V(G)\cup E(G)$, let $\operatorname{dim_m}(G)$ denote the minimum cardinality of a mixed metric generator of G, let...

conjecture | open | MathDB 页面 | 原文线索

V2-053450 | MATHDB-376391

Exponential relation between maximal-clique deficiency and layered-tree parameter

摘录 / EXCERPT: Let $f(n,k)$ denote the maximal-clique deficiency parameter and let $c(n,k)$ denote the layered-tree parameter for k-uniform hypergraphs, as defined in the paper. Exponential rel...

conjecture | open | MathDB 页面 | 原文线索

V2-053451 | MATHDB-376392

Frankl's quadratic-threshold conjecture for critical intersecting hypergraphs

摘录 / EXCERPT: Frankl's conjecture. There exists a constant c such that, whenever $k>c d^2$,

conjecture | open | MathDB 页面 | 原文线索

V2-053452 | MATHDB-376393

Bradač-Janzer-Sudakov-Tomon conjecture for Cartesian products of trees

摘录 / EXCERPT: For graphs F and G, their Cartesian product $F\Box G$ has vertex set $V(F)\times V(G)$, with (u,v) adjacent to (u',v') if and only if either $u=u'$ and $vv'\in E(G)$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-053453 | MATHDB-376407

Semicircle-law conjecture for the large-degree Watts-Strogatz spectral measure

摘录 / EXCERPT: Let K,p be fixed parameters, and suppose that the empirical spectral distribution of the adjacency matrix of $WS_n(K,p)$, scaled by $\sqrt{Kp(2-p)}$, converges to a limiting prob...

conjecture | open | MathDB 页面 | 原文线索

V2-053454 | MATHDB-376408

Limiting-moments conjecture for Watts-Strogatz adjacency matrices

摘录 / EXCERPT: Let $K\ge1$ be an even integer and $p\in[0,1]$. For each $n\in\mathbb{N}$, let A_n denote the adjacency matrix of $WS_n(K,p)$. Let $F(K,p)$ be the limiting rooted locally finite...

conjecture | open | MathDB 页面 | 原文线索

V2-053455 | MATHDB-376423

Fan-Raspaud conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. Fan-Raspaud conjecture. The graph G contains three perfect matchings M_1,M_2,M_3 such that no edge is covered by all three of them. The Ber...

conjecture | open | MathDB 页面 | 原文线索

V2-053456 | MATHDB-376424

Mazzuoccolo's bipartite-complement conjecture for cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. Mazzuoccolo's conjecture. There exist two perfect matchings M_1,M_2 such that the graph obtained by deleting their union,...

conjecture | open | MathDB 页面 | 原文线索

V2-053457 | MATHDB-376425

Erdős-Sós conjecture for trees

摘录 / EXCERPT: Let T be a tree on t vertices. For a graph G, write $e(G)$ for its number of edges, and call G T-free if it contains no subgraph isomorphic to T. Erdős-Sós conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-053458 | MATHDB-376426

Erdős-Sós order-form conjecture for bipartite host graphs

摘录 / EXCERPT: Every tree is bipartite. Write $T_{a,b}=(A,B)$ when the two parts of T satisfy ... For a bipartite graph H, let $\operatorname{ex}_{\mathrm{bip}}(N;H)$ be the maximum number of e...

conjecture | open | MathDB 页面 | 原文线索

V2-053459 | MATHDB-376427

Erdős-Sós fixed-part conjecture for bipartite host graphs

摘录 / EXCERPT: Let $T_{a,b}=(A,B)$ be a tree whose bipartition satisfies ... For a bipartite graph H, let $\operatorname{ex}_{\mathrm{bip}}(m,n;H)$ be the maximum number of edges in an H-free b...

conjecture | open | MathDB 页面 | 原文线索

V2-053460 | MATHDB-376428

The minimum-degree conjecture for colour-biased loose Hamilton cycles

摘录 / EXCERPT: Let H be a red/blue coloured 3-graph on n vertices, and let $\delta_1(H)$ denote its minimum vertex degree. A loose Hamilton cycle is a cyclic ordering of the vertices in whi...

conjecture | open | MathDB 页面 | 原文线索

V2-053461 | MATHDB-376742

The rationality conjecture for the Neveu-Schwarz minimal series vertex operator superalgebras

摘录 / EXCERPT: Rationality conjecture. The vertex operator superalgebra $V_{c_{p,q}}$ is rational. Moreover, the minimal series modules $L_{c,h_{r,s}}$, with $0<r<p$, $0<s<q$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-053462 | MATHDB-376854

Uniform spectral convergence for operators on abstract quasicrystal graphs

摘录 / EXCERPT: Let $G(V,E)$ be a connected infinite graph with bounded vertex degrees. Assume that G is amenable and is an abstract quasicrystal graph: for every radius r and every r-patter...

conjecture | open | MathDB 页面 | 原文线索

V2-053463 | MATHDB-376872

Occurrence of interior spectral edges for periodic Schrödinger operators

摘录 / EXCERPT: Consider a planar embedding of a periodic graph with equal angles between the tangent lines of the edges meeting at each vertex, and let Ω_ε be the associated fatt...

conjecture | open | MathDB 页面 | 原文线索

V2-053464 | MATHDB-376923

Finite-dimensionality conjecture for homology of degenerate minimal models

摘录 / EXCERPT: Let p and q be parameters, define ... and define ... Let ${\mathcal L}_s$ be the Lie superalgebra introduced above, and let $L(c_{p,q},h_{p,q}^{m,n})$ denote the corresponding irre...

conjecture | open | MathDB 页面 | 原文线索

V2-053465 | MATHDB-376946

Adjacency conjecture for partition m-hemimetrics

摘录 / EXCERPT: Let V_n be the ground set, and let $\alpha(S_1,\dots,S_{m+1})$ and $\alpha(T_1,\dots,T_{m+1})$ be two partition m-hemimetrics on V_n. Two rays are adjacent when they generate...

conjecture | open | MathDB 页面 | 原文线索

V2-053466 | MATHDB-376947

Induced-ridge-graph conjecture for m-hemimetric cones

摘录 / EXCERPT: Let HM_n^m, NHM_n^m, and P_n^m be the corresponding cones of m-hemimetrics, and let the ridge graph of a cone have the facets as vertices, with adjacency when two facets me...

conjecture | open | MathDB 页面 | 原文线索

V2-053467 | MATHDB-376948

Ridge-graph structure conjecture for NHM m-hemimetric cones

摘录 / EXCERPT: Let NHM_n^m be the cone of nonnegative m-hemimetrics. Denote by $T_{i_1\dots i_{m+1},i_{m+2}}$ an $(m+1)$-simplex facet and by $N_{i_1\dots i_{m+1}}$ a nonnegativity facet. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-053468 | MATHDB-376982

Freeness conjecture for the equivariant cohomology of a graph

摘录 / EXCERPT: Let Γ be a graph with axial function α, let $H(\Gamma,\alpha)$ be its equivariant cohomology ring, and let ${\mathbb S}({\mathfrak g}^*)$ be the polynomial ring occur...

conjecture | open | MathDB 页面 | 原文线索

V2-053469 | MATHDB-376983

Thom-class conjecture for the equivariant cohomology of a graph

摘录 / EXCERPT: Let Γ be a graph with axial function α. For each vertex p, let σ_p be its index, let F_p be the set of vertices that can be joined to p by an ascending p...

conjecture | open | MathDB 页面 | 原文线索

V2-053470 | MATHDB-377092

Conjecture on sublinear chromatic-root bounds for series-parallel graphs

摘录 / EXCERPT: Series-parallel chromatic-root bound. There exists a universal constant $C<\infty$ such that, for every series-parallel graph G of maximum degree k, every chromatic root z li...

conjecture | open | MathDB 页面 | 原文线索

V2-053471 | MATHDB-377093

Maxmaxflow conjecture for bounded chromatic roots

摘录 / EXCERPT: Maxmaxflow conjecture. There exist universal constants $C(k)<\infty$ such that every chromatic root z of any graph G with $\Lambda(G)=k$ lies in the disc

conjecture | open | MathDB 页面 | 原文线索

V2-053472 | MATHDB-377104

Kontsevich's polynomial-counting conjecture for graph hypersurface complements

摘录 / EXCERPT: Let G be a finite graph with vertex set $V(G)$) and edge set $E(G)$. For each edge e, let x_e be a variable, and define the Kirchhoff polynomial ... where the sum runs over spa...

conjecture | open | MathDB 页面 | 原文线索

V2-053473 | MATHDB-377114

Finite-dimensional homology conjecture for degenerate minimal models

摘录 / EXCERPT: Let p,q be positive integers, set ... and for integers m,n set ... For the Lie superalgebra ${\mathcal L}_s$ and the irreducible module $L(c_{p,q},h_{p,q}^{m,n})$, finite-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-053474 | MATHDB-377132

Barnette's Hamiltonian cycle conjecture for even simple 3-polytopes

摘录 / EXCERPT: Let P be an even simple 3-polytope, meaning that every facet of P has an even number of vertices. Its vertex-edge graph is the graph whose vertices and edges are those of P...

conjecture | open | MathDB 页面 | 原文线索

V2-053475 | MATHDB-377197

Isoperimetric criterion for bounded percolation thresholds

摘录 / EXCERPT: Let G be a finite graph and define ... Here ∂S is the vertex boundary used in the paper, and p^G is the percolation threshold. Isoperimetric-threshold conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-053476 | MATHDB-377246

The planar graph algebraic-connectivity conjecture

摘录 / EXCERPT: Let G be a planar graph, and let $a(G)$ denote its algebraic connectivity, the second-smallest eigenvalue of its Laplacian. The graphs K_4 and $2K_1\dot{+}C_4$ are the complete...

conjecture | open | MathDB 页面 | 原文线索

V2-053477 | MATHDB-377247

The planar bichromatic graph algebraic-connectivity conjecture

摘录 / EXCERPT: Let G be a planar bichromatic graph, meaning a planar graph whose vertices can be coloured with two colours so that adjacent vertices have different colours. Let $a(G)$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-053478 | MATHDB-377248

The surface-independence conjecture for asymptotic algebraic connectivity

摘录 / EXCERPT: Fix a surface S, and let $\mathcal{A}^{\infty}(S)$ denote the asymptotic algebraic connectivity associated with graphs on S. The double wheel graph is a graph whose algebraic c...

conjecture | open | MathDB 页面 | 原文线索

V2-053479 | MATHDB-377269

Hamiltonian-path distance congruence conjecture for Cartesian products of directed cycles

摘录 / EXCERPT: Let X be the Cartesian product of k directed cycles of lengths $m_1,m_2,\ldots,m_k$, where $k\geq 3$ and each $m_i\geq 2$. For vertices u and v of X, let $d_X(u,v)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-053480 | MATHDB-377316

Asymptotic conjecture for the domatic number of almost regular graphs

摘录 / EXCERPT: Asymptotic domatic-number conjecture. For every fixed real number $C\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-053481 | MATHDB-377323

The order-three conjecture for rational conformal vectors

摘录 / EXCERPT: Order-three conjecture. Under this assumption, $\tau_e\tau_f$ has order three.

conjecture | open | MathDB 页面 | 原文线索

V2-053482 | MATHDB-377384

The asymptotic lower-bound conjecture for edge-coloring components

摘录 / EXCERPT: Asymptotic lower-bound conjecture. For $k\geq r+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-053483 | MATHDB-377412

Toric Residue Mirror Conjecture

摘录 / EXCERPT: Toric Residue Mirror Conjecture. The Laurent expansion of $R_P(a)$ at the vertex $v_{\mathcal T}\in\operatorname{Sec}(A)$ corresponding to ${\mathcal T}$ coincides with

conjecture | open | MathDB 页面 | 原文线索

V2-053484 | MATHDB-377413

Toric Residue Mirror Conjecture for the first Hirzebruch surface

摘录 / EXCERPT: Toric Residue Mirror Conjecture for the first Hirzebruch surface. The series expansion at the vertex $v_{\mathcal T}\in\operatorname{Sec}(A)$ of the toric residue

conjecture | open | MathDB 页面 | 原文线索

V2-053485 | MATHDB-377440

The conjecture that planar-move graphs differ from knot-move graphs

摘录 / EXCERPT: The conjecture that ${\cal C}_2$ and ${\cal C}_3$ differ.

conjecture | open | MathDB 页面 | 原文线索

V2-053486 | MATHDB-377441

The conjecture that knot-generated graphs differ from graphs with polynomial one

摘录 / EXCERPT: The conjecture that ${\cal C}_3$ and ${\cal C}_4$ differ.

conjecture | open | MathDB 页面 | 原文线索

V2-053487 | MATHDB-377529

Weak convergence conjecture for bounded-degree planar triangulations

摘录 / EXCERPT: Let $c4_N$ denote the distribution of the relevant finite planar triangulations, conditioned to have all vertex degrees bounded by M. Weak convergence conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-053488 | MATHDB-377583

Circuit-partition unimodality conjecture for weakly Eulerian graphs and digraphs

摘录 / EXCERPT: Let a weakly Eulerian graph or digraph be given, and for each integer k let the number of its partitions into k circuits be counted. Circuit-partition unimodality conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-053489 | MATHDB-377601

Finiteness conjecture for genera of rational vertex operator algebras

摘录 / EXCERPT: Finiteness conjecture. The set $\operatorname{gen}(V)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-053490 | MATHDB-377617

The three-edge balanced-circle conjecture for abelian gain graphs

摘录 / EXCERPT: Let (Γ,g,G) be an abelian gain graph, where G is a torsion-free abelian group, possibly 2-divisible. Let $Z_1(\Gamma)$ be the integral cycle group, and suppose it has a...

conjecture | open | MathDB 页面 | 原文线索

V2-053491 | MATHDB-377653

The independence-polynomial characterization of well-covered trees

摘录 / EXCERPT: Independence-polynomial characterization conjecture. If T is a well-covered tree and $I(T;x)=I(G;x)$, then G is well-covered.

conjecture | open | MathDB 页面 | 原文线索

V2-053492 | MATHDB-377721

Isoperimetric conjecture for finite vertex-transitive graphs

摘录 / EXCERPT: Let G be a finite, connected, vertex-transitive graph. Write $\operatorname{diam}(G)$ for its diameter and, for a vertex set S, let ∂S denote its edge boundary. Isope...

conjecture | open | MathDB 页面 | 原文线索

V2-053493 | MATHDB-377731

The list coloring conjecture for line graphs

摘录 / EXCERPT: Let G be a graph. Its choice number $ch(G)$ is the least integer k such that every assignment of a k-element list of colors to each vertex admits a proper coloring from the a...

conjecture | open | MathDB 页面 | 原文线索

V2-053494 | MATHDB-377760

Absolute convergence conjecture for sewn supergeometric tubes

摘录 / EXCERPT: Let $Q_1\in SK(m)$ and $Q_2\in SK(n)$ be supergeometric data of the forms ... and ... Here $SK(k)$ denotes the space of supergeometric configurations with k tubes, and Γ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053495 | MATHDB-377768

Existence and finite-dimensionality conjecture for Griess algebras

摘录 / EXCERPT: Griess algebra realization conjecture. There is an OZ vertex algebra V such that

conjecture | open | MathDB 页面 | 原文线索

V2-053496 | MATHDB-377776

Curvature-zero propagation conjecture for eternal triod blow-ups

摘录 / EXCERPT: Let $\mathbb T^\infty_{\mathfrak t}$ be an eternal evolution by curvature of an unbounded embedded triod, curve, or curve with a single endpoint, and let its curvature be defined a...

conjecture | open | MathDB 页面 | 原文线索

V2-053497 | MATHDB-377777

Translation conjecture for eternal triod blow-ups

摘录 / EXCERPT: Let $\mathbb T^\infty_{\mathfrak t}$ be an eternal evolution by curvature of an unbounded embedded triod, curve, or curve with a single endpoint. Translation conjecture. The triods...

conjecture | open | MathDB 页面 | 原文线索

V2-053498 | MATHDB-377778

No-singularity conjecture for embedded triod flow with positive length bounds

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^2$ be a bounded, strictly convex domain, and let $\mathbb T_t$ be the flow by curvature of an embedded triod in Ω. Assume that the lengths of the...

conjecture | open | MathDB 页面 | 原文线索

V2-053499 | MATHDB-377831

The covering-number characterization of S-GQPQs

摘录 / EXCERPT: The covering-number converse conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053500 | MATHDB-377891

Berge's characterization of minimally imperfect graphs

摘录 / EXCERPT: Let G be a finite simple graph. A minimally imperfect graph is a graph that is not perfect but whose proper induced subgraphs are all perfect. A hole is a chordless cycle of leng...

conjecture | open | MathDB 页面 | 原文线索

V2-053501 | MATHDB-377910

Well-covered tree determination by the independence polynomial

摘录 / EXCERPT: Independence-polynomial conjecture. Then G is a well-covered tree. This concerns whether a well-covered tree is characterized, among connected graphs, by its independence polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-053502 | MATHDB-378019

The planar graph Markov width conjecture

摘录 / EXCERPT: Planar graph Markov width conjecture. There is a universal constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-053503 | MATHDB-378020

The tree-width conjecture for Markov width

摘录 / EXCERPT: Tree-width conjecture. The invariant $mu(G)$ is a function only of the tree width of G.

conjecture | open | MathDB 页面 | 原文线索

V2-053504 | MATHDB-378022

The edge-deletion monotonicity conjecture for Markov width

摘录 / EXCERPT: Edge-deletion monotonicity conjecture. If H is obtained from G by deleting an edge, then

conjecture | open | MathDB 页面 | 原文线索

V2-053505 | MATHDB-378023

The vertex-transitive graph lower-bound conjecture for loop-erased random walk

摘录 / EXCERPT: Loop-erased random walk lower-bound conjecture. The expected length of the loop-erasure of R satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053506 | MATHDB-378080

The Duval-Reiner conjecture for family-pair Laplacian spectra

摘录 / EXCERPT: Let $(\mathcal K,\mathcal K')$ be a family pair, with relative Laplacian spectrum $\mathbf{s}(\mathcal K,\mathcal K')$ and transposed relative generalized degree sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-053507 | MATHDB-378090

Sphericality conjecture for the pinched braid poset

摘录 / EXCERPT: Let ... -complex, with vertex set the pinched braids having precisely one regular strand. Sphericality conjecture. ... is large enough. The paper presents this as evidence toward under...

conjecture | open | MathDB 页面 | 原文线索

V2-053508 | MATHDB-378313

Layer-monotonicity conjecture for Richardson infection times

摘录 / EXCERPT: Let G) be a graph with distinguished vertex x, and let $H=G\times\{0,1\}$ have two copies of G, with corresponding vertices joined by an edge. Run Richardson's model on H f...

conjecture | open | MathDB 页面 | 原文线索

V2-053509 | MATHDB-378319

Separation of survival and root-reinfection thresholds on strongly exponentially growing trees

摘录 / EXCERPT: Let T be an infinite tree. Call a vertex v essential if $T\setminus\{v\}$ has at least two infinite components. Say that T has strongly exponential growth if there is some in...

conjecture | open | MathDB 页面 | 原文线索

V2-053510 | MATHDB-378330

Negative association for up-events and edges in uniform spanning trees

摘录 / EXCERPT: Let G be a finite connected graph, let ${\bf T}$ be a uniform spanning tree, let e be an edge, and let A be an up-event, meaning an upwardly closed event in the space of subg...

conjecture | open | MathDB 页面 | 原文线索

V2-053511 | MATHDB-378333

The disjoint-path comparison conjecture for graded graphs

摘录 / EXCERPT: Let G be a graded graph with root ρ, levels $V_0,\ldots,V_N$, and IID vertex labels with common distribution μ. For an event B of label sequences, let $P(G;B)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-053512 | MATHDB-378344

The dimension and vertex-number conjecture for complementary weak pseudomanifolds

摘录 / EXCERPT: A complementary weak pseudomanifold is a weak pseudomanifold whose simplicial complex satisfies complementarity. Let M be a d-dimensional complementary weak pseudomanifold with...

conjecture | open | MathDB 页面 | 原文线索

V2-053513 | MATHDB-378373

Winding partitions conjecture for drawings of complete graphs

摘录 / EXCERPT: Let $K_{3(q-1)+1}$ be the complete graph on $3(q-1)+1$ vertices, and consider any drawing of this graph in the plane. For a triangle of edges and a point not on it, its winding num...

conjecture | open | MathDB 页面 | 原文线索

V2-053514 | MATHDB-378381

Neighborly vertex-transitive triangulations of balanced sphere products

摘录 / EXCERPT: Neighborly sphere-product conjecture. There is a centrally $(\lfloor d/2\rfloor+1)$-neighborly combinatorial triangulation of every product...

conjecture | open | MathDB 页面 | 原文线索

V2-053515 | MATHDB-378386

Nikiforov's asymptotic cycle-complete graph Ramsey conjecture

摘录 / EXCERPT: Let $r(C_p,K_r)$ denote the Ramsey number for a cycle C_p versus a complete graph K_r, where p, r, and k are positive integers. Nikiforov's asymptotic Ramsey conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-053516 | MATHDB-378394

Chung's singular-value discrepancy conjecture for regular graphs

摘录 / EXCERPT: For a graph G on n vertices, write $e(G)$ for its number of edges and set ... For nonempty vertex subsets $X,Y\subseteq V(G)$, let ... Let $\sigma_{2}(G)$ denote the second-largest...

conjecture | open | MathDB 页面 | 原文线索

V2-053517 | MATHDB-378417

Lins triality formulas for prisms and antiprisms

摘录 / EXCERPT: Let ${\cal M}$ be a map, and denote the Euler characteristics of ${\cal M}$, $phial({\cal M})$, and $skew({\cal M})$ by $\chi({\cal M})$, $\chi_p({\cal M})$, and $\chi_s({\cal M})$...

conjecture | open | MathDB 页面 | 原文线索

V2-053518 | MATHDB-378465

Clarke-Hochberg-Hurlbert connectivity conjecture for graph pebbling

摘录 / EXCERPT: Let $d\ge 1$. For a finite graph $\mathbf{G}$, let its diameter be the maximum distance between two vertices, let its connectivity be the minimum number of vertices whose removal d...

conjecture | open | MathDB 页面 | 原文线索

V2-053519 | MATHDB-378536

Fusion-limit conjecture for finitized principal subspaces

摘录 / EXCERPT: Let $A_{(m)}(n)$ be the subalgebra generated by $\xi_0,\ldots,\xi_{n-1}$ in $A_{(m)}$, and let $I_{(m)}(n)$ be its defining ideal. For a family...

conjecture | open | MathDB 页面 | 原文线索

V2-053520 | MATHDB-378556

The asymptotic Ramsey-function conjecture for mean Ramsey-Turán numbers

摘录 / EXCERPT: Mean Ramsey-function conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053521 | MATHDB-378557

The complete-graph conjecture for 2-mean Ramsey-Turán numbers

摘录 / EXCERPT: Complete-graph 2-mean conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053522 | MATHDB-378607

Integral-level semi-infinite cohomology and the Verlinde algebra

摘录 / EXCERPT: The paper considers the affine Lie algebra $\hat{\mathfrak g}$, its central element $\mathbf k$, and truncated vertex operator algebras $\hat{\mathbf F}_{k+h^\vee}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053523 | MATHDB-378613

Sub-polynomial boundary-growth conjecture for fire containment in d-dimensional grids

摘录 / EXCERPT: Let $\mathbb{L}^{d}$ be the infinite d-dimensional square grid, whose vertices are the points of $\mathbb{R}^{d}$ with integer coordinates, with adjacency defined by Euclidean di...

conjecture | open | MathDB 页面 | 原文线索

V2-053524 | MATHDB-378653

The conjecture that every graph is good

摘录 / EXCERPT: Let $G=(V,E)$ be a graph. A distribution of pebbles on G is an initial arrangement of pebbles on a subset of vertices, and its support is that subset. Let $\gamma(G)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-053525 | MATHDB-378654

The acylindrical PD3-pair splitting conjecture

摘录 / EXCERPT: Acylindrical vertex-pair conjecture. Either $s_i=1$ (and hence $s_j=1$), or $i=j$ and $g^{-1}H_rg=H_r$.

conjecture | open | MathDB 页面 | 原文线索

V2-053526 | MATHDB-378666

The longest induced cycle conjecture for unitary Cayley graphs

摘录 / EXCERPT: Let X_n be the unitary Cayley graph considered in the paper, and let $m(r)$ denote the maximum length of an induced cycle in X_n when the underlying modulus has r distinct pr...

conjecture | open | MathDB 页面 | 原文线索

V2-053527 | MATHDB-378696

The Colin de Verdière parameter and algebraic shifting

摘录 / EXCERPT: Let G be a graph, let $\mu(G)$ denote the Colin de Verdière parameter of G, and let k be a positive integer. Let $\Delta(G)$ denote either the symmetric or exterior algebraic...

conjecture | open | MathDB 页面 | 原文线索

V2-053528 | MATHDB-378713

Haines's refinement for triangles with sums of minuscule weights

摘录 / EXCERPT: Haines's refinement. The assumption that one vertex of T is special can be replaced by

conjecture | open | MathDB 页面 | 原文线索

V2-053529 | MATHDB-378815

Seymour's strengthening of Hadwiger's conjecture for graphs with no antitriangle

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with no stable set of size 3, and let a prevertex be a connected subgraph contracted to one vertex in a minor. Seymour's strengthening. The graph G has...

conjecture | open | MathDB 页面 | 原文线索

V2-053530 | MATHDB-378927

Simultaneous multipartite subhypergraph conjecture

摘录 / EXCERPT: Simultaneous multipartite subhypergraph conjecture. There exists a partition of V into r classes $V_1,\dots,V_r$ such that, for every $i=1,\dots,\ell$, at least

conjecture | open | MathDB 页面 | 原文线索

V2-053531 | MATHDB-378944

Nonnegative derivatives of the chromatic polynomial at maxmaxflow

摘录 / EXCERPT: Let G be a loopless graph with maxmaxflow Λ, and let $P_G(q)$ be its chromatic polynomial. Maxmaxflow derivative-positivity conjecture. The polynomial $P_G(q)/q$ and all...

conjecture | open | MathDB 页面 | 原文线索

V2-053532 | MATHDB-378945

Positivity of the chromatic polynomial above maxmaxflow

摘录 / EXCERPT: Let G be a loopless graph with maxmaxflow Λ, and let $P_G(q)$ be its chromatic polynomial. Maxmaxflow chromatic positivity conjecture. For every real $q>\Lambda$, ... In pa...

conjecture | open | MathDB 页面 | 原文线索

V2-053533 | MATHDB-379068

Conjecture on sharp eigenvalue deviation bounds for large graphs

摘录 / EXCERPT: Let G be a graph of order n and size m, and let $\mu(G)$ denote its spectral radius and $s(G)$ its degree deviation parameter. Sharp eigenvalue deviation conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-053534 | MATHDB-379069

Asymptotic conjecture for the spectral Nordhaus-Gaddum function $f_1(n)$

摘录 / EXCERPT: Let $f_1(n)$ denote the spectral Nordhaus-Gaddum function defined in the paper for graphs of order n. Asymptotic conjecture for $f_1(n)$. ... The preceding construction gives the l...

conjecture | open | MathDB 页面 | 原文线索

V2-053535 | MATHDB-379070

Asymptotic conjecture for the spectral Nordhaus-Gaddum function $f_n(n)$

摘录 / EXCERPT: Let $f_n(n)$ denote the spectral Nordhaus-Gaddum function defined in the paper for graphs of order n. Asymptotic conjecture for $f_n(n)$. ... The paper proves the lower bound...

conjecture | open | MathDB 页面 | 原文线索

V2-053536 | MATHDB-379084

Kühnel-Kalai conjecture on vertex bounds for even-dimensional manifolds

摘录 / EXCERPT: Kühnel-Kalai conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053537 | MATHDB-379085

Kühnel's Betti-number bounds for triangulated manifolds

摘录 / EXCERPT: Kühnel's conjecture. For $j=0,\dots,\lfloor(d-1)/2\rfloor$,

conjecture | open | MathDB 页面 | 原文线索

V2-053538 | MATHDB-379086

Kühnel-Lassmann minimality conjecture for combinatorial tori

摘录 / EXCERPT: Kühnel-Lassmann torus conjecture. The Kühnel-Lassmann series of combinatorial d-tori T^d with $2^{d+1}-1$ vertices is vertex-minimal.

conjecture | open | MathDB 页面 | 原文线索

V2-053539 | MATHDB-379087

Kühnel-Lassmann uniqueness conjecture for the 15-vertex 3-torus

摘录 / EXCERPT: Kühnel-Lassmann 3-torus conjecture. The Kühnel-Lassmann triangulation of T^3 with 15 vertices is vertex-minimal and is unique with

conjecture | open | MathDB 页面 | 原文线索

V2-053540 | MATHDB-379091

Kühnel's componentwise minimality conjecture for the twisted sphere product

摘录 / EXCERPT: Kühnel's componentwise minimality conjecture. The f-vector

conjecture | open | MathDB 页面 | 原文线索

V2-053541 | MATHDB-379094

The extremal configuration conjecture for the pi-pebbling function

摘录 / EXCERPT: Extremal configuration conjecture. There exists a non-t-solvable configuration C of $k-1$ pebbles on G such that, for every vertex i, c_i is either 0 or p_i-1, except...

conjecture | open | MathDB 页面 | 原文线索

V2-053542 | MATHDB-379103

Uniqueness conjecture for the 15-vertex triangulation of a non-orientable 4-manifold

摘录 / EXCERPT: Let $M=(S^3\text{$timeshspace{-1.62ex}hspace{-.4ex}hspace{.7ex}$}S^1)\#({\mathbb C}{\bf P}^{\,2})^{\# 5}$, and let $\text{}^4\hspace{.3pt}15^{\,4}_{\,1}$ denote the specified combi...

conjecture | open | MathDB 页面 | 原文线索

V2-053543 | MATHDB-379113

Tutte's three-edge-coloring conjecture for cubic graphs without a Petersen minor

摘录 / EXCERPT: Let G be a 2-connected cubic graph with no Petersen minor. A proper three-edge-coloring assigns one of three colors to each edge so that edges incident with the same vertex rec...

conjecture | open | MathDB 页面 | 原文线索

V2-053544 | MATHDB-379114

The matroid strong-map spectral recursion conjecture

摘录 / EXCERPT: Let N and N' be matroids, and let $N \rightarrow N'$ be a strong map. Write $IN(N)$ and $IN(N')$ for their independence complexes, and let $(IN(N),IN(N'))$ denote the resulting...

conjecture | open | MathDB 页面 | 原文线索

V2-053545 | MATHDB-379155

Spectrum regularity conjecture for rainbow cycle lengths

摘录 / EXCERPT: Let S be a spectrum of rainbow cycle lengths, namely the set of cycle lengths occurring as rainbow cycles under a fixed edge-coloring. Spectrum regularity conjecture. The asympto...

conjecture | open | MathDB 页面 | 原文线索

V2-053546 | MATHDB-379199

The spectrum-polygon conjecture for standardized Laplacian matrices

摘录 / EXCERPT: Let n be a positive integer, let a standardized Laplacian matrix of order n be a matrix of the type considered in the paper, and let S be the polygon defined earlier as the r...

conjecture | open | MathDB 页面 | 原文线索

V2-053547 | MATHDB-379275

The separating-set characterization of unlinkable webs

摘录 / EXCERPT: Let Γ be a web, and let A and B be its distinguished vertex sets. An A--B-separating set S meets every A--B path; S is linkable into A in...

conjecture | open | MathDB 页面 | 原文线索

V2-053548 | MATHDB-379294

The cover-to-domination cover pebbling ratio conjecture

摘录 / EXCERPT: Let G be a graph with more than one vertex. Write $\lambda(G)$ for its cover pebbling number and $\psi(G)$ for its domination cover pebbling number. Cover-to-domination cover peb...

conjecture | open | MathDB 页面 | 原文线索

V2-053549 | MATHDB-379313

The free multiplicative convolution conjecture for graph wreath products

摘录 / EXCERPT: Let X and Y be colored graphs. Write $X*Y$ for the colored graph obtained by putting a copy of X at each vertex of Y, let $\mu(Z)$ denote the spectral measure of the charac...

conjecture | open | MathDB 页面 | 原文线索

V2-053550 | MATHDB-379317

Proper-subset conjecture for classical spatial graphs in virtual spatial graph theory

摘录 / EXCERPT: A classical spatial graph is a spatial graph represented by a classical graph diagram, and two such graphs are virtually equivalent or classically equivalent according to the virtu...

conjecture | open | MathDB 页面 | 原文线索

V2-053551 | MATHDB-379338

The principal conjecture on clique coverings and stable sets in B-graphs

摘录 / EXCERPT: Let G be a B-graph, meaning that $sigma_v(G)=alpha(G)$, and suppose that G has no isolated vertices. Let $omega_e(G)$ be the greatest natural number such that every edge of G...

conjecture | open | MathDB 页面 | 原文线索

V2-053552 | MATHDB-379339

The uniform hypergraph bound for the vertex-stable-set invariant

摘录 / EXCERPT: Let ${\cal H}=(V,{cal E})$ be a hypergraph with $|V|=n$, no isolated vertices, and every edge $h\in{cal E}$ of size r. Let $sigma_v({\cal H})$ and $alpha({\cal H})$ be the stated...

conjecture | open | MathDB 页面 | 原文线索

V2-053553 | MATHDB-379340

Ore's connected-graph product conjecture for edge invariants

摘录 / EXCERPT: Let G be a graph with n vertices and no isolated vertices. Let $alpha(G)$ be its stability number, $sigma_e(G)=omega_e(\overline G)$ its edge-stable-set invariant, $omega(G)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-053554 | MATHDB-379433

The Solidity Conjecture for universal graphs with a forbidden graph

摘录 / EXCERPT: Let C be a finite or countable graph with no isolated vertices. Decompose C into its blocks, where each block is a 2-connected component, and let $\tilde C$ be the underlying f...

conjecture | open | MathDB 页面 | 原文线索

V2-053555 | MATHDB-379434

The Reduction Conjecture for universal graphs

摘录 / EXCERPT: Let C be a finite or countable graph with no isolated vertices, and decompose it into blocks, its 2-connected components. Let $\tilde C$ be the underlying tree of C: its vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-053556 | MATHDB-379439

Plesník's chromatic decomposition conjecture for complete graphs

摘录 / EXCERPT: A k-decomposition of the complete graph K_n is a decomposition into k spanning subgraphs $G_1,\ldots,G_k$, and for a graph parameter p let ... Here χ denotes chromatic n...

conjecture | open | MathDB 页面 | 原文线索

V2-053557 | MATHDB-379440

Mixed chromatic-clique decomposition bound

摘录 / EXCERPT: For $0\leq m\leq k$, define ... Thus $\chi_0(k;K_n)=\omega(k;K_n)$ and $\chi_k(k;K_n)=\chi(k;K_n)$. Mixed decomposition conjecture. Let m and $n\geq 1$ be non-negative integers. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-053558 | MATHDB-379451

Karp's evasiveness conjecture for monotone graph properties

摘录 / EXCERPT: Let a graph property be a partition of the unlabeled graphs on a fixed number of vertices into graphs with and without the property. The property is monotone if it is preserved und...

conjecture | open | MathDB 页面 | 原文线索

V2-053559 | MATHDB-379539

The component-isomorphism conjecture for polynomial digraphs

摘录 / EXCERPT: Let $\Phi(x,y)$ be a standard polynomial of partial degree d. Let $G(\Phi)$ be its associated polynomial digraph, and let $G(\Phi)^*$ denote the nonsingular part of this digraph....

conjecture | open | MathDB 页面 | 原文线索

V2-053560 | MATHDB-379540

Brualdi-Mena conjecture on components of graphs from standard pairings

摘录 / EXCERPT: Let (H,Φ) be a standard pairing, where H is a finite connected regular graph and $G(\Phi)^*$ denotes the graph associated with the pairing after removing singular components...

conjecture | open | MathDB 页面 | 原文线索

V2-053561 | MATHDB-379566

The constant-term-one conjecture for F-polynomials

摘录 / EXCERPT: Fix an initial exchange matrix B^0, an initial vertex t_0, and a vertex t. Let $F_{\ell;t}^{B^0;t_0}(y_1,\dots,y_n)$ be the associated F-polynomial of the cluster variable...

conjecture | solved | MathDB 页面 | 原文线索

V2-053562 | MATHDB-379572

The sign-coherence conjecture for denominator vectors

摘录 / EXCERPT: Let $\mathbf{d}_{i;t}^{B^0;t_0}$ be the denominator vectors of the cluster variables at a vertex t relative to the initial vertex t_0. A collection of vectors is sign-coherent...

conjecture | open | MathDB 页面 | 原文线索

V2-053563 | MATHDB-379573

The denominator-vector injectivity conjecture for cluster monomials

摘录 / EXCERPT: Let Σ_{t_0} be an initial seed, and let $\mathbf{d}_{\mathbf{a};t}^{B^0;t_0}$ denote the denominator vector of a cluster monomial at vertex t. Denominator-vector injectivi...

conjecture | open | MathDB 页面 | 原文线索

V2-053564 | MATHDB-379584

Parity edge-coloring numbers of balanced complete bipartite graphs

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with n vertices in each part. Let $p(G)$ and $\widehat{p}(G)$ denote the minimum numbers of colors in a parity edge-coloring and a s...

conjecture | open | MathDB 页面 | 原文线索

V2-053565 | MATHDB-379585

Equality of parity edge-coloring numbers for bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph. Let $p(G)$ and $\widehat{p}(G)$ denote the minimum numbers of colors in a parity edge-coloring and a strong parity edge-coloring, respectively. The bi...

conjecture | open | MathDB 页面 | 原文线索

V2-053566 | MATHDB-379591

The Dynkin-presentation conjecture for discriminant-complement groups

摘录 / EXCERPT: Let f be a singular function with Milnor number $4mu$, and let a Dynkin diagram of f have vertex cardinality $4mu$. Write $t_1,cdots,t_4mu$ for generators indexed by its vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-053567 | MATHDB-379622

Positive-coefficient conjecture for adjacent edges in independence polynomials

摘录 / EXCERPT: Let $\mathsf K_n$ be the complete graph and let $Z(\mathbf l\mathsf K_n;\mathbf y)$ be the partition function of its independent-set system. For edges e,f, let $\Delta Z\{e,f\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-053568 | MATHDB-379623

K4-minor conjecture for negative coefficients of graph independence Rayleigh differences

摘录 / EXCERPT: Let $\mathsf G$ be a graph and let e,f be edges of $\mathsf G$. Write $\Delta Z(\mathbf l\mathsf G)\{e,f\}$ for the Rayleigh difference of the independence-set partition function...

conjecture | open | MathDB 页面 | 原文线索

V2-053569 | MATHDB-379624

Rayleigh conjecture for graph component-size weights

摘录 / EXCERPT: Let $\mathsf G=(V,E)$ be a finite connected graph. For each independent edge set S of $\mathsf G$, let $\omega_G(S)$ be the product of the sizes of the connected components of...

conjecture | open | MathDB 页面 | 原文线索

V2-053570 | MATHDB-379669

Rationality and Zhu algebra semisimplicity conjecture for vertex operator algebras

摘录 / EXCERPT: Rationality-semisimplicity conjecture. V is rational if and only if $A(V)$ is semisimple.

conjecture | open | MathDB 页面 | 原文线索

V2-053571 | MATHDB-379672

Minimal layered-triangulation conjecture for handlebodies

摘录 / EXCERPT: Let H_g be a genus-g handlebody, and let τ be a one-vertex triangulation of its boundary. A $[\tau]$-layered-triangulation is a layered-triangulation of H_g extending...

conjecture | open | MathDB 页面 | 原文线索

V2-053572 | MATHDB-379716

The conjecture relating Whitney homology and hyperarboreal Lie algebras

摘录 / EXCERPT: Let $\mathcal{HA}_n$ be the poset of hyperarbres, and let $\operatorname{WH}_i(\mathcal{HA}_n)$ denote its Whitney homology groups. Define the symmetric-function generating series...

conjecture | open | MathDB 页面 | 原文线索

V2-053573 | MATHDB-379764

De Simone-Körner conjecture on hereditary normality of graphs

摘录 / EXCERPT: De Simone-Körner conjecture. A graph G is hereditarily normal if and only if neither G nor its complement contains a 5-cycle or a 7-cycle as an induced subgraph.

conjecture | open | MathDB 页面 | 原文线索

V2-053574 | MATHDB-379843

Thomassen's conjecture for nonorientable surface triangulations

摘录 / EXCERPT: Nonorientable analogue of Thomassen's conjecture. Every such triangulation contains an NSC such that the two surfaces separated by the NSC have Euler genera h and $g-h$, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-053575 | MATHDB-379844

Conjecture on nonseparating cycle types in nonorientable triangulations

摘录 / EXCERPT: Conjecture on nonseparating cycle types. If $g \geq 3$ is odd, every triangulation of N_g has a nonseparating cycle that is one-sided and orientable-leaving, one that is one-side...

conjecture | open | MathDB 页面 | 原文线索

V2-053576 | MATHDB-379864

The spectral reconstruction conjecture for symmetric matrices

摘录 / EXCERPT: Let A be a real $n \times n$ symmetric matrix. For each i, write A_i for the matrix obtained by deleting the ith row and column, and let $\operatorname{eigen}(A)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-053577 | MATHDB-379865

The twisted-product refinement of the spectral reconstruction conjecture

摘录 / EXCERPT: Let S_n be the permutation group on n letters and let $\mathbb Z_2^n$ denote the group of diagonal sign matrices. Twisted-product refinement. The group $G(A)$ in the spectral r...

conjecture | open | MathDB 页面 | 原文线索

V2-053578 | MATHDB-379866

The permutation reconstruction conjecture for simple eigenvectors of hypomorphic matrices

摘录 / EXCERPT: Let A and B be two hypomorphic matrices, meaning that $B=\Sigma(A)$ for a hypomorphism Σ as in the surrounding reconstruction setting. Let λ_i be a simple eigenv...

conjecture | open | MathDB 页面 | 原文线索

V2-053579 | MATHDB-379934

Minimal-rank completion conjecture for bipartite chordal patterns

摘录 / EXCERPT: Let a partial matrix have a bipartite graph whose vertices represent its rows and columns and whose edges represent specified entries. A fully specified submatrix is one in which e...

conjecture | open | MathDB 页面 | 原文线索

V2-053580 | MATHDB-379940

The two-dimensional kings graph half-dependent density conjecture

摘录 / EXCERPT: Two-dimensional kings graph density conjecture. The half-dependent limiting density is

conjecture | open | MathDB 页面 | 原文线索

V2-053581 | MATHDB-379964

Dantzig conjecture for the deformation graph of Dantzig figures

摘录 / EXCERPT: Dantzig conjecture. The graph $E(d)$ is strongly connected: for any vertices x,y of $E(d)$, there is an oriented path from x to y.

conjecture | open | MathDB 页面 | 原文线索

V2-053582 | MATHDB-379965

Strong Dantzig conjecture for fundamental deformations

摘录 / EXCERPT: Strong Dantzig conjecture. Every fundamental deformation of a d-dimensional Dantzig figure is good; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-053583 | MATHDB-379966

3-geodesic conjecture for Dantzig figures

摘录 / EXCERPT: 3-geodesic conjecture. For every d-dimensional Dantzig figure (P,x,y),

conjecture | open | MathDB 页面 | 原文线索

V2-053584 | MATHDB-379969

Spectral Turán conjecture for the two largest adjacency eigenvalues

摘录 / EXCERPT: Let G be a graph with m edges that contains no complete subgraph K_{r+1}. Write $\mu_1(G)\geq \mu_2(G)\geq\cdots$ for the eigenvalues of its adjacency matrix. Spectral Turán...

conjecture | open | MathDB 页面 | 原文线索

V2-053585 | MATHDB-379982

Factor-criticality conjecture for K_{1,5}-free 3-vertex-critical graphs

摘录 / EXCERPT: Let G be a finite simple graph. It is factor-critical if, for every vertex $v\in V(G)$, the graph $G-v$ has a perfect matching. The graph G is $K_{1,5}$-free if it has no induc...

conjecture | open | MathDB 页面 | 原文线索

V2-053586 | MATHDB-379986

Injectivity conjecture for classical spatial graphs in virtual spatial graph theory

摘录 / EXCERPT: Injectivity conjecture. If two classical spatial graphs are virtually equivalent, then they are classically equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-053587 | MATHDB-380037

Uniqueness conjecture for the Rudvalis enhanced VOA with prescribed local algebra

摘录 / EXCERPT: Let $(U,Y,{\bf 1},\Omega)$ be a nice rational enhanced VOA with local algebra $\mathcal{A}(\Omega)\cong\mathcal{A}(\Omega_{\operatorname{\textsl{Ru}}})$. Assume that U has rank...

conjecture | open | MathDB 页面 | 原文线索

V2-053588 | MATHDB-380040

Archimedes configuration conjecture for uniform sorting networks

摘录 / EXCERPT: Archimedes configurations. For all $t\in(0,1)$, the scaled configuration converges in distribution to the Archimedes measure in the vague topology for random Borel measures on...

conjecture | open | MathDB 页面 | 原文线索

V2-053589 | MATHDB-380041

Great-circle conjecture for uniform sorting networks

摘录 / EXCERPT: Let ω_n be an n-element uniform sorting network, let ${\mathbb S}_n$ denote the permutahedron, and let d_∞ be the distance from a sorting network to a great circle...

conjecture | open | MathDB 页面 | 原文线索

V2-053590 | MATHDB-380096

The asymptotic shape conjecture for effective-resistance balls in the square lattice

摘录 / EXCERPT: Asymptotic shape conjecture. There exists a nonempty compact subset B_0 of $\mathbb{R}^2$ such that, for every positive number ε,

conjecture | open | MathDB 页面 | 原文线索

V2-053591 | MATHDB-380097

The bounded-variance conjecture for effective resistance on the square lattice

摘录 / EXCERPT: Bounded-variance conjecture. As v tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-053592 | MATHDB-380098

The inverse-square variance conjecture for left-right resistance on square grids

摘录 / EXCERPT: Inverse-square variance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053593 | MATHDB-380160

Lian-Zuckerman's lifting conjecture for topological vertex algebras

摘录 / EXCERPT: Lian-Zuckerman's lifting conjecture. The product and bracket extend to a G_∞ structure on V. The conjecture is presented as an open problem in the source, with no resoluti...

conjecture | open | MathDB 页面 | 原文线索

V2-053594 | MATHDB-380165

Sarnak's conjecture on Ramanujan random regular graphs

摘录 / EXCERPT: Let $G_{n,d}$ be a uniformly random simple d-regular graph, and write ... for the maximum absolute value of its nontrivial adjacency eigenvalues. A d-regular graph is Ramanujan w...

conjecture | open | MathDB 页面 | 原文线索

V2-053595 | MATHDB-380177

Square-root diameter bound for random homomorphism ranges

摘录 / EXCERPT: Let G_n be a family of vertex-transitive bipartite graphs with $|G_n|\to\infty$. Let Δ_n denote the diameter of G_n, and let f_n be uniformly random in...

conjecture | open | MathDB 页面 | 原文线索

V2-053596 | MATHDB-380201

The fan embedding conjecture for complete bipartite graphs

摘录 / EXCERPT: Let $K_{m,n}$ be a complete bipartite graph with independent vertex sets $\{a_1,\dots,a_m\}$ and $\{b_1,\dots,b_n\}$. Place a_i at $(i,0)$ on the x-axis and b_j at $(0,j)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-053597 | MATHDB-380203

Tracy-Widom limit and Ramanujan proportions for random regular graphs

摘录 / EXCERPT: Tracy-Widom and proportion conjecture. The distribution of $\lambda_\pm(G)$, normalized in this way, converges as $N\to\infty$ to the $\beta=1$ Tracy-Widom distribution rather than...

conjecture | open | MathDB 页面 | 原文线索

V2-053598 | MATHDB-380222

Submodularity conjecture for influence in social networks

摘录 / EXCERPT: Influence submodularity conjecture. The function σ is submodular.

conjecture | open | MathDB 页面 | 原文线索

V2-053599 | MATHDB-380248

The nonexistence conjecture for planar covers of the displayed cover of K_4

摘录 / EXCERPT: Nonexistence conjecture for the displayed cover. For every finite planar cover

conjecture | open | MathDB 页面 | 原文线索

V2-053600 | MATHDB-380292

The adjacent vertex distinguishing chromatic conjecture

摘录 / EXCERPT: Zhang's conjecture. If $G\neq C_5$ and $G\neq K_2$, where C_5 is the cycle of size 5, then the avd-chromatic number of G is at most

conjecture | open | MathDB 页面 | 原文线索

V2-053601 | MATHDB-380297

Subdivision invariance conjecture for graph Brill-Noether numbers

摘录 / EXCERPT: Let G be a graph, and let Γ be the associated $\mathbb Q$-graph. For an integer $r\geq 1$, let $D(G,r)$ and $D(\Gamma,r)$ be the minimal degrees of a g^r_d on G and...

conjecture | open | MathDB 页面 | 原文线索

V2-053602 | MATHDB-380329

The primitive pure graph conjecture

摘录 / EXCERPT: Let a pure graph G (or its parity class $[G]$) be primitive when, for every family $\mathcal{F}$ of pure classes, $[G]\in\overline{\mathcal{F}}$ implies $[G]\in\mathcal{F}$. A pu...

conjecture | open | MathDB 页面 | 原文线索

V2-053603 | MATHDB-380331

The invertibility conjecture for self-complementary symmetric graphs

摘录 / EXCERPT: Let G be a self-complementary symmetric graph of order n, let $\overline{G}$ denote its complement, and let G^{-1} denote its inverse under the graph operation used in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-053604 | MATHDB-380333

The complementation-set correspondence conjecture

摘录 / EXCERPT: Let G be a bicoloured graph with no isolated vertices and no twins, where twins are vertices having identical relevant neighbourhoods. Let $[G]$ denote the parity class of G, a...

conjecture | open | MathDB 页面 | 原文线索

V2-053605 | MATHDB-380343

The smallest-eigenspace dimension conjecture for qualitative independence graphs

摘录 / EXCERPT: Smallest-eigenspace dimension conjecture. For all positive integers k, the eigenspace corresponding to this smallest eigenvalue has dimension ...

conjecture | open | MathDB 页面 | 原文线索

V2-053606 | MATHDB-380346

The generalized eigenspace-dimension conjecture for uniform qualitative independence graphs

摘录 / EXCERPT: Generalized eigenspace-dimension conjecture. For all positive integers c,k, this eigenspace has dimension ...

conjecture | open | MathDB 页面 | 原文线索

V2-053607 | MATHDB-380412

The hereditary speed conjecture for ordered hypergraphs

摘录 / EXCERPT: Let $\mathcal{P}$ be a hereditary property of ordered hypergraphs. For every constant $c>0$, suppose there exists an $N=N(c)\in\mathbb{N}$ such that $|\mathcal{P}_N|>c^N$. The orde...

conjecture | open | MathDB 页面 | 原文线索

V2-053608 | MATHDB-380417

Conjecture on the exact value of m(n,r,1,k)

摘录 / EXCERPT: Exact-value conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-053609 | MATHDB-380418

Conjecture on the six-colour two-colour connectivity function

摘录 / EXCERPT: The conjecture. The maximum order of a k-connected subgraph using at most two colours in every 6-colouring of K_n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053610 | MATHDB-380419

Odd-colour phase-transition conjecture for multicoloured connectivity

摘录 / EXCERPT: Odd-colour phase-transition conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-053611 | MATHDB-380434

The Hom-complex realization conjecture for diameter-one graphs

摘录 / EXCERPT: Hom-complex realization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053612 | MATHDB-380489

Linear chromatic-root bound in maxmaxflow

摘录 / EXCERPT: Maxmaxflow chromatic-root conjecture. There exist universal constants $C(\Lambda)<\infty$ such that every chromatic root of every loopless graph of maxmaxflow Λ lies in

conjecture | open | MathDB 页面 | 原文线索

V2-053613 | MATHDB-380579

Amicability conjecture for vertex pairs in networks

摘录 / EXCERPT: Amicability conjecture. For every network G and every pair of distinct vertices v and w, the pair $\{v,w\}$ is amicable in G.

conjecture | open | MathDB 页面 | 原文线索

V2-053614 | MATHDB-380634

The Conway-Kinoshita-Terasaka chord-diagram inequivalence conjecture

摘录 / EXCERPT: Conway-Kinoshita-Terasaka conjecture. The chord diagrams C and KT are not equivalent.

conjecture | refuted | MathDB 页面 | 原文线索

V2-053615 | MATHDB-380635

The elementary-transformation generation conjecture for chord diagrams

摘录 / EXCERPT: Elementary-transformation generation conjecture. The group of elementary transformations is generated by transformations of the following two types: reflecting a share across anoth...

conjecture | open | MathDB 页面 | 原文线索

V2-053616 | MATHDB-380662

The Master Family degeneration conjecture for mirror-symmetric hypersurfaces

摘录 / EXCERPT: Master Family degeneration conjecture. Vertex algebras arising from Mirror Symmetry for hypersurfaces defined by Δ and Δ^* are degenerations of the Master Family of v...

conjecture | open | MathDB 页面 | 原文线索

V2-053617 | MATHDB-380740

Lian-Zuckerman homotopy Gerstenhaber conjecture for topological vertex operator algebras

摘录 / EXCERPT: A topological vertex operator algebra is the algebraic structure considered in the paper, and a homotopy Gerstenhaber algebra is the corresponding higher-algebraic structure introd...

conjecture | solved | MathDB 页面 | 原文线索

V2-053618 | MATHDB-380741

Kimura-Voronov-Zuckerman completion conjecture for topological vertex operator algebras

摘录 / EXCERPT: Let V be a topological vertex operator algebra, and let an appropriate topological completion mean the completion of V required in the cited construction. A genus-zero topologi...

conjecture | open | MathDB 页面 | 原文线索

V2-053619 | MATHDB-380758

The rational vertex operator algebra decomposition conjecture

摘录 / EXCERPT: Let V be a rational vertex operator algebra, and let $W_{1},\dots,W_{r}$ be a complete set of representatives of equivalence classes of irreducible V-modules. Here $U(V)$ is th...

conjecture | open | MathDB 页面 | 原文线索

V2-053620 | MATHDB-380767

Classification conjecture for irreducible modules of $W_{1+\infty,-N}$

摘录 / EXCERPT: Let $W_{1+\infty,-N}$ be the vertex algebra at central charge $-N$, let $\mathfrak h$ be the parameter space used to construct the modules $V(\lambda,-N)$, and let $V(\lambda,-N)$...

conjecture | open | MathDB 页面 | 原文线索

V2-053621 | MATHDB-380788

The list-chromatic-index criticality conjecture

摘录 / EXCERPT: List-chromatic-index criticality conjecture. Every χ'_ℓ-critical graph is χ'-critical.

conjecture | open | MathDB 页面 | 原文线索

V2-053622 | MATHDB-380808

The maximum-spanning-tree conjecture for Moore graphs

摘录 / EXCERPT: Let G be a simple graph, and let $\tau(G)$ denote its number of spanning trees. A k-regular graph of girth g with the minimum possible number of vertices is called a (k,g)-...

conjecture | open | MathDB 页面 | 原文线索

V2-053623 | MATHDB-380809

The lattice-quotient conjecture for extremal surface graphs

摘录 / EXCERPT: For $3\leq k\leq 6$, let $\mathcal{X}_{n,k,g}$ be the class of graphs considered in the source and define ... where g is fixed sufficiently large and $\tau(G)$ is the number of spa...

conjecture | open | MathDB 页面 | 原文线索

V2-053624 | MATHDB-380850

The discrete Möbius flattening conjecture for non-contractible polygons

摘录 / EXCERPT: Let P be an embedded non-contractible closed polygon in $\mathbb{RP}^2$, and call a vertex configuration a flattening when it has the polygonal analogue of an inflection point. D...

conjecture | open | MathDB 页面 | 原文线索

V2-053625 | MATHDB-380851

The discrete Segre flattening conjecture for spherical polygons

摘录 / EXCERPT: Let P be an embedded closed polygon in S^2 that bisects the area, and call a vertex configuration a flattening when it has the polygonal analogue of an inflection point. Discre...

conjecture | open | MathDB 页面 | 原文线索

V2-053626 | MATHDB-380852

The hyperplane-intersection flattening conjecture for strictly convex projective polygons

摘录 / EXCERPT: Let P be a strictly convex polygon in $\mathbb{RP}^d$. Suppose that P intersects a hyperplane with multiplicity k, where $k\geq d+2$. Call a vertex configuration a flattening...

conjecture | open | MathDB 页面 | 原文线索

V2-053627 | MATHDB-380924

Hyperbolicity violation conjecture for high-dimensional neural networks

摘录 / EXCERPT: Let f be a mapping (neural network) as defined in the paper, with sufficiently high dimension d. A hyperbolicity violation conjecture asserts that there exists at least one bif...

conjecture | open | MathDB 页面 | 原文线索

V2-053628 | MATHDB-380925

Existence of a codimension epsilon bifurcation set conjecture

摘录 / EXCERPT: Let f be a mapping (neural network) with sufficiently high dimension d, and let U be a bifurcation chain set as in the hyperbolicity violation conjecture. The existence of a...

conjecture | open | MathDB 页面 | 原文线索

V2-053629 | MATHDB-380926

Periodic-window probability conjecture for high-dimensional neural networks

摘录 / EXCERPT: Let f be a mapping (neural network) and let U be a bifurcation chain set as in the hyperbolicity violation conjecture. The periodic-window probability conjecture asserts that,...

conjecture | open | MathDB 页面 | 原文线索

V2-053630 | MATHDB-380927

Non-generality of structural stability conjecture for high-dimensional systems

摘录 / EXCERPT: Let f be a mapping (neural network) with sufficiently high dimension d, let U be a bifurcation chain set as in the hyperbolicity violation conjecture, and let V be the chai...

conjecture | open | MathDB 页面 | 原文线索

V2-053631 | MATHDB-380928

Robust chaos conjecture for high-dimensional bounded nonlinear systems

摘录 / EXCERPT: A robust chaos conjecture asserts that chaos is a robust, high-probability behavior for high-dimensional, bounded, nonlinear dynamical systems. The paper reports near-unity experim...

conjecture | open | MathDB 页面 | 原文线索

V2-053632 | MATHDB-380941

The abelian intertwining algebra conjecture for the sum of G-simple currents

摘录 / EXCERPT: Let V be a vertex operator algebra and let G be a group of automorphisms of finite order of V. A G-simple current is an irreducible weak g-twisted V-module whose tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-053633 | MATHDB-380946

The homotopy Gerstenhaber structure conjecture for topological vertex operator algebras

摘录 / EXCERPT: Let $V^\bullet[\cdot]$ be a topological vertex operator algebra (TVOA), with dot product and bracket product defined by the vertex-operator operations above. The homotopy Gerstenha...

conjecture | open | MathDB 页面 | 原文线索

V2-053634 | MATHDB-380982

The growth-rate conjecture for triangulations with at least six triangles at each internal vertex

摘录 / EXCERPT: Let a_n be the number of triangulations of a fixed convex n-gon such that at least six triangles meet at each internal vertex. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-053635 | MATHDB-381001

The one-entangled-edge and common-vertex conjecture for graph density matrices

摘录 / EXCERPT: Let G be a graph with $|V|=pq$, and let $entangled edges$ mean edges whose associated pure states are entangled. One-entangled-edge and common-vertex conjecture. If G has only...

conjecture | open | MathDB 页面 | 原文线索

V2-053636 | MATHDB-381002

The entanglement-of-formation averaging conjecture for graph density matrices

摘录 / EXCERPT: Let G be a graph with $|V|=pq$ and m edges. For each edge, let $\sigma(k)$ denote the pure density matrix associated with the k-th edge. If $\sigma(G)$ is entangled in...

conjecture | open | MathDB 页面 | 原文线索

V2-053637 | MATHDB-381003

The star graph maximizer conjecture for entanglement of formation

摘录 / EXCERPT: Let $\mathcal{G}_{n}^{c}$ be the set of all connected graphs on n vertices, and let $G\in\mathcal{G}_{n}^{c}$ with $|V|=pq$. Star graph maximizer conjecture. ... This asserts that...

conjecture | open | MathDB 页面 | 原文线索

V2-053638 | MATHDB-381027

MED decomposition conjecture for subcubic 2-connected graphs

摘录 / EXCERPT: MED decomposition conjecture. Every 2-connected graph with maximum degree 3 has a MED decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-053639 | MATHDB-381046

The d-step conjecture for simple polytopes

摘录 / EXCERPT: Let P be a simple d-polytope with $2d$ facets, and let u and v be two complementary vertices, meaning vertices not lying in a common facet. The d-step conjecture. There i...

conjecture | open | MathDB 页面 | 原文线索

V2-053640 | MATHDB-381066

Conjectures on classification of Hamiltonian operators of arbitrary odd order

摘录 / EXCERPT: Classification conjectures. (1) Every Hamiltonian operator is equivalent to one whose leading coefficient is 1. For $N=2n+3\geq 3$, this is equivalent to an operator with leading...

conjecture | open | MathDB 页面 | 原文线索

V2-053641 | MATHDB-381068

NP-completeness of deciding whether the covering number is at most k

摘录 / EXCERPT: Let G be a graph, and let $\sigma(G)$ denote the minimum number of equivalence relations needed to cover the line graph structure under the paper's definition. For an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-053642 | MATHDB-381075

Approximation of arbitrary vertex couplings by magnetic Schrödinger operators

摘录 / EXCERPT: Let H_ε^ω be the magnetic Schrödinger operator obtained by replacing the approximating graph by a network with fat edge width ε, replacing the δ...

conjecture | open | MathDB 页面 | 原文线索

V2-053643 | MATHDB-381103

Graham's tree reconstruction conjecture

摘录 / EXCERPT: Graham's tree reconstruction conjecture. For each sequence of natural numbers $a_0,a_1,a_2,\ldots$, all the conditions $|L^{(j)}(G)|=a_j$ for $j\geq 0$ are satisfied by at most one...

conjecture | open | MathDB 页面 | 原文线索

V2-053644 | MATHDB-381126

The global linkage characterization conjecture for rigid graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a rigid graph and let $u,v\in V$. A pair of vertices is globally linked if every generic realisation of G determines the distance between its two vertices. Globa...

conjecture | open | MathDB 页面 | 原文线索

V2-053645 | MATHDB-381129

Equality of susceptibility and component-size thresholds for size rules

摘录 / EXCERPT: Let $ell geqslant 2$ and let ${\mathcal R}$ be an ell-vertex size rule. Define $t_{\mathrm{b}}^{\mathcal R}$ as the supremum of the set of $t \geqslant 0$ for which the suscept...

conjecture | open | MathDB 页面 | 原文线索

V2-053646 | MATHDB-381147

The smallest cyclically 4-connected snark with oddness 4

摘录 / EXCERPT: A snark is a connected cubic graph with no proper 3-edge-colouring. A snark is cyclically 4-connected if no edge-cut of size at most 3 separates two subgraphs each containing a c...

conjecture | open | MathDB 页面 | 原文线索

V2-053647 | MATHDB-381155

Erdős-Hajnal conjecture for rainbow triangles

摘录 / EXCERPT: Let r and t be positive integers, and let $r(t)$ denote the two-color Ramsey number: the least integer m such that every 2-coloring of the edges of the complete graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-053648 | MATHDB-381160

Ilić-Stevanović's spectral-moment conjecture for Volkmann trees

摘录 / EXCERPT: Ilić-Stevanović's conjecture. The Volkmann tree has maximum spectral moment

conjecture | open | MathDB 页面 | 原文线索

V2-053649 | MATHDB-381163

The semi-arc k-visibility graph edge-maximization conjecture

摘录 / EXCERPT: A semi-arc k-visibility graph is a graph represented by semi-arcs, with visibility allowing at most k obstructing semi-arcs; let n be the number of vertices. Semi-arc k-vis...

conjecture | open | MathDB 页面 | 原文线索

V2-053650 | MATHDB-381171

Du and Shi's minimum spectral radius conjecture for connected graphs

摘录 / EXCERPT: Let P_α be a path of order α, and form a graph by blowing up each vertex of P_α to a clique of order k. Consider connected graphs of order $k\alpha$ with ind...

conjecture | open | MathDB 页面 | 原文线索

V2-053651 | MATHDB-381172

Mkrtchyan-Petrosyan-Vardanyan conjecture on maximum matchings

摘录 / EXCERPT: Mkrtchyan-Petrosyan-Vardanyan conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-053652 | MATHDB-381178

Price-of-symmetrisation conjecture for average transmission

摘录 / EXCERPT: Price-of-symmetrisation conjecture. For $2\leq n\leq 10$,

conjecture | open | MathDB 页面 | 原文线索

V2-053653 | MATHDB-381181

Lukoťka-Máčajová-Mazák-Škoviera conjecture on the order of cubic graphs of given oddness

摘录 / EXCERPT: Let G be a 2-edge-connected cubic graph, and let $\omega(G)$ denote its oddness, the minimum number of odd circuits in a 2-factor of G. Lukoťka-Máčajová-Mazák-Škoviera conj...

conjecture | open | MathDB 页面 | 原文线索

V2-053654 | MATHDB-381182

Component spanning trees avoiding any prescribed label

摘录 / EXCERPT: Let $\hat{\mathcal{G}}_{2n+1}$ be the graph whose components A have labeled edges, and let η be a label of a component A. A component spanning tree of A is a spanning tr...

conjecture | open | MathDB 页面 | 原文线索

V2-053655 | MATHDB-381186

Zero forcing and path number under vertex sums

摘录 / EXCERPT: Let G and H be graphs with an identified vertex v. Write $G\,\makebox[0ex][l]{$+$}\raisebox{-.75ex}{\makebox[2ex]{${v}$}}\,H$ for their vertex sum at v. Suppose that...

open_problem | open | MathDB 页面 | 原文线索

V2-053656 | MATHDB-381191

The neighborhood characterization conjecture for distance antimagic graphs

摘录 / EXCERPT: Let G be a graph. A D-neighborhood of a vertex is the set of vertices at distance at most D from it. Neighborhood characterization conjecture. A graph is D-distance antimag...

conjecture | open | MathDB 页面 | 原文线索

V2-053657 | MATHDB-381200

The bounded-degree almost sure period two conjecture for majority dynamics

摘录 / EXCERPT: Let G be an infinite graph of bounded degree, and choose the initial opinions independently and uniformly from $\{-1,1\}$. Under majority dynamics, each node's opinion almost sur...

conjecture | open | MathDB 页面 | 原文线索

V2-053658 | MATHDB-381203

Akbari et al.'s upper-bound conjecture for edge Roman domination

摘录 / EXCERPT: Akbari et al.'s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053659 | MATHDB-381212

Expected meeting time conjecture for random walks on regular graphs

摘录 / EXCERPT: Let G) be a connected d-regular graph with adjacency matrix A and n vertices. Let $\xi_1,\xi_2,\ldots,\xi_n$ be orthogonal eigenvectors of A, with the conditions stated b...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053660 | MATHDB-381217

Erdős-Sós extremal coloring conjecture for rainbow triangles

摘录 / EXCERPT: Let K_n be the complete graph on n vertices with its edges colored using three colors. Let $F(G)$ be the number of rainbow triangles in a coloring G, and let ... A coloring is...

conjecture | open | MathDB 页面 | 原文线索

V2-053661 | MATHDB-381220

Deletion-triangle subdivision conjecture for Hilbert bases of cuts

摘录 / EXCERPT: Let G be a graph in the class $\mathscr{H}$ of graphs whose cut semigroup has the Hilbert basis property, and let $f\in E(G)$ be an edge. Write $G\setminus f$ for deletion of f...

conjecture | open | MathDB 页面 | 原文线索

V2-053662 | MATHDB-381233

Characterization of strongly Tutte-descriptive activities

摘录 / EXCERPT: Characterization of strongly Tutte-descriptive activities. For any graph G, an activity ψ is strongly Tutte-descriptive if and only if there exists a decision tree Δ...

conjecture | open | MathDB 页面 | 原文线索

V2-053663 | MATHDB-381241

A strict improvement for the asymptotic Ky Fan eigenvalue sum

摘录 / EXCERPT: For a graph G, let $\tau_k(n)$ denote the maximum, over graphs on n vertices, of the sum of the k largest eigenvalues of G, and let ... for fixed $k\geq 1$. The asymptotic Ky...

conjecture | open | MathDB 页面 | 原文线索

V2-053664 | MATHDB-381243

The vertex bound conjecture for critical graphs of tree-depth k

摘录 / EXCERPT: Let G be a graph of tree-depth k, meaning that k is the least number of labels in a vertex ranking of G such that every path joining two vertices with the same label contai...

conjecture | open | MathDB 页面 | 原文线索

V2-053665 | MATHDB-381261

Stephenson's conjecture on electrical-network approximation of Riemann mappings

摘录 / EXCERPT: Let a simply-connected, bounded Jordan domain be approximated by a sequence of electrical networks, for instance networks arising from a sequence of disk packings. Choose conductan...

conjecture | open | MathDB 页面 | 原文线索

V2-053666 | MATHDB-381265

Equality of Krull-dimension bounds for Khovanov-Rozansky homology

摘录 / EXCERPT: Let G be a directed graph and let E be a subset of its edge set $E(G)$. Write $\mathscr{H}_0^\mathbb{Q}(G)$ for the degree-zero Khovanov-Rozansky homology over $\mathbb{Q}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053667 | MATHDB-381291

Small blow-up conjecture for triangle covers

摘录 / EXCERPT: Let G be a graph, and for a positive integer t let $G(t)$ denote the graph obtained by replacing every vertex of G with an independent set of size t, with edges between rep...

conjecture | open | MathDB 页面 | 原文线索

V2-053668 | MATHDB-381294

Harary-Schwenk-Scott conjecture for locally finite trees

摘录 / EXCERPT: Let G be a locally finite tree. A graph G is reconstructible if every graph hypomorphic to G is isomorphic to G, where hypomorphism means that there is a bijection between...

conjecture | open | MathDB 页面 | 原文线索

V2-053669 | MATHDB-381304

Irredundant broadcast equality conjecture for graphs with no strength-one vertices

摘录 / EXCERPT: Let G be a graph. A broadcast f has positive-support vertices $V_f^+=\{v\in V(G):f(v)>0\}$ and strength-one vertices $V_f^1=\{v\in V(G):f(v)=1\}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-053670 | MATHDB-381305

Nearly platonic graph two-exceptional-faces conjecture

摘录 / EXCERPT: Let G be a vertex-regular planar graph, and suppose that all but two faces of G have the same degree. Nearly platonic graph conjecture. The remaining two faces must have the sa...

conjecture | open | MathDB 页面 | 原文线索

V2-053671 | MATHDB-381315

The classification of digraphs of second minimal odd-period orbits

摘录 / EXCERPT: Second minimal odd-orbit digraph conjecture. For every integer $k\geq 3$, the digraph of any second minimal $(2k+1)$-orbit has one of $4k-3$ possible types up to inverses.

conjecture | open | MathDB 页面 | 原文线索

V2-053672 | MATHDB-381345

Asymptotic ratio conjecture for spanning trees of suitable graphs

摘录 / EXCERPT: Asymptotic ratio conjecture. The limit exists:

conjecture | open | MathDB 页面 | 原文线索

V2-053673 | MATHDB-381347

Polynomial weak colouring-number conjecture for $K^*_{s,t}$-minor-free graphs

摘录 / EXCERPT: Polynomial weak colouring-number conjecture. There exists a function f such that for every $K^*_{s,t}$-minor-free graph G and every $r\geqslant1$,

conjecture | open | MathDB 页面 | 原文线索

V2-053674 | MATHDB-381350

Hajós' cycle decomposition conjecture for Eulerian graphs

摘录 / EXCERPT: Hajós' conjecture. Every simple Eulerian graph G has a cycle decomposition with at most

conjecture | open | MathDB 页面 | 原文线索

V2-053675 | MATHDB-381351

Lin et al.'s vertex-count conjecture for ABC-minimal trees

摘录 / EXCERPT: Let t be the number of leaves of a tree, and call a tree t-minimal if it has t leaves and no other tree with the same number of leaves has a smaller ABC-index, where ... for a...

conjecture | open | MathDB 页面 | 原文线索

V2-053676 | MATHDB-381371

Rao-Rao k-factor conjecture for graphic degree sequences

摘录 / EXCERPT: Let π_1 be a graphic degree sequence, and let $\mathbf{k}$ denote the degree sequence in which every term is k. Two graphic degree sequences pack when they have realizations...

conjecture | open | MathDB 页面 | 原文线索

V2-053677 | MATHDB-381375

Dynamical splitting conjecture for shifted states on star graphs

摘录 / EXCERPT: Dynamical splitting conjecture. In the case $K=1$, the shifted state with $a<0$ leads to a solitary wave that moves towards the vertex point at $x=0$ along the only incoming edge,...

conjecture | open | MathDB 页面 | 原文线索

V2-053678 | MATHDB-381383

Nested-clique extremal conjecture for the inverse Turán problem

摘录 / EXCERPT: For positive integers $r_1\geq\dots\geq r_\ell$, let $K(r_1,\dots,r_\ell)$ be the multigraph consisting of nested copies of K_{r_i}: on vertex set $[r_1]$, overlay a copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-053679 | MATHDB-381384

Spectral-network pre-slicing and stability-condition conjecture

摘录 / EXCERPT: Spectral-network stability-condition conjecture. The collection $\mathcal{P}$ forms a pre-slicing of the standard $\mathbb{V}_{\mathcal{S}}$-collection, and $(Z,\mathcal{P})$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-053680 | MATHDB-381396

Unsplittability conjecture for cyclic configurations of rank at least four

摘录 / EXCERPT: A cyclic (n_k) configuration is a cyclic configuration in which every point lies on k lines and every line contains k points. Unsplittability conjecture. All cyclic (n_k) c...

conjecture | open | MathDB 页面 | 原文线索

V2-053681 | MATHDB-381399

The stable fixed point conjecture for target-free cliques in combinatorial threshold-linear networks

摘录 / EXCERPT: Stable fixed point conjecture. A subset $\sigma\subseteq[n]$ is the support of a stable fixed point if and only if σ is a target-free clique.

conjecture | open | MathDB 页面 | 原文线索

V2-053682 | MATHDB-381405

The truncation-curve conjecture for the universal even spin b[W]_-infinity algebra

摘录 / EXCERPT: Let $n\geq 2$ and let $\mathcal F^k(n)$ be the universal vertex algebra considered in the paper. A truncation curve in the (c,λ)-plane is parametrized by ... ... where ... ... Le...

conjecture | open | MathDB 页面 | 原文线索

V2-053683 | MATHDB-381412

The truncate-and-relax algorithm's threshold conjecture for hypergraph community recovery

摘录 / EXCERPT: The truncate-and-relax threshold conjecture. If $I_2(\alpha,\beta) > 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-053684 | MATHDB-381417

Persistence and robustness of oscillations in the mixed-mechanism phosphorylation system

摘录 / EXCERPT: Oscillation-persistence conjecture. There exist values of P_{tot} and S_{tot} such that, for K_{tot} arbitrarily large, the unique steady state in...

conjecture | open | MathDB 页面 | 原文线索

V2-053685 | MATHDB-381430

Cheeger constant conjecture for distance-regular graphs

摘录 / EXCERPT: Cheeger constant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053686 | MATHDB-381433

Cardoso-Luz conjecture on adverse graphs

摘录 / EXCERPT: Cardoso-Luz conjecture. All adverse graphs are $\mathcal{Q}$-graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-053687 | MATHDB-381451

Alfaro's conjecture on shortest directed networks

摘录 / EXCERPT: Alfaro's conjecture. If $A=B$, then a shortest (A,B)-network has no Steiner points and is a union of cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-053688 | MATHDB-381452

Akbari-Cameron-Khosravi conjecture on the maximum order of reduced graphs of given rank

摘录 / EXCERPT: Akbari-Cameron-Khosravi conjecture. For every integer $r\geqslant2$, the maximum order of any reduced graph of rank r is

conjecture | open | MathDB 页面 | 原文线索

V2-053689 | MATHDB-381454

Weak convergence conjecture for imitation dynamics

摘录 / EXCERPT: Let $\mathbf{A}$ be a game matrix and let α be a learning rate. Suppose that the maximal set of $I_G(t)$ is constant for all $t\geq t_0$. Let $\mathcal{X}^*$ denote the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-053690 | MATHDB-381455

Strong convergence conjecture for imitation dynamics

摘录 / EXCERPT: Let $\mathbf{A}$ be a game matrix and let α be a learning rate. Let $I_G(t)$ denote the relevant maximal-set quantity in the imitation dynamics, and let α^* be a lear...

conjecture | open | MathDB 页面 | 原文线索

V2-053691 | MATHDB-381469

The edge domination conjecture for connected regular graphs

摘录 / EXCERPT: Let G be a finite, simple, connected, Δ-regular graph of order n, where $\Delta\geq 3$. The edge domination conjecture asserts that ... Equality holds if and only if G h...

conjecture | open | MathDB 页面 | 原文线索

V2-053692 | MATHDB-381475

Fixed-size independent-set conjecture for highly connected chromatic graphs

摘录 / EXCERPT: Let G be an n-vertex k-chromatic ℓ-connected graph, and let $i_t(G)$ denote the number of independent sets of size t in G. Fixed-size independent-set conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-053693 | MATHDB-381487

Equality of critical parameters for loop and Bernoulli percolation at diverging degree

摘录 / EXCERPT: Let $(G_n)_n=(V_n,E_n)_n$ be a sequence of finite, connected, vertex-transitive graphs with vertex degree $\bDelta_n$ diverging as $n\to\infty$. For a fixed sequence and fixed u,...

conjecture | open | MathDB 页面 | 原文线索

V2-053694 | MATHDB-381493

Strengthened lower-bound conjecture for the circular flow number of snarks

摘录 / EXCERPT: Strengthened circular-flow conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053695 | MATHDB-381495

The period-two formula for the affine-quiver quantity $\tilde{K}_n$

摘录 / EXCERPT: Let $\tilde{K}_n$ be the determinant introduced above, namely ... Here the variables a_n and h_n are the cluster variables associated with the extending vertex and its neighborin...

conjecture | open | MathDB 页面 | 原文线索

V2-053696 | MATHDB-381496

The period-two formula for the affine-quiver quantity $\tilde{K}_n$ in type $\tilde{E}_8$

摘录 / EXCERPT: Let a_n be the cluster variables associated with the extending vertex, and let $\tilde{K}_n$ denote the quantity whose period-two behavior is being conjectured. Period-two formul...

conjecture | open | MathDB 页面 | 原文线索

V2-053697 | MATHDB-381502

The 1-2-3 Conjecture for neighbour-sum-distinguishing edge weightings

摘录 / EXCERPT: Let G be a connected graph with at least three vertices. An edge weighting assigns a weight from $\{1,2,3\}$ to every edge of G, and the sum at a vertex is the sum of the weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-053698 | MATHDB-381505

The lower-bound conjecture for the induced-forest ratio

摘录 / EXCERPT: Lower-bound conjecture. If G has minimum degree at least 2, then

conjecture | open | MathDB 页面 | 原文线索

V2-053699 | MATHDB-381508

Chromatic symmetric homology distinguishes the graph pairs in Figure 1

摘录 / EXCERPT: Chromatic symmetric homology distinction conjecture. Each such pair is distinguished by chromatic symmetric homology.

conjecture | open | MathDB 页面 | 原文线索

V2-053700 | MATHDB-381517

The rank-+1 hypergraph conjecture for quantum entropy inequalities

摘录 / EXCERPT: Rank-$n+1$ hypergraph conjecture. If an n-party inequality contracts on $(n+1)$-graphs, then it is a valid inequality on all hypergraphs of finite rank.

conjecture | open | MathDB 页面 | 原文线索

V2-053701 | MATHDB-381525

Shift-invariance conjecture for stochastic colored vertex-model probabilities

摘录 / EXCERPT: Shift-invariance conjecture. If $\\{\alpha,\beta\\}$ is not an inversion of w, then the three probability ratios satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-053702 | MATHDB-381534

Asymptotic planar Turán conjecture for cycles of length at least seven

摘录 / EXCERPT: Let G be an n-vertex C_{ℓ}-free plane graph, meaning that G contains no cycle of length ℓ, where $\ell\geq 7$. Write $e(G)$ for the number of edges of G. Asymptot...

conjecture | open | MathDB 页面 | 原文线索

V2-053703 | MATHDB-381552

The extremal characterization of global amoeba graphs of minimum degree one

摘录 / EXCERPT: Extremal global amoeba conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053704 | MATHDB-381554

Infinitesimal rigidity of hinged spanning cycles on the hypercube

摘录 / EXCERPT: Let the n-Roberts graph encode the facet adjacencies of the n-cube, and let a hinged spanning cycle be a spanning cycle of this graph whose corresponding $2n$ facet adjacencies...

conjecture | open | MathDB 页面 | 原文线索

V2-053705 | MATHDB-381560

Unitarity conjecture for exceptional W-algebras of type d535_k(\mathfrak{sp}_4,f_{subreg})

摘录 / EXCERPT: Let k be one of the four levels ... The vertex algebra under consideration is the exceptional W-algebra associated with the subregular nilpotent element f_{subreg} of...

conjecture | open | MathDB 页面 | 原文线索

V2-053706 | MATHDB-381562

Hofmeister-Joret-Wood conjecture on tree densities in minor-closed graph classes

摘录 / EXCERPT: Let s,t be integers with $t\geqslant s\geqslant 1$, and let $\mathcal{B}_{s,t}$ be the class of graphs containing no $K_{s,t}$ minor. For a fixed graph H, define an s-separat...

conjecture | open | MathDB 页面 | 原文线索

V2-053707 | MATHDB-381564

The heroic-set conjecture for oriented forests

摘录 / EXCERPT: Heroic-set conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-053708 | MATHDB-381578

Kim's Italian domination conjecture for Cartesian products of directed cycles

摘录 / EXCERPT: Let C_n denote the directed cycle of order n, and let $b3_I(D)$ be the Italian domination number of a digraph D. For an odd integer n, Kim's conjecture. ... This conjecture c...

conjecture | open | MathDB 页面 | 原文线索

V2-053709 | MATHDB-381579

Capacity region of the linear deterministic broadcast channel with a diamond message set

摘录 / EXCERPT: The capacity-region claim. The capacity region of the K-user linear deterministic broadcast channel for this diamond message set is the set of non-negative rate tuples satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-053710 | MATHDB-381597

The DGA$_\alpha$-spectral determination conjecture for graphs in $\mathfrak{F}_n$

摘录 / EXCERPT: DGA$_\alpha$-spectral determination conjecture. If $G\in\mathfrak{F}_n$, n is even, and $c_\alpha\geqslant 3$ is odd, then G is DGA$_\alpha$S.

conjecture | open | MathDB 页面 | 原文线索

V2-053711 | MATHDB-381599

The strong product conjecture for Grundy domination

摘录 / EXCERPT: Let G and H be graphs. Their strong product $G\boxtimes H$ has vertex set $V(G)\times V(H)$, with two vertices (g_1,h_1) and (g_2,h_2) adjacent when either $g_1=g_2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-053712 | MATHDB-381600

The graph dissimilarity metric conjecture

摘录 / EXCERPT: Let G_1 and G_2 be graphs with the same number of vertices, and let $Delta$ be the dissimilarity measure obtained from the collections of horizontal homologies partitioned by t...

conjecture | open | MathDB 页面 | 原文线索

V2-053713 | MATHDB-381625

The trichotomy conjecture for the cohomology of simple modules under the linearisation map

摘录 / EXCERPT: Trichotomy conjecture. For every vertex $i\neq 0$, there is a unique $k(i)\in\\{-1,0\\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-053714 | MATHDB-381626

The Type III stochastic-matrix graph conjecture

摘录 / EXCERPT: Let A be a stochastic matrix of order n whose characteristic polynomial is Type III: ... where $n=qd+y$, $1\leq y\leq q-1$, and $d\geq 2$. Type III stochastic-matrix graph conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-053715 | MATHDB-381630

The general-graph extension of the estimation-error lower-bound lemma

摘录 / EXCERPT: Let G satisfy the paper's general-graph regularity assumption, and let $\\{V_n\\}_{n \ge 1}$ satisfy the candidate-set assumption. For each initial source v, write...

conjecture | open | MathDB 页面 | 原文线索

V2-053716 | MATHDB-381640

Faudree-Schelp-Gyárfás-Tuza conjecture on strong edge-coloring of subcubic graphs

摘录 / EXCERPT: Faudree-Schelp-Gyárfás-Tuza conjecture. The following bounds hold:

conjecture | open | MathDB 页面 | 原文线索

V2-053717 | MATHDB-381643

Existence of graphs outside the 2-OR-PCG class

摘录 / EXCERPT: Existence conjecture. There exist graphs that are not 2-OR-PCGs.

conjecture | open | MathDB 页面 | 原文线索

V2-053718 | MATHDB-381662

Convergence conjecture for pseudo-q-traces of products of intertwining operators

摘录 / EXCERPT: Let V be a vertex operator algebra satisfying the conditions in the earlier conjecture on products of twisted intertwining operators, and let G be a finite group of automorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-053719 | MATHDB-381671

The conjectured formula for maximal chordal subgraphs

摘录 / EXCERPT: The maximal chordal subgraph conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-053720 | MATHDB-381678

Asymptotic upper-bound conjecture for the minimum size of k-dicritical oriented graphs

摘录 / EXCERPT: Asymptotic upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053721 | MATHDB-381681

Manifold decomposition conjecture for Kuramoto graph equilibria

摘录 / EXCERPT: Manifold decomposition conjecture. For every graph, the set of equilibria of the Kuramoto network is a finite union of manifolds. The question arises because the equilibrium set is...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053722 | MATHDB-381687

Strong Hadwiger's conjecture

摘录 / EXCERPT: Let G be a graph with chromatic number $[?]{}$ and let $S\subseteq V(G)$ be colorful if every proper $\chi(G)$-coloring of G assigns all $\chi(G)$ colors to vertices in S. An...

conjecture | open | MathDB 页面 | 原文线索

V2-053723 | MATHDB-381692

The multi-robber damage conjecture on saving three vertices

摘录 / EXCERPT: Multi-robber damage conjecture. For all $s>2$, if

conjecture | open | MathDB 页面 | 原文线索

V2-053724 | MATHDB-381703

The decomposition conjecture for class II graphs

摘录 / EXCERPT: Let G be a graph with maximum degree Δ and edge-chromatic number $\chi'(G)=\Delta+k$, and let p,q be positive integers satisfying ... with $p,q\leq\Delta$. A graph is clas...

conjecture | open | MathDB 页面 | 原文线索

V2-053725 | MATHDB-381704

The maximum matching conjecture for maximum edge-colorable subgraphs

摘录 / EXCERPT: Let G be an r-regular graph with maximum degree Δ and edge-chromatic number $\chi'(G)=\Delta+1$. A maximum Δ-colorable subgraph is a subgraph with as many edges a...

conjecture | open | MathDB 页面 | 原文线索

V2-053726 | MATHDB-381720

The $C_{3\ell+1}$-free clique extremal conjecture

摘录 / EXCERPT: Let $k\geq 2$ and let n be sufficiently large. Let G be an n-vertex graph containing no cycle $C_{3\ell+1}$ for any integer $\ell\geq k$. For $r\geq 2$, write $N_r(G)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-053727 | MATHDB-381727

McCuaig-Ota conjecture on contractible sets in 3-connected graphs

摘录 / EXCERPT: Let G be a 3-connected graph. A set $W\subset V(G)$ is contractible if $G(W)$ is connected and $G-W$ is 2-connected; a contractible set with k vertices is called a k-cont...

conjecture | open | MathDB 页面 | 原文线索

V2-053728 | MATHDB-381747

The order divisibility conjecture for (anti)palindromic graphs

摘录 / EXCERPT: Let a graph be (anti)palindromic if it has the palindromic or antipalindromic symmetry defined in the paper, and let its order be its number of vertices. Order divisibility conject...

conjecture | open | MathDB 页面 | 原文线索

V2-053729 | MATHDB-381751

The three-colour exception conjecture for split matchings

摘录 / EXCERPT: Let G be a graph on $2n$ vertices whose edge set is decomposed into perfect matchings M_1, M_2, and M_3, and let a_1, a_2, and a_3 be non-negative integers satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-053730 | MATHDB-381764

Guiduli-Mohar conjecture on minimum algebraic connectivity with prescribed minimum degree

摘录 / EXCERPT: Guiduli-Mohar conjecture. If $n\equiv0\pmod{d+1}$, then the μ-minimal graph on n vertices with $\delta=d$ is the graph displayed in Figure Mohar.

conjecture | open | MathDB 页面 | 原文线索

V2-053731 | MATHDB-381765

Guiduli-Mohar asymptotic structure conjecture for minimum-degree extremal graphs

摘录 / EXCERPT: Guiduli-Mohar conjecture. The graph G is path-like, and, except for some blocks near each end, it has the same structure as Figure Mohar.

conjecture | open | MathDB 页面 | 原文线索

V2-053732 | MATHDB-381766

Conjecture on the structure of minimum-algebraic-connectivity regular graphs

摘录 / EXCERPT: Regular extremal-structure conjecture. For every integer $d\ge3$, there exist constants C_1 and C_2 such that any μ-minimal d-regular graph G of order at least C_1 is...

conjecture | open | MathDB 页面 | 原文线索

V2-053733 | MATHDB-381767

Conjecture that asymptotically minimum algebraic connectivity implies maximum diameter

摘录 / EXCERPT: Minimum-connectivity-maximum-diameter conjecture. If Γ_n has asymptotically minimum algebraic connectivity, then its diameter is asymptotically maximum, namely

conjecture | open | MathDB 页面 | 原文线索

V2-053734 | MATHDB-381778

Cycle-complement conjecture for dense uniformly most reliable graphs

摘录 / EXCERPT: Cycle-complement conjecture. For any $n\geq 5$ and $e=n(n-3)/2$, the graph

conjecture | open | MathDB 页面 | 原文线索

V2-053735 | MATHDB-381779

Regular-graph reformulation of Boesch's maximum-girth conjecture

摘录 / EXCERPT: Regular Boesch conjecture. If an $(n,kn/2)$ UMRG exists and has girth g, then it has maximum girth among all k-regular $(n,kn/2)$-graphs and, among the k-regular $(n,kn/2)$-g...

conjecture | open | MathDB 页面 | 原文线索

V2-053736 | MATHDB-381789

Girã o-Lewis-Popielarz conjecture on rainbow saturation of complete graphs

摘录 / EXCERPT: Let K_r be the complete graph on r vertices, let $\operatorname{sat}(n,\mathcal{R}(K_r))$ denote the minimum number of edges in an n-vertex edge-colored graph containing no r...

conjecture | open | MathDB 页面 | 原文线索

V2-053737 | MATHDB-381790

Exact formula conjecture for the rainbow-saturation coefficient

摘录 / EXCERPT: For each integer $k\geq 4$, let $f(k)$ be the auxiliary extremal function from the paper and let α_{k+2} be the coefficient in the asymptotic formula for rainbow saturation,...

conjecture | open | MathDB 页面 | 原文线索

V2-053738 | MATHDB-381803

The vanishing proportion conjecture for trees attaining the Laplacian lower bound

摘录 / EXCERPT: Let T be a tree of order n, let d denote its diameter, and let $m_T[0,1)$ be the number of Laplacian eigenvalues of T in the interval $[0,1)$. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-053739 | MATHDB-381804

The almost-all-trees lower-bound conjecture for small Laplacian eigenvalues

摘录 / EXCERPT: Let T be a tree with diameter d, and let $m_T[0,1)$ denote the number of Laplacian eigenvalues of T less than 1. Almost-all-trees lower-bound conjecture. Almost all trees h...

conjecture | open | MathDB 页面 | 原文线索

V2-053740 | MATHDB-381822

Reduction-by-stages conjecture for affine W-algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a semisimple Lie algebra, let $k\in\mathbf{C}$, and let f_1,f_2 be nilpotent elements satisfying the hypotheses of the source's Main Theorem on freeness. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-053741 | MATHDB-381843

Erdős et al.'s Ramsey-Turán density conjecture for odd cliques

摘录 / EXCERPT: Let $r(3,q)$ denote the two-color Ramsey number for a triangle and a clique of order q. For $q\geq 2$, the Ramsey-Turán density $\rho(3,2q-1)$ is defined using graphs admitting a...

conjecture | open | MathDB 页面 | 原文线索

V2-053742 | MATHDB-381844

Kim-Kim-Liu's local Ramsey-Turán density conjecture for (K_3,K_6)

摘录 / EXCERPT: For $\delta>0$, let $\rho(3,6,\delta)$ be the limiting maximum edge density of an n-vertex graph with independence number at most δn that admits a red/blue coloring with...

conjecture | open | MathDB 页面 | 原文线索

V2-053743 | MATHDB-381845

The joint Ramsey-Turán density conjecture for (K_3,K_7)

摘录 / EXCERPT: For $\delta>0$, let $\rho(3,7,\delta)$ be the limiting maximum edge density of an n-vertex graph with independence number at most δn that admits a red/blue coloring with...

conjecture | open | MathDB 页面 | 原文线索

V2-053744 | MATHDB-381860

The square-grid error-correcting locating-dominating set density conjecture

摘录 / EXCERPT: Let SQ denote the infinite square grid, and let $\textrm{ERR:LD}(SQ)$ denote its error-correcting locating-dominating set density. Square-grid density conjecture. ... The authors s...

conjecture | open | MathDB 页面 | 原文线索

V2-053745 | MATHDB-381873

Bang-Jensen et al.'s tightness conjecture for the cycle-transversal bound

摘录 / EXCERPT: Bang-Jensen et al.'s tightness conjecture. For every positive integer n, there exists an oriented graph D such that

conjecture | open | MathDB 页面 | 原文线索

V2-053746 | MATHDB-381874

Bang-Jensen et al.'s NP-completeness conjecture for bounded inversion number

摘录 / EXCERPT: Bang-Jensen et al.'s complexity conjecture. Deciding whether a given digraph D has inversion number at most k is NP-complete for any fixed positive integer k.

conjecture | open | MathDB 页面 | 原文线索

V2-053747 | MATHDB-381910

Equality between k-Grundy Z-domination and k-forcing number

摘录 / EXCERPT: Equality conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053748 | MATHDB-381913

The Log-Concavity Genus Distribution conjecture for graph genus polynomials

摘录 / EXCERPT: Log-Concavity Genus Distribution conjecture. For every graph G, the genus polynomial is log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-053749 | MATHDB-381914

Stahl's real-rootedness conjecture for graph genus polynomials

摘录 / EXCERPT: Stahl's real-rootedness conjecture. For every graph G, the genus polynomial $\Gamma_G(x)$ has only real roots.

conjecture | open | MathDB 页面 | 原文线索

V2-053750 | MATHDB-381928

Open problems on cyclic cycle decompositions of nearly complete 3-uniform hypergraphs

摘录 / EXCERPT: Let $K_v^{(3)}$ be the complete 3-uniform hypergraph on vertex set $\mathbb{Z}_v$, and let I consist of the missing triplets $(i,i+v/3,i+2v/3)$ whenever $3\mid v$. A decomposit...

open_problem | open | MathDB 页面 | 原文线索

V2-053751 | MATHDB-381929

The conjectural determinant asymptotics for the 6V' model

摘录 / EXCERPT: Determinant-asymptotics conjecture. The leading asymptotics is given by

conjecture | open | MathDB 页面 | 原文线索

V2-053752 | MATHDB-381944

The asymptotic vanishing-ratio conjecture for higher diversities

摘录 / EXCERPT: For each k, let $m_{\ell}(k)$ denote the extremal quantity corresponding to the ℓ-diversity introduced in the paper. Asymptotic vanishing-ratio conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-053753 | MATHDB-381947

Logarithmic bridge-girth stabilization conjecture for beta and beta-star

摘录 / EXCERPT: Logarithmic bridge-girth stabilization conjecture. For all n and p,

open_problem | open | MathDB 页面 | 原文线索

V2-053754 | MATHDB-381950

A vertex whose deletion decreases inversion number by at most one

摘录 / EXCERPT: Vertex-deletion conjecture. There exists $v\in V(D)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-053755 | MATHDB-381962

Structure of non-degenerate subgraphs of curvature-flow limits

摘录 / EXCERPT: Non-degenerate-subgraph conjecture. The subgraph G_W coincides with the subgraph of G induced by W, all transition rates from W to $V\setminus W$ are zero, and (G_W,P_W)...

conjecture | open | MathDB 页面 | 原文线索

V2-053756 | MATHDB-381964

Spectral stability conjecture for simple random walks on complete graphs

摘录 / EXCERPT: Complete-graph spectral conjecture. The simple random walk P^s is asymptotically stable, and the eigenvalues of $DF(P^s)$ are

conjecture | open | MathDB 页面 | 原文线索

V2-053757 | MATHDB-381965

Spectral instability conjecture for simple random walks on hypercubes

摘录 / EXCERPT: Hypercube spectral conjecture. The simple random walk P^s is unstable, all eigenvalues of the corresponding linearized curvature-flow matrix are real, and

conjecture | open | MathDB 页面 | 原文线索

V2-053758 | MATHDB-381983

Planar Turán conjecture for two disjoint 4-cycles

摘录 / EXCERPT: Let $ex_{\mathcal{P}}(n,2C_4)$ denote the maximum number of edges in an n-vertex planar graph containing no two vertex-disjoint copies of the cycle C_4. Planar Turán conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-053759 | MATHDB-381984

Axiom for the small quasi-kernel conjecture

摘录 / EXCERPT: Let D be a non-empty digraph. For a vertex $u\in V(D)$, let $d^+(u)$ denote its out-degree, let $N^+[u]$ denote its closed out-neighborhood, and call a vertex a source if it has...

conjecture | open | MathDB 页面 | 原文线索

V2-053760 | MATHDB-381985

Global surface-area maximizer conjecture for seven-vertex inscribed polytopes

摘录 / EXCERPT: Let $\mathcal{I}_7$ denote the class of seven-vertex polytopes inscribed in the unit sphere $\mathbb{S}^2$, and let $S(P)$ be the surface area of P. Let e_3 be the third standa...

conjecture | open | MathDB 页面 | 原文线索

V2-053761 | MATHDB-381992

Nearly Ramanujan conjecture for the graphs A(n, q)

摘录 / EXCERPT: Let $A(n,q)$ be the algebraically defined graph introduced in the paper, and let its second largest eigenvalue mean the second largest eigenvalue of its adjacency matrix. Nearly Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-053762 | MATHDB-382017

Asymptotic extremal conjecture for ordered 3-uniform tight paths

摘录 / EXCERPT: Asymptotic extremal conjecture for ordered 3-uniform tight paths. The constructions described in the source are asymptotically optimal:

conjecture | open | MathDB 页面 | 原文线索

V2-053763 | MATHDB-382040

The depth-drop conjecture for powers of edge ideals

摘录 / EXCERPT: Let ${\Gamma}$ be a connected non-bipartite graph, let R be the polynomial ring associated to its vertices, and let I be the edge ideal of ${\Gamma}$. Depth-drop conjecture. If...

open_problem | open | MathDB 页面 | 原文线索

V2-053764 | MATHDB-382042

Joint asymptotic normality of the DFS variables

摘录 / EXCERPT: Let T, B, F, and C be the four random variables defined in the depth-first search analysis of the random digraph. Joint asymptotic normality conjecture. All four variables...

conjecture | open | MathDB 页面 | 原文线索

V2-053765 | MATHDB-382058

Rainbow tight Hamilton cycle conjecture at the minimum degree threshold

摘录 / EXCERPT: Rainbow minimum-threshold conjecture. There exists n_0 such that, for every $n\geq n_0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-053766 | MATHDB-382066

The strongly chordal graph conjecture for graphic arrangements

摘录 / EXCERPT: For a simple graph G, let $\mathscr{A}_G$ denote its graphic arrangement. A graph is strongly chordal if it has no induced n-sun for $n\geq 3$. An arrangement is flag-accurate...

conjecture | open | MathDB 页面 | 原文线索

V2-053767 | MATHDB-382067

The flip-width subpolynomial collapse conjecture

摘录 / EXCERPT: Flip-width subpolynomial collapse conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-053768 | MATHDB-382074

Connectedness conjecture for friends-and-strangers graphs of complete bipartite graphs

摘录 / EXCERPT: Let Y be a graph on $n\ge 2k\ge 4$ vertices. A k-bridge is a set of k edges whose deletion disconnects the graph, and it is non-trivial when neither resulting component is an...

conjecture | open | MathDB 页面 | 原文线索

V2-053769 | MATHDB-382075

Two-component conjecture for friends-and-strangers graphs with a complete bipartite factor

摘录 / EXCERPT: Let Y be a connected bipartite graph on $n\ge 5$ vertices with no non-trivial cut edge and $Y\not=C_n$. The graph $\mathsf{FS}(K_{2,n-2},Y)$ is the friends-and-strangers graph as...

conjecture | open | MathDB 页面 | 原文线索

V2-053770 | MATHDB-382089

Alternating path covering conjecture for matroid circuits

摘录 / EXCERPT: Alternating path covering conjecture. There exists a path that alternates between $A\setminus B$ and $B\setminus A$ and covers $\mathcal C[A\cup B]$.

conjecture | open | MathDB 页面 | 原文线索

V2-053771 | MATHDB-382104

DT/PT orbifold topological vertex correspondence for the cyclic group

摘录 / EXCERPT: DT/PT $\mathbb{Z}_{n}$-orbifold vertex correspondence. If ν is multi-regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-053772 | MATHDB-382120

Exponent improvement conjecture for even prisms

摘录 / EXCERPT: For each integer $k\geq2$, consider the Cartesian product $C_{2k}\square P_2$ of the cycle C_{2k} and the path P_2. The notation $\operatorname{ex}(n,H)$ denotes the maximum nu...

conjecture | open | MathDB 页面 | 原文线索

V2-053773 | MATHDB-382130

The local reversal edge conjecture for characteristic imset polytopes

摘录 / EXCERPT: Let $\mathcal{G}$ be a directed acyclic graph with skeleton G, and let i be a vertex. Define $\mathcal{G}_{\downarrow i}$ by orienting every edge between i and a neighbor tow...

conjecture | open | MathDB 页面 | 原文线索

V2-053774 | MATHDB-382141

The odd-order three-partition conjecture for cycle-generated Cayley graphs

摘录 / EXCERPT: Let $n\geq 7$ be odd and let I satisfy ... For a partition λ of n, write λ_λ^I for the eigenvalue associated with the corresponding irreducible representati...

conjecture | open | MathDB 页面 | 原文线索

V2-053775 | MATHDB-382144

Lee, Loh and Sudakov's directed-cut bisection conjecture

摘录 / EXCERPT: Lee, Loh and Sudakov's conjecture. Every digraph D with minimum outdegree at least $d+1$ admits a bipartition $V(D)=V_1\cup V_2$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-053776 | MATHDB-382163

The 3/8 conjecture for 2-tight components in 2-colored 3-graphs

摘录 / EXCERPT: The 3/8 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053777 | MATHDB-382165

The minimal-class conjecture for sparse hereditary graph classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a sparse hereditary class of graphs with unbounded tree-width. A hereditary graph class is minimal of unbounded tree-width if it has unbounded tree-width, whil...

conjecture | open | MathDB 页面 | 原文线索

V2-053778 | MATHDB-382178

Conjecture on finite-multiplicity representations of minimal W-algebras

摘录 / EXCERPT: Let $\mathfrak g$ be the Lie superalgebra under consideration, let k lie in its unitarity range, and set ... to be the subcategory of weight $W_k^{\min}(\mathfrak g)$-modules with...

conjecture | open | MathDB 页面 | 原文线索

V2-053779 | MATHDB-382201

Polynomial bound conjecture for 3-uniform clique versus monotone-path Ramsey numbers

摘录 / EXCERPT: Polynomial bound conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-053780 | MATHDB-382203

Polynomial-time branchwidth conjecture for graphs embeddable in the torus and projective plane

摘录 / EXCERPT: Polynomial-time branchwidth conjecture. Branchwidth can be computed in polynomial time on H-minor-free graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-053781 | MATHDB-382216

The q-zero forcing conjecture for the Cartesian product of a complete bipartite graph and K_2

摘录 / EXCERPT: Let $K_{n,m}$ be the complete bipartite graph with parts of sizes n and m, and let K_2 be the complete graph on two vertices. Write $Z_q(G)$ for the q-analogue of the zero...

conjecture | open | MathDB 页面 | 原文线索

V2-053782 | MATHDB-382217

The q-zero forcing conjecture for complete multipartite graphs

摘录 / EXCERPT: Let $G_{n,\ell}:=K_{\underbrace{n,n,\ldots,n}_{\ell}}$ be the complete multipartite graph with ℓ parts, each of size n, and let $Z_q(G)$ denote the q-analogue of the zero...

conjecture | open | MathDB 页面 | 原文线索

V2-053783 | MATHDB-382220

The genus conjecture for the prefix-reversal graph on four symbols

摘录 / EXCERPT: Let $\mathbb{P}_4$ be the prefix-reversal graph on the permutations of $\{1,2,3,4\}$, and let $\gamma(\mathbb{P}_4)$ denote its orientable genus. A graph is a pretzel graph if it a...

conjecture | open | MathDB 页面 | 原文线索

V2-053784 | MATHDB-382229

Berge-Sauer conjecture on regular subgraphs

摘录 / EXCERPT: A subgraph $H\leq G$ of a graph G is a graph $H=(V,E)$ such that $V\subseteq V(G)$ and $E\subseteq E(G)$. A graph is k-regular if every vertex has degree k. Berge-Sauer conje...

conjecture | open | MathDB 页面 | 原文线索

V2-053785 | MATHDB-382230

Linear-size Borel independent complete sections conjecture

摘录 / EXCERPT: Linear-size Borel independent complete sections conjecture. There exist constants $C,0>0$ with the following property: if the maximum degree of G_1 is at most 0 and every compo...

conjecture | open | MathDB 页面 | 原文线索

V2-053786 | MATHDB-382235

Polynomial-time conjecture for Hamiltonian cycles after adding fixed vertices

摘录 / EXCERPT: Conjecture on fixed-vertex extensions. For every fixed integer k, there exists a polynomial-time algorithm for deciding whether such a digraph has a Hamiltonian cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-053787 | MATHDB-382240

Veselovac's stackability conjecture for complete binary trees

摘录 / EXCERPT: Veselovac's conjecture. The tree T_h is stackable for all $h\ge 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-053788 | MATHDB-382247

Nonexistence of planar semi-covers of $K_{2,2,2,1}$ under specified conditions

摘录 / EXCERPT: A planar semi-cover of $K_{2,2,2,1}$ is a planar graph G' equipped with the projection structure described in the paper. Suppose G' satisfies the following conditions: if...

conjecture | open | MathDB 页面 | 原文线索

V2-053789 | MATHDB-382248

Eoh-Choi's corona-free characterization conjecture for graph grabbing

摘录 / EXCERPT: Let G be a graph with a weight function $w\colon V(G)\to\mathbb{R}_{\geq 0}$ whose codomain is $\{0,1\}$. Let $\mathcal{C}_\text{odd}$ be the family of corona products of an odd...

conjecture | open | MathDB 页面 | 原文线索

V2-053790 | MATHDB-382256

Shangguan-Tamo conjecture on limits for hypergraph configurations

摘录 / EXCERPT: Let $f^{(r)}(n;s,k)$ be the maximum number of edges in an n-vertex r-uniform hypergraph containing no collection of k edges spanning at most s vertices. For positive intege...

conjecture | open | MathDB 页面 | 原文线索

V2-053791 | MATHDB-382260

Gago-Coroničová-Hurajová-Madaras conjecture on low-betweenness uniform graphs

摘录 / EXCERPT: Let G be a low-betweenness BUG, meaning a betweenness-uniform graph whose betweenness value is less than 1. Write $\overline{G}$ for its complement. Gago-Coroničová-Hurajová-Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-053792 | MATHDB-382268

Converse of Sun's characterization of rainbow-common graphs

摘录 / EXCERPT: Converse of Sun's characterization. H is rainbow-common if and only if H is a forest.

conjecture | open | MathDB 页面 | 原文线索

V2-053793 | MATHDB-382271

Distinct spectral radii for Type 5 digraphs

摘录 / EXCERPT: Distinct-root conjecture. One has $\rho_1\neq\rho_2$. This asserts that distinct admissible parameter triples determine distinct spectral radii for the corresponding Type 5 digra...

conjecture | open | MathDB 页面 | 原文线索

V2-053794 | MATHDB-382304

Polynomial formulas for Betti numbers of squared-path cut complexes

摘录 / EXCERPT: Polynomial formula conjecture. For $k=4$ and $k=5$, the Betti numbers are conjectured to satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-053795 | MATHDB-382313

The strong attacking-cop-number conjecture

摘录 / EXCERPT: Strong attacking-cop-number conjecture. For all integers $k \geq 4$ there exists a graph H such that $\operatorname{c}(H)=k$ and

conjecture | open | MathDB 页面 | 原文线索

V2-053796 | MATHDB-382314

Minimal-order conjecture for graphs with large attacking-cop-number gap

摘录 / EXCERPT: Minimal-order conjecture for the attacking-cop-number gap. If

conjecture | open | MathDB 页面 | 原文线索

V2-053797 | MATHDB-382315

Geodesic Norine conjecture for antipodal edge-colorings of the cube

摘录 / EXCERPT: Geodesic Norine conjecture. For $n\ge 2$, every antipodal edge-coloring of Q_n contains antipodal vertices u and $\bar u$ joined by a monochromatic geodesic path.

conjecture | open | MathDB 页面 | 原文线索

V2-053798 | MATHDB-382326

The single-interval characterization of d-convex graph-nonedges

摘录 / EXCERPT: Let $d\geq 1$, let G be a graph, and let F be a set of nonedges of G. For each $f\in F$, write $(G\cup(F\setminus\{f\},f))$ for the graph-nonedge pair obtained by adding the...

conjecture | open | MathDB 页面 | 原文线索

V2-053799 | MATHDB-382342

The chemical antiregular graph conjecture for total σ-irregularity

摘录 / EXCERPT: Chemical antiregular graph conjecture. The same graphs as in Theorem $ have maximum value of $\sigma_t^{f(n)}$ even if $f(n)$ is a constant in the interval $(0,1)$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053800 | MATHDB-382349

Exact WKB convergence conjecture for opers

摘录 / EXCERPT: Let C) be a Riemann surface with a compatible $\mathbb{CP}^1$-structure and a spin structure. Let $u=(u_2,\ldots,u_n)\in\bigoplus_{i=2}^n \mathrm{H}^0(C,K^{\otimes i})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-053801 | MATHDB-382366

The minimum-degree four conjecture for majority edge list colourings

摘录 / EXCERPT: Minimum-degree four conjecture. Every graph with minimum degree at least 4 admits a majority edge colouring from lists of size 3.

conjecture | open | MathDB 页面 | 原文线索

V2-053802 | MATHDB-382369

The Merino-Welsh conjecture for bipartite graphs of minimum degree at least two

摘录 / EXCERPT: Let H be a bipartite graph, and let $\widetilde{T}_H(x,y)$ denote its normalized Tutte polynomial. Assume that every vertex of H has degree at least 2. Merino-Welsh conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-053803 | MATHDB-382376

Sazdanovic-Scofield homological span conjecture for chromatic homology

摘录 / EXCERPT: Sazdanovic-Scofield conjecture. The homological span satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-053804 | MATHDB-382391

The planar 3-connected graph conjecture for Speyer's polynomial

摘录 / EXCERPT: A graph G is 3-connected if it remains connected after the deletion of any set of at most two vertices, and it is planar if it can be drawn in the plane without edge crossings. L...

conjecture | open | MathDB 页面 | 原文线索

V2-053805 | MATHDB-382392

The circulant graph conjecture for Speyer's polynomial

摘录 / EXCERPT: For integers $n\geq 1$ and k, let $C^n_{1,k}$ be the graph on vertices identified with $\{1,\ldots,n\}$, obtained from the cycle by adding the edges $\{i,i+k\}$, with vertex labe...

conjecture | open | MathDB 页面 | 原文线索

V2-053806 | MATHDB-382393

The zigzag circulant closed-form conjecture for Speyer's polynomial

摘录 / EXCERPT: For $n\geq 5$, let $C^n_{1,2}$ be the circulant graph obtained from the cycle on n vertices by adding the edges joining each vertex to the vertex two steps away, with indices tak...

conjecture | open | MathDB 页面 | 原文线索

V2-053807 | MATHDB-382394

The complete graph recurrence conjecture for Speyer's polynomial

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $g_{K_n}(t)$ denote its Speyer polynomial, and let $g'_{K_{n-1}}(t)$ denote the derivative of $g_{K_{n-1}}(t)$ with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-053808 | MATHDB-382400

The unique-obstruction conjecture for Hamilton decompositions of regular tripartite tournaments

摘录 / EXCERPT: Let n be sufficiently large, and let G be a regular tripartite tournament on $3n$ vertices. Denote by ${\mathcal T}_\triangle$ the known regular tripartite tournament obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-053809 | MATHDB-382402

So's spectral characterization conjecture for integral circulant graphs

摘录 / EXCERPT: Let $n\in\mathbb{N}$, and let $S,T\subseteq\mathbb{Z}_n$ be integral symbols, so that $\Gamma_S=Cay(\mathbb{Z}_n,S)$ and $\Gamma_T=Cay(\mathbb{Z}_n,T)$ are integral circulant graph...

conjecture | open | MathDB 页面 | 原文线索

V2-053810 | MATHDB-382404

Nonexistence conjecture for 8-regular K3-irregular graphs

摘录 / EXCERPT: Nonexistence conjecture. There are no 8-regular K_3-irregular graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-053811 | MATHDB-382409

The average-degree rainbow path conjecture for subgraphs of hypercubes

摘录 / EXCERPT: Average-degree rainbow path conjecture. If G has average degree at least d, then every proper edge-coloring of G contains a rainbow copy of the path on d edges.

conjecture | open | MathDB 页面 | 原文线索

V2-053812 | MATHDB-382413

Smith normal form equivalence for rooted-tree products

摘录 / EXCERPT: Let T be the tree in Fig., and let T be any nontrivial tree with root r. Define $G_1=T(r) \odot \mathsf{T}(4)$ and $G_2=T(r) \odot \mathsf{T}(7)$. Smith normal form...

conjecture | open | MathDB 页面 | 原文线索

V2-053813 | MATHDB-382414

The characterization of perpane graphs as perfect cancellation graphs

摘录 / EXCERPT: Perpane-perfect cancellation conjecture. A graph is perpane if and only if it is a perfect cancellation graph.

conjecture | open | MathDB 页面 | 原文线索

V2-053814 | MATHDB-382429

Cohen-Macaulayness conjecture for r-co-connected complexes

摘录 / EXCERPT: Cohen-Macaulayness conjecture. The complex $\Sigma_r(G)$ is Cohen-Macaulay if and only if $\operatorname{Con}_r(G)$ is co-chordal.

conjecture | open | MathDB 页面 | 原文线索

V2-053815 | MATHDB-382445

Henning-Oellermann-Swart conjecture on the Steiner k-diameter and Steiner k-radius

摘录 / EXCERPT: Henning-Oellermann-Swart conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-053816 | MATHDB-382450

Polynomial-time 3-colorability of P_t-free graphs with one induced odd cycle length

摘录 / EXCERPT: For any integer t and any odd integer k, let $\mathcal{G}_{t,k}$ be the class of graphs that are P_t-free and whose induced odd cycles all have length k. Polynomial-time 3-...

conjecture | open | MathDB 页面 | 原文线索

V2-053817 | MATHDB-382453

Goedgebeur and Schaudt's conjecture on 4-vertex-critical \(\{P_7,C_3\}\)-free graphs

摘录 / EXCERPT: Let P_7 denote the path on seven vertices and let C_3 denote the cycle on three vertices. A graph is 4-vertex-critical if its chromatic number is 4 and deleting any vertex lo...

conjecture | open | MathDB 页面 | 原文线索

V2-053818 | MATHDB-382458

The logarithmic-degree conjecture for cycles with linearly many chords

摘录 / EXCERPT: Cycle-with-chords conjecture. If G has average degree at least $C\log\log(n)$, then G contains a cycle C on $[202~\ell$ vertices with at least $[203~\ell/2$ chords, for some...

conjecture | open | MathDB 页面 | 原文线索

V2-053819 | MATHDB-382471

The asymptotic coloring conjecture for oriented graphs with bounded cycle lengths

摘录 / EXCERPT: Asymptotic coloring conjecture. As $s\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-053820 | MATHDB-382474

Determinantal factorization conjecture for R-disjoint graphs

摘录 / EXCERPT: Let G be an R-disjoint graph. Write $B(G)$ for its bipartite part, and for each odd cycle C of G, let $R(C)$ denote the subgraph associated with C. Determinantal factoriz...

conjecture | open | MathDB 页面 | 原文线索

V2-053821 | MATHDB-382476

Polynomial-time computability of the core and corona in 2-bicritical graphs

摘录 / EXCERPT: Let G be a 2-bicritical graph with two odd cycles. The polynomial-time computability claim. The sets core $\core G$ and corona $\corona G$ can be computed in polynomial time. T...

conjecture | open | MathDB 页面 | 原文线索

V2-053822 | MATHDB-382479

The antenna-vertex conjecture for the partition graph

摘录 / EXCERPT: Antenna-vertex conjecture. For every $n\geq 4$, the only vertices of local simplex dimension 1 are the two antenna vertices:

conjecture | open | MathDB 页面 | 原文线索

V2-053823 | MATHDB-382487

The sharp formula for non-trivially t-intersecting families

摘录 / EXCERPT: For integers $r\ge 3$ and $r-2\ge t\ge 1$, let $\iota_0(t,n;r)$ denote the maximum size of a t-intersecting family in an r-partite r-uniform hypergraph with all parts of size...

conjecture | open | MathDB 页面 | 原文线索

V2-053824 | MATHDB-382488

The extremal characterization conjecture for the coefficient $f_r(s)$

摘录 / EXCERPT: Extremal characterization conjecture. $f_r(s)$ achieves the maximum for either $\mathcal{K}_{s+r-1}^r$, or the hypergraph $\mathcal{G}$ on $s+r$ vertices obtained by this minimum-e...

conjecture | open | MathDB 页面 | 原文线索

V2-053825 | MATHDB-382494

The universal isolated-clique conjecture for extremal uniform hypergraphs

摘录 / EXCERPT: Let G be an extremal ℓ-uniform hypergraph for Problem 13 on n vertices, and let K_{t+1}^{ℓ} denote the complete ℓ-uniform hypergraph on $t+1$ vertices. An isolat...

conjecture | open | MathDB 页面 | 原文线索

V2-053826 | MATHDB-382496

The jump threshold conjecture for 3-uniform hypergraphs

摘录 / EXCERPT: Jump threshold conjecture. All numbers in $[0,4/9)$ are jumps and all numbers in $[4/9,1)$ are non-jumps.

conjecture | open | MathDB 页面 | 原文线索

V2-053827 | MATHDB-382498

The upper-bound conjecture for strong majority edge-colorings of admissible graphs

摘录 / EXCERPT: Let G be an admissible graph, and let ${\mathrm Maj'}(G)$ denote the minimum number of colors in a strong majority edge-coloring of G. Upper-bound conjecture. If G is an admi...

conjecture | open | MathDB 页面 | 原文线索

V2-053828 | MATHDB-382504

The odd-cycle terminating-game nimber conjecture

摘录 / EXCERPT: Let C_n be the cycle graph on vertices $v_1,\ldots,v_n$, where $n\geq 4$ and each v_i is adjacent to v_{i+1} with indices taken modulo n. Let $\operatorname{TER}(C_n)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-053829 | MATHDB-382505

Vizing's average degree conjecture for edge-Δ-critical simple graphs

摘录 / EXCERPT: Let G be a finite, undirected, loopless simple graph, let $|G|$ denote its number of vertices, let $\Delta=\Delta(G)$ be its maximum degree, and let $\overline{d}(G)$ be its aver...

conjecture | open | MathDB 页面 | 原文线索

V2-053830 | MATHDB-382516

The optimality conjecture for induced odd cycles in planar graphs

摘录 / EXCERPT: Let $k\geq 3$, and let $h_k(n)$ denote the number of induced copies of C_{2k+1} obtained by evenly blowing up k pairwise non-adjacent vertices in a C_{2k+1} on n vertices....

conjecture | open | MathDB 页面 | 原文线索

V2-053831 | MATHDB-382544

Converse characterization of Property R graphs

摘录 / EXCERPT: Converse Property R graph conjecture. For any configuration ${\cal G}$ of stars ${\cal S}_j$ and isolated points as in the characterization theorem, there exists a group G such t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053832 | MATHDB-382560

Nonexistence conjecture for {k, k}-equivelar wnp maps

摘录 / EXCERPT: Let K be a wnp map, meaning a polyhedral map attaining equality in the lower bound $n\geq (k-1)^2$ for an n-vertex $\{k,k\}$-equivelar polyhedral map. Nonexistence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-053833 | MATHDB-382562

The low-degree Ehrhart coefficient conjecture for lattice-face polytopes

摘录 / EXCERPT: Let P be a d-polytope with vertex set V, and let $d'\le d-1$. Suppose that for every integer k with $0\le k\le d'$, condition (dfcon) is satisfied. Write $i(P,m)$ for the E...

conjecture | open | MathDB 页面 | 原文线索

V2-053834 | MATHDB-382571

Higher-dimensional max-flow min-cut law for random networks

摘录 / EXCERPT: Higher-dimensional max-flow min-cut conjecture. There exists a map μ on the unit sphere such that, for every convex set A with 0 in its interior,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053835 | MATHDB-382873

Modular edge-coloring conjectures

摘录 / EXCERPT: For an integer $k\geq 2$ and a graph G, let $\chi'_k(G)$ be the minimum number of colors in an edge-coloring of G such that, for every color and every vertex, the degree in the...

conjecture | solved | MathDB 页面 | 原文线索

V2-053836 | MATHDB-382954

Lei-Bai conjecture on 5-regular Lin-Lu-Yau Ricci-flat graphs

摘录 / EXCERPT: Conjecture 1 (Lei and Bai). If G is a Ricci-flat 5-regular graph, then G is either isomorphic to RF_{72}^{5} or is a Cartesian product of the Petersen graph, the Triplex gr...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053837 | MATHDB-382970

Claw-free regular-graph conjecture for generalized power domination

摘录 / EXCERPT: For all integers $k\geq \ell\geq 1$, every connected claw-free $(k+\ell+1)$-regular graph G of order n satisfies $\gamma_{P,k}(G)\leq \frac{n}{k+\ell+2}$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053838 | MATHDB-382983

Albertson-Berman conjecture on induced forests in planar graphs

摘录 / EXCERPT: For every simple planar graph G with $n=|V(G)|$ vertices, if $a(G)$ denotes the maximum number of vertices in an induced forest of G, then $a(G)\geq \frac{n}{2}$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053839 | MATHDB-383001

Aigner-Horev-Levy problem and Araújo-Piga-Schacht question on tight Hamiltonian cycles

摘录 / EXCERPT: For every $d>0$ and every $\alpha>0$, do there exist $\rho>0$ and n_0 such that every (ρ,d)-dense 3-uniform hypergraph H on $n\ge n_0$ vertices with minimum codegree...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053840 | MATHDB-383002

Pavez-Signé conjecture and length-control question for spanning H-subdivisions

摘录 / EXCERPT: For every $\varepsilon>0$, there exists a constant $C_0>0$ such that, for every digraph H with h arcs and no isolated vertices, every n-vertex digraph D satisfying...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053841 | MATHDB-383030

Bauer-van den Heuvel-Schmeichel conjecture on triangle-free graphs and 2-factors

摘录 / EXCERPT: For every $\varepsilon>0$, there exists a finite triangle-free graph G such that $\tau(G)\ge 2-\varepsilon$ and G has no 2-factor, where a 2-factor is a spanning 2-regula...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053842 | MATHDB-383031

Dybizbański-Furmańczyk-Mkrtchyan gap-one conjecture

摘录 / EXCERPT: For every finite block graph G, define $L(G)=\max\left\{\omega(G),\left\lceil\frac{|V(G)|+1}{\alpha_{\min}(G)+1}\right\rceil\right\}$, where $\omega(G)$ is the clique number and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053843 | MATHDB-383053

Do Shannon capacity and its reversible variant coincide?

摘录 / EXCERPT: 2403.10985 introduced the notion of reversible capacity of a graph and showed that it is at most the Shannon capacity. It is conjectured there that the two values are equal. At the...

open_problem | open | Timeroot | MathDB 页面 | 原文线索

V2-053844 | MATHDB-383163

Badakhshian-Falgas-Ravry-Sharifzadeh traversal-density conjecture

摘录 / EXCERPT: Conjecture 1.1 ([1], Conjecture 1.17). With $\pi_r(\mathcal{F})$ denoting the supremum of the values $\alpha\in[0,1]$ for which there exists an r-partite graph G with...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053845 | MATHDB-383194

Chen-Raspaud conjecture

摘录 / EXCERPT: For every integer $k\ge 2$, every graph G with odd girth $\operatorname{og}(G)\ge 2k+1$ and maximum average degree $\operatorname{mad}(G)<2+1/k$ admits a $(2k+1:k)$-coloring; tha...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053846 | MATHDB-383198

Mizzi’s conjecture on unstable graphs

摘录 / EXCERPT: For every nontrivially unstable graph G, there exists an odd positive integer k such that G contains cycles isomorphic to both C_k and C_{2k}.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053847 | MATHDB-383201

Isaak’s Hamiltonian-path problem for P(n,n-2)

摘录 / EXCERPT: For $n\ge 4$, let $P(n,n-2)$ be the directed overlap graph whose vertices are the $(n-2)$-permutations of $[n]=\{1,\ldots,n\}$, with an arc from $(a_1,\ldots,a_{n-2})$ to...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053848 | MATHDB-383202

Odd Hadwiger conjecture for graphs with independence number two

摘录 / EXCERPT: For every integer $n\ge 1$ and every n-vertex graph G with independence number $\alpha(G)=2$, the graph G contains $K_{\lceil n/2\rceil}$ as an odd minor.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053849 | MATHDB-383204

Graph parking-space extension conjecture

摘录 / EXCERPT: For every finite simple graph G with distinguished vertex v, let $\operatorname{Aut}_v(G)\leq \operatorname{Aut}(G)$ be the subgroup fixing v. Then the conjecture asserts tha...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053850 | MATHDB-383223

Kozerenko-Skochko imbalance conjecture

摘录 / EXCERPT: Let G be a finite simple graph, and for each edge $uv\in E(G)$ define its imbalance by $\operatorname{imb}_G(uv)=|d_G(u)-d_G(v)|$. Suppose that $\operatorname{imb}_G(uv)>0$ for e...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053851 | MATHDB-383224

Blanco-Buehrle generalized pancake expander conjecture

摘录 / EXCERPT: For every fixed integer $n\ge 2$, the family of undirected generalized pancake graphs $\{P(m,n)\}_{m\ge 2}$ is an expander family; equivalently, there exists a constant $c_n>0$ suc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053852 | MATHDB-383240

Domination-packing inequality $\gamma(G)\le2\rho(G)+1$

摘录 / EXCERPT: For every connected graph G with maximum degree $\Delta(G)\le 3$, the domination number satisfies $\gamma(G)\le 2\rho(G)+1$, where $\rho(G)$ is the maximum cardinality of a set o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053853 | MATHDB-383242

Erdős-Gallai-Tuza triangle independence-covering problem

摘录 / EXCERPT: For a graph G, let $\alpha_1(G)$ be the maximum cardinality of an edge set $F\subseteq E(G)$ such that $|F\cap E(T)|\le 1$ for every triangle T of G, and let $\tau_1(G)$ be t...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053854 | MATHDB-383261

Bickle’s conjecture on collapsible graphs

摘录 / EXCERPT: For every pair of integers $k\ge 3$ and $n\ge k+1$, the minimum of $\left|\{v\in V(G):\deg_G(v)=k\}\right|$ over all graphs G of order n satisfying $\delta(G)=k$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053855 | MATHDB-383296

Rainbow Lehel conjecture

摘录 / EXCERPT: ...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053856 | MATHDB-383297

Erves-Tepeh (1,2)-domination conjectures

摘录 / EXCERPT: Let G be a cubic graph of order n, and let $c_{\mathrm{ind}}(G)$ be the maximum cardinality of a vertex set whose induced subgraph is 2-regular. Let $\gamma_{1,2}(G)$ denote...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053857 | MATHDB-383300

Erdős four-edge intersection problem

摘录 / EXCERPT: For an n-vertex graph G and a permutation σ of its vertex set, define $I_G(\sigma)=|E(G)\cap E(\sigma(G))|$, and let $f(n,4)$ be the minimum possible value of $|E(G)|$ o...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053858 | MATHDB-383302

Albertson-Berman conjecture

摘录 / EXCERPT: For every simple planar graph G with $n=|V(G)|$ vertices, there exists a set $S\subseteq V(G)$ such that the induced subgraph $G[S]$ is a forest and $|S|\ge n/2$. Equivalently, i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053859 | MATHDB-383304

Koolen-Hayat-Iqbal distance-spectrum problem

摘录 / EXCERPT: For all connected graphs G and H, if their distance matrices $D(G)$ and $D(H)$ are cospectral, meaning that $\operatorname{Spec}(D(G))=\operatorname{Spec}(D(H))$, then G is b...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053860 | MATHDB-383319

Sedlar-Škrekovski mixed metric-dimension conjecture

摘录 / EXCERPT: For every connected graph G that is not a cycle, the mixed metric dimension satisfies $\operatorname{mdim}(G)\leq \ell(G)+2c(G)$, where $\ell(G)$ is the number of leaves and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053861 | MATHDB-383322

Kashima’s pancyclicity analogue conjecture

摘录 / EXCERPT: For every integer $n\ge 4$, every 2-connected graph G of order n with minimum degree $\delta(G)\ge \sqrt{3n}$ contains a 2-connected subgraph of every order...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053862 | MATHDB-383341

Wang-Xu conjecture on spectral determination of complements of T-shape trees

摘录 / EXCERPT: For positive integers $\ell_1\leq \ell_2\leq \ell_3$, let $T(\ell_1,\ell_2,\ell_3)$ be the tree with a unique degree-three vertex v such that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053863 | MATHDB-383342

Simonovits Product Conjecture

摘录 / EXCERPT: For every finite family $\mathcal L$ of ordinary forbidden graphs, let $p(\mathcal L)=\min\{\chi(H):H\in\mathcal L\}-1$. The Simonovits Product Conjecture asserts that, for all suf...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053864 | MATHDB-383343

Pavez-Signé’s question on similarly sized paths in spanning H-subdivisions

摘录 / EXCERPT: For a fixed graph H with $h\ge 3$ edges and no isolated vertices, determine whether every sufficiently large n-vertex graph G satisfying the relevant minimum-degree hypothesi...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053865 | MATHDB-383344

Akbari et al.’s edge-path matrix Conjectures 1 and 2

摘录 / EXCERPT: Let G be a graph of order n, and let $EP(G)=(p_{ij})$ be its edge-path matrix, where p_{ij} is the maximum number of pairwise edge-disjoint paths between distinct vertices...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053866 | MATHDB-383346

Hua-Peng problem on the saturation number of the kite graph

摘录 / EXCERPT: Let K be the kite graph obtained from K_4 by deleting one edge and then attaching a pendant edge to one of the two vertices of degree 2 in the resulting graph. For every inte...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053867 | MATHDB-383347

Brualdi-Hoffman-Turán problem on odd wheels

摘录 / EXCERPT: Let G be a graph with $m=|E(G)|$ edges, let $\rho(G)$ denote its adjacency spectral radius, and let $W_{2k+1}=K_1\vee C_{2k}$. Determine the sharp upper bound for $\rho(G)$, incl...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053868 | MATHDB-383402

WOWII Graph Conjecture 143

摘录 / EXCERPT: For every finite simple connected graph G containing a cycle, define $t(G)=\max\{|S|:S\subseteq V(G)\text{ and }G[S]\text{ is a tree}\}$, let $g(G)$ be the length of a shortest c...

conjecture | solved | MathDB 页面 | 原文线索

V2-053869 | MATHDB-383405

Unavoidable claws in tournaments

摘录 / EXCERPT: For each positive integer n, let $u(n)$ be the largest integer d such that every n-vertex claw with at most d branches occurs as a subtournament of every tournament on n...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053870 | MATHDB-383406

Strong majority edge-colouring conjecture

摘录 / EXCERPT: For every graph G with minimum degree $cdelta(G)\ge 4$, there exists an edge-colouring $c:E(G)\to\{1,2,3\}$ such that, for every edge $e\in E(G)$ and every colour...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053871 | MATHDB-383420

Tratnik et al.’s nanotube resonance-graph conjecture

摘录 / EXCERPT: The conjecture asserts that for every matchable nanotube N, its resonance graph $R_6(N)$ is disconnected. Here $R_6(N)$ is the graph whose vertices are the perfect matchings of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053872 | MATHDB-383434

Morrison-Nir-Norin-Rzążewski-Wesolek Turán-goodness questions

摘录 / EXCERPT: For a graph H and integer $r\ge 2$, say that H is K_{r+1}-Turán-good if, for every sufficiently large n, the Turán graph $T_r(n)$ maximizes the number of copies of H amon...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053873 | MATHDB-383476

Wang et al.’s conjecture on maximal dissociation sets in trees

摘录 / EXCERPT: Let T be a tree of order $n\geq 3$. A set $S\subseteq V(T)$ is a dissociation set if every vertex of the induced subgraph $T[S]$ has degree at most 1, and it is maximal if it i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053874 | MATHDB-383477

Elphick-Farber-Goldberg-Wocjan square-energy conjecture

摘录 / EXCERPT: For every connected graph G on n vertices, let $\lambda_1,\ldots,\lambda_n$ be the eigenvalues of its adjacency matrix, and define $s^+(G)=\sum_{\lambda_i>0}\lambda_i^2$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053875 | MATHDB-383495

Caro-Petrusevski-Skrekovski-Tuza odd independence conjecture for Queen graphs

摘录 / EXCERPT: Let Q_n be the Queen graph on an $n\times n$ chessboard: its vertices are the board squares, and two distinct vertices are adjacent if they lie in a common row, column, or diagon...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053876 | MATHDB-383500

Bougard-Joret conjecture

摘录 / EXCERPT: For integers n,α,k satisfying $n\le k\alpha$, $n\ge 2\alpha$, $n\ge \alpha+k$, $\alpha\ge 2$, and $k\ge 3$, define $f(n,\alpha,k)$ to be the minimum of $|E(G)|$ over all k...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053877 | MATHDB-383504

Ding-Lamaison-Liu-Wang-Yang layered-graph conjecture

摘录 / EXCERPT: For every 3-uniform hypergraph F, $\pi_{\mathrm{co}}(F)=0$ if and only if F is layered and $\pi(F)=0$, where $\pi(F)$ is the uniform Turán density and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053878 | MATHDB-383507

Ma’s saturation-number conjecture

摘录 / EXCERPT: Let $\mathcal{C}_{[4,r]}=\{C_4,C_5,\ldots,C_r\}$. For every integer $r\ge 5$, there exists an integer $n_0(r)$ such that, for every $n\ge n_0(r)$,...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053879 | MATHDB-383510

Teng-Tian connectivity-preserving spanning path question

摘录 / EXCERPT: For integers $k\ge 2$ and $t\ge 3$, let G be a k-connected graph of order n. Is it true that, whenever $\delta(G)\ge \left\lceil\frac{n+t}{2}\right\rceil$ and either...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053880 | MATHDB-383511

Lin-Zhao graph Steklov eigenvalue problem

摘录 / EXCERPT: Does there exist a universal constant $C>0$ such that, for every connected graph $G=(V,E)$ with boundary B, maximum degree $d_{\max}$, and genus g, one has...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053881 | MATHDB-383536

Bang-Jensen-Gutin Prescribed-Endpoint Path Problems

摘录 / EXCERPT: Given a semicomplete digraph D and distinct vertices $x,y\in V(D)$, determine a longest directed (x,y)-path. Given a locally semicomplete digraph D and distinct vertices...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053882 | MATHDB-383539

Dong-Luo Conjecture on Tight (k,0)-Stable Graphs

摘录 / EXCERPT: There exists an integer k_0 such that, for every integer $k\ge k_0$, every graph G of order n satisfying $\alpha(G-S)\ge \alpha(G)$ for every subset $S\subseteq V(G)$ with...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053883 | MATHDB-383553

Stanley’s tree-isomorphism conjecture

摘录 / EXCERPT: For all finite trees T_1 and T_2, if their chromatic symmetric functions are equal, $X_{T_1}=X_{T_2}$, then the trees are isomorphic, $T_1\cong T_2$. Here, for a finite graph...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053884 | MATHDB-383582

Square-energy conjecture

摘录 / EXCERPT: Let G be a connected simple graph of order n, with adjacency eigenvalues $\lambda_1,\ldots,\lambda_n$. Define $s^{+}(G)=\sum_{\lambda_i>0}\lambda_i^2$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053885 | MATHDB-383650

Bollobás-Erdős-Tuza conjecture

摘录 / EXCERPT: For every constant $c>0$, there exists a function $f_c(n)=o(n)$ such that every graph G on n vertices with independence number $\alpha(G)\ge cn$ satisfies $h(G)\le f_c(n)$, whe...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053886 | MATHDB-383681

Montgomery’s Hamiltonicity conjecture for regular sublinear expanders

摘录 / EXCERPT: For every sufficiently large integer d and every sufficiently large n, every n-vertex d-regular sublinear expander contains a Hamiltonian cycle.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053887 | MATHDB-383684

Fractional coloring conjecture for triply efficient shadow tomography

摘录 / EXCERPT: Let $\mathcal{P}_n$ denote the set of n-qubit Pauli observables. For every n-qubit quantum state ϱ and every $\epsilon\in(0,1]$, define...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053888 | MATHDB-383685

Broadcast domination conjecture

摘录 / EXCERPT: For every connected graph G, the broadcast domination number is at most twice the multipacking number: $\gamma_b(G)\leq 2\operatorname{mp}(G)$, where $\gamma_b(G)$ is the minimum...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053889 | MATHDB-383687

Lichiardopol’s conjecture

摘录 / EXCERPT: ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053890 | MATHDB-383688

Bandeira-Kireeva-Maillard-Rödder Kuramoto conjecture

摘录 / EXCERPT: For every $\varepsilon>0$, there exists a finite simple graph G on n vertices such that $\delta(G)\geq (3/4-\varepsilon)n$ and G is not globally synchronizing for the Kuramot...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053891 | MATHDB-383715

WOWII Graph Conjecture 36

摘录 / EXCERPT: The supplied source identifies this as a conjecture about finite graphs, with the relevant configuration involving a connected graph G of radius 3 having exactly one diametrica...

conjecture | refuted | MathDB 页面 | 原文线索

V2-053892 | MATHDB-383716

WOWII Graph Conjecture 160

摘录 / EXCERPT: The supplied source identifies a source-corrected Written on the Wall II (WOWII) Graph Conjecture 160 and refers to an exact corrected statement, but does not provide that statemen...

conjecture | open | MathDB 页面 | 原文线索

V2-053893 | MATHDB-383732

Four-colour conjecture for generalized total colourings of planar graphs

摘录 / EXCERPT: Every finite planar graph G admits a colouring $c:V(G)\cup E(G)\to\{1,2,3,4\}$ such that: (i) any two adjacent vertices, adjacent edges, or incident vertex-edge pairs receive dis...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053894 | MATHDB-383747

Golden-mean lower-bound conjecture for connective constants of infinite vertex-transitive cubic graphs

摘录 / EXCERPT: For every infinite, connected, vertex-transitive, cubic graph G, let $c_n(G,v)$ denote the number of self-avoiding walks of length n starting at a vertex v, and let the conne...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053895 | MATHDB-383762

Henning-Yeo conjecture on identifying vertex covers

摘录 / EXCERPT: For every finite graph G with order $n=|V(G)|$, size $m=|E(G)|$, and maximum degree $\Delta=\Delta(G)$, does the identifying vertex cover number of G satisfy the upper-bound in...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053896 | MATHDB-383763

Balanced complete-bipartite graph likelihood conjecture

摘录 / EXCERPT: Let G be a finite simple undirected graph on n vertices, and let $\mathcal{L}(G)$ denote the probability that the uniform sequential attachment process outputs a graph isomorph...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053897 | MATHDB-384379

Combinatorics and Graph Theory (problem 1)

摘录 / EXCERPT: begin{enumerate} item Is it true that ... item Estimate μ_{2n}. end{enumerate}

open_problem | open | MathDB 页面 | 原文线索

V2-053898 | MATHDB-384380

Is there a decent multivariate extension of the Chudnovsky-Seymour Theorem that the vertex independent set polynomial of a claw-free graph has only real roots?

摘录 / EXCERPT: Is there a decent multivariate extension of the Chudnovsky-Seymour Theorem that the vertex independent set polynomial of a claw-free graph has only real roots?

open_problem | open | MathDB 页面 | 原文线索

V2-053899 | MATHDB-384381

Combinatorics and Graph Theory (problem 3)

摘录 / EXCERPT: Generalize the Chudnovsky-Seymour (Heilmann-Lieb) Theorem. Extend the common interlacing property to several variables.

open_problem | open | MathDB 页面 | 原文线索

V2-053900 | MATHDB-384700

How do we measure the curvature of real networks?

摘录 / EXCERPT: How do we measure the curvature of real networks?

open_problem | open | MathDB 页面 | 原文线索

V2-053901 | MATHDB-384704

Problem session (problem 5)

摘录 / EXCERPT: What is the curvature of a network? How to define it?

open_problem | open | MathDB 页面 | 原文线索

V2-053902 | MATHDB-384710

Complete hypergraphs (problem 1)

摘录 / EXCERPT: This problem is just a placeholder. Replace it bu a real problem, or we can jsut delete it later.

open_problem | open | MathDB 页面 | 原文线索

V2-053903 | MATHDB-384711

Complete hypergraphs (problem 2)

摘录 / EXCERPT: $\pi(K_m^3)=1-\left(\frac{2}{m-1}\right)^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-053904 | MATHDB-384712

Complete hypergraphs (problem 3)

摘录 / EXCERPT: Does $k(1-\pi(K_{k+1}^k))$ tend to ∞ as $k\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-053905 | MATHDB-384718

Paths in 3-colorings of 6-chromatic graphs

摘录 / EXCERPT: Is it true that in any 3-edge-coloring of a graph of chromatic number at least 6, there is a monochromatic path of length 3?

open_problem | open | MathDB 页面 | 原文线索

V2-053906 | MATHDB-384719

Hypergraph size Ramsey numbers

摘录 / EXCERPT: (Dudek--R"odl) Is $\hat r_3(n,n) = \binom{r_3(n,n)}{3}$? In particular, Dudek--R"odl) Let $N = r_3(n,n)$. Is it true that every 2-edge-coloring of $K_N^{(3)}-e$ contains a monochro...

open_problem | open | MathDB 页面 | 原文线索

V2-053907 | MATHDB-384721

Erd\H{o}s--Hajnal for tournaments

摘录 / EXCERPT: Does there exist $\epsilon > 0$ such that if T_n is an n-vertex tournament not containing T^* as a subtournament, then T_n contains a transitive tournament with...

open_problem | open | MathDB 页面 | 原文线索

V2-053908 | MATHDB-384727

Burr--Erd\H{o}s for hypergraphs

摘录 / EXCERPT: (Conlon--Fox--Sudakov) Is $r_f(H,H) \le c_k(d) n$ for all n-vertex d-degenerate (in the above sense) k-uniform hypergraph H?

open_problem | open | MathDB 页面 | 原文线索

V2-053909 | MATHDB-384732

Directed paths in Eulerian digraphs

摘录 / EXCERPT: Does every Eulerian digraph of average degree d have a directed path of length $\Omega(d)$?

open_problem | open | MathDB 页面 | 原文线索

V2-053910 | MATHDB-384798

Can we use forts/havens, hyper graphs and clutters to improve wavefront?

摘录 / EXCERPT: Can we use forts/havens, hyper graphs and clutters to improve wavefront? A fort is a subgraph such that every vertex outside of the fort connects to at least two vertices inside th...

open_problem | open | MathDB 页面 | 原文线索

V2-053911 | MATHDB-384801

Problem List (problem 23)

摘录 / EXCERPT: Given a set of zero forcing sets can we transform one set into another using basic operations (vertex swaps, adding/deleting, token sliding and jumping, etc.) and stay within the s...

open_problem | open | MathDB 页面 | 原文线索

V2-053912 | MATHDB-384806

Problem List (problem 28)

摘录 / EXCERPT: If G is a connected, r-uniform hypergraph, with $n\ge k+r$ vertices, then the k-power domination number is less than or equal to $\dfrac{n}{k+r}$. True if $r =2$.

open_problem | open | MathDB 页面 | 原文线索

V2-053913 | MATHDB-384808

Vertex weighted forcing, let ... be the weight function, i.e.

摘录 / EXCERPT: Vertex weighted forcing, let w be the weight function, i.e. $w(v)$ is the weight of vertex v. Then we want to minimize the cost where the cost is the sum of the weights of a ze...

open_problem | open | MathDB 页面 | 原文线索

V2-053914 | MATHDB-384816

Adding loops to SBM

摘录 / EXCERPT: Real-world networks are likely to have many small loops (e.g. friends introduce their friends to each other) but this feature is not captured by the SBM. One proposed model to capt...

open_problem | open | MathDB 页面 | 原文线索

V2-053915 | MATHDB-384819

Iterative plurality

摘录 / EXCERPT: Let $G \sim \mathrm{SBM}$. Consider the following algorithm. Initialize each vertex to a random color from $[k]$. Repeatedly update each node's color to the plurality of its neighb...

open_problem | open | MathDB 页面 | 原文线索

V2-053916 | MATHDB-384891

Ehrhart $h^\ast$-polynomial: Unimodality and Real-Rootedness (problem 20)

摘录 / EXCERPT: A textbf{spinal graph} G is a connected digraph on vertex set $V=\{1,2,...,n\}$ with edge multiset E, where all edges are directed from smaller to larger vertices and E conta...

open_problem | open | MathDB 页面 | 原文线索

V2-053917 | MATHDB-384897

Cases of Harassment in the Community

摘录 / EXCERPT: How can we support people who have been harassed? How do we as a community handle working with people who have been accused (including by way of warnings over the "whisper network"...

open_problem | open | MathDB 页面 | 原文线索

V2-053918 | MATHDB-384914

How badly can Sidorenko fail for hypergraphs

摘录 / EXCERPT: Let H be a 3-uniform, 3-partite hypergraph. Then any n-vertex 3-uniform hypergraph with pn^3 edges contains at least $p^{Ce(H)}n^{V(H)}$ copies of H.

open_problem | open | MathDB 页面 | 原文线索

V2-053919 | MATHDB-384917

Turan exponents of hypergraphs

摘录 / EXCERPT: There exists $c>0$ such that the following holds. If H is an n-vertex, 3-uniform hypergraph with at least $n^{3-c/d}$ edges, then H contains some/every 3-uniform, 3-partite l...

open_problem | open | MathDB 页面 | 原文线索

V2-053920 | MATHDB-384922

Vertex distinguishing achromatic edge coloring

摘录 / EXCERPT: Let $n \ge 4$. The maximum number of colors in a minimal vertex distinguishing edge labeling is $\lfloor \frac{2n}{3}\rfloor +1 $.

open_problem | open | MathDB 页面 | 原文线索

V2-053921 | MATHDB-384928

Random graph labeling

摘录 / EXCERPT: Given a graph G with random vertex label, determine the probability that G is radio labeled.

open_problem | open | MathDB 页面 | 原文线索

V2-053922 | MATHDB-384964

Graph Theory Motivated Questions (problem 3)

摘录 / EXCERPT: Let $G = (V,E)$ be a graph of maximum degree Δ. Let $\chi(t)$ be the chromatic polynomial (i.e. $\chi(q)$ is the number of ways of properly coloring the graph G with q c...

open_problem | open | MathDB 页面 | 原文线索

V2-053923 | MATHDB-385511

Rigidity (problem 5)

摘录 / EXCERPT: Is it possible to strengthen a theorem of Walter Whiteley (in A matroid on hypergraphs with applications in scene analysis and geometry, 1989) on body-pin frameworks in $d=2$, to s...

open_problem | open | MathDB 页面 | 原文线索

V2-053924 | MATHDB-385703

Is there a dichotomy characterizing which Borel graphs admit Borel orientations for which every vertex has finite out degree?

摘录 / EXCERPT: Is there a dichotomy characterizing which Borel graphs admit Borel orientations for which every vertex has finite out degree?

open_problem | open | MathDB 页面 | 原文线索

V2-053925 | MATHDB-386373

Is there a mean-field limit for ResNet as the depth of the network goes to infinity, with an appropriate scaling?

摘录 / EXCERPT: Is there a mean-field limit for ResNet as the depth of the network goes to infinity, with an appropriate scaling?

open_problem | open | MathDB 页面 | 原文线索

V2-053926 | MATHDB-386375

Is there a comparison of NTK and neural networks using Monte Carlo bounds?

摘录 / EXCERPT: Is there a comparison of NTK and neural networks using Monte Carlo bounds? Are there regimes in which one is generically better than the other?

open_problem | open | MathDB 页面 | 原文线索

V2-053927 | MATHDB-386377

When we use deep neural networks we are trying to move away from the overparameterization scenario.

摘录 / EXCERPT: When we use deep neural networks we are trying to move away from the overparameterization scenario. In order to pull this off, you need to do very smart architecture design. But we...

open_problem | open | MathDB 页面 | 原文线索

V2-053928 | MATHDB-386378

Questions (problem 9)

摘录 / EXCERPT: How can we extend the theory of fully-connected neural networks (i.e. the problems in section 1) to convolutional neural networks? Can the geometry of the data (e.g. the spectral s...

open_problem | open | MathDB 页面 | 原文线索

V2-053929 | MATHDB-386379

Questions (problem 10)

摘录 / EXCERPT: How does the generalization error for a convolutional neural network depend on the input dimension (e.g. the number of pixels in the image). (Here we fixed the width of the channel...

open_problem | open | MathDB 页面 | 原文线索

V2-053930 | MATHDB-386418

#5 Combinatorics of splittability on K_n

摘录 / EXCERPT: There is also a $\frac{1}{\text{poly}(n)}$ lower bound for the probability that a uniformly random spanning tree of an n-vertex complete graph is k-splittable. If you plot 2-sp...

open_problem | open | MathDB 页面 | 原文线索

V2-053931 | MATHDB-386791

Algebraic K-theory of digraphs

摘录 / EXCERPT: Compute $K_0(*)$ and check if it agrees with what one might expect from spaces, i.e., an isomorphism $K_0(*) \cong \mathbb{Z}$, given by taking a class $[X] \in K_0(*)$ to the Eule...

open_problem | open | MathDB 页面 | 原文线索

V2-053932 | MATHDB-386953

Classify fracton models constructed from ... -foliated topological defect networks.

摘录 / EXCERPT: Classify fracton models constructed from 1-foliated topological defect networks.

open_problem | open | MathDB 页面 | 原文线索

V2-053933 | MATHDB-388116

Well-balanced orientations of hypergraphs

摘录 / EXCERPT: When can we characterize hypergraphs that have an orientation satisfying a prescribed symmetric local edge-connectivity requirement? Special case: can we characterize hypergraphs t...

open_problem | open | MathDB 页面 | 原文线索

V2-053934 | MATHDB-389233

Conjecture 4.12

摘录 / EXCERPT: Let $\underline{\theta}=(\theta_1,\ldots,\theta_m)$ be a m-tuple of complex numbers. Define ... and assume $t\ge 1$. There exist positive constants c_1 and c_2 with the followi...

open_problem | open | MathDB 页面 | 原文线索

V2-053935 | MATHDB-389238

Question 4.17 - Mazur

摘录 / EXCERPT: Assume that $K=\mathbb{Q}$ and that $V(\mathbb{Q})$ is Zariski dense; is Z a union of connected components of $V(\mathbb{R})$?

open_problem | open | MathDB 页面 | 原文线索

V2-053936 | MATHDB-390435

Chromatic-polynomial/list-color-function persistence conjecture

摘录 / EXCERPT: For every finite graph G and every integer $k\ge 1$, if $P(G,k)=P_{\ell}(G,k)>0$, then $P(G,k+1)=P_{\ell}(G,k+1)$, where $P(G,k)$ is the number of proper k-colorings of G and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053937 | MATHDB-390449

Generalized rational exponents conjecture

摘录 / EXCERPT: For every rational number $\alpha\ge 1$, there exist fixed finite simple graphs H_α and F_α such that, as $n\to\infty$, the maximum number of copies of H_α in...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053938 | MATHDB-390450

Triangle-density bound for C4-free graphs

摘录 / EXCERPT: For $(\varepsilon,\kappa)\in[0,1]^2$, determine the asymptotic minimum and maximum triangle densities among sequences of finite C_4-free graphs G_n satisfying...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053939 | MATHDB-390491

Vizing’s list edge-colouring conjecture

摘录 / EXCERPT: For every loopless multigraph G, the list edge-chromatic number equals the edge-chromatic number: $\operatorname{ch}'(G)=\chi'(G)$. Equivalently, if each edge of G is assigned...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053940 | MATHDB-390602

Kahn-Lovász theorem extension problem for graph factors

摘录 / EXCERPT: For a fixed connected graph F, let G be a graph with $|V(F)|\mid |V(G)|$, and let $\Phi_F(G)$ denote the number of spanning subgraphs of G that are disjoint unions of...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053941 | MATHDB-390617

Kohayakawa’s clique-cover conjecture

摘录 / EXCERPT: For each integer $s\ge 1$, let G_s be the bipartite graph with parts $\binom{[2s]}{s}$ and $\binom{[2s]}{s-1}$, where an s-subset is adjacent to an $(s-1)$-subset if and only i...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053942 | MATHDB-391905

Jamison’s questions on the number and average length of subpaths

摘录 / EXCERPT: For every connected graph G with n vertices, let $\mathcal{P}(G)$ be the set of its simple subpaths, including the one-vertex paths, and define...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053943 | MATHDB-391906

Open questions on the boxicity of compressed zero-divisor graphs of Z_N

摘录 / EXCERPT: For every positive integer $N=\prod_{i=1}^{a}p_i^{n_i}$, where the p_i are distinct primes and $n_i\geq 1$, determine the boxicity $\operatorname{box}(\Gamma_E(\mathbb{Z}_N))$ of...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053944 | MATHDB-391912

Albertson-Berman induced-forest conjecture

摘录 / EXCERPT: Every planar graph G contains a vertex subset $S\subseteq V(G)$ such that the induced subgraph $G[S]$ is a forest and $|S|\geq \frac{|V(G)|}{2}$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053945 | MATHDB-391927

Gerbner-Győri-Methuku-Vizer generalized Turán conjecture

摘录 / EXCERPT: For all integers $l>k\ge 2$, let $\mathscr{C}_{2k}=\{C_3,C_4,\ldots,C_{2k}\}$. Then ... where $\operatorname{ex}(n,H,\mathcal{F})$ is the maximum number of copies of H in an n-ve...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053946 | MATHDB-391928

Tutte’s three-edge-coloring conjecture

摘录 / EXCERPT: Every 2-connected cubic graph with no Petersen-graph minor admits a proper edge-coloring with 3 colors; equivalently, its edges can be partitioned into 3 matchings.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053947 | MATHDB-391939

Gao-Wu-Xue treewidth conjecture for generalized Turán problems

摘录 / EXCERPT: For every integer $r\ge 3$ and every finite graph F satisfying $\chi(F)=r$ and $\operatorname{tw}(F)\ge r$, there exist constants $c_F>0$ and $n_0(F)$ such that, for every intege...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053948 | MATHDB-391950

Bowler-Emde-Gut question on Maker-Breaker games on infinite graphs

摘录 / EXCERPT: Let $B=K_{\aleph_0}$, and let $G_1,\dots,G_k\subseteq B$ be finitely many infinite subgraphs. In the color-preserving Maker-Breaker game on B, Maker and Breaker alternately claim...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053949 | MATHDB-392091

Parameterized complexity of Lp-norm maximization over zonotopes

摘录 / EXCERPT: For each fixed rational $p\in(1,\infty)$, given vectors $v_1,\ldots,v_m\in\mathbb{Q}^d$ defining the zonotope $Z=\left\{\sum_{i=1}^m\lambda_i v_i: -1\leq\lambda_i\leq 1\right\}$, d...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053950 | MATHDB-392112

Instance-optimality of bidirectional Dijkstra on simple weighted graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a simple graph, directed or undirected, with positive real-valued edge weights, and let $s,t\in V$. When the graph is accessed through vertex and incident-edge que...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053951 | MATHDB-392115

Spectral Erdős-Ko-Rado problem for uniform hypergraphs

摘录 / EXCERPT: For integers $n\ge k$ and $1\le t<k$, determine ... where $\rho(\mathcal F)$ is the spectral radius of the adjacency tensor of the k-uniform hypergraph with edge set $\mathcal F$....

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053952 | MATHDB-392116

Maximum number of vertices in every metric basis

摘录 / EXCERPT: For a connected nontrivial graph G of order n, let a metric basis be a resolving set of minimum cardinality, let $\dim(G)$ denote the metric dimension, and define...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053953 | MATHDB-392117

Existence of H-induced-saturated graphs for cycles

摘录 / EXCERPT: For each integer $n\ge 4$, does there exist a finite simple graph G such that G has no induced subgraph isomorphic to C_n, but, for every pair of distinct vertices...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053954 | MATHDB-392127

Open arc-decomposition problems of Bang-Jensen, Bessy, Gonçalves, and Picasarri-Arrieta

摘录 / EXCERPT: Given a digraph $D=(V,A)$, determine whether there is a partition $A=A_1\mathbin{\dot\cup}A_2$ such that one of the following holds: (i) $D[A_1]$ is a perfect matching and $D[A_2]$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053955 | MATHDB-392267

Clique-saturating non-edges problem of He, Ma, Ma and Ye

摘录 / EXCERPT: For an integer $p\ge 3$, let $\operatorname{ex}(n,K_r)$ denote the maximum number of edges in an n-vertex K_r-free graph. For integers n and m, define $f_{p+1}(n,m)$ to be...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053956 | MATHDB-392268

McKay-Praeger conjecture

摘录 / EXCERPT: Let $\mathcal{V}_n$ and $\mathcal{D}_n$ denote, respectively, the sets of labelled vertex-transitive graphs and vertex-transitive digraphs on $[n]=\{1,\ldots,n\}$, and let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053957 | MATHDB-392272

Erdős-Sós conjecture

摘录 / EXCERPT: For every integer $t\ge 2$, every finite graph G with average degree greater than $t-2$ contains every tree T on t vertices as a subgraph; equivalently, if T is any tree on...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053958 | MATHDB-392280

3-Decomposition Conjecture

摘录 / EXCERPT: Every finite connected loopless multigraph $G=(V,E)$ such that $\deg_G(v)=3$ for every $v\in V$ has an edge partition $E=E_T\mathbin{\dot\cup}E_C\mathbin{\dot\cup}E_M$ such that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053959 | MATHDB-392304

Acharya-Mütze-Verciani connectivity conjecture for cyclically colored triangulations

摘录 / EXCERPT: Let $\mathcal{H}_N$ be the twist reconfiguration graph whose vertices are the valid triangulations of a cyclically 3-colored convex N-gon, with edges corresponding to valid twi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053960 | MATHDB-392461

Verstraëte's conjecture on consecutive even cycle lengths

摘录 / EXCERPT: For every integer $k\ge 1$ and every n-vertex graph G, if $e(G)>\frac{(2k+1)(n-1)}{2}$, then there exists an integer $r\ge 2$ such that G contains cycles of every length in t...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-053961 | MATHDB-392462

Sharp quadratic χ-binding for powers of bipartite graphs

摘录 / EXCERPT: For every integer $r\ge 2$, does there exist a constant $c_r>0$ and an infinite family of bipartite graphs B_n such that $\omega(B_n^r)\to\infty$ and...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053962 | MATHDB-392468

Boxicity of Zero Divisor Graphs

摘录 / EXCERPT: For every finite commutative ring R, let $\Gamma(R)$ be the graph whose vertices are the nonzero zero divisors of R, with distinct vertices x,y adjacent if and only if $xy=0$...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-053963 | MATHDB-392473

Adm-Fallat-Meagher-Nasserasr-Plosker-Yang question on MB(\overline{P_n})

摘录 / EXCERPT: For a graph G, let $\mathcal{S}(G)$ be the set of real symmetric matrices whose off-diagonal zero-nonzero pattern is the adjacency pattern of G. Define $MB(G)$ to be the minimu...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053964 | MATHDB-392474

Spielman-Teng conjecture on bounded-genus graph eigenvalues

摘录 / EXCERPT: $\exists C>0\ \forall g\ge 1\ \forall n\ge 1\ \forall G$, if G is an n-vertex graph embeddable on an orientable surface of genus g and has maximum degree Δ, then...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053965 | MATHDB-392481

Packing and covering cycles through prescribed vertices

摘录 / EXCERPT: For every finite simple graph G and every set $S\subseteq V(G)$, let $\tau(G,S)$ be the minimum cardinality of a vertex set $X\subseteq V(G)$ such that every cycle C of G wit...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-053966 | MATHDB-392482

Hamiltonicity of high-degree regular sublinear expanders

摘录 / EXCERPT: $\forall\varepsilon>0\,\exists C=C(\varepsilon)>0\,\exists n_0\,\forall n\ge n_0\,\forall d\ge C\log^2 n$, every n-vertex d-regular (ε,d)-expander contains a Hamilt...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-053967 | MATHDB-392625

Maximum-degree bound for connected claw-free graphs containing a stable triple

摘录 / EXCERPT: Let G be a connected, claw-free graph and assume that G contains a stable set of size three. Then every vertex of G has degree at most $4\omega(G)$.

open_problem | open | MathDB 页面 | 原文线索

V2-053968 | MATHDB-392627

Turán’s tetrahedron-density conjecture

摘录 / EXCERPT: Let $\operatorname{ex}_3(n,F)$ be the maximum number of edges in an n-vertex 3-uniform hypergraph containing no subhypergraph isomorphic to F, and let $K_4^{(3)}$ be the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-053969 | MATHDB-392628

Sós's conjecture for the Fano-plane-free triple-system extremal number

摘录 / EXCERPT: Let F be the Fano configuration, the 3-uniform hypergraph on vertices $\{x_1,x_2,x_3,a,b,c,d\}$ with edges...

conjecture | open | MathDB 页面 | 原文线索

V2-053970 | MATHDB-392629

Turán’s conjecture for the extremal size of a $K_5^{(3)}$-free triple system

摘录 / EXCERPT: For every even integer n, the 2-colorable triple system H_n is conjectured to be the largest $K_5^{(3)}$-free 3-uniform hypergraph. Here $H_n=H(X,\bar X)$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-053971 | MATHDB-392631

Exact degree and Ore-type conditions forcing a graph to be k-linked

摘录 / EXCERPT: For $k\ge 2$, let $D(n,k)$ be the minimum positive integer d such that every n-vertex graph with minimum degree at least d is k-linked, and let $R(n,k)$ be the minimum posi...

open_problem | open | MathDB 页面 | 原文线索

V2-053972 | MATHDB-392642

Removal theorem for copies of complete uniform hypergraphs

摘录 / EXCERPT: Suppose an n-vertex k-uniform hypergraph H contains only $o(n^{\ell})$ copies of $K^{(k)}_{\ell}$. Then one can delete $o(n^k)$ edges of H to make it $K^{(k)}_{\ell}$-free.

open_problem | open | MathDB 页面 | 原文线索

V2-053973 | MATHDB-392645

Extremal number of Fano-plane-free 3-uniform hypergraphs

摘录 / EXCERPT: The maximum number of triples in a 3-uniform hypergraph on n vertices containing no copy of the Fano plane is attained by partitioning the vertices into two equal parts and takin...

open_problem | open | MathDB 页面 | 原文线索

V2-053974 | MATHDB-392646

Frankl’s conjecture for the expansion of the triangle

摘录 / EXCERPT: Let $C_r^{(2k)}$ be the $2k$-uniform hypergraph formed from pairwise disjoint k-sets $P_1,\ldots,P_r$, with edges $P_i\cup P_j$ for $i\ne j$. Partition the vertex set into...

conjecture | open | MathDB 页面 | 原文线索

V2-053975 | MATHDB-392733

Multicolour Beck-Fiala discrepancy bound

摘录 / EXCERPT: For a finite hypergraph H and c colors, the multicolour discrepancy satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-053976 | MATHDB-392736

Discrepancy bounds for the multidimensional declustering hypergraph

摘录 / EXCERPT: For positive integers N and d, let ... where $[N]=\{1,\ldots,N\}$ and $[x_i..y_i]=\{x_i,x_i+1,\ldots,y_i\}$. For a coloring $\chi:[N]^d\to[M]$, define ... and ... Let q_1 be the sm...

open_problem | open | MathDB 页面 | 原文线索

V2-053977 | MATHDB-392820

Vertex-stabilizer or hyperbolicity conjecture for vertex-transitive graphs

摘录 / EXCERPT: Let Γ be an undirected, connected, locally finite graph, let G be a vertex-transitive group of automorphisms of Γ, and let $x\in V(\Gamma)$. Then at least one of th...

conjecture | open | MathDB 页面 | 原文线索

V2-053978 | MATHDB-392845

Hardness of Approximating Maximum Subgraphs with Hereditary Properties

摘录 / EXCERPT: For every nontrivial hereditary graph property π and every $\epsilon>0$, the maximum-subgraph-with-property-π problem-given a graph G, find a maximum vertex-induced subgr...

open_problem | open | MathDB 页面 | 原文线索

V2-053979 | MATHDB-392849

Asymptotic approximability of Asymmetric k-Center

摘录 / EXCERPT: In the Asymmetric k-Center problem, the input is an integer k and a complete digraph on n points with nonnegative distances c_{uv} satisfying the directed triangle inequali...

open_problem | open | MathDB 页面 | 原文线索

V2-053980 | MATHDB-392854

Linear-time approximation for the weighted matching problem

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with edge-weight function $w:E\to\mathbb{R}_+$. A matching is a subset $M\subseteq E$ such that no two edges in M are incident to the same vertex, and it...

open_problem | open | MathDB 页面 | 原文线索

V2-053981 | MATHDB-392860

Polynomial-time approximation for minimum multi-multiway cut

摘录 / EXCERPT: For a weighted undirected graph $G=(V,E)$ with edge-weight function $w:E\to\mathbb{R}_+$ and vertex sets $S_1,S_2,\ldots,S_k$, the minimum multi-multiway cut problem asks for a min...

open_problem | open | MathDB 页面 | 原文线索

V2-053982 | MATHDB-392913

Scarf-type resolutions for the deformed ideals J_G

摘录 / EXCERPT: For the restricted class of saturated digraphs G, does J_G have a “Scarf type” resolution?

open_problem | open | MathDB 页面 | 原文线索

V2-053983 | MATHDB-393120

Spectral transition for the irreversible quantum graph below the critical coupling

摘录 / EXCERPT: Let Γ be the star graph with d infinite edges, and let A_α be the self-adjoint operator in $L^2(\Gamma\times\mathbb{R})$ associated with ... subject to the vertex con...

open_problem | open | MathDB 页面 | 原文线索

V2-053984 | MATHDB-393191

Erdős-Menger Conjecture for possibly infinite digraphs

摘录 / EXCERPT: Given two sets of vertices, A and B, in a possibly infinite digraph, there exists a family $\mathcal{P}$ of disjoint A-B paths and an A-B-separating set S of vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-053985 | MATHDB-393202

Simultaneous cuts in two graphs

摘录 / EXCERPT: Suppose that G_1 and G_2 are two graphs on the same vertex set V, where G_i has m_i edges. Then there is a bipartition of V into two classes A and B such that, for...

open_problem | open | MathDB 页面 | 原文线索

V2-053986 | MATHDB-393203

Almost-perfect matchings in 3-partite 3-uniform hypergraphs

摘录 / EXCERPT: Every 3-uniform 3-partite hypergraph H whose three vertex classes have size n and whose minimum degree $\[2m]\delta_2(H)\u001b[0m$ is at least $n/3$ has a matching which covers...

open_problem | open | MathDB 页面 | 原文线索

V2-053987 | MATHDB-393205

Pfaffian labelings and signs of edge colorings (Theorem 5)

摘录 / EXCERPT: A multigraph G admits a Pfaffian labeling if and only if, for all k, all the k-edge colorings of every k-regular multigraph G' with the same underlying simple graph as...

open_problem | open | MathDB 页面 | 原文线索

V2-053988 | MATHDB-393586

Spectral-measure formula for free products of colored graphs

摘录 / EXCERPT: Conjecture 4. We have an equality of spectral measures ... where X,Y are colored graphs, $X * Y$ is obtained by “putting a copy of X at each vertex of Y,” and $\boxtimes$ is Vo...

open_problem | open | MathDB 页面 | 原文线索

V2-053989 | MATHDB-393663

Network design and facility location with service capacities

摘录 / EXCERPT: Given a metric space (V,c), a set $D\subseteq V$ of terminals/customers, demands $d:D\to\mathbb{R}_+$, a facility opening cost $f\in\mathbb{R}_+$, and a capacity...

open_problem | open | MathDB 页面 | 原文线索

V2-053990 | MATHDB-393664

Efficient computation of earliest-arrival transshipments with multiple sources and one sink

摘录 / EXCERPT: Given a network with arc capacities $u_e\ge 0$ and transit times $\tau_e\ge 0$, multiple source nodes with prescribed positive supplies, and a single sink with the corresponding de...

open_problem | open | MathDB 页面 | 原文线索

V2-053991 | MATHDB-393669

Characterization of multicast-rearrangeable symmetric Clos networks

摘录 / EXCERPT: For a symmetric three-stage Clos network with parameters n, r, and m, determine necessary and sufficient conditions on n, r, and m for the network to be multicast-rearr...

open_problem | open | MathDB 页面 | 原文线索

V2-053992 | MATHDB-393702

An analogue of the Tian gauge for vertex-algebra deformation theory

摘录 / EXCERPT: Can one find an analogue of the Tian gauge?

open_problem | open | MathDB 页面 | 原文线索

V2-053993 | MATHDB-393733

Quadratic-size Sidon-type construction for equivelar surfaces

摘录 / EXCERPT: Given the cyclic construction on the vertex set $\mathbb Z_N$ with face vertices given by the cyclic translates of $0,A_1,A_2,\ldots,A_{2m}$ and successive differences...

open_problem | open | MathDB 页面 | 原文线索

V2-053994 | MATHDB-393734

Conjecture 7: a 4-uniform linear hypergraph configuration

摘录 / EXCERPT: If a 4-uniform linear hypergraph H on n vertices does not contain four distinct edges e_1,e_2,e_3,e_4, where $e_1=\{a,b,c,\ast\}$, $e_2=\{a,d,e,\ast\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-053995 | MATHDB-393735

Frankl-Rödl conjecture on 4-uniform linear hypergraphs

摘录 / EXCERPT: If a 4-uniform linear hypergraph H on n vertices does not contain four distinct edges e_1,e_2,e_3,e_4 such that ... then the number of edges of H is $o(n^2)$. Here the symbol...

conjecture | open | MathDB 页面 | 原文线索

V2-053996 | MATHDB-393739

Asymptotic behavior of the maximum degree in a random planar graph

摘录 / EXCERPT: Let R_n be a uniformly random simple planar graph on the vertex set $[U+007F]{1,[U+007F]ldots,n[U+007F]}$. Determine the asymptotic behavior of its maximum degree.

open_problem | open | MathDB 页面 | 原文线索

V2-053997 | MATHDB-393741

Exact Turán numbers for generalized triangles T_i

摘录 / EXCERPT: Determine the exact Turán function $\operatorname{ex}(n,T_i)$ for every integer $i\ge 5$, where T_i is the i-uniform hypergraph with edges $\{1,\ldots,i\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-053998 | MATHDB-393744

Hamiltonian cycles in Dirac graphs

摘录 / EXCERPT: Let G be an n-vertex Dirac graph, meaning that its minimum degree is at least $n/2$, and let $\Psi(G)$ denote the number of Hamiltonian cycles in G. Theorem 1 asserts that ...

open_problem | open | MathDB 页面 | 原文线索

V2-053999 | MATHDB-393746

Erdős-T. Sós conjecture on trees

摘录 / EXCERPT: The Erdős-T. Sós conjecture asserts that, for any k-vertex tree T_k, $\operatorname{ext}(n,T_k)\le \frac{(k-2)n}{2}$, where $\operatorname{ext}(n,L)$ is the maximum number of e...

conjecture | open | MathDB 页面 | 原文线索

V2-054000 | MATHDB-393747

Erdős-Sós conjecture for sufficiently large trees

摘录 / EXCERPT: There exists an integer k_0 such that, for every integer $k>k_0$, every n-vertex graph G_n containing no k-vertex tree T_k satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-054001 | MATHDB-393822

Necessity of a condition corresponding to (1) for loopless digraphs

摘录 / EXCERPT: Considering Theorems 2 and 3, is a condition corresponding to (1), namely $\forall i\in\{1,\ldots,n\}: v_i\notin C_i$, needed in the case of loopless digraphs?

open_problem | open | MathDB 页面 | 原文线索

V2-054002 | MATHDB-393826

Sufficient conditions for arbitrarily vertex decomposable graphs beyond traceability

摘录 / EXCERPT: Find other sufficient conditions for graphs to be arbitrarily vertex decomposable that do not imply traceability.

open_problem | open | MathDB 页面 | 原文线索

V2-054003 | MATHDB-393840

Dichotomy conjecture for digraph homomorphism

摘录 / EXCERPT: For every fixed digraph H, the decision problem that, given an input digraph G, asks whether there is a homomorphism from G to H is either polynomial-time solvable or NP-co...

conjecture | open | MathDB 页面 | 原文线索

V2-054004 | MATHDB-393862

Localization for radial metric trees with random vertex δ-potentials

摘录 / EXCERPT: Let T be a radial metric tree in which the edge lengths are $\ell_e=\ell_n$ and the vertex degrees satisfy $d(v)=d_n\ge 3$, with both depending only on the generation n. At eac...

open_problem | open | MathDB 页面 | 原文线索

V2-054005 | MATHDB-393867

Existence of a 27-vertex triangulation of a nonspherical combinatorial 16-manifold

摘录 / EXCERPT: Does there exist a simplicial complex triangulating a combinatorial 16-manifold with 27 vertices which is not a sphere?

open_problem | open | MathDB 页面 | 原文线索

V2-054006 | MATHDB-393877

Degree-5 and degree-7 triangulation of the torus

摘录 / EXCERPT: Does there exist a triangulation T of the torus with exactly one vertex of degree 5, exactly one vertex of degree 7, and every other vertex of degree 6?

open_problem | open | MathDB 页面 | 原文线索

V2-054007 | MATHDB-393879

Four-dimensional characterization of regular vertices in piecewise-linear Morse theory

摘录 / EXCERPT: The report proposes the following conjectural extension to four dimensions: the link of a vertex is a 3-sphere, and, for a regular vertex, it is necessary and sufficient that the...

open_problem | open | MathDB 页面 | 原文线索

V2-054008 | MATHDB-394005

Uniformization of the cross-ratio parameter space for one-vertex circle packings

摘录 / EXCERPT: Let τ be a graph on Σ_g, with $g\geq 2$, having one vertex and lifting to an honest triangulation of the universal cover $\widetilde{\Sigma}_g$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-054009 | MATHDB-394099

Characterization of multiterminal networks permitting ideal inverse-problem reduction

摘录 / EXCERPT: Characterize those multiterminal networks that allow the “ideal” situation in which the complicated nonlinear equations for recovering the unknown Markovian routing strategy from i...

open_problem | open | MathDB 页面 | 原文线索

V2-054010 | MATHDB-394203

Decentralized path construction in line-of-sight networks

摘录 / EXCERPT: Let $p = C\ln n/\omega$ for a constant C to be specified below. There is a decentralized algorithm that, given s and t, with high probability constructs an s-t path with...

open_problem | open | MathDB 页面 | 原文线索

V2-054011 | MATHDB-394206

Rainbow Turán problem for all cycles: an $O(n\\log n)$ bound

摘录 / EXCERPT: Could there be an $O(n\log n)$ upper bound for the maximum number of edges in an n-vertex graph admitting a proper edge-colouring with no rainbow cycle, where a rainbow cycle is...

open_problem | open | MathDB 页面 | 原文线索

V2-054012 | MATHDB-394214

Hamiltonian cycles in uniform hypergraphs at the Dirac codegree threshold

摘录 / EXCERPT: Conjecture. Let $k\ge 2$, and let H be a k-uniform hypergraph on n vertices. If every $(k-1)$-element set of vertices of H belongs to at least...

open_problem | open | MathDB 页面 | 原文线索

V2-054013 | MATHDB-394215

Complexity of dense hypergraph perfect matching at the threshold

摘录 / EXCERPT: For integers $k \ge 2$ and $0 \le r \le k-1$ and $0<c<1$, let $PM(k,r,c)$ be the problem of deciding whether, for an input k-graph H with $|V(H)|=n\equiv r\pmod{k}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-054014 | MATHDB-394350

Regularity inequalities for Stanley-Reisner ideals

摘录 / EXCERPT: Let K be a field, let $S=K[x_1,\ldots,x_n]$, and let X and Y be simplicial complexes on the same vertex set $[n]=\{1,\ldots,n\}$. For a simplicial complex Z on $[n]$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-054015 | MATHDB-394423

Hadwiger’s conjecture for quasi-line graphs

摘录 / EXCERPT: Every finite quasi-line graph G with chromatic number $\cchi(G)$ has a clique minor of size $\cchi(G)$. Here, a quasi-line graph is a graph in which, for every vertex v, the ne...

conjecture | open | MathDB 页面 | 原文线索

V2-054016 | MATHDB-394425

Conjecture on feedback arc sets in 3-free digraphs

摘录 / EXCERPT: Let G be a finite digraph with no parallel edges, and call G 3-free if it has no directed cycle of length at most 3. Let $\gamma(G)$ be the number of unordered pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-054017 | MATHDB-394440

Seymour’s openly disjoint circuits question for regular digraphs

摘录 / EXCERPT: Does every r-regular digraph D contain a vertex x with $\\kappa_D(x,x)=r$? Here the report assumes finite digraphs without loops or multiple edges of the same direction; r-...

open_problem | open | MathDB 页面 | 原文线索

V2-054018 | MATHDB-394442

Bonato-Janssen uniqueness claim for infinite locally random graphs

摘录 / EXCERPT: For finite graphs G and H, let $\uparrow G$ and $\uparrow H$ denote the limit graphs obtained by iterating the pure-extension construction, in which for every vertex x and ev...

open_problem | open | MathDB 页面 | 原文线索

V2-054019 | MATHDB-394477

Threshold for online vertex colorings of random graphs

摘录 / EXCERPT: Determine, for a fixed graph F and r available colors, the threshold $p_0(F,r,n)$ for the online vertex-coloring game on $G_{n,p}$: vertices are revealed one at a time together...

open_problem | open | MathDB 页面 | 原文线索

V2-054020 | MATHDB-394478

Generalization of the online edge-coloring threshold to arbitrarily many colors

摘录 / EXCERPT: The generalization of Theorem 2.1 to an arbitrary number of colors $r\ge 2$ remains a challenging open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-054021 | MATHDB-394481

Avoiding a monochromatic K_4 in the planar online Ramsey game

摘录 / EXCERPT: Let $H=K_4$. Builder chooses some vertex set and, on each turn, selects an edge so that the graph consisting of all edges selected so far is planar. Painter immediately colors the...

open_problem | open | MathDB 页面 | 原文线索

V2-054022 | MATHDB-394882

Exact finite-\n extremal function for graphs excluding a $K_{2,t}$ minor

摘录 / EXCERPT: For integers $t\ge 2$ and n such that $t\nmid(n-1)$, determine the maximum number of edges in an n-vertex graph with no $K_{2,t}$ minor. The report proves the upper bound...

open_problem | open | MathDB 页面 | 原文线索

V2-054023 | MATHDB-394885

Average sizes of convex sets in acyclic digraphs

摘录 / EXCERPT: Let D be an acyclic digraph of order n. A non-empty vertex set X is convex if no directed path in D between vertices of X contains a vertex outside X; it is connected i...

open_problem | open | MathDB 页面 | 原文线索

V2-054024 | MATHDB-394886

Optimality of the running time for enumerating connected convex sets

摘录 / EXCERPT: Is $O(n\,\mathrm{cc}(D))$ asymptotically the best running time for computing all connected convex sets of a digraph D of size n?

open_problem | open | MathDB 页面 | 原文线索

V2-054025 | MATHDB-394888

Minimum semidegree forcing a cycle of any fixed length through every vertex

摘录 / EXCERPT: For every integer $\ell\ge 4$, there exists an integer n_0 such that whenever G is an oriented graph on $n\ge n_0$ vertices with minimum semidegree...

open_problem | open | MathDB 页面 | 原文线索

V2-054026 | MATHDB-394893

Dirac-type conjecture for tight Hamiltonian cycles in uniform hypergraphs

摘录 / EXCERPT: Let H be a k-graph with $n\ge k+1\ge 4$ vertices. If every $(k-1)$-element set of vertices is contained in at least $\left\lfloor\frac{n-k+3}{2}\right\rfloor$ edges, then H h...

conjecture | open | MathDB 页面 | 原文线索

V2-054027 | MATHDB-394894

Generalized induced removal lemma for k-uniform hypergraphs

摘录 / EXCERPT: For every (possibly infinite) family $\mathcal F$ of k-uniform hypergraphs and every $\eta>0$, there exist constants $c>0$, $C>0$, and n_0 such that the following holds. Suppos...

open_problem | open | MathDB 页面 | 原文线索

V2-054028 | MATHDB-394895

Hypergraph removal lemma conjecture (Erdős-Frankl-Rödl Problem 6.1)

摘录 / EXCERPT: For every integer $k\ge 2$, every k-uniform hypergraph $F^{(k)}$ on ℓ vertices, and every $\eta>0$, there exist constants $c>0$ and n_0 such that, whenever $H^{(k)}$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-054029 | MATHDB-394908

Thresholds for perfect factors in random graphs and hypergraphs

摘录 / EXCERPT: Estimate the threshold $\operatorname{th}_H(n)$ of the property that an Erdős-Rényi random graph or uniform hypergraph on n points contains an H-factor, for a general fixed H...

open_problem | open | MathDB 页面 | 原文线索

V2-054030 | MATHDB-394911

Shamir’s problem: threshold for perfect matchings in random uniform hypergraphs

摘录 / EXCERPT: The threshold for a random r-uniform hypergraph to contain a perfect matching is $\Theta\!\left(\frac{\log n}{n^{r-1}}\right)$.

open_problem | open | MathDB 页面 | 原文线索

V2-054031 | MATHDB-395297

Turán bound for triangulations of the torus

摘录 / EXCERPT: Every n-vertex two-dimensional simplicial complex with $\Omega(n^{5/2})$ simplices contains a triangulation of the torus.

open_problem | open | MathDB 页面 | 原文线索

V2-054032 | MATHDB-395300

Chromatic number of point-rectangle hypergraphs

摘录 / EXCERPT: For a finite set of n points $P\subset\mathbb{R}^2$, form the hypergraph with vertex set P and hyperedges $P\cap R$, where R ranges over axis-parallel rectangles and...

open_problem | open | MathDB 页面 | 原文线索

V2-054033 | MATHDB-395391

Conjecture 2 on subtree descriptions of mixed-integer vertex-cover polyhedra

摘录 / EXCERPT: Let $G=(U\cup V,E)$ be a bipartite graph, let $I\subseteq U\cup V$, and let $b_{ij}\in\mathbb{Q}$ for each $ij\in E$. Define ... Let $\mathcal{T}$ be the family of subtrees T of...

conjecture | open | MathDB 页面 | 原文线索

V2-054034 | MATHDB-395394

Tree-routing property of the symmetric Virtual Private Network Design problem

摘录 / EXCERPT: For every simple, undirected, connected graph $G=(V,E)$, nonnegative edge-cost vector $c\in\mathbb{R}_+^E$, and nonnegative integer maximum-cumulative-demand vector...

open_problem | open | MathDB 页面 | 原文线索

V2-054035 | MATHDB-395395

Tree-routing property for the concave symmetric VPN design problem

摘录 / EXCERPT: Every instance of the concave symmetric Virtual Private Network Design (cVPND) problem, with a simple undirected connected graph $G=(V,E)$, edge-cost vector $c\in\mathbb{R}_+^E$, m...

open_problem | open | MathDB 页面 | 原文线索

V2-054036 | MATHDB-395399

Gallai’s conjecture on circuit covers of strong digraphs

摘录 / EXCERPT: Every strongly connected directed graph D with stability number α is spanned by the union of α directed circuits.

conjecture | open | MathDB 页面 | 原文线索

V2-054037 | MATHDB-395435

The boundary of an infinite resistance network

摘录 / EXCERPT: Let G be the connected weighted graph in the report, let (G_n) be an exhaustion by finite subnetworks, and define ... with ... The boundary term is defined by ... With respect to wha...

open_problem | open | MathDB 页面 | 原文线索

V2-054038 | MATHDB-395486

Problem 9: Relations among signed areas in a triangular partition

摘录 / EXCERPT: We are given a partition of the unit square into T triangles (not necessarily a triangulation). A vertex that lies on an edge of some triangle or of the bounding square has 1 d...

open_problem | open | MathDB 页面 | 原文线索

V2-054039 | MATHDB-395490

Equivariant flat PL metrics on abstract lattice triangulations

摘录 / EXCERPT: Given an abstract lattice triangulation of d-space with a transitive vertex action by $G\cong\mathbb{Z}^d$, and an associated G-equivariant PL metric in which every simplex is...

open_problem | open | MathDB 页面 | 原文线索

V2-054040 | MATHDB-395491

Neighborly triangulations of the 3-torus with unbounded vertex-transitive symmetry

摘录 / EXCERPT: Does there exist an unbounded number of neighborly triangulations of the 3-torus with vertex-transitive $\mathbb Z_n$-symmetry, for $n\ge 15$, where “neighborly” means that any two...

open_problem | open | MathDB 页面 | 原文线索

V2-054041 | MATHDB-395627

Robust stability radius under parametric uncertainty in biochemical reaction networks

摘录 / EXCERPT: Given nominal parameter values p_0, compute a robust stability radius ψ^* such that, for every $x\in\mathbb{R}^n$ and $p\in\mathbb{R}^q$ satisfying $Sv(x,p)=0$ and ... the Jac...

open_problem | open | MathDB 页面 | 原文线索

V2-054042 | MATHDB-395639

Pisot conjecture: vertex sets of Pisot substitution tilings are model sets

摘录 / EXCERPT: One formulation of the Pisot conjecture states that the vertex set Λ of a Pisot substitution tiling is always a model set: there exists a window set $\Omega\subset W$, equa...

conjecture | open | MathDB 页面 | 原文线索

V2-054043 | MATHDB-395750

F-free hypergraph chromatic-number conjecture

摘录 / EXCERPT: Let F be a k-graph. There is a constant c_F depending only on F such that every F-free k-graph with maximum degree Δ has chromatic number at most...

conjecture | open | MathDB 页面 | 原文线索

V2-054044 | MATHDB-395752

Threshold for homological connectivity over the integers

摘录 / EXCERPT: For the random simplicial complex $X_d(n,p)$ on vertex set $[n]$, with a full $(d-1)$-dimensional skeleton and each d-dimensional face included independently with probability p...

open_problem | open | MathDB 页面 | 原文线索

V2-054045 | MATHDB-395763

Asymptotic VCG cost for a random minimum spanning arborescence

摘录 / EXCERPT: For a minimum spanning arborescence in the complete digraph $\vec K_n$, with i.i.d. edge weights having either the uniform $[0,1]$ distribution or the exponential distribution with...

open_problem | open | MathDB 页面 | 原文线索

V2-054046 | MATHDB-395766

Off-diagonal 3-uniform hypergraph Ramsey lower bound

摘录 / EXCERPT: There is a constant $c>0$ such that ... for all $4\le s\le n$.

open_problem | open | MathDB 页面 | 原文线索

V2-054047 | MATHDB-395767

Three-colour 3-uniform hypergraph Ramsey lower bound

摘录 / EXCERPT: There is a constant $c>0$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-054048 | MATHDB-396208

The k edge-disjoint paths problem in digraphs with bounded independence number

摘录 / EXCERPT: For every fixed natural number k and every fixed constant $\b1$, the k edge-disjoint paths problem is solvable in polynomial time for digraphs with independence number at most...

open_problem | open | MathDB 页面 | 原文线索

V2-054049 | MATHDB-396210

Seymour-Goldberg conjecture for multigraph edge-colouring

摘录 / EXCERPT: For every multigraph G, the chromatic index satisfies ... where ...

conjecture | open | MathDB 页面 | 原文线索

V2-054050 | MATHDB-396656

Straightening robot arms in Euclidean space

摘录 / EXCERPT: A robot arm is a path without self-intersections in $\mathbb{R}^d$ consisting of finitely many segments (bars), with two bars intersecting only at a vertex. Is it always possible t...

open_problem | open | MathDB 页面 | 原文线索

V2-054051 | MATHDB-396669

Optimal average queue-size scaling for switched networks

摘录 / EXCERPT: Identify a quantity $\chi_1(S)$ depending on the scheduling set S such that, under the best policy with a well-defined stationary distribution, for any $\rho\in(0,1)$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-054052 | MATHDB-396670

Optimal exponential-tail scaling for switched-network queue sizes

摘录 / EXCERPT: For a switched network with scheduling constraints S and $\alpha>0$, identify a quantity $\chi_2(S,\alpha)$ depending on S and α such that, under the best scheduling pol...

open_problem | open | MathDB 页面 | 原文线索

V2-054053 | MATHDB-396671

Simultaneous capacity, queue-size, and computational efficiency for switched-network scheduling

摘录 / EXCERPT: Is it possible, for an arbitrary scheduling set (or conflict graph) in the switched-network model, to construct an implementation that is (i) capacity achieving, (ii) has queue siz...

open_problem | open | MathDB 页面 | 原文线索

V2-054054 | MATHDB-396673

Nonparametric inference of service-time distributions in partially observed stochastic networks

摘录 / EXCERPT: Develop a nonparametric statistical theory and estimators for the service-time distributions at the nodes of stochastic networks with an arbitrary fixed number of nodes, where exte...

open_problem | open | MathDB 页面 | 原文线索

V2-054055 | MATHDB-396813

Asymptotic minimum size of 5-cycle-semisaturated graphs

摘录 / EXCERPT: Conjecture 2. The minimum number of edges in an n-vertex C_5-semisaturated graph satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-054056 | MATHDB-396817

Local out-neighbour chromatic number versus global tournament chromatic number

摘录 / EXCERPT: For all $k \ge 0$ there exists c such that, if G is a tournament in which the set of out-neighbours of each vertex has chromatic number at most k, then $\chi(G) \le c$.

open_problem | open | MathDB 页面 | 原文线索

V2-054057 | MATHDB-396818

Analogues for the edge-coloring setting

摘录 / EXCERPT: Do analogues of Theorem 1 and Theorem 2 hold in other similar settings, in particular for the edge-coloring setting first studied by Friedgut et al. [2]?

open_problem | open | MathDB 页面 | 原文线索

V2-054058 | MATHDB-396819

Vertex-disjoint directed paths in semicomplete digraphs

摘录 / EXCERPT: For every fixed integer k, there is a polynomial-time algorithm that, given a semicomplete digraph G, vertices $s_1,t_1,\ldots,s_k,t_k$ of G, and integers...

open_problem | open | MathDB 页面 | 原文线索

V2-054059 | MATHDB-396823

Asymptotic enumeration of sparse 2-connected graphs

摘录 / EXCERPT: Let $T(n,m)$ denote the number of simple 2-connected graphs on vertex set $[n]=\{1,\ldots,n\}$ with m edges. The report states that, uniformly for $m=O(n\log n)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-054060 | MATHDB-396824

Asymptotic enumeration of strongly connected digraphs

摘录 / EXCERPT: Let $S(n,m)$ denote the number of labelled strongly connected simple digraphs on n vertices with m arcs, where loops are permitted but multiple arcs are forbidden. Uniformly fo...

open_problem | open | MathDB 页面 | 原文线索

V2-054061 | MATHDB-396833

The pagoda extremal problem for three-dimensional pagodas

摘录 / EXCERPT: Let V be a finite vertex set. A 3-dimensional pagoda over V consists of sets $V_1,V_2,V_3,V_4\subseteq V$, edge sets $E_{ij}\subseteq\binom{V}{2}$, triple sets...

open_problem | open | MathDB 页面 | 原文线索

V2-054062 | MATHDB-396836

Quasi-randomness of hypergraph cut properties

摘录 / EXCERPT: Let D_1 denote the property that a k-uniform hypergraph $H=(V,E)$ satisfies $e(U)=\frac{p}{k!}|U|^k\pm o(n^k)$ for every $U\subseteq V(H)$. For a vector...

open_problem | open | MathDB 页面 | 原文线索

V2-054063 | MATHDB-396841

Alon's 100-dominating directed graph problem

摘录 / EXCERPT: Call a directed graph $D=(V,E)$ s-dominating if, for any subset U of at most s vertices of V, there is a vertex v such that (v,u) is a directed edge for every $u\in U$....

open_problem | open | MathDB 页面 | 原文线索

V2-054064 | MATHDB-396847

Equality of the extremal functions $f_1(k)$ and $f_2(k)$

摘录 / EXCERPT: A (k,d)-tree is a binary tree in which every vertex has either two or zero children, every leaf has depth at least k, and every vertex has at most d leaf-descendants at dista...

open_problem | open | MathDB 页面 | 原文线索

V2-054065 | MATHDB-396848

Triangle-free versus bipartite subgraphs above minimum degree $3n/4$

摘录 / EXCERPT: Let G be an n-vertex graph with minimum degree larger than $3n/4$. Is it true that the largest triangle-free subgraph of G is bipartite?

open_problem | open | MathDB 页面 | 原文线索

V2-054066 | MATHDB-396947

Quiver-coefficient positivity and finiteness conjecture

摘录 / EXCERPT: Let Q be a quiver without oriented cycles, with vertex set $[U+007F]{1,2,\ldots,n\}$, let $e=(e_1,\ldots,e_n)$ be a dimension vector, set $E_i=\mathbb C^{e_i}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054067 | MATHDB-396962

Joyal’s conjecture on the free locally cartesian closed quasi-category generated by a graph

摘录 / EXCERPT: For a graph G, let $\mathrm{MLTT}(G)$ be the type system obtained by adjoining to intensional Martin-Löf type theory a type $T(a)$ for each vertex $a\in G$ and a term...

conjecture | open | MathDB 页面 | 原文线索

V2-054068 | MATHDB-397050

Structure of the projective-free part of the Lie module

摘录 / EXCERPT: Let $n=p^k$. Is the projective-free part of $\\operatorname{Lie}(n)$ indecomposable? Does it have an endo-permutation module as a source? Is its vertex a regular elementary abelian...

open_problem | open | MathDB 页面 | 原文线索

V2-054069 | MATHDB-397212

Refined cohomological interpretation of the truncated quiver polynomials

摘录 / EXCERPT: For the one-vertex quiver with g loops, with g 0, and for a partition λ of n, the conjecture is that $A_\lambda(1)$ equals the number of connected components of type λ of the natur...

open_problem | open | MathDB 页面 | 原文线索

V2-054070 | MATHDB-397253

Semiregular automorphisms of finite vertex-transitive graphs

摘录 / EXCERPT: Does every finite vertex-transitive graph have a semiregular automorphism, that is, an automorphism whose cycles all have the same length $\ll>1$?

open_problem | open | MathDB 页面 | 原文线索

V2-054071 | MATHDB-397319

Asymptotics of t-union-free r-uniform families

摘录 / EXCERPT: Given integers $r \ge t \ge 1$, find an asymptotic formula for $UF_t(n,r)$, the maximum size of a t-union-free r-uniform family on an n-element vertex set.

open_problem | open | MathDB 页面 | 原文线索

V2-054072 | MATHDB-397342

2-colorability of shift-chain hypergraphs

摘录 / EXCERPT: For $A\subset [n]$, let a_i be the ith smallest element of A. For two k-element sets $A,B\subset [n]$, define $A\le B$ if $a_i\le b_i$ for every i. A k-uniform hypergra...

open_problem | open | MathDB 页面 | 原文线索

V2-054073 | MATHDB-397348

Convex partitions of 3-space with large cell-vertex incidence counts

摘录 / EXCERPT: Is there, for every c, a locally finite (perhaps even triply periodic) partition of 3-space into convex cells such that every cell is incident to at least c vertices and every...

open_problem | open | MathDB 页面 | 原文线索

V2-054074 | MATHDB-397427

Max 2-VDP-free Induced Subgraph

摘录 / EXCERPT: Input: A graph $G=(V,E)$) and distinct terminals $s_1,t_1,s_2,t_2\in V$. Problem: Find a vertex set $V_0\subseteq V$ of maximum cardinality such that...

open_problem | open | MathDB 页面 | 原文线索

V2-054075 | MATHDB-397428

Integrality gap of the undirected hypergraphic Steiner-tree relaxation

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with terminal set $R\subseteq V$ and nonnegative edge costs $c:E\to\mathbb{R}_+$. Let $\mathcal{K}$ be the set of all full components of the instance, wher...

open_problem | open | MathDB 页面 | 原文线索

V2-054076 | MATHDB-397508

Rare-event simulation uniformly in network size and rarity

摘录 / EXCERPT: Design rare-event simulation algorithms for the described risk-network setting whose performance is uniform both in the size of the network and in the underlying large-deviations p...

open_problem | open | MathDB 页面 | 原文线索

V2-054077 | MATHDB-397549

Synchronization in networks with time-delay coupling

摘录 / EXCERPT: Can systems in networks show synchronous behaviour even in the presence of time delays?

open_problem | open | MathDB 页面 | 原文线索

V2-054078 | MATHDB-397618

Turán's conjecture for the complete 3-graph on four vertices

摘录 / EXCERPT: Let $K_4^{(3)}$ denote the complete 3-graph on 4 vertices, and let $\\operatorname{ex}(n,F)$ be the maximum number of edges in a k-uniform hypergraph on n vertices containi...

conjecture | open | MathDB 页面 | 原文线索

V2-054079 | MATHDB-397619

General hypergraph Turán problem

摘录 / EXCERPT: Given a family of r-graphs $\mathcal{F}$, compute $\operatorname{ex}(n,\mathcal{F})$, or, failing this, find the Turán density $\pi(\mathcal{F})$.

open_problem | open | MathDB 页面 | 原文线索

V2-054080 | MATHDB-397620

Decidability of the density function

摘录 / EXCERPT: Is $\pi(F) \le \alpha$ decidable, where the input is a finite family F of r-uniform hypergraphs and $\alpha \in \mathbb{Q}$?

open_problem | open | MathDB 页面 | 原文线索

V2-054081 | MATHDB-397622

Erdős's conjecture on the size of a hypergraph with bounded matching number

摘录 / EXCERPT: Every k-uniform hypergraph H on n vertices with matching number $\nu(H)<t\le \frac{n}{k}$ satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-054082 | MATHDB-397625

Triangle-clique hypergraph Ramsey number

摘录 / EXCERPT: Conjecture: $R(C_3,K_t^3)=o(t^{3/2})$ as $t\to\infty$. Here C_3 is the loose 3-cycle, K_t^3 is the complete 3-uniform hypergraph on t vertices, and $R(C_3,K_t^3)$ is the leas...

open_problem | open | MathDB 页面 | 原文线索

V2-054083 | MATHDB-397626

Chung-Graham rationality conjecture for finite hypergraph Turán densities

摘录 / EXCERPT: 4Pi{mathrm{fin}}^{k} consists of rational numbers only.

conjecture | open | MathDB 页面 | 原文线索

V2-054084 | MATHDB-397627

Minimum monochromatic K_4 density in two-colourings

摘录 / EXCERPT: Determine the minimum number of monochromatic copies of K_4 in a red-blue edge-colouring of the complete graph K_n, asymptotically as $n\to\infty$ (equivalently, determine the...

open_problem | open | MathDB 页面 | 原文线索

V2-054085 | MATHDB-397629

Erdős-Simonovits conjecture on bipartite Turán numbers

摘录 / EXCERPT: For a family $\mathcal F$ of bipartite graphs, let $\operatorname{ex}(n,\mathcal F)$ be the maximum number of edges in an n-vertex graph containing no member of $\mathcal F$ as a...

conjecture | open | MathDB 页面 | 原文线索

V2-054086 | MATHDB-397631

Kohayakawa-Łuczak-Rödl conjecture for 2-balanced graphs

摘录 / EXCERPT: Let H be a fixed 2-balanced graph. For positive parameters ε and p, let $\mathcal G(H,n,m,p,\varepsilon)$ be the family of graphs whose vertex set is partitioned...

conjecture | open | MathDB 页面 | 原文线索

V2-054087 | MATHDB-397632

Asymptotic Turán number for 3-uniform expansions of forests

摘录 / EXCERPT: Given a forest T with at least one edge, define its 3-uniform expansion $T^{(3)}$ by replacing every edge xy of T with a 3-set containing x,y and one new vertex, with d...

open_problem | open | MathDB 页面 | 原文线索

V2-054088 | MATHDB-397689

Evasiveness conjecture for vertex-homogeneous simplicial complexes

摘录 / EXCERPT: Conjecture: If K is a finite vertex-homogeneous simplicial complex and K is non-evasive, then K is a simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-054089 | MATHDB-398062

Monodromies of complexified Shabat-polynomial mappings

摘录 / EXCERPT: Let Γ be an isotopy class of finite vertex-bicollared planar trees, and let $f_\Gamma:\mathbb{C}\to\mathbb{C}$ be the associated polynomial with critical values in $\{0,1\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-054090 | MATHDB-398195

The k-disjoint paths problem in directed planar graphs

摘录 / EXCERPT: Given a directed planar graph G with n vertices and k prescribed terminal pairs $(s_1,t_1),\dots,(s_k,t_k)$, decide whether there are pairwise vertex-disjoint directed paths...

open_problem | open | MathDB 页面 | 原文线索

V2-054091 | MATHDB-398196

Limits of real edge-coloring models

摘录 / EXCERPT: Let $\mathbb{R}[x_1,x_2,\ldots]$ be the set of polynomials in infinitely many variables, with each polynomial involving only finitely many variables. Let B be the set of all...

open_problem | open | MathDB 页面 | 原文线索

V2-054092 | MATHDB-398203

Common vertex in diameter simplices

摘录 / EXCERPT: For any $d \ge 4$, every pair of $(d-1)$-dimensional simplices in a graph of diameters in $\mathbb{R}^d$ share at least one vertex.

open_problem | open | MathDB 页面 | 原文线索

V2-054093 | MATHDB-398208

Ramsey goodness of hypercubes

摘录 / EXCERPT: For every integer $s\ge 1$, is the family of hypercubes $\{Q_n\}_{n\ge 1}$ s-good? Here Q_n has vertex set $\{0,1\}^n$, with two vertices adjacent exactly when they differ in o...

open_problem | open | MathDB 页面 | 原文线索

V2-054094 | MATHDB-398221

Loebl-Komlós-Sós conjecture

摘录 / EXCERPT: Let G be an n-vertex graph with at least $n/2$ vertices of degree at least k. Let T be any tree on $k+1$ vertices. Then $T\subseteq G$.

conjecture | open | MathDB 页面 | 原文线索

V2-054095 | MATHDB-398319

Forcing conjecture for bipartite graphons

摘录 / EXCERPT: Let $\Omega=(X,\mathcal B,\mu)$ be a measure space of total measure 1, let $W:X^2\to[0,1]$ be a symmetric measurable graphon, and let F be a bipartite graph with vertex set...

conjecture | open | MathDB 页面 | 原文线索

V2-054096 | MATHDB-398348

Erdős-Hajnal-type clique/independent-set bounds for hypergraphs

摘录 / EXCERPT: For every integer $k\ge 4$, there exist a constant c_k, a k-uniform hypergraph H, and a sequence (G_n) of H-free k-uniform hypergraphs with n vertices such that the l...

open_problem | open | MathDB 页面 | 原文线索

V2-054097 | MATHDB-398354

Absolute-density high-girth 3-uniform hypergraphs

摘录 / EXCERPT: Is there an absolute constant $c>0$ and n-vertex 3-uniform hypergraphs with cn^2 hyperedges and arbitrarily large girth?

open_problem | open | MathDB 页面 | 原文线索

V2-054098 | MATHDB-398355

Independence numbers of sparse powers of Kneser graphs

摘录 / EXCERPT: Let K_n^t be the graph with vertex set ... where two distinct vertices $(A_1,\ldots,A_t)$ and $(B_1,\ldots,B_t)$ are adjacent if and only if there is an index i, $1\le i\le t$, s...

open_problem | open | MathDB 页面 | 原文线索

V2-054099 | MATHDB-398356

Hypergraph extension of the clique blow-up theorem

摘录 / EXCERPT: Is it true that every 3-uniform hypergraph on n vertices which contains cn^4 copies of $K_4^{(3)}$ also contains a blow-up of $K_4^{(3)}$ with each of the 4 sets of size...

open_problem | open | MathDB 页面 | 原文线索

V2-054100 | MATHDB-398366

Sharp edge count for the triangle-free process

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let $(G_m)_{m\in\mathbb{N}}$ be the random graph process on vertex set $[n]=\{1,\ldots,n\}$ that starts with the empty graph G_0 and, at each step, add...

open_problem | open | MathDB 页面 | 原文线索

V2-054101 | MATHDB-398580

Local regularity theorem for the network flow

摘录 / EXCERPT: Let $(\gamma_t)_{t\in[0,T)}$ be a smooth, regular network flow which reaches the point x_0 at time $t_0\in(0,T]$. Let $0<\varepsilon,\eta<1$. There exists $C=C(\varepsilon,\eta)$...

open_problem | open | MathDB 页面 | 原文线索

V2-054102 | MATHDB-398598

Compactness under a lower Ricci-curvature bound for a fully nonlinear Yamabe equation

摘录 / EXCERPT: Let (f,Γ) satisfy: 1. $\Gamma\subset\mathbb{R}^n$ is an open, convex, symmetric cone with vertex at the origin and ... 2. $f\in C^\infty(\Gamma)\cap C^0(\overline{\Gamma})$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-054103 | MATHDB-398777

Comb Conjecture for the square-root comb

摘录 / EXCERPT: For $k=\sqrt{n}$ (so n is a square), let Comb_n be the tree consisting of an (n/k)-vertex path P together with disjoint k-vertex paths beginning at the vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-054104 | MATHDB-398782

Inducibility of the 4-vertex path

摘录 / EXCERPT: Determine the inducibility $\limsup_{|G|\to\infty} p(P,G)$ of the 4-vertex path P, where $p(P,G)$ is the probability that a uniformly random set of four vertices of G induces...

open_problem | open | MathDB 页面 | 原文线索

V2-054105 | MATHDB-398788

Bondy’s conjecture on generating cycles by long circuits

摘录 / EXCERPT: For every $d\in\mathbb{N}$ and every vertex-3-connected graph G with $|G|\ge 2d$ and minimum degree $\delta(G)\ge d$, every cycle in G can be constructed as the symmetric dif...

conjecture | open | MathDB 页面 | 原文线索

V2-054106 | MATHDB-398791

Lagrangian of intersecting uniform hypergraphs

摘录 / EXCERPT: Let $r\ge 4$, and let $\mathcal F$ be the family of intersecting r-uniform hypergraphs. For an r-uniform hypergraph G on $[n]$, define ... and define...

open_problem | open | MathDB 页面 | 原文线索

V2-054107 | MATHDB-398795

Extremal number of a tight path in a uniform hypergraph

摘录 / EXCERPT: A k-uniform hypergraph is a hypergraph in which all edges have size k. A k-uniform tight path P^k_ℓ is given by an ordering of ℓ vertices with every k consecutive...

open_problem | open | MathDB 页面 | 原文线索

V2-054108 | MATHDB-398799

Spanning structured subgraph in dense odd-order graphs

摘录 / EXCERPT: For every odd integer $n \ge 7$, does every n-vertex graph with minimum degree at least $\lceil n/2 \rceil$ contain a spanning copy of the graph obtained as follows: start with t...

open_problem | open | MathDB 页面 | 原文线索

V2-054109 | MATHDB-398800

Blow-ups of fixed subhypergraphs in dense uniform hypergraphs

摘录 / EXCERPT: For which r-uniform hypergraphs F does the following hold: for every $c>0$ there exists $\gamma>0$ such that, for all sufficiently large n, every n-vertex r-uniform hyper...

open_problem | open | MathDB 页面 | 原文线索

V2-054110 | MATHDB-398801

Large tripartite blow-ups from many triangles

摘录 / EXCERPT: Is it true that every n-vertex graph with $n^3/\exp(C\sqrt{\log n})$ triangles contains a t-blow-up of K_3 with $t\ge \gamma\sqrt{\log n}$ for some $\gamma=\gamma(C)$?

open_problem | open | MathDB 页面 | 原文线索

V2-054111 | MATHDB-398802

Bollobás-Erdős-Simonovits blow-up conjecture for hypergraphs

摘录 / EXCERPT: For every r-uniform hypergraph F and every $\epsilon>0$, there exists a $\gamma>0$ such that, for sufficiently large n, every n-vertex r-uniform hypergraph H with at le...

conjecture | open | MathDB 页面 | 原文线索

V2-054112 | MATHDB-399265

Bichromaticity of hypergraphs with the $(2r,2r-1)$-property

摘录 / EXCERPT: Is every r-graph satisfying the $(2r,2r-1)$-property bichromatic (bipartite)? Equivalently, must such a hypergraph satisfy $\chi(G)=2$?

open_problem | open | MathDB 页面 | 原文线索

V2-054113 | MATHDB-399276

Asymptotic ratio of geodesic to all simplicial spheres

摘录 / EXCERPT: For fixed dimension d, where $s_d(n)$ denotes the number of n-vertex simplicial d-spheres and $g_d(n)$ denotes the number of n-vertex geodesic d-spheres, is ... zero or pos...

open_problem | open | MathDB 页面 | 原文线索

V2-054114 | MATHDB-399380

Size of the $2.5+1.5$-core in a random graph with sliders

摘录 / EXCERPT: Let $G=G(n,c/n)$ be the random graph described on the preceding pages, with each vertex independently of type 1 with probability $1-q$ and of type 2 with probability q, where...

open_problem | open | MathDB 页面 | 原文线索

V2-054115 | MATHDB-399398

Computing the minimum cover in a hereditary hypergraph

摘录 / EXCERPT: Let $H=(V,E)$ be a hereditary hypergraph, meaning that if $e\in E$, then every subset of e also belongs to E. Compute the minimum size ρ of a cover of V by members of...

open_problem | open | MathDB 页面 | 原文线索

V2-054116 | MATHDB-399493

Inverse problem for alternating-current networks

摘录 / EXCERPT: For a connected alternating-current network with b boundary vertices-namely, a graph whose edges xy have complex conductances c_{xy} with $\operatorname{Re} c_{xy}>0$-let the...

open_problem | open | MathDB 页面 | 原文线索

V2-054117 | MATHDB-399496

Responses of series-parallel alternating-current networks

摘录 / EXCERPT: Describe the set of responses of all series-parallel networks.

open_problem | open | MathDB 页面 | 原文线索

V2-054118 | MATHDB-399528

Every unitary VOA is strongly local

摘录 / EXCERPT: Every unitary vertex operator algebra (VOA) is strongly local.

open_problem | open | MathDB 页面 | 原文线索

V2-054119 | MATHDB-399529

Monoidal equivalence for the symplectic-fermion VOA

摘录 / EXCERPT: The representation category of the C_2-cofinite vertex operator algebra constructed out of symplectic fermions is conjecturally monoidally equivalent to...

open_problem | open | MathDB 页面 | 原文线索

V2-054120 | MATHDB-399616

Reid’s dream for Calabi-Yau threefolds

摘录 / EXCERPT: “Reid’s dream” says that all Calabi-Yau threefolds may be connected by a network of conifold transitions.

open_problem | open | MathDB 页面 | 原文线索

V2-054121 | MATHDB-399779

Asymptotic size of the non-frozen core in critical scale-free Boolean networks

摘录 / EXCERPT: For scale-free Boolean networks with power-law in- and outdegree exponents $\tau_{\mathrm{in}},\tau_{\mathrm{out}}>2$, asymptotically independent indegree and outdegree, canonical...

open_problem | open | MathDB 页面 | 原文线索

V2-054122 | MATHDB-399868

Asymptotic ferromagnetic ordering of energy levels on large lattice boxes

摘录 / EXCERPT: For each $d,L\in\{1,2,\ldots\}$, let $B_d(L)$ be the graph with vertex set $\{1,\ldots,L\}^d\subset\mathbb Z^d$ and the induced edge set inherited from $\mathbb Z^d$. For each...

open_problem | open | MathDB 页面 | 原文线索

V2-054123 | MATHDB-399890

Eight-vertex 3-sphere triangulations and collapsibility

摘录 / EXCERPT: Theorem 1. There are 17 triangulations of the 3-sphere with eight vertices such that all spanning trees lead to a collapsible 2-complex. The remaining 22 eight-vertex triangulation...

open_problem | open | MathDB 页面 | 原文线索

V2-054124 | MATHDB-399899

Determine the 3-chromatic number of the 3-sphere

摘录 / EXCERPT: What is the value of $\chi_3(S^3)$? Here vertex colorings of triangulations of S^3 allow monochromatic triangles but forbid monochromatic tetrahedra; more generally, χ_i all...

open_problem | open | MathDB 页面 | 原文线索

V2-054125 | MATHDB-399913

Characterization of acyclic quivers with linearly recurrent frieze sequences

摘录 / EXCERPT: Let $Q=(Q_0,Q_1)$ be a finite acyclic quiver. For each vertex $v\in Q_0$, define a sequence $(v(n))_{n\in\mathbb N}$ by $v(0)=1$ and ... Can one characterize the finite acyclic quive...

open_problem | open | MathDB 页面 | 原文线索

V2-054126 | MATHDB-399992

Randomized triangle detection in the vertex-assigned multiparty model

摘录 / EXCERPT: What is the communication complexity of randomized triangle detection in a model where each vertex of an undirected graph is assigned to some player, and that player receives all e...

open_problem | open | MathDB 页面 | 原文线索

V2-054127 | MATHDB-400154

Error bounds for regularized maximum-likelihood estimation of a sparse Poisson autoregressive network

摘录 / EXCERPT: For observations $(X_t)_{t=0}^{T}$ from the log-linear vector Poisson autoregressive model $X_{t+1}\mid X_t\sim\operatorname{Poisson}(e^{\nu-A^{*}X_t})$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-054128 | MATHDB-400178

Directed Grid Theorem

摘录 / EXCERPT: There is a function $f : \mathbb{N} \to \mathbb{N}$ such that every digraph of directed tree width at least $f(k)$ contains a cylindrical grid of order k as a butterfly minor.

open_problem | open | MathDB 页面 | 原文线索

V2-054129 | MATHDB-400182

Asymptotic minimum triangle density at fixed edge density

摘录 / EXCERPT: The report states that there is an optimal function $f:[0,1]\to[0,1]$ such that, if an n-vertex graph G contains at least $\alpha\binom{n}{2}$ edges, then G contains at least...

open_problem | open | MathDB 页面 | 原文线索

V2-054130 | MATHDB-400183

Erdős-Sós conjecture on uniformly dense 3-uniform hypergraphs

摘录 / EXCERPT: Every sufficiently large uniformly dense 3-uniform hypergraph of density greater than $1/4$ contains four vertices spanning at least three hyperedges.

conjecture | open | MathDB 页面 | 原文线索

V2-054131 | MATHDB-400233

Triangle decomposition of tridivisible typical graphs

摘录 / EXCERPT: Let G be a graph on n vertices, with density $d(G)=|G|/\binom{n}{2}$. Call G c-typical if every vertex has $(1\pm c)d(G)n$ neighbours and every pair of vertices has...

open_problem | open | MathDB 页面 | 原文线索

V2-054132 | MATHDB-400238

Hardness of approximating the clique number

摘录 / EXCERPT: For every $\varepsilon>0$, it is NP-hard to approximate the clique number of an n-vertex graph within a factor of $n^{1-\varepsilon}$.

open_problem | open | MathDB 页面 | 原文线索

V2-054133 | MATHDB-400245

Diameter of Geometric Power-Law Random Graphs

摘录 / EXCERPT: Fix parameters $2<\beta<3$, $d\in\mathbb{N}$, and $\alpha>1$. For each n, form the geometric power-law random graph on n vertices by assigning independently to each vertex v...

open_problem | open | MathDB 页面 | 原文线索

V2-054134 | MATHDB-400251

Structure of extremal graphs minimizing the number of odd-cycle edges

摘录 / EXCERPT: Suppose that G is an n-vertex graph with e edges, such that $e>n^2/4$ and it has the minimum number of C_{2k+1}-edges, $k\ge 3$, $n>n_k$. Then G is connected and has two...

open_problem | open | MathDB 页面 | 原文线索

V2-054135 | MATHDB-400253

Erdős-Simonovits conjecture on supersaturation of 4-cycles

摘录 / EXCERPT: Whether an n-vertex graph with $\\operatorname{ex}(n,C_4)+1$ edges has at least two copies of C_4.

conjecture | open | MathDB 页面 | 原文线索

V2-054136 | MATHDB-400257

Determine the Turán density of the tetrahedron

摘录 / EXCERPT: Determine the Turán density $\pi\!\left(K_4^{(3)}\right)$ of the complete 3-uniform hypergraph $K_4^{(3)}$ on four vertices. The report gives the bounds...

open_problem | open | MathDB 页面 | 原文线索

V2-054137 | MATHDB-400348

Control of road e to remove the Braess paradox

摘录 / EXCERPT: For the traffic network described in the report, can one enforce a control on road e-in particular, a suitable speed limit-such that the Braess paradox does not occur? The report...

open_problem | open | MathDB 页面 | 原文线索

V2-054138 | MATHDB-400657

Cutoff for random walks from a typical vertex of the giant component

摘录 / EXCERPT: Let $G(n,p)$ be the Erdős-Rényi random graph with $p=c/n$ for fixed $c>1$, and let the random walk start from a uniformly chosen vertex of its giant component. For a fixed total-va...

open_problem | open | MathDB 页面 | 原文线索

V2-054139 | MATHDB-400659

Finite cover time under a majorizing-measure entropy condition

摘录 / EXCERPT: Let (V,E,F, ν) be a compact measured resistance network, and let $X=(X_t)_{t\ge 0}$ be the associated ν-symmetric, V-valued Feller process. Assume that there exists a pro...

open_problem | open | MathDB 页面 | 原文线索

V2-054140 | MATHDB-400776

Minimum vertex-degree condition for tight Hamiltonian cycles in 3-uniform hypergraphs

摘录 / EXCERPT: For every $\b1>0$ there exists an integer n_0 such that every 3-uniform hypergraph H with $n\e n_0$ vertices and minimum vertex degree...

open_problem | open | MathDB 页面 | 原文线索

V2-054141 | MATHDB-400924

Flag-algebra upper bound for a specified forbidden-family Turán density

摘录 / EXCERPT: Let $\mathcal L$ be the family of the following nine 3-uniform hypergraphs: ... Here $\operatorname{ex}(n,\mathcal L)$ is the maximum number of edges in a 3-uniform hypergraph on n...

open_problem | open | MathDB 页面 | 原文线索

V2-054142 | MATHDB-400940

Rationality of a four-vertex nested polytope in three dimensions

摘录 / EXCERPT: Given rational polytopes $P\subset Q\subset\mathbb{R}^3$, if there exists a polytope Δ with four vertices such that $P\subseteq\Delta\subseteq Q$, must there also exist such...

open_problem | open | MathDB 页面 | 原文线索

V2-054143 | MATHDB-401073

Existence of small-mass ground states for the NLS with potential on a metric graph

摘录 / EXCERPT: Let G be a finite, connected metric graph with at least one external edge, let V be its vertex set, and let $W=W_+-W_-$ satisfy $W7"). Hmm invalid. Need regenerate clean. 7 acc...

open_problem | open | MathDB 页面 | 原文线索

V2-054144 | MATHDB-401396

Phase transition for SIS epidemics on completely general networks

摘录 / EXCERPT: Consider a sequence of finite connected edge-weighted graphs and vertex recovery-rate functions, indexed by n, with the SIS dynamics described on the source page. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-054145 | MATHDB-401430

Sharpness of the lower bound on sparse connectivity

摘录 / EXCERPT: For a given class of functions C, does there exist a neural network such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-054146 | MATHDB-401432

Monotonicity of variational networks

摘录 / EXCERPT: It would be helpful to show that variational networks decrease in each iteration some (Bregman) distance to the ground truth solution.

open_problem | open | MathDB 页面 | 原文线索

V2-054147 | MATHDB-401450

Depth growth for optimal L^∞ approximation by ReLU networks

摘录 / EXCERPT: If one insists on L^∞ approximation with the same “network size to approximation error” relation as above, it is unknown whether one can avoid this growth of the depth.

open_problem | open | MathDB 页面 | 原文线索

V2-054148 | MATHDB-401703

Weak-recovery threshold for the two-group symmetric stochastic block model

摘录 / EXCERPT: Consider the two-group symmetric stochastic block model on n vertices: the vertex set is partitioned uniformly into two subsets S_+ and S_- of size $n/2$, and, conditional on...

open_problem | open | MathDB 页面 | 原文线索

V2-054149 | MATHDB-401871

Closed saddle connection on the dodecahedron

摘录 / EXCERPT: Does the dodecahedron admit a closed saddle connection, meaning a saddle connection-a geodesic trajectory joining two vertices with no vertex in its interior-that starts and ends a...

open_problem | open | MathDB 页面 | 原文线索

V2-054150 | MATHDB-401929

Determining the quickest-transshipment horizon without parametric search

摘录 / EXCERPT: Can one determine, for every quickest transshipment problem (N,b) on a dynamic network $N=(D=(V,A),u,\tau,S^+,S^-)$ with integral supplies and demands b, the minimal feasible t...

open_problem | open | MathDB 页面 | 原文线索

V2-054151 | MATHDB-401930

Minimal feasible time horizon for quickest transshipment with a single sink

摘录 / EXCERPT: Given a dynamic network $N=(D=(V,A),u,\tau,S^+,t)$ with a single sink t, and a supply/demand function b, Algorithm 1 returns the minimal feasible time horizon T for the quick...

open_problem | open | MathDB 页面 | 原文线索

V2-054152 | MATHDB-401933

Continuity of the Shannon capacity on circular graphs

摘录 / EXCERPT: Let $C_{k,n}$, for $k,n\in\mathbb N$ with $n/k\ge 2$, be the graph with vertex set $V(C_n)$ in which distinct vertices u,v are adjacent exactly when their distance in the circuit...

open_problem | open | MathDB 页面 | 原文线索

V2-054153 | MATHDB-401942

Optimization over degree sequences

摘录 / EXCERPT: Given k, n, m, and univariate functions $f_1,\ldots,f_n$, consider the problem of finding a k-hypergraph on $[n]$ with m edges maximizing $\sum_{i=1}^n f_i(d_i)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-054154 | MATHDB-401945

Rainbow paths in edge-coloured graphs

摘录 / EXCERPT: Each simple k-regular properly k-edge-coloured graph has a rainbow path of length $k-1$, where a rainbow path is a path whose edges have pairwise distinct colours.

open_problem | open | MathDB 页面 | 原文线索

V2-054155 | MATHDB-402015

Asymptotic equality between orthogonal-representation dimension and chromatic number

摘录 / EXCERPT: Let $G\sim G(n,0.5)$, and let $d(G)$ be the minimum dimension d for which one can assign a unit vector $x(v)\in\mathbb{R}^d$ to every vertex v such that $x(u)$ and $x(v)$ are o...

open_problem | open | MathDB 页面 | 原文线索

V2-054156 | MATHDB-402019

Chung-Diaconis-Graham universal-cycle conjecture

摘录 / EXCERPT: For every $k\in\mathbb{N}$, there exists $n_0\in\mathbb{N}$ such that for all $n\ge n_0$, the complete k-uniform hypergraph $K_n^{(k)}$ has an Euler tour whenever k divides...

conjecture | open | MathDB 页面 | 原文线索

V2-054157 | MATHDB-402026

Chi-boundedness Conjecture for proper vertex-minor-closed graph classes

摘录 / EXCERPT: Let $\mathcal{F}$ be any proper class of graphs that is closed under isomorphism and taking vertex minors. Conjecture (Chi-boundedness Conjecture): There is a function f so that...

conjecture | open | MathDB 页面 | 原文线索

V2-054158 | MATHDB-402027

Max-Clique Testing Conjecture

摘录 / EXCERPT: Let $\mathcal F$ be any proper class of graphs closed under isomorphism and taking vertex minors. There is a polynomial-time algorithm that, given a graph $G\in\mathcal F$, compute...

conjecture | open | MathDB 页面 | 原文线索

V2-054159 | MATHDB-402030

Conlon-Fox-Sudakov conjecture: Question (b) for sparse H-free graphs

摘录 / EXCERPT: Is it true that for every graph H there exists $\[1m\varepsilon>0\[0m$ such that every n-vertex $\[1m\varepsilon\[0m$-sparse graph G with $n>1$ and containing no induced copy...

conjecture | open | MathDB 页面 | 原文线索

V2-054160 | MATHDB-402031

Sparse graphs with forbidden induced subgraphs: Question (b)

摘录 / EXCERPT: For which graphs H does there exist $\epsilon>0$ such that every n-vertex ϵ-sparse H-free graph G with $n>1$ contains disjoint vertex sets A,B satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-054161 | MATHDB-402032

Characterization of graphs with the sparse strong Erdős-Hajnal property

摘录 / EXCERPT: For which graphs H does there exist $\[1m\varepsilon>0\[0m$ such that every n-vertex $\[1m\varepsilon\[0m$-sparse H-free graph G with $n>1$ contains two sets A,B of verti...

open_problem | open | MathDB 页面 | 原文线索

V2-054162 | MATHDB-402126

Distinct degrees in induced subgraphs of Ramsey graphs

摘录 / EXCERPT: For every fixed constant $C>0$, every N-vertex graph G satisfying $\operatorname{hom}(G)\le C\log N$ has ... where $f(G)$ is the maximum number of distinct degrees in an induced...

open_problem | open | MathDB 页面 | 原文线索

V2-054163 | MATHDB-402127

Universal containment of oriented Hamilton cycles in a random digraph

摘录 / EXCERPT: If $p=(\log n+\omega(1))/n$, then the random digraph $D(n,p)$ almost surely contains a copy of every oriented n-vertex cycle.

open_problem | open | MathDB 页面 | 原文线索

V2-054164 | MATHDB-402130

Decomposition of the complete graph into bounded-degree regular graphs

摘录 / EXCERPT: For all $\Delta \in \mathbb{N}$, there exists an $n_0 \in \mathbb{N}$ such that the following holds for all $n \ge n_0$. Let $F_1, \ldots, F_t$ be n-vertex graphs such that F_i...

open_problem | open | MathDB 页面 | 原文线索

V2-054165 | MATHDB-402136

Large-girth approximate Steiner triple systems

摘录 / EXCERPT: For which integers $\ell \ge 4$ and constants $c \in (0,1/6)$ do there exist, for some $n_0(\ell,c)$ and every $n \ge n_0(\ell,c)$, an n-vertex partial Steiner triple system with...

open_problem | open | MathDB 页面 | 原文线索

V2-054166 | MATHDB-402137

Erdős's conjecture on high-girth Steiner triple systems

摘录 / EXCERPT: Let $\ll \ge 4$. Does there exist $n_0=n_0(\ll)$ such that, for every $n\ge n_0$ with $n\\\ 1,3 \pmod 6$, there exists an n-vertex Steiner triple system with girth greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-054167 | MATHDB-402139

Does vertex percolation preserve max cut?

摘录 / EXCERPT: Let $G=G_n$ be a sequence of graphs with $e(G)/v(G)\to\infty$. Let S be a random subset of $V(G)$ in which every vertex is included independently with probability $1/2$. Prove or...

open_problem | open | MathDB 页面 | 原文线索

V2-054168 | MATHDB-402142

Hypergraph cuts above the average: the $m^{2/3}$ excess conjecture

摘录 / EXCERPT: Fix integers $k\ge r\ge 2$ such that $r\ge 3$ or $k\ge 4$. Let G be a k-uniform hypergraph with m edges. An r-cut of G is a partition of its vertex set into r parts, an...

conjecture | open | MathDB 页面 | 原文线索

V2-054169 | MATHDB-402311

Collapsibility of almost all higher-dimensional hypertrees

摘录 / EXCERPT: For $d \ge 2$, only a vanishingly small (in n) fraction of the n-vertex d-hypertrees are collapsible.

open_problem | open | MathDB 页面 | 原文线索

V2-054170 | MATHDB-402318

Removing the rationality hypothesis for the polytope face-hypergraph connectivity theorem

摘录 / EXCERPT: Is the rationality hypothesis in Corollary 1 unnecessary; that is, for an arbitrary (not necessarily rational) d-dimensional polytope P, is its k-face hypergraph $(d-k)$-conn...

conjecture | open | MathDB 页面 | 原文线索

V2-054171 | MATHDB-402322

Spanning trees of bipartite graphs: determinant inequality

摘录 / EXCERPT: Let G be a finite bipartite graph with vertex bipartition $V(G)=X\sqcup Y$, where $X=[n]$. For each nonempty $S\subseteq[n]$, let $X_S=\#\{y\in Y:N(y)=S\}$. Define the...

open_problem | open | MathDB 页面 | 原文线索

V2-054172 | MATHDB-402326

Same-vertex-set extension of a combinatorial sphere

摘录 / EXCERPT: Given a simplicial complex S that is a combinatorial d-sphere with $n \ge d+3$ vertices, does there always exist a combinatorial $(d+1)$-sphere S' with the same vertex set an...

open_problem | open | MathDB 页面 | 原文线索

V2-054173 | MATHDB-402416

Quantum operations with a prescribed hypergroup relation

摘录 / EXCERPT: Problem 9. For A a conformal net associated with a lattice [DX06] or a strongly local vertex operator algebra [CKLW18], find $\\phi\in\operatorname{QuOp}(A)$ with ... for small pos...

open_problem | open | MathDB 页面 | 原文线索

V2-054174 | MATHDB-402417

Reconstruction of anyon models by vertex operator algebras

摘录 / EXCERPT: It is conjectured that every anyon model can be reconstructed as the representation category of a vertex operator algebra (VOA).

open_problem | open | MathDB 页面 | 原文线索

V2-054175 | MATHDB-402423

Equivalence of Hilbert-space and locally convex completions for unitary chiral CFTs

摘录 / EXCERPT: If the chiral conformal field theory is unitary, then the Hilbert space completions and the locally convex topological completions of modules for the vertex operator algebra are th...

open_problem | open | MathDB 页面 | 原文线索

V2-054176 | MATHDB-402614

Brown-Erdős-Sós conjecture

摘录 / EXCERPT: For every integer $e\ge 3$, ... Here $f_r(n,v,e)$ denotes the largest number of edges in an r-uniform hypergraph on n vertices containing no (v,e)-configuration, i.e. no e ed...

conjecture | open | MathDB 页面 | 原文线索

V2-054177 | MATHDB-402623

Spread-hypergraph formulation of the fractional expectation-threshold theorem

摘录 / EXCERPT: There is a universal constant K such that, for any $\[1m\ell$-bounded, $\[1m\kappa$-spread hypergraph $\[1m\mathcal H$ on a finite set X, a uniformly random...

open_problem | open | MathDB 页面 | 原文线索

V2-054178 | MATHDB-402630

Cycles of length divisible by four in tournaments

摘录 / EXCERPT: Let $d(\ell,n)$ be the ratio of the maximum number of directed cycles of length ℓ in an n-vertex tournament to the expected number of such cycles in a random n-vertex tour...

open_problem | open | MathDB 页面 | 原文线索

V2-054179 | MATHDB-402634

Majority 3-coloring conjecture for digraphs

摘录 / EXCERPT: Every digraph D admits a coloring $c:V(D)\to\{1,2,3\}$ such that, for every vertex $v\in V(D)$, at most half of the out-neighbors $w\in N^+(v)$ satisfy $c(w)=c(v)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054180 | MATHDB-402804

Finite-dimensionality of completed Jacobi algebras

摘录 / EXCERPT: For the one-vertex quiver $Q(l)$ with l loops and a formal potential W, determine precisely when the completed Jacobi algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-054181 | MATHDB-402817

Conjectural network-chart formula for cluster characters

摘录 / EXCERPT: With the notation of the report, for any $M\in\operatorname{CM}(B)$, conjecture the network-chart formula ... where the sum is over $X\subseteq\operatorname{Hom}_B(T,M)$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-054182 | MATHDB-402820

Hamiltonicity of the 1-Delaunay graph

摘录 / EXCERPT: It is conjectured that the 1-Delaunay graph is Hamiltonian. Here, for a planar point set S, the k-Delaunay graph has vertex set S and an edge pq whenever some closed disk w...

open_problem | open | MathDB 页面 | 原文线索

V2-054183 | MATHDB-402838

Hereditary polychromatic k-colorings

摘录 / EXCERPT: For a hypergraph H, let m_k be the smallest number such that the vertices can be k-colored so that every hyperedge of size at least m_k contains all k colors. Let m_k^*...

open_problem | open | MathDB 页面 | 原文线索

V2-054184 | MATHDB-402902

Liouville integrability of electrical networks

摘录 / EXCERPT: Are electrical networks Liouville integrable?

open_problem | open | MathDB 页面 | 原文线索

V2-054185 | MATHDB-403293

Universality characterization for deep complex-valued neural networks

摘录 / EXCERPT: Let $sigma:\mathbb C\to\mathbb C\$ be locally bounded, and suppose that the closure of its set of discontinuities, ${z\in\mathbb C:\sigma\text{ is discontinuous at }z}\$, has Lebes...

open_problem | open | MathDB 页面 | 原文线索

V2-054186 | MATHDB-403321

Minimal vertex-and-edge coloured graph representing a permutation group

摘录 / EXCERPT: Let $G \leq \operatorname{Sym}(\Omega)$. A vertex- and edge-coloured graph Γ with vertex set $V \supseteq \Omega$ and edge set E represents G if and only if its automorp...

open_problem | open | MathDB 页面 | 原文线索

V2-054187 | MATHDB-403508

Randomized online edge coloring with nearly Δ colors

摘录 / EXCERPT: In the regime $\Delta = \omega(\log n)$, [3] conjectured that the online algorithm using $\Delta + O\!\left(\sqrt{\Delta\log n}\right)$ colors, which samples a color independently...

open_problem | open | MathDB 页面 | 原文线索

V2-054188 | MATHDB-403514

Approximating Symmetric Submodular Functions by Hypergraphs

摘录 / EXCERPT: Is there a fixed constant c such that every symmetric submodular function can be approximated by a hypergraph cut function within a factor of c?

open_problem | open | MathDB 页面 | 原文线索

V2-054189 | MATHDB-403515

Iterative rounding for feedback vertex set

摘录 / EXCERPT: For the LP ... is it true that every extreme solution x^* has a vertex v with $x^*(v)\ge \tfrac12$?

open_problem | open | MathDB 页面 | 原文线索

V2-054190 | MATHDB-403536

Reconstruction from higher-genus partition functions

摘录 / EXCERPT: Conjecture: A vertex algebra is uniquely determined, up to isomorphism, by the collection of its partition functions $\{\mathbf{1}_g(V)\}_{g\ge 1}$.

open_problem | open | MathDB 页面 | 原文线索

V2-054191 | MATHDB-403541

Quantum-group/VOA ribbon tensor equivalences at [U+007F]$q=e^{\pi i/p}$

摘录 / EXCERPT: For the triplet, singlet, and Virasoro vertex operator algebras W_p, M_p, and $Vir_p=Vir(c_p)$, with $c_p=1-6(1-p)^2/p$ and $q=e^{\pi i/p}$, the conjectural ribbon tensor equiv...

open_problem | open | MathDB 页面 | 原文线索

V2-054192 | MATHDB-403555

Maximum number of copies of a graph in a graph class

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph class, and let H be a graph. What is the maximum number $\operatorname{ex}(H,\mathcal{G},n)$ of H-subgraphs contained in an n-vertex graph in...

open_problem | open | MathDB 页面 | 原文线索

V2-054193 | MATHDB-403559

Rainbow Turán number of a fixed even cycle

摘录 / EXCERPT: For every integer $k\ge 2$, the rainbow Turán number satisfies ... Here $\operatorname{ex}^*(n,C_{2k})$ is the maximum number of edges in a properly edge-coloured graph on n vertic...

open_problem | open | MathDB 页面 | 原文线索

V2-054194 | MATHDB-403562

Rainbow Turán threshold for subdivisions of a clique

摘录 / EXCERPT: Fix an integer $m\ge 2$. What is the smallest c such that, for all sufficiently large n, every properly edge-coloured graph G on n vertices with at least...

open_problem | open | MathDB 页面 | 原文线索

V2-054195 | MATHDB-403563

Linear rank-width of graphs excluding a tree as a vertex-minor

摘录 / EXCERPT: For every tree T, does the class of graphs having no T-vertex-minor have bounded linear rank-width?

open_problem | open | MathDB 页面 | 原文线索

V2-054196 | MATHDB-403567

Compressibility of acyclic digraphs

摘录 / EXCERPT: For an acyclic oriented graph F, define the compressibility of F as the smallest integer k such that F is homomorphic to any tournament on k vertices. For what graphs F...

open_problem | open | MathDB 页面 | 原文线索

V2-054197 | MATHDB-403625

Braided tensor structures on categories of modules of admissible affine vertex algebras

摘录 / EXCERPT: For a simple Lie algebra $\mathfrak g$ (or $\mathfrak g=\mathfrak{osp}(1|2n)$) and an admissible level k, does the category $\mathcal C_k(\mathfrak g)$ of the relevant, not-neces...

open_problem | open | MathDB 页面 | 原文线索

V2-054198 | MATHDB-403679

Sós-Verstraëte conjecture on hypergraphs with no tight cycles

摘录 / EXCERPT: For an integer $r\ge 3$, let $f_r(n)$ be the maximum number of edges in an r-uniform hypergraph on n vertices containing no tight cycle, where a tight cycle is a sequence of at...

conjecture | open | MathDB 页面 | 原文线索

V2-054199 | MATHDB-403684

Weakly Saturated Hypergraphs and a Conjecture of Tuza

摘录 / EXCERPT: For every r-graph H, let $s=\operatorname{sp}(H)$ be the smallest size of a vertex set $W\subseteq V(H)$ contained in precisely one edge of H. Then there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-054200 | MATHDB-403689

Turán inequality for partially directed uniform hypergraphs

摘录 / EXCERPT: Conjecture 4. Fix $k\ge 2$. For all sufficiently large n (depending on k), every n-vertex partially directed k-uniform hypergraph (k-PDG) without $\vec T_k$ as a subgraph...

open_problem | open | MathDB 页面 | 原文线索

V2-054201 | MATHDB-403693

Standard clique edge-weighting conjecture and stability

摘录 / EXCERPT: Let G be an n-vertex graph. For an edge $e\in E(G)$, let r be the order of the largest clique in G containing e, and assign it weight $w(e)=\frac{r}{2(r-1)}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-054202 | MATHDB-403694

Counting Petersen graphs in pseudorandom C_4-free graphs

摘录 / EXCERPT: Let G be a graph on vertex set $[n]$, let $p=e(G)/\binom{n}{2}$ with $p\gtrsim n^{-1/2}$, and suppose that G is C_4-free and ... for all $A,B\subseteq[n]$. Is the number of cop...

open_problem | open | MathDB 页面 | 原文线索

V2-054203 | MATHDB-403843

Global margin optimization by gradient descent for deep networks

摘录 / EXCERPT: For the deep-network optimization problem ... when does gradient descent globally maximize equation (1)?

open_problem | open | MathDB 页面 | 原文线索

V2-054204 | MATHDB-404013

Erdős-Hajnal conjecture for the five-vertex path

摘录 / EXCERPT: There exists a constant $c>0$ such that every P_5-free graph G contains a clique or an independent set of order at least $|G|^c$, where P_5 denotes the path on five vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-054205 | MATHDB-404024

Rainbow matchings in uniform hypergraphs

摘录 / EXCERPT: Let $M_1,\ldots,M_N$ be matchings of size t in some r-uniform hypergraph, where the matchings are regarded as having distinct colors. How large must N be, as a function of...

open_problem | open | MathDB 页面 | 原文线索

V2-054206 | MATHDB-404026

Growth of rainbow-matching obstruction numbers in the 3-uniformity-parameter regime

摘录 / EXCERPT: Let $f(r,t)$ be the maximum number of matchings of size t in an r-partite r-uniform hypergraph that has no rainbow matching of size t. Determine the growth of $f(r,3)$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-054207 | MATHDB-404027

Exponential upper bound for rainbow-matching numbers

摘录 / EXCERPT: Let $f(r,t)$ denote the maximum possible number of matchings of size t in an r-partite r-uniform hypergraph without a rainbow matching of size t. Can $f(r,t)$ be upper-boun...

open_problem | open | MathDB 页面 | 原文线索

V2-054208 | MATHDB-404029

Optimal K_s-free pseudorandom graphs

摘录 / EXCERPT: For each fixed integer $s\ge 4$, does there exist an optimal K_s-free (n,d,λ)-graph, meaning an n-vertex d-regular graph whose nontrivial adjacency-matrix eigenvalues...

open_problem | open | MathDB 页面 | 原文线索

V2-054209 | MATHDB-404048

Conjecture 3 on finite surviving components in infinite WARM graphs

摘录 / EXCERPT: For an infinite graph in the standing WARM setting, with bounded vertex degrees and firing rates bounded above, assume that the firing rates satisfy that $p(v)$ is bounded away fro...

conjecture | open | MathDB 页面 | 原文线索

V2-054210 | MATHDB-404243

A question about second Neumann eigenfunctions of triangular domains

摘录 / EXCERPT: Let T be an acute triangle in the plane. For any second Neumann eigenfunction u of the Laplacian on T, does u have at most one critical point that is not a vertex of T?

open_problem | open | MathDB 页面 | 原文线索

V2-054211 | MATHDB-404417

Move classification of 110-equivalence for countable vertex-finite directed graphs

摘录 / EXCERPT: For the class of countable vertex-finite directed graphs, the 110-equivalence relation is generated by Moves (O), (I^+), and (K^+): ...

open_problem | open | MathDB 页面 | 原文线索

V2-054212 | MATHDB-404428

Graph-isomorphism formulas with non-CFI Sherali-Adams degree lower bounds

摘录 / EXCERPT: Find n-vertex graphs G_n and H_n with $\operatorname{vc}(G_n)<0.51n$ and $\operatorname{vc}(H_n)>0.99n$ such that the Sherali-Adams degree of refuting $\mathrm{ISO}(G_n,H_n)$...

open_problem | open | MathDB 页面 | 原文线索

V2-054213 | MATHDB-404552

Learning the probing functions for direct sampling methods for limited-aperture data

摘录 / EXCERPT: Construct and train a probing-function network $NN_\vartheta(z,\hat x)$ for limited-aperture far-field data $u^\infty|_\Gamma$ so that ... for any far-field data u^∞, preferab...

open_problem | open | MathDB 页面 | 原文线索

V2-054214 | MATHDB-404553

Training-data reconstruction from trained neural-network parameters

摘录 / EXCERPT: Given a trained neural network with final weights $\[1m\theta^*\u001b[0m$ and training examples $(x_i,y_i)\in\mathbb{R}^K\times\mathbb{R}$ for $i\in\{1,\ldots,m\}$, where ... the rep...

open_problem | open | MathDB 页面 | 原文线索

V2-054215 | MATHDB-404574

Morell-Skutella conjecture for single-source unsplittable flows

摘录 / EXCERPT: For the single-source unsplittable flow problem on acyclic digraphs, any fractional flow $(x_e)_{e\in E}$ can be expressed as a convex combination of unsplittable flows...

conjecture | open | MathDB 页面 | 原文线索

V2-054216 | MATHDB-404576

Schrijver’s partition conjecture for strengthening sets

摘录 / EXCERPT: Let $D=(V,A)$ be a digraph whose underlying graph is 2-edge-connected, and let τ be the minimum size of a dicut of D. A strengthening set is an arc set whose reversal make...

conjecture | open | MathDB 页面 | 原文线索

V2-054217 | MATHDB-404578

Bérczi-Chandrasekaran hypergraph orientation conjecture

摘录 / EXCERPT: Let $H=(V,\b3mathcal{E})$ be a $\b3mathit{\tau}$-uniform hypergraph such that every nonempty proper vertex subset is crossed by at least $\b3mathit{\tau}$ hyperedges. Then H has...

conjecture | open | MathDB 页面 | 原文线索

V2-054218 | MATHDB-404579

Polylogarithmic approximation for Directed Steiner Tree

摘录 / EXCERPT: Given a directed graph with edge costs, a root vertex r, and a set of k terminals T, does there exist a polynomial-time algorithm that outputs a minimum-cost-feasible rooted...

open_problem | open | MathDB 页面 | 原文线索

V2-054219 | MATHDB-404584

Maximum balanced independent sets in the Boolean cube

摘录 / EXCERPT: For the Boolean cube Q_r with vertex set $\\{0,1\\}^r$ and edges joining pairs at Hamming distance 1, let $\operatorname{Ev}(r)=\{x\in\{0,1\}^r:|x|\text{ is even}\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-054220 | MATHDB-404622

GSOM mean-field games on large-scale networks

摘录 / EXCERPT: How can the generalized second-order traffic-flow mean-field game (GSOM-MFG) be formulated on large-scale networks, where cars choose routing decisions, and solved efficiently?

open_problem | open | MathDB 页面 | 原文线索

V2-054221 | MATHDB-404707

Efficiency of refined omni-absorbers for clique decompositions

摘录 / EXCERPT: For all integers $q>r\ge 1$, there exist an integer $C\ge 1$ and a real number $\varepsilon\in(0,1)$ such that the following holds. Let G be an r-uniform hypergraph on n vert...

open_problem | open | MathDB 页面 | 原文线索

V2-054222 | MATHDB-404709

Lovász conjecture on Hamiltonicity of vertex-transitive graphs

摘录 / EXCERPT: Every connected vertex-transitive graph contains a Hamilton path, and, except for five known examples, a Hamilton cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-054223 | MATHDB-404749

Vertex stabilizer amalgams for locally characteristic-[U+007F]p graphs

摘录 / EXCERPT: Let Δ be a thick, locally finite, locally s-arc transitive G-graph with $s\ge 4$, of local characteristic p but not of pushing up type. Is its vertex stabilizer amalga...

open_problem | open | MathDB 页面 | 原文线索

V2-054224 | MATHDB-404780

Common-neighbour conjecture for generalised Saxl graphs

摘录 / EXCERPT: Let G be a primitive permutation group on a finite set $\d4aa$, with base size $b(G)\ge 2$. Define the generalised Saxl graph $\Sigma(G)$ to have vertex set $\d4aa$, with a 2-s...

conjecture | open | MathDB 页面 | 原文线索

V2-054225 | MATHDB-404884

Uniqueness of the moonshine vertex operator algebra

摘录 / EXCERPT: If $V=\bigoplus_{n\in\\mathbb{Z}_{\ge 0}} V_n$ is a holomorphic vertex operator algebra with central charge $c=24$ and weight-one subspace $V_1=\{0\}$, then V is isomorphic to th...

open_problem | open | MathDB 页面 | 原文线索

V2-054226 | MATHDB-404927

Conditional impossibility of higher-order quadrature for neural-network integrands

摘录 / EXCERPT: The report claims that, assuming the Strong Exponential Time Hypothesis (SETH), no higher-order quadrature algorithm exists for approximating ... where Φ is a neural network,...

open_problem | open | MathDB 页面 | 原文线索

V2-054227 | MATHDB-404949

Conlon-Fox-Kwan-Sudakov surplus conjecture for hypergraph cuts

摘录 / EXCERPT: Conjecture 2 (Conlon-Fox-Kwan-Sudakov). For any integers $2\le r\le k$ with either $k\ge 4$ or $r\ge 3$, every k-uniform hypergraph with m edges has an r-cut whose surplus is...

conjecture | open | MathDB 页面 | 原文线索

V2-054228 | MATHDB-404957

Characterization of harmonious labellings of trees by finite abelian groups

摘录 / EXCERPT: Let G be an abelian group of order n, and let T be a tree on n vertices. A labelling $\f0a5:V(T)\to G$ is harmonious if the vertex-labelling $\f0a5$ and the induced edge-la...

open_problem | open | MathDB 页面 | 原文线索

V2-054229 | MATHDB-404958

Graham-Sloane harmonious labelling conjecture

摘录 / EXCERPT: For every tree T on n vertices, there exists a labelling $\fvarphi:V(T)\to \mathbb{Z}/(n-1)\mathbb{Z}$ such that exactly one vertex-label is repeated and all edge-labels...

conjecture | open | MathDB 页面 | 原文线索

V2-054230 | MATHDB-404966

Maximum diameter of simplicial complexes (Santos upper-bound conjecture)

摘录 / EXCERPT: For every integer $d\ge 2$, there exists an integer $n_0>0$ such that, for every $n>n_0$, the maximum diameter $H_s(n,d)$ of a strongly connected simplicial d-complex on vertex s...

conjecture | open | MathDB 页面 | 原文线索

V2-054231 | MATHDB-404967

Rainbow directed paths in properly edge-coloured regular digraphs

摘录 / EXCERPT: Let a directed graph be properly edge-coloured if no two edges with a common start-vertex and no two edges with a common end-vertex have the same colour. Does every d-regular dig...

open_problem | open | MathDB 页面 | 原文线索

V2-054232 | MATHDB-405046

Long directed cycles in connected vertex-transitive digraphs

摘录 / EXCERPT: Theorem 2. In any connected vertex-transitive digraph D on $n\ge 2$ vertices, there is a directed cycle of length at least $\Omega(n^{1/3})$.

open_problem | open | MathDB 页面 | 原文线索

V2-054233 | MATHDB-405047

Unbounded perimeter gaps in connected vertex-transitive digraphs

摘录 / EXCERPT: Alspach asked whether, for every constant C, there exists a connected vertex-transitive digraph whose perimeter gap-the number of vertices minus the length of its longest directe...

open_problem | open | MathDB 页面 | 原文线索

V2-054234 | MATHDB-405060

Asymptotically optimal independent sets in linear uniform hypergraphs

摘录 / EXCERPT: For any $n,r,d\ge 1$, as $d\to\infty$, any linear $(r+1)$-uniform n-vertex hypergraph with maximum degree d contains an independent set of size at least ...

open_problem | open | MathDB 页面 | 原文线索

V2-054235 | MATHDB-405061

Small maximal independent sets in sparse hypergraphs

摘录 / EXCERPT: It seems reasonable to conjecture that each δ-sparse n-vertex hypergraph with degree profile d contains a maximal independent set of size...

open_problem | open | MathDB 页面 | 原文线索

V2-054236 | MATHDB-405062

DeBiasio-Kamel-McCourt-Sheats bounded-diameter strengthening of Ryser’s conjecture

摘录 / EXCERPT: For all $\alpha \ge 1$, there exists $d=d(\alpha)$ such that, for all $r\ge 2$, if G is a graph with $\alpha(G)=\alpha$, then in every r-edge-coloring of G there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-054237 | MATHDB-405063

Local clique removal lemma

摘录 / EXCERPT: For every $\b5>0$ and integer $k\ge 2$, there exists $\b4=\b4(\u0003b5,k)>0$ such that every graph on n vertices either has a vertex contained in at least $\b4 n^{k-1}$ copies of...

open_problem | open | MathDB 页面 | 原文线索

V2-054238 | MATHDB-405067

Generalized Matthews-Sumner conjecture for induced-$K_{1,k}$-free graphs

摘录 / EXCERPT: There exists a function f such that every $f(k)$-vertex-connected, induced-$K_{1,k}$-free graph is Hamiltonian.

conjecture | open | MathDB 页面 | 原文线索

V2-054239 | MATHDB-405071

Estimating the number of colors avoiding repeated triangles

摘录 / EXCERPT: Question 14. Let $g(n)$ be the smallest number of colors necessary to edge-color K_n such that there are no two vertex-disjoint color-isomorphic triangles. Estimate $g(n)$. It is...

open_problem | open | MathDB 页面 | 原文线索

V2-054240 | MATHDB-405078

Mathematical foundations of spiking neural networks

摘录 / EXCERPT: Establish robust mathematical foundations for spiking neural networks, particularly their expressivity and function-approximation capabilities compared with classical neural networ...

open_problem | open | MathDB 页面 | 原文线索

V2-054241 | MATHDB-405079

ReLU depth required for CPWL functions

摘录 / EXCERPT: Does there exist a continuous piecewise-linear (CPWL) function that genuinely requires more than two hidden layers in a ReLU-network representation?

open_problem | open | MathDB 页面 | 原文线索

V2-054242 | MATHDB-405144

Euler Sombor index for trees with given diameter

摘录 / EXCERPT: For a finite tree T with vertex degrees $d(v)$, define its Euler Sombor index by $EUS(T)=\sum_{uv\in E(T)}\sqrt{d(u)^2+d(v)^2+d(u)d(v)}$. For trees with a prescribed diameter d...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054243 | MATHDB-405161

Baswana-Bhanja-Pandey open problem on dynamic exact flow/min-cut sensitivity oracles

摘录 / EXCERPT: Given a directed graph G, vertices $s,t\in V(G)$, and an integer $f\ge 1$, construct a sensitivity oracle that, after preprocessing G, answers exactly for every set U of at m...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054244 | MATHDB-405169

de Klerk-Pasechnik conjecture

摘录 / EXCERPT: For every finite simple graph G, the semidefinite-hierarchy bound satisfies $\vartheta^{(\alpha(G)-1)}(G)=\alpha(G)$, where $\alpha(G)$ is the stability number of G.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054245 | MATHDB-405171

Ramsey threshold for trees versus odd cycles

摘录 / EXCERPT: For an odd integer $m\ge 3$, determine the least integer $f(m)$ such that, for every integer $n\ge f(m)$ and every tree T_n on n vertices, the Ramsey number satisfies...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054246 | MATHDB-405186

Model-RB frb100-40 maximum-independent-set challenge

摘录 / EXCERPT: Let $G_{\mathrm{frb100\text{-}40}}=(V,E)$ be the publicly specified $4{,}000$-vertex Model-RB benchmark graph. Determine its independence number...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054247 | MATHDB-405199

Dunbar-Haynes-Teschner-Volkmann planar bondage conjecture

摘录 / EXCERPT: For every nontrivial planar graph G, the bondage number satisfies $b(G)\leq \Delta(G)+1$, where $b(G)=\min\{\lvert F\rvert:F\subseteq E(G),\ \gamma(G-F)>\gamma(G)\}$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054248 | MATHDB-405201

Class edge-reconstruction number of maximal planar graphs

摘录 / EXCERPT: Let $\mathcal{M}$ be the class of finite maximal planar graphs. For $G\in\mathcal{M}$, an edge card is a graph of the form $G-e$, where $e\in E(G)$. Define...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054249 | MATHDB-405202

Benjamini-Tzalik maximum-shortest-paths conjecture

摘录 / EXCERPT: For every finite multigraph G of maximum degree at most Δ, and every pair of vertices x,y with graph distance $d_G(x,y)=t$, the number $n_G(x,y)$ of shortest x-y pat...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054250 | MATHDB-405204

Erdős’s nested-cycle problem

摘录 / EXCERPT: For every fixed integer $k\ge 2$, there exists a constant C_k such that every graph G on n vertices with at least $C_k n$ edges contains pairwise edge-disjoint cycles...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054251 | MATHDB-405210

Logarithmic basis number conjecture for graphs

摘录 / EXCERPT: There exists a universal constant $C>0$ such that every finite simple graph G with $n=|V(G)|\ge 2$ satisfies $\mathrm{bn}(G)\le C\log n$, where $\mathrm{bn}(G)$ is the minimum, o...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054252 | MATHDB-405233

Li-Ma-Xie conjecture on weak rainbow saturation of cycles

摘录 / EXCERPT: For every integer $\ell\ge 4$, the weak rainbow saturation number of the cycle satisfies $\operatorname{rwsat}(n,C_\ell)=\frac{3}{2}n+O_\ell(1)$ as $n\to\infty$; equivalently, ther...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054253 | MATHDB-405242

The planar scramble-number bound problem

摘录 / EXCERPT: There exists a constant $C>0$ such that, for every integer $n\geq 1$ and every simple planar graph G with $|V(G)|=n$, its scramble number satisfies...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054254 | MATHDB-405246

The conjecture implying θ(p_c)=0 for Bernoulli bond percolation on Z^d

摘录 / EXCERPT: For every $\delta>0$ there exists $\varepsilon>0$ such that the following holds uniformly over every finite graph G, every percolation parameter $p\in[0,1]$, every origin vertex...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054255 | MATHDB-405248

Structural classification of graphs with γ(G)=γ_t(G)=2

摘录 / EXCERPT: Given a finite graph G with no isolated vertices and domination number $\gamma(G)=2$, determine exactly which graphs satisfy $\gamma_t(G)=2$; equivalently, characterize all finit...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054256 | MATHDB-405251

Marion et al.’s correlated-initialization conjecture for deep residual networks

摘录 / EXCERPT: For deep residual networks whose weights are initialized with correlations across layers, the infinite-depth limiting dynamics should interpolate continuously between the stochasti...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054257 | MATHDB-405253

Characterization of graphs determined by their spectra (DGSS problem)

摘录 / EXCERPT: Let $\Sigma=(G,\sigma)$ be an oriented graph with skew-adjacency matrix S, and let J denote the all-ones matrix. Determine all oriented graphs Σ such that, for every ori...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-054258 | MATHDB-405254

Scheinerman-Ullman fractional Hamiltonicity conjecture and Devriendt's resistance-positivity conjecture

摘录 / EXCERPT: Every finite graph $G=(V,E)$ that is 2-tough is fractionally Hamiltonian. Here, G is 2-tough if, for every vertex set $S\subseteq V$ with $c(G-S)>1$, one has $|S|\ge 2c(G-S)$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054259 | MATHDB-405255

Adjacent vertex distinguishing total coloring conjecture

摘录 / EXCERPT: For every simple graph G, the adjacent vertex distinguishing total chromatic number satisfies $\chi''_{a}(G)\leq \Delta(G)+3$. Here, $\chi''_{a}(G)$ is the least integer k such...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054260 | MATHDB-405258

Aldous's spectral gap conjecture

摘录 / EXCERPT: For every finite connected weighted graph $G=(V,E,w)$, the spectral gap of the interchange process on G equals the spectral gap of the continuous-time random walk on G:...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-054261 | MATHDB-405265

Gallai’s conjecture on certain families of bipartite graphs

摘录 / EXCERPT: For every connected graph G with $n=|V(G)|$ vertices, the edges of G admit a decomposition into at most $\left\lceil\frac{n}{2}\right\rceil$ paths.

conjecture | claimed_progress | MathDB 页面 | 原文线索

B12 概率 / Probability | 2,397 条

V2-054262 | MATHDB-315488

Near-T^{-1} convergence for stochastic multi-gradient descent

摘录 / EXCERPT: Under the standard smoothness and bounded-variance assumptions, how fast can vanilla stochastic multi-gradient descent drive the squared Pareto-stationarity measure toward zero?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054263 | MATHDB-315490

Dimension-free consistency of empirical spatial depth

摘录 / EXCERPT: Theorem 2.1. For a sample of iid random elements $X_{1},\ldots,X_{n}\sim P^{X}$ with values in $\mathbb{R}^{d}$, the empirical spatial distribution estimator and the plug-in spatia...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054264 | MATHDB-315507

Gaussian completely monotone conjecture

摘录 / EXCERPT: Conjecture 1 (Gaussian Completely Monotone Conjecture). Let $Y_t=X+Z_t$, where X and Z_t are independent and $Z_t\sim\mathcal{N}(0,t)$. Then $I(Y_t)$ is completely monotone in...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054265 | MATHDB-315508

Log-convexity of Fisher information along heat flow

摘录 / EXCERPT: For every smooth positive density f on $\mathbb{R}^d$, must the Fisher information $t\mapsto I(f*\gamma_t)$ be log-convex along the heat flow?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054266 | MATHDB-315593

Weighted Bernoulli sums above their mean

摘录 / EXCERPT: Conjecture 1. Let $0\leq p\leq 1/3$ be a fixed real number. Let $w\in\mathbb{R}_{>0}^{m}$ satisfy $\sum_{i=1}^{m}w_i=1$, and let $v\in\{0,1\}^{m}$ be a random vector whose coordina...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-054267 | MATHDB-315661

Feige’s $1/e$ conjecture

摘录 / EXCERPT: Conjecture 1 (Feige). Let $X_1,\ldots,X_n$ be arbitrary nonnegative independent random variables with expectations $\mu_1,\ldots,\mu_n$, where $\mu_i\leq 1$ for every i, and let...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054268 | MATHDB-315664

Sharp thin-shell inequality for log-concave measures

摘录 / EXCERPT: Suppose μ is log-concave and isotropic. Is it true that ... for a universal constant $C>0$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054269 | MATHDB-315683

Signed BAR conjecture

摘录 / EXCERPT: Open problem 1. Prove Proposition 1 with $\pi,\nu_1,\ldots,\nu_d$ are finite measures replaced by $\pi,\nu_1,\ldots,\nu_d$ are finite signed measures. To resolve this open problem...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054270 | MATHDB-315705

Gaussian Moments Conjecture in dimensions $n\geq3$

摘录 / EXCERPT: If all positive moments of a complex polynomial in independent standard Gaussian variables vanish, must every polynomially weighted moment eventually vanish as well?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054271 | MATHDB-315712

Gaussian Moments Conjecture in dimension two

摘录 / EXCERPT: Let X,Y be independent standard real Gaussian random variables. Suppose that $P\\in\\mathbb{C}[X,Y]$ satisfies ... for every integer $m\\geq 1$. Must it follow that, for every...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054272 | MATHDB-315722

Particle-swarm stagnation law is non-Gaussian

摘录 / EXCERPT: Can the limiting position distribution of the canonical one-dimensional particle-swarm optimizer during stagnation be Gaussian?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054273 | MATHDB-315772

Return probability for a lamplighter walk

摘录 / EXCERPT: Let $d\ge 3$, let T_d be the infinite d-regular tree, and let $p_n(e,e)$ be the return probability of the switch--walk--switch lamplighter walk over T_d with lamp group...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054274 | MATHDB-315773

Total variation for a lamplighter walk

摘录 / EXCERPT: Let P_t^x denote the law at time t of the discrete-time switch-walk-switch walk on $\mathbb{Z}_2\wr\mathbb{Z}$, where each departure and arrival lamp is resampled independently...

open_problem | open | MathDB 页面 | 原文线索

V2-054275 | MATHDB-315782

Strong log-concavity of Chernoff’s density

摘录 / EXCERPT: Is the density f of $\operatorname{argmax}_t\{W(t)-t^2\}$, for two-sided Brownian motion W, strongly log-concave?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054276 | MATHDB-315783

Benjamini-Hochberg under correlated Gaussian tests

摘录 / EXCERPT: Does the Benjamini-Hochberg procedure always control false-discovery rate at its nominal level for correlated two-sided Gaussian p-values?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054277 | MATHDB-316044

Ibragimov-Iosifescu conjecture for φ-mixing sequences

摘录 / EXCERPT: Let $(\Omega,\mathcal{F},\mu)$ be a probability space and let $X:=(X_k, k\in\mathbb{Z})$ be a strictly stationary sequence of real-valued random variables on it. For...

conjecture | open | MathDB 页面 | 原文线索

V2-054278 | MATHDB-316229

Gaussian correlation inequality

摘录 / EXCERPT: Let $n\ge 1$ and let μ be a centered Gaussian probability measure on $\mathbb{R}^n$, i.e. the distribution of a random vector $X=(X_1,\dots,X_n)$ having a multivariate normal l...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-054279 | MATHDB-316291

The uniqueness-threshold conjecture for Ising Glauber dynamics on sparse random graphs

摘录 / EXCERPT: Uniqueness-threshold conjecture. If $d \tanh(\beta) > 1$ then with high probability over $G(n,d/n)$ the mixing time of the Gibbs sampler is $\exp(\Omega(n))$. If $d > 1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054280 | MATHDB-316316

The critical-value conjecture for the one-dimensional continuum random-cluster model

摘录 / EXCERPT: The one-dimensional continuum random-cluster critical-value conjecture. For $q\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-054281 | MATHDB-316339

Classical large-sample consistency conjecture for the proposed signal-count estimator

摘录 / EXCERPT: Classical consistency conjecture. With n fixed and $m\to\infty$, $\widehat{k}$ is a consistent estimator of the true number of signals k.

conjecture | open | MathDB 页面 | 原文线索

V2-054282 | MATHDB-316340

Large-system consistency conjecture for the effective signal-count estimator

摘录 / EXCERPT: Large-system consistency conjecture. In this joint limit, $\widehat{k}$ is a consistent estimator of $k_{\mathrm{eff}}(c\mid {\bf R})$.

conjecture | open | MathDB 页面 | 原文线索

V2-054283 | MATHDB-316393

The Brownian-sheet small deviation conjecture

摘录 / EXCERPT: Brownian-sheet small deviation conjecture. For $d\ge2$, as $\epsilon\downarrow0$,

conjecture | open | MathDB 页面 | 原文线索

V2-054284 | MATHDB-316421

Successive intermittency thresholds in dimensions at least four

摘录 / EXCERPT: Threshold conjecture. In $d\geq4$, there exists a strictly increasing sequence $0<\kappa_2<\kappa_3<\ldots$ such that, for $p=2,3,\ldots$, the system is p-intermittent if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-054285 | MATHDB-316463

Nielsen's characterization of catalytic majorization by power sums

摘录 / EXCERPT: Let P_d be the set of d-dimensional probability vectors, let $T_d(y)$ and $M_d(y)$ denote the sets associated with the relations $prec_T$ and $prec_M$, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054286 | MATHDB-316468

Asymptotic game chromatic number conjecture for random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the random graph on n vertices in which each edge is present independently with probability p, and let $\chi_g(G_{n,p})$ denote its game chromatic number. For...

conjecture | open | MathDB 页面 | 原文线索

V2-054287 | MATHDB-316469

Conjecture on the tail density of the integrated Brownian supremum

摘录 / EXCERPT: Let $B(t)$ be standard Brownian motion, let $M(t)=\sup_{0\leq s\leq t}B(s)$, and let $\mathcal A=\int_0^1 M(t)\,dt$. Write $f_{\mathcal A}$ for the density function of $\mathcal A$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054288 | MATHDB-316502

Wagner's Rayleigh-to-ULC Big Conjecture

摘录 / EXCERPT: Let $\mathfrak{P}_n$ be the probability measures on subsets of $[n]$, and let a Rayleigh measure be a measure satisfying the Rayleigh negative-dependence inequalities. Let ULC mean...

conjecture | open | MathDB 页面 | 原文线索

V2-054289 | MATHDB-316503

Pemantle's CNA+-to-ULC conjecture

摘录 / EXCERPT: Let $\mathfrak{P}_n$ be the probability measures on subsets of $[n]$, and let CNA+ denote the strongest negative-association property used in the source. Let ULC mean that the rank...

conjecture | open | MathDB 页面 | 原文线索

V2-054290 | MATHDB-316505

Pemantle's CNA+ truncation stochastic-domination conjecture

摘录 / EXCERPT: Let $\mu\in\mathfrak{P}_n$ be a probability measure on subsets of $[n]$, and for $0\leq k\leq n$ let μ_k be its conditional measure given that the sampled set has size k. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-054291 | MATHDB-316583

Percolation and circle-packing modulus conjecture for periodic triangulations

摘录 / EXCERPT: Let T^* be a triangulation of the torus, and let $\hat T^*$ be its universal cover. For $\alpha\in\mathbb{H}$, write $\hat T^*_{\alpha}$ for the corresponding realization in the...

conjecture | open | MathDB 页面 | 原文线索

V2-054292 | MATHDB-316602

The oriented-percolation critical crossing summability conjecture

摘录 / EXCERPT: Oriented-percolation critical crossing summability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054293 | MATHDB-316707

Cardy's formula in Carleson's version for critical percolation

摘录 / EXCERPT: Let D be a simply connected domain in the complex plane, and let a,b,c,x be boundary points in counterclockwise order. Suppose that D is conformally equivalent to the equilat...

conjecture | open | MathDB 页面 | 原文线索

V2-054294 | MATHDB-316842

The nonsingularity conjecture for random symmetric matrices with bounded atom probabilities

摘录 / EXCERPT: The bounded-atom nonsingularity conjecture. Then A is almost surely nonsingular. The conjecture extends sparse nonsingularity results beyond the case in which all entries have th...

conjecture | open | MathDB 页面 | 原文线索

V2-054295 | MATHDB-316856

The partition-based updating conjecture for minimax decision rules

摘录 / EXCERPT: Let ${\cal P}$ be an arbitrary set of distributions on ${\cal X}\times{\cal Y}$, let L be an arbitrary loss function on ${\cal Y}\times{\cal A}$, and let δ be the minimax...

conjecture | refuted | MathDB 页面 | 原文线索

V2-054296 | MATHDB-316947

Pemantle's NA+ implies NMP conjecture

摘录 / EXCERPT: Let μ be a probability measure on $\{0,1\}^n$. Write μ_k for the conditional measure given $\sum_i\eta_i=k$, and say that μ has the normalized matching property (NMP) w...

conjecture | open | MathDB 页面 | 原文线索

V2-054297 | MATHDB-316948

Pemantle's stochastic covering conjecture for NA+ measures

摘录 / EXCERPT: Let μ be a probability measure on $\{0,1\}^n$. Say that μ has the stochastic covering property (SCP) if, for every coordinate i, the conditional law given $\eta_i=0$ stoc...

conjecture | open | MathDB 页面 | 原文线索

V2-054298 | MATHDB-316986

Power-series identity relating the Leutwyler-Smilga series

摘录 / EXCERPT: Power-series identity conjecture. The two series are identical:

conjecture | open | MathDB 页面 | 原文线索

V2-054299 | MATHDB-317009

Ballistic central-limit conjecture for the infinite Edwards model

摘录 / EXCERPT: Infinite Edwards-model conjecture. Under $\mathbb{Q}^{\beta}$, the process X is transient and

conjecture | open | MathDB 页面 | 原文线索

V2-054300 | MATHDB-317023

Uniform infimum convolution inequality for symmetric log-concave measures

摘录 / EXCERPT: Uniform infimum convolution conjecture. Any symmetric log-concave probability measure satisfies $\mathrm{IC}(C)$ with a uniform constant C.

conjecture | open | MathDB 页面 | 原文线索

V2-054301 | MATHDB-317024

Uniform concentration inequality for symmetric log-concave measures

摘录 / EXCERPT: Uniform concentration inequality conjecture. Any symmetric log-concave probability measure satisfies $\mathrm{CI}(C)$ for some universal constant C.

conjecture | open | MathDB 页面 | 原文线索

V2-054302 | MATHDB-317025

Comparable weak and strong moments for symmetric log-concave measures

摘录 / EXCERPT: Comparable weak and strong moments conjecture. Every symmetric log-concave probability measure on $\mathbb R^n$ satisfies $\mathrm{CWSM}(C)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054303 | MATHDB-317046

Polynomiality of free-cumulant coefficients for Jack character polynomials

摘录 / EXCERPT: Polynomiality conjecture. The coefficients of K_μ are polynomials in α with integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-054304 | MATHDB-317047

Positivity of the transformed Jack character polynomials

摘录 / EXCERPT: Let $\beta=1-\alpha$. Given the polynomials K_μ, define polynomials $\widetilde K_\mu$ by ... where the inner sum runs over decompositions of the parts of μ into k disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-054305 | MATHDB-317048

Nonnegative free-cumulant expansion of the Jack Kerov polynomial

摘录 / EXCERPT: Let r be a positive integer, let $\beta=1-\alpha$, and write $R_\rho=\prod_iR_{\rho_i}$. Nonnegative expansion conjecture. For $r\leq20$, the computations support ... where the coe...

conjecture | open | MathDB 页面 | 原文线索

V2-054306 | MATHDB-317049

Monomial positivity of the symmetric functions governing Jack Kerov components

摘录 / EXCERPT: Let f_{ij} be the inhomogeneous symmetric functions in the uniform-components conjecture. Monomial-positivity conjecture. They can be written as ... where all coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-054307 | MATHDB-317050

Q-positivity of Jack Kerov polynomial components

摘录 / EXCERPT: Let Q_k be the formal polynomials in the free cumulants defined by ... for $n\geq2$, with $Q_0=1$ and $Q_1=0$. Let $K_r^{(i,j)}$ be the coefficient of $\alpha^{r-i}\beta^j$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-054308 | MATHDB-317051

Nonnegative free-cumulant expansion of the transformed Jack character polynomial

摘录 / EXCERPT: Transformed-polynomial positivity conjecture. The highest weight of $\widetilde K_\mu$ is $|\mu|-l(\mu)+2$, and

conjecture | open | MathDB 页面 | 原文线索

V2-054309 | MATHDB-317065

The limsup law for normalized long-range dependent processes

摘录 / EXCERPT: Let $p\geq 1$, and let $Y_{n,p}$ and $\sigma_{n,p}$ denote the process and normalizing sequence defined in the paper. Limsup conjecture. ... This conjecture would identify the optima...

conjecture | open | MathDB 页面 | 原文线索

V2-054310 | MATHDB-317101

Nordenstam's scaling-limit conjecture for the Aztec diamond particle process

摘录 / EXCERPT: Let $(\mathpzc{X}(t))_{t=0,1,\dots}$ be the particle process arising from the shuffling algorithm for random tilings of the Aztec diamond, and let $(\mathbf{X}(t))_{t\geq 0}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-054311 | MATHDB-317180

Reversibility of chordal SLE with force points

摘录 / EXCERPT: Reversibility of chordal SLE with force points. After a time-change, the reversal of $(W(\beta(t)))$ has the same distribution as a chordal SLE$(\kappa;\rho_-,\rho_+)$ trace starte...

conjecture | open | MathDB 页面 | 原文线索

V2-054312 | MATHDB-317201

Cufaro Petroni's odd-degrees-of-freedom convolution conjecture

摘录 / EXCERPT: Let $Z_N=\frac{1}{N}\boldsymbol{?}$ be the normalized sum of N independent Student t-distributed random variables, each having $f=2n+1$ degrees of freedom with...

conjecture | open | MathDB 页面 | 原文线索

V2-054313 | MATHDB-317202

Cufaro Petroni's convolution-tail asymptotic conjecture for Student t-distributions

摘录 / EXCERPT: For $\nu>0$, let $f_\nu(x)=A_\nu(1+x^2)^{-(\nu+1/2)}$ be the Student t-density, and for $c>0$ let f_ν^{*c} denote its c-fold convolution, defined through the characteristic...

conjecture | open | MathDB 页面 | 原文线索

V2-054314 | MATHDB-317203

Cufaro Petroni's non-integer convolution mixture conjecture

摘录 / EXCERPT: Let $f_{\nu_0}^{*c}$ be the c-fold convolution of the Student t-density f_{ν_0}, where $\nu_0>0$ and $c>0$. Cufaro Petroni's conjecture. For all $\nu_0>0$ and all $c>0$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-054315 | MATHDB-317228

Critical-value conjecture for giant components in partially structured random graphs

摘录 / EXCERPT: Critical-value conjecture. In the random graph defined by these edge probabilities, $c=1$ is the critical value for the emergence of a giant component if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-054316 | MATHDB-317273

Convexity conjecture for the secrecy graph percolation radius

摘录 / EXCERPT: Convexity conjecture. The percolation radius is convex, and consequently $\lambda_c(r)$ is concave:

conjecture | open | MathDB 页面 | 原文线索

V2-054317 | MATHDB-317291

The circular law conjecture for non-symmetric random matrices

摘录 / EXCERPT: Let ξ be a random variable with mean 0 and variance 1, and let M_n be the random matrix whose entries are independent identically distributed copies of ξ. Circular la...

conjecture | open | MathDB 页面 | 原文线索

V2-054318 | MATHDB-317313

The Laguerre sum-rule conjecture for the rate function

摘录 / EXCERPT: Let ν be a probability measure on $(0,\infty)$, let MP denote the Marchenko-Pastur measure, let ${\mathcal K}(MP\!\ |\!\ \nu)$ denote the reversed Kullback-Leibler divergence...

conjecture | open | MathDB 页面 | 原文线索

V2-054319 | MATHDB-317314

The Jacobi sum-rule conjecture for the rate function

摘录 / EXCERPT: Let $\widehat{J\beta E}(N,\kappa_1N,\kappa_2N)$ denote the Jacobi beta-ensemble model with parameters κ_1N and κ_2N, let ν be a probability measure on $[0,1]$, le...

conjecture | open | MathDB 页面 | 原文线索

V2-054320 | MATHDB-317413

Conjecture on the asymptotic covariance polynomials in subcritical random graphs

摘录 / EXCERPT: Asymptotic covariance conjecture. The polynomial $\hat P_{k,l}$ equals the polynomial $P_{k,l}$ defined in the covariance formula.

conjecture | open | MathDB 页面 | 原文线索

V2-054321 | MATHDB-317420

Sharpness of the integral criterion for transience of conditioned Brownian motion

摘录 / EXCERPT: Let $f:[0,\a0\to\bar{\infty})$ be nondecreasing and positive, let $\mathcal C$ be the space of continuous sample paths, and let $\mathbb{W}_t$ be Wiener measure conditioned on the...

conjecture | open | MathDB 页面 | 原文线索

V2-054322 | MATHDB-317421

Sublinear local time under the recurrent limiting measure

摘录 / EXCERPT: Let f satisfy ... and let $\mathbb{P}$ be any weak limit of the conditioned Wiener measures $\mathbb{W}_t$. Let L_t denote the local time at the origin at time t. Local-time se...

conjecture | open | MathDB 页面 | 原文线索

V2-054323 | MATHDB-317454

The lower-threshold conjecture for standard bootstrap percolation

摘录 / EXCERPT: Lower-threshold conjecture. For any $\epsilon>0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-054324 | MATHDB-317473

Concentration conjecture for the verification decoder's edge-type counts

摘录 / EXCERPT: Let t denote the discrete decoding time, let $R_{i,j,k}(t)$ be the number of edges connected to IVNs of type $r_{i,j,k}$, and let $\bar{R}_{i,j,k}(t)=E[R_{i,j,k}(t)\mid H_t]$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-054325 | MATHDB-317490

Existence of a braided extension of free probability

摘录 / EXCERPT: The paper considers square-root presentations of free groups and their associated braided random sequences. Braided extension conjecture. There exists a braided extension of free p...

conjecture | open | MathDB 页面 | 原文线索

V2-054326 | MATHDB-317566

Reversibility conjecture for chordal SLE$(\kappa;\rho_+,\rho_-)$ traces

摘录 / EXCERPT: Reversibility conjecture. After a time-change, $(W_0(\beta_0(1/t)))$, $0<t<\infty$, has the same distribution as $(\beta_0(t))$, $0<t<\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-054327 | MATHDB-317567

Small-ball asymptotic conjecture for line visibility in the hyperbolic plane

摘录 / EXCERPT: Fix a point $o\in\mathbb{H}^2$. For each $r>0$, let p_r be the least $p\in[0,1]$ such that every random closed set $\mathcal Z\subset\mathbb{H}^2$ with isometry-invariant law and...

conjecture | open | MathDB 页面 | 原文线索

V2-054328 | MATHDB-317667

The local-mass asymptotics conjecture for spatial branching processes with self-interaction

摘录 / EXCERPT: Let Z_n be the population measure at generation n, let $g\in C_c^+\left(\mathbb R^d\right)$, and write $\langle Z_n,g\rangle$ for its integral against g. Let γ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-054329 | MATHDB-317676

Pathwise uniqueness for stochastic heat equations with coloured noise

摘录 / EXCERPT: Consider the stochastic heat equation ... alpha<2(2gamma-1), ... This strengthens the preceding pathwise-uniqueness result in the white-noise case, where the argument gives the condi...

conjecture | open | MathDB 页面 | 原文线索

V2-054330 | MATHDB-317691

The equivalence between discrete spectrum and stochastic incompleteness

摘录 / EXCERPT: A complete noncompact Riemannian manifold is stochastically incomplete if its heat kernel $p(x,y,t)$ satisfies ... for some, equivalently any, $(x,t)\in M\times(0,+\infty)$; otherwis...

conjecture | open | MathDB 页面 | 原文线索

V2-054331 | MATHDB-317692

Conjecture on interior equality of quenched and averaged rate functions

摘录 / EXCERPT: Consider nearest-neighbor random walk in a random environment on $\mathbb{Z}^d$ with an i.i.d. uniformly elliptic environment. Let I_q and I_a denote the quenched and averaged...

conjecture | open | MathDB 页面 | 原文线索

V2-054332 | MATHDB-317819

Mézard-Montanari's Kesten-Stigum conjecture for the Potts model

摘录 / EXCERPT: Mézard-Montanari's conjecture. If $q\leq 4$ and $d<d_{\max}$, then there is reconstruction if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-054333 | MATHDB-317854

The circular law conjecture for iid random matrices

摘录 / EXCERPT: Let M_n be the $n \times n$ random matrix whose entries are iid complex random variables with mean 0 and variance 1. The empirical spectral distribution (ESD) of...

conjecture | open | MathDB 页面 | 原文线索

V2-054334 | MATHDB-317857

Equivalence of finiteness for cutpoints and strong cutpoints

摘录 / EXCERPT: Let $\{X_n\}$ be the transient nearest-neighbor random walk described above, and call a site a cutpoint or a strong cutpoint according to the definitions in the paper. A strong cut...

conjecture | open | MathDB 页面 | 原文线索

V2-054335 | MATHDB-317893

Complete monotonicity conjecture for relative entropy in Poisson approximation

摘录 / EXCERPT: Let $D(P)$ denote the relative entropy of a probability distribution P from the Poisson distribution with the same mean. Write $bi(n,p)$ for the binomial distribution with parame...

conjecture | open | MathDB 页面 | 原文线索

V2-054336 | MATHDB-317898

Fundamental sequences and convergence of relativistic diffusion boundary data

摘录 / EXCERPT: Let $(\mathbb{M},q)$ be the spacetime under consideration, let L be the generator, and let $\{\mathbf e_n\}_{n\geqslant 0}$ be a sequence of points of the orthonormal frame bundl...

conjecture | open | MathDB 页面 | 原文线索

V2-054337 | MATHDB-317899

The relativistic H-theorem for the $(V,\mathfrak z)$-diffusion

摘录 / EXCERPT: Relativistic H-theorem conjecture. The $\mathbb{H}\mathbb{M}$-valued $(V,\mathfrak z)$-diffusion process $(\gamma_.,\gamma'_.)=\{(m_s,\mathbf g_s^0)\}_{s\geqslant 0}$ almost surely...

conjecture | open | MathDB 页面 | 原文线索

V2-054338 | MATHDB-317926

The directed Bunkbed conjecture for the D_3^T model

摘录 / EXCERPT: Directed Bunkbed conjecture. For every $u,v\in V(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-054339 | MATHDB-317937

Quintic-kernel conjecture for the process at touching Pearcey cusps

摘录 / EXCERPT: The quintic kernel is defined by ... where the z- and $\tilde z$-integration paths are appropriate subpaths of the paths shown in the source figure, with the indicated orientation....

conjecture | open | MathDB 页面 | 原文线索

V2-054340 | MATHDB-317955

The ASEP almost-sure speed convergence conjecture

摘录 / EXCERPT: Let $\pin(1/2,1]$ and write $\rho=2p-1$. Consider the asymmetric simple exclusion process (ASEP), with particles jumping right at rate p and left at rate $1-p$, and let $X_0(t)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-054341 | MATHDB-317972

Durrett's mixing-time conjecture for lazy random walk on random 3-regular graphs

摘录 / EXCERPT: Let $G\sim\mathcal{G}(n,3)$ be a random 3-regular graph, and consider the lazy random walk on G. Its mixing time is measured in total variation and the lazy walk stays in place...

conjecture | open | MathDB 页面 | 原文线索

V2-054342 | MATHDB-317982

Conditional distribution conjecture for the Askey-Wilson Markov process

摘录 / EXCERPT: Conditional distribution conjecture. The conditional distribution of Y_t given $\mathcal{F}_{s,u}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-054343 | MATHDB-318008

Diffeomorphism conjecture for pair frequencies in the RN+YpR model

摘录 / EXCERPT: For $x\in\{C,A\}$, let $(x,x)(t)$ and x,x denote the corresponding pair-frequency functions in the RN+YpR model, and let $(x)_*$ be the stationary frequency of x. Diffeomorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-054344 | MATHDB-318034

Mean field frozen percolation scaling conjecture

摘录 / EXCERPT: Mean field frozen percolation scaling conjecture. For every $T^g<t$,

conjecture | open | MathDB 页面 | 原文线索

V2-054345 | MATHDB-318053

Moment bounds for vector martingale quadratic forms without the convergence assumption

摘录 / EXCERPT: Let (ε_n) be a martingale difference sequence satisfying the homogeneity condition ... and assumption (H_p) introduced in the preceding theorem, for some integer...

conjecture | open | MathDB 页面 | 原文线索

V2-054346 | MATHDB-318064

Conjecture on the necessity of a positive Kolmogorov exponent

摘录 / EXCERPT: Necessity of a positive Kolmogorov exponent. Theorem is not true if hypothesis (2), namely the condition ${\mathbb E}\,|X_t-X_s|^\beta\le C|t-s|^{1+\alpha}$ with $\alpha>0$, is rep...

conjecture | open | MathDB 页面 | 原文线索

V2-054347 | MATHDB-318124

Change-of-variable conjecture for the entropic measure under diffeomorphisms

摘录 / EXCERPT: Let M be the underlying multidimensional space, let $\mathcal{P}$ denote the space of probability measures on M, and let $\mathbb{P}^\beta$ be the entropic measure with paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-054348 | MATHDB-318153

The converse measurement-rate conjecture for two-sensor distributed compressive sensing

摘录 / EXCERPT: Converse measurement-rate conjecture. The following conditions on the measurement rates are necessary for recovery with probability one:

conjecture | open | MathDB 页面 | 原文线索

V2-054349 | MATHDB-318159

The conditional-coordinate-tail conjecture for isotropic down-monotone logconcave distributions

摘录 / EXCERPT: Let f be an isotropic down-monotone logconcave function in $\mathbb{R}^m$, and let X have density f. Let $1>p>0$. The conditional-coordinate-tail conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-054350 | MATHDB-318181

Monotonicity and zero-velocity conjecture for excited random walks with competing drifts

摘录 / EXCERPT: Monotonicity and zero-velocity conjecture. For all $d\geq2$ and $(\mu,\beta)\in[0,1]^2$, the velocity v_1 exists, is monotone increasing in β for fixed μ, and is decrea...

conjecture | open | MathDB 页面 | 原文线索

V2-054351 | MATHDB-318187

Aizenman's conformal invariance and Cardy formula conjecture for critical planar percolation

摘录 / EXCERPT: Let Ω be a simply connected open domain in the plane, and let A,B,C,D be four boundary points of Ω in clockwise order. For critical percolation on the $1/n$-scaled...

conjecture | open | MathDB 页面 | 原文线索

V2-054352 | MATHDB-318190

The Aldous-Lyons conjecture on limits of bounded-degree graphs

摘录 / EXCERPT: A graphing is a measure-preserving graph limit object for bounded-degree graphs, and a unimodular distribution is a distribution on rooted countable graphs satisfying the unimodula...

conjecture | open | MathDB 页面 | 原文线索

V2-054353 | MATHDB-318222

Factorization conjecture for concentrated multilinear forms

摘录 / EXCERPT: Let k be a fixed positive integer. Let $y_1,\dots,y_k$ be independent vectors uniformly chosen from $\{-1,1\}^n$, let A be a k-multilinear form whose coefficients are all non...

conjecture | open | MathDB 页面 | 原文线索

V2-054354 | MATHDB-318223

Factorization conjecture for concentrated homogeneous polynomials

摘录 / EXCERPT: Let $x_1,\dots,x_n$ be independent and uniformly chosen from $\{-1,1\}$, and let ... be a degree-k homogeneous polynomial with at least mn^{k-1} nonzero coefficients. Polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-054355 | MATHDB-318236

The sign conjecture for Parrondo's paradox in the absence of bias

摘录 / EXCERPT: Let games A and B be as in Theorem 4, with the bias parameter absent. For positive integers r and s, let $\mu_{[r,s]}(0)$ denote the expected gain per turn for the periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-054356 | MATHDB-318331

Measure-theoretic determinant conjecture

摘录 / EXCERPT: Let ${\mathcal R}$ be a standard equivalence relation on a standard probability space. Its groupoid ring is ${\mathbb Z}{\mathcal R}$, and ${\mathcal N}({\mathcal R})$ is the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-054357 | MATHDB-318344

Class-B and Class-C worst-attack symmetry conjecture for probabilistic traitor tracing

摘录 / EXCERPT: Let $f(p)$ be a symmetric pdf, and let θ_σ^* denote the Class-B collusion parameter for $[...].$ For collusion size c, write $R_{\text{simple}}^{B}(c)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054358 | MATHDB-318374

Equivalence of transience and ballisticity in random environment

摘录 / EXCERPT: Let $d\ge 2$ and let $l\in\mathbb S^{d-1}$. A random walk in a uniformly elliptic i.i.d. random environment is transient in the direction l if ... P_0-almost surely, and it is ba...

conjecture | open | MathDB 页面 | 原文线索

V2-054359 | MATHDB-318376

The entropy non-reduction conjecture for the Markov-chain transformation

摘录 / EXCERPT: Entropy non-reduction conjecture. The transformation described above cannot reduce, on average, the entropy of a realization of the stochastic process below the average entropy of...

conjecture | open | MathDB 页面 | 原文线索

V2-054360 | MATHDB-318439

Thinned entropy power inequality

摘录 / EXCERPT: Let X and Y be independent random variables with ultra-log-concave (ULC) probability mass functions on $\mathbf{Z}_+$. Let $T_\alpha X$ denote thinning of X with parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-054361 | MATHDB-318516

Polynomial-factor concentration conjecture for determinants of random matrices

摘录 / EXCERPT: Let $c \le C$ be positive constants. Let A be the random matrix of size n whose entries are independent random variables with mean zero and variances between c and C. Polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-054362 | MATHDB-318518

The mother-group zero-entropy random-walk conjecture

摘录 / EXCERPT: Let d be the degree of a mother group, and consider random walks whose step distribution is supported on a finite generating set. The mother-group zero-entropy conjecture. The mo...

conjecture | open | MathDB 页面 | 原文线索

V2-054363 | MATHDB-318519

The bounded-support zero-entropy conjecture for polynomial automaton groups

摘录 / EXCERPT: Let G be a polynomial automaton group of degree at most 2, and let a random walk on G have a step distribution with bounded support. The bounded-support zero-entropy conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-054364 | MATHDB-318594

Three sign changes in the relative covariance for random orientations of $G(n,p)$

摘录 / EXCERPT: The three-sign-change conjecture. For $n\ge 27$, the relative covariance $R_n(p)$ changes sign at three critical probabilities $p_1(n)<p_2(n)<p_3(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054365 | MATHDB-318595

Asymptotics of the first two covariance sign changes

摘录 / EXCERPT: Let $p_1(n)<p_2(n)<p_3(n)$ be the three critical probabilities from the preceding conjecture for the relative covariance in the randomly oriented graph $G(n,p)$. Asymptotic sign-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-054366 | MATHDB-318596

Positivity at one-half for the relative covariance

摘录 / EXCERPT: Half-probability positivity conjecture. For $p=\frac12$ and $n\ge 6$, the relative covariance $R_n(p)$ is positive.

conjecture | open | MathDB 页面 | 原文线索

V2-054367 | MATHDB-318601

Conjecture on four sufficient conditions for Parrondo's paradox in periodic patterns

摘录 / EXCERPT: Four-condition conjecture. The conclusion of Theorem 5 continues to hold if the assumption that $r+s$ divides J is replaced by any one of the following conditions:

conjecture | open | MathDB 页面 | 原文线索

V2-054368 | MATHDB-318763

Conjecture on the asymptotic mean absolute self-linking number of confined random walks and polygons

摘录 / EXCERPT: Let SL denote the self-linking number of an oriented uniform random walk or polygon in confined space, and let E denote expectation. Mean absolute self-linking conjecture. As t...

conjecture | open | MathDB 页面 | 原文线索

V2-054369 | MATHDB-318799

Convergence to a stable equilibrium for random initial opinions

摘录 / EXCERPT: Consider the continuous-time opinion-dynamics system with n agents, and choose the agents' initial opinions independently and randomly according to a bounded probability density...

conjecture | open | MathDB 页面 | 原文线索

V2-054370 | MATHDB-318810

Gaussian cycle-count conjecture for permutations with averaged cycle weights

摘录 / EXCERPT: Let $\vec{\sigma}=(\sigma_1,\sigma_2,\ldots)$ be a sequence of nonnegative real numbers with mean α, meaning that ... Permutations of $[n]$ selected according to the weights...

conjecture | open | MathDB 页面 | 原文线索

V2-054371 | MATHDB-318865

The scaling conjecture for Lyapunov exponents of the voter model on a catalyst

摘录 / EXCERPT: Let $p_t(x,y)$ be the transition probability at time t for the rate-one random walk with kernel $p(,)$, and let $\blambda_p(kappa)$ denote the associated Lyapunov exponent. Suppo...

conjecture | open | MathDB 页面 | 原文线索

V2-054372 | MATHDB-318997

Ewens distribution cycle-count conjecture

摘录 / EXCERPT: Ewens cycle-count conjecture. The expected number of such cycles approaches

conjecture | open | MathDB 页面 | 原文线索

V2-054373 | MATHDB-319040

The central point theorem for k-flats

摘录 / EXCERPT: Let μ be an absolutely continuous probabilistic measure with compact support on the set of all affine k-flats in $\mathbb R^d$. For a subset X of $\mathbb R^d$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-054374 | MATHDB-319096

Storage-efficiency conjecture for load-balancing modulation codes

摘录 / EXCERPT: Storage-efficiency conjecture. If $l=2$ and $q-1=c\ln n$, then the load-balancing modulation code has storage efficiency $\gamma=k$ with probability $1-o(1)$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-054375 | MATHDB-319118

Strict bounds for the critical density of activated random walkers

摘录 / EXCERPT: Let $d\geqslant 1$, let the particles perform any translation-invariant random walk, and let $\lambda>0$. For i.i.d. Poisson initial data of density ζ, let $\zeta_c(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-054376 | MATHDB-319152

Growth exponent conjecture for one-dimensional long-range diffusion-limited aggregation

摘录 / EXCERPT: Let R be a symmetric random walk on $\mathbb{Z}$ with step distribution satisfying ... for some $0<\alpha<1$, and let D_n denote the diameter of the aggregate after n particles...

conjecture | open | MathDB 页面 | 原文线索

V2-054377 | MATHDB-319153

Capacity sparsity conjecture for transient random-walk aggregation

摘录 / EXCERPT: Let R be a transient random walk on $\mathbb{Z}$, and let A_n be the aggregate after n particles. Write $\operatorname{Cap}(A_n)$ for its capacity. Capacity sparsity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-054378 | MATHDB-319181

Instability conjecture for the stochastic Deterministic First Chunk system

摘录 / EXCERPT: Deterministic First Chunk instability conjecture. With λ large enough, the stochastic Deterministic First Chunk system is unstable: almost surely, either X or Y escapes...

conjecture | open | MathDB 页面 | 原文线索

V2-054379 | MATHDB-319182

Stability conjecture for the stochastic Enforced Friedman system

摘录 / EXCERPT: Enforced Friedman stability conjecture. The stochastic Enforced Friedman system is stable for every input rate λ.

conjecture | open | MathDB 页面 | 原文线索

V2-054380 | MATHDB-319193

General form conjecture for the coefficients of multidimensional q-Normal distributions

摘录 / EXCERPT: Let $A_{r,k}^{(n)}$ be the coefficients appearing in the conditional expectation expansion for the multidimensional q-Normal distribution, with parameters ρ_{i-1}, ρ_i,...

conjecture | open | MathDB 页面 | 原文线索

V2-054381 | MATHDB-319197

The fast-embedding quadrant traversal-time asymptotic

摘录 / EXCERPT: Let $(A(t),B(t))$ be the fast continuous-time embedding of the simple harmonic urn, let ... and write $\mathbb{E}_n$ for expectation conditional on $A(0)=n$ and $B(0)=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054382 | MATHDB-319279

Critical cylinder exponent conjecture for first-passage percolation

摘录 / EXCERPT: Let $a_n(h)$ be the first-passage percolation time across a cylinder of length n and half-height h, and define ... For distributions satisfying the stated moment and admissibilit...

conjecture | open | MathDB 页面 | 原文线索

V2-054383 | MATHDB-319280

Thin-cylinder variance-order conjecture for first-passage percolation

摘录 / EXCERPT: Let $a_n(h)$ denote the first-passage percolation time across a cylinder of length n and half-height h, and let $d\ge 2$. Thin-cylinder variance-order conjecture. The variance...

conjecture | open | MathDB 页面 | 原文线索

V2-054384 | MATHDB-319317

Factorization conjecture for limits of nonlinear random matrix statistics

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\dots,\lambda_M)$ be a partition, and let $\mathcal{L}_\lambda$ denote the large-N limit associated with the statistic indexed by λ. Factorization...

conjecture | open | MathDB 页面 | 原文线索

V2-054385 | MATHDB-319403

Conditional negative association for partitioned urn indicators

摘录 / EXCERPT: Partitioned-urn conjecture. The variables xi_{ij} are negatively associated conditional on

open_problem | open | MathDB 页面 | 原文线索

V2-054386 | MATHDB-319422

Andrews' asymptotic conjecture for generalized bootstrap percolation

摘录 / EXCERPT: Andrews' conjecture. For each $k\geq 2$, there is a positive constant c_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-054387 | MATHDB-319435

The Jack-measure coefficient independence conjecture

摘录 / EXCERPT: Let F be a symmetric function of degree k, let α be a positive real number, and let $\mu\vdash k$ be a partition. Denote by $\mathcal{A}^{(\alpha)}_\mu(F,n)$ the coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-054388 | MATHDB-319468

Density conjecture for the square-grid sandpile

摘录 / EXCERPT: Let $b6_c$ be the threshold density of the fixed-energy sandpile on the square grid, and let $b6_s$ be the stationary density of the classical abelian sandpile on $b6^2$. Density c...

conjecture | open | MathDB 页面 | 原文线索

V2-054389 | MATHDB-319476

The local-volatility minimum conjecture for options on realized variance

摘录 / EXCERPT: Let S follow a continuous stochastic-volatility model and let $V_T=\langle\log S\rangle_T$ be its realized variance. Let $\tilde S$ be the corresponding local-volatility model ca...

conjecture | open | MathDB 页面 | 原文线索

V2-054390 | MATHDB-319555

Non-amenability conjecture for traces of branching random walks

摘录 / EXCERPT: Non-amenability conjecture. Let μ be the measure of the trace of a transient BRW on a URG. Then μ is non-amenable.

conjecture | open | MathDB 页面 | 原文线索

V2-054391 | MATHDB-319556

Continuity conjecture for the Hausdorff dimension of branching-random-walk limit sets

摘录 / EXCERPT: Hausdorff-dimension continuity conjecture. The Hausdorff dimension of the limit set is continuous for $m\neq 1/\rho(P)$ and discontinuous at $m=1/\rho(P)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054392 | MATHDB-319559

Strong recurrence conjecture for branching random walks on traces

摘录 / EXCERPT: Trace strong-recurrence conjecture. Every BRW on almost every trace of a transient BRW is strongly recurrent.

conjecture | open | MathDB 页面 | 原文线索

V2-054393 | MATHDB-319689

Non-ordering conjecture for fuzzy Potts measures on the lattice

摘录 / EXCERPT: Let $q\geq 3$, $r\in \\{1,\dots,q-1\\}$, and consider the fuzzy Potts model on $\mathbb{Z}^d$. For $J>0$, write $\nu_{q,J,r}^{\mathbb{Z}^d,+}$ for its plus-state fuzzy Potts measur...

conjecture | open | MathDB 页面 | 原文线索

V2-054394 | MATHDB-319690

Distinct-domination-sets conjecture for fuzzy Potts phases

摘录 / EXCERPT: Let $J>0$, $q\geq 3$, and $r\in \\{1,\dots,q-1\\}$. Consider the fuzzy Potts model on the torus $\mathbb{T}^d$, and let γ_p be Bernoulli product measure of density p. Defi...

conjecture | open | MathDB 页面 | 原文线索

V2-054395 | MATHDB-319691

Monotonicity conjecture for fuzzy Potts plus states beyond criticality

摘录 / EXCERPT: Let $q\geq 3$, $r\in \\{1,\dots,q-1\\}$, and consider the fuzzy Potts model on the torus $\mathbb{T}^d$. Denote by J_c the critical value corresponding to non-uniqueness of Gibbs...

conjecture | open | MathDB 页面 | 原文线索

V2-054396 | MATHDB-319692

Probabilistic computation of Jacobson form from a diagonal form

摘录 / EXCERPT: Let R be a simple Euclidean domain and let M be a square matrix in diagonal form ... The quantity $\sum\deg(m_i)$ is an invariant of the module $R^{r\times r}/M$. Choose polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-054397 | MATHDB-319737

Asymptotic factorization conjecture for normalized power sums

摘录 / EXCERPT: Asymptotic factorization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054398 | MATHDB-319761

Criterion for moderate infection under increasing division rates

摘录 / EXCERPT: Let X_t be the total quantity of parasites at time t, let V_t be the set of cells at time t, let $N_t=\#V_t$, and let X_t^i denote the parasite quantity in cell...

conjecture | open | MathDB 页面 | 原文线索

V2-054399 | MATHDB-319788

Chavel's conjecture on Neumann heat-kernel monotonicity

摘录 / EXCERPT: Let $D_1,D_2\subset \mathbb{R}^d$ be smooth bounded convex domains, with $d\geq 1$. For $i=1,2$, let $p_{D_i}(t,x,y)$ denote the Neumann heat kernel in D_i. If $D_2\subset D_1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-054400 | MATHDB-319832

Capacity-matrix probabilistic recovery bound

摘录 / EXCERPT: Let $\boldsymbol{Q}$ be the capacity matrix, let $\boldsymbol{Q}^{V}$ be the vector of values $Q_{i,j}$ with $i<j$ in Ω, and let E_Q and σ_Q^2 denote its mean and v...

conjecture | open | MathDB 页面 | 原文线索

V2-054401 | MATHDB-319853

Optimal first transmission conjecture for Boolean threshold functions

摘录 / EXCERPT: Let $X^n=(X_1,\ldots,X_n)$ be the block of Boolean observations, let $\Pi_{n-k}(X^n)$ denote the Boolean threshold function, and suppose the probability distribution satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-054402 | MATHDB-319861

Sznitman's quenched atypical-exit estimate conjecture

摘录 / EXCERPT: Let $d\geq2$, let $\mathbb{P}$ be uniformly elliptic, and let $l\in\mathbb{S}^{d-1}$. Assume condition $(T')|l$ holds. For $\beta\in(0,1)$ and $L>0$, define the slab ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054403 | MATHDB-319877

Non-necessity of the recurrence criterion in dimension two

摘录 / EXCERPT: Non-necessity conjecture. This condition, which is sufficient for recurrence of the two-dimensional random walk, is not necessary. The claim would show that recurrence can hold wit...

conjecture | open | MathDB 页面 | 原文线索

V2-054404 | MATHDB-319878

Failure of the central limit theorem in high dimensions

摘录 / EXCERPT: A random walk on a discrete point process is the model studied in the paper, and a central limit theorem means convergence of its appropriately normalized position to a Gaussian li...

conjecture | open | MathDB 页面 | 原文线索

V2-054405 | MATHDB-319944

The correction-limit conjecture for asymmetric Burgers discretisations

摘录 / EXCERPT: Let u solve the stochastic Burgers equation with viscosity $ u$, noise amplitude σ, and asymmetric finite-difference parameters a,b. Let u^{ε} denote the sol...

conjecture | open | MathDB 页面 | 原文线索

V2-054406 | MATHDB-319945

The finite-difference correction conjecture for stochastic Burgers equation

摘录 / EXCERPT: Let u^N be the solution of the finite-difference approximation with an even number N of modes, and let $\nu>0$ and σ denote the viscosity and noise parameters. Finite-di...

conjecture | open | MathDB 页面 | 原文线索

V2-054407 | MATHDB-319946

The corrected-limit conjecture for multidimensional stochastic Burgers nonlinearities

摘录 / EXCERPT: Multidimensional correction conjecture. As $\varepsilon\to0$, u^{ε} converges to the solution of

conjecture | open | MathDB 页面 | 原文线索

V2-054408 | MATHDB-319947

The discretisation-independence conjecture for classically ill-posed stochastic equations

摘录 / EXCERPT: Discretisation-independence conjecture. As $\varepsilon\to0$, these two equations converge to the same limit. The conjecture proposes a limiting notion of solution for equations wh...

conjecture | open | MathDB 页面 | 原文线索

V2-054409 | MATHDB-319948

The multiplicative-noise correction conjecture for stochastic Burgers discretisations

摘录 / EXCERPT: Multiplicative-noise correction conjecture. As $\varepsilon\to0$, its solution converges to the solution of

conjecture | open | MathDB 页面 | 原文线索

V2-054410 | MATHDB-319949

The two-regime conjecture for finite-difference approximations of viscous Burgers equation

摘录 / EXCERPT: Let u be governed by the regularised Burgers equation with viscosity parameter ε, and let δ be the spatial mesh size. Write $c\in\{1,0,-1\}$ according as the d...

conjecture | open | MathDB 页面 | 原文线索

V2-054411 | MATHDB-319969

Tightness conjecture for the centered maximum of the two-dimensional discrete Gaussian free field

摘录 / EXCERPT: Tightness conjecture. The sequence of random variables $\{Y_N\}_{N\geq 1}$ is tight.

conjecture | open | MathDB 页面 | 原文线索

V2-054412 | MATHDB-320047

The threshold conjecture for concavity of stable distributions

摘录 / EXCERPT: A strictly stable probability measure is a stable law, and a probability measure is convex when it satisfies the paper's convexity property for measures. In finite dimensions, fix...

conjecture | open | MathDB 页面 | 原文线索

V2-054413 | MATHDB-320179

Contractible-domain conjecture for shy co-adapted couplings

摘录 / EXCERPT: Let D be a bounded contractible domain in a Euclidean space of arbitrary dimension, and consider reflecting Brownian motions in D together with co-adapted couplings. A coupling...

conjecture | open | MathDB 页面 | 原文线索

V2-054414 | MATHDB-320210

The Wiener sausage volume minimization conjecture

摘录 / EXCERPT: Let $W_g(t)$ denote the Wiener sausage generated up to time t by a function g in ${\mathbb R}^d$, and let $W_0(t)$ be the Wiener sausage generated by the zero function. For a s...

conjecture | open | MathDB 页面 | 原文线索

V2-054415 | MATHDB-320251

Super-strong separation lemma for critical percolation interfaces

摘录 / EXCERPT: Super-strong separation lemma. Conditioned on at least four of these interfaces reaching $B(R_1)$, the set of interfaces reaching R_1 has positive quality with positive probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-054416 | MATHDB-320328

Universality of the mild convergence condition for PEPA models

摘录 / EXCERPT: PEPA mild-condition conjecture. The formula $$ in Theorem holds for any PEPA model.

open_problem | open | MathDB 页面 | 原文线索

V2-054417 | MATHDB-320335

The concentration conjecture for random stable commutator length

摘录 / EXCERPT: Let (G,μ,Y) be nondegenerate in the sense of Definition, and let g be obtained by an even-length random walk on G, conditioned to lie in $[G,G]$. Concentration conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-054418 | MATHDB-320336

The norm conjecture for random stable commutator length subspaces

摘录 / EXCERPT: Let (G,μ,Y) be nondegenerate in the sense of Definition. Let $g_1,\ldots,g_k$ be obtained by independent even-length random walks on G of lengths $\ell_1n,\ldots,\ell_kn$, co...

conjecture | open | MathDB 页面 | 原文线索

V2-054419 | MATHDB-320338

The finite-dimensionality conjecture for quasimorphisms

摘录 / EXCERPT: Let G be a finitely presented group, and let g_n be obtained by a length-n random walk, conditioned to lie in $[G,G]$. Suppose that, for every $\epsilon>0$, there is a positi...

conjecture | open | MathDB 页面 | 原文线索

V2-054420 | MATHDB-320385

Brown's converse conjecture on concave renewal functions and DFR distributions

摘录 / EXCERPT: Let $F(t)$ be a distribution function on $\mathbf{R}_+=[0,\infty)$ with $F(0)=0$, and let $M(t)$ be the renewal function satisfying ... A distribution on $\mathbf{R}_+$ is DFR (decre...

conjecture | open | MathDB 页面 | 原文线索

V2-054421 | MATHDB-320395

Functional limit law for partial match queries in quadtrees

摘录 / EXCERPT: Let $N_t(x)$ denote the partial match query cost at position $x\in[0,1]$, let $\beta^\ast=\sqrt{2}-1$, and let $\mathbb{D}([0,1])$ be the space of càdlàg functions on $[0,1]$ equip...

conjecture | open | MathDB 页面 | 原文线索

V2-054422 | MATHDB-320472

Brownian-motion lower bound for random walks on self-similar groups

摘录 / EXCERPT: Let G be a self-similar group and let $\mathcal{X}$ be its limit space. Let W_t be Brownian motion on $\mathcal{X}$, let $d(0,W_t)$ denote its displacement from the base point,...

conjecture | open | MathDB 页面 | 原文线索

V2-054423 | MATHDB-320523

Size conjecture for the emerging $(3+2)$-core

摘录 / EXCERPT: Size conjecture for the emerging $(3+2)$-core. With high probability, the size of the $(3+2)$-core when it emerges is qn. This predicts the asymptotic size of the core at emergen...

conjecture | open | MathDB 页面 | 原文线索

V2-054424 | MATHDB-320559

Intermittency conjecture for the parabolic Anderson model with moving catalysts

摘录 / EXCERPT: Fix $n\in\mathbb{N}$. Let d be the spatial dimension, let $\kappa\geq 0$ be the diffusion constant of the catalyst, and let $\kappa_p^{(n)}$ denote the critical values governing...

conjecture | open | MathDB 页面 | 原文线索

V2-054425 | MATHDB-320560

Distinct critical diffusion constants conjecture

摘录 / EXCERPT: Fix $n\geq 1$. For each p, let $\kappa_p^{(n)}$ denote the critical diffusion constant associated with p-intermittency. Distinct critical values conjecture. For all fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-054426 | MATHDB-320604

Four-dimensional subdiffusive heat-kernel decay conjecture

摘录 / EXCERPT: Let $d=4$, and let $\mathbb P$ be any i.i.d. environment law satisfying $\mathbb P(0<\omega_b\leq 1)=1$. Write $\cmss P_\omega^{2n}(0,0)$ for the quenched return probability at tim...

conjecture | open | MathDB 页面 | 原文线索

V2-054427 | MATHDB-320641

Ghirlanda-Guerra extremes conjecture for stochastically stable ROSts

摘录 / EXCERPT: Let the Ghirlanda-Guerra identities be imposed on the laws of random overlap structures, and consider the convex set of laws of stochastically stable ROSts. Ghirlanda-Guerra extrem...

conjecture | open | MathDB 页面 | 原文线索

V2-054428 | MATHDB-320716

Optimal precoding matrices conjecture for slow-fading MIMO channels

摘录 / EXCERPT: Let $\Gamma(\mathbf{Q},R)$ denote the goodput-to-power ratio, let $P_{\mathrm{out}}(\mathbf{Q},R)$ denote the outage probability, and let $\overline{P}$ be the available transmit-p...

conjecture | open | MathDB 页面 | 原文线索

V2-054429 | MATHDB-320731

Stationary distribution conjecture for two reflected Brownian motions in dimension three

摘录 / EXCERPT: Stationary distribution conjecture. If $d=3$, then there is $r_0>0$ such that for $r\leq r_0$, the process (X,Y) has a stationary distribution Q which does not charge the diago...

conjecture | open | MathDB 页面 | 原文线索

V2-054430 | MATHDB-320732

Many-hole domain conjecture for reflected Brownian motions

摘录 / EXCERPT: Many-hole domain conjecture. If k is sufficiently large and

conjecture | open | MathDB 页面 | 原文线索

V2-054431 | MATHDB-320733

Small-hole stationary-measure convergence conjecture

摘录 / EXCERPT: Small-hole convergence conjecture. The measures Q_r converge to $\operatorname{Unif}((D_1)^2)$ when $r\to0$.

conjecture | open | MathDB 页面 | 原文线索

V2-054432 | MATHDB-320734

Lagrangian measure convergence conjecture for reflected Brownian flow

摘录 / EXCERPT: Lagrangian measure convergence conjecture. The measures ${\bf Q}_{r,t}$ converge to a random measure ${\bf Q}_r$ on $D_1\setminus{\mathcal B}(0,r)$ as $t\to\infty$, in the sense of...

conjecture | open | MathDB 页面 | 原文线索

V2-054433 | MATHDB-320838

Poisson boundary conjecture for bounded-range lamplighter walks on oriented trees

摘录 / EXCERPT: Let $\mathcal{T}$ be an oriented tree with a fixed end, and let (Z_n) be a random walk with bounded range on $\mathbb{Z}_2\wr\mathcal{T}$. Write (X_n) for its projection to...

conjecture | open | MathDB 页面 | 原文线索

V2-054434 | MATHDB-320949

Mean-square consistency for Gaussian stationary processes

摘录 / EXCERPT: Gaussian consistency conjecture. If X is a Gaussian stationary process with $\|RX\|_1<\infty$, then $S_J[\mathcal P_J]X$ is a mean-square consistent estimator of $\overline S X$....

conjecture | open | MathDB 页面 | 原文线索

V2-054435 | MATHDB-320970

The singularity probability conjecture for random Bernoulli matrices

摘录 / EXCERPT: Singularity probability conjecture. The singularity probability satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054436 | MATHDB-320971

The singularity probability conjecture for random symmetric Bernoulli matrices

摘录 / EXCERPT: Symmetric-matrix singularity conjecture. The singularity probability satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054437 | MATHDB-321034

Wigner-Dyson-Mehta bulk universality conjecture for Wigner matrices

摘录 / EXCERPT: Wigner-Dyson-Mehta bulk universality conjecture. For fixed k, u, and atom distributions, $\rho_{n,u}^{(k)}$ converges to $\rho_{\operatorname{Sine}}^{(k)}$ as...

conjecture | solved | MathDB 页面 | 原文线索

V2-054438 | MATHDB-321035

Linear overhead conjecture for greedy random walk on complete graphs

摘录 / EXCERPT: Linear overhead conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054439 | MATHDB-321040

Finite intersection property for critical percolation crossings

摘录 / EXCERPT: Let Q be a quad and let α be a smooth curve. A crossing of Q is a percolation crossing between the designated boundary arcs of the quad. Finite intersection property. In...

conjecture | open | MathDB 页面 | 原文线索

V2-054440 | MATHDB-321041

Tightness of intersections for critical percolation crossings

摘录 / EXCERPT: Let Q be a quad and let α be a smooth curve. For a discrete percolation measure μ_η, let $N_\eta=N_\eta(Q,\alpha)$ be the minimal possible number of intersections w...

conjecture | open | MathDB 页面 | 原文线索

V2-054441 | MATHDB-321072

Strict hierarchy of V-causality conditions for probabilistic cellular automata

摘录 / EXCERPT: Strict V-causality hierarchy conjecture. For every k, there exists a V^kBox which is V^k-causal but not V^{k+1}-causal.

conjecture | open | MathDB 页面 | 原文线索

V2-054442 | MATHDB-321080

Queue-area tail asymptotics for a stable GI/GI/1 queue

摘录 / EXCERPT: Queue-area tail conjecture. If $F\in\mathcal{S}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054443 | MATHDB-321081

Workload-area tail asymptotics for a stable GI/GI/1 queue

摘录 / EXCERPT: Workload-area conjecture. If $\bar F\in\mathcal{S}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054444 | MATHDB-321082

Utility-functional tail conjecture for workload in a stable GI/GI/1 queue

摘录 / EXCERPT: Utility-functional tail conjecture. Assume that $\bar F\in\mathcal{S}$ and $f(x)=x^k$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-054445 | MATHDB-321175

Refuted asymptotic-direction conjecture for random affine Weyl group walks in type A

摘录 / EXCERPT: Let $W=S_n$ be the finite Weyl group, let ρ be the usual Weyl vector regarded as a point of the reflection representation V, and let $\psi\in V$ be the asymptotic direction...

conjecture | refuted | MathDB 页面 | 原文线索

V2-054446 | MATHDB-321232

Bruno-Rota-Torney conjecture on maximal binary cumulants

摘录 / EXCERPT: Let $\b7$?

conjecture | open | MathDB 页面 | 原文线索

V2-054447 | MATHDB-321256

The EG2 conjecture for good configurations

摘录 / EXCERPT: Let $p\in[\frac12,1]$ and $\varepsilon\in[0,1)$ be the parameters of the random walk, and let $\mathbb{B}$ be the set of bad configurations. A configuration is good when it is not...

conjecture | open | MathDB 页面 | 原文线索

V2-054448 | MATHDB-321272

The singularity probability conjecture for random sign matrices

摘录 / EXCERPT: Let A be chosen uniformly from the set of $n\times n$ matrices with entries in $\{-1,+1\}$. Equivalently, $A\in\{-1,+1\}^{[n]^2}$, and singularity means $\det(A)=0$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-054449 | MATHDB-321385

Signed-measure converse conjecture for stationary distributions of diffusions

摘录 / EXCERPT: Stronger converse conjecture. Then π is a nonnegative measure and consequently it is a stationary distribution of X.

conjecture | open | MathDB 页面 | 原文线索

V2-054450 | MATHDB-321406

Zero-distribution conjecture for the regularized two-dimensional stochastic Allen-Cahn equation

摘录 / EXCERPT: Let u_N be the regularizations of the two-dimensional stochastic Allen-Cahn equation, and let $\langle\cdot,\cdot\rangle$ denote the L^2-duality pairing. For every $t>0$ and ev...

conjecture | open | MathDB 页面 | 原文线索

V2-054451 | MATHDB-321409

Small-time percolation conjecture for perturbed hexagonal packing

摘录 / EXCERPT: Let Π_0 be the hexagonal circle-packing point process, let Π_t be obtained by moving its nodes independently according to Brownian motion for time t, and let $R(\Pi_t)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-054452 | MATHDB-321410

Nonpercolation conjecture for Brownian-perturbed square packing

摘录 / EXCERPT: Let Π_0^s be the vertices of the square lattice with side length 1, and let Π_ϵ^s be obtained by moving those nodes independently according to Br...

conjecture | open | MathDB 页面 | 原文线索

V2-054453 | MATHDB-321412

Critical-radius ordering conjecture for transitive, Gaussian-analytic, and Poisson processes

摘录 / EXCERPT: Let Π_L be any transitive point process of intensity 1, let Π_GAF be the zero set of a Gaussian analytic function with intensity 1, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054454 | MATHDB-321508

Optimal input process achieves capacity up to fourth order for the deletion channel

摘录 / EXCERPT: Let d be the deletion probability, let C denote the capacity of the deletion channel, and let $\mathbb{X}^\dagger$ denote the input process constructed in the paper. The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-054455 | MATHDB-321582

Novel-partition approximation conjecture for Bernoulli singularity

摘录 / EXCERPT: Let $M=M_n$ be a random $n\times n$ Bernoulli matrix, let S be the event that M is singular, and let $\lambda(1),\lambda(2),\ldots$ be any enumeration of the set of novel integ...

conjecture | open | MathDB 页面 | 原文线索

V2-054456 | MATHDB-321635

Long-power trace bound conjecture for Artin-Schreier L-functions

摘录 / EXCERPT: Assume $(d,p)=1$. Let Ψ be the set of nontrivial additive characters of $\mathbb{F}_p$, let $\mathcal{F}_d$ be the relevant family, and write ... for the r-th power sum of the...

conjecture | open | MathDB 页面 | 原文线索

V2-054457 | MATHDB-321636

Odd Artin-Schreier symplectic trace conjecture

摘录 / EXCERPT: Odd Artin-Schreier symplectic trace conjecture. There exists a constant $\delta>0$ such that, for every $\varepsilon>0$,

conjecture | solved | MathDB 页面 | 原文线索

V2-054458 | MATHDB-321661

Matching upper-bound conjecture for the Bohman-Frieze process near criticality

摘录 / EXCERPT: Let $C_1(t)$ denote the largest connected component of the Bohman-Frieze process at time t, and let t_c be its critical time. Here $\frac{1}{n}\frac{d}{dt}$ is measured in the...

conjecture | open | MathDB 页面 | 原文线索

V2-054459 | MATHDB-321697

The planar loop-erased random walk neighbor probability conjecture

摘录 / EXCERPT: Let $x \in \mathbb{Z}^2$ be a neighbor of the origin, and let p_N be the probability that x lies on the loop-erased random walk from o to ∂G_N. Loop-erased random w...

conjecture | open | MathDB 页面 | 原文线索

V2-054460 | MATHDB-321699

Monotonicity conjecture for separability of random induced states

摘录 / EXCERPT: Let $d\geqslant 2$ and let $\pi_{d,s}$ denote the probability that a random state on $\mathbf{C}^d\otimes\mathbf{C}^d$ with distribution $\mu_{d^2,s}$ is separable. Monotonicity co...

conjecture | open | MathDB 页面 | 原文线索

V2-054461 | MATHDB-321805

Conjecture on the diffusion and analytic structure of the Strichartz hexacarpet

摘录 / EXCERPT: Hexacarpet diffusion conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-054462 | MATHDB-321859

Samuels's conjecture on sums of independent nonnegative random variables

摘录 / EXCERPT: Samuels's conjecture. For all admissible values of $\mu_1,\ldots,\mu_l$,

conjecture | open | MathDB 页面 | 原文线索

V2-054463 | MATHDB-321906

The WMS divergence conjecture for regular LDPC codes

摘录 / EXCERPT: Let (d_v,d_c)-regular LDPC codes have girth $\Omega\left(\log n\right)$, and transmit them over a binary symmetric channel with cross-over probability p. Let p^* be the bit-e...

conjecture | open | MathDB 页面 | 原文线索

V2-054464 | MATHDB-321968

Gurvits's doubly stochastic permanent inequality conjecture

摘录 / EXCERPT: Let ϕ be a doubly stochastic $n\times n$ matrix, so that its entries are non-negative and every row and column sums to 1. The permanent is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-054465 | MATHDB-321970

The asymptotic-direction conjecture for minimizing geodesics in Euclidean first-passage percolation

摘录 / EXCERPT: Let $B^{\mathrm{E}}_n=B^{\mathrm{E}}(0,n)$ be the Euclidean ball of radius n, let $\mathcal V_n\subseteq S^{d-1}$ be the set of initial velocities of minimizing unit-speed geodes...

conjecture | open | MathDB 页面 | 原文线索

V2-054466 | MATHDB-321974

Noise sensitivity of the Poisson Boolean model with random radii

摘录 / EXCERPT: Random-radius noise-sensitivity conjecture. For every $R>0$ and μ_R, the Poisson Boolean model with random radii chosen according to μ_R is noise sensitive at criticality.

conjecture | open | MathDB 页面 | 原文线索

V2-054467 | MATHDB-321975

Noise sensitivity of Poisson Boolean models with bounded simply connected shapes

摘录 / EXCERPT: Shape-noise-sensitivity conjecture. If S is bounded and simply connected, then the Poisson Boolean model for S is noise sensitive at criticality.

conjecture | open | MathDB 页面 | 原文线索

V2-054468 | MATHDB-321976

Noise sensitivity of Voronoi percolation

摘录 / EXCERPT: Voronoi noise-sensitivity conjecture. Voronoi percolation is noise sensitive at criticality.

conjecture | open | MathDB 页面 | 原文线索

V2-054469 | MATHDB-322046

Existence and universality of arm exponents for self-dual bond percolation

摘录 / EXCERPT: Arm-exponent universality conjecture. There exists $\rho=\rho(\sigma,\mathbf{p})>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054470 | MATHDB-322070

The strong acyclicity characterization of initial segment complexes

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\Delta\subseteq 2^{[n]}$ be a simplicial complex. Define the Winder desirability relation $\leq_W$ on $2^{[n]}$ by $A\leq_W B$ if and only if, for...

conjecture | open | MathDB 页面 | 原文线索

V2-054471 | MATHDB-322071

The extension conjecture for the Winder existential ordering

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let $\Delta\subseteq2^{[n]}$ be a strongly acyclic simplicial complex. Define the Winder desirability relation $\leq_W$ on $2^{[n]}$ by $A\leq_W B$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-054472 | MATHDB-322119

Strict separation of critical intensities for mobile-node detection

摘录 / EXCERPT: Let d be the spatial dimension, and let λ_c denote the critical intensity for the relevant space-time percolation transition and λ_t the critica...

conjecture | open | MathDB 页面 | 原文线索

V2-054473 | MATHDB-322120

Equality of past-information and ordinary detection thresholds

摘录 / EXCERPT: Let d be the spatial dimension. Let λ_c be the critical intensity for the relevant space-time percolation transition, and let λ_past be the smal...

conjecture | open | MathDB 页面 | 原文线索

V2-054474 | MATHDB-322143

Equally spaced drifts maximize the speed of one-dimensional random walks

摘录 / EXCERPT: Consider random walks in environments on ${\mathbb{Z}}$ composed of two types, $(p,1-p)$ and $(q,1-q)$, where $\frac{1}{2}<p<q<1$. Equally spaced drifts conjecture. An environment...

conjecture | open | MathDB 页面 | 原文线索

V2-054475 | MATHDB-322159

Nienhuis's BKT conjecture for O(n) models

摘录 / EXCERPT: Consider the $O(n)$ model on the hexagonal lattice with loop-weight $n\in(0,2)$ and edge-weight $x>0$, and define ... Nienhuis's BKT conjecture. For $x<x_c(n)$, the probability that...

conjecture | open | MathDB 页面 | 原文线索

V2-054476 | MATHDB-322160

The simplex conjecture for covering shadows

摘录 / EXCERPT: Let $K,L\in\mathcal{K}_n$ be convex bodies, and for a unit vector $u\in\mathbb{R}^n$ let K_u and L_u denote their projections onto the hyperplane orthogonal to u. Suppose tha...

conjecture | open | MathDB 页面 | 原文线索

V2-054477 | MATHDB-322161

Iwaniec-Luo-Sarnak Zero Density Conjecture for modular and quadratic-character families

摘录 / EXCERPT: Let $\phi(x)$ be a Schwartz-class function on $\mathbb{R}$ whose Fourier transform has compact support. Define ... For $f\in H_k$, let $\Lambda(s,f)$ be its completed L-function, a...

conjecture | open | MathDB 页面 | 原文线索

V2-054478 | MATHDB-322190

Infinite-dimensional sharpness conjecture for Orthomin(k)

摘录 / EXCERPT: Let $\mathbb{S}^1$ be the unit circle with Haar probability measure $d\mu_0$, and let $U:L^2(\mathbb{S}^1,d\mu_0)\to L^2(\mathbb{S}^1,d\mu_0)$ be the multiplication operator ... Cons...

conjecture | open | MathDB 页面 | 原文线索

V2-054479 | MATHDB-322277

Almost-sure persistence under mutual invasibility

摘录 / EXCERPT: Consider the two-species Markov chain ... is almost surely persistent. Mutual invasibility is known to imply persistence in probability under the monotonicity assumptions; the conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-054480 | MATHDB-322278

Persistence in probability for general multispecies systems

摘录 / EXCERPT: Let ... is persistent in probability. The cited theorem proves almost-sure persistence when every invariant boundary measure has a species with positive invasion rate, equivalently u...

conjecture | open | MathDB 页面 | 原文线索

V2-054481 | MATHDB-322279

Persistence for structured population models without monotonicity

摘录 / EXCERPT: Consider the structured stochastic difference equation ... is almost surely persistent and persistent in probability. A theorem establishes this conclusion under the stronger monoton...

conjecture | open | MathDB 页面 | 原文线索

V2-054482 | MATHDB-322285

Critical branching-rate conjecture for coexistence in the symbiotic branching model

摘录 / EXCERPT: Let $d\geq 3$ and let $\varrho>0$. In the two-type symbiotic branching model $\textrm{SBM}_\gamma(\varrho)$ on $\mathbb{Z}^d$, coexistence means that there are summable initial con...

conjecture | open | MathDB 页面 | 原文线索

V2-054483 | MATHDB-322312

Refined approximation conjecture for the rescaled momentum

摘录 / EXCERPT: Refined approximation conjecture. The law of $\lambda^{\frac12}P_{\cdot/\lambda}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054484 | MATHDB-322316

Asymptotic constant and monotonicity conjecture for two-dimensional anisotropic bootstrap percolation

摘录 / EXCERPT: Asymptotic constant conjecture. If $a<b$, then there is a constant $C_{a,b}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054485 | MATHDB-322317

Sharp asymptotics and monotonicity conjecture for three-dimensional anisotropic bootstrap percolation

摘录 / EXCERPT: Three-dimensional threshold conjecture. There is a constant $C_{a,b,c}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054486 | MATHDB-322331

The integrability conjecture for quadratic harnesses

摘录 / EXCERPT: Let $(X_t)_{t\in(1-\delta,1+\delta)}$ be a quadratic harness on $(1-\delta,1+\delta)$ for some $\delta>0$, with parameters $(\eta,\theta,\sigma,\tau,\gamma)$. For a fixed t in th...

conjecture | open | MathDB 页面 | 原文线索

V2-054487 | MATHDB-322337

The universal critical-slope conjecture for copolymers near a selective interface

摘录 / EXCERPT: Let $h_c^\mathrm{que}(\beta)$ be the quenched critical curve and let ... when this limit exists, for a copolymer model whose renewal exponent is $\alpha>1$. Critical-slope conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-054488 | MATHDB-322365

Random-cluster critical-point conjecture for the discontinuity set

摘录 / EXCERPT: Let $d\geq 2$ and $q\geq 1$. For the random-cluster model on $\mathbb{L}^d$, let $\phi_{p,q}^0$ and $\phi_{p,q}^1$ be the infinite-volume random-cluster measures with free and wire...

conjecture | open | MathDB 页面 | 原文线索

V2-054489 | MATHDB-322433

The central limit theorem under finite second moments

摘录 / EXCERPT: Finite-second-moment conjecture. Theorem $$ remains true under the weaker assumption that only the second moments are finite.

conjecture | open | MathDB 页面 | 原文线索

V2-054490 | MATHDB-322434

Examples with a central limit theorem but infinite second moments

摘录 / EXCERPT: Infinite-second-moment examples conjecture. There exist random walks on discrete point processes that satisfy a Central Limit Theorem but do not have all of their second moments fi...

conjecture | open | MathDB 页面 | 原文线索

V2-054491 | MATHDB-322435

Invariance principle for random walks on discrete point processes

摘录 / EXCERPT: Brownian-motion conjecture. Under assumptions ... , Theorem $$ can be strengthened so that the random walk on a discrete point process, under appropriate scaling, converges to Browni...

conjecture | open | MathDB 页面 | 原文线索

V2-054492 | MATHDB-322437

Conjecture on the limiting Cheeger constant of the supercritical percolation cluster

摘录 / EXCERPT: Limiting Cheeger constant conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-054493 | MATHDB-322475

Pansu-Grigorchuk gap conjectures for heat kernels, Følner functions and spectral density

摘录 / EXCERPT: Let G be an amenable finitely generated group, let μ be a symmetric finitely supported probability measure generating G, let M be the associated Markov operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054494 | MATHDB-322476

Pansu-Grigorchuk gap conjectures with parameter beta

摘录 / EXCERPT: Let G, μ, $P(n)$, $F(n)$ and $\mathcal N(\lambda)$ be as in the corresponding gap conjectures. Pansu-Grigorchuk gap conjectures with parameter β. There exists a number...

conjecture | open | MathDB 页面 | 原文线索

V2-054495 | MATHDB-322570

Density classification conjecture for the GKL, Kari traffic, and majority-traffic rules

摘录 / EXCERPT: Let a cellular automaton or probabilistic cellular automaton evolve configurations in the binary state space on the one-dimensional lattice $\mathbb{Z}$. The GKL cellular automaton...

conjecture | open | MathDB 页面 | 原文线索

V2-054496 | MATHDB-322660

Komlós' singularity probability conjecture for random Bernoulli matrices

摘录 / EXCERPT: Let A_n be an n by n random matrix whose entries are independent Bernoulli random variables. Komlós' singularity probability conjecture. The probability that A_n is singula...

conjecture | open | MathDB 页面 | 原文线索

V2-054497 | MATHDB-322859

Strong reinforcement conjecture for simultaneous drawing urns

摘录 / EXCERPT: Strong reinforcement conjecture. If $s_{\infty}<\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054498 | MATHDB-322886

Beta-gamma characterization conjecture for sums of i.i.d. random variables

摘录 / EXCERPT: Let $n\geqslant2$ and $\textbf{Y}=(Y_1,Y_2,\ldots,Y_n)$ be a vector of i.i.d. random variables. Let G_a denote a gamma random variable with parameter a, and let $B_{a,b}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-054499 | MATHDB-322890

The stochastic-process interpretation conjecture for gradient-flow structures

摘录 / EXCERPT: The diffusion equation can be viewed as a gradient flow in several distinct ways, including as the L^2-gradient flow of the Dirichlet integral, as the H^{-1}-gradient flow of t...

conjecture | open | MathDB 页面 | 原文线索

V2-054500 | MATHDB-322956

General sampling formula conjecture for Lambda-coalescents

摘录 / EXCERPT: Let Λ be a finite measure on $[0,1]$ with $\Lambda(\{1\})=0$, and let $\psi(q)=\int_{[0,1]}(e^{-qx}-1+qx)x^{-2}\Lambda(dx)$. For a sample of size n, let X_n denote eith...

conjecture | open | MathDB 页面 | 原文线索

V2-054501 | MATHDB-322987

The intersection property for binomial and Poisson distributions

摘录 / EXCERPT: Let M be a binomially distributed or Poisson-distributed random variable, let P_0 be its distribution, and let $G(M)$ be its G-transform, defined by assigning the signed squa...

conjecture | open | MathDB 页面 | 原文线索

V2-054502 | MATHDB-322988

The increasing property for binomial and Poisson distributions

摘录 / EXCERPT: Let M be a binomially or Poisson-distributed random variable with G-transform $G(M)$, and let Φ denote the standard Gaussian distribution function. The increasing property...

conjecture | open | MathDB 页面 | 原文线索

V2-054503 | MATHDB-322996

Decelle-Krzakala-Moore-Zdeborová parameter-estimation threshold conjecture

摘录 / EXCERPT: Let $\mathcal{G}(n,a/n,b/n)$ be the sparse planted partition model with two equal-sized clusters and parameters $a>b>0$. A consistent estimator of a and b is an estimator whose...

conjecture | open | MathDB 页面 | 原文线索

V2-054504 | MATHDB-322998

Decelle-Krzakala-Moore-Zdeborová Potts-model conjecture for k communities

摘录 / EXCERPT: Let $\mathcal{G}(n,a/n,b/n)$ be the planted partition model with k communities, where $a>b$. Let $c(k)$ be a threshold depending only on k. Decelle-Krzakala-Moore-Zdeborová's P...

conjecture | open | MathDB 页面 | 原文线索

V2-054505 | MATHDB-323006

Cantelli's conjecture on Gaussian mixtures

摘录 / EXCERPT: Cantelli's conjecture. The random variable $X+\varphi(X)\cdot Y$ has a Gaussian distribution if and only if φ is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-054506 | MATHDB-323077

The ag-ensemble probability bound

摘录 / EXCERPT: Let (Γ,ν) be the ag-ensemble, with parameters α, γ, and D, and let X^i_n denote the relevant random variable for the ensemble. The ag-ensemble probabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-054507 | MATHDB-323080

The $1/4$ subdiffusivity exponent conjecture for simple random walk on the UIPQ

摘录 / EXCERPT: Let Q_∞ be the uniform infinite planar quadrangulation (UIPQ), rooted at ρ, and let $(X_k)_{k\geq 0}$ be simple random walk on Q_∞ started at ρ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-054508 | MATHDB-323081

Recurrence conjecture for the uniform infinite planar quadrangulation

摘录 / EXCERPT: Let Q_∞ be the uniform infinite planar quadrangulation (UIPQ), rooted at ρ, and consider simple random walk on its vertices. A graph is recurrent when simple random wal...

conjecture | open | MathDB 页面 | 原文线索

V2-054509 | MATHDB-323083

Blanket-cover time conjecture for graphs

摘录 / EXCERPT: Let $G=(V,E)$) be a graph. For a random walk on G, let $\text{{\bf COV}}[G]$ denote its cover time and let $\text{{\bf BCOV}}[G]$ denote the blanket-cover time, the expected firs...

conjecture | open | MathDB 页面 | 原文线索

V2-054510 | MATHDB-323102

Gap conjectures for heat kernel, Følner function, isoperimetric profile and spectral density

摘录 / EXCERPT: Let G be a finitely generated amenable group, with the heat-kernel function $P(n)$, Følner function $\mathcal F(n)$, isoperimetric profile $\Lambda(n)$, and spectral density...

conjecture | open | MathDB 页面 | 原文线索

V2-054511 | MATHDB-323103

The weak gap conjecture for HK, Følner, isoperimetric and spectral-density types

摘录 / EXCERPT: Let G be a finitely generated amenable group, and let $P(n)$, $\mathcal F(n)$, $\Lambda(n)$ and $\mathcal N(\lambda)$ denote respectively the heat-kernel, Følner, isoperimetric-p...

conjecture | open | MathDB 页面 | 原文线索

V2-054512 | MATHDB-323141

Quantitative strong log-concavity conjecture for Chernoff's density

摘录 / EXCERPT: Let Z_1 be a standard Chernoff random variable with density f_{Z_1}. Let $\varphi(x)=(2\uppi)^{-1/2}\exp(-x^2/2)$ be the standard normal density. Quantitative strong log-concav...

conjecture | open | MathDB 页面 | 原文线索

V2-054513 | MATHDB-323203

Transience conjecture for self-interacting random walks in dimension at least four

摘录 / EXCERPT: Let $\mu_1,\ldots,\mu_{d-1}$ be d-dimensional measures in $\mathbb{R}^d$, with $d\geq 4$, zero mean, and finite $(2+\beta)$-moments for some $\beta>0$, and let ℓ be an arbit...

conjecture | open | MathDB 页面 | 原文线索

V2-054514 | MATHDB-323233

Uniqueness conjecture for k-symmetric multiplicatively stable distributions

摘录 / EXCERPT: Uniqueness conjecture. The measures σ^k_β are the only k-symmetric $\boxtimes$-stable distributions.

conjecture | open | MathDB 页面 | 原文线索

V2-054515 | MATHDB-323234

Domain-of-attraction characterization of free stable laws

摘录 / EXCERPT: Domain-of-attraction conjecture. Assume that $\mu\in\mathcal{M}_k$ is not a point mass. Then ν is $\boxplus$-stable if and only if the free domain of attraction of ν is not...

conjecture | open | MathDB 页面 | 原文线索

V2-054516 | MATHDB-323280

The persistence exponent conjecture for integrated stable Lévy processes

摘录 / EXCERPT: Let Z be a strictly α-stable Lévy process, let $A_t=\int_0^tZ_s\,\mathrm{d}s$, and define $T_1=\inf\{t\ge0:A_t>1\}$. The stable Lévy persistence conjecture. If ... then ther...

conjecture | open | MathDB 页面 | 原文线索

V2-054517 | MATHDB-323281

The persistence exponent conjecture for general integrated Lévy processes

摘录 / EXCERPT: Let Z be a Lévy process, let $A_t=\int_0^tZ_s\,\mathrm{d}s$, and for $x>0$ define $T_x=\inf\{t\ge0:A_t>x\}$. The general Lévy persistence conjecture. If ... then for every $x>0$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-054518 | MATHDB-323282

The persistence asymptotic conjecture for fractional Brownian motion

摘录 / EXCERPT: Let B^H be fractional Brownian motion with Hurst parameter $H\in(0,1)$, let $A_t^H=\int_0^tB_s^H\,\mathrm{d}s$, and define $T_1^H=\inf\{t\ge0:A_t^H>1\}$. The fractional Brownian...

conjecture | open | MathDB 页面 | 原文线索

V2-054519 | MATHDB-323283

Convexity of the persistence exponent for fractionally integrated Lévy processes

摘录 / EXCERPT: For each $\beta\ge0$, let $\theta(\beta)$ denote the persistence exponent of a β-fractionally integrated Lévy process, when defined. The convexity conjecture. The function .....

conjecture | open | MathDB 页面 | 原文线索

V2-054520 | MATHDB-323284

Universality of persistence exponents for fractionally integrated Lévy processes

摘录 / EXCERPT: For each $\beta\ge0$, consider a β-fractionally integrated Lévy process and let $\theta(\beta)$ be its persistence exponent. The Lévy universality conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-054521 | MATHDB-323285

Khokhlov-Molchan conjecture for integrated fractional Brownian motion

摘录 / EXCERPT: Khokhlov-Molchan conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-054522 | MATHDB-323286

Universality of weighted-sum persistence for finite-variance increments

摘录 / EXCERPT: Weighted-sum persistence conjecture. If μ has finite variance and $\sigma_n\asymp n^p$ for some $p>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054523 | MATHDB-323288

Caravenna-Deuschel conjecture for integrated random bridges

摘录 / EXCERPT: Let (X_i) be centered i.i.d. increments with law μ and finite variance, let S_n and A_n be the associated random walk and integrated walk, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-054524 | MATHDB-323305

Universality conjecture for arm-event exponents on isoradial graphs

摘录 / EXCERPT: Let G be an isoradial graph with the bounded-angles property, let $k\in\mathbb{N}$, and let $\sigma\in\{0,1\}^k$. For a translated annulus $\mathcal{A}^u(N,n)$ centred at u,...

conjecture | open | MathDB 页面 | 原文线索

V2-054525 | MATHDB-323330

Second-order strong convergence for the bounded-domain SPDE discretisation

摘录 / EXCERPT: Let $U_T=u(T,\cdot)$ be the solution of the initial-boundary value problem at time T for a given Brownian path, and let $\widehat{U}_T$ be its numerical approximation with spatia...

conjecture | open | MathDB 页面 | 原文线索

V2-054526 | MATHDB-323375

Dyson's bulk gap probability asymptotic conjecture

摘录 / EXCERPT: Let $E_\beta^{\rm bulk}(0;(0,s))$ denote the probability that an interval of length s in the bulk contains no eigenvalues for the beta ensemble. Dyson's conjecture. ... This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-054527 | MATHDB-323376

Chen-Manning hard-edge gap probability asymptotic conjecture

摘录 / EXCERPT: Let $E_\beta^{\rm hard}(0;(0,s);a)$ denote the hard-edge probability that $(0,s)$ contains no eigenvalues, with hard-edge parameter a. Chen-Manning's conjecture. ... This predictio...

conjecture | open | MathDB 页面 | 原文线索

V2-054528 | MATHDB-323377

Dyson-Fogler-Shklovskii conditioned bulk gap asymptotic conjecture

摘录 / EXCERPT: Let $E_\beta^{\rm bulk}(n;(0,s))$ denote the probability that $(0,s)$ contains exactly n eigenvalues. Dyson-Fogler-Shklovskii's conjecture. For $0\ll n\ll s$, ... The authors furth...

conjecture | open | MathDB 页面 | 原文线索

V2-054529 | MATHDB-323378

Forrester-Witte conditioned hard- and soft-edge gap asymptotic conjecture

摘录 / EXCERPT: Let $E_\beta^{\rm hard}(n;(0,s);\beta a/2)$ and $E_\beta^{\rm soft}(n;(s,\infty))$ denote the probabilities of exactly n eigenvalues in the indicated hard- and soft-edge interval...

conjecture | open | MathDB 页面 | 原文线索

V2-054530 | MATHDB-323379

Gaussian number-fluctuation conjecture for large intervals

摘录 / EXCERPT: Let $E_\beta^{(\cdot)}(n;J)$ denote the probability of exactly n particles in J, let n_J be the corresponding particle number, and let $\langle n_J\rangle$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054531 | MATHDB-323380

Forrester's hard-edge conditioned gap constant conjecture

摘录 / EXCERPT: Forrester's conjecture. As $s\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-054532 | MATHDB-323381

Diluted-spectrum gap probability asymptotic conjecture

摘录 / EXCERPT: Let $E_\beta^{(\cdot)}(J;\xi)$ be the generating function for the probability that no retained eigenvalues lie in J after independent deletion with probability $1-\xi$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054533 | MATHDB-323395

The subcritical-family conjecture for two-dimensional bootstrap percolation

摘录 / EXCERPT: Let $\mathcal{U}$ be a two-dimensional bootstrap percolation update family, with closure process $[A]$ started from the random initially infected set A under the product measure...

conjecture | open | MathDB 页面 | 原文线索

V2-054534 | MATHDB-323486

Worst-case $T^{1/3}$ correction conjecture for time-inhomogeneous branching Brownian motion

摘录 / EXCERPT: For one-dimensional branching Brownian motion with a smooth diffusivity profile $\sigma:[0,1]\to(0,\infty)$ whose image is contained in a compact subset of $(0,\infty)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-054535 | MATHDB-323536

Almost-sure local-global convergence of growing random regular graphs

摘录 / EXCERPT: Local-global convergence conjecture. A growing sequence of random d-regular graphs is local-global convergent with probability one.

conjecture | open | MathDB 页面 | 原文线索

V2-054536 | MATHDB-323540

Classification of recurrent groups for finite-alphabet group-extended Markov systems

摘录 / EXCERPT: Let $\left(\Sigma\times G,\sigma\rtimes\Psi\right)$ be an irreducible group-extended Markov system with finite alphabet I, and let $\varphi:\Sigma\rightarrow\mathbb{R}$ be a Höld...

conjecture | open | MathDB 页面 | 原文线索

V2-054537 | MATHDB-323584

Dirichlet distribution conjecture for asymptotic sector widths of the radial spanning tree

摘录 / EXCERPT: Let m be the number of unbounded trees, and condition on $m\in\{2,3,4,5\}$. For cyclically labeled interfaces, define the sector widths ... with the conventions...

conjecture | open | MathDB 页面 | 原文线索

V2-054538 | MATHDB-323659

Semi-general asymptotic formula for fixed-gain amplify-and-forward multihop outage probability

摘录 / EXCERPT: Semi-general asymptotic formula. The outage probability satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054539 | MATHDB-323675

Universality of random kernel matrix spectra for isotropic log-concave distributions

摘录 / EXCERPT: Let X_i be iid random vectors from a normalized isotropic log-concave distribution. Universality conjecture. The results of Theorem hold for these random vectors. This conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-054540 | MATHDB-323691

The shape fluctuation exponent conjecture for random Riemannian metrics

摘录 / EXCERPT: Let $B_g(t)$ be the random metric ball at time t, and let $B(t/\mu)$ be the corresponding rescaled limiting shape, where μ is the time constant. The fluctuations of these two...

conjecture | open | MathDB 页面 | 原文线索

V2-054541 | MATHDB-323765

The second-order asymptotic conjecture for ring-graph jigsaw percolation

摘录 / EXCERPT: Let $p_c(n)$ denote the critical probability for jigsaw percolation on the ring puzzle graph with an Erdős-Rényi people graph. Second-order asymptotic conjecture. There exist const...

conjecture | open | MathDB 页面 | 原文线索

V2-054542 | MATHDB-323766

The critical-constant conjecture for ring-graph jigsaw percolation

摘录 / EXCERPT: Let $p_c(n)$ denote the critical probability for jigsaw percolation on the ring puzzle graph with an Erdős-Rényi people graph. Critical-constant conjecture. The leading constant in...

conjecture | open | MathDB 页面 | 原文线索

V2-054543 | MATHDB-323767

The sharp-threshold conjecture for ring-graph jigsaw percolation

摘录 / EXCERPT: For the ring puzzle, let $p_{\epsilon}(n)$ be the unique value of p such that the probability that the jigsaw process solves the puzzle is $f0a0\epsilon$, for fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-054544 | MATHDB-323811

Trapping-set criterion for high-probability correction by decoder diversity

摘录 / EXCERPT: Let $\mathcal{C}$ be an LDPC code, let $\mathcal{D}$ be a decoder diversity set, and let $\mathcal{E}^{[t]}(\Lambda^{(A,B)})$ be the union of the error sets associated with all tra...

conjecture | open | MathDB 页面 | 原文线索

V2-054545 | MATHDB-323929

General large deviations principle for stationary reflected Brownian motion in the octant

摘录 / EXCERPT: General large deviations principle. For every measurable $A\subset\mathbb{R}^d_+$,

conjecture | open | MathDB 页面 | 原文线索

V2-054546 | MATHDB-323937

Forrester-Frankel hard-edge gap-ratio asymptotic conjecture

摘录 / EXCERPT: Hard-edge gap-ratio asymptotic conjecture. As $s\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-054547 | MATHDB-323938

Large-argument asymptotic conjecture for a multivariate hypergeometric function

摘录 / EXCERPT: Large-argument hypergeometric asymptotic conjecture. For large values of the arguments $\{s_j\}_{j=0,\dots,n}$, the generalized hypergeometric function has the asymptotic form give...

conjecture | open | MathDB 页面 | 原文线索

V2-054548 | MATHDB-323982

Schramm's FK interface convergence conjecture

摘录 / EXCERPT: Fix $q\in[0,4]$. A critical FK interface is the exploration interface in a Dobrushin domain for FK percolation with cluster-weight q. Schramm's conjecture. The law of critical FK...

conjecture | open | MathDB 页面 | 原文线索

V2-054549 | MATHDB-324135

Conjectures on the moment polynomials for Brownian motion with parabolic drift

摘录 / EXCERPT: Let V denote the location of the maximum of Brownian motion with parabolic drift, and for each nonnegative integer n write its moment in the form ... where ... for even n, with...

conjecture | open | MathDB 页面 | 原文线索

V2-054550 | MATHDB-324151

Closure of the set of Markov stopping-time measures under convolution and convex combination

摘录 / EXCERPT: Let $\mathfrak{P}$ denote the set of measures considered in the paper that arise directly or indirectly, after extension, from a Markov stopping time. Closure conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-054551 | MATHDB-324152

Realization of every measure in \mathfrak{P} by the cited theorem

摘录 / EXCERPT: Let $\mathfrak{P}$ be the set of measures considered in the paper that arise directly or indirectly, after extension, from a Markov stopping time, and let Theorem be the cited cons...

conjecture | open | MathDB 页面 | 原文线索

V2-054552 | MATHDB-324192

Geometric limit conjecture for the largest family sizes in the critical case

摘录 / EXCERPT: Geometric limit conjecture. As $t\to\infty$, on the survival event, $L_t(x_t)$ converges in distribution to a non-degenerate geometric random variable. This conjecture concerns the...

conjecture | open | MathDB 页面 | 原文线索

V2-054553 | MATHDB-324216

Complete convergence conjecture for voter model perturbations

摘录 / EXCERPT: Let $\alpha=(\alpha_0,\alpha_1)$ be in the coexistence region of Theorem 1.10 of CDP11, with each $\alpha_i<1$ and sufficiently close to 1. Let $\nu_{1/2}$ denote the nontrivial...

conjecture | open | MathDB 页面 | 原文线索

V2-054554 | MATHDB-324254

Equality of the bistochastic and correlation-matrix parameters

摘录 / EXCERPT: Let λ_n be the smallest nonnegative number such that ... and let ρ_n be the smallest nonnegative number such that ... Here $\mathcal{B}_n^{sym}$ is the set of n by n s...

conjecture | open | MathDB 页面 | 原文线索

V2-054555 | MATHDB-324299

Criticality conjecture for multi-particle reinforced interacting random walks

摘录 / EXCERPT: Let D be the state space of pairs of occupation measures, let $x(n),y(n)$ denote the two occupation-measure processes, let $z(n)=(x(n),y(n))$, and let U denote the uniform occu...

conjecture | open | MathDB 页面 | 原文线索

V2-054556 | MATHDB-324374

Vázquez's conjecture on the expected waiting time in the priority model

摘录 / EXCERPT: Let $L>2$ be the list length, let $0\leq p\leq 1$ be the probability of selecting the highest-priority task, and let τ denote a task's waiting time in stationarity. Vázquez's...

conjecture | open | MathDB 页面 | 原文线索

V2-054557 | MATHDB-324448

Diagonal dominance conjecture for squared components of a truncated normal vector

摘录 / EXCERPT: Let $X\sim\mathcal{N}_{v}(0,\Lambda)$, where $\Lambda=\operatorname{diag}(\lambda)$, and let ... be the Euclidean ball of radius ρ. Consider the covariance matrix of the vector...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054558 | MATHDB-324463

Variance bound for downcrossings in feasible martingale processes

摘录 / EXCERPT: Downcrossing variance conjecture. For any $\boldsymbol{0}$-feasible process,

conjecture | open | MathDB 页面 | 原文线索

V2-054559 | MATHDB-324497

Finite and macroscopic loop decomposition conjecture

摘录 / EXCERPT: Let $(\Lambda,\mathcal E)$ be a finite periodic box, let $\beta>0$, and let $L_1(\omega),L_2(\omega),\ldots$ be the loop lengths in decreasing order. The normalized loop lengths ar...

conjecture | open | MathDB 页面 | 原文线索

V2-054560 | MATHDB-324549

Universality conjecture for the Meijer random field in Cauchy matrix models

摘录 / EXCERPT: Consider Cauchy two-matrix models with positive-definite matrices M_1,M_2 and measure ... where V_i is analytic near the origin, and use the scaling $x\mapsto x n^{-3}$. Universa...

conjecture | open | MathDB 页面 | 原文线索

V2-054561 | MATHDB-324550

The sharp threshold conjecture for rank deficiency in sparse random GF[2] matrices

摘录 / EXCERPT: Let W_n be the row-weight random variables, assume that the sequence (W_n) is uniformly bounded, and suppose ... for every n. Let T_n denote the rank deficiency and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054562 | MATHDB-324559

The Unification Conjecture for mutual free information and free entropy

摘录 / EXCERPT: Unification Conjecture. The mutual free information satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054563 | MATHDB-324560

The projection Unification Conjecture for mutual free information

摘录 / EXCERPT: Projection Unification Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-054564 | MATHDB-324564

Conjecture on generating a lattice with n+1 random vectors

摘录 / EXCERPT: Lattice-generation conjecture. For every $n\in\mathbb{N}$, there exists a constant $0<c_n<1$ and a rational function $f_n\in\mathbb{R}(x,y)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-054565 | MATHDB-324567

A universal upper bound for pair discrepancy

摘录 / EXCERPT: Let $D({\bf p})$ denote the total variation discrepancy between the color distributions of a pair selected by the two matching procedures, and define ... where the supremum ranges ov...

conjecture | open | MathDB 页面 | 原文线索

V2-054566 | MATHDB-324571

Equal-distribution pair discrepancy bound

摘录 / EXCERPT: Let $D({\bf p},{\bf q})$ denote the discrepancy between the color distributions of the matching left-right pair when the left and right shoes have distributions ${\bf p}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054567 | MATHDB-324572

Asymptotic extremum for equal-distribution pair discrepancy

摘录 / EXCERPT: Let $D({\bf p},{\bf q})$ be the discrepancy for matching left-right pairs, and let $\ell(a,b)$ denote the limiting discrepancy for ${\bf p}={\bf p}(n,a/\sqrt n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054568 | MATHDB-324582

Macroscopic-loop support conjecture for the zero-curvature loop measure

摘录 / EXCERPT: Let ${\mathbb P}_{LC^0}$ be the limiting loop measure associated with the zero-curvature scaling, and let Ω_1 denote the event that the loop has nonzero area. Macroscopic-lo...

conjecture | open | MathDB 页面 | 原文线索

V2-054569 | MATHDB-324625

Conjecture for moments of second derivatives of quadratic Dirichlet L-functions

摘录 / EXCERPT: Let $\mathcal{D}(X):=\{d: |d|<X,\ d\text{ is a fundamental discriminant}\}$, and let $L(s,\chi_d)$ denote the quadratic Dirichlet L-function associated with the fundamental discr...

conjecture | open | MathDB 页面 | 原文线索

V2-054570 | MATHDB-324663

Classification conjecture for free Bogoljubov crossed products in the almost periodic case

摘录 / EXCERPT: Let $(\rho_{\mathbb R},H_{\mathbb R})$ be an infinite-dimensional, faithful, almost periodic orthogonal representation of ${\mathbb Z}$. Its complexification has a subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-054571 | MATHDB-324772

Conjecture on singular symmetric Bernoulli matrices with sharp exponential rate

摘录 / EXCERPT: Let M_n^{sym} be a random symmetric matrix whose upper-diagonal entries are iid Bernoulli variables, and let p_n^{sym} denote its probability of being singula...

conjecture | open | MathDB 页面 | 原文线索

V2-054572 | MATHDB-324826

Bounded-dimensional orthostochasticity conjecture

摘录 / EXCERPT: Let ${\mathcal B}_n$ denote the set of bistochastic $n\times n$ matrices, and let $d_{\min}(n)$ be the smallest value of d such that every matrix in ${\mathcal B}_n$ is d-ortho...

conjecture | open | MathDB 页面 | 原文线索

V2-054573 | MATHDB-324881

Convergence of planar loop-erased random walk to SLE_2

摘录 / EXCERPT: Let D be a bounded, simply connected domain containing the origin whose boundary is a Jordan curve, and let U,V be disjoint, closed subarcs of ∂D. For the lattice app...

conjecture | open | MathDB 页面 | 原文线索

V2-054574 | MATHDB-324896

Asymptotic Gaussianity conjecture for crossings in set partitions

摘录 / EXCERPT: Let X_n and Y_n be the random variables defined in the paper, representing the relevant crossing statistics on set partitions. Asymptotic Gaussianity conjecture. The distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-054575 | MATHDB-324926

Continuity conjecture for Kuksin measures

摘录 / EXCERPT: Continuity conjecture. Kuksin measures are in fact supported on continuous vorticities.

conjecture | open | MathDB 页面 | 原文线索

V2-054576 | MATHDB-324941

Bell-shape conjecture for positive self-decomposable densities

摘录 / EXCERPT: A positive random variable X is self-decomposable when its Laplace transform is ... where $k:(0,+\infty)\to\mathbb{R}^+$ is non-increasing, after taking $\gamma_0=0$. Let f_k be...

conjecture | open | MathDB 页面 | 原文线索

V2-054577 | MATHDB-325027

Measure-theoretic defect conjecture for Butson matrices

摘录 / EXCERPT: Measure-theoretic defect conjecture. For $H\in C_N(s)$, with s minimal,

conjecture | open | MathDB 页面 | 原文线索

V2-054578 | MATHDB-325028

Main conjecture on recovering the defect from its associated measure

摘录 / EXCERPT: Main conjecture. For $H\in C_N(s)$, with $s\in\mathbb N$ chosen to be minimal, $d(H)$ can be recaptured from the knowledge of the associated measure μ.

conjecture | open | MathDB 页面 | 原文线索

V2-054579 | MATHDB-325034

Almost-sure LD-convergence of Erdős-Rényi and random regular graph sequences

摘录 / EXCERPT: Let $\mathcal{G}_n$ be the space of simple graphs on n vertices, and equip $\prod_n\mathcal{G}_n$ with the product probability measure induced by independent $\mathbb{G}(n,c/n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-054580 | MATHDB-325091

Cubic-lattice conjecture on the critical threshold after deleting infinite clusters

摘录 / EXCERPT: Let ω_p be Bernoulli site percolation on $\mathbb{Z}^d$, and let $C^\infty(\omega_p)$ be the set of vertices belonging to infinite open clusters. For a graph obtained by del...

conjecture | open | MathDB 页面 | 原文线索

V2-054581 | MATHDB-325102

Formal-series correlation inequality for FKG probability measures

摘录 / EXCERPT: Let X be a finite set, let μ be an FKG probability measure on 2^X, and let $p(A)=p_1(A)t+p_2(A)t^2+\cdots$ be a formal series whose coefficients are nonnegative monotone fu...

conjecture | open | MathDB 页面 | 原文线索

V2-054582 | MATHDB-325235

The rationality phenomenon for submatching events

摘录 / EXCERPT: The rationality phenomenon for submatching events. For any $\pi_0\in\textnormal{NC}_{n}$, the probability of the associated submatching event is a dyadic rational number that is co...

conjecture | open | MathDB 页面 | 原文线索

V2-054583 | MATHDB-325236

The rationality phenomenon for finite submatching events

摘录 / EXCERPT: The rationality phenomenon for finite submatching events. For $n\ge k$, there is an even polynomial $Q_{\pi_0}(n)$ of degree $k(k+1)$ with dyadic rational coefficients such that

conjecture | open | MathDB 页面 | 原文线索

V2-054584 | MATHDB-325246

Conjecture on balances generated by strongly coherent betting systems

摘录 / EXCERPT: Balance-equivalence conjecture. Every balance generated by Π and P is a version of a balance generated by betting system 2 and P.

conjecture | open | MathDB 页面 | 原文线索

V2-054585 | MATHDB-325253

Selmer p^ℓ-rank distribution conjecture for all elliptic curves

摘录 / EXCERPT: Selmer-rank distribution conjecture. For a positive integer ℓ, $k\in\mathbb N$, and $\delta\in\{0,1\}$, the probability over all elliptic curves $E/\mathbb Q$, ordered by cond...

conjecture | open | MathDB 页面 | 原文线索

V2-054586 | MATHDB-325265

The three-dimensional Brownian-path principal eigenvalue asymptotic

摘录 / EXCERPT: Let $\beta[0,s]$ be the path of Brownian motion on the three-dimensional torus ${\mathbb T}^3$ up to time s, and let Δ_s be the Laplacian on...

conjecture | open | MathDB 页面 | 原文线索

V2-054587 | MATHDB-325320

Moment lower-bound conjecture for bootstrap percolation on Galton-Watson trees

摘录 / EXCERPT: Let T_ξ be a Galton-Watson tree with offspring distribution ξ. Moment lower-bound conjecture. For every $r\geq 2$ and $\alpha\in(0,r-1]$, there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-054588 | MATHDB-325329

Trust-region radius convergence conjecture for probabilistically fully quadratic models

摘录 / EXCERPT: Trust-region radius convergence conjecture. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-054589 | MATHDB-325383

The loop-erased random-walk up-to-constants passage estimate

摘录 / EXCERPT: Let $z\in\mathbb{D}$, let $\epsilon>0$ be sufficiently small, and let Z, W^w, ξ_x, and $\sigma_{\mathbb{D}}$ have the meanings used in the loop-erased-random-walk estimates...

conjecture | open | MathDB 页面 | 原文线索

V2-054590 | MATHDB-325445

Monomial Wald statistic distribution conjecture

摘录 / EXCERPT: Monomial distribution conjecture. The Wald variable satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054591 | MATHDB-325446

Stochastic lower-bound conjecture for quadratic Wald statistics

摘录 / EXCERPT: Let f be a nonzero quadratic form and let Σ be a nonzero positive semidefinite matrix. Write $W_{f,\Sigma}$ for the corresponding Wald random variable. Quadratic lower-bou...

conjecture | open | MathDB 页面 | 原文线索

V2-054592 | MATHDB-325447

Cauchy distribution conjecture for weighted ratios of correlated normals

摘录 / EXCERPT: Cauchy ratio conjecture. The random variable

conjecture | open | MathDB 页面 | 原文线索

V2-054593 | MATHDB-325448

Inverse-chi-square quadratic-form conjecture for weighted normal reciprocals

摘录 / EXCERPT: Inverse-chi-square conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-054594 | MATHDB-325467

The 4-free-path exponential bound conjecture

摘录 / EXCERPT: Let the exponential bound in Theorem be the bound denoted by, and interpret a diagonal path and a 4-free path as the corresponding path notions in the cellular automaton model. 4-f...

conjecture | open | MathDB 页面 | 原文线索

V2-054595 | MATHDB-325511

Conjecture on free infinite divisibility of the t-distribution

摘录 / EXCERPT: Let $\mathbf{t}_q$ denote the t-distribution with parameter q. A probability distribution is freely infinitely divisible (FID) if it is infinitely divisible with respect to free...

conjecture | open | MathDB 页面 | 原文线索

V2-054596 | MATHDB-325554

Monotonicity criterion for the interspecies correlation coefficient

摘录 / EXCERPT: Let $V^{(n)}_{a}(\alpha,0)$ and $V^{(n)}_{c}(\alpha)$ denote the anagenetic and cladogenetic contributions to the variance, and let $C^{(n)}_{a}(\alpha,0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054597 | MATHDB-325592

Decomposition conjecture for empirical-process suprema

摘录 / EXCERPT: Let $(X_i)_{i\leq N}$ be independent identically distributed random variables and let ${\cal F}$ be a countable class of measurable functions. Define ... and let $d_p(f,g)=\|f-g\|_p$...

conjecture | open | MathDB 页面 | 原文线索

V2-054598 | MATHDB-325662

Sublinear growth of Lipschitz harmonic functions with trivial Poisson boundary

摘录 / EXCERPT: Let G be an amenable group, let μ be a probability measure on G with finite second moment and trivial Poisson boundary, and let $u:G\to\mathbb R$ be a Lipschitz μ-harmo...

conjecture | open | MathDB 页面 | 原文线索

V2-054599 | MATHDB-325768

Universality of the stochastic Airy operator at nonregular spectral edges

摘录 / EXCERPT: Universality conjecture. The operator $H_{n,k}$ converges in the sense of Theorem (i) to

conjecture | open | MathDB 页面 | 原文线索

V2-054600 | MATHDB-325774

Sublinear zero-server conjecture for GRAND

摘录 / EXCERPT: Let $p\in(0,1)$. Consider a sequence of systems operating under the GRAND(Z^p) algorithm, with ... For each r, let $\boldsymbol{x}^r(\infty)$ denote the random state of the fluid...

conjecture | open | MathDB 页面 | 原文线索

V2-054601 | MATHDB-325775

Asymptotic optimality conjecture for GRAND(0)

摘录 / EXCERPT: Consider a sequence of systems operating under the GRAND(0) algorithm. For each r, let $\boldsymbol{x}^r(\infty)$ denote the random state of the fluid-scaled process in the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-054602 | MATHDB-325799

Uninformativeness of bulk singular vectors below the effective rank

摘录 / EXCERPT: Let X be an $m\times n$ noise matrix with singular values $\sigma_i(X)$, and let b be the right spectral edge of its limiting singular-value distribution with D-transform...

conjecture | open | MathDB 页面 | 原文线索

V2-054603 | MATHDB-325800

Complete delocalization of bulk singular vectors

摘录 / EXCERPT: Let X be an $m\times n$ noise matrix with $q=\min(m,n)$ and singular values $\sigma_i(X)$, and let b be the right edge of its limiting singular-value distribution with D-transf...

conjecture | open | MathDB 页面 | 原文线索

V2-054604 | MATHDB-325891

Monotonicity and critical-limit conjecture for the divide and color model

摘录 / EXCERPT: Let $\mathcal{L}\in\{\mathbb{Z}^2,\mathbb{H}\}$ be either the square or hexagonal lattice, let $p_c^{\mathcal{L}}$ denote its critical percolation parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054605 | MATHDB-326018

Finite expected number of convex-minorant segments for Vervaat bridges

摘录 / EXCERPT: Let $V(B^{\lambda,br})$ be a Vervaat bridge with parameter $\lambda<0$, and consider its convex minorant, whose segments are the linear pieces between consecutive vertices. Finite-...

conjecture | open | MathDB 页面 | 原文线索

V2-054606 | MATHDB-326100

Uniqueness conjecture for infinitely divisible solutions of the size-bias equation

摘录 / EXCERPT: Let $c>1$, and let U_c be the Choquet simplex of probability distributions solving the size-bias scaling relation associated with c. Write $c=\exp(\sigma^2/2)$, and let Z be...

conjecture | open | MathDB 页面 | 原文线索

V2-054607 | MATHDB-326112

The conjectural Fredholm determinant formula for the q-PushASEP q-Laplace transform

摘录 / EXCERPT: Let q satisfy $0<q<1$, let $x_n(t)$ denote the position of particle n at time t, and define ... For all $\zeta\in\mathbb{C}\setminus\mathbb{R}_{>0}$, the conjectural Fredholm d...

conjecture | open | MathDB 页面 | 原文线索

V2-054608 | MATHDB-326122

Existence of a fixed point for the stochastic Goodwin-Lotka-Volterra return map

摘录 / EXCERPT: Fixed-point conjecture. The map $\mathcal S$ has at least one fixed point in $(0,\tilde y)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054609 | MATHDB-326179

Monotonicity conjecture for average singular values of Gaussian matrices

摘录 / EXCERPT: Let $\alpha_{\mathbb{R}}(d)$ and $\alpha_{\mathbb{C}}(d)$ be the average singular value of a $d\times d$ matrix with random i.i.d. real-valued and complex-valued entries, respectiv...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054610 | MATHDB-326270

Erschler-Tóth-Werner localization conjecture for stuck walks

摘录 / EXCERPT: Let $\alpha>1/3$, define $\alpha_1:=+\infty$ and, for $L\ge2$, ... For the walk, let R' be the set of sites visited infinitely often, ... Erschler-Tóth-Werner's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-054611 | MATHDB-326280

Critical freezing-rate conjecture for percolation with constant freezing

摘录 / EXCERPT: Let $d\geq 2$ and consider the percolation with constant freezing (PCF) process on $\mathbb{Z}^d$, where edges open at rate 1 and clusters freeze at rate $\alpha>0$. Critical fr...

conjecture | open | MathDB 页面 | 原文线索

V2-054612 | MATHDB-326281

Multiple infinite clusters conjecture for percolation with constant freezing

摘录 / EXCERPT: Let $d\geq 3$ and consider the percolation with constant freezing (PCF) process on $\mathbb{Z}^d$, with edge-opening rate 1 and freezing rate $\alpha>0$. Multiple infinite clust...

conjecture | open | MathDB 页面 | 原文线索

V2-054613 | MATHDB-326303

Gaussian decay conjecture for stochastic routing probabilities

摘录 / EXCERPT: Gaussian decay conjecture. There exists a constant $\rho>0$ such that, for sufficiently small h,

conjecture | open | MathDB 页面 | 原文线索

V2-054614 | MATHDB-326369

Benford behavior for additive decompositions stopped at evens or primes

摘录 / EXCERPT: Consider an additive decomposition process starting from a stick of integer length L: choose uniformly an integer cut point $c\in[1,L-1]$, replace the stick by pieces of lengths...

conjecture | open | MathDB 页面 | 原文线索

V2-054615 | MATHDB-326370

Failure of Benford behavior for polynomial, exponential, and Fibonacci stopping sequences

摘录 / EXCERPT: Fix a monotonically increasing sequence $\{a_n\}$. In the additive decomposition process, a stick whose length belongs to $\{a_n\}$ does not decompose further; sticks of length 1...

conjecture | open | MathDB 页面 | 原文线索

V2-054616 | MATHDB-326400

Approximate Nash equilibrium from a mean-field optimal-control minimizer

摘录 / EXCERPT: Let $(m,m\mathbf v)$ be a minimizer of the dual optimal-control problem, and let η be the induced probability measure on continuous trajectories $\Gamma=C([0,T];\mathbb T^N)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-054617 | MATHDB-326417

Antal and Krapivsky's growth conjecture for multistage epidemics

摘录 / EXCERPT: Antal and Krapivsky's conjecture. For each $k=1,\ldots,K$, the expected number of individuals of type k satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054618 | MATHDB-326447

The weak thin shell conjecture for log-concave random vectors

摘录 / EXCERPT: Weak thin shell conjecture. There exists $c>0$ such that for every log-concave random vector X,

conjecture | open | MathDB 页面 | 原文线索

V2-054619 | MATHDB-326448

The weak and strong moments conjecture for log-concave random vectors

摘录 / EXCERPT: Weak and strong moments conjecture. There exist $C,c>0$ such that for any log-concave random vector X and any norm $\|\cdot\|$,

conjecture | open | MathDB 页面 | 原文线索

V2-054620 | MATHDB-326608

General-domain hole probability conjecture for the Gaussian entire function

摘录 / EXCERPT: Let f be the Gaussian entire function, let $K\subset\mathbb{C}$ be a connected compact set with non-empty interior, and let rK denote the homothety of K by r. Write ... and l...

conjecture | open | MathDB 页面 | 原文线索

V2-054621 | MATHDB-326629

Unit-circle zero conjecture for Gaussian β-ensemble moment coefficients

摘录 / EXCERPT: Let the coefficients of the moments m_{2l} be regarded as polynomials in the ensemble parameter κ, after extracting their stated powers of N and κ. The coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-054622 | MATHDB-326724

The density-of-the-closure conjecture for subcritical $d53c$-bootstrap percolation

摘录 / EXCERPT: Density-of-the-closure conjecture. For every $p\in[0,1]$, there exists a constant $\delta(p)$ such that $D(n)$ converges in probability to $\delta(p)$ as $n\rightarrow\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-054623 | MATHDB-326725

The higher-dimensional subcriticality characterization conjecture for $d53c$-bootstrap percolation

摘录 / EXCERPT: Fix an integer $d\geqslant2$ and let $\mathcal{U}$ be a d-dimensional update family. For $u\in S^{d-1}$, define the half-space ... and the stable set ... Let μ be Lebesgue measur...

conjecture | open | MathDB 页面 | 原文线索

V2-054624 | MATHDB-326732

The universal Sudakov minoration conjecture for log-concave random vectors

摘录 / EXCERPT: Let X be a d-dimensional log-concave random vector. The universal Sudakov minoration conjecture. Every d-dimensional log-concave random vector satisfies the Sudakov-minoratio...

conjecture | open | MathDB 页面 | 原文线索

V2-054625 | MATHDB-326881

Universality conjecture for simple random walk on growing domains

摘录 / EXCERPT: Let $d\ge3$, and let $\bmathbb{D}_t\subseteq\mathbb{Z}^d$ be connected and non-decreasing, with...

conjecture | open | MathDB 页面 | 原文线索

V2-054626 | MATHDB-326961

James's uniqueness conjecture for the James function

摘录 / EXCERPT: Let $P(a,b)$ be a function of two team winning percentages satisfying the six proto-James conditions: $P(a,a)=\frac12$; $P(a,\frac12)=a$; if $a>b$ then $P(a,b)>\frac12$, and if...

conjecture | open | MathDB 页面 | 原文线索

V2-054627 | MATHDB-327004

Moment boundedness for hybrid linear systems with state resetting

摘录 / EXCERPT: Consider the hybrid linear system with jumps ... where W_t is a Wiener process, N_t is a counting process, and L_t satisfies a moment condition of order Q. Moment-boundedness...

conjecture | open | MathDB 页面 | 原文线索

V2-054628 | MATHDB-327040

Conjectural diagonal distribution for Werner's measure

摘录 / EXCERPT: Let ν_0 be the probability measure induced by Werner's loop measure on the triangular factorization parameter a of a welding homeomorphism, and let $\beta_0<\frac{5\pi^2}{4}$...

conjecture | open | MathDB 页面 | 原文线索

V2-054629 | MATHDB-327086

Conjecture on common non-zero zeros of Hermite polynomials

摘录 / EXCERPT: Let $\mathit{He}_n(x)$ denote the probabilists' Hermite polynomial of degree n, defined by ... The preceding properties imply that consecutive Hermite polynomials have no common ze...

conjecture | open | MathDB 页面 | 原文线索

V2-054630 | MATHDB-327100

Universality conjecture for low-lying zeros of automorphic families

摘录 / EXCERPT: Let $\mathfrak{F}$ be a rank 0 essentially homogeneous family. Write the zeros of each completed L-function as $\frac12+i\gamma_j^{(\pi)}$ and normalize them by...

conjecture | open | MathDB 页面 | 原文线索

conjecture | open | MathDB 页面 | 原文线索

V2-054631 | MATHDB-327150

Conjecture on open paths with backsteps in the hypercube

摘录 / EXCERPT: Open-path conjecture. For $x<x^*_1$,

conjecture | open | MathDB 页面 | 原文线索

V2-054632 | MATHDB-327151

Generalized open-path conjecture for the hypercube

摘录 / EXCERPT: Generalized open-path conjecture. When $x<x^*_{\alpha}$,

conjecture | open | MathDB 页面 | 原文线索

V2-054633 | MATHDB-327176

Mixed-state extension of the pairwise error bound

摘录 / EXCERPT: Let r be the number of hypotheses, let $A_i\ge0$ satisfy ... and let Γ^* denote the optimal success probability for the multiple-state discrimination problem. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054634 | MATHDB-327177

Converse decoupling for i.i.d. versus averaged i.i.d. discrimination

摘录 / EXCERPT: Let $0<p<1$, let ρ and $\sigma_1,\ldots,\sigma_r$ be quantum states, and let $q_1,\ldots,q_r$ be a probability distribution. For n copies, compare the i.i.d. hypothesis...

conjecture | open | MathDB 页面 | 原文线索

V2-054635 | MATHDB-327291

Poghosyan-Poghosyan-Priezzhev-Ruelle conjecture on threshold and stationary sandpile densities

摘录 / EXCERPT: Let $\zeta_{\tau}(h)$ denote the threshold density of the fixed-energy sandpile with constant initial condition h, and let ζ_s denote the stationary density of Dhar's abeli...

conjecture | open | MathDB 页面 | 原文线索

V2-054636 | MATHDB-327296

Central limit conjecture for quantum Markov-chain fluctuations

摘录 / EXCERPT: Let $X\in\mathcal{A}\otimes\mathcal{B}$ be a selfadjoint operator with $\langle X\rangle_{ss}^{out}=0$. Let $\mathbb{F}_n(X)$ denote the associated fluctuations operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054637 | MATHDB-327332

The cyclicity probability conjecture for Jacobians of random graphs

摘录 / EXCERPT: Let $G(n,q)$ be the Erdős-Rényi random graph on n vertices, with each edge included independently with fixed probability $0<q<1$. The cyclicity probability conjecture. The probab...

conjecture | open | MathDB 页面 | 原文线索

V2-054638 | MATHDB-327336

Uniform distribution conjecture for the generalized coupon collector time

摘录 / EXCERPT: Let $n\geq 1$, let $c\in\{1,\ldots,n\}$, and let $p=(p_1,\ldots,p_n)\in(0,1)^n$ satisfy $p_1+\cdots+p_n\leq 1$. Write ... Here $T_{c,n}(p)$ denotes the time needed in the generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-054639 | MATHDB-327377

Stability conjecture for random initial opinions in the Hegselmann-Krause model

摘录 / EXCERPT: Let the initial opinions $x_1(0),\dots,x_N(0)$ be chosen independently and uniformly from $[0,L]$, and let $q_{L,N}$ be the probability that the resulting equilibrium is stable, me...

conjecture | open | MathDB 页面 | 原文线索

V2-054640 | MATHDB-327492

The non-cycling conjecture for 4-free Fibonacci sequences

摘录 / EXCERPT: A 4-free Fibonacci sequence is a Fibonacci-type sequence defined by the paper's 4-free divisibility rule. Non-cycling conjecture. With probability 1, a 4-free Fibonacci sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-054641 | MATHDB-327545

Conjecture on the potential energy of a biased random walk on a tree

摘录 / EXCERPT: Let $(X_n)_{n\geq 0}$ be the random walk in a random environment on a tree, let $V(X_n)$ denote its potential energy, and let $\mathbb{P}$ be the governing probability measure. Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-054642 | MATHDB-327546

Conjecture on the displacement of a biased random walk on a tree

摘录 / EXCERPT: Let $(X_n)_{n\geq 0}$ be the random walk in a random environment on a tree, let $|X_n|$ denote the distance of the walk from the root, and let $\mathbb{P}$ be the governing probabi...

conjecture | open | MathDB 页面 | 原文线索

V2-054643 | MATHDB-327644

The heuristic all-colluders-caught estimate for fingerprinting

摘录 / EXCERPT: All-colluders-caught estimate. With probability at least $1-\varepsilon_1$, no innocent users are accused, and with probability at least $1-\varepsilon_2$, all colluders are caught...

conjecture | open | MathDB 页面 | 原文线索

V2-054644 | MATHDB-327645

Van den BergMaes conjecture on critical percolation

摘录 / EXCERPT: Critical percolation conjecture. For any lattice ${\mathcal L}$ in $d\geq 2$ dimensions,

conjecture | open | MathDB 页面 | 原文线索

V2-054645 | MATHDB-327683

Eigenvalue attraction explains low density near the real axis

摘录 / EXCERPT: Eigenvalue-attraction conjecture. The low density of eigenvalues of real random matrices near the real axis is the result of the large imaginary components of the acceleration into...

conjecture | open | MathDB 页面 | 原文线索

V2-054646 | MATHDB-327688

Slide dimension limits for uniform point processes

摘录 / EXCERPT: Let $U=(u_1,u_2,\dots)$ be a sequence of points chosen uniformly at random from $[0,1]^m$, and let ... Here ρ_n denotes the nth slide statistic. Uniform point-process limit co...

conjecture | open | MathDB 页面 | 原文线索

V2-054647 | MATHDB-327689

Sign conjecture for slide statistics of continuous random variables

摘录 / EXCERPT: Let X be a continuous real-valued random variable, and suppose that $\rho_n(X)$ exists. Sign conjecture. Then ... The conjecture is supported in the paper by the observed negativit...

conjecture | open | MathDB 页面 | 原文线索

V2-054648 | MATHDB-327708

The pseudo-threshold conjecture for homomorphisms to odd cycles

摘录 / EXCERPT: Let $G_{n,\frac{c}{n}}$ be the random graph with edge probability $\frac{c}{n}$, let $C_{2\ell+1}$ be the cycle of length $2\ell+1$, and let odd-girth denote the length of the shor...

conjecture | open | MathDB 页面 | 原文线索

V2-054649 | MATHDB-327709

The density conjecture for circular chromatic numbers of random graphs

摘录 / EXCERPT: Let $G_{n,\frac{c}{n}}$ be the random graph with edge probability $\frac{c}{n}$, and let $\chi_{\mathrm{c}}(G)$ denote its circular chromatic number. Density conjecture. There are...

conjecture | open | MathDB 页面 | 原文线索

V2-054650 | MATHDB-327729

The recurrence-transience conjecture for the frog model on rooted trees

摘录 / EXCERPT: Recurrence-transience conjecture. The frog model on $\mathbb{T}_d$ is recurrent almost surely for $d=3$ and transient almost surely for $d=4$.

conjecture | open | MathDB 页面 | 原文线索

V2-054651 | MATHDB-327734

Broadcasting-with-revealed-nodes boundary irrelevance conjecture

摘录 / EXCERPT: Let T be an infinite tree with root ρ. Assign the root a uniformly random label in $\{\pm1\}$ and propagate labels along edges, retaining the parent label with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-054652 | MATHDB-327747

Lam's direction conjecture for reduced affine Weyl group walks

摘录 / EXCERPT: Let $\tilde A_{n-1}$ be the affine Weyl group of type A_{n-1}, let ψ be the direction vector associated with a reduced random walk, and let ρ be the sum of all positive...

conjecture | open | MathDB 页面 | 原文线索

V2-054653 | MATHDB-327748

Increasing nearest-neighbour correlation conjecture for the multispecies TASEP

摘录 / EXCERPT: Let $E_{i_1,\dots,i_r}$ denote the joint probability of seeing particle types i_a in positions a, for $a=1,\dots,r$. Increasing-correlation conjecture. If ... then ... This extends...

conjecture | open | MathDB 页面 | 原文线索

V2-054654 | MATHDB-327749

Far-particle factorization conjecture for multispecies TASEP correlations

摘录 / EXCERPT: Let $E_{i_1,\dots,i_r}$ denote the joint probability of seeing particle types i_a in positions a, for $a=1,\dots,r$. Far-particle factorization conjecture. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-054655 | MATHDB-327750

One-apart independence conjecture for multispecies TASEP correlations

摘录 / EXCERPT: Let $w=(w_a)$ denote the cyclic multispecies TASEP word, and let $E_{i_1,\dots,i_r}$ denote the joint probability of seeing i_a in position a, for $a=1,\dots,r$. One-apart inde...

conjecture | open | MathDB 页面 | 原文线索

V2-054656 | MATHDB-327751

Two-block independence conjecture for multispecies TASEP correlations

摘录 / EXCERPT: Let $w=(w_a)$ denote the cyclic multispecies TASEP word, and let $E_{i_1,\dots,i_r}$ and $E_{j_{r+1},\dots,j_{r+s}}$ denote the corresponding nearest-neighbour block probabilities....

conjecture | open | MathDB 页面 | 原文线索

V2-054657 | MATHDB-327766

Random groups contain subgroups commensurable with orbifold geodesic-spine groups

摘录 / EXCERPT: Random-spine subgroup conjecture. For any fixed geodesic spine K, with overwhelming probability G contains a subgroup commensurable with the orbifold fundamental group of some...

conjecture | open | MathDB 页面 | 原文线索

V2-054658 | MATHDB-327801

Marić's selection principle for the Fleming-Viot random walk

摘录 / EXCERPT: Let p and q be the right- and left-jump rates, respectively, with $q>p$, and let ρ^N be the equilibrium density of the Fleming-Viot process driven by the nearest-neighbor...

conjecture | open | MathDB 页面 | 原文线索

V2-054659 | MATHDB-327811

Uniqueness of Brownian Gibbs line ensembles with prescribed mean height

摘录 / EXCERPT: Uniqueness conjecture. For every $c\in\mathbb{R}$, there is a unique Brownian Gibbs line ensemble $\mathcal{L}\in\mathscr{G}_{\Theta}$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-054660 | MATHDB-327830

Witte-Forrester conjecture on Gaussian beta-ensemble one-point correlators

摘录 / EXCERPT: Witte-Forrester conjecture. For even $l\leq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-054661 | MATHDB-327854

Hinz-Młotkowski conjecture on free Poisson double convolution powers

摘录 / EXCERPT: Hinz-Młotkowski conjecture. The double power $\tilde{\bm{\pi}}_{st}$ is a probability measure if and only if $\max(s,t)\geq1$. Equivalently, the sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-054662 | MATHDB-327868

Multiple-SLE crossing-probability formula

摘录 / EXCERPT: Let $\kappa\in(0,8)$, and consider a multiple-SLE$_\kappa$ process growing $2N$ curves in the upper half-plane from points $x_1<x_2<\ldots<x_{2N}$, with SLE$_\kappa$ partition func...

conjecture | open | MathDB 页面 | 原文线索

V2-054663 | MATHDB-328007

Sznitman's conjecture that condition (T') is equivalent to ballisticity

摘录 / EXCERPT: Sznitman's conjecture. Condition (T') is equivalent to ballisticity.

conjecture | open | MathDB 页面 | 原文线索

V2-054664 | MATHDB-328132

Bounded-probe conjecture for the probing simple random walk

摘录 / EXCERPT: Consider a probing simple random walk (PSRW) $\{K_t\}$ on growing domains $\{\mathbb{D}_t\}$ in $\mathbb{Z}^d$, $d\ge2$, starting at $K_0=0$ with $\mathbb{D}_0=\{0\}$. At time t,...

conjecture | open | MathDB 页面 | 原文线索

V2-054665 | MATHDB-328143

Equivalence of conditions $(T)|l$ and $(T')|l$ for uniformly elliptic RWRE

摘录 / EXCERPT: Consider a random walk in a uniformly elliptic random environment in dimension $d\ge 2$ and a direction $l\in\mathbb S^{d-1}$. For $\gamma\in(0,1]$, condition $(T)_\gamma|l$ requir...

open_problem | open | MathDB 页面 | 原文线索

V2-054666 | MATHDB-328159

Uniqueness of distribution conjecture for independent copies spanning an Orlicz sequence space

摘录 / EXCERPT: Uniqueness conjecture. If there is a unique (up to equivalence near 0) mean zero function f whose independent copies are equivalent in L_p to the unit vector basis in l_M,...

conjecture | open | MathDB 页面 | 原文线索

V2-054667 | MATHDB-328162

Quenched central limit theorem for biased random walks on Galton-Watson trees with leaves

摘录 / EXCERPT: Let β-biased random walk be the random walk on a Galton-Watson tree with leaves, and let β_c denote the critical bias. Quenched central limit theorem conjecture. A quen...

conjecture | open | MathDB 页面 | 原文线索

V2-054668 | MATHDB-328163

Quenched central limit theorem for randomly biased random walks

摘录 / EXCERPT: Let α be the exponent defined in the paper's theorem on randomly biased random walks. Quenched central limit theorem conjecture. For randomly biased random walks, a quenched...

conjecture | open | MathDB 页面 | 原文线索

V2-054669 | MATHDB-328164

Stable fluctuations for biased random walks on Galton-Watson trees

摘录 / EXCERPT: Consider a β-biased random walk with exponent $\alpha\in(1,2)$, where α is defined at the paper's equation labelled α_{tree}. Let Δ_n denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-054670 | MATHDB-328165

Aging for randomly biased random walks

摘录 / EXCERPT: Let $\alpha\in(0,1)$ be the exponent governing the randomly biased random walk. Aging conjecture. In the case of randomly biased random walks, aging similar to the result establish...

conjecture | open | MathDB 页面 | 原文线索

V2-054671 | MATHDB-328166

Stable laws and aging for randomly biased walks on leafless Galton-Watson trees

摘录 / EXCERPT: Consider randomly biased random walks on Galton-Watson trees without leaves, and impose a non-lattice condition on $\log A$, where A is the random bias variable in the model. Sta...

conjecture | open | MathDB 页面 | 原文线索

V2-054672 | MATHDB-328167

Stable fluctuations for randomly biased walks on leafless Galton-Watson trees

摘录 / EXCERPT: Consider randomly biased random walks on Galton-Watson trees without leaves, with exponent $\alpha\in(1,2)$, and assume a non-lattice condition. Stable-fluctuation conjecture. It w...

open_problem | open | MathDB 页面 | 原文线索

V2-054673 | MATHDB-328168

Quenched central limit theorem for biased walks on supercritical percolation clusters

摘录 / EXCERPT: Let X_n be a biased random walk on the infinite supercritical percolation cluster of parameter $p>p_c$, with bias direction $\vec{\ell}$. Quenched central limit theorem conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-054674 | MATHDB-328169

Scaling limits for biased walks on supercritical percolation clusters

摘录 / EXCERPT: Let X_n be a biased random walk on a supercritical percolation cluster, let $\vec{\ell}$ be the bias direction, let α be the displacement exponent, and let Δ_n be t...

conjecture | open | MathDB 页面 | 原文线索

V2-054675 | MATHDB-328171

Armstrong's conjecture on the longest k-alternating subsequence

摘录 / EXCERPT: Let π be uniformly distributed in ${\mathcal S}_n$, and let $L_{n,k}=L_{n,k}(\pi)$ denote the length of its longest k-alternating subsequence, where a subsequence is k-alte...

conjecture | open | MathDB 页面 | 原文线索

V2-054676 | MATHDB-328226

Conjecture on the maximal Nazarov-Sodin constant for symmetric measures

摘录 / EXCERPT: Let $\mathcal{P}_{Symm}$ be the family of symmetric probability measures on $\mathcal{S}^{1}$, and let $d_{\max}$ denote the maximal value attained by the Nazarov-Sodin constant...

conjecture | open | MathDB 页面 | 原文线索

V2-054677 | MATHDB-328227

Conjecture on the maximal Nazarov-Sodin constant for planar spectral measures

摘录 / EXCERPT: Let $\mathcal{P}$ be the collection of probability measures on the unit ball $B(1)\subseteq\mathbb{R}^{2}$, and let $c_{\max}$ denote the maximal value attained by the Nazarov-Sodi...

conjecture | open | MathDB 页面 | 原文线索

V2-054678 | MATHDB-328346

Uniqueness and analyticity conjecture for equilibrium measures

摘录 / EXCERPT: Equilibrium-measure conjecture. The equilibrium measure is unique, and its density away from the edges of its support is a real-analytic function of $u>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-054679 | MATHDB-328348

Off-criticality conjecture for the Seifert matrix model

摘录 / EXCERPT: Off-criticality conjecture. Off-criticality holds for all values of $u>0$, and consequently the displayed asymptotic expansion holds for all such u.

conjecture | open | MathDB 页面 | 原文线索

V2-054680 | MATHDB-328419

Convergence under eventual convexity and a unique single-support minimizer

摘录 / EXCERPT: Let V be a C^2 potential function, and let $K_1,K_2,r>0$ satisfy ... and ... Suppose that Σ_V has a unique minimizer μ_V whose support is a single compact set. Convergen...

conjecture | open | MathDB 页面 | 原文线索

V2-054681 | MATHDB-328420

Biane-Speicher's convergence conjecture for a weak double-well potential

摘录 / EXCERPT: Let V be the double-well potential ... where g is a negative constant sufficiently close to zero, and let μ_t and μ_V denote the empirical measure process and the equilib...

conjecture | open | MathDB 页面 | 原文线索

V2-054682 | MATHDB-328486

Regularity dichotomy and spectral atom conjecture for polynomial variables

摘录 / EXCERPT: Let M be a tracial von Neumann algebra and let $y_{1},\dots,y_{n}\in M$ be self-adjoint elements such that $M=W^{*}(y_{1},\dots,y_{n})$. Define Δ_{M} as in the surrounding...

conjecture | open | MathDB 页面 | 原文线索

V2-054683 | MATHDB-328502

Scaling-limit conjecture for the oil and water odometer in higher dimensions

摘录 / EXCERPT: Scaling-limit conjecture. There exists a deterministic function $w:\mathbb R^d\to\mathbb R$ such that, almost surely, for every $\xi\in\mathbb R^d-\{0\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-054684 | MATHDB-328503

Reversible Markov-chain surprise bound

摘录 / EXCERPT: Let X be a reversible Markov chain with finite state space, and let π be the stationary distribution of X. For states x and y, write $\tau(y)$ for the hitting time of...

conjecture | open | MathDB 页面 | 原文线索

V2-054685 | MATHDB-328504

Holroyd's interval hitting-time conjecture for finite Markov chains

摘录 / EXCERPT: Let X be a Markov chain with finite state space S, where $|S|=n$, and let x and y be any two states. Holroyd's interval hitting-time conjecture. For all $t>n$, ... This conje...

conjecture | open | MathDB 页面 | 原文线索

V2-054686 | MATHDB-328603

The conjecture on SDP recovery for k-means at separation greater than $2+\epsilon$

摘录 / EXCERPT: SDP recovery conjecture. The SDP relaxation recovers the clusters with high probability whenever

conjecture | open | MathDB 页面 | 原文线索

V2-054687 | MATHDB-328659

The converse to Kleiner's theorem for harmonic functions on groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a finitely generated group, and let μ be a symmetric probability measure on $\mathbb{G}$ with finite support that generates $\mathbb{G}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054688 | MATHDB-328660

Liouville invariance for courteous measures

摘录 / EXCERPT: Let $\mathbb{G}$ be a finitely generated group, and let μ and ν be courteous probability measures on $\mathbb{G}$. Liouville invariance conjecture. The random-walk space...

conjecture | open | MathDB 页面 | 原文线索

V2-054689 | MATHDB-328662

Lipschitz-to-linear harmonic-function finiteness conjecture

摘录 / EXCERPT: Let $\mathbb{G}$ be a finitely generated group and let μ be a courteous probability measure on $\mathbb{G}$. Let $\mathsf{LHF}(\mathbb{G},\mu)$ denote the space of Lipschitz...

conjecture | open | MathDB 页面 | 原文线索

V2-054690 | MATHDB-328666

Conjecture on decomposing CD-affine needles under a zero-mean constraint

摘录 / EXCERPT: Let $f535$ be a probability measure on $f535$ that is a $CD(\kappa,N+1)$-needle, where $N\in(0,+\infty]$ and $\kappa\geq0$. Let $\varphi:\mathbb R\to\mathbb R$ be continuous and...

conjecture | open | MathDB 页面 | 原文线索

V2-054691 | MATHDB-328768

The refined-term-order conjecture for I_{S_n}

摘录 / EXCERPT: Let S_n be the polytope of symmetric doubly-stochastic matrices, let T_{S_n} be the polynomial ring with variables t_M indexed by the relevant matrices M, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054692 | MATHDB-328807

Positive anchored expansion for higher-dimensional hyperbolic Poisson-Voronoi dual graphs

摘录 / EXCERPT: Let $\mathbb{H}^d$ denote d-dimensional hyperbolic space, and let Π^λ be a Poisson process with invariant intensity measure. The dual graph of the Voronoi tessellation...

conjecture | open | MathDB 页面 | 原文线索

V2-054693 | MATHDB-328808

Singularity of harmonic measure for the hyperbolic Poisson-Voronoi tessellation

摘录 / EXCERPT: Let $\mathscr{V}^\lambda$ be the hyperbolic Poisson-Voronoi graph, and let ν_0 be the harmonic measure on S^1 obtained from the almost-sure limit of simple random walk on...

conjecture | open | MathDB 页面 | 原文线索

V2-054694 | MATHDB-329012

Sublinear variance conjecture for direct products of infinite finitely generated groups

摘录 / EXCERPT: Sublinear variance conjecture. The estimate $$ is satisfied for all Cayley graphs of G.

conjecture | open | MathDB 页面 | 原文线索

V2-054695 | MATHDB-329027

Talagrand's Gaussian anti-concentration conjecture for diffusion

摘录 / EXCERPT: Let $n\geq 1$ and let γ_n be the standard Gaussian probability measure on $\mathbb R^n$. For $t>0$, let U_t denote the Gaussian diffusion operator, and for a measurable no...

conjecture | solved | MathDB 页面 | 原文线索

V2-054696 | MATHDB-329028

Talagrand's discrete-cube anti-concentration conjecture

摘录 / EXCERPT: Let $n\geq 1$ and $t\geq 0$, let $\mu_t=\frac{1-e^{-t}}{2}\delta_{-1}+\frac{1+e^{-t}}{2}\delta_1$, and let $\mu=\mu_\infty$ be the uniform probability measure on $\{-1,1\}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-054697 | MATHDB-329036

Very strong disorder conjecture for quenched and annealed free energies

摘录 / EXCERPT: Assume (Dir), (Ind), (Loc), $V\in\mathcal{L}$, and $\Lambda_q(V)<\infty$. Let $\underline h_\infty^\lambda$ and $\bar h_\infty^\lambda$ denote the lower and upper limiting free-ene...

conjecture | open | MathDB 页面 | 原文线索

V2-054698 | MATHDB-329048

Conjecture on the asymptotic curvature of conjugacy-class random walks

摘录 / EXCERPT: Let κ_c denote the curvature quantity associated with the random walk studied in the paper, and let $\theta(c)$ be the function appearing in the preceding curvature estimate...

conjecture | open | MathDB 页面 | 原文线索

V2-054699 | MATHDB-329171

Doob-random Schnorr randomness does not imply e.c.u. randomness

摘录 / EXCERPT: Let $\omega\in 2^{\mathbb{N}\times\mathbb{N}}$. A real is Doob random if it satisfies the paper's characterization in terms of computable martingales, and it is e.c.u. random if it...

conjecture | open | MathDB 页面 | 原文线索

V2-054700 | MATHDB-329220

General cyclic-system conjecture for contextuality

摘录 / EXCERPT: Let $\left(\mathfrak{S},\mathfrak{C}\right)$ be a system with ... where π is a circular permutation of $\{1,\ldots,n\}$, and ... Here $\Delta_0(\mathfrak{C})$ is the sum of minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-054701 | MATHDB-329234

GOE fluctuation conjecture for flat initial conditions

摘录 / EXCERPT: Let P be the pressure function, let β be the slope of flat initial conditions, and set $\lambda=\gamma P''(\beta)\neq 0$ and $A=-P'(\beta)$. For the particle system with fla...

conjecture | open | MathDB 页面 | 原文线索

V2-054702 | MATHDB-329238

Kerov's stochastic monotonicity conjecture for Jordan types over finite fields

摘录 / EXCERPT: Kerov's stochastic monotonicity conjecture. If $r<i$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054703 | MATHDB-329301

Asymptotic formula for the traveling salesperson tour in randomly embedded random graphs

摘录 / EXCERPT: Let $G_{{\mathcal X},p}$ be the randomly embedded random graph on n points in $[0,1]^d$, with edge-probability p, and let $T(G_{{\mathcal X},p})$ denote the length of a minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-054704 | MATHDB-329306

The tail-mass conjecture for percolation in Poisson graphs

摘录 / EXCERPT: Let $G=G_1(\mu)$ be the one-dimensional Poisson graph generated by a degree distribution μ on $\mathbb{N}$. Tail-mass conjecture. For every $\varepsilon>0$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-054705 | MATHDB-329314

Linear chemical distances in scale-free Gilbert graphs

摘录 / EXCERPT: Let $G(X^{(n)})$ be the scale-free Gilbert graph on the point set $X^{(n)}$ in the torus $\mathbb{T}_n$, let $\mathsf{e}_1=(1,0,\ldots,0)$, and let $q(x)$ denote the closest point...

conjecture | open | MathDB 页面 | 原文线索

V2-054706 | MATHDB-329383

The random-walk pattern probability characterization conjecture

摘录 / EXCERPT: Let S_n be the symmetric group, let $X_1,X_2,\ldots,X_{n-1}$ be independent identically distributed random variables, and let $p(X_1,X_2,\ldots,X_{n-1})$ denote the permutation p...

conjecture | open | MathDB 页面 | 原文线索

V2-054707 | MATHDB-329387

Equality conjecture for the hard edge tacnode temperature-derivative functions

摘录 / EXCERPT: Equality conjecture. The functions g_σ and h_σ are the same. The statement is presented without proof in the context of the temperature derivative theorem; the suppli...

conjecture | open | MathDB 页面 | 原文线索

V2-054708 | MATHDB-329528

Shape conjecture for the function G

摘录 / EXCERPT: Let $G:(0,\infty)\to\mathbb{R}$ be the function ... where p_0 and p_1 are the invariant-density functions defined in the Lyapunov-exponent theorem. Shape conjecture for G. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-054709 | MATHDB-329607

Latała's moment conjecture for norms of log-concave vectors

摘录 / EXCERPT: Latała's moment conjecture. There exists a universal constant C such that, for every log-concave vector X with values in a finite-dimensional normed space $(F,\|\ \|)$ and ever...

conjecture | open | MathDB 页面 | 原文线索

V2-054710 | MATHDB-329637

The converse recurrence criterion for anisotropic random walks on the plane

摘录 / EXCERPT: Let $(p_j)_{j\in\mathbb Z}$ be the transition-probability parameters of the anisotropic walk, with $p_j>0$. The walk is called transient if it visits each state only finitely often...

conjecture | open | MathDB 页面 | 原文线索

V2-054711 | MATHDB-329677

Polylogarithmic gaps conjecture for the 6-GP-free process

摘录 / EXCERPT: Let T be the sequence generated by the 6-GP-free process, and interpret a gap of size $O((\log x)^C)$ as the existence of a constant implicit in the big-O bound, independent...

conjecture | open | MathDB 页面 | 原文线索

V2-054712 | MATHDB-329689

The continuous one-away TASEP density conjecture

摘录 / EXCERPT: Let $g_w(q_1,\dots,q_n)$ denote the stationary probability density for the continuous TASEP configuration with permutation w, and let w_0 be the longest permutation. For...

conjecture | open | MathDB 页面 | 原文线索

V2-054713 | MATHDB-329690

The continuous multi-away TASEP density conjecture

摘录 / EXCERPT: Let $g_w(q_1,\dots,q_n)$ denote the stationary probability density for the continuous TASEP configuration with permutation w, and let $\mathbf{k}=(k_1\ge\cdots\ge k_r\ge 1)$ sati...

conjecture | open | MathDB 页面 | 原文线索

V2-054714 | MATHDB-329691

The upper-right two-point correlation conjecture for continuous TASEP

摘录 / EXCERPT: For a continuous TASEP on a ring with n particles, let ... be the stationary two-point correlation for an adjacent pair, where a is arbitrary. The upper-right correlation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-054715 | MATHDB-329706

Martin boundary conjecture for reversible random planar maps

摘录 / EXCERPT: Let (M,ρ) be a reversible random planar map. The Martin boundary of M is the boundary describing asymptotic behavior of positive harmonic functions for random walk on M. A...

conjecture | open | MathDB 页面 | 原文线索

V2-054716 | MATHDB-329707

Conjecture on localization of self-reinforced diffusions with trigonometric interaction

摘录 / EXCERPT: Self-reinforced diffusion conjecture. If $\sum_{k=1}^{n}k^2a_k<0$, then X_t converges almost surely; if $\sum_{k=1}^{n}k^2a_k>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054717 | MATHDB-329716

Corrected Aldous conjecture for regular graphs

摘录 / EXCERPT: Fix $d\in\mathbb{N}$ and let G be a d-regular graph. Let p be the parameter of the constrained Ising process on G, and let $\tau_{\mathrm{mix}}^{\mathrm{RW}}$ denote the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-054718 | MATHDB-329717

Coalescence-corrected mixing-time conjecture for regular graphs

摘录 / EXCERPT: Fix $d\in\mathbb{N}$ and let G be a d-regular graph. Let p be the parameter of the constrained Ising process on G, let $\tau_{\mathrm{mix}}^{\mathrm{RW}}$ denote the mixing...

conjecture | open | MathDB 页面 | 原文线索

V2-054719 | MATHDB-329774

Sharpness of the stabilization bound for divisible sandpiles in cones

摘录 / EXCERPT: Sharpness conjecture. The divisible sandpile s_a stabilizes if and only if $\frac{2ma}{1+a^2}\leq 1$. Furthermore, s_0 stabilizes if and only if $k\leq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-054720 | MATHDB-329918

The alternative conjecture on the spectral norm of Gaussian random matrices

摘录 / EXCERPT: Let X be the $d\times d$ symmetric random matrix with entries $X_{ij}=b_{ij}g_{ij}$, where $\{g_{ij}:i\ge j\}$ are independent standard Gaussian random variables and...

conjecture | open | MathDB 页面 | 原文线索

V2-054721 | MATHDB-329930

Preferential attachment graphs approaching the Rado graph

摘录 / EXCERPT: Rado graph conjecture. With probability 1, the infinite limit of ${\textrm{GPA}_f}(G')$ is the Rado graph.

conjecture | open | MathDB 页面 | 原文线索

V2-054722 | MATHDB-329971

The bounded-degree graph conjecture for Bernoulli CMP

摘录 / EXCERPT: Let G be an infinite graph with bounded degrees. Bernoulli CMP on G has a non-trivial phase transition, meaning that its critical parameter satisfies $0<p_c<1$. Bounded-degree...

conjecture | open | MathDB 页面 | 原文线索

V2-054723 | MATHDB-329972

The Galton-Watson conjecture for degree-weighted CMP

摘录 / EXCERPT: Let $\alpha\geq 1$, and let G be a Galton-Watson tree with reproduction law B satisfying ... for every $c>0$. Galton-Watson degree-weighted CMP conjecture. Degree-weighted CMP on...

conjecture | open | MathDB 页面 | 原文线索

V2-054724 | MATHDB-330035

High-temperature Poisson variational limit conjecture for directed polymers in region R_5

摘录 / EXCERPT: Let $R_5:=\{(\gamma_c,\gamma_a): \gamma_a>1/2,\ \max\{0,2/\gamma_a-1,(\gamma_a-5)/(\gamma_a-2)\}<\gamma_c<3/(2\gamma_a)\}$. Consider the directed-polymer partition function...

conjecture | open | MathDB 页面 | 原文线索

V2-054725 | MATHDB-330047

Generalized Shepp-Olkin conjecture for Rényi and Tsallis entropy

摘录 / EXCERPT: Generalized Shepp-Olkin conjecture. There are critical values q_R^* and q_T^* such that:

conjecture | open | MathDB 页面 | 原文线索

V2-054726 | MATHDB-330065

The Gaussian Product Conjecture

摘录 / EXCERPT: Let $d\geq 2$, let $(G_1,\ldots,G_d)$ be a d-dimensional real-valued centered Gaussian vector, and let $m\geq 1$ be an integer. Gaussian Product Conjecture. One has ... This is an...

open_problem | open | MathDB 页面 | 原文线索

V2-054727 | MATHDB-330137

Uniqueness and KPZ limits for stationary fractional Burgers energy solutions

摘录 / EXCERPT: Consider the fractional stochastic Burgers equation (ec1.5.3). Let $\mathcal{Y}_t^{m,\rho}$ denote a stationary energy solution when it is unique, and let $\mathcal{X}_t^\rho$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-054728 | MATHDB-330177

The limiting maximum-value distribution for the GUE characteristic polynomial

摘录 / EXCERPT: Let H be an $N\times N$ Hermitian matrix from the Gaussian Unitary Ensemble, with characteristic polynomial $p_N(x)=\operatorname{det}(xI-H)$. Define ... Let Γ denote the Eu...

conjecture | open | MathDB 页面 | 原文线索

V2-054729 | MATHDB-330186

Universality conjecture for leading-order small-maturity forward smile asymptotics

摘录 / EXCERPT: Universality conjecture. The leading-order small-maturity forward smile asymptotics generated from the stochastic-volatility model given by equation are equivalent to those given i...

conjecture | open | MathDB 页面 | 原文线索

V2-054730 | MATHDB-330200

Boundary scaling limit conjecture for frozen-boundary diffusion

摘录 / EXCERPT: Boundary scaling limit conjecture. For every $\alpha\in(0,1)$, there exists $\ell_{\alpha}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054731 | MATHDB-330201

Weak-limit conjecture for frozen-boundary diffusion

摘录 / EXCERPT: Weak-limit conjecture. There exists a probability measure $\mu_{\infty}(\alpha)$ on $\mathbb R$ such that, as $t\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-054732 | MATHDB-330292

Unique minimum conjecture for the proportion of problematic stations

摘录 / EXCERPT: Unique minimum conjecture. For every $\frac{\lambda}{\mu}>0$, P_b has a unique minimum P_b^*, attained at $\rho_V=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-054733 | MATHDB-330293

Monotonicity conjecture for the proportion of problematic stations

摘录 / EXCERPT: Monotonicity conjecture. For every $s\in\mathopen{[}0,K\mathclose{[}$, the proportion of problematic stations $P_b^{+}(s)$ is increasing as the traffic λ/μ increases.

conjecture | open | MathDB 页面 | 原文线索

V2-054734 | MATHDB-330298

The transitive Boolean functions conjecture on diverging pivotal counts

摘录 / EXCERPT: Let $\{f_n\}_{n\ge 1}$ be a sequence of Boolean functions, with parameters $\{p_n\}$, that is nondegenerate with respect to $\{p_n\}$. Let C_n denote the number of pivotal bits f...

conjecture | open | MathDB 页面 | 原文线索

V2-054735 | MATHDB-330310

Recurrence of nonadaptive random walk in changing environment on subgraphs of the square lattice

摘录 / EXCERPT: Consider a labyrinth evolving by adding edges between adjacent vertices of $\mathbb{Z}^2$, without removing existing edges, and let Theseus perform the random walk on the resulting...

conjecture | open | MathDB 页面 | 原文线索

V2-054736 | MATHDB-330311

Recurrence and transience criteria for nonadaptive random walks in changing environments

摘录 / EXCERPT: A nonadaptive random walk in changing environment (nonadaptive RWCE) is a random walk whose edge weights are chosen in advance and therefore do not depend on the walker's location....

conjecture | open | MathDB 页面 | 原文线索

V2-054737 | MATHDB-330319

The law of large numbers and velocity threshold for non-nearest-neighbor excited random walks

摘录 / EXCERPT: Velocity conjecture. There exists a deterministic constant $v_0\in[0,\infty)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054738 | MATHDB-330422

Poisson-Dirichlet conjecture for macroscopic loop lengths

摘录 / EXCERPT: Let $\ell_1,\ell_2,\dots$ be the loop lengths ordered decreasingly, let $m(\beta)$ be the long-loop mass, and let u be the model parameter. Define ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-054739 | MATHDB-330529

Concavity conjecture for the speed of the maximum of frogs with drift

摘录 / EXCERPT: For $p\in[0,1]$, let the frog model with drift parameter p have maximum speed, namely the limiting speed of its rightmost frog, denoted by $v(p)$. Concavity conjecture. The speed...

open_problem | open | MathDB 页面 | 原文线索

V2-054740 | MATHDB-330565

Bounded-density conjecture for the critical random Dirichlet series

摘录 / EXCERPT: Let $\beta=1$ and let $4mu4$ denote the distribution of the random Dirichlet series considered in the paper. In the critical case $s=1$, its density is in L^q for every...

conjecture | open | MathDB 页面 | 原文线索

V2-054741 | MATHDB-330595

Pair-distribution conjecture for the two-nut caching game

摘录 / EXCERPT: Assume $k=j=2$, so the hider caches two nuts and the searcher aims to find both. A hiding strategy is represented by pairs (y_1,y_2) of depths, with the hole placements randomize...

conjecture | open | MathDB 页面 | 原文线索

V2-054742 | MATHDB-330608

Zhang's finiteness conjecture for the critical passage-time exponent

摘录 / EXCERPT: Let F_a be the distribution defined by ... where p_c is the critical probability for Bernoulli bond percolation on $\mathbb{Z}^2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054743 | MATHDB-330615

Strong-solution convergence conjecture for one-dimensional coarsening

摘录 / EXCERPT: Let $(\Omega,\mathbb P)$ be the probability space of initial configurations, and let $\Psi_t^{(L)}(N)$ denote the evolution at time t in the finite-volume system of size L. Str...

conjecture | open | MathDB 页面 | 原文线索

V2-054744 | MATHDB-330616

Poisson attraction conjecture for one-dimensional coarsening

摘录 / EXCERPT: Let ν be a probability measure on $\mathbb R$ with finite mean $m<\infty$, and let Q_ν be the stationary renewal process on $\mathbb R$ with renewal distribution ν. A w...

conjecture | open | MathDB 页面 | 原文线索

V2-054745 | MATHDB-330658

Brown measure convergence conjecture for polynomials in Wigner matrices

摘录 / EXCERPT: Brown measure convergence conjecture. The eigenvalue distributions μ_{A_N} converge to the Brown measure μ_x of x.

conjecture | open | MathDB 页面 | 原文线索

V2-054746 | MATHDB-330659

Convergence conjecture for discrete spans of simple random walks

摘录 / EXCERPT: Let $(RW_k)_{k\in\mathbb{N}}$ be a simple random walk and let ... be its spans through time N. Let $\spp^{[0,1]}(1)$ denote the corresponding Brownian spans on $[0,1]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054747 | MATHDB-330660

Knight-coupling convergence conjecture for Brownian spans

摘录 / EXCERPT: Fix $n\in\mathbb{N}$, let $(RW_k^{(n)})_{k\in\mathbb{N}}$ be the simple random walk obtained from Brownian motion by the stopping-time construction in the source, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054748 | MATHDB-330683

Strong transience criterion for the conditioned right step

摘录 / EXCERPT: Let P be a distribution on environments satisfying Assumption asmiid, and suppose that ... Here, $\mathfrak{R}$ denotes the first return time to the origin, and X_1 is the first...

conjecture | open | MathDB 页面 | 原文线索

V2-054749 | MATHDB-330721

Flatness conjecture for consecutive random polygon subdivisions

摘录 / EXCERPT: Let L_n be the random polygon sequence generated by the consecutive random subdivision, and suppose that $ u$ satisfies ... and that the support of ξ contains at least two dist...

conjecture | open | MathDB 页面 | 原文线索

V2-054750 | MATHDB-330727

The orbit-closure conjecture for Gaussian moment-vanishing polynomials

摘录 / EXCERPT: Let $X=(X_1,\dots,X_n)$ be a jointly Gaussian random vector, let $P(x)\in\mathbb{C}[x]$, and let $\operatorname{O}_n(\mathbb{C})$ act by linear changes of variables. A one-paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-054751 | MATHDB-330746

The zero-one law for Besov regularity of periodic Lévy noises

摘录 / EXCERPT: Let w be a periodic noise on $\mathbb{T}^d$, let $\tau\bin\tyu0005\mathbb{R}$, and let $0<p,q\textless\textless=\infty$. The zero-one law. For every such noise and every such Bes...

conjecture | open | MathDB 页面 | 原文线索

V2-054752 | MATHDB-330809

Distribution-free convex-hull avoidance probabilities in higher dimensions

摘录 / EXCERPT: Distribution-free convex-hull avoidance conjecture. For every $d\geq 3$ and every $n\geq 1$, the probability

conjecture | open | MathDB 页面 | 原文线索

V2-054753 | MATHDB-330810

Universal one-third collision-before-trapping conjecture

摘录 / EXCERPT: Let P be the transition matrix of a finite irreducible and reversible Markov chain. Let X,Y,Z be independent continuous-time Markov chains with transition matrix P and speeds...

conjecture | open | MathDB 页面 | 原文线索

V2-054754 | MATHDB-330867

The Gaussian Correlation Conjecture

摘录 / EXCERPT: Let γ_n be the Gaussian probability measure on $\mathbb{R}^n$, and let K_1 and K_2 be symmetric convex bodies in $\mathbb{R}^n$. The Gaussian Correlation Conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-054755 | MATHDB-330965

Coram-Diaconis self-similarity conjecture for CUE eigenvalues

摘录 / EXCERPT: Let U be an $N\times N$ random matrix from the circular unitary ensemble with eigenvalues $\{e^{i\theta_j}\}_{1\le j\le N}$, where $0\le \theta_1\le\dots\le\theta_N<2\pi$. Choose...

conjecture | open | MathDB 页面 | 原文线索

V2-054756 | MATHDB-330987

Necessary and sufficient conditions for the strong renewal theorem for random walks

摘录 / EXCERPT: Let S be an aperiodic random walk whose increment distribution belongs to $D(\alpha,\rho)$, where $\alpha\in(0,1)$ and $\rho>0$. Let and denote the conditions stated in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-054757 | MATHDB-331059

The discontinuity conjecture for free compound Poisson distributions

摘录 / EXCERPT: Discontinuity conjecture. The preceding assertion holds for every such probability measure $ u$.

conjecture | open | MathDB 页面 | 原文线索

V2-054758 | MATHDB-331060

The classical-free unimodality equivalence conjecture

摘录 / EXCERPT: Classical-free unimodality equivalence conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-054759 | MATHDB-331061

Existence of non-unimodal measures in the free Jurek class

摘录 / EXCERPT: Free Jurek-class non-unimodality conjecture. There exists a non-unimodal probability measure in ${\rm U}(\boxplus)$.

conjecture | open | MathDB 页面 | 原文线索

V2-054760 | MATHDB-331062

Eventual unimodality under free convolution powers

摘录 / EXCERPT: Eventual unimodality conjecture. The free convolution powers $\mu^{\boxplus s} are unimodal for every $sgeq T$.

conjecture | open | MathDB 页面 | 原文线索

V2-054761 | MATHDB-331091

Local belief propagation optimality conjecture with noisy side information

摘录 / EXCERPT: Under the binary symmetric stochastic block model with $alpha$-noisy side information, let $p_{G_n}(\widehat{\sigma}_{\rm BP}^t)$ be the estimation accuracy of belief propagation a...

conjecture | open | MathDB 页面 | 原文线索

V2-054762 | MATHDB-331092

Uniqueness conjecture for the belief-propagation fixed-point equation

摘录 / EXCERPT: In the large-degree regime, let μ and α be the parameters appearing in the scalar density-evolution recursion, and let $h(v)$ be the function defined by that recursion....

conjecture | open | MathDB 页面 | 原文线索

V2-054763 | MATHDB-331185

Uniform relativization conjecture for computable randomness under layerwise maps

摘录 / EXCERPT: Assume that $\mathbb{X}$ and $\mathbb{Y}$ are computable metric spaces, that $\{\mu_a\}_{a\in\mathbb{N}^{\mathbb{N}}}$ is a uniformly computable family of probability measures on...

conjecture | open | MathDB 页面 | 原文线索

V2-054764 | MATHDB-331187

A distributional convergence conjecture for DCPD deflation

摘录 / EXCERPT: DCPD distributional convergence conjecture. There exists at least one absolutely continuous probability measure μ on tensors $\boldsymbol{\mathcal{T}}$ in $\mathscr{T}^{(R)}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-054765 | MATHDB-331267

The heterogeneous infinite-server GRAND conjecture

摘录 / EXCERPT: Consider the heterogeneous infinite-server service system with server types s, packing-constraint sets ${\cal K}^s$, and total number of customers Z. Let ${\cal X}^*$ be the op...

conjecture | open | MathDB 页面 | 原文线索

V2-054766 | MATHDB-331268

The stationary convergence conjecture for finite-server GRAND-F

摘录 / EXCERPT: Suppose Assumption holds. Consider a sequence of finite-server systems under GRAND-F, indexed by r, and let $\boldsymbol{x}^r(\infty)$ denote the random state of the fluid-scaled...

conjecture | open | MathDB 页面 | 原文线索

V2-054767 | MATHDB-331290

Joint speed-entropy exponent conjecture for random walks on groups

摘录 / EXCERPT: Let β and δ be speed and entropy exponents for a random walk, so that the speed and entropy functions have forms $n^{\beta+o(1)}$ and $n^{\delta+o(1)}$, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-054768 | MATHDB-331328

The mixture representation conjecture for stationary gap distributions

摘录 / EXCERPT: Let the diffusion coefficients satisfy $\sigma_n=1$ for every $n\in\mathbb{Z}$, and assume the conditions of Theorem 2.1 hold. For $(a,b)\in\mathbb{R}^2$, define ... and let ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-054769 | MATHDB-331354

The unique-fixed-point conjecture for local belief propagation

摘录 / EXCERPT: Unique-fixed-point conjecture. If $mu=nu$, then this recursion has a unique fixed point for every

conjecture | open | MathDB 页面 | 原文线索

V2-054770 | MATHDB-331394

Vanishing probability of a Latin-square autotopism component

摘录 / EXCERPT: For $n>0$, let S_n be the symmetric group on n symbols, let $\mathrm{Atp}(n)$ denote the autotopism group of Latin squares of order n, and let $\mathbb{P}(n)$ be the probabil...

conjecture | open | MathDB 页面 | 原文线索

V2-054771 | MATHDB-331447

The unimodular density conjecture for monic rectangular polynomial matrices

摘录 / EXCERPT: Unimodular density conjecture. The probability that a uniformly random element of $M_{n,k}({\mathbb F}_q[x];m)$ is unimodular is given by

conjecture | open | MathDB 页面 | 原文线索

V2-054772 | MATHDB-331448

Noise sensitivity of bootstrap percolation for more than two neighbours

摘录 / EXCERPT: Noise sensitivity conjecture. For any $d greater than or equal to k > 2$, the event $\mathcal{C}_n$ is noise sensitive.

conjecture | open | MathDB 页面 | 原文线索

V2-054773 | MATHDB-331451

Non-free-infinite-divisibility conjectures for powers of semicircular variables

摘录 / EXCERPT: Let S have the centered semicircular distribution with variance parameter 1, written $S\sim\mathcal{S}(0,1)$. Let r be real. Semicircular power conjectures. The following ass...

open_problem | open | MathDB 页面 | 原文线索

V2-054774 | MATHDB-331544

Inversion invariance conjecture for full-plane CLE6 and k-arm IIC limits

摘录 / EXCERPT: Let $\mathbb{C}$ be the full plane, let μ_k denote the scaling limit of the k-arm IIC for $k\in\{1,2,4\}$, and consider inversion $z\mapsto 1/z$. Inversion-invariance conject...

conjecture | open | MathDB 页面 | 原文线索

V2-054775 | MATHDB-331545

Four-arm IIC winding-number variance and central-limit conjecture

摘录 / EXCERPT: Let $\mathcal{A}^{\eta}_4(\eta,1)$ be the four-arm event at mesh size η, let ν_4^{η} be the four-arm IIC measure, and let $\tilde{\theta}_{\eta}$ be the winding number...

conjecture | open | MathDB 页面 | 原文线索

V2-054776 | MATHDB-331565

Existence of a strong phase transition for the p-adic Ising-Vannimenus model

摘录 / EXCERPT: Strong phase-transition conjecture. Under these assumptions, the displayed intersection is nonempty.

conjecture | open | MathDB 页面 | 原文线索

V2-054777 | MATHDB-331581

Sharpness conjecture for the threshold of containing the origin

摘录 / EXCERPT: Sharpness conjecture. For every $\varepsilon>0$ there exist $\delta=\delta(\varepsilon)\in(0,1)$ and $C=C(\varepsilon)>0$ such that, for all $n<\mathrm{e}^{d/(u+\varepsilon)}$,

conjecture | open | MathDB 页面 | 原文线索

V2-054778 | MATHDB-331610

The universal crossing conjecture for square tilings

摘录 / EXCERPT: Universal crossing conjecture. There is a universal constant $c>0$ such that the probability of a black left-to-right crossing is greater than c.

open_problem | open | MathDB 页面 | 原文线索

V2-054779 | MATHDB-331713

Decelle et al.'s partial-recovery threshold conjecture for the stochastic block model

摘录 / EXCERPT: Throughout, let $G(n,a/n,b/n)$ be the two-community stochastic block model with two roughly equal-sized communities, within-community connection probability a/n, between-communit...

conjecture | open | MathDB 页面 | 原文线索

V2-054780 | MATHDB-331721

Correct-path survival conjecture for split-reduced successive cancellation list decoding

摘录 / EXCERPT: Correct-path survival conjecture. As $P_e(u_i)$ approaches zero, with high probability the current path survives at u_i without splitting and u_i is correctly decoded. If the s...

conjecture | open | MathDB 页面 | 原文线索

V2-054781 | MATHDB-331812

The free transportation inequality at coefficient one-half for the zero potential

摘录 / EXCERPT: Let $Q\equiv0$ on the circle, so that the equilibrium measure μ_Q is the Haar measure. Write $T(\rho)$ for the free transportation inequality ... for every probability measure...

conjecture | open | MathDB 页面 | 原文线索

V2-054782 | MATHDB-331871

The Dirichlet-weighted Arcsin conjecture for power semicircle distributions

摘录 / EXCERPT: Dirichlet-weighted Arcsin conjecture. The random variable S_n has the power semicircle distribution on $(-a,a)$ with $\lambda=\frac{n-1}{2}$ and density

conjecture | open | MathDB 页面 | 原文线索

V2-054783 | MATHDB-331948

Mean-field limit conjecture for the bias voter model on lattices

摘录 / EXCERPT: Let θ be the bias parameter, $\theta<\text{?}$, and let θ and θ be the parameters appearing in the bias voter model on $\d^d$; for $p\a0in (0,1)$ and $t>0$, co...

conjecture | open | MathDB 页面 | 原文线索

V2-054784 | MATHDB-331949

Weak-convergence conjecture for the bias voter model on trees and lattices

摘录 / EXCERPT: Let η_t be the bias voter model on S, where $S=\mathbb{T}^d$ or $S=\mathbb{Z}^d$ with $d\geq 1$, started from the product initial distribution μ_p with $p\in (0,1)$. Wea...

conjecture | open | MathDB 页面 | 原文线索

V2-054785 | MATHDB-331958

Furstenberg-Kesten conjecture on bigeodesics in two-dimensional first-passage percolation

摘录 / EXCERPT: Let (t_e) be i.i.d. edge weights on the nearest-neighbor edges of $\mathbb{Z}^2$, with a continuous distribution. A bigeodesic is a doubly infinite path whose finite segments are...

conjecture | open | MathDB 页面 | 原文线索

V2-054786 | MATHDB-332057

Uniqueness conjecture for the Brownian path decomposition

摘录 / EXCERPT: Uniqueness conjecture. If, for each $i\in\{1,2\}$, the union of K_i and all loops from BS intersecting K_i has the same law as the trace of Brownia...

conjecture | open | MathDB 页面 | 原文线索

V2-054787 | MATHDB-332065

Conjecture on non-Gaussian minimizers in the variational problems

摘录 / EXCERPT: Non-Gaussian minimizer conjecture. For every $p\in(2,\infty)$ and $w\in(-1,1)\setminus\{0\}$, the minimizer in is not μ_2. Similarly, for every $\gamma\in\mathcal{T}_p$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054788 | MATHDB-332077

The entropic form of the hyperplane conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and let $f^{(n)}$ be a log-concave probability density on $\mathbb{R}^{n}$. Let $D(f^{(n)})$ denote its relative entropy with respect to the Gaussian density h...

conjecture | open | MathDB 页面 | 原文线索

V2-054789 | MATHDB-332160

The update-stable zero-one conjecture for wired and free spanning forests

摘录 / EXCERPT: Update-stable zero-one conjecture. For every update-stable Borel set $\mathscr A\subseteq \{0,1\}^E$, the probability that $\mathfrak F$ belongs to $\mathscr A$ is either zero or o...

conjecture | open | MathDB 页面 | 原文线索

V2-054790 | MATHDB-332174

Decelle et al.'s computational impossibility conjecture below the Kesten-Stigum threshold

摘录 / EXCERPT: Decelle et al.'s computational impossibility conjecture. Irrespective of k, if $\operatorname{SNR}<1$, it is impossible to detect communities in polynomial time. This is the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-054791 | MATHDB-332262

Negative-binomial conjecture for the number of geodesic paths

摘录 / EXCERPT: Let $G(n,\pi)$ be the unit disk graph, and let two vertices be displaced by r_{xy}. Write $\sigma_{r_{xy}}$ for the number of geodesic paths between them. Negative-binomial conje...

conjecture | open | MathDB 页面 | 原文线索

V2-054792 | MATHDB-332264

The Birkhoff copula scaling-limit conjecture

摘录 / EXCERPT: A Birkhoff copula is a matrix with real entries satisfying the defining conditions of a discrete copula; equivalently, it corresponds to a doubly stochastic matrix. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054793 | MATHDB-332334

The exponential maximal-excursion tail control for random-environment diffusions

摘录 / EXCERPT: the exponential maximal-excursion tail control asserts

conjecture | open | MathDB 页面 | 原文线索

V2-054794 | MATHDB-332369

Gamma-shape conjecture for the condensation wave

摘录 / EXCERPT: Let $\Xi_t(f,g)$ denote the mass distribution of families with fitness in the interval determined by f and g, and let α, β, and γ be the model parameters. U...

conjecture | open | MathDB 页面 | 原文线索

V2-054795 | MATHDB-332375

Conjecture on the Plancherel limiting distribution of partition eigenvalues

摘录 / EXCERPT: Let $\kappa \vdash n$ be a partition with Plancherel measure, and let λ_{κ} be the quantity defined in with parameter $\alpha>0$. Let F_2 denote the distribution fun...

conjecture | open | MathDB 页面 | 原文线索

V2-054796 | MATHDB-332381

Hard-edge universality conjecture for products of two coupled real Gaussian matrices

摘录 / EXCERPT: Consider the real counterpart of the joint probability density function in the paper's equation (matrixpdf), with the same parameters and scalings as in Theorem (hardlimits). Real...

conjecture | open | MathDB 页面 | 原文线索

V2-054797 | MATHDB-332491

Faithfulness and inner faithfulness of the universal flat matrix representation

摘录 / EXCERPT: Let S_N^+ be the quantum permutation group, let X_N be the associated matrix-model space, and let ... be the universal $N\times N$ flat matrix representation. Flat matrix represe...

conjecture | open | MathDB 页面 | 原文线索

V2-054798 | MATHDB-332496

The inflection-point conjecture for the α-norm of a parametrized probability vector

摘录 / EXCERPT: Inflection-point conjecture. There exists an inflection point $\chi_n(\alpha)\in\bigl(H_{\text{\scriptsize \boldmath }v_n}(\mathrm{e}^{-n}),\ln n\bigr)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-054799 | MATHDB-332566

Generalized Lam-Williams positivity conjecture for multispecies TASEP components

摘录 / EXCERPT: Let $\boldsymbol\tau$ and $\boldsymbol\nu$ be the model parameters, let $\psi_{\bf w}(\boldsymbol\tau,\boldsymbol\nu)$ denote the component associated with a particle configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-054800 | MATHDB-332578

Uniform asymptotic kasha distribution on the cube

摘录 / EXCERPT: Let six dragons occupy the six faces of a cube, and let kasha be redistributed by the dragon fights described in the paper. A steady state is a kasha distribution that remains unch...

conjecture | open | MathDB 页面 | 原文线索

V2-054801 | MATHDB-332584

Localized Hölder contraction estimate for stochastic homogenization

摘录 / EXCERPT: Localized Hölder contraction estimate. For each $f\in C^{0,\beta}(\mathbb{R}^d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-054802 | MATHDB-332585

Exponential maximal-excursion estimate for diffusions in random environments

摘录 / EXCERPT: Fix $x\in\mathbb{R}^d$, $\omega\in\Omega$, and $n\geq 0$. Let $P_{y,\omega}$ denote the law of the diffusion started at y in environment ω, and let $X^*_{L_n^2}$ be its ma...

conjecture | open | MathDB 页面 | 原文线索

V2-054803 | MATHDB-332590

Conjecture on the scaling of the critical height

摘录 / EXCERPT: Let $ Gamma_n^\star$ denote the energy barrier and let $\mathbb{P}_n$ be the probability law of the random graph. Scaling conjecture for the critical height. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-054804 | MATHDB-332591

Conjecture on the limiting product of critical-set size and transition probability

摘录 / EXCERPT: Let $\mathcal{C}_n^\star$ denote the critical set of configurations, let $K_n^\star$ denote the associated quantity governing the transition from the critical set, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054805 | MATHDB-332686

The critical-window conjecture for range-R bond percolation

摘录 / EXCERPT: Let $V(R)$ be the number of potential range-R neighbors and let $p_c(R)$ be the critical bond-percolation probability. For dimensions $d=2$ or 3, let $\theta_c=\theta_c(d)>0$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-054806 | MATHDB-332765

Polynomial rejection under square-root rational approximation

摘录 / EXCERPT: Let ${\mathscr{S}}$ be a step multiset and let ${\textsc{H}}_{\theta}({\mathscr{S}})$ be a half-plane model, with $h_n(\theta)$ denoting the number of walks of length n in this m...

conjecture | open | MathDB 页面 | 原文线索

V2-054807 | MATHDB-332918

Perturbed Brownian-motion limit for recurrent ERW with Markovian cookie stacks

摘录 / EXCERPT: Let X_n be an excited random walk in a cookie environment with Markovian cookie stacks, and let θ and $\tilde\theta$ denote the associated parameters. A...

conjecture | open | MathDB 页面 | 原文线索

V2-054808 | MATHDB-332940

Wang-Madiman generalized Gaussian maximizer conjecture for Rényi entropy power

摘录 / EXCERPT: For $p>\frac{d}{d+2}$, define β_p by ... Let $Z^{(p)}$ be a random vector drawn from g_{β_p}. Let $X_1,\ldots,X_n$ be independent random vectors in $\mathbb{R}^d$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-054809 | MATHDB-333003

Edge-transitive graph conjecture for weak concentration and unique giant components

摘录 / EXCERPT: Edge-transitive graph conjecture. One can always rescale the edge-weight so that the background assumption holds; after such rescaling, both the weak concentration and unique giant...

conjecture | open | MathDB 页面 | 原文线索

V2-054810 | MATHDB-333004

Unique giant component implies weak concentration conjecture

摘录 / EXCERPT: Unique giant component implies weak concentration conjecture. Under the background assumption, the unique giant component property implies the weak concentration property.

conjecture | open | MathDB 页面 | 原文线索

V2-054811 | MATHDB-333077

Failure of the square-root pattern for the three-legged spider constant

摘录 / EXCERPT: Let C_n denote the spider constant for a spider with n legs, so that the explicitly computed values are $C_0=1$, $C_1=\sqrt{2}$, and $C_2=\sqrt{3}$. Failure of the square-root...

conjecture | open | MathDB 页面 | 原文线索

V2-054812 | MATHDB-333087

Relative mutual-information loss conjecture for shuffled stochastic block models

摘录 / EXCERPT: Let $(G_1,G_2)\sim \rho$-SBM$(K,\vec{n},b,\Lambda)$, with K and Λ fixed in n, and let $\boldsymbol{\sigma}$ be uniformly distributed on $\Pi(n)$. Relative information-l...

conjecture | open | MathDB 页面 | 原文线索

V2-054813 | MATHDB-333095

The determinantal-complexity lower-bound conjecture for the constructed family

摘录 / EXCERPT: Constructed-family determinantal lower-bound conjecture. If $a>0$ is small enough, then, with high probability, f_n cannot be expressed as a symbolic determinant of size at most...

conjecture | open | MathDB 页面 | 原文线索

V2-054814 | MATHDB-333098

Boundary-mass vanishing conjecture for the two-dimensional generalized divisible sandpile

摘录 / EXCERPT: Boundary-mass vanishing conjecture. The boundary mass satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054815 | MATHDB-333136

Sharp clustering conjecture for the radial spectral sample

摘录 / EXCERPT: Let S be the radial spectral sample associated with the one-arm event f_R, let $\alpha_1(R)$ denote the one-arm probability, and let $\alpha_4(r)$ and $\alpha_4(r,r_0)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-054816 | MATHDB-333137

Clustering conjecture for left-right crossing spectral samples

摘录 / EXCERPT: Let g_n be the indicator of the left-right crossing event of $[-n,n]^2$, and let S be its spectral sample. Let $\alpha_4(r)$ and $\alpha_4(r,n)$ denote the relevant four-arm pr...

conjecture | open | MathDB 页面 | 原文线索

V2-054817 | MATHDB-333164

Small free Stein discrepancy implies free transport to the semicircle law

摘录 / EXCERPT: Let $X=(x_1,\ldots,x_n)$ be an n-tuple with joint law μ_X, and let $A=A^\dagger$ be a free Stein kernel for X relative to the potential V_1. Write $\Sigma^*(X\mid V_1)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-054818 | MATHDB-333202

Monotonicity conjecture for geometric-scaled sums

摘录 / EXCERPT: Let ${\boldsymbol X} \equiv \{X_i\}$ be i.i.d. arbitrarily distributed random variables with mean λ, and let ... The preceding central limit theorem states that the p.m.f. of...

conjecture | open | MathDB 页面 | 原文线索

V2-054819 | MATHDB-333234

Exact simultaneous and sequential limits for the scaled LR statistic

摘录 / EXCERPT: Let $LR_{0,p,T}$ be Johansen's likelihood-ratio statistic for testing no cointegrating relationships, with $p,T\to\infty$ and $p/T\to c<1/2$ in the simultaneous regime. Scaled LR l...

conjecture | open | MathDB 页面 | 原文线索

V2-054820 | MATHDB-333344

Conjecture on fluctuation regimes for the proportion of heads

摘录 / EXCERPT: Fluctuation-regime conjecture. (i) If $E(N,2)=\mathcal{O}(1/N)$ as $N\to\infty$, then the proportion of heads obeys a central limit theorem. (ii) If $E(N,2)=\mathcal{O}(1/N)$ fails...

conjecture | open | MathDB 页面 | 原文线索

V2-054821 | MATHDB-333414

Universal bounded-degree graph conjecture

摘录 / EXCERPT: Universal bounded-degree graph conjecture. Let $\Delta\in\mathbb N$ and $p=\omega\left(n^{-1}\log^{1/\Delta}n\right)^{\frac{2}{\Delta+1}}$. Then, with high probability, $G_{n,p}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-054822 | MATHDB-333445

Lakshminarayan's real-eigenvalue conjecture for products of random matrices

摘录 / EXCERPT: Lakshminarayan's conjecture. The probability that A_n has all real eigenvalues tends to 1:

conjecture | open | MathDB 页面 | 原文线索

V2-054823 | MATHDB-333511

Cycle-lexicographic likelihood-order conjecture for the involution walk

摘录 / EXCERPT: Let S_n be the symmetric group, and consider the involution walk whose step distribution depends on a parameter p. Order the conjugacy classes by cycle lexicographic order. Cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-054824 | MATHDB-333512

Alternating-sum separation bound for the involution walk at p = 1/2

摘录 / EXCERPT: Let n be odd, let $p=\frac{1}{2}$, and let $\operatorname{sep}(t)$ denote the separation distance of the involution walk after t steps. Separation-distance conjecture. ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-054825 | MATHDB-333518

Integrality conjecture for time-delay matrix expansion coefficients

摘录 / EXCERPT: Let $\tau_{k,g}^{(\beta)}$ denote the coefficients in the large-N expansion of the time-delay matrix moments, with symmetry parameter $\beta\in\{1,2\}$ and indices k and g. I...

conjecture | open | MathDB 页面 | 原文线索

V2-054826 | MATHDB-333539

Rumanov's boundary-value conjecture for the beta-six Bloemendal-Virág equation

摘录 / EXCERPT: Rumanov's boundary-value conjecture. The function above determines the solution of the boundary-value problem for the Bloemendal-Virág equation if (q_2,α) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054827 | MATHDB-333553

Eulerian percolation threshold conjecture on the square lattice

摘录 / EXCERPT: Let $p_{c,\mathrm{even}}$ denote the percolation threshold for Eulerian percolation on $\mathbb{Z}^2$, let μ_p be the corresponding Eulerian percolation measure, let...

conjecture | open | MathDB 页面 | 原文线索

V2-054828 | MATHDB-333591

Concentration conjecture for sampling the sphere by mutually orthogonal subspaces

摘录 / EXCERPT: Let $A\subseteq S^{n-1}$ be a measurable subset with $\sigma_{n-1}(A)\geq e^{-c\sqrt{n}}$. For a subspace $H\subseteq\mathbb{R}^n$ of dimension $\lfloor n/2\rfloor$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-054829 | MATHDB-333617

Bordenave-Caputo-Chafaï-Tikhomirov conjecture on the spectral radius without a fourth moment

摘录 / EXCERPT: Bordenave-Caputo-Chafaï-Tikhomirov conjecture. The convergence in probability

conjecture | open | MathDB 页面 | 原文线索

V2-054830 | MATHDB-333644

The chi-square characterization conjecture for the normal distribution

摘录 / EXCERPT: Let $X_1,X_2,\dots,X_n$ be non-degenerate, independently and identically distributed classical random variables with finite non-zero variance σ^2, and let $\overline{X}$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-054831 | MATHDB-333701

First-order statistics determine the complete modeling theory

摘录 / EXCERPT: Let $\mathbf A$ be a modeling, and for every first-order formula ϕ let $\langle\phi,\mathbf A\rangle$ denote its satisfaction probability in $\mathbf A$. Modeling-theory conje...

conjecture | open | MathDB 页面 | 原文线索

V2-054832 | MATHDB-333706

The Airy process limit for the L-shaped Aztec diamond

摘录 / EXCERPT: Let X_N^R denote the top line of an Aztec diamond of size N with a square removed from the top so that its lower tip is at height $N/\sqrt{2}+2^{-5/6}N^{1/3}R$. Define ... For an...

conjecture | open | MathDB 页面 | 原文线索

V2-054833 | MATHDB-333707

The extended-kernel limit for the pentagonal Aztec diamond

摘录 / EXCERPT: Let X_N^R denote the top line of an Aztec diamond of size N with a triangle removed from the top corner at height $N/\sqrt{2}+2^{-5/6}N^{1/3}R$. Define ... For given observation...

conjecture | open | MathDB 页面 | 原文线索

V2-054834 | MATHDB-333743

Asymptotic normality conjecture for distinct-part core-partition sizes

摘录 / EXCERPT: For each positive integer s, let X_s be the size of a uniformly chosen $(s,s+1)$-core partition with distinct parts. Let $\mathbb{E}[X_s]$ and $\operatorname{Var}(X_s)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-054835 | MATHDB-333763

The greedy-Haar thinning discrepancy conjecture

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and let $\beta\in(0,1]$. A greedy-Haar $(1+\beta)$-thinning strategy produces a sequence $Z=(Z_n)_{n\geq 1}$, and write $Z^n=(Z_1,\ldots,Z_n)$ for its first...

conjecture | open | MathDB 页面 | 原文线索

V2-054836 | MATHDB-333767

Benjamini-Lyons-Peres-Schramm conjecture on bushes along a uniform spanning forest ray

摘录 / EXCERPT: Let T_{0} be the component of the origin in the uniform spanning forest on $\mathbb{Z}^{d}$, and let ξ be the unique infinite ray from 0 in T_{0}. The Benjamini-Lyons-Per...

conjecture | open | MathDB 页面 | 原文线索

V2-054837 | MATHDB-333780

Conjecture on optimal quantizers for the uniform distribution on a disc

摘录 / EXCERPT: Let the unit disc be equipped with the uniform probability distribution. For an optimal set of n-means, consider the associated Voronoi regions. Conjecture on optimal quantizers...

conjecture | open | MathDB 页面 | 原文线索

V2-054838 | MATHDB-333781

Conjecture on optimal quantizers for the uniform distribution on a square

摘录 / EXCERPT: Let the unit square be equipped with the uniform probability distribution, and let an optimal set of n-means have associated quantizer points. Conjecture on optimal quantizers fo...

conjecture | open | MathDB 页面 | 原文线索

V2-054839 | MATHDB-333804

Steiner symmetrization conjecture for the largest random angle

摘录 / EXCERPT: Let K be a convex subset of ${\mathbb R}^d$, let u be a unit vector in ${\mathbb R}^d$, and let S_uK denote the Steiner symmetrization of K along u. For...

conjecture | open | MathDB 页面 | 原文线索

V2-054840 | MATHDB-333872

Stability conjecture for the Brascamp-Lieb and reverse Brascamp-Lieb inequalities

摘录 / EXCERPT: Let f be an even probability density on $\mathbb{R}$ with variance 1, and let ... be the standard normal density. Let μ be an even isotropic measure on S^{n-1} supported at...

conjecture | open | MathDB 页面 | 原文线索

V2-054841 | MATHDB-333916

Vanishing conjecture for the hard-edge tacnode kernel

摘录 / EXCERPT: Let $\alpha>-1$, let $x,y\in\mathbb{R}_+$ be fixed, and let $\tau\in\mathbb{R}$ be fixed. The function $K^{\operatorname{tac},(\alpha)}(x,y;s,\tau)$ is the hard-edge tacnode kernel...

conjecture | open | MathDB 页面 | 原文线索

V2-054842 | MATHDB-333955

Grimmett's conjecture on infinite directed paths in biased square-lattice orientations

摘录 / EXCERPT: For $0 \leq p \leq 1$, let $\vec{\mathbb{Z}}^2(p)$ be the random orientation of the square lattice in which each horizontal edge is oriented rightwards with probability p and lef...

conjecture | open | MathDB 页面 | 原文线索

V2-054843 | MATHDB-333967

Monotonicity conjecture for avoiding increasing patterns in the Mallows distribution

摘录 / EXCERPT: Let m be a positive integer, let n be a positive integer, and let $P_n(12\dots m,q)$ denote the probability that a permutation of size n drawn from the Mallows distribution w...

conjecture | open | MathDB 页面 | 原文线索

V2-054844 | MATHDB-333983

Determinantal formula for the largest-root distribution of non-central Wishart matrices

摘录 / EXCERPT: Determinantal formula. There is a function $C(x)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054845 | MATHDB-334011

Decelle-Krzakala-Moore-Zdeborová conjecture for the planted partition model

摘录 / EXCERPT: Decelle-Krzakala-Moore-Zdeborová conjecture. If $(a-b)^2>2(a+b)$, one can almost surely find a bisection positively correlated with the original clusters; if $(a-b)^2<2(a+b)$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-054846 | MATHDB-334014

Weighted-average tail conjecture for independent random variables

摘录 / EXCERPT: Let X and Y be independent random variables on $\mathbb{R}_+$ that are not compactly supported, and let $\lambda\in(0,1)$. Weighted-average tail conjecture. ... This conjecture i...

conjecture | open | MathDB 页面 | 原文线索

V2-054847 | MATHDB-334093

Conjecture for moments of the partial Hadamard product

摘录 / EXCERPT: Let $\mathcal{H}_{2g+1}$ be the set of monic square-free polynomials of degree $2g+1$ over $\mathbb{F}_q[x]$, and write ... For the partial Hadamard product, set...

conjecture | open | MathDB 页面 | 原文线索

V2-054848 | MATHDB-334152

Kozma's rough-isometry conjecture for L_∞ mixing times

摘录 / EXCERPT: Let G and H be finite graphs that are K-roughly isometric, and suppose both have maximal degree at most d. Denote by $\tau_{\infty}(G)$ and $\tau_{\infty}(H)$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-054849 | MATHDB-334156

Sankar-Teng-Spielman least singular value conjecture for subgaussian random matrices

摘录 / EXCERPT: Let ξ be a mean-zero, variance-at-least-1, subgaussian random variable, and let N_n be an $n\times n$ random matrix with iid entries distributed as ξ. The least singula...

conjecture | open | MathDB 页面 | 原文线索

V2-054850 | MATHDB-334197

The critical-density recurrence conjecture for the frog model on non-amenable graphs

摘录 / EXCERPT: Let G be a connected non-amenable graph of bounded degree, and let λ be the frog-model particle density. Critical-density recurrence conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-054851 | MATHDB-334256

Long-term behavior dichotomy for branching diffusion with interactions

摘录 / EXCERPT: Long-term behavior dichotomy. The following dichotomy holds:

conjecture | open | MathDB 页面 | 原文线索

V2-054852 | MATHDB-334281

Fixed-point and spinodal-curve conjecture for asymmetric community detection

摘录 / EXCERPT: Fixed-point and spinodal-curve conjecture. (i) If $\lambda>1$, then G has two fixed points, 0 and $\alpha>0$; moreover, 0 is unstable and α is stable. (ii) For...

conjecture | open | MathDB 页面 | 原文线索

V2-054853 | MATHDB-334295

The odd periodic highest-coefficient conjecture for the loop model

摘录 / EXCERPT: Let $F_{2n+1}(x)$ be the generating polynomial associated with the odd-sized periodic ground state, let ${\rm A}_n$ denote the number of $n\times n$ alternating sign matrices, and...

conjecture | open | MathDB 页面 | 原文线索

V2-054854 | MATHDB-334296

The odd periodic constant-term conjecture for the loop model

摘录 / EXCERPT: Let $F_{2n+1}(x)$ be the generating polynomial for the odd-sized periodic ground state, and let ${\rm AHT}_{m}$ denote the corresponding alternating-sign-triangle sum rule. Odd per...

conjecture | open | MathDB 页面 | 原文线索

V2-054855 | MATHDB-334297

The odd periodic special-value conjecture for the loop model

摘录 / EXCERPT: Let $F_{2n+1}(x)$ be the generating polynomial associated with the odd-sized periodic ground state, and let ${\rm AV}_{2n+1}$ and ${\rm AHT}_{2n+1}$ denote the corresponding altern...

conjecture | open | MathDB 页面 | 原文线索

V2-054856 | MATHDB-334308

Benjamini's conjecture on infinite friend clusters in finite time

摘录 / EXCERPT: Let G be an infinite connected graph of bounded degree. Assume that for some $0<p<1$, Bernoulli bond percolation on G with survival probability p has an infinite connected co...

conjecture | open | MathDB 页面 | 原文线索

V2-054857 | MATHDB-334337

The structured Gaussian matrix norm conjecture

摘录 / EXCERPT: Let $X=(X_{ij})_{1\leq i,j\leq n}$ be the centered Gaussian random matrix considered in this section, with entry standard deviations b_{ij}, and assume without loss of generality...

conjecture | open | MathDB 页面 | 原文线索

V2-054858 | MATHDB-334338

Latała's conjecture on the norm of structured random matrices

摘录 / EXCERPT: Let $X=(X_{ij})_{1\leq i,j\leq n}$ be the random matrix in the setting of the section, with independent centered Gaussian entries as specified there, and let $\|X\|$ denote its ope...

conjecture | open | MathDB 页面 | 原文线索

V2-054859 | MATHDB-334339

The noncommutative Gaussian norm conjecture

摘录 / EXCERPT: Let $X=\sum_{k=1}^s g_kA_k$ be the Gaussian matrix model from the noncommutative Khintchine theorem, where g_k are independent standard Gaussian variables and A_k are the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-054860 | MATHDB-334375

One-survivor conjecture for multitype growth on the square lattice

摘录 / EXCERPT: For each integer $s\geqslant2$, the s-type growth model on ${\mathbb Z}^2$ allows s colours, and an uncoloured site is infected at rate 0, 1, or ∞ by a colour accord...

conjecture | open | MathDB 页面 | 原文线索

V2-054861 | MATHDB-334376

One-survivor conjecture for multitype urns on cycles

摘录 / EXCERPT: Let C_k be the cycle graph of length k, and consider the s-type urn model on C_k, with balls of s colours and each urn monochromatic. At each step, a uniformly chosen bal...

conjecture | open | MathDB 页面 | 原文线索

V2-054862 | MATHDB-334377

The r-neighbour threshold conjecture for multitype growth

摘录 / EXCERPT: For integers $d,r,s\geqslant2$, define the r-neighbour s-type growth model on ${\mathbb Z}^d$ by assigning an uncoloured site colour i at rate 0 when it has no neighbours o...

conjecture | open | MathDB 页面 | 原文线索

V2-054863 | MATHDB-334387

Strong-forcing invariant-measure conjecture for the stochastic Ginzburg-Landau equations

摘录 / EXCERPT: Consider a stochastically forced version of the full Gor'kov-Eliashberg equations ... For each $\varepsilon>0$, let the forcing be strong enough that the resulting system admits an i...

conjecture | open | MathDB 页面 | 原文线索

V2-054864 | MATHDB-334469

Witness-tree probability conjecture for fully orderable witness trees

摘录 / EXCERPT: A witness tree is built from bad-event labels, with children of a node labeled B required to have distinct labels $B_1,\dots,B_s$ whose set is orderable to B. For a tree-struct...

conjecture | open | MathDB 页面 | 原文线索

V2-054865 | MATHDB-334489

Benjamini's conjecture on the asymptotic Cheeger constant of the giant percolation component

摘录 / EXCERPT: Let $d\geq 2$ and $p>p_c(d)$. For the giant component $\textbf{C}_n$ of supercritical bond percolation in $[-n,n]^d$, let $\Phi_{\textbf{C}_n}$ denote its Cheeger constant. Benjami...

conjecture | open | MathDB 页面 | 原文线索

V2-054866 | MATHDB-334500

The SOS-Bernoulli Ferrari-Spohn scaling conjecture

摘录 / EXCERPT: Fix $\delta>0$ and $c>0$, and consider the probability distribution $\mathbb Q_{\beta,\delta,c,N}^{+,n}$ on n ordered level-line paths $\eta_1,\ldots,\eta_n$ with weights combini...

conjecture | open | MathDB 页面 | 原文线索

V2-054867 | MATHDB-334633

Surjectivity conjecture for rectangular random integer matrices

摘录 / EXCERPT: Rectangular surjectivity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-054868 | MATHDB-334678

Rivin's conjecture on injectivity radii of random mapping tori

摘录 / EXCERPT: Let S be a closed connected orientable surface of genus at least 2, let $\operatorname{Mod}(S)$ be its mapping class group, and let (w_n) be an appropriate random walk on...

conjecture | open | MathDB 页面 | 原文线索

V2-054869 | MATHDB-334736

The first-order exit-distribution conjecture for symmetric smart kinetic walks

摘录 / EXCERPT: Let D be a simply connected domain containing the origin, and let δ be the lattice spacing of the square lattice. Let $\mu_\delta(0,|dz|;D)$ be the discrete exit distribut...

conjecture | open | MathDB 页面 | 原文线索

V2-054870 | MATHDB-334777

Rivest's optimality conjecture for correlated sampling

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathcal{U}(A)$ denote the uniform distribution on a finite set A, and let (f,g) be probabilistic strategies satisfying the correctness condition...

conjecture | open | MathDB 页面 | 原文线索

V2-054871 | MATHDB-334801

Hodge-GUE correspondence for the even GUE partition function

摘录 / EXCERPT: Hodge-GUE correspondence. The even GUE partition function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054872 | MATHDB-334833

Conjecture on random Kneser graph chromatic number

摘录 / EXCERPT: Let $KG_{n,k}$ be the Kneser graph on the k-element subsets of $[n]$, and let $KG_{n,k}(p)$ be its random subgraph obtained by retaining each edge independently with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-054873 | MATHDB-334907

Stochastic ordering conjecture for needy patients in Erlang-R and closed ward models

摘录 / EXCERPT: Let $Q_1^b(\infty)$, $Q_1^h(\infty)$ and $Q_1^c(\infty)$ denote the stationary numbers of needy patients in the Erlang-R models with blocking and holding and in the closed ward mod...

conjecture | open | MathDB 页面 | 原文线索

V2-054874 | MATHDB-334948

Non-constant positive harmonic functions on parabolic unimodular random maps

摘录 / EXCERPT: The positive harmonic functions conjecture. Then M does not admit any non-constant positive harmonic functions almost surely.

conjecture | open | MathDB 页面 | 原文线索

V2-054875 | MATHDB-334968

Conjecture on null boundaries of limiting colored regions

摘录 / EXCERPT: Let t be a time, let $\Xi_{\le t}$ be the set of particles arriving by time t, and let $A(t,\xi)$ denote the measurable region associated with $\xi\in\Xi_{\le t}$. The regions...

conjecture | open | MathDB 页面 | 原文线索

V2-054876 | MATHDB-335006

The volume-tail conjecture for critical percolation clusters on the UIPT

摘录 / EXCERPT: Let $\mathcal{C}$ be the cluster of the origin for a critical site percolation on the uniform infinite planar triangulation (UIPT), and let $|\mathcal{C}|$ denote its number of ver...

conjecture | open | MathDB 页面 | 原文线索

V2-054877 | MATHDB-335007

Surjectivity implies bijectivity for PSOs

摘录 / EXCERPT: Let a PSO be a polynomial stochastic operator on the simplex. Surjectivity-bijectivity conjecture. Any surjective PSO is bijective. The statement proposes that surjectivity alone f...

conjecture | open | MathDB 页面 | 原文线索

V2-054878 | MATHDB-335140

Variable-length feedback coding error-exponent lower-bound conjecture

摘录 / EXCERPT: Consider a variable-length feedback code with $M=2^K$ messages, target error probability P_e, expected decoding time $\mathbb{E}[T]$, rate $\overline{R}$, channel capacity C, f...

conjecture | open | MathDB 页面 | 原文线索

V2-054879 | MATHDB-335171

The equal-radius simplex conjecture for Gaussian error probability

摘录 / EXCERPT: Let $P(v_1,\ldots,v_N)=\int_{\mathbb R^n}\max_i e^{-|x-v_i|^2}\,dx$ for a code consisting of N vectors in $\mathbb R^n$. The simplex conjecture. For $N=n+1$ and fixed...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-054880 | MATHDB-335240

Monotonicity conjecture for the speed of once-reinforced walk on regular trees

摘录 / EXCERPT: Let d be fixed, and consider the once-reinforced walk on a d-regular tree with forward-edge weights u_0 before reinforcement and u_1 after reinforcement. Monotonicity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-054881 | MATHDB-335319

Two-scale asymptotic normality for distinct-part core-partition sizes

摘录 / EXCERPT: Two-scale normality conjecture. For each fixed n, the distribution of $X_{d,n}$ is not asymptotically normal; in fact, $(X_{d,n}-\mu_{d,n})/\sigma_{d,n}$ tends to some abnormal d...

conjecture | open | MathDB 页面 | 原文线索

V2-054882 | MATHDB-335321

Asymptotic normality for fixed d

摘录 / EXCERPT: Fixed-d normality conjecture. For fixed d, the distribution of $X_{d,n}$ is asymptotically normal. That is,

conjecture | open | MathDB 页面 | 原文线索

V2-054883 | MATHDB-335401

Global infinite-oscillation conjecture for parabolic SPDEs

摘录 / EXCERPT: Global infinite-oscillation conjecture. There exist such correlation functions f for which

conjecture | open | MathDB 页面 | 原文线索

V2-054884 | MATHDB-335430

Spatial asymptotic conjecture for the stochastic heat equation with compactly supported initial data

摘录 / EXCERPT: Spatial asymptotic conjecture. In the particular case of space-time white noise,

conjecture | open | MathDB 页面 | 原文线索

V2-054885 | MATHDB-335449

Ising crossing-probability conjecture for alternating boundary conditions

摘录 / EXCERPT: Ising crossing-probability conjecture. The crossing probabilities converge according to

conjecture | open | MathDB 页面 | 原文线索

V2-054886 | MATHDB-335490

Hypergeometric SLE reversibility conjecture

摘录 / EXCERPT: Let $\kappa\in(0,8)$, let $\nu>(-4)\vee(\kappa/2-6)$, and let $(\Omega;x_1,x_2,x_3,x_4)$ be a quad. Let η be an $\operatorname{hSLE}_{\kappa}(\nu)$ in Ω from x_1 to...

conjecture | open | MathDB 页面 | 原文线索

V2-054887 | MATHDB-335565

Uniform finite-dimensional invariance implies infinite-dimensional invariance

摘录 / EXCERPT: Uniform finite-dimensional invariance conjecture. The ε-invariance of all finite-dimensional distributions with a uniform δ should suffice for invariance with re...

conjecture | open | MathDB 页面 | 原文线索

V2-054888 | MATHDB-335572

Extinction and convergence conjecture for stochastic Lotka-Volterra food chains

摘录 / EXCERPT: Extinction and convergence conjecture. The predators $(X_{j^*+1},\ldots,X_n)$ go extinct, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-054889 | MATHDB-335580

Feige's lower-tail probability conjecture

摘录 / EXCERPT: Let $n \in \mathbb{Z}_{\ge 1}$, let $\mu_1 \ge \dotsc \ge \mu_n > 0$ satisfy ... and let $X_1,\dotsc,X_n$ be independent non-negative random variables with $EX_i=1$ for every i. De...

conjecture | open | MathDB 页面 | 原文线索

V2-054890 | MATHDB-335607

Phase transition conjecture for PCA subspace and score recovery

摘录 / EXCERPT: Let $n,d\to\infty$ with $n/d\to c>0$ and $n_\ell/n\to p_\ell$ for $\ell\in\{1,\dots,L\}$. Let A and β_i be the quantities defined in the paper, and let $\hat{u}_i$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-054891 | MATHDB-335699

Mazur-Rubin's Gaussian distribution conjecture for normalized modular symbols

摘录 / EXCERPT: Mazur-Rubin's Gaussian distribution conjecture. The data above has limit the standard normal distribution. The conjecture predicts a central limit law for normalized raw modular sy...

conjecture | open | MathDB 页面 | 原文线索

V2-054892 | MATHDB-335708

The Kesten-Stigum conjecture for symmetric stochastic block models

摘录 / EXCERPT: Kesten-Stigum conjecture. For any $k\geq 2$, if $\mathrm{SNR}>1$, it is possible to detect communities in polynomial time. If $k\geq 4$, it is possible to detect communities inform...

conjecture | open | MathDB 页面 | 原文线索

V2-054893 | MATHDB-335709

The polynomial-time barrier below the Kesten-Stigum threshold

摘录 / EXCERPT: Let $k\in\mathbb{Z}_+$, let $p\in(0,1)^k$ be a probability distribution, and let Q be a symmetric $k\times k$ matrix with nonnegative entries. Let P be the diagonal matrix with...

conjecture | open | MathDB 页面 | 原文线索

V2-054894 | MATHDB-335743

Large deviation principle for averages of Jordan projections

摘录 / EXCERPT: Jordan-projection LDP conjecture. The sequence $\frac{1}{n}\lambda(Y_n)$ satisfies a large deviation principle with the same rate function $I:\mathfrak{a}^{+}\to[0,\infty]$ as the...

conjecture | open | MathDB 页面 | 原文线索

V2-054895 | MATHDB-335773

The scout-number conjecture for effective exploration of integer grids

摘录 / EXCERPT: The scout-number conjecture. For $d\geq 3$, any effective scout protocol on $\mathbb{Z}^d$ requires at least $d+1$ scouts.

conjecture | open | MathDB 页面 | 原文线索

V2-054896 | MATHDB-335781

Ball-Nayar-Tkocz-Madiman-Melbourne-Xu norm conjecture for Rényi entropy

摘录 / EXCERPT: Recall that a probability measure on $\mathbb{R}^n$ is κ-concave if it satisfies the Brunn-Minkowski-type inequality associated with κ; a random vector is κ-co...

conjecture | open | MathDB 页面 | 原文线索

V2-054897 | MATHDB-335844

Sharp covering-decay conjecture for short paths

摘录 / EXCERPT: Sharp covering-decay conjecture. There are an $\epsilon>0$ and a $\hat P_d\in(0,1)$ such that, for every $d\ge4$ and every such path satisfying $|\mathcal{P}|\le N^{1+\epsilon}$,

conjecture | open | MathDB 页面 | 原文线索

V2-054898 | MATHDB-335845

Axis-minimization conjecture for monotonic path covering

摘录 / EXCERPT: Axis-minimization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054899 | MATHDB-335846

Strong Hall's conjecture for triangle-shape distributions

摘录 / EXCERPT: Let S be a convex region, let $F_S(\phi)$ denote the distribution function for the shape of a random triangle formed by three points chosen from S, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-054900 | MATHDB-335902

Subexponential end time conjecture for diffusion limited aggregation in the Boolean lattice

摘录 / EXCERPT: Let t_{end} denote the time at which the diffusion limited aggregation process reaches its terminal stage in the Boolean lattice on n elements. Subexponential end time...

conjecture | open | MathDB 页面 | 原文线索

V2-054901 | MATHDB-335907

Extended general structure conjecture for the projected drift

摘录 / EXCERPT: Extended general structure conjecture. The projected coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054902 | MATHDB-335973

Symmetric-group monochromatic eigenvalue statistics conjecture

摘录 / EXCERPT: Symmetric-group eigenvalue statistics conjecture. As $n\to\infty$, for almost all choices of the generators, the eigenvalue distribution converges weakly to the Kesten-McKay law; t...

conjecture | open | MathDB 页面 | 原文线索

V2-054903 | MATHDB-335992

Rademacher polynomial linear-factor conjecture

摘录 / EXCERPT: Let $g_{\pm1,d}(x)=x^d+a_{d-1}x^{d-1}+\dots+a_0$ be a monic polynomial of odd positive degree d, where each a_i is independently $+1$ or -1 with probability $\frac{1}{2}$. Ra...

conjecture | open | MathDB 页面 | 原文线索

V2-054904 | MATHDB-335993

Irreducibility conjecture for random Rademacher polynomials

摘录 / EXCERPT: Let $g_{\pm1,d}(x)=x^d+a_{d-1}x^{d-1}+\dots+a_0$ be a monic polynomial of positive degree d, where each a_i is independently $+1$ or -1 with probability $\frac{1}{2}$. Radema...

conjecture | open | MathDB 页面 | 原文线索

V2-054905 | MATHDB-335994

Binomial-coefficient random polynomial reducibility conjecture

摘录 / EXCERPT: Let $h_{\mathrm{Bin},K,d}(x)=x^d+a_{d-1}x^{d-1}+\dots+a_1x+a_0$ be a random monic polynomial whose coefficients $a_0,\dots,a_{d-1}$ are independently chosen from $[0,K]$ according...

conjecture | open | MathDB 页面 | 原文线索

V2-054906 | MATHDB-336000

Exponential decay of the covering probability for three-dimensional random walk

摘录 / EXCERPT: Let $X_{3,n}$ be the three-dimensional simple random walk, and let $[... truncated...$ be a nearest-neighbor path in $[... truncated...$ connecting 0 to $[... truncated...$. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-054907 | MATHDB-336001

Exponential covering bound for the embedded one-dimensional random walk

摘录 / EXCERPT: Let $X_{3,n}=(X_{3,n}^1,X_{3,n}^2,X_{3,n}^3)$ be the three-dimensional simple random walk. Define the successive return times $[... truncated...$ of the projected walk...

conjecture | open | MathDB 页面 | 原文线索

V2-054908 | MATHDB-336008

Conjecture on the size of a critical percolation cluster in a Boltzmann triangulation

摘录 / EXCERPT: Let $T^{(\ell)}$ be a critical random Boltzmann triangulation of the ℓ-gon, with its simple boundary colored black, and let $\mathfrak{C}^{(\ell)}$ be the boundary cluster of...

conjecture | open | MathDB 页面 | 原文线索

V2-054909 | MATHDB-336009

Conjecture on the size of the largest critical percolation cluster in a uniform triangulation

摘录 / EXCERPT: Let T_n be a uniform triangulation with n faces, and perform a critical site or bond percolation on T_n. For random variables, write $X_n\approx n^{\alpha}$ when, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-054910 | MATHDB-336024

Critical conditional geometric-limit conjecture for the Derrida-Retaux recursion

摘录 / EXCERPT: Conditional geometric-limit conjecture. Conditionally on $X_n\ne0$, the random variable X_n converges weakly to a random variable Y_∞ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-054911 | MATHDB-336025

Critical moment-generating-function asymptotics conjecture for the Derrida-Retaux recursion

摘录 / EXCERPT: Critical generating-function asymptotics conjecture. As $n\to\infty$, for $s\in(0,m)$ with $s\ne1$,

conjecture | open | MathDB 页面 | 原文线索

V2-054912 | MATHDB-336050

Backbone scaling-limit conjecture for the IIC random walk

摘录 / EXCERPT: Recall that $(\mathcal{B}_k)_{k\in\mathbb{N}}$ are the branches of the IIC, and that $E_{\mathcal{B}_k}:\mathcal{B}_k\to\ell_1$ are isometric embeddings. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-054913 | MATHDB-336056

DLA growth exponent conjecture for lattice aggregates

摘录 / EXCERPT: DLA growth exponent conjecture. There exists $\tfrac{1}{d}<\alpha<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-054914 | MATHDB-336057

Strictly superminimal growth of DLA radius on lattices

摘录 / EXCERPT: DLA superminimal growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054915 | MATHDB-336109

Mean-dominates-standard-deviation conjecture for the FK coupling time

摘录 / EXCERPT: Mean-dominates-standard-deviation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054916 | MATHDB-336110

Gumbel limiting-distribution conjecture for the FK coupling time

摘录 / EXCERPT: Gumbel limiting-distribution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054917 | MATHDB-336201

Theorem on the scaling limit for the pushed front

摘录 / EXCERPT: Let $\widehat{R}$ denote the front of the modified system in which, when multiple absorptions are possible, exactly one particle is absorbed, the front advances one step, and all e...

conjecture | open | MathDB 页面 | 原文线索

V2-054918 | MATHDB-336216

Constancy conjecture for the long-range percolation asymptotic function

摘录 / EXCERPT: Constancy conjecture. The function $\mathit\phi$ above is constant for each $\beta>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-054919 | MATHDB-336270

Conjecture on the clique number in the intermediate regime

摘录 / EXCERPT: Let G_{n,r_n} be the split-and-drift random graph, let $_n$ denote its parameter, and let κ_n be its clique number. In the intermediate regime, $r_n \to +\infty$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-054920 | MATHDB-336315

Conjecture on explosive distances in scale-free percolation

摘录 / EXCERPT: Let $d_L(x,y)$ denote the L-distance between vertices $x,y\in\mathbb{Z}^d$, and let V_x be the explosion time from x. Under the condition of the distance theorem, these quant...

conjecture | open | MathDB 页面 | 原文线索

V2-054921 | MATHDB-336316

Double-exponential escape conjecture for shortest exploding paths

摘录 / EXCERPT: Let $\gamma_{0}^\star=(0,v_1^\star,v_2^\star,\dots):=\arg\min\{\lvert\gamma_L\rvert\}$ be the shortest exploding path starting at 0, where $\lvert\gamma_L\rvert$ denotes its L-...

conjecture | open | MathDB 页面 | 原文线索

V2-054922 | MATHDB-336360

Critical Hawkes process existence and uniqueness conjecture

摘录 / EXCERPT: Let F be an absolutely continuous distribution function with $F(0)=0$ and density f, let $\tilde{F}$ be the symmetrized version of F, and let $\lambda>0$. A critical Hawkes p...

conjecture | open | MathDB 页面 | 原文线索

V2-054923 | MATHDB-336362

Palm construction conjecture for critical Hawkes processes

摘录 / EXCERPT: Let F be a displacement distribution and let N_0 be the random counting measure constructed from the renewal and random-walk occupation measures in the paper, with distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-054924 | MATHDB-336455

The susceptibility upper-bound conjecture for anisotropic percolation

摘录 / EXCERPT: Consider a bond percolation model on $\mathbb{Z}^d\times\mathbb{Z}^{s}$ with parameters (p,q), where $p<p_c(d)$. Let $q_c(p)$ be the critical value of q, and let $\chi_d(p)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-054925 | MATHDB-336456

Classification conjecture for finite irreducible affine processes without autonomous components

摘录 / EXCERPT: Classification conjecture. Every such process is, up to an invertible affine transformation, the process constructed in Example, allowing λ to take values in $[0,\infty)$ r...

conjecture | open | MathDB 页面 | 原文线索

V2-054926 | MATHDB-336468

Existence of arbitrarily many metastable critical points of g_{ξ}

摘录 / EXCERPT: For integers $m,n_1,\dots,n_m$ and parameters $\nu_{1}^{(1)},\dots,\nu_{n_1}^{(1)},\nu_{1}^{(2)},\dots,\nu_{n_2}^{(2)},\dots,\nu_{n_m}^{(m)}$, consider the functions g_{ξ} aris...

conjecture | open | MathDB 页面 | 原文线索

V2-054927 | MATHDB-336495

Stochastic ordering of needy patients in Erlang-R and closed ward models

摘录 / EXCERPT: Let $Q^b_1(\infty)$, $Q_1^h(\infty)$ and $Q_1^J(\infty)$ denote the stationary number of needy patients in the Erlang-R model with blocking, holding and the closed ward, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-054928 | MATHDB-336524

Mild random-walk conjecture for one-factorizations

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $\mathcal{G}_n$ be the graph whose vertices are all colorings of $E(K_n)$ with colors ${1,\ldots,n-1}$, with two colorings...

conjecture | open | MathDB 页面 | 原文线索

V2-054929 | MATHDB-336528

Critical percolation has no infinite cluster

摘录 / EXCERPT: Let β_c denote the critical parameter for Bernoulli percolation on $\mathbb{Z}^d$, and let $\mathbb{P}_{p_c}$ be the corresponding critical percolation measure. Critical non-...

conjecture | open | MathDB 页面 | 原文线索

V2-054930 | MATHDB-336548

Constant-community approximation conjecture for subgraph-counting functions

摘录 / EXCERPT: Constant-community approximation conjecture. There is a constant c independent of N, but dependent on the weights β_i, such that every $X\in\mathcal{X}_f$ is $o(n)$-close...

conjecture | open | MathDB 页面 | 原文线索

V2-054931 | MATHDB-336613

The doubly stochastic versus permutation binary X-ray conjecture

摘录 / EXCERPT: Let D be a doubly stochastic matrix of order n, meaning that it has nonnegative real entries and every row and column sums to 1. Its binary diagonal X-ray is the vector...

conjecture | open | MathDB 页面 | 原文线索

V2-054932 | MATHDB-336678

The biquadratic-curve one-level density conjecture

摘录 / EXCERPT: The biquadratic-curve one-level density conjecture. As $g\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-054933 | MATHDB-336697

Schramm's locality conjecture for random-cluster models

摘录 / EXCERPT: Let $q>0$ and let $(\mathbb{G}_n)$ be a sequence of infinite transitive graphs converging locally to an infinite transitive graph $\mathbb{G}$. For an infinite transitive graph...

conjecture | open | MathDB 页面 | 原文线索

V2-054934 | MATHDB-336737

The SC-ML dominance conjecture for polar codes under list decoding

摘录 / EXCERPT: Consider two polar codes A and B with the same code length n and message length k. Let ${\rm FER_{SC}}(\cdot)$ and ${\rm FER_{ML}}(\cdot)$ denote their frame-error rates un...

conjecture | open | MathDB 页面 | 原文线索

V2-054935 | MATHDB-336794

One-force-one-solution in random Hamilton-Jacobi equations

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and let $F_\omega(t,x)$ have exponential decay of space-time correlations. For every $b\in\mathbb{R}^d$, consider global solutions of the random Hamilton-Jacob...

conjecture | open | MathDB 页面 | 原文线索

V2-054936 | MATHDB-336822

Fixed-pair generator probability conjecture for Erdős-Rényi graph Jacobians

摘录 / EXCERPT: Fixed-pair generator probability conjecture. For each graph, fix vertices x and y. Then

conjecture | open | MathDB 页面 | 原文线索

V2-054937 | MATHDB-336845

The conjectured lower bound for the centre-of-mass escape rate

摘录 / EXCERPT: Let X be the increment random variable of the random walk in $\mathbb{R}^d$, and let G_n denote its centre of mass after n steps. Assume that the support of X is not contai...

conjecture | open | MathDB 页面 | 原文线索

V2-054938 | MATHDB-336891

Sharp small-bias asymptotics for the critical curve

摘录 / EXCERPT: Let $d\geq 2$, let δ denote the charge bias, and let ε_δ and κ_d be the quantities defined in the paper through the small-δ expansion of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-054939 | MATHDB-336892

Weakly self-avoiding walk free-energy correction

摘录 / EXCERPT: For $d\geq 3$, let $f^{\mathrm{wsaw}}(u)$ be the weakly self-avoiding walk free energy and let λ_d be the constant defined from the simple random walk Green function. Weakl...

conjecture | open | MathDB 页面 | 原文线索

V2-054940 | MATHDB-336893

Trimmed-range large deviations for simple random walk

摘录 / EXCERPT: For $n\in\mathbb N$ and $A\in\mathbb N$, define the set of sites visited between one and A times and its associated local time by ... Let $\lambda_d(A)$ and $J(A,s)$ denote the qua...

conjecture | open | MathDB 页面 | 原文线索

V2-054941 | MATHDB-337008

Lakshminarayan-Puchala-Życzkowski conjecture for moments of squared unimodular random matrices

摘录 / EXCERPT: Lakshminarayan-Puchala-Życzkowski conjecture. For k and N positive integers, it holds that

conjecture | open | MathDB 页面 | 原文线索

V2-054942 | MATHDB-337099

Existence and uniqueness conjecture for specialized non-symmetric Macdonald polynomials

摘录 / EXCERPT: Existence and uniqueness conjecture. There exists a unique composition ν for which this specialization is well defined and

conjecture | open | MathDB 页面 | 原文线索

V2-054943 | MATHDB-337241

Conjecture on transience of the randomly switched predator-prey process

摘录 / EXCERPT: Let $X_t=(x_t,y_t)$ be the process defined by the Lotka-Volterra PDMP in the paper, with initial condition $X_0=(x_0,y_0)\in\mathbb{R}_{++}^2$. Here, $\mathbb{R}_{++}^2$ denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-054944 | MATHDB-337242

Reflecting-boundary strategy for unbounded harvesting

摘录 / EXCERPT: Consider the population process with unbounded harvesting governed by ... where $(Z_t)_{t\geq0}$ is a non-negative, increasing, right-continuous process adapted to...

conjecture | open | MathDB 页面 | 原文线索

V2-054945 | MATHDB-337348

André-Oort equidistribution conjecture for products of modular curves

摘录 / EXCERPT: Equidistribution conjecture. Under the stated finiteness condition, the Galois-orbit measures μ_i converge weak-$*$ to the uniform probability measure on X.

conjecture | open | MathDB 页面 | 原文线索

V2-054946 | MATHDB-337394

Conjecture on Peng-Wei exact recovery for the generalized stochastic ball model

摘录 / EXCERPT: Exact-recovery conjecture. The Peng-Wei relaxation achieves exact recovery with high probability if

conjecture | open | MathDB 页面 | 原文线索

V2-054947 | MATHDB-337447

Independence conjecture for cycle indicators of b_r-regular permutations

摘录 / EXCERPT: Independence conjecture. For b_r-regular permutations, any two cycle indicators are independent if and only if they are $r-1$-separated.

conjecture | open | MathDB 页面 | 原文线索

V2-054948 | MATHDB-337454

Symmetric free convolution preserves unimodality

摘录 / EXCERPT: Unimodality conjecture. The distribution $\mu\boxplus\nu$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-054949 | MATHDB-337455

Large-time unimodality for stable convolutions

摘录 / EXCERPT: Stable-process unimodality conjecture. The convolution $\mu\ast S_t$ is unimodal for sufficiently large $t>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-054950 | MATHDB-337562

Feynman-Kac conjecture for the stochastic Bessel operator

摘录 / EXCERPT: Let $\mathcal{B}$ be the stochastic Bessel operator, let $f\in L^2[0,1]$, and let X be the stochastic process formally satisfying ... with functional ... Let A be the event...

conjecture | open | MathDB 页面 | 原文线索

V2-054951 | MATHDB-337589

Outage Probability Conjecture for MIMO channels

摘录 / EXCERPT: Outage Probability Conjecture. Every covariance matrix attaining the infimum has the stated form: power is allocated equally among some k transmit dimensions and is zero on the r...

conjecture | open | MathDB 页面 | 原文线索

V2-054952 | MATHDB-337591

Outage Probability Conjecture for TIMO channels

摘录 / EXCERPT: Outage Probability Conjecture for TIMO. The pairs (q_1,q_2) minimizing $f(q_1,q_2)$ over $\mathcal{D}$ belong to

conjecture | open | MathDB 页面 | 原文线索

V2-054953 | MATHDB-337610

Continuity and strict monotonicity conjecture for the tilted critical parameter

摘录 / EXCERPT: Let G be a connected, locally finite graph, and let $\Gamma\subseteq\operatorname{Aut}(G)$ be quasi-transitive and nonunimodular. For each λ, let $p_c(\lambda)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-054954 | MATHDB-337627

Spectral characterization of stationarity for universal flat models

摘录 / EXCERPT: Let $G\subset S_N^+$ be a transitive closed subgroup. Let $\chi=\sum_i u_{ii}$ be its main character, and let μ be the distribution of χ. The universal flat model of...

conjecture | open | MathDB 页面 | 原文线索

V2-054955 | MATHDB-337629

Fourier characterization of stationary Hadamard matrix models

摘录 / EXCERPT: Let a Hadamard matrix model be a matrix model arising from a complex Hadamard matrix, and call it stationary on its image when its associated integration state has the stationarity...

conjecture | open | MathDB 页面 | 原文线索

V2-054956 | MATHDB-337668

The critical threshold conjecture for Bargmann-Fock percolation

摘录 / EXCERPT: Let $\mathcal{D}_p$ denote the excursion domain at level p associated with a centered, normalized, non-degenerate, sufficiently smooth, stationary, isotropic, and positively corr...

conjecture | open | MathDB 页面 | 原文线索

V2-054957 | MATHDB-337680

Recurrence conjecture for the critical biased random walk on self-avoiding-walk trees

摘录 / EXCERPT: Critical-walk recurrence conjecture. The biased random walk RW_{λ_c} on $T_{\mathbb H}$ or $T_{\mathbb Z^2}$ is recurrent.

conjecture | open | MathDB 页面 | 原文线索

V2-054958 | MATHDB-337681

Conjectured convergence diagram for half-plane self-avoiding-walk measures

摘录 / EXCERPT: Convergence-diagram conjecture. The following convergence diagram holds:

conjecture | open | MathDB 页面 | 原文线索

V2-054959 | MATHDB-337725

Sharp entropy-at-infinity upper bound for uniformly continuous potentials

摘录 / EXCERPT: Let (M,g) be a pinched negatively curved manifold. Let $(\mu_n)_n$ be a sequence of invariant probability measures converging vaguely to μ, and let $F\in C_b(T^1M)$ be a unif...

conjecture | open | MathDB 页面 | 原文线索

V2-054960 | MATHDB-337766

Monotonicity conjecture for the even-indexed stopping probabilities

摘录 / EXCERPT: Let p_n denote the sequence of stopping probabilities for the random walk considered in the paper. Even-index monotonicity conjecture. ... The paper proves the corresponding inequa...

conjecture | open | MathDB 页面 | 原文线索

V2-054961 | MATHDB-337767

Limit conjecture for the stopping probabilities of a random walk

摘录 / EXCERPT: Let p_n denote the sequence of stopping probabilities for the random walk considered in the paper. Limit conjecture. ... This is presented as the main question of the problem. The...

conjecture | open | MathDB 页面 | 原文线索

V2-054962 | MATHDB-337788

Extension of the Brownian-motion moment theorem to real α

摘录 / EXCERPT: Let X^* be the process considered in Theorem, and let α be the parameter appearing in the regular-variation assumption. The theorem currently assumes that α is an i...

conjecture | open | MathDB 页面 | 原文线索

V2-054963 | MATHDB-337816

Universality conjectures for random knot models

摘录 / EXCERPT: Let K_n be a random knot sampled from the Random Jump, Petaluma, or Grid model. Random-knot universality conjectures. The following properties are expected: with high probability...

conjecture | open | MathDB 页面 | 原文线索

V2-054964 | MATHDB-337825

The Kleene-expression probability conjecture for Karnofsky-Rhodes expansions

摘录 / EXCERPT: Let S be a finite semigroup with generating set A, let $(\mathsf{KR}(S,A),A)$ be its Karnofsky-Rhodes expansion, and let $K(\mathsf{KR}(S,A))$ be a Kleene expression for its no...

conjecture | open | MathDB 页面 | 原文线索

V2-054965 | MATHDB-337869

Large-activity monochromaticity conjecture for the non-integrable Widom-Rowlinson model

摘录 / EXCERPT: Large-activity monochromaticity conjecture. For activities z large enough, every Widom-Rowlinson measure is monochromatic. The paper proves the existence of polychromatic phases...

conjecture | open | MathDB 页面 | 原文线索

V2-054966 | MATHDB-337886

Linear upper-bound conjecture for the SIA-index

摘录 / EXCERPT: Let n be a positive integer and let $\mathcal{S}$ be an SIA set of $n \times n$ stochastic matrices. Its SIA-index is the length of the shortest SIA product formed from matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-054967 | MATHDB-337940

The critical percolation threshold conjecture on high-dimensional lattices

摘录 / EXCERPT: Critical-percolation conjecture.

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-054968 | MATHDB-337949

The universal approval conjecture for the Tangle

摘录 / EXCERPT: Universal approval conjecture. The probability of universal eventual approval satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054969 | MATHDB-337984

Asymptotically deterministic coupling conjecture

摘录 / EXCERPT: Asymptotically deterministic coupling conjecture. Assume $H(X)=H(Y)$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-054970 | MATHDB-337985

Deterministic coupling conjecture

摘录 / EXCERPT: Deterministic coupling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-054971 | MATHDB-338050

Conjecture on sublinear transition-zone width in all dimensions

摘录 / EXCERPT: Let u_ω be the solution from Definition, under the hypotheses of Theorem (i) but without its limitation on the dimension d, and let $L_{u_\omega,\eta}(t)$ denote the trans...

conjecture | open | MathDB 页面 | 原文线索

V2-054972 | MATHDB-338068

Alon and Bilu-Linial conjectures for random and two-covers of hyperbolic surfaces

摘录 / EXCERPT: Alon and Bilu-Linial conjectures. (1) For every $\epsilon>0$, with probability $1-o(1)$ as $n\to\infty$, every new eigenvalue λ of an n-cover X' satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-054973 | MATHDB-338143

Sharp asymptotics for unbalanced alpha-unrooted critical update families

摘录 / EXCERPT: Let ${\mathcal U}$ be an α-unrooted and unbalanced critical two-dimensional update family with difficulty α. Let q denote the vacancy density and let τ_0 be th...

conjecture | open | MathDB 页面 | 原文线索

V2-054974 | MATHDB-338165

Benjamini-Kalai-Schramm entropy conjecture for percolation distributions

摘录 / EXCERPT: Let f be the Boolean function representing the crossing event in planar percolation. Let ${\mathcal S}(f)$ and ${\mathcal P}(f)$ denote its spectral and pivotality distributions,...

open_problem | open | MathDB 页面 | 原文线索

V2-054975 | MATHDB-338191

Monotonicity conjecture for the number of spectral components

摘录 / EXCERPT: Let F_t be the limiting spectral distribution of the noisy sample covariance matrix at noise power $t\geq 0$, and let $N(t)$ denote the number of connected components of its supp...

conjecture | open | MathDB 页面 | 原文线索

V2-054976 | MATHDB-338192

Convexity conjecture for the spectral separation utility

摘录 / EXCERPT: Utility-function conjecture. $\mathcal{U}$ is non-increasing and convex in t. Numerical evidence supports this stronger claim, which would describe not only when the information/...

conjecture | open | MathDB 页面 | 原文线索

V2-054977 | MATHDB-338215

Percolation conjecture for a single current above the Ising critical point

摘录 / EXCERPT: Let β_c be the critical inverse temperature, and let ${\bf P}_{\mathbb Z^2,\beta}^{\emptyset}$ denote the infinite-volume sourceless random-current measure on $\mathbb Z^2$....

conjecture | open | MathDB 页面 | 原文线索

V2-054978 | MATHDB-338282

Global uniqueness conjecture for physical solutions of the McKean-Vlasov equation

摘录 / EXCERPT: Global uniqueness conjecture. Solutions satisfying the physical jump condition are unique. Between jump times, the solution is continuously differentiable, and for every $t_0>0$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-054979 | MATHDB-338309

Friedgut-Kalai sharp-threshold conjecture for monotone graph properties

摘录 / EXCERPT: Friedgut-Kalai conjecture. There is an absolute constant c such that, if $ u_p^n(P)>\epsilon$, then

open_problem | open | MathDB 页面 | 原文线索

V2-054980 | MATHDB-338321

The three-dimensional Babai error-probability bound

摘录 / EXCERPT: Let P_e denote the error probability of a Babai partition. Consider a three-dimensional lattice and a Babai partition constructed from the QR decomposition associated with an obt...

conjecture | open | MathDB 页面 | 原文线索

V2-054981 | MATHDB-338322

The densest-lattice conjecture for worst error probability

摘录 / EXCERPT: Let n be the dimension of a lattice, and let the error probability refer to the approximation produced by the Babai partition. The densest-lattice conjecture. The worst error pro...

conjecture | open | MathDB 页面 | 原文线索

V2-054982 | MATHDB-338341

The maximal-subset-property conjecture for blind MIMO optimization

摘录 / EXCERPT: Maximal-subset-property conjecture. If k is slightly larger than n, then with high probability the only optima of the optimization problem are

conjecture | open | MathDB 页面 | 原文线索

V2-054983 | MATHDB-338369

Connectivity conjecture for regular hyperbolic tilings

摘录 / EXCERPT: Let F be the graph of a regular tiling of the hyperbolic plane. For each $r\in\mathbb{N}$, call (F,r) connectivity-forcing when $G_n(F,r)$ is connected with high probability. C...

conjecture | open | MathDB 页面 | 原文线索

V2-054984 | MATHDB-338440

The word-measure rigidity conjecture for free groups

摘录 / EXCERPT: Let $\mathbf{F}_r$ be the free group of rank r, and let $w\in\mathbf{F}_r$ induce a probability measure on each compact group via its word map. Word-measure rigidity conjecture....

conjecture | solved | MathDB 页面 | 原文线索

V2-054985 | MATHDB-338468

The probability interpretation conjecture for normalized solution lengths

摘录 / EXCERPT: Probability interpretation conjecture. This quantity equals the real probability that the solution occurs.

conjecture | open | MathDB 页面 | 原文线索

V2-054986 | MATHDB-338481

Drift classification conjecture for Markov-modulated diffusions

摘录 / EXCERPT: Let $\mathcal{A}$ be a manifold in $\mathbb{R}^d$ without boundary, and let (X_t,η_t) be a diffusion process on $\mathbb{R}\times\mathcal{A}$. Assume that η_t has station...

conjecture | open | MathDB 页面 | 原文线索

V2-054987 | MATHDB-338482

Recurrence conjecture for the one-dimensional centre of mass

摘录 / EXCERPT: Let X be a one-dimensional random-walk increment with zero mean, and let G_n denote the centre of mass at time n. Recurrence conjecture. Suppose $d=1$. If ... then G_n is rec...

conjecture | open | MathDB 页面 | 原文线索

V2-054988 | MATHDB-338483

Lower-growth conjecture for the multidimensional centre of mass

摘录 / EXCERPT: Let X be a random-walk increment in $\mathbb{R}^d$, and let G_n denote the centre of mass at time n. Lower-growth conjecture. Suppose that $\operatorname{supp}X$ is not conta...

conjecture | open | MathDB 页面 | 原文线索

V2-054989 | MATHDB-338542

Convergence of uniform 231-avoiding permutations to the Brownian binary tree

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let S_n be a uniform 231-avoiding permutation of $[n]$, and define its normalized graph by ... View $\operatorname{gr}(S_n)$ as a random compact subset...

conjecture | open | MathDB 页面 | 原文线索

V2-054990 | MATHDB-338556

Chas-Li-Maskit Gaussian limit conjecture for geometric lengths of loops

摘录 / EXCERPT: Let Xcong $\Gamma \backslash \mathbb H$ be a pair of pants equipped with an arbitrary hyperbolic metric ρ whose boundary is geodesic. Fix a choice of free generators for...

conjecture | open | MathDB 页面 | 原文线索

V2-054991 | MATHDB-338570

Neuhauser-Pacala coexistence conjecture for the symmetric model

摘录 / EXCERPT: Let the symmetric Neuhauser-Pacala model be the competing-species system on the lattice $\mathbb Z^d$, with coexistence meaning that both species persist in the long term. Neuhause...

conjecture | open | MathDB 页面 | 原文线索

V2-054992 | MATHDB-338620

The exponential field-size conjecture for random set systems

摘录 / EXCERPT: Exponential field-size conjecture. Assume that such a $k\times n$ matrix exists over a field $\mathbb{F}$. Then, with high probability over the random set system, there is an absol...

conjecture | open | MathDB 页面 | 原文线索

V2-054993 | MATHDB-338631

The q-moment conjecture for isotropic probability masses

摘录 / EXCERPT: The q-moment conjecture. If μ is an isotropic probability mass on $\mathbb{R}^k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-054994 | MATHDB-338730

The Arakelov class-group distribution conjecture

摘录 / EXCERPT: Let G, A, S, R, and V be as above, with $\sum_{g\in G}g=0$ in A. Let K be a number field, let $\bar{K}$ be an algebraic closure of K, and let $\mathcal{F}$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-054995 | MATHDB-338735

First universality conjecture for large-cap equity return distributions

摘录 / EXCERPT: Let the log-return distributions of large-cap equities be viewed as probability distributions, with scale and location allowed to vary. First Universality Conjecture. The distribut...

conjecture | open | MathDB 页面 | 原文线索

V2-054996 | MATHDB-338736

Spectral universality conjecture for asset-price trajectory matrices

摘录 / EXCERPT: For each asset-price time series, form its trajectory matrix by the maximal trajectory-matrix construction used in the source, and consider the matrix's spectral distribution and s...

conjecture | open | MathDB 页面 | 原文线索

V2-054997 | MATHDB-338812

Continuity conjecture for projection eigenfunctions in fractal percolations

摘录 / EXCERPT: Let F_α be the operator defined in the paper, let $\mathcal{I}$ be the index set of the construction, let $p_{\mathbf{i}}$ be the corresponding probabilities, and let M de...

conjecture | open | MathDB 页面 | 原文线索

V2-054998 | MATHDB-338914

One-arm conjecture for diffusion-limited aggregation in narrow wedges

摘录 / EXCERPT: Let A_∞ be the limiting diffusion-limited aggregation cluster in a wedge, and interpret its arms as the ends of the graph A_∞. One-arm conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-054999 | MATHDB-338915

Small-angle growth-rate conjecture for diffusion-limited aggregation in wedges

摘录 / EXCERPT: Let $(A_n)_{n\geq 0}$ be the DLA cluster process in the wedge W_{θ_1,θ_2}, and define its growth rate by ... where diam denotes Euclidean diameter, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-055000 | MATHDB-338926

Uniqueness conjecture for equality in the hook-integral inequality

摘录 / EXCERPT: The equality conjecture. Equality holds only when the shapes are VKLS shapes. This would characterize the equality case in the hook-integral inequality and remains open.

conjecture | open | MathDB 页面 | 原文线索

V2-055001 | MATHDB-338962

The strict inequality conjecture for critical and unification probabilities on non-amenable graphs

摘录 / EXCERPT: Strict inequality conjecture. If Γ is non-amenable and quasi-transitive, then

conjecture | open | MathDB 页面 | 原文线索

V2-055002 | MATHDB-338966

Derrida-Hilhorst regular-expansion conjecture for random matrix characteristic exponents

摘录 / EXCERPT: Derrida-Hilhorst conjecture. Suppose first that there exists $\alpha\in(0,+\infty)$ such that $\mathrm{E}[Z^\alpha]=1$. If $\alpha\notin\\{1,2,\ldots\\}$, then, as $\epsilon\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-055003 | MATHDB-338969

The replicate-to-all request service distribution probability conjecture

摘录 / EXCERPT: In the replicate-to-all system, let f_j denote the probability of the type-j request service-time distribution S_j, where the possible distributions are the trimmed-down fami...

conjecture | open | MathDB 页面 | 原文线索

V2-055004 | MATHDB-339053

Triangle condition conjecture for nonamenable quasi-transitive graphs

摘录 / EXCERPT: Triangle condition conjecture. One has $\nabla_{p_c}<\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-055005 | MATHDB-339054

Strict q-to-q threshold conjecture for nonamenable quasi-transitive graphs

摘录 / EXCERPT: Strict q-to-q threshold conjecture. For every $q\in(1,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-055006 | MATHDB-339129

Generalized gamma convolution threshold conjecture for ${\bf K}_\alpha$

摘录 / EXCERPT: Let ${\bf K}_\alpha$ be the random variable defined in the paper, with parameter $\alpha\in(0,1)$. A positive random variable has a generalized gamma convolution (GGC) law if its d...

conjecture | open | MathDB 页面 | 原文线索

V2-055007 | MATHDB-339155

The degeneracy-order conjecture for the online fixed graph process

摘录 / EXCERPT: Degeneracy-order conjecture. With high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-055008 | MATHDB-339242

Density of quasi-infinitely divisible random measures

摘录 / EXCERPT: Density conjecture. Real-valued QID random measures are dense in the space of independently scattered real-valued random measures, regarded as random elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-055009 | MATHDB-339324

Greedy Dictator optimality conjecture for guessing efficiency on the binary erasure channel

摘录 / EXCERPT: Let X^n be uniformly distributed over $\\{0,1\\}^n$, let Y^n be obtained by passing X^n through a memoryless binary erasure channel with erasure probability ϵ, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-055010 | MATHDB-339386

Crossover-regime mixing-time conjecture for the weakly asymmetric simple exclusion process

摘录 / EXCERPT: Let b_N and k_N satisfy ... for some $\beta\in(0,\infty)$, with $\lim_{N\to\infty}k_N/N=0$. Write $T_{\rm mix}^{N,k_N}(\varepsilon)$ for the mixing time of the weakly asymmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-055011 | MATHDB-339393

Taranenko's positivity conjecture for even-order line-stochastic tensors

摘录 / EXCERPT: Let A be an $n\times\cdots\times n$ tensor of order d that is line-stochastic, meaning that every line sum is 1. Taranenko's conjecture. If d is even, then ... The paper pres...

conjecture | open | MathDB 页面 | 原文线索

V2-055012 | MATHDB-339399

The conjecture on convergence of the closure-map pushforward measure

摘录 / EXCERPT: Let $\operatorname{Arm}(n,d,w)$ be the space of open n-edge arms in $\mathbb{R}^d$ with fixed edgelength w, and let $\operatorname{Pol}(n,d,w)$ be the corresponding space of cl...

conjecture | open | MathDB 页面 | 原文线索

V2-055013 | MATHDB-339400

The local-knot frequency conjecture for random arms and polygons

摘录 / EXCERPT: In dimension 3, let $K^\text{Arm}(n,w,k,r)$ be the probability that a length-k arc of a random open arm is an r-local knot, and let $K^\text{Pol}(n,w,k,r)$ be the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-055014 | MATHDB-339483

The $N^{0.634}$ fluctuation conjecture for the height of $\mathbb{C}_0(t)$

摘录 / EXCERPT: Let $\mathbb{C}_0(t)$ denote the relevant boundary cluster at time t, and let N be the scaling parameter used for the Poisson percolation model. Fluctuation conjecture. Fluctua...

conjecture | open | MathDB 页面 | 原文线索

V2-055015 | MATHDB-339485

Large-population approximation for the variable-vector epidemic model

摘录 / EXCERPT: Let n be the host population size, let $I_h^n(t)$ denote the number of infected hosts at time t, and let $\mathcal{R}_0$ be the basic reproduction number. When...

conjecture | open | MathDB 页面 | 原文线索

V2-055016 | MATHDB-339486

Quasi-stationary distributions in fixed and random vector-population epidemic models

摘录 / EXCERPT: Let n be the population-size scaling parameter, let $I_h^n(t)$ and $I_v^n(t)$ denote infected hosts and infected vectors, and let $\mathcal{R}_0$ be the basic reproduction number...

conjecture | open | MathDB 页面 | 原文线索

V2-055017 | MATHDB-339556

Magnetic gap-labelling conjecture for principal solenoidal tori

摘录 / EXCERPT: Let Σ be a Cantor set with a minimal action of $\mathbb{Z}^p$ that preserves a Borel probability measure μ. Let σ be the multiplier on $\mathbb{Z}^p$ associated t...

conjecture | open | MathDB 页面 | 原文线索

V2-055018 | MATHDB-339562

Exponential limit conjecture for the rescaled Schur off-diagonal entry

摘录 / EXCERPT: Let X be a random matrix satisfying the assumptions of Theorem, and let T be its Schur form. Define $Y=n|t_{12}|^2$, where t_{12} is the relevant off-diagonal entry of T, a...

conjecture | open | MathDB 页面 | 原文线索

V2-055019 | MATHDB-339603

Sixth-moment conjecture for high-dimensional oriented percolation

摘录 / EXCERPT: Consider sufficiently spread-out oriented percolation in dimension $d>4$, with spread parameter L sufficiently large, and let $\mathcal {T}_n$ denote the set of vertices connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-055020 | MATHDB-339689

Uniqueness of the critical probability for \(\mathcal{U}\)-bootstrap percolation

摘录 / EXCERPT: For the $\mathcal{U}$-bootstrap percolation model, let q_{c} denote its critical probability and let $\tilde q_{\mathrm{c}}$ denote the supremum of the directional criti...

conjecture | open | MathDB 页面 | 原文线索

V2-055021 | MATHDB-339724

Unique stationary measure for totally non-stable context trees

摘录 / EXCERPT: Let $({\mathscr T},q)$ be a non-null probabilised context tree. A context tree is totally non-stable when its set of infinite branches ${\mathscr C}^i$ has no shift-invariant subse...

conjecture | open | MathDB 页面 | 原文线索

V2-055022 | MATHDB-339791

Uniform finite-exceptional-spectrum conjecture for dense generators of compact Lie groups

摘录 / EXCERPT: Let G be a simple compact Lie group, let $k\geq 4$, and let S be a symmetric set of size k generating a dense subgroup of G. Let Δ^S denote the associated averaging...

conjecture | open | MathDB 页面 | 原文线索

V2-055023 | MATHDB-339842

The barrier-cone characterization of bounded cluster-growth directions

摘录 / EXCERPT: Barrier-cone characterization conjecture. For every $p\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-055024 | MATHDB-339847

Conjecture on the emergence of macroscopic cycles in spatial random permutations

摘录 / EXCERPT: Emergence of macroscopic cycles conjecture. The longest-cycle length has the asymptotic scales above, with macroscopic cycles in the regimes where $C_\beta^N=\Theta(N)$. The predic...

conjecture | open | MathDB 页面 | 原文线索

V2-055025 | MATHDB-339932

Non-asymptotic convergence conjecture for spiked matrix estimation

摘录 / EXCERPT: Let $c_1,c_2,c_3,d_0>0$ be universal constants. For ... , let ... denote the distributions and random variables specified in the conjecture, and let $\operatorname{MMSE}_{d,\lambda}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-055026 | MATHDB-339972

Maximum permanent conjecture for stochastic matrices of bounded rank

摘录 / EXCERPT: Let $\mathcal{M}_n^+$ denote the set of $n\times n$ nonnegative real matrices. Let $R_n^{\overline{k}}$ and $L_n^{\overline{k}}$ be the sets of row-stochastic and column-stochastic...

conjecture | open | MathDB 页面 | 原文线索

V2-055027 | MATHDB-340174

Conjecture on optimal allocation under probabilistic access

摘录 / EXCERPT: Let p be the access failure probability for each node, let α be the allocation parameter, and let $\mu_s(\alpha)$ denote the average service rate under probabilistic acces...

conjecture | open | MathDB 页面 | 原文线索

V2-055028 | MATHDB-340185

Measure-zero conjecture for the nonminimal Jeandel-Rao tilings

摘录 / EXCERPT: Let Ω_0 be the Jeandel-Rao Wang shift and let $X_0\subset\Omega_0$ be the minimal aperiodic subshift constructed in the paper. A shift-invariant probability measure on...

conjecture | open | MathDB 页面 | 原文线索

V2-055029 | MATHDB-340189

CLT for the final outcome of an epidemic on an MR graph with dropping

摘录 / EXCERPT: Let $\tilde{\boldsymbol{W}}^N(t)$ be the time-changed effective-degree epidemic process, let $\tilde{\tau}^N$ be its random terminal time, and let $\tilde{\boldsymbol{w}}(t)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055030 | MATHDB-340207

Schramm's locality conjecture for critical probabilities

摘录 / EXCERPT: Schramm's locality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055031 | MATHDB-340239

The two-dimensional Poisson SINR percolation upper-bound conjecture

摘录 / EXCERPT: Let X^λ be a homogeneous Poisson point process on $\mathbb{R}^2$, so that $\Lambda\equiv |\cdot|$, and let $g_{(\gamma,N_0,\tau)}(X^\lambda)$ be the SINR graph with noise p...

conjecture | open | MathDB 页面 | 原文线索

V2-055032 | MATHDB-340273

Nemirovski's matrix infinitely divisible series tail conjecture

摘录 / EXCERPT: Nemirovski's conjecture. Whenever $t=O\big[\sqrt{\ln(M+N)}\big]$, there are absolute constants θ_1 and θ_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-055033 | MATHDB-340274

Uniqueness of subsequential scaling limits for the two-dimensional KPZ equation

摘录 / EXCERPT: Let $ u$, λ, and D satisfy the effective-coupling condition from the paper, and let $(h^\varepsilon)_{\varepsilon>0}$ denote the rescaled KPZ fields, with...

conjecture | open | MathDB 页面 | 原文线索

V2-055034 | MATHDB-340347

The predicted two-dimensional growth rates for colliding-particle models

摘录 / EXCERPT: Two-dimensional growth-rate prediction. For both the $\mathrm{M1}_\varepsilon$ and $\mathrm{M2}_\varepsilon$ models, as $t\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055035 | MATHDB-340353

Stationary fixed-point conjecture for the Rath-Toth forest fire equations

摘录 / EXCERPT: Let $v_k^n(t)$ be the random proportion of vertices at time t that belong to clusters of size k under the stationary law $\mathbb{P}_{\mathrm{stat}}^n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055036 | MATHDB-340362

Regularization conjecture for random matrix operator norms

摘录 / EXCERPT: Let A be an $n \times n$ random matrix with i.i.d. mean-zero entries satisfying ... For a suitable L_m-norm, let $\tilde{A}$ be obtained from A by zeroing out every row and col...

conjecture | open | MathDB 页面 | 原文线索

V2-055037 | MATHDB-340385

The probability of Simpson conversion in four-dimensional binary tables

摘录 / EXCERPT: Let $p_{ijk\ell}$ be entries of a $2\times2\times2\times2$ table sampled uniformly from the probability simplex, with $i,j,k,\ell\in\{0,1\}$ and $p_{ijk\ell}\geq0$. Consider the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-055038 | MATHDB-340394

Powered spectral methods achieve weak recovery in the stochastic block model

摘录 / EXCERPT: Powered spectral recovery conjecture. Choose p and Q such that vertices from every community have the same expected degree and there is an efficient algorithm that solves weak...

conjecture | open | MathDB 页面 | 原文线索

V2-055039 | MATHDB-340396

Powered adjacency spectral clustering achieves weak recovery above the SBM threshold

摘录 / EXCERPT: Powered adjacency recovery conjecture. Let $a,b\geq 0$ satisfy $(a-b)^2>2(a+b)$, let $r=\Theta(\sqrt{\log n})$, and let $G\sim\mathrm{SBM}(n,a,b)$. If w is the eigenvector of...

conjecture | open | MathDB 页面 | 原文线索

V2-055040 | MATHDB-340405

Binomial approximation and large deviations for defaults

摘录 / EXCERPT: Let D_n be the total number of defaults among n banks, and let p be the corresponding asymptotic default probability. Default-count conjecture. The random variable D_n shou...

conjecture | open | MathDB 页面 | 原文线索

V2-055041 | MATHDB-340410

Closed-form expression for the minimum redundancy of binary Huffman codes

摘录 / EXCERPT: Let X be a source alphabet with $|X|=2$, consisting of symbols x_1 and x_2, with probabilities $\text{p}(x_1)$ and $\text{p}(x_2)$. Let β_T, β_0,...

conjecture | open | MathDB 页面 | 原文线索

V2-055042 | MATHDB-340498

Kahn-Komlós-Szemerédi conjecture on singularity of random Bernoulli matrices

摘录 / EXCERPT: Kahn-Komlós-Szemerédi conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055043 | MATHDB-340557

Integrality conjecture for complex Wishart cumulant moments

摘录 / EXCERPT: Let $W=(NQ)^{-1}$ be the Wishart matrix described above, and define $c_0(\mu_1,\dots,\mu_\ell)$ by ... Here $\ell\geq 1$, $(\mu_1,\dots,\mu_\ell)\in\mathbb{N}^{\ell}$, and C_ℓ d...

conjecture | open | MathDB 页面 | 原文线索

V2-055044 | MATHDB-340577

Lipschitz-interface conjecture for isoperimetric sets of strictly log-concave measures

摘录 / EXCERPT: Let $\mu=e^{-V}dx$ be a strictly log-concave probability measure on $\mathbb{R}^n$, and let E be an open isoperimetric subset with $\mu(E)=\alpha$. Lipschitz-interface conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-055045 | MATHDB-340580

Symmetric convex-subset conjecture for log-concave isoperimetric sets

摘录 / EXCERPT: Let $0<\alpha<1$, let $\mu=w(x)dx$ be a strictly log-concave probability measure supported on all of $\mathbb{R}^n$, with centrally symmetric density $w(-x)=w(x)$, and let E be a...

conjecture | open | MathDB 页面 | 原文线索

V2-055046 | MATHDB-340686

Low- and high-temperature behavior of the interacting path process

摘录 / EXCERPT: Let Υ be the space of polygonal paths equipped with the push-forward a priori probability measure p, let f be the potential ... where $J(n)\geq 0$ tends to zero sufficie...

conjecture | open | MathDB 页面 | 原文线索

V2-055047 | MATHDB-340713

The extra-power epsilon-longer-board conjecture for Maker-Breaker crossing games

摘录 / EXCERPT: Let $S_{m\times n}$ denote the $m\times n$ square grid, and let the (p,q)-crossing game be the Maker-Breaker game on this grid in which Maker claims p edges for every q edges...

conjecture | open | MathDB 页面 | 原文线索

V2-055048 | MATHDB-340714

The balanced-power narrower-board conjecture for Maker-Breaker crossing games

摘录 / EXCERPT: Let $S_{m\times n}$ denote the $m\times n$ square grid, and let the (p,p)-crossing game be the Maker-Breaker game in which Maker and Breaker claim equal numbers of edges per roun...

conjecture | open | MathDB 页面 | 原文线索

V2-055049 | MATHDB-340734

The monotonicity and unimodality conjecture for the derivative of F_z

摘录 / EXCERPT: Let $F_z(s)$ be the function arising from the limiting gap distribution in the paper, defined for $s\in(0,1)$, and let F'_z denote its derivative. The monotonicity and unimodalit...

conjecture | open | MathDB 页面 | 原文线索

V2-055050 | MATHDB-340754

Green's formula for the non-weighted pluricomplex energy

摘录 / EXCERPT: Let dV be a probability measure on $\mathbb{C}^{n}$ satisfying the assumptions stated before the conjecture, and let $E(dV)$ be its non-weighted pluricomplex energy. Energy Green...

conjecture | open | MathDB 页面 | 原文线索

V2-055051 | MATHDB-340813

Finite-limit-set conjecture for discounted stochastic-game values

摘录 / EXCERPT: Let v_λ denote the value vector associated with the parameter $\lambda>0$ in the stochastic game, and let V be a finite set. Finite-limit-set conjecture. There exists a f...

conjecture | open | MathDB 页面 | 原文线索

V2-055052 | MATHDB-340822

Central limit conjecture for degree growth of tame automorphisms

摘录 / EXCERPT: Let Q be the affine quadric threefold and let G be its tame automorphism group. Let μ be a symmetric atomic measure on G, let $\nu_n=\mu^{*n}$ be its n-fold convolution...

conjecture | open | MathDB 页面 | 原文线索

V2-055053 | MATHDB-340866

The probability ordering conjecture for regular prisms and their inscribed and circumscribed cylinders

摘录 / EXCERPT: For any integer $n\ge 3$, let P_n be a regular n-gon, and let C_n and c_n be the circumscribed and inscribed circles of P_n, respectively. Normalize the circumscribed cir...

conjecture | open | MathDB 页面 | 原文线索

V2-055054 | MATHDB-340883

The phase-transition conjecture for weighted PCA recovery

摘录 / EXCERPT: Phase-transition conjecture. If $A(\beta_i)\leq 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-055055 | MATHDB-340910

Conformal-block expansion conjecture for the CUE moment function

摘录 / EXCERPT: Let $f_k(x)$ be the function defined in equation (a5), let N be the parameter appearing there, and let $\mathcal{B}_{\lambda,\mu}$ denote the conformal blocks defined in equation...

conjecture | open | MathDB 页面 | 原文线索

V2-055056 | MATHDB-340911

CUE joint-moment coefficient formula via Young diagrams

摘录 / EXCERPT: Let $F(h,k)$ be the large-N limit ... where $F_N(h,k)$ is the joint moment considered in the paper. For a Young diagram λ, write $|\lambda|$ for its size, λ_1 for i...

conjecture | open | MathDB 页面 | 原文线索

V2-055057 | MATHDB-340920

Banach-space Hanson-Wright moment conjecture

摘录 / EXCERPT: Let $(a_{ij})_{i,j e n}$ be a symmetric matrix with values in a normed space $(F,\|\cdot\|)$, let $g_1,\ldots,g_n$ be independent standard Gaussian random variables, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055058 | MATHDB-340921

Triple-indexed Gaussian supremum conjecture

摘录 / EXCERPT: Let $p\geq 1$, let n,m be positive integers, let $(a_{ijk})_{i,j\leq n,k\leq m}$ be a triple-indexed real matrix, let $T\subset\mathbb R^m$ be bounded and nonempty, and let g_i...

conjecture | open | MathDB 页面 | 原文线索

V2-055059 | MATHDB-340978

Univalence of Cauchy transforms for infinitesimal arrays

摘录 / EXCERPT: Let ${\rm \bf Univ}(\mathbb{R})$ be the class of probability measures on $\mathbb{R}$ with univalent Cauchy transforms, and let ${\rm\bf IA}(\rhd)$ be the class of weak limits of m...

conjecture | open | MathDB 页面 | 原文线索

V2-055060 | MATHDB-340979

Weak closedness of monotonically infinitely divisible distributions

摘录 / EXCERPT: Let ${\rm \bf ID}(\rhd)$ denote the class of monotonically infinitely divisible probability measures on $\mathbb{R}$, and equip the space of probability measures with the weak topo...

conjecture | open | MathDB 页面 | 原文线索

V2-055061 | MATHDB-340980

Khintchine-type conjecture for monotone convolution

摘录 / EXCERPT: Let $\mu,\mu_n\in {\rm \bf P}(\mathbb{R})$, where $n\in\mathbb{N}$, and let $\{k_n\}_{n\geq1}$ be a sequence of strictly increasing natural numbers. Monotone Khintchine conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-055062 | MATHDB-340995

Majorization characterization conjecture for unimodal jump distributions

摘录 / EXCERPT: Let p and q be symmetric unimodal weight functions, meaning that $i\mapsto p(i)$ and $i\mapsto q(i)$ are symmetric and decreasing for $i\geq0$. Let $p\preceq_Mq$ denote majoriz...

conjecture | open | MathDB 页面 | 原文线索

V2-055063 | MATHDB-341048

Extension of the law of iterated logarithm from basis vectors to all vectors

摘录 / EXCERPT: Let T be the random weighted shift associated with independent weights $\{X_n\}_{n=1}^{\infty}$ on a Hilbert space $\mathcal{H}$, and let $\{e_k\}_{k=1}^{\infty}$ be its associat...

conjecture | open | MathDB 页面 | 原文线索

V2-055064 | MATHDB-341069

The q-Hahn PushTASEP Fredholm determinant conjecture

摘录 / EXCERPT: The q-Hahn PushTASEP Fredholm determinant conjecture. For the step initial configuration,

conjecture | open | MathDB 页面 | 原文线索

V2-055065 | MATHDB-341120

Conjecture on the conditioned critical mass distribution

摘录 / EXCERPT: Let X_t be the mass variable of the continuous-time Derrida-Retaux model started from μ_0, and let $\mathcal{L}(X_t\mid X_t>0)$ denote its conditional law given positivity. C...

conjecture | open | MathDB 页面 | 原文线索

V2-055066 | MATHDB-341128

Conjecture on the limiting distribution of normalized interface-curve length

摘录 / EXCERPT: Let Γ be the interface curve joining opposite corners of a three-dimensional cube of size N, and let L_N be its length, defined as the number of tetrahedra crossed by th...

conjecture | open | MathDB 页面 | 原文线索

V2-055067 | MATHDB-341129

Conjecture on the limiting distribution of normalized Alexander-polynomial logarithms

摘录 / EXCERPT: Alexander-invariant limit conjecture. For $t=-1$ (respectively, $t=i$), there exist numbers c_t, γ_t, and α_t such that the sequence of random variables

conjecture | open | MathDB 页面 | 原文线索

V2-055068 | MATHDB-341156

Adaptive local mesh-size conjecture for penultimate histograms

摘录 / EXCERPT: Let PEN be a finite set of interior histograms and, for each $y\in PEN$, define the local mesh size $LMS(y)$ by ... For a target histogram $G\in\mathcal{H}^{\circ}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055069 | MATHDB-341157

Interior geodesics conjecture for nearly-minimal broken paths

摘录 / EXCERPT: Let $\mathcal{H}$ be the histogram space and consider two interior histograms. Interior geodesics conjecture. For N and $1/m$ sufficiently large, nearly-minimal broken geodesics...

conjecture | open | MathDB 页面 | 原文线索

V2-055070 | MATHDB-341230

The Airy sheet extension conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be the Airy line ensemble, and define $\mathcal{H}:\mathbb{R}^2\to\mathbb{R}$ deterministically from $\mathcal{A}$ by the formulas in the source on...

conjecture | open | MathDB 页面 | 原文线索

V2-055071 | MATHDB-341231

Absolute continuity conjecture for directed-geodesic increments

摘录 / EXCERPT: Let $\Pi=\Pi_{0,0}$ be the directed geodesic from $(0,0)$, and define the increment process ... For $0\le t<u<1/2$, absolute continuity conjecture. The laws of η_t and η_u...

conjecture | open | MathDB 页面 | 原文线索

V2-055072 | MATHDB-341236

Symmetry conjecture for the conditional gambler's ruin duration

摘录 / EXCERPT: Let the gambler's ruin process evolve on $\mathbb{E}=\{0,\ldots,N\}$, with birth probabilities $p(i)$ and death probabilities $q(i)$, and let $W_{0:i:N}$ denote its duration condit...

conjecture | open | MathDB 页面 | 原文线索

V2-055073 | MATHDB-341339

The log-concave canonical-process decomposition conjecture

摘录 / EXCERPT: Let $(\xi_i)_{i\geqslant 1}$ be independent log-concave random variables with mean 0 and variance 1. For a set $T\subset\ell^2$, write $X_t=\sum_i t_i\xi_i$ and let $S_X(T)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-055074 | MATHDB-341364

Conjecture on the optimal strategy for dependent Bernoulli variables

摘录 / EXCERPT: Optimal-strategy conjecture. After observing $I_{k}=1$, the player whose turn it is should give up his turn to his opponent if and only if $p_{i,k}<\frac{1}{2}$ for all $i>k$. This...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055075 | MATHDB-341377

Steady-state phase-difference conjecture for the two-community noisy Kuramoto model

摘录 / EXCERPT: Let θ denote the phase on the circle, let θ_{1} and θ_{2} be the average phases of the two communities, and write their phase difference as ... In the disorder-f...

conjecture | open | MathDB 页面 | 原文线索

V2-055076 | MATHDB-341418

Characterization of zero asymptotic variance for component functionals

摘录 / EXCERPT: Let ψ be a graph functional satisfying the paper's growth condition, let p_T denote the relevant probability associated with a finite tree T, and let σ^2_ψ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-055077 | MATHDB-341493

Cohen-Lenstra random-matrix conjecture over complete discrete valuation rings

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be a complete discrete valuation ring with residue field $R/\mathfrak{m}=\mathbb{F}_{q}$. Let $P_{1}(t),\ldots,P_{r}(t)\in R[t]$ be monic polynomials whose r...

conjecture | open | MathDB 页面 | 原文线索

V2-055078 | MATHDB-341514

Small-gap limiting law for Gaussian beta ensembles

摘录 / EXCERPT: Let $\beta>0$, and let the Gaussian beta ensemble GβE have joint density ... Let $t^n_{k,\beta}$ be its k-th smallest gap. Small-gap conjecture. There exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-055079 | MATHDB-341524

Poisson distribution conjecture for the number of mmWave retransmissions

摘录 / EXCERPT: Let N^* denote the random variable counting the number of transmissions required in the intermittent mmWave-link model, let A be the model parameter, and let $R_{\min}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055080 | MATHDB-341591

The high-dimensional zero-breaking record fraction conjecture

摘录 / EXCERPT: Zero-breaking record fraction conjecture. There exist constants $p_d\in(0,1)$ such that, almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-055081 | MATHDB-341620

Critical-SNR conjecture for the 2-WSBM

摘录 / EXCERPT: In the 2-WSBM, let N be the number of vertices, k the community size, and let γ denote the signal-to-noise ratio; assume $k=o(\log N)$. For $k=2$, the model is equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-055082 | MATHDB-341655

Extension conjecture for theories with well-behaved thermodynamics

摘录 / EXCERPT: Extension conjecture. Every theory with a “well-behaved” thermodynamics can be extended to a sharp theory with purification.

conjecture | open | MathDB 页面 | 原文线索

V2-055083 | MATHDB-341656

The Glivenko-Cantelli conjecture for bivariate record breaks

摘录 / EXCERPT: Glivenko-Cantelli conjecture. The empirical fractions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-055084 | MATHDB-341657

The fixed-break strong law conjecture

摘录 / EXCERPT: For each $M\geq1$ and $k\geq0$, let $\tilde p_{M,k}$ be the fraction of the first M records that break precisely k remaining records. Strong law conjecture. For every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-055085 | MATHDB-341658

The variance bound conjecture for fixed-break record counts

摘录 / EXCERPT: Variance bound conjecture. For each fixed $k\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-055086 | MATHDB-341659

The variance bound conjecture for records without coordinate records

摘录 / EXCERPT: Let $R^{(0)}_{n,k}$ count, through time n, records that break precisely k remaining records and do not set a record in either coordinate. Reduced variance bound conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-055087 | MATHDB-341725

Correct upper-tail behaviour for stars without a lower bound on epsilon

摘录 / EXCERPT: Correct upper-tail behaviour. Theorem remains valid without the assumption $\varepsilon\geq n^{-\alpha}$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-055088 | MATHDB-341728

The asymptotic critical branching-rate conjecture for the general contact process

摘录 / EXCERPT: Critical branching-rate conjecture. In the general model,

conjecture | open | MathDB 页面 | 原文线索

V2-055089 | MATHDB-341800

The curvature coefficient conjecture for the geometric stochastic heat equation

摘录 / EXCERPT: Curvature coefficient conjecture. The universal constant is

conjecture | open | MathDB 页面 | 原文线索

V2-055090 | MATHDB-341810

Absence of an infinite cluster at criticality for Bernoulli percolation on the lattice

摘录 / EXCERPT: Critical percolation conjecture. For every $d\ge2$,

conjecture | open | MathDB 页面 | 原文线索

V2-055091 | MATHDB-341813

Limiting-configuration percolation-threshold conjecture for majority dynamics

摘录 / EXCERPT: For majority dynamics on $\mathbb{Z}^2$, let η_∞ be the pointwise limiting configuration and define ... Let $p_c(t)$ be the critical percolation probability at time t. Li...

conjecture | open | MathDB 页面 | 原文线索

V2-055092 | MATHDB-341824

External DLA radius growth conjecture on integer lattices

摘录 / EXCERPT: Let $\mathcal{E}_n$ be the external DLA cluster on $\mathbb{Z}^d$ started from $\mathcal{E}_0=\{0\}$, and let $r(\mathcal{E}_n)$ denote its radius. External DLA radius-growth conje...

conjecture | open | MathDB 页面 | 原文线索

V2-055093 | MATHDB-341831

Monotonicity conjecture for quantization coefficients of regular polygons

摘录 / EXCERPT: Polygonal quantization conjecture. The quantization coefficient for the uniform distribution on the boundary of a regular m-sided polygon inscribed in a circle is an increasing f...

conjecture | open | MathDB 页面 | 原文线索

V2-055094 | MATHDB-341852

The dimension-conserving projections conjecture for near-regular measures

摘录 / EXCERPT: Let $ u$ be a near-d-regular measure on $\mathbb{R}^{2}$, let $0\leq s<d$, and let $\mathsigma$ be a Borel probability measure on S^{1}. Suppose that for $\mathsigma$-almost ev...

conjecture | open | MathDB 页面 | 原文线索

V2-055095 | MATHDB-341904

Friedgut's continuous-cube resilience conjecture

摘录 / EXCERPT: Let $f\colon [0,1]^n \to \{0,1\}$ be a measurable function, and let $I_S^b(f)$ denote the corresponding resilience quantity for a set $S\subseteq[n]$ and $b\in\{0,1\}$. Friedgut's...

conjecture | open | MathDB 页面 | 原文线索

V2-055096 | MATHDB-341950

Monotonicity conjecture for the finite-memory ballistic speed

摘录 / EXCERPT: For integers $d\geq 2$ and $k\geq d-1$, consider the random walk in ${\mathbb R}^d$ whose step distribution avoids the convex hull of the origin and its most recent k positions,...

conjecture | open | MathDB 页面 | 原文线索

V2-055097 | MATHDB-341951

Continuity conjecture for finite- and infinite-memory ballistic speeds

摘录 / EXCERPT: For integers $d\geq 2$ and $k\geq d-1$, let $v_{d,k}$ be the asymptotic speed of the finite-memory convex-hull-avoiding walk, and let $v_{d,\infty}$ be the speed associated with th...

conjecture | open | MathDB 页面 | 原文线索

V2-055098 | MATHDB-341983

The normalized SRLP conjecture for birth-death processes

摘录 / EXCERPT: Normalized SRLP conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-055099 | MATHDB-342113

Positive speed conjecture for tagged particles on Galton-Watson trees

摘录 / EXCERPT: Let a Galton-Watson tree be supercritical and condition it on survival. Consider the simple exclusion process on this tree in the constant speed model and in the variable speed mod...

conjecture | open | MathDB 页面 | 原文线索

V2-055100 | MATHDB-342179

Independent coherent opinions bound for radically different opinions

摘录 / EXCERPT: Let (X,Y) be a coherent random vector with values in $[0,1]^2$, and suppose that X and Y are independent. Independent-opinions bound. Then ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-055101 | MATHDB-342180

Extreme coherent laws are 2 by 2 laws

摘录 / EXCERPT: Let $\mathcal{C}$ be the compact convex set of coherent probability laws of (X,Y) on $[0,1]^2$, and let $\operatorname{ext}(\mathcal{C})$ denote its extreme points. A law is a...

conjecture | open | MathDB 页面 | 原文线索

V2-055102 | MATHDB-342204

Transience conjecture for simple random walk on square subdivision graphs

摘录 / EXCERPT: Let ${\mathbf{c}_{\mathrm M}}\in(1,25)$, and let $\mathcal S_h^1$ be the corresponding square subdivision graph. Consider simple random walk on $\mathcal S_h^1$. Transience conject...

conjecture | open | MathDB 页面 | 原文线索

V2-055103 | MATHDB-342281

Ballistic deposition height-growth conjecture on a strip

摘录 / EXCERPT: Ballistic deposition height-growth conjecture. With probability one, for all $j\in\mathbb N$,

conjecture | open | MathDB 页面 | 原文线索

V2-055104 | MATHDB-342324

Maximal-eigenvalue conjecture for the integral means spectrum of whole-plane LLEs

摘录 / EXCERPT: Let a whole-plane LLE be driven by a Lévy process without drift, and suppose there is an integer N such that ... in the unbounded version of LLE, or ... in the bounded version. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055105 | MATHDB-342325

Infinite-matrix limit conjecture for the integral means spectrum of whole-plane LLEs

摘录 / EXCERPT: Let a whole-plane LLE be driven by a Lévy process without drift. For each N, let B_N be the N-dimensional matrix obtained from the three-diagonal matrix B in the unbounded...

conjecture | open | MathDB 页面 | 原文线索

V2-055106 | MATHDB-342346

The improved sample-independent MGF bound for binomial KL divergence

摘录 / EXCERPT: The improved binomial MGF conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-055107 | MATHDB-342416

The triangle condition for critical percolation on nonamenable graphs

摘录 / EXCERPT: Triangle-condition conjecture. For every $v\in V$,

conjecture | open | MathDB 页面 | 原文线索

V2-055108 | MATHDB-342417

The L^2 boundedness conjecture for nonamenable quasi-transitive graphs

摘录 / EXCERPT: L^2 boundedness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055109 | MATHDB-342418

The exponential connectivity decay conjecture for nonamenable graphs

摘录 / EXCERPT: Exponential connectivity decay conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055110 | MATHDB-342463

Robin semigroup trace convergence conjecture

摘录 / EXCERPT: Let $\hat K^w_n(t)$ and $\hat K^w(t)$ be the random and limiting Robin semigroups from Theorem. For every $t_1,\ldots,t_k>0$, Trace convergence conjecture. ... in joint distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-055111 | MATHDB-342464

Bridge coupling conjecture for split Skorokhod couplings

摘录 / EXCERPT: Let T^{x^n} and $\tilde X^{x^n}$ denote the processes appearing in Theorem, and let $T^{x^n,x^n}_{\vartheta_t}$ and $\tilde X^{x^n,x^n}_t$ be the corresponding bridge processes. Br...

conjecture | open | MathDB 页面 | 原文线索

V2-055112 | MATHDB-342466

Madiman-Wang conjecture on Rényi entropy of sums

摘录 / EXCERPT: For $\beta>0$, let the generalized Gaussian have density ... where β and α are chosen so that it integrates to one. Let X_j, $j=1,\dots,n$, be independent random variab...

conjecture | open | MathDB 页面 | 原文线索

V2-055113 | MATHDB-342473

The higher-dimensional pointwise error expansion conjecture for Zakai equations

摘录 / EXCERPT: Let $d\geq 3$, and let v solve the d-dimensional Zakai SPDE. For a spatial grid with spacings $h_{x_i}>0$, let $V_{i_1,\ldots,i_d}^N$ denote the implicit finite-difference appr...

conjecture | open | MathDB 页面 | 原文线索

V2-055114 | MATHDB-342474

The higher-dimensional sparse-grid mixed-moment conjecture for Zakai equations

摘录 / EXCERPT: Let $d\geq 3$, let $h_{x_i}=h_0\cdot 2^{-l_i}$ for $i=1,\ldots,d$, and let $\Delta=\Delta_1\otimes\cdots\otimes\Delta_d$ be the tensor product of the first-order difference operato...

conjecture | open | MathDB 页面 | 原文线索

V2-055115 | MATHDB-342492

Universality conjecture for critical update families with infinitely many stable directions

摘录 / EXCERPT: Let ${\mathcal U}$ be a critical update family with an infinite number of stable directions. Let q denote the vacancy probability, let τ_0 be the infection time at the origi...

conjecture | open | MathDB 页面 | 原文线索

V2-055116 | MATHDB-342529

The isoperimetric-profile conjecture for finite supercritical clusters

摘录 / EXCERPT: Isoperimetric-profile conjecture. For every $p_c<p<1$ there exist positive constants c_p and C_p such that

conjecture | open | MathDB 页面 | 原文线索

V2-055117 | MATHDB-342530

Finite truncated susceptibility conjecture for supercritical percolation

摘录 / EXCERPT: Truncated susceptibility conjecture. This quantity is finite for every $p_c<p\leq1$.

open_problem | open | MathDB 页面 | 原文线索

V2-055118 | MATHDB-342531

Exponential radius-tail conjecture for finite supercritical clusters

摘录 / EXCERPT: Let $G=(V,E)$ be an infinite, connected, locally finite, transitive graph, let K_v be the cluster of v, and let $B(v,n)$ be the graph-distance ball of radius n around v. Ex...

conjecture | open | MathDB 页面 | 原文线索

V2-055119 | MATHDB-342650

Persistence exponent conjecture for fixed-size truncations of random orthogonal matrices

摘录 / EXCERPT: Persistence exponent conjecture. The persistence probability satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-055120 | MATHDB-342651

Large-size asymptotic conjecture for the persistence exponent

摘录 / EXCERPT: Let $\theta(\ell)$ be the persistence decay exponent appearing in the fixed-ℓ persistence asymptotic, with ℓ a positive integer. Large-size persistence exponent conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-055121 | MATHDB-342661

High-density flux conjecture for TASEP-H with l-perturbations

摘录 / EXCERPT: Let ρ be the particle density, let h be the critical density, and let ϕ^* denote the maximal-equilibrium-state flux. Consider TASEP-H with l-perturbations. High-density...

conjecture | open | MathDB 页面 | 原文线索

V2-055122 | MATHDB-342662

Sub-exponential perturbation-rate conjecture for TASEP-H

摘录 / EXCERPT: Let ρ be the particle density, let h be the critical density, let $\phi(\rho;n)$ be the finite-n typical flux, and let ϕ^* be the maximal-equilibrium-state flux. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055123 | MATHDB-342663

Sub-exponential slowing conjecture for the zero-range model

摘录 / EXCERPT: Let ρ be the particle density, let h be the critical density, let $\phi_{\pi(n)}(\rho;n)$ be the finite-n flux in the zero-range model with parameter $\pi(n)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055124 | MATHDB-342677

Extremum-count conjecture for average normalized Penrose-Banzhaf power indices

摘录 / EXCERPT: Extremum-count conjecture. For every $k=1,\ldots,n$, the function $q\mapsto\mathbb{E}({\Greekmath 010C}_k^{\downarrow})$ has exactly $k-1$ local extrema on $(1/2,1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-055125 | MATHDB-342782

Universality conjecture for normalized collision times

摘录 / EXCERPT: Let T be the total time considered above, with expectation and variance denoted by $\mathbb E[T]$ and $\operatorname{Var}(T)$, and let N and M be the parameters satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-055126 | MATHDB-342784

Berestycki's coupon-collector mixing conjecture for random transpositions

摘录 / EXCERPT: Berestycki's coupon-collector mixing conjecture. The random permutation is asymptotically well mixed at the stopping time τ_n. At time τ_n-1 there is at least one fixed p...

conjecture | open | MathDB 页面 | 原文线索

V2-055127 | MATHDB-342785

Berestycki's conditional uniformity conjecture on the giant component

摘录 / EXCERPT: Let X_t be the random-transposition permutation and let $\mathfrak C_t$ be the giant component of the associated random graph. Suppose ... Given $\mathfrak C_t$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055128 | MATHDB-342854

The GUE hypothesis for local correlations of zeta zeros

摘录 / EXCERPT: GUE hypothesis. For $n\geq 1$ and any $\varphi\in\mathcal{S}(\mathbb{R}^n)$,

conjecture | open | MathDB 页面 | 原文线索

V2-055129 | MATHDB-342856

Alon-Hefetz-Krivelevich-Tyomkyn Poisson anti-concentration conjecture for graph edge statistics

摘录 / EXCERPT: Alon-Hefetz-Krivelevich-Tyomkyn's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-055130 | MATHDB-342902

The compact-group word-measure orbit conjecture

摘录 / EXCERPT: Let $\mathbf{F}$ be the free group, and let $w_{1},w_{2}\in\mathbf{F}$. For a compact group G, write μ_{w} for the probability measure on G obtained by evaluating w on a...

conjecture | open | MathDB 页面 | 原文线索

V2-055131 | MATHDB-342923

Exceptional-time dimensions for Ornstein-Uhlenbeck Bargmann-Fock percolation

摘录 / EXCERPT: Exceptional-time conjecture. Almost surely, exceptional times at which $\mathcal{C}_t$ contains an unbounded component exist, and the Hausdorff dimension of the random set of such...

conjecture | open | MathDB 页面 | 原文线索

V2-055132 | MATHDB-342924

Exceptional-plane dimensions for the three-dimensional Bargmann-Fock field

摘录 / EXCERPT: Let F be a three-dimensional Bargmann-Fock field on $\mathbb{R}^3$. For each $t\in\mathbb{R}$, let $f^{\mathrm{hor}}(t)$ be the two-dimensional Bargmann-Fock field obtained by re...

conjecture | open | MathDB 页面 | 原文线索

V2-055133 | MATHDB-342938

Nonlinear p-stable projection conjecture for $1<p<2$

摘录 / EXCERPT: Nonlinear p-stable projection conjecture. With probability at least $1-N^{-(c-2)}$, for all $x,y\in P$, if $\lVert x-y\rVert_p=O(\epsilon^2)$ then

conjecture | open | MathDB 页面 | 原文线索

V2-055134 | MATHDB-342939

The general first-order correction conjecture for extreme eigenvalues of the Jacobi unitary ensemble

摘录 / EXCERPT: Let α_1 and α_2 be the Jacobi unitary ensemble parameters, let ϕ_n and ϕ_1 denote the appropriately scaled largest and smallest eigenvalues, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055135 | MATHDB-342962

Measure-independence conjecture for finite-dimensional spaces of linear-growth harmonic functions

摘录 / EXCERPT: Finite-dimensionality characterization conjecture. The group G has polynomial growth if and only if $$ is finite dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-055136 | MATHDB-342963

Measure-independence conjecture for dimensions of polynomial-growth harmonic functions

摘录 / EXCERPT: Harmonic-function dimension conjecture. For every $k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-055137 | MATHDB-343061

Conjecture on the gap between Poisson Stein-factor constants

摘录 / EXCERPT: Stein-factor gap conjecture. For all $k>\lambda$,

conjecture | open | MathDB 页面 | 原文线索

V2-055138 | MATHDB-343117

Weak moderate deviations for the halo volume

摘录 / EXCERPT: Let R satisfy $R\in(2,\frac12L)$, and let B_R be the disc of radius R, with halo volume $V(\gamma)$ under the probability measure μ_β. Write $I(B_R)$ for the halo rat...

conjecture | open | MathDB 页面 | 原文线索

V2-055139 | MATHDB-343235

Arithmetic conjecture for expected internal-angle sums

摘录 / EXCERPT: Let $\beta\geq -1$ be an integer or half-integer, and let $n\in\mathbb{N}$ and $k\in\{1,\ldots,n\}$. The quantity $\mathbb{J}_{n,k}(\beta)$ denotes the expected internal-angle sum...

conjecture | open | MathDB 页面 | 原文线索

V2-055140 | MATHDB-343236

Arithmetic conjecture for dual expected internal-angle sums

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and $k\in\{1,\ldots,n\}$. Let $\beta>(n-1)/2$ be an integer or half-integer, and let $\tilde{\mathbb{J}}_{n,k}(\beta)$ denote the corresponding dual expected i...

conjecture | open | MathDB 页面 | 原文线索

V2-055141 | MATHDB-343237

Arithmetic conjecture for expected face numbers of Poisson-Voronoi polytopes

摘录 / EXCERPT: Let $d\in\mathbb{N}$ and $k\in\{0,\ldots,d-1\}$. Write $\mathcal{V}_d$ for the Poisson-Voronoi polytope in dimension d, and let $\mathbb{E}f_k(\mathcal{V}_d)$ denote its expected...

conjecture | open | MathDB 页面 | 原文线索

V2-055142 | MATHDB-343250

Painlevé I hierarchy conjecture for odd-valence recurrence coefficients

摘录 / EXCERPT: Painlevé I hierarchy conjecture. The recurrence coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-055143 | MATHDB-343284

Central limit theorem for Benford errors

摘录 / EXCERPT: Let $a>0$ be a real number satisfying the relevant condition, and write $\alpha=\log_{10}a$. Suppose that the continued fraction expansion of α satisfies ... For a digit...

conjecture | open | MathDB 页面 | 原文线索

V2-055144 | MATHDB-343285

Nonnormal distribution of Benford errors for integer sequences a^n

摘录 / EXCERPT: Let $a\geq 2$ be an integer that is not a power of 10, let $d\in\{1,2,\dots,9\}$, and let $E_d(N,\{a^n\})$ denote the Benford error. Nonnormal distribution conjecture for Benford...

conjecture | open | MathDB 页面 | 原文线索

V2-055145 | MATHDB-343297

Hydrodynamic limit conjecture for the BHP

摘录 / EXCERPT: Let $\rho_0: \mathbb{R}\rightarrow[0,1]$ be an initial density profile, and let $ \{\nu^{\epsilon}\}_{\epsilon\in\mathbb{R}_{>0}}$ be probability measures on the configuration spac...

conjecture | open | MathDB 页面 | 原文线索

V2-055146 | MATHDB-343353

All-degrees elliptope conjecture for the Montanari-Sen witness

摘录 / EXCERPT: Let $\alpha,\delta \in (0,1)$ and let $d \in 2\mathbb{N}$. The random matrix $\bm{M}^{(\alpha,\delta)}(\bm{W})$ is the proposed degree-d witness, and $\mathscr{E}_d^N$ is the deg...

conjecture | open | MathDB 页面 | 原文线索

V2-055147 | MATHDB-343357

Conjecture on convergence from free to Boolean maximum powers

摘录 / EXCERPT: Free-to-Boolean maximum convergence conjecture. Then there exists $G\in\Delta_+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055148 | MATHDB-343381

The Minkowski inequality for the discriminant of finite free convolutions

摘录 / EXCERPT: Let p and q be degree-d polynomials, let $p\boxplus_d q$ denote their finite free convolution, and let $\operatorname{Dis}(p)$ denote the discriminant of p. Minkowski inequ...

conjecture | open | MathDB 页面 | 原文线索

V2-055149 | MATHDB-343461

The Boundary Conjecture for doubly stochastic single eigenvalues

摘录 / EXCERPT: Let DS_n be the set of complex numbers that occur as eigenvalues of n-by-n doubly stochastic matrices, and let a convex combination of permutation matrices mean a matrix of t...

conjecture | open | MathDB 页面 | 原文线索

V2-055150 | MATHDB-343462

The Perfect-Mirsky conjecture for dimensions 6 through 11

摘录 / EXCERPT: Let DS_n be the set of complex numbers that occur as eigenvalues of n-by-n doubly stochastic matrices, and let PM_n be the union of the convex hulls of the k-th roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-055151 | MATHDB-343484

Clancy-Leake-Payne conjecture on paired Sylow subgroups of random graph critical groups

摘录 / EXCERPT: Clancy-Leake-Payne conjecture. As n tends to infinity, the probability that $K(G(n,q))_{(p)}$ with its monodromy pairing is isomorphic to $(\Gamma,\langle\cdot,\cdot\rangle)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-055152 | MATHDB-343563

Existence of a Markov category of measurable spaces with conditional expectations

摘录 / EXCERPT: Conditional-expectation Markov category conjecture. Such a Markov category exists, with conditionals in the sense of the paper's definition, and these conditionals are given by con...

conjecture | open | MathDB 页面 | 原文线索

V2-055153 | MATHDB-343568

Critical-frontier conjecture for the independent model

摘录 / EXCERPT: Independent-model phase-diagram conjecture. There exists $\lambda_{\mathrm{c}}(p,d)\in(0,1)$ such that $\lambda_{\mathrm{c}}(\mathord{\cdot},d)$ is continuous and strictly increasi...

conjecture | open | MathDB 页面 | 原文线索

V2-055154 | MATHDB-343577

Phase-transition conjecture for unimodality of Fuss-Catalan distributions

摘录 / EXCERPT: Let $p>1$ and let $\mu(p,r)$ denote the Fuss-Catalan distribution with parameters p and r. Phase-transition conjecture. For every $p>1$ there exists $r_{0}(p)$ satisfying ... suc...

conjecture | open | MathDB 页面 | 原文线索

V2-055155 | MATHDB-343614

Scaling-limit conjecture for the wetting-model dynamics

摘录 / EXCERPT: Let $K=\{h\in H:h\geq 0\ \text{a.e.}\}$, where $H=L^2(0,1)$, and let μ be the law on K of a reflecting Brownian motion on $[0,1]$. Let $(u_t)_{t\geq 0}$ be the reversible Mar...

conjecture | open | MathDB 页面 | 原文线索

V2-055156 | MATHDB-343615

Sequential zero-width limit conjecture for the sticky wetting dynamics

摘录 / EXCERPT: For $\eta,\epsilon>0$, let $(u_t^{\eta,\epsilon})_{t\geq 0}$ be the stationary K-valued Markov process associated with the Dirichlet form $\mathcal{E}^{1,\eta,\epsilon}$ and star...

conjecture | open | MathDB 页面 | 原文线索

V2-055157 | MATHDB-343626

Uniqueness conjecture for Pólya ensembles on Hermitian matrix spaces

摘录 / EXCERPT: Let $H_m^{(n)}$ denote the relevant space of Hermitian matrices, and let a polynomial ensemble on $H_m^{(n)}$ have weights ... as in Eq. (pol.H). A random matrix g is drawn from a...

conjecture | open | MathDB 页面 | 原文线索

V2-055158 | MATHDB-343636

Poisson multinomial decomposition conjecture for floors of strongly Rayleigh variables

摘录 / EXCERPT: Let X be a strongly Rayleigh random variable, and let $j,k\geq 1$. Write $\left\lfloor jX/k\right\rfloor$ for the integer-valued floor of jX/k. Poisson multinomial decompositio...

conjecture | open | MathDB 页面 | 原文线索

V2-055159 | MATHDB-343646

Discreteness conjecture for non-even p-frame energy minimizers

摘录 / EXCERPT: Let $p>0$ with $p\notin 2\mathbb N$, and let μ be a probability measure on $\mathbb S^{d-1}$. Its p-frame energy is ... A finite discrete measure is a measure supported on fini...

conjecture | open | MathDB 页面 | 原文线索

V2-055160 | MATHDB-343667

Three-dimensional critical-family universality conjecture for bootstrap percolation

摘录 / EXCERPT: Critical-family universality conjecture. As $p\rightarrow 0$, either

conjecture | open | MathDB 页面 | 原文线索

V2-055161 | MATHDB-343733

Conjecture on existence of surface tension for the exponential interaction

摘录 / EXCERPT: Let $exp}$ denote the exponential interaction, and for each N let σ_N^{exp} be its finite surface tension. Exponential surface-tension conjecture. The limit ... exi...

conjecture | open | MathDB 页面 | 原文线索

V2-055162 | MATHDB-343734

Conjecture on limit shapes for bead and exponential stochastic interfaces

摘录 / EXCERPT: Let $S=[0,1]^2$, and let ϕ^N be a stochastic interface model on S_N with interaction potential $V=\mathsf{bead}$ or $V=\mathsf{exp}$. Suppose its weight functions are ... wher...

conjecture | open | MathDB 页面 | 原文线索

V2-055163 | MATHDB-343764

Sine-law conjecture for intersection angles of closed geodesics

摘录 / EXCERPT: Intersection-angle distribution conjecture. As $D\rightarrow\infty$, the multiset $\theta_q(D)$ tends towards the probability distribution with density

conjecture | open | MathDB 页面 | 原文线索

V2-055164 | MATHDB-343773

The exact overlap conjecture for self-similar sets and measures

摘录 / EXCERPT: Let Φ be an iterated function system and let $(p_i)_{i\in I}$ be a probability vector. For a finite word $\mathbf{a}=(a_1,\ldots,a_n)\in I^n$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055165 | MATHDB-343774

The explicit heavy-traffic limit conjecture with mobility

摘录 / EXCERPT: Explicit heavy-traffic limit conjecture. As $\rho\uparrow1$,

conjecture | open | MathDB 页面 | 原文线索

V2-055166 | MATHDB-343783

Hard edge statistics conjecture for products involving Pólya ensembles

摘录 / EXCERPT: Hard edge statistics conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055167 | MATHDB-343829

Conjecture on the asymptotic distribution of the longest expanded cycle

摘录 / EXCERPT: Let ... be the spatial period parameter and $r\geq 1$ the neighborhood radius, and consider the limit as $n\to\infty$. The asymptotic cycle-length conjecture. If ... n^{-}Cnto 1 ... is e...

conjecture | open | MathDB 页面 | 原文线索

V2-055168 | MATHDB-343893

Quantitative convergence conjecture for Green's functions on the percolation cluster

摘录 / EXCERPT: Quantitative Green's-function convergence conjecture. There exists a positive constant $C<\infty$, depending on d, $\mathfrak{p}$, λ and s, such that for every $t>0$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-055169 | MATHDB-343984

The cyclic cubic and quintic unit signature probability conjecture

摘录 / EXCERPT: Cyclic cubic and quintic unit signature conjecture. As K varies over cyclic number fields of degree ℓ, the probability that the unit signature rank is 1 is

conjecture | open | MathDB 页面 | 原文线索

V2-055170 | MATHDB-344014

O'Rourke-Wei central limit conjecture for von Neumann entropy

摘录 / EXCERPT: Let m and n be the dimensions of the two subsystems, let $c=m/n$, and let X be the standardized von Neumann entropy. Consider the fixed-trace ensemble of reduced density matr...

conjecture | open | MathDB 页面 | 原文线索

V2-055171 | MATHDB-344029

The threshold-exponent conjecture for Hamming-graph percolation

摘录 / EXCERPT: Let $V=[n]^d$, let G_0 be the Cartesian product of d complete graphs on $[n]$, namely the d-dimensional Hamming graph, and let p_c be the critical probability for saturatio...

conjecture | open | MathDB 页面 | 原文线索

V2-055172 | MATHDB-344030

The threshold-exponent conjecture for K_d-completion

摘录 / EXCERPT: Consider the K_d-completion dynamics on $[n]$, in which copies of K_d missing one edge are iteratively completed, with initial graph G_0 having edges $i\leftrightarrow j$ whe...

conjecture | open | MathDB 页面 | 原文线索

V2-055173 | MATHDB-344041

MTP$_2$ distributions satisfy Constraint A

摘录 / EXCERPT: Let Q be an MTP$_2$ distribution, meaning a distribution whose density satisfies the multivariate total positivity of order two condition. Let Constraint A denote the constraint...

conjecture | open | MathDB 页面 | 原文线索

V2-055174 | MATHDB-344108

The sparse block-model coupling conjecture below the square-root threshold

摘录 / EXCERPT: Let $0 \leq \delta \leq c$. Define $\mathbf{Q}=G(n,\kappa_1)$ for the constant kernel $\kappa_1\equiv c$, and let $\mathbf{P}=G(n,\kappa_2)$, where κ_2 equals $c+\delta$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-055175 | MATHDB-344113

Distribution of the cotangent of the SLE tip argument

摘录 / EXCERPT: Let $\kappa<8$, and consider the backward SLE_{κ} flow and its SLE_{κ} trace. At any fixed deterministic time $t>0$, let the tip of the trace be viewed through the co...

conjecture | open | MathDB 页面 | 原文线索

V2-055176 | MATHDB-344185

The probabilistic scaling conjecture for nonlinear Schrödinger equations

摘录 / EXCERPT: Let r and d be positive integers, and let $s_p=-\frac{1}{2r}$ be the critical threshold for probabilistic scaling. Consider the nonlinear Schrödinger equation referred to as...

conjecture | open | MathDB 页面 | 原文线索

V2-055177 | MATHDB-344297

Normal-distribution and variance conjecture for modular symbols

摘录 / EXCERPT: Modular-symbol distribution conjecture. As m tends to infinity, the distribution of the values $\{[a/m]^+/\sqrt{\log(m)}:a\in(\mathbb{Z}/m\mathbb{Z})^\times\}$ converges to a nor...

conjecture | open | MathDB 页面 | 原文线索

V2-055178 | MATHDB-344387

Exponential convergence-rate conjecture for the invading species

摘录 / EXCERPT: Exponential convergence-rate conjecture. For every initial population state x with $x^{i_0}\neq 0$ and every $s\in E$,

conjecture | open | MathDB 页面 | 原文线索

V2-055179 | MATHDB-344388

Persistence threshold conjecture for species and environments

摘录 / EXCERPT: Persistence threshold conjecture. Persistence of the system is possible if and only if the number of environments is at least the number of species.

conjecture | open | MathDB 页面 | 原文线索

V2-055180 | MATHDB-344391

Conjecture on weakening the intensity-growth condition for limit theorems

摘录 / EXCERPT: Let t denote the intensity parameter and let d tend to infinity. The Gaussian and Poisson limit theorems above require growth conditions on t because of dimension-dependent f...

conjecture | open | MathDB 页面 | 原文线索

V2-055181 | MATHDB-344442

Conjectured centered fluctuation limit for the Cramér-von Mises statistic

摘录 / EXCERPT: Let H be the random matrix and let β and the statistic $\mathcal{A}_N$ be as in Theorem. Let Z_k, $k\geqslant 1$, denote the random variables appearing in that theorem, a...

conjecture | open | MathDB 页面 | 原文线索

V2-055182 | MATHDB-344535

Dichotomy conjecture for the directed edge-reinforced random walk

摘录 / EXCERPT: Dichotomy conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-055183 | MATHDB-344572

The unimodality conjecture for median-dynamics marginals

摘录 / EXCERPT: Let p_t^x denote the density of the marginal law of the median-dynamics opinion $\eta_t(x)$ on $[0,1]$. Unimodality conjecture. The density p_t^x is symmetric around $\frac12$...

conjecture | open | MathDB 页面 | 原文线索

V2-055184 | MATHDB-344598

The power of one conjecture for majority dynamics on random graphs

摘录 / EXCERPT: Let $\rho(k,n)$ be the probability that Red wins when Red's camp initially has size $n/2+k$, with edge probability $p=.5$. The power of one conjecture. There is a constant $b4>0$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-055185 | MATHDB-344599

Conjecture on decreasing marginal value of defectors in majority dynamics

摘录 / EXCERPT: Let $c1(k,n)$ be the probability that Red wins when Red's camp initially has size $n/2+k$, with $p=.5$, and define the value of the ith defector by ... with $c1(n,0)=1/2$. Values o...

conjecture | open | MathDB 页面 | 原文线索

V2-055186 | MATHDB-344683

Containment probability conjecture for random permutations

摘录 / EXCERPT: Let k be a positive integer. Let π be a permutation of length k, let $n=1000k^2$, and let σ be a uniformly random permutation of length n. Containment probability...

conjecture | open | MathDB 页面 | 原文线索

V2-055187 | MATHDB-344684

Fixed-pattern containment conjecture at the universality threshold

摘录 / EXCERPT: Let $\mathcal{S}_k$ denote the permutations of $[k]$. Fix $\varepsilon>0$ and set $n=(1+\varepsilon)k^2/4$. For $\pi\in\mathcal{S}_k$, let σ be a uniformly random permutatio...

open_problem | open | MathDB 页面 | 原文线索

V2-055188 | MATHDB-344722

Exponential stabilization conjecture for coupled spin-J systems with state feedback

摘录 / EXCERPT: Exponential stabilization conjecture. The first controller almost surely exponentially stabilizes $(\rho_t,\hat{\rho}_t)$ to $(\boldsymbol\rho_{\bar n},\boldsymbol\rho_{\bar n})$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-055189 | MATHDB-344737

Positive correlation of three-fluid percolation between opposite sides

摘录 / EXCERPT: Let M_n be the finite honeycomb-lattice region with four-colored cells, and let $B_{k,n}\subset\Omega$ be the event that fluid k, for $k=1,2,3$, percolates between opposite sid...

conjecture | open | MathDB 页面 | 原文线索

V2-055190 | MATHDB-344738

Positive correlation of three fluids percolating from the center

摘录 / EXCERPT: Let M_n be the finite honeycomb-lattice region, let O be its central cell, and let $A_{k,n}\subset\Omega$ be the event that fluid k, for $k=1,2,3$, percolates from O to the...

conjecture | open | MathDB 页面 | 原文线索

V2-055191 | MATHDB-344740

Positive correlation for arbitrary three-fluid crossing events

摘录 / EXCERPT: Let M_n be the finite honeycomb-lattice region. For $k=1,2,3$, choose sets $A_k,B_k\subset M_n$, and let U_k be the event that fluid k percolates between A_k and B_k. Gen...

conjecture | open | MathDB 页面 | 原文线索

V2-055192 | MATHDB-344769

Shift-invariance conjecture for the KPZ equation

摘录 / EXCERPT: Fix $t>0$, $\Delta>0$, two segments $[X,X']$ and $[Y,Y']$ in $\mathbb R$, and real numbers $(x_1,\dots,x_n)$, $(y_1,\dots,y_n)$, $(\tilde x_1,\dots,\tilde x_k)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055193 | MATHDB-344820

High-probability convergence of the subspace power method

摘录 / EXCERPT: Let $\{a_i\}_{i=1}^R$ be R independent standard Gaussian vectors in $\mathbb{R}^L$, let $c_n<\frac{1}{n!}$ be a constant, and consider the power method iterations given by the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-055194 | MATHDB-344822

Convergence of the distributed stochastic preconditioned forward-backward algorithm with stochastic approximation

摘录 / EXCERPT: Let Assumptions ... , and ... , using the approximation in ... . Convergence conjecture. The generated sequence converges to a v-SGNE of the game. The conjecture concerns the stochastic-...

conjecture | open | MathDB 页面 | 原文线索

V2-055195 | MATHDB-344845

Arboreal gas phase-transition conjecture in dimensions at least three

摘录 / EXCERPT: Let the arboreal gas be defined on $\mathbb{Z}^d$ with $d\geq 3$, let B_r be the radius-r ball, and let T_0 denote the tree containing the origin. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-055196 | MATHDB-344846

Goldstone-mode decay conjecture for the arboreal gas

摘录 / EXCERPT: Let the arboreal gas be defined on $\mathbb{Z}^d$ with $d\geq 3$, and write $\mathbb{P}_{\beta}[i\leftrightarrow j]$ for the probability that i and j lie in the same tree. Gold...

conjecture | open | MathDB 页面 | 原文线索

V2-055197 | MATHDB-344847

Two-dimensional arboreal gas mass-gap conjecture

摘录 / EXCERPT: Let the arboreal gas be defined on $\mathbb{Z}^2$, let $\beta>0$, and write $\mathbb{P}_{\beta}[i\leftrightarrow j]$ for the probability that i and j lie in the same tree. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055198 | MATHDB-344848

Negative association conjecture for the arboreal gas

摘录 / EXCERPT: Let $\beta>0$, and for distinct edges write $\mathbb{P}_{\beta}[e_1,\ldots,e_k]$ for the probability that all listed edges occur in the forest. Negative association conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-055199 | MATHDB-344891

Algebraic tangles have no more figure-eight than trefoil descendants

摘录 / EXCERPT: A resultant knot probability is the proportion of crossing assignments of a free knot diagram that produce a specified knot. An algebraic tangle is a tangle obtained through the al...

conjecture | open | MathDB 页面 | 原文线索

V2-055200 | MATHDB-344892

The 2n diagrams attain the upper bound on unknots

摘录 / EXCERPT: A free $2n$ knot is the free knot diagram referred to in the source, with $n+2$ crossings. A minimal free knot diagram is one with the minimum crossing number for its knot. $2n$-un...

conjecture | open | MathDB 页面 | 原文线索

V2-055201 | MATHDB-344895

Tangle symmetries preserve resultant-knot probability

摘录 / EXCERPT: A free tangle is a tangle whose crossings are free crossings, and its four anchor points remain fixed on the boundary. The resultant knot probability is the proportion of crossing...

conjecture | open | MathDB 页面 | 原文线索

V2-055202 | MATHDB-344896

Nontrivial knots have resultant probability below one half

摘录 / EXCERPT: Let a free knot diagram be given, and let a nontrivial knot be a knot other than the unknot. The resultant knot probability is the proportion of crossing assignments of the free kn...

conjecture | open | MathDB 页面 | 原文线索

V2-055203 | MATHDB-344909

Optimality conjecture for the dyadic grand coupling of Brownian motions

摘录 / EXCERPT: Let $\{\{\tilde{X}_{\alpha,t}\}_{t}\}_{\alpha\in\mathbb{R}}$ be a grand coupling of Brownian motions $\{\mathrm{BM}(\alpha)\}_{\alpha\in\mathbb{R}}$. A failure probability bound is...

conjecture | open | MathDB 页面 | 原文线索

V2-055204 | MATHDB-344923

The disjoint-interval density conjecture

摘录 / EXCERPT: Let μ be an absolutely continuous probability measure supported on $[0,1]$ with density f. Assume there exist $C>0$ and $p>1$ such that, for all $0<r<R<1$ and x in the supp...

conjecture | open | MathDB 页面 | 原文线索

V2-055205 | MATHDB-344950

Dimensional crossover critical exponent conjecture for anisotropic bond percolation

摘录 / EXCERPT: Let $q_c:[0,1]\to[0,1]$ be the critical function for anisotropic bond percolation, and let $p_c(d)$ denote the critical parameter for d-dimensional percolation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-055206 | MATHDB-344951

Susceptibility threshold conjecture for anisotropic bond percolation

摘录 / EXCERPT: Consider an anisotropic bond percolation process on ${\mathbb {Z}}^d\times{\mathbb {Z}}^s$, with $d,s\geq1$, parameters (p,q), and $p_c(d)-p>0$ sufficiently small. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055207 | MATHDB-344952

Lattice-valued random-vector concentration conjecture

摘录 / EXCERPT: Let $X_1,\ldots, X_n$ be iid random vectors in $\mathbb{Z}^d$. A choice of weights $w_i\in \\{-1,1\\}$ should exist such that, for all non-zero $a_i\in \mathbb{R}$ and all...

conjecture | open | MathDB 页面 | 原文线索

V2-055208 | MATHDB-344953

General concentration conjecture for bounded-atom random vectors

摘录 / EXCERPT: Let $X_1,\ldots, X_n$ be iid random vectors in $\mathbb{R}^d$ satisfying ... A choice of weights $w_i\in \\{-1,1\\}$ should exist such that, for all non-zero $a_i\in \mathbb{R}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055209 | MATHDB-344969

The monotonicity conjecture for service-type probabilities in FJ-FA systems with locality two

摘录 / EXCERPT: In the FJ-FA model with availability codes of locality two, let f_i denote the probability that a request has service type i, where i is the number of recovery groups at whic...

conjecture | open | MathDB 页面 | 原文线索

V2-055210 | MATHDB-344973

Random Weil-Petersson surface eigenfunction sup-norm conjecture

摘录 / EXCERPT: Random-surface eigenfunction sup-norm conjecture. With probability tending to 1 as $g\to+\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055211 | MATHDB-345017

Free decomposability conjecture for random matrix models

摘录 / EXCERPT: Free decomposability conjecture. There is a von Neumann subalgebra $\mathcal{N}\leq\mathcal{M}$ and an atomic subalgebra $\mathcal{D}\leq\mathcal{M}$ such that $\mathcal{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055212 | MATHDB-345063

The local weak limit conjecture for random intersection graphs

摘录 / EXCERPT: Construct the locally finite connected rooted random graph $G^\bullet$ from the partially labeled rooted random tree $T^\bullet$ as described in the source: even-generation vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-055213 | MATHDB-345103

The exact overlaps conjecture for self-similar measures

摘录 / EXCERPT: Let $\Phi=\{\varphi_j(x)=\lambda_jx+t_j\}_{j=0}^m$ be a self-similar iterated function system on the real line, let $p=(p_j)_{j=0}^m$ be a probability vector, and let μ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-055214 | MATHDB-345244

The conditions (S1) and (S2) imply hypergroup structure

摘录 / EXCERPT: Let (Γ,w_0) be a pointed graph satisfying the conditions $(S1)$ and $(S2)$. Say that it produces a hypergroup when its associated distance distribution yields a hypergroup.…

conjecture | open | MathDB 页面 | 原文线索

V2-055215 | MATHDB-345245

Metastable winding-number random-walk conjecture for the periodic XY chain

摘录 / EXCERPT: Metastable random-walk conjecture. (i) The metastable time satisfies $\tau^{(N)}=C(J_N,\sigma_N)e^{J_N/\sigma_N^2-\log N}$, where $C(J,\sigma)$ is polynomial in J and σ. (...

conjecture | open | MathDB 页面 | 原文线索

V2-055216 | MATHDB-345257

The polynomial tail lower-bound conjecture for spiral-like domains

摘录 / EXCERPT: Polynomial tail conjecture. For every $p>{\rm H}(W)$, there exists a constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055217 | MATHDB-345279

The uniform permutation-walk arcsine law

摘录 / EXCERPT: Let a random walk of length $2n+1$ be generated from a uniform random permutation, and let $2k$ be the number of its edges lying above the origin. Write $\alpha_{2n,2k}$ for the co...

conjecture | open | MathDB 页面 | 原文线索

V2-055218 | MATHDB-345317

Nonconstant potential-field conjecture for strongly extendable perforated domains

摘录 / EXCERPT: Let $\boldsymbol{\mathrm{P}}\subset\Omega$ be a corresponding prototype with the strong extension property, and let $\mathcal{V}_{\mathrm{pot}}^{p}(\boldsymbol{\mathrm{P}})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-055219 | MATHDB-345326

Benjamini's square-tiling crossing conjecture

摘录 / EXCERPT: Tile the unit square by, possibly infinitely many, squares of varying sizes, with at most three squares meeting at any corner. Color each square black or white independently, each...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-055220 | MATHDB-345327

The critical one-half conjecture for circle packings

摘录 / EXCERPT: Let $4\mathcal{S}4$ be a circle packing and let $s_0\in\mathcal{S}$. For $0<p<1$, let $E_{\mathcal{S},p}(s_0,r)$ denote the event that s_0 is connected to distance r by open di...

conjecture | open | MathDB 页面 | 原文线索

V2-055221 | MATHDB-345328

Uniform exponential decay below one-half for circle packings

摘录 / EXCERPT: Let $E_{\mathcal{S},p}(s_0,r)$ be the event that s_0 is connected to distance r by open disks in site percolation with parameter p. Uniform exponential decay conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-055222 | MATHDB-345329

The critical one-half conjecture for bounded-aspect-ratio ellipse packings

摘录 / EXCERPT: Let $M\geq1$, and let $\mathcal{S}$ be a packing of closed elliptical disks whose aspect ratios are at most M. Let $s_0\in\mathcal{S}$. Ellipse-packing critical and exponential-d...

conjecture | open | MathDB 页面 | 原文线索

V2-055223 | MATHDB-345345

Conjectured distributional limit for the quenched CLT error in one-dimensional random environments

摘录 / EXCERPT: Distributional-limit conjecture. If $\kappa\in(2,3)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-055224 | MATHDB-345385

Wright's Friedman conjecture for random hyperbolic surfaces

摘录 / EXCERPT: Let X be a random compact hyperbolic surface of genus g, sampled according to the Weil-Petersson probability measure. Write $N_X^\Delta(0,a)$ for the number of eigenvalues of t...

conjecture | open | MathDB 页面 | 原文线索

V2-055225 | MATHDB-345409

Monotonicity conjecture for the median of a weighted sum of gamma variables

摘录 / EXCERPT: Gamma-sum median conjecture. The function $\alpha\mapsto\operatorname{Med}(Z_\alpha)$ is non-decreasing for $\alpha\in(0,1)$. This is one of two conjectures concerning medians of s...

conjecture | open | MathDB 页面 | 原文线索

V2-055226 | MATHDB-345410

Median monotonicity and bounds conjecture for the variance-gamma distribution

摘录 / EXCERPT: Variance-gamma median conjecture. The function $\sigma\mapsto\operatorname{Med}(V_{r,\theta,\sigma})$ is strictly decreasing for $\sigma\in(0,\infty)$, and consequently

conjecture | open | MathDB 页面 | 原文线索

V2-055227 | MATHDB-345411

Median monotonicity and bounds conjecture for the McKay Type I distribution

摘录 / EXCERPT: McKay Type I median conjecture. The function $\phi\mapsto\operatorname{Med}(Z_{m,c,\phi})$ is strictly increasing for $c\in(1,\infty)$. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-055228 | MATHDB-345412

Lower-bound conjecture for the median of the generalized hyperbolic distribution

摘录 / EXCERPT: Generalized hyperbolic median conjecture. The median satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-055229 | MATHDB-345434

Extension of the lower information-divergence bound beyond integer np

摘录 / EXCERPT: Let n and p be the parameters of the binomial and hypergeometric distributions considered in Theorem ... should hold without the condition that np is an integer. This would ext...

conjecture | open | MathDB 页面 | 原文线索

V2-055230 | MATHDB-345504

Conjectured pure-jump process limit for total length of Beta-coalescents

摘录 / EXCERPT: Total-length process conjecture. The sequence of stationary processes

conjecture | open | MathDB 页面 | 原文线索

V2-055231 | MATHDB-345537

Csóka's averaging conjecture for convex combinations of Bernoulli variables

摘录 / EXCERPT: Csóka's conjecture. For every p and t there exists a $k\in\mathbb N$ such that $\pi(p,t)$ is realized by $c_i=\frac1k$ for $i\leq k$ and $c_i=0$ for $i>k$; equivalently, the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-055232 | MATHDB-345538

Samuels' conjecture on sums of independent non-negative random variables

摘录 / EXCERPT: Samuels' conjecture. The supremum of $\mathbf P(X_1+\cdots+X_l\geq1)$ over all such collections is a maximum attained by $X_j=c_j$ for $j\leq k$ and $X_j=(1-b)\beta_j$ for $j>k$.

conjecture | open | MathDB 页面 | 原文线索

V2-055233 | MATHDB-345539

Feige's conjecture on lower tails of sums of independent non-negative variables

摘录 / EXCERPT: Feige's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-055234 | MATHDB-345638

Four- and five-dimensional Brownian loop-soup gluing conjecture

摘录 / EXCERPT: For $d\geq4$, consider the Brownian loop-soup in $(0,1)^d$ and the operation of contracting each loop, so that percolation can be viewed as a mechanism for gluing loops into cluste...

conjecture | open | MathDB 页面 | 原文线索

V2-055235 | MATHDB-345652

Transience conjecture for the frog model on supercritical Galton-Watson trees

摘录 / EXCERPT: Transience conjecture. There exists a constant $c=c(\mathsf{GW})>0$ such that $\mathrm{FM}(\mathsf{T},\eta,\mathsf{SRW})$ is transient for GW-almost all infinite realiza...

conjecture | open | MathDB 页面 | 原文线索

V2-055236 | MATHDB-345683

Conjecture on the stationary density and critical threshold of the discrete Bak-Sneppen model

摘录 / EXCERPT: Bak-Sneppen stationary-density conjecture. There exists a function $\nu(p)=\lim_{n\to\infty}\nu^{(n)}(p)\in[0,1]$ that is monotone increasing on $[0,1]$. Moreover, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-055237 | MATHDB-345721

Fractional-moment conjecture for cutpoints of non-homogeneous random walks

摘录 / EXCERPT: Let $\mathcal C$ be the set of cutpoints of the random walk, and let α be a real number. Under the conditions of Theorem, fractional-moment conjecture. ... for all $\alpha<1$....

conjecture | open | MathDB 页面 | 原文线索

V2-055238 | MATHDB-345737

Ising-eroding conjecture for critical families

摘录 / EXCERPT: Let $\mathcal U$ be a finite family defining two-dimensional $\mathcal U$-Ising dynamics. A finite set $\mathcal T$ of stable directions determines a $(\mathcal T,L)$-droplet when...

conjecture | open | MathDB 页面 | 原文线索

V2-055239 | MATHDB-345738

The imaginary-axis zeros conjecture for Gaussian-unitary ensemble Schur averages

摘录 / EXCERPT: Imaginary-axis zeros conjecture. For every partition μ, $\mathbb{E}^{(H)}_N[P_\mu]$ is a polynomial in N whose zeros lie on the line $\operatorname{Re}(N)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-055240 | MATHDB-345781

Asymptotic independence characterization in the two envelopes problem

摘录 / EXCERPT: Let X be a positive random variable, and write ... where $\lfloor\log_2 X\rfloor$ and $\{\log_2 X\}$ are respectively the whole and fractional parts of $\log_2 X$. Let A and...

conjecture | open | MathDB 页面 | 原文线索

V2-055241 | MATHDB-345848

Baik-Corwin-Guaraco-Romik distributional identity for oriented swap and last-passage vectors

摘录 / EXCERPT: Let $\mathbf{U}_N=(U_N(1),\dots,U_N(N-1))$ be the vector of swap-passage times in the oriented swap process, and let $\mathbf{V}_N=(V_N(1),\dots,V_N(N-1))$ be the vector of last-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-055242 | MATHDB-345901

Infinite-dimensional limit of the finite Riccati matrix

摘录 / EXCERPT: Let $\Phi^N(t)$ be the finite-dimensional matrix-valued function defined by the Riccati system in the source, with entries $\Phi^{N,i,j}(t)$. Consider the associated functions...

conjecture | open | MathDB 页面 | 原文线索

V2-055243 | MATHDB-345923

Conjecture on the largest Fiedler-vector entry in the stochastic block model

摘录 / EXCERPT: Largest-entry conjecture. The largest entry of $\mathbf{v_2}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-055244 | MATHDB-346020

Conjecture on improving the Poisson approximation bound in Wasserstein distance

摘录 / EXCERPT: Conjecture on improving the Poisson approximation bound. The order of the upper bound in the displayed inequality can be significantly improved.

conjecture | open | MathDB 页面 | 原文线索

V2-055245 | MATHDB-346029

Spin Whittaker function orthogonality conjecture

摘录 / EXCERPT: Spin Whittaker orthogonality conjecture. For all $\underline{L}_N,\underline{L}'_Ninmathpzc{W}_N$,

conjecture | open | MathDB 页面 | 原文线索

V2-055246 | MATHDB-346031

Entropy criterion for finiteness of the weak-disorder critical point

摘录 / EXCERPT: Entropy criterion conjecture. The weak-disorder critical point is finite if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-055247 | MATHDB-346047

Conjecture on stationary distributions of the centered particle system

摘录 / EXCERPT: Let $f^n(\cdot,t)$ denote the empirical distribution of the locations of an n-particle stochastic system, and let $f^n_*(\cdot,t)$ be its recentered version whose median is at...

conjecture | open | MathDB 页面 | 原文线索

V2-055248 | MATHDB-346065

Invariant-property characterization of ρ-irregularity

摘录 / EXCERPT: Invariant-property conjecture. For any pair (γ,ρ), there exists a property $\mathcal{P}$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-055249 | MATHDB-346072

Symmetrized Li-Médard conjecture

摘录 / EXCERPT: Let X be uniformly distributed on $\{0,1\}^n$, let Y be obtained from X through a binary symmetric channel with crossover probability $0<p<\frac{1}{2}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055250 | MATHDB-346160

The decoupled Markov group conjecture for column-stochastic matrix semigroups

摘录 / EXCERPT: Decoupled Markov group conjecture. For every real number $M\geq 1$, there exists a real number $C=C(M)\geq0$ such that, for every $d\in\mathbb{N}$ and every matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-055251 | MATHDB-346202

Right-eigenvalue conjecture for the context-tree matrix

摘录 / EXCERPT: Right-eigenvalue conjecture. If, for every $\alpha s\in{\mathscr S}$, the sequence $\left(\kappa_{\alpha s}(n)\right)_n$ converges to 0, then Q admits 1 as a right eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-055252 | MATHDB-346203

Geometric convergence conjecture for cascade series

摘录 / EXCERPT: Geometric convergence conjecture. If the tree has no infinite shift-stable subtree, then all its cascade series converge with geometric rates.

conjecture | open | MathDB 页面 | 原文线索

V2-055253 | MATHDB-346204

Vanishing-of-cascades conjecture for sigma-finite invariant measures

摘录 / EXCERPT: Vanishing-of-cascades conjecture. For every $\alpha s\in{\mathscr S}$, the sequence $\left(\kappa_{\alpha s}(n)\right)_n$ tends to 0 as n tends to infinity. Consequently, Q i...

conjecture | open | MathDB 页面 | 原文线索

V2-055254 | MATHDB-346292

The cactus-graph probability conjecture for sparse random graphs

摘录 / EXCERPT: Let $A(n,m)$ denote the random graph with n vertices and m edges. For $1/2<\alpha<1$, set $m=\alpha n$. Cactus-graph probability conjecture. The probability that $A(n,m)$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-055255 | MATHDB-346293

The non-cactus conjecture for near-linear sparse random graphs

摘录 / EXCERPT: Let $A(n,m)$ denote the random graph with n vertices and m edges, and write $m=n+t$ with $t=o(n)$. Non-cactus conjecture. With high probability, $A(n,m)$ is not a cactus graph....

conjecture | open | MathDB 页面 | 原文线索

V2-055256 | MATHDB-346305

Expected-time asymptotics for binary border aggregation

摘录 / EXCERPT: Expected-time asymptotics conjecture. The expectation satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-055257 | MATHDB-346312

Dow-Gibson's modified van der Waerden conjecture for 1-polystochastic matrices

摘录 / EXCERPT: Let $\Omega_1(d,n)$ denote the set of 1-polystochastic d-dimensional matrices of order n, let $\operatorname{per_1}(A)$ denote the permanent, and let J_n^d be the d-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-055258 | MATHDB-346313

Taranenko's local-extrema conjecture for 1-polystochastic permanents

摘录 / EXCERPT: Let $\Omega_1(d,n)$ be the polytope of 1-polystochastic d-dimensional matrices of order n, and let a local extremum mean a local maximum or local minimum of the permanent on...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055259 | MATHDB-346364

The oriented swap process-last passage percolation distributional identity

摘录 / EXCERPT: Let $\bm{U}_n$ denote the vector of oriented swap process finishing times and let $\bm{V}_n$ denote the point-to-line last passage percolation vector. The oriented swap process-LPP...

conjecture | open | MathDB 页面 | 原文线索

V2-055260 | MATHDB-346389

Weiss's singularity conjecture for symmetric random sign matrices

摘录 / EXCERPT: Weiss's singularity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055261 | MATHDB-346390

The k-core nonsingularity conjecture for random graphs

摘录 / EXCERPT: k-core nonsingularity conjecture. There is a constant k_0 such that, for every $k\geq k_0$, the adjacency matrix of the k-core of $Giant(n,p)$ is nonsingular with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-055262 | MATHDB-346393

The symmetric determinant magnitude conjecture for random sign matrices

摘录 / EXCERPT: Symmetric determinant magnitude conjecture. With probability $1-o(1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-055263 | MATHDB-346394

The symmetric permanent magnitude conjecture for random sign matrices

摘录 / EXCERPT: Symmetric permanent magnitude conjecture. With probability $1-o(1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-055264 | MATHDB-346395

The super-exponential permanent singularity conjecture

摘录 / EXCERPT: Permanent singularity conjecture. The probability that

conjecture | open | MathDB 页面 | 原文线索

V2-055265 | MATHDB-346397

The normality probability conjecture for random sign matrices

摘录 / EXCERPT: Normality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055266 | MATHDB-346398

The integral-spectrum probability conjecture for symmetric random sign matrices

摘录 / EXCERPT: Integral-spectrum conjecture. The probability that all eigenvalues of M_n^{sym} are integers is

conjecture | open | MathDB 页面 | 原文线索

V2-055267 | MATHDB-346399

The Gaussian-integral-spectrum conjecture for random sign matrices

摘录 / EXCERPT: Gaussian-integral-spectrum conjecture. The probability that all eigenvalues of M_n are Gaussian integers is

conjecture | open | MathDB 页面 | 原文线索

V2-055268 | MATHDB-346400

The irreducible characteristic-polynomial conjecture for random sign matrices

摘录 / EXCERPT: Irreducibility conjecture. With probability $1-o(1)$, the characteristic polynomial of M_n is irreducible.

conjecture | open | MathDB 页面 | 原文线索

V2-055269 | MATHDB-346402

The real-eigenvalue count conjecture for random sign matrices

摘录 / EXCERPT: Real-eigenvalue count conjecture. With high probability, M_n has $\Theta(\sqrt n)$ real eigenvalues.

conjecture | open | MathDB 页面 | 原文线索

V2-055270 | MATHDB-346403

The rank-resilience conjecture for random sign matrices

摘录 / EXCERPT: Rank-resilience conjecture. With probability $1-o(1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-055271 | MATHDB-346404

The stubborn-matrix conjecture for random sign matrices

摘录 / EXCERPT: Stubborn-matrix conjecture. With probability $1-o(1)$, M_n is stubborn.

conjecture | open | MathDB 页面 | 原文线索

V2-055272 | MATHDB-346516

Extension of the Hermite-process limit theorem to higher-order integrals

摘录 / EXCERPT: Higher-order limit conjecture. The conclusion of Theorem nCLT continues to hold if $p\geq 3$; that is, as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055273 | MATHDB-346518

Conjecture on error probability near the broadcasting threshold

摘录 / EXCERPT: Let $P_e(\delta)$ denote the optimal reconstruction error probability in the binary-tree broadcasting model, let δ_c be the critical channel parameter, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-055274 | MATHDB-346520

The Secret Leakage Planted Clique Conjecture

摘录 / EXCERPT: Let ρ be a distribution on k-subsets of $[n]$, and let $p_\rho(s)=\mathbb P_{S,S'\sim\rho^{\otimes 2}}[|S\cap S'|=s]$ be the intersection-size probability mass function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055275 | MATHDB-346528

The one-third-two-thirds conjecture for sorting probability

摘录 / EXCERPT: The one-third-two-thirds conjecture. For every finite poset $P=(X,\prec)$ that is not a chain, we have

conjecture | open | MathDB 页面 | 原文线索

V2-055276 | MATHDB-346546

Szarek-type singular-value bound for random matrices satisfying the general assumption

摘录 / EXCERPT: Let $\boldsymbol{M}_n$ be a real random matrix satisfying the paper's Assumption with parameter $K>0$, possibly with some moment assumptions on its entries. For a deterministic mat...

conjecture | open | MathDB 页面 | 原文线索

V2-055277 | MATHDB-346595

Strengthened varentropy conjecture for finite positive monotone concave sequences

摘录 / EXCERPT: Strengthened varentropy conjecture. The function K is log-concave: $\log K(t)$ is concave on $(-\gamma,+\infty)$.

conjecture | open | MathDB 页面 | 原文线索

V2-055278 | MATHDB-346636

Outer-perimeter proportion conjecture for high-dimensional Eden models

摘录 / EXCERPT: Outer-perimeter proportion conjecture. There exists a number $\operatorname{per}_d>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055279 | MATHDB-346659

Soberón's colorful Tverberg tolerance conjecture

摘录 / EXCERPT: Let r,d be fixed positive integers, let N be a positive integer, and let p_r be the probability that a random permutation of r numbers has at least one fixed point. For N...

conjecture | open | MathDB 页面 | 原文线索

V2-055280 | MATHDB-346684

The L^2-critical stochastic NLS blow-up conjecture

摘录 / EXCERPT: Let $u_0 \in H^1(\mathbb R)$ and let $u(t)$, $t>0$, be an evolution of the stochastic nonlinear Schrödinger equation with $\sigma=2$ and the specified space-time white-noise coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-055281 | MATHDB-346685

The L^2-supercritical stochastic NLS blow-up conjecture

摘录 / EXCERPT: Let $u_0\in H^1(\mathbb R)$ and let $u(t)$ be an evolution of the stochastic nonlinear Schrödinger equation with $\sigma>2$ and the specified space-time white-noise coefficient. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-055282 | MATHDB-346711

The transience-condition (T) conjecture for strongly mixing random environments

摘录 / EXCERPT: The transience-condition (T) conjecture. Directional transience along an open set $\mathcal U_{\ell}$ is equivalent to condition $(T)|\ell$.

conjecture | open | MathDB 页面 | 原文线索

V2-055283 | MATHDB-346846

The modified transitive Boolean-function tameness conjecture

摘录 / EXCERPT: Let $(p_n)_{n\geq 1}$ be a sequence of biases and let $(f_n)_{n\geq 1}$ be Boolean functions, with C_n the number of changes of $f_n(X_t^{(n)})$ during $(0,1)$. A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-055284 | MATHDB-346864

The two-dimensional moving-polymer scaling conjecture

摘录 / EXCERPT: Let $J>0$ be the intrinsic length of a moving polymer taking values in $\mathbf{R}^2$. Because the path $x\mapsto\mathbf{u}(t,x)$ behaves like a two-dimensional Brownian motion, it...

conjecture | open | MathDB 页面 | 原文线索

V2-055285 | MATHDB-346867

Convergence of uniform measures on finite sphere triangulations

摘录 / EXCERPT: Let $\mathcal{D}$ be trivial, and let ν_k be the uniform probability measure on sphere triangulations with at most k faces. The measures are regarded as transverse measures,...

conjecture | open | MathDB 页面 | 原文线索

V2-055286 | MATHDB-346986

The probability upper-bound conjecture for balanced non-transitive dice

摘录 / EXCERPT: Probability upper-bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055287 | MATHDB-347001

Hydrodynamic limit for the boundary driven zero-range process

摘录 / EXCERPT: Let I_N be the lattice site set, let Ω_N be the configuration space, and let $\{\mu^N\}_{N\in\mathbb N}$ be probability measures on Ω_N satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-055288 | MATHDB-347016

Majorization conjecture for strongly Rayleigh processes

摘录 / EXCERPT: Let $\mathfrak{X}$ be a strongly Rayleigh process on $[n]$, with kernel $g_{\mathfrak{X}}$, and let $\lambda_1,\dots,\lambda_n$ be the roots of $\overline{g}_{\mathfrak{X}}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-055289 | MATHDB-347028

Dominance of the continuous spectrum for random hyperbolic surfaces of large genus

摘录 / EXCERPT: Let $\mathbb{P}_{g,k(g)}$ be the probability measure on hyperbolic surfaces of genus g with $k(g)$ cusps, and let $X \in \mathcal{M}_{g,k(g)}$ be a random surface. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-055290 | MATHDB-347029

The coefficient-of-variation bound for fast mRNA-microRNA dynamics

摘录 / EXCERPT: Let $\rho_\text{fast}^{\delta,\gamma,p}$ denote the fast-regime distribution and let ρ_0^{δ} denote the corresponding free-mRNA distribution, with CV their coe...

conjecture | open | MathDB 页面 | 原文线索

V2-055291 | MATHDB-347075

Uniform Berry-Esseen conjecture for Rademacher sums

摘录 / EXCERPT: Let $x_1,\ldots,x_n$ be independent random variables uniformly distributed in $\{-1,1\}$, and let ... be a Rademacher sum with $\operatorname{Var}(X)=1$ and $0<a_i\leq a_1$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-055292 | MATHDB-347140

The cycle-factor conjecture for random hypercube subgraphs

摘录 / EXCERPT: Let $\mathcal{Q}^n$ be the n-dimensional hypercube, and let $\mathcal{Q}^n_p$ be the random subgraph obtained by retaining each edge independently with probability p. For fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-055293 | MATHDB-347167

Nonzero limiting endpoint distance for hyperbolic self-avoiding walks at a power-law step size

摘录 / EXCERPT: Consider continuous n-step self-avoiding walks on the hyperbolic space $\mathbb{H}^d$, with steps of length ε and with the distance between every pair of vertices con...

conjecture | open | MathDB 页面 | 原文线索

V2-055294 | MATHDB-347184

The 3/2 conjecture for triangle covering and packing in random graphs

摘录 / EXCERPT: The 3/2 random-graph conjecture. For all $C>3/2$ and $G=G(n,m)$, with high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-055295 | MATHDB-347190

Strong uniform mod-Poisson convergence in quadratic-differential strata

摘录 / EXCERPT: Let $\mathcal{C}$ be a non-hyperelliptic connected component of a stratum of holomorphic quadratic differentials, and let $p_{\mathcal{C}}(k)$ be the probability that a random quad...

conjecture | open | MathDB 页面 | 原文线索

V2-055296 | MATHDB-347200

Non-reversible condensate movement conjecture

摘录 / EXCERPT: Let S be the finite state space of the underlying random walk, with jump rates $r(x,y)$, and let $S_{\star}$ be the set of sites on which the condensate may reside. Define the Ma...

conjecture | open | MathDB 页面 | 原文线索

V2-055297 | MATHDB-347201

Reversible-scale conjecture for symmetric condensate transitions

摘录 / EXCERPT: Let $S_{\star}$ be the set of sites on which the condensate may reside, and let $r(x,y)$ be the jump rates of the underlying random walk. Let $Y^{\mathrm{rv}}(\cdot)$ be the Markov...

conjecture | open | MathDB 页面 | 原文线索

V2-055298 | MATHDB-347213

Detailed-balance characterization for the discrete KdV map

摘录 / EXCERPT: Detailed-balance characterization for the discrete KdV map. The random variables U and V should be independent if and only if there exist $\lambda,c_1,c_2>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055299 | MATHDB-347214

Ultra-discrete KdV detailed-balance characterization on a half-infinite domain

摘录 / EXCERPT: Ultra-discrete KdV detailed-balance conjecture. The variables U and V should be independent if and only if there exist $\lambda,c_1,c_2>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055300 | MATHDB-347237

The capacity conjecture for binary erasure channels

摘录 / EXCERPT: Let $\eta\in(0,1)$ be an erasure probability, and let $\mathsf{BEC}(\eta)$ denote the binary erasure channel with erasure probability η. Let C_{perm} denote noisy p...

conjecture | open | MathDB 页面 | 原文线索

V2-055301 | MATHDB-347345

Quotienting conjecture for Picard groups of double covers of punctured curves

摘录 / EXCERPT: Let C be a smooth projective curve over $\mathbb F_q$, let $S\subset C$ be a reduced effective divisor, and consider smooth projective double covers $D\to C$ unramified over S....

conjecture | open | MathDB 页面 | 原文线索

V2-055302 | MATHDB-347358

Critical Hausdorff dimension conjecture for branching random walks on hyperbolic groups

摘录 / EXCERPT: Let Γ be a nonelementary hyperbolic group with word metric d, let $\partial\Gamma$ have a visual metric d_a for some parameter $a>1$, and let μ be an admissible, sup...

conjecture | solved | MathDB 页面 | 原文线索

V2-055303 | MATHDB-347488

Babai-Vu-Wood irreducibility conjecture for random characteristic polynomials

摘录 / EXCERPT: Babai-Vu-Wood conjecture. φ is irreducible with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055304 | MATHDB-347555

Asymptotic rarity conjecture for non-simple tile labels

摘录 / EXCERPT: Asymptotic label conjecture. There exists an index $j\notin I$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055305 | MATHDB-347593

Final-size law for epidemics with preventive rewiring

摘录 / EXCERPT: Let $\bar{\boldsymbol{X}}^{(n)}(0)$ denote the vector of rescaled epidemic variables, let $x(t)=(s(t),i(t),i_E(t),w(t))$ be the solution of the deterministic limiting equations, an...

conjecture | open | MathDB 页面 | 原文线索

V2-055306 | MATHDB-347606

Sufficient conditions for Nash equilibria in non-zero-sum Dynkin games

摘录 / EXCERPT: Let X^1,X^2,Y^1,Y^2,Z^1,Z^2 be the payoff processes of a non-zero-sum Dynkin game, with payoff functions ... ... Define the regions...

conjecture | open | MathDB 页面 | 原文线索

V2-055307 | MATHDB-347657

Generality conjecture for coalescent-walk processes

摘录 / EXCERPT: For $[? Let $[? no

conjecture | open | MathDB 页面 | 原文线索

V2-055308 | MATHDB-347710

Strict inclusion conjecture for causal information integration and causal information structure

摘录 / EXCERPT: Let $\mathcal{M}_{CII}$ denote the model of causal information integration and let $\mathcal{M}_{CIS}$ denote the model of causal information structure, with...

conjecture | open | MathDB 页面 | 原文线索

V2-055309 | MATHDB-347733

The noisy equitable stochastic block model detection conjecture

摘录 / EXCERPT: Let $\bT_\delta$ be the noise operator that rewires $\lfloor\delta n\rfloor$ swaps of a d-regular graph, and let $\sG_{n,d,k,\eta}^{\,\mathrm{eq}}$ be the equitable stochastic bl...

conjecture | open | MathDB 页面 | 原文线索

V2-055310 | MATHDB-347735

The GUE L^∞-to-L^1 tensor-norm conjecture

摘录 / EXCERPT: Fix an integer $r\geq 2$. Let $X_1^{(k)},\ldots,X_r^{(k)},Y_1^{(k)},\ldots,Y_r^{(k)}$ be independent self-adjoint $k\times k$ GUE matrices, and write $X^{(k)}=(X_j^{(k)})_{j=1}^r$...

conjecture | open | MathDB 页面 | 原文线索

V2-055311 | MATHDB-347848

The random-regular-graph MaxCut pseudoexpectation conjecture

摘录 / EXCERPT: Random-regular-graph MaxCut conjecture. For some $\delta>0$, with high probability there is a degree-n^δ pseudoexpectation operator $\widetilde{\mathop{\varmathbb{E}}}$ on B...

conjecture | open | MathDB 页面 | 原文线索

V2-055312 | MATHDB-347905

Exact earth mover's correlation formula for bivariate normal variables

摘录 / EXCERPT: Let (X,Y) be bivariate normal random variables, and write $\varrho(X,Y)=\varrho$ for their correlation. The bivariate normal earth mover's correlation conjecture. One has ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-055313 | MATHDB-347906

The optimal coupling conjecture for earth mover's covariance

摘录 / EXCERPT: The optimal coupling conjecture. Under “general conditions”, the infimum can be attained or computed among couplings satisfying $X=X'$ or $Y=Y'$. The text presents this as a conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-055314 | MATHDB-347912

The lattice-avoidance criterion for the substitution property

摘录 / EXCERPT: Let f be the characteristic function of a probability measure with a nontrivial absolutely continuous component, and let φ be the density function of this component. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055315 | MATHDB-347944

Uniqueness without a hard-core assumption for repulsive Gibbs point processes

摘录 / EXCERPT: Let ϕ be a pair potential satisfying assumptions (A2) and (A3), with activity z and inverse temperature β. A Gibbs measure is a probability measure describing the equi...

conjecture | open | MathDB 页面 | 原文线索

V2-055316 | MATHDB-347947

The maximal-index conjecture for sums of matrix entries from a probability vector

摘录 / EXCERPT: Maximal-index conjecture. The maximal integer $h(d)$ for which $A+G\leq d$ holds for every probability vector p is

conjecture | open | MathDB 页面 | 原文线索

V2-055317 | MATHDB-348058

Asymptotic concentration conjecture for the malnormality statistic

摘录 / EXCERPT: For each n, let $\mathcal J_n$ denote the random matrix ensemble considered in the paper, let $\operatorname{mal}(\mathcal J_n)$ be its malnormality statistic, and let $\mu(n)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-055318 | MATHDB-348105

Rank conjecture for random sign matrices

摘录 / EXCERPT: Let B_n be an $n\times n$ random sign matrix with independent entries taking the values 1 and -1 with probability $1/2$ each, and let k be any positive integer. Rank conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-055319 | MATHDB-348169

The NAND 2D regular-grid broadcasting conjecture

摘录 / EXCERPT: Let $P_{X_k}^{+}$ and $P_{X_k}^{-}$ denote the distributions of the observations at layer k conditioned on the source bit being 1 and 0, respectively. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-055320 | MATHDB-348170

The superharmonic-function conjecture for the NAND 2D regular grid

摘录 / EXCERPT: Let $\mathcal{Y}^*=\bigcup_{k\in\mathbb{N}\setminus\{0\}}\mathcal{Y}^k$ be the set of nonempty finite strings over $\mathcal{Y}$, let $\{Y_k:k\in\mathbb{N}\}$ be the Markov chain a...

conjecture | open | MathDB 页面 | 原文线索

V2-055321 | MATHDB-348183

The aging conjecture for the running maximum of the Mott random walk

摘录 / EXCERPT: Aging conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055322 | MATHDB-348203

Gnutzmann-Smilansky-Weber quantum-graph nodal-statistics conjecture

摘录 / EXCERPT: Let a quantum graph be a metric graph with edge lengths, and let the nodal domains of an eigenfunction be the connected components of the complement of its zero set. Consider well-...

conjecture | open | MathDB 页面 | 原文线索

V2-055323 | MATHDB-348204

Al-Obeid-Band-Berkolaiko nodal-surplus Gaussian-limit conjecture

摘录 / EXCERPT: Let ... be its nodal surplus random variable. Write $\operatorname{Var}(\sigma)$ for its variance. Al-Obeid-Band-Berkolaiko's conjecture. As $\to\infty$, ... where the convergence is i...

conjecture | open | MathDB 页面 | 原文线索

V2-055324 | MATHDB-348205

Gnutzmann-Smilansky-Weber universal nodal-surplus Gaussian conjecture

摘录 / EXCERPT: Let $\{\Gamma_{\vec{l}}^{(\beta)}\}_{\beta\nearrow\infty}$ be any sequence of standard graphs parameterized by their first Betti numbers, with each graph having rationally independ...

conjecture | open | MathDB 页面 | 原文线索

V2-055325 | MATHDB-348212

Homogenization conjecture for weakly reinforced Pólya urns on countable graphs

摘录 / EXCERPT: Let G be a countable graph with uniformly bounded degrees. Let N_t be the system of edge weights initialized at 1 and evolving according to the continuous-time weakly reinfor...

conjecture | open | MathDB 页面 | 原文线索

V2-055326 | MATHDB-348223

Continuity conjecture for the minimal loss function under additive perturbations

摘录 / EXCERPT: Continuity conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-055327 | MATHDB-348237

The homogeneous magnetic-field direction-reversal probability conjecture

摘录 / EXCERPT: Let $a_1(x,t,u)$ denote the first component of the Feynman checker amplitude at position x and time t in the external field u, and define $u(x+1/2,t+1/2)=-1$ if both x and...

conjecture | open | MathDB 页面 | 原文线索

V2-055328 | MATHDB-348238

The limiting probability conjecture for the modified Feynman checker

摘录 / EXCERPT: Let $b_1(x,t)$ denote the first component of the modified Feynman checker amplitude at position x and time t. Modified Feynman checker limiting-probability conjecture. The tota...

conjecture | open | MathDB 页面 | 原文线索

V2-055329 | MATHDB-348278

The local obstruction conjecture for Bernoulli matrix singularity

摘录 / EXCERPT: Let $B_n(p)$ be an $n\times n$ random matrix with independent and identically distributed Bernoulli entries of mean p, and let $q_n(p)=\mathbb{P}[B_n(p)\text{ is singular}]$. Loc...

conjecture | open | MathDB 页面 | 原文线索

V2-055330 | MATHDB-348279

Litvak-Tikhomirov sparse Bernoulli singularity conjecture

摘录 / EXCERPT: Litvak-Tikhomirov conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055331 | MATHDB-348280

Nguyen's singularity conjecture for random matrices with fixed row sums

摘录 / EXCERPT: Let Q_n be a random matrix whose independent rows are chosen uniformly from the vectors in $\{0,1\}^n$ having sum exactly $\lfloor n/2\rfloor$. Nguyen's conjecture. ... This model...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055332 | MATHDB-348281

Borot-Eynard-Majumdar-Nadal asymptotic conjecture for the generalized Tracy-Widom distribution

摘录 / EXCERPT: Borot-Eynard-Majumdar-Nadal conjecture. The left-tail asymptotic expansion is

conjecture | open | MathDB 页面 | 原文线索

V2-055333 | MATHDB-348296

The limit shape conjecture for minimal membranes

摘录 / EXCERPT: Limit shape conjecture for minimal membranes. There exists a unique surface $S\subset\mathbb R^n$ with boundary $\langle L\rangle$ such that, for every $\epsilon>0$, there is $N>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-055334 | MATHDB-348297

The limiting density measure conjecture for minimal membranes

摘录 / EXCERPT: Limiting density measure conjecture for minimal membranes. As $t\to\infty$, the measures $\mu_{L,t}$ converge to a limit measure

conjecture | open | MathDB 页面 | 原文线索

V2-055335 | MATHDB-348325

Density conjecture for QID distributions with finite quasi-Lévy measure

摘录 / EXCERPT: Density conjecture. The class of QID distributions on $\mathbb{R}^{d}$ with finite quasi-Lévy measure and zero Gaussian variance is dense in the space of probability distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-055336 | MATHDB-348343

Post-first-step monotonicity conjecture for majority dynamics

摘录 / EXCERPT: Let $G_{n,p}$ be a random graph sampled independently of an arbitrary initial red-blue coloring of its vertices, and let $\varepsilon>0$. Assume ... After the first step of majority...

conjecture | open | MathDB 页面 | 原文线索

V2-055337 | MATHDB-348353

Critical constraint conjecture for hypercubic lattices

摘录 / EXCERPT: Let $\kappa_{\mathrm{c}}(d)$ be the critical constraint, defined by ... Here $t_{\mathrm{c}}^{\kappa}(d)$ is the critical time for constrained-degree percolation with constraint...

conjecture | open | MathDB 页面 | 原文线索

V2-055338 | MATHDB-348366

Expectation asymptotic for the analysis-to-synthesis operator-norm ratio

摘录 / EXCERPT: Let A be a random matrix in $\mathbb R^{m\times k}$ with independently and identically normally distributed entries, and let L be the synthesis operator appearing in the formul...

conjecture | open | MathDB 页面 | 原文线索

V2-055339 | MATHDB-348415

Conjectured sharp joint density bound for near-unit-circle roots

摘录 / EXCERPT: Let $\mathbb{T}_0$ be the domain of the ordered k-tuples used in the paper, and let $p_k(\mathbf{x},\mathbf{y})$ denote the corresponding joint density. Sharp density conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-055340 | MATHDB-348434

Singularity conjecture for random matrices with anti-concentrated entries

摘录 / EXCERPT: Let $\mathcal{D}$ be a distribution over $\mathbb{R}$ and set $p=\|\mathcal{D}\|_{\infty}$, where $\|\mathcal{D}\|_{\infty}=\max_x \mathcal{D}(x)$. Let M be a random $n \times n$...

conjecture | open | MathDB 页面 | 原文线索

V2-055341 | MATHDB-348438

Scaling-limit conjecture for directed polymers on fractal graphs

摘录 / EXCERPT: Let the underlying random walk satisfy the local limit theorem in the sense discussed by Croydon and Hambly. For $T>0$, let Z^0_{Tn,β_n} be the directed-polymer partition fun...

conjecture | open | MathDB 页面 | 原文线索

V2-055342 | MATHDB-348468

Unique-infinite-cluster conjecture for finitary random interlacement

摘录 / EXCERPT: Let $d\ge 3$ and $u>0$, and let $\mathcal{FI}^{u,T}_d$ be finitary random interlacement with parameters u and T. Unique-infinite-cluster conjecture. There is a critical value...

conjecture | open | MathDB 页面 | 原文线索

V2-055343 | MATHDB-348469

Asymptotic critical-value conjecture for finitary random interlacement

摘录 / EXCERPT: For each $d\ge 3$, let $T_c(u)$ be the critical value from the preceding conjecture. Asymptotic critical-value conjecture. There is a constant c_d such that ... The conjecture is b...

conjecture | open | MathDB 页面 | 原文线索

V2-055344 | MATHDB-348505

Infinitesimal monotonicity of matrices

摘录 / EXCERPT: Suppose that $(\mathcal{M},\varphi,\varphi')$ is an infinitesimal probability space, and let ... be sets that are infinitesimally monotone in $\mathcal{M}$. For each r, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055345 | MATHDB-348572

Equality of inner and outer densities for activated random walkers

摘录 / EXCERPT: Density equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055346 | MATHDB-348594

Converse to the logical and probabilistic exclusivity implication

摘录 / EXCERPT: LEP--PEP converse conjecture. If PEP holds, then LEP holds:

conjecture | open | MathDB 页面 | 原文线索

V2-055347 | MATHDB-348690

The scaling and Brownian approximation conjecture for gambler's ruin probabilities

摘录 / EXCERPT: For positive integers A,B,C, let $P_{A,B,C}(\sigma)$ denote the first hitting probability of the permutation $\sigma\in S_3$ in the gambler's ruin walk, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055348 | MATHDB-348751

The random-matrix Markov-Krein correspondence conjecture

摘录 / EXCERPT: Let M_N be a sequence of unitarily invariant random matrices, let $\lambda=(\lambda_1,\ldots,\lambda_N)$ be the eigenvalue vector of M_N, and let $\mu=\pi_{N,N-1}\lambda$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-055349 | MATHDB-348773

Binomial condition (S_2) conjecture

摘录 / EXCERPT: Binomial (S_2) conjecture. X satisfies Condition (S_2).

conjecture | open | MathDB 页面 | 原文线索

V2-055350 | MATHDB-348774

Poisson condition (S_2) conjecture

摘录 / EXCERPT: Poisson (S_2) conjecture. X satisfies Condition (S_2).

conjecture | open | MathDB 页面 | 原文线索

V2-055351 | MATHDB-348868

Menon-Srinivasan conjecture on preservation of Feller processes for scalar conservation laws

摘录 / EXCERPT: Let $\rho(x,t)$ be the entropy solution of the scalar conservation law with initial data ξ. The initial data is either a white noise derived from a spectrally positive Lévy pro...

conjecture | open | MathDB 页面 | 原文线索

V2-055352 | MATHDB-348934

The simultaneous-crossing independence conjecture for the 4-state Potts model

摘录 / EXCERPT: Take a polygon Ω whose boundary is composed of edges of a hexagonal lattice of mesh h, and let A_1,A_2,A_3,A_4,A_5,A_6 be boundary vertices. Assign each hexagon one of f...

conjecture | open | MathDB 页面 | 原文线索

V2-055353 | MATHDB-348935

The conformal-limit interface probability conjecture for critical site percolation

摘录 / EXCERPT: Let Ω_h be a polygon whose boundary consists of edges of a hexagonal lattice of meshsize h, with distinct side midpoints a_h,b_h and lattice-edge midpoints u_h,v_h. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-055354 | MATHDB-348964

Optimal strategy upper-bound conjecture for card guessing with Yes/No feedback

摘录 / EXCERPT: Let m and n denote the card multiplicity and number of card types, respectively, and let $\mathbf{S}_{m,n}$ be the score under an optimal strategy with Yes/No feedback. Optimal...

conjecture | open | MathDB 页面 | 原文线索

V2-055355 | MATHDB-348982

Cutoff for biased one-sided transposition shuffles on the hyperoctahedral group

摘录 / EXCERPT: Let $\operatorname{OST}_{n,w}$ be the biased one-sided transposition shuffle, where $w(j)/j$ is monotonically decreasing, and let $N_w(n)$ denote its normalising quantity. Cutoff c...

conjecture | open | MathDB 页面 | 原文线索

V2-055356 | MATHDB-348983

Cutoff scale for biased one-sided transposition shuffles with general weights

摘录 / EXCERPT: Let $\operatorname{OST}_{n,w}$ be the biased one-sided transposition shuffle, where $w(j)/j$ is monotonically decreasing, and let $N_w(n)$ and $N'_w(n)$ be the quantities used in t...

conjecture | open | MathDB 页面 | 原文线索

V2-055357 | MATHDB-348984

Separation cutoff for biased one-sided transposition shuffles

摘录 / EXCERPT: Let $\operatorname{OST}_{n,\alpha}$ be the biased one-sided transposition shuffle with weight parameter $\alpha\in(0,\infty)$, and let $t_{n,\alpha}$ be the total-variation cutoff...

conjecture | open | MathDB 页面 | 原文线索

V2-055358 | MATHDB-348987

Random transposition cutoff on generalized symmetric groups

摘录 / EXCERPT: Let $G_{r,n}\cong\mathbb{Z}_r\wr S_n$ be the generalized symmetric group, and let the random transposition shuffle be defined as in the source. Random-transposition cutoff conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-055359 | MATHDB-348988

Biased one-sided transposition cutoff on generalized symmetric groups

摘录 / EXCERPT: Let $G_{r,n}\cong\mathbb{Z}_r\wr S_n$ be the generalized symmetric group, let $w(j)=j^\alpha$, and let $t_{n,\alpha}$ be the cutoff scale defined in the source. Biased one-sided cu...

conjecture | open | MathDB 页面 | 原文线索

V2-055360 | MATHDB-349174

Dihedral commutant conjecture for dual-unitary qubit circuits

摘录 / EXCERPT: Let $\mathcal M'_T$ be the commutant of the algebra generated by the operators M_a and $M_{ab,\iota}+R_{2t}M_{ab,\iota}R_{2t}$ on the periodic chain of $2t$ spins, where R_{2t}...

conjecture | open | MathDB 页面 | 原文线索

V2-055361 | MATHDB-349267

Average-case hardness of approximating probabilities for FLO circuits

摘录 / EXCERPT: Let V be a passive or active fermionic linear-optics circuit initialized in the state $\left|\Psi_{\mathrm{in}}\right>$, let $\mathbf{x}_0$ be a fiducial outcome, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055362 | MATHDB-349268

The critical orthogonal-group moment-of-moments conjecture

摘录 / EXCERPT: Critical orthogonal-group conjecture. For every $k>1$ with $s=1/\sqrt{k}$,

conjecture | open | MathDB 页面 | 原文线索

V2-055363 | MATHDB-349269

The critical symplectic-group moment-of-moments conjecture

摘录 / EXCERPT: Critical symplectic-group conjecture. For these parameters,

conjecture | open | MathDB 页面 | 原文线索

V2-055364 | MATHDB-349302

Small deviation conjecture for geodesics in semi-discrete Brownian last passage percolation

摘录 / EXCERPT: Let $\{B_i(\cdot)\}_{i\in\mathbb{Z}}$ be a sequence of two-sided standard Brownian motions on $\mathbb{R}$, and let Π_n be the almost surely unique geodesic from $(0,0)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-055365 | MATHDB-349363

The computational hardness conjecture for constant discrepancy

摘录 / EXCERPT: Let A be drawn from the $(m,n,1/2)$-Bernoulli ensemble, where C is a universal constant and $n\geq C m\log m$. Computational hardness conjecture. With high probability, no effi...

conjecture | open | MathDB 页面 | 原文线索

V2-055366 | MATHDB-349491

Reversibility conjecture for Fleming-Viot diffusions

摘录 / EXCERPT: Let $\tt FV(\alpha,\theta)$ be the Fleming-Viot diffusion with parameters $\alpha\in(0,1)$ and $\theta>-\alpha$, and let $\tt PDRM(\alpha,\theta)$ denote its stationary Poisson-Dir...

conjecture | open | MathDB 页面 | 原文线索

V2-055367 | MATHDB-349540

Asymptotic unfairness under convergent weight distributions

摘录 / EXCERPT: Let $m^{(n)}$ be a sequence of weight distributions with corresponding probability distributions $(P^{(n)})_{n \in \mathbb{N}}$, where $P^{(n)}=(p_i^{(n)})_{i \in \mathbb{N}}$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-055368 | MATHDB-349541

Monoid conjecture for bracelet matrices

摘录 / EXCERPT: Let $\mathcal{L}_d$ be the set of bracelet matrices of dimension d. A monoid is a set closed under matrix multiplication and containing an identity element $\mathbb{I}_d$. Monoid...

conjecture | open | MathDB 页面 | 原文线索

V2-055369 | MATHDB-349542

Circulant unistochastic matrices and circulant unitary matrices conjecture

摘录 / EXCERPT: Let $\mathcal{C}_d$ denote the set of circulant matrices, $\mathcal{U}_d$ the set of unistochastic matrices, and $\mathcal{C}_d^2$ the set of doubly circulant unistochastic matrice...

conjecture | open | MathDB 页面 | 原文线索

V2-055370 | MATHDB-349569

Weak spatial mixing conjecture for BMS surveys

摘录 / EXCERPT: Consider the model on a regular tree of degree parameter d, with broadcast parameter θ, and a BMS survey channel W. Write $P_e(W)$ for the survey error probability, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055371 | MATHDB-349576

Period equality conjecture for the hunger game

摘录 / EXCERPT: Let a finite irreducible Markov chain have rational transition probabilities, stationary distribution π, and let n be the least common denominator of the entries of π. A...

conjecture | open | MathDB 页面 | 原文线索

V2-055372 | MATHDB-349641

The entropic-time conjecture for L_2 cutoff on random Cayley graphs

摘录 / EXCERPT: The entropic-time conjecture. Under conditions similar to those in the paper's total-variation cutoff result, with high probability the random walk on G_k exhibits cutoff in the...

conjecture | open | MathDB 页面 | 原文线索

V2-055373 | MATHDB-349757

Linear variance conjecture for zero-free probability generating functions

摘录 / EXCERPT: Let $X\in\{0,\ldots,n\}$ be a random variable with $p_0p_n>0$ and probability generating function f_X. Let $\delta>0$ and $R\geqslant1$. If every zero ζ of f_X satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-055374 | MATHDB-349777

Global reversibility classification conjecture for involutive walks

摘录 / EXCERPT: Let P^λ be an $n\times n$ stochastic matrix of the form P^λ. Let $J(n)$ denote the $n\times n$ matrix whose entries are all equal to $1/n$, and let the matrices in...

conjecture | open | MathDB 页面 | 原文线索

V2-055375 | MATHDB-349778

Strong anti-diagonal eigenvalue property conjecture for stochastic matrices

摘录 / EXCERPT: Let A be a stochastic matrix with distinct eigenvalues. Say that A has the strong anti-diagonal eigenvalue property when it satisfies the stronger anti-diagonal eigenvalue cond...

conjecture | open | MathDB 页面 | 原文线索

V2-055376 | MATHDB-349889

Multiplet probability conjecture for 3-class rank one

摘录 / EXCERPT: Let f be a regular conductor, let $d>0$ be a quadratic fundamental discriminant, and consider 3-admissible pairs (f,d) with 3-class rank $\varrho=1$. Let $(L_1,\ldots,L_m)$...

conjecture | open | MathDB 页面 | 原文线索

V2-055377 | MATHDB-349951

Tail statistics of a sum of heavy-tailed random matrix ensembles

摘录 / EXCERPT: Let $X_1,\ldots,X_L$ be independently (not necessarily identically) distributed random matrices with heavy tails and unitary invariance, meaning that their eigenvalues and eigenvec...

conjecture | open | MathDB 页面 | 原文线索

V2-055378 | MATHDB-349952

Poisson tail-statistics conjecture for heavy-tailed stable random matrices

摘录 / EXCERPT: Let X be a heavy-tailed stable random matrix with unitary invariance, meaning that its eigenvalues and eigenvectors are uncorrelated. Suppose that the largest eigenvalues are con...

conjecture | open | MathDB 页面 | 原文线索

V2-055379 | MATHDB-349964

Quenched law of large numbers for the final ranking in multi-player leagues

摘录 / EXCERPT: Let $(s_i)_{i\u2208 [2n-1]_0}$ be a sequence of initial strengths, and let $(\tilde T_i)_{i\in [2n-1]_0}$ be the teams reordered from weakest to strongest according to these streng...

conjecture | open | MathDB 页面 | 原文线索

V2-055380 | MATHDB-349973

Random sampling conjecture for high-order spectra

摘录 / EXCERPT: Random sampling conjecture. The orbit of a generic signal $x\in\mathbb{C}^N$ can be recovered from $O(N)$ random samples of its high-order spectra with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055381 | MATHDB-349982

General finite-support random matrix corank conjecture

摘录 / EXCERPT: Let ξ be a random variable supported on finitely many points. For $k\geq 1$, let $\xi_1,\ldots,\xi_{k+1}$ be independent samples of ξ, and let $M_n(\xi)$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-055382 | MATHDB-350077

The extreme-point and convex-hull conjecture for laws of distinct values

摘录 / EXCERPT: Let $(X_1,X_2,\ldots)$ be an infinite exchangeable sequence, and let K_n be the number of distinct values among its first n terms. For $m\in\{1,2,\ldots,\infty\}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055383 | MATHDB-350131

Conjecture on fluctuations of singular derivative Gibbs functionals

摘录 / EXCERPT: Let $Z_t(F)$ be the functional of the derivative Gibbs measure associated with branching Brownian motion, and suppose that $F(x)\sim x^{-\alpha}$ as $x\to0$ for some α satis...

conjecture | open | MathDB 页面 | 原文线索

V2-055384 | MATHDB-350191

The random-matrix conjecture for the Hadamard-product mean square

摘录 / EXCERPT: Let $\boldsymbol{\ell}=(\ell_\chi)$ be a tuple of nonnegative integers indexed by characters modulo q, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055385 | MATHDB-350194

Mass-point conjecture for steady states of dual Markov chains

摘录 / EXCERPT: Let $\boldsymbol{\kappa}_\lambda$ be the nonnegative left eigenvector of the dual multiple stochastic matrices associated with a measure μ and parameter λ. A mass point...

conjecture | open | MathDB 页面 | 原文线索

V2-055386 | MATHDB-350196

Null recurrence conjecture for Jacobi--Piñeiro Markov chains

摘录 / EXCERPT: Let the dual Jacobi--Piñeiro random walks be the Markov chains associated with the Jacobi--Piñeiro multiple orthogonal polynomials, with parameter γ. In the recurrent parame...

conjecture | open | MathDB 页面 | 原文线索

V2-055387 | MATHDB-350198

The Zigzag Conjecture for the chromatic number of random graphs

摘录 / EXCERPT: Zigzag Conjecture. Set $p=\frac12$. There is a sequence of intervals of length $n^{\lambda+o(1)}$ containing $\chi(G_{n,1/2})$ with high probability. However, for every fixed...

conjecture | open | MathDB 页面 | 原文线索

V2-055388 | MATHDB-350232

The basic ℓ^2-distance conjecture for Ramanujan graphs

摘录 / EXCERPT: Let X be a sequence of Ramanujan graphs, let n denote the number of vertices, let d be their common degree, and set $p=d-1$. Let $d_2(t)$ be the average squared ℓ^2 dist...

conjecture | open | MathDB 页面 | 原文线索

V2-055389 | MATHDB-350235

Signed clique expectation decay conjecture

摘录 / EXCERPT: Let k and p be fixed, let $\mathcal{G}(n,p,d)$ denote the noisy high-dimensional random geometric graph model without edge subsampling, and let $\tau_{[k]}$ be the signed k-c...

conjecture | open | MathDB 页面 | 原文线索

V2-055390 | MATHDB-350236

Signed cycle expectation decay conjecture

摘录 / EXCERPT: Let C^0 be a cycle of length k with its signed cycle statistic κ_{C^0}, and let $\mathcal{G}(n,p,d)$ denote the noisy high-dimensional random geometric graph model witho...

conjecture | open | MathDB 页面 | 原文线索

V2-055391 | MATHDB-350262

Bell-number bounds for normalized raw moments of the Poisson distribution

摘录 / EXCERPT: Let $B(k,\mu)=\sum_i {k\brace i}\mu^i$ denote the Bell polynomial, and write $B_x=B(x,1)$. For a Poisson random variable with mean μ, its kth raw moment is $B(k,\mu)$. Bell-n...

conjecture | open | MathDB 页面 | 原文线索

V2-055392 | MATHDB-350277

Eigenvector delocalization conjecture for random graph walks

摘录 / EXCERPT: Let G be the random graph and let $\nu_1,\dots,\nu_n$ be the eigenvectors appearing in Theorem 1 of the paper, with p as in that theorem. For a vector v, write $\|v\|_\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-055393 | MATHDB-350317

Conjecture on projected Wiener dynamics for mass-conserving Allen-Cahn interfaces

摘录 / EXCERPT: Projected Wiener dynamics conjecture. For $T_\varepsilon\leq c\varepsilon\eta_\varepsilon^{-1}\wedge\tau$, one expects

conjecture | open | MathDB 页面 | 原文线索

V2-055394 | MATHDB-350435

The Laplace-transform criterion for \(L_m(s)=1/(mse^s+1)\)

摘录 / EXCERPT: Laplace-transform criterion. The function L_m is the Laplace transform of some random variable if and only if $m=0$, corresponding to a trivial random variable, or $m\ge e$.

conjecture | open | MathDB 页面 | 原文线索

V2-055395 | MATHDB-350436

The age-of-information ordering conjecture for finite-buffer systems

摘录 / EXCERPT: Best-systems conjecture. Regardless of the message-size distribution,

conjecture | open | MathDB 页面 | 原文线索

V2-055396 | MATHDB-350539

Central limit theorem for infinitesimal gradient boosting

摘录 / EXCERPT: Let $\hat{F}^{\lambda}_{[t/\lambda]}$ be the stochastic gradient boosting process, let $(\hat{F}_t)_{t\geq 0}$ be its infinitesimal gradient boosting limit, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055397 | MATHDB-350555

Monotonicity conjecture for principal determinants of the two-season system matrix

摘录 / EXCERPT: Let $\overline{D}_n$ denote the principal determinant of the system matrix in (2x2). The system matrix arises in the linear equations for the finite-horizon ruin probabilities of t...

conjecture | open | MathDB 页面 | 原文线索

V2-055398 | MATHDB-350614

Janson's height and width conjectures for simply generated trees

摘录 / EXCERPT: Let ${\mathrm{w}}=(w_k,k \ge 0)$ be a weight sequence with $w_0>0$ and with $w_k>0$ for some $k \ge 2$. For $n \ge 0$ with $Z_n({\mathrm{w}})>0$, let ${\mathcal{T}}_n$ be a simply...

conjecture | open | MathDB 页面 | 原文线索

V2-055399 | MATHDB-350620

Positivity conjecture for the multivariate Bessel addition law

摘录 / EXCERPT: Positivity conjecture. For every such $\a$ and $\b$, there exists a probability measure on N-tuples $\c=(c_1\le\dots\le c_N)$ satisfying the identity for all...

conjecture | open | MathDB 页面 | 原文线索

V2-055400 | MATHDB-350766

Large-degree density conjecture for low-lying eigenphases of hyperelliptic curves

摘录 / EXCERPT: Density Conjecture. For a fixed q, one has

conjecture | open | MathDB 页面 | 原文线索

V2-055401 | MATHDB-350793

Zooming-in conjecture at the maximal distance from the origin for Lévy processes

摘录 / EXCERPT: Assume that the Lévy process X has finite lifetime $\zeta\in(0,\infty)$. Let ... be the last time at which the Euclidean norm is maximal. Set $M:=X_\tau$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-055402 | MATHDB-350888

Strong mixing and uniform density conjecture for LLL coefficients

摘录 / EXCERPT: LLL mixing and density conjecture. (i) The sequence $(|\mu_{k(i)}|)_{i=1,2,\ldots}$ is strongly mixing as a stochastic process. (ii) Each $|\mu_{k(i)}|$ is contained in a compact s...

conjecture | open | MathDB 页面 | 原文线索

V2-055403 | MATHDB-350889

Bounded output-density conjecture for LLL and LLL-SP

摘录 / EXCERPT: Bounded output-density conjecture. For every such $\mathcal{D}$, the probability density function of $\mathcal{D}^{\circ}$ is bounded from above by a constant C that depends only...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-055404 | MATHDB-350904

Rademacher matrix norm conjecture

摘录 / EXCERPT: Rademacher matrix norm conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-055405 | MATHDB-350905

General independent-entry matrix norm conjecture

摘录 / EXCERPT: General independent-entry matrix norm conjecture. Then $$

conjecture | open | MathDB 页面 | 原文线索

V2-055406 | MATHDB-350906

The first-return asymptotic conjecture for transient transitive graphs

摘录 / EXCERPT: The first-return asymptotic conjecture. One has $f_n\asymp u_n$, and more precisely

conjecture | open | MathDB 页面 | 原文线索

V2-055407 | MATHDB-350907

The one-big-excursion conjecture for conditioned return times

摘录 / EXCERPT: Let $G=(V,E)$ be a locally finite, connected, transitive, transient graph with spectral radius ρ, fix $o\in V$, and let $(X_n)_{n\geq 0}$ be simple random walk started at o....

conjecture | open | MathDB 页面 | 原文线索

V2-055408 | MATHDB-351008

Existence of a finite dimension attaining the limiting 1-independent percolation threshold

摘录 / EXCERPT: For $d\geq 3$, let $p_{1,c}(\mathbb{Z}^d)$ denote the critical probability for 1-independent percolation on the integer lattice. High-dimensional percolation conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-055409 | MATHDB-351009

Two-state component-size conjecture for 1-independent percolation on the hypercube

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, let $p\in (\frac{1}{2},1]$ be fixed, and let $\mathcal{M}_{1,\geq p}(Q_n)$ denote the 1-independent percolation measures on Q_n whos...

conjecture | open | MathDB 页面 | 原文线索

V2-055410 | MATHDB-351033

Third-moment Kolmogorov bound conjecture for constrained U-statistics

摘录 / EXCERPT: Consider the setting of Theorem Trate: a constrained or unconstrained U-statistic built from the random sequence and kernel in the paper, with Kolmogorov distance d_K and the the...

conjecture | open | MathDB 页面 | 原文线索

V2-055411 | MATHDB-351067

Conjecture on the limiting stochastic fixed point of the branching-process generating function

摘录 / EXCERPT: Let S be the environment state space, let $f(M)=\sum P_nM^n$ be the matrix generating function, and let E be the extinction matrix. Write I for the identity matrix, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055412 | MATHDB-351107

Universality of critical exponents on transitive graphs of polynomial growth

摘录 / EXCERPT: Let Bernoulli percolation and lattice trees be defined on transitive graphs of polynomial growth, and let the volume-growth dimension denote the dimension governing the graph's pol...

conjecture | open | MathDB 页面 | 原文线索

V2-055413 | MATHDB-351263

Necessity of the zero-pattern condition for surjective DQSOs

摘录 / EXCERPT: Necessity conjecture. If V is surjective on S, then there exists a sequence $j_n_{n\geq1}\subset\mathbb N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055414 | MATHDB-351270

Random Rodrigues descendants converge to deterministic measures

摘录 / EXCERPT: Let $\x03be$ be a probability measure compactly supported in $\x1d53d$, let $P_n=\x220f_{i=1}^n(x-\x03be_i)$ have roots independently sampled according to $\x03be$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055415 | MATHDB-351292

Conjecture that exact community recovery is possible near the matching threshold

摘录 / EXCERPT: Let G_1 and G_2 be correlated stochastic block model graphs with community assignment $\boldsymbol{\boldsymbol{\sigma}}$, parameters α, β, and correlation paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-055416 | MATHDB-351356

The biased-separable representation conjecture for skew Brownian permutons

摘录 / EXCERPT: Biased-separable representation conjecture. For every $p\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-055417 | MATHDB-351357

The biased-sign construction conjecture for the Tanaka flow

摘录 / EXCERPT: Biased-sign construction conjecture. For $\rho=1$ and every $p\in[0,1]$, the solutions $\bm{\mathscr Z}_{1,1-p}^{(u)}$ can be constructed using independent biased signs satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-055418 | MATHDB-351358

The existence and uniqueness conjecture for the skew Brownian SDE flow

摘录 / EXCERPT: Existence and uniqueness conjecture. Existence and pathwise uniqueness for the solutions of these SDEs hold for every

conjecture | open | MathDB 页面 | 原文线索

V2-055419 | MATHDB-351359

The well-definedness conjecture for skew Brownian permutons

摘录 / EXCERPT: Skew Brownian permuton well-definedness conjecture. The skew Brownian permuton is well-defined for every parameter pair

conjecture | open | MathDB 页面 | 原文线索

V2-055420 | MATHDB-351361

The LQG-SLE mating-map conjecture

摘录 / EXCERPT: LQG-SLE mating-map conjecture. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-055421 | MATHDB-351365

Sahi's generalized FKG conjecture for multilinear functionals

摘录 / EXCERPT: Let L be a distributive lattice equipped with a probability measure μ satisfying ... for all $a,b\in L$; such an L is an FKG poset. Let $f_1,\ldots,f_n$ be positive monotone d...

conjecture | open | MathDB 页面 | 原文线索

V2-055422 | MATHDB-351366

Sahi's positivity conjecture for coefficients of the geometric-mean expansion

摘录 / EXCERPT: Sahi's coefficient-positivity conjecture. For every $n\geq 1$, the coefficient c_n satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-055423 | MATHDB-351421

The random-graph clone conjecture for total difference labeling

摘录 / EXCERPT: Random-graph clone conjecture. For every $\epsilon>0$, with probability 1,

conjecture | open | MathDB 页面 | 原文线索

V2-055424 | MATHDB-351482

Training-based equivalence of asymptotic symbol error probabilities

摘录 / EXCERPT: Training-based error-probability equivalence conjecture. For the system with B input-output training pairs (X_B,Y_B),

conjecture | open | MathDB 页面 | 原文线索

V2-055425 | MATHDB-351542

Permanent inequality for rank-at-most-two matrices

摘录 / EXCERPT: Rank-two permanent inequality. For any matrix T with rank at most 2,

conjecture | open | MathDB 页面 | 原文线索

V2-055426 | MATHDB-351660

Sarnak's unbounded-component conjecture for monochromatic random waves

摘录 / EXCERPT: Sarnak's conjecture. There exists $\delta>0$ such that, for every $\ell\in[-\delta,\delta]$, almost surely $\{f=\ell\}\cap\mathbb{R}^3$ contains an unbounded component.

conjecture | open | MathDB 页面 | 原文线索

V2-055427 | MATHDB-351727

Survival-regime conjecture for competing first passage percolation on lattices

摘录 / EXCERPT: Survival-regime conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055428 | MATHDB-351800

Probabilistic Yau-Tian-Donaldson conjecture for Fano manifolds

摘录 / EXCERPT: Let X be a Fano manifold. Let $N_k:=\dim H^0(X,-kK_X)$, and let the corresponding canonical point processes be defined by the Gibbs measures associated with determinants of bases...

conjecture | open | MathDB 页面 | 原文线索

V2-055429 | MATHDB-351940

The probabilistic-variational correspondence for negative-temperature point processes

摘录 / EXCERPT: Let X be a Fano manifold with volume form dV, let $\beta_0<0$, and let β range over the real numbers satisfying $\beta>\beta_0$. For each N, let $\mathcal{Z}_N(\beta)$...

conjecture | open | MathDB 页面 | 原文线索

V2-055430 | MATHDB-352034

Simon's dimension conjecture for self-similar measures on the line

摘录 / EXCERPT: Let μ be the self-similar measure on $\mathbb{R}$ associated to an IFS $\{\varphi_i:i\in\Lambda\}$ without exact overlaps and a probability vector $\{p_i\}$. Let λ_i be...

conjecture | open | MathDB 页面 | 原文线索

V2-055431 | MATHDB-352183

Minimality conjecture for the dense mPH subclass

摘录 / EXCERPT: Let $\mathbb{R}_+^d$ denote the positive orthant, and let MPH* distributions have general phase-type (PH) margins. The mPH minimality conjecture. The mPH class is the smalles...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055432 | MATHDB-352256

Poisson zero-cell face-probability limit conjecture

摘录 / EXCERPT: Fix $k\in\{2,3,\ldots\}$ and sample k vertices uniformly at random from the vertices of the random polytope $\operatorname{conv}\Pi_{d,1}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055433 | MATHDB-352258

The NFM exact-recovery impossibility conjecture under Dirichlet noise

摘录 / EXCERPT: NFM exact-recovery impossibility conjecture. No algorithm can exactly recover the true clusters in G with probability not converging to 0 as $n\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-055434 | MATHDB-352261

The NFM strong-node containment conjecture for ℓ_2-norm-diag

摘录 / EXCERPT: NFM strong-node containment conjecture. With probability not converging to 0 as $n\to\infty$, ℓ_2-norm-diag returns exactly k disjoint clusters $V_1',\dots,V_k'$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-055435 | MATHDB-352307

Central moment conjecture for continuously increasing subsequences

摘录 / EXCERPT: Central moment conjecture. For all $r\ge 1$, if n is sufficiently large in terms of m, then

conjecture | open | MathDB 页面 | 原文线索

V2-055436 | MATHDB-352308

Non-central moment conjecture for continuously increasing subsequences

摘录 / EXCERPT: Non-central moment conjecture. For all $r\ge 1$, if n is sufficiently large in terms of m, then

conjecture | open | MathDB 页面 | 原文线索

V2-055437 | MATHDB-352338

A sufficient covariance condition for the Gaussian correlation inequality

摘录 / EXCERPT: Under Assumption ... sigma^Y{i,j}geq sigma^X{i,j}geq 0 ... operatorname{Cov}(log SN({mathbf G}),pi({mathbf G})pj({mathbf G}))leq 0. ... ; its resolution is not indicated in the supplied...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055438 | MATHDB-352346

Almost no percolation/almost full percolation dichotomy in the Bradonjić-Saniee model

摘录 / EXCERPT: Let (a,p,θ) be fixed with $a>1$, and let $G_{n,r}^2$ be the random geometric graph in the Bradonjić-Saniee bootstrap percolation model, with initially infected set A_0, fi...

conjecture | open | MathDB 页面 | 原文线索

V2-055439 | MATHDB-352347

Symmetric local growth threshold conjecture for the Bradonjić-Saniee model

摘录 / EXCERPT: Let (a,p,θ) be fixed with $a>1$, and let $\theta_{\mathrm{local}}(a,p)$ denote the threshold for the appearance of large local, symmetrically distributed infectious outbreak...

conjecture | open | MathDB 页面 | 原文线索

V2-055440 | MATHDB-352348

Full percolation threshold conjecture for the Bradonjić-Saniee model

摘录 / EXCERPT: Let (a,p,θ) be fixed with $a>1$, and let T_n^2 be the two-dimensional torus. In the Bradonjić-Saniee bootstrap percolation model on the random geometric graph $G_{n,r}^2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055441 | MATHDB-352372

Square-spiral conjecture for the least positive harmonic measure

摘录 / EXCERPT: Square-spiral conjecture. Asymptotically, the square spiral realizes the least positive value of harmonic measure, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-055442 | MATHDB-352375

Cutoff for Activated Random Walk at the stationary density

摘录 / EXCERPT: Let u be the uniform driving sequence on $B_r = B(0,r) \cap \mathbb{Z}^d$. Let $0 < \lambda < \infty$ be a constant sleep rate, and let P be the simple random walk on B_r wit...

conjecture | open | MathDB 页面 | 原文线索

V2-055443 | MATHDB-352415

Conjecture on Grammians of independent symmetric rank-one tensors

摘录 / EXCERPT: Grammians conjecture. There exists a constant $\kappa_n>0$ and an increasing function $\gamma_n:(0,\kappa_n)\to(0,1)$, both depending only on n, such that $\gamma_n(\kappa)\to1$...

conjecture | open | MathDB 页面 | 原文线索

V2-055444 | MATHDB-352546

Poisson limit-profile conjecture for lazy conjugacy-class shuffles on the symmetric group

摘录 / EXCERPT: Let P be the transition matrix of a lazy random walk on S_n generated by a conjugacy class or a union of conjugacy classes having $n-o(n)$ fixed points. Suppose that P exhibi...

conjecture | open | MathDB 页面 | 原文线索

V2-055445 | MATHDB-352560

Small-intensity lower bound for hyperbolic Poisson-Voronoi percolation

摘录 / EXCERPT: Let $p_c(bb)$ denote the critical value for Poisson-Voronoi percolation on the hyperbolic plane. Small-intensity lower-bound conjecture. There exists a $bb_0>0$ such that ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-055446 | MATHDB-352561

Benjamini-Schramm monotonicity conjecture for hyperbolic Poisson-Voronoi percolation

摘录 / EXCERPT: Let $p_c(bb)$ denote the critical value for Poisson-Voronoi percolation on the hyperbolic plane as a function of the intensity bb. Benjamini-Schramm monotonicity conjecture. ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-055447 | MATHDB-352562

Differentiability conjecture for hyperbolic Poisson-Voronoi percolation

摘录 / EXCERPT: Let $p_c(bb)$ denote the critical value for Poisson-Voronoi percolation on the hyperbolic plane as a function of the intensity bb. Differentiability conjecture. ... is differentiab...

conjecture | open | MathDB 页面 | 原文线索

V2-055448 | MATHDB-352563

Small-intensity face-count conjecture for hyperbolic Poisson-Voronoi percolation

摘录 / EXCERPT: For each dimension d, consider Poisson-Voronoi percolation on d-dimensional hyperbolic space ${\mathbb H}^d$, with intensity bb, and let the typical cell be the typical Poiss...

conjecture | open | MathDB 页面 | 原文线索

V2-055449 | MATHDB-352564

High-intensity Euclidean-limit conjecture for hyperbolic Poisson-Voronoi percolation

摘录 / EXCERPT: For each dimension d, let $p_c^{\mathbb H^d}(bb)$ and $p_c^{\mathbb R^d}(bb)$ denote the critical values for Poisson-Voronoi percolation on d-dimensional hyperbolic space and E...

conjecture | open | MathDB 页面 | 原文线索

V2-055450 | MATHDB-352595

Giambelli compatibility implies determinantal structure

摘录 / EXCERPT: Let E be the underlying space, let $\mathbb{P}$ be a point process on E, and let ρ^R be an L^1-specialization. The pair $(\mathbb{P},\rho^R)$ is Giambelli compatible whe...

conjecture | open | MathDB 页面 | 原文线索

V2-055451 | MATHDB-352600

Uniform minimization conjecture for light-traffic coefficients

摘录 / EXCERPT: Uniform minimization conjecture. For every $q\geq 0$, the coefficient under the uniform policy equals the minimum over all edge-selection probability vectors:

conjecture | open | MathDB 页面 | 原文线索

V2-055452 | MATHDB-352644

Exact variance conjecture for fermionic Gaussian-state entanglement entropy

摘录 / EXCERPT: Let m and n be subsystem dimensions with $m\leq n$. Let S denote the von Neumann entanglement entropy of a fermionic Gaussian state, and let ψ_0 and ψ_1 denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-055453 | MATHDB-352678

Euclidean annulus dimension conjecture

摘录 / EXCERPT: Let d be the ambient dimension, and let $0<\underline{d}\leq\overline{d}\leq d$. For a probability measure μ supported inside $[0,1]^d$, define $E_\mu(\delta',\eta)$ as in th...

conjecture | open | MathDB 页面 | 原文线索

V2-055454 | MATHDB-352730

Alternative q-analogue probability measures for partitions in a rectangle

摘录 / EXCERPT: Let n and k be positive integers, let λ range over all partitions contained in an $n\times k$ rectangle, and write $\overline{\lambda}$ for the complementary partition...

conjecture | open | MathDB 页面 | 原文线索

V2-055455 | MATHDB-352734

Tightness conjecture for approximating Yang-Mills measures

摘录 / EXCERPT: Let G be a Lie group satisfying the condition given at the beginning of Section 3.1, and let $\mathcal{X}$ be the state space of Yang-Mills heat-flow trajectories. For $t>0$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-055456 | MATHDB-352735

Free-field-behavior conjecture for Yang-Mills approximations

摘录 / EXCERPT: Let G be a Lie group satisfying the condition given at the beginning of Section 3.1. Let $\mathfrak{g}$ be its Lie algebra, and let $\{\mathbf{A}^n\}_{n\geq 1}$ be a sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-055457 | MATHDB-352776

Liouville property conjecture for degree-2 mother groups and automaton groups

摘录 / EXCERPT: Let an automaton group have degree 2, meaning that every element g has activity $a_n(g)=O(n^2)$, and let the associated degree-2 mother groups be the specific universal autom...

conjecture | open | MathDB 页面 | 原文线索

V2-055458 | MATHDB-352790

Monotonicity of inversion probability in skew Brownian permutons

摘录 / EXCERPT: Let $\bm \mu_{\rho,q}$ be a skew Brownian permuton with parameters $(\rho,q)\in(-1,1]\times[0,1]$, and define ... Here 21 is the inversion pattern of size two. Inversion monotonici...

conjecture | open | MathDB 页面 | 原文线索

V2-055459 | MATHDB-352792

Random projection of norm rate of invariance

摘录 / EXCERPT: Let X be a random vector in $\mathbb{R}^n$ with independent and identically distributed entries, ... Assume that $\mathbb{E}x_1=0$, $\mathbb{E}x_1^2=1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055460 | MATHDB-352847

The exponential lower-bound conjecture for sphere percolation thresholds

摘录 / EXCERPT: Exponential lower-bound conjecture. For every dimension $d\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-055461 | MATHDB-352850

Weak-disorder moment conjecture for two-dimensional Gaussian polymers

摘录 / EXCERPT: Weak-disorder moment conjecture. For all $\delta\in(0,1)$ and $\hat\beta\in(0,1)$, uniformly for all $q<(1-\delta)q_\star(N,\hat\beta)$, as $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055462 | MATHDB-352923

Critical-value conjecture for chase-escape on the configuration model

摘录 / EXCERPT: Configuration-model critical-value conjecture. If $\mathcal G$ is sampled from the configuration model with an independent D-distributed degree sequence satisfying the displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-055463 | MATHDB-352970

General hitting-probability conjecture for SIMO systems with uniformly circular absorbing receivers

摘录 / EXCERPT: Let an SIMO molecular communication system have a point source emitting an information molecule (IM) and N fully absorbing receivers (FARs) arranged as a uniform circular array (...

conjecture | open | MathDB 页面 | 原文线索

V2-055464 | MATHDB-353019

Braverman-Milatovic-Shubin conjecture on the L^2-positivity preserving property

摘录 / EXCERPT: Braverman-Milatovic-Shubin conjecture. If (M,g) is a geodesically complete Riemannian manifold, then the L^2-positivity preserving property holds.

conjecture | open | MathDB 页面 | 原文线索

V2-055465 | MATHDB-353067

Limit conjecture for expected modularity above the Erdős-Rényi connectivity threshold

摘录 / EXCERPT: For a constant $c>0$, let $G_{n,c/n}$ be the binomial Erdős-Rényi random graph with edge-probability c/n, and define ... For $0<c\leq1$, the source states that $\bar{q}(n,c)\to1$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-055466 | MATHDB-353115

Asymptotic convergence-rate conjecture for iterative hard thresholding in matrix completion

摘录 / EXCERPT: Let $\bm H_n=\bm W_{pq}^{n}(\bm W_{pq}^{n})^{\top}$ be the matrix from Theorem, where $\bm W_q^n=\bm V_{\perp}^{n_2}\otimes\bm U_{\perp}^{n_1}$ forms a concentrated sequence of sem...

conjecture | open | MathDB 页面 | 原文线索

V2-055467 | MATHDB-353311

Full-space Beta RWRE Gamma-distribution conjecture

摘录 / EXCERPT: Let $t\in\mathbb Z$ and consider the full-space Beta random walk in random environment. For each $x\in\mathbb Z$, let $\mathsf Z_t^{\mathbb Z}(x)$ denote the full-space point-to-li...

conjecture | open | MathDB 页面 | 原文线索

V2-055468 | MATHDB-353312

Half-space Beta RWRE Gamma-distribution conjecture

摘录 / EXCERPT: Fix $t\in\mathbb Z$. In the half-space Beta random walk in random environment, define ... for $x\in\mathbb Z_{\geqslant 0}$. Half-space Gamma-distribution conjecture. The limit exist...

conjecture | open | MathDB 页面 | 原文线索

V2-055469 | MATHDB-353313

Tracy-Widom GUE fluctuations for half-space Beta RWRE point-to-point probabilities

摘录 / EXCERPT: Tracy-Widom GUE fluctuation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-055470 | MATHDB-353355

Higher-dimensional universality conjecture for critical bootstrap percolation families

摘录 / EXCERPT: Let $\mathcal U$ be a critical d-dimensional family, meaning that its stable directions satisfy the criticality condition, and let $L_c(\mathcal U,p)$ be the critical length at i...

conjecture | open | MathDB 页面 | 原文线索

V2-055471 | MATHDB-353387

Leader-Radcliffe's variance-matched concentration conjecture for Bernoulli random variables

摘录 / EXCERPT: Leader-Radcliffe's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055472 | MATHDB-353388

Uniqueness conjecture for the joint fixed point of the consistency equation

摘录 / EXCERPT: Let (U,X) be a random pair satisfying the consistency equation ... where $(U_k,X_k)_{k\geq 1}$ are independent copies of (U,X), $(\xi_k)_{k\geq 1}$ and $(\zeta_k)_{k\geq 1}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-055473 | MATHDB-353633

Universality of the likelihood-ratio statistic under stable finite-order VAR dynamics

摘录 / EXCERPT: Fix $k\in\mathbb{N}$ and $C>0$. Suppose that the data-generating process satisfies ... where the innovations are independent and identically distributed as $\mathcal{N}(0,\Lambda)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055474 | MATHDB-353636

A strengthened truncated-mean inequality for sums of uniform random variables

摘录 / EXCERPT: Strengthened truncated-mean inequality. For every $-1<t<1$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055475 | MATHDB-353638

L1 convergence for greedy lattice paths

摘录 / EXCERPT: L^1 convergence conjecture. There exists a constant $M\in(-\infty,+\infty)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055476 | MATHDB-353674

Weak convergence conjecture for the pro-ell random-group measures

摘录 / EXCERPT: Weak convergence conjecture. On ${\mathcal P}_{\Gamma,q,\ell}$, the measures $nu_{q,\ell,n}$ converge weakly, as $n\to\infty$, to the measure $mu_{\Gamma,q}$ from the existence and...

conjecture | open | MathDB 页面 | 原文线索

V2-055477 | MATHDB-353774

Incomplete PIIDG orthogonal-transform conjecture

摘录 / EXCERPT: Let $\bm{U}$ be an orthogonal matrix and let $\bm{U}_1$ be an $N\times(N-m)$ block of $\bm{U}$. Let $\bm{b}_1\in\mathbb{R}^{N-m}$ be a PIIDG vector, meaning a partially independent...

conjecture | open | MathDB 页面 | 原文线索

V2-055478 | MATHDB-353775

PIIDG extension of the orthogonality lemma for separable-IID functions

摘录 / EXCERPT: PIIDG orthogonality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055479 | MATHDB-353887

The light-tailed ballistic deposition universality conjecture

摘录 / EXCERPT: A deposition process is a mixture of random deposition (RD) and ballistic deposition (BD), with a fixed positive amount of BD, and its block heights have light-tailed distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-055480 | MATHDB-353920

Edge universality for non-trivial random regular graphs

摘录 / EXCERPT: Edge universality conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-055481 | MATHDB-353926

Krzakala et al.'s TAP prediction for the free energy of Bayes linear regression

摘录 / EXCERPT: Let $\beta=(\beta_1,\ldots,\beta_p)$ have product prior $\pi_0=\pi^{\otimes p}$, where π is a probability distribution on $[-1,1]$. Let $\mathcal{Z}_p$ denote the linear-regres...

conjecture | open | MathDB 页面 | 原文线索

V2-055482 | MATHDB-354037

Asymptotic separation conjecture for mixtures of two uniform distributions

摘录 / EXCERPT: Asymptotic separation conjecture. For every probability vector $(p,1-p)$, there exists a positive integer N such that, for all $n eg N$, the optimal set α_n contains point...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055483 | MATHDB-354057

Higher-moment equivalence for identical-letter pair counts

摘录 / EXCERPT: Let $X^{(n)}_1$ be the number of distinct adjacent pairs of identical letters in a geometrically distributed word of length n, and let ... where the independent random variables...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055484 | MATHDB-354058

Limiting-distribution equivalence for identical-letter pair counts

摘录 / EXCERPT: Let $X^{(n)}_1$ be the number of distinct adjacent pairs of identical letters in a geometrically distributed word of length n, and let $\xi^{(n)}$ be the sum of independent rando...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055485 | MATHDB-354059

Gaussian limiting distribution for the three-letter pair count

摘录 / EXCERPT: Let $X^{(n)}_3$ denote the number of distinct adjacent pairs of distinct letters in a geometrically distributed word of length n. Gaussian-limit conjecture. The asymptotic distri...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055486 | MATHDB-354060

Boundedness of cumulant correction terms for the three-letter pair count

摘录 / EXCERPT: Let T_r denote the correction to the rth cumulant of $X^{(n)}_3$ arising from dependence among the pair indicators. Cumulant-correction conjecture. ... The surrounding argument e...

conjecture | open | MathDB 页面 | 原文线索

V2-055487 | MATHDB-354081

Tilde-theta algorithmic-threshold conjecture for the symmetric binary perceptron

摘录 / EXCERPT: Let $kappa>0$ be the SBP parameter and let the algorithmic threshold be the largest constraint density up to which efficient algorithms can find a solution with high probability. T...

conjecture | open | MathDB 页面 | 原文线索

V2-055488 | MATHDB-354096

Continuity and strict monotonicity conjecture for the interspike-time limit distribution

摘录 / EXCERPT: Let (ϑ,τ,σ) be parameter values, and let $H^{(\vartheta,\tau,\sigma)}$ be the limit distribution of the empirical distribution functions of the interspike times,...

conjecture | open | MathDB 页面 | 原文线索

V2-055489 | MATHDB-354097

Benchmark parameter existence conjecture for rhythmic Hodgkin-Huxley spiking

摘录 / EXCERPT: Let $H^{(\vartheta,\tau,\sigma)}$ be the limit distribution of interspike times for parameter triplet (ϑ,τ,σ). Define the quantile spread ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-055490 | MATHDB-354099

Critical extinction conjecture for the contact process in a dynamic percolation

摘录 / EXCERPT: Critical extinction conjecture. For every finite nonempty C and every admissible background configuration B, one has

conjecture | open | MathDB 页面 | 原文线索

V2-055491 | MATHDB-354101

Exponential estimates conjecture for the contact process in dynamic percolation

摘录 / EXCERPT: Exponential estimates conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-055492 | MATHDB-354166

Omega-Kakeya maximal conjecture

摘录 / EXCERPT: Ω-Kakeya maximal conjecture. Fix any probability measure μ on $\mathbb{S}^{n-1}$ satisfying, for some $d\in[0,n-1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-055493 | MATHDB-354190

Fish-Reyzin's coupled-walk meeting conjecture for random DFAs

摘录 / EXCERPT: Let $(V,\mathcal C,\{f_x\}_{x\in V})$ be a random DFA with $|V|=n$. For a fixed $t\in\mathbb N$, let $\mathbf Q=\mathbf Q_t$ be the uniform distribution over words in...

conjecture | open | MathDB 页面 | 原文线索

V2-055494 | MATHDB-354438

Single-trace realization conjecture for one-dimensional Dirac ensembles

摘录 / EXCERPT: Let D be the Dirac operator in a formal Dirac ensemble of type $(1,0)$ or $(0,1)$, with coupling constants t_j and partition function ... Here $\mathcal{G}$ is the relevant Dirac...

conjecture | open | MathDB 页面 | 原文线索

V2-055495 | MATHDB-354450

Zero-loss conjecture for conflict-free joint decisions

摘录 / EXCERPT: Let M be the number of players and N the number of arms. For each player i, let $S_i^\dagger$ denote the corresponding aggregate preference-satisfaction value, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055496 | MATHDB-354474

Gaussian equivalence conjecture for the empirical conjugate kernel matrix

摘录 / EXCERPT: Let $\tilde{\boldsymbol X}$ and $\tilde{\boldsymbol y}$ be new training data and labels independent of $\boldsymbol W_1$. Assume the activation σ is odd, Assumption 1 holds,...

conjecture | open | MathDB 页面 | 原文线索

V2-055497 | MATHDB-354491

Subgaussian critical-window fluctuations for the symmetric perceptron

摘录 / EXCERPT: Let A be the random matrix and let $\mathrm{disc}(A)$ denote its discrepancy, with K_c the critical threshold. A random variable is subgaussian at scale $1/n^2$ if it has subga...

conjecture | open | MathDB 页面 | 原文线索

V2-055498 | MATHDB-354547

Approximate Bayesian inversions in probabilistic quasi-Borel spaces

摘录 / EXCERPT: Let $\mathbf{QBS}$ be the category of quasi-Borel spaces, and let $\mathbb{P}:\mathbf{QBS}\to\mathbf{QBS}$ denote the probabilistic construction. For objects A and B, a morphis...

conjecture | open | MathDB 页面 | 原文线索

V2-055499 | MATHDB-354585

Finite-size correction conjecture for longest increasing subsequences

摘录 / EXCERPT: Let $l_N^\square$ be the length of the longest increasing subsequence in the finite N model, and let $E_2^{\rm soft}(0;(t,\infty))$ denote the soft-edge limiting distribution. Se...

conjecture | open | MathDB 页面 | 原文线索

V2-055500 | MATHDB-354680

The complexity bound conjecture for weakly mixing subshifts

摘录 / EXCERPT: Let $p(q)$ denote the word complexity of a subshift, namely the number of distinct words of length q, and suppose the subshift admits a weakly mixing probability measure. The qua...

conjecture | open | MathDB 页面 | 原文线索

V2-055501 | MATHDB-354772

Type B distributional identity for oriented swap absorption times

摘录 / EXCERPT: Let $T_0,\dots,T_{n-1}$ be the absorption times in the type B oriented swap process, and let $L((0,0),(u,v))$ denote the corresponding half-space LPP passage time. Type B distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-055502 | MATHDB-354791

The abrupt-or-strongly-eroded dichotomy for infinite variation Lévy processes

摘录 / EXCERPT: Dichotomy conjecture. Every infinite variation Lévy process is either abrupt or strongly eroded.

conjecture | open | MathDB 页面 | 原文线索

V2-055503 | MATHDB-354811

Stability conjecture for lattice systems with elastic collisions

摘录 / EXCERPT: Let $N+1$ particles form a lattice model with elastic collisions. Define ... so that U_k is the speed of the centre of mass of the leftmost k particles, while U_{N+1} is the sp...

conjecture | open | MathDB 页面 | 原文线索

V2-055504 | MATHDB-354821

Critical intensity conjecture for the Manhattan grid model

摘录 / EXCERPT: Let $I_c>0$ denote the critical intensity in the phase transition theorem for $I=(\mu+\mu)\cdot\lambda$, and consider the Boolean model generated by a homogeneous Poisson point pro...

conjecture | open | MathDB 页面 | 原文线索

V2-055505 | MATHDB-354925

Flip-flop conjecture for long-term averages of the stochastic logistic map

摘录 / EXCERPT: Let ρ be a positive integer, let $\bar{\lambda}$ be the average parameter value, and let $p_i(\bar{\lambda})$ for $1\leq i\leq 2^\rho$ denote the stable period-2^ρ points...

conjecture | open | MathDB 页面 | 原文线索

V2-055506 | MATHDB-355022

The monotonicity conjecture for wired percolation on the Kertész line

摘录 / EXCERPT: For $q\in[1,\infty)$, let $p_c(q,h)$ be the critical edge parameter and let $\phi^1_{p,q,h,\mathbb{Z}^d}$ denote the wired random-cluster measure. The wired-percolation monotonicit...

conjecture | open | MathDB 页面 | 原文线索

V2-055507 | MATHDB-355023

The random-current Kertész-line monotonicity conjecture

摘录 / EXCERPT: Let $P_h^\emptyset$ denote the random-current measure and let $P_h^{\otimes2}$ denote the sourceless double random-current measure on $\mathbb{Z}^d$. The random-current monotonicit...

conjecture | open | MathDB 页面 | 原文线索

V2-055508 | MATHDB-355054

Conjectured improvements for concentration bounds in definable groups

摘录 / EXCERPT: Let G be a definable group in an o-minimal structure, let $S\subset G$ be a definable subset avoiding unboundedly large arithmetic progressions, and let $k=\dim(S)$. Consider the...

conjecture | open | MathDB 页面 | 原文线索

V2-055509 | MATHDB-355128

Finite $4/3$-variation conjecture for the zero energy part

摘录 / EXCERPT: Let A denote the zero energy part of the median process, and let $0=t_0<t_1<\dots<t_n=T$ be a sequence of partitions of $[0,T]$ whose grid size tends to 0. For $p=\frac{4}{3}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055510 | MATHDB-355137

The Edge-Wilson 25% speed-up conjecture for transitive graphs

摘录 / EXCERPT: Edge-Wilson speed-up conjecture. If Wilson's algorithm has this property on G, then the expected time taken by Edge-Wilson(G) is asymptotically 25% less than the expected ti...

conjecture | open | MathDB 页面 | 原文线索

V2-055511 | MATHDB-355296

The probability-measure cover conjecture for metric trees

摘录 / EXCERPT: Let $r>0$, let T be a metric tree with total length $|T|\geq 2r$, and let $\mathcal{C}(T,0)$ denote the relevant space of covers of T by radius-zero centers. Let $E\nu$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-055512 | MATHDB-355456

Conjectured asymptotic bounds for the probability of alternating Galois group

摘录 / EXCERPT: Fix $n\geq4$. Let f be a degree-n polynomial whose coefficients are chosen independently and uniformly from $[-L,L]\cap\mathbb{Z}$, and let G_f denote its Galois group. The a...

conjecture | open | MathDB 页面 | 原文线索

V2-055513 | MATHDB-355478

Coordinatewise monotonicity of connectivity on the integer lattice

摘录 / EXCERPT: Let $d\geq 1$ and $p\in(0,1)$. On the graph $\mathbbm{Z}^d$, whose vertices at Euclidean distance 1 are joined by an edge, use the componentwise order $u\leq v$ defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-055514 | MATHDB-355480

Layered-graph monotonicity conjecture for Bernoulli connectivity

摘录 / EXCERPT: Let $G=(V,E)$ be a finite connected graph. Its layered graph $G^\times$ is the Cartesian product $G\times\mathbbm{Z}$, with horizontal edges within each layer and vertical edges jo...

conjecture | open | MathDB 页面 | 原文线索

V2-055515 | MATHDB-355481

Intersection Heuristics for anti-cyclotomic extensions

摘录 / EXCERPT: Intersection Heuristics. The probability that

conjecture | open | MathDB 页面 | 原文线索

V2-055516 | MATHDB-355484

Bogomolny-Schmit conjecture for random-plane-wave nodal geometry

摘录 / EXCERPT: Let the random plane wave (RPW) be the stationary Gaussian field in the plane discussed in the paper, and consider its nodal lines and nodal domains at level zero. Let critical per...

conjecture | open | MathDB 页面 | 原文线索

V2-055517 | MATHDB-355485

Large deviation conjecture for Gelfand-Tsetlin interfaces

摘录 / EXCERPT: Let μ be a probability measure, and let $\rho_\mu:[0,1] \to \mathbb{R}$ denote the right-continuous inverse function of $F_\mu(s):= \int_{-\infty}^s \mu(\mathrm{d}x)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055518 | MATHDB-355527

Uniqueness and iteration conjecture for the fixed point of \(\hat{U}^{\dagger,0}\)

摘录 / EXCERPT: Let $\hat{U}^{\dagger,0}:\mathbb{R}^N\to\mathbb{R}^N$ be the operator defined for the unknown stochastic shortest path problem, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055519 | MATHDB-355528

Procedure conjecture for the fixed point of \(\hat{L}_{\pi}^{\dagger,0}\)

摘录 / EXCERPT: Let $P_0,P_1,P_2,P_{11},P_{12},P_{21},P_{22}$ be the transition matrices defining the active pieces of $\hat{L}_{\pi}^{\dagger,0}$, let $x_0,x_1,x_2,x_{11},x_{12},x_{21},x_{22}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-055520 | MATHDB-355727

Conjecture on volume growth of the neighbourhood of the invasion percolation cluster

摘录 / EXCERPT: Let $T(\clock)$ be the branching process tree, and let ... be the k-neighbourhood of the invasion percolation cluster. Under the conditions of the volume-growth theorem, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055521 | MATHDB-355769

Conjecture on the optimal Berry-Esseen rate for sample medians

摘录 / EXCERPT: Let M_n be the sample median, let θ and σ be the quantities in Theorem, and suppose the setting of that theorem applies. Assume that $D_i=O(1)$ for $i\in\{1,2,3\}$....

conjecture | open | MathDB 页面 | 原文线索

V2-055522 | MATHDB-355797

The limiting generating-function conjecture for slowly controlled Markovian queues

摘录 / EXCERPT: Let Q be the queue length and S the server speed, and let $ u$ be the measurement-rate parameter tending to zero. Define ... the generating function of the scaled pair...

conjecture | open | MathDB 页面 | 原文线索

V2-055523 | MATHDB-355814

Generalized Noise-Outsourcing Lemma for Borel data structures

摘录 / EXCERPT: Let E and D be Borel data structures, let R denote the randomization Borel data structure with laws $\boldsymbol{\upsilon}$, and let $\operatorname{SYM}(E)$ denote the exchan...

conjecture | open | MathDB 页面 | 原文线索

V2-055524 | MATHDB-355833

Law of large numbers for roots of finite free multiplicative convolution

摘录 / EXCERPT: Let p_d be a monic polynomial of degree d with non-negative real roots ... Let $\mu\in\mathcal{P}_{+,c}$, and suppose that the empirical root distributions of p_d converge weak...

conjecture | open | MathDB 页面 | 原文线索

V2-055525 | MATHDB-355845

Quenched localization conjecture for the rescaled running supremum

摘录 / EXCERPT: Quenched localization conjecture. There exists an environment-measurable process $(\Gamma_t)_{t\geq 0}$ with $\Gamma_t\subseteq\omega$ and

conjecture | open | MathDB 页面 | 原文线索

V2-055526 | MATHDB-355859

Generalization of the joint-probability result for geometric path integral quantization

摘录 / EXCERPT: Let ${\bf M}$ be a complete Kähler manifold satisfying the condition of Proposition as a Riemannian manifold, and let its Kähler form be the symplectic form. Assume that ${\bf M}$...

conjecture | open | MathDB 页面 | 原文线索

V2-055527 | MATHDB-355913

Three-dimensional Gaussian product inequality conjecture

摘录 / EXCERPT: Let $m_1,m_2,m_3\in\mathbb{N}$, and let (X_1,X_2,X_3) be a centered Gaussian random vector. Three-dimensional Gaussian product inequality conjecture. ... The equality sign is asser...

conjecture | open | MathDB 页面 | 原文线索

V2-055528 | MATHDB-355918

Normal-distribution conjecture for additive twists on Fuchsian groups

摘录 / EXCERPT: Let Γ be a Fuchsian group of the first kind with cusps, and let ϕ be a Maaß cusp form for Γ of integer weight. For $\Gamma=\operatorname{SL}(2,\mathbb{Z})$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-055529 | MATHDB-356000

Strongly connected tournament orientation-counting conjecture

摘录 / EXCERPT: Let $\vec{H}$ be a strongly connected tournament with $k:=v(H)\geqslant 4$, and let $p\gg n^{-2/(k+1)}$. Tournament orientation-counting conjecture. Then, with high probability, ……

conjecture | open | MathDB 页面 | 原文线索

V2-055530 | MATHDB-356015

Small-set isoperimetric conjecture for high-dimensional product graphs

摘录 / EXCERPT: Let G be a high-dimensional product graph whose base graphs are connected, regular, and of bounded size, and let d be the degree of G. For a subset $X\subseteq V(G)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055531 | MATHDB-356033

Asymptotic mutual-information conjecture for the sparse stochastic block model

摘录 / EXCERPT: Let f denote the unique viscosity solution of the infinite-dimensional Hamilton-Jacobi equation governing the enriched free energy, and let $\mathbf{G}_N$ and σ^* denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-055532 | MATHDB-356150

Regret bound for the adaptive online policy in Algorithm 5

摘录 / EXCERPT: Let a_t be an independent and identically distributed process, and let r_i follow either a linear regression model with white noise or a weighted random-walk model. Let π_5...

conjecture | open | MathDB 页面 | 原文线索

V2-055533 | MATHDB-356173

Pointwise kernel domination conjecture for probabilistic discrete Riesz transforms

摘录 / EXCERPT: For $d\geq2$, let $K_{\mathbb{H}^{(k)}}$ and $K_{R_\mathrm{dis}^{(k)}}$ be the kernels of the probabilistic and Calderón-Zygmund discrete Riesz transforms, respectively, with...

conjecture | open | MathDB 页面 | 原文线索

V2-055534 | MATHDB-356320

Bernoulli polynomial approximation conjecture for the discrete differences

摘录 / EXCERPT: Let b_n be the factorially normalized Bernoulli polynomial of degree n, and let $\delta_{k:2n}$ denote the difference between the uniform distribution on $\{1,\dots,2n\}$ and t...

conjecture | open | MathDB 页面 | 原文线索

V2-055535 | MATHDB-356321

Positive Bernstein expansion conjecture for Bernoulli polynomial densities

摘录 / EXCERPT: For each positive integer n, let $b_n(u)$ denote the factorially normalized Bernoulli polynomial of degree n, and consider the polynomial probability density $1-2^n b_n(u)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-055536 | MATHDB-356339

Sandwich conjecture for random regular graphs

摘录 / EXCERPT: Let N be the number of vertices, let $d=d(N)$, and let $\bm{G}_{N,d}$ denote a uniformly random d-regular graph. For edge probabilities p_* and p^*, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055537 | MATHDB-356413

Optimal stochastic homogenization rate under the CFS condition

摘录 / EXCERPT: Let $\mathbb{P}$ satisfy the condition $\mathsf{CFS}(\beta,\Psi)$, where $\beta\in(\frac12,1)$, and let ${\mathbf{b}}_r(x)$ and $\bar{\mathbf{a}}$ denote the corresponding rescaled...

conjecture | open | MathDB 页面 | 原文线索

V2-055538 | MATHDB-356444

Strong concentration conjecture for deletion-channel trace ambiguity

摘录 / EXCERPT: Let X be uniformly distributed on $\{0,1\}^n$, let $Y=\operatorname{BDC}_d(X)$ be the output of the binary deletion channel with deletion probability d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055539 | MATHDB-356445

Uniform deletion-channel capacity remains positive up to deletion probability one

摘录 / EXCERPT: Uniform-capacity threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055540 | MATHDB-356470

Sahi's conjecture for three increasing events

摘录 / EXCERPT: Let Ω be a product space, let μ be a product measure on Ω, and let ${\mathcal A},{\mathcal B},{\mathcal C}\subseteq\Omega$ be increasing events. Sahi's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-055541 | MATHDB-356477

Fast-update limit conjecture for the critical infection rate

摘录 / EXCERPT: Fast-update limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055542 | MATHDB-356504

Irreducible curvature bounds of few-relator complexes

摘录 / EXCERPT: Irreducible curvature bounds of few-relator complexes. The probability that $\rho_+(X)=\kappa(X)$ tends to 1 as $l\to\infty$. In particular, with high probability, $\rho_+(X)<0$...

conjecture | open | MathDB 页面 | 原文线索

V2-055543 | MATHDB-356505

Surface curvature of random complexes in the density model

摘录 / EXCERPT: Surface curvature of random complexes in the density model. The probability that $\sigma_+(X)<0$ tends to 1 as $l\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-055544 | MATHDB-356524

A two-sided local-limit conjecture for the diffusion approximation of tree profiles

摘录 / EXCERPT: Two-sided local-limit conjecture. The conditioned discrete process converges in distribution in $D([0,t_1],\mathbb{R}^3)$ to $\xi^\star$, started at (δ_0,m_0,s_0) and condit...

conjecture | open | MathDB 页面 | 原文线索

V2-055545 | MATHDB-356525

The zero-boundary extension of the two-sided local-limit conjecture

摘录 / EXCERPT: Zero-boundary extension conjecture. The two-sided local-limit conjecture also holds with the cutoff parameter set to $\epsilon=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-055546 | MATHDB-356526

Equispaced Fourier Matérn matrix error bound

摘录 / EXCERPT: Equispaced Fourier Matérn matrix error bound. With high probability as $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055547 | MATHDB-356540

Baird-Epstein-Flint-Miller Gaussianity conjecture for random Zeckendorf games

摘录 / EXCERPT: Baird-Epstein-Flint-Miller Gaussianity conjecture. As $N\to\infty$, the distribution of the game length converges to a Gaussian distribution, with expectation and variance approxim...

conjecture | open | MathDB 页面 | 原文线索

V2-055548 | MATHDB-356548

Borsuk-Ulam conjecture for random complexes with free involution

摘录 / EXCERPT: Borsuk-Ulam conjecture. With high probability, for every continuous map $f\colon\widetilde Z\to\mathbb{R}^{2d+1}$, there exists an $x\in|\widetilde Z|$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055549 | MATHDB-356562

Giant rainbow tree size conjecture near the phase transition

摘录 / EXCERPT: Let $G_c(n,p)$ be the random graph on n vertices whose edges are independently present with probability p and uniformly coloured with c colours. Let $b1>0$, set $c=b1 n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055550 | MATHDB-356736

A sharp monotonicity and upper-bound conjecture for binomial tail ratios

摘录 / EXCERPT: Let $B_{n,k}(p)$ be the binomial tail probability, $b_{n,k}(p)$ the corresponding binomial point probability, and $L(n,k,p)$ the lower bound from the paper's Theorem. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-055551 | MATHDB-356776

Uniform-edge conjecture for the expected maximum matching in random bipartite graphs

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with n vertices on each side. A fractional bipartite matching is a collection of edge probabilities $x=(x_{uv})$ satisfying the vert...

conjecture | open | MathDB 页面 | 原文线索

V2-055552 | MATHDB-356777

Weighted uniform-edge conjecture for random bipartite matchings

摘录 / EXCERPT: Let $A=(a_{ij})$ be a doubly stochastic $n\times n$ matrix, and let $W=(w_{ij})$ be a matrix of edge weights satisfying … Let R be the random bipartite graph in which edge...

conjecture | open | MathDB 页面 | 原文线索

V2-055553 | MATHDB-356834

Nonzero fixed-point conjecture for the hierarchical percolation renormalization map

摘录 / EXCERPT: Let $cP(cell^2_\downarrow)$ be the space of probability measures on $cell^2_\downarrow$, and let $\mathscr R=\mathscr R_{d,L,\alpha}$ be the renormalization map obtained by indepen...

conjecture | open | MathDB 页面 | 原文线索

V2-055554 | MATHDB-356866

The strong Gaussian product conjecture for even exponents

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and let $p_1,\dots,p_n\in 2\mathbb{N}$. For a real centered Gaussian vector $(X_1,\dots,X_n)$, the statement $\mathbf{GPI}_n(p_1,\dots,p_n)$ means ... with equal...

conjecture | open | MathDB 页面 | 原文线索

V2-055555 | MATHDB-356873

A free fourth moment theorem for fully symmetric chaotic eigenfunctions

摘录 / EXCERPT: Let $\mathcal{M}$ be the tracial von Neumann algebra in the abstract derivation setting, with generator Δ and carré du champ operator Γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055556 | MATHDB-356924

The minimum-degree threshold conjecture for giant components in high-dimensional product graphs

摘录 / EXCERPT: Let G be a high-dimensional product graph whose isoperimetric constant is at least inverse polynomial in the dimension. Let $\epsilon>0$, let $\delta\coloneqq\delta(G)$ be the mi...

conjecture | open | MathDB 页面 | 原文线索

V2-055557 | MATHDB-356976

Binomial-law conjecture for class and norm factors in cyclic p-extensions

摘录 / EXCERPT: Let L/K be any cyclic p-extension of degree p^N, with $N \geq 1$, and Galois group G. Assume that L/K is ramified at $r \geq 1$ places of K, totally ramified in L/K....

conjecture | open | MathDB 页面 | 原文线索

V2-055558 | MATHDB-357004

Non-singularity conjecture for the shifted random-walk system matrix

摘录 / EXCERPT: Non-singularity conjecture. The system's matrix in the referenced linear system is non-singular for all $\kappa,N\in\mathbb{N}$. In particular, if $N=3$ and $\kappa=1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-055559 | MATHDB-357026

The Minkowski-measurability asymptotic conjecture for coalescing Brownian motions

摘录 / EXCERPT: Let Λ be a Minkowski measurable set with dimension $\delta\in[0,1]$ and Minkowski content $\kappa\in(0,\infty)$. Let $Z_t(\mathbb R)$ denote the total mass at time t in t...

conjecture | open | MathDB 页面 | 原文线索

V2-055560 | MATHDB-357078

Tóth's infinite-cycle conjecture for the interchange process

摘录 / EXCERPT: Tóth's conjecture. Almost surely, π_β has only finite cycles when $d=2$, for every $\beta>0$, whereas when $d\geq3$ it has infinite cycles for every $\beta>\beta_c$.

conjecture | open | MathDB 页面 | 原文线索

V2-055561 | MATHDB-357101

No-infinite-path conjecture for the uniform infinite meandric system

摘录 / EXCERPT: Let $\mathcal L$ and $\mathcal R$ be independent two-sided simple random walks on $\mathbb Z$ with $\mathcal L_0=\mathcal R_0=0$, and use them to construct the two infinite non-cro...

conjecture | open | MathDB 页面 | 原文线索

V2-055562 | MATHDB-357103

UIMS origin-loop tail exponent conjecture

摘录 / EXCERPT: Let $(\mathcal M,\Gamma)$ be the planar map and loop collection associated with the UIMS, and let ℓ_0 be the loop or infinite path in Γ passing through $0\in\mathbb Z$....

conjecture | open | MathDB 页面 | 原文线索

V2-055563 | MATHDB-357125

A discriminant-volume conjecture for compact torus orbits

摘录 / EXCERPT: Let X_n be the ambient homogeneous space, and let $Ax_i\subset X_n$ be a sequence of compact orbits corresponding to orders R_i. Write μ_{Ax_i} for the invariant probabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-055564 | MATHDB-357197

The $N^{3/2}$ random Artin-group probability conjecture

摘录 / EXCERPT: Random Artin-group probability conjecture. For $f(N)=N^{3/2}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-055565 | MATHDB-357253

Critical-noise-level conjecture for full intermittency

摘录 / EXCERPT: Let $U\subset\mathbb{R}^d$ be a bounded domain satisfying the hypotheses of either Theorem or Theorem, and let $u(t,x)$ be the corresponding solution to the stochastic heat equatio...

conjecture | open | MathDB 页面 | 原文线索

V2-055566 | MATHDB-357275

The trace conjecture for banded substochastic matrices

摘录 / EXCERPT: Let A be a nonnegative substochastic matrix, meaning ... for all i and j. Suppose that above the diagonal, A has nonzero entries only at distance 1 from the diagonal, while...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055567 | MATHDB-357305

Fuchsian singularity conjecture for hitting measures

摘录 / EXCERPT: Let (X_n) be a finite-range admissible random walk generated by a probability measure μ on a cocompact Fuchsian group Γ, and let ν_μ be its hitting measure on th...

conjecture | open | MathDB 页面 | 原文线索

V2-055568 | MATHDB-357310

Dimension drop conjecture for harmonic measure on Fuchsian limit sets

摘录 / EXCERPT: Let G be the group of isometries of the hyperbolic plane $\mathbb{H}^2$. Let μ be a finitely supported probability on G whose support generates, as a semigroup, a discrete...

conjecture | open | MathDB 页面 | 原文线索

V2-055569 | MATHDB-357385

Limiting empirical spectral distribution conjecture for randomly permuted sparse matrices

摘录 / EXCERPT: Fix $\alpha \in [0,1)$. Let $A=PLQ$ be an $n\times n$ random matrix, where P and Q are independent uniformly chosen permutation matrices and L is independent of P and Q....

conjecture | open | MathDB 页面 | 原文线索

V2-055570 | MATHDB-357399

Threshold conjecture for the scaling limit of TASEP with a long caterpillar

摘录 / EXCERPT: In the setting of the section, consider right Bernoulli TASEP with sequential updates and an additional caterpillar of length $b\varepsilon^{-1/2}$ placed at distance...

conjecture | open | MathDB 页面 | 原文线索

V2-055571 | MATHDB-357486

Stable-regime overlap conjecture for perturbed geodesics

摘录 / EXCERPT: Let T_0 and T_t be the passage times at perturbation times 0 and t, and let π_0 and π_t be the corresponding geodesics. Write $\operatorname{Var}(T)$ for the varian...

conjecture | open | MathDB 页面 | 原文线索

V2-055572 | MATHDB-357550

Discrete-to-continuum quantum-disk welding conjecture for self-avoiding bubbles

摘录 / EXCERPT: For $\ell>0$, let $\overline{\mathrm{MD}}_{0,2}^n(\cdot,[\ell n^{1/2}])^\#$ and $\underline{\mathrm{MD}}_{0,1}^n([\ell n^{1/2}])^\#$ be the probability measures on the two types of...

conjecture | open | MathDB 页面 | 原文线索

V2-055573 | MATHDB-357624

The asymptotic constant for maximum leaf height in the critical beta-splitting random tree

摘录 / EXCERPT: Maximum-leaf-height constant. The constant in the asymptotic relation $D_n^* \sim c\log n$ in probability is

conjecture | open | MathDB 页面 | 原文线索

V2-055574 | MATHDB-357628

Conjecture on the dimension of the earthworm's set of holes

摘录 / EXCERPT: Let S_n denote the number of holes created by the earthworm after n steps, and interpret the dimension of the set of holes through the growth exponent of its expected size. Dim...

conjecture | open | MathDB 页面 | 原文线索

V2-055575 | MATHDB-357641

Ben-Naim-Krapivsky moment-growth conjecture for the Ulam-Kac adder

摘录 / EXCERPT: Let $(X_n)_{n\geq 0}$ be the Ulam-Kac adder sequence, and write $f(n)\sim g(n)$ when $\lim_{n\to\infty}f(n)/g(n)=1$. For each positive integer m, let c_m denote the constant go...

conjecture | open | MathDB 页面 | 原文线索

V2-055576 | MATHDB-357642

Central-limit conjecture for the logarithm of the Ulam-Kac adder

摘录 / EXCERPT: Let $(X_n)_{n\geq 0}$ be the Ulam-Kac adder sequence, and let convergence in law be denoted by $\to$. Central-limit conjecture. There are positive constants μ and σ such...

open_problem | open | MathDB 页面 | 原文线索

V2-055577 | MATHDB-357643

Generalized Ulam-Kac moment-bound conjecture

摘录 / EXCERPT: Consider the generalized Ulam-Kac adder ... where γ is the parameter in the source's generalized recurrence. Let c_m denote the corresponding moment-growth constants, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-055578 | MATHDB-357668

Ball-shaped scaling limit conjecture for sandpiles on the supercritical percolation cluster

摘录 / EXCERPT: Let $\mathfrak{p}\in(\mathfrak{p}_c(d),1)$, condition the origin to lie in the infinite cluster $\mathscr{C}_{\infty}$, and let s_n be the stabilized sandpile obtained from n c...

conjecture | open | MathDB 页面 | 原文线索

V2-055579 | MATHDB-357669

Existence of a deterministic sandpile scaling limit on the supercritical percolation cluster

摘录 / EXCERPT: Let $\mathfrak{p}\in(\mathfrak{p}_c(d),1)$, condition the origin to lie in the infinite cluster $\mathscr{C}_{\infty}$, and let $\overline{s}_n$ be the rescaled sandpiles defined f...

conjecture | open | MathDB 页面 | 原文线索

V2-055580 | MATHDB-357686

The low-degree polynomial conjecture on computational indistinguishability

摘录 / EXCERPT: Let $\mathbb{P}_n$ and $\mathbb{Q}_n$ denote a sequence of probability measures with sample space $\mathbb{R}^{n^k}$, where $k=O(1)$. Suppose that every polynomial f of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-055581 | MATHDB-357703

Gaunt-Merkle conjectured bounds for the median of the variance-gamma distribution

摘录 / EXCERPT: Gaunt-Merkle median bounds conjecture. For $r>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-055582 | MATHDB-357752

The local Bernoulli limit conjecture for random tiles of $\mathbb{Z}^d$

摘录 / EXCERPT: Fix an integer k, and pick $x \in [n]^d$ uniformly at random. Let $\mathcal{S}$ be a random tile contained in $[n]$, and let B_k be the k-ball around x. Local Bernoulli lim...

conjecture | open | MathDB 页面 | 原文线索

V2-055583 | MATHDB-357758

The space-time conductance IIP conjecture

摘录 / EXCERPT: The space-time conductance IIP conjecture. An IIP holds for the continuous-time Markov chain on $\mathbb Z^d$ with generator

conjecture | open | MathDB 页面 | 原文线索

V2-055584 | MATHDB-357818

Diffusive scaling conjecture for generalized excited random walks

摘录 / EXCERPT: Let X) be a p_n-ERW in direction $ell in \mathbb{S}^{d-1}$ on $\mathbb{Z}^d$, where $dgeq 3$ and $p_n=\mathcal{C}n^{-1/2}wedge 1$. Let $\widehat{B}^n$ denote the associated res...

conjecture | open | MathDB 页面 | 原文线索

V2-055585 | MATHDB-357847

Metastability scaling conjecture for critical two-dimensional update families

摘录 / EXCERPT: Let ${\mathcal U}$ be a critical two-dimensional update family. Its difficulty is ... where a critical update family is balanced if there are no two opposite directions $v,-v\in S^1$...

conjecture | open | MathDB 页面 | 原文线索

V2-055586 | MATHDB-357907

Discrete-continuum exponent conjecture for separable permutations and cographs

摘录 / EXCERPT: Discrete-continuum exponent conjecture. With probability tending to 1 as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055587 | MATHDB-357908

Longest increasing subsequence exponent conjecture for skew Brownian permutons

摘录 / EXCERPT: Skew Brownian LIS exponent conjecture. There exists a function $\ell:[-1,1]\times(0,1)\to[1/2,1)$ such that, with probability tending to 1 as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055588 | MATHDB-357935

High-probability rank growth conjecture for generalized Moore matrices

摘录 / EXCERPT: Let $\bm{M} \in \mathbb{F}_{q^m}^{k \times n}$ be a randomly chosen matrix with full $\mathbb{F}_{q^m}$-rank such that each block $\bm{M}^{(i)}$, for $i=1,\dots,\ell$, has full col...

conjecture | open | MathDB 页面 | 原文线索

V2-055589 | MATHDB-357992

Typical near-optimal spectral gap conjecture for Weil-Petersson random hyperbolic surfaces

摘录 / EXCERPT: Let $\mathbb{P}_g^{\mathrm{\scriptsize{WP}}}$ denote the Weil-Petersson probability measure on compact hyperbolic surfaces of genus g, and let $\lambda_1>0$ be the smallest non-z...

conjecture | open | MathDB 页面 | 原文线索

V2-055590 | MATHDB-358088

Maximum-criterion conjecture for linear-fragmentation processes

摘录 / EXCERPT: A linear-fragmentation process is a continuous process with finite mean, variance, and third moment. Maximum-criterion conjecture. Every linear-fragmentation process satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-055591 | MATHDB-358167

Conjecture on products of tilted stochastic matrices

摘录 / EXCERPT: Product conjecture. The product of the u_i-tilted matrices

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055592 | MATHDB-358282

The initial-means conjecture for discrete geometric distributions

摘录 / EXCERPT: Let $k\geq 2$, let $(p_1,p_2,\ldots,p_{k-1})$ be a permutation of $\{1/2,1/2^2,\ldots,1/2^{k-1}\}$, and define the probability measure ... on $\mathbb{R}$, supported on $\mathbb{N}$....

conjecture | open | MathDB 页面 | 原文线索

V2-055593 | MATHDB-358316

The signed-triangle threshold conjecture for spherical random geometric graphs

摘录 / EXCERPT: Let $\tau_{p,d}$ be chosen so that the spherical threshold connection...

conjecture | open | MathDB 页面 | 原文线索

V2-055594 | MATHDB-358336

Very sparse integrality-gap conjecture for random hitting set

摘录 / EXCERPT: Let m be the number of subsets, let p be the inclusion probability, and let n be the number of elements. Write $\operatorname{val}_{\mathrm{GR}}$ for the value produced by th...

conjecture | open | MathDB 页面 | 原文线索

V2-055595 | MATHDB-358362

Simis' normality conjecture for doubly stochastic monomial ideals

摘录 / EXCERPT: Simis' conjecture. If A has entries in $\{0,1\}$ and $\det(A)=\pm d$, then the Rees algebra $S[It]$ is normal.

conjecture | open | MathDB 页面 | 原文线索

V2-055596 | MATHDB-358400

First-order free group factor alternative

摘录 / EXCERPT: For $r\in(1,\infty]$, let $L(\mathbb F_r)$ be the interpolated free group factor, and let the first-order fundamental group of $L(\mathbb F_r)$ be the group of positive scalars t...

conjecture | open | MathDB 页面 | 原文线索

V2-055597 | MATHDB-358440

The nodal-domain count conjecture for eigenvectors of random signed graphs

摘录 / EXCERPT: Nodal-domain count conjecture. There are constants $c_1(\epsilon,p,q)$ and $c_2(\epsilon,p,q)$ such that, with probability $1-o_N(1)$, the ith eigenvector $\bm{\varphi}$ satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-055598 | MATHDB-358450

Forcing-independence conjecture for the large deviations principle

摘录 / EXCERPT: Forcing-independence conjecture. The large deviations principle for the system holds independently of the choice of the forcing $f\in L^2(0,T;H_0)$; equivalently, if an LDP holds f...

conjecture | open | MathDB 页面 | 原文线索

V2-055599 | MATHDB-358505

Variation comparison conjecture for the F-distribution

摘录 / EXCERPT: Let $d_1\in\mathbb{N}$ and $d_2\in\mathbb{N}$ with $d_2\geq 5$. Let X_{d_1,d_2} be an F-random variable with numerator and denominator degrees of freedom d_1 and d_2, respe...

conjecture | open | MathDB 页面 | 原文线索

V2-055600 | MATHDB-358519

Weighted random matrix model conjecture for cubic characters

摘录 / EXCERPT: Weighted random matrix model conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055601 | MATHDB-358521

Logarithmic-dimensional convergence of mean-field BBVI

摘录 / EXCERPT: Let BBVI denote black-box variational inference, and let the mean-field variational family be the class of variational distributions whose coordinates are independent. Under mild a...

conjecture | open | MathDB 页面 | 原文线索

V2-055602 | MATHDB-358544

Infinite-family criterion for pivotal exact e-variables

摘录 / EXCERPT: Let $\mathcal{P}$ and $\mathcal{Q}$ be families of probability measures on $\mathfrak{X}$. Suppose there exist $P_0\in\mathcal{P}$ and $Q_0\in\mathcal{Q}$ such that $P\ll P_0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-055603 | MATHDB-358564

Closed-form conjecture for the rectangular optimal shrinker

摘录 / EXCERPT: Let $\Upsilon_{\gamma,\beta}(y)$ be the shrinker defined in equation (Upsilon), and let $\mathsf{z}^{+}_{\gamma,\beta}$ and $\mathsf{z}^{-}_{\gamma,\beta}$ be the associated spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-055604 | MATHDB-358581

Optimality of the ERM error bound with upper isometry remainder

摘录 / EXCERPT: Let n be the sample size, let $\cF$ be a function class, and let $\mathbb P$ be a probability distribution. Write $\widehat f_n$ for the empirical risk minimizer,...

conjecture | open | MathDB 页面 | 原文线索

V2-055605 | MATHDB-358616

Palette-sparsification conjecture for arbitrary palettes

摘录 / EXCERPT: Palette-sparsification conjecture. The graph G is L-colorable with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055606 | MATHDB-358617

Palette-sparsification conjecture at the local degree threshold

摘录 / EXCERPT: Local palette-sparsification conjecture. The graph G is L-colorable with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055607 | MATHDB-358707

Critical polynomial decay conjecture for the loop O(1) model on higher-dimensional lattices

摘录 / EXCERPT: Critical polynomial decay conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-055608 | MATHDB-358708

Sharp percolative transition conjecture for the single random current on \(\mathbb{Z}^d\)

摘录 / EXCERPT: Random-current sharp-transition conjecture. The single random current on $\mathbb{Z}^d$ has a unique sharp percolative phase transition, namely

conjecture | open | MathDB 页面 | 原文线索

V2-055609 | MATHDB-358774

Airy-process limit conjecture for the rescaled Bernoulli-Exponential passage time

摘录 / EXCERPT: Let T_h be the limiting passage time defined by the preceding convergence result, let $x\in\mathbb R$ be fixed, and let $\mathcal A(x)$ denote the stationary Airy process. Airy-p...

conjecture | open | MathDB 页面 | 原文线索

V2-055610 | MATHDB-358872

Uniform-entropy cutoff conjecture for random walks with weighted random matching

摘录 / EXCERPT: Let (G_n) be a sequence of graphs with uniformly bounded degrees and diverging sizes. Let (ε_n) be a sequence of constants in $(0,1)$, and let (G_n^*) be as in the...

conjecture | open | MathDB 页面 | 原文线索

V2-055611 | MATHDB-358908

Dimension monotonicity conjecture for directed k-neighbor percolation

摘录 / EXCERPT: Let $\theta^{\rm D}(k,d)$ denote the probability that the origin belongs to an infinite directed cluster in the directed k-neighbor graph on $\mathbb{Z}^d$. Dimension monotonicit...

conjecture | open | MathDB 页面 | 原文线索

V2-055612 | MATHDB-358909

Two-dimensional directed 2-neighbor percolation conjecture

摘录 / EXCERPT: Let $\theta^{\rm D}(k,d)$ denote the probability that the origin belongs to an infinite directed cluster in the directed k-neighbor graph on $\mathbb{Z}^d$. Directed 2-neighbor p...

conjecture | open | MathDB 页面 | 原文线索

V2-055613 | MATHDB-358910

Two-neighbor directed percolation in all dimensions conjecture

摘录 / EXCERPT: For each $d\in\mathbb{N}$, let the 2-DnG be the directed 2-neighbor graph on $\mathbb{Z}^d$, and let $\theta^{\rm D}(2,d)$ be its percolation probability. Two-neighbor directed...

conjecture | open | MathDB 页面 | 原文线索

V2-055614 | MATHDB-358911

Dimension monotonicity conjecture for directed k-neighbor percolation at fixed k

摘录 / EXCERPT: Let $\theta^{\rm D}(k,d)$ denote the percolation probability of the directed k-neighbor graph on $\mathbb{Z}^d$. Fixed-k dimension monotonicity conjecture. For all $k,d\geq 2$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-055615 | MATHDB-358912

Bidirectional k-neighbor percolation threshold conjecture

摘录 / EXCERPT: Let the k-BnG be the bidirectional k-neighbor graph on $\mathbb{Z}^d$, and let $\theta^{\rm B}(k,d)$ denote its percolation probability. Bidirectional threshold conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-055616 | MATHDB-358913

Strict monotonicity conjecture for X-neighbor percolation

摘录 / EXCERPT: Let $\theta^{\rm X}(k,d)$ denote the percolation probability of the k-X-neighbor graph on $\mathbb{Z}^d$. X-neighbor monotonicity conjecture. For fixed d, the map...

conjecture | open | MathDB 页面 | 原文线索

V2-055617 | MATHDB-358914

Percolation threshold conjecture for the 2-X-neighbor graph

摘录 / EXCERPT: For $d\in\mathbb{N}$, let $\theta^{\rm X}(2,d)$ denote the percolation probability of the 2-X-neighbor graph on $\mathbb{Z}^d$. Two-neighbor X-graph conjecture. The 2-XnG perco...

conjecture | open | MathDB 页面 | 原文线索

V2-055618 | MATHDB-358915

In- and out-percolation equivalence conjecture for directed neighbor graphs

摘录 / EXCERPT: Let $\theta^{\rm D}(k,d)$ be the probability of an infinite directed path starting at the origin in the directed k-neighbor graph on $\mathbb{Z}^d$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055619 | MATHDB-358992

Maximality conjecture for the universal renewal polynomial

摘录 / EXCERPT: Maximality conjecture. The polynomial Q_k is a maximal element of $\mathop{\mathtt{A}}_k$ and is the largest element of $\widehat{\mathop{\mathtt{A}}}_k$ for every $k\geq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-055620 | MATHDB-359017

The law of large numbers for the ballistic random walk

摘录 / EXCERPT: Let $(Z_n)_{n\geq 0}$ be the random walk defined in Section 2, and let $\mathbb{P}$ denote its probability measure. Law of large numbers conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-055621 | MATHDB-359027

Shifted Bernoulli full-shift probability conjecture

摘录 / EXCERPT: Shifted Bernoulli full-shift probability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055622 | MATHDB-359051

Gravitational-field variational-shape conjecture for the Ising phase separation

摘录 / EXCERPT: Gravitational-field variational-shape conjecture. The preceding convergence in probability holds. The supplied context says that the solution of this variational problem is known i...

conjecture | open | MathDB 页面 | 原文线索

V2-055623 | MATHDB-359061

Absorbing-game limit values of maximal algebraic order

摘录 / EXCERPT: Let $m\geq 1$. A rational absorbing game with m actions per player is an absorbing game whose data are rational and in which each player has m available actions. Maximal-degree...

conjecture | open | MathDB 页面 | 原文线索

V2-055624 | MATHDB-359062

Algebraic representability by rational absorbing games

摘录 / EXCERPT: An absorbing game has a limit value, and a rational absorbing game is one whose payoff and transition data are rational. A number is algebraic if it is a root of a nonzero polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-055625 | MATHDB-359092

The strict inequality conjecture for the polymer moment threshold

摘录 / EXCERPT: Strict inequality conjecture. There exists no $\beta\in\mathbb R_+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055626 | MATHDB-359233

The linear Green function sphere-sum conjecture for finitely generated nilpotent groups

摘录 / EXCERPT: Let Γ be a finitely generated nilpotent group, let μ be a symmetric admissible finitely supported probability measure, and let S be a finite generating set. Writing...

conjecture | open | MathDB 页面 | 原文线索

V2-055627 | MATHDB-359234

Nondegenerate scaling limits for Wick powers of the loop-soup field

摘录 / EXCERPT: Let $\theta \in (0,1/2)$, let $h_{\theta,N}$ be the discrete loop-soup field, and define its Wick powers by ... where $G(x,x)$ is the relevant Green function and...

conjecture | open | MathDB 页面 | 原文线索

V2-055628 | MATHDB-359238

Saunderson-Chandrasekaran-Parrilo-Willsky ellipsoid-fitting conjecture

摘录 / EXCERPT: Saunderson-Chandrasekaran-Parrilo-Willsky conjecture. For arbitrarily small $\varepsilon>0$, if $n<(\frac14-\varepsilon)\cdot d^2$, then with high probability over $z_1,\ldots,z_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-055629 | MATHDB-359244

Boundary-determined maximal regularity conjecture for Gaussian stochastic fields

摘录 / EXCERPT: A Gaussian stochastic field is a stochastic field whose finite-dimensional distributions are Gaussian. Its boundary behaviour means the behaviour of the field when restricted to th...

conjecture | open | MathDB 页面 | 原文线索

V2-055630 | MATHDB-359349

Stochastic codimension conjecture for Gaussian-process images

摘录 / EXCERPT: Stochastic codimension conjecture. For every Borel set F in the state space,

conjecture | open | MathDB 页面 | 原文线索

V2-055631 | MATHDB-359360

Failure of tensor-product compatibility for supports in Markov categories

摘录 / EXCERPT: Let C be a Markov category with supports, and let p and q be morphisms in C. Write S_f for the support object of a morphism f. Support tensor-product...

conjecture | open | MathDB 页面 | 原文线索

V2-055632 | MATHDB-359439

Positive critical Hausdorff measure conjecture for random Cantor sets

摘录 / EXCERPT: Positive critical Hausdorff measure conjecture. With probability one with respect to $\mu_{\mathbf{p}}$,

conjecture | open | MathDB 页面 | 原文线索

V2-055633 | MATHDB-359456

Nonlinear majorization monotone for blockwise bistochastic quantum measurements

摘录 / EXCERPT: Nonlinear majorization monotone conjecture. If there exists a blockwise bistochastic matrix $\bm{B}\in\mathcal{B}_{n,d}$ such that $\bm{Q}=\bm{B}*\bm{P}$, then for any ordering of...

conjecture | open | MathDB 页面 | 原文线索

V2-055634 | MATHDB-359508

Conjecture on Benford behavior of perimeters in rectangle fragmentation

摘录 / EXCERPT: Rectangle-perimeter conjecture. The perimeters of the sub-rectangles from a rectangle undergoing an unrestricted fragmentation process converge to Benford behavior.

conjecture | open | MathDB 页面 | 原文线索

V2-055635 | MATHDB-359509

Conjecture on Benford behavior of lower-dimensional volumes

摘录 / EXCERPT: Lower-dimensional volume conjecture. A d-dimensional volume for an m-dimensional object, where $d<m$, undergoing a fragmentation process follows the Benford distribution for th...

conjecture | open | MathDB 页面 | 原文线索

V2-055636 | MATHDB-359530

The conjectured inequality for the fractional Brownian variation constant

摘录 / EXCERPT: Variation-constant conjecture. For $H\neq 1/2$, the $(1/H)$-th variation along deterministic partitions differs from that along uniform Lebesgue partitions, more precisely,

conjecture | open | MathDB 页面 | 原文线索

V2-055637 | MATHDB-359547

Renner-Wolf conjecture on reduced intrinsic information and secret-key rate

摘录 / EXCERPT: Let P_{XYZ} be a probability distribution. The Renner-Wolf conjecture. ... The reduced intrinsic information is a known upper bound on the secret-key rate and can be strictly small...

conjecture | open | MathDB 页面 | 原文线索

V2-055638 | MATHDB-359594

General i.i.d. coefficient extension of the harmonic Kac asymptotic

摘录 / EXCERPT: Let the coefficients of the harmonic polynomial model be independent and identically distributed random variables with mean zero, positive variance, and finite moments. The i.i.d....

conjecture | open | MathDB 页面 | 原文线索

V2-055639 | MATHDB-359625

Universality conjecture for heat-evolved polynomial zeros

摘录 / EXCERPT: Let ν be a rotationally invariant probability measure on $\mathbb{C}\backslash\{0\}$ satisfying $\int_{0}^{\varepsilon} \nu(\{|z| < r\})r^{-1}\,\operatorname{d}r<\infty$ for so...

conjecture | open | MathDB 页面 | 原文线索

V2-055640 | MATHDB-359633

Fully packed $O(2)$ map model scaling-limit conjecture

摘录 / EXCERPT: For $k\in\mathbb N$, let (M_k,L_k) be sampled from the probability measure on triangulations of the disk decorated by fully packed $O(2)$ loop configurations, at the critical par...

conjecture | open | MathDB 页面 | 原文线索

V2-055641 | MATHDB-359692

Square-free modulus conjecture for polynomial zero counts

摘录 / EXCERPT: Let $m=p_1\cdots p_r$ be a square-free product of r distinct primes, and let $K_{n,d}^m$ denote the number of distinct zeros of a random polynomial in...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055642 | MATHDB-359784

General-number-of-parts conjecture for discrete stick fragmentation

摘录 / EXCERPT: Fix an integer $k\geq2$. In the discrete stick-fragmentation process, break each stick into k pieces by choosing $k-1$ cut points recursively according to the uniform distributio...

conjecture | open | MathDB 页面 | 原文线索

V2-055643 | MATHDB-359813

High-dimensional Brownian motion conformal-dimension conjecture

摘录 / EXCERPT: Let B be n-dimensional Brownian motion. High-dimensional Brownian motion conjecture. For all sufficiently large n, B is not minimal almost surely; moreover, for all suffici...

conjecture | open | MathDB 页面 | 原文线索

V2-055644 | MATHDB-359817

The alternating two-player urn distribution conjecture

摘录 / EXCERPT: Alternating two-player urn distribution conjecture. The random variable $Y\in[0,1]$ has a density with respect to Lebesgue measure on $[0,1]$, is not Beta-distributed, and has a di...

conjecture | open | MathDB 页面 | 原文线索

V2-055645 | MATHDB-359818

The biased edge-reinforced random walk transience conjecture

摘录 / EXCERPT: Biased edge-reinforced random walk transience conjecture. The $\lambda^\ast$-biased edge-reinforced random walk is transient whenever $\lambda\neq1$.

conjecture | open | MathDB 页面 | 原文线索

V2-055646 | MATHDB-359864

The generalized WAX decomposition validity conjecture

摘录 / EXCERPT: Generalized WAX decomposition validity conjecture. A randomly chosen matrix ${\boldsymbol{H}}$ admits WAX decomposition with probability 1 for the given ${\boldsymbol{A}}$ if

conjecture | open | MathDB 页面 | 原文线索

V2-055647 | MATHDB-359883

Monotone first-marginal conjecture for separable supermodular VMOT

摘录 / EXCERPT: Let $d \ge 2$, let $\mu_i \preceq_c \nu_i$ for $i=1,\ldots,d$, and let $\vec\mu=(\mu_1,\ldots,\mu_d)$. A monotone coupling of $\vec\mu$ is the unique monotone probability measure w...

conjecture | open | MathDB 页面 | 原文线索

V2-055648 | MATHDB-359893

Poisson order-k median offset bound

摘录 / EXCERPT: Median offset bound conjecture. For fixed $k>1$ and $n\ge\kappa$,

conjecture | open | MathDB 页面 | 原文线索

V2-055649 | MATHDB-359963

High-dimensional asymptotic ordering conjecture for the Hyper-Sphere random connection model

摘录 / EXCERPT: Let d denote the dimension, let q_φ be the integral of the connection function, let λ_c be the critical intensity, and let $\varphi^{\star k}(\mathbf{0})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055650 | MATHDB-360019

The tableau-process weight conjecture for general multiline diagrams

摘录 / EXCERPT: Tableau-process weight conjecture. This weight is proportional to the weight derived from the tableau process of Ayyer et al.

conjecture | open | MathDB 页面 | 原文线索

V2-055651 | MATHDB-360063

The finite cosine-Kronecker-delta formula for polygonal mean-distance moments

摘录 / EXCERPT: Let K be a regular n-sided polygon with $n\geq 3$, circumscribed by a circle of unit radius. For a positive even integer p, let $L_{22}^{(p)}$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-055652 | MATHDB-360089

Conjectured monotonicity and asymptotics of the critical drift for the frog model on trees

摘录 / EXCERPT: Let $\operatorname{FM}(d,p,\nu)$ be the frog model on the rooted d-ary tree with drift parameter p and initial particle-distribution parameter ν. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055653 | MATHDB-360113

Polynomial-time indistinguishability conjecture for planted and unplanted number partitioning

摘录 / EXCERPT: Polynomial-time indistinguishability conjecture. Fix any $c>1$. There exists no polynomial-time algorithm that distinguishes the p-texttt{NPP} from the u-texttt{NPP} with probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-055654 | MATHDB-360355

Ethier and Lee's Parrondo-effect conjecture for two-armed slot machines

摘录 / EXCERPT: Ethier and Lee's conjecture. If the assumption that $r+s$ divides J is replaced by the assumption that $J=2$, with $r\geq 1$ and $s\geq 1$, then the same conclusion holds: the Pa...

conjecture | open | MathDB 页面 | 原文线索

V2-055655 | MATHDB-360365

Nonzero limiting value conjecture for the trapping-probability solution

摘录 / EXCERPT: Nonzero-limit conjecture. The limit L exists and is different from zero.

conjecture | open | MathDB 页面 | 原文线索

V2-055656 | MATHDB-360370

Cheong and Kaplan's polynomial push-forward cokernel conjecture

摘录 / EXCERPT: Let $P(t) \in \mathbb{Z}_p[t]$ be a non-constant monic square-free polynomial, written as ... where each $P_j(t)\in\mathbb{Z}_p[t]$ is monic and its reduction $\bar{P}_j(t)$ modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-055657 | MATHDB-360388

Uniform finite-size criterion for percolation on non-one-dimensional transitive graphs

摘录 / EXCERPT: Let $\mathcal G_d^*$ be the class of infinite transitive graphs of degree d that are not one-dimensional. For a graph G with origin o, let S_r be the sphere of radius r a...

conjecture | open | MathDB 页面 | 原文线索

V2-055658 | MATHDB-360390

Unimodular finite-size criterion for subexponential connection decay

摘录 / EXCERPT: Let $\mathcal U_d^*$ be the class of infinite unimodular transitive graphs of degree d that are not one-dimensional. For a graph G with origin o, let B_n be the radius-n...

conjecture | open | MathDB 页面 | 原文线索

V2-055659 | MATHDB-360410

Survival via long spread in the long-range contact process

摘录 / EXCERPT: Let $q^{(t)},q^{(x)},p\in(0,1)$ be given, and let $\alpha>1$. The long-range contact process (LoRaC) is the percolation model described in the paper, with edges whose length distri...

conjecture | open | MathDB 页面 | 原文线索

V2-055660 | MATHDB-360411

Extinction due to sparse environment in the long-range contact process

摘录 / EXCERPT: Let $q^{(t)},p\in(0,1)$ be given and let $\alpha>1$. Let $q^{(x)}$ denote the parameter controlling the sparse spatial environment in the long-range contact process (LoRaC). Extinc...

conjecture | open | MathDB 页面 | 原文线索

V2-055661 | MATHDB-360412

Fewer edges and phase transition in the long-range contact process

摘录 / EXCERPT: Consider the long-range contact process (LoRaC) on the spatial model of the paper, with edge length $|y-x|$ and a parameter $\alpha>0$. Fewer edges. The LoRaC has a phase transitio...

conjecture | open | MathDB 页面 | 原文线索

V2-055662 | MATHDB-360490

O. Nguyen's intermediate real-root count conjecture for random polynomials

摘录 / EXCERPT: O. Nguyen's conjecture. Given $\alpha\in(0,1)$, there exists a bounded sequence (c_k) of real numbers such that

conjecture | open | MathDB 页面 | 原文线索

V2-055663 | MATHDB-360586

Conjectured asymptotics for characteristic-polynomial moments of the real Ginibre ensemble

摘录 / EXCERPT: Let G_N be the $N\times N$ real Ginibre matrix, let γ vary in a fixed compact subset of $(-\frac{1}{2},\infty)$, and let x vary in a fixed compact subset of $(-1,1)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-055664 | MATHDB-360590

The Gaussian product inequality for disjoint principal minors of Wishart matrices

摘录 / EXCERPT: Let $p\in\mathbb{N}$, let $\alpha\in(p-1,\infty)$, and let $\Sigma\in\mathcal{S}_{++}^p$. Let $\mathfrak{X}\sim\mathcal{W}_p(\alpha,\Sigma)$ be a $p\times p$ Wishart random matrix,...

conjecture | open | MathDB 页面 | 原文线索

V2-055665 | MATHDB-360591

The Gaussian correlation inequality for disjoint Wishart principal minors

摘录 / EXCERPT: Let $\mathfrak{X}$ be the Wishart random matrix in the setting of the cited theorem, partitioned into diagonal blocks $\mathfrak{X}_{ii}$ of widths $p_1,\ldots,p_d$. Wishart princi...

conjecture | open | MathDB 页面 | 原文线索

V2-055666 | MATHDB-360608

Joint tail distribution conjecture for overlap times in identical tandem queues

摘录 / EXCERPT: Let j be the number of stations, and let $O^{(i)}_{n,n+k}$ denote the overlap time at station i for the n^{th} and $(n+k)^{th}$ customers in an $(M/M/\infty)^j$ tandem queue...

conjecture | open | MathDB 页面 | 原文线索

V2-055667 | MATHDB-360660

Separation-distance conjecture for least positive harmonic measures

摘录 / EXCERPT: For $d\geq3$, integers $n\geq2$ and $m\geq3$, define ... where $\|x_1-x_2\|$ is graph distance. Separation-distance conjecture. (1) There exist $C(d,m),c(d,m)>0$ such that, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-055668 | MATHDB-360763

Random-selection analogue of the alternating-walker limit theorem

摘录 / EXCERPT: Consider the two-walker model in which the walker to move is selected randomly, and let $M^{\rm rd}_\infty$ denote the limiting proportion of the left edge weight. The alternating-...

conjecture | open | MathDB 页面 | 原文线索

V2-055669 | MATHDB-360773

High-dimensional envy-free convex partition conjecture

摘录 / EXCERPT: Let n,d be positive integers, and let $(\mu^1_1,\dots,\mu^1_n),\dots,(\mu^d_1,\dots,\mu^d_n)$ be d tuples of n absolutely continuous probability measures on $\mathbb{R}^d$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-055670 | MATHDB-360833

The determinant-scaling conjecture for Stratonovich signatures of linear Brownian motions

摘录 / EXCERPT: Let $(B_t)_{t\geq 0} = (B_t^1, \dots, B_t^d)_{t\geq 0}$ be a d-dimensional Brownian motion and let V be an invertible $d\times d$ matrix. For $X=VB$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055671 | MATHDB-360875

Balayage density exponent conjecture for planar Coulomb gases

摘录 / EXCERPT: Let μ be a probability measure and let $S:=\operatorname{supp}\mu$. Assume that S is compact, U satisfies Assumption, ∂U is piecewise smooth, and $U\subset S$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-055672 | MATHDB-360885

Polynomial concentration conjecture for the number of real zeros of Kac polynomials

摘录 / EXCERPT: Concentration conjecture. Under mild conditions, the probability that the number of real zeros deviates from its expectation by at least $\varepsilon\log n$ is bounded above and be...

conjecture | open | MathDB 页面 | 原文线索

V2-055673 | MATHDB-360929

General-jump extension of exponential traveling-wave and speed-bound results

摘录 / EXCERPT: Let J denote the independent jump-size distribution, and let v_{**} be the minimum selected speed. A mean-field limit (MFL) is the deterministic limiting particle-system dynami...

conjecture | open | MathDB 页面 | 原文线索

V2-055674 | MATHDB-360946

The conjecture that word measures separate automorphism orbits

摘录 / EXCERPT: Let $\mathbb{\mathbf{F}}$ be a free group and let $w_{1},w_{2}\in\mathbb{\mathbf{F}}$. For each word, consider the induced probability measure on every compact group. Word-measure...

conjecture | open | MathDB 页面 | 原文线索

V2-055675 | MATHDB-360963

Benign-landscape conjecture for rank-two and rank-three synchronization factorizations

摘录 / EXCERPT: Let p denote the factorization rank in the Burer-Monteiro factorization ... is benign with high probability in each of the following settings: the high-dimensional Kuramoto model,...

conjecture | open | MathDB 页面 | 原文线索

V2-055676 | MATHDB-360991

Unimodality conjecture for pattern occurrences in Mallows permutations

摘录 / EXCERPT: Mallows unimodality conjecture. If π is any consecutive permutation pattern, then the function $p\mapsto\mathbb{E}_{n,p}(\pi)$ is unimodal.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055677 | MATHDB-361068

Joint-moment conjecture for derivatives of quadratic Dirichlet L-functions

摘录 / EXCERPT: Let $\mathcal{D}(X)$ be the set of fundamental discriminants d with $|d|<X$, and let $L(s,\chi_d)$ be the Dirichlet L-function attached to the quadratic character χ_d. For...

conjecture | open | MathDB 页面 | 原文线索

V2-055678 | MATHDB-361150

Conjecture on the Kertész-line asymptotic for the two-dimensional Ising model

摘录 / EXCERPT: Consider the two-dimensional Ising model and let $h_c(p)$ denote the Kertész line as a function of the parameter p, with $p_c=p_c(2,0)$ the zero-field critical point. Kertész-lin...

conjecture | open | MathDB 页面 | 原文线索

V2-055679 | MATHDB-361180

Existence of a highly normal number that is not KL-stochastic

摘录 / EXCERPT: Existence conjecture. There is a highly normal number that is not KL-stochastic.

open_problem | open | MathDB 页面 | 原文线索

V2-055680 | MATHDB-361232

The general-potential extension of the conditioned Brownian-motion limit

摘录 / EXCERPT: General-potential conditioning conjecture. As $T\to\infty$, the probability measures

conjecture | open | MathDB 页面 | 原文线索

V2-055681 | MATHDB-361272

The friendship paradox inequality for Binomial, Geometric and Poisson variables

摘录 / EXCERPT: The friendship paradox inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-055682 | MATHDB-361353

Universality conjecture for the final particle count

摘录 / EXCERPT: Let $N\geq 1$ be the initial number of particles, and let $\tilde{X}_N$ denote the random number of particles remaining after the perfectly plastic collision process. A random vari...

conjecture | open | MathDB 页面 | 原文线索

V2-055683 | MATHDB-361354

Distributional equivalence conjecture for physical and non-physical particle counts

摘录 / EXCERPT: Let $N\geq 1$, let $\tilde{X}_N$ be the random number of particles remaining in the physical perfectly plastic collision process, and let $\tilde{Z}_N$ be the corresponding random...

conjecture | open | MathDB 页面 | 原文线索

V2-055684 | MATHDB-361380

Johansson's Latin-square threshold conjecture

摘录 / EXCERPT: For each n, independently include each symbol in each list $L_{ij}\subseteq\{1,\dots,n\}$ with probability p, and let $\mathcal{F}_{\mathrm{Latin}}$ be the property that there...

conjecture | open | MathDB 页面 | 原文线索

V2-055685 | MATHDB-361381

Random k-SAT satisfiability threshold conjecture

摘录 / EXCERPT: For given n, $k\geq2$, and p, let $F_k(n,p)$ be the random k-SAT formula obtained by including each of the $\binom{2n}{k}$ possible k-clauses independently with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-055686 | MATHDB-361506

The Flory scaling conjecture for the barycentric excluded-volume random walk

摘录 / EXCERPT: Let (W_n,G_n) be the barycentric excluded-volume random walk described in the paper, with $W_n\in\mathbb R^2$ its position, and let $\|\cdot\|$ denote Euclidean norm. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055687 | MATHDB-361558

Giro et al.'s conjecture on reconstructing a positive fraction of points

摘录 / EXCERPT: Giro et al.'s reconstruction conjecture. For every $\varepsilon>0$ and every $p\geq(1+\varepsilon)/n$, with high probability there exists a subset $V'\subseteq[n]$ of size...

conjecture | open | MathDB 页面 | 原文线索

V2-055688 | MATHDB-361559

The conjecture that random 4-regular graphs are globally rigid in the real line

摘录 / EXCERPT: Random 4-regular graph rigidity conjecture. A random 4-regular graph with n vertices is globally rigid in $\mathbb{R}$ with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055689 | MATHDB-361585

The post-FKG conjecture for connection probabilities

摘录 / EXCERPT: Let G be a finite graph with arbitrary probabilities on its edges. Let $0,b\in G$ be vertices and let $A\subset G$ be a set of vertices. The post-FKG conjecture. ... This is an FKG...

conjecture | open | MathDB 页面 | 原文线索

V2-055690 | MATHDB-361586

The pre-FKG conjecture for connection probabilities

摘录 / EXCERPT: Let G be a finite graph with arbitrary probabilities on its edges. Let $0,b\in G$ be vertices and let $A\subset G$ be a set of vertices. The pre-FKG conjecture. ... This is a forma...

conjecture | open | MathDB 页面 | 原文线索

V2-055691 | MATHDB-361587

The uniform continuity consequence of the pre-FKG conjecture

摘录 / EXCERPT: Let G be a graph, let $A\subset G$, and let $0,b\in G$ be vertices. The uniform continuity conjecture. For every $\varepsilon>0$ there exists $\delta>0$ such that, whenever ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-055692 | MATHDB-361630

Krauth-Mézard conjecture for the capacity of the binary perceptron

摘录 / EXCERPT: Let $A\in\mathbb{R}^{\alpha N\times N}$ have independent standard Gaussian entries, and consider the set of $\sigma\in\{-1,+1\}^N$ satisfying $A\sigma>0$ entrywise. The largest sol...

conjecture | open | MathDB 页面 | 原文线索

V2-055693 | MATHDB-361633

Kedlaya-set optimality conjecture for product-free subsets

摘录 / EXCERPT: Let G be a compact group equipped with its Haar probability measure, and let a Kedlaya set be a set of the form ... where G acts on a set X, $B\subseteq X$, and $x\in X$. A mea...

conjecture | open | MathDB 页面 | 原文线索

V2-055694 | MATHDB-361635

Matrix p-Chebyshev inequality

摘录 / EXCERPT: Let $\mathbf{X}_1,\dots,\mathbf{X}_n$ be i.i.d. matrices with common mean matrix $\mathbb{E}\mathbf{X}_1=\mathbf{M}$ and pth central moment matrix ... For any...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055695 | MATHDB-361664

Limiting distribution conjecture for the optimal objective value

摘录 / EXCERPT: Let m, n, and $A(n)$ be as in Theorem, and for each n let $c=c(n)$ be a uniformly random unit cost vector independent of $A(n)$. Write z^* for the optimal objective value....

conjecture | open | MathDB 页面 | 原文线索

V2-055696 | MATHDB-361673

The asymptotic Jensen-gap formula for neighbor-independent spanning-tree families

摘录 / EXCERPT: Let $\mathcal{F}=\{T_n\subset G(n_n)\mid n\in\mathbb{N}\}$ be a bounded, convergent, neighbor-independent family, and let f be its associated power series. For each n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-055697 | MATHDB-361689

Recovery threshold conjecture for the K-means linear-programming relaxation

摘录 / EXCERPT: Recovery-threshold conjecture. If $\Delta > 2.14$, then Problem recovers the planted clusters with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-055698 | MATHDB-361765

The time-inhomogeneous branching random walk asymptotic conjecture

摘录 / EXCERPT: Time-inhomogeneous branching random walk asymptotic conjecture. If $L(T)\sim\alpha T^{1/3}$ for $\alpha\in\mathbb{R}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-055699 | MATHDB-361778

Optimal low-rank covariance approximation conjecture for probabilistic projection methods

摘录 / EXCERPT: Optimal low-rank covariance approximation conjecture. For almost any positive definite matrix A, for every iteration $m\leq n$, there exist $d(m)\ll n$ and $r(d)\leq d(m)$ such t...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-055700 | MATHDB-361878

Conjecture on the longest increasing subsequence of conjugation-invariant permutations

摘录 / EXCERPT: Let $(\tau_n)_n$ be a sequence of conjugation-invariant random permutations, where τ_n has size n and random cycle type $t^{(n)}$. Suppose that ... in probability. Conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-055701 | MATHDB-361891

Vu's global rank resilience conjecture for Rademacher matrices

摘录 / EXCERPT: Let $M\in\{\pm1\}^{n\times m}$ with $m\geq n$, and let $\operatorname{Res}(M)$ be the least number of entry-flips needed to produce from M a matrix whose rank is strictly less th...

conjecture | open | MathDB 页面 | 原文线索

V2-055702 | MATHDB-361928

Extension of the empirical Wasserstein barycenter rate to the regime $1\leq\beta-\alpha<2$

摘录 / EXCERPT: Let P be the probability measure and let $\bar P$ be its Wasserstein barycenter. Let ${\bar P}_n$ denote the empirical Wasserstein barycenter based on n observations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055703 | MATHDB-361971

Asymptotic approximation of population covariances by linearized recursions

摘录 / EXCERPT: Asymptotic covariance approximation conjecture. The population covariances can be asymptotically approximated by the last iterates of the corresponding linear recursions, with the...

conjecture | open | MathDB 页面 | 原文线索

V2-055704 | MATHDB-362052

The limiting row-membership probability conjecture for QUBO minimizers and maximizers

摘录 / EXCERPT: Limiting row-membership probability conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055705 | MATHDB-362066

The vanishing WLP probability conjecture for random Artinian monomial algebras

摘录 / EXCERPT: Let S be a polynomial ring in n variables, let $I\sim\mathcal{I}(n,D,p)$, and consider the random Artinian algebra ... where $n\geq 3$. Set $p=1/D^2$. The vanishing WLP probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-055706 | MATHDB-362213

Polytope conjecture for achievable derivatives in the spin-spin system

摘录 / EXCERPT: Let Δ^3 denote the probability simplex, let $\mathsf{derv}(\lambda)$ be the set of achievable derivatives at λ, let S_4 be the permutation group on four elements,...

conjecture | open | MathDB 页面 | 原文线索

V2-055707 | MATHDB-362219

Distributional convergence conjecture for final infection sizes

摘录 / EXCERPT: Consider the competing long-range infection model and the notation of the theorem concerning absence of coexistence, including the final-size quantity M_n and the parameter Z_n...

conjecture | open | MathDB 页面 | 原文线索

V2-055708 | MATHDB-362220

Coexistence conjecture at the critical long-range exponent

摘录 / EXCERPT: Consider the competing long-range infection model on $\mathbb T_n^d$ with long-range parameters $\alpha_\ominus(n)$ and $\alpha_\oplus(n)$, infection-rate ratio $\lambda(n)$, and c...

conjecture | open | MathDB 页面 | 原文线索

V2-055709 | MATHDB-362221

Competition conjecture on the long-range percolation cluster

摘录 / EXCERPT: Consider the long-range percolation graph on $\mathbb T_n^d$ with all nearest-neighbour edges and each non-nearest-neighbour edge included independently with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-055710 | MATHDB-362275

Conjecture on nonconvergence and asymptotic uniformity for vanishing-duration partitions

摘录 / EXCERPT: Let $T_\infty=(h_1,h_2,\ldots)$ be a fixed partition with $h_1+h_2+\cdots=+\infty$, and let $v_{T_\infty,\lambda}(p)$ denote the value of the game from Theorem with discount parame...

conjecture | open | MathDB 页面 | 原文线索

V2-055711 | MATHDB-362286

Spectral transition conjecture for censored additive spiked matrices

摘录 / EXCERPT: Let $\bm{x}$ be random as in the paper's spiked-prior assumption, let $p(y)$ be a density defining an additive noise channel $\mathcal{P}$, and let $F_{\mathcal{P}}$ denote its Fis...

conjecture | open | MathDB 页面 | 原文线索

V2-055712 | MATHDB-362331

Noncyclicity conjecture for stationary-measure solutions under the backward shift

摘录 / EXCERPT: Let μ be a probability measure satisfying the Blaschke condition, with support generating a discrete subgroup of $PSU(1,1)$. Let $f_\nu(z)$ be a solution of the equation referr...

conjecture | open | MathDB 页面 | 原文线索

V2-055713 | MATHDB-362358

Aldous-Diaconis universality of cutoff conjecture for random Cayley graphs

摘录 / EXCERPT: Let G be a group, let k satisfy $k \gg \log |G|$ and $\log k \ll \log |G|$, and let $\mathcal{G}_k$ be the Cayley graph of G generated by k independently and uniformly chos...

conjecture | open | MathDB 页面 | 原文线索

V2-055714 | MATHDB-362392

Triangle-free graph criterion for discontinuity in lower-tail probabilities

摘录 / EXCERPT: Triangle-free graph discontinuity conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-055715 | MATHDB-362401

The influential-pair obstruction conjecture for monochromatic subgraph counts

摘录 / EXCERPT: Influential-pair obstruction conjecture. If for every N there is some $n>N$ such that G_n has an influential pair of vertices ${\mathbf{w}}=(v,w)\in V(G_n)_2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-055716 | MATHDB-362512

General-distribution nearest-neighbor optimality conjecture

摘录 / EXCERPT: Let X and Z be independent draws from arbitrary unknown probability distributions with densities f_X and f_Z, satisfying … Let Y be drawn from the mixture distribution ……

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055717 | MATHDB-362513

Nearest-neighbor optimality for the general one-dimensional classification setting

摘录 / EXCERPT: Let x, y, and z be three points on the real line, where x and z are generated independently from two distinct distributions and y is generated from their 50--50 mixture...

conjecture | open | MathDB 页面 | 原文线索

V2-055718 | MATHDB-362514

Integral inequality for nearest-neighbor classification

摘录 / EXCERPT: Let $X\sim f_X$ and $Z\sim f_Z$ be independent random variables, where f_X and f_Z are densities with cumulative distribution functions F_X and F_Z, respectively. Integral...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055719 | MATHDB-362697

The reduction conjecture from temporal cliques to Erdős-Rényi cliques

摘录 / EXCERPT: Let ${\cal G}_{n,\delta}$ denote the Erdős-Rényi random graph model with edge probability δ, and let (K_n,λ) be a random simple temporal graph. A δ-clique is...

conjecture | open | MathDB 页面 | 原文线索

V2-055720 | MATHDB-362698

The uniform-distribution conjecture for temporal-clique label intervals

摘录 / EXCERPT: Let $G_{n,\delta}$ be an Erdős-Rényi random graph, and construct a temporal instance ${\cal I}(G_{n,\delta})$ by assigning labels to the edges of $G_{n,\delta}$ independently and u...

conjecture | open | MathDB 页面 | 原文线索

V2-055721 | MATHDB-362699

The logarithmic-corrections conjecture for critical long-range percolation

摘录 / EXCERPT: Let $d\geq 1$, and let $J:\mathbb{Z}^d\times\mathbb{Z}^d\to[0,\infty)$ be a symmetric, integrable, translation-invariant kernel. Suppose that there are constants $0<\alpha<d$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055722 | MATHDB-362728

Small two-terminal connection probabilities force a negligible three-way connection

摘录 / EXCERPT: Let a, b, and c be vertices, and let $\mathbf{P}(ab\\|c)$, $\mathbf{P}(ac\\|b)$, $\mathbf{P}(abc)$, and $\mathbf{P}(a\\|b\\|c)$ denote the corresponding bond-percolation even...

conjecture | open | MathDB 页面 | 原文线索

V2-055723 | MATHDB-362729

Approximate factorization from a small two-terminal connection probability

摘录 / EXCERPT: Let a, b, and c be vertices, and let $\mathbf{P}(ab\\|c)$, $\mathbf{P}(abc)$, $\mathbf{P}(ac)$, and $\mathbf{P}(bc)$ denote the corresponding bond-percolation event probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-055724 | MATHDB-362733

The finer first-passage-time asymptotic conjecture for branching random walks

摘录 / EXCERPT: Let B_x be the shifted unit ball at Euclidean distance x from the origin, let τ_x denote the first passage time of the branching random walk to B_x, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055725 | MATHDB-362760

Asymptotic logarithmic periodicity of the Hungarian roulette probabilities

摘录 / EXCERPT: Asymptotic logarithmic periodicity conjecture. The sequence p_n does not converge as $n\to\infty$ over the positive integers, but it does along every subsequence (n_j) for whic...

conjecture | open | MathDB 页面 | 原文线索

V2-055726 | MATHDB-362796

Blockwise Fourier cancellation conjecture for Syracuse paths

摘录 / EXCERPT: Let $\lambda(k_{j-1},k_j,p_j,p_{[j+1,r]},\xi/2^p)$ be the normalized exponential sum associated with a block of Syracuse-sequence coordinates, and let $k_0=0$, $k_j\ge1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055727 | MATHDB-362817

Conjecture on the critical intensity thresholds in the soft Boolean model

摘录 / EXCERPT: Let $\beta_c(\gamma,0,\delta)$ be the critical intensity and let $\widetilde{\beta}_c$ be the critical intensity of Theorem. Critical-intensity equality conjecture. … The conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-055728 | MATHDB-362862

Two-dimensional critical stochastic heat flow black-noise conjecture

摘录 / EXCERPT: Let the two-dimensional critical stochastic heat flow be the flow of random measures on $\mathbb R^2$ constructed as the scaling limit of intermediate-disorder directed-polymer par...

conjecture | open | MathDB 页面 | 原文线索

V2-055729 | MATHDB-362863

Directed landscape two-dimensional black-noise conjecture

摘录 / EXCERPT: A two-dimensional noise is a probability space $(\Omega,\mathcal F,\mathbb P)$ equipped with sub-σ-algebras associated with open rectangles in $\mathbb R^2$ and measure-pres...

conjecture | open | MathDB 页面 | 原文线索

V2-055730 | MATHDB-362897

Existence of cyclic 2-designs in every dimension at least 3

摘录 / EXCERPT: Let a cyclic t-design in dimension d be a finite orbit of a unitary acting on quantum states whose decohered probability vectors form a simplex t-design. Cyclic 2-design exis...

conjecture | open | MathDB 页面 | 原文线索

V2-055731 | MATHDB-362898

Existence of cyclic 2-designs in an infinite family of dimensions

摘录 / EXCERPT: A cyclic 2-design is a finite unitary orbit whose associated decohered probability vectors form a simplex 2-design. Infinite-family conjecture. Cyclic 2-designs exist in an infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-055732 | MATHDB-362940

Poincaré-Korn rigidity conjecture for strongly log-concave measures

摘录 / EXCERPT: Let μ be a centered and isotropic probability measure on $\mathbb{R}^n$ with density $d\mu=e^{-\phi}dx$, meaning ... Let γ denote the standard Gaussian measure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055733 | MATHDB-363024

Vu's distinct singular values conjecture for symmetric Rademacher matrices

摘录 / EXCERPT: Let M_n^{sym} be a symmetric random matrix whose entries are independent Rademacher variables. Vu's distinct singular values conjecture. With probability $1-o(1)$, all the singul...

conjecture | open | MathDB 页面 | 原文线索

V2-055734 | MATHDB-363042

Second-order limit conjecture for the renewal covering

摘录 / EXCERPT: Let $(P_n)_n$ be a renewal covering of the natural numbers as defined in the paper. Let Z be a random variable. Second-order limit conjecture. ... where the convergence is in distr...

conjecture | open | MathDB 页面 | 原文线索

V2-055735 | MATHDB-363066

Monotonicity conjecture for the normalized Cramér-transform exponential functional

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let X_n be the rotationally invariant random vector denoted by $\mathcal{E}$ in $\mathbb{R}^n$, and let $\Lambda_{X_n}^\ast$ be its Cramér transform. N...

conjecture | open | MathDB 页面 | 原文线索

V2-055736 | MATHDB-363265

The universal profile conjecture for high-dimensional percolation

摘录 / EXCERPT: Let W^* be the Brownian excursion of length 1, let $\Psi(x)=\mathbb{E}\exp\left[x\int_0^1W^*(t)\,\mathrm dt\right]$ be the moment-generating function of its area, and define .....

conjecture | open | MathDB 页面 | 原文线索

V2-055737 | MATHDB-363285

Cogrowth formula for non-negative functions on biregular trees

摘录 / EXCERPT: Cogrowth formula for non-negative functions. Fix $2\le c\le d$ and $3\le d$, and let $0\ne f\colon V(\mathbb{T}_{c,d})\to\mathbb{R}_{\ge 0}$ be a non-negative function on the verti...

conjecture | open | MathDB 页面 | 原文线索

V2-055738 | MATHDB-363293

Beta-limit conjecture for stationary two-island and seed-bank diffusions

摘录 / EXCERPT: Beta-limit conjecture. As $c\to\infty$, in the two-island case both X and Y converge in distribution to the beta distribution with parameters (a_1+b_1,a_2+b_2); in the seed-b...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-055739 | MATHDB-363331

Diaconis-Freedman conjecture for transient partially exchangeable processes

摘录 / EXCERPT: Diaconis-Freedman conjecture. The partially exchangeable σ-algebra of X is generated up to null sets by the transition numbers M.

conjecture | open | MathDB 页面 | 原文线索

V2-055740 | MATHDB-363407

Particle filter optimization parameter-selection conjectures

摘录 / EXCERPT: Parameter-selection conjecture. It is likely to reach a better solution with lower k_{max} and λ when N and Q are high. The results suggest a correlation between...

conjecture | open | MathDB 页面 | 原文线索

V2-055741 | MATHDB-363479

Connection-probability conjecture for critical loop $O(n)$ models

摘录 / EXCERPT: Consider the loop $O(n)$ model with $n\in(0,2]$ and boundary condition $\beta\in\mathrm{LP}_N$. Let ς^{δ} be the random connectivity in LP_N induced by t...

conjecture | open | MathDB 页面 | 原文线索

V2-055742 | MATHDB-363492

Majority Dynamics conjecture for oscillating vertices in random regular graphs

摘录 / EXCERPT: Majority Dynamics conjecture. For every $varepsilon>0$, with high probability the number of oscillating vertices lies in

conjecture | open | MathDB 页面 | 原文线索

V2-055743 | MATHDB-363494

The retention-probability conjecture for Gibbs point processes

摘录 / EXCERPT: Retention-probability conjecture. The function $p_{\lambda,\alpha}(\cdot)$ in the thinning representation is

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055744 | MATHDB-363619

Asymptotic constant conjecture for vertices of polystochastic matrix polytopes

摘录 / EXCERPT: For integers $d,n\geq 2$, let Ω_n^d be the polytope of d-dimensional polystochastic matrices, and let $V(n,d)$ denote its number of vertices. For fixed d, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055745 | MATHDB-363655

Gaussian block random matrix determinant conjecture

摘录 / EXCERPT: Let $m\in\mathbb{N}$ and $a_i\in M_m(\mathbb{Z})$ be given. Define the completely positive map ... Assume that η is rank non-decreasing. For n independent GUE random matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-055746 | MATHDB-363951

The isolation-by-distance central limit conjecture under varying population size

摘录 / EXCERPT: Let $t\geq 0$ and let $\psi_1,\psi_2\in\mathcal{D}(\mathbb{T}^d)$. Write η_N for the scaling factor, $P_t^N(\psi_1,\psi_2)$ for the isolation-by-distance quantity, Z_t for t...

conjecture | open | MathDB 页面 | 原文线索

V2-055747 | MATHDB-363990

Percolation conjecture for the infinitely reinforced planar 2-out model

摘录 / EXCERPT: Percolation conjecture for $d=k=2$. Assume that $d=k=2$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-055748 | MATHDB-363991

Finite-reinforcement percolation conjecture for planar reinforced 2-out percolation

摘录 / EXCERPT: Finite-reinforcement percolation conjecture. Assume that $d=k=2$. Then there exists $\alpha_0>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055749 | MATHDB-364002

Entropy factorization conjecture for cone measures on ℓ_p-spheres

摘录 / EXCERPT: For $p>0$, let ${\mathbb{S}}^{n-1}_p=\{x\in\mathbb{R}^n:\|x\|_p=1\}$, where $\|x\|_p=(\sum_{i=1}^n|x_i|^p)^{1/p}$. Let $\mathbf{P}$ be the cone measure on ${\mathbb{S}}^{n-1}_p$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-055750 | MATHDB-364026

The logarithmic maximum-entry conjecture for tame random matrices

摘录 / EXCERPT: Logarithmic maximum-entry conjecture. With high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-055751 | MATHDB-364038

Teheranchi's strong Gaussian correlation conjecture for Gaussian rectangles

摘录 / EXCERPT: Let $X_1,\dots, X_n$ be zero-mean jointly Gaussian real random variables, and let $s_1,\dots, s_n, t_1,\dots, t_n \in [0,\infty)\cup \{\infty\}$. Teheranchi's strong Gaussian corre...

conjecture | open | MathDB 页面 | 原文线索

V2-055752 | MATHDB-364039

Teheranchi's strong Gaussian correlation conjecture for symmetric convex sets

摘录 / EXCERPT: Let $d>0$ and let $K,T\subseteq \mathbb R^d$ be origin-symmetric convex sets, and let γ_d denote the usual Gaussian probability measure. Teheranchi's strong Gaussian correla...

conjecture | open | MathDB 页面 | 原文线索

V2-055753 | MATHDB-364082

Order-preservation of realizable bases and their intervals

摘录 / EXCERPT: Let n and k be fixed, let $\mathcal{R}=\{I_1,I_2,\ldots,I_N\}$ be the lexicographically ordered set of realizable bases, and let $L:\mathcal{R}\to\mathcal{L}$ map each realizab...

conjecture | open | MathDB 页面 | 原文线索

V2-055754 | MATHDB-364083

Simultaneous-change conjecture for exceptional basis transitions

摘录 / EXCERPT: Let I_i and I_{i+1} be consecutive realizable bases at a transition point. Simultaneous-change conjecture. If several elements $i_1,i_2,\ldots,i_r$ leave I_i simultaneously,...

conjecture | open | MathDB 页面 | 原文线索

V2-055755 | MATHDB-364084

Rationality conjecture for exceptional points

摘录 / EXCERPT: For fixed n and k, an exceptional point is a parameter value at which several basis elements leave simultaneously and the transition to the next realizable basis is exceptional...

conjecture | open | MathDB 页面 | 原文线索

V2-055756 | MATHDB-364085

Symmetry conjecture for realizable bases and intervals

摘录 / EXCERPT: Symmetry conjecture. For $1\leq i\leq N$,

conjecture | open | MathDB 页面 | 原文线索

V2-055757 | MATHDB-364086

Reflection conjecture for exceptional points

摘录 / EXCERPT: Fix n and k, and let d be an exceptional point, meaning a parameter value where several basis elements leave simultaneously during a transition between consecutive realizable...

conjecture | open | MathDB 页面 | 原文线索

V2-055758 | MATHDB-364087

Monotonicity conjecture for the maximum spread of realizable bases

摘录 / EXCERPT: Monotonicity conjecture. For each fixed k, $\operatorname{Msp}(n,k)$ is a monotonic non-decreasing function of n. The conjecture is motivated by numerical tables of maximum spr...

conjecture | open | MathDB 页面 | 原文线索

V2-055759 | MATHDB-364088

Square-root formula conjecture for maximum spread when $k=4$

摘录 / EXCERPT: For $k=4$, let $\operatorname{Msp}(n,4)$ denote the maximum spread among all realizable bases for the parameter n. Square-root formula conjecture. For every $n\geq6$, ... This is a...

conjecture | open | MathDB 页面 | 原文线索

V2-055760 | MATHDB-364140

The dimension conjecture for the product-measure locus in the permutation simplex

摘录 / EXCERPT: Dimension conjecture. For all m, the dimension of P_m is exactly $C(m)$.

conjecture | open | MathDB 页面 | 原文线索

V2-055761 | MATHDB-364165

The distributional embedding conjecture for intelligent language models

摘录 / EXCERPT: Let $\mathcal{S}$ be the set of sentences, let $\mathscr{L}$ be a causal language model, and let ... map each sentence to its output-probability distribution, where...

conjecture | open | MathDB 页面 | 原文线索

V2-055762 | MATHDB-364271

The no-positive-equal-coordinates conjecture for z-measures on the Thoma simplex

摘录 / EXCERPT: Let $mathcal M_{z,z';\theta}$ be the boundary z-measure on the Thoma simplex $Omega$, whose points are written $(\alpha;\beta)$ with nonincreasing nonnegative coordinate sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-055763 | MATHDB-364273

Wishart-prior matrix generalized linear model conjecture

摘录 / EXCERPT: Let $\Phi_d^{(G)}$ be the free entropy of the Gaussian-equivalent matrix generalized linear model, with $\bS^\star\sim\mcW_{m,d}$, and let Q_0, α, J_q, and Ψ have t...

conjecture | open | MathDB 页面 | 原文线索

V2-055764 | MATHDB-364373

The asymmetric three-cluster connection conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a locally finite connected simple graph, let a, b, and c be distinct vertices, and use $\mathbf{P}(abc)$ for the probability that all three are in one cluste...

conjecture | open | MathDB 页面 | 原文线索

V2-055765 | MATHDB-364374

Monotonicity conjecture for implied Poisson parameters

摘录 / EXCERPT: The implied-Poisson-parameter conjecture. The sequence $\{\lambda_k\}$ is decreasing. This is proposed as a strengthening of the paper's earlier conjecture on the probabilities of...

conjecture | open | MathDB 页面 | 原文线索

V2-055766 | MATHDB-364404

The unimodal dispersion comparison for Cramér and average value metric distances

摘录 / EXCERPT: Let F and G be unimodal distributions in the sense that their quantile functions are differentiable almost everywhere, with derivatives $(F^{-1})'$ and $(G^{-1})'$ decreasing f...

conjecture | open | MathDB 页面 | 原文线索

V2-055767 | MATHDB-364422

Erdős's odd-dimensionality-adjusted reverse Littlewood-Offord conjecture

摘录 / EXCERPT: Let $v_1,\ldots,v_n$ be unit vectors in $\mathbb{R}^2$, where n is odd, and let $\epsilon_1,\ldots,\epsilon_n$ be independent Rademacher random variables. Odd-n Erdős conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-055768 | MATHDB-364465

Gersho's conjecture for empirical quantization constants

摘录 / EXCERPT: Gersho's conjecture. The identity $q_{p,d}=\tilde q_{p,d}$ holds for every $p\geq 1$ and $d\in\mathbb{N}_1$. These constants govern the high-resolution asymptotics of ordinary and...

conjecture | solved | MathDB 页面 | 原文线索

V2-055769 | MATHDB-364466

Critical percolation distributional complexity exponent conjecture

摘录 / EXCERPT: Critical percolation complexity conjecture. As $m\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055770 | MATHDB-364487

The Erdős-Gimbel cochromatic gap conjecture

摘录 / EXCERPT: Let G be sampled from the binomial random graph $G_{n,1/2}$, and let $\chi(G)$ and $\zeta(G)$ denote its chromatic and cochromatic numbers, respectively. Here, “with high probabi...

conjecture | open | MathDB 页面 | 原文线索

V2-055771 | MATHDB-364526

Large- t asymptotic conjecture for the Painlev\e9 c3\mathrm{PIII}' transcendent

摘录 / EXCERPT: Painlevé asymptotic conjecture. For $0\leq\omega<\pi$, up to an order not yet determined,

conjecture | open | MathDB 页面 | 原文线索

V2-055772 | MATHDB-364577

Convergence to the maximal-current stationary measure for half-line open TASEP

摘录 / EXCERPT: Convergence conjecture. For $\alpha>\frac12$, the half-line open TASEP with step initial condition η_t^{step} converges to the probability measure...

conjecture | open | MathDB 页面 | 原文线索

V2-055773 | MATHDB-364633

The 0-1 law for uniformly repetitive rhombus tilings

摘录 / EXCERPT: Let $G=(V,E)$ be the adjacency graph of a uniformly repetitive (or uniformly recurrent) rhombus tiling. Let F be a percolation process, monotone and freezing, on configurations...

conjecture | open | MathDB 页面 | 原文线索

V2-055774 | MATHDB-364665

The cumulant variance conjecture for the periodic directed polymer

摘录 / EXCERPT: Let $\kappa_n(t)$ be the quenched n-th cumulant of the endpoint of the point-to-line directed polymer, and let $R_\beta^{(\theta)}$ be the covariance function defined in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-055775 | MATHDB-364666

The scaling-limit conjecture for periodic polymer Markov chains

摘录 / EXCERPT: Let N be fixed, let γ denote the parameters of the coupled Markov chain, and consider the coupled chain defined by, its dual chain defined by, the system of SDEs, and its...

conjecture | open | MathDB 页面 | 原文线索

V2-055776 | MATHDB-364709

Rough local smoothing conjecture for Ahlfors-David regular measures

摘录 / EXCERPT: Suppose that μ is an Ahlfors--David regular probability measure supported on $[1,2]$ and satisfying ... for $r<\operatorname{diam}(\operatorname{supp}\mu)$. Rough local smoothing...

conjecture | open | MathDB 页面 | 原文线索

V2-055777 | MATHDB-364793

The determinant identity for the half-space and full-space TASEP kernels

摘录 / EXCERPT: Let K_1 be the product of the first three matrices in the displayed factorization of the TASEP kernel, and let K_2 be the operator defined by ... Here the operators act on the in...

conjecture | open | MathDB 页面 | 原文线索

V2-055778 | MATHDB-364830

Probabilistic representer conjecture under integrability of feature maps

摘录 / EXCERPT: Probabilistic representer conjecture. There exists an $\mathbb{F}$-measure ν on X, with support contained in the support of μ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-055779 | MATHDB-364847

The interface-erosion scaling-limit conjecture on square-tiled surfaces

摘录 / EXCERPT: Interface-erosion scaling-limit conjecture. As $N\to\infty$, the processes $(D_{N^2t}^{i,N},gamma_{N^2t}^{i,N})$ converge in probability towards a solution $(D^i_t,gamma^i_t)$ to t...

conjecture | open | MathDB 页面 | 原文线索

V2-055780 | MATHDB-364933

The asymptotic Christoffel-like function conjecture for tensor-product polynomial spaces

摘录 / EXCERPT: Let $\boldsymbol{\Omega}$ be the compact support of a positive Borel probability measure μ on $\mathbb{R}^d$, and let f be its positive, continuous density with respect to Le...

conjecture | open | MathDB 页面 | 原文线索

V2-055781 | MATHDB-364938

Limit-profile conjecture for the strictly biased random transposition shuffle

摘录 / EXCERPT: Let P be the transition matrix of the biased random transposition shuffle on N cards, let U be the uniform distribution on permutations, let d_{TV} denote to...

conjecture | open | MathDB 页面 | 原文线索

V2-055782 | MATHDB-364941

Integral Fejes Tóth conjecture for projective spaces

摘录 / EXCERPT: Integral Fejes Tóth conjecture. The maximum of $I_1(\mu)$ over all Borel probability measures on $\mathbb S^d$ is achieved by μ_{ONB}. The discrete conjecture remains open for...

conjecture | open | MathDB 页面 | 原文线索

V2-055783 | MATHDB-364994

Monotonicity conjecture for favorable longer-word pairs

摘录 / EXCERPT: Monotonicity conjecture. The number of pairs (v,w) satisfying

conjecture | refuted | MathDB 页面 | 原文线索

V2-055784 | MATHDB-364995

Conjecture on arbitrarily oscillating win-probability derivatives

摘录 / EXCERPT: Oscillation conjecture. There exists a constant $C>0$ such that, for every positive integer k, there is a pair of binary words (v,w) with lengths at most $\exp(Ck)$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-055785 | MATHDB-365070

Symmetric domain Markov property conjecture for infinite self-avoiding walk

摘录 / EXCERPT: Let $d\geq1$, and let $\operatorname{SAW}_\infty$ be the space of infinite self-avoiding walks. For a walk η, write $\eta[i,\infty)$ for its tail and $\eta[0,k]=\zeta'$ for it...

conjecture | open | MathDB 页面 | 原文线索

V2-055786 | MATHDB-365071

Unique escape-weighted measure conjecture for infinite self-avoiding walk

摘录 / EXCERPT: Let μ be the connective constant, let SAW_k be the set of self-avoiding paths of length k, and write $\eta\text{ escapes }\zeta$ for the event that the infin...

conjecture | open | MathDB 页面 | 原文线索

V2-055787 | MATHDB-365098

Free Cramér conjecture for strictly convex free Gibbs measures

摘录 / EXCERPT: Let $ u_{\phi}$ and $ u_{\psi}$ be free Gibbs measures associated with strictly $\mathcal{C}^2$ convex potentials $\phi,\psi:\mathbb{R}\rightarrow\mathbb{R}$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-055788 | MATHDB-365099

Fathi-Nelson free transport conjecture

摘录 / EXCERPT: Let $X=(X_1,\ldots,X_n)$ be a noncommutative random vector with joint law μ_X, let V_0 and V_1 be potentials, and let $A=A^{\dagger}$ be a free Stein kernel for X with re...

conjecture | open | MathDB 页面 | 原文线索

V2-055789 | MATHDB-365102

The barycenter conjecture for one-dimensional free Gibbs measures

摘录 / EXCERPT: Let $f:\mathbb{R}\rightarrow\textbackslash \mathbb{R}$ be continuous and satisfy the assumptions denoted by (gibbs). For $\lambda\textbackslash in\mathbb{R}$, let the fr...

open_problem | open | MathDB 页面 | 原文线索

V2-055790 | MATHDB-365126

Ouaki-Pitman conjecture on the concave-majorant transform of Brownian motion

摘录 / EXCERPT: Let B be a one-dimensional Brownian motion on $[0,\infty)$ and let K be its almost surely existing concave majorant on $[0,\infty)$. Ouaki-Pitman conjecture. The process $2K-B$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-055791 | MATHDB-365185

Conjecture on convergence of integer-valued Lipschitz functions under general boundary conditions

摘录 / EXCERPT: Convergence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055792 | MATHDB-365208

Strict ordering of the high-order critical coupling constants

摘录 / EXCERPT: Let $d\geq 3$, and let $(\beta_{L^N})_{N\geq 2}$ denote the critical coupling constants associated with the N-th moments of the stochastic heat equation. Strict-ordering conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-055793 | MATHDB-365258

Almost-sure divergence of denominators in p-adic continued fractions

摘录 / EXCERPT: Let $p\geq 5$, and let $\{|B_n|\}_{n\in\mathbb{N}}$ be the sequence of denominators of convergents for Browkin I, Browkin II and Algorithm MR p-adic continued fractions. Denominato...

conjecture | open | MathDB 页面 | 原文线索

V2-055794 | MATHDB-365302

Shlyakhtenko-Tao entropy conjecture for the bead process

摘录 / EXCERPT: Let μ be a probability measure and let $[\mu]_\tau$ denote its free compression for $\tau\in[0,1]$. Define the sub-probability measure μ_τ of total mass τ by...

conjecture | open | MathDB 页面 | 原文线索

V2-055795 | MATHDB-365349

The conformal-invariance conjecture for FK percolation

摘录 / EXCERPT: Conformal-invariance conjecture. The collections Γ_n converge in distribution to Γ with respect to the metric $d_{\mathfrak{L}}$. This conjecture describes the expect...

conjecture | open | MathDB 页面 | 原文线索

V2-055796 | MATHDB-365406

Angel-Holmes equalization conjecture for the three-player betting game

摘录 / EXCERPT: Angel-Holmes equalization conjecture. For $x,y,z>0$ with $x<y$,

conjecture | open | MathDB 页面 | 原文线索

V2-055797 | MATHDB-365460

Conjecture on the extensive metastable energy barrier

摘录 / EXCERPT: Let $\Gamma_N^\star$ denote the metastable energy barrier for the dynamics on the random graph with N vertices, and let $\mathbb{P}_N$ be the corresponding probability law. Conje...

conjecture | open | MathDB 页面 | 原文线索

V2-055798 | MATHDB-365463

Kuperberg-Lalin symplectic variance conjecture for quadratic character sums

摘录 / EXCERPT: Let k be a positive integer, let r be a positive fundamental discriminant, and let χ_r be the primitive quadratic character modulo r. Define ... For $y>0$, define the vari...

conjecture | open | MathDB 页面 | 原文线索

V2-055799 | MATHDB-365488

The log-Gamma polymer lower-tail large deviation rate function conjecture

摘录 / EXCERPT: The log-Gamma polymer large deviation conjecture. There exists $\theta_0>0$ such that uniformly for $0<\theta<\theta_0$ and $\epsilon<s<-\theta\psi(\theta)$ for any $\epsilon>0$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-055800 | MATHDB-365577

The 2SRWB large-deviation rate-function ansatz

摘录 / EXCERPT: 2SRWB rate-function ansatz. For each $x\in[0,1]$, there is a regular function I^{2SRWB} such that

conjecture | open | MathDB 页面 | 原文线索

V2-055801 | MATHDB-365863

Airy edge universality for general invariant ensemble additions

摘录 / EXCERPT: Airy edge universality conjecture. Under some mild conditions on the summands, the upper edge limit of $Vec{\lambda}(N)$ near mu_+ is still given by $\operatorname{Airy}(\beta)$...

conjecture | open | MathDB 页面 | 原文线索

V2-055802 | MATHDB-365997

Universality conjecture for diagonal resolvent entries of rotationally invariant random matrices

摘录 / EXCERPT: Let $\vb{M}$ be a rotationally invariant random matrix of size N with a non-pathological limiting spectral density. For z outside the spectrum of $\vb{M}$, define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055803 | MATHDB-366070

Scaling limit conjecture for first-passage percolation geodesics on planar maps

摘录 / EXCERPT: Let $a\from (3/2,5/2)$, let $\bigl(f_i\bigr)_{i\geq 1}$ be the sampled marked vertices or faces, let $d^\dagger_\mathrm{fpp}$ denote the first-passage percolation distance, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055804 | MATHDB-366105

Logarithmic passage-time conjecture for the two-point critical edge distribution

摘录 / EXCERPT: Let $T_n(1,2)$ be the passage time between vertices 1 and 2 in the critical random graph model, and suppose the edge passage time has distribution ... Thus each edge has passage...

conjecture | open | MathDB 页面 | 原文线索

V2-055805 | MATHDB-366144

The two-dimensional martingale-field convergence conjecture for the voter model

摘录 / EXCERPT: Let $T>0$, let ν_p be the product Bernoulli measure of density p, and let $\hat{\mathcal{Y}}^N$ and $\hat{\mathcal{Y}}$ be the discrete and limiting random fields defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-055806 | MATHDB-366154

The percolation critical scaling profile conjecture

摘录 / EXCERPT: Percolation scaling-profile conjecture. There exist constants $a,b>0$ such that, as $V\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055807 | MATHDB-366158

Basu-Sapozhnikov quasi-multiplicativity conjecture for Bernoulli percolation

摘录 / EXCERPT: Basu-Sapozhnikov's quasi-multiplicativity conjecture. For any $3\le d\le 5$ and $\delta>0$, there exists $c=c(d,\delta)>0$ such that, for every $p\in[p_c(d),p_c(d)+\delta]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055808 | MATHDB-366200

Limiting probability conjecture for the symmetric mirrored gambler's ruin problem

摘录 / EXCERPT: Consider a gambler's ruin process on the integer interval from 0 to N, with absorption at the endpoints. From an interior position x, the particle moves to $x-1$ with probabi...

conjecture | open | MathDB 页面 | 原文线索

V2-055809 | MATHDB-366249

ASEP backward-geodesic endpoint convergence conjecture

摘录 / EXCERPT: ASEP backward-geodesic endpoint conjecture. For any $N\in\mathbb{Z}$,

conjecture | open | MathDB 页面 | 原文线索

V2-055810 | MATHDB-366250

ASEPsc particle-position Tracy-Widom GOE conjecture

摘录 / EXCERPT: ASEPsc particle-position conjecture. For any $N\in\mathbb{Z}$ and every $s\in\mathbb{R}$,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-055811 | MATHDB-366317

The $2/5$ conjecture for the Beat the Average game

摘录 / EXCERPT: Let X_1,X_2,X_3,X_4 be independent and identically distributed random variables with common distribution μ, and let the supremum range over all probability distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-055812 | MATHDB-366351

Stochastic-integral representation conjecture for the renormalised Amperean area

摘录 / EXCERPT: Let W be the Brownian motion and let G be the Green kernel appearing in the definition of the renormalised Amperean area. Write I_W for the recentred self-intersection local...

conjecture | open | MathDB 页面 | 原文线索

V2-055813 | MATHDB-366490

Superdiffusive limit and recurrence for locally perturbed infinite-horizon Lorentz processes

摘录 / EXCERPT: Infinite-horizon Lorentz-process conjecture. As $N\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055814 | MATHDB-366491

Local limit theorem and recurrence for a random walk with unbounded jumps

摘录 / EXCERPT: Unbounded-jump random-walk conjecture. In this setup, the local version of the limit law for $V_n(1)$ is also true, and the random walk defined above is recurrent.

conjecture | open | MathDB 页面 | 原文线索

V2-055815 | MATHDB-366601

Tóth's regeneration dominance conjecture for random walks on trees

摘录 / EXCERPT: Consider a discrete-time random walk of finite length n on a tree. A regeneration occurs if an edge is traversed exactly once. Let T_3 be the 3-regular tree, and let T be a...

open_problem | open | MathDB 页面 | 原文线索

V2-055816 | MATHDB-366621

Invariant-probability dichotomy for planar Dirichlet environments

摘录 / EXCERPT: Let X be a random walk in Dirichlet environment on ${\mathbb Z}^2$ with positive parameters $(\alpha_1,\ldots,\alpha_4)$. Define ... A local accelerating function γ is a pos...

conjecture | open | MathDB 页面 | 原文线索

V2-055817 | MATHDB-366734

Infinite mean offspring number in the I2 case

摘录 / EXCERPT: Let $0<\alpha\leq 1$ and consider the allometric setting, with $m_{x_0,R}(\xi_0)$ denoting the mean number of offspring from initial state x_0, radius parameter R, and initial...

conjecture | open | MathDB 页面 | 原文线索

V2-055818 | MATHDB-366777

Quantitative Benjamini conjecture for heat-kernel expansion

摘录 / EXCERPT: Let $\mathcal{G}$ be an infinite, irreducible, bounded-degree graph, let $\nu_{L,o}$ denote the L-step heat-kernel measure rooted at o, and let ∂A be the boundary of...

conjecture | open | MathDB 页面 | 原文线索

V2-055819 | MATHDB-366837

Non-iid Gaussian comparison conjecture for minimum eigenvalues

摘录 / EXCERPT: Let ... where the positive-semidefinite random matrices $\bm{W}_i$ are independent, and let ... where the independent random matrices $\bm{X}_i$ satisfy ... Here $\mathsf{Mom}[\bm{W}_i]$...

conjecture | open | MathDB 页面 | 原文线索

V2-055820 | MATHDB-366948

Threshold conjecture for the EpiSLFV reproduction number

摘录 / EXCERPT: Threshold conjecture. For all such γ and ν: if $\mathrm{R}_{0}(\gamma,\nu)<1$, then the (γ,ν)-EpiSLFV process goes extinct; if $\mathrm{R}_{0}(\gamma,\nu)>1$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-055821 | MATHDB-366963

Uniqueness of near-critical Ising scaling limits

摘录 / EXCERPT: Let $\lambda\neq 0$ be fixed and let (ω_n,σ_n) be an FK-Ising coupling of an Ising model at $\beta=\beta_c+\frac{\lambda}{n}$ in $[-1,1]^2\cap\frac{1}{n}\mathbb{Z}^2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055822 | MATHDB-366964

Absolute continuity of near-critical 3-Potts and 4-Potts FK scaling limits

摘录 / EXCERPT: Consider FK percolation for the 3-state and 4-state Potts models, with inverse temperatures $\beta^{q=3}$ and $\beta^{q=4}$ satisfying ... These shifts correspond to the expected cor...

conjecture | open | MathDB 页面 | 原文线索

V2-055823 | MATHDB-366965

Hutchcroft's weighted-amenability threshold conjecture

摘录 / EXCERPT: Let G be a graph, let Γ be a closed subgroup of $\operatorname{Aut}(G)$ acting transitively on G, and let $\mathbf{w}$ be the induced relative weight function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055824 | MATHDB-367014

Locally uniform convergence conjecture for almost periodic stochastic processes

摘录 / EXCERPT: Let $f\in B_2\cap C(\mathbb{R},\mathbb{C})$ satisfy the linear independence condition and assume either the condition ensuring continuous sample paths of $\mathcal{M}_f$, or the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-055825 | MATHDB-367066

The Gompertz approximation conjecture for ties in the M&M Game

摘录 / EXCERPT: Let $P_{\text{tie}}(\lambda)$ denote the probability of a tie in the Mtextup{&}M Game, where the probability of flipping heads evolves according to an exponential distribution, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055826 | MATHDB-367115

Hopkins's low-degree conjecture for hypothesis testing

摘录 / EXCERPT: Consider asymptotic hypothesis-testing problems between probability measures $mathbb P$ and $mathbb Q$. For degree at most D, define the low-degree advantage by ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-055827 | MATHDB-367120

The Markov-chain extension of the most-informative Boolean function conjecture

摘录 / EXCERPT: Let $X^n\sim\operatorname{Unif}(\mathbb{H}^n)$, let Y^n be obtained by passing X^n through a binary symmetric channel with crossover probability $p\in[0,1]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055828 | MATHDB-367223

The non-primitivity conjecture for random countable Boolean algebras

摘录 / EXCERPT: Let W be the Cantor set $\mathscr{D}_{1}$ and let $R=\operatorname{Co}(W)$ be the countable atomless Boolean algebra. Let I be the set of finite binary strings, including the e...

conjecture | open | MathDB 页面 | 原文线索

V2-055829 | MATHDB-367261

Feasibility of partial alignment in the Erdős-Rényi model above the threshold

摘录 / EXCERPT: Feasibility conjecture. Partial alignment is feasible: there exists an estimator $\hat{\boldsymbol{\pi}}$ and some $\varepsilon>0$ such that, with high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-055830 | MATHDB-367281

Polylogarithmic non-giant components conjecture for the square graph

摘录 / EXCERPT: Polylogarithmic non-giant components conjecture. Suppose

conjecture | open | MathDB 页面 | 原文线索

V2-055831 | MATHDB-367288

Tight lower-bound conjecture for segregation in the Offended Voter Model

摘录 / EXCERPT: Let $q_N=q\in(0,1)$ be constant, let $Z_0^{N,\textnormal{min}}=\lfloor N/2\rfloor$, and let $E^N\subset\mathbf{Z}^N_0$. Let $\mathcal{A}_N$ denote the segregation event, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055832 | MATHDB-367289

Monotonicity conjecture for final-state probabilities in the Offended Voter Model

摘录 / EXCERPT: Let $\mathcal{A}_N$ and $\mathcal{C}_N$ be the final states defined in the model's classification, and write $\mathbb{P}_q$ for probability under parameter q. Monotonicity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-055833 | MATHDB-367338

Joint tracial moment conjecture for the limiting spectral distribution of data-matrix commutators

摘录 / EXCERPT: Let Σ_1 and Σ_2 be the covariance matrices, and let H be their joint limiting spectral distribution. For any $n_1,n_2\in\mathbb{N}$ and every selection of non-negat...

conjecture | open | MathDB 页面 | 原文线索

V2-055834 | MATHDB-367357

Chan-Pak conjecture on the log-concavity of minimum positions in linear extensions

摘录 / EXCERPT: Chan-Pak conjecture. For every $A\subseteq P$, the sequence of probabilities of the minimum position is log-concave:

conjecture | open | MathDB 页面 | 原文线索

V2-055835 | MATHDB-367364

Nonconstancy conjecture for the stable-domain-of-attraction coefficient

摘录 / EXCERPT: Let $\Lambda>1$ and let $\alpha\in(0,2)$. For each $\theta>0$, let c_θ denote the coefficient defined in Proposition corresponding to the asymptotics of the characteristic f...

conjecture | open | MathDB 页面 | 原文线索

V2-055836 | MATHDB-367416

Mean-field approximation conjecture for upper tails of strictly balanced subgraph counts

摘录 / EXCERPT: Let H be a strictly balanced graph, with v_H vertices, e_H edges, maximum degree Δ, and parameter α_H^*. Let $\Psi_{p,H}(\delta)$ denote the mean-field variatio...

conjecture | open | MathDB 页面 | 原文线索

V2-055837 | MATHDB-367445

Optimal fourth-moment deficit in Khintchine's inequality

摘录 / EXCERPT: Optimal fourth-moment deficit conjecture. For every $p\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-055838 | MATHDB-367446

Stability conjecture for the third Khintchine moment

摘录 / EXCERPT: Third-moment stability conjecture. There exists a universal constant $C_3>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055839 | MATHDB-367447

Distributional stability for Khintchine inequalities

摘录 / EXCERPT: Let G be a standard Gaussian random variable, let $X_1,\ldots,X_n$ be independent identically distributed symmetric random variables with ... and let $a_1,\ldots,a_n$ satisfy the n...

open_problem | open | MathDB 页面 | 原文线索

V2-055840 | MATHDB-367469

Large-loop log-Gamma distribution conjecture for phi-four periods

摘录 / EXCERPT: For each loop order L, consider the period Feynman integrals of ϕ^4 theory and sample them with probability inversely proportional to the symmetry factor of their associated...

conjecture | open | MathDB 页面 | 原文线索

V2-055841 | MATHDB-367477

The MERW transition conjecture for the growing pyramidal model

摘录 / EXCERPT: MERW transition conjecture. The unique maximal entropy random walk for the growing pyramidal model depicted by the Kreweras-walk correspondence has transition probabilities

conjecture | open | MathDB 页面 | 原文线索

V2-055842 | MATHDB-367478

The limit-shape conjecture for the MERW pyramidal process

摘录 / EXCERPT: Let $(P_n)_{n\geq 0}$ be the suitably scaled maximal entropy random walk pyramidal process. Define ... and let $\mathcal{H}$ be the graph of Ω. Set ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-055843 | MATHDB-367521

The symmetric-point maximum conjecture for random geometric graph entropy

摘录 / EXCERPT: Let $H_t(\mathcal{G})$ denote the entropy of the random geometric graph distribution at parameter t, and let $\bar{p}_{\max}$ be the corresponding edge-probability parameter at w...

conjecture | open | MathDB 页面 | 原文线索

V2-055844 | MATHDB-367531

Ostrowski automaticity conjecture for records of $S_n(2\xi)$

摘录 / EXCERPT: For an irrational number ξ, define the deterministic random walk by ... Call r a record if $S_r(2\xi)$ has not occurred among the preceding partial sums, including...

conjecture | open | MathDB 页面 | 原文线索

V2-055845 | MATHDB-367602

Equality of the tail constants for maximal displacement

摘录 / EXCERPT: Equality of the tail constants. The constants C_1 and C_2 are equal. This would sharpen the established two-sided asymptotic-order estimate for the tail probability of the whol...

conjecture | open | MathDB 页面 | 原文线索

V2-055846 | MATHDB-367611

Generic uniqueness of the artery in the Handel-Mosher axis bundle

摘录 / EXCERPT: Generic uniqueness conjecture. For a reasonable random walk on $\operatorname{Out}(F_r)$, having that $\mathcal{A}_\varphi$ has a unique artery is generic.

conjecture | open | MathDB 页面 | 原文线索

V2-055847 | MATHDB-367740

Monotonicity conjecture for the expected beta-content function

摘录 / EXCERPT: Let d and n be as in the beta-polytope model, and fix parameters $\beta_1,\ldots,\beta_n\geq -1$. Write $\mathscr{P}_{n,d}^{\beta_1,\ldots,\beta_n}$ for the corresponding beta...

conjecture | open | MathDB 页面 | 原文线索

V2-055848 | MATHDB-367889

Bilyk-Glazyrin-Matzke-Park-Vlasiuk's discreteness conjecture for p-frame minimizers

摘录 / EXCERPT: Let $p>0$, and for a probability measure μ on $\mathbb{S}^{d-1}$ define the p-frame energy by ... A minimizer is a probability measure attaining the minimum of this energy. Bil...

conjecture | open | MathDB 页面 | 原文线索

V2-055849 | MATHDB-367894

Conjecture on stochastic invasion of type 2

摘录 / EXCERPT: Let $(n_{1a}^*,n_{1i}^*,n_3^*)$ be an asymptotically stable equilibrium of the three-dimensional virus system, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055850 | MATHDB-367916

Poisson-integral convergence and conditionality on the impaired-integrability space

摘录 / EXCERPT: Let N be a Poisson random measure and let L^2_{imp} be the space of bounded, square-integrable functions for which the relevant improper integrals exist. For a f...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055851 | MATHDB-367917

Lower-moment norm equivalence for renewal partial sums

摘录 / EXCERPT: Let (X_k) be independent copies of a nondegenerate nonnegative random variable, let $S_n=X_1+\cdots+X_n$, and set $Z_n=\sum_{k=1}^n e^{\imath S_k}$. For $p>0$, write...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055852 | MATHDB-367919

Almost-s sure summability of Poisson-arrival double sequences

摘录 / EXCERPT: Let $f\in L^2_{\operatorname{imp}}$, let (A_k) be pairwise disjoint Borel subsets of $\mathbb R_+$, and define $f_k=f\mathbf 1_{A_k}$. Let (S_n) be the arrival times of a Poiss...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055853 | MATHDB-367938

LFE-MLFE asymptotic equivalence conjecture

摘录 / EXCERPT: Let ρ_t and μ_t denote, respectively, the laws at time $t \geq 0$ of solutions to the local-field equation (LFE) and the Markov local-field equation (MLFE). Let d be a s...

conjecture | open | MathDB 页面 | 原文线索

V2-055854 | MATHDB-367939

Sparse free energy Lyapunov conjecture for the local-field equation

摘录 / EXCERPT: Let ν be a probability measure on $\mathbb{R}^3$, let $\bar{\nu}$ be its $(0,1)$-marginal, and let π and $\bar{\pi}$ denote the invariant measure of the Markov local-field...

conjecture | open | MathDB 页面 | 原文线索

V2-055855 | MATHDB-367989

Conjecture on comparable probabilities of essential and influential edges

摘录 / EXCERPT: Comparability conjecture. There exists a constant C independent of n such that

conjecture | open | MathDB 页面 | 原文线索

V2-055856 | MATHDB-367990

Conjecture on the total second-order derivative mass in first-passage percolation

摘录 / EXCERPT: Second-order mass conjecture. There exists a constant C independent of N such that

conjecture | open | MathDB 页面 | 原文线索

V2-055857 | MATHDB-367991

Conjecture on nonnegative expected higher-order derivatives

摘录 / EXCERPT: Nonnegative-expectation conjecture. For most sets S, the expected value of the higher-order derivative is non-negative:

conjecture | open | MathDB 页面 | 原文线索

V2-055858 | MATHDB-367992

Positive-drift conjecture for higher-order environment derivatives

摘录 / EXCERPT: Positive-drift conjecture. There exists $\theta_k\in(0,1)$ such that, for most reasonable k-edge sets S and every positive m,

conjecture | open | MathDB 页面 | 原文线索

V2-055859 | MATHDB-367993

Higher-order derivative probability-ratio conjecture in the torus model

摘录 / EXCERPT: Derivative probability-ratio conjecture. There exists a constant $\theta\in(0,d/2)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055860 | MATHDB-368043

The asymptotic number of probabilistic compositional graphoids

摘录 / EXCERPT: Asymptotic counting conjecture. The number of compositional graphoids which are representable by n discrete random variables is asymptotically

conjecture | open | MathDB 页面 | 原文线索

V2-055861 | MATHDB-368138

Quasirandomness conjecture for constant-degree tests of stochastic block models

摘录 / EXCERPT: Quasirandomness conjecture. There exists a constant-degree polynomial test that distinguishes these hypotheses with high probability if and only if one of the signed subgraph count...

conjecture | open | MathDB 页面 | 原文线索

V2-055862 | MATHDB-368155

Sharp norm conjecture for probabilistic continuous second-order Riesz transforms

摘录 / EXCERPT: Let $d\ge 2$, $1<p<\infty$, and let $\mathbf{R}^{(jk)}$ be the probabilistic continuous second-order Riesz transforms on $\mathbb{R}^d$. Let $p^*=\max\\{p,p/(p-1)\\}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055863 | MATHDB-368180

Stationary-distribution approximation for lounge-from-queue systems

摘录 / EXCERPT: Let $ u$ be the transition rate from the lounge to the queue, let β satisfy ... where $\eta>0$ is constant, and let $A=\lfloor\eta\mu/\alpha\rfloor$. For each $ u$, write...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055864 | MATHDB-368206

The conjecture on global synchrony with negative edges

摘录 / EXCERPT: Let A be an $n\times n$ random matrix with zero diagonal and independent off-diagonal entries satisfying ... A matrix is globally synchronizing when the only local minima of its Ku...

conjecture | open | MathDB 页面 | 原文线索

V2-055865 | MATHDB-368217

Conjectured asymptotic dependence of truncated martingale tails on the starting position

摘录 / EXCERPT: Let $W_\infty^{(x)}$ and $D_\infty^{(x)}$ denote the truncated martingales started from position x, and let κ be the parameter appearing in the tail behavior of...

conjecture | open | MathDB 页面 | 原文线索

V2-055866 | MATHDB-368220

Representability of maxoids by positive discrete distributions

摘录 / EXCERPT: Representability conjecture. Every maxoid is representable as the conditional-independence structure of a positive discrete distribution.

conjecture | open | MathDB 页面 | 原文线索

V2-055867 | MATHDB-368302

Conjecture on fifth-order weighted central limit approximation

摘录 / EXCERPT: Weighted fifth-order central limit conjecture. There exists such a weighting function $\bar w$ and a universal constant $C_3>0$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-055868 | MATHDB-368321

Long-cycle conjecture for the percolated hypercube above the supercritical threshold

摘录 / EXCERPT: Long-cycle conjecture. Let $\varepsilon>0$ be a constant, and let

conjecture | open | MathDB 页面 | 原文线索

V2-055869 | MATHDB-368401

Conjecture on finite-population traveling waves and propagation speed

摘录 / EXCERPT: Finite-population traveling-wave conjecture. There exists a propagation speed σ^K such that, in an appropriate convergence sense,...

conjecture | open | MathDB 页面 | 原文线索

V2-055870 | MATHDB-368463

Sigma-Laplacian recovery conjecture for the Gaussian planted submatrix model

摘录 / EXCERPT: Consider the Gaussian Planted Submatrix model in the setting of Theorem, with signal parameter β, signal vector $\bm{x}$, and leading eigenvector...

conjecture | open | MathDB 页面 | 原文线索

V2-055871 | MATHDB-368464

Transfer conjecture from Gaussian planted submatrices to planted clique

摘录 / EXCERPT: The Gaussian Planted Submatrix problem has signal parameter β, while the Planted Clique problem is obtained from a centered Erdős-Rényi adjacency matrix with an inserted cliq...

conjecture | open | MathDB 页面 | 原文线索

V2-055872 | MATHDB-368532

Kaimanovich-Masur singularity conjecture for mapping class groups

摘录 / EXCERPT: Kaimanovich-Masur singularity conjecture. If m has finite support, then the m-stationary measure ν is singular to every Busemann Patterson-Sullivan measur...

conjecture | open | MathDB 页面 | 原文线索

V2-055873 | MATHDB-368533

Singularity conjecture for stationary measures on Furstenberg boundaries

摘录 / EXCERPT: Singularity conjecture. If m has finite support, then the m-stationary measure ν is singular to the Lebesgue measure class on $\operatorname{\mathcal{F}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-055874 | MATHDB-368534

Generalized singularity conjecture for coarse Iwasawa Patterson-Sullivan measures

摘录 / EXCERPT: Generalized singularity conjecture. If m has finite support, then the m-stationary measure ν is singular to every coarse Iwasawa Patterson-Sullivan measur...

conjecture | open | MathDB 页面 | 原文线索

V2-055875 | MATHDB-368571

Finite-sample consistency conjecture for empirical kernel quantile embeddings with p greater than 1

摘录 / EXCERPT: Let $\mathcal{X} \subseteq \mathbb{R}^d$, let ν have a density, and let P and Q be measures on $\mathcal{X}$ with densities bounded away from zero, satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-055876 | MATHDB-368593

Nam-Nguyen-Sly critical-value conjecture for configuration graphs

摘录 / EXCERPT: Let μ be a probability distribution on the non-negative integers satisfying ... Let $(\mathscr{G}_n)_{n\in\mathbb{N}}$ be configuration graphs on n vertices derived from μ,...

conjecture | open | MathDB 页面 | 原文线索

V2-055877 | MATHDB-368597

Contiguity conjecture for unions of random perfect matchings

摘录 / EXCERPT: Random matching decomposition conjecture. The event that the disjoint union belongs to A_n holds with high probability if and only if the event that a uniform random d-regular...

conjecture | open | MathDB 页面 | 原文线索

V2-055878 | MATHDB-368665

Conjecture on the vanishing probability of a fixed linking number in large grid models

摘录 / EXCERPT: Let k be a fixed number, and consider links chosen randomly from links represented by grids whose size tends to infinity. Vanishing fixed-linking-number conjecture. The probabili...

conjecture | open | MathDB 页面 | 原文线索

V2-055879 | MATHDB-368666

The typical-cell width convergence conjecture for high-dimensional Poisson-Voronoi cells

摘录 / EXCERPT: Let $\mathcal V_{\text{typ}}$ be the typical cell of a Poisson-Voronoi tessellation in dimension d, with intensity $\lambda=\lambda(d)>0$, and let B denote the unit ball in...

conjecture | open | MathDB 页面 | 原文线索

V2-055880 | MATHDB-368697

The multivariable random unitary pencil determinant limit conjecture

摘录 / EXCERPT: For a g-tuple of matrices $\mathcal X=(X_1,\dots,X_g)$, define its outer spectral radius by ... where $\operatorname{rad}(X)$ is the ordinary spectral radius. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055881 | MATHDB-368798

Ekström-Persson conjecture on dimensions of random limsup sets

摘录 / EXCERPT: Let $ u$ be a probability measure on $\mathbb{R}^d$, let $(\omega_k)_{k=1}^{\infty}$ be independent random points distributed according to ν, and for $\alpha>0$ define ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-055882 | MATHDB-368812

Bombelli's conjecture on Poisson reconstruction of distinguishing spacetimes

摘录 / EXCERPT: Bombelli's conjecture. If, for every finite poset C, the probabilities that the samples from M and N are order-isomorphic to C coincide, then M and N are isometric.

conjecture | open | MathDB 页面 | 原文线索

V2-055883 | MATHDB-368914

Optimality of the generalized pairwise testing algorithm for ordered probabilities

摘录 / EXCERPT: GPTA optimality conjecture. Among all nested testing procedures that preserve the order $u_1,u_2,\ldots,u_n$, the GPTA is an optimal nested procedure. It is not necessarily uniquel...

conjecture | open | MathDB 页面 | 原文线索

V2-055884 | MATHDB-368946

Tracy-Widom limit conjecture for the centered statistic $T_1(Y)$

摘录 / EXCERPT: Let $T_1(Y)$ be the statistic in the separability test, and let γ_0 and E_+ be the quantities defined in the paper's equation (sec4.3.eq3). Let TW_1 denote the Tracy-Wid...

conjecture | open | MathDB 页面 | 原文线索

V2-055885 | MATHDB-368980

Connectedness threshold conjecture for reflected random-walk ranges

摘录 / EXCERPT: Let $Q=\sqrt{({\mathbf{c}}-1)/6}$ and let X^k be the continuous-time random walk on $\mathcal M_k$ reflected off infinity, stopped when it first hits the boundary at time...

conjecture | open | MathDB 页面 | 原文线索

V2-055886 | MATHDB-368983

Critical percolation phase-transition conjecture for supercritical random planar maps

摘录 / EXCERPT: Let $Q\in(0,2)$ and let $p_c(Q)\in(0,1)$ be the critical probability from the non-trivial percolation transition conjecture. Consider critical site percolation on $\mathcal M_k$ wi...

conjecture | open | MathDB 页面 | 原文线索

V2-055887 | MATHDB-369121

Asymptotic disintegrability of compact connected Riemannian manifolds

摘录 / EXCERPT: Let $(\mathfrak{X},d,\mu)$ be a metric probability space, where $\mathfrak{X}$ is a compact, connected, D-dimensional Riemannian manifold, d is its geodesic distance, and μ...

conjecture | open | MathDB 页面 | 原文线索

V2-055888 | MATHDB-369122

Equivalence of asymptotic average-case approximability and asymptotic disintegrability

摘录 / EXCERPT: Let $(\mathfrak{X},d,\mu)$ be a metric Polish probability space. Equivalence conjecture. The space $(\mathfrak{X},d,\mu)$ is asymptotically average-case approximable if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-055889 | MATHDB-369127

Extension of Theorem 3 to bivariate distributions with densities

摘录 / EXCERPT: Let F be a bivariate distribution admitting a density, and let the iteration map and the conclusion of Theorem 3 be understood as in the source paper. Extension conjecture. Theor...

conjecture | open | MathDB 页面 | 原文线索

V2-055890 | MATHDB-369128

Multivariate extension of Theorem 3 for distributions with densities

摘录 / EXCERPT: Let F be a multivariate distribution admitting a density, with the relevant multivariate iteration map and a suitably reformulated version of Theorem 3. Multivariate extension co...

conjecture | open | MathDB 页面 | 原文线索

V2-055891 | MATHDB-369135

Non-isomorphism of ultrapowers for structures with the independence property

摘录 / EXCERPT: Let M be a structure whose first-order theory has the independence property witnessed by an affine quantifier-free formula. Let (Ω,μ) be a diffuse probability space, and...

conjecture | open | MathDB 页面 | 原文线索

V2-055892 | MATHDB-369170

Joint-growth height conjecture for wreath and Kronecker products

摘录 / EXCERPT: Joint-growth height conjecture. As $\min(n,m)\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055893 | MATHDB-369190

Monotonicity conjecture for balanced Radon partitions of Gaussian points

摘录 / EXCERPT: Let $N,dfin\mathbb{N}$ satisfy $N\geq d+2$. For a partition $[N]=A\cup B$, let $\mathbb{P}_d(A,B)$ denote the probability that (A,B) is a Radon partition among N independent ra...

conjecture | open | MathDB 页面 | 原文线索

V2-055894 | MATHDB-369261

Random-cluster negative-correlation conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $\boldsymbol{x}=x_e:e otin E$ be positive edgeweights, and let the random cluster model on G have parameter q. For distinct edges $e,f otin E$, wr...

conjecture | open | MathDB 页面 | 原文线索

V2-055895 | MATHDB-369282

The FDH optimality conjecture for m-shelf shuffles

摘录 / EXCERPT: Let H_{2m} be the $2m$th harmonic number, and let $F^{\mathbf{G}}(n,m)$ denote the expected number of correct guesses made by the strategy $\mathbf{G}$ for a deck of n cards su...

conjecture | open | MathDB 页面 | 原文线索

V2-055896 | MATHDB-369287

Stochastic diameter monotonicity conjecture for random graphs with fixed surplus

摘录 / EXCERPT: Stochastic diameter monotonicity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-055897 | MATHDB-369296

The negative partial-sum conjecture for normalization of aggregated Markov-chain distributions

摘录 / EXCERPT: Negative partial-sum conjecture. If there are pairwise different indices $i_1, \dots, i_{n-1}$ such that

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055898 | MATHDB-369366

Busy Beaver and Kolmogorov-random-string conjectures imply Feige's Hypothesis

摘录 / EXCERPT: Let Feige's Hypothesis be the assertion that no efficient algorithm can prove the unsatisfiability of a random 3-SAT formula with high probability, even when the formula has a sm...

conjecture | open | MathDB 页面 | 原文线索

V2-055899 | MATHDB-369373

Asymptotic conjugation-invariance conjecture for regular word values

摘录 / EXCERPT: Let S_n be the symmetric group and let $\mathbb F_r$ be the free group on r generators. For a word with constants $w\in S_n\ast\mathbb F_r$, let $\mu_w^{S_n}$ be the distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-055900 | MATHDB-369403

Asymptotic difference conjecture for the two labeled coupon collector problems

摘录 / EXCERPT: Asymptotic difference conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055901 | MATHDB-369404

Asymptotic expectation conjecture for the Type II labeled coupon collector problem

摘录 / EXCERPT: Type II expectation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055902 | MATHDB-369461

The critical two-dimensional KPZ scaling-limit conjecture

摘录 / EXCERPT: Critical KPZ scaling-limit conjecture. There exists a non-random function $b(\varepsilon)>0$ with $b(\varepsilon)\to\infty$ as $\varepsilon\to0$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-055903 | MATHDB-369518

Conjecture on the optimal speed of bulk gap convergence for Wigner matrices

摘录 / EXCERPT: Let (H_N) be a generalized Wigner ensemble with uniform p-support, and let $ \mu_1\leq\cdots\leq\mu_N$ be the eigenvalues of a GOE matrix while...

conjecture | open | MathDB 页面 | 原文线索

V2-055904 | MATHDB-369519

Conjecture on the smallest bulk gap for real Wigner matrices

摘录 / EXCERPT: Fix $ \alpha>0$. Let $ \mu_1\leq\cdots\leq\mu_N$ be the eigenvalues of a GOE matrix, and let $ \lambda_1\leq\cdots\leq\lambda_N$ be the eigenvalues of a real generalized Wigner mat...

conjecture | open | MathDB 页面 | 原文线索

V2-055905 | MATHDB-369607

The hiding conjecture for Gaussian boson sampling

摘录 / EXCERPT: Let U_{NK} be the top-left $N\times K$ submatrix of an $M\times M$ Haar-random unitary matrix. Let $Z=Z_{N,K}$ be a random matrix distributed according to either…

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055906 | MATHDB-369680

Conditional law of large numbers for directed last passage percolation

摘录 / EXCERPT: Conditional law of large numbers conjecture. The convergence in the conditional law of large numbers holds with

conjecture | open | MathDB 页面 | 原文线索

V2-055907 | MATHDB-369681

Conditional fluctuation conjecture for directed last passage percolation

摘录 / EXCERPT: Conditional fluctuation conjecture. The following hold. For $(x,y)\in\Omega_1$, the centered passage time has Tracy-Widom TW_2 fluctuations on the $N^{1/3}$ scale…

conjecture | open | MathDB 页面 | 原文线索

V2-055908 | MATHDB-369682

Conditional geodesic fluctuation conjecture

摘录 / EXCERPT: Conditional geodesic fluctuation conjecture. Conditional on $\mathcal L(aN,bN)=\ell N$,

conjecture | open | MathDB 页面 | 原文线索

V2-055909 | MATHDB-369694

The singularity conjecture for stationary measures on the Furstenberg boundary

摘录 / EXCERPT: Let $\operatorname{\mathsf{G}}$ be a connected semisimple Lie group without compact factors and with finite center. Suppose $\Gamma<\operatorname{\mathsf{G}}$ is a Zariski dense di…

conjecture | open | MathDB 页面 | 原文线索

V2-055910 | MATHDB-369748

The intermediate-dimension interpolation conjecture for connected components of Mandelbrot percolation

摘录 / EXCERPT: Intermediate-dimension interpolation conjecture. There exists a monotone function $\alpha(s)\colon[0,1]\to[0,\infty)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055911 | MATHDB-369749

The 2R-Conjecture for cluster gaps in the Hegselmann-Krause model

摘录 / EXCERPT: Let $R>0$ be the interaction radius, and consider the Hegselmann-Krause dynamics in the restricted setting with N points on a compact interval. A fixed point consists of clusters…

conjecture | open | MathDB 页面 | 原文线索

V2-055912 | MATHDB-369750

The Strong 2R-Conjecture for Hegselmann-Krause cluster locations

摘录 / EXCERPT: For each $\lambda>0$, let $\Xi_{\lambda}^{(0)}$ be a simple stationary renewal process on $\mathbb{R}$ satisfying the functional strong law of large numbers, and let…

conjecture | open | MathDB 页面 | 原文线索

V2-055913 | MATHDB-369759

Lovász-Szegedy Borel representation conjecture for Lebesgue graphs

摘录 / EXCERPT: Let $\mathbf L$ be a Lebesgue graph, meaning a graph on the standard probability space $[0,1]$ whose edge set is Lebesgue-measurable. A finite graph F has a density in a graph gi…

conjecture | open | MathDB 页面 | 原文线索

V2-055914 | MATHDB-369824

The Maximum Criterion conjecture for linear box fragmentation processes

摘录 / EXCERPT: Let $\mathfrak{B}_0\supset\mathfrak{B}_1\supset\cdots$ be an m-dimensional linear-fragmentation process, and for $1\leq d\leq m$ let $\mathfrak{m}_d^{(N)}$ denote the maximum pr…

conjecture | open | MathDB 页面 | 原文线索

V2-055915 | MATHDB-370032

Convergence of SVS-SPRING for linear least-quadratics

摘录 / EXCERPT: Convergence of SVS-SPRING for LLQ. Under these assumptions and the SVS(J,k,λ) sampling distribution, SPRING should satisfy the displayed convergence bound, where the hidden…

conjecture | open | MathDB 页面 | 原文线索

V2-055916 | MATHDB-370239

Subquadratic minimal generation conjecture for random hypertree fundamental groups

摘录 / EXCERPT: Subquadratic generation conjecture. The normalized minimal number of generators converges to zero in probability:

conjecture | open | MathDB 页面 | 原文线索

V2-055917 | MATHDB-370246

The empirical limit measure dimension conjecture for branching random walks on hyperbolic groups

摘录 / EXCERPT: Let Γ be a non-elementary hyperbolic group generated by a finite set S, and let ν be a probability measure on S with $\operatorname{supp}(\nu)=S$. Consider the branc…

conjecture | open | MathDB 页面 | 原文线索

V2-055918 | MATHDB-370337

The automorphism-invariant positive-association percolation criterion

摘录 / EXCERPT: Percolation criterion. There are an $\epsilon>0$ and an $N \in \mathbb{N}$ such that, whenever $\mathbb{P}$ is an automorphism-invariant positively associated percolation measure o…

conjecture | open | MathDB 页面 | 原文线索

V2-055919 | MATHDB-370377

Non-universality conjecture for the first non-universal term in Kac polynomial real zeros

摘录 / EXCERPT: Let $P_n(x)=\sum_{k=0}^{n}\xi_kx^k$ be a Kac random polynomial, where the coefficients ξ_k are independent and identically distributed copies of a random variable ξ. Assume…

conjecture | open | MathDB 页面 | 原文线索

V2-055920 | MATHDB-370425

Chen-Hermite balancing conjecture for Gaussian single increments

摘录 / EXCERPT: Chen-Hermite Gaussianity conjecture. Then $(X_t)_{t\in[0,T]}$ must be a Gaussian process. This would establish joint Gaussianity from Gaussian single-increment laws and the Chen-He...

conjecture | open | MathDB 页面 | 原文线索

V2-055921 | MATHDB-370480

Perimeter approximation conjecture for the convex hull of multiple random walks

摘录 / EXCERPT: Let L_n be the perimeter of the convex hull of the multiple random walks, and let ${\mathcal G}_n$ and ${\mathcal E}^{\mathrm{f}}_n$ be the sets defined in the paper. For each re...

conjecture | open | MathDB 页面 | 原文线索

V2-055922 | MATHDB-370556

The conjecture on the SBBS full-process scaling limit

摘录 / EXCERPT: Let d be the number of balls in the system, and fix an error probability $ to$?

conjecture | open | MathDB 页面 | 原文线索

V2-055923 | MATHDB-370564

IIC hitting conjecture for p-capacity

摘录 / EXCERPT: IIC hitting conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055924 | MATHDB-370565

Two-set p-capacity factorisation conjecture for the IIC

摘录 / EXCERPT: Two-set p-capacity factorisation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055925 | MATHDB-370581

The dimension formula conjecture for self-similar measures

摘录 / EXCERPT: Let Φ be an iterated function system, let p be a probability vector, and let μ be the corresponding self-similar measure. The entropy rate is ... where $X_1,X_2,\ldots$ ar...

conjecture | open | MathDB 页面 | 原文线索

V2-055926 | MATHDB-370597

Gumbel separation-profile conjecture for the random transposition shuffle

摘录 / EXCERPT: Consider, for $n\geq 2$, the random transposition shuffle $(X_t^{(n)})_{t\geq 0}$ on the symmetric group $\mathfrak{S}_n$, in discrete or continuous time, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-055927 | MATHDB-370778

Non-concavity conjecture for the critical curve of long-range percolation

摘录 / EXCERPT: Non-concavity conjecture. There exists $\delta=\delta(k,d)>0$ such that $p\mapsto q_c(p)$ is not concave on $[0,\delta]$.

conjecture | open | MathDB 页面 | 原文线索

V2-055928 | MATHDB-370780

Pythagorean-like inequality for isoperimetric profiles of product measures

摘录 / EXCERPT: Let μ_{1} and μ_{2} be log-concave probability measures on Riemannian manifolds $\mathcal{M}_{1}$ and $\mathcal{M}_{2}$, respectively. For a in the domain of the isoperim...

conjecture | open | MathDB 页面 | 原文线索

V2-055929 | MATHDB-370812

Divergence of the expected odometer at criticality in activated random walk

摘录 / EXCERPT: Let $m(0)$ denote the odometer at the origin for fixed-energy activated random walk with i.i.d. initial particle density ρ, and let $\rho_c(\lambda)$ be its critical density f...

conjecture | open | MathDB 页面 | 原文线索

V2-055930 | MATHDB-370867

The maximizer conjecture for the Trail of Lost Pennies

摘录 / EXCERPT: Let $\lambda_{\rm max}:(0,1]^2\to[1,\infty)$ be the function measuring the maximum value associated with the game. Maximizer conjecture. The maximum value of...

conjecture | open | MathDB 页面 | 原文线索

V2-055931 | MATHDB-371047

Convergence to the Massive Modified Arratia Flow

摘录 / EXCERPT: Convergence conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-055932 | MATHDB-371112

Stationary-measure decomposition for kinetically constrained models

摘录 / EXCERPT: Stationary-measure decomposition conjecture. If μ is a stationary measure of a KCM, then there exists a probability measure μ_{*} on $\mathcal{E}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055933 | MATHDB-371201

Weight preservation for Fibonacci recurrences

摘录 / EXCERPT: Let F_m be the set of periodic sequences from the Fibonacci recurrence or its parity transform with initial condition $(a_0,a_1)\in(\mathbb Z/m\mathbb Z)^2$, taken modulo m. Fo...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055934 | MATHDB-371230

Asymptotic probability formula for conditioned convex hulls

摘录 / EXCERPT: Let $\mathbf{K}$ be a compact convex set of area 1, let $\lambda>0$, and let $C^\star$ be the unique element of $\operatorname{argmax}\Phi_{\lambda,\mathbf{K}}$, where ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-055935 | MATHDB-371361

The uniform dimension conjecture for parabolic SPDEs on the torus

摘录 / EXCERPT: Uniform dimension conjecture. The uniform dimension theorem stated for $p\geqslant4$ is valid whenever $p\geqslant2$, provided that

conjecture | open | MathDB 页面 | 原文线索

V2-055936 | MATHDB-371362

The multiplicative-noise extension conjecture for uniform dimension theorems

摘录 / EXCERPT: Multiplicative-noise extension conjecture. Theorem (ii) and (iii) for the additive-noise equation remain valid in the presence of multiplicative noise, provided that

conjecture | open | MathDB 页面 | 原文线索

V2-055937 | MATHDB-371372

KLS conjecture for log-concave probability measures

摘录 / EXCERPT: KLS conjecture. There exists a universal constant $\mathbf{C}_{\mathrm{univ}}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-055938 | MATHDB-371380

Critical-window scaling conjecture for the weakly coupled $|\varphi|^4$ model

摘录 / EXCERPT: Let $\operatorname{\bf NG}_{s}$ be the probability measure whose moment generating function is specified by ... Under the assumptions of Theorem, let w_N denote the critical-window...

conjecture | open | MathDB 页面 | 原文线索

V2-055939 | MATHDB-371455

Support conjecture for Brown measures of multiplicative Brownian products

摘录 / EXCERPT: Let $(\mathscr{A},\tau)$ be a W^*-probability space and let b_0 be an invertible operator in it. For $z\in\mathbb{C}$, define ... For any operator a, write $|a|^2=a^*a$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-055940 | MATHDB-371528

Payoff imbalance in Nash equilibria of the imbalanced $(m,2k+1)$-RPS game

摘录 / EXCERPT: Consider the given imbalanced $(m,2k+1)$-RPS game and a Nash equilibrium N. Let $P_i(S)$ be the probability that player i plays S, and let $P_i(T_p)$ be the probability that...

conjecture | open | MathDB 页面 | 原文线索

V2-055941 | MATHDB-371587

Non-atomicity conjecture for distances between typical CLE carpet points

摘录 / EXCERPT: Let X be the carpet of a non-nested $\operatorname{CLE}_\kappa$. Given X, let x and y be independent samples from the canonical conformally covariant probability measure...

conjecture | open | MathDB 页面 | 原文线索

V2-055942 | MATHDB-371612

Hambly-Jordan conjecture on critical resistance divergence

摘录 / EXCERPT: Let R_n denote the resistance in the random resistance model, and let $p_c=\frac12$. Weak convergence is understood for the random variables under consideration. Hambly and Jorda...

conjecture | open | MathDB 页面 | 原文线索

V2-055943 | MATHDB-371613

Addario-Berry et al.'s critical resistance scaling conjecture

摘录 / EXCERPT: Let R_n denote the resistance in the random resistance model, let $p_c=\frac12$, and let $y\in\mathbb R$. Addario-Berry et al.'s conjecture. If $p=p_c=\frac12$, there exists a co...

conjecture | open | MathDB 页面 | 原文线索

V2-055944 | MATHDB-371614

Derrida's conjecture for the critical resistance scaling constant

摘录 / EXCERPT: Let R_n be the resistance in the critical resistance model, let $p_c=\frac12$, and let c_{eff} be the constant in the critical scaling conjecture for $\log R_n$. Derri...

conjecture | open | MathDB 页面 | 原文线索

V2-055945 | MATHDB-371621

Polynomial runtime scaling conjecture for reluctant dynamics in the Sherrington-Kirkpatrick model

摘录 / EXCERPT: Let μ be a “sufficiently nice” probability measure on $\mathbb{R}$ with $\mathbb{E}_{X \sim \mu}X=0$ and $\mathbb{E}_{X \sim \mu}X^2=1$, let $\lambda\in[0,+\infty]$, and suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-055946 | MATHDB-371622

Universality conjecture for greedy dynamics runtime exponents

摘录 / EXCERPT: Let μ be a “sufficiently nice” probability measure with mean 0 and variance 1, and let $\beta(\mu,0)$ denote the scaling exponent of the runtime of 0-reluctant, or greedy...

conjecture | open | MathDB 页面 | 原文线索

V2-055947 | MATHDB-371623

Universality conjecture for positive-discrepancy entry distributions

摘录 / EXCERPT: Let μ be a probability measure with mean 0 and variance 1, and define its discrepancy by ... Let $\beta(\mu,\infty)$ denote the scaling exponent of the runtime of ∞-re...

conjecture | open | MathDB 页面 | 原文线索

V2-055948 | MATHDB-371624

Non-universality conjecture for zero-discrepancy entry distributions

摘录 / EXCERPT: Let μ_1,μ_2,μ_3 be probability measures with mean 0 and variance 1, and let $\Delta(\mu)$ be the discrepancy of a probability measure. Let $\beta(\mu,\infty)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-055949 | MATHDB-371652

Talagrand's convolution conjecture for the Boolean hypercube

摘录 / EXCERPT: Consider the Boolean hypercube $\left\{-1,1\right\}^n$ with its uniform probability measure μ. For $t\geq 0$, let ξ_t have independent coordinates satisfying ... and define t...

conjecture | open | MathDB 页面 | 原文线索

V2-055950 | MATHDB-371793

Random permanent zero-probability conjecture over finite fields

摘录 / EXCERPT: Random permanent zero-probability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-055951 | MATHDB-371813

Polynomial growth and diffusive random walk for non-negatively curved graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $dgeq 1$, and suppose that G has non-negative Ollivier--Ricci curvature and degrees bounded by d. Write $B(x,r)$ for the ball of radius r center...

conjecture | open | MathDB 页面 | 原文线索

V2-055952 | MATHDB-371861

Scaling-limit conjecture for the first detachment time

摘录 / EXCERPT: Let $\hat\tau^{(n)}$ denote the first detachment time in the n-detachment process. Scaling-limit conjecture. The random variables ... have a limiting distribution as $n\to\infty$....

conjecture | open | MathDB 页面 | 原文线索

V2-055953 | MATHDB-371862

Diffusive-scaling conjecture for the detachment-counting process

摘录 / EXCERPT: Diffusive-scaling conjecture. The laws of the processes $\mathcal{M}^{(m)}$, as $m\to\infty$, have a limit. The process has monotone non-decreasing paths, and the conjecture asks f...

conjecture | open | MathDB 页面 | 原文线索

V2-055954 | MATHDB-371880

Existence conjecture for the frog-model phase transition on transitive graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a transitive graph with superlinear growth. For $x uin V$, let $B_x(n)$ denote the ball of radius n centered at x, and write $g_x(n)=|B_x(n)|$. The graph has s...

conjecture | open | MathDB 页面 | 原文线索

V2-055955 | MATHDB-371883

Trivial-critical-parameter conjecture for graphs with $p_{\mathrm{c}}(G)=1$

摘录 / EXCERPT: Let G be a fixed graph, and let $p_{\mathrm{c}}(G)$ denote its Bernoulli percolation critical parameter. Assume $p_{\mathrm{c}}(G)=1$, and let $\lambda_{\mathrm{c}}(t)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-055956 | MATHDB-371958

Super-diffusive drift conjecture for bounded-degree Cayley graphs of symmetric groups

摘录 / EXCERPT: Let d_{T_n} be the word metric on the Cayley graph $\Gamma(\operatorname{Sym}_n,T_n)$. For the simple random walk $(W_t)_{t=1}^{\infty}$ on this graph started at $W_0=1$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-055957 | MATHDB-372030

Tropicalization conjecture for non-Archimedean lattices

摘录 / EXCERPT: Tropicalization conjecture. For every lattice $L\subset K^n$, there exists $\alpha_L\in\mathbb R_{>0}$ such that, for every $\alpha\geq\alpha_L$, $Q_{L,\alpha}$ is the survival fun...

conjecture | open | MathDB 页面 | 原文线索

V2-055958 | MATHDB-372034

Moment lower-bound conjecture for weighted sums of exponential random variables

摘录 / EXCERPT: Let $q>0$ be real, let $a_1,\dots,a_n$ be non-negative real numbers satisfying ... and let $X_1,\dots,X_n$ be independent identically distributed standard exponential random variable...

conjecture | open | MathDB 页面 | 原文线索

V2-055959 | MATHDB-372096

Conjecture on the emptiness of the critical slow-point set

摘录 / EXCERPT: Let θ_c be the critical parameter and let $ \mathfrak S(\theta_c)$ denote the corresponding random set of points of slow growth. Write $ \operatorname{dim_{_{\rm H}}}$ for H...

conjecture | open | MathDB 页面 | 原文线索

V2-055960 | MATHDB-372097

Conjectured hitting-probability threshold for slow-growth points

摘录 / EXCERPT: Let $K\subset\mathbb{R}$ be a nonrandom compact set, let $\mathfrak S(\theta)$ be the random set of points of slow growth at parameter θ, and let $\lambda(\theta)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-055961 | MATHDB-372104

The universal property of Baire measurable spaces

摘录 / EXCERPT: Let $\mathsf {BaireMeas}$ be the category of Baire measurable spaces, and let $\mathsf {BaireStoch}$ be its associated Markov category of kernels. A Markov functor is understood to...

conjecture | open | MathDB 页面 | 原文线索

V2-055962 | MATHDB-372197

Zartash-Robeva conjecture on Gaussian convolution and log-supermodularity

摘录 / EXCERPT: Zartash-Robeva conjecture. The convolution $f*g$ is log-supermodular.

conjecture | open | MathDB 页面 | 原文线索

V2-055963 | MATHDB-372407

Random matrix distribution conjecture for normalized matrix Kloosterman sums

摘录 / EXCERPT: Let $y \in \operatorname{GL}_n(\mathbb{F}_q)$ be a fixed regular elliptic element. Let $\widetilde{\operatorname{Kl}}_n(\bm{\alpha}, \psi, y)$ denote the normalized matrix Klooster...

conjecture | open | MathDB 页面 | 原文线索

V2-055964 | MATHDB-372453

Equality of the critical thresholds for three-dimensional Brownian loop soup

摘录 / EXCERPT: Let $\lpha_c$, $\lpha_u$, $\lpha_{\rm tr}$, and $\lpha_{\mathop{\rm b2b}}$ denote the thresholds defined in the paper for, respectively, percolation, uniqueness, truncated-model pe...

conjecture | open | MathDB 页面 | 原文线索

V2-055965 | MATHDB-372465

The stability-to-noise-sensitivity transition conjecture for geometric last-passage percolation

摘录 / EXCERPT: Consider planar last-passage percolation with geometric weights. Let $T_n(\omega)$ and $T_n(\tilde\omega^t)$ denote the last-passage times in the original and site-resampled enviro...

conjecture | open | MathDB 页面 | 原文线索

V2-055966 | MATHDB-372478

Prokhorenkova et al.'s modularity conjecture for preferential attachment graphs

摘录 / EXCERPT: Prokhorenkova et al.'s modularity conjecture. With high probability,

conjecture | open | MathDB 页面 | 原文线索

V2-055967 | MATHDB-372546

The replica conjecture for the non-Hermitian R-transform of a unitary sum

摘录 / EXCERPT: Let $\mathbf{A}$ and $\mathbf{B}$ be two large independent random matrices, and let $\mathbf{U}$ be a Haar-distributed unitary random matrix. For large $z\in\mathbb{C}$ or large...

conjecture | open | MathDB 页面 | 原文线索

V2-055968 | MATHDB-372547

Self-consistent R-transform equations for rotationally invariant random matrices

摘录 / EXCERPT: Let $\mathbf{B}$ be a large rotationally invariant random matrix, with $\mathfrak{g}_{1,\mathbf{B}}$ and $\mathfrak{g}_{2,\mathbf{B}}$ denoting the corresponding transform componen...

conjecture | open | MathDB 页面 | 原文线索

V2-055969 | MATHDB-372566

Benjamini-Schramm Conjecture 8 on critical site percolation in planar graphs

摘录 / EXCERPT: Let G be a planar graph, and consider i.i.d. Bernoulli site percolation at parameter $p=\tfrac{1}{2}$. Benjamini-Schramm Conjecture 8. If percolation occurs almost surely, meanin...

conjecture | open | MathDB 页面 | 原文线索

V2-055970 | MATHDB-372580

Conjecture on the perfect-recovery threshold for unregularized gradient flow

摘录 / EXCERPT: Perfect-recovery-threshold conjecture. The perfect-recovery threshold is

conjecture | open | MathDB 页面 | 原文线索

V2-055971 | MATHDB-372767

Asymptotic uniformity conjecture for the smallest maximum

摘录 / EXCERPT: Let $\mathbf{X}^{(1)},\mathbf{X}^{(2)},\ldots$ be independent and identically distributed d-dimensional observations with independent $\operatorname{Exponential}(1)$ coordinates,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055972 | MATHDB-372941

Uniform spectral gap conjecture for random covers of Anosov surfaces

摘录 / EXCERPT: Uniform spectral gap conjecture. For any $\varepsilon>0$, with probability tending to 1 as $n\to+\infty$, the cover $M_n\to M$ has no new resonance in the half-space

conjecture | open | MathDB 页面 | 原文线索

V2-055973 | MATHDB-372992

Ashkin-Teller magnetisation-interface percolation conjecture

摘录 / EXCERPT: Ashkin-Teller magnetisation-interface percolation conjecture. The law of the outermost interfaces of the Ashkin-Teller magnetisation field is the law of the outermost outer boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-055974 | MATHDB-372996

Conjecture on the limiting measure and skew-adjoint loop operators

摘录 / EXCERPT: Limiting-measure and operator conjecture. The family $\hat\nu_\kappa^\sigma$ converges weakly as $\sigma\to0$ to a measure $\hat\nu_\kappa$ on $\hat\mathcal A$. Moreover, the opera...

conjecture | open | MathDB 页面 | 原文线索

V2-055975 | MATHDB-373005

Sharp one-sided concentration conjecture for standardized log-concave distributions

摘录 / EXCERPT: Let L denote the set of all real random variables X with a log-concave probability density function such that $EX=0$ and $EX^2=1$. Let Y have the standard exponential distrib...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-055976 | MATHDB-373032

Clay's position-matrix eigenvalue conjectures for the single-shelf shuffle

摘录 / EXCERPT: Clay's position-matrix eigenvalue conjectures. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-055977 | MATHDB-373044

Asymptotics for higher moments of random polyhedral cone angles

摘录 / EXCERPT: Higher-moment asymptotics. For every fixed $k\in \{2,3,\ldots\}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-055978 | MATHDB-373045

Conditional non-degeneracy conjecture for spherical simplices

摘录 / EXCERPT: Conditional non-degeneracy conjecture. For fixed $k\geq 2$, the conditional distribution of $\alpha(\mathcal W_{d+k,d})$ converges weakly as $d\to\infty$ to a limit law not concent...

conjecture | open | MathDB 页面 | 原文线索

V2-055979 | MATHDB-373085

Finite free convolution moment comparison conjecture

摘录 / EXCERPT: Let p and q be real-rooted polynomials of degree d, and let $\f$ be in $C^{4}(\mathbb{R})$ with $f^{(4)}\geq 0$. Write μ_p and μ_q for the probability measures assign...

conjecture | open | MathDB 页面 | 原文线索

V2-055980 | MATHDB-373086

Monotonicity conjecture for c-convolution

摘录 / EXCERPT: Let μ and ν be compactly supported probability measures, let $c\in[0,\infty]$, and let $\mu\oplus_c\nu$ denote their c-convolution. Let $f\in C^{4}(\mathbb{R})$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-055981 | MATHDB-373114

Linear-latency contention resolution in the GlobalClock model

摘录 / EXCERPT: GlobalClock contention-resolution conjecture. In the finite setting, there exists a protocol with latency $O(n)$ with high probability against an adaptive adversary. In the unbound...

conjecture | open | MathDB 页面 | 原文线索

V2-055982 | MATHDB-373127

The maximum-running-time comparison conjecture for infection rules

摘录 / EXCERPT: Maximum-running-time comparison conjecture. For all infection rules H,

conjecture | open | MathDB 页面 | 原文线索

V2-055983 | MATHDB-373128

The tree conjecture for maximum running time

摘录 / EXCERPT: Tree conjecture. For all $t\in\mathbb N$,

conjecture | open | MathDB 页面 | 原文线索

V2-055984 | MATHDB-373129

The sublinear running-time dichotomy conjecture

摘录 / EXCERPT: Let H be a connected infection rule, and let $M_H(n)$ denote its maximum running time. Sublinear dichotomy conjecture. If ... then ... The paper describes the classification of conne...

conjecture | open | MathDB 页面 | 原文线索

V2-055985 | MATHDB-373130

The tree-width two running-time conjecture

摘录 / EXCERPT: Let H be a graph, let ${\mathrm{tw}}(H)$ denote its tree-width, and let $M_H(n)$ denote the maximum running time of the H-process. Tree-width two conjecture. Every graph H wi...

conjecture | open | MathDB 页面 | 原文线索

V2-055986 | MATHDB-373131

The $K_{3,3}$ and cube running-time exponent conjecture

摘录 / EXCERPT: Let $K_{3,3}$ be the complete bipartite graph with parts of size three, let Q_3 be the three-dimensional cube graph, and let $M_H(n)$ denote the maximum running time of the H-p...

conjecture | open | MathDB 页面 | 原文线索

V2-055987 | MATHDB-373132

The polynomial-exponent limit conjecture for running times

摘录 / EXCERPT: Let H be an infection rule and let $M_H(n)$ denote its maximum running time. Polynomial-exponent limit conjecture. The limit ... exists for every infection rule H. The paper does...

conjecture | open | MathDB 页面 | 原文线索

V2-055988 | MATHDB-373149

Steady-state convergence and moment conjecture for general-objective SGD

摘录 / EXCERPT: Let $f:cmathbb{R}ctocmathbb{R}$ be convex and of class $C^h(cmathbb{R})$ for an even integer $h>0$, with unique minimizer x^*, satisfying $\lim_{x\to x^*}f^{(k)}(x)=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-055989 | MATHDB-373157

GRD monotonicity for insertion depth and tree height

摘录 / EXCERPT: Let $T_n(f)$ be the preferential attachment tree on $[n]$ generated by the nondecreasing attachment function $f:{\mathbb{N}}_0\to\mathbb{R}_{>0}$. Write $D_n(f)$ for the graph dist...

conjecture | open | MathDB 页面 | 原文线索

V2-055990 | MATHDB-373194

Quadratic optimal-time conjecture for the quantum Ornstein-Uhlenbeck semigroup

摘录 / EXCERPT: For $2\leq p<8$, let t'_p denote the optimal time and let β be the parameter of the quantum Ornstein-Uhlenbeck semigroup. There are positive constants...

conjecture | open | MathDB 页面 | 原文线索

V2-055991 | MATHDB-373217

Bousseyroux-Potters R-transform conjecture for rotationally invariant matrix deformations

摘录 / EXCERPT: Let $\vb{A}$ be a large deterministic matrix and $\vb{B}$ a rotationally invariant random matrix. Define the matrix-valued transforms $\mathcal{G}_{\vb{A}+\vb{B}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-055992 | MATHDB-373221

Particle-conjugation symmetry for multispecies inhomogeneous t-PushTASEP probabilities

摘录 / EXCERPT: Particle-conjugation symmetry conjecture. The following equality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-055993 | MATHDB-373228

The least-atom principle for optimal hypercontractivity of finite probability spaces

摘录 / EXCERPT: Least-atom principle. For all $1<p<q<\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-055994 | MATHDB-373269

High-density scaling identity for the half-space log-Gamma polymer when $\tilde{\beta}<\tilde{t}$

摘录 / EXCERPT: Let $\tilde{\beta}$, $\tilde{t}$, α, and A be the parameters from the cited high-density asymptotic theorem, and let $\widetilde{Q}_{\tilde{\beta}<\tilde{t}}^{High}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-055995 | MATHDB-373300

Diffusive-limit conjecture for stochastic transport by heavy-tailed jump processes

摘录 / EXCERPT: Diffusive-limit conjecture. As η goes to zero, the processes $({\mathbf{X}}^\eta_t({\mathbf{x}}))_{t\ge 0}$ converge in law to the 2-dimensional process...

conjecture | open | MathDB 页面 | 原文线索

V2-055996 | MATHDB-373657

Percolation transition for height-three vertices

摘录 / EXCERPT: Height-three percolation conjecture. There exists $p_c\in(0,1)$ such that, for all $p<p_c$,

conjecture | open | MathDB 页面 | 原文线索

V2-055997 | MATHDB-373658

Shape theorem for single-source stochastic sandpiles

摘录 / EXCERPT: Consider stochastic sandpiles with uniform topplings on $\mathbb{Z}$. Let δ_0 denote the unit mass at the origin, and let D_n be the set of vertices toppled during stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-055998 | MATHDB-373710

Conjecture on improving the weak-noise threshold for Hölder regularity

摘录 / EXCERPT: Fix $\theta\in\mathbb{R}$, and let $ u_\theta$ be the invariant measure associated with the solution $V^{(\theta)}$ of the SPDE started from the constant initial profile θ....

open_problem | open | MathDB 页面 | 原文线索

V2-055999 | MATHDB-373759

The MDS probability convergence conjecture

摘录 / EXCERPT: MDS probability convergence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-056000 | MATHDB-373903

The balanced-event probability conjecture for the hard-core model on the hypercube

摘录 / EXCERPT: Balanced-event probability conjecture. There exist $C,d_{0}>0$ such that for all dimensions $d\ge d_{0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-056001 | MATHDB-373909

Large-temperature Poisson limit for the limiting point process

摘录 / EXCERPT: Poisson-limit conjecture. When $a\to\infty$, the rescaled point process $\mathcal{M}_a$ converges towards a Poisson point process.

conjecture | open | MathDB 页面 | 原文线索

V2-056002 | MATHDB-373910

Matching conjecture for the stochastic Bessel limiting point process

摘录 / EXCERPT: Matching conjecture. For all $a\geq 0$, the joint distributions of $(\hat{\mu}_a^{\infty}(i),\;i\geq 1)$ and $(\mu_a^{\infty}(i),\;i\geq 1)$ are equal. As a corollary, using the ex...

conjecture | open | MathDB 页面 | 原文线索

V2-056003 | MATHDB-373924

Telek's conjecture on Markovian representations of rational arrival processes

摘录 / EXCERPT: Let a rational arrival process (RAP) have joint densities on $(0,\infty)^k$ for every $k\ge 1$, and let $\bm{G}_0$ be the matrix in a RAP representation governing the evolution bet...

conjecture | open | MathDB 页面 | 原文线索

V2-056004 | MATHDB-374149

Geometric spectrum conjecture for the finite free coupling matrix

摘录 / EXCERPT: Let $n\geq 3$, let $h\in\mathbb{R}^n$ be the normalized Hermite root vector, and define ... Let $W=\mathbf{1}^\perp=\{u\in\mathbb{R}^n:u^\top\mathbf{1}=0\}$ be the mean-zero subspace...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056005 | MATHDB-374150

Finite free p-Stam phase transition conjecture

摘录 / EXCERPT: For each $n\geq 3$, consider the p-Stam inequality for monic real-rooted pairs of degree n, with critical exponent denoted by p^*. p-Stam phase transition conjecture. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-056006 | MATHDB-374206

Critical Pool-model linear-growth conjecture

摘录 / EXCERPT: Let $\mathcal{E}_t$ denote the pool radius or size at time t in the Pool model, and let λ be the particle density. Critical Pool-model linear-growth conjecture. At the cr...

conjecture | open | MathDB 页面 | 原文线索

V2-056007 | MATHDB-374207

Non-explosion conjecture for the critical Brownian Pool model

摘录 / EXCERPT: Consider the Brownian Pool model obtained by replacing the continuous-time and continuous-space random walks in the Pool model by Brownian motion, and let $\lambda=1$ be the critic...

conjecture | open | MathDB 页面 | 原文线索

V2-056008 | MATHDB-374208

Linear-growth conjecture for the critical annihilating Pool model

摘录 / EXCERPT: Consider the annihilating version of the Pool model, in which excess particles at an attached location are annihilated, and take the particle density at criticality, $\lambda=1$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-056009 | MATHDB-374209

Engulfing MDLA threshold and limiting-shape conjecture

摘录 / EXCERPT: Consider an engulfing version of multi-particle diffusion-limited aggregation (MDLA) with particle density λ and internal random-walk rate γ, where $\gamma=\infty$ m...

conjecture | open | MathDB 页面 | 原文线索

V2-056010 | MATHDB-374263

The uniform approximation conjecture for the Boţ-Nguyen coefficients

摘录 / EXCERPT: Let $calpha>2$ and let $cbeta$ be the parameter defining the coefficient arrays $(c_{n,k})$ and $(b_{n,k})$. Uniform approximation conjecture. If ... then ... as $n\to\infty$. Such an...

conjecture | open | MathDB 页面 | 原文线索

V2-056011 | MATHDB-374264

The unimodality conjecture for the Boţ-Nguyen coefficients

摘录 / EXCERPT: Let $calpha>2$, and for each n consider the coefficient sequence $(c_{n,k})_{k\in\{1,2,\ldots,n\}}$. Unimodality conjecture. For every choice of $calpha>2$, the sequence ... is uni...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056012 | MATHDB-374266

Asymptotic variance-duality conjecture for beta equals 1 and 4

摘录 / EXCERPT: Let ${\mathcal N}(L)$ be the counting function, let λ_L be the L-th ordered eigenvalue in the infinite ordered spectrum, and let ${\rm var}_\beta$ denote variance in the...

conjecture | open | MathDB 页面 | 原文线索

V2-056013 | MATHDB-374267

Conjecture on the beta-dependent variance of ordered eigenvalues

摘录 / EXCERPT: Eigenvalue-variance conjecture. As $L\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-056014 | MATHDB-374268

Conjectured beta-dependent sum rule for level-spacing autocovariances

摘录 / EXCERPT: Level-spacing sum-rule conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056015 | MATHDB-374272

Low-degree heuristic for computational detection

摘录 / EXCERPT: Let $\mathbb P$ and $\mathbb Q$ be probability measures for a high-dimensional hypothesis-testing problem based on $\mathsf Y\in\mathbb R^N$, with $N=N_n$ and parameters scaling wi...

conjecture | open | MathDB 页面 | 原文线索

V2-056016 | MATHDB-374275

Strong pruning conjecture for exact finite-horizon quantile Kelly computation

摘录 / EXCERPT: Let p and q be the event-probability and pricing data, let n be the horizon, and let α be the quantile parameter. Suppose the exact finite-horizon problem is represent...

conjecture | open | MathDB 页面 | 原文线索

V2-056017 | MATHDB-374441

Hermitian Wigner universality for level crossings

摘录 / EXCERPT: Fix a real-valued probability distribution μ with mean zero and finite moments, and a complex-valued probability distribution ν with mean zero, unit variance, finite moment...

conjecture | open | MathDB 页面 | 原文线索

V2-056018 | MATHDB-374530

Cosine covariance stochastic domination conjecture for Gaussian minima

摘录 / EXCERPT: Cosine covariance conjecture. For every $n\geq 1$, every correlation matrix Σ, and every $t\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-056019 | MATHDB-374563

Symbolic pattern emergence in LAWS expert classes

摘录 / EXCERPT: Let $\mathcal{M}$ be a sufficiently capable base model trained on a corpus containing code, mathematics, and structured data, and let n^* be a PLT trie node with probability...

conjecture | open | MathDB 页面 | 原文线索

V2-056020 | MATHDB-374633

Median-preserving conjecture for the stopped-random-walk tree dynamics

摘录 / EXCERPT: Tree-asymmetry conjecture. For every $v\in\{L,R\}^{\star}$,

conjecture | open | MathDB 页面 | 原文线索

V2-056021 | MATHDB-374749

Monotonicity conjecture for norms of uniformly supported random walks

摘录 / EXCERPT: Let $d\ge 1$. Suppose $D(x)$ has finite support $\mathcal{U}$. If $D(x)$ is uniformly distributed on $\mathcal{U}$, and if every $x\in \mathcal{U}$ is on the boundary of...

conjecture | open | MathDB 页面 | 原文线索

V2-056022 | MATHDB-374915

Milman's contracting Gaussian transport conjecture under the CD condition

摘录 / EXCERPT: Let d be a positive integer, let $(\mathbb{R}^d,g,\mu)$ satisfy $\operatorname{CD}(\rho,\infty)$ for some $\rho>0$, and define the centered Gaussian probability measure ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-056023 | MATHDB-375005

Hurwitz-stable flip--repair mixing conjecture

摘录 / EXCERPT: Let μ be a Hurwitz-stable distribution supported on one parity class. Consider the two-step parity flip--repair graph and its walk on each connected component: the walk flips a...

conjecture | open | MathDB 页面 | 原文线索

V2-056024 | MATHDB-375035

Single-round KL contraction conjecture for the binary hypercube Gaussian-location channel

摘录 / EXCERPT: Let Y be the output of the binary hypercube Gaussian-location channel, let V denote its binary-hypercube input, and let P_Y and $P_{V\mid Y}$ be the corresponding distributio...

conjecture | open | MathDB 页面 | 原文线索

V2-056025 | MATHDB-375069

The random matrix strong convergence conjecture for tensor actions

摘录 / EXCERPT: Let $r\in\mathbb{N}$, let $\alpha\in\mathbb{C}[\mathbb{F}_r]\otimes\mathbb{C}[\mathbb{F}_r]$, and let u be an r-tuple of free Haar unitaries. For almost every sequence of Haar-...

conjecture | open | MathDB 页面 | 原文线索

V2-056026 | MATHDB-375174

Choda-Dykema conjecture on purely infinite reduced free products

摘录 / EXCERPT: Let (A,φ) and (B,ψ) be nontrivial C^*-probability spaces inducing faithful GNS representations. Form their reduced free product $(A,\varphi)*(B,\psi)$. Choda-Dykema...

conjecture | open | MathDB 页面 | 原文线索

V2-056027 | MATHDB-375225

Global PCL-indexability of discounted single-project problems

摘录 / EXCERPT: Let p_{01}, ρ, κ, and β be parameters satisfying ... A discounted single-project problem is PCL-indexable when the conditions $\textup{(PCLI1)--(PCLI3)}$ hold on...

conjecture | open | MathDB 页面 | 原文线索

V2-056028 | MATHDB-375271

Busy-probability comparison between the true system and mean-field approximations

摘录 / EXCERPT: Mean-field busy-probability comparison. For $<$,

conjecture | open | MathDB 页面 | 原文线索

V2-056029 | MATHDB-375317

Spiked matrix model eigenvector fluctuation conjecture

摘录 / EXCERPT: Let $\theta>1$, let $d_0\geq 1$ be fixed, and use the spiked matrix model notation of Theorem, assuming and. Let $\bm R=\bm R^{(d)}\in\mathbb{R}^{d_0\times d}$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-056030 | MATHDB-375349

Variance-sensitive Matrix Spencer conjecture

摘录 / EXCERPT: Let $A_1,\ldots,A_n\in\mathbb R^{n\times n}$ be symmetric matrices satisfying $\|A_i\|_{\mathrm{op}}\le 1$. A coloring is a vector $x\in\{\pm1\}^n$. Variance-sensitive Matrix Spenc...

conjecture | open | MathDB 页面 | 原文线索

V2-056031 | MATHDB-375391

Vinzant's refined conjecture on random complex phase retrieval

摘录 / EXCERPT: Let $N=4M-5$ and let $A\in\mathbb C^{N\times M}$ have entries drawn independently from the standard complex Gaussian distribution, where a standard complex Gaussian has the law of...

conjecture | open | MathDB 页面 | 原文线索

V2-056032 | MATHDB-375417

Product conjecture for Patnaik-Pearson dimension of random matrix products

摘录 / EXCERPT: Let $c{lambda}$ and $c{phi}$ be regularly varying random variables, with tail exponents $calpha$ and $cbeta$, respectively. Let A and B be random matrices in general position,...

conjecture | open | MathDB 页面 | 原文线索

V2-056033 | MATHDB-375418

Diagonal stability conjecture for Khintchine-type inequalities

摘录 / EXCERPT: Diagonal stability conjecture. There exist positive constants $C_p>0$ such that, if (r_k) is log-concave, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056034 | MATHDB-375511

The mode-one conjecture for the largest exponential m-spacing

摘录 / EXCERPT: Let $m\geq 2$ be a fixed integer. For independent exponential random variables Y_k with rates k, define the limiting maximizing index ... where ... and write...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056035 | MATHDB-375607

The simplex-only representation conjecture for symmetric W-way DPI-additive divergences

摘录 / EXCERPT: Let W be a positive integer and let D be a symmetric, W-way DPI-additive divergence on bounded W-tuples of probability distributions. Write ... and let C_{α}...

conjecture | open | MathDB 页面 | 原文线索

V2-056036 | MATHDB-375736

Aspelmeier's critical variance conjecture for the SK free energy

摘录 / EXCERPT: Aspelmeier's conjecture. The variance satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-056037 | MATHDB-375771

Non-invariance of circular turning-angle distributions under sub-sampling

摘录 / EXCERPT: Let $f_\circ(\theta;\lambda_1,\ldots,\lambda_n)$ be a circular distribution, let $f^{\{r\}}_\circ$ denote the turning-angle distribution obtained after sub-sampling every r steps...

conjecture | open | MathDB 页面 | 原文线索

V2-056038 | MATHDB-375839

Diagonal monitored-transport limit

摘录 / EXCERPT: Diagonal monitored-transport limit. If $L_N/N\to\tau$, then

conjecture | open | MathDB 页面 | 原文线索

V2-056039 | MATHDB-375904

Universal Type I behavior for asymmetric initial densities

摘录 / EXCERPT: Universal Type I conjecture. The model $I(G,p)$ is of Type $\mathcal{I}$ for every initial density $p\in(0,1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-056040 | MATHDB-375946

Fulman-Kaplan-Singhal-Warnaar conjecture on ranks of Sylow sandpile groups

摘录 / EXCERPT: Let $G_{\alpha}(n,u)=G(n,\lceil\alpha n\rceil,u)$ be the Erdős-Rényi random bipartite graph with edge probability u, let $S_{G_{\alpha}(n,u)}$ be its sandpile group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056041 | MATHDB-375984

Non-singularity conjecture for the partial-sums coefficient matrix

摘录 / EXCERPT: Let S_N be the sum of the discrete random variables considered in the system, and let $\mathbb{E}S_N$ denote its expectation. The coefficient matrix of is the matrix determining...

conjecture | open | MathDB 页面 | 原文线索

V2-056042 | MATHDB-375986

Champagnat-Bovier conjecture on fast evolutionary branching

摘录 / EXCERPT: Champagnat-Bovier's fast-branching conjecture. If $\partial_{22}f(x^*,x^*)>\partial_{11}f(x^*,x^*)>0$, then there is fast evolutionary branching: for some $\varepsilon_0>0$, every...

conjecture | open | MathDB 页面 | 原文线索

V2-056043 | MATHDB-376005

Critical behavior conjecture at the RLD threshold

摘录 / EXCERPT: Consider the random location deaths (RLD) process: species are born with probability p, die with probability $q=1-p$, and, on a death event, the least fit species is removed with...

conjecture | open | MathDB 页面 | 原文线索

V2-056044 | MATHDB-376006

RLD high-fitness population-density conjecture

摘录 / EXCERPT: Consider the random location deaths (RLD) process with birth probability p, death probability $q=1-p$, and parameter r. Let $L_n(a)$ be the number of species alive at time n...

conjecture | open | MathDB 页面 | 原文线索

V2-056045 | MATHDB-376040

Nearest-neighbor critical-radius conjecture for the square-lattice disc model

摘录 / EXCERPT: Let $R_c^{\texttt{NN}}(p)$ denote the critical radius for the nearest-neighbor model with parameter p, and let $R_c^{\texttt{dec}}$ denote the critical radius of the decorrelated...

conjecture | open | MathDB 页面 | 原文线索

V2-056046 | MATHDB-376145

Ellipsoidal separation conjecture for totally positive matrix families

摘录 / EXCERPT: Ellipsoidal separation conjecture. For every fixed $\alpha\in(0,1)$, there exists a dimension $n_0(\alpha)$ such that, for all $n\ge n_0(\alpha)$, the three ellipsoids...

conjecture | open | MathDB 页面 | 原文线索

V2-056047 | MATHDB-376151

The minimax regret conjecture for stochastic bandit convex optimization

摘录 / EXCERPT: Let d be the dimension, let $\calA$ denote the action space, and let $\mathfrak R_T^\star(d;\calA)$ be the minimax expected regret after T rounds in stochastic bandit convex op...

conjecture | open | MathDB 页面 | 原文线索

V2-056048 | MATHDB-376176

Quantitative spectral rate for the normal-scores covariance

摘录 / EXCERPT: Let $\widehat{\bm{S}}_n$ be the normal-scores covariance, and let m_n and m_F denote its empirical Stieltjes transform and the limiting generalized Marčenko--Pastur Stieltjes t...

conjecture | open | MathDB 页面 | 原文线索

V2-056049 | MATHDB-376178

Spectral equivalence of normal-scores and Kendall rank inputs

摘录 / EXCERPT: Let $\widehat{\bm{S}}_n$ be the normal-scores covariance and let $\widehat{\bm{\Sigma}}^{\tau}$ be the sin-transformed Kendall matrix. Assume $p/n\to\gamma$. Spectral equivalence o...

conjecture | open | MathDB 页面 | 原文线索

V2-056050 | MATHDB-376226

Bubeck-Ding-Eldan-Rácz high-dimensional indistinguishability conjecture

摘录 / EXCERPT: Let $G(n,d,p)$ be the spherical random geometric graph and let $G(n,p)$ be the Erdős-Rényi random graph. For probability distributions on a finite set, define total variation dista...

conjecture | open | MathDB 页面 | 原文线索

V2-056051 | MATHDB-376227

Boundedness of the proxy approximation between TD error and value gradients

摘录 / EXCERPT: Boundedness of the proxy approximation. There exists a constant $C>0$ such that, for a sufficiently fine state-space discretization,

conjecture | open | MathDB 页面 | 原文线索

V2-056052 | MATHDB-376300

Universality of the oriented minimum spanning tree limit for positive continuous weights

摘录 / EXCERPT: Let $(W_{\vec e})_{\vec e\in\vec E}$ be i.i.d. weights with continuous distributions supported in $(0,\infty)$, rather than i.i.d. Exponential(1) weights. Let the main theorem as...

conjecture | open | MathDB 页面 | 原文线索

V2-056053 | MATHDB-376370

Hutchcroft-Münch conjecture on polynomial growth and diffusive displacement

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with non-negative Ollivier-Ricci curvature and degrees bounded by $d<\infty$. Write $B(x,r)$ for the ball of radius r centred at x, let...

conjecture | open | MathDB 页面 | 原文线索

V2-056054 | MATHDB-376557

Prähofer-Spohn conjecture on TASEP height fluctuation distributions

摘录 / EXCERPT: Let ρ_- and ρ_+ be the left and right Bernoulli densities for the TASEP, let $\bar h(y)$ be the hydrodynamic limiting height profile, and let $h_t(j)$ be the height at ti...

conjecture | open | MathDB 页面 | 原文线索

V2-056055 | MATHDB-376573

Korepin-Izergin-Baxter ferromagnetic-string probability formula

摘录 / EXCERPT: For the even-site ground state $|\Psi^{(2n)}\rangle$, define ... and set $\alpha_i=(1+\sigma_i^z)/2$. Korepin-Izergin-Baxter ferromagnetic-string formula. The probabilities of format...

conjecture | open | MathDB 页面 | 原文线索

V2-056056 | MATHDB-376618

Constant-iteration and constraint conjecture for adaptive LP decoding

摘录 / EXCERPT: Adaptive LP decoding conjecture. As n increases, the algorithm converges with probability arbitrarily close to 1 in at most $b1$ iterations and with at most $b2$ final parity-c...

conjecture | open | MathDB 页面 | 原文线索

V2-056057 | MATHDB-376622

Chi-squared distribution conjecture for the squared norm of effective channels

摘录 / EXCERPT: Let ${\bf h}^{\textrm{eff}}_i$ denote the effective channel vector for user i in a system with M transmit dimensions and an effective-channel subspace of dimension N. Chi-squ...

conjecture | open | MathDB 页面 | 原文线索

V2-056058 | MATHDB-376626

The Lovász-Vempala thin-shell variance conjecture

摘录 / EXCERPT: Let K be an isotropic convex body, and let X be a random point from K. Lovász-Vempala variance conjecture. ... Under either definition of isotropy used in the source, this conj...

conjecture | open | MathDB 页面 | 原文线索

V2-056059 | MATHDB-376692

The two-story space-time hypothesis for quantum effects

摘录 / EXCERPT: Two-story space-time hypothesis. The weaker and more stochastic communication between the various lumps, beneath the macroscopic or quasiclassical surface structure, is responsible...

conjecture | open | MathDB 页面 | 原文线索

V2-056060 | MATHDB-376769

Faceted-droplet hypothesis for the low-temperature three-dimensional Ising model

摘录 / EXCERPT: Faceted-droplet hypothesis. As $l\to\infty$, with probability tending to 1, there exist six distinct two-dimensional planes $L_i=L_i(\sigma)\subset\mathbb{T}^3$, $i=1,\ldots,6$,...

conjecture | open | MathDB 页面 | 原文线索

V2-056061 | MATHDB-376805

Refined flat-facet conjecture for three-dimensional random crystals

摘录 / EXCERPT: Refined flat-facet conjecture. The event that there are such planes for which $L_i\cap\Gamma(\sigma)$ are flat facets in the following sense has probability tending to 1 as...

conjecture | open | MathDB 页面 | 原文线索

V2-056062 | MATHDB-376806

A recursive upper-bound conjecture for graph partition weights

摘录 / EXCERPT: Recursive upper-bound conjecture. There exists a geometric constant $C<\infty$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-056063 | MATHDB-376823

Polynomial degree-distribution conjecture for random triangulations

摘录 / EXCERPT: Let n denote the size of a triangulation, and let the average number of its nodes of degree d be divided by n, where d is an integer larger than 2. Polynomial degree-dist...

conjecture | open | MathDB 页面 | 原文线索

V2-056064 | MATHDB-376855

Stochastic Airy operator soft-edge eigenvalue conjecture

摘录 / EXCERPT: Let k be a positive integer, and let β be the parameter of the stochastic Airy operator and of the soft edge distribution. Stochastic Airy soft-edge conjecture. The kth l...

conjecture | open | MathDB 页面 | 原文线索

V2-056065 | MATHDB-376856

Stochastic Bessel operator hard-edge singular-value conjecture

摘录 / EXCERPT: Let $\mathcal{J}_{a}^{\beta}$ denote the stochastic Bessel operator in its original form, and let $\tilde{\mathcal{J}}_{a}^{\beta}$ denote its Liouville normal form. Consider type...

conjecture | open | MathDB 页面 | 原文线索

V2-056066 | MATHDB-376862

GOE Dyson Brownian motion conjecture for the Airy one process

摘录 / EXCERPT: Let $\lambda_{\max}^{(N)}(t)$ be the largest eigenvalue of an $N\times N$ matrix undergoing Dyson's Brownian motion with parameter $\beta=1$. Let ${\cal A}_{\rm 1}$ denote the Airy...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056067 | MATHDB-376874

Equality of relative-entropy, Riemannian and geodesic contraction coefficients

摘录 / EXCERPT: Equality conjecture. For each fixed $g\in{\cal G}$,

conjecture | open | MathDB 页面 | 原文线索

V2-056068 | MATHDB-376929

Scaling-limit conjecture for longest increasing subsequences

摘录 / EXCERPT: Scaling-limit conjecture. As $n\to\infty$, the following limits hold. If $0\leq\alpha_+<\tilde\alpha_+$ and $0\leq\alpha_-<\tilde\alpha_-$ are fixed, then

conjecture | open | MathDB 页面 | 原文线索

V2-056069 | MATHDB-377052

Random-graph conjecture on bounded expected nontrivial invariant factors

摘录 / EXCERPT: Let $\EuScript{G}(n,p)$ be a random simple undirected graph with n vertices and edge-probability $0\leq p\leq 1$. For $c>1$, assume $c\log(n)/n<p(n)<1-o(\log(n)/n)$. Bounded-expe...

conjecture | open | MathDB 页面 | 原文线索

V2-056070 | MATHDB-377053

Random-graph conjecture on square-free spanning-tree counts

摘录 / EXCERPT: Let $\EuScript{G}(n,p)$ be a random simple undirected graph with n vertices and edge-probability $0\leq p\leq 1$. For $c>1$, assume $c\log(n)/n<p(n)<1-o(\log(n)/n)$. Square-free...

conjecture | open | MathDB 页面 | 原文线索

V2-056071 | MATHDB-377098

The critical-exponent conjecture for long-range percolation cluster diameter

摘录 / EXCERPT: Critical-exponent conjecture. (A) If $s=2$, then the diameter's order of magnitude is N^δ, where δ is a function of β. (B) If $1<s<2$, then the diameter is...

conjecture | open | MathDB 页面 | 原文线索

V2-056072 | MATHDB-377143

Asymptotical freeness conjecture for the polar part and modulus of a biinvariant ensemble

摘录 / EXCERPT: Asymptotical freeness conjecture. The sequences

conjecture | open | MathDB 页面 | 原文线索

V2-056073 | MATHDB-377179

Extension of the uniqueness theorem beyond ${\mathcal A}$

摘录 / EXCERPT: Let ${\mathcal A}$ be the class of probability measures appearing in Theorem, let μ be an invariant probability measure, and let the theorem's part b) assert uniqueness under t...

conjecture | open | MathDB 页面 | 原文线索

V2-056074 | MATHDB-377195

Diameter criterion for bounded finite-graph percolation thresholds

摘录 / EXCERPT: Let G be a finite transitive graph, let $|G|$ denote its number of vertices, and let p^G be its percolation threshold. Diameter criterion conjecture. There is a constant $C<1$...

conjecture | open | MathDB 页面 | 原文线索

V2-056075 | MATHDB-377196

Exponential-intersection-tail conjecture for finite transitive graphs

摘录 / EXCERPT: Let G be a finite transitive graph. Say that G has the $EIT(c)$ property, for $0<c<1$, if for every pair of vertices $u,v\in G$ there is a set of paths from v to u equipped...

conjecture | open | MathDB 页面 | 原文线索

V2-056076 | MATHDB-377198

Giant-component conjecture for growing-girth cubic graphs

摘录 / EXCERPT: Let G_n be a sequence of finite 3-regular graphs with growing girth, converging locally to the 3-regular tree T_3, and assume that the thresholds p^{G_n} remain bounded away...

conjecture | open | MathDB 页面 | 原文线索

V2-056077 | MATHDB-377199

SLE8 trace continuity conjecture

摘录 / EXCERPT: Let γ be the trace of chordal $\textrm{SLE}_\kappa$ in the upper half-plane. SLE8 trace continuity conjecture. For $\kappa=8$, the trace γ is almost surely a continuo...

conjecture | open | MathDB 页面 | 原文线索

V2-056078 | MATHDB-377200

Uniform grid-path convergence to radial and chordal SLE8

摘录 / EXCERPT: Let $\mathbb{U}$ be the unit disk and consider a square grid of mesh ϵ. Let the uniform measure be taken on simple grid paths from 0 to $\partial\mathbb{U}$, and separat...

conjecture | open | MathDB 页面 | 原文线索

V2-056079 | MATHDB-377201

Random-cluster interface convergence to SLE

摘录 / EXCERPT: Let G_n be the partially wired square grid described in the source, let ω have the critical random-cluster measure with $q\in(0,4)$, let $\omega^\dagger$ be its dual confi...

conjecture | open | MathDB 页面 | 原文线索

V2-056080 | MATHDB-377202

SLE trace reversibility conjecture

摘录 / EXCERPT: Let D be a simply connected domain, let $a,b\in\partial D$ be distinct boundary points, and consider the chordal $\textrm{SLE}_\kappa$ trace from a to b, modulo reparameteriz...

conjecture | open | MathDB 页面 | 原文线索

V2-056081 | MATHDB-377281

Andrews-Curtis random-walk conjecture for black box normal subgroups

摘录 / EXCERPT: Andrews-Curtis black-box conjecture. A random walk on the Andrews-Curtis graph $\Delta_k(G,N)$ provides a good' black box for N.

conjecture | open | MathDB 页面 | 原文线索

V2-056082 | MATHDB-377284

AC-replacement algorithm conjecture for black box groups

摘录 / EXCERPT: Let G be a black box group and let $N\lhd G$. The Andrews--Curtis graph $\Delta_k(G,N)$ supports a nearest-neighbour random walk, and the AC-replacement algorithm $ACR_k(G,N)$ ru...

conjecture | open | MathDB 页面 | 原文线索

V2-056083 | MATHDB-377468

The moment formula conjecture for powers of Voiculescu's circular operator

摘录 / EXCERPT: Let T be Voiculescu's circular operator, let τ denote the trace, and let $k,n\in{\mathbf N}$. Moment formula conjecture. ... The formula is proved for $k=1$, and it is also el...

conjecture | open | MathDB 页面 | 原文线索

V2-056084 | MATHDB-377503

Uniqueness of the maximum of the Airy process minus a parabola

摘录 / EXCERPT: Uniqueness conjecture. For each $T>0$, $H(t)$ has a unique point of maximum in $[-T,T]$ almost surely.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056085 | MATHDB-377550

Strong stochastic stability of multimodal Collet-Eckmann maps

摘录 / EXCERPT: A multimodal Collet-Eckmann map is a multimodal interval map satisfying the Collet-Eckmann condition, and strong stochastic stability means stability of its statistical behaviour u...

conjecture | open | MathDB 页面 | 原文线索

V2-056086 | MATHDB-377563

Conjecture on the distributional convergence of rescaled UIPT ball volumes

摘录 / EXCERPT: Let B_r be the ball of radius r around the root in the type II uniform infinite planar triangulation, and let $\overline{B}_r$ be its hull, obtained by adjoining to B_r all f...

conjecture | open | MathDB 页面 | 原文线索

V2-056087 | MATHDB-377611

The convolution-ratio conjecture for probability density functions

摘录 / EXCERPT: Let f be a probability density function supported on $[-\frac14,\frac14]]$. The convolution $f\ast f$ has supremum norm $\|f\ast f\|_\infty$ and L^2 norm $\|f\ast f\|_2$. Convo...

conjecture | open | MathDB 页面 | 原文线索

V2-056088 | MATHDB-377675

Barma-Dhar critical-bias conjecture for biased random walk on percolation clusters

摘录 / EXCERPT: Let $p\in(p_c,1)$, where $p_c=\frac{1}{2}$ is the critical probability for bond percolation on $\mathbb{Z}^2$. For the biased random walk on the infinite percolation cluster, let...

conjecture | open | MathDB 页面 | 原文线索

V2-056089 | MATHDB-377878

Recovery of finitely many biases for mean-zero non-symmetric random walks

摘录 / EXCERPT: Recovery conjecture. The bias function η can be recovered up to a shift only.

conjecture | open | MathDB 页面 | 原文线索

V2-056090 | MATHDB-377934

Cardy's annular crossing probability formula

摘录 / EXCERPT: Let $a=\log q$ for $q\in(0,1)$, and let $\mathbb{P}(N)(q)$ denote the probability of the annular crossing event N described above. Set ... Here η denotes Dedekind's eta functi...

conjecture | open | MathDB 页面 | 原文线索

V2-056091 | MATHDB-378024

The critical-dimension conjecture for alpha-weighted Laplacian loop-erased random walk

摘录 / EXCERPT: Critical-dimension conjecture. The critical dimension is

conjecture | open | MathDB 页面 | 原文线索

V2-056092 | MATHDB-378025

The logarithmic-correction conjecture for loop-erased random walk in dimension four

摘录 / EXCERPT: Four-dimensional logarithmic-correction conjecture. The accurate upper-bound scale for the length of L is

conjecture | open | MathDB 页面 | 原文线索

V2-056093 | MATHDB-378192

Subcritical largest-component asymptotics near the hypercube scaling window

摘录 / EXCERPT: Let $p_c(n)$ be the critical probability for the random subgraph of the n-cube, let $V=2^n$, and write ... Assume $\epsilon\rightarrow0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056094 | MATHDB-378265

Uniform sign-change conjecture for lattice random-walk averages

摘录 / EXCERPT: Let d be fixed. For a finitely supported function $\xi:{\mathbb Z}^d\to{\mathbb R}$, let $W_{2n}({\bf v},\xi)$ denote the associated lattice random-walk quantity at ${\bf v}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056095 | MATHDB-378291

Improvement of the $b8$-log-convexity threshold for MaxwellBoltzmann convolutions

摘录 / EXCERPT: Let (μ_n) be a θ-log-convex Stieltjes moment sequence, and let $(\mu_n^{\circ t})$ denote its Maxwell-Boltzmann convolution semigroup for $t\in\mathbf{R}_+$. The precedi...

conjecture | open | MathDB 页面 | 原文线索

V2-056096 | MATHDB-378317

The graded-graph path-count conjecture

摘录 / EXCERPT: Graded-graph path-count conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-056097 | MATHDB-378320

Continuity of the contact-process survival probability on trees

摘录 / EXCERPT: Let T be an infinite tree, and consider the contact process on T started from a single infection. Write λ for the infection rate and let the survival probability be the...

conjecture | open | MathDB 页面 | 原文线索

V2-056098 | MATHDB-378325

Equivalence of CNA+, JNRD+ and h-NLC+

摘录 / EXCERPT: The properties CNA+, JNRD+ and h-NLC+ are stronger versions of CNA, JNRD and hereditary NLC, respectively, requiring the corresponding property after every imposition of an externa...

conjecture | open | MathDB 页面 | 原文线索

V2-056099 | MATHDB-378326

The ULC conjecture for negatively associated measures

摘录 / EXCERPT: For a measure on a Boolean lattice, its rank sequence is the sequence of probabilities of the possible values of the total number of occupied coordinates. It is ULC (ultra-log-conc...

conjecture | open | MathDB 页面 | 原文线索

V2-056100 | MATHDB-378327

ULC rank sequences in the random-cluster, competing-urns and exclusion models

摘录 / EXCERPT: In the random-cluster (RC) model and the competing urns model, let X_e denote the indicator associated with an edge or site, and let S be any subset of the relevant index set....

conjecture | open | MathDB 页面 | 原文线索

V2-056101 | MATHDB-378328

ULC+ implies negative association

摘录 / EXCERPT: Let μ be a measure on a Boolean lattice. It is ULC+ when every measure obtained from it by imposing an external field and projecting is ULC. ULC implication conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-056102 | MATHDB-378329

The rank-cover conjecture for CNA+ measures

摘录 / EXCERPT: Let μ be a CNA+ measure, and condition it on the total rank $\sum_eX_e$. Rank-cover conjecture. For each k, the conditional distribution given total rank $k+1$ stochastically...

conjecture | open | MathDB 页面 | 原文线索

V2-056103 | MATHDB-378331

One-endedness of essential spanning-forest components in high dimensions

摘录 / EXCERPT: Let G be the infinite nearest-neighbor graph on the integer lattice in d dimensions, and let ${\bf T}$ be the uniform spanning forest obtained as a distributional limit of unif...

open_problem | open | MathDB 页面 | 原文线索

V2-056104 | MATHDB-378332

The converse capacity conjecture for quasi-Bernoulli tree-indexed processes

摘录 / EXCERPT: Let (Ω,S,B,μ) be a tree-indexed process. Define ... so that $f(n)$ is the probability that a sequence of n IID picks from μ extends to some sequence in B. The conver...

conjecture | open | MathDB 页面 | 原文线索

V2-056105 | MATHDB-378334

The bouncing-ray conjecture for tree-indexed random walks

摘录 / EXCERPT: Bouncing-ray conjecture. If μ has mean zero and finite variance and Γ almost surely has bouncing rays, then Γ almost surely has escaping rays. This is presented a...

conjecture | open | MathDB 页面 | 原文线索

V2-056106 | MATHDB-378422

The unoriented percolation incipient infinite cluster scaling conjecture

摘录 / EXCERPT: Unoriented percolation scaling conjecture. There exist constants A,V,v and $\delta\in(0,1)$ such that the scaling relation (rhoscal) holds for $\rho^{(\mathrm{pe})}$.

conjecture | open | MathDB 页面 | 原文线索

V2-056107 | MATHDB-378424

The invasion percolation scaling conjecture to infinite canonical super-Brownian motion

摘录 / EXCERPT: Invasion percolation scaling conjecture. For each $r\geq2$, $\vec t\in\mathbb{R}^{r-1}$ and $\vec k\in\mathbb{R}^{d(r-1)}$, there exist constants A,V,v and $\delta\in(0,1)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-056108 | MATHDB-378469

Uniqueness conjecture for viscosity solutions of the mixed stochastic differential game

摘录 / EXCERPT: Let V^+ and V^- be the upper and lower value functions of the stochastic differential game of mixed type, defined by ... and ... Here $\mathcal{A}_1$ and $\mathcal{A}_2$ are the ad...

conjecture | open | MathDB 页面 | 原文线索

V2-056109 | MATHDB-378473

Conjecture on the total cost of Quick-Find-Weighted

摘录 / EXCERPT: Let $C^{QFW}_{n,m}$ denote the total cost of the Quick-Find-Weighted algorithm through its first m mergers, and let $\build{\longrightarrow}{}_{\mathcal L_2}$ denote convergence...

conjecture | open | MathDB 页面 | 原文线索

V2-056110 | MATHDB-378495

Census non-solvability at the Kesten-Stigum threshold

摘录 / EXCERPT: Let M be a channel with second eigenvalue $\lambda_2(M)$, and let T_b be the b-ary tree. The reconstruction problem is census-solvable when the root can be reconstructed from...

conjecture | open | MathDB 页面 | 原文线索

V2-056111 | MATHDB-378496

Potts-model reconstruction below the Kesten-Stigum bound

摘录 / EXCERPT: Consider the symmetric Potts channel M on $q\geq 3$ symbols, with b-ary tree branching factor b and second eigenvalue $\lambda_2(M)$. Potts-model below-bound conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-056112 | MATHDB-378497

Monotonicity of Potts reconstruction with the number of symbols

摘录 / EXCERPT: Potts monotonicity conjecture. If reconstruction is solvable for M_1, then it is also solvable for M_2.

conjecture | open | MathDB 页面 | 原文线索

V2-056113 | MATHDB-378498

Robust reconstruction conjecture below the Kesten-Stigum threshold

摘录 / EXCERPT: Let M be a channel and T_b the b-ary tree. For n and m, let $\sigma_{n,m}$ be obtained from the level-n configuration σ_n by applying M^m independently to each...

conjecture | open | MathDB 页面 | 原文线索

V2-056114 | MATHDB-378578

Kuperberg's mixed-base moment-matching conjecture

摘录 / EXCERPT: Kuperberg's mixed-base conjecture. There are unique constants

conjecture | open | MathDB 页面 | 原文线索

V2-056115 | MATHDB-378592

Supercritical case of the upper-bound conjecture for chemical distance

摘录 / EXCERPT: Supercritical upper-bound conjecture. The displayed limsup is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-056116 | MATHDB-378710

The dominant-pair conjecture for singular random sign matrices

摘录 / EXCERPT: Dominant-pair conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056117 | MATHDB-378726

SLE duality conjecture for boundary traces and conditional laws

摘录 / EXCERPT: Let $\kappa>4$ and set $\kappa'=16/\kappa$. In the configuration $(\mathbb{H},x,y,z,\infty)$ with $x<y<z$, let (K_t) be the Loewner chain obtained by running an...

conjecture | open | MathDB 页面 | 原文线索

V2-056118 | MATHDB-378738

Baik-Rains interlacing conjecture for GOE and GSE eigenvalues

摘录 / EXCERPT: Let $F_{\beta}(s,m)$ denote the distribution of the m^{th} largest eigenvalue in the appropriate scaling limit for the Gaussian orthogonal ensemble ($\beta=1$) and Gaussian sympl...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056119 | MATHDB-378805

Positive critical regrowth parameter for self-organized forest fires

摘录 / EXCERPT: Positive critical regrowth conjecture. The critical regrowth parameter is positive:

conjecture | open | MathDB 页面 | 原文线索

V2-056120 | MATHDB-378812

Reversed-speed states are least likely in multispecies exclusion

摘录 / EXCERPT: Least-likely-state conjecture. In the stationary distribution, the least likely states are precisely those that are reversed-speed ordered.

conjecture | open | MathDB 页面 | 原文线索

V2-056121 | MATHDB-378825

The density conjecture for the integrated superBrownian excursion

摘录 / EXCERPT: Let μ_{ISE} be the integrated superBrownian excursion measure on $\mathbb{R}$, and let $Y(\lambda)$ be the non-negative random variable whose law is specified by the l...

conjecture | open | MathDB 页面 | 原文线索

V2-056122 | MATHDB-378898

The hyperbolicity and linear expected-volume conjecture for random Heegaard splittings

摘录 / EXCERPT: Let N_ϕ be the 3-manifold obtained from a random Heegaard splitting determined by a mapping class ϕ, and let L denote the random-walk length. The hyperbolicity and line...

conjecture | open | MathDB 页面 | 原文线索

V2-056123 | MATHDB-378940

Theorem 5.1 in all dimensions at least two

摘录 / EXCERPT: Theorem 5.1 conjecture. Theorem 5.1 is true in all $d\ge2$.

conjecture | open | MathDB 页面 | 原文线索

V2-056124 | MATHDB-378941

High-dimensional tightness conjecture for the percolation corrector

摘录 / EXCERPT: High-dimensional corrector tightness conjecture. Let $d\gg1$. Then for each $\epsilon>0$ there exists $K<\infty$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-056125 | MATHDB-378942

Gaussian free field scaling conjecture for the percolation corrector

摘录 / EXCERPT: Corrector Gaussian free field conjecture. Let $d\ge1$. Then the law of

conjecture | open | MathDB 页面 | 原文线索

V2-056126 | MATHDB-378943

Stable-process scaling conjecture for long-range percolation

摘录 / EXCERPT: Long-range percolation scaling conjecture. If $\alpha\in(0,2)$, then the corresponding random walk scales to a symmetric α-stable Lévy process in $\mathbb R^d$.

conjecture | open | MathDB 页面 | 原文线索

V2-056127 | MATHDB-379115

Coexistence conjecture for competing Bernoulli percolation infections

摘录 / EXCERPT: Consider the competition process on $\mathbb{Z}^d$ with blue and yellow infections, whose transmission parameters are p_{b} and p_{y}, respectively. Let p_c be...

conjecture | open | MathDB 页面 | 原文线索

V2-056128 | MATHDB-379216

The quasi-loop theorem for Euclidean nets

摘录 / EXCERPT: Quasi-loop theorem conjecture. Theorem 4.6 should hold for any Euclidean net. Extending the result would give quasi-loop control for loop-erased random walks on all Euclidean nets,...

conjecture | open | MathDB 页面 | 原文线索

V2-056129 | MATHDB-379217

The isotropy conjecture for stitchings of lattices

摘录 / EXCERPT: Consider a graph obtained by a non-trivial stitching of the lattices $\mathbb{Z}^{d}$ and $2\mathbb{Z}^{d}$. Stitching isotropy conjecture. Any such graph is an isotropic graph. If...

conjecture | open | MathDB 页面 | 原文线索

V2-056130 | MATHDB-379219

Necessary and sufficient summability for almost sure convergence of stochastic difference equations

摘录 / EXCERPT: Let $(\xi_n)_{n\geq 1}$ be i.i.d. random variables satisfying ... and let $\alpha>0$ satisfy ... For a given sequence $(S_n)_{n\geq 0}$, consider the stochastic difference equation ... H...

conjecture | open | MathDB 页面 | 原文线索

V2-056131 | MATHDB-379280

The square-exponential tail conjecture for the excited random walk's maximum

摘录 / EXCERPT: Let $V(t)$ denote the random variable measuring the quantity studied in the preceding estimates for the excited random walk at time t, and let $C,c>0$ be constants. Square-expone...

conjecture | open | MathDB 页面 | 原文线索

V2-056132 | MATHDB-379315

Random-walk tunnel number one 3-manifolds do not fiber

摘录 / EXCERPT: Let H be the handlebody used to construct tunnel number one 3-manifolds, let ∂H be its boundary, and let $\mathcal{MCG}(\partial H)$ be its mapping class group. Choose...

conjecture | open | MathDB 页面 | 原文线索

V2-056133 | MATHDB-379316

Algebraic fibering is generic for random two-generator one-relator groups with commutator relator

摘录 / EXCERPT: Let $\mathcal{G}'(r)$ denote the random model of two-generator one-relator groups whose defining relation is a commutator, and let p_r be the probability that a group...

conjecture | open | MathDB 页面 | 原文线索

V2-056134 | MATHDB-379357

Spencer's conjecture on infinitely recurring bin orderings

摘录 / EXCERPT: Consider a balls-in-bins process $(I_m(i))_{i\in[B],m\geq 0}$ with feedback function f in the almost-balanced regime. Here $[B]=\{1,\ldots,B\}$, and Π ranges over all permuta...

conjecture | open | MathDB 页面 | 原文线索

V2-056135 | MATHDB-379362

Conjecture on all occupation times of unit-ball surfaces

摘录 / EXCERPT: Let $d\geq 3$, let $\mathcal{S}(1)=\{\mathbf{z}\in\mathcal{Z}_d:\|\mathbf{z}\|=1\}$, and define the occupation time of the unit-ball surface centered at $\mathbf{z}$ by ... Writing...

conjecture | open | MathDB 页面 | 原文线索

V2-056136 | MATHDB-379458

Conjecture on asymptotic stabilization of the two-species regions

摘录 / EXCERPT: Let $R(t)$ and $B(t)$ be the red and blue regions of the two-species competition model, and let $\mathcal{S}$ be the Richardson-model limit shape. For a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-056137 | MATHDB-379469

Arnold's average-density conjecture for numerical semigroups

摘录 / EXCERPT: Arnold's average-density conjecture. For large ${\bf d}^m$, with overwhelming probability, this fraction is asymptotically $1/m$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-056138 | MATHDB-379588

Near-field scaling conjecture for the diffusion-off boundary

摘录 / EXCERPT: Let $v(x)$ be an antisymmetric velocity field with $v(x)>0$ for $x>0$, and suppose ... Let $b(x)$ denote the diffusion-off boundary. Near-field scaling conjecture. Then ... This predic...

conjecture | open | MathDB 页面 | 原文线索

V2-056139 | MATHDB-379619

Negative association conjecture for Rayleigh measures

摘录 / EXCERPT: Let $Z(\omega;\mathbf y)$ be a partition function on a finite ground set E, and let $\mathsf A_1,\mathsf A_2$ be increasing events with disjoint support. Negative association con...

conjecture | open | MathDB 页面 | 原文线索

V2-056140 | MATHDB-379702

Equal-maximal-rate poisoning conjecture

摘录 / EXCERPT: Let n and d be arbitrary, and let $p_1,p_2,\ldots,p_n$ denote the type-dependent poisoning parameters of the particle system. Starting from the all-zero configuration, suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-056141 | MATHDB-379703

Monotonicity conjecture for the process of 1's

摘录 / EXCERPT: Consider two systems on the same lattice dimension d with the same number of types n, having parameter vectors $(p_1,p_2,\ldots,p_n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056142 | MATHDB-380161

The growth-random-walk equivalence for finitely generated quantum groups

摘录 / EXCERPT: Let (A,u) be a finitely generated quantum group, and let b_k and p_k denote the growth and random-walk quantities constructed in the paper. Here $f_k\approx g_k$ means that t...

conjecture | open | MathDB 页面 | 原文线索

V2-056143 | MATHDB-380163

The converse of the one-dimensional infinite-cluster criterion

摘录 / EXCERPT: Consider the GN$(1,\underline{\alpha})$ model on a Poisson process on $\mathbb{R}$, where α_i are the connection parameters and clusters are formed according to the model's...

conjecture | open | MathDB 页面 | 原文线索

V2-056144 | MATHDB-380164

High-dimensional critical threshold conjecture for GN_k continuum percolation

摘录 / EXCERPT: Let $\alpha_c^{(k)}(d)$ denote the critical value of the parameter α for percolation in the GN$_k(d,\alpha)$ model in dimension d, with integer $k\geq 1$. High-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-056145 | MATHDB-380206

Sapir's asymptotic-cone conjecture for critical exponents

摘录 / EXCERPT: Let G and G' be groups with Cayley graphs whose asymptotic cones are isometric. A group is non-elementary hyperbolic if it is hyperbolic and is not virtually cyclic. Sapir's co...

conjecture | open | MathDB 页面 | 原文线索

V2-056146 | MATHDB-380264

Conjectured differentiability and sequence-independence of fluid limits

摘录 / EXCERPT: Let z be a fluid limit with initial condition $z(0)=x$, where $x=\lim_n x^n/|x^n|$, and let V_i denote the neighborhood associated with coordinate i. The right derivative at...

conjecture | open | MathDB 页面 | 原文线索

V2-056147 | MATHDB-380358

Moment equivalence conjecture for stable-process passage times and functionals

摘录 / EXCERPT: Let $Z=(Z_t)_{t\geq 0}$ be the stable process considered in the paper, let T be the associated stopping time, and let α be its stability index. For every $k>0$, the moment...

conjecture | open | MathDB 页面 | 原文线索

V2-056148 | MATHDB-380383

Limits of rescaled renormalized diffusion matrices

摘录 / EXCERPT: Let $D\subset\mathbb R^d$ be the domain of a renormalization class $\mathcal W$, let $(c_k)_{k\geq0}$ be positive migration constants, and define $s_n=\sum_{k=0}^{n-1}c_k^{-1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056149 | MATHDB-380425

Penrose-Ong conjecture on central limit theorems for the on-line nearest-neighbour graph

摘录 / EXCERPT: Penrose-Ong conjecture. Suppose $d\in\mathbb{N}$. The limit theorems for $\d835α\in(0,d/4)$, namely the variance limits and the corresponding Gaussian convergence for the binomial...

conjecture | open | MathDB 页面 | 原文线索

V2-056150 | MATHDB-380426

Critical logarithmic central limit conjecture for the on-line nearest-neighbour graph

摘录 / EXCERPT: Critical logarithmic CLT conjecture. Let $d\in\mathbb{N}$. There exists a constant $σ_{d,d/2}^2\in(0,\infty)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-056151 | MATHDB-380560

Local lozenge probability conjecture for boxed plane partitions

摘录 / EXCERPT: Let α,β,γ be the normalized side lengths of a hexagon, let (x,y) be a normalized location in this hexagon, and let $\mathcal P_{\alpha,\beta,\gamma}(x,y)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-056152 | MATHDB-380625

Catalan product conjecture for Young tableau faces of the Birkhoff polytope

摘录 / EXCERPT: Let $n\ge 2$, and let F_n be the $n\times n$ matrix whose (i,j) entry is 1 when $j\le i+1$ and 0 otherwise. The matrix F_n is a union of permutation matrices correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-056153 | MATHDB-380847

Quantum central limit theorem for self-adjoint non-commutative polynomials

摘录 / EXCERPT: Let $(\mathcal{M},\rho)$ be a quantum probability space, and let $A_1,\ldots,A_k$ be self-adjoint elements with mean 0 and symmetric covariance matrix. For each t, write ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-056154 | MATHDB-380848

Cyclic Markov-chain GUE conjecture for random-word shapes

摘录 / EXCERPT: Let M be an indecomposable, doubly stochastic matrix indexed by an alphabet of size k, satisfying ... where $k+1\equiv1$. A word of length N is generated by the Markov chain wi...

conjecture | open | MathDB 页面 | 原文线索

V2-056155 | MATHDB-380854

The asymptotic singularity conjecture for random 0/1-matrices

摘录 / EXCERPT: Let P_d be the probability that a random $0/1$-matrix of size $(d+1)\times(d+1)$ is singular; equivalently, in the associated ±1 model, let P_d denote the probability that...

conjecture | open | MathDB 页面 | 原文线索

V2-056156 | MATHDB-380993

The enumerable probability coding conjecture for P^E

摘录 / EXCERPT: Let $B^{\sharp}$ be the domain of finite binary strings under consideration, and let $P^E(x)$ and $K^E(x)$ denote the corresponding enumerable measure and complexity functions. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-056157 | MATHDB-380994

The generalized probability coding conjecture for P^G

摘录 / EXCERPT: Let $B^{\sharp}$ be the domain of finite binary strings under consideration, and let $P^G(x)$ and $K^G(x)$ denote the corresponding generalized probability and complexity functions...

conjecture | open | MathDB 页面 | 原文线索

V2-056158 | MATHDB-381014

Sub-dominant eigenvalue and spectral gap conjecture for the Dirichlet Markov Ensemble

摘录 / EXCERPT: Let $\mathbf{M}$ be the random matrix from theorem $\mathrm{th:sym}$, with eigenvalues ordered so that $\lambda_1(\mathbf{M})$ is the dominant eigenvalue and...

conjecture | open | MathDB 页面 | 原文线索

V2-056159 | MATHDB-381031

Purity conjecture for the Potts measure on random trees

摘录 / EXCERPT: Let q be the number of Potts symbols, let $\theta=\tanh(\beta)$, and let d_0 denote the random offspring degree under the tree measure Q. Reconstruction is the problem of rec...

conjecture | open | MathDB 页面 | 原文线索

V2-056160 | MATHDB-381042

The third critical probability conjecture for relative covariance in random orientations

摘录 / EXCERPT: Let $p_3(n)$ denote the third critical probability at which the relative covariance of paths in $\vec G(n,p)$ changes sign, where $p_1(n)<p_2(n)<p_3(n)$ are the three critical prob...

conjecture | open | MathDB 页面 | 原文线索

V2-056161 | MATHDB-381055

Pemantle's conjecture that CNC implies CNA

摘录 / EXCERPT: Pemantle's conjecture. CNC implies CNA. In particular, the equivalent formulation considered here is that R^+ implies CNA.

conjecture | open | MathDB 页面 | 原文线索

V2-056162 | MATHDB-381107

Existence of transient two-dimensional random walks on discrete point processes

摘录 / EXCERPT: A two-dimensional random walk on a discrete point process is a random walk in dimension 2 whose state space is a discrete point process. Existence conjecture. There are transient...

conjecture | open | MathDB 页面 | 原文线索

V2-056163 | MATHDB-381108

Failure of the central limit theorem for high-dimensional point-process random walks

摘录 / EXCERPT: A high-dimensional random walk on a discrete point process is a random walk on such a point process in sufficiently large dimension. Failure conjecture. There are random walks on d...

conjecture | open | MathDB 页面 | 原文线索

V2-056164 | MATHDB-381130

Conjecture on the range of the percolation-time result

摘录 / EXCERPT: Let T denote the percolation time in the r-neighbour bootstrap percolation model on $\mathbb{T}_n^d$, and let $t=t(n)$ be a time parameter. Range conjecture. Theorem holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-056165 | MATHDB-381131

Conjecture on the staircase and stabilization of five-species coefficients

摘录 / EXCERPT: Staircase and stabilization conjecture. The coefficients satisfy the following properties. First, for $j\geqslant 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-056166 | MATHDB-381223

Uniqueness conjecture for unmixed-symplectic spectral factors

摘录 / EXCERPT: Let $\Phi(z)\in\mathbb{R}(z)^{n\times n}$ be a spectrum of normal rank $\mathrm{rk}(\Phi)=r\neq 0$. Let $\mathscr{A}_p$ and $\mathscr{A}_z$ be two unmixed-symplectic sets. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056167 | MATHDB-381245

Existence and uniqueness conjecture for the random-horizon BSDE

摘录 / EXCERPT: Existence and uniqueness conjecture. The BSDE admits a unique solution such that Y^b is bounded and $\int_0^{\cdot}Z_s^{b}\,dW_s$ is a $\operatorname{BMO}(\mathbb{G})$-martingale...

conjecture | open | MathDB 页面 | 原文线索

V2-056168 | MATHDB-381247

Kipnis-Varadhan quenched functional CLT conjecture

摘录 / EXCERPT: Let $(X_n)_{n\in\mathbb{Z}}$ be a stationary reversible Markov chain with transition operator Q, invariant distribution π, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056169 | MATHDB-381268

Coexistence conjecture for two-type growth models

摘录 / EXCERPT: Let $d\geq 2$. For $i=1,2$, let A_i be the event that infection type i reaches sites arbitrarily far from the origin, and let $A=A_1\cap A_2$ be the coexistence event. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056170 | MATHDB-381269

Häggström-Pemantle coexistence conjecture for the two-type Richardson model

摘录 / EXCERPT: Häggström-Pemantle's conjecture. In every dimension $d\geq 2$, infinite coexistence has positive probability if and only if the infection intensities are equal:

conjecture | solved | MathDB 页面 | 原文线索

V2-056171 | MATHDB-381270

Monotonicity conjecture for coexistence probabilities in the two-type Richardson model

摘录 / EXCERPT: Monotonicity conjecture. For $\lambda<\lambda'\in(0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-056172 | MATHDB-381320

The quasi-polynomiality conjecture for vanishing probabilities of Schur functions

摘录 / EXCERPT: Let λ be a partition, and let $P(s_\lambda\mapsto0)$ denote the probability that the Schur function s_λ evaluates to zero over the finite field under consideration....

conjecture | open | MathDB 页面 | 原文线索

V2-056173 | MATHDB-381324

Polynomial mixing-time conjecture for the flip walk on sphere triangulations

摘录 / EXCERPT: Let $(T_n(k))_{k \geq 0}$ denote the flip walk on triangulations of the sphere with n vertices, and let its mixing time be measured with respect to the uniform distribution on su...

conjecture | open | MathDB 页面 | 原文线索

V2-056174 | MATHDB-381339

The GUE corners limit conjecture in the diverging-temperature regime

摘录 / EXCERPT: Let Assumption $\mathrm{(function)}$ hold, and set $q=q(N)=e^{-\gamma(N)/N}$, where $\gamma(N)\to+\infty$ and $\gamma(N)\ll N$. Let $\lambda^r_j(N)$ denote the parts of the random...

conjecture | open | MathDB 页面 | 原文线索

V2-056175 | MATHDB-381359

The shell-based inpainting Gaussian-blur conjecture

摘录 / EXCERPT: The shell-based inpainting Gaussian-blur conjecture. If u_0 is independent of its y-coordinate, then

conjecture | open | MathDB 页面 | 原文线索

V2-056176 | MATHDB-381410

Approximation conjecture for the ARMA point process by INARMA processes

摘录 / EXCERPT: Approximation conjecture. There exists $\delta\in(0,\infty)$ such that, for every $\Delta\in(0,\delta)$, the process above is a stationary INARMA process, and the family of point p...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056177 | MATHDB-381445

The invariant-measure conjecture for geometric stochastic heat equations

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be a Riemannian manifold, let A be a smooth vector field, and let μ be the loop measure determined by the diffusion with generator ... Assume that the dif...

conjecture | open | MathDB 页面 | 原文线索

V2-056178 | MATHDB-381447

Polynomial mixing conjecture for repelling random walks

摘录 / EXCERPT: Let n particles evolve according to the Attracting Random Walks (ARW) model on an arbitrary graph, with interaction parameter $\beta<0$. The mixing time is measured with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-056179 | MATHDB-381449

Unimodality under free multiplicative convolution on the unit circle

摘录 / EXCERPT: Unimodality conjecture. If bc and bd are symmetric unimodal distributions on $a4$, then $bc\boxtimesbd$ is also unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-056180 | MATHDB-381494

Optimality conjecture for shifted IDLA mixing time on quasi-regular graphs

摘录 / EXCERPT: Optimality conjecture. Under Assumption, t_γ is optimal up to logarithmic factors.

conjecture | open | MathDB 页面 | 原文线索

V2-056181 | MATHDB-381537

Nonexistence conjecture for nonnegative jump amplitudes satisfying diffusion-scaling moment conditions

摘录 / EXCERPT: Let $(J_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative random variables, and let $(\lambda_n)_{n\in\mathbb{N}}$ satisfy the limiting conditions denoted by ... -- ... -- ... does not e...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056182 | MATHDB-381544

The martingale existence conjecture for attraction-repulsion processes

摘录 / EXCERPT: Let f be an interaction function satisfying the conditions of Theorem 2, and let n be finite. A function ... should exist such that $\{h(\Phi_t)\}_{t\in\mathbb{N}}$ is a non-nega...

conjecture | open | MathDB 页面 | 原文线索

V2-056183 | MATHDB-381545

The generalized Sato-Tate conjecture for generic curves

摘录 / EXCERPT: Let C be a smooth projective curve of genus g. Assume that C is in the generic family. For each $k=1,2,\dots,g$, let $a_k(m;g)$ denote the m-th moment of the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-056184 | MATHDB-381546

Scaling-limit conjecture for the breadth-first walk on critical Erdős-Rényi graphs

摘录 / EXCERPT: Let θ_n satisfy the scaling condition in the source, and let $X_n=(X_n(k);k=0,1,\dotsm)$ be the breadth-first walk on the random graph $\mathscr{G}_n^\theta$ constructed for...

conjecture | open | MathDB 页面 | 原文线索

V2-056185 | MATHDB-381563

Global rigidity lower bounds for the Pearcey process

摘录 / EXCERPT: Global rigidity lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056186 | MATHDB-381574

Absence of unbounded nodal components at criticality for planar Gaussian fields

摘录 / EXCERPT: Critical nodal-set conjecture. The set $\{f=0\}$ has no unbounded connected components.

conjecture | open | MathDB 页面 | 原文线索

V2-056187 | MATHDB-381585

Uniqueness of the positive harmonic function for the Laplacian operator

摘录 / EXCERPT: Let the Laplacian operator be the operator defined by the jump conditions in --, and let h_1 denote the function in Theorem. Uniqueness conjecture. The unique, up to multiplicati...

conjecture | open | MathDB 页面 | 原文线索

V2-056188 | MATHDB-381615

Good D implies good ϱ

摘录 / EXCERPT: Let $G\in\mathbb{F}_q^{\ell\times\ell}$ be any kernel, and let $\mathbb{P}\{\mathsf{D}_1=1\}$ denote the probability that the first distance-process value equals 1. Good D impl...

conjecture | open | MathDB 页面 | 原文线索

V2-056189 | MATHDB-381650

Conjecture on the range and rate of convergence for the sticky-diffusion approximation

摘录 / EXCERPT: Convergence-rate conjecture. Theorem ... and $$ increase with α.

conjecture | open | MathDB 页面 | 原文线索

V2-056190 | MATHDB-381653

Existence of non-smooth processes satisfying only one smoothness assumption

摘录 / EXCERPT: Let w be a process and let the three assumptions in Definition be the smooth-expectation condition, the relative-fitness-density condition, and the local-change-density condition...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056191 | MATHDB-381660

The non-necessity of the core condition for characterizing exponential distributions

摘录 / EXCERPT: Let X be a random variable, and let $n\geq 3$. Suppose the characterization theorem is formulated under condition. Non-necessity conjecture. For $n\geq 3$, condition is an artifa...

conjecture | open | MathDB 页面 | 原文线索

V2-056192 | MATHDB-381679

Pointwise convergence conjecture for hook-length mass functions

摘录 / EXCERPT: For $t \geq 4$, let $f_{t;n}(x)$ be the scaled mass function associated with the distribution of $\hat{Y}_t(n)$, and let $g_t(x)$ be the corresponding limiting function. Pointwise...

conjecture | open | MathDB 页面 | 原文线索

V2-056193 | MATHDB-381686

The general CNF-DNF correlation inequality for random points in a hypercube

摘录 / EXCERPT: General CNF-DNF correlation conjecture. Under these assumptions, the inequality referred to in the source as the CNF-DNF inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-056194 | MATHDB-381739

Berry-Esseen dependency-graph conjecture

摘录 / EXCERPT: Berry-Esseen dependency-graph conjecture. The bound

conjecture | open | MathDB 页面 | 原文线索

V2-056195 | MATHDB-381761

Decelle-Krzakala-Moore-Zdeborová conjecture on stochastic block model detection

摘录 / EXCERPT: Decelle-Krzakala-Moore-Zdeborová conjecture. For the block model: (I) for all q, it is possible to detect communities better than random if $d\lambda^2>1$; (II) for $q\leq 4$, it...

conjecture | open | MathDB 页面 | 原文线索

V2-056196 | MATHDB-381781

The thermomajorization characterization by fixed-point stochastic maps

摘录 / EXCERPT: Let $x,y,d\in\mathbb R^n$ with $d\geq 0$ be given. A matrix $A\in\mathbb R_+^{n\times n}$ is column-stochastic when $\mathbbm e^\top A=\mathbbm e^\top$; it fixes d when $Ad=d$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-056197 | MATHDB-381783

Conjecture on the levels and adjacency of infinite connected components

摘录 / EXCERPT: Consider the infinite connected components of the corner-percolation configuration, each assigned its integer-valued level. Level-bijection conjecture. The function which maps each...

conjecture | open | MathDB 页面 | 原文线索

V2-056198 | MATHDB-381891

Four-root-of-unity anti-concentration conjecture

摘录 / EXCERPT: Four-root-of-unity anti-concentration conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-056199 | MATHDB-381893

A linear upper bound for the Gaussian Simes-test error probability

摘录 / EXCERPT: Gaussian Simes bound. There exists $c\geqslant 1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-056200 | MATHDB-381945

Kabluchko's almost-sure convergence conjecture for critical points

摘录 / EXCERPT: Let μ be a probability measure on $\mathbb{C}$, let $X_1,X_2,\ldots$ be independent and identically distributed with law μ, and define $P_n(z)=\prod_{j=1}^n(z-X_j)$. Let .....

conjecture | open | MathDB 页面 | 原文线索

V2-056201 | MATHDB-381987

Scaling conjecture for clustering in finite spatial inhomogeneous random graphs

摘录 / EXCERPT: Scaling conjecture for finite graphs. Consider the finite SIRG $\mathbb{G}^{(n)}$. If $\phi(k_n)=o(\psi(n))$, then

conjecture | open | MathDB 页面 | 原文线索

V2-056202 | MATHDB-382038

Matúš's rational distribution conjecture for conditional independence models

摘录 / EXCERPT: Matúš's conjecture. There exists a distribution p in the zero set of $J_{\mathcal C}$ such that all joint probabilities of p are rational.

conjecture | open | MathDB 页面 | 原文线索

V2-056203 | MATHDB-382121

Conjectured log-partition function for the generalized rectangular spherical integral

摘录 / EXCERPT: Log-partition function conjecture. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-056204 | MATHDB-382226

The DMPK limiting-density conjecture for ballistic initial conditions

摘录 / EXCERPT: DMPK limiting-density conjecture. The density $\rho(x;\mathbb{M})$ is related to $\psi(x)$ by

conjecture | open | MathDB 页面 | 原文线索

V2-056205 | MATHDB-382228

Asymptotic independence of extreme eigenvalues in Gaussian beta-ensembles

摘录 / EXCERPT: Asymptotic independence conjecture. For every $x,y\in\mathbb{R}$,

conjecture | open | MathDB 页面 | 原文线索

V2-056206 | MATHDB-382269

Existence of an independent factor for fractional operators

摘录 / EXCERPT: Existence conjecture. Under certain conditions on W, there exists a random variable V, independent of W and with support in $[0,1]$, such that holds. In terms of...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-056207 | MATHDB-382297

The bunkbed conjecture for transitive tournaments

摘录 / EXCERPT: The bunkbed conjecture for transitive tournaments. The statement

conjecture | open | MathDB 页面 | 原文线索

V2-056208 | MATHDB-382347

The non-extreme threshold criterion for alpha-preservation invalidity

摘录 / EXCERPT: Non-extreme threshold conjecture. If Γ is α-satisfiable and Δ is not α-tautologous, and if there are no $\gamma \in \Gamma$ and $\delta \in \Delta$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-056209 | MATHDB-382356

Stable-limit conjecture for finite free Appell polynomials

摘录 / EXCERPT: Let f be in the Laguerre--Pólya class and satisfy the regular-variation condition on its roots. Let ... and set $\rho=\frac{1}{\alpha}-1$. Write μ_p for the empirical root dist...

conjecture | open | MathDB 页面 | 原文线索

V2-056210 | MATHDB-382383

Vu's distinct singular values conjecture for random symmetric Bernoulli matrices

摘录 / EXCERPT: Vu's distinct singular values conjecture. With probability $1-o(1)$, all the singular values of M_n^{sym} are distinct.

conjecture | open | MathDB 页面 | 原文线索

V2-056211 | MATHDB-382384

Ellipsoid-fitting threshold conjecture

摘录 / EXCERPT: Let $n,d\geq 1$, and let $x_1,\ldots,x_n$ be independent random vectors with ... For an ellipsoid $\mathcal{E}$, say that it is an ellipsoid fit to $(x_i)_{i=1}^n$ if every point...

conjecture | open | MathDB 页面 | 原文线索

V2-056212 | MATHDB-382443

Carpentier's motif-counting conjecture for community recovery in the stochastic block model

摘录 / EXCERPT: Carpentier's motif-counting conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-056213 | MATHDB-382528

Uniform boundedness conjecture for the density of states at regular energies

摘录 / EXCERPT: Let ρ be a probability density with moments of all orders, meaning that ... for every $q\geq 1$. Let $E_o\in\mathbb R$. Suppose there is a $\delta>0$ such that, on ... the symbol...

conjecture | open | MathDB 页面 | 原文线索

V2-056214 | MATHDB-382586

Exact finite-energy fundamental limit for deterministic no-feedback AWGN coding

摘录 / EXCERPT: Consider the real additive white Gaussian noise sequence channel ... where the noise coordinates are independent $N(0,N_0/2)$. An (E,M,ϵ) code has M equiprobable messages,...

conjecture | solved | Abhijeet Mulgund | MathDB 页面 | 原文线索

V2-056215 | MATHDB-382868

Ross’s conjecture

摘录 / EXCERPT: For every stochastic transition matrix $M=(m_{ij})_{i,j\in\{1,2\}}$, every $\alpha_1,\alpha_2\in[0,1)$, and every $C_1,C_2>0$, the two-site moving-target search problem has an opti...

conjecture | solved | MathDB 页面 | 原文线索

V2-056216 | MATHDB-382869

Online discrepancy conjecture

摘录 / EXCERPT: Determine whether there exist universal constants $C,c>0$ and a polynomial-time online signing algorithm such that, for every dimension $n\ge 1$ and every finite horizon T, when...

conjecture | partial | MathDB 页面 | 原文线索

V2-056217 | MATHDB-383195

Talagrand’s convolution conjecture

摘录 / EXCERPT: With μ_a the biased-coin product probability measure on the product group G defined in Section 3, Conjecture 6 states: Given $a>0$, there exist $C(a)>0$ such that for...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056218 | MATHDB-383260

Lesniewski-Ruskai Conjecture 4.9

摘录 / EXCERPT: For every $n\ge 1$, every unital stochastic map $\Phi:M_n\to M_n$, and every normalized monotone Riemannian metric specified by an operator-monotone function κ, the Riemanni...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056219 | MATHDB-383294

O’Donnell-Schramm Sherali-Adams question

摘录 / EXCERPT: For random Boolean k-ary constraint satisfaction problems, particularly for odd k, does the Sherali-Adams hierarchy achieve refutation guarantees comparable to those obtained b...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-056220 | MATHDB-383433

Gilmer-Kopparty local central limit theorem conjecture

摘录 / EXCERPT: Let H be a fixed connected graph, let $G_{n,p}$ be the Erdős-Rényi random graph, and let X_H denote the number of copies of H in $G_{n,p}$. Define...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056221 | MATHDB-383494

Krivelevich-Samotij percolation conjecture

摘录 / EXCERPT: Let G be a finite graph, let $\lambda(G)$ denote the spectral radius of its adjacency matrix, and let G_p be obtained from G by retaining each edge independently with probabi...

conjecture | open | MathDB 页面 | 原文线索

V2-056222 | MATHDB-383611

Last-iterate convergence conjecture in definable optimization

摘录 / EXCERPT: Let (x_k) be the iterates generated by the mini-batch stochastic approximation framework under the hypotheses stated in Remark 12 of Bolte and Pauwels (2021), with definable pote...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056223 | MATHDB-383682

Spearman rho-footrule region problem

摘录 / EXCERPT: Let $\mathcal{C}$ be the set of bivariate copulas, equivalently the set of joint laws of random variables $U,V\sim\mathcal{U}(0,1)$. Define Spearman's rho and Spearman's footrule b...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-056224 | MATHDB-384395

Miscellaneous Problems (problem 3)

摘录 / EXCERPT: Let μ be a probability measure on $\{0,1\}^n$. Consider ... and ... What property of g is equivalent to the stability of f?

open_problem | open | MathDB 页面 | 原文线索

V2-056225 | MATHDB-384455

About uniqueness

摘录 / EXCERPT: Find a suitable notion of "probabilistic uniqueness" for solutions of SQG equations.

open_problem | open | MathDB 页面 | 原文线索

V2-056226 | MATHDB-384460

The cubic Szeg\H{o} equation with random initial data (problem 1)

摘录 / EXCERPT: Assume $\mathbb{T}^\infty$ is embedded with the uniform probability measure. Find a suitable fixed set of actions $\{I_k\}\in\Omega $ for which the H^s-norms of solutions to the...

open_problem | open | MathDB 页面 | 原文线索

V2-056227 | MATHDB-384560

Nonlocal interaction equation on graphs

摘录 / EXCERPT: The gradient flow of the nonlocal interaction energy on graphs, with respect to nonlocal Wasserstein distance is the following ... where μ_n is the uniform probability measure on...

open_problem | open | MathDB 页面 | 原文线索

V2-056228 | MATHDB-384609

In 2D, formulate the random vortex methods analog to the model of a flow past an obstacle (Stochastic ODE).

摘录 / EXCERPT: In 2D, formulate the random vortex methods analog to the model of a flow past an obstacle (Stochastic ODE). Is there a stability transition (Hopf bifurcation) in the Reynolds numbe...

open_problem | open | MathDB 页面 | 原文线索

V2-056229 | MATHDB-384656

Random Matrix Theory (problem 1)

摘录 / EXCERPT: Study the expansion of partition functions of normal matrix models for ... and ... where $z=t\sqrt{d}$. Study the "mad criticality" that happens here.

open_problem | open | MathDB 页面 | 原文线索

V2-056230 | MATHDB-385042

Log-Minkowski inequality

摘录 / EXCERPT: Show that if K and L symmetric convex bodies in ${\\mathbb{R}}^n$, then ... where $\bar{V}_L$ is the cone-volume probability measure of L.

open_problem | open | MathDB 页面 | 原文线索

V2-056231 | MATHDB-385209

The Sublinearly Morse boundary and the Martin boundary

摘录 / EXCERPT: If G is a Hierarchically Hyperbolic Space, and μ is a symmetric non-degenerate probability measure on G with finite support, then the Sublinearly Morse Boundary of G embe...

open_problem | open | MathDB 页面 | 原文线索

V2-056232 | MATHDB-385211

Dimension of Harmonic Measure over Moduli Space

摘录 / EXCERPT: Fix some measure μ on $\pi_1(M)$, where M is a manifold of curvature -1. Construct the random walk on hyperbolic n-space by picking a representation...

open_problem | open | MathDB 页面 | 原文线索

V2-056233 | MATHDB-385212

Witnessing a Conformal Measure

摘录 / EXCERPT: Let G be a δ-hyperbolic group. Let ν be a conformal measure on the Gromov Boundary $\partial_{\text{Gromov}} G$. does there exist some probability measure μ on G...

open_problem | open | MathDB 页面 | 原文线索

V2-056234 | MATHDB-386148

Prove existence of most probable escape paths in non-gradient systems.

摘录 / EXCERPT: Prove existence of most probable escape paths in non-gradient systems.

open_problem | open | MathDB 页面 | 原文线索

V2-056235 | MATHDB-386186

Statistics of KPZ equation in $1+1$ dimension:

摘录 / EXCERPT: Compute statistics for different initial data including $\mathcal{Z}_0(x)=1$ (flat) and $\mathcal{Z}_0(x)=e^{B(x)}$ for $B(x)$ a two-sided Brownian motion (equilibrium).

open_problem | open | MathDB 页面 | 原文线索

V2-056236 | MATHDB-386187

Compute multi-point (spatial) distribution for KPZ with various initial data.

摘录 / EXCERPT: Compute multi-point (spatial) distribution for KPZ with various initial data. For example, with $\mathcal{Z}_0(x)=\delta_{x=0}$, let ... What is the distribution $F(\xi_1, \xi_2)$ su...

open_problem | open | MathDB 页面 | 原文线索

V2-056237 | MATHDB-386189

Higher dimensions

摘录 / EXCERPT: Make rigorous sense of the anisotropic KPZ equation ... and show that the Gaussian free field is invariant. Is there a way of getting this equation out of the 2d Schur process dynami...

open_problem | open | MathDB 页面 | 原文线索

V2-056238 | MATHDB-386194

Is there such an array for ASEP or for some sort of transformed version of ASEP?

摘录 / EXCERPT: Is there such an array for ASEP or for some sort of transformed version of ASEP?

open_problem | open | MathDB 页面 | 原文线索

V2-056239 | MATHDB-386195

Is there an analog of the Karlin-McGregor formula when the non-intersecting conditioning is replaced by a softer form of conditioning?

摘录 / EXCERPT: Is there an analog of the Karlin-McGregor formula when the non-intersecting conditioning is replaced by a softer form of conditioning? Does this explain the solvability and occurre...

open_problem | open | MathDB 页面 | 原文线索

V2-056240 | MATHDB-386196

Expand the universality of the KPZ equation to:

摘录 / EXCERPT: Growth model such as ballistic deposition.

open_problem | open | MathDB 页面 | 原文线索

V2-056241 | MATHDB-386200

Prove that the fixed point is attractive (i.e., universal in some class of models).

摘录 / EXCERPT: Prove that the fixed point is attractive (i.e., universal in some class of models).

open_problem | open | MathDB 页面 | 原文线索

V2-056242 | MATHDB-386201

For example, prove universality of LPP and polymer (with respect to weight distributions).

摘录 / EXCERPT: For example, prove universality of LPP and polymer (with respect to weight distributions).

open_problem | open | MathDB 页面 | 原文线索

V2-056243 | MATHDB-386203

Exact solvability

摘录 / EXCERPT: Use the Macdonald processes formulas to prove the Baik-Ben Arous-P'{e}ch'{e} transition for the semi-discrete polymer with a finite number of non-zero drifts for the underlying Bro...

open_problem | open | MathDB 页面 | 原文线索

V2-056244 | MATHDB-386213

Polymer universality

摘录 / EXCERPT: Consider intermediate scaling for polymer with $\beta=n^{-1/4}c(n)$ where $c(n)$ is any function which tends to infinity with n. Show universality of the F_{GUE} fluctuations w...

open_problem | open | MathDB 页面 | 原文线索

V2-056245 | MATHDB-386215

Universality of initial data for (T)ASEP.

摘录 / EXCERPT: Consider two initial conditions for (T)ASEP corresponding to height functions $h_1(x;t=0)$ and $h_2(x;t=0)$ which can be random, but are independent of each other. Hydrodynamic the...

open_problem | open | MathDB 页面 | 原文线索

V2-056246 | MATHDB-386221

Two matrix model with non symmetric potentials

摘录 / EXCERPT: Study eigenvalue properties (in particular, the limiting density of eigenvalues) of M_1 in a two matrix model given by the probability density ... In this context, $V(x)=x^2/2$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-056247 | MATHDB-386222

Coupled random matrix model with negative potentials

摘录 / EXCERPT: Analysis of the coupled matrix model similar to problem, but with potentials ... Is there any new critical behavior for some value of a?

open_problem | open | MathDB 页面 | 原文线索

V2-056248 | MATHDB-386224

Random matrices with external source. More general cases

摘录 / EXCERPT: Study the random matrix model with probability measure ... where A is an $N\times N$ diagonal matrix with eigenvalues $\pm a$ (each with equal multiplicity $N/2$, assuming N is e...

open_problem | open | MathDB 页面 | 原文线索

V2-056249 | MATHDB-386226

Normal matrix model and d-bar problems

摘录 / EXCERPT: Is it possible to use the theory of $\overline{\partial}$-problems to analyze the normal matrix models?

open_problem | open | MathDB 页面 | 原文线索

V2-056250 | MATHDB-386227

S-curves for complex potentials in the scalar case

摘录 / EXCERPT: Analysis of existence and properties of S-curves in the scalar case when the potential is $\phi(z)=\textrm{Re}\, z^p$, for $p\geq 3$, and we take τ as the set of continua th...

open_problem | open | MathDB 页面 | 原文线索

V2-056251 | MATHDB-386229

S-curves with piecewise harmonic external field

摘录 / EXCERPT: Consider the external potential ... where $k_1,k_2>0$. Is there a continuum with the S-property connecting the points $z_1=-\varepsilon_1+i$ and $z_2=\varepsilon_2-i$ in the comple...

open_problem | open | MathDB 页面 | 原文线索

V2-056252 | MATHDB-386230

Green's potential

摘录 / EXCERPT: Existence theory for S-curves in a general sense, if the logarithmic potential is replaced by Green's potential.

open_problem | open | MathDB 页面 | 原文线索

V2-056253 | MATHDB-386231

In a general sense, is it possible to find some explicit connection between the notions of string equations, spectral curve, Lax pair, WKB approximation, etc, and the vector equilibrium problems that

摘录 / EXCERPT: In a general sense, is it possible to find some explicit connection between the notions of string equations, spectral curve, Lax pair, WKB approximation, etc, and the vector equili...

open_problem | open | MathDB 页面 | 原文线索

V2-056254 | MATHDB-386235

Fast mixing for anti-ferromagnetic Ising at high temperature

摘录 / EXCERPT: Take a (random) d-regular graph, and consider the anti-ferromagnetic Ising model, textit{i.e.} $J_{ij}=-1$ for all i,j. Can we show show fast mixing $O(n\log n)$ up to ...

open_problem | open | MathDB 页面 | 原文线索

V2-056255 | MATHDB-386238

Diagnostics

摘录 / EXCERPT: 1. Consider the ferromagnetic Ising model on a general graph G. Which two $x,y\in\Omega$ maximize $\|P^t(x,\cdot)-P^t(y,\cdot)\|_{TV}$? More generally, is it true that in any mon...

open_problem | open | MathDB 页面 | 原文线索

V2-056256 | MATHDB-386240

Edge-flip chain on dyadic tilings and rectangular dissections (Sarah Cannon)

摘录 / EXCERPT: For $\lambda=1$, can we establish a polynomial upper-bound (fast-mixing) for the mixing time of the edge-flip chain on dyadic tilings or on rectangular dissections?

open_problem | open | MathDB 页面 | 原文线索

V2-056257 | MATHDB-386245

Cycle + Erdos-Renyi

摘录 / EXCERPT: Prove that for $p=\frac \epsilon n$, $\epsilon>0$ fixed, the nonreversible chain has $t_{\text{mix}}\asymp \log n$. Prove that when $p=\epsilon n^{-\frac 32}$ the nonreversible cha...

open_problem | open | MathDB 页面 | 原文线索

V2-056258 | MATHDB-386246

Nonreversible vs. reversible chains

摘录 / EXCERPT: Is it always true that $t_{\text{mix}}^P\lesssim \tilde t_{\text{mix}}$? Is the same also true of the cover time?

open_problem | open | MathDB 页面 | 原文线索

V2-056259 | MATHDB-386248

Comparison of interchange to SRW and Exclusion

摘录 / EXCERPT: Consider the interchange process with n particles on n vertices. Can you say the following? ...

open_problem | open | MathDB 页面 | 原文线索

V2-056260 | MATHDB-386251

Cutoff for random-to-random shuffle

摘录 / EXCERPT: The random-to-random card shuffle on S_n has cutoff at $\frac {3n}4 \log n$.

open_problem | open | MathDB 页面 | 原文线索

V2-056261 | MATHDB-386254

Kac's random walk

摘录 / EXCERPT: Does Kac's random walk on $\mathcal S^n$ exhibit cutoff? Does the analogously defined random walk on $SO(n)$ exhibit cutoff?

open_problem | open | MathDB 页面 | 原文线索

V2-056262 | MATHDB-386255

Metropolis on $[0,1]^2$

摘录 / EXCERPT: Obtain a good bound on the mixing time of the Metropolis algorithm for this stationary distribution in terms of A. Generalize this to $[0,1]^d$.

open_problem | open | MathDB 页面 | 原文线索

V2-056263 | MATHDB-386256

Random walk on monotone subsets of $\{0,1\}^n$

摘录 / EXCERPT: Can we establish an upper bound in $O(n\log n)$ for the mixing time ?

open_problem | open | MathDB 页面 | 原文线索

V2-056264 | MATHDB-386257

Function-specific mixing time and concentration

摘录 / EXCERPT: Does there exist an absolute constant $c>0$ such that, for all $\varepsilon>0$ and $n\geq 1$, ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-056265 | MATHDB-386258

Prove, for some reasonable class of edge weights, that the asymptotic shape is strictly convex.

摘录 / EXCERPT: Prove, for some reasonable class of edge weights, that the asymptotic shape is strictly convex. Is the boundary of the asymptotic shape differentiable?

open_problem | open | MathDB 页面 | 原文线索

V2-056266 | MATHDB-386259

Prove the conjectured asymptotics for two dimensional first-passage percolation predicted by the KPZ equation.

摘录 / EXCERPT: Prove the conjectured asymptotics for two dimensional first-passage percolation predicted by the KPZ equation. That is, show that $\chi=1/3$ and $\xi=2/3$, and that travel times (p...

open_problem | open | MathDB 页面 | 原文线索

V2-056267 | MATHDB-386260

Main open problems (problem 3)

摘录 / EXCERPT: Prove that, in two dimensions, $0<\chi<1/2$ and $1/2<\xi<1$.

open_problem | open | MathDB 页面 | 原文线索

V2-056268 | MATHDB-386261

Main open problems (problem 4)

摘录 / EXCERPT: Come up with a convincing heuristic for the behaviour of χ and ξ for higher dimensions, and support the heuristic with rigorous bounds.

open_problem | open | MathDB 页面 | 原文线索

V2-056269 | MATHDB-386262

Main open problems (problem 5)

摘录 / EXCERPT: Rule out the existence of bigeodesics for $d=2$, or for d small. That is, prove that the probability that the origin is visited by a bigeodesic is zero.

open_problem | open | MathDB 页面 | 原文线索

V2-056270 | MATHDB-386263

The time constant and asymptotic shape (problem 1)

摘录 / EXCERPT: For $d=2$, show that Ball is not a square.

open_problem | open | MathDB 页面 | 原文线索

V2-056271 | MATHDB-386264

The time constant and asymptotic shape (problem 2)

摘录 / EXCERPT: Consider Bernoulli distributed edge weights with parameter p. Let $g(p)$ denote the time constant in the coordinate direction for this weight distribution. Is $g(p)$ differentiab...

open_problem | open | MathDB 页面 | 原文线索

V2-056272 | MATHDB-386265

The time constant and asymptotic shape (problem 3)

摘录 / EXCERPT: How does $\mu({\bf e}_1)$ behave as a function of d?

open_problem | open | MathDB 页面 | 原文线索

V2-056273 | MATHDB-386266

The time constant and asymptotic shape (problem 4)

摘录 / EXCERPT: Assume that the edge weights $\{\omega_e\}$ are bounded and that $1/\omega_e$ has polynomial tails. Prove that $\mu({\bf e}_1)\neq0$, or find a counterexample.

open_problem | open | MathDB 页面 | 原文线索

V2-056274 | MATHDB-386267

Fluctuations and scaling relations (problem 1)

摘录 / EXCERPT: Let Γ_n denote the geodesic of winding number 1 on the d-dimensional torus of side length n. Prove that ${\rm Var}(T(\Gamma_n))$ is divergent.

open_problem | open | MathDB 页面 | 原文线索

V2-056275 | MATHDB-386268

Fluctuations and scaling relations (problem 2)

摘录 / EXCERPT: Assume finite exponential moment of the weight distribution. Does there exists a constant $c>0$ such that for $\lambda>0$ ...

open_problem | open | MathDB 页面 | 原文线索

V2-056276 | MATHDB-386269

Fluctuations and scaling relations (problem 3)

摘录 / EXCERPT: For $d=2$, prove anything about the correlation between $T(0,x)$ and $T(0,y)$, for x and y that are close, but far from the origin, and use that to say something about the rati...

open_problem | open | MathDB 页面 | 原文线索

V2-056277 | MATHDB-386270

Can one obtain a comparison principle for ... ?

摘录 / EXCERPT: Can one obtain a comparison principle for ${\rm Var}(T(0,n{\bf e}_1))$?

open_problem | open | MathDB 页面 | 原文线索

V2-056278 | MATHDB-386271

Fluctuations and scaling relations (problem 5)

摘录 / EXCERPT: In the inhomogeneous model for which the pair (F_-,F_+) produces a pyramid' on the asymptotic shape, does that imply that ${\rm Var}(T(0,n{\bf e}_2))$ is order n, and that...

open_problem | open | MathDB 页面 | 原文线索

V2-056279 | MATHDB-386272

Structure of geodesics (problem 1)

摘录 / EXCERPT: Does $\mathbb{P}(0\in{\rm Geo}(-n{\bf e}_1,n{\bf e}_1))\to0$ as $n\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-056280 | MATHDB-386273

Prove that for a given unit vector ... , there is almost surely no bigeodesic with one end directed in that direction.

摘录 / EXCERPT: Prove that for a given unit vector v, there is almost surely no bigeodesic with one end directed in that direction.

open_problem | open | MathDB 页面 | 原文线索

V2-056281 | MATHDB-386274

Structure of geodesics (problem 3)

摘录 / EXCERPT: Show that there exists a geodesic $g=(v_0,v_1,\ldots)$ and a unit direction v such that $v_k/|v_k|$ tends to v as $k\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-056282 | MATHDB-386275

Prove that this empirical measure converges (weakly) to some probability measure.

摘录 / EXCERPT: Prove that this empirical measure converges (weakly) to some probability measure. Is this measure singular with respect to the original law on the edge weights?

open_problem | open | MathDB 页面 | 原文线索

V2-056283 | MATHDB-386276

Structure of geodesics (problem 5)

摘录 / EXCERPT: Show that $\lim_{n\to\infty}\frac{1}{n}|{\rm Geo}(0,n{\bf e}_1)|$ exists almost surely.

open_problem | open | MathDB 页面 | 原文线索

V2-056284 | MATHDB-386277

Structure of geodesics (problem 6)

摘录 / EXCERPT: Let $(e_1,e_2,\ldots,e_m)$ be an enumeration of the edges of the geodesic from the origin to $n{\bf e}_1$. How does $\max_{1\le k\le m}\omega_{e_k}$ scale with n?

open_problem | open | MathDB 页面 | 原文线索

V2-056285 | MATHDB-386278

Structure of geodesics (problem 7)

摘录 / EXCERPT: How does the geodesic ${\rm Geo}(0,n{\bf e}_1)$ change when each edge weight is resampled independently with probability $\varepsilon>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-056286 | MATHDB-386279

Prove (or disprove) that in sufficiently high dimensions the limit … does not exist.

摘录 / EXCERPT: Prove (or disprove) that in sufficiently high dimensions the limit $\lim_{n\to\infty}\big[T(0,n{\bf e}_1)-T({\bf e}_1,n{\bf e}_1)\big]$ does not exist.

open_problem | open | MathDB 页面 | 原文线索

V2-056287 | MATHDB-386282

Discrete Entropy Power Inequalities (problem 1)

摘录 / EXCERPT: Open problems: begin{enumerate} item textbf{Entropy power inequality} … item textbf{Monotonicity in the CLT} … end{enumerate}

open_problem | open | MathDB 页面 | 原文线索

V2-056288 | MATHDB-386283

Entropy Power Inequality Extensions (problem 1)

摘录 / EXCERPT: Consider independent random vectors, … with X and Y uncorrelated. The problem is to find a function f universal over the laws of the vectors in the above such that … The prob…

open_problem | open | MathDB 页面 | 原文线索

V2-056289 | MATHDB-386284

Entropy Power Inequality Extensions (problem 2)

摘录 / EXCERPT: Let $X_1 \sim p_1(X_1; \theta_1)$ defined for all $\theta_1 \in \mathbb{R}$ and $X_2 \sim p_2(X_2; \theta_2)$ defined for all $\theta_2 \in \mathbb{R}$ be random elements of a gene…

open_problem | open | MathDB 页面 | 原文线索

V2-056290 | MATHDB-386285

Entropy Power Inequality Extensions (problem 3)

摘录 / EXCERPT: For X, Y, Z independent vectors in $\mathbb{R}^d$ having densities, we define the standard multivariate entropy power as $N(X) \triangleq \exp\left(\frac{2}{d} h(X)\right)$. The…

open_problem | open | MathDB 页面 | 原文线索

V2-056291 | MATHDB-386286

Entropy Power Inequality Extensions (problem 4)

摘录 / EXCERPT: By Zamir-Feder, for any matrix A with … with components X_i independent and $\tilde{X}_i$ i.i.d. Gaussian such that $h(X_i) = h(\tilde{X}_i)$ for $1 \leq i \leq n$, then … The…

open_problem | open | MathDB 页面 | 原文线索

V2-056292 | MATHDB-386287

Entropy Power Inequality Extensions (problem 5)

摘录 / EXCERPT: Let $[n] = \{1, 2, \ldots, n\}$. We define a d-dimensional entropy power region as follows: … Then we let the entropy power region be … Let $\mathcal{G}$ be a collection of nonem…

open_problem | open | MathDB 页面 | 原文线索

V2-056293 | MATHDB-386288

Geometric Inequalities (problem 1)

摘录 / EXCERPT: Let $K, L \subset \mathbb{R}^n$ be convex bodies. Question 1: If γ is a standard Gaussian measure and $0 \leq \lambda \leq 1$, is it true that … if K and L are origin-sy…

open_problem | open | MathDB 页面 | 原文线索

V2-056294 | MATHDB-386295

Miscellaneous (problem 7)

摘录 / EXCERPT: We know that the following sub-modularity property of the entropy is true: if $X, Y,$ and Z are independent, real-valued random variables, then … We consider a conjecture relatin…

open_problem | open | MathDB 页面 | 原文线索

V2-056295 | MATHDB-386296

Miscellaneous (problem 8)

摘录 / EXCERPT: Let $X_1, X_2, \ldots X_n$ be i.i.d. with mean 0 and variance 1. Define … have density f_n and let the standard normal density be ϕ. If X_i has finite Poincar'{e} cons…

open_problem | open | MathDB 页面 | 原文线索

V2-056296 | MATHDB-386297

Bounds on ground state enery in the Sherrington-Kirkpatrick model

摘录 / EXCERPT: Give a polynomial time algorithm to certify better upper or lower bounds on this problem.

open_problem | open | MathDB 页面 | 原文线索

V2-056297 | MATHDB-386298

Random CSP phase transitions for refutation via semidefinite programs

摘录 / EXCERPT: Show that there exists some constant $\epsilon > 0$ such that, for random NAE-3-SAT with $\alpha > \alpha_s + \epsilon$, the basic SDP relaxation still has value 1.

open_problem | open | MathDB 页面 | 原文线索

V2-056298 | MATHDB-386299

Algorithms for NAE-k-SAT below α_d

摘录 / EXCERPT: Are there polynomial-time algorithms for NAE-k-SAT for large k when $\alpha \le \alpha_d$?

open_problem | open | MathDB 页面 | 原文线索

V2-056299 | MATHDB-386300

Bridging the gap for planted NAE-k-SAT solutions

摘录 / EXCERPT: Bridge the gap, by demonstrating an algorithm for $\alpha_{\ell} <\alpha < \alpha_u$. Better yet, demonstrate a single algorithm which works at any density.

open_problem | open | MathDB 页面 | 原文线索

V2-056300 | MATHDB-386301

Greedy algorithms for Planted Clique

摘录 / EXCERPT: Prove that, when $k \le \sqrt{n}$, the algorithm fails with high probability.

open_problem | open | MathDB 页面 | 原文线索

V2-056301 | MATHDB-386302

Recovering planted solutions to Random CSPs

摘录 / EXCERPT: Is there any planted random CSP for which there is an algorithm that finds a solution in time $n \log^{O(1)} n$ that is closer to the planted solution (as measured in expected Hamm...

open_problem | open | MathDB 页面 | 原文线索

V2-056302 | MATHDB-386303

Planted clique with bounded edge queries

摘录 / EXCERPT: What is the largest clique we can find if we only look at $O(n\log^{O(1)} n)$ edges, but can use an arbitrary amount of time? (By modifying the greedy algorithm, we can get a cliqu...

open_problem | open | MathDB 页面 | 原文线索

V2-056303 | MATHDB-386304

Balanced colorings past criticality

摘录 / EXCERPT: Does there exist a feasible coloring for the final graph, G_T, which is imbalanced, so that one color class has size at least $(\frac{1}{3} + \epsilon)n$ for some constant...

open_problem | open | MathDB 页面 | 原文线索

V2-056304 | MATHDB-386305

Is the satisfiability threshold equal to the reconstruction threshold on trees for your favorite model or random CSP?

摘录 / EXCERPT: Is the satisfiability threshold equal to the reconstruction threshold on trees for your favorite model or random CSP?

open_problem | open | MathDB 页面 | 原文线索

V2-056305 | MATHDB-386306

Spin Glasses (problem 3)

摘录 / EXCERPT: Suppose that A is the adjacency matrix of a d-regular random graph, and let $X \subset \{-1,1\}^n$ be the subset of balanced sign vectors,...

open_problem | open | MathDB 页面 | 原文线索

V2-056306 | MATHDB-386307

Solution Geometry (problem 1)

摘录 / EXCERPT: Prove that the condensation threshold for 3-colorability on an Erdos-Renyi graph $G(n,p)$ is $p = 4/n$.

open_problem | open | MathDB 页面 | 原文线索

V2-056307 | MATHDB-386308

In the Sherrington-Kirkpatrick model, is the overlapping distribution on the instance almost surely atomic?

摘录 / EXCERPT: In the Sherrington-Kirkpatrick model, is the overlapping distribution on the instance almost surely atomic? The same for max-CSP at large density, or at low but finite temperature.

open_problem | open | MathDB 页面 | 原文线索

V2-056308 | MATHDB-386316

Two approaches to generalize Stein's method to non-Gaussian distributions are the following: ...

摘录 / EXCERPT: Two approaches to generalize Stein's method to non-Gaussian distributions are the following: begin{itemize} item[(i)] Stein coefficients: for a random variable Y, the emph{Stein...

open_problem | open | MathDB 页面 | 原文线索

V2-056309 | MATHDB-386319

Is $RBM_{4,3}$ a universal approximator?

摘录 / EXCERPT: Does the closure of $RBM_{4,3}$ fill the simplex $\Delta_{15},$ making it a textit{universal approximator}?

open_problem | open | MathDB 页面 | 原文线索

V2-056310 | MATHDB-386320

Determine the maximum divergence of $RBM_{3,1}.$

摘录 / EXCERPT: Determine the maximum divergence of $RBM_{3,1}.$

open_problem | open | MathDB 页面 | 原文线索

V2-056311 | MATHDB-386321

What kind of distributions can be represented by an RBM as opposed to directed models?

摘录 / EXCERPT: What kind of distributions can be represented by an RBM as opposed to directed models?

open_problem | open | MathDB 页面 | 原文线索

V2-056312 | MATHDB-386327

Given a fixed connectivity structure, how many inference functions can an RBM model compute?

摘录 / EXCERPT: How many inference functions can an RBM model compute, constrained by a fixed connectivity structure?

open_problem | open | MathDB 页面 | 原文线索

V2-056313 | MATHDB-386339

Deterministic criteria for asymptotics of real roots

摘录 / EXCERPT: For any $\epsilon>0$, give sufficient, explicit (efficiently checkable) conditions, holding with probability $\ge 1-\epsilon$ for a sequence of iid standard Gaussians, on a determi...

open_problem | open | MathDB 页面 | 原文线索

V2-056314 | MATHDB-386340

Almost sure convergence for Kac polynomials

摘录 / EXCERPT: Let $N_n(\mathbb R)$ be the number of real zeros for the Kac polynomial of degree n with iid standard normal coefficients. It is known that $N_n(\mathbb R) / \log n \to 2/\pi$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-056315 | MATHDB-386341

Sub-leading order for $\mathbb E N_n(\mathbb R)$

摘录 / EXCERPT: Let $N_n(\mathbb R)$ be the number of real zeros for a Kac polynomial with coefficients that are iid copies of a real variable ξ. Show...

open_problem | open | MathDB 页面 | 原文线索

V2-056316 | MATHDB-386342

Number of real roots (problem 4)

摘录 / EXCERPT: For the standard Gaussian Kac polynomial $P_n(z)$ with real coefficients, let $p_n(k)$ be the probability that the k roots of largest modulus are all real. i) Show...

open_problem | open | MathDB 页面 | 原文线索

V2-056317 | MATHDB-386343

Variance and CLT for real roots of random orthogonal polynomials

摘录 / EXCERPT: Let p_k be the orthonormal polynomials for a measure ν on $\mathbb R$ that is absolutely continuous with respect to Lebesgue measure and whose support is a compact interval....

open_problem | open | MathDB 页面 | 原文线索

V2-056318 | MATHDB-386344

Number of real roots (problem 6)

摘录 / EXCERPT: For various sequences of polynomials $P_n(z)$ with independent real coefficients we have ... Give general conditions on sequences of coefficients (in particular on the sequence of th...

open_problem | open | MathDB 页面 | 原文线索

V2-056319 | MATHDB-386345

Intermediate growth of $\mathbb E N_n(\mathbb R)$?

摘录 / EXCERPT: Fix $\epsilon\in (0,1/2)$. Find a bounded sequence $c_k\in \mathbb R$ such that the expected number of real roots of $\sum_{0\le k\le n} c_k \xi_k z^k$ is $n^{\epsilon+o(1)}$, wher...

open_problem | open | MathDB 页面 | 原文线索

V2-056320 | MATHDB-386346

Other properties of the zero measures (problem 1)

摘录 / EXCERPT: Let $P_n(z) = \sum_{0\le k\le n} a_k R_k^{(n)}(z)$, where the a_k are iid random variables satisfying $\mathbb E \log(1+|a_0|)<\infty$, and $R_k^{(n)}$ are orthonormal under the...

open_problem | open | MathDB 页面 | 原文线索

V2-056321 | MATHDB-386347

Other properties of the zero measures (problem 2)

摘录 / EXCERPT: For a sequence of random orthogonal polynomials P_n, fix a set $E\subset \mathbb C$ compactly contained in the complement of the limiting support of the empirical measure of zero...

open_problem | open | MathDB 页面 | 原文线索

V2-056322 | MATHDB-386348

Other properties of the zero measures (problem 3)

摘录 / EXCERPT: Let $P_n(z)$ be a random Kac polynomial with iid coefficients that are not necessarily Gaussian. Consider the rescaled point process of zeros in a region within $O(1/n)$ of the uni...

open_problem | open | MathDB 页面 | 原文线索

V2-056323 | MATHDB-386349

Universality for local statistics of zeros in the bulk

摘录 / EXCERPT: Consider an array of polynomials $\{p_{n,k}\}_{1\le k\le n<\infty }$ where for each n, $\{p_{n,k}\}_{1\le k\le n}$ is an orthonormal set with respect to the inner product...

open_problem | open | MathDB 页面 | 原文线索

V2-056324 | MATHDB-386350

Other properties of the zero measures (problem 5)

摘录 / EXCERPT: Let $P_n(z) = \sum_{0\le k\le n} \xi_k p_{n,k}(z)$, where the ξ_k are iid non-degenerate random variables (with some additional conditions), and $\{p_{n,k}\}_{1\le k\le n}$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-056325 | MATHDB-386351

Minimum modulus for Kac polynomials

摘录 / EXCERPT: Consider random Kac polynomials $P_n(z) = \sum_{0\le k\le n} \xi_k z^k$ with ξ_k iid standard Gaussians (real or complex), and put $m_n:= \min_{|z|=1} |P_n(z)|$. Does...

open_problem | open | MathDB 页面 | 原文线索

V2-056326 | MATHDB-386353

Maximum modulus for log-correlated polynomials on the circle

摘录 / EXCERPT: Let $P_n(z) = \sum_{0\le k\le n} \frac1{\sqrt{k}} \xi_k z^k$ with ξ_k iid centered random variables of variance one and sub-exponential tails, and put...

open_problem | open | MathDB 页面 | 原文线索

V2-056327 | MATHDB-386355

Is there a way to have more than one walk together form a partially exchangeable process?

摘录 / EXCERPT: Is there a way to have more than one walk together form a partially exchangeable process? - this might not be the best hypothesis. [ref: R. Dan Mauldin]

open_problem | open | MathDB 页面 | 原文线索

V2-056328 | MATHDB-386359

For which sigma models (choice of target space ...) is there an isomorphism theorem involving a self-interacting walk which is a process?

摘录 / EXCERPT: For which sigma models (choice of target space M) is there an isomorphism theorem involving a self-interacting walk which is a process? Known examples are hemisphere, hyperbolic...

open_problem | open | MathDB 页面 | 原文线索

V2-056329 | MATHDB-386366

Does a spin system on another symmetric space have a random walk representation?

摘录 / EXCERPT: Does a spin system on another symmetric space have a random walk representation?

open_problem | open | MathDB 页面 | 原文线索

V2-056330 | MATHDB-386368

Understand once-reinforced random walk and/or a martingale variant.

摘录 / EXCERPT: Understand once-reinforced random walk and/or a martingale variant.

open_problem | open | MathDB 页面 | 原文线索

V2-056331 | MATHDB-386369

Cover, relaxation, and hitting time for reinforced walks.

摘录 / EXCERPT: Cover, relaxation, and hitting time for reinforced walks.

open_problem | open | MathDB 页面 | 原文线索

V2-056332 | MATHDB-386370

Questions (problem 1)

摘录 / EXCERPT: Quantify the convergence rate (in terms of the Wasserstein distance, or KL-divergence) for the two-layer mean-field limit diffusion equation ... to its steady-state solution as...

open_problem | open | MathDB 页面 | 原文线索

V2-056333 | MATHDB-386371

How can we extend the mean-field model from two layers to three or more layers?

摘录 / EXCERPT: How can we extend the mean-field model from two layers to three or more layers?

open_problem | open | MathDB 页面 | 原文线索

V2-056334 | MATHDB-386372

What can we say about truncations of the neural tangent hierarchy?

摘录 / EXCERPT: What can we say about truncations of the neural tangent hierarchy? See https://arxiv.org/abs/1909.11304.

open_problem | open | MathDB 页面 | 原文线索

V2-056335 | MATHDB-386374

How do we quantify the intrinsic difficulty of a classification problem (besides by how big the margin of the set of interest is).

摘录 / EXCERPT: How do we quantify the intrinsic difficulty of a classification problem (besides by how big the margin of the set of interest is).

open_problem | open | MathDB 页面 | 原文线索

V2-056336 | MATHDB-386376

Questions (problem 7)

摘录 / EXCERPT: Is there a phase transition between lazy learning (NTK regime, $\alpha=1$) and feature-based regimes ($\alpha=n^{-1/2}$)? Can we characterize at which value of α we switch b...

open_problem | open | MathDB 页面 | 原文线索

V2-056337 | MATHDB-386380

How does data distribution/geometry/group invariances impinge on the questions we are raising above?

摘录 / EXCERPT: How does data distribution/geometry/group invariances impinge on the questions we are raising above? The work of Stephane Mallat may be relevant.

open_problem | open | MathDB 页面 | 原文线索

V2-056338 | MATHDB-386383

Questions (problem 14)

摘录 / EXCERPT: How can we design architecture and optimization to train well emph{in practice}?

open_problem | open | MathDB 页面 | 原文线索

V2-056339 | MATHDB-386384

Do solutions of elliptic/parabolic PDEs live in Barron spaces?

摘录 / EXCERPT: Do solutions of elliptic/parabolic PDEs live in Barron spaces? What about solutions to Burgers?

open_problem | open | MathDB 页面 | 原文线索

V2-056340 | MATHDB-386385

How does uncertainty (noise in labels or nonexact solutions) affect these results?

摘录 / EXCERPT: How does uncertainty (noise in labels or nonexact solutions) affect these results?

open_problem | open | MathDB 页面 | 原文线索

V2-056341 | MATHDB-386386

How to use deep learning to solve PDEs in the viscosity sense (Hamilton-Jacobi-Bellman and Isoacs equations, e.g.) where solutions are not classical and need a mechanism to select viscosity solution?

摘录 / EXCERPT: How to use deep learning to solve PDEs in the viscosity sense (Hamilton-Jacobi-Bellman and Isoacs equations, e.g.) where solutions are not classical and need a mechanism to select...

open_problem | open | MathDB 页面 | 原文线索

V2-056342 | MATHDB-386387

Questions (problem 18)

摘录 / EXCERPT: Solving hyperbolic conservation laws with NN? How to preserve symmetry of the physic? Also invariance with respect to symmetrcies.

open_problem | open | MathDB 页面 | 原文线索

V2-056343 | MATHDB-386388

Questions (problem 19)

摘录 / EXCERPT: How to design+ train NN to be adversarial robust? How to select training data to improve robustness?

open_problem | open | MathDB 页面 | 原文线索

V2-056344 | MATHDB-386389

2-dimensional ASEP

摘录 / EXCERPT: begin{itemize} item Study the two-dimensional asymmetric simple exclusion process (ASEP) on the lattice $\mathbb{Z}^2$, where particles attempt hops to the four cardinal directions...

open_problem | open | MathDB 页面 | 原文线索

V2-056345 | MATHDB-386390

Isotropic 2d KPZ

摘录 / EXCERPT: The isotropic two-dimensional KPZ problem remains largely open. A central challenge is to identify the fluctuation exponents and to characterize the fixed point process that would...

open_problem | open | MathDB 页面 | 原文线索

V2-056346 | MATHDB-386391

Anisotropic models

摘录 / EXCERPT: begin{itemize} item In contrast to the isotropic case, anisotropic KPZ models,, are expected to exhibit different scaling behaviors. A key question is whether these models converge...

open_problem | open | MathDB 页面 | 原文线索

V2-056347 | MATHDB-386392

There has been success in finding exact formulas for PNG with arbitrary initial conditions.

摘录 / EXCERPT: There has been success in finding exact formulas for PNG with arbitrary initial conditions. Can this be extended to multilayer PNG? What class of initial conditions can we consider...

open_problem | open | MathDB 页面 | 原文线索

V2-056348 | MATHDB-386393

Find the scaling limit of the multilayer PNG in the spirit of KPZ FP ,.

摘录 / EXCERPT: Find the scaling limit of the multilayer PNG in the spirit of KPZ FP ,.

open_problem | open | MathDB 页面 | 原文线索

V2-056349 | MATHDB-386394

More generally, consider multilayer models such as the multi-path partition functions in the O'Connell-Yor semi-discrete directed Brownian polymer, see Theorem~3.1 and subsequent SDEs in, Definition~4

摘录 / EXCERPT: More generally, consider multilayer models such as the multi-path partition functions in the O'Connell-Yor semi-discrete directed Brownian polymer, see Theorem3.1 and subsequent SD...

open_problem | open | MathDB 页面 | 原文线索

V2-056350 | MATHDB-386395

Log-concavity (problem 1)

摘录 / EXCERPT: Consider continuous-time TASEP with step initial condition. There is an associated transition probability function $f(a)$ into the configuration a. A question arises whether...

open_problem | open | MathDB 页面 | 原文线索

V2-056351 | MATHDB-386396

The density of the Tracy--Widom random variable is itself log-concave.

摘录 / EXCERPT: The density of the Tracy--Widom random variable is itself log-concave. For a large permutation of size n, the longest increasing subsequence length L_n often (after suitable ce...

open_problem | open | MathDB 页面 | 原文线索

V2-056352 | MATHDB-386397

Compute the large-time current fluctuations for the XXZ spin- … chain on a 1D lattice.

摘录 / EXCERPT: Compute the large-time current fluctuations for the XXZ spin-$1/2$ chain on a 1D lattice. Some of the progress is the following: The edge fluctuations for $\Delta=0$ with domain wa...

open_problem | open | MathDB 页面 | 原文线索

V2-056353 | MATHDB-386398

Can one derive an analogue of the TASEP speed process, for colored particle systems (TASEP or ASEP, for starters) evolving on a ring?

摘录 / EXCERPT: Can one derive an analogue of the TASEP speed process, for colored particle systems (TASEP or ASEP, for starters) evolving on a ring?

open_problem | open | MathDB 页面 | 原文线索

V2-056354 | MATHDB-386399

Does a color-position symmetry exist in particle systems on a ring?

摘录 / EXCERPT: Does a color-position symmetry exist in particle systems on a ring? (Model particular case: Half-open systems?) For models on the full space or in half-space geometries, such symme...

open_problem | open | MathDB 页面 | 原文线索

V2-056355 | MATHDB-386400

Can the dynamics of colored particles on a ring be recast in terms of an (affine) Hecke algebra structure?

摘录 / EXCERPT: Can the dynamics of colored particles on a ring be recast in terms of an (affine) Hecke algebra structure?

open_problem | open | MathDB 页面 | 原文线索

V2-056356 | MATHDB-386401

Is there an analogue of color-position symmetry within the framework of LPP models?

摘录 / EXCERPT: Is there an analogue of color-position symmetry within the framework of LPP models? Recent works on “hidden invariance” cited above hint at deep invariance properties that might ex...

open_problem | open | MathDB 页面 | 原文线索

V2-056357 | MATHDB-386402

Within the known stationary measures of ASEP, can we investigate correlations?

摘录 / EXCERPT: Within the known stationary measures of ASEP, can we investigate correlations? Consider the asymmetric simple exclusion process (ASEP) on a large ring with two particle colors, wit...

open_problem | open | MathDB 页面 | 原文线索

V2-056358 | MATHDB-386403

Longest common subsequence (problem 1)

摘录 / EXCERPT: Consider two independent random sequences (strings) of i.i.d. Bernoulli(0,1) variables, or more generally, i.i.d. variables taking values in a finite alphabet. The main question is...

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V2-056359 | MATHDB-386404

Longest common subsequence (problem 2)

摘录 / EXCERPT: One central object of study is the almost-sure growth rate (or emph{time constant}) of the LCS as the string lengths grow. Determining this constant precisely remains an open probl...

open_problem | open | MathDB 页面 | 原文线索

V2-056360 | MATHDB-386405

Take a vertical slice of the KPZ line ensemble, how close is the permutation of ... to the identity permutation?

摘录 / EXCERPT: Take a vertical slice of the KPZ line ensemble, how close is the permutation of $\mathbb{Z}_{\ge1}$ to the identity permutation? Is the permutation finite at all? In the Airy line...

open_problem | open | MathDB 页面 | 原文线索

V2-056361 | MATHDB-386406

The Airy line ensemble and its Gibbs property play a central role in integrable probability.

摘录 / EXCERPT: The Airy line ensemble and its Gibbs property play a central role in integrable probability. One intriguing question is what happens when random path segments are resampled using t...

open_problem | open | MathDB 页面 | 原文线索

V2-056362 | MATHDB-386407

Beyond the original construction via Brownian last passage percolation, alternate descriptions of the DL have begun to emerge, leveraging tools such as Busemann functions, Airy line ensembles, and sta

摘录 / EXCERPT: Beyond the original construction via Brownian last passage percolation, alternate descriptions of the DL have begun to emerge, leveraging tools such as Busemann functions, Airy lin...

open_problem | open | MathDB 页面 | 原文线索

V2-056363 | MATHDB-386408

Can one can relax the iid Bernoulli random walks that underlie the nontrivial structure in integrable models like PNG or TASEP, in formulas around the KPZ FP, ?

摘录 / EXCERPT: Can one can relax the iid Bernoulli random walks that underlie the nontrivial structure in integrable models like PNG or TASEP, in formulas around the KPZ FP, ? Another example inc...

open_problem | open | MathDB 页面 | 原文线索

V2-056364 | MATHDB-386409

Characterizing stochastic processes that one can put into formulas instead of random walks.

摘录 / EXCERPT: Characterizing stochastic processes that one can put into formulas instead of random walks.

open_problem | open | MathDB 页面 | 原文线索

V2-056365 | MATHDB-386410

Find an explicit formula or expression for large deviation principles for current fluctuation in the case of ASEP on general graphs.

摘录 / EXCERPT: Find an explicit formula or expression for large deviation principles for current fluctuation in the case of ASEP on general graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-056366 | MATHDB-386412

#7 MST ReCom

摘录 / EXCERPT: Running ReCom with random minimal spanning trees (i.e., drawing trees using Kruskal's algorithm with random weights): (a) Can we modify the algorithm so that we get a nice (approxi...

open_problem | open | MathDB 页面 | 原文线索

V2-056367 | MATHDB-386414

#19 ReCom rapid mixing on restricted graph families

摘录 / EXCERPT: Can we prove ReCom mixes rapidly on some families of graphs? Triangular/square lattice? Complete graphs? Dense random graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-056368 | MATHDB-386415

#20 Cycle walk rapid mixing on restricted graph families

摘录 / EXCERPT: Can we prove the cycle walk mixes rapidly on some families of graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-056369 | MATHDB-386417

#4 Splittability and duality on 3D grids

摘录 / EXCERPT: Can we extend the polynomial splittability bound to 3D grids? Specifically, can we get a $\frac{1}{\text{poly}(n)}$ lower bound on splittability into $k = 2$ components of exactly...

open_problem | open | MathDB 页面 | 原文线索

V2-056370 | MATHDB-386419

#6 Component sizes on the infinite grid

摘录 / EXCERPT: Consider UST on the infinite grid lattice. If you remove any edge, with probability 1 you get a finite and an infinite component. What does the distribution over finite component s...

open_problem | open | MathDB 页面 | 原文线索

V2-056371 | MATHDB-386420

#11 What makes a tree or graph splittable?

摘录 / EXCERPT: Star graphs are not splittable, paths are. What other structures of trees make it splittable or not splittable? What determines whether a graph is splittable or not splittable? Is...

open_problem | open | MathDB 页面 | 原文线索

V2-056372 | MATHDB-386421

#21 Splittability with non-constant weights

摘录 / EXCERPT: Are trees still splittable with polynomial probability when node populations differ by more than a constant amount asymptotically?

open_problem | open | MathDB 页面 | 原文线索

V2-056373 | MATHDB-386422

#23 MST vs UST splittability

摘录 / EXCERPT: For a given family of graphs (like grids, complete graphs), what is the assymptotic probability a uniformly random spanning tree versus a random minimal spanning tree is 2-splittab...

open_problem | open | MathDB 页面 | 原文线索

V2-056374 | MATHDB-386423

#32 Improving bound from $1/N^2$ to $1/N$ on grid graphs

摘录 / EXCERPT: In a uniformly random spanning tree of a grid graph with N vertices, we know the probability of having an exactly balanced split edge is at least $\frac{1}{N^2}$. Can we improve...

open_problem | open | MathDB 页面 | 原文线索

V2-056375 | MATHDB-386424

#33 ReCom transition function complexity

摘录 / EXCERPT: Does the ReCom transition function run in polynomial time? Or can it get stuck in configurations where it is very unlikely a random tree will be splittable? Can we at least show th...

open_problem | open | MathDB 页面 | 原文线索

V2-056376 | MATHDB-386426

#8 UST speedup

摘录 / EXCERPT: Is there a faster algorithm to draw an (approximately) uniformly random spanning tree, perhaps on well-behaved graph families?

open_problem | open | MathDB 页面 | 原文线索

V2-056377 | MATHDB-386427

#10 Uniform partition sampling

摘录 / EXCERPT: Can we sample a uniformly random balanced graph partition into two pieces in polynomial time on well-behaved graphs? Specifically, can we do this on grid graphs? Wesley Pegden has...

open_problem | open | MathDB 页面 | 原文线索

V2-056378 | MATHDB-386428

#22 Recursive splitting balance constraints

摘录 / EXCERPT: When recursively splitting trees to build a balanced partition, what population balance constraints should be enforced in the earlier steps to get approximate balance at the end?

open_problem | open | MathDB 页面 | 原文线索

V2-056379 | MATHDB-386429

#31 Most likely MST on grids

摘录 / EXCERPT: Can we find the MST probability maximizer (with iid random weights) in, say, an 8×8 grid graph? Can we approximately optimize this numerically?

open_problem | open | MathDB 页面 | 原文线索

V2-056380 | MATHDB-386432

#14 Partition similarity via spanning forests

摘录 / EXCERPT: Can the collection of spanning forests on a pair of graph partitions be used to make define a notion of distance between them or characterize their similarity?

open_problem | open | MathDB 页面 | 原文线索

V2-056381 | MATHDB-386433

#15 Inference from adjacency data

摘录 / EXCERPT: What properties of the geography or political geography can be read off the graph adjacency data alone?

open_problem | open | MathDB 页面 | 原文线索

V2-056382 | MATHDB-386434

#16 Computational tools to explain collections of real maps

摘录 / EXCERPT: Given a collection of redistricting maps, how can you find a distribution or family of distributions explaining these maps? (a) Can we find a good regularization to make the proble...

open_problem | open | MathDB 页面 | 原文线索

V2-056383 | MATHDB-386435

#17 Models of graphs arising in real geographic data

摘录 / EXCERPT: - (a) What is a good generative model? (b) Perturbed Delaunay triangulations of random point sets seem to work very well. You add edges with preferential attachment then delete edg...

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V2-056384 | MATHDB-386436

#18 Beyond redistricting: processor assignment application

摘录 / EXCERPT: Is there a many-body problem where these kinds of partitioning algorithms can give a speedup when used for processor assignment? This is a dynamic problem where processors can only...

open_problem | open | MathDB 页面 | 原文线索

V2-056385 | MATHDB-386437

#24 SLE for ReCom

摘录 / EXCERPT: Is there a scaling limit for the distribution over partitions generated from running ReCom on finer and finer grids on the unit square.

open_problem | open | MathDB 页面 | 原文线索

V2-056386 | MATHDB-386438

#25 Perimeter in the limit

摘录 / EXCERPT: Fix $P_1, P_2, \dots, P_L$ partitions of the unit square into 2 pieces. Compute ... where $\tau_n(P)$ is the number of tree-weighted partitions of the $n \times n$ grid graph that lo...

open_problem | open | MathDB 页面 | 原文线索

V2-056387 | MATHDB-387859

Research Problems in Function Theory - Problem 7.6

摘录 / EXCERPT: We consider the range of the random function ... (F chosen at random in the natural way) defined in $\mathbb{D}$, where $\sum|a_n|^2=\infty$. Is the image of $w=F(z)$ with probabil...

open_problem | open | MathDB 页面 | 原文线索

V2-056388 | MATHDB-390597

Oliveira’s Kac-walk mixing conjecture

摘录 / EXCERPT: Let $\mu_{n,T}$ be the law on $\mathrm{SO}(n)$ of Kac's walk after T steps, and let Haar_n denote Haar measure on $\mathrm{SO}(n)$. The conjecture asks whether,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056389 | MATHDB-391916

Conjectured uniqueness of Brownian loop measure for SLE restriction

摘录 / EXCERPT: For every admissible central charge c, the Brownian-loop measure is the unique loop measure whose associated restriction function satisfies the relevant conformal covariance, res...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056390 | MATHDB-391923

Technical-condition question for asymptotic optimality in two-stage sample robust optimization

摘录 / EXCERPT: ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-056391 | MATHDB-391929

Ancheta-Massey problem

摘录 / EXCERPT: Let $X^n=(X_1,5e5c,5e5c,X_n)$ have independent coordinates distributed as $\operatorname{Bernoulli}(p)$, with $0<p<\tfrac12$, and let distortion be normalized Hamming distortion. D...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056392 | MATHDB-392101

Breiman’s 1965 conjecture

摘录 / EXCERPT: Let $(X_i)_{i\ge1}$ be iid centered integrable random variables, let $(Y_i)_{i\ge1}$ be iid nonnegative random variables independent of (X_i), and set $S_n=\sum_{i=1}^nY_i$. If,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056393 | MATHDB-392142

Minimax-optimal alternating regret problem

摘录 / EXCERPT: Determine the minimax-optimal alternating-regret rates for online learning: (i) for online linear optimization over the probability simplex Δ_d over a time horizon T, dete...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-056394 | MATHDB-392269

Optimal leading coefficient problem for AWGN spherical codes

摘录 / EXCERPT: For every pair of integers n,k satisfying $3\leq k\leq n-1$, the optimal leading coefficient in the high-SNR maximum-likelihood error-probability expansion for SNR-dependent...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056395 | MATHDB-392278

Xiao’s open question on Hausdorff measures of collision sets

摘录 / EXCERPT: Let $X=(X_t)_{t\ge 0}$ and $Y=(Y_t)_{t\ge 0}$ be independent Markov processes taking values in a common metric measure space (E,d,μ). Define the collision-time and collision-po...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-056396 | MATHDB-392300

Yu-Tan conjectures on exact common information and exact channel synthesis

摘录 / EXCERPT: Let (X,Y) be jointly Gaussian with $X,Y\sim\mathcal{N}(0,1)$ and correlation coefficient $\rho\in(0,1)$. The exact common information of (X,Y) is conjectured to satisfy...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056397 | MATHDB-392301

Fyodorov’s conjecture on permanental roots of Gaussian random matrices

摘录 / EXCERPT: Let H_N be an $N\times N$ random matrix from the standard Gaussian unitary ensemble or Gaussian orthogonal ensemble, and let $p_N(z)=\operatorname{Per}(zI_N-H_N)$. Define the nor...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056398 | MATHDB-392463

Ellipsoid fitting conjecture

摘录 / EXCERPT: Let $x_1,\ldots,x_n\in\mathbb{R}^d$ be independent standard Gaussian vectors, and let $n=n(d)$ as $d\to\infty$. The ellipsoid fitting conjecture predicts a sharp threshold at...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056399 | MATHDB-392466

Geman-Horowitz question on local time of Weierstrass functions

摘录 / EXCERPT: Let $W_{a,b}(t)=\sum_{n=0}^{\infty}a^n\cos(b^n t)$ be a classical Weierstrass function, with $0<a<1$ and $b>1$ in the usual nowhere-differentiable regime. Determine for which param...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-056400 | MATHDB-392654

Testing the joint random-walk and no-GARCH reduction

摘录 / EXCERPT: For the model ... where the ε_i are independent and identically distributed with expectation 0, variance 1, and finite third and fourth moments, test the null hypothe...

open_problem | open | MathDB 页面 | 原文线索

V2-056401 | MATHDB-392748

Probability of disputed elections under direct election versus the Electoral College

摘录 / EXCERPT: Is it the case that ... Here, $V_{1,N}$ and $V_{2,N}$ are the nationwide popular-vote totals of the leading candidate and runner-up; $\{s_j\}$ is a set of states with one or more mem...

open_problem | open | MathDB 页面 | 原文线索

V2-056402 | MATHDB-392825

Optimal stopping for powers of a random walk’s positive part

摘录 / EXCERPT: Let $n\ge 1$, let $(\xi_1,\xi_2,\ldots)$ be i.i.d. random variables with $\mathbb{E}\xi_1<0$, and define $X_k=x+\xi_1+\cdots+\xi_k$. For the gain function $g(x)=(x^+)^n$, consider...

open_problem | open | MathDB 页面 | 原文线索

V2-056403 | MATHDB-392864

Extension of the boosted-sampling framework to multiple inflation parameters

摘录 / EXCERPT: Can the boosted-sampling framework be extended to handle stochastic problems with multiple inflation parameters, where different parts of the problem change in different ways?

open_problem | open | MathDB 页面 | 原文线索

V2-056404 | MATHDB-392873

Existence of a continuous optimal transport for the Monge problem in $\mathbb{R}^2$

摘录 / EXCERPT: Let μ^+ and μ^- be absolutely continuous probability measures on $\mathbb{R}^2$, with compact, convex, disjoint supports K^+ and K^-. Assume that the densities of...

open_problem | open | MathDB 页面 | 原文线索

V2-056405 | MATHDB-392921

Effect of interval information on asymptotic minimax loss

摘录 / EXCERPT: Assume that the unknown Bernoulli probability satisfies $p\in[a,b]$ with $0\le a<b\le 1$. The report states that this prior information does not diminish the asymptotic minimax los...

open_problem | open | MathDB 页面 | 原文线索

V2-056406 | MATHDB-392996

Existence and uniqueness for a max-type stochastic fixed-point equation

摘录 / EXCERPT: Let $s>0$, let $A_r,b_r\in L^s$, let $\mu_0\in M$, and define $\zeta_s:=\mathbb{E}\sum_{r=1}^k|A_r|^s<1$. If $\ell_s(\mu_0,T\mu_0)<\infty$, then the stochastic equation ... where the...

open_problem | open | MathDB 页面 | 原文线索

V2-056407 | MATHDB-393014

Realizability above density $1/4$ for a lattice pair-correlation function

摘录 / EXCERPT: Let $g:\d4a4\to\{0,1\}$ be given by ... Under the exclusion condition that each site of $\mathbb Z$ contains at most one particle, ρ is the one-site occupation probability and...

open_problem | open | MathDB 页面 | 原文线索

V2-056408 | MATHDB-393369

Recurrence of linearly edge-reinforced random walk on integer lattices

摘录 / EXCERPT: Is the linearly edge-reinforced random walk on $\mathbb{Z}^d$ recurrent for all $d>1$?

open_problem | open | MathDB 页面 | 原文线索

V2-056409 | MATHDB-393370

Triangle problem for strongly edge-reinforced random walk

摘录 / EXCERPT: Let G be a triangle and let $(I_n,(X_n^e)_{e\in E})_{n\ge 0}$ be the edge-W-reinforced random walk with reinforcement weights $W:\mathbb{N}_0\to(0,\infty)$ and bounded init...

open_problem | open | MathDB 页面 | 原文线索

V2-056410 | MATHDB-393638

Consistency of Bayes factors based on chi-squared statistics for multinomial data

摘录 / EXCERPT: With the notation and regularity conditions of Johnson (2005), let p be a multinomial probability vector satisfying the null hypothesis, and suppose that under the alternative hy...

open_problem | open | MathDB 页面 | 原文线索

V2-056411 | MATHDB-393651

Supercritical percolation limit for a discrete membrane with randomly distributed defects

摘录 / EXCERPT: For the random bond-percolation model described by the discrete membrane energy on a bounded open set $\Omega\subset\mathbb{R}^2$, if $p>1/2$, then the Γ-limit F^ω is...

open_problem | open | MathDB 页面 | 原文线索

V2-056412 | MATHDB-393656

Nearly optimal complexity for stochastic diffusion moment computation

摘录 / EXCERPT: For the stochastic diffusion problem $-\operatorname{div}(a(\cdot,\omega)\nabla u(\cdot,\omega))=f(\cdot,\omega)$ in $H^{-1}(D)$, with homogeneous boundary conditions, and for accu...

open_problem | open | MathDB 页面 | 原文线索

V2-056413 | MATHDB-393665

Provably near-optimal dual-balancing policies for stochastic inventory control models

摘录 / EXCERPT: For the broad class of stochastic inventory-control models considered in the report-including the periodic-review problem with a finite horizon, correlated and non-stationary evolv...

open_problem | open | MathDB 页面 | 原文线索

V2-056414 | MATHDB-393756

Conjecture on the singularity probability of random Bernoulli matrices

摘录 / EXCERPT: Let M_n be an $n\times n$ random matrix whose entries are independent and identically distributed, each taking the values -1 and 1 with probability $1/2$. Let p_n denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-056415 | MATHDB-393784

Unimodality of the Ehrhart h-vector of the Birkhoff polytope

摘录 / EXCERPT: For the Birkhoff polytope P of real doubly stochastic $n\times n$ matrices, define $E(P,m)=|mP\cap\mathbb{Z}^{n\times n}|$ for $m>0$ and $E(P,0)=1$, and write its Ehrhart series...

open_problem | open | MathDB 页面 | 原文线索

V2-056416 | MATHDB-394158

Alexander-Orbach behavior for spread-out oriented percolation in dimensions 5 and 6

摘录 / EXCERPT: For the IIC of spread-out oriented percolation on $\mathbb{Z}^d\times\mathbb{Z}_+$, does the conclusion $d_s(\widetilde C)=4/3$ remain valid in the unresolved dimensions $d=5,6$ (m...

open_problem | open | MathDB 页面 | 原文线索

V2-056417 | MATHDB-394159

Hypothesis A for the IIC of ordinary percolation

摘录 / EXCERPT: Prove that the incipient infinite cluster (IIC) for ordinary percolation satisfies Hypothesis A: there exist constants $C_2>0$ and $q>0$ such that, for every $r\in[1,\infty)$ and e...

conjecture | open | MathDB 页面 | 原文线索

V2-056418 | MATHDB-394160

Universality of critical percolation on other two-dimensional lattices

摘录 / EXCERPT: Does the scaling limit of critical percolation in other two-dimensional lattices exist and coincide with the known scaling limit, and, in particular, does Cardy’s formula hold in g...

open_problem | open | MathDB 页面 | 原文线索

V2-056419 | MATHDB-394162

Uniform positive enhancement gap for planar self-destructive percolation

摘录 / EXCERPT: For self-destructive site percolation on the square lattice, there exists a $\[1m\[0m\delta>0\[0m$ such that $\[1m\[0m\theta(p,\delta)=0\[0m$ for every $p>p_c$.

open_problem | open | MathDB 页面 | 原文线索

V2-056420 | MATHDB-394164

Determining the critical inverse temperature for directed polymers

摘录 / EXCERPT: For the directed polymer model with simple random walk on $\mathbb{Z}^d$ and i.i.d., non-constant environment variables $\eta(n,x)$ having finite exponential moments, the free ener...

open_problem | open | MathDB 页面 | 原文线索

V2-056421 | MATHDB-394199

Sharp transition for bootstrap percolation on finite grids

摘录 / EXCERPT: For fixed integers d and k with $2\le k\le d$, let $p_\rho(n)$ be the infimum of p for which k-neighbour bootstrap percolation on the d-dimensional grid $[1,n]^d$, starte...

open_problem | open | MathDB 页面 | 原文线索

V2-056422 | MATHDB-394364

Orthogonal bisection of a convex body and probability measures

摘录 / EXCERPT: Let $K \subseteq \mathbb{R}^2$ be a convex body of area 1, and let $0<t<1/4$. Do there exist two orthogonal lines that subdivide K into four regions of areas t, t, $1/2-t$,...

open_problem | open | MathDB 页面 | 原文线索

V2-056423 | MATHDB-394393

A multifactor generalisation of the Olivier-Smith model for stochastic mortality

摘录 / EXCERPT: Give a more detailed and flexible specification of the joint distribution of the dependent Gamma random variables $G(t+1,T,x)$ in the generalized Olivier-Smith model, subject to ……

open_problem | open | MathDB 页面 | 原文线索

V2-056424 | MATHDB-394472

Breaking point of the accelerated diameter-3 game

摘录 / EXCERPT: We suspect that, for the diameter-3 game $D_3(3:b)$, the breaking point should be $b_0 \approx n^{2/3}\times\operatorname{polylog}(n)$, in accordance with the probabilistic intuiti...

open_problem | open | MathDB 页面 | 原文线索

V2-056425 | MATHDB-394548

Deterministic normalization of the critical branching-random-walk martingale

摘录 / EXCERPT: Consider a branching random walk on the real line, with particle positions $V(u)$, satisfying $\\sup_{|u|=1}|V(u)|\le C$ for some $C>0$ and … for some $\delta>0$. Define … Assume...

open_problem | open | MathDB 页面 | 原文线索

V2-056426 | MATHDB-394551

Absolute continuity of near-critical percolation scaling limits with respect to SLE(6)

摘录 / EXCERPT: For fixed $\varepsilon>0$, define $p^*(\varepsilon,N)=\inf\{p:R(p,N)>1/2+\varepsilon\}$ and take $p=p^*(\varepsilon,N)$. For possible subsequential scaling limits of the interface...

open_problem | open | MathDB 页面 | 原文线索

V2-056427 | MATHDB-394553

Aging probability for the p-spin trap-model dynamics

摘录 / EXCERPT: Let $A_N^{\varepsilon}(t,s)=\{R_N(\sigma_N(te^{\gamma N}),\sigma_N((t+s)e^{\gamma N}))\ge 1-\varepsilon\}$. Under the hypothesis of Theorem 1, for every $\varepsilon\in(0,1)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-056428 | MATHDB-394580

Characterization of testable graph properties via regularity

摘录 / EXCERPT: Let a graph property P be called testable if there is a randomized algorithm that, for every $\[1m\varepsilon>0\[0m$, distinguishes with probability at least $2/3$ between graphs...

open_problem | open | MathDB 页面 | 原文线索

V2-056429 | MATHDB-394588

The probability that a small perturbation of a numerical analysis problem is difficult

摘录 / EXCERPT: Theorem 1. Let C be a conic condition number with set of ill-posed inputs $ ... and ... Here $zin B(a,sigma)$ is uniformly distributed in the spherical cap of radius $arcsinsigma...

open_problem | open | MathDB 页面 | 原文线索

V2-056430 | MATHDB-394786

RSW theorem for anisotropic percolation

摘录 / EXCERPT: Is there a RSW theorem for anisotropic percolation?

open_problem | open | MathDB 页面 | 原文线索

V2-056431 | MATHDB-394787

Power-law behavior of the percolation probability near criticality

摘录 / EXCERPT: For independent site or bond percolation on a periodic graph in two dimensions, let $\theta(p)=\mathbb{P}_p(|C_0|=\infty)$ and $p_c=\sup\{p:\theta(p)=0\}$. Is it true that, as...

open_problem | open | MathDB 页面 | 原文线索

V2-056432 | MATHDB-394788

Triangle condition for nearest-neighbor percolation above the upper critical dimension

摘录 / EXCERPT: For bond percolation on the nearest-neighbor lattice $\mathbb{Z}^d$ with $d>6$, the critical triangle diagram ... is finite, where $\mathit{p_c}$ is the critical probability and...

open_problem | open | MathDB 页面 | 原文线索

V2-056433 | MATHDB-394789

Nonvanishing critical Voronoi rectangle-crossing probability

摘录 / EXCERPT: In planar Voronoi percolation at the critical parameter $p=1/2$, let $h(\rho,s)$ be the probability that there is a black path crossing the rectangle $[0,\rho s]\times[0,s]$ horizo...

open_problem | open | MathDB 页面 | 原文线索

V2-056434 | MATHDB-394790

Percolation in the L-hierarchical model

摘录 / EXCERPT: Let $L\ge 2$ be an integer. Define $\widetilde{\xi}^{L}_j:=\max\{k\in\mathbb Z_{+}:L^k\mid j\}$, and obtain the L-hierarchical binary sequence $(\xi^L_j)_j$ by replacing each val...

open_problem | open | MathDB 页面 | 原文线索

V2-056435 | MATHDB-394793

RSW-type a priori square-crossing bound for anisotropic critical bond percolation

摘录 / EXCERPT: Consider independent bond percolation on the square lattice with horizontal-edge parameter p_h and vertical-edge parameter p_v, where $p_h+p_v=1$. For $H(a,b)$ denoting the eve...

open_problem | open | MathDB 页面 | 原文线索

V2-056436 | MATHDB-394795

Can the anisotropic percolation rectangle bound yield a nontrivial limiting aspect ratio?

摘录 / EXCERPT: Consider bond percolation on the square lattice with horizontal and vertical edge parameters p_h and p_v, respectively, satisfying $p_h+p_v=1$. Let $H(a,b)$ be the event that t...

open_problem | open | MathDB 页面 | 原文线索

V2-056437 | MATHDB-394796

Anisotropic two-point connectivity inequality

摘录 / EXCERPT: Consider an anisotropic independent bond percolation model on $\mathbb{Z}_+^2$ in which each vertical edge is open with probability p and each horizontal edge is open with probab...

open_problem | open | MathDB 页面 | 原文线索

V2-056438 | MATHDB-395063

Strong coupling from a Stein coefficient

摘录 / EXCERPT: Suppose W is a random variable with $\\mathbb{E}(W)=0$ and $\\mathbb{E}(W^2)<\\infty$. Let T be another random variable, defined on the same probability space as W, such that...

open_problem | open | MathDB 页面 | 原文线索

V2-056439 | MATHDB-395147

Reverse Wasserstein contraction inequality for the geodesic Radon transform

摘录 / EXCERPT: For the geodesic Radon transform R on probability measures on S^{n-1}, is there a uniform inequality in the direction opposite to the contraction estimate-equivalently, a facto...

open_problem | open | MathDB 页面 | 原文线索

V2-056440 | MATHDB-395150

Quantitative equidistribution for random walks on the torus

摘录 / EXCERPT: Theorem 3 (Quantitative). Let ν and $\Gamma=\langle\operatorname{supp}(\nu)\rangle<\mathrm{SL}_d(\mathbb Z)$ be as in Theorem 2. Then there exist $c>0$ and $k<\infty$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-056441 | MATHDB-395328

Permanent of doubly stochastic matrices with bounded Frobenius norm

摘录 / EXCERPT: Let gammage 1 be a fixed constant. Is it true that ... provided that $A=(a_{ij})$ is an $n\times n$ doubly stochastic matrix satisfying ... Here...

open_problem | open | MathDB 页面 | 原文线索

V2-056442 | MATHDB-395378

Singleton random attractor for a stochastic porous media equation

摘录 / EXCERPT: In the setup of Theorem 3.1, let $O\subset\mathbb{R}^d$ be open and bounded with $d\ge 1$, let $p\in(1,\infty)$, set $\Psi(s)=s|s|^{p-1}$, and fix $m\in\mathbb{N}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-056443 | MATHDB-395434

Scaling limits for min-driven clustering

摘录 / EXCERPT: Let $t_0>0$, let F_{t_0} be an arbitrary probability measure on $(0,\infty)$ with positive minimum, and let F_t ($t\ge t_0$) be the associated measure-valued solution of (2).nn...

open_problem | open | MathDB 页面 | 原文线索

V2-056444 | MATHDB-395496

Unique stationary distribution for the limiting Aldous-move diffusion

摘录 / EXCERPT: The uniform distribution on metric probability measure spaces will turn out to be the unique stationary distribution of the limit diffusion under construction.

open_problem | open | MathDB 页面 | 原文线索

V2-056445 | MATHDB-395569

The isoperimetric inequality for Riesz capacities for $2<\alpha<d$

摘录 / EXCERPT: Let $d\ge 2$ and let $2<\alpha<d$. For every compact set $A\subset\mathbb{R}^d$, define its α-Riesz capacity by ... where the infimum is over all probability Borel measures su...

open_problem | open | MathDB 页面 | 原文线索

V2-056446 | MATHDB-395696

Brenier's solution to the optimal transport problem in the Euclidean case

摘录 / EXCERPT: For probability measures μ and ν on $\mathbb{R}^d$ with finite second moment, and cost $c(x,y)=\frac{\lVert x-y\rVert^2}{2}$, there exists a unique solution to the associat...

open_problem | open | MathDB 页面 | 原文线索

V2-056447 | MATHDB-395744

Expectation characterization for monotone distributions on the continuous cube

摘录 / EXCERPT: Let D be a probability distribution on $[0,1]^n$ with density function f, and suppose D is monotone: whenever $x_i\le y_i$ for every i, one has $f(x)\le f(y)$. If D is...

open_problem | open | MathDB 页面 | 原文线索

V2-056448 | MATHDB-395749

Erdős-Palka conjecture on induced trees in sparse random graphs

摘录 / EXCERPT: For any constant $c>1$, the random graph $G(n,c/n)$ would with high probability contain an induced tree of order $\b3(c)n$.

conjecture | open | MathDB 页面 | 原文线索

V2-056449 | MATHDB-395753

Cutoff for simple random walk on a random regular graph

摘录 / EXCERPT: Let $G\sim\mathcal{G}(n,d)$ be a random regular graph for fixed $d\ge 3$. Then, with high probability, the simple random walk on G exhibits cutoff at $\frac{d}{d-2}\log_{d-1} n$...

open_problem | open | MathDB 页面 | 原文线索

V2-056450 | MATHDB-395757

Critical threshold for two-neighbour bootstrap percolation on the two-dimensional grid

摘录 / EXCERPT: There exist constants $C>c>0$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-056451 | MATHDB-395758

Improve the upper bound for zero-temperature Glauber dynamics on \(\mathbb Z^2\)

摘录 / EXCERPT: The report records the best-known bounds for the critical probability of zero-temperature Glauber dynamics on $\mathbb Z^2$: ... Improve the upper bound.

open_problem | open | MathDB 页面 | 原文线索

V2-056452 | MATHDB-395761

A constructive version of the Lovász Local Lemma

摘录 / EXCERPT: Let $\mathcal P=\{P_1,P_2,\ldots,P_n\}$ be a finite set of independent random variables over a probability space, and let $\mathcal A=\{A_1,A_2,\ldots,A_m\}$ be a collection of eve...

open_problem | open | MathDB 页面 | 原文线索

V2-056453 | MATHDB-395769

Convergence time of random better-response dynamics in uncoordinated two-sided markets

摘录 / EXCERPT: There exists a family of two-sided markets in which the random better-response dynamics takes, with high probability, exponentially many steps in the size of the graph to reach a s...

open_problem | open | MathDB 页面 | 原文线索

V2-056454 | MATHDB-395942

Precise conditions for ballisticity of random walks in random environment

摘录 / EXCERPT: What are the precise conditions that ensure ballisticity for random walks in random environment (RWRE)?

open_problem | open | MathDB 页面 | 原文线索

V2-056455 | MATHDB-395946

Gyration-radius bound for long-range self-avoiding walk and oriented percolation

摘录 / EXCERPT: Let $\alpha>0$, and let D be the $\mathbb Z^d$-symmetric one-step distribution described on pages 26-27, satisfying ... for some $v_\alpha\in(0,\infty)$ and $\varepsilon>0$. For th...

open_problem | open | MathDB 页面 | 原文线索

V2-056456 | MATHDB-395947

Critical percolation with free boundary conditions on finite high-dimensional boxes

摘录 / EXCERPT: What is the behavior of critical bond percolation at $p=p_c$ on a finite d-dimensional box with free boundary conditions?

open_problem | open | MathDB 页面 | 原文线索

V2-056457 | MATHDB-395948

Critical percolation torus results with the sharp threshold dimension $d_0=6$

摘录 / EXCERPT: For critical Bernoulli bond percolation on the d-dimensional torus $T_r^d=(\mathbb Z/r\mathbb Z)^d$, the conclusions of Theorems 1-3, namely the estimates (1)-(4) for the sizes,...

open_problem | open | MathDB 页面 | 原文线索

V2-056458 | MATHDB-395950

Equivalence of ballisticity conditions for RWRE

摘录 / EXCERPT: Consider a random walk in a uniformly elliptic i.i.d. random environment in dimensions $d\ge 2$. For every $\gamma\in(0,1)$ and every direction $l\in S^{d-1}$, are...

open_problem | open | MathDB 页面 | 原文线索

V2-056459 | MATHDB-396090

Stability conjecture for the Prékopa-Leindler inequality

摘录 / EXCERPT: Fix n and $\[?]$. Let $f,g,m:\mathbb{R}^n\to[0,\infty)$ be integrable, with f and g probability distributions having zero mean, and suppose ... Define ... Is there a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-056460 | MATHDB-396134

Does the closure characterization remain valid for multiple-copy majorization?

摘录 / EXCERPT: Let P_d be the set of probability vectors in dimension d, and let $x,y\in P_d$ have nonzero coordinates. Write $x\prec y$ for majorization, and write $x\prec_M y$ if there exis...

open_problem | open | MathDB 页面 | 原文线索

V2-056461 | MATHDB-396412

Volume-growth criterion for stochastic completeness on general Riemannian manifolds

摘录 / EXCERPT: For a connected Riemannian manifold M, let $B(x,r)$ be the closed metric ball, $V(x,r)=\operatorname{vol}_n(B(x,r))$, and $A(x,r)=\operatorname{vol}_{n-1}(\partial B(x,r))$. Grig...

open_problem | open | MathDB 页面 | 原文线索

V2-056462 | MATHDB-396476

Regularity of optimal transport maps on compact Riemannian manifolds

摘录 / EXCERPT: Let M be a compact n-dimensional Riemannian manifold with positive bounded measurable probability densities ρ and $\bar\rho$, and let $T:M\to M$ be the unique optimal map...

open_problem | open | MathDB 页面 | 原文线索

V2-056463 | MATHDB-396663

Spectral-gap equality for the interchange and single-particle chains

摘录 / EXCERPT: Let n be positive, and let transition probabilities $(\alpha_{ij})_{1\le i<j\le n}$ satisfy $\alpha_{ij}\ge 0$ and $\sum_{1\le i<j\le n}\alpha_{ij}=1$. Consider the Markov chain...

open_problem | open | MathDB 页面 | 原文线索

V2-056464 | MATHDB-396665

Completion time of the ADVOCATE protocol

摘录 / EXCERPT: Starting with each user having one unique piece, the ADVOCATE protocol operating under soft constraints finishes in n time steps with probability converging to one as...

open_problem | open | MathDB 页面 | 原文线索

V2-056465 | MATHDB-396666

Universality classes for first-passage percolation on the complete graph

摘录 / EXCERPT: What are the universality classes for first-passage percolation (FPP) on the complete graph?

open_problem | open | MathDB 页面 | 原文线索

V2-056466 | MATHDB-396667

Stability of the stochastic Deterministic First Chunk system

摘录 / EXCERPT: For the two-chunk “Deterministic First Chunk” peer-to-peer file-sharing system described in the report, the stochastic system is conjectured, on the basis of simulation evidence, t...

open_problem | open | MathDB 页面 | 原文线索

V2-056467 | MATHDB-396668

Stability of the stochastic Enforced Friedman file-sharing system

摘录 / EXCERPT: For the two-chunk peer-to-peer file-sharing system described in the report, prove stability of the stochastic system using the Enforced Friedman algorithm: a peer contacts three pe...

open_problem | open | MathDB 页面 | 原文线索

V2-056468 | MATHDB-396672

Harmonic functions for heavy-tailed ordered random walks

摘录 / EXCERPT: Let $W=\{x\in\mathbb{R}^k:x_1<x_2<\cdots<x_k\}$ and let S be a k-dimensional random walk with independent identically distributed coordinates. In the heavy-tailed regime...

open_problem | open | MathDB 页面 | 原文线索

V2-056469 | MATHDB-396710

Convergence of finite-difference approximations to stochastic Burgers-type SPDEs

摘录 / EXCERPT: Let $\nu,\sigma>0$, let ξ be space-time white noise, and impose periodic boundary conditions on $[0,2\pi]$. For $a,b\ge 0$, define ... For the approximating equation ... the report...

open_problem | open | MathDB 页面 | 原文线索

V2-056470 | MATHDB-396711

Correction-limit conjecture for finite-difference approximations of a stochastic Burgers-type SPDE

摘录 / EXCERPT: For the periodic SPDE on $[0,2\pi]$ with $\nu,\sigma>0$ and space-time white noise ξ, consider the approximation ... where $a,b\ge 0$. The conjecture is that, as...

conjecture | open | MathDB 页面 | 原文线索

V2-056471 | MATHDB-396715

Boundary terms for vanishing-viscosity stochastic perturbations

摘录 / EXCERPT: The bulk part J^0 of the dynamic functional ([17]) governs large deviations for a vanishing-viscosity stochastic perturbation of equation (2). Can one add boundary conditions to...

open_problem | open | MathDB 页面 | 原文线索

V2-056472 | MATHDB-396718

Efficiently handling metastability in non-equilibrium steady states

摘录 / EXCERPT: How can one efficiently deal with metastability in non-equilibrium steady states, namely probability distributions defined as the stationary measure of a stochastic dynamics?

open_problem | open | MathDB 页面 | 原文线索

V2-056473 | MATHDB-396829

Right-convergence of sparse Erdős-Rényi graphs

摘录 / EXCERPT: For every fixed $c>0$, does the sequence of sparse Erdős-Rényi graphs $G(n,c/n)$ right-converge with high probability; that is, for every finite weighted graph H with a symmetric...

open_problem | open | MathDB 页面 | 原文线索

V2-056474 | MATHDB-396865

Idempotent probability measures on arbitrary binary systems

摘录 / EXCERPT: Conjecture 1. If $(S,\ast)$ is any binary system, then there is an idempotent probability measure on S.

open_problem | open | MathDB 页面 | 原文线索

V2-056475 | MATHDB-396978

TCC implies existence of the p-adic quantum-field-theory measure

摘录 / EXCERPT: For the p-adic BMS model, one conjectures that the transverse convergence criterion ... implies that the limiting probability measure exists, namely ...

open_problem | open | MathDB 页面 | 原文线索

V2-056476 | MATHDB-397058

Unimodal category of a uniform Dyck path

摘录 / EXCERPT: Let $f:[0,2n]\to\u000a\mathbb{R}_+$ be the linearly interpolated simple random walk with n steps $+1$ and n steps -1, conditioned to stay positive (a Dyck path with uniform m...

open_problem | open | MathDB 页面 | 原文线索

V2-056477 | MATHDB-397064

Invertibility and eigenvalue separation for symmetric Wigner matrices

摘录 / EXCERPT: Let H be a real symmetric $n\times n$ random matrix whose above-diagonal entries are independent and identically distributed, with mean zero, unit variance, and sub-Gaussian tail...

open_problem | open | MathDB 页面 | 原文线索

V2-056478 | MATHDB-397066

Sampling inequality for intersections with random subspaces

摘录 / EXCERPT: Let $A\subseteq S^{n-1}$ be a fixed measurable set, and let $E\subseteq\mathbb R^n$ be a random k-dimensional subspace. Write σ_E for the uniform probability measure on...

open_problem | open | MathDB 页面 | 原文线索

V2-056479 | MATHDB-397069

Hyperplane (slicing) conjecture for isotropic log-concave densities

摘录 / EXCERPT: Does there exist a universal constant C such that, for any isotropic log-concave probability density f in $\\mathbb{R}^n$, $f(0)^{1/n} \le C$?

conjecture | open | MathDB 页面 | 原文线索

V2-056480 | MATHDB-397081

Submatrix norm bound without unconditionality

摘录 / EXCERPT: Let A be an $n\times N$ random matrix whose rows are independent isotropic log-concave random vectors in $\mathbb{R}^N$. For $k\le n$ and $m\le N$, define ... The report conjecture...

open_problem | open | MathDB 页面 | 原文线索

V2-056481 | MATHDB-397082

Circular law for random matrices with independent isotropic log-concave rows

摘录 / EXCERPT: Let A_n be a sequence of $n\times n$ random matrices with independent rows $X_1^{(n)},\ldots,X_n^{(n)}$, defined on the same probability space. Assume that, for each n and...

open_problem | open | MathDB 页面 | 原文线索

V2-056482 | MATHDB-397087

Simplicity of the Kontsevich-Zorich spectrum for other invariant measures

摘录 / EXCERPT: Does the Kontsevich-Zorich conjecture-that the Lyapunov spectrum is simple-remain true for other $SL(2,\mathbb R)$-invariant probability measures?

open_problem | open | MathDB 页面 | 原文线索

V2-056483 | MATHDB-397143

Universality of critical arm exponents for selected two-dimensional bond-percolation models

摘录 / EXCERPT: Conditional on the existence of the relevant exponents, the exponents ρ and ρ_k for $k\ge 1$ are constant within the class of inhomogeneous bond-percolation models on the...

open_problem | open | MathDB 页面 | 原文线索

V2-056484 | MATHDB-397277

Behavior of stochastic KdV solitons at the time scale $t=\varepsilon^{-2}$

摘录 / EXCERPT: What is the behavior, at time exactly $t=\varepsilon^{-2}$, of solutions of the stochastic Korteweg-de Vries equations (2) perturbed by space-time noise, including the additive and...

open_problem | open | MathDB 页面 | 原文线索

V2-056485 | MATHDB-397284

Sublinear-time approximation for linear classification

摘录 / EXCERPT: For given $\\delta\in(0,1)$, the new algorithm takes $O(\varepsilon^{-2}(n+d)(\log n)\log(n/\delta))$ time to return an $\\varepsilon$-approximate solution with probability at leas...

open_problem | open | MathDB 页面 | 原文线索

V2-056486 | MATHDB-397307

Optimal covering probability for three independently distributed points in the plane

摘录 / EXCERPT: For three independent random points in $\mathbb{R}^2$, with the points allowed to have different absolutely continuous probability distributions, determine the best universal value...

open_problem | open | MathDB 页面 | 原文线索

V2-056487 | MATHDB-397483

Extrapolation of a stationary stochastic process from a finite time interval

摘录 / EXCERPT: Let $X=(X_t)_{t\\in\mathbb R}$ be a second-order weakly continuous, weakly stationary stochastic process on $(\Omega,\mathcal A,P)$ with $EX_t=0$. For $a>0$, suppose X is observe...

open_problem | open | MathDB 页面 | 原文线索

V2-056488 | MATHDB-397496

Random invariant measures and relative exactness for randomly chosen quadratic maps

摘录 / EXCERPT: Let $T_\lambda:[0,1]\to[0,1]$ be defined by $T_\lambda(x)=\lambda x(1-x)$ for $\lambda\in[1,4]$. Assume that P is a probability measure on $[1,4]$ absolutely continuous with resp...

open_problem | open | MathDB 页面 | 原文线索

V2-056489 | MATHDB-397501

Uniqueness and Monge form for the two-marginal Coulomb optimal-transport problem

摘录 / EXCERPT: For the optimal-transport problem (1), minimizing the Coulomb cost over probability measures on $\mathbb{R}^6$ with both marginals equal to ρ_1, the report states that the sol...

open_problem | open | MathDB 页面 | 原文线索

V2-056490 | MATHDB-397504

Non-exit probability from a growing region for a random walk among random conductances

摘录 / EXCERPT: Let $B\subset\mathbb R^d$ be finite, convex, and open, with $0\in B$, and let α_t be a sufficiently regular scaling function satisfying $1\ll\alpha_t\ll t^{1/(d+2)}$. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-056491 | MATHDB-397815

Learning sets with separating kernels

摘录 / EXCERPT: Given a probability space (X,ρ), a measurable reproducing kernel K on X, and i.i.d. samples $x_1,\ldots,x_n$ drawn from ρ, construct from the samples an estimator...

open_problem | open | MathDB 页面 | 原文线索

V2-056492 | MATHDB-398023

Equality of the small-intensity threshold and the vacant-set critical parameter

摘录 / EXCERPT: For the vacant set V^u of random interlacements on $\mathbb{Z}^d$ with $d\ge 3$, let u^* be the percolation critical parameter and let $\bar u$ be the supremum of intensities...

open_problem | open | MathDB 页面 | 原文线索

V2-056493 | MATHDB-398025

Intensity of loop-erased random walk at $(1,0)$ on the square lattice

摘录 / EXCERPT: For the loop-erased random walk on $\Z^2$ from $(0,0)$ to infinity, the probability that its path passes through $(1,0)$ is ...

open_problem | open | MathDB 页面 | 原文线索

V2-056494 | MATHDB-398027

Percolation of the vacant set of Poisson cylinders in three dimensions

摘录 / EXCERPT: Let a0$\omega=\sum_{i\ge 0}\delta_{\ell_i}$ be a Poisson point process on the space of lines in $\mathbb{R}^3$ with intensity measure $u\mu$, where $u>0$ and μ is, up to a mult...

open_problem | open | MathDB 页面 | 原文线索

V2-056495 | MATHDB-398029

Counting self-avoiding paths on a supercritical percolation cluster

摘录 / EXCERPT: Let $(\omega_e)_{e\in\mathbb E^d}\$ be iid Bernoulli bond percolation on $\mathbb Z^d\$ with $\mathbb P(\omega_e=1)=p>p_c\$. Let $Z_N\$ be the number of self-avoiding nearest-neigh...

open_problem | open | MathDB 页面 | 原文线索

V2-056496 | MATHDB-398032

Benjamini’s isoperimetric Cheeger-constant conjecture for supercritical percolation

摘录 / EXCERPT: For Bernoulli bond percolation on $\mathbb{Z}^d$ with $d\ge 2$ and $p>p_c(d)$, let $\mathbf{C}^n$ be the giant component of the percolation configuration restricted to...

conjecture | open | MathDB 页面 | 原文线索

V2-056497 | MATHDB-398151

High-probability preservation of the variance conjecture under random orthogonal composition

摘录 / EXCERPT: Let X be an isotropic log-concave random vector in $\\mathbb{R}^n$ verifying the variance conjecture with constant C_1. Let $T\in GL(n)$, and let ... where $(\lambda_i)_{i=1}^n$...

conjecture | open | MathDB 页面 | 原文线索

V2-056498 | MATHDB-398166

Distribution of the ψ_2-norm of linear functionals on isotropic convex bodies

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be an isotropic convex body, and define ... where σ is the rotationally invariant probability measure on S^{n-1} and...

open_problem | open | MathDB 页面 | 原文线索

V2-056499 | MATHDB-398216

Characterization of the set of 3-local profiles of large graphs

摘录 / EXCERPT: For a graph G and $i\in\{0,1,2,3\}$, let $p_i(G)$ be the probability that three distinct vertices chosen uniformly at random from G span exactly i edges, and write...

open_problem | open | MathDB 页面 | 原文线索

V2-056500 | MATHDB-398273

A comprehensive multiscale stochastic model for squall-line convection

摘录 / EXCERPT: Develop a reasonably comprehensive mathematical model for the main precipitation band of a strong storm front (squall line), combining boundary-layer and related process descriptio...

open_problem | open | MathDB 页面 | 原文线索

V2-056501 | MATHDB-398340

Turán’s theorem for random graphs

摘录 / EXCERPT: Given an integer $r\ge 3$, for what values of $p=p(n)$ is it true that, with high probability, the maximal size of a K_r-free subgraph of $G_{n,p}$ is equal to the maximal size o...

open_problem | open | MathDB 页面 | 原文线索

V2-056502 | MATHDB-398344

Second-order terms for the largest triangular submatrix of a random matrix

摘录 / EXCERPT: Let $X_n=(x_{ij})$ be an $n\times n$ random matrix with independent entries taking values in $\{0,1\}$ and satisfying $\Pr(x_{ij}=0)=p_0>0$. Set $Q=1/p_0>1$, and let T_n denote t...

open_problem | open | MathDB 页面 | 原文线索

V2-056503 | MATHDB-398349

Whether the critical time t_b equals the percolation threshold

摘录 / EXCERPT: The report conjectures that the critical time t_b coincides with the percolation threshold.

open_problem | open | MathDB 页面 | 原文线索

V2-056504 | MATHDB-398367

Turán’s Theorem for random graphs: equality threshold

摘录 / EXCERPT: For a fixed integer r, for what functions $p=p(n)$ is it true with high probability that $t_r(G_{n,p})=b_r(G_{n,p})$, where $t_r(G)$ is the maximum number of edges in a K_r-fre...

open_problem | open | MathDB 页面 | 原文线索

V2-056505 | MATHDB-398378

Aldous’ critical killed-branching-random-walk conjecture

摘录 / EXCERPT: Consider a branching system in which particles reaching a specified killing subset of $\mathbb{R}^d$-in particular, the negative half-line when $d=1$-are immediately killed and rem...

conjecture | open | MathDB 页面 | 原文线索

V2-056506 | MATHDB-398380

Moderate deviations for the minimum of a branching random walk

摘录 / EXCERPT: For a discrete-time branching random walk with point process $\Theta=\sum_{|u|=1}\delta_{V(u)}$ in the boundary case ... let $M_n=\min_{|u|=n}V(u)$, with $\inf\varnothing=\infty$, an...

open_problem | open | MathDB 页面 | 原文线索

V2-056507 | MATHDB-398381

The Aldous conjecture on a killed branching random walk

摘录 / EXCERPT: Consider a one-dimensional discrete-time branching random walk on $\mathbb{R}$ started from one particle at 0, with first-generation point process L, genealogical tree T, and...

conjecture | open | MathDB 页面 | 原文线索

V2-056508 | MATHDB-398382

Aldous' subcritical killed branching random walk tail conjecture

摘录 / EXCERPT: In the subcritical case, if $\psi'(\rho^*)<0$, then there exists a constant $b>1$ such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-056509 | MATHDB-398387

Near-critical survival probability for branching Brownian motion in a strip

摘录 / EXCERPT: Consider branching Brownian motion in the interval $(0,K)$ in which particles perform standard Brownian motion, are killed on hitting 0 or K, and branch into two offspring at r...

open_problem | open | MathDB 页面 | 原文线索

V2-056510 | MATHDB-398413

Torsion of tightly confined equilateral random walks

摘录 / EXCERPT: The conjecture is that, for an equilateral random walk (an open curve) with n edges of unit length inside a sphere of radius $r\ge 1/2$, the unsigned total torsion tends to...

open_problem | open | MathDB 页面 | 原文线索

V2-056511 | MATHDB-398415

Quenched invariance principle for random walks on conductance clusters in half-spaces and orthants

摘录 / EXCERPT: Fix $d_1,d_2\in\mathbb{Z}_+$ such that $d_1\ge 1$ and $d:=d_1+d_2\ge 2$. Let $L:=\mathbb{Z}_+^{d_1}\times\mathbb{Z}^{d_2}$ and $E_L:=\{\{x,y\}:x,y\in L,\ |x-y|=1\}$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-056512 | MATHDB-398597

Partial regularity for optimal transport maps

摘录 / EXCERPT: Let $X,Y\subset\mathbb{R}^n$ be bounded open sets, and let $f:X\to\mathbb{R}_+$ and $g:Y\to\mathbb{R}_+$ be continuous probability densities, respectively bounded away from zero an...

open_problem | open | MathDB 页面 | 原文线索

V2-056513 | MATHDB-398648

Furstenberg measure rigidity conjecture for multiplication maps on the circle

摘录 / EXCERPT: Let $p,q\ge 2$ be multiplicatively independent integers, let $\mathbb T=\mathbb R/\mathbb Z$, and define $T_r(x)=rx\pmod 1$. Is every Borel probability measure on $\mathbb T$ invar...

conjecture | open | MathDB 页面 | 原文线索

V2-056514 | MATHDB-398682

The Ornstein-Weiss question on Bernoulli shifts

摘录 / EXCERPT: Are all Bernoulli shifts over a countable nonamenable group measurably conjugate? The report says no. More specifically, for a free group G of finite rank and standard probabilit...

open_problem | open | MathDB 页面 | 原文线索

V2-056515 | MATHDB-398709

Equivalence of non-Gibbsianness and multiple minimizing trajectories

摘录 / EXCERPT: Consider the trajectory of the empirical distribution $(\pi_s)_{s\in[0,t]}$ of an interacting particle system, conditioned on ending at $\pi_t=\nu$, where ν is a probability me...

open_problem | open | MathDB 页面 | 原文线索

V2-056516 | MATHDB-398711

Perturbative quenched central limit theorem for small-disorder RWRE in dimension $d\ge 3$

摘录 / EXCERPT: For the nearest-neighbor random walk in random environment on $\mathbb Z^d$, let $\Omega=\mathcal P^{\mathbb Z^d}$, where $\mathcal P$ is the set of probability distributions on th...

open_problem | open | MathDB 页面 | 原文线索

V2-056517 | MATHDB-398716

Quenched functional central limit theorem for planar random walks in random sceneries

摘录 / EXCERPT: Let [?m$\xi=(\xi(x))_{x\in\mathbb Z^d}$ be an i.i.d. random scenery, let $S=(S_n)_{n\ge 0}$ be an independent random walk in $\mathbb Z^d$, and define $Z_n=\sum_{k=1}^n\xi(S_k)$. A...

open_problem | open | MathDB 页面 | 原文线索

V2-056518 | MATHDB-398717

Hydrodynamic limit for the anharmonic chain with artificial viscosity

摘录 / EXCERPT: For the anharmonic chain with artificial-viscosity stochastic dynamics ... where the two Wiener-process families are independent and ... the hyperbolic scaling limit is expected to be...

open_problem | open | MathDB 页面 | 原文线索

V2-056519 | MATHDB-398718

Uniqueness of the hydrodynamic limit for the stochastic anharmonic chain

摘录 / EXCERPT: For the stochastic anharmonic-chain model and empirical process $(u_\varepsilon,v_\varepsilon)$ described on pp. 67-68, under the report’s hypotheses on V (strict convexity, the...

open_problem | open | MathDB 页面 | 原文线索

V2-056520 | MATHDB-398775

Cycle factors and renewal theory

摘录 / EXCERPT: Let G be a random 3-regular graph on n vertices and define ... If $k\ge K_0(n)$ is a divisor of n, then G contains a k-cycle factor with high probability; whereas if ... th...

open_problem | open | MathDB 页面 | 原文线索

V2-056521 | MATHDB-398776

Asymptotics of the fixed-\k cycle-factor threshold in $G(n,p)$

摘录 / EXCERPT: For fixed k, determine the asymptotics of the threshold $p_c(n)$ at which a random ErdősRényi graph $G(n,p)$ contains a k-cycle factor with probability $1/2$. It is known that...

open_problem | open | MathDB 页面 | 原文线索

V2-056522 | MATHDB-398781

Inferring global graph properties from a local profile

摘录 / EXCERPT: Given the probability distribution $\pi_k(G)$, which global conclusions can you draw about the graph G?

open_problem | open | MathDB 页面 | 原文线索

V2-056523 | MATHDB-398803

At least two real eigenvalues of a random sign matrix

摘录 / EXCERPT: Let M_n be a random $\71$ matrix of size n. Prove-or, as conjectured in the report, show-that with probability $1-o(1)$, M_n has at least two real eigenvalues.

open_problem | open | MathDB 页面 | 原文线索

V2-056524 | MATHDB-398820

Pinned cardinal of measure equivalence

摘录 / EXCERPT: Let E be the measure equivalence (connecting probability measures if they have the same ideal of null sets). What is $\boldsymbol{\kappa}(E)$?

open_problem | open | MathDB 页面 | 原文线索

V2-056525 | MATHDB-398944

How many random sumsets have a finite intersection?

摘录 / EXCERPT: Let M be a random set of integers greater than 1, containing each integer n independently with probability $1/n$, and let $S(M)$ be the set of all sums of subsets of M. How...

open_problem | open | MathDB 页面 | 原文线索

V2-056526 | MATHDB-399132

Cyclicity conjecture for Jacobians of Erdős-Rényi random graphs

摘录 / EXCERPT: For any fixed $0<q<1$, the probability that the Jacobian of the Erdős-Rényi random graph $G(n,q)$ is cyclic tends, as n tends to infinity, to ... which is roughly 0.7935.

conjecture | open | MathDB 页面 | 原文线索

V2-056527 | MATHDB-399241

Probable Proposition 2 on a two-sheeted orbifold cover

摘录 / EXCERPT: Let $O=\mathbb{H}^3/\Gamma$ be an orientable hyperbolic 3-orbifold. Suppose that every finite subgroup of Γ is cyclic. Then O has a two-sheeted orbifold cover O' such...

open_problem | open | MathDB 页面 | 原文线索

V2-056528 | MATHDB-399292

Bayes estimator for Bhattacharyya loss on the probability simplex

摘录 / EXCERPT: Let ... For the loss ... the Bayes estimator is ... where $\vec a=(a_0,a_1,\ldots,a_{K-1})$ is the eigenvector associated with the maximal eigenvalue of the matrix whose entries are ... wi...

open_problem | open | MathDB 页面 | 原文线索

V2-056529 | MATHDB-399341

Unstable invariance of accumulation points in the multiple-mixing argument

摘录 / EXCERPT: Let X be a metric space, let $f:X\to X$ be a Borel invertible transformation preserving a probability measure μ, and let W^s,W^u denote the stable and unstable distribution...

open_problem | open | MathDB 页面 | 原文线索

V2-056530 | MATHDB-399440

Correct order of the optimal entropic Ricci-curvature bound for the random transposition model

摘录 / EXCERPT: For the simple random walk on the symmetric group S_n obtained by joining two permutations when they differ by exactly one transposition, with uniform reversible measure...

open_problem | open | MathDB 页面 | 原文线索

V2-056531 | MATHDB-399543

van der Waerden’s permanent conjecture

摘录 / EXCERPT: For an $n\times n$ doubly stochastic matrix A, $\operatorname{per}(A)\ge \frac{n!}{n^n}$, with equality if and only if all entries of A are equal to $\frac{1}{n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-056532 | MATHDB-399548

Anari-Oveis-Gharan theorem for homogeneous strongly Rayleigh measures

摘录 / EXCERPT: Theorem 2 (AO). Let μ be a homogeneous strongly Rayleigh probability measure on $2^{[m]}$ satisfying $\Pr_{S\sim\mu}[i\in S]\le \varepsilon_1$ for every $i=1,\ldots,m$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-056533 | MATHDB-399594

Maximal probability of joint default under a nonsingular transformation

摘录 / EXCERPT: Let F_1 and F_2 be continuous distribution functions. Set $T=F_2\circ F_1^{-1}$. Let ... where the supremum is taken over all random pairs with the given univariate marginals. If...

open_problem | open | MathDB 页面 | 原文线索

V2-056534 | MATHDB-399660

Spectral norm conjecture for structured Gaussian random matrices

摘录 / EXCERPT: Let X be an $n\times n$ symmetric random matrix with independent Gaussian entries $X_{ij}\sim N(0,b_{ij}^{2})$. Define $\sigma^{2}:=\max_i\sum_j b_{ij}^{2}$. The conjecture is ...

conjecture | open | MathDB 页面 | 原文线索

V2-056535 | MATHDB-399666

Best constant in the Lieb-Oxford inequality

摘录 / EXCERPT: Determine the best constant in the Lieb-Oxford inequality ... where E_{Ind} is the indirect Coulomb energy associated with an N-particle symmetric probability distributi...

open_problem | open | MathDB 页面 | 原文线索

V2-056536 | MATHDB-399671

Constrained logarithmic-energy minimization in the Douglas-Kazakov transition

摘录 / EXCERPT: For $T>0$, find the minimum of ... over all Borel probability measures on $\mathbb{R}$ that are absolutely continuous with respect to Lebesgue measure and have density bounded above...

open_problem | open | MathDB 页面 | 原文线索

V2-056537 | MATHDB-399695

Goodness of the Green metric on a hyperbolic group

摘录 / EXCERPT: Let Γ be a hyperbolic group, and let the random walk on Γ be symmetric and supported on a finite generating subset. Define $F(x,y)$ as the probability that the walk s...

open_problem | open | MathDB 页面 | 原文线索

V2-056538 | MATHDB-399733

Local isotropy probability for random quadratic forms over \(\mathbb{Q}_p\)

摘录 / EXCERPT: Let $Q(X_1,\ldots,X_n)=\sum_{1\le i\le j\le n}a_{ij}X_iX_j$ be an n-ary quadratic form with coefficients $a_{ij}\in\mathbb{Z}_p$, chosen with respect to the normalized additive H...

open_problem | open | MathDB 页面 | 原文线索

V2-056539 | MATHDB-399770

Wave-kinetic-equation limit for the stochastic NLS

摘录 / EXCERPT: For the stochastic equation (5), with the scaling $\rho=L^{1/2}\rho_0$ and $b_k=L^{-d/2}b_k^0$ from (7), does the double limit $\lim_{L\to\infty}\lim_{\nu\to0}$ yield weak converge...

open_problem | open | MathDB 页面 | 原文线索

V2-056540 | MATHDB-399773

Shape of the condensing wave in a branching process with selection and mutation

摘录 / EXCERPT: Consider the continuous-time branching-process model in which the mutation probability is $\beta\in[0,1]$, the mutant-fitness law is a probability measure μ on $(0,1)$ with ess...

open_problem | open | MathDB 页面 | 原文线索

V2-056541 | MATHDB-399778

Stochastic homogenization of Ising energies on stationary random lattices

摘录 / EXCERPT: Let $D\subset\mathbb{R}^n$ be bounded and open, let L be an admissible stationary stochastic lattice, and let the interaction coefficients satisfy the bounds and summability assu...

open_problem | open | MathDB 页面 | 原文线索

V2-056542 | MATHDB-399780

Limit shapes for interacting random walks beyond the planar low-temperature regime

摘录 / EXCERPT: For the continuous-time simple random walk on $\mathbb{Z}^d$ with jump rate $2d$, let $R(t)$ be the hull of the vertices visited by time t. For $\beta>0$, equip paths with the Gi...

open_problem | open | MathDB 页面 | 原文线索

V2-056543 | MATHDB-399862

Recurrence of the edge-reinforced random walk on the two-dimensional lattice

摘录 / EXCERPT: The edge-reinforced random walk on $\mathbb{Z}^2$ with constant initial edge weights is recurrent for every choice of the constant initial weight.

open_problem | open | MathDB 页面 | 原文线索

V2-056544 | MATHDB-399866

Quenched crossing probabilities for Voronoi percolation

摘录 / EXCERPT: Let $S=[0,1]^2$, and let $\[1m\eta\[0m$ be a set of n points in S, each chosen independently and uniformly. Form the Voronoi tiling associated with $\[1m\eta\[0m$ inside S, c...

open_problem | open | MathDB 页面 | 原文线索

V2-056545 | MATHDB-399880

Conjugacy problem for measure-preserving transformations

摘录 / EXCERPT: Given two measure-preserving transformations T_1 and T_2, defined on probability spaces $(X_1,\mathcal B_1,m_1)$ and $(X_2,\mathcal B_2,m_2)$, respectively, determine whether t...

open_problem | open | MathDB 页面 | 原文线索

V2-056546 | MATHDB-399988

Feige’s R3SAT refutation hypothesis

摘录 / EXCERPT: There is no efficient algorithm that, for a random 3SAT instance with $m=n^{1.01}$ clauses, produces a refutation with high probability.

conjecture | open | MathDB 页面 | 原文线索

V2-056547 | MATHDB-400105

Asymptotic order of spherical quermassintegral deficits for random polytopes in a halfsphere

摘录 / EXCERPT: Let K be a closed halfsphere in d-dimensional spherical space. Let $X_1,\ldots,X_n$ be stochastically independent random points uniformly distributed in K, and let P_n be t...

open_problem | open | MathDB 页面 | 原文线索

V2-056548 | MATHDB-400244

Alon’s 1000k² bound for random k-universal permutations

摘录 / EXCERPT: Let $f(k)$ be the minimum integer n such that a sequence of n distinct real numbers chosen independently and uniformly from $[0,1]$ is k-universal with high probability, wher...

open_problem | open | MathDB 页面 | 原文线索

V2-056549 | MATHDB-400247

A three-up-set correlation inequality

摘录 / EXCERPT: Let A, B, and C be up-sets (increasing events) in a finite product probability space. Is the inequality … always satisfied?

open_problem | open | MathDB 页面 | 原文线索

V2-056550 | MATHDB-400248

Babai’s simple-spectrum conjecture for $G(n,1/2)$

摘录 / EXCERPT: For the Erdős-Rényi random graph $G(n,1/2)$, the probability that its adjacency matrix has simple spectrum is $1-o(1)$ as $n\to\infty$. Here, a matrix has simple spectrum if all it...

conjecture | open | MathDB 页面 | 原文线索

V2-056551 | MATHDB-400259

The binomial conjecture for degree sequences of random graphs

摘录 / EXCERPT: Let A_n and B_n be probability spaces with the same underlying set. Call them asymptotically quite equivalent (a.q.e.) if, for every sequence of events H_n whose probability...

conjecture | open | MathDB 页面 | 原文线索

V2-056552 | MATHDB-400377

Deviations and convergence rates for the empirical distance-to-measure function

摘录 / EXCERPT: Given a probability distribution P in $[U+007F]\mathbb{R}^d$ and parameters $m\in[0,1]$ and $r\ge 1$, define … where … For observations $X_1,\ldots,X_n$ sampled from P, with empirical...

open_problem | open | MathDB 页面 | 原文线索

V2-056553 | MATHDB-400381

Estimation in tournaments under strong stochastic transitivity

摘录 / EXCERPT: For a tournament of n teams, observe jointly independent $y_{ij}\\sim\operatorname{Bern}(\theta_{ij})$ for $1\le i\le j\le n$, with $y_{ij}=1-y_{ji}$ for $j<i$,...

open_problem | open | MathDB 页面 | 原文线索

V2-056554 | MATHDB-400436

Success probability of Bray’s algorithm for strong involutions

摘录 / EXCERPT: Let G be an n-dimensional finite classical group of odd characteristic, let $t\in G$ be a strong involution, and let $\varepsilon=\pm$. Prove that there is a constant c' such...

open_problem | open | MathDB 页面 | 原文线索

V2-056555 | MATHDB-400482

Finite generators without invariant probability measures for arbitrary Polish actions

摘录 / EXCERPT: Does every continuous action of an arbitrary countable group G on a Polish space X that admits no G-invariant Borel probability measure admit a finite Borel generating partit...

open_problem | open | MathDB 页面 | 原文线索

V2-056556 | MATHDB-400769

Probabilistic forbidden-intersection theorem

摘录 / EXCERPT: Let $0<n^{-1},\delta\ll\kappa,\gamma,\varepsilon,\alpha_1,\alpha_2,\beta_1,\beta_2$. Suppose that μ_q is a κ-bounded product measure on $({\{0,1\}}\times{\{0,1\}})^n$ wi...

open_problem | open | MathDB 页面 | 原文线索

V2-056557 | MATHDB-400787

Kannan-Lovász-Simonovits (KLS) conjecture

摘录 / EXCERPT: There exists a universal constant $c>0$ such that, for every dimension n and every isotropic log-concave probability measure $\bc$ on $\bb^n$, for every $S\subset\u0003bb^n$, … w...

conjecture | open | MathDB 页面 | 原文线索

V2-056558 | MATHDB-400850

Extrapolating the volatility smile to resolve stochasticity from market data

摘录 / EXCERPT: It is an open question whether or not the volatility smile can be extrapolated from visible strikes robustly enough to permit accurate resolution of stochasticity from market data.

open_problem | open | MathDB 页面 | 原文线索

V2-056559 | MATHDB-400854

Critical exponent for finite-time blowup of a stochastic wave equation

摘录 / EXCERPT: For the one-dimensional stochastic wave equation … with two-parameter white noise $\dot W$ and initial conditions … determine whether there is a critical exponent governing finite-...

open_problem | open | MathDB 页面 | 原文线索

V2-056560 | MATHDB-400857

Hedging with price impact

摘录 / EXCERPT: The report considers hedging a European contingent claim in a financial model with temporary price impact. It formulates the hedging task as a linear-quadratic stochastic control p...

open_problem | open | MathDB 页面 | 原文线索

V2-056561 | MATHDB-400860

Relaxing the condition $w(u^+)=w(u^-)$ in affine stochastic-volatility large deviations

摘录 / EXCERPT: Understand whether and how the condition $w(u^+)=w(u^-)$, assumed in Theorems 1 and 2 for an affine stochastic-volatility model with essentially smooth h and $J=[u^-,u^+]$, may b...

open_problem | open | MathDB 页面 | 原文线索

V2-056562 | MATHDB-400883

Conjecture 1 on the largest zero of a mixed characteristic polynomial

摘录 / EXCERPT: Let $P\in D_{m,d}$, where $D_{m,d}$ is the space of homogeneous, doubly stochastic, stable polynomials in $\mathbb{R}[x_1,\ldots,x_m]$ of degree d. Here, doubly stochastic means...

conjecture | open | MathDB 页面 | 原文线索

V2-056563 | MATHDB-400902

Registration and recovery of randomly deformed point processes in dimension $d>1$

摘录 / EXCERPT: Let $\Π_1,\dots,\Π_n$ be iid point processes on $[0,1]^d$, with mean measure $\μ=τ\λ$, where $\λ$ is a probability measure and $τ>0$. Let $T_i:[0,1]^d\to[0,1]^d$ be unobservable ra...

open_problem | open | MathDB 页面 | 原文线索

V2-056564 | MATHDB-400958

Minimal hidden-unit count for universal approximation by binary RBMs

摘录 / EXCERPT: For each number n of visible binary variables, determine the smallest integer m such that every probability distribution on $\\{0,1\\}^n$ can be approximated arbitrarily well b...

open_problem | open | MathDB 页面 | 原文线索

V2-056565 | MATHDB-401062

Equality of the probabilistic LCFT three-point structure constants with the DOZZ formula

摘录 / EXCERPT: The three-point structure constants of the probabilistic construction of Liouville Conformal Field Theory are given by the DOZZ formula.

open_problem | open | MathDB 页面 | 原文线索

V2-056566 | MATHDB-401064

Optimal large-scale corrector estimate in quantitative stochastic homogenization

摘录 / EXCERPT: Let a be a stationary random field on $[U+007F]\mathbb{R}^d$ with values in the symmetric positive-definite $d\times d$ matrices, having finite range of dependence and satisfying, almos...

open_problem | open | MathDB 页面 | 原文线索

V2-056567 | MATHDB-401410

Continuity of the total cost for relativistic transport costs

摘录 / EXCERPT: Let $\mu\ne\nu$ be probability measures with compact support in $\mathbb{R}^N$, let $c_t(x,y)=h\!\left(\frac{y-x}{t}\right)$ be a relativistic cost function as defined in the repor...

open_problem | open | MathDB 页面 | 原文线索

V2-056568 | MATHDB-401425

Stochastic stabilization of nonlinear systems over communication channels

摘录 / EXCERPT: Given the nonlinear system $x_{t+1}=f(x_t,u_t,w_t)$, where $x_t\in\\mathbb{R}^N$, u_t is the control action, and (w_t) is an i.i.d. noise process, controlled over a noisy chann...

open_problem | open | MathDB 页面 | 原文线索

V2-056569 | MATHDB-401436

Two-sided composite-composite testing threshold in a Gaussian mixture model

摘录 / EXCERPT: Let ν be a probability distribution supported on $\mathbb{R}$. Observe n i.i.d. data with distribution $X_i\sim \nu*\mathcal{N}(0,1)$. For $\rho>0$ and $p\in[0,1)$, consider...

open_problem | open | MathDB 页面 | 原文线索

V2-056570 | MATHDB-401449

Optimal weighted least-squares approximations

摘录 / EXCERPT: For a general domain $X\subset\mathbb{R}^d$, a probability measure ρ, an unknown function $u\in L^2(X,\rho)$, and a linear approximation space V_m of dimension m, can inst...

open_problem | open | MathDB 页面 | 原文线索

V2-056571 | MATHDB-401451

The Mathematical Learning Problem

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^d$, let $(\Omega,\mathcal G,\mathbb P)$ be a probability space, and let $X:\Omega\to K$ and $Y:\Omega\to\mathbb{R}^n$ be random vectors. For a Borel measu...

open_problem | open | MathDB 页面 | 原文线索

V2-056572 | MATHDB-401595

Analyticity of the percolation density above the critical point

摘录 / EXCERPT: Is $\theta(p)$ analytic above p_c for nearest-neighbour Bernoulli percolation on $\mathbb{Z}^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-056573 | MATHDB-401614

3-cores of random planar graphs

摘录 / EXCERPT: Show that the 3-core of a random planar graph has a connected component of size cn, for some constant $c>0$, with high probability (probability tending to 1 as $n\to\infty$). H...

open_problem | open | MathDB 页面 | 原文线索

V2-056574 | MATHDB-401819

Analytical properties of probust chance-constraint probability functions

摘录 / EXCERPT: For the probust chance constraint ... where X is a Banach space and ξ is a finite-dimensional continuously distributed random vector, infer analytical properties of ϕ-incl...

open_problem | open | MathDB 页面 | 原文线索

V2-056575 | MATHDB-401820

Statistical optimality for Learning Enabled Optimization

摘录 / EXCERPT: Is there an appropriate analogue, say statistical optimality, which can quantify the probability of a solution being ϵ-optimum?

open_problem | open | MathDB 页面 | 原文线索

V2-056576 | MATHDB-401821

Appropriateness of deterministic optimality in Learning Enabled Optimization

摘录 / EXCERPT: Is deterministic optimality as suggested in (4) an appropriate goal, given that the composition of estimation and stochastic optimization creates a model which may be far too deman...

open_problem | open | MathDB 页面 | 原文线索

V2-056577 | MATHDB-401822

From Data to Decisions: Distributionally Robust Optimization Is Optimal

摘录 / EXCERPT: For stochastic programs in which the decision-maker observes only finitely many independent samples from an unknown distribution of exogenous uncertainties, the report states that...

open_problem | open | MathDB 页面 | 原文线索

V2-056578 | MATHDB-401827

Convergence of projected stochastic gradients in Banach spaces and the nonconvex case

摘录 / EXCERPT: Under what conditions can convergence be shown in Banach spaces and in the nonconvex case?

open_problem | open | MathDB 页面 | 原文线索

V2-056579 | MATHDB-401842

Applicability of the Lipschitz surface construction to percolation clusters

摘录 / EXCERPT: It is however not clear if the Lipschitz surface construction can be applied to percolation clusters or other non-uniformly elliptic settings.

open_problem | open | MathDB 页面 | 原文线索

V2-056580 | MATHDB-401843

Accumulation points of general bootstrap random walks

摘录 / EXCERPT: Let $(X_n)_{n\ge 0}$ be a symmetric simple random walk with increments $\xi_1,\xi_2,\ldots$, where $\mathbb P(\xi_1=\pm1)=1/2$. Let (η_n) be a law-preserving, non-anticipative...

open_problem | open | MathDB 页面 | 原文线索

V2-056581 | MATHDB-401866

Transversal exponent for directed polymers with a heavy-tailed environment when \(\alpha>5\)

摘录 / EXCERPT: Consider the directed polymer in dimension $1+1$ based on a nearest-neighbor simple symmetric random walk S on $\mathbb Z$, with i.i.d. environment...

open_problem | open | MathDB 页面 | 原文线索

V2-056582 | MATHDB-401893

On the singularity probability of random symmetric matrices

摘录 / EXCERPT: Let M_n be a random symmetric $n\times n$ matrix whose diagonal and upper-diagonal entries are independent and identically distributed Bernoulli random variables, each taking the...

open_problem | open | MathDB 页面 | 原文线索

V2-056583 | MATHDB-401910

Vinzant’s Conjecture

摘录 / EXCERPT: Draw $\operatorname{range}(\Phi^*)$ uniformly from $\operatorname{Gr}(\mathbb{C}^{4d-5},d)$, and let p_d denote the probability that the map $[x]\mapsto |\Phi^*x|^2$ is injective...

conjecture | open | MathDB 页面 | 原文线索

V2-056584 | MATHDB-401916

Classification of μ-stationary probability measures

摘录 / EXCERPT: Describe all the μ-stationary measures ν on X.

open_problem | open | MathDB 页面 | 原文线索

V2-056585 | MATHDB-401938

Diameter Bounds for Well-Conditioned Polytopes

摘录 / EXCERPT: Prove a diameter bound for a random polytope of the form $P=\{x\in\mathbb{R}^d:Ax\leq 1\}$, where $A\in\mathbb{R}^{n\times d}$ is a random matrix with independent identically distr...

open_problem | open | MathDB 页面 | 原文线索

V2-056586 | MATHDB-401978

Kernels of sums of selfadjoint variables free over an Abelian algebra

摘录 / EXCERPT: Let $(\mathcal A,E,L,\tau)$ be a tracial operator-valued W^*-noncommutative probability space, with $L\subseteq\mathcal A$ an Abelian von Neumann subalgebra and...

open_problem | open | MathDB 页面 | 原文线索

V2-056587 | MATHDB-402004

Removing the balance assumption for measure-valued Pólya processes

摘录 / EXCERPT: Extend Theorem 2 for measure-valued Pólya processes by removing the balance assumption (b), namely that the family of replacement measures $(R_x)_{x\in P}$ is a probability kernel...

open_problem | open | MathDB 页面 | 原文线索

V2-056588 | MATHDB-402123

Non-concentration of the chromatic number of a random graph

摘录 / EXCERPT: For any constant $\varepsilon>0$, there is no sequence of intervals of length $n^{1/4-\varepsilon}$ which contain $\chi(G_{n,1/2})$ with high probability.

open_problem | open | MathDB 页面 | 原文线索

V2-056589 | MATHDB-402125

Maximum running time of the K_5-bootstrap percolation process

摘录 / EXCERPT: For the graph-bootstrap process on K_n in which, at each round, every edge whose addition creates a new copy of K_5 is infected simultaneously, define $G_0=(V(K_n),E_0)$ and .....

open_problem | open | MathDB 页面 | 原文线索

V2-056590 | MATHDB-402129

Upper-tail probability for triangle counts in sparse random graphs

摘录 / EXCERPT: Let X be the number of triangles in $G_{n,p}$. For fixed $\delta>0$, define ... Then, for every $p=p(n)$ satisfying $n^{-1}\log n\ll p\ll 1$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-056591 | MATHDB-402132

Asymptotics of triangle-count tail probabilities in random graphs

摘录 / EXCERPT: For every $[U+007F]delta>0$, determine the asymptotics of the logarithmic tail probabilities ... when $X=N_{K_3}(G_{n,p})$, the number of triangles in the random graph $G_{n,p}$.

open_problem | open | MathDB 页面 | 原文线索

V2-056592 | MATHDB-402159

Whether the minimax Wasserstein estimation rate can improve on $n^{-1/d}$

摘录 / EXCERPT: Let P be the set of probability distributions supported on $[0,1]^d$, and let ... where $\widehat W$ is constructed from n independent samples from μ and n independent samp...

open_problem | open | MathDB 页面 | 原文线索

V2-056593 | MATHDB-402199

Estimation of influential time points in nonparametric regression with functional predictors

摘录 / EXCERPT: Given an i.i.d. sample (X_i,Y_i), $i=1,\ots,n$, where $X_i=\{X_i(t):t\in[a,b]\}$ is a stochastic process satisfying $E\!\left(\int_a^b X_i(t)^2\,dt\right)<\infty$,...

open_problem | open | MathDB 页面 | 原文线索

V2-056594 | MATHDB-402200

Estimating the Bayes error probability at a rate faster than the plug-in estimate

摘录 / EXCERPT: For the d-dimensional feature vector X and binary label Y, the Bayes error probability is the smallest achievable error probability in binary classification. The problem is e...

open_problem | open | MathDB 页面 | 原文线索

V2-056595 | MATHDB-402347

Aldous’s frozen-percolation question on the binary tree

摘录 / EXCERPT: The frozen percolation process on the binary tree is not determined by the local random variables that intuitively dictate its local dynamics; extra randomness enters from the boun...

open_problem | open | MathDB 页面 | 原文线索

V2-056596 | MATHDB-402351

KPZ universality of the positive-temperature β-ballistic deposition model

摘录 / EXCERPT: For the -Ballistic Deposition (-BD) process defined by independent rate-1 Poisson clocks at each site, with transition probabilities ... where $C(x)$ is chosen so that...

open_problem | open | MathDB 页面 | 原文线索

V2-056597 | MATHDB-402590

Explicit representation of microphysical feedback in Reynolds-averaged equations

摘录 / EXCERPT: Whether the Reynolds-averaged equations are able to represent this feedback explicitly without the help of an additional stochastic sub-model remains an open question.

open_problem | open | MathDB 页面 | 原文线索

V2-056598 | MATHDB-402627

Upper-tail probability for triangle counts in sparse Erdős-Rényi graphs

摘录 / EXCERPT: Let X be the number of triangles in $G_{n,p}$, and define ... For every fixed positive constant δ and every $p=p(n)$ satisfying $n^{-1}\log n\ll p\ll 1$, determine the upper...

open_problem | open | MathDB 页面 | 原文线索

V2-056599 | MATHDB-402644

Continuous two-sided stochastic description of the Schonmann projection

摘录 / EXCERPT: Does the Schonmann projection of the plus-phase of the two-dimensional Ising model onto a one-dimensional line possess a continuous two-sided stochastic description in an almost su...

open_problem | open | MathDB 页面 | 原文线索

V2-056600 | MATHDB-402645

Everywhere continuity of the one-sided description of the Schonmann projection

摘录 / EXCERPT: For the Schonmann projection of the plus phase of the two-dimensional Ising model in the phase-transition regime, is the one-sided stochastic description by a g-function continuo...

open_problem | open | MathDB 页面 | 原文线索

V2-056601 | MATHDB-402650

Phase diagram for quasilocality of one-dimensional random walk in random scenery

摘录 / EXCERPT: For random walk in random scenery in one dimension, where the walk stays put with probability ϵ, moves right with probability $p(1-\epsilon)$, and moves left with probabil...

open_problem | open | MathDB 页面 | 原文线索

V2-056602 | MATHDB-402933

Removal of regularization in fermionic stochastic quantization

摘录 / EXCERPT: The first problem is the removal of the regularization in the fermionic stochastic quantization equation.

open_problem | open | MathDB 页面 | 原文线索

V2-056603 | MATHDB-402955

Singularity conjecture for sparse Bernoulli matrices

摘录 / EXCERPT: For each sequence $p_n\in(0,1/2]$, let M_n be an $n\times n$ matrix whose entries are independent $0/1$ Bernoulli random variables with probability p_n of being 1. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056604 | MATHDB-402958

Floating-body approximation for κ-concave random polytopes

摘录 / EXCERPT: Let $\b5$ be a Borel probability measure on $\d835\dc3^d$ that is absolutely continuous with respect to Lebesgue measure and has convex support. Do there exist constants...

open_problem | open | MathDB 页面 | 原文线索

V2-056605 | MATHDB-403030

Interpretation of time-oscillatory signals as mass or probability measures

摘录 / EXCERPT: The interpretation of time-oscillatory signals in terms of mass or probability measures remains largely open.

open_problem | open | MathDB 页面 | 原文线索

V2-056606 | MATHDB-403041

Scaling equations underlying the Sinkhorn algorithm

摘录 / EXCERPT: Given probability vectors $a\in\mathbb{R}_{+}^{m}$ and $b\in\mathbb{R}_{+}^{n}$ and a matrix $K\in\mathbb{R}_{+}^{m\times n}$, with $\min(m,n)\ge 2$, find vectors...

open_problem | open | MathDB 页面 | 原文线索

V2-056607 | MATHDB-403042

A priori globally convergent and faster overrelaxation for the Sinkhorn algorithm

摘录 / EXCERPT: Let $a\in\mathbb{R}^m_{+}$ and $b\in\mathbb{R}^n_{+}$ be probability vectors, and let $K\in\mathbb{R}^{m\times n}_{+}$ have strictly positive entries. The relaxed Sinkhorn iteratio...

open_problem | open | MathDB 页面 | 原文线索

V2-056608 | MATHDB-403122

An appropriate probability measure on \(\operatorname{Sub}(\mathbb Q)\)

摘录 / EXCERPT: Determine an appropriate probability measure on $\operatorname{Sub}(\mathbb Q)$ for these purposes.

open_problem | open | MathDB 页面 | 原文线索

V2-056609 | MATHDB-403163

Optimal spectral gaps for Weil-Petersson random hyperbolic surfaces

摘录 / EXCERPT: For all $\varepsilon>0$, the Weil-Petersson probability that a surface of genus g has $\lambda_1<\frac14-\varepsilon$ goes to zero as $g\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-056610 | MATHDB-403292

Near-optimal adaptive stochastic Galerkin approximation with tree-structured spatial discretizations

摘录 / EXCERPT: The report states an affirmative theorem rather than an unresolved problem. For the uniformly elliptic affine-parametric problem on $D\subset\mathbb{R}^d$, with $f\in L^2(D)$, coef...

open_problem | open | MathDB 页面 | 原文线索

V2-056611 | MATHDB-403350

Strict separation of survival and percolation thresholds for the nearest-neighbor contact process

摘录 / EXCERPT: For the classical nearest-neighbor contact process on $\bb{Z}^d$ (with $d\ge 2$), does the strict inequality $\bb{lambda}_c<\lambda_p$ hold?

open_problem | open | MathDB 页面 | 原文线索

V2-056612 | MATHDB-403351

Continuity of the long-range percolation phase transition for all \alpha>0 in dimensions d\ge 2

摘录 / EXCERPT: Conjecture. For long-range percolation on $\mathbb{Z}^d$ with power-law intensity $J(x)\sim A\lVert x\rVert^{-d-\alpha}$ for $A>0$ and $\alpha>0$, the phase transition is continuou...

open_problem | open | MathDB 页面 | 原文线索

V2-056613 | MATHDB-403352

Universal lower bound for planar crossing probabilities

摘录 / EXCERPT: For every $\rho\ge 1$, there exists a homeomorphism $\psi_\rho:[0,1]\to[0,1]$ such that, for every bond-percolation measure $\mathbb P$ on $\mathbb Z^2$ invariant under the symmetr...

open_problem | open | MathDB 页面 | 原文线索

V2-056614 | MATHDB-403353

Universality of the critical exponents for long-range-correlated percolation

摘录 / EXCERPT: Conjecture: The values of the critical exponents listed in Table 1 should hold not only for the Gaussian-free-field percolation model considered in the report, but for a universali...

open_problem | open | MathDB 页面 | 原文线索

V2-056615 | MATHDB-403520

Removing the dimensional factor in noncommutative Khintchine inequalities

摘录 / EXCERPT: For a general Gaussian random matrix $X=\\sum_{i=1}^n g_iA_i$, where the g_i are i.i.d. standard Gaussian variables and the A_i are matrix coefficients, does noncommutativity c...

open_problem | open | MathDB 页面 | 原文线索

V2-056616 | MATHDB-403532

KLS conjecture

摘录 / EXCERPT: There exists a universal constant $c>0$ such that, for every log-concave probability density p on $\\mathbb{R}^d$, its Cheeger isoperimetric coefficient satisfies ... where ... ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-056617 | MATHDB-403590

Relaxing assumption (H) for functional convergence

摘录 / EXCERPT: Can assumption (H) be relaxed to, for example, an $X\log X$ condition, while still obtaining Theorem 1, namely convergence in probability of $(W_n(\lambda))_{\lambda\in I}$ to...

open_problem | open | MathDB 页面 | 原文线索

V2-056618 | MATHDB-403595

Which simplicial-complex random walk should be used for PageRank?

摘录 / EXCERPT: Since the two random walks associated with the up-Laplacian and down-Laplacian have different invariant measures, which one should be preferred to obtain an equivalent of PageRank...

open_problem | open | MathDB 页面 | 原文线索

V2-056619 | MATHDB-403604

Universal compression for larger block sizes in stochastic block models

摘录 / EXCERPT: For block sizes $k > \\sqrt{\delta}\log n$, find a universal compressor for large or unknown alphabets, where n is the size of the adjacency matrix.

open_problem | open | MathDB 页面 | 原文线索

V2-056620 | MATHDB-403612

Precise reconstruction threshold for the symmetric planted partition model with q colors

摘录 / EXCERPT: Suppose that n vertices are randomly colored using q colors. Given the coloring, construct a random graph by connecting each pair of vertices independently with probability...

open_problem | open | MathDB 页面 | 原文线索

V2-056621 | MATHDB-403674

Clique-factor hitting-time question

摘录 / EXCERPT: Let $r\ge 4$, and let $G=(G_t)_{t=0}^{\binom n2}$ be the random graph process. Define ... If $n\in r\mathbb{Z}$, does G_{T_G} with high probability contain an r-clique factor (a...

open_problem | open | MathDB 页面 | 原文线索

V2-056622 | MATHDB-403675

Hitting time for triangle factors in the random graph process

摘录 / EXCERPT: For the random graph process $G=(G_t)_{t=0}^{\binom{n}{2}}$, define ... Is it true that, for $r=3$ and $3\mid n$, with high probability G_{T_G} contains an r-clique factor (that...

open_problem | open | MathDB 页面 | 原文线索

V2-056623 | MATHDB-403731

Invariance of the Yang-Mills measure for the gauge-orbit Markov process

摘录 / EXCERPT: Prove that the Yang-Mills measure is invariant for the Markov process on a suitable space of gauge orbits obtained by projecting the solution of the stochastic Yang-Mills equation.

open_problem | open | MathDB 页面 | 原文线索

V2-056624 | MATHDB-403761

Characterization of bounded harmonic functions for finitely supported random walks

摘录 / EXCERPT: Given a countable group G and a non-degenerate finitely supported probability measure μ on G, characterize when every bounded μ-harmonic function $f:G\to\mathbb{R}$ sat...

open_problem | open | MathDB 页面 | 原文线索

V2-056625 | MATHDB-403844

Laplace-transform relationship for stochastic gradient descent

摘录 / EXCERPT: Does the Laplace transform reveal an analogous relationship for stochastic gradient descent, described by the infinitesimal dynamics ... where W_t is a standard Brownian diffusion...

open_problem | open | MathDB 页面 | 原文线索

V2-056626 | MATHDB-403854

Limit law for the maximal leaf local time of a random walk on a regular tree

摘录 / EXCERPT: Let $b\ge 2$, let $\mathbb{T}_n$ be a rooted b-ary tree of depth n with leaf set $\mathbb{L}_n$, and let $\{X_t:t\ge 0\}$ be a continuous-time random walk on $\mathbb{T}_n$ wit...

open_problem | open | MathDB 页面 | 原文线索

V2-056627 | MATHDB-403855

Limit law for the most visited site of planar simple random walk

摘录 / EXCERPT: Let $X=(X_0,X_1,\ldots)$ be simple random walk on $\mathbb Z^2$, let $\ell_n(x):=\sum_{k=0}^n\mathbf 1_{\{X_k=x\}}$, and let $M_n:=\max_{x\in\mathbb Z^2}\ell_n(x)$. It is expected...

open_problem | open | MathDB 页面 | 原文线索

V2-056628 | MATHDB-403920

Type-I generalized Bose-Einstein condensation for soft Poissonian obstacles

摘录 / EXCERPT: Prove the occurrence, in probability, of type-I generalized Bose-Einstein condensation for noninteracting Bose gases in the presence of soft Poissonian obstacles in any dimension.

open_problem | open | MathDB 页面 | 原文线索

V2-056629 | MATHDB-404039

Kalikow’s zero-one law in dimensions $d\ge 3$

摘录 / EXCERPT: For a uniformly elliptic i.i.d. nearest-neighbor random walk in random environment on $\mathbb Z^d$, with $d\ge 3$, and with $A^+=\{X_n\cdot e_1\to+\infty\}$ under the annealed law...

open_problem | open | MathDB 页面 | 原文线索

V2-056630 | MATHDB-404041

Phase transition in the vacant set of a random walk on a discrete torus

摘录 / EXCERPT: For the vacant set left by a random walk run up to time uN^d on a d-dimensional discrete torus, with $d\ge 3$ and $u>0$, it was conjectured that there is a non-degenerate criti...

open_problem | open | MathDB 页面 | 原文线索

V2-056631 | MATHDB-404046

Deterministic limits for the trace-reinforced ants process on finite graphs

摘录 / EXCERPT: For the algorithm that adds one to the weights of all edges crossed by each random walk on its way from the nest N to the food F, prove the conjecture that, on any finite graph...

open_problem | open | MathDB 页面 | 原文线索

V2-056632 | MATHDB-404050

Density of the superdiffusive elephant random-walk limit

摘录 / EXCERPT: For the one-dimensional elephant random walk in the superdiffusive regime $p>3/4$, where $S_n/n^{2p-1}\to L$ almost surely, the limiting random variable L has a density supported...

open_problem | open | MathDB 页面 | 原文线索

V2-056633 | MATHDB-404257

Mean-field convergence for the intermediate interacting particle system

摘录 / EXCERPT: “When does the mean-field convergence in probability $\mu^\eta \to u^\eta$ hold true?” The report notes that this is known when the potential has the form $V(x)=|x|^{-\alpha}$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-056634 | MATHDB-404345

Analytic or stochastic interpretations of the multi-index grouping of trees

摘录 / EXCERPT: Do multi-indices, which encode linear combinations of trees and are used as a bookkeeping device in rough paths and regularity structures, admit any analytic or stochastic interpre...

open_problem | open | MathDB 页面 | 原文线索

V2-056635 | MATHDB-404397

Positivity and bound preservation in gPC-based stochastic methods

摘录 / EXCERPT: Another open question in gPC-based stochastic methods is the representation of strictly positive quantities such as gas density or water depth and/or the enforcement of discrete...

open_problem | open | MathDB 页面 | 原文线索

V2-056636 | MATHDB-404554

Geometric renormalisation of stochastic geometric heat equations in the full subcritical regime

摘录 / EXCERPT: Given the stochastic geometric heat equation on $\mathbb{R}_+\times\mathbb{T}$, with smooth symmetric Christoffel symbols $\Gamma^\alpha_{\beta\gamma}$, smooth vector-field coeffic...

open_problem | open | MathDB 页面 | 原文线索

V2-056637 | MATHDB-404615

Logarithmic correction to the critical one-arm probability in dimension six

摘录 / EXCERPT: For the critical level-set percolation of the Gaussian free field on the metric graph of $\mathbb{Z}^d$, with $\theta_d(N):=\mathbb{P}(0\xrightarrow{\,\ge 0\,}\partial B(N))$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-056638 | MATHDB-404616

One-arm probability for the metric-graph Gaussian free field at criticality

摘录 / EXCERPT: For the critical level-set percolation of the Gaussian free field on the metric graph over $\mathbb{Z}^3$, the one-arm probability has matching upper and lower bounds up to multipl...

open_problem | open | MathDB 页面 | 原文线索

V2-056639 | MATHDB-404617

Erdős-Révész question on the number of favorite sites

摘录 / EXCERPT: Let $K^{(d)}(n):=\{x\in\mathbb{Z}^d:\xi(x,n)=\xi^*(n)\}$ be the set of favorite sites of a discrete-time simple random walk on $\mathbb{Z}^d$. Can $\#K^{(d)}(n)=r$ occur infinitely...

open_problem | open | MathDB 页面 | 原文线索

V2-056640 | MATHDB-404619

Time-continuous limit of the stochastic multi-agent system in Bures-Wasserstein space

摘录 / EXCERPT: An open problem is to consider the limit $\Delta t \to 0$ and derive a time-continuous description of the stochastic multi-agent system in Bures-Wasserstein space.

open_problem | open | MathDB 页面 | 原文线索

V2-056641 | MATHDB-404620

Convergence of Wasserstein-2 gradient flows for min-max measure games

摘录 / EXCERPT: Determine whether Wasserstein-2 gradient flows for min-max problems over spaces of probability measures converge to a Nash equilibrium.

open_problem | open | MathDB 页面 | 原文线索

V2-056642 | MATHDB-404621

Convergence of Wasserstein-2 gradient descent-ascent for min-max games over measures

摘录 / EXCERPT: For displacement-convex-concave functionals over unbounded sets, does the Wasserstein-2 gradient descent-ascent flow for a min-max problem over spaces of probability measures conve...

open_problem | open | MathDB 页面 | 原文线索

V2-056643 | MATHDB-404756

Extension of second-order uncertainty results to credal sets

摘录 / EXCERPT: To what extent do the results for second-order distributions carry over to credal sets-sets of probability distributions on Y-as an alternative second-order representation?

open_problem | open | MathDB 页面 | 原文线索

V2-056644 | MATHDB-404758

Optimal scaling of random walk-like Metropolis algorithms on the sphere

摘录 / EXCERPT: Explain why numerical experiments for classical random walk-like Metropolis algorithms on the sphere indicate that the integrated autocorrelation time grows like $O(d^5)$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-056645 | MATHDB-404759

Optimal scalings of random walk-like Metropolis algorithms on high-dimensional spheres

摘录 / EXCERPT: What are the optimal scalings of random walk-like Metropolis algorithms on S^{d-1} as $d\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-056646 | MATHDB-404776

Logarithmic diffusivity for the critical two-dimensional stochastic Burgers equation

摘录 / EXCERPT: For the critical two-dimensional stochastic Burgers equation (1), the diffusion coefficient has large-time growth of order $(\log t)^{2/3}$ as $t\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-056647 | MATHDB-404778

CFT interpretations of backbone-related percolation quantities

摘录 / EXCERPT: Find conformal field theory (CFT) interpretations for the backbone-related quantities in critical two-dimensional percolation, namely the backbone exponent and its corresponding an...

open_problem | open | MathDB 页面 | 原文线索

V2-056648 | MATHDB-404942

Above-capacity list-decodability of random linear codes

摘录 / EXCERPT: For a random linear code over $[U+007F]\mathbb{F}_q$ of rate $R=1-h_q(\rho)+\varepsilon$, is it true that, with high probability, every ball of radius ρ contains at most...

open_problem | open | MathDB 页面 | 原文线索

V2-056649 | MATHDB-405053

The 1/3-2/3 Conjecture for finite non-chain posets

摘录 / EXCERPT: Let P be a finite poset that is not a chain. For distinct $x,y\in P$, let ... where the probability is taken over a uniformly random linear extension of P, and set ... The conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-056650 | MATHDB-405085

Asymptotic convergence of the sliced Wasserstein flow

摘录 / EXCERPT: For probability measures ρ and ν on $\mathbb{R}^d$, define $\rho^\theta=(\pi_\theta)_\#\rho$ and $\nu^\theta=(\pi_\theta)_\#\nu$, where $\pi_\theta(x)=x\cdot\theta$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-056651 | MATHDB-405143

Percolation phase-transition puzzle

摘录 / EXCERPT: For every infinite, locally finite, transitive graph $G=(V,E)$, let $\mathbb{P}_p$ denote Bernoulli bond percolation with parameter p, let $o\in V$, and define...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056652 | MATHDB-405150

Sheffield’s localization conjecture for sloped height functions

摘录 / EXCERPT: For every dimension $d\ge 3$, every prescribed slope $\theta\in\mathbb{R}^d$, and every general convex interaction, the corresponding finite-volume integer-valued $\nabla\phi$ heig...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056653 | MATHDB-405160

Circular-law conjecture for non-Hermitian random band matrices

摘录 / EXCERPT: For non-Hermitian random band matrices of size N and bandwidth W_N, the empirical spectral distribution converges to the circular law whenever $W_N\to\infty$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056654 | MATHDB-405185

Damron-Gravner-Junge-Lyu-Sivakoff parking-model questions

摘录 / EXCERPT: Consider the parking process on $\mathbb{Z}^d$ in which each site initially contains a car with probability p and a parking spot with probability $1-p$, independently, and cars p...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-056655 | MATHDB-405197

Polynomial-time coding theorem for time-bounded quantum Kolmogorov complexity

摘录 / EXCERPT: Conjecture: For every quantum polynomial-time sampler S and every classical string x that S outputs with probability δ, there is a quantum program for x of length at...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056656 | MATHDB-405219

Fathi-Mikulincer-Shenfeld conjecture on Caffarelli estimates under Lipschitz perturbations

摘录 / EXCERPT: For every $L\ge 0$, there exists a finite constant $C(L)$, independent of the dimension d, such that for every $d\ge 1$ and every globally L-Lipschitz function...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056657 | MATHDB-405224

Endpoint fractional Riesz estimate conjecture of Naor, Eskenazis, and Ivanisvili

摘录 / EXCERPT: For every $1<p<2$, there exists a constant $C_p<\infty$ such that, for every $n\in\mathbb{N}$ and every function $f:\Omega_n\to\mathbb{C}$, where $\Omega_n=\{-1,1\}^n$, the discret...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-056658 | MATHDB-405244

Banerjee et al. (2015) open problem on restricted eigenvalues under heavy-tailed designs

摘录 / EXCERPT: Let $A\subseteq\mathbb{S}^{p-1}$ and let $X\in\mathbb{R}^p$ be an isotropic random vector satisfying a uniform small-ball condition: there exist constants $\kappa,q>0$ such that...

open_problem | claimed_solved | MathDB 页面 | 原文线索

B13 集合论 / Set Theory | 475 条

V2-056659 | MATHDB-316042

Vaught conjecture

摘录 / EXCERPT: Let L be a first-order language with countably many symbols, and let T be a complete L-theory having infinite models. For a cardinal α, let $I(T,\alpha)$ denote the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-056660 | MATHDB-316138

generalized continuum hypothesis

摘录 / EXCERPT: Work in Zermelo-Fraenkel set theory with the axiom of choice. For a set S, let $|S|$ denote its cardinality, and for a cardinal κ let $2^{\kappa}=|\mathcal{P}(K)|$ where...

open_problem | open | MathDB 页面 | 原文线索

V2-056661 | MATHDB-316299

The admissible Borel hierarchy embedding conjecture

摘录 / EXCERPT: Admissible Borel hierarchy embedding conjecture. It is relatively consistent with ZF that, for every pair of admissible functions f and g,

conjecture | open | MathDB 页面 | 原文线索

V2-056662 | MATHDB-316306

Slaman-Woodin conjecture on the complexity of the c.e. orbit relation

摘录 / EXCERPT: Let $(W_i)_{i\in\mathbb{N}}$ be an effective enumeration of the computably enumerable sets, and write $W_i\approx W_j$ when the two sets lie in the same orbit under automorphisms o...

conjecture | open | MathDB 页面 | 原文线索

V2-056663 | MATHDB-316374

Shelah's conjecture on increasing chains modulo small-set ideals

摘录 / EXCERPT: Let θ and μ be cardinals with $\mu>\theta$. Consider the partial order of functions ${}^{(\theta^{+3})}\mu$ ordered modulo the ideal $[\theta^{+3}]^{\leq\theta}$ of subs...

conjecture | open | MathDB 页面 | 原文线索

V2-056664 | MATHDB-317108

The coincidence of the weak index and Borel hierarchies for weakly recognizable languages

摘录 / EXCERPT: Weak index-Borel hierarchy conjecture. For weakly recognizable languages, the weak index hierarchy and the Borel hierarchy coincide; equivalently, every weakly recognizable...

conjecture | open | MathDB 页面 | 原文线索

V2-056665 | MATHDB-317124

A depth inequality for Boolean algebra ultraproducts at singular cardinals

摘录 / EXCERPT: The depth inequality conjecture. In ZFC,

conjecture | open | MathDB 页面 | 原文线索

V2-056666 | MATHDB-317896

The decomposability-or-regularity conjecture for ultrafilters

摘录 / EXCERPT: Let $ obreak\break obreak\lambda<\mu$ be cardinals, and let D be a (λ^+,μ)-regular ultrafilter. Decomposability-or-regularity conjecture. D is either...

conjecture | open | MathDB 页面 | 原文线索

V2-056667 | MATHDB-318270

The dichotomy conjecture for conjugacy problems of automorphism groups

摘录 / EXCERPT: Conjugacy-complexity dichotomy conjecture. If $\mathcal M$ is the countable model in an $\aleph_0$-categorical theory, then $\mathcal C_{\mathcal M}$ is either smooth or Borel comp...

conjecture | open | MathDB 页面 | 原文线索

V2-056668 | MATHDB-318437

Fiber Conjecture for characteristic-polynomial fibers of block companion matrices

摘录 / EXCERPT: Fiber Conjecture. For every $f(X)\in {\EuScript P}(mn;q)$, the cardinality of $\Psi^{-1}(f(X))$ is independent of the choice of $f(X)$ and equals

conjecture | open | MathDB 页面 | 原文线索

V2-056669 | MATHDB-318648

The cardinality conjecture for spaces generated by sigma-sets

摘录 / EXCERPT: Let $\mathbb{N}=\{1,2,3,\ldots\}$, let $\mathbb{N}^{\star}=\{1^{\ast},2^{\ast},3^{\ast},\ldots\}$, and let $\mathbb{N}_{\Theta}=\{1_{\Theta},2_{\Theta},3_{\Theta},\ldots\}$. For a set S, write 2^S for...

conjecture | open | MathDB 页面 | 原文线索

V2-056670 | MATHDB-318649

The commutative loop conjecture for integer spaces

摘录 / EXCERPT: The commutative loop conjecture. For all $X\in 2^{\mathbb{N}}$, $(3^X,\oplus)$ satisfies: (1) $(\forall A,B\in 3^X)(A\oplus B\in 3^X)$; (2)...

conjecture | open | MathDB 页面 | 原文线索

V2-056671 | MATHDB-318650

The partial-order conjecture for integer spaces

摘录 / EXCERPT: The partial-order conjecture. For all $X\in 2^{\mathbb{N}}$, $(3^X,\leq)$ satisfies: (1) $(\forall A\in 3^X)(A\leq A)$; (2) $(\forall A,B\in 3^X)(A\leq B\wedge B\leq A\rightarrow A=B)$; an...

conjecture | open | MathDB 页面 | 原文线索

V2-056672 | MATHDB-318713

Binary-operation conjecture for the ternary power set of a sigma-set

摘录 / EXCERPT: Ternary power-set operation conjecture. The operation $\oplus$ is a binary operation on 3^{A}.

conjecture | open | MathDB 页面 | 原文线索

V2-056673 | MATHDB-318930

Borel incomparability of isomorphism and quasi-isomorphism for p-local groups

摘录 / EXCERPT: Let $m,n\geq3$ and let p,q be primes. Write $\oiso_{m,p}$ for isomorphism equivalence on the space of p-local torsion-free abelian groups of rank m, and $\oqiso_{n,q}$ for qu...

conjecture | open | MathDB 页面 | 原文线索

V2-056674 | MATHDB-318973

The straightening theorem for irreducible continuous lattices with minimal elements

摘录 / EXCERPT: Let L be an irreducible and continuous lattice with minimal elements, and let n be the cardinality of one of its affine references. An involutory anti-automorphism is a map...

conjecture | open | MathDB 页面 | 原文线索

V2-056675 | MATHDB-319453

Representation conjecture for dense weakly selective ideals

摘录 / EXCERPT: Representation conjecture. If J is a dense $\mathbf{F}_{\sigma\delta}$ weakly selective ideal on ω, then there exists a Polish space with a countable base $\mathcal{O}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-056676 | MATHDB-320166

Shelah's conjecture on models at the first uncountable beth cardinal

摘录 / EXCERPT: Shelah's conjecture. Every ${\mathcal{L}_{\omega_1,\omega}}$-sentence with a model of cardinality $\aleph_{\omega_1}$ has a model of cardinality $2^{\aleph_0}$.

conjecture | open | MathDB 页面 | 原文线索

V2-056677 | MATHDB-320477

Nonexistence of weakly tight families at ${\aleph}_1<\mathfrak{s}=\mathfrak{c}=\aleph_2$

摘录 / EXCERPT: Nonexistence conjecture. There is a model of

conjecture | open | MathDB 页面 | 原文线索

V2-056678 | MATHDB-320478

Existence of Sacks-indestructible MAD families below ${\aleph}_{\omega}$

摘录 / EXCERPT: Sacks-indestructibility conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-056679 | MATHDB-320576

The odd-exponent formula for the Erdős-Ginzburg-Ziv constant

摘录 / EXCERPT: Let $\mathsf{s}(G)$ be the least length forcing a zero-sum subsequence of length $\exp(G)$, and let $\mathsf{g}(G)$ be the least length forcing such a subsequence in every squarefr...

conjecture | open | MathDB 页面 | 原文线索

V2-056680 | MATHDB-320857

McLaughlin's conjecture on countable arithmetic subsets of Baire space

摘录 / EXCERPT: An arithmetic subset of Baire space is a subset of $\mathbb{N}^{\mathbb{N}}$ definable by an arithmetic formula, and an arithmetic singleton is a sequence that is the unique elemen...

conjecture | refuted | MathDB 页面 | 原文线索

V2-056681 | MATHDB-321589

Classical pure-state cardinality conjecture

摘录 / EXCERPT: Classical pure-state cardinality conjecture. The set of pure states of the classical system cannot have lower cardinality than the set of pure states of the generic system. In the...

conjecture | open | MathDB 页面 | 原文线索

V2-056682 | MATHDB-321894

The standard-example conjecture for high-dimensional posets

摘录 / EXCERPT: A standard example of dimension m is the poset formed by the one-element subsets and the $(m-1)$-element subsets of an m-element set, ordered by inclusion. Standard-example con...

conjecture | open | MathDB 页面 | 原文线索

V2-056683 | MATHDB-322013

The locally conical reflection arrangement conjecture

摘录 / EXCERPT: Let W be a well-generated reflection group acting on a complex vector space V. A generating set R is well-generating when it generates W and has cardinality $\dim V$. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-056684 | MATHDB-322024

Minimal non-face conjecture for multi-cluster complexes

摘录 / EXCERPT: Minimal non-face conjecture. Every minimal non-face of $\Delta_c^k(W)$ has cardinality $k+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-056685 | MATHDB-322059

Asymptotic cardinality conjecture for universal sets in the Gaussian integers

摘录 / EXCERPT: Asymptotic cardinality conjecture. The minimal cardinality satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-056686 | MATHDB-322169

Uniformization conjecture for Borel homomorphisms from arithmetic to Turing equivalence

摘录 / EXCERPT: Let $\equiv_A$ denote arithmetic equivalence and $\equiv_T$ Turing equivalence on 2^ω. A homomorphism from $\equiv_A$ to $\equiv_T$ preserves equivalence, and uniformity mea...

conjecture | open | MathDB 页面 | 原文线索

V2-056687 | MATHDB-322333

Martin's maximum conjecture for forcing extensions of $H_{\aleph_2}$

摘录 / EXCERPT: Let V be a model of ${\text{{\sf MM}}}^{++}$ together with large cardinals, and let $P\in V$ preserve ${\text{{\sf MM}}}^{++}$. Martin's maximum conjecture. For every such P, ...

conjecture | open | MathDB 页面 | 原文线索

V2-056688 | MATHDB-322334

The maximality conjecture for $\Gamma_{\aleph_3}$

摘录 / EXCERPT: Let $\Gamma_{\aleph_3}$ be a class of partial orders, and let maximality with respect to a theory mean the five conditions given in the preceding definition: inclusion of the relev...

conjecture | open | MathDB 页面 | 原文线索

V2-056689 | MATHDB-322493

The optimal sampling bound conjecture for random subsets and RIP

摘录 / EXCERPT: Optimal sampling bound conjecture. A random subset $\Omega\subset\{0,1,\ldots,n-1\}$ of average cardinality

conjecture | open | MathDB 页面 | 原文线索

V2-056690 | MATHDB-323015

Stanley's strict rank growth conjecture for differential posets

摘录 / EXCERPT: Let r be a positive integer, let P be an r-differential poset, let P_n denote its nth rank, and write $p_n:=|P_n|$ for the rank cardinalities. Stanley's conjecture. An...

conjecture | open | MathDB 页面 | 原文线索

V2-056691 | MATHDB-323090

Shelah's saturation characterization conjecture for dependent theories

摘录 / EXCERPT: Let T be a first-order theory, let M be a model of T, and let κ be an infinite cardinal. A sequence $\langle a_\alpha:\alpha<\delta\rangle$ in M is indiscernible whe...

conjecture | open | MathDB 页面 | 原文线索

V2-056692 | MATHDB-323317

Universality of local core models

摘录 / EXCERPT: Assume there is no proper class inner model with a Woodin cardinal. Let δ be an uncountable regular cardinal, and let $\mathbf{M}$ be a proper class inner model such that...

conjecture | open | MathDB 页面 | 原文线索

V2-056693 | MATHDB-323388

The non-smoothness conjecture for Lascar equivalence on KP-classes

摘录 / EXCERPT: Let X be an E_{KP}-class. The relations E_L and E_0 are defined on spaces of types, where $E_L\!\upharpoonright\!X$ denotes the restriction of Lascar equivalence to X, an...

conjecture | open | MathDB 页面 | 原文线索

V2-056694 | MATHDB-323389

The non-smoothness conjecture for Lascar groups

摘录 / EXCERPT: Let T be a first-order theory, and let $Gal_0(T)$ denote the kernel of the canonical map from the Lascar group $Gal_L(T)$ to the Kim--Pillay group $Gal_{KP}(T)$. Lascar-group non...

conjecture | open | MathDB 页面 | 原文线索

V2-056695 | MATHDB-323390

The group quotient non-smoothness conjecture for G^{000}/G^{00}

摘录 / EXCERPT: Let G be a group definable or $*$-definable over $\emptyset$ in a monster model, and let E_G be the associated Borel equivalence relation coding the quotient G^{00}/G^{000}....

conjecture | open | MathDB 页面 | 原文线索

V2-056696 | MATHDB-323391

The Lascar-group upper-bound conjecture for countable tuples

摘录 / EXCERPT: Let T be a theory with monster model ${\EuFrak C}$, let $\overline{a}$ be a countable tuple from ${\EuFrak C}$, and let $E_L^{\overline a}$ and $F_L^{\overline a}$ denote respect...

conjecture | open | MathDB 页面 | 原文线索

V2-056697 | MATHDB-323392

The monotonicity conjecture for Lascar equivalence of countable tuples

摘录 / EXCERPT: Let $\overline a$ and $\overline b$ be countable tuples from a monster model, with $\overline b$ containing $\overline a$. Let $E_L^{\overline a}$ and $E_L^{\overline b}$ be the co...

conjecture | open | MathDB 页面 | 原文线索

V2-056698 | MATHDB-323411

Forcing axioms and trivial automorphisms of corona algebras

摘录 / EXCERPT: Forcing-axiom conjecture. Forcing axioms imply that $Q(A)$ has only trivial automorphisms.

conjecture | open | MathDB 页面 | 原文线索

V2-056699 | MATHDB-323521

The cofinality characterization of non-homogeneously characterizable cardinals

摘录 / EXCERPT: Cofinality conjecture. For κ a characterizable cardinal, κ is not homogeneously characterizable if and only if

open_problem | open | MathDB 页面 | 原文线索

V2-056700 | MATHDB-323600

Graph complexity conjecture for finite-level decomposable functions

摘录 / EXCERPT: Let X and Y be separable metrizable spaces with X analytic, and let $\widetilde{Y}$ denote the completion used for the graph formulation. For $1<n<\omega$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-056701 | MATHDB-323650

The PFA conjecture

摘录 / EXCERPT: PFA conjecture. The following theories are equiconsistent:

conjecture | open | MathDB 页面 | 原文线索

V2-056702 | MATHDB-323651

The consistency conjecture for HOD large-cardinal axioms

摘录 / EXCERPT: Consistency conjecture for HOD large-cardinal axioms. For each large-cardinal axiom ϕ, S_ϕ is consistent relative to some large-cardinal axiom. In particular, if ϕ a...

conjecture | open | MathDB 页面 | 原文线索

V2-056703 | MATHDB-323652

The Generation of Closed Pointclasses

摘录 / EXCERPT: Generation of closed pointclasses conjecture. If there is a Suslin cardinal $\kappa>w(\Gamma)$ and $L(\Gamma,\mathbb R)\vDash MC$, then for some hod pair $(\mathcal P,\Sigma)$,

conjecture | open | MathDB 页面 | 原文线索

V2-056704 | MATHDB-323653

The LST-hod mouse equiconsistency conjecture

摘录 / EXCERPT: LST-hod mouse equiconsistency conjecture. The following theories are equiconsistent:

conjecture | open | MathDB 页面 | 原文线索

V2-056705 | MATHDB-323654

The PFA equiconsistency conjecture

摘录 / EXCERPT: Let PFA denote the Proper Forcing Axiom, and let a supercompact cardinal be a cardinal with the usual supercompactness property. PFA equiconsistency conjecture. PFA is equiconsiste...

conjecture | open | MathDB 页面 | 原文线索

V2-056706 | MATHDB-323655

The singular square failure conjecture for subcompact inner models

摘录 / EXCERPT: Let $ckappa$ be a singular cardinal, and write $csquare_ckappa$ for the square principle at $ckappa$. Singular square failure conjecture. If ... holds, then there is an inner model w...

conjecture | open | MathDB 页面 | 原文线索

V2-056707 | MATHDB-324024

The SOP_2 characterization of maximality in Keisler's order

摘录 / EXCERPT: Let EC be an elementary class with the strict order property SOP_2, meaning that there are a model $M\in EC$, a first-order formula $\varphi(x;y)$, and parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-056708 | MATHDB-324836

The folklore conjecture on Gorenstein rings of countable Cohen-Macaulay representation type

摘录 / EXCERPT: A Gorenstein ring is a commutative Noetherian local ring whose canonical module is isomorphic to the ring itself, and its Cohen-Macaulay representation type describes the cardinali...

conjecture | open | MathDB 页面 | 原文线索

V2-056709 | MATHDB-325045

Excellence under forcing for atomic classes

摘录 / EXCERPT: Let ${\mathbb P}$ be a c.c.c. forcing notion of cardinality λ such that forcing with ${\mathbb P}$ forces Martin's axiom and $2^{\aleph_0}=\lambda$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-056710 | MATHDB-325048

Stationarity dichotomy conjecture for abstract elementary classes

摘录 / EXCERPT: Stationarity dichotomy conjecture. The classes S_1 and S_2 are not both stationary. A weaker version replaces S_2 by the class of $\lambda\in C^{\rm fp}_{\aleph_0}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-056711 | MATHDB-325049

Large-cardinal dichotomy for abstract elementary classes

摘录 / EXCERPT: Let ${\mathfrak k}$ be an abstract elementary class. Large-cardinal dichotomy conjecture. For a closed unbounded class C of cardinals, the alternatives stated in the source shoul...

conjecture | open | MathDB 页面 | 原文线索

V2-056712 | MATHDB-325215

Instability conjecture for the number of models of a theory

摘录 / EXCERPT: Instability conjecture. If $\lambda>|T|$ is strong limit, $\lambda>\theta>\operatorname{cf}(\lambda)$, and T is not definably stable, then

conjecture | open | MathDB 页面 | 原文线索

V2-056713 | MATHDB-325271

Finite-level decomposability conjecture for Borel functions

摘录 / EXCERPT: Let $m,n\in\omega$. For a class of Borel functions, $\mathbf{dec}_{n+1}\mathcal{F}$ denotes functions decomposable into countably many $\mathcal{F}$-functions with...

conjecture | open | MathDB 页面 | 原文线索

V2-056714 | MATHDB-325610

Brendle's correctness-preservation conjecture for Suslin ccc posets

摘录 / EXCERPT: A Suslin ccc poset is a poset that is Suslin and satisfies the countable chain condition. It is correctness preserving if, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-056715 | MATHDB-325654

The equilateral dimension conjecture for finite-dimensional normed spaces

摘录 / EXCERPT: Let X be an n-dimensional normed space. The equilateral dimension conjecture. ... Here $e(X)$ denotes the maximal cardinality of an equilateral set in X, that is, a set of pair...

conjecture | open | MathDB 页面 | 原文线索

V2-056716 | MATHDB-325719

Nowakowski's independent domination conjecture for categorical graph products

摘录 / EXCERPT: Let G and H be graphs, and let $i(G)$ denote the smallest cardinality of an independent dominating set in G, equivalently the cardinality of a smallest maximal independent se...

conjecture | open | MathDB 页面 | 原文线索

V2-056717 | MATHDB-325874

Invariance of transfinite jumps under lower-level piecewise operators

摘录 / EXCERPT: Let β and α be ordinals with $\beta < \alpha$, and let $\mathbf{X}$ be a represented space. Write $\mathbf{X}^{(\alpha)}$ for its α-th jump and $\chi^{(\beta)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-056718 | MATHDB-327117

The multiboard Gale-Stewart determinacy conjecture

摘录 / EXCERPT: Let $\lambda\geq 2$ be a cardinal and let $A\subseteq{}^{\omega}\lambda$. In the multiboard Gale-Stewart game ... the players play the ordinary Gale-Stewart game on κ boards s...

conjecture | open | MathDB 页面 | 原文线索

V2-056719 | MATHDB-328036

Large ultrapowers without $\mathbf L_{\omega_1\omega}$-compact cardinals

摘录 / EXCERPT: Large-ultrapower conjecture. If the existence of measurable cardinals is not refutable in ZFC, then such a model of ZFC exists. This asks whether arbitrarily large ultrapowers with...

conjecture | open | MathDB 页面 | 原文线索

V2-056720 | MATHDB-328124

The finite common-zero cardinality conjecture for centrohermitian matrices

摘录 / EXCERPT: Let A be a centrohermitian matrix of size $n\times(n+m)$, and let $P_I(z_0,\ldots,z_n)$ be the associated generalized characteristic polynomials indexed by sets I with $|I|=m$....

conjecture | open | MathDB 页面 | 原文线索

V2-056721 | MATHDB-328417

The Goldman classification conjecture for closed surfaces

摘录 / EXCERPT: Goldman classification conjecture. The classification in Goldman is a Borel reduction into $\cong_G$, and hence

conjecture | open | MathDB 页面 | 原文线索

V2-056722 | MATHDB-328633

Weak representability conjecture for measurable preimage functions

摘录 / EXCERPT: Let n and m be the parameters for the corresponding Borel measurability levels, and let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be Polish represented spaces. Write $\mathfrak{b}$...

conjecture | open | MathDB 页面 | 原文线索

V2-056723 | MATHDB-328634

Strong representability conjecture for measurable preimage functions

摘录 / EXCERPT: Let n and m be the parameters for the corresponding Borel measurability levels, and let $\mathbf{X}$ and $\mathbf{Y}$ be Polish represented spaces. Write $\mathfrak{b}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-056724 | MATHDB-328695

Hrušák's dichotomy conjecture for fragmented ideals

摘录 / EXCERPT: Let $\mathcal{I}$ be a fragmented ideal. The associated Rothberger number $\mathfrak{b}(\mathcal{I})$ is the least size of a family witnessing the corresponding Rothberger property...

conjecture | open | MathDB 页面 | 原文线索

V2-056725 | MATHDB-328953

The Full Decomposability Conjecture

摘录 / EXCERPT: Let ${\mathcal A}$ be an analytic subset of a Polish space and let ${\mathcal Y}$ be separable metrizable. For a function $f:{\mathcal A}\to{\mathcal Y}$, interpret...

conjecture | open | MathDB 页面 | 原文线索

V2-056726 | MATHDB-328991

The generic pair conjecture for dependent diagrams

摘录 / EXCERPT: Fix a dependent diagram D and a strongly compact cardinal $\theta>|T|$. Let $\theta<\lambda=\lambda^{<\lambda}$ and $\lambda^{+}=2^{\lambda}$. Let ... be an increasing continuous s...

conjecture | open | MathDB 页面 | 原文线索

V2-056727 | MATHDB-329572

Törnquist's universality conjecture for isomorphism of factors

摘录 / EXCERPT: Törnquist's conjecture. The isomorphism relation for separably acting factors is universal, with respect to Borel reducibility, for orbit equivalence relations induced by a continu...

conjecture | open | MathDB 页面 | 原文线索

V2-056728 | MATHDB-329847

Weakly homogeneous forcing changing the cofinality of the first inaccessible

摘录 / EXCERPT: Consistency conjecture. It is consistent, relative to the existence of large cardinals, that there is a weakly homogeneous forcing which changes the cofinality of the first inacces...

conjecture | open | MathDB 页面 | 原文线索

V2-056729 | MATHDB-329848

Reflective forcing at the first ω_1-Mahlo cardinal

摘录 / EXCERPT: Consistency conjecture. It is consistent, relative to the existence of large cardinals, that there is (κ,ω_1)-reflective forcing where κ is the first ω_1-...

conjecture | open | MathDB 页面 | 原文线索

V2-056730 | MATHDB-329965

The cardinality conjecture for gossamer numbers

摘录 / EXCERPT: Let $*G$ denote the gossamer numbers and let $d53f$ denote the real numbers. Cardinality conjecture. The cardinality of the gossamer numbers is larger than the cardinality of the r...

conjecture | open | MathDB 页面 | 原文线索

V2-056731 | MATHDB-330245

The large-rank almost synchronizing conjecture for primitive groups

摘录 / EXCERPT: Let G be a primitive permutation group of degree n, and let a map have rank r, meaning that its image has cardinality r. Large-rank almost synchronizing conjecture. If ... th...

conjecture | open | MathDB 页面 | 原文线索

V2-056732 | MATHDB-330534

Greenberg's conjecture on cyclotomic Iwasawa modules

摘录 / EXCERPT: For the cyclotomic Iwasawa module X, let X^+ and X^- be the eigenspaces for complex conjugation. Greenberg's conjecture. For every N, X^+ has finite cardinality. In parti...

conjecture | open | MathDB 页面 | 原文线索

V2-056733 | MATHDB-331167

The interval conjecture for uniqueness of limit models in stable AECs

摘录 / EXCERPT: Interval conjecture. This set is a non-trivial interval of regular cardinals. Moreover, its minimum is an important measure of the complexity of $\mathcal{K}$.

conjecture | open | MathDB 页面 | 原文线索

V2-056734 | MATHDB-331694

The rational-point invariance conjecture for tropical linear series

摘录 / EXCERPT: Let Δ be a tropical complex and let D be a Weil divisor on Δ. Define $h^0(\Delta,D)$ as the smallest cardinality of a set of rational points that is not contained i...

conjecture | open | MathDB 页面 | 原文线索

V2-056735 | MATHDB-332096

The equivalence between unique generic filters and OTM-parameter-recognizable sets

摘录 / EXCERPT: Assume that $V=L$. A forcing notion $\mathbb{P}$ is said to add an OTM-parameter-recognizable set if, in a forcing extension, it adds a set recognizable by an ordinal Turing machi...

conjecture | open | MathDB 页面 | 原文线索

V2-056736 | MATHDB-332118

The cyclic-generator conjecture for closed systems of invertible maps

摘录 / EXCERPT: Let $A=\\{1,\dots,k\\}$ be a finite alphabet of even cardinality, and let $n\geq 2$. Write $B_n(A)$ for the group of invertible maps on A^n, let $(1,\dots,k)$ denote the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-056737 | MATHDB-332134

The conjecture on orbit equivalence relations of non-Archimedean abelian Polish groups

摘录 / EXCERPT: Orbit-relation conjecture. Every G-orbit equivalence relation is Borel reducible to

conjecture | open | MathDB 页面 | 原文线索

V2-056738 | MATHDB-332135

The conjecture that essentially countable subrelations of abelian Polish-group orbit relations are essentially hyperfinite

摘录 / EXCERPT: Essential hyperfiniteness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-056739 | MATHDB-332300

Alon-Stav conjecture on the optimal constant for identifiable-parent-property codes

摘录 / EXCERPT: Let $i_t(N,q)$ denote the maximal cardinality of a t-identifiable-parent-property code of length N over an alphabet of size q, and let ... Alon and Stav proved...

conjecture | open | MathDB 页面 | 原文线索

V2-056740 | MATHDB-332715

Convergence conjecture for the modified sphere-packing auxiliary functions

摘录 / EXCERPT: Let f_k and $\widehat f_k$ be the auxiliary functions obtained by forcing the modified roots described in the paper, viewed as analytic functions of a radial variable. Convergenc...

conjecture | open | MathDB 页面 | 原文线索

V2-056741 | MATHDB-332742

Generalized Giuga conjecture for finite commutative rings

摘录 / EXCERPT: Let R be a finite commutative unital ring, and let $|R|$ denote its cardinality. Generalized Giuga conjecture. ... if and only if R is a field. This generalizes the classical Giu...

conjecture | open | MathDB 页面 | 原文线索

V2-056742 | MATHDB-333086

Integer-root conjecture for total domination polynomials

摘录 / EXCERPT: Let $G=(V,E)$ be a simple graph, let ... be its total domination polynomial, where n is the order of G and $d_t(G,i)$ counts the total dominating sets of cardinality i. A root...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-056743 | MATHDB-333309

Uniform universality characterization for countable Borel equivalence relations

摘录 / EXCERPT: Uniform universality characterization. A countable Borel equivalence relation is universal if and only if it is uniformly universal with respect to every way it can be generated.

conjecture | open | MathDB 页面 | 原文线索

V2-056744 | MATHDB-333311

Optimistic uniform universality conjecture for universal countable Borel equivalence relations

摘录 / EXCERPT: Optimistic uniform universality conjecture. If E is a universal countable Borel equivalence relation, then E is uniformly universal with respect to every way of generating E.

conjecture | open | MathDB 页面 | 原文线索

V2-056745 | MATHDB-333312

Nonreducibility of the universal treeable relation to Turing equivalence

摘录 / EXCERPT: Treeable nonreducibility conjecture. The universal treeable countable Borel equivalence relation is not Borel reducible to Turing equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-056746 | MATHDB-333313

Nonuniversality conjecture for recursive isomorphism on 2^ω

摘录 / EXCERPT: Nonuniversality conjecture. Recursive isomorphism on 2^ω is not a universal countable Borel equivalence relation.

conjecture | open | MathDB 页面 | 原文线索

V2-056747 | MATHDB-333399

The linear greedy defining set conjecture for Latin squares

摘录 / EXCERPT: Let g_n be the cardinality of a smallest greedy defining set among all $n\times n$ Latin squares, with greedy defining sets taken in lexicographic order. Linear greedy defining s...

conjecture | open | MathDB 页面 | 原文线索

V2-056748 | MATHDB-333711

The consistency-strength conjecture for the Halpern-Läuchli theorem at a measurable cardinal

摘录 / EXCERPT: Let $de 2$ be finite, let $appa$ be measurable, and let $\sigma<appa$. Write $\operatorname{HL}(d,\sigma,appa)$ for the Halpern-Läuchli property for d many regular trees of heigh...

conjecture | open | MathDB 页面 | 原文线索

V2-056749 | MATHDB-333712

The strict-strength conjecture for the Halpern-Läuchli theorem and antichain colorings

摘录 / EXCERPT: Let $d\ge 2$ be finite. Let $\operatorname{HL}(d,\sigma,\kappa)$ denote the Halpern-Läuchli property for d many regular trees at a measurable cardinal κ, and let the compa...

conjecture | open | MathDB 页面 | 原文线索

V2-056750 | MATHDB-333835

Consistency of LSA from the consistency of T and PFA

摘录 / EXCERPT: Let $\textsf{LSA}$ denote the theory “$\textsf{AD}^+ + \theta_{\beta+1}+\theta_\beta$ is the largest Suslin cardinal.” Here $\operatorname{Con}(T)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056751 | MATHDB-333956

The maximum spherical two-distance set conjecture

摘录 / EXCERPT: Let $g(n)$ denote the maximum cardinality of a spherical two-distance set in $\mathbb{R}^n$. Maximum spherical two-distance set conjecture. ... The conjecture reflects the observed p...

conjecture | open | MathDB 页面 | 原文线索

V2-056752 | MATHDB-334314

Reconstruction of finite abelian groups from zero-sum graph properties

摘录 / EXCERPT: For a finite abelian group $mathcal G$, let $Z(mathcal G)$ denote the class of graphs that are zero-forcing for $mathcal G$. Zero-sum graph reconstruction conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056753 | MATHDB-334653

Lonc's conjecture on almost tiling the Boolean lattice with copies of a poset

摘录 / EXCERPT: Lonc's conjecture. If n is sufficiently large and $|P|$ divides 2^n-2, then

conjecture | open | MathDB 页面 | 原文线索

V2-056754 | MATHDB-334654

Almost-tiling conjecture for the Boolean lattice with copies of a poset

摘录 / EXCERPT: Almost-tiling conjecture. There exists a constant $c=c(P)$ such that for every positive integer n, there is a set $S\subseteq 2^{[n]}$ with $|S|\leq c$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-056755 | MATHDB-334856

Aichholzer-Aurenhammer-Hurtado-Krasser compatible triangulations conjecture

摘录 / EXCERPT: Let P and Q be point sets in general position, meaning that no three points are collinear, with the same cardinality and the same number of points on their convex hulls. A tria...

conjecture | open | MathDB 页面 | 原文线索

V2-056756 | MATHDB-335564

The ordinal-condition characterization of the finite-model subtheory of MLSP

摘录 / EXCERPT: Let ψ be an $\textnormal{\textsf{MLSP}}$-formula, let $\mathrm{Vars}(\psi)$ be its set of variables, and let M be a model of ψ. The subtheory...

conjecture | open | MathDB 页面 | 原文线索

V2-056757 | MATHDB-335601

Recursive coding conjecture for convexly generated subsets of Euclidean space

摘录 / EXCERPT: Recursive convexly generated coding conjecture. Every subset $A\subseteq\mathbb{R}^N$ is Δ^1_1 and convexly generated exactly when it admits a recursive convexly generated c...

conjecture | open | MathDB 页面 | 原文线索

V2-056758 | MATHDB-335841

The non-implication from I3 towers deciding their critical sequence to Jonssonness

摘录 / EXCERPT: Non-implication conjecture. The existence of an I3 tower of height $\aleph_\omega$ that decides its critical sequence does not imply that $\aleph_\omega$ is Jonsson.

conjecture | open | MathDB 页面 | 原文线索

V2-056759 | MATHDB-336224

Conjectural characterization of the order on κ_3^x by eventual successor-cardinal comparison

摘录 / EXCERPT: κ_3 comparison conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056760 | MATHDB-336508

The MA$_{\aleph_1}$ non-implication for uncountable strongly surjective real suborders

摘录 / EXCERPT: Let $\textmd{MA}_{\aleph_1}$ denote Martin's axiom for partially ordered sets satisfying the countable chain condition at cardinality $\aleph_1$. An uncountable strongly surjective...

open_problem | open | MathDB 页面 | 原文线索

V2-056761 | MATHDB-336519

The linear upper-bound conjecture for almost-equidistant sets

摘录 / EXCERPT: Let $f_{\mathrm{ae}}(d)$ denote the maximum cardinality of an almost-equidistant set in $\mathbb R^d$, meaning a set in which among every three points, some pair is at distance 1...

conjecture | open | MathDB 页面 | 原文线索

V2-056762 | MATHDB-336764

Rényi divergence and generalized channel capacity relation

摘录 / EXCERPT: Let Q be a non-uniform distribution and U the uniform distribution on a support of cardinality 2^n. Let T be the matrix with two rows, Q and U, and 2^n columns. The q...

conjecture | open | MathDB 页面 | 原文线索

V2-056763 | MATHDB-337109

The strict Borel reducibility conjecture for positive-rank torsion-free abelian groups

摘录 / EXCERPT: Let $\bigsqcup_{n\geq 1} \cong_n^+$ denote the disjoint union of the isomorphism relations $\cong_n^+$ on the corresponding spaces of torsion-free abelian groups, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056764 | MATHDB-337110

The Borel reducibility conjecture for Bratteli diagrams of exact rank

摘录 / EXCERPT: For each $n\geq 1$, let $\mathcal{BD}^*_n$ be the class of Bratteli diagrams of exact rank n, and let BD^*_n be the equivalence relation induced by equivalence of Bratteli diag...

conjecture | open | MathDB 页面 | 原文线索

V2-056765 | MATHDB-337111

The Borel bireducibility conjecture for consecutive-rank torsion-free abelian groups

摘录 / EXCERPT: For each $n\geq 1$, let $\cong_n$ be the isomorphism relation on the space of torsion-free abelian groups of rank n, and let $\cong_{n+1}^+$ be the corresponding relation with th...

conjecture | open | MathDB 页面 | 原文线索

V2-056766 | MATHDB-337199

The measurable repeat-point consistency conjecture for mutual stationarity

摘录 / EXCERPT: Let $\langle \kappa_n \mid n < \omega\rangle$ be a sequence of infinitely many measurable cardinals. For each n, suppose that κ_n carries an (ω,κ_{n-1}^+) repe...

conjecture | open | MathDB 页面 | 原文线索

V2-056767 | MATHDB-337322

Ultimate L conjecture under the Ultrapower and Ground Axioms

摘录 / EXCERPT: Assume the Ultrapower Axiom, the Ground Axiom, and that there exists a supercompact cardinal. Ultimate L conjecture. Then ... This conjecture proposes that the Ultrapower Axiom, toge...

conjecture | open | MathDB 页面 | 原文线索

V2-056768 | MATHDB-337758

Bailey's ring-theoretic characterization of the Generalized Continuum Hypothesis

摘录 / EXCERPT: Let GCH denote the assertion that for every infinite cardinal κ, there is no cardinal properly between κ and 2^κ. For an infinite cardinal number κ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-056769 | MATHDB-337851

The cardinal-invariant lower-bound conjecture for mad families of subspaces

摘录 / EXCERPT: Let F be a countable field, let $\mathrm{non}(\mathcal{M})$ denote the uniformity of the meager ideal, and let $\mathfrak{a}_{\mathrm{vec},F}$ be the least cardinality of a mad...

conjecture | open | MathDB 页面 | 原文线索

V2-056770 | MATHDB-337852

The diamond principle conjecture for mad families of block subspaces

摘录 / EXCERPT: Let F be a countable field, and let $\diamondsuit(\omega^\omega,=^\infty)$ denote the parameterized diamond principle appearing in the source. Diamond principle conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-056771 | MATHDB-338092

The closure conjecture for the first omitting cardinal of Magidority

摘录 / EXCERPT: Closure conjecture. If $\theta<\alpha$, then

conjecture | open | MathDB 页面 | 原文线索

V2-056772 | MATHDB-338245

The torus exponential domination lower-bound conjecture

摘录 / EXCERPT: Let C_m and C_n be cycle graphs, and let $C_m\square C_n$ be their Cartesian product. Write $\gamma^*_e(G)$ for the minimum cardinality of an exponential dominating set in a gr...

conjecture | open | MathDB 页面 | 原文线索

V2-056773 | MATHDB-338246

The King grid exponential domination lower-bound conjecture

摘录 / EXCERPT: Let $\mathcal{K}_n=P_n\boxtimes P_n$ be the King grid, where P_n is the path on n vertices. Write $\gamma^*_e(G)$ for the minimum cardinality of an exponential dominating set i...

conjecture | open | MathDB 页面 | 原文线索

V2-056774 | MATHDB-338523

Better-quasi-order conjecture for continuous rational-valued functions

摘录 / EXCERPT: Let $[\omega]^2_+$ denote the relevant domain of ordered pairs of natural numbers, let $C([\omega]^2_+,\mathbb{Q})$ be the space of continuous functions from this domain to...

conjecture | open | MathDB 页面 | 原文线索

V2-056775 | MATHDB-338955

Equiconsistency of $\textbackslash kappa$-compactness and $\textbackslash Pi^1_1$-subcompactness

摘录 / EXCERPT: Equiconsistency conjecture. The existence of $\textbackslash kappa$ which is $\textbackslash kappa$-compact is equiconsistent with the existence of a cardinal...

conjecture | open | MathDB 页面 | 原文线索

V2-056776 | MATHDB-339068

The analytic non-orbit-equivalence conjecture for E_1

摘录 / EXCERPT: The analytic non-orbit-equivalence conjecture. The equivalence E_1 is not the least analytic equivalence relation which is not Borel reducible to an orbit equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-056777 | MATHDB-339548

The nonexistence of a set of all sets in the category of sets and functions

摘录 / EXCERPT: In the category of sets and functions, whose objects are sets and whose morphisms are functions between sets, the nonexistence conjecture for a set of all sets. The category does n...

conjecture | open | MathDB 页面 | 原文线索

V2-056778 | MATHDB-339549

The nonexistence of a set of all functions in the category of sets and functions

摘录 / EXCERPT: Let the category of sets and functions on sets have sets as objects and functions on sets as morphisms. The nonexistence conjecture for a set of all functions. The category does no...

conjecture | open | MathDB 页面 | 原文线索

V2-056779 | MATHDB-339605

Consistency of separating the generalized null ideal from Cohen forcing

摘录 / EXCERPT: Consistency conjecture. There exists a model $\bo\mathbf V$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-056780 | MATHDB-339606

Consistency of separating generalized null additivity from partial-slalom additivity

摘录 / EXCERPT: Partial-slalom separation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056781 | MATHDB-339634

The reduced-product rigidity conjecture for II$_1$-factors

摘录 / EXCERPT: Assume $\mathrm{OCA}_\infty+\mathrm{MA}_{\aleph_1}$. Let (M_i) and (N_i) be sequences of separably representable type II$_1$ factors, and let $\bigoplus_\tau M_i$ be the ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-056782 | MATHDB-340061

Grossberg's eventual amalgamation conjecture for abstract elementary classes

摘录 / EXCERPT: Let λ be a cardinal. For an abstract elementary class (AEC) $\boldsymbol{K}$, write $\operatorname{LS}(\boldsymbol{K})$ for its Löwenheim-Skolem number, and say that...

conjecture | open | MathDB 页面 | 原文线索

V2-056783 | MATHDB-340591

The forcing obstruction to embedding large abelian C*-algebras into the reduced product

摘录 / EXCERPT: Forcing obstruction conjecture. The poset $\mathbb{E}_A$ forces that A does not embed into ℓ_∞/c_0.

conjecture | solved | MathDB 页面 | 原文线索

V2-056784 | MATHDB-340592

The nonseparable subalgebra nonliftability conjecture for the generic embedding

摘录 / EXCERPT: Nonseparable subalgebra nonliftability conjecture. The poset $\mathbb{E}_A$ forces that, whenever B is a nonseparable subalgebra of the image of A under the generic embedding,...

conjecture | solved | MathDB 页面 | 原文线索

V2-056785 | MATHDB-340697

Marks's conjecture on uniform universality of countable Borel equivalence relations

摘录 / EXCERPT: A countable Borel equivalence relation is an equivalence relation on a standard Borel space whose classes are countable and which is Borel as a subset of the product space. Marks's...

conjecture | open | MathDB 页面 | 原文线索

V2-056786 | MATHDB-340749

Hjorth-Kechris-Louveau strictness conjecture for Borel equivalence-relation hierarchies

摘录 / EXCERPT: Hjorth-Kechris-Louveau strictness conjecture. The following Borel reducibilities are strict:

conjecture | open | MathDB 页面 | 原文线索

V2-056787 | MATHDB-341039

Strong Sperner conjecture for weak orders on finite Coxeter groups

摘录 / EXCERPT: Let C be a finite Coxeter group, equipped with its weak order. A ranked poset is strongly Sperner if, for every positive integer r, the maximum size of a union of r antichain...

conjecture | open | MathDB 页面 | 原文线索

V2-056788 | MATHDB-341252

Coskey-Farah conjecture on trivial homeomorphisms of Čech-Stone remainders

摘录 / EXCERPT: Let X and Y be second countable, locally compact, noncompact spaces. Write $X^*=\beta X\setminus X$ and $Y^*=\beta Y\setminus Y$, where βX and βY are their Čech-S...

conjecture | open | MathDB 页面 | 原文线索

V2-056789 | MATHDB-341585

Gorenstein* poset nonnegativity and Boolean-lattice lower-bound conjectures

摘录 / EXCERPT: A poset is Gorenstein when it is Cohen--Macaulay and Eulerian. Its cd-index is denoted by $\Phi_P(c,d)$; coefficientwise comparison means comparison of the coefficients of corres...

conjecture | open | MathDB 页面 | 原文线索

V2-056790 | MATHDB-342052

The forcing-set decomposition conjecture for bipartite planar graphs

摘录 / EXCERPT: Forcing-set decomposition conjecture. The matching M can be decomposed into g pairwise disjoint forcing sets.

conjecture | open | MathDB 页面 | 原文线索

V2-056791 | MATHDB-342317

Woodin's Axiom of Choice conjecture

摘录 / EXCERPT: Let V be a model of set theory with a large cardinal, for example an extendible cardinal. Woodin's Axiom of Choice conjecture. If V has a large cardinal, e.g., an extendible ca...

conjecture | open | MathDB 页面 | 原文线索

V2-056792 | MATHDB-342332

Equiconsistency of supercompactness of ω_1 with determinacy and a Woodin limit

摘录 / EXCERPT: Equiconsistency conjecture. The three displayed theories are equiconsistent.

conjecture | open | MathDB 页面 | 原文线索

V2-056793 | MATHDB-342333

Determinacy equivalence under supercompactness of ω_1

摘录 / EXCERPT: Determinacy equivalence conjecture. Assuming that ω_1 is supercompact, $\sf{AD}$ is equivalent to $\sf{AD}^+$.

conjecture | open | MathDB 页面 | 原文线索

V2-056794 | MATHDB-342334

Determinacy of Suslin co-Suslin sets from supercompactness of ω_1

摘录 / EXCERPT: Suslin co-Suslin determinacy conjecture. If ω_1 is supercompact, then every Suslin co-Suslin set A is determined.

conjecture | open | MathDB 页面 | 原文线索

V2-056795 | MATHDB-342412

Type B Hurwitz orbit cardinality conjecture for grouped Coxeter factorizations

摘录 / EXCERPT: Fix simple reflections $s_0,s_1,\ldots,s_{kn-1}$ in the hyperoctahedral group of type B_{kn}, with $(s_0s_1)^4=1$, and let $\mathbf{t}$ be the grouped factorization ... The Hurwitz...

conjecture | open | MathDB 页面 | 原文线索

V2-056796 | MATHDB-342460

Tannakian reformulation of Matzat's freeness conjecture

摘录 / EXCERPT: Let k be an algebraically closed field of characteristic zero, set $K=k(x)$ with derivation d/dx, and let F_X be the abstract free group on a set X of cardinality $|k|$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-056797 | MATHDB-342706

The de Jongh property conjecture for intuitionistic set theory

摘录 / EXCERPT: Let IZF denote Intuitionistic Zermelo-Fraenkel set theory, and let an intermediate logic be a propositional logic extending intuitionistic propositional logic. The...

conjecture | open | MathDB 页面 | 原文线索

V2-056798 | MATHDB-342801

Jaegers' recursive monotone fixed-point conjecture

摘录 / EXCERPT: Let ${\mathcal{N}}$ denote Baire space. A partial ordering $\preceq$ on ${\mathcal{N}}$ is Π^0_1, every $\preceq$-chain has a least upper bound, and...

conjecture | open | MathDB 页面 | 原文线索

V2-056799 | MATHDB-342842

Claw-free regular graph power domination conjecture

摘录 / EXCERPT: Let $l\ge 1$ and $k\ge l$ be integers, and let G be a connected claw-free $(k+l+1)$-regular graph of order n. Let $\gamma_{p,k}(G)$ denote the minimum cardinality of a k-powe...

conjecture | open | MathDB 页面 | 原文线索

V2-056800 | MATHDB-343123

Equiconsistency conjecture for KP and real determinacy

摘录 / EXCERPT: Let ${\mathsf{KP}}$ denote Kripke-Platek set theory, let ${\mathsf{AD}}_\mathbb{R}$ denote the axiom of determinacy for games on reals, and let ${\mathsf{AD}}$ denote the axiom of...

conjecture | open | MathDB 页面 | 原文线索

V2-056801 | MATHDB-343172

Level-by-level determinacy equivalence for games on the reals

摘录 / EXCERPT: Let $n\in\mathbb{N}$. A game on $\mathbb{R}$ is Π^1_{n+1}-determinacy when every game whose payoff set is in the pointclass Π^1_{n+1} is determined. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056802 | MATHDB-343520

Non-reversibility of the implication from SCC^cof to SCC^split

摘录 / EXCERPT: Let SCC^{cof} and $\text{SCC}^{\text{split}}$ denote the two set-theoretic principles introduced in the surrounding discussion. Non-reversibility conjecture. The im...

conjecture | open | MathDB 页面 | 原文线索

V2-056803 | MATHDB-343521

Saturation conjecture for ideals extending the nonstationary ideal

摘录 / EXCERPT: Let $\kappa=\nu^+$, and let θ be a regular infinite cardinal less than κ with $\theta\ne\operatorname{cf}(\nu)$. Let $NS_\kappa\vert E^\kappa_\theta$ denote the restr...

conjecture | open | MathDB 页面 | 原文线索

V2-056804 | MATHDB-343644

Steel's conjecture on critical points in decreasing Mitchell-order sequences

摘录 / EXCERPT: Steel's conjecture. The critical points of the extenders are cofinal in λ:

conjecture | open | MathDB 页面 | 原文线索

V2-056805 | MATHDB-343677

The tropical-polynomial characterization of tropicalized local field Gaussians

摘录 / EXCERPT: Let K be a local field with residue-field cardinality q, let L be a lattice in K^d, and let V be the tropicalization of a full-dimensional Gaussian in K^d with lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-056806 | MATHDB-343712

Base-size conjecture for definably primitive groups of finite Morley rank

摘录 / EXCERPT: Let (G,X) be a faithful definable permutation group of finite Morley rank, let G be connected and definably primitive on X, and let $b(G)$ denote the minimal cardinality of a...

conjecture | open | MathDB 页面 | 原文线索

V2-056807 | MATHDB-344360

The $2^{\mathrm{rk}(\Gamma)}$ chain-length consequence

摘录 / EXCERPT: Let Γ be a vector space of infinite rank, and let $\mathrm{rk}(\Gamma)$ denote its rank. A chain of subspaces is a totally ordered collection under inclusion, and its length...

conjecture | open | MathDB 页面 | 原文线索

V2-056808 | MATHDB-344453

Woodin's HOD conjecture

摘录 / EXCERPT: HOD conjecture. There is a proper class of regular uncountable cardinals κ which are not ω-strongly measurable in HOD.

conjecture | open | MathDB 页面 | 原文线索

V2-056809 | MATHDB-344781

The C-sequence number and infinite productivity of the Knaster property

摘录 / EXCERPT: Let κ be a regular uncountable cardinal. Write $\chi(\kappa)$ for the C-sequence number of κ, and call a poset $\mathbb P$ κ-Knaster if every subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-056810 | MATHDB-345396

The consistency-strength conjecture for ZFR with dependent choice

摘录 / EXCERPT: Let ZFR denote the set-theoretic theory under discussion, and let DC denote the principle of dependent choice. Consistency-strength conjecture. The theory...

conjecture | open | MathDB 页面 | 原文线索

V2-056811 | MATHDB-345820

The finite sparse-forcing conjecture

摘录 / EXCERPT: Let $\mathcal{S}$ be a set of graphs. It is sparse forcing if, whenever graphs G_n satisfy $|G_n|\to\infty$, have edge density $p_n=|G_n|^{-o(1)}$, and the limits ... exist for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-056812 | MATHDB-346044

Steel's conjecture for $(\equiv_T,\equiv_m)$-invariant functions

摘录 / EXCERPT: Let 2^{ω} be Cantor space. A function $f\colon 2^{\omega}\to 2^{\omega}$ is $(\equiv_T,\equiv_m)$-invariant if Turing-equivalent inputs have many-one-equivalent outputs; it...

conjecture | open | MathDB 页面 | 原文线索

V2-056813 | MATHDB-346045

Steel's cone-equivalence conjecture for $(\equiv_T,\equiv_m)$-invariant functions

摘录 / EXCERPT: Let 2^{ω} be Cantor space. A function $f\colon 2^{\omega}\to 2^{\omega}$ is $(\equiv_T,\equiv_m)$-invariant, and a function is uniformly invariant when its many-one equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-056814 | MATHDB-346046

Pointed-perfect-set conjecture for increasing Borel approximations to Turing equivalence

摘录 / EXCERPT: Let $(E_n)_{n\in\omega}$ be Borel equivalence relations on 2^{ω} such that $E_n\subseteq E_{n+1}$ for every n, and suppose $\equiv_T=\bigcup_{n\in\omega}E_n$. A pointed pe...

conjecture | open | MathDB 页面 | 原文线索

V2-056815 | MATHDB-346449

Initial minors conjecture for elliptic curve discrete logarithms

摘录 / EXCERPT: Initial minors conjecture. For a matrix $\mathcal{A}$ coming from the above approach to solve ECDLP, there exists a set of initial minors whose cardinality is bounded by a sub-expo...

conjecture | open | MathDB 页面 | 原文线索

V2-056816 | MATHDB-346598

Consistency conjecture for the Copernican multiverse theory

摘录 / EXCERPT: Consistency conjecture. $\mathsf{GR}^\omega(\mathsf{ZFC})$ interprets...

conjecture | open | MathDB 页面 | 原文线索

V2-056817 | MATHDB-346728

Prediction about consistency of the Reinhardt cardinal theory

摘录 / EXCERPT: Let κ be the least Reinhardt cardinal, meaning the least critical point of a nontrivial elementary embedding from the universe to itself in a model of ZF. Consist...

conjecture | open | MathDB 页面 | 原文线索

V2-056818 | MATHDB-346784

Unique normal ultrafilters in an extendible-cardinal model

摘录 / EXCERPT: A cardinal κ is measurable if it carries a nonprincipal κ-complete ultrafilter, and a normal ultrafilter is one satisfying the usual normality condition. Uniqueness c...

conjecture | open | MathDB 页面 | 原文线索

V2-056819 | MATHDB-346785

Linearity of the Mitchell order with an extendible cardinal

摘录 / EXCERPT: For normal ultrafilters U and W, the Mitchell order is defined by $U\mathbin{\triangleleft}W$ when U belongs to the ultrapower of the universe by W. Mitchell-order linearit...

conjecture | open | MathDB 页面 | 原文线索

V2-056820 | MATHDB-346786

The Inner Model Problem for an extendible cardinal

摘录 / EXCERPT: An inner model is a transitive proper-class model of set theory containing the relevant ordinals and satisfying the indicated axioms. An extendible cardinal is a cardinal with the...

open_problem | open | MathDB 页面 | 原文线索

V2-056821 | MATHDB-346787

Consistency of the Ultrapower Axiom with an extendible cardinal

摘录 / EXCERPT: The Ultrapower Axiom is the assertion that any pair of ultrapower embeddings has an internal ultrapower comparison. An extendible cardinal is a cardinal satisfying the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-056822 | MATHDB-346788

Consistency of the Ultrapower Axiom with a regular pathological cardinal

摘录 / EXCERPT: A cardinal λ is regular if $\operatorname{cf}(\lambda)=\lambda$, and it is pathological in the sense used in the source when it witnesses the failure of the relevant superc...

conjecture | open | MathDB 页面 | 原文线索

V2-056823 | MATHDB-346812

The Kinna-Wagner conjecture for intermediate generic extensions

摘录 / EXCERPT: Let V be a model satisfying the Kinna-Wagner principle KWP, let G be a V-generic filter, and let M be an intermediate model with $V\subseteq M\subseteq V[G]$. Ki...

conjecture | open | MathDB 页面 | 原文线索

V2-056824 | MATHDB-347419

The consistency-strength conjecture for bounded symbiosis and upwards reflection

摘录 / EXCERPT: Consistency-strength conjecture. ${\mathcal{USR}}(\mathrm{PwSt})$ and ${\sf ULST}_\infty({\mathcal{L}}^2)$ are defined if and only if there exists an extendible cardinal.

conjecture | open | MathDB 页面 | 原文线索

V2-056825 | MATHDB-348835

The natural consistency-strength conjecture for extensions of ZFC

摘录 / EXCERPT: Natural consistency-strength conjecture. Any natural extension of ZFC is either equiconsistent with ZFC or equiconsistent with $\mathsf{ZFC}+A$, where A is...

conjecture | open | MathDB 页面 | 原文线索

V2-056826 | MATHDB-349081

The minimal mediated sequence cardinality conjecture

摘录 / EXCERPT: For integers p and q with $0<q<p$, let $N(q/p)$ denote the number of elements in a minimal $(0,p)$-mediated sequence containing q. A sequence of integers...

conjecture | open | MathDB 页面 | 原文线索

V2-056827 | MATHDB-349386

Černý-Pin rank conjecture for complete automata

摘录 / EXCERPT: Let $\mathrsfs{A}$ be an n-state complete automaton, and let r be the minimal rank of any word, where the rank of a word is the cardinality of its image on the state set. Rank...

conjecture | open | MathDB 页面 | 原文线索

V2-056828 | MATHDB-349466

Extension conjecture for orbit equivalences of cocompact cross sections

摘录 / EXCERPT: Extension conjecture. There exists a smooth equivalence $\xi: \Omega_{1} \to \Omega_{2}$ that extends ζ. This would establish the possibility of extending orbit equivalences...

conjecture | open | MathDB 页面 | 原文线索

V2-056829 | MATHDB-349601

The Iwasawa-theoretic Mazur-Rubin-Sano conjecture

摘录 / EXCERPT: Let K_∞/k be the relevant $\mathds{Z}_p$-extension, let Γ be its pro-p direction, let $W=V'\setminus V$ have cardinality e, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056830 | MATHDB-349699

Goldberg's conjecture on products of ultrafilters

摘录 / EXCERPT: Let U and W be ultrafilters. Their Cartesian product $U\times W$ is the filter on the product of their underlying sets generated by rectangles $A\times B$, where $A\in U$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056831 | MATHDB-349700

Kunen-Blass equivalence conjecture for countably complete ultrafilters

摘录 / EXCERPT: Let U and W be countably complete ultrafilters. Write $j_U:V\to M_U$ and $j_W:V\to M_W$ for their ultrapower embeddings, and let $j_{j_W(U)}$ and $j_{j_U(W)}$ denote the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-056832 | MATHDB-349986

Consistency conjecture for adjacent distance graphs

摘录 / EXCERPT: Let $n\geq 1$ be a number. Let Δ_n be the graph on $\mathbb{R}^n$ connecting points of rational Euclidean distance. The adjacent distance-graph consistency conjecture. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-056833 | MATHDB-350316

Sperner conjecture for the posets $P_{n,l}$

摘录 / EXCERPT: Let $P_{n,l}$ be the poset defined in the paper, with parameters n and l satisfying ... A poset is Sperner if its largest antichain has cardinality equal to the largest rank size...

conjecture | open | MathDB 页面 | 原文线索

V2-056834 | MATHDB-350778

The intermediate-model classification conjecture for Magidor-Radin forcing

摘录 / EXCERPT: Let $G\subseteq \mathbb{M}[\vec{U}]$ be a V-generic filter, where ... Let $V\subseteq M\subseteq V[G]$ be a transitive ZFC model. Intermediate-model classification conjecture. Ei...

conjecture | open | MathDB 页面 | 原文线索

V2-056835 | MATHDB-350874

Halting-realizability conjecture for cardinal-inequality \texorpdfstring{$\otimes$}{⊗}-graphs

摘录 / EXCERPT: Let $\mathcal{A}$ be an algorithm and x an input. A $\otimes$-graph with cardinal inequalities is an $\otimes$-graph $\mathcal{G}$ equipped with cardinal inequalities between its...

conjecture | open | MathDB 页面 | 原文线索

V2-056836 | MATHDB-350958

Martin's conjecture on degree-invariant functions

摘录 / EXCERPT: Martin's conjecture. Under these assumptions: (I) if f is degree-invariant and is not increasing a.e., then f is constant a.e.; and (II) $\leq_m$ pre-well-orders the set of deg...

conjecture | open | MathDB 页面 | 原文线索

V2-056837 | MATHDB-350997

The inner mantle conjecture for limit ordinals

摘录 / EXCERPT: Let $d702$ be a limit ordinal, and let $(\mathrm{M}^{\alpha})_{\alpha\in\mathrm{Ord}}$ denote the sequence of inner mantles, with M^{α} its stage at index α....

conjecture | open | MathDB 页面 | 原文线索

V2-056838 | MATHDB-350998

The inner mantle non-definability conjecture for limit ordinals

摘录 / EXCERPT: Let γ be a limit ordinal, and let $(\mathrm{M}^{\alpha})_{\alpha\in\mathrm{Ord}}$ denote the sequence of inner mantles, with M^{α} its stage at index...

conjecture | open | MathDB 页面 | 原文线索

V2-056839 | MATHDB-351256

Ineffability in the constructible universe from non-amenability

摘录 / EXCERPT: Let ba be a regular uncountable cardinal, and let $c3\textup{A}_ba$ denote the class defined in the paper. Non-amenability implies ineffability in $c3\textup{L}$. If ... then ba...

conjecture | open | MathDB 页面 | 原文线索

V2-056840 | MATHDB-351701

Grossberg's two-cardinal categoricity conjecture for AECs

摘录 / EXCERPT: Grossberg's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-056841 | MATHDB-351830

Debs-Saint Raymond rank characterization conjecture for analytic ideals

摘录 / EXCERPT: Debs-Saint Raymond conjecture. For every analytic ideal $\mathcal{I}$ and $0<\alpha<\omega_1$,

conjecture | open | MathDB 页面 | 原文线索

V2-056842 | MATHDB-351852

Model completeness conjecture for the universally Baire Chang model

摘录 / EXCERPT: Model completeness conjecture. The theory of $L_\Theta(\mathsf{UB}^{\omega_1})$ is model complete for the signature

conjecture | open | MathDB 页面 | 原文线索

V2-056843 | MATHDB-351853

AMC-spectrum conjecture for model companions of set-theoretic theories

摘录 / EXCERPT: AMC-spectrum conjecture. If $S+T_{\in,A}$ has a model companion, then

conjecture | open | MathDB 页面 | 原文线索

V2-056844 | MATHDB-352120

Alikhani-Peng conjecture on unimodality of domination polynomials

摘录 / EXCERPT: Let G be any graph, and let $D(G)$ denote its domination polynomial, whose coefficient sequence is the sequence of numbers of dominating sets of each cardinality. Alikhani-Peng c...

conjecture | open | MathDB 页面 | 原文线索

V2-056845 | MATHDB-352147

The PFA superstrong-cardinal conjecture for HOD

摘录 / EXCERPT: Let $\force$ denote the Proper Forcing Axiom, let $\theta\text{-}\rm{UB}$ denote the relevant universally Baire framework, and let ${\mathbb{R}}_g$ and Γ^∞_g be the rea...

conjecture | open | MathDB 页面 | 原文线索

V2-056846 | MATHDB-352148

The equivalence conjecture for sealing principles

摘录 / EXCERPT: Assume that there are unboundedly many Woodin cardinals and that the class of measurable cardinals is stationary. The principles $\sf{Sealing}$, $\sf{Sealing}^+$,...

conjecture | open | MathDB 页面 | 原文线索

V2-056847 | MATHDB-352202

Covering with Chang models

摘录 / EXCERPT: Assume ${\sf{NLE}}$ and suppose there are unboundedly many Woodin cardinals and strong cardinals. Let κ be a limit of Woodin cardinals and strong cardinals such that either...

conjecture | open | MathDB 页面 | 原文线索

V2-056848 | MATHDB-352207

The covering conjecture for iteration sets at Suslin cardinals

摘录 / EXCERPT: Covering conjecture. For every such λ, A, and B, there is an A-iteration set $X\in{\wp}_\kappa(\lambda)$ such that $B\subseteq X$. This asserts a covering property fo...

conjecture | open | MathDB 页面 | 原文线索

V2-056849 | MATHDB-352442

Consistency of SOCA at the aleph-two level

摘录 / EXCERPT: SOCA($\aleph_2$) consistency conjecture. The statement $\textbf{SOCA}(\aleph_2)$ is consistent.

conjecture | open | MathDB 页面 | 原文线索

V2-056850 | MATHDB-352471

Conjecture on decidable properties of the space of marked groups

摘录 / EXCERPT: Decidability-clopenness conjecture. Every Λ_{WP}-decidable property of $\mathcal{G}_{WP}$ is clopen.

conjecture | open | MathDB 页面 | 原文线索

V2-056851 | MATHDB-352573

Non-reducibility of NIN to HBU for Baire class 2 functions

摘录 / EXCERPT: Let $\textup{\textsf{NIN}}$ denote the problem of finding a function witnessing non-injectivity, and let $\textup{\textsf{HBU}}$ denote the Heine-Borel theorem for covers of...

conjecture | open | MathDB 页面 | 原文线索

V2-056852 | MATHDB-352610

Minimal-model conjecture under LSA

摘录 / EXCERPT: Assume AD^+ and that there is a largest Suslin cardinal which is a member of the Solovay sequence. A transitive model M of ZF is a minimal model of $\Theta strong$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-056853 | MATHDB-352611

LSA conjecture for a strong-cardinal model with a Woodin cardinal

摘录 / EXCERPT: Let LSA abbreviate the assertion that AD^+ holds and there is a largest Suslin cardinal belonging to the Solovay sequence. Let $\vec{E}$ be an extender sequence and write...

conjecture | open | MathDB 页面 | 原文线索

V2-056854 | MATHDB-352712

The ccc forcing limitation on clubsuit at the first uncountable cardinal

摘录 / EXCERPT: Let $V=L$. Let $\mathbb{P}$ be a ccc forcing notion and let $G\subseteq\mathbb{P}$ be generic. Write $\clubsuit_S$ for the clubsuit principle associated with a stationary set...

conjecture | open | MathDB 页面 | 原文线索

V2-056855 | MATHDB-352926

Invariance of Hausdorff null-set cardinal invariants for compact Polish spaces

摘录 / EXCERPT: Invariance conjecture. For every compact Polish space X with

conjecture | open | MathDB 页面 | 原文线索

V2-056856 | MATHDB-352963

Conjecture on mixing models for Boolean-satisfiable infinitary theories

摘录 / EXCERPT: Let L be a relational ω-signature. A theory in $\mathrm{L}_{\infty\infty}$ is Boolean satisfiable if it has a Boolean-valued model satisfying the theory. A Boolea...

conjecture | open | MathDB 页面 | 原文线索

V2-056857 | MATHDB-353055

A generic-classification dichotomy for analytic equivalence relations

摘录 / EXCERPT: Generic-classification dichotomy. Exactly one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-056858 | MATHDB-353104

Single HCP equivalence class conjecture for $D^2=2\ell^2$ divisible by 3

摘录 / EXCERPT: Single-class HCP conjecture. All EPGMs are generated by PCs from a single equivalence class. This class has cardinality $16\ell^3$ and consists of 8 D-HCP configurations and th...

conjecture | open | MathDB 页面 | 原文线索

V2-056859 | MATHDB-353164

The countability conjecture for threshold sets

摘录 / EXCERPT: Let D be the discrete Molchanov-Vainberg Laplacian, let $\boldsymbol{\Theta}_{m,\kappa}(D)$ and $\boldsymbol{\Theta}_{\kappa}(D)$ be the threshold sets defined in the paper, and...

conjecture | open | MathDB 页面 | 原文线索

V2-056860 | MATHDB-353423

The meta-conjecture for Borel quotient structures

摘录 / EXCERPT: Meta-conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-056861 | MATHDB-353424

The forcing-axiom rigidity conjecture for quotients over Borel ideals

摘录 / EXCERPT: Forcing-axiom rigidity conjecture. CH implies that if there is an isomorphism between

conjecture | open | MathDB 页面 | 原文线索

V2-056862 | MATHDB-353427

The forcing-absoluteness conjecture for non-isomorphic Borel quotients

摘录 / EXCERPT: Non-isomorphism absoluteness conjecture. If ${\mathcal M}/E$ and ${\mathcal N}/F$ are non-isomorphic in some forcing extension, then they are non-isomorphic in every forcing extens...

conjecture | open | MathDB 页面 | 原文线索

V2-056863 | MATHDB-353776

Nonexistence of iterated countable meets and joins in the nowhere centered quotient

摘录 / EXCERPT: Let $\beta=\bigwedge_{n_0<\omega}\bigvee_{n_1<\omega}\cdots\omega^{k-1}\times A_{(n_0,n_1,\ldots,n_{k-1})}$ be the indicated iterated operation on a family of sets...

conjecture | open | MathDB 页面 | 原文线索

V2-056864 | MATHDB-353952

Shelah's generalized categoricity conjecture for $L_{\lambda^+,\omega}$

摘录 / EXCERPT: Let ψ be a sentence of $L_{\lambda^+,\omega}$ in a language of size λ. Categoricity means having exactly one model, up to isomorphism, in a given cardinality. Shelah's...

conjecture | open | MathDB 页面 | 原文线索

V2-056865 | MATHDB-354111

Effectivizability conjecture for finite Borel CSPs

摘录 / EXCERPT: Effectivizability conjecture. For any finite structure $\mathcal{D}$, $\mathrm{CSP}_B(\mathcal{D})$ is in $\mathbf{\Pi}^1_1$ if and only if it is effectivizable.

conjecture | open | MathDB 页面 | 原文线索

V2-056866 | MATHDB-355505

Strong dualizability conjecture for Borel equivalence relations

摘录 / EXCERPT: Let X be a standard Borel space, and let E be a Borel equivalence relation on X. Strong dualizability conjecture. Every such Borel equivalence relation is strongly dualizable...

conjecture | open | MathDB 页面 | 原文线索

V2-056867 | MATHDB-355531

General recursion formula for derivatives of local representation densities

摘录 / EXCERPT: Let q be the residue-field cardinality, and let $A_{\alpha\beta\gamma\delta}$ denote the relevant diagonal hermitian form with $\alpha>\beta\geq\gamma\geq\delta\geq0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-056868 | MATHDB-355974

Matzat's conjecture on absolute differential Galois groups

摘录 / EXCERPT: Matzat's conjecture. The absolute differential Galois group of F is the free proalgebraic group on a set of cardinality $|F|$.

conjecture | open | MathDB 页面 | 原文线索

V2-056869 | MATHDB-355975

Shelah's definability-function cardinal conjecture

摘录 / EXCERPT: Shelah's conjecture. If for some $\lambda\geq|\tau_1|$, $\operatorname{Df}(\lambda)>\operatorname{Ded}(\lambda)$, then for every λ,...

conjecture | open | MathDB 页面 | 原文线索

V2-056870 | MATHDB-355976

Shelah's consistency conjecture for Specker orders and Aronszajn trees

摘录 / EXCERPT: Let a Specker order and the orders I,J be as described immediately before the statement. Let T be an Aronszajn tree, let $c:T\to 2$, and write $\leq_T$ for its tree order. Shel...

conjecture | solved | MathDB 页面 | 原文线索

V2-056871 | MATHDB-356021

The Alternate Thomassé conjecture on sibling numbers of relations

摘录 / EXCERPT: The Alternate Thomassé conjecture. For every relation R of arbitrary cardinality,

conjecture | open | MathDB 页面 | 原文线索

V2-056872 | MATHDB-356110

Exact count conjecture for time-invariant Nash equilibria

摘录 / EXCERPT: Let $Q:(0,\infty)\to\mathbb{N}$ denote the maximum cardinality of a set of mutually shift-inequivalent time-invariant Nash equilibria for the game $\operatorname{Standard}(x)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-056873 | MATHDB-356317

Shelah's chain-length conjecture for eventual domination

摘录 / EXCERPT: Shelah's chain-length conjecture. For every $\mu\geq\theta$, there are no chains in

conjecture | open | MathDB 页面 | 原文线索

V2-056874 | MATHDB-356517

Sargsyan's regular Suslin unreachability conjecture

摘录 / EXCERPT: Assume AD^+. Let κ be a regular Suslin cardinal. Sargsyan's conjecture. The successor cardinal κ^+ is $S(\kappa)$-unreachable. The abstract presents this as the ana...

conjecture | open | MathDB 页面 | 原文线索

V2-056875 | MATHDB-356518

Steel's lightface Levy-hierarchy mouse set conjecture

摘录 / EXCERPT: Assume $ZF + AD + DC + V=L(\mathbb{R})$. A mouse set is a set $X\subseteq\mathbb{R}$ for which there is an ω_1+1-iterable premouse M with $X=M\cap\mathbb{R}$. Let Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-056876 | MATHDB-356519

The relativized mouse set conjecture for Levy-hierarchy pointclasses

摘录 / EXCERPT: Assume $ZF + AD + DC + V=L(\mathbb{R})$. Let $\alpha\in ON$ and $n\in\omega$. For each $x\in\mathbb{R}$, define Γ_x to be the collection of all pointsets A such that for s...

conjecture | open | MathDB 页面 | 原文线索

V2-056877 | MATHDB-356655

Regularity of successor thorns under large-cardinal assumptions

摘录 / EXCERPT: Let $\textit{\th}_0=\omega$, let $\textit{\th}_{\alpha+1}=\textup{\TH}(\textit{\th}_\alpha^\omega)$, and for limit γ let...

conjecture | open | MathDB 页面 | 原文线索

V2-056878 | MATHDB-356673

Directed forcing conjecture for oriented graphs

摘录 / EXCERPT: Let B be an oriented graph, let $\overline{B}$ be its underlying undirected graph obtained by forgetting edge directions, and call B directed-forcing when it has the directed f...

conjecture | open | MathDB 页面 | 原文线索

V2-056879 | MATHDB-357263

Non-finite generation of semirings of p-cardinalities

摘录 / EXCERPT: Non-finite-generation conjecture. For every such complexity class $\mathcal{C}$, the semiring of p-cardinalities of languages in $\mathcal{C}$ is not finitely generated. The same h...

conjecture | open | MathDB 页面 | 原文线索

V2-056880 | MATHDB-357354

The relative consistency conjecture for Baire's grand theorem with dependent choice

摘录 / EXCERPT: The principles $\mathsf{GBT+DC}$ and ZF are considered as set-theoretic theories, where GBT denotes Baire's grand theorem and DC denotes the axiom...

conjecture | open | MathDB 页面 | 原文线索

V2-056881 | MATHDB-358074

The cofinality and Sealing conjecture for $\mathcal{B}^\infty$

摘录 / EXCERPT: Suppose κ is a supercompact cardinal and there is a proper class of inaccessible limits of Woodin cardinals. Let $g\subseteq\operatorname{Col}(\omega,2^{2^\kappa})$ be V-g...

conjecture | open | MathDB 页面 | 原文线索

V2-056882 | MATHDB-358252

Decidability of the monadic theory of orders of size at most $\aleph_1$

摘录 / EXCERPT: Small-order decidability conjecture. The monadic theory of orders of cardinality at most $\aleph_1$ is decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-056883 | MATHDB-358253

Decidability of Boolean algebras with ideal quantification

摘录 / EXCERPT: Boolean-algebra decidability conjecture. The theory of Boolean algebras of cardinality less than λ, or the first-order theory of Boolean algebras expanded by quantification...

conjecture | open | MathDB 页面 | 原文线索

V2-056884 | MATHDB-358254

Consistency of the Boolean-algebra splitting property

摘录 / EXCERPT: Consistency conjecture. The property $(*)$ is consistent with ZFC.

conjecture | open | MathDB 页面 | 原文线索

V2-056885 | MATHDB-358310

The consistency-strength conjecture for $d4a2\mathfrak{Sp}^* = \omega_1$

摘录 / EXCERPT: Let $\mathfrak{Sp}^*$ denote the spectrum introduced in the surrounding discussion, and let a cardinal κ be virtually strong if for every $\gamma \geq \kappa$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-056886 | MATHDB-358458

Converse of the internal-automorphism criterion for Frucht's theorem

摘录 / EXCERPT: Let M be a model of set theory satisfying the Axiom of Choice, and let $M'=M/\mathcal{F}$ be a permutation model inside M. Suppose that Frucht's Theorem fails in M'. An inter...

conjecture | open | MathDB 页面 | 原文线索

V2-056887 | MATHDB-358475

Finite Clifford spectrum cardinality conjecture

摘录 / EXCERPT: Let $A_{1},\dots,A_{d}$ be Hermitian matrices in $\boldsymbol{M}_{n}(\mathbb{C})$, and let $\Lambda(A_{1},\dots,A_{d})$ denote their Clifford spectrum. Finite-cardinality conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-056888 | MATHDB-358724

Part 1 of Martin's conjecture for Turing invariant functions

摘录 / EXCERPT: Part 1 of Martin's conjecture. Assuming $\mathsf{ZF}+\mathsf{AD}$, if $f\colon 2^\omega\to 2^\omega$ is a Turing invariant function, then either

conjecture | refuted | MathDB 页面 | 原文线索

V2-056889 | MATHDB-358976

The covering-with-Chang-models conjecture

摘录 / EXCERPT: Covering with Chang Models conjecture. The asserted transitive model M exists with all the listed properties. This conjecture is intended to overcome limitations of the core mode...

conjecture | open | MathDB 页面 | 原文线索

V2-056890 | MATHDB-359188

The generalized derived model theorem for regularity of Θ

摘录 / EXCERPT: Let δ be a Woodin cardinal that is a limit of Woodin cardinals. Let $g\text{?}$ be V-generic for $\mathrm{Col}(\omega,{<}\delta)$, and let $L(\Gamma^*_g,\mathbb{R}^*_g)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-056891 | MATHDB-359189

Conjecture on the height of the derived cardinal

摘录 / EXCERPT: Let $(\mathcal{V},\Omega)$ be an excellent least branch hod pair with $\mathcal{V}\models\mathsf{ZFC}$, let δ be a Woodin limit of Woodin cardinals in $\mathcal{V}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056892 | MATHDB-359239

The bounded generic-base-gap conjecture for finite Morley rank actions

摘录 / EXCERPT: Let (G,S) be a definably primitive permutation group of finite Morley rank. Let $b(G,S)$ be the minimal cardinality of a base, and let $b^1(G,S)$ be the least positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-056893 | MATHDB-359801

Kechris's universality conjecture for Turing equivalence

摘录 / EXCERPT: A countable Borel equivalence relation is an equivalence relation on a Borel subset of 2^ω whose equivalence classes are countable and whose relation is Borel. A Borel reduc...

conjecture | open | MathDB 页面 | 原文线索

V2-056894 | MATHDB-359802

Non-universality conjecture for height-one locally countable Borel quasi-orders

摘录 / EXCERPT: A quasi-order $(P,\leq_P)$ is a reflexive and transitive relation, and its height is the length of the longest strictly decreasing chain, equivalently the height of the partial ord...

conjecture | open | MathDB 页面 | 原文线索

V2-056895 | MATHDB-359809

Removal of the cardinality restriction in weak naturality

摘录 / EXCERPT: A uniformisable construction is a construction of the kind considered by Hodges and Shelah in which the relevant lifting can be chosen uniformly. Their result constructs a transiti...

conjecture | open | MathDB 页面 | 原文线索

V2-056896 | MATHDB-360279

The structural reflection characterization conjecture for large cardinals

摘录 / EXCERPT: Structural reflection characterization conjecture. Every known large cardinal notion is equivalent to some form of SRP.

conjecture | open | MathDB 页面 | 原文线索

V2-056897 | MATHDB-360280

The Structural Reflection characterization of least large cardinals

摘录 / EXCERPT: Least-cardinal structural reflection conjecture. A cardinal $kappa$ is the least cardinal satisfying some large-cardinal notion if and only if $kappa$ is the least cardinal satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-056898 | MATHDB-360680

Conjecture on realizing minimal-base cardinalities for transitive permutation groups

摘录 / EXCERPT: Let $X\subseteq\mathbb{N}$ with $1\notin X$. For a permutation group G acting transitively on a domain Ω, let $\mathcal{M}(G,\Omega)$ denote the set of cardinalities of mi...

conjecture | open | MathDB 页面 | 原文线索

V2-056899 | MATHDB-360681

Conjecture that minimal-base cardinalities form an interval for primitive groups

摘录 / EXCERPT: Let G be a primitive permutation group on a domain Ω, and let $\mathcal{M}(G,\Omega)$ denote the set of cardinalities of minimal bases of G on Ω. Primitive interv...

conjecture | open | MathDB 页面 | 原文线索

V2-056900 | MATHDB-360682

Conjecture on forbidden intervals of irredundant-base cardinalities for primitive groups

摘录 / EXCERPT: Let $\mathcal{I}(G,\Omega)$ denote the set of cardinalities of irredundant bases of a permutation group G on Ω. Forbidden-interval conjecture. There exists an interval X...

conjecture | open | MathDB 页面 | 原文线索

V2-056901 | MATHDB-360695

Higher-cardinality universal optimality conjecture for rectangular flat tori

摘录 / EXCERPT: For $\beta>0$, let ... and, for $m\geq 2$, define ... A configuration is L^m_β-universally optimal if it minimizes every admissible periodic energy among $2m$-point configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-056902 | MATHDB-360821

Invariance of reflecting ordinals under iterated Sacks, Miller, and Laver forcing

摘录 / EXCERPT: Let L be the constructible universe, and let θ range over the reflecting ordinals σ^1_n and π^1_n. Consider forcing over L with countable-support iterations o...

conjecture | open | MathDB 页面 | 原文线索

V2-056903 | MATHDB-360970

Unreachability below $\bf{\Delta^1_3}$

摘录 / EXCERPT: Let $\bf{\Gamma}$ be a pointclass, and let κ be $\bf{\Gamma}$-reachable when there is a sequence of distinct $\bf{\Gamma}$-sets of length κ; otherwise, κ is...

conjecture | open | MathDB 页面 | 原文线索

V2-056904 | MATHDB-360971

Existence of pointclasses realizing intermediate least unreachable cardinals

摘录 / EXCERPT: Let λ be a cardinal with $\bf{\delta^1_4}\leq\lambda\leq\bf{\delta^1_6}$, and suppose $\operatorname{cof}(\lambda)>\omega$. A cardinal is $\bf{\Gamma}$-unreachable when the...

conjecture | open | MathDB 页面 | 原文线索

V2-056905 | MATHDB-361154

The condensed Whitehead conjecture for complete Boolean algebras

摘录 / EXCERPT: The condensed Whitehead conjecture. Suppose that A is an abelian group, $\mathbb{B}$ is a complete Boolean algebra, and

conjecture | open | MathDB 页面 | 原文线索

V2-056906 | MATHDB-361657

Conjecture on the Cantor-Bendixson structure of spaces of shifts

摘录 / EXCERPT: Cantor-Bendixson structure conjecture. For all $d>1$, the Cantor-Bendixson rank of $\mathcal{S}^d$ is ω. Furthermore, the Cantor-Bendixson residue is the closure of the set...

conjecture | open | MathDB 页面 | 原文线索

V2-056907 | MATHDB-361704

Solovay-sequence conjecture on club HOD-Berkeley cardinals

摘录 / EXCERPT: Solovay-sequence conjecture. Under these assumptions, $V_\zeta^{{\sf{HOD}}_X}$ is a model of

conjecture | open | MathDB 页面 | 原文线索

V2-056908 | MATHDB-362570

Archer-Geary conjecture on chain-avoiding permutations

摘录 / EXCERPT: Let $C_n(231,1432: 231)$ be the set of permutations in S_n that avoid the chain $(231,1432: 231)$, and let $c_n(231,1432: 231)$ denote its cardinality. Thus, the permutation avoi...

conjecture | open | MathDB 页面 | 原文线索

V2-056909 | MATHDB-362604

Working hypothesis on embedding partial orders into the Rudin-Keisler ordering of P-points

摘录 / EXCERPT: Let $\mathsf{MA}(sigma\text{-centered})$ hold, and let $(P,<)$ be a partial order of size at most $2^{\mathfrak{c}}$ such that every $p\text{in}P$ has at most $\mathfrak{c}$ many p...

conjecture | open | MathDB 页面 | 原文线索

V2-056910 | MATHDB-363217

The countable invariant uniformization characterization of reducibility to countable

摘录 / EXCERPT: Countable invariant uniformization conjecture. Let E be a Borel equivalence relation on a Polish space X. Then the following are equivalent: (a) E is reducible to countable;...

conjecture | open | MathDB 页面 | 原文线索

V2-056911 | MATHDB-363289

The finite Friedman-Stanley jump spectrum conjecture

摘录 / EXCERPT: Spectrum conjecture. For each countable ordinal α, $=^{+\alpha}$ is in the spectrum of the meager ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-056912 | MATHDB-363386

Order ideal cardinality homomesy for odd uniform self-dual fences

摘录 / EXCERPT: Let a be a positive integer, let t be odd, and let $F=\breve{F}(a^t)$ be the uniform self-dual fence with t segments. Write $n=\#F$ for its number of elements, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056913 | MATHDB-363585

Woodin's Ultimate-L conjecture that Ultimate-L implies the Extender Embedding Axiom

摘录 / EXCERPT: Let EEA denote the Extender Embedding Axiom, and let $V=\mathrm{Ultimate-}L$ denote the corresponding Ultimate-L axiom. Woodin's Ultimate-L conjecture. The theory .....

conjecture | open | MathDB 页面 | 原文线索

V2-056914 | MATHDB-364157

DMS's conjecture on intervals of irredundant base cardinalities for primitive groups

摘录 / EXCERPT: DMS's conjecture. There exists an interval X of positive integers, not containing 1, such that no primitive group G acting on any domain Ω satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-056915 | MATHDB-364457

The amenability-hyperfiniteness conjecture for countable Borel equivalence relations

摘录 / EXCERPT: Amenability-hyperfiniteness conjecture. For countable Borel equivalence relations, Borel amenability is equivalent to hyperfiniteness.

conjecture | open | MathDB 页面 | 原文线索

V2-056916 | MATHDB-364458

Borel almost finiteness conjecture for uniformly Borel amenable Følner graphs

摘录 / EXCERPT: Borel Almost Finiteness Conjecture. Every uniformly Borel amenable Følner graph is Borel almost finite.

conjecture | open | MathDB 页面 | 原文线索

V2-056917 | MATHDB-364471

Griggs's chain decomposition conjecture for the Boolean algebra

摘录 / EXCERPT: Let B_n be the Boolean algebra of all subsets of $[n]$, ordered by inclusion. A chain decomposition of B_n has a type, namely the partition obtained by rearranging its chain si...

conjecture | open | MathDB 页面 | 原文线索

V2-056918 | MATHDB-364472

Lonc-Elzobi's nice-property conjecture for products of chains

摘录 / EXCERPT: For positive integers $n_1,\ldots,n_r$, let $\mathbf{n_i}$ denote a chain of length n_i, and let a lattice be nice when every stable-partition type dominates only types that are...

conjecture | open | MathDB 页面 | 原文线索

V2-056919 | MATHDB-364873

The complete-spectrum conjecture for k-avoiding sets

摘录 / EXCERPT: Let $\mathrm{PG}(2,q)$ be the Desarguesian projective plane of order q, and let $\mathrm{Sp}(k,q)$ be the set of cardinalities of point sets avoiding k-secants. Complete-spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-056920 | MATHDB-365065

Erdős's similarity conjecture

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}$ be a measure universal set, meaning that every Lebesgue measurable subset of $\mathbb{R}$ with positive Lebesgue measure contains an affine copy of X....

conjecture | open | MathDB 页面 | 原文线索

V2-056921 | MATHDB-365068

Full measure universality conjecture

摘录 / EXCERPT: A subset $X\subseteq\mathbb{R}$ is full measure universal if every Lebesgue measurable subset of $\mathbb{R}$ whose complement has Lebesgue measure zero contains an affine copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-056922 | MATHDB-365069

Full measure universality and strong meagerness conjecture

摘录 / EXCERPT: For $X\subseteq\mathbb{R}$, call X strongly meager if $X+M\neq\mathbb{R}$ for every Lebesgue measure zero set $M\subseteq\mathbb{R}$. Full measure universality characterization c...

conjecture | open | MathDB 页面 | 原文线索

V2-056923 | MATHDB-365536

Equality of density cardinals for singleton targets in the unit interval

摘录 / EXCERPT: For $r\in(0,1)$, let ${\mathfrak{dd}}_{\{r\},{\mathsf{all}}}$ denote the corresponding density cardinal; the same notation is used when ${\mathsf{all}}$ is replaced by $[0,1]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-056924 | MATHDB-365537

Invariance of density cardinals under changes outside the open unit interval

摘录 / EXCERPT: Let ${\mathsf{all}}$ be the ambient parameter set, and let $X,X'\subseteq{\mathsf{all}}$ satisfy ... Let Y be arbitrary, and write ${\mathfrak{dd}}_{X,Y}$ for the associated densit...

conjecture | open | MathDB 页面 | 原文线索

V2-056925 | MATHDB-365851

Ultraexacting-above-extendible rank-Berkeley model conjecture

摘录 / EXCERPT: Ultraexacting-above-extendible rank-Berkeley model conjecture. There is a set model of ZF with a rank-Berkeley cardinal.

conjecture | open | MathDB 页面 | 原文线索

V2-056926 | MATHDB-365925

Optimal second-leading-term conjecture for stable orbital integrals on general linear Lie algebras

摘录 / EXCERPT: Let F be a non-Archimedean local field, let $\boldsymbol{o}$ be its ring of integers with residue field $\boldsymbol{\text{}}$ of cardinality q, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056927 | MATHDB-366018

The local largeness conjecture for continuous images of urepa trees

摘录 / EXCERPT: Let $T\subseteq {^{<\omega_2}}\omega_2$ be an ω_2-Kurepa tree, and let $[T]$ denote its set of cofinal branches. For $\sigma\in T$, let N_σ be the basic neighborhood...

conjecture | open | MathDB 页面 | 原文线索

V2-056928 | MATHDB-366041

Finite invariant-set conjecture for exactly solvable operators

摘录 / EXCERPT: Let $T: \mathbb{C}_n[x] \to \mathbb{C}_n[x]$ be an invertible linear operator and let $S \in {{\mathcal I}_{n}^T}$ contain at least two points and have finite cardinality. A finite...

conjecture | open | MathDB 页面 | 原文线索

V2-056929 | MATHDB-366106

TxGraffiti's claw-free graph zero forcing conjecture

摘录 / EXCERPT: Let G be a graph. The parameters $Z(G)$ and $Z_+(G)$ denote the standard and positive semidefinite zero forcing numbers of G, respectively. A graph is claw-free if it has no in...

conjecture | open | MathDB 页面 | 原文线索

V2-056930 | MATHDB-366193

The Chang-type uB-powerset sealing conjecture

摘录 / EXCERPT: Chang-type uB-powerset sealing conjecture. Moreover, the $\sf{Sealing}$ Theorem holds for $\mathsf{C-uBp}$ in $V[g]$.

conjecture | open | MathDB 页面 | 原文线索

V2-056931 | MATHDB-366194

The PFA-to-superstrong-cardinal conjecture in the uB derived model

摘录 / EXCERPT: PFA-to-superstrong-cardinal conjecture. Under these hypotheses, there is a superstrong cardinal in $\operatorname{HOD}^{L({{\sf{uB}}}_g,\mathbb{R}_g)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-056932 | MATHDB-366243

The full-measure universality and strongly meager sets conjecture

摘录 / EXCERPT: A set $X\subset\mathbb{R}$ is full measure universal if every set of full Lebesgue measure contains an affine copy $\lambda X+t$ with $\lambda\in\mathbb{R}\setminus\{0\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-056933 | MATHDB-366280

Blass's conjecture on ultrafilters in Solovay models

摘录 / EXCERPT: Blass's conjecture. If $\mathnormal{\mathbf{L}(\mathbb R)}$ is a Solovay model and p is a selective ultrafilter on ω, then the ultrafilters belonging to...

conjecture | open | MathDB 页面 | 原文线索

V2-056934 | MATHDB-366456

The cardinality conjecture for integer spaces of sigma-sets

摘录 / EXCERPT: Cardinality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056935 | MATHDB-366457

The cardinality conjecture for meta-spaces of sigma-sets

摘录 / EXCERPT: Meta-space cardinality conjecture. For all such A and B,

conjecture | open | MathDB 页面 | 原文线索

V2-056936 | MATHDB-366605

Wilson's conjecture on the first Solovay ordinal of a derived model

摘录 / EXCERPT: Let V satisfy PFA, let $\nobreak\bkappa$ be a limit of Woodin cardinals, and let $D(V,\kappa)$ denote the derived model at κ. Write $\Theta_0^{D(V,\kappa)}$ for the first...

conjecture | open | MathDB 页面 | 原文线索

V2-056937 | MATHDB-366606

Wilson's conjecture on the Solovay ordinal of a derived model

摘录 / EXCERPT: Let V satisfy PFA, let κ be a limit of Woodin cardinals, and let $D(V,\kappa)$ denote the derived model at κ. Let $\Theta^{D(V,\kappa)}$ be its Solovay ordinal. Wil...

conjecture | open | MathDB 页面 | 原文线索

V2-056938 | MATHDB-366611

The bounded-width boundary conjecture for Borel homomorphism problems

摘录 / EXCERPT: Bounded-width boundary conjecture. A homomorphism problem is $\mathbf{\Pi}^1_1$ iff it is not $\mathbf{\Sigma}^1_2$-complete iff it is bounded width.

conjecture | open | MathDB 页面 | 原文线索

V2-056939 | MATHDB-366698

Proof-theoretic dilator conjecture for $0^\sharp$

摘录 / EXCERPT: Proof-theoretic dilator conjecture. The proof-theoretic ordinals satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-056940 | MATHDB-366699

Characterization conjecture for Σ^1_2-altitude

摘录 / EXCERPT: Σ^1_2-altitude characterization conjecture. $\mathsf{Alt}_{\Sigma^1_2}(R)$ is equal to the least height of such a transitive model M.

conjecture | open | MathDB 页面 | 原文线索

V2-056941 | MATHDB-366880

The Chang-model cardinal-regularity conjecture

摘录 / EXCERPT: Cardinal-regularity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-056942 | MATHDB-366881

The Solovay-sequence Nairian-model conjecture

摘录 / EXCERPT: Solovay-sequence conjecture. Then $N\vDash\mathsf{ZF}$, $\Theta^N=\theta_\alpha$, N satisfies that every cardinal $\eta\geq\Theta$ has regular successor η^+, and, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-056943 | MATHDB-367124

Classification conjecture for Baumgartner axioms on perfect Polish spaces

摘录 / EXCERPT: Let X be a perfect Polish space, meaning a separable completely metrizable space with no isolated points. Let $cmathsf{BA}(X)$ denote the Baumgartner axiom parametrized by X, a...

conjecture | open | MathDB 页面 | 原文线索

V2-056944 | MATHDB-367434

Nonexistence of a continuum-separable envelope for infinite-dimensional Lebesgue measure

摘录 / EXCERPT: Let μ be the infinite-dimensional Lebesgue measure on $\mathbb{R}^{\mathbb{N}}$, and let $\mathfrak{c}$ denote the cardinality of the continuum. A $\mathfrak{c}$-separable enve...

conjecture | open | MathDB 页面 | 原文线索

V2-056945 | MATHDB-367515

The mod-3 power-sum conjecture for real multisets

摘录 / EXCERPT: Mod-3 power-sum conjecture. A real multiset X of cardinality n is uniquely determined by the n values

conjecture | open | MathDB 页面 | 原文线索

V2-056946 | MATHDB-367568

Loquacious highness characterization of isomorphism completeness for classes of structures

摘录 / EXCERPT: Loquacious highness characterization. If every degree loquaciously high for isomorphism for K is uniformly high for isomorphism, then the isomorphism problem for K is...

open_problem | open | MathDB 页面 | 原文线索

V2-056947 | MATHDB-367884

Polynomiality conjecture for type-A_n lower Bruhat interval sizes

摘录 / EXCERPT: Type-A_n polynomiality conjecture. The cardinality of the lower Bruhat interval is a polynomial in the coordinates $\lambda_1,\dots,\lambda_n$ of degree n:

conjecture | open | MathDB 页面 | 原文线索

V2-056948 | MATHDB-368279

The back-and-forth complexity tradeoff conjecture

摘录 / EXCERPT: Back-and-forth complexity tradeoff conjecture. For the indicated parity conditions, there are structures with the following exact complexities:

conjecture | open | MathDB 页面 | 原文线索

V2-056949 | MATHDB-368287

The Farah conjecture on random twins of unstable theories

摘录 / EXCERPT: Let T be a first-order countable unstable theory. Two models M and N of T are random twins if they are non-isomorphic but become isomorphic after the universe is extended b...

conjecture | open | MathDB 页面 | 原文线索

V2-056950 | MATHDB-368314

Signed-cardinality homomesy conjecture for interval-closed sets of products of two chains

摘录 / EXCERPT: Let $[m]\times[n]$ be the product of two chains, and let an interval-closed set mean a subset that is interval-closed in this poset. For $x\in[m]\times[n]$, define the signed cardi...

conjecture | open | MathDB 页面 | 原文线索

V2-056951 | MATHDB-368750

Hjorth-Kechris dichotomy conjecture for Borel equivalence relations

摘录 / EXCERPT: Let E be a Borel equivalence relation, and let E_1 denote eventual agreement on countable sequences of reals. For equivalence relations E and F, write $E\leq_B F$ when E...

conjecture | open | MathDB 页面 | 原文线索

V2-056952 | MATHDB-368751

Dichotomy conjecture for idealistic Borel equivalence relations

摘录 / EXCERPT: Let E be a Borel equivalence relation, and let E_1 denote eventual agreement on countable sequences of reals. Write $E\leq_B F$ when E is Borel reducible to F. An orbit equ...

conjecture | open | MathDB 页面 | 原文线索

V2-056953 | MATHDB-369290

Propagation-time interval conjecture for minimum zero forcing sets of hypercubes

摘录 / EXCERPT: Propagation-time interval conjecture. For every $d\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-056954 | MATHDB-369339

Generalized Gaifman conjecture on non-structure over a predicate

摘录 / EXCERPT: Generalized Gaifman conjecture. The theory T has 2^λ models of cardinality μ that are pairwise non-isomorphic over P.

conjecture | open | MathDB 页面 | 原文线索

V2-056955 | MATHDB-369340

Higher instability non-structure conjecture over a predicate

摘录 / EXCERPT: Higher instability conjecture. The theory T has 2^λ models of cardinality μ that are pairwise non-isomorphic over P.

conjecture | open | MathDB 页面 | 原文线索

V2-056956 | MATHDB-369341

Finite-level instability non-structure conjecture over a predicate

摘录 / EXCERPT: Finite-level non-structure conjecture. The theory T has 2^κ models of cardinality μ that are pairwise non-isomorphic over P.

conjecture | open | MathDB 页面 | 原文线索

V2-056957 | MATHDB-369510

The regular-cardinal conjecture for sunflowerable dense linear orderings

摘录 / EXCERPT: The regular-cardinal conjecture for sunflowerable dense linear orderings. If κ is regular, then every κ-dense linear ordering is sunflowerable.

conjecture | open | MathDB 页面 | 原文线索

V2-056958 | MATHDB-369726

Brimkov's propagation time interval conjecture for hypercubes

摘录 / EXCERPT: Brimkov's conjecture. For every integer $d\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-056959 | MATHDB-369727

Cameron's fort-number lower-bound conjecture for maximum nullity

摘录 / EXCERPT: Cameron's conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-056960 | MATHDB-369924

The consistency of non-CH in the second-order constructible universe with Woodin cardinals

摘录 / EXCERPT: Let $C^2(\alpha9omega)$ denote the inner model obtained from second-order definability over $\alpha9omega$, let CH denote the continuum hypothesis, and let M_n denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-056961 | MATHDB-369925

The inclusion of the second-order constructible universe in the first-order constructible universe

摘录 / EXCERPT: Let $C(\mathop{\text{aa}})$ and $C^2(\omega)$ denote the inner models defined in the paper, with $C(\mathop{\text{aa}})$ the model generated using the relevant class of structures...

conjecture | open | MathDB 页面 | 原文线索

V2-056962 | MATHDB-370889

Lower-bound conjecture for irreducible λ-quidditiy length in characteristic two

摘录 / EXCERPT: Let $\mathbb{K}$ be a finite field of characteristic 2, let $\operatorname{card}(\mathbb{K})$ denote its cardinality, and let $\ell_{\mathbb{K}}$ denote the invariant defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-056963 | MATHDB-370926

Martin's conjecture that real Blackwell determinacy implies determinacy

摘录 / EXCERPT: Let ${\mathsf{AD}}$ denote the axiom of determinacy and let ${\mathsf{Bl}\text{-}\mathsf{AD}}$ denote the axiom of Blackwell determinacy. Martin's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-056964 | MATHDB-371076

The minor-excluded-cover-graph dim-boundedness conjecture

摘录 / EXCERPT: Let K_t denote the complete graph on t vertices, and let a graph exclude K_t as a minor if it has no graph minor isomorphic to K_t. The dimension of a poset P, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-056965 | MATHDB-371607

Rank-two word conjecture for irreducible automata

摘录 / EXCERPT: Let $\mathcal{A}=(Q,\Sigma,\delta)$ be an irreducible automaton. For a word $u\in\Sigma^*$, let $\operatorname{rk}(u)$ denote the cardinality of the image of Q under u. Rank-tw...

conjecture | open | MathDB 页面 | 原文线索

V2-056966 | MATHDB-371688

The SL(2) infinite-series conjecture for unitals

摘录 / EXCERPT: Let k be the cardinality of a finite field, let $G=SL(2,\mathbb F_k)$, and let a G-space split into one orbit of size $|G|$ and $k+1$ orbits of size 1. The SL(2) infinite-ser...

conjecture | open | MathDB 页面 | 原文线索

V2-056967 | MATHDB-371781

The dense digit-pair conjecture for kernel-expandability

摘录 / EXCERPT: Let b be the base and let $D_1,D_2\subseteq\{0,1,\ldots,b-1\}$ be digit sets, with $\#D_1$ and $\#D_2$ denoting their cardinalities. A pair (D_1,D_2) is called degenerate or Ke...

conjecture | open | MathDB 页面 | 原文线索

V2-056968 | MATHDB-371867

Tower Sealing in generic extensions of hod pairs with reflecting strong cardinals

摘录 / EXCERPT: Let $({\mathcal{P}},\Psi)$ be a hod pair such that ... Let κ be the least strong cardinal which reflects the class of strong cardinals, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-056969 | MATHDB-372363

The dichotomy conjecture for conjugacy of discrete subgroups of connected Lie groups

摘录 / EXCERPT: Let G be a connected Lie group, let $ath.E(G,\mathcal{D}(G))$ denote the conjugacy relation on the space $\mathcal{D}(G)$ of discrete subgroups of G, and call an equivalence re...

conjecture | open | MathDB 页面 | 原文线索

V2-056970 | MATHDB-372520

The large-canopy dimension conjecture for alternating-move games

摘录 / EXCERPT: Let $T\subseteq \mathbb{Y}$ be a subtree, and write ... For $d>\frac{1}{2}$ and $\delta\in\left[\frac{1}{2},d\right]$, consider win-lose alternating-move games on T with payoff set...

conjecture | open | MathDB 页面 | 原文线索

V2-056971 | MATHDB-372540

Standard Manin conjecture over global function fields

摘录 / EXCERPT: Let $\pi:\mathcal X\to B$ be a good Fano fibration, let q be the cardinality of the constant field, and suppose that $\mathcal X_{\eta}(K(B))$ is not thin. With...

conjecture | open | MathDB 页面 | 原文线索

V2-056972 | MATHDB-372549

The conjecture on the possible numbers of one-element commutation classes

摘录 / EXCERPT: Let $n\geq 1$ and let $\sigma\in\mathfrak{S}_{n+1}$. Let $R_{\bullet}(\sigma)$ denote the set of one-element commutation classes of reduced words of σ. The possible-cardinal...

conjecture | open | MathDB 页面 | 原文线索

V2-056973 | MATHDB-372646

Maximum cardinality of the center of distances of a finite ultrametric space

摘录 / EXCERPT: Let $n\geq1$ be an integer, let $\log_2(n)$ denote the binary logarithm of n, and let $\lfloor\log_2(n)\rfloor$ be its integer part. For an ultrametric space (X,d), let $C(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-056974 | MATHDB-373174

Axiom of Choice equivalence for colourings of infinite connected graphs

摘录 / EXCERPT: In ZF, let a cardinal number be defined as an equivalence class under equinumerosity. For a graph, its chromatic index is the least cardinal number of colours in a proper edge-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-056975 | MATHDB-373216

PFA conjecture on derived models at limits of Woodin cardinals

摘录 / EXCERPT: Let $\mathcal{D}(V,\kappa)$ denote the derived model at κ, let $\Theta^{D(V,\kappa)}$ be its supremum of ordinals surjected by its reals, and let κ^+ be the successor...

conjecture | open | MathDB 页面 | 原文线索

V2-056976 | MATHDB-373344

Dualization conjecture for the Raisonnier ideal and the null ideal

摘录 / EXCERPT: Dualization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-056977 | MATHDB-373360

The Trichotomy Conjecture for untranscendable linear orders

摘录 / EXCERPT: Let φ be an untranscendable linear order. A linear order is strongly indecomposable when [the source context does not provide a definition]. An uncountable real type is the...

conjecture | open | MathDB 页面 | 原文线索

V2-056978 | MATHDB-374779

Adler's conjecture on spectra of two-cardinal counting functions

摘录 / EXCERPT: Let T be a countable theory and let $f_T(kappa,lambda)$ be the associated two-cardinal counting function for infinite cardinals $kappaleqlambda$. Adler's counting-spectrum conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-056979 | MATHDB-375313

Forcing equivalence of Cohen forcings for non-trivial Fraïssé classes

摘录 / EXCERPT: Let X be the underlying set and let $\AgeK_0$ and $\AgeK_1$ be non-trivial Fraïssé classes, each with at most $|X|$ many elements up to isomorphism. Write $\CohenX$ for the Cohen...

conjecture | open | MathDB 页面 | 原文线索

V2-056980 | MATHDB-375314

Universality of random generic extensions for invariant measures

摘录 / EXCERPT: Let X be a set, let $\operatorname{\mathscr{L}}$ be a language, and let κ be the relevant cardinal. Let μ_0 and μ_1 be invariant measures on...

conjecture | open | MathDB 页面 | 原文线索

V2-056981 | MATHDB-375365

The tropical equilateral-dimension conjecture

摘录 / EXCERPT: Let $d4a3^n$ be equipped with the tropical metric, and let $\mathrm{e}_{\mathrm{tr}}(\mathbb{R}^{n})$ denote the maximal cardinality of a tropical equilateral subset of $d4a3^n$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-056982 | MATHDB-375576

Optimal support cardinality conjecture for the discrete Schrödinger equation

摘录 / EXCERPT: Support cardinality conjecture. For $d\geq 1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-056983 | MATHDB-376251

The TxGraffiti zero forcing versus independence conjecture for subcubic graphs

摘录 / EXCERPT: TxGraffiti's conjecture. If $G \not\cong K_4$ and $\Delta(G)\leq 3$, then

conjecture | refuted | MathDB 页面 | 原文线索

V2-056984 | MATHDB-376613

Smallest $(r,r-1)$-erasure correcting set conjecture

摘录 / EXCERPT: For each $r\geq 2$, let $A_{r,r-1}$ be the explicitly constructed $(r,r-1)$-erasure correcting set, and let $F(r,r-1)$ be the minimum cardinality of any such set. Smallest-set conj...

conjecture | open | MathDB 页面 | 原文线索

V2-056985 | MATHDB-377243

Rationality conjecture for Mellin transforms of p-adic Whittaker functions

摘录 / EXCERPT: Let $I_{\lambda_0,f}(\lambda)$ denote the Mellin transform associated with the parameters λ_0, f, and λ, and let q be the residue-field cardinality. Rationality...

conjecture | open | MathDB 页面 | 原文线索

V2-056986 | MATHDB-377643

The maximum-feasible-subset scaling conjecture for random K-LSAT

摘录 / EXCERPT: Consider the random K-LSAT linear program with n variables and m constraints, with all auxiliary variables ψ set to zero. Let $N(n,m)$ be the maximum cardinality of a feas...

conjecture | open | MathDB 页面 | 原文线索

V2-056987 | MATHDB-377723

The CH conjecture for MAD sigma-sets

摘录 / EXCERPT: A MAD family is a maximal almost disjoint family of infinite subsets of omega, and a subset $X\subseteq 2^\omega$ is a sigma-set if for every Borel set...

conjecture | open | MathDB 页面 | 原文线索

V2-056988 | MATHDB-377724

The CH conjecture for concentrated MAD families

摘录 / EXCERPT: A MAD family is a maximal almost disjoint family of infinite subsets of omega. A family is concentrated on a countable subset of itself if there is a countable subfamily s...

conjecture | open | MathDB 页面 | 原文线索

V2-056989 | MATHDB-378108

The decoration conjecture for forcing relations of horseshoe orbits

摘录 / EXCERPT: The decoration conjecture. If $q<q'$ and $w\succcurlyeq w'$, then P_q^w forces $P_{q'}^{w'}$. Moreover, P_q^w and $P_{q'}^{w'}$ have the same braid type if and only if $q=q'$ a...

conjecture | refuted | MathDB 页面 | 原文线索

V2-056990 | MATHDB-378199

Finite powers of small strong measure zero sets

摘录 / EXCERPT: Let X be a set of reals with strong measure zero, and suppose that $|X|<\mathfrak{b}$, where $\mathfrak{b}$ is the bounding number. A finite power of X is a Cartesian product...

conjecture | open | MathDB 页面 | 原文线索

V2-056991 | MATHDB-378467

Consistency strength of indestructible supercompact maximality

摘录 / EXCERPT: Consistency-strength claim. The consistency strength of the theory $

conjecture | open | MathDB 页面 | 原文线索

V2-056992 | MATHDB-378489

The lower-bound conjecture for equilateral sets in normed spaces

摘录 / EXCERPT: Let X be an n-dimensional normed space, and let $e(X)$ denote the maximum cardinality of an equilateral subset of X. Lower-bound conjecture. If $\dim X=n$, then ... This is the...

conjecture | open | MathDB 页面 | 原文线索

V2-056993 | MATHDB-378523

Analytic trace ideals for proper forcing with continuous reading of names

摘录 / EXCERPT: Let I be a $c3$-ideal on 2^ω or ω^ω, and let P_I be the corresponding factor poset. Assume that P_I is proper and has continuous reading of names. Trace-id...

conjecture | open | MathDB 页面 | 原文线索

V2-056994 | MATHDB-378693

The odd-square count conjecture for primitive square-tiled surfaces in \(\mathcal{H}(2)\)

摘录 / EXCERPT: Let n be an odd integer with $n\geqslant 5$. Let a_n and b_n denote the cardinalities of the two discrete orbits of primitive n-square-tiled surfaces in ${\mathcal{H}}(2)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-056995 | MATHDB-378846

Todorčević's Tukey basis conjecture for binary relations

摘录 / EXCERPT: Let R be a binary relation. Write $\fulldiamond$ for the proper forcing axiom appearing in the source, and let $\beta\to(\beta_1;\beta_2)^2_2$ denote the corresponding partition...

conjecture | open | MathDB 页面 | 原文线索

V2-056996 | MATHDB-378848

Galvin's partition relation conjecture at omega one

摘录 / EXCERPT: For a coloring of unordered pairs of ω_1 with two colors, write $\omega_1\rightarrow(\omega_1;\omega_1)^2_2$ for the stated polarized partition relation. Galvin's partition...

conjecture | open | MathDB 页面 | 原文线索

V2-056997 | MATHDB-378851

Shelah's five-element basis conjecture for uncountable linear orders

摘录 / EXCERPT: Let X be a set of reals of cardinality $\aleph_1$, let ω_1 be the first uncountable ordinal, let ω_1^* be its converse order, and let C be a Countryman line, mean...

conjecture | open | MathDB 页面 | 原文线索

V2-056998 | MATHDB-378852

The Mapping Reflection Principle cardinal-arithmetic conjecture

摘录 / EXCERPT: Let θ be a regular cardinal with $\theta \geq \omega_2$. The Mapping Reflection Principle, MRP, is a forcing axiom concerning open stationary set mappings on coun...

conjecture | open | MathDB 页面 | 原文线索

V2-056999 | MATHDB-379367

The block-subspace dichotomy and classification conjecture

摘录 / EXCERPT: Let X be a Banach space with an unconditional basis. A block-subspace is a closed subspace generated by a block-sequence of that basis, and E_0 is the equivalence relation of e...

open_problem | open | MathDB 页面 | 原文线索

V2-057000 | MATHDB-379368

The block-sequence permutative-equivalence conjecture

摘录 / EXCERPT: Let X be a Banach space with a basis, respectively an unconditional basis, which is not equivalent to the canonical basis of c_0 or ℓ_p for $1\leq p<+\infty$. A normalized...

conjecture | open | MathDB 页面 | 原文线索

V2-057001 | MATHDB-379515

The Aronszajn-null game characterization conjecture

摘录 / EXCERPT: Let X be a separable Banach space over $\mathbb{R}$, let $A\subset X$ be Borel and Aronszajn-null, and let $\mathcal{A}_{\mathsf{G}}$ be the game in which player $\mathbf{I}$ ch...

conjecture | open | MathDB 页面 | 原文线索

V2-057002 | MATHDB-379516

The auxiliary Aronszajn-null game characterization conjecture

摘录 / EXCERPT: Let X be a separable Banach space, let $A\subseteq X$ be Borel, and let $\mathbb{A}_{\mathsf{G}}$ be the auxiliary game referred to in the paper, with $\mathbb{A}$ denoting the c...

conjecture | open | MathDB 页面 | 原文线索

V2-057003 | MATHDB-379615

The power-of-two conjecture for skew-equivalence classes

摘录 / EXCERPT: Power-of-two conjecture. Every skew-equivalence class of skew diagrams has cardinality a power of 2.

conjecture | open | MathDB 页面 | 原文线索

V2-057004 | MATHDB-379689

Regularity limit problem for singular cardinals

摘录 / EXCERPT: Let λ be a singular cardinal, let $\mu<\lambda$, and let D be an ultrafilter that is ... -complete and (μ,κ)-regular for every $\kappa<\lambda$. Regularity limit p...

open_problem | open | MathDB 页面 | 原文线索

V2-057005 | MATHDB-379778

Milner-Sauer conjecture on antichains in posets of singular cofinality

摘录 / EXCERPT: Milner-Sauer conjecture. If $\operatorname{cf}(P,\le)=\lambda>\operatorname{cf}(\lambda)=\kappa$, then P contains an antichain of size κ.

conjecture | open | MathDB 页面 | 原文线索

V2-057006 | MATHDB-380484

Existence of reflexive abelian groups of arbitrarily large cardinalities

摘录 / EXCERPT: A reflexive abelian group is an abelian group naturally isomorphic to its double dual. Existence conjecture. There are reflexive abelian groups of arbitrarily large cardinalities....

conjecture | open | MathDB 页面 | 原文线索

V2-057007 | MATHDB-380514

Consistency of the internal forcing schema IFS

摘录 / EXCERPT: Let IFS be the axiom schema asserting that for every weakly absolutely definable set X and every partial order P definable in $L(X)$, if ... then there is an $L(X)$-generic sub...

conjecture | open | MathDB 页面 | 原文线索

V2-057008 | MATHDB-380533

Mouse-set conjecture for projective-like ordinals

摘录 / EXCERPT: Let $A_{(\alpha,n)}$ be the sets defined from $J_{\alpha}(\mathbb{R})$, and let a mouse set mean a set $A\subseteq\mathbb{R}$ of the form $A=\mathbb{R}\cap\mathcal{M}$ for some cou...

conjecture | open | MathDB 页面 | 原文线索

V2-057009 | MATHDB-380795

Shelah's pcf boundedness conjecture for inaccessible cardinals

摘录 / EXCERPT: Let ${\mathfrak a}$ be a set of regular cardinals such that every member of ${\mathfrak a}$ is greater than $|{\mathfrak a}|$. Shelah's pcf boundedness conjecture. For no inaccessi...

conjecture | open | MathDB 页面 | 原文线索

V2-057010 | MATHDB-380796

Shelah's consistency conjecture for large sets of singular cardinals

摘录 / EXCERPT: Let θ be any uncountable cardinal. Shelah's consistency conjecture for pseudopowers. It is consistent that, for some cardinal λ, both of the following hold: (A)...

conjecture | open | MathDB 页面 | 原文线索

V2-057011 | MATHDB-380797

Shelah's pcf conjecture for limits of limit cardinals

摘录 / EXCERPT: Let μ be a limit of limit cardinals, and let $\lambda<\mu$ range over arbitrarily large regular cardinals. Write $\forall^*$ for “for every large enough.” Shelah's pcf conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-057012 | MATHDB-380798

Shelah's first-fix-point pseudopower conjecture

摘录 / EXCERPT: Let $\aleph_\delta$ be the first fix point, meaning the first ordinal index satisfying $\aleph_\delta=\delta$, and hence having cofinality $\aleph_0$. Shelah's first-fix-point pseu...

conjecture | open | MathDB 页面 | 原文线索

V2-057013 | MATHDB-380807

Equiconsistency conjecture for indestructible weakly compact cardinals

摘录 / EXCERPT: Equiconsistency conjecture. The existence of an indestructible weakly compact cardinal is equiconsistent over ZFC with the existence of a supercompact cardinal.

conjecture | open | MathDB 页面 | 原文线索

V2-057014 | MATHDB-381005

The Clifford bound conjecture for unitary 2-designs

摘录 / EXCERPT: Clifford bound conjecture. The cardinality of any unitary 2-design is at least

conjecture | open | MathDB 页面 | 原文线索

V2-057015 | MATHDB-381015

Shelah's diamond principle conjecture for stationary subsets

摘录 / EXCERPT: Let λ and κ be cardinals, with $\lambda=\lambda^{<\lambda}>\kappa^+$, $\kappa=\operatorname{cf}(\kappa)$, and $\lambda\ne\aleph_1$. The weaker alternative allows...

open_problem | open | MathDB 页面 | 原文线索

V2-057016 | MATHDB-381069

Characterizability and homomorphic characterizability at singular cardinals

摘录 / EXCERPT: Characterizability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057017 | MATHDB-381603

Kechris's first conjecture on projective unreachable cardinals

摘录 / EXCERPT: Let ${\mathbf{\delta}}^1_{2n+2}$ denote the supremum of the lengths of ${\mathbf{\Delta}}^1_{2n+2}$ pre-well-orderings of ${\mathbb{R}}$. Assume ${\sf{ZF+AD+DC}}$. Kechris's first...

conjecture | open | MathDB 页面 | 原文线索

V2-057018 | MATHDB-381604

Kechris's second conjecture on projective unreachable cardinals

摘录 / EXCERPT: Let ${\mathbf{\delta}}^1_{2n+2}$ denote the supremum of the lengths of ${\mathbf{\Delta}}^1_{2n+2}$ pre-well-orderings of ${\mathbb{R}}$. Assume ${\sf{ZF+AD+DC}}$. Kechris's second...

conjecture | open | MathDB 页面 | 原文线索

V2-057019 | MATHDB-381605

The Suslin-cardinal unreachable successor conjecture

摘录 / EXCERPT: Assume ${\sf{AD}}^+$. Let κ be a regular Suslin cardinal, and let ${\mathbf{\Gamma}}$ be the pointclass of κ-Suslin sets. Suslin-cardinal unreachable successor conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-057020 | MATHDB-381677

Existence of Shelah-Steprāns almost disjoint families in the constructed model

摘录 / EXCERPT: Existence conjecture. There are Shelah-Steprāns almost disjoint families in ${\mathbf{V}}[{G}_{{\omega}_{2}}]$, necessarily of size ${\aleph}_{2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057021 | MATHDB-381688

Shelah's three cardinal and model-existence claims

摘录 / EXCERPT: Shelah's three cardinal and model-existence claims. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-057022 | MATHDB-381713

Barnabei-Bonetti-Castronuovo-Silimbani's prefix-exchange conjecture for alternating involutions

摘录 / EXCERPT: Let $I_3=123$ and $J_3=321$. For a permutation pattern τ, prefix exchange over alternating involutions means equality of the avoidance-class cardinalities after adjoining any...

conjecture | open | MathDB 页面 | 原文线索

V2-057023 | MATHDB-381811

Bounded-degree enclosure conjecture for proper minor-closed classes

摘录 / EXCERPT: Let ${\mathcal C}_0$ be a proper minor-closed class of countable graphs. For a minor-closed graph family ${\mathcal C}$, let $\Delta({\mathcal C})$ be the smallest cardinal k suc...

conjecture | open | MathDB 页面 | 原文线索

V2-057024 | MATHDB-382102

Optimality of Neeman's continuously coded length determinacy theorem

摘录 / EXCERPT: Let $G_{\operatorname{cont}}(\nu,A)$ be the game of continuously coded length, where ν is a partial function from reals to natural numbers and A is a payoff set of sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-057025 | MATHDB-382103

Optimality of Woodin's equiconsistency theorem for ordinal-definable games

摘录 / EXCERPT: An ordinal definable game of length ω_1 on natural numbers with real parameters is a game in which the payoff and defining data are ordinal definable, with real parameters a...

conjecture | open | MathDB 页面 | 原文线索

V2-057026 | MATHDB-382173

Correctness preservation for shattered iterations with Borel or Suslin ccc forcing

摘录 / EXCERPT: Correctness-preservation conjecture. The diagram $\bar{\mathbb E}$ is correct.

conjecture | open | MathDB 页面 | 原文线索

V2-057027 | MATHDB-382174

Preservation of cardinal characteristics in shattered Cohen iterations

摘录 / EXCERPT: Let $\mathfrak b$, $\mathfrak d$, and $\mathfrak c$ denote the bounding number, dominating number, and continuum, respectively. Assume ... and let κ,λ be regular cardina...

conjecture | open | MathDB 页面 | 原文线索

V2-057028 | MATHDB-382175

Splitting and reaping numbers in shattered Cohen or Hechler iterations

摘录 / EXCERPT: Let $\mathfrak s$ and $\mathfrak r$ denote the splitting and reaping numbers, and let $\mathfrak c$ denote the continuum. Consider models obtained by the shattered iteration of eit...

conjecture | open | MathDB 页面 | 原文线索

V2-057029 | MATHDB-382435

The regular-equilibrium forcing conjecture for glacial cycles

摘录 / EXCERPT: Consider the piecewise-smooth glacial cycles system, with a regular equilibrium or two regular equilibria located close to the discontinuity boundary. Regular-equilibrium forcing c...

conjecture | open | MathDB 页面 | 原文线索

V2-057030 | MATHDB-382473

Friedman-Hyttinen-Weinstein generalized Borel-reducibility main gap problem

摘录 / EXCERPT: Let $\mathcal T_1$ be a classifiable theory and $\mathcal T_2$ a non-classifiable theory. A generalized Borel-reducibility main gap problem. asks whether there is a Borel reduction...

open_problem | open | MathDB 页面 | 原文线索

V2-057031 | MATHDB-382984

Kusner’s conjecture on equilateral sets in ℓ_p^n

摘录 / EXCERPT: For every integer $n\geq 1$ and every real p satisfying $2<p<\infty$, the maximum cardinality of an equilateral set in ℓ_p^n is $e(\ell_p^n)=n+1$. Equivalently, every set...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057032 | MATHDB-384689

Partial Embeddings into Trees (problem 1)

摘录 / EXCERPT: Given an input metric space M and distortion $c<\infty$, can we efficiently approximate the maximal number m such that there exists a subset of M of cardinality m that embe...

open_problem | open | MathDB 页面 | 原文线索

V2-057033 | MATHDB-385006

Is there a model of set theory in which the Calkin algebra has a not approximately inner automorphism preserving K-theory?

摘录 / EXCERPT: Is there a model of set theory in which the Calkin algebra has a not approximately inner automorphism preserving K-theory?

open_problem | open | MathDB 页面 | 原文线索

V2-057034 | MATHDB-385011

Tensor products (problem 3)

摘录 / EXCERPT: Does $\otimes _{\min }$ or $\otimes _{\max }$ commute with forcing?

open_problem | open | MathDB 页面 | 原文线索

V2-057035 | MATHDB-385728

HOD (problem 1)

摘录 / EXCERPT: begin{enumerate} item Is it consistent that every uncountable regular cardinal is supercompact in HOD? item What about κ^+ is supercompact in HOD whenever κ is singul...

open_problem | open | MathDB 页面 | 原文线索

V2-057036 | MATHDB-385731

Uniform Indiscernibles (problem 1)

摘录 / EXCERPT: begin{enumerate} item Is $u_2(\omega_1)=\omega_3$ consistent with (sufficiently) large cardinals? item How to increase $u_2(\omega_1)$ without collapsing ω_1 or ω_2?...

open_problem | open | MathDB 页面 | 原文线索

V2-057037 | MATHDB-385734

Forcing Axioms (problem 1)

摘录 / EXCERPT: begin{enumerate} item Does Con(ZFC) imply Con(MM for posets of size $\aleph_1$)? item Is there a stationary set preserving poset of size $\aleph_1$ which is not semiproper? end{enu...

open_problem | open | MathDB 页面 | 原文线索

V2-057038 | MATHDB-386898

Problems in Low Forcing (problem 1)

摘录 / EXCERPT: Is it consistent (with the continuum hypothesis) that there is a minimal $\aleph_2$-Countryman line-that is, an $\aleph_2$-Countryman line that order-embeds into all others?

open_problem | open | MathDB 页面 | 原文线索

V2-057039 | MATHDB-386899

Is it consistent with the continuum hypothesis that any two … -Countryman lines are near or co-near?

摘录 / EXCERPT: Is it consistent with the continuum hypothesis that any two $\aleph_2$-Countryman lines are near or co-near?

open_problem | open | MathDB 页面 | 原文线索

V2-057040 | MATHDB-386901

Problems in Low Forcing (problem 4)

摘录 / EXCERPT: Is it consistent to have $\lim^p \mathbb{A} = 0$ for all p?

open_problem | open | MathDB 页面 | 原文线索

V2-057041 | MATHDB-386907

Problems in Low Forcing (problem 10)

摘录 / EXCERPT: Do any of the analogs of PFA at $\aleph_2$ give us $|X| \le 2^{\aleph_1}$ for every Hausdorff space with no discrete subspaces of size $\aleph_2$?

open_problem | open | MathDB 页面 | 原文线索

V2-057042 | MATHDB-386908

Problems in Low Forcing (problem 11)

摘录 / EXCERPT: Suppose X is a compact Hausdorff space of tightness $\aleph_1$ (i.e. there are no ω_2-free sequences) and density $\aleph_1$. Is it true that $|X| \le 2^{\aleph_1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-057043 | MATHDB-386910

Problems in Low Forcing (problem 13)

摘录 / EXCERPT: Can $\omega_2 \rightarrow (\omega_2,\alpha)^2$ be deduced from one of the forcing axioms at $\aleph_2$.

open_problem | open | MathDB 页面 | 原文线索

V2-057044 | MATHDB-386913

Problems in High Forcing (problem 2)

摘录 / EXCERPT: Is it consistent to have the failure of both SCH and $\square_{\aleph_\omega}^\ast$?

open_problem | open | MathDB 页面 | 原文线索

V2-057045 | MATHDB-386914

Problems in High Forcing (problem 3)

摘录 / EXCERPT: Is $\neg \square_\kappa^\ast + \neg SCH_\kappa + GCH_{<\kappa}$ consistent for κ singular?

open_problem | open | MathDB 页面 | 原文线索

V2-057046 | MATHDB-386915

Problems in High Forcing (problem 4)

摘录 / EXCERPT: Is it consistent that there is a poset turning $\aleph_{\omega+1}$ into $\aleph_2$?

open_problem | open | MathDB 页面 | 原文线索

V2-057047 | MATHDB-386919

Is it possible to obtain the super tree property at a successor of a singular cardinal?

摘录 / EXCERPT: Is it possible to obtain the super tree property at a successor of a singular cardinal?

open_problem | open | MathDB 页面 | 原文线索

V2-057048 | MATHDB-386922

Problems in High Forcing (problem 11)

摘录 / EXCERPT: Is it consistent to obtain $TP_{\aleph_2}$ together with the existence of a saturated ideal on $\aleph_2$?

open_problem | open | MathDB 页面 | 原文线索

V2-057049 | MATHDB-386926

Problems in the Overlap of High and Low Forcing (problem 4)

摘录 / EXCERPT: Is the forcing axiom for proper posets of size $\aleph_1$ with $\aleph_2$-many dense sets consistent? If so, what if we replace the size-$\aleph_2$ requirement with the $\aleph_2$-...

open_problem | open | MathDB 页面 | 原文线索

V2-057050 | MATHDB-386927

Can this forcing be made proper using side conditions?

摘录 / EXCERPT: Can this forcing be made proper using side conditions? Does this preserve the $\aleph_2$-chain condition?

open_problem | open | MathDB 页面 | 原文线索

V2-057051 | MATHDB-387575

Research Problems in Function Theory - Problem 2.66

摘录 / EXCERPT: Given a countable number of entire functions, one can find an entire function growing faster than any of these. Without making any assumption about the Continuum Hypothesis, can on...

open_problem | open | MathDB 页面 | 原文线索

V2-057052 | MATHDB-392121

Hayut's HOD power-set disagreement question and optimality of Magidor's Covering Theorem

摘录 / EXCERPT: Can there exist a model V of ZFC in which $\aleph_\omega$ is the least cardinal κ such that $\mathcal{P}^V(\kappa)\neq\mathcal{P}^{\mathrm{HOD}^V}(\kappa)$; tha...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057053 | MATHDB-392838

PTAS for maximum independent set in 2-dimensional box intersection graphs

摘录 / EXCERPT: Given a set R of n axis-parallel 2-dimensional boxes, find a maximum-cardinality subset $R^*\subseteq R$ of pairwise disjoint boxes. Does this problem admit a polynomial-time...

open_problem | open | MathDB 页面 | 原文线索

V2-057054 | MATHDB-392839

Efficient approximability of Maximum Independent Set for axis-parallel rectangles

摘录 / EXCERPT: For a set R of n axis-parallel rectangles in the plane, let $R^*\subseteq R$ be a maximum-cardinality subset of pairwise disjoint rectangles. Determine the efficient approximab...

open_problem | open | MathDB 页面 | 原文线索

V2-057055 | MATHDB-392840

APX-hardness of Maximum Independent Set for axis-parallel d-boxes

摘录 / EXCERPT: For every fixed integer $d\ge 3$, the Maximum Independent Set problem for a collection of axis-parallel d-dimensional boxes-finding a maximum-cardinality subcollection of pairwis...

open_problem | open | MathDB 页面 | 原文线索

V2-057056 | MATHDB-392922

Exact total-variation reconstruction from incomplete Fourier samples

摘录 / EXCERPT: The report states the following typical result: suppose a one-dimensional discrete signal f has ... Then, for nearly all sets Ω of cardinality $|\Omega|$, f can be reconst...

open_problem | open | MathDB 页面 | 原文线索

V2-057057 | MATHDB-393182

Kissing number in dimension 4

摘录 / EXCERPT: For a finite spherical code $C\subset S^3$ with $z=\frac12$, meaning $x_i^{\mathsf T}x_j\le\frac12$ for all distinct code points, the maximum possible cardinality is $|C|=24$; equi...

open_problem | open | MathDB 页面 | 原文线索

V2-057058 | MATHDB-393271

Isomorphism rigidity for Kac-Moody groups

摘录 / EXCERPT: Let K,K' be algebraically closed or finite fields of cardinality at least 4. If there is an isomorphism $\varphi:G(D,K)\to G(D',K')$ between Kac-Moody groups of types D and...

open_problem | open | MathDB 页面 | 原文线索

V2-057059 | MATHDB-393291

Local recognition of line-hyperline graphs

摘录 / EXCERPT: For every possibly infinite cardinal number $n\ge 5$ and every division ring F, if Γ is a connected graph that is locally $L_n(F)$, then Γ is isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-057060 | MATHDB-393714

A covering property, PFA and SCH

摘录 / EXCERPT: The Proper Forcing Axiom implies the Singular Cardinal Hypothesis: ...

open_problem | open | MathDB 页面 | 原文线索

V2-057061 | MATHDB-393717

Whether failure of SCH implies the Approachability Property

摘录 / EXCERPT: Does failure of the Singular Cardinals Hypothesis at a cardinal κ imply that the Approachability Property holds at κ? The report answers negatively: assuming κ...

open_problem | open | MathDB 页面 | 原文线索

V2-057062 | MATHDB-393718

Martin’s Maximum and an inner model with a \+-supercompact cardinal

摘录 / EXCERPT: Does Martin’s Maximum imply the existence of an inner model with a cardinal κ that is κ^+-supercompact?

open_problem | open | MathDB 页面 | 原文线索

V2-057063 | MATHDB-393720

Possible finite numbers of near-coherence classes of ultrafilters

摘录 / EXCERPT: For each finite cardinal $n\ge 3$, can there be a model of ZFC in which the nonprincipal ultrafilters on ω have exactly n equivalence classes under near coherence?

open_problem | open | MathDB 页面 | 原文线索

V2-057064 | MATHDB-393721

Finite near-coherence classes of ultrafilters

摘录 / EXCERPT: Which finite cardinals can be the number of equivalence classes of ultrafilters on [?] with respect to near coherence?

open_problem | open | MathDB 页面 | 原文线索

V2-057065 | MATHDB-393946

Consistency of Large Condensation with an \c9-superstrong cardinal

摘录 / EXCERPT: If $\ba$ is $\c9$-superstrong, then it remains $\c9$-superstrong in some forcing extension satisfying Large Condensation.

open_problem | open | MathDB 页面 | 原文线索

V2-057066 | MATHDB-393948

Conjecture 3 on the approachability ideal at a singular cardinal

摘录 / EXCERPT: Under an appropriate large cardinal hypothesis, there is a generic extension with a singular cardinal κ such that, for each uncountable cardinal $\gamma<\kappa$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-057067 | MATHDB-393950

Mouse Set Conjecture (MSC)

摘录 / EXCERPT: Assume AD^+ and that there is no $\d714_1$-iteration strategy for a mouse with a superstrong cardinal. Let $\varphi(v)=\exists A\subseteq\mathbb{R}\,\psi(A,v)$ be a Σ^1_2...

conjecture | open | MathDB 页面 | 原文线索

V2-057068 | MATHDB-393951

Consistency of the Strong Inner Model Hypothesis

摘录 / EXCERPT: Is SIMH consistent, relative to some large cardinal hypothesis?

conjecture | open | MathDB 页面 | 原文线索

V2-057069 | MATHDB-394106

Convergence of a level-set evolution for simultaneous reservoir-shape and permeability reconstruction

摘录 / EXCERPT: Find forcing terms f,h_i,h_e in the evolution laws ... such that the evolution converges to the desired solution of the inverse problem: find a level-set function $\psi(x)$ and int...

open_problem | open | MathDB 页面 | 原文线索

V2-057070 | MATHDB-394242

Finitistic consistency of Elementary Constructive Set Theory

摘录 / EXCERPT: Conjecture. The consistency of Elementary Constructive Set Theory (ECST) is provable in Primitive Recursive Arithmetic (PRA); equivalently, ECST has a finitistic consistency proof.

open_problem | open | MathDB 页面 | 原文线索

V2-057071 | MATHDB-394378

Minimum number of shared values determining an entire function from its derivative

摘录 / EXCERPT: Determine the least cardinality of a set $S\subset\mathbb C$ such that, for every non-constant entire function f, the condition $E(S,f)=E(S,f')$ implies $f\equiv f'$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-057072 | MATHDB-394554

Effective velocity for mean-curvature flow with periodic forcing

摘录 / EXCERPT: For the forced mean-curvature-flow equation ... where V_x is the normal velocity and κ_x is the mean curvature of the interface $\Sigma(t)$ at x, is there, for every direc...

open_problem | open | MathDB 页面 | 原文线索

V2-057073 | MATHDB-394555

Effective velocity for forced mean-curvature flow with weak periodic forcing

摘录 / EXCERPT: For the periodic forced mean-curvature flow ... where the forcing $f(x)+F$ is sufficiently small in C^1 but is not required to have a fixed sign, does every normal direction...

open_problem | open | MathDB 页面 | 原文线索

V2-057074 | MATHDB-394871

Exactness of the optimized tropical lower bound for toric secant dimensions

摘录 / EXCERPT: Let $A\subseteq\mathbb N^m$ have cardinality n and lie on an affine hyperplane, let $f=(x^\alpha)_{\alpha\in A}$, and let $C=\overline{\operatorname{im} f}\subseteq\mathbb C^n$....

open_problem | open | MathDB 页面 | 原文线索

V2-057075 | MATHDB-394889

Minimum semidegree forcing fixed-length cycles in oriented graphs

摘录 / EXCERPT: For each integer $ ell\ge 4$ divisible by 3, determine the minimum semidegree threshold, as a function of the order n, that forces an oriented graph G on n vertices to cont...

open_problem | open | MathDB 页面 | 原文线索

V2-057076 | MATHDB-394921

A uniform-filter coloring on a regular Jónsson cardinal

摘录 / EXCERPT: Suppose κ is a regular Jónsson cardinal. Can there be a function $F:[\kappa]^{<\kappa}\to\kappa$ and a uniform filter $\mathcal{F}$ on κ such that, for any...

open_problem | open | MathDB 页面 | 原文线索

V2-057077 | MATHDB-394924

Question 1: Collapse-forcing-invariant models of ZFC

摘录 / EXCERPT: Does there exist a model N of ZFC such that, for every cardinal $\\lambda$ of N and every generic subset H of $\\operatorname{Col}(\\omega,\\lambda)$, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-057078 | MATHDB-394930

Chang transfer for a singular cardinal of countable cofinality

摘录 / EXCERPT: Is it consistent that $(\kappa^+,\kappa)\twoheadrightarrow(\aleph_2,\aleph_1)$ for a singular cardinal κ of countable cofinality?

open_problem | open | MathDB 页面 | 原文线索

V2-057079 | MATHDB-395386

Even Cut Isomorphism Problem

摘录 / EXCERPT: Let (G,T) and (G',T') be grafts, where $T\subseteq V(G)$ and $T'\subseteq V(G')$ have even cardinality. A cut $\delta_G(U)$ is even if $|U\cap T|$ is even, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-057080 | MATHDB-395397

Feasible pairings for crossing families of odd sets

摘录 / EXCERPT: Let V be a ground set of even cardinality, and let $\mathcal F$ be a crossing family of odd subsets of V. A pairing M of V is a set of unordered pairs in which every elemen...

open_problem | open | MathDB 页面 | 原文线索

V2-057081 | MATHDB-395510

Main Theorem on Hilbertian extensions with free absolute Galois group

摘录 / EXCERPT: Let K be a PAC field of characteristic p and cardinality m, and let x be a variable. Let $\mathcal F$ be the set of all monic irreducible polynomials in $K[x]$. For each...

open_problem | open | MathDB 页面 | 原文线索

V2-057082 | MATHDB-395839

Possible “fake CM” forcing even rank

摘录 / EXCERPT: The elliptic curve of discriminant -114 given by ... has everywhere good reduction over $K=\mathbb{Q}(\sqrt[12]{-11})$. Its global root number is $+1$ over every finite extension o...

open_problem | open | MathDB 页面 | 原文线索

V2-057083 | MATHDB-396252

Rate-independent hysteresis from heterogeneous forcing

摘录 / EXCERPT: Assuming a viscous law for the relation between the driving force and the velocity of the interface, an important question is whether rate independent hysteresis can emerge from th...

open_problem | open | MathDB 页面 | 原文线索

V2-057084 | MATHDB-396531

Bounded size of independent systems of word equations

摘录 / EXCERPT: Is the cardinality of the maximal independent system of equations on n unknowns bounded by a function of n?

open_problem | open | MathDB 页面 | 原文线索

V2-057085 | MATHDB-396826

Forcing conjecture for bipartite graphs with two complete vertices

摘录 / EXCERPT: The forcing conjecture holds for every bipartite graph H which has two vertices in one part complete to the other part, which has at least two vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-057086 | MATHDB-396855

Strong P-points and Mathias-Prikry forcing

摘录 / EXCERPT: Is every strong P-point U such that the corresponding Mathias-Prikry forcing M_U does not add a dominating real?

open_problem | open | MathDB 页面 | 原文线索

V2-057087 | MATHDB-396856

Canjar’s question on P-points and Mathias-Prikry forcing

摘录 / EXCERPT: Is every P-point U without rapid Rudin-Keisler predecessors such that the corresponding Mathias-Prikry forcing M_U does not add a dominating real?

open_problem | open | MathDB 页面 | 原文线索

V2-057088 | MATHDB-396874

Todorcevic’s higher-cardinal rectangle-coloring question

摘录 / EXCERPT: Is it consistent that there exists a function $f:\cspan\cspan\omega_3\times\omega_3\to\omega$ which is not constant on any $A\times B\subseteq\omega_3\times\omega_3$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-057089 | MATHDB-397172

Arithmetic Fundamental Lemma for unitary groups

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}_F$ and residue field of cardinality q, and let E/F be an unramified quadratic extension with...

open_problem | open | MathDB 页面 | 原文线索

V2-057090 | MATHDB-397411

Conservativity of weak intuitionistic constructive set theories over arithmetic

摘录 / EXCERPT: For every weak set theory T in question and every arithmetical statement A, $T \vdash A \Longleftrightarrow HA + TI(<\lvert T\rvert) \vdash A$, where $TI(<\lvert T\rvert)$ deno...

open_problem | open | MathDB 页面 | 原文线索

V2-057091 | MATHDB-397526

Computational power of the cardinality function for K-triviality levels

摘录 / EXCERPT: Fix a universal prefix-free machine and let K be its prefix-free Kolmogorov complexity. For each $e\in\mathbb{N}$, let K_e be the class of sets $A\subseteq\mathbb{N}$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-057092 | MATHDB-397975

Mixed packings of balls

摘录 / EXCERPT: Assume that balls of two sizes, with radii r_1 and r_2, are packed in Euclidean space. Assume also that the sizes are distributed, in terms of the limit cardinalities of the se...

open_problem | open | MathDB 页面 | 原文线索

V2-057093 | MATHDB-398082

Ruzsa’s conjecture on van der Corput, recurrence, and forcing-continuity sets

摘录 / EXCERPT: Given the implications $(FC^+) \Rightarrow (vdC) \Rightarrow (P)$, Ruzsa conjectured that the inverse implications also hold; the report states that Bourgain disproved this conject...

conjecture | open | MathDB 页面 | 原文线索

V2-057094 | MATHDB-398133

Strong uniquely solvable puzzles of near-optimal cardinality

摘录 / EXCERPT: Conjecture 3 (strong uniquely solvable puzzles). There exist strong uniquely solvable puzzles of cardinality $\binom{n}{n/3}^{1-o(1)}$. In the report, a uniquely solvable puzzle is...

open_problem | open | MathDB 页面 | 原文线索

V2-057095 | MATHDB-398134

Classical sunflower conjecture for 3-sunflowers

摘录 / EXCERPT: For every positive integer s, there exists a constant $c>0$ such that every collection of s-element subsets having cardinality at least c^s contains a 3-sunflower.

conjecture | open | MathDB 页面 | 原文线索

V2-057096 | MATHDB-398184

Different cardinalities of maximal independent generic sets

摘录 / EXCERPT: The report states that there is a finitely generated model of the theory of the free group-specifically, the free product of $\mathbb{Z}$ with the fundamental group of the connecte...

open_problem | open | MathDB 页面 | 原文线索

V2-057097 | MATHDB-398225

Theorem 41: Edge connectivity forcing immersions

摘录 / EXCERPT: There exists a function g satisfying the following. Let $k\ge 4$ and $n\ge 1$ be positive integers. Then, for all graphs H with maximum degree k on n vertices and for all...

open_problem | open | MathDB 页面 | 原文线索

V2-057098 | MATHDB-398676

Well-separated spherical designs of asymptotically minimal cardinality

摘录 / EXCERPT: For each integer $d\ge 2$, there exist positive constants C_d and λ_d, depending only on d, such that for every $t\in\mathbb N$ and every integer $N>C_dt^d$, there is a...

open_problem | open | MathDB 页面 | 原文线索

V2-057099 | MATHDB-398807

Universality number of graphs at a singular cardinal

摘录 / EXCERPT: Can it be consistent with ZFC that there is a singular cardinal κ such that $u_\kappa=2^\kappa>\kappa$? Here u_κ denotes the least size of a family of graphs on...

open_problem | open | MathDB 页面 | 原文线索

V2-057100 | MATHDB-398808

Maximal universality number at a singular cardinal

摘录 / EXCERPT: Let ba be a cardinal with ba e . For graphs on ba, let fba be the least size of a family of graphs such that every graph on ba embeds into a member of the family, where the embeddi...

open_problem | open | MathDB 页面 | 原文线索

V2-057101 | MATHDB-398810

Violating GCH at every infinite cardinal by adding one real

摘录 / EXCERPT: Assume the consistency of a κ^{+3}-strong cardinal κ. Then there exist models W and V of ZFC such that: (a) W and V have the same cardinals; (b) GCH holds in...

open_problem | open | MathDB 页面 | 原文线索

V2-057102 | MATHDB-398813

Necessity of the cardinal-arithmetic hypothesis in Theorem 6

摘录 / EXCERPT: Let d45ed45e be an infinite cardinal satisfying $\kappa^{<\kappa}=\kappa$. Is this condition necessary for the conclusion of Theorem 6, namely that embeddability on generalized tre...

conjecture | open | MathDB 页面 | 原文线索

V2-057103 | MATHDB-398814

Strong coloring under a nonreflecting stationary set

摘录 / EXCERPT: If λ and χ are regular cardinals, $\lambda>\chi$, and $E^{\lambda}_{\geq\chi}$ admits a nonreflecting stationary set, then $\Pr_1(\lambda,\lambda,\lambda,\chi)$ holds.

open_problem | open | MathDB 页面 | 原文线索

V2-057104 | MATHDB-398815

Strong coloring at successors of singular cardinals

摘录 / EXCERPT: For every singular cardinal λ, does ... hold? The source also explicitly asks whether at least the weaker assertion ... holds for every singular cardinal λ. Here...

open_problem | open | MathDB 页面 | 原文线索

V2-057105 | MATHDB-398818

Antichain number of Silver forcing

摘录 / EXCERPT: Is $a(\text{Si})=\mathfrak c$? Here $a(\text{Si})$ is the minimal size of an uncountable maximal antichain of Silver forcing, and $\mathfrak c$ denotes the cardinality of the conti...

open_problem | open | MathDB 页面 | 原文线索

V2-057106 | MATHDB-399109

Asymptoticity of one-sided minimizers for randomly kicked Burgers dynamics

摘录 / EXCERPT: In the fully noncompact space-time stationary Burgers model with kick forcing, distinct one-sided minimizers are known not to coalesce. It remains open whether any two such minimiz...

open_problem | open | MathDB 页面 | 原文线索

V2-057107 | MATHDB-399174

Endpoint inhomogeneous Strichartz estimate

摘录 / EXCERPT: Is the endpoint estimate ... valid for generic forcing terms, where $u(t,x)$ solves ... for $t\in\mathbb{R}$, $x\in\mathbb{R}^n$, and F is supported on $[0,1]\times\mathbb{R}^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-057108 | MATHDB-399390

Packing odd T-joins with at most two terminals

摘录 / EXCERPT: Let (G,Σ,T) be an Eulerian signed graft with $|T|\le 2$. Here, a signed graft consists of a graph G, an edge set $\Sigma\subseteq E(G)$), and an even-cardinality terminal...

open_problem | open | MathDB 页面 | 原文线索

V2-057109 | MATHDB-399411

Feferman’s conjecture on CH in a semi-intuitionistic set theory

摘录 / EXCERPT: For the specified semi-intuitionistic set theory T, T does not prove $\\mathrm{CH} \\vee \\neg\\mathrm{CH}$; equivalently, CH is not definite in T.

conjecture | open | MathDB 页面 | 原文线索

V2-057110 | MATHDB-400136

Minimal number of conics determined by subsets of a finite point set

摘录 / EXCERPT: In analogy to the Green-Tao Theorem, it would be interesting to know what is the minimal number of conics determined by subsets of P in dependence on the cardinality of the set...

open_problem | open | MathDB 页面 | 原文线索

V2-057111 | MATHDB-400714

Intermediate value theorem for κ-computable functions on Rκ

摘录 / EXCERPT: Let $\ba$ be an uncountable cardinal such that $\ba^{<\ba}=\ba$, and let $R_{\ba}$ be the Cauchy completion of $No_{\ba}$. Elements of $R_{\ba}$ can be coded by binary sequences of...

open_problem | open | MathDB 页面 | 原文线索

V2-057112 | MATHDB-400804

Existence of a coloring graph with gap two

摘录 / EXCERPT: Is it consistent that, for some infinite cardinal θ, there exists a (θ,θ^{+2})-coloring graph?

open_problem | open | MathDB 页面 | 原文线索

V2-057113 | MATHDB-400818

Radin forcing and special Aronszajn trees

摘录 / EXCERPT: Is there a Radin forcing which adds a club $C\subseteq\kappa$ such that for all cardinals $\mu\in C$, there are no special μ^+-Aronszajn trees?

open_problem | open | MathDB 页面 | 原文线索

V2-057114 | MATHDB-400830

Consistency of a \[?25l\[?25h$\[?25l\[?25h$\[?25l\[?25h\Delta^1_1$\[?25l\[?25h reduction from a Vaught counterexample to $F_{\[?25l\[?25h\omega_1}$\[?25l\[?25h$\[?25l\[?25h$

摘录 / EXCERPT: Assume that counterexamples to Vaught’s conjecture exist. Is it consistent, possibly relative to $0^\sharp$ or to a measurable cardinal, that there is some counterexample to Vaught...

open_problem | open | MathDB 页面 | 原文线索

V2-057115 | MATHDB-400833

A maximal generator for the approachability ideal

摘录 / EXCERPT: For an uncountable cardinal κ, is it consistent that the approachability ideal $I[\kappa^+]$ does not have a single generator over the nonstationary ideal on κ^+ (equ...

open_problem | open | MathDB 页面 | 原文线索

V2-057116 | MATHDB-400835

The approachability ideal without a maximal set

摘录 / EXCERPT: Assuming the consistency of a greatly Mahlo cardinal, it is consistent that the approachability ideal $I[\\omega_2]$ does not have a maximal set modulo clubs; equivalently,...

open_problem | open | MathDB 页面 | 原文线索

V2-057117 | MATHDB-400836

Ultrafilter over a measurable cardinal with infinitely many Rudin-Frolik predecessors

摘录 / EXCERPT: Is there a measurable cardinal (kappa) carrying a κ-complete ultrafilter with infinitely many predecessors in the Rudin-Frolik ordering?

open_problem | open | MathDB 页面 | 原文线索

V2-057118 | MATHDB-400837

Consistency strength of the perfect set property for universally Baire sets

摘录 / EXCERPT: Modulo ZFC, the following are equiconsistent: there is a cardinal κ that is Shelah for remarkability, and every universally Baire set of reals $A\subseteq\omega^\omega$ has...

open_problem | open | MathDB 页面 | 原文线索

V2-057119 | MATHDB-400941

Determining the maximum size of almost-equidistant sets

摘录 / EXCERPT: Determine the maximum cardinality $f(d)$ of an almost-equidistant set in $\r\mathbb{R}^d$, where a set $V\subseteq\mathbb{R}^d$ is almost equidistant if among any three points of...

open_problem | open | MathDB 页面 | 原文线索

V2-057120 | MATHDB-401939

Approximation and Integrality Gap for Linear k-Cut

摘录 / EXCERPT: Given a directed graph $G=(V,E)$ and k distinct terminal nodes $s_1,s_2,\c,s_k$, Linear-k-Cut asks for a minimum-cardinality set of edges whose removal ensures that s_i has n...

open_problem | open | MathDB 页面 | 原文线索

V2-057121 | MATHDB-402290

Farah’s Theorem 4 on ultrapowers and asymptotic sequence algebras

摘录 / EXCERPT: Assume that the Continuum Hypothesis holds and let B be a unital and separable C^*-algebra. If K denotes the Cantor space and $ u$ is a nonprincipal ultrafilter on...

open_problem | open | MathDB 页面 | 原文线索

V2-057122 | MATHDB-402365

Local Langlands in families in depth zero: mild conjecture

摘录 / EXCERPT: Assume the setup of the report: F is a non-archimedean local field with finite residue field of cardinality q, G is the F-points of an F-quasi-split reductive group, $\$...

conjecture | open | MathDB 页面 | 原文线索

V2-057123 | MATHDB-402612

Uniform dimension for forcing a red unit pair or a blue copy

摘录 / EXCERPT: Erdős et al. asked whether, for every natural number d, there exists a natural number n depending only on d such that, for every $K\subset E^d$, every red/blue-coloring of...

open_problem | open | MathDB 页面 | 原文线索

V2-057124 | MATHDB-402661

Noncommutative Bézout bounds for finite fibers

摘录 / EXCERPT: Given a noncommutative polynomial map $f:M_k(\mathbb{C})_{\mathrm{sa}}^n\to M_k(\mathbb{C})_{\mathrm{sa}}^n$, find bounds for the cardinalities of its finite fibers that improve on...

open_problem | open | MathDB 页面 | 原文线索

V2-057125 | MATHDB-402672

Rigidity conjecture for coronas of separable non-unital C*-algebras

摘录 / EXCERPT: Forcing axioms imply that all isomorphisms between coronas of separable, non-unital C^*-algebras are trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-057126 | MATHDB-402675

Equality of the higher pseudo-intersection and tower numbers

摘录 / EXCERPT: For every regular uncountable cardinal κ, is $\mathfrak p(\kappa)=\mathfrak t(\kappa)$? Here $\mathfrak p(\kappa)$ is the least cardinality of a family of subsets of...

open_problem | open | MathDB 页面 | 原文线索

V2-057127 | MATHDB-402676

Provability that the first almost strongly compact cardinal is strongly compact

摘录 / EXCERPT: Is it provable outright that the first almost strongly compact cardinal is strongly compact?

open_problem | open | MathDB 页面 | 原文线索

V2-057128 | MATHDB-403566

Bollobás-Erdős-Tuza conjecture on hitting maximum independent sets

摘录 / EXCERPT: For a graph $G=(V,E)$ on n vertices, let $\alpha(G)$ denote its independence number, and let $h(G)$ denote the minimum cardinality of a set S of vertices that intersects all ma...

conjecture | open | MathDB 页面 | 原文线索

V2-057129 | MATHDB-404492

Existence of spherical t-designs with optimal-order cardinality

摘录 / EXCERPT: The historical open problem was to prove the existence of spherical t-designs on S^d having $O_d(t^d)$ points, matching the order of the lower bound as $t\to\infty$ for fixed...

open_problem | open | MathDB 页面 | 原文线索

V2-057130 | MATHDB-404567

Erdős-Turán three-term arithmetic-progression problem

摘录 / EXCERPT: What is the maximum cardinality ...

open_problem | open | MathDB 页面 | 原文线索

V2-057131 | MATHDB-404793

Immediate descendants of the groups $G_{E,P}(\f_p)$

摘录 / EXCERPT: For which pairs (E,P) does there exist a central extension ... where G has nilpotency class 3 and cardinality p^{10}? Here E and P range over the elliptic-curve/point dat...

open_problem | open | MathDB 页面 | 原文线索

V2-057132 | MATHDB-404964

Maximum size of r-cover-free families

摘录 / EXCERPT: For positive integers n and r, determine the maximum cardinality of a family $\mathcal F\subseteq 2^{[n]}$ such that no member of $\mathcal F$ is contained in the union of r...

open_problem | open | MathDB 页面 | 原文线索

V2-057133 | MATHDB-404985

Sharpness of the realization-cardinality lower bound for monomial ideals

摘录 / EXCERPT: The report conjectures that, for every realizable monomial ideal I of $k[x_1,x_2]$, the lower bound $|D|\geq \max\{q:\operatorname{SH}^{q-1}(I\subset O)\neq 0\}$ for the cardinal...

open_problem | open | MathDB 页面 | 原文线索

B14 逻辑 / Logic | 402 条

V2-057134 | MATHDB-316305

Harrington's conjecture on representatives of c.e. set orbits

摘录 / EXCERPT: Let A be a computably enumerable set and let $\boldsymbol{d}$ be a Turing degree. Write A' for the Turing jump of A, and let $\mathcal{L}^*(A)$ denote the lattice structure a...

conjecture | open | MathDB 页面 | 原文线索

V2-057135 | MATHDB-316328

The classifiability and wandering-types characterization of the Schröder-Bernstein property

摘录 / EXCERPT: The classifiability and wandering-types conjecture. The complete theory T has the Schröder-Bernstein property if and only if it is classifiable and has no wandering types.

conjecture | open | MathDB 页面 | 原文线索

V2-057136 | MATHDB-316329

The definable-set lifting conjecture for the Schröder-Bernstein property

摘录 / EXCERPT: Definable-set lifting conjecture. If T' does not have the Schröder-Bernstein property, then T does not have the Schröder-Bernstein property.

conjecture | open | MathDB 页面 | 原文线索

V2-057137 | MATHDB-316330

The naming-constants conjecture for the Schröder-Bernstein property

摘录 / EXCERPT: Naming-constants conjecture. (1) There is a small set $A\subseteq\mathfrak{C}$ such that $\operatorname{Th}_A(\mathfrak{C})$ has the Schröder-Bernstein property if and only if T...

conjecture | open | MathDB 页面 | 原文线索

V2-057138 | MATHDB-316331

The infinite-family conjecture for failure of the Schröder-Bernstein property

摘录 / EXCERPT: Infinite-family conjecture. If T does not have the Schröder-Bernstein property, then there is an infinite collection of models of T that are pairwise nonisomorphic and pairwise...

conjecture | open | MathDB 页面 | 原文线索

V2-057139 | MATHDB-316400

Solvable-by-finite conjecture for superrosy definably amenable groups of thorn-rank 2

摘录 / EXCERPT: Solvable-by-finite conjecture. Then G is solvable-by-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-057140 | MATHDB-316401

Algebraic-or-real-closed conjecture for superrosy NIP fields

摘录 / EXCERPT: Algebraic-or-real-closed field conjecture. The field K is either algebraically closed or real closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057141 | MATHDB-316909

Soundness, completeness, and effective completeness of CL11

摘录 / EXCERPT: The logic in question is $\textbf{CL11}$, whose derivability relation is written $\textbf{CL11}\vdash F$, and whose formulas are interpreted as computational problems. An HPM is a...

conjecture | open | MathDB 页面 | 原文线索

V2-057142 | MATHDB-317452

Non-recursive enumerability for effective graph models

摘录 / EXCERPT: Effective graph-model conjecture. All the effective graph models have non-r.e. equational theories.

conjecture | open | MathDB 页面 | 原文线索

V2-057143 | MATHDB-317453

Non-recursive enumerability for effective models in continuous semantics

摘录 / EXCERPT: Continuous-semantics conjecture. All the effective models living in the continuous semantics have non-r.e. equational theories.

conjecture | open | MathDB 页面 | 原文线索

V2-057144 | MATHDB-317506

Halupczok's tree-level conjecture for definable p-adic sets

摘录 / EXCERPT: Let $X \subset \mathbb{Q}_p^n$ be a definable set in the language of fields. For each $\lambda \geq 0$, let X_λ be the image of $X \cap \mathbb{Z}_p^n$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-057145 | MATHDB-318790

Zilber's theorem for continuous logic, naïve version

摘录 / EXCERPT: Zilber's theorem for continuous logic, naïve version. Every $\aleph_0$-categorical $\aleph_0$-stable theory is SFB.

conjecture | open | MathDB 页面 | 原文线索

V2-057146 | MATHDB-318791

Zilber's theorem for continuous logic

摘录 / EXCERPT: Zilber's theorem for continuous logic. Whenever T is an $\aleph_0$-categorical $\aleph_0$-stable theory (in a countable language), T_P is $\aleph_0$-categorical up to arbitrari...

conjecture | open | MathDB 页面 | 原文线索

V2-057147 | MATHDB-318964

Conjecture on deep Ricci-flat Kähler minima in Calabi-Yau moduli space

摘录 / EXCERPT: Let $\mathop{\rm AI}(M)$ denote the relevant space of metrics, and let c_1 and n be the diameter-bound parameters for a fixed Ricci-flat Kähler metric ω_0. A local minim...

conjecture | open | MathDB 页面 | 原文线索

V2-057148 | MATHDB-319060

The separable-algebraicity genericity conjecture for stable fields

摘录 / EXCERPT: Separable-algebraicity genericity conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057149 | MATHDB-319061

The algebraic genericity characterization of algebraically closed stable fields

摘录 / EXCERPT: Algebraic genericity characterization. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057150 | MATHDB-319562

No represented space characterizes limit computability by closed choice

摘录 / EXCERPT: Let (X,δ) be a represented space, let ${\rm C}_X$ denote closed choice on X, and let $\lim$ denote the limit operation on Baire-space names. Non-characterization conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-057151 | MATHDB-319902

Generalized unprovability of Herbrand consistency for ${\rm IDelta}_0$

摘录 / EXCERPT: Let ${\rm IDelta}_0$ be the bounded-induction theory of arithmetic, let ${\rm Omega}_n$ denote the corresponding family of axioms, and let ${\rm HCon}^*({\rm IDelta}_0)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-057152 | MATHDB-320126

The NIP conjecture for omega-categorical groups

摘录 / EXCERPT: NIP conjecture for omega-categorical groups. Every ω-categorical group with NIP is nilpotent-by-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-057153 | MATHDB-320127

The NIP conjecture for omega-categorical rings

摘录 / EXCERPT: NIP conjecture for omega-categorical rings. Every ω-categorical ring with NIP is nilpotent-by-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-057154 | MATHDB-320439

Finitarity and definability of the closure operator

摘录 / EXCERPT: Finitarity and definability conjecture. cl is always finitary. Moreover, cl is definable: for every small $A\subseteq\mathfrak C$, the set...

conjecture | open | MathDB 页面 | 原文线索

V2-057155 | MATHDB-320440

Extension of an independence relation to the lovely-pair structure

摘录 / EXCERPT: Extension conjecture for lovely pairs. There exists an independence relation $mathop{\mathpalette\Ind{}^{\!\!\!\!\!\rlap{$\scriptscriptstyletextnormal{2}$}\,\,\,\,}}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-057156 | MATHDB-320680

The dichotomy conjecture for Polish structures

摘录 / EXCERPT: Dichotomy conjecture. There is a dichotomy: either the model is similar to categorical theories, or there are “many complicated models”.

conjecture | open | MathDB 页面 | 原文线索

V2-057157 | MATHDB-321055

Cut admissibility for translated geometric frame-axiom rules

摘录 / EXCERPT: Let the method in Definition translate geometric frame axioms into hypersequent rules, and let $\mathrm{HG3ipm}$-like calculi be augmented with the lin rule. Cut-admissi...

conjecture | open | MathDB 页面 | 原文线索

V2-057158 | MATHDB-321420

The restrained-structure dichotomy conjecture

摘录 / EXCERPT: A restrained structure is a DC expansion of an ordered field such that, for every definable discrete set $D \subseteq \mathbb K^n$ and every definable function $g:D\to\mathbb K$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-057159 | MATHDB-321421

The dimension-function conjecture for restrained structures

摘录 / EXCERPT: Let $\mathbb K$ be restrained. A dimension function is a function d assigning to every definable set a number in $\mathbb N\cup\{-\infty\}$ and satisfying the following axioms fo...

conjecture | open | MathDB 页面 | 原文线索

V2-057160 | MATHDB-321422

The dimension inequality for definable images of \(\mathcal D_\Sigma\) sets

摘录 / EXCERPT: Let $C\subseteq\mathbb K^m$ be a $\mathcal D_\Sigma$ set, and let $f:C\to\mathbb K^n$ be definable, without assuming that f is continuous. The dimension-image conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-057161 | MATHDB-321423

The i-minimal open-core conjecture

摘录 / EXCERPT: Let $\mathbb K$ be a DC structure, and let its open core be the structure whose definable sets are generated by the open definable subsets of $\mathbb K^n$. The i-minimal open-core...

conjecture | open | MathDB 页面 | 原文线索

V2-057162 | MATHDB-321496

Recursive-spectrum trichotomy conjecture for strongly minimal theories

摘录 / EXCERPT: Recursive-spectrum trichotomy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057163 | MATHDB-321497

Positive-dimensional recursive model conjecture for strongly minimal theories

摘录 / EXCERPT: Positive-dimensional recursive model conjecture. If some positive-dimensional model $M\models T$ is recursive, then all models of T are recursively presentable.

conjecture | open | MathDB 页面 | 原文线索

V2-057164 | MATHDB-321599

Soundness conjecture for the recurrence operators in CL15

摘录 / EXCERPT: Let $\textbf{CL15}$ be the cirquent calculus system with recurrence operators $\text{\raisebox{-0.05cm}{$circ$}\hspace{-0.13cm}\raisebox{0.16cm}{\tiny mid}\hspace{2pt}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-057165 | MATHDB-321644

Thomas's finite reducts conjecture for finite-language homogeneous structures

摘录 / EXCERPT: Thomas's conjecture. The structure Δ has only finitely many reducts up to first-order interdefinability.

conjecture | open | MathDB 页面 | 原文线索

V2-057166 | MATHDB-321861

The embeddability-complexity conjecture for finitely presented groups

摘录 / EXCERPT: Embeddability-complexity conjecture. The set K is Σ^0_3-complete. Equivalently, deciding whether $\overline{P}_i$ embeds into $\overline{P}_j$ for finite presentations...

conjecture | open | MathDB 页面 | 原文线索

V2-057167 | MATHDB-321945

Inner mantle non-stabilization conjecture

摘录 / EXCERPT: Let V be a model of ${\rm ZFC}$, and let $\mathord{\rm M}^\alpha$ denote the αth inner mantle obtained by starting with V, taking the mantle at successor stages, and int...

conjecture | open | MathDB 页面 | 原文线索

V2-057168 | MATHDB-321946

Iterated mantle and HOD coincidence conjecture

摘录 / EXCERPT: Let V be a model of ${\rm ZFC}$ and let $\alpha\leq\mathop{\rm ORD}$. In another model $\overline{V}$ of ${\rm ZFC}$, write $({\mathord{\rm M}}^\alpha)^{\overline{V}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-057169 | MATHDB-322194

Polynomial-time computability of Julia sets without critical points in the postcritical set

摘录 / EXCERPT: Computability conjecture. If Ω_f does not contain any critical points, then J_f is polynomial-time computable by a Turing machine with an oracle for the coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-057170 | MATHDB-322195

Polynomial-time computability of Julia sets under the critical-orbit dichotomy

摘录 / EXCERPT: Computability conjecture. Under these two conditions, J_f is polynomial-time computable by a Turing machine with an oracle for the coefficients of f.

conjecture | open | MathDB 页面 | 原文线索

V2-057171 | MATHDB-322196

Almost-everywhere polynomial-time computability of Julia sets

摘录 / EXCERPT: Almost-everywhere computability conjecture. For almost all rational maps f of degree $d\geqslant 2$, J_f is polynomial-time computable.

conjecture | open | MathDB 页面 | 原文线索

V2-057172 | MATHDB-322201

Transfer-principle conservation conjecture for Shoenfield-normal formulas

摘录 / EXCERPT: Transfer-principle conservation conjecture. Under these hypotheses, the displayed internal consequence follows. The claim is introduced as a consequence that would establish the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-057173 | MATHDB-322517

The norm-one btt obstruction for intrinsically n-approximable distance functions

摘录 / EXCERPT: Let G be a computable connected graph whose distance function is intrinsically n-approximable from above, where $n>0$. A function is n-approximable from above if it has an ap...

conjecture | open | MathDB 页面 | 原文线索

V2-057174 | MATHDB-322666

Frank Wagner's Baldwin-Saxl conjecture for type definable groups

摘录 / EXCERPT: Frank Wagner's conjecture. There is some n such that, for every finite set $v\subseteq\omega$, the intersection

conjecture | open | MathDB 页面 | 原文线索

V2-057175 | MATHDB-322761

The model-completeness conjecture for the differential field of transseries

摘录 / EXCERPT: Let $\mathbb{T}$ be the ordered valued differential field of transseries. Let H-fields be ordered differential fields equipped with the associated H-field structure, and let sm...

conjecture | open | MathDB 页面 | 原文线索

V2-057176 | MATHDB-322762

The $\mathbb{T}$-conjecture on model completeness

摘录 / EXCERPT: Consider $\mathbb{T}$ as an ordered valued differential field. The $\mathbb{T}$-conjecture. The field $\mathbb{T}$ is model complete. Equivalently, in the language of differential...

conjecture | open | MathDB 页面 | 原文线索

V2-057177 | MATHDB-322763

The refined $\mathbb{T}$-conjecture for existentially closed H-fields

摘录 / EXCERPT: Let $\mathbb{T}$ be the differential field of transseries, and let an H-field with small derivation be an ordered differential field satisfying the relevant H-field axioms and...

conjecture | open | MathDB 页面 | 原文线索

V2-057178 | MATHDB-322764

The characterization conjecture for existentially closed H-fields

摘录 / EXCERPT: Let K be an H-field with small derivation. Let $\imag=\sqrt{-1}$, and let $K[\der]$ and $K[\imag][\der]$ denote the corresponding rings of differential operators. The character...

conjecture | open | MathDB 页面 | 原文线索

V2-057179 | MATHDB-322811

Local-to-global cyclic polymorphism conjecture for omega-categorical cores

摘录 / EXCERPT: Let ${\mathfrak B}$ be a countable ω-categorical model-complete core, and let f be an n-ary polymorphism of ${\mathfrak B}$. Cyclic polymorphism conjecture. If, for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-057180 | MATHDB-322812

Cyclic polymorphism conjecture for omega-categorical model-complete cores

摘录 / EXCERPT: Let ${\mathfrak B}$ be a countable ω-categorical model-complete core. A polymorphism is an operation preserving all relations of ${\mathfrak B}$, and an endomorphism is a un...

conjecture | open | MathDB 页面 | 原文线索

V2-057181 | MATHDB-322908

The connected-component inclusion conjecture for group extensions

摘录 / EXCERPT: Connected-component inclusion conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-057182 | MATHDB-322909

The type-definable subgroup inclusion conjecture for group extensions

摘录 / EXCERPT: Let $(\widetilde{G^*},A^*,G^*)$ be in Situation (0), and let B be the parameter set used to define the connected components. Suppose ... Let H be a B-invariant subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-057183 | MATHDB-323091

The Unique Limit Model Conjecture for dependent theories

摘录 / EXCERPT: Let T be dependent, let $|T|<\lambda=\lambda^{<\lambda}$, let $\lambda^+=2^\lambda$, and let $\sigma=\operatorname{cf}(\sigma)<\lambda$. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-057184 | MATHDB-323311

Sub-additivity of burden in NTP2 theories

摘录 / EXCERPT: In a first-order theory T, let $\operatorname{bdn}(a)$ denote the burden of a tuple a, and say that T is NTP$_2$ when it has no inp-pattern of depth ω in a single vari...

conjecture | open | MathDB 页面 | 原文线索

V2-057185 | MATHDB-323327

Conjecture on quantity structures of special relativity with approximate motion

摘录 / EXCERPT: Let n be the spacetime dimension, let $\mathsf{SpecRel}$ be the theory of special relativity, and let $\mathsf{AxThExp^-}$ be the axiom expressing that inertial observers can mov...

conjecture | open | MathDB 页面 | 原文线索

V2-057186 | MATHDB-324125

The ordered-field conjecture for special relativity with approximate expansion

摘录 / EXCERPT: Let $d\ge2$. Write $\operatorname{Num}(\mathsf{Th})$ for the class of quantity structures of models of a theory Th, and let $\mathsf{SpecRel}_d$ be the axiom system for...

conjecture | open | MathDB 页面 | 原文线索

V2-057187 | MATHDB-324309

Extensible definability conjecture for non-forking formulas

摘录 / EXCERPT: Extensible definability conjecture. There is $\theta(y)\in \operatorname{tp}(d/M)$ such that the partial type

conjecture | open | MathDB 页面 | 原文线索

V2-057188 | MATHDB-324310

Definable extension conjecture for non-forking formulas in dp-minimal theories

摘录 / EXCERPT: Definable extension conjecture. The formula $\phi(x;b)$ extends to an M-definable type.

conjecture | open | MathDB 页面 | 原文线索

V2-057189 | MATHDB-324311

Countable coheir detection conjecture for commutativity

摘录 / EXCERPT: Countable coheir detection conjecture. There is a countable subset $Q\subseteq S_y(\mathcal U)$ of coheirs of q such that, for every invariant type p, if p commutes over M...

conjecture | open | MathDB 页面 | 原文线索

V2-057190 | MATHDB-324312

Invariant-type quotient conjecture

摘录 / EXCERPT: Invariant-type quotient conjecture. There are $b\in\mathcal U$ and a formula $\psi(z;y)$ such that $\operatorname{tp}(b/N)$ is finitely satisfiable in M and

conjecture | open | MathDB 页面 | 原文线索

V2-057191 | MATHDB-324313

Finite-pattern definability conjecture for invariant types

摘录 / EXCERPT: Finite-pattern definability conjecture. There is a finite set Δ' of formulas such that, for every $\phi(x;y)\in p$ and $b,b'\in\mathcal U$, if

conjecture | open | MathDB 页面 | 原文线索

V2-057192 | MATHDB-324348

The no-interpretable-fields conjecture for torsion-free hyperbolic groups

摘录 / EXCERPT: Let G be a torsion-free hyperbolic group. An interpretable field is a field interpretable in a model of $Th(G)$. No-interpretable-fields conjecture. No infinite field is interpre...

conjecture | open | MathDB 页面 | 原文线索

V2-057193 | MATHDB-324406

Canonicity of thorn-independence in simple theories

摘录 / EXCERPT: Let T be a simple theory, and let $\thind$ denote thorn-independence in T. Canonicity conjecture. In any simple theory, $\thind$ is canonical. This conjecture proposes that tho...

conjecture | open | MathDB 页面 | 原文线索

V2-057194 | MATHDB-324742

NTP2 conjecture for pseudo real closed and pseudo p-adically closed fields

摘录 / EXCERPT: PRC and PpC NTP2 conjecture. A PRC field is NTP_{2} if and only if it is bounded. Similarly, a PpC field is NTP_{2} if and only if it is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-057195 | MATHDB-324743

NTP2 conjecture for Kaplansky algebraically maximal valued fields

摘录 / EXCERPT: Valued-field NTP2 conjecture. Then (K,v) is NTP_{2} (respectively, strong) if and only if k is NTP_{2} (respectively, strong).

conjecture | open | MathDB 页面 | 原文线索

V2-057196 | MATHDB-325016

The highness-tracing correspondence for computable limit ordinals

摘录 / EXCERPT: Let α be a computable limit ordinal, possibly with additional closure properties such as closure under addition. For an oracle A, let...

conjecture | open | MathDB 页面 | 原文线索

V2-057197 | MATHDB-325030

Conjecture on units in Grothendieck rings of modules

摘录 / EXCERPT: Let M be a nonzero module, and let $K_0(M)$ denote its model-theoretic Grothendieck ring. A unit of $K_0(M)$ is an element having a multiplicative inverse. Unit conjecture for mo...

conjecture | open | MathDB 页面 | 原文线索

V2-057198 | MATHDB-325032

The geometric triviality-unimodularity conjecture for strongly minimal sets

摘录 / EXCERPT: Let Y be a strongly minimal set in a differentially closed field. It is unimodular if, whenever tuples $\overline{y}$ and $\overline{y}'$ of the same length are interalgebraic ov...

conjecture | open | MathDB 页面 | 原文线索

V2-057199 | MATHDB-325033

Full range of game values in infinite chess

摘录 / EXCERPT: For infinite chess, let $\omega_1^{\mathfrak{Ch}}$ be the supremum of game values for positions with at most finitely many pieces, let $\omega_1^{{\mathfrak{Ch}},c}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-057200 | MATHDB-325044

Shelah's excellence conjecture for atomic classes

摘录 / EXCERPT: Let T be a countable first-order theory and let K_T be its class of atomic models. Excellence conjecture. If K_T is categorical in every $\beth_n$ for $n<\beth_0$, then K_T...

conjecture | open | MathDB 页面 | 原文线索

V2-057201 | MATHDB-325046

Categoricity transfer conjecture for abstract elementary classes

摘录 / EXCERPT: Let ${\mathfrak k}=(K_{\mathfrak k},\leq_{\mathfrak k})$ be an abstract elementary class, with Löwenheim-Skolem-Tarski number ${\rm LST}_{\mathfrak k}$. Categoricity transfer conje...

conjecture | open | MathDB 页面 | 原文线索

V2-057202 | MATHDB-325047

Main gap and super-limit spectrum conjecture for abstract elementary classes

摘录 / EXCERPT: Let ${\mathfrak k}$ be an abstract elementary class. Main gap and super-limit conjecture. (1) The main gap theorem holds for $K_{\mathfrak k}$ for all sufficiently large λ....

conjecture | open | MathDB 页面 | 原文线索

V2-057203 | MATHDB-325115

Grossberg-VanDieren's categoricity-to-tameness conjecture for abstract elementary classes

摘录 / EXCERPT: Grossberg-VanDieren's categoricity-to-tameness conjecture. Suppose K is an AEC. If K is categorical in some $\lambda \geq Hanf(LS(K))$ (or some other value depending only on...

conjecture | open | MathDB 页面 | 原文线索

V2-057204 | MATHDB-325461

The Lascar-rank conjecture for countable models of superstable theories

摘录 / EXCERPT: Lascar-rank conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-057205 | MATHDB-325462

The generic-type conjecture for simple superstable groups

摘录 / EXCERPT: Generic-type conjecture. The generic type of G is ESN.

conjecture | open | MathDB 页面 | 原文线索

V2-057206 | MATHDB-326124

The superrosy NIP field conjecture

摘录 / EXCERPT: A field is NIP if its theory has the non-independence property. The superrosy NIP field conjecture. Every infinite superrosy field with NIP is either algebraically closed or real c...

conjecture | open | MathDB 页面 | 原文线索

V2-057207 | MATHDB-326125

The superrosy field conjecture via PRC fields

摘录 / EXCERPT: A field is PRC (pseudo real closed) if it satisfies the PRC property; its absolute Galois group is small if it has only finitely many closed subgroups of each finite index. The sup...

conjecture | open | MathDB 页面 | 原文线索

V2-057208 | MATHDB-326126

The Hasson-Shelah definable valuation conjecture for NIP fields

摘录 / EXCERPT: Let K be an infinite field with NIP such that, for every finite extension L of K and every natural number $n>0$, the index $[L^*:(L^*)^n]$ is finite. The Hasson-Shelah defina...

conjecture | open | MathDB 页面 | 原文线索

V2-057209 | MATHDB-326127

The common valuation conjecture for superrosy fields

摘录 / EXCERPT: Let K be a field such that for every finite extension L of K and every natural number $n>0$, the index $[L^*:(L^*)^n]$ is finite. If $\operatorname{char}(K)=p>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-057210 | MATHDB-326128

The valuation-residue-field conjecture for superrosy fields

摘录 / EXCERPT: Let K be a superrosy field and let v be a non-trivial valuation on K. Its value group is the ordered abelian group of values, and its residue field is the quotient field indu...

conjecture | open | MathDB 页面 | 原文线索

V2-057211 | MATHDB-326132

Universality-number conjecture for Banach spaces and Boolean algebras

摘录 / EXCERPT: Let ... with isomorphic embeddings and Boolean algebras of size ... with isomorphic embeddings is the same as the universality number of the class of Boolean algebras of size ... , even...

conjecture | open | MathDB 页面 | 原文线索

V2-057212 | MATHDB-326619

MSO$_2$-orderability conjecture for VR-equational classes with property SEP

摘录 / EXCERPT: Let $\mathcal{C}$ be a VR-equational class of graphs. Say that $\mathcal{C}$ has property SEP when it satisfies the separation property defined in the surrounding theory...

conjecture | open | MathDB 页面 | 原文线索

V2-057213 | MATHDB-326907

Conjecture on the complexity of definable orbits of c.e. sets

摘录 / EXCERPT: Let A be a c.e. set whose orbit is properly Δ^0_{α} for a finite $\alpha>8$, as in Theorem 3.2, and let $\mathcal{L}_{\omega_1,\omega}$ denote the infinitary logic use...

conjecture | open | MathDB 页面 | 原文线索

V2-057214 | MATHDB-326908

Soare's effective automorphism conjecture for semilow sets

摘录 / EXCERPT: Let A be a c.e. set. Its complement $\overline{A}$ is semilow when the corresponding semilowness property holds, and A is low when $A'\equiv_T\mathbf{0}'$. Two c.e. sets are au...

conjecture | open | MathDB 页面 | 原文线索

V2-057215 | MATHDB-326909

Cholak-Epstein conjecture on outer splitting and automorphism to low-two sets

摘录 / EXCERPT: Let A be a c.e. set. It has the outer splitting property if there are computable functions f,h such that, for every e, $W_e=W_{f(e)}\sqcup W_{h(e)}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-057216 | MATHDB-327138

Definable (p,q)-theorem for NIP formulas

摘录 / EXCERPT: Assume that the formula $\phi(x;y)$ is NIP. Let M be a model and let $\phi(x;b)\in L(\mathcal U)$ be non-dividing over M. Definable (p,q)-conjecture. There is a formula...

conjecture | open | MathDB 页面 | 原文线索

V2-057217 | MATHDB-327139

Definable (p,q)-conjecture for NIP theories

摘录 / EXCERPT: Let T be an NIP theory and let $M\models T$. Let $\phi(x;d)\in L(\mathcal U)$ be a formula that is non-forking over M. Definable (p,q)-conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-057218 | MATHDB-327140

Definability conjecture for non-forking formulas in dp-minimal theories

摘录 / EXCERPT: Assume that T is dp-minimal. If the formula $\phi(x;b)$ does not fork over M, dp-minimal definability conjecture. it extends to an M-definable type. The paper states this as...

conjecture | open | MathDB 页面 | 原文线索

V2-057219 | MATHDB-327141

Finite-parameter determination of invariant types

摘录 / EXCERPT: Let $p(x)$ be a global M-invariant type and let Δ be a finite set of formulas. Finite determination conjecture. There is a finite set Δ' of formulas such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-057220 | MATHDB-327155

MSO limit laws for unlabelled addable minor-closed classes

摘录 / EXCERPT: Let $\mathcal{G}$ be an addable, minor-closed class and let $\mathcal{U}\mathcal{G}$ be its corresponding collection of unlabelled graphs. Let $C_n \in_u \mathcal{U}\mathcal{C}_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-057221 | MATHDB-327218

Zilber's strong exponential-algebraic closedness conjecture in one variable

摘录 / EXCERPT: Let $k\subset\mathbb{C}$ be a finitely generated field, and let $p(x,y)\in k[x,y]$ be irreducible with $\frac{\partial p}{\partial x},\frac{\partial p}{\partial y}\neq 0$. Zilber's...

conjecture | open | MathDB 页面 | 原文线索

V2-057222 | MATHDB-327432

The bounded expansion characterization for first-order definable colorings

摘录 / EXCERPT: Bounded expansion characterization. The following properties are equivalent: for every integer p there is a non-bipartite graph H_p of odd-girth strictly greater than $2p+1$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-057223 | MATHDB-327518

Countability of id-jump traceable reals

摘录 / EXCERPT: Countability conjecture. There are only countably many id-jump traceable reals.

conjecture | open | MathDB 页面 | 原文线索

V2-057224 | MATHDB-328298

Uniform Vaught's conjecture

摘录 / EXCERPT: Let phi be an $\mathcal{L}_{omega_{1},omega}$-sentence of quantifier depth $alpha$, and let Scott rank refer to the Scott rank of a model of phi. Uniform Vaught's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-057225 | MATHDB-328494

Doubly-exponential computability of the minimum variable-degree function

摘录 / EXCERPT: Let $\operatorname{minnonmer}:\mathbb{N}\to\mathbb{N}$ denote the minimum variable-degree function for minimally unsatisfiable clause-sets indexed by deficiency, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057226 | MATHDB-328547

Computable presentation of compact congruences of an upper semilattice

摘录 / EXCERPT: An upper semilattice with least element is a partially ordered structure with binary joins and a least element. A presentation is computably enumerable when its atomic diagram is c...

conjecture | open | MathDB 页面 | 原文线索

V2-057227 | MATHDB-328549

Dobrinen-Simpson conjecture on high degrees and almost everywhere domination

摘录 / EXCERPT: Let A be a set of natural numbers, let A' denote its Turing jump, and let $0”$ denote the second Turing jump of the computable degree. A set is almost everywhere dominating whe...

conjecture | open | MathDB 页面 | 原文线索

V2-057228 | MATHDB-328575

The lowness conjecture for lattice initial segments of the Turing degrees

摘录 / EXCERPT: Lowness conjecture. Under these assumptions, $[\mathbf a,\mathbf g]$ is $\Sigma^0_1(\mathbf x)$-presentable.

conjecture | open | MathDB 页面 | 原文线索

V2-057229 | MATHDB-328579

Valued-field version of Peterzil's conjecture

摘录 / EXCERPT: Valued-field version of Peterzil's conjecture. Then M interprets a field.

conjecture | open | MathDB 页面 | 原文线索

V2-057230 | MATHDB-329821

The odd-order limit-law conjecture for metric spaces

摘录 / EXCERPT: Let $r\geq 3$ be an odd integer, and let $M_r(n)$ denote the class of finite metric spaces on $[n]$ with distances in the relevant r-element distance set. Let $\mathcal{L}_r$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-057231 | MATHDB-329827

Effective finiteness conjecture for rational matrix families

摘录 / EXCERPT: Let $\mathcal F$ be any finite set of square matrices with rational entries, and let $\overline\rho_k(\mathcal F)$ denote the maximum normalized spectral radius among products of l...

conjecture | open | MathDB 页面 | 原文线索

V2-057232 | MATHDB-329901

Generic compact domination conjecture for definably amenable NIP groups

摘录 / EXCERPT: Generic compact domination conjecture. For any definable set X, the set E_X has Haar measure 0. This is the compact-domination property needed in the approach to the Ellis gr...

conjecture | open | MathDB 页面 | 原文线索

V2-057233 | MATHDB-330154

The non-definability conjecture for asymptotic couples in precontraction groups

摘录 / EXCERPT: Non-definability conjecture. In no nontrivial precontraction group (Γ,χ) can one define, even allowing parameters, a function $\psi:\Gamma^{\neq}\to\Gamma$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-057234 | MATHDB-330261

The hyperimaginary elimination conjecture for Urysohn monoids

摘录 / EXCERPT: Hyperimaginary elimination conjecture. The theory $\operatorname{Th}(\mathcal{U}_{\mathcal{R}})$ has elimination of hyperimaginaries if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-057235 | MATHDB-330463

The measurable-function representation conjecture for QF-limits of binary functional structures

摘录 / EXCERPT: Measurable-function representation conjecture. The limit of every QF-convergent sequence of finite σ-structures can be represented by a measurable function

conjecture | open | MathDB 页面 | 原文线索

V2-057236 | MATHDB-330485

The propositional theory conjecture for the Muchnik lattices at levels 2 and 3

摘录 / EXCERPT: Propositional theory conjecture. The propositional theories of $\mathcal{M}_2$ and $\mathcal{M}_3$ are Jan; in fact, there are principal factors of $\mathcal{M}_2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-057237 | MATHDB-330986

Standard-model conjecture for finitely satisfiable Gödel theories

摘录 / EXCERPT: Let T be a theory in additive Gödel logic. A theory is finitely satisfiable if every finite subset of T has a standard model, where a standard model is a model whose truth valu...

conjecture | open | MathDB 页面 | 原文线索

V2-057238 | MATHDB-331203

Nonexistence of definable Skolem functions in strictly weakly o-minimal non-valuational expansions

摘录 / EXCERPT: Let M be a strictly weakly o-minimal non-valuational expansion of an ordered group. A definable Skolem function is a definable choice function for every definable family of nonem...

conjecture | open | MathDB 页面 | 原文线索

V2-057239 | MATHDB-331220

Finite-rank conjecture for grouplike types in $\mathbb{Q}$ACFA

摘录 / EXCERPT: Finite-rank conjecture. Suppose that p_1 is minimal in the sense of L_1 and ACFA. If p_1 is grouplike in the sense of L_1 and ACFA, then p has finite rank in the sense of...

conjecture | open | MathDB 页面 | 原文线索

V2-057240 | MATHDB-331221

Independence characterization for sufficient reducts

摘录 / EXCERPT: Independence characterization conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-057241 | MATHDB-331237

Fixed-point boundedness and the expressive power of infinitary logic

摘录 / EXCERPT: Let K be a class of finite structures, let $t(n)$ be polynomially bounded, and let $r(s, \mathcal{A})$ denote the relevant iteration-depth parameter for a structure $\mathcal{A}$...

conjecture | open | MathDB 页面 | 原文线索

V2-057242 | MATHDB-331480

The Gurarij space's WEMI and transitive-space conjecture

摘录 / EXCERPT: Gurarij space conjecture. The Gurarij space $\mathbf{G}$ has WEMI, and every transitive complete $\operatorname{Aut}(\mathbf{G})$-space is a singleton.

conjecture | open | MathDB 页面 | 原文线索

V2-057243 | MATHDB-331574

The NSOP$_1$ conjecture for pseudofinite countably categorical theories

摘录 / EXCERPT: NSOP$_1$ conjecture. Every pseudofinite countably categorical theory is NSOP$_1$.

conjecture | open | MathDB 页面 | 原文线索

V2-057244 | MATHDB-332270

The conformal-equivalence conjecture for GenRel without localized time symmetry

摘录 / EXCERPT: Let $GenRel$ be the first-order axiom system for general relativity, let $AxSymTime^-$ denote the localized time-symmetry axiom, and let Lorentzian manifolds be the standard models...

conjecture | open | MathDB 页面 | 原文线索

V2-057245 | MATHDB-332301

The reduced-power maximality conjecture for SOP3

摘录 / EXCERPT: Reduced-power maximality conjecture. (T,Δ) is $\trianglelefteq_{\rm rp}$-maximal if and only if (T,Δ) has the ${\rm SOP}_3$.

conjecture | open | MathDB 页面 | 原文线索

V2-057246 | MATHDB-332302

Atomic saturation conjecture for stable theories without the finite cover property

摘录 / EXCERPT: Atomic saturation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057247 | MATHDB-332412

The definable (p,q)-theorem for distal theories

摘录 / EXCERPT: Let $M\prec N$ be NIP L-structures. Let $\varphi(x,y)$ be an L_M-formula and let $b\in N^{|y|}$. Assume $\varphi(x,b)$ does not divide over M. Chernikov-Simon conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-057248 | MATHDB-332773

The lower-bounds phase-transition conjecture

摘录 / EXCERPT: Lower-bounds phase-transition conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-057249 | MATHDB-332774

The lower-bounds sharpening conjecture

摘录 / EXCERPT: Lower-bounds sharpening conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-057250 | MATHDB-332852

Koponen's conjecture on primitive binary finitely homogeneous supersimple structures

摘录 / EXCERPT: A structure is primitive if no nontrivial equivalence relation on its domain is invariant under its automorphism group; it is binary if its language has relation symbols of arity a...

conjecture | open | MathDB 页面 | 原文线索

V2-057251 | MATHDB-333700

Aldous-Lyons-type characterization of finite-graph limits

摘录 / EXCERPT: Let $\mathbf L$ be a modeling. Assume: (i) the theory of $\mathbf L$ has the finite model property; (ii) every interpretation of $\mathbf L$ satisfies the finitary mass transport p...

conjecture | open | MathDB 页面 | 原文线索

V2-057252 | MATHDB-334121

Classification conjecture for ultraproducts of matrix algebras

摘录 / EXCERPT: Let $(n_i)_{i\to\theta}$ and $(m_j)_{j\to\theta'}$ be families of natural numbers, and let θ and θ' be ultrafilters. Write τ for the corresponding ultralimit of...

conjecture | open | MathDB 页面 | 原文线索

V2-057253 | MATHDB-334685

The partial general search conjecture for subset sum

摘录 / EXCERPT: Partial general search conjecture. Each instance S of the subset sum problem can be solved by a partial general search routine g_k working on a bounded domain in exponential ti...

conjecture | open | MathDB 页面 | 原文线索

V2-057254 | MATHDB-334867

Separation property for the calculus E

摘录 / EXCERPT: Separation property for $\mathbf{E}$. Any formula derivable in $\mathbf{E}$ is also derivable using only the axioms in group $(\mathbf{i})$ and those groups among $(\mathbf{c})$-...

conjecture | open | MathDB 页面 | 原文线索

V2-057255 | MATHDB-334868

Proper intermediate normal extension of E with admissible weakening

摘录 / EXCERPT: Intermediate-extension conjecture. There is a proper normal extension $\mathbf{L}$ of $\mathbf{E}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057256 | MATHDB-334923

Weak definable amenability conjecture for discrete definable groups

摘录 / EXCERPT: Weak definable amenability conjecture. If G is weakly definably amenable, then

conjecture | open | MathDB 页面 | 原文线索

V2-057257 | MATHDB-335327

Elementary-equivalence conjecture for component graphs of uniform spanning forests

摘录 / EXCERPT: Let $\mathbb G_1$ and $\mathbb G_2$ be d-dimensional transitive graphs, let $r_1,r_2\geq 1$, and let $\mathfrak F_1$ and $\mathfrak F_2$ be the uniform spanning forests of...

conjecture | open | MathDB 页面 | 原文线索

V2-057258 | MATHDB-335394

Khoussainov's conjectures on quotient presentations of nonstandard arithmetic models

摘录 / EXCERPT: Khoussainov's conjectures. (1) No nonstandard model of arithmetic admits a computable quotient presentation by a computably enumerable equivalence relation on the natural numbers....

conjecture | open | MathDB 页面 | 原文线索

V2-057259 | MATHDB-335459

Projectivity conjecture for absolute Galois groups of definably closed PAC substructures

摘录 / EXCERPT: Projectivity conjecture. The absolute Galois group of N is projective.

conjecture | open | MathDB 页面 | 原文线索

V2-057260 | MATHDB-335625

Top-computability conjecture for arrangements and free locally quasi-homogeneous divisors

摘录 / EXCERPT: Let f be the defining polynomial of either an arrangement ${\mathcal A}:f=0$ of d hyperplanes in $\mathbb{P}^n$, or a free locally quasi-homogeneous divisor $V:f=0$ of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-057261 | MATHDB-336018

Generalized quasiminimality conjecture for countably many elliptic exponentials

摘录 / EXCERPT: Let $\exp$ denote complex exponentiation, and let $\{E_i\}$ be any countable set of complex elliptic curves, with associated maps $\exp_{E_i}$. Generalized quasiminimality conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-057262 | MATHDB-336223

Conjecture on eventual indiscernibility of the indiscernibles of $M_{1,\infty}$

摘录 / EXCERPT: Eventual indiscernibility conjecture. Suppose $A\subseteq u_\omega$ is in $M_{1,\infty}$. Then there is $m<\omega$ such that either

conjecture | open | MathDB 页面 | 原文线索

V2-057263 | MATHDB-337016

The expansion conjecture for trivial Lascar group

摘录 / EXCERPT: Let M be an ω-categorical structure. Its Lascar group is the quotient of $\operatorname{Aut}(M)$ by the subgroup of Lascar strong automorphisms. Expansion conjecture. Any...

conjecture | open | MathDB 页面 | 原文线索

V2-057264 | MATHDB-337273

Conjecture on the impossibility of universally ultra-tight oracle-use bounds

摘录 / EXCERPT: Let X and Y be infinite binary sequences, and let an oracle computation of X by Y have an oracle-use function measuring how many bits of Y are queried to compute the firs...

conjecture | open | MathDB 页面 | 原文线索

V2-057265 | MATHDB-337371

Braun's Primitivity Conjecture for homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: Braun's Primitivity Conjecture. Every primitive homogeneous finite-dimensional permutation structure can be constructed by this procedure, yielding a fully generic structure, possi...

conjecture | open | MathDB 页面 | 原文线索

V2-057266 | MATHDB-337372

Genericity conjecture for homogeneous permutation structures realizing all 3-types

摘录 / EXCERPT: Genericity conjecture. Then 93 is fully generic.

conjecture | open | MathDB 页面 | 原文线索

V2-057267 | MATHDB-337990

Wilkie's local definability conjecture for holomorphic functions

摘录 / EXCERPT: Let $\mathcal{A}$ be a family of complex holomorphic functions. Let $z\in\mathbb{C}^{n}$, let U be an open neighbourhood of z, and let $f:U\rightarrow\mathbb{C}$ be holomorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-057268 | MATHDB-338949

Conjecture that the class of all mazes is not solvable

摘录 / EXCERPT: Let $\mathcal{M}$ denote the class of all mazes under consideration. All-maze nonsolvability conjecture. There is no algorithm that solves the class $\mathcal{M}$ of all mazes. The...

conjecture | open | MathDB 页面 | 原文线索

V2-057269 | MATHDB-338950

Fixed-distance nonsolvability conjecture for maze classes

摘录 / EXCERPT: For each k, let $\mathcal{M}_k\subseteq\mathcal{M}$ be the set of mazes whose destination is at distance k from the origin. Fixed-distance nonsolvability conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-057270 | MATHDB-338952

The $k=10$ fixed-distance maze nonsolvability conjecture

摘录 / EXCERPT: Let $\mathcal{M}_{10}\subseteq\mathcal{M}$ be the class of mazes whose destination is at distance 10 from the origin. The $k=10$ conjecture. Conjecture that there exists a k fo...

conjecture | open | MathDB 页面 | 原文线索

V2-057271 | MATHDB-339142

FO model checking on FO interpretations of nowhere dense classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a nowhere dense graph class, and let $\mathcal{D}$ be a graph class FO interpretable in $\mathcal{C}$. FO model-checking conjecture. The class $\mathcal{D}$ ha...

conjecture | open | MathDB 页面 | 原文线索

V2-057272 | MATHDB-339143

Nowhere FO dense conjecture for FO model checking

摘录 / EXCERPT: Let $\mathcal{D}$ be a graph class, and for every FO formula $\psi(x,y)$ let F_ψ be a graph such that F_ψ is not an induced subgraph of any member of...

conjecture | open | MathDB 页面 | 原文线索

V2-057273 | MATHDB-339202

The simple fields conjecture

摘录 / EXCERPT: Let K be an infinite (super)-simple field. The simple fields conjecture. K is a (perfect) PAC field. Duret showed that this conjecture implies the stable fields conjecture, but...

conjecture | open | MathDB 页面 | 原文线索

V2-057274 | MATHDB-339203

The dependent fields henselian valuation conjecture

摘录 / EXCERPT: Let K be an infinite dependent field that is neither real closed nor separably closed. The dependent fields henselian valuation conjecture. K admits a non-trivial henselian val...

conjecture | open | MathDB 页面 | 原文线索

V2-057275 | MATHDB-339204

The definable valuation conjecture for dependent fields

摘录 / EXCERPT: Let K be an infinite dependent field. The definable valuation conjecture. K is real closed, separably closed, or admits a definable non-trivial valuation. The source notes that...

conjecture | open | MathDB 页面 | 原文线索

V2-057276 | MATHDB-339205

The strongly dependent field valuation classification conjecture

摘录 / EXCERPT: Let K be a field. A valued field (K,v) is henselian if v is henselian, and defectless if every finite valued field extension satisfies $[L:K]=[vL:vK][Lv:Kv]$. The residue fie...

conjecture | open | MathDB 页面 | 原文线索

V2-057277 | MATHDB-339212

Classification conjecture for homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: Classification conjecture. Every homogeneous finite-dimensional permutation structure with lattice of $\emptyset$-definable equivalence relations isomorphic to Λ is interde...

conjecture | open | MathDB 页面 | 原文线索

V2-057278 | MATHDB-339213

Primitivity Conjecture for homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: Primitivity Conjecture. Every primitive homogeneous finite-dimensional permutation structure can be constructed as follows: identify certain orders, up to reversal; then take the F...

conjecture | open | MathDB 页面 | 原文线索

V2-057279 | MATHDB-339214

3-type realization conjecture for homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: 3-type realization conjecture. If Γ realizes all 3-types, then Γ is fully generic.

conjecture | open | MathDB 页面 | 原文线索

V2-057280 | MATHDB-339215

3-type extension conjecture for primitive homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: 3-type extension conjecture. All 3-types involving realized 2-types are realized in Γ.

conjecture | open | MathDB 页面 | 原文线索

V2-057281 | MATHDB-339216

Homogeneity of equivalence classes after naming another class

摘录 / EXCERPT: Homogeneity-after-naming conjecture. The class C_2 remains homogeneous after naming C_1.

conjecture | open | MathDB 页面 | 原文线索

V2-057282 | MATHDB-339217

Completeness conjecture for the construction of homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: Construction completeness conjecture. This construction produces all homogeneous finite-dimensional permutation structures.

conjecture | open | MathDB 页面 | 原文线索

V2-057283 | MATHDB-339248

C$_2$MS model-checking algorithm for graphs of bounded rank-width

摘录 / EXCERPT: Let G be a graph, let $\phi(\mathcal{X})$ be a C$_2$MS formula, and let α be an assignment to its free variables. Write $\operatorname{qr}(\phi)$ for the quantifier rank o...

conjecture | open | MathDB 页面 | 原文线索

V2-057284 | MATHDB-339640

Equality of rank-dimension and coarse dimension for quantifier-free types

摘录 / EXCERPT: Let $(F,\operatorname{Frob})\in\mathcal{S}$. For any countable set $A\subseteq F$ and complete quantifier-free $\mathcal{L}_\sigma$-type $p(x)\in S^{qf}_n(A)$, the two dimensions a...

conjecture | open | MathDB 页面 | 原文线索

V2-057285 | MATHDB-339876

Structural conjectures for finitely homogeneous NIP structures

摘录 / EXCERPT: Let M be a finitely homogeneous NIP structure. Structural conjecture. The following hold: 1. The automorphism group $\operatorname{Aut}(M)$ acts oligomorphically on the space of...

conjecture | open | MathDB 页面 | 原文线索

V2-057286 | MATHDB-340237

Nonuniform computability conjecture for special fan functionals and Pincherle realisers

摘录 / EXCERPT: Nonuniform computability conjecture. There is an M_{0}^{3} satisfying $\textup{\textsf{PR}}(M_{0})$ such that no Θ^{3} satisfying $\textup{\textsf{SCF}}(\Theta)$ is comput...

conjecture | open | MathDB 页面 | 原文线索

V2-057287 | MATHDB-340690

Failure of Boolean-ultrapower saturation invariance without distributivity

摘录 / EXCERPT: Boolean-ultrapower saturation conjecture. There exist such $\cappa$, $\cB$, U, $\cM$, and $\cN$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-057288 | MATHDB-340904

Equivalence of boundedness, dimensionality, and surjectivity of the restriction map

摘录 / EXCERPT: Equivalence conjecture. The following are equivalent: T is bounded; T is dimensional; and $\mathfrak e$ is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-057289 | MATHDB-341014

Finite monomorphic decomposition sibling conjecture

摘录 / EXCERPT: Let R be a relational structure with a finite monomorphic decomposition, and let siblings be counted up to isomorphism. Finite monomorphic decomposition conjecture. Under this as...

conjecture | open | MathDB 页面 | 原文线索

V2-057290 | MATHDB-341140

Equivalence of amalgamation, type amalgamation, coloring, and pattern conditions

摘录 / EXCERPT: Let T be a simple theory and let $3 \leq k \leq \aleph_0$. Consider the following conditions: (A) T has $\mathcal{P}^-(k)$-amalgamation of models; (B) T has $<k$-type amalgam...

conjecture | open | MathDB 页面 | 原文线索

V2-057291 | MATHDB-341141

Saturation characterizations for suitable ultrafilter sequences

摘录 / EXCERPT: Let T be a complete countable theory, let $\mathbf{s}$ be a suitable sequence, and let $\mathcal{U}_{\mathbf{s}}$ be the associated ultrafilter. Write $k_*(\mathbf{s})$ for the s...

conjecture | open | MathDB 页面 | 原文线索

V2-057292 | MATHDB-341421

The conjecture that unstable NIP theories interpret infinite linear orders

摘录 / EXCERPT: Unstable NIP conjecture. If T is an unstable NIP theory, then T interprets an infinite linear order.

conjecture | open | MathDB 页面 | 原文线索

V2-057293 | MATHDB-341704

Pillay-Yıldırım conjecture on dfg groups over the p-adics

摘录 / EXCERPT: Let G be a definable group over ${\mathbb Q}_p$ with dfg, meaning that G is definably amenable and has a global definable f-generic type. Then there exist a trigonalizable al...

conjecture | open | MathDB 页面 | 原文线索

V2-057294 | MATHDB-341870

The conjecture that simple fields are pseudo-algebraically closed

摘录 / EXCERPT: Simple-field conjecture. Every simple field is pseudo-algebraically closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057295 | MATHDB-341871

The simple-field conjecture via strongly IAC and VAC fields

摘录 / EXCERPT: Simple-field reduction conjecture. 1. K is strongly IAC; and 2. If K is VAC, then K is pseudo-algebraically closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057296 | MATHDB-342254

The ideality conjecture for M-infinitesimals in good Bezout domains

摘录 / EXCERPT: Let K be a field, let $R\subseteq K$ be a good Bezout domain, and let I_M denote the set of M-infinitesimals. The M-infinitesimals I_M are an ideal in R. This would pro...

conjecture | open | MathDB 页面 | 原文线索

V2-057297 | MATHDB-342622

Classification conjecture for monotone proof-theoretic operators

摘录 / EXCERPT: Classification conjecture for monotone proof-theoretic operators. Suppose $\mathfrak{g}$ is monotone, non-constant, and recursive, with $\mathfrak{g}(\varphi)\in\Pi_1$ for every se...

conjecture | open | MathDB 页面 | 原文线索

V2-057298 | MATHDB-342707

Finite burden for fields with finitely many valuation rings

摘录 / EXCERPT: Finite-burden conjecture. The structure $(K,\mathcal{O}_1,\ldots,\mathcal{O}_n)$ has finite burden.

conjecture | open | MathDB 页面 | 原文线索

V2-057299 | MATHDB-342708

Finite burden for finite dp-rank fields with finitely many valuation rings

摘录 / EXCERPT: Finite-dp-rank valuation conjecture. The structure $(K,\mathcal{O}_1,\ldots,\mathcal{O}_n)$ has finite burden.

conjecture | open | MathDB 页面 | 原文线索

V2-057300 | MATHDB-342709

Finite burden after adding a valuation ring to a field of finite burden

摘录 / EXCERPT: Valuation-ring finite-burden conjecture. The structure $(K,\mathcal{O})$ has finite burden.

conjecture | open | MathDB 页面 | 原文线索

V2-057301 | MATHDB-342845

Flexible Atom Conjecture for finite integral relation algebras

摘录 / EXCERPT: Let A be a finite integral relation algebra (RA), and suppose that A has a flexible atom. A relation algebra is representable over a finite cyclic group if it embeds into a Com...

conjecture | open | MathDB 页面 | 原文线索

V2-057302 | MATHDB-342846

Existence of a dfs measure that is not finitely approximated

摘录 / EXCERPT: Let T be a theory, let $\mathcal{U}$ be a monster model of T, and let $\mu\in\mathfrak{M}_x(\mathcal{U})$ be a Keisler measure. A measure is dfs if it is definable and finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-057303 | MATHDB-343390

The undecidability conjecture for the Sigma-1 theory of Young's lattice with all constants

摘录 / EXCERPT: Let $\mathcal P$ be the set of partitions, and consider Young's lattice with a constant symbol for every partition, $\left\langle \mathcal P,\leq,\pi:\pi\in\mathcal P\right\rangle$...

conjecture | open | MathDB 页面 | 原文线索

V2-057304 | MATHDB-343391

The undecidability and non-finite-axiomatizability conjecture for nontrivial differential posets

摘录 / EXCERPT: Let $\textbf P$ be a differential poset, meaning a graded locally finite poset with a unique least element whose up- and down-degree operators satisfy Stanley's differential-poset...

conjecture | open | MathDB 页面 | 原文线索

V2-057305 | MATHDB-343863

The conjecture that no finitely axiomatized consistent sequential theory is Enayat

摘录 / EXCERPT: Let U be a sequential theory. It is Enayat when it interprets itself together with the Tarski biconditionals for its language. Sequential Enayat-theory conjecture. There are no c...

conjecture | open | MathDB 页面 | 原文线索

V2-057306 | MATHDB-343864

The conjecture that no finitely axiomatized consistent Vaught theory is Enayat

摘录 / EXCERPT: Let U be a Vaught theory, meaning a theory with the relevant direct interpretation properties described in the paper. Call U Enayat when it interprets itself together with the...

conjecture | open | MathDB 页面 | 原文线索

V2-057307 | MATHDB-343865

The conjecture on extensions of ${\sf S}^1_2$ mutually interpretable with Tarski extensions

摘录 / EXCERPT: Let A be finitely axiomatized, consistent, and sequential, and let $N:{\sf S}^1_2\lhd A$. Write $A+{\sf TB}^{-}_N$ for the corresponding Tarski-biconditional extension. Mutual-in...

conjecture | open | MathDB 页面 | 原文线索

V2-057308 | MATHDB-343866

The conjecture that Tarski biconditionals imply uniform biconditionals under Fujimoto interpretability

摘录 / EXCERPT: Let U be a Vaught theory and let $N:{\sf R}\lhd U$. For theories or axiom sets α and Γ, write $\alpha\mathrel{\text{\textcolor{gray}{$ blacktriangleright...

conjecture | open | MathDB 页面 | 原文线索

V2-057309 | MATHDB-343867

The conjecture on Fujimoto-interpretable finite approximations to Tarski biconditionals

摘录 / EXCERPT: Let A be a finitely axiomatized Vaught theory in signature Θ_0, let $N:A\rhd{\sf R}$, and suppose $\top\rhd_A{\sf TB}^{-}_N$. Finite-approximation conjecture. There is a...

conjecture | open | MathDB 页面 | 原文线索

V2-057310 | MATHDB-344090

Conjecture that subexponential exponential growth is equivalent to monadic NIP

摘录 / EXCERPT: Monadic NIP growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057311 | MATHDB-344122

The multi-valuation ideal conjecture for unstable dp-finite fields

摘录 / EXCERPT: Multi-valuation ideal conjecture. There is a small submodel $K \preceq \mathbb{K}$ for which the infinitesimals I_K contain a nonzero multi-valuation ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-057312 | MATHDB-344123

Henselianity conjecture for NIP valued fields

摘录 / EXCERPT: Henselianity conjecture. The valuation ring $\mathcal{O}$ is henselian.

conjecture | open | MathDB 页面 | 原文线索

V2-057313 | MATHDB-344124

Valuation-type conjecture for dp-finite fields

摘录 / EXCERPT: Valuation-type conjecture. If $\mathbb{K}$ is a dp-finite field, then $\mathbb{K}$ is stable or has valuation type.

conjecture | open | MathDB 页面 | 原文线索

V2-057314 | MATHDB-344125

Stabilizer-ring conjecture for saturated unstable dp-finite fields

摘录 / EXCERPT: Stabilizer-ring conjecture. There is a non-zero bounded subgroup $J \subseteq (\mathbb{K},+)$ for which R_J is a multi-valuation ring on $\mathbb{K}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057315 | MATHDB-344237

Noiselessness of the open core without definable integers

摘录 / EXCERPT: Let ${\mathscr{R}}$ be an expansion of $(\mathbb{R},<,+,\times)$, and let ${\mathscr{R}}^\circ$ be its open core. The structure ${\mathscr{R}}$ is type A when it has no dense defin...

conjecture | open | MathDB 页面 | 原文线索

V2-057316 | MATHDB-344238

Generic local o-minimality of the open core without monadic second-order arithmetic

摘录 / EXCERPT: Let ${\mathscr{R}}$ be an expansion of $(\mathbb{R},<,+)$, and let ${\mathscr{R}}^\circ$ denote its open core. The structure $({\mathcal P}(\mathbb{N}),\mathbb{N},\in,s)$ is the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-057317 | MATHDB-344239

The field-type conjecture for NIP expansions by closed sets

摘录 / EXCERPT: Let ${\mathscr{R}}$ be an NIP expansion of $(\mathbb{R},<,+)$ by closed subsets of Euclidean space. Call ${\mathscr{R}}$ field-type when it satisfies condition (1): there is a none...

conjecture | open | MathDB 页面 | 原文线索

V2-057318 | MATHDB-344240

No infinite zero-dimensional fields in NIP expansions by closed sets

摘录 / EXCERPT: Let ${\mathscr{R}}$ be an NIP expansion of $(\mathbb{R},<,+)$ by closed subsets of Euclidean space. A definable set has dimension zero when its dimension in the paper's dimension t...

conjecture | open | MathDB 页面 | 原文线索

V2-057319 | MATHDB-344241

The ordered-field IP conjecture on the natural numbers

摘录 / EXCERPT: Let $\oplus$ and $\otimes$ be binary operations on $\mathbb{N}$ such that $(\mathbb{N},\oplus,\otimes)$ is a field. Ordered-field IP conjecture. The expansion...

conjecture | open | MathDB 页面 | 原文线索

V2-057320 | MATHDB-344242

The directed-family IP conjecture for expansions of infinite fields

摘录 / EXCERPT: Let ${\mathscr{F}}$ be an expansion of an infinite field F. A family ${\mathcal X}$ of subsets of F is directed when any two members have a common member containing their union...

conjecture | open | MathDB 页面 | 原文线索

V2-057321 | MATHDB-344243

NIP preservation for the completion and original structure

摘录 / EXCERPT: Let $(R,<,+)$ be a dense archimedean ordered abelian group, let ${\mathscr{R}}$ be an expansion of it, and let ${\mathscr{R}}^{\square}$ be the completion structure used in the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-057322 | MATHDB-344244

Non-field-type completions do not interpret infinite fields

摘录 / EXCERPT: Let $(R,<,+)$ be a dense archimedean ordered abelian group, let ${\mathscr{R}}$ be a strongly dependent and noiseless expansion, and let ${\mathscr{N}}$ be a highly saturated eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-057323 | MATHDB-344259

Conjecture on the number of bi-embeddability classes of models of an age

摘录 / EXCERPT: Let $\mathfrak A$ be an age in a finite relational language, and let $\operatorname{Mod}(\mathfrak A)/{\equiv}$ denote the bi-embeddability classes of countable structures with age...

conjecture | open | MathDB 页面 | 原文线索

V2-057324 | MATHDB-344399

Pillay-Yao conjecture on almost periodicity of global f-generic types

摘录 / EXCERPT: Let G be a group definable in a NIP structure. Say that G is dfg if it has a definable global f-generic type, and call a global f-generic type almost periodic when its orbit un...

conjecture | open | MathDB 页面 | 原文线索

V2-057325 | MATHDB-344466

The valuation conjecture for dp-finite fields

摘录 / EXCERPT: Valuation conjecture. The group I_K is an ideal in a valuation ring on $\mathbb{K}$; equivalently, $\operatorname{Stab}(I_K)$ is a valuation ring.

conjecture | open | MathDB 页面 | 原文线索

V2-057326 | MATHDB-344467

The axiomatization conjecture for extended directories

摘录 / EXCERPT: Axiomatization conjecture for extended directories. Then D is an extended directory: there is an object A in a K_0-linear abelian category such that

conjecture | open | MathDB 页面 | 原文线索

V2-057327 | MATHDB-344712

Undecidability of doubly periodic Gardens of Eden in the Game of Life

摘录 / EXCERPT: A doubly periodic configuration in the Game of Life is a configuration with two independent spatial periods, and a Garden of Eden is a configuration with no preimage. Undecidabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-057328 | MATHDB-344752

The conjecture that there are no strictly n-dependent fields

摘录 / EXCERPT: No strictly n-dependent fields conjecture. There are no strictly n-dependent fields for $n\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-057329 | MATHDB-345886

Finite homomorphism basis conjecture for first-order equivalence

摘录 / EXCERPT: Let $k\ge 1$. Fix a set Γ of colours and consider graphs G whose colour interpretations satisfy $\operatorname{rg}(\gamma^G)\subseteq\Gamma$. For graphs G,G' and a finit...

conjecture | open | MathDB 页面 | 原文线索

V2-057330 | MATHDB-345904

Valuation conjecture for unstable dp-finite fields

摘录 / EXCERPT: Valuation conjecture. If K is an unstable dp-finite field, then K is valuation type.

conjecture | open | MathDB 页面 | 原文线索

V2-057331 | MATHDB-345924

The double commutant conjecture for w-spectral gap subfactors

摘录 / EXCERPT: Let M be an embeddable II$_1$ factor that is existentially closed among embeddable factors, and let $N\subseteq M$ be a w-spectral gap subfactor. Double commutant conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-057332 | MATHDB-346100

Continuum many separable non-embeddable II$_1$ factors with the generalized Jung property

摘录 / EXCERPT: Generalized Jung factor abundance conjecture. There are continuum many non-isomorphic separable non-embeddable II$_1$ factors with the generalized Jung property.

conjecture | open | MathDB 页面 | 原文线索

V2-057333 | MATHDB-346101

Conjecture on the complexity of theories of infinitely generic operator algebras

摘录 / EXCERPT: Infinitely generic theories conjecture. Second-order arithmetic Turing reduces to both theories

conjecture | open | MathDB 页面 | 原文线索

V2-057334 | MATHDB-346102

Conjecture on arithmetic complexity of theories of existentially closed II$_1$ factors

摘录 / EXCERPT: Arithmetic-complexity conjecture. First-order arithmetic Turing reduces to the theory of any (embeddable) e.c. II$_1$ factor.

conjecture | open | MathDB 页面 | 原文线索

V2-057335 | MATHDB-346174

The generic smooth expansion conjecture for semialgebraic Mordell-Lang circle groups

摘录 / EXCERPT: Generic smooth expansion conjecture. There is a comeager subset $\Lambda\subseteq C^\infty(I)$, possibly depending on α, such that for every $f\in\Lambda$: 1....

conjecture | open | MathDB 页面 | 原文线索

V2-057336 | MATHDB-346175

The p-adic completion conjecture for dp-minimal expansions

摘录 / EXCERPT: p-adic completion conjecture. The structure induced on $\mathbb{Z}$ by ${\mathscr{Z}}^{\square}$ is interdefinable with ${\mathscr{Z}}^{\mathrm{Sh}}$, and every...

conjecture | open | MathDB 页面 | 原文线索

V2-057337 | MATHDB-346176

The finite-rank subgroup conjecture for p-adic elliptic curves

摘录 / EXCERPT: Finite-rank subgroup conjecture. The expansion $(\mathbb{Q}_p,+,\times,A)$ is NIP, $\operatorname{Th}(\mathbb{Q}_p,+,\times)$ is an open core of...

conjecture | open | MathDB 页面 | 原文线索

V2-057338 | MATHDB-346575

Existential closedness conjecture for the modular j-function

摘录 / EXCERPT: Let $\operatorname{\mathbb{H}}$ be the complex upper half-plane, let $j:\operatorname{\mathbb{H}}\to\operatorname{\mathbb{C}}$ be the modular j-function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057339 | MATHDB-346576

Existential closedness conjecture for the modular j-function and its derivatives

摘录 / EXCERPT: Let $J:\operatorname{\mathbb{H}}\to\operatorname{\mathbb{C}}^3$ be defined by ... and extend it coordinatewise to $\operatorname{\mathbb{H}}^n$, with graph...

conjecture | open | MathDB 页面 | 原文线索

V2-057340 | MATHDB-346701

The definable-isomorphism conjecture for infinite fields interpretable in p-adic fields

摘录 / EXCERPT: Definable-isomorphism conjecture. Any infinite field interpretable in K is K-definably isomorphic to some finite extension of K.

conjecture | open | MathDB 页面 | 原文线索

V2-057341 | MATHDB-347319

The conjecture that the theory of the hyperfinite II$_1$ factor does not admit quantifier simplification

摘录 / EXCERPT: Let $\mathcal R$ denote the hyperfinite II$_1$ factor, and let $\operatorname{Th}(\mathcal R)$ denote its theory. We say that $\operatorname{Th}(\mathcal R)$ admits quantifier simp...

conjecture | open | MathDB 页面 | 原文线索

V2-057342 | MATHDB-347511

Bi-interpretability characterization for irreducible lattices

摘录 / EXCERPT: Bi-interpretability characterization. The group Δ is bi-interpretable with the ring of integers if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-057343 | MATHDB-347651

The characterization of minimal distal valuation expansions of the integers

摘录 / EXCERPT: Characterization conjecture. Let $(\mathbb{Z},+,0,1,v)$ be distal. Then the following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057344 | MATHDB-347679

The distal expansion conjecture for ordered abelian groups

摘录 / EXCERPT: Let G be an ordered abelian group. Distal expansion conjecture. Every ordered abelian group admits a distal expansion. This proposes that distality can always be obtained by expa...

conjecture | open | MathDB 页面 | 原文线索

V2-057345 | MATHDB-347680

The henselian residue-field conjecture for distal fields

摘录 / EXCERPT: Let K be a field. A structure is distal if it has the model-theoretic distality property. Henselian residue-field conjecture. Suppose K has a distal expansion; then K has a h...

conjecture | open | MathDB 页面 | 原文线索

V2-057346 | MATHDB-347681

The interpretable-fields conjecture for logarithmic-exponential transseries

摘录 / EXCERPT: Let $\mathbb T$ be the differential field of logarithmic-exponential transseries, and let $\mathbb R$ be the real field. Write $\mathbb T[\imag]$ and $\mathbb C=\mathbb R[\imag]$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-057347 | MATHDB-347963

Weaker form of Shelah's categoricity conjecture for accessible categories

摘录 / EXCERPT: Let $\mathcal{A}$ be a nice accessible category with directed colimits having just one model of presentability rank κ. Weaker conjecture. Then $\mathcal{A}$ must be categori...

conjecture | open | MathDB 页面 | 原文线索

V2-057348 | MATHDB-348109

Equivalence of almost canonical covers and simplicial finite-cover groupoids

摘录 / EXCERPT: Equivalence conjecture. The categories AC_A and $SFCG_A$ are equivalent. Moreover, this equivalence can be proved by extending the functor $C: Iso(SFCG_A) \rightarrow Iso(AC_A)$...

conjecture | open | MathDB 页面 | 原文线索

V2-057349 | MATHDB-348195

Definable Tietze extension conjecture for DCULOAS structures

摘录 / EXCERPT: Let M be a definably complete uniformly locally o-minimal structure of the second kind, abbreviated DCULOAS, and suppose that M enjoys the definable Tietze extension property....

conjecture | open | MathDB 页面 | 原文线索

V2-057350 | MATHDB-348198

The pathwidth-treewidth incomparability conjecture

摘录 / EXCERPT: For each positive integer n, let $_n$ be the class of graphs of pathwidth at most n, and let $_n$ be the class of graphs of treewidth at most n. Two graph classes are non-com...

conjecture | open | MathDB 页面 | 原文线索

V2-057351 | MATHDB-348428

Shelah-Hasson conjecture for strongly NIP ordered fields

摘录 / EXCERPT: Let F be a strongly NIP ordered field. Such a field is almost real closed if it admits a henselian valuation whose residue field is real closed. Shelah-Hasson conjecture for orde...

conjecture | open | MathDB 页面 | 原文线索

V2-057352 | MATHDB-348443

Dilatorial characterization conjecture for beta-models of strong systems

摘录 / EXCERPT: Let T be a strong system with an ordinal analysis based on ordinal representation systems OT. These systems give rise to functors ... that send dilators to ordinal repre...

conjecture | open | MathDB 页面 | 原文线索

V2-057353 | MATHDB-348489

Classification conjecture for strongly NIP ordered fields

摘录 / EXCERPT: Classification conjecture. Any strongly NIP ordered field $(K,<)$ is almost real closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057354 | MATHDB-348733

The stable-field and simple-field conjectures

摘录 / EXCERPT: Stable-field and simple-field conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-057355 | MATHDB-349474

Hyperarithmetical pseudojump inversion conjecture

摘录 / EXCERPT: Let $e\in\mathbb{N}$ and let A be a real such that ... For a real B, define the e-th pseudo-hyperjump by $\operatorname{HJ}_e(B)=B\oplus V_e^B$, where V_e^B is the e-th mem...

conjecture | open | MathDB 页面 | 原文线索

V2-057356 | MATHDB-349561

The ordered-abelian-group conjecture on trace-defining fields

摘录 / EXCERPT: Ordered-abelian-group conjecture. An ordered abelian group cannot trace define an infinite field.

conjecture | open | MathDB 页面 | 原文线索

V2-057357 | MATHDB-349562

The countability conjecture for finitely homogeneous structures

摘录 / EXCERPT: Countability conjecture. There are $\aleph_0$ finitely homogeneous structures up to trace equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-057358 | MATHDB-349649

Fixed-parameter tractability conjecture for first-order model checking on hereditary classes

摘录 / EXCERPT: Let $\mathscr C$ be a hereditary class of structures. A class is monadically NIP if it remains NIP after arbitrary unary predicates are added. Hereditary first-order model-checking...

conjecture | open | MathDB 页面 | 原文线索

V2-057359 | MATHDB-350068

Pillay's conjecture on almost quantifier elimination and largeness

摘录 / EXCERPT: Let K be a field. Say that K has almost quantifier elimination if every formula $\phi(x)$, with $x=(x_1,\ldots,x_m)$, is equivalent to a formula $\exists y\,\theta(x,y)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-057360 | MATHDB-350431

Conjecture on exponent-one distal cell decompositions for locally modular geometric structures

摘录 / EXCERPT: Exponent-one locally modularity conjecture. Every distal locally modular geometric structure admits distal cell decompositions of exponent 1.

conjecture | open | MathDB 页面 | 原文线索

V2-057361 | MATHDB-350882

Computability of the essential spectrum of differential forms

摘录 / EXCERPT: Let (M^n,g) be a complete noncompact Riemannian manifold with asymptotically nonnegative Ricci curvature. For $0\leq k\leq n$, write $\sigma_{\rm ess}(k,\Delta,M)$ for the essent...

conjecture | open | MathDB 页面 | 原文线索

V2-057362 | MATHDB-350926

Elementary classification by cores for graph products in the specified classes

摘录 / EXCERPT: Let $\mathcal G(\Gamma,\{G_v\}_{v\in\Gamma})$ and $\mathcal G(\Gamma',\{G'_v\}_{v\in\Gamma'})$ belong to the specified class $\mathcal G(\mathbf F_N,\mathbf C^r_\Phi)$ with $N>0$....

conjecture | open | MathDB 页面 | 原文线索

V2-057363 | MATHDB-350959

The monotone-operator classification conjecture

摘录 / EXCERPT: Monotone-operator classification conjecture. For some $\beta\preceq\alpha$ and some true sentence φ, for every ψ such that $T+\psi\vdash\varphi$,

conjecture | open | MathDB 页面 | 原文线索

V2-057364 | MATHDB-350960

The sound reflection hierarchy classification conjecture

摘录 / EXCERPT: Sound reflection hierarchy classification conjecture. For every $n\in\mathbb{N}$, both of the following hold: (1) there exists a $0^{(n)}$-recursive sequence $(T_n)_{n<\omega}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-057365 | MATHDB-351153

Finite-vocabulary obstruction to elimination of imaginaries in Hrushovski constructions

摘录 / EXCERPT: Finite-vocabulary conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-057366 | MATHDB-351762

The pure-field NIPn conjecture

摘录 / EXCERPT: Pure-field NIP$_n$ conjecture. For every $n\geqslant 2$, no strictly NIP$_n$ pure fields exist; equivalently, a pure field is NIP$_n$ if and only if it is NIP.

conjecture | open | MathDB 页面 | 原文线索

V2-057367 | MATHDB-352165

Gap-preserving compression for commuting-operator strategies

摘录 / EXCERPT: Let $\mathscr{G}=(G_n)_{n\in\mathbb{N}}$ be a sequence of nonlocal games, and let the complexity of a sequence refer to the complexity of its nth game. Gap-preserving compression...

conjecture | open | MathDB 页面 | 原文线索

V2-057368 | MATHDB-352212

Existence of a c.e. set intermediate for Sigma-Feiner and low for Delta-Feiner

摘录 / EXCERPT: C.e. Sigma-intermediate/Delta-low conjecture. There is a computably enumerable set that is intermediate for Σ-Feiner and low for Δ-Feiner.

conjecture | open | MathDB 页面 | 原文线索

V2-057369 | MATHDB-352476

Non-semi-decidability of isolated groups from finite presentations

摘录 / EXCERPT: Finite-presentation isolated-groups conjecture. The set of isolated groups is not Λ_{FP}-semi-decidable.

conjecture | open | MathDB 页面 | 原文线索

V2-057370 | MATHDB-352478

Finite representation property conjecture for relation algebra reduct signatures

摘录 / EXCERPT: Finite representation property conjecture. The signature τ has the finite representation property if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-057371 | MATHDB-352565

The existence conjecture for data-compressing effective ad hoc descriptions

摘录 / EXCERPT: A number is compressed by a non-constructive unique effective description when the validity of the description can be checked effectively given the number, but the number cannot be...

conjecture | open | MathDB 页面 | 原文线索

V2-057372 | MATHDB-353238

The bigstar classification-theoretic property for fusions

摘录 / EXCERPT: Bigstar property conjecture. There is a classification-theoretic property $\bigstar$ such that: (1) there is a property P of set systems for which T has $\bigstar$ if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-057373 | MATHDB-353564

Approximate categoricity of beautiful pairs of randomized theories

摘录 / EXCERPT: Let Q be an $\aleph_0$-categorical, $\aleph_0$-stable theory in a countable language, and let Q_P denote the theory of beautiful pairs of models of Q. Approximate categoricit...

conjecture | open | MathDB 页面 | 原文线索

V2-057374 | MATHDB-353565

Approximate categoricity for randomized theories with disintegrated geometry

摘录 / EXCERPT: Let $\mathcal{T}$ be the class of $\aleph_0$-categorical, $\aleph_0$-stable classical theories in countable languages, and let $\mathcal{D}\subseteq\mathcal{T}$ consist of those th...

conjecture | open | MathDB 页面 | 原文线索

V2-057375 | MATHDB-353656

Conjecture on CB ranks of Ziegler spectra of Prüfer domains

摘录 / EXCERPT: Let R be a Prüfer domain and let λ be a limit ordinal. The Ziegler-spectrum CB-rank conjecture. There does not exist a Prüfer domain whose Ziegler spectrum has Cantor-Ben...

conjecture | open | MathDB 页面 | 原文线索

V2-057376 | MATHDB-353953

Shelah's Hanf-number threshold conjecture for categoricity transfer in AECs

摘录 / EXCERPT: Let ${\bf K}$ be an abstract elementary class and let $\lambda=\operatorname{LS}({\bf K})$, its Löwenheim-Skolem number. Shelah's Hanf-number threshold conjecture. The threshold fo...

conjecture | open | MathDB 页面 | 原文线索

V2-057377 | MATHDB-354184

Conjecture on quasi Frobenius groups in o-minimal structures

摘录 / EXCERPT: Let $C<G$ be a connected quasi Frobenius group definable in an o-minimal structure, with C definable and connected. Assume that G contains involutions, and that ... contains all...

conjecture | open | MathDB 页面 | 原文线索

V2-057378 | MATHDB-354185

Burdges-Nesin conjecture on splitting ranked Frobenius groups

摘录 / EXCERPT: Let $C<G$ be a ranked Frobenius group, where a Frobenius group has a proper subgroup C that is its Frobenius complement. Burdges-Nesin conjecture. The group G is split: there i...

conjecture | open | MathDB 页面 | 原文线索

V2-057379 | MATHDB-354448

The dfg conjecture for zero-dimensional interpretable groups over the p-adadics

摘录 / EXCERPT: The dfg conjecture. If G is a $\mathbb{Q}_p$-interpretable zero-dimensional, definably amenable group, then G has dfg.

conjecture | open | MathDB 页面 | 原文线索

V2-057380 | MATHDB-354449

The degree-spectrum characterization of successor recoverability for computable block functions

摘录 / EXCERPT: Degree-spectrum conjecture. If the degree spectrum of f on $(\omega,<)$ is equal to all c.e. degrees, then the successor is recoverable from f on $(\omega,<)$.

conjecture | open | MathDB 页面 | 原文线索

V2-057381 | MATHDB-354654

Proof-system domination by conservative extensions conjecture

摘录 / EXCERPT: Let Q be a propositional proof system. A theory $\mathcal{T}$ is a conservative extension of a base theory when it extends that theory without proving any new sentences in the ba...

conjecture | open | MathDB 页面 | 原文线索

V2-057382 | MATHDB-354783

Definable (p,q)-conjecture for formulas of VC-codensity less than q

摘录 / EXCERPT: Let $q\geq 2$ be an integer, M be an L-structure, and let $\varphi(x,y)$ be an $L(M)$-formula with dual shatter function $\pi_{\varphi}^*(n)\in o(n^q)$. A family of instances...

conjecture | open | MathDB 页面 | 原文线索

V2-057383 | MATHDB-355182

Finiteness conjecture for chains of definable subgroups of abelian interpretable groups

摘录 / EXCERPT: Let X be an abelian interpretable group. By Corollary, there is an increasing chain of textit{dfg} subgroups $(Y_i:i<\alpha)$ with $Y_0=0$, whose successive quotients have dp-ran...

conjecture | open | MathDB 页面 | 原文线索

V2-057384 | MATHDB-355950

Restricted Trichotomy Conjecture in concrete form

摘录 / EXCERPT: Let K be an algebraically closed field, and let $\mathcal M=(M,...)$ be a non-locally modular strongly minimal structure. Assume that the universe M of $\mathcal M$, as well as...

conjecture | open | MathDB 页面 | 原文线索

V2-057385 | MATHDB-356141

Weakly generic and almost periodic types in definably amenable groups

摘录 / EXCERPT: Let G be a definable, definably amenable group defined over either an o-minimal structure or a p-adically closed field. Say that G is stationary when it has the stationarit...

conjecture | open | MathDB 页面 | 原文线索

V2-057386 | MATHDB-356142

Badness and non-stationarity for dfg groups

摘录 / EXCERPT: Let H be a group definable in an o-minimal structure or a p-adically closed field, and assume that H is a dfg group. Say that H is bad when it has the badness...

conjecture | open | MathDB 页面 | 原文线索

V2-057387 | MATHDB-356197

The monadic stability characterization conjecture for hereditary graph classes

摘录 / EXCERPT: Monadic stability characterization conjecture. A class of graphs is monadically stable if and only if it is a first-order transduction of a nowhere dense class of graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057388 | MATHDB-356198

The stable hereditary graph class characterization conjecture

摘录 / EXCERPT: Stable hereditary graph class characterization conjecture. The following properties are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057389 | MATHDB-356436

Pseudo-Siggers dichotomy conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Let $\mathbf{A}$ be the core of a reduct of a finitely bounded homogeneous structure. A pseudo-Siggers identity is an identity of the form ... where s is a 6-ary polymorphism and...

conjecture | open | MathDB 页面 | 原文线索

V2-057390 | MATHDB-356643

Popa's Factorial Commutant Embedding Problem

摘录 / EXCERPT: Let N be a separable II$_1$ factor that embeds into $\mathcal R^\mathcal U$. Suppose further that N has property (T). Popa's Factorial Commutant Embedding Problem. Then there i...

open_problem | open | MathDB 页面 | 原文线索

V2-057391 | MATHDB-356692

The one-dimensional definable strongly minimal non-locally modular field interpretation claim

摘录 / EXCERPT: Let $\mathbb K$ and $\mathcal K$ be the ambient field and structure, and let N be a $\mathbb K$-definable set of dimension one. Let $\mathcal N=(N,\ldots)$ be a strongly minimal,...

conjecture | open | MathDB 页面 | 原文线索

V2-057392 | MATHDB-357715

The conjecture that every model of curve-excluding fields has SOP$_3$

摘录 / EXCERPT: Let $C\mathrm{XF}$ be the theory of fields excluding the curve C over the base field K_0, and let M be a model of $C\mathrm{XF}$. The property SOP_3 is the strict...

conjecture | open | MathDB 页面 | 原文线索

V2-057393 | MATHDB-357928

The nonnegative formal-sum model satisfies induction with inequality

摘录 / EXCERPT: Induction-with-inequality hypothesis. The introduced structure $d4c^+$ is a model of $d400(\ne)$.

conjecture | open | MathDB 页面 | 原文线索

V2-057394 | MATHDB-358073

The square-sequence conjecture for $L(\mathcal{A}^\infty)$

摘录 / EXCERPT: Let $\Theta^\infty=\Theta^{L(\Gamma^\infty,\mathbb{R})}$ and let $L(\mathcal{A}^\infty)$ be the model generated by $\mathcal{A}^\infty$. Square-sequence conjecture. ... The claim is...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-057395 | MATHDB-358089

Preservation of elementary equivalence under tensoring with type I C*-algebras

摘录 / EXCERPT: Tensor-product preservation conjecture. If B and C are elementarily equivalent C^*-algebras, then

conjecture | open | MathDB 页面 | 原文线索

V2-057396 | MATHDB-358126

Basis conjecture for dual vector spaces under definable Stone-Čech compactification

摘录 / EXCERPT: Let $\mathcal{A}$ be an ω-categorical structure, let X range over definable sets, and let $\mathbf{K}^X$ denote the associated dual vector space. Let $\overline{X}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-057397 | MATHDB-358156

The quantifier-elimination characterization for theories of tracial von Neumann algebras

摘录 / EXCERPT: Quantifier-elimination conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057398 | MATHDB-358157

The model-completeness characterization for theories of tracial von Neumann algebras

摘录 / EXCERPT: Model-completeness conjecture. The theory T is model complete if and only if N is of type I.

conjecture | open | MathDB 页面 | 原文线索

V2-057399 | MATHDB-358246

The A_1 conjecture for dihedral quasi-Frobenius configurations

摘录 / EXCERPT: Let $(C < G)$ be a definable, connected, quasi-Frobenius pair of finite Morley rank. Suppose that G is $U_2^\perp$ with involutions, and that the configuration is dihedral. The...

conjecture | open | MathDB 页面 | 原文线索

V2-057400 | MATHDB-358250

Monadic characterization of the rational order

摘录 / EXCERPT: Rational-order conjecture. There is a monadic sentence ψ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057401 | MATHDB-358251

Non-isomorphic monadically equivalent order to the rational order

摘录 / EXCERPT: Non-isomorphic equivalent-order conjecture. There is an order M such that M and R have the same monadic theory, but M is not isomorphic to R.

conjecture | open | MathDB 页面 | 原文线索

V2-057402 | MATHDB-358283

Non-existence of unicorn theories

摘录 / EXCERPT: Non-existence conjecture. There are no unicorn theories.

conjecture | open | MathDB 页面 | 原文线索

V2-057403 | MATHDB-358493

Existence of aperiodic triangulated disk packings

摘录 / EXCERPT: Aperiodic disk-packing conjecture. There exists a finite set of disk sizes that admits a triangulated packing other than the hexagonal compact packing, while no such packing is per...

conjecture | open | MathDB 页面 | 原文线索

V2-057404 | MATHDB-358725

Zilber's isomorphism conjecture for the complex exponential field

摘录 / EXCERPT: Let $\mathbb{C}_{\mathrm{exp}}$ denote the complex exponential field and let $\mathbb{B}_{\mathrm{exp}}$ denote Zilber's quasiminimal excellent exponential field constructed by the...

conjecture | open | MathDB 页面 | 原文线索

V2-057405 | MATHDB-358977

The L[vec E]-capturing conjecture

摘录 / EXCERPT: Let $ obreak\textsf{AD}^+$ denote the strengthened axiom of determinacy, and let $ obreak\textsf{NLE}$ be the no-long-extender assertion. Let $ obreak\textsf{LEC}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-057406 | MATHDB-358979

The generic-generator non-badness conjecture

摘录 / EXCERPT: Let $ obreak\textsf{AD}^+$ denote the strengthened axiom of determinacy. For a function F as in the paper's Theorem nx, call F pe-bad when, for a cone of reals x, the supremu...

conjecture | open | MathDB 页面 | 原文线索

V2-057407 | MATHDB-358980

The DLIH and BDL conjecture for generic generators

摘录 / EXCERPT: Let $ obreak\textsf{AD}_{\mathbb R}$ denote determinacy for real games. Suppose that for every set $B\subseteq\mathbb R$ there is a function F as in Theorem nx such that F is n...

conjecture | open | MathDB 页面 | 原文线索

V2-057408 | MATHDB-358981

The common-iterate conjecture for generic generators

摘录 / EXCERPT: Let V satisfy $ obreak\textsf{AD}_{\mathbb R}+ obreak\textsf{NLE}+V=L({\wp}(\mathbb R))$. Let $ obreak\Delta\subseteq{\wp}(\mathbb R)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057409 | MATHDB-359103

Generic stability conjecture for definable groups over valuation rings

摘录 / EXCERPT: Let ${\mathcal O}_\mathbb{K}$ be the valuation ring of a field with residue map res, and let $G\subseteq {\mathcal O}_\mathbb{K}^n$ be a $\emptyset$-definable grou...

conjecture | open | MathDB 页面 | 原文线索

V2-057410 | MATHDB-359196

Weaker Rudominer-Steel conjecture with an existing omega-one

摘录 / EXCERPT: Assume $\mathrm{ZF}+\mathsf{AD}^{L(\mathbb R)}$. Let M be a $(0,\omega_1)$-iterable countable ω-small premouse satisfying “ω_1 exists”. Then there are γ,...

conjecture | open | MathDB 页面 | 原文线索

V2-057411 | MATHDB-359240

Geometric Borovik-Cherlin conjecture for differential fields

摘录 / EXCERPT: Let $\Delta=\{\delta_1,\dots,\delta_m\}$ be a set of m commuting derivations in characteristic zero, and let G be a strongly connected differential-algebraic group of Δ-...

conjecture | open | MathDB 页面 | 原文线索

V2-057412 | MATHDB-359729

Zilber's trichotomy principle relative to a theory

摘录 / EXCERPT: Zilber's trichotomy principle relative to T. If $\mathcal P$ is strongly minimal and non-locally modular, then there is an infinite field interpretable in $\mathcal P$.

conjecture | open | MathDB 页面 | 原文线索

V2-057413 | MATHDB-359951

Feder-Vardi tractability conjecture for infinite-domain CSPs

摘录 / EXCERPT: Feder-Vardi tractability conjecture. If ${\mathfrak B}$ does not interpret primitively positively a graph which is homomorphically equivalent to K_3, then

conjecture | open | MathDB 页面 | 原文线索

V2-057414 | MATHDB-360040

The large-characteristic conjecture for simple Lie rings of finite Morley rank

摘录 / EXCERPT: Let $\mathfrak{g}$ be an infinite simple Lie ring of finite Morley rank. Assume that its characteristic is sufficiently large. Large-characteristic conjecture. Then $\mathfrak{g}$...

conjecture | open | MathDB 页面 | 原文线索

V2-057415 | MATHDB-360123

Fractional-polymorphism classification conjecture for homogeneous reducts

摘录 / EXCERPT: Fractional-polymorphism classification conjecture. If $(\{0,1\};\operatorname{OIT})$ has no pp-construction in Γ, then $\operatorname{VCSP}(\Gamma)$ is in P...

conjecture | open | MathDB 页面 | 原文线索

V2-057416 | MATHDB-360234

Nonexistence of finitely generated standard elementary extensions of free lattices

摘录 / EXCERPT: Let $\mathbf F_n$ be the free lattice on n generators. An elementary extension $\mathbf K\succ\mathbf F_n$ is standard if the associated map $h_{\mathbf K}$ has range in...

conjecture | open | MathDB 页面 | 原文线索

V2-057417 | MATHDB-360422

The isomorphism conjecture for notational and ordinal normal forms

摘录 / EXCERPT: Isomorphism conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-057418 | MATHDB-360567

Pillay-Yao conjecture on definably amenable NIP groups

摘录 / EXCERPT: Let G be a definably amenable NIP group. An fsg group is a group with finitely satisfiable generics, and a dfg group is a group with definable f-generics. Pillay-Yao's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-057419 | MATHDB-360702

Thomas's conjecture on reducts of countable ultrahomogeneous structures

摘录 / EXCERPT: Let $\mathstrut M$ be a countable ultrahomogeneous structure in a finite relational language. A reduct of $\mathstrut M$ is a structure on the same domain whose relations and funct...

conjecture | open | MathDB 页面 | 原文线索

V2-057420 | MATHDB-360748

Quadratic EPPA lower bound outside finite homogeneous subgraphs

摘录 / EXCERPT: Quadratic EPPA conjecture. The graph G is either a subgraph of a finite homogeneous graph, or every EPPA-witness of G has at least

conjecture | open | MathDB 页面 | 原文线索

V2-057421 | MATHDB-361066

The computable-subfield characterization of random infinite extensions

摘录 / EXCERPT: Computable-subfield characterization. Then K is random.

conjecture | open | MathDB 页面 | 原文线索

V2-057422 | MATHDB-361251

Axiomatization conjecture for Büchi arithmetic $_2$

摘录 / EXCERPT: Axiomatization conjecture. The axioms and schemata (1)-(17) axiomatize $_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-057423 | MATHDB-361445

The NIPn Fields Conjecture

摘录 / EXCERPT: Let $n\geqslant 2$. A pure field is a field considered only in the language of rings, and a theory is strictly NIP$_n$ when it is NIP$_n$ but has IP$_{n-1}$. The NIP$_n$ Fields Con...

conjecture | open | MathDB 页面 | 原文线索

V2-057424 | MATHDB-361446

Jahnke's NIPn henselian expansion conjecture

摘录 / EXCERPT: Let K be a field whose pure field theory is NIP$_n$, and let v be a henselian valuation on K. Write (K,v) for the resulting valued field. Jahnke's NIP$_n$ henselian expansi...

conjecture | open | MathDB 页面 | 原文线索

V2-057425 | MATHDB-361649

Pilipczuk-Toruńczyk's rank conjecture for elementarily-fpt model checking

摘录 / EXCERPT: Let $ C$ be a hereditary graph class. For $d\in\mathbb{N}$, let $ T_d$ denote the class of all trees of depth d. Pilipczuk-Toruńczyk's rank conjecture. The class $ C$ has element...

conjecture | open | MathDB 页面 | 原文线索

V2-057426 | MATHDB-361650

Bounded-rank characterization by bounded alternation rank

摘录 / EXCERPT: Let $ C$ be a hereditary graph class. Bounded-rank alternation conjecture. The class $ C$ has bounded rank if and only if there is $k\in\mathbb{N}$ such that every first-order form...

conjecture | open | MathDB 页面 | 原文线索

V2-057427 | MATHDB-361908

Classification conjecture for dp-minimal expansions of the integers

摘录 / EXCERPT: Let $\mathcal{Z}$ be a dp-minimal proper expansion of $(\mathbb{Z},+,0,1)$. A classification conjecture for dp-minimal expansions of the integers. Exactly one of the following hold...

conjecture | open | MathDB 页面 | 原文线索

V2-057428 | MATHDB-362417

Jin's conjecture on almost internal logarithmic-differential pullbacks

摘录 / EXCERPT: Jin's conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-057429 | MATHDB-362472

The finite-by-null conjecture for omega-categorical pseudofinite rings

摘录 / EXCERPT: Let R be a ring that is both omega-categorical, meaning that its theory has a unique countable model up to isomorphism, and pseudofinite, meaning that it is infinite and every fi...

conjecture | open | MathDB 页面 | 原文线索

V2-057430 | MATHDB-362821

Characterization of strongly minimal hyperfields

摘录 / EXCERPT: Strong minimality conjecture. A hyperfield is strongly minimal if and only if it is either a strongly minimal field, that is, an algebraically closed field, or a hyperfield whose h...

conjecture | open | MathDB 页面 | 原文线索

V2-057431 | MATHDB-362871

Onshuus-Pillay conjecture on definable groups in P-minimal structures

摘录 / EXCERPT: Let M be a highly saturated P-minimal structure over a p-adic field K, and let G be an n-dimensional definable group. Recall that G^{00} is the smallest type-definabl...

conjecture | open | MathDB 页面 | 原文线索

V2-057432 | MATHDB-362872

Modified Onshuus-Pillay conjecture with a small-model-definable subgroup

摘录 / EXCERPT: Let M be a highly saturated P-minimal structure over a p-adic field K, let G be a definable group, and set $n=\dim(G)$. Let $M_0\prec M$ be a small model defining G. Re...

conjecture | open | MathDB 页面 | 原文线索

V2-057433 | MATHDB-363085

Kim's pseudolinearity conjecture

摘录 / EXCERPT: Kim's pseudolinearity conjecture. If $\mathrm{SU}(p)=1$ and p is k-linear for some $k<\omega$, then p is linear.

conjecture | open | MathDB 页面 | 原文线索

V2-057434 | MATHDB-363523

Peterzil's trichotomy conjecture for o-minimal relics

摘录 / EXCERPT: Let $\mathcal R$ be an o-minimal expansion of a real closed field, and let $\mathcal M$ be a non-locally modular strongly minimal $\mathcal R$-relic, meaning a strongly minimal str...

conjecture | open | MathDB 页面 | 原文线索

V2-057435 | MATHDB-363642

The conjecture that all o-minimal structures are istem

摘录 / EXCERPT: O-minimal istem conjecture. All o-minimal structures are istem.

conjecture | open | MathDB 页面 | 原文线索

V2-057436 | MATHDB-364181

The homogeneous-structure characterization of measurable ultraproduct classes

摘录 / EXCERPT: Homogeneous-structure conjecture. (i) Let M be a homogeneous structure over a finite relational language L. Then there is an m.e.c. with ultraproduct elementarily equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-057437 | MATHDB-364692

Model-companion conjecture for globally valued fields

摘录 / EXCERPT: Model-companion conjecture. The theory GVF_e has a model companion for every Archimedean error $e\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-057438 | MATHDB-365576

Peterzil's Zilber trichotomy conjecture for strongly minimal relics of o-minimal structures

摘录 / EXCERPT: Peterzil's conjecture. Every strongly minimal ${\mathcal R}$-relic is either locally modular or interprets an algebraically closed field.

conjecture | open | MathDB 页面 | 原文线索

V2-057439 | MATHDB-366052

The low$_2$ supersets lattice conjecture

摘录 / EXCERPT: Let A be a computably enumerable set. Write $\mathcal{E}^*$ for the lattice of all computably enumerable sets modulo finite sets, and let $L^*(A)$ denote the lattice of supersets...

conjecture | open | MathDB 页面 | 原文线索

V2-057440 | MATHDB-366424

Chernikov's field conjecture on n-dependence and dependence

摘录 / EXCERPT: A field is viewed as a structure in the pure ring language, and n-dependence is understood in the model-theoretic sense. Chernikov's field conjecture. If a field is n-dependent...

conjecture | open | MathDB 页面 | 原文线索

V2-057441 | MATHDB-366425

Non-abelian generalization of the G-infinity theorem for k-dependent groups

摘录 / EXCERPT: Let T be a k-dependent theory, let A be a small parameter set, and let G be a type-definable group over A. Let $\mathcal{M} \supseteq A$ be a small model and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057442 | MATHDB-366442

The Ryll-Nardzewski conjecture for biquadratic spaces

摘录 / EXCERPT: Let V be a $\boldsymbol{\lambda}$-space of dimension at most $\a0\aleph_0$. A group of automorphisms of V is linearly oligomorphic when it has finitely many orbits on each fini...

conjecture | open | MathDB 页面 | 原文线索

V2-057443 | MATHDB-366559

Fixed-parameter tractability of first-order model checking on monadically dependent classes

摘录 / EXCERPT: The monadic dependence model-checking conjecture. For every monadically dependent class $\mathscr{C}$ there is a constant $c\in\mathbb N$ and an algorithm that, given a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-057444 | MATHDB-366679

Chernikov's VC-density conjecture for NIP structures

摘录 / EXCERPT: Chernikov's VC-density conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-057445 | MATHDB-366688

Prawitz's conjecture on completeness of intuitionistic logic for the theory of grounds

摘录 / EXCERPT: An operational type is a specification of the premises and conclusion of a rule together with the domains and co-domains governing its discharged assumptions. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-057446 | MATHDB-366689

Full-completeness conjecture for universal operations on grounds

摘录 / EXCERPT: Let Λ be over the base structure $\mathfrak{B}_1$, and let den^*_1 be a denotation relative to Λ such that, for some operational symbol ϕ of Λ,...

conjecture | open | MathDB 页面 | 原文线索

V2-057447 | MATHDB-366742

Generic stability and connected-component equality for definably compact groups over the Laurent series field

摘录 / EXCERPT: Conjecture on definably compact groups. The group G is generically stable, and

conjecture | open | MathDB 页面 | 原文线索

V2-057448 | MATHDB-366743

Equality of connected components and bounded-index components over $\mathbb{C}((t))$

摘录 / EXCERPT: Connected-component conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-057449 | MATHDB-366744

Virtual split-solvability of dfg groups over $\mathbb{C}((t))$

摘录 / EXCERPT: Split-solvability conjecture. The group G is virtually a $\mathbb{C}((t))$-split solvable algebraic group.

conjecture | open | MathDB 页面 | 原文线索

V2-057450 | MATHDB-366745

Definable compact quotient by a definable dfg subgroup over $\mathbb{C}((t))$

摘录 / EXCERPT: Definable compact quotient conjecture. There is a $\mathbb{C}((t))$-definable dfg subgroup H of G such that G/H is definable, rather than merely interpretable, and definably...

conjecture | open | MathDB 页面 | 原文线索

V2-057451 | MATHDB-367262

The omega-categoricity conjecture for second-order strongly minimal sets

摘录 / EXCERPT: The ω-categoricity conjecture. The set $\mathcal{X}$ is ω-categorical if and only if there is a positive integer k such that, for every such tuple $x_1,\ldots,x_n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-057452 | MATHDB-367309

Gaifman's conjecture on categoricity over a predicate and nulldimensionality

摘录 / EXCERPT: Let T be a countable complete first-order theory with a distinguished predicate P. Say that T is categorical over P when it has at most one model, up to isomorphism over...

conjecture | open | MathDB 页面 | 原文线索

V2-057453 | MATHDB-367626

Domination decomposition conjecture for invariant types in o-minimal theories

摘录 / EXCERPT: Let T be an o-minimal theory and let $\mathfrak U\models T$ be a monster model. A type over U is invariant over a small set if it is invariant over some small subset of the mon...

conjecture | open | MathDB 页面 | 原文线索

V2-057454 | MATHDB-367682

Prawitz's conjecture on logically valid inference rules

摘录 / EXCERPT: Let $\mathscr{R}$ be an inference rule, let $\mathfrak{J}$ be a justification structure, and let logical validity relative to $\mathfrak{J}$ mean validity relative to...

conjecture | open | MathDB 页面 | 原文线索

V2-057455 | MATHDB-367683

Completeness of intuitionistic logic over the paper's logical consequence relation

摘录 / EXCERPT: Let $\models$ be the logical-consequence relation defined in the paper's monotonic proof-theoretic semantics, and let $\texttt{IL}$ denote intuitionistic logic. Completeness over...

conjecture | open | MathDB 页面 | 原文线索

V2-057456 | MATHDB-367910

Countability and finite arity conjecture for finitely homogeneous structures modulo trace equivalence

摘录 / EXCERPT: Countability and finite arity conjecture. There are only countably many finitely homogeneous structures modulo trace equivalence. More strongly, for every $k65 1$ there are only fi...

conjecture | open | MathDB 页面 | 原文线索

V2-057457 | MATHDB-367911

Binary finitely homogeneous trace-equivalence conjecture

摘录 / EXCERPT: Binary trace-equivalence conjecture. Every structure admitting quantifier elimination in a finite binary relational language is trace equivalent to either the trivial structure,...

conjecture | open | MathDB 页面 | 原文线索

V2-057458 | MATHDB-367912

Finite-arity conjecture for finitely homogeneous structures modulo trace equivalence

摘录 / EXCERPT: Finite-arity conjecture. For every $d\in\mathbb{N}_{\ge 1}$, there are only finitely many d-ary finitely homogeneous structures modulo trace equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-057459 | MATHDB-367914

Ordered abelian groups and projective planes conjecture

摘录 / EXCERPT: Ordered-group projective-plane conjecture. An ordered abelian group cannot locally trace define an infinite projective plane.

conjecture | open | MathDB 页面 | 原文线索

V2-057460 | MATHDB-367915

O-minimal trace-definable field conjecture

摘录 / EXCERPT: O-minimal field conjecture. Any infinite field trace definable in an o-minimal structure is real closed or algebraically closed of characteristic zero.

conjecture | open | MathDB 页面 | 原文线索

V2-057461 | MATHDB-368029

The generalized henselianity conjecture for NIP integral domains

摘录 / EXCERPT: Generalized henselianity conjecture. The ring R is a henselian local ring.

conjecture | open | MathDB 页面 | 原文线索

V2-057462 | MATHDB-368269

Stable transduction conjecture for hereditary graph classes

摘录 / EXCERPT: Stable transduction conjecture. A hereditary class of graphs is stable if and only if it is a transduction of a nowhere dense class of graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057463 | MATHDB-368763

Shelah's conjecture for NIP real fields

摘录 / EXCERPT: Shelah's conjecture for NIP real fields. Any NIP real field is almost real closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057464 | MATHDB-368764

Archimedean specialization of Shelah's conjecture for NIP real fields

摘录 / EXCERPT: Archimedean specialization of Shelah's conjecture. Then K is real closed.

conjecture | open | MathDB 页面 | 原文线索

V2-057465 | MATHDB-369002

Conjecture on definability and model companions for Γ-commutative operator fields

摘录 / EXCERPT: Definability and model-companion conjecture. (1) There is an $\mathcal L_{\underline{\mathcal D}}(F)$-formula which uniformly describes this set in any...

conjecture | open | MathDB 页面 | 原文线索

V2-057466 | MATHDB-369014

Axiomatizability of existentially closed globally valued fields

摘录 / EXCERPT: Axiomatizability conjecture. There exists a theory in the language of globally valued fields whose models are exactly the existentially closed globally valued fields of archimedean...

conjecture | open | MathDB 页面 | 原文线索

V2-057467 | MATHDB-369187

The MSO counting-logic conjecture for bounded arithmetic

摘录 / EXCERPT: Let $\mathsf{MSO}^\phi(\sharp)$ be monadic second-order logic with the Härtig quantifier, and call an arithmetical predicate directly definable when it is defined by the logic in t...

conjecture | open | MathDB 页面 | 原文线索

V2-057468 | MATHDB-369240

Schmitz's path characterization of labelled well-quasi-orderings

摘录 / EXCERPT: Schmitz's conjecture. The following are equivalent: (i) the class of structures in $\mathcal{C}$ labelled with any well-quasi-ordered set $(Y,\leq)$ is itself well-quasi-ordered un...

conjecture | open | MathDB 页面 | 原文线索

V2-057469 | MATHDB-369536

The superstability conjecture for Artin groups

摘录 / EXCERPT: Superstability conjecture. An Artin group is superstable if and only if it is abelian.

conjecture | open | MathDB 页面 | 原文线索

V2-057470 | MATHDB-369550

Elimination of imaginaries for generic derivations

摘录 / EXCERPT: Let T be an algebraically bounded theory, and let $T^{\bar \delta,?}_g$ be its theory of generic derivations. Elimination of imaginaries modulo T^{eq} means that imaginaries in...

conjecture | open | MathDB 页面 | 原文线索

V2-057471 | MATHDB-369551

Definability in families of Kolchin invariants

摘录 / EXCERPT: Let $\langle \mathfrak C,\bar \delta\rangle$ be a monster model of $T^{\bar \delta}_g$. For an $L^{\bar \delta}$-definable set X, let ω_X be the supremum of the relevant K...

conjecture | open | MathDB 页面 | 原文线索

V2-057472 | MATHDB-369553

Model completion for trace-monoid derivations

摘录 / EXCERPT: Let Λ be a partial commutative monoid generated by a tuple $\bar\delta$; equivalently, let Λ be a trace monoid (Cartier-Foata monoid). Let T^Λ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-057473 | MATHDB-369556

Decidability of the Membership Problem for class $\mathscr{C}$

摘录 / EXCERPT: Let $\mathscr{C}$ be the class of hypergeometric sequences considered in the paper, and let the Membership Problem ask whether a given sequence belongs to that class. Decidability...

conjecture | open | MathDB 页面 | 原文线索

V2-057474 | MATHDB-369583

The idempotent generically stable type conjecture

摘录 / EXCERPT: Let p be a generically stable type, and call it idempotent when it is idempotent under the convolution product; call it generically transitive when it is concentrated on the type...

conjecture | open | MathDB 页面 | 原文线索

V2-057475 | MATHDB-369761

Monadic NIP extension conjecture for modeling FO-limits

摘录 / EXCERPT: A countable signature σ and a class $\mathscr C$ of σ-structures are given. A unary expansion of $\mathscr C$ is obtained by adding unary relation symbols. Monadic NI...

conjecture | open | MathDB 页面 | 原文线索

V2-057476 | MATHDB-369792

Podewski's conjecture on infinite model-theoretically minimal fields

摘录 / EXCERPT: Let K be an infinite field. It is model-theoretically minimal if every definable subset of K is finite or cofinite. Podewski's conjecture. If K is model-theoretically minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-057477 | MATHDB-371378

The Cherlin-Zilber conjecture for definable simple Lie rings

摘录 / EXCERPT: A definable simple Lie ring of finite Morley rank is a Lie ring $\mathfrak{g}$ definable in a structure of finite Morley rank. Its characteristic is the characteristic associated w...

conjecture | solved | MathDB 页面 | 原文线索

V2-057478 | MATHDB-371379

The log-Cherlin-Zilber conjecture for absolutely simple Lie rings

摘录 / EXCERPT: Let T be a finite-dimensional theory, meaning that it carries a dimension function on interpretable sets satisfying the usual invariance, finiteness, union, and fiber-dimension a...

conjecture | open | MathDB 页面 | 原文线索

V2-057479 | MATHDB-371424

The extensional ESO NP-intermediate problem conjecture

摘录 / EXCERPT: An extensional ESO sentence Φ defines a decision problem by asking whether the sentence holds on the input structure. Extensional ESO NP-intermediate conjecture. There is an e...

conjecture | open | MathDB 页面 | 原文线索

V2-057480 | MATHDB-371475

Conjectured reflection classifications for labelled Kruskal's theorem

摘录 / EXCERPT: Let $\operatorname{KT}_{\ell}(\omega)$ denote labelled Kruskal's theorem with labels drawn from ω, and let $\operatorname{KT}_{\ell}(n)$ denote its restriction to labels bel...

conjecture | open | MathDB 页面 | 原文线索

V2-057481 | MATHDB-371477

Reflection-principle classifications for labelled Kruskal's theorem

摘录 / EXCERPT: Pakhomov-Freund conjectures. The following two equivalences are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-057482 | MATHDB-371543

Bi-interpretability conjecture for the internal language of derived Γ-geometry

摘录 / EXCERPT: Let the internal language of derived Γ-geometry be the language associated with its Γ-topoi, and let dependent type theory be extended by a ternary connective modelin...

conjecture | open | MathDB 页面 | 原文线索

V2-057483 | MATHDB-371685

The Cherlin-Zilber conjecture for infinite simple Lie rings of finite Morley rank

摘录 / EXCERPT: Let $\mathfrak{g}$ be an infinite simple Lie ring of finite Morley rank, and suppose that its characteristic is sufficiently large. Cherlin-Zilber conjecture for Lie rings. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-057484 | MATHDB-371724

The idempotent measure conjecture for definable groups

摘录 / EXCERPT: Let $G=G(\mathfrak{C})$ be a definable group, and let $\mu\in\mathfrak{M}^{\operatorname{def}}_G(\mathfrak{C},M)$ be fim over M. Its right stabilizer, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-057485 | MATHDB-371735

Scoped full-termination conjecture for pure recursive calculi

摘录 / EXCERPT: Let R be an operator-only Pure Recursive Calculus with a recursor rule of the form ... The step argument s is unrestricted, and an internally definable measure means a measure sa...

conjecture | open | MathDB 页面 | 原文线索

V2-057486 | MATHDB-372249

Braunfeld's monadic NIP orbit-growth conjecture

摘录 / EXCERPT: Braunfeld's monadic NIP orbit-growth conjecture. For a countable $c9$-categorical structure $c4$, $c4$ is monadically NIP if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-057487 | MATHDB-372250

The slowest-growth conjecture outside the order-interpretable class

摘录 / EXCERPT: Slowest-growth conjecture. If an $c9$-categorical structure $c4$ is not interpretable in $(\mathbb Q;<)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-057488 | MATHDB-372251

The primitive intermediate-growth conjecture for the circular order

摘录 / EXCERPT: Primitive intermediate-growth conjecture. If a primitive $c9$-categorical structure $c4$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-057489 | MATHDB-372257

The FO model-checking conjecture for monadically dependent hereditary graph classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary class of graphs. FO model-checking conjecture. There is an FPT first-order model-checking algorithm for graphs in $\mathcal{C}$ if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-057490 | MATHDB-372777

Lopez's path-transduction conjecture for non-2-well-quasi-ordered graph classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary class of finite graphs. Lopez's path-transduction conjecture. If $\mathcal{C}$ is not 2-well-quasi-ordered, then $\mathcal{C}$ existentially tra...

conjecture | open | MathDB 页面 | 原文线索

V2-057491 | MATHDB-372990

Hodges's conjecture on relative categoricity and the Gaifman property

摘录 / EXCERPT: Let T be a complete theory with quantifier elimination in a relational language L, and let P be a distinguished unary predicate symbol in L. For a model $M\models T$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-057492 | MATHDB-373418

Transition-computability conjecture for the Yu-Gi-Oh! TCG

摘录 / EXCERPT: Transition-computability conjecture. The function mapping a game configuration together with a legal move to the consequent configuration in the Yu-Gi-Oh! TCG is computable.

conjecture | open | MathDB 页面 | 原文线索

V2-057493 | MATHDB-373472

Triviality over a finite invariant base for non-definable invariant types

摘录 / EXCERPT: Let T be a countable weakly quasi-o-minimal theory with few countable models. Let $\mathcal P$ be the collection of all global 1-types that are invariant over some finite param...

conjecture | open | MathDB 页面 | 原文线索

V2-057494 | MATHDB-374448

NTP2 transfer conjecture for henselian valued fields of positive characteristic

摘录 / EXCERPT: Let (K,v) be a henselian valued field of positive characteristic. A valued field is tame if it satisfies the standard tameness conditions for henselian valued fields. NTP2 transf...

conjecture | open | MathDB 页面 | 原文线索

V2-057495 | MATHDB-374579

The n-determined product conjecture for generically stable measures

摘录 / EXCERPT: Let T be an m-distal theory, and let $n>m>0$. For global generically stable measures $\mu_0(x_0),\ldots,\mu_{n-1}(x_{n-1})$, write ... A tuple of measures is m-determined when...

conjecture | open | MathDB 页面 | 原文线索

V2-057496 | MATHDB-374580

The smoothness conjecture for invariant generically stable measures in n-distal groups

摘录 / EXCERPT: Assume $2\leq k\in\omega$, let T be a k-distal NIP theory, and let G be a definable group. Let $\mu\in\mathfrak{M}_{G}(\mathbb{M})$ be a generically stable, G-invariant mea...

conjecture | open | MathDB 页面 | 原文线索

V2-057497 | MATHDB-374581

The n-distal field conjecture

摘录 / EXCERPT: Let K be a field in the ring language, and let n be a positive integer. A theory is n-distal when it satisfies the corresponding n-ary distality condition. n-distal field con...

conjecture | open | MathDB 页面 | 原文线索

V2-057498 | MATHDB-374776

The optimal $d+1$ bound for fractional Helly in ultraproducts of p-adics

摘录 / EXCERPT: Let K be an ultraproduct of the fields $\boldsymbol{Q}_{p_i}$ or $\boldsymbol{F}_{p_i}((t))$ over a non-principal ultrafilter, and let $varphi(x,y)$ be a partitioned formula with...

conjecture | open | MathDB 页面 | 原文线索

V2-057499 | MATHDB-374777

ULCFS and strong ULCFS conjectures for NTP$_2$ and resilient formulas

摘录 / EXCERPT: Let $varphi(x,y)$ be a partitioned formula. A formula is NTP$_2$ when it does not have the tree property of the second kind, and it is resilient in the sense used in the paper. ULC...

conjecture | open | MathDB 页面 | 原文线索

V2-057500 | MATHDB-374883

The GEI conjecture for generic derivations over rosy algebraically bounded theories

摘录 / EXCERPT: Generic-derivation GEI conjecture. The theories

conjecture | open | MathDB 页面 | 原文线索

V2-057501 | MATHDB-375154

A C-minimal analogue with a nowhere differentiable power function

摘录 / EXCERPT: C-minimal analogue conjecture. There is some $\alpha\in\mathbb{C}_p$ and an algebraically closed subfield $M\subseteq\mathbb{C}_p$ closed under f_α such that the structure...

conjecture | open | MathDB 页面 | 原文线索

V2-057502 | MATHDB-375240

Conjecture on properness and incomparability of positive free logic systems

摘录 / EXCERPT: The paper considers the classical positive free logic systems appearing in Theorem ... are proper. (b) $\mathbf{CPF}^{\begin{sideways}\begin{sideways}$iota...

conjecture | open | MathDB 页面 | 原文线索

V2-057503 | MATHDB-375414

The dependence conjecture for first-order model checking on hereditary graph classes

摘录 / EXCERPT: Let $\mathscr C$ be a hereditary class of graphs. A class is dependent if it has the model-theoretic non-independence property, and first-order model checking is fixed parameter tr...

conjecture | open | MathDB 页面 | 原文线索

V2-057504 | MATHDB-375602

The substitution lemma for Scott complexity of finite-rank trees

摘录 / EXCERPT: Let $m\geq 1$. Write $\mathcal{A}_{m+1}$ for the canonical tree of rank $m+1$, and let $B^{(m)}_j:=[m+1:j]$ be the tree obtained by deleting all but j of the level-one subtrees o...

conjecture | open | MathDB 页面 | 原文线索

V2-057505 | MATHDB-375901

The maximality conjecture for nm-independence on Polish groups

摘录 / EXCERPT: The maximality conjecture. The relation $\mathrel{\raise0.2ex\text{\ooalign{\hidewidth$vert$\hidewidth\cr\raise-0.9ex\text{$smile$}}}}^{nm}$ is the maximum strongly admissible geom...

conjecture | open | MathDB 页面 | 原文线索

V2-057506 | MATHDB-375925

The stable forking conjecture

摘录 / EXCERPT: Stable forking conjecture. If $a \mathrel{\mathop{\mathpalette\notind{}}_{C}} b$, then there is a formula $\varphi(x,\overline{b})\in\operatorname{tp}(a/Cb)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-057507 | MATHDB-375926

The stable forking conjecture over a base

摘录 / EXCERPT: Stable forking conjecture over a base. If $a \mathrel{\mathop{\mathpalette\notind{}}_{C}} b$, there is a formula $\varphi(x,b)\in\operatorname{tp}(a/Cb)$ which forks over C, such...

conjecture | open | MathDB 页面 | 原文线索

V2-057508 | MATHDB-376170

The vertical non-uniqueness conjecture for PC connectives outside PU

摘录 / EXCERPT: Vertical non-uniqueness conjecture. For every connective $\texttt{\#}$, if $\texttt{\#}\in\texttt{PC}$ but $\texttt{\#}\notin\texttt{PU}$, then $\texttt{\#}$ is not vertically inte...

conjecture | open | MathDB 页面 | 原文线索

V2-057509 | MATHDB-376171

The vertical non-uniqueness conjecture for special cases

摘录 / EXCERPT: Special-case vertical non-uniqueness conjecture. Special cases are not vertically unique.

conjecture | open | MathDB 页面 | 原文线索

V2-057510 | MATHDB-376284

Projective categoricity conjecture for compact convex bodies

摘录 / EXCERPT: Projective categoricity conjecture. $K\equiv_{\mathcal{L}_{\mathrm{B}}}K'$ if and only if K and K' are projectively equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-057511 | MATHDB-376285

Projective categoricity conjecture for the ball

摘录 / EXCERPT: Ball categoricity conjecture. $K\equiv_{\mathcal{L}_{\mathrm{B}}}\overline{B}{}^{\,n}$ if and only if ∂K is a quadric, equivalently, if and only if K is projectively eq...

conjecture | open | MathDB 页面 | 原文线索

V2-057512 | MATHDB-376286

Projective categoricity conjecture for open properly convex sets

摘录 / EXCERPT: Open-domain projective categoricity conjecture. Two such sets U and V are $\mathcal{L}_{\mathrm{B}}$-elementarily equivalent if and only if they are projectively equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-057513 | MATHDB-377763

Interpolation conjecture for the calculus of structures

摘录 / EXCERPT: Interpolation conjecture. For every derivation $T\vdash_{\mathsf{SKS}}R$, there is a structure P and derivations

conjecture | open | MathDB 页面 | 原文线索

V2-057514 | MATHDB-377764

Separation conjecture for core and non-core rules in the calculus of structures

摘录 / EXCERPT: Separation conjecture. For every derivation $T\vdash_{\mathsf{SKS}}R$, there is a derivation of the form

conjecture | open | MathDB 页面 | 原文线索

V2-057515 | MATHDB-377827

The benignity conjecture for subsets of models of stable theories

摘录 / EXCERPT: Let T be a stable theory, let M be a model of T, and let $A\subseteq M$. The set A is benign in M if equality of types over A implies equality of the corresponding type...

conjecture | open | MathDB 页面 | 原文线索

V2-057516 | MATHDB-377828

The weak benignity conjecture for subsets of models of stable theories

摘录 / EXCERPT: Let T be a stable theory, let M be a model of T, and let $A\subseteq M$ be arbitrary. The set A is weakly benign in M if equality of strong types over A implies equalit...

conjecture | open | MathDB 页面 | 原文线索

V2-057517 | MATHDB-377854

The small-index conjecture for subgroups of the Lascar strong automorphism group

摘录 / EXCERPT: Small-index conjecture. If H is a subgroup of $Autf(M)$ with index strictly less than $2^{\aleph_{0}}$, then there exists a finite subset $A\subset M$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057518 | MATHDB-378687

Scott-set characterization of transcendence for countable recursively saturated models

摘录 / EXCERPT: Let M be a countable recursively saturated model of PA, and let $\operatorname{SSy}(M)$ denote its standard system. Transcendence characterization conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-057519 | MATHDB-379305

Shelah's eventual model-counting bound for abstract elementary classes

摘录 / EXCERPT: Let $\operatorname{\mathcal{K}}$ be an abstract elementary class (AEC), and let ... Assume that $\operatorname{\mathcal{K}}$ has at least one model but fewer than the maximal number...

conjecture | open | MathDB 页面 | 原文线索

V2-057520 | MATHDB-379853

The p-adic splitting conjecture for definable sets

摘录 / EXCERPT: Let K be a p-adically closed field and let A be an infinite definable subset of K^n that is open in its closure. Let $(B_k)_{k\leq q}$ be a finite collection of closed defi...

conjecture | open | MathDB 页面 | 原文线索

V2-057521 | MATHDB-380407

The generic pair conjecture for dependent theories

摘录 / EXCERPT: Assume that T is a first-order theory and that ... with $2^\lambda=\lambda^+$. Let $\langle M_\alpha:\alpha<\lambda^+\rangle$ be a $\prec$-increasing continuous sequence with...

conjecture | open | MathDB 页面 | 原文线索

V2-057522 | MATHDB-380493

Tameness of imaginaries for finite definable equivalence relations

摘录 / EXCERPT: Let ${\mathcal T}_{\infty}$ be the theory under consideration, and let $\sim$ be a definable equivalence relation over ${\mathcal T}_{\infty}$. The relation is finite if its equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-057523 | MATHDB-380534

Stable-tree decomposition conjecture for finite models

摘录 / EXCERPT: Stable-tree decomposition conjecture. If $\langle K,\prec_K\rangle$ satisfies $P_1,\ldots,P_n$, then for every sufficiently large $M\in K$ there exist a finite tree...

conjecture | open | MathDB 页面 | 原文线索

V2-057524 | MATHDB-381083

Petrykowski's bounded-orbit conjecture for definable groups

摘录 / EXCERPT: Petrykowski's conjecture. If G has a bounded orbit, then G is definably amenable.

conjecture | open | MathDB 页面 | 原文线索

V2-057525 | MATHDB-382183

The NIP characterization of tractable model-checking on hereditary classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary class of relational structures. Here, model-checking on $\mathcal{C}$ is tractable if there is an algorithm that, on input a structure...

conjecture | open | MathDB 页面 | 原文线索

V2-057526 | MATHDB-382206

Monadic-theory characterization conjecture for orders in La08uchli's class

摘录 / EXCERPT: Let $\underline{M}$ be the class of orders generated from the one-point order by finite sums, products with ω and ω^*, and the specified dense rational-indexed sums....

conjecture | open | MathDB 页面 | 原文线索

V2-057527 | MATHDB-383164

Generalized Gaifman Conjecture

摘录 / EXCERPT: Let T be a countable complete theory and let P be a distinguished unary predicate in its vocabulary. Assume that T fails the Gaifman property; that is, assume that there exis...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057528 | MATHDB-386930

What is the right metric to determine computability?

摘录 / EXCERPT: In problems where mathematical analysis precedes a computer assisted step: How can we ensure that the formulation of problem is correctly posed so that computability and non-comput...

open_problem | open | MathDB 页面 | 原文线索

V2-057529 | MATHDB-393674

Computability of the convex hull of the two-row mixed-integer set

摘录 / EXCERPT: For the set ... where I is the 2×2 identity matrix, A is a nonsingular 2×2 matrix, B and C are $2\times p$ and $2\times q$ matrices, respectively, and...

open_problem | open | MathDB 页面 | 原文线索

V2-057530 | MATHDB-397527

Computability of constraint sets after adjoining a differential-transcendental element

摘录 / EXCERPT: Conjecture 5. Let K be a computable differential field, and let z be an element differentially transcendental over K within some larger computable differential field. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-057531 | MATHDB-399936

Polynomial-time computability of the k-Frobenius number

摘录 / EXCERPT: Let $a=(a_1,\ldots,a_n)^T\in\mathbb{Z}_{>0}^n$ satisfy $\gcd(a_1,\ldots,a_n)=1$. For a positive integer k, let $F_k(a)$ be the largest integer that cannot be represented in at l...

open_problem | open | MathDB 页面 | 原文线索

V2-057532 | MATHDB-400717

Model theory of the surreal exponential field enriched by omega-series functions

摘录 / EXCERPT: It is conjectured that the exponential field $(\mathbf{No},\exp)$, enriched with all functions $\hat f:\mathbf{No}_{>\mathbb{R}}\to\mathbf{No}$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-057533 | MATHDB-402503

Rationality and computability of multipoint Seshadri constants

摘录 / EXCERPT: Let $Z \subset \mathbb{P}^2$ be a finite set of points. Let $\pi_Z:X_Z\to\mathbb{P}^2$ be the blow-up of Z, with E_Z the exceptional locus, and let $L'=\pi_Z^*L$ for a general...

open_problem | open | MathDB 页面 | 原文线索

V2-057534 | MATHDB-402508

Rationality and computability of Waldschmidt constants of singular points of plane curves

摘录 / EXCERPT: Let $C\subset \mathbb{P}^2$ be a reduced curve, let $Z=Z_C=\operatorname{Sing}(C)$, and let $\widehat{\alpha}(Z)$ denote the Waldschmidt constant of Z. Must $\widehat{\alpha}(Z)$...

open_problem | open | MathDB 页面 | 原文线索

V2-057535 | MATHDB-405164

Composition conjecture for n-dependence

摘录 / EXCERPT: For all positive integers n and k, if M is an n-dependent structure, $R(y_1,\ldots,y_m)$ is a relation definable in M, and $f_i:X_i^{r_i}\to M$ are arbitrary functions wi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

B15 理论计算机科学 / Computer Science | 923 条

V2-057536 | MATHDB-315469

Superlinear circuit lower bounds for the permanent

摘录 / EXCERPT: Can one prove a superlinear-in-the-input-size lower bound for unrestricted arithmetic circuits computing the permanent exactly?

open_problem | open | MathDB 页面 | 原文线索

V2-057537 | MATHDB-315471

Polynomial-factor hardness of Euclidean closest vector

摘录 / EXCERPT: Is Euclidean closest vector NP-hard to approximate within n^c for some fixed constant $c>0$ under a deterministic reduction?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057538 | MATHDB-315485

Linear-time approximation of product-distribution TV distance

摘录 / EXCERPT: Theorem 1 (cited paper, Section 2). Let $q,n>0$ be integers, and let $P=\bigotimes_{i=1}^{n}P_i$ and $Q=\bigotimes_{i=1}^{n}Q_i$ be product distributions on $[q]^n$, given explicit...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057539 | MATHDB-315494

Optimal exponential approximation of positive-semidefinite permanents

摘录 / EXCERPT: What is the optimal exponential approximation ratio achievable in deterministic polynomial time for the permanent of a Hermitian positive-semidefinite matrix?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057540 | MATHDB-315510

Exhaustive AdaBoost cycling question

摘录 / EXCERPT: Let $M\in\{-1,+1\}^{m\times N}$ be any finite sign matrix representing a finite hypothesis class that is closed under negation. Starting from the AdaBoost example-weight distributi...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057541 | MATHDB-315511

Optimal online discrepancy in linear time

摘录 / EXCERPT: Theorem 2. Fix a sequence $v_{1},\dots,v_{T}\in\mathbb{R}^{d}$ such that $\|v_{t}\|_{2}\leq 1$ for all $t\in[T]$. At time step $t\in[T]$, Algorithm 2 receives v_{t} and chooses a...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057542 | MATHDB-315522

Nearly uniform sampling of directed Eulerian tours

摘录 / EXCERPT: Does the proposed flip-repair Markov chain mix rapidly enough to yield a nearly uniform directed-Eulerian-tour sampler in $\widetilde O(m^{3/2})$ worst-case time?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057543 | MATHDB-315525

Last-iterate rate for anchored gradient descent-ascent

摘录 / EXCERPT: Under the standard assumptions in Section 2 of the cited paper, namely monotonicity, K-Lipschitz continuity of the saddle operator G, and existence of a saddle point $z^\star$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057544 | MATHDB-315591

Nonuniform definability of automorphisms

摘录 / EXCERPT: Does there exist a computably represented countable structure A that is computably AUT-countable on a cone, but for every finite parameter tuple $\bar a$ has an automorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-057545 | MATHDB-315665

Near-quadratic value-oracle lower bound for convex optimization

摘录 / EXCERPT: How many function-value queries are necessary for deterministic optimization of a Lipschitz convex function in high dimension?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057546 | MATHDB-315668

Polynomial-time low-degree conjecture

摘录 / EXCERPT: Does failure of every low-degree polynomial test imply computational indistinguishability for broad permutation-invariant average-case problems?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057547 | MATHDB-315671

Möbius-Bridge attack on seven-round AES-128

摘录 / EXCERPT: Can the best known attack on seven-round AES-128 be improved at the standard chosen-plaintext data budget?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-057548 | MATHDB-315673

KINDI CCA-security proof

摘录 / EXCERPT: Is KINDI CCA secure without a re-encryption check, as claimed by its published uniqueness lemma?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057549 | MATHDB-315674

Kemeny rank aggregation for three voters

摘录 / EXCERPT: Given a set A of alternatives, three linear orders $\succ_{1},\succ_{2},\succ_{3}$ over A, and a number b, does there exist a linear order $\succ$ over A such that...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057550 | MATHDB-315684

SS-RS-GD inequalities

摘录 / EXCERPT: For well-conditioned symmetric quadratic losses, must the expected single-shuffle, random-reshuffle, and gradient-descent operators satisfy...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057551 | MATHDB-315685

Three-server heterogeneous-queue threshold policy

摘录 / EXCERPT: Does the threshold-policy structure of the Lin-Kumar two-server queue remain optimal once a third heterogeneous server is introduced?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-057552 | MATHDB-315686

Fixed-k graph coloring below 2^n time

摘录 / EXCERPT: Can k-coloring be solved in time $O^{*}((2-\varepsilon_k)^n)$, for some $\varepsilon_k>0$, for every k?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057553 | MATHDB-315687

The 4^k barrier for the k-distinct language

摘录 / EXCERPT: Can the k-distinct language be recognized by an acyclic NFA of size $c^kn^{O(1)}$ for some $c<4$?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057554 | MATHDB-315720

Single-pass semi-streaming matching

摘录 / EXCERPT: Can a single-pass semi-streaming algorithm beat the naive greedy $1/2$ approximation for maximum matching?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057555 | MATHDB-315721

Terminal-only Manhattan cost-radius spanning trees

摘录 / EXCERPT: Given Manhattan terminals, a root, a total-length budget, and a source-radius budget, is deciding whether a spanning tree meets both bounds polynomial-time solvable?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057556 | MATHDB-315745

Černý conjecture for one-cluster automata

摘录 / EXCERPT: Conjecture 1 (Černý). An n-state synchronizing automaton admits a synchronizing word of length at most $(n-1)^2$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057557 | MATHDB-315746

Online Spencer vector-balancing question

摘录 / EXCERPT: Can online vector balancing in the Spencer setting achieve the optimal order of prefix discrepancy by an efficient algorithm?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-057558 | MATHDB-315750

Minimum sparsity of S-decoding polynomials

摘录 / EXCERPT: Can an S-decoding polynomial modulo a suitable product of k primes attain the lower-bound minimum of $k+1$ nonzero coefficients?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-057559 | MATHDB-315752

Copying-versus-moving conjecture for online submodular welfare

摘录 / EXCERPT: Is the expected marginal gain from copying an item to the end of a random-order stream always at most the gain from moving its original occurrence there?

open_problem | open | MathDB 页面 | 原文线索

V2-057560 | MATHDB-315753

Avidor-Zwick Max-Cut question for triangle-strengthened SDP

摘录 / EXCERPT: For every fixed integer d, there exists a randomized polynomial-time algorithm such that, given a feasible d-dimensional solution $\{v_i\in\mathbb{R}^d:\|v_i\|_2=1\}_{i\in V}$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057561 | MATHDB-315764

Minimum edge-outerplanar embedding

摘录 / EXCERPT: Can the minimum edge-outerplanarity of a finite loopless planar graph-minimized over all planar embeddings-be computed in polynomial time?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-057562 | MATHDB-315778

Bicriteria submodular maximization over p-systems

摘录 / EXCERPT: Does the greedy algorithm for monotone submodular maximization over a p-system achieve a $(1-\varepsilon,\lceil\log_{p+1}(1/\varepsilon)\rceil)$ bicriteria approximation? If not,...

open_problem | open | MathDB 页面 | 原文线索

V2-057563 | MATHDB-316217

Sensitivity conjecture

摘录 / EXCERPT: In computational complexity, the sensitivity theorem, proved by Hao Huang in 2019, states that the sensitivity of a Boolean function $f\colon \{0,1\}^{n}\to \{0,1\}$ is at least th...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-057564 | MATHDB-316424

Nikolaev's conjecture relating arithmetic complexity to the rank of a CM elliptic curve

摘录 / EXCERPT: Let E_{CM} be an elliptic curve with complex multiplication, let ${\theta}_{E_{CM}}$ be the real quadratic irrational associated with E_{CM}, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057565 | MATHDB-317011

The square-root query-complexity conjecture for testing H-minor freeness

摘录 / EXCERPT: Square-root query-complexity conjecture. For every H, being H-minor free can be tested with one-sided error using $\tilde{O}(\sqrt{n})$ queries, where n is the number of vert...

conjecture | open | MathDB 页面 | 原文线索

V2-057566 | MATHDB-317500

Dobbertin's conjecture on APN permutations in even dimension

摘录 / EXCERPT: Let $m\geq 1$ be a natural number, let $A=({\mathbb F}_2)^m$, and let an APN function be a vectorial Boolean function $F:A\to A$ with differential uniformity 2, meaning that ... fo...

conjecture | open | MathDB 页面 | 原文线索

V2-057567 | MATHDB-317582

The orbit algorithm conjecture for automorphisms of right-angled Artin groups

摘录 / EXCERPT: Let A_Γ be a right-angled Artin group associated to a finite graph Γ, and let $u,v\in A_\Gamma$. An automorphism orbit problem asks whether some...

conjecture | open | MathDB 页面 | 原文线索

V2-057568 | MATHDB-318138

Polynomial-time conjecture for quadrilateral-to-standard solution set conversion

摘录 / EXCERPT: Let n be the number of tetrahedra and let k' denote the size of the standard solution set produced by Algorithm for converting a quadrilateral solution set to a standard soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-057569 | MATHDB-318154

Optimal ordering conjecture for divided decoding split sizes

摘录 / EXCERPT: Let $\hat{\mathbf{s}}$, $\hat{\mathbf{s}}_M$, or $\hat{\mathbf{s}}_{\epsilon}$ be approximated by divided decoding with sphere decoding according to a splitting index set...

conjecture | open | MathDB 页面 | 原文线索

V2-057570 | MATHDB-318617

Barvinok's concentration conjecture for the Frobenius permanent estimator

摘录 / EXCERPT: Let A) be an $n\times n$ matrix with entries in $\{0,1\}$. For a matrix A, define $M(A)$ by ... where the ρ_{ij} are chosen independently from the Gaussian distribution on...

conjecture | open | MathDB 页面 | 原文线索

V2-057571 | MATHDB-318618

Exponential critical-ratio conjecture for the symmetrized Frobenius estimator

摘录 / EXCERPT: Exponential critical-ratio conjecture. There is a sequence of constants θ_d with

conjecture | open | MathDB 页面 | 原文线索

V2-057572 | MATHDB-319123

The variable bound conjecture for minimally unsatisfiable k-DNF sets

摘录 / EXCERPT: Variable bound conjecture. Suppose that ${\mathbb{D}}$ is a minimally unsatisfiable k-DNF set for some arbitrary but fixed positive integer k. Then the number of variables in...

conjecture | open | MathDB 页面 | 原文线索

V2-057573 | MATHDB-319237

The exponential-time physical-oracle conjecture

摘录 / EXCERPT: A physical theory is called “realistic” when it satisfies the intended assumptions described in the paper, and an experiment is used as a physical oracle by a Turing machine throug...

conjecture | open | MathDB 页面 | 原文线索

V2-057574 | MATHDB-319238

The computational-power conjecture for exponential-time physical oracles

摘录 / EXCERPT: Consider a Turing machine using an experiment as a physical oracle through an exponential-time protocol. For infinite and unbounded precision, the relevant class is $P/log*$; for f...

conjecture | open | MathDB 页面 | 原文线索

V2-057575 | MATHDB-319262

Finiteness conjecture for algorithm A

摘录 / EXCERPT: Let algorithm A be the factorization procedure that applies the local commutation rules to a sequence of elementary matrices and their inverses, with the choices between rules...

conjecture | open | MathDB 页面 | 原文线索

V2-057576 | MATHDB-319263

Finiteness conjecture for algorithm A along valuations

摘录 / EXCERPT: Let a valuation be a ray in ${\mathbb{R}}^3$ generated by a vector v with rationally independent coordinates. In the local algorithm, the valuation ray determines the cone chosen...

conjecture | open | MathDB 页面 | 原文线索

V2-057577 | MATHDB-319456

Cusick-Stănică conjecture on the nonlinearity of a cubic rotation-symmetric Boolean function

摘录 / EXCERPT: Let F_3^n be the Boolean function on n variables defined by ... Here, the weight of a Boolean function is the number of inputs on which it takes the value 1, and its nonlineari...

conjecture | open | MathDB 页面 | 原文线索

V2-057578 | MATHDB-319743

Polynomial-time cyclic-sliding conjecture for essential reduction curves

摘录 / EXCERPT: Cyclic-sliding polynomial-bound conjecture. There is a polynomial bound on the number of times one has to apply cyclic sliding in order to decrease the complexity of any essential...

conjecture | open | MathDB 页面 | 原文线索

V2-057579 | MATHDB-319823

Bulatov-Jeavons-Krokhin conjecture for finite cores

摘录 / EXCERPT: Let Γ be a finite core, and let ... A primitive positive interpretation of Δ in Γ is an interpretation using primitive positive formulas. Bulatov-Jeavons-Krokhin...

conjecture | open | MathDB 页面 | 原文线索

V2-057580 | MATHDB-319985

F5 termination at the F5 rejection degree

摘录 / EXCERPT: Let d denote the current degree in the F5 algorithm, and let $d_{\textrm{FR}}$ be the degree at which all Gröbner-basis-critical (GB-critical) pairs have been rejected by the F5...

conjecture | open | MathDB 页面 | 原文线索

V2-057581 | MATHDB-320037

The communication lower-bound conjecture for multi-value Byzantine agreement

摘录 / EXCERPT: Communication lower-bound conjecture. In order to achieve agreement on l bits, at least

conjecture | open | MathDB 页面 | 原文线索

V2-057582 | MATHDB-320038

The disagreement-detection lower-bound conjecture for multi-value Byzantine agreement

摘录 / EXCERPT: Disagreement-detection lower-bound conjecture. If the k initial values are not identical, at least one node can detect this only after at least

conjecture | open | MathDB 页面 | 原文线索

V2-057583 | MATHDB-320218

The polynomial bound conjecture for cyclic slidings

摘录 / EXCERPT: Polynomial bound conjecture. The integer t is bounded by a polynomial in ℓ and n.

conjecture | open | MathDB 页面 | 原文线索

V2-057584 | MATHDB-320365

Fixed-rank NP-hardness conjecture for nonnegative rank

摘录 / EXCERPT: For a nonnegative matrix M, its nonnegative rank $\operatorname{rank}_+(M)$ is the least inner dimension of a factorization of M into two nonnegative matrices. The ordinary ran...

conjecture | open | MathDB 页面 | 原文线索

V2-057585 | MATHDB-320662

Sturmfels-Sullivant phylogenetic complexity conjecture

摘录 / EXCERPT: Sturmfels-Sullivant's conjecture. For every abelian group G,

conjecture | open | MathDB 页面 | 原文线索

V2-057586 | MATHDB-320887

The strong-duality polynomial-solvability conjecture for optimization problems

摘录 / EXCERPT: Let P_a and P_b be optimization problems such that one is the dual of the other, so that they exhibit strong duality. Strong-duality polynomial-solvability conjecture. Both...

conjecture | open | MathDB 页面 | 原文线索

V2-057587 | MATHDB-321067

Asymptotic approximation conjecture for colouring claw-free graphs

摘录 / EXCERPT: Asymptotic approximation conjecture. There is an algorithm that, given any claw-free graph G, returns a proper $(1+o(1))\chi(G)$-colouring of G in polynomial time.

conjecture | open | MathDB 页面 | 原文线索

V2-057588 | MATHDB-321143

Polynomial rank-bound conjecture for depth-four identities

摘录 / EXCERPT: Let K be a field of characteristic $\operatorname{ch}(K)$, and let $R_{\delta}(k,s)$ strictly upper-bound the rank of any minimal and simple $\Sigma\Pi_{\delta}\Sigma\Pi(k,s,n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-057589 | MATHDB-321313

Polynomial-time satisfiability conjecture for multihitting generalised clause-sets

摘录 / EXCERPT: Multihitting satisfiability conjecture. Satisfiability decision for multihitting generalised clause-sets can be done in polynomial time.

conjecture | open | MathDB 页面 | 原文线索

V2-057590 | MATHDB-321314

Polynomial-time conjecture for generalised clause-sets of bounded hermitian rank

摘录 / EXCERPT: Bounded-hermitian-rank satisfiability conjecture. Satisfiability decision for generalised clause-sets can be done in polynomial time for bounded hermitian rank.

conjecture | open | MathDB 页面 | 原文线索

V2-057591 | MATHDB-321500

The pyramid path-search equality for triangular query budgets

摘录 / EXCERPT: Pyramid path-search conjecture. If $s_l=1+2+\ldots+l$ for some l, then

conjecture | open | MathDB 页面 | 原文线索

V2-057592 | MATHDB-321513

The encoding conjecture for convex-formulation languages

摘录 / EXCERPT: Encoding conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057593 | MATHDB-321967

Gurvits's exponential BP upper-bound conjecture for the permanent

摘录 / EXCERPT: Gurvits's BP upper-bound conjecture. For any non-negative p, $f(n)$ is asymptotic to $\sqrt{2}^n$. This conjecture concerns the worst-case multiplicative gap between the permanen...

conjecture | open | MathDB 页面 | 原文线索

V2-057594 | MATHDB-322023

Conjecture on activized empirical risk minimization for VC classes

摘录 / EXCERPT: Activized empirical risk minimization conjecture. For any VC class, there exists an active meta-algorithm $\mathcal A_a$ and an empirical risk minimization algorithm $\mathcal A_p$...

conjecture | open | MathDB 页面 | 原文线索

V2-057595 | MATHDB-322081

The classification conjecture for APN power functions with exponent 2^t-1

摘录 / EXCERPT: Let ${\mathbb F}_{2^n}$ be the finite field with 2^n elements, and for $2\leq t\leq n-1$ define ... A function is APN if its differential uniformity is at most 2. The classificat...

conjecture | open | MathDB 页面 | 原文线索

V2-057596 | MATHDB-322324

Degree-complexity conjecture for two-variable monomial toric ideals

摘录 / EXCERPT: Let $\alpha=(\alpha_1,\alpha_2)$ with $\gcd(\alpha_1,\alpha_2)=1$ and suppose $\alpha_1\geq\alpha_2$. Let I_n be the ideals in the chain of non-Laurent toric ideals induced by th...

conjecture | open | MathDB 页面 | 原文线索

V2-057597 | MATHDB-322652

González-Meneses-Wiest polynomial bound for cyclic sliding periods

摘录 / EXCERPT: Let B_n be the braid group on n strands, let $x\in B_n$ have canonical length r, and let $\mathfrak{s}$ denote the cyclic sliding operation. Let t be the minimal positive i...

conjecture | open | MathDB 页面 | 原文线索

V2-057598 | MATHDB-322858

Bremner-Felipe-Sánchez-Ortega conjecture relating the KP and BSO algorithms

摘录 / EXCERPT: Bremner-Felipe-Sánchez-Ortega conjecture. If $\mathbb{F}$ has characteristic 0 or $p>d$, then

conjecture | open | MathDB 页面 | 原文线索

V2-057599 | MATHDB-323049

Without-replacement matrix-product norm conjectures

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\bm{A}_1,\ldots,\bm{A}_n$ be positive semidefinite matrices, and let k be a positive integer. For a function of an ordered k-tuple, define ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-057600 | MATHDB-323197

Mulmuley-Sohoni's strengthened permanent versus determinant conjecture

摘录 / EXCERPT: Mulmuley-Sohoni's strengthened conjecture. For each $c>0$ and infinitely many n, $\mathcal{P}erm_n^{n^c}\not\subseteq\mathcal{D}et_{n^c}$. This is a central geometric-complexity-...

conjecture | open | MathDB 页面 | 原文线索

V2-057601 | MATHDB-323551

Log Approximate Monomial Complexity Conjecture for symmetric Boolean functions

摘录 / EXCERPT: Let $f:\{0,1\}^n\to\{\pm1\}$ be symmetric, and let $\operatorname{mon}_{1/3}(f)$ denote its $1/3$-approximate monomial complexity. Let $r(f)$ be the parameter defined earlier in th...

conjecture | open | MathDB 页面 | 原文线索

V2-057602 | MATHDB-323552

Log Approximation Rank Conjecture

摘录 / EXCERPT: Let F be any two-party communication problem. Let M_F be its communication matrix, let $\operatorname{rank}_{\epsilon'}(M_F)$ be the minimum rank of a matrix that ϵ'-a...

conjecture | open | MathDB 页面 | 原文线索

V2-057603 | MATHDB-323760

Nisan and Szegedy's Sensitivity Conjecture

摘录 / EXCERPT: Nisan and Szegedy's Sensitivity Conjecture. There exist constants c and k such that for every Boolean function f,

conjecture | open | MathDB 页面 | 原文线索

V2-057604 | MATHDB-323763

Li's conjecture on the complexity of deciding k-connectivity

摘录 / EXCERPT: Let $k\geq 3$ be fixed, let G be a graph, and let $\ell\geq 2$ be an integer. Li's conjecture. Deciding whether $\kappa_k(G)\geq\ell$ is $\mathcal{N}\mathcal{P}$-complete. The so...

conjecture | open | MathDB 页面 | 原文线索

V2-057605 | MATHDB-323856

Commutative coherent configurations suffice for matrix multiplication exponent two

摘录 / EXCERPT: Commutative coherent configuration conjecture. There exist commutative coherent configurations ${\cal C}_n$ realizing $\langle n,n,n\rangle$ and of rank $n^{2+o(1)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057606 | MATHDB-323993

Finite-family Hermite matrix conjecture for cubic irrational pairs

摘录 / EXCERPT: Finite-family Hermite matrix conjecture. There is a family of multidimensional continued fractions spanned by finitely many multidimensional continued fraction algorithms such that

conjecture | open | MathDB 页面 | 原文线索

V2-057607 | MATHDB-324050

The S-adic conjecture on sub-linear complexity

摘录 / EXCERPT: Let $f53f$ be a sequence over a finite alphabet. An S-adic sequence is one admitting an S-adic representation for some set S of morphisms; in particular, the representation i...

conjecture | open | MathDB 页面 | 原文线索

V2-057608 | MATHDB-324051

The morphic-sequence complexity spectrum conjecture

摘录 / EXCERPT: Let $f53f$ be a morphic sequence, and let $p_{\mathbf{w}}(n)$ denote its factor-complexity function. Morphic-sequence complexity spectrum conjecture. The function $p_{\mathbf{w}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-057609 | MATHDB-324185

Conjecture on explicitness of categorical quotients and invariant rings

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, let G be a connected reductive algebraic group of the form $\prod G_i$, with each factor a torus or a classical s...

conjecture | open | MathDB 页面 | 原文线索

V2-057610 | MATHDB-324401

NP-completeness of higher graph powers of girth $2k+1$

摘录 / EXCERPT: Let k and r be positive integers. The problem textsc{k^{th}Power of a Graph With Girth r} asks, given a graph G, whether there exists a graph H of girth r such that...

conjecture | open | MathDB 页面 | 原文线索

V2-057611 | MATHDB-324429

The delay-information duality conjecture for lookahead queues

摘录 / EXCERPT: Let $p\text{ be fixed in }(0,1)$, let $w(\lambda)$ denote the size of the lookahead window, and let $C^*_{\Pi_{w(\lambda)}}(p,\lambda)$ be the best achievable delay under policies...

conjecture | open | MathDB 页面 | 原文线索

V2-057612 | MATHDB-324821

Polynomial Alon-Saks-Seymour conjecture

摘录 / EXCERPT: Let G be a graph. Its bipartite packing $\mathbf{bp}(G)$ is the minimum number of edge-disjoint complete bipartite graphs needed to partition the edges of G, and $\chi(G)$ is i...

conjecture | open | MathDB 页面 | 原文线索

V2-057613 | MATHDB-325150

Conjecture on the nonexistence of homogeneous rotation symmetric bent functions

摘录 / EXCERPT: A Boolean function is homogeneous if all its monomials have the same degree, and rotation symmetric if it is invariant under cyclic permutations of its variables. A Boolean functio...

conjecture | open | MathDB 页面 | 原文线索

V2-057614 | MATHDB-325209

Constant-congestion integrality-gap conjecture for maximum edge-disjoint paths in minor-closed families

摘录 / EXCERPT: Let ${\cal G}$ be any proper minor-closed family of graphs. The constant-congestion integrality-gap conjecture. The integrality gap of the flow LP for maximum edge-disjoint paths i...

conjecture | open | MathDB 页面 | 原文线索

V2-057615 | MATHDB-325288

The instruction-cost efficiency conjecture for Yaroslavskiy's algorithm

摘录 / EXCERPT: Let Yaroslavskiy's algorithm and classic Quicksort be implementations of Quicksort, and consider computational models that assign constant cost contributions to individual instruct...

conjecture | open | MathDB 页面 | 原文线索

V2-057616 | MATHDB-325315

The Hatami-Lovász-Szegedy conjecture for local independent sets

摘录 / EXCERPT: Hatami-Lovász-Szegedy conjecture. There exists a sequence of r-local independence functions f_r, $r\geq 1$, such that almost surely $I(f_r,n)$ is an independent set in...

conjecture | open | MathDB 页面 | 原文线索

V2-057617 | MATHDB-325547

Bulatov-Jeavons-Krokhin algebraic dichotomy conjecture

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite relational structure, and let $\operatorname{CSP}(\mathbb{A})$ denote the problem of deciding whether a finite input structure has a homomorphism to...

conjecture | open | MathDB 页面 | 原文线索

V2-057618 | MATHDB-325558

NP-hardness of the Fooling-Set-Submatrix problem

摘录 / EXCERPT: The Fooling-Set-Submatrix problem takes integers n,m and an $m\times m$ $0/1$-matrix A as input, and asks whether A contains a fooling-set submatrix of size n. NP-hardness...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057619 | MATHDB-325690

NP-completeness threshold conjecture for dense partial Latin-square completion

摘录 / EXCERPT: Let $\epsilon>0$, and consider the decision problem of completing an arbitrary ϵ-dense partial Latin square, where an $n\times n$ partial Latin square is ϵ-dense...

conjecture | open | MathDB 页面 | 原文线索

V2-057620 | MATHDB-325724

Large zero rectangle conjecture for low-rank sparse matrices

摘录 / EXCERPT: Let M be an $n \times n$ real matrix with $\operatorname{rank}(M)=r$ and such that $M_{i,j} \ne 0$ for at most εn^2 entries. A large zero rectangle conjecture assert...

conjecture | open | MathDB 页面 | 原文线索

V2-057621 | MATHDB-325749

Polynomial-time support-recovery barrier for sparse PCA

摘录 / EXCERPT: Consider the single-spike model with dimension p, sample size n, spike support size k, and signal strength β. Let $\mathbf{z}$ denote the sparse spike, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057622 | MATHDB-326003

Complex almost-injective intensity measurements have NP-complete consistency

摘录 / EXCERPT: Let $\mathcal{F}=\{\Phi_M\}_{M=2}^\infty$ be a family of ensembles $\Phi_M=\{\varphi_{M;n}\}_{n=1}^{2M}\subseteq\mathbb{C}^M$ which yield almost injective intensity measurements an...

conjecture | open | MathDB 页面 | 原文线索

V2-057623 | MATHDB-326070

Arborization conjecture for forest forms of cluster monomials

摘录 / EXCERPT: Arborization conjecture. The forest form of a cluster monomial can be found from the web form via the arborization algorithm.

conjecture | open | MathDB 页面 | 原文线索

V2-057624 | MATHDB-326075

First-order necessary conditions under constant Jacobian rank for structured low-rank approximation

摘录 / EXCERPT: Let (P_*,L_*) be a feasible limit point of the sequence generated by the structured low-rank approximation algorithm, and let the Jacobian of the constraint mapping at (P,L) ha...

conjecture | open | MathDB 页面 | 原文线索

V2-057625 | MATHDB-326241

CoNP-completeness of balanced skew partition detection

摘录 / EXCERPT: Let BSP denote the decision problem whose instance is a graph G and whose question is whether G has a balanced skew partition. The balanced skew partition conj...

conjecture | open | MathDB 页面 | 原文线索

V2-057626 | MATHDB-326242

Polynomial-time detection of balanced skew partitions in Berge graphs

摘录 / EXCERPT: A Berge graph is a finite simple graph with no odd hole and no odd antihole, and a balanced skew partition is a skew partition satisfying the balance conditions for its anticomplet...

conjecture | open | MathDB 页面 | 原文线索

V2-057627 | MATHDB-326275

Sparsity conjecture for the cryo-EM covariance matrix blocks

摘录 / EXCERPT: Let $\hat L^{k_1,k_2}$ be a block of the matrix $\hat L$ arising in the covariance-based cryo-electron microscopy algorithm, with k_1,k_2 its associated frequency indices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-057628 | MATHDB-326374

Polynomial-time computation of the hull number of atomistic lattices

摘录 / EXCERPT: Atomistic-lattice hull-number conjecture. The minimum size of a subset $H\subseteq A(\mathcal{L})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057629 | MATHDB-326375

Gem-complexity bound for closed orientable 3-manifolds

摘录 / EXCERPT: Gem-complexity bound. One has

conjecture | open | MathDB 页面 | 原文线索

V2-057630 | MATHDB-326376

Gem-complexity formula for lens spaces

摘录 / EXCERPT: Lens-space gem-complexity conjecture. For every $p\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-057631 | MATHDB-326377

Equality between gem-complexity and Matveev complexity for lens spaces

摘录 / EXCERPT: Lens-space complexity equality conjecture. For every $p\ge 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-057632 | MATHDB-326379

The nontrivial-invariant analogue of Valiant's conjecture

摘录 / EXCERPT: Let $\mathbf{k}=\mathbb{R}$ or $\mathbb{C}$, and let $K_0=K_0(\operatorname{sa}_\mathbb{R})$ when $\mathbf{k}=\mathbb{R}$ and $K_0=K_0(\operatorname{Var}_\mathbb{C})$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-057633 | MATHDB-326381

Strictness conjecture for constructible-sheaf quantifier levels

摘录 / EXCERPT: For each positive integer p, let $\boldsymbol{\Lambda}^{(p)}\boldsymbol{\mathcal{P}}_\mathbb{R}$ denote the p-th sheaf-theoretic quantifier level built from...

conjecture | open | MathDB 页面 | 原文线索

V2-057634 | MATHDB-326382

Sheaf-theoretic Toda conjecture

摘录 / EXCERPT: For a sequence of constructible sheaves in the real sheaf-theoretic polynomial hierarchy $\boldsymbol{\mathcal{PH}}_\mathbb{R}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-057635 | MATHDB-326438

The tree-width computation problem for switch cographs

摘录 / EXCERPT: A switch cograph is a graph in the class discussed above, obtained through the switch-cograph decomposition; let its tree-width be the minimum width of a tree decomposition of the...

conjecture | open | MathDB 页面 | 原文线索

V2-057636 | MATHDB-326472

NP-hardness conjecture for continuous-time switching stabilizability

摘录 / EXCERPT: A continuous-time system with switching uncertainty consists of a system whose uncertain matrix may switch among a given set of matrices. The fixed-uncertainty reduction uses the m...

conjecture | open | MathDB 页面 | 原文线索

V2-057637 | MATHDB-326610

Pisot eigenvalue conjecture for periods of the multidimensional slow continued fraction algorithm

摘录 / EXCERPT: Let K be a real cubic field, and let $\alpha^{(0)},\alpha^{(1)},\alpha^{(2)}$ be its positive $\mathbb{Q}$-basis. Set ... Let $\{\varepsilon_n\}_{n=0}^{\infty}$ be the expansion of...

conjecture | open | MathDB 页面 | 原文线索

V2-057638 | MATHDB-326671

NP-hardness of testing IC-planarity and NIC-planarity

摘录 / EXCERPT: A graph is IC-planar if it has a plane drawing in which no two crossings share an endpoint, and NIC-planar if it has a plane drawing in which any two crossings share at most one en...

conjecture | open | MathDB 页面 | 原文线索

V2-057639 | MATHDB-326673

The anticipation-scaling conjecture for wager sets

摘录 / EXCERPT: Let $A,B \subset \mathbb R_+$ be sets of nonnegative real numbers. The set B anticipates A when the regular gamblers' wagers from B can be evaded against gambler 0's wagers...

conjecture | open | MathDB 页面 | 原文线索

V2-057640 | MATHDB-327232

Computational hardness of submatrix localization below the convexified MLE threshold

摘录 / EXCERPT: Let Y^* denote the true support matrix in the submatrix localization model. Assume $\mu\le 1$, $n_L=n_R=n$, $K_L=K_R=K$, and $K\ge c_1\log n$. Submatrix localization computationa...

conjecture | open | MathDB 页面 | 原文线索

V2-057641 | MATHDB-327388

The quarter-bias conjecture for bit-commitment coin-flipping protocols

摘录 / EXCERPT: Let a strong coin-flipping protocol based on bit-commitment be any protocol in the family formally defined in Section, and let its bias be the maximum excess of either party's chea...

conjecture | open | MathDB 页面 | 原文线索

V2-057642 | MATHDB-327570

Large zero rectangles in low-rank sparse matrices

摘录 / EXCERPT: Let M be an $n\times n$ real matrix with $\operatorname{rank}(M)=r$, and suppose that at most εn^2 entries are nonzero. Large-zero-rectangle conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-057643 | MATHDB-327575

Product-criterion coverage conjecture for same-index signatures

摘录 / EXCERPT: Let $\mathcal{R}^m$ be the module of labeled polynomials, let s denote the signature, and let $\boldsymbol{e}_k$ be the kth module basis vector. Assume the posit...

conjecture | open | MathDB 页面 | 原文线索

V2-057644 | MATHDB-327586

Mulmuley's #P conjecture for Kronecker coefficients

摘录 / EXCERPT: For partitions λ, μ, and ν of the same integer, let $g(\lambda,\mu,\nu)$ denote the Kronecker coefficient, and let ${\textsc{Kron}}$ be the problem of computing thi...

conjecture | open | MathDB 页面 | 原文线索

V2-057645 | MATHDB-327587

Strong #P-hardness conjecture for Kronecker and Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $g(\lambda,\mu,\nu)$ denote Kronecker coefficients and $c^\alpha_{\beta\gamma}$ denote Littlewood-Richardson coefficients. Strong hardness conjecture. The Kronecker coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-057646 | MATHDB-327717

The dichotomy conjecture for directed graph homomorphism problems

摘录 / EXCERPT: Dichotomy conjecture. The H-colouring problem is polynomial-time solvable or NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057647 | MATHDB-327718

Bang-Jensen-Hell conjecture for directed graph homomorphism problems

摘录 / EXCERPT: Bang-Jensen-Hell conjecture. If the core of H is a directed cycle, then the H-colouring problem is polynomial-time solvable. Otherwise, it is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057648 | MATHDB-327843

Sparse Gaussian mean estimation communication-risk conjecture

摘录 / EXCERPT: Sparse estimation tradeoff conjecture. If some protocol estimates the mean for any distribution $P\in\mathcal{P}_s$ with mean-squared loss R and communication cost C, then

conjecture | open | MathDB 页面 | 原文线索

V2-057649 | MATHDB-327914

The symmetric Strassen additivity conjecture for Waring rank

摘录 / EXCERPT: Symmetric Strassen additivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057650 | MATHDB-328067

PSPACE-completeness conjecture for the on-line chromatic number

摘录 / EXCERPT: Let G be a graph and let $k\in\mathbb{N}$. The on-line chromatic number conjecture. Deciding whether ... is textsc{PSPACE}-complete. The complexity of deciding whether the on-line...

conjecture | open | MathDB 页面 | 原文线索

V2-057651 | MATHDB-328199

Larose-Tesson characterization conjecture for logspace CSPs

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite relational structure that is a core, and let $\operatorname{CSP}(\mathbb{A})$ denote the problem of deciding whether an input structure admits a homomo...

conjecture | open | MathDB 页面 | 原文线索

V2-057652 | MATHDB-328296

The CSP dichotomy conjecture

摘录 / EXCERPT: The CSP dichotomy conjecture. For every finite relational structure $\mathbb A$, $\mathrm{CSP}(\mathbb A)$ is in P or NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057653 | MATHDB-328533

Non-adaptive simulation impossibility conjecture for irregular communication orders

摘录 / EXCERPT: Let Pi be an interactive protocol with a sufficiently non-regular, for example pseudo-random, communication order, and let Pi' be a non-adaptive protocol whose communication or...

conjecture | open | MathDB 页面 | 原文线索

V2-057654 | MATHDB-328568

The strong-versus-Hippocrates energy randomness conjecture

摘录 / EXCERPT: A real is strongly $gamma$-random if it is strongly random with parameter $gamma$, and Hippocrates $gamma$-energy random if it satisfies the Hippocrates energy-randomness notion wi...

conjecture | open | MathDB 页面 | 原文线索

V2-057655 | MATHDB-328698

The right upper bound conjecture for automatic complexity structure functions

摘录 / EXCERPT: Let $h_x(m)$ be the minimum number of states of a nondeterministic finite automaton accepting x and at most b^m strings of length $\lvert x\rvert$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-057656 | MATHDB-328792

The Blum-Floyd-Pratt-Rivest-Tarjan selection conjecture for groups of 3 or 4

摘录 / EXCERPT: Consider the deterministic selection algorithm introduced by Blum et al., which repeatedly partitions the input into groups of a fixed size and recursively selects a pivot from the...

conjecture | open | MathDB 页面 | 原文线索

V2-057657 | MATHDB-328908

Rivest--Vuillemin's generalized Aanderaa--Rosenberg conjecture

摘录 / EXCERPT: Let E be a finite ground set, let $\mathcal{F}\subseteq 2^E$ be a set system, and let $G\subseteq \operatorname{Symm}_E$ be a symmetry group acting transitively on E. Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-057658 | MATHDB-329149

Unique-lineage conjecture for the two-dimensional iterated function system

摘录 / EXCERPT: Let α be the parameter described in the preceding construction, and consider the branching of Algorithm 2d IFS, whose children are labelled by L, R, and M. A child is...

conjecture | open | MathDB 页面 | 原文线索

V2-057659 | MATHDB-329332

Universal upper bound conjecture for slack matrices of polygons

摘录 / EXCERPT: Let S_n be the slack matrix of any n-gon, and let $\lfloor x\rfloor$ denote the largest integer not exceeding x. Polygon slack-matrix upper-bound conjecture. ... and equality h...

conjecture | open | MathDB 页面 | 原文线索

V2-057660 | MATHDB-329377

Algorithmic conjecture for edge-dominating cycles in ℓK_2-free graphs

摘录 / EXCERPT: Algorithmic edge-dominating-cycle conjecture. For every fixed $\ell\geq 2$, there is a polynomial-time algorithm that finds an edge-dominating cycle in every $(\ell-1)$-connected,...

conjecture | open | MathDB 页面 | 原文线索

V2-057661 | MATHDB-329563

Existence of a universal lower-semicomputable semi-density matrix

摘录 / EXCERPT: Universal semi-density matrix conjecture. There is a lower-semicomputable semi-density matrix μ dominating every other such matrix: for every lower-semicomputable semi-density...

conjecture | open | MathDB 页面 | 原文线索

V2-057662 | MATHDB-329564

Entropy representation by prefix Kolmogorov complexity

摘录 / EXCERPT: Entropy-complexity conjecture. For $H=\overline{H}$ or $H=\underline{H}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-057663 | MATHDB-329754

Logarithmic lower-bound conjecture for monotonicity testing in the EVAL model

摘录 / EXCERPT: Logarithmic lower-bound conjecture. Monotonicity testing in the ${\sf EVAL}$ model has query complexity

conjecture | open | MathDB 页面 | 原文线索

V2-057664 | MATHDB-329823

The compilation-equivalence characterization conjecture for microcosms

摘录 / EXCERPT: Compilation-equivalence characterization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-057665 | MATHDB-329995

Cycle conjecture for exact skew-symmetric permutations

摘录 / EXCERPT: Cycle conjecture. The elements n and s_n are in the same cycle of π.

conjecture | open | MathDB 页面 | 原文线索

V2-057666 | MATHDB-330210

Conjecture that the two box-move algorithms produce the same sequence

摘录 / EXCERPT: Let X and Z be LR-fillings of the same type, with the skew shape $\beta\setminus\gamma$ a rook strip and $Z<_{\sf dom}X$. The first and second algorithms produce LR-fillings...

conjecture | open | MathDB 页面 | 原文线索

V2-057667 | MATHDB-330396

Polynomial randomness implies rational normality

摘录 / EXCERPT: A real x is polynomially random if it satisfies the polynomial-time randomness notion used in the source, and it is rationally normal if it is normal in every rational base $r>1$...

conjecture | open | MathDB 页面 | 原文线索

V2-057668 | MATHDB-330397

Polynomial randomness implies uniform distribution for polynomial-time integer sequences

摘录 / EXCERPT: Let $(a_i)_{i\in\mathbb{N}}$ be a sequence of distinct integers such that the function mapping the unary input 1^j to a_j is polynomial-time computable. For a real y, write...

conjecture | open | MathDB 页面 | 原文线索

V2-057669 | MATHDB-330457

The forbidden-subgraph characterization conjecture for dominating induced matchings

摘录 / EXCERPT: Forbidden-subgraph characterization conjecture. Unless $P=NP$, the dominating induced matching problem is polynomial-time solvable in the class of M-free graphs if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-057670 | MATHDB-330481

Bounded deficiency and unbounded price conjecture for nondeterministic automatic complexity

摘录 / EXCERPT: Let A_N denote nondeterministic automatic complexity. For a random infinite binary sequence, let D_n be the nondeterministic automatic complexity deficiency of its length-n p...

conjecture | open | MathDB 页面 | 原文线索

V2-057671 | MATHDB-330575

Parity problem for pattern-avoidance sequences

摘录 / EXCERPT: For a finite set of permutation patterns $\mathcal{F}$, let $C_n(\mathcal{F})$ denote the number of permutations in S_n avoiding all patterns in $\mathcal{F}$. Parity problem. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-057672 | MATHDB-330868

The dichotomy conjecture for constraint satisfaction problems of reducts

摘录 / EXCERPT: Dichotomy conjecture. $\operatorname{CSP}(\Gamma)$ is either in P or NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057673 | MATHDB-330989

Tarjan's deque conjecture for online binary search trees

摘录 / EXCERPT: Let ${\mathcal{A}}$ be an online binary search tree algorithm. Starting with any initial tree T with n elements, consider inserting or deleting the current minimum or maximum e...

conjecture | open | MathDB 页面 | 原文线索

V2-057674 | MATHDB-330990

Lucier's split conjecture for online binary search trees

摘录 / EXCERPT: Let ${\mathcal{A}}$ be an online binary search tree algorithm. Starting with any initial tree T with n elements, consider any sequence of n splits. A split at an element x...

conjecture | open | MathDB 页面 | 原文线索

V2-057675 | MATHDB-330991

Path conjecture for preorder access in binary search trees

摘录 / EXCERPT: Let ${\mathcal{A}}$ be an online binary search tree algorithm. Starting with any initial tree T with n elements, let X be the preorder sequence of a binary search tree T' t...

conjecture | open | MathDB 页面 | 原文线索

V2-057676 | MATHDB-331242

Effective virtual Haken conjecture

摘录 / EXCERPT: Let N be a closed hyperbolic 3-manifold. A finite-sheeted Haken covering of N is a finite-sheeted covering space of N that is Haken. Effective virtual Haken conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-057677 | MATHDB-331243

ER mutual-refinement conjecture for triangulations of closed 3-manifolds

摘录 / EXCERPT: Let M be a closed 3-manifold, and let τ_1 and τ_2 be triangulations of M. A mutual refinement is a triangulation refining both τ_1 and τ_2. ER mutual-refine...

conjecture | open | MathDB 页面 | 原文线索

V2-057678 | MATHDB-331477

Residual-error conjecture for the recursive tridiagonal inversion algorithm

摘录 / EXCERPT: Let $A\in\mathbb{R}^{n\times n}$ be a nonsingular matrix whose inverse has only nonzero elements in the lower triangle. Let X be the inverse computed numerically by algorithm (KW...

conjecture | open | MathDB 页面 | 原文线索

V2-057679 | MATHDB-331596

The core-free finite tractability conjecture via height-1 clone homomorphisms

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite relational structure, and let $\mathscr{A}$ be its polymorphism clone. Let $\mathbf{1}$ be the clone of projections on a two-element set, and let an h1...

conjecture | open | MathDB 页面 | 原文线索

V2-057680 | MATHDB-331621

Feder-Vardi dichotomy conjecture for constraint satisfaction problems

摘录 / EXCERPT: Feder-Vardi dichotomy conjecture. For any relational structure T, $\textsc{Csp}(T)$ is either NP-complete or polynomial-time solvable.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057681 | MATHDB-331622

Full dichotomy conjecture for signed graph s-homomorphisms

摘录 / EXCERPT: Full dichotomy conjecture for signed graph s-homomorphisms. Let (H,Π) be a connected signed graph. Then, $\textsc{s-Hom}(H,\Pi)$ is polynomial-time solvable if the s-core of...

conjecture | open | MathDB 页面 | 原文线索

V2-057682 | MATHDB-331717

Asymptotic best-variable branching conjecture for MVB

摘录 / EXCERPT: Asymptotic best-variable branching conjecture. For each instance of MVB, there exists a gap H such that, for all gaps greater than H, variable i is always branched on at the...

conjecture | open | MathDB 页面 | 原文线索

V2-057683 | MATHDB-331966

Termination conjecture for the refined Disk Tree algorithm

摘录 / EXCERPT: Let G be a graph with tree-length $\mathrm{tl}(G)$, and let (T_i,ϕ_i), H_i, S_i, $B_k(x_i)$, and property P_i be as in the (k,ℓ)-Disk Tree algorithm. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-057684 | MATHDB-331990

Kernelization hardness for crossing number with prescribed rotation systems

摘录 / EXCERPT: Let G be a graph with a given rotation system, and let $k\geq 1$ be an integer. A drawing of G respecting the prescribed rotation system is one in which the clockwise order of...

conjecture | open | MathDB 页面 | 原文线索

V2-057685 | MATHDB-332031

The VP_s versus VNP separation conjecture

摘录 / EXCERPT: Let $\textup{\textsf{VP$s$}}$ denote the class of polynomial-size algebraic circuits with bounded degree, and let $\textup{\textsf{VNP}}$ denote Valiant's class of efficiently defi...

conjecture | open | MathDB 页面 | 原文线索

V2-057686 | MATHDB-332103

A local evacuation-shuffling algorithm for arbitrary skew tableaux

摘录 / EXCERPT: Let T be any (semi)standard skew tableau and let $\scalebox{.5}{\young(x)}$ be an inner co-corner of T. Write esh for evacuation-shuffling,...

open_problem | open | MathDB 页面 | 原文线索

V2-057687 | MATHDB-332132

Residual subgradient norm bound for the proximal point algorithm

摘录 / EXCERPT: Let $\left\{\alpha_k\right\}_k$ be a sequence of positive step sizes and let x_0 be an initial iterate satisfying ... for some optimal point x_*. Let $\left\{x_k\right\}_k$ be an...

conjecture | open | MathDB 页面 | 原文线索

V2-057688 | MATHDB-332212

Consistency-provability strengthening conjecture

摘录 / EXCERPT: Let $\mathcal T$ be the class of theories under consideration, and let Con_S mean that S is consistent. Consistency-provability strengthening conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-057689 | MATHDB-332213

Nonuniform reflection incompleteness conjecture

摘录 / EXCERPT: Let $\mathcal T$ be the class of theories under consideration, and let $\Sigma^b_1RFN_T(\bar n)$ denote the finite Σ^b_1 reflection principle for T. Nonuniform reflection...

conjecture | open | MathDB 页面 | 原文线索

V2-057690 | MATHDB-332214

TFNP non-completeness conjecture

摘录 / EXCERPT: Let $\mathcal T$ be the class of theories under consideration. A TFNP problem is a total polynomial search problem, and $\operatorname{TFNP}(T)$ is the class of such problems prova...

conjecture | open | MathDB 页面 | 原文线索

V2-057691 | MATHDB-332215

No complete TFNP problem conjecture

摘录 / EXCERPT: A TFNP problem is a total polynomial search problem, and polynomial reduction is the usual polynomial-time reduction between search problems. No complete TFNP problem conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-057692 | MATHDB-332216

Herbrand consistency search form of the TFNP conjecture

摘录 / EXCERPT: Let $\mathcal T$ be the class of theories under consideration. For a consistent universal sentence Φ, let $HCS(\Phi)$ be its Herbrand Consistency Search problem: given finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-057693 | MATHDB-332218

No complete disjoint NP-pair conjecture

摘录 / EXCERPT: A disjoint NP pair is a pair (A,B) of disjoint languages in NP, and polynomial reduction between pairs maps the first component to the first and the second to the second. No comp...

conjecture | open | MathDB 页面 | 原文线索

V2-057694 | MATHDB-332219

No complete disjoint coNP-pair conjecture

摘录 / EXCERPT: A disjoint coNP pair is a pair (A,B) of disjoint languages in coNP, with polynomial reductions defined componentwise. No complete disjoint coNP-pair conjecture. There exist no co...

conjecture | open | MathDB 页面 | 原文线索

V2-057695 | MATHDB-332220

No p-optimal propositional proof system conjecture

摘录 / EXCERPT: A p-optimal proof system is a propositional proof system that polynomially simulates every propositional proof system. No p-optimal proof system conjecture. There exists no p-optim...

conjecture | open | MathDB 页面 | 原文线索

V2-057696 | MATHDB-332221

No complete UP set conjecture

摘录 / EXCERPT: The class UP consists of languages accepted by polynomial-time nondeterministic Turing machines having a unique accepting computation on every accepted input. No complete UP set co...

conjecture | open | MathDB 页面 | 原文线索

V2-057697 | MATHDB-332222

No complete NP-intersection-coNP set conjecture

摘录 / EXCERPT: A language is in $\mathbf{NP}\cap\mathbf{coNP}$ when it belongs to both NP and coNP, and completeness is with respect to polynomial-time many-one reductions. No complete...

conjecture | open | MathDB 页面 | 原文线索

V2-057698 | MATHDB-332490

Convergence of the Sinkhorn orthogonalization algorithm

摘录 / EXCERPT: Convergence conjecture. The maps Φ and Ψ increase the volume and, after an infinite number of steps, respectively land in K_N and X_N. This conjecture would imply con...

conjecture | open | MathDB 页面 | 原文线索

V2-057699 | MATHDB-332712

A 2-plus-epsilon approximation for precedence-constrained single-machine scheduling with release dates

摘录 / EXCERPT: Approximation conjecture. For any $\varepsilon>0$, there is a $(2+\varepsilon)$-approximation algorithm for

conjecture | open | MathDB 页面 | 原文线索

V2-057700 | MATHDB-332756

The GW effective-algorithm conjecture for quintic threefolds

摘录 / EXCERPT: GW effective-algorithm conjecture. By choosing d_0 and d_∞ positive, these relations provide an effective algorithm to determine all genus GW invariants $N_{g,d}$, provide...

conjecture | open | MathDB 页面 | 原文线索

V2-057701 | MATHDB-332757

The FJRW effective-algorithm conjecture for quintic singularities

摘录 / EXCERPT: FJRW effective-algorithm conjecture. The polynomial relations among the FJRW invariants $\Theta_{g,k}$ obtained from these localization relations, with

conjecture | open | MathDB 页面 | 原文线索

V2-057702 | MATHDB-332758

The low-degree GW algorithm conjecture for quintic threefolds

摘录 / EXCERPT: Low-degree GW algorithm conjecture. Applying the localization vanishing to this data determines $N_{g,d}$ provided that

conjecture | open | MathDB 页面 | 原文线索

V2-057703 | MATHDB-333030

Linear extension complexity conjecture for spanning tree polytopes of proper minor-closed graph families

摘录 / EXCERPT: Let $\mathcal{C}$ be a proper minor-closed family of graphs, and let $G=(V,E)$ be a connected graph in $\mathcal{C}$. Let $\operatorname{P_{\mathrm{sp.trees}}}(G)$ denote the conve...

conjecture | open | MathDB 页面 | 原文线索

V2-057704 | MATHDB-333155

Criterion of NP-hardness for properly posed problems

摘录 / EXCERPT: Let $({\cal P})$ be a properly posed problem, and let ${\cal S}^+_c$ denote the positive canonical dual feasible set; write $\operatorname{int}{\cal S}^+_c$ for its interior. Crite...

conjecture | open | MathDB 页面 | 原文线索

V2-057705 | MATHDB-333626

Non-context-free co-language conjecture for maximally complex words

摘录 / EXCERPT: Let L_3 be the language of maximally complex words for nondeterministic automatic complexity over the alphabet ${0,1,2}$, and let coCFL denote the complements of...

conjecture | open | MathDB 页面 | 原文线索

V2-057706 | MATHDB-333672

The BoxContZonotope NP-hardness conjecture

摘录 / EXCERPT: Let $a_1,\ldots,a_m\in\mathbb{Q}^n$ and let $\nu\in[0,\infty)\cap\mathbb{Q}$. BoxContZonotope conjecture. Deciding whether there exists a rectangular box B such that ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-057707 | MATHDB-333725

Polynomial-time computation of fixed points and spectral norms for nonsingular symmetric tensors

摘录 / EXCERPT: Fix $n\in\mathord{\mathbb N}$ with $2<n$, and let ${\cal S}\in\mathrm{S}^d\mathord{\mathbb C}^n$ be nonsingular. Let $\mathbf H$ denote the polynomial map whose fixed points are us...

conjecture | open | MathDB 页面 | 原文线索

V2-057708 | MATHDB-333990

Valiant's VP versus VNP conjecture

摘录 / EXCERPT: Let $\mathfrak S_m$ denote the permutation group on m elements, let y^i_j be linear coordinates on $\mathbb C^{m^2}$, and define the permanent by ... A polynomial-size circuit is...

conjecture | open | MathDB 页面 | 原文线索

V2-057709 | MATHDB-334506

The bounded width dichotomy conjecture for constraint satisfaction problems

摘录 / EXCERPT: Let $\mathbb{A}=\langle A;\mathscr{R}\rangle$ be a finite core relational structure of finite relational signature. A polymorphism of $\mathbb{A}$ is an operation that is a homomor...

conjecture | open | MathDB 页面 | 原文线索

V2-057710 | MATHDB-334557

Algebraic CSP dichotomy conjecture

摘录 / EXCERPT: Let $\bA$ be an algebra with a Taylor term in its clone, and let R be a finite set of relations compatible with $\bA$. Write $CSP(R)$ for the constraint satisfaction problem over...

conjecture | open | MathDB 页面 | 原文线索

V2-057711 | MATHDB-334579

The hard-but-detectable conjecture for sparse PCA

摘录 / EXCERPT: Let X have prior distribution P_0, and consider the rank-one symmetric matrix estimation model ... where $Z_{i,j}$ are standard Gaussian noises. Assume $\mathbb{E}_{P_0}[X]=0$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-057712 | MATHDB-334580

The conjectured optimal efficient mean square error for symmetric low-rank estimation

摘录 / EXCERPT: Let X have prior distribution P_0, and consider the symmetric rank-one matrix estimation model with signal-to-noise parameter λ. For $\eta>0$, define ... and set ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-057713 | MATHDB-334887

The extended half-space algorithm conjecture for NET maps

摘录 / EXCERPT: Extended half-space algorithm conjecture. The extended half-space algorithm decides, in finite time, whether or not f is equivalent to a rational function.

conjecture | open | MathDB 页面 | 原文线索

V2-057714 | MATHDB-334890

The large-scale performance conjecture for linear superiorization

摘录 / EXCERPT: Large-scale performance conjecture. There exists a size level of huge problems above which LinSup will perform better than linear minimization algorithms.

conjecture | open | MathDB 页面 | 原文线索

V2-057715 | MATHDB-334920

The homomorphic-equivalence CSP dichotomy conjecture

摘录 / EXCERPT: Homomorphic-equivalence CSP dichotomy conjecture. One of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-057716 | MATHDB-334936

The nowhere-dense domination-set kernelization dichotomy

摘录 / EXCERPT: The nowhere-dense domination-set kernelization dichotomy. If $\mathcal{C}$ is nowhere dense, then for each $r\in\mathbb{N}$, the textsc{Distance-r Dominating Set} problem admits...

conjecture | open | MathDB 页面 | 原文线索

V2-057717 | MATHDB-334937

The neighborhood-complexity characterization of nowhere-dense classes

摘录 / EXCERPT: The neighborhood-complexity characterization. The class $\mathcal{C}$ is nowhere dense if and only if there exists a function $f_{\mathrm{nei}}(r,\varepsilon)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057718 | MATHDB-335039

The shifted Coxeter-Knuth insertion conjecture for involution words

摘录 / EXCERPT: Let an involution word be a word in the generators of the relevant involution-Hecke monoid, and let shifted Coxeter-Knuth equivalence be the equivalence relation generated by the f...

conjecture | open | MathDB 页面 | 原文线索

V2-057719 | MATHDB-335154

The Kakeya conjecture for polynomial Wolff axioms

摘录 / EXCERPT: Kakeya conjecture for the polynomial Wolff axioms. For every dimension n, there is a complexity E such that, whenever $\mathbb{T}$ is a set of δ-tubes in $\mathbb{R}^n$...

conjecture | open | MathDB 页面 | 原文线索

V2-057720 | MATHDB-335377

Monotonicity conjecture for communication complexity under user-set changes

摘录 / EXCERPT: Let $R_{\op{S}}$ be the minimum communication rate required to achieve the secrecy capacity in the secret-key agreement model. A trusted helper is a user whose source is trusted an...

conjecture | open | MathDB 页面 | 原文线索

V2-057721 | MATHDB-335817

The linear-time conjecture for detecting rooted $K_4(X)$ or $W_4(X)$ minors

摘录 / EXCERPT: Let G be a graph and let $K_4(X)$- and $W_4(X)$-minors denote the rooted minor notions used in the source. The rooted $K_4(X)$-$W_4(X)$ recognition conjecture. There is a linear...

conjecture | open | MathDB 页面 | 原文线索

V2-057722 | MATHDB-336200

Convergence conjecture for the coordinate-descent PARAFAC algorithm

摘录 / EXCERPT: Let $\{\mathbf{A}^k\}$ be the sequence generated by Algorithm, where $\mathbf{A}^k$ is the solution $(\mathbf{A}^{(1)},\mathbf{A}^{(2)},\ldots,\mathbf{A}^{(N)})$ in the k-th iter...

conjecture | open | MathDB 页面 | 原文线索

V2-057723 | MATHDB-336526

Polynomial-time convergence conjecture for non-increasing edge recoloring

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let Φ be the potential function on colorings of $E(K_n)$ with $n-1$ colors. Consider the process that starts from a random...

conjecture | open | MathDB 页面 | 原文线索

V2-057724 | MATHDB-336750

Nonexistence conjecture for intermediate-degree monomial GAPN functions

摘录 / EXCERPT: Nonexistence conjecture. For sufficiently large n, there are no monomial GAPN functions f_d on $\mathbb{F}_{p^n}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-057725 | MATHDB-337315

The linear parallel-runtime conjecture for the step-wise odd-even algorithm

摘录 / EXCERPT: Consider the step-wise odd-even algorithm depicted in Figure, and let N denote the number of elements being sorted. Its sequential worst-case runtime is compared with Bubble Sort...

conjecture | open | MathDB 页面 | 原文线索

V2-057726 | MATHDB-337757

Conjecture on an optimal nonuniform partition for Pascal matrix recursion

摘录 / EXCERPT: Let $n \gg k > 2$, and let $\lambda_1,\ldots,\lambda_k \in \mathbb{Q}$ satisfy $\lambda_i>0$ for each i and ... Set $n_i=\lambda_i n$ for each i, so that the λ_i determin...

conjecture | open | MathDB 页面 | 原文线索

V2-057727 | MATHDB-337762

Characterization of inverse M-matrix interval systems

摘录 / EXCERPT: Inverse M-matrix characterization. $\mathbf A$ is an inverse M-matrix if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-057728 | MATHDB-337838

NP-hardness conjecture for deciding deep holes of Gabidulin codes

摘录 / EXCERPT: Let a Gabidulin code be given together with a received word. A received word is a deep hole if its distance from the code equals the covering radius. NP-hardness conjecture. Decidi...

conjecture | open | MathDB 页面 | 原文线索

V2-057729 | MATHDB-337920

Conjectured linear complexity of generalized cyclotomic binary sequences

摘录 / EXCERPT: Linear-complexity conjecture. If $2^{p-1}\not\equiv 1\pmod {p^2}$, then the linear complexity L of s^∞ is

conjecture | open | MathDB 页面 | 原文线索

V2-057730 | MATHDB-338051

Inverse-free Solovay-Kitaev theorem

摘录 / EXCERPT: Let $d\geq 2$ be fixed, and let $\mathcal{G}\subset \mathrm{SU}(d)$ be a finite gate set that densely generates $\mathrm{SU}(d)$. An inverse-free Solovay-Kitaev theorem asserts tha...

conjecture | open | MathDB 页面 | 原文线索

V2-057731 | MATHDB-338062

The depth bound conjecture for conflict-free connection colorings of trees

摘录 / EXCERPT: Let T be a tree. Let $D(T)$ be the minimum depth produced by Algorithm 1 over all possible priorities for removing edges of T: ... Here $cfc(T)$ denotes the conflict-free conne...

conjecture | open | MathDB 页面 | 原文线索

V2-057732 | MATHDB-338166

Benjamini-Kalai-Schramm noise-stability conjectures for threshold circuits

摘录 / EXCERPT: Let f be a Boolean function, and let M and D denote the size and depth of a threshold circuit representing it. A circuit is monotone threshold when its threshold gates have n...

conjecture | open | MathDB 页面 | 原文线索

V2-057733 | MATHDB-338167

Low-influence approximation conjecture for Boolean functions

摘录 / EXCERPT: For a Boolean function f, define its total influence by ... A Boolean function g ϵ-approximates f when $\mathbb E(f-g)^2\le\epsilon$. Low-influence approximation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-057734 | MATHDB-338396

The additive-complexity regularity conjecture for automatic sequences

摘录 / EXCERPT: Additive-complexity regularity conjecture. The sequence $(\rho^+_{\mathbf{w}}(n))_{n\geq 1}$ is k-regular.

conjecture | open | MathDB 页面 | 原文线索

V2-057735 | MATHDB-338667

Quadratic-time partition conjecture for noncomplete connected graphs

摘录 / EXCERPT: Let G be a connected graph with maximum degree $k \geq 3$, distinct from K_{k+1}. For integers $s \geq 2$ and $p_1,\ldots,p_s \geq 0$ satisfying ... a $(p_1,\ldots,p_s)$-partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-057736 | MATHDB-338731

The dense subgraph detection conjecture

摘录 / EXCERPT: The dense subgraph detection conjecture. For any sequence of randomized polynomial-time tests $(\psi_n)_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-057737 | MATHDB-338999

The conjecture on the order types of initial segments of addition chain defects

摘录 / EXCERPT: Order-type conjecture for addition chain defects. For each whole number k, the set $\mathscr{D}^\ell\cap[0,k]$ has order type ω^k.

conjecture | open | MathDB 页面 | 原文线索

V2-057738 | MATHDB-339062

Waldschmidt's conjecture on effective complexity bounds for numeration expansions

摘录 / EXCERPT: Let G and H be expansions whose corresponding recurrence relations have multiplicatively independent dominant roots. Let $H_G(n)$ and $H_H(n)$ denote the relevant expansion com...

conjecture | open | MathDB 页面 | 原文线索

V2-057739 | MATHDB-339197

Undecidability conjecture for three-block Presburger arithmetic with non-quadratic scalar multiplication

摘录 / EXCERPT: Let α be a non-quadratic scalar. Undecidability conjecture. α-Presburger arithmetic sentences with three alternating blocks of quantifiers are undecidable. The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-057740 | MATHDB-339403

NP-completeness conjecture for the strong geodetic set problem on complete multipartite graphs

摘录 / EXCERPT: Strong geodetic set complexity conjecture. The strong geodetic set problem restricted to complete multipartite graphs is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057741 | MATHDB-339648

Subdivision conjecture for the MTP2 log-concave maximum likelihood estimator

摘录 / EXCERPT: Let $X\subseteq\mathbb{R}^d$ be the sample configuration, let $X'\supseteq X$ and let Δ' be the finite set and bimonotone subdivision produced by the paper's algorithm. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-057742 | MATHDB-339726

The more-choice conjecture for two-thinning

摘录 / EXCERPT: Let n bins receive balls in the two-thinning setting, where the overseer may iteratively reject up to k suggested allocations for each ball. More-choice conjecture. The asympto...

conjecture | open | MathDB 页面 | 原文线索

V2-057743 | MATHDB-339727

The more-balls conjecture for two-thinning

摘录 / EXCERPT: Let $m=\Omega(n)$ balls be allocated to n bins in the two-thinning setting, and let the overseer two-thin the allocations. More-balls conjecture. The asymptotically optimal maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-057744 | MATHDB-339737

Finite termination of the extended simplex procedure for non-completely-positive matrices

摘录 / EXCERPT: Finite-termination conjecture. For $A\notin{\mathcal{CP}}^n$, Procedure with a suitable pivot rule in Step2(b) ends after finitely many iterations with a separating witness W.

conjecture | open | MathDB 页面 | 原文线索

V2-057745 | MATHDB-339896

Hereditary NP-completeness of cubic-graph coloring targets

摘录 / EXCERPT: Hereditary-class conjecture. The class $\mathcal{C}_{NP}$ is a hereditary class of cubic graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057746 | MATHDB-339897

The P_{12} dichotomy conjecture for coloring problems

摘录 / EXCERPT: P_{12} dichotomy conjecture. If H admits a P_{12}-coloring, then the H-problem is NP-complete; if H does not admit a P_{12}-coloring, then the H-problem is polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-057747 | MATHDB-340082

Asymptotic gap between monotone and non-monotone formula complexity

摘录 / EXCERPT: Let $a_0=2-1$ and, for $n\geq 0$, let ... Consider formula encodings whose gates are restricted to $\{+,\times,\text{^}\}$ and whose inputs are restricted to $\{-1,1\}$. The asymptot...

conjecture | open | MathDB 页面 | 原文线索

V2-057748 | MATHDB-340168

Gap-ETH

摘录 / EXCERPT: Gap-ETH. There exist constants $\epsilon,c>0$ such that no algorithm running in time $O(2^{cn})$ can distinguish between a satisfiable 3-SAT formula and a 3-SAT formula with at...

conjecture | open | MathDB 页面 | 原文线索

V2-057749 | MATHDB-340253

Polynomial-time recognition of relating edges in graphs with no 6-cycles

摘录 / EXCERPT: Relating-edge recognition conjecture. The following problem is polynomially solvable: given a graph $G \in \mathcal{G}(\widehat{C}_{6})$ and an edge $xy \in E(G)$, determine whethe...

conjecture | open | MathDB 页面 | 原文线索

V2-057750 | MATHDB-340354

Non-unicyclicity conjecture for even-degree compositions

摘录 / EXCERPT: Let n be the degree parameter, and let $\sigma=\sigma_{n-1}\cdots\sigma_0$ be the composition of the permutations associated with the construction. A permutation is unicyclic if...

conjecture | open | MathDB 页面 | 原文线索

V2-057751 | MATHDB-340355

Positive liminf conjecture for unicyclic permutation ratios

摘录 / EXCERPT: Let $\mathcal{I}_n$ be the set of irreducible polynomials of degree n, and let $\mathcal{J}_n$ be the number of irreducible polynomials of degree n such that, for a fixed pertu...

conjecture | open | MathDB 页面 | 原文线索

V2-057752 | MATHDB-340390

Weak recovery threshold in the geometric block model

摘录 / EXCERPT: Weak recovery threshold conjecture. Let $s,t\geq 0$ and suppose $\mathrm{GBM}(n,s,t)$ has a giant component. Weak recovery is efficiently solvable in $\mathrm{GBM}(n,s,t)$ if and o...

conjecture | open | MathDB 页面 | 原文线索

V2-057753 | MATHDB-340391

Powered adjacency eigenvectors achieve weak recovery in the geometric block model

摘录 / EXCERPT: Powered adjacency conjecture. Let $s,t\geq 0$ and suppose $\mathrm{GBM}(n,s,t)$ has a giant component. Taking the second-largest eigenvector of $A^{(r)}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-057754 | MATHDB-340392

Nonbacktracking spectral clustering is not optimally accurate in the geometric block model

摘录 / EXCERPT: Nonbacktracking failure conjecture. Computing the eigenvector of B with the second-largest eigenvalue and dividing the vertices into those with above-median and below-median sums...

conjecture | open | MathDB 页面 | 原文线索

V2-057755 | MATHDB-340393

Powered adjacency spectral clustering is optimally accurate in the geometric block model

摘录 / EXCERPT: Powered adjacency conjecture. For all $m>0$, $s>0$, $t>0$, and $0<r'<1/2$, the algorithm that finds the eigenvector of $A^{(r'd)}$ with the second-largest eigenvalue and then divid...

conjecture | open | MathDB 页面 | 原文线索

V2-057756 | MATHDB-340395

Subpolynomial-power spectral methods fail on dense geometric block models

摘录 / EXCERPT: Subpolynomial-power failure conjecture. There exists $s_0>0$ such that, for all $m>0$, $s\geq s_0$, $t>0$, and $r=n^{o(1)}$, the algorithm that finds the second-largest-eigenvalue...

conjecture | open | MathDB 页面 | 原文线索

V2-057757 | MATHDB-340397

The spectral meta-algorithm is optimally accurate in the geometric block model

摘录 / EXCERPT: Spectral meta-algorithm conjecture. For any s and r such that $\mathrm{GBM}(n,s,t)$ has a giant component, there exists $\delta>1/2$ such that the spectral meta-algorithm recov...

conjecture | open | MathDB 页面 | 原文线索

V2-057758 | MATHDB-340695

Informal supremacy conjecture for random quantum circuits

摘录 / EXCERPT: Informal supremacy conjecture. Approximating to $1/\operatorname{poly}(n)$ multiplicative error most amplitudes of most quantum circuits is a $\#P$-hard problem.

conjecture | open | MathDB 页面 | 原文线索

V2-057759 | MATHDB-340696

Quantum supremacy conjecture for random circuit sampling

摘录 / EXCERPT: Quantum supremacy conjecture. There is no classical randomized algorithm that performs RCS to inverse-polynomial total variation-distance error.

conjecture | open | MathDB 页面 | 原文线索

V2-057760 | MATHDB-340729

Average-case hardness conjecture for approximating the Sherrington-Kirkpatrick partition function

摘录 / EXCERPT: Average-case hardness conjecture. (a) If

open_problem | open | MathDB 页面 | 原文线索

V2-057761 | MATHDB-340930

Existence of a competitive online selector for convex body chasing

摘录 / EXCERPT: Let $d4a6$ denote the family of convex sets in $d53d^d$, let $d4a6^*$ be the set of finite strings with alphabet $d4a6$, and let an online selector be a map $S:d4a6^*\tod53d^d$ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-057762 | MATHDB-341017

The monotonicity conjecture for recognizing generating complete bipartite subgraphs

摘录 / EXCERPT: Let p,q be positive integers with $p\geq 1$ and $q\geq 2$, and let $i\leq p$ and $j\leq q$. Let Ψ be a family of graphs for which recognizing generating subgraphs isomorphic...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057763 | MATHDB-341041

Strassen's support-functional conjecture for tight tensors

摘录 / EXCERPT: Strassen's support-functional conjecture. The asymptotic spectrum of the class of tight tensors coincides with the set of support functionals $\widehat{\zeta}^{\theta}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057764 | MATHDB-341052

Approximation-ratio conjecture for randomized binary sequence design

摘录 / EXCERPT: Approximation-ratio conjecture. The ratio satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-057765 | MATHDB-341242

The asymptotic complexity conjecture for arithmetic expressions

摘录 / EXCERPT: Asymptotic complexity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057766 | MATHDB-341460

The existence of a PSAKS for the Rural Postman Problem parameterized by c

摘录 / EXCERPT: RPP PSAKS conjecture. RPP has a PSAKS with respect to the parameter c.

conjecture | open | MathDB 页面 | 原文线索

V2-057767 | MATHDB-341605

The algorithmic termination conjecture for rational relation numbers

摘录 / EXCERPT: Algorithmic termination conjecture. For all $s,r\in\mathbb{N}$ satisfying $s/r<4$, there exists $M>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057768 | MATHDB-341617

Symplectic Hecke insertion conjecture for FPF-involution words

摘录 / EXCERPT: Let v and w be FPF-involution words, and let $\overset{\textsf{Sp}}\sim$ denote the symplectic Hecke equivalence relation. The symplectic Hecke insertion algorithm assigns to e...

conjecture | open | MathDB 页面 | 原文线索

V2-057769 | MATHDB-341618

Shifted Hecke insertion conjecture for involution words

摘录 / EXCERPT: Let v and w be involution words, and let $\overset{\textsf{O}}\sim$ denote the shifted Hecke equivalence relation. The shifted Hecke insertion algorithm assigns to each word...

conjecture | open | MathDB 页面 | 原文线索

V2-057770 | MATHDB-341642

Existence of cyclotomic aperiodic substitution tilings with dense tile orientations

摘录 / EXCERPT: For each integer $n\geq 2$, consider cyclotomic aperiodic substitution tilings (CASTs) with dense tile orientations (DTO), finite local complexity (FLC), the minimal inflation mult...

conjecture | open | MathDB 页面 | 原文线索

V2-057771 | MATHDB-341900

Informal low-degree likelihood-ratio conjecture

摘录 / EXCERPT: Informal low-degree conjecture. For “nice” distributions $\mathbb{P}_n$ and $\mathbb{Q}_n$, if

conjecture | open | MathDB 页面 | 原文线索

V2-057772 | MATHDB-341929

Subexponential pattern matching for permutations avoiding a fixed pattern

摘录 / EXCERPT: Fixed-pattern avoidance conjecture. The permutation pattern matching problem with text t and pattern π can be solved in time $2^{o(n)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057773 | MATHDB-342002

Improved upper-bound DNF compression conjecture

摘录 / EXCERPT: A DNF is a disjunction of conjunctive terms; its width is the maximum number of literals in a term, its size is its number of terms, and two Boolean functions are ε-clo...

conjecture | open | MathDB 页面 | 原文线索

V2-057774 | MATHDB-342024

Orbit-by-orbit inequality for genomic tableaux in the orthogonal Grassmannian

摘录 / EXCERPT: Let ω be the permutation governing the switching algorithm, and let $\mathcal{O}\in\operatorname{Orb}(\omega)$ be one of its orbits. A genomic tableau is a tableau counted b...

conjecture | open | MathDB 页面 | 原文线索

V2-057775 | MATHDB-342377

Log-free performance conjecture for the even-order Kikuchi algorithms

摘录 / EXCERPT: Fix an integer p and let ℓ be constant independently of n. Log-free performance conjecture. There exists a constant $c_p(\ell)>0$, with $c_p(\ell)\to 0$ as $\ell\to\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-057776 | MATHDB-342378

Odd-order Kikuchi Hessian optimality conjecture

摘录 / EXCERPT: Let p be odd and fix an integer $\ell\in[\lfloor p/2\rfloor,n-\lceil p/2\rceil]$. For $S\subseteq[n]$ with $|S|=\ell$ and $T\subseteq[n]$ with $|T|=\ell+1$, define the rectangula...

conjecture | open | MathDB 页面 | 原文线索

V2-057777 | MATHDB-342510

Hardness conjecture for approximate 3-coloring

摘录 / EXCERPT: Let G be a 3-colorable graph on n nodes. Hardness of approximate 3-coloring. For some fixed $\gamma > 0$, there is no polynomial time algorithm that, given G, returns a v...

conjecture | open | MathDB 页面 | 原文线索

V2-057778 | MATHDB-342533

The induced-subgraph counting hardness conjecture

摘录 / EXCERPT: Induced-subgraph counting hardness conjecture. Then $\#{\textsc{IndSub}}(\Phi)$ is $\#{\mathsf{W[1]}}$-hard.

conjecture | open | MathDB 页面 | 原文线索

V2-057779 | MATHDB-342590

Nonexistence conjecture for non-linear q-nearly bent Boolean functions

摘录 / EXCERPT: Let $n>2$ be any integer and let $q\in {\cal B}_n$. A Boolean function is q-nearly bent if it has the nearly bent property with respect to the q-transform, and $wt(q)$ denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-057780 | MATHDB-342644

Inverse bound conjecture for arbitrary-norm rank-revealing QR factorization

摘录 / EXCERPT: Let $A\in\mathbb{R}^{m \times m}$, let $\\|\cdot\\|$ be a norm on $\mathbb{R}^m$, and let P,Q,R be the factors output by the arbitrary-norm rank-revealing QR algorithm described...

conjecture | open | MathDB 页面 | 原文线索

V2-057781 | MATHDB-342877

Unprovability of P = NP in PV

摘录 / EXCERPT: Let f be a polynomial-time algorithm, viewed as a function symbol of PV, and let ... where $\psi_{\mathsf{SAT}}(x,y)$ expresses that y satisfies the Boolean formula en...

conjecture | open | MathDB 页面 | 原文线索

V2-057782 | MATHDB-342905

Universal limiting profile for successive minimum spanning tree forests

摘录 / EXCERPT: For each $k\geqslant1$, let $F_k(t)$ be the forest produced by Kruskal's algorithm at time t, and let $\rho_k(t)$ be the limiting fraction of vertices in its largest component. T...

conjecture | open | MathDB 页面 | 原文线索

V2-057783 | MATHDB-343107

Chen Conjecture for quantified constraint satisfaction problems

摘录 / EXCERPT: Let Γ be a finite constraint language with all constants. Its polymorphism clone $\operatorname{Pol}(\Gamma)$ has the polynomially generated powers (PGP) property if, for so...

conjecture | open | MathDB 页面 | 原文线索

V2-057784 | MATHDB-343127

The PCSP hardness conjecture for non-bipartite loop-less graphs

摘录 / EXCERPT: The PCSP hardness conjecture. $\operatorname{PCSP}(G,H)$ is NP-hard for every non-bipartite loop-less pair of graphs G,H. Equivalently, every loop-less graph is left-hard, and ev...

conjecture | open | MathDB 页面 | 原文线索

V2-057785 | MATHDB-343143

The coNP conjecture for three-sphere recognition

摘录 / EXCERPT: A triangulation $\mathcal{T}$ is a finite triangulation of a closed 3-manifold, and S^3 recognition asks whether $\mathcal{T}$ triangulates the 3-sphere. The complexity class...

conjecture | open | MathDB 页面 | 原文线索

V2-057786 | MATHDB-343351

The low-degree likelihood-ratio conjecture for computational indistinguishability

摘录 / EXCERPT: Low-degree likelihood-ratio conjecture. If $\|L_n^{\leq D(n)}\|_{L^2(\mathbb{Q}_n)}$ remains bounded as $n\to\infty$ whenever $D(n)\leq t(n)\cdot\operatorname{polylog}(n)$, then th...

conjecture | open | MathDB 页面 | 原文线索

V2-057787 | MATHDB-343352

The refined low-degree conjecture for polynomial-time distinguishability

摘录 / EXCERPT: Refined low-degree conjecture. If there exists a polynomial-time algorithm that strongly distinguishes (P_n) and (Q_n), then

conjecture | open | MathDB 页面 | 原文线索

V2-057788 | MATHDB-343389

The NP-completeness conjecture for the Sigma-1 theory of Young's lattice

摘录 / EXCERPT: Let $\textbf{Y}=\left\langle \mathcal P,\leq\right\rangle$ be Young's lattice, where $\mathcal P$ is the set of partitions ordered by inclusion of their Young diagrams. The Sigma-1...

conjecture | open | MathDB 页面 | 原文线索

V2-057789 | MATHDB-343403

Schalekamp-Williamson-van Zuylen conjecture on the subtour-LP integrality gap

摘录 / EXCERPT: Consider the metric traveling salesperson problem and its subtour linear programming relaxation. A feasible solution is half-integral if every edge variable satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-057790 | MATHDB-343529

Guth-Zahl polynomial Wolff axiom maximal inequality conjecture

摘录 / EXCERPT: Guth-Zahl conjecture. Let $p\geqslant\frac{n}{n-1}$. For every $\varepsilon>0$, there are a complexity $D=D_{\varepsilon,n}\in\mathbb{N}$ and a constant $C_{\varepsilon,n}>0$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-057791 | MATHDB-343545

The k-partite planted clique conjecture

摘录 / EXCERPT: Fix a constant $p\in(0,1)$. Let n_t and k_t be increasing sequences of positive integers such that $k_t=o(\sqrt{n_t})$ and k_t divides n_t, and let ... be a sequence of rando...

conjecture | open | MathDB 页面 | 原文线索

V2-057792 | MATHDB-343640

The Schur-vector INDEC tractability conjecture

摘录 / EXCERPT: Let (Γ,Ω) be a quiver, let $\alpha\in\Delta_+$ be a dimension vector, and let ... A Schur vector is a dimension vector for which the corresponding representations have th...

conjecture | open | MathDB 页面 | 原文线索

V2-057793 | MATHDB-343691

The MVB conjecture on asymptotically optimal branching variables

摘录 / EXCERPT: An MVB instance consists of variables $(l_i,r_i)_{1\leq i\leq n}$, with recurrence ... Its variable ratios are denoted by φ_i. The MVB conjecture. For each instance of MVB, t...

conjecture | refuted | MathDB 页面 | 原文线索

V2-057794 | MATHDB-343908

Termination conjecture for the SatSAGBI procedure

摘录 / EXCERPT: SatSAGBI termination conjecture. If there is a finite σ-SAGBI basis of $S:a_0^\infty$, then Procedure terminates in a finite number of iterations, hence it is an algorithm....

conjecture | open | MathDB 页面 | 原文线索

V2-057795 | MATHDB-344036

The involution-word bumping conjecture for queer supercrystals

摘录 / EXCERPT: Involution-word bumping conjecture. For every $i\in[n]$, one has $\tilde w_i-w_i\in\{0,1\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057796 | MATHDB-344037

The fixed-point-free involution bumping conjecture

摘录 / EXCERPT: Fixed-point-free involution bumping conjecture. For every $i\in[n]$, one has $\tilde w_i-w_i\in\{0,1,2\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-057797 | MATHDB-344111

Khot's Unique Games Conjecture in transversal-polynomial form

摘录 / EXCERPT: Let X^{α} be an r-fold cover of a graph X, where $\xi(X^{\alpha},t)$ is the transversal polynomial and $\deg(\xi(X^{\alpha},t))$ is the maximum number of edges satisfied...

conjecture | open | MathDB 页面 | 原文线索

V2-057798 | MATHDB-344250

Heegaard disc-subset distance conjecture for triangulation complexity

摘录 / EXCERPT: Let M be a closed orientable manifold with a Heegaard surface S that divides M into handlebodies V and W. Let $\mathrm{Sp}(S)$ be the spine graph of S, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057799 | MATHDB-344289

Uniqueness of worst-case clusterings under a generalized exclusive-element condition

摘录 / EXCERPT: Generalized uniqueness conjecture. The ground-truth clustering ${\cal C}$ is the only valid clustering consistent with the entire query matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-057800 | MATHDB-344393

Barrington-Straubing-Thérien conjecture on CC-circuit lower bounds

摘录 / EXCERPT: Barrington-Straubing-Thérien conjecture. There exists $q>0$ such that C_n has size

conjecture | open | MathDB 页面 | 原文线索

V2-057801 | MATHDB-344472

Banica's convergence conjecture for the quantum Sinkhorn algorithm

摘录 / EXCERPT: For n a positive integer, let the alternating normalization procedure be the row and column normalization algorithm in Algorithm, initialized by a random complex Gaussian tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-057802 | MATHDB-344497

Quadratic-time algorithm for overlapping Hankel block low-rank completion

摘录 / EXCERPT: Consider the overlapping Hankel block low-rank completion problem, with matrix-size parameter N, and let $\hat{X}$ denote a candidate completed matrix. Quadratic-time completion...

conjecture | open | MathDB 页面 | 原文线索

V2-057803 | MATHDB-344597

NP-completeness of the hull number problem for oriented chordal graphs

摘录 / EXCERPT: Oriented chordal hull-number conjecture. Determining whether

open_problem | open | MathDB 页面 | 原文线索

V2-057804 | MATHDB-344711

Co-NP-completeness of finite-support Gardens of Eden and rectangular orphans

摘录 / EXCERPT: For the Game of Life, consider finite-support Gardens of Eden and rectangular orphans, with the encodings specified in the paper and polynomial-time many-one reductions. Co-NP-comp...

conjecture | open | MathDB 页面 | 原文线索

V2-057805 | MATHDB-344762

The butterfly permutation parameter conjecture

摘录 / EXCERPT: Let $q=2^n$ with n odd and $\gcd(i,n)=1$. For ... set ... Let Γ be the parameter set defined in equation. Butterfly permutation parameter conjecture. If V_i is a permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-057806 | MATHDB-344861

The fast self-correcting algorithm conjecture for computing π

摘录 / EXCERPT: Fast self-correcting pi algorithm conjecture. Algorithm runs in $O(M(n)\log n)$ bit operations.

conjecture | open | MathDB 页面 | 原文线索

V2-057807 | MATHDB-344866

Nisan-Szegedy's sensitivity conjecture for Boolean functions

摘录 / EXCERPT: Nisan-Szegedy's sensitivity conjecture. There exists an absolute constant $C>0$ such that, for every Boolean function f,

conjecture | open | MathDB 页面 | 原文线索

V2-057808 | MATHDB-344974

Vector-contraction conjecture for Rademacher complexities in $\p$-norms

摘录 / EXCERPT: Let $\mathcal{X}$ be a set, let $n\in\mathbb{N}$ and $(x_1,\ldots,x_n)\in\mathcal{X}^n$. Let $\mathcal{F}$ be a class of functions $f:\mathcal{X}\to\ell_p$ for $0<p\leq 2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057809 | MATHDB-345056

Schleimer-Thompson conjecture on the complexity of fibered knots

摘录 / EXCERPT: Schleimer-Thompson conjecture. For any three-manifold M, there is a constant $t(M)$ such that if $K\subset M$ is a fibered knot, then the monodromy of K has complexity at most...

conjecture | open | MathDB 页面 | 原文线索

V2-057810 | MATHDB-345112

Evasiveness conjecture for monotone transitive Boolean functions

摘录 / EXCERPT: A Boolean function is a function $f:\{0,1\}^n\to\{0,1\}$. Its decision-tree complexity $D(f)$ is the minimum number of input bits an adaptive algorithm must query to determine f,...

conjecture | open | MathDB 页面 | 原文线索

V2-057811 | MATHDB-345423

The NP versus co-NP conjecture

摘录 / EXCERPT: NP versus co-NP conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057812 | MATHDB-345512

NP-completeness of deciding path-chromatic number at most two

摘录 / EXCERPT: For a graph G, let $\mathsf{path}\textnormal{-}\chi(G)$ denote its path-chromatic number. Path-chromatic two-colourability conjecture. It is NP-complete to decide if...

conjecture | open | MathDB 页面 | 原文线索

V2-057813 | MATHDB-345687

Polynomial-time root-counting conjecture for fixed-arity sparse polynomials over p-adic fields

摘录 / EXCERPT: Let p and t be fixed, and let $f\in\mathbb{Z}[x_1]$ be an input t-nomial, meaning a univariate integer polynomial with at most t monomial terms. Root-counting conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-057814 | MATHDB-345870

NP-hardness conjecture for Grothendieck d-norms

摘录 / EXCERPT: Let $d,m,n\in\mathbb{N}$. The real symmetric spaces $\mathbb{S}^n(\mathbb{R})$, real rectangular spaces $\mathbb{R}^{m\times n}$, complex symmetric spaces...

conjecture | open | MathDB 页面 | 原文线索

V2-057815 | MATHDB-345962

Chen's complexity conjecture for the surjective constraint satisfaction problem

摘录 / EXCERPT: Let A be a finite domain and let Γ be a constraint language on A. Write $\operatorname{CSP}(\Gamma)$ for the ordinary constraint satisfaction problem and...

conjecture | open | MathDB 页面 | 原文线索

V2-057816 | MATHDB-346144

Polynomial consistency-speedup conjecture for bounded-arithmetic theories

摘录 / EXCERPT: Polynomial consistency-speedup conjecture. For every sound $\Delta^b_1(S^1_2)$ theory S and every $l\in\omega$, there exists a sound $\Delta^b_1(S^1_2)$ theory T such that

conjecture | open | MathDB 页面 | 原文线索

V2-057817 | MATHDB-346157

Termination conjecture for the pentagon contact representation algorithm

摘录 / EXCERPT: Termination conjecture. The algorithm terminates with a non-negative solution for every graph G which is an inner triangulation of a 5-gon, and for every initial five color for...

conjecture | open | MathDB 页面 | 原文线索

V2-057818 | MATHDB-346178

Equidistribution conjecture for the class and regulator data in the Greenberg algorithm

摘录 / EXCERPT: Equidistribution conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-057819 | MATHDB-346180

Output-polynomial enumeration for incomparability graphs of S_t-free posets

摘录 / EXCERPT: The S_t-free incomparability conjecture. For every t, there is an output-polynomial time algorithm for $\textsc{Dom-Enum}$ in incomparability graphs of S_t-free posets.

conjecture | open | MathDB 页面 | 原文线索

V2-057820 | MATHDB-346181

Output-polynomial enumeration for $2K_p$-free incomparability graphs

摘录 / EXCERPT: The $2K_p$-free incomparability conjecture. For every p, there is an output-polynomial time algorithm for $\textsc{Dom-Enum}$ in $2K_p$-free incomparability graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057821 | MATHDB-346182

Output-polynomial enumeration for H-free incomparability graphs

摘录 / EXCERPT: The co-bipartite forbidden-subgraph conjecture. For any co-bipartite H, there is an output-polynomial time algorithm for $\textsc{Dom-Enum}$ in H-free incomparability graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057822 | MATHDB-346233

The PcN power-exponent conjecture over seventh-degree finite fields

摘录 / EXCERPT: PcN power-exponent conjecture. For all $0\leq j\leq 6$, the only possible values of d are the elements of

conjecture | open | MathDB 页面 | 原文线索

V2-057823 | MATHDB-346319

Planar support graph conjecture for shrinking rules C1, C2, and C3

摘录 / EXCERPT: Let G be a graph, and let $(y,x)\in P^G_A$ be a vector. The support graph $\bar{G}$ of (y,x) is the graph associated with this vector. Planar support graph conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-057824 | MATHDB-346320

Isomorphism conjecture for graphs obtained by fixed shrinking strategies

摘录 / EXCERPT: Let G be a graph, let $(y,x)\in P^G_A$ be a vector, and let ... be a fixed shrinking strategy. Shrinking-strategy isomorphism conjecture. The graphs obtained by applying...

conjecture | open | MathDB 页面 | 原文线索

V2-057825 | MATHDB-346336

The sensitivity-weighted variable bound for Boolean functions

摘录 / EXCERPT: Sensitivity-weighted variable conjecture. For every Boolean function f,

conjecture | open | MathDB 页面 | 原文线索

V2-057826 | MATHDB-346521

Specific hardness assumptions for planted clique variants

摘录 / EXCERPT: Let m and n be polynomial in one another. The problems $k\textsc{-pc}(n,k,1/2)$, $\textsc{bpc}(m,n,k_m,k_n,1/2)$, $k\textsc{-bpc}(m,n,k_m,k_n,1/2)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-057827 | MATHDB-346522

Hopkins's formal low-degree conjecture for symmetric almost-independent distributions

摘录 / EXCERPT: Let Ω be finite or $\mathbb R$, let k be fixed, and let $N=\binom nk$. Let ν be a product distribution on Ω^N, let μ be another distribution on Ω^N,...

conjecture | open | MathDB 页面 | 原文线索

V2-057828 | MATHDB-346601

The low-degree conjecture for high-dimensional testing

摘录 / EXCERPT: Let $t: \mathbb{N} \to \mathbb{N}$. Consider a “natural” high-dimensional testing problem specified by distributions $\mathbb{P}_N$ and $\mathbb{Q}_N$ on an observation space...

conjecture | open | MathDB 页面 | 原文线索

V2-057829 | MATHDB-346688

Rejmer's conjecture on the indefinite continuation of his coloring algorithm

摘录 / EXCERPT: Let $c(1)=+1$ and construct a perfectly balanced multiplicative coloring recursively by the greedy procedure described in the paper: at each step color the next integer while prese...

conjecture | open | MathDB 页面 | 原文线索

V2-057830 | MATHDB-346712

Triangle Conjecture

摘录 / EXCERPT: Consider the word-RAM model of computation with word length $O(\log b)$ for inputs of length b. An undirected graph G has m edges. Triangle Conjecture. There exists a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-057831 | MATHDB-346713

ℓ-layered reachability oracle conjecture

摘录 / EXCERPT: Let ℓ be an integer, and let G be an ℓ-layered directed acyclic graph with layers $L_1,\ldots,L_\ell$, where every edge goes from L_i to L_{i+1}. An ℓ-layered...

conjecture | open | MathDB 页面 | 原文线索

V2-057832 | MATHDB-346942

Runtime lower-bound conjecture for sparse PCA in the hard regime

摘录 / EXCERPT: Sparse PCA runtime conjecture. For any λ in the hard regime

conjecture | open | MathDB 页面 | 原文线索

V2-057833 | MATHDB-347731

The complexity trichotomy for circular-colouring mixing

摘录 / EXCERPT: Let $G_{p,q}$ denote the graph used as the target for (p,q)-colourings, and let the textsc{G{p,q}}-Mixing problem ask whether the reconfiguration graph of (p,q)-colourings of a...

conjecture | open | MathDB 页面 | 原文线索

V2-057834 | MATHDB-347988

The agreement property for independently trained classifiers

摘录 / EXCERPT: Let $\mathcal{F}$ be a classifier family and let $\mathcal{D}$ be a distribution. For independent train sets $S_1,S_2\sim\mathcal{D}^n$, let f_1,f_2 be classifiers trained on...

conjecture | open | MathDB 页面 | 原文线索

V2-057835 | MATHDB-347989

Generalized feature calibration for distinguishable features

摘录 / EXCERPT: Let $\mathcal{D}$ be a natural distribution over examples, let n be the number of samples, let $\mathcal{F}$ be a family of interpolating models, and let $\varepsilon\geq0$. A pa...

conjecture | open | MathDB 页面 | 原文线索

V2-057836 | MATHDB-348060

Memory-length conjecture for tight IQC convergence rates

摘录 / EXCERPT: Let a first-order algorithm have T steps of memory. For each integer $n\geq 1$, an off-by-n pointwise IQC is the quadratic constraint associated with a pair of iterates...

conjecture | open | MathDB 页面 | 原文线索

V2-057837 | MATHDB-348241

NP-hardness conjecture for $\mathrm{PCSP}(\mathbf{1in3},\mathbf{\check{C}^+})$

摘录 / EXCERPT: Let $\mathbf{1in3}$ and $\mathbf{\check{C}^+}$ be the finite relational templates used in the paper, with a homomorphism $\mathbf{1in3}\to\mathbf{\check{C}^+}$. The associated prom...

conjecture | open | MathDB 页面 | 原文线索

V2-057838 | MATHDB-348269

Triangle Detection Conjecture

摘录 / EXCERPT: Let G be an input graph with n vertices and m edges. An algorithm runs in the word RAM model with word size $O(\log n)$ bits. Triangle Detection Conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-057839 | MATHDB-348355

The tight log-rank conjecture for AND-functions

摘录 / EXCERPT: Let $f: \{0,1\}^n \to \{0,1\}$ be a Boolean function, and let $\operatorname{spar}(f)=r$ be its sparsity, namely the number of nonzero coefficients in its unique multilinear polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-057840 | MATHDB-348416

Dekking's conjecture on the subword complexity of the Fibonacci-Thue-Morse sequence

摘录 / EXCERPT: Let $({\bf ftm}[n])_{n\geq 0}$ be the Fibonacci-Thue-Morse sequence, obtained by summing modulo 2 the bits in the Fibonacci representation of n. Let $\rho_{\bf ftm}(n)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-057841 | MATHDB-348480

Induced subgraph isomorphism hardness conjecture for bipartite permutation graphs

摘录 / EXCERPT: Induced subgraph isomorphism hardness conjecture. Unless $\mathrm{P}=\mathrm{NP}$, the induced subgraph isomorphism problem is NP-hard in $\mathcal X$ if and only if $\mathcal X$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-057842 | MATHDB-348590

Steinke-Zakynthinou's linear CMI conjecture for realizable proper learning

摘录 / EXCERPT: Let $\mathcal H \subseteq \mathcal X\to\mathcal Y$ be a VC class of dimension d, and let $\mathcal A$ be a proper learning algorithm. For a distribution $\mathcal D$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-057843 | MATHDB-348721

The quadratic-time word problem conjecture for special monoids with virtually free units

摘录 / EXCERPT: Let M be a finitely presented special monoid, generated by A, with virtually free group of units. For inputs $u,v\in A^*$, write $n=|u|+|v|$. Quadratic-time word problem conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-057844 | MATHDB-348732

Accelerated convergence conjecture for Algorithm HOT-2

摘录 / EXCERPT: Let $\bar{L}$ be a continuously differentiable, smooth, and strongly convex function, and consider constant regressors. Let $\{\theta_k\}_{k=0}^{\infty}$ denote the iterates genera...

conjecture | open | MathDB 页面 | 原文线索

V2-057845 | MATHDB-348924

Subexponential hardness of certifying non-negative PCA

摘录 / EXCERPT: Let $\bW \in \mathbb{R}^{n \times n}_{\mathrm{sym}}$ be drawn from the Gaussian orthogonal ensemble, and let $\lambda^+(\bW)$ denote the non-negative principal component analysis o...

conjecture | open | MathDB 页面 | 原文线索

V2-057846 | MATHDB-348925

Subexponential hardness of strong detection below the BBP transition

摘录 / EXCERPT: Fix constants $\gamma > 0$, $\beta > -1$, and $\rho \in (0,1)$. Let $N=N(n)$ satisfy $n/N \to \gamma$ as $n \to \infty$. In the spiked Wishart model, under $\mathbb{Q}$ the observa...

conjecture | open | MathDB 页面 | 原文线索

V2-057847 | MATHDB-349001

The tractable-or-NP-hard dichotomy conjecture for generic combinations

摘录 / EXCERPT: Tractable-or-NP-hard dichotomy conjecture. If either

conjecture | open | MathDB 页面 | 原文线索

V2-057848 | MATHDB-349323

The approximability frontier conjecture for sparse Diameter

摘录 / EXCERPT: Let sparse graphs have $m=\tilde{O}(n)$ edges, and consider the four variants obtained by choosing directed or undirected graphs and weighted or unweighted edges. For...

conjecture | open | MathDB 页面 | 原文线索

V2-057849 | MATHDB-349496

Lovász-Saks log-rank conjecture for Boolean functions

摘录 / EXCERPT: Log-rank conjecture. For every function $f: \mathcal X \times \mathcal Y \to \{0,1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-057850 | MATHDB-349498

Parallel 2-partitioned configuration conjecture

摘录 / EXCERPT: Parallel 2-partition conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-057851 | MATHDB-349524

The sampling-algorithm cybernetics conjecture

摘录 / EXCERPT: Consider the subcategory of $\mathbf{BayesLens}$ spanned by finite-dimensional smooth manifolds, with morphisms generated under sequential and parallel composition by the generaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-057852 | MATHDB-349567

Internal-solution convergence conjecture for the data-driven Lanczos ROM

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{C}$ be the set of spectral points, and let V_0 and Q_0 be the solution basis and Lanczos matrix corresponding to the background $q_0=0$. Let T be th...

conjecture | open | MathDB 页面 | 原文线索

V2-057853 | MATHDB-349770

Gaussian equivalent model conjecture for generic feature maps

摘录 / EXCERPT: Let $\{\boldsymbol{x}^{\mu}\}_{\mu=1}^{n}$ be independent samples from a data distribution on $\mathcal{X}\subset\mathbb{R}^{D}$, and let the centred teacher and student features b...

conjecture | open | MathDB 页面 | 原文线索

V2-057854 | MATHDB-349830

Order-wise sample-complexity conjecture for monotone test functions

摘录 / EXCERPT: Sample-complexity conjecture. For any monotone test function $f(\cdot)$,

conjecture | open | MathDB 页面 | 原文线索

V2-057855 | MATHDB-349918

Pseudodistribution product inequality for semidefinite quadratic forms

摘录 / EXCERPT: Pseudodistribution product inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-057856 | MATHDB-350047

Conjecture on strict Valiant-rigidity of known Hadamard matrices

摘录 / EXCERPT: A Hadamard matrix is a square matrix whose entries are $+1$ and -1 and whose rows are mutually orthogonal. A matrix is strictly Valiant-rigid when its strict rigidity does not ad...

conjecture | open | MathDB 页面 | 原文线索

V2-057857 | MATHDB-350113

Termination of the minimum expected cost algorithm

摘录 / EXCERPT: Let Algorithm be the algorithm defined in the paper, let M denote its termination parameter, and let $\epsilon>0$. Termination conjecture. Algorithm will terminate with...

conjecture | open | MathDB 页面 | 原文线索

V2-057858 | MATHDB-350248

Strongest type-2 constant conjecture for operator-space tensor products

摘录 / EXCERPT: Let n, k, and $\tilde k'$ be the parameters in the construction, let $\mathcal{S}_1^{\tilde k'\!, n}$ denote the corresponding trace-class space, let...

conjecture | open | MathDB 页面 | 原文线索

V2-057859 | MATHDB-350249

Weaker type-2 growth conjecture for the cryptographic tensor-product space

摘录 / EXCERPT: Weaker-form conjecture. There exists a dimension-independent constant $\beta<1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057860 | MATHDB-350250

Existence of a valid norm with sublinear type-2 growth

摘录 / EXCERPT: Let X be a norm on the algebraic tensor product $\mathcal{S}_1^{\tilde k'\!, n}\otimes\mathcal{S}_1^{\tilde k'\!, n}\otimes\ell_2^{k\tilde k'}$. Call X valid if ... for every x...

conjecture | open | MathDB 页面 | 原文线索

V2-057861 | MATHDB-350251

Sublinear type-2 growth for the identity into the smallest valid norm

摘录 / EXCERPT: Let X be a valid norm, meaning a norm on $\mathcal{S}_1^{\tilde k'\!, n}\otimes\mathcal{S}_1^{\tilde k'\!, n}\otimes\ell_2^{k\tilde k'}$ satisfying ... for every x, together with...

conjecture | open | MathDB 页面 | 原文线索

V2-057862 | MATHDB-350365

The Sylow q-subquasigroup generator conjecture

摘录 / EXCERPT: Let g generate the maximal multiplicative subgroupoid $\mathbb{E}^*_{(p-1)^2}$ of an entropoid, with $p\geq 11$. Define ... The Sylow generator conjecture. The element g_q genera...

conjecture | open | MathDB 页面 | 原文线索

V2-057863 | MATHDB-350490

Finite nuclear dimension implies finite complexity relative to subhomogeneous algebras

摘录 / EXCERPT: Relative complexity conjecture. Every separable C^*-algebra with finite nuclear dimension has finite complexity relative to the class of subhomogeneous C^*-algebras.

conjecture | open | MathDB 页面 | 原文线索

V2-057864 | MATHDB-350504

Orthogonal Sagan-Worley insertion conjecture for orthogonal Knuth operators

摘录 / EXCERPT: Orthogonal Sagan-Worley insertion conjecture. For every $i\in\mathbb{Z}$,

conjecture | open | MathDB 页面 | 原文线索

V2-057865 | MATHDB-350505

Orthogonal Knuth equivalence characterization by Sagan-Worley insertion

摘录 / EXCERPT: Orthogonal Knuth equivalence conjecture. Two primed words satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-057866 | MATHDB-350506

Symplectic Sagan-Worley recording-tableau conjecture for modified Knuth operators

摘录 / EXCERPT: Symplectic Sagan-Worley recording-tableau conjecture. If $i>0$ and a is any primed word, then

conjecture | open | MathDB 页面 | 原文线索

V2-057867 | MATHDB-350569

Hardness conjecture from zeros of the complex-valued Ising partition function

摘录 / EXCERPT: Let Δ be an integer with $\Delta \ge 3$ and let $\beta\in \mathbb{C}_{\mathbb{A}}$ with $\beta \not \in \mathbb{R}\cup \{i, -i\}$. Let G be a graph with maximum degree at...

conjecture | open | MathDB 页面 | 原文线索

V2-057868 | MATHDB-350589

Subexponential growth conjecture for equal-length Whitehead orbit elements

摘录 / EXCERPT: Let F_r be a free group, and let $w\in F_r$ be a word whose length cannot be reduced by any elementary Whitehead automorphism; in particular, w is cyclically reduced. Consider...

conjecture | open | MathDB 页面 | 原文线索

V2-057869 | MATHDB-350637

NP-completeness of exact tournament inversion number

摘录 / EXCERPT: Exact inversion-number complexity conjecture. Given a tournament and an integer k, deciding whether

conjecture | open | MathDB 页面 | 原文线索

V2-057870 | MATHDB-350638

NP-completeness of k-Inversion for every positive k

摘录 / EXCERPT: Universal k-Inversion complexity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057871 | MATHDB-350643

Decidability of the word problem for Gray categories and braided monoidal categories

摘录 / EXCERPT: Let the word problem for 3-cells of Gray categories be the problem of deciding whether two such cells are equal, and likewise let the word problem for cells of braided monoidal cat...

conjecture | open | MathDB 页面 | 原文线索

V2-057872 | MATHDB-350886

The low-degree polynomial hardness threshold conjecture for random k-SAT

摘录 / EXCERPT: Low-degree polynomial threshold conjecture. Theorem (and Theorem) holds for all $\kappa > 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-057873 | MATHDB-350927

The strengthened Fourier-class inclusion conjecture for PQC-based models

摘录 / EXCERPT: Strengthened Fourier-class inclusion conjecture. For any PQC-based model, one has

conjecture | open | MathDB 页面 | 原文线索

V2-057874 | MATHDB-351042

Conjecture on Thue-Morse maximum order complexity along polynomial subsequences

摘录 / EXCERPT: Let $\mathcal{T}_P$ denote the Thue-Morse sequence along a polynomial subsequence, where P is a polynomial of degree d. Write $M(\mathcal{T}_P,N)$ for its maximum order complex...

conjecture | open | MathDB 页面 | 原文线索

V2-057875 | MATHDB-351043

Conjecture on Zeckendorf sum-of-digits maximum order complexity along polynomial subsequences

摘录 / EXCERPT: Let $s_Z(n)$ denote the sum of digits of n in Zeckendorf base, let $\mathcal{S}_Z=(s_Z(n)\bmod 2)_n$, and let $\mathcal{S}_{Z,P}$ be the sequence obtained by restricting...

conjecture | open | MathDB 页面 | 原文线索

V2-057876 | MATHDB-351044

Conjecture on successive-step ratios for Thue-Morse and Zeckendorf complexity

摘录 / EXCERPT: For a sequence $\mathcal{S}$, let a step be an integer N at which its maximum order complexity changes from $M(\mathcal{S},N-1)$ to $M(\mathcal{S},N)$, and define the ratio of su...

conjecture | open | MathDB 页面 | 原文线索

V2-057877 | MATHDB-351061

The individual-testing optimality conjecture above the threshold γ_1

摘录 / EXCERPT: Let x be the prevalence of a condition in a homogeneous population, let γ_1 be the threshold at which the elementary algorithm $\mathcal A_1$ becomes dominant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057878 | MATHDB-351285

The local-distance conjecture for autoencoder generalization

摘录 / EXCERPT: Let $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$ be the training and test subsets of the dataset, and let $D_{model}\circ E_{model}$ be a trained autoencoder. Near each training...

conjecture | open | MathDB 页面 | 原文线索

V2-057879 | MATHDB-351530

The asymptotic Gotsman-Linial conjecture

摘录 / EXCERPT: Asymptotic Gotsman-Linial conjecture. There is an upper bound

conjecture | open | MathDB 页面 | 原文线索

V2-057880 | MATHDB-351531

The superweak Gotsman-Linial conjecture

摘录 / EXCERPT: Superweak Gotsman-Linial conjecture. For some function g depending only on d,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057881 | MATHDB-351532

The limiting Gotsman-Linial conjecture

摘录 / EXCERPT: Limiting Gotsman-Linial conjecture. The average sensitivity satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-057882 | MATHDB-351533

Nonexistence of even-dimensional degree-two hypersensitive functions

摘录 / EXCERPT: Even-n, degree-two consequence. For every even n, $(n,2)$-HSFs do not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-057883 | MATHDB-351621

Mulmuley-Sohoni's orbit-closure conjecture for permanent versus determinant

摘录 / EXCERPT: Let ℓ be a linear coordinate on $\mathbb C^1$, and define the projective orbit closures ... ... Let $n(m)$ be a polynomial. Mulmuley-Sohoni's orbit-closure conjecture. For all suf...

conjecture | open | MathDB 页面 | 原文线索

V2-057884 | MATHDB-351651

Boyle-Ishai-Pass-Wootters toy conjecture on permuted Reed-Muller puzzles

摘录 / EXCERPT: Toy conjecture. The distributions $(S_1,\ldots,S_m)$ and $(T_1,\ldots,T_m)$ are computationally indistinguishable.

conjecture | refuted | MathDB 页面 | 原文线索

V2-057885 | MATHDB-351652

New toy conjecture on randomly sampled permuted Reed-Muller puzzles

摘录 / EXCERPT: New toy conjecture. The distributions $(S_1,\ldots,S_m)$ and $(T_1,\ldots,T_m)$ are computationally indistinguishable.

conjecture | refuted | MathDB 页面 | 原文线索

V2-057886 | MATHDB-351753

Rapid mixing for standard set-valued tableaux with bounded diagonal complexity

摘录 / EXCERPT: Bounded-complexity rapid-mixing conjecture. The Markov chain $\mathcal{MC}_{\sf SVT}(\lambda,N,S)$ is rapidly mixing for every $(\lambda,N,S)\in\mathcal{F}_d$.

conjecture | open | MathDB 页面 | 原文线索

V2-057887 | MATHDB-351854

Quantum communication upper-bound conjecture for Hidden Hypermatching

摘录 / EXCERPT: Quantum Hidden Hypermatching upper-bound conjecture. If $t,r>2$, there is a protocol for $r\text{-}{\mathsf{HH}}(\alpha,t,n)$ using

conjecture | open | MathDB 页面 | 原文线索

V2-057888 | MATHDB-351932

The greedy L_2 discrepancy conjecture for the sequence generated by Algorithm 1

摘录 / EXCERPT: Greedy L_2 discrepancy conjecture. The sequence $\mathcal{S}^*$ should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-057889 | MATHDB-351933

The greedy star-discrepancy conjecture for the sequence generated by Algorithm 1

摘录 / EXCERPT: Greedy star-discrepancy conjecture. The sequence $\mathcal{S}^*$ should be a low-discrepancy sequence, meaning that

conjecture | open | MathDB 页面 | 原文线索

V2-057890 | MATHDB-351973

Online Banaszczyk conjecture for online vector balancing

摘录 / EXCERPT: Let $v_1,\ldots,v_T\in\mathbb{R}^m$ be a sequence of vectors with Euclidean norm at most 1. In the oblivious adversarial online setting, the algorithm receives v_t at time t...

conjecture | open | MathDB 页面 | 原文线索

V2-057891 | MATHDB-351994

The Plücker-coordinate volume relation for polytopes

摘录 / EXCERPT: Let $G(n-d,n)$ denote the Grassmannian of $(n-d)$-dimensional subspaces of an n-dimensional space, and let $\bm{p}$ be a Plücker-coordinate vector in $\bigcup\limits_n G(n-d,n)$....

conjecture | open | MathDB 页面 | 原文线索

V2-057892 | MATHDB-352435

Sparse average-case k-SUM conjecture

摘录 / EXCERPT: In the (k,N,r)-SUM problem, the input consists of r elements $z_1,\ldots,z_r$, independently and uniformly chosen from $\{-N,\ldots,N\}$. The goal is to find an ordered k-tup...

conjecture | open | MathDB 页面 | 原文线索

V2-057893 | MATHDB-352440

NP-completeness conjecture for APUD(k,m) recognition

摘录 / EXCERPT: APUD recognition conjecture. Recognition of $\mathrm{APUD}(k,m)$ is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057894 | MATHDB-352446

Arias's degree-one defect classification conjecture

摘录 / EXCERPT: Arias's conjecture. The set A has finite symmetric difference with E, and the one-sided difference $A\setminus E$ is a finite subset of

conjecture | open | MathDB 页面 | 原文线索

V2-057895 | MATHDB-352519

Kawasaki dynamics mixing conjecture for the fixed-magnetization Ising model

摘录 / EXCERPT: Kawasaki dynamics mixing conjecture. For $\beta<\beta_c(\Delta)$, the Kawasaki dynamics mix in time polynomial in n for any fixed magnetization and any graph G of maximum degre...

conjecture | open | MathDB 页面 | 原文线索

V2-057896 | MATHDB-352538

Transfinite sequences describing the greatest numbers of each complexity class modulo 3

摘录 / EXCERPT: Let $\|n\|$ denote the complexity of a natural number, and let ξ be an ordinal. Consider three transfinite sequences $(a_\alpha)_{\alpha<\xi}$, $(b_\alpha)_{\alpha<\xi}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-057897 | MATHDB-352541

Limit relations between successive blocks of the transfinite complexity sequences

摘录 / EXCERPT: Let $(a_\alpha)_{\alpha<\xi}$, $(b_\alpha)_{\alpha<\xi}$, and $(c_\alpha)_{\alpha<\xi}$ be the transfinite sequences associated with the three complexity classes modulo 3. Block-...

conjecture | open | MathDB 页面 | 原文线索

V2-057898 | MATHDB-352951

The UWSCP characterization by bounded virtual-endomorphism growth

摘录 / EXCERPT: Let f be a hyperbolic rational map, and let ϕ be its virtual endomorphism. Write $N[\phi^n]$ for the corresponding complexity quantity at the nth iterate, and say that it...

conjecture | open | MathDB 页面 | 原文线索

V2-057899 | MATHDB-353263

Exponential Time Hypothesis

摘录 / EXCERPT: Exponential Time Hypothesis. There is no $2^{o(n)}$-time algorithm which can determine whether a given 3SAT formula on n variables is satisfiable.

conjecture | open | MathDB 页面 | 原文线索

V2-057900 | MATHDB-353753

Quadratic subword-complexity conjecture for the Thue-Morse two-block fixed point

摘录 / EXCERPT: Let $x^{(00)}$ be the fixed point of the Thue-Morse two-block substitution, and let $p^{(00)}(n)$ denote its subword complexity, namely the number of distinct factors of length n...

conjecture | open | MathDB 页面 | 原文线索

V2-057901 | MATHDB-354082

Stable-algorithm existence conjecture for the symmetric binary perceptron

摘录 / EXCERPT: Let $kappa>0$ and let the symmetric binary perceptron (SBP) have constraint density $alpha$. A search algorithm is stable when it satisfies the stability notion used in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-057902 | MATHDB-354083

Two-sided limiting-threshold conjecture for the symmetric binary perceptron

摘录 / EXCERPT: Two-sided limiting-threshold conjecture. For every $epsilon>0$, there exists $kappa^*(epsilon)>0$ such that for every $kappalekappa^*(epsilon)$: no polynomial-time search algorithm...

conjecture | open | MathDB 页面 | 原文线索

V2-057903 | MATHDB-354117

Aldous's conjecture on instability of queue-free backoff protocols

摘录 / EXCERPT: Aldous's conjecture. In the queue-free setting, no backoff protocol is stable for any positive value of $lambda$.

conjecture | open | MathDB 页面 | 原文线索

V2-057904 | MATHDB-354245

Convergence and complexity of the magnitude-matrix balancing algorithm

摘录 / EXCERPT: Balancing-algorithm conjecture. Algorithm always converges. Moreover, the number of iterations required to reach a given tolerance is $O(\log(n))$ for fixed ϵ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-057905 | MATHDB-354418

Binomial basis conjecture for #P closure properties

摘录 / EXCERPT: Let φ be a multivariate polynomial closure property of $\#P$, meaning that applying φ to polynomially many $\#P$ functions yields a function in the indicated class....

conjecture | open | MathDB 页面 | 原文线索

V2-057906 | MATHDB-354419

Univariate binomial basis conjecture for #P closure properties

摘录 / EXCERPT: Let φ be a univariate polynomial closure property of $\#P$, and say that it relativizes when the corresponding closure holds for every oracle version $\#P^A$. Univariate bi...

conjecture | open | MathDB 页面 | 原文线索

V2-057907 | MATHDB-354642

Larose-Tesson conjecture on Kearnes-Kiss chains and NL

摘录 / EXCERPT: Let $\mathbb{H}$ be a finite structure with finite relational signature, and let its polymorphisms be the edge-preserving operations on its domain. A Kearnes-Kiss chain is the chai...

conjecture | open | MathDB 页面 | 原文线索

V2-057908 | MATHDB-354643

Egri-Larose-Tesson conjecture on Noname chains and L

摘录 / EXCERPT: Let $\mathbb{H}$ be a finite structure with finite relational signature, and let its polymorphisms be the operations preserving the relations of $\mathbb{H}$. A Noname chain is the...

conjecture | open | MathDB 页面 | 原文线索

V2-057909 | MATHDB-354644

Bulín's easy-trees-cannot-count conjecture

摘录 / EXCERPT: A tree is an orientation of a finite undirected tree. A tree has the ability to count if its constraint satisfaction problem can encode solving systems of linear equations over...

conjecture | open | MathDB 页面 | 原文线索

V2-057910 | MATHDB-354645

Asymptotic NL-hardness conjecture for core trees

摘录 / EXCERPT: For each positive integer n, consider the finite set of core trees with n vertices, meaning oriented trees whose only endomorphisms are automorphisms. Let the fraction of these...

conjecture | open | MathDB 页面 | 原文线索

V2-057911 | MATHDB-354652

Dense hard-sequence conjecture for true unprovable coTHEOREMS

摘录 / EXCERPT: Let H' be the set comprising H and other dense sets of true sentences unprovable in $\mathcal{T}$, including the specified sets associated with universal Turing machines and Tu...

conjecture | open | MathDB 页面 | 原文线索

V2-057912 | MATHDB-354755

NP-completeness of the planar graph square-root problem

摘录 / EXCERPT: Planar square-root conjecture. The $\mathcal{H}$-square-root problem is NP-complete for the class $\mathcal{H}$ of planar graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057913 | MATHDB-354850

Polynomial-time Maximum Independent Set in graphs with bounded induced cycle packing

摘录 / EXCERPT: Polynomial-time independent-set conjecture. $\textsc{Maximum Independent Set}$ is solvable in polynomial time in $\mathcal{O}_k$-free graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057914 | MATHDB-355003

Unique-solvability conjecture for the input-queued switch functional equation

摘录 / EXCERPT: Consider an input-queued switch operating under a scheduling algorithm that achieves state space collapse. Let $L(\boldsymbol\theta)$ and $\mathbf M(\boldsymbol\theta)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-057915 | MATHDB-355340

Efficient computation of the degree of a neural ideal

摘录 / EXCERPT: Efficient degree computation conjecture. There is a polynomial-time algorithm that, given the codewords of $\mathcal C$ as binary vectors of length n, determines whether every ps...

conjecture | open | MathDB 页面 | 原文线索

V2-057916 | MATHDB-355352

Asymptotically optimal time-and-budget conjecture for k-connectivity

摘录 / EXCERPT: Let k be a positive integer, let $\varepsilon>0$, and let a (t,b)-strategy B of Builder be a strategy that observes at most t arriving edges and purchases at most b of th...

conjecture | open | MathDB 页面 | 原文线索

V2-057917 | MATHDB-355510

General asymptotic conjecture for the normalized linear complexity of composed arrays

摘录 / EXCERPT: General asymptotic conjecture. As the size of $\text{\bf A}$ increases, ${\cal L}_n(\text{\bf A})$ approaches ${\cal L}_n(\text{\bf C})$ or ${\cal L}_n(\text{\bf F})$, according as...

conjecture | open | MathDB 页面 | 原文线索

V2-057918 | MATHDB-355594

Planar graph NP-hardness conjecture for higher divisorial gonality

摘录 / EXCERPT: For a fixed positive integer r, let the r^{th} Divisorial Gonality problem ask whether a graph has divisorial gonality at most a given integer, and let the r^{th} Stable Divi...

conjecture | open | MathDB 页面 | 原文线索

V2-057919 | MATHDB-355595

NP-completeness conjecture for stable divisorial gonality

摘录 / EXCERPT: Fix a positive integer r. The r^{th} Stable Divisorial Gonality problem asks whether a graph has stable divisorial gonality at most a given integer. Stable divisorial gonality...

conjecture | open | MathDB 页面 | 原文线索

V2-057920 | MATHDB-355596

NP-completeness conjecture for metric divisorial gonality

摘录 / EXCERPT: Fix a positive integer r. For a metric graph $\Gamma=(G,l)$ with rational edge lengths, the r^{th} Metric Divisorial Gonality problem asks whether...

conjecture | open | MathDB 页面 | 原文线索

V2-057921 | MATHDB-355709

Polynomial-time complexity of distance-d independent set reconfiguration on trees under token sliding

摘录 / EXCERPT: Let $d\geq 3$. In distance-d independent set reconfiguration, denoted by $\textsc{D}d\textsc{ISR}$, configurations are distance-d independent sets, and under the token-sliding...

open_problem | open | MathDB 页面 | 原文线索

V2-057922 | MATHDB-355713

Smale's conjecture on the complexity of Hilbert's Nullstellensatz over the complex numbers

摘录 / EXCERPT: Let $\{f_i\}_{i=1}^k$ be a sequence of polynomials in $\mathscr{P}_n$. The Hilbert Nullstellensatz decision problem over $\mathbb{C}$ asks whether these polynomials have a common z...

conjecture | open | MathDB 页面 | 原文线索

V2-057923 | MATHDB-355731

Constructive automata conjecture for Ising-model languages

摘录 / EXCERPT: Constructive automata conjecture. For every context free ${\sf L}_{\mathcal{M}}$ there exists a constructive PDA that decides it. For every context sensitive...

conjecture | open | MathDB 页面 | 原文线索

V2-057924 | MATHDB-355746

Optimal first-order oracle complexity for heterogeneous Hölder-smooth sums

摘录 / EXCERPT: Let $\mathcal{J}$ be a finite index set, and for each $j\in\mathcal{J}$ let f_j be a convex (M_j,v_j)-Hölder smooth function. Let x_0 be the initial point, x^* an optimizer...

conjecture | open | MathDB 页面 | 原文线索

V2-057925 | MATHDB-355841

Existence of a p-time generator whose range intersects every infinite P set

摘录 / EXCERPT: Let g be a polynomial-time function that maps inputs of length n to outputs of length $n+1$, and let $rng(g)$ denote its range. Proof search conjecture. There exists such a fun...

conjecture | open | MathDB 页面 | 原文线索

V2-057926 | MATHDB-355957

Minimality of the cones produced by the Newton-Puiseux algorithm

摘录 / EXCERPT: Let $\mathbb{K}$ be the coefficient field, let $\bold{x}$ denote the tuple of auxiliary variables, and let $p\in\mathbb{K}[\bold{x}][y]$ be a primitive polynomial. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-057927 | MATHDB-356106

Decay-threshold conjecture for random information on weighted finite-dimensional ell_p balls

摘录 / EXCERPT: Decay-threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057928 | MATHDB-356196

Standard simplex conjecture, bilinear positive-correlation version

摘录 / EXCERPT: Let $m\geq 3$ and $0<\rho<1$. Consider two measurable partitions $\Omega_1,\ldots,\Omega_m$ and $\Omega'_1,\ldots,\Omega'_m$ of $\mathbb{R}^{n+1}$ with...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-057929 | MATHDB-356239

The conjecture that PM-VC-Sym is as hard as exact perfect matching

摘录 / EXCERPT: Hardness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057930 | MATHDB-356421

The GI-hardness conjecture for complementary subgraph partitioning on long-hole-free graphs

摘录 / EXCERPT: GI-hardness conjecture. $\textsc{Comp-Sub}(\mathcal{PM})$ is $\GI$-hard on $C_{k\geq 6}$-free graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-057931 | MATHDB-356432

Feder-Vardi bounded-width conjecture for CSPs

摘录 / EXCERPT: A constraint satisfaction problem (CSP) is a computational problem of deciding whether a finite set of variables can be assigned values from a finite domain so that all specified c...

conjecture | open | MathDB 页面 | 原文线索

V2-057932 | MATHDB-356435

Subpower membership conjecture for algebras with few subpowers

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite algebra, and say that it has few subpowers when the equivalent conditions in the source's Few Subpowers theorem hold. Given a finite subset...

conjecture | open | MathDB 页面 | 原文线索

V2-057933 | MATHDB-356437

NP-hardness conjecture for promise graph coloring

摘录 / EXCERPT: For integers $k\geq l\geq 3$, $\operatorname{PCSP}(K_l,K_k)$ is the promise problem of coloring a graph with k colors under the promise that it is l-colorable. Promise-coloring...

conjecture | open | MathDB 页面 | 原文线索

V2-057934 | MATHDB-356465

Conjecture on the complexity of the ranking algorithm

摘录 / EXCERPT: Let m be the number of labels, and let Algorithm be the algorithm described in the paper for checking combinations of pairwise label preferences and finding...

conjecture | open | MathDB 页面 | 原文线索

V2-057935 | MATHDB-356762

The balanced and efficient numbers counting conjecture

摘录 / EXCERPT: The balanced and efficient numbers counting conjecture. For all x,

conjecture | open | MathDB 页面 | 原文线索

V2-057936 | MATHDB-356826

The iteration bound for the algorithm cref{alg:Px}

摘录 / EXCERPT: Let n be the dimension parameter, and let (C,c) be the randomly sampled input used by algorithm. The algorithm repeatedly applies Step. The iteration-bound conjecture. terminat...

conjecture | open | MathDB 页面 | 原文线索

V2-057937 | MATHDB-356828

NP-completeness of maximal independent queen domination on polyominoes

摘录 / EXCERPT: Maximal independent queen domination conjecture. The maximal independent queen domination problem on polyominoes is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057938 | MATHDB-356829

NP-completeness of minimal rook domination on convex polyominoes

摘录 / EXCERPT: Convex-polyomino rook domination conjecture. The minimal domination problem for attacking or non-attacking rooks on convex polyominoes is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057939 | MATHDB-356830

NP-completeness of minimal queen domination on convex polyominoes

摘录 / EXCERPT: Convex-polyomino queen domination conjecture. The minimal domination problem for attacking, or non-attacking, queens on convex polyominoes is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057940 | MATHDB-356831

NP-completeness of maximal non-attacking queen domination on convex polyominoes

摘录 / EXCERPT: Convex-polyomino maximal queen domination conjecture. The maximal domination problem for non-attacking queens on convex polyominoes is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057941 | MATHDB-356832

NP-completeness of minimal queen completion on chessboards

摘录 / EXCERPT: Chessboard queen completion conjecture. The minimal chessboard domination queen completion problem is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-057942 | MATHDB-356984

Polynomial-time solvability conjecture for MOTJ and SOC

摘录 / EXCERPT: Let MOTJ denote the problem of deciding, for an undirected graph $G=(V,E)$ with rational edge weights $w:E\to\mathbb{Q}$, a set $T\subseteq V$, and $k\in\mathbb{Q}$, whether there...

conjecture | open | MathDB 页面 | 原文线索

V2-057943 | MATHDB-357177

Tight subgradient-norm convergence conjecture for the proximal point algorithm

摘录 / EXCERPT: Subgradient-norm convergence conjecture. For every iterate x_N, there exists a subgradient $g_N\in\partial f(x_N)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-057944 | MATHDB-357189

The seven-hole conjecture for reduced clique graphs

摘录 / EXCERPT: Seven-hole conjecture. There is no chordal graph G such that $C_R(G)$ contains an induced cycle with seven or more vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-057945 | MATHDB-357190

The polynomial-time recognition conjecture for reduced clique graphs of chordal graphs

摘录 / EXCERPT: Reduced clique graph recognition conjecture. There is a polynomial-time algorithm for deciding whether a given graph is isomorphic to $C_R(G)$ for some chordal graph G.

conjecture | open | MathDB 页面 | 原文线索

V2-057946 | MATHDB-357210

The exhaustive-search conjecture for proofs of Kolmogorov randomness

摘录 / EXCERPT: Let R be the set of Kolmogorov-random strings, and let t be a proof-length bound for statements asserting that no proof of $x\in R$ has length at most t. Exhaustive-search co...

conjecture | open | MathDB 页面 | 原文线索

V2-057947 | MATHDB-357211

The isomorphism conjecture for true unprovable sentences

摘录 / EXCERPT: Let $\mathcal{S}$ and $\mathcal{T}$ be consistent theories, with $\mathcal{T}$ strictly stronger than $\mathcal{S}$, and let $\mathfrak{I}$ be a collection of sentences unprovable...

conjecture | open | MathDB 页面 | 原文线索

V2-057948 | MATHDB-357264

Optimality conjecture for serial and parallel clique-complex construction algorithms

摘录 / EXCERPT: Let G be a graph and let k be a nonnegative integer. Consider the serial algorithm and parallel algorithm for constructing the k-skeleta of the clique complex of G describe...

conjecture | open | MathDB 页面 | 原文线索

V2-057949 | MATHDB-357293

Generalized complexity conjecture for generalized log Calabi-Yau pairs

摘录 / EXCERPT: Generalized complexity conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-057950 | MATHDB-357438

The z-permanent anti-concentration conjecture

摘录 / EXCERPT: Let $\mathbb{T}$ denote the unit circle, let X be sampled from the complex Gaussian ensemble $\mathcal{G}=\mathcal{N}_{\mathbb{C}}(0,1)^{n\times n}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057951 | MATHDB-357498

NP-hardness of deciding Nash equilibrium existence in single-peaked Schelling games

摘录 / EXCERPT: Let $\Lambda\in(0,1)$ be a peak. A Single-Peaked Jump Schelling Game and a Single-Peaked Swap Schelling Game are instances of the respective models, and NE denotes...

conjecture | open | MathDB 页面 | 原文线索

V2-057952 | MATHDB-357517

NP-hardness conjecture for recognizing generic-radii sphere-packing contact graphs

摘录 / EXCERPT: A graph is a generic-radii sphere-packing contact graph if it is the contact graph of a sphere packing whose radii are generic. NP-hardness conjecture. Determining whether a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-057953 | MATHDB-357617

Low-Degree Conjecture for hypothesis testing

摘录 / EXCERPT: Let $\P_N$ and $\Q_N$ be distributions on $\mathbb{R}^N$, and let $\chi^2(\P_N\|\Q_N):=\mathbb{E}_{x\sim\Q_N}\left[\frac{d\P_N(x)}{d\Q_N(x)}\right]^2-1$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-057954 | MATHDB-357779

The automorphism-group conjecture for algorithms computing matrix multiplication

摘录 / EXCERPT: Let $\mathcal{A}$ be an algorithm of length r computing the matrix multiplication tensor $\langle m,n,p\rangle$. Its automorphism group is denoted by $Aut(\mathcal{A})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-057955 | MATHDB-358057

Fourier completely bounded polynomial influential-variable conjecture

摘录 / EXCERPT: Let $p:\{-1,1\}^n\to\mathbb{R}$ be a polynomial of degree at most d. Its Fourier completely bounded d-norm is denoted by $\|p\|_{\operatorname{fcb},d}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-057956 | MATHDB-358092

NP-hardness of the CONVEX coordinate-projection problem for rotation matrices

摘录 / EXCERPT: Let $S\subseteq [n]\times[n]$. Define the decision problem $\texttt{CONVEX}$ to output TRUE if the projection $\pi_S(\operatorname{SO}(n))$ is convex, and FALSE otherwise. NP-hardn...

conjecture | open | MathDB 页面 | 原文线索

V2-057957 | MATHDB-358098

Daniely-Shalev-Shwartz conjecture on DS dimension and risk complexity

摘录 / EXCERPT: Let $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ be a hypothesis class with Daniely-Shalev-Shwartz dimension d_{DS}, and let $(d_n)_{n \in \mathbb{N}}$ be the...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-057958 | MATHDB-358206

Linear-time emulation conjecture for the tree-path intersection algorithm

摘录 / EXCERPT: Linear-time emulation conjecture. $$ can be emulated in $\mathcal{O}(n)$ time.

conjecture | open | MathDB 页面 | 原文线索

V2-057959 | MATHDB-358216

Hardness conjecture for maximizing polytope sections, projections, and halfspace sections

摘录 / EXCERPT: Hardness conjecture. (i) It is NP-hard to find a hyperplane H which maximizes the number of i-dimensional faces of $P\cap H$. (ii) It is $\#P$-hard to find a hyperplane H w...

conjecture | open | MathDB 页面 | 原文线索

V2-057960 | MATHDB-358459

Omega sequences have linear complexity near half the sequence length

摘录 / EXCERPT: Omega linear-complexity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057961 | MATHDB-358461

Legendre sequence has near-maximal 2-adic complexity profile

摘录 / EXCERPT: Legendre 2-adic-complexity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-057962 | MATHDB-358540

Bounded tree-independence conjecture for (even hole, diamond)-free graphs

摘录 / EXCERPT: An (even hole, diamond)-free graph is a graph containing neither an even hole nor a diamond as an induced subgraph. Its tree independence number is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-057963 | MATHDB-358728

Lutz-De Panafieu-Stein-Scott active clustering query complexity conjecture

摘录 / EXCERPT: Active clustering query complexity conjecture. The average number of membership queries required is $(1+o(1))\sum_{i=1}^k i p_i n$. This conjecture concerns the optimal average que...

conjecture | open | MathDB 页面 | 原文线索

V2-057964 | MATHDB-358731

Completeness conjecture for five-dimensional reflexive polytopes with $h^{1,1}=1$

摘录 / EXCERPT: A five-dimensional reflexive polytope is a lattice polytope whose only interior lattice point is the origin and whose polar dual is also a lattice polytope. Such a polytope gives r...

conjecture | open | MathDB 页面 | 原文线索

V2-057965 | MATHDB-358960

Valiant's conjecture that VBP differs from VNP

摘录 / EXCERPT: Let VBP be the class of polynomial families computable by polynomial-size algebraic branching programs, and let VNP be Valiant's class of polynomially verifia...

conjecture | open | MathDB 页面 | 原文线索

V2-057966 | MATHDB-358961

Valiant's determinantal-complexity conjecture for the permanent

摘录 / EXCERPT: For a polynomial f, let $\mathrm{dc}(f)$ denote the least size of a matrix of affine linear forms whose determinant equals f. Let per_m denote the permanent of an...

conjecture | open | MathDB 页面 | 原文线索

V2-057967 | MATHDB-358964

Panova's equivalence conjecture for #P computation of Kronecker coefficients

摘录 / EXCERPT: Let $\textsc{ComputeKron}$ be the problem of computing $g(\lambda,\mu,\nu)$ from partitions λ,μ,ν, and let $\ell(\lambda)$ denote the number of parts of λ. Pano...

conjecture | open | MathDB 页面 | 原文线索

V2-057968 | MATHDB-358966

The Kronecker-coefficient obstruction conjecture in geometric complexity theory

摘录 / EXCERPT: Let $g(\lambda,n^d,n^d)$ be a rectangular Kronecker coefficient, and let $\gamma_{\lambda,d,n,m}$ be the multiplicity of the irreducible representation indexed by λ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-057969 | MATHDB-358967

The planted clique detection conjecture

摘录 / EXCERPT: Let $p\in(0,1)$ be constant. For each n, let $A_n:G_n\to\{0,1\}$ be a randomized polynomial-time algorithm, and let k_n be a sequence of positive integers satisfying ... For the...

conjecture | open | MathDB 页面 | 原文线索

V2-057970 | MATHDB-358968

The planted dense subgraph recovery conjecture

摘录 / EXCERPT: the planted dense subgraph recovery conjecture. No polynomial-time algorithm $\mathcal{A}:G\to{[n]\choose k_n}$ can achieve exact recovery asymptotically.

conjecture | open | MathDB 页面 | 原文线索

V2-057971 | MATHDB-358969

The k-planted clique detection conjecture

摘录 / EXCERPT: the k-planted clique conjecture. If G is an instance of $k\textsc{-PC}_D(n,k_n,p)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-057972 | MATHDB-358999

Linear-time positive definite Hankel factorization conjecture

摘录 / EXCERPT: Let $\mathbf{H} \in \mathbb{R}^{n\times n}$ be a positive definite Hankel matrix with bit complexity ℓ. A symmetric positive factorization algorithm should find a representati...

conjecture | open | MathDB 页面 | 原文线索

V2-057973 | MATHDB-359000

Linear-time inverse positive definite Hankel factorization conjecture

摘录 / EXCERPT: Let $\mathbf{H} \in \mathbb{R}^{n\times n}$ be a positive definite Hankel matrix with bit complexity ℓ and condition number bounded by 2^ℓ. An inverse symmetric positive...

conjecture | open | MathDB 页面 | 原文线索

V2-057974 | MATHDB-359121

Extension of the refined convergence-rate bound to regular graphs

摘录 / EXCERPT: Regular-graph extension conjecture. The bound given in Theorem remains valid in the simple symmetric case, for example for regular graphs without transitivity.

conjecture | open | MathDB 页面 | 原文线索

V2-057975 | MATHDB-359158

The Factor Conjecture for arithmetic circuit complexity

摘录 / EXCERPT: Factor Conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-057976 | MATHDB-359203

SETH-based lower-bound conjecture for Induced Matching on graphs of bounded pathwidth

摘录 / EXCERPT: Let (G,ℓ) be an instance of textsc{Induced Matching}, together with a path decomposition of G of width pw. The Strong Exponential Time Hypothesis (SETH) ass...

conjecture | open | MathDB 页面 | 原文线索

V2-057977 | MATHDB-359265

Hardness conjecture for the spiked Wishart model

摘录 / EXCERPT: Let $\beta \in [-1,\infty)$ and $\gamma>0$. For each $n\in\mathbb N$, set $N=\lceil n/\gamma\rceil$, and let $\mathbb P$ and $\mathbb Q$ be the spiked Wishart distributions defined...

conjecture | open | MathDB 页面 | 原文线索

V2-057978 | MATHDB-359521

Conjecture on the Weyl group optimization step in GIT algorithms

摘录 / EXCERPT: The algorithms use a Weyl group W and a subset $W'\subset W$ in their final optimization step to remove nonmaximal states. Weyl group optimization conjecture. At the last step of...

conjecture | open | MathDB 页面 | 原文线索

V2-057979 | MATHDB-359548

The bound-secrecy conjecture

摘录 / EXCERPT: Let X,Y,Z be jointly distributed random variables, and let $S(X:Y||Z)$ denote the secret-key rate and $I(X:Y \downarrow Z)$ the intrinsic information. The bound-secrecy conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-057980 | MATHDB-359549

The binarization criterion for bound secrecy

摘录 / EXCERPT: Let X,Y,Z be jointly distributed, and let $\overline{X}$ and $\overline{Y}$ be binary outputs of arbitrary binary output channels applied to X and Y. For $N\geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-057981 | MATHDB-359550

Independence-inducing binarizations for the three-symbol distribution

摘录 / EXCERPT: Let X,Y,Z have the distribution specified in the source, with Z determined by X and Y according to the displayed table. For $N\geq1$, let $\overline{Y^N}$ be any binary out...

conjecture | open | MathDB 页面 | 原文线索

V2-057982 | MATHDB-359551

The recursive target-value construction for two copies

摘录 / EXCERPT: Let $a_{ij}:=P_{\overline{Y^2}|Y^2}(0,rs)$ for $r,s\in\{1,2,3\}$, and define τ_2 on the indicated set by ... where $j=\left\lceil i/2\right\rceil$ for $1\leq i\leq6$, and ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-057983 | MATHDB-359552

Binarization and conditional independence for the Renner-Wolf candidate distribution

摘录 / EXCERPT: Let XYZ be the distribution displayed in the source, depending on a parameter a. The candidate-distribution binarization conjecture. There exists a value a such that, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-057984 | MATHDB-359582

The height-two poset linear-extension almost-completeness conjecture

摘录 / EXCERPT: Let $e(P)$ be the number of linear extensions of a finite poset P, and let e' denote its restriction to posets of height two. Write $\mathcal T_{e'}$ for the set of values atta...

conjecture | open | MathDB 页面 | 原文线索

V2-057985 | MATHDB-359584

The concise contingency-table realization conjecture

摘录 / EXCERPT: Let $\mathbf a\in\mathbb N^r$ and $\mathbf b\in\mathbb N^s$ have common size $n=|\mathbf a|=|\mathbf b|$, and let $\operatorname{CT}(\mathbf a,\mathbf b)$ be the number of continge...

conjecture | open | MathDB 页面 | 原文线索

V2-057986 | MATHDB-359587

The concise degree-sequence graph-counting conjecture

摘录 / EXCERPT: Let $\mathbf d=(d_1,\ldots,d_n)\in\mathbb N^n$ satisfy $0\le d_i\le n-1$, and let $c(\mathbf d)$ be the number of simple graphs on vertices $v_1,\ldots,v_n$ with $\deg(v_i)=d_i$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-057987 | MATHDB-359589

The concise tree-counting conjecture for planar graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a simple planar graph, and let $tr(G)$ be the number of trees in G of all sizes. Planar tree-counting conjecture. The function tr is concise: every positive in...

conjecture | open | MathDB 页面 | 原文线索

V2-057988 | MATHDB-359612

Constrained mixed-integer transfer conjecture

摘录 / EXCERPT: Let ℓ be a lower bound on the information complexity, with respect to a fixed oracle, for a family of continuous, constrained convex optimization instances. Mixed-integer tran...

conjecture | open | MathDB 页面 | 原文线索

V2-057989 | MATHDB-359613

Exponential mixed-integer lower bound for general binary oracles

摘录 / EXCERPT: Let n be the number of integer variables, let d be the continuous dimension, and let M, R, ρ, and ε be the problem parameters. A first-order chart...

conjecture | open | MathDB 页面 | 原文线索

V2-057990 | MATHDB-359614

Transfer from full-information first-order to general binary oracles

摘录 / EXCERPT: Let $u(n,d,R,\rho,M,\mathcal{G})$ denote the worst-case information complexity of a query strategy under the full-information first-order oracle based on a first-order chart...

conjecture | open | MathDB 页面 | 原文线索

V2-057991 | MATHDB-359861

The Tokareva-Shapovalov conjecture on balanced Boolean functions as bent derivatives

摘录 / EXCERPT: Tokareva-Shapovalov conjecture. The function f is a derivative of a bent function.

conjecture | open | MathDB 页面 | 原文线索

V2-057992 | MATHDB-360020

The critical-region equation for randomized graph coloring experiments

摘录 / EXCERPT: Critical-region conjecture. The critical regions are characterized by the equation

conjecture | open | MathDB 页面 | 原文线索

V2-057993 | MATHDB-360071

The discrepancy-maximization conjecture for the middle slice

摘录 / EXCERPT: Let $A\in\binom{[n]}{n/2}$, and define $\textit{Disc-max-}d(A)=1$ if and only if ... Equivalently, if $disc(j)$ is the signed discrepancy of the prefix $[j]$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-057994 | MATHDB-360122

Resilience tractability conjecture for unions of connected conjunctive queries

摘录 / EXCERPT: Resilience tractability conjecture. If the structure $(\{0,1\};\operatorname{OIT})$ has no pp-construction in Γ, then Γ has a fractional polymorphism of arity...

conjecture | open | MathDB 页面 | 原文线索

V2-057995 | MATHDB-360129

The alignment conjecture for the padded permanent and determinant

摘录 / EXCERPT: Let f be the form under consideration, let f_0 be its leading form, let $L(\lambda)$ be the Levi subgroup associated with the special one-parameter subgroup λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-057996 | MATHDB-360243

Radical Sylvester-Gallai conjecture

摘录 / EXCERPT: Let $\rK$ be the base field. For $d\in\N$, an $\rsg{d}$ configuration is a configuration $\cF$ of polynomials as defined in the cited formulation of Gupta's conjecture; its transce...

conjecture | open | MathDB 页面 | 原文线索

V2-057997 | MATHDB-360244

Strong radical Sylvester-Gallai conjecture

摘录 / EXCERPT: Let $\rK$ be the base field, let $d\in\N$, and let $\cF$ be an $\rsg{d}$ configuration. Write $\Kspan{\cF}$ for its $\rK$-linear span. Strong radical Sylvester-Gallai conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-057998 | MATHDB-360330

Completeness conjecture for matrix and universal tracial satisfiability

摘录 / EXCERPT: Completeness conjecture. In parts (a) and (b) of the cited dichotomy theorem, $\operatorname{SAT}_{matrix}(\mathcal{L})$ is RE-complete, and...

conjecture | open | MathDB 页面 | 原文线索

V2-057999 | MATHDB-360457

Charikar-Liu-Liu-Vuong conjecture on balanced grid partitions

摘录 / EXCERPT: Let the $m \times n$ grid graph be partitioned into k connected pieces, and consider the spanning tree distribution on partitions, in which a partition has weight proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-058000 | MATHDB-360467

The oblivious adversary conjecture for Oja's algorithm

摘录 / EXCERPT: Oja's linear-regret conjecture. There exists such a sequence $\boldsymbol{A}_1,\dots,\boldsymbol{A}_T$, chosen obliviously and independently of Oja's initial random vector, for whi...

conjecture | open | MathDB 页面 | 原文线索

V2-058001 | MATHDB-360593

The low-degree runtime correspondence conjecture

摘录 / EXCERPT: Consider a testing problem in the asymptotic setting, with degree-D polynomial tests used as a proxy for computationally efficient algorithms. Strong separation means ... while wea...

conjecture | open | MathDB 页面 | 原文线索

V2-058002 | MATHDB-360603

Brakensiek-Guruswami's approximate graph homomorphism hardness conjecture

摘录 / EXCERPT: Let $\mathbf{A}$ and $\mathbf{B}$ be non-bipartite loopless undirected graphs such that $\mathbf{A}\to\mathbf{B}$. The Brakensiek-Guruswami conjecture asserts that ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-058003 | MATHDB-360623

Non-membership conjecture for the sign-imbalance problem in #P

摘录 / EXCERPT: For a poset P, its sign imbalance is the absolute difference between the numbers of even and odd linear extensions. The computational problem Sign Imbalance takes a poset P as...

conjecture | open | MathDB 页面 | 原文线索

V2-058004 | MATHDB-360631

The Chvátal-Reed satisfiability threshold conjecture for random k-SAT

摘录 / EXCERPT: Chvátal-Reed conjecture. For all $k\geq 2$ there exists a value $\alpha_\mathrm{sat}(k)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058005 | MATHDB-360636

Polynomial-time conjecture for counting constrained linear extensions

摘录 / EXCERPT: Let P be a finite poset with fixed prescribed values represented by $\mathbf z,\mathbf c$, and let $N_{\mathbf z,\mathbf c}(P)$ denote the number of linear extensions satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-058006 | MATHDB-360638

Height-two linear-extension universality conjecture

摘录 / EXCERPT: Let m be a positive integer. Height-two universality conjecture. For all sufficiently large m, there exists a poset P of height two such that $e(P)=m$. The source states that...

conjecture | open | MathDB 页面 | 原文线索

V2-058007 | MATHDB-360733

The SDBV near-hardness conjecture for the $(1+1)$-EA

摘录 / EXCERPT: Let SDBV (Switching Dynamic Binary Value) be the dynamic monotone function defined in the paper, and let $T_f(n)$ denote the expected number of generations required by the $(1+1)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-058008 | MATHDB-360767

Sharp o-minimal complexity conjecture for self-dual flux vacua

摘录 / EXCERPT: Let the self-dual flux-vacuum locus be the locus of flux vacua satisfying the self-duality conditions, and let (F,D) denote its sharp o-minimal format and degree. Sharp o-minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-058009 | MATHDB-360997

Euclidean Steiner tree conjecture for regular simplicial complexes

摘录 / EXCERPT: Let $r\in(0,1)$ and $\alpha\in(0,1/r-1)$. A regular, unit simplicial complex on m vertices has constituent simplices that are regular and unit. Euclidean Steiner tree conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-058010 | MATHDB-361056

The bipartite MaxLA proportion conjecture for free trees

摘录 / EXCERPT: Strong bipartition conjecture. The limit exists and is a positive constant:

conjecture | open | MathDB 页面 | 原文线索

V2-058011 | MATHDB-361057

The positive bipartite MaxLA proportion conjecture for free trees

摘录 / EXCERPT: Weak bipartition conjecture. The same limiting assertion as in the strong bipartition conjecture holds with merely $c>0$; equivalently, the asymptotic proportion is positive.

conjecture | open | MathDB 页面 | 原文线索

V2-058012 | MATHDB-361058

The slow-decay conjecture for bipartite MaxLA on free trees

摘录 / EXCERPT: Slow-decay conjecture. The proportion decays according to

conjecture | open | MathDB 页面 | 原文线索

V2-058013 | MATHDB-361059

The odd-bridge conjecture for 2-linear trees

摘录 / EXCERPT: Odd-bridge conjecture. If the bridge of T has odd length, then no maximal non-bipartite arrangement is a maximum arrangement; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-058014 | MATHDB-361394

The second-step complexity conjecture for multivariable Chebyshev subdivision

摘录 / EXCERPT: Let n be the dimension and d the degree. Consider a system of n degree-d Chebyshev polynomials in n variables, and suppose that d is sufficiently large. Second-step com...

conjecture | open | MathDB 页面 | 原文线索

V2-058015 | MATHDB-361503

The 2-DIOMEGA-ORDERING NP-completeness conjecture

摘录 / EXCERPT: Let T be a tournament, and let $\operatorname{\overrightarrow{\omega}}(T)$ denote the minimum, over all orderings of $V(T)$, of the clique number of the corresponding backedge gr...

conjecture | open | MathDB 页面 | 原文线索

V2-058016 | MATHDB-361557

PSPACE-completeness conjecture for Fast-Strategy on bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph, and let t denote the number of guards in the eternal domination game. The decision problem $\textsc{Fast-Strategy}$ asks whether the guards have a s...

conjecture | open | MathDB 页面 | 原文线索

V2-058017 | MATHDB-361606

Positive-characteristic validity of the singularity classification algorithm

摘录 / EXCERPT: Let $\overline{K}$ be an algebraically closed field of characteristic p, and let the paper's singularity-classification algorithm be denoted by Algorithm 1. Positive-characterist...

conjecture | open | MathDB 页面 | 原文线索

V2-058018 | MATHDB-361675

Papadimitriou-Yannakakis' completeness conjecture for minimum generating sets of quasigroups

摘录 / EXCERPT: Papadimitriou-Yannakakis' conjecture. The minimum generating set problem for quasigroups is $\exists^{\log^2 n}\mathsf{P}$-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-058019 | MATHDB-361759

Color class count function equivalence conjecture for partial colorings

摘录 / EXCERPT: Let G_i' be the graph associated with a node X_i of the smooth tree-decomposition, and let f and g be partial colorings of G_i'. For each tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-058020 | MATHDB-361895

Approximation conjecture for the ordered clique number of tournaments

摘录 / EXCERPT: Approximation conjecture. There is a function f such that for every integer k, there is a polynomial-time algorithm which, given a tournament T, correctly concludes that...

conjecture | open | MathDB 页面 | 原文线索

V2-058021 | MATHDB-362285

The informal low degree conjecture for computational thresholds

摘录 / EXCERPT: Let $(\mathcal{X}_n,\mathcal{P})$ be a sequence of CLVMs, giving rise to sequences of measures $\mathbb{P}_n$ and $\mathbb{Q}_n$ on Ω^N, where $N=N(n)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058022 | MATHDB-362314

Polynomial-time Hermite normal form conjecture for pseudo-matrices over A_{fi}

摘录 / EXCERPT: Let A_{fi} be the function-field ring considered in the paper, and let a pseudo-matrix over A_{fi} represent a module by its coefficient ideals and matrix. Polynomial-time Herm...

conjecture | open | MathDB 页面 | 原文线索

V2-058023 | MATHDB-362344

Geometric interpolation conjecture for EXTRA performance guarantees

摘录 / EXCERPT: Geometric interpolation conjecture. For all $\mathcal{K}>0$, $\kappa_1,\kappa_2>0$, and $\theta\in[0,1]$, the mixed-conditioning performance satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058024 | MATHDB-362385

Ollinger's conjecture on undecidability of 8-, 9-, and 10-polyomino tiling

摘录 / EXCERPT: For a fixed positive integer k, the k-polyomino tiling problem asks whether there is an algorithm deciding whether a set of k polyominoes can tile the plane by translated cop...

conjecture | open | MathDB 页面 | 原文线索

V2-058025 | MATHDB-362450

Exact finite-iteration convergence rates for DCA regimes p_1 through p_4

摘录 / EXCERPT: Finite-iteration tightness conjecture. The DCA rates corresponding to regimes p_1, p_2, p_3 and p_4 are tight for any number of iterations N.

conjecture | open | MathDB 页面 | 原文线索

V2-058026 | MATHDB-362451

Asymptotic exactness of DCA rates for regimes p_5 and p_6

摘录 / EXCERPT: Asymptotic-rate conjecture. The exact sublinear rates for regimes p_5 and p_6 correspond to

conjecture | open | MathDB 页面 | 原文线索

V2-058027 | MATHDB-362539

The binary analogue of Brauer's addition-chain bound

摘录 / EXCERPT: Let $\|n\|_{2}$ denote the binary complexity of the positive integer n. Let r be a positive integer and let s be a nonnegative integer, with $(r,s)\ne(1,0)$. The binary analo...

conjecture | open | MathDB 页面 | 原文线索

V2-058028 | MATHDB-362615

The mighty tuple II conjecture for quantified constraint satisfaction

摘录 / EXCERPT: Let A be a finite domain, let Γ be a constraint language on A, and let $\QCSP(\Gamma)$ denote the quantified constraint satisfaction problem for Γ. A mighty tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-058029 | MATHDB-362673

Rational dlt strata and spherical dual complex imply birational toricity

摘录 / EXCERPT: Let (X,B) be a log Calabi--Yau pair of dimension n. Write $4\mathcal{D}(X,B)4$ for its dual complex, and let $c_{\rm bir}(X,B)$ denote its birational complexity. Let Σ^n...

conjecture | open | MathDB 页面 | 原文线索

V2-058030 | MATHDB-362688

Extended asymptotic rank conjecture for concise tensors

摘录 / EXCERPT: Let $\mathbb{F}$ be the underlying field, and consider concise tensors in ... Write $\sigma(d)$ for the worst-case exponent of this space. Extended asymptotic rank conjecture. For al...

conjecture | open | MathDB 页面 | 原文线索

V2-058031 | MATHDB-362795

The pp-construction conjecture for hard finite-domain PCSPs over homogeneous reducts

摘录 / EXCERPT: Pp-construction conjecture. If $\operatorname{PCSP}(G,H)$ is NP-hard, then K_3 has a pp-construction in S.

conjecture | open | MathDB 页面 | 原文线索

V2-058032 | MATHDB-362838

Learning parity with noise computational hardness conjecture

摘录 / EXCERPT: Let the secret key be $\mathbf{s}\sim_{\operatorname{unif}}\mathbb{F}_2^k$ and let the noise rate be $\eta=0.05$. A noisy observation is ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-058033 | MATHDB-362864

Hardness conjecture for certifying lift-monotone properties of random regular graphs

摘录 / EXCERPT: Let H be a d-regular multigraph on k vertices, and let $\varepsilon>0$. Write $\mathcal{L}_{m}(H)$ for an m-sheeted lift of H. The operators...

conjecture | open | MathDB 页面 | 原文线索

V2-058034 | MATHDB-362936

Pak-Yang's three-rectangle NP-completeness conjecture

摘录 / EXCERPT: A rectangle tiling set is a finite set R of rectangles, and a simply connected region is a region in the plane without holes that is to be tiled using translated copies of rectan...

conjecture | open | MathDB 页面 | 原文线索

V2-058035 | MATHDB-362955

Panyushev's coisotropy conjecture for complexity one

摘录 / EXCERPT: Let G/H be an affine homogeneous space with H connected reductive, let ${\mathfrak m}$ be its isotropy representation, and let $c_G(G/H)$ denote the complexity of G/H. Panyus...

conjecture | open | MathDB 页面 | 原文线索

V2-058036 | MATHDB-363128

The admissible-cycle filling complexity conjecture for spherical Artin complexes

摘录 / EXCERPT: Let A_S be an irreducible spherical Artin group, let Δ_S be its associated Artin complex, and let $\mathsf C_S$ be its associated Coxeter complex with the natural action o...

conjecture | open | MathDB 页面 | 原文线索

V2-058037 | MATHDB-363233

Uniform distribution conjecture for the interlaced Halton sequence

摘录 / EXCERPT: Uniform distribution conjecture. The d-dimensional interlaced Halton sequence in Algorithm is uniformly distributed.

conjecture | open | MathDB 页面 | 原文线索

V2-058038 | MATHDB-363756

Subexponential optimization and sampling conjecture for the low-temperature CREM

摘录 / EXCERPT: Consider a continuous random energy model (CREM) with concave covariance function A, inverse temperature $\beta>\beta_c$, $\tilde\text{ and }$?, and ?. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058039 | MATHDB-363776

Strong Exponential Time Hypothesis

摘录 / EXCERPT: The iterated logarithm function is defined recursively by ... For a positive integer k, k-SAT asks whether a Boolean formula with n variables has a satisfying assignment. Stron...

conjecture | open | MathDB 页面 | 原文线索

V2-058040 | MATHDB-363777

Orthogonal Vectors Conjecture

摘录 / EXCERPT: For positive integers n and m, let $\emph{\textsf{OV}}(n,m)$ denote the Orthogonal Vectors problem: given sets $A=\{x^{(1)},\ldots,x^{(n)}\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-058041 | MATHDB-363822

Minimal-space characterization conjecture for perfect incremental samplers

摘录 / EXCERPT: Let $\mathbf{x} \in \mathbb{R}_+^n$ be updated by an incremental stream. Let G be a nonnegative function on the coordinates, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-058042 | MATHDB-364022

Conjecture on the sum-freedom of the multiplicative inverse function

摘录 / EXCERPT: Sum-freedom conjecture for the multiplicative inverse function. For even n, F_{inv} is not kth order sum-free for $2\le k\le n-2$. For odd n,...

conjecture | open | MathDB 页面 | 原文线索

V2-058043 | MATHDB-364051

The maxmin-ω eigenvector extraction conjecture

摘录 / EXCERPT: Let $A\in\mathbb{R}^{n\times n}$ have rational entries, let $p\in[n]$, and set $\omega=p/n$. Let λ_p be the eigenvalue produced by the refinement algorithm, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058044 | MATHDB-364071

The linear-complexity spectrum conjecture for de Bruijn sequences

摘录 / EXCERPT: Let $n\geq 3$, let c be an integer satisfying ... and suppose $c\neq 2^{n-1}+n+1$. A span n de Bruijn sequence is a cyclic binary sequence in which every binary word of length...

conjecture | open | MathDB 页面 | 原文线索

V2-058045 | MATHDB-364099

Single-bubble conjecture for Clifford-Euclidean orders

摘录 / EXCERPT: Let $\mathcal{O}$ be an order in the relevant Clifford algebra, and suppose that it is Clifford-Euclidean, meaning that it admits the Clifford-Euclidean division property used to c...

conjecture | open | MathDB 页面 | 原文线索

V2-058046 | MATHDB-364236

NP-hardness of approximating ultrafilter width under connectivity expansion

摘录 / EXCERPT: Let (X,f) be a finite connectivity system. Its ultrafilter width is ... provided that at least one such ultrafilter exists. Connectivity-system SSE-type hardness conjecture. Under...

conjecture | open | MathDB 页面 | 原文线索

V2-058047 | MATHDB-364335

The converse to uniform complexity implying dimension equality for sofic self-affine fractals

摘录 / EXCERPT: Let $S \subset D_2^{\mathbb{N}}$ be a sofic system with primitive adjacency matrix. Let X be the associated self-affine fractal, and say that S has uniform complexity when the...

conjecture | open | MathDB 页面 | 原文线索

V2-058048 | MATHDB-364449

The Kronecker-product conjecture for persistent tensors

摘录 / EXCERPT: Let $\mathcal{P}_1\in U_1\otimes\cdots\otimes U_n$ and $\mathcal{P}_2\in V_1\otimes\cdots\otimes V_n$ be persistent tensors, where $U_1,\ldots,U_n$ and $V_1,\ldots,V_n$ are vector...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058049 | MATHDB-364467

Eventual multiplicativity of algorithmic complexity under parity-perturbed composition

摘录 / EXCERPT: Composition multiplicativity conjecture. For every $n,m\ge1$, there exists $k_0(n,m)\in\mathbb N$ such that, for every Boolean functions f and g on n and m bits, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-058050 | MATHDB-364610

XQUATH (Linear Cross-Entropy Quantum Threshold Assumption)

摘录 / EXCERPT: Let U be a quantum circuit drawn from a distribution $\mathcal{D}$, and let ... A classical algorithm is polynomial-time if it runs in time polynomial in the circuit description an...

conjecture | open | MathDB 页面 | 原文线索

V2-058051 | MATHDB-364726

Wqo-based decidability conjecture for the joint embedding property

摘录 / EXCERPT: Let $(\mathcal{F},\le)$ be a well-quasi-order (wqo), and suppose there are algorithms for basic problems related to $\le$, including deciding whether $X\le Y$ and, given X, findi...

conjecture | open | MathDB 页面 | 原文线索

V2-058052 | MATHDB-364746

Hanauer's feedback arc set conjecture for degree-five oriented multigraphs

摘录 / EXCERPT: Hanauer's conjecture. If $Delta\leq 5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-058053 | MATHDB-364747

Hanauer et al.'s feedback arc set conjecture for strongly connected degree-five oriented graphs

摘录 / EXCERPT: Hanauer et al.'s conjecture. If $Delta\leq 5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-058054 | MATHDB-364771

Functional monotonicity conjecture for contraction distance to minor-closed classes

摘录 / EXCERPT: Let $\mathcal{H}$ be a minor-closed graph class. For a graph G, let $\mathsf{id}_{\mathcal{H}}(G)$ denote the minimum number of edge identifications needed to transform G into...

conjecture | open | MathDB 页面 | 原文线索

V2-058055 | MATHDB-364788

Extrapolated multilevel computational-work conjecture

摘录 / EXCERPT: Let $\prescript{\mathrm{Ext}}{}{\Phi}^{\ell}$ and $\prescript{\mathrm{Ext}}{}{\Psi}^{\ell}$ denote the extrapolated fine- and coarse-level quantities at level ℓ. Extrapolated...

conjecture | open | MathDB 页面 | 原文线索

V2-058056 | MATHDB-364800

The complexity dichotomy conjecture for backbone coloring of degree-4 graphs

摘录 / EXCERPT: Let G be a graph with maximum degree $Delta(G) \le 4$, let $\lambda \ge 2$, and let H be a backbone. For an integer l, $BBC_{\lambda}(G,H) \le l$ denotes the λ-backbo...

conjecture | open | MathDB 页面 | 原文线索

V2-058057 | MATHDB-364812

Fixed-dimensional hardness conjecture for MAX-EMST-ratio

摘录 / EXCERPT: For a point set in Euclidean space, the k-MAX-EMST-ratio problem restricts the input to k dimensions. Fixed-dimensional hardness conjecture. The k-MAX-EMST-ratio problem for...

conjecture | open | MathDB 页面 | 原文线索

V2-058058 | MATHDB-364813

Bipartite-coloring approximation conjecture for planar MAX-EMST-ratio

摘录 / EXCERPT: Let $P\subseteq\mathbb{R}^2$ be a finite point set. A bipartite coloring is obtained by choosing a Euclidean minimum spanning tree of P and partitioning $P=R\cup B$ so that every...

conjecture | open | MathDB 页面 | 原文线索

V2-058059 | MATHDB-364814

NP-hardness conjecture for the maximum chromatic crossing number

摘录 / EXCERPT: For a finite point set in the plane and a red-blue partition, the chromatic crossing number is the number of crossings between an edge of $\operatorname{EMST}(R)$ and an edge of...

conjecture | open | MathDB 页面 | 原文线索

V2-058060 | MATHDB-365088

The complexity bound conjecture for S(e,23,e)-adic languages

摘录 / EXCERPT: Complexity bound conjecture. The complexity function $p_\mathcal{L}$ for any $S(e,23,e)$-adic language $\mathcal{L}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058061 | MATHDB-365113

The quantum GapSVP hardness conjecture

摘录 / EXCERPT: Quantum GapSVP hardness conjecture. There is no polynomial-time quantum algorithm that solves GapSVP to within polynomial factors.

conjecture | open | MathDB 页面 | 原文线索

V2-058062 | MATHDB-365260

Real convergence of Browkin-type continued fractions

摘录 / EXCERPT: Let $\alpha^{(p)}$ be a quadratic irrational in $\mathbb{Q}_p$ that has an image in $\mathbb{R}$. Consider its continued fraction obtained by Browkin I, Browkin II or Algorithm MR....

conjecture | open | MathDB 页面 | 原文线索

V2-058063 | MATHDB-365280

The hypercube-decomposition conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Hypercube-decomposition conjecture. For any hypercube decomposition $[u,z]$ of $[u,v]$,

conjecture | open | MathDB 页面 | 原文线索

V2-058064 | MATHDB-365295

Sum-freeness conjecture for the multiplicative inverse function

摘录 / EXCERPT: Sum-freeness conjecture. For odd $n\ge 7$,

conjecture | open | MathDB 页面 | 原文线索

V2-058065 | MATHDB-365445

The Fourier-Hahn-Lévy conjecture for tractable integer-valued moments

摘录 / EXCERPT: Fourier-Hahn-Lévy conjecture. If $f:\mathbb{Z}\to\mathbb{Z}$ is tractable, then the f-moment can be estimated to within a constant factor with a Lévy-Tower, either directly or af...

conjecture | open | MathDB 页面 | 原文线索

V2-058066 | MATHDB-365683

Truncation-loss conjecture for quasi-regular distributions

摘录 / EXCERPT: Let $\boldsymbol{F}=\{F_i\}_{i\in[n]}$ be $n\geq1$ asymmetric quasi-regular valuation distributions, let $\varepsilon\in(0,1)$, and define truncated distributions...

conjecture | open | MathDB 页面 | 原文线索

V2-058067 | MATHDB-365874

Impagliazzo-Paturi-Zane's Strong Exponential Time Hypothesis

摘录 / EXCERPT: Let q-CNFSAT denote the satisfiability problem for conjunctive normal form formulas whose clauses have at most q literals, and let n be the number of variables in the input f...

conjecture | open | MathDB 页面 | 原文线索

V2-058068 | MATHDB-365882

Lovett's sparse low-rank matrix conjecture

摘录 / EXCERPT: Lovett's conjecture. The matrix M contains an all-zero square submatrix of size at least

conjecture | open | MathDB 页面 | 原文线索

V2-058069 | MATHDB-365910

The Abelian embedding conjecture for approximability of satisfiable k-CSPs

摘录 / EXCERPT: Abelian embedding conjecture. For a distribution μ on Σ^k, Conclusion $$ holds if and only if μ admits no Abelian embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-058070 | MATHDB-366019

The low-degree conjecture for strong detection

摘录 / EXCERPT: Let $\mathcal{Q}$ and $\mathcal{P}$ be sufficiently nice distributions, and let $\operatorname{Adv}_{\leq D}(\mathcal{Q},\mathcal{P})$ denote the degree-D low-degree advantage fo...

conjecture | open | MathDB 页面 | 原文线索

V2-058071 | MATHDB-366060

Schubert-vanishing non-membership conjecture for coNP

摘录 / EXCERPT: Let $\operatorname{sc}_{\mathrm{SchubertVanishing}}$ be the decision problem asking whether a Schubert coefficient vanishes, for Schubert coefficients in the classical types A,...

conjecture | open | MathDB 页面 | 原文线索

V2-058072 | MATHDB-366061

coNP-hardness conjecture for Schubert vanishing

摘录 / EXCERPT: Let $\operatorname{sc}_{\mathrm{SchubertVanishing}}$ be the decision problem asking whether a Schubert coefficient vanishes. Schubert-vanishing hardness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-058073 | MATHDB-366189

Random graph-state overlap conjecture for Pauli-diagonal product states

摘录 / EXCERPT: Let G be sampled uniformly from graph states, and let $d4af$ be the set of eigenstates of the operators $X+Y$ and $X-Y$. For an n-qubit product state, write...

conjecture | open | MathDB 页面 | 原文线索

V2-058074 | MATHDB-366346

Bertsimas-Grigni universal TSP lower-bound conjecture on the plane

摘录 / EXCERPT: Bertsimas-Grigni conjecture. For every linear order $\leq$ on the unit square,

conjecture | open | MathDB 页面 | 原文线索

V2-058075 | MATHDB-366460

The constant-playability relaxation of the playable 1-2-3 conjecture

摘录 / EXCERPT: Constant-playability relaxation. Let $G=(V,E)$ be a graph without isolated edges. Then there exists a constant $c\in\mathbb{N}\cup\{0\}$ such that G is f-unbalanceable, where

open_problem | open | MathDB 页面 | 原文线索

V2-058076 | MATHDB-366871

Finite-dimensional linear minimization is no slower than projection

摘录 / EXCERPT: Let $\mathcal H$ be the Hilbert space underlying the compact convex set considered, and let P denote the optimal runtime complexity of its projection operator, while $L(0)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-058077 | MATHDB-366944

Average puncture bound for covering with Reed-Solomon codes

摘录 / EXCERPT: Let C be an $[n,k,d]_q$ generalized Reed-Solomon (GRS) code, and consider the covering algorithm that repeatedly punctures the received word and applies a GRS decoder. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058078 | MATHDB-366976

NP-completeness conjecture for M-identifiability

摘录 / EXCERPT: Let $G=(V \cup \mathcal{H},D)$ be a factor analysis graph. M-identifiability and extended M-identifiability are the two identifiability properties defined by the matching criterion...

conjecture | open | MathDB 页面 | 原文线索

V2-058079 | MATHDB-367263

The tractability conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: The tractability conjecture. If ${\mathbb A}$ does not pp-construct ${\mathbb K}_3$, then $\operatorname{CSP}({\mathbb A})$ is polynomial-time solvable.

conjecture | open | MathDB 页面 | 原文线索

V2-058080 | MATHDB-367264

Restricted CSP tractability conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Restricted CSP tractability conjecture. If the restricted CSP-template does not rpp-construct $({\mathbb K}_3,{\mathbb L})$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-058081 | MATHDB-367284

The bounded blocky-matrix complexity conjecture for bounded max-norm

摘录 / EXCERPT: Let M be a Boolean matrix, meaning a matrix with entries in $\{0,1\}$. Its factorization max-norm is denoted by $\gamma_2(M)$, and let $\operatorname{bl}(M)$ be the minimum numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-058082 | MATHDB-367285

The large constant submatrix conjecture for bounded max-norm matrices

摘录 / EXCERPT: Let M be an $m\times n$ binary matrix, with entries in $\{0,1\}$, and let $\gamma_2(M)$ denote its factorization max-norm. A submatrix is obtained by restricting M to selected...

conjecture | open | MathDB 页面 | 原文线索

V2-058083 | MATHDB-367305

Block reduction conjecture for graph covers

摘录 / EXCERPT: Block reduction conjecture. The problem ${\sc H\text{-}Cover}$ for simple input graphs polynomially reduces to ${\sc H'\text{-}Cover}$ for simple input graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-058084 | MATHDB-367409

Index-equating conjecture for UCB algorithms

摘录 / EXCERPT: Consider a K-armed bandit model. For arm k, let $N_{k,T}$ be the number of pulls by time T, let $\bar\mu_{k,T}$ be its empirical mean reward, and let $I(\cdot)$ be the index...

conjecture | open | MathDB 页面 | 原文线索

V2-058085 | MATHDB-367433

Equality conjecture for the complexity bounds of the 42 magic-manifold knot families

摘录 / EXCERPT: Complexity equality conjecture. The inequality in the bounded-complexity result can be upgraded to equality for all 42 families of knots in this paper.

conjecture | open | MathDB 页面 | 原文线索

V2-058086 | MATHDB-367455

The constant-degree sum-of-squares barrier conjecture for random matrices

摘录 / EXCERPT: Fix $1<\alpha<2$, set $T=N^\alpha$, and let M_{ran} be a random $T\times N$ Gaussian matrix. Let $v\in\mathbb R^N$ satisfy $\|v\|_\infty\leq1$, and suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-058087 | MATHDB-367459

Randomized Exponential Time Hypothesis

摘录 / EXCERPT: Let φ be a [?]

conjecture | open | MathDB 页面 | 原文线索

V2-058088 | MATHDB-367633

The bounded realization problem is a2^P-complete

摘录 / EXCERPT: The bounded realization problem asks whether a valid realization exists subject to the specified bounds on the weights of the cycles. Bounded realization conjecture. The bounded re...

conjecture | open | MathDB 页面 | 原文线索

V2-058089 | MATHDB-367786

Linear-time computation of the tetra-separation decomposition

摘录 / EXCERPT: Let G be a 4-connected finite graph with n vertices and m edges. The linear-time decomposition algorithm conjecture. There is an algorithm that returns the tetra-separation d...

conjecture | open | MathDB 页面 | 原文线索

V2-058090 | MATHDB-367875

NP-hardness of maximal angle computation for the nonnegative orthant

摘录 / EXCERPT: Let $Q\subseteq\mathbb R^n$ be a polyhedral cone, and let ${\rm MA}(P,Q)$ denote the problem of computing the maximal angle between cones P and Q. Orthant-restriction conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-058091 | MATHDB-367896

Polynomial-time solvability of Max Partial H-Coloring on (bull, E)-free graphs

摘录 / EXCERPT: Let H be a simple graph. In $\textrm{Max Partial }H\textrm{-Coloring}\$, the input is a graph G with a revenue function $\mathsf{rev}:V(G)\times V(H)\to\mathbb{Q}_{\geq 0}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-058092 | MATHDB-367905

The Decay and Fission algorithm for good quivers

摘录 / EXCERPT: Let Q be a good 3d $\mathcal{N}=4$ quiver theory, let $\mathcal{M}_C(Q)$ denote its Coulomb branch, and let $\mathcal{L}$ be the set of decay and fission products of Q, ordered...

conjecture | open | MathDB 页面 | 原文线索

V2-058093 | MATHDB-367906

The Decay and Fission algorithm for generalized quivers

摘录 / EXCERPT: A generalized quiver is a triple (V,A,K), where V is finite, $A:V\times V\to\mathbb{Z}$, and $K:V\to\mathbb{Z}_{>0}$, satisfying the diagonal and symmetrizability conditions st...

conjecture | open | MathDB 页面 | 原文线索

V2-058094 | MATHDB-367977

The first-order-transduction characterization of bounded merge-width

摘录 / EXCERPT: For a graph class $\mathcal{C}$, say that it has bounded merge-width when its merge-width is bounded by a constant. A first-order transduction is an interpretation of graphs from s...

conjecture | open | MathDB 页面 | 原文线索

V2-058095 | MATHDB-368035

The low-degree polynomial hardness conjecture for independent sets in dense random graphs

摘录 / EXCERPT: Let $\mathbb{G}(n,1/2)$ be the Erdős-Rényi random graph on n vertices, and let a degree-r polynomial algorithm mean a polynomial of degree at most r used to find an independe...

conjecture | open | MathDB 页面 | 原文线索

V2-058096 | MATHDB-368126

Near-threshold exponential resolution lower-bound conjecture for Term Coding

摘录 / EXCERPT: Near-threshold resolution lower-bound conjecture. For an instance size n with no solution, $n\notin S$, if

conjecture | open | MathDB 页面 | 原文线索

V2-058097 | MATHDB-368252

Carlet's conjecture on the sum-freedom of the binary multiplicative inverse function

摘录 / EXCERPT: Let n be a positive integer and let $f_{\text{inv}}:\mathbb F_{2^n}\to\mathbb F_{2^n}$ be defined by ... A function is kth order sum-free if the sum of its values is nonzero on e...

conjecture | solved | MathDB 页面 | 原文线索

V2-058098 | MATHDB-368267

Undecidability conjecture for the two-variable, two-unary-predicate fragment of QS5

摘录 / EXCERPT: Let $\mathbf{QS5}$ be the quantified modal logic QS5, and consider its fragment using two individual variables and two unary predicate letters. Undecidability conjecture. The fragm...

conjecture | open | MathDB 页面 | 原文线索

V2-058099 | MATHDB-368268

Decidability conjecture for the two-variable, one-unary-predicate fragment of QS5

摘录 / EXCERPT: Let $\mathbf{QS5}$ be the quantified modal logic QS5, and consider its fragment using two individual variables and a single unary predicate letter. Decidability conjecture. The fra...

conjecture | open | MathDB 页面 | 原文线索

V2-058100 | MATHDB-368317

NP-completeness of mutual-visibility coloring

摘录 / EXCERPT: Mutual-visibility coloring conjecture. The mutual-visibility coloring decision problem is NP-complete.

conjecture | open | MathDB 页面 | 原文线索

V2-058101 | MATHDB-368518

Hendrych's NP-hardness conjecture for Bayesian A-optimal design

摘录 / EXCERPT: Hendrych's NP-hardness conjecture. The Bayesian AOD problem is NP-hard. This conjecture is resolved in the source: the paper proves NP-hardness for its regularized A-optimal design...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-058102 | MATHDB-368734

The FP-versus-symmetric subtraction conjecture for monotone 2SAT counting

摘录 / EXCERPT: FP-versus-symmetric subtraction conjecture. $[\mathrm{FP}-\text{\normalfont\textsc{\#mon2sat}}]^{\log}$ is strictly contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-058103 | MATHDB-368936

The analyzability lower-bound conjecture for propositional proof systems

摘录 / EXCERPT: Analyzability requires lower bounds. For every propositional proof system Q, if Q is analyzable, then Q is not optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-058104 | MATHDB-368937

The p-optimality obstruction conjecture for analyzable proof systems

摘录 / EXCERPT: Analyzability requires lower bounds or proofs that are hard to find. For every propositional proof system Q, if Q is analyzable, then Q is not p-optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-058105 | MATHDB-369092

Linear monochromatic submatrix conjecture for bounded randomized communication

摘录 / EXCERPT: Let M be an $n\times n$ Boolean matrix, and let randomized communication complexity mean the public-coin randomized communication complexity of M. A submatrix is monochromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-058106 | MATHDB-369093

Bounded $cgamma_2$-norm decomposition conjecture for Boolean matrices

摘录 / EXCERPT: Let M be an $m\times n$ Boolean matrix, and let $\gamma_2(M)$ denote its γ_2-norm. A blocky matrix is a blow-up of a permutation matrix. Bounded γ_2-norm decomposit...

conjecture | open | MathDB 页面 | 原文线索

V2-058107 | MATHDB-369108

Strong Dichotomy Conjecture for graph covers

摘录 / EXCERPT: For every graph H, consider the problem $\textsc{H-Cover}$ of deciding whether an input graph covers H. Strong Dichotomy Conjecture. For every graph H, the problem...

conjecture | open | MathDB 页面 | 原文线索

V2-058108 | MATHDB-369149

The absence of a closed-form expression for iterated powers

摘录 / EXCERPT: No-closed-form-expression hypothesis. There is no general closed-form expression for the components of a^n in terms of a fixed polynomial formula with finitely many terms whose s...

conjecture | open | MathDB 页面 | 原文线索

V2-058109 | MATHDB-369150

Hardness of the discrete iteration problem

摘录 / EXCERPT: Hardness conjecture for the discrete iteration problem. Given a and a^n, it is computationally hard to recover n for general parameter values, input vectors, and sufficiently...

conjecture | open | MathDB 页面 | 原文线索

V2-058110 | MATHDB-369152

Radius-selection conjecture for weak-form system identification

摘录 / EXCERPT: Radius-selection conjecture. The stagnation radius satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058111 | MATHDB-369252

The tractability conjecture for CSPs of reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Tractability conjecture. If $\mathfrak{B}$ does not pp-construct $(\{0,1\};\operatorname{NAE})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-058112 | MATHDB-369285

Steiner Tree complexity dichotomy for 1-colorful minor-closed classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a 1-colorful minor-closed class, meaning a class of 1-colorful graphs closed under taking colorful minors. A 1-colorful rainbow $(k\times k)$-grid is the...

conjecture | open | MathDB 页面 | 原文线索

V2-058113 | MATHDB-369364

Exponential proof-size conjecture for unprovable consistency extensions

摘录 / EXCERPT: Let $\mathcal{S}$ be a theory, let ϕ be a sentence, and let n be a natural number. Write $Con_{\mathcal{S}+\phi}(n)$ for the bounded consistency statement for...

conjecture | open | MathDB 页面 | 原文线索

V2-058114 | MATHDB-369365

Kolmogorov-random-string consistency proof-size conjecture

摘录 / EXCERPT: Let R be the set of Kolmogorov-random binary strings defined using the fixed universal Turing machine U, and let k be the threshold supplied by Chaitin's incompleteness theor...

conjecture | open | MathDB 页面 | 原文线索

V2-058115 | MATHDB-369367

Hardness of proving lattice shortest-vector threshold statements

摘录 / EXCERPT: Let a lattice be a discrete subgroup of Euclidean space, and let its shortest nonzero integer vector length be the relevant lattice parameter. Lattice shortest-vector hardness conj...

conjecture | open | MathDB 页面 | 原文线索

V2-058116 | MATHDB-369437

Infinite-domain CSP dichotomy conjecture

摘录 / EXCERPT: Let $\mathfrak{B}$ be a first-order reduct of a finitely bounded homogeneous structure in a finite relational signature. The constraint satisfaction problem...

conjecture | open | MathDB 页面 | 原文线索

V2-058117 | MATHDB-369438

The coRE-hardness conjecture for distinguishing commuting quantum game values

摘录 / EXCERPT: CoRE-hardness conjecture. Deciding which of these two cases holds is coRE-hard; equivalently, $\mathrm{MIP}^{\mathrm{co}}=\mathrm{coRE}$.

conjecture | open | MathDB 页面 | 原文线索

V2-058118 | MATHDB-369457

Strassen's direct sum conjecture for tensor rank

摘录 / EXCERPT: Let A_1,B_1,C_1,A_2,B_2,C_2 be finite-dimensional vector spaces over a field $\mathbb{K}$, and let $T_1\in A_1\otimes B_1\otimes C_1$ and $T_2\in A_2\otimes B_2\otimes C_2$. Thei...

conjecture | open | MathDB 页面 | 原文线索

V2-058119 | MATHDB-369635

Degree-three invariant recovery in the cryo-EM model

摘录 / EXCERPT: Cryo-EM sample-complexity conjecture. If $R\geq 3$ is large enough, then the minimal number of observations required for accurate recovery of the $\operatorname{O}(n)$ orbit of f...

conjecture | solved | MathDB 页面 | 原文线索

V2-058120 | MATHDB-369671

Termination of the point-to-edge matching uncrossing algorithm

摘录 / EXCERPT: Let $Q_1,\ldots,Q_n$ be points and let $B_1,\ldots,B_n$ be edges of a convex hull, with each point matched to one edge and the resulting triangles considered as in the preceding he...

conjecture | open | MathDB 页面 | 原文线索

V2-058121 | MATHDB-369972

Lower bound for the geodesic complexity of the dodecahedron

摘录 / EXCERPT: Let the geodesic complexity of a convex polyhedron mean the minimum number of sets needed to cover its geodesic motion-planning problem, so that a complexity of at least four means...

conjecture | open | MathDB 页面 | 原文线索

V2-058122 | MATHDB-369973

Upper bound for the geodesic complexity of the icosahedron

摘录 / EXCERPT: Let the geodesic complexity of a convex polyhedron mean the minimum number of sets needed to cover its geodesic motion-planning problem, so that a complexity of at most four means...

conjecture | open | MathDB 页面 | 原文线索

V2-058123 | MATHDB-369998

NP-hardness of computing lattice diameters of lattice polytopes

摘录 / EXCERPT: Let $d \geq 3$ and let P be a lattice d-polytope. Lattice-polytope diameter conjecture. Computing a lattice diameter of P is an $\operatorname{\mathsf{NP}}$-hard problem. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-058124 | MATHDB-370017

Polynomial-time hardness conjecture for balanced independent sets in dense random bipartite graphs

摘录 / EXCERPT: Let $G_{\mathrm{bip}}(n,p)$ be the random bipartite graph in the online model considered in the source, let γ be the balance parameter, and let $\alpha_{\rm COMP}$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-058125 | MATHDB-370018

Future-edge algorithm conjecture for balanced independent sets

摘录 / EXCERPT: Let G be the graph in the paper's online model, let $\mathcal{A}(G)$ be the set selected by an online algorithm $\mathcal{A}$, and let E be the set of all future edges ever rev...

conjecture | open | MathDB 页面 | 原文线索

V2-058126 | MATHDB-370058

High-order discrete adiabatic convergence under boundary cancellation

摘录 / EXCERPT: High-order discrete adiabatic convergence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058127 | MATHDB-370102

The optimal expected online discrepancy conjecture for sparse random vectors

摘录 / EXCERPT: Optimal expected online discrepancy conjecture. For all $2\leq d\leq n/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-058128 | MATHDB-370231

Unbounded strength of the determinant

摘录 / EXCERPT: For each form P, let $\str(P)$ denote the least number of reducible forms whose sum is P. Let det_n be the determinant viewed as a homogeneous polynomial in the rows...

conjecture | open | MathDB 页面 | 原文线索

V2-058129 | MATHDB-370432

XQAOA level-1 limiting performance conjecture for the Binary Paint Shop Problem

摘录 / EXCERPT: Consider Binary Paint Shop Problem instances encoded as Ising models, and let Δ_C be the number of paint swaps. Let XQAOA$_1$ denote the level-1 XQAOA ansatz, with its param...

conjecture | open | MathDB 页面 | 原文线索

V2-058130 | MATHDB-370672

The proposed functional form for collinearity as a function of correlation and dimension

摘录 / EXCERPT: Let c denote the collinearity parameter, Δ the correlation parameter, and p the number of variables in the latent linear-regression framework described above. Proposed f...

conjecture | open | MathDB 页面 | 原文线索

V2-058131 | MATHDB-370730

NP-hardness of Circuit Distance for polygons

摘录 / EXCERPT: Circuit Distance conjecture. The Circuit Distance problem is NP-hard for polygons.

conjecture | open | MathDB 页面 | 原文线索

V2-058132 | MATHDB-370748

The monochromatic-rectangle formulation of the log-rank conjecture

摘录 / EXCERPT: Let M be a Boolean matrix, and let a monochromatic rectangle be a submatrix all of whose entries have the same value. Let ${\mathrm{r}}(M)$ denote the rank of M over the reals,...

conjecture | open | MathDB 页面 | 原文线索

V2-058133 | MATHDB-370885

NP-hardness conjecture for sandwich problems with forbidden trees and cliques

摘录 / EXCERPT: For a set $\mathcal T$ of trees, a graph is $(\mathcal T\cup\{K_k\})$-free if it contains no member of $\mathcal T$ and no copy of K_k as an induced subgraph. Sandwich-problem ha...

conjecture | open | MathDB 页面 | 原文线索

V2-058134 | MATHDB-371097

Chudnovsky-Scott-Seymour's Burling-control conjecture

摘录 / EXCERPT: A graph class is Burling-controlled when its chromatic complexity is controlled by the Burling graphs, as defined in the source. An induced subdivision of a graph F is a subdivis...

conjecture | open | MathDB 页面 | 原文线索

V2-058135 | MATHDB-371100

The model-free string-graph coloring algorithm conjecture

摘录 / EXCERPT: Let G be a string graph, let n be its size, and let $\omega(G)$ and $\beta(G)$ denote the parameters used by the source. An $f(\omega(G),\beta(G))$-coloring is a coloring with...

conjecture | open | MathDB 页面 | 原文线索

V2-058136 | MATHDB-371600

Grochow's determinantal ideal closure conjecture

摘录 / EXCERPT: Let X be an $n\times n$ matrix of indeterminates, and let I_n be the ideal generated by the $\tfrac{n}{2}\times\tfrac{n}{2}$ minors of X. For a nonzero polynomial $f\in I_n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-058137 | MATHDB-371683

Polynomial-time acyclic k-dicolouring conjecture for tournaments

摘录 / EXCERPT: Tournament acyclic dicolouring algorithm conjecture. For every fixed $k\geq3$, it is polynomial-time decidable whether a tournament T satisfies

open_problem | open | MathDB 页面 | 原文线索

V2-058138 | MATHDB-371765

Quadratic quantum sample lower bounds for entropy, trace distance, and fidelity estimation

摘录 / EXCERPT: Quadratic sample-complexity conjecture. The quantum sample complexities of von Neumann entropy estimation, trace distance estimation, and fidelity estimation are

conjecture | open | MathDB 页面 | 原文线索

V2-058139 | MATHDB-371811

The conjecture on polynomial-size join/union expressions for permutation cycles

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathfrak{S}_n$ be the set of permutations of $[n]$, and let $\mathfrak{C}_n$ be the set of n-cycles in $\mathfrak{S}_n$. A join/union expression i...

conjecture | open | MathDB 页面 | 原文线索

V2-058140 | MATHDB-371933

Computational hardness conjecture for the spiked Wishart hypothesis test

摘录 / EXCERPT: Let $\beta\geq -1$ and $\gamma>0$, and for $n\in\mathbb{N}$ set $N=\lceil n/\gamma\rceil$. Under μ_0, draw $\mathbf{z}_1,\ldots,\mathbf{z}_N\sim N(0,I_n)$ independently. Under...

conjecture | open | MathDB 页面 | 原文线索

V2-058141 | MATHDB-372132

The unambiguity conjecture for substitution-system refinement

摘录 / EXCERPT: A substitution system is a tiling system with finitely many tile types and a substitution rule, and Algorithm is the paper's refinement procedure that augments each tile type with...

conjecture | open | MathDB 页面 | 原文线索

V2-058142 | MATHDB-372191

Subpolynomial treewidth conjecture for induced-minor-free graphs

摘录 / EXCERPT: Let $W_{t\times t}$ be the t-by-t hexagonal grid and let $K_{t,t}$ be the complete bipartite graph with both sides of the bipartition of size t. For a positive integer t, l...

conjecture | open | MathDB 页面 | 原文线索

V2-058143 | MATHDB-372284

Conjecture on puncture complexity and covering radius for fixed-rate GRS codes

摘录 / EXCERPT: Fixed-rate puncture conjecture. For an $[n,k,d]_q$ GRS code ${\mathcal C}$ of fixed rate R, the average number of punctures needed for $\operatorname{GRS\text{-}cover}$ to succee...

conjecture | open | MathDB 页面 | 原文线索

V2-058144 | MATHDB-372319

Universality-class extension conjecture for OAMP/VAMP

摘录 / EXCERPT: Let $\mathscr{U}$ be the universality class used in the paper, let $\mathscr{Q}$ be a matrix class, let OAMP/VAMP denote the stated algorithms, and let SE denote their state evolut...

conjecture | open | MathDB 页面 | 原文线索

V2-058145 | MATHDB-372464

The VCSP tractability dichotomy conjecture for resilience templates

摘录 / EXCERPT: Let cmu be a union of connected conjunctive queries over a signature $ctau$, let $cDelta_cmu$ be the associated valued structure, and let $cmathfrak A$ be a finitely bounded homo...

conjecture | open | MathDB 页面 | 原文线索

V2-058146 | MATHDB-372515

Strassen's asymptotic rank conjecture for tight concise tensors

摘录 / EXCERPT: Let $\mathbb{k}$ be the underlying field, and let $T\in\mathbb{k}^a\otimes\mathbb{k}^b\otimes\mathbb{k}^c$ be a tensor that is tight and concise. Here, concise means that the flatt...

conjecture | open | MathDB 页面 | 原文线索

V2-058147 | MATHDB-372586

SBP algorithmic threshold conjecture

摘录 / EXCERPT: Let $G\nin\mathbb{R}^{m\times n}$ have independent standard-normal entries. For constraint density $\alpha=\lim_{n\to\infty}m/n$ and $\kappa\in\mathbb{R}_+$, consider the statistic...

conjecture | open | MathDB 页面 | 原文线索

V2-058148 | MATHDB-372587

Parametric fl-RDT algorithmic conjecture

摘录 / EXCERPT: Assume the setup of the paper's fl-RDT theorem, including the quantities ψ_{rp}, ψ_{rd}, $\hat{\mathbf p}$, $\hat{\mathbf q}$, and $\hat{\mathbf c}$. Define the algorithm...

conjecture | open | MathDB 页面 | 原文线索

V2-058149 | MATHDB-372816

NP-hardness conjecture for the bicolored MST crossing number

摘录 / EXCERPT: Let P be a finite point set in the plane, and let $\operatorname{cr-MST}(P)$ denote its bicolored minimum spanning tree crossing number. The NP-hardness conjecture. Finding...

conjecture | open | MathDB 页面 | 原文线索

V2-058150 | MATHDB-372900

The shift-additivity conjecture for periodic binary sequences

摘录 / EXCERPT: Let n be a positive integer, let $c\geq \left\lceil\frac{n}{2}\right\rceil$, and let $d=n-c$. Let $\mathcal{B}(n,c,d)$ denote the specified class of binary sequences, let...

conjecture | open | MathDB 页面 | 原文线索

V2-058151 | MATHDB-373011

Morell-Skutella-type conjecture for unsplittable transshipments

摘录 / EXCERPT: Morell-Skutella-type conjecture. Given a b-transshipment x, one can efficiently compute an unsplittable b-transshipment y such that

conjecture | open | MathDB 页面 | 原文线索

V2-058152 | MATHDB-373108

TournamentClique completeness conjecture for the second level of the polynomial hierarchy

摘录 / EXCERPT: A tournament is a directed graph in which exactly one of the two possible directions is chosen for every pair of distinct vertices. For a tournament T, its directed clique number...

conjecture | open | MathDB 页面 | 原文线索

V2-058153 | MATHDB-373303

Aldous's instability conjecture for backoff protocols

摘录 / EXCERPT: Consider a multiple-access channel in discrete time with Poisson arrivals of mean birth rate $\lambda>0$. A backoff protocol is specified by a send sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-058154 | MATHDB-373326

Polynomial-time clustering threshold for Gaussian mixtures

摘录 / EXCERPT: Let n, d, and K be positive integers with $d\geq K$, and let Δ_* denote the separation parameter for the Gaussian-mixture clustering model. Write $a\gtrsim_{\log} b$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-058155 | MATHDB-373500

The skewed coloring sampling conjecture

摘录 / EXCERPT: Skewed sampling conjecture. There should be a polynomial-time algorithm that approximately samples uniformly from the set of $\vec{n}$-colorings of G whenever

conjecture | open | MathDB 页面 | 原文线索

V2-058156 | MATHDB-373997

Equality of the lower and upper support functionals on the boundary of the parameter simplex

摘录 / EXCERPT: Boundary support-functional conjecture. For every $\theta\in\Theta_{(\varkappa)}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-058157 | MATHDB-373999

Alon et al.'s VC-dimension conjecture for completions of gap Hamming distance

摘录 / EXCERPT: Let GHD_k^n be the partial sign matrix for the gap Hamming distance problem, and let A be any total sign matrix obtained by replacing every $*$ entry of...

conjecture | open | MathDB 页面 | 原文线索

V2-058158 | MATHDB-374060

Hard generator conjecture for propositional proof systems

摘录 / EXCERPT: A generator is a map $g:\{0,1\}^*\rightarrow\{0,1\}^*$ whose restriction to inputs of length n maps into strings of length $m(n)>n$ and is computed by circuits of size polynomial in...

conjecture | open | MathDB 页面 | 原文线索

V2-058159 | MATHDB-374234

The NP-hardness conjecture for lower oriented general position number two

摘录 / EXCERPT: Let G be a graph, and let ${\rm gp}_{\rightarrow}(G)$ denote the minimum general position number over all orientations of G. The lower-number-two hardness conjecture. It is NP-...

conjecture | open | MathDB 页面 | 原文线索

V2-058160 | MATHDB-374333

The annealed complexity formula for TAP critical points

摘录 / EXCERPT: The annealed complexity formula. There exists an open subset $D\subset\mathbb{R}$ containing $[\inf_\zeta\mathcal{P}\mathrm{arisi}(\zeta),+\infty)$ such that, for every $f\in D$,

conjecture | open | MathDB 页面 | 原文线索

V2-058161 | MATHDB-374334

The quenched complexity formula for TAP critical points

摘录 / EXCERPT: The quenched complexity formula. There exists an open subset $D\subset\mathbb{R}$ such that, for every $f\in D$,

conjecture | open | MathDB 页面 | 原文线索

V2-058162 | MATHDB-374335

Ultrametric organization of TAP states

摘录 / EXCERPT: Ultrametric organization conjecture. The TAP critical points at free-energy level f are organized in an ultrametric tree; uniformly sampled overlaps between two such states take...

conjecture | open | MathDB 页面 | 原文线索

V2-058163 | MATHDB-374485

Pudlák's conjecture on non-simulation of consistency extensions

摘录 / EXCERPT: Let $\mathcal S$ be the base arithmetic theory under consideration, and let $Con_{\mathcal S}(n)$ denote its bounded consistency statement at length n. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-058164 | MATHDB-374486

Kolmogorov hardness conjecture for random-axiom extensions

摘录 / EXCERPT: Fix a universal Turing machine U, a constant c_U, and let R be the set of strings x satisfying $K_U(x)\geq |x|-c_U$, where $K_U(x)$ is the plain Kolmogorov complexity of...

conjecture | open | MathDB 页面 | 原文线索

V2-058165 | MATHDB-374487

SETH-K-Finite conjecture on finite random-axiom consistency proofs

摘录 / EXCERPT: Fix a base theory $\mathcal S$ and let R be the set of Kolmogorov-random strings defined using the fixed universal machine and additive constant. Let $N_R^{\mathcal S}$ be the ma...

conjecture | open | MathDB 页面 | 原文线索

V2-058166 | MATHDB-374753

Cameron's Greedy Conjecture for primitive permutation groups

摘录 / EXCERPT: Let $G\leqslant\operatorname{Sym}(\Omega)$ be a finite permutation group. A base for G is a sequence of points of Ω with trivial pointwise stabiliser, and $b(G)$ denotes t...

conjecture | open | MathDB 页面 | 原文线索

V2-058167 | MATHDB-375106

Quantum advantage under spectral discretization for analytic Young-measure densities

摘录 / EXCERPT: Let $d\geq 2$, let $\nu_{x,y}(\xi)$ be a Young-measure density analytic in x and y, discretize ξ algebraically at $N_\xi\sim\varepsilon^{-1}$, and suppose R_1 is polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-058168 | MATHDB-375152

Valiant's conjecture for iterated matrix multiplication

摘录 / EXCERPT: Valiant's conjecture. Valiant's conjecture for R is

conjecture | open | MathDB 页面 | 原文线索

V2-058169 | MATHDB-375184

Nivat's conjecture on low rectangular complexity

摘录 / EXCERPT: Let $\mathcal A$ be a finite alphabet with at least two elements, and let $\eta\in\mathcal A^{\mathbb Z^2}$ be a two-dimensional configuration. For $k,n\in\mathbb N$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-058170 | MATHDB-375270

Bruhat-order monotonicity conjecture for the function f_n

摘录 / EXCERPT: Let $\mathfrak A_n$ denote the alternating group, let f_n be the function defined by the paper's Algorithm 1, and let $\leq$ denote the Bruhat order on $\mathfrak A_n$. Bruhat-or...

conjecture | open | MathDB 页面 | 原文线索

V2-058171 | MATHDB-375367

Low-degree conjecture for computational indistinguishability

摘录 / EXCERPT: Low-degree conjecture. If there exist $\epsilon>0$ and $D=D(n)\geq (\log n)^{1+\epsilon}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058172 | MATHDB-375555

Linear-time certifying algorithm conjecture for star coloring of split graphs

摘录 / EXCERPT: Linear-time certifying algorithm conjecture. For every fixed positive integer k, there is a certifying algorithm that runs in time

conjecture | open | MathDB 页面 | 原文线索

V2-058173 | MATHDB-375889

Gartland and Lokshtanov's induced-minor-free graph algorithm conjecture

摘录 / EXCERPT: Gartland and Lokshtanov's conjecture. For every fixed r and CMSO$_2$ formula ψ, $(\operatorname{tw} \leq r,\psi)$-MWIS and $\operatorname{lcol}(3)$ can be solved in polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-058174 | MATHDB-375940

Polynomial-time MIS conjecture for graphs forbidding independent planar minors

摘录 / EXCERPT: Independent-planar-minor MIS conjecture. For every planar H and every k, MIS is polynomial-time solvable on the class of kH-free graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-058175 | MATHDB-376020

Linear-time balanced four-coloring conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph with $n\geq 3$ vertices. A balanced four-coloring is a proper 4-coloring in which each color is used on fewer than $n/2$ vertices. Linear-time balanced...

conjecture | open | MathDB 页面 | 原文线索

V2-058176 | MATHDB-376045

Space-entropy lower-bound conjecture for online random sampling

摘录 / EXCERPT: Let $p_1,p_2,\dots$ be a sequence of arbitrary discrete distributions presented adaptively over time. An online random sampling algorithm generates exact samples from these distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-058177 | MATHDB-376125

Fixed-degree NP-completeness conjecture for non-synchronizing colorings

摘录 / EXCERPT: Fixed-degree counting conjecture. The counting problem ... is $\#P$-complete on the class of primitive k-out graphs, and remains $\#P$-complete for fixed out-degree k, already fo...

conjecture | open | MathDB 页面 | 原文线索

V2-058178 | MATHDB-376261

Extension of the lower bounds to local oracles

摘录 / EXCERPT: Extension conjecture. Theorem and Corollary remain valid (in substance) even under local oracles.

conjecture | open | MathDB 页面 | 原文线索

V2-058179 | MATHDB-376262

Nonexistence of dimension-free algorithms for Goldstein approximate SOSPs

摘录 / EXCERPT: Let f be an L-smooth function. A Goldstein approximate second-order stationary point (SOSP) is an approximate second-order stationary point in the Goldstein sense, as defined i...

conjecture | open | MathDB 页面 | 原文线索

V2-058180 | MATHDB-376301

Expander and wired-tree local-limit conjecture for the minimum spanning algorithm

摘录 / EXCERPT: Let G_n be an expander sequence converging locally to the d-regular tree. Consider the local limits associated with item c., a sequence of d-regular expanders with n vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-058181 | MATHDB-376601

Constant expected search complexity for CCL-coded fast forward permutations

摘录 / EXCERPT: Let $(a_0,a_1,\dots,a_l)$ be a sequence generated by the CCL procedure, and let π be the fast forward permutation coded by this sequence. Choose $x\in\{0,\dots,N-1\}$ uniformly...

conjecture | open | MathDB 页面 | 原文线索

V2-058182 | MATHDB-376605

Universal convergence bound for MDL predictions of Bernoulli sequences

摘录 / EXCERPT: Let θ_0 be the true Bernoulli parameter, let θ^x denote the MDL prediction based on the observed sequence, let $K(\theta_0)$ be the prefix Kolmogorov complexity of...

conjecture | open | MathDB 页面 | 原文线索

V2-058183 | MATHDB-376608

Error-resistant protocols for generalized Ulam's game

摘录 / EXCERPT: Let $X_1,\ldots,X_k$ and Y be sets, let $f\colon X_1\times\cdots\times X_k\to Y$ be a function, and let ${\cal G}\left<f,q,l\right>$ denote the generalized Ulam game in which Pau...

conjecture | open | MathDB 页面 | 原文线索

V2-058184 | MATHDB-376614

Finite termination of min-sum auction I for unique maximum weight matchings

摘录 / EXCERPT: Finite-termination conjecture. If π^* is unique, then the min-sum auction I algorithm terminates after finitely many iterations when this condition is removed from step (4).

conjecture | open | MathDB 页面 | 原文线索

V2-058185 | MATHDB-376636

Convergence conjecture for adaptive LP decoding

摘录 / EXCERPT: Let a random parity-check code have length n, $m=n(1-R)$ parity checks, and arbitrary degree distributions. The adaptive LP decoding algorithm is the iterative algorithm that add...

conjecture | open | MathDB 页面 | 原文线索

V2-058186 | MATHDB-376672

Minimal quotient complexity characterizes quotients of the shift algebra

摘录 / EXCERPT: Let A be a finite-dimensional, indecomposable, unital commutative operator algebra of minimal quotient complexity. Here, minimal quotient complexity means that every r...

conjecture | open | MathDB 页面 | 原文线索

V2-058187 | MATHDB-377819

Well-behaved tori conjecture for the complexity invariants c_n

摘录 / EXCERPT: For a marked torus T or $T\times I$, let $c_n(M)$ denote the complexity invariant introduced in the paper, and let $c(M)=c_1(M)$. Well-behaved tori conjecture. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-058188 | MATHDB-377872

Generic Whitehead minimality conjecture

摘录 / EXCERPT: Let F_r be a free group of rank $r\geq 2$, and interpret “almost all” using asymptotic density on the set of elements of F_r ordered by word length. An element is Whitehead min...

conjecture | open | MathDB 页面 | 原文线索

V2-058189 | MATHDB-377873

Average length-reduction complexity conjecture for Whitehead descent

摘录 / EXCERPT: Let $F=F_n$ be a free group of rank n, and let $NMin_l\subset F$ be the set of all non-minimal elements of length l. Let $LR(w)$ denote the length-reducing complexity associate...

conjecture | open | MathDB 页面 | 原文线索

V2-058190 | MATHDB-377874

Exponential word-length growth with Whitehead complexity

摘录 / EXCERPT: Exponential word-length growth conjecture. There exists a constant $c_r>1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058191 | MATHDB-377875

Generic monotonicity conjecture for restricted Whitehead descent

摘录 / EXCERPT: Let F_r be a free group, and let $T=T_r$ be the restricted set of Whitehead automorphisms described in the source. For each length l, let $NMin_l\subset F_r$ be the set of non-...

conjecture | open | MathDB 页面 | 原文线索

V2-058192 | MATHDB-377876

Polynomial Whitehead-descent time-complexity conjecture

摘录 / EXCERPT: Let F_r be a free group, let $NMin\subset F_r$ be the set of non-minimal elements, and let $WC(w)$ denote Whitehead complexity. Polynomial Whitehead-descent time-complexity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-058193 | MATHDB-377894

The phase-transition conjecture for phylogenetic reconstruction

摘录 / EXCERPT: Let mutation matrices be governed by a single order parameter θ, and let $\theta(e)$ denote the parameter of the mutation matrix M^e on edge e. Consider the Markov rando...

conjecture | open | MathDB 页面 | 原文线索

V2-058194 | MATHDB-377963

Maximal root-state-complexity conjecture for binary languages

摘录 / EXCERPT: Let T_n be the full transformation monoid on an n-element set, and let $U_{k,l}$ be the monoid defined above. For a language L over an alphabet of size 2, write...

conjecture | open | MathDB 页面 | 原文线索

V2-058195 | MATHDB-378031

Termination of the balanced pair algorithm under rewriting

摘录 / EXCERPT: Let φ be a Pisot substitution on n letters. Let $\tilde\varphi$ be a rewriting of φ such that, for some letters b,e in the rewritten alphabet,...

conjecture | open | MathDB 页面 | 原文线索

V2-058196 | MATHDB-378224

Bounded Markov complexity for independent subsets

摘录 / EXCERPT: Bounded Markov complexity conjecture. If $\{1,2,\ldots,j\}$ is an independent subset of the underlying graph of Δ, then the preceding bounded-format conclusion holds. This w...

conjecture | open | MathDB 页面 | 原文线索

V2-058197 | MATHDB-378409

Frigerio-Martelli-Petronio conjecture on minimal volume in fixed boundary type

摘录 / EXCERPT: Let g and k be nonnegative integers, and let ${\mathcal{M}}_{g,k}$ be the set of orientable hyperbolic manifolds of complexity $g+k$ with connected geodesic boundary of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-058198 | MATHDB-378478

Finite algorithm conjecture for stable graph representations

摘录 / EXCERPT: A stable representation of a graph is an ${\mathcal{M}}$-representation that is a local minimum with respect to the ordering relation defining stability. Finite algorithm conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-058199 | MATHDB-378834

The dihedral and bicyclic Galois-group feasibility conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be the family of polynomial systems F such that $Z(F)$ is finite and the Galois group of F over $\mathbb{Q}$ is dihedral or bicyclic. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058200 | MATHDB-379009

Conjecture that every non-special plane system admits a reduction algorithm

摘录 / EXCERPT: Let $D\subset\mathbb{N}^{2}$ be finite, and consider a system $\mathcal{L}_{D}(m_{1},\dots,m_{r})$. The system admits a reduction algorithm if there is a sequence of sets ... that sa...

conjecture | open | MathDB 页面 | 原文线索

V2-058201 | MATHDB-379603

The strengthened universal MDL convergence bound

摘录 / EXCERPT: Let K be prefix Kolmogorov complexity, let $\boldsymbol{\vartheta}_0$ be the true parameter, and let $\Delta(k)$ denote the complexity difference used in the MDL setup. Strengthe...

conjecture | open | MathDB 页面 | 原文线索

V2-058202 | MATHDB-379657

The optimality conjecture for the hyperbolic-polynomial approximation factor

摘录 / EXCERPT: The optimality conjecture. The particular polynomial $S_{n,n-m}$ is optimal in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-058203 | MATHDB-379691

Asymptotic Nielsen-reduction conjecture for non-minimal free-group elements

摘录 / EXCERPT: Let F be a free group. For each positive integer n, let U_n be the set of all non-minimal elements in F of length n, and let $NU_n\subset U_n$ be the subset consisting of...

conjecture | open | MathDB 页面 | 原文线索

V2-058204 | MATHDB-379811

The high-complexity torus-gluing conjecture for unstabilized Heegaard splittings

摘录 / EXCERPT: Let X and Y be 3-manifolds, each with a single incompressible torus boundary component, and let M be obtained by gluing them along their boundaries. There is a complexity on...

conjecture | open | MathDB 页面 | 原文线索

V2-058205 | MATHDB-379813

The bounded-genus high-complexity conjecture for higher-genus gluings

摘录 / EXCERPT: Let X and Y each have a single incompressible boundary component of genus at least two, let $\varphi\colon\partial X\to\partial Y$ be a gluing map, and let $M=X\cup_\varphi Y$....

conjecture | open | MathDB 页面 | 原文线索

V2-058206 | MATHDB-380336

The computational hardness threshold conjecture for weighted independent sets

摘录 / EXCERPT: Let $d\geq 3$ be the maximum degree, let ℓ be the activity, and let $\ell_c(d)=(d-1)^{d-1}/(d-2)^d$ be the critical activity for decay of correlations on the infinite d-regu...

conjecture | open | MathDB 页面 | 原文线索

V2-058207 | MATHDB-380492

Conjectured sharp bounds for the expected linear complexity of multisequences

摘录 / EXCERPT: Let q be the alphabet size, let n be the length, and let m be the number of component sequences. Write $E_n^{(m)}$ for the expected linear complexity of an m-multisequence...

conjecture | open | MathDB 页面 | 原文线索

V2-058208 | MATHDB-380529

The random greedy algorithm conjecture for covering designs

摘录 / EXCERPT: Let k and t be fixed, and let $D=\binom{k}{t}-1$. Consider a covering produced by the random greedy algorithm, and define its density as the number of blocks divided by the asy...

conjecture | open | MathDB 页面 | 原文线索

V2-058209 | MATHDB-380621

NP-completeness of coloring finite distance graphs

摘录 / EXCERPT: Let D be a finite set of positive integers, and let $\chi(D)$ denote the chromatic number of the distance graph with distance set D. For a fixed integer $k\ge3$, consider the d...

conjecture | open | MathDB 页面 | 原文线索

V2-058210 | MATHDB-380992

The monotone complexity coding conjecture for P^M

摘录 / EXCERPT: Let $B^{\sharp}$ be the domain of finite binary strings under consideration, and let $P^M(x)$ and $K^M(x)$ denote the corresponding measure and complexity functions. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-058211 | MATHDB-381079

Polynomial-time solvability of integer knapsacks in the diagonal Frobenius cone

摘录 / EXCERPT: Let A be an integer matrix with columns ${\boldsymbol v}_1,\ldots,{\boldsymbol v}_n\in\mathbb{Z}^m$, let ... and set ${\boldsymbol v}={\boldsymbol v}_1+\cdots+{\boldsymbol v}_n$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-058212 | MATHDB-381176

Polynomial-time computation of poset dimension from a linear extension graph

摘录 / EXCERPT: Polynomial-time dimension conjecture. The dimension of a poset given its linear extension graph can be determined in polynomial time.

conjecture | open | MathDB 页面 | 原文线索

V2-058213 | MATHDB-381210

Polynomial-time computation of autarkies beyond the lean minimum-degree bound

摘录 / EXCERPT: Let F be a clause-set with deficiency surplus $\operatorname{\sigma}(F) \ge 1$. Write $\mu\!\operatorname{vd}(F)$ for its minimum variable-degree and $\operatorname{nM}(k)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-058214 | MATHDB-381254

Polynomial-time k-colouring conjecture for graphs with maximal local edge-connectivity k

摘录 / EXCERPT: Polynomial-time k-colouring conjecture. There is a polynomial-time algorithm that, given G, finds a k-colouring of G, or determines that none exists.

conjecture | open | MathDB 页面 | 原文线索

V2-058215 | MATHDB-381416

Polynomial-time computation of the well-covered weight space for graphs with no 6- or 7-cycles

摘录 / EXCERPT: Polynomial-time computation conjecture. The following problem can be solved in polynomial time: given a graph $G \in \mathcal{G}(\widehat{C}_{6},\widehat{C}_{7})$, output $WCW(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-058216 | MATHDB-381448

Conjecture that every region in the fixed-multiplicity algorithm is empty

摘录 / EXCERPT: Let $m\geq 3$ and consider the regions arising in Algorithm for testing numerical semigroups of multiplicity m. Empty-region conjecture. For each $m\geq 3$, every region consider...

conjecture | open | MathDB 页面 | 原文线索

V2-058217 | MATHDB-381467

Bounded-memory impossibility conjecture for belief propagation reconstruction

摘录 / EXCERPT: Let ρ) be the root of a tree, let $ u$ denote the broadcast labeling on its vertices, and consider message-passing algorithms whose messages take values in an alphabet...

conjecture | open | MathDB 页面 | 原文线索

V2-058218 | MATHDB-381509

Mulmuley's efficient generators conjecture for invariant rings

摘录 / EXCERPT: Let $W=(W_1,W_2,\ldots)$ be a sequence of representations, with W_i an SL_{n_i}-representation of dimension N_i. An arithmetic circuit succinctly encodes the g...

conjecture | open | MathDB 页面 | 原文线索

V2-058219 | MATHDB-381588

A faster zigzag sampling algorithm for strongly log-concave distributions

摘录 / EXCERPT: Let U satisfy Assumption, let μ be the target distribution, let $\rho(X)$ denote the law of an algorithm's output X, let μ_0 be the initial distribution, let κ b...

conjecture | open | MathDB 页面 | 原文线索

V2-058220 | MATHDB-381624

NP-hardness conjecture for recognizing f-vectors of fixed-dimensional polytopes

摘录 / EXCERPT: NP-hardness conjecture. It is NP-hard to decide if a given N-bit vector $f=(1,f_0,\ldots,f_{d-1})$ of positive integers is the f-vector of a d-polytope.

conjecture | open | MathDB 页面 | 原文线索

V2-058221 | MATHDB-381637

Polynomial-time recognition conjecture for APUD(1,1)

摘录 / EXCERPT: APUD(1,1) recognition conjecture. Given a graph G, it can be determined whether $G\in\mathrm{APUD}(1,1)$ in polynomial time.

conjecture | open | MathDB 页面 | 原文线索

V2-058222 | MATHDB-381663

Polynomial-time computation of the ν robustness measure

摘录 / EXCERPT: Polynomial-time computation conjecture. There exists a polynomial-time algorithm to compute ν within arbitrary precision.

conjecture | open | MathDB 页面 | 原文线索

V2-058223 | MATHDB-382307

Budaghyan-Carlet-Helleseth conjecture on the algebraic degree of APN functions

摘录 / EXCERPT: Budaghyan-Carlet-Helleseth conjecture. No APN function $F \colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ has algebraic degree n for all $n \geq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-058224 | MATHDB-382337

Tightness of Deletion Bound I under the cover and swap conditions

摘录 / EXCERPT: Deletion Bound I tightness conjecture. If set $\mathcal{J}$ satisfies both cover condition and swap condition, then Algorithm (Deletion Bound I) can find the exact value of...

conjecture | open | MathDB 页面 | 原文线索

V2-058225 | MATHDB-382398

Cluster robustness conjecture for randomized small-block Lanczos

摘录 / EXCERPT: Let $\Lambda_{1},\dotsc,\Lambda_{d}\in\mathbb{R}^{b\times b}$ be real diagonal matrices, and assume that Λ_i and Λ_j have disjoint spectra for $1\leq i<j\leq d$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-058226 | MATHDB-383149

Dihedral Coset Problem

摘录 / EXCERPT: For $N=2^n$, let $s\in\mathbb{Z}_N$ be hidden. Given polynomially many independent copies of the mixed coset state...

open_problem | partial | MathDB 页面 | 原文线索

V2-058227 | MATHDB-383451

Reed-Solomon mutual correlated agreement problem

摘录 / EXCERPT: ...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058228 | MATHDB-383491

Worst-case growth factor in Gaussian elimination

摘录 / EXCERPT: For a pivoting rule P and dimension n, let $g_n^{P}(\mathbb{R})$ denote the supremum, over all nonsingular matrices $A\in\mathbb{R}^{n\times n}$, of the ratio between the large...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058229 | MATHDB-383492

Grossi-Vitter constant-time suffix-array representation question

摘录 / EXCERPT: Does there exist a family of data structures which, for every binary text $T\in\{0,1\}^n$, uses $O(n)$ bits of space and answers suffix-array access queries...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058230 | MATHDB-383508

Kasami APN-function conjecture

摘录 / EXCERPT: For every pair of positive integers n,k with $\gcd(k,n)=1$, let $F:\mathbb{F}_{2^n}\to\mathbb{F}_{2^n}$ be defined by $F(x)=x^{4^k-2^k+1}$, and let...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058231 | MATHDB-383597

Entangled parallel repetition conjecture

摘录 / EXCERPT: Let G be any finite two-player, one-round entangled game with entangled value $\omega^*(G)=1-\varepsilon<1$. The conjecture asserts that there exists a constant $c=c(G)>0$ such t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-058232 | MATHDB-383677

Aaronson-Ambainis conjecture

摘录 / EXCERPT: For every T-query quantum algorithm A, there is a polynomial p such that, for every $\epsilon,\delta>0$, there exists a classical randomized query algorithm C making at mos...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058233 | MATHDB-384638

Are there any known inequalities for Kolmogorov complexity that imply interesting results in geometric measure theory via the point-to-set principle?

摘录 / EXCERPT: Are there any known inequalities for Kolmogorov complexity that imply interesting results in geometric measure theory via the point-to-set principle?

open_problem | open | MathDB 页面 | 原文线索

V2-058234 | MATHDB-384972

What witnesses are there of Goppa decodability or non-decodability?

摘录 / EXCERPT: What witnesses are there of Goppa decodability or non-decodability?

open_problem | open | MathDB 页面 | 原文线索

V2-058235 | MATHDB-384975

Concrete Costs (problem 1)

摘录 / EXCERPT: Given: $x_0,\dots, x_{2^n-1} \in \{0,1\}$. What is the real-world cost of ... where $i = i_0 + 2i_1 + \dots + 2^{n-1}i_{n-1}$. Can error correction be done efficiently for this probl...

open_problem | open | MathDB 页面 | 原文线索

V2-058236 | MATHDB-384976

What are the concrete costs of quantum sieving and quantum enumeration within BKZ?

摘录 / EXCERPT: What are the concrete costs of quantum sieving and quantum enumeration within BKZ? Go beyond asymptotics.

open_problem | open | MathDB 页面 | 原文线索

V2-058237 | MATHDB-384977

How fast are approximate SVP attacks via hidden shift algorithms?

摘录 / EXCERPT: How fast are approximate SVP attacks via hidden shift algorithms?

open_problem | open | MathDB 页面 | 原文线索

V2-058238 | MATHDB-384981

Hidden Shift (problem 5)

摘录 / EXCERPT: Hidden shift on $\mathbb Z^d$ where f_0 and f_1 are periodic under unary oracle cost.

open_problem | open | MathDB 页面 | 原文线索

V2-058239 | MATHDB-384990

Parameter determination for post-quantum hash functions

摘录 / EXCERPT: In the design of the hash function one has control over two parameters; the dimension of the special linear group considered, and the characteristic of the underlying field. We wou...

open_problem | open | MathDB 页面 | 原文线索

V2-058240 | MATHDB-384991

Cryptanalysis of SDLP

摘录 / EXCERPT: On the quantum side the question of quantum equivalence of SDLP and the CDH-type problem underpinning semidirect product key exchange (SCDH) remains unproven, though we have strong...

open_problem | open | MathDB 页面 | 原文线索

V2-058241 | MATHDB-384995

Creation of challenge instances

摘录 / EXCERPT: Across the various group-based protocols there is a general lack of precision on which parameter specification. As the field matures we should make available challenge instances of...

open_problem | open | MathDB 页面 | 原文线索

V2-058242 | MATHDB-384996

Implementation of group-based cryptography

摘录 / EXCERPT: In general very little work exists on efficient implementation of group-based protocols; over time we would like to build up a software library of such implementations.

open_problem | open | MathDB 页面 | 原文线索

V2-058243 | MATHDB-385033

Borel complexity (problem 10)

摘录 / EXCERPT: Assume A and B are C-algebras and E_A is Borel-reducible to E_B. What does this fact imply about the relation between A and B?

open_problem | open | MathDB 页面 | 原文线索

V2-058244 | MATHDB-385678

Can we implement algorithms in M2 to do computations over non-commutative algebras?

摘录 / EXCERPT: Can we implement algorithms in M2 to do computations over non-commutative algebras?

open_problem | open | MathDB 页面 | 原文线索

V2-058245 | MATHDB-385704

What is the complexity of the set of pmp graphs admitting a measurable 3-coloring?

摘录 / EXCERPT: What is the complexity of the set of pmp graphs admitting a measurable 3-coloring?

open_problem | open | MathDB 页面 | 原文线索

V2-058246 | MATHDB-386935

Define Space complexity!

摘录 / EXCERPT: Does there exist a notion of space complexity?

open_problem | open | MathDB 页面 | 原文线索

V2-058247 | MATHDB-386939

Smales' 9th problem.

摘录 / EXCERPT: Does there exist an algorithm for linear programming over the reals done by an algorithm that is independent of the bit-size. (in the framework of the sci-hierarchy).

open_problem | open | MathDB 页面 | 原文线索

V2-058248 | MATHDB-387001

What is the complexity of the theory of the Borel equivalence relation under Borel reducibility?

摘录 / EXCERPT: What is the complexity of the theory of the Borel equivalence relation under Borel reducibility?

open_problem | open | MathDB 页面 | 原文线索

V2-058249 | MATHDB-387002

What is the complexity of the theory of the smooth equivalence relations under continuous reducibility?

摘录 / EXCERPT: What is the complexity of the theory of the smooth equivalence relations under continuous reducibility?

open_problem | open | MathDB 页面 | 原文线索

V2-058250 | MATHDB-388829

Boundaries of Groups and Kleinian Groups - Problem 97

摘录 / EXCERPT: Consider finite cell complexes X. Is there an algorithm to determine if X is contractible?

open_problem | open | MathDB 页面 | 原文线索

V2-058251 | MATHDB-391937

Spectral density estimation under local graph access

摘录 / EXCERPT: Given local random-neighbor access to an unknown unweighted graph G with n vertices, let N_G denote its normalized adjacency matrix and let...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058252 | MATHDB-391952

Second law of quantum complexity

摘录 / EXCERPT: For a generic chaotic finite-dimensional quantum system evolving unitarily from a low-complexity state, let $C(t)$ denote its quantum circuit complexity and let t_{eq} b...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-058253 | MATHDB-392093

Yamakawa-Zhandry certifiable-randomness conjecture

摘录 / EXCERPT: For every publicly computable function H used in the Yamakawa-Zhandry quantum random-oracle certifiable-randomness protocol, every quantum prover that succeeds in the protocol mu...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058254 | MATHDB-392104

Ground-state energy estimation with a guiding state

摘录 / EXCERPT: Given a unitary U with largest eigenphase $\theta_{\max}$ and a guiding state $|\psi\rangle$ having overlap at least γ with the eigenspace corresponding to $\theta_{\max}$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058255 | MATHDB-392484

Parameter-free first-order gradient minimization in ℓp geometry

摘录 / EXCERPT: Fix $1<p<\infty$ and let $q=p/(p-1)$. Given a first-order local value--gradient oracle for an objective $f:\mathbb{R}^n\to\mathbb{R}$ whose gradient is L-Lipschitz from...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058256 | MATHDB-392500

Randomness-certification question in energy-constrained semi-device-independent scenarios

摘录 / EXCERPT: Given a finite energy bound $E<\infty$, does there exist a prepare-and-measure semi-device-independent protocol such that the observed data certify strictly positive extractable ra...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058257 | MATHDB-392648

Lower bound for deterministic online algorithms for Dsca

摘录 / EXCERPT: For the problem Dsca with $d_j=\infty$ and $p_j=b_j$ for each request σ_j, the competitive ratio of each deterministic competitive algorithm is km, where k denotes the n...

open_problem | open | MathDB 页面 | 原文线索

V2-058258 | MATHDB-392739

Polynomial-time construction of low-star-discrepancy point sets

摘录 / EXCERPT: Does there exist an algorithm running in time polynomial in n and d that, for every $n,d\in\mathbb N$, constructs a set $M_n\subset[0,1]^d$ satisfying ... where $c>0$ is the cons...

open_problem | open | MathDB 页面 | 原文线索

V2-058259 | MATHDB-392768

Reducing the running-time order for multi-level lot-sizing algorithms

摘录 / EXCERPT: The running times of the presented algorithms are of relatively high order, namely $O(T^5)$ to $O(T^7)$ for the two-level cases. It would be interesting to determine whether the or...

open_problem | open | MathDB 页面 | 原文线索

V2-058260 | MATHDB-392837

A polynomial-time $3/4$-approximation algorithm for MaxATSP$(0,1)$

摘录 / EXCERPT: For the problem MaxATSP$(0,1)$-given a directed complete loopless graph with edge weights in $[U+007F]{0,1}$, compute a maximum-weight TSP tour-the report states that there is a polynomia...

open_problem | open | MathDB 页面 | 原文线索

V2-058261 | MATHDB-392841

Approximate counting of vertices of arbitrary transportation polytopes

摘录 / EXCERPT: Do there exist algorithms-by Markov-chain Monte Carlo or any other method-that approximately count the vertices of arbitrary transportation polytopes?

open_problem | open | MathDB 页面 | 原文线索

V2-058262 | MATHDB-392842

Sublinear-time approximation for metric uniform Minimum Facility Location

摘录 / EXCERPT: For the Minimum Facility Location problem on an n-point metric space, with uniform costs and uniform demands, there is a randomized algorithm running in $O(n\log^2 n)$ time that...

open_problem | open | MathDB 页面 | 原文线索

V2-058263 | MATHDB-392844

Theorem 3: PCPs with bounded free-bit and query complexity

摘录 / EXCERPT: For any integer $f\ge 2$, any positive integer $t\le f(f-1)/2$, and any constant $\epsilon>0$, there is a PCP for NP with free-bit complexity f, query complexity $f+t$, completen...

open_problem | open | MathDB 页面 | 原文线索

V2-058264 | MATHDB-392855

Width-independent algorithm for abstract mixed fractional packing and covering

摘录 / EXCERPT: Find an efficient, width-independent Lagrangian-relaxation algorithm for the abstract mixed packing and covering problem: find $x\in B$ such that $Px\le (1+\epsilon)a$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-058265 | MATHDB-392856

Improved approximation algorithm for the mixed fractional packing and covering problem

摘录 / EXCERPT: Let $f:B\to\mathbb{R}_+^M$ be a vector of M nonnegative continuous convex functions, let $g:B\to\mathbb{R}_+^M$ be a vector of M nonnegative continuous concave functions, let...

open_problem | open | MathDB 页面 | 原文线索

V2-058266 | MATHDB-392863

Complexity of approximating polynomials on the unit sphere

摘录 / EXCERPT: The complexity of approximating a polynomial over the unit sphere is not known.

open_problem | open | MathDB 页面 | 原文线索

V2-058267 | MATHDB-392865

WSEPT expected competitive ratio for exponential processing times

摘录 / EXCERPT: For m identical parallel machines with exponentially distributed processing times, the WSEPT algorithm satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-058268 | MATHDB-392866

Characterization of polynomial smoothed complexity for binary optimization problems

摘录 / EXCERPT: A binary optimization problem has a polynomial smoothed complexity if and only if it has a pseudopolynomial complexity.

open_problem | open | MathDB 页面 | 原文线索

V2-058269 | MATHDB-393153

Efficiently finding bounded solutions to summation-polynomial equations

摘录 / EXCERPT: For a natural number $n\ge 2$, a small $\varepsilon>0$, and a random residue $x\pmod p$, does there exist an auxiliary algorithm that finds integers $x_1,\ldots,x_n$ satisfying ... w...

open_problem | open | MathDB 页面 | 原文线索

V2-058270 | MATHDB-393154

Deterministic polynomial-time factorization over finite fields, even under GRH

摘录 / EXCERPT: Does there exist a deterministic algorithm that factors polynomials over finite fields in time polynomial in the input size, even assuming the Generalized Riemann Hypothesis (GRH)?

open_problem | open | MathDB 页面 | 原文线索

V2-058271 | MATHDB-393188

Improving the complexity bound for a full-Newton-step infeasible interior-point method

摘录 / EXCERPT: “We conjecture, however, that a more careful analysis will reduce this bound by a factor of $\sqrt n$.” Here “this bound” refers to the preceding claim that the described full-Newt...

open_problem | open | MathDB 页面 | 原文线索

V2-058272 | MATHDB-393198

Polynomial-time recognition of graphs of bounded clique-width

摘录 / EXCERPT: For fixed integer $k>3$, is there a polynomial-time algorithm that decides whether an input graph has clique-width at most k?

open_problem | open | MathDB 页面 | 原文线索

V2-058273 | MATHDB-393301

Least complexity class proving the matrix principle

摘录 / EXCERPT: Determine the least complexity class in the sequence $AC^0 \subset AC^0(2) \subset TC^0 \subset NC^1 \subset L \subset P$ whose associated minimal theory proves the matrix principl...

open_problem | open | MathDB 页面 | 原文线索

V2-058274 | MATHDB-393356

Convergence of the local-global iterative upscaling algorithm for tensor coefficients

摘录 / EXCERPT: Study the convergence of the global-local iterative algorithm in the case of tensor coefficients.

open_problem | open | MathDB 页面 | 原文线索

V2-058275 | MATHDB-393375

Convergence-rate claim for the domain decomposition solver

摘录 / EXCERPT: For an idealized version of the domain decomposition algorithm used in the variant parallel adaptive-meshing paradigm, the rate of convergence is independent of both the global pro...

open_problem | open | MathDB 页面 | 原文线索

V2-058276 | MATHDB-393385

Classical reduction for the learning-with-errors problem

摘录 / EXCERPT: Can the reduction from the learning problem to quantum algorithms for SVP and SIVP be made classical (i.e. non-quantum)?

open_problem | open | MathDB 页面 | 原文线索

V2-058277 | MATHDB-393387

Polynomial-factor hardness of SVP and SIVP

摘录 / EXCERPT: It is conjectured that there is no classical polynomial-time algorithm that approximates the shortest vector problem (SVP) or the shortest independent vectors problem (SIVP) to wit...

open_problem | open | MathDB 页面 | 原文线索

V2-058278 | MATHDB-393393

Approximation complexity of undirected Edge-Disjoint Paths

摘录 / EXCERPT: Given an undirected graph G and a collection of source-sink pairs $(s_1,t_1),\ldots,(s_k,t_k)$, determine the optimal polynomial-time approximation ratio for the Edge-Disjoint Pa...

open_problem | open | MathDB 页面 | 原文线索

V2-058279 | MATHDB-393493

Polynomial-time computation of torsion divisors with prescribed Hecke action

摘录 / EXCERPT: Develop algorithms, provably running in polynomial time, for computing torsion divisors with prescribed Hecke action on modular curves-for example, divisor classes in the linear sp...

open_problem | open | MathDB 页面 | 原文线索

V2-058280 | MATHDB-393551

Optimal-complexity realization of the adaptive refinement condition

摘录 / EXCERPT: For hierarchical refinement of a tree of admissible meshes, let $V_N=\{V\text{ admissible}:\dim(V)=N\}$, and let $\phi:\bigcup_{N\in\mathbb N}V_N\to\mathbb R$ be a quasi-monotone e...

open_problem | open | MathDB 页面 | 原文线索

V2-058281 | MATHDB-393554

Convergence of the adaptive finite-element algorithm for rough and conforming obstacles

摘录 / EXCERPT: Suppose that the initial triangulation is subordinated to the lower obstacle. Then the indicated adaptive algorithm converges in a finite number of steps, or produces an infinite s...

open_problem | open | MathDB 页面 | 原文线索

V2-058282 | MATHDB-393616

Algorithmic problems for finitely presented subgroups of direct products of surface groups

摘录 / EXCERPT: The conjugacy problem and the membership problem are solvable for every finitely presented subgroup of a direct product of surface groups.

open_problem | open | MathDB 页面 | 原文线索

V2-058283 | MATHDB-393629

Decidability of the rank problem for torsion-free word-hyperbolic Kleinian groups

摘录 / EXCERPT: There exists an algorithm which, given a finite presentation of a torsion-free word-hyperbolic Kleinian group G, determines the rank of G.

open_problem | open | MathDB 页面 | 原文线索

V2-058284 | MATHDB-393661

Complexity of the three-terminal triangle case of Blocking Shortest Paths

摘录 / EXCERPT: Given an undirected graph, with or without edge costs, and three given vertices, find a minimum (cost) edge set whose deletion increases the distance of any two of the three given...

open_problem | open | MathDB 页面 | 原文线索

V2-058285 | MATHDB-393715

Lowest Levy complexity of a parameter-free definable well-order of the reals

摘录 / EXCERPT: What is the lowest possible degree of logical complexity for which there is a formula $\Phi(x,y)$, without parameters, with that complexity and with the property that one can pass...

open_problem | open | MathDB 页面 | 原文线索

V2-058286 | MATHDB-393813

Hilbert’s Tenth Problem over the rationals

摘录 / EXCERPT: Does there exist an algorithm which, given an arbitrary polynomial equation in several variables with coefficients in $\mathbb{Q}$, decides whether the equation has a solution in...

open_problem | open | MathDB 页面 | 原文线索

V2-058287 | MATHDB-393823

Complexity of 4-Coloring on P_5-free graphs

摘录 / EXCERPT: What is the computational complexity of 4-Coloring on P_5-free graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-058288 | MATHDB-393825

NC approximation for maximum-weight matching

摘录 / EXCERPT: For every $\varepsilon>0$, there exists an NC algorithm that, given a graph $G=(V,E)$ with positive edge weights, finds a matching M satisfying $w(M)\ge(1-\varepsilon)w(M^*)$, wh...

open_problem | open | MathDB 页面 | 原文线索

V2-058289 | MATHDB-393832

Lower bound for coloring an unoriented cycle with one-bit communication

摘录 / EXCERPT: For the n-node cycle, if only one bit can be sent along each edge in each round, then every distributed coloring algorithm in which all nodes have the same initial state and init...

open_problem | open | MathDB 页面 | 原文线索

V2-058290 | MATHDB-393835

Strict and group-strict cost sharing for Steiner forest

摘录 / EXCERPT: There exists a primal-dual 2-approximation algorithm for the Steiner forest problem that is 3-strict and 4-group-strict.

open_problem | open | MathDB 页面 | 原文线索

V2-058291 | MATHDB-393837

Complexity of MAX TUTTE SET for 3-connected planar graphs

摘录 / EXCERPT: What is the complexity of $\textsc{MAX TUTTE SET}$ for 3-connected planar graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-058292 | MATHDB-393977

Algorithm for deciding strong freeness beyond four generators

摘录 / EXCERPT: Develop an algorithm for deciding whether a sequence of quadratic Lie polynomials $\rho_1,\ldots,\rho_m$ in $m>4$ variables is strongly free.

open_problem | open | MathDB 页面 | 原文线索

V2-058293 | MATHDB-394013

Complexity and algebraicity of higher-genus \SL_2(\mathbb R) orbit closures

摘录 / EXCERPT: For genus $g\ge 3$, determine whether the closures of $\mathrm{SL}_2(\mathbb R)$-orbits in the one-form bundle $\Omega\mathcal M_g$ are complex manifolds, and ideally algebraic man...

open_problem | open | MathDB 页面 | 原文线索

V2-058294 | MATHDB-394049

Computing the order of a Mealy element

摘录 / EXCERPT: Is there an algorithm that computes the order of a Mealy element, with the output in $\mathbb{N}\cup\{\infty\}$?

open_problem | open | MathDB 页面 | 原文线索

V2-058295 | MATHDB-394105

Optimality of the fully discrete filtered backprojection algorithm in 2D

摘录 / EXCERPT: Under approximation properties (2) and (3), with $\beta_{\max},\tau_{\max}\geq \frac{3}{2}$, one has ... where $\alpha_{\min}=\max\{1,\beta_{\min}-\tfrac12\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-058296 | MATHDB-394221

Simulated annealing versus the Metropolis algorithm at fixed temperature

摘录 / EXCERPT: Is there a natural example where SA outperforms MA for all fixed temperatures?

open_problem | open | MathDB 页面 | 原文线索

V2-058297 | MATHDB-394222

Expected runtime of ACO algorithms on simple problems

摘录 / EXCERPT: Determine the expected runtime of Ant Colony Optimization (ACO) algorithms on simple problems in a similar fashion to what has been done for evolutionary algorithms (EAs).

open_problem | open | MathDB 页面 | 原文线索

V2-058298 | MATHDB-394336

Changing the inpainting order to address the tangency singularity

摘录 / EXCERPT: The authors conjecture that changing the order by distance in Telea’s algorithm will address the singular case of tangency. They have not changed this order so far. They suggest th...

open_problem | open | MathDB 页面 | 原文线索

V2-058299 | MATHDB-394443

3-colorability of triangle-free graphs on a fixed surface

摘录 / EXCERPT: For every surface Σ there exists a polynomial-time algorithm that, given an input triangle-free graph G embedded in Σ, correctly determines whether G is 3-color...

open_problem | open | MathDB 页面 | 原文线索

V2-058300 | MATHDB-394589

Compressibility of SAT instances

摘录 / EXCERPT: Does there exist an efficient algorithm and a polynomial p such that, given a formula consisting of m clauses over n variables, it produces an equivalent formula (with respec...

open_problem | open | MathDB 页面 | 原文线索

V2-058301 | MATHDB-394836

Numerical Implicitization

摘录 / EXCERPT: Problem: Numerical Implicitization. Most existing algorithms for implicitization are not numerical: when the input parametrization has coefficients given by floats and are subject...

open_problem | open | MathDB 页面 | 原文线索

V2-058302 | MATHDB-395126

Efficient algorithms for computing the rank invariant

摘录 / EXCERPT: Develop efficient algorithms for computing the rank invariant.

open_problem | open | MathDB 页面 | 原文线索

V2-058303 | MATHDB-395273

Decision problem for equations and inequations over hyperbolic groups

摘录 / EXCERPT: Given a hyperbolic group G (possibly with torsion), there exists an algorithm which decides whether a given system of equations and inequations has a solution in G.

open_problem | open | MathDB 页面 | 原文线索

V2-058304 | MATHDB-395288

Rigidity of the curve complex

摘录 / EXCERPT: Let $S=S_{g,b}$ be a connected, compact, orientable surface with complexity $\xi(S)=3g-3+b\ge 2$. Then every quasi-isometry of the curve complex $\mathcal{C}(S)$ is a bounded dista...

open_problem | open | MathDB 页面 | 原文线索

V2-058305 | MATHDB-395305

Union complexity of infinite cylinders in three-dimensional space

摘录 / EXCERPT: Let $\mathcal{K}$ be a collection of n infinite cylinders in $\mathbb{R}^3$, let $\mathcal{A}(\mathcal{K})$ be the arrangement induced by their boundaries, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-058306 | MATHDB-395307

Union complexity of three-dimensional cigars of arbitrary radii

摘录 / EXCERPT: For a collection of n cigars in $\[3$-space-where a cigar is the Minkowski sum of a line segment and a ball, and the cigars may have arbitrary radii-the report states that the co...

open_problem | open | MathDB 页面 | 原文线索

V2-058307 | MATHDB-395309

Computational complexity of embedding 2-dimensional simplicial complexes in $\mathbb{R}^3$

摘录 / EXCERPT: Determine the computational complexity of $\mathrm{EMBED}_{2\to3}$: given a finite simplicial complex K of dimension at most 2, decide whether K admits a piecewise-linear emb...

open_problem | open | MathDB 页面 | 原文线索

V2-058308 | MATHDB-395385

Polynomial-time computation of the Alcuin number for perfect graphs

摘录 / EXCERPT: Is there a polynomial-time algorithm for computing the Alcuin number of a perfect graph?

open_problem | open | MathDB 页面 | 原文线索

V2-058309 | MATHDB-395392

Polynomial-time computation of the crossing number for almost planar graphs

摘录 / EXCERPT: Is there a polynomial-time algorithm for computing the crossing number $\operatorname{cr}(G)$ for almost planar graphs? Here, an almost planar graph is a non-planar graph $G=(V,E)$...

open_problem | open | MathDB 页面 | 原文线索

V2-058310 | MATHDB-395606

The other direction of the UGA VIGRE conjecture on Specht-module complexity

摘录 / EXCERPT: Resolve the other direction of Conjecture 1 (UGA VIGRE): for a Specht module S^λ in a block of weight w, determine whether λ not being $p\times p$ implies that th...

conjecture | open | MathDB 页面 | 原文线索

V2-058311 | MATHDB-395633

Rauzy fractals associated with Brun expansions

摘录 / EXCERPT: How to associate a Rauzy fractal with a Brun expansion? If we know that the a.e. exponential convergence of Brun’s algorithm gives us convergence toward a Rauzy fractal, can we use...

open_problem | open | MathDB 页面 | 原文线索

V2-058312 | MATHDB-395743

Algorithmic solution of random k-SAT beyond the $2^k/k$ density scale

摘录 / EXCERPT: Let Φ be a random k-SAT formula on variables $V=\{x_1,\ldots,x_n\}$, formed by choosing m clauses uniformly and independently from all possible k-clauses. Is there an ef...

open_problem | open | MathDB 页面 | 原文线索

V2-058313 | MATHDB-395810

Stable Borel conjecture for finite-decomposition-complexity groups

摘录 / EXCERPT: The stable Borel conjecture holds for a closed aspherical manifold M whose fundamental group has finite decomposition complexity. Precisely, if another closed manifold M' is ho...

conjecture | open | MathDB 页面 | 原文线索

V2-058314 | MATHDB-395902

Algorithm for Dirichlet domains, finite presentations, and the word problem

摘录 / EXCERPT: Let d4d7 be the group in the Shimura-curve construction and let d49fd49f be a point in the upper half-plane with trivial stabilizer, d4d7d49fd49f = {1}. Define ... There exists an al...

open_problem | open | MathDB 页面 | 原文线索

V2-058315 | MATHDB-396084

Nearly-linear algebraic multivariate multipoint evaluation

摘录 / EXCERPT: Give a nearly-linear algebraic algorithm for multivariate multipoint evaluation.

open_problem | open | MathDB 页面 | 原文线索

V2-058316 | MATHDB-396086

Adaptive versus non-adaptive query complexity of graph-property testing

摘录 / EXCERPT: For every integer $t>2$, there exists a graph property in the adjacency-matrix model such that, if q denotes the query complexity of general (adaptive) testing of the property an...

open_problem | open | MathDB 页面 | 原文线索

V2-058317 | MATHDB-396087

Geometric Complexity Theory orbit-closure separation conjecture

摘录 / EXCERPT: For every positive polynomial p and every $m_0\in\mathbb{N}$, there exists an $m\ge m_0$ such that there is no $GL_{p(m)^2}$-equivariant surjection ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-058318 | MATHDB-396088

Whether BPP equals P

摘录 / EXCERPT: Is $\mathrm{BPP}=\mathrm{P}$; that is, can every polynomial-time randomized algorithm be simulated by a polynomial-time deterministic algorithm?

open_problem | open | MathDB 页面 | 原文线索

V2-058319 | MATHDB-396317

Isomorphism problem for relatively hyperbolic groups with universally parabolic peripheral groups

摘录 / EXCERPT: Theorem 2. Let $\mathcal C$ be a recursively enumerable class of finitely presented groups that are residually finite and universally parabolic. There is an algorithm which, given...

open_problem | open | MathDB 页面 | 原文线索

V2-058320 | MATHDB-396319

Isomorphism problem for toral relatively hyperbolic groups

摘录 / EXCERPT: The isomorphism problem is solvable for the class of toral relatively hyperbolic groups: there is an algorithm that, given finite presentations of two groups in this class, decides...

open_problem | open | MathDB 页面 | 原文线索

V2-058321 | MATHDB-396526

S-adic characterization of infinite words with linear complexity

摘录 / EXCERPT: Find a characterization, in S-adic terms, of infinite words having linear complexity.

open_problem | open | MathDB 页面 | 原文线索

V2-058322 | MATHDB-396538

Question 31: Complexity of factor universality for Kleene stars

摘录 / EXCERPT: Given a finite set S of finite words over an alphabet Σ, what is the computational complexity of deciding whether $\operatorname{Fact}(S^*)=\Sigma^*$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-058323 | MATHDB-396595

The Atiyah problem for a group: determining the \2-complexity

摘录 / EXCERPT: Let G be a countable discrete group. Define its $\2$-complexity by ... What is the set $C(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-058324 | MATHDB-396600

Isomorphism problem for nilpotent groups

摘录 / EXCERPT: There is an algorithm that decides whether two nilpotent groups, given by finite presentations, are isomorphic.

open_problem | open | MathDB 页面 | 原文线索

V2-058325 | MATHDB-396793

Algorithm for invariant rings of reductive groups on nonreduced algebras

摘录 / EXCERPT: Find an algorithm for computing invariant rings of reductive groups acting on nonreduced algebras.

open_problem | open | MathDB 页面 | 原文线索

V2-058326 | MATHDB-396805

Validity of linear inequalities between subgraph densities

摘录 / EXCERPT: Given fixed simple graphs $F_1,\ldots,F_n$ and coefficients $a_1,\ldots,a_n$ (arbitrary real coefficients, or rational coefficients in an algorithmic setting), determine whether .....

open_problem | open | MathDB 页面 | 原文线索

V2-058327 | MATHDB-396830

Deterministic versions of faster randomized SAT algorithms

摘录 / EXCERPT: Do the faster randomized algorithms for k-SAT-namely the PPSZ algorithm or an amalgamation of PPSZ with Schöning’s algorithm-which outperform pure Schöning search, admit determin...

open_problem | open | MathDB 页面 | 原文线索

V2-058328 | MATHDB-396963

Unique algorithm generating the 15 rows of Plimpton 322

摘录 / EXCERPT: Are the 15 rows the result of a unique algorithm?

open_problem | open | MathDB 页面 | 原文线索

V2-058329 | MATHDB-397061

Historical growth conjecture for partial-match queries in random quadtrees

摘录 / EXCERPT: For a two-dimensional random quadtree built from independent uniformly distributed points in $[0,1]^2$, let $C_n(s)$ be the complexity of the partial-match query $(s,*)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058330 | MATHDB-397077

Computational complexity of learning Boolean k-juntas under the uniform distribution

摘录 / EXCERPT: Determine the computational complexity of learning a Boolean function of k relevant variables in an n-dimensional space from labeled examples drawn from the uniform distributio...

open_problem | open | MathDB 页面 | 原文线索

V2-058331 | MATHDB-397227

Algorithm for enumerating relative cubic extensions over imaginary quadratic fields

摘录 / EXCERPT: Let K be an imaginary quadratic number field with class number $h_K=1$. For $X>0$, let ... There exists an algorithm that lists all cubic extensions in $\mathcal F_{K,3}(X)$ in tim...

open_problem | open | MathDB 页面 | 原文线索

V2-058332 | MATHDB-397259

Algorithmic generalisation of small cancellation theory

摘录 / EXCERPT: Let Γ be a small groupoid, and let ... be the alphabet. Let A^* be the monoid of finite words in A under concatenation. The multiplication in Γ defines a terminatin...

open_problem | open | MathDB 页面 | 原文线索

V2-058333 | MATHDB-397260

Polynomial-time explicit isomorphism algorithms with growing algebra dimension

摘录 / EXCERPT: Can one obtain polynomial-time ff-algorithms for the explicit isomorphism problem when the dimension of the algebra over the number field K is allowed to grow?

open_problem | open | MathDB 页面 | 原文线索

V2-058334 | MATHDB-397261

Explicit isomorphisms for non-split central simple algebras

摘录 / EXCERPT: The existence of ff-algorithms with similar time bounds for finding an explicit isomorphism of a non-split central simple algebra with a suitable algebra of matrices over a skewfie...

open_problem | open | MathDB 页面 | 原文线索

V2-058335 | MATHDB-397434

Existence of 0/1 polytopes with exponential extension complexity

摘录 / EXCERPT: For every $n\in\mathbb{N}$, there exists a set $X\subseteq\{0,1\}^{n}$ such that ... where $\operatorname{xc}(P)$ is the minimum number of inequalities needed to describe a polytope...

open_problem | open | MathDB 页面 | 原文线索

V2-058336 | MATHDB-397435

Deterministic approximation for the metric s-t path TSP

摘录 / EXCERPT: Given a symmetric metric cost on n vertices and two prescribed endpoints s and t, there is a deterministic $\frac{1+\sqrt{5}}{2}$-approximation algorithm for finding a shorte...

open_problem | open | MathDB 页面 | 原文线索

V2-058337 | MATHDB-397519

Complexity of maximal and strongly maximal chains in computable well partial orders

摘录 / EXCERPT: If P is a computable wpo, how complicated must maximal and strongly maximal chains in P be?

open_problem | open | MathDB 页面 | 原文线索

V2-058338 | MATHDB-397532

Quasi-optimal and λ-robust adaptive Crouzeix-Raviart FEM for pure displacement elasticity

摘录 / EXCERPT: For the approximation class $A_s:=\{(u,f)\mid |(u,f)|_{A_s}<\infty\}$ with $s>0$, the report states that the adaptive algorithm A_{crfem}, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-058339 | MATHDB-397650

Algorithmic determination of polynomial decompositions

摘录 / EXCERPT: Is there a method (i.e. an algorithm) to find, for a given polynomial f, all decompositions? More generally, is it possible to describe all composite polynomials f and all thei...

open_problem | open | MathDB 页面 | 原文线索

V2-058340 | MATHDB-397674

Polynomial-time complexity of 3-sphere recognition

摘录 / EXCERPT: For the decision problem with instance a triangulated 3-dimensional manifold M and question whether M is homeomorphic to the 3-sphere, does there exist an algorithm with poly...

open_problem | open | MathDB 页面 | 原文线索

V2-058341 | MATHDB-397687

Polynomial-time test for relative torsion over all pure subcomplexes

摘录 / EXCERPT: For a finite simplicial complex K of dimension greater than p, there is a polynomial-time algorithm to test whether $H_p(L,L_0)$ has torsion for some pair of pure subcomplexes...

open_problem | open | MathDB 页面 | 原文线索

V2-058342 | MATHDB-397704

Polynomial-time random sampling of closed 3-manifold triangulations

摘录 / EXCERPT: There are finitely many closed 3-manifold triangulations with N tetrahedra. Is there a polynomial-time algorithm to pick one at random? What if we restrict to 3-spheres?

open_problem | open | MathDB 页面 | 原文线索

V2-058343 | MATHDB-397786

Convergence of a generalized iterative algorithm for constrained sparse recovery

摘录 / EXCERPT: For the constrained problem ... consider the iteration ... with arbitrary initial values x^0,w^0,v^0, where $P_\lambda=\operatorname{Id}-S_\lambda$ is the projection onto the...

open_problem | open | MathDB 页面 | 原文线索

V2-058344 | MATHDB-397787

Block-iterative method for TV-penalized least-squares minimization

摘录 / EXCERPT: Develop a block-iterative algorithm for the minimization of a total-variation (TV)-penalized least functional that does not approximate the TV function, but uses the same proximal...

open_problem | open | MathDB 页面 | 原文线索

V2-058345 | MATHDB-397792

Rigorous conditions for good mean-square-error performance of hashing algorithms

摘录 / EXCERPT: Under what conditions can we rigorously expect hashing algorithms to produce good mean-square-error results?

open_problem | open | MathDB 页面 | 原文线索

V2-058346 | MATHDB-398132

Effect of using the third tensor power of the Coppersmith-Winograd algorithm

摘录 / EXCERPT: What happens if one uses A^3 as the basic algorithm (the “base case”) instead?

open_problem | open | MathDB 页面 | 原文线索

V2-058347 | MATHDB-398224

6-approximation for minimum-cost k-connectivity augmentation

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected graph with at least $k^3(k-1)+k$ nodes. Given nonnegative edge costs $c:\binom{V}{2}\to\mathbb{R}_+$, there is a polynomial-time algorithm that finds...

open_problem | open | MathDB 页面 | 原文线索

V2-058348 | MATHDB-398231

Sample complexity for discrete least-squares projection from random evaluations

摘录 / EXCERPT: The report’s question about the required sample size is answered as follows: for any $\gamma>0$, if $M\bracket\log M\ge C(1+\gamma)|P_\Lambda|^2$, then, for N uniformly distribut...

open_problem | open | MathDB 页面 | 原文线索

V2-058349 | MATHDB-398290

Efficient direct recovery of measures and polyhedra

摘录 / EXCERPT: Develop efficient algorithms for direct recovery of μ and P, in particular in the case of non-convex P.

open_problem | open | MathDB 页面 | 原文线索

V2-058350 | MATHDB-398321

Optimality of the 43.52% independent-set density for local algorithms

摘录 / EXCERPT: Is the density of an independent set containing 43.52% of the vertices the maximum that can be obtained by a local algorithm in d-regular large-girth graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-058351 | MATHDB-398361

Approximation algorithms for higher-dimensional planar random assignment

摘录 / EXCERPT: For $D>3$, are there polynomial-time algorithms yielding solutions within logarithmic or $n^{o(1)}$ factors for the D-dimensional planar random assignment problem?

open_problem | open | MathDB 页面 | 原文线索

V2-058352 | MATHDB-398412

Critical unknots via complexity theory

摘录 / EXCERPT: For energy functionals on the space of loops in S^3, can ideas from computational complexity be applied to show the existence of nontrivial local minima for interesting knot ener...

open_problem | open | MathDB 页面 | 原文线索

V2-058353 | MATHDB-398459

Conjugacy problem in $\operatorname{Out}(F_n)$

摘录 / EXCERPT: For a free group F_n of finite rank n, does there exist an algorithm that, given two elements of $\operatorname{Out}(F_n)$, decides whether they are conjugate?

open_problem | open | MathDB 页面 | 原文线索

V2-058354 | MATHDB-398673

Polynomial tractability under a nonvanishing profile derivative

摘录 / EXCERPT: For the sampling problem for ridge functions on the Euclidean unit ball, with deterministic adaptive algorithms using function values and error measured in L_∞, let...

open_problem | open | MathDB 页面 | 原文线索

V2-058355 | MATHDB-398677

Nearly optimal complexity of cubature for weighted infinite-variate function spaces

摘录 / EXCERPT: For the function spaces considered in the report, with univariate regularity degree r and product order-dependent weights $\gamma_u=(|u|!)^\alpha\prod_{j\in u}c_j^\beta$ satisfyi...

open_problem | open | MathDB 页面 | 原文线索

V2-058356 | MATHDB-398786

4-coloring of P_6-free graphs

摘录 / EXCERPT: Determine the computational complexity of 4-coloring P_6-free graphs, where P_6 is the path on six vertices.

open_problem | open | MathDB 页面 | 原文线索

V2-058357 | MATHDB-398999

Decidability of strict stability for the joint spectral radius

摘录 / EXCERPT: Is there an algorithm that can take as input m matrices of dimension n with rational entries, halt in finite time, and output the correct yes-no answer as to whether their join...

open_problem | open | MathDB 页面 | 原文线索

V2-058358 | MATHDB-399077

Construction of H²-matrix approximations

摘录 / EXCERPT: Can products, inverses, and factorizations of H^2-matrices be approximated by algorithms structurally similar to those for standard hierarchical matrices? The report answers yes:...

open_problem | open | MathDB 页面 | 原文线索

V2-058359 | MATHDB-399088

Exact complexity of the lens spaces $L(n,1)$

摘录 / EXCERPT: It was conjectured that $c(L(n,1))=n-3$ for every integer $n>3$, where $c(L(n,1))$ denotes the (pseudo-simplicial) complexity of the lens space $L(n,1)$.

open_problem | open | MathDB 页面 | 原文线索

V2-058360 | MATHDB-399089

Improving the coefficients in the linear Cheeger-Gromov bounds

摘录 / EXCERPT: Find optimal or improved (smaller) coefficients in the linear universal bounds of Theorems 2, 3, and 4, namely the bounds for the simplicial complexity, Heegaard-Lickorish complexi...

open_problem | open | MathDB 页面 | 原文线索

V2-058361 | MATHDB-399140

Effective determination of all-genus quintic GW and FJRW invariants

摘录 / EXCERPT: The moduli of MSP-fields will provide an effective algorithm to determine all genus FJRW invariants of the quintic polynomial w_5 of multiple insertions $-\frac{2}{5}$ and all ge...

open_problem | open | MathDB 页面 | 原文线索

V2-058362 | MATHDB-399281

Complete problems for \(\mathrm{NP}^{\mathrm{SSQRT}}\)

摘录 / EXCERPT: Find complete problems for the complexity class $\mathrm{NP}^{\mathrm{SSQRT}}$.

open_problem | open | MathDB 页面 | 原文线索

V2-058363 | MATHDB-399282

Decidability of embeddability of a triangulated 3-manifold in the 3-sphere

摘录 / EXCERPT: There is an algorithm that, given a triangulated 3-manifold X, decides whether X embeds in the 3-sphere S^3.

open_problem | open | MathDB 页面 | 原文线索

V2-058364 | MATHDB-399307

Recognition of elliptic manifolds is in NP

摘录 / EXCERPT: For triangulated three-manifolds, the decision problem of recognizing whether the underlying manifold is elliptic lies in the complexity class NP. Here, an elliptic manifold is a q...

open_problem | open | MathDB 页面 | 原文线索

V2-058365 | MATHDB-399389

Polynomial-time solvability of bin packing with a fixed number of item types

摘录 / EXCERPT: For every fixed constant d, there is a polynomial-time algorithm for the one-dimensional cutting-stock/bin-packing problem with d item types: type i has size $s_i\in[0,1]$ an...

open_problem | open | MathDB 页面 | 原文线索

V2-058366 | MATHDB-399391

Constant-factor approximation for the gluttonous Steiner-forest algorithm

摘录 / EXCERPT: The report identifies as a long-standing question whether the gluttonous algorithm for Steiner Forest-connecting the closest two yet-unsatisfied terminals, regardless of whether th...

open_problem | open | MathDB 页面 | 原文线索

V2-058367 | MATHDB-399399

Computational complexity of computing \(\beta\)

摘录 / EXCERPT: What is the complexity of computing β?

open_problem | open | MathDB 页面 | 原文线索

V2-058368 | MATHDB-399476

Almost-sure convergence of a distributed AIMD optimization algorithm

摘录 / EXCERPT: Let $n\in\mathbb{N}$, $C>0$, and let $f_i:[0,C]\to\mathbb{R}$, $i=1,\ldots,n$, be strictly convex and continuously differentiable. Consider ... Assume that there are constants...

open_problem | open | MathDB 页面 | 原文线索

V2-058369 | MATHDB-399546

Constructive selection algorithm for the Akemann-Weaver Lyapunov-type theorem

摘录 / EXCERPT: Given a finite Bessel sequence $\Phi=\{u_j\}_{j\in J}\subset\mathbb C^n$ with Bessel bound $B=1$, meaning $\sum_{j\in J}u_ju_j^*\le I$, suppose that $\|u_j\|^2\le\varepsilon$ for e...

open_problem | open | MathDB 页面 | 原文线索

V2-058370 | MATHDB-399827

Study of the Weak Chebyshev Greedy Algorithm for bases

摘录 / EXCERPT: “The main message of the talk is that it is time to conduct a deep and thorough study of the WCGA with respect to bases in a style of the corresponding study of the TGA.”

open_problem | open | MathDB 页面 | 原文线索

V2-058371 | MATHDB-399828

Which bases are suitable for the Thresholding Greedy Algorithm?

摘录 / EXCERPT: Which bases are suitable for the use of the Thresholding Greedy Algorithm (TGA)?

open_problem | open | MathDB 页面 | 原文线索

V2-058372 | MATHDB-399892

Decidability of even-Euler-genus non-orientable surface embeddability

摘录 / EXCERPT: Does there exist an algorithm, with no complexity requirement, that, given an even Euler genus g and a 3-manifold M, decides whether a non-orientable surface of Euler genus...

open_problem | open | MathDB 页面 | 原文线索

V2-058373 | MATHDB-399902

Complexity of integer quadratic optimization in dimension 3

摘录 / EXCERPT: For the problem ... where P is a rational polytope, the problem is solvable in polynomial time when $n=2$. What is its complexity when $n=3$?

open_problem | open | MathDB 页面 | 原文线索

V2-058374 | MATHDB-399904

Membership of the completely positive cone in NP

摘录 / EXCERPT: Is checking whether a given matrix satisfies $A\in CP^n$ in the complexity class NP?

open_problem | open | MathDB 页面 | 原文线索

V2-058375 | MATHDB-399989

Feige’s R3SAT Hypothesis

摘录 / EXCERPT: For random 3SAT with $m \ge Cn$ for all sufficiently large constants C, there is no efficient algorithm that can prove that a random instance is at most $(1-\varepsilon)$-satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-058376 | MATHDB-400017

Metric entropy of unconditional convex bodies

摘录 / EXCERPT: Pisier asked whether a statement analogous to the doubly-exponential complexity estimate for the Minkowski compactum holds for the set of all unconditional convex bodies.

open_problem | open | MathDB 页面 | 原文线索

V2-058377 | MATHDB-400240

Algorithmic hardness beyond the overlap-gap threshold for large-average Gaussian submatrices

摘录 / EXCERPT: For an $n\times n$ matrix C_n with i.i.d. standard Gaussian entries, let $\alpha^*=\frac{5\sqrt{2}}{3\sqrt{3}}\approx 1.3608$. The report conjectures that the overlap gap propert...

open_problem | open | MathDB 页面 | 原文线索

V2-058378 | MATHDB-400379

Consistency of gradient descent by local linear regression for flow-line estimation

摘录 / EXCERPT: Show that the gradient descent by local linear regression algorithm can consistently estimate the flow line as pixel resolution increases.

open_problem | open | MathDB 页面 | 原文线索

V2-058379 | MATHDB-400469

Finite-propagation characterization of the uniform Roe algebra without finite decomposition complexity

摘录 / EXCERPT: Let (X,d) be a uniformly discrete metric space and let $a\in B(\ell^2(X))$. The report proves that the following are equivalent: (i) for every $\varepsilon>0$, there is $R>0$ suc...

open_problem | open | MathDB 页面 | 原文线索

V2-058380 | MATHDB-400559

Undecidability of algebraic equations for analytic functions without differentiation

摘录 / EXCERPT: Is there an algorithm which, given an algebraic equation (in one or more variables), with coefficients in $\mathbb{Z}[z]$, decides whether the equation has or does not have a solut...

open_problem | open | MathDB 页面 | 原文线索

V2-058381 | MATHDB-400561

Decidability of algebraic equations with entire-function solutions

摘录 / EXCERPT: Does there exist an algorithm which, given an algebraic equation (in one or more variables) with coefficients in $\mathbb{Z}[z]$, decides whether the equation has a solution which...

open_problem | open | MathDB 页面 | 原文线索

V2-058382 | MATHDB-400616

A codimension-one Sharbly resolution and algorithm for symplectic groups

摘录 / EXCERPT: Construct, in general, a suitable analogue of the Sharbly complex for the symplectic group-namely, a resolution of the Steinberg module that reflects the cellular structure in MacP...

open_problem | open | MathDB 页面 | 原文线索

V2-058383 | MATHDB-400707

Dehn’s solution of the word problem for surface groups

摘录 / EXCERPT: For the surface groups, namely the fundamental groups of orientable 2-manifolds, there is a finite algorithm that determines, for any finite product of generators and inverse gener...

open_problem | open | MathDB 页面 | 原文线索

V2-058384 | MATHDB-400766

Applicability of the improved random K-SAT algorithm to random NAE-\4K\4-SAT

摘录 / EXCERPT: Can a similar idea to the better algorithm known for random 4K4-SAT be applied in the random NAE-4K4-SAT setting?

open_problem | open | MathDB 页面 | 原文线索

V2-058385 | MATHDB-400901

Polynomial-time exact recovery of low-rank tensors at nuclear-norm sample complexity

摘录 / EXCERPT: Do there exist polynomial-time algorithms that recover a low-rank tensor exactly with sample-size requirements similar to those obtained by tensor nuclear-norm minimization?

open_problem | open | MathDB 页面 | 原文线索

V2-058386 | MATHDB-400908

Basin of attraction for Lloyd’s algorithm in symmetric two-component spherical Gaussian mixtures

摘录 / EXCERPT: Assume $r\ge C$ and $n\ge C$ for a sufficiently large constant C. For symmetric, two-component spherical Gaussian mixtures with centers θ^* and $-\theta^*\in\mathbb{R}^d$,...

open_problem | open | MathDB 页面 | 原文线索

V2-058387 | MATHDB-400909

Linear sample complexity for polynomial approximation on suitable irregular domains

摘录 / EXCERPT: The report conjectures that, for suitable domains, the sample complexity of the polynomial approximation method may in fact be linear in N, the dimension of the polynomial space,...

open_problem | open | MathDB 页面 | 原文线索

V2-058388 | MATHDB-400910

Conjectural linear sample complexity for polynomial approximation on suitable domains

摘录 / EXCERPT: The sample complexity estimate for hyperbolic-cross polynomial approximation is conjectured to be linear in $N=|P_n^{HC}|$ for suitable domains.

open_problem | open | MathDB 页面 | 原文线索

V2-058389 | MATHDB-400912

Existence of algorithms for accurate minimisation with inexact input

摘录 / EXCERPT: Question 1 (Existence of algorithms). Given any of the four problems in (1)-Basis Pursuit, Constrained Lasso, Unconstrained Lasso, or Basis Pursuit with noise-where the input may b...

open_problem | open | MathDB 页面 | 原文线索

V2-058390 | MATHDB-400913

Existence of algorithms for subclasses of regularisation problems

摘录 / EXCERPT: Given any of the problems in (1), which subclasses Ω of inputs A and y will provide a positive answer to Question 1?

open_problem | open | MathDB 页面 | 原文线索

V2-058391 | MATHDB-400915

Existence of algorithms for subclasses

摘录 / EXCERPT: Question 3 (Existence of algorithms for subclasses). (i) Given any of the problems in (1), which subclasses Ω of inputs A and y will provide a positive answer to Questio...

open_problem | open | MathDB 页面 | 原文线索

V2-058392 | MATHDB-400929

Computational complexity of the colorful Carathéodory problem

摘录 / EXCERPT: Determine the computational complexity of the colorful Carathéodory problem (Ccp) in arbitrary dimension, where the input consists of point sets...

open_problem | open | MathDB 页面 | 原文线索

V2-058393 | MATHDB-400956

Membership algorithm for the triangle causal model

摘录 / EXCERPT: Assume the three observed random variables in the triangle scenario are binary, and the three hidden variables have a fixed alphabet size d. Let C_d be the set of observable di...

open_problem | open | MathDB 页面 | 原文线索

V2-058394 | MATHDB-400957

Algorithmic membership in the infinite-hidden-alphabet triangle model

摘录 / EXCERPT: In the triangle scenario with three binary observed random variables, let C_d be the set of observable distributions compatible with the structure when each of the three hidden v...

open_problem | open | MathDB 页面 | 原文线索

V2-058395 | MATHDB-400996

Effective Pila-Wilkie bounds for restricted Pfaffian surfaces

摘录 / EXCERPT: Let $B>0$, let $U\subseteq\mathbb{R}^2$ be open with $[0,1]^2\subseteq U$, and suppose that $f:U\to\mathbb{R}$ is Pfaffian of complexity at most B. For every $\varepsilon>0$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-058396 | MATHDB-401169

Hard graph families for Hilbert-Nullstellensatz 3-colouring algorithms

摘录 / EXCERPT: There are explicit families of non-3-colourable graphs such that the algorithms based on Hilbert’s Nullstellensatz over $\mathrm{GF}(2)$ in [4, 5] need to find certificates of line...

open_problem | open | MathDB 页面 | 原文线索

V2-058397 | MATHDB-401180

Graph-complexity formulation of Lehmer’s question

摘录 / EXCERPT: Given $ ...

open_problem | open | MathDB 页面 | 原文线索

V2-058398 | MATHDB-401321

Rigorous universality for more general numerical algorithms

摘录 / EXCERPT: Prove universality rigorously for more general numerical algorithms, including: (a) eigenvalue algorithms such as shifted QR and the Jacobi algorithm, including algorithms for comp...

open_problem | open | MathDB 页面 | 原文线索

V2-058399 | MATHDB-401398

Closing the factor-two algorithmic gap for maximum cliques in random graphs

摘录 / EXCERPT: For the random graph $G(n,p)$, the largest clique has asymptotic size $2\log_{1/p} n$, whereas the best stated constructive algorithm (Greedy) produces only a clique of size...

open_problem | open | MathDB 页面 | 原文线索

V2-058400 | MATHDB-401400

Information-theoretic and algorithmic recovery in the intermediate sample-size regime

摘录 / EXCERPT: In the sparse high-dimensional linear regression model $Y=X\beta^*+W$, with $X\in\mathbb{R}^{n\times p}$ and $W\in\mathbb{R}^n$ having independent standard Gaussian entries, and wi...

open_problem | open | MathDB 页面 | 原文线索

V2-058401 | MATHDB-401657

Algorithmic comparison of closed spraiges

摘录 / EXCERPT: Does there exist an algorithm that, given two closed spraiges, decides whether they are admissibly equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-058402 | MATHDB-401824

Efficiency of the basic Graver-basis augmentation algorithm

摘录 / EXCERPT: Is the basic augmentation algorithm (A)-starting from a feasible point x_0 and repeatedly choosing any $g\in\mathcal G(E)$ with $c^Tg>0$ such that $x+g$ is feasible-an efficient...

open_problem | open | MathDB 页面 | 原文线索

V2-058403 | MATHDB-401923

Locality beyond exponentially decaying interactions in machine-learning interatomic potentials

摘录 / EXCERPT: Beyond simple settings in which the interaction strength decays exponentially in space, what is a good way to account for locality when partitioning quantum-mechanical total energi...

open_problem | open | MathDB 页面 | 原文线索

V2-058404 | MATHDB-401934

A $(6+\b5)$-approximation algorithm for the Santa Claus problem

摘录 / EXCERPT: The Santa Claus problem admits a $(6+\b5)$-approximation algorithm in time $n^{\Theta_{\varepsilon}(1)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-058405 | MATHDB-401941

Sums of powers

摘录 / EXCERPT: Given integers $r_1,r_2,\ldots,r_n$, is there an algorithm running in time polynomial in the input size, $O\!\left(\sum_{i=1}^n \log(|r_i|+1)\right)$, that decides whether ...

open_problem | open | MathDB 页面 | 原文线索

V2-058406 | MATHDB-401954

Does randomized logspace equal deterministic logspace?

摘录 / EXCERPT: Does $RL=L$ hold, i.e. can every problem solvable by a randomized logspace algorithm with one-sided error be solved by a deterministic logspace algorithm?

open_problem | open | MathDB 页面 | 原文线索

V2-058407 | MATHDB-401982

Smoothed complexity under sparse perturbations

摘录 / EXCERPT: Can we give smoothed complexity bounds when only a random ε-fraction of entries of (A,b) get perturbed?

open_problem | open | MathDB 页面 | 原文线索

V2-058408 | MATHDB-402034

Complexity of coloring on the class GUT

摘录 / EXCERPT: Determine the computational complexity of the graph coloring problem restricted to the class GUT of graphs that may contain, among Truemper configurations, only twin wheels and u...

open_problem | open | MathDB 页面 | 原文线索

V2-058409 | MATHDB-402059

Low-complexity optimal channel-resolvability codes

摘录 / EXCERPT: Can one construct optimal channel-resolvability codes with lower encoding complexity, especially with linear complexity?

open_problem | open | MathDB 页面 | 原文线索

V2-058410 | MATHDB-402122

Sample complexity of VDF testing for extendable hereditary graph properties

摘录 / EXCERPT: Is it true that every extendable hereditary graph property $\mathcal{P}$ can be tested in the VDF model with the same (or close to the same) sample complexity as in the standard de...

open_problem | open | MathDB 页面 | 原文线索

V2-058411 | MATHDB-402131

Markov-chain algorithms for approximate counting and sampling at low temperatures

摘录 / EXCERPT: Develop Markov-chain-based algorithms for approximate counting and sampling at low temperatures, and analyze the most natural such algorithms. The report notes that initial steps h...

open_problem | open | MathDB 页面 | 原文线索

V2-058412 | MATHDB-402186

Complexity of Hurwitz matrix completion

摘录 / EXCERPT: What is the complexity class-whether in $\[1mP\[0m$ or NP-hard in the Turing model-of the Hurwitz matrix completion problem? A real $n\times n$ matrix is Hurwitz (or stable) if all...

open_problem | open | MathDB 页面 | 原文线索

V2-058413 | MATHDB-402191

Efficient global algorithms for multi-stage robust optimization

摘录 / EXCERPT: The development of efficient global algorithms for multi-stage robust problems, in particular with non-convex stages that might involve integral decisions.

open_problem | open | MathDB 页面 | 原文线索

V2-058414 | MATHDB-402319

Precise complexity of embeddability in $\mathbb{R}^3$

摘录 / EXCERPT: For the decision problem $\mathrm{EMBED}_{k\to d}$-where $k\le d$ are fixed positive integers and, given a finite simplicial complex K of dimension k, one asks whether K admi...

open_problem | open | MathDB 页面 | 原文线索

V2-058415 | MATHDB-402354

Whether critical points cause the difficulty of efficient algorithms

摘录 / EXCERPT: Is the difficulty in designing efficient counting and sampling algorithms for discrete statistical-mechanics models due to the behaviour of the models precisely at their critical p...

open_problem | open | MathDB 页面 | 原文线索

V2-058416 | MATHDB-402656

Optimal degrees of output formulas in real-algebraic algorithms

摘录 / EXCERPT: Identify the best possible degrees of the output formulas for one-block quantifier elimination by relating them to the degrees of fundamental geometric objects.

open_problem | open | MathDB 页面 | 原文线索

V2-058417 | MATHDB-402692

Splitting-complexity interpretation of the L^2-Euler characteristic for free-by-cyclic groups

摘录 / EXCERPT: Conjecture. Let G be a free-by-cyclic group. For a character $\\varphi:G\to\mathbb Z$, the negative L^2-Euler characteristic $-\chi^{(2)}(\ker\varphi)$ coincides with the minim...

open_problem | open | MathDB 页面 | 原文线索

V2-058418 | MATHDB-402728

Algorithmic determination of rational points on curves of genus at least two

摘录 / EXCERPT: Let F be a number field and let C be a smooth, geometrically irreducible, projective curve of genus $g\ge 2$ defined over F. Suppose C is presented in some projective space...

open_problem | open | MathDB 页面 | 原文线索

V2-058419 | MATHDB-402798

Algorithmic computation of the complete derived-invariant data from a gentle algebra

摘录 / EXCERPT: Given only the quiver with relations $(Q(\Delta), I(\Delta))$ defining the finite-dimensional gentle algebra $A(\Delta)$ associated with a dissected surface (S,M,P,Δ) with...

open_problem | open | MathDB 页面 | 原文线索

V2-058420 | MATHDB-402829

Lower bound for the maximum complexity of the middle level

摘录 / EXCERPT: For an arrangement L of n simple, nonvertical lines in $\mathbb{R}^2$, let $f_k(n)$ be the maximum size of the k-level over all such arrangements, and define...

open_problem | open | MathDB 页面 | 原文线索

V2-058421 | MATHDB-402830

Determine the asymptotically tight complexity of higher-dimensional k-levels

摘录 / EXCERPT: For arrangements $\mathcal A$ of n hyperplanes in $\mathbb R^d$ such that no $d+1$ hyperplanes intersect in a common point and no hyperplane is parallel to the x_d-axis, let...

open_problem | open | MathDB 页面 | 原文线索

V2-058422 | MATHDB-402831

Average complexity of the $(\le k)$-level in a great-circle arrangement

摘录 / EXCERPT: Let C be a simple arrangement of great-circles. The expected size of the $(\le k)$-level is at most $16e\,(k+2)^2$ when the south pole is chosen uniformly at random among the cel...

open_problem | open | MathDB 页面 | 原文线索

V2-058423 | MATHDB-403004

Mulmuley-Sohoni border determinantal-complexity conjecture

摘录 / EXCERPT: Let $\operatorname{perm}_m=\sum_{\pi\in S_m}\prod_{i=1}^{m}x_{i,\pi(i)}$. The sequence of natural numbers $\operatorname{dc}(\operatorname{perm}_m)$, where $\operatorname{dc}(f)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-058424 | MATHDB-403008

Representation-theoretic obstructions for permanent-versus-determinant lower bounds

摘录 / EXCERPT: There exist representation-theoretic obstructions that show superpolynomial lower bounds on the determinantal complexity of the permanent, i.e. on $dc(\mathrm{perm})$.

open_problem | open | MathDB 页面 | 原文线索

V2-058425 | MATHDB-403010

Superpolynomial trace-of-powers complexity of the permanent

摘录 / EXCERPT: The report defines $\operatorname{Pow}_m^n=\operatorname{tr}(X^m)$, where $X=(X_{i,j})$, and defines $\operatorname{pc}(\operatorname{perm}_m)$ as the smallest n such that...

open_problem | open | MathDB 页面 | 原文线索

V2-058426 | MATHDB-403021

Valiant’s VBP versus VNP conjecture via determinantal complexity

摘录 / EXCERPT: Valiant’s $\mathrm{VBP}\ne\mathrm{VNP}$ conjecture is equivalent to the conjecture that $dc'(\mathrm{perm})$ grows superpolynomially.

conjecture | open | MathDB 页面 | 原文线索

V2-058427 | MATHDB-403096

Convergent evolving surface finite element algorithm for Willmore flow

摘录 / EXCERPT: Construct a convergent evolving surface finite element algorithm for the Willmore flow of closed two-dimensional surfaces in three-dimensional space.

open_problem | open | MathDB 页面 | 原文线索

V2-058428 | MATHDB-403173

Alternative methods for computing tautological cycle classes from Hurwitz spaces

摘录 / EXCERPT: Can the cycle classes $(\phi,\delta)_*([H])$ be accessed by means other than the virtual-localization algorithm of Faber-Pandharipande, whose implementation is described as intract...

open_problem | open | MathDB 页面 | 原文线索

V2-058429 | MATHDB-403186

Complexity reduction for space-time DG approximation of acoustic waves

摘录 / EXCERPT: For the two-dimensional acoustic wave equation with conical spatial singularities, after suitable spatial mesh grading and a suitable relation between the spatial and temporal mesh...

open_problem | open | MathDB 页面 | 原文线索

V2-058430 | MATHDB-403273

Algorithm for computing the p-class tower group

摘录 / EXCERPT: Give an algorithm for computing $G_{\varnothing}$, say by giving a presentation of it as a pro-p group.

open_problem | open | MathDB 页面 | 原文线索

V2-058431 | MATHDB-403314

Exact complexity-hierarchy position of Norm-Sym

摘录 / EXCERPT: Where exactly does the problem Norm-Symcomputing $N_{S_n}(H)$ from a subgroup $H\leq S_n$sit in Luks's complexity hierarchy?

open_problem | open | MathDB 页面 | 原文线索

V2-058432 | MATHDB-403318

Complexity and implementation of the commutator decision algorithm

摘录 / EXCERPT: Assuming the answer to Question 1 is positive-namely, that for an infinite linear group such as $\mathrm{SL}(n,\mathbb{Z})$ or $\mathrm{GL}(n,\mathbb{Z})$, with $n\ge 2$, one can d...

open_problem | open | MathDB 页面 | 原文线索

V2-058433 | MATHDB-403384

Global convergence of the PRP conjugate-gradient algorithm under WWP line search

摘录 / EXCERPT: For nonconvex functions, establish the global convergence of the Polak-Ribière-Polyak conjugate-gradient algorithm under the weak Wolfe-Powell line-search conditions ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-058434 | MATHDB-403460

Quantum complexity of approximating row sums of a dense matrix

摘录 / EXCERPT: Given a dense non-negative matrix A whose entries sum to at most 1, determine the complexity of finding an $\\\varepsilon$-ℓ_1-approximation of its row-sum vector...

open_problem | open | MathDB 页面 | 原文线索

V2-058435 | MATHDB-403511

A (Slightly) Improved Approximation Algorithm for Metric TSP - Theorem 1

摘录 / EXCERPT: For some absolute constant $\varepsilon > 10^{-36}$, there is a randomized algorithm that outputs a tour with expected cost at most $(\frac{3}{2}-\varepsilon)\cdot c(x)$, where x...

open_problem | open | MathDB 页面 | 原文线索

V2-058436 | MATHDB-403512

Extension complexity of polytopes with vertices on a common sphere

摘录 / EXCERPT: For every fixed dimension $d\ge 2$, is every d-dimensional polytope with n vertices, all of whose vertices lie on a common sphere, guaranteed to have extension complexity at mo...

open_problem | open | MathDB 页面 | 原文线索

V2-058437 | MATHDB-403573

Query complexity of finding fixed subgraphs in hidden Erdős-Rényi graphs

摘录 / EXCERPT: Let $G(\mathbb{N},p)$ be a hidden Erdős-Rényi random graph with $p>0$ small, and let H be a fixed target graph. Let $f(H,p)$ be the number of adjacency queries needed to reveal a...

open_problem | open | MathDB 页面 | 原文线索

V2-058438 | MATHDB-403575

Query complexity of finding a copy of K_6 in a hidden Erdős-Rényi graph

摘录 / EXCERPT: Let G be an infinite hidden Erdős-Rényi random graph $G(\N,p)$, where $p>0$ is small, and let $f(H,p)$ be the number of adjacency queries needed to reveal a copy of a fixed targe...

open_problem | open | MathDB 页面 | 原文线索

V2-058439 | MATHDB-403931

Convergence of neighbourhood coordinate descent

摘录 / EXCERPT: If $\operatorname{rank}(S)\ge n$ and the n-core of G is empty, then the NCD algorithm converges to the MLE.

open_problem | open | MathDB 页面 | 原文线索

V2-058440 | MATHDB-403989

LOCAL complexity of $(\Delta+1)$-coloring

摘录 / EXCERPT: What is the LOCAL complexity of $(\Delta+1)$-coloring?

open_problem | open | MathDB 页面 | 原文线索

V2-058441 | MATHDB-403990

Complexity classes for finitary factors of iid

摘录 / EXCERPT: Describe all complexity classes in the finitary factor of iid setting, e.g., speed-up results and new examples of local problems.

open_problem | open | MathDB 页面 | 原文线索

V2-058442 | MATHDB-404006

Theorem 3: Equidecomposition by translations with controlled boundaries and complexity

摘录 / EXCERPT: If $k\geq 1$ and $A,B\subseteq\mathbb{R}^k$ are bounded sets such that $\lambda(A)=\lambda(B)>0$, $\dim^{\square}(\partial A)<k$, and $\dim^{\square}(\partial B)<k$, then A and...

open_problem | open | MathDB 页面 | 原文线索

V2-058443 | MATHDB-404207

Complexity of the robust Bilevel Selection Problem under interval uncertainty

摘录 / EXCERPT: Given finite leader and follower item sets E_l and E_f, an integer $b\in\{0,\ldots,|E_l\cup E_f|\}$, leader values $c:E_l\cup E_f\to\mathbb{Q}$, and an interval uncertainty set...

open_problem | open | MathDB 页面 | 原文线索

V2-058444 | MATHDB-404499

Average-case versus worst-case complexity for random 3-SAT

摘录 / EXCERPT: Are random 3-SAT formulas computationally easier than worst-case 3-SAT formulas; in particular, do random 3-SAT instances admit efficient refutation algorithms comparable to the kn...

open_problem | open | MathDB 页面 | 原文线索

V2-058445 | MATHDB-404549

Linear sample-complexity growth for strongly non-normal operators

摘录 / EXCERPT: The report conjectures that, for the adjoint-free operator-learning algorithm applied to uniformly elliptic PDEs perturbed away from self-adjointness by lower-order terms, the appr...

open_problem | open | MathDB 页面 | 原文线索

V2-058446 | MATHDB-404570

Computational complexity of minimum transitive covers

摘录 / EXCERPT: Given a directed graph G, what is the computational complexity of computing $\tc(G)$, the minimum number of edges in a transitive cover of G?

open_problem | open | MathDB 页面 | 原文线索

V2-058447 | MATHDB-404572

Existence of a strongly polynomial algorithm for linear programming

摘录 / EXCERPT: Does there exist a strongly polynomial algorithm for linear programming?

open_problem | open | MathDB 页面 | 原文线索

V2-058448 | MATHDB-404702

Shortest paths and cycles with prescribed length modulo m

摘录 / EXCERPT: Let ℓ and m be nonnegative integers with $m \ge 2$.nn1. Does there exist a polynomial-time algorithm which either finds a shortest path in an input graph between two specifi...

open_problem | open | MathDB 页面 | 原文线索

V2-058449 | MATHDB-404703

Shortest path of prescribed length modulo m

摘录 / EXCERPT: Let ℓ and m be nonnegative integers with $m \ge 2$. Does there exist a polynomial-time algorithm which either finds a shortest path in an input graph between two specified v...

open_problem | open | MathDB 页面 | 原文线索

V2-058450 | MATHDB-404785

Polynomial-time group isomorphism for multiplication-table inputs

摘录 / EXCERPT: For groups given by multiplication tables, is there an algorithm for the group isomorphism problem that runs in time polynomial in the orders of the given groups?

open_problem | open | MathDB 页面 | 原文线索

V2-058451 | MATHDB-404826

Batch verification for NISZK proofs

摘录 / EXCERPT: Any problem in NISZK has a non-interactive statistical zero-knowledge batch-verification protocol with communication complexity $\operatorname{poly}(n,\log k)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-058452 | MATHDB-404828

Average-case computational threshold for uniform number partitioning

摘录 / EXCERPT: For continuous i.i.d. inputs $a_1,\\ldots,a_n\leftarrow[0,1]$, determine the average-case computational complexity of finding a vector $x\in\{-1,1\}^n$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-058453 | MATHDB-405037

Conditions guaranteeing polynomial-time automatability of Sum-of-Squares proofs

摘录 / EXCERPT: Let $P=\{p_1=0,\ldots,p_m=0\}$ be a system of polynomial equations in n variables, and let r be an input polynomial of degree at most d, with the bit complexity needed to rep...

open_problem | open | MathDB 页面 | 原文线索

V2-058454 | MATHDB-405038

Complexity of computing the diameter of graph associahedra

摘录 / EXCERPT: For graph associahedra, is the diameter computation solvable in polynomial time, or is it NP-hard?

open_problem | open | MathDB 页面 | 原文线索

V2-058455 | MATHDB-405173

Polynomial kernel for 2-Club Cluster Edge Deletion

摘录 / EXCERPT: Given a graph $G=(V,E)$ and an integer k, determine whether there exists an edge set $F\subseteq E$ with $|F|\le k$ such that every connected component of $G-F$ has diameter at m...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058456 | MATHDB-405182

Watrous’s no-disentanglers conjecture; QMA versus QMA(2) relative to an oracle

摘录 / EXCERPT: For every pair of constants $\varepsilon,\delta\ge 0$ satisfying $\varepsilon+\delta<1$, every (ε,δ)-disentangler with output space K and input space H must sa...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058457 | MATHDB-405215

Finite classical-communication simulation problem for quantum communication and Bell nonlocality

摘录 / EXCERPT: For each integer $d\ge 2$, determine whether there exists a classical protocol using a finite amount of classical communication and unlimited shared randomness that exactly reprodu...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058458 | MATHDB-405243

Aaronson’s Open Question 11 on forward-only quantum query complexity

摘录 / EXCERPT: For each n, let $X=\mathbb{F}_2^n$ and $N=|X|=2^n$. Let $r=(r_x)_{x\in X}$ be a fixed table of n-bit strings, and define $f_{h,r}:X\to X\times X\times\{0,1\}^n$ by...

open_problem | claimed_solved | MathDB 页面 | 原文线索

B16 数学物理 / Mathematical Physics | 1,526 条

V2-058459 | MATHDB-315470

Quantum parallel repetition for all two-player games

摘录 / EXCERPT: For every finite two-player entangled game G with $\omega^*(G)<1$, must $\omega^*(G^{\otimes n})$ decay exponentially in n?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058460 | MATHDB-315484

Full replica-symmetry breaking at zero temperature in the SK model

摘录 / EXCERPT: Theorem 1.3 (Zero-temperature structure). For the zero-field Sherrington--Kirkpatrick model at zero temperature, there exists a nonnegative function...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058461 | MATHDB-315486

Fixed-order incompleteness of atom-centered structural descriptors

摘录 / EXCERPT: Are five-body trispectrum descriptors, or any fixed finite order of atom-centered density correlations, universally sufficient to distinguish noncongruent three-dimensional atomic...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058462 | MATHDB-315491

Existence of an $\operatorname{AME}(11,5)$ graph state

摘录 / EXCERPT: Does an absolutely maximally entangled state of eleven local systems of dimension five exist?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058463 | MATHDB-315492

Explicit circulant construction of an $\operatorname{AME}(11,4)$ state

摘录 / EXCERPT: Can $\operatorname{AME}(11,4)$ be realized by an explicit circulant graph-state construction?

open_problem | open | MathDB 页面 | 原文线索

V2-058464 | MATHDB-315497

Non-local photonic SWAP, CNOT, Toffoli, and Fredkin gates

摘录 / EXCERPT: Can essential multiphoton gates be implemented non-locally between spatially separated photons without sending the photons to one location, pre-sharing entanglement, or performing...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058465 | MATHDB-315498

Meta-solutions for quantum-state experiment families

摘录 / EXCERPT: Can one automatically discover a human-readable program that generates correct quantum-optical experiments for every size in a state family, including families for which no constru...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058466 | MATHDB-315499

High-dimensional multipartite quantum transformations

摘录 / EXCERPT: How can one manipulate and control the general case of n-photon, d-dimensional quantum states with experimentally meaningful photonic transformations?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058467 | MATHDB-315501

SAT synthesis of photonic state-preparation experiments

摘录 / EXCERPT: Can the search for an interpretable photonic experiment that prepares a prescribed quantum state be formulated and solved exactly, rather than relying only on continuous optimizati...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058468 | MATHDB-315513

Single-minus gluon tree amplitudes

摘录 / EXCERPT: In the special kinematic region $\mathcal{R}_1$ for the single-minus decay, with particle 1 the ingoing minus-helicity gluon and particles $2,\ldots,n$ the outgoing plus-helicity g...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058469 | MATHDB-315606

FullRSB jamming identity $a+b=1$

摘录 / EXCERPT: Can the numerically observed fullRSB critical-exponent relation $a+b=1$ at the jamming transition be derived analytically from the scaling equations?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058470 | MATHDB-315662

Sharp quantum conditional-entropy continuity bound

摘录 / EXCERPT: Let ρ_{AB} and σ_{AB} be quantum states, let $d=\dim A\geq 2$, and suppose that $T(\rho_{AB},\sigma_{AB})\leq\delta\leq 1$, where T is the trace distance. Then ... If...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058471 | MATHDB-315675

Bipartite bound information exists

摘录 / EXCERPT: Does bipartite classical bound information exist: correlations with positive secrecy cost but zero distillable secret-key rate?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058472 | MATHDB-315703

Maxwell conjecture for point-charge equilibria

摘录 / EXCERPT: Conjecture 1.3 (Maxwell; [11], see also Section 4). The total number of points of equilibrium, all assumed nondegenerate, of any configuration with l charges in $\mathbb{R}^{3}$...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058473 | MATHDB-315707

Long-range quantum Gibbs-state separability threshold

摘录 / EXCERPT: Do long-range quantum spin systems exhibit a system-size-independent, constant-temperature transition above which every Gibbs state is separable, and what is the sharp dependence o...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058474 | MATHDB-315708

Higher-dimensional quantum Gibbs correlation decay

摘录 / EXCERPT: Does high-temperature quantum correlation decay extend to higher-dimensional geometrically local systems throughout the zero-free regime without imposing logarithmic separation bet...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058475 | MATHDB-315709

Unital-qubit incompatibility-breaking criterion

摘录 / EXCERPT: Let $\mathcal{N}_{D}$ be a unital qubit channel whose Heisenberg adjoint has Bloch matrix D. The channel is incompatibility breaking for arbitrary POVMs, meaning that for every f...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058476 | MATHDB-315710

Non-Gaussian communication beyond the thermal threshold

摘录 / EXCERPT: For the thermal attenuator $\Phi_{\eta,\nu}$, is the coherent information optimized over all input states equal to that optimized over the set G_{1} of single-mode Gauss...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058477 | MATHDB-315711

Optimality of the Keyl-Werner spectrum estimator

摘录 / EXCERPT: Let $\rho\in\mathbb{C}^{d\times d}$ be an unknown quantum state with spectrum $\alpha=(\alpha_{1},\ldots,\alpha_{d})$. Given copies of ρ, estimate α to total variation...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058478 | MATHDB-315728

Fixed-cost quantum transport metrics

摘录 / EXCERPT: Can a single fixed cost operator turn coupling-based quantum optimal transport into a metric, or square-root metric, on the full quantum state space?

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058479 | MATHDB-315729

Pauli encodings for unclonable encryption

摘录 / EXCERPT: Can efficient one-bit quantum encryption prevent an adversary from producing two systems that both reveal the plaintext after the key is disclosed?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-058480 | MATHDB-315738

Factorial asymptotics of the Matryoshka numbers

摘录 / EXCERPT: Let $(a_n)_{n\ge 1}$ be the Matryoshka numbers of Definition 1.1 in Liu's note. Then there exists a real number S with $0<S<\infty$ such that ... ... and ... Moreover, ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058481 | MATHDB-315747

Type-D ASEP Tracy-Widom marginals

摘录 / EXCERPT: Let $x_i(m,t)$ denote the position at time t of the m-th particle of class i from the left. Let $\tau=t\left(q^{1-2n}+q^{2n-1}\right)$. We conjecture that, for the type-D ASE...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058482 | MATHDB-315749

Two-copy distillability of Werner states

摘录 / EXCERPT: For every local dimension $d\geq 2$, the Werner state ρ_{α} is two-copy undistillable if and only if $-\frac{1}{2}\leq\alpha\leq 1$. Equivalently, ρ_{α} is two-...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058483 | MATHDB-315763

FGG conjecture for QAOA on the ring of disagrees

摘录 / EXCERPT: For an even cycle of size N and depth p satisfying $2p+2\leq N$, is the optimal QAOA approximation ratio for MaxCut exactly ...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058484 | MATHDB-315776

Cosmic-string radiation integral

摘录 / EXCERPT: Evaluate, for arbitrary loop angle α and integer harmonic N, ... the singular integral determining the cosmic-string radiation power $P_N=32G\mu^2 I(N,\alpha)/(\pi^3N^2)$.

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-058485 | MATHDB-316280

Enomoto-Kashiwara's global crystal-basis conjecture for $V_\theta(\lambda)$

摘录 / EXCERPT: Let $V_\theta(\lambda)$ be the symmetric-crystal representation for a dominant integral weight λ fixed by θ, and let $L_\theta(\lambda)$ be its crystal lattice with...

conjecture | open | MathDB 页面 | 原文线索

V2-058486 | MATHDB-316281

Locally analytic quantum group Hopf and Fréchet-Stein conjecture

摘录 / EXCERPT: Let H be the open maximal compact subgroup of a quantum group, let H_r be a sufficiently small compact open subgroup, and let $C^{la}(H,K)_q$ and $D^{la}(H,K)_q$ denote respect...

conjecture | open | MathDB 页面 | 原文线索

V2-058487 | MATHDB-316333

The forward-backward correlation conjecture for matrix product states

摘录 / EXCERPT: Forward-backward correlation conjecture. There exists a function $f(D_{eff})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058488 | MATHDB-316355

The power-of-two Pfaffian conjecture for labeled Cauchon diagrams

摘录 / EXCERPT: Power-of-two Pfaffian conjecture. The absolute value of the Pfaffian satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058489 | MATHDB-316371

Dual global-basis balancedness conjecture for the symmetric-crystal module

摘录 / EXCERPT: Let $L_\theta(\lambda)$ be the crystal lattice, let $c(L_\theta(\lambda))$ denote its bar-conjugate lattice, and let $V_\theta(\lambda)^{\mathrm{up}}_{\mathbf A}$ be the dual latti...

conjecture | open | MathDB 页面 | 原文线索

V2-058490 | MATHDB-316385

The Kazhdan-Lusztig-factorizable-sheaves equivalence

摘录 / EXCERPT: Let G_1 be a simple group, let G_1 be its Langlands dual group, and let $kappa_1$ be the invariant form used to define the Kazhdan-Lusztig category $KL^{kappa_1}(G_1)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058491 | MATHDB-316386

Quantum geometric Langlands equivalence for twisted Whittaker and Kazhdan-Lusztig categories

摘录 / EXCERPT: Let G be a reductive group with Langlands dual group G, and let $kappa$ and $check\kappa$ be invariant forms on their Lie algebras such that, for the mutually dual Cartan subal...

conjecture | open | MathDB 页面 | 原文线索

V2-058492 | MATHDB-316387

Stoyanovsky's quantum geometric Langlands conjecture

摘录 / EXCERPT: Let X be a global curve, let G and G be Langlands-dual groups, and let Bun_G and Bun_{G} be their stacks of principal bundles on X. For twisting parameters c and...

conjecture | open | MathDB 页面 | 原文线索

V2-058493 | MATHDB-316395

Flatness conjecture for braided symmetric algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a reductive Lie algebra, let V^q be a finite-dimensional $U_q(\mathfrak{g})$-module, and let V be its classical limit. The reduced symmetric algebra is...

conjecture | open | MathDB 页面 | 原文线索

V2-058494 | MATHDB-316407

The equality characterization for the N-volume quantum Fisher information conjecture

摘录 / EXCERPT: Equality characterization conjecture. Equality in the N-volume inequality holds if and only if $A_1{}_0,\ldots,A_N{}_0$ are linearly dependent.

conjecture | open | MathDB 页面 | 原文线索

V2-058495 | MATHDB-316408

The monotonicity conjecture for N-volume quantum Fisher information

摘录 / EXCERPT: Monotonicity conjecture. For any $f,g\in{\cal F}_{op}^{\,r}$,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058496 | MATHDB-316446

The quantum-correction gluing conjecture for mirrors of anticanonical complements

摘录 / EXCERPT: Let X be a Kähler surface with anticanonical divisor D, and let M be the complexified moduli space of special Lagrangian tori in $X\setminus D$. A Maslov index 0 wall is a re...

conjecture | open | MathDB 页面 | 原文线索

V2-058497 | MATHDB-316530

Nienhuis's critical-value conjecture for the O(n) loop model

摘录 / EXCERPT: Let $n\in[0,2]$ be the loop weight and $x>0$ the edge weight in the hexagonal-lattice $O(n)$ loop model, whose configuration weights are proportional to...

conjecture | open | MathDB 页面 | 原文线索

V2-058498 | MATHDB-316544

Existence of a ground state for quantum layers with integrable Gaussian curvature

摘录 / EXCERPT: Let $p:\varSigma\to\mathbb R^3$ be an embedded, orientable, complete, non-compact surface that is not totally geodesic and whose second fundamental form tends to zero at infinity....

conjecture | open | MathDB 页面 | 原文线索

V2-058499 | MATHDB-316649

Quantum spectral conjecture for finite alternating Hilbert operators

摘录 / EXCERPT: Quantum spectral conjecture. The displayed formula gives the complete spectrum of the quantum alternating Hilbert matrix, with separate forms according to the parity of N. The so...

conjecture | open | MathDB 页面 | 原文线索

V2-058500 | MATHDB-316786

All-time existence conjecture for Ricci Yang-Mills flow with nontrivial curvature power

摘录 / EXCERPT: All-time existence conjecture. The solution to the Ricci Yang-Mills flow with initial condition (g,A) exists for all time.

conjecture | open | MathDB 页面 | 原文线索

V2-058501 | MATHDB-316787

All-time existence conjecture for odd-dimensional Ricci Yang-Mills flow

摘录 / EXCERPT: Odd-dimensional all-time existence conjecture. The solution to the Ricci Yang-Mills flow exists for all time.

conjecture | open | MathDB 页面 | 原文线索

V2-058502 | MATHDB-316998

Invariance of the coherent master-space component under toric Seiberg duality

摘录 / EXCERPT: Coherent-component invariance conjecture. Quivers which are toric (Seiberg) duals have the same coherent component of the master space.

conjecture | open | MathDB 页面 | 原文线索

V2-058503 | MATHDB-317170

The conjecture that general-relativistic gravitation is effective quantum-states thermodynamics

摘录 / EXCERPT: Let a General Relativistic spacetime determine a gravitational field, and let its associated quantum states admit an effective thermodynamic description. Gravitational thermodynami...

conjecture | open | MathDB 页面 | 原文线索

V2-058504 | MATHDB-317216

Geometric realization conjecture for the diagrammatic rings $R(\nu)$

摘录 / EXCERPT: Let $Gamma$ be a graph and $nu\in\mathbbm N[I]$. Let $R(\nu)$ be the diagrammatically defined graded ring, let ... and let L be the sum of the simple perverse sheaves L_b for all...

conjecture | open | MathDB 页面 | 原文线索

V2-058505 | MATHDB-317222

Gram-matrix parity conjecture for the dual canonical basis

摘录 / EXCERPT: Gram-matrix parity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-058506 | MATHDB-317266

Nilpotency conjecture for the composition structure of the D-particle field category

摘录 / EXCERPT: Let ${\mathfrak{I}}$ denote the composition structure on the category ${\rm ker}\,\mathbf{Q}$, where the propagators of open string fields are defined from the Poisson-type sigma m...

conjecture | open | MathDB 页面 | 原文线索

V2-058507 | MATHDB-317341

Continuous and discrete knot-type equivalence for quantum knots

摘录 / EXCERPT: Let K_1 and K_2 be knot n-mosaics, and let $\lvert K_1\rangle$ and $\lvert K_2\rangle$ be the corresponding quantum knots. Let $\underset{n}{\approx}$ denote equivalence unde...

conjecture | open | MathDB 页面 | 原文线索

V2-058508 | MATHDB-317397

Bray-Ilmanen conjecture on outermost generalized apparent horizons

摘录 / EXCERPT: Let (M^n,g,p) be a complete asymptotically flat initial data set containing a generalized trapped surface, with $2\leq n\leq 7$. A hypersurface $\Sigma^{n-1}\subset M^n$ is calle...

conjecture | open | MathDB 页面 | 原文线索

V2-058509 | MATHDB-317401

Freeness conjecture for fusion semirings of free quantum algebras

摘录 / EXCERPT: Let A be a free quantum algebra, meaning a Hopf algebra satisfying ... whose tensor category is spanned by partitions. Let $R^+(A)$ denote its fusion semiring. Freeness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-058510 | MATHDB-317402

Dixmier-property conjecture for reduced free quantum algebras

摘录 / EXCERPT: Let A be a free quantum algebra, and let A_{red} be its reduced algebra, obtained by quotienting A by the null ideal of its Haar functional. Dixmier-property conject...

conjecture | open | MathDB 页面 | 原文线索

V2-058511 | MATHDB-317471

The noncommutative Shephard-Todd-Chevalley conjecture for quantum polynomial rings

摘录 / EXCERPT: Let B be a quantum polynomial ring and let G be a finite subgroup of $\operatorname{Aut_{gr}}(B)$. Noncommutative Shephard-Todd-Chevalley conjecture. The fixed subring B^G ha...

conjecture | open | MathDB 页面 | 原文线索

V2-058512 | MATHDB-317472

The mystic reflection group conjecture for quantum polynomial rings

摘录 / EXCERPT: Let B be a quantum polynomial ring and let G be a finite subgroup of $\operatorname{Aut_{gr}}(B)$ such that the fixed subring B^G has finite global dimension. For a positive...

conjecture | open | MathDB 页面 | 原文线索

V2-058513 | MATHDB-317487

Kashiwara-Enomoto conjecture on crystal bases of symmetric-crystal modules

摘录 / EXCERPT: Let I be the index set of a symmetrizable Cartan datum, let θ be an involutive Dynkin-diagram automorphism with no fixed point, and let $V_\theta(\lambda)$ be the irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-058514 | MATHDB-317626

Scaling-limit conjecture for random quadrangulations and Liouville quantum gravity

摘录 / EXCERPT: Let M_n be a uniformly chosen simply connected planar quadrangulation with n quadrilaterals, let M_∞ be its infinite-volume limit with a distinguished square, and confor...

conjecture | open | MathDB 页面 | 原文线索

V2-058515 | MATHDB-317641

Semiclassical asymptotic conjecture for the quantum Chern-Simons invariant

摘录 / EXCERPT: Let X be a closed oriented 3-manifold with a finite set $\mathcal{M}_X$ of equivalence classes of flat $SU(2)$-connections, and let $\mathcal{M}_X^0\subseteq\mathcal{M}_X$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-058516 | MATHDB-317664

Area bound in one-dimensional gapped models

摘录 / EXCERPT: Area bound conjecture. There exists such a function f for every gapped one-dimensional model, so the entanglement entropy is bounded in terms of the ratio of the Lieb-Robinson ve...

conjecture | open | MathDB 页面 | 原文线索

V2-058517 | MATHDB-317729

K-theory continuation conjecture for global quantum D-modules

摘录 / EXCERPT: Let $\mathcal{X}_1$ and $\mathcal{X}_2$ be K-equivalent spaces connected by a global quantum D-module, and let γ be a path in the Kähler moduli space from a point of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-058518 | MATHDB-317760

Additivity-operator convexity conjecture for unitarily covariant channels

摘录 / EXCERPT: Let $\mathcal{U}$ be the set of functions that are additive for every pair of unitarily covariant quantum channels (Φ,Ω). A function is operator convex when it satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-058519 | MATHDB-317997

Jonsson's partition-function conjecture for odd square-grid cylinders

摘录 / EXCERPT: Let $C_{m,n}$ be the square-grid cylinder, let $Z(C_{m,n})$ be the hard-particle partition function at activity -1, and suppose that n is odd. Jonsson's cylinder partition-func...

conjecture | open | MathDB 页面 | 原文线索

V2-058520 | MATHDB-318010

The $T(\lambda)$-$M(\lambda)$-$L(\gb\lambda)$ conjecture for minimal affinizations

摘录 / EXCERPT: Let $\lambda\in P^+$ be such that its extended support $\overline{\operatorname{supp}}(\lambda)$ does not contain a subdiagram of type D_4. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058521 | MATHDB-318011

The $T_k(\lambda)$-$M_k(\lambda)$ conjecture for regular multiple minimal affinizations

摘录 / EXCERPT: Let $\lie g$ be of type D or E, let i_0 be its unique triply connected node, and let $\lambda\in P^+$ be supported on the three connected components of $I\setminus\{i_0\}$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-058522 | MATHDB-318012

The tensor-product realization conjecture for irregular minimal affinizations

摘录 / EXCERPT: Let $\lie g$ be of type D or E, with the decomposition used in the paper into subdiagrams I_1, J, and I_2, where J is of type D_4. Let $\lambda\in P^+$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058523 | MATHDB-318049

McNamara-Sagan conjectures for quantum binomial coefficients

摘录 / EXCERPT: Let q be a variable and define the quantum integers and quantum binomial coefficients by ... Call a sequence of Laurent polynomials infinitely q-log-concave when repeated applica...

conjecture | open | MathDB 页面 | 原文线索

V2-058524 | MATHDB-318186

Clustering bound for truncated correlations in unbounded spin systems

摘录 / EXCERPT: Let $\psi^{\sharp}$ denote either spin field, let $\lambdaup>0$ be sufficiently small, and let $\Lambdaup\subset\mathbb{Z}^d$ be a finite volume. For even $n\geq 2$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-058525 | MATHDB-318208

Affine Langlands character conjecture for projected simple characters

摘录 / EXCERPT: Affine Langlands character conjecture. The polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-058526 | MATHDB-318233

Givental conjecture for the Landau-Ginzburg A-model of a singularity

摘录 / EXCERPT: Let ${\mathscr D}_{W,G}$ be the tau-function of the Landau-Ginzburg A-model for the pair (W,G), and let ${\mathscr D}_{0,W,G}$ be its genus 0 tau-function. Suppose that the Fro...

conjecture | open | MathDB 页面 | 原文线索

V2-058527 | MATHDB-318250

The multiple-black-hole Penrose inequality

摘录 / EXCERPT: Multiple-black-hole Penrose conjecture. The mass satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058528 | MATHDB-318296

The arbitrary-level monoidal categorification conjecture

摘录 / EXCERPT: Arbitrary-level monoidal categorification conjecture. The assignment

conjecture | open | MathDB 页面 | 原文线索

V2-058529 | MATHDB-318428

Future causal geodesic completeness conjecture for symmetric cosmological spacetimes

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be the maximal Cauchy development of T^2-symmetric or $k=-1$ surface-symmetric initial data in the vacuum or with Vlasov matter and with $\Lambda \ge 0$. As...

conjecture | open | MathDB 页面 | 原文线索

V2-058530 | MATHDB-318485

Nakajima's monoidal categorification conjecture for ADE cluster algebras

摘录 / EXCERPT: Nakajima's monoidal categorification conjecture. The cluster algebra $\mathscr A$ has a monoidal categorification, given by the monoidal subcategory $\mathscr C_1$.

conjecture | open | MathDB 页面 | 原文线索

V2-058531 | MATHDB-318486

Symmetric product-state conjecture for maximal overlap of permutation-invariant states

摘录 / EXCERPT: Let a permutation-invariant multipartite state be given, and consider its maximal overlap with product states. The question concerns whether the product state attaining this maxima...

open_problem | open | MathDB 页面 | 原文线索

V2-058532 | MATHDB-318524

The Penrose conjecture for future apparent horizons

摘录 / EXCERPT: Let (M^3,g,k) be complete Cauchy data satisfying the nonnegative energy density condition $4\mu \ge |J|4$, and suppose it is Schwarzschild at infinity with total mass m in a ch...

conjecture | open | MathDB 页面 | 原文线索

V2-058533 | MATHDB-318525

Ilmanen's existence conjecture for outermost generalized apparent horizons

摘录 / EXCERPT: Let (M^n,g,k) be complete asymptotically flat Cauchy data with $3\le n\le 7$, and let Σ^{n-1} be a generalized trapped surface. Ilmanen's conjecture. There exists a unique...

conjecture | open | MathDB 页面 | 原文线索

V2-058534 | MATHDB-318527

The outermost case of the generalized Penrose conjecture

摘录 / EXCERPT: Let (M^3,g,k) be complete Cauchy data satisfying $\mu\ge |J|$, Schwarzschild at infinity with total mass m in a chosen end. Let $\Sigma^2=\partial U^3\in\mathcal S$ be a strict...

conjecture | open | MathDB 页面 | 原文线索

V2-058535 | MATHDB-318528

Existence conjecture for the Jang-zero-divergence system

摘录 / EXCERPT: Let (M,g,k) be asymptotically flat Cauchy data with an outermost generalized apparent horizon boundary Σ. Define ... ... where $v=\phi\nabla f/(1+\phi^2|df|_g^2)^{1/2}$. Jang-...

open_problem | open | MathDB 页面 | 原文线索

V2-058536 | MATHDB-318656

Even-dimensional Hopf algebra square conjecture

摘录 / EXCERPT: Let A be a semisimple factorizable Hopf algebra over a field K of characteristic zero. Let $n=\dim A$, let u be its Drinfel'd element, and let χ_R be the character of it...

conjecture | open | MathDB 页面 | 原文线索

V2-058537 | MATHDB-318691

Bray and Khuri's generalized Penrose inequality

摘录 / EXCERPT: Let $(\Sigma,\gamma_{ij},A_{ij})$ be an asymptotically Euclidean initial data set that is Schwarzschild at infinity, with total mass M_{ADM} in a chosen end, and satisfying the d...

conjecture | open | MathDB 页面 | 原文线索

V2-058538 | MATHDB-318704

Integral-kernel formulas for affine Toda Q-operators

摘录 / EXCERPT: Affine Toda Q-operator conjecture. The integral kernels are given by the eight compositions displayed in the source, namely those for $A^{(2)}_{2n}$, $A^{(2)}_{2n-1}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-058539 | MATHDB-318708

Squier's intersection conjecture for braid-group power subgroups

摘录 / EXCERPT: Let B_n be the braid group on n strands, and for each positive integer k let $B_n[k]$ be the normal subgroup generated by g_j^k for $1\leq j\leq n-1$. Squier's intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-058540 | MATHDB-318793

Geometric interpretation conjecture for twisted monstrous Lie algebras

摘录 / EXCERPT: Let $\mathbb M$ be the Monster. For each $g\in\mathbb M$, let $\mathfrak m_g$ be the corresponding twisted monstrous Lie algebra and let $\mathcal I_g$ be its instanton space. Assu...

conjecture | open | MathDB 页面 | 原文线索

V2-058541 | MATHDB-318916

Quantum infinitesimal Weierstrass equation conjecture

摘录 / EXCERPT: Let ${}^{\bullet}\wp^{\mathrm{nc}}_{\mu,\theta}$ and ${}^{\bullet}\wp'_{\mu,\theta}$ be the quantum nonstandard Weierstrass function and its derivative, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058542 | MATHDB-318926

The Conformal Conjecture for asymptotically flat three-manifolds with boundary

摘录 / EXCERPT: Conformal Conjecture. The function u is harmonic with respect to h and tends to 1 at infinity; in the metric h_0, the areas of $\widetilde \Sigma$ and Σ differ by at m...

conjecture | open | MathDB 页面 | 原文线索

V2-058543 | MATHDB-318927

Equality characterization for the Riemannian ZAS inequality

摘录 / EXCERPT: Equality characterization conjecture. Then (M,g) is isometric to a Schwarzschild metric with finitely many points deleted when $-\infty<m<0$, or to the flat metric on...

conjecture | open | MathDB 页面 | 原文线索

V2-058544 | MATHDB-318928

Mixed Riemannian Penrose-ZAS inequality

摘录 / EXCERPT: Mixed Riemannian ZAS inequality conjecture. The ADM mass satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058545 | MATHDB-318963

Acharya-Douglas minimal-distance conjecture for the string Landscape

摘录 / EXCERPT: Let $\mathcal{C}$ be the configuration space of string vacua, and let physically distinct vacua be points of this space. Acharya-Douglas minimal-distance conjecture. There exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-058546 | MATHDB-318974

The separable straightening theorem for irreducible continuous lattices

摘录 / EXCERPT: Let L be an irreducible, complete and continuous lattice such that every element is the supremum of a family of minimal elements and 1 is the supremum of a countable family of...

conjecture | open | MathDB 页面 | 原文线索

V2-058547 | MATHDB-319090

Additivity conjecture for minimum output entropy of quantum channels

摘录 / EXCERPT: Let a quantum channel be a linear completely positive trace-preserving map $\Phi:\mathcal{M}_n(\mathbb{C})\to\mathcal{M}_k(\mathbb{C})$. Its minimum output entropy is ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-058548 | MATHDB-319204

Rosengren's positivity conjecture for the polynomials p_n

摘录 / EXCERPT: Rosengren's positivity conjecture. The polynomials p_n have positive coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-058549 | MATHDB-319207

Rosengren's free-energy formula conjecture for the three-colour model

摘录 / EXCERPT: Rosengren's free-energy conjecture. The free energy per volume is

conjecture | open | MathDB 页面 | 原文线索

V2-058550 | MATHDB-319284

The zero-location conjecture for fused transfer-matrix functions

摘录 / EXCERPT: Let u be sufficiently small, with $|u|\ll 1$, let s be a positive integer, and let $a\in\{1,2,\dots,s\}$. For each positive integer m, let $T^{(a)}_{m}(v)$ denote the transfe...

conjecture | open | MathDB 页面 | 原文线索

V2-058551 | MATHDB-319479

The graded-equivalence conjecture for isomorphic quantum projective spaces

摘录 / EXCERPT: Let $\mathbb{P}^{n-1}_{\omega}$ and $\mathbb{P}^{n-1}_{\omega'}$ be quantum projective spaces with associated graded algebras S_{ω} and $S_{\omega'}$, respectively. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-058552 | MATHDB-319493

Mochizuki's formula with partition-function coefficients

摘录 / EXCERPT: Mochizuki's conjecture. Mochizuki's result holds for X, up to sign, when $\widetilde{\mathcal A}(\xi_1,y;a)$ is replaced by the coefficients of $\mathcal B(\xi_1,\xi;a)$ in the p...

open_problem | open | MathDB 页面 | 原文线索

V2-058553 | MATHDB-319528

The projector relation for braided trace powers

摘录 / EXCERPT: Let ${\rm Tr}_R L^k\in {\cal L}^k(R)$ be the braided trace power, and let $P_+^{(k)}$ and $P_{+2}^{(k-1)}$ be the symmetrizer operators acting on the corresponding tensor powers. P...

conjecture | open | MathDB 页面 | 原文线索

V2-058554 | MATHDB-319706

The NPT bound-entanglement conjecture

摘录 / EXCERPT: NPT bound-entanglement conjecture. There exist bipartite NPT states which are not distillable.

conjecture | open | MathDB 页面 | 原文线索

V2-058555 | MATHDB-319707

The Werner-state nondistillability conjecture

摘录 / EXCERPT: Werner-state nondistillability conjecture. None of the Werner states $\rho_W(t)$, $t\in(1/d,1/2]$, is distillable.

conjecture | open | MathDB 页面 | 原文线索

V2-058556 | MATHDB-319708

The 2-distillability conjecture for Werner states

摘录 / EXCERPT: 2-distillability conjecture for Werner states. None of the Werner states $\rho_W(t)$, $t\in(1/d,1/2]$, is 2-distillable.

conjecture | open | MathDB 页面 | 原文线索

V2-058557 | MATHDB-319709

The positivity conjecture for the Werner-state matrix H

摘录 / EXCERPT: Positivity conjecture for the Werner-state matrix H. The matrix $H(X,Y)$ is positive semidefinite for all $X,Y\in M_d$.

conjecture | open | MathDB 页面 | 原文线索

V2-058558 | MATHDB-319710

The generic determinant conjecture for the Werner-state matrix H

摘录 / EXCERPT: Generic determinant conjecture for the Werner-state matrix H. $D(X,Y)\ne0$ for generic (X,Y) and $d\ge3$.

conjecture | open | MathDB 页面 | 原文线索

V2-058559 | MATHDB-319736

The general lifting procedure conjecture for Nichols algebras

摘录 / EXCERPT: Let $C(L)$ and $D(L)$ be the indexing sets of relations for the PBW basis $[L]$ of the Nichols algebra, and let I_U be the ideal generated by all relations with leading super wor...

conjecture | open | MathDB 页面 | 原文线索

V2-058560 | MATHDB-319751

Haldane's conjecture on translation invariant symmetric polynomials

摘录 / EXCERPT: Let p be a symmetric polynomial in n variables. Write each symmetrized monomial as a multiset $[l_1,\ldots,l_n]$, and order symmetrized monomials by squeezing: $s>t$ when t c...

conjecture | refuted | MathDB 页面 | 原文线索

V2-058561 | MATHDB-319765

The Mixmaster conjecture

摘录 / EXCERPT: Let $n+1\leq 10$, and consider vacuum space-times. The conjecture concerns inextendible geodesics along which the curvature is unbounded and allows unspecified natural geometric va...

conjecture | open | MathDB 页面 | 原文线索

V2-058562 | MATHDB-319846

High-density ground-state and infinite-density ground-state existence conjecture

摘录 / EXCERPT: High-density ground-state and infinite-density ground-state conjecture. If u is superstable and strongly tempered, then high-density GSCs and IDGSs in infinite space exist. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-058563 | MATHDB-319850

Non-localization conjecture for discrete-time quantum walks with continuous spectrum

摘录 / EXCERPT: Let WC be the quantum walk operator of a discrete-time quantum walk (DTQW), and let its distribution be the associated position distribution. The operator has only continuous spe...

conjecture | open | MathDB 页面 | 原文线索

V2-058564 | MATHDB-319873

Pointedness conjecture for non-semisimple Hopf algebras of dimension pq^2

摘录 / EXCERPT: Pointedness conjecture. The Hopf algebra H or its dual H^* is pointed.

conjecture | open | MathDB 页面 | 原文线索

V2-058565 | MATHDB-319937

The Distillability Conjecture for bipartite NPT states

摘录 / EXCERPT: Let ρ be a bipartite quantum state. It is NPT if its partial transpose is not positive semidefinite, and indistillable if no tensor power of ρ is 1-distillable. Distill...

conjecture | open | MathDB 页面 | 原文线索

V2-058566 | MATHDB-319938

The Generalized Distillability Conjecture for critical Werner states

摘录 / EXCERPT: Generalized Distillability Conjecture. All generalized critical Werner states $\rho_{(d_1,\ldots,d_m)}^W$, $m\geq1$, are 1-indistillable.

conjecture | open | MathDB 页面 | 原文线索

V2-058567 | MATHDB-319939

The generalized Cauchy-Bunyakovsky-Schwarz inequality conjecture for hypermatrices

摘录 / EXCERPT: Let $m,n\geq1$, let ${\bf d}=(d_1,\ldots,d_m)$, and let $\mathcal{M}_{\bf d}$ be the space of complex hypermatrices of type $d_1\times\cdots\times d_m$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-058568 | MATHDB-319983

Optimal thermal clustering distinction between massless and massive theories

摘录 / EXCERPT: Optimal thermal clustering conjecture. In the massless case, the decay has the form

conjecture | open | MathDB 页面 | 原文线索

V2-058569 | MATHDB-320080

Determination of a standard half-sided modular inclusion by its algebraic inclusion

摘录 / EXCERPT: Determination conjecture. There exists a unitary operator W such that

conjecture | open | MathDB 页面 | 原文线索

V2-058570 | MATHDB-320216

Moore-Witten conjecture for SU(2) Donaldson invariants of CP^2

摘录 / EXCERPT: Consider the complex projective plane $\mathbb{C}\mathrm{P}^2$ with gauge group $\mathrm{SU}(2)$. Its regularized u-plane integral has a contribution from the cusp at...

conjecture | open | MathDB 页面 | 原文线索

V2-058571 | MATHDB-320299

The quantum-group si]-system scalar-splitting conjecture

摘录 / EXCERPT: Let $\mathcal{U}_\varpi=U_\varpi(\mathfrak{g})$ be a quantized universal enveloping algebra at an odd root of unity, and let $\mathcal{B}^{\pm}_\varpi$ be its Borel Hopf subalgebra...

conjecture | open | MathDB 页面 | 原文线索

V2-058572 | MATHDB-320310

Quantum lifespan conjecture for the homogeneous complex Monge-Ampère equation

摘录 / EXCERPT: Let $S_T=[0,T]\times\mathbb{R}$ and let φ_∞ be the quantum analytic continuation potential for the Cauchy data $(\omega_{\varphi_0},\dot{\varphi}_0)$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-058573 | MATHDB-320324

Freidlin-Wentzell/semi-classical limit for quantum momentum distributions

摘录 / EXCERPT: Freidlin-Wentzell/semi-classical limit. For $\lambda\ll 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-058574 | MATHDB-320325

Conjectured Ornstein-Uhlenbeck scaling limit for the quantum momentum process

摘录 / EXCERPT: Ornstein-Uhlenbeck scaling-limit conjecture. As $\lambda\to0$, with convergence in law for the Skorokhod metric,

conjecture | open | MathDB 页面 | 原文线索

V2-058575 | MATHDB-320341

Biased Glauber dynamics conjecture for even-degree regular trees

摘录 / EXCERPT: Let T be a regular tree of even degree, and let the initial spin configuration have a product distribution whose Bernoulli marginals are biased. Consider zero-temperature Glauber...

conjecture | open | MathDB 页面 | 原文线索

V2-058576 | MATHDB-320379

Bray-Iga conjecture on the Riemannian Penrose inequality for conformally flat manifolds

摘录 / EXCERPT: Bray-Iga conjecture. The Riemannian Penrose inequality holds for arbitrary-dimensional conformally flat, asymptotically flat manifolds with nonnegative scalar curvature.

conjecture | open | MathDB 页面 | 原文线索

V2-058577 | MATHDB-320392

The primitive-spectrum and symplectic-leaf conjecture for quantized function algebras

摘录 / EXCERPT: Let G be a complex semisimple algebraic group, and let $\mathcal{O}_q(G)$ be the corresponding quantized function algebra. Let $\mathbb{C}[G]$ carry the standard Poisson structur...

conjecture | open | MathDB 页面 | 原文线索

V2-058578 | MATHDB-320399

Tóth's phase-transition conjecture for spontaneous magnetization

摘录 / EXCERPT: Let $m(\beta)$ denote the spontaneous magnetization of the quantum Heisenberg ferromagnet, represented through the interchange process on $[-r,r]^3$ at inverse temperature β....

conjecture | open | MathDB 页面 | 原文线索

V2-058579 | MATHDB-320410

Unbounded-critical-point conjecture for non-planar Potts-model zeros

摘录 / EXCERPT: Consider a family of non-planar connected graphs $G_{n,L}$ with n identical layers of width L and periodic boundary conditions in the n-direction, together with a curve...

conjecture | open | MathDB 页面 | 原文线索

V2-058580 | MATHDB-320479

Bethe-equation formula for transfer-matrix eigenvalues

摘录 / EXCERPT: Let $T^{(a)}_m(u)$ be the transfer matrix normalized as in, and consider the sector with weight $\sum_{k=1}^N s_k\omega_{r_k}-\sum_{a=1}^r n_a\alpha_a$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-058581 | MATHDB-320480

Root-of-unity restricted Q-system conjecture

摘录 / EXCERPT: Fix a level ℓ and let $Q^{(a)}_m=\dim_q\mathrm{res}\,W^{(a)}_m$ be the root-of-unity quantum dimensions of the restricted Kirillov-Reshetikhin modules, with $a\in I$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-058582 | MATHDB-320499

Quantization of birational symplectomorphisms of algebraic tori

摘录 / EXCERPT: Let ${\bf k}$ be an algebraically closed field of characteristic zero, let $\mathbb G_{m,\bf k}^{2n}$ carry the standard symplectic form ... and let $BirSympl_{n,\bf k}$ be the group...

conjecture | open | MathDB 页面 | 原文线索

V2-058583 | MATHDB-320505

Weak ultimate Schwarzschild conjecture for the spacetime of Theorem maintheorem

摘录 / EXCERPT: The spacetime under consideration satisfies the hypotheses of Theorem ... is weakly ultimately Schwarzschildean to order $k+2$. This conjecture is motivated by techniques that obtain...

conjecture | open | MathDB 页面 | 原文线索

V2-058584 | MATHDB-320542

Simple product conjecture for dual canonical basis elements

摘录 / EXCERPT: Let $\mathscr B(\infty)$ be the crystal indexing the dual canonical basis. For basis elements $G^{\operatorname{up}}(b_i)$, write $b_i\bot b_j$ for the compatibility relation used...

conjecture | open | MathDB 页面 | 原文线索

V2-058585 | MATHDB-320548

The integral-form conjecture for the double Ringel-Hall algebra

摘录 / EXCERPT: Let ${\mathcal Z}=\mathbb Z[v,v^{-1}]$, let ${\boldsymbol{\mathfrak D}_{\!\vartriangle\!}}(n)$ be the double Ringel-Hall algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058586 | MATHDB-320549

Affine Schur-Weyl reciprocity over a non-root-of-unity specialization

摘录 / EXCERPT: Let n and r be as in the affine quantum Schur-Weyl construction, let ${\boldsymbol{\mathcal H}_{\!\vartriangle\!}(r)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-058587 | MATHDB-320551

The realization conjecture for the affine quantum group

摘录 / EXCERPT: Let ${\boldsymbol{\mathfrak A}}_{\!\vartriangle\!}(n)$ be the $\mathbb Q(v)$-space constructed from the affine BLM-type algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058588 | MATHDB-320555

The quantum-classical partition-function inequality

摘录 / EXCERPT: Let $Z_q(\beta,h)=\sum_{n=1}^{\infty}e^{-\beta E_n(h)}$ be the quantum statistical sum, where $E_n(h)$ are the energy eigenvalues, and let ... be the classical statistical sum for th...

conjecture | open | MathDB 页面 | 原文线索

V2-058589 | MATHDB-320556

The quantum-classical mean-energy inequality

摘录 / EXCERPT: Quantum-classical mean-energy inequality. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-058590 | MATHDB-320557

The high-temperature quantum-classical asymptotics conjecture

摘录 / EXCERPT: Let $Z_q(\beta,h)$ and $Z_c(\beta)$ be the quantum and classical statistical sums, and let $E_q(\beta,h)$ and $E_c(\beta)$ be the corresponding quantum and classical mean energies....

conjecture | open | MathDB 页面 | 原文线索

V2-058591 | MATHDB-320558

The monotonicity conjecture for the quantum statistical sum

摘录 / EXCERPT: Let $Z_q(\beta,h)=\sum_{n=1}^{\infty}e^{-\beta E_n(h)}$ be the quantum statistical sum for the Schrödinger system, where $h>0$ is Planck's constant, $\beta>0$, and N is the dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-058592 | MATHDB-320633

Existence of renormalized Mattis-model fermion correlation functions in the QFT limit

摘录 / EXCERPT: Let $q_j,r_j,s_j=7$, ${\bf x}_j\in{\mathbb R}^2$, and $t_j\in{\mathbb C}$ for $j=1,2,\ldots,N$, with $N\in{\mathbb N}$. Let ... be the renormalized fermion operators, where K is th...

conjecture | open | MathDB 页面 | 原文线索

V2-058593 | MATHDB-320642

Independent smoothing conjecture for one-shot multiparty state merging

摘录 / EXCERPT: One-shot multiparty state-merging conjecture. There exists a multiparty state-merging protocol of error ϵ whenever

conjecture | open | MathDB 页面 | 原文线索

V2-058594 | MATHDB-320648

Bugrij-Lisovyy conjecture for isotropic Ising-model matrix elements

摘录 / EXCERPT: Bugrij-Lisovyy conjecture. The matrix elements of $D^{-\tau}$, BD^{-1}, and $D^{-1}C$ are proportional to, respectively,

conjecture | open | MathDB 页面 | 原文线索

V2-058595 | MATHDB-320649

The exceptional-element slice conjecture for algebraic groups

摘录 / EXCERPT: Let G be the algebraic group under consideration, let $g\in G$ be an exceptional element, and let T_s be a slice indexed by a Weyl group element s. Let $Z_G(g)$ denote the ce...

conjecture | open | MathDB 页面 | 原文线索

V2-058596 | MATHDB-320775

The monomer-dimer entropy maximization conjecture for sequences of finite subsets of \mathbb{Z}^d

摘录 / EXCERPT: Let $\mathcal{S} = \left\{S_i, i \in \mathbb{N}\right\}$ be an increasing sequence of finite sets in $\mathbb{Z}^d$, and let $\lambda_{\mathcal{S}}(p)$ denote its p monomer-dimer...

conjecture | open | MathDB 页面 | 原文线索

V2-058597 | MATHDB-320776

The monomer-dimer entropy bound for sequences of increasing boxes

摘录 / EXCERPT: Let $d \in \mathbb{N}$, let $[\mathbf{m}_i]$ be boxes with $\mathbf{m}_i=(m_{1,i},\dots,m_{d,i})\in\mathbb{N}^d$, and suppose that $\lim_{i\to\infty}m_{k,i}=\infty$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-058598 | MATHDB-320796

The normalization-factor formula for the disordered asymmetric Glauber model

摘录 / EXCERPT: Normalization-factor conjecture. The normalization factor is

conjecture | open | MathDB 页面 | 原文线索

V2-058599 | MATHDB-320802

The structural physical approximation conjecture for optimal entanglement witnesses

摘录 / EXCERPT: Let W be a normalized entanglement witness with ${\rm Tr}\,W=1$, and define ... The structural physical approximation (SPA) of W is an operator $\widetilde{W}(p)$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-058600 | MATHDB-320876

Poynting-vector tilt conjecture for Euler-Heisenberg effective light cones

摘录 / EXCERPT: Poynting-vector tilt conjecture. For any given carrier field, the corresponding effective light cones tilt into the direction of the Poynting vector of the field. Moreover, the hig...

conjecture | open | MathDB 页面 | 原文线索

V2-058601 | MATHDB-320929

Decomposition conjecture for linear metric perturbations

摘录 / EXCERPT: Decomposition conjecture. There exist a tensor field ${\cal H}_{ab}$ and a vector field X^a such that

conjecture | open | MathDB 页面 | 原文线索

V2-058602 | MATHDB-320948

The conformal conjecture for asymptotically flat manifolds with boundary

摘录 / EXCERPT: Let (M^n,g), $n \ge 3$, be a complete, asymptotically flat, smooth n-manifold with nonempty compact boundary $\Sigma=\partial M$ smooth with respect to both the smooth structur...

conjecture | open | MathDB 页面 | 原文线索

V2-058603 | MATHDB-320962

Quantum invariant detection conjecture for the A-ideal

摘录 / EXCERPT: Quantum invariant detection conjecture. Under these hypotheses, $\limsup_{r\rightarrow\infty}\frac{1}{r}\lvert Z_r(M,f)\rvert$ defines a seminorm on $K_{-1}(T)$, and its radical is...

conjecture | open | MathDB 页面 | 原文线索

V2-058604 | MATHDB-321139

Existence of a three-sender quantum simultaneous decoder

摘录 / EXCERPT: Existence of a three-sender quantum simultaneous decoder. There exists a decoding POVM $\{\Lambda_{l,m,k}\}_{l,m,k}$ such that, for every $\epsilon>0$ and sufficiently large n, t...

conjecture | open | MathDB 页面 | 原文线索

V2-058605 | MATHDB-321140

The algebra-object decomposition conjecture for the _1-_3 subfactor

摘录 / EXCERPT: Algebra-object decomposition conjecture. The algebra object satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058606 | MATHDB-321145

Three-sender quantum simultaneous decoding conjecture for multiple access channels

摘录 / EXCERPT: Let $\mathcal{C}_{\textrm{3MAC}}$ denote the capacity region of a ccc-q multiple access channel with three senders, whose inputs x_1,x_2,x_3 produce the quantum state...

conjecture | open | MathDB 页面 | 原文线索

V2-058607 | MATHDB-321166

Kontsevich-Soibelman wall-crossing conjecture for quivers with potential

摘录 / EXCERPT: Let Q be a finite quiver, let W be a polynomial potential on Q, and let $\operatorname{\mathsf{nil}}(\mathcal P(Q,W))$ be the category of finite-dimensional right modules ove...

conjecture | open | MathDB 页面 | 原文线索

V2-058608 | MATHDB-321264

Milnor-invariant criterion for the universal sl_2 invariant of bottom tangles

摘录 / EXCERPT: Milnor-invariant conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-058609 | MATHDB-321340

Uniqueness conjecture for complete static Einstein solutions

摘录 / EXCERPT: Let (Σ,g,N) be a complete solution of the Static Einstein equations with regular but possibly disconnected (non-empty) horizon $\partial\Sigma$. Suppose that the conformal m...

conjecture | open | MathDB 页面 | 原文线索

V2-058610 | MATHDB-321394

The integral-form isomorphism conjecture for quantum cluster algebras and quantum coordinate rings

摘录 / EXCERPT: Let w be an element of the Weyl group, let ${\mathcal C}_w$ be the associated category, and let $\mathcal{A}_{\mathbb{A}}({\mathcal C}_w)$ and $A_{\mathbb{A}}({\mathfrak n}(w))$...

conjecture | open | MathDB 页面 | 原文线索

V2-058611 | MATHDB-321395

The canonical-basis conjecture for quantum cluster monomials

摘录 / EXCERPT: Let w be an element of the Weyl group, let ${\mathcal C}_w$ be the associated category, and let ${\mathcal B}(w)$ be the canonical basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-058612 | MATHDB-321416

Critical inverse-temperature conjecture for macroscopic cycles and loops

摘录 / EXCERPT: Consider the cycle or loop model on the cubic lattice $\Lambda_n=\{1,\dots,n\}^d$ in $\mathbb{Z}^d$. Let $\lambda^{(i)}$ be the length of the ith-longest cycle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058613 | MATHDB-321417

Poisson-Dirichlet law conjecture for normalized cycle lengths

摘录 / EXCERPT: Under the preceding critical-temperature conjecture, let $\xi\in\Delta_1$ be the weak limit of the ordered normalized cycle-length vector. Poisson-Dirichlet law conjecture. The dis...

conjecture | open | MathDB 页面 | 原文线索

V2-058614 | MATHDB-321577

The generic-basis recurrence for regular simples in nonhomogeneous tubes

摘录 / EXCERPT: Generic-basis recurrence. With these notations, one has

conjecture | open | MathDB 页面 | 原文线索

V2-058615 | MATHDB-321588

Minimal classical pure-state model conjecture

摘录 / EXCERPT: Minimal classical pure-state model conjecture. The simplest classical physical system capable of reproducing the statistical phenomenology of a generic physical statistical system,...

conjecture | open | MathDB 页面 | 原文线索

V2-058616 | MATHDB-321590

Disjoint classical purifications conjecture

摘录 / EXCERPT: Disjoint classical purifications conjecture. The sets A and B have neither pure nor mixed states in common.

conjecture | open | MathDB 页面 | 原文线索

V2-058617 | MATHDB-321614

Optimality conjecture for the family of positive maps $\Phi[\alpha]$

摘录 / EXCERPT: Optimality conjecture for $\Phi[\alpha]$. For $\frac{\pi}{3}\leq\alpha\leq\frac{5}{3\pi}$, the positive maps $\Phi[\alpha]$ are optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-058618 | MATHDB-321804

Symplectic-capacity conjecture for Fermi ellipsoids

摘录 / EXCERPT: Let Ψ be a quantum state such that the Fermi equation $g_{\mathrm{F}}(x,p)$ defines a hypersurface in phase space bounding a compact set Ω_{F}. A quantum blob...

conjecture | open | MathDB 页面 | 原文线索

V2-058619 | MATHDB-321837

Rotating-state conjecture for the driven N-clock model

摘录 / EXCERPT: Rotating-state conjecture. For every $N\geq\bar N$ and $\beta>\beta_N^{cr}$, there exists a critical drift $d_{cr}(\beta,N)>0$ such that, if $|d|<d_{cr}(\beta,N)$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-058620 | MATHDB-321838

Continuum of rotating states conjecture for the driven N-clock model

摘录 / EXCERPT: Continuum rotating-states conjecture. For every sufficiently large β, there exists $N=N(\beta)$ such that, for every $N\geq N(\beta)$ and every $d>0$, the driven N-clock mo...

conjecture | open | MathDB 页面 | 原文线索

V2-058621 | MATHDB-321842

Strong Censorship Conjecture for maximal globally hyperbolic vacuum extensions

摘录 / EXCERPT: Strong Censorship Conjecture. The MGHVE of a "generic" initial data set (Σ,g,k) is locally inextendible as a $\mathcal{C}^{1,1}$ Lorentzian manifold.

conjecture | open | MathDB 页面 | 原文线索

V2-058622 | MATHDB-321847

Outermost apparent horizon conjecture for black holes

摘录 / EXCERPT: Outermost apparent horizon conjecture. There exists a nonempty asymptotically connected component ... of i^+, $$

conjecture | open | MathDB 页面 | 原文线索

V2-058623 | MATHDB-321848

Continuous extendibility conjecture at the Cauchy horizon

摘录 / EXCERPT: Continuous extendibility conjecture. The Cauchy horizon … .

conjecture | open | MathDB 页面 | 原文线索

V2-058624 | MATHDB-321849

Cauchy-horizon curvature blow-up conjecture

摘录 / EXCERPT: Cauchy-horizon curvature blow-up conjecture. Among all the data admissible from Theorem 1, there exists a generic subclass such that, if the maximal future development has … .

conjecture | open | MathDB 页面 | 原文线索

V2-058625 | MATHDB-321869

The rho-invariant stationary-phase conjecture for Witten-Reshetikhin-Turaev invariants

摘录 / EXCERPT: Rho-invariant stationary-phase conjecture. In the Atiyah 2-framing, as $k\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-058626 | MATHDB-321879

The three-party Kirchberg conjecture for full free-group C*-algebras

摘录 / EXCERPT: Three-party Kirchberg conjecture. The minimal and maximal tensor products of three copies of $C^*(\mathbb{F}_{\infty})$ should coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-058627 | MATHDB-322132

Batchelor-Yung surface fugacity conjecture for the honeycomb O(n) loop model

摘录 / EXCERPT: For the $O(n)$ loop model on the semi-infinite hexagonal lattice with $n\in[-2,2]$, associate a fugacity $x_{\rm c}(n) = 1/\sqrt{2+\sqrt{2-n}}$ with occupied vertices and an additi…

conjecture | open | MathDB 页面 | 原文线索

V2-058628 | MATHDB-322265

Goswami's conjecture on quantum isometries of manifolds

摘录 / EXCERPT: Let M be a compact connected Riemannian manifold. A genuine quantum isometry means a quantum isometry not arising from an ordinary classical isometry group. Goswami's conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-058629 | MATHDB-322266

Conjectural correspondence between Hadamard families and quantum-group matrix models

摘录 / EXCERPT: Let $\mathbb T$ denote the unit circle, let $\{H^q\mid q\in\mathbb T^r\}$ be an affine family of complex Hadamard matrices, and let $G^q=G(H^q)$ be the corresponding family of quan…

conjecture | open | MathDB 页面 | 原文线索

V2-058630 | MATHDB-322287

Pearce-Rasmussen selection-rule conjecture for loop-model spectra

摘录 / EXCERPT: Let N be the strip width, d the number of defects, and $A^M_{m,n}$ the sets of admissible two-column configurations specified by the preceding selection rules, with choices of…

conjecture | open | MathDB 页面 | 原文线索

V2-058631 | MATHDB-322293

Conjecture on the low-temperature radius of convergence of the Mayer series

摘录 / EXCERPT: Suppose that v satisfies the stated pair-potential and uniformity assumptions. Let $R^\mathsf{May}(\beta)$ denote the radius of convergence of the Mayer series, and let…

conjecture | open | MathDB 页面 | 原文线索

V2-058632 | MATHDB-322294

Rupel's positivity conjecture for counting polynomials of rigid representations

摘录 / EXCERPT: Rupel's positivity conjecture. The polynomial $P_\mathbf{e}^\mathbf{v}(q)$ has nonnegative integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-058633 | MATHDB-322295

Efimov's unimodularity conjecture for counting polynomials of rigid representations

摘录 / EXCERPT: Efimov's unimodularity conjecture. In the hypotheses of the positivity conjecture, the polynomial $P_\mathbf{e}^\mathbf{v}(q)$ is unimodular.

conjecture | open | MathDB 页面 | 原文线索

V2-058634 | MATHDB-322336

Constructing PTSQM pseudometrics from sparse solutions of the Dieudonné equation

摘录 / EXCERPT: Sparse-pseudometric construction proposal. In place of choosing the sufficiently elementary Krein-space pseudometric $\mathcal{P}$ in advance, it may prove more efficient to start…

conjecture | open | MathDB 页面 | 原文线索

V2-058635 | MATHDB-322457

Dynamic instability conjecture for Kerr-AdS spacetimes

摘录 / EXCERPT: Let Kerr-AdS spacetimes be the rotating asymptotically anti-de Sitter black-hole solutions of the Einstein equations, and consider generic perturbations of these spacetimes. Dynami…

conjecture | open | MathDB 页面 | 原文线索

V2-058636 | MATHDB-322458

Quantum-layer bound-state conjecture for integrable Gauss curvature

摘录 / EXCERPT: Quantum-layer bound-state conjecture. There exists $\alpha=\alpha(\varSigma)$ such that, for every $a\in(0,\alpha)$, the discrete spectrum of the quantum layer over $\varSigma$ of…

conjecture | open | MathDB 页面 | 原文线索

V2-058637 | MATHDB-322459

Thin quantum-layer conjecture under bounded geometry and integrable Gauss curvature

摘录 / EXCERPT: Let $\varSigma$ be a surface satisfying the hypotheses on the embedded quantum layer, and let B_∞ denote the supremum of the norm of its second fundamental form. Assume the…

conjecture | open | MathDB 页面 | 原文线索

V2-058638 | MATHDB-322460

Riemannian inequality conjecture for positive total Gauss curvature

摘录 / EXCERPT: Let $\varSigma$ be a complete surface in $\mathbb R^3$ satisfying the stated hypotheses, and let κ be its Gauss curvature. Assume κ satisfies the stated integrability...

conjecture | open | MathDB 页面 | 原文线索

V2-058639 | MATHDB-322464

Global validity conjecture for the quantum Ising quasicrystal spectrum

摘录 / EXCERPT: Let $J_0,J_1>0$, and let $\{\sigma_k\}$ be the sequence of spectra whose Hausdorff limit is studied in the paper. Let $B_\infty(J_0,J_1)$ denote the limiting spectrum when this lim...

conjecture | open | MathDB 页面 | 原文线索

V2-058640 | MATHDB-322542

The conjecture on the validity of relation (6.24') for one-dimensional systems

摘录 / EXCERPT: Let C and $ u$ be the parameters appearing in relation (6.24'), and let L_s denote the function used there. For $n=1$ and $ u<0$, assume ... Relation (6.24') is the inequality es...

conjecture | open | MathDB 页面 | 原文线索

V2-058641 | MATHDB-322550

Analytic-root conjecture for rational Floquet operators

摘录 / EXCERPT: Analytic-root conjecture. Each root of $P(t,z)$ is an analytic function of t with period 1.

conjecture | open | MathDB 页面 | 原文线索

V2-058642 | MATHDB-322590

Donaldson's positive-definite connection conjecture

摘录 / EXCERPT: Let X be a compact 4-manifold admitting a positive definite connection, meaning a connection whose curvature components form an oriented frame for Λ^+. Donaldson's positi...

conjecture | open | MathDB 页面 | 原文线索

V2-058643 | MATHDB-322717

Denef-Moore D4-D6 generating-series conjecture

摘录 / EXCERPT: Denef-Moore conjecture. (i) The invariant $\operatorname{DT}_H(0,mH,-\beta,-n)$ vanishes unless

conjecture | open | MathDB 页面 | 原文线索

V2-058644 | MATHDB-322819

Smolin-Verstraete-Winter asymptotic quantum Birkhoff conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a finite-dimensional Hilbert space, let $\mathrm{T}(\mathcal{H})$ denote the set of quantum channels on $\mathcal{H}$, and let $\mathbb{U}(\mathcal{H})$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-058645 | MATHDB-322826

Ori's conjecture on perturbations of two-ended Kerr data

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be the maximal vacuum Cauchy development of sufficiently small perturbations of asymptotically flat two-ended Kerr data corresponding to parameters $0<|a|<M$....

conjecture | open | MathDB 页面 | 原文线索

V2-058646 | MATHDB-322827

Generic lower-bound conjecture for scalar-field tails on Reissner-Nordström interiors

摘录 / EXCERPT: Consider the data and maximal development covered by the stability theorem for the two-ended Reissner-Nordström interior, and let the additional lower bounds on the scalar field al...

conjecture | open | MathDB 页面 | 原文线索

V2-058647 | MATHDB-322912

Dimension-free trace commutator inequality

摘录 / EXCERPT: Let A and B be positive definite matrices satisfying $\operatorname{Tr}(A+B)=1$. Trace commutator conjecture. There exists a constant c, independent of the dimensions of A...

conjecture | open | MathDB 页面 | 原文线索

V2-058648 | MATHDB-323012

Existence of a measurement realizing finite quantum relative entropy

摘录 / EXCERPT: Let ω_1 and ω_2 be states, let $S(\omega_1\Vert\omega_2)$ denote their quantum relative entropy, and let $\mathrm{M}(\omega_1\precomega_2)$ be the set of measurement schem...

conjecture | open | MathDB 页面 | 原文线索

V2-058649 | MATHDB-323148

Finiteness conjecture for braid group representations from $(d,3,1)$-gYBE objects

摘录 / EXCERPT: A generalized Yang-Baxter equation object (gYBE object) is an object in a unitary ribbon fusion category that produces braid group representations through the associated $(d,3,1)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-058650 | MATHDB-323174

The Plancherel-type decomposition conjecture for positive representations

摘录 / EXCERPT: Plancherel-type decomposition conjecture. The space $L^2(G_q)$ is decomposed into a direct integral of tensor products of the positive representations under the left and right regu...

conjecture | open | MathDB 页面 | 原文线索

V2-058651 | MATHDB-323175

The characterization conjecture for positive representations

摘录 / EXCERPT: Characterization conjecture. The positivity and transcendental relations completely characterize the class of positive representations for $U_{q\tilde{q}}(\mathfrak{g}_\mathbb{R})$...

conjecture | open | MathDB 页面 | 原文线索

V2-058652 | MATHDB-323224

Ontological single pure state conjecture

摘录 / EXCERPT: Ontological single pure state conjecture. A quantized system is in a state corresponding to a two-valued measure on a single definite context; ontologically, there does not exist a...

conjecture | open | MathDB 页面 | 原文线索

V2-058653 | MATHDB-323346

Threshold conjecture for proper bosonic atomic ground states

摘录 / EXCERPT: Threshold conjecture. A proper N-body ground-state wave function $\Psi^{(Z,N)}_{\mathrm{Bose}}$ exists whenever $Z>Z_*(N)$.

conjecture | open | MathDB 页面 | 原文线索

V2-058654 | MATHDB-323347

Monotonic binding conjecture for bosonic electron numbers

摘录 / EXCERPT: Monotonic binding conjecture. A proper N-body ground-state wave function $\Psi^{(Z,N)}_{\mathrm{Bose}}$ exists for every integer N satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-058655 | MATHDB-323348

Monotonicity conjecture for the normalized bosonic binding threshold

摘录 / EXCERPT: Monotonicity conjecture. The map $N\mapsto\Lambda_*(N)$ is monotonically increasing.

conjecture | open | MathDB 页面 | 原文线索

V2-058656 | MATHDB-323477

Positive energy-momentum conjecture with rigidity for asymptotically hyperbolic manifolds

摘录 / EXCERPT: Let (M,g) be a complete, n-dimensional Riemannian asymptotically hyperbolic manifold whose conformal infinity is the $(n-1)$-unit sphere. Let R_g denote the scalar curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-058657 | MATHDB-323587

Duplantier-Sheffield conjecture for scaling limits of random planar maps

摘录 / EXCERPT: Let $\mathbf{m}_n\in\mathbf{M}_n^p$ be sampled according to a statistical physics model in the γ-universality class. Let $\widehat{\mathbf{m}}_n\subset\mathbb{S}^2$ be any n...

conjecture | open | MathDB 页面 | 原文线索

V2-058658 | MATHDB-323603

The observable-function decomposition conjecture for normal operators

摘录 / EXCERPT: Let V be the Hilbert space underlying the quantum system, let $d4c(P(V))$ be the relevant poset of contexts, and let $\hat{A}=\hat{C}+i\hat{B}$ be a normal operator with self-adj...

conjecture | open | MathDB 页面 | 原文线索

V2-058659 | MATHDB-323756

Equivariant subobject-classifier conjecture for the contextuality topos

摘录 / EXCERPT: Let $\mathcal{V}(\mathcal{H})$ be the poset of contexts of the Hilbert space $\mathcal{H}$, let $\mathcal{V}_f(\mathcal{H})$ denote the corresponding context category with the grou...

conjecture | open | MathDB 页面 | 原文线索

V2-058660 | MATHDB-323869

The asymptotic electron-number and kinetic-energy conjecture for the Hartree equation

摘录 / EXCERPT: Let $u(t)$ be the unique solution to the Hartree equation for some $u_0\in H^1(\mathbb{R}^3)$, let $\mathscr{A}_Z$ be the global attractor, and let γ_c be the constant appea...

conjecture | open | MathDB 页面 | 原文线索

V2-058661 | MATHDB-323890

Categorification conjecture for the positive half of a small quantum group

摘录 / EXCERPT: Let cGamma be a Cartan datum of ADE type. Let $(R(\mathrm{cGamma}),\partial)$ be the associated p-DG algebra, with the parameters of the differential ∂ specified in the source'...

conjecture | open | MathDB 页面 | 原文线索

V2-058662 | MATHDB-323961

The degeneracy-ratio conjecture for triangulated three-manifolds

摘录 / EXCERPT: Degeneracy-ratio conjecture. For any $\mathcal{A}_6<x<\mathcal{A}_{4.5}$, the limit

conjecture | open | MathDB 页面 | 原文线索

V2-058663 | MATHDB-324175

Conjectured existence of complexifier-based semi-classical states

摘录 / EXCERPT: Complexifier conjecture. There exists a choice of complexifier giving a sequence $\phi^\hbar=\{\phi_n^\hbar\}$ of semi-classical states satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-058664 | MATHDB-324176

Conjectured continuum Kasparov module from the graph sequence

摘录 / EXCERPT: Continuum Kasparov-module conjecture. The infinite sequence

conjecture | open | MathDB 页面 | 原文线索

V2-058665 | MATHDB-324177

Conjectured semi-finite spectral triple in the continuum limit

摘录 / EXCERPT: Continuum spectral-triple conjecture. The Kasparov module

conjecture | open | MathDB 页面 | 原文线索

V2-058666 | MATHDB-324269

Three-site tomography conjecture for open-boundary matrix product states

摘录 / EXCERPT: Let Ψ be a translation-invariant N-qubit matrix product state with open boundary conditions and bond and physical dimensions $D=d=2$. A reduced density operator on a chain o...

conjecture | open | MathDB 页面 | 原文线索

V2-058667 | MATHDB-324324

AJ-type conjecture for colored HOMFLY-PT recurrences and the augmentation polynomial

摘录 / EXCERPT: Let $\{P_{K;n}(a,q)\}$ be the colored HOMFLY-PT polynomials of K, colored by the n-th symmetric power of the fundamental representation. Define operators by ... Suppose these pol...

conjecture | open | MathDB 页面 | 原文线索

V2-058668 | MATHDB-324332

Equality of the charge(3) and even model capacities

摘录 / EXCERPT: Let the charge(3) model consist of configurations of $\oplus,\ominus$ spins on $\mathbb{Z}^2$ such that, for every finite horizontal or vertical line segment, the sum of the spins...

conjecture | open | MathDB 页面 | 原文线索

V2-058669 | MATHDB-324335

Conjecture on partial symmetries in dimension three

摘录 / EXCERPT: Let ${\mathfrak D}$ denote the set of block-positive symmetries under consideration, and let $s\in{\mathfrak D}$ be a partial symmetry for $n=3$. Partial-symmetry conjecture. The r...

conjecture | open | MathDB 页面 | 原文线索

V2-058670 | MATHDB-324351

The primitive-ideal and symplectic-leaf conjecture for compatible cluster pairs

摘录 / EXCERPT: Compatible-pair conjecture. There exists a homeomorphism between the space of primitive ideals of $\mathfrak{A}_q$ and the symplectic leaves on X defined by Λ.

conjecture | open | MathDB 页面 | 原文线索

V2-058671 | MATHDB-324353

The supertoric cluster stratification conjecture

摘录 / EXCERPT: Supertoric cluster stratification conjecture. There exists a supertoric cluster ${\bf x}$ with COS of depth r such that, up to a suitable permutation of indices, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-058672 | MATHDB-324355

The cluster-stratification Dixmier map conjecture

摘录 / EXCERPT: Let $\mathfrak{A}_q$ be the cluster algebra structure on $R_q[G]$, let ${\bf x}$ be any cluster, and let a COS stratification be a stratification of the type used in the paper. Clu...

conjecture | open | MathDB 页面 | 原文线索

V2-058673 | MATHDB-324445

Parabolic compatibility of quantum Langlands for the Siegel parabolic

摘录 / EXCERPT: Parabolic compatibility conjecture. There is an automorphism $\delta:\operatorname{Bun}_n\widetilde\to\operatorname{Bun}_n$ such that the diagram

conjecture | open | MathDB 页面 | 原文线索

V2-058674 | MATHDB-324447

Properties of the quantum Langlands image of automorphic sheaves

摘录 / EXCERPT: Whittaker and theta-lifting conjecture. (i) $\mathcal K_E$ is a perverse sheaf, each of whose $\mathbb Z/2\mathbb Z$-parity pieces is irreducible, and

conjecture | open | MathDB 页面 | 原文线索

V2-058675 | MATHDB-324458

Quantized generalized Causality for locally covariant quantum field theories

摘录 / EXCERPT: Let ${*}\text{-}\mathfrak{Alg}$ be the category of non-commutative, associative, $*$-algebras, and let $\mathfrak{Bkgr}_c$ and $\mathfrak{SpBkgr}_c$ be the categories of spacetime...

conjecture | open | MathDB 页面 | 原文线索

V2-058676 | MATHDB-324553

Centrality preservation by braided Laplace operators

摘录 / EXCERPT: Let ${\cal N}$ be the algebra equipped with the braided differential operators defined by the matrix D, and let $Z({\cal N})$ denote its center. For each positive integer k, de...

conjecture | open | MathDB 页面 | 原文线索

V2-058677 | MATHDB-324675

Equality of the quantum Galois deformation group in the first case

摘录 / EXCERPT: Equality conjecture. If q is not a root of unity, the inclusion $\mathcal{NCF}_{L/k}(A)\hookrightarrow\widehat{G}_{II}(A)$ is an equality.

conjecture | open | MathDB 页面 | 原文线索

V2-058678 | MATHDB-324676

Equality for commutative deformations of the logarithmic q-difference extension

摘录 / EXCERPT: Equality conjecture. If q is not a root of unity, the injection $\mathcal{F}_{L/k}(A)\rightarrow\widehat{G}_{III}$ is an equality.

conjecture | open | MathDB 页面 | 原文线索

V2-058679 | MATHDB-324677

Quantum Galois group as the third quantum formal group

摘录 / EXCERPT: Quantum Galois group conjecture. If q is not a root of unity, the injection in Proposition 6.8o is an equality, so

conjecture | open | MathDB 页面 | 原文线索

V2-058680 | MATHDB-324692

Classification of one-dimensional gapped phases by boundary-state dimensions

摘录 / EXCERPT: Let Φ_0 and Φ_1 be two translation-invariant finite-range interactions. For $i=0,1$, suppose there are integers l_i and r_i such that ... Here ${\mathcal S}_i^\Gamma$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-058681 | MATHDB-324704

Undecidability conjecture for ALLOWS_QUANTUM

摘录 / EXCERPT: Let H be a contextuality scenario and let $C^*(H)$ denote its universal unital C^*-algebra. The decision problem $\mathtt{ALLOWS_QUANTUM}$ asks whether a given contextuality s...

conjecture | open | MathDB 页面 | 原文线索

V2-058682 | MATHDB-324711

The depth-two integrality conjecture for finite-dimensional elementary Nichols algebras

摘录 / EXCERPT: The depth-two integrality conjecture. All finite-dimensional elementary Nichols algebras are t-integral of depth two. In particular, every such Nichols algebra is bg-equivalent t...

conjecture | open | MathDB 页面 | 原文线索

V2-058683 | MATHDB-324815

Runde's operator amenability conjecture for locally compact quantum groups

摘录 / EXCERPT: Runde's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058684 | MATHDB-324875

Geiss-Leclerc-Schröer conjecture on quantum cluster monomials

摘录 / EXCERPT: Let $\mathbf U_q^-(w)$ be the quantum unipotent subgroup associated with w, let $\mathscr B^{\operatorname{up}}(w)$ denote its upper global, or dual canonical, basis, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058685 | MATHDB-324878

Nakajima's singular-support uniqueness conjecture for quantum cluster monomials

摘录 / EXCERPT: Let T be a reachable $\mathcal C_w$-maximal rigid module and let $R\in\operatorname{add}(T)$. Let Λ_R be the closure of the orbit through R, and let b_R be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-058686 | MATHDB-324965

The pivot time-scale conjecture for observables depending on an internal distance

摘录 / EXCERPT: Pivot time-scale conjecture. The integrated autocorrelation time of this observable is of the same order as the time required to make successful pivots at all length scales with re...

conjecture | open | MathDB 页面 | 原文线索

V2-058687 | MATHDB-325022

General Small Incremental Mixing conjecture

摘录 / EXCERPT: Small Incremental Mixing conjecture. For every such ensemble,

conjecture | open | MathDB 页面 | 原文线索

V2-058688 | MATHDB-325035

The fractional Fourier transform refinement of Wright's conjecture

摘录 / EXCERPT: Let F denote the $M\times M$ discrete fractional Fourier transform defined in Candan, Kutay, and Ozaktas (2000), and let I be the $M\times M$ identity matrix. The columns of a...

conjecture | refuted | MathDB 页面 | 原文线索

V2-058689 | MATHDB-325132

Finite-order generalized R-matrix simulation conjecture

摘录 / EXCERPT: Let $R_X:\mathbb C^m\otimes\mathbb C^m\longrightarrow\mathbb C^m\otimes\mathbb C^m$ be a unitary (generalized) R-matrix of finite order, used to define braid group representation...

conjecture | open | MathDB 页面 | 原文线索

V2-058690 | MATHDB-325143

Penrose's Gibbons-Penrose inequality for surfaces in Minkowski spacetime

摘录 / EXCERPT: Let Σ be a two-dimensional spacelike closed embedded orientable surface in Minkowski spacetime, diffeomorphic to S^2. Let $T_0\in\mathbb{R}^{3,1}$ be a fixed future timeli...

conjecture | open | MathDB 页面 | 原文线索

V2-058691 | MATHDB-325144

Gibbons-Penrose inequality for surfaces in Schwarzschild spacetime

摘录 / EXCERPT: Let Σ be a spacelike 2-surface in Schwarzschild spacetime, and let $\vec{J}$ denote its dual mean curvature vector. Suppose that the past null hypersurface generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-058692 | MATHDB-325216

Klainerman-Machedon space-time bound for the quantum BBGKY hierarchy

摘录 / EXCERPT: Let the interaction parameter satisfy $\beta\in(0,1]$, and let $\{\Gamma_N\}_{N=1}^{\infty}$ be a sequence of quantum BBGKY hierarchy solutions satisfying the energy condition ... fo...

conjecture | open | MathDB 页面 | 原文线索

V2-058693 | MATHDB-325249

Divisibility conjecture for semisimple Hopf actions on AS regular algebras

摘录 / EXCERPT: Divisibility conjecture. The dimension $\dim H$ is divisible by some integer d' satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-058694 | MATHDB-325370

Conjecture on optimal input and measurement angles for Pauli channel tomography

摘录 / EXCERPT: Let $\underline{\lambda}$ denote the channel contraction parameters and let N be the number of channel uses. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058695 | MATHDB-325398

The classification conjecture for quasi-constant c_2-invariants

摘录 / EXCERPT: Let Γ be a completed primitive graph, let $c_2[\Gamma]_q$ denote its c_2-invariant, and call Γ quasi-constant when this invariant is one of the quasi-constant funct...

conjecture | open | MathDB 页面 | 原文线索

V2-058696 | MATHDB-325418

The classification conjecture for irreducible subfactor planar algebras of index between 5 and $3+\sqrt{5}$

摘录 / EXCERPT: Let a subfactor planar algebra mean the planar algebra associated with a finite-index subfactor, and let A_∞ denote the infinite-depth principal graph. Consider irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-058697 | MATHDB-325463

Distinct Steenrod extensions of categorified quantum groups

摘录 / EXCERPT: Distinct-extension conjecture. Any such extension decategorifies an extension of the ground field as above, and if $\mathcal{A}$ and $\mathcal{A}'$ are distinct sub-Hopf algebras,...

conjecture | open | MathDB 页面 | 原文线索

V2-058698 | MATHDB-325579

Linear Toffoli-gate bound for simple periodic-function synthesis

摘录 / EXCERPT: Toffoli-gate bound conjecture. To synthesize the circuit for S_p, one needs at most n Toffoli gates:

conjecture | open | MathDB 页面 | 原文线索

V2-058699 | MATHDB-325767

Area-angular momentum inequality for multiple black holes

摘录 / EXCERPT: Let (M,g,k) be a simply connected, axially symmetric, maximal initial data set with multiple ends, one (M_{end}^{1}) asymptotically flat and the others...

conjecture | open | MathDB 页面 | 原文线索

V2-058700 | MATHDB-325782

The conjecture on weak null singularities from polynomially decaying scattering data

摘录 / EXCERPT: Let smooth vacuum scattering data be as in the paper's main theorem, but suppose they settle down to a Kerr solution on $\mathcal{H}^+$ and $\mathcal{I}^+$ at a suitably fast inver...

conjecture | open | MathDB 页面 | 原文线索

V2-058701 | MATHDB-325803

The cosmic no-hair conjecture for generic Einstein developments

摘录 / EXCERPT: Cosmic no-hair conjecture. Generic elements of $\mathcal{A}$ yield maximal globally hyperbolic developments that are future asymptotically de Sitter like.

conjecture | open | MathDB 页面 | 原文线索

V2-058702 | MATHDB-325854

Equality of smooth and almost periodic elements for locally compact quantum groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a locally compact quantum group, and let $\mathcal{P}^\infty(\mathbb{G})$ denote its smooth elements and $\mathcal{AP}(\mathbb{G})$ its almost periodic elements...

conjecture | open | MathDB 页面 | 原文线索

V2-058703 | MATHDB-325855

Admissibility of finite-dimensional unitary corepresentations

摘录 / EXCERPT: Let $\mathbb{G}$ be a locally compact quantum group, and let $C_0(\mathbb{G})$ be its reduced C^*-algebra. A finite-dimensional unitary corepresentation of $C_0(\mathbb{G})$ is c...

conjecture | open | MathDB 页面 | 原文线索

V2-058704 | MATHDB-325904

General form conjecture for semiclassical moment generating functions

摘录 / EXCERPT: Let g be a positive integer, let $\beta=1$ in the orthogonal case and $\beta=2$ in the unitary case, and let ξ and s be the variables used in the generating functions...

conjecture | open | MathDB 页面 | 原文线索

V2-058705 | MATHDB-325906

Quantum positivity conjecture for quantum cluster algebras

摘录 / EXCERPT: Quantum positivity conjecture. In the Laurent phenomenon theorem, the coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-058706 | MATHDB-325959

Existence conjecture for the Dirac-Jang system

摘录 / EXCERPT: Let (M,g,k,E) be a strongly asymptotically flat initial data set for the Einstein-Maxwell equations. The Dirac-Jang existence conjecture. The Dirac-Jang system has a solution...

conjecture | open | MathDB 页面 | 原文线索

V2-058707 | MATHDB-325960

Charged positive mass theorem with rigidity

摘录 / EXCERPT: Let (M,g,k,E) be a strongly asymptotically flat initial data set for the Einstein-Maxwell equations satisfying the charged dominant energy condition, with total mass m and tota...

conjecture | open | MathDB 页面 | 原文线索

V2-058708 | MATHDB-326015

Distinct odd-bubble basis conjecture for odd categorification of quantum $\mathfrak{sl}(2)$

摘录 / EXCERPT: Let $\epsilon,\epsilon'\in\mathrm{Seq}$, let λ be a weight, and let $B_{\epsilon,\epsilon',\lambda}^{\pi}$ be the diagram set with fixed Π-data. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058709 | MATHDB-326016

Identity 2-endomorphism space conjecture for odd categorification of quantum $\mathfrak{sl}(2)$

摘录 / EXCERPT: Let Ξ be the graded $\mathbb{Z}$-module appearing in the map ... defined by the paper's construction. Here $\mathbb{1}_{\lambda}$ is the identity 1-morphism at weight λ...

conjecture | open | MathDB 页面 | 原文线索

V2-058710 | MATHDB-326020

Ahlswede's randomness-assisted capacity conjecture for arbitrarily varying quantum channels

摘录 / EXCERPT: Let $(\mathcal{N}_t)_{t\in\theta}$ be an arbitrarily varying quantum channel. Its entanglement generating capacity is denoted by $A((\mathcal{N}_t)_{t\in\theta})$, and the corresponding capacity u...

conjecture | open | MathDB 页面 | 原文线索

V2-058711 | MATHDB-326051

The simple-module assumption for affine Kac-Moody algebras of simply laced type

摘录 / EXCERPT: The simple-module assumption. For any affine Kac-Moody algebra of type $A^{(1)}_n$ with $n\geqslant 1$, $D^{(1)}_n$ with $n\geqslant 4$, $E^{(1)}_6$, $E^{(1)}_7$, or $E^{(1)}_8$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-058712 | MATHDB-326059

Bryan-Gholampour quantum McKay correspondence for surface singularities

摘录 / EXCERPT: Let G be a finite subgroup of $SU(2)$, let $\widehat{\mathbb{C}^{2}/G}$ be the crepant resolution of $\mathbb{C}^{2}/G$, and let $[\mathbb{C}^{2}/G]$ be the corresponding orbifol...

conjecture | open | MathDB 页面 | 原文线索

V2-058713 | MATHDB-326062

Conjecture on a continuum of massless Gibbs measures for high-dimensional clock models

摘录 / EXCERPT: Consider the nearest-neighbor q-state clock model in spatial dimension $d\geq 3$, at inverse temperature below the critical-temperature threshold and with q sufficiently large....

conjecture | open | MathDB 页面 | 原文线索

V2-058714 | MATHDB-326063

Conjecture on the temperature location of the intermediate clock-model regime

摘录 / EXCERPT: Let β be the inverse temperature, let β_c denote the critical inverse-temperature threshold, and let q be the number of clock states. The notation $O(q^2)$ denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-058715 | MATHDB-326105

Berenstein-Zelevinsky's quantum twist automorphism conjecture

摘录 / EXCERPT: Berenstein-Zelevinsky's twist conjecture. Under the hypotheses of the commutation lemma, the restriction of $\hat\eta_\mathbf{i}$ to $\mathcal U_\mathbf{i}$ is an isomorphism of al...

conjecture | open | MathDB 页面 | 原文线索

V2-058716 | MATHDB-326119

The modular-operad isomorphism conjecture for the quantum open operad

摘录 / EXCERPT: Modular-operad isomorphism conjecture. There is a modular operad isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-058717 | MATHDB-326200

Banica-Vergnioux freeness conjecture for free unitary easy quantum groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a free unitary easy quantum group. A free fusion semiring is obtained from a set S with an involution and a binary law by taking the free monoid $R(S)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-058718 | MATHDB-326201

Banica-Vergnioux refined freeness conjecture for easy quantum groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a free unitary easy quantum group without any non-trivial one-dimensional representation. Let $\mathcal{C}^{\circ,\bullet}$ be its category of noncrossing color...

conjecture | open | MathDB 页面 | 原文线索

V2-058719 | MATHDB-326202

Simplicity and unique trace conjecture for quantum groups satisfying freeness

摘录 / EXCERPT: Let $\mathbb{G}$ be a compact quantum group, and suppose that its fusion semiring satisfies the freeness conjecture. Simplicity and unique trace conjecture. The reduced C^*-algeb...

conjecture | open | MathDB 页面 | 原文线索

V2-058720 | MATHDB-326265

Collective-excitation conjecture for many-particle dynamics

摘录 / EXCERPT: Consider a many-particle system placed in a heat bath and represent its equilibrium state as a Hilbert-space vector. The authors assume that the many-particle system and its dynami...

conjecture | open | MathDB 页面 | 原文线索

V2-058721 | MATHDB-326397

The fibre-bundle conjecture for fault-tolerant logical gates

摘录 / EXCERPT: Fibre-bundle conjecture. Fault-tolerant logical gates can always be expressed as arising from monodromies of an appropriate fibre bundle with a flat projective connection.

conjecture | open | MathDB 页面 | 原文线索

V2-058722 | MATHDB-326677

Lagrangianity conjecture for non-Abelian Yang-Mills boundary data

摘录 / EXCERPT: Lagrangianity conjecture. The submanifold L_M is Lagrangian for non-Abelian Yang-Mills theory.

conjecture | open | MathDB 页面 | 原文线索

V2-058723 | MATHDB-326714

Weak null singularity conjecture for perturbed Kerr black holes

摘录 / EXCERPT: Let $(\mathcal M,g)$ be the development of smooth characteristic initial data for the vacuum Einstein equations $Ric_{\mu\nu}=0$ on a pair of null hypersurfaces H_0 and...

conjecture | open | MathDB 页面 | 原文线索

V2-058724 | MATHDB-326733

Gallavotti's equivalence conjecture for reversible and irreversible thermostatted ensembles

摘录 / EXCERPT: Let N particles occupy a volume V, with density $d=N/V$. Let $\mu_{\d,T}$ be the SRB distribution of the reversible Gaussian-thermostatted model at kinetic-energy parameter T,...

conjecture | open | MathDB 页面 | 原文线索

V2-058725 | MATHDB-326832

The positive definite connection sphere conjecture

摘录 / EXCERPT: Let A be a positive definite connection over a compact 4-manifold, and let the associated symplectic Fano be the symplectic manifold obtained from A. Positive definite connecti...

conjecture | open | MathDB 页面 | 原文线索

V2-058726 | MATHDB-326834

The positive definite connection deformation conjecture

摘录 / EXCERPT: Let A be a positive definite connection over a compact 4-manifold, and let S be the volume functional on such connections. Positive definite connection deformation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-058727 | MATHDB-326840

Higher-rank tensor product decomposition conjecture for positive representations

摘录 / EXCERPT: Let $\mathcal{U}_{q\widetilde{q}}(\mathfrak{g}_\mathbb{R})$ be the modular-double quantum group and let $\mathcal{P}_{\alpha}$ and $\mathcal{P}_{\beta}$ be positive representations...

conjecture | open | MathDB 页面 | 原文线索

V2-058728 | MATHDB-326898

AdS instability conjecture for black-hole formation

摘录 / EXCERPT: Let AdS_{d+1} denote the Anti-de Sitter space in dimension $d+1$, with $d\geq 3$, and consider the spherically symmetric Einstein-scalar system with negative cosmo...

conjecture | open | MathDB 页面 | 原文线索

V2-058729 | MATHDB-326962

Simultaneous min-entropy smoothing conjecture

摘录 / EXCERPT: Let $m\in\mathbb{N}$, let $A=A_1\cdots A_m$ be an m-party system, and let $\rho\in\mathcal{S}_{\leq}(A)$. For a nonempty subsystem $S\subset\{A_1,\dotsc,A_m\}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-058730 | MATHDB-327043

The quantum loop algebra conjecture for observables on the cylinder

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra used to define the gauge theory, let $\operatorname{Obs}^q({\mathbb{C}}^\times \times {\mathbb{R}}^2)$ be the resulting quantized E_2-algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-058731 | MATHDB-327077

The KLR character conjecture for rigid quiver representations

摘录 / EXCERPT: Let $(Q,\mathbf{d})$ be a valued quiver, let Ω be the quantum shuffle character, and let $[V]^*$ denote the dual Hall--Ringel algebra element associated with a representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-058732 | MATHDB-327078

The quantum cluster structure conjecture for quantum unipotent cells

摘录 / EXCERPT: Let W be the Weyl group of a symmetrizable Cartan datum, let $w\in W$, and let $\mathcal{A}_v[N^w]$ denote the corresponding quantum unipotent cell. Quantum cluster structure con...

conjecture | open | MathDB 页面 | 原文线索

V2-058733 | MATHDB-327098

The weak-integrality conjecture for braid group representations

摘录 / EXCERPT: Weak-integrality conjecture. These braid group representations have finite image for all $n>0$ if and only if $\operatorname{FPdim}(X)^2\in\mathbb Z$.

conjecture | open | MathDB 页面 | 原文线索

V2-058734 | MATHDB-327126

Direct-observation conjecture for conditional mixing

摘录 / EXCERPT: Let $(X_v,Y_v)_{v\in\mathbb{Z}^2}$ be a hidden Markov field with $E=F=\{-1,1\}$. Suppose ... where $(\xi_v)_{v\in\mathbb{Z}^2}$ are independent and identically distributed, independe...

conjecture | open | MathDB 页面 | 原文线索

V2-058735 | MATHDB-327283

Large-data local BV scattering conjecture for the Einstein-scalar field system

摘录 / EXCERPT: Large-data local BV scattering conjecture. There exists initial data of arbitrarily large BV norm whose maximal global development scatters locally in the BV norm.

conjecture | open | MathDB 页面 | 原文线索

V2-058736 | MATHDB-327284

Non-spherical stability conjecture for locally BV-scattering Einstein-scalar field solutions

摘录 / EXCERPT: Non-spherical stability conjecture. Spherically symmetric locally BV-scattering dispersive solutions to the Einstein-scalar field equations are stable against non-spherically symme...

conjecture | open | MathDB 页面 | 原文线索

V2-058737 | MATHDB-327298

Congruent skein relations for colored Jones polynomials of links

摘录 / EXCERPT: Congruent skein relations. For any such link,

conjecture | open | MathDB 页面 | 原文线索

V2-058738 | MATHDB-327398

Higher-order gauge-transformation conjecture for metric perturbations

摘录 / EXCERPT: Higher-order gauge-transformation conjecture. There exists a vector field $\hat\sigma_{(n)}^a$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058739 | MATHDB-327435

Volume Conjecture for hyperideal polyhedra

摘录 / EXCERPT: Let $\mathcal{P}$ be a hyperideal polyhedron with one-skeleton $\Gamma=\mathcal{P}^{(1)}$, edge set E, and exterior dihedral angles $\gamma:E\to(0,\pi)$. Define ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058740 | MATHDB-327530

Monotonicity conjecture for Rényi generalizations of conditional quantum mutual information

摘录 / EXCERPT: Let $\rho_{ABC}\in\mathcal{S}(\mathcal{H}_{ABC})_{++}$, $\tau_{AC}\in\mathcal{S}(\mathcal{H}_{AC})_{++}$, $\theta_{BC}\in\mathcal{S}(\mathcal{H}_{BC})_{++}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-058741 | MATHDB-327531

Asymptotically hyperbolic Penrose inequality

摘录 / EXCERPT: Suppose that (M,g) is a compact perturbation of Schwarzschild-anti-de Sitter of mass $\boldsymbol{m}\text{?}$ and has scalar curvature $R_g\geq -6$. If $m_{\partial M}\geq 0$ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-058742 | MATHDB-327603

Global equivalence conjecture for cloning and state estimation

摘录 / EXCERPT: Let M denote the number of output clones, and let $F_{\mathrm{clon}}[N\to M]$ and $F_{\mathrm{est}}[N\to M]$ be, respectively, the global fidelities of quantum cloning and measur...

conjecture | open | MathDB 页面 | 原文线索

V2-058743 | MATHDB-327732

Toda mirror conjecture for symplectic induction

摘录 / EXCERPT: Let $L\subset G$ be a Levi subgroup, let $G^\vee$ and $L^\vee$ be the corresponding dual groups, and let $T(G^\vee)$ and $T(L^\vee)$ denote their Toda spaces. Via the Toda isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-058744 | MATHDB-327859

Uniqueness of pointed elements in strongly positive bases

摘录 / EXCERPT: Let $\mathcal{A}_v(b,c)$ be a rank 2 quantum cluster algebra, and let a strongly positive basis $\mathcal{B}$ mean a basis whose elements are bar-invariant, independent of the init...

conjecture | open | MathDB 页面 | 原文线索

V2-058745 | MATHDB-327860

Upper strongly positive basis conjecture for rank 2 quantum cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}_v(b,c)$ be a rank 2 quantum cluster algebra, and let $Y\le Z$ mean that $Z-Y$ is a Laurent polynomial with coefficients in $\mathbb{Z}_{\ge0}[v^{\pm1}]$. For a str...

conjecture | open | MathDB 页面 | 原文线索

V2-058746 | MATHDB-327861

Lower strongly positive basis conjecture for divisible rank 2 quantum cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}_v(b,c)$ be a rank 2 quantum cluster algebra, and suppose that $b\mid c$ or $c\mid b$. Let $Y\le Z$ mean that $Z-Y$ is a Laurent polynomial with coefficients in...

conjecture | open | MathDB 页面 | 原文线索

V2-058747 | MATHDB-327862

Positivity of expansions in the quantum greedy basis

摘录 / EXCERPT: Let $\mathcal{A}_v(b,c)$ be a rank 2 quantum cluster algebra, let $C[a_1,a_2]$ be a triangular basis element, and write its expansion in the quantum greedy basis as ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-058748 | MATHDB-328003

Existence conjecture for quantum Schrödinger systems

摘录 / EXCERPT: Let ${\mathcal E}_{0:T}^\dagger$ be a positivity-improving Kraus map, meaning a Kraus map satisfying strict positivity, and let ρ_0 and ρ_T be density matrices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058749 | MATHDB-328176

Logarithmic-derivative conjecture for torus and cylinder generating functions

摘录 / EXCERPT: For an $L_v\times L_h$ lattice at $z=-1$, let $G^{CC}_{L_h}$ and $G^{CF}_{L_h}$ be the generating functions along the periodic direction for the torus and cylinder, respectively, a...

conjecture | open | MathDB 页面 | 原文线索

V2-058750 | MATHDB-328239

Stieltjes-property conjecture for the diagonal two-point function

摘录 / EXCERPT: Stieltjes-property conjecture. The function $a\mapsto G_{aa}$ is a Stieltjes function for

conjecture | open | MathDB 页面 | 原文线索

V2-058751 | MATHDB-328250

Conjecture on decomposing admissible pair potentials

摘录 / EXCERPT: Ruelle-potential decomposition conjecture. Any stable and tempered pair potential can always be written as a sum of a non-negative potential plus an absolutely summable stable pote...

conjecture | open | MathDB 页面 | 原文线索

V2-058752 | MATHDB-328271

Support-coupling characterization of (D)-isometric quantum actions

摘录 / EXCERPT: Support-coupling conjecture. A faithful quantum action of A on (X,d) is (D)-isometric if, for every $x,y\in X$ and $\psi\in S(A)$, there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-058753 | MATHDB-328272

Linear-span conjecture for Lipschitz-isometric states

摘录 / EXCERPT: Linear-span conjecture. The states contained in this weak*-closed linear span coincide with $S_p(A)$.

conjecture | open | MathDB 页面 | 原文线索

V2-058754 | MATHDB-328669

Quantum Drinfeld-Sokolov reduction for class S punctures

摘录 / EXCERPT: qDS reduction conjecture. The chiral algebra associated to the class $\mathcal{S}$ theory with a puncture of type Λ is obtained by performing qDS reduction with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-058755 | MATHDB-328681

He-Vidal monogamy conjecture for the negativity of entanglement

摘录 / EXCERPT: Let $\lvert\psi\rangle$ be a normalised complex vector in the tensor product Hilbert space ... For a subsystem, Tr_B and Tr_C denote the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-058756 | MATHDB-328726

Integral trace conjecture for categorified quantum \(\mathfrak{sl}_n\)

摘录 / EXCERPT: Integral trace conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-058757 | MATHDB-328736

Global existence and convergence conjecture for Yang-Mills gradient flow

摘录 / EXCERPT: Global existence and convergence conjecture. There is a smooth solution $A(t)$ for $t\in[0,\infty)$ to this flow for the Yang-Mills energy functional with respect to the L^2 Riem...

conjecture | open | MathDB 页面 | 原文线索

V2-058758 | MATHDB-328776

The surjectivity conjecture for D-coloured graph representations

摘录 / EXCERPT: Let D be odd or even, and let $\widetilde{\mathfrak{C}}_{D,coloured}$ be the set of labelled loopless D-regular boundary graphs whose edges can be coloured with D colours so...

conjecture | open | MathDB 页面 | 原文线索

V2-058759 | MATHDB-328796

The non-distillation conjecture for noisy Popescu-Rohrlich boxes

摘录 / EXCERPT: For $1/2<\eta_1<\eta_2<1$, let $\operatorname{PR}_{\eta_1}$ and $\operatorname{PR}_{\eta_2}$ denote the corresponding noisy Popescu-Rohrlich boxes. Non-distillation conjecture. Two...

conjecture | open | MathDB 页面 | 原文线索

V2-058760 | MATHDB-328990

The generalized cluster algebra conjecture for the Grothendieck ring of \(\mathcal{C}_{\varepsilon^\mathbb{Z}}\)

摘录 / EXCERPT: Let $l\geq 2$, and let $\mathcal{C}_{\varepsilon^\mathbb{Z}}$ be the category whose Grothendieck ring is under consideration. Write $\mathcal G_l$ for a generalized cluster algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-058761 | MATHDB-329050

The charged Riemannian Penrose inequality for Einstein-Maxwell-Dilaton black holes

摘录 / EXCERPT: EMD charged Riemannian Penrose inequality. If the charge densities $\operatorname{div}(e^{-2\phi}E)$ and $\operatorname{div}B$ are compactly supported, then

conjecture | open | MathDB 页面 | 原文线索

V2-058762 | MATHDB-329062

Conjecture that the quantum measure determines the harmonic extension off the set

摘录 / EXCERPT: Quantum-measure determination conjecture. In the setting described above, the measure $cmu_{X,h}$ almost surely determines the harmonic extension of h off X.

conjecture | open | MathDB 页面 | 原文线索

V2-058763 | MATHDB-329100

Crystal-basis conjecture for the positive quantum affinization

摘录 / EXCERPT: Let $L{A}$ be the subring of $\mathbb{C}(v)$ consisting of rational functions without a pole at $v=0$. Let $L{L}$ be the $L{A}$-submodule of $U^+_v(Lg{g})$ generated by the prescri...

conjecture | open | MathDB 页面 | 原文线索

V2-058764 | MATHDB-329122

Liouville quantum gravity duality analytic-continuation conjecture

摘录 / EXCERPT: Let $=2/\gamma+\gamma/2$, let ... , and let $_{\gamma,\mu}^{(z_i\alpha_i)_i}(\hat g,1)$ denote the LQG partition function. Analytic-continuation conjecture. The function ... is analyti...

conjecture | open | MathDB 页面 | 原文线索

V2-058765 | MATHDB-329194

Double canonical basis invariance under the star involution

摘录 / EXCERPT: Let $\mathbf B_{\mathfrak{n}_\pm}$ be the canonical bases of the positive and negative parts, let $\mathbf K$ be the group of Cartan elements, and let $x\mapsto x^*$ be the star an...

conjecture | open | MathDB 页面 | 原文线索

V2-058766 | MATHDB-329299

Duncan--Frenkel twisted chiral gravity conjecture

摘录 / EXCERPT: Let Γ_g be the group associated to $g\in\mathbb{M}$, and let $T^{(-1)}_{\Gamma_g}$ denote the normalized Rademacher sum attached to Γ_g. Duncan--Frenkel conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-058767 | MATHDB-329550

Leclerc's first conjecture on products of upper global basis elements

摘录 / EXCERPT: Leclerc's first conjecture. The expansion of b_1b_2 with respect to ${\mathbf{B}}^\upper$ has the form

conjecture | open | MathDB 页面 | 原文线索

V2-058768 | MATHDB-329553

Spectral instability conjecture for stationary states on the tadpole graph

摘录 / EXCERPT: Let $(u_{n,\omega}^{\pm},0)$ be the stationary standing-wave branch indexed by $n\in\mathbb{N}$, and let the spectral stability problem be the linearized eigenvalue problem for the...

conjecture | open | MathDB 页面 | 原文线索

V2-058769 | MATHDB-329578

Global existence conjecture for vacuum Einstein equations with a spacelike translational Killing field

摘录 / EXCERPT: Let (Σ,q,K) be asymptotically flat Cauchy data for the $3+1$-dimensional vacuum Einstein equations admitting a spacelike translational Killing field, and let its maximal Cau...

conjecture | open | MathDB 页面 | 原文线索

V2-058770 | MATHDB-329587

Symmetric-skew mirror symmetry conjecture for colored HOMFLY-PT polynomials

摘录 / EXCERPT: Let B be a colored, oriented braid, and let $P(B)$ be the invariant of braid conjugacy classes with values in ... where $\mathcal{I}$ is the ideal of relations between colored circ...

conjecture | open | MathDB 页面 | 原文线索

V2-058771 | MATHDB-329672

Quantum unipotent-subgroup canonical-basis conjecture

摘录 / EXCERPT: Let $(C,\mathbf i)$ be of type (i), let $\mathcal A(\mathbf i)$ be the associated quantum cluster algebra, and let $\{\mathbf S(w)\}$ be the bar-invariant basis obtained from the d...

conjecture | open | MathDB 页面 | 原文线索

V2-058772 | MATHDB-329673

Integral-form isomorphism conjecture for quantum unipotent subgroups

摘录 / EXCERPT: Assume $(C,\mathbf i)$ is of type (i), and let κ be the algebra isomorphism from the associated quantum cluster algebra $\mathcal A^\dagger(\mathbf i)$ to the quantum unipot...

conjecture | open | MathDB 页面 | 原文线索

V2-058773 | MATHDB-329674

Hernandez-Leclerc adaptable-word monoidal categorification conjecture

摘录 / EXCERPT: Let $\mathbf i$ be an adaptable word, let $\mathcal A^\dagger(\mathbf i)$ be the associated quantum cluster algebra, and let κ be the map to the relevant Grothendieck ring w...

conjecture | open | MathDB 页面 | 原文线索

V2-058774 | MATHDB-329702

Nonlinear instability of the Reissner-Nordström Cauchy horizon under spherical perturbations

摘录 / EXCERPT: Consider smooth, spherically symmetric initial data with two asymptotically flat ends for the Einstein-Maxwell-scalar field equation, and suppose the data are globally close to Rei...

conjecture | open | MathDB 页面 | 原文线索

V2-058775 | MATHDB-329710

Bartnik's Hilbert-manifold conjecture for the constraint map

摘录 / EXCERPT: Let $\mathcal{F}$ be the space of initial data satisfying the prescribed boundary conditions, with $\mathcal{G}$ the space of metrics and $\mathcal{K}$ the space of momentum tensor...

conjecture | open | MathDB 页面 | 原文线索

V2-058776 | MATHDB-329711

Yang-Mills characterization conjecture for regular Markovian holonomy fields

摘录 / EXCERPT: Let ${\sf HF}$ be a regular Markovian holonomy field, and let an admissible Lévy process determine a Yang-Mills field through the Yang-Mills-fields construction. Yang-Mills charact...

conjecture | open | MathDB 页面 | 原文线索

V2-058777 | MATHDB-329881

Bubbling for Yang-Mills connections

摘录 / EXCERPT: Bubbling for Yang-Mills. Either $T_{+}=\infty$, or there are such sequences for which $A_{(n)}$ converges, through its curvature, to a Lorentz transform of a nontrivial finite-ener...

conjecture | open | MathDB 页面 | 原文线索

V2-058778 | MATHDB-329891

No-improvement conjecture for annihilation-only coherent controllers

摘录 / EXCERPT: No-improvement conjecture. The coherent-classical estimation cost satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058779 | MATHDB-329892

Optimal-angle no-improvement conjecture for coherent-classical estimation

摘录 / EXCERPT: Optimal-angle no-improvement conjecture. At the optimal homodyne angle,

conjecture | open | MathDB 页面 | 原文线索

V2-058780 | MATHDB-329893

Feedback coherent-classical improvement conjecture

摘录 / EXCERPT: Feedback coherent-classical improvement conjecture. If there exists a homodyne angle θ_i such that

conjecture | open | MathDB 页面 | 原文线索

V2-058781 | MATHDB-329909

The Yokota-type volume conjecture for plane graphs

摘录 / EXCERPT: Let Γ be an oriented plane graph, and let S_{Γ} be a hyperbolic convex polyhedron whose one-skeleton is Γ. Take sequences of integer colors of Γ as in T...

conjecture | open | MathDB 页面 | 原文线索

V2-058782 | MATHDB-329913

Drinfeld's quantization conjecture for Lie bialgebras

摘录 / EXCERPT: A Lie bialgebra is a vector space with a Lie bracket and a compatible Lie cobracket. Drinfeld's quantization conjecture. Any Lie bialgebra can be quantized. This conjecture asks wh...

conjecture | open | MathDB 页面 | 原文线索

V2-058783 | MATHDB-329923

Binary-extension QMDS existence conjecture

摘录 / EXCERPT: For $q=2^m$, consider quantum MDS codes over systems of dimension q, with parameters written as $[[n,n-6,4]]_{2^m}$. Binary-extension QMDS existence conjecture. If $q=2^m$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-058784 | MATHDB-329954

The upper global basis conjecture for quantum cluster monomials

摘录 / EXCERPT: Let $\mathfrak{g}$ be a symmetric Kac-Moody algebra, let w be an element of its Weyl group, and let $A_q(\mathfrak{n}(w))$ be the associated quantum unipotent coordinate algebra....

conjecture | open | MathDB 页面 | 原文线索

V2-058785 | MATHDB-329979

The equivalence conjecture for the functors from quiver Hecke modules

摘录 / EXCERPT: Equivalence conjecture. For $t=1,2$, the functor

conjecture | open | MathDB 页面 | 原文线索

V2-058786 | MATHDB-330055

Existence and Hadamard property of states of low energy

摘录 / EXCERPT: Let $f\in C_0^\infty(I)$ be a smearing function, where $I\subset\mathbb{R}$ is the domain of conformal time, and consider a scalar field of mass m on a cosmological spacetime wit...

conjecture | open | MathDB 页面 | 原文线索

V2-058787 | MATHDB-330260

Resolution-of-unity conjecture for coherent states on Lie groups

摘录 / EXCERPT: Resolution-of-unity conjecture. For sufficiently small $t>0$, the displayed resolution of unity and the stated asymptotic scaling of C_t hold for an arbitrary Lie group G. If...

conjecture | open | MathDB 页面 | 原文线索

V2-058788 | MATHDB-330387

Completeness of the higher-level Bethe-equation parametrization at low temperature

摘录 / EXCERPT: Let the XXZ chain be in the antiferromagnetic massive regime at finite magnetic field and sufficiently low temperature. The quantum transfer matrix has eigenstates, and the higher-...

conjecture | open | MathDB 页面 | 原文线索

V2-058789 | MATHDB-330555

Open quantum Kirwan map conjecture for disk vortices

摘录 / EXCERPT: Open quantum Kirwan map conjecture. Counting affine vortices over the upper half-plane $\mathbb H$ defines an A_∞ morphism

conjecture | open | MathDB 页面 | 原文线索

V2-058790 | MATHDB-330584

The brane conjecture for open-closed BCOV theory and \(6\)-dimensional \((1,0)\) theories

摘录 / EXCERPT: Consider type IIA string theory on $\mathbb{R}^{10}$ with one $NS5$ brane on $\mathbb{R}^6\subset\mathbb{R}^{10}$, where $x_6,\ldots,x_9=0$, together with k semi-infinite $D6$ br...

conjecture | open | MathDB 页面 | 原文线索

V2-058791 | MATHDB-330671

The quantum-double realization conjecture for completely rational conformal nets

摘录 / EXCERPT: Let $\mathcal{F}$ be a unitary fusion category, and let $Z(\mathcal{F})$ denote its Drinfeld center. For a finite-depth subfactor $N\subset M$, let $D(N\subset M)$ denote its quant...

conjecture | open | MathDB 页面 | 原文线索

V2-058792 | MATHDB-330676

Fluid-phase repulsion conjecture for hard-sphere and hard-core models

摘录 / EXCERPT: Let the hard-sphere model or hard-core model be at density α. The hard-sphere model consists of nonoverlapping spheres, while the hard-core model consists of particles occup...

conjecture | open | MathDB 页面 | 原文线索

V2-058793 | MATHDB-330749

Quaternionic gate-net covering conjecture for $PSU(2)$

摘录 / EXCERPT: Let $t\in\mathbb{N}^*$, let $S^3\subset\mathbb{R}^4$ be the unit 3-sphere, and let $\nu(5^t)$ denote the integer solutions of ... For $x\in\nu(5^t)\cup\nu(5^{t-1})$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-058794 | MATHDB-330812

Kim's localization conjecture for quantum unipotent subgroups

摘录 / EXCERPT: Localization conjecture. After localizing both algebras at the multiplicative set generated by $\\{\Delta_{w\lambda,\lambda}\mid\lambda\in P_{+}\\}$, there is an induced isomorphis...

conjecture | open | MathDB 页面 | 原文线索

V2-058795 | MATHDB-330813

Quantum cluster algebra conjecture for triple intersections

摘录 / EXCERPT: Quantum cluster intersection conjecture. If $w_{1}\leq w_{2}$ in the Bruhat order, this triple intersection contains a quantum cluster algebra, and its quantum cluster monomials ar...

conjecture | open | MathDB 页面 | 原文线索

V2-058796 | MATHDB-330997

Malec-O Murchadha conjecture on the TSS-CMC foliation of the Kruskal extension

摘录 / EXCERPT: Malec-O Murchadha conjecture. Given any constant mean curvature H, the Kruskal extension can be foliated by the TSS-CMC hypersurfaces $\{\Sigma_H\}$. The paper presents this as t...

conjecture | open | MathDB 页面 | 原文线索

V2-058797 | MATHDB-331049

ADO inequality for combining three quantum states

摘录 / EXCERPT: Let $\mathcal{D}(d)$ denote the set of d-dimensional quantum states. Let $f:\mathcal{D}(d)\to\mathbb{R}$ be concave and symmetric, let $\rho_1,\rho_2,\rho_3\in\mathcal{D}(d)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-058798 | MATHDB-331090

Non-negativity conjecture for the mass functional of GB initial data

摘录 / EXCERPT: Non-negativity conjecture. Consider a GB initial data set. Then $\mathcal{M}\left(v,\lambda',Y\right)$ is a non-negative functional for any orbit space.

conjecture | open | MathDB 页面 | 原文线索

V2-058799 | MATHDB-331111

Zauner-Appleby conjecture on Weyl-Heisenberg SIC fiducials

摘录 / EXCERPT: Let $U_{\mathcal{Z}}$ be the Zauner unitary associated with the Zauner symplectic matrix ... A Weyl-Heisenberg SIC fiducial is a fiducial vector generating a symmetric informationall...

conjecture | open | MathDB 页面 | 原文线索

V2-058800 | MATHDB-331132

Optimality conjecture for contraction coefficients of non-unital CQ channels

摘录 / EXCERPT: Optimality conjecture. Equality holds in the three displayed bounds above.

conjecture | open | MathDB 页面 | 原文线索

V2-058801 | MATHDB-331224

Conjecture on the thermodynamic limit of the effective reduced density-matrix series

摘录 / EXCERPT: Effective-series thermodynamic-limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058802 | MATHDB-331229

The perturbative holomorphic-gauge asymptotic conjecture for Wilson loop expectations

摘录 / EXCERPT: Let S^2 be the two-sphere equipped with an area form, let G be a compact gauge group, let f specify the Wilson loop observable, and let β be the coupling parameter. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-058803 | MATHDB-331255

The quantum Grothendieck-Springer isomorphism for the big cell

摘录 / EXCERPT: Quantum Grothendieck-Springer conjecture. The map $\widetilde{\xi}$ is an algebra isomorphism. This is the quantum analogue of the corresponding classical Cartesian isomorphism. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-058804 | MATHDB-331256

The quantum Gelfand-Kirillov property for the localized quantum group

摘录 / EXCERPT: Let U be the quantum group, Z its center, U_0 its Cartan subalgebra, $F_l(U)$ the relevant left fraction algebra, and let $\mathcal O_q[G/H]$ and T be the quantum homogeneo...

conjecture | open | MathDB 页面 | 原文线索

V2-058805 | MATHDB-331257

Haag duality conjecture for Kitaev models from quasi-triangular Hopf-* algebras

摘录 / EXCERPT: Let H be a quasi-triangular Hopf-$*$ algebra, and consider the corresponding Kitaev model on the plane. Haag duality conjecture. The model satisfies Haag duality for cones. This...

conjecture | open | MathDB 页面 | 原文线索

V2-058806 | MATHDB-331285

K-theoretic derived Satake conjecture for minuscule weights

摘录 / EXCERPT: Let G be a semisimple group, $G^\vee$ its Langlands dual group, and let $\underline{\lambda}$ and $\underline{\mu}$ be sequences of minuscule dominant weights. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058807 | MATHDB-331286

Main K-theoretic convolution equivalence conjecture

摘录 / EXCERPT: Let G be the group under consideration, $G^\vee$ its Langlands dual group, $KConv^{\widetilde{G^\vee}}(Gr)$ and $KConv^{\widetilde{G^\vee}\times\mathbb C^\times}(Gr)$ the equivar...

conjecture | open | MathDB 页面 | 原文线索

V2-058808 | MATHDB-331288

Generic quantum affine Hecke kernel conjecture

摘录 / EXCERPT: Let $\widehat{\mathcal H}_m(q)$ be the affine Hecke algebra over $\mathbb C(q)$, let $R^m_{n,(q)}$ be the corresponding generic-q module, and let $e_j^{(i)}$ be defined recursive...

conjecture | open | MathDB 页面 | 原文线索

V2-058809 | MATHDB-331289

Integral quantum affine Hecke kernel conjecture

摘录 / EXCERPT: Let $\widehat{\mathcal H}_m[q^{\pm}]$ be the affine Hecke algebra over $\mathbb C[q^{\pm}]$, let $R^m_{n,[q^{\pm}]}$ be the corresponding integral module, and let $e_j^{(i)}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-058810 | MATHDB-331294

Product-state conjecture for tensor-product unitary operators

摘录 / EXCERPT: Product-state conjecture. For every state ρ, there exists a product state $\rho_1\otimes\rho_2$ with the same pair of entropies.

conjecture | open | MathDB 页面 | 原文线索

V2-058811 | MATHDB-331295

Fourier-matrix decomposition conjecture for entropy diagrams

摘录 / EXCERPT: Fourier-matrix decomposition conjecture. Replacing U_d^F by $U_{d_1}^F\otimes U_{d_2}^F$ does not change the entropy diagram.

conjecture | open | MathDB 页面 | 原文线索

V2-058812 | MATHDB-331296

Conjecture on independence of optimal states from the dual Rényi indices

摘录 / EXCERPT: Independence conjecture. If ρ is optimal for U and one dual pair (α,β), then ρ is also optimal for every other dual pair.

conjecture | open | MathDB 页面 | 原文线索

V2-058813 | MATHDB-331297

Real-real symmetric-state conjecture for Fourier observables

摘录 / EXCERPT: Real-real symmetric-state conjecture. There exists a real-real symmetric state σ with the same entropy pair as ρ.

conjecture | open | MathDB 页面 | 原文线索

V2-058814 | MATHDB-331299

The positive mass conjecture

摘录 / EXCERPT: Positive mass conjecture. The ADM mass of (M,g) is nonnegative. Moreover, if the mass is zero, then (M,g) is isometric to Euclidean space $(\mathbb{R}^n,\delta)$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-058815 | MATHDB-331338

Metaplectic Casselman-Shalika conjecture

摘录 / EXCERPT: Let G be the reductive group with coweight lattice Λ, let $\Lambda^{\sharp,+}$ be the dominant metaplectic coweights, and let $\gamma\in\Lambda^{\sharp,+}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-058816 | MATHDB-331396

Efficient simulation with partial quantum Fourier transforms and quadratic phase gates

摘录 / EXCERPT: Partial-QFT and quadratic-phase simulation conjecture. There exist nontrivial families of abelian hypergroups for which the normalizer circuits of the paper's simulation theorem re...

conjecture | open | MathDB 页面 | 原文线索

V2-058817 | MATHDB-331663

Uniqueness conjecture for the Kerr family

摘录 / EXCERPT: Uniqueness of the Kerr family. Any sufficiently regular, asymptotically flat, and stationary solution to $\operatorname{Ric}(g)=0$ which is bounded by a non-degenerate event horizo...

conjecture | open | MathDB 页面 | 原文线索

V2-058818 | MATHDB-331687

Asymptotic volume conjecture with Reidemeister-torsion prefactor for hyperbolic knots

摘录 / EXCERPT: Let $\mathcal K$ be a hyperbolic knot. Let $U\subset\mathbb C$ be a neighborhood of the origin, let $u\in U-\pi i\mathbb Q$, and let χ_u be the character associated with Thurs...

conjecture | open | MathDB 页面 | 原文线索

V2-058819 | MATHDB-331718

The l_1-coherence dominance conjecture for relative entropy of coherence

摘录 / EXCERPT: l_1-coherence dominance conjecture. For all states ρ,

conjecture | open | MathDB 页面 | 原文线索

V2-058820 | MATHDB-331767

Joint interface-function convergence for bipolar-oriented planar maps

摘录 / EXCERPT: Joint interface-function convergence conjecture. The joint law of both NW/SE and NE/SW interface functions of a random bipolar-oriented planar map converges to the joint law of bot...

conjecture | open | MathDB 页面 | 原文线索

V2-058821 | MATHDB-331773

The Picard modular U-duality conjecture for CM-type rigid Calabi-Yau threefolds

摘录 / EXCERPT: Picard modular U-duality conjecture. For type IIB string theory compactified on X, the U-duality group is

conjecture | open | MathDB 页面 | 原文线索

V2-058822 | MATHDB-331909

Maximal conjecture on free parameters in HOMFLY polynomials of virtual knots

摘录 / EXCERPT: Maximal conjecture. An infinite number of free parameters, namely values of q-dimensions for dessins that cannot be decomposed using the $[N-1]$-, $[N-2]$-, $[2]$-, 1-, and...

conjecture | open | MathDB 页面 | 原文线索

V2-058823 | MATHDB-331988

Characterization of real weak values for qunits

摘录 / EXCERPT: Let $|\varphi_0\rangle$ and $|\psi_0\rangle$ be distinct, nonorthogonal unit vectors in $\mathbb{C}^n$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058824 | MATHDB-332139

The $SU(n)$ colored-invariant volume conjecture

摘录 / EXCERPT: Let $\mathcal{K}$ be a hyperbolic knot and let n be a positive integer. The $SU(n)$ volume conjecture. For $a=1,2,\ldots,n-1$, ... This is presented as the conjecture from the cite...

conjecture | open | MathDB 页面 | 原文线索

V2-058825 | MATHDB-332394

The hexagon equation of the non-compact quantum dilogarithm

摘录 / EXCERPT: Let P, Q, and V be as in Lem.. The hexagon equation of the non-compact quantum dilogarithm. ... This identity is presented as the equation corresponding to Fock-Goncharov's...

conjecture | open | MathDB 页面 | 原文线索

V2-058826 | MATHDB-332395

The octagon equation of the non-compact quantum dilogarithm

摘录 / EXCERPT: Let P, Q, and V be as in Lem.. The octagon equation of the non-compact quantum dilogarithm. ... This identity is presented as the equation corresponding to Fock-Goncharov's...

conjecture | open | MathDB 页面 | 原文线索

V2-058827 | MATHDB-332419

Asymptotic coincidence of Hamming bounds for unrestricted data-syndrome codes

摘录 / EXCERPT: Asymptotic Hamming-bound conjecture. For any d there exists $n(d)$ such that for $n\geq n(d)$ the Hamming bound in equation (HBnondeg) holds for unrestricted data-syndrome codes.

conjecture | open | MathDB 页面 | 原文线索

V2-058828 | MATHDB-332492

Monotonicity of the function F_p under orthogonalization

摘录 / EXCERPT: F_p-monotonicity conjecture. For every $x\in U_N^N$ and every $p\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-058829 | MATHDB-332509

The classification conjecture for surjective rank-k projector maps when n=2k

摘录 / EXCERPT: Let $\mathcal{H}$ be an n-dimensional Hilbert space, and let $\mathcal{P}_k(\mathcal{H})$ denote the set of rank-k orthogonal projectors on $\mathcal{H}$. Assume that $n=2k$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-058830 | MATHDB-332536

The Riemannian Horowitz-Myers conjecture

摘录 / EXCERPT: Riemannian Horowitz-Myers conjecture. The Wang mass-energy m of (M,g) satisfies $m\ge m_0$, where m_0 is the mass of the Horowitz-Myers geon having least mass among all Horow...

conjecture | open | MathDB 页面 | 原文线索

V2-058831 | MATHDB-332561

Logarithmic entanglement scaling for arbitrary blocks in the Motzkin spin chain

摘录 / EXCERPT: Logarithmic entanglement-scaling conjecture. The entanglement entropy of any such block, to leading order, scales as

conjecture | open | MathDB 页面 | 原文线索

V2-058832 | MATHDB-332576

The Hermitian structure conjecture for irreducible quantum flag manifolds

摘录 / EXCERPT: Let $\mathbb{C}_q[G_0/L_0]$ be an irreducible quantum flag manifold, with its Heckenberger--Kolb differential calculus $\Omega^{(\bullet,\bullet)}$ and the left-coinvariant form...

conjecture | open | MathDB 页面 | 原文线索

V2-058833 | MATHDB-332577

The spectral triple conjecture for irreducible quantum flag manifolds

摘录 / EXCERPT: Let $\mathbb{C}_q[G/L_S]$ be an irreducible quantum flag manifold, and let $\Omega^{(0,\bullet)}$ be the subcomplex of its Heckenberger--Kolb calculus. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-058834 | MATHDB-332660

Volume conjecture for prism-truncated hyperbolic tetrahedra

摘录 / EXCERPT: Prism-truncated Volume Conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-058835 | MATHDB-332676

Sanpera-Bruß-Lewenstein conjecture on Schmidt number of 3×3 PPT bound entangled states

摘录 / EXCERPT: Sanpera-Bruß-Lewenstein conjecture. All bound entangled states with positive partial transpose in $\mathbb{C}^3\otimes\mathbb{C}^3$ have Schmidt number 2.

conjecture | open | MathDB 页面 | 原文线索

V2-058836 | MATHDB-332763

The amenability conjecture for compact quantum groups

摘录 / EXCERPT: Let $G\subset U_N^+$ be a compact quantum group. Write $C_{max}(G)$ and $C_{min}(G)$ for its maximal and minimal Hopf C^*-algebras; G is coamenable when the canonical quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-058837 | MATHDB-332764

The polynomial growth conjecture for discrete quantum groups

摘录 / EXCERPT: Let $G\subset U_N^+$ be a compact quantum group, and let $\widehat{G}$ denote its discrete dual. For each $Q\in U_N$, let Γ_Q be the associated discrete quantum group. Polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-058838 | MATHDB-332934

The universal flat representation conjecture for quantum subgroups of the free symmetric group

摘录 / EXCERPT: Universal flat representation conjecture. Under these suitable uniformity properties, π_G is inner faithful; moreover, if the dual quantum group $\widehat{G}$ is amenable, then...

conjecture | open | MathDB 页面 | 原文线索

V2-058839 | MATHDB-332969

The physicists' conjecture on strong OPE bounds for reasonable quantum field theories

摘录 / EXCERPT: Physicists' conjecture. Any reasonable quantum field theory satisfies the operator product expansion in the strong sense expressed by the EFNNB, without the $\frac{d}{2}$ restricti...

conjecture | open | MathDB 页面 | 原文线索

V2-058840 | MATHDB-332996

The finite-image conjecture for unitary finite-order Yang-Baxter solutions

摘录 / EXCERPT: Let V be a finite-dimensional complex vector space, and let R be a unitary solution to the Yang-Baxter equation on $V^{\otimes 2}$. Suppose that R has finite projective order...

conjecture | open | MathDB 页面 | 原文线索

V2-058841 | MATHDB-332997

The localizability conjecture for simple objects in braided fusion categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a braided fusion category and let X be a simple object of $\mathcal{C}$. Denote by ρ_X the braid-group representations on...

conjecture | open | MathDB 页面 | 原文线索

V2-058842 | MATHDB-333007

The positive mass conjecture for asymptotically flat manifolds

摘录 / EXCERPT: Let N^n be an asymptotically flat Riemannian manifold of dimension $n\geq 3$ and order $p>\frac{n-2}{2}$, with integrable scalar curvature R_g and, when ∂N is noncomp...

conjecture | open | MathDB 页面 | 原文线索

V2-058843 | MATHDB-333008

The positive mass conjecture with a noncompact boundary

摘录 / EXCERPT: Let N^n be an asymptotically flat Riemannian manifold of dimension $n\geq 3$ and order $p>\frac{n-2}{2}$, with integrable scalar curvature R_g and integrable mean curvature...

conjecture | open | MathDB 页面 | 原文线索

V2-058844 | MATHDB-333048

Global monomial-set conjecture for simple Lie algebras

摘录 / EXCERPT: Let $\boldsymbol{d}:\Delta_+\rightarrow\mathbb{N}$ be a degree function for $U(\mathfrak n^-)$, and let $\mathcal{S}_{\operatorname*{gm}}$ be the set of degree functions such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-058845 | MATHDB-333088

Rigidity conjecture for four-dimensional Ricci-flat Brinkmann spacetimes

摘录 / EXCERPT: Rigidity conjecture. Every strongly causal, Ricci-flat 4-dimensional Brinkmann spacetime satisfying certain completeness condition is isometric to (a quotient of) a gravitational...

conjecture | open | MathDB 页面 | 原文线索

V2-058846 | MATHDB-333105

Brinkmann rigidity conjecture for complete Ricci-flat four-dimensional spacetimes

摘录 / EXCERPT: Brinkmann rigidity conjecture. Every strongly causal, Ricci-flat 4-dimensional Brinkmann spacetime satisfying certain completeness condition is isometric to a quotient of a plane...

conjecture | open | MathDB 页面 | 原文线索

V2-058847 | MATHDB-333138

Equivalence of quantum and classical periodicity for generalized quantum Y-seed mutations

摘录 / EXCERPT: Periodicity equivalence conjecture. The sequence of quantum mutations is σ-periodic if and only if the corresponding sequence of classical mutations is σ-periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-058848 | MATHDB-333282

The graph-distance scaling-limit conjecture for mated-CRT maps

摘录 / EXCERPT: Graph-distance scaling-limit conjecture. The appropriately normalized graph distances in the graphs $\mathcal G^\epsilon$ converge in the scaling limit to a metric on γ-LQG....

conjecture | open | MathDB 页面 | 原文线索

V2-058849 | MATHDB-333283

The small-parameter graph-distance exponent conjecture for LQG

摘录 / EXCERPT: Small-parameter graph-distance exponent conjecture. There exists $\gamma_*\in(\sqrt2,\sqrt{8/3}]$ such that, for every $\gamma\in(0,\gamma_*]$,

conjecture | open | MathDB 页面 | 原文线索

V2-058850 | MATHDB-333284

The planar-map diameter exponent conjecture for mating-of-trees maps

摘录 / EXCERPT: Planar-map diameter exponent conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058851 | MATHDB-333404

Triangular-basis agreement for acyclic skew-symmetrizable seeds

摘录 / EXCERPT: Let $\mathcal A^q$ be a quantum cluster algebra with a seed t, and let $\mathbf L^t$ denote the triangular basis introduced for that seed. Let C denote Berenstein-Zelevinsky's...

conjecture | open | MathDB 页面 | 原文线索

V2-058852 | MATHDB-333470

Quantum geometric Langlands conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, and let $G^{\vee}$ be its Langlands dual group. Let $\mathfrak{det}$ denote the determinant line...

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V2-058853 | MATHDB-333474

Panzer's Hepp-period identity conjecture

摘录 / EXCERPT: Panzer's Hepp-period identity conjecture. Equality of Feynman periods is equivalent to equality of Hepp periods:

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V2-058854 | MATHDB-333475

Graphical-function asymptotic expansion conjecture

摘录 / EXCERPT: Graphical-function asymptotic expansion conjecture. Whenever the right-hand side exists, the asymptotic expansion at $z=z_0$ is

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V2-058855 | MATHDB-333535

Minimum trace-distance contraction conjecture for qubit channels over a fixed classical channel

摘录 / EXCERPT: Let $a,f\in[0,1]$ and let $\mathcal{Q}_{\mathbb{C}}(a,f)$ denote the set of qubit channels over the classical channel ... For a qubit channel Q, let $\eta^{\operatorname{Tr}}(Q)$ b...

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V2-058856 | MATHDB-333549

Integer multiplicity current conjecture for Seiberg-Witten blow-up sets

摘录 / EXCERPT: A blow-up set Z for the Seiberg-Witten equations with two spinors on a three-dimensional manifold is a closed subset arising as the blow-up set of a sequence of solutions. An int...

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V2-058857 | MATHDB-333614

The infinitesimal minimum output entropy conjecture for single-mode gauge-covariant Gaussian channels

摘录 / EXCERPT: Let ... with entropy $S=S_0>0$, let $^{\operatorname{th}}_{g^{-1}(S_0)}$ denote the thermal state with the same entropy. Infinitesimal minimum output entropy conjecture. The entropy...

conjecture | open | MathDB 页面 | 原文线索

V2-058858 | MATHDB-333653

The noncommutative arithmetic progression conjecture for compact quantum groups

摘录 / EXCERPT: Let $\mathbb{G}$ be the compact quantum group under consideration, let $\widehat{\mathbb{G}}$ be its set of irreducible representations, and let $\Sigma=\{F_n\}$ be a sequence of f...

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V2-058859 | MATHDB-333658

Bethe Ansatz for the unit-mass sector of the quantised Kadomtsev-Petviashvili equation

摘录 / EXCERPT: Let $\mathcal{F}_M^{(1\ldots1)}$ be the subsector containing M particles of unit mass, let $\mathfrak{S}_{[1,M]}$ be the permutation group of $(1,\ldots,M)$, let...

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V2-058860 | MATHDB-333682

Polyakov triangulation conjecture for Liouville measure on higher-genus surfaces

摘录 / EXCERPT: Let M be a surface of genus ${\bf g}$, let $\mathcal{T}_{N,\mathbf{g}}$ denote the set of triangulations of size N, and let ν_a be the random measure defined by ... for posit...

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V2-058861 | MATHDB-333741

Second-gap and Weyl-law conjecture for special-alphabet open quantum maps

摘录 / EXCERPT: Let $d4d0\text{?}$ be an alphabet satisfying the special-alphabet condition, with $0,M-1 otin\fumathcal A$, and let $d712$ be a cutoff equal to 1 near the Cantor set $d54a_d3f$....

conjecture | open | MathDB 页面 | 原文线索

V2-058862 | MATHDB-333805

Subtheory conjecture for sharp theories with purification

摘录 / EXCERPT: A physical theory is said to have a well-behaved thermodynamics when its thermodynamic structure satisfies the relevant well-behavedness requirements. Subtheory conjecture. Every t...

conjecture | open | MathDB 页面 | 原文线索

V2-058863 | MATHDB-333824

Conjectural action of the spectral-dual generators on integral generalized Macdonald polynomials

摘录 / EXCERPT: Conjectural action of $X^{(1)}_{\pm1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-058864 | MATHDB-333839

Georgiou-Pavon conjecture for positive quantum channels

摘录 / EXCERPT: Let $Q:\mathbb{C}^{n\times n}\to\mathbb{C}^{n\times n}$ be a positive quantum channel, and let ρ_0,ρ_T be density matrices. Let $\mathrm{H}_{n,++}$ denote the positive defi...

conjecture | open | MathDB 页面 | 原文线索

V2-058865 | MATHDB-333848

Descriptive proximity form of gauge-string duality

摘录 / EXCERPT: Gauge-string duality conjecture. The duality asserting that gauge theory in 4 dimensions is dual to string theory in 10 dimensions has descriptive proximity form.

conjecture | open | MathDB 页面 | 原文线索

V2-058866 | MATHDB-333883

Module extension conjecture for finite-dimensional irreducible \widetilde \square_q-modules

摘录 / EXCERPT: Let V be a finite-dimensional irreducible $\widetilde \square_q$-module, where $\widetilde \square_q$ and $\boxtimes_q$ are the algebras appearing in the paper, with generators...

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V2-058867 | MATHDB-333884

Quantum affine cube module conjecture for tridiagonal pairs of q-Racah type

摘录 / EXCERPT: Let A,B be a tridiagonal pair over $\mathbb F$ having q-Racah type in the sense of Bockting-Conrad, and let V be its underlying vector space. Let $\square_q$ denote the quant...

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V2-058868 | MATHDB-334255

Affine Macdonald evaluation conjecture

摘录 / EXCERPT: Let $J_{\mu,k,\mathsf{k}}(q,\lambda,\omega)=\chi_{\mu,k,\mathsf{k}}(q,\lambda,\omega)/\chi_{0,0,\mathsf{k}}(q,\lambda,\omega)$ be the affine Macdonald polynomial for...

conjecture | open | MathDB 页面 | 原文线索

V2-058869 | MATHDB-334268

Gauge-rigidity conjecture for the Calderón problem for connections

摘录 / EXCERPT: Let (M,g) be a Riemannian manifold, let E be a Hermitian vector bundle over M, and let A and B be unitary connections on E. Denote by Λ_A and Λ_B the Di...

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V2-058870 | MATHDB-334358

Van Leeuwen's extended Runge-Gross conjecture for interaction replacement

摘录 / EXCERPT: Let a system with initial wave function ψ_0 evolve under interaction potential w and external potential v. For a different interaction potential w', there should be a un...

conjecture | open | MathDB 页面 | 原文线索

V2-058871 | MATHDB-334419

Photon-information-efficiency conjecture for collective Dolinar receivers

摘录 / EXCERPT: Consider a K-mode receiver that applies adaptive passive linear-optical transformations and coherent displacements to a multimode coherent-state input, detecting the resulting ou...

conjecture | open | MathDB 页面 | 原文线索

V2-058872 | MATHDB-334450

The Potts occupancy conjecture for regular graphs

摘录 / EXCERPT: Potts occupancy conjecture. For every d-regular graph G, every integer $q\ge 3$, and every $\beta>0$,

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V2-058873 | MATHDB-334589

Modified noncommutative Borsuk-Ulam conjecture of type II

摘录 / EXCERPT: Let A be a unital C-algebra equipped with a free action of a non-trivial compact quantum group (H,Δ), and let $A\circledast^\delta H$ be the equivariant noncommutative joi...

conjecture | open | MathDB 页面 | 原文线索

V2-058874 | MATHDB-334614

Witten's conjecture relating Donaldson and Seiberg-Witten invariants

摘录 / EXCERPT: Witten's conjecture. Then X has simple type; the basic classes of X are precisely the Seiberg-Witten basic classes of X; and there is a nonzero constant $c(X)$ depending only...

conjecture | open | MathDB 页面 | 原文线索

V2-058875 | MATHDB-334702

Non-surjectivity of the static vacuum boundary map near Schwarzschild data

摘录 / EXCERPT: Let $\gamma\in\operatorname{Met}^{m,\alpha}(S^2)$ have non-constant Gauss curvature, and let $C_+^{m-1,\alpha}(S^2)$ denote the positive mean-curvature data. Non-surjectivity of th...

conjecture | open | MathDB 页面 | 原文线索

V2-058876 | MATHDB-334846

The geometric Razumov-Stroganov conjecture for conormal Schubert sheaves

摘录 / EXCERPT: Geometric Razumov-Stroganov conjecture. The pushforward of σ_r to $\operatorname{Mat}_<(N)$ in T-equivariant K-theory is equal, up to normalization, to

conjecture | open | MathDB 页面 | 原文线索

V2-058877 | MATHDB-334874

Lusztig's isomorphism conjecture for quiver mutations

摘录 / EXCERPT: Let T_i denote the Lusztig isomorphisms associated with the simple roots of the quantum group, and let quiver mutations act on the corresponding quantum cluster structure. Luszti...

conjecture | open | MathDB 页面 | 原文线索

V2-058878 | MATHDB-334885

Curvature blow-up formulation of Strong Cosmic Censorship

摘录 / EXCERPT: Curvature blow-up conjecture. For generic initial data to Einstein's equations, the Kretschmann scalar $R_{\alpha\beta\gamma\delta}R{}^{\alpha\beta\gamma\delta}$ is unbounded in th...

conjecture | open | MathDB 页面 | 原文线索

V2-058879 | MATHDB-334978

The modified Moore-Mariño conjecture for the ${\rm U}(3)$ series

摘录 / EXCERPT: Let X be a four-manifold of ${\rm U}(3)$-simple type, let $\{K_i\}$ be its Kronheimer-Mrowka basic classes, and let $\mathfrak s_i$ be chosen so that the basic class associated t...

conjecture | open | MathDB 页面 | 原文线索

V2-058880 | MATHDB-335023

The saddle-point characterization of the Curie-Weiss critical curve

摘录 / EXCERPT: Let $f_\beta(u)$ be the function whose saddle-point equation is ... and let $u^*(\beta)$ be the branch of solutions defined in the source, satisfying $u^*(1)=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-058881 | MATHDB-335049

Complex compactification conjecture for stable bundle moduli on Kähler manifolds

摘录 / EXCERPT: Complex compactification conjecture. The natural complex space structure of ${\mathcal M}^\mathrm{st}_{\mathcal D}(E)$, induced from ${\mathcal M}^\mathrm{si}_{\mathcal D}(E)$, ext...

conjecture | open | MathDB 页面 | 原文线索

V2-058882 | MATHDB-335061

Li-Batchelor conjecture on common roots at half-integer bias

摘录 / EXCERPT: Let $P^{(N,\frac12)}_N(x)$ and $\tilde{P}^{(N+1,\frac12)}_{N+1}(x)$ be the constraint polynomials associated with the asymmetric quantum Rabi model at $\epsilon=\frac12$. Li-Batche...

conjecture | open | MathDB 页面 | 原文线索

V2-058883 | MATHDB-335062

Li-Batchelor conjecture on polynomial divisibility for half-integer bias

摘录 / EXCERPT: Let $\ell\in\mathbb{Z}_{\geq0}$ and $N\in\mathbb{Z}_{>0}$, and let $P^{(N,\ell/2)}_N(x,\Delta^2)$ and $\tilde{P}^{(N+\ell,\ell/2)}_{N+\ell}(x,\Delta^2)$ be the corresponding constr...

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V2-058884 | MATHDB-335067

The uniqueness conjecture for universal perfect state transfer among circulants with property $\mathbb{T}$

摘录 / EXCERPT: A complex unit gain graph is a graph whose adjacency matrix has complex entries, with each nonzero entry of unit magnitude. A graph has property $\mathbb{T}$ if all nonzero coeffic...

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V2-058885 | MATHDB-335132

Gukov-Putrov-Vafa conjecture on the categorification of WRT invariants

摘录 / EXCERPT: Categorification conjecture. There are series $\widehat Z_b(q)$, convergent for $|q|<1$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-058886 | MATHDB-335134

Knot-decorated WRT decomposition conjecture

摘录 / EXCERPT: Knot-decorated decomposition conjecture. The WRT invariant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058887 | MATHDB-335324

The C^k formulation of strong cosmic censorship

摘录 / EXCERPT: Let asymptotically flat initial data for the Einstein-Maxwell-(real)-scalar field system be given, and let their maximal globally hyperbolic future development be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-058888 | MATHDB-335325

The generic $s=3$ lower-bound conjecture for spherical Einstein-Maxwell-scalar field evolution

摘录 / EXCERPT: Consider spherically symmetric admissible Cauchy initial data with non-vanishing charge and compactly supported initial scalar field, and let their maximal globally hyperbolic futu...

conjecture | open | MathDB 页面 | 原文线索

V2-058889 | MATHDB-335415

Whether the behavioural special relativity principle implies the body-observer special relativity principle

摘录 / EXCERPT: Let ${{\mathsf{SPR}}}\xspace^+$ denote the behavioural formulation of the special principle of relativity, and let ${{\mathsf{SPR}_{{{\mathit{B}}}}\xspace}}\xspace$ denote the body...

open_problem | open | MathDB 页面 | 原文线索

V2-058890 | MATHDB-335505

The $p\to q$ norm conjecture for Gaussian quantum-limited amplifiers

摘录 / EXCERPT: The $p\to q$ norm conjecture. For any $\kappa\ge1$ and any $p,q\ge1$, every quantum state $\hat\rho$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058891 | MATHDB-335552

Zauner's conjecture on Weyl-Heisenberg covariant SICs

摘录 / EXCERPT: Zauner's conjecture. In all finite dimensions there exists a fiducial vector for a Weyl-Heisenberg covariant SIC that is an eigenvector of Z.

conjecture | open | MathDB 页面 | 原文线索

V2-058892 | MATHDB-335577

Heralded-channel near-additivity conjecture

摘录 / EXCERPT: A flagged channel randomly selects one channel from a fixed ensemble and reports the selected channel along with the output state. A channel is strongly additive when its use toget...

conjecture | open | MathDB 页面 | 原文线索

V2-058893 | MATHDB-335590

The maximal-Abelian-subgroup conjecture for SIC Galois groups

摘录 / EXCERPT: Maximal-Abelian-subgroup conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058894 | MATHDB-335591

The type-a Galois-group conjecture for SIC fiducials

摘录 / EXCERPT: Type-a Galois-group conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058895 | MATHDB-335664

The conjectural expansion of images of quantum Grothendieck polynomials

摘录 / EXCERPT: Image expansion conjecture. For every $w\in S_n$, there is a polynomial $\widetilde g_w\in\Lambda_{(n)}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058896 | MATHDB-335665

The longest-element factorization conjecture for dual stable Grothendieck functions

摘录 / EXCERPT: Longest-element factorization conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-058897 | MATHDB-335733

Polynomiality conjecture for generalized Macdonald operators

摘录 / EXCERPT: Let ${\mathcal M}_{a_1,\ldots,a_{\alpha}}$ denote the generalized Macdonald operators indexed by nonnegative integers $a_1,\ldots,a_{\alpha}$, and let ${\mathcal M}_p$ denote the c...

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V2-058898 | MATHDB-335735

Homogeneity and stationarity of universal quasi-flat models

摘录 / EXCERPT: Let $G\subset S_N^+$ be quasi-transitive, with all orbits of size K, and write $\Gamma=\widehat{G}$. Let X_G be the universal quasi-flat model space and let ... be the correspond...

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V2-058899 | MATHDB-335736

Inner faithfulness of the universal quasi-flat model under uniformity

摘录 / EXCERPT: Let $G\subset S_N^+$ be quasi-transitive, with all orbits of size K, and let ... be its universal quasi-flat model. The uniformity conjecture. If G additionally satisfies a suita...

conjecture | open | MathDB 页面 | 原文线索

V2-058900 | MATHDB-335793

PBW basis conjecture for the q-Onsager algebra

摘录 / EXCERPT: PBW basis conjecture. These vectors induce a Poincare-Birkhoff-Witt-type basis of $\tilde{{\cal A}}_q^{\{\delta\}}$, called the WG-basis.

conjecture | open | MathDB 页面 | 原文线索

V2-058901 | MATHDB-335794

Bijection conjecture for the q-Onsager algebra presentation

摘录 / EXCERPT: Bijection conjecture. The map Ψ is a bijection.

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V2-058902 | MATHDB-335840

Price's law conjecture for charged massive scalar fields

摘录 / EXCERPT: Let (M,g,ϕ,F) be a spherically symmetric solution of the Einstein-Maxwell-Klein-Gordon system that perturbs a Reissner-Nordström background of mass M and charge e, with...

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V2-058903 | MATHDB-335847

Frenkel-Reshetikhin conjecture on q-character substitutions for transfer-matrix eigenvalues

摘录 / EXCERPT: Frenkel-Reshetikhin conjecture. For every eigenvalue λ_j of $\mathcal{T}_V(z)$ on W, the preceding substitution in $\chi_q(V)$ yields λ_j; here each $Q_{i,j}(z)$...

conjecture | open | MathDB 页面 | 原文线索

V2-058904 | MATHDB-335856

Inverse spectral conjecture for quantum integrable systems with non-degenerate singularities

摘录 / EXCERPT: Let $\mathscr{N\!D}_n$ be the class of completely integrable systems on a $2n$-dimensional symplectic manifold with non-degenerate singularities. Let $Q(\mathscr{N\!D}_n)$ be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-058905 | MATHDB-335948

The ADE quantum field theory algebra and symplectic reflection algebra conjecture

摘录 / EXCERPT: Let X be a resolution of an ADE singularity, with deformation $\mathscr O_{\epsilon,\lambda_1,\dots,\lambda_r}(X)$ of its algebra of functions, where r is the ADE rank, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058906 | MATHDB-335949

The double-loop algebra B-algebra Yangian conjecture

摘录 / EXCERPT: Yangian degeneration conjecture. There is an isomorphism

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V2-058907 | MATHDB-335950

The shifted Yangian conjecture for cocharacter B-algebras

摘录 / EXCERPT: Let ρ be a cocharacter of SL_N, and use the combined $\mathbb C^\times$-action on $U^{comb}_\delta(\mathscr O_\epsilon(\mathbb C^2)\otimes\mathfrak{gl}_N)$ obtained from the...

conjecture | open | MathDB 页面 | 原文线索

V2-058908 | MATHDB-335951

The quantum-field-theoretic deformation of differential-operator enveloping algebras

摘录 / EXCERPT: Let Σ be an affine algebraic curve, and let $\operatorname{Diff}(\Sigma)$ be its algebra of polynomial differential operators. The five-dimensional quantum field theory on...

conjecture | open | MathDB 页面 | 原文线索

V2-058909 | MATHDB-335990

The K-Peterson conjecture for equivariant quantum K-theory

摘录 / EXCERPT: Let G) be a simple and simply-connected complex Lie group with Borel subgroup B, maximal torus T, Weyl group W, affine Weyl group $W_{\mathrm{af}}=W\ltimes Q^\vee$, and cor...

conjecture | open | MathDB 页面 | 原文线索

V2-058910 | MATHDB-336105

The stationary K-model conjecture for quantum subgroups of S_N^+

摘录 / EXCERPT: Stationary K-model conjecture. Under these assumptions, the following are equivalent:

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V2-058911 | MATHDB-336106

The homogeneous model-space conjecture for uniform group duals

摘录 / EXCERPT: Homogeneous model-space conjecture. Under the stated strong uniformity assumption:

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V2-058912 | MATHDB-336122

Simple-sequence characterization of the dual PBW and upper global bases

摘录 / EXCERPT: Let w_0 be the longest element, let $[\widetilde{w}_0]$ denote a commutation class of w_0, let $P_{[\widetilde{w}_0]}$ be the associated dual PBW-basis, and let $B(\infty)$ be...

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V2-058913 | MATHDB-336123

Langlands equivalence conjecture for the quiver Hecke and quantum affine representation categories

摘录 / EXCERPT: Langlands equivalence conjecture. The functor $\mathcal{F}_{\mathscr{Q}}\colon \operatorname{Rep}(R)\to\mathscr{C}_{\mathscr{Q}}$ is an equivalence of categories. This is proposed...

conjecture | open | MathDB 页面 | 原文线索

V2-058914 | MATHDB-336132

Bethe ansatz conjecture for quantum toroidal $\mathcal E_2$ transfer-matrix spectra

摘录 / EXCERPT: Quantum toroidal $\mathcal E_2$ Bethe ansatz conjecture. For each such eigenvector, there exist polynomials Q_0 and Q_1 whose roots satisfy the two displayed Bethe ansatz equat...

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V2-058915 | MATHDB-336133

Affine Gaudin Bethe ansatz conjecture for qKdV non-local integrals of motion

摘录 / EXCERPT: Affine Gaudin Bethe ansatz conjecture. Every eigenvalue of $\boldsymbol G_n$ in the level-N subspace has the form $r_n(\bar s,\bar t)$ for a solution of the affine Gaudin Bethe a...

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V2-058916 | MATHDB-336208

Conjecture on small complex Hadamard matrices with noncommutative entries

摘录 / EXCERPT: Small-size classification conjecture. The matrices in Proposition 1.8 are the only 4×4 Hadamard matrices, and there is no 5×5 Hadamard matrix with self-adjoint entrie...

conjecture | open | MathDB 页面 | 原文线索

V2-058917 | MATHDB-336479

BQP-completeness conjecture for CFT unitary evolution

摘录 / EXCERPT: For finite-Fourier-series functions $f^{(1)},\ldots,f^{(k)}$, with coefficients supplied as inputs and with n_j denoting the largest nonzero Fourier index of $f^{(j)}$, define th...

conjecture | open | MathDB 页面 | 原文线索

V2-058918 | MATHDB-336499

Quantum Mrs. Gerber's Lemma

摘录 / EXCERPT: Let X_1 and X_2 be binary classical random variables with quantum side-information systems B_1 and B_2, respectively, and let $\rho^{X_1B_1}$ and $\rho^{X_2B_2}$ be classic...

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V2-058919 | MATHDB-336529

Discontinuity conjecture for the Potts model in dimensions at least three

摘录 / EXCERPT: Let $d\ge 3$ and $q\ge 3$. Consider the nearest-neighbor Potts model on $\mathbb{Z}^d$. Discontinuity conjecture. The phase transition of this model is discontinuous. The claim is...

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V2-058920 | MATHDB-336533

Non-triviality conjecture for four-dimensional Ising theory

摘录 / EXCERPT: Define the renormalized coupling constant by ... where U_4 is the connected four-point function, $\mathit{\chi(\beta)}$ is the susceptibility, and $\mathit{\xi(\beta)}$ is the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-058921 | MATHDB-336549

Gauge symmetry breaking conjecture for vacuum-restricted geometric Langlands categories

摘录 / EXCERPT: Let G be a reductive group, let $\mathfrak h$ be a Cartan subalgebra with Weyl group W, and let $v\in\mathfrak h^*/W$. Let $x\in\mathfrak g^*$ be a semisimple lift of v, and...

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V2-058922 | MATHDB-336585

The no-additional-components conjecture for outermost regions

摘录 / EXCERPT: Let M' be an outermost region of a geometrostatic manifold, written as ... where each Ω_i is diffeomorphic to a three-ball, has stable minimal boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-058923 | MATHDB-336993

Algebraicity of the radial symmetry operators in the n-body problem

摘录 / EXCERPT: Algebraicity conjecture. All symmetry operators K_{ij} are algebraic differential operators in the variables ρ_{ij}, with coefficients linear in the ρ_{ij}.

conjecture | open | MathDB 页面 | 原文线索

V2-058924 | MATHDB-336994

The Cayley-Menger factorization conjecture for the radial metric determinant

摘录 / EXCERPT: Cayley-Menger factorization conjecture. The determinant $\det g^{\mu\nu}$ is homogeneous of degree M in the M variables ρ_{ij} and factors as

conjecture | open | MathDB 页面 | 原文线索

V2-058925 | MATHDB-337120

Equivalence conjecture for complex and quaternionic quantum mechanics

摘录 / EXCERPT: A quaternionic quantum system consists of a quaternionic Hilbert space with observables and a unitary time-evolution semigroup. A quaternionic system is an external quaternionifica...

conjecture | open | MathDB 页面 | 原文线索

V2-058926 | MATHDB-337123

The cosmic censorship hypothesis for maximal Cauchy developments

摘录 / EXCERPT: Let (Σ, h_{ab}, k_{ab}) be an initial data set with Σ three-dimensional. A spacetime is strongly causal if every point p and neighborhood O of p have a neighbor...

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V2-058927 | MATHDB-337124

Earman's hole-free cosmic censorship conjecture

摘录 / EXCERPT: Let (M,g_{ab}) be an inextendible, hole-free, vacuum solution. A spacetime is strongly causal if every point p and neighborhood O of p have a neighborhood $V\subseteq O$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-058928 | MATHDB-337166

The associative deformation conjecture for the operator product expansion

摘录 / EXCERPT: Associative deformation conjecture. The $\hbar$-derivative of the operator product expansion defines an associative deformation of the OPE algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-058929 | MATHDB-337302

The ray-and-line criterion for CMC Cauchy surfaces

摘录 / EXCERPT: Let $\mathcal{V}$ be a globally hyperbolic, cosmological spacetime. A future ray or past ray is a complete timelike path in the corresponding time direction, and a line is a comple...

conjecture | open | MathDB 页面 | 原文线索

V2-058930 | MATHDB-337305

The open-set conjecture for initial data without CMC slices

摘录 / EXCERPT: Let (M,g,K) be an initial data set, equipped with the $C^{2,\alpha}$ norm, and let its associated spacetime be the evolution determined by the data. Open-set conjecture. There is...

conjecture | open | MathDB 页面 | 原文线索

V2-058931 | MATHDB-337306

The stability of future rays under perturbation of vacuum initial data

摘录 / EXCERPT: Let (M,g,K) be a compact, vacuum initial data set whose evolution is a non-static spacetime. Suppose there is a family of future rays intersecting the initial Cauchy surface in a...

conjecture | open | MathDB 页面 | 原文线索

V2-058932 | MATHDB-337402

Allegretti-Kim's structure-coefficient positivity conjecture for quantum elements

摘录 / EXCERPT: Structure-coefficient positivity conjecture. Only finitely many structure coefficients $c^q(\ell,\ell';\ell'')$ are nonzero, and every one belongs to $\mathbb{Z}_{\geq 0}[q,q^{-1}]$...

conjecture | open | MathDB 页面 | 原文线索

V2-058933 | MATHDB-337436

Non-degeneracy conjecture for Riemannian Schwarzschild-anti de Sitter metrics

摘录 / EXCERPT: Non-degeneracy conjecture. P_L has no L^2-kernel except if $K=1$ and μ is given by the exceptional value above.

conjecture | open | MathDB 页面 | 原文线索

V2-058934 | MATHDB-337437

Absence of scalar master modes in the L^2-kernel of the linearised Einstein operator

摘录 / EXCERPT: Scalar-mode conjecture. There are no non-trivial scalar master modes associated with the L^2-kernel of the operator P_L.

conjecture | open | MathDB 页面 | 原文线索

V2-058935 | MATHDB-337448

The EGNO conjecture on generation of pointed finite tensor categories

摘录 / EXCERPT: EGNO's conjecture. A pointed finite tensor category is tensor generated by objects of length 2.

conjecture | open | MathDB 页面 | 原文线索

V2-058936 | MATHDB-337498

The dimension criterion for generalized and quasi-localizability

摘录 / EXCERPT: Dimension criterion for localizability. Any simple $X\in\mcC$ is generalized or quasi-localizable if, and only if,

conjecture | open | MathDB 页面 | 原文线索

V2-058937 | MATHDB-337628

Twisting conjecture for quantum groups from Weyl matrix models

摘录 / EXCERPT: Let a Weyl matrix model be a matrix model constructed from Weyl matrices, and let PE denote the associated quantum group appearing in the untwisted comparison model. Weyl-model t...

conjecture | open | MathDB 页面 | 原文线索

V2-058938 | MATHDB-337656

Delocalization conjecture for two-dimensional integer-valued random surfaces with general potentials

摘录 / EXCERPT: Let $U:\mathbb{R}\to\mathbb{R}$ be smooth and satisfy $U(x)=U(-x)$ and ... for some $\alpha>0$. For each $M>0$, define the interaction sequence $u^M=(u^M_n)$ by $u^M_n:=U(n/M)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-058939 | MATHDB-337722

Dafermos-Holzegel conjecture on naked singularity formation for Eguchi-Hanson-AdS perturbations

摘录 / EXCERPT: Dafermos-Holzegel conjecture. Small perturbations of Eguchi-Hanson-AdS spacetimes have a Penrose diagram as depicted in Figure. Moreover, $\mathcal I$ is future incomplete.

conjecture | open | MathDB 页面 | 原文线索

V2-058940 | MATHDB-337723

Clarkson-Mann minimal-mass conjecture for Eguchi-Hanson-AdS data

摘录 / EXCERPT: Clarkson-Mann conjecture. The mass satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-058941 | MATHDB-337872

The Cosmic No-Hair conjecture for scalar fields with positive constant potential

摘录 / EXCERPT: Let $\mathcal A$ denote the class of initial data such that the corresponding maximal globally hyperbolic developments are future causally geodesically complete solutions to the Ei...

conjecture | open | MathDB 页面 | 原文线索

V2-058942 | MATHDB-337905

Canonical quantization conjecture for the Alekseev-Meinrenken dual exponential map

摘录 / EXCERPT: Canonical quantization conjecture. The isomorphism φ is a canonical quantization of the Alekseev-Meinrenken dual exponential map

conjecture | open | MathDB 页面 | 原文线索

V2-058943 | MATHDB-337922

Factorization and positivity conjecture for asymmetric quantum Rabi constraint polynomials

摘录 / EXCERPT: Factorization and positivity conjecture. For $\ell,N\in\mathbb{Z}_{\geq 0}$, there exists a polynomial $A^\ell_N(x,y)\in\mathbb{Z}[x,y]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-058944 | MATHDB-337942

Marin-Wagner's dimension conjecture for Links-Gould centralizer algebras

摘录 / EXCERPT: Fix $b1 enewcommand$?

conjecture | open | MathDB 页面 | 原文线索

V2-058945 | MATHDB-338018

Equivalence of ribbon twist-braided module categories for inner-equivalent involutions

摘录 / EXCERPT: Let $\mathfrak{u}$ be a compact semisimple Lie algebra with $*$-compatible Chevalley generators $\{e_r,f_r,h_r\mid r\in I\}$ of $\mathfrak{g}=\mathfrak{u}^{\mathbb{C}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058946 | MATHDB-338083

The quantum symplecticity conjecture for the canonical solution

摘录 / EXCERPT: Quantum symplecticity conjecture. For any $u\in\mathcal C$, the function $H_{\hbar}(z,u)$ is valued in the group $L^{(2)}GL(h)_\hbar$.

conjecture | open | MathDB 页面 | 原文线索

V2-058947 | MATHDB-338110

The noncommutative Borsuk-Ulam conjecture for equivariant joins

摘录 / EXCERPT: Noncommutative Borsuk-Ulam conjecture. There does not exist a $\mathbb{G}$-equivariant $*$-homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-058948 | MATHDB-338112

Klyachko's maximal-spectrum minimal-rank conjecture

摘录 / EXCERPT: Let ρ_{AB} be a quantum state on $\mathcal{H}_A\otimes\mathcal{H}_B$ with marginal states ρ_A and ρ_B. Let $\mathcal{C}(\rho_A,\rho_B)$ be the set of states with the...

conjecture | open | MathDB 页面 | 原文线索

V2-058949 | MATHDB-338168

Feynman's efficient quantum simulation conjecture for QED and QCD

摘录 / EXCERPT: QED is quantum electrodynamics and QCD is quantum chromodynamics. Feynman's quantum-simulation conjecture. Computations in high-energy physics, especially computations in QED and Q...

conjecture | open | MathDB 页面 | 原文线索

V2-058950 | MATHDB-338169

Kalai-Kindler noise-impossibility conjecture for quantum devices

摘录 / EXCERPT: Consider noisy non-interacting bosons, quantum circuits, and other quantum devices operating without quantum error correction. Kalai-Kindler noise-impossibility conjecture. (i) The...

conjecture | open | MathDB 页面 | 原文线索

V2-058951 | MATHDB-338171

Quantization problem for Lagrangian torus fibrations

摘录 / EXCERPT: Let (X,L) be a polarized compact Kähler manifold, with L a pre-quantum bundle, and let $\pi:X\rightarrow B$ be a Lagrangian torus fibration. Let $\Gamma_{X/B}(X,L)$ denote the...

open_problem | open | MathDB 页面 | 原文线索

V2-058952 | MATHDB-338218

Berenstein-Zelevinsky's q-commuting upper global basis conjecture

摘录 / EXCERPT: Let $\mathfrak g$ be a quantum group of type A_n, and let $\mathbf B^\mathrm{up}$ be its upper global basis. For $b_1,b_2\in\mathbf B^\mathrm{up}$, call them q-commuting when ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-058953 | MATHDB-338242

Emms-Hancock-Severini-Wilson conjecture on positive supports of Grover walks

摘录 / EXCERPT: Emms-Hancock-Severini-Wilson conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058954 | MATHDB-338293

Fano-type conjecture for covariant Kähler structures on quantum flag manifolds

摘录 / EXCERPT: Let ${\mathbb C}_q[G_0/L_0]$ be an irreducible quantum flag manifold. Let $(\Omega^{(\bullet,\bullet)},\kappa)$ consist of the Heckenberger-Kolb complex structure and a left-coinva...

conjecture | open | MathDB 页面 | 原文线索

V2-058955 | MATHDB-338469

The symmetry characterization conjecture for stable solutions

摘录 / EXCERPT: Stable-symmetry characterization conjecture. Stable solutions are characterized by geometrical symmetries: there is a group of symmetries acting on the space that preserves the sol...

conjecture | open | MathDB 页面 | 原文线索

V2-058956 | MATHDB-338470

The symmetry-group correspondence conjecture for stable solutions

摘录 / EXCERPT: Symmetry-group correspondence conjecture. There is a one-to-one correspondence between symmetry groups acting on (M,A,f) and stable solutions for ψ in (M,A,f).

conjecture | open | MathDB 页面 | 原文线索

V2-058957 | MATHDB-338471

The boundary information inequality conjecture

摘录 / EXCERPT: Boundary information inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-058958 | MATHDB-338661

Dafermos-Holzegel-Rodnianski restricted nonlinear stability conjecture for Schwarzschild

摘录 / EXCERPT: Let $(\Sigma_M,\bar{g}_M,K_M)$ be the induced data on a spacelike asymptotically flat slice of the Schwarzschild solution of mass M, crossing the future horizon and bounded by a...

conjecture | open | MathDB 页面 | 原文线索

V2-058959 | MATHDB-338725

The quantum Grothendieck ring evaluation conjecture

摘录 / EXCERPT: Let $\mathcal{Y}$ be the monomial ring, let m be a dominant monomial in $\mathcal{Y}$, let $L_t(\underline{m})$ denote the (q,t)-character of the simple module $L(m)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058960 | MATHDB-338840

Entangling-power edge-point conjecture for normalized two-qubit operations

摘录 / EXCERPT: Let U be a normalized two-qubit operation with canonical parameters x,y,z and coefficients c_0,c_1,c_2,c_3 as in the paper, satisfying $c_2=c_3$, equivalently $y=z$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058961 | MATHDB-338845

Baum-Dąbrowski-Hajac noncommutative Borsuk-Ulam conjectures

摘录 / EXCERPT: Let A be a unital C^*-algebra with a free coaction δ of a nontrivial compact quantum group (H, Δ). Equip the join $A \circledast^{\delta} H$ with the induced free...

conjecture | open | MathDB 页面 | 原文线索

V2-058962 | MATHDB-339047

Nonvanishing conjecture for commutators of quantum root vectors

摘录 / EXCERPT: Nonvanishing conjecture. If $\beta_i,\beta_j\in\Phi^+$ and $\beta_i+\beta_j=\beta_k\in\Phi^+$, then the coefficient $m_{(a_{i+1},\ldots,a_{j-1})}$ is nonzero for the vector...

conjecture | open | MathDB 页面 | 原文线索

V2-058963 | MATHDB-339136

The diagonal Eigenstate Thermalization Hypothesis

摘录 / EXCERPT: Eigenstate Thermalization Hypothesis. The diagonal expectation values satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-058964 | MATHDB-339137

The off-diagonal Eigenstate Thermalization Hypothesis

摘录 / EXCERPT: Eigenstate Thermalization Hypothesis. The local statistics of the off-diagonal matrix elements satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-058965 | MATHDB-339223

Zauner's weak conjecture on symmetric informationally complete measurements

摘录 / EXCERPT: For each positive integer $d\geq 2$, a SIC POVM is a set of d^2 rank-one positive operators on $\mathbb{C}^d$ that forms an informationally complete symmetric positive operator-v...

conjecture | open | MathDB 页面 | 原文线索

V2-058966 | MATHDB-339224

Zauner's strong conjecture on Weyl-Heisenberg SIC POVMs

摘录 / EXCERPT: Let $d\geq 2$, let $\mathbb{Z}_d=\{0,1,\ldots,d-1\}$, and let $W_{i,j}=U^iV^j$ be the discrete Weyl matrices on $\mathbb{C}^d$, with U the cyclic shift and V the diagonal matri...

conjecture | open | MathDB 页面 | 原文线索

V2-058967 | MATHDB-339272

Non-Boltzmann limiting dynamics for periodic quantum transport

摘录 / EXCERPT: Let $L(t)$ denote the limiting family of linear operators arising in the scaling limit where the quantum wavelength and scattering radius satisfy $h=r\to0$, with observables evolve...

conjecture | open | MathDB 页面 | 原文线索

V2-058968 | MATHDB-339317

Positive-integrality conjecture for time-delay matrix cumulants

摘录 / EXCERPT: Positive-integrality conjecture. For $\beta\in\{1,2\}$, the expansion coefficients satisfy

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-058969 | MATHDB-339390

Minimum output entropy conjecture for multi-mode Gaussian quantum channels

摘录 / EXCERPT: Let $n\in\mathbb{N}$, and let $\hat{\rho}$ be a finite-entropy state of an n-mode Gaussian quantum system. Let g be the entropy function for a single-mode thermal Gaussian stat...

conjecture | open | MathDB 页面 | 原文线索

V2-058970 | MATHDB-339494

Audenaert-Datta conjecture on joint convexity of an α-z trace functional

摘录 / EXCERPT: Audenaert-Datta conjecture. If $1\leq p\leq2$ and $-1\leq q<0$, then this map is jointly convex. The conjecture specifies the parameter regime in which the trace functional underly...

conjecture | open | MathDB 页面 | 原文线索

V2-058971 | MATHDB-339495

Audenaert-Datta conjecture on the full joint convexity range

摘录 / EXCERPT: Audenaert-Datta conjecture. If $1\leq p\leq2$, $-1\leq q<0$, and $s\geq1/(p+q)$, then this map is jointly convex. These conditions complement the necessary conditions stated in the...

conjecture | open | MathDB 页面 | 原文线索

V2-058972 | MATHDB-339686

Uniqueness and algebraic characterization of analytic tangent cones

摘录 / EXCERPT: Let B be a ball in $\mathbb C^n$ and let $\mathcal E$ be a reflexive sheaf over B with an isolated singularity at 0. For an admissible Hermitian-Yang-Mills connection A on...

conjecture | open | MathDB 页面 | 原文线索

V2-058973 | MATHDB-339717

Cuntz's associativity conjecture for the absolute fusion algebra

摘录 / EXCERPT: Cuntz's conjecture. This multiplication is associative.

conjecture | open | MathDB 页面 | 原文线索

V2-058974 | MATHDB-339720

Epimorphism from the Drinfeld Yangian of the queer Lie superalgebra

摘录 / EXCERPT: The epimorphism conjecture. The map defined by

conjecture | open | MathDB 页面 | 原文线索

V2-058975 | MATHDB-339732

The PPT squared conjecture

摘录 / EXCERPT: Let $\mathcal{M}_d$ denote the algebra of complex $d\times d$ matrices, and let $T:\mathcal{M}_d\rightarrow\mathcal{M}_d$ be a linear map. A map is completely positive if all its a...

conjecture | open | MathDB 页面 | 原文线索

V2-058976 | MATHDB-339733

Sanpera-Bruß-Lewenstein Schmidt-number conjecture for PPT maps

摘录 / EXCERPT: Let $\mathcal{M}_d$ denote the algebra of complex $d\times d$ matrices, and let $T:\mathcal{M}_d\rightarrow\mathcal{M}_d$ be a linear map. Its Choi matrix is denoted by C_T, and...

conjecture | open | MathDB 页面 | 原文线索

V2-058977 | MATHDB-339734

The PPT squared conjecture via tensor products of PPT maps

摘录 / EXCERPT: Let $\mathcal{M}_{d_i}$ denote the algebra of complex $d_i\times d_i$ matrices. Let ... be linear maps, each completely positive and completely copositive. For a positive matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-058978 | MATHDB-339735

The PPT squared conjecture via decomposability of compositions

摘录 / EXCERPT: Let $\mathcal{M}_{d_i}$ denote the algebra of complex $d_i\times d_i$ matrices. Let $T:\mathcal{M}_{d_1}\rightarrow\mathcal{M}_{d_2}$ be completely positive and completely copositi...

conjecture | open | MathDB 页面 | 原文线索

V2-058979 | MATHDB-339745

Factorization Kazhdan-Lusztig conjecture for the Jacquet functor

摘录 / EXCERPT: Let $\hat{\mathfrak{g}}_{\kappa}\textup{-mod}^{G(\mathcal{O})}$ be the category of $G(\mathcal{O})$-equivariant modules for the affine Kac-Moody algebra at level κ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-058980 | MATHDB-339755

The generalized Satake diagram classification conjecture for reflection-equation solutions

摘录 / EXCERPT: Generalized Satake diagram classification conjecture. 1. If $K\in\operatorname{GL}(V)$ is a solution of the matrix reflection equation, then there exists a generalized Satake diagr...

conjecture | open | MathDB 页面 | 原文线索

V2-058981 | MATHDB-339866

Renormalized leaky-wire limit conjecture

摘录 / EXCERPT: Let Γ be a graph with some semi-infinite, asymptotically straight edges, and suppose ... Consider the renormalized operator $-\Delta_{\alpha,\Gamma}+\frac14\alpha^2$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-058982 | MATHDB-339868

BZ positivity conjecture for quantum cluster algebras

摘录 / EXCERPT: Let $mathcal A_q$ be a quantum cluster algebra and t be a seed. A quantum cluster variable is a cluster variable obtained by iterated mutation, and write $X(t)^{\vec a}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-058983 | MATHDB-339879

Two-types conjecture for finite actions of quizzy quantum groups

摘录 / EXCERPT: Two-types conjecture. There are two types of possible actions of the quizzy quantum groups $H_N\subset G\subset O_N^+$ on finite spaces: those coming from...

conjecture | open | MathDB 页面 | 原文线索

V2-058984 | MATHDB-339880

Three-orbital distinction conjecture for quizzy quantum groups

摘录 / EXCERPT: Three-orbital distinction conjecture. The quantum groups $H_N\subset\bar{O}_N$ are distinguished by their 3-orbitals.

conjecture | open | MathDB 页面 | 原文线索

V2-058985 | MATHDB-339898

Strict diameter-based eigenvalue bound for non-tree quantum graphs

摘录 / EXCERPT: Let $\mathcal{G}$ be any connected, compact graph with total length L and diameter D, and let $\mu_k(\mathcal{G})$ denote its k-th eigenvalue. Assume that $L/k>D/2$. Strict d...

conjecture | open | MathDB 页面 | 原文线索

V2-058986 | MATHDB-339923

Mirror symmetry conjecture for categories of line operators

摘录 / EXCERPT: Let $\calT$ be a three-dimensional $N=4$ theory and let $\calT^*$ denote its mirror. Write $\calC_C(\calT)$ and $\calC_H(\calT)$ for the Coulomb and Higgs categories of line operat...

conjecture | open | MathDB 页面 | 原文线索

V2-058987 | MATHDB-339927

Hecke-category realization conjecture for the theory $\calT[G]$

摘录 / EXCERPT: Let G be a reductive group, let $\Gr_G$ be its affine Grassmannian, and let $D(\Gr_G)^{\operatorname{Hecke}}$ denote the derived category of Hecke eigen-modules, namely D-modul...

conjecture | open | MathDB 页面 | 原文线索

V2-058988 | MATHDB-339929

Local geometric Langlands compatibility conjecture for S-duality

摘录 / EXCERPT: Let G be a group with Langlands dual group $G^{\vee}$, let $\calT$ be a theory with a G-action, and let $\calT^{\vee}$ be its S-dual theory with a $G^{\vee}$-action. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-058989 | MATHDB-339936

Removing hypothesis (ii) from the main theorem

摘录 / EXCERPT: The paper considers the quantum and classical cluster-algebra constructions ${\mathcal A}_q(\Lambda,\widetilde{B})$, ${\mathcal U}_q(\Lambda,\widetilde{B})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-058990 | MATHDB-340034

Conjecture on analytic tangent cones of admissible Hermitian-Yang-Mills connections

摘录 / EXCERPT: Let A be an admissible Hermitian-Yang-Mills connection on $(\mathcal{E},B)$, and let $\hat{\mathcal{E}}$ be any chosen optimal extension of $\mathcal{E}$ at 0. An analytic tang...

conjecture | open | MathDB 页面 | 原文线索

V2-058991 | MATHDB-340201

Banica's universal flat matrix model conjecture for quantum permutation groups

摘录 / EXCERPT: Let S_N^+ be the quantum permutation group, and let X_N be the compact space of $N\times N$ matrices $P=(P_{ij})$ whose entries are rank-one projections in $M_N(\mathbb C)$ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-058992 | MATHDB-340278

King's one-way LOCC distinguishability conjecture for three-dimensional subspaces

摘录 / EXCERPT: Let ${\mathcal C}$ be a three-dimensional subspace of ${\mathbb C}^3 \otimes {\mathbb C}^n$, with $n \ge 3$. A basis of states is reliably distinguished using one-way LOCC when mea...

conjecture | open | MathDB 页面 | 原文线索

V2-058993 | MATHDB-340279

Commuting-compression formulation of King's conjecture

摘录 / EXCERPT: Let ${\mathcal P}$ be the projection onto a three-dimensional subspace of ${\mathbb C}^3 \otimes {\mathbb C}^n$, with $n \ge 3$. Commuting-compression formulation of King's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-058994 | MATHDB-340377

Perez-Garcia-Verstraete-Wolf-Cirac stabilization conjecture

摘录 / EXCERPT: Let L be a linear space of $D\times D$ matrices, and let L^k denote the linear space spanned by all products of k matrices in L. If ... for some k, then ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-058995 | MATHDB-340445

Compactness conjecture for all neutral molecular isomerizations

摘录 / EXCERPT: Let $N=|Z|$ be the neutral case, let $\bar Y_0$ and $\bar Y_1$ be two local minima of $Y\mapsto E_N(Y,Z)$, and define ... Assume $c>\max\{E_N(\bar Y_0,Z),E_N(\bar Y_1,Z)\}$. Compactn...

conjecture | open | MathDB 页面 | 原文线索

V2-058996 | MATHDB-340603

Conjecture that entanglement-assisted Shannon capacity equals the Lovász theta function

摘录 / EXCERPT: Let G be a graph, and let $\Theta_*(G)$ be its entanglement-assisted Shannon capacity. Let $\vartheta(G)$ be the Lovász theta function. Entanglement-assisted capacity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-058997 | MATHDB-340604

Conjecture that quantum and entanglement-assisted Shannon capacities coincide

摘录 / EXCERPT: Let G be a graph, and let $\Theta_q(G)$ and $\Theta_*(G)$ denote its quantum and entanglement-assisted Shannon capacities, respectively. Quantum capacity conjecture. For every gr...

open_problem | open | MathDB 页面 | 原文线索

V2-058998 | MATHDB-340712

The necklace braid group image-size conjecture for the Ising representation

摘录 / EXCERPT: Let V be the two-dimensional vector space and let ... with ρ^R the associated representation of the necklace braid group $\mathcal{NB}_n$. For $n\geq 3$, the necklace braid gr...

conjecture | open | MathDB 页面 | 原文线索

V2-058999 | MATHDB-340755

Negative-temperature tropical Gibbs-measure conjecture for convex bodies

摘录 / EXCERPT: Let $P\subset\mathbb{R}^{n}$ be a convex body containing 0 in its interior, and let $R_P\in(0,1]$ be the invariant of P defined in the source. For fixed $\beta<0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059000 | MATHDB-340835

Bounded L^2-curvature conjecture for the Einstein scalar field equations

摘录 / EXCERPT: Bounded L^2-curvature conjecture. The Einstein scalar field equations admit local Cauchy developments for initial data sets (Σ,g,k,u_{0},u_{1}) with locally finite L^{2}...

conjecture | open | MathDB 页面 | 原文线索

V2-059001 | MATHDB-340938

Carlen-Frank-Lieb conjecture on joint convexity of trace functions

摘录 / EXCERPT: Carlen-Frank-Lieb conjecture. If $1\leq p\leq 2$, $-1\leq q<0$, $(p,q)\ne(1,-1)$, and $s\geq\frac{1}{p+q}$, then $\Psi_{p,q,s}(A,B)$ is jointly convex in (A,B). This is the stron...

conjecture | open | MathDB 页面 | 原文线索

V2-059002 | MATHDB-340971

The polyadic convolution conjecture for multiplace multivalued linear maps

摘录 / EXCERPT: Let $\scalebox{1.15}{\textsf{\upshape A}}^{\left(n\right)}$ and $\scalebox{1.15}{\textsf{\upshape C}}^{\left(n'\right)}$ be a polyadic associative algebra and a coassociative coalg...

conjecture | open | MathDB 页面 | 原文线索

V2-059003 | MATHDB-341026

Numerical criterion for solvability of the deformed Hermitian-Yang-Mills equation

摘录 / EXCERPT: Let (X,ω) be a Kähler manifold, let L be a line bundle or let $[\alpha]$ be a real $(1,1)$ class, and let $Z_V(L)$ denote the central charge associated to each analytic su...

conjecture | open | MathDB 页面 | 原文线索

V2-059004 | MATHDB-341077

The positive mass conjecture for asymptotically hyperbolic manifolds with boundary

摘录 / EXCERPT: Let (M,g,Σ) be an asymptotically hyperbolic manifold with scalar curvature R_g and boundary mean curvature H_g satisfying ... For any admissible chart F, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059005 | MATHDB-341210

Integral truncation-ideal conjecture for shifted quantum affine Coulomb branches

摘录 / EXCERPT: For a dominant coweight $\unl{\lambda}$ and a coweight μ, let $\fU^{\ad,\mu}_\vv[\sz^{\pm1}_1,\ldots,\sz^{\pm1}_N]$ be the integral form of the shifted quantum affine algebra,...

conjecture | open | MathDB 页面 | 原文线索

V2-059006 | MATHDB-341224

Feigin-Frenkel ODE/IM correspondence for higher quantum KdV states

摘录 / EXCERPT: Let a quantum $\widehat{\mathfrak{g}}$-KdV oper be an oper of the form ... with the additional regular singularities chosen to have trivial monodromy for every value of λ. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-059007 | MATHDB-341237

Scaling exponent conjecture for the mixed threshold FA model

摘录 / EXCERPT: Let τ_{0} be the relaxation time at the origin in the mixed threshold FA model, let q be the vacancy density, let ν denote the law of the random environment, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059008 | MATHDB-341367

Multiplicative geometric Langlands pseudo-conjecture

摘录 / EXCERPT: Let C be the curve, D the puncture divisor, and $\omega^\vee$ the chosen coweight data. Let G be a Langlands self-dual group. Multiplicative geometric Langlands pseudo-conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-059009 | MATHDB-341382

Agrachev's universal gap conjecture for quantum control systems

摘录 / EXCERPT: Let Σ be a finite-dimensional quantum control system on $\mathbb{C}^n$ of the form ... where A and the B_i are skew-adjoint matrices, and normalize the drift by...

conjecture | open | MathDB 页面 | 原文线索

V2-059010 | MATHDB-341476

Vanishing defect measure conjecture for finite-time Yang-Mills flow

摘录 / EXCERPT: Finite-time defect measure conjecture. At time $\tau=T$, the singular set satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059011 | MATHDB-341504

Conjecture on the Mayer-series radius after removing collapsing even coefficients

摘录 / EXCERPT: Let b_{2j}, $j=1,\ldots,n$, be the leading even coefficients of the Mayer series, and let β_{2n} and $\beta_{2(n+1)}$ denote the corresponding inverse-temperature threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-059012 | MATHDB-341612

Diffusion conjecture for the almost Mathieu-Markovian equation

摘录 / EXCERPT: Almost Mathieu diffusion conjecture. For almost every $\theta,\alpha\in[0,1]$, the equation has a diffusion constant $D(g,\lambda)\in(0,\infty)$ that is smooth for all...

conjecture | open | MathDB 页面 | 原文线索

V2-059013 | MATHDB-341690

Exact TISGM count conjecture for the ferromagnetic hard-core and soft-core Widom-Rowlinson model

摘录 / EXCERPT: Let $k\geq 4$ and consider the ferromagnetic model with critical values $\theta'_{\rm c}(k)$, $\lambda'_{\rm cr}(k)$, $\theta_{\rm c}(k)$, and $\lambda_{\rm cr}(k)$ as defined in t...

conjecture | open | MathDB 页面 | 原文线索

V2-059014 | MATHDB-341840

Sanda-Shamoto's Dubrovin conjecture for Fano manifolds

摘录 / EXCERPT: Sanda-Shamoto's Dubrovin conjecture. There exists an irregular perverse schober whose nearby cycle is $D^b(X)$ and whose Hochschild decategorification gives Stokes data that is the...

conjecture | open | MathDB 页面 | 原文线索

V2-059015 | MATHDB-341851

The isolated-singularity conjecture for cyclic invariants of quantum affine space

摘录 / EXCERPT: Let $A:=\Bbbk_{-1}[\mathbf{x}]$ and let $G:=C_n$ act as in the paper. A graded isolated singularity means that the invariant ring has the isolated-singularity property in the grade...

conjecture | open | MathDB 页面 | 原文线索

V2-059016 | MATHDB-341959

The parallel-source conjecture for classical de Sitter solutions

摘录 / EXCERPT: Parallel-source conjecture. There is no classical de Sitter solution with parallel sources.

conjecture | open | MathDB 页面 | 原文线索

V2-059017 | MATHDB-341960

The instability conjecture for classical de Sitter solutions with intersecting sources

摘录 / EXCERPT: Intersecting-source instability conjecture. Classical de Sitter solutions with intersecting sources are unstable.

conjecture | open | MathDB 页面 | 原文线索

V2-059018 | MATHDB-341961

The scale-separation conjecture for classical de Sitter solutions

摘录 / EXCERPT: Scale-separation conjecture. Classical de Sitter solutions cannot have at the same time a large internal volume, a small string coupling, a bounded number of orientifolds and quant...

conjecture | open | MathDB 页面 | 原文线索

V2-059019 | MATHDB-341995

Existence and uniqueness of double quantum affinizations for untwisted affine Kac-Moody algebras

摘录 / EXCERPT: Let $\dot{\mathfrak g}$ be an untwisted affine Kac-Moody Lie algebra, and let $\ddot{\mathrm{U}}_q(\dot{\mathfrak g})$ denote its double quantum affinization. Existence and uniquen...

conjecture | open | MathDB 页面 | 原文线索

V2-059020 | MATHDB-341996

The elliptic Hall algebra realization of the positive Cartan subalgebra

摘录 / EXCERPT: Let $\mathcal E_{q_1,q_2,q_3}$ be the elliptic Hall algebra with $q_1q_2q_3=1$, and let $\mathcal E_{q^{-4},q^2,q^2}$ be its specialization at (q^{-4},q^2,q^2). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059021 | MATHDB-341997

The bicontinuous elliptic Hall algebra isomorphism for type a1

摘录 / EXCERPT: Let $\mathcal E_{q^{-4},q^2,q^2}$ be the specialized elliptic Hall algebra, let $\widehat{\mathcal E_{q^{-4},q^2,q^2}}$ denote its completion, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059022 | MATHDB-342012

Emergence of the renormalized Gibbs measure

摘录 / EXCERPT: Emergence of the renormalized Gibbs measure. In the limit $\lambda=T^{-1}\to0$, one obtains the renormalized Gibbs measure in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-059023 | MATHDB-342166

The Kähler structure conjecture for irreducible quantum flag manifolds

摘录 / EXCERPT: Let $\mathcal{O}_q(G/L_S)$ be an irreducible quantum flag manifold, with its Heckenberger--Kolb calculus $Omega^{(bullet,bullet)}$. A covariant Kähler structure consists of the pai...

conjecture | open | MathDB 页面 | 原文线索

V2-059024 | MATHDB-342169

The Dirac-Dolbeault spectral triple conjecture for weak Gelfand-type quantum flag manifolds

摘录 / EXCERPT: Let $\mathcal{O}_q(G/L_S)$ be an irreducible quantum flag manifold of weak Gelfand type, and let $(Omega^{(bullet,bullet)},kappa)$ be its covariant Kähler structure, unique up to r...

conjecture | open | MathDB 页面 | 原文线索

V2-059025 | MATHDB-342203

Distance function scaling limit for square subdivision graphs

摘录 / EXCERPT: Let ${\mathbf{c}_{\mathrm M}}\in(-\infty,25)$, and let $\mathcal S_h^\epsilon$ be the square subdivision graph associated with the field h, equipped with its graph distance. Dist...

conjecture | open | MathDB 页面 | 原文线索

V2-059026 | MATHDB-342243

Conjecture on nonempty parameter sets for PPT entangled edge states

摘录 / EXCERPT: For each integer $n\ge 3$, let $(\alpha_1,\ldots,\alpha_n,\beta_1,\ldots,\beta_n)$ be the parameters used in the paper's construction of PPT states in $M_n\otimes M_n$, and let the...

conjecture | open | MathDB 页面 | 原文线索

V2-059027 | MATHDB-342246

Large-\N critical values for quantum minimal surfaces in flat tori

摘录 / EXCERPT: Let M be a symplectic surface and let S_{cl} be the classical functional for maps from M to a flat torus $(S^1)^n$. For each N, let $\Phi_1,\dots,\Phi_n\in U(N)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059028 | MATHDB-342452

Injectivity conjecture for the horizontal quantum affine subalgebra

摘录 / EXCERPT: Let $U_q\,\widehat{\mathfrak{sl}}_{m|n}$ be the quantum affine algebra and let ${\mathcal E}_{m|n}$ be the quantum toroidal algebra. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-059029 | MATHDB-342540

Continuum-limit conjecture for discretized higher gauge connections

摘录 / EXCERPT: Let M be a manifold and let ${\cal D}$ range over all discretizations of M. Denote by $\boldsymbol{\operatorname{Conn}} /\!/ \boldsymbol{\operatorname{Gauge}}(M,\mathcal{G})$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-059030 | MATHDB-342597

The LQG metric-random planar map convergence conjecture

摘录 / EXCERPT: For $gamma in (0,2)$, let $gamma$-Liouville quantum gravity (LQG) surfaces be equipped with their $gamma$-LQG metric, and let random planar maps in the $gamma$-LQG universality cla...

conjecture | open | MathDB 页面 | 原文线索

V2-059031 | MATHDB-342668

Irrationality conjecture for the Chern-Simons invariant of a hyperbolic surgery

摘录 / EXCERPT: Let 5_2^* be the mirror of the knot 5_2 in Rolfsen’s knot table, and let r_s be the invariant used to study the $\mathit{SU}(2)$-Chern-Simons functional. Irrationality conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-059032 | MATHDB-342728

Conjecture A on graded character-shifted quantum Schubert cells

摘录 / EXCERPT: Let W be the Weyl group, let $w\in W$, and let ϕ be a character. Give $U^-_q(\mathfrak{g})$ the $\mathbb{Z}$-grading determined by $\deg(E_{\alpha_i})=1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-059033 | MATHDB-342729

Conjecture B on triangular Borel subalgebras of quantum groups

摘录 / EXCERPT: Let C be a triangular right coideal subalgebra of the form ... Here $w_+,w_-\in W$, ϕ_+ and ϕ_- are characters, L is the associated lattice, S is the antipode, and...

conjecture | open | MathDB 页面 | 原文线索

V2-059034 | MATHDB-342742

Asymptotic sector-length bound for quantum states

摘录 / EXCERPT: Let A_k denote the k-sector length of an n-qubit quantum state. Asymptotic sector-length conjecture. For all k there exists an n_0 such that, for all $n\geq n_0$, every...

conjecture | open | MathDB 页面 | 原文线索

V2-059035 | MATHDB-342769

Period invariance of the graph-theoretic c_2-invariant

摘录 / EXCERPT: For primitive graphs G_1 and G_2, let $P(G_i)$ denote their periods and let $c_2^{(q)}(G_i)$ denote the reduction modulo q of the graph point-count divided by q^2. Period-i...

conjecture | open | MathDB 页面 | 原文线索

V2-059036 | MATHDB-342783

Constrained minimum output entropy conjecture for Gaussian quantum-limited amplifiers

摘录 / EXCERPT: Constrained minimum output entropy conjecture. For every $n\in\mathbb{N}$, ρ and $\kappa\ge1$,

conjecture | open | MathDB 页面 | 原文线索

V2-059037 | MATHDB-342894

Existence and profile covariance of the modified QED scattering matrix and interacting fields

摘录 / EXCERPT: Let η be a profile, let v be a four-velocity, let $C\in\{F^{\mu\nu},j^\mu\}$ be an observable field, and let $\Psi,\Psi'\in\mathcal{D}_2$. The BRST charge is denote...

conjecture | open | MathDB 页面 | 原文线索

V2-059038 | MATHDB-343066

DELL affine-Laumon wavefunction eigenfunction conjecture

摘录 / EXCERPT: Let $\mathbf{x}=(x_1,\ldots,x_N)$ be the position vector, and let $\mathscr{Z}(p,\mathbf{x})$ be a properly normalized equivariant elliptic genus of the affine Laumon space in the...

conjecture | open | MathDB 页面 | 原文线索

V2-059039 | MATHDB-343067

Ruijsenaars-Schneider defect-partition-function eigenvalue conjecture

摘录 / EXCERPT: Let $\mathbf{x}=(x_1,\ldots,x_N)$ be the position vector of the elliptic Ruijsenaars-Schneider model, and define its wavefunction by ... Let $\vec{\omega}_i$ be the i-th fundamenta...

conjecture | open | MathDB 页面 | 原文线索

V2-059040 | MATHDB-343083

Hyper-Kähler structure conjecture for moduli spaces of contact instantons on 3-Sasakian 7-manifolds

摘录 / EXCERPT: Let (M,g) be a 3-Sasakian 7-manifold, with Sasakian structures satisfying the quaternionic relations ... Let $\mathcal{M}^\ast$ be the moduli space under consideration. The tra...

conjecture | open | MathDB 页面 | 原文线索

V2-059041 | MATHDB-343139

Asymptotic Hamming bound for unrestricted data-syndrome codes

摘录 / EXCERPT: Let d be the distance parameter, let n denote the block length, and let unrestricted DS codes be quantum data-syndrome codes without a non-degeneracy restriction. Write the Ham...

conjecture | open | MathDB 页面 | 原文线索

V2-059042 | MATHDB-343168

Gaitsgory-Lurie spherical Whittaker-affine Lie algebra equivalence conjecture

摘录 / EXCERPT: Let G be a reductive group, let $\check{G}$ be its Langlands dual group, and let κ be a nondegenerate level for G with dual level $\check{\kappa}$ for $\check{G}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059043 | MATHDB-343325

Fukaya's bounding-cochain conjecture for immersed Lagrangians

摘录 / EXCERPT: Let M be a three-manifold with a cylindrical end isometric to $\Sigma \times [0,+\infty)$, and let $P \to M$ be an $SO(3)$-bundle whose restriction to every connected component o...

conjecture | open | MathDB 页面 | 原文线索

V2-059044 | MATHDB-343326

The refined $SO(3)$ Atiyah-Floer conjecture

摘录 / EXCERPT: Let M be a closed three-manifold with a suitable $SO(3)$-bundle $P\to M$ and a suitable splitting $M=M^-\cup M^+$. Let $I(M,P)$ denote the instanton Floer invariant and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059045 | MATHDB-343367

Parity criterion for the Meixner-Pollaczek denominator polynomial

摘录 / EXCERPT: Let a be the Meixner-Pollaczek parameter and let $d_1,\ldots,d_M$ satisfy ... The condition ... is ... This is supported by numerical calculations for small M and d_j, but no analy...

conjecture | open | MathDB 页面 | 原文线索

V2-059046 | MATHDB-343400

The Alexander-polynomial formula for the universal invariant of long knots

摘录 / EXCERPT: Universal-invariant conjecture. The universal invariant associated to B has the form

conjecture | open | MathDB 页面 | 原文线索

V2-059047 | MATHDB-343841

The eigenvalue conjecture for Racah matrices

摘录 / EXCERPT: Let R be a representation, and consider Racah matrices obtained from the tensor product $R\otimes R\otimes R$. Normalize each corresponding $\hat{\mathcal{R}}$-matrix so that the...

conjecture | open | MathDB 页面 | 原文线索

V2-059048 | MATHDB-343842

The eigenvalue conjecture for link Racah matrices

摘录 / EXCERPT: Let $L=(\mathcal{\hat{R}}_1,\mathcal{\hat{R}}_2,\mathcal{\hat{R}}_3)$ and $L'=(\mathcal{\hat{R}}'_1,\mathcal{\hat{R}}'_2,\mathcal{\hat{R}}'_3)$ be triples of $\mathcal{\hat{R}}$-ma...

conjecture | open | MathDB 页面 | 原文线索

V2-059049 | MATHDB-344031

Conjecture on ultraviolet regularization for the holonomy-diffeomorphism algebra

摘录 / EXCERPT: Ultraviolet-regularization conjecture. In either case, it is possible to choose p sufficiently large so that the inner product $$ and the expectation values of the...

conjecture | open | MathDB 页面 | 原文线索

V2-059050 | MATHDB-344079

The handlebody skein-representation composition-factor conjecture

摘录 / EXCERPT: Let Σ_g be a closed genus-g surface and H_g a genus-g handlebody. The reduced skein module $\mathcal{S}_q^{\mathrm{red}}(H_g)$ carries the representation of the skein...

conjecture | open | MathDB 页面 | 原文线索

V2-059051 | MATHDB-344097

Joint scaling limit of the spanning-tree Peano curve and area measure

摘录 / EXCERPT: In the setting of the spanning-tree LQG scaling-limit conjecture, let $\lambda^{\infty,n}$ be the image in $\mathbb C$ of the Peano curve λ^∞ of T^∞ under the u...

conjecture | open | MathDB 页面 | 原文线索

V2-059052 | MATHDB-344235

Denominator formula conjecture for KR modules in type E

摘录 / EXCERPT: Let $d_{l^p,k^m}(z)$ denote the denominator of the normalized R-matrix between the relevant Kirillov-Reshetikhin modules in untwisted affine type $E_n^{(1)}$. Denominator formula...

conjecture | open | MathDB 页面 | 原文线索

V2-059053 | MATHDB-344307

Lefschetz-type conjecture for quantum theta basis coefficients

摘录 / EXCERPT: Let $L_{\\s{V}}$ be the lattice associated to the quantum cluster algebra, let $p\in L_{\\s{V}}$, and let $\mathcal{Q}\in L_{\\s{V},\mathbb{R}}$ be generic. For the quantum theta f...

conjecture | open | MathDB 页面 | 原文线索

V2-059054 | MATHDB-344308

Lefschetz-type preservation for scattering diagrams

摘录 / EXCERPT: Let $\\f{D}_{\operatorname{in}}$ be an initial scattering diagram of the form specified in the source, with incoming wall functions determined by Laurent polynomials $p_i(t)$. A sc...

conjecture | open | MathDB 页面 | 原文线索

V2-059055 | MATHDB-344337

Positive definiteness conjecture for Heckenberger-Kolb Kähler structures

摘录 / EXCERPT: Positive definiteness conjecture. For $q\in\mathbb{R}\setminus\{-1,0\}$, the pair $(\Omega^{(\bullet,\bullet)},\kappa)$ is a positive definite covariant Kähler structure.

conjecture | open | MathDB 页面 | 原文线索

V2-059056 | MATHDB-344338

Dolbeault-Dirac spectral triple conjecture for irreducible quantum flag manifolds

摘录 / EXCERPT: Dolbeault-Dirac spectral triple conjecture. The point spectrum of $D_{\overline{\partial}}$ has finite multiplicity and tends to infinity. Consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-059057 | MATHDB-344411

Lamberti et al.'s metric conjecture for the quantum Jensen-Shannon divergence

摘录 / EXCERPT: Lamberti et al.'s conjecture. $\operatorname{QJSD}^{1/2}$ is a metric on $\mathbb{P}_d$.

conjecture | open | MathDB 页面 | 原文线索

V2-059058 | MATHDB-344483

K-theoretic Hall algebra-quantum affine algebra isomorphism conjecture

摘录 / EXCERPT: Let Q be a quiver, and let $(\widetilde{Q},\widetilde{W})$ be the associated tripled quiver with potential. Let $\mathfrak{g}_Q$ be the Lie algebra associated to Q in the Okoun...

conjecture | open | MathDB 页面 | 原文线索

V2-059059 | MATHDB-344608

Uniqueness conjecture for stationary charged black holes

摘录 / EXCERPT: Let $(\mathcal{M},g_{ab},H_{ab})$ be a stationary Einstein-Maxwell space-time, where $\mathcal{M}$ is a four-dimensional orientable smooth manifold with Lorentzian metric g_{ab},...

conjecture | open | MathDB 页面 | 原文线索

V2-059060 | MATHDB-344638

Superfluity of the third condition in the degeneration criterion

摘录 / EXCERPT: Superfluity conjecture. In the proposition above, condition (3) is superfluous.

conjecture | open | MathDB 页面 | 原文线索

V2-059061 | MATHDB-344663

Central limit conjecture for the Neumann surplus

摘录 / EXCERPT: Neumann surplus central limit conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-059062 | MATHDB-344756

Galois invariance of quantum-dimension distributions in FC-set classes

摘录 / EXCERPT: Galois distribution conjecture. The distribution of quantum dimensions, counted with multiplicity, is the same in C and $galg{\mathsf{C}}{ell}$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-059063 | MATHDB-344774

Maximal-abelianity conjecture for the Galois symmetry group of SIC-POVMs

摘录 / EXCERPT: Let S be the symmetry group of the overlap map of a fiducial, let $C(S)$ be its centralizer in $\mathrm{GL}_2(\mathbb{Z}_{\bar d})$, and let M be a subgroup of $C(S)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-059064 | MATHDB-344778

Coherent categorification conjecture for the affine $SL(2)$ quantum loop algebra

摘录 / EXCERPT: Coherent categorification conjecture. There is an equivalence of triangulated graded monoidal categories

conjecture | open | MathDB 页面 | 原文线索

V2-059065 | MATHDB-344779

Localization conjecture comparing categorifications of the quantum open unipotent cell

摘录 / EXCERPT: Localization equivalence conjecture. The functor $\widetilde\Psi_w$ is a graded monoidal equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-059066 | MATHDB-344934

The quantum A-polynomial conjecture for the knot series

摘录 / EXCERPT: Let $K\subset S^3$ be a knot, let $f_K(x,q)$ be its rescaled knot series, let $\widehat A$ be the quantum polynomial of K, and let $\Delta_K(x)$ be its Alexander polynomial. Defi...

conjecture | open | MathDB 页面 | 原文线索

V2-059067 | MATHDB-345006

Existence of a fundamental algebraic condition for smooth forms theories

摘录 / EXCERPT: A smooth forms theory is a classical theory (Conf,S) whose configuration space Conf is a class of algebra-valued smooth forms. The fundamental-con...

conjecture | open | MathDB 页面 | 原文线索

V2-059068 | MATHDB-345007

Existence of an optimal fundamental algebraic condition for smooth forms theories

摘录 / EXCERPT: A smooth forms theory is a classical theory (Conf,S) whose configuration space Conf is a class of algebra-valued smooth forms. An algebraic condit...

conjecture | open | MathDB 页面 | 原文线索

V2-059069 | MATHDB-345049

Gauge-invariant decomposition conjecture for metric perturbations

摘录 / EXCERPT: Gauge-invariant decomposition conjecture. There exist a tensor field ${\cal H}_{ab}$ and a vector field X^a such that

conjecture | open | MathDB 页面 | 原文线索

V2-059070 | MATHDB-345096

Equal-cost conjecture for truth and falsity of experimental quantum propositions

摘录 / EXCERPT: Let P be an experimental atomic proposition referring to a quantum system in a pure state. Let the cost of a computation be the time required to solve the relevant computational...

conjecture | open | MathDB 页面 | 原文线索

V2-059071 | MATHDB-345097

Quantum parallelism efficiency conjecture

摘录 / EXCERPT: Consider a quantum parallel computation performed by a quantum parallel random-access machine, or QPRAM, and let its efficiency be the ratio of the serial work to the product of th...

conjecture | open | MathDB 页面 | 原文线索

V2-059072 | MATHDB-345098

Valuation symmetry for incompatible quantum propositions

摘录 / EXCERPT: Let $|\Psi\rangle$ be a quantum state, let $\hat{P}$ be the operator associated with an experimental proposition, and let v be the function assigning images to truth values. Valu...

conjecture | open | MathDB 页面 | 原文线索

V2-059073 | MATHDB-345315

Bögli-Kennedy-Lang conjecture on complex Robin eigenvalues

摘录 / EXCERPT: Bögli-Kennedy-Lang conjecture. If $\operatorname{Re}\alpha \to -\infty$, then there exists an infinite family of analytic branches of absolutely divergent eigenvalues such that, if...

conjecture | open | MathDB 页面 | 原文线索

V2-059074 | MATHDB-345324

Polynomiality and commutativity conjecture for the Q-operator

摘录 / EXCERPT: Let $\mathcal Q(z)$ be the Q-operator constructed for the open spin chain, with matrix entries acting on the quantum space and depending on the spectral parameter z. Polynomialit...

conjecture | open | MathDB 页面 | 原文线索

V2-059075 | MATHDB-345333

The C^0 Strong Cosmic Censorship conjecture for charged scalar field black holes

摘录 / EXCERPT: Let g be the maximal globally hyperbolic development of generic regular data for the Einstein-Maxwell-Klein-Gordon system ... where $D=\nabla+iq_0A$. C^0 Strong Cosmic Censorship...

conjecture | open | MathDB 页面 | 原文线索

V2-059076 | MATHDB-345336

Weak Cosmic Censorship conjecture for the Einstein-Maxwell-Klein-Gordon model

摘录 / EXCERPT: Let $\mathcal{I}^+$ denote future null infinity, and consider the admissible data in the one-ended spherically symmetric Einstein-Maxwell-Klein-Gordon model. Weak Cosmic Censorship...

conjecture | open | MathDB 页面 | 原文线索

V2-059077 | MATHDB-345339

The cube-term conjecture for the Hidden Kernel Problem

摘录 / EXCERPT: Let $\mathbb{A}$ be a finite algebra, and let a cube term be a term operation $t(x_1,\dots,x_n)$ such that, for every $i\in[n]$, there is a choice of $u_1,\dots,u_n\in\{x,y\}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-059078 | MATHDB-345340

The antisymmetrizer quotient conjecture for tensor Hecke algebras

摘录 / EXCERPT: Let N and n be integers with $n>N$. Let $H_{\boldsymbol{k},n}(q)$ be the tensor Hecke algebra, let $\overline{H}^N_{\boldsymbol{k},n}(q)$ be the corresponding centraliser quoti...

conjecture | open | MathDB 页面 | 原文线索

V2-059079 | MATHDB-345341

Centrality conjecture for the tensor antisymmetrizer

摘录 / EXCERPT: Let $H_{\boldsymbol{k},N+1}(q)$ be the tensor Hecke algebra on $N+1$ ellipses, and let $\mathrm{AS}_{\boldsymbol{k},N+1}(q)$ be its antisymmetrizer element. Centrality conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-059080 | MATHDB-345426

Donald's conjecture on G_2-instantons on the standard 7-sphere

摘录 / EXCERPT: Donald's conjecture. A is equivalent, modulo the action of ${\mathrm{Spin}}(7)$ and ${\mathscr G}_E$, to the pullback of a HermitianYangMills connection on...

conjecture | open | MathDB 页面 | 原文线索

V2-059081 | MATHDB-345443

Stability conjecture for near-biprojections in subfactor planar algebras

摘录 / EXCERPT: Suppose $\mathscr{P}_{\bullet,\pm}$ is an irreducible subfactor planar algebra. For $x\in\mathscr{P}_{2,\pm}$, let $\mathfrak{F}(x)$ denote its Fourier transform, and let P,Q be...

conjecture | open | MathDB 页面 | 原文线索

V2-059082 | MATHDB-345444

Quantum central limit conjecture for the block map

摘录 / EXCERPT: Let $n\geqslant 1$, let $f\in L^{\infty}(\mathbb{R}^n)\cap L^{1}(\mathbb{R}^n)\cap L^{2}(\mathbb{R}^n)$, and let B_{λ} be the block map. Quantum central limit conjecture. U...

conjecture | open | MathDB 页面 | 原文线索

V2-059083 | MATHDB-345445

Hirschman-Beckner entropy monotonicity conjecture for the block map

摘录 / EXCERPT: Let $n\geqslant 1$, let $f\in L^{\infty}(\mathbb{R}^n)\cap L^{1}(\mathbb{R}^n)\cap L^{2}(\mathbb{R}^n)$, and let B_{λ} be the block map. Hirschman-Beckner entropy conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-059084 | MATHDB-345501

Linear-length measurement sequence conjecture for fault-tolerant error correction

摘录 / EXCERPT: Linear-length measurement sequence conjecture. For any such family, there exists a fault-tolerant error-correction sequence consisting of $O(d)$ parity-check measurements.

conjecture | open | MathDB 页面 | 原文线索

V2-059085 | MATHDB-345547

Conjecture on canonical q-oper coordinates and generalized Baxter TQ-relations for Miura-Plücker opers

摘录 / EXCERPT: Let G be a simply laced group, let $T_j(z)$ be a canonical coordinate of a (G,q)-oper, and let $Q^i_+(z)$ and $\Lambda_i(z)$, for $i=1,\ldots,r$, be the corresponding polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-059086 | MATHDB-345597

PPT-square conjecture for block Schur products

摘录 / EXCERPT: Let M_A be a matrix algebra, and let $\varrho_1,\varrho_2\in M_A\otimes M_A$ be states. Let $\varrho_1\square\varrho_2\in M_A\otimes(M_A\otimes M_A)$ denote their block Schur pro...

conjecture | open | MathDB 页面 | 原文线索

V2-059087 | MATHDB-345722

Hawking's chronology protection conjecture

摘录 / EXCERPT: Hawking's chronology protection conjecture. The laws of physics do not allow the appearance of closed timelike curves.

conjecture | open | MathDB 页面 | 原文线索

V2-059088 | MATHDB-345764

Nekrasov's semiclassical-limit conjecture for noncommutative HYM solutions

摘录 / EXCERPT: Let $\hbar$ be the deformation parameter, and let ... denotes the usual Hermitian Yang-Mills equation with parameter λ. Nekrasov's semiclassical-limit conjecture. In the limi...

conjecture | open | MathDB 页面 | 原文线索

V2-059089 | MATHDB-346033

Finite-component conjecture for global hot spots

摘录 / EXCERPT: Finite-component conjecture. The set M always has finitely many connected components; in particular, it is either finite or uncountable.

conjecture | open | MathDB 页面 | 原文线索

V2-059090 | MATHDB-346034

Generic two-hot-spots conjecture for quantum graphs

摘录 / EXCERPT: Generic two-hot-spots conjecture. Generically, $mu_2(amma)$ is simple and the corresponding eigenfunction has exactly one minimum and one maximum, so that $|M|=2$.

conjecture | open | MathDB 页面 | 原文线索

V2-059091 | MATHDB-346035

Rationally independent edge-length hot-spots conjecture

摘录 / EXCERPT: Rationally independent edge-length conjecture. There exists a graph $gamma$ whose edge lengths are pairwise rationally independent and for which $|M|=3$; more generally, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-059092 | MATHDB-346036

Doubly connected graphs with arbitrarily close hot spots conjecture

摘录 / EXCERPT: Close hot-spots conjecture. For every $epsilon>0$, there exists a doubly connected graph $gamma$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059093 | MATHDB-346037

Pumpkin hot-spots diameter conjecture

摘录 / EXCERPT: Pumpkin hot-spots conjecture. The set M has exactly two points and

conjecture | open | MathDB 页面 | 原文线索

V2-059094 | MATHDB-346132

Thomas-Yau Bridgeland stability conjecture for dHYM and special Lagrangians

摘录 / EXCERPT: Let X be a Calabi-Yau manifold, let $L\rightarrow X$ be a holomorphic line bundle, and let $L\hookrightarrow X$ be a Lagrangian submanifold. The objects L are viewed respective...

conjecture | open | MathDB 页面 | 原文线索

V2-059095 | MATHDB-346133

Converse conjecture to the dHYM stability inequalities for Kähler threefolds

摘录 / EXCERPT: Let (X,ω) be a Kähler threefold and let $L\rightarrow X$ be a holomorphic line bundle. Suppose that the Chern number inequality holds, the lifted angle is well-defined with...

conjecture | open | MathDB 页面 | 原文线索

V2-059096 | MATHDB-346145

The Witten-style conjecture for finite-order mapping tori

摘录 / EXCERPT: Let X be a smooth closed oriented 4-manifold such that ... where $\widetilde X$ is the universal abelian cover of X. The invariants $\lambda_{\,\rm{SW}}(X)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059097 | MATHDB-346281

Multiplicative reachability conjecture for dual canonical bases

摘录 / EXCERPT: Let w be a Weyl group element, let ${\mathsf{A_q}}[N_{-}(w)]$ be the corresponding quantum unipotent subgroup, and let $\mathsf{B}^{{\mathsf{up}}}(w)$ be its dual canonical basis...

conjecture | open | MathDB 页面 | 原文线索

V2-059098 | MATHDB-346315

The fluid ball conjecture for asymptotically flat static perfect fluid spacetimes

摘录 / EXCERPT: Let an asymptotically flat static perfect fluid spacetime be a solution satisfying the equations denoted by and. Fluid ball conjecture. Every such solution must be spherically symm...

conjecture | open | MathDB 页面 | 原文线索

V2-059099 | MATHDB-346345

Schur-Weyl duality for roots of unity and twisted Lawrence-Krammer-Bigelow representations

摘录 / EXCERPT: Let ξ be a primitive $4m$th root of 1, and let the mth twisted Lawrence-Krammer-Bigelow representation be the representation referred to below. Schur-Weyl duality conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-059100 | MATHDB-346462

The ADO limit formula for $\widehat Z$-invariants

摘录 / EXCERPT: Let K be a knot, let p be a positive integer, and let $\operatorname{ADO}_p(x;K)$ denote the Akutsu-Deguchi-Ohtsuki polynomial invariant associated with the quantum group...

conjecture | open | MathDB 页面 | 原文线索

V2-059101 | MATHDB-346654

The (a,t)-deformed F_K interpolation conjecture for knots

摘录 / EXCERPT: The (a,t)-deformed F_K conjecture. There should exist a four-variable function satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-059102 | MATHDB-346655

The t-deformed ADO polynomial limit conjecture for F_K

摘录 / EXCERPT: The t-deformed ADO limit conjecture. The preceding expression should be a polynomial and, at $t=-1$, should equal the usual p-th ADO polynomial of K for $SU(2)$. More generally...

conjecture | open | MathDB 页面 | 原文线索

V2-059103 | MATHDB-346719

Nonlinear stability conjecture for supersymmetric compactifications in three spatial dimensions

摘录 / EXCERPT: Let $(\mathbb{R}^{1+3}\times K,\hat{g})$ be a spacetime with a supersymmetric compactification. Nonlinear stability conjecture. Spacetimes with a supersymmetric compactification an...

conjecture | open | MathDB 页面 | 原文线索

V2-059104 | MATHDB-346745

Parity congruences for hard-hexagon partition functions

摘录 / EXCERPT: Parity-coefficient conjecture. For $s\in\{2,4\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-059105 | MATHDB-346791

The rank-one projection conjecture at the exceptional parameter

摘录 / EXCERPT: Let $d\geq 3$ be an integer, let $|x\rangle$ be a vector on the unit sphere in $\mathbf R^{d-1}$, and construct matrices A_m using $V_m=2|x\rangle\langle x|-I_{d-1}$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-059106 | MATHDB-347118

Projective-injective splitting conjecture for extension modules

摘录 / EXCERPT: Splitting conjecture. There exist R and $\mathbb{G}$ such that $\operatorname{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}))$ is injective as an $R[\mathbb{G}]$-module, ever...

conjecture | open | MathDB 页面 | 原文线索

V2-059107 | MATHDB-347259

Generalized presentation conjecture for quantum symmetric pair coideals

摘录 / EXCERPT: Let $X \subseteq I$ be of finite type and let $\tau \in \operatorname{Aut}_X(A)$ satisfy $\tau^2=\operatorname{id}_I$ and $\tau|_X=\operatorname{id}_X$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-059108 | MATHDB-347260

Parameter-inversion conjecture for universal K-matrices

摘录 / EXCERPT: Parameter-inversion conjecture. There exists $g\in(U_q\mathfrak h)^{\mathcal W,\times}$ such that this relation is satisfied. Consequently, $k_{Y,\eta}$ specializes to a solution o...

conjecture | open | MathDB 页面 | 原文线索

V2-059109 | MATHDB-347443

The conjecture on differentiability of the Ising magnetization away from criticality

摘录 / EXCERPT: Let G be an infinite Cayley graph. The magnetization $m^+(\beta,h)$ is the plus-state magnetization, with $\beta,h\in[0,\infty)$ and β_c the critical inverse temperature. Di...

conjecture | open | MathDB 页面 | 原文线索

V2-059110 | MATHDB-347467

Hayashi's quiver quotient conjecture for finite-dimensional weak bialgebras

摘录 / EXCERPT: Hayashi's quiver quotient conjecture. Every finite-dimensional weak bialgebra H with commutative counital subalgebras is isomorphic to a weak bialgebra quotient of $\hay$ for som...

conjecture | open | MathDB 页面 | 原文线索

V2-059111 | MATHDB-347544

Lienardy-Stéphan conjecture for logarithmic bipartite fidelity scalar products

摘录 / EXCERPT: Let $1\leqslant m\leqslant N$ and let $N_1,\dots,N_m\geqslant 1$ be integers satisfying $N=N_1+\dots+N_m$. For the ground-state vectors of the open XXZ chain, define ... The scalar p...

conjecture | open | MathDB 页面 | 原文线索

V2-059112 | MATHDB-347545

Conjecture relating XXZ-chain component sums to totally symmetric alternating sign matrices

摘录 / EXCERPT: Component-sum conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-059113 | MATHDB-347570

The Askey-Wilson quotient conjecture for the centralizer of three $U_q(\mathfrak{sl}_2)$ representations

摘录 / EXCERPT: Let j_1,j_2,j_3 be representation labels, let $AW(3)$ be the centrally extended Askey-Wilson algebra, and let $\mathcal{C}_{j_1,j_2,j_3}$ be the centralizer under consideration....

conjecture | open | MathDB 页面 | 原文线索

V2-059114 | MATHDB-347571

The direct-sum Askey-Wilson quotient conjecture for the centralizer

摘录 / EXCERPT: Let j_1,j_2,j_3 be representation labels and let $\mathcal{J}(j_1,j_2,j_3)$ be the set of admissible intermediate labels. For each $k\in\mathcal{J}(j_1,j_2,j_3)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059115 | MATHDB-347645

Lusztig braid-group cluster-map conjecture for parabolic positive representations

摘录 / EXCERPT: Let $\mathcal{P}_\lambda^J$ be a parabolic positive representation, let $\mathcal{X}_q^{\mathbf{D}(\overline{\mathbf{i}})}$ be its associated doubled quantum torus, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059116 | MATHDB-347704

Quantum geometric Langlands correspondence

摘录 / EXCERPT: Quantum geometric Langlands correspondence. There is an equivalence of DG categories

conjecture | open | MathDB 页面 | 原文线索

V2-059117 | MATHDB-347814

Collins-J-Yau conjecture for the deformed Hermitian-Yang-Mills equation

摘录 / EXCERPT: Collins-J-Yau conjecture. The class $[\alpha]$ admits a solution to the deformed Hermitian-Yang-Mills equation with supercritical phase if and only if $Z(X)\neq 0$ and, for every a...

conjecture | open | MathDB 页面 | 原文线索

V2-059118 | MATHDB-347864

Collins's chiral-ring K-stability conjecture

摘录 / EXCERPT: Let (X,ζ) be a polarized ring with symmetry or Reeb vector field ζ. The ring is the chiral ring of a four-dimensional $\mathcal{N}=1$ superconformal field theory (SCFT)...

conjecture | open | MathDB 页面 | 原文线索

V2-059119 | MATHDB-347942

Spectral gap for the BGNST piston model

摘录 / EXCERPT: Let $N\in\mathbb Z^+$ be the number of ball-piston pairs aligned along a one-dimensional chain, and consider the associated piston model. Spectral-gap conjecture. The spectral gap...

conjecture | open | MathDB 页面 | 原文线索

V2-059120 | MATHDB-347951

The simple type conjecture for mod 2 Seiberg-Witten invariants

摘录 / EXCERPT: Let X be a closed, connected, oriented, smooth 4-manifold with $b_2^+(X)>1$. Simple type conjecture. X has Seiberg-Witten simple type. This conjecture asserts a fundamental c...

conjecture | open | MathDB 页面 | 原文线索

V2-059121 | MATHDB-348009

Reverse Burnett's conjecture on approximating Einstein-massless Vlasov solutions by vacuum spacetimes

摘录 / EXCERPT: Reverse Burnett's conjecture. Any solution to the Einstein-massless Vlasov system arises as a limit of solutions to the Einstein vacuum equations in the sense described above.

conjecture | open | MathDB 页面 | 原文线索

V2-059122 | MATHDB-348051

Dequantization conjecture for quantum spectral curves

摘录 / EXCERPT: Let ${\mathcal W}$ be a point of the Sato Grassmannian with wave function Ψ and quantum spectral curve operator ${\mathtt P}_{\mathcal W}$, so that ... For a tau-function with a...

conjecture | open | MathDB 页面 | 原文线索

V2-059123 | MATHDB-348098

Extension of the double-twist knot formulas to genus-one knots

摘录 / EXCERPT: Theorems concerning the ADO invariant and the relationship between CGP and WRT invariants are stated for double twist knots. A Seifert genus 1 knot is a knot whose Seifert genus is...

conjecture | open | MathDB 页面 | 原文线索

V2-059124 | MATHDB-348128

The weak BLZ conjecture for monster potentials

摘录 / EXCERPT: Weak BLZ conjecture. For every α and L, the number of solutions of the BLZ system is at most $p(N)N!$. For generic values of α and L, the number of solutions is...

conjecture | open | MathDB 页面 | 原文线索

V2-059125 | MATHDB-348189

The charge-refined q-series identity for the principal subspace of type B

摘录 / EXCERPT: Charge-refined q-series conjecture. The following identity holds:

conjecture | open | MathDB 页面 | 原文线索

V2-059126 | MATHDB-348314

Maximum-entropy extension conjecture for locally consistent marginals

摘录 / EXCERPT: Consider a set of locally consistent marginals satisfying the entropy conditions given by Eqs. … . Here, $S(X)$ denotes the entropy of the marginal on region X, and A, B, and...

conjecture | open | MathDB 页面 | 原文线索

V2-059127 | MATHDB-348485

The generic MPDO conjecture on exponential decay of conditional mutual information

摘录 / EXCERPT: Generic MPDO conjecture. For a generic' MPDO, there exists a constant $c>0$ such that for any tripartition ABC in which B separates A from C by distance ℓ,

conjecture | open | MathDB 页面 | 原文线索

V2-059128 | MATHDB-348486

Completely contractive data-processing conjecture for conditional mutual information

摘录 / EXCERPT: Completely contractive data-processing conjecture for conditional mutual information. For any channel $\mathcal{E}:C\rightarrow C'$ with trivial correctable algebra, there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-059129 | MATHDB-348487

Completely contractive data-processing conjecture for trace-norm conditional mutual information

摘录 / EXCERPT: Completely contractive data-processing conjecture for trace-norm conditional mutual information. If

conjecture | open | MathDB 页面 | 原文线索

V2-059130 | MATHDB-348529

Conjugate-tuple characterization of Clifford hierarchy gates

摘录 / EXCERPT: Conjugate-tuple conjecture. Gates of the $k+1$-th level of the Clifford hierarchy, up to phase, are in bijective correspondence with conjugate tuples of k-th level gates.

conjecture | open | MathDB 页面 | 原文线索

V2-059131 | MATHDB-348595

Converse to the state-extension theorem for logical exclusivity

摘录 / EXCERPT: State-extension conjecture. If state ν of A satisfies PEP, then there is a state $\hat{\nu}$ of $A[\perp]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059132 | MATHDB-348610

Composite quantum Chernoff conjecture for finitely many hypotheses

摘录 / EXCERPT: Composite quantum Chernoff conjecture. The limit $p_e(\vec \varrho,\{\vec\sigma_1,\ldots,\vec\sigma_r\})$ exists, and

conjecture | open | MathDB 页面 | 原文线索

V2-059133 | MATHDB-348653

Typicality conjecture for 2-independent sets in random 3-regular graphs

摘录 / EXCERPT: Let α^*_3 denote the threshold appearing in the definition of 2-independent-set-typicality. For $\alpha \in \left(0,\alpha^*_3\right)$ and $\varepsilon>0$, a 3-regular graph...

conjecture | open | MathDB 页面 | 原文线索

V2-059134 | MATHDB-348840

Quantum Laurent integrality conjecture for the ${\rm SL}_3$ duality map

摘录 / EXCERPT: Let Δ be an ideal triangulation of a triangulable generalized marked surface $\frak{S}$, let $\mathbb{I}^q_\Delta$ be the proposed quantum duality map, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059135 | MATHDB-348847

Quantum Laurent coefficient positivity conjecture

摘录 / EXCERPT: Let Δ be an ideal triangulation of $\frak{S}$ and let $\mathbb{I}^q_\Delta$ be the proposed quantum duality map. Quantum Laurent coefficient positivity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-059136 | MATHDB-348848

Quantum structure-constant positivity conjecture

摘录 / EXCERPT: Let $\mathbb{I}^q_\Delta$ be the quantum duality map and let $c^q(\ell,\ell';\ell”)$ denote the coefficients in the product expansion of its basis elements from the stated quantum-...

conjecture | open | MathDB 页面 | 原文线索

V2-059137 | MATHDB-349003

The full-support eigenfunction conjecture for compact metric graphs

摘录 / EXCERPT: Full-support eigenfunction conjecture. There exists a choice of eigenfunctions ψ_n forming an orthonormal basis of $L^2(\mathcal{G})$ and a subsequence...

conjecture | open | MathDB 页面 | 原文线索

V2-059138 | MATHDB-349011

Variant of the quantum Singleton bound for hybrid stabilizer codes

摘录 / EXCERPT: An $\left[\!\left[n,k\!:\!m,d\!:\!c\right]\!\right]_{q}$ hybrid stabilizer code has length n, quantum dimension k, classical dimension m, quantum minimum distance d, and cl...

conjecture | open | MathDB 页面 | 原文线索

V2-059139 | MATHDB-349280

The graphical-basis conjecture for quantum cluster monomials

摘录 / EXCERPT: Let $\mathsf{BWeb}_{\mathfrak{sl}_{3},{\Sigma}}$ be the graphical basis of the quantum $\mathfrak{sl}_3$ skein algebra of Σ, and let cluster monomials be monomials formed fr...

conjecture | open | MathDB 页面 | 原文线索

V2-059140 | MATHDB-349288

Local Debye-potential generation conjecture for linearized gravity on vacuum type II spacetimes

摘录 / EXCERPT: Debye-potential generation conjecture. All real solutions of the linearized Einstein vacuum equations on a vacuum type II background can be locally obtained, up to gauge, as the re...

conjecture | open | MathDB 页面 | 原文线索

V2-059141 | MATHDB-349296

Nonvanishing of minimizers of the approximate Levy-Lieb functional

摘录 / EXCERPT: Nonvanishing-minimizer conjecture. Any minimizer ψ of $F^{\boldsymbol{\alpha},(0)}(r)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059142 | MATHDB-349301

The uniqueness conjecture for thermal averages in two-dimensional disordered spin systems

摘录 / EXCERPT: Fix a disordered spin system of the type considered in the paper in dimension $d=2$. Let $\beta>0$ be the inverse temperature, $\tau_1,\tau_2\text{?}$ be boundary conditions in...

conjecture | open | MathDB 页面 | 原文线索

V2-059143 | MATHDB-349453

The quintic quantum K-theory formula in terms of Gopakumar-Vafa invariants

摘录 / EXCERPT: The quintic quantum K-theory conjecture. The small J-function of the quintic is expressed linearly in terms of the GV-invariants by

conjecture | open | MathDB 页面 | 原文线索

V2-059144 | MATHDB-349523

The double-optics doubling conjecture

摘录 / EXCERPT: Let X,A,Y,B be objects of the underlying actegories, and let $((X,A),(Y,B))$ be an object of the category of double lenses over the optic category. A context on $((X,A),(Y,B))$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-059145 | MATHDB-349681

Completeness conjecture for generalized controlled-not identities on Boolean isomorphisms

摘录 / EXCERPT: Completeness conjecture for generalized controlled-not identities. These identities are complete for $\operatorname{Iso}(\mathsf{FinSet}_2)$: every equality of Boolean isomorphisms...

conjecture | open | MathDB 页面 | 原文线索

V2-059146 | MATHDB-349752

Non-existence conjecture for universal Lie bialgebroid quantizations

摘录 / EXCERPT: Let a universal quantization of Lie bialgebroids mean a quantization whose formulas are given by expansions in terms of graphs with universal coefficients, and let a quantum groupo...

conjecture | open | MathDB 页面 | 原文线索

V2-059147 | MATHDB-349758

Monotonic decrease of information leakage with memory-input correlation

摘录 / EXCERPT: Let the memory and input state be correlated, and quantify the information leakage associated with the physical context, with or without memory. Monotonicity conjecture. With incre...

conjecture | open | MathDB 页面 | 原文线索

V2-059148 | MATHDB-349852

The leading-word conjecture for Lyndon loop root vectors

摘录 / EXCERPT: Let Δ^+ be the set of positive roots, let $d\in\mathbb{Z}$, and let $\ell(\alpha,d)$ denote the Lyndon loop word of degree (α,d). Let Φ^L be the homomorphism fro...

conjecture | open | MathDB 页面 | 原文线索

V2-059149 | MATHDB-349933

The Virasoro conjecture for admissible Landau-Ginzburg A-model pairs

摘录 / EXCERPT: Let (W,G) be an admissible Landau-Ginzburg pair, and let $\mathcal{A}^{\clubsuit}_{W,G}$ denote its total ancestor potential in FJRW, Polishchuk-Vaintrob, or Kiem-Li theory. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059150 | MATHDB-349976

The shifted Yangian identification for $Y^{\mu}(\mathfrak{g})$

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra under consideration, let μ be the corresponding shift, and let $Y^{\mu}(\mathfrak{g})$ be the associative algebra defined by the RLL relat...

conjecture | open | MathDB 页面 | 原文线索

V2-059151 | MATHDB-350230

Asymptotic expansion conjecture for discrete Fourier transforms of quantum $6j$-symbols

摘录 / EXCERPT: Let (I,J) partition $\{1,\dots,6\}$, and let $\{(\mathbf b_I^{(r)},\mathbf a_J^{(r)})\}$ be colorings defining deeply truncated tetrahedra $\Delta^{(r)}$ with dihedral angles...

conjecture | open | MathDB 页面 | 原文线索

V2-059152 | MATHDB-350286

Irreducibility and genus conjecture for quantum KdV spectral curves

摘录 / EXCERPT: For each $k\geq 1$, let $\Sigma_{k,m}$, for $m\geq 1$, be the weight-k spectral curve ... Here $\mathscr{P}_k$ denotes the set of partitions of weight k, and $\ell(\lambda)$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-059153 | MATHDB-350304

Future timelike completeness conjecture for CMC Cauchy surfaces

摘录 / EXCERPT: Let (M,g) be a spacetime with compact Cauchy surfaces. CMC existence conjecture. If (M,g) is future timelike geodesically complete and satisfies the strong energy condition, th...

conjecture | open | MathDB 页面 | 原文线索

V2-059154 | MATHDB-350305

Dilts-Holst conjecture on CMC Cauchy surfaces

摘录 / EXCERPT: Let (M,g) be a spacetime with compact Cauchy surfaces satisfying the strong energy condition. Dilts-Holst conjecture. If (M,g) has a Cauchy surface of constant signed mean curv...

conjecture | open | MathDB 页面 | 原文线索

V2-059155 | MATHDB-350307

The split field-redefinition conjecture for direct products of CYMH gauge theories

摘录 / EXCERPT: Let N be a smooth manifold such that its tangent bundle admits a CYMH GT, and let $K\to S$ be a Lie algebroid bundle over a smooth manifold S which also admits a CYMH GT. Set .....

conjecture | open | MathDB 页面 | 原文线索

V2-059156 | MATHDB-350609

Global signed-crystal structure conjecture for the quantum alcove model

摘录 / EXCERPT: Let λ be an arbitrary weight and let Γ be a reduced chain of roots corresponding to λ. Write $\mathcal{A}(\Gamma)$ for the associated quantum alcove model, a...

conjecture | open | MathDB 页面 | 原文线索

V2-059157 | MATHDB-350687

Non-standard central limit conjecture for the edge-triangle model

摘录 / EXCERPT: Let S_n be the number of non-zero elements of the random adjacency matrix under $\mathbb{P}_{n;\alpha,h}$, let (α_c,h_c) be the critical point, and let $u^*(\alpha_c,h_c)$...

conjecture | open | MathDB 页面 | 原文线索

V2-059158 | MATHDB-350713

Generic failure of conformal smoothness at future null infinity for Einstein vacuum scattering

摘录 / EXCERPT: Let the Einstein vacuum equations be posed as a scattering problem with conformally regular data on an ingoing null hypersurface and no incoming radiation from past null infinity....

conjecture | open | MathDB 页面 | 原文线索

V2-059159 | MATHDB-350830

Monotonicity conjecture for contextual fraction under n-contextuality

摘录 / EXCERPT: Let a contextuality model be evaluated in the hierarchy of n-contextuality, where increasing n permits inclusion of larger simplices in the underlying simplicial complex, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-059160 | MATHDB-350849

Finite-regularity logarithmic Price's law conjecture for higher angular modes

摘录 / EXCERPT: Consider solutions ϕ of the wave equation on a fixed Schwarzschild background. Let Γ be the timelike initial hypersurface, let t be a time coordinate, let u and v...

conjecture | open | MathDB 页面 | 原文线索

V2-059161 | MATHDB-350930

Regularized $+1$-surgery formula

摘录 / EXCERPT: Let K be a knot and consider the $+1$-surgery formula of Gukov and Manolescu for the invariant $\hat{Z}$. Regularized $+1$-surgery conjecture. When the $+1$-surgery formula conve...

conjecture | open | MathDB 页面 | 原文线索

V2-059162 | MATHDB-350931

Regularized $+1/r$-surgery formula

摘录 / EXCERPT: Let K be a knot, let r be the surgery parameter, and write ... Let ... be the Ramanujan theta function. Regularized $+1/r$-surgery conjecture. When the $+1/r$-surgery formula of Gu...

conjecture | open | MathDB 页面 | 原文线索

V2-059163 | MATHDB-350955

Second-order stationary-point conjecture for quantum compiling

摘录 / EXCERPT: Let $V_{\mathsf{ct}}(\cdot)$ be the circuit-to-unitary map, let U be a target unitary, and define ... A stationary point ${\mathbold{\theta}}$ is a point where the gradient vanishe...

conjecture | open | MathDB 页面 | 原文线索

V2-059164 | MATHDB-350956

Surjectivity conjecture for sequential and spin CNOT layouts

摘录 / EXCERPT: For n qubits, let $\mathsf{sequ}(L)$ denote the layout consisting of L CNOT units in sequence, and let $\mathsf{spin}(L)$ denote the alternating spin layout. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059165 | MATHDB-350968

The generalized Atiyah-Jones conjecture for four-manifolds

摘录 / EXCERPT: Let X be a 4-dimensional real manifold. Let $\mathcal{M}_k$ be the moduli space of $SU(2)$ instantons on X of charge k, let P_k be a principal bundle with $c_2(P_k)=k$, l...

conjecture | open | MathDB 页面 | 原文线索

V2-059166 | MATHDB-351015

Polynomial identity conjecture for asymmetric quantum Rabi model constraint polynomials

摘录 / EXCERPT: Polynomial identity conjecture. For all natural numbers $N,\ell\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-059167 | MATHDB-351016

Density conjecture for degenerate parameter curves in the asymmetric quantum Rabi model

摘录 / EXCERPT: For a fixed integer bias parameter ℓ, define ... The inclusion ... expresses that the parameter values producing degeneracies do not exhaust the parameter plane. Density conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-059168 | MATHDB-351039

Okounkov's monodromy formula for wall operators

摘录 / EXCERPT: Okounkov's wall-operator conjecture. The wall operator is given by

conjecture | open | MathDB 页面 | 原文线索

V2-059169 | MATHDB-351057

The polynomial-factor hardness conjecture for the Shortest Vector Problem

摘录 / EXCERPT: Let SVP denote the Shortest Vector Problem on lattices, and let a polynomial-factor approximation mean an approximation within a factor bounded by a polynomial in the la...

conjecture | open | MathDB 页面 | 原文线索

V2-059170 | MATHDB-351094

Conjecture that the BMS group is not locally exponential

摘录 / EXCERPT: Let the BMS group be the semidirect product $\operatorname{BMS}=\mathcal{S}\rtimes_\sigma \mathcal{M}$, where $\mathcal{S}$ is the supertranslation group and $\mathcal{M}$ is the Mö...

conjecture | open | MathDB 页面 | 原文线索

V2-059171 | MATHDB-351140

Generalized Riemannian Penrose inequality with charge and cosmological constant

摘录 / EXCERPT: Let (M,γ,E,Λ) be an electrically charged Riemannian manifold of dimension $n+1\geq3$ which is geodesically complete up to its closed, strictly outward minimizing minim...

conjecture | open | MathDB 页面 | 原文线索

V2-059172 | MATHDB-351195

The proposed vector space basis for the quantum symplectic sphere

摘录 / EXCERPT: Let ${\mathcal O}(S_q^{4n-1})$ be the coordinate algebra of the quantum symplectic sphere, with generators x_j and y_j and their involutive partners x_j^* and y_j^*. For no...

conjecture | open | MathDB 页面 | 原文线索

V2-059173 | MATHDB-351312

Leclerc's conjecture on products of dual canonical basis elements

摘录 / EXCERPT: Leclerc's conjecture. The product has the form

conjecture | open | MathDB 页面 | 原文线索

V2-059174 | MATHDB-351457

The KHA-affine quantum algebra isomorphism for quivers

摘录 / EXCERPT: Let Q be a quiver, with doubled quiver Q^d and tripled quiver $\widetilde{Q}$. Let the torus $\mathbb{C}^*$ scale the linear maps corresponding to edges of Q^d with weight...

conjecture | open | MathDB 页面 | 原文线索

V2-059175 | MATHDB-351468

The link-level root-of-unity relation for CGP invariants

摘录 / EXCERPT: Let L be a framed link in S^3 with V components, let B be its linking matrix, and let L_{α} denote the link with labels...

conjecture | open | MathDB 页面 | 原文线索

V2-059176 | MATHDB-351577

The linear H^1-formulation of Strong Cosmic Censorship for AdS black holes

摘录 / EXCERPT: Let ψ be a linear perturbation solving the wave equation on Reissner--Nordström--AdS or Kerr--AdS, arising from generic Cauchy data with Dirichlet boundary conditions imposed...

conjecture | open | MathDB 页面 | 原文线索

V2-059177 | MATHDB-351578

Kehle's boundedness conjecture for Lebesgue-generic Kerr--AdS black holes

摘录 / EXCERPT: Let $\mathcal P$ be the parameter space of dimensionless Kerr--AdS black hole parameters $(\mathfrak m,\mathfrak a):=(M\sqrt{-\Lambda},a\sqrt{-\Lambda})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059178 | MATHDB-351613

Al-Rashed-Zegarliński monotonicity conjecture below Schatten exponent two

摘录 / EXCERPT: Al-Rashed-Zegarliński's conjecture. If $p\in[1,2)$, then for every $X\in B(H)$ and every unital completely positive trace-preserving map ϕ on $B(H)$,

conjecture | open | MathDB 页面 | 原文线索

V2-059179 | MATHDB-351614

Al-Rashed-Zegarliński conjecture on monotonicity of the Schatten quadratic form

摘录 / EXCERPT: Al-Rashed-Zegarliński's quadratic-form conjecture. For every completely positive trace-preserving map ϕ on $B(H)$ and all $A,B,D\in B(H)$,

conjecture | open | MathDB 页面 | 原文线索

V2-059180 | MATHDB-351638

Top-row conjecture for the refined quantum modularity matrix

摘录 / EXCERPT: Let $r+1$ be the size of the matrix $\mathbf J$, and let $\mathrm{DJ}^{(m)}(q)$ denote the descendant invariant. The matrix $\mathbf J$ is the matrix of the conjectured knot invari...

conjecture | open | MathDB 页面 | 原文线索

V2-059181 | MATHDB-351639

Kashaev's matrix-valued descendant colored Jones conjecture

摘录 / EXCERPT: Let K be a knot, let N be a strictly positive integer, let n be an integer parameter, and let ζ be a primitive N-th root of unity. Write $\langle K\rangle_{N,n}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-059182 | MATHDB-351682

The initial-data-set rigidity conjecture for Euclidean prisms

摘录 / EXCERPT: Let $n\geq 3$, let P^n be a Euclidean prism, $P_0\times[0,1]^{n-2}$, and let (Ω^n,g,k) be an initial data set admitting a degree-one map onto P. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-059183 | MATHDB-351689

The quantization conjecture for quantum unipotent subgroups

摘录 / EXCERPT: Quantization conjecture. Up to scalar multiples in $q^{\frac{1}{2}}\mathbb{Z}}$, the quantum unipotent subgroup ${\mathsf{A_q}}[N_{-}(w)]$ is a quantum cluster algebra, and its du...

conjecture | open | MathDB 页面 | 原文线索

V2-059184 | MATHDB-351691

The positivity conjecture for cluster expansions

摘录 / EXCERPT: Positivity conjecture. The coefficients c_n for cluster monomials are contained in $\mathbb{N}$ in the classical case, or in $\mathbb{N}[v^{\pm}]$ in the quantum case.

conjecture | open | MathDB 页面 | 原文线索

V2-059185 | MATHDB-351724

Exponential convergence conjecture for the sixth-order quantum-diffusion equation

摘录 / EXCERPT: Let u be one of the weak solutions to the sixth-order quantum-diffusion equation constructed in the proof of the existence theorem, let u^0 be its initial datum, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059186 | MATHDB-351746

Quantum Markov conjecture for thermal states

摘录 / EXCERPT: Quantum Markov conjecture. For arbitrary quantum Gibbs states,

conjecture | open | MathDB 页面 | 原文线索

V2-059187 | MATHDB-351771

The differential-operator description of spherical operators for holomorphic twists

摘录 / EXCERPT: Let G be a reductive algebraic group. Let $R_2=H^\bu(\mathbb{A}^2,\mathcal{O}_{\mathbb{A}^2}^{\mathrm{alg}})$, let $\mathfrak{g}$ be the Lie algebra of G, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059188 | MATHDB-351792

Seiberg-duality invariance of the Newton polynomial and Mahler measure

摘录 / EXCERPT: Consider Seiberg-dual dimer models with canonical edge weights, and let $P_1(z,w)$ and $P_2(z,w)$ be their Newton polynomials. Let n_i be the number of edges, equivalently multip...

conjecture | open | MathDB 页面 | 原文线索

V2-059189 | MATHDB-351793

Seiberg-duality invariance of the Ronkin function

摘录 / EXCERPT: Consider Seiberg-dual dimer models, with the canonical weight choice, and let $R(x,y)$ be the Ronkin function associated with their Newton polynomial. Ronkin Seiberg-duality conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-059190 | MATHDB-351908

Reichardt diagonalizing sequence convergence conjecture

摘录 / EXCERPT: Diagonalizing sequence conjecture. For every $U_0\in\operatorname{U}(2)$, the sequences satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-059191 | MATHDB-351942

Paracausal equivalence criterion for globally hyperbolic spacetimes

摘录 / EXCERPT: Paracausal equivalence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059192 | MATHDB-352138

Undecidability of tensor stable positive maps

摘录 / EXCERPT: Tensor stable positivity conjecture. The set of yes-instances of tensor stable positivity, denoted $\textsc{tsp}$, is not recursively enumerable.

conjecture | open | MathDB 页面 | 原文线索

V2-059193 | MATHDB-352182

Conjectural two-sided bounds for harmonic measure in the Liouville quantum gravity setting

摘录 / EXCERPT: Let $\widehat{q}(y)$ be the quantity defined in the paper from the harmonic measure of the embedded graph, and let y tend to infinity in the Euclidean embedding. Harmonic-measure...

conjecture | open | MathDB 页面 | 原文线索

V2-059194 | MATHDB-352395

Exponentially strong converse conjecture for coherent quantum channel discrimination

摘录 / EXCERPT: Exponentially strong converse conjecture. Any coherent strategy achieving a Type II error exponent larger than the regularized channel divergence...

conjecture | open | MathDB 页面 | 原文线索

V2-059195 | MATHDB-352502

Strong-converse entanglement cost conjecture for the two-qutrit state ω_3

摘录 / EXCERPT: Strong-converse entanglement cost conjecture. The state ω_3 satisfies

open_problem | open | MathDB 页面 | 原文线索

V2-059196 | MATHDB-352511

Unitary-equivalence conjecture for three-copy SIC estimation fidelity

摘录 / EXCERPT: Let $\mathscr{A}$ and $\mathscr{B}$ be SICs in dimension 3, and let $F(\mathscr{A}^{\otimes 3})$ and $F(\mathscr{B}^{\otimes 3})$ denote their three-copy estimation fidelities. S...

conjecture | open | MathDB 页面 | 原文线索

V2-059197 | MATHDB-352775

The simple type conjecture for smooth four-manifolds

摘录 / EXCERPT: Simple type conjecture. Every closed, connected, oriented and smooth four-manifold with $b_2^+\ge 2$ is of simple type.

conjecture | open | MathDB 页面 | 原文线索

V2-059198 | MATHDB-352954

The geometric and algebraic R-matrix correspondence for the rank-one Fock space

摘录 / EXCERPT: R-matrix correspondence conjecture. The endomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-059199 | MATHDB-352955

Hopf algebra isomorphism between the quantum toroidal algebra and the FRT algebra

摘录 / EXCERPT: Hopf algebra isomorphism conjecture. The map Υ induces a Hopf algebra isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-059200 | MATHDB-353061

NSA for inverse temperatures between saturation and criticality

摘录 / EXCERPT: Let s be a direction and let $\beta_{\rm sat}(s)$ and β_c denote the saturation and critical inverse-temperature thresholds, respectively. NSA conjecture. NSA holds for any...

conjecture | open | MathDB 页面 | 原文线索

V2-059201 | MATHDB-353110

Commuting-limits conjecture for the regularized one-dimensional point interaction

摘录 / EXCERPT: Let β be the expansion parameter, let a be the regularization parameter, and let $\psi_{a,\beta}(x)$ denote the function given by Theorem. At fixed a and x, consider an...

conjecture | open | MathDB 页面 | 原文线索

V2-059202 | MATHDB-353111

The master pentagon identity for Ding-Iohara-Miki algebra generators

摘录 / EXCERPT: Master pentagon identity. The generating functions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-059203 | MATHDB-353237

Fractional-statistics emergence conjecture for quasihole tracer particles

摘录 / EXCERPT: Fractional-statistics emergence conjecture. As $N\to+\infty$, the identity

conjecture | open | MathDB 页面 | 原文线索

V2-059204 | MATHDB-353282

Singer's master-field conjecture for Yang-Mills holonomies

摘录 / EXCERPT: Let Σ be a closed, connected, orientable, two dimensional, Riemannian manifold, the Euclidean plane $\mathbb{R}^2$, or a disc in $\mathbb{R}^2$. Let G_N be a classical gro...

conjecture | open | MathDB 页面 | 原文线索

V2-059205 | MATHDB-353284

Large-\N master-field convergence conjecture for Yang\Mills theory on surfaces

摘录 / EXCERPT: Let G_N be a classical compact matrix Lie group of size N, and let Σ be a compact surface, the Euclidean plane $\mathbb{R}^2$, or the Poincare9 disc...

conjecture | open | MathDB 页面 | 原文线索

V2-059206 | MATHDB-353337

The fixed-point conjecture for Coulomb branches

摘录 / EXCERPT: Let $\mathcal M(\lambda,\mu)$ be the Coulomb branch associated with a quiver gauge theory, let $\chi\colon\mathbb C^\times\to\pi_1(\mathbf G)^\wedge$ be the one-parameter subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-059207 | MATHDB-353338

Geometric Satake conjecture for Kac-Moody Coulomb branches

摘录 / EXCERPT: Let $\mathcal M(\lambda,\mu)$ be the Coulomb branch associated with a quiver gauge theory, let χ be the one-parameter subgroup defined by a character of the gauge group, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-059208 | MATHDB-353434

Compatibility conjecture for sequences satisfying the length condition

摘录 / EXCERPT: Compatibility conjecture. For any ${\widetilde{w}}$ satisfying this condition, the pair $(\Lambda^{\widetilde{w}},\widetilde{\mathsf{B}}^{\widetilde{w}})$ is compatible, namely

conjecture | open | MathDB 页面 | 原文线索

V2-059209 | MATHDB-353497

Kundt's conjecture on curvature alignment and degenerate Kundt spacetimes

摘录 / EXCERPT: Kundt's conjecture. If all curvature tensors of (M,g) are of aligned type II or more special with respect to some null congruence, then (M,g) is degenerate Kundt: for each poin...

conjecture | open | MathDB 页面 | 原文线索

V2-059210 | MATHDB-353503

All-order resurgent asymptotics for the third-order resonant renormalon

摘录 / EXCERPT: All-order resurgent asymptotics. The coefficients satisfy the asymptotic expansion

conjecture | open | MathDB 页面 | 原文线索

V2-059211 | MATHDB-353504

Converse positivity conjecture for rank-2 quantum cluster scattering diagrams

摘录 / EXCERPT: Consider a rank-2 cluster algebra whose exchange matrix has the form ... Let $G^{\leq 4}$ denote the truncation through degree 4 used in the source, and let Theorem be the source's...

conjecture | open | MathDB 页面 | 原文线索

V2-059212 | MATHDB-353556

The third-derivative conjecture for the first band function at its minimum

摘录 / EXCERPT: Let μ_{1} be the first band function, let ξ_{0} denote its minimum point, and write $\mu_{1}^{(3)}(\xi_{0})$ for its third derivative there. Third-derivative conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-059213 | MATHDB-353661

Complete anisotropy conjecture for the small quantum group of \(\mathfrak{sl}_2\)

摘录 / EXCERPT: Complete anisotropy conjecture. The category $\mathcal{C}$ is completely anisotropic: the only rigid Frobenius algebra in $\mathcal{C}$ is the unit object.

conjecture | open | MathDB 页面 | 原文线索

V2-059214 | MATHDB-353678

Sorkin's quantum-field singularity conjecture for Morse geometries

摘录 / EXCERPT: Let $\mathcal{M}$ be a compact cobordism of dimension n, h a Riemannian metric, $\zeta>1$ a constant, and f a Morse function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059215 | MATHDB-353694

Quantum environmental entropy and invertibility conjecture

摘录 / EXCERPT: Let $\mathcal{W}$ be a quantum process with quantum environmental entropy S_{EC}, dispersive entropy S_{dis}, and mixing entropy...

conjecture | open | MathDB 页面 | 原文线索

V2-059216 | MATHDB-353720

Conjectured bounds relating quantum and private capacities

摘录 / EXCERPT: Capacity bounds conjecture. For a quantum channel ${\cal N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-059217 | MATHDB-353844

Existence of positive tensor rank decompositions for unital symmetric real tensors

摘录 / EXCERPT: Let P_{abc} be a unital symmetric real tensor, meaning that its indices are symmetric and its unit conditions are $P_{1ab}=P_{a1b}=P_{ab1}=\delta_{ab}$. A positive tensor rank de...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-059218 | MATHDB-353845

Pointwise minimal decomposition conjecture for unital tensors with covering systems

摘录 / EXCERPT: Let P_{abc} be a unital tensor that admits a covering system of partial algebras. A pointwise decomposition is a positive tensor rank decomposition ... such that every factor p^i...

conjecture | open | MathDB 页面 | 原文线索

V2-059219 | MATHDB-353910

Positivity conjecture for generalized quantum cluster algebras

摘录 / EXCERPT: Let $\mathcal{A}(X,\mathbf{h},\Lambda,\widetilde{B})$ be a generalized quantum cluster algebra with initial extended cluster $\{X_1,\ldots,X_m\}$. A Laurent polynomial in this clus...

conjecture | open | MathDB 页面 | 原文线索

V2-059220 | MATHDB-353986

Geometric representation-theoretic equivalence from formality

摘录 / EXCERPT: Formality-implies-equivalence conjecture. If the Formality Conjecture holds, then this non-monoidal pushforward restricts, via the Kempf embedding, to an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-059221 | MATHDB-354016

The degree-eight algebraic boundary conjecture for the rough disordered region

摘录 / EXCERPT: Let τ and ξ be the parameters describing the model, and let the rough disordered region denote the region in the (τ,ξ)-parameter plane associated with rough disorde...

conjecture | open | MathDB 页面 | 原文线索

V2-059222 | MATHDB-354043

Connectedness conjecture for the equivariant gauge group

摘录 / EXCERPT: Let $Q\simeq\mathbb{R}^4$ carry a linear action of $\hat{\Gamma}$, with nontrivial $\Gamma\subset SU(2)$. Let $T^{\vee}_{\Lambda}$ be the relevant dual space, let R_{Γ} be t...

conjecture | open | MathDB 页面 | 原文线索

V2-059223 | MATHDB-354062

Tensor-product conjecture for regular positive representations

摘录 / EXCERPT: Tensor-product conjecture. The tensor product of regular positive representations is again regular, and it can be decomposed into a direct integral of regular positive representati...

conjecture | open | MathDB 页面 | 原文线索

V2-059224 | MATHDB-354133

Large-coupling absence of ground states in the infrared-critical spin boson model

摘录 / EXCERPT: Large-coupling absence conjecture. There exists $\lambda_{\mathsf c}>0$ such that $H(\lambda,0)$ has no ground state for

conjecture | open | MathDB 页面 | 原文线索

V2-059225 | MATHDB-354227

Sharp quantum reverse Young inequality for irreducible planar algebras

摘录 / EXCERPT: Let $\mathscr{P}_{\bullet}$ be an irreducible planar algebra with index δ^2, where $\delta>0$. For positive operators $x,y\in\mathscr{P}_{2,\pm}$ and parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-059226 | MATHDB-354275

Non-trivial phase conjecture for RFP MPDOs from the Lee-Yang C*-weak Hopf algebra

摘录 / EXCERPT: Let the Lee-Yang C-weak Hopf algebra be the algebra described in Example, and let the associated RFP MPDOs be the renormalization fixed-point matrix product density operators const...

conjecture | open | MathDB 页面 | 原文线索

V2-059227 | MATHDB-354328

The tropical parametrization conjecture for the quantum lamination basis

摘录 / EXCERPT: Tropical parametrization conjecture. The basis $\mathbb{I}_{\mathcal{X}}^q(\mathcal{L}_{\mathfrak{sl}_3}^p(\Sigma,\mathbb{Z}))$ is parametrized by tropical points. Namely, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-059228 | MATHDB-354352

GHPL convergence of unmarked loop O(n) planar maps to the LQG disk

摘录 / EXCERPT: Unmarked planar-map scaling-limit conjecture. There exist unique constants $d(\kappa)>0$ and $c,c'>0$ such that the rescaled spaces converge in law to...

conjecture | open | MathDB 页面 | 原文线索

V2-059229 | MATHDB-354393

Quantum Fourier entropy-influence conjecture for qubit systems

摘录 / EXCERPT: Quantum Fourier entropy-influence conjecture. There is a constant c independent of n such that

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059230 | MATHDB-354394

Small incremental magic conjecture

摘录 / EXCERPT: Small incremental magic conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-059231 | MATHDB-354501

Instability conjecture for Kerr-anti-de Sitter spaces

摘录 / EXCERPT: Let Kerr-anti-de Sitter spaces be solutions of the Einstein equations with cosmological constant, considered as an initial boundary value problem with Dirichlet-type boundary condi...

conjecture | open | MathDB 页面 | 原文线索

V2-059232 | MATHDB-354502

Brandão's generalised quantum Stein's lemma

摘录 / EXCERPT: Let $(\mathcal{M}_n)_n$ be a family of sets of quantum states satisfying Axioms 1--5 in Section, and let $D^\infty_{\mathcal{M}}(\rho)$ denote the corresponding regularised relativ...

conjecture | open | MathDB 页面 | 原文线索

V2-059233 | MATHDB-354664

ADM mass and surface-gravity inequality for stationary black holes

摘录 / EXCERPT: Let M be the ADM mass, κ the surface gravity, and $\mathcal{A}$ the event-horizon area of a 4-dimensional asymptotically flat, static or axisymmetric stationary black hole...

conjecture | open | MathDB 页面 | 原文线索

V2-059234 | MATHDB-354668

Canonical Higgsing conjecture for generic quiver qq-characters

摘录 / EXCERPT: Let Γ be a generic quiver and let $w=(w_i)_{i\in\Gamma_0}\in\mathbb{Z}_{\ge 0}^{\operatorname{rk}\Gamma}$ be a generic weight dimension vector. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059235 | MATHDB-354669

Higgsing conjecture for Kirillov-Reshetikhin qq-characters

摘录 / EXCERPT: Let Γ be a finite-type quiver and fix $i\in\Gamma_0$. Consider a weight dimension vector with a single nonzero entry, $w_i\neq 0$ and $w_j=0$ for $j\neq i$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059236 | MATHDB-354686

Exact catalytic entropy conjecture

摘录 / EXCERPT: Exact catalytic entropy conjecture. Such a density matrix σ and unitary U exist if and only if $H(\rho)<H(\rho')$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059237 | MATHDB-354723

The iterative surface-Hadamard conjecture for the semiclassical Yukawa model

摘录 / EXCERPT: Let χ be the classical scalar field, let $\omega_{\mathcal C}$ be an initial state on an initial surface $\mathcal C$, and consider the three 3-smeared initial two-point funct...

conjecture | open | MathDB 页面 | 原文线索

V2-059238 | MATHDB-354724

The surface-Hadamard well-posedness conjecture for the semiclassical Yukawa model

摘录 / EXCERPT: Let $\mathcal C$ be an initial constant-time surface, let $(\chi_{\rm in},\dot\chi_{\rm in})$ be classical initial data on $\mathcal C$, and let $\omega_{\mathcal C}$ be a state on...

conjecture | open | MathDB 页面 | 原文线索

V2-059239 | MATHDB-354726

The semiclassical-gravity iterative surface-Hadamard conjecture

摘录 / EXCERPT: Let $\mathcal C$ be an initial surface, let $\omega_{\mathcal C}$ be an initial state, and let $g^{(r)}$ denote the rth time derivative of the metric on $\mathcal C$. Semiclassic...

conjecture | open | MathDB 页面 | 原文线索

V2-059240 | MATHDB-354727

The surface-Hadamard well-posedness conjecture for semiclassical gravity

摘录 / EXCERPT: Let $\mathcal C$ be a 3-manifold with classical initial data $(g^{\rm in}_{ij},\dot g^{\rm in}_{ij},g^{(2)\,\rm in}_{ij},g^{(3)\,\rm in}_{ij})$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-059241 | MATHDB-354728

The order-$\hbar^0$ initial-value conjecture for semiclassical gravity

摘录 / EXCERPT: Let $\mathcal C$ be a 3-manifold with $\hbar$-dependent initial metric $g^{\rm in}_{ij}$ and extrinsic curvature $K^{\rm in}_{ij}$, and let $\omega_{\mathcal C}$ be an $\hbar$-depe...

conjecture | open | MathDB 页面 | 原文线索

V2-059242 | MATHDB-354769

The singularity-infinity separation conjecture

摘录 / EXCERPT: Let (M,g) be a smooth Lorentzian manifold with a physically relevant choice of curve family $\mathcal{C}$, such as geodesics with affine parameter, C^1 curves with generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-059243 | MATHDB-354770

The black-region exclusion conjecture for points at infinity

摘录 / EXCERPT: Let (M,g) be a spacetime, let $I^+(p)$ be the chronological future of $p\in M$, and let $\mathscr{B}$ denote the black region. Black-region exclusion conjecture. If $p\in M$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-059244 | MATHDB-354771

Global weak cosmic censorship conjecture

摘录 / EXCERPT: Let (M,g) be a generic, maximal, physically admissible spacetime admitting a complete spacelike hypersurface Σ. Let $D^+(\Sigma)$ denote its future domain of dependence, l...

conjecture | open | MathDB 页面 | 原文线索

V2-059245 | MATHDB-354844

Collins-Yau's Nakai-Moishezon criterion for the dHYM equation

摘录 / EXCERPT: Collins-Yau's conjecture. The following are equivalent:

conjecture | refuted | MathDB 页面 | 原文线索

V2-059246 | MATHDB-355130

The ADO-3-Links-Gould specialization conjecture for links

摘录 / EXCERPT: ADO-3-Links-Gould specialization conjecture. For every link L,

conjecture | open | MathDB 页面 | 原文线索

V2-059247 | MATHDB-355156

Convergence conjecture for Hermitian-Yang-Mills metrics under divisor degeneration

摘录 / EXCERPT: Let $\overline{E}$ be a holomorphic vector bundle on $\overline{X}$ such that $\overline{E}|_D$ is $c_1(N_D)$-polystable and $\overline{E}$ is $(c_1(D),[\omega_0])$-stable. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059248 | MATHDB-355202

Rationality and upper bound conjecture for adjacent AKLT edge projectors

摘录 / EXCERPT: Let S_1, S_2, and S_3 be positive integers or half-integers. For adjacent edges $e=\{1,2\}$ and $e'=\{2,3\}$, let $P_{12}=P_{S_1+S_2}(\vec{S}_1+\vec{S}_2)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059249 | MATHDB-355261

The IPT-squared conjecture for real and complex matrix algebras

摘录 / EXCERPT: Let $\mathbb{K}\in\{\mathbb{R},\mathbb{C}\}$, and let $\Phi:M_n(\mathbb{K})\rightarrow M_n(\mathbb{K})$ be an IPT map, meaning that its Choi matrix is invariant under partial trans...

conjecture | open | MathDB 页面 | 原文线索

V2-059250 | MATHDB-355271

Orthosymplectic Satake conjecture for quantum orthosymplectic supergroups

摘录 / EXCERPT: Let n and k be nonnegative integers, let $c\in\mathbb{C}^{\times}\setminus\mathbb{Q}^{\times}$, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059251 | MATHDB-355272

Gaiotto's conjecture for quantum general linear supergroups

摘录 / EXCERPT: Let $0<K<N-1$ be positive integers, let $c\in\mathbb{C}^{\times}\setminus\mathbb{Q}^{\times}$, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059252 | MATHDB-355273

Exceptional Lie superalgebra F(4) Satake conjecture

摘录 / EXCERPT: Let $e\in\mathfrak{sp}(6)$ be a nilpotent element of Jordan type $(3,3)$, let $\operatorname{SL}(2)$ be a maximal reductive subgroup of its centralizer, and choose an...

conjecture | open | MathDB 页面 | 原文线索

V2-059253 | MATHDB-355351

The collimation-chamber conjecture for affine-root-lattice theories

摘录 / EXCERPT: Consider five-dimensional theories with physical moduli spaces containing highly fine-tuned regions and with underlying affine root lattices. A collimation chamber is a region in w...

conjecture | open | MathDB 页面 | 原文线索

V2-059254 | MATHDB-355512

Instability conjecture for three-dimensional CSC ALH manifolds with thin conformal-infinity necks

摘录 / EXCERPT: A three-dimensional constant-scalar-curvature asymptotically locally hyperbolic (CSC ALH) manifold is a three-dimensional asymptotically locally hyperbolic manifold with constant s...

conjecture | open | MathDB 页面 | 原文线索

V2-059255 | MATHDB-355597

The free easy quantum group factor conjecture

摘录 / EXCERPT: Free easy quantum group factor conjecture. The associated von Neumann algebra $L^\infty(G)$ is a factor.

conjecture | open | MathDB 页面 | 原文线索

V2-059256 | MATHDB-355598

The abstractly free quantum group factor conjecture

摘录 / EXCERPT: Abstractly free factor conjecture. The associated von Neumann algebra $L^\infty(G)$ is a factor.

conjecture | open | MathDB 页面 | 原文线索

V2-059257 | MATHDB-355599

The free complexification factor conjecture

摘录 / EXCERPT: Free complexification factor conjecture. The von Neumann algebra $A=L^\infty(\widetilde{G})$ is a factor.

conjecture | open | MathDB 页面 | 原文线索

V2-059258 | MATHDB-355600

The affine homogeneous space free complexification factor conjecture

摘录 / EXCERPT: Affine homogeneous space factor conjecture. The algebra $A=L^\infty(\widetilde{X})$ is a factor.

conjecture | open | MathDB 页面 | 原文线索

V2-059259 | MATHDB-355643

Hohenberg-Kohn conjecture for density-potential determination

摘录 / EXCERPT: Hohenberg-Kohn conjecture. For every $\rho\in\mathcal{D}^N$, either

conjecture | open | MathDB 页面 | 原文线索

V2-059260 | MATHDB-355765

Coates-Corti-Iritani-Tseng's quantum Lefschetz conjecture without convexity

摘录 / EXCERPT: Let G be an abelian group acting in a CI-GIT presentation, and let $Y\mathord{/{\mkern-6mu/}}G$ be the corresponding quotient with projective coarse moduli space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059261 | MATHDB-355798

Parameter-generality conjecture for the mixed tetrahedron equations

摘录 / EXCERPT: The 3D R-operators in the equations under consideration are constructed from the representations π_O and π_Z, with parameters assigned to the six tensor components as des...

conjecture | open | MathDB 页面 | 原文线索

V2-059262 | MATHDB-355936

Cancellation-free identity for graded characters of level-zero Demazure submodules in type C

摘录 / EXCERPT: Cancellation-free identity. There exists some $m\leq l\leq n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059263 | MATHDB-356075

Christodoulou's trapped-surface conjecture for terminal indecomposable past sets

摘录 / EXCERPT: Let (M,g) be the maximal future development of generic asymptotically flat initial data. A terminal indecomposable past set is a past set $\mathcal{P}$ in M of the type describ...

conjecture | open | MathDB 页面 | 原文线索

V2-059264 | MATHDB-356107

Quantum Bohnenblust-Hille inequality for low-degree operators

摘录 / EXCERPT: Quantum Bohnenblust-Hille conjecture. Fix $d\ge 1$. There exists $C_d>0$, depending only on d, such that for all $n\ge 1$ and every degree-at-most-d operator...

conjecture | open | MathDB 页面 | 原文线索

V2-059265 | MATHDB-356108

The quantum Borel invariant conjecture for the SO(3) WRT invariant

摘录 / EXCERPT: Let q be a primitive root of unity of odd order N. For every closed oriented framed 3-manifold M with $b_1(M)=0$, let $Z(M,f;u_q(\mathfrak{sl}_2^+))$ denote the invariant a...

conjecture | open | MathDB 页面 | 原文线索

V2-059266 | MATHDB-356135

S-duality conjecture for supergroup superquiver gauge theories

摘录 / EXCERPT: S-duality conjecture. S-duality relates supergroup superquiver gauge theories by rotating their brane-web diagrams by 90 degrees. In particular, supergroup ordinary gauge theory...

conjecture | open | MathDB 页面 | 原文线索

V2-059267 | MATHDB-356136

Order independence of equal-rank superquiver partition functions

摘录 / EXCERPT: Equal-rank order-independence conjecture. The partition function Z does not depend on the order of the NS5-branes. In particular, the resulting partition functions are related to...

conjecture | open | MathDB 页面 | 原文线索

V2-059268 | MATHDB-356204

The bi-Galois conjecture for the twist between Manin's universal quantum groups

摘录 / EXCERPT: Let V be a 4-dimensional vector space over $\Bbbk$ with basis $\{x_0,x_1,x_2,x_3\}$. Let $H=\underline{\operatorname{aut}}^l(\Bbbk[x_0,x_1,x_2,x_3])$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059269 | MATHDB-356338

Trivial LGBs are associated with pre-classical theories

摘录 / EXCERPT: Trivial LGBs are associated with pre-classical theories. There exists a $\lambda \in \Omega^1(M;\mathcal{g})$ such that $\widetilde{\nabla^{\mathrm{YM}}}^{\lambda}$ is flat.

conjecture | open | MathDB 页面 | 原文线索

V2-059270 | MATHDB-356481

Conjecture that inflation-NPA convergence yields commutator bilocality

摘录 / EXCERPT: Let $\vec{Q}=\{q(abc|xyz)\}$ be a distribution in the bilocal scenario. Say that $\vec{Q}$ passes the bilocal inflation-NPA hierarchy when it satisfies the hierarchy of compatible...

conjecture | open | MathDB 页面 | 原文线索

V2-059271 | MATHDB-356677

UV line operators and the Fueter endomorphism category conjecture

摘录 / EXCERPT: Let $\mathcal{T}$ be a four-dimensional $\mathcal{N}=2$ field theory with Seiberg-Witten integrable system (M,Ω_I) and holomorphic Lagrangian section S. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059272 | MATHDB-356687

Predictivity conjecture for effective quantum general relativity

摘录 / EXCERPT: Predictivity conjecture. (Effective) Quantum General Relativity, possibly coupled to matter from the Standard Model, is predictive.

conjecture | open | MathDB 页面 | 原文线索

V2-059273 | MATHDB-356688

Transversality criterion for quantum gauge theories

摘录 / EXCERPT: Transversality criterion. The condition

conjecture | open | MathDB 页面 | 原文线索

V2-059274 | MATHDB-356689

Graviton density-family independence conjecture for perturbative amplitudes

摘录 / EXCERPT: Graviton density-family conjecture. The perturbative expansion with respect to $\mathfrak{h}^{\{\omega\}}_{\mu\nu}$ depends on ω only through the external-leg structure. Con...

conjecture | open | MathDB 页面 | 原文线索

V2-059275 | MATHDB-356895

Free-energy formula for the one-dimensional 3-permaspin model

摘录 / EXCERPT: Let $c=e^{-2\beta H}$ and $d=e^{-2\beta J}$, and consider the one-dimensional 3-permaspin model with couplings J and H. Let z be the unique root of the cubic ... whose argument...

conjecture | open | MathDB 页面 | 原文线索

V2-059276 | MATHDB-356983

Non-periodicity conjecture for toroidal grids

摘录 / EXCERPT: Let $n,m \geq 3$ with $(n,m) \neq (4,4)$. A toroidal (n,m)-grid is the toroidal grid with parameters n and m, and periodic means periodic under the discrete-time quantum walk...

conjecture | open | MathDB 页面 | 原文线索

V2-059277 | MATHDB-357030

Many-copy robustness equality conjecture for Hoggar states

摘录 / EXCERPT: Let $\ket{\rm Hog}$ be a 3-qubit Hoggar state, obtained from the fiducial state $\ket{\rm Hog_0} \propto (-1+2i,1,1,1,1,1,1,1)^T$ by the action of the Pauli group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059278 | MATHDB-357035

The vacuum third law of black hole thermodynamics

摘录 / EXCERPT: Let g be a spacetime metric satisfying the vacuum Einstein equations ... A Cauchy datum is regular and one-ended when it is regular and has one asymptotically flat end, and gravita...

conjecture | open | MathDB 页面 | 原文线索

V2-059279 | MATHDB-357185

Mittleman's conjecture on the Dirac-Fock ground state energy

摘录 / EXCERPT: Let e_q denote the physical ground state energy of the electron-positron Hartree-Fock model, obtained by maximizing the model's ground state energy over all allowed one-particle...

conjecture | open | MathDB 页面 | 原文线索

V2-059280 | MATHDB-357186

Weak Mittleman's conjecture for the Dirac-Fock ground state energy

摘录 / EXCERPT: Weak Mittleman's conjecture. The Dirac-Fock model is an approximation of the electron-positron Hartree-Fock model in the precise sense of the displayed inequality. This weaker stat...

conjecture | open | MathDB 页面 | 原文线索

V2-059281 | MATHDB-357287

The linear-factor conjecture for Wie-length

摘录 / EXCERPT: Let S be a finite generating system of $M_n(\mathbb{C})$, with $\ell(S)$ its ordinary length and $\operatorname{Wie}\ell(S)$ its Wie-length. Linear-factor conjecture for Wie-leng...

conjecture | open | MathDB 页面 | 原文线索

V2-059282 | MATHDB-357288

The quadratic injectivity-threshold conjecture

摘录 / EXCERPT: Let $S=\{A_1,\ldots,A_g\}\subseteq M_n(\mathbb{C})$ satisfy condition $C1$, meaning that words of some common length span $M_n(\mathbb{C})$, and let Γ_L be the associated ma...

conjecture | open | MathDB 页面 | 原文线索

V2-059283 | MATHDB-357448

Quantum chi-independence for quasimap generating series

摘录 / EXCERPT: For a fixed rank ${\mathsf r}$, define the generating series obtained by summing over the flux index,...

conjecture | open | MathDB 页面 | 原文线索

V2-059284 | MATHDB-357451

Quantum chi-independence for Gromov-Witten invariants

摘录 / EXCERPT: Let $\check{\mathsf{GW}}_{{\mathsf d},{\mathsf w}}^{\bullet}$ and $\hat{\mathsf{GW}}_{{\mathsf d},{\mathsf w}}^{\bullet}$ denote the corresponding Gromov-Witten invariants. Quantum...

conjecture | open | MathDB 页面 | 原文线索

V2-059285 | MATHDB-357549

Generalized SLE-bubble welding conjecture with a boundary insertion

摘录 / EXCERPT: Fix $W_1\geq\frac{\gamma^2}{2}$ and $W>2$. Let $\mathrm{Bubble}_{\mathbb H}(p)$ denote the space of bubbles in $\mathbb H$ rooted at p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059286 | MATHDB-357569

Skein-module Langlands duality for closed oriented 3-manifolds

摘录 / EXCERPT: Skein-module Langlands duality conjecture. There is a linear isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-059287 | MATHDB-357604

Converse characterization of controlled Markovian quantum generators

摘录 / EXCERPT: The preceding theorem concerns a collection of generators for controlled Markovian quantum dynamics, with the relevant notation and generator class defined there. Converse characte...

conjecture | open | MathDB 页面 | 原文线索

V2-059288 | MATHDB-357698

Generalised Rayleigh-Normal conjecture for quantum dichotomy transformations

摘录 / EXCERPT: Generalised Rayleigh-Normal conjecture. The optimal rate for transforming quantum dichotomies in the small deviation regime is given by Eq. ... .

conjecture | open | MathDB 页面 | 原文线索

V2-059289 | MATHDB-357699

Achievability of optimality bounds for noncommuting quantum dichotomies

摘录 / EXCERPT: Achievability conjecture. All of these optimality bounds are achievable for general states satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-059290 | MATHDB-357813

Coproduct triangularity conjecture for the generators $a_{i,7$

摘录 / EXCERPT: Coproduct triangularity conjecture. For any sign $\pm\in\{+,-\}$ and any $i\in I$,

conjecture | open | MathDB 页面 | 原文线索

V2-059291 | MATHDB-357827

Stability of Kerr conjecture

摘录 / EXCERPT: Let $K(a,M)$ be a Kerr spacetime with $0\leq |a|\leq M$, and let vacuum initial data sets be sufficiently close to Kerr initial data. Stability of Kerr conjecture. Their maximal de...

conjecture | open | MathDB 页面 | 原文线索

V2-059292 | MATHDB-357832

The braid-group preservation conjecture for higher-term quantum Plücker relations

摘录 / EXCERPT: Braid-group preservation conjecture. The maps σ_i and σ_i^{-1} also preserve r-term quantum Plücker relations for $r\geq 4$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-059293 | MATHDB-357906

Good qubit LDPC codes yield good quantum LDPC rotor codes

摘录 / EXCERPT: Rotor-code conjecture. Under these assumptions, $\phi_\infty(\mathcal{S})$ represents a good quantum LDPC rotor code.

conjecture | open | MathDB 页面 | 原文线索

V2-059294 | MATHDB-358063

KR-polynomial and canonical-basis conjecture

摘录 / EXCERPT: Let (i,p) and (i,s) be vertices of the indicated folded Dynkin diagram with $p<s$. Let $m^{(i)}[p,s]$ be the corresponding dominant monomial, let $\underline{m^{(i)}[p,s]}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-059295 | MATHDB-358064

Positivity conjecture for real canonical-basis elements

摘录 / EXCERPT: Let $m\in\mathcal{M}_+$, and let $L_q(m)$ be the corresponding canonical-basis element of the quantum Grothendieck ring. Call $L_q(m)$ real if, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-059296 | MATHDB-358065

q-commutativity conjecture for KR-polynomials

摘录 / EXCERPT: Let $(m^{(i)}_{k,r},m^{(j)}_{l,t})$ be a pair of KR-monomials, and let $F_q(\underline{m}^{(i)}_{k,r})$ and $F_q(\underline{m}^{(j)}_{l,t})$ be the associated KR-polynomials. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059297 | MATHDB-358066

Cluster-variable recognition conjecture for KR-polynomials

摘录 / EXCERPT: Let $s\in\mathbb{Z}$, let $\mathfrak{S}_s$ be the indicated quantum seed, and let ${}^{s}\mathfrak{u}_{i,p}$ denote its seed variables. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059298 | MATHDB-358078

Charged-multiplet formula for elliptic Calabi-Yau threefolds

摘录 / EXCERPT: Charged-multiplet conjecture. The charged multiplets H_{ch} in the massless spectrum satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-059299 | MATHDB-358090

Vacuum Schwarzschild-to-Kerr characteristic gluing conjecture

摘录 / EXCERPT: Vacuum Schwarzschild-to-Kerr characteristic gluing conjecture. The Schwarzschild symmetry sphere of mass M_i and radius R_i can be characteristically glued to any such non-anti...

conjecture | open | MathDB 页面 | 原文线索

V2-059300 | MATHDB-358106

The Penrose inequality for asymptotically flat initial data

摘录 / EXCERPT: Let (M,g,k) be an asymptotically flat initial data set satisfying appropriate fall-off conditions and the dominant energy condition, with boundary ∂M consisting of an o...

conjecture | open | MathDB 页面 | 原文线索

V2-059301 | MATHDB-358107

Radiality conjecture for solutions of the coupled generalized Jang equation and zero divergence system

摘录 / EXCERPT: Let (M,g,k) be a spherically symmetric initial data set, and consider solutions (f,ϕ) of the coupled system … … with the appropriate asymptotics for application to the Penro...

conjecture | open | MathDB 页面 | 原文线索

V2-059302 | MATHDB-358192

Sufficiency conjecture for minimal quantum Rényi divergences under positive maps

摘录 / EXCERPT: Minimal Rényi sufficiency conjecture. The family $D^{\alpha}_{\min}$ provides a sufficient family of divergences for interconversion of finite-dimensional density matrices via posi...

conjecture | open | MathDB 页面 | 原文线索

V2-059303 | MATHDB-358229

McNamara-Vafa cobordism conjecture

摘录 / EXCERPT: Let $c9$ be a tangential structure, and suppose a consistent n-dimensional theory of quantum gravity sums over spacetime backgrounds carrying a $c9$-structure. The associated bor...

conjecture | open | MathDB 页面 | 原文线索

V2-059304 | MATHDB-358263

Quantum strong cosmic censorship conjecture

摘录 / EXCERPT: Let S) be a strictly partial Cauchy surface of the (not necessarily globally hyperbolic) spacetime (M,g_{ab}), and let $D(S)$ be its domain of dependence. Regard...

conjecture | open | MathDB 页面 | 原文线索

V2-059305 | MATHDB-358265

Quantum very strong cosmic censorship conjecture

摘录 / EXCERPT: Let (M,g_{ab}), S, $D(S)$, $\mathscr{A}$, $\mathscr{B}$ and ω be as in the quantum strong cosmic censorship conjecture, and suppose that $(D(S),\omega)$ satisfies the se...

conjecture | open | MathDB 页面 | 原文线索

V2-059306 | MATHDB-358289

The BSC-PSC bound for sandwiched Rényi conditional entropy

摘录 / EXCERPT: Let X_1,X_2 be binary random variables with quantum side information B_1,B_2, let ρ_1,ρ_2 denote their respective classical-quantum states, and let X_1+X_2 be their s...

conjecture | open | MathDB 页面 | 原文线索

V2-059307 | MATHDB-358290

The BEC bound for sandwiched Rényi conditional entropy

摘录 / EXCERPT: Let X_1,X_2 be binary random variables with quantum side information B_1,B_2, and let τ denote the product construction in which the output is X_1+X_2. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059308 | MATHDB-358306

Existence of detectably non-Hausdorff manifolds without bifurcate curves

摘录 / EXCERPT: Let M be a manifold. A bifurcate curve in M is a pair of smooth injective curves that agree up to a parameter value and then disagree at that value and thereafter. A manifold i...

conjecture | open | MathDB 页面 | 原文线索

V2-059309 | MATHDB-358413

Rigid-object and real-simple-module bijection conjectures

摘录 / EXCERPT: Let ξ be an arbitrary height function and let $\ell\geq 2$. Let $Q^{\leq \xi}_\ell$ be the principal quiver, let $A^{\leq \xi}_\ell$ be its Jacobian algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059310 | MATHDB-358414

The truncated q-character formula for real simple modules

摘录 / EXCERPT: Let ξ be a height function and let $\ell\geq 2$. Let $\mathscr{C}^{\leq \xi}_\ell$ be the corresponding module subcategory, and let $L(m)\in\mathscr{C}^{\leq \xi}_\ell$ be a re...

conjecture | open | MathDB 页面 | 原文线索

V2-059311 | MATHDB-358472

The high-degree free-boundary state nonequilibrium conjecture

摘录 / EXCERPT: High-degree free-boundary state nonequilibrium conjecture. For every $\varepsilon>0$, for all sufficiently large r, if $J\geq(1+\varepsilon)J_{uniq}(r)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-059312 | MATHDB-358486

Loop-constrained regular polygon maximization conjecture for quantum dot families

摘录 / EXCERPT: Let Y be a finite set of points on a loop Γ of fixed length in $\mathbb{R}^{\nu}$, where $\nu=2,3$, with the points equidistant in the arc-length variable, and suppose tha...

conjecture | open | MathDB 页面 | 原文线索

V2-059313 | MATHDB-358490

The bimodule characterization of morphisms between quantum currents

摘录 / EXCERPT: Bimodule characterization. The morphisms between $(Q,\beta_{Q,-})$ and $(X,\beta_{X,-})$ in the Drinfeld center are in bijection with the...

conjecture | open | MathDB 页面 | 原文线索

V2-059314 | MATHDB-358552

Volume version of Penrose's singularity theorem

摘录 / EXCERPT: Let (M,g) be a globally hyperbolic spacetime satisfying the strong energy condition, and let $S\subset M$ be a future trapped surface. Volume version of Penrose's singularity the...

conjecture | open | MathDB 页面 | 原文线索

V2-059315 | MATHDB-358553

Volumetric WCC for future incomplete causal geodesics

摘录 / EXCERPT: Let (M,g) be a spacetime and define ... A causal geodesic is future incomplete if it cannot be extended to arbitrarily large future parameter values. Volumetric WCC. In a generic p...

conjecture | open | MathDB 页面 | 原文线索

V2-059316 | MATHDB-358609

The higher-dimensional sharp p-Penrose inequality

摘录 / EXCERPT: Let (M^n,g) be a complete, asymptotically flat manifold with non-negative scalar curvature and non-empty minimal boundary Σ, where $3\leq n\leq 7$, and assume ... Here C_p...

conjecture | open | MathDB 页面 | 原文线索

V2-059317 | MATHDB-358638

Burns-space scattering amplitudes and defect-free chiral-algebra correlators

摘录 / EXCERPT: Let WZW$_4$ plus Mabuchi gravity be the bulk theory on Burns space, and let the large N chiral algebra be the chiral algebra obtained from the compactification...

conjecture | open | MathDB 页面 | 原文线索

V2-059318 | MATHDB-358639

Burns-space self-dual Yang-Mills splitting-function/OPE correspondence

摘录 / EXCERPT: Consider self-dual Yang-Mills theory on Burns space with gauge group $\operatorname{SO}(8)$, and let the large N chiral algebra be the celestial chiral algebra described in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-059319 | MATHDB-358723

Collins-Yau conjecture on dHYM solutions and Bridgeland stability

摘录 / EXCERPT: Let X be the compact Kähler manifold under consideration, and let L be a holomorphic line bundle on X. A metric solving the deformed Hermitian-Yang-Mills equation is a metric...

conjecture | open | MathDB 页面 | 原文线索

V2-059320 | MATHDB-358752

True-solution conjecture for gluing small black holes along timelike geodesics

摘录 / EXCERPT: Let $V\subset M$ be a precompact open set. For sufficiently small $\epsilon>0$, let g_ϵ be the formal solution on M_ϵ obtained by gluing small Kerr black holes al...

conjecture | open | MathDB 页面 | 原文线索

V2-059321 | MATHDB-358873

Entanglement-entropy count conjecture for Dicke-state orbits

摘录 / EXCERPT: Let $\ket{D^N_1}$ denote the N-qubit Dicke state with one excitation, and let $(HC)_{1,2}$ denote circuits generated by Hadamard and controlled-not gates acting on qubits 1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-059322 | MATHDB-358926

Bray's conformal conjecture for asymptotically flat 3-manifolds

摘录 / EXCERPT: Let (M,g) be an asymptotically flat 3-manifold with smooth, non-empty, compact boundary Σ, and let $\mathcal{H}(g)$ denote the class of conformal metrics under considerati...

conjecture | open | MathDB 页面 | 原文线索

V2-059323 | MATHDB-359013

Koszul duality for observables in holomorphically twisted AdS/CFT

摘录 / EXCERPT: Let $\mathbb{X}$ and $\mathbb{Y}$ be field theories, and let $\mathsf{Obs}_{\mathbb{X}}$ and $\mathsf{Obs}_{\mathbb{Y}}$ denote their observables. For an $\mathbb{E}_n$-algebra A...

conjecture | open | MathDB 页面 | 原文线索

V2-059324 | MATHDB-359097

The Kac conjecture for Liouville quantum gravity

摘录 / EXCERPT: Let D be a fixed bounded domain, let h be the Gaussian free field on D, and let $(\boldsymbol{\lambda}_n)_{n\ge 0}$ be the eigenvalues of Liouville Brownian motion associated...

conjecture | open | MathDB 页面 | 原文线索

V2-059325 | MATHDB-359098

The stronger Kac conjecture for Liouville quantum gravity surfaces

摘录 / EXCERPT: Let $(\boldsymbol{\lambda}_n)_{n\ge 0}$ be the eigenvalue sequence of Liouville Brownian motion on a random surface (D,h), where D is the domain and h is the Gaussian free fi...

conjecture | open | MathDB 页面 | 原文线索

V2-059326 | MATHDB-359100

GOE level-spacing conjecture for Liouville quantum gravity

摘录 / EXCERPT: Let $\gamma\in(0,2)$, let μ_γ be the Liouville measure on D, and let $(\boldsymbol{\lambda}_j)_{j\ge1}$ be the LQG eigenvalues. Write $F_{\mathrm{GOE}}(x)$ for the GOE l...

conjecture | open | MathDB 页面 | 原文线索

V2-059327 | MATHDB-359180

The Ricci flow conjecture

摘录 / EXCERPT: Ricci flow conjecture. Any Ricci-flow fixed point at infinite distance should be accompanied by an infinite tower of additional fields with a flow-dependent mass threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-059328 | MATHDB-359182

The AdS distance conjecture

摘录 / EXCERPT: Consider a one-parameter family of D-dimensional Anti-de Sitter metrics $g_{\mu\nu}(\Lambda)$ arising as quantum-gravity low-energy effective theories, labelled by their cosmolog...

conjecture | open | MathDB 页面 | 原文线索

V2-059329 | MATHDB-359212

Collins-Jacob-Yau's stability conjecture for the dHYM equation

摘录 / EXCERPT: Let (X,ω) be a compact Kähler manifold of complex dimension n, let $\alpha^{\mathbb{C}}=\alpha+\mathrm{i}\beta$ be a complexified Kähler class, and let $\hat{\vartheta}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-059330 | MATHDB-359213

C^{1,1} geodesic conjecture for the coupled dHYM system

摘录 / EXCERPT: Let $(X,\alpha^{\mathbb{C}})$ be a complexified Kähler manifold in the hypercritical range, and let $\mathcal{H}$ denote the space of admissible potentials. C^{1,1} geodesic conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-059331 | MATHDB-359277

Nam's conjecture on the binding energy of a trapped Bose gas

摘录 / EXCERPT: Let $E(N,\lambda v)$ and $E(N-1,\lambda v)$ denote the ground-state energies of the trapped Bose gas with interaction scaling λ, and let T and v satisfy appropriate condit...

conjecture | open | MathDB 页面 | 原文线索

V2-059332 | MATHDB-359292

Low-temperature Bethe root pattern conjecture for the antiferromagnetic massive regime

摘录 / EXCERPT: Low-T Bethe root pattern conjecture. All excitations of the quantum transfer matrix at low T and sufficiently large N can be parametrised in this way, and, up to corrections...

conjecture | open | MathDB 页面 | 原文线索

V2-059333 | MATHDB-359451

The BPS Cardy conjecture for three-dimensional N=2 superconformal field theories

摘录 / EXCERPT: BPS Cardy conjecture. In every three-dimensional superconformal field theory with $\mathcal{N}=2$ supersymmetry, the spectrum of supersymmetric (BPS) states obeys this Cardy asympt...

conjecture | open | MathDB 页面 | 原文线索

V2-059334 | MATHDB-359452

The refined BPS growth conjecture for three-dimensional N=2 theories

摘录 / EXCERPT: Let a_n be the coefficients of a supersymmetric partition function for a three-dimensional $\mathcal{N}=2$ theory. Let $r\in\mathbb{Q}/\mathbb{Z}$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059335 | MATHDB-359453

The root-of-unity asymptotics conjecture for three-dimensional supersymmetric partition functions

摘录 / EXCERPT: Root-of-unity asymptotics conjecture. Supersymmetric partition functions have the asymptotic behavior

conjecture | open | MathDB 页面 | 原文线索

V2-059336 | MATHDB-359458

Koszul duality conjecture for mirror dual symplectic singularities

摘录 / EXCERPT: Let the Higgs and Coulomb branches of a 3d $\mathcal{N}=4$ supersymmetric gauge theory be mirror dual symplectic singularities, and let $\mathcal{O}$ denote the category associated...

conjecture | open | MathDB 页面 | 原文线索

V2-059337 | MATHDB-359634

Nontermination conjecture for the supercritical fully packed $O(2)$ iteration

摘录 / EXCERPT: Fix $s>1$. In the modified iteration procedure, each hole of perimeter m is filled using a ring whose inner boundary has length either $\lfloor sm\rfloor$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-059338 | MATHDB-359635

Supercritical fully packed $O(2)$ maps scaling-limit conjecture

摘录 / EXCERPT: Fix $s>1$, and let (M_k',L_k') be sampled from the modified iteration procedure. Let $\mathbf c_{\mathrm L}\in(1,25)$ be the central charge parameter determined by s through ……

conjecture | open | MathDB 页面 | 原文线索

V2-059339 | MATHDB-359679

Non-Abelian large-N convergence conjecture for the Chern-Simons matrix model

摘录 / EXCERPT: Let $p\geq 2$, set $\epsilon_1=1$, $\epsilon_2=k+p$, and define the rescaled operators ... Let $\mathcal B$ be an appropriate energy eigenbasis for the Hilbert space. Non-Abelian lar...

conjecture | open | MathDB 页面 | 原文线索

V2-059340 | MATHDB-359680

Large-N commutation-relation conjecture for the Chern-Simons matrix model

摘录 / EXCERPT: Large-N commutation-relation conjecture. In the large N limit, these operators satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-059341 | MATHDB-359735

Subquotient conjecture for quantum groups under edge contractions

摘录 / EXCERPT: Subquotient conjecture. The explicit embedding for quantum groups in Theorem is a section of a subquotient. This conjecture is motivated by the corresponding result for Hall algebr...

conjecture | open | MathDB 页面 | 原文线索

V2-059342 | MATHDB-359736

Isomorphism conjecture for the edge-contraction quantum-group subquotient

摘录 / EXCERPT: Isomorphism conjecture. The map Φ is an isomorphism, hence $\mathbf U_I$ is a split subquotient of $\mathbf U_{\widehat I}$. This makes the conjectural subquotient statement p...

conjecture | open | MathDB 页面 | 原文线索

V2-059343 | MATHDB-360256

Main flatness conjecture for G-Fock bundles

摘录 / EXCERPT: Let (P,Φ,σ) be a G-Fock bundle. A hermitian structure ρ is compatible when it commutes with σ, and positive when the associated positivity condition for the...

conjecture | open | MathDB 页面 | 原文线索

V2-059344 | MATHDB-360257

Gauge-theoretic flatness conjecture for Fock fields

摘录 / EXCERPT: Let (P,Φ_0,σ) be a G-Fock bundle over S equipped with a compatible hermitian structure ρ. Let $\eta\in\Omega^0(S,\mathfrak{g}_P^{\sigma,-\rho})$ be a hermitian e...

conjecture | open | MathDB 页面 | 原文线索

V2-059345 | MATHDB-360258

Small-covector flatness conjecture for G-Fock bundles

摘录 / EXCERPT: Let (P,Φ) be a G-Fock bundle over S with hermitian structure ρ such that $\Phi\in\mathcal{P}$ is positive. Let μ be the induced $\mathfrak{g}$-complex structure o...

conjecture | open | MathDB 页面 | 原文线索

V2-059346 | MATHDB-360274

Universal state-inversion positivity conjecture for multipartite quadratic forms

摘录 / EXCERPT: Universal state-inversion positivity conjecture. For every $C\in L(\mathscr{H})$ and every $v\in\\{0,1\\}^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-059347 | MATHDB-360339

BAE from gauge origami on a toric Calabi-Yau three-fold times a line

摘录 / EXCERPT: Gauge-origami Bethe ansatz conjecture. The Bethe ansatz equations are

conjecture | open | MathDB 页面 | 原文线索

V2-059348 | MATHDB-360341

BPS-crystal expansion of D6 qq-characters

摘录 / EXCERPT: D6 qq-character BPS-crystal conjecture. For $i\in Q_0$, the qq-characters are

conjecture | open | MathDB 页面 | 原文线索

V2-059349 | MATHDB-360360

High-temperature exponential clustering conjecture for short-range interactions

摘录 / EXCERPT: High-temperature exponential clustering conjecture. There exist β^*, C_Cov, $c_\mathrm{Cov}>0$, and $n\in\mathbb{N}$ such that, for every $\beta<\beta^*$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-059350 | MATHDB-360374

Parity-density conjecture for the asymmetric quantum Rabi model

摘录 / EXCERPT: For $\ell\in\mathbb{Z}$, set $\epsilon=\ell/2$. Let $d_\ell(N;g,\Delta)$ be the proportion of consecutive eigenvalue pairs up to N having the same parity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059351 | MATHDB-360375

The G-conjecture on eigenvalue distribution for the quantum Rabi model

摘录 / EXCERPT: Fix a parity sector of the quantum Rabi model, and for each integer n consider the interval $[n,n+1)$. G-conjecture. The number of eigenvalues in $[n,n+1)$ is either 0, 1, or...

conjecture | open | MathDB 页面 | 原文线索

V2-059352 | MATHDB-360399

The spanning-tree and GFF scaling-limit conjecture for random triangulations

摘录 / EXCERPT: The spanning-tree and GFF scaling-limit conjecture. Under an appropriate scaling as $n\to\infty$, the 4-tuple (M,P,ϕ_1,ϕ_2) converges in law to a unit boundary length...

conjecture | open | MathDB 页面 | 原文线索

V2-059353 | MATHDB-360428

A Penrose-like lower bound for ADM momentum

摘录 / EXCERPT: Let $(M,g,K) fin \mathcal{M}_{\textnormal{AF}}$ and let $\{\Sigma_r\}_{r\in[r_0,\infty)}\subset\mathcal{M}$, where ... Suppose that the inner boundary Σ_{r_0} is a generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-059354 | MATHDB-360538

Threshold conjecture for higher-moment star graph gaps

摘录 / EXCERPT: Let $G_{\star}$ be a star graph with $n_{\star}$ vertices, local dimension q, and moment k, and let $\Delta(G_{\star},n_{\star},k)$ denote its spectral gap. Higher-moment star-...

conjecture | open | MathDB 页面 | 原文线索

V2-059355 | MATHDB-360606

The Kac type characterization conjecture for second countable compact quantum groups

摘录 / EXCERPT: Kac type characterization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-059356 | MATHDB-360718

The scaling-limit conjecture for critical Fortuin-Kasteleyn planar maps

摘录 / EXCERPT: Let $q\text{-FK}$ planar maps denote the critical Fortuin-Kasteleyn planar maps with parameter q, and let $gamma$ be a parameter in $[sqrt 2,2)$ satisfying ... Scaling-limit conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-059357 | MATHDB-360783

Quantum-group realization conjecture for admissible-level affine $mathfrak{sl}_2$ weight modules

摘录 / EXCERPT: Let ell be an admissible level and set ... Let $L_{ell}(\mathfrak{sl}_2)^A\operatorname{-wtmod}$ be the indicated braided tensor subcategory of...

open_problem | open | MathDB 页面 | 原文线索

V2-059358 | MATHDB-360790

State-dependent contextuality implies state-independent AvN in a partial closure

摘录 / EXCERPT: Partial-closure AvN conjecture. Any measurement cover $\mathcal{M}$ that realizes a contextual empirical model for some state ψ is state-independently AvN in a part...

conjecture | open | MathDB 页面 | 原文线索

V2-059359 | MATHDB-360874

Non-spin extension of the tilted spacetime positive mass theorem

摘录 / EXCERPT: Non-spin extension conjecture. Theorem 3.1 holds even if M is not spin; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-059360 | MATHDB-361024

Distance-matrix characterization of triangular operators

摘录 / EXCERPT: Let $\mathcal{H}$ be a Hilbert space with an orthonormal basis $(\mathbf{u}_i)_{i\in[n]}$, and let $\mathbf{E}$ be an operator satisfying ... for some $d_{ij}\geq 0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059361 | MATHDB-361081

Gauge uniqueness conjecture for the nonabelian twisted X-ray transform

摘录 / EXCERPT: Gauge uniqueness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-059362 | MATHDB-361249

Uniqueness conjecture for two-sided quantum tangent spaces

摘录 / EXCERPT: Let $\mathbf{j}$ and $\mathbf{j}'$ be two reduced decompositions of the longest element of the Weyl group of $\mathfrak{sl}_{n+1}$. For each reduced decomposition, take the span of...

conjecture | open | MathDB 页面 | 原文线索

V2-059363 | MATHDB-361250

Maximal-prolongation dimension conjecture for Lusztig root-vector calculi

摘录 / EXCERPT: Let $\mathbf{j}$ and $\mathbf{j}'$ be two reduced decompositions of the longest element of the Weyl group of $\mathfrak{sl}_{n+1}$. For each reduced decomposition, consider the ass...

conjecture | open | MathDB 页面 | 原文线索

V2-059364 | MATHDB-361324

DELL eigenfunction conjecture for affine Laumon-space elliptic genera

摘录 / EXCERPT: Let $N-1$ degrees of freedom of the quantum double elliptic (DELL) system be described by the current $\mathcal{O}(z)$ and let $\mathcal{Z}^{DELL}_{\text{inst}}(p,x_1,\dots, x_N)$...

conjecture | open | MathDB 页面 | 原文线索

V2-059365 | MATHDB-361390

Thermalization or resonant-relaxation conjecture for tagged particles

摘录 / EXCERPT: Consider a conservative long-range interacting particle system at thermal equilibrium with a tagged particle whose mean-field evolution is trivial. Let the linearized mean-field op...

conjecture | open | MathDB 页面 | 原文线索

V2-059366 | MATHDB-361391

Weak cosmic censorship conjecture

摘录 / EXCERPT: Let asymptotically flat initial data for Einstein's field equations be given, and let its maximal development be the corresponding spacetime. A future null infinity $\mathcal{I}^+$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-059367 | MATHDB-361406

The generalized quantum Stein's lemma for resource theories

摘录 / EXCERPT: Generalized quantum Stein's lemma. For any such family,

conjecture | open | MathDB 页面 | 原文线索

V2-059368 | MATHDB-361429

Nonexistence conjecture for nonpositive mass-aspect types

摘录 / EXCERPT: Let ρ be the energy density and J the momentum density on a smooth, conformally compact, geometrically finite general relativistic initial data set satisfying the energy con...

conjecture | open | MathDB 页面 | 原文线索

V2-059369 | MATHDB-361439

Generic non-optimality of constant potentials on non-path metric trees

摘录 / EXCERPT: Generic non-optimality conjecture. The set of edge-length vectors and values $M>0$ for which the constant potential minimizes the spectral gap in $C_{T,M}$ is of the first Baire ca...

conjecture | open | MathDB 页面 | 原文线索

V2-059370 | MATHDB-361552

The derivation conjecture for quantum grassmannians

摘录 / EXCERPT: Let K be a field of characteristic zero, let q be a nonzero element of K that is not a root of unity, and let $\mathcal{O}_{q}(G(k,n))$ be the quantum grassmannian with...

conjecture | open | MathDB 页面 | 原文线索

V2-059371 | MATHDB-361616

Convolutional strong subadditivity conjecture

摘录 / EXCERPT: Convolutional strong subadditivity. One has

conjecture | open | MathDB 页面 | 原文线索

V2-059372 | MATHDB-361738

Quantum sufficiency conjecture for the quantum exceptional category

摘录 / EXCERPT: Quantum sufficiency conjecture. If every polynomial in $\mathbf P$ is invertible in R, then the space of n-boundary-point diagrams in $\mathsf{QExc}_{R,v,w}$ is finitely genera...

conjecture | open | MathDB 页面 | 原文线索

V2-059373 | MATHDB-361739

Quantum consistency conjecture for the quantum exceptional category

摘录 / EXCERPT: Let R be a ring and let $\mathsf{QExc}_{R,v,w}$ be the quantum exceptional diagram category, with $\mathbb Q(v,w)$-valued closed-diagram evaluations defined by the stated relatio...

conjecture | open | MathDB 页面 | 原文线索

V2-059374 | MATHDB-361740

Quantum semisimplicity conjecture for the exceptional category

摘录 / EXCERPT: Let U be a subset of $\mathbb C^2$, and let the Cauchy completion denote the additive and idempotent completion. Quantum semisimplicity conjecture. There is an open dense subset...

conjecture | open | MathDB 页面 | 原文线索

V2-059375 | MATHDB-361773

Necessary and sufficient conditions for full solvability of UQSD for unitarily related mixtures

摘录 / EXCERPT: Necessary and sufficient conditions for full solvability of UQSD for URM. The UQSD optimization problem induced by the partition $\bigcup_{i=1}^{N}\Omega_i$ is fully solvable as...

conjecture | open | MathDB 页面 | 原文线索

V2-059376 | MATHDB-361840

Frobenius light-cone conjecture for the shift unitary

摘录 / EXCERPT: Frobenius light-cone conjecture. The bound above can be replaced by

conjecture | open | MathDB 页面 | 原文线索

V2-059377 | MATHDB-361842

Quantum positivity and canonical-basis equality for Kirillov-Reshetikhin monomials

摘录 / EXCERPT: Let $m^{(i)}[p,s]$ be a Kirillov-Reshetikhin monomial, and let $F_q(m^{(i)}[p,s])$ and $L_q(m^{(i)}[p,s])$ denote the associated KR-polynomial and canonical-basis element. KR-polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-059378 | MATHDB-361951

De Palma-Trevisan conjecture on quantum Wasserstein divergences

摘录 / EXCERPT: De Palma-Trevisan conjecture. Up to non-degeneracy assumptions on the A_j ensuring definiteness, the function $d_{\mathcal{A}}$ is a true metric on the set of states on...

conjecture | open | MathDB 页面 | 原文线索

V2-059379 | MATHDB-362051

The limiting block-structure conjecture for QUBO minimizers and maximizers

摘录 / EXCERPT: Limiting block-structure conjecture. There exist constants $\alpha_1,\alpha_2,\alpha_3,\alpha_4$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059380 | MATHDB-362171

Cederbaum-Sakovich conjecture on convergence of the Riemannian center of mass

摘录 / EXCERPT: Let (M^3,g) be an asymptotically Euclidean manifold of order $\tau\in(-1,-\frac{1}{2})$ with respect to a structure of infinity Φ_z and coordinates $\{z^i\}_{i=1}^3$. Assume...

conjecture | open | MathDB 页面 | 原文线索

V2-059381 | MATHDB-362175

Conjectured tight weighted Frobenius norm bounds for matrix commutators

摘录 / EXCERPT: Conjectured tight bounds. The tight bounds for cases (i), (ii), and (iv) are respectively

conjecture | open | MathDB 页面 | 原文线索

V2-059382 | MATHDB-362393

The outgoing black-hole approximation problem for asymptotically flat vacuum solutions

摘录 / EXCERPT: An asymptotically flat solution of the vacuum Einstein equations is said to approximate n outgoing black holes in a neighbourhood of future timelike infinity i_+. Outgoing blac...

open_problem | open | MathDB 页面 | 原文线索

V2-059383 | MATHDB-362419

The defect half-index interpretation conjecture

摘录 / EXCERPT: Let M_3 be a weakly negative plumbed three-manifold with defects labelled by highest weights $\nu_v\vec\omega$ at nodes $v\in V_W$, and let $\hat Z_b(M_3;W_{\{\nu_v\}};\tau)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-059384 | MATHDB-362597

Choquet-Bruhat's weak cosmic censorship conjecture

摘录 / EXCERPT: Let $(\Sigma,\leftidx{^\Sigma}{\mathbf{g}}{},\leftidx{^\Sigma}{\mathbf{K}}{},\mu,J)$ be generic asymptotically flat initial data, and let $(M,\mathbf{g})$ be its maximal globally h...

conjecture | open | MathDB 页面 | 原文线索

V2-059385 | MATHDB-362717

Burnett's backward conjecture on realizing Einstein-massless Vlasov solutions

摘录 / EXCERPT: Burnett's backward conjecture. Every sufficiently regular solution to the Einstein--massless Vlasov system arises as a high-frequency limit of vacuum spacetimes.

conjecture | open | MathDB 页面 | 原文线索

V2-059386 | MATHDB-362718

Scalar-flat approximation conjecture for nonnegative scalar curvature metrics

摘录 / EXCERPT: Scalar-flat approximation conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-059387 | MATHDB-362790

Shifted-quantum-affine-algebra factorization conjecture

摘录 / EXCERPT: Let $bpsi$ be the highest l-weight of an irreducible l-highest-weight module $L(bpsi)$ over the quantum affine Borel algebra $\mathcal{U}_qmathfrak{b}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059388 | MATHDB-362792

Transversal Clifford impossibility conjecture for logical C-QSK circuits

摘录 / EXCERPT: Transversal Clifford impossibility conjecture. There exists no stabilizer code where transversal Clifford gates realize the logical C-QSK circuit for any non-trivial exponentiated...

conjecture | open | MathDB 页面 | 原文线索

V2-059389 | MATHDB-362798

Scaling-limit factorization conjecture for the non-integrable Ising model

摘录 / EXCERPT: Scaling-limit factorization conjecture. There exists $\lambda_0=\lambda_0(U)>0$ such that, for all $|\lambda|\leq\lambda_0$, there exists $\beta^*=\beta^*(\lambda)$ for which, for...

conjecture | open | MathDB 页面 | 原文线索

V2-059390 | MATHDB-363032

B-side Frobenius action correspondence for matrix factorizations

摘录 / EXCERPT: Let (A,f) be a formal smooth commutative algebra of dimension $d<p$ over $\mathbf{k}$, with superpotential $f\in A$. Assume the HKR-type quasi-isomorphisms ... and ... Under the iden...

conjecture | open | MathDB 页面 | 原文线索

V2-059391 | MATHDB-363082

Boundary Casimir connection monodromy conjecture

摘录 / EXCERPT: Let $\mathfrak{h}^{\mathrm{reg}}$ be the regular locus for the Cartan subalgebra and let the boundary Casimir connection descend to the quotient $\mathfrak{h}^{\mathrm{reg}}/W$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-059392 | MATHDB-363104

Second-level collapse conjecture for the PPT entanglement-cost hierarchies

摘录 / EXCERPT: Let $\rho=\rho_{AB}$ be a state. The quantities $E_{\chi,p}(\rho)$ are the χ-quantities defined by the hierarchy at level p, and $E_{\kappa,p}(\rho)$ are the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-059393 | MATHDB-363191

Subsonic-or-sonic criterion for spectral stability of QHD wave profiles

摘录 / EXCERPT: Subsonic-or-sonic stability conjecture. A standing or traveling wave profile for the QHD systems with linear or nonlinear viscosity is spectrally stable if and only if its velocity...

conjecture | open | MathDB 页面 | 原文线索

V2-059394 | MATHDB-363192

Gibbons-type conjecture for abelian Higgs critical points

摘录 / EXCERPT: Gibbons-type conjecture. The critical point is necessarily two-dimensional: it is the pullback under the projection $\mathbb{R}^n\to\mathbb{R}^2$ of the standard degree ±1 solu...

conjecture | open | MathDB 页面 | 原文线索

V2-059395 | MATHDB-363205

Steps toward the modular functor conjecture for quantized geodesic lengths

摘录 / EXCERPT: Let $\mathfrak{S} = \Sigma \setminus \mathcal{P}$ be a triangulable marked surface, let Δ be an ideal triangulation, and let $\hbar \in \mathbb{R}_{>0}$. For an essential si...

conjecture | open | MathDB 页面 | 原文线索

V2-059396 | MATHDB-363206

Dehn twist-length operator conjecture for quantized Teichmüller space

摘录 / EXCERPT: Let $\mathfrak{S}$ be a triangulable marked surface and let Δ be its ideal triangulation. Let γ be an essential non-peripheral simple loop, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059397 | MATHDB-363262

Yang-Mills scattering form conjecture for the spinor scattering form

摘录 / EXCERPT: Yang-Mills scattering form conjecture. The $(2n-4)$-form Υ_n coincides with the $\mathcal{N}=4$ SYM scattering form defined by HZ.

conjecture | open | MathDB 页面 | 原文线索

V2-059398 | MATHDB-363263

The non-Gaussian limit and universal profile conjecture for the high-dimensional Ising model

摘录 / EXCERPT: Non-gaussian limit and universal profile conjecture. As $r\to\infty$, $h_r^{-1}S_r$ converges in distribution under $\langle\cdot\rangle_{\beta_c+sw_r}^{\mathbb{T}_r}$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-059399 | MATHDB-363329

Reshetikhin's projector asymptotics conjecture for the nested Bethe ansatz

摘录 / EXCERPT: Let V^{aux} be the auxiliary representation embedded in $V^{J+1}\otimes (V^{J})^*$, and let $D^{J+1,J}(u)$ be the corresponding operator. Write $\Pi^{J+1,\overline{J}}$ for the p...

conjecture | open | MathDB 页面 | 原文线索

V2-059400 | MATHDB-363369

Self-consistent initial-data conjecture for semiclassical gravity

摘录 / EXCERPT: Self-consistent initial-data conjecture. Self-consistent initial data for the fourth-order time derivatives of the metric can be obtained from these classical initial data by solvi...

conjecture | open | MathDB 页面 | 原文线索

V2-059401 | MATHDB-363408

Berenstein-Zelevinsky's positivity conjecture for quantum cluster algebras

摘录 / EXCERPT: Berenstein-Zelevinsky's positivity conjecture. For any quantum cluster variable X of $\mathcal A_v$,

conjecture | open | MathDB 页面 | 原文线索

V2-059402 | MATHDB-363481

The SOP connective-constant conjecture

摘录 / EXCERPT: Let w_{2n} be the number of self-overlapping polygons of length $2n$ in the SOP model, counted with multiplicity factor μ_γ. SOP connective-constant conjecture. The limi...

conjecture | open | MathDB 页面 | 原文线索

V2-059403 | MATHDB-363483

The SOP critical-exponent conjecture

摘录 / EXCERPT: Let ν and ϑ be the critical exponents governing the continuum limit and the zero-cosmological-constant partition function of the SOP model. SOP critical-exponent conj...

conjecture | open | MathDB 页面 | 原文线索

V2-059404 | MATHDB-363484

The zero-curvature JT area-tail conjecture with conformal matter

摘录 / EXCERPT: Let $\rho^0_{\beta_{\mathrm q}}(A)$ be the area distribution of zero-curvature JT gravity coupled to conformal matter with $c\leq0$, and let ν be the critical exponent specifie...

conjecture | open | MathDB 页面 | 原文线索

V2-059405 | MATHDB-363485

The negative-curvature JT area-tail conjecture

摘录 / EXCERPT: Let $\rho^-_{\beta_{\mathrm q}}(A)$ be the area distribution of JT gravity with negative curvature $-2/L^2$, coupled to conformal matter with $c\leq0$, and let ν be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-059406 | MATHDB-363486

The exact negative-curvature area exponent conjecture

摘录 / EXCERPT: Let χ^-_ν be the exponent governing the large-area tail of the negative-curvature JT area distribution. Exact negative-curvature exponent conjecture. ... This agrees with the...

conjecture | open | MathDB 页面 | 原文线索

V2-059407 | MATHDB-363513

The test-function convergence conjecture for discrete curvature on LQG maps

摘录 / EXCERPT: Let $\{M_n\}_{n\geq1}$ be a sequence of infinite random planar maps believed to lie in the universality class of γ-LQG, such as uniform infinite triangulations when...

conjecture | open | MathDB 页面 | 原文线索

V2-059408 | MATHDB-363673

Weak separation characterizes simplicity of tensor products of fundamental modules

摘录 / EXCERPT: Let J,J' be k-element subsets of $[n]$, and let $L(M_J)$ and $L(M_{J'})$ be the corresponding simple $U_q(\widehat{\mathfrak{sl}_k})$-modules. Weak-separation tensor-product co...

conjecture | open | MathDB 页面 | 原文线索

V2-059409 | MATHDB-363675

The quark-confinement conjecture for the Yang-Mills theory

摘录 / EXCERPT: Let G be the gauge group and let the nontrivial Yang-Mills theory on Minkowskian spacetime $\mathbf{R}^{1,3}$ be the theory whose existence and quantization are asserted by the Y...

conjecture | open | MathDB 页面 | 原文线索

V2-059410 | MATHDB-363676

The weak-coupling Yang-Mills mass gap conjecture

摘录 / EXCERPT: Weak-coupling Yang-Mills mass gap conjecture. For any bare coupling constant $g\in[0,g_0[$, the spectrum of H^g contains 0 as a simple eigenvalue for the vacuum eigenstate and...

conjecture | open | MathDB 页面 | 原文线索

V2-059411 | MATHDB-363741

Folded skew Howe duality for spin centralizer algebras

摘录 / EXCERPT: Let $n\geq 1$ and $m\geq 2$. Write $K_0^\tau(-)$ for the weighted Grothendieck group of a category with involution, let $({\EuScript{B}}_{m}^{n})^\tau$ be the equivariant category,...

conjecture | open | MathDB 页面 | 原文线索

V2-059412 | MATHDB-363742

Equivariant categorification of the quantum orthogonal algebra

摘录 / EXCERPT: Let $m\geq 2$, let $U'_{-q^2}(\mathfrak{so}_{m})$ be the quantum orthogonal algebra with elements $\mathsf{x}_i^{(k)}$, and let ${\EuScript B}_{m}$ be the relevant monoidal categor...

conjecture | open | MathDB 页面 | 原文线索

V2-059413 | MATHDB-363743

Higher spin Serre relations

摘录 / EXCERPT: Let A_m^n be a unital ${\mathbb C}(q)$-algebra, and let $\{\mathsf X_r^{(i)}\}_{1\leq r\leq m-1,\,0\leq i\leq n}\subset A_m^n$ satisfy $\mathsf X_r^{(0)}=1$ together with the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-059414 | MATHDB-363780

The global crystal basis conjecture for the common triangular basis of quantum coordinate rings

摘录 / EXCERPT: Global crystal basis conjecture. Up to $q^{\frac{\mathbb{Z}}{2}}$ multiples, the common triangular basis ${{\mathbf L}}$ for $\mathcal{O}_q[G]$ coincides with the global crystal ba...

conjecture | open | MathDB 页面 | 原文线索

V2-059415 | MATHDB-363834

Approximate quantum Markov property for Gibbs states

摘录 / EXCERPT: Let ρ_β be an arbitrary quantum Gibbs state on a system partitioned as $\Lambda=A\sqcup B\sqcup C$, and let $d_{A,C}$ denote the distance between A and C. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059416 | MATHDB-364177

Nakamura's gauge-invariant decomposition conjecture for linear metric perturbations

摘录 / EXCERPT: Let ${{\mathscr M}}_{\epsilon}$ be a physical spacetime, let ${{\mathscr M}}$ be its background spacetime, and let h_{ab} be a perturbative metric tensor pulled back from...

conjecture | open | MathDB 页面 | 原文线索

V2-059417 | MATHDB-364241

Fused-current commutation conjecture for minimal pairs

摘录 / EXCERPT: Let Δ^+ be the positive roots of a finite-type ADE Lie algebra with a fixed total convex order, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059418 | MATHDB-364347

Nonnegative coefficients in the symmetric-type case

摘录 / EXCERPT: The positivity conjecture. The coefficient satisfies $g_{\boldsymbol{u},\boldsymbol{u}'}\in q\mathbb{Z}_{\geqslant 0}[q]$. This predicts positivity of the triangular coefficients i...

conjecture | open | MathDB 页面 | 原文线索

V2-059419 | MATHDB-364381

Hexagonal-lattice conjecture for simultaneous quantum optimization

摘录 / EXCERPT: Let Λ range over periodic configurations of arbitrary fixed density in phase space, and consider the associated quantum packing, covering, and paving problems. The hexagona...

conjecture | open | MathDB 页面 | 原文线索

V2-059420 | MATHDB-364382

Leech-lattice conjecture for quantum paving in dimension 24

摘录 / EXCERPT: Let Λ be a Leech lattice in phase space of dimension 24, with arbitrary fixed density greater than 1. The Leech-lattice conjecture. There exists a Leech lattice that so...

conjecture | open | MathDB 页面 | 原文线索

V2-059421 | MATHDB-364516

Christandl-Ferrara-Lancien continuity conjecture for filtered relative entropy

摘录 / EXCERPT: Christandl-Ferrara-Lancien's conjecture. There should exist a universal constant κ and a function $g:[0,1]\to\mathbb{R}_+$ with

conjecture | open | MathDB 页面 | 原文线索

V2-059422 | MATHDB-364521

Quantum graph homomorphism-strategy correspondence conjecture

摘录 / EXCERPT: Let $G:X\to X$ and $H:A\to A$ be quantum graphs. A quantum homomorphism from G to H is a quantum function $E:X\to_{\mathcal K}A$ satisfying the quantum graph homomorphism condi...

conjecture | open | MathDB 页面 | 原文线索

V2-059423 | MATHDB-364522

Quantum graph isomorphism-strategy correspondence conjecture

摘录 / EXCERPT: Let $G:X\to X$ and $H:A\to A$ be irreflexive quantum graphs. The quantum graph isomorphism game is ... with ... A quantum graph isomorphism from G to H is the corresponding structu...

conjecture | open | MathDB 页面 | 原文线索

V2-059424 | MATHDB-364553

Decay of correlations conjecture for the massless hierarchical Liouville model

摘录 / EXCERPT: Decay of correlations conjecture. The events satisfy $\nu_{L,m,\epsilon}^\mathrm{Lv}(\mathcal A_{x,r,L})\to1$ as first $\epsilon\to0$, then $m\to0$, and finally $L\to\infty$, and,...

conjecture | open | MathDB 页面 | 原文线索

V2-059425 | MATHDB-364562

Naidu-Rowell Property F conjecture for weakly integral braided fusion categories

摘录 / EXCERPT: Naidu-Rowell Property F conjecture. The braid group representations associated with any object in a weakly integral braided fusion category have finite images.

conjecture | open | MathDB 页面 | 原文线索

V2-059426 | MATHDB-364566

Equivalence of the seven quantum lens space isomorphism conditions in dimension at most five

摘录 / EXCERPT: Let r and d be the parameters defining the quantum lens spaces and let (I)--(VII) denote the seven conditions introduced in the paper for their equivariant isomorphism. Equival...

conjecture | open | MathDB 页面 | 原文线索

V2-059427 | MATHDB-364567

Equivalence of the four basic quantum lens space isomorphism conditions

摘录 / EXCERPT: Let r and d be the parameters defining the quantum lens spaces and let (I)--(IV) denote the four basic conditions introduced in the paper for equivariant isomorphism. Equivalen...

conjecture | open | MathDB 页面 | 原文线索

V2-059428 | MATHDB-364676

Complete contraction and expansion conjecture for amplitude damping channels

摘录 / EXCERPT: Let $\mathcal A_{\gamma}$ denote the amplitude damping channel with damping parameter γ, and for two channels define the complete contraction and expansion coefficients by .....

conjecture | open | MathDB 页面 | 原文线索

V2-059429 | MATHDB-364920

The quantum K-theoretic abelian/non-abelian correspondence conjecture

摘录 / EXCERPT: Let $V//G$ be a GIT quotient satisfying the source's mild hypotheses, including that the G-unstable locus has codimension two. Let T_G be a maximal torus of G, let W be its...

conjecture | open | MathDB 页面 | 原文线索

V2-059430 | MATHDB-365117

Po's general LSM constraint conjecture for crystallographic lattices

摘录 / EXCERPT: Let a crystallographic group G act on a lattice, and let a symmetric short-range-entangled phase (sym-SRE phase) mean a phase preserving the symmetry and having no intrinsic long...

conjecture | open | MathDB 页面 | 原文线索

V2-059431 | MATHDB-365136

The monoidal reachability conjecture

摘录 / EXCERPT: Let $\mathscr{C}_\ell$ be the Hernandez-Leclerc category for $\ell\in\mathbb{Z}_{\geq 1}$, let $\mathcal{P}_\ell^+$ be its monoid of dominant monomials, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059432 | MATHDB-365153

Asymptotic stability conjecture for anti-de Sitter spacetime with dissipative boundary conditions

摘录 / EXCERPT: The pure anti-de Sitter spacetime has metric ... The boundary conditions are optimally dissipative. Asymptotic stability conjecture. Anti-de Sitter spacetime is asymptotically stable...

conjecture | open | MathDB 页面 | 原文线索

V2-059433 | MATHDB-365155

Holzegel-Smulevici conjecture on nonlinear instability of Kerr-AdS

摘录 / EXCERPT: The Kerr-AdS spacetimes are solutions of the Einstein equations considered as an initial-boundary value problem with Dirichlet boundary conditions. Holzegel-Smulevici conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-059434 | MATHDB-365364

Asymptotic maximal-eigenvalue conjecture for Haar-wavelet matrices

摘录 / EXCERPT: Let $A(N,K)$ be the $(N+1)K\times(N+1)K$ matrix whose entries are the integrals of products of Haar wavelets defined in the paper, and let $\lambda_{\max}(A(N,K))$ denote its maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-059435 | MATHDB-365365

Haar-wavelet solvability conjecture for near-Tsirelson Bell-CHSH violations

摘录 / EXCERPT: Haar-wavelet solvability conjecture. For every $\eta\in(\sqrt2-1,1)$, a resolution $\{N_0,N_1,K\}$ can be found such that this finite system admits a solution of the prescribed Haa...

conjecture | open | MathDB 页面 | 原文线索

V2-059436 | MATHDB-365378

Existence of anti-self-dual connections with small instanton number

摘录 / EXCERPT: Let X be a closed, connected, oriented, smooth four-dimensional manifold with $b^+(X)>1$ and odd $b^+(X)-b_1(X)$, possessing a Seiberg-Witten basic class. Let (E,H) be a comple...

conjecture | open | MathDB 页面 | 原文线索

V2-059437 | MATHDB-365379

Existence of anti-self-dual connections with small instanton number on the smooth blow-up

摘录 / EXCERPT: Let X satisfy the hypotheses of the Bogomolov-Miyaoka-Yau conjecture for four-manifolds with a Seiberg-Witten basic class, and let ... be its smooth blow-up. Let (E,H) be a compl...

conjecture | open | MathDB 页面 | 原文线索

V2-059438 | MATHDB-365386

Gaussian optimality conjecture for squashed entanglement of bosonic Gaussian channels

摘录 / EXCERPT: Let $\N$ be a bosonic Gaussian channel, and let the optimal squashing channel in Equation ... is always a bosonic Gaussian channel. This conjecture would reduce the optimization defi...

conjecture | open | MathDB 页面 | 原文线索

V2-059439 | MATHDB-365422

Regularity conjecture for the stable moduli hypersurfaces

摘录 / EXCERPT: Let $\mathfrak M$ be a Banach-space moduli space of initial data, and for $\mathfrak r\in[0,1]$ let $\mathfrak M^\mathfrak r_\mathrm{stab}\subset\mathfrak M$ consist of initial dat...

conjecture | open | MathDB 页面 | 原文线索

V2-059440 | MATHDB-365426

CMC existence conjecture for future timelike geodesically complete cosmological spacetimes

摘录 / EXCERPT: CMC existence conjecture. If (M,g) is future timelike geodesically complete and satisfies the strong energy condition, then (M,g) contains a CMC Cauchy surface.

conjecture | open | MathDB 页面 | 原文线索

V2-059441 | MATHDB-365558

The affine infinite-product factorization conjecture for universal K-matrices

摘录 / EXCERPT: Affine factorization conjecture. In affine type, Υ admits an infinite-product factorization analogous to such a factorization of the quasi-R-matrix of quantum affine alg...

conjecture | open | MathDB 页面 | 原文线索

V2-059442 | MATHDB-365559

The meromorphicity conjecture for spectral K-matrices

摘录 / EXCERPT: Meromorphicity conjecture. For all $V\in\mathcal{C}$, not necessarily irreducible, and for generic quantum symmetric pair parameters, $K_V(z)$ is the Laurent series expansion of a...

conjecture | open | MathDB 页面 | 原文线索

V2-059443 | MATHDB-365560

The classification conjecture for K-matrices on Kirillov-Reshetikhin modules

摘录 / EXCERPT: Kirillov-Reshetikhin K-matrix conjecture. Symmetrizable invertible solutions of the generalized spectral reflection equation in tensor products of Kirillov-Reshetikhin modules, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-059444 | MATHDB-365672

Robust product self-testing conjecture for synchronous games

摘录 / EXCERPT: Let $\mcG_1$ and $\mcG_2$ be two synchronous games. Robust product self-testing conjecture. The product game $\mcG_1\times\mcG_2$ is a robust self-test for its perfect strategies i...

conjecture | open | MathDB 页面 | 原文线索

V2-059445 | MATHDB-365712

The iSWAP optimality conjecture for 2-local unitary circuit ensembles

摘录 / EXCERPT: Let $n\geq 3$, let $\mathcal{G}$ be a fixed connected graph, and let $\mathcal{E}$ be a 2-local unitary circuit ensemble. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059446 | MATHDB-365713

The iSWAP equality conjecture for complete-graph gadgets

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and consider a gadget associated with a family of 2-local gates having parameter $a=\frac{5}{9}$ in the notation of the paper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059447 | MATHDB-365714

The iSWAP optimality conjecture for the brickwork model

摘录 / EXCERPT: Let n be the number of qubits in the brickwork model, let t denote the circuit depth or number of layers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059448 | MATHDB-365726

Link between trivial gauge transformations and power law weak fall-offs

摘录 / EXCERPT: Let $(M,\boldsymbol{g})$ be a D-dimensional Lorentzian manifold with conformal boundary ∂M_c. Let (J, M, G) be an abelian gauge theory with gauge group G and field...

conjecture | open | MathDB 页面 | 原文线索

V2-059449 | MATHDB-365869

Quantum KKL conjecture

摘录 / EXCERPT: Quantum KKL conjecture. There exists a universal constant $C>0$ such that for each $n\in\mathbb N$ and quantum Boolean function T,

conjecture | open | MathDB 页面 | 原文线索

V2-059450 | MATHDB-365935

Quantum regularity inherited from an associated classical model

摘录 / EXCERPT: Quantum regularity conjecture. If $(q(x),p(x\mid\theta))$ is regular, then $(\rho,\sigma(\theta))$ is also quantum regular.

conjecture | open | MathDB 页面 | 原文线索

V2-059451 | MATHDB-366048

Regularizing adiabatic Fredholm family conjecture for the gauge-theoretic adiabatic limit

摘录 / EXCERPT: Adiabatic Fredholm family conjecture. The data

conjecture | open | MathDB 页面 | 原文线索

V2-059452 | MATHDB-366064

Haar measure conjecture for symplectically integral lattices of fixed type

摘录 / EXCERPT: Let $D=\operatorname{diag}(d_1,\ldots,d_n)$ be a type, and let $\mathcal{L}\subseteq\mathcal{L}^{\perp}\subseteq\mathbb{R}^{2n}$ be a symplectically integral lattice of type D. H...

conjecture | open | MathDB 页面 | 原文线索

V2-059453 | MATHDB-366072

Large-N holography for the self-dual integrable $operatorname{Sp}(K)$ theory

摘录 / EXCERPT: Large-N holography conjecture. The large N limit of the open string algebra $\mathcal{A}^{N,K}$ is the celestial chiral algebra for the self-dual integrable...

conjecture | open | MathDB 页面 | 原文线索

V2-059454 | MATHDB-366073

The flavour-background quotient for collinear singularities

摘录 / EXCERPT: Flavour-background quotient conjecture. Collinear singularities in this theory are given by the quotient of $\mathcal{A}^{N,K}_{\mathrm{Defect}}$ by the relation that sets all bulk...

conjecture | open | MathDB 页面 | 原文线索

V2-059455 | MATHDB-366216

Large-loop asymptotic conjecture for the primitive beta function

摘录 / EXCERPT: Primitive beta-function asymptotic conjecture. The leading growth of the primitive beta function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059456 | MATHDB-366218

Universal finite-size scaling profiles on the torus

摘录 / EXCERPT: Universal-profile conjecture. The preceding susceptibility and two-point-function asymptotics hold for every listed model, with the same profile function for its SR and LR versions...

conjecture | open | MathDB 页面 | 原文线索

V2-059457 | MATHDB-366237

A lower-bound conjecture for the tripartite nullity

摘录 / EXCERPT: Let V_1,V_2,V_3 be the local qudit vector spaces with dimensions d_1,d_2,d_3, and let n_a be the corresponding one-party nullities. For a tripartite state...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059458 | MATHDB-366270

Uniform boundedness conjecture for the suboptimality ratio of projective measurements

摘录 / EXCERPT: Let $D(\mathcal{S})$ denote the density operators supported on an r-dimensional subspace $\mathcal{S}\subseteq\mathbb{C}^n$, and let $K_{n,r}$ be the suboptimality ratio defined...

conjecture | open | MathDB 页面 | 原文线索

V2-059459 | MATHDB-366329

Variance formula for sums of independent quantum computing errors

摘录 / EXCERPT: Let X_1 and X_2 be independent quantum computing errors in an n-qubit computation, with variances $V_1(X_1)$ and $V_2(X_2)$, respectively. Variance formula conjecture. The va...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059460 | MATHDB-366378

Conjecture on large strictly stable STCMC surfaces in asymptotically Schwarzschildean lightcones

摘录 / EXCERPT: Let $\mathcal{N}$ be an asymptotically Schwarzschildean lightcone. Let Σ be a strictly stable STCMC surface with area radius ρ, and let $B_\sigma(B_1,B_2,B_3)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-059461 | MATHDB-366379

Gauge-independent uniqueness conjecture for STCMC foliations

摘录 / EXCERPT: Let $\mathcal{N}$ be an asymptotically Schwarzschildean lightcone. An asymptotic foliation by strictly stable STCMC surfaces is a foliation of the asymptotic region of...

conjecture | open | MathDB 页面 | 原文线索

V2-059462 | MATHDB-366387

Lower-bound conjecture for the effective squeezing parameter

摘录 / EXCERPT: Effective-squeezing lower-bound conjecture. The effective squeezing satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059463 | MATHDB-366445

Seiberg-Witten solutions converge to gradient-flow t-orbits

摘录 / EXCERPT: Let Y^+ be a cylindrical-end 3-manifold with harmonic Morse function f^+, let Z be the set of its index-1 and index-2 critical points, and let a t-orbit be a weighted c...

conjecture | open | MathDB 页面 | 原文线索

V2-059464 | MATHDB-366461

Kashaev-Murakami-Murakami volume conjecture for knots and links

摘录 / EXCERPT: Let K be a framed knot or link in S^3. Let $V_N(K)$ be the colored Jones polynomial corresponding to the $(N+1)$-dimensional irreducible representation of the quantum group...

conjecture | open | MathDB 页面 | 原文线索

V2-059465 | MATHDB-366463

Fourier transform conjecture for descendant 1-loop invariants

摘录 / EXCERPT: Fix a cusped hyperbolic 3-manifold M, with longitude λ and a choice of meridian μ. For a root of unity ζ of odd order n, let $\tau_{M,\lambda,m}(\zeta)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059466 | MATHDB-366464

Baseilhac-Benedetti and 1-loop invariant correspondence conjecture

摘录 / EXCERPT: Let M be a cusped hyperbolic 3-manifold with geometric representation, and let n be odd. Consider the normalized 1-loop invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-059467 | MATHDB-366507

Unfairness-witness conjecture for symmetric two-mode mixtures

摘录 / EXCERPT: Unfairness-witness conjecture. The following inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-059468 | MATHDB-366545

Extension of the trace-representation formula for quantum Hellinger divergences

摘录 / EXCERPT: Let ρ and σ be the states appearing in the definition of the quantum Hellinger divergence $H_\alpha(\rho\|\sigma)$, and let $\alpha>1$. Trace-representation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-059469 | MATHDB-366609

Conjecture on the tensor structure of type II_{∞} von Neumann algebras in quantum gravity

摘录 / EXCERPT: Quantum-gravity tensor-structure conjecture. The tensor-product structure above describes the first steps away from the semiclassical approximation toward a full theory of quantum...

conjecture | open | MathDB 页面 | 原文线索

V2-059470 | MATHDB-366658

Classification of two-qudit Clifford hierarchy gates

摘录 / EXCERPT: Classification of two-qudit Clifford hierarchy gates. Every hierarchy gate of two qudits is either a Clifford gate or can be uniquely expressed as

conjecture | open | MathDB 页面 | 原文线索

V2-059471 | MATHDB-366659

Monotonicity conjecture for sets of quantum p-cost operators

摘录 / EXCERPT: For a separable Hilbert space $\mathcal{H}$, let $\mathbf{C}_p(\mathcal{H})$ denote the set of all possible p-cost operators. Monotonicity conjecture for quantum p-cost operato...

conjecture | open | MathDB 页面 | 原文线索

V2-059472 | MATHDB-366682

Asymptotic distance bound conjecture for $[[n,1,d]]$ M_3-codes

摘录 / EXCERPT: Distance bound conjecture. The distance of an $[[n,1,d]]$ M_3-code satisfies $d<d_{\max}$, with

conjecture | open | MathDB 页面 | 原文线索

V2-059473 | MATHDB-366720

De Sitter realization conjecture for worldsheet $(0,1)$ GLSMs

摘录 / EXCERPT: A GLSM is a gauged linear sigma model, and worldsheet $(0,1)$-supersymmetry is supersymmetry with this chirality structure. De Sitter realization conjecture. GLSMs with worldsheet...

conjecture | open | MathDB 页面 | 原文线索

V2-059474 | MATHDB-366812

Extreme black hole uniqueness conjecture for multiple degenerate black holes

摘录 / EXCERPT: Extreme black hole uniqueness conjecture. At least one tangent map parameter must be nonzero:

conjecture | open | MathDB 页面 | 原文线索

V2-059475 | MATHDB-366845

Categorical Grothendieck-ring conjecture for the shifted quantum affine cluster algebra

摘录 / EXCERPT: Let $\mathcal{A}$ be the infinite-rank cluster algebra and let ... be the injective ring morphism described above. Let $\mathcal{O}^{\mathrm{sh,f}}_{\mathbb{Z}}$ be the full subcateg...

conjecture | open | MathDB 页面 | 原文线索

V2-059476 | MATHDB-366911

Quantitative spectral-projector bound for quantum completely integrable manifolds

摘录 / EXCERPT: Quantum complete integrability conjecture. There exists $\kappa=\kappa(N,d)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059477 | MATHDB-366949

Lexicographic quasi-idempotent Schur duality for polynomial quantum wreath products

摘录 / EXCERPT: Assume that $B\wr \cH_d$ is a polynomial quantum wreath product satisfying conditions ... , and write $\bfH_d=B\wr \cH_d$. For each composition $\lambda\vDash d$, let K_λ be...

conjecture | open | MathDB 页面 | 原文线索

V2-059478 | MATHDB-366962

Strong contextuality conjecture for universally transversal code-switching protocols

摘录 / EXCERPT: Let $\mathcal{Q}_1$ and $\mathcal{Q}_2$ be subsystem stabilizer codes, and consider a fault-tolerant code-switching protocol between them. Let the set of transversal gates for each...

conjecture | open | MathDB 页面 | 原文线索

V2-059479 | MATHDB-367026

Compatibility conjecture for the sandwiched Rényi perspective

摘录 / EXCERPT: Compatibility conjecture. For $\alpha\in[2,\infty)$, the function $-\Psi_\alpha$ is $(\mathbb{H}^n_+,(2\alpha-1)/3)$-compatible with the domain $\mathbb{H}^n_+\times\mathbb{H}^n_+$...

conjecture | open | MathDB 页面 | 原文线索

V2-059480 | MATHDB-367154

No-naked-singularity conjecture for cosmological spacetimes

摘录 / EXCERPT: Let $(\widetilde{M}, \widehat{g})$ be a smooth, time-oriented, globally hyperbolic Lorentzian $(3+1)$-dimensional spacetime with $\widetilde{M} \cong M \times \mathbb{R}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-059481 | MATHDB-367212

The PPT-squared conjecture for entanglement-breaking quantum channels

摘录 / EXCERPT: Let $\matrices$ denote the matrix algebra in the paper. A linear map $\psi:\matrices\to\matrices$ is PPT if it is completely positive and completely copositive, meaning that both...

conjecture | open | MathDB 页面 | 原文线索

V2-059482 | MATHDB-367234

Quantum Bochner-Riesz conjecture

摘录 / EXCERPT: Let d be a positive integer, let m_δ be the Bochner-Riesz multiplier from the classical formulation, and let $\mathfrak{M}_p({\mathbb R}^{2d})$ denote the class of multipl...

conjecture | open | MathDB 页面 | 原文线索

V2-059483 | MATHDB-367293

Derrida-Griffiths-Higgs self-averaging and weak freezing transition conjecture

摘录 / EXCERPT: Derrida-Griffiths-Higgs conjecture. The following assertions hold: (i) the specific heat, namely the derivative of the finite-volume free energy, converges $\mathbb P$-almost surel...

conjecture | open | MathDB 页面 | 原文线索

V2-059484 | MATHDB-367554

Conjecture on smooth parametrization by initial, boundary, and corner data

摘录 / EXCERPT: Let ${\mathcal E}$ be the space of solutions modulo the relevant gauge equivalence, let ${\mathcal B}={\mathcal B}_C$ denote the conformal-mean curvature boundary data space, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059485 | MATHDB-367573

Weak-star density of Killing-vector derivations in local observables

摘录 / EXCERPT: Let (M,g) be the spacetime under consideration and let $ \mathcal{O}\in\mathcal{K}(M,g)$. Write $\mathrm{QuantLie}$ for the set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059486 | MATHDB-367641

Thermodynamic-limit conjecture for the extended Ising model

摘录 / EXCERPT: Pressure per particle of the extended Ising model. There exists $\varphi^{\scriptscriptstyle \rm elsing}(\beta,k,h)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059487 | MATHDB-367642

Thermodynamic-limit conjecture for the random-cluster partition function

摘录 / EXCERPT: Thermodynamic limit of the random-cluster partition function. As $n\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-059488 | MATHDB-367744

Locally compatible quantum d-to-1 pseudo-telepathy conjecture

摘录 / EXCERPT: Fix $d\geq 2$. A quantum assignment to a label-cover instance is k-compatible when the measurements associated with variables in every subinstance of Gaifman-graph diameter at mo...

conjecture | open | MathDB 页面 | 原文线索

V2-059489 | MATHDB-367790

Raising and lowering symmetry operators of the periodic Motzkin chain

摘录 / EXCERPT: Raising and lowering operator conjecture. There exist raising and lowering operators $\Sigma^\pm$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-059490 | MATHDB-367825

CMI decay at arbitrary temperatures

摘录 / EXCERPT: Let A, B, and C be regions of a quantum system ρ_β, with ... where Λ denotes the entire system. For a tripartite state, define the conditional mutual informati...

conjecture | open | MathDB 页面 | 原文线索

V2-059491 | MATHDB-367826

CMI decay at high temperatures

摘录 / EXCERPT: Let A, B, and C be regions of a quantum system, and let ρ_β be a quantum Gibbs state. Suppose ... The conditional mutual information is ... where $S_{\rho_\beta}(X)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-059492 | MATHDB-367827

Non-convergence of the cluster expansion for conditional mutual information

摘录 / EXCERPT: Let ρ_β be a reduced density matrix on a subsystem, and consider the cluster-expansion method applied to the logarithm of reduced density matrices in order to compute cond...

conjecture | open | MathDB 页面 | 原文线索

V2-059493 | MATHDB-367897

Bartnik Splitting Conjecture for globally hyperbolic spacetimes

摘录 / EXCERPT: Let M be a globally hyperbolic spacetime with compact Cauchy surfaces satisfying the strong energy condition. Bartnik Splitting Conjecture. If M is timelike geodesically comple...

conjecture | open | MathDB 页面 | 原文线索

V2-059494 | MATHDB-367924

Low-degree conjecture for color factors of non-oriented vacuum graphs

摘录 / EXCERPT: Let G be a non-oriented vacuum graph with $v_G\geq4$, and let its color factor be a polynomial or Laurent polynomial in N whose low degree is the smallest exponent of N with...

conjecture | open | MathDB 页面 | 原文线索

V2-059495 | MATHDB-368140

Optimality conjecture for the two-mode squeezed vacuum state in transmittance change detection

摘录 / EXCERPT: Let $\mathcal{E}^{\bar{n}_{\rm B},\eta_s}$ be the channel with background noise $\bar{n}_{\rm B}$ and transmittance η_s. For two-mode Gaussian input states with a fixed photon...

conjecture | open | MathDB 页面 | 原文线索

V2-059496 | MATHDB-368172

Born's rule conjecture for spacetime observables

摘录 / EXCERPT: Born's rule conjecture. The result of Proposition 3.1 of Finster (2013) should apply to pull back the spacetime observable of $\mathcal{U}$ to an observable on B.

conjecture | open | MathDB 页面 | 原文线索

V2-059497 | MATHDB-368179

Optimality of the noise threshold for three-outcome simulation of two binary measurements

摘录 / EXCERPT: Optimality conjecture. The value $t\leq 1/3$ in the preceding result is optimal: pairs of noisy binary measurements with noise parameter $t>1/3$ cannot, in general, be simulated by...

conjecture | open | MathDB 页面 | 原文线索

V2-059498 | MATHDB-368247

The quantum Wielandt conjecture for matrix sets

摘录 / EXCERPT: Let $S=\{v_i\}_{i=1}^g\subseteq\matrices$ be a collection of matrices, and define its Wielength by ... where $S^k=\{v_\sigma:\sigma\in\{1,\dots,g\}^k\}$,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059499 | MATHDB-368446

The generic singular-vector spanning conjecture

摘录 / EXCERPT: Let w be a word, let $H(w)$ be its space of ℓ-singular vectors of weight 0, and let $\mathrm{Conf}(w)$ be the set of arc configurations of w. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-059500 | MATHDB-368447

The standard-configuration singular-vector spanning conjecture

摘录 / EXCERPT: Let w be a word, let $H(w)$ be its space of ℓ-singular vectors of weight 0, and let C be the standard configuration of w. Let $v_C(u)$ denote the associated generic si...

conjecture | open | MathDB 页面 | 原文线索

V2-059501 | MATHDB-368519

Links-Gould invariant conjecture for braided exterior algebras

摘录 / EXCERPT: Let V be a vector space of dimension N, let ρ be the R-matrix-induced structure on the braided exterior algebra $[...].$ For a long knot, let J_{ρ} denote the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-059502 | MATHDB-368537

The global hyperbolicity conjecture for physical spacetimes

摘录 / EXCERPT: A strongly asymptotically predictable spacetime $(M,g_{\mu\nu})$ has black hole region ... where $J^-(\mathscr{I}^+)$ is the causal past of future null infinity and ψ is the ass...

conjecture | open | MathDB 页面 | 原文线索

V2-059503 | MATHDB-368579

Optimal correlational bound in the highly correlated regime

摘录 / EXCERPT: Let $Phi\in\mathfrak{h}\wedge\mathfrak{h}$ be normalized with canonical form ... Write $\lambda_{\max}=\sup_{k}\lambda_k$. Highly correlated regime conjecture. There exists a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-059504 | MATHDB-368640

The quantum cluster structure conjecture for bosonic extension subalgebras

摘录 / EXCERPT: Quantum cluster structure conjecture. The algebra $\widehat{\mathcal{A}}(b)$ is a quantum cluster algebra with initial seed $\mathbf{t}_{\mathbf{i}}$, and its cluster monomials are...

conjecture | open | MathDB 页面 | 原文线索

V2-059505 | MATHDB-368645

Quantum Riemannian limit conjecture for collapsing CFTs

摘录 / EXCERPT: Consider the collapsing family of unitary conformal field theories described above, with λ_{min} the minimal eigenvalue of the non-negative operator $L_0+\overline{L}_0$. R...

conjecture | open | MathDB 页面 | 原文线索

V2-059506 | MATHDB-368646

Precompactness conjecture for quantum Riemannian 1-spaces

摘录 / EXCERPT: Fix real numbers C, a, and b. Consider quantum Riemannian 1-spaces with measure whose associated Hilbert space comes from a unital involutive algebra and a state, with non-...

conjecture | open | MathDB 页面 | 原文线索

V2-059507 | MATHDB-368780

TRAP conjecture for distribution-valued meromorphic germs

摘录 / EXCERPT: Let $\mathcal{M}_{\mathrm{lin}}(\mathbb{C}^n,\mathcal{D}'_{\mathrm{reg}}((\mathbb{R}^d)^{\otimes N}))$ denote the space of distribution-valued meromorphic germs with linear poles,...

conjecture | open | MathDB 页面 | 原文线索

V2-059508 | MATHDB-368782

Acceleration conjecture for asymptotically free QFTs

摘录 / EXCERPT: Let $G(\Lambda,a)$ be the formal series for the two-point function of an asymptotically free quantum field theory, where Λ is the kinematical parameter. Suppose its β...

conjecture | open | MathDB 页面 | 原文线索

V2-059509 | MATHDB-368947

Quantum variance conjecture for compact hyperbolic manifolds

摘录 / EXCERPT: Quantum variance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-059510 | MATHDB-368979

Singleton end-image conjecture for Tutte embeddings

摘录 / EXCERPT: Let $\mathcal M_k$ be the supercritical random planar map and let H_k be its Tutte embedding into $\overline{\mathbb D}$. For an end ω of $\mathcal M_k$, define its image...

conjecture | open | MathDB 页面 | 原文线索

V2-059511 | MATHDB-368981

FUSF connectedness threshold conjecture for supercritical random planar maps

摘录 / EXCERPT: Let $\mathcal M_k$ be the supercritical random planar map, and let its free uniform spanning forest (FUSF) be the forest obtained from the free boundary condition. FUSF connectedne...

conjecture | open | MathDB 页面 | 原文线索

V2-059512 | MATHDB-369096

The Grassmannian Coulomb-branch relations conjecture

摘录 / EXCERPT: Grassmannian relations conjecture. Symmetric combinations of the equations

conjecture | open | MathDB 页面 | 原文线索

V2-059513 | MATHDB-369098

The level-dependent decomposition conjecture for quantum K-theory of gerbes

摘录 / EXCERPT: Gerbe decomposition conjecture. For the Ruan-Zhang/Chern-Simons level corresponding to ordinary quantum K-theory,

conjecture | open | MathDB 页面 | 原文线索

V2-059514 | MATHDB-369103

The coideal conjecture for the twisted affine Yangian of type D

摘录 / EXCERPT: Let $TY_\hbar(\widehat{\mathfrak{so}}(2n))$ be the associative algebra generated by ... with the defining relations specified in the paper. Let the affine Yangian associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-059515 | MATHDB-369107

The quantum toroidal superalgebra isomorphism for gl_{n|m}

摘录 / EXCERPT: Quantum toroidal superalgebra isomorphism. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-059516 | MATHDB-369125

Minuscule or cominuscule periodicity conjecture for quantum twist automorphisms

摘录 / EXCERPT: Minuscule or cominuscule periodicity conjecture. Let $t\in I$. Then $\PD(x_t)$ is finite if and only if $t\in I_{\mathrm{MC}}$. Moreover, if $t\in I_{\mathrm{MC}}$, then the follow...

conjecture | open | MathDB 页面 | 原文线索

V2-059517 | MATHDB-369130

Connes-Rovelli thermal time conjecture for gravitational clock rates

摘录 / EXCERPT: Let $g_{ik}(x)$ be a spacetime metric, let $dt=dx^0$ be coordinate time, and consider an ideal clock fixed at spatial position x^i, $i=1,2,3$. Its proper time satisfies ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059518 | MATHDB-369131

Sakharov's emergent-gravity conjecture

摘录 / EXCERPT: Let the vacuum fluctuations of matter fields be the quantum fluctuations of the matter degrees of freedom in the vacuum state. Sakharov's conjecture. General Relativity arises as a...

conjecture | open | MathDB 页面 | 原文线索

V2-059519 | MATHDB-369132

Conjecture on real-space creation and annihilation operators

摘录 / EXCERPT: Let $a^{(+)}(k)$ denote real-space creation or annihilation variables, and let $a^{(+)}(k,\beta)$ and $\tilde{a}^{(+)}(k,\beta)$ denote temperature-dependent primordial variables....

conjecture | open | MathDB 页面 | 原文线索

V2-059520 | MATHDB-369182

Conjecture on the quantization matrix for double Bruhat cells

摘录 / EXCERPT: Let G be a simply-connected, simple, complex algebraic group with Lie algebra $\mathfrak{g}$ of simply-laced type. Let $\mathcal{A}_{t,w_0}$ be the quantum cluster algebra struct...

conjecture | open | MathDB 页面 | 原文线索

V2-059521 | MATHDB-369207

The CM-field and Hermitianity conjecture for fusion categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a modular, ribbon or braided fusion category over $\mathbb{C}$. A category is Hermitian over a field K when it admits the corresponding Hermitian structure d...

conjecture | open | MathDB 页面 | 原文线索

V2-059522 | MATHDB-369210

Frenkel-Hernandez positivity conjecture for affine Langlands branching

摘录 / EXCERPT: Let $\mathcal{U}_q(\hat{\mathfrak{g}})$ and $\mathcal{U}_q(\leftindex^L{\hat{\mathfrak{g}}})$ be Langlands-dual quantum affine algebras, let P' be the specified sublattice of the...

conjecture | open | MathDB 页面 | 原文线索

V2-059523 | MATHDB-369314

The orbit-equivalence conjecture for prequantum implicit representations

摘录 / EXCERPT: Let $\mathcal{O}$ be a $\operatorname{Diff}_0(M)$-orbit in the explicit shape space $\mathcal{X}$, let $\mathcal{F}_{\mathcal{O}}\coloneqq \Pi^{-1}\mathcal{O}$ be the entire fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-059524 | MATHDB-369435

The Faber-Krahn conjecture for the first positive quantum-dot Dirac eigenvalue

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded domain with $\mathcal C^2$ boundary. Let $D\subset\mathbb{R}^2$ be a disk with the same area as Ω. For $a>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-059525 | MATHDB-369468

Fredkin-chain conjecture on the anti-adjoint representation of the raising and lowering operators

摘录 / EXCERPT: Anti-adjoint representation conjecture. There exist coefficients γ_k such that

conjecture | open | MathDB 页面 | 原文线索

V2-059526 | MATHDB-369469

Fredkin-chain conjecture on the B- and C-type symmetry algebra

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra generated by the raising and lowering operators $\Sigma^\pm$ of the periodic Fredkin chain with N sites, and set $n=\lceil N/2\rceil$. B- an...

conjecture | open | MathDB 页面 | 原文线索

V2-059527 | MATHDB-369470

Fredkin-chain conjecture on the central extension by total spin

摘录 / EXCERPT: Let $\mathfrak{g}$ be the Lie algebra generated by the raising and lowering operators of the periodic Fredkin chain with N sites. Let S^z be the third component of the...

conjecture | open | MathDB 页面 | 原文线索

V2-059528 | MATHDB-369475

The quantum dot Dirac eigenvalue minimization conjecture

摘录 / EXCERPT: Assume that $m\geq0$. For a bounded domain $\Omega\subset\mathbb{R}^2$ with C^2 boundary, let $\lambda_\Omega(\theta)$ denote the first nonnegative eigenvalue of the quantum dot...

conjecture | open | MathDB 页面 | 原文线索

V2-059529 | MATHDB-369522

Maximal incompatibility conjecture for random balanced dichotomic PVMs

摘录 / EXCERPT: Let $g\geq 2$. For each d, let $P_1,\ldots,P_g$ be g independent Haar-random projections of rank $d/2$ on $\mathbb C^d$, and let $A_1,\ldots,A_g$ be their associated dichotomic...

conjecture | open | MathDB 页面 | 原文线索

V2-059530 | MATHDB-369593

Conjecture on the listed poly-depth dualities between commuting Pauli models and classical interaction models

摘录 / EXCERPT: Poly-depth duality conjecture. The poly-depth dualities listed in the table hold in the sense of the paper's definition of duality.

conjecture | open | MathDB 页面 | 原文线索

V2-059531 | MATHDB-369894

Uniqueness conjecture for the Slater state in the ranges of the symmetric projections

摘录 / EXCERPT: Let d parties share the multiqudit Slater state $\ket{\Psi_{\mathrm S}}$, and let S_μ be the symmetric projections defined from the measurement construction. Uniqueness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-059532 | MATHDB-369895

Symmetric-operator commutant conjecture for the multiqudit Slater construction

摘录 / EXCERPT: Symmetric-operator commutant conjecture. The operators T_μ generate all the symmetric algebra $\mathfrak A_f$. Equivalently, for

conjecture | open | MathDB 页面 | 原文线索

V2-059533 | MATHDB-369901

Inverse inequality conjecture for quantum uniformity and hierarchy-overlap measures

摘录 / EXCERPT: Let U be an n-qudit unitary, and let $\left|\left|U\right|\right|_{Q^k}$ and $\left|\left|U\right|\right|_{q^k}$ denote its quantum uniformity norm and hierarchy-overlap measur...

conjecture | open | MathDB 页面 | 原文线索

V2-059534 | MATHDB-369964

Gaitsgory-Lysenko-Drinfeld-Stoyanovsky quantum geometric Langlands conjecture

摘录 / EXCERPT: Let G be a reductive group, let $\check{G}$ be its Langlands dual group, and let κ be a Kac-Moody level for G. Denote by $\check{\kappa}$ the dual level for $\check{G}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-059535 | MATHDB-370152

Fock-Goncharov's algebraic modular functor conjecture for PGL_{n+1}

摘录 / EXCERPT: Let $G=\operatorname{PGL}_{n+1}$, let S be a marked surface, and let c be an essential simple closed curve on S. Let S' be obtained by cutting S open along c, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059536 | MATHDB-370173

Strict hierarchy of critical couplings for the fermionic multi-particle model

摘录 / EXCERPT: Strict-hierarchy conjecture. If $q<2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-059537 | MATHDB-370209

Kerr-de Sitter rigidity conjecture for stationary vacuum black holes

摘录 / EXCERPT: Let $(\mathcal{M},\mathbf{g})$ be a regular, stationary (appropriately defined), four-dimensional vacuum black hole solution of the Einstein vacuum equations with positive cosmolog...

conjecture | open | MathDB 页面 | 原文线索

V2-059538 | MATHDB-370373

The alpha-SE operator approximation conjecture

摘录 / EXCERPT: The α-SE operator approximation conjecture. For any quantum operator satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-059539 | MATHDB-370426

Positivity conjecture for quantizability of simple modules

摘录 / EXCERPT: Positivity conjecture. Every simple module is quantizable. The supplied status evidence says that this conjecture is proved for reachable simple modules in affine types $C_n^{(1)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-059540 | MATHDB-370429

Triangle inequality conjecture for distance-matrix-induced pure-state distances

摘录 / EXCERPT: Triangle inequality conjecture. For any pure states $\textup{\textbf{x}},\textup{\textbf{y}},\textup{\textbf{z}}\in\mathbb{P}(\mathbb{C}^n)$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059541 | MATHDB-370446

Infirri's formality conjecture for Hermitian Yang-Mills moduli spaces

摘录 / EXCERPT: Let $M^{\bullet,\bullet}=\bigwedge^{\bullet,\bullet}V^*\otimes\operatorname{End}(R)$, let $\alpha\in\mathcal{N}_0\cap\mu^{-1}(\zeta)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059542 | MATHDB-370448

The fibrewise Hermitian Yang-Mills conjecture for the universal connection

摘录 / EXCERPT: Let $\mathfrak{p}_\zeta:\mathbb{P}\to\mathbb{M}$ be the universal bundle with connection $\mathbb{A}$, and let $p_\zeta:P_\zeta\to M_\zeta$ denote its restriction to a fibre. The f...

conjecture | open | MathDB 页面 | 原文线索

V2-059543 | MATHDB-370507

The theta specialization conjecture for the Rozansky-Overbay invariant

摘录 / EXCERPT: Theta specialization conjecture. The invariant θ dominates ρ_1; in particular, under the specialization $T_1\to T$ and $T_2\to 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-059544 | MATHDB-370508

The two-loop polynomial identification conjecture for theta

摘录 / EXCERPT: Two-loop polynomial identification conjecture. Up to conventions and normalizations, θ equals that two-loop polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-059545 | MATHDB-370509

The theta-$\mathfrak{sl}_3$ Rozansky invariant conjecture

摘录 / EXCERPT: Theta-$\mathfrak{sl}_3$ identification conjecture. Up to conventions and normalizations,

conjecture | open | MathDB 页面 | 原文线索

V2-059546 | MATHDB-370560

Hernandez-Leclerc's quiver-moduli formula for q-characters

摘录 / EXCERPT: Let ${\mathfrak{g}}$ be of finite type and let ${\boldsymbol{\psi}}$ be a reachable highest ℓ-weight. Write ${\boldsymbol{\psi}}$ in terms of vectors ${\mathbf{l}}=(l_{i,s})$...

conjecture | open | MathDB 页面 | 原文线索

V2-059547 | MATHDB-370570

Conjecture on conformal-gravity observables and the inclusion of Einstein observables

摘录 / EXCERPT: Let $d4d4_g$ and $d4d4$ denote the linearized conformal-gravity and general-relativity theories, respectively, and let $d4d0\mathrm{bs}^{\mathrm{cl}}(-,d4d4)$ denote their classica...

conjecture | open | MathDB 页面 | 原文线索

V2-059548 | MATHDB-370585

The non-root-of-unity extension conjecture for skein-module finiteness and holonomicity

摘录 / EXCERPT: Let $q\in\mathbb{C}^\times$ be a quantum parameter, and consider the finiteness and holonomicity assertions for skein modules stated in Theorems and. Non-root-of-unity extension co...

conjecture | open | MathDB 页面 | 原文线索

V2-059549 | MATHDB-370631

Ehlers-Kundt conjecture for complete Ricci-flat pp-waves

摘录 / EXCERPT: Let $(\mathbb{R}^4,g)$ be a Ricci-flat pp-wave with metric ... where $H\colon\mathbb{R}\times\mathbb{R}^2\to\mathbb{R}$. Ehlers-Kundt conjecture. The metric is geodesically complete...

conjecture | open | MathDB 页面 | 原文线索

V2-059550 | MATHDB-370634

Koszul duality for the geometric embedding and Kazhdan-Laumon category O

摘录 / EXCERPT: Let G be a semisimple group with Langlands dual group $G^\vee$, let $\mathfrak{u}_q$ be the small quantum group of $G^\vee$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059551 | MATHDB-370796

Spin alignment conjecture for entropy minimization

摘录 / EXCERPT: Let n be a positive integer, let $sigma$ be a density operator, and let M^c denote the complement of M in $\{1,2,\cdots,n\}$. For a subset M, write $\sigma^{\otimes M^c}$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-059552 | MATHDB-370851

Dimension-independent robustness conjecture for Gaussian symmetry tests

摘录 / EXCERPT: Let ψ be a pure even-parity state for Tests 1 or 2, or a pure state for Tests 4 or 5. In the two-copy cases set $p_{\operatorname{acc}}=\operatorname{Tr}[\psi^{\otimes 2}P]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-059553 | MATHDB-370852

Generic coexistence of null and spacelike singularities in gravitational collapse

摘录 / EXCERPT: Let a black hole form from generic gravitational collapse, and denote its terminal boundary's Cauchy horizon by $\mathcal{CH}_{i^+}$ and its spacelike singularity by $\mathcal{S}$....

conjecture | open | MathDB 页面 | 原文线索

V2-059554 | MATHDB-370924

Odd-length denominator formula for type C affine quantum algebras

摘录 / EXCERPT: Let $m,p\in 2\mathbb{Z}_{\geq 1}+1$, let $k<n$, and let $\mathfrak{g}=C_n^{(1)}$. Write $d_{k^m,k^p}(z)$ for the denominator polynomial of the corresponding normalized R-matrix,...

conjecture | open | MathDB 页面 | 原文线索

V2-059555 | MATHDB-370925

Higher Dorey's rule for odd Kirillov-Reshetikhin modules in type B

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type $B_n^{(1)}$, let $1\leq l<n-1$, and let $m\in 2\mathbb{Z}_{\geq 1}+1$. Let $\mathtt{V}(\boxed{i^r})_a$ denote the corresponding Kirillov-Reshetikhin m...

conjecture | open | MathDB 页面 | 原文线索

V2-059556 | MATHDB-371023

The scaling limit conjecture for random planar maps to Liouville quantum gravity

摘录 / EXCERPT: Let M_n be a random planar map with n edges whose law is given, for each fixed map $\mathfrak m$ with n edges, by ... where $\Delta_{\mathfrak m}$ is the graph Laplacian and...

conjecture | open | MathDB 页面 | 原文线索

V2-059557 | MATHDB-371024

The conformal embedding convergence conjecture for random planar maps

摘录 / EXCERPT: For each $n\in{\bf N}$, let M_n be a random planar map sampled as in the planar-map model above, and embed it into $\mathbbm S^2$ or a subset of $\mathbbm C$ by a discrete confor...

conjecture | open | MathDB 页面 | 原文线索

V2-059558 | MATHDB-371095

Conjectured maximum purity of absolutely PPT qutrit-qudit states

摘录 / EXCERPT: Let $\rho\in M_3\otimes M_n$ be an absolutely PPT state with $n\geq 2$, and let $\lambda_1,\ldots,\lambda_{3n}$ be its eigenvalues. Maximum-purity conjecture. The maximum purity of...

conjecture | open | MathDB 页面 | 原文线索

V2-059559 | MATHDB-371096

Conjectured Hilbert-Schmidt radius for absolutely PPT qutrit-qudit states

摘录 / EXCERPT: Let $M_3\otimes M_n$ be equipped with the Hilbert-Schmidt norm, and let $I/(3n)$ denote the maximally mixed state. Consider the smallest Euclidean ball centered at $I/(3n)$ that co...

conjecture | open | MathDB 页面 | 原文线索

V2-059560 | MATHDB-371222

The directed LQG metric existence and scaling-limit conjecture

摘录 / EXCERPT: For $\gamma\in(0,2)$ and $\widetilde\theta\in(0,\pi)$, let $\mathfrak L_\Phi^{\operatorname{LDP}}$ and $\mathfrak L_\Phi^{\operatorname{SDP}}$ denote candidate random functions obt...

conjecture | open | MathDB 页面 | 原文线索

V2-059561 | MATHDB-371248

The affine Laumon partition function conjecture for the non-stationary difference equation

摘录 / EXCERPT: Let N be a positive integer, let x_i, b_i, d_i, and $\overline{d}_i$ be parameters, and let $\mathcal{H}^{\widehat{\mathfrak{gl}}_N}(x_i;b_i,d_i,\overline{d}_i,q,\kappa)$...

conjecture | open | MathDB 页面 | 原文线索

V2-059562 | MATHDB-371303

Freeness conjecture for the positive preprojective K-theoretic Hall algebra

摘录 / EXCERPT: Let Q be a quiver, let ... is a free R-module. The conjecture is used to obtain the integral-form isomorphism with the positive half of the Maulik-Okounkov quantum loop group; th...

conjecture | open | MathDB 页面 | 原文线索

V2-059563 | MATHDB-371389

Uniqueness conjecture for faithful realizations of the Schrödinger algebra in the skew-hermitian Weyl algebra

摘录 / EXCERPT: Uniqueness conjecture. The Schrödinger algebra $\mathcal{S}$ admits a unique faithful realization as a subalgebra of $\hat{A}_1$, up to automorphism. This realization is the one ex...

conjecture | open | MathDB 页面 | 原文线索

V2-059564 | MATHDB-371391

The coproduct estimate conjecture for twisted Yangian generators

摘录 / EXCERPT: Let ${}^{\imath}\mathscr{Y}$ be regarded as a right coideal subalgebra of the Yangian $\mathscr{Y}$. For generating series ... let $\xi_i(u)$ be the corresponding Yangian Cartan gene...

conjecture | open | MathDB 页面 | 原文线索

V2-059565 | MATHDB-371463

Well-definedness conjecture for the Sinh-Gordon bootstrap correlation functions

摘录 / EXCERPT: Let $k\in\mathbb{N}$, let $G\in\mathcal{S}\big((\mathbb{R}^{1,1})^k\big)$, and let $\alpha_1,\dots,\alpha_k$ be labels of solutions to the system a)-d). For each...

conjecture | open | MathDB 页面 | 原文线索

V2-059566 | MATHDB-371502

The UV-finite theory conjecture on effective quantum master equations

摘录 / EXCERPT: In the BV formalism, let I be a local interaction for a UV-finite theory, and let $r\to 0$ denote the removal of the regularization. Write $I_n^\hbar$ for the multilinear operati...

conjecture | open | MathDB 页面 | 原文线索

V2-059567 | MATHDB-371511

Mass-systole conjecture for generalized asymptotically flat manifolds

摘录 / EXCERPT: Mass-systole conjecture. If (M,g,E) has nonnegative scalar curvature, then

conjecture | open | MathDB 页面 | 原文线索

V2-059568 | MATHDB-371542

Γ-geometric correspondence conjecture for quantized fields

摘录 / EXCERPT: Let X_Γ be a derived Γ-space, let $\mathcal D_{\mathrm{coh}}(X_\Gamma)$ be its coherent derived category, and let Phys_Γ be the category of Γ-p...

conjecture | open | MathDB 页面 | 原文线索

V2-059569 | MATHDB-371575

Universal K-matrix specialization conjecture for the q-Onsager algebra

摘录 / EXCERPT: Let ${\bf K}^{(\frac{1}{2})}(u)$ and ${\cal K}^{(\frac{1}{2})}(u)$ be, respectively, the spin-$\frac{1}{2}$ specialization of a universal K-matrix and the fused K-operator, with...

conjecture | open | MathDB 页面 | 原文线索

V2-059570 | MATHDB-371588

CLE carpet and stable-carpet identification conjecture

摘录 / EXCERPT: Set $\alpha \defeq 1/2+4/\kappa$. Let $(\mathcal{D},\Phi,0,\infty)$ be a γ-Liouville quantum gravity disk, and let Γ be an independent non-nested...

conjecture | open | MathDB 页面 | 原文线索

V2-059571 | MATHDB-371589

Weyl-scaling exponent conjecture for the LQG metric

摘录 / EXCERPT: Let D_Φ be the γ-Liouville quantum gravity metric on a γ-LQG surface, and let ζ denote its Weyl scaling exponent. Weyl-scaling exponent conjecture. The exp...

conjecture | open | MathDB 页面 | 原文线索

V2-059572 | MATHDB-371686

De Palma-Trevisan's triangle inequality conjecture for the quantum Wasserstein divergence

摘录 / EXCERPT: Let $W_{2,\mathrm{DPT}}$ be the quantum Wasserstein divergence defined from the De Palma-Trevisan quantum Wasserstein distance by ... Here $D(\rho,\sigma)$ is the original quadratic...

conjecture | open | MathDB 页面 | 原文线索

V2-059573 | MATHDB-371692

Conjecture on the minimum size of uniqueness sets for the Ising model

摘录 / EXCERPT: Let $u(k,q)$ denote the size of the smallest set of uniqueness for the positive function class $\left(\mathcal{B}^{k}_{q}\right)_+$. In particular, consider the case $q=2$. Minimum...

conjecture | open | MathDB 页面 | 原文线索

V2-059574 | MATHDB-371733

Intermediate Macdonald polynomial conjecture for transitive Weyl actions

摘录 / EXCERPT: Let $(\mathbf{U},\mathbf{B},\gamma)$ be a commutative triple, and suppose that the action of W_Σ on $\mathfrak{B}(\gamma)$ is transitive and that one stabiliser W_J is a...

conjecture | open | MathDB 页面 | 原文线索

V2-059575 | MATHDB-371737

Tsirelson's conjecture on quantum correlation sets

摘录 / EXCERPT: For integers $n,k\ge 2$, let $C_{qa}(n,k)$ and $C_{qc}(n,k)$ denote the corresponding approximately quantum and commuting quantum correlation sets, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059576 | MATHDB-371779

Conjecture on the distributional pointwise heat-kernel asymptotics of Liouville Brownian motion

摘录 / EXCERPT: Let $\textbackslash varphi$ be the Gaussian free field defining Liouville quantum gravity, let $\textbackslash mu_\textbackslash varphi$ be the associated Liouville measure, and sa...

conjecture | open | MathDB 页面 | 原文线索

V2-059577 | MATHDB-371877

Christandl's PPT^2 conjecture for entanglement-breaking compositions

摘录 / EXCERPT: Let $\Phi_1,\Phi_2\in\mathcal{T}_n(\mathbb{C})$ be linear maps. A map Φ is PPT when both Φ and $\Phi\circ\mathsf{T}$ are completely positive, and it is entanglement break...

conjecture | open | MathDB 页面 | 原文线索

V2-059578 | MATHDB-371885

The Kerr-Newman extension of the angular-momentum Penrose inequality

摘录 / EXCERPT: Consider initial data satisfying the appropriate energy conditions, with horizon area A, angular momentum J, and electric charge Q. Kerr-Newman extension conjecture. The ADM...

conjecture | open | MathDB 页面 | 原文线索

V2-059579 | MATHDB-371887

The generalized charged Penrose inequality for non-rotating black holes

摘录 / EXCERPT: Generalized charged Penrose conjecture. The ADM mass should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-059580 | MATHDB-371888

The Hawking mass bound for stable marginally outer trapped surfaces

摘录 / EXCERPT: Let Σ be a two-surface with area A and mean curvature H, and define its Hawking mass by ... Suppose that Σ is a stable marginally outer trapped surface (MOTS) with...

conjecture | open | MathDB 页面 | 原文线索

V2-059581 | MATHDB-371891

The Bekenstein bound for systems falling into black holes

摘录 / EXCERPT: Consider a system of energy E and size R falling into a black hole, and let ΔS_{BH} be the resulting change in black-hole entropy. Bekenstein bound conjecture. The seco...

conjecture | open | MathDB 页面 | 原文线索

V2-059582 | MATHDB-371893

The multipole bound for axisymmetric black holes

摘录 / EXCERPT: Let M be the mass, J the angular momentum, and M_2 the mass quadrupole of an axisymmetric black hole. Multipole bound conjecture. The quadrupole should satisfy ... The bound is...

conjecture | open | MathDB 页面 | 原文线索

V2-059583 | MATHDB-371934

The preprojective K-theoretic Hall algebra conjecture for quantum affine algebras

摘录 / EXCERPT: Let Q be a quiver of symmetric Kac-Moody type. Let $\mathrm{KHA}^{\mathbb{C}^*}_{\Pi_Q}$ denote the preprojective K-theoretic Hall algebra of the preprojective algebra Π_Q, f...

conjecture | open | MathDB 页面 | 原文线索

V2-059584 | MATHDB-372072

The BF blowup simple type and homogeneous type conjecture

摘录 / EXCERPT: Let X be a closed, smooth, oriented 4-manifold with $b_2^+(X)\geq 2$. The Bauer-Furuta invariant has BF blowup simple type and X is of BF homogeneous type. This conjecture ex...

conjecture | open | MathDB 页面 | 原文线索

V2-059585 | MATHDB-372099

Positivity conjecture for dual canonical basis coefficients and coproducts

摘录 / EXCERPT: Positivity conjecture. For every $b\in\mathbf{C}^\imath$, all coefficients $f_c(v)$, $g_{b',c}(v)$ and $h_{c,c'}(v)$ belong to $\mathbb{N}[v^{\frac12},v^{-\frac12}]$. In particular...

conjecture | open | MathDB 页面 | 原文线索

V2-059586 | MATHDB-372163

Complex compatibility-degree conjecture for one dichotomic and one $(k+1)$-outcome measurement

摘录 / EXCERPT: Let $k\in\mathbb N$ and $d\geq 2$. The minimum compatibility degree $s_{\mathbb C}(d,2,(2,k+1))$ concerns one dichotomic measurement and one measurement with $k+1$ outcomes on a...

conjecture | open | MathDB 页面 | 原文线索

V2-059587 | MATHDB-372164

Four-dichotomic-qubit compatibility-degree conjecture

摘录 / EXCERPT: Let $s_{\mathbb C}(2,4)$ denote the minimum compatibility degree of four dichotomic measurements on a complex qubit. Four-dichotomic-qubit compatibility-degree conjecture. ... The va...

conjecture | open | MathDB 页面 | 原文线索

V2-059588 | MATHDB-372218

Mainiero's generalized GHZ Poincaré polynomial conjecture

摘录 / EXCERPT: Let $n\geq 3$ and let $\vec{d}$ be a dimension vector. Write $P^{\mathrm{GHZ}}_n(y)=P\left(\lvert\mathrm{GHZ}_{n,\vec{d}}\rangle;y\right)$. Mainiero's generalized GHZ Poincaré poly...

conjecture | open | MathDB 页面 | 原文线索

V2-059589 | MATHDB-372219

Mainiero's generalized W Poincaré polynomial conjecture

摘录 / EXCERPT: Let $n\geq 3$ and let $\vec{d}$ be a dimension vector. Write $P^{\mathrm{W}}_n(y)=P\left(\lvert\mathrm{W}_{n,\vec{d}}\rangle;y\right)$. Mainiero's generalized W Poincaré polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-059590 | MATHDB-372255

The coproduct conjecture for the algebras $\mathcal{A}_{M,N}$

摘录 / EXCERPT: Let $M,N_1,N_2\in\mathbb{Z}$. The algebras $\mathcal{A}_{M,N}$ contain a subalgebra $\mathcal{K}_{M,N}$ extending the commuting quantum toroidal $\mathfrak{gl}_1$ algebras, and...

conjecture | open | MathDB 页面 | 原文线索

V2-059591 | MATHDB-372281

Coutts et al.'s spectral-calculus optimality conjecture for quantum channel optimization

摘录 / EXCERPT: Let $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ be finite-dimensional complex Hilbert spaces, let $\rho \in \mathsf{D}(\mathcal{X}\otimes \mathcal{Z})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059592 | MATHDB-372304

The Hopf-invariant formulas for fibered knots

摘录 / EXCERPT: Let K be a knot, let $\genusK$ denote its genus, let $\hopfK$ denote its Hopf invariant, let $\el(K)$ denote the integer-valued exponent invariant in the leading term of F_K, l...

conjecture | open | MathDB 页面 | 原文线索

V2-059593 | MATHDB-372309

The ADO radial-limit formula for nice knots

摘录 / EXCERPT: Let K be a nice knot, let $F_K(x,q)$ be its F_K series, let $N_p(K,x)$ be its ADO polynomial, let $\Delta_K(x)$ be its Alexander polynomial, and let $\zeta_p=e^{2\pi i/p}$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-059594 | MATHDB-372310

The fibered-knot leading-term conjecture for F_K

摘录 / EXCERPT: Let K be a fibered knot, let $\genusK$ be its genus, let $\hopfK$ be its Hopf invariant, and let F_K have leading term determined by the exponent $\el(K)$. Fibered-knot leading...

conjecture | open | MathDB 页面 | 原文线索

V2-059595 | MATHDB-372312

The roots-of-unity MMR expansion conjecture for F_K

摘录 / EXCERPT: Let ω be a primitive p-th root of unity with $p\geq 1$, let K be a nice knot, and let $F_K(x,q)$ be its F_K series. Let $P_{K,\omega}^{(j)}(x)$ be the polynomials occu...

conjecture | open | MathDB 页面 | 原文线索

V2-059596 | MATHDB-372313

The stability-series formula for strongly quasinegative knots

摘录 / EXCERPT: Let K be a strongly quasinegative knot. Assume that its stratified sum converges to a power series $F_K\in\mathop{\mathrm{FKring}}$, and let $\Phi_K(x,q)$ denote the stability se...

conjecture | open | MathDB 页面 | 原文线索

V2-059597 | MATHDB-372388

Conjecture on folding identities for arbitrary dominant monomials

摘录 / EXCERPT: Let $\mathfrak g$ be simply laced, let σ be the folding data, and let $\phi^\sigma_{\mathfrak g}$ and $\Phi^\sigma_{\mathfrak g}$ be the monomial and Grothendieck-ring foldi...

conjecture | open | MathDB 页面 | 原文线索

V2-059598 | MATHDB-372404

Two-body escape/merger threshold conjecture for prescribed ADM parameters

摘录 / EXCERPT: Let $(\mathbf{E}_1,\mathbf{P}_1,\mathbf{J}_1)$ and $(\mathbf{E}_2,\mathbf{P}_2,\mathbf{J}_2)$ lie in $\mathbb{R}_+\times\mathbb{R}^3\times\mathbb{R}^3$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-059599 | MATHDB-372422

Quantum uncertainty conjecture for portfolio risk and returns

摘录 / EXCERPT: Let $c3_{risk}$ and $c3_{returns}$ denote the standard deviations of portfolio risk and returns, respectively, as defined in modern portfolio theory. Let $b6$ be a lower bound infl...

conjecture | open | MathDB 页面 | 原文线索

V2-059600 | MATHDB-372511

The quantum-number positivity conjecture for quantum dilogarithm factorisations

摘录 / EXCERPT: Let Q be the quiver with two vertices $0,1$, no loops, and $k\in\mathbb{Z}_{\geq 1}$ arrows from 0 to 1. For $n\in\mathbb{Z}_{\geq 1}$, define the signed quantum numbers ... Us...

conjecture | open | MathDB 页面 | 原文线索

V2-059601 | MATHDB-372602

The classification conjecture for 1-super-transitive étale algebras in type A

摘录 / EXCERPT: Let N and k range over positive integers, and let $\mathcal{C}(\mathfrak{sl}_N,k)$ denote the corresponding categories. Let $\mathcal{A}$ be the set of all non-pointed étale al...

conjecture | open | MathDB 页面 | 原文线索

V2-059602 | MATHDB-372603

Uniform boundedness conjecture for super-transitivity of type A étale algebras

摘录 / EXCERPT: Let N and k range over positive integers, and consider all non-pointed étale algebra objects in the categories $\mathcal{C}(\mathfrak{sl}_N,k)$. Their super-transitivity is the...

conjecture | open | MathDB 页面 | 原文线索

V2-059603 | MATHDB-372775

Conjecture on Brownian motion and cutoff on the reflection quantum group H_N^{\infty+}

摘录 / EXCERPT: Brownian motion and cutoff conjecture. For every $t\geq 0$, the state φ_t is not absolutely continuous with respect to the Haar state. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-059604 | MATHDB-372865

Non-full exactness conjecture for vector spaces over general small quantum groups

摘录 / EXCERPT: Let $\mathcal{C}$ be the braided tensor category of finite-dimensional modules over $u_q(\mathfrak{g})$, where q is a root of unity of odd order and the category is equipped with...

conjecture | open | MathDB 页面 | 原文线索

V2-059605 | MATHDB-373079

Positive-PPT composition decomposability conjecture

摘录 / EXCERPT: Positive-PPT composition conjecture. The composition

conjecture | open | MathDB 页面 | 原文线索

V2-059606 | MATHDB-373092

The conjecture for categories associated with the subregular nilpotent in G_2

摘录 / EXCERPT: Let $l\geq 7$ be an integer not divisible by 3, and let q be the parameter defining the braided finite tensor category $\mathcal{C}(G_2,G_2(a_1),l,q)$. Write $T(\omega_1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059607 | MATHDB-373093

The Frobenius-Perron dimension conjecture for distinguished nilpotent categories

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $e\in\mathfrak{g}$ be distinguished nilpotent, and let $Q=Q(e)$ be the finite centralizer of an sl_2-triple associated with e in the si...

conjecture | open | MathDB 页面 | 原文线索

V2-059608 | MATHDB-373118

The no-enclosing-minimal-surfaces conjecture for maximal capacity extensions

摘录 / EXCERPT: Let (Σ,γ,H) be Bartnik data, and let (M,g) be a maximal capacity extension of (Σ,γ,H). No-enclosing-minimal-surfaces conjecture. The manifold M has no m...

conjecture | open | MathDB 页面 | 原文线索

V2-059609 | MATHDB-373210

The QCA-Witt group homomorphism conjecture

摘录 / EXCERPT: Let k be a field, let $\mathbf Q(\mathbb Z^3)$ be the QCA space in dimension three, and let Witt_k denote the Witt group of braided fusion k-linear categories. The Q...

conjecture | open | MathDB 页面 | 原文线索

V2-059610 | MATHDB-373382

Borel-category realization of the generalized QQ-system

摘录 / EXCERPT: Assume that ${\mathfrak{g}}$ is of finite type. Let I index the simple roots, let W be the Weyl group, and let $\mathcal{Q}_{w(\omega_i^\vee),a}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059611 | MATHDB-373383

Frenkel-Hernandez character expansion conjecture

摘录 / EXCERPT: Assume that ${\mathfrak{g}}$ is of finite type. Let W be the Weyl group, let $i\in I$, let $a\in\mathbb{C}^*$, and let $\boldsymbol{\psi}_{w(\omega_i^\vee),a}$ be the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-059612 | MATHDB-373400

Min-Oo conjecture for metrics on the hemisphere

摘录 / EXCERPT: Let g) be a smooth Riemannian metric on the closed hemisphere $\mathbb{S}^3_+$ satisfying the following conditions: its scalar curvature obeys ${\mathrm R}_g\geq 6$; its induced...

conjecture | open | MathDB 页面 | 原文线索

V2-059613 | MATHDB-373431

Berenstein-Greenstein positivity conjecture for coproduct structure constants

摘录 / EXCERPT: Let $\widetilde{{\mathbf U}}$ be the quantum group with double (equivalently, dual) canonical basis, and let its coproduct be the corresponding bialgebra coproduct. Berenstein-Gree...

conjecture | open | MathDB 页面 | 原文线索

V2-059614 | MATHDB-373469

The CFT conjectural limit for two-dimensional statistical mechanics models

摘录 / EXCERPT: CFT scaling-limit conjecture. The scaling limits of two-dimensional statistical mechanics models undergoing a continuous phase transition are given by the correlation functions of...

conjecture | open | MathDB 页面 | 原文线索

V2-059615 | MATHDB-373537

Equisingularity conjecture for adapted HYM metrics

摘录 / EXCERPT: Let X be a compact Kähler manifold and let $lpha$ be a nef and big class on X. Let $T=\theta+dd^c\boldsymbol{varphi}$ be an adapted closed positive $(1,1)$-current with minimal...

conjecture | open | MathDB 页面 | 原文线索

V2-059616 | MATHDB-373550

Minimal block-length conjecture for qubit permutation-invariant codes

摘录 / EXCERPT: Let a qubit permutation-invariant quantum error-correcting code correct errors of weight at most t, and let $n_{\rm min}(t)$ denote the smallest block length among such codes. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-059617 | MATHDB-373551

Mirror and phase-flip symmetry conjecture for real minimal qubit PI codes

摘录 / EXCERPT: Let a minimal qubit permutation-invariant code have real codeword coefficients, written as α_i and β_i, with minimal block length $n_{\rm min}$ for error weight t. T...

conjecture | open | MathDB 页面 | 原文线索

V2-059618 | MATHDB-373552

Manifold-dimension conjecture for real minimal qubit permutation-invariant codes

摘录 / EXCERPT: Fix an error weight t, set $n=n_{\rm min}(t)$, and consider the family of real minimal qubit permutation-invariant quantum error-correcting codes with parameters (n,t). Real mi...

conjecture | open | MathDB 页面 | 原文线索

V2-059619 | MATHDB-373614

General conjecture for the stationary behavior of multi-floor TASEP

摘录 / EXCERPT: Consider a finite multi-floor TASEP with $c\ge 1$ floors and attempted particle arrival and departure rates $\alpha>0$ and $\beta>0$. Let ... and let ... Assume condition (a): L^* co...

conjecture | open | MathDB 页面 | 原文线索

V2-059620 | MATHDB-373692

Instability of the Kerr Cauchy horizon

摘录 / EXCERPT: Consider generic initial data giving rise to a perturbation of a Kerr black hole governed by the Einstein vacuum equations, and let the perturbed spacetime have a Cauchy horizon. I...

conjecture | open | MathDB 页面 | 原文线索

V2-059621 | MATHDB-373700

The open-closed holography partition-function conjecture

摘录 / EXCERPT: Let $\mathcal{P}_{\mathcal{C},\lambda,h,s}=\{\mathcal{C},\lambda,h,s\}$ be a holography package consisting of a smooth cyclic A_∞-category $\mathcal{C}$, an object...

conjecture | open | MathDB 页面 | 原文线索

V2-059622 | MATHDB-373770

Banica-Bichon's maximality conjecture for classical permutation groups

摘录 / EXCERPT: Let S_N be the classical permutation group and S_N^+ Wang's quantum permutation group, with $S_N\leq S_N^+$ denoting the inclusion as a quantum subgroup. The Maximality Conject...

conjecture | open | MathDB 页面 | 原文线索

V2-059623 | MATHDB-373823

Spin alignment phenomenon for partially erasing channels

摘录 / EXCERPT: Let $mathcal N_0^{A_0\to B_0B_1}$ and $mathcal N_1^{A_1\to B_0B_1}$ be the partially erasing channels defined by fixed states $sigma_0\forin\fomathcal D(B_0)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059624 | MATHDB-373844

Balanced triples attaining finite-degree GV

摘录 / EXCERPT: Let (j_Z,j_X,k) be any balanced triple satisfying ... The corresponding HA-side and MN-side classical constituent codes are the codes associated with this triple, and a balanced tr...

conjecture | open | MathDB 页面 | 原文线索

V2-059625 | MATHDB-373876

Linear-kernel dimension conjecture for HGP codes based on LPS graphs

摘录 / EXCERPT: Fix t, and let n be large enough that the parameters $m_2,\ldots,m_t$ defined by ... where $c\in(-1,1)$ is chosen so that $\sum_i m_i=n+1$. Let H be the Tanner-code parity-chec...

conjecture | open | MathDB 页面 | 原文线索

V2-059626 | MATHDB-373946

The generic quantum group Drinfeld center conjecture

摘录 / EXCERPT: Let $\mathfrak g$ be a semisimple Lie algebra with weight and root lattices P and Q, respectively. For $q\in\mathbb{C}$, let $U_q(\mathfrak g)\mathbf{-mod}$ be the category of...

conjecture | open | MathDB 页面 | 原文线索

V2-059627 | MATHDB-374158

Nonlinear instability of Schwarzschild-AdS

摘录 / EXCERPT: Nonlinear instability of Schwarzschild-AdS. For any $s\in\mathbb N$, there exists an open and dense set $J^{\mathrm{EVE}}_s\subset(0,+\infty)$ of black-hole masses M such that th...

conjecture | open | MathDB 页面 | 原文线索

V2-059628 | MATHDB-374210

Semiclassical Weyl law for flat quantum tori

摘录 / EXCERPT: Let $\mathbb{T}^n_\theta$ be the quantum torus, let Δ be its flat Laplacian, let $q>0$, set $r=2nq^{-1}$, and suppose that either $r\neq 1$ and $r'=\max(r,1)$, or $r=1<r'$....

conjecture | open | MathDB 页面 | 原文线索

V2-059629 | MATHDB-374247

Symmetry correspondence conjecture for effective physical models and correlation geometries

摘录 / EXCERPT: Let an effective physical model be a model featuring an isometry, and let all fields be invariant under this isometry. Suppose the model is mapped by a reference system S satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-059630 | MATHDB-374370

Commutativity conjecture for the tetrahedral X-operators

摘录 / EXCERPT: For a positive integer N, let $X_j^{(N)}(z)$, for $j=0,\dots,N$, be the tetrahedral X-operators depending on the spectral parameter z. Commutativity conjecture. For an arbitr...

conjecture | open | MathDB 页面 | 原文线索

V2-059631 | MATHDB-374446

Yau's variant of the Penrose conjecture

摘录 / EXCERPT: Let $\alpha\geq 0$, and let (M^3,g) be an asymptotically flat manifold with nonnegative scalar curvature containing an outermost minimal surface Σ with induced metric...

conjecture | open | MathDB 页面 | 原文线索

V2-059632 | MATHDB-374449

The SR-selection conjecture for Haag-Kastler nets

摘录 / EXCERPT: Let $\{\mathcal{A}(\mathcal{O})\}$ be a Haag-Kastler net on a spacetime $\mathcal{M}$ satisfying axioms (HK1)-(HK4), with (HK2) interpreted as a sharper-than-equal-time commutation...

conjecture | open | MathDB 页面 | 原文线索

V2-059633 | MATHDB-374465

Conjecture on algebraic representatives of long-range charge densities

摘录 / EXCERPT: Let $\mathcal{R}$ be the R-matrix, and for $k\geq 2$ let $\mathcal{Q}^{(k)}$ denote the algebraic charge densities obtained by differentiating the R-matrix with respect to the...

conjecture | open | MathDB 页面 | 原文线索

V2-059634 | MATHDB-374696

Localized stated skein algebra and quantum cluster algebras conjecture

摘录 / EXCERPT: Localized skein-cluster conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-059635 | MATHDB-374715

Coproduct positivity conjecture for the embedded dual canonical basis

摘录 / EXCERPT: Let $\widetilde{{\mathbf U}}^\imath$ be a right coideal subalgebra of $\widetilde{{\mathbf U}}$, with embedding ι, and let $\mathcal{B}^\imath$ and $\mathcal{B}$ be the respe...

conjecture | open | MathDB 页面 | 原文线索

V2-059636 | MATHDB-374716

Positivity conjecture for dual canonical bases in type AIII

摘录 / EXCERPT: Consider the $\imath$quantum group of type ${\rm AIII}_{2r}$ and its dual canonical basis. Type-${\rm AIII}_{2r}$ positivity conjecture. The positivity results for the dual canonic...

conjecture | open | MathDB 页面 | 原文线索

V2-059637 | MATHDB-374751

Hikami's conjectural identities for Bailey-pair q-series and Eichler integrals

摘录 / EXCERPT: For $p\geq 1$, define the q-series $H_p^{(2)}(q),\ldots,H_p^{(7)}(q)$ by the formulas given in the source, and let ζ_N denote an Nth root of unity. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-059638 | MATHDB-374755

Weak Cosmic Censorship conjecture in the regularity class \(\mathcal{B}\)

摘录 / EXCERPT: Consider spherically symmetric initial data for the Einstein-scalar field equations in the regularity class $\mathcal{B}$, and let their maximal globally hyperbolic development (MG...

conjecture | open | MathDB 页面 | 原文线索

V2-059639 | MATHDB-374764

Uniform boundedness conjecture for scattering states of bounded compactly supported potentials

摘录 / EXCERPT: Uniform boundedness conjecture. There is a universal constant B_{V_0,R_0} such that

conjecture | open | MathDB 页面 | 原文线索

V2-059640 | MATHDB-374954

The generalized Penrose conjecture for asymptotically flat initial data sets

摘录 / EXCERPT: Let $(M^3,g,\mathbf{k})$ be a complete connected asymptotically flat initial data set satisfying the dominant energy condition. Choose an end with ADM mass m, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059641 | MATHDB-375016

Linear independence conjecture for the quantum unitary group $U_{q,\Theta}(n)$

摘录 / EXCERPT: Let $C(U_{q,\Theta}(n))$ be the universal C^*-algebra generated by $\{\mathscr{D}^{-1}\}\cup\{V_{i,j}:1\leq i,j\leq n\}$ satisfying the defining relations above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059642 | MATHDB-375065

Folding conjecture for q-characters of quantum affine algebras

摘录 / EXCERPT: Folding conjecture. The q-characters of the simple modules $L(\boldsymbol{\psi})$ of $U_q(L{\mathfrak{g}})$ and $L(\rho(\boldsymbol{\psi}))$ of $U_q(L{\mathfrak{g}}^\sigma)$ sati...

conjecture | open | MathDB 页面 | 原文线索

V2-059643 | MATHDB-375273

Kaledin-Verbitsky harmonic metric conjecture for NHYM bundles

摘录 / EXCERPT: Let (E,∇) be a bundle with a non-Hermitian Yang-Mills connection ∇. A bundle is ∇-stable if it is stable with respect to the connection ∇, and a harmoni...

conjecture | open | MathDB 页面 | 原文线索

V2-059644 | MATHDB-375290

Monodromy generation conjecture for the trigonometric Casimir connection

摘录 / EXCERPT: Monodromy generation conjecture. The monodromy representation is generated by the operator $\mathbf{B}_{\mathbf{m}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-059645 | MATHDB-375291

Slope subalgebra conjecture for finite-type quantum loop groups

摘录 / EXCERPT: Slope subalgebra conjecture. The slope subalgebra $\mathcal{B}_{\mathbf{m}}$ is isomorphic to the quantum group

conjecture | open | MathDB 页面 | 原文线索

V2-059646 | MATHDB-375357

Gleitz's generalized cluster categorification conjecture for restricted quantum loop algebra modules

摘录 / EXCERPT: Let $cal C_{varepsilon,xi}$ be the full subcategory of finite-dimensional modules of the restricted quantum loop algebra $U_varepsilon^{{\rm res}}(Lmathfrak{sl}_k)$ whose compositi...

conjecture | open | MathDB 页面 | 原文线索

V2-059647 | MATHDB-375358

Fraser's upper generalized cluster algebra conjecture for quantum affine module categories

摘录 / EXCERPT: Let $cal C_{varepsilon,xi}$ be the full subcategory of finite-dimensional modules of the restricted quantum loop algebra $U_varepsilon^{{\rm res}}(Lmathfrak{sl}_k)$ determined by t...

conjecture | open | MathDB 页面 | 原文线索

V2-059648 | MATHDB-375516

Global monotonicity conjecture for the Kubo-Mori update

摘录 / EXCERPT: Global monotonicity conjecture. The Kubo-Mori update can only increase the log-likelihood:

conjecture | open | MathDB 页面 | 原文线索

V2-059649 | MATHDB-375595

Quantum-disc heat-trace asymptotic conjecture

摘录 / EXCERPT: Let (Ω,h) be a quantum disc as above, and let $\mathbf{H}(t)$ denote its heat trace, with c_γ the leading Weyl-law constant and $\mu(\Omega)$ its quantum area. Quantu...

conjecture | open | MathDB 页面 | 原文线索

V2-059650 | MATHDB-375596

Quantum-disc KPZ exponent conjecture for the heat content

摘录 / EXCERPT: Let (Ω,h) be a quantum disc as defined in the cited work, and let $\mathbf{K}(t)$ denote the corresponding heat-content quantity. Quantum-disc heat-content conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-059651 | MATHDB-375618

Full polyhomogeneity of the spacetime metric

摘录 / EXCERPT: Let (γ,k) be initial data as in Theorem, let $\mathcal E_0$ be an index set with $\min\operatorname{Re}\mathcal E_0>1$, and suppose ... Let g be the spacetime metric produce...

conjecture | open | MathDB 页面 | 原文线索

V2-059652 | MATHDB-375619

Bondi mass limits and mass loss for boosted Kerr stability

摘录 / EXCERPT: Let (γ,k) be the initial data from Theorem, with ADM mass $\mathfrak m_0$ and vanishing linear momentum, let g be the resulting spacetime metric, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-059653 | MATHDB-375676

Relative quantumness data-processing conjecture for CoP channels

摘录 / EXCERPT: Relative quantumness data-processing conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-059654 | MATHDB-375680

Sharpness and uniqueness conjecture for the average confidence-width uncertainty constant

摘录 / EXCERPT: Let x) and p be the canonical position and momentum operators, and let E_0 be the ground-state energy of the self-adjoint operator $|x|+|p|$, namely ... The optimal constant...

conjecture | open | MathDB 页面 | 原文线索

V2-059655 | MATHDB-375714

Matter-sector proximity scaling in the general case

摘录 / EXCERPT: Let (M,g) be a closed Riemannian manifold with $n=\dim M$, let $\xi^*\in H^1(M,\mathbb{Z}_2)$ be a flat bundle class selected by the variational principle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059656 | MATHDB-375749

Mass gap conjecture for Potts lattice gauge theory

摘录 / EXCERPT: Let $\nu=\nu(\beta,q,d)$ be Potts lattice gauge theory on $\mathbb{Z}^d$. Let $\sigma_N=[0,1]^2\times\{N\}\times\{0\}$ be a plaquette, and for a plaquette σ let...

conjecture | open | MathDB 页面 | 原文线索

V2-059657 | MATHDB-375873

Penrose's weak cosmic censorship conjecture

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be a $3+1$-dimensional Lorentzian spacetime solving the Einstein field equations, arising as the maximal future development of generic, regular, asymptoticall...

conjecture | open | MathDB 页面 | 原文线索

V2-059658 | MATHDB-375874

Strong cosmic censorship conjecture

摘录 / EXCERPT: Consider generic initial data for a suitable Einstein-matter system or for the Einstein vacuum equations, and let the maximal Cauchy development be the spacetime arising from those...

conjecture | open | MathDB 页面 | 原文线索

V2-059659 | MATHDB-375937

Quantized universal enveloping algebra conjecture for the Neumann boundary condition

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie bialgebra with cobracket δ, let $U(\mathfrak{g})$ denote its universal enveloping algebra, and let $A^{!}_{\mathcal{N}}$ be the E_1 Koszul dual...

conjecture | open | MathDB 页面 | 原文线索

V2-059660 | MATHDB-375938

Chern-Simons universal line defect algebra conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be a Lie algebra, let $U(\mathfrak{g})[[{\mathchar'26\mkern-9muh}]]$ be its formal power-series enveloping algebra with standard coproduct Δ_0, let Φ...

conjecture | open | MathDB 页面 | 原文线索

V2-059661 | MATHDB-376023

Rotation symmetry conjecture for type B webs

摘录 / EXCERPT: Let $\mathsf{X}^{(k)}$ be the web in $\mathbf{Web}(\mathfrak{so}_{2n+1})$ defined in the surrounding construction, for $0\leq k\leq n$. Rotation symmetry conjecture. The 90°...

conjecture | open | MathDB 页面 | 原文线索

V2-059662 | MATHDB-376024

Nonclassical iota-canonical basis conjecture

摘录 / EXCERPT: Let $(V,\mathbb{B}_V)$ be a based U_{-q^2}-module, for example a finite-dimensional irreducible type I representation with Lusztig canonical basis $\mathbb{B}_V$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059663 | MATHDB-376025

Type B web divided-power exchange relation conjecture

摘录 / EXCERPT: Let $\mathsf{X}_i^{(k)}$ denote the nonclassical divided-power web at position i in $\mathbf{Web}(\mathfrak{so}_{2n+1})$, with $1\leq k\leq n$. Divided-power exchange relation co...

conjecture | open | MathDB 页面 | 原文线索

V2-059664 | MATHDB-376032

Positivity of the quadratic overshoot coefficients in the NPA hierarchy

摘录 / EXCERPT: For each finite NPA level k, let a_k be the coefficient in the small-s expansion ... where the coefficients satisfy $a_k\downarrow0$ as k increases. Positivity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-059665 | MATHDB-376133

Stability and commutation relations for quantum Volterra hierarchies

摘录 / EXCERPT: Let $\mathcal J$ and $\hat{\mathcal J}$ be the ideals defining the quotient algebras $\mathcal A_{\mathcal J}$ and $\mathcal A_{\hat{\mathcal J}}$, and let ∂_{t_ℓ},...

conjecture | open | MathDB 页面 | 原文线索

V2-059666 | MATHDB-376181

The factorized B'_2-Leonard pair conjecture for the quantum loop algebra construction

摘录 / EXCERPT: Let $V_{2j_1}(u)\tensor V_{2j_2}(u)$ be the tensor-product vector space, and let $(\mathcal{A}_2,\mathcal{A}_{12};\mathcal{B}_{12},\mathcal{B}_1)$ be the indicated ordered 4-tuple...

conjecture | open | MathDB 页面 | 原文线索

V2-059667 | MATHDB-376235

Ruskai-Audenaert decomposition conjecture for quantum channels

摘录 / EXCERPT: Let $P\in \operatorname{Herm}(\mathbb{C}^{d_1}\otimes \mathbb{C}^{d_2})$ be the Choi matrix of a completely positive trace-preserving map...

conjecture | open | MathDB 页面 | 原文线索

V2-059668 | MATHDB-376348

Conjectures on dual-unitary involutive R-matrices and pair creation

摘录 / EXCERPT: Let R be a unitary involutive R-matrix. An invertible matrix α is assumed to satisfy the relation referred to as Eq. (Ralphadef-graphical), and θ denotes the part...

conjecture | open | MathDB 页面 | 原文线索

V2-059669 | MATHDB-376469

Iwahori-equivariant convolution conjecture for the affine Grassmannian

摘录 / EXCERPT: Let G be a semisimple group, let $I\subset L^+G$ be an Iwahori subgroup, and let Gr be the affine Grassmannian. Let $P_I(Gr)$ be the category of I-equivariant perverse sheave...

conjecture | open | MathDB 页面 | 原文线索

V2-059670 | MATHDB-376477

Harmonic metric conjecture for non-Hermitian Yang-Mills connections

摘录 / EXCERPT: Harmonic metric conjecture. There exists a harmonic metric h on ${\cal B}$ if and only if ${\cal B}$ is a direct sum of ∇-stable bundles. If ${\cal B}$ itself is ∇-...

conjecture | open | MathDB 页面 | 原文线索

V2-059671 | MATHDB-376556

Ground-state transverse matrix-element formula conjecture

摘录 / EXCERPT: Transverse matrix-element conjecture. The first-order splitting matrix element satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059672 | MATHDB-376561

The quantum-invariant XXZ-chain 1-sum conjecture

摘录 / EXCERPT: The quantum-invariant-chain 1-sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059673 | MATHDB-376582

Zero-location conjecture for fused transfer-matrix eigenvalues

摘录 / EXCERPT: Let u be the parameter of the quantum transfer matrix, let $r\geq 1$, and let $a\in\{1,2,\dots,r\}$ and $m\in\mathbb{Z}_{\geq 1}$. Denote by $\widetilde{T}^{(a)}_{m}(v)$ the fuse...

conjecture | open | MathDB 页面 | 原文线索

V2-059674 | MATHDB-376585

Generalized ten Martini conjecture for magnetic Schrödinger operators on Fuchsian groups

摘录 / EXCERPT: Let Γ be a cocompact Fuchsian group and let σ be a multiplier on Γ. Let θ denote the flux associated with σ, and let $H_{\sigma,V}$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-059675 | MATHDB-376590

Odd-width square-lattice strip conjecture for the critical crossing

摘录 / EXCERPT: Let $q_0(L)$ denote the maximum real value in the limiting curve for a square-lattice strip of width L with fully periodic boundary conditions. Odd-width square-lattice conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-059676 | MATHDB-376591

Even-width square-lattice strip conjecture for the thermodynamic crossing

摘录 / EXCERPT: Let $q_0(L)$ denote the maximum real value in the limiting curve for a square-lattice strip of width L with fully periodic boundary conditions, and let $q_{\rm c}(\mathrm{sq})=3$...

conjecture | open | MathDB 页面 | 原文线索

V2-059677 | MATHDB-376593

Triangular-lattice strip conjecture for the 3p ± 1 widths

摘录 / EXCERPT: Let $q_0(L)$ denote the relevant real crossing for a triangular-lattice strip of width L with fully periodic boundary conditions. Triangular-lattice 3p ± 1 conjecture. For all wi...

conjecture | open | MathDB 页面 | 原文线索

V2-059678 | MATHDB-376600

Finiteness of open gaps in the local Iwatsuka model

摘录 / EXCERPT: Consider the local Iwatsuka model with a nonzero magnetic field b satisfying minimal regularity assumptions such as those formulated in Section 2.1. Finiteness conjecture. The nu...

conjecture | open | MathDB 页面 | 原文线索

V2-059679 | MATHDB-376657

Gluing conjecture for Donaldson invariants along surfaces

摘录 / EXCERPT: Let (X_1, Σ_1) and (X_2,Σ_2) be permissible 4-manifolds with $b_1(X_i)=0$, and let $\tilde X_i=X_i\#_{\Sigma_i}B_g$. Write ... Choose an identification ϕ, set...

conjecture | open | MathDB 页面 | 原文线索

V2-059680 | MATHDB-376678

CMC foliation conjecture for globally hyperbolic spacetimes

摘录 / EXCERPT: A spacetime is maximally extended if it admits no further extension, and it is globally hyperbolic if it has a Cauchy surface. A constant mean curvature (CMC) hypersurface is a spa...

conjecture | open | MathDB 页面 | 原文线索

V2-059681 | MATHDB-376679

Bray's multiple-end Penrose conjecture

摘录 / EXCERPT: Let (M^3,g,h) be complete, asymptotically flat Cauchy data with $\mu \ge |J|$ and more than one end. Choose one end as special, and let A be the minimum area required to enclos...

conjecture | open | MathDB 页面 | 原文线索

V2-059682 | MATHDB-376680

Equality of total inner and outer quasi-local masses

摘录 / EXCERPT: Let (M^3,g,h) be asymptotically flat with total ADM mass m_{ADM}. Define the total inner mass m_{inner}^{total} and total outer mass m_{outer}^{total} as the suprema of the...

conjecture | open | MathDB 页面 | 原文线索

V2-059683 | MATHDB-376681

Equality of total inner and outer masses for exhausting legal surfaces

摘录 / EXCERPT: Let (M^3,g,h) satisfy $\mu \ge |J|$, and suppose there is a nested sequence of connected legal surfaces ... with $\bigcup_iD_i=M$ and $\lim |\Sigma_i|=\infty$. Here legal means out...

conjecture | open | MathDB 页面 | 原文线索

V2-059684 | MATHDB-376682

Large-scale geometric reconstruction conjecture for stably causal Lorentzian manifolds

摘录 / EXCERPT: Let M be a stably causal Lorentzian manifold whose sectional curvatures are much less than 1, and let γ be a timelike curve from s to t in M whose radius of curvat...

conjecture | open | MathDB 页面 | 原文线索

V2-059685 | MATHDB-376691

Achronality of late marginally trapped spheres

摘录 / EXCERPT: Let $\mathcal{A}$ be the set of spherically symmetric marginally trapped spheres, let v be the null coordinate used in the paper, and let V be sufficiently large. Achronality c...

conjecture | open | MathDB 页面 | 原文线索

V2-059686 | MATHDB-376695

The conjectural string partition function product formula

摘录 / EXCERPT: Let $\mathcal{Z}$ be the string partition function, let x be the string coupling parameter, and let $F_0^{(0)}$ and $F_1^{(0)}$ be the classical genus-zero and genus-one potentia...

conjecture | open | MathDB 页面 | 原文线索

V2-059687 | MATHDB-376706

Hopf-kernel characterization of confinement walls

摘录 / EXCERPT: Let $\mathcal{T}$ be the residual symmetry algebra, let $\mathcal{U}$ be the non-confined algebra, and let $\Gamma:\mathcal{T}\to\mathcal{U}$ be the Hopf map. Define the left and r...

conjecture | open | MathDB 页面 | 原文线索

V2-059688 | MATHDB-376714

Toric duality as Seiberg duality for toric moduli spaces

摘录 / EXCERPT: Consider ${\cal N}=1$ theories whose moduli spaces admit toric descriptions. Toric-duality conjecture. Toric duality is Seiberg duality for ${\cal N}=1$ theories with toric moduli...

conjecture | open | MathDB 页面 | 原文线索

V2-059689 | MATHDB-376715

Universality conjecture for split localization entropy on a halfline

摘录 / EXCERPT: The transverse symmetry and absence of correlations reduce the area-density problem to a chiral theory on a lightray, which is mapped by a Cayley transform to a chiral theory on th...

conjecture | open | MathDB 页面 | 原文线索

V2-059690 | MATHDB-376720

Affine theta-function expansion conjecture for the A_{N-1} blow-up formula

摘录 / EXCERPT: Affine theta-function expansion conjecture. The A_{N-1} theta function in the blow-up formula satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059691 | MATHDB-376730

Jordan decomposition conjecture for factorizable ribbon quantum groups

摘录 / EXCERPT: Let H be a factorizable ribbon quantum group, with ribbon element having multiplicative Jordan decomposition. The decomposition gives $SL(2,\mathbb{Z})$-representations $\bar\pi$...

conjecture | open | MathDB 页面 | 原文线索

V2-059692 | MATHDB-376734

The Clebsch-Gordan coefficient property for nilpotent representations of quantum groups

摘录 / EXCERPT: The paper considers nilpotent representations of $U_q(g)$, where g is a simple Lie algebra, satisfying $(X_i^{\pm})^N=0$ and $(K_i)^N=q^{2Np_i}$ for $p_i\in\mathbf{C}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-059693 | MATHDB-376735

Feigin's quotient-ring conjecture for quantum nilpotent algebras

摘录 / EXCERPT: Feigin's conjecture. The map $\Phi(i_1,\ldots,i_N)$ is an embedding, and, for at least one special reduced decomposition of w_0, it extends to an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-059694 | MATHDB-376736

The quantum Gel'fand-Kirillov field-isomorphism conjecture

摘录 / EXCERPT: Quantum Gel'fand-Kirillov field-isomorphism conjecture. The map ρ should provide an isomorphism between a quotient field of $\widehat{U}_q(\mathfrak{sl}_{n+1})$ and a quotient...

conjecture | open | MathDB 页面 | 原文线索

V2-059695 | MATHDB-376744

Braided tangent vector-field equations for quantum orbits

摘录 / EXCERPT: Braided tangent-field conjecture. Equations (4) hold if we replace U_i by U_i^q and $f_{\beta,i}$ by $f_{\beta,i}(q)$.

conjecture | open | MathDB 页面 | 原文线索

V2-059696 | MATHDB-376752

Modular cocycle conjecture for massive double cone algebras

摘录 / EXCERPT: Let $4\mathcal{A}(\mathcal{O})4$ be the algebra of a massive theory associated with a double cone $4\mathcal{O}4$, and let $4\omega_{m=0}4$ denote the conformally invariant vacuum...

conjecture | open | MathDB 页面 | 原文线索

V2-059697 | MATHDB-376754

Plekton-chiral conformal field theory correspondence conjecture

摘录 / EXCERPT: Let “free” plektons denote the class of free plektonic quantum field theories in $2+1$ dimensions, and let chiral conformal quantum field theories be conformal theories on a one-di...

conjecture | open | MathDB 页面 | 原文线索

V2-059698 | MATHDB-376756

Exact U-duality Eisenstein-series conjecture for toroidal type II R^4 couplings

摘录 / EXCERPT: Let type II string theory be compactified on T^d, equivalently M-theory on T^{d+1}, and let $E_{d+1(d+1)}(\mathbb{Z})$ be the U-duality group. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-059699 | MATHDB-376758

Genus-g spinor Eisenstein-series conjecture for lattice integrals

摘录 / EXCERPT: Let I_d^g be the integral of the (d,d) lattice partition function over the fundamental domain of the moduli space of genus-g Riemann surfaces, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059700 | MATHDB-376759

Type IIA R^4H^{4g-4} constrained-Eisenstein-series conjecture

摘录 / EXCERPT: Let type IIA string theory be compactified on T^d with $d\leq 4$, let ϕ be the T-duality-invariant dilaton, let H_{RR} denote the Ramond three-form field strengths, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-059701 | MATHDB-376761

Liouville fusion-transform conjecture for conformal blocks

摘录 / EXCERPT: Let $mathbb S$ be the contour of intermediate Liouville momenta, and let ${{\mathcal F}}^s_{\alpha_{21}}$ and ${{\mathcal F}}^t_{\alpha_{32}}$ denote the s- and t-channel conformal...

conjecture | open | MathDB 页面 | 原文线索

V2-059702 | MATHDB-376773

Large-spin Bose-gas limit conjecture for the XXZ spectral gap

摘录 / EXCERPT: Large-spin Bose-gas conjecture. There is a function $\gamma_\infty:(1,\infty)\times\mathbb{R}\to\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059703 | MATHDB-376787

Boundary-data conjecture for finite-energy monopole differences

摘录 / EXCERPT: Let ${\mathcal M}$ be the monopole moduli space, let ${\mathcal M}^{\infty}$ be the moduli space of boundary data at infinity, and let ... be the boundary map. For monopoles...

conjecture | open | MathDB 页面 | 原文线索

V2-059704 | MATHDB-376797

The ground-state energy ordering conjecture for Heisenberg XXZ and XY models

摘录 / EXCERPT: Let L be the sector label and let ε_m denote the quantity introduced in the paper for sector m. Ground-state energy ordering conjecture. ... for all m such that...

conjecture | open | MathDB 页面 | 原文线索

V2-059705 | MATHDB-376801

Lieb's Wehrl entropy lower-bound conjecture for spin states

摘录 / EXCERPT: Let ${\mathcal F}_j$ be the spin-j function space, let $\lVert\cdot\rVert_2$ be the normalized norm, and let ... be the Wehrl entropy of an $f\in{\mathcal F}_j$ normalized by...

conjecture | open | MathDB 页面 | 原文线索

V2-059706 | MATHDB-376804

Petz's monotonicity conjecture for the scalar curvature of the Kubo-Mori metric

摘录 / EXCERPT: Let D_1 and D_2 be states, and write $D_1\prec D_2$ when D_1 is more mixed than D_2 in the majorization order. Let $\operatorname{Scal}(D)$ denote the scalar curvature of t...

conjecture | open | MathDB 页面 | 原文线索

V2-059707 | MATHDB-376814

The noncommutative alpha-geometry conjecture on scalar-curvature Schur monotonicity

摘录 / EXCERPT: Noncommutative alpha-geometry conjecture. Suppose $n\geq2$. If $p\in(1,2)$, then Scal_p is strictly Schur-decreasing; if $p\in(2,+\infty]$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-059708 | MATHDB-376815

The WYD scalar-curvature Schur-increasing conjecture near the BKM metric

摘录 / EXCERPT: WYD scalar-curvature conjecture. There exists $\varepsilon>0$ such that, for every $p\in I$, the scalar curvature of the WYD(p) metric is a Schur-increasing function.

conjecture | open | MathDB 页面 | 原文线索

V2-059709 | MATHDB-376817

The twisted Weyl-vector formula for twisted loop algebras

摘录 / EXCERPT: Let $\widetilde{\mathfrak{g}}$ be a twisted loop algebra with $\epsilon(\mathfrak{h})=0$. Let $\widehat{\rho}$ be the Weyl vector determined by the simple roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-059710 | MATHDB-376822

The Zeno product formula conjecture for admissible functions

摘录 / EXCERPT: Zeno product formula conjecture. If assumption in Theorem II.6a is dropped, then for every $f\in{\mathfrak h}$, $T>0$, and arbitrary admissible function ϕ,

conjecture | open | MathDB 页面 | 原文线索

V2-059711 | MATHDB-376850

Ferromagnetic Ordering of Energy Levels conjecture

摘录 / EXCERPT: Ferromagnetic Ordering of Energy Levels conjecture. All ferromagnetic Heisenberg models have the FOEL property: if $S'<S$, then

conjecture | open | MathDB 页面 | 原文线索

V2-059712 | MATHDB-376870

The irrational-angle counterexample conjecture for weak rotational symmetry

摘录 / EXCERPT: Irrational-angle counterexample conjecture. It is impossible to find a counterexample satisfying $\varphi_0\notin 2\pi\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-059713 | MATHDB-376871

The extended local-permutation conjecture for block-shift matrices

摘录 / EXCERPT: Extended local-permutation conjecture. Every element of $E(\mathfrak{t}_{\mathrm{loc}})$ is similar to a block-shift matrix via an element of $\Pi_{\mathrm{loc}}^{\mathrm{ex}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-059714 | MATHDB-376904

The unipotency conjecture for twisted function algebras

摘录 / EXCERPT: Unipotency conjecture. The operator

conjecture | open | MathDB 页面 | 原文线索

V2-059715 | MATHDB-376949

Volume conjecture for closed three-manifolds

摘录 / EXCERPT: Volume conjecture for closed three-manifolds.

conjecture | open | MathDB 页面 | 原文线索

V2-059716 | MATHDB-376950

Stationary-phase conjecture for figure-eight knot surgeries

摘录 / EXCERPT: For an integer p, define ... Let M_p be the closed three-manifold obtained from the three-sphere by p-surgery along the figure-eight knot. Stationary-phase conjecture for figur...

conjecture | open | MathDB 页面 | 原文线索

V2-059717 | MATHDB-376985

The 10/8-conjecture for non-spin 4-manifolds

摘录 / EXCERPT: The $10/8$-conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-059718 | MATHDB-376986

Carter-Marsh conjecture on simplicial regions of Lusztig's piecewise-linear function

摘录 / EXCERPT: Suppose $\mathbf{g}$ is of type A_n, and let $\mathbf{i}$ be a reduced expression for the longest Weyl-group element w_0. Let $\mathbf{k}=(1,3,5,\ldots,2,4,6,\ldots)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-059719 | MATHDB-376995

The Lusztig-cone region-of-linearity conjecture for type A canonical bases

摘录 / EXCERPT: Let w_0 be the longest element of the Weyl group, let $\mathbf{i}$ and $\mathbf{j}$ be reduced expressions of w_0, let $C_{\mathbf{i}}$ be the Lusztig cone, and let ... be the tr...

conjecture | open | MathDB 页面 | 原文线索

V2-059720 | MATHDB-377003

The local edge-density conjecture for domino tilings

摘录 / EXCERPT: In the dimer model on an $n \times n$ torus, let p_a, p_b, p_c, and p_d denote the limiting densities of the four edge types in the thermodynamic limit as...

conjecture | open | MathDB 页面 | 原文线索

V2-059721 | MATHDB-377024

The GGS conjecture for the GGS R-matrix

摘录 / EXCERPT: Let R_{GGS} be the matrix defined by ... where the polynomials $P_{i,j,k,l}$ are uniquely determined by ... Here $q=e^{\hbar}$, and the quantum Yang-Baxter equation is ... while...

conjecture | open | MathDB 页面 | 原文线索

V2-059722 | MATHDB-377171

Virtual-crystal realization conjecture for the four affine types

摘录 / EXCERPT: Let $4\mathfrak{g}4$ be of type $D_{n+1}^{(2)}$, $A_{2n}^{(2)}$, $A_{2n}^{(2)\dagger}$, or $C_n^{(1)}$, let $1\leq r\leq n$ and $s\geq1$, and let $V^{r,s}$ be the virtual crystal w...

conjecture | open | MathDB 页面 | 原文线索

V2-059723 | MATHDB-377194

Finiteness conjecture for regular weak Hopf algebras

摘录 / EXCERPT: Finiteness conjecture. In any given dimension, the number of non-isomorphic weak Hopf algebras H with this property is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-059724 | MATHDB-377286

The intersection-sheaf characterization of the canonical basis

摘录 / EXCERPT: Let ${\mathcal{P}}$ be the category of semisimple perverse sheaves on ${\mathfrak{T}(\mathbf{d})}'$ constructible with respect to the stratification by the...

conjecture | open | MathDB 页面 | 原文线索

V2-059725 | MATHDB-377287

Verdier duality corresponds to the canonical-basis involution

摘录 / EXCERPT: Let ${\mathcal{P}}$ be the category of semisimple perverse sheaves on ${\mathfrak{T}(\mathbf{d})}'$, let $\mathcal{D}:{\mathcal{P}}\to{\mathcal{P}}$ be Verdier duality, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059726 | MATHDB-377291

Isomorphism conjecture for the q-deformed multiple-zeta-value and quantum Grothendieck-Teichmüller representations

摘录 / EXCERPT: Representation isomorphism conjecture. The corresponding representations of $\mathcal{Z}_q$ and $\widetilde{\mathcal{H}_{GT}}$ on quantum field theories on $S^{4,\theta}$ are isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-059727 | MATHDB-377374

Two-term expansion conjecture for products involving a real dual canonical basis vector

摘录 / EXCERPT: Let $b_1,b_2\in{\bf B}^*$, where b_1 is real, meaning $b_1^2\in q^{\mathbb Z}{\bf B}^*$. Suppose that $b_1b_2\notin q^{\mathbb Z}{\bf B}^*$. Two-term expansion conjecture. The ex...

conjecture | open | MathDB 页面 | 原文线索

V2-059728 | MATHDB-377386

Nakajima's monomial realization conjecture for highest weight crystals

摘录 / EXCERPT: Let ${\mathcal M}$ be the set of monomials in variables $Y_i(n)$, with $i\in I$ and $n\in\mathbb{Z}$, equipped with the crystal structure defined by the functions and operators in...

conjecture | open | MathDB 页面 | 原文线索

V2-059729 | MATHDB-377488

The quantum Hall algebra isomorphism conjecture for the positive loop algebra

摘录 / EXCERPT: Let $\mathbf{U}^+$ be the $\mathbb{C}_v$-subalgebra of $\mathbf{U}_v(L\mathfrak{gl}_p)$ generated by $\mathbf{U}_0^+$ and $h_{0,l}$ for $l>0$, where $\mathbf{U}_0^+$ is generated b...

conjecture | open | MathDB 页面 | 原文线索

V2-059730 | MATHDB-377517

The center-trace equality conjecture for factorizable ribbon Hopf algebras

摘录 / EXCERPT: Center-trace equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059731 | MATHDB-377537

The q-index characteristic-polynomial conjecture for braided orbits

摘录 / EXCERPT: The q-index characteristic-polynomial conjecture. For every $m\ge 2$, the polynomial ${\cal CH^{\chi}}_{(m)}(t)$ has degree

conjecture | open | MathDB 页面 | 原文线索

V2-059732 | MATHDB-377570

Berenstein-Zelevinsky's multiplicativity conjecture for dual canonical basis vectors

摘录 / EXCERPT: Let Q be a Dynkin quiver of type cA_n, cD_n or $cE_{6,7,8}$, let U_v^- be the negative part of the quantized enveloping algebra of the corresponding Lie algebra, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-059733 | MATHDB-377623

Explicit projector formula for spinor tensor products

摘录 / EXCERPT: The explicit projector formula. The idempotent projecting $\lambda\otimes S$ onto the simple object indexed by b is

conjecture | open | MathDB 页面 | 原文线索

V2-059734 | MATHDB-377634

The RLL relation for the elliptic quantum group L-operator

摘录 / EXCERPT: Let $\widehat{L}^+(v)\in\operatorname{End}(\mathbb{C}^N)\otimes U_{q,p}(\widehat{\mathfrak{sl}}_N)$ be the L-operator constructed from the half currents, and let $R^+(v,P)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-059735 | MATHDB-377655

The double-limit volume conjecture for cusped manifolds

摘录 / EXCERPT: Let M be a cusped manifold, and let (W_n,L_n,ρ_n) be a sequence of compact hyperbolic Dehn fillings of M converging to M, with L_n the link of short simple geodesics f...

conjecture | open | MathDB 页面 | 原文线索

V2-059736 | MATHDB-377860

Commutativity conjecture for powers of the raising and lowering maps

摘录 / EXCERPT: Let U_0 be the subspace used in the paper's decomposition associated with a tridiagonal pair, and let L and R denote the corresponding lowering and raising maps. Commutativit...

conjecture | open | MathDB 页面 | 原文线索

V2-059737 | MATHDB-377866

Uniqueness conjecture for static asymptotically flat extensions

摘录 / EXCERPT: Uniqueness conjecture for static extensions. (Ω,g) determines a unique static asymptotically flat manifold (S,g) with boundary $\Sigma\simeq\partial\Omega$ satisfying thes...

conjecture | open | MathDB 页面 | 原文线索

V2-059738 | MATHDB-377884

The critical Yang-Mills global well-posedness conjecture

摘录 / EXCERPT: Let $\mathcal G$ be a compact Lie group with Lie algebra $\mathbf g$, let $s_0=\frac{n-2}{2}$ be the scaling index, and consider the Yang-Mills equation ... for a connection with cur...

conjecture | open | MathDB 页面 | 原文线索

V2-059739 | MATHDB-377941

Local structure conjecture for the compactified instanton moduli space

摘录 / EXCERPT: Local structure conjecture. (1) There exists a proper continuous map

conjecture | open | MathDB 页面 | 原文线索

V2-059740 | MATHDB-377952

Quantum hyperbolic asymptotics and Dehn-filling compatibility conjecture

摘录 / EXCERPT: Let M be a cusped manifold, define ... and let $H_N(M)$ denote the proposed quantum hyperbolic invariant. Here ${\rm CS}(M)$ and ${\rm Vol}(M)$ are respectively the metric Chern-Si...

conjecture | open | MathDB 页面 | 原文线索

V2-059741 | MATHDB-377953

Quantum hyperbolic Dehn-filling convergence conjecture

摘录 / EXCERPT: Let M be a cusped manifold, and let (W_n,L_n,ρ_n) be a sequence of compact hyperbolic Dehn fillings of M converging geometrically to M, where L_n consists of the simpl...

conjecture | open | MathDB 页面 | 原文线索

V2-059742 | MATHDB-377954

The quantum dilogarithmic scissors-congruence conjecture

摘录 / EXCERPT: Let M be a weakly-gentle cusped manifold, and let $H_N(M)$ be its quantum dilogarithmic invariant. Let $\mathfrak c_{\mathcal I_{fc}}(M)$ denote its ${\mathcal I}_{fc}$-scissors...

conjecture | open | MathDB 页面 | 原文线索

V2-059743 | MATHDB-378084

Type C_n versus type A_{n-1} Kostka-Foulkes comparison conjecture

摘录 / EXCERPT: Let Λ_k^A and Λ_{n-k}^A be the k-th and $(n-k)$-th fundamental weights of $U_q(\mathfrak{sl}_n)$, and set ... Let $K_{\Lambda_{2k},0}(q)$ be the type C_n Kostka-F...

conjecture | open | MathDB 页面 | 原文线索

V2-059744 | MATHDB-378136

Invertibility conjecture for the hyperkähler Nahm transform

摘录 / EXCERPT: Hyperkähler Nahm transform conjecture. Then M is diffeomorphic to N, and the hyperkähler Nahm transform is invertible.

conjecture | open | MathDB 页面 | 原文线索

V2-059745 | MATHDB-378170

The CDSW conjecture for simple Lie algebras

摘录 / EXCERPT: Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra over $\mathbb{C}$, let g be its dual Coxeter number, and let A be the quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-059746 | MATHDB-378295

Conjecture on the thermodynamic limit of quadri-tiling Boltzmann measures

摘录 / EXCERPT: Let ${\mathbb T}$ be a periodically tiled torus with lattice of periods Λ, and let ${\mathbb T}_n={\mathbb T}/n\Lambda$ be such that ${\mathbb T}_n^*$ is bipartite. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059747 | MATHDB-378390

Specialization formula for the adjoint HOMFLY invariant

摘录 / EXCERPT: Specialization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059748 | MATHDB-378581

The q-tetrahedron algebra realization conjecture for tridiagonal pairs

摘录 / EXCERPT: Assume $\mathbb K$ is algebraically closed. Let V be a finite-dimensional vector space over $\mathbb K$ of positive dimension, and let A,A^* be a tridiagonal pair on V. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059749 | MATHDB-378631

Reconstruction conjecture for commutative measured quantum groupoids

摘录 / EXCERPT: Let $(N,M,\alpha,\beta,\Gamma,\mu,T_L,T_R)$ be a measured quantum groupoid, and suppose that M is commutative. Let G denote a locally compact groupoid, with unit space...

conjecture | open | MathDB 页面 | 原文线索

V2-059750 | MATHDB-378757

Higher Cayley-Hamilton conjecture for 1-generic noncommutative orbits

摘录 / EXCERPT: Higher Cayley-Hamilton conjecture. On ${\cal L}_{{\hbar},q}^{\chi}$, the matrix $L_{(m)}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059751 | MATHDB-378792

Right-splitting morphism conjecture for Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Right-splitting morphism conjecture. The stated right-splitting morphism and its extension after tensoring with any such B exist with the asserted injectivity and preservation of...

conjecture | open | MathDB 页面 | 原文线索

V2-059752 | MATHDB-378830

X=M=K conjecture for stable one-dimensional sums

摘录 / EXCERPT: Let $\overline{K}^{\diamondsuit}_{L,\lambda}(t)$ and $\overline{M}^{\diamondsuit}_{L,\lambda}(t)$ be the K- and fermionic formulas associated with an allowed diagram shape...

conjecture | open | MathDB 页面 | 原文线索

V2-059753 | MATHDB-378856

Nakai-Nakanishi conjecture for type C_n tableau formulas

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type C_n, let λ/μ be a skew diagram, and let $\chi_{\lambda/\mu,a}$ be the associated Jacobi-Trudi determinant for $a\in\mathbb{C}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059754 | MATHDB-378888

Lascoux-Leclerc-Thibon's Schur-positivity conjecture for ribbon-tableau functions

摘录 / EXCERPT: Let $\mathcal G^{(n)}_\lambda(q,x)$ be the symmetric functions arising from ribbon tableaux and Fock-space representations of the quantum affine algebra...

conjecture | open | MathDB 页面 | 原文线索

V2-059755 | MATHDB-378915

Kirillov-Reshetikhin crystal-base conjecture for quantum affine algebras

摘录 / EXCERPT: Kirillov-Reshetikhin crystal-base conjecture. For some $\zeta^{(i)}_s\in\mathbb{C}(q)^\times$, the module $W^{(i)}_s(\zeta^{(i)}_s)$ has a crystal base $\mathcal{B}^{i,s}$ that is...

conjecture | open | MathDB 页面 | 原文线索

V2-059756 | MATHDB-378954

Commutative adapted measured quantum groupoid reconstruction conjecture

摘录 / EXCERPT: Let $(N,M,\alpha,\beta,\Gamma,\mu,T_L,T_R)$ be an adapted measured quantum groupoid, meaning a measured quantum groupoid with the stated adaptedness property, and suppose that M...

conjecture | open | MathDB 页面 | 原文线索

V2-059757 | MATHDB-378977

Coset Virasoro realization conjecture for the Fock spaces

摘录 / EXCERPT: Let ... where $\mathcal{F}_{M;m,\ell,n}$ are the Fock-space components in the decomposition of the $U_{x,p}(\widehat{\mathfrak{sl}}_2)$-module, and let a satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-059758 | MATHDB-379096

The critical-surface asymptotic conjecture for quantum invariants of figure-8 knot surgeries

摘录 / EXCERPT: Let p/q be a rational surgery coefficient, let d be the inverse of p modulo q, let $(a,b)\in\{0,1\}^2$, $n\in\mathbb{Z}/|q|\mathbb{Z}$, and $(\mu,\nu)\in\{\pm1\}^2$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-059759 | MATHDB-379323

Unique factorization conjecture for indecomposable dual canonical basis elements

摘录 / EXCERPT: Let B^* be the dual canonical basis of the quantum matrix algebra $O_q(M(n))$. Call $b\in B^*$ decomposable if there exist $m\in\mathbb Z$ and $b_1,b_2\in B^*$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-059760 | MATHDB-379497

Existence of ground states for quantum layers over asymptotically flat surfaces

摘录 / EXCERPT: Let Σ be an embedded asymptotically flat surface in R^3 that is not totally geodesic, and suppose its Gauss curvature is integrable. Consider the quantum layer built from...

conjecture | open | MathDB 页面 | 原文线索

V2-059761 | MATHDB-379557

Injectivity conjecture for the positive quantum affine subalgebra of the q-tetrahedron algebra

摘录 / EXCERPT: Let ${\mathcal A}_q$ be the unital associative algebra generated by x,y subject to the two cubic q-Serre relations ... ... For each $i\in\mathbb Z_4$, let the homomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-059762 | MATHDB-379973

The physicists' conjecture on global Coulomb gauge fixing

摘录 / EXCERPT: Physicists' conjecture. Fix a connection $\nabla_A\in\mathcal{C}$. Then for every $\nabla_{A'}\in\mathcal{C}$, there exists a unique $g\in\mathcal{G}$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-059763 | MATHDB-380183

The precise crepant resolution conjecture for analytic big A-model VSHS

摘录 / EXCERPT: Let $\mathcal{X}$ be an orbifold with projective coarse moduli space X, and let $\pi:Y\to X$ be a crepant resolution. Suppose that the big quantum products $\bullet_\tau$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-059764 | MATHDB-380195

S. Zhang's embedding conjecture for Hall algebra Drinfeld doubles

摘录 / EXCERPT: Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be finitary hereditary connected $\mathbb{F}_q$-linear categories, and let $\widetilde{\mathbf{D}}\mathbf{H}_{\mathcal{C}_i}$ denote their r...

conjecture | open | MathDB 页面 | 原文线索

V2-059765 | MATHDB-380196

Derived invariance conjecture for reduced Drinfeld doubles

摘录 / EXCERPT: Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be finitary hereditary connected $\mathbb{F}_q$-linear categories. Their reduced Drinfeld doubles are denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-059766 | MATHDB-380197

Functoriality conjecture for reduced Drinfeld doubles under friendly functors

摘录 / EXCERPT: Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be finitary hereditary connected $\mathbb{F}_q$-linear categories. An exact functor $F:D^b(\mathcal{C}_1)\to D^b(\mathcal{C}_2)$ is friendly...

conjecture | open | MathDB 页面 | 原文线索

V2-059767 | MATHDB-380244

Conjecture on the determinant of the resonance-relation system

摘录 / EXCERPT: Let D be the determinant of the system of resonance relations, with parameters x and $\Lambda_2,\dots,\Lambda_n$, and let $[z]$ denote the quantum-number factor used in the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-059768 | MATHDB-380245

Conjecture on uniqueness of conformal-block solutions of the Macdonald-Ruijsenaars equations

摘录 / EXCERPT: Let $\vec\Lambda\in\mathbb Z_{\ge0}^n[2{\mathfrak m}]$, and let $\mathbb M_\Theta$ be the Macdonald-Ruijsenaars operator acting on $L_{\vec\Lambda}[0]$-valued trigonometric polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-059769 | MATHDB-380315

Discrete spectrum conjecture for quantum layers over asymptotically flat surfaces

摘录 / EXCERPT: Quantum-layer discrete-spectrum conjecture. If $\varSigma$ is not the plane and K is integrable on $\varSigma$, then the Dirichlet Laplacian on Ω has non-empty discrete sp...

open_problem | open | MathDB 页面 | 原文线索

V2-059770 | MATHDB-380584

Numerical zero-location conjecture for the dominant fusion QTM eigenvalues

摘录 / EXCERPT: Zero-location conjecture. All the zeros of $T^{(1)}_{n-1}(u,v)$ are located on the almost straight lines

conjecture | open | MathDB 页面 | 原文线索

V2-059771 | MATHDB-380585

Numerical zero-location conjecture for the third QTM eigenvalue

摘录 / EXCERPT: Zero-location conjecture. For $n<p_0$, $T^{(3)}_{n-1}(u,v)$ has two real zeros $\pm\zeta^{(3)}_{n-1}$ for some $\zeta^{(3)}_{n-1}\in\mathbf R_{>0}$, while for $n=p_0$ it has a doub...

conjecture | open | MathDB 页面 | 原文线索

V2-059772 | MATHDB-380665

The scalar proportionality conjecture for the qKZB heat kernel

摘录 / EXCERPT: Let $U({\vec{z}},\tau,p)$ be the function defined in Corollary 1, namely ... Here u is the function appearing in the qKZB heat equation, α is multiplication by the function...

conjecture | open | MathDB 页面 | 原文线索

V2-059773 | MATHDB-380688

Finiteness conjecture for Woronowicz algebras with a fixed fusion-list invariant

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many Woronowicz algebras having a given $(R^+,list)$ invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-059774 | MATHDB-380689

Anti-deformation conjecture for Woronowicz algebras

摘录 / EXCERPT: Anti-deformation conjecture. Any Woronowicz algebra is a dimension-preserving R^+-deformation of a Woronowicz-Kac algebra.

conjecture | open | MathDB 页面 | 原文线索

V2-059775 | MATHDB-380703

Faithfulness conjecture for the lattice representation of the Hopf algebra

摘录 / EXCERPT: Faithfulness conjecture. The homomorphism κ is faithful, that is, injective.

conjecture | open | MathDB 页面 | 原文线索

V2-059776 | MATHDB-380705

The conjectural Kirillov-Reshetikhin module family

摘录 / EXCERPT: Conjectural Kirillov-Reshetikhin module family. For every (r,s), there exists an irreducible finite-dimensional $U'_q(X_n^{(1)})$-module $W^{(r)}_s$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-059777 | MATHDB-380706

The homogeneous X=M fermionic formula conjecture

摘录 / EXCERPT: X=M conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-059778 | MATHDB-380707

The inhomogeneous X=M fermionic formula conjecture

摘录 / EXCERPT: Inhomogeneous X=M conjecture. There exists $b_0\in B_0$ with $\varphi(b_0)=k\Lambda_0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-059779 | MATHDB-380708

The level-restricted X=M fermionic formula conjecture

摘录 / EXCERPT: Level-restricted X=M conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-059780 | MATHDB-380720

The twisted presentation conjecture for the quantum affine algebra of \(\mathfrak{sl}_2\)

摘录 / EXCERPT: Let $U_q(\widehat{\mathfrak{sl}_2})$ denote the quantum affine algebra. Let the twisted algebra be generated by a_{1m} for $m\in 2\mathbb Z+1$, by γ, and by $x_{1n}^{\pm}$...

conjecture | open | MathDB 页面 | 原文线索

V2-059781 | MATHDB-380721

GGS conjecture for explicit quantum R-matrices of \(\mathfrak{sl}_n\)

摘录 / EXCERPT: Let (Γ_1,Γ_2,τ) be an admissible triple and let $r^0\in\mathfrak h\otimes\mathfrak h$ satisfy the equations referenced in the source. Define $\tilde r^0=r^0-P^0/2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-059782 | MATHDB-380765

The d-algebra structure conjecture for translation-dilation invariant QFTs

摘录 / EXCERPT: Let $V=\mathbb{R}^d$, let Φ be the bundle of local fields of a quantum field theory, and let Φ_0 be its fiber at the origin. Let G_d be the group of parallel translatio...

conjecture | open | MathDB 页面 | 原文线索

V2-059783 | MATHDB-380842

Extensibility conjecture for doubly-periodic anti-self-dual instantons

摘录 / EXCERPT: Let A be an anti-self-dual connection on a bundle $E\rightarrow T\times\mathbb{C}$, with curvature satisfying $|F_A|\sim O(|w|^{-2})$. A holomorphic structure...

conjecture | open | MathDB 页面 | 原文线索

V2-059784 | MATHDB-380885

The separation conjecture for asymptotic values of small-mass hyperbolic monopoles

摘录 / EXCERPT: Let the parameter space of hyperbolic monopoles contain a compact subset and a disjoint subset, and let the monopoles have sufficiently small mass. Small-mass asymptotic separation...

conjecture | open | MathDB 页面 | 原文线索

V2-059785 | MATHDB-380940

Resolution conjecture for generalized Kauffman brackets with multiple double points

摘录 / EXCERPT: Resolution conjecture. The generalized Kauffman bracket is given by

conjecture | open | MathDB 页面 | 原文线索

V2-059786 | MATHDB-380968

The pole-location conjecture for normalized R-matrices

摘录 / EXCERPT: Let I_0 index the fundamental representations, let $R^{{\operatorname{nor}}}_{ij}(x,y)$ be the normalized R-matrix for $i,j\in I_0$, and let γ be the parameter defined i...

conjecture | open | MathDB 页面 | 原文线索

V2-059787 | MATHDB-380985

Gerstenhaber-Giaquinto-Schack conjecture for Belavin-Drinfeld triples

摘录 / EXCERPT: Let $\mathfrak{g}=\mathfrak{sl}(n)$, let τ be a Belavin-Drinfeld triple, and suppose that $r^0=t^0/2+\tilde{r}^0\in\mathfrak{h}\otimes\mathfrak{h}$ is τ-admissible. Defin...

conjecture | open | MathDB 页面 | 原文线索

V2-059788 | MATHDB-380995

CHSH optimality conjecture for correlated 2×2×2 Bell proofs

摘录 / EXCERPT: CHSH optimality conjecture. Among all $2\times2\times2$ proofs, and allowing correlated setting distributions, CHSH is best.

conjecture | open | MathDB 页面 | 原文线索

V2-059789 | MATHDB-380996

GHZ optimality conjecture for correlated 3×2×2 Bell proofs

摘录 / EXCERPT: GHZ optimality conjecture. Among all $3\times2\times2$ proofs, and allowing correlated setting distributions, GHZ is best.

conjecture | open | MathDB 页面 | 原文线索

V2-059790 | MATHDB-380997

CHSH optimality conjecture for the Bell singlet state

摘录 / EXCERPT: CHSH optimality conjecture. The best experiment with the Bell singlet state is the CHSH experiment.

conjecture | open | MathDB 页面 | 原文线索

V2-059791 | MATHDB-380998

Optimal 2×2×3 nonlocality-strength conjecture

摘录 / EXCERPT: Optimal 2×2×3 strength conjecture. The strongest possible $2\times2\times3$ nonlocality proof has statistical strength 0.077, and it uses the bipartite state

conjecture | open | MathDB 页面 | 原文线索

V2-059792 | MATHDB-380999

Superadditivity conjecture for 2×4×4 experiments on Bell singlet pairs

摘录 / EXCERPT: Bell-pair superadditivity conjecture. There is an experiment on pairs of Bell singlets, of the $2\times4\times4$ type, more than twice as strong as CHSH, and involving joint measur...

conjecture | open | MathDB 页面 | 原文线索

V2-059793 | MATHDB-381000

No-improvement conjecture for coincidence post-selection

摘录 / EXCERPT: Let δ and ϵ be functions producing arbitrary real-valued outputs, and write ... Variant Task 2 accepts a pair when $|S-T|<c$ for a fixed constant c, rather than con...

conjecture | open | MathDB 页面 | 原文线索

V2-059794 | MATHDB-381054

Conjecture on the non-integer BP solution for the permanent

摘录 / EXCERPT: Non-integer BP solution conjecture. The non-integer solution extends beyond the small $T_c+\varepsilon$ vicinity of T_c and transitions smoothly as $T\to\infty$ into the fully ho...

conjecture | open | MathDB 页面 | 原文线索

V2-059795 | MATHDB-381097

Two-transition conjecture for the three-dimensional symmetric N-clock model

摘录 / EXCERPT: Two-transition conjecture. There exists $\bar N$ such that, for every $N\geq\bar N$, the model undergoes two phase transitions: for

conjecture | open | MathDB 页面 | 原文线索

V2-059796 | MATHDB-381098

Non-vanishing charge conjecture for generic black holes

摘录 / EXCERPT: Consider smooth developments of the Einstein-Maxwell-Klein-Gordon system as in Theorem 1.1, with coupling parameter $\mathfrak{e}\neq 0$. Let $\mathcal{Q}^+$ be the quotient spacet...

conjecture | open | MathDB 页面 | 原文线索

V2-059797 | MATHDB-381114

The automorphism-group conjecture for quantum matrices

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $q\in\mathbb{K}^*$ be not a root of unity, and let ${\mathcal O}_q(M_n)$ be the quantisation of the ring of regular functions on $n\times n$ matric...

conjecture | open | MathDB 页面 | 原文线索

V2-059798 | MATHDB-381161

Naturality and decomposability conjectures for the categorified ribbon element

摘录 / EXCERPT: Naturality and decomposability conjectures. The chain maps κ_X and η_X are natural, namely, they commute with all 2-morphisms of $\dot{\mathcal{U}}$. Moreover, for...

conjecture | open | MathDB 页面 | 原文线索

V2-059799 | MATHDB-381199

Universal positivity conjecture for quantum greedy elements when one exchange parameter divides the other

摘录 / EXCERPT: Let b,c be positive integers, and let $X[a_1,a_2]$ denote the quantum greedy element of the rank 2 quantum cluster algebra $\mathcal{A}_v(b,c)$ associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-059800 | MATHDB-381209

Generic transverse blow-up conjecture for scalar waves on Reissner-Nordström interiors

摘录 / EXCERPT: Generic transverse blow-up conjecture. For generic data on Σ, the transverse derivatives of ϕ blow up on $\mathcal C\mathcal H^+$; in fact, ϕ is not in...

conjecture | open | MathDB 页面 | 原文线索

V2-059801 | MATHDB-381286

The real quadratic ray class field conjecture for SICs in every dimension

摘录 / EXCERPT: Real quadratic ray class field conjecture. The statement that, for the dimensions listed in Proposition 1, there is a SIC orbit E with $\mathbb{Q}(E)=\mathfrak{R}$, while every o...

conjecture | open | MathDB 页面 | 原文线索

V2-059802 | MATHDB-381335

The ASD moduli-space construction conjecture for the bounding cochain

摘录 / EXCERPT: Let $(M,\mathcal P_M)$ be as in Theorem 61, let Σ be the cylindrical end of M, and let $R(M;\mathcal P_M)$ and $R(\Sigma;\mathcal P_\Sigma)$ denote the corresponding repre...

conjecture | open | MathDB 页面 | 原文线索

V2-059803 | MATHDB-381456

The uniqueness conjecture for the exceptional Potts-model solution

摘录 / EXCERPT: Let $q=3$, let $\theta\le\tilde{\theta}_1$, and let $\alpha=\alpha_1(\theta)$. Consider the system of equations in the variables (u,v) given by. Uniqueness conjecture. The system...

conjecture | open | MathDB 页面 | 原文线索

V2-059804 | MATHDB-381457

The universal 2^q-1 bound for Potts-model boundary-law solutions

摘录 / EXCERPT: Let $q\ge2$, $k\ge2$, $\theta\ge1$, and $\alpha\in\mathbb{R}$. Let $\nu(\theta,\alpha)$ denote the number of positive solutions of the system of equations. Universal bound conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-059805 | MATHDB-381484

The core-triviality conjecture for finite-dimensional cocommutative cosemisimple Yetter-Drinfel'd Hopf algebras

摘录 / EXCERPT: Let K be an algebraically closed field of characteristic zero, let G be a finite abelian group, and let ... Let A be a finite-dimensional cocommutative cosemisimple Yetter-Drin...

conjecture | open | MathDB 页面 | 原文线索

V2-059806 | MATHDB-381510

Order \b14 conjecture for boundary operators on surfaces

摘录 / EXCERPT: Let Σ be a smooth surface with smooth boundary, endowed with a Riemannian metric. Let $\varkappa$ denote the boundary operator and D_Σ the Dirichlet-to-Neumann operat...

conjecture | open | MathDB 页面 | 原文线索

V2-059807 | MATHDB-381523

Nekrasov's uniqueness conjecture for noncommutative Hermitian Yang-Mills solutions

摘录 / EXCERPT: Let Equation ... has a unique solution with a given appropriate boundary condition at infinity. This is an informal conjecture attributed to Nekrasov; the source does not specify the...

conjecture | open | MathDB 页面 | 原文线索

V2-059808 | MATHDB-381529

Kimura's quantization conjecture for quantum unipotent subgroups

摘录 / EXCERPT: Let $\mathfrak{g}$ be a symmetrizable Kac-Moody algebra, let W be its Weyl group, and let $w\in W$. Denote by ${\mathsf{A_q}}[N_{-}(w)]$ the corresponding quantum unipotent subgr...

conjecture | open | MathDB 页面 | 原文线索

V2-059809 | MATHDB-381644

Soliton resolution conjecture for the radial Yang-Mills toy model

摘录 / EXCERPT: Soliton resolution conjecture. Any smooth, finite-energy solution $W(t,r)$ tends as $t\to\infty$, modulo sign, either to the half-kink or to a rescaled half-kink together with an a...

conjecture | open | MathDB 页面 | 原文线索

V2-059810 | MATHDB-381645

Uniqueness of the Gibbs measure at activity squared four

摘录 / EXCERPT: Let D denote the activity parameter of the high-density hard-core model on $\mathbb{Z}^3$, and let a Gibbs measure be an infinite-volume equilibrium measure for this model. Uniqu...

conjecture | open | MathDB 页面 | 原文线索

V2-059811 | MATHDB-381670

Nayak and Yuen's rigidity conjecture for superdense coding

摘录 / EXCERPT: Let $(\tau,(W_i))$ be a d-dimensional superdense coding protocol. Let $\mathcal{H}_{A'}$, $\mathcal{H}_{A”}$, $\mathcal{H}_{B'}$, and $\mathcal{H}_{B”}$ be the Hilbert spaces in...

conjecture | open | MathDB 页面 | 原文线索

V2-059812 | MATHDB-381806

The generalized semi-Clifford conjecture for the qubit Clifford hierarchy

摘录 / EXCERPT: A gate in $U(2^n)$ is generalized semi-Clifford if it can be expressed as ... where C_1,C_2 are n-qubit Clifford gates, P is a permutation matrix in $U(2^n)$, and the Z_j are...

conjecture | open | MathDB 页面 | 原文线索

V2-059813 | MATHDB-381807

The commuting-Toffoli characterization of third-level permutation gates

摘录 / EXCERPT: Let $\mathcal{CH}_k$ denote the kth level of the qubit Clifford hierarchy, and let a permutation gate be a unitary that permutes the computational-basis states. A Clifford permut...

conjecture | open | MathDB 页面 | 原文线索

V2-059814 | MATHDB-381808

The inverse-closure conjecture for permutation gates in the Clifford hierarchy

摘录 / EXCERPT: Let $\mathcal{CH}_k$ denote the kth level of the qubit Clifford hierarchy, and let π be a permutation gate on the computational-basis states. The inverse-closure conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-059815 | MATHDB-381958

Monoidal recovery conjecture for the principal block

摘录 / EXCERPT: Let $\widetilde{\mathcal{N}}$ be the Springer resolution, let $u(G_q)$ be the small quantum group, let $\operatorname{PrinBlock}_{\mathrm{dg}}(u(G_q))$ be its principal block, and...

conjecture | open | MathDB 页面 | 原文线索

V2-059816 | MATHDB-382002

The JMGS conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: The JMGS conjecture. The small J-function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-059817 | MATHDB-382056

Spacelike singularity conjecture for generic asymptotically flat black holes

摘录 / EXCERPT: Let a gravitational-collapse spacetime arise from one-ended initial data on a hypersurface Σ. For a point p on a spacelike singularity $\mathcal{S}$, say that its causal p...

conjecture | open | MathDB 页面 | 原文线索

V2-059818 | MATHDB-382057

BKL proposal for dynamics near generic spacelike cosmological singularities

摘录 / EXCERPT: Let $\mathcal{S}$ be a generic spacelike initial cosmological singularity, and let $J^+(p)$ denote the causal future of a point $p\in\mathcal{S}$. BKL proposal. Sufficiently close...

conjecture | open | MathDB 页面 | 原文线索

V2-059819 | MATHDB-382059

Ooguri-Vafa integrality conjecture for the chromatic Lagrangian superpotential

摘录 / EXCERPT: Let L be a Lagrangian with boundary S_Γ, let A be a choice of phase and framing, and let $W_\Gamma^{(A)}$ be a primitive function whose graph of the differential gives t...

conjecture | open | MathDB 页面 | 原文线索

V2-059820 | MATHDB-382060

Higher-genus Ooguri-Vafa integrality conjecture for chromatic Lagrangian wavefunctions

摘录 / EXCERPT: Let L be a Lagrangian with boundary S_Γ, let A be a choice of phase and framing, and let $\Psi_\Gamma^{(A)}$ be the corresponding algebraic wavefunction. For coordinates...

conjecture | open | MathDB 页面 | 原文线索

V2-059821 | MATHDB-382078

The higher-order determination conjecture for Bar-Natan-van der Veen polynomials

摘录 / EXCERPT: Higher-order determination conjecture. The value of $Z_\mathbb{D}(K)$ is completely determined by the Alexander polynomial Δ_K and the family of polynomials $\rho_K^{i,0}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-059822 | MATHDB-382080

The equivalence conjecture for the universal invariant of a knot

摘录 / EXCERPT: Universal-invariant equivalence conjecture. The datum of the universal invariant $Z_\mathbb{D}(K)$ is equivalent to any of the three pieces of data in the preceding theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-059823 | MATHDB-382081

The conjectural identification of the first Bar-Natan-van der Veen polynomial

摘录 / EXCERPT: First-polynomial identification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059824 | MATHDB-382082

The identification of the first Bar-Natan-van der Veen polynomial with the reduced 2-loop polynomial

摘录 / EXCERPT: Reduced 2-loop identification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-059825 | MATHDB-382093

Stronger bound for impure distance-three stabilizer codes

摘录 / EXCERPT: An impure $[[n,k,3]]$ stabilizer code is a quantum stabilizer code encoding k logical qubits into n physical qubits, with minimum distance 3 and with nontrivial stabilizer el...

conjecture | open | MathDB 页面 | 原文线索

V2-059826 | MATHDB-382098

Velocity-scaling conjecture for quantum walks in periodic fields

摘录 / EXCERPT: Let $t\in(0,1)$ and $n\in\mathord{\mathbb N}$. Let $\mathcal{D}_n$ be the corresponding local n-periodic field, and let $U_{t,n}$ be the associated quantum walk. Its velocity is...

conjecture | open | MathDB 页面 | 原文线索

V2-059827 | MATHDB-382168

Strong-subsolution conjecture for the higher-rank deformed Hermitian Yang-Mills equation

摘录 / EXCERPT: Let $E\to(X,\omega)$ be a Hermitian vector bundle, and let h be a Hermitian metric on E. A strong subsolution for the Z-critical equation is a metric satisfying ... as an...

conjecture | open | MathDB 页面 | 原文线索

V2-059828 | MATHDB-382188

The good GKP codes conjecture for random NTRU lattices

摘录 / EXCERPT: Good GKP codes conjecture. The code is likely to be good, encodes $k=n$ qubits, and has distance

conjecture | open | MathDB 页面 | 原文线索

V2-059829 | MATHDB-382219

The integral factorisation conjecture for the fused Hecke algebra

摘录 / EXCERPT: Let $\mathcal{A}_{k+1}$ be the algebra generated by the generators and relations of Definition of the paper, and let $E_{k+1}^{(q,\alpha_1)}$ be the corresponding element. Speciali...

conjecture | open | MathDB 页面 | 原文线索

V2-059830 | MATHDB-382238

Penrose's inequality for asymptotically Euclidean manifolds

摘录 / EXCERPT: Let (M^d,g) be a complete asymptotically Euclidean manifold of real dimension d with non-negative scalar curvature and an outermost minimal hypersurface Σ. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-059831 | MATHDB-382242

Uniform tight continuity bound for von Neumann entropy

摘录 / EXCERPT: Consider a Hilbert space ${\mathcal{H}}$ of finite dimension $d=|{\mathcal{H}}|$. Let ρ and σ be states on $\mathcal{H}$. Define ... The function $f_d(\varepsilon,\nu)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-059832 | MATHDB-382296

The correspondence between the renormalized r-matrix and the quantum R-matrix

摘录 / EXCERPT: Renormalized r-matrix correspondence conjecture. Under the equivalence of Proposition E1, the renormalized r-matrix corresponds to the lowest non-trivial loop degree part of the...

conjecture | open | MathDB 页面 | 原文线索

V2-059833 | MATHDB-382350

Montonen-Olive duality conjecture for pure Yang-Mills theory

摘录 / EXCERPT: Let G be a gauge group with gauge coupling τ, and let $G^\vee$ be a group whose Lie algebra satisfies $\operatorname{Lie}(G^\vee)=\mathfrak{g}=\operatorname{Lie}(G)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-059834 | MATHDB-382451

Universality conjecture for the dimension of three-stars in Liouville quantum gravity

摘录 / EXCERPT: Fix a whole-plane Gaussian free field h and let D_h be the associated γ-Liouville quantum gravity metric for $\gamma\in(0,2)$. Let $\mathfrak{S}_3\subseteq\mathbb{C}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-059835 | MATHDB-382452

Non-singleton intersection conjecture for pairs of Liouville quantum gravity geodesics

摘录 / EXCERPT: Fix $\gamma\in(0,2)$, a whole-plane Gaussian free field h, and distinct points $z_1,z_2,z_3,z_4\in\mathbb{C}$. Let Γ_{z_i,z_j} denote a geodesic from z_i to z_j in the...

conjecture | open | MathDB 页面 | 原文线索

V2-059836 | MATHDB-382457

Stability conjecture for the Kerr-de Sitter and Kerr families

摘录 / EXCERPT: Stability conjecture. The family $\mathcal{B}_\Lambda$ is a stable family of solutions to the Einstein vacuum equations with $\Lambda\geq 0$: the evolution of the small perturbatio...

conjecture | open | MathDB 页面 | 原文线索

V2-059837 | MATHDB-382495

Bahturin-Regev invariance conjecture for minimal quantum commutative decompositions

摘录 / EXCERPT: Bahturin-Regev conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-059838 | MATHDB-382526

Conjecture on finite-time blowup and universal self-similar profile in dimension five

摘录 / EXCERPT: Consider equation ... corresponding to sufficiently large initial data should blow up in finite time, in the sense that $w_{rr}(t,0)$ diverges as $t\nearrow T$ for some $T>0$, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-059839 | MATHDB-382527

Conjecture on finite-time blowup and instanton profile in dimension four

摘录 / EXCERPT: Consider equation ... with sufficiently large energy should blow up in finite time, in the sense that $w_{rr}(t,0)$ diverges as $t\nearrow T$ for some $T>0$. More precisely, there sh...

conjecture | open | MathDB 页面 | 原文线索

V2-059840 | MATHDB-383293

Classification problem for order-six complex Hadamard matrices

摘录 / EXCERPT: Classify, up to standard equivalence, all matrices $H\in\mathbb{C}^{6\times 6}$ satisfying $|H_{ij}|=1$ for every i,j and $HH^{*}=6I_{6}$. The conjecture asserts that every such...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059841 | MATHDB-383690

Categorical characterization of the Petz recovery map

摘录 / EXCERPT: Let R be a retrodiction, or quantum Bayesian inversion, assignment for normal quantum processes between von Neumann algebras equipped with faithful states, satisfying th...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-059842 | MATHDB-383746

Hod’s conjecture

摘录 / EXCERPT: For every black-hole spacetime with Hawking temperature T_H, the least-damped quasinormal mode with frequency ω_0 satisfies...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-059843 | MATHDB-386961

Where to look for counterexamples to the conjecture that: a pseudounitary category is unitary.

摘录 / EXCERPT: Where to look for counterexamples to the conjecture that: a pseudounitary category is unitary. For a pseudounitary category the quantum dimensions of all objects are positive.

open_problem | open | MathDB 页面 | 原文线索

V2-059844 | MATHDB-386967

How are QCAs module {finite depth Quantum circuits} related to Witt group of Modular Tensor Categories.

摘录 / EXCERPT: How are QCAs module {finite depth Quantum circuits} related to Witt group of Modular Tensor Categories.

open_problem | open | MathDB 页面 | 原文线索

V2-059845 | MATHDB-386968

List of all question (problem 26)

摘录 / EXCERPT: Are $(3+1)$D Quantum Cellular Automata (QCA) coming from $(3+1)$D TQFT states beyond cobordisms?

open_problem | open | MathDB 页面 | 原文线索

V2-059846 | MATHDB-390490

Quantum Weyl conjecture

摘录 / EXCERPT: ...

conjecture | partial | MathDB 页面 | 原文线索

V2-059847 | MATHDB-391909

Kerr black hole uniqueness conjecture

摘录 / EXCERPT: Every regular, stationary, asymptotically flat, vacuum black-hole spacetime in four-dimensional general relativity is isometric to a member of the Kerr family, parametrized by mass...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059848 | MATHDB-392266

Gibbons-Hartnoll-Pope conjecture on Böhm-metric instability

摘录 / EXCERPT: For every Böhm Einstein metric g on a manifold of the form $S^{k+1}\times S^l$ or S^{k+l+1}, the Lichnerowicz Laplacian on transverse-traceless symmetric 2-tensors has a nega...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-059849 | MATHDB-392458

QGN stiffness conjecture

摘录 / EXCERPT: For every quantum geometric nesting (QGN) model with an exact frustration-free ground state and a spontaneously broken continuous symmetry, the exact stiffness associated with that...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059850 | MATHDB-393042

Quantum statistical mechanical system for imaginary quadratic fields

摘录 / EXCERPT: Construct a quantum statistical mechanical system that recovers the explicit class field theory construction for an imaginary quadratic field K, starting from the noncommutative...

open_problem | open | MathDB 页面 | 原文线索

V2-059851 | MATHDB-393048

Admissible parameters for the quantum ax+b group

摘录 / EXCERPT: Let $\hbar\in(0,2\pi]$. Then $\hbar$ is admissible if and only if ...

open_problem | open | MathDB 页面 | 原文线索

V2-059852 | MATHDB-393049

Perturbative renormalisability of the noncommutative ϕ^4 theory on Moyal $\mathbb{R}^4$

摘录 / EXCERPT: For a real scalar field ϕ on the four-dimensional Moyal plane, with Euclidean metric and $\widetilde{x}_\mu:=2(\theta^{-1})_{\mu\nu}x^\nu$, the quantum field theory associated...

open_problem | open | MathDB 页面 | 原文线索

V2-059853 | MATHDB-393100

Small-data global regularity for the Yang-Mills equations in high-dimensional Minkowski space

摘录 / EXCERPT: Let G be a compact semisimple Lie group, let $M=\mathbb{R}^{n+1}$ be Minkowski space with $n\geq 6$, and consider the Yang-Mills equations $D^\alpha F_{\alpha\beta}=0$ with smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-059854 | MATHDB-393215

Relation between optimal mixed-state estimation and quantum transformations

摘录 / EXCERPT: Determine the relation between optimal estimation procedures and other kinds of quantum transformations, such as cloning and transposition, for mixed states.

open_problem | open | MathDB 页面 | 原文线索

V2-059855 | MATHDB-393221

Drinfeld generators for affine quantum groups via Ringel-Hall algebras

摘录 / EXCERPT: For the affine Lie algebra $\widehat{\mathfrak{sl}}_n$, with the Ringel-Hall algebra of the cyclic quiver C_n having vertices $1,\ldots,n$ and arrows $i\to i+1\pmod n$, understan...

open_problem | open | MathDB 页面 | 原文线索

V2-059856 | MATHDB-393222

Relation between derived Hall algebras and quantum groups

摘录 / EXCERPT: It remains to be investigated if and how the derived Hall algebra of the category of representations of a quiver $\vec{\Delta}$ over a finite field is related to the quantum group...

open_problem | open | MathDB 页面 | 原文线索

V2-059857 | MATHDB-393278

Efficient quantum solution of the Hidden Subgroup Problem for the symmetric group

摘录 / EXCERPT: Solve the Hidden Subgroup Problem efficiently for nonabelian groups, in particular for the symmetric group S_n, using quantum computation.

open_problem | open | MathDB 页面 | 原文线索

V2-059858 | MATHDB-393305

Spectral-measure formula for free wreath products

摘录 / EXCERPT: Under mild assumptions on quantum groups (or the corresponding objects) A and B, conjecture that the associated spectral measure satisfies ... Here $*_w$ denotes the free wreath...

open_problem | open | MathDB 页面 | 原文线索

V2-059859 | MATHDB-393338

Controllability notions and criteria for infinite-dimensional quantum control equations

摘录 / EXCERPT: The report leaves unanswered the question of identifying appropriate notions of controllability for infinite-dimensional equations, together with easily implementable criteria for...

open_problem | open | MathDB 页面 | 原文线索

V2-059860 | MATHDB-393386

Classicality of the LWE-to-lattice reduction

摘录 / EXCERPT: Can the reduction from worst-case lattice problems such as SVP and SIVP to the learning-with-errors problem be made classical (i.e., nonquantum), rather than quantum?

open_problem | open | MathDB 页面 | 原文线索

V2-059861 | MATHDB-393647

Explicit solution in the fully symmetric quantum-field-theory limit

摘录 / EXCERPT: Find the explicit solution of the bosonic quantum field theory model with external field $B_{\bc\bd}$ and a four-point interaction defined using the Moyal product, satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-059862 | MATHDB-393779

Quantum marginal problem for three pure-state marginals

摘录 / EXCERPT: Let $\psi=[\psi_{\alpha\beta\gamma}]$ be a complex cubic matrix of format $\ell\times m\times n$, and let ρ_A, ρ_B, and ρ_C be the Hermitian matrices of orders...

open_problem | open | MathDB 页面 | 原文线索

V2-059863 | MATHDB-393780

Quantum marginal problem for spectra of a bipartite state and its reductions

摘录 / EXCERPT: Given spectra $\lambda^{AB}=\operatorname{Spec}(\rho_{AB})$, $\lambda^A=\operatorname{Spec}(\rho_A)$, and $\lambda^B=\operatorname{Spec}(\rho_B)$, all constraints relating them are...

open_problem | open | MathDB 页面 | 原文线索

V2-059864 | MATHDB-393931

A complete renormalization-group trajectory between two fixed points

摘录 / EXCERPT: For the three-dimensional $\phi^4_{3,\varepsilon}$ Euclidean quantum field theory with sufficiently small $\varepsilon>0$, construct a double-sided sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-059865 | MATHDB-393960

Homeomorphism between primitive ideals and symplectic leaves for quantum SL_n

摘录 / EXCERPT: For the generic standard quantized coordinate ring $O_q(\mathrm{SL}_n(\mathbb{C}))$, with $q\in\mathbb{C}^{\times}$ not a root of unity, is the Hodges-Levasseur bijection ... a homeo...

open_problem | open | MathDB 页面 | 原文线索

V2-059866 | MATHDB-394189

Bulk (thermodynamic) limit for a quantum-mechanical crystal

摘录 / EXCERPT: Does the ground-state energy per unit volume of a large finite crystal converge as the size tends to infinity, and can the limit energy be computed by solving a simpler (e.g. “unit...

open_problem | open | MathDB 页面 | 原文线索

V2-059867 | MATHDB-394254

Existence of a stationary quantum BGK solution with Bose-Einstein condensation

摘录 / EXCERPT: For some n_0 and any $n>n_0$, and for $\[1m\eta\u001b[0m$ sufficiently small, there exists a distribution function...

open_problem | open | MathDB 页面 | 原文线索

V2-059868 | MATHDB-394258

Zero-arrow completion conjecture for quantum Demazure crystal graphs

摘录 / EXCERPT: The crystal graph of the quantum Demazure module $D_q(m,\lambda^\vee)$ is a subgraph of the crystal graph of the corresponding irreducible $U_q(\widehat{\mathfrak g})$-representati...

conjecture | open | MathDB 页面 | 原文线索

V2-059869 | MATHDB-394590

Quantum-circuit construction of histories

摘录 / EXCERPT: Let B be the basis of the Hilbert space of n qubits consisting of the 2^n strings of 0-1 bits. Let $K=(U_1, \ldots,U_t)$ be a string of unitary operators. For...

open_problem | open | MathDB 页面 | 原文线索

V2-059870 | MATHDB-394709

Fourier integral operator structure of the quantum scattering map at integer indicial-gap parameters

摘录 / EXCERPT: For a de Sitter-like Lorentzian manifold with boundary (X,g) satisfying (A1)-(A2), let $P(\bb)=\square_g-\lambda$ and let ... When $s_+(\lambda)-s_-(\lambda)$ is not an integer, th...

open_problem | open | MathDB 页面 | 原文线索

V2-059871 | MATHDB-395045

First-principles derivation of the AdResS force-interpolation formula

摘录 / EXCERPT: Derive the AdResS force-interpolation formula from first principles of statistical mechanics, ... where w is the molecular switching function, X_α and X_β are molecula...

open_problem | open | MathDB 页面 | 原文线索

V2-059872 | MATHDB-395202

Classification of quantum groups between S_n and O_n^+

摘录 / EXCERPT: Classify the quantum groups satisfying $S_n\subseteq G\subseteq O_n^+$.

open_problem | open | MathDB 页面 | 原文线索

V2-059873 | MATHDB-395591

Free de Finetti theorem for quantum exchangeable sequences

摘录 / EXCERPT: The report states the following theorem: for an infinite sequence of random variables I, the following are equivalent: (a) I is quantum exchangeable; (b) I is identically dis...

open_problem | open | MathDB 页面 | 原文线索

V2-059874 | MATHDB-395595

Does spreadability imply braidability?

摘录 / EXCERPT: For sequences of quantum random variables, does spreadability imply braidability?

open_problem | open | MathDB 页面 | 原文线索

V2-059875 | MATHDB-395654

Cluster variables and simple objects in quantum affine representation categories

摘录 / EXCERPT: For a simple Lie algebra $\g$ of type A_n, D_n, or E_n, and the subcategory $\C_\ell$ and Grothendieck ring R_ℓ introduced in the report, every cluster variable obtained...

open_problem | open | MathDB 页面 | 原文线索

V2-059876 | MATHDB-395963

Local formula for the Connes-Chern character of a twisted spectral triple

摘录 / EXCERPT: Can the Connes-Chern character of a twisted spectral triple be expressed in local terms as in [3], by means of residue integrals that eliminate all quantum infinitesimal perturbati...

open_problem | open | MathDB 页面 | 原文线索

V2-059877 | MATHDB-395969

The Baum-Connes conjecture for the free orthogonal quantum group $A_o(n)$

摘录 / EXCERPT: Let $n\in\mathbb{N}$, and let $A_o(n)$ be the universal C^*-algebra generated by self-adjoint elements u_{ij}, $1\leq i,j\leq n$, satisfying ... With comultiplication...

conjecture | open | MathDB 页面 | 原文线索

V2-059878 | MATHDB-395970

Nontrivial projections in the reduced free orthogonal quantum-group algebra

摘录 / EXCERPT: For $n>2$, the reduced C^*-algebra $A_o(n)_{\mathrm{red}}$ does not contain nontrivial projections.

open_problem | open | MathDB 页面 | 原文线索

V2-059879 | MATHDB-396067

Asymptotic sharpness of the fractal upper bound for open quantum maps

摘录 / EXCERPT: For the open quantum maps described in the report, is the fractal upper bound ... (an estimate stated for any small $\varepsilon,\varepsilon'>0$ and sufficiently small $h>0$) actuall...

open_problem | open | MathDB 页面 | 原文线索

V2-059880 | MATHDB-396133

Deterministic derandomization of almost randomizing quantum channels

摘录 / EXCERPT: How can one deterministically find a single quantum channel using $O(d)$ Kraus operators with the (generic) property of being ε-randomizing?

open_problem | open | MathDB 页面 | 原文线索

V2-059881 | MATHDB-396135

Additivity of minimum-output Rényi entropy for p > 2

摘录 / EXCERPT: For every $p>2$, there exist quantum channels for which the minimum-output Rényi entropy is nonadditive.

open_problem | open | MathDB 页面 | 原文线索

V2-059882 | MATHDB-396136

Additivity of minimum output entropy for quantum channels

摘录 / EXCERPT: Whether the minimum output entropy of quantum channels is additive.

open_problem | open | MathDB 页面 | 原文线索

V2-059883 | MATHDB-396139

Maximal output p-norm multiplicativity conjecture

摘录 / EXCERPT: For every $p>1$ and every pair of quantum channels $\\Phi,\\Psi$, one has ... Here a quantum channel is a completely positive trace-preserving linear map between matrix algebras,...

conjecture | open | MathDB 页面 | 原文线索

V2-059884 | MATHDB-396237

Minimum output Rényi entropy additivity counterexamples

摘录 / EXCERPT: For a given $p\ge 1$, do there exist two quantum channels Φ_1,Φ_2 such that $H_p^{\mathrm{min}}(\Phi_1\otimes\Phi_2)<H_p^{\mathrm{min}}(\Phi_1)+H_p^{\mathrm{min}}(\Phi_2)$?...

open_problem | open | MathDB 页面 | 原文线索

V2-059885 | MATHDB-396245

Nontrivial idempotents in the reduced free orthogonal quantum-group C*-algebra

摘录 / EXCERPT: For every integer $n>2$, the reduced C-algebra $A_o(n)_{\mathrm{red}}$ does not contain nontrivial idempotents.

open_problem | open | MathDB 页面 | 原文线索

V2-059886 | MATHDB-396269

Extension of the quantum-Bruhat/Kirillov-Reshetikhin crystal bijection to types B and D

摘录 / EXCERPT: Theorem 4 establishes that, in types A and C, there are bijections ... where $B_\mu:=\bigotimes_{i=1}^{\mu_1}B_{\mu_i',1}$ and μ' is the conjugate partition to μ. It is conje...

open_problem | open | MathDB 页面 | 原文线索

V2-059887 | MATHDB-396303

Polynomiality of the quantum K-theoretic product

摘录 / EXCERPT: Let G be a semisimple algebraic group over $\mathbb{C}$ and let P be a maximal parabolic subgroup, so that $\operatorname{Pic}(G/P)\cong\mathbb{Z}$. Is the quantum K-theoretic...

open_problem | open | MathDB 页面 | 原文线索

V2-059888 | MATHDB-396414

Geometric quantization for proper moment maps (Vergne conjecture)

摘录 / EXCERPT: Let (M,ω) be a possibly noncompact symplectic manifold with a prequantum line bundle (L,h^L,∇^L), and let J be an almost-complex structure such that...

conjecture | open | MathDB 页面 | 原文线索

V2-059889 | MATHDB-396630

Natural \-involution on the reduced quantum algebra

摘录 / EXCERPT: Let $\mathcal A=(C^\infty(M)[[\lambda]],\star)$ be a Hermitian star-product algebra, let $\mathcal J\subseteq\mathcal A$ be a left ideal, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-059890 | MATHDB-396633

Deformation of the Fredholm family to an affine quantum-group-covariant family

摘录 / EXCERPT: Is it possible to deform the family of Fredholm operators transforming covariantly under the loop group to a family of operators transforming covariantly under an affine quantum gr...

open_problem | open | MathDB 页面 | 原文线索

V2-059891 | MATHDB-396634

Construction of interacting four-dimensional quantum field theories beyond perturbation theory

摘录 / EXCERPT: Construct interacting quantum field theories in four space-time dimensions beyond the perturbative level.

open_problem | open | MathDB 页面 | 原文线索

V2-059892 | MATHDB-396958

Categorification of tensor products of finite-dimensional quantum \(\mathfrak{sl}_2\)-representations

摘录 / EXCERPT: For the finite-dimensional $U_q(\mathfrak{sl}_2)$-representations $V_{d_1}\otimes V_{d_2}\otimes\cdots\otimes V_{d_r}$, is it possible to categorify these representations?

open_problem | open | MathDB 页面 | 原文线索

V2-059893 | MATHDB-396979

Convergence of OPE coefficients in the coupling constants near a UV fixed point

摘录 / EXCERPT: It has been conjectured that the OPE coefficients have a convergent expansion in the coupling constants near an ultraviolet fixed point, i.e. around a conformally invariant quantum...

open_problem | open | MathDB 页面 | 原文线索

V2-059894 | MATHDB-397065

One-round quantum communication protocol with exponential classical-quantum separation

摘录 / EXCERPT: Can the same exponential separation as in Ran Raz’s result be achieved with a quantum protocol using only one round of communication-namely, for a partial function with a quantum p...

open_problem | open | MathDB 页面 | 原文线索

V2-059895 | MATHDB-397067

Spectrum and PPT threshold for partial transposes of random quantum states

摘录 / EXCERPT: Let ρ be a random state on $\mathbb{C}^d\otimes\mathbb{C}^d$, distributed according to $\mu_{d^2,p}$, where $\mu_{n,p}$ is the distribution of...

open_problem | open | MathDB 页面 | 原文线索

V2-059896 | MATHDB-397164

Bad primes for quantum mapping-class-group representations

摘录 / EXCERPT: For given genus g and parameter p, find the bad primes q in the sense of the Nori-Weisfeiler theorem, namely those for which the homomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-059897 | MATHDB-397354

Strong Baum-Connes property for free quantum groups

摘录 / EXCERPT: Theorem 2. Let ... where $P_j\in GL_{m_j}(\mathbb C)$, $m_j>2$, $P_j\overline{P_j}=\pm 1$, and $Q_j\in GL_{n_j}(\mathbb C)$. Then G has the strong Baum-Connes property: the localiz...

open_problem | open | MathDB 页面 | 原文线索

V2-059898 | MATHDB-397984

Finiteness of the exponential growth rate of quantum hyperbolic invariants

摘录 / EXCERPT: For a cusped hyperbolic 3-manifold M, determine whether the exponential growth rate of the sequence $(H_N(M))_N$, where $H_N(M)$ is defined for each odd integer $N\ge 3$, is fi...

open_problem | open | MathDB 页面 | 原文线索

V2-059899 | MATHDB-398298

Extending nonlinear quantum time-evolutions to composite systems

摘录 / EXCERPT: How can nonlinear quantum time-evolutions be extended to composite systems, with or without the “no signal” property?

open_problem | open | MathDB 页面 | 原文线索

V2-059900 | MATHDB-398602

Yang-Mills Plateau minimization in dimension five

摘录 / EXCERPT: For every boundary datum $\bar A\in\mathcal A^{SU(2)}(S^4)$, the Yang-Mills Plateau problem on B^5 admits a minimizer: there exists a weak-connection class...

open_problem | open | MathDB 页面 | 原文线索

V2-059901 | MATHDB-398735

Statistical mechanics on the objective molecular-dynamics invariant manifold

摘录 / EXCERPT: Find a “statistical mechanics” on this invariant manifold.

open_problem | open | MathDB 页面 | 原文线索

V2-059902 | MATHDB-398764

Cluster algebra structures on quantum double Bruhat cells

摘录 / EXCERPT: For any complex simple Lie group G and a pair of Weyl group elements (w,v), the quantum double Bruhat cell algebra $R_q[G^{w,v}]$ possesses a canonical structure of quantum clu...

open_problem | open | MathDB 页面 | 原文线索

V2-059903 | MATHDB-399027

Classification of quantum homogeneous spaces for quantized enveloping algebras

摘录 / EXCERPT: Let $\mathfrak g$ be a simple complex Lie algebra and let $U_q(\mathfrak g)$ be its quantized enveloping algebra. A right coideal subalgebra $C\subseteq U_q(\mathfrak g)$ is a quan...

open_problem | open | MathDB 页面 | 原文线索

V2-059904 | MATHDB-399033

Extensions of enveloping algebras by finite-dimensional pointed Hopf algebras

摘录 / EXCERPT: Question 5 (S. Lentner). The Lusztig quantum groups of divided powers are infinite-dimensional Hopf algebras that contain a finite-dimensional pointed Hopf algebra (the small quant...

open_problem | open | MathDB 页面 | 原文线索

V2-059905 | MATHDB-399080

Berenstein-Zelevinsky quantum cluster algebra conjecture

摘录 / EXCERPT: The Berenstein-Zelevinsky conjecture postulates the existence of quantum cluster algebra structures on all quantized coordinate rings of double Bruhat cells in simple Lie groups.

conjecture | open | MathDB 页面 | 原文线索

V2-059906 | MATHDB-399294

Rank-free rigorous confidence-region protocol for quantum state estimation

摘录 / EXCERPT: A rigorous protocol that depends at no point on a rough estimate for the rank has not yet been given, even though the coverage of the confidence regions of [5] does not depend on t...

open_problem | open | MathDB 页面 | 原文线索

V2-059907 | MATHDB-399295

Quantum Gaussian optimizers conjecture

摘录 / EXCERPT: For Bosonic Gaussian channels, the optimizers belong to the class of pure Bosonic Gaussian states.

conjecture | open | MathDB 页面 | 原文线索

V2-059908 | MATHDB-399296

Noncommutative Gaussian maximizers for quantum \(q\to p)\u0000-norms

摘录 / EXCERPT: Investigate a possible development of such an approach to obtain noncommutative generalizations of the classical “Gaussian maximizers” results for $(q\to p)$-norms.

open_problem | open | MathDB 页面 | 原文线索

V2-059909 | MATHDB-399297

Quantum Gaussian optimizers for gauge-covariant and contravariant channels

摘录 / EXCERPT: Let Φ be a gauge-covariant or gauge-contravariant Bosonic Gaussian channel, as defined by equations (2)-(5) on PDF pages 29-30, and let $f:[0,1]\to\mathbb{R}$ be concave with...

open_problem | open | MathDB 页面 | 原文线索

V2-059910 | MATHDB-399401

GapSVP dimension discrepancy between quantum and classical LWE reductions

摘录 / EXCERPT: Remove or explain the GapSVP dimension discrepancy between the quantum and classical reductions.

open_problem | open | MathDB 页面 | 原文线索

V2-059911 | MATHDB-399460

Universal solutions of the reflection equation for other quantum symmetric pairs

摘录 / EXCERPT: Is it possible to extend this program to other quantum symmetric pairs, and is it possible to prove (c) and (d) in those cases as well?

open_problem | open | MathDB 页面 | 原文线索

V2-059912 | MATHDB-399462

Relating quantum and Poisson homogeneous-space classifications

摘录 / EXCERPT: Relate the classification results for quantum homogeneous spaces and for quantum symmetric pairs to the classification of Poisson homogeneous spaces in terms of Lagrangian Lie suba...

open_problem | open | MathDB 页面 | 原文线索

V2-059913 | MATHDB-399466

Relation between distinct solution constructions for boundary qKZ equations

摘录 / EXCERPT: For the boundary quantum Knizhnik-Zamolodchikov equations associated with the type-$C_n^\$ affine Hecke algebra and the spin representation $(\mathbb{C}^2)^{\otimes n}$, understand...

open_problem | open | MathDB 页面 | 原文线索

V2-059914 | MATHDB-399467

Replacing the augmented q-Onsager coideal in boundary qKZ integral solutions

摘录 / EXCERPT: Can the coideal subalgebra used in the construction of solutions of the boundary quantum Knizhnik-Zamolodchikov equations be replaced by the q-Onsager algebra associated with the a...

open_problem | open | MathDB 页面 | 原文线索

V2-059915 | MATHDB-399468

Classification of general right coideal subalgebras of a quantum group

摘录 / EXCERPT: For generic q, classify all right coideal subalgebras $C\subset U_q(\mathfrak g)$, without the additional assumptions $U^0\subset C$ or $C\subset U_q(\mathfrak g)^{\ge 0}$.

open_problem | open | MathDB 页面 | 原文线索

V2-059916 | MATHDB-399646

Frenkel-Reshetikhin conjecture on spectra of quantum integrable systems

摘录 / EXCERPT: Let $\mathfrak g$ be an untwisted affine Kac-Moody algebra and let $q\in\mathbb C^*$ be not a root of unity. For simplicity, assume that $\mathfrak g$ is simply laced. Let V be a...

conjecture | open | MathDB 页面 | 原文线索

V2-059917 | MATHDB-399649

Quantum K-theoretic geometric Satake correspondence

摘录 / EXCERPT: Let G be a semisimple complex group, let $G^{\\vee}$ be its Langlands dual group, and let $Gr=G^{\\vee}((t))/G^{\\vee}[[t]]$ be the affine Grassmannian. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-059918 | MATHDB-399665

Existence of periodic time-reversal-symmetric Bloch frames in three dimensions

摘录 / EXCERPT: For a gapped periodic quantum system in three dimensions with fermionic time-reversal symmetry, determine whether there exists a global smooth Bloch frame that is both periodic and...

open_problem | open | MathDB 页面 | 原文线索

V2-059919 | MATHDB-399712

Quantum representations as subrepresentations of the constructed type

摘录 / EXCERPT: Is every quantum representation of the mapping class group-namely, one arising from the theory of conformal blocks-obtained as a complex subrepresentation of a representation of th...

open_problem | open | MathDB 页面 | 原文线索

V2-059920 | MATHDB-399917

Dual canonical basis conjecture for quantum cluster algebras

摘录 / EXCERPT: The dual canonical basis conjecture states that a localization of $\Psi(A_v)$ carries a quantum cluster algebra structure and that the quantum cluster monomials are contained in th...

conjecture | open | MathDB 页面 | 原文线索

V2-059921 | MATHDB-400298

Finite-time singularities for Yang-Mills flow in dimension four

摘录 / EXCERPT: Can Yang-Mills flow in dimension four develop singularities in finite time?

open_problem | open | MathDB 页面 | 原文线索

V2-059922 | MATHDB-400503

Can quantum dynamics be described by the density alone?

摘录 / EXCERPT: Given a prescribed initial wave-function ψ_0, does the time-dependent one-particle density $\rho(x,t)$ determine the external potential $V(x,t)$ up to an additive function...

open_problem | open | MathDB 页面 | 原文线索

V2-059923 | MATHDB-400744

Existence of quantum curves for admissible spectral curves

摘录 / EXCERPT: Let (Σ,x,y) be an admissible spectral curve in the report's sense: Σ is a Torelli-marked compact Riemann surface, x and y are meromorphic functions on Σ, t...

open_problem | open | MathDB 页面 | 原文线索

V2-059924 | MATHDB-400748

WKB quantum periods and the twisted superpotential for the T_3 theory

摘录 / EXCERPT: For a compact Riemann surface C with punctures $\{z_1,\ldots,z_m\}$ and $G=SU(N)$, let ... The proposed construction associates to each datum in B a family of G-opers on C, a...

open_problem | open | MathDB 页面 | 原文线索

V2-059925 | MATHDB-400862

Removal of the ultraviolet and infrared regularization in the four-dimensional interacting model

摘录 / EXCERPT: For the scalar quantum-field model with spatial dimension $s=3$ (so $d=s+1=4$), potential $V(y)=\frac{|y|^2}{2}+\lambda W(y)$, coupling constant $\lambda\ge 0$, and continuous real...

open_problem | open | MathDB 页面 | 原文线索

V2-059926 | MATHDB-400863

Truncated moment problem for quantum-field-theoretic functionals

摘录 / EXCERPT: For the relativistic and Euclidean quantum-field settings described in the report, determine the family of possible linear functionals W on the Borchers-Uhlmann algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-059927 | MATHDB-400983

Spectra of quantum integrable systems via generalized Baxter polynomials

摘录 / EXCERPT: Theorem 2 [Frenkel-Hernandez 2015, conjectured by Frenkel-Reshetikhin 1998]. Let V,W be as for an XXZ-type model above. The eigenvalues $\b$lambdak$ of $tV(z)$ on $W...

open_problem | open | MathDB 页面 | 原文线索

V2-059928 | MATHDB-401196

Geometric interpretation of complex parameters in the quantum $6j$-symbol model

摘录 / EXCERPT: For the quantum invariant of knotted graphs in S^3, when the solution of the saddle-point equation consists of complex numbers, and the parameter is complex rather than real or p...

open_problem | open | MathDB 页面 | 原文线索

V2-059929 | MATHDB-401240

Lower bound for the interacting quantum backflow operator

摘录 / EXCERPT: Suppose that V satisfies $\int dx\,(1+|x|)|V(x)|<\infty$. Then, for every $f\in\mathcal S(\mathbb R)$ with $f\ge 0$, there exists a constant $c_{V,f}>0$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-059930 | MATHDB-401394

Quantum optimal-transport formulation of the semiclassical pseudometric

摘录 / EXCERPT: Could the pseudometric introduced here, or its analogue considered in [3], be formulated in terms of some quantum variant of optimal transport-for example, is there an analogue of...

open_problem | open | MathDB 页面 | 原文线索

V2-059931 | MATHDB-401440

Rigorous density-potential mapping via the quantum-fluid fixed-point equation

摘录 / EXCERPT: For the many-particle time-dependent Schrödinger system described in the report, give a rigorous treatment of the fixed-point proof of the density-potential mapping based on ... inco...

open_problem | open | MathDB 页面 | 原文线索

V2-059932 | MATHDB-401495

Counterterm equation for the renormalized many-body problem

摘录 / EXCERPT: For dimensions $d=2,3$, the counterterm problem is formulated as the nonlinear integral equation ... where $\rho_{\tau}^{\kappa+u}$ is the free quantum density associated with...

open_problem | open | MathDB 页面 | 原文线索

V2-059933 | MATHDB-401575

Closure of the quantum-correlation set $C_q(4,2)$

摘录 / EXCERPT: Is the set of quantum correlations $C_q(4,2)$ closed?

open_problem | open | MathDB 页面 | 原文线索

V2-059934 | MATHDB-401581

Deterministic violation of minimum-output-entropy additivity

摘录 / EXCERPT: Does there exist a deterministic quantum channel for which the minimum output entropy (MOE) violates additivity?

open_problem | open | MathDB 页面 | 原文线索

V2-059935 | MATHDB-401582

Removing the logarithmic Kraus-rank factor for general quantum channels

摘录 / EXCERPT: Let 4 0<varepsilon<1 and let $N:L(A)\to L(B)$ be a CPTP map between finite-dimensional Hilbert spaces, with Kraus rank $|E|\ge |A|,|B|$. Theorem 1 gives a completely positive map...

open_problem | open | MathDB 页面 | 原文线索

V2-059936 | MATHDB-401589

Normality criterion for the meet of normal idempotent states

摘录 / EXCERPT: Let G be a locally compact quantum group and let $\omega,\mu\in\operatorname{Idem}^{\mathrm{nor}}(G)$. Then ... In this case, ...

open_problem | open | MathDB 页面 | 原文线索

V2-059937 | MATHDB-401594

Rank inequality for reduced states of a tripartite quantum state

摘录 / EXCERPT: For a tripartite density matrix ρ_{ABC}, does the inequality ... hold, where $r_{XY}=\operatorname{rank}(\rho_{XY})$?

open_problem | open | MathDB 页面 | 原文线索

V2-059938 | MATHDB-401958

Classical verification of quantum computations

摘录 / EXCERPT: For any language $L\in\mathrm{BQP}$, there is a four-message argument system for L with a classical polynomial-time verifier such that: (i) there is a quantum polynomial-time pro...

open_problem | open | MathDB 页面 | 原文线索

V2-059939 | MATHDB-401967

Hernandez's conjectural Kazhdan-Lusztig formula for quantum affine modules

摘录 / EXCERPT: Conjecture 1 (Hernandez, 2004). For monomial labels $m,m'\in(\mathcal M,\leq)$, one has ... and the polynomials $P_{m,m'}(t)$ are positive. Here $M(m)$ is a standard module, $L(m)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-059940 | MATHDB-402281

Finite-dimensional ancillas for factorizable quantum channels

摘录 / EXCERPT: Does every factorizable channel admit a finite-dimensional ancilla?

open_problem | open | MathDB 页面 | 原文线索

V2-059941 | MATHDB-402339

Detecting inequivalent reflection-symmetric gapped phases

摘录 / EXCERPT: Let $\mathcal P_\Gamma$ be the set of uniformly bounded, finite-range, reflection-invariant interactions on the quantum spin chain with a unique gapped ground state. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-059942 | MATHDB-402343

Matching lower bound for the two-dimensional quantum Heisenberg ferromagnet free energy

摘录 / EXCERPT: The report gives the upper bound ... and states that proving a lower bound matching (6) remains an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-059943 | MATHDB-402345

Quasi-classical evolution of a coupled quantum-field system

摘录 / EXCERPT: For the microscopic states $\Gamma^\varepsilon\in\mathcal L^1_+(\mathcal H\otimes\mathcal K_\varepsilon)$ in the model defined on pages 53-54, assume they are normalized and satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-059944 | MATHDB-402387

Classification of broader classes of quantum groups

摘录 / EXCERPT: Extend the classification known for easy (or partition) quantum groups to broader classes, including those arising in [TaWe16] and [SpWe16], and describe their structure comprehens...

open_problem | open | MathDB 页面 | 原文线索

V2-059945 | MATHDB-402389

Baum-Connes conjecture for complex semisimple quantum groups

摘录 / EXCERPT: Let $q\in(0,1)$ and let G_q be a complex semisimple quantum group. Then the assembly map ... is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-059946 | MATHDB-402391

Characterisation of the Haagerup property by proper cocycles

摘录 / EXCERPT: A discrete quantum group $\Gamma=\widehat{G}$ has the Haagerup property if and only if $C\Gamma\cong\operatorname{Pol}(G)$ admits a proper cocycle.

open_problem | open | MathDB 页面 | 原文线索

V2-059947 | MATHDB-402393

Free decompositions and free fusion semirings

摘录 / EXCERPT: When does there exist a free decomposition of a compact quantum group (G,u)? Is this the case if and only if G has a free fusion semiring?

open_problem | open | MathDB 页面 | 原文线索

V2-059948 | MATHDB-402394

Existence of compact quantum isometry groups for compact metric spaces

摘录 / EXCERPT: Does every compact metric space (X,d) have a compact quantum isometry group?

open_problem | open | MathDB 页面 | 原文线索

V2-059949 | MATHDB-402414

Equivalence of the two module-category constructions for quantum symmetric spaces

摘录 / EXCERPT: Are the module categories constructed from cyclotomic Knizhnik-Zamolodchikov equations and from Letzter-Kolb coideal subalgebras equivalent, thereby inducing equivalent representat...

open_problem | open | MathDB 页面 | 原文线索

V2-059950 | MATHDB-402421

Fusion rules of the exceptional quantum subgroups $C_{N,0,1}$

摘录 / EXCERPT: Determine the fusion rules of the exceptional quantum subgroups $C_{N,0,1}$ for $N\in\mathbb{N}_{+}$.

open_problem | open | MathDB 页面 | 原文线索

V2-059951 | MATHDB-402432

Module categories and subfactors from quantum symmetric-pair coideal algebras

摘录 / EXCERPT: Construct a general theory of module categories with a C^*-structure and of subfactors arising from coideal algebras $U'_q\mathfrak g^\theta\subset U_q\mathfrak g$, extending the...

open_problem | open | MathDB 页面 | 原文线索

V2-059952 | MATHDB-402516

Existence and uniqueness of the Liouville quantum gravity metric for $\gamma\in(0,2)$

摘录 / EXCERPT: Fix $\gamma\in(0,2)$, let h be a whole-plane Gaussian free field normalized so that its average over the unit circle is zero, and let $h_\varepsilon=h*p_\varepsilon$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-059953 | MATHDB-402517

Quantum optimal transport for qubits of arbitrary size

摘录 / EXCERPT: For all $j\in\mathbb N$, an j-qubit is a state on the finite-dimensional C^*-algebra $A_j:=\bigotimes_{n=1}^{j}M_2(\mathbb C)$. The arbitrary-size setting is the CAR algebra...

open_problem | open | MathDB 页面 | 原文线索

V2-059954 | MATHDB-403227

Quantum analogue of Birkhoff’s conjecture for ellipses

摘录 / EXCERPT: Ellipses E_e of any eccentricity e are uniquely determined by their spectrum $\operatorname{Sp}(E_e)$.

conjecture | open | MathDB 页面 | 原文线索

V2-059955 | MATHDB-403362

The characteristic gluing problem of general relativity

摘录 / EXCERPT: Given two spacelike 2-spheres S_1 and S_2 in spacetimes (M_1,g_1) and (M_2,g_2), respectively, is it possible to glue (M_1,g_1) and (M_2,g_2) along an outgoing null hyp...

open_problem | open | MathDB 页面 | 原文线索

V2-059956 | MATHDB-403376

Classification of generic Big Bang singularities in vacuum general relativity

摘录 / EXCERPT: Give a complete classification of generic Big Bang singularities in $1+3$-dimensional vacuum general relativity, including the possible oscillatory and inhomogeneous behaviors sugg...

open_problem | open | MathDB 页面 | 原文线索

V2-059957 | MATHDB-403405

Factoriality of von Neumann algebras of universal orthogonal quantum groups

摘录 / EXCERPT: For a matrix $F\\in M_n(\\mathbb C)$ satisfying $F\\overline{F}=\\pm I_n$, is the von Neumann algebra M associated with the universal orthogonal quantum group $A_o(F)$ a factor?

open_problem | open | MathDB 页面 | 原文线索

V2-059958 | MATHDB-403412

Langlands duality for skein modules of 3-manifolds

摘录 / EXCERPT: For a group G, let G_L denote its Langlands dual group, and let $\\operatorname{Sk}_G(M)$ be the skein module of a closed oriented 3-manifold M associated with the quantum-...

open_problem | open | MathDB 页面 | 原文线索

V2-059959 | MATHDB-403413

Isomorphism classes of quantum automorphism group von Neumann algebras

摘录 / EXCERPT: For a finite-dimensional C^*-algebra B with its canonical Plancherel trace, let $G^+(B)$ be its quantum automorphism group and let $n=\dim B$. In the regime $n\ge 5$, can Theor...

open_problem | open | MathDB 页面 | 原文线索

V2-059960 | MATHDB-403418

Lévy-Khinchine decomposition for the twisted quantum group $SU_q(n)$

摘录 / EXCERPT: For the twisted algebraic quantum group $SU_q(n)$, with $q\\in(0,1)$ and $n\\ge 3$, the (LK) property holds: every generating functional admits a Lévy-Khinchine decomposition int...

open_problem | open | MathDB 页面 | 原文线索

V2-059961 | MATHDB-403419

Extension of the Hunt/Lévy-Khintchine method to other quantum groups

摘录 / EXCERPT: The author conjectures that the method used for the quantum groups $SU_q(n)$ and $U_q(n)$ can be further extended to q-deformations G_q of other simple compact Lie groups.

open_problem | open | MathDB 页面 | 原文线索

V2-059962 | MATHDB-403423

Residual nilpotence of the augmentation ideal of the free orthogonal quantum group

摘录 / EXCERPT: Determine $I^\\infty$ for $\mathcal O(O_N^+)$, where I is the augmentation ideal and $I^\infty=\bigcap_{n\in\mathbb N} I^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-059963 | MATHDB-403449

Subquadratic-sample quantum eigenvalue estimation

摘录 / EXCERPT: For an unknown quantum mixed state $\rho\in\mathbb{C}^{d\times d}$, can its eigenvalues p be learned to sorted-total-variation distance ε using $n=o(d^2)$ samples?

open_problem | open | MathDB 页面 | 原文线索

V2-059964 | MATHDB-403450

Quantum mixedness testing

摘录 / EXCERPT: Testing whether an unknown quantum mixed state $\rho\in\mathbb{C}^{d\times d}$ is the maximally mixed state I/d, or else ε-far from it in trace distance, can be done...

open_problem | open | MathDB 页面 | 原文线索

V2-059965 | MATHDB-403452

Entropy constraints for quantum and classical marginal entropies

摘录 / EXCERPT: What are the constraints that the von Neumann entropies of the 2^n possible marginals of an n-party quantum state have to obey? Similarly, what are the constraints for the Shan...

open_problem | open | MathDB 页面 | 原文线索

V2-059966 | MATHDB-403456

Efficiency and phase transition of SEBD and Patching for random shallow 2D quantum circuits

摘录 / EXCERPT: Conjecture 1 (Informal). SEBD and Patching are asymptotically efficient for any family of 2D quantum circuits with Haar-random gates of sufficiently low depth, but experience a com...

open_problem | open | MathDB 页面 | 原文线索

V2-059967 | MATHDB-403497

Gaiotto’s quantum geometric Satake conjecture for \(GL(M|N)\)

摘录 / EXCERPT: For fixed $M<N$, let $U_{M,N}\subset GL(N)$ be the unipotent subgroup normalized by the embedded $GL(M)$, and let $\chi_{M,N}:U_{M,N}\to\mathbb G_a$ be a character fixed by the adj...

conjecture | open | MathDB 页面 | 原文线索

V2-059968 | MATHDB-403503

Conjecture on the monoidal realization of the quantum Grothendieck ring

摘录 / EXCERPT: In the setup of the report, where $\mathfrak g$ is an arbitrary complex simple Lie algebra, $U_q(\widehat{\mathfrak g})$ is its untwisted quantum affine algebra at generic q,...

conjecture | open | MathDB 页面 | 原文线索

V2-059969 | MATHDB-403519

Quantum causal polynomial optimization

摘录 / EXCERPT: Given a causal structure, a polynomial function f_0 on quantum states, and a countable set $f_1,f_2,\ldots$ of polynomial functions that are non-negative on states compatible wit...

open_problem | open | MathDB 页面 | 原文线索

V2-059970 | MATHDB-403730

Construction of the three-dimensional Yang-Mills measure

摘录 / EXCERPT: For a fixed compact Lie group G, construct the Yang-Mills Euclidean quantum-field measure in dimension $d=3$, formally given on $ \mathfrak g$-valued 1-forms A over...

open_problem | open | MathDB 页面 | 原文线索

V2-059971 | MATHDB-403732

Volume measure for LQG in the matter-central-charge phase $(1,25)$

摘录 / EXCERPT: It remains an open problem to construct the volume measure for Liouville quantum gravity in the phase $c_M\in(1,25)$, extending the corresponding feature from the regime...

open_problem | open | MathDB 页面 | 原文线索

V2-059972 | MATHDB-403807

Quantum restoration of strong cosmic censorship at the Reissner-Nordström-de Sitter Cauchy horizon

摘录 / EXCERPT: For a free scalar quantum field on a Reissner-Nordström-de Sitter spacetime, in any state Ψ that is Hadamard across the event and cosmological horizons, the renormalized stres...

open_problem | open | MathDB 页面 | 原文线索

V2-059973 | MATHDB-404082

Decidability of perfect quantum strategies for 3XOR games

摘录 / EXCERPT: Given a 3XOR game, determine whether it does or does not have a “perfect” quantum strategy.

open_problem | open | MathDB 页面 | 原文线索

V2-059974 | MATHDB-404236

Multiplicity of the discrete spectrum of a soft quantum waveguide

摘录 / EXCERPT: Consider, in two dimensions, the soft quantum waveguide operator $H_{\Gamma,V}=-\Delta+V(\operatorname{dist}(x,\Gamma))$, where Γ is an infinite, appropriately regular (say,...

open_problem | open | MathDB 页面 | 原文线索

V2-059975 | MATHDB-404472

Positivity of quantum greedy elements in rank-2 quantum cluster algebras

摘录 / EXCERPT: For the rank-2 quantum cluster algebra without coefficients with initial exchange matrix $B=\begin{pmatrix}0&-\delta_1\\ \delta_2&0\end{pmatrix}$, the positivity property of the qu...

open_problem | open | MathDB 页面 | 原文线索

V2-059976 | MATHDB-404540

Convergence of the fixed-point iteration for quantum signal-processing phase factors

摘录 / EXCERPT: For a target polynomial $f(x)=\sum_{j=0}^{d}\hat f_jT_{2j}(x)$ satisfying $\lVert\hat f\rVert_1\le 0.861$, there exists a reduced phase-factor vector Ψ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-059977 | MATHDB-404754

Type A invariance of the center of the small quantum group

摘录 / EXCERPT: In type A, one has an isomorphism of algebras ...

open_problem | open | MathDB 页面 | 原文线索

V2-059978 | MATHDB-404806

Closed expression for the genus-1 divisor invariant of \(\operatorname{Hilb}^d(\mathbb{C}^2)\)

摘录 / EXCERPT: For every positive integer d, the genus-1 invariant satisfies ... where Tr_e is the trace of quantum multiplication by the divisor D, and...

open_problem | open | MathDB 页面 | 原文线索

V2-059979 | MATHDB-405080

Categorical connection for the two-dimensional quantum lattice model example

摘录 / EXCERPT: By direct calculation we also found an example in a 2d quantum lattice model, but here the connection with categories remains an open problem. Could a 2-category be involved?

open_problem | open | MathDB 页面 | 原文线索

V2-059980 | MATHDB-405106

Quantum automorphism groups and the limiting quantum isometry group

摘录 / EXCERPT: For the primitive adjacency matrix $A\in M_N(\{0,1\})$ considered in the report, when $QAut(A)\ne Aut(A)$, is $QAut(A)$ a quantum subgroup of G^∞_A?

open_problem | open | MathDB 页面 | 原文线索

V2-059981 | MATHDB-405109

Semidefinite relaxation of quantum equilibrium behaviours

摘录 / EXCERPT: How can the explicit tensor product be replaced by some kind of commutation condition so that general methods of semidefinite programming relaxation of noncommutative polynomial op...

open_problem | open | MathDB 页面 | 原文线索

V2-059982 | MATHDB-405163

Conjecture that all gapped ground states satisfy approximate Haag duality

摘录 / EXCERPT: $\text{Every gapped ground state satisfies approximate Haag duality.}$

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-059983 | MATHDB-405194

Perfect fusion strategies for fusion-based quantum computation

摘录 / EXCERPT: Given a quantum error-correcting code encoding k logical qubits, characterize the codes and fusion strategies for which the induced logical fusion succeeds for every pattern in w...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-059984 | MATHDB-405232

Arai’s classification question for infrared-renormalized van Hove-Miyatake models

摘录 / EXCERPT: Given two infrared-renormalized van Hove-Miyatake models, determine necessary and sufficient conditions for them to be unitarily equivalent; that is, characterize when there exists...

open_problem | claimed_solved | MathDB 页面 | 原文线索

B17 其他 / Miscellaneous | 20,553 条

V2-059985 | MATHDB-63

Kuznetsov's categorical rationality conjecture for cubic fourfolds

摘录 / EXCERPT: Kuznetsov's conjecture. The fourfold X is rational if and only if ... for an ordinary, untwisted projective K3 surface S.

conjecture | open | Vihaan Dheer | MathDB 页面 | 原文线索

V2-059986 | MATHDB-315853

P versus NP problem

摘录 / EXCERPT: Fix the alphabet $\{0,1\}$, and call a set $L\subseteq\{0,1\}^{*}$ a language. For $x\in\{0,1\}^{*}$ let $|x|$ denote its length, and let...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-059987 | MATHDB-315926

Rendezvous problem

摘录 / EXCERPT: Fix an integer $n\ge 3$. There are n locations, indexed $1,\dots ,n$, and two players. Time is discrete, $t=1,2,\dots$; at each time each player occupies one location, and in one...

open_problem | open | MathDB 页面 | 原文线索

V2-059988 | MATHDB-315947

Borsuk's problem

摘录 / EXCERPT: For a nonempty bounded set $A \subset \mathbb{R}^n$ let ... where $\|\cdot\|$ is the Euclidean norm, and set $\operatorname{diam} \emptyset = 0$. For every integer $n \geq 1$ and eve...

open_problem | open | MathDB 页面 | 原文线索

V2-059989 | MATHDB-315950

disk covering problem

摘录 / EXCERPT: For a point $c\in\mathbb{R}^2$ and $r>0$ write $B(c,r)=\{x\in\mathbb{R}^2:\lVert x-c\rVert\le r\}$, and let $D=B(0,1)$ be the closed unit disk. For each integer $n\ge 1$ put ... Dete...

open_problem | open | MathDB 页面 | 原文线索

V2-059990 | MATHDB-315952

Reinhardt's conjecture

摘录 / EXCERPT: For a compact convex set $K\subset\mathbb{R}^{2}$ with nonempty interior, let ... where the supremum is over all families $\{K_{i}\}_{i\in I}$ of congruent copies of K with pairwis...

conjecture | open | MathDB 页面 | 原文线索

V2-059991 | MATHDB-315953

Square packing

摘录 / EXCERPT: For a real number $a \ge 1$, let $s(a)$ denote the maximum number of unit squares that can be packed into a square of side a: that is, the largest N for which there exist N c...

open_problem | open | MathDB 页面 | 原文线索

V2-059992 | MATHDB-315955

Tammes problem

摘录 / EXCERPT: Let $S^{2}=\{x\in\mathbb{R}^{3}:\|x\|=1\}$ be the unit sphere with the Euclidean norm $\|\cdot\|$, and let $N\ge 2$ be an integer. For a configuration $X=(x_{1},\dots,x_{N})$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-059993 | MATHDB-316007

Albertson conjecture

摘录 / EXCERPT: For a finite simple graph G, let $\operatorname{cr}(G)$ denote its crossing number, i.e. the minimum number of pairwise crossings of edges over all drawings of G in the plane i...

conjecture | open | MathDB 页面 | 原文线索

V2-059994 | MATHDB-316009

GNRS conjecture

摘录 / EXCERPT: For a finite metric space (X,d_X), a map $f\colon X\to \ell_1$, where ℓ_1 carries the norm $\|x\|_1=\sum_i |x_i|$, has distortion at most D if there are constants $c,C>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-059995 | MATHDB-316010

Harborth's conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite planar graph. A straight-line drawing of G is an injective map $p\colon V\to\mathbb{R}^2$ such that, when each edge $uv\in E$ is drawn as the closed str...

conjecture | open | MathDB 页面 | 原文线索

V2-059996 | MATHDB-316012

strong Papadimitriou-Ratajczak conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a finite simple graph. A map $f\colon V\to\mathbb{R}^2$ is a greedy embedding of G if f is injective and for every ordered pair of distinct vertices $x,t\in V$...

conjecture | open | MathDB 页面 | 原文线索

V2-059997 | MATHDB-316040

Hilbert's tenth problem

摘录 / EXCERPT: Fix an effective encoding of polynomials with integer coefficients in finitely many unknowns by finite strings over a finite alphabet (so that, for each $n\ge 1$, every...

open_problem | open | MathDB 页面 | 原文线索

V2-059998 | MATHDB-316043

Decidability (logic)

摘录 / EXCERPT: In logic, a true/false decision problem is decidable if there exists an effective method for deriving the correct answer. Logical systems are decidable if membership in their set o...

open_problem | open | MathDB 页面 | 原文线索

V2-059999 | MATHDB-316045

Büchi's problem

摘录 / EXCERPT: For an integer $M\ge 3$, consider tuples $(x_0,x_1,\dots,x_{M-1})\in\mathbb{Z}^M$ satisfying the system ... that is, all tuples of integers for which ... equivalently all tuples for wh...

open_problem | open | MathDB 页面 | 原文线索

V2-060000 | MATHDB-316047

Catalan-Dickson conjecture

摘录 / EXCERPT: For a positive integer k, let $\sigma_1(m)=\sum_{d\mid m} d$ denote the sum of the divisors of m, and let $s(m)=\sigma_1(m)-m$ denote the sum of the proper divisors of m. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-060001 | MATHDB-316051

Hall's conjecture

摘录 / EXCERPT: For integers x,y write $D(x,y)=|y^{2}-x^{3}|$. Strong form: there exists a constant $C>0$ such that for all integers x,y with $y^{2}\neq x^{3}$, ... Weak form: for every real...

conjecture | open | MathDB 页面 | 原文线索

V2-060002 | MATHDB-316055

Newman's conjecture

摘录 / EXCERPT: For a nonnegative integer n, let $p(n)$ denote the number of partitions of n, i.e. the number of ways of writing n as a sum of positive integers without regard to order, with...

conjecture | open | MathDB 页面 | 原文线索

V2-060003 | MATHDB-316056

Scholz conjecture

摘录 / EXCERPT: For a positive integer x, an addition chain for x is a finite sequence of positive integers ... such that for every k with $1 \le k \le r$ there exist indices i, j with...

conjecture | open | MathDB 页面 | 原文线索

V2-060004 | MATHDB-316100

Schanuel's conjecture

摘录 / EXCERPT: For a field K with $\mathbb{Q}\subseteq K\subseteq\mathbb{C}$, let $\operatorname{trdeg}_{\mathbb{Q}}K$ denote the transcendence degree of K over $\mathbb{Q}$, that is, the car...

conjecture | open | MathDB 页面 | 原文线索

V2-060005 | MATHDB-316173

Willmore conjecture

摘录 / EXCERPT: Let M be a compact orientable smooth surface and let $v\colon M\to\mathbb{R}^3$ be a smooth immersion. Equip M with the Riemannian metric induced by v, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060006 | MATHDB-316174

Erdős distinct distances problem

摘录 / EXCERPT: For a finite set $P\subset\mathbb{R}^2$ let ... be its set of distinct Euclidean distances, and for an integer $n\geq 2$ put ... the minimum being over all sets of n distinct points...

open_problem | open | MathDB 页面 | 原文线索

V2-060007 | MATHDB-316175

Heterogeneous tiling conjecture (squaring the plane)

摘录 / EXCERPT: Squaring the square is the problem of tiling an integral square using only other integral squares. The name was coined in a humorous analogy with squaring the circle. Squaring the...

conjecture | open | MathDB 页面 | 原文线索

V2-060008 | MATHDB-316181

Double bubble conjecture

摘录 / EXCERPT: Work in $\mathbb{R}^3$ with Lebesgue measure $|\cdot|$ and 2-dimensional Hausdorff measure $\mathcal{H}^2$. For disjoint sets $R_1,R_2\subset\mathbb{R}^3$ of finite perimeter (eq...

conjecture | open | MathDB 页面 | 原文线索

V2-060009 | MATHDB-316182

Honeycomb theorem

摘录 / EXCERPT: The honeycomb theorem, formerly the honeycomb conjecture, states that a regular hexagonal grid or honeycomb has the least total perimeter of any subdivision of the plane into regio...

open_problem | open | MathDB 页面 | 原文线索

V2-060010 | MATHDB-316183

Lange's conjecture

摘录 / EXCERPT: Let C be a smooth projective curve of genus $g \ge 2$, and for integers $r \ge 1$ and d let $M(r,d)$ denote the moduli space of semistable vector bundles on C of rank r and...

conjecture | open | MathDB 页面 | 原文线索

V2-060011 | MATHDB-316185

Kepler conjecture

摘录 / EXCERPT: The Kepler conjecture, named after the 17th-century mathematician and astronomer Johannes Kepler, is a mathematical theorem about sphere packing in three-dimensional Euclidean spac...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060012 | MATHDB-316204

André-Oort conjecture

摘录 / EXCERPT: Let $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\mathbb{G}_{m}$ and let $\mathbb{A}_f$ denote the ring of finite adeles of $\mathbb{Q}$. A Shimura datum is a pair (G,X) whe...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060013 | MATHDB-316211

Catalan's conjecture

摘录 / EXCERPT: For all integers x, y, a, b with $x > 0$, $y > 0$, $a > 1$, $b > 1$, the equation ... holds only for $x = 3$, $a = 2$, $y = 2$, $b = 3$; that is, $3^{2} - 2^{3} = 1$ is the unique...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060014 | MATHDB-316216

Boolean Pythagorean triples problem

摘录 / EXCERPT: Call a triple (a,b,c) of positive integers a Pythagorean triple if $a^{2}+b^{2}=c^{2}$; such triples are not required to be primitive, i.e. $\gcd(a,b,c)$ may exceed 1. For a ma...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-060015 | MATHDB-316228

Erdős discrepancy problem

摘录 / EXCERPT: Let $(x_1, x_2, x_3, \ldots)$ be a sequence with $x_i \in \{+1, -1\}$ for every positive integer i, and let C be an integer. Then there exist positive integers k and d such...

open_problem | open | MathDB 页面 | 原文线索

V2-060016 | MATHDB-316232

Lax conjecture

摘录 / EXCERPT: Peter David Lax was a Hungarian-born American mathematician and Abel Prize laureate working in the areas of pure and applied mathematics.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060017 | MATHDB-316233

Fundamental lemma (Langlands program)

摘录 / EXCERPT: Let F be a nonarchimedean local field with ring of integers $\mathcal{O}_F$, and let G be an unramified group over F, i.e. a quasi-split connected reductive F-group that sp...

open_problem | open | MathDB 页面 | 原文线索

V2-060018 | MATHDB-316236

Kato's conjecture

摘录 / EXCERPT: Let $n\ge 1$ and let $A:\mathbb{R}^n\to M_n(\mathbb{C})$ be a measurable matrix-valued function for which there exist constants $\lambda>0$ and $\Lambda<\infty$ such that, for almo...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060019 | MATHDB-316240

The hafnian analogue of Marcus's inequality for positive semidefinite matrices

摘录 / EXCERPT: Hafnian inequality. The hafnian of this block matrix is at least

conjecture | open | MathDB 页面 | 原文线索

V2-060020 | MATHDB-316241

Armstrong's F=M conjecture for generalised non-crossing partitions

摘录 / EXCERPT: Let Φ be a finite root system of rank n, let m be a positive integer, and let $F^m_\Phi(x,y)$ be the F-triangle of the generalised cluster complex. Let $NC^m(\Phi)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-060021 | MATHDB-316242

Higher-dimensional characterization of non-uniqueness sets for the spherical mean transform

摘录 / EXCERPT: Let d be the ambient dimension and let $S\subset\mathbb{R}^d$. A set S is a non-uniqueness set for the spherical mean transform in the space of compactly supported functions wh...

conjecture | open | MathDB 页面 | 原文线索

V2-060022 | MATHDB-316246

The Gaussian perimeter limit-law conjecture for self-avoiding polygons

摘录 / EXCERPT: Gaussian perimeter limit-law conjecture. The mean and standard deviation satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-060023 | MATHDB-316258

Batyrev-Manin conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let $S\subseteq\mathbb{P}^{d}$ be a non-singular del Pezzo surface of degree d, let $U\subset S$ be the Zariski open subset obtained by deleting the finitely many lines on S, a...

conjecture | open | MathDB 页面 | 原文线索

V2-060024 | MATHDB-316261

The vanishing conjecture for homogeneous Hessian-nilpotent polynomials

摘录 / EXCERPT: Let $z=(z_1,\ldots,z_n)$ be commuting variables, let ... be the Laplace operator, and call a formal power series $P(z)$ Hessian nilpotent if its Hessian matrix ... is nilpotent. Vanish...

conjecture | open | MathDB 页面 | 原文线索

V2-060025 | MATHDB-316265

The orbifold Virasoro constraints

摘录 / EXCERPT: Let $\mathcal{X}$ be a compact orbifold, let $\mathcal{D}_\mathcal{X}$ denote its total descendant Gromov-Witten potential, and let $\mathcal{L}_m$ be the Virasoro operators define...

conjecture | open | MathDB 页面 | 原文线索

V2-060026 | MATHDB-316266

The genus-one invariant conjecture for orbifolds

摘录 / EXCERPT: Let $\mathcal{X}$ be an orbifold, and let $I\!I\mathcal{X}$ be its double inertia orbifold, whose points are triples (x,g,h) with $x\in\mathcal{X}$ and commuting elements...

conjecture | open | MathDB 页面 | 原文线索

V2-060027 | MATHDB-316276

DPC optimality conjecture for the K-user quasi-static fading broadcast channel

摘录 / EXCERPT: DPC optimality conjecture. If the K decoders are ordered such that

conjecture | open | MathDB 页面 | 原文线索

V2-060028 | MATHDB-316282

Hubbard's existence conjecture for type-3 parameters of the complex Hénon map

摘录 / EXCERPT: Let $(a,c)\in\mathcal{H}^{\mathbb{C}}\cap\mathbb{R}^2$ be a real parameter in the hyperbolic locus of the complex Hénon map. Such a parameter is of type-1 if...

conjecture | open | MathDB 页面 | 原文线索

V2-060029 | MATHDB-316283

Hubbard's surjectivity conjecture for the Hénon-map monodromy homomorphism

摘录 / EXCERPT: Let $\mathcal{H}^{\mathbb{C}}_0$ be the component of the complex Hénon hyperbolic locus containing HOV, let $(a_0,c_0)\in\mathrm{DN}$ be a basepoint, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060030 | MATHDB-316293

The isometric-copy translation conjecture for self-affine sponges

摘录 / EXCERPT: Isometric-copy translation conjecture. There exists a translation t such that

conjecture | open | MathDB 页面 | 原文线索

V2-060031 | MATHDB-316295

The uniqueness conjecture for tetrahedrally symmetric monopoles

摘录 / EXCERPT: Uniqueness conjecture. For a symmetric monopole, the denominator of $Q_0(z)$ has $2(|n_1|-1)$ zeros, and consequently the tetrahedrally symmetric monopole is the only monopole in t...

conjecture | open | MathDB 页面 | 原文线索

V2-060032 | MATHDB-316296

Conjecture on existence of limits for Poincaré functions of exceptional polynomials

摘录 / EXCERPT: Let p be the polynomial and f its Poincaré function appearing in Theorem, and let the limits in that theorem refer to the limits associated with the measure and exponent introd...

conjecture | open | MathDB 页面 | 原文线索

V2-060033 | MATHDB-316297

The slope conjecture for effective divisors on the moduli space of stable curves

摘录 / EXCERPT: Let $\overline{\mathcal M}_{g}$ be the Deligne-Mumford moduli space of genus g stable curves. For an effective divisor $D=a\lambda-\sum_{i=0}^{\lfloor g/2\rfloor}b_i\delta_i$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-060034 | MATHDB-316301

Higher truncation functors and factorization conjecture for weight structures

摘录 / EXCERPT: Let $(\underline{C},w)$ be a triangulated category with a weight structure, let $N\geq 0$, and write $\underline{C}^{[0,N]}$ for the corresponding weight range. The strong weight c...

conjecture | open | MathDB 页面 | 原文线索

V2-060035 | MATHDB-316307

Directed star arboricity bound by maximum degree

摘录 / EXCERPT: Maximum-degree directed star arboricity conjecture. If $\Delta\geq 3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-060036 | MATHDB-316308

Directed star arboricity under bounded in- and outdegree

摘录 / EXCERPT: Bounded-degree directed star colouring conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060037 | MATHDB-316318

The non-unit gradient degree conjecture for hypersurfaces with isolated singularities

摘录 / EXCERPT: Let $f\in \mathbb{C}[x_0,\ldots,x_n]$ be a reduced homogeneous polynomial, and let $V(f)$ be its associated projective hypersurface. Write $d=\operatorname{deg}(f)$ and let $d(f)$...

conjecture | open | MathDB 页面 | 原文线索

V2-060038 | MATHDB-316322

Conjecture on the limiting degree distribution of random planar graphs

摘录 / EXCERPT: For each integer $k\geq 1$, let $Z^{(k)}_n$ denote the proportion of vertices of degree k in a uniformly random planar graph with n vertices. Limiting degree-distribution conje...

conjecture | open | MathDB 页面 | 原文线索

V2-060039 | MATHDB-316325

Noether-Lefschetz generation conjecture for the Picard group of K3 moduli

摘录 / EXCERPT: Let $\mathcal{M}_l$ be the moduli space of quasi-polarized $K3$ surfaces, and let ... be the subspace spanned by the Noether-Lefschetz divisors $D_{h,d}$. Noether-Lefschetz generatio...

conjecture | open | MathDB 页面 | 原文线索

V2-060040 | MATHDB-316336

Integral-basis conjecture for Euler sums

摘录 / EXCERPT: Integral-basis conjecture. There exist $\mathbb{Q}$-linearly independent Euler sums of weight n such that every Euler sum of weight n is a $\mathbb{Z}$-linear combination of th...

conjecture | open | MathDB 页面 | 原文线索

V2-060041 | MATHDB-316346

Thomas-Yong's minuscule jeu de taquin rule for K-theoretic Schubert calculus

摘录 / EXCERPT: Thomas-Yong's conjecture. For any minuscule G/P,

conjecture | open | MathDB 页面 | 原文线索

V2-060042 | MATHDB-316347

The product-differences positivity conjecture for Schubert structure constants

摘录 / EXCERPT: Product-differences conjecture. For every ν,

conjecture | open | MathDB 页面 | 原文线索

V2-060043 | MATHDB-316348

Existence of a fibre with infinitely many rational points for quartic even polynomials

摘录 / EXCERPT: Let $a,b,d\in\mathbb{Z}$ and let $f(t)=at^4+bt^2+d$. For each $t_0\in\mathbb{Q}$, write E_{t_0} for the curve $y^2=x^3+f(t_0)x$. Existence conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-060044 | MATHDB-316349

Existence of a fibre with infinitely many rational points for sextic even polynomials

摘录 / EXCERPT: Let $a,c,e\in\mathbb{Z}$ and let $g(t)=t^6+at^4+ct^2+e$. For each $t_0\in\mathbb{Q}$, write E^{t_0} for the curve $y^2=x^3+g(t_0)$. Existence conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-060045 | MATHDB-316350

Existence of a fibre with infinitely many rational points in a non-constant elliptic family

摘录 / EXCERPT: Let $\{\mathcal{E}_t\}_{t\in\mathbb{Q}}$ be a non-constant family of elliptic curves. Non-constant-family conjecture. There exists $t\in\mathbb{Q}$ such that the curve...

conjecture | open | MathDB 页面 | 原文线索

V2-060046 | MATHDB-316357

Liu-Xu conjecture for psi-class intersection numbers

摘录 / EXCERPT: Let $g\geq2$, let n be a positive integer, and let $d_j\geq1$ satisfy ... For the psi-class intersection numbers $\langle\tau_{a_1}\cdots\tau_{a_m}\rangle_h$ on moduli spaces of st...

conjecture | open | MathDB 页面 | 原文线索

V2-060047 | MATHDB-316358

Liu-Xu Hodge-integral conjecture

摘录 / EXCERPT: Let $g\geq2$, let n be a positive integer, and let $d_j\geq1$ satisfy ... Let λ_i denote the Chern classes of the Hodge bundle, let τ_d denote a psi-class insertion, l...

conjecture | open | MathDB 页面 | 原文线索

V2-060048 | MATHDB-316362

The bounded-denominator conjecture for inverse shuffle-products of binary polynomials

摘录 / EXCERPT: Let σ be the quadratic form on $\overline{\mathbb F_2}[[X]]$, restricting to a bijection on $1+X\overline{\mathbb F_2}[[X]]$. For $k\in\mathbb N$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060049 | MATHDB-316377

Infinite-mean initial-degree conjecture for the PARID model

摘录 / EXCERPT: Let $r_k_{k\geq 1}$ be the distribution of the initial degrees in the PARID model, and suppose that it is a power-law distribution with exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-060050 | MATHDB-316378

Conjecture on double-logarithmic diameter and typical distance in preferential attachment graphs

摘录 / EXCERPT: Let $G_{m,\delta}(t)$ be the preferential attachment graph with parameters m and δ, and let $V_1,V_2\in [t]$ be two uniformly chosen independent vertices. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-060051 | MATHDB-316389

Nested Dyck path conjecture for the Schur expansion of nabla

摘录 / EXCERPT: Nested Dyck path conjecture. For every partition λ,

conjecture | open | MathDB 页面 | 原文线索

V2-060052 | MATHDB-316415

The duality conjecture for tridiagonal pairs

摘录 / EXCERPT: The duality conjecture. The tridiagonal pairs A,A^* and B,B^* are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-060053 | MATHDB-316418

The trace formula conjecture for split sequences of tridiagonal pairs

摘录 / EXCERPT: Let A,A^* be a tridiagonal pair whose shape vector satisfies $\rho_0=1$, and let $(\lbrace\theta_i\rbrace_{i=0}^d;\lbrace\theta_i^*\rbrace_{i=0}^d;\lbrace\zeta_i\rbrace_{i=0}^d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-060054 | MATHDB-316419

The product trace formula conjecture for tridiagonal pairs

摘录 / EXCERPT: Let A,A^* be a tridiagonal pair with shape vector satisfying $\rho_0=1$, and let $(\lbrace\theta_i\rbrace_{i=0}^d;\lbrace\theta_i^*\rbrace_{i=0}^d;\lbrace\zeta_i\rbrace_{i=0}^d)$...

conjecture | open | MathDB 页面 | 原文线索

V2-060055 | MATHDB-316420

The split-sequence symmetry conjecture for tridiagonal pairs

摘录 / EXCERPT: Let A,A^* be a tridiagonal pair. Let $\lbrace\theta_i\rbrace_{i=0}^d$ and $\lbrace\theta_i^*\rbrace_{i=0}^d$ be standard orderings of the eigenvalues of A and A^*, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-060056 | MATHDB-316422

Monotonicity, convexity and limiting value of Lyapunov exponents for the voter model

摘录 / EXCERPT: Voter-model Lyapunov-exponent conjecture. On $[0,\infty)$, $\kappa\mapsto\lambda_p(\kappa)$ is strictly decreasing and convex, with

conjecture | open | MathDB 页面 | 原文线索

V2-060057 | MATHDB-316423

Large-diffusion asymptotics for voter-model Lyapunov exponents

摘录 / EXCERPT: Large-diffusion asymptotic conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-060058 | MATHDB-316431

Minimal splitting-type conjecture for Weyl translates of a line

摘录 / EXCERPT: Splitting-type minimization conjecture. The pair (a_C,b_C) is the solution (a,b) satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-060059 | MATHDB-316441

Deligne-Ribet generation conjecture for Stark elements

摘录 / EXCERPT: Let k be a totally real number field, let k_∞/k be its cyclotomic $\mathbb{Z}_p$-extension, let L/k be the abelian extension and χ the character in the paper, and w...

conjecture | open | MathDB 页面 | 原文线索

V2-060060 | MATHDB-316454

The bilinear adjoint restriction conjecture for transverse paraboloid pieces

摘录 / EXCERPT: Let $n\ge 3$, and let S_1 and S_2 be smooth compact non-empty subsets of the paraboloid in ${\mathbf{R}}\times{\mathbf{R}}^{n-1}$ whose unit normals are separated by at least a...

conjecture | open | MathDB 页面 | 原文线索

V2-060061 | MATHDB-316457

The GES conjecture on controlled Følner pairs

摘录 / EXCERPT: GES conjecture. All GES groups have controlled Følner pairs and therefore satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-060062 | MATHDB-316467

Explicit formula conjecture for the Poisson kernel F_γ

摘录 / EXCERPT: Let γ be a non-negative integer, and let F_γ be the Poisson kernel under consideration. For $1\leq\beta\leq\gamma+2$, let $f_\beta(x)$ be polynomials. Explicit...

conjecture | open | MathDB 页面 | 原文线索

V2-060063 | MATHDB-316470

Generalized Reed-type bound for r-bounded colorings

摘录 / EXCERPT: Generalized Reed-type conjecture. Every such graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-060064 | MATHDB-316475

Herzog-Srinivasan Taylor multiplicity bound conjecture

摘录 / EXCERPT: Herzog-Srinivasan's Taylor multiplicity bound conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-060065 | MATHDB-316477

The predicted orbifold Gromov-Witten potential formula for ADE quotient singularities

摘录 / EXCERPT: Let $G\subset SU(2)$ be a finite subgroup, let $\mathcal{X}=[\mathbb{C}^{2}/G]$, and set the unit parameter x_0 equal to zero. Let $F_{\mathcal{X}}(x_1,\dotsc,x_n)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-060066 | MATHDB-316478

The distribution-at-ones conjecture for multiple polylogarithm values

摘录 / EXCERPT: Distribution-at-ones conjecture. All distribution relations in with $x_j=1$ for every j are consequences of regularized double shuffle relations (RDS) for MPVs of level d. This...

conjecture | open | MathDB 页面 | 原文线索

V2-060067 | MATHDB-316479

The weight-two regularized distribution conjecture

摘录 / EXCERPT: Weight-two RDT conjecture. All RDT in weight two are consequences of the RDS and FDT. The source reports numerical evidence through level $N=49$ for this completeness claim.

conjecture | open | MathDB 页面 | 原文线索

V2-060068 | MATHDB-316480

The level-three and level-four binary-index conjecture for MPVs

摘录 / EXCERPT: Binary-index conjecture. If $N=3$ or $N=4$, every MPV of level N is a linear combination of MPVs

conjecture | open | MathDB 页面 | 原文线索

V2-060069 | MATHDB-316482

The weight-two RDT generation conjecture

摘录 / EXCERPT: Weight-two RDT generation conjecture. The RDT in follow from the seeded relations, the RDS in and, and the FDT in and. The source bases this on computation and does not state a pro...

conjecture | open | MathDB 页面 | 原文线索

V2-060070 | MATHDB-316488

Random Cayley graph Moore-bound conjecture

摘录 / EXCERPT: Let $\{G_n\}$ be a sequence of groups, and consider the d-regular random Cayley graph of G_n. Moore-bound conjecture. As $n\to\infty$, almost surely the girth is at most ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-060071 | MATHDB-316493

Intermediate-girth exponent conjecture for symmetric two-groups

摘录 / EXCERPT: Let W_n be the symmetric 2-group, and consider its random Cayley graph generated by k random generators. Intermediate-girth exponent conjecture. There is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-060072 | MATHDB-316501

Negative association under symmetric exclusion evolution

摘录 / EXCERPT: Let a symmetric exclusion process start from a deterministic initial configuration. Write η_t for its configuration at time t, and say that a distribution is negatively asso...

conjecture | open | MathDB 页面 | 原文线索

V2-060073 | MATHDB-316504

Pemantle's h-NLC and Rayleigh equivalence conjecture

摘录 / EXCERPT: Let CNA and CNA+ denote the corresponding conditional negative-association properties, and let h-NLC and h-NLC+ denote the hereditary negative-lattice-condition properties; Rayleig...

conjecture | open | MathDB 页面 | 原文线索

V2-060074 | MATHDB-316507

The identification of type $\widehat{A_1}$ generalized minors with block-Toeplitz minors

摘录 / EXCERPT: Generalized-minor identification conjecture. The minors $\Delta^{(i)}_{m,n}$ are precisely the generalized minors of Fomin-Zelevinsky.

conjecture | open | MathDB 页面 | 原文线索

V2-060075 | MATHDB-316509

The exceptional-index conjecture for Stirling numbers S(n,5)

摘录 / EXCERPT: For $n\in\mathbb{N}$, let $q_n=\nu_{2}(S(n,5))$, and call i exceptional when $q_{4i}\neq q_{4i+3}$. The observed exceptional indices begin with $7,39,71,103,\ldots$. The exceptio...

conjecture | open | MathDB 页面 | 原文线索

V2-060076 | MATHDB-316510

The second approximation conjecture for $\nu_2(S(m,5))$

摘录 / EXCERPT: The second approximation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060077 | MATHDB-316511

The third approximation claim for $\nu_2(S(m,5))$

摘录 / EXCERPT: The third approximation claim. For most $m\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-060078 | MATHDB-316526

Neumann heat kernel monotonicity along hyperbolic radii in symmetric convex domains

摘录 / EXCERPT: Let Ω be a bounded convex domain in the plane symmetric with respect to the x-axis, and let $p^N_{\Omega}(t,z,w)$ be its Neumann heat kernel. Let $f:\mathbb{B}\to\Omega$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-060079 | MATHDB-316527

Ground-state-normalized Dirichlet heat kernel monotonicity in the ball

摘录 / EXCERPT: Let $\mathbb{B}$ be the unit ball in $\mathbb{R}^d$, with $d\geq 1$, let $p^D_{\mathbb{B}}(t,x,y)$ be the Dirichlet heat kernel, and let $\varphi_1(x)$ be the ground-state eigenfun...

conjecture | open | MathDB 页面 | 原文线索

V2-060080 | MATHDB-316531

Fortuin-Kasteleyn SLE scaling-limit conjecture

摘录 / EXCERPT: Let $q\in[0,4]$, and consider the Fortuin-Kasteleyn random-cluster model on the square lattice at the self-dual value $p_{\mathrm{sd}}=\sqrt q/(\sqrt q+1)$, with Dobrushin boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-060081 | MATHDB-316533

Vanishing of the solid angle function at the origin

摘录 / EXCERPT: Let $\mathcal{P}$ be a real convex d-polytope, where d is any dimension. Vanishing conjecture. The associated function satisfies ... This conjecture generalizes the preceding two...

conjecture | open | MathDB 页面 | 原文线索

V2-060082 | MATHDB-316534

Burns-Flach rationality conjecture for equivariant elliptic-curve L-values

摘录 / EXCERPT: Burns-Flach rationality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060083 | MATHDB-316540

Naserasr's planar graph power conjecture

摘录 / EXCERPT: Let k be a positive integer, and let $\mathcal{P}_{2k+1}$ be the class of planar graphs with odd-girth at least $2k+1$. For a graph G, write $G^{(2k-1)}$ for its $(2k-1)$st pow...

conjecture | open | MathDB 页面 | 原文线索

V2-060084 | MATHDB-316541

The massive essential spectrum conjecture for singular integral operators on variable Lebesgue spaces

摘录 / EXCERPT: Massive essential spectrum conjecture. There exists a measurable function $p:\mathbb{T}\to(1,\infty)$ such that S is bounded on $L^{p(\cdot)}(\mathbb{T})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-060085 | MATHDB-316547

Simpson stability conjecture for simple bundles on elliptic-curve degenerations

摘录 / EXCERPT: Let E be a reduced projective curve with trivial dualizing sheaf and m irreducible components, and let $(n,\mathbbm{d})\in\mathbb{Z}_+\times\mathbb{Z}^m$ satisfy ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060086 | MATHDB-316549

F-threshold multiplicity inequality for parameter ideals

摘录 / EXCERPT: F-threshold multiplicity conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-060087 | MATHDB-316550

The stronger minimum pseudocodeword weight bound for expander codes

摘录 / EXCERPT: Stronger pseudocodeword weight conjecture. The resulting LDPC code has

conjecture | open | MathDB 页面 | 原文线索

V2-060088 | MATHDB-316551

The extended multiplier-index conjecture for polynomial maps

摘录 / EXCERPT: Extended multiplier-index conjecture. The converse of the emptiness criterion also holds when $d\geq 8$. Moreover, if

conjecture | open | MathDB 页面 | 原文线索

V2-060089 | MATHDB-316552

The extended fixed-point index conjecture for polynomial maps

摘录 / EXCERPT: Extended fixed-point index conjecture. The map $\widetilde{\Phi}_d$ is finite, and the analogous results to the main theorems hold for $\widetilde{\Phi}_d$ for every parameter valu...

conjecture | open | MathDB 页面 | 原文线索

V2-060090 | MATHDB-316562

The mixed symmetric-skew-symmetric DDVV matrix inequality

摘录 / EXCERPT: Let $A_1,\ldots,A_{m_1}$ be symmetric $n\times n$ matrices and let $A_{m_1+1},\ldots,A_{m_1+m_2}$ be skew-symmetric $n\times n$ matrices. For the commutator...

conjecture | solved | MathDB 页面 | 原文线索

V2-060091 | MATHDB-316564

Eventual shift conjecture for characteristic polynomials of Coxeter multiarrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a Coxeter arrangement with Coxeter number h, and let $m:\mathcal{A}\rightarrow\mathbb{Z}_{\geq 0}$ be a multiplicity. The characteristic polynomial of a mult...

conjecture | open | MathDB 页面 | 原文线索

V2-060092 | MATHDB-316566

Universality conjecture for random-cluster divide-and-colour scaling limits

摘录 / EXCERPT: Let $\tilde X(p,q,r)$ be the divide-and-colour model associated with a random-cluster measure with parameters $q>0$ and $p<p_c(q)$, and consider it on the triangular lattice at...

conjecture | open | MathDB 页面 | 原文线索

V2-060093 | MATHDB-316570

Hinkkanen-Martin conjecture on Julia sets of polynomial semigroups

摘录 / EXCERPT: Hinkkanen-Martin's modified conjecture. For all $f,g\in G$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-060094 | MATHDB-316573

Uniform 2-partition conjecture for the second mixed eigenvalue of the disk

摘录 / EXCERPT: Let the unit disk carry mixed Dirichlet--Neumann boundary conditions, with the Dirichlet part varying among boundary arrangements; the uniform 2--partition is the arrangement con...

conjecture | open | MathDB 页面 | 原文线索

V2-060095 | MATHDB-316586

Finiteness conjecture for similarity classes of reflective IV-type lattices

摘录 / EXCERPT: Let T be a lattice of IV type, meaning a lattice whose associated type-IV domain ... and all ground fields. This is the type-IV analogue of the finiteness statement for reflective...

conjecture | open | MathDB 页面 | 原文线索

V2-060096 | MATHDB-316588

The Deligne-Ihara-type kernel conjecture for the thrice-punctured projective line

摘录 / EXCERPT: Let a be the tangential base point $(0,\partial/\partial t_{|0})$ for $X:=\mathbb{P}^1_{\mathbb{Q}}\setminus\{0,1,\infty\}$. Let $G_{BK}(\mathbb{Q})$ be the relevant Bloch-Kato m...

conjecture | open | MathDB 页面 | 原文线索

V2-060097 | MATHDB-316589

Moment conjecture for the family of all elliptic curves

摘录 / EXCERPT: Let $\mathcal{F}(X)$ be the family of elliptic curves $E_{a,b}$ ordered by the coefficient box in the source, and let $L(s,E)$ be the associated L-function. Let $G(s)$ be the Bar...

conjecture | open | MathDB 页面 | 原文线索

V2-060098 | MATHDB-316590

Precise moment conjecture for the family of all elliptic curves

摘录 / EXCERPT: Let $\mathcal{F}(X)$ be the family of elliptic curves, let N_E be the conductor of E, and let $X_E(s)$ be defined by the functional equation ... For a nonnegative integer k, de...

conjecture | open | MathDB 页面 | 原文线索

V2-060099 | MATHDB-316591

Moment conjecture for derivatives in the positive-rank family

摘录 / EXCERPT: Positive-rank moment conjecture. For any $k\in\mathbb{C}$ with $\operatorname{Re}k>-\tfrac12$, there exists $a_k'\ne0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-060100 | MATHDB-316592

Precise derivative-moment conjecture for the positive-rank family

摘录 / EXCERPT: Let $\mathcal{F}'(X)$ be the positive-rank family of curves E_{a,b^2}, let N_E be the conductor, and let $X_E(s)$ satisfy ... For a nonnegative integer k, define $Q_k(N)$ by .....

conjecture | open | MathDB 页面 | 原文线索

V2-060101 | MATHDB-316593

First shifted moment conjecture for the positive-rank family

摘录 / EXCERPT: Let $\mathcal{F}'(X)$ be the positive-rank family, let N_E be the conductor, and let $A'(\alpha)$ be the arithmetical factor arising from the shifted-moment calculation. First sh...

conjecture | open | MathDB 页面 | 原文线索

V2-060102 | MATHDB-316605

Hausel-Thaddeus-Langlands stringy mirror symmetry conjecture

摘录 / EXCERPT: Let $\M^d_{\DR}(\SL_n)$ and ${\mathcal M}^e_{\DR}(\PGL_n)$ be the corresponding de Rham moduli spaces, with $d,e\in\mathbb Z$ satisfying $(d,n)=(e,n)=1$. Let B^e and $\hat B^d$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-060103 | MATHDB-316608

Double clustering conjecture for bounded-doubling graph families

摘录 / EXCERPT: Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two families of graphs with bounded doubling dimension, not necessarily with the same constants. For any two graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-060104 | MATHDB-316610

Bogomolov's rational curves conjecture for K3 surfaces

摘录 / EXCERPT: Bogomolov's conjecture. Every $\overline{k}$-rational point on X lies on some rational curve $C\subset X$ defined over $\overline{k}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060105 | MATHDB-316614

Deficiency growth conjecture for Bestvina-Brady approximations

摘录 / EXCERPT: Deficiency growth conjecture. If Σ is not simply connected, then

conjecture | open | MathDB 页面 | 原文线索

V2-060106 | MATHDB-316625

Stable-degree conjecture for permutation representations

摘录 / EXCERPT: Stable-degree conjecture. There exists a stably equivalent permutation representation $\rho'\colon G\to S_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-060107 | MATHDB-316633

Equilibrium conjecture for infinitely repeated games with incomplete information

摘录 / EXCERPT: Equilibrium conjecture. For all signalling functions Λ, the corresponding infinitely repeated game of incomplete information on one side has an equilibrium, meaning a pair...

conjecture | open | MathDB 页面 | 原文线索

V2-060108 | MATHDB-316636

Local L_1 wave propagator conjecture for uniformly elliptic approximations

摘录 / EXCERPT: Local L_1 wave propagator conjecture. If $\alpha>1/2$, then there exists a constant $C_{\alpha,c}$ such that, for all $\epsilon\leq1/4$ and sufficiently small t,

conjecture | open | MathDB 页面 | 原文线索

V2-060109 | MATHDB-316638

Hirsch equality conjecture for non-degenerate axial transportation polytopes

摘录 / EXCERPT: Let P be a non-degenerate $l\times m\times n$ axial transportation polytope with $l,m,n\geq 3$. Its dimension is ... Let f be the number of facets of P, and let $G(P)$ be its g...

conjecture | open | MathDB 页面 | 原文线索

V2-060110 | MATHDB-316641

Conjecture on non-stabilized boundary stabilizations for the T(7,17,2,10m-4) family

摘录 / EXCERPT: Let $K_m=T(7,17,2,10m-4)$, and let (V,W) be the unique minimal genus-two Heegaard splitting of its exterior $E(K_m)=S^3-N(K_m)$. Let (V',W') be the boundary stabilization of...

conjecture | open | MathDB 页面 | 原文线索

V2-060111 | MATHDB-316647

The 1/2 conjecture for simple extension classes of fix-free codes

摘录 / EXCERPT: Let a P_γ-simple extension class be the class notion defined in the survey for which the extension property holds with parameter γ. Simple-extension conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-060112 | MATHDB-316667

The Björner-Swartz conjecture for g-vectors of doubly Cohen-Macaulay complexes

摘录 / EXCERPT: Björner-Swartz conjecture. The g-vector of any doubly Cohen-Macaulay complex is an M-sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-060113 | MATHDB-316678

McMullen's characterization conjecture for simplicial polytopes

摘录 / EXCERPT: McMullen's conjecture. The sequence $(h_0,h_1,\dots,h_d)$ is the h-vector of the boundary of a simplicial d-polytope if and only if $h_0=1$,...

conjecture | solved | MathDB 页面 | 原文线索

V2-060114 | MATHDB-316694

Exchange-related braids have equal first-order Q-polynomials

摘录 / EXCERPT: Exchange-vanishing conjecture. The polynomials ${\bf Q}_{\widehat\beta,1}(x)$ of exchange-related braids vanish in the same way as the Fiedler polynomial; in particular, the first-...

conjecture | open | MathDB 页面 | 原文线索

V2-060115 | MATHDB-316704

Strange Duality morphisms conjecture for moduli spaces on abelian surfaces

摘录 / EXCERPT: Let A be the abelian surface and let v and w be Mukai vectors satisfying Assumptions and, with the associated moduli spaces and determinant line bundles as in the paper. Deno...

conjecture | open | MathDB 页面 | 原文线索

V2-060116 | MATHDB-316705

Theta line-bundle variation conjecture for moduli spaces on abelian surfaces

摘录 / EXCERPT: Let v be a Mukai vector, and let F_{1} and F_{2} be sheaves with the same Mukai vector orthogonal to v. Let Θ_{F_i} denote the corresponding theta line bundles on th...

conjecture | open | MathDB 页面 | 原文线索

V2-060117 | MATHDB-316713

Cappell-Miller conjecture on torsion and Reidemeister torsion

摘录 / EXCERPT: Cappell-Miller conjecture. The Cappell-Miller torsion satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-060118 | MATHDB-316714

Supermodularity conjecture for common information

摘录 / EXCERPT: Let m_1, m_2, and m_3 be information elements, and write m_{ij} and m_{123} for their common information elements indexed by the corresponding subsets. Supermodularity co...

conjecture | open | MathDB 页面 | 原文线索

V2-060119 | MATHDB-316722

Goulden-Rattan's numerical conjecture for the k=3 coefficients c_rho

摘录 / EXCERPT: Let $c_\rho^{(3)}$ be the coefficients in the monomial expansion of f_3. Goulden-Rattan's numerical conjecture. For $k=3$, the values of $2\cdot 6!\cdot 8!\,c_\rho^{(3)}$ are giv...

conjecture | open | MathDB 页面 | 原文线索

V2-060120 | MATHDB-316723

Kerov's Q-positivity conjecture for Kerov polynomial components

摘录 / EXCERPT: Define $Q_0=1$, $Q_1=0$, and for $n\ge 2$ let ... where $l(\mu)$ is the length of μ and $\mathcal{ R}_\mu=\prod_{i\ge2}((i-1)R_i)^{m_i(\mu)}/m_i(\mu)!$. Kerov's Q-positivity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-060121 | MATHDB-316724

Kerov's numerical conjecture for the k=3 coefficients a_rho

摘录 / EXCERPT: Let $a_\rho^{(3)}$ be the coefficients in the monomial expansion of g_3. Kerov's numerical conjecture. For $k=3$, the values of $500\cdot5!\cdot7!\,a_\rho^{(3)}$ are given by the...

conjecture | open | MathDB 页面 | 原文线索

V2-060122 | MATHDB-316725

The realization conjecture for integer surgeries on lens spaces

摘录 / EXCERPT: Realization conjecture. Then $L(p,q)$ is realized by integer surgery on a Berge knot.

conjecture | open | MathDB 页面 | 原文线索

V2-060123 | MATHDB-316729

Strange duality for Mukai moduli spaces on K3 and abelian surfaces

摘录 / EXCERPT: Mukai strange-duality conjecture. The duality morphism $\mathsf {D}$ is either an isomorphism or zero. Numerical symmetry and vanishing results motivate this dichotomy, while the p...

conjecture | open | MathDB 页面 | 原文线索

V2-060124 | MATHDB-316736

Maximal CM labelling conjecture for subdivided polygons

摘录 / EXCERPT: Let an n-gon have a subdivision with k additional edges and no additional vertices. A CM monomial labelling is a monomial labelling whose cellular complex gives a Cohen-Macaula...

conjecture | open | MathDB 页面 | 原文线索

V2-060125 | MATHDB-316744

Finite-coefficient conjecture for degenerations of surfaces with Kodaira dimension zero

摘录 / EXCERPT: Finite-coefficient conjecture. There exists a finite subset ${\cal C}\subset \mathbb{Q}$ such that, for every such degeneration, $c_p^{\ast}\in {\cal C}$. This conjecture is propos...

conjecture | open | MathDB 页面 | 原文线索

V2-060126 | MATHDB-316747

Caselli's nonnegativity conjecture for Kazhdan-Lusztig coefficients

摘录 / EXCERPT: Caselli's conjecture. For each Bruhat interval $[u,v]$ and each $i=0,1,\dots,\lfloor n/2\rfloor$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-060127 | MATHDB-316748

Sign conjecture for the complete cd-index of Bruhat intervals

摘录 / EXCERPT: Sign conjecture. For every ${\bf cd}$-word w, one has

conjecture | open | MathDB 页面 | 原文线索

V2-060128 | MATHDB-316749

The slice-measure inequalities conjecture for two-variable weighted shifts

摘录 / EXCERPT: Slice-measure inequalities conjecture. The subnormality of $\mathbf{T}$ is determined by a countable collection of inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-060129 | MATHDB-316788

Kollár-Szabó extension conjecture for curves on low-degree hypersurfaces

摘录 / EXCERPT: Let $X\subset \mathbb{P}^{n+1}$ be a smooth hypersurface of degree at most $n+1$ defined over a finite field $\mathbb{F}_q$. Let C be a smooth projective curve and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060130 | MATHDB-316791

Andrews' smallest-parts moment inequality

摘录 / EXCERPT: Andrews' smallest-parts moment inequality. For k even, $k\geq2$, and $n\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-060131 | MATHDB-316805

The virtual Hodge polynomial generating-function conjecture for stable-bundle moduli spaces

摘录 / EXCERPT: Virtual Hodge polynomial conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060132 | MATHDB-316819

The GG conjecture on singularities after resolving the discriminant locus

摘录 / EXCERPT: GG conjecture. For every non-torsion primitive Hodge class ζ, there is an integer k and a resolution $f:S\to P=|\mathcal L^k|$ of the discriminant locus such that, for any...

conjecture | open | MathDB 页面 | 原文线索

V2-060133 | MATHDB-316827

Logarithmic upper-bound conjecture for the trellis-treewidth gap

摘录 / EXCERPT: Let ${\mathcal C}_i$, $i\geq 1$, be a sequence of codes over a finite field ${\mathbb F}$, with lengths n_i. Let $\kappa({\mathcal C}_i)$ denote treewidth and...

conjecture | open | MathDB 页面 | 原文线索

V2-060134 | MATHDB-316828

Asymptotic goodness obstruction for logarithmic-treewidth code families

摘录 / EXCERPT: Let ${\mathcal C}_i$, $i\geq 1$, be a sequence of $[n_i,k_i,d_i]$ codes, where n_i is the block length, k_i the dimension, and d_i the minimum distance. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060135 | MATHDB-316833

Woodall's modified positivity conjecture for bipartite planar graphs

摘录 / EXCERPT: Woodall's modified conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060136 | MATHDB-316834

Weak positivity conjecture for bipartite planar graphs above three

摘录 / EXCERPT: Let G be a bipartite planar graph, and let $P_G(q)$ denote its chromatic polynomial. The weakened bipartite-planar positivity conjecture. ... for every real $q\geq 3$. This is expl...

conjecture | open | MathDB 页面 | 原文线索

V2-060137 | MATHDB-316835

Positivity conjecture for plane quadrangulations above three

摘录 / EXCERPT: Let G be a plane quadrangulation, meaning a plane graph whose faces are bounded by four edges, and let $P_G(q)$ denote its chromatic polynomial. The quadrangulation positivity co...

conjecture | open | MathDB 页面 | 原文线索

V2-060138 | MATHDB-316838

The defect bound for terminal Gorenstein double sextics

摘录 / EXCERPT: A double sextic is a double cover of $\mathbb{P}^3$ ramified along a sextic surface. Suppose its singularities are terminal Gorenstein, and let its defect denote the relevant defec...

conjecture | open | MathDB 页面 | 原文线索

V2-060139 | MATHDB-316843

The sparsified-matrix nonsingularity conjecture

摘录 / EXCERPT: Let W be a symmetric matrix containing at least m nonzero entries in each row. Let $Q(W,p)$ be the symmetric matrix obtained by independently retaining each entry of W with p...

conjecture | open | MathDB 页面 | 原文线索

V2-060140 | MATHDB-316846

The uniform-personalization conjecture for optimal PageRank link structures

摘录 / EXCERPT: Let $\mathcal{E}_\mathrm{in}(\mathcal{I})\neq\emptyset$ and $\mathcal{E}_{\overline{\mathcal{I}}}$ be given. Let $\mathcal{E}_{\mathcal{I}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-060141 | MATHDB-316847

The qQTFPL enumeration product formula

摘录 / EXCERPT: Let $A(N)$, $A_{\textsc{HT}}(N)$, and $A_{\textsc{QT}}(N)$ denote the numbers of FPLs, HTFPLs, and qQTFPLs of the indicated sizes, respectively. A qQTFPL has size congruent to 2...

conjecture | open | MathDB 页面 | 原文线索

V2-060142 | MATHDB-316848

The refined qQTFPL enumeration product formula

摘录 / EXCERPT: For each size n, let $A(n;y)$, $A_{\textsc{HT}}(n;y)$, and $A_{\textsc{QT}}(n;y)$ denote the enumerating polynomials of FPLs, HTFPLs, and qQTFPLs, respectively. Give an object we...

conjecture | open | MathDB 页面 | 原文线索

V2-060143 | MATHDB-316850

The quarter-turn FPL link-pattern distribution conjecture

摘录 / EXCERPT: Let $A_{\textsc{QT}}(4n;w)$ denote the number of quarter-turn-invariant FPLs of size $4n$ with link pattern w, where w is a bilateral Dyck word of length $2n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060144 | MATHDB-316851

The quasi-quarter-turn FPL link-pattern distribution conjecture

摘录 / EXCERPT: Let $A_{\textsc{QT}}(4n+2;w)$ denote the number of qQTFPLs of size $4n+2$ with the induced half-turn-invariant link pattern w of length $2n+1$, and let $A_{\textsc{HT}}(2n+1;w)$...

conjecture | open | MathDB 页面 | 原文线索

V2-060145 | MATHDB-316852

The residual torsion-free nilpotence conjecture for pure braid groups of reflection arrangements

摘录 / EXCERPT: Residual torsion-free nilpotence conjecture. The fundamental group of the complement

conjecture | open | MathDB 页面 | 原文线索

V2-060146 | MATHDB-316854

The discriminant formula conjecture for the reflection group $G(e,e,r)$

摘录 / EXCERPT: Discriminant formula conjecture. If e is odd, the discriminant is

conjecture | open | MathDB 页面 | 原文线索

V2-060147 | MATHDB-316858

Orbit representatives conjecture for short roots of type C

摘录 / EXCERPT: Let Δ be of type C_n, let $\Delta_s^+(C_n)$ be its short positive roots, and let $\mathfrak X_s$ be the reverse operator on their antichains. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060148 | MATHDB-316872

Generic Newton slope conjecture for dominant affine Weyl elements

摘录 / EXCERPT: Let $G=SL_n(F)$, let $x=\pi^{\mu}w\in\widetilde{W}$, and suppose that μ is dominant, meaning ... for every simple root α_i in $\operatorname{Lie}(G)$, equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-060149 | MATHDB-316874

Soficity preservation under the induction operation

摘录 / EXCERPT: Let A, G, Γ, and J satisfy the hypotheses of Lemma, and let $\iota_J:A^G\to A^\Gamma$ denote the induction operation. A subshift $X\subseteq A^G$ is sofic if it is the...

conjecture | open | MathDB 页面 | 原文线索

V2-060150 | MATHDB-316888

Characterization of reciprocal functions by independence of interpolation orders

摘录 / EXCERPT: Let $M_{f,m_{1},m_{2}}$ be the means defined in the paper, with m_{1} and m_{2} denoting the interpolation orders, and let f be a function on the relevant domain. Reciprocal-...

open_problem | open | MathDB 页面 | 原文线索

V2-060151 | MATHDB-316892

The cross-number equality conjecture for finite Abelian groups

摘录 / EXCERPT: Let G be a finite Abelian group. For a sequence $S=(g_1,\ldots,g_l)$ over G, define its cross number by ... Let $\mathsf{k}(G)$ be the maximum of $\mathsf{k}(S)$ over zero-sumfre...

conjecture | open | MathDB 页面 | 原文线索

V2-060152 | MATHDB-316905

The transitive-graph cut-off conjecture for Glauber dynamics

摘录 / EXCERPT: Let (G_n) be a sequence of transitive graphs, and let $t_{{\rm mix}}(n)$ denote the mixing time of Glauber dynamics on G_n. A sequence of chains has cut-off if there are times...

conjecture | open | MathDB 页面 | 原文线索

V2-060153 | MATHDB-316911

Finiteness conjecture for modular tensor categories

摘录 / EXCERPT: Finiteness conjecture. If the rank of MTCs is fixed, then there are only finitely many equivalence classes of MTCs.

conjecture | open | MathDB 页面 | 原文线索

V2-060154 | MATHDB-316918

Fibonacci dimension conjecture for generalized polylogarithm values

摘录 / EXCERPT: For each weight w, let $\mathcal{L}_w$ be the $\mathbb{Q}$-linear space generated by the values $\mathop{\rm Li}\nolimits_{\vec{s}}(1/2)$ with $w(\vec{s})=w$, and let $f_0=f_1=1$...

conjecture | open | MathDB 页面 | 原文线索

V2-060155 | MATHDB-316919

Fibonacci-word basis conjecture for generalized polylogarithms

摘录 / EXCERPT: Fibonacci-word basis conjecture. As a basis of $\mathcal{L}_w$, it is possible to take any one of the following sets:...

conjecture | open | MathDB 页面 | 原文线索

V2-060156 | MATHDB-316920

Independence conjecture for polylogarithms at one half

摘录 / EXCERPT: For a positive integer w, let ... The numbers $\mathop{\rm Li}\nolimits_{\vec{s}}(1/2)$ are generalized polylogarithm values indexed by $\vec{s}\in\mathcal{B}_w$. Independence conj...

conjecture | open | MathDB 页面 | 原文线索

V2-060157 | MATHDB-316921

Reduction conjecture for polylogarithms at one half

摘录 / EXCERPT: Let w be the weight of a vector $\vec{s}_0$, and define ... The values $\mathop{\rm Li}\nolimits_{\vec{s}}(1/2)$ are generalized polylogarithm values at one half. Reduction conject...

conjecture | open | MathDB 页面 | 原文线索

V2-060158 | MATHDB-316927

Petri-type injectivity conjecture for general stable bundles on curves

摘录 / EXCERPT: Petri-type injectivity conjecture. The Petri map μ_N is injective.

open_problem | open | MathDB 页面 | 原文线索

V2-060159 | MATHDB-316932

The MUB maximum-moment conjecture

摘录 / EXCERPT: Let X be a normalized uniformly distributed random vector. For mutually unbiased bases, let $m_1,m_2,\ldots,m_k$ denote the relevant maximum second-moment quantities, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060160 | MATHDB-316937

Uniqueness of the stationary distribution for attractive spin systems in a unique stationary environment

摘录 / EXCERPT: Let the background process have a unique stationary distribution. For each environment state $i\in\{1,2\}$, let $c_i(x,\eta)$ be the flip rate at site $x\in\mathbb{Z}$ for configur...

conjecture | open | MathDB 页面 | 原文线索

V2-060161 | MATHDB-316941

Very ampleness bound conjecture for generalized theta linear series

摘录 / EXCERPT: Let X be a smooth projective curve, and let $SU_X(r,d)$ denote the moduli space of semistable vector bundles of rank r and degree d, with generalized theta line bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-060162 | MATHDB-316943

The equivalence of CNC and CNA and of NC+ and NA+

摘录 / EXCERPT: Let μ be a measure on a finite Boolean cube. The properties CNC, CNA, NC+, and NA+ are the correlation properties defined in the paper, with “+” denoting closure under external...

conjecture | open | MathDB 页面 | 原文线索

V2-060163 | MATHDB-316944

The NA+ conjecture for random cluster measures with q<1

摘录 / EXCERPT: Let G be a finite graph and consider a random cluster measure on its edge sets with parameter $q<1$. Random-cluster conjecture. Every such random cluster measure is NA+. Equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-060164 | MATHDB-316946

The ULC conjecture for competing urn measures

摘录 / EXCERPT: Let m balls be assigned independently to urns $1,\ldots,n$, and let X_i indicate whether urn i is occupied. The law of $(X_1,\ldots,X_n)$ is a competing urn measure. Its rank...

conjecture | open | MathDB 页面 | 原文线索

V2-060165 | MATHDB-316950

The phase-transition conjecture for the smallest eigenvalue rate function

摘录 / EXCERPT: Let $I_k(\alpha)$ denote the rate function for the smallest eigenvalue in the k-dimensional model, and let ... For each k and all $\alpha\geq 0$, there exists $\alpha_k^*>0$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-060166 | MATHDB-316960

The pwo approximation conjecture for permutation-class growth rates

摘录 / EXCERPT: Pwo approximation conjecture. Every upper, lower, or proper growth rate of a permutation class is achieved by a pwo permutation class. This conjecture would support approximating a...

conjecture | open | MathDB 页面 | 原文线索

V2-060167 | MATHDB-316961

The rational-generating-function conjecture below κ

摘录 / EXCERPT: Rational-generating-function conjecture. Every permutation class with growth rate less than κ has a rational generating function. Classes of growth rate below 2 are known...

conjecture | open | MathDB 页面 | 原文线索

V2-060168 | MATHDB-316970

Stabilization conjecture for the arc operad

摘录 / EXCERPT: Let $\mathcal{A}rc$ be the arc operad, and let ${}^{\rm Fat}{\mathcal S}t{\mathcal{GT}ree}$ denote the stabilized fat graph operad. A suitable stabilization conjecture for the arc...

conjecture | open | MathDB 页面 | 原文线索

V2-060169 | MATHDB-316971

Bi-crossed product conjecture for stabilized graph-tree operads

摘录 / EXCERPT: Let ${\mathcal{GT}ree}(1)$ be the arity-one graph-tree operad, let ${\mathcal{LT}ree}$ be the operad of linear trees, and let ${\mathcal S}t{\mathcal{LG}Tree}^{(n)}$ denote the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-060170 | MATHDB-316994

Integral-group and Ehrhart characterization conjecture for uniform clutters

摘录 / EXCERPT: Let $\mathcal C$ be a uniform clutter with edge incidence vectors $v_1,\ldots,v_q$, and suppose its packing property holds. Define ... as the matrix whose columns are the augmented v...

conjecture | open | MathDB 页面 | 原文线索

V2-060171 | MATHDB-317007

Characterisation of proof translations by retractability

摘录 / EXCERPT: Retractability characterisation. A circuit is the translation of a proof if and only if it is retractable with respect to Maieli's $R_1,\ldots,R_4$ (dropping R_5).

conjecture | open | MathDB 页面 | 原文线索

V2-060172 | MATHDB-317015

Existence of equivariant bases for chord-diagram spaces

摘录 / EXCERPT: Let ${\cal C}^m_n$ denote the space of chord diagrams with m distinguishable components and degree n, modulo the 4T-relation. The symmetric group on the components acts on this...

conjecture | open | MathDB 页面 | 原文线索

V2-060173 | MATHDB-317020

Refined Lang-Trotter conjecture for varying Frobenius traces

摘录 / EXCERPT: Refined Lang-Trotter conjecture. Uniformly in this range,

conjecture | open | MathDB 页面 | 原文线索

V2-060174 | MATHDB-317030

Nonexistence of symmetric self-complementary 4-inflatable permutations at 64

摘录 / EXCERPT: Let $4\pi\in\mathfrak{S}_{64}4$ be a permutation that is 4-inflatable, meaning that it has the equal-pattern-count property required for the paper's 4-inflatable permutations. Say...

conjecture | open | MathDB 页面 | 原文线索

V2-060175 | MATHDB-317037

Asymptotic conjecture for the bottom of the spectrum with hypersurface magnetic wells

摘录 / EXCERPT: Let H^h be the operator acting in $L^2(M)$, let $\lambda_0(H^h)$ denote the bottom of its spectrum, let k be the order of vanishing of the magnetic field under the current assu...

conjecture | open | MathDB 页面 | 原文线索

V2-060176 | MATHDB-317043

Periodic points in minimally supported surface-group graph subshifts

摘录 / EXCERPT: Let Σ be a surface group, let V be a finite set, and let $X\subset V^{\Sigma}$ be the graph subshift defined by a finite graph $\mathcal G$. A point $x\in X$ is periodic w...

conjecture | open | MathDB 页面 | 原文线索

V2-060177 | MATHDB-317045

Conjecture on elliptic root systems for Weyl group Jacobi orbit spaces

摘录 / EXCERPT: Let W be a Weyl group, let Ω be the relevant domain, and let $J(W)$ be the Jacobi group acting on it. Write $\mathcal{R}_W$ for the root system of W, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060178 | MATHDB-317052

Polynomiality of Jack character polynomials in content power sums

摘录 / EXCERPT: Content-polynomiality conjecture. The quantity $\vartheta^\lambda_\mu(\zeta,\eta)$ is a polynomial in $p_0(C_\lambda)$ and the power sums $p_k(C_\lambda)$ for $k\geq1$. After repla...

conjecture | open | MathDB 页面 | 原文线索

V2-060179 | MATHDB-317053

Equivalence of the spherical Hecke category with the semisimple Weil category for m greater than n

摘录 / EXCERPT: Let m,n be integers with $m>n$. Let $_{-a}\operatorname{Sph}_{\mathbb G}$ be the spherical category for $\mathbb G$ with parameter $-a$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060180 | MATHDB-317057

The generation conjecture for three-dimensional polynomial automorphisms fixing one variable

摘录 / EXCERPT: Let $\mathbb{K}$ be the field appearing in the polynomial automorphism group $\operatorname{GA}_3(\mathbb{K})$. An automorphism fixing one variable is a polynomial automorphism tha...

conjecture | open | MathDB 页面 | 原文线索

V2-060181 | MATHDB-317058

The linearizable generation conjecture for polynomial automorphism groups

摘录 / EXCERPT: Let $\mathbb{K}$ be a field and let n denote the number of variables. A polynomial automorphism is linearizable if it can be transformed into a linear automorphism by a polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-060182 | MATHDB-317059

Separability conjecture for relatively quasi-convex subgroups of relatively hyperbolic groups

摘录 / EXCERPT: Let G be a group hyperbolic relative to the peripheral system $\mathcal{P}=\{P_1,\ldots,P_n\}$. Suppose that hyperbolic groups are residually finite, and that each P_i is finit...

conjecture | open | MathDB 页面 | 原文线索

V2-060183 | MATHDB-317071

Extension conjecture for automorphisms of plane-curve complements

摘录 / EXCERPT: Let $C\subset\mathbb{P}^2_{\mathbb{C}}$ be an irreducible curve. A curve is of type I if there exists a point $a\in C$ such that $C\backslash a$ is isomorphic to the affine line; a...

conjecture | open | MathDB 页面 | 原文线索

V2-060184 | MATHDB-317093

Nonnegative $(q-1)$-coefficients conjecture for counting polynomials of simple quiver representations

摘录 / EXCERPT: Let Q be a quiver and d a dimension vector, and let $P_{M_d^{\rm simp}(Q)}(q)$ denote the counting polynomial associated with the moduli of simple representations of dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-060185 | MATHDB-317109

D-finiteness conjecture for shuffle closures of regular languages

摘录 / EXCERPT: Let $\mathcal{L}$ be a regular language, and let its shuffle closure be the union of all finite shuffles of $\mathcal{L}$: ... If $L(z)$ is the ordinary generating function of this l...

conjecture | open | MathDB 页面 | 原文线索

V2-060186 | MATHDB-317120

Kapustka's modularity conjecture for nonrigid Calabi--Yau threefolds with five singular fibers

摘录 / EXCERPT: Let S_1 and S_2 be two non-isogenous elliptic surfaces with section defined over $\mathbb{Q}$, with semistable fibers, such that their fiber product ... has exactly five singular...

conjecture | open | MathDB 页面 | 原文线索

V2-060187 | MATHDB-317163

Shaping-loss conjecture for the Gaussian doubly-dirty MAC

摘录 / EXCERPT: Let the Gaussian doubly-dirty multiple access channel have high signal-to-noise ratio and strong interference. Denote its sum capacity by C^{sum} and the sum rate of the best kno...

conjecture | open | MathDB 页面 | 原文线索

V2-060188 | MATHDB-317164

Standard rank conjectures for elliptic curves over number fields

摘录 / EXCERPT: The standard rank conjectures. The following assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-060189 | MATHDB-317165

Geck-Lam conjecture on Kazhdan-Lusztig cells and r-cycle equivalence

摘录 / EXCERPT: Let a,b be the parameters defining the relevant Kazhdan-Lusztig cells, let $r\geq 1$, and let signed permutations lie in B_n. Two signed permutations are r-cycle equivalent (...

conjecture | open | MathDB 页面 | 原文线索

V2-060190 | MATHDB-317169

The general-\p formula for annealed Lyapunov exponents

摘录 / EXCERPT: Let ξ be the random potential satisfying the essential-supremum and bounded independent-identically-distributed assumptions, let $\kappa>0$ be the diffusion parameter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060191 | MATHDB-317176

Second-moment conjecture for central symmetric-square L-values

摘录 / EXCERPT: Let H_k be the set of normalized holomorphic Hecke eigenforms of level one and weight k, and let $L(s,\operatorname{sym}^2 f)$ be the associated symmetric square L-function....

conjecture | open | MathDB 页面 | 原文线索

V2-060192 | MATHDB-317182

Sufficiency conjecture for isostatic frameworks with C_{2v} or C_{3v} symmetry

摘录 / EXCERPT: Let $G=\{J,B\}$ be the connectivity of a framework, with $j=|J|$ joints and $b=|B|$ bars, and let p be a plane configuration generic with symmetry group ${\cal G}$. Write j_c f...

conjecture | open | MathDB 页面 | 原文线索

V2-060193 | MATHDB-317195

Reid's recipe conjecture for geometric McKay transforms

摘录 / EXCERPT: Let $G\subset\operatorname{SL}_3(\mathbb{C})$ be a finite abelian group, let $Y=G\text{-Hilb}(\mathbb{C}^3)$, and let $\Psi(\mathcal{O}_0\otimes\chi)$ be the inverse Fourier-Mukai...

conjecture | open | MathDB 页面 | 原文线索

V2-060194 | MATHDB-317204

Nonlinear scattering conjecture for finite-energy Gross-Pitaevskii solutions

摘录 / EXCERPT: Let $\psi=1+u$ be a global solution of the Gross-Pitaevskii equation in $C(\mathbb{R};1+F_1)$, and let E_1 denote its renormalized energy. Fix the threshold $\mathcal{E}_0>0$ fro...

conjecture | open | MathDB 页面 | 原文线索

V2-060195 | MATHDB-317211

The threshold conjecture for factors in random graphs

摘录 / EXCERPT: Let H be a fixed graph with $v=v(H)$ vertices, and let $G(n,p)$ be the random graph. Write $th_H(n)$ for a threshold for the property that $G(n,p)$ has an H-factor. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060196 | MATHDB-317214

A sectorial-projection criterion for weak unique continuation

摘录 / EXCERPT: Sectorial-projection criterion. There is a criterion for weak unique continuation for A in terms of the range of $P_+(B_0)$.

conjecture | open | MathDB 页面 | 原文线索

V2-060197 | MATHDB-317215

Vanishing of the inner index for first-order elliptic operators

摘录 / EXCERPT: Vanishing of the inner-index conjecture. The inner index vanishes:

conjecture | open | MathDB 页面 | 原文线索

V2-060198 | MATHDB-317231

The normalized inertia conjecture for partial Hessians of symmetric polynomials

摘录 / EXCERPT: Let q be the partial Hessian of a symmetric polynomial, let $Z(0,A)$ be the associated middle matrix evaluated at $A\in\mathbb S_n(\mathbb R^g)$, let $\mu_\pm$ denote the number...

conjecture | open | MathDB 页面 | 原文线索

V2-060199 | MATHDB-317247

Functorial S^{1}-equivariant trace for rigid symmetric monoidal $\mathbb{S}$-categories

摘录 / EXCERPT: Functorial trace conjecture. There exists a morphism

conjecture | solved | MathDB 页面 | 原文线索

V2-060200 | MATHDB-317268

Generalization of Beardon’s theorem to tame rational functions

摘录 / EXCERPT: Let $\mathbb{K}$ be a field of characteristic $\operatorname{char}(\mathbb{K})$, and let f be a rational function over $\mathbb{K}$ whose degree is not a multiple of...

conjecture | open | MathDB 页面 | 原文线索

V2-060201 | MATHDB-317279

The rationality conjecture for the Tamagawa number conjecture

摘录 / EXCERPT: Let M be one of the motives $M(F)$ or $M(\eta)$. Let $\Xi(M)$ be the fundamental line ... and let $\vartheta_\infty:E_\infty\simeq\Xi(M)\otimes_\mathbb{Q}\mathbb{R}$ be the isomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-060202 | MATHDB-317285

Extension of the Gaussian free field correlation theorem without ordering assumptions

摘录 / EXCERPT: Let $N=1,2,\dots$, let $\varkappa_j=(\nu_j,\eta_j,\tau_j)\in\mathcal D$ be distinct triples, and let $\Omega_j=\Omega(\nu_j,\eta_j,\tau_j)$. The correlation formula of Theorem 1.17...

conjecture | open | MathDB 页面 | 原文线索

V2-060203 | MATHDB-317299

Conjecture on eventual regular unimodular triangulations of lattice polytopes

摘录 / EXCERPT: Eventual and uniform unimodular triangulation conjectures. (a) For every lattice polytope P, there exists an integer m such that nP admits a regular unimodular triangulation...

conjecture | open | MathDB 页面 | 原文线索

V2-060204 | MATHDB-317301

Sum-rate conjecture for the binary skew-symmetric broadcast channel

摘录 / EXCERPT: Let (U,V) be auxiliary random variables and let X,Y_1,Y_2 be the input and outputs of the binary skew-symmetric channel with $p=\frac{1}{2}$, such that...

conjecture | open | MathDB 页面 | 原文线索

V2-060205 | MATHDB-317302

Hästö-Diening characterization for the Cauchy singular integral on Carleson curves

摘录 / EXCERPT: Let Γ be a simple Carleson curve, let $w:\Gamma\to[0,\infty]$ be a weight, and let $p:\Gamma\to(1,\infty)$ be continuous and satisfy the Dini-Lipschitz condition. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060206 | MATHDB-317317

The GLND generation conjecture

摘录 / EXCERPT: GLND generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060207 | MATHDB-317318

The locally finite automorphism generation conjecture

摘录 / EXCERPT: Locally finite generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060208 | MATHDB-317319

Shpilrain's generalized Shestakov-Umirbaev generation conjecture

摘录 / EXCERPT: Shpilrain's generation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060209 | MATHDB-317320

The GLIN generation conjecture

摘录 / EXCERPT: GLIN generation conjecture. If $\operatorname{char}(k)\ne2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-060210 | MATHDB-317323

Generic simple-spectrum conjecture for heteroclinic orbits

摘录 / EXCERPT: Let $H(t)$ denote the operator whose eigenvalues are considered along a heteroclinic orbit u of … such that, for every heteroclinic orbit u, all eigenvalues of $H(t)$ are simpl…

conjecture | open | MathDB 页面 | 原文线索

V2-060211 | MATHDB-317337

The so(3) volume conjecture for knots and links

摘录 / EXCERPT: so(3) volume conjecture. For all knots and links L,

conjecture | open | MathDB 页面 | 原文线索

V2-060212 | MATHDB-317338

The so(3) volume conjecture for knotted trivalent graphs

摘录 / EXCERPT: so(3) volume conjecture. The so(3) volume conjecture holds for all knotted trivalent graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-060213 | MATHDB-317344

Sullivan's Liouville conjecture for complete properly embedded minimal surfaces

摘录 / EXCERPT: Sullivan's conjecture. A complete, properly embedded minimal surface admits no non-constant, positive harmonic functions.

conjecture | open | MathDB 页面 | 原文线索

V2-060214 | MATHDB-317346

The ACC conjecture for smooth log canonical thresholds

摘录 / EXCERPT: ACC conjecture. For any n, there is no infinite increasing subsequence in $\mathcal{HT}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-060215 | MATHDB-317353

Inverse-closure conjecture for sign-determined Riordan matrices

摘录 / EXCERPT: Inverse-closure conjecture. The inverse satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-060216 | MATHDB-317354

Closure under shifted minors for sign-determined Riordan matrices

摘录 / EXCERPT: Shifted-minor closure conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060217 | MATHDB-317355

Goettsche's conjecture on effective Chow motives

摘录 / EXCERPT: Let k be a field, and let Mot^{eff}_k be the category of effective Chow motives over k with rational coefficients. For objects M,N of Mot^{eff}_k, write $[M]$ and $[N]$ f…

conjecture | open | MathDB 页面 | 原文线索

V2-060218 | MATHDB-317358

Douady-Hubbard local connectivity conjecture for the Mandelbrot set

摘录 / EXCERPT: Douady-Hubbard's local connectivity conjecture. The set $\mathcal{B}$ is locally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-060219 | MATHDB-317359

Triviality of fibers for quadratic polynomials and exponential maps

摘录 / EXCERPT: Trivial-fibers conjecture. In the parameter spaces of quadratic polynomials and exponential maps, all fibers are trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-060220 | MATHDB-317364

Connectivity conjecture for the infinite Schaeffer forest

摘录 / EXCERPT: Connectivity conjecture. The graph ${\rm Sch}_\infty(Q)$ is connected almost surely.

conjecture | open | MathDB 页面 | 原文线索

V2-060221 | MATHDB-317365

Local convergence conjecture for Schaeffer trees in large quadrangulations

摘录 / EXCERPT: Local convergence conjecture. The pair

conjecture | open | MathDB 页面 | 原文线索

V2-060222 | MATHDB-317366

Galton-Watson distribution conjecture for the infinite Schaeffer tree

摘录 / EXCERPT: Distribution conjecture. The pair $({\rm Sch}_\infty,\Delta h)$ is distributed as a Galton-Watson tree with geometric-$1/2$ offspring distribution, conditioned to be infinite, with…

conjecture | open | MathDB 页面 | 原文线索

V2-060223 | MATHDB-317368

Schur positivity and unimodality of refined Eulerian quasisymmetric functions

摘录 / EXCERPT: Schur-positivity conjecture. For all $j=0,1,\dots,n-1$, $Q_{\lambda,j}$ is Schur positive. Moreover, $Q_{\lambda,j}-Q_{\lambda,j-1}$ is Schur positive if $1\leq j\leq\frac{n-k}{2}$…

conjecture | open | MathDB 页面 | 原文线索

V2-060224 | MATHDB-317369

t-unimodality of refined Eulerian polynomials by cycle type

摘录 / EXCERPT: Cycle-type t-unimodality conjecture. The polynomial $A_{\lambda}^{{\rm maj},{\rm des},{\rm exc}}(q,p,q^{-1}t)$ is t-unimodal with center of symmetry $(n-k)/2$. Consequently, for…

conjecture | open | MathDB 页面 | 原文线索

V2-060225 | MATHDB-317382

The connectivity threshold conjecture for uniform random intersection graphs

摘录 / EXCERPT: Connectivity threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060226 | MATHDB-317388

Prokhorov's smoothing conjecture for cDV Fano threefolds

摘录 / EXCERPT: Prokhorov's smoothing conjecture. The threefold X has a smoothing by a flat deformation.

conjecture | open | MathDB 页面 | 原文线索

V2-060227 | MATHDB-317395

Tableau-sum conjecture for supersymmetric transfer matrices

摘录 / EXCERPT: Tableau-sum conjecture. For $m_{1},m_{2},m_{3},m_{1}-m_{2}\in\mathbb Z_{\ge 0}$ and $(m_{2}+m_{3})\ne 0$, the expressions can be written as

conjecture | open | MathDB 页面 | 原文线索

V2-060228 | MATHDB-317398

Bessel-polynomial form conjecture for principal angular-integral terms

摘录 / EXCERPT: Let n and β be parameters, let ${\hat{\cal I}}_{\beta,n}$ be the principal term of the angular integral, and let ... For each pair $i<j$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060229 | MATHDB-317404

MNOP-Pandharipande-Thomas rationality conjecture for DT and stable-pair series

摘录 / EXCERPT: MNOP-Pandharipande-Thomas rationality conjecture. For fixed $\x5cbeta$, the series $I_{\x5cbeta}(q)$ and $P_{\x5cbeta}(q)$ are Laurent expansions of rational functions of q invar...

conjecture | open | MathDB 页面 | 原文线索

V2-060230 | MATHDB-317405

GW-DT-PT correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: GW-DT-PT correspondence. After the variable change $q=-e^{i\lambda}$,

conjecture | open | MathDB 页面 | 原文线索

V2-060231 | MATHDB-317427

Cubical approximation conjecture

摘录 / EXCERPT: Let P and Q be two cubes, and let $f:V(P)\rightarrow V(Q)$. A cubical map is a map between the corresponding cubical face complexes that sends each cube into a cube. Cubical ap...

conjecture | open | MathDB 页面 | 原文线索

V2-060232 | MATHDB-317432

The freeness conjecture for packing clutters and the associated lattice group

摘录 / EXCERPT: Let $\cal C$ be a clutter with incidence vectors $v_1,\dots,v_q$, and suppose every minor $\cal C'$ satisfies $\tau({\cal C}')=\nu({\cal C}')$. If $x^{v_1},\dots,x^{v_q}$ have degr...

conjecture | open | MathDB 页面 | 原文线索

V2-060233 | MATHDB-317433

The packing-Mengerian equivalence for 2-partitionable clutters

摘录 / EXCERPT: Let $\cal C$ be a 2-partitionable clutter. It has the packing property when every minor has the König property, and it is Mengerian when the max-flow min-cut equality holds for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-060234 | MATHDB-317434

Weak separation purity conjecture for compatibility complexes

摘录 / EXCERPT: Let $[m]=\{1,\ldots,m\}$, and let ${\bf m}^{(ws)}_{k \leq l}$ be the compatibility complex whose vertices are the subsets of $[m]$ having at least k and at most l elements, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-060235 | MATHDB-317435

Polytopality conjecture for non-crossing compatibility complexes

摘录 / EXCERPT: Let L be a partial staircase shape, and let $2^{\bf m}_L$ be the associated collection of sets, with 2_L^* obtained by excluding the solid elements. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060236 | MATHDB-317440

Guivarc'h-Kaimanovich-Ledrappier singularity conjecture for stationary measures

摘录 / EXCERPT: Let $\Gamma<\operatorname{PSL}_2(\mathbb{R})$ be a lattice, and let m be a finitely supported measure whose support generates Γ. Let the corresponding stationary measure b...

conjecture | open | MathDB 页面 | 原文线索

V2-060237 | MATHDB-317442

The single-letter characterization of words with trivial fixed-point excess

摘录 / EXCERPT: Let $\mathbf{F}_k$ be the free group on generators $g_1,\ldots,g_k$, let $X_w^{(n)}$ be the number of fixed points of the word map associated with w, and let $\sim$ denote the eq...

conjecture | open | MathDB 页面 | 原文线索

V2-060238 | MATHDB-317446

Λ-Cannings convergence to tree-valued Λ-resampling dynamics

摘录 / EXCERPT: Let $\Lambda\in\mathcal M_f([0,1])$ satisfy the dust-free condition ... For the corresponding tree-valued Λ-Cannings dynamics $\mathcal V^{N,\Lambda}$, suppose there is a ran...

conjecture | open | MathDB 页面 | 原文线索

V2-060239 | MATHDB-317447

Continuous paths characterization for tree-valued Λ-resampling dynamics

摘录 / EXCERPT: Let $\mathcal V^\Lambda$ be the tree-valued Λ-resampling dynamics associated with a finite measure $\Lambda\in\mathcal M_f([0,1])$. Continuous-paths conjecture. The followi...

conjecture | open | MathDB 页面 | 原文线索

V2-060240 | MATHDB-317448

Atomlessness of tree-valued Λ-resampling dynamics at almost every time

摘录 / EXCERPT: Let $\mathcal V^\Lambda=(\mathcal V_t^\Lambda)_{t\geq 0}$ be the tree-valued Λ-resampling dynamics. Atomlessness conjecture. For Lebesgue almost every t, the metric measu...

conjecture | open | MathDB 页面 | 原文线索

V2-060241 | MATHDB-317449

Compactness and coming down from infinity for Λ-resampling dynamics

摘录 / EXCERPT: Let $\Lambda\in\mathcal M_f([0,1])$, and let $\mathcal V^\Lambda_t$ be the tree-valued Λ-resampling dynamics at time t. Write $\mathbb U_c$ for the space of compact metri...

conjecture | open | MathDB 页面 | 原文线索

V2-060242 | MATHDB-317463

q-specialization conjecture for q,t-Fuss-Catalan numbers

摘录 / EXCERPT: Let W be a well-generated complex reflection group of rank l, with fundamental-invariant degrees $d_1\leq\cdots\leq d_l$, Coxeter number $h=d_l$, and $N:=\sum_{i=1}^l(d_i-1)$....

conjecture | open | MathDB 页面 | 原文线索

V2-060243 | MATHDB-317464

Type B area conjecture for q,t-Fuss-Catalan numbers

摘录 / EXCERPT: For $n\geq 1$, let W_{B_n} be the reflection group of type B_n. A type B Catalan path of length n is a lattice path of $2n$ north and east steps from $(0,0)$ staying above...

conjecture | open | MathDB 页面 | 原文线索

V2-060244 | MATHDB-317467

Filtration conjecture for generalized (q,t)-Catalan numbers

摘录 / EXCERPT: For positive integers n and k, let $C_n^{(k)}(q,t)$ denote the generalized (q,t)-Catalan numbers of level k, and let $C_n(q,t)$ be the usual (q,t)-Catalan number. Filtrat...

conjecture | open | MathDB 页面 | 原文线索

V2-060245 | MATHDB-317468

Recursive formula conjecture for level-2 generalized (q,t)-Catalan numbers

摘录 / EXCERPT: Let $C_n^{(2)}(q,t)$ denote the generalized (q,t)-Catalan number of level 2. Level-2 recursion conjecture. For $n>1$, ... The formula is presented as a conjectural special-case d...

conjecture | open | MathDB 页面 | 原文线索

V2-060246 | MATHDB-317478

Infinite log-concavity conjecture for longest-increasing-subsequence polynomials

摘录 / EXCERPT: For each $n\geq 1$, let $P_n(x)=\sum_k P_{n,k}x^k$, where $P_{n,k}$ is the number of permutations of $[n]=\{1,2,\ldots,n\}$ whose longest increasing subsequence has length k. A p...

conjecture | open | MathDB 页面 | 原文线索

V2-060247 | MATHDB-317480

Infinite q-log-convexity conjecture for matching crossing-number polynomials

摘录 / EXCERPT: For each $n\geq 1$, let $M_{2n}(x)=\sum_k M_{2n,k}x^k$, where $M_{2n,k}$ is the number of matchings on $[2n]$ with crossing number k. A polynomial sequence is infinitely q-log-co...

conjecture | open | MathDB 页面 | 原文线索

V2-060248 | MATHDB-317488

The dual cross-number conjecture for long zero-sumfree sequences

摘录 / EXCERPT: Dual cross-number conjecture. A zero-sumfree sequence whose cross number is at least the lower bound $\mathsf{k}^*(G)$ has length at most the corresponding maximal lower-bound leng...

conjecture | open | MathDB 页面 | 原文线索

V2-060249 | MATHDB-317495

The separable bottom theorem for random extensions of Hilbertian fields

摘录 / EXCERPT: Bottom theorem conjecture. For almost all $\text{\boldmath$sigma$}\in\operatorname{Gal}(K)^e$, the field $K_s(\text{\boldmath$sigma$})$ is a finite separable extension of no proper...

open_problem | open | MathDB 页面 | 原文线索

V2-060250 | MATHDB-317497

The squarefree twin-complete numbers conjecture

摘录 / EXCERPT: Squarefree twin-complete numbers conjecture. The complete list of squarefree numbers that can be written as a sum of at most two positive squares, but not as a sum of three positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-060251 | MATHDB-317498

The odd-dimensional twin conjecture for integral vectors

摘录 / EXCERPT: Odd-dimensional twin conjecture. Suppose that the dimension $n\geq 5$ is odd. Then every non-odd vector has a twin.

conjecture | open | MathDB 页面 | 原文线索

V2-060252 | MATHDB-317499

The congruence twin conjecture for odd-dimensional integral vectors

摘录 / EXCERPT: Congruence twin conjecture. Suppose that the dimension $n\geq 5$ is odd and $d\not\equiv n\pmod{8}$. Then every vector of length $\sqrt{d}$ has a twin.

conjecture | open | MathDB 页面 | 原文线索

V2-060253 | MATHDB-317502

Rallis inner-product formula conjecture for unitary theta lifts

摘录 / EXCERPT: Rallis inner-product formula conjecture. Under these conditions, the Eisenstein series is holomorphic in a neighborhood of 0, and there is a nonzero constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-060254 | MATHDB-317525

Conjecture that constant image Milnor number implies excellence of unfoldings

摘录 / EXCERPT: Excellence conjecture. If μ_I is constant, then the unfolding is excellent.

conjecture | open | MathDB 页面 | 原文线索

V2-060255 | MATHDB-317527

A family of bijections preserving generalized equivalence classes

摘录 / EXCERPT: Let w and u be permutations, and let $\theta^{(k)}$ denote a bijection on permutations for each k. The bijections are required to satisfy the properties stated in Proposition...

conjecture | open | MathDB 页面 | 原文线索

V2-060256 | MATHDB-317529

Morton-Silverman lower-bound conjecture for canonical heights

摘录 / EXCERPT: Morton-Silverman canonical-height conjecture. There is a positive constant $M'=M'(d)>0$ such that, for every polynomial $\phi\in\mathbb{Q}[z]$ of degree d and every point...

conjecture | open | MathDB 页面 | 原文线索

V2-060257 | MATHDB-317532

Calderón-Moreno et al.'s conjecture that LCT implies strong Euler homogeneity for free divisors

摘录 / EXCERPT: Let D be a free divisor, and suppose that the logarithmic comparison theorem (LCT) holds for D, meaning that the inclusion ... is a quasi-isomorphism. A free divisor is strongly...

conjecture | open | MathDB 页面 | 原文线索

V2-060258 | MATHDB-317543

Depth-zero base change compatibility for unramified extensions

摘录 / EXCERPT: Let F be a local field, let E/F be an unramified extension, and let $\underline{G}$ be a reductive group over F. For a point x in the building of $\underline{G}(F)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060259 | MATHDB-317544

Depth-zero base change compatibility for tamely ramified extensions

摘录 / EXCERPT: Let F be a local field, let E/F be a cyclic tamely ramified extension, and let $\underline{G}$ be a reductive group over F. For a point x in the building of...

conjecture | open | MathDB 页面 | 原文线索

V2-060260 | MATHDB-317545

Positivity conjecture for Chern-Schwartz-MacPherson classes of graph hypersurfaces

摘录 / EXCERPT: Let $X_\Gamma=\{\Psi_\Gamma=0\}$ be the graph hypersurface associated with a graph Γ, embedded in its ambient projective space, and let H denote the hyperplane class. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-060261 | MATHDB-317550

Characterization of non-empty energy spaces for continuous magnetic fields

摘录 / EXCERPT: Let $B\in C(\mathbb R^2)$ be a continuous magnetic field, and let $\mathcal E_B$ denote the energy space associated with B. The notation $B=\operatorname{curl} A'$ means that the...

conjecture | open | MathDB 页面 | 原文线索

V2-060262 | MATHDB-317552

Casselman's regularity conjecture for cell languages

摘录 / EXCERPT: Let W be a Coxeter group with generating set S, and let $C\subset W$ be a cell. Write $\mathtt{Red}(C)$ for the language of reduced expressions of elements of C. Casselman's...

conjecture | open | MathDB 页面 | 原文线索

V2-060263 | MATHDB-317553

The affine hyperplane arrangement conjecture for left cells

摘录 / EXCERPT: Let Φ^+ be the set of positive roots, with short and long roots distinguished, and define ... by $\nu(\alpha)=0$ if α is short and $\nu(\alpha)=1$ if α is long. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-060264 | MATHDB-317561

The least accumulation-point conjecture for permutation-class growth rates

摘录 / EXCERPT: Let $\xi\approx 2.30524$ be the unique positive root of $x^5-2x^4-x^2-x-1$, and let the accumulation points from above of a set of real numbers be limits approached by larger eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-060265 | MATHDB-317565

List-colouring extension of Wegner's conjecture for nice graph families

摘录 / EXCERPT: List-colouring extension of Wegner's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060266 | MATHDB-317572

Central limit conjecture for mean-field interacting boson random point fields

摘录 / EXCERPT: Let $ u_{\kappa,\beta,\mu}$ be the distribution of the random point measure $4\xi_{\kappa}4$ on $4\mathbb{R}^d4$ for $4\kappa>04$, and let $4\mathfrak{L}4$ denote Lebesgue measure...

conjecture | open | MathDB 页面 | 原文线索

V2-060267 | MATHDB-317575

Conjecture that every type I geometric transition has a conifold degeneration

摘录 / EXCERPT: Type I conifold-degeneration conjecture. Every type I geometric transition admits a conifold degeneration. In particular, the canonical model of a type I geometric transition is a...

conjecture | open | MathDB 页面 | 原文线索

V2-060268 | MATHDB-317577

Conjecture on canonical models of small geometric transitions

摘录 / EXCERPT: Small-transition decomposition conjecture. Every small geometric transition which does not admit any conifold degeneration is equivalent to a composition of conifold transitions. C...

conjecture | open | MathDB 页面 | 原文线索

V2-060269 | MATHDB-317585

Finite-dimensional classifying-space conjecture for IA-automorphisms of right-angled Artin groups

摘录 / EXCERPT: Let Γ be a graph and let A_Γ be the associated right-angled Artin group. Let $\mathrm{IAut}\,A_\Gamma$ denote the subgroup of automorphisms acting trivially on the ab...

conjecture | open | MathDB 页面 | 原文线索

V2-060270 | MATHDB-317587

The Ferrers-diagram rank-metric code dimension conjecture

摘录 / EXCERPT: Let ${\cal F}$ be a Ferrers diagram and let δ be a positive integer. For each i with $0\leq i\leq\delta-1$, let ν_i be the number of dots in ${\cal F}$ outside the fir...

conjecture | open | MathDB 页面 | 原文线索

V2-060271 | MATHDB-317601

Johnson-Holroyd-Stahl conjecture on the circular chromatic number of Kneser graphs

摘录 / EXCERPT: Let ${\rm KG}(m,n)$ denote the Kneser graph, whose vertices are the n-element subsets of an m-element set, with two vertices adjacent when the corresponding subsets are disjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-060272 | MATHDB-317610

The conjecture for the dynamics group of rule 57

摘录 / EXCERPT: Let G_{57} be the dynamics group generated by the local permutations associated with Wolfram rule 57 on binary configurations of length n. Rule 57 conjecture. For all $n>4$,...

conjecture | open | MathDB 页面 | 原文线索

V2-060273 | MATHDB-317611

The conjecture for the dynamics group of rule 54

摘录 / EXCERPT: Let G_{54} be the dynamics group generated by the local permutations associated with Wolfram rule 54 on binary configurations of length n, with the fixed all-zero configurati...

conjecture | open | MathDB 页面 | 原文线索

V2-060274 | MATHDB-317627

The presentation conjecture for the monoid of labelled cubes

摘录 / EXCERPT: Presentation conjecture. Proposition $$ gives a presentation by generators and relations of the monoid $\overline{\square}([n],[n])$ for every $n\geqslant 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-060275 | MATHDB-317629

Lyle's conjecture on Mullineux maps and regularisation

摘录 / EXCERPT: Let $e\geqslant2$ and let λ be a partition. Write $G\lambda$ for the e-regularisation of λ, $T\lambda$ for the transpose of λ, and $M\lambda$ for the Mull...

conjecture | open | MathDB 页面 | 原文线索

V2-060276 | MATHDB-317631

Characterization of simulation rates for generalized Lipschitz functions

摘录 / EXCERPT: Let $\alpha\in\mathbb{N}$, and let $f\in {C^\alpha}^*[0,1]$ be a function bounded strictly between 0 and 1. Here ${C^\alpha}^*[0,1]$ consists of functions that are...

conjecture | open | MathDB 页面 | 原文线索

V2-060277 | MATHDB-317632

A codimension-3 tautological relation on the moduli space of compact-type curves of genus 5

摘录 / EXCERPT: Codimension-3 relation. The displayed linear combination of decorated-graph classes in $R^3(\mathcal{M}^{\operatorname{ct}}_5)$, with coefficients exactly as in the source statemen...

conjecture | open | MathDB 页面 | 原文线索

V2-060278 | MATHDB-317634

Sapko's annihilator conjecture for Buchsbaum tangent cones

摘录 / EXCERPT: Sapko's annihilator conjecture. If $\operatorname{gr}_{\mathfrak m}(R)$ is Buchsbaum, then for some $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-060279 | MATHDB-317643

The backward-orbit relative integrality conjecture

摘录 / EXCERPT: Let K be a number field, let S be a finite set of places of K, and let φ be a rational map. For $\alpha\in\mathbb P^1(K)$ and $\beta\in\mathbb P^1(\overline K)$, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-060280 | MATHDB-317669

Stable tameness conjecture for polynomial automorphisms

摘录 / EXCERPT: Let k be a field and let $F\in \operatorname*{GA}_n(k)$ be a polynomial automorphism of affine n-space. The automorphism F is stably tame if, after adjoining finitely many ne...

conjecture | open | MathDB 页面 | 原文线索

V2-060281 | MATHDB-317674

Dominance conjecture for derived-category components of Fano threefolds

摘录 / EXCERPT: Dominance conjecture. There is a correspondence $Z_d\subset\mathcal{M}\!\mathcal{F}^2_d\times\mathcal{M}\!\mathcal{F}^1_{4d+2}$ which is dominant over each factor and such that for...

conjecture | open | MathDB 页面 | 原文线索

V2-060282 | MATHDB-317675

Generalized correspondence conjecture for moduli of Fano threefold bundles

摘录 / EXCERPT: For $1\leq d\leq 5$, let $\mathcal{M}\!\mathcal{F}\mathcal{I}^2_d(k)$ be the moduli space of pairs (Y,E), where Y is a Fano threefold of index 2 and degree d, and E is an...

conjecture | open | MathDB 页面 | 原文线索

V2-060283 | MATHDB-317680

The equivalence of avoiding and enumerating generating functions for binary tree patterns

摘录 / EXCERPT: Avoiding-enumerating equivalence conjecture. If s and t are avoiding-equivalent, then they are also enumerating-equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-060284 | MATHDB-317681

Replacement-bijection conjecture for equivalent binary tree patterns

摘录 / EXCERPT: Replacement-bijection conjecture. If s and t are equivalent, then there is a sequence of top-down replacements, bottom-up replacements, and left--right reflections that produce...

conjecture | open | MathDB 页面 | 原文线索

V2-060285 | MATHDB-317682

The spectral-bound conjecture for large skew-symmetric perturbations of the harmonic oscillator

摘录 / EXCERPT: Spectral-bound conjecture. For sufficiently small $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-060286 | MATHDB-317686

Mutation-equivalence conjecture for canonical TP-bases of positroid cells

摘录 / EXCERPT: Canonical TP-basis conjecture. For any positroid cell, the canonical TP-basis can be transformed by Plücker mutations into a TP-basis consisting of variables coming from unique pat...

conjecture | open | MathDB 页面 | 原文线索

V2-060287 | MATHDB-317687

Uniqueness conjecture for TP-bases of positroid cells

摘录 / EXCERPT: TP-basis mutation-equivalence conjecture. For any positroid cell, all TP-bases are mutation equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-060288 | MATHDB-317688

Universal subtraction-free Laurent expansion conjecture for TP-bases

摘录 / EXCERPT: Universal TP-basis conjecture. For any cell and any TP-basis, every other Plücker variable in the cell can be expressed as a subtraction-free Laurent polynomial in the variables of...

conjecture | open | MathDB 页面 | 原文线索

V2-060289 | MATHDB-317694

Bijectivity and compactness conjecture for primitive polynomial straightening maps

摘录 / EXCERPT: Let $f_0\in\mathcal{C}(d)$ be a primitive polynomial, meaning that for all distinct and bounded Fatou components U,V of f_0, ... Assume that f_0 is hyperbolic and post-critical...

conjecture | open | MathDB 页面 | 原文线索

V2-060290 | MATHDB-317695

Campana's Kobayashi pseudo-distance conjecture for the core

摘录 / EXCERPT: Campana's core-distance conjecture. The Kobayashi pseudo-distance is the pullback of the orbifold Kobayashi pseudo-distance on the core:

conjecture | open | MathDB 页面 | 原文线索

V2-060291 | MATHDB-317702

Belbachir et al.'s ray-unimodality conjecture for Pascal's triangle

摘录 / EXCERPT: Let $\binom{n}{k}$ be a fixed element of Pascal's triangle, and let a ray cross a sequence of entries in the triangle. Ray-unimodality conjecture. The sequence of binomial coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-060292 | MATHDB-317703

Monotonicity of the modes of shifted non-decreasing polynomials

摘录 / EXCERPT: Mode-monotonicity conjecture. If $0<d_1<d_2$, then

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V2-060293 | MATHDB-317710

Koszulness and freeness conjectures for compatible structures

摘录 / EXCERPT: Koszulness and freeness conjecture. If $\mathrsfs{O}$ is a Koszul operad, then:

conjecture | open | MathDB 页面 | 原文线索

V2-060294 | MATHDB-317712

The volume-form preservation conjecture for automorphisms of $(\mathbb{C}^*)^2$

摘录 / EXCERPT: Volume-form preservation conjecture. Every automorphism F of $\mathbb{C}^*\times\mathbb{C}^*$ preserves this form:

conjecture | open | MathDB 页面 | 原文线索

V2-060295 | MATHDB-317715

The low-value Kazhdan-Lusztig polynomial conjecture for permutations

摘录 / EXCERPT: Low-value Kazhdan-Lusztig conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060296 | MATHDB-317726

Bloch-Beilinson pull-back conjecture for K3 surfaces

摘录 / EXCERPT: Let X be a smooth projective K3 surface over a number field $K\subset\mathbb{C}$, or over $\bar{\mathbb{Q}}$, and let $X_{\mathbb{C}}$ be its base change to $\mathbb{C}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060297 | MATHDB-317728

Binder-DeMarco's dimension conjecture for equilibrium measures of endomorphisms of projective space

摘录 / EXCERPT: Let f be a holomorphic endomorphism of $\mathbb{C}\mathbb{P}^k$ of degree $d\geq 2$, and let μ be its equilibrium measure. Write $\lambda_1\geq\cdots\geq\lambda_k$ for the Ly...

conjecture | open | MathDB 页面 | 原文线索

V2-060298 | MATHDB-317732

Localization formula conjecture for generating functions

摘录 / EXCERPT: Localization-form conjecture. If part (a) of the generating-function conjecture holds, then

conjecture | open | MathDB 页面 | 原文线索

V2-060299 | MATHDB-317737

Lyapunov-exponent formulation of the Kotani-Last conjecture

摘录 / EXCERPT: Lyapunov-exponent formulation of the Kotani-Last conjecture. If the potentials V_ω are not almost periodic, then

conjecture | open | MathDB 页面 | 原文线索

V2-060300 | MATHDB-317738

The universal spectrum conjecture for lattice tilings

摘录 / EXCERPT: Universal spectrum conjecture. Under these assumptions, Γ has a universal spectrum of the form

conjecture | open | MathDB 页面 | 原文线索

V2-060301 | MATHDB-317744

Goncharov's Grassmannian polylogarithm period conjecture

摘录 / EXCERPT: Goncharov's period conjecture. The maximal period of the $\mathbb{R}$-mixed Hodge structure attached to $\mathcal{G}(v_0,\dots,v_{2r-1})$ is the right-hand side of (600).

conjecture | open | MathDB 页面 | 原文线索

V2-060302 | MATHDB-317757

The hyperlogarithm-to-polylogarithm reduction conjecture

摘录 / EXCERPT: Let ${\cal H}_n(E)$ and ${\cal P}_n(E)$ be the hyperlogarithmic and polylogarithmic spaces, respectively, with coproduct δ_n, and let $\mathbb{R}H_n$ and $\mathbb{R}P_n$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-060303 | MATHDB-317758

The weight-four hyperlogarithm reduction conjecture

摘录 / EXCERPT: Let E be a field, let ${\cal H}_4(E)$ be the weight-four hyperlogarithmic space, and let $\delta_{2,2}:{\cal H}_4(E)\to\Lambda^2{\cal H}_2(E)$ be the component of δ_4 in...

conjecture | open | MathDB 页面 | 原文线索

V2-060304 | MATHDB-317759

Goncharov's weight-four element conjecture

摘录 / EXCERPT: Goncharov's conjecture. The element $B(x,y;z)$ belongs to ${\cal P}_4(E)$; equivalently, it is a linear combination of 4-logarithms. The construction gives an element killed by...

conjecture | open | MathDB 页面 | 原文线索

V2-060305 | MATHDB-317794

Gromov-Zimmer's quasi-affine rigidity conjecture

摘录 / EXCERPT: Let D be a semisimple Lie group all of whose factors have property (T), or a lattice in such a Lie group. A rigid geometric structure is a geometric structure of the rigid type...

conjecture | open | MathDB 页面 | 原文线索

V2-060306 | MATHDB-317796

The weakly hyperbolic toral-action conjecture

摘录 / EXCERPT: Let Γ be as in the uniformly partially hyperbolic quasi-affine conjecture, and let $\mathbb T^n$ be the n-torus. A toral action is weakly hyperbolic when it has the weak h...

conjecture | open | MathDB 页面 | 原文线索

V2-060307 | MATHDB-317797

The arithmetic-quotient fundamental-group conjecture for lattice actions

摘录 / EXCERPT: Let G be a semisimple group whose simple factors have real rank at least 2, let Γ be a lattice in G, and let Γ act faithfully and preserving volume on a compact...

conjecture | open | MathDB 页面 | 原文线索

V2-060308 | MATHDB-317800

Cutoff conjecture for the supermarket model's jump chain

摘录 / EXCERPT: Supermarket jump-chain cutoff conjecture. The sequence $d_n(\epsilon,t)$ has a cutoff in the sense of Definition, with window size n. The conjecture concerns mixing from good ini...

conjecture | open | MathDB 页面 | 原文线索

V2-060309 | MATHDB-317813

Conjecture on positive-rank fibres of the associated elliptic surface

摘录 / EXCERPT: Let $c,f\in\mathbb{Z}$ with $c\neq f$, and consider the elliptic surface ... For $t\in\mathbb{Q}$, write $\mathcal{E}_t$ for the fibre at t. Positive-rank fibre conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-060310 | MATHDB-317815

The exclusion-voter passage-time moment conjecture

摘录 / EXCERPT: The passage-time moment conjecture. Suppose that $p\leq1/2$ and $\beta\in[0,1]$. For any $\varepsilon>0$ and $S_0\in\mathcal{D}\setminus\{\mathcal{D}_0\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-060311 | MATHDB-317816

The recurrence conjecture for the positive-voter mixture

摘录 / EXCERPT: Let ξ be the exclusion-voter mixture process with parameters β,p and state space $\mathcal{D}$. A process is recurrent from S_0 when...

conjecture | open | MathDB 页面 | 原文线索

V2-060312 | MATHDB-317818

The equality of the sharp and recursive Potts constants

摘录 / EXCERPT: Equality conjecture. We believe that

conjecture | open | MathDB 页面 | 原文线索

V2-060313 | MATHDB-317840

Full-move connectivity conjecture for positive-marginal fibers in multiple logistic regression

摘录 / EXCERPT: Let m covariates index combinations of levels, and let $\Bj=(j_1,\dots,j_m)$ denote such a combination. Let $\bm{e}_{\Bj}$ be the array with 1 at cell $(1,\Bj)$ and -1 at cel...

conjecture | open | MathDB 页面 | 原文线索

V2-060314 | MATHDB-317841

Restricted-move connectivity conjecture for positive-marginal fibers in multiple logistic regression

摘录 / EXCERPT: Let m covariates index combinations of levels, and let ${\cal B}_{\Lambda(A_1 \otimes \dots \otimes A_m)}$ be the set of moves ... Consider the subset for which every element of...

conjecture | open | MathDB 页面 | 原文线索

V2-060315 | MATHDB-317849

Conjecture on isomorphism types of commutative A-loops of order p^3

摘录 / EXCERPT: Isomorphism-type conjecture. The loops

conjecture | open | MathDB 页面 | 原文线索

V2-060316 | MATHDB-317871

The bucketing-code exponent inequality for the statistical nearest-neighbor problem

摘录 / EXCERPT: Bucketing-code exponent inequality. The following inequality should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-060317 | MATHDB-317881

Regularity conjecture for the left-Garside category of self-distributive terms

摘录 / EXCERPT: Let $\mathcal{LD}^{\scriptscriptstyle+}$ be the left-Garside category of self-distributive terms. Regularity conjecture. The category $\mathcal{LD}^{\scriptscriptstyle+}$ is regula...

conjecture | open | MathDB 页面 | 原文线索

V2-060318 | MATHDB-317883

The stabilized inscribed-square conjecture for immersed plane curves

摘录 / EXCERPT: Let T be an immersed curve in the plane, and let St, J^+, and J^- denote its associated invariants. An inscribed square of a curve is a square whose vertices lie on the cur...

conjecture | open | MathDB 页面 | 原文线索

V2-060319 | MATHDB-317892

Torsion-square divisibility conjecture for quadratic twists

摘录 / EXCERPT: Let E be an optimal elliptic curve over $\mathbf{Q}$ of conductor N, and let $-D$ be a negative fundamental discriminant such that $(E,-D)$ satisfies hypothesis $(**)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060320 | MATHDB-317897

Integrality conjecture for coarse-moduli BPS functions

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold over $\mathbb C$, and let $(\tau,G,\leqslant)$ be a generic Gieseker stability condition. For each $\alpha\in C(\operatorname{coh}(X))$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060321 | MATHDB-317927

The converse of the WLP criterion for level monomial ideals

摘录 / EXCERPT: Let K be a field of characteristic zero, let $R=K[x,y,z]$, and let $I_{alpha,beta,gamma,t}$ denote the level monomial almost complete intersection used in the preceding criterion...

conjecture | open | MathDB 页面 | 原文线索

V2-060322 | MATHDB-317928

Characterization of the explicit path formula for Kostka-Macdonald polynomials

摘录 / EXCERPT: Path-formula conjecture. The displayed explicit formula for the Kostka-Macdonald polynomial is valid if and only if at least one of the following conditions holds: (i)...

conjecture | open | MathDB 页面 | 原文线索

V2-060323 | MATHDB-317929

Generalized tau generating-function conjecture for rectangular crystal paths

摘录 / EXCERPT: Let $R=\{(r_i,s_i)_{i=1}^L\}$ be a sequence of rectangles, and let $\mathcal{P}_R$ be the set of all paths, including non-highest elements, in the tensor product ... For a path p,...

conjecture | open | MathDB 页面 | 原文线索

V2-060324 | MATHDB-317930

Parabolic Kostka product duality conjecture

摘录 / EXCERPT: Parabolic Kostka duality conjecture. There is a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-060325 | MATHDB-317931

Rectangular-path energy generating-function conjecture

摘录 / EXCERPT: Let $R=\{(r_i,s_i)_{i=1}^L\}$ be a sequence of rectangles, and let $\mathcal{P}_R$ be the set of paths in the tensor product ... For a path p, let $\operatorname{wt}(p)$ be its wei...

conjecture | open | MathDB 页面 | 原文线索

V2-060326 | MATHDB-317934

The Asai local Langlands compatibility conjecture

摘录 / EXCERPT: Let $Fsubset K$ be a quadratic extension of local fields, let $G_n(K)=GL(n,K)$, and let W'_K and W'_F be the Weil-Deligne groups. For a finite-dimensional representation ρ...

conjecture | open | MathDB 页面 | 原文线索

V2-060327 | MATHDB-317940

Ring-generation conjecture for shifted plactic skew Schur P-functions

摘录 / EXCERPT: Ring-generation conjecture. $\mathcal{P}_{\lambda/\mu}$ belongs to the ring generated by the shifted plactic Schur P-functions.

conjecture | open | MathDB 页面 | 原文线索

V2-060328 | MATHDB-317944

Monotonicity and vanishing-minimum-energy conjecture for discrete cavity flow

摘录 / EXCERPT: Minimum-energy conjecture. For every $R\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-060329 | MATHDB-317948

Uniform expansion conjecture for the Julia set of the post-critically finite endomorphism

摘录 / EXCERPT: Let f be the post-critically finite endomorphism of $\mathbb{P}\mathbb{C}^2$ considered above, with Julia set J. An orbispace metric is a metric on the orbifold-like space obta...

conjecture | open | MathDB 页面 | 原文线索

V2-060330 | MATHDB-317957

The equality characterization conjecture for ribbon Schur Q-functions

摘录 / EXCERPT: Equality characterization conjecture. For ribbons $alpha,beta$, one has $rho_alpha=rho_beta$ if and only if there exist j,k,ell such that

conjecture | open | MathDB 页面 | 原文线索

V2-060331 | MATHDB-317959

Minimal transitive-subset conjecture for permutation encoding of input-symmetric degraded broadcast channels

摘录 / EXCERPT: Let k be the alphabet size of the channel input, and let l_s be the alphabet size of the code for User 2 in a permutation encoding approach. For an input-symmetric degraded bro...

conjecture | open | MathDB 页面 | 原文线索

V2-060332 | MATHDB-317960

The classification conjecture for good braid-group representations

摘录 / EXCERPT: Classification conjecture. Every good non-cyclic transitive homomorphism $B_k\to S_n$ is canonical when $n=k$, standard when $k\ne n$ and $k\mid n$, or derived from a canonical or...

conjecture | open | MathDB 页面 | 原文线索

V2-060333 | MATHDB-317968

The conjecture that higher-rank Clifford indices equal the classical Clifford index

摘录 / EXCERPT: Let C be a smooth projective curve, and for each integer $n\geq3$ let γ_n' denote the rank-n Clifford index defined using semistable vector bundles, while γ_1 is...

conjecture | open | MathDB 页面 | 原文线索

V2-060334 | MATHDB-317971

Homogeneous SHGH conjecture for plane linear systems

摘录 / EXCERPT: Homogeneous SHGH conjecture. If $k\geq10$, a system $\mathcal{L}(d;m^k)$ is never special.

conjecture | open | MathDB 页面 | 原文线索

V2-060335 | MATHDB-317977

Conjecture on the affine full-dimensionality of D-optimal factorial designs

摘录 / EXCERPT: Let r denote the number of runs, and distinguish affinely full-dimensional factorial designs from subset fractional factorial designs. The D-optimal design conjecture. The D-op...

conjecture | open | MathDB 页面 | 原文线索

V2-060336 | MATHDB-317979

Large-genus asymptotics of mixed Weil-Petersson intersection numbers

摘录 / EXCERPT: Let $V_{g,n;d}$ denote the mixed intersection number associated with a fixed set $d=(d_1,\dots,d_n)$ of non-negative integers, and let $V_{g,n}$ denote the corresponding Weil-Peter...

conjecture | open | MathDB 页面 | 原文线索

V2-060337 | MATHDB-317981

Censoring symmetry conjecture for Glauber dynamics on transitive graphs

摘录 / EXCERPT: Let $\{G_n\}$ be a sequence of transitive graphs, and consider the Glauber dynamics for the Ising model on these graphs. Let β_c denote the critical temperature, let δ...

conjecture | open | MathDB 页面 | 原文线索

V2-060338 | MATHDB-317983

Schur-log-concavity conjecture for Eulerian quasisymmetric functions

摘录 / EXCERPT: Let $Q_{n,j}$ and $Q_{n,j,k}$ denote the symmetric functions defined in the paper. A sequence is log-concave if each interior term satisfies the corresponding quadratic inequality....

conjecture | solved | MathDB 页面 | 原文线索

V2-060339 | MATHDB-318000

The homogeneity conjecture for supergravity backgrounds

摘录 / EXCERPT: Homogeneity conjecture. All supergravity backgrounds with

conjecture | open | MathDB 页面 | 原文线索

V2-060340 | MATHDB-318001

The surface \ce 3-system list-colouring conjecture

摘录 / EXCERPT: Let S be a fixed surface. For a graph G embeddable on S with a Σ-system, define ... and let $\Delta(G;\Sigma)=\max_{v}|\Sigma(v)|$. The surface Σ-system list-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-060341 | MATHDB-318014

Finite-interval quasi-classical decomposition conjecture for fermionic sums

摘录 / EXCERPT: Let C be the matrix defining the fermionic sums, let m be a multiplicity vector, let $r\leq s$, and let $W_{C,m}$ denote the quadratic weight appearing in the fermionic sum. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-060342 | MATHDB-318018

Conjecture on root-counting measures for generalized Heun equations

摘录 / EXCERPT: Consider a generalized Heun equation ... where $\deg Q_{k+1}(z)=k+1$ and $\deg Q_i(z)\leq i$ for $i=2,3,\ldots,k$. For each sufficiently large positive integer n, let μ_n be th...

conjecture | open | MathDB 页面 | 原文线索

V2-060343 | MATHDB-318030

Cauchy conjecture for the partition metric of sparse random graphs

摘录 / EXCERPT: Let $ obreak\kappa$ be a bounded kernel, and let $ obreakd_\mathrm{part}$ denote the partition metric. Partition-metric Cauchy conjecture. The random sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-060344 | MATHDB-318031

Partition-metric separation conjecture for non-equivalent sparse random graph models

摘录 / EXCERPT: Let $p=1/n$, and let $ obreak\kappa_1$ and $ obreak\kappa_2$ be bounded kernels such that the models $G_{1/n}(n, obreak\kappa_1)$ and $G_{1/n}(n, obreak\kappa_2)$ are not essential...

conjecture | open | MathDB 页面 | 原文线索

V2-060345 | MATHDB-318032

Square-root threshold conjecture for two-type sparse random graphs

摘录 / EXCERPT: Let $c>1$ and $-c\leq\delta\leq c$ be constants, and let $ obreak\kappa_{c+ obreak\delta,c- obreak\delta}$ be the two-type kernel defined in the source. Square-root threshold conje...

conjecture | open | MathDB 页面 | 原文线索

V2-060346 | MATHDB-318033

Fibonacci recurrence-coefficient conjecture for 2-distant and 3-distant noncrossing partitions

摘录 / EXCERPT: Fibonacci recurrence-coefficient conjecture. If $\mu_n=NCP_2(n)$, then $b_0=b_1=\lambda_1=1$, and for $n\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-060347 | MATHDB-318039

Conjectured degree sequence for systems of parameters of graph invariant rings

摘录 / EXCERPT: Degree-sequence conjecture. For every $n\ge3$, there exists a system of parameters for $\mathcal{I}_n$ with degrees

conjecture | open | MathDB 页面 | 原文线索

V2-060348 | MATHDB-318040

Conjectured explicit system of parameters for graph invariant rings

摘录 / EXCERPT: Explicit parameter-system conjecture. If $n\ge3$, the following invariants form a system of parameters for $\mathcal{I}_n$:

conjecture | open | MathDB 页面 | 原文线索

V2-060349 | MATHDB-318041

Conjecture on the maximal degree of minimal generators of graph invariant rings

摘录 / EXCERPT: Maximal-generator-degree conjecture. For every $n\ge4$,

conjecture | open | MathDB 页面 | 原文线索

V2-060350 | MATHDB-318042

Unimodality conjecture for secondary invariants of permutation groups

摘录 / EXCERPT: Secondary-invariant unimodality conjecture. The generating series of the secondary invariants is unimodal, except possibly at $d=0$ and $d=e_t$.

conjecture | open | MathDB 页面 | 原文线索

V2-060351 | MATHDB-318050

McNamara-Sagan-Stanley real-rootedness conjecture for the log-concavity transform

摘录 / EXCERPT: Let $\{a_k\}$ be a finite sequence of positive real numbers, and define ... Let $\mathfrak{L}(\{a_k\})=\{a_k^2-a_{k-1}a_{k+1}\}$. McNamara-Sagan-Stanley real-rootedness conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-060352 | MATHDB-318051

Nested-stratification conjecture for de Rham and Higgs moduli

摘录 / EXCERPT: Nested-stratification conjecture. The stratifications of $M_{\rm DR}(X,r)$ and $M_{\rm H}(X,r)$ are nested, and their strata have the same arrangement under the stated identificati...

conjecture | open | MathDB 页面 | 原文线索

V2-060353 | MATHDB-318052

The interpretation conjecture for the action of profinite words on subsets

摘录 / EXCERPT: Let M be the finite monoid under consideration, let X be its generating alphabet, let R be the relevant set on which $ Omega(X)$ acts, and let $P(M)$ denote the power set of...

conjecture | open | MathDB 页面 | 原文线索

V2-060354 | MATHDB-318054

The graded hat-harmonics conjecture

摘录 / EXCERPT: Let $\widehat{D}_k=\sum_{i=1}^n\partial_i^{k+1}x_i$ for $k\geq1$, and define the space of hat-harmonics by ... The symmetric group $\mathfrak S_n$ acts by permuting variables, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-060355 | MATHDB-318055

The dimension conjecture for weighted diagonal harmonics

摘录 / EXCERPT: Let $\mathcal D_n^{\mathbf a}$ be the common polynomial zero space in $\mathbb C[\mathbf{x},\mathbf{y}]$ of the operators ... for all $k,j\in\mathbb N$ with $k+j>0$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-060356 | MATHDB-318068

Uniqueness conjecture for the minimizer of the ground-state energy

摘录 / EXCERPT: Let $\lambda_0(\alpha)$ be the ground state energy of the operator defined in the paper, and let $\hat{\nu}$ be its minimum value. A minimizer is a point...

conjecture | open | MathDB 页面 | 原文线索

V2-060357 | MATHDB-318072

Linearity of Cappell-Shaneson L-class transformations

摘录 / EXCERPT: Let $\Omega(-)$ denote the domain of Cappell--Shaneson's L-class transformations, and consider a natural transformation ... without the normalization condition. Let ${L_*^{CS}}_i$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-060358 | MATHDB-318077

Tucker's perfect-graph sequential colorability conjecture

摘录 / EXCERPT: Tucker's conjecture. For perfect graphs, unique t-colorability is equivalent to sequential t-colorability.

conjecture | open | MathDB 页面 | 原文线索

V2-060359 | MATHDB-318098

Su and Wang's real-rootedness conjecture for binomial coefficient sequences

摘录 / EXCERPT: Su and Wang's real-rootedness conjecture. The polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-060360 | MATHDB-318099

Su and Wang's log-concavity-to-log-convexity conjecture for binomial coefficient sequences

摘录 / EXCERPT: Let n, k, a, and b be integers such that $n\geq k\geq 0$ and $a>b>0$. The sequence ... is defined by these binomial coefficients. Su and Wang's log-concavity-to-log-convexity...

conjecture | open | MathDB 页面 | 原文线索

V2-060361 | MATHDB-318104

Global minimality conjecture for positive minimal-growth solutions of p-Laplacian-type equations

摘录 / EXCERPT: Let $1<p<\infty$, let Q_V be the functional under consideration on a domain Ω, and let $\Omega_1\Subset\Omega$ be a smooth open set. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-060362 | MATHDB-318109

The finite-dimensionality conjecture for type spaces of finitely forcible graphons

摘录 / EXCERPT: Let $T(W)$ denote the type space associated with a graphon W. Finite-dimensionality conjecture. If W is finitely forcible, then $T(W)$ is finite dimensional. The source intenti...

conjecture | open | MathDB 页面 | 原文线索

V2-060363 | MATHDB-318112

Baker-Rumely lower bound conjecture for the tau constant

摘录 / EXCERPT: Baker-Rumely's lower bound conjecture. There is a universal constant $C>0$ such that, for every metrized graph Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-060364 | MATHDB-318113

Strict 1/108 lower bound conjecture for the tau constant

摘录 / EXCERPT: Strict $1/108$ lower bound conjecture. For every metrized graph Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-060365 | MATHDB-318114

Nair-Wang information inequality for the binary skew-symmetric broadcast channel

摘录 / EXCERPT: We consider the binary skew-symmetric broadcast channel with input X and outputs Y_1,Y_2. Let U,V be random variables such that ... forms a Markov chain. Nair-Wang information...

conjecture | open | MathDB 页面 | 原文线索

V2-060366 | MATHDB-318117

Lam's alternating-sign conjecture for K-theoretic k-Schur functions

摘录 / EXCERPT: Lam's conjecture. The following hold: each $g^{(k)}_\lambda$ is a finite sum of k-Schur functions with positive integer coefficients;...

conjecture | open | MathDB 页面 | 原文线索

V2-060367 | MATHDB-318118

Lam's alternating-sign conjecture for affine stable Grothendieck polynomials

摘录 / EXCERPT: Lam's conjecture. Every G_w is a finite alternating linear combination of the $G^{(k)}_\lambda$; every $G^{(k)}_\lambda$ is an alternating integer linear combination of the...

conjecture | open | MathDB 页面 | 原文线索

V2-060368 | MATHDB-318119

The determinantal-component dimension conjecture for generalized diagonal coinvariants

摘录 / EXCERPT: Let W be a well-generated complex reflection group, let $D\!R^{(m)}(W)$ be the space of generalized diagonal coinvariants, and let $\mathbf{e}_{\epsilon}$ denote the idempotent p...

conjecture | open | MathDB 页面 | 原文线索

V2-060369 | MATHDB-318120

The reciprocal-specialization formula for q,t-Fuss-Catalan numbers

摘录 / EXCERPT: Let W be a well-generated complex reflection group of rank ℓ, acting on $\mathbb{C}[V\oplus V^*]$. Let d_i and d_i^* be the invariant and codegrees appearing in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-060370 | MATHDB-318121

The positive-region coheight formula for q,t-Fuss-Catalan numbers

摘录 / EXCERPT: Let W be a crystallographic reflection group, let $\mathsf{Shi}^{(m)}(W)$ be the extended Shi arrangement, and let $\operatorname{coh}(R)$ be the coheight of a positive region...

conjecture | open | MathDB 页面 | 原文线索

V2-060371 | MATHDB-318123

Bonnafé's semicontinuity conjecture for Kazhdan-Lusztig cells

摘录 / EXCERPT: Let W be a Coxeter group with generating set S, and identify weight functions with integer points in a Euclidean space V whose coordinates are their values on the conjugacy c...

conjecture | open | MathDB 页面 | 原文线索

V2-060372 | MATHDB-318125

The cusp connection conjecture for limit PK arrangements

摘录 / EXCERPT: Let (L_j,β_j) be a weighted arrangement in $\mathbb{CP}^2$ satisfying all conditions of Theorem existence apart from one inequality, and suppose it has a multiple point x,...

conjecture | open | MathDB 页面 | 原文线索

V2-060373 | MATHDB-318126

The asphericity conjecture for complements of PK arrangements

摘录 / EXCERPT: Let an arrangement be a collection of weighted lines in $\mathbb{CP}^2$ satisfying the conditions of Theorem existence. Its complement is the space obtained by removing the arrange...

conjecture | open | MathDB 页面 | 原文线索

V2-060374 | MATHDB-318155

Komlós-Sós conjecture on degree-sequence conditions for tree embeddings

摘录 / EXCERPT: Let G be a graph on n vertices, and let k be a positive integer. A tree with k edges has $k+1$ vertices. Komlós-Sós conjecture. If at least $n/2$ vertices of G have degre...

conjecture | open | MathDB 页面 | 原文线索

V2-060375 | MATHDB-318157

The enriched-category model conjecture for cartesian model categories

摘录 / EXCERPT: Let C be a small category, let $ ... and suppose that M is a cartesian model category. For an object X of $M\operatorname{-Cat}$, let $hX...

conjecture | open | MathDB 页面 | 原文线索

V2-060376 | MATHDB-318158

The isolated-vertices conjecture for logconcave random graphs

摘录 / EXCERPT: Let F be a distribution in the positive orthant with a down-monotone logconcave density. For each coordinate e, write ${\sf E}(X_e^2)=\sigma_e^2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060377 | MATHDB-318176

A mixed Bruhat chain conjecture for Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $v\le_k w$ be permutations, let $u(\lambda,k)$ be a Grassmannian permutation, and let Δ be a mixed Bruhat chain from the identity to v. A mixed Bruhat chain is a conca...

conjecture | open | MathDB 页面 | 原文线索

V2-060378 | MATHDB-318178

Akita's integral Riemann-Roch conjecture for surface bundles

摘录 / EXCERPT: Let $\pi\colon E\to B$ be an oriented surface bundle with closed fibers of genus g, equipped with a fiberwise metric. Let $\mathcal{H}(E)$ be the associated Hodge bundle, with fi...

conjecture | open | MathDB 页面 | 原文线索

V2-060379 | MATHDB-318180

Bers' slice non-local-connectivity conjecture

摘录 / EXCERPT: Let S be a surface, and let a Bers' slice for S be the corresponding quasifuchsian deformation slice. Bers' slice conjecture. If S is not a punctured torus or a 4-times pun...

conjecture | open | MathDB 页面 | 原文线索

V2-060380 | MATHDB-318182

El Soufi's index conjecture for minimal hypersurfaces of spheres

摘录 / EXCERPT: El Soufi's index conjecture. Either

conjecture | open | MathDB 页面 | 原文线索

V2-060381 | MATHDB-318189

The cut-distance comparison conjecture for graphs

摘录 / EXCERPT: Let G and G' be graphs on n nodes. The quantities $\hat\delta_\square(G,G')$ and $\delta_\square(G,G')$ are the cut distances defined in the paper. Cut-distance comparison co...

conjecture | open | MathDB 页面 | 原文线索

V2-060382 | MATHDB-318191

The Littlewood-Richardson rule for Λ-cominuscule elements

摘录 / EXCERPT: Let G be a Kac-Moody group, let P be a parabolic subgroup, and let $X=G/P$. Let W be the Weyl group, and let Λ be the dominant weight associated with P. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060383 | MATHDB-318193

Reflectionless measures are critical measures

摘录 / EXCERPT: Reflectionless-measure conjecture. For any positive reflectionless measure μ with finite energy, there exists a polar set $\mathcal A$ on $\mathbb{C}$ such that μ is...

conjecture | open | MathDB 页面 | 原文线索

V2-060384 | MATHDB-318198

The Laplace-transform formulation of the Image Conjecture

摘录 / EXCERPT: Let ${\mathcal M}$ be the subspace of polynomials $f(\xi,z)\in{\mathbb C}[\xi,z]$ such that ${\mathcal Z}(f)$ has no holomorphic part, equivalently such that the Laplace transforma...

conjecture | open | MathDB 页面 | 原文线索

V2-060385 | MATHDB-318199

The generalized image conjecture for orthogonal-polynomial differential operators

摘录 / EXCERPT: The generalized image conjecture. The subspace $\operatorname{Im}'\Lambda$ is a Mathieu subspace of $\mathbb C[z]$. The paper notes that this generalizes the image conjecture and p...

conjecture | open | MathDB 页面 | 原文线索

V2-060386 | MATHDB-318201

Tay-Whiteley Molecular Conjecture for body-and-hinge frameworks

摘录 / EXCERPT: Molecular Conjecture. G can be realized as an infinitesimally rigid body-and-hinge framework in $\mathbb{R}^d$ if and only if G can be realized as an infinitesimally rigid pane...

conjecture | open | MathDB 页面 | 原文线索

V2-060387 | MATHDB-318203

Ginzburg-Jiang-Soudry lowest occurrence conjecture

摘录 / EXCERPT: Ginzburg-Jiang-Soudry lowest occurrence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060388 | MATHDB-318204

Holomorphy conjecture for normalized local zeta integrals

摘录 / EXCERPT: Holomorphy conjecture for normalized local zeta integrals. The function $Z_v^*(s,f_{1,v},f_{2,v},Phi_v,pi_v)$ is holomorphic for all $s\in\mathbb{C}$ and for every choice of...

conjecture | open | MathDB 页面 | 原文线索

V2-060389 | MATHDB-318205

Weak holomorphy conjecture for normalized local zeta integrals

摘录 / EXCERPT: Weak holomorphy conjecture for normalized local zeta integrals. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-060390 | MATHDB-318206

Strong second term identity conjecture

摘录 / EXCERPT: Strong second term identity conjecture. The weak second term identity and the weak first term identity on the boundary can be extended to every

conjecture | open | MathDB 页面 | 原文线索

V2-060391 | MATHDB-318213

The Wronskian power criterion for bases of top powers

摘录 / EXCERPT: Let $U\in\mathcal{G}_k\mathcal{P}_n$ admit a basis of top powers, and let $W(U)$ denote its Wronskian covariant. The notation $U^{(\ell-1)}$ denotes the associated derived subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-060392 | MATHDB-318219

Ideal clique clutters of perfect graphs have the MFMC property

摘录 / EXCERPT: Let G be a perfect graph, and let $\mathcal{C}$ be its clique clutter, whose edges are the maximal cliques of G. The clutter $\mathcal{C}$ is ideal when its covering polyhedron...

conjecture | open | MathDB 页面 | 原文线索

V2-060393 | MATHDB-318220

Varley's conjecture on Thom-Boardman symbols of polynomial multiplication maps

摘录 / EXCERPT: Varley's conjecture. For every $n\geq r$,

conjecture | open | MathDB 页面 | 原文线索

V2-060394 | MATHDB-318226

The large-volume-limit conjecture for mirror hypersurfaces

摘录 / EXCERPT: Let $(X^\vee,W)$ be mirror to X, let D be an anticanonical divisor in X, and assume that $-K_X$ is nef. Large-volume-limit conjecture. As $t\to\infty$, the family of hypersur...

conjecture | open | MathDB 页面 | 原文线索

V2-060395 | MATHDB-318258

Conjecture that the higher Newton polygon family is a p-integral basis

摘录 / EXCERPT: Higher Newton polygon basis conjecture. The following family is a p-integral basis of K:

conjecture | open | MathDB 页面 | 原文线索

V2-060396 | MATHDB-318265

Divonne's higher-order jet-differential ampleness conjecture

摘录 / EXCERPT: Higher-order jet-differential ampleness conjecture. The bundle map

conjecture | open | MathDB 页面 | 原文线索

V2-060397 | MATHDB-318269

The 132-avoiding principal-interval Möbius conjecture

摘录 / EXCERPT: Principal-interval Möbius conjecture. If τ avoids the pattern 132 (equivalently, 312, 213, or 231), then

conjecture | open | MathDB 页面 | 原文线索

V2-060398 | MATHDB-318273

Stoll's uniform polynomial height-bound conjecture for genus-2 curves

摘录 / EXCERPT: Let ${\mathcal C}_N$ be the family of genus-2 curves defined by integral binary sextics with coefficients bounded in absolute value by N. For a rational point P on such a cur...

conjecture | open | MathDB 页面 | 原文线索

V2-060399 | MATHDB-318276

Hain's conjecture on NMHS structures on relative Malcev completions

摘录 / EXCERPT: Hain's conjecture. The relative completion $\mathscr G$ should carry a nonabelian mixed Hodge structure in general. This structure should be a quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-060400 | MATHDB-318292

Garvan's conjecture on crank and rank moments

摘录 / EXCERPT: For an integer k, let $M_k(n)$ and $N_k(n)$ denote the k-th crank and rank moments, respectively. Thus, for even indices, $M_{2k}(n)$ and $N_{2k}(n)$ are the corresponding even...

conjecture | open | MathDB 页面 | 原文线索

V2-060401 | MATHDB-318293

Bringmann-Mahlburg refined conjecture on crank and rank moments

摘录 / EXCERPT: For an integer $k\geq 1$, let $M_{2k}(n)$ and $N_{2k}(n)$ be the $2k$-th crank and rank moments, respectively, let $p(n)$ be the partition function, and define ... The constants sati...

conjecture | open | MathDB 页面 | 原文线索

V2-060402 | MATHDB-318299

Density hypothesis for multiplicities in principal congruence quotients

摘录 / EXCERPT: Let ${\sf G}(\mathbb{Z})$ be cocompact, let $\Gamma(N)$ be the principal congruence subgroup modulo N, and let π be an irreducible non-trivial representation. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-060403 | MATHDB-318306

Concavity conjecture for the logarithm of a matrix permanent

摘录 / EXCERPT: Concavity conjecture. The function $f(\boldsymbol\lambda)$ is concave on $\mathcal{D}^{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060404 | MATHDB-318309

Helton-McCullough conjecture on convex positivity domains

摘录 / EXCERPT: Helton-McCullough conjecture. If p is irreducible and $\mathcal D_p$ is convex, then $\mathcal D_p$ is a ball.

conjecture | open | MathDB 页面 | 原文线索

V2-060405 | MATHDB-318320

The anticyclotomic Iwasawa nonvanishing conjecture for definite Hida families

摘录 / EXCERPT: Let $f_\mathfrak p^\dagger$ be a non-exceptional arithmetic specialization of weight $k_\mathfrak p\ge2$, let χ be a finite-order character of G_∞, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060406 | MATHDB-318321

The indefinite Heegner-class non-torsion conjecture

摘录 / EXCERPT: Let K be the imaginary quadratic field, let $\mathcal R$ be the Hida-family coefficient ring, and let $\mathfrak Z_0\in\widetilde H^1_f(K,\mathbf T^\dagger)$ be the corestriction...

conjecture | open | MathDB 页面 | 原文线索

V2-060407 | MATHDB-318322

Strictification of lax morphisms between bipermutative categories

摘录 / EXCERPT: Strictification conjecture. There is a functor on the bipermutative category level that replaces lax morphisms by strict morphisms, in a sense analogous to the corresponding result...

conjecture | open | MathDB 页面 | 原文线索

V2-060408 | MATHDB-318323

Interval separation conjecture for the Schur cone

摘录 / EXCERPT: Let $\mathcal{SP}_N$ be the collection of multisets of partitions with total size N, let $\phi(A)=\lambda$ be the associated partition, and let λ^{++} be defined by ... A s...

conjecture | open | MathDB 页面 | 原文线索

V2-060409 | MATHDB-318324

Refined interval separation conjecture for even partitions

摘录 / EXCERPT: Let $A\in\mathcal{SP}_N$ with associated partition $\phi(A)=\lambda$. For a partition $\mu=(\mu_1\geq\cdots\geq\mu_m>0)$, define ... Say that μ is $\downarrow$-invertible when...

conjecture | open | MathDB 页面 | 原文线索

V2-060410 | MATHDB-318325

The rational orthogonal realization conjecture for rational zero-patterns

摘录 / EXCERPT: Let X be an $m\times n$ matrix over a field. Its zero-pattern (or support) is the $(0,1)$-matrix $\underline{X}$ defined by ... Let $O(n)$ denote the real orthogonal $n\times n$ ma...

conjecture | open | MathDB 页面 | 原文线索

V2-060411 | MATHDB-318326

The symmetric rational orthogonal zero-pattern conjecture

摘录 / EXCERPT: The symmetric rational orthogonal zero-pattern conjecture. For any hermitian $X\in U(n)$ there exists a symmetric $Z\in O_n(\mathbb{Q})$ such that $\underline{X}=\underline{Z}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-060412 | MATHDB-318330

Uniform measure equivalence invariance conjecture for L^2-torsion

摘录 / EXCERPT: Let G and H be countable groups such that all the L^2-Betti numbers of G and H vanish. Assume that both G and H admit finite CW-models for their classifying spaces. T...

conjecture | open | MathDB 页面 | 原文线索

V2-060413 | MATHDB-318340

Generalized Zariski component conjecture for branch-curve strata

摘录 / EXCERPT: For $ ue 3$, define the numerical invariants ... ... Let $V(d,c,n)$ be the space of irreducible plane curves of degree d with c cusps and n nodes as their only singularities, a...

conjecture | open | MathDB 页面 | 原文线索

V2-060414 | MATHDB-318345

Tardos-code convergence and orthogonal-projection conjecture for worst collusion attacks

摘录 / EXCERPT: Tardos worst-attack convergence conjecture. As the collusion size increases, the worst collusion attack makes $\Pr[Y=1\mid P=p]$ converge to $q^{\mathrm{conv}}(p)$. More specifical...

conjecture | open | MathDB 页面 | 原文线索

V2-060415 | MATHDB-318350

The mortar-and-brick conjecture for K_2-QPDSs in the triangular lattice

摘录 / EXCERPT: Let Δ be the triangular lattice. A K_2-QPDS S in Δ has an associated hexagonal structure ${\mathcal P}(S)$ whose hexagons have types 1, 2, and 3. Mortar-and...

conjecture | open | MathDB 页面 | 原文线索

V2-060416 | MATHDB-318352

Generalized Greenberg conjecture for the cyclotomic tower

摘录 / EXCERPT: Let K be a number field, and let L'_{∞} and T_{∞} be the fields appearing in the source's Iwasawa-theoretic construction. Generalized Greenberg conjecture. For an a...

conjecture | open | MathDB 页面 | 原文线索

V2-060417 | MATHDB-318357

Congruent number conjecture for angles pi/3 and 2pi/3

摘录 / EXCERPT: Congruent number conjecture for angles $\pi/3$ and $2\pi/3$. If

conjecture | open | MathDB 页面 | 原文线索

V2-060418 | MATHDB-318359

The nilpotent-by-abelian FP_m conjecture

摘录 / EXCERPT: Let G be a finitely generated nilpotent-by-abelian group. Let N be a nilpotent normal subgroup with $Q=G/N$ abelian, and consider the factors of the lower central series of N...

conjecture | open | MathDB 页面 | 原文线索

V2-060419 | MATHDB-318362

Ellipticity implies hypoellipticity for operators on p.c.f. fractals

摘录 / EXCERPT: Let X be a p.c.f. fractal, let P be a polynomial, and let $P(Delta)$ denote the associated operator on the product space X^m. The operator $P(Delta)$ is elliptic when...

conjecture | open | MathDB 页面 | 原文线索

V2-060420 | MATHDB-318363

Hörmander-type sufficient condition for hypoellipticity on p.c.f. fractals

摘录 / EXCERPT: Let P be a polynomial in m variables and let $P(\Delta)$ be the corresponding operator on a product of p.c.f. fractals. For a multi-index κ, write $D^{\kappa}P$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-060421 | MATHDB-318364

Quasiellipticity implies hypoellipticity on p.c.f. fractals

摘录 / EXCERPT: Let $P(\Delta)$ be an operator on a product of p.c.f. fractals, and let P_0 be the principal part of P. The operator is quasielliptic if there is a constant $c>0$ such that ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-060422 | MATHDB-318375

The less-than-one-bit space-saving conjecture for the invertible transformation

摘录 / EXCERPT: Space-saving conjecture. The space saving using this transformation is on average less than one bit, even when applied to high-entropy input.

conjecture | open | MathDB 页面 | 原文线索

V2-060423 | MATHDB-318381

Alexander's perimeter conjecture for intersections of congruent disks

摘录 / EXCERPT: Alexander's conjecture. Under arbitrary contraction of the center points of finitely many congruent disks in the Euclidean plane, the perimeter of the intersection of the disks can...

conjecture | open | MathDB 页面 | 原文线索

V2-060424 | MATHDB-318397

Conjecture on disjoint duplexes in Latin squares

摘录 / EXCERPT: Duplex partition conjecture. Every Latin square has the maximum possible number of disjoint duplexes. In particular, every Latin square of even order has a 2-partition, and every...

conjecture | open | MathDB 页面 | 原文线索

V2-060425 | MATHDB-318398

Conjecture on Latin squares without transversals but with a 3-plex

摘录 / EXCERPT: 3-plex existence conjecture. For every even $n>4$, there exists a Latin square of order n that has no transversal but does contain a 3-plex.

conjecture | open | MathDB 页面 | 原文线索

V2-060426 | MATHDB-318401

Douglas's asymptotic Euler-characteristic conjecture for Kronecker moduli spaces

摘录 / EXCERPT: Douglas's conjecture. There exists a continuous function $f:[m_1,m_2]\subset\mathbb{R}\to\mathbb{R}$ such that, for every $r\in[m_1,m_2]$ and every $\varepsilon>0$, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-060427 | MATHDB-318407

The tangledness-reduction conjecture for braid diagrams

摘录 / EXCERPT: Tangledness-reduction conjecture. The product $\beta\beta_+$ has smaller τ-length than β.

conjecture | open | MathDB 页面 | 原文线索

V2-060428 | MATHDB-318429

Reversibility conjecture for whole-plane SLE

摘录 / EXCERPT: Whole-plane SLE reversibility conjecture. The whole-plane SLE$_\kappa$ trace satisfies reversibility for $\kappa\in(0,4]$.

conjecture | open | MathDB 页面 | 原文线索

V2-060429 | MATHDB-318432

Ideal monotonicity conjecture for the polygamma Lehmer-mean difference

摘录 / EXCERPT: Ideal monotonicity conjecture. The function $f_{i,p,s,t}$ is increasing with respect to x if and only if $p\le -(i+1)$, and decreasing with respect to x if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-060430 | MATHDB-318433

Labastida-Mariño-Ooguri-Vafa integrality conjecture for colored HOMFLY invariants of knots

摘录 / EXCERPT: Let ${\cal K}$ be a knot, let R be a representation, and let $\hat f_R(t,\nu)$ be the reformulated colored HOMFLY invariant obtained by applying the inverse matrix M^{-1} to th...

conjecture | open | MathDB 页面 | 原文线索

V2-060431 | MATHDB-318434

Labastida-Mariño-Ooguri-Vafa integrality conjecture for colored HOMFLY invariants of links

摘录 / EXCERPT: Let ${\cal L}$ be an unoriented link with L components, colored by representations $R_1,\ldots,R_L$. Let $\hat f_{R_1,\ldots,R_L}(t,\nu)$ denote the reformulated colored HOMFLY i...

conjecture | open | MathDB 页面 | 原文线索

V2-060432 | MATHDB-318435

Integrality conjecture for reformulated colored Kauffman invariants of knots

摘录 / EXCERPT: Let ${\cal K}$ be a knot, let R be a representation, and let $\hat h_R(t,\nu)$ and $\hat g_R(t,\nu)$ be the reformulated colored Kauffman invariants obtained by applying the inve...

conjecture | open | MathDB 页面 | 原文线索

V2-060433 | MATHDB-318436

Integrality conjecture for reformulated colored Kauffman invariants of links

摘录 / EXCERPT: Let ${\cal L}$ be an unoriented link with L components, colored by representations $R_1,\ldots,R_L$. Let $\hat h_{R_1,\ldots,R_L}(t,\nu)$ and $\hat g_{R_1,\ldots,R_L}(t,\nu)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-060434 | MATHDB-318438

Shepp-Olkin conjecture on concavity of Bernoulli-sum entropy

摘录 / EXCERPT: Let X_i and Y_i be independent Bernoulli random variables with parameters a_i and b_i, respectively, for $i=1,\ldots,n$. Write $H(a_1,\ldots,a_n)$ for the entropy of...

conjecture | open | MathDB 页面 | 原文线索

V2-060435 | MATHDB-318440

The spectral gap conjecture for multi-species ASEP

摘录 / EXCERPT: Let $m=(m_1,\,\ldots,m_n)\in\mathcal M$ be a sector, let ${\rm Spec}(m)$ denote the spectrum in that sector, and let $E_j^{\pm}(m)=E^{\pm}(m_1+\cdots+m_j, m_{j+1}+\cdots+m_n)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-060436 | MATHDB-318441

The eigen-polynomial conjecture for the multi-species ASEP transfer matrix

摘录 / EXCERPT: Let $T(\lambda)$ be the transfer matrix acting on a sector $V(m)$, let $d=\dim V(m)$, and suppose that $p\neq q$ are generic. Eigen-polynomial conjecture. For any sector $V(m)$, in...

conjecture | open | MathDB 页面 | 原文线索

V2-060437 | MATHDB-318442

Completeness conjecture for the nested Bethe ansatz of the multi-species ASEP

摘录 / EXCERPT: Suppose that $p\neq q$ are generic. Let $T(\lambda)$ be the transfer matrix, let $\Lambda_1(\lambda),\ldots,\Lambda_d(\lambda)$ be its eigen-polynomials in a sector with...

conjecture | open | MathDB 页面 | 原文线索

V2-060438 | MATHDB-318443

Fenn-Kauffman-Manturov bracket-polynomial conjecture for virtual knots

摘录 / EXCERPT: Fenn-Kauffman-Manturov conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060439 | MATHDB-318447

The admissible-order picture-crystal bijection conjecture

摘录 / EXCERPT: Let λ,μ,ν be Young diagrams. Let A be an admissible order on $\nu\setminus\lambda$ and let A' be an admissible order on μ. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-060440 | MATHDB-318465

Asymptotic growth-rate conjecture for the HUM control operator

摘录 / EXCERPT: Let $M_{T,\omega}$ be the frequency-dependent HUM control operator, let $\|\cdot\|_H$ denote its norm, and let T and U be as in the preceding discussion. For a generic choice o...

conjecture | open | MathDB 页面 | 原文线索

V2-060441 | MATHDB-318466

Conjecture on the distribution of quadratic twists by 2-Selmer rank

摘录 / EXCERPT: Let K be a number field and E an elliptic curve over K. For $X\in\mathbf{R}^+$, let $N_r(E,X)$ count quadratic extensions F/K with $d_2(E^F/K)=r$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-060442 | MATHDB-318467

Shafarevich-Tate 2-parity conjecture for elliptic curves

摘录 / EXCERPT: Let K be a number field and E an elliptic curve over K, and let $\text{\cyrr Sh}(E/K)$ denote its Shafarevich-Tate group. Shafarevich-Tate 2-parity conjecture. For every elli...

conjecture | open | MathDB 页面 | 原文线索

V2-060443 | MATHDB-318497

The differential equivariant K-theory pushforward conjecture

摘录 / EXCERPT: Differential equivariant pushforward conjecture. The equality holds as class functions modulo $R(\Gamma)$. This conjectures that the index-theoretic description of the ordinary equ...

conjecture | open | MathDB 页面 | 原文线索

V2-060444 | MATHDB-318506

Symmetry conjecture for spectra of logarithmic Brieskorn lattices of reductive linear free divisors

摘录 / EXCERPT: Let $D\subset V=\mathbb{C}^n$ be a reductive linear free divisor, with defining equation h. Its logarithmic Brieskorn lattice is $(G_0(h),\nabla)$, and its spectrum at $\theta=0$...

conjecture | open | MathDB 页面 | 原文线索

V2-060445 | MATHDB-318510

Asymptotic optimality of the interleaving attack for fingerprinting capacity

摘录 / EXCERPT: Asymptotic optimality conjecture. When $k \rightarrow \infty$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-060446 | MATHDB-318533

Chen et al.'s unit-circle conjecture for Potts partition zeros

摘录 / EXCERPT: Let G be a finite planar self-dual lattice, or let G be the square lattice with free or periodic boundary conditions in the thermodynamic limit. For $q\geq 1$, introduce v by...

conjecture | refuted | MathDB 页面 | 原文线索

V2-060447 | MATHDB-318534

Generalized Graham conjecture for sets of pebble distributions

摘录 / EXCERPT: Let G and H be graphs, and let $\mathcal{S}_G$ and $\mathcal{S}_H$ be sets of pebble distributions on G and H. For distributions D_g on G and D_h on H, define their...

conjecture | open | MathDB 页面 | 原文线索

V2-060448 | MATHDB-318535

Parameterized pebbling conjecture for target vertices

摘录 / EXCERPT: Let G and H be graphs, let s,t be positive integers, and let $x\in V(G)$ and $y\in V(H)$. Write $\pi_r(G,x)$ for the r-pebbling number targeting x, and similarly for H...

conjecture | open | MathDB 页面 | 原文线索

V2-060449 | MATHDB-318544

The width bound for orientable surface knots

摘录 / EXCERPT: Let $F \subset \mathbf{R}^4$ be an orientable surface knot. Here $w(F)$ denotes the width of F, and $\operatorname{Braid}(F)$ denotes its braid index. Width bound conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-060450 | MATHDB-318554

The linear t-pebbling bound conjecture

摘录 / EXCERPT: Linear t-pebbling bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060451 | MATHDB-318555

The diameter-three t-pebbling conjecture

摘录 / EXCERPT: Diameter-three t-pebbling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060452 | MATHDB-318561

Kawanaka's Macdonald polynomial conjecture

摘录 / EXCERPT: Kawanaka's conjecture. The following formal identity holds:

conjecture | open | MathDB 页面 | 原文线索

V2-060453 | MATHDB-318568

The Birch-Swinnerton-Dyer formula via a Heegner point

摘录 / EXCERPT: Let E be an elliptic curve of analytic rank zero or one, let K be a quadratic imaginary field satisfying the splitting and discriminant hypotheses in the construction, let N...

conjecture | open | MathDB 页面 | 原文线索

V2-060454 | MATHDB-318583

Uniqueness conjecture for Schottky projective structures

摘录 / EXCERPT: Fix a marking of the projective structure Ω/Γ on S with Schottky holonomy ρ. Let $\mathcal{P}_\rho$ be the collection of projective structures on S with holono...

conjecture | open | MathDB 页面 | 原文线索

V2-060455 | MATHDB-318588

The principal ideal conjecture for factorizations of totally positive loop-group elements

摘录 / EXCERPT: Let Ω be the space of totally positive loop-group elements. For $X\in\Omega$, let $I(X)$ be the set of equivalence classes $[\mathbf i]$ of infinite reduced words...

conjecture | open | MathDB 页面 | 原文线索

V2-060456 | MATHDB-318589

Injectivity conjecture for minimal infinite reduced words

摘录 / EXCERPT: Let $\mathbf i$ be an infinite reduced word, and let $e_{\mathbf i}$ be its associated factorization map. Say that $\mathbf i$ is minimal in its block when it is minimal in the rel...

conjecture | open | MathDB 页面 | 原文线索

V2-060457 | MATHDB-318590

Conjecture that factorization-set inclusions characterize limit weak order

摘录 / EXCERPT: Let $E_{[\mathbf i]}$ denote the subset associated with the equivalence class $[\mathbf i]$ of an infinite reduced word, and let $\leq$ be the limit weak order. Inclusion character...

conjecture | open | MathDB 页面 | 原文线索

V2-060458 | MATHDB-318591

Join conjecture for factorizations in limit weak order

摘录 / EXCERPT: Let Ω be the space of totally positive loop-group elements, and let $E_{\mathbf i}$ be the subset associated with an infinite reduced word $\mathbf i$. Join conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-060459 | MATHDB-318592

Principal ideal conjecture for greedy factorizations

摘录 / EXCERPT: Let Ω be the space of totally positive loop-group elements. For $X\in\Omega$, let $J(X)$ be the set of equivalence classes $[\mathbf i]$ of infinite reduced words...

conjecture | open | MathDB 页面 | 原文线索

V2-060460 | MATHDB-318597

A single covariance sign change in random orientations of $G(n,m)$

摘录 / EXCERPT: Single-sign-change conjecture. For any fixed $n\ge 5$, the covariance in $G(n,m)$ changes sign only once between two values of m.

conjecture | open | MathDB 页面 | 原文线索

V2-060461 | MATHDB-318602

FJRW-Saito-Givental mirror symmetry conjecture

摘录 / EXCERPT: Let W be a non-degenerate invertible potential, let G^{max} be its maximal group of diagonal symmetries, and let W^T denote the transpose potential. The FJRW-Saito-Giv...

conjecture | open | MathDB 页面 | 原文线索

V2-060462 | MATHDB-318610

Large-coupling equally spaced level conjecture for spin chains

摘录 / EXCERPT: Large-coupling level-spacing conjecture. For $\overline\beta\gg N$, the eigenvalues cluster around an equally spaced set of levels of the form $j\,\overline\beta$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-060463 | MATHDB-318611

Gaussian level-density conjecture for spin chains

摘录 / EXCERPT: Gaussian level-density conjecture. For $N\gg1$, the level density follows a Gaussian distribution.

conjecture | open | MathDB 页面 | 原文线索

V2-060464 | MATHDB-318612

Polynomial dependence conjecture for spin-chain energy moments

摘录 / EXCERPT: Polynomial energy-moments conjecture. The average energy $\mu_{\pm}$ and its squared standard deviation $\sigma_{\pm}^2$ depend polynomially on N.

conjecture | open | MathDB 页面 | 原文线索

V2-060465 | MATHDB-318622

Kramer-Tunnell conjecture for elliptic curves

摘录 / EXCERPT: Kramer-Tunnell conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060466 | MATHDB-318623

The local p-parity formula for elliptic curves with a p-isogeny

摘录 / EXCERPT: Local p-parity formula. If $p\ne2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-060467 | MATHDB-318627

The general Clausen identity for a bivariate hypergeometric function

摘录 / EXCERPT: Let a, b, and c be general complex parameters, and let ${}_{2}\operatorname{F}_{1}$ and $F^{2:1;1}_{1:1;1}$ denote the hypergeometric functions appearing below. Generalized C...

conjecture | open | MathDB 页面 | 原文线索

V2-060468 | MATHDB-318631

T-system and fusion-function correspondence conjecture for supersymmetric spin chains

摘录 / EXCERPT: T-system and fusion-function correspondence conjecture. For the non-skew Young diagram μ, the relations

conjecture | open | MathDB 页面 | 原文线索

V2-060469 | MATHDB-318632

Conjecture on block-size ratios for existence of the limiting spectral distribution

摘录 / EXCERPT: Block-size ratio conjecture. If the limiting spectral distribution (LSD) of $\mathbf{A}_n$ exists, then

conjecture | open | MathDB 页面 | 原文线索

V2-060470 | MATHDB-318637

Projection-volume conjecture for finite reflection groups

摘录 / EXCERPT: Let $\mathcal{W}$ be a finite reflection group acting on $\mathbb{R}^n$, with fundamental chamber $\mathcal{C}$ and characteristic polynomial $\chi(t)$ of its associated reflection...

conjecture | open | MathDB 页面 | 原文线索

V2-060471 | MATHDB-318642

Eli's descent-set conjecture for zero-marked cyclic permutations

摘录 / EXCERPT: Eli's descent-set conjecture. For any n and any $I\subseteq[n-1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-060472 | MATHDB-318644

Minimal-energy blowup solution conjecture for wave maps

摘录 / EXCERPT: Let $\dot{\mathcal H}^1$ be the wave-map energy space, and let an energy-class solution be a restriction of a maximal Cauchy development to a subinterval. Such a solution...

conjecture | open | MathDB 页面 | 原文线索

V2-060473 | MATHDB-318646

Comparison principle without strict extrema for the infinity Laplace equation

摘录 / EXCERPT: Let Ω be the domain under consideration. Assume that $u,-v\in\operatorname{USC}(\bar\Omega)$ and $f\in C(\Omega)$ satisfy ... Suppose that either u has no strict local maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-060474 | MATHDB-318653

The joint value-distribution conjecture for phi and sigma fibers

摘录 / EXCERPT: Joint fiber conjecture. For every $k\geqslant 1$ and $l\geqslant 2$, there are integers n such that

conjecture | open | MathDB 页面 | 原文线索

V2-060475 | MATHDB-318658

The degenerate stability conjecture for discrete faithful free-group representations

摘录 / EXCERPT: Let F_n be the free group of rank n, and let $\rho:F_n\to {\operatorname{PSL}_2(\mathbb C)}$ be a discrete faithful representation without parabolics. A representation is primi...

conjecture | open | MathDB 页面 | 原文线索

V2-060476 | MATHDB-318659

The ending-lamination criterion for primitive stability

摘录 / EXCERPT: Let F_n be the free group of rank n, and let $\rho:F_n\to {\operatorname{PSL}_2(\mathbb C)}$ be a discrete faithful representation. Its ending lamination may be disconnected, w...

conjecture | open | MathDB 页面 | 原文线索

V2-060477 | MATHDB-318675

Mukai's conjecture on semi-homogeneous presentations of stable sheaves

摘录 / EXCERPT: Let X be an abelian surface with $\operatorname{NS}(X)=\mathbb{Z} H$ and let v be a Mukai vector with ... Suppose that v has at least one solution of the numerical equation. Am...

conjecture | open | MathDB 页面 | 原文线索

V2-060478 | MATHDB-318684

Sturmfels-Sullivant's normality conjecture for cut polytopes

摘录 / EXCERPT: Sturmfels-Sullivant's conjecture. The cut polytope ${\rm Cut}^\square(G)$ is normal if and only if G has no K_5 minor.

conjecture | open | MathDB 页面 | 原文线索

V2-060479 | MATHDB-318685

Normality conjecture for cut polytopes of 4-connected plane triangulations

摘录 / EXCERPT: The 4-connected plane triangulation conjecture. The cut polytope ${\rm Cut}^\square(G)$ is normal if G is a 4-connected plane triangulation.

conjecture | open | MathDB 页面 | 原文线索

V2-060480 | MATHDB-318686

Normality conjecture for cut polytopes of grid graphs

摘录 / EXCERPT: The grid-graph conjecture. The cut polytope ${\rm Cut}^\square(G)$ is normal if G is a grid graph.

conjecture | open | MathDB 页面 | 原文线索

V2-060481 | MATHDB-318687

Eliahou-Kryuchkov signed diagonal flip conjecture for polygon triangulations

摘录 / EXCERPT: Eliahou-Kryuchkov conjecture. Given two triangulations of the same polygon, there exist signs for them such that it is possible to transform one to the other by a finite sequence o...

conjecture | open | MathDB 页面 | 原文线索

V2-060482 | MATHDB-318688

Eliahou-Kryuchkov lift conjecture for signed associahedra

摘录 / EXCERPT: Eliahou-Kryuchkov conjecture. For any pair of vertices on $\mathcal{A}_{n-2}$ there exists a path connecting them that can be lifted to a path on the graph $\mathcal{A}_{n-2}^{s}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060483 | MATHDB-318694

Positivity of the affine Grothendieck expansion of Schur functions

摘录 / EXCERPT: Let k be a positive integer, let $mathcal P^k$ denote the set of k-bounded partitions, and let s_λ and $G_\mu^{(k)}$ denote the Schur function and affine Grothendieck p...

conjecture | open | MathDB 页面 | 原文线索

V2-060484 | MATHDB-318695

Alternating-sign positivity for k-K-Schur expansions in $(k+1)$-K-Schur functions

摘录 / EXCERPT: Alternating-sign K-Schur transition conjecture. For all $\lambda,\mu\in\mathcal P^k$, the integer coefficient $(-1)^{|\lambda|+|\mu|}b_{\lambda\mu}^{k}$ is non-negative. The clai...

conjecture | open | MathDB 页面 | 原文线索

V2-060485 | MATHDB-318696

GW/NCDT correspondence conjecture

摘录 / EXCERPT: Let (Q,W) be a quiver with superpotential function associated to a crepant resolution $\pi:Y\to X$, where X is an affine Calabi--Yau 3-fold with an isolated singularity. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060486 | MATHDB-318701

Lusztig's extension-by-zero conjecture for the semisimple character sheaf

摘录 / EXCERPT: Let G^1 be the relevant component, let $\overline{G^1}^{ss}$ be its semistable locus, and let S' be the simple perverse sheaf on $\overline{G^1}^{ss}$ obtained from the semisim...

conjecture | open | MathDB 页面 | 原文线索

V2-060487 | MATHDB-318705

Mixed Tate motive conjecture for residues of Feynman integrals

摘录 / EXCERPT: A residue of a Feynman integral is obtained by removing the divergent Gamma factor; the remaining integral may still require renormalization. A mixed Tate motive is a motive built...

conjecture | open | MathDB 页面 | 原文线索

V2-060488 | MATHDB-318707

Squier's kernel conjecture for Burau representations at roots of unity

摘录 / EXCERPT: Let B_n be the braid group on n strands with standard generators $g_1,dots,g_{n-1}$, and let $B_n[k]$ be the normal subgroup generated by g_j^k for $1\leq j\leq n-1$. For a p...

conjecture | open | MathDB 页面 | 原文线索

V2-060489 | MATHDB-318712

Closed formula conjecture for the invariant Hilbert-Poincaré series of the standard representation

摘录 / EXCERPT: Closed formula conjecture. The ordinary Hilbert-Poincaré series is

conjecture | open | MathDB 页面 | 原文线索

V2-060490 | MATHDB-318718

Wang's finiteness conjecture for modular categories of fixed rank

摘录 / EXCERPT: Wang's finiteness conjecture. There are finitely many modular categories of rank n for any fixed $n\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060491 | MATHDB-318721

Modified Sziklai conjecture for plane curves over finite fields

摘录 / EXCERPT: Modified Sziklai conjecture. Unless C is defined over ${\mathbb F}_4$ and is projectively equivalent to this exceptional curve, one should have

conjecture | open | MathDB 页面 | 原文线索

V2-060492 | MATHDB-318728

Robertson's Hadwiger-type conjecture for degree sequences

摘录 / EXCERPT: Let D be a graphic degree sequence, and let $\mathcal{R}(D)$ be the set of its simple graph realizations. Define $\chi(D)$ as the maximum chromatic number over graphs in...

conjecture | open | MathDB 页面 | 原文线索

V2-060493 | MATHDB-318729

Higher-order constraint conjecture for elastic shell models

摘录 / EXCERPT: Let S be a midsurface, let $E^h(u^h)$ denote the elastic energy of a shell of thickness h, and suppose ... For $i\geq 1$, write ... when $i>1$, and let $\mathcal{V}_N$ denote the s...

conjecture | open | MathDB 页面 | 原文线索

V2-060494 | MATHDB-318732

Felsner-Reuter conjecture for the linear extension diameter of Boolean lattices

摘录 / EXCERPT: Felsner-Reuter conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060495 | MATHDB-318737

The supercritical NLS standing-ring blowup conjecture

摘录 / EXCERPT: Let psi be a singular standing-ring solution of the NLS equation with $d>1$ and $sigma>2$. Let $L(t)$ be its ring-region similarity scale, $r_{max}(t)$ its ring radius, and...

conjecture | open | MathDB 页面 | 原文线索

V2-060496 | MATHDB-318738

The supercritical NLS standing-vortex blowup conjecture

摘录 / EXCERPT: Let $psi(t,r,theta)=A(t,r)e^{imtheta}$ be a singular standing-vortex solution of the two-dimensional NLS with $sigma>2$, and let psi_F denote the radial self-similar profile. Sup...

conjecture | open | MathDB 页面 | 原文线索

V2-060497 | MATHDB-318739

The supercritical BNLS standing-ring blowup conjecture

摘录 / EXCERPT: Let psi be a singular standing-ring solution of the BNLS with $d>1$ and $sigma>4$. Let $L(t)$ be the ring-region similarity scale, $r_{max}(t)$ its radius, and psi_B the ring p...

conjecture | open | MathDB 页面 | 原文线索

V2-060498 | MATHDB-318740

The biharmonic nonlinear heat equation standing-ring conjecture

摘录 / EXCERPT: Consider the radially symmetric biharmonic nonlinear heat equation ... and let $L(t)$ and $r_{\max}(t)$ denote the ring similarity scale and radius. Let $v_B(t,x)$ be a singular peak...

conjecture | open | MathDB 页面 | 原文线索

V2-060499 | MATHDB-318750

The linear-logarithmic conjecture for torsion on elliptic curves over number fields

摘录 / EXCERPT: For $d\in\mathbb{Z}^+$, let $T(d)$ be the supremum of the orders of the groups $E(K)[\operatorname{tors}]$ as K ranges over number fields of degree d and E ranges over ellipt...

conjecture | open | MathDB 页面 | 原文线索

V2-060500 | MATHDB-318753

Zudilin's integrality conjecture for generalized canonical coordinates

摘录 / EXCERPT: Zudilin's conjecture. For any positive integers $N_1,N_2,\dots,N_k$,

conjecture | open | MathDB 页面 | 原文线索

V2-060501 | MATHDB-318756

The bottom theorem for separable cofinite subfields of random extensions

摘录 / EXCERPT: Bottom theorem. For almost all $\boldsymbol{\sigma}\in\operatorname{Gal}(K)^e$ and every proper subfield $N\subsetneqq K_s(\boldsymbol{\sigma})$, if the extension...

conjecture | open | MathDB 页面 | 原文线索

V2-060502 | MATHDB-318757

Song's q=t Schröder identity conjecture

摘录 / EXCERPT: Song's conjecture. For the $q=t$ specialization,

conjecture | open | MathDB 页面 | 原文线索

V2-060503 | MATHDB-318766

Uniform eigenvector conditioning conjecture for the force-based quasicontinuum operator

摘录 / EXCERPT: Let V be the matrix whose columns are eigenvectors of the force-based quasicontinuum operator $\mathcal{P}_{\mathcal{U}}L_{{\rm qcf},F}$, and let A_F denote the stability param...

conjecture | open | MathDB 页面 | 原文线索

V2-060504 | MATHDB-318767

Uniform Fourier-Mukai kernel bound conjecture equivalent to Kawamata's finiteness conjecture

摘录 / EXCERPT: Uniform Fourier-Mukai kernel bound conjecture. The line bundles $\mathcal A_i$ and equivalences F_i can be chosen so that there exists a finitely supported function...

conjecture | open | MathDB 页面 | 原文线索

V2-060505 | MATHDB-318774

The Mathieu-subspace conjecture for generalized Laguerre polynomials

摘录 / EXCERPT: Let ${\bf k}\in\mathbb N^n$, and let $L^{[\bf k]}_\alpha(z)$ denote the generalized Laguerre polynomials for multi-indices $\alpha\in\mathbb N^n$. Let $\mathcal M$ be the subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-060506 | MATHDB-318781

Nagata's numerical inequality for effective classes

摘录 / EXCERPT: Nagata's numerical inequality. If

conjecture | open | MathDB 页面 | 原文线索

V2-060507 | MATHDB-318785

The retractability and multipermutation conjecture for finite square-free solutions

摘录 / EXCERPT: Retractability and multipermutation conjecture. Every finite square-free solution (X,r) is retractable. Moreover, every finite square-free solution (X,r) of finite order n is...

conjecture | open | MathDB 页面 | 原文线索

V2-060508 | MATHDB-318786

The logarithmic multipermutation-level conjecture for square-free solutions

摘录 / EXCERPT: Logarithmic multipermutation-level conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060509 | MATHDB-318789

Harbourne-Gimigliano-Hirschowitz conjecture for plane linear systems

摘录 / EXCERPT: Harbourne-Gimigliano-Hirschowitz conjecture. The linear system $|D|$ has expected dimension

conjecture | open | MathDB 页面 | 原文线索

V2-060510 | MATHDB-318794

Edoukou's bound for intersections with a degenerate quadric

摘录 / EXCERPT: Let $\mathcal{X}=\Pi_{n-r}\mathcal{Q}_{r-1}$ be a degenerate quadric of rank $r(\mathcal{X})=r$, and let $\mathcal{Q}$ be another degenerate quadric. Here $EH(r-1,q)$ denotes the p...

conjecture | open | MathDB 页面 | 原文线索

V2-060511 | MATHDB-318796

Cobham-type conjecture for independent substitutions

摘录 / EXCERPT: Independent-substitution conjecture. Then x is ultimately periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-060512 | MATHDB-318797

King's tilting bundle conjecture for smooth complete geometric quotients

摘录 / EXCERPT: Let V be a vector space, let G be a reductive group acting linearly on V, and let $U\subseteq V$ be a Zariski-open subset such that the geometric quotient U/G is a smooth c...

conjecture | open | MathDB 页面 | 原文线索

V2-060513 | MATHDB-318806

Betti sequences of pure resolutions are simplicial-complex f-vectors

摘录 / EXCERPT: Let I be a monomial ideal with a pure minimal free resolution, and let its Betti sequence be the sequence of Betti numbers associated with that resolution. Betti-sequence conject...

conjecture | open | MathDB 页面 | 原文线索

V2-060514 | MATHDB-318809

The SRS weak-tiling conjecture

摘录 / EXCERPT: Let $\mathbf{r}=(r_0,\ldots,r_{d-1})\in\operatorname{int}(\mathcal{D}_d)$ with $r_0\ne0$. Weak-tiling conjecture. The collection ... is a weak tiling, meaning a weak 1-tiling: ever...

conjecture | open | MathDB 页面 | 原文线索

V2-060515 | MATHDB-318811

Spacetime bounds for the mass-critical generalized KdV equation

摘录 / EXCERPT: Let u solve the mass-critical generalized KdV equation on ${\mathbb R}\times{\mathbb R}$, let $u_0\in L^2_x({\mathbb R})$ be its initial data, let $M(u_0)$ denote its mass, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-060516 | MATHDB-318812

Spacetime bounds for the mass-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Let v solve the mass-critical nonlinear Schrödinger equation on ${\mathbb R}\times{\mathbb R}$, let $v_0\in L^2_x({\mathbb R})$ be its initial data, let $M(v_0)$ denote its mass,...

conjecture | open | MathDB 页面 | 原文线索

V2-060517 | MATHDB-318816

The skew Littlewood-Richardson rule for products of skew Schur functions

摘录 / EXCERPT: Skew Littlewood-Richardson conjecture. For all skew shapes λ/μ and σ/τ,

conjecture | open | MathDB 页面 | 原文线索

V2-060518 | MATHDB-318821

Conjecture on covariant differentiation for piecewise smooth data

摘录 / EXCERPT: Let U^i be a piecewise smooth vector field and let $\Gamma^{\mu}_{\nu\alpha}\in\Gamma^{3}_{S}(M)$. Piecewise-smooth covariant-derivative conjecture. Covariant differentiation alo...

conjecture | open | MathDB 页面 | 原文线索

V2-060519 | MATHDB-318836

The realizability conjecture for cutsets in polytopes

摘录 / EXCERPT: Let P be a d-polytope with graph G_P, and let a cutset be a set of vertices selected for removal by a hyperplane. Properties 1 and 2 of Proposition are connectedness of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-060520 | MATHDB-318838

The contiguous-path conjecture for translation sequences

摘录 / EXCERPT: Let F_N be the new facet in a translation sequence of a polytope, and let x,y be the vertices defining the paths under consideration. Contiguous-path conjecture. Every edge adj...

conjecture | open | MathDB 页面 | 原文线索

V2-060521 | MATHDB-318839

The facet-connectedness conjecture for cutsets

摘录 / EXCERPT: Let P be a polytope, let F be a facet of P, and let C be a cutset. Facet-connectedness conjecture. The cutset C cannot contain vertices u and v lying in a common face...

conjecture | open | MathDB 页面 | 原文线索

V2-060522 | MATHDB-318840

Kalai's separator conjecture for simple polytopes

摘录 / EXCERPT: Let P be a simple d-polytope with m vertices, and let G_P be its graph. Kalai's separator conjecture. There exists a subset V' of vertices of P such that ... and removing...

conjecture | open | MathDB 页面 | 原文线索

V2-060523 | MATHDB-318867

The classification conjecture for sharp tridiagonal systems

摘录 / EXCERPT: Classification conjecture. There exists a sharp tridiagonal system Φ over $\mathbb{K}$ with this parameter array if and only if the following conditions hold: (i)...

conjecture | open | MathDB 页面 | 原文线索

V2-060524 | MATHDB-318868

Conjecture on the number of bright images produced by isothermal elliptical galaxies

摘录 / EXCERPT: Image-count conjecture. (i) The number of bright images lensed by an isothermal elliptical galaxy without shear is at most 6. (ii) In the presence of a shear, the number of brigh...

conjecture | open | MathDB 页面 | 原文线索

V2-060525 | MATHDB-318869

Local Langlands conjecture for generic supercuspidal representations of $PGSp_6$

摘录 / EXCERPT: Let $PGSp_6$ be the split adjoint group, and let $\operatorname{lrr}_g^\circ(PGSp_6)$ denote the set of isomorphism classes of its generic supercuspidal irreducible representations...

conjecture | open | MathDB 页面 | 原文线索

V2-060526 | MATHDB-318870

Coherence for planar traced categories

摘录 / EXCERPT: Let a planar traced category be a monoidal category equipped with right and left traces satisfying the three additional axioms of interchange, left pivoting, and right pivoting, an...

conjecture | open | MathDB 页面 | 原文线索

V2-060527 | MATHDB-318871

Coherence for spacial traced categories

摘录 / EXCERPT: Let a spacial traced category be a planar traced category satisfying the spacial and spherical axioms, and consider the graphical language in which diagrams are identified up to is...

conjecture | open | MathDB 页面 | 原文线索

V2-060528 | MATHDB-318875

The theta-function vanishing conjecture for tetrahedrally symmetric monopoles

摘录 / EXCERPT: Theta-function vanishing conjecture. For each such pair (m,n), $H(y)$ vanishes $2(|n|-1)$ times on the interval $\lambda\in(0,2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-060529 | MATHDB-318878

Last's absolutely continuous spectrum conjecture for decaying Jacobi matrices

摘录 / EXCERPT: Let J be the discrete Schrödinger operator on the half-lattice with bounded real potential (v_n), represented by the Jacobi matrix with diagonal entries v_n and off-diagonal...

conjecture | open | MathDB 页面 | 原文线索

V2-060530 | MATHDB-318879

Conjectural skew Littlewood-Richardson rule for Schur Q-functions

摘录 / EXCERPT: Conjectural skew Littlewood-Richardson rule. The product of skew Schur Q-functions should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-060531 | MATHDB-318900

Capacity lower-bound conjecture for compact non-pluripolar sets

摘录 / EXCERPT: Capacity lower-bound conjecture. There exists a constant $C_{a,n}>0$ such that, for every compact non-pluripolar set $E\bigsubset A$,

conjecture | open | MathDB 页面 | 原文线索

V2-060532 | MATHDB-318904

Technical conjecture on truncated divisor sums near rational approximants

摘录 / EXCERPT: Technical divisor-sum conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-060533 | MATHDB-318913

The (q,t)-deformed hook product formula for connected d-complete posets

摘录 / EXCERPT: Let P be a connected d-complete poset with maximum element v_0, rank function $r:P\to\mathbb{N}$, top tree T, and d-complete coloring $c:P\to T$. For a P-partition...

conjecture | open | MathDB 页面 | 原文线索

V2-060534 | MATHDB-318915

Bounded-symbol conjecture for truncated Toeplitz operators

摘录 / EXCERPT: Let Θ be an inner function. A bounded-symbol conjecture asserts that any bounded truncated Toeplitz operator has a bounded symbol if and only if Θ belongs to (CLS)....

conjecture | open | MathDB 页面 | 原文线索

V2-060535 | MATHDB-318918

Binary-weight divisibility conjecture for even binomial moments

摘录 / EXCERPT: Let $n,r\geqslant 1$. Define $\alpha(N)$ to be the number of 1's in the binary expansion of N. Binary-weight divisibility conjecture. The sum ... is divisible by ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-060536 | MATHDB-318919

Odd Catalan-triangle power congruence conjecture

摘录 / EXCERPT: Let $B_{n,k}$ denote the Catalan-triangle quantities used in the paper, and let $n,r\geqslant 1$. Odd-power Catalan-triangle congruence conjecture. ... if and only if $n=2^a-2^b$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-060537 | MATHDB-318920

Odd-exponent Catalan-triangle congruence conjecture with a power-of-four threshold

摘录 / EXCERPT: Let $B_{n,k}$ denote the Catalan-triangle quantities used in the paper, let $r\geqslant 0$ and $s\geqslant 1$, and assume $n\geqslant 4^s-1$. Power-of-four threshold conjecture. ......

conjecture | open | MathDB 页面 | 原文线索

V2-060538 | MATHDB-318921

Even-exponent Catalan-triangle congruence conjecture

摘录 / EXCERPT: Let $B_{n,k}$ denote the Catalan-triangle quantities used in the paper, and let $n,r,s\geqslant 1$. Even-exponent Catalan-triangle congruence conjecture. ... This is a complementary...

conjecture | open | MathDB 页面 | 原文线索

V2-060539 | MATHDB-318922

Mixed Catalan-triangle product divisibility conjecture

摘录 / EXCERPT: Let $B_{m,k}$ and $B_{n,k}$ denote the Catalan-triangle quantities used in the paper. Let $m,n,s,t\geqslant 1$ and $r\geqslant 0$, with $r+s+t\equiv 1\pmod 2$. Mixed-product divisi...

conjecture | open | MathDB 页面 | 原文线索

V2-060540 | MATHDB-318923

Triple Catalan-triangle product divisibility conjecture for rows n, $2n$, and $3n$

摘录 / EXCERPT: Let $B_{n,k}$ denote the Catalan-triangle quantities used in the paper. For positive integers r,s,t satisfying $r+s+t\equiv 1\pmod 2$, triple-product divisibility conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-060541 | MATHDB-318925

Product structure conjecture for the generalized Somos-4 recurrence tree

摘录 / EXCERPT: Let T be the recurrence tree generated by the recurrence referred to as (Som4Ord3), and let the generalized Somos-4 sequence be the sequence associated with that recurrence. Prod...

conjecture | open | MathDB 页面 | 原文线索

V2-060542 | MATHDB-318931

Monotonicity conjecture for optimal Variant II CPC compositions

摘录 / EXCERPT: Monotonicity conjecture. If $J>1$ and $E[\eta_\ell]$ is convex in ℓ, then the optimal multiplicities n_i for Variant II CPCs increase monotonically with i.

conjecture | open | MathDB 页面 | 原文线索

V2-060543 | MATHDB-318932

Monotonicity conjecture for optimal Variant I CPC compositions

摘录 / EXCERPT: Variant I monotonicity conjecture. If $J>1$, then the optimal n_i for Variant I CPCs increase monotonically with $i\in\mathcal{S}_1$ and decrease monotonically with...

conjecture | open | MathDB 页面 | 原文线索

V2-060544 | MATHDB-318934

The presentation conjecture for invariant rank-two parabolic arrangements

摘录 / EXCERPT: Let W be a finite real reflection group, and let $\mathscr{P}$ be a collection of rank 2 parabolic subgroups of W closed under conjugation. Set ... Define a Coxeter group W'...

conjecture | open | MathDB 页面 | 原文线索

V2-060545 | MATHDB-318966

Bj"orner-Swartz inequalities and M-sequence conjecture

摘录 / EXCERPT: Let Δ be a $(d-1)$-dimensional simplicial complex that is doubly Cohen-Macaulay over k. Write ${\mathfrak h}_i(\Delta)$ for its ${\mathfrak h}$-vector and...

conjecture | open | MathDB 页面 | 原文线索

V2-060546 | MATHDB-318970

Non-equivalence of strong and weak Morrey quasiconvexity

摘录 / EXCERPT: Non-equivalence conjecture. If $m,n>1$ and f is lower semicontinuous, then strong Morrey quasiconvexity and weak Morrey quasiconvexity are not equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-060547 | MATHDB-318975

Eulerian descent-set distribution conjecture for marked cycles

摘录 / EXCERPT: Eulerian descent-set distribution conjecture. For any n and any subset $D\subseteq\{1,2,\dots,n-1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-060548 | MATHDB-318976

Divisibility by six conjecture for realized permutations

摘录 / EXCERPT: Divisibility by six conjecture. For every $n,N\geq 3$, $a_{n,N}$ is divisible by 6.

conjecture | open | MathDB 页面 | 原文线索

V2-060549 | MATHDB-318979

The étale index equals the index over iterated Laurent series fields

摘录 / EXCERPT: Étale-index conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060550 | MATHDB-318995

The connectivity conjecture for globally rigid k-chains

摘录 / EXCERPT: Connectivity conjecture. Any $(d+1)$-connected k-chain in $\mathbb{R}^d$ with more than $\binom{d+2}{2}$ vertices is generically globally rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-060551 | MATHDB-318996

The blow-up conjecture for globally rigid graph constructions

摘录 / EXCERPT: Blow-up conjecture. For any connected graph G with $v > 1$, there exists some $a_1, a_2, \ldots, a_v$ and some d such that if we replace each v_i with an independent set of s...

conjecture | open | MathDB 页面 | 原文线索

V2-060552 | MATHDB-319005

Adams's stable-range theta-packet exhaustion conjecture for unitary groups

摘录 / EXCERPT: Adams's stable-range conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060553 | MATHDB-319015

Bolthausen-Sznitman scaling-limit conjecture for the ancestral partition process

摘录 / EXCERPT: For each fixed sample size $k\ge1$, let $(\Pi_t^{k,N},t\ge0)$ be the ancestral partition process of a population of size N, restricted to the sample, and let $(\Pi_t^k,t\ge0)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-060554 | MATHDB-319025

The monoidal equivalence conjecture for the rational matrix bi-factorisation category

摘录 / EXCERPT: Let d be a positive integer. Let $(\mathcal{P}_{d})_{0}$ be the full subcategory of the R-charge-zero subcategory of matrix bi-factorisations of (x^d,x^d) generated under direc...

conjecture | solved | MathDB 页面 | 原文线索

V2-060555 | MATHDB-319028

Asymptotic square-root conjecture for the optimal shrinkage parameter

摘录 / EXCERPT: Let m denote the number of areas, let $\boldsymbol{\theta}_{(,)}$ be the ordered random effects, and let $\boldsymbol{\theta}^{[2]}_{(,)}(\boldsymbol{ ,})$ denote the predictor...

conjecture | open | MathDB 页面 | 原文线索

V2-060556 | MATHDB-319029

The Littelmann pattern formula for coefficients of type D Weyl group multiple Dirichlet series

摘录 / EXCERPT: Let $N(\mathbf{x};\ell)=\sum_{\lambda}a_{\lambda}\mathbf{x}^{\lambda}$ be the ϖ-part constructed by averaging for the Weyl group multiple Dirichlet series...

conjecture | open | MathDB 页面 | 原文线索

V2-060557 | MATHDB-319037

Kähler-Einstein convergence conjecture for minimal models of general type

摘录 / EXCERPT: General-type convergence conjecture. As $s\to\infty$, the flow converges in the Gromov-Hausdorff sense to the unique Kähler-Einstein metric ω_{KE} on X_{can}. Weak distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-060558 | MATHDB-319038

Ricci-flat convergence conjecture for minimal models with Kodaira dimension zero

摘录 / EXCERPT: Ricci-flat convergence conjecture. As $t\to\infty$, the flow converges in the Gromov-Hausdorff sense to the unique Ricci-flat Kähler metric in $[H]$. Smooth convergence is known wh...

conjecture | open | MathDB 页面 | 原文线索

V2-060559 | MATHDB-319047

Binary-alphabet conjecture for the optimal auxiliary random variable

摘录 / EXCERPT: Let U be the auxiliary random variable in the feedback-based collaborative secrecy encoding problem over binary symmetric channels. The preceding result establishes that U can...

conjecture | open | MathDB 页面 | 原文线索

V2-060560 | MATHDB-319049

The stronger q-four functions conjecture

摘录 / EXCERPT: The stronger q-four functions conjecture. For every $A,B\transport \mathcal{P}(n)$,

conjecture | open | MathDB 页面 | 原文线索

V2-060561 | MATHDB-319050

Integration conjecture for virtual-knot weight systems

摘录 / EXCERPT: For virtual knots, finite type invariants are defined by the schematic difference relation ... A weight system is a linear functional on virtual-knot arrow diagrams satisfying the re...

conjecture | open | MathDB 页面 | 原文线索

V2-060562 | MATHDB-319051

Integration conjecture for generalized virtual-knot theories

摘录 / EXCERPT: Consider virtual-knot theories with skeleton equal to a circle (round), a line (long), or a line with all real crossings descending; with R2 and R3 treated in the standard, braid-l...

conjecture | open | MathDB 页面 | 原文线索

V2-060563 | MATHDB-319057

Gliozzi-Tateo functional dilogarithm identities for simply laced Y-systems

摘录 / EXCERPT: Let X_r and $X'_{r'}$ be simply laced Dynkin diagrams of finite type, with index sets I and I', ranks r and r', and Coxeter numbers h and h'. Let $Y_{ii'}(u)$ be a po...

conjecture | open | MathDB 页面 | 原文线索

V2-060564 | MATHDB-319064

Planar unbounded Tate-defect conjecture for primitive-divergent phi-four graphs

摘录 / EXCERPT: Unbounded Tate-defect conjecture. There exists a sequence of primitive-divergent graphs G_i in ϕ^4 such that

conjecture | open | MathDB 页面 | 原文线索

V2-060565 | MATHDB-319072

Millson's filtration conjecture for conformal block and invariant rings

摘录 / EXCERPT: Millson's conjecture. Give the ring $$ the associated term-order filtration from HMM and HMSV, with associated graded ring isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-060566 | MATHDB-319074

Viehweg's variation conjecture for fibre spaces

摘录 / EXCERPT: Let $f:X\to S$ be a fibre space X/S whose generic geometric fibre has Kodaira dimension at least 0. Let $\omega_{X/S}$ denote the relative canonical sheaf, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060567 | MATHDB-319075

Viehweg's induced inequality for relative Kodaira dimension

摘录 / EXCERPT: Let $f:X\to S$ be a fibre space X/S, let $X_{\bar{\eta}}$ be its generic geometric fibre, and let $\omega_{X/S}$ and $\omega_{X_{\bar{\eta}}}$ denote the relative and generic-fib...

conjecture | open | MathDB 页面 | 原文线索

V2-060568 | MATHDB-319080

Factorization conjecture for the characteristic polynomial of the asymmetric annihilation Markov matrix

摘录 / EXCERPT: Characteristic-polynomial factorization conjecture. The characteristic polynomial and its successive ratios are

conjecture | open | MathDB 页面 | 原文线索

V2-060569 | MATHDB-319081

Diagonal-entry formula conjecture for the characteristic polynomial of the asymmetric annihilation Markov matrix

摘录 / EXCERPT: Let M_L be the Markov matrix of the asymmetric annihilation model, with rows and columns indexed by $i=0,\ldots,2^L-1$. For an integer m, define $n_1(m)$ to be (-1) raised to...

conjecture | open | MathDB 页面 | 原文线索

V2-060570 | MATHDB-319086

Gupta-Newman-Rabinovich-Sinclair conjecture on minor-closed flow gaps

摘录 / EXCERPT: Gupta-Newman-Rabinovich-Sinclair conjecture. For every family of finite graphs $\mathcal F$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-060571 | MATHDB-319089

The single-expression implication for identities of $f(x,y)=x^2+y^3$

摘录 / EXCERPT: Let $f(x,y)=x^2+y^3$. An expression is formed by recursively applying f to a set of variables. Suppose that $G(x)$ is an expression formed by recursively applying f to the vari...

conjecture | open | MathDB 页面 | 原文线索

V2-060572 | MATHDB-319091

Barnes-function conjecture for the hypergeometric-kernel asymptotic constant

摘录 / EXCERPT: Assume the hypotheses of Proposition. Let C be the constant in the asymptotic expansions of the hypergeometric-kernel determinant $D(t)$ as $t\to1$, and define ... where G is the...

conjecture | open | MathDB 页面 | 原文线索

V2-060573 | MATHDB-319092

Barnes-function conjecture for the confluent hypergeometric-kernel constant

摘录 / EXCERPT: Assume that $0\leq\operatorname{Re}(z+z')<1$ and $z,z',w,z+z'+w\notin\mathbb{Z}$, as in Proposition. Let C_L be the constant in the small-s asymptotics of the confluent hyperge...

conjecture | open | MathDB 页面 | 原文线索

V2-060574 | MATHDB-319097

The force-based QC spectral equivalence conjecture

摘录 / EXCERPT: Let $N\geq4$, let $1\leq K\leq N-2$, and let $F>0$. Denote by $L^{\rm qcf}}_F$ and $L^{\rm qnl}}_F$ the force-based and quasinonlocal QC operators, respectively. Spectral equival...

conjecture | open | MathDB 页面 | 原文线索

V2-060575 | MATHDB-319098

The force-based QC eigenvector conditioning conjecture

摘录 / EXCERPT: Let V denote the matrix of eigenvectors for the force-based QC operator $L^{\rm qcf}}_F$. Assume $A_F>0$. Eigenvector conditioning conjecture. As $N\to\infty$, ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-060576 | MATHDB-319099

The preconditioned force-based QC spectral equivalence conjecture

摘录 / EXCERPT: Let $N\geq4$, let $1\leq K\leq N-2$, and let $F>0$. Let ${L}^{-1}$ be the inverse of the discrete Laplacian operator L, and consider the preconditioned operators...

conjecture | open | MathDB 页面 | 原文线索

V2-060577 | MATHDB-319100

The symmetrically preconditioned force-based QC eigenvector conditioning conjecture

摘录 / EXCERPT: Let ${\widetilde W}$ denote the matrix of eigenvectors for the symmetrically preconditioned force-based QC operator ${L}^{-1/2}L^{\rm qcf}}_F{L}^{-1/2}$. Assume $A_F>0$. Symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-060578 | MATHDB-319104

Conway-Sloane's non-isometry conjecture for tetralattice pairs

摘录 / EXCERPT: Let $L_{1;a,b,c,d}$ and $L_{2;a,b,c,d}$ be the two lattices in Conway and Sloane's real four-parameter family, with parameters $(a,b,c,d)\in\mathbb R^4$. Conway-Sloane's conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-060579 | MATHDB-319109

Hovey's conjecture that all trees are k-cordial

摘录 / EXCERPT: Hovey's conjecture. All trees are k-cordial for all k.

conjecture | open | MathDB 页面 | 原文线索

V2-060580 | MATHDB-319115

Possible numbers of projective zeros of a mixed homogeneous polynomial

摘录 / EXCERPT: Let q and r be nonnegative integers, let $f\in\mathcal M(q+2r,q;2)$, and let ... Write $\alpha=\sharp V$ for the number of points of V. The possible-zero-count claim. The possi...

conjecture | open | MathDB 页面 | 原文线索

V2-060581 | MATHDB-319120

Weak lower semicontinuity conjecture for homogenized elliptic energies

摘录 / EXCERPT: Let Ω be the domain, let A_{ε} and B_{ε} be the coefficient fields from the preceding boundary-value problem, and let $B^\sharp$ be the homogenized c...

conjecture | open | MathDB 页面 | 原文线索

V2-060582 | MATHDB-319133

A q-analogue of the Strauss-Shallit-Zagier congruence

摘录 / EXCERPT: Let $a\geq1$ and $m\geq1$, let $\Phi_r(q)$ denote the rth cyclotomic polynomial, and let q be an indeterminate. The q-analogue conjecture. The congruences ... and ... hold. The fir...

conjecture | open | MathDB 页面 | 原文线索

V2-060583 | MATHDB-319148

The no-elliptic-ramification conjecture for Modular Tower components

摘录 / EXCERPT: Assume the Main Conjecture hypotheses, and let $\bar O_k$ be any component at level k. Elliptic ramification from $\bar O_k$ to a component at level $k+1$ means the ramification...

conjecture | open | MathDB 页面 | 原文线索

V2-060584 | MATHDB-319151

The integrality conjecture for numerical Donaldson-Thomas invariants

摘录 / EXCERPT: Integrality conjecture. For a generic central charge Z, all numbers $\Omega(\gamma)$, $\gamma\in\Gamma\setminus\{0\}$, are integers. This predicts that the rational numerical Don...

conjecture | open | MathDB 页面 | 原文线索

V2-060585 | MATHDB-319160

Goldfeld's density conjecture for quadratic twists

摘录 / EXCERPT: Goldfeld's conjecture. For every elliptic curve $E/\mathbb Q$, those $\chi\rein \Sigma_2$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-060586 | MATHDB-319162

Astala's conjecture on the sharp dimension of quasicircles

摘录 / EXCERPT: Let $0<k<1$, and let a k-quasicircle be a quasicircle with parameter k. Astala's conjecture. There exists a k-quasicircle Γ with Hausdorff dimension ... Astala's upper e...

conjecture | open | MathDB 页面 | 原文线索

V2-060587 | MATHDB-319169

Monotonic-zero conjecture for truncated Riemann Xi-functions

摘录 / EXCERPT: Let Q denote the first quadrant of the complex plane, and for $N\in\mathbb N$ let $\Xi_N(z)$ be the truncation of Riemann's uniformly convergent series for the Riemann Ξ-func...

conjecture | open | MathDB 页面 | 原文线索

V2-060588 | MATHDB-319170

Monotonic-zero conjecture for truncated Ramanujan Xi-functions

摘录 / EXCERPT: Let Q denote the first quadrant of the complex plane, and let $\Xi_{\Delta,N}(z)$ be the truncation after N terms of the hyperbolic-gamma-function series defining the Ramanujan...

conjecture | open | MathDB 页面 | 原文线索

V2-060589 | MATHDB-319171

Monotonic-zero conjecture for incomplete gamma functions

摘录 / EXCERPT: Let $a>0$ be a fixed real number, and let Q denote the first quadrant of the complex plane. A function has monotonic zeros in Q when its zeros there, listed by increasing real...

conjecture | open | MathDB 页面 | 原文线索

V2-060590 | MATHDB-319172

Monotonic motion conjecture for zeros of interpolated Xi-approximates

摘录 / EXCERPT: Let $\Xi_k(z)=\sum_{j=1}^{k}\Phi_j(z)$ and let $\Phi_{k+1}(z)$ be the next summand in the truncated Riemann Xi-function series. For $k\geq1$ and $0\leq t\leq1$, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-060591 | MATHDB-319175

A large-deviation principle for non-parametric estimators of Loynes' exponent

摘录 / EXCERPT: Let $X(1),\ldots,X(n)$ be a sequence of consecutive increments, or alternatively let $W(1),\ldots,W(n)$ be observed workloads, and let θ^* denote Loynes' exponent. Consider...

conjecture | open | MathDB 页面 | 原文线索

V2-060592 | MATHDB-319180

The GW/DT/PT correspondence for smooth Calabi-Yau threefolds

摘录 / EXCERPT: For a smooth Calabi-Yau threefold Y, let $N_{g,\beta}$, $I_{n,\beta}$, and $P_{n,\beta}$ denote its Gromov-Witten, Donaldson-Thomas, and Pandharipande-Thomas invariants, respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-060593 | MATHDB-319188

Mühlherr's twist conjecture for finite-rank Coxeter systems

摘录 / EXCERPT: Twist conjecture. If $S'\subseteq S^W$ and S is sharp-angled with respect to S', then (W,S) is twist equivalent to (W,S').

conjecture | open | MathDB 页面 | 原文线索

V2-060594 | MATHDB-319210

Infinite logconcavity conjecture for the polynomials $A_n(x)$

摘录 / EXCERPT: Let $A_n(x)$ be the polynomial defined earlier in the paper, and let a sequence be infinite logconcave when every iterate of the operator $\mathfrak{L}$ produces a positive sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-060595 | MATHDB-319211

Conjectured Γ-limit for higher-order phase transitions with boundary line tension

摘录 / EXCERPT: Higher-order line-tension conjecture. Under the hypotheses of Theorem, the sequence $\{\mathcal{F}_{\varepsilon}\}_{\varepsilon>0}$ Γ-converges as $\varepsilon\to0^+$ to

conjecture | open | MathDB 页面 | 原文线索

V2-060596 | MATHDB-319212

The Leonardness conjecture for completely regular codes

摘录 / EXCERPT: Let C be a completely regular code in a Q-polynomial distance-regular graph, and let its covering radius be sufficiently large. Leonardness conjecture. The code C is a Leonar...

conjecture | open | MathDB 页面 | 原文线索

V2-060597 | MATHDB-319214

Adelic extension conjecture for commensurated subgroups

摘录 / EXCERPT: Adelic extension conjecture. The homomorphism φ extends continuously to a homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-060598 | MATHDB-319216

Durrett-Rogers law of large numbers conjecture for Brownian polymers

摘录 / EXCERPT: Durrett-Rogers law of large numbers conjecture. Under these assumptions, $X(t)/t\to0$ almost surely. This conjecture concerns the long-time displacement of a one-dimensional self-i...

conjecture | open | MathDB 页面 | 原文线索

V2-060599 | MATHDB-319217

Diffusivity exponent conjecture for self-repelling Brownian polymers

摘录 / EXCERPT: Let $E(t)={\mathbf E}(X(t)^2)$, and suppose that for some $-1<\alpha<1$ the infrared bounds ... hold. Diffusivity exponent conjecture. Under these conditions, the true asymptotic ord...

conjecture | open | MathDB 页面 | 原文线索

V2-060600 | MATHDB-319225

Bressan's compactness conjecture for nearly incompressible flows

摘录 / EXCERPT: Bressan's compactness conjecture. Under these assumptions, the sequence $\{X_k\}$ is strongly precompact in $L^1_{\operatorname{loc}}([0,+\infty[\times\mathbb{R}^d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-060601 | MATHDB-319234

The coherent-family limit conjecture for the ultra-discretization of the affine geometric crystal

摘录 / EXCERPT: Coherent-family limit conjecture. The crystal ${\mathcal UD}(\chi,T,\theta)$ is the limit of a coherent family of perfect crystals of type $\TY(G,1,2)$ in the cited work.

conjecture | open | MathDB 页面 | 原文线索

V2-060602 | MATHDB-319235

Conjecture on the involutory growth rate of permutation patterns

摘录 / EXCERPT: Let $ath\pi$ be a permutation pattern. The involutory growth-rate conjecture. The limit ... exists and satisfies ... Here $I_n(\pi)$ is the set of π-avoiding involutions of $[n]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-060603 | MATHDB-319236

Conjecture on the central limit law for factorizations into random involutions

摘录 / EXCERPT: Let π be a uniformly random permutation of $[n]$, let $F(\pi)$ denote the number of factorizations of π into two involutions, and let Φ be the distribution function of...

conjecture | open | MathDB 页面 | 原文线索

V2-060604 | MATHDB-319257

Flat-rank one subgroups in hyperbolic totally disconnected groups

摘录 / EXCERPT: Flat-rank one subgroup conjecture. Every flat subgroup H of flat-rank 1 in G has a limit set containing two elements, both fixed by H. Moreover, H_1 is relatively compact...

conjecture | open | MathDB 页面 | 原文线索

V2-060605 | MATHDB-319264

Rigidity conjecture at the ground-state mass for mass-critical NLS

摘录 / EXCERPT: Rigidity conjecture at the ground-state mass. If $M(u_0)=M(Q)$, then only the following cases can occur: if u blows up in finite time, it coincides with $Pc(Q)$ up to symmetries...

conjecture | open | MathDB 页面 | 原文线索

V2-060606 | MATHDB-319265

Bers's conjecture that every b-group is a boundary group

摘录 / EXCERPT: Let G be a Fuchsian group of the first kind, and let G^ϕ be a b-group arising from a point ϕ in the relevant deformation space. A group G^ϕ is called a boundary gr...

conjecture | open | MathDB 页面 | 原文线索

V2-060607 | MATHDB-319274

Shevelev's bound for the normalized maximal minor increment

摘录 / EXCERPT: Write ... A normalized-increment bound conjecture asserts that, for every $i\geq6$, ... The paper uses this proposed bound to derive the stated sufficient condition for infinitely many...

conjecture | open | MathDB 页面 | 原文线索

V2-060608 | MATHDB-319290

Bhargava-Malle conjecture for marked squarefree extensions over function fields

摘录 / EXCERPT: Fix $n\geq 1$. Let $N(k)$ be the number of isomorphism classes of marked degree-n extensions of $\mathbb{F}_q(t)$ with squarefree discriminant of degree k, totally split above...

conjecture | open | MathDB 页面 | 原文线索

V2-060609 | MATHDB-319293

Asymptotic comparison conjecture for tautological dimensions

摘录 / EXCERPT: For $g\geq 2$, let R_g denote the relevant tautological dimension and let $\omega_g=\omega(g-2)$, where $\omega(n)$ is the coefficient sequence defined earlier in the paper. Dime...

conjecture | open | MathDB 页面 | 原文线索

V2-060610 | MATHDB-319294

Coefficientwise asymptotic conjecture for tautological dimensions

摘录 / EXCERPT: Let $p(n)$ and $p_{\omega}(n)$ be the coefficient sequences associated with the dimension series, and let $a(n)$ and $a_{\omega}(n)$ be the corresponding auxiliary coefficient sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-060611 | MATHDB-319295

Coefficientwise rank conjecture for Faber's intersection matrix

摘录 / EXCERPT: Let $g\geq 2$ and $0\leq k\leq g-2$. Let R_g^k and ω_g^k denote the degree-k entries in the two dimension decompositions considered in the paper. Coefficientwise rank co...

conjecture | open | MathDB 页面 | 原文线索

V2-060612 | MATHDB-319298

Strong Euler homogeneity conjecture for free divisors

摘录 / EXCERPT: Let D be a free divisor. The logarithmic comparison theorem holds for D when the de Rham morphism ... is a quasi-isomorphism, where $j\colon U=X\setminus D\hookrightarrow X$ is t...

conjecture | open | MathDB 页面 | 原文线索

V2-060613 | MATHDB-319300

Kaiser-Král'-King conjecture on fractional total coloring of high-girth graphs

摘录 / EXCERPT: Let $\Delta\ge5$ be an odd integer. Let the fractional total chromatic number be the fractional analogue of the total chromatic number, and let the girth of a graph be the length o...

conjecture | open | MathDB 页面 | 原文线索

V2-060614 | MATHDB-319303

Relaxation of the parameter condition in the Lieb-type inequalities

摘录 / EXCERPT: Let A be a positive semi-definite Hermitian matrix, with the block partition and inequalities referred to in the source's Lieb-type theorem and its corollary. The theorem assumes...

conjecture | open | MathDB 页面 | 原文线索

V2-060615 | MATHDB-319310

The superlinear crossing-family conjecture for geometric complete graphs

摘录 / EXCERPT: Let $t(n)$ be the minimum number of colours needed to colour the edges of some geometric drawing of K_n so that every pair of edges with the same colour crosses; each colour clas...

conjecture | open | MathDB 页面 | 原文线索

V2-060616 | MATHDB-319318

Supercritical BNLS peak-type singularity conjecture

摘录 / EXCERPT: Let ψ be a peak-type singular solution of the supercritical BNLS, let $L(t)$ be its collapsing length scale, let $T_{\rm c}$ be the collapse time, and set $\rho=r/L(t)$. Defin...

conjecture | open | MathDB 页面 | 原文线索

V2-060617 | MATHDB-319321

Modified Cohen-Lenstra-type distribution conjecture for relative class groups

摘录 / EXCERPT: Modified class-group distribution conjecture. A given finite p-group H occurs as the Sylow p-subgroup of a relative class group $\operatorname{Cl}(K/K_0)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-060618 | MATHDB-319323

Orthogonal invariant-tensor character conjecture

摘录 / EXCERPT: Let $n\ge 3$, let $\mathfrak{g}=\mathfrak{so}(n)$, and let V be the vector representation. For a partition μ with at most n parts and an integer $p\ge 0$, define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060619 | MATHDB-319331

The amicable-pair counting conjecture for elliptic curves

摘录 / EXCERPT: Let $E/\bb{Q}$ be an elliptic curve. Define the amicable-pair counting function ... Assume that there are infinitely many primes p such that $\#\widetilde E_p(\mathbb{F}_p)$ is pri...

conjecture | open | MathDB 页面 | 原文线索

V2-060620 | MATHDB-319333

The CM amicable-pair ratio conjecture for elliptic curves with nonzero j-invariant

摘录 / EXCERPT: The CM amicable-pair ratio conjecture. Either $\mathcal N_E(X)$ is bounded, or

conjecture | open | MathDB 页面 | 原文线索

V2-060621 | MATHDB-319334

The CM j=0 Type-1 amicable-pair ratio conjecture

摘录 / EXCERPT: Let k be an integer for which the curve $E:y^2=x^3+k$ is defined, and let $\mathcal N_k$ be the set of primes $p\ge5$ of good reduction such that...

conjecture | open | MathDB 页面 | 原文线索

V2-060622 | MATHDB-319338

The higher-dimensional characterization of non-uniqueness sets in photoacoustic tomography

摘录 / EXCERPT: Let n be a positive integer, let $S\subset\mathbb{R}^n$, and let f be compactly supported. A set S is a non-uniqueness set for compactly supported functions f if the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-060623 | MATHDB-319342

Solomon's annihilation conjecture for class groups

摘录 / EXCERPT: Solomon's conjecture. The element sol_F annihilates

conjecture | open | MathDB 页面 | 原文线索

V2-060624 | MATHDB-319343

The Spherical Law for products of random matrices

摘录 / EXCERPT: The Spherical Law. The spectral densities of $A_N B_N^{-1}$ converge to the uniform density on the Riemann sphere as N tends to infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-060625 | MATHDB-319369

2- and 3-adic expansion conjecture for the gamma coefficients as alpha-polynomials

摘录 / EXCERPT: 2- and 3-adic expansion conjecture. The polynomials $\gamma_{-\frac{1}{2}+r}^{\ast\ast}(a)$ possess 2- and 3-adic expansions to all orders:

conjecture | open | MathDB 页面 | 原文线索

V2-060626 | MATHDB-319372

Equality-case integral-closure conjecture for multiplicity bounds

摘录 / EXCERPT: Let R, $\mathfrak{a}$ and J be as in the relevant corollary, with positive integers $a_1,\ldots,a_d,b_1,\ldots,b_d$ supplied by that setup. Suppose that ... Put...

conjecture | open | MathDB 页面 | 原文线索

V2-060627 | MATHDB-319375

Shevelev's main-increment conjecture for the dual sequence

摘录 / EXCERPT: Let $\tilde{c}(n)$ be the sequence defined by ... An increment is called main if it occurs at one of the distinguished fundamental points described in the paper. Shevelev's main-incr...

conjecture | open | MathDB 页面 | 原文线索

V2-060628 | MATHDB-319379

The reducing property for Artin-Tits presentations

摘录 / EXCERPT: Let $(\mathcal{S},\mathcal{R})$ be an Artin-Tits presentation, and let denote the reducing equivalence defined earlier in the paper. Reducing conjecture. Every Artin-Tits presentat...

conjecture | open | MathDB 页面 | 原文线索

V2-060629 | MATHDB-319385

Total nonnegativity criterion for finite Hurwitz matrices

摘录 / EXCERPT: Total nonnegativity criterion. The finite Hurwitz matrix $\mathcal H_k(p,q)$ is totally nonnegative if and only if $\widetilde p$ and $\widetilde q$ have only nonpositive zeros,...

conjecture | open | MathDB 页面 | 原文线索

V2-060630 | MATHDB-319392

Bridgeland's kernel conjecture for autoequivalences of K3 surfaces

摘录 / EXCERPT: Let X be a complex projective K3 surface. Let ${\rm Aut}({\rm D}^{\rm b}(X))$ be the group of $\mathbb{C}$-linear exact autoequivalences of ${\rm D}^{\rm b}(X)$, and let ... be the...

conjecture | open | MathDB 页面 | 原文线索

V2-060631 | MATHDB-319401

The half-integrality conjecture for resolved-conifold coefficients

摘录 / EXCERPT: Let $\Psi^{(a)}_{0,1}(t;x)$ be the resolved-conifold open-string one-point function, and set $Q=-e^{-t}$. Half-integrality conjecture. There are half-integers $e^{(a)}_{k,m}$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-060632 | MATHDB-319414

The unified even-modulus eta-function identity

摘录 / EXCERPT: Let k,N be positive integers and set $n=\lfloor N/2\rfloor$. For $1\leq a,b\leq N-1$, let C_{ab} be the Cartan integers of A_{N-1}, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-060633 | MATHDB-319433

The sharp formula conjecture for the harmonic mapping modulus bound

摘录 / EXCERPT: Sharp formula conjecture. The function in this bound is

conjecture | open | MathDB 页面 | 原文线索

V2-060634 | MATHDB-319436

The second-order Jack-measure coefficient independence conjecture

摘录 / EXCERPT: Let F be a homogeneous symmetric function of degree k, let α be a positive real number, and let $\mu\vdash k-1$ be a partition. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-060635 | MATHDB-319442

Existence of non-degenerate balanced configurations for two ends

摘录 / EXCERPT: Existence conjecture. There exists a non-degenerate balanced configuration $\{p_{k,i}\}$ with $N=2$, $n_1=2$, $n_2=2k-1$, and $T=0$ for every $k\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060636 | MATHDB-319443

Labastida-Mariño integrality conjecture for colored HOMFLY invariants

摘录 / EXCERPT: Labastida-Mariño integrality conjecture. For every coloring R,

conjecture | open | MathDB 页面 | 原文线索

V2-060637 | MATHDB-319459

Baker-Forrester constant-term conjecture for prescribed symmetry

摘录 / EXCERPT: Let $D_1(n_0;n_1;a,b;q)$ be the constant term of the Laurent polynomial ... Here CT denotes the constant term, $(x;q)_k$ is the q-shifted factorial, and Γ_q...

conjecture | open | MathDB 页面 | 原文线索

V2-060638 | MATHDB-319460

Baker-Forrester ratio conjecture for multiblock constant terms

摘录 / EXCERPT: Let $D_p(n_1,\ldots,n_p;n_0;a,b,k;q)$ be the constant term ... where $\widetilde J_\alpha$ is determined by the prescribed block sizes. Write $[r]_q!$ for the q-factorial and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060639 | MATHDB-319464

The metric cotype gap conjecture

摘录 / EXCERPT: Let (X,d) be a metric space, and let q_X denote its metric cotype exponent. Metric cotype gap conjecture. There is no metric space (X,d) for which ... The preceding results est...

conjecture | open | MathDB 页面 | 原文线索

V2-060640 | MATHDB-319473

Conjecture that random graphs of fixed genus are asymptotically almost surely 4-chromatic

摘录 / EXCERPT: Chromatic-number conjecture. The chromatic number of a random graph of fixed genus g with n vertices is asymptotically almost surely equal to 4.

conjecture | open | MathDB 页面 | 原文线索

V2-060641 | MATHDB-319483

Boyd's conductor 30 Mahler measure conjectures

摘录 / EXCERPT: Let $F(b,c)$ denote the lattice sums used in the paper, and let $m(\alpha)$ and $g(\alpha)$ be the Mahler measures defined by ... and ... where $y=e^{2\pi i\theta_1}$ and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-060642 | MATHDB-319484

The sharp normality conjecture for iterates

摘录 / EXCERPT: Let $D\subset\mathbb C$ be a domain, let $n\geq 2$, and let $\varepsilon>0$. Let $\mathcal F$ be the family of all holomorphic functions $f:D\to\mathbb C$ such that ... for every fix...

conjecture | open | MathDB 页面 | 原文线索

V2-060643 | MATHDB-319485

The degree-sharp polynomial iterate multiplier conjecture

摘录 / EXCERPT: Let p be a polynomial of degree $d\geq 2$, and let $n\geq 2$. Degree-sharp multiplier conjecture. Then p^n has a fixed point ξ satisfying … The source states that this is s...

conjecture | open | MathDB 页面 | 原文线索

V2-060644 | MATHDB-319486

Shrinking-ring blowup conjecture for the biharmonic nonlinear Schrödinger equation

摘录 / EXCERPT: Shrinking-ring conjecture. The solution is quasi self-similar, $\psi\sim\psi_{Q_{\mathrm B}}$, for $r-r_{\rm max}=\mathcal{O}(L)$; it is a shrinking ring, with...

conjecture | open | MathDB 页面 | 原文线索

V2-060645 | MATHDB-319487

Equal-rate blowup conjecture for critical biharmonic nonlinear Schrödinger rings

摘录 / EXCERPT: Equal-rate blowup conjecture. The solution undergoes equal-rate collapse,

conjecture | open | MathDB 页面 | 原文线索

V2-060646 | MATHDB-319492

The loop-space TMF to twisted K-theory map conjecture

摘录 / EXCERPT: Loop-space TMF to twisted K-theory conjecture. There is a map

conjecture | open | MathDB 页面 | 原文线索

V2-060647 | MATHDB-319501

The complexified Volume Conjecture

摘录 / EXCERPT: Let K be a hyperbolic knot, meaning that its complement $S^3\setminus K$ possesses a complete hyperbolic structure. Let $J_N(K;q)$ be the N-dimensional colored Jones polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-060648 | MATHDB-319505

Invariance of the symbolic dynamics under the complex symbolic space

摘录 / EXCERPT: Let f be the rational function considered in the paper. Let $\mathcal{A}$ be the alphabet whose symbols I_i correspond to the intervals produced by pa on the real axis, and I...

conjecture | open | MathDB 页面 | 原文线索

V2-060649 | MATHDB-319511

Ablowitz-Ladik conjecture for the resolved conifold

摘录 / EXCERPT: Let X_1 be the resolved conifold $\mathcal{O}_{\mathbb{P}^1}(-1)\oplus\mathcal{O}_{\mathbb{P}^1}(-1)$ equipped with the anti-diagonal action, and consider its equivariantly Calab...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-060650 | MATHDB-319516

The characteristic-polynomial conjecture for cross section lattices

摘录 / EXCERPT: Let Λ^* be the cross section lattice of a J-irreducible monoid M, and let J_0 and n have the meanings established in the preceding theorem. Characteristic-polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-060651 | MATHDB-319529

The ring class field conjecture for Néron-Severi groups of singular Enriques surfaces

摘录 / EXCERPT: Let Y be an Enriques surface whose universal cover X is a singular K3 surface. Let $d<0$ denote the discriminant of X, and let $H(d)$ and $H(4d)$ denote the corresponding rin...

conjecture | open | MathDB 页面 | 原文线索

V2-060652 | MATHDB-319533

The lower-bound conjecture for the order-bound threshold in numerical semigroups

摘录 / EXCERPT: Let S be a numerical semigroup with multiplicity e, conductor c, subconductor c', dominant d, and greatest element d' of S preceding c'. Let s_m be the first elem...

conjecture | open | MathDB 页面 | 原文线索

V2-060653 | MATHDB-319534

Goldman-Wentworth-Labourie uniqueness conjecture for conformal structures

摘录 / EXCERPT: Let Σ be a surface, let G^τ be a split real simple Lie group, and let θ be a representation in the Hitchin component. For a conformal structure on Σ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060654 | MATHDB-319535

Properness and non-degeneracy conjecture for Hitchin-component energy

摘录 / EXCERPT: Let Σ be a surface, let G be a split real simple Lie group, and let θ be a representation in a Hitchin component. Let E_θ be the associated energy function o...

conjecture | open | MathDB 页面 | 原文线索

V2-060655 | MATHDB-319547

Strong Shamai-Laroia conjecture for additive Gaussian interference

摘录 / EXCERPT: Let $h,p,v\in\mathbb{R}_+$, let $X_1^G\sim\mathcal{N}(0,p)$ and let $Z\sim\mathcal{N}(0,v)$ be independent of X_1^G. Let X and X_1 be i.i.d. random variables with mean 0 an...

conjecture | open | MathDB 页面 | 原文线索

V2-060656 | MATHDB-319550

The twisted functional equation conjecture for elliptic-curve L-functions

摘录 / EXCERPT: Twisted functional equation conjecture. The function $\Lambda(E/k,\tau,s)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-060657 | MATHDB-319551

The elliptic-curve parity conjecture for irreducible representations

摘录 / EXCERPT: Parity conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-060658 | MATHDB-319566

Casselman's reflection-invariance conjecture for Kazhdan-Lusztig intervals

摘录 / EXCERPT: Let $\hat{\Phi}$ be a simply-laced root system, let $u<v$ in the Bruhat order, and let $P_{u,v}$ be the Kazhdan-Lusztig polynomial. For $\beta\in\hat{\Phi}^+$, write r_β for...

conjecture | open | MathDB 页面 | 原文线索

V2-060659 | MATHDB-319569

Almost-every-direction trace conjecture for prevalent Besov functions

摘录 / EXCERPT: Let $D=d+d'$ and let $B^{s}_{p,q}(\mathbb{R}^D)$ be the Besov space under the hypotheses on p,q,s of Theorem. For a d-dimensional subspace $\mathcal{H}$ of $\mathbb{R}^D$, writ...

open_problem | open | MathDB 页面 | 原文线索

V2-060660 | MATHDB-319577

Ehrhart's multivariate quasi-polynomial conjecture

摘录 / EXCERPT: Let $A(n_1,\dots,n_r)$ be an $m\times k$ matrix and $b(n_1,\dots,n_r)$ be a column vector of length m, with all entries linear functions of $n_1,\dots,n_r$ having integer coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-060661 | MATHDB-319578

Schilling-Shimozono fermionic formula conjecture for affine crystals

摘录 / EXCERPT: Schilling-Shimozono conjecture. For $\diamondsuit\in\{(1),(2),(1,1)\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-060662 | MATHDB-319580

Free-energy asymptotic expansion conjecture for the square-lattice Potts eigenvalues

摘录 / EXCERPT: Free-energy expansion conjecture. The two displayed asymptotic expansions hold. This has been verified for $m\le 10$ using the explicitly computed polynomials $C_1,\ldots,C_{10}$....

conjecture | open | MathDB 页面 | 原文线索

V2-060663 | MATHDB-319581

Linear-polynomial coefficient conjecture for Potts-model eigenvalue expansions

摘录 / EXCERPT: Linear-coefficient conjecture. Both displayed expansions hold. The coefficients have been checked computationally in the ranges reported in the paper, and the relations among the a...

conjecture | open | MathDB 页面 | 原文线索

V2-060664 | MATHDB-319592

Real-root conjecture for the tau-deformed groundstate polynomials

摘录 / EXCERPT: Let π be a matching, and consider the bivariate polynomial $\Psi_\pi(\tau,t)$ as a polynomial in t with coefficients in $\mathbb{Q}[\tau]$. Let $d(\pi)$ and $m_i(\pi)$ have t...

conjecture | open | MathDB 页面 | 原文线索

V2-060665 | MATHDB-319593

Tau-deformed negative-integer factorization conjecture

摘录 / EXCERPT: Let π be a matching and let p be an integer with $1\leq p\leq|\pi|-1$ such that $m_p(\pi)=0$. Write $\pi=\alpha\circ\beta$ with $|\alpha|=p$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-060666 | MATHDB-319594

Positivity and sum-rule conjecture for tau-deformed special values

摘录 / EXCERPT: For a matching π, define $G_\pi(\tau)=\Psi_\pi(\tau,-|\pi|)$. Let $d(\pi)$ be the degree parameter used in the source, and let $\Psi_\pi(-\tau)$ denote the specialization...

conjecture | open | MathDB 页面 | 原文线索

V2-060667 | MATHDB-319595

Coefficient-positivity conjecture for the tau-deformed polynomials

摘录 / EXCERPT: Let π be a matching, let $d(\pi)$ be the degree parameter used in the source, and let $P_\pi(\tau,t)$ be the associated bivariate polynomial. Tau-deformed coefficient-positivit...

conjecture | open | MathDB 页面 | 原文线索

V2-060668 | MATHDB-319618

Shioda-Artin supersingularity conjectures for K3 surfaces

摘录 / EXCERPT: Shioda-Artin supersingularity conjectures. For K3 surfaces, Shioda-supersingularity implies unirationality; Artin-supersingularity implies unirationality; and Artin-supersingularit...

conjecture | open | MathDB 页面 | 原文线索

V2-060669 | MATHDB-319619

Artin's Frobenius-pullback conjecture for supersingular elliptic K3 surfaces

摘录 / EXCERPT: Artin's Frobenius-pullback conjecture. The elliptic fibration on X arises via Frobenius pullback from a rational elliptic surface.

conjecture | open | MathDB 页面 | 原文线索

V2-060670 | MATHDB-319626

Asymptotic sharp-slope conjecture for effective divisors on moduli spaces of curves

摘录 / EXCERPT: For an effective divisor $D=a\lambda-b\delta$ on $\overline{\mathcal M}_g$ with $a,b>0$, define its slope by ... Here g tends to infinity. Asymptotic sharp-slope conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-060671 | MATHDB-319640

The type C Weyl group multiple Dirichlet series conjecture

摘录 / EXCERPT: Type C multiple-Dirichlet-series conjecture. The series $Z_\Psi(\mathbf{s};\mathbf{m})$ should have meromorphic continuation to $\mathbb{C}^r$ and satisfy a group of functional equ...

conjecture | open | MathDB 页面 | 原文线索

V2-060672 | MATHDB-319643

Newton approximation fails for 100% of the primes

摘录 / EXCERPT: Let f be a polynomial of degree $d\geq 2$ with coefficients in a number field K, and let $x_0\in K$. Define the Newton map ... and, for each $n\geq 0$, set ... Assume that the Newt...

conjecture | open | MathDB 页面 | 原文线索

V2-060673 | MATHDB-319650

Categorical strange duality for invertible polynomials

摘录 / EXCERPT: Categorical strange duality conjecture. There should exist quivers with relations (Q,I) and (Q',I') such that there are triangulated equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-060674 | MATHDB-319652

Finite-variable equality conjecture for transposed staircase foundations

摘录 / EXCERPT: For $n\geq 1$, $k\geq 0$, and a partition λ such that the skew diagrams $\mathcal{S}(\lambda,k,n)$ and $\mathcal{S}(\lambda^t,k,n)$ are connected, write...

conjecture | open | MathDB 页面 | 原文线索

V2-060675 | MATHDB-319663

Adjacent-case configuration lifting conjecture for highly palindromic Toeplitz matrices

摘录 / EXCERPT: Adjacent-case configuration lifting conjecture. All configurations of the $(2k+2)$nd moment obtained by adding a pair of adjacent entries to a fixed configuration of the $2k$th mom...

conjecture | open | MathDB 页面 | 原文线索

V2-060676 | MATHDB-319672

Conjecture on automatic multiplicative functions at the primes

摘录 / EXCERPT: Let $k\geq 2$ and let $f:\mathbb{N}\to\mathbb{C}$ be a k-automatic multiplicative function. A function is eventually periodic if there exist integers $N\geq 0$ and $d\geq 1$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-060677 | MATHDB-319680

Cohn-Kumar universal optimality conjecture for the hexagonal, E8, and Leech lattices

摘录 / EXCERPT: Cohn-Kumar conjecture. For $n=2$, 8, or 24, the linear programming bounds for potential energy minimization in $\mathbb{R}^n$ are sharp for every completely monotonic potential...

conjecture | open | MathDB 页面 | 原文线索

V2-060678 | MATHDB-319686

McKenna's uniqueness conjecture for large solutions

摘录 / EXCERPT: McKenna's uniqueness conjecture. Under these assumptions, there exists a unique large solution. This conjecture concerns uniqueness for boundary blow-up problems under the conditio...

conjecture | open | MathDB 页面 | 原文线索

V2-060679 | MATHDB-319687

Brezis' radial symmetry conjecture for boundary blow-up solutions

摘录 / EXCERPT: Let $f\in C^1(\mathbb R)$ satisfy the positivity and Keller-Osserman conditions, and let u be a solution of the boundary blow-up problem studied in the source on the unit ball...

conjecture | open | MathDB 页面 | 原文线索

V2-060680 | MATHDB-319721

Gorenstein Fano Ehrhart root narrow-strip conjecture

摘录 / EXCERPT: Let P be a Gorenstein Fano polytope of dimension D, and let α be a root of its Ehrhart polynomial. Gorenstein Fano Ehrhart root narrow-strip conjecture. All roots...

conjecture | open | MathDB 页面 | 原文线索

V2-060681 | MATHDB-319738

The limiting shift conjecture for the Gamma-function expansion

摘录 / EXCERPT: Let $(v_n)_{n\geq 0}$ be the sequence of shifts appearing in the asymptotic expansion ... The sequence begins with $v_1=\frac{23}{90}$, $v_2=\frac{1792627}{7289730}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-060682 | MATHDB-319742

Generalization of Proposition 3punctures to all simple closed curves in the four-punctured disk

摘录 / EXCERPT: Generalization conjecture. Proposition 3punctures can be generalized to every simple closed curve in D_4, including those surrounding two punctures, using the same kind of argume...

conjecture | open | MathDB 页面 | 原文线索

V2-060683 | MATHDB-319744

Round-or-almost-round image conjecture for reduction curves

摘录 / EXCERPT: Round-or-almost-round image conjecture. There is some i between 1 and r such that the curve $\mathcal C$ is sent to a round or almost round curve by the braid

conjecture | open | MathDB 页面 | 原文线索

V2-060684 | MATHDB-319749

Rudnick-du Sautoy conjecture on meromorphic continuation of Euler products

摘录 / EXCERPT: Let $h(X_1,X_2)\in\mathbb{Z}[X_1,X_2]$ and define ... A polynomial $g(U)$ is cyclotomic here if it divides $(1-U^{m})^{n}$ for some positive integers m and n. Rudnick-du Sautoy c...

conjecture | open | MathDB 页面 | 原文线索

V2-060685 | MATHDB-319750

Natural-boundary conjecture for non-cyclotomic Euler products

摘录 / EXCERPT: Natural-boundary conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-060686 | MATHDB-319756

Novosibirsk 3-color conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph. Two cycles are adjacent if they have an edge in common. Novosibirsk 3-color conjecture. If G has no pair of adjacent 3-cycles and no pair consisting...

conjecture | open | MathDB 页面 | 原文线索

V2-060687 | MATHDB-319759

Rigidity conjecture for the families $(\varphi_\alpha\Phi_n)_{\alpha\in U}$

摘录 / EXCERPT: Rigidity conjecture. Then the family $(\varphi_\alpha\Phi_n)_{\alpha\in U}$ is holomorphically trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-060688 | MATHDB-319773

Lehmer-type shell design conjecture for two-dimensional Euclidean lattices

摘录 / EXCERPT: Shell design conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-060689 | MATHDB-319777

The maximum-degree conjecture for extreme points of the Held-Karp relaxation

摘录 / EXCERPT: Consider extreme points of the Held-Karp relaxation on n vertices, and let the maximum degree of an extreme point mean the maximum degree of its support graph. The maximum-degree...

conjecture | open | MathDB 页面 | 原文线索

V2-060690 | MATHDB-319781

The ℓ^1 Bass conjecture

摘录 / EXCERPT: The ℓ^1 Bass conjecture. The image of HS^1 is contained in

conjecture | solved | MathDB 页面 | 原文线索

V2-060691 | MATHDB-319795

Classification conjecture for surfaces with the Keum-Naie fundamental group

摘录 / EXCERPT: Let Γ be the fundamental group of a Keum-Naie surface, and let S be a minimal smooth projective surface satisfying ... with $\pi_1(S)\cong\Gamma$. Keum-Naie classification c...

conjecture | open | MathDB 页面 | 原文线索

V2-060692 | MATHDB-319805

MMSE transform injectivity conjecture up to reflection

摘录 / EXCERPT: Let X and Z be zero-mean random variables, and for $\rho\in[0,\infty)$ let ${\mathsf{mmse}}(X,\rho)$ denote the minimum mean-square error in estimating X from a Gaussian-nois...

conjecture | open | MathDB 页面 | 原文线索

V2-060693 | MATHDB-319807

The rationality and reciprocity conjecture for Darmon points

摘录 / EXCERPT: Let $\mathcal{O}$ be the quadratic order used to define the Darmon points, let H be its narrow ring class field over the real quadratic field K, and let $J^{(\ell)}_\pm$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-060694 | MATHDB-319815

The dimer model realization conjecture for Laurent polynomials

摘录 / EXCERPT: Dimer model realization conjecture. For a suitable choice of W and distinguished basis, there is a bicolored graph Y on $W^{-1}(0)$ such that an edge of Y corresponds to an i...

conjecture | open | MathDB 页面 | 原文线索

V2-060695 | MATHDB-319824

The Laface-Ugaglia conjecture on special classes in blow-ups of projective 3-space

摘录 / EXCERPT: Laface-Ugaglia conjecture. begin{enumerate} item If $q(D)\leq 0$, then $h^0(D)=h^0(D-Q)$. item If $q(D)>0$, then D is special if and only if $d<m_1+m_2-1$, and in that case

conjecture | open | MathDB 页面 | 原文线索

V2-060696 | MATHDB-319825

The smallest eigenvalue of a circulant B-spline matrix

摘录 / EXCERPT: Let C_m^d be the circulant associated with the degree-d B-spline collocation matrix, with Toeplitz symbol coefficients $t_0^d,\ldots,t_r^d$, and let $\lambda_\mu(C_m^d)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-060697 | MATHDB-319826

Refined multicover conjecture for Higgs sheaf invariants

摘录 / EXCERPT: Refined multicover conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060698 | MATHDB-319827

Doubly refined wallcrossing and multicover conjecture

摘录 / EXCERPT: Doubly refined wallcrossing and multicover conjecture. Under the same conditions as the preceding refined wallcrossing conjecture, there exist doubly refined invariants in...

conjecture | open | MathDB 页面 | 原文线索

V2-060699 | MATHDB-319829

Doubly refined asymptotic ADHM generating-function conjecture

摘录 / EXCERPT: Doubly refined asymptotic ADHM conjecture. Under the conditions of the singly refined asymptotic conjecture,

conjecture | open | MathDB 页面 | 原文线索

V2-060700 | MATHDB-319830

Independence of refined Higgs invariants from the degree

摘录 / EXCERPT: Degree-independence conjecture. For fixed r, both $\overline H(r,e)(y)$ and $\overline H(r,e)(u,v)$ are independent of e; equivalently, they have the same value for all pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-060701 | MATHDB-319833

Unirationality of the moduli space of genus 36 polarized K3 surfaces

摘录 / EXCERPT: Let $\mathcal{K}_{36}$ denote the moduli space of primitively polarized smooth $\mathrm{K3}$ surfaces of genus 36. Unirationality conjecture. The moduli space $\mathcal{K}_{36}$...

conjecture | open | MathDB 页面 | 原文线索

V2-060702 | MATHDB-319835

The degenerate singular-locus conjecture for Verra threefolds

摘录 / EXCERPT: Let $(J(T),\Xi)$ be the intermediate Jacobian of a general Verra threefold T, let Ξ be its theta divisor, and let e be the canonical extension class; write Ξ_e for the...

conjecture | open | MathDB 页面 | 原文线索

V2-060703 | MATHDB-319843

Identification of the F-different with the different

摘录 / EXCERPT: Let R be an F-finite normal local ring of characteristic $p>0$, let $X:=\operatorname{Spec} R$, and let $D+B$ be an effective $\mathbb{Q}$-divisor on X, where D is a normal...

conjecture | open | MathDB 页面 | 原文线索

V2-060704 | MATHDB-319864

Capacity conjecture for degraded broadcast relay channels with common relay

摘录 / EXCERPT: Let a broadcast relay channel with common relay have input alphabets represented by X,X_1,X_2 and output alphabets Y_1,Z_1,Y_2,Z_2. In the degraded case, the channel satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-060705 | MATHDB-319865

Finite generation conjecture for adjoint rings with a pseudo-effective canonical divisor

摘录 / EXCERPT: Let (X,Δ) be a projective klt pair such that $K_X+\Delta$ is pseudo-effective, and let A be an ample $\mathbb{Q}$-divisor on X. The finite generation conjecture. The adj...

conjecture | open | MathDB 页面 | 原文线索

V2-060706 | MATHDB-319868

Faber--Hain--Looijenga's Gorenstein conjecture for tautological rings

摘录 / EXCERPT: Faber--Hain--Looijenga's Gorenstein conjecture. The tautological ring $R^*(\overline{\mathcal{M}}_{g,n})$ is a Poincaré duality ring, with socle in top degree $3g-3+n$.

conjecture | open | MathDB 页面 | 原文线索

V2-060707 | MATHDB-319871

The maximal monotone mapping relative-boundary image conjecture

摘录 / EXCERPT: Let $F\colon\mathbb{R}^n\rightrightarrows\mathbb{R}^n$ be a maximal monotone mapping. Then the maximal monotone mapping relative-boundary image conjecture. The image of the map...

conjecture | open | MathDB 页面 | 原文线索

V2-060708 | MATHDB-319874

The fold-map cobordism conjecture for powers of two

摘录 / EXCERPT: Fold-map cobordism conjecture. There exists a 1-dimensional linear subspace $\mathfrak G^1\leq\mathfrak N_n$ such that, if M admits a fold map into a stably parallelizable...

conjecture | open | MathDB 页面 | 原文线索

V2-060709 | MATHDB-319875

Continuity conjecture for similarities of the almost Mathieu rational butterfly

摘录 / EXCERPT: Let the full butterfly be the parameterized spectrum of the almost Mathieu operators, including irrational values of the vertical parameter θ, and let $S=S_{M,r,\pm}$ be a s...

conjecture | open | MathDB 页面 | 原文线索

V2-060710 | MATHDB-319887

Nilpotent approximate subgroup conjecture for general linear groups

摘录 / EXCERPT: Let α be a symmetric subset of $GL(n,\mathbb{F}_p)$ containing 1 and satisfying ... for some $K\ge 1$. Folklore conjecture. There are subgroups $S\trianglerighteq P$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-060711 | MATHDB-319888

Negative extended-Shi conjectures for negative powers of the nabla operator

摘录 / EXCERPT: Let $D^{mn-1}(n)$ be the dilated simplex associated with bounded chambers of the m-extended Shi arrangement, and let ${\sf shi}^m(A)$ and ${\sf ish}^{-1}(A)$ be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-060712 | MATHDB-319893

Generic hyperbolicity conjecture for complex Hénon maps with dominated splitting

摘录 / EXCERPT: Generic hyperbolicity conjecture. Generically, Kupka-Smale complex Hénon maps with dominated splitting are hyperbolic.

conjecture | open | MathDB 页面 | 原文线索

V2-060713 | MATHDB-319897

The spherical restricted-surface-spline Lagrange-basis conjecture

摘录 / EXCERPT: Let Ξ be a quasiuniform set of centers on the sphere, with mesh ratio ρ. For restricted surface splines φ_m and the associated conditionally positive-definite kern...

conjecture | open | MathDB 页面 | 原文线索

V2-060714 | MATHDB-319898

The general finite-smoothness kernel Lagrange-basis conjecture

摘录 / EXCERPT: Let Ξ be a quasiuniform center set with mesh ratio ρ, and let $\phi_{\beta}=G_{\beta}+G_{\beta}*\psi$ with $\beta>d$. Let $S(\phi_{\beta},\Xi)$ be the associated kernel sp...

conjecture | open | MathDB 页面 | 原文线索

V2-060715 | MATHDB-319899

The surface-spline Lagrange-basis conjecture on $SO(3)$

摘录 / EXCERPT: Let Ξ be a quasiuniform subset of $SO(3)$ with mesh ratio ρ. For the kernel k_m on $SO(3)$ and its associated space $S(k_m,\Xi)$, let χ_{ξ} denote the Lagrange fu...

conjecture | open | MathDB 页面 | 原文线索

V2-060716 | MATHDB-319901

Odd-dimensional generalized finite-field distance-set conjecture

摘录 / EXCERPT: Generalized finite-field distance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060717 | MATHDB-319903

Simple-zero conjecture for Wronskians of Hermite polynomials

摘录 / EXCERPT: Let λ be a partition, and let $W_{\lambda}(z)$ denote the Wronskian of Hermite polynomials associated with λ. Simple-zero conjecture. For every partition λ,...

conjecture | open | MathDB 页面 | 原文线索

V2-060718 | MATHDB-319904

Real and imaginary zero conjecture for Wronskians of doubled Hermite partitions

摘录 / EXCERPT: Let $\nu=(\mu_1^2,\dots,\mu_n^2)$ be a doubled partition with distinct parts, where μ_i^2 denotes that the part μ_i occurs twice, and let $W_{\nu}(z)$ be its associated Wro...

conjecture | open | MathDB 页面 | 原文线索

V2-060719 | MATHDB-319905

Possible polarization dichotomy for modified mutual information quantities

摘录 / EXCERPT: Possible polarization dichotomy. Fix a $0<delta<1$. There exists a partition of $\mathcal{S}_n$ into two sets $\mathcal{S}'_n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-060720 | MATHDB-319907

Physical rigidity conjecture for Kloosterman sheaves

摘录 / EXCERPT: Let k be the finite base field, let $\check{G}$ be the dual group, and let $\operatorname{Kl}_{\check{G}}(\phi,\chi)$ be the Kloosterman sheaf associated with a generic linear fu...

conjecture | open | MathDB 页面 | 原文线索

V2-060721 | MATHDB-319908

Strong rigidity characterization conjecture for Kloosterman sheaves

摘录 / EXCERPT: Strong rigidity characterization conjecture. There exists a generic linear function $\phi:I(1)/I(2)\to{\mathbb G}_a$ such that $L\cong\operatorname{Kl}_{\check{G}}(\phi,\chi)$ up t...

conjecture | open | MathDB 页面 | 原文线索

V2-060722 | MATHDB-319909

Functoriality conjecture for Kloosterman sheaves

摘录 / EXCERPT: Let G and G' be split almost simple groups over k whose dual groups satisfy $\check{G}\supset\check{G}'$ and appear in the same line of the paper's global-monodromy table. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-060723 | MATHDB-319910

Finite-dimensional determination conjecture for two-dimensional conformal field theories

摘录 / EXCERPT: Let two two-dimensional conformal field theories have central charge c, spectrum, and three-point functions for their operators. Finite-dimensional determination conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-060724 | MATHDB-319917

The bulk eigenvalue asymptotic conjecture for Wigner matrices

摘录 / EXCERPT: Bulk eigenvalue asymptotic conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060725 | MATHDB-319918

The Hessian conjecture on polynomiality of modular-tree generating functions

摘录 / EXCERPT: Let $a_{g,N}=0$ for $g>0$ and let $a_{0,N}=0$ for $|N|>d$. Let $H(X)$ be the matrix defined by $H(X)=\nabla_XF(X)$ in the Hessian formulation. Suppose that $H(X)$ is nilpotent. Hes...

conjecture | open | MathDB 页面 | 原文线索

V2-060726 | MATHDB-319919

Precise L^p spectrum conjecture for non-compact arithmetic locally symmetric spaces

摘录 / EXCERPT: Precise L^p spectrum conjecture. For every $p\in(1,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-060727 | MATHDB-319925

Yang's test-map discreteness conjecture for subgroups of $SL(2,\mathbb C)$

摘录 / EXCERPT: Let G be a non-elementary subgroup of $SL(2,\mathbb C)$ containing elliptic elements, and let f be a loxodromic or elliptic transformation. For each elliptic element $g\in G$,...

conjecture | open | MathDB 页面 | 原文线索

V2-060728 | MATHDB-319926

Necessity of finite return-intersection sum for positive critical temperature

摘录 / EXCERPT: Let $\alpha\in(0,\infty)$, let $\chi:=\sum_{n\in\mathbb{N}}[\textbf{P}(S_n=0)]^2$, and let β_c denote the critical inverse temperature separating the annealed and quenched cr...

conjecture | open | MathDB 页面 | 原文线索

V2-060729 | MATHDB-319934

Alternating-permutation equivalence implies Wilf-equivalence

摘录 / EXCERPT: For a pattern p, let $A_m(p)$ be the set of alternating permutations of length m that avoid p. Patterns p and q are equivalent for alternating permutations of a parity if...

conjecture | open | MathDB 页面 | 原文线索

V2-060730 | MATHDB-319941

The weak rearrangement conjecture for generalized factor order

摘录 / EXCERPT: Let P be the positive integers with their usual order, let $\mathbb{P}^*$ denote the set of finite words over P, and let $u\backsim v$ mean that the weight generating functions...

conjecture | open | MathDB 页面 | 原文线索

V2-060731 | MATHDB-319942

The strong rearrangement bijection conjecture for generalized factor order

摘录 / EXCERPT: Let $\mathbb{P}^*$ be the set of finite words over the positive integers, let $\mathcal{F}(u)$ denote the set of words containing u as a factor in generalized factor order, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-060732 | MATHDB-319956

Batyrev-Nill polynomiality conjecture for stringy E-functions

摘录 / EXCERPT: Let P be a Gorenstein polytope of index r and dimension d, and let $E_{st}(P;u,v)$ denote its stringy E-function. Batyrev-Nill polynomiality conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-060733 | MATHDB-319957

Batyrev-Nill conjecture on Calabi-Yau dimensions of stringy E-functions

摘录 / EXCERPT: Let P be a Gorenstein polytope of Calabi-Yau dimension n, so that $E_{st}(P;u,v)$ is its stringy E-function. Batyrev-Nill conjecture. The following assertions hold: 1....

conjecture | open | MathDB 页面 | 原文线索

V2-060734 | MATHDB-319959

Krasikov's discrete Laguerre inequality for polynomials

摘录 / EXCERPT: Let $\mathcal{L}\text{-}\mathcal{P}_n$ denote the real polynomials of degree n with only real zeros. For such a polynomial p, let $\mu(p)$ be the minimum spacing between neighb...

conjecture | open | MathDB 页面 | 原文线索

V2-060735 | MATHDB-319960

Rescaled discrete Laguerre inequality

摘录 / EXCERPT: Let $\mathcal{L}\text{-}\mathcal{P}_n$ denote the real polynomials of degree n with only real zeros, let $\mu(p)$ be their mesh size, and define ... Rescaled discrete Laguerre conj...

conjecture | open | MathDB 页面 | 原文线索

V2-060736 | MATHDB-319961

Zero-spacing preservation under a unit difference

摘录 / EXCERPT: Let $\mathcal{L}\text{-}\mathcal{P}_n$ denote the real polynomials of degree n with only real zeros, and let $\mu(p)$ be the minimum spacing between neighboring zeros. For p, d...

conjecture | open | MathDB 页面 | 原文线索

V2-060737 | MATHDB-319964

The quasi-projectivity conjecture for general type Artin groups

摘录 / EXCERPT: Let A_{Γ} be the Artin group associated with a general type Artin graph Γ. Quasi-projectivity conjecture. All general type Artin groups A_{Γ} are not quasi-pro...

conjecture | open | MathDB 页面 | 原文线索

V2-060738 | MATHDB-319967

Necessary lower bound for radial harmonic mappings of annuli with regular metrics

摘录 / EXCERPT: The regular-metric annulus conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-060739 | MATHDB-319970

Pattern-avoidance formulation of the Riemann Hypothesis

摘录 / EXCERPT: Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060740 | MATHDB-319973

Convexity conjecture for the core

摘录 / EXCERPT: Convexity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060741 | MATHDB-319977

Stern and twisted Stern generating-series factorization conjecture

摘录 / EXCERPT: Generating-series factorization conjecture. The series $A(z)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-060742 | MATHDB-319994

Kelly, Kühn and Osthus's short-cycle conjecture for oriented graphs

摘录 / EXCERPT: Kelly, Kühn and Osthus's conjecture. There exists an integer $n_0=n_0(\ell)$ such that every oriented graph G on $n\geq n_0$ vertices with

conjecture | open | MathDB 页面 | 原文线索

V2-060743 | MATHDB-320000

Symmetric tensegrity analogue of Connelly's theorem

摘录 / EXCERPT: Symmetric tensegrity analogue of Connelly's theorem. If $(\hat{G},p)$ has a cable or strut and no symmetry-preserving finite flex, then it has a fully symmetric, non-zero, proper s...

conjecture | open | MathDB 页面 | 原文线索

V2-060744 | MATHDB-320001

Fatou direction conjecture along the Julia filtration

摘录 / EXCERPT: Let $f$colon ${\mathbb{P}^k}\to{\mathbb{P}^k}$ be a holomorphic endomorphism, let T be its Green current, and write $J_q=\operatorname{Supp}(T^q)$. For an integer q with...

conjecture | open | MathDB 页面 | 原文线索

V2-060745 | MATHDB-320010

The duplicate-leaf parse-word conjecture for binary trees

摘录 / EXCERPT: Let T_1' and T_2' be $(n-1)$-leaf binary trees. Let $1 \leq i \leq n-1$, and let T_1 and T_2 be the n-leaf trees obtained from T_1' and T_2' by duplicating leaf i....

conjecture | open | MathDB 页面 | 原文线索

V2-060746 | MATHDB-320011

The path-tree parse-word and level-restriction conjecture

摘录 / EXCERPT: Let $n \geq 4$, and let T_1 and T_2 be n-leaf path trees such that leaf 1 is on level 1 in T_1 and leaf n is on level 1 in T_2. A pair is mutually crooked if it c...

conjecture | open | MathDB 页面 | 原文线索

V2-060747 | MATHDB-320012

The double-zero parse-word conjecture for path trees

摘录 / EXCERPT: Let $n \geq 4$. A path tree is the path-tree object used in the paper, and a parse word is a word that parses a binary tree. Double-zero parse-word conjecture. Every pair of n-le...

conjecture | open | MathDB 页面 | 原文线索

V2-060748 | MATHDB-320013

Global well-posedness and scattering conjecture for the defocusing mass-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Global well-posedness and scattering conjecture. For $d\geq 1$, the defocusing, mass-critical nonlinear Schrödinger initial value problem $$ is globally well-posed for...

conjecture | open | MathDB 页面 | 原文线索

V2-060749 | MATHDB-320017

The torsion conjecture for the Selmer group of a CM elliptic curve

摘录 / EXCERPT: Let E be the elliptic curve and F_∞/F the extension in the paper. Let $\mathcal{G}=\operatorname{Gal}(F_\infty/F)$, let $\mathcal{H}$ be the closed normal subgroup with...

conjecture | open | MathDB 页面 | 原文线索

V2-060750 | MATHDB-320018

Existence conjecture for the non-commutative p-adic L-function of a CM elliptic curve

摘录 / EXCERPT: Let E/F be the CM elliptic curve, let F_∞/F be the p-adic Lie extension, and put $\mathcal{G}=\operatorname{Gal}(F_\infty/F)$. Let S^* be the Ore set used to localize...

conjecture | open | MathDB 页面 | 原文线索

V2-060751 | MATHDB-320019

The non-commutative main conjecture for a CM elliptic curve

摘录 / EXCERPT: In the notation above, let $\mathcal{L}_E\ \epsilon\ K_1(\Lambda(\mathcal{G})_{S^*})$ be the conjectural p-adic L-function, let $X(E/F_\infty)$ be the Pontryagin dual of the Se...

conjecture | open | MathDB 页面 | 原文线索

V2-060752 | MATHDB-320021

Stable-limit conjecture for the Dirichlet eigenvalue counting function of stable trees

摘录 / EXCERPT: Let $N^D(\lambda)$ be the Dirichlet eigenvalue counting function for an α-stable tree, let $m(\infty)$ be the limiting first-order coefficient, and let Z_α be an...

conjecture | open | MathDB 页面 | 原文线索

V2-060753 | MATHDB-320024

The fundamental gap conjecture for convex Schrödinger operators

摘录 / EXCERPT: Fundamental gap conjecture. The eigenvalues satisfy the inequality above.

conjecture | open | MathDB 页面 | 原文线索

V2-060754 | MATHDB-320035

Hyperplane Diameter Conjecture for bounded cells of arrangements

摘录 / EXCERPT: Hyperplane Diameter Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-060755 | MATHDB-320036

Hirsch-sharpness conjecture for 3-way transportation polytopes

摘录 / EXCERPT: Hirsch-sharpness conjecture for 3-way transportation polytopes.

conjecture | open | MathDB 页面 | 原文线索

V2-060756 | MATHDB-320039

The sharp $5/2$ inverse-degree diameter conjecture for planar graphs

摘录 / EXCERPT: Sharp $5/2$ diameter conjecture. For any planar graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-060757 | MATHDB-320041

Convex-ear decomposition inheritance for rank-selected face posets

摘录 / EXCERPT: Let Σ be a $(d-1)$-dimensional complex admitting a convex-ear decomposition, and let $S\subseteq [d]$. Write P_Σ for the face poset of Σ, and $(P_\Sigma)_S$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-060758 | MATHDB-320063

The order-four subsemigroup conjecture for finite non-adequate amiable semigroups

摘录 / EXCERPT: Subsemigroup conjecture. If S is not adequate, then S contains a subsemigroup isomorphic to M.

conjecture | open | MathDB 页面 | 原文线索

V2-060759 | MATHDB-320069

The completion-invariance conjecture for the graph invariant c_2

摘录 / EXCERPT: Let G be a primitive-divergent graph in phi^4 theory, and let $c_2(G)$ denote its graph invariant. The completion relation identifies graphs whose completions are isomorphic; w...

conjecture | open | MathDB 页面 | 原文线索

V2-060760 | MATHDB-320070

Period-determines-c_2 conjecture for primitive-divergent graphs

摘录 / EXCERPT: Let G_1 and G_2 be primitive-divergent graphs in phi^4 theory, let I_{G_i} denote their residues, and let $c_2(G_i)$ be their graph invariants. Period-determines-c_2 conj...

conjecture | open | MathDB 页面 | 原文线索

V2-060761 | MATHDB-320074

Generation conjecture for orthogonal semi-invariants of symmetric quivers

摘录 / EXCERPT: Generation conjecture for orthogonal semi-invariants. The ring $OSI(Q,\beta)$ is generated by the semi-invariants

conjecture | open | MathDB 页面 | 原文线索

V2-060762 | MATHDB-320076

The d-complete poset hook-length bijection conjecture

摘录 / EXCERPT: The d-complete poset hook-length bijection conjecture. If $G\in\mathcal{G}_c^0$, then $s(G)=c$ and $e(G)=G$; if $G\in\mathcal{G}_c^k$ for $k\geq1$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-060763 | MATHDB-320078

Segre's nonspeciality conjecture for fat point series on very general surfaces

摘录 / EXCERPT: Let $S\subset\mathbb{P}^3$ be a very general surface of degree at least 4, and let $p_1,\ldots,p_r$ be general points on S. For multiplicities $m_1,\ldots,m_r$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-060764 | MATHDB-320079

Classification conjecture for special fat point series on very general surfaces

摘录 / EXCERPT: Let $S\subset\mathbb{P}^3$ be a very general surface of degree $d\geq4$, and let $\mathscr{L}=\mathscr{L}_e^S(m_1,\ldots,m_r)$ be a general fat point series. Classification conject...

conjecture | open | MathDB 页面 | 原文线索

V2-060765 | MATHDB-320082

The nonstandard weighted fundamental lemma for metaplectic groups

摘录 / EXCERPT: Lemme fondamental pondéré non standard. Si $Y_1\in\mathfrak{m}_{1,G_1-\mathrm{reg}}(F)$ et $Y_2\in\mathfrak{m}_{2,G_2-\mathrm{reg}}(F)$ se correspondent, alors

conjecture | open | MathDB 页面 | 原文线索

V2-060766 | MATHDB-320086

Robust-regime conjecture for the splitting model

摘录 / EXCERPT: Robust-regime conjecture. Theorem $$ holds for all

conjecture | open | MathDB 页面 | 原文线索

V2-060767 | MATHDB-320087

Polygonal limiting-shape conjecture for the splitting automaton

摘录 / EXCERPT: Polygonal limiting-shape conjecture. For every

conjecture | open | MathDB 页面 | 原文线索

V2-060768 | MATHDB-320089

The AVV conjecture on Ramanujan's gamma-function remainder

摘录 / EXCERPT: AVV conjecture. The function H is increasing from $(1,\infty)$ into $(1/100,1/30)$. This conjecture is attributed to the authors' earlier work and is motivated by Ramanujan's rec...

conjecture | open | MathDB 页面 | 原文线索

V2-060769 | MATHDB-320093

Uniform distribution conjecture for convex periodic potentials

摘录 / EXCERPT: Uniform distribution conjecture. For every such f, $M_f(\theta_1,\dots,\theta_n)$ is maximised when the points are uniformly distributed on $[0,2\pi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-060770 | MATHDB-320098

The equivalence conjecture for irreducible finite-dimensional sheaves and pure motives

摘录 / EXCERPT: Let ${\mathbb B}^{\bullet}$ be the fully faithful functor from pure k-motives to semisimple sheaves of finite length of finite-dimensional graded ${\mathbb Q}$-vector spaces, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-060771 | MATHDB-320099

The embedding conjecture for simple A1-invariant sheaves

摘录 / EXCERPT: Let k be the base field, and let ${\mathcal F}$ be a simple ${\mathbb A}^1$-invariant sheaf. Write $\Omega^{\bullet}_{k(Y)|k}$ for the graded differential forms of the function f...

conjecture | open | MathDB 页面 | 原文线索

V2-060772 | MATHDB-320101

The semisimplicity conjecture for Vbullet

摘录 / EXCERPT: Let $V^{\bullet}$ be the sheaf constructed in the paper from closed differential forms and related quotients. Semisimplicity conjecture. The sheaf $V^{\bullet}$ is semisimple. The...

conjecture | open | MathDB 页面 | 原文线索

V2-060773 | MATHDB-320106

Ball-complement outer-bound conjecture for Aloha stability regions

摘录 / EXCERPT: Ball-complement outer-bound conjecture. The set $\Lambda_{\rm bc}$ is an outer bound for Λ, namely

conjecture | open | MathDB 页面 | 原文线索

V2-060774 | MATHDB-320107

Ellipsoidal inner and outer bounds conjecture for Aloha stability regions

摘录 / EXCERPT: Let $n\geq 2$, let Λ be the stability region, and let $\mathcal{S}$ be the unit simplex. Define ellipsoids $\mathcal{E}_I$ and $\mathcal{E}_O$ with center...

conjecture | open | MathDB 页面 | 原文线索

V2-060775 | MATHDB-320118

Integral Swan class conjecture

摘录 / EXCERPT: Let U and V be as in the paper, let $\mathcal F$ be the locally constant constructible sheaf under consideration, and let ζ_{p^∞} denote the group of all p-power r...

conjecture | open | MathDB 页面 | 原文线索

V2-060776 | MATHDB-320120

Montassier et al.'s bounded-degree forest decomposition conjecture

摘录 / EXCERPT: Montassier et al.'s conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060777 | MATHDB-320122

Majorization minimality conjecture for joint-potential minimizers

摘录 / EXCERPT: Let $(m,\mathbf{k},d)$ be parameters and let $\mathbf{v}\in\mathbb{R}_{>0}^d$. Theorem provides a vector $\lambda_\mathbf{v}(m,\mathbf{k},d)$ and a feasible set...

conjecture | open | MathDB 页面 | 原文线索

V2-060778 | MATHDB-320123

The conjecture that unimodular lattices attain secrecy gain at y=1

摘录 / EXCERPT: Let Λ be an n-dimensional unimodular lattice, and let $\Xi_{\Lambda}(y)$ be its secrecy function for $y>0$. The secrecy gain is...

conjecture | open | MathDB 页面 | 原文线索

V2-060779 | MATHDB-320129

Schramm-Loewner evolution scaling-limit conjecture for self-avoiding walks

摘录 / EXCERPT: Self-avoiding-walk SLE conjecture. For $x=x_c$, the law of γ_δ in $(\Omega_\delta,a_\delta,b_\delta)$ converges as $\delta\to0$ to chordal Schramm-Loewner Evolution wit...

conjecture | open | MathDB 页面 | 原文线索

V2-060780 | MATHDB-320130

Wenger's conjecture on the chromatic number of squares of planar graphs

摘录 / EXCERPT: Let G be a planar graph with maximum degree Δ and let G^2 denote its square, in which two vertices are adjacent whenever their distance in G is at most 2. Wenger's c...

conjecture | open | MathDB 页面 | 原文线索

V2-060781 | MATHDB-320134

BKMP conjecture for open Gromov-Witten potentials of toric Calabi-Yau threefolds

摘录 / EXCERPT: BKMP conjecture. These integrated correlators are equal to the A-model framed open Gromov-Witten potentials of (X,L), where L is the mirror brane to $\Gamma_X\subset\widehat X$...

conjecture | open | MathDB 页面 | 原文线索

V2-060782 | MATHDB-320135

Open orbifold mirror symmetry conjecture for Gromov-Witten potentials

摘录 / EXCERPT: Open orbifold mirror symmetry conjecture. The open orbifold potentials $F_{g,h}^{(\mathfrak{X},\mathcal{L})}$ are given, after applying the orbifold open and closed mirror maps, by

conjecture | open | MathDB 页面 | 原文线索

V2-060783 | MATHDB-320137

Symmetry-preserving contraction conjecture for block and hole polyhedra

摘录 / EXCERPT: Let an abstract block and hole polyhedron have graph symmetry group $\Phi(\mathcal{S})$, with no vertices or edges fixed. Symmetry-preserving contraction conjecture. There is a sym...

conjecture | open | MathDB 页面 | 原文线索

V2-060784 | MATHDB-320138

Cut-cycle criterion for isostatic block and hole polyhedra

摘录 / EXCERPT: Let a block and hole polyhedron $\mathcal{P}$ have cut cycles $\mathcal{C}$ of size c, and let $\mathcal{P}_{C}'$ denote each corresponding component, with block sizes b_i' and...

conjecture | open | MathDB 页面 | 原文线索

V2-060785 | MATHDB-320141

The isomorphism problem for groups enveloped by right-angled Artin groups

摘录 / EXCERPT: A group is enveloped by a right-angled Artin group if it embeds in a right-angled Artin group. The enveloped-group isomorphism conjecture. The isomorphism problem for groups envelo...

conjecture | open | MathDB 页面 | 原文线索

V2-060786 | MATHDB-320159

Row-sum conjecture for inverse double-dimer matrices

摘录 / EXCERPT: Let M^{-1} be the inverse matrix indexed by paths, and let λ be a row of order n. For a chord c of λ, write $|c|$ for its length, and let $f_{\lambda/\mu}(q)$...

conjecture | open | MathDB 页面 | 原文线索

V2-060787 | MATHDB-320160

Column-sum conjecture for inverse double-dimer matrices

摘录 / EXCERPT: Let M^{-1} be the inverse matrix indexed by paths, and let μ be a column index. For each chord c under μ, let its height be the number specified by the diagram, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060788 | MATHDB-320161

Almost-sure BIC consistency for Property 2

摘录 / EXCERPT: In the setting of the preceding proposition, consider a countable class of maps and the BIC penalty. BIC consistency conjecture. We almost surely have Property 2 from the propositi...

conjecture | open | MathDB 页面 | 原文线索

V2-060789 | MATHDB-320177

Compact-type perfect-pairing and freeness conjecture for tautological rings

摘录 / EXCERPT: Compact-type perfect-pairing and freeness conjecture. (C) These pairings are perfect for $k\geq 0$. (D) In addition to (C), $R_c^*(M_{g,n}^{ct})$ is a free $R^*(M_{g,n}^{ct})$-modu...

conjecture | refuted | MathDB 页面 | 原文线索

V2-060790 | MATHDB-320184

Strang's bandwidth factorization conjecture for permutation matrices

摘录 / EXCERPT: Let P be a finite permutation matrix of bandwidth $w>0$, where bandwidth 1 means tridiagonal. A bandwidth--1 permutation matrix has nonzero entries only on the main diagonal an...

conjecture | open | MathDB 页面 | 原文线索

V2-060791 | MATHDB-320185

Greedy bubble factorization conjecture for permutation matrices

摘录 / EXCERPT: Let P be a finite permutation matrix of bandwidth $w>0$. A greedy bubble matrix of P is a product of reducing and overtaking swaps of P such that P and the resulting matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-060792 | MATHDB-320189

Conjecture PC5 on hereditary maximum parsimony trees

摘录 / EXCERPT: Let $S:=f_1f_2\ldots f_n$ be a sequence of characters (an alignment) on a set X of taxa, with $|X|=m$, and let $\mathcal{T}$ be a maximum parsimony tree for S. For a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-060793 | MATHDB-320195

Tight extension conjecture for valuations

摘录 / EXCERPT: Tight extension conjecture. The valuation nu admits at least one tight extension $hat\nu_-$. It can be chosen so that, for rings R' sufficiently far in $cal T$,

conjecture | open | MathDB 页面 | 原文线索

V2-060794 | MATHDB-320205

The k-Schur filtration conjecture

摘录 / EXCERPT: The k-Schur filtration conjecture. There are polynomials $\tilde b_{\mu\lambda}^{(k\to k')}(t)\in\mathbb{Z}_{\geq 0}[t]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-060795 | MATHDB-320206

The charged branching-polynomial conjecture for k-Schur functions

摘录 / EXCERPT: Charged branching-polynomial conjecture. For all $\lambda\in\mathcal{C}^{k+1}$ and $\mu\in\mathcal{C}^k$,

conjecture | open | MathDB 页面 | 原文线索

V2-060796 | MATHDB-320209

The $(k-1,1)$ minor conjecture

摘录 / EXCERPT: $(k-1,1)$ minor conjecture. Every k-chromatic graph has either a K_k minor or a

conjecture | open | MathDB 页面 | 原文线索

V2-060797 | MATHDB-320222

Convexity conjecture for the minimal energy of doubly connected domains

摘录 / EXCERPT: Let Ω^* be a bounded doubly connected domain in $\mathbb{C}$, and let $\mathsf{E}(\tau,\Omega^*)$ denote the infimum of the Dirichlet energy over deformations from a doubly...

conjecture | open | MathDB 页面 | 原文线索

V2-060798 | MATHDB-320223

Sharp modulus inequality for energy-minimal diffeomorphisms

摘录 / EXCERPT: Let Ω and Ω^* be bounded doubly connected domains in $\mathbb{C}$, and suppose that there is an energy-minimal diffeomorphism ... Write $\operatorname{Mod}\Omega$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-060799 | MATHDB-320226

The compositional shuffle conjecture for B_α

摘录 / EXCERPT: Let $\alpha\models n$ be a composition, let $B_\alpha[X;q]$ be the generalized Hall-Littlewood symmetric function indexed by α, let $e_n[X]$ be the elementary symmetric func...

conjecture | open | MathDB 页面 | 原文线索

V2-060800 | MATHDB-320227

The compositional shuffle conjecture for C_α

摘录 / EXCERPT: Let $\alpha\models n$ be a composition, let $C_\alpha[X;q]$ be the generalized Hall-Littlewood symmetric function indexed by α, let $e_n[X]$ be the elementary symmetric func...

conjecture | open | MathDB 页面 | 原文线索

V2-060801 | MATHDB-320228

The compositional shuffle conjecture for ∇C_α

摘录 / EXCERPT: Let $\alpha\models n$ be a composition, let $C_\alpha[X;q]$ be the generalized Hall-Littlewood symmetric function indexed by α, let $\mathcal{W}P_D$ denote the set of words...

conjecture | open | MathDB 页面 | 原文线索

V2-060802 | MATHDB-320231

The normal-subgroup conjecture for quasiparabolic subgroups

摘录 / EXCERPT: Let W be a Coxeter group and let $H\subset W$ be a quasiparabolic subgroup. Let N be the normal subgroup associated to H in Theorem. Normal-subgroup conjecture. The subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-060803 | MATHDB-320240

Spectral-band conjecture for the one-dimensional Bernoulli displacement model

摘录 / EXCERPT: Spectral-band conjecture. For every $\lambda\neq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-060804 | MATHDB-320253

The exceptional-value conjecture for Engel trace maps

摘录 / EXCERPT: Exceptional-value conjecture. Unless either $a=1$ and $\sqrt{2}\notin\mathbb{F}_q$, or the triple (q,a,n) occurs in one of the explicitly listed exceptional cases, the map...

conjecture | open | MathDB 页面 | 原文线索

V2-060805 | MATHDB-320261

Finite arithmetic intersection conjecture for special cycles

摘录 / EXCERPT: Let L be a lattice with discriminant group L'/L, let $\bar\rho_L$ be its associated representation, and let $\mathcal{X}_K$ be a regular projective flat integral model over...

conjecture | open | MathDB 页面 | 原文线索

V2-060806 | MATHDB-320262

Faltings-height formula conjecture for arithmetic special cycles

摘录 / EXCERPT: Let d be the dimension parameter used in the paper, let $f\in H_{1-d,\bar\rho_L}$, and let $\hat{\mathcal{Z}}(f)$ and $\mathcal{Z}(W)$ be the associated arithmetic divisor and cy...

conjecture | open | MathDB 页面 | 原文线索

V2-060807 | MATHDB-320263

BP-collapse conjecture for good codes

摘录 / EXCERPT: Let ${\mathcal{C}}$ be a code drawn from a sequence of “good” codes, and let ${\bf Y}_2$ be the channel output at the relay. Let ${\bf Y}^{\textrm{BP}}_2$ be the output of belief p...

conjecture | open | MathDB 页面 | 原文线索

V2-060808 | MATHDB-320264

Necessity conjecture for achievable soft-DF-BP rates

摘录 / EXCERPT: Let soft-DF-BP operate with codes confined to sequences of “good” codes. Let the conditions of the source's Theorem formalize sufficient conditions for a rate to be achievable with...

conjecture | open | MathDB 页面 | 原文线索

V2-060809 | MATHDB-320268

Wegner's conjecture on good covers

摘录 / EXCERPT: Let a good cover in $\mathbb{R}^d$ be a collection of open sets such that every subcollection has intersection either empty or homeomorphic to an open d-ball. The nerve of a coll...

conjecture | open | MathDB 页面 | 原文线索

V2-060810 | MATHDB-320270

Poonen-Stoll conjecture on rational points on genus-two curves

摘录 / EXCERPT: Consider the density obtained by ordering genus-two curves of the form $y^2=f(x)$, where $f=f_6x^6+\dots+f_1x+f_0\in{\mathbb Z}[x]$, by the condition $\max\{|f_j|\}\leq N$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060811 | MATHDB-320271

Finite descent conjecture for curves

摘录 / EXCERPT: Let C be a curve over ${\mathbb Q}$, and let $\operatorname{Sel}^{(n)}(C)$ denote the n-Selmer set, consisting of the n-coverings of C with points everywhere locally. Finit...

conjecture | open | MathDB 页面 | 原文线索

V2-060812 | MATHDB-320273

Coset Mordell-Weil sieve conjecture

摘录 / EXCERPT: Let C be a curve over ${\mathbb Q}$ with Jacobian J, let $\iota:C\to J$ be the embedding used by the Mordell-Weil sieve, let $N\geq1$, and let $Q\in J({\mathbb Q})$. A coset of...

conjecture | open | MathDB 页面 | 原文线索

V2-060813 | MATHDB-320280

Ayyer-Mallick's maximal-degeneracy conjecture for asymmetric annihilation matrices

摘录 / EXCERPT: Let M_L be the Markov matrix of the asymmetric annihilation process on a system of size L. The characteristic polynomial has only $2L$ distinct eigenvalues out of a possible...

conjecture | open | MathDB 页面 | 原文线索

V2-060814 | MATHDB-320289

Zhu's unimodality conjecture for vertebrated graph independence polynomials

摘录 / EXCERPT: Let $V_n^{(m)}$ be the vertebrated graph, and let $I(V_n^{(m)};x)$ denote its independence polynomial. Zhu's conjecture. For $n,m\text{ with }n,m\text{ nonnegative integers}$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-060815 | MATHDB-320292

NLKG scattering conjecture below the ground-state threshold

摘录 / EXCERPT: Let $\mu=\pm1$. Let $(u_0,u_1)\in H^1_x\times L^2_x$. In the focusing case, assume additionally that $M(u_0)<M(Q)$ and $E(u_0,u_1)<E(Q)$, where Q is the ground state. A solution...

conjecture | open | MathDB 页面 | 原文线索

V2-060816 | MATHDB-320293

Mass-critical cubic NLS scattering conjecture

摘录 / EXCERPT: Let $\mu=\pm1$, and let $w_0\in L^2_x(\mathbb{R}^2)$. In the focusing case assume $M(w_0)<M(Q)$. A solution scatters if there are $w_\pm\in L^2_x$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-060817 | MATHDB-320302

Guo-Zeng's mixed ballot-number congruences

摘录 / EXCERPT: Let $A_{n,k}$ be the ballot numbers ... For $r\in\mathbb{N}$, $s,t\in\mathbb{Z}^+$ with $r+s+t\equiv1\pmod2$, $n\in\mathbb{Z}^+$, and $\varepsilon=\pm1$, Guo-Zeng's adjacent-index co...

conjecture | open | MathDB 页面 | 原文线索

V2-060818 | MATHDB-320305

Guo-Zeng's three-factor ballot-number congruences

摘录 / EXCERPT: Let $A_{n,k}$ be the ballot numbers ... For all $n,r,s,t\in\mathbb{Z}^+$ with $r+s+t\equiv1\pmod2$, and with $\varepsilon=\pm1$, Guo-Zeng's three-factor ballot-number conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-060819 | MATHDB-320306

Guo-Zeng's general product ballot-number congruence

摘录 / EXCERPT: Let $A_{n,k}$ be the ballot numbers ... For all $n,r_1,\ldots,r_m\in\mathbb{Z}^+$ such that $r_1+\cdots+r_m\equiv1\pmod2$, and with $\varepsilon=\pm1$, Guo-Zeng's general ballot-prod...

conjecture | open | MathDB 页面 | 原文线索

V2-060820 | MATHDB-320311

Wagner's sum-of-squares conjecture for spanning-forest Rayleigh differences

摘录 / EXCERPT: Let $\mathcal G$ be a graph, and let e and f be distinct edges. Write $G=F(\mathcal G;\mathbf y)$ for the generating polynomial of its spanning forests, and let $\mathscr S$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-060821 | MATHDB-320316

Regularity and two-class conjecture for isolated singularities of p-Laplacian equations

摘录 / EXCERPT: Let $Q'(u)=0$ be the p-Laplacian type equation in a domain Ω, let $\zeta\in\{0,\infty\}$ be an isolated singular point, and let $\mathcal{G}_\zeta$ denote the germs of posit...

conjecture | open | MathDB 页面 | 原文线索

V2-060822 | MATHDB-320318

The Image Conjecture for regular sequences

摘录 / EXCERPT: Image Conjecture. The image of $\mathcal D$ is a Mathieu subspace of B.

conjecture | open | MathDB 页面 | 原文线索

V2-060823 | MATHDB-320330

The orbifold C_{n,m} conjecture

摘录 / EXCERPT: Let $(X\vert\Delta)$ be a smooth pair and let $f:X\to Y$ be a neat fibration, with generic fibre $(X_y\vert\Delta_y)$ and orbifold base $(Y\vert\Delta_{(f\vert\Delta)})$. Orbifold...

conjecture | open | MathDB 页面 | 原文线索

V2-060824 | MATHDB-320337

The local limit conjecture for stable commutator length

摘录 / EXCERPT: Let G be any finitely generated group, let S be a finite symmetric generating set, and suppose that the space $Q(G)$ of homogeneous quasimorphisms modulo homomorphisms is finit...

conjecture | open | MathDB 页面 | 原文线索

V2-060825 | MATHDB-320344

The linear recurrence conjecture for Jack-polynomial coefficients

摘录 / EXCERPT: Let α be fixed, let ρ be a partition, and let $a_{\rho}^{k,(\alpha)}$ be the coefficients defined by ... For a partition ρ, write $\rho\cup(m)$ for adjoining a part...

conjecture | open | MathDB 页面 | 原文线索

V2-060826 | MATHDB-320348

Hyperelliptic Jacobian Selmer-distribution conjecture

摘录 / EXCERPT: Fix a global field k of characteristic not 2, a positive integer g, and let $f\in k[x]$ range over separable polynomials of degree $2g+1$ with coefficients of height bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-060827 | MATHDB-320349

Even-genus hyperelliptic Jacobian 2-Selmer conjecture

摘录 / EXCERPT: Fix a global field k of characteristic not 2, and fix an even positive integer g. Let $f\in k[x]$ range over polynomials of degree $2g+2$ with coefficients of height bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-060828 | MATHDB-320364

Random linear Euclidean distance matrix conjecture

摘录 / EXCERPT: Let M be a random linear Euclidean distance matrix of dimension n, so that M is an $n\times n$ matrix of pairwise squared distances generated by randomly chosen points on a l...

conjecture | open | MathDB 页面 | 原文线索

V2-060829 | MATHDB-320370

Goodwillie-Whitehead conjecture: weak form

摘录 / EXCERPT: Let $E_r^{k,t}(S^1)$ denote the pages of the Goodwillie spectral sequence (GSS) for S^1. Goodwillie-Whitehead conjecture, weak form. The GSS for S^1 collapses at the E_2-page...

conjecture | open | MathDB 页面 | 原文线索

V2-060830 | MATHDB-320371

Kadell's first constant-term conjecture for the Dyson product

摘录 / EXCERPT: Let $I=\{i_1,\ldots,i_m\}$ be a proper subset of $\{0,1,\ldots,n\}$ and let $J=\{j_1,\ldots,j_m\}$ be a multi-subset of $\{0,1,\ldots,n\}\setminus I$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-060831 | MATHDB-320372

Matsaev's conjecture for contractions on commutative L^p-spaces

摘录 / EXCERPT: Let $1<p<\infty$ with $p\ne 2$, let Ω be a measure space, and let T be a contraction on $L^p(\Omega)$. Let S be the right shift on ℓ^p, defined by ... For a complex p...

conjecture | open | MathDB 页面 | 原文线索

V2-060832 | MATHDB-320383

Finite-defect numerical-range conjecture for unitary dilations

摘录 / EXCERPT: Suppose $T\in {\mathcal L}({\mathcal H})$ is a contraction with ... Let ${\mathcal U}$ be the set of unitary N-dilations of T to ${\mathcal H}\oplus \mathbb{C}^N$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-060833 | MATHDB-320387

Integrability conjecture for the canonical multidimensional system associated with system (64)

摘录 / EXCERPT: The integrability conjecture. The corresponding $(n+4)$-dimensional system

conjecture | open | MathDB 页面 | 原文线索

V2-060834 | MATHDB-320388

Generation conjecture for integrable (2+1)-dimensional chains

摘录 / EXCERPT: Let an integrable $(2+1)$-dimensional chain be of the form ... where the corresponding $(1+1)$-dimensional chain arises from a two-parameter linear family with parameters γ_1...

conjecture | open | MathDB 页面 | 原文线索

V2-060835 | MATHDB-320415

Positivity conjecture for Schur functions under a hypergeometric specialization

摘录 / EXCERPT: Let $x>0$, and specialize the complete symmetric functions by ... For a partition λ, let s_λ denote the corresponding Schur function. Schur-function positivity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-060836 | MATHDB-320419

The conjecture on local gradient control for continuously graded composites

摘录 / EXCERPT: Let $S\subset\Omega$ be an open set with closure contained in Ω. Consider continuously graded composites with inclusions having $C^{1,\alpha}$ boundaries, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060837 | MATHDB-320425

The Lie group conjecture on Weyl-compatible configuration maps

摘录 / EXCERPT: Lie group conjecture. There exists a continuous map

conjecture | open | MathDB 页面 | 原文线索

V2-060838 | MATHDB-320428

Uniform boundedness conjecture for reflection length in Coxeter groups

摘录 / EXCERPT: Let W be a Coxeter group, and let ℓ_R denote its reflection length function. The group W is of spherical type if it is finite, and of affine type if it is an affine Coxete...

conjecture | open | MathDB 页面 | 原文线索

V2-060839 | MATHDB-320432

The one-component characterization for bounded symbols of truncated Toeplitz operators

摘录 / EXCERPT: One-component characterization conjecture. The equivalent conditions of Theorem 3 are fulfilled if and only if θ is one-component.

conjecture | open | MathDB 页面 | 原文线索

V2-060840 | MATHDB-320436

Schwarz potential conjecture for a three-dimensional Neumann oval

摘录 / EXCERPT: Three-dimensional Neumann-oval conjecture. W can be analytically continued throughout $\Omega\setminus B$. This is a proposed higher-dimensional analogue of the Schwarz potential...

conjecture | open | MathDB 页面 | 原文线索

V2-060841 | MATHDB-320438

Elliptic Schwarz potential conjecture

摘录 / EXCERPT: Let α be an entire function. On a nonsingular analytic surface, let u solve the Cauchy problem for ... with entire data. Let q solve ... and let v solve the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-060842 | MATHDB-320445

The 2-isogeny local root-number formula

摘录 / EXCERPT: Let F be a local field of characteristic zero, and let E/F be an elliptic curve with a 2-isogeny $\phi:E\to E'$ over F. In the notation of the source, let $a,b\in F$ be the p...

conjecture | open | MathDB 页面 | 原文线索

V2-060843 | MATHDB-320447

Parity over the biquadratic field $\mathbb Q(i,\sqrt{17})$

摘录 / EXCERPT: Let $K=\mathbb Q(\sqrt{-1},\sqrt{17})$. Biquadratic-field parity conjecture. Every elliptic curve $E/\mathbb Q$ has even Mordell-Weil rank over K: ... The proposed statement follow...

conjecture | open | MathDB 页面 | 原文线索

V2-060844 | MATHDB-320448

Positive rank over the eighth cyclotomic field

摘录 / EXCERPT: Let $E/\mathbb Q$ be an elliptic curve with split multiplicative reduction at 2. Cyclotomic positive-rank conjecture. The curve has infinitely many rational points over...

conjecture | open | MathDB 页面 | 原文线索

V2-060845 | MATHDB-320450

Conforti-Johnson's Petersen-minor T-join min-max conjecture

摘录 / EXCERPT: Let G be a graph, and let T be the set of its odd-degree vertices. A T-join is a subgraph whose odd-degree vertices are precisely the vertices in T, and a T-cut is a cut...

conjecture | open | MathDB 页面 | 原文线索

V2-060846 | MATHDB-320460

Brändén's real-rootedness conjecture for the second derived Boros-Moll polynomial

摘录 / EXCERPT: Brändén's conjecture. The polynomial $R_m(x)$ has only real zeros for every positive integer m. The source states that the corresponding conjectures on derived polynomials imply...

conjecture | open | MathDB 页面 | 原文线索

V2-060847 | MATHDB-320465

The Generalized Secant Conjecture for Schubert problems on Grassmannians

摘录 / EXCERPT: Let $G(k,n)$ be a Grassmannian, let γ be a rational normal curve, and let $\lambda^1,\dotsc,\lambda^m$ be a Schubert problem on $G(k,n)$. A generalized secant subspace to...

conjecture | open | MathDB 页面 | 原文线索

V2-060848 | MATHDB-320467

The almost reducibility conjecture for analytic almost periodic Jacobi cocycles

摘录 / EXCERPT: Let (β,A_ϵ^E) be an analytic almost periodic Jacobi cocycle, and suppose that its determinant is bounded away from zero. Almost reducibility conjecture for Jacobi cocy...

conjecture | open | MathDB 页面 | 原文线索

V2-060849 | MATHDB-320474

Threshold-policy entropy optimality conjecture in the special case

摘录 / EXCERPT: Threshold-policy entropy optimality conjecture. In the special case, the infimum of entropy attainable under threshold policies is the same as that under general policies.

conjecture | open | MathDB 页面 | 原文线索

V2-060850 | MATHDB-320475

Threshold-policy conjecture for the greedy observation policy

摘录 / EXCERPT: Greedy threshold-policy conjecture. The greedy policy is always a threshold policy.

conjecture | open | MathDB 页面 | 原文线索

V2-060851 | MATHDB-320481

Bipartite finite-field Falconer distance conjecture

摘录 / EXCERPT: Bipartite finite-field Falconer distance conjecture. If $|A||B|\geq Cq^d$, with $C>0$ large enough, then

conjecture | open | MathDB 页面 | 原文线索

V2-060852 | MATHDB-320482

Generalized S-genus identity for ternary quadratic forms over totally real fields

摘录 / EXCERPT: Let F be a totally real number field of class number one in which 2 is inert, and let A_F be its ring of integers. Suppose that $2S(A_F^\times)^2$ is the squarefree determina...

conjecture | open | MathDB 页面 | 原文线索

V2-060853 | MATHDB-320493

Inoue-surface conjecture for non-elliptic surfaces with multiplicatively automorphic pluriharmonic functions

摘录 / EXCERPT: Let X be a non-elliptic compact complex surface admitting a positive, multiplicatively automorphic, non-constant, pluriharmonic function on a ${\mathbb Z}$-covering. Inoue-surfac...

conjecture | open | MathDB 页面 | 原文线索

V2-060854 | MATHDB-320509

Extended Bertram-Feinberg-Mukai conjecture for canonical coherent systems

摘录 / EXCERPT: Extended Bertram-Feinberg-Mukai conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060855 | MATHDB-320518

Gross-Siebert central-fiber and large-complex-limit conjecture for Hitchin moduli spaces

摘录 / EXCERPT: Consider the Gross-Siebert toric degeneration of Hitchin's moduli spaces and the large complex degeneration of the same moduli space. The central fiber of the first is the special...

conjecture | open | MathDB 页面 | 原文线索

V2-060856 | MATHDB-320522

Laman-spanning conjecture for the $(3+2)$-core of a random graph

摘录 / EXCERPT: Let $G(n,c/n)$ be an Erdős-Rényi random graph, and let c_2 be the threshold above which the $(3+2)$-core emerges. A subgraph is Laman-spanning if it contains a spanning Laman gra...

conjecture | open | MathDB 页面 | 原文线索

V2-060857 | MATHDB-320524

Spinor-genus correspondence conjecture for ternary quadratic-form genera

摘录 / EXCERPT: Spinor-genus correspondence conjecture. Suppose that G_1 and G_2 each have exactly two spinor genera. Then G_1 has spinor exceptional integers if and only if G_2 has spinor...

conjecture | open | MathDB 页面 | 原文线索

V2-060858 | MATHDB-320525

Spinor-genus preservation conjecture for genus-correspondences

摘录 / EXCERPT: Spinor-genus preservation conjecture. Suppose that G_1 and G_2 have exactly the same number, namely some power 2^j, of spinor genera. Then the genus-correspondence respects s...

conjecture | open | MathDB 页面 | 原文线索

V2-060859 | MATHDB-320536

The two-endomorphism nilpotency conjecture for Schur-finite objects

摘录 / EXCERPT: Let A be an object with endomorphisms π_1 and π_2. Set ... and ... Assume also that ... for all i and j. Let ν be the hook partition occurring in the definition of the...

conjecture | open | MathDB 页面 | 原文线索

V2-060860 | MATHDB-320537

Crosshatch-pair conjecture for equivalence classes of Kazhdan-Lusztig pairs

摘录 / EXCERPT: Given a composition $\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_k)\models n$, define the permutation ... Let $X_n=\{x_\alpha:\alpha\models n\}$. A crosshatch pair is a pair $x\leq w$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-060861 | MATHDB-320539

Non-maximal peaks conjecture for Legendrian mountain ranges

摘录 / EXCERPT: For a Legendrian knot, write $tb(L)$ for its Thurston--Bennequin number and $r(L)$ for its rotation number. The maximal possible tb means the largest Thurston--Bennequin number a...

conjecture | open | MathDB 页面 | 原文线索

V2-060862 | MATHDB-320547

Finite realizability conjecture for Julia sets of rational functions

摘录 / EXCERPT: Finite realizability conjecture. The Julia set $J(f)$ of any complex rational function f is finitely realizable.

conjecture | open | MathDB 页面 | 原文线索

V2-060863 | MATHDB-320561

Kottwitz's trace formula conjecture for regular discrete series representations

摘录 / EXCERPT: Let G be the reductive group under consideration, let π be a regular discrete series representation of $G(\mathbb{R})$, and let e_{π} denote the stable cuspidal test func...

conjecture | open | MathDB 页面 | 原文线索

V2-060864 | MATHDB-320568

Infinite induced-system freeness conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be a possibly infinite set of systems of induced equations, and let a Boolean function be $\mathcal{F}$-free when it contains no induced solution of any system in...

conjecture | open | MathDB 页面 | 原文线索

V2-060865 | MATHDB-320569

Semi-stable presentations are minimal presentations

摘录 / EXCERPT: Let φ be a self map of the projective line and let Φ be a semi-stable presentation of φ. A presentation is called minimal when it is minimal at every place. Mi...

conjecture | open | MathDB 页面 | 原文线索

V2-060866 | MATHDB-320570

The discriminant conjecture for elliptic curves over number fields

摘录 / EXCERPT: Let K be a number field, and let E be an elliptic curve over K. Let $D(K)$ denote the discriminant of K, let $\Delta(E)$ denote the norm of the minimal discriminant of E,...

conjecture | open | MathDB 页面 | 原文线索

V2-060867 | MATHDB-320571

The resultant conjecture for self maps of the projective line over function fields

摘录 / EXCERPT: Let K be a function field over a field k, and let φ be a self map of degree d of $\mathbb{P}^1_K$. Let $R(\varphi)$ be the minimal resultant and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060868 | MATHDB-320572

Rank-two conjecture for the tiny zero-sum constant

摘录 / EXCERPT: Let m,n be positive integers with $m\mid n$. For a finite Abelian group G, let $\mathsf{t}(G)$ be the smallest integer such that every sequence over G of length at least...

conjecture | open | MathDB 页面 | 原文线索

V2-060869 | MATHDB-320573

Equality of tiny and short zero-sum constants for homocyclic groups

摘录 / EXCERPT: For positive integers r,n, let C_n^r be the direct sum of r copies of the cyclic group C_n. Let $\mathsf{t}(G)$ be the smallest integer such that every sufficiently long se...

conjecture | open | MathDB 页面 | 原文线索

V2-060870 | MATHDB-320577

cscK blowup conjecture with moment-map perturbation

摘录 / EXCERPT: cscK blowup conjecture with moment-map perturbation. If $m>2$, there exist $\delta_0,\varepsilon_0>0$ such that, whenever $\mu(p)+\delta\Delta\mu(p)=0$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-060871 | MATHDB-320586

Plantholt-Tipnis multigraph 1-factorization conjecture

摘录 / EXCERPT: Plantholt-Tipnis multigraph 1-factorization conjecture. If the degree of G is at least rn, then G is 1-factorizable.

conjecture | open | MathDB 页面 | 原文线索

V2-060872 | MATHDB-320588

The blue-edge density conjecture for the diamond-free process

摘录 / EXCERPT: Blue-edge density conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-060873 | MATHDB-320594

The arithmetic-to-spectrum conjecture for L-cuspidal lattices

摘录 / EXCERPT: The arithmetic-to-spectrum conjecture. The lattices Γ_{K_1} and Γ_{K_2} are L-equivalent in $G_{1,\infty}\simeq G_{2,\infty}$.

conjecture | open | MathDB 页面 | 原文线索

V2-060874 | MATHDB-320599

The limiting growth-rate conjecture for threshold languages

摘录 / EXCERPT: Let ${\sf L}(k,\beta)$ be the k-ary threshold language, and let the growth rate of a language mean its exponential growth rate. Limiting growth-rate conjecture. The sequence of g...

conjecture | open | MathDB 页面 | 原文线索

V2-060875 | MATHDB-320600

The asymptotic growth-rate conjecture below exponent 2

摘录 / EXCERPT: Let $\alpha(k,\beta)$ denote the growth rate of the k-ary β-free language. For fixed integers n,k with $k>n\geqslant3$, consider the exponents $n/(n-1)$ and its right-han...

conjecture | open | MathDB 页面 | 原文线索

V2-060876 | MATHDB-320601

The limiting growth-rate conjecture near the threshold exponent

摘录 / EXCERPT: Limiting growth-rate conjecture. For any integer $n\geqslant0$, these limits exist. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-060877 | MATHDB-320609

Periodic-plus-localized structure of optimal potentials

摘录 / EXCERPT: For increasing support widths $0<a_1<a_2<\dots<a_m$, let $V_{\mathrm{opt},a}$ denote locally optimal potentials for the potential design problem on the corresponding admissible cla...

conjecture | open | MathDB 页面 | 原文线索

V2-060878 | MATHDB-320613

Stronger transcendence conclusion for equal exponential powers

摘录 / EXCERPT: Stronger transcendence conclusion. Under the hypotheses of Proposition $(i)RT=TR=A$, both s and t are transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-060879 | MATHDB-320614

Transcendence of solutions to a nested exponential equation

摘录 / EXCERPT: Nested exponential transcendence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060880 | MATHDB-320618

The complete Segal space comparison conjecture for $Theta_n$-spaces

摘录 / EXCERPT: Complete Segal space comparison conjecture. There are Quillen equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-060881 | MATHDB-320619

The iterated complete Segal space model conjecture for (∞,n)-categories

摘录 / EXCERPT: Iterated complete Segal space model conjecture. There is a chain of Quillen equivalences

conjecture | open | MathDB 页面 | 原文线索

V2-060882 | MATHDB-320624

Boissy-Lanneau's equivalence conjecture for extended Rauzy classes

摘录 / EXCERPT: Boissy-Lanneau's equivalence conjecture. The sets $\mathfrak{R}_{ex}(\pi)$ and $\mathfrak{R}'_{ex}(\pi)$ coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-060883 | MATHDB-320625

The Rauzy-class decomposition conjecture for extended Rauzy classes

摘录 / EXCERPT: Rauzy-class decomposition conjecture. The extended Rauzy class $\mathfrak{R}_{ex}$ is a disjoint union of exactly k nonempty Rauzy classes.

conjecture | open | MathDB 页面 | 原文线索

V2-060884 | MATHDB-320627

Classification of the automorphic loops constructed from anisotropic planes

摘录 / EXCERPT: Classification conjecture. The loops $Q(A)$ and $Q(B)$ are isomorphic if and only if A and B are of the same type.

conjecture | open | MathDB 页面 | 原文线索

V2-060885 | MATHDB-320634

The two-dimensional integer-hull conjecture without the linear-growth assumption

摘录 / EXCERPT: Two-dimensional integer-hull conjecture. In dimension $d=2$, Theorem holds without the $O(n)$ assumption.

conjecture | open | MathDB 页面 | 原文线索

V2-060886 | MATHDB-320638

Classification conjecture for non-negative initial-minor q-Toeplitz matrices

摘录 / EXCERPT: Classification conjecture. The matrix $d[i,j]$ coincides with one of the matrices from Proposition 1, namely

conjecture | open | MathDB 页面 | 原文线索

V2-060887 | MATHDB-320646

Rigidity conjecture for the mirror of the canonical coisotropic brane

摘录 / EXCERPT: Let Y be the complexified target and let $\widetilde Y$ be its mirror. Assume the canonical coisotropic brane on Y has mirror brane $\widetilde{\mathcal B}_{cc}$, obtained as t...

conjecture | open | MathDB 页面 | 原文线索

V2-060888 | MATHDB-320651

Integrality conjecture for normalized orthogonal group moments

摘录 / EXCERPT: Let $a\geq b\geq c$ be nonnegative integers, let n denote the dimension parameter in the normalized orthogonal-group integral J, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060889 | MATHDB-320665

The polynomial fourth-integral conjecture for the half-integer parameter family

摘录 / EXCERPT: Let H_1 and I_2 be the first integrals in case 2 of Theorem, let $d=-(2p-1)/2$ with $p\in\mathbb{N}$, and let $\widetilde{H}_{3,p}$ denote the additional first integral. Polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-060890 | MATHDB-320669

Deza-Frankl conjecture on maximum k-intersecting families of permutations

摘录 / EXCERPT: Deza-Frankl conjecture. For any $k\in\mathbb{N}$ and any n sufficiently large depending on k, if $I\subset S_n$ is k-intersecting, then

conjecture | open | MathDB 页面 | 原文线索

V2-060891 | MATHDB-320671

The Bernoulli-operator Riemann hypothesis for the function $\xi(B+s)$

摘录 / EXCERPT: Let B denote the Bernoulli operator, defined formally by $e^{-Bx}=\frac{x}{e^x-1}$, and let ξ be the Riemann xi function. The expression $\sin \pi B\cdot\xi(B+s)$ is defined...

conjecture | open | MathDB 页面 | 原文线索

V2-060892 | MATHDB-320676

The six-color conjecture for star chromatic index of subcubic graphs

摘录 / EXCERPT: Six-color conjecture. If G is a subcubic graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-060893 | MATHDB-320678

Coincidence conjecture for maximal bi-invariant orders on Hermitian Lie groups

摘录 / EXCERPT: Let G be the Hermitian Lie group considered above, with maximal bi-invariant order, maximal continuous bi-invariant order, and, when G is of tube type, geometric order. Coincid...

conjecture | open | MathDB 页面 | 原文线索

V2-060894 | MATHDB-320679

The finite-hyperplane approximation conjecture for graph-statistics limit objects

摘录 / EXCERPT: Let $\mathbf{F}$ be a vector of graphs, and let $P_{\mathbf{F};\infty}$ be its limit object, a compact convex set whose volume is measured in its ambient finite-dimensional space....

conjecture | open | MathDB 页面 | 原文线索

V2-060895 | MATHDB-320681

The q-deformed diagonal harmonics conjecture for G(m,n)

摘录 / EXCERPT: Let $W=G(m,n)$, let $\ell\in\mathbb{N}$, and let q be formal. Define $\mathcal{D}W^{(\ell)}_q$ as the space of polynomials annihilated by all operators $D_{q,\mathbf{d}}$ for whi...

conjecture | open | MathDB 页面 | 原文线索

V2-060896 | MATHDB-320682

The q-harmonic coincidence conjecture for G(m,p,n)

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, let p divide m, and let $G(m,p,n)$ be the complex reflection group contained in $G(m,n)$. Its q-harmonic space is defined using the q-deformed opera...

conjecture | open | MathDB 页面 | 原文线索

V2-060897 | MATHDB-320684

Nash's conjecture for normal surface singularities

摘录 / EXCERPT: Let (X,O) be a normal surface singularity, and let ... be its minimal resolution, with exceptional divisor $E=\bigcup_{k=1}^r E_k$. For each exceptional component E_k, let ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-060898 | MATHDB-320686

Stable-pairs rationality conjecture for descendent partition functions

摘录 / EXCERPT: Stable-pairs rationality conjecture. The partition function $\mathsf Z_{\beta}^X\left(\prod_{j=1}^k \tau_{i_j}(\gamma_j)\right)$ is the Laurent expansion of a rational function in...

conjecture | open | MathDB 页面 | 原文线索

V2-060899 | MATHDB-320687

Denominator conjecture for local-curve stable-pairs descendent partition functions

摘录 / EXCERPT: Denominator conjecture. The denominators of these partition functions are products of factors of the form q^k and

conjecture | open | MathDB 页面 | 原文线索

V2-060900 | MATHDB-320688

The free traced symmetric monoidal category conjecture for the trace functional

摘录 / EXCERPT: Let T be the underlying structure and let $\operatorname{TSMC}(T)$ denote the category constructed from T using the trace functional as in the paper. A traced symmetric monoida...

conjecture | open | MathDB 页面 | 原文线索

V2-060901 | MATHDB-320689

Universal rigidity conjecture for generically globally rigid graphs

摘录 / EXCERPT: Universal rigidity conjecture. Every d-GGR graph has a d-GUR framework.

conjecture | open | MathDB 页面 | 原文线索

V2-060902 | MATHDB-320690

The local-field fewnomial root-count conjecture

摘录 / EXCERPT: Let $ Omega_L(n,k)$ denote the maximum number of non-degenerate roots in a local field L of a system of n polynomials in n variables having at most k monomials in total. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-060903 | MATHDB-320692

Conjecture on permanent tracking-error control by online adjustment of alpha

摘录 / EXCERPT: Permanent tracking-control conjecture. Under certain conditions, still to be defined, the Euler-Lagrange equation applied to the variational model-free Lagrangian establishes the p...

conjecture | open | MathDB 页面 | 原文线索

V2-060904 | MATHDB-320694

Fine's existence conjecture for the extended g-vector

摘录 / EXCERPT: Let X be a convex polytope, and let the extended g-vector be the vector whose components are the linear functions on flag vectors specified by the axioms in the previous sectio...

conjecture | open | MathDB 页面 | 原文线索

V2-060905 | MATHDB-320695

Fine's broadness conjecture for 01-polytopes and their polars

摘录 / EXCERPT: Let $P_{01,d}$ be the set of all d-dimensional 01-polytopes, together with their polars. A set P of d-polytopes is broad when every word-set bb_s that is nonnegative on e...

conjecture | open | MathDB 页面 | 原文线索

V2-060906 | MATHDB-320701

Conjectured semidegree threshold for perfect transitive-triangle packings

摘录 / EXCERPT: Let G be an oriented graph on n vertices, where 3 divides n. Write $\delta^0(G)$ for the minimum of the minimum outdegree and minimum indegree of G, and let T_3 denote...

conjecture | open | MathDB 页面 | 原文线索

V2-060907 | MATHDB-320717

Quasi-concavity conjecture for the UPA goodput-to-power ratio

摘录 / EXCERPT: Let $\mathbf{D}$ be the transmit covariance matrix under uniform power allocation, with $\mathbf{D}=\frac{p}{n_t}\mathbf{I}_{n_t}$, where p is the allocated power and...

conjecture | open | MathDB 页面 | 原文线索

V2-060908 | MATHDB-320722

Rimányi's nonnegativity conjecture for Thom-polynomial coefficients

摘录 / EXCERPT: Rimányi's conjecture. For every such multiindex, the coefficient is nonnegative:

conjecture | open | MathDB 页面 | 原文线索

V2-060909 | MATHDB-320723

A coefficient-ratio bound for Thom-polynomial coefficients

摘录 / EXCERPT: Thom-polynomial coefficient-ratio conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-060910 | MATHDB-320724

Deformation equivalence of singular moduli spaces of sheaves on K3 surfaces

摘录 / EXCERPT: Let m be a positive integer. For $i=1$ or 2, let X_i be a projective K3 surface, let $v_i=(r_i,c_i,a_i)$ be a primitive positive Mukai vector, and let H_i be a polarization...

conjecture | open | MathDB 页面 | 原文线索

V2-060911 | MATHDB-320739

Existence of a nontrivial self-similar biharmonic NLS profile

摘录 / EXCERPT: Let σ lie in the L^2-supercritical and H^2-subcritical regime, namely ... Consider the radial nonlinear eigenvalue problem for $\tilde S_{\mathrm B}(\rho)$ with $\nu=1$: .....

conjecture | open | MathDB 页面 | 原文线索

V2-060912 | MATHDB-320740

Universal quartic-root blowup conjecture for peak-type biharmonic NLS solutions

摘录 / EXCERPT: Let ψ be a peak-type singular solution of the supercritical biharmonic nonlinear Schrödinger equation. Let $S_{\mathrm B}(\rho)$ be a radial self-similar profile satisfying ……

conjecture | open | MathDB 页面 | 原文线索

V2-060913 | MATHDB-320745

Quenched large deviation principle for the SK perceptron empirical measures

摘录 / EXCERPT: Let $L_{N,\alpha}=\frac{1}{N}\sum_{i=1}^{N}\delta_{X_{\mathbf{\sigma},i}}$ be the empirical measure associated with the configurations in the SK perceptron, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-060914 | MATHDB-320746

OMRS sufficiency conjecture for small i-codebooks

摘录 / EXCERPT: Let $\mathcal{C}$ be a small i-codebook, and let an OMRS codebook mean an i-codebook whose multiple-relay selection vectors are pairwise orthogonal in the source's coordinatewise s...

conjecture | open | MathDB 页面 | 原文线索

V2-060915 | MATHDB-320749

Kapovich-Weidmann conjecture on higher stabilization for Nielsen equivalence

摘录 / EXCERPT: Kapovich-Weidmann higher-stabilization conjecture. For every $1\le t<m$, the generating $(k+t)$-tuple $(a_1,\ldots,a_k,1,\ldots,1)$ is not equivalent to a $(k+t)$-tuple of type...

conjecture | open | MathDB 页面 | 原文线索

V2-060916 | MATHDB-320752

Al-Salam-Carlitz moment Pfaffian conjecture

摘录 / EXCERPT: Let $U_n^{(a)}(x;q)$ be the Al-Salam-Carlitz polynomials, and let $G_n(a;q)=\sum_{k=0}^n\left[{{n}\atop{k}}\right]_q a^k$ be their moment sequence, where...

conjecture | open | MathDB 页面 | 原文线索

V2-060917 | MATHDB-320758

The equivalence of strongly reduced and reduced quiver dimension vectors

摘录 / EXCERPT: Let Q be a quiver and let (Q,α) denote a dimension vector. Say that (Q,α) is reduced if there is a weight σ_β such that φ_E is not an isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-060918 | MATHDB-320759

The weak-core conjecture for reduced quiver dimension vectors

摘录 / EXCERPT: Let Q be a quiver with dimension vector α. If (Q,α) is reduced, meaning that there is a weight σ_β for which φ_E is not an isomorphism for any ex...

conjecture | open | MathDB 页面 | 原文线索

V2-060919 | MATHDB-320760

Classification of two-dimensional GIT quotients in the quiver setting

摘录 / EXCERPT: Let Q be a quiver, let α be a dimension vector, and consider the GIT moduli spaces $\operatorname{Mod}_\alpha^{\sigma_\beta}(Q)$ of representations for weights...

conjecture | open | MathDB 页面 | 原文线索

V2-060920 | MATHDB-320761

Infinitude conjecture for near-perfect numbers with a prescribed redundant divisor

摘录 / EXCERPT: A positive integer n is near-perfect if it is the sum of all its proper divisors except one proper divisor, called its redundant divisor. For a fixed integer $k\geq 1$, the redun...

conjecture | open | MathDB 页面 | 原文线索

V2-060921 | MATHDB-320778

The random-vector model conjecture for eigenvectors of sparse random graphs

摘录 / EXCERPT: Let $G(n,p)$ be an Erdős-Rényi random graph. Assume ... for some constant $\epsilon>0$. Let v be a random unit vector uniformly distributed on the $(n-1)$-dimensional unit sphere,...

conjecture | open | MathDB 页面 | 原文线索

V2-060922 | MATHDB-320779

Conjecture on quadratic twists of optimal elliptic curves

摘录 / EXCERPT: The quadratic-twist divisibility conjecture. The quantity $|E_{-D}(\mathbb{Q})|^2$ divides

conjecture | open | MathDB 页面 | 原文线索

V2-060923 | MATHDB-320781

Rayleigh-quotient monotonicity for spectral mass

摘录 / EXCERPT: Let Δ and Δ' be two Laplacians whose Rayleigh quotients satisfy ... For each sign, let $M^{\pm}(\lambda)$ and $(M^{\pm})'(\lambda)$ denote the corresponding upper and l...

conjecture | open | MathDB 页面 | 原文线索

V2-060924 | MATHDB-320782

Spectral-mass asymptotics on the infinite Sierpinski gasket

摘录 / EXCERPT: Let SG denote the Sierpinski gasket and SG_{∞} its infinite version, and let $M(\lambda)$ denote the spectral-mass function. Infinite-gasket asymptotics conjecture. The fu...

conjecture | open | MathDB 页面 | 原文线索

V2-060925 | MATHDB-320787

Distinguished involutions conjecture for rigid conjugates

摘录 / EXCERPT: Let W be a Coxeter system with distinguished involutions ${\mathcal D}$, and let ${\mathcal D}_f^\bullet={\mathcal D}_f\smallsetminus(S\cup\{1\})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-060926 | MATHDB-320788

Basic equivalences conjecture for right cells

摘录 / EXCERPT: Let (W,S) be a Coxeter system, with ${\mathcal M}(w)$ the set of maximal finite-parabolic subwords of w, ${\mathcal D}$ the distinguished involutions, a Lusztig's a-functio...

conjecture | open | MathDB 页面 | 原文线索

V2-060927 | MATHDB-320791

Genus-like expansion conjecture for D2- and M2-brane path integrals

摘录 / EXCERPT: Let Y be a target space-time, and let $Z_{D2}(Y)$ and $Z_{M2}(Y)$ denote the path integrals of perturbative D2-brane and M2-brane theories, respectively. The relevant worldvolume...

conjecture | open | MathDB 页面 | 原文线索

V2-060928 | MATHDB-320795

Finiteness conjecture for ultimate dimension under semi-orthogonal decompositions

摘录 / EXCERPT: Ultimate-dimension finiteness conjecture. If the ultimate dimensions of $\mathcal A$ and $\mathcal B$ are finite, then the ultimate dimension of $\mathcal T$ is finite. The source...

conjecture | open | MathDB 页面 | 原文线索

V2-060929 | MATHDB-320803

Anti-directed 2-factor conjecture for dense directed graphs

摘录 / EXCERPT: Anti-directed 2-factor conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060930 | MATHDB-320810

The nonnegativity conjecture for Littlewood-Richardson coefficients of reflection groups

摘录 / EXCERPT: Let $A=(a_{ij})$ be a quasi-Cartan matrix compatible with a Coxeter group $W=langle s_i,i i I$; thus $a_{ii}=2$, $a_{ij}=0$ if and only if $a_{ji}=0$, and W acts by the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-060931 | MATHDB-320811

The strong positivity conjecture for Littlewood-Richardson polynomials

摘录 / EXCERPT: Let A be a quasi-Cartan matrix compatible with a Coxeter group W, and let $p_{u,v}^{\mathbf{i}}({\bf A})$ be the universal Littlewood-Richardson polynomial. Define ... where Id...

conjecture | open | MathDB 页面 | 原文线索

V2-060932 | MATHDB-320812

Invariance of Littlewood-Richardson polynomials under commutativity

摘录 / EXCERPT: Let A be a quasi-Cartan matrix compatible with a Coxeter group W, and let $p_{u,v}^{\mathbf{i}}(t)$ be the polynomial obtained from the universal Littlewood-Richardson polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-060933 | MATHDB-320817

The Integral Conjecture for polynomial moment subspaces

摘录 / EXCERPT: Integral conjecture. The subspace $V_B(\sigma)$ is a Mathieu subspace of $\mathbb C[t]$. This is one of the one-dimensional Integral and Image Conjectures; the source records speci...

conjecture | open | MathDB 页面 | 原文线索

V2-060934 | MATHDB-320820

Bruhat-Tits boundary characterization conjecture for totally disconnected spherical buildings

摘录 / EXCERPT: Bruhat-Tits boundary conjecture. Then B is the Tits boundary of a locally finite simplicial Bruhat-Tits building.

conjecture | open | MathDB 页面 | 原文线索

V2-060935 | MATHDB-320824

Convergence of critical component sizes for random graphs with a given degree sequence

摘录 / EXCERPT: Let ν satisfy the paper's standing condition denoted by $\text{(thevrairevivalprepaa)}$. Let $\boldsymbol{\mathcal{SC}}^{\nu}_n$ be the ordered sequence of component sizes of...

conjecture | open | MathDB 页面 | 原文线索

V2-060936 | MATHDB-320834

Andrews' conjecture on a ${}_4\phi_3$ summation

摘录 / EXCERPT: Let $n\geq 0$, and let the basic hypergeometric series ${}_4\phi_3$ and the q-shifted factorials be defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-060937 | MATHDB-320854

The toroidal cusp-form criterion via the central \(L\)-value

摘录 / EXCERPT: Let F be the global function field, let π be an irreducible unramified cuspidal subrepresentation of the space of automorphic forms, and let $f\in\pi$ be a cusp form. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-060938 | MATHDB-320855

The dimension conjecture for unramified toroidal cuspidal forms

摘录 / EXCERPT: Let $\mathcal{A}_{0,\operatorname{tor}}^K$ denote the space of K-invariant toroidal cuspidal automorphic forms. Let an irreducible unramified cuspidal representation π be a c...

conjecture | open | MathDB 页面 | 原文线索

V2-060939 | MATHDB-320861

Derived-category identification for the blowup of a nodal quartic double solid

摘录 / EXCERPT: Let V and W be four-dimensional vector spaces, let $s:W\subset S^2V^\vee$ be a generic embedding, and let $D\subset\mathbb{P}(W)\cong\mathbb{P}^3$ be the associated quartic sym...

conjecture | open | MathDB 页面 | 原文线索

V2-060940 | MATHDB-320865

Nodal-line width conjecture for eigenfunctions of the two-dimensional torus

摘录 / EXCERPT: Let $\mathbb T^2$ be the two-dimensional torus, let φ be an eigenfunction with eigenvalue $E=\lambda^2$, and let $C\subset\mathcal N_\varphi$ be a regular arc. For a convex...

conjecture | open | MathDB 页面 | 原文线索

V2-060941 | MATHDB-320870

Asymptotic part-allocation conjecture for the B-q-Plancherel measure

摘录 / EXCERPT: Let $(\lambda^{(1)},\lambda^{(2)})$ be a random bipartition under the B-q-Plancherel measure, and let $\lambda=\lambda^{(1)}\sqcup\lambda^{(2)}$. Asymptotic part-allocation conje...

conjecture | open | MathDB 页面 | 原文线索

V2-060942 | MATHDB-320874

Baez-Breen-Dolan stabilization hypothesis for weak n-categories

摘录 / EXCERPT: Let $n\ge 0$ be fixed. For $k\ge 0$, define recursively the dendroidal set $wCat_k^n$ of weak k-monoidal n-categories by ... and, for $k>0$, ... A dendrex of shape η in...

conjecture | open | MathDB 页面 | 原文线索

V2-060943 | MATHDB-320880

Burkholder-function integral conjecture for the Beurling-Ahlfors inequality

摘录 / EXCERPT: Let U be the Burkholder function appearing in the source, let ∂ and $\overline{\partial}$ denote the complex derivatives, and let $f\in C_0^{\infty}(\mathbb{C})$. Burkho...

conjecture | open | MathDB 页面 | 原文线索

V2-060944 | MATHDB-320889

Universality conjecture for inversion symmetry of WDVV tau functions

摘录 / EXCERPT: Let $F(v)$ be a solution of the WDVV equations, let $\hat F(\hat v)$ be its inversion, and let $\tilde{\mathcal F}_g$ and $\hat{\mathcal F}_g$ be the corresponding genus-g free e...

conjecture | open | MathDB 页面 | 原文线索

V2-060945 | MATHDB-320894

Convexity conjecture for degraded CMLOBC capacity regions

摘录 / EXCERPT: Convexity conjecture. For the degraded CMLOBCs described above, the capacity region is strictly concave ($\cap$) if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-060946 | MATHDB-320896

Fontaine-Perrin-Riou extension of the Bloch-Kato conjecture for Rankin-Selberg motives

摘录 / EXCERPT: Fontaine-Perrin-Riou extension of the Bloch-Kato conjecture. The following equality of fractional ideals of $O_K[1/S]$ holds:

conjecture | open | MathDB 页面 | 原文线索

V2-060947 | MATHDB-320909

Conjecture on the critical p-frame potential for N=d+1

摘录 / EXCERPT: Critical p-frame-potential conjecture. If $N=d+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-060948 | MATHDB-320913

Conjecture on the monodromy Hermitian form for A-hypergeometric functions

摘录 / EXCERPT: Let ${\cal I}$ index the vectors and transition-matrix columns X_I arising from the Mellin-Barnes bases, let H be a Hermitian matrix preserved by all monodromy matrices, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-060949 | MATHDB-320923

The asymptotic enumeration conjecture for proper colorings of hypercubes

摘录 / EXCERPT: Let Q_d be the d-dimensional hypercube, and let $\mathcal C_q(Q_d)$ denote its proper q-colorings. The hypercube coloring enumeration conjecture. For every $q\in\mathbb N$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-060950 | MATHDB-320927

Unproved elliptic-integral moment identity involving derivatives

摘录 / EXCERPT: Let $K(x)$ and $E(x)$ denote the complete elliptic integrals of the first and second kinds, respectively, and let $K'(x)$ and $E'(x)$ denote their complementary counterparts. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-060951 | MATHDB-320928

Conjectural evaluation of the cubic complementary elliptic-integral moment

摘录 / EXCERPT: Let $K(x)$ be the complete elliptic integral of the first kind, and let $K'(x)$ denote its complementary counterpart. The integral is taken over $0\leq x\leq 1$. Cubic elliptic-int...

conjecture | open | MathDB 页面 | 原文线索

V2-060952 | MATHDB-320930

Panyushev's conjectures on the complement of nonnesting partitions

摘录 / EXCERPT: Let W be a finite Weyl group of rank r, with h its Coxeter number, and let ${\sf Pan}$ be the Panyushev complement on antichains in the associated positive-root poset...

conjecture | open | MathDB 页面 | 原文线索

V2-060953 | MATHDB-320931

Conjectured asymptotics for regularized cartoon functions

摘录 / EXCERPT: Regularization asymptotics conjecture. There exist d positive constants $C(k)$, $k\in\{0,\ldots,d-1\}$, depending on φ,p,m,d, such that

conjecture | open | MathDB 页面 | 原文线索

V2-060954 | MATHDB-320939

Effective reduced-density-matrix series conjecture

摘录 / EXCERPT: Let $\rho_N(x,t)$ be the finite-volume reduced density matrix. Set ${\mathcal B}_L=\{n\in\mathbb Z:-w_L<n<w_L\}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-060955 | MATHDB-320940

Biliotti's conjecture on polar actions on higher-rank compact symmetric spaces

摘录 / EXCERPT: Let M be an irreducible compact Riemannian symmetric space of rank greater than one, and let a connected Lie group act nontrivially and polarly on M. A polar action is one admi...

conjecture | open | MathDB 页面 | 原文线索

V2-060956 | MATHDB-320942

Moduli conjecture for del Pezzo orders of type F^{2+}_5

摘录 / EXCERPT: Let C be the canonically embedded complete-intersection curve of degree 8 and genus 5 arising as the base locus of a net of quadrics in $\mathbb{P}^4$. Fix ramification data...

conjecture | open | MathDB 页面 | 原文线索

V2-060957 | MATHDB-320950

Bounded-edge reduction conjecture for 5-connected graphs

摘录 / EXCERPT: Bounded-edge reduction conjecture. There exist integers b,h such that every 5-connected graph on at least b vertices is reducible to a smaller one by contracting at most h...

conjecture | open | MathDB 页面 | 原文线索

V2-060958 | MATHDB-320951

Mader's cubic-order conjecture for 3-critically connected graphs

摘录 / EXCERPT: Mader's cubic-order conjecture. There exists a constant $c>0$ such that every 3-critically k-connected graph has at most

conjecture | open | MathDB 页面 | 原文线索

V2-060959 | MATHDB-320952

Quadratic average-degree conjecture for contractible edges

摘录 / EXCERPT: Quadratic average-degree conjecture. There exists a constant c such that every finite k-connected graph of average degree at least $c k^2$ admits a k-contractible edge.

conjecture | open | MathDB 页面 | 原文线索

V2-060960 | MATHDB-320953

Mader's nonexistence conjecture for clique-critical graphs

摘录 / EXCERPT: Mader's conjecture. There is no clique-critical k-connected graph.

conjecture | open | MathDB 页面 | 原文线索

V2-060961 | MATHDB-320954

McCuaig-Ota conjecture on connected subgraph deletion

摘录 / EXCERPT: McCuaig-Ota conjecture. For every $ellgeq1$ there exists an $f(ell)$ such that every 3-connected graph on at least $f(ell)$ vertices contains a connected subgraph H on ell ve...

conjecture | open | MathDB 页面 | 原文线索

V2-060962 | MATHDB-320956

Tokunaga and Horton self-similarity conjecture for fractional Brownian motions

摘录 / EXCERPT: Fractional Brownian motion self-similarity conjecture. The tree $\operatorname{shape}(\operatorname{tree}(B^H))$ is Tokunaga self-similar with

conjecture | open | MathDB 页面 | 原文线索

V2-060963 | MATHDB-320957

Konyagin's lacunary Walsh-Fourier convergence conjecture

摘录 / EXCERPT: Let $\{n_j:j\ge 1\}$ be a lacunary sequence of integers, meaning that ... For a function f on $\mathbb T$, let $S_{n_j}f$ denote its Walsh-Fourier partial sums, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-060964 | MATHDB-320961

The asymptotic conjecture for minimal pseudo-Anosov translation lengths on punctured surfaces

摘录 / EXCERPT: Let $S_{g,n}$ be a surface of genus g with n punctures, let $\operatorname{Mod}(S_{g,n})$ be its mapping class group, and let $L_{\mathcal C}(\operatorname{Mod}(S_{g,n}))$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-060965 | MATHDB-320963

Krivelevich-Yuster conjecture on rainbow connection and minimum degree

摘录 / EXCERPT: Krivelevich-Yuster conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-060966 | MATHDB-320972

Balister-Bollobás-Sarkar-Walters sharpness conjecture for k-nearest-neighbour graphs

摘录 / EXCERPT: Let $S_n=[0,\sqrt{n}]\times[0,\sqrt{n}]\subset\mathbb{R}^2$, place points in S_n according to a Poisson process of intensity 1, and join each point to its k nearest neighbour...

conjecture | open | MathDB 页面 | 原文线索

V2-060967 | MATHDB-320985

The realization-map classification conjecture for Whitney tower filtrations

摘录 / EXCERPT: Let $\mathcal{T}_n$ be the tree group of order n, let $\text{\sf W}_n$ be the corresponding graded group of link concordance classes, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-060968 | MATHDB-320989

The rank-two isometry conjecture for stable commutator length

摘录 / EXCERPT: Rank-two isometry conjecture. For every $g\in F_2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-060969 | MATHDB-320990

The interval conjecture for stable commutator length in a rank-two free group

摘录 / EXCERPT: Interval conjecture. The set of values

conjecture | open | MathDB 页面 | 原文线索

V2-060970 | MATHDB-320993

Monomial Harbourne-Hirschowitz conjecture

摘录 / EXCERPT: Let $S=\mathbb P^2$, let $I=(y,x^r)^m$ with r and m natural numbers and $r>9$, and let Y be a general member of ${\rm Hilb}_I S$. For a divisor $D=dH$ on $\mathbb P^2$, the m...

conjecture | open | MathDB 页面 | 原文线索

V2-060971 | MATHDB-320996

Maximal-rank restriction conjecture for linear systems on blow-ups

摘录 / EXCERPT: Let X be the blow-up of $\mathbb P^2$ at r general points, let D be an effective divisor on X, and let E be a (-1)-curve. Consider the restriction map ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-060972 | MATHDB-321003

Bárány's conjecture on extrema and Turán bounds for Bessel functions

摘录 / EXCERPT: Bessel-function conjecture. The equation $\Phi_\nu'(x)=0$ has infinitely many roots, with

conjecture | open | MathDB 页面 | 原文线索

V2-060973 | MATHDB-321005

Last's periodic approximation conjecture for almost periodic Schrödinger operators

摘录 / EXCERPT: Last's periodic approximation conjecture. For any $\alpha\notin\mathbb{Q}$, $\mathcal{A}(\alpha)=\lim_{p/q\to\alpha}S_-(p/q)$, in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-060974 | MATHDB-321025

Faber-Pandharipande Gorenstein conjecture for the rational-tail tautological ring

摘录 / EXCERPT: Faber-Pandharipande's rational-tail Gorenstein conjecture. The ring $R^*(M_{g,n}^{\mathrm{rt}})$ is Gorenstein with socle in degree

conjecture | open | MathDB 页面 | 原文线索

V2-060975 | MATHDB-321026

Faber-Pandharipande Gorenstein conjecture for the compactified tautological ring

摘录 / EXCERPT: Faber-Pandharipande's compactified Gorenstein conjecture. The ring $R^*(\overline{M}_{g,n})$ is Gorenstein with socle in degree

conjecture | open | MathDB 页面 | 原文线索

V2-060976 | MATHDB-321027

Faber's compact-support conjecture for the rational-tail tautological ring

摘录 / EXCERPT: Faber's rational-tail compact-support conjecture. (C) These intersection pairings are perfect for every $k\geq 0$. (D) In addition to (C), $R_c^*(M_{g,n}^{\mathrm{rt}})$ is a free...

conjecture | open | MathDB 页面 | 原文线索

V2-060977 | MATHDB-321028

The skew Hall-Littlewood rule for multiplication by a Schur function

摘录 / EXCERPT: Let λ and μ be partitions with $\mu \subseteq \lambda$, and let $r\geq 0$. For a skew shape α/β, write $\operatorname{sk}(\alpha/\beta)$ for the coefficient...

conjecture | open | MathDB 页面 | 原文线索

V2-060978 | MATHDB-321029

The skew Hall-Littlewood rule for multiplication by an elementary symmetric function

摘录 / EXCERPT: Let λ and μ be partitions with $\mu \subseteq \lambda$, and let $r\geq 0$. Write $e_r=P_{1^r}$. For a vertical strip α/β, define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-060979 | MATHDB-321032

Modified Lichtenbaum-Gross conjecture

摘录 / EXCERPT: Let S be a finite set of places of k containing S_∞ and all places ramified in F/k. Let $\lambda:Y_r\to X_r$ be a homomorphism with finite kernel and cokernel, let...

conjecture | open | MathDB 页面 | 原文线索

V2-060980 | MATHDB-321036

Chartrand's existence conjecture for prescribed rainbow connection numbers

摘录 / EXCERPT: Let G be a connected graph, and let $rc(G)$ and $src(G)$ denote its rainbow connection number and strong rainbow connection number, respectively. Let a and b be positive inte...

conjecture | open | MathDB 页面 | 原文线索

V2-060981 | MATHDB-321037

Chartrand's spanning-subgraph conjecture for strong rainbow connection

摘录 / EXCERPT: Let H be a connected spanning subgraph of a nontrivial connected graph G, and let $src(G)$ denote the strong rainbow connection number of G. Chartrand's conjecture. One has .....

conjecture | open | MathDB 页面 | 原文线索

V2-060982 | MATHDB-321039

Chakraborty-Fischer-Matsliah-Yuster conjecture on bounded strong rainbow connection

摘录 / EXCERPT: Let G be a connected graph with n vertices, let $\delta(G)$ denote its minimum degree, and let $src(G)$ denote its strong rainbow connection number. For a fixed positive consta...

conjecture | open | MathDB 页面 | 原文线索

V2-060983 | MATHDB-321052

The cd-index flag f-vector conjecture for regular CW-spheres

摘录 / EXCERPT: Let P be an $(n-1)$-dimensional regular CW-sphere, or more generally a Gorenstein poset of rank $n+1$. Write its homogeneous cd-index as ... where w ranges over cd-monomials of d...

conjecture | open | MathDB 页面 | 原文线索

V2-060984 | MATHDB-321065

Quasi-embeddings of polyhedra and compacta in products of dendrites

摘录 / EXCERPT: Let P be a compact n-polyhedron. A quasi-embedding of P in a product of n dendrites means the approximation notion used in the source, namely the existence of arbitrarily f...

conjecture | open | MathDB 页面 | 原文线索

V2-060985 | MATHDB-321089

The deformation conjecture for Du Bois singularities

摘录 / EXCERPT: Deformation conjecture for Du Bois singularities. Then X has only Du Bois singularities in a neighborhood of x.

conjecture | solved | MathDB 页面 | 原文线索

V2-060986 | MATHDB-321090

The rationality conjecture for a Du Bois divisor

摘录 / EXCERPT: Rationality conjecture for a Du Bois divisor. Then X has rational singularities, and in particular is Cohen-Macaulay.

conjecture | open | MathDB 页面 | 原文线索

V2-060987 | MATHDB-321092

Made's weak genericity conjecture for minimizing measures

摘录 / EXCERPT: Made's weak conjecture. For a generic Lagrangian L, there is an open dense subset U of $H^1(M,\mathbb R)$ such that, for every $c\in U$, there is a unique (L,c)-minimizing me...

conjecture | open | MathDB 页面 | 原文线索

V2-060988 | MATHDB-321095

The stable commutator length formula for a family of words in F_2

摘录 / EXCERPT: Let $p,q,r\in\mathbb{N}$ and set $n=p+q+r$. Stable commutator length formula. Then ... The paper presents this as an unrelated conjecture at its conclusion. No resolution or partial...

conjecture | open | MathDB 页面 | 原文线索

V2-060989 | MATHDB-321096

Schoen's Bernstein conjecture for stable minimal hypersurfaces in four-dimensional Euclidean space

摘录 / EXCERPT: Let $\Sigma^3\subset\mathbb{R}^4$ be a complete immersed 2-sided stable minimal hypersurface. Schoen's Bernstein conjecture. Σ is flat. The conjecture extends the Bernstei...

conjecture | open | MathDB 页面 | 原文线索

V2-060990 | MATHDB-321102

Maximum Möbius value conjecture for separable permutations

摘录 / EXCERPT: Let ... be separable permutations, and suppose that ... and ... of length n is ... This maximum is attained by the specified permutation ... of the same form and the lengths prescribed in...

conjecture | open | MathDB 页面 | 原文线索

V2-060991 | MATHDB-321112

The positive-capacity conjecture for completely invariant sets

摘录 / EXCERPT: Let $f:\overline{{\mathbb R}^n}\to\overline{{\mathbb R}^n}$ be quasiregular with $\operatorname{deg}(f)>K_I(f)$. Let $E(f)$ be the exceptional set. A set X is completely invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-060992 | MATHDB-321116

Almost-planar Cayley complex conjecture

摘录 / EXCERPT: A Cayley complex is the 2-dimensional complex associated with a Cayley graph and its presentation; an almost planar Cayley complex is one admitting the paper's generalized planar r...

conjecture | open | MathDB 页面 | 原文线索

V2-060993 | MATHDB-321117

Planar presentation characterization conjecture

摘录 / EXCERPT: A planar presentation is a group presentation satisfying the paper's planar condition, namely admitting a suitable assignment of spin flags so that relations do not cross. Planar p...

conjecture | open | MathDB 页面 | 原文线索

V2-060994 | MATHDB-321127

Gap-one conjecture for the listed singular threefold examples

摘录 / EXCERPT: Consider the Artin--Mumford example, V_{10} with 10 singular points, the double covering of a quadric ramified in an octic with 20 nodal singular points, and the double solid...

conjecture | open | MathDB 页面 | 原文线索

V2-060995 | MATHDB-321130

Uyemura-Reyes' integrality conjecture for noninversion operators

摘录 / EXCERPT: Let $W=\mathfrak{S}_n$ and let $\nu_{(k,1^{n-k})}$ be the matrix whose (u,v)-entry is the k-noninversion number of $v^{-1}u$, for $k=1,2,\ldots,n$. Integrality conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-060996 | MATHDB-321131

Commutativity and integrality classification for partition-type operators

摘录 / EXCERPT: Let λ and γ be distinct partitions of n, both different from (1^n) and (n). For a partition λ, let ν_λ be the operator associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-060997 | MATHDB-321137

Markov degree bound for triangulated spheres

摘录 / EXCERPT: Let Γ be a triangulation of a sphere of dimension n. The Markov basis of $I_{\Gamma, {\bf 2}}$ is a finite set of binomials associated to the simplicial complex Γ....

conjecture | open | MathDB 页面 | 原文线索

V2-060998 | MATHDB-321146

The firefighter number conjecture for planar graphs

摘录 / EXCERPT: Let $\mathcal{P}$ be the class of planar graphs, and let $\mathit{ff}(\mathcal{P})$ denote its firefighter number. Firefighter number conjecture for planar graphs. ... The paper note...

conjecture | open | MathDB 页面 | 原文线索

V2-060999 | MATHDB-321147

The firefighter number conjecture for planar graphs of girth at least 5

摘录 / EXCERPT: Let $\mathcal{P}_5$ be the class of planar graphs of girth at least 5, and let $\mathit{ff}(\mathcal{P}_5)$ denote its firefighter number. Firefighter number conjecture for plana...

conjecture | open | MathDB 页面 | 原文线索

V2-061000 | MATHDB-321157

Laszkiewicz et al.'s convergence conjecture for the scalar Padé iteration

摘录 / EXCERPT: Laszkiewicz et al.'s convergence conjecture. If $k\ge m-1$ and x_0 lies in $L^{(\operatorname{Pad\acute{e}})}_p$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061001 | MATHDB-321159

Effectivity of the jumping divisor for curves in the moduli space

摘录 / EXCERPT: Let $f:T\to\overline{\mathcal M}_g$ be a morphism from a projective curve whose image is not contained in Δ_0, and let J_T denote the associated jumping divisor. Weak jump...

conjecture | open | MathDB 页面 | 原文线索

V2-061002 | MATHDB-321171

Conjecture on error density estimation for long-memory regression errors

摘录 / EXCERPT: Error-density estimation conjecture. For each fixed x,

conjecture | open | MathDB 页面 | 原文线索

V2-061003 | MATHDB-321173

Injectivity conjecture for the ultradiscretized mapping

摘录 / EXCERPT: Let $X\in\mathbb{R}^{M\times N}$ and let $\mathrm{UD}(\eta)$ denote the ultradiscretization of the mapping η constructed from the divisor component and the ultradiscretized ma...

conjecture | open | MathDB 页面 | 原文线索

V2-061004 | MATHDB-321176

Monomial-factor conjecture for the stationary distribution

摘录 / EXCERPT: Let S_n be the symmetric group and let $\zeta(w)$ be the normalized stationary-distribution component defined by the nullspace of P^T-I. Assume that each $\zeta(w)$ is a polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-061005 | MATHDB-321177

Special-value conjecture for the identity stationary component

摘录 / EXCERPT: Let $\zeta(w)$ be the normalized stationary-distribution component for $w\in S_n$, and let $\mathfrak S_u$ denote the Schubert polynomial indexed by u. Special-value conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-061006 | MATHDB-321178

Special Schubert-factor conjecture for adjacent cyclic strings

摘录 / EXCERPT: Let $w\in S_n$ and let $\zeta(w)$ be its stationary-distribution component. Let $\mathfrak S_u$ denote the Schubert polynomial indexed by u. Special Schubert-factor conjecture. C...

conjecture | open | MathDB 页面 | 原文线索

V2-061007 | MATHDB-321180

The degree-of-irrationality conjecture for generic hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a generic hypersurface of degree $d\geq 2n-1$, and let $\mathrm{d}_r(X)$ denote its degree of irrationality. A dominant rational map is a rational ma...

conjecture | open | MathDB 页面 | 原文线索

V2-061008 | MATHDB-321189

Beck-De Loera-Develin-Pfeifle-Stanley conjecture on roots of Ehrhart polynomials

摘录 / EXCERPT: Let P be a lattice D-polytope, and let its Ehrhart polynomial have complex roots. For a root α, write $\Re(\alpha)$ for its real part. Beck-De Loera-Develin-Pfeifle-Stan...

conjecture | open | MathDB 页面 | 原文线索

V2-061009 | MATHDB-321199

The rational-root conjecture for Stern polynomials

摘录 / EXCERPT: Let $B_n(t)$ denote the Stern polynomial associated with the positive integer n. A rational number a is a zero of $B_n(t)$ if $B_n(a)=0$ for some positive integer n. The rati...

conjecture | open | MathDB 页面 | 原文线索

V2-061010 | MATHDB-321200

The characterization conjecture for equal consecutive Stern degrees

摘录 / EXCERPT: The characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061011 | MATHDB-321203

Transcendence conjecture for the Stern-polynomial power series

摘录 / EXCERPT: Let $e(n)$ denote the Stern sequence, and for a nonnegative integer k define ... The function $\mathcal{E}_{k}(x)$ is transcendental for $k\geq 2$. This extends the known facts tha...

conjecture | open | MathDB 页面 | 原文线索

V2-061012 | MATHDB-321207

The radius and boundary singularity conjecture for mirror maps

摘录 / EXCERPT: Let $N_1,N_2,\ldots,N_k\ge2$ be integers with $\Phi_{\mathbf{N}}\ge4$, and let ${\bf z}_{\bf N}(q)$ be the mirror map with associated function ${\bf q}_{\bf N}(z)$ and constant...

conjecture | open | MathDB 页面 | 原文线索

V2-061013 | MATHDB-321208

The sign-pattern conjecture for coefficients of mirror maps

摘录 / EXCERPT: Let $N_1,N_2,\ldots,N_k$ be positive integers, all at least 2, and let ${\bf z}_{\bf N}(q)$ be the corresponding mirror map. Sign-pattern conjecture. 1. If $\Phi_{\mathbf N}\le3$...

conjecture | open | MathDB 页面 | 原文线索

V2-061014 | MATHDB-321209

The coefficient-positivity conjecture for transformed mirror maps

摘录 / EXCERPT: Let $N_1,N_2,\ldots,N_k\ge2$ be integers, let n be a positive integer, and let ${\bf q}_{\bf N}(z)$ be the function associated with the mirror map. For a power series, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061015 | MATHDB-321214

A quadratic upper-bound conjecture for the van der Waerden numbers $w(2;3,t)$

摘录 / EXCERPT: Let $w(2;3,t)$ denote the van der Waerden number for 2-colourings with monochromatic progressions of lengths 3 and t. Quadratic upper-bound conjecture. For $t\geqslant 3$, ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-061016 | MATHDB-321234

Sodin's conjecture on asymptotic dependence of local lengths

摘录 / EXCERPT: Let $x\in \mathcal{S}^{2}$ and, for $t\in \mathbb{R}$, let $\mathcal{L}_{l;x}^{t}$ denote the random length of the unique component of $f_l^{-1}(t)$ that contains x inside, or...

conjecture | open | MathDB 页面 | 原文线索

V2-061017 | MATHDB-321238

Three-point bounds for the square antiprism code in the sphere

摘录 / EXCERPT: Square-antiprism conjecture. For every completely monotonic potential function f on $(0,4]$, some square antiprism minimizes the energy E_f among all eight-point codes in S^2...

conjecture | open | MathDB 页面 | 原文线索

V2-061018 | MATHDB-321239

Virtual class wall-crossing conjecture for stable quasi-maps

摘录 / EXCERPT: Let $\tau:{\mathcal C}_+\to \mathfrak{Q}_{g,m,d,n}^{\delta_+}$ be the universal curve, with universal family of quasi-maps and wall-crossing morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-061019 | MATHDB-321241

Generic equality of the true and counting synthesis filter lengths

摘录 / EXCERPT: Counting-length conjecture. Generically, a length-m_v FIR synthesis bank achieving perfect reconstruction exists if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-061020 | MATHDB-321249

Convergence of principal-eigenvector rankings to HodgeRank rankings

摘录 / EXCERPT: Let $X^{(k)}$ be the family of pairwise-comparison matrices obtained by varying the parameter $k>0$, and consider the rankings induced by the Principal Eigenvector and HodgeRank me...

conjecture | open | MathDB 页面 | 原文线索

V2-061021 | MATHDB-321251

Conjectural base-change isomorphism for arithmetic differential operators

摘录 / EXCERPT: Conjectural base-change isomorphism. There is a canonical isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-061022 | MATHDB-321261

Positivity conjecture for stability-dependent Donaldson-Thomas invariants

摘录 / EXCERPT: Let Q be a symmetric quiver, let $\theta\in\mathbb{R}^{Q_0}$ be a stability parameter, and let $\boldsymbol{\alpha}\in\mathbb{N}^{Q_0}$. Positivity conjecture for stability-depen...

conjecture | open | MathDB 页面 | 原文线索

V2-061023 | MATHDB-321262

Exponential positivity conjecture for nonnegative matrices

摘录 / EXCERPT: Exponential positivity conjecture. Then $b_{\boldsymbol{\alpha}}(q)\in\mathbb{N}[q]$ for every $\boldsymbol{\alpha}\in\mathbb{N}^r$. The conjecture was proved by Markus Reineke for...

conjecture | open | MathDB 页面 | 原文线索

V2-061024 | MATHDB-321266

Tau-function representation conjecture for vessel scattering matrices

摘录 / EXCERPT: Let $S(\lambda,x)$ be the vessel matrix and let $\tau(x)$ be the associated tau-function. The entries of $S(\lambda,x)$ are understood with respect to some p-norm on $\mathrm I$....

conjecture | open | MathDB 页面 | 原文线索

V2-061025 | MATHDB-321267

Emerton's conjecture on locally analytic vectors in unitary principal series

摘录 / EXCERPT: Let L be a finite extension of ${\mathbb Q}_p$, let $\mathscr{S}_{\mathrm{irr}}$ parameterize the two-dimensional irreducible trianguline representations, and for...

conjecture | open | MathDB 页面 | 原文线索

V2-061026 | MATHDB-321268

Breuil-Bergdall conjecture on locally analytic vectors for crystalline trianguline representations

摘录 / EXCERPT: Let $\mathscr{S}_*^{\mathrm{cris}}$ denote the crystalline parameter space used in the paper. For $s\in\mathscr{S}_*^{\mathrm{cris}}$ non-exceptional, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061027 | MATHDB-321278

The z-Parity Alexander polynomial span conjecture for virtual knots

摘录 / EXCERPT: z-Parity Alexander polynomial span conjecture. The exponents satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-061028 | MATHDB-321281

Evaluation of the third logarithmic Mahler measure of $1+x+y$

摘录 / EXCERPT: Let ... denote the third logarithmic Mahler measure of the polynomial $1+x+y$, and let Ls_4 and Cl_s denote the log-sine and Clausen functions, resp...

conjecture | open | MathDB 页面 | 原文线索

V2-061029 | MATHDB-321287

The conjectural leading term for optimal spherical $\mathbb{L}_2$-discrepancy

摘录 / EXCERPT: The discrepancy asymptotic conjecture. If the fundamental conjecture for optimal Riesz energy on spheres holds, then

conjecture | open | MathDB 页面 | 原文线索

V2-061030 | MATHDB-321293

Cyclotomic factorization conjecture for tatami tiling generating polynomials

摘录 / EXCERPT: Cyclotomic factorization conjecture. The generating polynomial $T(n,z)$ has the factorization

conjecture | open | MathDB 页面 | 原文线索

V2-061031 | MATHDB-321294

Self-reciprocity conjecture for rectangular tatami tiling generating functions

摘录 / EXCERPT: Self-reciprocity conjecture. The denominator satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061032 | MATHDB-321295

Quadratic denominator-degree conjecture for rectangular tatami tiling generating functions

摘录 / EXCERPT: Denominator-degree conjecture. The degree of the denominator is

conjecture | open | MathDB 页面 | 原文线索

V2-061033 | MATHDB-321296

The conjecture on a non-elementary mock automorphism over the field with four elements

摘录 / EXCERPT: Mock automorphism conjecture. The map F is not tamely equivalent to any map of the form

conjecture | open | MathDB 页面 | 原文线索

V2-061034 | MATHDB-321297

The classification conjecture for degree-two mock automorphisms over finite fields of characteristic two

摘录 / EXCERPT: Classification conjecture. (i) In the case of $\mathbb{F}_4$, F is tamely equivalent to $(P(x),y,z)$, where

conjecture | open | MathDB 页面 | 原文线索

V2-061035 | MATHDB-321298

The degree-two homogeneous perturbation conjecture for polynomial automorphisms

摘录 / EXCERPT: Degree-two tameness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061036 | MATHDB-321302

The cubic normality conjecture for the moduli space of SL_2 bundles

摘录 / EXCERPT: Cubic normality conjecture. For a general curve C, the multiplication map η is surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-061037 | MATHDB-321325

The finite-facet conjecture for Kazhdan-Lusztig cell partitions

摘录 / EXCERPT: Let V' be the space of weight functions for a Coxeter system (W,S), and let $\mathfrak H$ be a finite set of rational hyperplanes. For a weight function L, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061038 | MATHDB-321328

MNOP rationality conjecture for reduced Donaldson-Thomas series

摘录 / EXCERPT: Let X) be a Calabi-Yau 3-fold, let β be a nonzero effective curve class, and let $\mathop{\rm DT}\nolimits'_{\beta}(X)$ denote the coefficient of t^β in the reduced...

conjecture | open | MathDB 页面 | 原文线索

V2-061039 | MATHDB-321329

MNOP Gromov-Witten/Donaldson-Thomas correspondence

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold. Define the Gromov-Witten generating series by ... and the reduced Donaldson-Thomas series by...

conjecture | open | MathDB 页面 | 原文线索

V2-061040 | MATHDB-321330

Pandharipande-Thomas correspondence between reduced DT and PT series

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold, let $\beta>0$ be an effective curve class, and let $\mathop{\rm DT}\nolimits'_{\beta}(X)$ and $\mathop{\rm PT}\nolimits_{\beta}(X)$ denote the deg...

conjecture | open | MathDB 页面 | 原文线索

V2-061041 | MATHDB-321331

Katz product expansion conjecture for the PT generating series

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold, let $\beta>0$ range over effective curve classes, and let $\mathop{\rm PT}\nolimits(X)$ be the Pandharipande-Thomas generating series. Katz produc...

conjecture | open | MathDB 页面 | 原文线索

V2-061042 | MATHDB-321332

Multicover formula for the curve-counting invariants $N_{n,\beta}$

摘录 / EXCERPT: Let $N_{n,\beta}$ be the curve-counting invariants introduced through the PT product expansion, let $n\in\mathbb Z_{>0}$, and let β be an effective curve class. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-061043 | MATHDB-321334

Multiple-cover conjecture for K3 sheaf-counting invariants on \(\Gamma_0\)

摘录 / EXCERPT: Let S be a K3 surface, let Γ_0 be the lattice of Mukai classes used for the sheaf-counting invariants, and let (v,v) denote the Mukai pairing. For $k\mid v$, assume...

conjecture | open | MathDB 页面 | 原文线索

V2-061044 | MATHDB-321337

The cyclotomic ideal-power conjecture for Fermat-type elements

摘录 / EXCERPT: The cyclotomic ideal-power conjecture. This equation has no solution except in the trivial cases

conjecture | open | MathDB 页面 | 原文线索

V2-061045 | MATHDB-321342

Strong Bogomolov-Gieseker inequality for stable sheaves at minimal first Chern class

摘录 / EXCERPT: Stable-sheaf inequality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061046 | MATHDB-321343

Asymptotic sharpness of the Bernstein inequality constant for rational functions in Hardy spaces

摘录 / EXCERPT: Asymptotic sharpness conjecture. For $1\leq p\leq\infty$, the constant $(1+r)^{1/p}$ in the estimate

conjecture | open | MathDB 页面 | 原文线索

V2-061047 | MATHDB-321345

Comparison conjecture for slow and fast transfer operators of Hecke triangle groups

摘录 / EXCERPT: Comparison conjecture. The space of 1-eigenfunctions of the meromorphic extension of $\mathcal L_{H,s}^{+}$ is linearly isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-061048 | MATHDB-321348

Reid's derived McKay correspondence conjecture

摘录 / EXCERPT: Reid's derived McKay correspondence conjecture. There is an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-061049 | MATHDB-321369

The evolutionary Eisenbud-Mazur conjecture for point ideals

摘录 / EXCERPT: Let $K[\mathbb{P}^{N}]$ be the homogeneous coordinate ring, let M be its homogeneous maximal ideal, and let I be the ideal of a finite set of points $p_i\in\mathbb{P}^{N}$. Evo...

conjecture | open | MathDB 页面 | 原文线索

V2-061050 | MATHDB-321370

The evolutionary initial-degree conjecture for point ideals

摘录 / EXCERPT: Let $K[\mathbb{P}^{N}]$ be the homogeneous coordinate ring, let I be the ideal of a finite set of points in $\mathbb{P}^{N}$, and let $\alpha(I)$ denote its initial degree. Evolu...

conjecture | open | MathDB 页面 | 原文线索

V2-061051 | MATHDB-321372

Perrin-Riou's p-adic L-function conjecture for pure motives

摘录 / EXCERPT: Let $M_{/\a0\mathbb{Q}}$ be a pure motive with crystalline p-adic realization M_p, let $d_{-}=\operatorname{dim}M_p^{-}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061052 | MATHDB-321386

Gaussian stationary-tail conjecture for QED many-server queues

摘录 / EXCERPT: Gaussian-tail conjecture. (a) $N_{n}(\infty)$ exists for each n; (b) the sequence $\{\tilde{N}_{n}(\infty):n\in\mathbb{N}\}$ weakly converges to $\check{N}(\infty)$; and (c)

conjecture | open | MathDB 页面 | 原文线索

V2-061053 | MATHDB-321387

The extension conjecture for divisorial log terminal pairs

摘录 / EXCERPT: Let (X,Δ) be an n-dimensional projective divisorial log terminal pair such that Δ is a $\mathbb Q$-divisor, and write $\lfloor\Delta\rfloor=S$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-061054 | MATHDB-321388

The weak non-vanishing conjecture for projective log canonical pairs

摘录 / EXCERPT: Weak non-vanishing conjecture. There exists an effective $\mathbb R$-divisor D on X such that

conjecture | open | MathDB 页面 | 原文线索

V2-061055 | MATHDB-321398

Sottile-Mukhin discriminant sum-of-squares conjecture

摘录 / EXCERPT: Let p be a polynomial of degree n, with roots whose pairwise differences are the relevant variables, and let $\mathcal{D}_{n}$ denote the discriminant of its derivative. Sottil...

conjecture | open | MathDB 页面 | 原文线索

V2-061056 | MATHDB-321399

Square-graph cone conjecture for derivative discriminants

摘录 / EXCERPT: Let p be a polynomial of degree n, let $\mathcal{D}_{n,k}$ be the discriminant of its kth derivative, and let a square graph be a graph in which every edge has even multiplic...

conjecture | open | MathDB 页面 | 原文线索

V2-061057 | MATHDB-321400

Nikolaev's rank conjecture for CM elliptic curves and RM noncommutative tori

摘录 / EXCERPT: Let ${\cal E}_{CM}^{(-D,f)}$ be an elliptic curve with complex multiplication by the order of conductor f in the imaginary quadratic field, and let ${\cal A}_{RM}^{(D,f)}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-061058 | MATHDB-321402

Chromatic graph generalization of the Tree Packing Conjecture

摘录 / EXCERPT: Chromatic tree-packing conjecture. If G is a k-chromatic graph, then the set of trees $T_2,\dots,T_k$ has a packing into G.

conjecture | open | MathDB 页面 | 原文线索

V2-061059 | MATHDB-321403

Minimum-degree generalization of the Tree Packing Conjecture

摘录 / EXCERPT: Minimum-degree tree-packing conjecture. If G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061060 | MATHDB-321404

Average-degree generalization of the Tree Packing Conjecture

摘录 / EXCERPT: Average-degree tree-packing conjecture. If G has average degree at least $k-1$, equivalently at least

conjecture | open | MathDB 页面 | 原文线索

V2-061061 | MATHDB-321419

Jones's polynomial Cunningham-chain reducibility conjecture

摘录 / EXCERPT: Jones's polynomial Cunningham-chain reducibility conjecture. The polynomial is reducible over $\mathbb{Q}$ if and only if $j=t$. This conjecture proposes an explicit family of init...

conjecture | open | MathDB 页面 | 原文线索

V2-061062 | MATHDB-321424

Random stable commutator length conjecture for hyperbolic groups

摘录 / EXCERPT: Let G be a hyperbolic group with finite generating set S, and let λ be such that the number of elements of length n is $\Theta(\lambda^n)$. Let v be a random elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-061063 | MATHDB-321429

The strong-versus-weak F-regularity conjecture

摘录 / EXCERPT: Strong-versus-weak F-regularity conjecture. R is strongly F-regular if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-061064 | MATHDB-321430

The reduction-to-characteristic-p multiplier ideal conjecture

摘录 / EXCERPT: Reduction-to-characteristic-p multiplier ideal conjecture. For all sufficiently large primes p,

conjecture | open | MathDB 页面 | 原文线索

V2-061065 | MATHDB-321431

The Watanabe-Yoshida minimal Hilbert-Kunz multiplicity conjecture

摘录 / EXCERPT: Watanabe-Yoshida conjecture. If R is not regular, then

conjecture | open | MathDB 页面 | 原文线索

V2-061066 | MATHDB-321432

Conjecture on optimal beamforming vectors for the two-user broadcast channel

摘录 / EXCERPT: Let ${\bf\Sigma}_1$ and ${\bf\Sigma}_2$ denote the channel covariance matrices, let ρ be the signal-to-noise ratio, and let $\alpha^{\star}(\rho)$ and $\beta^{\star}(\rho)$ ma...

conjecture | open | MathDB 页面 | 原文线索

V2-061067 | MATHDB-321453

Yuan-Zhang-Zhang-type nonvanishing conjecture for Darmon's points

摘录 / EXCERPT: Yuan-Zhang-Zhang-type conjecture. There exists $b\in\widehat{B}^{\times}$ with $k(b,\omega_{\varphi})\neq0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061068 | MATHDB-321460

The weak DRY conjecture for stable bundles on Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a Calabi-Yau threefold with $pi_1(X)=0$. For $c\in H^4(X,\mathbf{Z})$, call c a DRY class if there exist an ample class $H\in H^2(X,\mathbf{R})$ and an integer...

conjecture | open | MathDB 页面 | 原文线索

V2-061069 | MATHDB-321464

Bapat's concavity conjecture for quotients of permanents

摘录 / EXCERPT: Let $b_0,b_1,\ldots,b_k$ be fixed vectors in $\mathbb{R}_{++}^n=(0,\infty)^n$, where $0\leq k<n$. For vectors $v_1,\ldots,v_n$, write $\operatorname{per}(v_1,\ldots,v_n)$ for the p...

conjecture | open | MathDB 页面 | 原文线索

V2-061070 | MATHDB-321465

The four-atom conjecture on the minimum Ingleton score

摘录 / EXCERPT: Four-atom conjecture. The lowest possible Ingleton score is approximately -0.089373. It is attained by a four-variable distribution with alphabet size two, given by

conjecture | open | MathDB 页面 | 原文线索

V2-061071 | MATHDB-321473

The primitive-word and measure-preservation conjecture for free groups

摘录 / EXCERPT: Let $\mathbf{F}_{k}$ be the free group of rank k. A nontrivial word $w\in\mathbf{F}_{k}$ is primitive if it belongs to some free basis. A finitely generated subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-061072 | MATHDB-321486

Surjectivity conjecture for the generic-point cycle class map

摘录 / EXCERPT: Generic-point cycle class conjecture. For all $r,m\geq 0$, statement (S3) holds. The supplied span does not reproduce statement (S3), so its precise mathematical content cannot be...

conjecture | open | MathDB 页面 | 原文线索

V2-061073 | MATHDB-321487

Integral surjectivity conjecture for the logarithmic cycle class map

摘录 / EXCERPT: Integral $d\log$ conjecture. The map $d\log_m$ is surjective. The rationalized map is the generic-point cycle class map $\lim(\textup{cl}_{m,m})$; the source notes that rational su...

conjecture | open | MathDB 页面 | 原文线索

V2-061074 | MATHDB-321519

Every positive harmonic ball is a subharmonic ball

摘录 / EXCERPT: Let K be a domain, let x^0 be a point, and let β be a positive measure. A positive harmonic ball is a set of the form $D(x^0,\alpha)$ satisfying the defining harmonic-bal...

conjecture | open | MathDB 页面 | 原文线索

V2-061075 | MATHDB-321528

Brouwer's 2-connectivity conjecture for strongly regular graphs

摘录 / EXCERPT: Let G be a connected (v,k,λ,μ)-strongly regular graph, and let S be a disconnecting set of G whose removal disconnects G into non-singleton components. Brouwer's...

conjecture | open | MathDB 页面 | 原文线索

V2-061076 | MATHDB-321531

Polynomial approximation conjecture for active exterior acoustic cloaking

摘录 / EXCERPT: Let ${\bf c}^*=(\beta,0)$ be as in the proof of Theorem. For $L>0$, define ... Let S_1 be any disk in the connected component of $D_{\beta,L}$ containing the origin, and let S_2...

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V2-061077 | MATHDB-321553

The maternal-generation conjecture for the meta-Fibonacci sequence μ(n)

摘录 / EXCERPT: Maternal-generation conjecture. The sequence $\mu(n)$ is defined for all positive integers n, and its maternal generation sequence is slow-growing. For each $g\geq 3$, the gth...

conjecture | open | MathDB 页面 | 原文线索

V2-061078 | MATHDB-321555

Bazzaro, Montassier and Raspaud's risky conjecture for planar triangle-free graphs

摘录 / EXCERPT: Bazzaro, Montassier and Raspaud's risky conjecture. If $Delta\geq3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061079 | MATHDB-321560

The support and nonvanishing conjecture for the change-of-basis coefficients $M_{\lambda\mu}$

摘录 / EXCERPT: Let $M_{\lambda\mu}\in\mathbb{Q}[a]$ be defined by ... Here $m_\mu[x-a_{>0}]$ is the basis of $\Lambda(x\|a)$ dual to the homogeneous symmetric functions...

conjecture | open | MathDB 页面 | 原文线索

V2-061080 | MATHDB-321565

Helleseth's conjecture on cross-correlation of maximal linear sequences

摘录 / EXCERPT: Helleseth's conjecture. If m is a power of 2, then $C_d(t)$ is not three-valued.

conjecture | open | MathDB 页面 | 原文线索

V2-061081 | MATHDB-321566

Contractibility conjecture for weighted barycentric spaces

摘录 / EXCERPT: Let Σ be a compact surface with singular points $p_1,\ldots,p_m$ and weights $-1<\alpha_1\leq\cdots\leq\alpha_m<0$. For $\rho>0$, let $\Sigma_{\rho,\underline{\alpha}}$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-061082 | MATHDB-321568

The affine sieve conjecture for perfect Zariski closures

摘录 / EXCERPT: Perfect-closure property (τ) conjecture. The group Γ has property (τ) with respect to the family $\{\Gamma(I)\}$ as I ranges over all finite-index ideals of R....

conjecture | open | MathDB 页面 | 原文线索

V2-061083 | MATHDB-321587

No invariant line fields conjecture for rational maps

摘录 / EXCERPT: Let R be a rational map of the Riemann sphere, and let its Julia set be the set on which the associated dynamics is chaotic. An invariant line field for R is a Beltrami form...

conjecture | open | MathDB 页面 | 原文线索

V2-061084 | MATHDB-321591

The fibration factorization conjecture for \(\overline{M}_{0,n}\)

摘录 / EXCERPT: Let $\overline{M}_{0,n}$ be the Deligne-Mumford moduli space of stable n-pointed rational curves. A fibration is a dominant morphism with connected fibers from...

conjecture | open | MathDB 页面 | 原文线索

V2-061085 | MATHDB-321598

Explicit power-sum formula for higher diagonal harmonics

摘录 / EXCERPT: Let $\mathscr{H}_n^{(r)}(\mathbf{w};1,1,1)$ denote the specialization of the graded Frobenius characteristic of the higher diagonal harmonics at $q_1=q_2=1$. For a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-061086 | MATHDB-321617

The Gog-Magog bijection conjecture

摘录 / EXCERPT: Gog-Magog bijection conjecture. There exists a bijection from Gog triangles of size n to Magog triangles of size n, which maps (n,k) Gog trapezoids to (n,k) Magog trapezoid...

conjecture | open | MathDB 页面 | 原文线索

V2-061087 | MATHDB-321620

Generalized localization equivalence conjecture for Yang-Baxter operators

摘录 / EXCERPT: Let X be a simple object in a fusion category $\mathcal{C}$ equipped with a Yang-Baxter operator (X,c). Let $(YB_{(X,c)},\rho^{(X,c)},\iota)$ denote the associated sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-061088 | MATHDB-321627

Ferromagnetic ordering of energy levels for symmetric spin chains

摘录 / EXCERPT: Ferromagnetic ordering conjecture. The spectral gap γ_k is independent of k; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-061089 | MATHDB-321629

Frequency-localized dispersive estimates for the wave equation on flat cones

摘录 / EXCERPT: Frequency-localized dispersive estimate. If

conjecture | open | MathDB 页面 | 原文线索

V2-061090 | MATHDB-321634

Short-arc equidistribution conjecture for Artin-Schreier L-functions

摘录 / EXCERPT: Let Ψ be the set of nontrivial additive characters of $\mathbb{F}_p$, let $\mathcal{F}_d$ be the family of polynomials under consideration, and let $L_{f,\psi}$ have normalize...

conjecture | open | MathDB 页面 | 原文线索

V2-061091 | MATHDB-321637

Low-degree factorization conjecture for polynomial congruences

摘录 / EXCERPT: Low-degree factorization conjecture. For all sufficiently large d, there exist $g_3,g_4\in\mathbb{F}_q[x]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061092 | MATHDB-321639

Sign-positivity conjecture for the Schur coefficients of the nabla operator

摘录 / EXCERPT: Let λ and μ be partitions of the same integer n. Write $\nabla_{\lambda\mu}(q,t)=\langle \nabla(S_\lambda(\mathbf{w})),S_\mu(\mathbf{w})\rangle$, and let k be the n...

conjecture | open | MathDB 页面 | 原文线索

V2-061093 | MATHDB-321640

Garsia-Haglund positivity conjecture for \nabla(\varepsilon_{n,j})

摘录 / EXCERPT: Let $\varepsilon_{n,j}(\mathbf{w};q)$ be the symmetric functions defined in the source, let ∇ be the nabla operator, and let $S_\mu(\mathbf{w})$ denote the Schur function in...

conjecture | open | MathDB 页面 | 原文线索

V2-061094 | MATHDB-321646

The v-adic period relation lifting conjecture for t-motives

摘录 / EXCERPT: Let $K:=\mathbb{F}_{q}(\theta)$ be the rational function field, let $K_{v(\theta)}$ be its completion at the place $v(\theta)$, and let $K(t)_{v}$ and $K^{\mathrm{sep}}[t]_{v}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-061095 | MATHDB-321653

Thinning-invariant partition structures from stationary random measures

摘录 / EXCERPT: Let $\rho\trimmed$ be a stationary random measure on $\mathbb{R}$ with intensity-one asymptotic density, let $\mathscr{K}$ be the disjoint union of...

conjecture | open | MathDB 页面 | 原文线索

V2-061096 | MATHDB-321654

Thinning-invariant gap distributions from stationary Cox processes

摘录 / EXCERPT: Let $\mathscr{K}'$ be the disjoint union of $\mathcal{M}_{+,1}(\mathbb{R})\times(0,\infty)$ and 0, and let $\mathcal{M}_{\mathrm{st}}(\mathscr{K}')$ denote measures invariant und...

conjecture | open | MathDB 页面 | 原文线索

V2-061097 | MATHDB-321666

Positivity conjecture for principal minors of the bivariate Bézier collocation matrix

摘录 / EXCERPT: The principal-minor positivity conjecture. For every such Γ, the matrix M_Γ is nonsingular and

conjecture | open | MathDB 页面 | 原文线索

V2-061098 | MATHDB-321667

Minimum determinant conjecture for bivariate Bézier collocation submatrices

摘录 / EXCERPT: Let d be fixed, let $\mathcal I_d$ be the set of weak 3-compositions of d, and for every nonempty subset $\Gamma\subset\mathcal I_d$ let M_Γ be the corresponding Berns...

conjecture | open | MathDB 页面 | 原文线索

V2-061099 | MATHDB-321668

Minimum determinant conjecture for bivariate Bézier matrix submatrices

摘录 / EXCERPT: Exact minimum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061100 | MATHDB-321669

Eigenvalue interlacing conjecture for the full bivariate Bézier collocation matrix

摘录 / EXCERPT: Let $M_{\mathcal I_d}$ be the full degree-d bivariate Bernstein collocation matrix, and denote its eigenvalues by $\lambda_{i,n}^{\langle d\rangle}$, where the second index recor...

conjecture | open | MathDB 页面 | 原文线索

V2-061101 | MATHDB-321677

Euler-transform coefficient conjecture for the partial theta function

摘录 / EXCERPT: Euler-transform coefficient conjecture. The sequence $(c_m)_{m=1}^{\infty}$ is strictly positive, increasing, strictly convex, and satisfies $\Delta^k c\geq0$ for $k=3,4$; explicit...

conjecture | open | MathDB 页面 | 原文线索

V2-061102 | MATHDB-321687

Full-weight conjecture for weighted pebbling numbers of trees

摘录 / EXCERPT: Let T be a tree, let $E(T)$ denote its edge set, and let $wp(T)$ be its weighted pebbling number. For a weighted graph, write $w_T(k)$ for the minimum total edge weight required...

conjecture | open | MathDB 页面 | 原文线索

V2-061103 | MATHDB-321688

Nonexistence of extended colorings in doubled Steiner triple systems

摘录 / EXCERPT: Nonexistence conjecture. It has no extended colorings for every $h\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-061104 | MATHDB-321691

Shuron's cotorsion conjecture for signed Selmer groups

摘录 / EXCERPT: Shuron's conjecture. Both $\mathcal{X}^{\sharp}(E/K_\infty)$ and $\mathcal{X}^{\flat}(E/K_\infty)$ are Λ-cotorsion.

conjecture | open | MathDB 页面 | 原文线索

V2-061105 | MATHDB-321692

Finite graph characterization conjecture for good slope sets

摘录 / EXCERPT: A good set of slopes is a set with which every cubic graph has a straight-line drawing. Finite graph characterization conjecture. There is a not necessarily connected finite graph...

conjecture | open | MathDB 页面 | 原文线索

V2-061106 | MATHDB-321698

Asymptotic conjecture for size-enumerated self-dual interval orders

摘录 / EXCERPT: Let r_n be the number of self-dual interval orders of size n, and let q_n be the number of primitive self-dual interval orders of size n. Size asymptotic conjecture. ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-061107 | MATHDB-321700

Stabilization-index formula for square-free monomial ideals

摘录 / EXCERPT: Let $I_G=(m_0,m_1,\ldots,m_k)\subseteq \Bbbk[x_1,\ldots,x_N]$ be a square-free monomial ideal with ... For a homogeneous ideal I equigenerated in degree r, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061108 | MATHDB-321704

Existence of symmetric optimal allocations

摘录 / EXCERPT: An allocation distributes storage across n nodes, and its recovery delay is denoted by D; the parameters are the node count n, request rate λ, and threshold T. An a...

conjecture | open | MathDB 页面 | 原文线索

V2-061109 | MATHDB-321705

Isaev's invariant-generation conjecture for homogeneous binary forms

摘录 / EXCERPT: Let X^n be the space of homogeneous binary forms of degree n, let ${\mathcal R}^n$ be the collection of invariant rational functions on X^n obtained by evaluating absolute cl...

conjecture | open | MathDB 页面 | 原文线索

V2-061110 | MATHDB-321708

Peak-optical-power optimality conjecture for single-subcarrier IM/DD constellations

摘录 / EXCERPT: Peak-optical-power conjecture. For any $Omega\thissubset Upsilon$,

conjecture | open | MathDB 页面 | 原文线索

V2-061111 | MATHDB-321728

Locating chromatic number of Kneser-type Cartesian products

摘录 / EXCERPT: Let $n\geq 3$ and define ... Here K_m denotes the complete graph, $\square$ the Cartesian product, and χ_L the locating chromatic number. Locating chromatic number conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-061112 | MATHDB-321729

Locating chromatic number conjecture for Cartesian products of complete graphs

摘录 / EXCERPT: Let K_m and K_n be complete graphs, with $\lceil (n+3)/2\rceil<m\leq n$, and let $\square$ denote the Cartesian product. Write χ_L for the locating chromatic number. Locat...

conjecture | open | MathDB 页面 | 原文线索

V2-061113 | MATHDB-321730

Locating chromatic numbers of Cartesian products with cycles and paths

摘录 / EXCERPT: Let K_m be the complete graph, and let C_n and P_n denote the cycle and path on n vertices, respectively. Write $\square$ for the Cartesian product and χ_L for the loc...

conjecture | open | MathDB 页面 | 原文线索

V2-061114 | MATHDB-321731

Locating chromatic number bound for Cartesian products of connected graphs

摘录 / EXCERPT: Let G and H be connected graphs, let $\square$ denote the Cartesian product, and write χ_L for the locating chromatic number. Locating chromatic number bound for Cartesian...

conjecture | open | MathDB 页面 | 原文线索

V2-061115 | MATHDB-321733

The length-five even-Wilf-equivalence conjectures

摘录 / EXCERPT: Even-Wilf-equivalence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061116 | MATHDB-321734

The odd-length direct-sum conjecture for even-Wilf-equivalence

摘录 / EXCERPT: Odd-length direct-sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061117 | MATHDB-321735

The Stankova-West even-shape-Wilf-equivalence conjecture

摘录 / EXCERPT: Stankova-West conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061118 | MATHDB-321736

The even-Wilf-equivalence refinement conjecture

摘录 / EXCERPT: Even-Wilf-equivalence refinement conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061119 | MATHDB-321738

Protasov's conjecture on spectrum-maximizing and spectrum-minimizing products

摘录 / EXCERPT: Let r be an odd integer, and let A_1,A_2 be the $(r-1)\times(r-1)$ matrices defined by ... and $(A_s)_{ij}=0$ otherwise, for $s=1,2$. A product is an s.m.p. if it is a spectrum-m...

conjecture | open | MathDB 页面 | 原文线索

V2-061120 | MATHDB-321739

Castryck-Cools generic gonality conjecture for lattice polygons

摘录 / EXCERPT: Let $\Delta\subset\mathbb{R}^2$ be a two-dimensional lattice polygon. For a Laurent polynomial $f\in\mathbb{C}[x^{\pm 1},y^{\pm 1}]$, write $\Delta(f)$ for its Newton polygon, and...

conjecture | open | MathDB 页面 | 原文线索

V2-061121 | MATHDB-321743

Verlinde bundle decomposition conjecture for abelian surfaces

摘录 / EXCERPT: Let A be an abelian surface, let v and w be Mukai vectors satisfying assumptions ${\bf (A.2)-(A.3)}$, let $\mathbf E(v,w)$ be the associated Verlinde bundle, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061122 | MATHDB-321750

Equivalence conjecture for the Stanley depths of powers of the maximal ideal and squarefree Veronese ideals

摘录 / EXCERPT: Let $R=K[x_1,\cdots, x_n]$ be a polynomial ring in n variables. Let $I_{n,d}$ be the squarefree Veronese ideal generated by all squarefree monomials of degree d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061123 | MATHDB-321752

Negative-root multiplicity conjecture for O(n) groundstate polynomials

摘录 / EXCERPT: Negative-root multiplicity conjecture. All real roots of $\psi_\pi(t)$ are negative integers, and $-p$ occurs with multiplicity $m_p(\pi)$. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-061124 | MATHDB-321764

Shareshian-Wachs's power-sum decomposition conjecture II

摘录 / EXCERPT: Let P be a natural unit interval order on $[n]$ with incomparability graph G. For a partition $\mu=(\mu_1\geq\cdots\geq\mu_l)\vdash n$, let z_μ be the standard symmetric-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-061125 | MATHDB-321777

The numerical-dimension conjecture for pseudo-effective divisors orthogonal to ample-normal curves

摘录 / EXCERPT: Numerical-dimension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061126 | MATHDB-321778

The codimension-one-nef formulation of the numerical-dimension conjecture

摘录 / EXCERPT: Codimension-one-nef formulation. If

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V2-061127 | MATHDB-321785

Necessity of approximate Kakutani structure for irreducible weighted shifts

摘录 / EXCERPT: Let T be an irreducible weighted shift with weights $\{\alpha_n\}_{n=1}^{\infty}$. For each $n\in\mathbb{N}$ and $\epsilon>0$, suppose there is an index $c_{n,\epsilon}\geq n$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-061128 | MATHDB-321814

The Mayer-Ruelle operator conjecture for complex multiplication lattices

摘录 / EXCERPT: Let $L_{CM}={\mathbb Z}+\tau_D{\mathbb Z}$ be a lattice with complex multiplication by the ring of integers in the imaginary quadratic field ${\mathbb Q}(\sqrt{-D})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061129 | MATHDB-321815

Extension of the lognormal multifractal limit theorem

摘录 / EXCERPT: Let M be the lognormal multifractal random measure with parameter γ^2, and let the extended singular integral be defined, for $x\in[0,1]$ and continuous f, by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061130 | MATHDB-321818

AdaBoost's quadratic-over-accuracy convergence conjecture

摘录 / EXCERPT: Let $\bm{\lambda}^*$ be any weight vector and let $B=\|\bm{\lambda}^*\|_1$. For exponential loss L and AdaBoost's iterates, quadratic convergence conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-061131 | MATHDB-321819

AdaBoost's near-optimal convergence conjecture for ternary feature matrices

摘录 / EXCERPT: Let m be the number of training examples and suppose the feature matrix has entries in $\{-1,0,+1\}$. Let L denote the exponential loss and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061132 | MATHDB-321823

Borg's large-label conjecture for strict t-stars of partial signed sets

摘录 / EXCERPT: Let $\mathcal{F}$ be a family and let $\mathcal{S}_{\mathcal{F},k}^*$ be its partial signed-set family. A family has the strict t-star property if every largest t-intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-061133 | MATHDB-321824

Cheliotis-den Hollander critical-temperature conjecture for pinning models

摘录 / EXCERPT: Cheliotis-den Hollander conjecture. For every μ_0,

conjecture | open | MathDB 页面 | 原文线索

V2-061134 | MATHDB-321826

Ruan's LG/CY correspondence conjecture for elliptic orbifold projective lines

摘录 / EXCERPT: Ruan's LG/CY correspondence conjecture. (1) There is a graded vector-space isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-061135 | MATHDB-321827

The half-duplex/full-duplex DMT equality for MIMO relay channels

摘录 / EXCERPT: Let an (m,k,n) half-duplex MIMO relay channel (HD-RC) have m, k, and n antennas at its three nodes, and let its diversity-multiplexing tradeoff (DMT) be compared with that...

conjecture | open | MathDB 页面 | 原文线索

V2-061136 | MATHDB-321828

The symmetric (n,k,n) half-duplex relay-channel DMT conjecture

摘录 / EXCERPT: Consider a symmetric (n,k,n) half-duplex MIMO relay channel, meaning that the source and destination each have n antennas and the relay has k antennas. Let $d^*(r)$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-061137 | MATHDB-321831

Pointwise stabilization equivalence for discrete-time stationary matrix-valued Markovian chains

摘录 / EXCERPT: Let $(\boldsymbol{S},\Xi)$ be a discrete-time stationary matrix-valued Markovian chain, with consistent convergence and consistent exponential convergence understood in the sense o...

conjecture | open | MathDB 页面 | 原文线索

V2-061138 | MATHDB-321844

Degree-truncated h-positivity conjecture for symmetric-group co-quasi-invariants

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathcal{Q}_{n}({\mathbf{q}})$ denote its Hilbert series, and write h_μ for products of complete homogeneous symmetric functions indexed...

conjecture | open | MathDB 页面 | 原文线索

V2-061139 | MATHDB-321876

The critical inverse-temperature characterization of path localization

摘录 / EXCERPT: Path-localization conjecture. The set of inverse temperatures for which path localization holds is

conjecture | open | MathDB 页面 | 原文线索

V2-061140 | MATHDB-321877

Odd-index reduction conjecture for the series H

摘录 / EXCERPT: For positive integers $b_1,\dots,b_s$, let $H_{b_1,\dots,b_s}(z)$ denote the series used in the paper, and let $H_{a_1,\dots,a_r}(z)$ be the corresponding series with indices a_i...

conjecture | open | MathDB 页面 | 原文线索

V2-061141 | MATHDB-321886

Characterization of images of the real line under reduced quasiconformal maps

摘录 / EXCERPT: Let $f\colon \mathbb{C}\to\mathbb{C}$ be a reduced quasiconformal map, and consider the image $f(\mathbb{R})$. A quasisymmetric graph is the image of $\mathbb{R}$ under a suitable...

conjecture | open | MathDB 页面 | 原文线索

V2-061142 | MATHDB-321890

The coniveau conjecture for complete intersections

摘录 / EXCERPT: Let $X\subset\mathbb{P}^n$ be a smooth complete intersection of r hypersurfaces of degrees $d_1\leq\cdots\leq d_r$, and let $H^{n-r}(X,\mathbb{Q})_{\mathrm{prim}}$ denote its pri...

conjecture | open | MathDB 页面 | 原文线索

V2-061143 | MATHDB-321895

The Hara-Yoshida reduction conjecture for log canonical and F-pure thresholds

摘录 / EXCERPT: Let $\mathfrak{a}\subset{\mathbf Z}[x_1,\ldots,x_n]$ be an ideal contained in $(x_1,\ldots,x_n)$, let $\mathfrak{a}_p=\mathfrak{a}\cdot{\mathbf F}_p[x_1,\ldots,x_n]$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-061144 | MATHDB-321897

The uniform reduction conjecture for test and multiplier ideals

摘录 / EXCERPT: Let $\mathfrak{a}\subset{\mathbf Z}[x_1,\ldots,x_n]$ be an ideal as above, let $\mathfrak{a}_p$ be its reduction modulo p, and let $\tau(\mathfrak{a}_p^\lambda)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061145 | MATHDB-321898

The quasi-monomiality conjecture for threshold-computing valuations

摘录 / EXCERPT: Quasi-monomiality conjecture. The valuation v is quasi-monomial. The source presents this as an expected strengthening of the existence theorem for a valuation computing the asym...

conjecture | open | MathDB 页面 | 原文线索

V2-061146 | MATHDB-321901

Finiteness conjecture for exceptional bundle slope vectors

摘录 / EXCERPT: Let Y be a smooth projective surface such that K_Y is ample, $H^2(\mathcal T_Y)=0$, $H^{2,0}(Y)=0$, and $\pi_1(Y)=0$. Let S be the set of slope vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-061147 | MATHDB-321903

The reversal-ratio conjecture for three linear extensions

摘录 / EXCERPT: Reversal-ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061148 | MATHDB-321910

The intensive-dust conjecture for initial types and Eves

摘录 / EXCERPT: Let Λ be the measure governing a Λ Fleming-Viot process, and consider the regime of intensive dust, meaning the regime in which the dust component is present. An Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-061149 | MATHDB-321911

The intensive-infinity conjecture for infinite sequences of Eves

摘录 / EXCERPT: Let Λ be the measure governing a Λ Fleming-Viot process, and consider the regime of intensive infinity, the regime denoted by $\textsc{Intensive}\ \infty$ in the so...

conjecture | open | MathDB 页面 | 原文线索

V2-061150 | MATHDB-321912

The CDI conjecture for infinite sequences of Eves

摘录 / EXCERPT: Let Λ be the measure governing a Λ Fleming-Viot process, and suppose the process is in the coming down from infinity regime (CDI). Let $\mathtt{E}$ denote the event...

conjecture | open | MathDB 页面 | 原文线索

V2-061151 | MATHDB-321929

Bethe permanent lifting bound conjecture

摘录 / EXCERPT: Let $\boldsymbol{\theta}$ be a non-negative $n\times n$ matrix. For $M\in\mathbb{Z}_{>0}$, let $\tilde{\Psi}_M$ be the set of degree-M graph-covering permutations and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061152 | MATHDB-321930

Coefficientwise Bethe permanent lifting conjecture

摘录 / EXCERPT: Let $\boldsymbol{\theta}$ be a matrix of indeterminates, let $M\in\mathbb{Z}_{>0}$, and let $\tilde{\Psi}_M$ be the set of degree-M graph-covering permutations. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061153 | MATHDB-321931

Bethe perm-vector pseudo-codeword conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a binary linear code with parity-check matrix ${\bm{H}}\in{\mathbb{F}}_2^{m\times n}$, where $m<n$. Let $\mathcal{I}({\bm{H}})$ be its column-index set, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061154 | MATHDB-321932

Bethe permanent kernel conjecture

摘录 / EXCERPT: Huang-Jebara Bethe permanent kernel conjecture. The function $\kappa_{\operatorname{perm}_{\mathrm{B}}}$ is a positive definite kernel on $\mathcal{X}^n\times\mathcal{X}^n$. The cl...

conjecture | open | MathDB 页面 | 原文线索

V2-061155 | MATHDB-321936

Pylyavskyy-first author conjecture on maximal connected skew shapes

摘录 / EXCERPT: Let N be a positive integer, let $\mathcal{P}_{N}$ be the poset of skew shapes of size N under Schur-positivity, and restrict to connected skew shapes. Two skew shapes are iden...

conjecture | open | MathDB 页面 | 原文线索

V2-061156 | MATHDB-321937

Conjecture on the minimal ribbon with prescribed row partition

摘录 / EXCERPT: Let N be fixed, let $\lambda\vdash N$, and consider the subposet of $\mathcal{P}_{N}$ consisting of ribbons R whose row partition is $\operatorname{rows}(R)=\lambda$. For a rib...

conjecture | open | MathDB 页面 | 原文线索

V2-061157 | MATHDB-321951

Uniform splitting conjecture for macroscopic cycles

摘录 / EXCERPT: Uniform splitting conjecture. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-061158 | MATHDB-321969

The fractional BP upper-bound conjecture for the permanent at gamma equals minus one-half

摘录 / EXCERPT: Let p be a non-negative matrix, and let $Z_{o\text{-}f}^{(\gamma)}(p)$ denote the fractional BP partition-function estimate at parameter γ. In particular, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061159 | MATHDB-321972

Periodic switching stability implies absolute stability for pairs in dimension at least four

摘录 / EXCERPT: Let $\boldsymbol{S}=\{S_1,S_2\}\subset\mathbb{R}^{d\times d}$, with $d\geq 4$, be an arbitrary pair satisfying condition (1.3a), namely that the matrices share a common non-strict...

open_problem | open | MathDB 页面 | 原文线索

V2-061160 | MATHDB-321980

Hertling's stabilizer conjecture for right-equivalence monodromy

摘录 / EXCERPT: Stabilizer conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061161 | MATHDB-321981

Hertling's canonical-structure Torelli conjecture for marked singularities

摘录 / EXCERPT: Canonical Brieskorn-lattice Torelli conjecture. The map BL is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-061162 | MATHDB-321986

The Hessian rank conjecture for Coxeter arrangements of types E6, E7, and E8

摘录 / EXCERPT: Let $\mathsc{A}$ be the Coxeter arrangement associated with a finite reflection group W. The Hessian rank conjecture. The Hessian rank condition holds for $\mathsc{A}$ if W is...

conjecture | open | MathDB 页面 | 原文线索

V2-061163 | MATHDB-321988

The regular-metric Nitsche bound for harmonic annulus maps

摘录 / EXCERPT: Nitsche bound. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061164 | MATHDB-321989

The MEX property for groups in On_p

摘录 / EXCERPT: MEX conjecture. If $\alpha,\beta\in On_p$ are groups, then the pair $\{\alpha,\beta\}$ has the MEX property.

conjecture | open | MathDB 页面 | 原文线索

V2-061165 | MATHDB-321990

The cross-characteristic field and transcendental conjecture for On_p

摘录 / EXCERPT: Cross-characteristic field and transcendental conjecture. If $\phi\in On_2$ is an infinite field, then ϕ is a field in On_p for all primes p. If $\phi\in On_2$ is a transc...

conjecture | open | MathDB 页面 | 原文线索

V2-061166 | MATHDB-322007

Bessis-Reiner cyclic sieving conjecture for nonnesting partitions

摘录 / EXCERPT: Let W be a finite crystallographic Coxeter group with Coxeter number h, let $J(\Phi^+(W))$ be the order ideals of its positive root poset, and let $d_1,d_2,\ldots,d_n$ be the d...

conjecture | open | MathDB 页面 | 原文线索

V2-061167 | MATHDB-322010

Uniqueness and least-gradient identification for the global median value problem

摘录 / EXCERPT: Global median value conjecture for strictly convex domains. There exists a unique continuous solution u with $u=g$ on $\partial\Omega$, and this solution satisfies $u=u^{*}$. Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-061168 | MATHDB-322011

The open-orbit vanishing conjecture for global integrals

摘录 / EXCERPT: Open-orbit vanishing conjecture. Suppose that the integral ... is zero unless $w=w_0$ and $\nu=\nu_0$. Moreover, w_0 can be chosen to be a Weyl element corresponding to the longest...

conjecture | open | MathDB 页面 | 原文线索

V2-061169 | MATHDB-322025

SIN-property characterization of multi-cluster complexes

摘录 / EXCERPT: SIN-property characterization conjecture. The subword complex $\Delta(Q,\pi)$ is isomorphic to a multi-cluster complex if and only if Q has the SIN-property and

conjecture | open | MathDB 页面 | 原文线索

V2-061170 | MATHDB-322026

Facet-maximality conjecture for multi-cluster complexes

摘录 / EXCERPT: Facet-maximality conjecture. The number of facets of $\Delta(Q,w_\circ)$ is at most the number of facets of $\Delta_c^k(W)$. Moreover, if the two numbers are equal, then Q has th...

conjecture | open | MathDB 页面 | 原文线索

V2-061171 | MATHDB-322027

Polytopality conjecture for multi-cluster complexes

摘录 / EXCERPT: Multi-cluster polytopality conjecture. The multi-cluster complex $\Delta_c^k(W)$ is the boundary complex of a simplicial polytope.

conjecture | open | MathDB 页面 | 原文线索

V2-061172 | MATHDB-322033

Generalized Brualdi conjecture on partial k-plexes in latin squares

摘录 / EXCERPT: Generalized Brualdi conjecture. Every latin square of order n has a partial k-plex of order $k(n-k)$ for each $k\le n$.

conjecture | open | MathDB 页面 | 原文线索

V2-061173 | MATHDB-322034

Stanley's LR conjecture for Jack functions

摘录 / EXCERPT: Let μ, ν, and λ be partitions, and let J_μ, J_ν, and J_λ be Jack functions. Define the Jack Littlewood-Richardson coefficient by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061174 | MATHDB-322039

Fullness conjecture for the minimal Lefschetz decomposition of Grassmannians

摘录 / EXCERPT: Let $X=\operatorname{Gr}(k,n)$, let $\mathcal U$ be the tautological subbundle, and let $\mathcal A_i$ be the full triangulated subcategories generated by the specified Schur power...

conjecture | open | MathDB 页面 | 原文线索

V2-061175 | MATHDB-322040

Minimality conjecture for the proposed Lefschetz decomposition of Grassmannians

摘录 / EXCERPT: Let $X=\operatorname{Gr}(k,n)$, let $\mathcal U$ be the tautological subbundle, and let $\mathcal A_i$ be the full triangulated subcategories defined in the source using the specif...

conjecture | open | MathDB 页面 | 原文线索

V2-061176 | MATHDB-322048

The exceptional-groups conjecture for short zero-sum subsequences

摘录 / EXCERPT: Let G be a non-cyclic finite abelian group. Define ... Here $D(G)$ is the Davenport constant, and a short zero-sum subsequence has length at most $\exp(G)$. The exceptional-groups...

conjecture | open | MathDB 页面 | 原文线索

V2-061177 | MATHDB-322049

The interval-location conjecture for $C_0(G)$

摘录 / EXCERPT: Let G be a finite non-cyclic abelian group, with $G\neq C_{2}\oplus C_{2m}$ for $m\in\mathbb N$. Let $C_0(G)$ be the set of integers t with $D(G)+1\leq t\leq\eta(G)-1$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-061178 | MATHDB-322050

Conjectured sharp quasiinvariance of the distance-ratio metric under ball automorphisms

摘录 / EXCERPT: Let $a\in\mathbb{B}^n$ and let $h:\mathbb{B}^n\to\mathbb{B}^n$ be a Möbius transformation of the unit ball onto itself satisfying $h(a)=0$. The distance-ratio metric is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-061179 | MATHDB-322051

Dimension-independent quasiconformal distortion conjecture for the hyperbolic distortion function

摘录 / EXCERPT: Let $K>1$, $n>2$, and $r\in(0,1)$. Define $\alpha=K^{1/(1-n)}$ and let $\varphi_{K,n}$ be the quasiconformal distortion function. Hyperbolic distortion conjecture. ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-061180 | MATHDB-322054

Stretchability conjecture for simplicial arrangements with at most 14 pseudolines

摘录 / EXCERPT: Stretchability conjecture. Every simplicial arrangement with at most 14 pseudolines is stretchable.

conjecture | open | MathDB 页面 | 原文线索

V2-061181 | MATHDB-322056

Hyperbolic torsion polynomial detection conjecture for hyperbolic knots

摘录 / EXCERPT: Hyperbolic torsion polynomial detection conjecture. The hyperbolic torsion polynomial $4\mathcal{T}_K4$ determines $x(K)$, or equivalently its genus. Moreover, K is fibered if an...

conjecture | open | MathDB 页面 | 原文线索

V2-061182 | MATHDB-322075

Structural conjecture for the small foliation of the moduli space of curves

摘录 / EXCERPT: Let ${\mathcal M}_{g,1}(2)$ be the moduli space equipped with the small foliation ${\mathcal S}$ obtained by intersecting the foliations defined by fixing the periods of the two di...

conjecture | open | MathDB 页面 | 原文线索

V2-061183 | MATHDB-322082

Open mirror-map formula for disk Gromov-Witten invariants of projective hypersurfaces

摘录 / EXCERPT: Let M_k^k be the degree-k projective hypersurface, let $\tau_k(x)$ and $F_o^k(x)$ be the functions introduced above, and define the mirror map ... Write $x(t)$ for its inverse an...

conjecture | open | MathDB 页面 | 原文线索

V2-061184 | MATHDB-322089

Random graphs have equal Hadwiger and fractional Hadwiger numbers

摘录 / EXCERPT: For a graph G, let $h(G)$ denote its Hadwiger number, and let $h_f(G)$ denote its fractional Hadwiger number. Let G be chosen uniformly at random from the graphs on n vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-061185 | MATHDB-322090

Gopakumar-Vafa product expansion conjecture for stable pair invariants

摘录 / EXCERPT: Gopakumar-Vafa product expansion conjecture. There are integers $n_g^{\beta}\in\mathbb{Z}$, for $g\geq 0$ and $\beta\in H_2(X,\mathbb{Z})$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-061186 | MATHDB-322091

Multiple cover formula for parabolic stable-pair invariants on a curve

摘录 / EXCERPT: Let C be the local curve considered in the paper, let $N_{n,\gamma}$ denote the corresponding generalized DT invariant for $(n,\gamma)\in\mathbb{Z}\oplus H_2(C,\mathbb{Z})$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-061187 | MATHDB-322092

Joyce-Song multiple cover conjecture for generalized Donaldson-Thomas invariants

摘录 / EXCERPT: Joyce-Song multiple cover conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-061188 | MATHDB-322093

Toda's local multiple cover conjecture for generalized Donaldson-Thomas invariants

摘录 / EXCERPT: Toda's local multiple cover conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-061189 | MATHDB-322094

Toda's multiple cover conjecture for Euler-characteristic invariants on K3 surfaces

摘录 / EXCERPT: Toda's K3 multiple cover conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-061190 | MATHDB-322099

The increasing-path conjecture for higher-order majorization

摘录 / EXCERPT: For integers n and k, let $s_1,\dots,s_{k-1}$ be real numbers and define ... Let $a,b\in S_k$ satisfy $a\succ_k b$. Increasing-path conjecture. There is a continuous function...

conjecture | open | MathDB 页面 | 原文线索

V2-061191 | MATHDB-322101

Reflection-order characterization and maximal-chain conjecture for biclosed sets

摘录 / EXCERPT: Reflection-order and maximal-chain conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061192 | MATHDB-322102

Ortholattice and closure conjecture for quasiparabolic root subsets

摘录 / EXCERPT: Quasiparabolic biclosed-set conjecture. The set $\mathrsfs{B}(\Lambda)$ is a complete ortholattice, with

conjecture | open | MathDB 页面 | 原文线索

V2-061193 | MATHDB-322103

Unipodal closure conjecture for positive roots

摘录 / EXCERPT: Unipodal closure conjecture. If $\Gamma\subseteq\Phi_{+}$ is unipodal, then $\overline{\Gamma}$ is biclosed in Φ_{+}.

conjecture | open | MathDB 页面 | 原文线索

V2-061194 | MATHDB-322104

Initial-section realization conjecture for closed quasipositive systems

摘录 / EXCERPT: Quasipositive initial-section conjecture. There is a total order $\preceq$ of Λ such that M consists of all initial sections of $\preceq$.

conjecture | open | MathDB 页面 | 原文线索

V2-061195 | MATHDB-322105

Bruhat-order description of joins of biclosed subsets

摘录 / EXCERPT: Biclosed-join conjecture. For $\Gamma,\Lambda\in\mathrsfs{B}(\Phi_{+})$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-061196 | MATHDB-322114

Bedford's conjecture on basins of attraction of non-autonomous automorphisms

摘录 / EXCERPT: Let $(\Phi_{n,n+1})_{n\in\mathbb N}$ be a family of automorphisms of $\mathbb C^q$. Assume that there exist $0<\nu\leq\mu<1$ and $r>0$ such that ... The basin of attraction ... is Bedf...

conjecture | open | MathDB 页面 | 原文线索

V2-061197 | MATHDB-322116

The factorization-closure conjecture for strictly singular operators on Lorentz sequence spaces

摘录 / EXCERPT: Let $d_{w,p}$ be the Lorentz sequence space. Let $\mathcal{SS}$ denote the ideal of strictly singular operators on $d_{w,p}$, let J^j be the ideal generated by the formal identit...

conjecture | open | MathDB 页面 | 原文线索

V2-061198 | MATHDB-322117

The conjecture that both bases are transcendental in the integer-power equation

摘录 / EXCERPT: Let $T,R\in\mathbb{R}^+$ satisfy $T^R=R^T=N\in\mathbb{N}$ and $T\neq R$. Suppose $N\neq 16$, so that at least one of T and R is transcendental. Both-bases transcendence conject...

conjecture | open | MathDB 页面 | 原文线索

V2-061199 | MATHDB-322118

The conjecture that both solutions are transcendental for rational tower bases

摘录 / EXCERPT: Let $Q\in\mathbb{Q}\cap\left(0,e^{-e}\right]$ with $Q\neq 1/16$, and let $h_o(Q)$ and $h_e(Q)$ denote the two solutions identified in the preceding proposition, with...

conjecture | open | MathDB 页面 | 原文线索

V2-061200 | MATHDB-322123

Free-factor formulation of Wiegold's conjecture

摘录 / EXCERPT: Wiegold's free-factor conjecture. If $n\geq 3$, then F_n has a proper free factor H such that $\Psi(H)=G$.

conjecture | open | MathDB 页面 | 原文线索

V2-061201 | MATHDB-322133

Conjectured proportionality factors for the two-boundary Temperley-Lieb eigenvector

摘录 / EXCERPT: Let L be the system size, let $z_1,\ldots,z_L$ be spectral parameters, let ζ_0 and ζ_L be boundary parameters, and let k and s denote the functions and parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-061202 | MATHDB-322134

Polynomial-over-normalisation conjecture for the expectation value

摘录 / EXCERPT: Let F_L be the expectation value defined from the two-boundary Temperley-Lieb ground-state eigenvectors, and let Z_L be the corresponding normalisation. Let P denote a polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-061203 | MATHDB-322135

Invariance conjecture for the proposed expression of the expectation value

摘录 / EXCERPT: Let Y_L be the quantity defined in the paper from the finite-size expectation-value calculation, and let q be the model parameter. Let w be its spectral variable. Invariance...

conjecture | open | MathDB 页面 | 原文线索

V2-061204 | MATHDB-322136

Karhunen-Loève basis optimality conjecture for nonlinear reconstruction

摘录 / EXCERPT: Let ${\bf Y}:=(Y_1,\ldots,Y_N)$ be an N-dimensional Gaussian vector with independent zero-mean components of variances $\sigma_i=E(Y_i^2)$, ordered so that...

conjecture | open | MathDB 页面 | 原文线索

V2-061205 | MATHDB-322151

Monotonicity conjecture for the moment of inertia along three-body brake orbits

摘录 / EXCERPT: Assume a brake orbit segment in the three-body problem, beginning at brake time and ending at its first syzygy. Let $I(t)$ denote the total moment of inertia of the three bodies. M...

conjecture | open | MathDB 页面 | 原文线索

V2-061206 | MATHDB-322152

Order-preservation conjecture for equal-mass three-body syzygies

摘录 / EXCERPT: Let three equal masses be released from rest in a scalene triangular configuration whose side lengths satisfy ... Let $r_{ij}(t)$ be the distance between masses i and j, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061207 | MATHDB-322156

Radiality of solutions on the complex ball

摘录 / EXCERPT: Let $\mathcal{B}\subset\mathbb{C}^{n}$ be the ball, and consider the equation referred to in the source as equation, with parameter γ. Let u_{γ} denote the radial sol...

conjecture | open | MathDB 页面 | 原文线索

V2-061208 | MATHDB-322162

Linear independence of Dirichlet zero ordinates

摘录 / EXCERPT: Let $L(s,\chi)$ range over Dirichlet L-functions attached to primitive characters, and consider the nonnegative imaginary parts of their nontrivial zeros. The linear-independence...

conjecture | open | MathDB 页面 | 原文线索

V2-061209 | MATHDB-322164

Nonvanishing of Dirichlet L-functions at the central point

摘录 / EXCERPT: Let χ be a Dirichlet character and let $L(s,\chi)$ denote its Dirichlet L-function. The central-point nonvanishing conjecture. ... for every Dirichlet character χ. The so...

conjecture | open | MathDB 页面 | 原文线索

V2-061210 | MATHDB-322165

Simplicity of zeros of Dirichlet L-functions

摘录 / EXCERPT: Let $L(s,\chi)$ range over Dirichlet L-functions, with zeros counted according to multiplicity. The simplicity conjecture. Every zero of every Dirichlet L-function is simple. T...

conjecture | open | MathDB 页面 | 原文线索

V2-061211 | MATHDB-322166

Dirichlet ordinates in arithmetic progressions

摘录 / EXCERPT: Given a Dirichlet character χ, call a real number γ a χ-ordinate if $L(\sigma+i\gamma,\chi)=0$ for some $0<\sigma<1$. Let a and b be positive real numbers. The...

conjecture | open | MathDB 页面 | 原文线索

V2-061212 | MATHDB-322167

No common rational multiple for all Dirichlet ordinates

摘录 / EXCERPT: Let a Dirichlet ordinate be a nonnegative real number that is a χ-ordinate for some Dirichlet character χ. Silberman's consequence of linear independence. There does not...

conjecture | open | MathDB 页面 | 原文线索

V2-061213 | MATHDB-322182

Gallego-Purnaprajna conjecture on very ampleness of twice a polarization

摘录 / EXCERPT: Let X be a smooth Calabi-Yau threefold and L a globally generated, ample line bundle. For a surface $S\in |L|$, write $|L\mathbin{\vert}_S|$ for the restriction of the linear s...

conjecture | open | MathDB 页面 | 原文线索

V2-061214 | MATHDB-322184

Depth criterion for squarefree lexsegment ideals

摘录 / EXCERPT: Depth criterion conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-061215 | MATHDB-322188

Passare-Rullgård's conjecture on maximally sparse Laurent polynomials

摘录 / EXCERPT: Passare-Rullgård's conjecture. A maximally sparse Laurent polynomial has a solid amoeba.

conjecture | open | MathDB 页面 | 原文线索

V2-061216 | MATHDB-322189

Sharpness conjecture for Orthomin(k) on root-of-unity diagonal systems

摘录 / EXCERPT: Let $k\in\mathbb{N}$. For each $d\in\mathbb{N}$, let $U=\operatorname{diag}(1,\zeta_d,\zeta_d^2,\dots,\zeta_d^{d-1})$, where $\zeta_d=\exp(2\pi\mathbf{i}/d)$ is a primitive root o...

conjecture | open | MathDB 页面 | 原文线索

V2-061217 | MATHDB-322203

Minimal critical discriminant conjecture for critically separable rational maps

摘录 / EXCERPT: Let K be a number field, let $d\geq 2$ be an integer, let $\lambda\in K^\times$, and let S_λ be the finite set of Archimedean places and places at which λ is not...

conjecture | open | MathDB 页面 | 原文线索

V2-061218 | MATHDB-322204

Wilson's weaker cone conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Wilson's weaker cone conjecture. If c_2 is strictly positive on the closure of the Kähler cone of a Calabi-Yau threefold, then the Kähler cone of Z is generated by finitely man...

conjecture | open | MathDB 页面 | 原文线索

V2-061219 | MATHDB-322206

The universal rho-degeneracy conjecture for knots

摘录 / EXCERPT: Let K be a knot in S^{3}. A knot is ρ-degenerate when its reduced spectral sequence converges at $\underline{E}_{1}$ and the induced filtration $\underline{\rho}$ is const...

conjecture | open | MathDB 页面 | 原文线索

V2-061220 | MATHDB-322209

Etzion's conjecture on diameter perfect Lee codes

摘录 / EXCERPT: Etzion's conjecture. There is no $DPL(n,d)$ code for $n\geq 3$ and $d>4$, with the exception of the pair $(n,d)=(3,6)$.

conjecture | open | MathDB 页面 | 原文线索

V2-061221 | MATHDB-322211

The rigged-configuration bijection for type D KR crystals

摘录 / EXCERPT: Let $B=B^{r_1,s_1}\otimes B^{r_2,s_2}\otimes\cdots\otimes B^{r_L,s_L}$ be a tensor product of Kirillov--Reshetikhin crystals, and let $\operatorname{RC}(B)$ be its set of rigged co...

conjecture | solved | MathDB 页面 | 原文线索

V2-061222 | MATHDB-322212

Compatibility of the rigged-configuration bijection with non-affine Kashiwara operators

摘录 / EXCERPT: Let Φ be the map from rigged configurations of a tensor product of type $D_n^{(1)}$ Kirillov--Reshetikhin crystals to the corresponding crystal, and let e_i and f_i denote...

conjecture | open | MathDB 页面 | 原文线索

V2-061223 | MATHDB-322214

The cocharge-energy conjecture for type D rigged configurations

摘录 / EXCERPT: Let (ν,J) be a rigged configuration and let $b=\Phi(\nu,J)$ be its image under the bijection to the corresponding tensor product of type $D_n^{(1)}$ Kirillov--Reshetikhin cryst...

conjecture | solved | MathDB 页面 | 原文线索

V2-061224 | MATHDB-322216

Conjecture on the $(\operatorname{asc},\operatorname{RLmin})$ distribution for 021-avoiding ascent sequences

摘录 / EXCERPT: Let ${\mathcal S}_{021}(n)$ be the set of ascent sequences of length n avoiding 021, and let asc and RLmin denote, respectively, the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-061225 | MATHDB-322217

Conjecture on 0012-avoiding ascent sequences and Catalan bistatistics

摘录 / EXCERPT: Let ${\mathcal S}_{0012}(n)$ be the set of ascent sequences of length n avoiding 0012, let $A_{0012}(n)=|{\mathcal S}_{0012}(n)|$, let fwd be the length of the...

conjecture | open | MathDB 页面 | 原文线索

V2-061226 | MATHDB-322218

Conjecture on 210-avoiding ascent sequences and 3-noncrossing partitions

摘录 / EXCERPT: Let $A_{210}(n)$ be the number of ascent sequences of length n avoiding 210. A set partition of $\{1,2,\ldots,n\}$ is 3-noncrossing if it has no 3-crossing. The 210 counting co...

conjecture | open | MathDB 页面 | 原文线索

V2-061227 | MATHDB-322219

Conjecture on 0123-avoiding ascent sequences and bounded-height Dyck paths

摘录 / EXCERPT: Let $A_{0123}(n)$ be the number of ascent sequences of length n avoiding 0123. A Dyck path of semilength n has height at most 5 if its maximum vertical height is at most...

conjecture | open | MathDB 页面 | 原文线索

V2-061228 | MATHDB-322220

Conjecture on Wilf equivalence of 0021 and 1012

摘录 / EXCERPT: For a pattern p, let $A_p(n)$ be the number of ascent sequences of length n avoiding p. Two patterns are Wilf equivalent if their avoidance sequences agree for every n. The...

conjecture | open | MathDB 页面 | 原文线索

V2-061229 | MATHDB-322221

Conjecture on modified ascent sequences and Bell-number enumerations

摘录 / EXCERPT: A modified ascent sequence is obtained from an ascent sequence by the modification process described in the source. For patterns 101, 0101, 1021, 1102, 1120, and 1210,...

conjecture | open | MathDB 页面 | 原文线索

V2-061230 | MATHDB-322232

Quadratic Gröbner basis conjecture for binomial ideals

摘录 / EXCERPT: Let I be an ideal generated by binomials, and suppose that I has a quadratic Gröbner basis with respect to every monomial order. Say that the generators of I are in pairwise...

conjecture | open | MathDB 页面 | 原文线索

V2-061231 | MATHDB-322238

Geometric Breuil-Mézard conjecture for two-dimensional representations

摘录 / EXCERPT: Let ${\bar{r}}:G_{\mathbb{Q}_p}\to\operatorname{GL}_2(\mathbb F)$ and the deformation rings and multiplicities $a_{m,n}$ and $a^{\operatorname{cr}}_{m,n}$ be as above. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-061232 | MATHDB-322239

Inertial local Langlands conjecture for GL_n with monodromy zero

摘录 / EXCERPT: Let K be a finite extension of $\mathbb Q_p$, let τ be an inertial type, and let $\widetilde\tau$ be a Frobenius-semisimple Weil-Deligne representation of W_K. Inertial lo...

conjecture | open | MathDB 页面 | 原文线索

V2-061233 | MATHDB-322240

Generalized Breuil-Mézard conjecture for GL_n

摘录 / EXCERPT: Let K be a finite extension of $\mathbb Q_p$, let ${\bar{r}}:G_K\to\operatorname{GL}_n(\mathbb F)$ be continuous, and let $R^\square_{{\bar{r}},\lambda,\tau}$ be the potentially...

conjecture | open | MathDB 页面 | 原文线索

V2-061234 | MATHDB-322247

Torisu's unknotting-number-one conjecture for Montesinos knots

摘录 / EXCERPT: Torisu's conjecture. The knot satisfies $u(K)=1$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-061235 | MATHDB-322249

Farkas's pure-resolution conjecture for nonspecial curves

摘录 / EXCERPT: Farkas's conjecture. A general smooth normal curve $C\subset\mathbb P^r$ of odd genus $g=2p+3$ and degree $d=2g$ has a pure resolution, equivalently satisfies N_p. A general smoo...

conjecture | solved | MathDB 页面 | 原文线索

V2-061236 | MATHDB-322251

The exceptional zero conjecture for p-adic L-functions of motives

摘录 / EXCERPT: Let M be a p-ordinary motive, let $L_p(s,M,\chi)$ be its analytic p-adic L-function, and let a_0 be a critical integer with character χ_0 such that the interpolation...

conjecture | open | MathDB 页面 | 原文线索

V2-061237 | MATHDB-322255

Nadler-Tanaka's point-generation conjecture for Lagrangian cobordisms

摘录 / EXCERPT: Nadler-Tanaka's point-generation conjecture. The object $pt\in\Lag_{pt}(pt)$ is a generator, every object of $\Lag_{pt}(pt)$ is a finite colimit of objects $pt[k]$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-061238 | MATHDB-322257

Nadler-Tanaka's cotangent-bundle conjecture for Lagrangian cobordisms

摘录 / EXCERPT: Nadler-Tanaka's cotangent-bundle conjecture. The object $T_x^*X\in\Lag_X(T^*X)$ is a generator; every object is a finite colimit of objects $T_x^*X[k]$, $k\in\mathbb Z$; and

conjecture | open | MathDB 页面 | 原文线索

V2-061239 | MATHDB-322258

Nadler-Tanaka's constructible-sheaf model for cotangent bundles

摘录 / EXCERPT: Nadler-Tanaka's constructible-sheaf conjecture. There is a canonical equivalence of stable ∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-061240 | MATHDB-322259

Nadler-Tanaka's stratified constructible-sheaf conjecture

摘录 / EXCERPT: Nadler-Tanaka's stratified constructible-sheaf conjecture. There is a canonical equivalence of stable ∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-061241 | MATHDB-322260

Nadler-Tanaka's cosheaf conjecture for local Lagrangian cobordism categories

摘录 / EXCERPT: Nadler-Tanaka's cosheaf conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-061242 | MATHDB-322262

Conjectural generating function for fibre-semistable sheaves on Hirzebruch surfaces

摘录 / EXCERPT: Let Σ_ℓ be a Hirzebruch surface with generic fibre f, let $r\geq 1$, and let $c_1\in H_2(\Sigma_\ell,\mathbb{Z})$. Write $H_{r,c_1}(z,\tau;f)$ for the generating functi...

conjecture | open | MathDB 页面 | 原文线索

V2-061243 | MATHDB-322263

Hodge-theoretic generalization of the fibre-restriction generating-function conjecture

摘录 / EXCERPT: Let $\Sigma_{g,\ell}$ be a ruled surface over a Riemann surface C_g of genus $g>0$, with fibre f. Let $r\geq 1$, c_1 be a first Chern class, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061244 | MATHDB-322267

Morrison's stability conjecture for Hilbert points of bicanonical curves

摘录 / EXCERPT: Let C be a smooth bicanonical curve of genus $g\geq 3$, and let m^{th} Hilbert point denote the Hilbert point associated to its bicanonical embedding at level m. Morrison's c...

conjecture | open | MathDB 页面 | 原文线索

V2-061245 | MATHDB-322304

Conjecture C for parabolic subgroups of finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group, let W_L be a proper parabolic subgroup of rank r, and let $\mathcal C_L$ represent the cuspidal conjugacy classes of W_L. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-061246 | MATHDB-322305

Stable Bernstein center formulation of the endoscopic character identity

摘录 / EXCERPT: Stable Bernstein center character identity conjecture. For any η-endoscopic transfer $h^{\mathbf{H}}$ of h, the function $z_\tau^{\mathbf{H}}\ast h^{\mathbf{H}}$ is a twiste...

conjecture | open | MathDB 页面 | 原文线索

V2-061247 | MATHDB-322306

Athanasiadis's freeness conjecture for coned deformations of braid arrangements

摘录 / EXCERPT: Athanasiadis's freeness conjecture. The coning $c\mathcal{A}_G$ is free if and only if G satisfies (A1) and (A2).

conjecture | open | MathDB 页面 | 原文线索

V2-061248 | MATHDB-322307

The upper bound conjecture for equivalence-class intersections in simplicial semigroup rings

摘录 / EXCERPT: Let B be the affine semigroup under consideration, let B_A denote its simplicial subsemigroup, and let $\sim$ be the equivalence relation on B. For $x,y\in B$, define ... For...

conjecture | refuted | MathDB 页面 | 原文线索

V2-061249 | MATHDB-322313

Segre's conjecture for generic K3 surfaces

摘录 / EXCERPT: Let Y be a generic $K3$ surface, and let $\mathcal{L}$ be a nonempty and reduced linear system on Y. Segre's conjecture for generic K3 surfaces. The linear system $\mathcal{L}$...

conjecture | open | MathDB 页面 | 原文线索

V2-061250 | MATHDB-322315

The list conjecture for negative curves on blown-up surfaces

摘录 / EXCERPT: Let $X=\operatorname{Bl}_r Y$ be the blow-up of a surface Y at r points. Let $C\subset X$ be a nonexceptional integral curve with $C^2<0$. List conjecture. There exist a positi...

conjecture | open | MathDB 页面 | 原文线索

V2-061251 | MATHDB-322327

Singularity confinement conjecture for the pentagram map in even dimension

摘录 / EXCERPT: Singularity confinement conjecture. Singularity confinement holds generically on X_S unless

conjecture | open | MathDB 页面 | 原文线索

V2-061252 | MATHDB-322338

Tame triangularization conjecture for locally nilpotent derivations

摘录 / EXCERPT: Tame triangularization conjecture. If $\exp D$ is tame, then D is tamely triangularizable.

conjecture | open | MathDB 页面 | 原文线索

V2-061253 | MATHDB-322339

Tame-coordinate conjecture for exponentials of locally nilpotent derivations

摘录 / EXCERPT: Tame-coordinate conjecture. If $\exp D$ is tame, then D kills a tame coordinate of $k[\bf x]$ over k.

conjecture | open | MathDB 页面 | 原文线索

V2-061254 | MATHDB-322340

Rank-three wildness conjecture for exponentials

摘录 / EXCERPT: Rank-three wildness conjecture. If $\operatorname{rank}D=3$, then $\exp D$ is wild.

conjecture | open | MathDB 页面 | 原文线索

V2-061255 | MATHDB-322341

Rank-one derivation conjecture for non-exponentially-wild coordinates

摘录 / EXCERPT: Rank-one derivation conjecture. There always exists $D\in\operatorname{LND}_k k[\bf x]$ with $\operatorname{rank}D=1$ such that $D(f)=0$ and D is tamely triangularizable.

conjecture | open | MathDB 页面 | 原文线索

V2-061256 | MATHDB-322360

Split exactness conjecture for maximal compact multigerm equivalence groups

摘录 / EXCERPT: Let f be a finitely $\mathcal{A}$-determined multigerm which is also a critical normalization. Write $D(f)=f(\Sigma(f))$ for its discriminant, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061257 | MATHDB-322364

Infinite representations conjecture for automatic presentations of the nonnegative rationals

摘录 / EXCERPT: Let $k\geq 2$ and let L be a regular language of base-k representations of pairs whose quotient set is ... A representation in L is a word encoding a pair of nonnegative intege...

conjecture | open | MathDB 页面 | 原文线索

V2-061258 | MATHDB-322366

A Pólya integral positivity conjecture for Gegenbauer polynomials

摘录 / EXCERPT: Integral positivity conjecture. The inequality $F_n^{\lambda,\delta}(t)>0$ holds for every $n\in\mathbb{N}_0$ and every $t\in(0,\pi]$ if and only if $\delta\geq\lambda+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-061259 | MATHDB-322371

Equality conjecture for invariants of homogeneous forms

摘录 / EXCERPT: Let X_m^n be the space of homogeneous forms of degree n in m variables, with $n\geq 3$. For $Q\in X_m^n$, let ${\mathcal R}_m^n$ be the collection of invariant rational funct...

conjecture | open | MathDB 页面 | 原文线索

V2-061260 | MATHDB-322373

Erdős-Rubin-Taylor multiplicative choosability conjecture

摘录 / EXCERPT: Erdős-Rubin-Taylor conjecture. If G is (a,b)-choosable, then G is (am,bm)-choosable for every integer $m\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-061261 | MATHDB-322374

McDiarmid-Reed conjecture for triangle-free induced subgraphs of the triangular lattice

摘录 / EXCERPT: McDiarmid-Reed conjecture. Every triangle-free induced subgraph of the triangular lattice is

conjecture | open | MathDB 页面 | 原文线索

V2-061262 | MATHDB-322379

Theta-correspondence characterization by distinction and L-functions

摘录 / EXCERPT: Let G be the relevant orthogonal group and let σ be an irreducible cuspidal automorphic representation of $G(\mathbb A)$ occurring as an irreducible subspace of...

conjecture | open | MathDB 页面 | 原文线索

V2-061263 | MATHDB-322397

The Łojasiewicz exponent equality conjecture for nondegenerate singularities

摘录 / EXCERPT: Łojasiewicz exponent equality conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-061264 | MATHDB-322413

Exponential 3-arboral coloring conjecture for toroidal graphs

摘录 / EXCERPT: Let G be a graph embeddable in the torus. An arboreal 3-coloring of G is a coloring with three colors such that each color class induces a forest. Exponential 3-arboral colorin...

conjecture | open | MathDB 页面 | 原文线索

V2-061265 | MATHDB-322416

Top-down polynomial conjecture for the partition coefficient $C_{0,1,N-r}(N)$

摘录 / EXCERPT: For integers N and $r>0$, let $C_{0,1,N-r}(N)$ denote the Rademacher coefficient indexed by the pole at $x=1$ and order $N-r$, and let $P_{0,1,N-r}(N)$ be the polynomial appearin...

conjecture | open | MathDB 页面 | 原文线索

V2-061266 | MATHDB-322422

Vanishing of finite-dimensional representations in infinite sofic groups

摘录 / EXCERPT: Let Γ be an infinite sofic group, let Σ be a sofic approximation, and let V be a finite-dimensional representation of Γ. The finite-dimensional vanishing con...

conjecture | open | MathDB 页面 | 原文线索

V2-061267 | MATHDB-322427

The fullness conjecture for the constructed exceptional collections

摘录 / EXCERPT: Let $\BG$ be a group of type BCD, and let $X=\BG/\BP$ be one of the homogeneous spaces considered in Theorem 1. The construction produces an exceptional collection of expected le...

conjecture | open | MathDB 页面 | 原文线索

V2-061268 | MATHDB-322429

The Grassmannian curve conjecture for semiorthogonal decompositions

摘录 / EXCERPT: Let C be a monotone curve in a $k\times l$ rectangle as described in the construction, with intersection points Q_i and associated integers a_i,b_i,c_i,d_i. Define blocks...

conjecture | open | MathDB 页面 | 原文线索

V2-061269 | MATHDB-322443

Nonintegrability conjecture for homogeneous potentials with small eigenvalues

摘录 / EXCERPT: Nonintegrability conjecture. There exists $k\in\mathbb{N}^*$ such that the variational equation at order k is not integrable.

conjecture | open | MathDB 页面 | 原文线索

V2-061270 | MATHDB-322444

Fifth- and seventh-order nonintegrability conjecture for homogeneous potentials

摘录 / EXCERPT: Fifth- and seventh-order nonintegrability conjecture. If n is odd, then the fifth variational equation is not integrable. If n is even and $n\geq 4$, then the seventh variation...

conjecture | open | MathDB 页面 | 原文线索

V2-061271 | MATHDB-322446

The mean-field conjecture for critical exponents of quasi-isometric trees

摘录 / EXCERPT: Let Γ be an infinite d-regular tree, and let Γ' be a graph quasi-isometric to Γ, including a decoration of Γ. A critical exponent is the power-law exp...

conjecture | open | MathDB 页面 | 原文线索

V2-061272 | MATHDB-322447

The mean-field conjecture for cell-wise first-wave critical exponents

摘录 / EXCERPT: Let Γ be an infinite d-regular tree, and let Γ' be a graph quasi-isometric to Γ, including a decoration of Γ. The cell-wise first-wave critical expone...

conjecture | open | MathDB 页面 | 原文线索

V2-061273 | MATHDB-322448

The singularity conjecture for the weighted-cluster generating function

摘录 / EXCERPT: Let Γ' be a decoration of a d-regular tree, and let $g(x)$ be the generating function for weighted clusters containing the origin. Let D denote the exponential growth fa...

conjecture | open | MathDB 页面 | 原文线索

V2-061274 | MATHDB-322470

Complete-series conjecture for linear stability and dual span bundles

摘录 / EXCERPT: Let C be a smooth curve and let ${\cal L}$ be a line bundle on C. Let $M_{\cal L}$ denote the kernel bundle of the evaluation map...

conjecture | open | MathDB 页面 | 原文线索

V2-061275 | MATHDB-322474

Grigorchuk's stronger conjecture for finitely presented groups

摘录 / EXCERPT: Let G be a finitely presented group. A free subsemigroup on two generators is a subsemigroup generated by two elements with no nontrivial relations. Grigorchuk's stronger conject...

conjecture | open | MathDB 页面 | 原文线索

V2-061276 | MATHDB-322481

Extension of the Chern-character filtration theorem to arbitrary-rank bundles

摘录 / EXCERPT: Chern-character filtration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061277 | MATHDB-322491

Braid group conjecture for acyclic quivers

摘录 / EXCERPT: Let Q be an acyclic quiver. Braid group conjecture. There is an isomorphism ... This is one of two conjectures proposed in the paper; the supplied text gives no resolution or furth...

conjecture | open | MathDB 页面 | 原文线索

V2-061278 | MATHDB-322492

Induced-path conjecture for weak equivalence classes

摘录 / EXCERPT: Let Q be a quiver, and let $\operatorname{\overrightarrow{\mathbf{P}}}(Q)$ be the directed path graph. Two directed paths are weakly equivalent when the unordered sets of their e...

conjecture | open | MathDB 页面 | 原文线索

V2-061279 | MATHDB-322494

The highest-coefficient conjecture for the polynomials W_r

摘录 / EXCERPT: Highest-coefficient conjecture. For a given $r>1$, the highest non-zero coefficient is $\gamma(\nu(r+1),r)$, and $\gamma(\nu(r+1),r)>0$. Together with the proved vanishing of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-061280 | MATHDB-322504

The quarter-integrality conjecture for harmonic Maass-form periods

摘录 / EXCERPT: Let Δ and r be as in the preceding notation, let E_Δ be the corresponding elliptic curve with minimal Weierstrass model W_Δ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061281 | MATHDB-322505

Coincidence threshold of even-dimensional Chebyshev hypersurfaces

摘录 / EXCERPT: Let ${\mathcal C}(n,d)$ be the Chebyshev hypersurface of degree d in $\mathbb{P}^n$, where $n=2n_1$ is even. Write $ct(D)$ for the coincidence threshold of a hypersurface D. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-061282 | MATHDB-322506

Defect formulas for Chebyshev surfaces and threefolds

摘录 / EXCERPT: Let $\mathcal C(n,d)$ denote the Chebyshev hypersurface of degree d in $\mathbb P^n$, and let $\mathcal N(n,d)$ be its set of nodes. For a finite node set $\mathcal N$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061283 | MATHDB-322509

Quadratic convergence of the refined Azuma-Hoeffding bound to the Bhattacharyya parameter

摘录 / EXCERPT: Let $\{U_k,\mathcal{F}_k\}_{k=0}^h$ be the martingale sequence introduced in the subsection, and let $Z_2^{(m)}$ denote the corresponding refined exponential base and...

conjecture | open | MathDB 页面 | 原文线索

V2-061284 | MATHDB-322511

The newform zeta-integral conjecture for unramified U(2,1)

摘录 / EXCERPT: Newform zeta-integral conjecture. There exists a newform v for π such that

conjecture | open | MathDB 页面 | 原文线索

V2-061285 | MATHDB-322514

Generalized Goldfeld conjecture for quadratic twists over number fields

摘录 / EXCERPT: Let K be a number field, let E/K be an elliptic curve, and let $\mathcal{C}(K,X)$ denote the set of quadratic characters of K of height at most X. Write δ for the de...

conjecture | open | MathDB 页面 | 原文线索

V2-061286 | MATHDB-322516

Crew's petites camarades conjecture for curves

摘录 / EXCERPT: Let k be a finite field of characteristic p, let U be a smooth geometrically connected curve over k, and let $\mathscr{F}$ be an irreducible smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-061287 | MATHDB-322538

The Acyclic System Conjecture

摘录 / EXCERPT: Acyclic System Conjecture. Any acyclic system of permutations on the edges of the simplex $n\Delta_{d-1}$ is achievable as the system of permutations of a fine mixed subdivision.

conjecture | refuted | MathDB 页面 | 原文线索

V2-061288 | MATHDB-322539

The weak Spread Out Simplices Conjecture

摘录 / EXCERPT: Weak Spread Out Simplices Conjecture. Any n spread out simplices in $n\Delta_{d-1}$ can be achieved as the simplices of an acyclic system of permutations.

conjecture | open | MathDB 页面 | 原文线索

V2-061289 | MATHDB-322543

Bounded ML and REML degrees for one-way variance component models with general mean structure

摘录 / EXCERPT: Consider the one-way layout ... where $i=1,,,q$, $j=1,,,n_i$, the random effects satisfy $_i\boldsymbol{sim}(0,)$, the errors satisfy $_{ij}\boldsymbol{sim}(0,)$, and all these varia...

conjecture | open | MathDB 页面 | 原文线索

V2-061290 | MATHDB-322545

Conjecture on near-optimal Hamilton coverings of random graphs

摘录 / EXCERPT: Hamilton covering conjecture. For any $p=\omega(\ln n)/n$, the random graph $G(n,p)$ a.a.s. admits a covering of its edges with at most

conjecture | open | MathDB 页面 | 原文线索

V2-061291 | MATHDB-322549

The critical-length conjecture for standard components of linear slices

摘录 / EXCERPT: Let ${\mathcal L}_c$ denote the linear slice at parameter c, and distinguish its standard component from its non-standard components. Critical-length conjecture. There exists a u...

conjecture | open | MathDB 页面 | 原文线索

V2-061292 | MATHDB-322563

Ratios Conjecture prediction for the 1-level density of Dirichlet L-functions

摘录 / EXCERPT: Let q be a modulus, let f be the test function used in the 1-level density, and let $D_{1;q}(\widehat{f})$ denote the corresponding density with scaling parameter $Q=q$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-061293 | MATHDB-322565

The laminarity conjecture for higher bifurcation currents

摘录 / EXCERPT: Let Λ be a parameter space of dimension D, let T_{bif} be the bifurcation current, and let $U\subset\Lambda$ be an open set where $D-k$ critical points, counte...

conjecture | open | MathDB 页面 | 原文线索

V2-061294 | MATHDB-322576

Frequency-measure conjecture for p-class tower groups

摘录 / EXCERPT: Let G be a finitely generated pro-p group. Let $\operatorname{Freq}(G)$ denote its limiting frequency among imaginary quadratic fields, and let $\operatorname{Meas}(G)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-061295 | MATHDB-322577

Periodic IPAD measure conjectures for 3-class tower groups

摘录 / EXCERPT: An IPAD is the invariant displayed in the source in the form of an abelianization together with the corresponding abelianizations of selected index subgroups. The notation...

conjecture | open | MathDB 页面 | 原文线索

V2-061296 | MATHDB-322595

Independence of conjugation-free presentations from geometric choices

摘录 / EXCERPT: Independence conjecture. The property of having a conjugation-free geometric presentation is independent of the choices of O, $cell$ and $u\incell$.

conjecture | open | MathDB 页面 | 原文线索

V2-061297 | MATHDB-322596

Conjugation-free presentation conjecture for adjoining a conic

摘录 / EXCERPT: Conjugation-free extension conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061298 | MATHDB-322608

Layered-pattern maximization conjecture for the Stanley-Wilf limit

摘录 / EXCERPT: For a permutation pattern τ, let $S_n(\tau)$ denote the number of permutations of length n avoiding τ, and let ... be its Stanley-Wilf limit. A pattern is layered if it i...

conjecture | open | MathDB 页面 | 原文线索

V2-061299 | MATHDB-322609

Quadratic upper-bound conjecture for the Stanley-Wilf limit

摘录 / EXCERPT: Let τ be a permutation pattern of length ℓ, and let $L(\tau)$ denote its Stanley-Wilf limit. Quadratic upper-bound conjecture. For any pattern τ of length ℓ, ...

conjecture | open | MathDB 页面 | 原文线索

V2-061300 | MATHDB-322610

Inversion-monotonicity conjecture for nonidentity permutation patterns

摘录 / EXCERPT: For a permutation pattern τ, let $\mathcal{S}_{n}^{k}(\tau)$ be the set of τ-avoiding permutations of length n with exactly k inversions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061301 | MATHDB-322625

The DAHA-Jones polynomial conjecture

摘录 / EXCERPT: Let $K=K_{r,s}$ be a torus knot with $r,s\in\mathbf{Z}$, let R be a root system, and let $b\in P_+$ be a dominant weight. Define ... Here P_b is the symmetric Macdonald polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-061302 | MATHDB-322626

DAHA super-polynomial evaluation conjecture

摘录 / EXCERPT: Let $b\in P_+$ correspond to a Young diagram μ_b with rows of lengths m_i, and let $b_i=m_i\omega_1$. Write ord for the order of μ_b. Evaluation conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-061303 | MATHDB-322628

Type-C hyper-polynomial hyper-duality conjecture

摘录 / EXCERPT: Identify dominant weights with Young diagrams and let b^{tr} correspond to the transposed diagram. Let $\mathcal{H}^{C}_{r,s}(b;q,t,u,a)$ be the type-C hyper-polynomial. Hyper-...

conjecture | open | MathDB 页面 | 原文线索

V2-061304 | MATHDB-322629

Type-B hyper-polynomial conjecture

摘录 / EXCERPT: Let $K=K_{r,s}$ be a knot with $r,s\in\mathbf{Z}$, and let b be a dominant weight for all B_m with $m\ge n\ge2$. Type-B hyper-polynomial conjecture. There exists a polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-061305 | MATHDB-322631

Constancy conjecture for cyclic solutions of the functional equations

摘录 / EXCERPT: Constancy conjecture. Then g is constant. The source presents this as a proposed extension of the assertion that non-constant outer solutions do not occur in the corresponding cl...

conjecture | open | MathDB 页面 | 原文线索

V2-061306 | MATHDB-322636

The totally symmetric polymorphism conjecture for omega-categorical CSPs

摘录 / EXCERPT: Let $Gamma$ be an $omega$-categorical structure with finite relational signature. A family of totally symmetric polymorphisms consists of polymorphisms of every arity whose value i...

conjecture | open | MathDB 页面 | 原文线索

V2-061307 | MATHDB-322643

The upper-tail exponent conjecture for cliques

摘录 / EXCERPT: Upper-tail exponent conjecture. For any H and $p>n^{-1/m_H}$,

conjecture | open | MathDB 页面 | 原文线索

V2-061308 | MATHDB-322653

Generic three-dimensional invariant-ring conjecture

摘录 / EXCERPT: Let $G=({\bf Z}/p)^r=\langle e_1,\ldots,e_r\rangle$ and let V be the generic three-dimensional representation over $\mathbf{k}=\mathbf{F}_p(x_{1j},x_{2j}\mid j=1,\ldots,r)$. Defi...

conjecture | open | MathDB 页面 | 原文线索

V2-061309 | MATHDB-322654

Faithful three-dimensional invariant-ring conjecture

摘录 / EXCERPT: Let $G=({\bf Z}/p)^r$ and let V be a faithful three-dimensional modular representation of G over a field of characteristic p. Its invariant ring is $\mathbf{k}[V]^G$. Faithfu...

conjecture | open | MathDB 页面 | 原文线索

V2-061310 | MATHDB-322657

Geometric-mean inequalities for generalized sine and hyperbolic sine

摘录 / EXCERPT: For $p,q>1$, define the generalized sine functions ${\rm sin}_{p,q}$ and ${\rm sinh}_{p,q}$ as the inverses of ${\rm arcsin}_{p,q}$ and ${\rm arsinh}_{p,q}$, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061311 | MATHDB-322658

Moment convergence conjecture for generation sizes of simply generated trees

摘录 / EXCERPT: Let ${\mathcal T}_n$ be the conditioned simply generated tree, let $\widehat{{\mathcal T}}$ be its limiting size-biased tree, and let $l_k(T)$ denote the number of vertices at dist...

conjecture | open | MathDB 页面 | 原文线索

V2-061312 | MATHDB-322659

Height scaling conjecture for infinite-variance simply generated trees

摘录 / EXCERPT: Let ${\mathcal T}_n$ be the conditioned simply generated tree, and let $H({\mathcal T}_n)$ denote its height. Assume that the condensation parameter satisfies $\nu=1$ and that the...

conjecture | open | MathDB 页面 | 原文线索

V2-061313 | MATHDB-322679

Conjectured asymptotic formula for the coherence of random sparse vectors

摘录 / EXCERPT: Fix N and M. Let $\boldsymbol{a} \in \mathbb{R}^N$ be a random K-sparse vector whose K nonzero entries, on an arbitrary support S, are independent and identically distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-061314 | MATHDB-322685

The entropy bound for optimally compressed chess endgame tablebases

摘录 / EXCERPT: Entropy-tablebase bound. The relation

conjecture | open | MathDB 页面 | 原文线索

V2-061315 | MATHDB-322702

The linearization maximum conjecture for set-system dimension

摘录 / EXCERPT: A set system is a family of sets, and its dimension $\dim \mathcal{X}$ is its learning-sequence order type. A linearization of a set system is a suitable set system preserving the...

open_problem | open | MathDB 页面 | 原文线索

V2-061316 | MATHDB-322703

The better elastic set-system closure conjecture

摘录 / EXCERPT: A better elastic set system (BESS) is a set system corresponding to a better quasi-ordering. For a set system $\mathcal{L}$, define ... and let $\hat{\mathcal{L}}$ be the class conta...

conjecture | open | MathDB 页面 | 原文线索

V2-061317 | MATHDB-322708

Adamović-Milas logarithmic constant term identity

摘录 / EXCERPT: Adamović-Milas logarithmic constant term conjecture. For odd positive integer n and nonnegative integer k, set $m=(n-1)/2$ and $K=2k+1$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061318 | MATHDB-322709

Complex BC_n constant term conjecture

摘录 / EXCERPT: Let $n\equiv\zeta\pmod 4$ with $\zeta\in\{0,1\}$, let $u\in\mathbb C$ satisfy ... and let τ_{ij} and σ_{ij} be signatures satisfying the paper's condition. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-061319 | MATHDB-322712

Existence conjecture for fullerene c-disks

摘录 / EXCERPT: Fullerene disk existence conjecture. Except for $(c,v)=(1,42)$, $(3,24)$, and $(5,22)$, a fullerene c-disk with $c\geq 1$ and v vertices exists if and only if v is even and

conjecture | open | MathDB 页面 | 原文线索

V2-061320 | MATHDB-322720

The run-transform conjecture for paths with upsteps, flatsteps and downsteps

摘录 / EXCERPT: Fix nonnegative integers j and m. Consider paths formed from j-upsteps $U_j=(j,j)$, flatsteps $F=(2,0)$, and downsteps $D=(1,-1)$, starting at $(0,m)$ and ending on the x-a...

conjecture | open | MathDB 页面 | 原文线索

V2-061321 | MATHDB-322734

The conjecture on three rational linear factors of trinomials

摘录 / EXCERPT: Three-linear-factor conjecture. There are no trinomials $x^n+A x+B$ defined over $\mathbb{Q}$ with reducibility type $(1,1,1,n-3)$.

conjecture | open | MathDB 页面 | 原文线索

V2-061322 | MATHDB-322740

Conjectural integrality, vanishing and multinomial distribution of small-phase-space r-spin numbers

摘录 / EXCERPT: On the small phase space, write $\langle\tau_{m_1}\cdots\tau_{m_s}\rangle$ for the corresponding r-spin correlation number. The conjectured properties concern integers...

conjecture | open | MathDB 页面 | 原文线索

V2-061323 | MATHDB-322742

The conjecture that subharmonic quadrature domains satisfy the equivalent conditions of Theorem 1

摘录 / EXCERPT: Let Ω be a subharmonic quadrature domain as above, with its associated function u and boundary decomposition as in the preceding setup. Subharmonic quadrature-domain conje...

conjecture | open | MathDB 页面 | 原文线索

V2-061324 | MATHDB-322758

Macdonald superpolynomial Hall-Littlewood positivity conjecture

摘录 / EXCERPT: Hall-Littlewood positivity conjecture. The coefficients $K_{\Lambda\Omega}(t)$ and $\bar K_{\Lambda\Omega}(t)$ defined by

conjecture | open | MathDB 页面 | 原文线索

V2-061325 | MATHDB-322759

Finite multiplication conjecture for stable Grothendieck P-functions

摘录 / EXCERPT: Finite multiplication conjecture. The direct sum

conjecture | open | MathDB 页面 | 原文线索

V2-061326 | MATHDB-322760

Finite multiplication conjecture for stable Grothendieck Q-functions

摘录 / EXCERPT: Finite multiplication conjecture. The direct sum

conjecture | open | MathDB 页面 | 原文线索

V2-061327 | MATHDB-322766

Matsui et al.'s root-bound conjecture for Ehrhart polynomials

摘录 / EXCERPT: Let ${\mathcal P} \subset {\mathbb R}^N$ be an integral convex polytope of dimension d, and let $i({\mathcal P},n)$ denote its Ehrhart polynomial. For a root α of...

conjecture | open | MathDB 页面 | 原文线索

V2-061328 | MATHDB-322767

Matsui et al.'s root-bound conjecture for Gorenstein Fano polytopes

摘录 / EXCERPT: Let ${\mathcal P} \subset {\mathbb R}^d$ be a Fano polytope, meaning an integral convex polytope of dimension d whose only interior integer point is the origin. Call...

conjecture | open | MathDB 页面 | 原文线索

V2-061329 | MATHDB-322770

Rigidity conjecture for type-preserving representations of 2-bridge links

摘录 / EXCERPT: Let $\rho:\pi_1(\text{\boldmath$T$})\to PSL(2,\mathbb{C})$ be a type-preserving representation such that its end-invariant set satisfies $\mathcal{E}(\rho)=\Lambda(\hat\Gamma_r)$....

conjecture | open | MathDB 页面 | 原文线索

V2-061330 | MATHDB-322772

The rationality conjecture for Siegel-Veech area constants

摘录 / EXCERPT: Let ${\mathcal M}_1$ be a regular $\operatorname{SL}(2,\mathbb R)$-invariant suborbifold in a stratum of Abelian differentials, and let $c_{\operatorname{area}}({\mathcal M}_1)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-061331 | MATHDB-322801

Original Gross-Prasad conjecture for special orthogonal groups

摘录 / EXCERPT: Let F be a number field with adèle ring $\mathbb{A}_F$, and let $V_n\subset V_{n+1}$ be quadratic spaces of dimensions n and $n+1$ over F, with $n\geq 2$ and V_n not a hype...

conjecture | open | MathDB 页面 | 原文线索

V2-061332 | MATHDB-322802

Ichino-Ikeda refined Gross-Prasad conjecture for special orthogonal groups

摘录 / EXCERPT: Let $G_n\subset G_{n+1}$ be the special orthogonal groups in the Gross-Prasad setting, and let π_n and π_{n+1} be the associated tempered cuspidal automorphic representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-061333 | MATHDB-322803

Calogero-Moser families conjecture

摘录 / EXCERPT: Let W be a complex reflection group with parameter function c, and let the Calogero-Moser families and Lusztig families be the corresponding partitions of the irreducible chara...

conjecture | open | MathDB 页面 | 原文线索

V2-061334 | MATHDB-322804

Calogero-Moser cells and Kazhdan-Lusztig cells conjecture

摘录 / EXCERPT: Let W be a complex reflection group with parameter function c, and let ${\mathfrak r}_c$ be the relevant ramification data, with a choice...

conjecture | open | MathDB 页面 | 原文线索

V2-061335 | MATHDB-322824

The cumulant conjecture for spectral moments of matrix gamma distributions

摘录 / EXCERPT: Cumulant conjecture. The right-hand side of this relation must be related to the pth cumulant of the pth moment of the mean spectral distribution of M, and this relation shou...

conjecture | open | MathDB 页面 | 原文线索

V2-061336 | MATHDB-322834

Positivity conjecture for variances of dashed-pattern statistics

摘录 / EXCERPT: Positivity conjecture. For any dashed pattern (τ,X), one has

conjecture | open | MathDB 页面 | 原文线索

V2-061337 | MATHDB-322837

Farkas's codimension conjecture for Brill-Noether strata of spin curves

摘录 / EXCERPT: Farkas's codimension conjecture. For $r\geq 1$ and $g\geq {r+2\choose 2}$, there exists a component of $\mathcal{S}_g^r$ having codimension ${r+1\choose 2}$ inside $\mathcal{S}_g$....

conjecture | open | MathDB 页面 | 原文线索

V2-061338 | MATHDB-322849

The centre conjecture for Mackey functors of a lextensive finite coproduct completion

摘录 / EXCERPT: Let $\mathcal{C}$ be a small category such that its finite coproduct completion $\operatorname{fam}\mathcal{C}$ is lextensive. Let $\operatorname{Aut}\mathcal{C}$ be the category o...

conjecture | open | MathDB 页面 | 原文线索

V2-061339 | MATHDB-322855

The strong soliton resolution conjecture for radial critical wave solutions

摘录 / EXCERPT: Consider the critical wave equation in three spatial dimensions, and let u be a radial energy class solution with energy close to the ground-state energy $E(W,0)$, where W deno...

conjecture | open | MathDB 页面 | 原文线索

V2-061340 | MATHDB-322857

Closure conjecture for multivariate q-holonomic sequences

摘录 / EXCERPT: Closure conjecture. Multivariate q-holonomic sequences are closed under twisting by complex roots of unity and under substitutions of the form $q\to q^{\alpha}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-061341 | MATHDB-322861

The non-discrete two-boundary-point stabiliser conjecture

摘录 / EXCERPT: Let T be a locally finite tree and let $G\leq\operatorname{Aut}(T)$ be closed, non-compact, and boundary-transitive. For distinct boundary points $\xi,\xi'\in\partial T$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061342 | MATHDB-322863

The directed bipartite graph directed 4-cycle conjecture

摘录 / EXCERPT: Let D be a directed bipartite graph with bipartition (A,B). Directed C_4 conjecture. If $d^{+}(u)>\frac{|B|}{3}$ for $u\in A$ and $d^{+}(v)\geq\frac{|A|}{3}$ for $v\in B$, or...

conjecture | open | MathDB 页面 | 原文线索

V2-061343 | MATHDB-322867

Uncorrelatedness of nonincident arcs in uniformly random DAGs

摘录 / EXCERPT: Let $G=(V,A)$ be a directed acyclic graph, with arcs represented by indicator random variables A_{ij} and covariance matrix Σ. Two arcs are incident when they share a comm...

conjecture | open | MathDB 页面 | 原文线索

V2-061344 | MATHDB-322868

Sparsity of the covariance matrix for uniformly random DAGs

摘录 / EXCERPT: Let $G=(V,A)$ be a directed acyclic graph, and let Σ denote the covariance matrix of its arc-indicator variables. Covariance-matrix sparsity conjecture. The covariance matri...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-061345 | MATHDB-322869

Monotonicity of arc covariance and correlation in uniformly random DAGs

摘录 / EXCERPT: Let $G=(V,A)$ be a directed acyclic graph, and consider two arcs that are incident on a common node. Let covariance and correlation refer to the corresponding arc-indicator random...

conjecture | open | MathDB 页面 | 原文线索

V2-061346 | MATHDB-322870

Bounds for covariance and correlation of incident arcs in uniformly random DAGs

摘录 / EXCERPT: Let $G=(V,A)$ be a directed acyclic graph, and consider two arcs incident on a common node. Incident-arc range conjecture. The covariance between the two arcs takes values in the i...

conjecture | open | MathDB 页面 | 原文线索

V2-061347 | MATHDB-322882

The directed out-star inducibility conjecture

摘录 / EXCERPT: The directed out-star conjecture. For every $k\geq3$,

conjecture | open | MathDB 页面 | 原文线索

V2-061348 | MATHDB-322889

Recursive tetris-type formula for powers of the Vandermonde determinant

摘录 / EXCERPT: Let $1\leq m\leq n$. Let $\lambda\vdash n(n-1)$ satisfy $n-1\leq\ell(\lambda)\leq n$ and ... Let $\mu\vdash n(n+1)$ satisfy ... and, if $m<n$, $\mu_j=\lambda_{j-1}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-061349 | MATHDB-322892

Conjecture on the diameter of the Hurwitz graph of the symmetric group

摘录 / EXCERPT: Let $G_T(n)$ be the Hurwitz graph whose vertices are reduced factorizations of the reverse permutation in S_n into transpositions, with edges given by local Hurwitz moves. Diamet...

conjecture | open | MathDB 页面 | 原文线索

V2-061350 | MATHDB-322893

Conjecture on the radius of the Hurwitz graph of type B_n

摘录 / EXCERPT: Let $H(B_n)$ be the Hurwitz graph of the finite Coxeter group of type B_n, whose vertices are reduced reflection factorizations of a Coxeter element and whose edges are local rig...

conjecture | open | MathDB 页面 | 原文线索

V2-061351 | MATHDB-322895

Conjecture on Wilf-equivalence of vincular patterns 2314 and 1234

摘录 / EXCERPT: A vincular pattern is a permutation pattern in which selected adjacent entries are required to occur consecutively; write $\sigma\sim\tau$ when two patterns are Wilf-equivalent, mea...

conjecture | open | MathDB 页面 | 原文线索

V2-061352 | MATHDB-322896

Conjectured filling-shape-Wilf-equivalences for consecutive patterns

摘录 / EXCERPT: For consecutive permutation patterns, write $\sigma\stackrel{fs}{\sim}\tau$ when σ and τ are filling-shape-Wilf-equivalent, meaning that they are avoided by fillings of ev...

conjecture | open | MathDB 页面 | 原文线索

V2-061353 | MATHDB-322910

Complementary McCarthy inequality beyond the known parameter ranges

摘录 / EXCERPT: Let A and B be positive operators and let $q\in\mathbb{R}$. Complementary McCarthy conjecture. ... for $-2<q<0$, while the reversed inequality should hold for $3\leq q$. The disp...

conjecture | open | MathDB 页面 | 原文线索

V2-061354 | MATHDB-322911

Monotonicity conjecture for weighted power means

摘录 / EXCERPT: Let X and Y be positive operators, and for $p\geq2$ define the weighted power mean $(tX^p+(1-t)Y^p)^{1/p}$ for $0\leq t\leq1$. Power-mean monotonicity conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-061355 | MATHDB-322913

General trace commutator inequality for functions

摘录 / EXCERPT: General trace commutator conjecture. There exists a constant c, independent of d, such that

conjecture | open | MathDB 页面 | 原文线索

V2-061356 | MATHDB-322914

Miao's majorisation conjecture for concave singular-value functions

摘录 / EXCERPT: Let $X,Y\in\mathbb{M}_{n,m}(\mathbb{C})$, and let $\sigma_i(X)$ denote the singular values in non-increasing order. Let $f:\mathbb{R}_+\to\mathbb{R}_+$ be concave with $f(0)=0$. Mi...

conjecture | open | MathDB 页面 | 原文线索

V2-061357 | MATHDB-322920

Generalized Pfaffian product formulas for Motzkin triangle entries

摘录 / EXCERPT: Let n and k be positive integers. Define the Motzkin-triangle entries by ... For a $2n\times 2n$ skew-symmetric matrix, let Pf denote its Pfaffian. Generalized M...

conjecture | open | MathDB 页面 | 原文线索

V2-061358 | MATHDB-322926

The parity conjecture for 2-torsion in the Tate-Shafarevich group

摘录 / EXCERPT: Let K be a number field and let E be an elliptic curve defined over K. The group of 2-torsion points in the Tate-Shafarevich group is $Ш(E/K)[2]$, viewed as an...

conjecture | open | MathDB 页面 | 原文线索

V2-061359 | MATHDB-322928

The rank bound conjecture for tree-automatic well-founded partial orders

摘录 / EXCERPT: Rank bound conjecture. Every tree-automatic well-founded partial order has rank below

conjecture | open | MathDB 页面 | 原文线索

V2-061360 | MATHDB-322932

Orbifold Cartan conjecture for ramified holomorphic curves

摘录 / EXCERPT: Orbifold Cartan conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061361 | MATHDB-322936

Fejes Toth's solidity conjecture for the triangular packing

摘录 / EXCERPT: Fejes Toth's conjecture. The triangle packing in the plane, with one packing disk removed, is solid.

conjecture | open | MathDB 页面 | 原文线索

V2-061362 | MATHDB-322937

Density Gap Conjecture for finite triangular-torus packings

摘录 / EXCERPT: Density Gap Conjecture. If $n+1$ is a triangular lattice number but n is not, then the most dense packing of n equal disks in the triangular torus has density

conjecture | open | MathDB 页面 | 原文线索

V2-061363 | MATHDB-322938

Density bound conjecture for collectively jammed triangular-torus packings

摘录 / EXCERPT: Collectively jammed density conjecture. If such a packing is collectively jammed and is not a triangular packing, then its density is at most

conjecture | open | MathDB 页面 | 原文线索

V2-061364 | MATHDB-322939

The normal-or-analytic conjecture for subnormal Toeplitz operators

摘录 / EXCERPT: Let T_ϕ be a scalar Toeplitz operator on the Hardy space whose self-commutator $[T_\phi^*,T_\phi]$ has finite rank. Normal-or-analytic conjecture. If T_ϕ is subnormal, th...

conjecture | open | MathDB 页面 | 原文线索

V2-061365 | MATHDB-322941

Semisimplicity conjecture for the complex Hodge bundle

摘录 / EXCERPT: Let ${\mathcal L}_1$ be a suborbifold in the moduli space of unit area Abelian differentials, or in the moduli space of unit area meromorphic quadratic differentials with at most s...

conjecture | open | MathDB 页面 | 原文线索

V2-061366 | MATHDB-322950

Geometric Langlands compatibility with geometric Satake

摘录 / EXCERPT: Let X be a smooth complete curve, let G be a reductive group, let $x\in X$, and let $\Sph(G,x)$ be the spherical Hecke category acting on $\Dmod(\Bun_G)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061367 | MATHDB-322957

Stanton's monotonicity conjecture for core partitions

摘录 / EXCERPT: Stanton's monotonicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061368 | MATHDB-322958

Positivity conjecture for self-conjugate 6-core partitions

摘录 / EXCERPT: Let $sc_6(n)$ denote the number of self-conjugate 6-core partitions of n, and let n be a positive integer. Positivity conjecture for self-conjugate 6-core partitions. ... excep...

conjecture | open | MathDB 页面 | 原文线索

V2-061369 | MATHDB-322959

Exceptional-value conjecture for self-conjugate 6- and 4-core counts

摘录 / EXCERPT: Let $sc_t(n)$ denote the number of self-conjugate t-core partitions of n, and let n be an integer with $n>15$. Exceptional-value conjecture for self-conjugate 6- and 4-core c...

conjecture | open | MathDB 页面 | 原文线索

V2-061370 | MATHDB-322962

Unimodality conjecture for normalized core-count increments

摘录 / EXCERPT: Let $p(n)$ be the number of partitions of n, let $c_t(n)$ be the number of t-core partitions of n, and define ... For fixed $n\geq 63$, consider the sequence indexed by t wit...

conjecture | open | MathDB 页面 | 原文线索

V2-061371 | MATHDB-322963

Unimodality conjecture for normalized self-conjugate core increments

摘录 / EXCERPT: Let $sc_t(n)$ be the number of self-conjugate t-core partitions of n, let $sc(n)$ be the number of self-conjugate partitions of n, and define ... For fixed $n\geq 139$, the seq...

conjecture | open | MathDB 页面 | 原文线索

V2-061372 | MATHDB-322964

Numerical inequalities for self-conjugate 9-core partitions

摘录 / EXCERPT: Let $sc_9(n)$ denote the number of self-conjugate 9-core partitions of n, and let n be a non-negative integer. Numerical inequalities for self-conjugate 9-core partitions. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-061373 | MATHDB-322968

Cook-Nagel and Constantinescu-Varbaro conjecture on flag-complex f-vectors

摘录 / EXCERPT: Cook-Nagel and Constantinescu-Varbaro conjecture. The following equality of sets holds:

conjecture | open | MathDB 页面 | 原文线索

V2-061374 | MATHDB-322973

Weak SHGH conjecture for nef linear systems

摘录 / EXCERPT: Let X_n be the blow-up of the projective plane at n general points, and let $\mathcal{L}$ be a nef linear system, meaning that $\mathcal{L}\cdot\mathcal{L}'\geq0$ for every eff...

conjecture | open | MathDB 页面 | 原文线索

V2-061375 | MATHDB-322975

Strong Nagata conjecture for positive-genus curves

摘录 / EXCERPT: Let X_n be the projective plane blown up at n general points, and let C be an irreducible curve of genus $g>0$ on X_n. Let $(3;1^9,0^{n-9})$ denote the indicated plane line...

conjecture | open | MathDB 页面 | 原文线索

V2-061376 | MATHDB-322976

The chromatic bound conjecture for additive coloring

摘录 / EXCERPT: Chromatic bound conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061377 | MATHDB-322977

The chromatic-plus-one conjecture for additive coloring

摘录 / EXCERPT: Chromatic-plus-one conjecture. Every graph G has an additive coloring modulo

conjecture | open | MathDB 页面 | 原文线索

V2-061378 | MATHDB-322978

The distinct-neighborhood antimagic labeling conjecture

摘录 / EXCERPT: Distinct-neighborhood antimagic labeling conjecture. There is a bijection $f:V(G)\rightarrow \{1,2,\ldots,n\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061379 | MATHDB-322979

Finite generation of adjoint divisorial rings

摘录 / EXCERPT: Finite-generation conjecture. The ring $R(X;D_1,\dots,D_k)$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-061380 | MATHDB-322981

The miracle of Atiyah bundles

摘录 / EXCERPT: Let $E/\mathbb F_q$ be an elliptic curve, and let I_r denote the indecomposable degree-zero Atiyah bundle of rank r. For a decomposition into multiplicities...

conjecture | open | MathDB 页面 | 原文线索

V2-061381 | MATHDB-322983

The factorization conjecture for Marton's inner bound

摘录 / EXCERPT: Factorization conjecture. For all product channels, all $\lambda\in[0,1]$, and all $p(x_1,x_2)$,

conjecture | solved | MathDB 页面 | 原文线索

V2-061382 | MATHDB-322984

The generalized factorization conjecture for Marton's inner bound

摘录 / EXCERPT: Generalized factorization conjecture. For all product channels, all $\lambda\in[0,1]$, all $\alpha\geq1$, and all $p(x_1,x_2)$,

conjecture | open | MathDB 页面 | 原文线索

V2-061383 | MATHDB-322985

The binary-alphabet concave-envelope conjecture for Marton's inner bound

摘录 / EXCERPT: Binary-alphabet concave-envelope conjecture. For all $\alpha\geq1$ and all such Markov chains,

conjecture | open | MathDB 页面 | 原文线索

V2-061384 | MATHDB-322986

Colored partition identity for modulus 25

摘录 / EXCERPT: Let $D_S(N)$ and $D_T(N)$ denote the numbers of partitions of N into parts from the multisets S and T, respectively. Warnaar's conjecture. Let S be the set containing one c...

conjecture | open | MathDB 页面 | 原文线索

V2-061385 | MATHDB-323002

The rational generating-function conjecture for simple permutations in strongly rational classes

摘录 / EXCERPT: A permutation class $\mathcal{C}$ is strongly rational if it has the property defined in the source context, and let the simple permutations in $\mathcal{C}$ be the permutations wi...

conjecture | open | MathDB 页面 | 原文线索

V2-061386 | MATHDB-323003

Cruz-Uribe-Pérez two-weight conjecture for Calderón-Zygmund operators

摘录 / EXCERPT: Let $1<p<\frac{\infty}{ }$, let A and B be Young functions, and let $\bar A$ and $\bar B$ denote their complementary Young functions. For a cube Q, write $\|\cdot\|_{A,Q}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-061387 | MATHDB-323005

Near-symmetry conjecture for generating functions of leafed-cycle cones of power-of-two length

摘录 / EXCERPT: Near-symmetry conjecture. The generating function has this form, and if $(a_0,\ldots,a_j)$ is the coefficient list of $f(q)$, then appending 0 to the end of this list and subtrac...

conjecture | open | MathDB 页面 | 原文线索

V2-061388 | MATHDB-323007

Tate's conjectural analytic continuation and functional equation for Artin-twisted elliptic-curve L-functions

摘录 / EXCERPT: Let E) be an elliptic curve and let ρ be an Artin representation, both defined over $\mathbb{Q}$. Define the twisted L-function by ... where ... $H_1^1(E)$ is the dual of the...

conjecture | open | MathDB 页面 | 原文线索

V2-061389 | MATHDB-323009

Irreducibility conjecture for the moduli space of stable bundles on a quintic surface

摘录 / EXCERPT: Let $X\subset \mathbb{P}^3$ be a very general quintic surface, and let $M_X(2,1,10)$ denote the moduli space of stable bundles on X of rank 2, degree 1, and second Chern clas...

conjecture | open | MathDB 页面 | 原文线索

V2-061390 | MATHDB-323017

Parabolic reduction for volumes and semistable masses

摘录 / EXCERPT: Let G be a split reductive group over a number field F, let B/F be a fixed Borel subgroup, and let $P=UM$ be a standard parabolic subgroup with Levi factor M. If...

conjecture | open | MathDB 页面 | 原文线索

V2-061391 | MATHDB-323019

A conjectured upper and diagonal bound for R-Wythoff Sprague-Grundy values

摘录 / EXCERPT: Let a and b be integers with $4\leq a\leq b$, and let $\mathcal{G}_{\mathcal{R}}(a,b)$ denote the Sprague-Grundy value of the position (a,b) in R-Wythoff. R-Wythoff bounds co...

conjecture | open | MathDB 页面 | 原文线索

V2-061392 | MATHDB-323020

Additive periodicity conjecture for R-Wythoff Sprague-Grundy sequences

摘录 / EXCERPT: Fix a nonnegative integer a, and let $\{\mathcal{G}_{\mathcal{R}}(a,i)\}_{i\geq 0}$ be the sequence of Sprague-Grundy values along the row with first coordinate a. A sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-061393 | MATHDB-323021

Diagonal Sprague-Grundy conjecture for E-Wythoff

摘录 / EXCERPT: Let $\mathcal{G}_{\mathcal{E}}(a,b)$ denote the Sprague-Grundy value of the position (a,b) in E-Wythoff. E-Wythoff diagonal conjecture. For integers $r\geq 0$ and $a\geq 2r$, ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-061394 | MATHDB-323022

E-Wythoff Sprague-Grundy values on a family of rows

摘录 / EXCERPT: Let $\mathcal{G}_{\mathcal{E}}(a,b)$ denote the Sprague-Grundy value of the position (a,b) in E-Wythoff. E-Wythoff row-value conjecture. For integers $a\geq 4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061395 | MATHDB-323024

The global rigidity characterisation conjecture for frameworks on a circular cylinder

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with $|V|\geq 5$, let ${\mathcal{M}}$ be a circular cylinder, and let p be generic for ${\mathcal{M}}$. The global rigidity characterisation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-061396 | MATHDB-323036

Toft's coloring conjecture for k-wheel-free graphs

摘录 / EXCERPT: Toft's conjecture. If G is a k-wheel-free graph, then G is k-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-061397 | MATHDB-323040

The conjectural norm formula for Macdonald superpolynomials

摘录 / EXCERPT: Let P_Λ be the Macdonald superpolynomial associated with a superpartition Λ of fermionic degree m, and let $h^\uparrow_\Lambda$ and $h^\downarrow_\Lambda$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-061398 | MATHDB-323041

The conjugation conjecture for fermionic-degree-one Macdonald superpolynomials

摘录 / EXCERPT: Conjugation conjecture. The map ψ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061399 | MATHDB-323042

The concatenation conjecture for fermionic-degree-one Macdonald superpolynomials

摘录 / EXCERPT: Concatenation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-061400 | MATHDB-323047

Orbifold Landau-Ginzburg mirror Hodge-number conjecture

摘录 / EXCERPT: Orbifold Landau-Ginzburg mirror conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061401 | MATHDB-323050

Higher-dimensional linear repetitivity conjecture for tiling spaces

摘录 / EXCERPT: Let A be a finite alphabet, and let a tiling in ${\mathbb R}^d$ have a suitably defined notion of linear repetitivity. Its transversal is the tiling space consisting of tilings l...

conjecture | open | MathDB 页面 | 原文线索

V2-061402 | MATHDB-323051

Generic non-embeddability conjecture for cut-and-project quasicrystal tiling spaces

摘录 / EXCERPT: Let $E_\parallel$ be a d-dimensional linear subspace of ${\mathbb R}^n$ with irrational orientation, and let $\mathcal L(E_\parallel)$ be the associated cut-and-project Delone se...

conjecture | open | MathDB 页面 | 原文线索

V2-061403 | MATHDB-323054

The four-object fusion category conjecture for cases (a)-(c)

摘录 / EXCERPT: Fusion category classification conjecture. Cases (a)-(c) are each realized by a unique fusion category, and each such category has the following properties: it has 4 invertible o...

conjecture | open | MathDB 页面 | 原文线索

V2-061404 | MATHDB-323055

Five-moment conjecture for GUE fluctuations of directed random polymers

摘录 / EXCERPT: Let ω be an i.i.d. random environment. For each $\beta>0$, let $Z_n^{\omega}(0;\beta)$ be the directed polymer partition function, and let F_{GUE} denote the Tracy-Widom G...

conjecture | open | MathDB 页面 | 原文线索

V2-061405 | MATHDB-323061

Path-connectedness of wall complements in Bridgeland stability spaces

摘录 / EXCERPT: Let X be a smooth complex projective surface and let $v\in N(X)$. Consider the space of good Bridgeland stability conditions for v, together with the boundary of its closure, a...

conjecture | open | MathDB 页面 | 原文线索

V2-061406 | MATHDB-323066

Piecewise separable-variables approximation for discontinuous conductivities

摘录 / EXCERPT: Piecewise separable-variables approximation claim. It is possible to approach $csigma$ by means of a function of this form. This function can be used to numerically construct a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-061407 | MATHDB-323086

Divisibility conjecture for locally definable connected abelian groups

摘录 / EXCERPT: Work in an o-minimal expansion of an ordered field, and let G be a locally definable connected abelian group. The group G is divisible if for every $n\geq 1$ the map $G\to G$,...

conjecture | open | MathDB 页面 | 原文线索

V2-061408 | MATHDB-323095

The pure cubic inhomogeneous-minimum conjecture

摘录 / EXCERPT: Pure cubic inhomogeneous-minimum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061409 | MATHDB-323097

Sudderth-Wainwright-Willsky conjecture on the Bethe partition function

摘录 / EXCERPT: Sudderth-Wainwright-Willsky conjecture. If f admits a pairwise, log-supermodular factorization over G, then

conjecture | open | MathDB 页面 | 原文线索

V2-061410 | MATHDB-323099

The Gap Conjecture with parameter beta

摘录 / EXCERPT: Let G be a finitely generated group with growth function $\gamma_G(n)$, and let β satisfy $0<\beta<1$. Gap Conjecture with parameter β. If ... then G has polynomial g...

conjecture | open | MathDB 页面 | 原文线索

V2-061411 | MATHDB-323100

The Weak Gap Conjecture for group growth

摘录 / EXCERPT: Let G be a finitely generated group with growth function $\gamma_G(n)$, and let β satisfy $0<\beta<1$. Weak Gap Conjecture. There exists such a β for which ... implies...

conjecture | open | MathDB 页面 | 原文线索

V2-061412 | MATHDB-323105

Normality conjecture for three-dimensional cyclic polytopes

摘录 / EXCERPT: A cyclic polytope is the convex hull of points on the moment curve; here, normal means that every lattice point in every positive dilate is a sum of lattice points of the polytope....

conjecture | open | MathDB 页面 | 原文线索

V2-061413 | MATHDB-323106

Characterization conjecture for normal four-dimensional cyclic polytopes

摘录 / EXCERPT: Let $C_4(\tau_1,\tau_2,\dotsc,\tau_n)$ be a cyclic polytope, and write $\Delta_{ij}=\tau_j-\tau_i$ for $i<j$. A lattice polytope is normal when every lattice point in every positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-061414 | MATHDB-323107

Monotonicity conjecture for normal cyclic polytopes

摘录 / EXCERPT: Let $C_d(\tau_1,\dotsc,\tau_n)$ and $C_d(\tau_1',\dotsc,\tau_n')$ be integral cyclic polytopes, and let Δ_{ij} denote the corresponding gaps used in the source. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-061415 | MATHDB-323108

Very-ampleness conjecture for integral cyclic polytopes

摘录 / EXCERPT: An integral cyclic polytope is a cyclic polytope with integral vertices. A lattice polytope is very ample when its associated semigroup is generated up to saturation in the usual t...

conjecture | open | MathDB 页面 | 原文线索

V2-061416 | MATHDB-323111

Higher-harmonic Frobenius characteristic conjecture for the trivariate specialization

摘录 / EXCERPT: Let $\mathscr{H}_n^{(r)}(\mathbf{w};q,1,1)$ denote the graded Frobenius characteristic of the trivariate diagonal higher-harmonic space, with $\Dyck{r}{n}$ the set of r-Dyck path...

conjecture | open | MathDB 页面 | 原文线索

V2-061417 | MATHDB-323112

The Shuffle Conjecture for bivariate diagonal higher harmonics

摘录 / EXCERPT: Let $\mathscr{H}_n^{(r)}(\mathbf{w};q,t)$ denote the graded Frobenius characteristic of the bivariate diagonal higher-harmonic space. Let $\delta_n^{(r)}$ be the relevant r-stair...

conjecture | open | MathDB 页面 | 原文线索

V2-061418 | MATHDB-323113

Trivariate Shuffle Conjecture for higher harmonics

摘录 / EXCERPT: Let $\mathscr{H}_n^{(r)}(\mathbf{w};q,t,1)$ denote the graded Frobenius characteristic of the trivariate diagonal higher-harmonic space. Let $\Dyck{r}{n}$ be the set of r-Dyck pa...

conjecture | open | MathDB 页面 | 原文线索

V2-061419 | MATHDB-323114

Bertolini-Darmon-Mazur conjecture on derived Selmer dimensions

摘录 / EXCERPT: Bertolini-Darmon-Mazur conjecture. In this situation,

conjecture | open | MathDB 页面 | 原文线索

V2-061420 | MATHDB-323115

The two-variable Heegner point Iwasawa main conjecture

摘录 / EXCERPT: The two-variable Heegner point main conjecture. Assuming that R is regular,

conjecture | open | MathDB 页面 | 原文线索

V2-061421 | MATHDB-323122

Dembélé's period conjecture for quadratic twists over $\mathbb{Q}(\sqrt{5})$

摘录 / EXCERPT: Dembélé's period conjecture. There is an integer c_χ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061422 | MATHDB-323123

Kaplanski's zero divisor conjecture for group rings

摘录 / EXCERPT: Let G be a torsion-free group and let R be an integral domain. A non-trivial zero divisor in the group ring $R[G]$ is a nonzero element $\alpha\in R[G]$ for which there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-061423 | MATHDB-323124

Incidentally periodic rigidity conjecture

摘录 / EXCERPT: Incidentally periodic rigidity conjecture. If $(\langle G,m\rangle,p)$ is infinitesimally rigid on the flexible torus, then it is infinitesimally rigid as an incidentally periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-061424 | MATHDB-323132

Relative stable pairs/Gromov-Witten descendent correspondence

摘录 / EXCERPT: Relative descendent GW/Pairs correspondence.

conjecture | open | MathDB 页面 | 原文线索

V2-061425 | MATHDB-323137

Coefficient-positivity conjecture for the multivariate type D Eulerian polynomial

摘录 / EXCERPT: For $n\geq 2$, let $D_n(\mathbold{x},\mathbold{y})$ be the multivariate refinement defined by ... Here A_{n-2} and B_n are the stable multivariate refinements of the type A and...

conjecture | open | MathDB 页面 | 原文线索

V2-061426 | MATHDB-323138

Stable multivariate refinement conjecture for affine type B Eulerian polynomials

摘录 / EXCERPT: For $n\geq 2$, let $\widetilde{B}_n(\mathbold{x},\mathbold{y})$ be the multivariate refinement defined by ... Write $\mathfrak{S}_{\mathbb{R}}[\mathbold{x},\mathbold{y}]$ for the cla...

conjecture | open | MathDB 页面 | 原文线索

V2-061427 | MATHDB-323139

Coefficient-positivity conjecture for the multivariate affine type B Eulerian polynomial

摘录 / EXCERPT: For $n\geq 2$, let $\widetilde{B}_n(\mathbold{x},\mathbold{y})$ be the multivariate refinement defined by ... Here $\mathbb{N}[\mathbold{x},\mathbold{y}]$ denotes the polynomials in...

conjecture | open | MathDB 页面 | 原文线索

V2-061428 | MATHDB-323143

Asymptotic growth conjecture for spectral and pseudospectral bounds of Oseen's vortex

摘录 / EXCERPT: Let $\Psi(\alpha)$ and $\Sigma(\alpha)$ denote, respectively, the pseudospectral and spectral bounds associated with the linearized operator $\mathcal{L}-\alpha\Lambda$ around Osee...

conjecture | open | MathDB 页面 | 原文线索

V2-061429 | MATHDB-323150

Ohno-Zudilin's two-one formula for multiple zeta-star values

摘录 / EXCERPT: For integers $n\geq 1$ and $j_1\geq 1$, $j_2,\ldots,j_n\geq 0$, let $\zeta^\star(k_1,\ldots,k_r)$ denote the multiple zeta-star value and let $\zeta^{1/2}(k_1,\ldots,k_r)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-061430 | MATHDB-323156

Karp-Sitnik coefficient-positivity conjecture for rising-factorial Toeplitz determinants

摘录 / EXCERPT: Let $\{f_k\}_{k=0}^{n}$ be a non-negative sequence, let $\alpha,\beta>0$, and define ... Here $(x)_k$ denotes the rising factorial. Karp-Sitnik's coefficient-positivity conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-061431 | MATHDB-323157

Hurwitz-stability conjecture for rising-factorial Toeplitz determinants

摘录 / EXCERPT: Let $\{f_k\}_{k=0}^{n}$ be a non-negative sequence, let $\alpha,\beta>0$, and define ... A real polynomial is Hurwitz stable when all its zeros have negative real part. The Hurwitz-s...

conjecture | open | MathDB 页面 | 原文线索

V2-061432 | MATHDB-323158

Real-negative-zero conjecture for Pólya-frequency Toeplitz polynomials

摘录 / EXCERPT: Let $\{f_k\}_{k=0}^{n}$ be a Pólya frequency sequence of infinite order, denoted PF_{∞}, meaning that all minors of the associated infinite Toeplitz matrix are non-negative....

conjecture | open | MathDB 页面 | 原文线索

V2-061433 | MATHDB-323159

Hurwitz-stability conjecture for higher Toeplitz determinant polynomials

摘录 / EXCERPT: Let PF_r be the class of sequences whose infinite Toeplitz matrix has all minors of order at most r non-negative. For $r\geq2$, define ... Here $(x)_k$ is the rising factorial, a...

conjecture | open | MathDB 页面 | 原文线索

V2-061434 | MATHDB-323162

Shellability of stacked triangulations of polytopes

摘录 / EXCERPT: Bagchi-Datta shellability conjecture. The $(r-1)$-stacked triangulation of P is shellable.

conjecture | open | MathDB 页面 | 原文线索

V2-061435 | MATHDB-323167

The refined Narayana Lie-idempotent conjecture

摘录 / EXCERPT: Let F_t be the parameterized series, let $\mathsf{F}_{n,t}$ be its homogeneous component of degree n, let $\operatorname{\mathbf{ca}}_{n,t}$ be the corresponding t-N...

conjecture | open | MathDB 页面 | 原文线索

V2-061436 | MATHDB-323169

Screening conjecture under directional derivative predictability

摘录 / EXCERPT: Let Z be a spatial process with spectral density f satisfying condition (f-cond), let N_ε be a set of observations approaching the origin, and let $b(r)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-061437 | MATHDB-323172

Congruence conjecture for non-commutative p-adic L-functions

摘录 / EXCERPT: Let σ and ρ be the Artin representations described above, and let $R=\mathbb Z_p[[T]]$. Let $H(\sigma,T)$ and $H(\rho,T)$ be the corresponding power series, with...

conjecture | open | MathDB 页面 | 原文线索

V2-061438 | MATHDB-323180

Monomial realization conjecture for generalized minors and Demazure crystals

摘录 / EXCERPT: Let I be the index set of the root system, let $\mathbf i=(i_1,\ldots,i_N)$ be a reduced expression of the longest Weyl-group element, and let $p=(p_{i,j})_{i\ne j}$ be the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-061439 | MATHDB-323182

CFKRS conjecture for the fourth moment of holomorphic cusp forms

摘录 / EXCERPT: CFKRS conjecture. As $k\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-061440 | MATHDB-323183

CFKRS conjecture for the fourth moment of Maass cusp forms

摘录 / EXCERPT: CFKRS conjecture. As $T\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-061441 | MATHDB-323187

The Douady-Sullivan criterion conjecture for non-locally-connected Fatou boundaries

摘录 / EXCERPT: Let f be a rational map whose Julia set is connected. A periodic Fatou component is a Fatou component eventually returning to itself under iteration, and a boundary is locally co...

conjecture | open | MathDB 页面 | 原文线索

V2-061442 | MATHDB-323188

The polynomial-Julia-set accumulation conjecture for critical Fatou boundaries

摘录 / EXCERPT: Let f be a rational map whose Julia set is connected. A periodic critical Fatou component is a periodic Fatou component containing a critical point, and a non-locally-connected p...

conjecture | open | MathDB 页面 | 原文线索

V2-061443 | MATHDB-323190

The non-local-connectedness criterion for infinitely satellite-renormalizable quadratics

摘录 / EXCERPT: Let $C>1$ and let $\mathscr{C}_C$ be the set of parameters defined by the conditions that $t_n=C|p_n/q_n|\in(0,1)$ and, for the associated maps φ_n and compositions...

conjecture | open | MathDB 页面 | 原文线索

V2-061444 | MATHDB-323191

The invariant residual-compact-set conjecture for infinitely satellite-renormalizable quadratics

摘录 / EXCERPT: Let $Q_c(z)=z^2+c$ be infinitely satellite renormalizable with rotation numbers $(1/q_n)_n$. Let K_∞ be the residual compact set of the model associated to the data...

conjecture | open | MathDB 页面 | 原文线索

V2-061445 | MATHDB-323193

Birch's conjecture on ratios of twist-family approximation functions

摘录 / EXCERPT: Birch's conjecture. The asymptotic behavior of

conjecture | open | MathDB 页面 | 原文线索

V2-061446 | MATHDB-323196

Approximation conjecture for L^2-torsion

摘录 / EXCERPT: Let X be a finite connected CW-complex and let $\overline{X} \to X$ be a G-covering. Approximation conjecture for L^2-torsion. If the G-CW-structure on $\overline{X}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-061447 | MATHDB-323212

Reconstruction conjecture from the jumping-line curve and line bundle

摘录 / EXCERPT: Let Y be the Fano threefold under consideration, let E be an instanton on Y, and let D_E be its curve of jumping lines. Let $\mathcal L_E$ be the associated line bundle on...

conjecture | open | MathDB 页面 | 原文线索

V2-061448 | MATHDB-323213

Semistable-net conjecture for nondegenerate quadrics

摘录 / EXCERPT: Let γ be a net of quadrics, meaning a three-dimensional linear system of quadratic forms, and suppose that γ is semistable. Semistable-net conjecture. If γ is...

conjecture | open | MathDB 页面 | 原文线索

V2-061449 | MATHDB-323229

Bridgeland's moduli-space birationality conjecture for K3 surfaces

摘录 / EXCERPT: Bridgeland's conjecture. Given a stability condition σ on a K3 surface and a numerical class v, there exists a coarse moduli space $M_{\sigma}(v)$ of σ-semistable c...

conjecture | open | MathDB 页面 | 原文线索

V2-061450 | MATHDB-323232

Power containment for the twisted cubic's covariant ideal

摘录 / EXCERPT: Assume $d=3$ and let r be arbitrary, with 3 not dividing r. Let $X\subseteq\mathbf P^3$ be the twisted cubic curve, let I_X be its defining ideal, and let $J_{r,3}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-061451 | MATHDB-323243

Gazeau et al.'s recurrence conjecture for exponential-polynomial coefficients

摘录 / EXCERPT: Gazeau et al.'s conjecture. These numbers satisfy the recurrence relation

conjecture | open | MathDB 页面 | 原文线索

V2-061452 | MATHDB-323258

Primary decomposition conjecture for powers of star-configuration ideals

摘录 / EXCERPT: Let $s>n$ and let $\mathcal H=\{H_1,\ldots,H_s\}$ be properly meeting hyperplanes in ${\bf P}^n$, defined by linear forms L_i. Let $N+1=s$, let $\mathcal H'$ be the coordinate hy...

conjecture | open | MathDB 页面 | 原文线索

V2-061453 | MATHDB-323261

Canonical representation conjecture for binary forms with a linear coefficient constraint

摘录 / EXCERPT: Let p be a general binary form of even degree $2s$, and let $\alpha_j,\beta_j\in\mathbb{C}$ for $1\leq j\leq s+1$. Canonical representation conjecture. The form p can be writte...

conjecture | open | MathDB 页面 | 原文线索

V2-061454 | MATHDB-323262

Non-D-finiteness conjecture for generating functions of colored set partitions

摘录 / EXCERPT: Let $C_n(r)$ denote the number of colored set partitions of $[n]$ with colors from a set of size r. A formal power series is D-finite if it satisfies a linear differential equati...

conjecture | open | MathDB 页面 | 原文线索

V2-061455 | MATHDB-323287

Aurell-Frisch-She conjecture for regular points of Burgers flow

摘录 / EXCERPT: Let $X=(X_x)_{x\in\mathbb R}$ be a self-similar process satisfying the growth condition needed to define the inviscid Burgers solution, and let $\mathcal L$ be the closure of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-061456 | MATHDB-323290

The d-dimensional body-bar orbit framework rigidity conjecture

摘录 / EXCERPT: Let $ \langle H,m\rangle$ be a body-bar orbit framework on the fixed torus $ \mathcal T_0^d$, and let $ \mathcal G_{\mathcal C}(Y)$ denote the collection of connected components as...

conjecture | open | MathDB 页面 | 原文线索

V2-061457 | MATHDB-323294

The m-step Darboux conjecture for exceptional orthogonal polynomial systems

摘录 / EXCERPT: m-step Darboux conjecture. Every X_m orthogonal polynomial system for any codimension m can be obtained by applying a sequence of at most m Darboux transformations t...

conjecture | open | MathDB 页面 | 原文线索

V2-061458 | MATHDB-323304

Conjecture on the composition law for relative information loss

摘录 / EXCERPT: Let $\mathbf{g}$ be a measurable function describing a system with input $\mathbf{X}$ and output $\mathbf{Y}$, and let $\mathbf{h}$ be another system function transforming input...

conjecture | open | MathDB 页面 | 原文线索

V2-061459 | MATHDB-323316

The q-Rodrigues conjecture for Askey-Wilson polynomials

摘录 / EXCERPT: q-Rodrigues conjecture. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-061460 | MATHDB-323332

Optimality of the extended coincidence-based test with nonzero doubleton weight

摘录 / EXCERPT: Extended-test optimality conjecture. If S_n^{*+} satisfies $\bar{l}<\infty$, $v_2>-2$, and $v_l\geq 0$ for all $3\leq l\leq\bar{l}$, then the test is optimal in terms of the gene...

conjecture | open | MathDB 页面 | 原文线索

V2-061461 | MATHDB-323343

The capacity comparison conjecture for the fractional Riesz transform

摘录 / EXCERPT: Let $d\geq 2$, let $0<s<d$ with $s\notin\mathbf{N}$, and let $E\subset\mathbf{R}^d$ be compact. For a gauge Φ, let $\operatorname{cap}_{\Phi,s}(E)$ denote the associated nonli...

conjecture | open | MathDB 页面 | 原文线索

V2-061462 | MATHDB-323344

Finiteness of nested polytope problem solutions in arbitrary rank

摘录 / EXCERPT: Let M satisfy ... and suppose that $\operatorname{conv}(\theta(\mathcal{P}(M)))$ has at least $k+1$ vertices. Let $\bar{\alpha}$ be defined as in Equation. The arbitrary-rank finit...

conjecture | open | MathDB 页面 | 原文线索

V2-061463 | MATHDB-323349

Greene's finiteness conjecture for quasi-alternating links

摘录 / EXCERPT: A quasi-alternating link is a link in the class referred to as quasi-alternating in the source, and $\det(L)$ denotes its determinant. Greene's conjecture. There exist only finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-061464 | MATHDB-323351

Universal MAP estimator MSE conjecture

摘录 / EXCERPT: Let $x\in\mathbb{R}^N$ be the signal, $y=\Phi x+z$ the measurement vector, and x^U_{MAP} the universal MAP estimator. Assume that $\Phi\in\mathbb{R}^{M\times N}$ is an i.i.d. Gau...

conjecture | open | MathDB 页面 | 原文线索

V2-061465 | MATHDB-323366

Characterization of non-trivial discrete multiplier sequences

摘录 / EXCERPT: Let $\{\alpha_i\}_{i=0}^\infty$ be a non-trivial sequence, meaning it is not zero for all but at most two indices, and suppose that $\alpha_i>0$ for some i. A discrete multiplier...

conjecture | open | MathDB 页面 | 原文线索

V2-061466 | MATHDB-323367

Finite-degree hyperbolicity-preserver conjecture for difference operators

摘录 / EXCERPT: Let T be a constant-coefficient difference operator ... It acts on hyperbolic polynomials of degree at most m whose mesh is at least one, where the mesh is the minimum distance b...

conjecture | open | MathDB 页面 | 原文线索

V2-061467 | MATHDB-323368

Finite-difference convolution conjecture for hyperbolic polynomials

摘录 / EXCERPT: For polynomials of degree at most d, define the forward difference operator by ... and define the product ... Here a hyperbolic polynomial has mesh at least one when consecutive root...

conjecture | open | MathDB 页面 | 原文线索

V2-061468 | MATHDB-323372

Dokos-Dwyer-Johnson-Sagan-Selsor coefficient-parity conjecture for 321-avoiding permutations

摘录 / EXCERPT: For each integer $k\geq 0$, let $M_{2^k-1}(321;q)$ denote the major-index generating polynomial associated with 321-avoiding permutations of length 2^k-1, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-061469 | MATHDB-323385

The rank-one criterion for abelian extensions

摘录 / EXCERPT: Let E be an elliptic curve over K, let $\mathfrak N$ be its conductor, and let H/K be an abelian extension unramified at the primes dividing $\mathfrak N$. Assume that the fu...

conjecture | open | MathDB 页面 | 原文线索

V2-061470 | MATHDB-323400

Fischermann et al.'s planar bondage bounds conjecture

摘录 / EXCERPT: Let G be a connected planar graph, let $b(G)$ be its bondage number, and let $g(G)$ denote its girth. Planar girth-bounds conjecture. ... Furthermore, ... The source states that thes...

conjecture | open | MathDB 页面 | 原文线索

V2-061471 | MATHDB-323415

Silverman's uniform boundedness conjecture for integer points in minimal rational-map orbits

摘录 / EXCERPT: Let $d\geq 2$, let $\phi\in\operatorname{Rat}_d(\mathbb{Q})$ be represented by a model $[F,G]\in M_d(\mathbb{Q})$, and let $[\phi]\in\mathcal{M}_d(\mathbb{Q})$ denote its moduli cl...

conjecture | open | MathDB 页面 | 原文线索

V2-061472 | MATHDB-323416

Exact maximal average sensitivity of nested canalizing functions

摘录 / EXCERPT: Exact maximal-average-sensitivity conjecture. The maximal value of s^f is

conjecture | open | MathDB 页面 | 原文线索

V2-061473 | MATHDB-323417

Conjecture on density of indices computing the limiting minimal log discrepancy

摘录 / EXCERPT: Density conjecture. The intersection $I_l\cap J$ is dense for every l.

conjecture | open | MathDB 页面 | 原文线索

V2-061474 | MATHDB-323429

Centralizer isomorphism conjecture for braid groups of reflection centralizers

摘录 / EXCERPT: Centralizer isomorphism conjecture. In the general case, the natural morphism

conjecture | open | MathDB 页面 | 原文线索

V2-061475 | MATHDB-323437

AMM refinement for Stirling numbers at k equal to 2^t plus 1

摘录 / EXCERPT: Let $k=2^t+1$ with $t\ge2$, and use the AMM-Conjecture's parameters M_k, μ_k, $\mathcal N_{k,m}$, ℓ, s, and $f_k(n)$. AMM refinement for $k=2^t+1$. The AMM-Conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-061476 | MATHDB-323440

Darmon's conjecture on ATR points and analytic rank

摘录 / EXCERPT: Darmon's conjecture. The point P_φ belongs to $E_0(H_{\mathfrak c})$, the point P_K belongs to $E_0(K)$, and P_K is non-torsion if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-061477 | MATHDB-323442

Bolognesi's large-charge limit conjecture for magnetic bags

摘录 / EXCERPT: Bolognesi's conjecture. For any magnetic bag (f,ϕ), there is a sequence $(A^{(n)},\Phi^{(n)})$ of charge-n solutions such that, as $n\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-061478 | MATHDB-323443

The page-number-2 conjecture for nonrepetitive graph coloring

摘录 / EXCERPT: A graph has page number k if its vertices can be arranged on a spine and its edges partitioned into k pages, with no two edges on the same page crossing in the spine order. A v...

conjecture | open | MathDB 页面 | 原文线索

V2-061479 | MATHDB-323447

Divisibility and sum condition for ceiling-function recursions

摘录 / EXCERPT: Let q, k, p_i, s_i, and a_{ij} define a recursion ... Suppose that $k=q$, $p_i=q-1$, and the ceiling function $\left\lceil\frac{n}{q}\right\rceil$ satisfies this recursion....

conjecture | open | MathDB 页面 | 原文线索

V2-061480 | MATHDB-323448

Only integer-slope ceiling solutions for nested recursions

摘录 / EXCERPT: Let r and q be the parameters in the general nested recursion, and let $\left\lceil\frac{rn}{q}\right\rceil$ denote the corresponding ceiling function. Integer-slope conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-061481 | MATHDB-323458

Boros-Moll infinite log-concavity conjecture for the Boros-Moll sequences

摘录 / EXCERPT: Let $P_n(x)=\sum_{i=0}^n d_i(n)x^i$ be the Boros-Moll polynomial, so that $\{d_i(n)\}_{i=0}^n$ is its coefficient sequence. A finite nonnegative sequence is infinitely log-concave...

conjecture | solved | MathDB 页面 | 原文线索

V2-061482 | MATHDB-323459

Brändén's real-rootedness conjecture for the normalized Boros-Moll polynomials R_n

摘录 / EXCERPT: Brändén's conjecture for R_n. For any $n\geq 1$, the polynomial $R_n(x)$ has only real zeros. The paper presents this as the second real-rootedness conjecture related to the Boro...

conjecture | open | MathDB 页面 | 原文线索

V2-061483 | MATHDB-323468

BKMP conjecture for toric Calabi-Yau threefolds

摘录 / EXCERPT: Let ${\mathfrak X}$ be a toric Calabi-Yau 3-fold, with a Lagrangian brane L and framing parameters $\mathbf t$. Let $\mathcal W_{g,n}({\mathfrak X},L,\mathbf t;x_1,\dots,x_n)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-061484 | MATHDB-323492

Groeneboom's cube-root-log local limit conjecture for interval-censoring case 2

摘录 / EXCERPT: Let F_0 and H be continuously differentiable at t_0 and (t_0,t_0), respectively, with strictly positive derivatives $f_0(t_0)$ and $h(t_0,t_0)$, where H is the distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-061485 | MATHDB-323493

Groeneboom's local limit conjecture for decreasing deconvolution mixtures

摘录 / EXCERPT: Groeneboom's deconvolution local limit conjecture. For the estimator $F_n^{(1)}$ appearing in the decreasing-mixture deconvolution model,

conjecture | open | MathDB 页面 | 原文线索

V2-061486 | MATHDB-323497

Gukov's generalized volume conjecture for colored Jones polynomials

摘录 / EXCERPT: Let $\mathfrak{K}$ be a knot in $\mathbb{S}_3$, let u lie in one of certain open domains $\mathcal{U}^{(\alpha)}\subseteq\mathbb{C}$, and consider $N\to\infty$, $\hbar\to0$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-061487 | MATHDB-323507

The alternating-circuit bound for balanced red-blue graphs

摘录 / EXCERPT: Let G be a balanced red-blue graph with n vertices and m edges. An alternating-circuit bound asserts that 1. there exists an alternating circuit of length at most $3n^2/m$; a...

conjecture | open | MathDB 页面 | 原文线索

V2-061488 | MATHDB-323510

Invariance of conclusion sizes in optimum bases of closure systems

摘录 / EXCERPT: Invariance conjecture. The value s does not depend on the choice of the optimum basis. In particular, in a UC-closure system, for every critical set C, the size $|B_C|$ of th...

conjecture | open | MathDB 页面 | 原文线索

V2-061489 | MATHDB-323519

The torsion-class criterion conjecture for preprojective ideals

摘录 / EXCERPT: Let Q be a quiver with compatible ordering of its vertices, let $c=s_1s_2\cdots s_n$ be the corresponding Coxeter element, and let W be its Weyl group. For $w\in W$, let I_w...

conjecture | open | MathDB 页面 | 原文线索

V2-061490 | MATHDB-323522

Critical-value conjecture for connectedness of planar self-affine sets

摘录 / EXCERPT: Let $A\in M_2(\mathbb{Z})$ be expanding with $|\det(A)|=3$, let f be its characteristic polynomial, and let ${\mathcal{D}}=\{0,1,b\}v$ with $b>1$ and $v\in\mathbb{R}^2$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-061491 | MATHDB-323525

Finiteness conjecture for nonreduced Hessenberg NRS-matrices

摘录 / EXCERPT: Let Ω be an arbitrary Hessenberg type, and let $NRS(\Omega)$ denote the set of NRS-matrices of type Ω. Call a matrix ς-reduced according to the paper's redu...

conjecture | open | MathDB 页面 | 原文线索

V2-061492 | MATHDB-323527

Quasimonomial valuation conjecture for graded sequences

摘录 / EXCERPT: Let R be an excellent regular domain of equicharacteristic zero, let $\mathfrak a_\bullet$ be a graded sequence of ideals in R, and let $\mathfrak q$ be a nonzero ideal in R....

conjecture | open | MathDB 页面 | 原文线索

V2-061493 | MATHDB-323529

Quasimonomial valuation conjecture for controlled-growth subadditive sequences

摘录 / EXCERPT: Let R be an excellent regular domain of equicharacteristic zero, let $\mathfrak b_\bullet$ be a subadditive sequence of ideals in R, and say that it has controlled growth when...

conjecture | open | MathDB 页面 | 原文线索

V2-061494 | MATHDB-323532

The Weak Conjecture for noncrossing parking spaces

摘录 / EXCERPT: Let W be a real reflection group with Coxeter number h, let $k\geq 1$, and let ${{\sf Park}}^{NC}_W(k)$ be the noncrossing parking space with its $W\times\mathbb{Z}_{kh}$-actio...

conjecture | open | MathDB 页面 | 原文线索

V2-061495 | MATHDB-323533

Generalized truncated Gauss identity for $J_{m,r}(n)$

摘录 / EXCERPT: Generalized truncated Gauss conjecture. There holds

conjecture | open | MathDB 页面 | 原文线索

V2-061496 | MATHDB-323534

Truncated Jacobi identity for the three-colour partition function

摘录 / EXCERPT: Let $t(n)$ be defined by ... Thus $t(n)$ counts partitions of n into three kinds of parts. Truncated Jacobi conjecture. For $n,k\geqslant 1$, there holds ... with strict inequality i...

conjecture | open | MathDB 页面 | 原文线索

V2-061497 | MATHDB-323544

The C3-C2 counting conjecture for finite-field polynomials

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, and let C_2 and C_3 be the conditions on a polynomial $f\in\mathbb{F}_q[X]$ that, respectively, for every $h\in\mathbb{F}_q^*$ one has ... a...

conjecture | open | MathDB 页面 | 原文线索

V2-061498 | MATHDB-323550

Approximate spectral norm conjecture for symmetric Boolean functions

摘录 / EXCERPT: For a Boolean function $f:\{0,1\}^n\to\{\pm1\}$, let $\|\widehat f\|_{1,\epsilon}$ be the smallest spectral norm of a function g satisfying ... The function f is symmetric when i...

conjecture | open | MathDB 页面 | 原文线索

V2-061499 | MATHDB-323553

Conjecture on the value of the fractional precoloring extension threshold at distance 7

摘录 / EXCERPT: Let $g(k,d)$ denote the fractional precoloring extension threshold for distance d as defined in the paper. Distance-7 value conjecture. For $k\in[2,2.5)$, ... while for...

conjecture | open | MathDB 页面 | 原文线索

V2-061500 | MATHDB-323554

Discontinuity conjecture for the fractional precoloring extension threshold

摘录 / EXCERPT: Let $g(k,d)$ be the function studied in the paper, with d a positive integer. Discontinuity conjecture. For a fixed integer $d\ge4$, the function $g(k,d)$ is discontinuous at...

conjecture | open | MathDB 页面 | 原文线索

V2-061501 | MATHDB-323565

Sernesi's conjecture on the Gieseker-Petri locus

摘录 / EXCERPT: For a smooth curve C of genus g, let ... where $\mu_0(L):H^0(C,L)\otimes H^0(C,K_C\otimes L^{\vee})\to H^0(C,K_C)$ is the Petri multiplication map. Sernesi's conjecture. The locu...

conjecture | open | MathDB 页面 | 原文线索

V2-061502 | MATHDB-323572

Tritronquée asymptotics for vortex filaments at gradient catastrophe

摘录 / EXCERPT: Vortex-filament tritronquée conjecture. The generic behavior near the gradient-catastrophe point is

conjecture | open | MathDB 页面 | 原文线索

V2-061503 | MATHDB-323575

Embedding invariance of the Dyck tilings tiles statistic

摘录 / EXCERPT: Let λ_1 and λ_2 be Dyck paths whose corresponding trees are isomorphic when their embeddings in the plane are ignored. For a Dyck tiling...

conjecture | open | MathDB 页面 | 原文线索

V2-061504 | MATHDB-323578

The kinetic equation of state auto-diffeomorphism conjecture

摘录 / EXCERPT: Kinetic equation of state auto-diffeomorphism conjecture. The map $(n,\beta)\leftrightarrow(\eta,\rho)$ given implicitly by these equations is an auto-diffeomorphism of...

conjecture | open | MathDB 页面 | 原文线索

V2-061505 | MATHDB-323579

The kinetic equation of state hyperbolicity and causality conjecture

摘录 / EXCERPT: Kinetic equation of state hyperbolicity and causality conjecture. Under this equation of state, there exists a smooth function $f_{\rm kinetic}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061506 | MATHDB-323580

Seyed Fakhari's Stanley depth conjecture for integrally closed monomial ideals

摘录 / EXCERPT: Let $S=\mathbb{K}[x_1,\dots,x_n]$ and let $I\subset S$ be an integrally closed monomial ideal. If $\ell(I)$ denotes the analytic spread of I, then Seyed Fakhari's conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-061507 | MATHDB-323588

Lam-Williams positivity conjecture for inhomogeneous multispecies TASEP weights

摘录 / EXCERPT: Let the inhomogeneous multispecies exclusion process have n classes of particles, with particles of class i jumping at rate x_i. Let $w(\boldsymbol{\rho})$ denote the station...

conjecture | open | MathDB 页面 | 原文线索

V2-061508 | MATHDB-323589

Multiline-queue Markov-chain conjecture for inhomogeneous multispecies TASEP weights

摘录 / EXCERPT: Let $\mathcal Q$ be a multiline queue with state space Ω^{FM}_m, and let Ω_m be the state space of the corresponding inhomogeneous multispecies exclusion process. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061509 | MATHDB-323602

Gamma-nonnegativity conjecture for inversions and excedances in the symmetric group

摘录 / EXCERPT: For a permutation $\sigma\in S_n$, let $\operatorname{inv}(\sigma)$ and $\operatorname{exc}(\sigma)$ denote its inversion number and excedance number, respectively, and define ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-061510 | MATHDB-323608

Localization transition conjecture for polymers in a Poissonian medium

摘录 / EXCERPT: Let $p(\beta,\nu)$ be the quenched free energy, let $\lambda(\beta)$ be the corresponding annealed free-energy function, let ${\cal D}(\nu)$ be the domain of admissible inverse tem...

conjecture | open | MathDB 页面 | 原文线索

V2-061511 | MATHDB-323618

Limiting law of the maximum of a log-correlated Gaussian field

摘录 / EXCERPT: Maximum-law conjecture. There is a constant $c>0$ and a limiting random variable G_d such that Y_t converges in law to G_d as $t\to\infty$, with

conjecture | open | MathDB 页面 | 原文线索

V2-061512 | MATHDB-323626

The classification conjecture for 3-chromatic distances in the three-dimensional integer lattice

摘录 / EXCERPT: Let m be a positive integer, and let $\chi(\mathbb{Z}^{3},\sqrt{m})$ denote the minimum number of colours needed to colour $\mathbb{Z}^{3}$ so that points at distance $\sqrt{m}$...

conjecture | open | MathDB 页面 | 原文线索

V2-061513 | MATHDB-323628

Lawler-Schramm-Werner scaling-limit conjecture for self-avoiding walks

摘录 / EXCERPT: Let Ω be a simply connected domain in $\mathbb{C}$ with boundary points a and b, and let $(\Omega_\delta,a_\delta,b_\delta)$ be discrete approximations. At the critical...

conjecture | open | MathDB 页面 | 原文线索

V2-061514 | MATHDB-323630

Smirnov's O(n)-model interface conjecture

摘录 / EXCERPT: Fix $n\in[0,2]$ and consider the loop $O(n)$ model with an interface between boundary points a and b. Smirnov's O(n)-model interface conjecture. For $z=1/\sqrt{2+\sqrt{2-n}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-061515 | MATHDB-323634

Conjecture on equivalence of irrational-angle tilings to triquadratic tilings

摘录 / EXCERPT: Let ABC be a triangle that is N-tiled, meaning it is tiled by N congruent triangles. Let the tile have angles α and β satisfying ... and suppose that α is n...

conjecture | open | MathDB 页面 | 原文线索

V2-061516 | MATHDB-323637

Conjecture that the fractional chromatic deficit equals one-half for degrees six through eight

摘录 / EXCERPT: For $\Delta\in\{6,7,8\}$, let G be a graph with maximum degree Δ and clique number at most $\Delta-1$. Fractional chromatic deficit conjecture. The fractional chromatic nu...

conjecture | open | MathDB 页面 | 原文线索

V2-061517 | MATHDB-323638

Monotonicity conjecture for the fractional chromatic deficit

摘录 / EXCERPT: For each $\Delta\geq 3$, define $f(\Delta)$ as the best universal fractional-coloring deficit for graphs of maximum degree Δ in the paper's setting. Monotonicity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-061518 | MATHDB-323657

Duncan-Steingrímsson's bistatistic equidistribution conjecture for 021-avoiding ascent sequences

摘录 / EXCERPT: An ascent sequence of length n is a nonnegative integer sequence $x=x_1x_2\ldots x_n$ with $x_1=0$ and $x_i\leq \operatorname{asc}(x_1x_2\ldots x_{i-1})+1$ for $1<i\leq n$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-061519 | MATHDB-323663

The list-homomorphism product conjecture

摘录 / EXCERPT: List-homomorphism product conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061520 | MATHDB-323664

Unimodality and log-concavity conjecture for random bipartite graphs

摘录 / EXCERPT: Random bipartite graph conjecture. For every function $p=p(n)$, almost surely the sequence $(i_t(G^p_n))_{t=0}^n$ is unimodal and, moreover, is log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-061521 | MATHDB-323666

Equal-rate convergence conjecture for the fast gradient method sequences

摘录 / EXCERPT: Let $x_0,x_1,\dots$ and $y_1,y_2,\dots$ be the main and auxiliary sequences, respectively, defined by the fast gradient method (FGM), and let f be the objective function in the c...

conjecture | open | MathDB 页面 | 原文线索

V2-061522 | MATHDB-323667

Blanco-Petersen's q-gamma expansion conjecture for inversions and excedances

摘录 / EXCERPT: Blanco-Petersen's conjecture. There exist polynomials $\gamma_{n,k}(q)$ with nonnegative integer coefficients such that

conjecture | open | MathDB 页面 | 原文线索

V2-061523 | MATHDB-323668

Majorization-minimizer conjecture for frame completions

摘录 / EXCERPT: Majorization-minimizer conjecture. There exists $\mu\in E(\mathcal F_0,\mathbf a)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-061524 | MATHDB-323679

Simple connectedness conjecture for evaluation spaces of bicategory objects

摘录 / EXCERPT: Let $t\in\Theta_2$ be an object, and let $h(t)\in\Theta_1$ denote its length. Let τ^- and τ^+ be evaluations of $h(t)$, and let $E_2(t,\tau^-,\tau^+)$ be the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-061525 | MATHDB-323680

The folklore irreducibility conjecture for unitary principal series

摘录 / EXCERPT: Let G be a split reductive group over ${\mathbb Q}_p$, with negative Borel subgroup $B^-({\mathbb Q}_p)$, torus $T({\mathbb Q}_p)$, and set of simple roots S. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061526 | MATHDB-323681

The socle-filtration conjecture for ordinary representations

摘录 / EXCERPT: Let $\rho:{\operatorname{Gal}}({\overline{\mathbb Q}}_p/{\mathbb Q}_p)\longrightarrow \widehat{B}_{C_\rho}(E)\subseteq\widehat{G}(E)$ be generic, with $C_\rho\subseteq R^{+\vee}$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-061527 | MATHDB-323683

The dominant-map invariant conjecture for products of general curves

摘录 / EXCERPT: Dominant-map invariant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061528 | MATHDB-323693

Conjecture on canonical forms for nonzero p by 2 by 2 tensors over F_2

摘录 / EXCERPT: Canonical-form conjecture. For $p \geq 4$, there are 10 canonical forms for nonzero $p \times 2 \times 2$ tensors over $\mathbb{F}_2$. For each orbit, the rank and the horizontal...

conjecture | open | MathDB 页面 | 原文线索

V2-061529 | MATHDB-323696

Descendent GW/Pairs correspondence for nonsingular projective 3-folds

摘录 / EXCERPT: Descendent GW/Pairs correspondence. Under the variable change $-q=e^{iu}$,

conjecture | open | MathDB 页面 | 原文线索

V2-061530 | MATHDB-323697

Relative descendent GW/Pairs correspondence

摘录 / EXCERPT: Relative descendent GW/Pairs correspondence. Under the variable change $-q=e^{iu}$,

conjecture | open | MathDB 页面 | 原文线索

V2-061531 | MATHDB-323698

Capped bi-relative GW/Pairs correspondence

摘录 / EXCERPT: Capped bi-relative GW/Pairs correspondence. For the theories specified in the source,

conjecture | open | MathDB 页面 | 原文线索

V2-061532 | MATHDB-323702

Deformation extension of log Calabi-Yau fiber spaces

摘录 / EXCERPT: Let $g_0:(X_0,\Delta_0)\to B_0$ be a log Calabi-Yau fiber space, where (X_0,Δ_0) is a proper klt pair and ... Let (X,Δ) be a klt pair and let $h:(X,\Delta)\to(0\in S)$...

conjecture | open | MathDB 页面 | 原文线索

V2-061533 | MATHDB-323707

Positive one-sided Lyapunov growth for sensitive cellular automata

摘录 / EXCERPT: Let F be a cellular automaton with no equicontinuous points, that is, a sensitive cellular automaton. For a point x, let $l_n^+(x)$ and $l_n^-(x)$ denote the right- and left-si...

conjecture | open | MathDB 页面 | 原文线索

V2-061534 | MATHDB-323716

Classification of normed-space angle classes

摘录 / EXCERPT: For positive real numbers $\alpha,\beta,\gamma,\delta$ satisfying ... let NORM_{ρ} denote the class of normed spaces having angle $\angle_{\rho}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061535 | MATHDB-323719

Perverse equivalence conjecture for cyclic-defect unipotent blocks

摘录 / EXCERPT: Let B be a unipotent block of kG with cyclic defect groups, and let B' be its Brauer correspondent. Let $\pi_{\kappa/d}$ be the perversity function defined from the generic d...

conjecture | open | MathDB 页面 | 原文线索

V2-061536 | MATHDB-323733

Modified Levenberg-Martin-Poletsky conjecture on pluripolar hulls of finely holomorphic graphs

摘录 / EXCERPT: Let U be a fine domain in $\mathbb{C}$, and let f be finely holomorphic on U. Denote by Γ_f^* the pluripolar hull of the graph of f, and by the maximal finely holomo...

conjecture | open | MathDB 页面 | 原文线索

V2-061537 | MATHDB-323738

Kauffman's conjecture on surgery curves of genus one slice knots

摘录 / EXCERPT: A knot is a smooth embedding $\mathbb{S}^1\hookrightarrow\mathbb{S}^3=\partial\mathbb{B}^4$. It is slice if it bounds a smoothly embedded disk in $\mathbb{B}^4$. A genus one Seifer...

conjecture | refuted | MathDB 页面 | 原文线索

V2-061538 | MATHDB-323741

Sturmfels-Sullivant degree-bound conjecture for group-based models

摘录 / EXCERPT: Sturmfels-Sullivant's group-based degree-bound conjecture. For any group G, the ideal of the group-based model is generated in degree at most $|G|$ for any tree T.

conjecture | open | MathDB 页面 | 原文线索

V2-061539 | MATHDB-323747

Macdonald's generalized Dyson conjecture for root systems

摘录 / EXCERPT: Macdonald's generalized Dyson conjecture. For all $s\in\mathbb C$ with $\operatorname{Re}(s)>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-061540 | MATHDB-323751

Nonvanishing of Rankin-Selberg central values for ring class characters

摘录 / EXCERPT: Let f be the eigenform, K the quadratic field, and let $\mathfrak{X}$ denote the relevant set of characters. Write a ring class character as $\rho=\rho_0\rho_w$, of any given...

conjecture | open | MathDB 页面 | 原文线索

V2-061541 | MATHDB-323754

The 2-3 conjecture for a transcendence basis of multiple zeta values

摘录 / EXCERPT: Let $\zeta(s_1,\ldots,s_k)$ denote a multiple zeta value of weight $s_1+\cdots+s_k$. A word in the alphabet $\{2,3\}$ is called Lyndon when it is strictly smaller than each of its...

conjecture | open | MathDB 页面 | 原文线索

V2-061542 | MATHDB-323755

Broadhurst-Kreimer conjecture on indecomposable multiple zeta values

摘录 / EXCERPT: Let $D_{n,k}$ denote the number of multiple zeta values of weight n and depth k that are not reducible to multiple zeta values of lesser depth. Broadhurst-Kreimer conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-061543 | MATHDB-323759

Aaronson's lattice variant of the Sensitivity Conjecture

摘录 / EXCERPT: Aaronson's lattice variant. There exist constants c and k such that for every non-trivial coloring C,

conjecture | open | MathDB 页面 | 原文线索

V2-061544 | MATHDB-323761

Li's conjecture on generalized connectivity of complete equipartite graphs

摘录 / EXCERPT: Li's conjecture. We have $\kappa_k(G)=\kappa(S)$. This is one of two conjectures proposed by W. Li in her Ph.D. thesis; the supplied text gives no resolution status.

conjecture | open | MathDB 页面 | 原文线索

V2-061545 | MATHDB-323762

Li's conjecture on generalized connectivity of complete multipartite graphs

摘录 / EXCERPT: Let G be a complete multipartite graph, let k be a positive integer, and let S be a k-subset of $V(G)$. Li's conjecture. We have $\kappa_k(G)=\kappa(S)$ when the numbers of...

conjecture | open | MathDB 页面 | 原文线索

V2-061546 | MATHDB-323764

Pixton's relations for the tautological ring of the moduli space of curves

摘录 / EXCERPT: Let g,n,r be integers, let $\boldsymbol{\tau}=(\sigma,a_1,\ldots,a_n)$ consist of a partition σ and integers a_i, and let $\mathcal{R}(g,n,r;\sigma,a_1,\ldots,a_n)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-061547 | MATHDB-323776

Delocalization conjecture for the two-dimensional discrete random Schrödinger operator

摘录 / EXCERPT: Let H_ω be the two-dimensional discrete random Schrödinger operator ... on $l^2(\mathbb{Z}^2)$, where the independent random variables ω_i are uniformly distributed in...

conjecture | open | MathDB 页面 | 原文线索

V2-061548 | MATHDB-323780

Equivariant K-theory Schubert coefficient formula via equivariant K-jeu de taquin

摘录 / EXCERPT: Let λ,μ,ν be partitions indexing equivariant K-theory Schubert classes, and let ${\tt EqINC}(\nu/\lambda,|\mu|)$ denote the relevant set of equivariant increasing tab...

conjecture | open | MathDB 页面 | 原文线索

V2-061549 | MATHDB-323783

Positivity conjecture for symmetric Askey-Wilson moments

摘录 / EXCERPT: Let $\tau_{2n}(a^2,c^2)=4^n(qa^2c^2;q^2)_n\mu_{2n}(a,-a,c,-c;q)/(1-q)^n$, where $\mu_{2n}(a,-a,c,-c;q)$ is the Askey-Wilson moment and τ_{2n} is a polynomial in a^2, c^2,...

conjecture | open | MathDB 页面 | 原文线索

V2-061550 | MATHDB-323784

Positivity conjecture for a specialization of Askey-Wilson moments

摘录 / EXCERPT: Let $\mu_n(a,b,c,d;q)$ denote the nth Askey-Wilson moment, let $[m]_q!$ be the q-factorial, and let i,j be the integer parameters occurring in the specialization $c=q^i/a$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-061551 | MATHDB-323805

Sharp \(\mathcal{SL}\)- and \(\mathcal{M}(\beta)\)-radii for the class \(\mathcal{F}_3\)

摘录 / EXCERPT: Let $\mathcal{F}_3$ be the class of functions considered in the paper, and let $R_{\mathcal{SL}}$ and $R_{\mathcal{M}(\beta)}$ denote the radii for the corresponding classes, with...

conjecture | open | MathDB 页面 | 原文线索

V2-061552 | MATHDB-323810

The local theta-lift nonvanishing criterion via normalized local zeta integrals

摘录 / EXCERPT: Local theta-lift nonvanishing conjecture. The local theta lift $\Theta_{n,r}(\pi_v)$ is nonzero if and only if $Z_v^*(s_{m,n})$ is nonzero on...

conjecture | open | MathDB 页面 | 原文线索

V2-061553 | MATHDB-323815

Boundedness of log-Ding and log-Mabuchi functionals under special degeneration

摘录 / EXCERPT: Let $\pi: ({\mathcal{X}}, {\mathcal{D}})\rightarrow \mathbb C$ be a special degeneration for (X,D). Suppose the central fiber $({\mathcal{X}}_0,{\mathcal{D}}_0)$ admits a singula...

conjecture | open | MathDB 页面 | 原文线索

V2-061554 | MATHDB-323816

Gromov-Hausdorff convergence of conical Kähler-Einstein metrics on the plane

摘录 / EXCERPT: Let ω_β be the conical Kähler-Einstein metric on $\mathbb P^2$ with cone angle $2\pi\beta$ along a smooth degree-2 curve. Convergence conjecture. As β tends to...

conjecture | open | MathDB 页面 | 原文线索

V2-061555 | MATHDB-323820

The existence of optimal solutions for incomplete optimization

摘录 / EXCERPT: An optimization problem is called incomplete when its formulation or relevant information is incomplete. Incomplete-optimization conjecture. Optimization problems with incomplete i...

conjecture | open | MathDB 页面 | 原文线索

V2-061556 | MATHDB-323822

Grytczuk's bounded Thue number conjecture for planar graphs

摘录 / EXCERPT: Grytczuk's conjecture. There is a constant c such that

conjecture | open | MathDB 页面 | 原文线索

V2-061557 | MATHDB-323829

The DDS optimality conjecture for book crossing numbers of complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $ u_k(K_n)$ denote its minimum number of crossings in a k-page book drawing, and let $Z_k(n)$ denote the number of crossings...

conjecture | open | MathDB 页面 | 原文线索

V2-061558 | MATHDB-323850

Generalized Mrs. Gerber's Lemma for finite abelian groups

摘录 / EXCERPT: Generalized MGL. If G is a finite abelian group, then $f_G(x,y)$ is convex in x for a fixed y, and by symmetry convex in y for a fixed x.

conjecture | open | MathDB 页面 | 原文线索

V2-061559 | MATHDB-323851

The Böttcher-Wenzel form for real Toeplitz matrices is a sum of squares

摘录 / EXCERPT: Böttcher-Wenzel sum-of-squares conjecture. The Böttcher-Wenzel form generated by two real Toeplitz matrices is a sum of squares of quadratic forms. If possible, this should admit a...

conjecture | open | MathDB 页面 | 原文线索

V2-061560 | MATHDB-323857

The conjectural optimal supremum-norm discrepancy bound

摘录 / EXCERPT: Let D_N be the discrepancy function of an N-point set $mathcal P_N\subset[0,1]^d$. Supremum-norm discrepancy conjecture. In dimensions $d\ge 3$ there holds ... This would improve...

conjecture | open | MathDB 页面 | 原文线索

V2-061561 | MATHDB-323865

Automorphic tensor decomposition conjecture for non-induced cuspidal representations

摘录 / EXCERPT: Let F be a number field and let π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbf{A}_F)$ that is not automorphically induced from any finite extension...

conjecture | open | MathDB 页面 | 原文线索

V2-061562 | MATHDB-323872

Langlands duality for the Hitchin fibration

摘录 / EXCERPT: Let C be the underlying smooth projective curve, let r and d be the rank and degree, let $M=\mathcal M(d,r,\omega_C)$ be the moduli space of S-equivalence classes of Higgs...

conjecture | open | MathDB 页面 | 原文线索

V2-061563 | MATHDB-323873

Fiberwise Langlands duality for the Hitchin fibration

摘录 / EXCERPT: Let $\mathcal A$ be the Hitchin base, let $H:M\to\mathcal A$ be the Hitchin fibration, and let $\overline P$ be the universal Poincaré sheaf from the global conjecture. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-061564 | MATHDB-323881

Type C and type D Chan-Robbins-Yuen volume conjecture

摘录 / EXCERPT: Let $CRYC_{n+1}$ and $CRYD_{n+1}$ denote the type C_{n+1} and type D_{n+1} analogues of the Chan-Robbins-Yuen polytope $CRYA_n$. For a nonnegative integer k, let ... be the k...

conjecture | open | MathDB 页面 | 原文线索

V2-061565 | MATHDB-323883

Type B, C and D flow-polytope volume relations

摘录 / EXCERPT: Let K_n^D, K_n^B and K_n^C be the signed complete graphs whose edges correspond to the positive roots of types D_n, B_n and C_n, respectively. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-061566 | MATHDB-323884

Convergence of quenched Lyapunov exponents to the annealed exponent at high mobility

摘录 / EXCERPT: Let $\lambda_p^{u_0}(\kappa)$ denote the quenched Lyapunov exponent of order $p\in\mathbb{N}$ for mobility parameter κ, and let $\lambda_0^{u_0}(\kappa)$ be the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-061567 | MATHDB-323888

General partial fraction decomposition conjecture for Gorenstein-symmetric rational functions

摘录 / EXCERPT: General partial fraction decomposition conjecture. Under these assumptions, $P(t)$ has a unique partial fraction decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-061568 | MATHDB-323891

Mod-p local-global compatibility conjecture for unitary groups

摘录 / EXCERPT: Let F be the relevant number field, let K^p be a tame level with split ramification, and let $x\in\mathbb{X}(L)$ be a regular classical point such that ρ_x is residually i...

conjecture | open | MathDB 页面 | 原文线索

V2-061569 | MATHDB-323892

Compatibility map conjecture for the mod-p and p-adic constructions

摘录 / EXCERPT: Let $B(\rho_x)^{\circ}$ and $\mathcal{B}_x^{\circ}$ be the lattices in the p-adic constructions, let M° be the specified lattice in...

conjecture | open | MathDB 页面 | 原文线索

V2-061570 | MATHDB-323897

The crepant resolution conjecture for orbifold Calabi-Yau threefolds

摘录 / EXCERPT: Let $\mathcal{X}$ be an orbifold Calabi-Yau threefold satisfying the hard Lefschetz condition. Let Y be the Calabi-Yau resolution of $\mathcal{X}$ given by ... Using the Fourier-Mu...

conjecture | open | MathDB 页面 | 原文线索

V2-061571 | MATHDB-323900

Finiteness conjecture for totally definite Euclidean quadratic forms

摘录 / EXCERPT: Let R range over the rings of integers of totally real number fields, and let q be a totally definite Euclidean quadratic form over R. Finiteness conjecture. There are only f...

conjecture | open | MathDB 页面 | 原文线索

V2-061572 | MATHDB-323915

Equivalence of k-stellated and k-stacked polytopal spheres

摘录 / EXCERPT: Let k and d be integers with $d\geq 2k$, and let S be a polytopal d-sphere, meaning a triangulated sphere isomorphic to the boundary complex of a simplicial polytope. A tri...

conjecture | open | MathDB 页面 | 原文线索

V2-061573 | MATHDB-323923

Coincidence conjecture for squarefree strongly stable ideals

摘录 / EXCERPT: Let $S=\mathbb{K}[x_1,\dots,x_n]$, and let $I\subset S$ be a squarefree strongly stable ideal generated by monomials of degree d. Let I^τ be the monomial ideal obtained from...

conjecture | open | MathDB 页面 | 原文线索

V2-061574 | MATHDB-323924

Monotonicity conjecture for the Hilbert depth of lexsegment ideals

摘录 / EXCERPT: Let $S=\mathbb{K}[x_1,\dots,x_n]$, and let $I\subset S$ be a lexsegment ideal generated by monomials of degree d. Write $\mu(I)$ for the number of minimal generators of I. Lexs...

conjecture | open | MathDB 页面 | 原文线索

V2-061575 | MATHDB-323925

The twisted Teichmüller curve uniqueness conjecture

摘录 / EXCERPT: Twisted Teichmüller curve uniqueness conjecture. Suppose $D\equiv 5\pmod 8$ is a fundamental discriminant with narrow class number $h_D^+=1$. For every $n\ind4aa_D$, all matrices

conjecture | open | MathDB 页面 | 原文线索

V2-061576 | MATHDB-323926

The generalized Sato-Tate conjecture for automorphic representations

摘录 / EXCERPT: Let F be a number field, let G be a reductive group, and let π be a cuspidal tempered automorphic representation of $G(\mathbb A_F)$. Assume that its conjectural global L...

conjecture | open | MathDB 页面 | 原文线索

V2-061577 | MATHDB-323927

The global Sato-Tate conjecture for automorphic representations

摘录 / EXCERPT: Let F be a number field, let G be a reductive group, and let π be a cuspidal tempered automorphic representation satisfying the hypothesis that its conjectural global L-p...

conjecture | open | MathDB 页面 | 原文线索

V2-061578 | MATHDB-323928

The guiding Sato-Tate principle for families of automorphic representations

摘录 / EXCERPT: Let F be a number field, let G be a reductive group, and let $\{\mathcal F_k\}_{k\geqslant1}$ be a “general” sequence of finite families of automorphic representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-061579 | MATHDB-323936

Ground-state domination conjecture for the Friedrichs spectrum

摘录 / EXCERPT: Ground-state domination conjecture. Then λ belongs to the spectrum of the Friedrichs extension of P on $L^2(\Omega,\mathrm{d}x)$. This conjecture links pointwise dominati...

conjecture | open | MathDB 页面 | 原文线索

V2-061580 | MATHDB-323939

The near-field conjecture for sharply 2-transitive groups

摘录 / EXCERPT: Let Γ be a sharply 2-transitive permutation group, meaning that it acts transitively and freely on ordered pairs of distinct points. A near field is a skew field that is d...

conjecture | open | MathDB 页面 | 原文线索

V2-061581 | MATHDB-323958

The conjecture for binomials x^{q-2}+t x^{q^2-q-1}

摘录 / EXCERPT: The conjecture for binomials $x^{q-2}+t x^{q^2-q-1}$. The polynomial f is a permutation polynomial of $\Bbb F_{q^2}$ if and only if one of the following holds: (i) $t=1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061582 | MATHDB-323960

Chalupnik's formality conjecture for derived Frobenius twists

摘录 / EXCERPT: Let k be a field of characteristic p, let r be a nonnegative integer, and let G belong to the category of homogeneous strict polynomial functors of degree d. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061583 | MATHDB-323977

Riemann-Hilbert uniqueness conjecture for Gaiotto-Moore-Neitzke coordinates

摘录 / EXCERPT: Let $\mathcal{Y}^{RH}_{\gamma}(\xi)$ be the piecewise holomorphic function on the ξ-plane obtained from the Fock-Goncharov coordinate data, with discontinuities on BPS...

conjecture | open | MathDB 页面 | 原文线索

V2-061584 | MATHDB-323978

Smooth splitting conjecture for the twisted bundle over Hitchin's moduli space

摘录 / EXCERPT: Let V_z be the twisted bundle obtained as the globally defined direct sum of the two locally defined bundles V_z^1 and V_z^2 over the relevant cover of the Hitchin moduli spa...

conjecture | open | MathDB 页面 | 原文线索

V2-061585 | MATHDB-323986

The type-D affine Eulerian real-rootedness conjecture

摘录 / EXCERPT: Type-D affine conjecture. The affine Eulerian polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-061586 | MATHDB-323987

The signed multiset-permutation and inversion-sequence descent-ascent equidistribution conjecture

摘录 / EXCERPT: Let $P^{\pm}(\{1,1,2,2,\ldots,n,n\})$ be the signed permutations of the multiset $\{1,1,2,2,\ldots,n,n\}$, and let $\mathfrak I_{2n}^{(1,4,3,8,\ldots,2n-1,4n)}$ be the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-061587 | MATHDB-323990

The f-vector comparison conjecture for order and chain polytopes

摘录 / EXCERPT: The f-vector comparison conjecture. One has $f_i\leq f'_i$ for all $1\leq i\leq d-1$. Moreover, if $f_i=f'_i$ for some $1\leq i\leq d-1$, then ${\mathcal O}(P)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061588 | MATHDB-323991

Nivat's nonexpansive-subspace reformulation

摘录 / EXCERPT: Let $\eta\in\mathcal{A}^{\mathbb{Z}^2}$, let $X_{\eta}:=\overline{\mathcal{O}(\eta)}$ be its orbit closure under the translation action, and let $P_{\eta}(n,k)$ be the number of di...

conjecture | open | MathDB 页面 | 原文线索

V2-061589 | MATHDB-324001

Period-Index Conjecture for Function Fields

摘录 / EXCERPT: Period-Index Conjecture for Function Fields. For every $\alpha\in\operatorname{Br}(k(X))$,

conjecture | open | MathDB 页面 | 原文线索

V2-061590 | MATHDB-324009

Local Reed's conjecture

摘录 / EXCERPT: Local Reed's conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061591 | MATHDB-324010

Clique-local fractional Reed conjecture

摘录 / EXCERPT: Clique-local fractional Reed conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061592 | MATHDB-324011

The signed left Pieri rule for the immaculate basis

摘录 / EXCERPT: The signed left Pieri rule. The product has the form

conjecture | open | MathDB 页面 | 原文线索

V2-061593 | MATHDB-324012

Positivity of immaculate structure constants for a partition factor

摘录 / EXCERPT: Immaculate product positivity conjecture. The coefficients ${\bf c}_{\alpha,\lambda}^\beta$ are non-negative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-061594 | MATHDB-324013

Positive Hall-Littlewood expansion in the immaculate basis

摘录 / EXCERPT: Hall-Littlewood immaculate positivity conjecture. The element $\mathcal{Q}'_\lambda$ expands in the immaculate basis with coefficients that are positive polynomials in q. More ex...

conjecture | open | MathDB 页面 | 原文线索

V2-061595 | MATHDB-324030

The induced-subposet characterization of strict pattern-avoidance inequality

摘录 / EXCERPT: Pattern-avoidance inequality conjecture. For every poset P,

conjecture | open | MathDB 页面 | 原文线索

V2-061596 | MATHDB-324046

Ichino-Ikeda-Harris refined Gan-Gross-Prasad conjecture for unitary groups

摘录 / EXCERPT: Let F be a number field, let E/F be a quadratic extension, and let G and H be the unitary groups in the source. Let $\pi=\pi_n\otimes\pi_{n+1}$ be a tempered cuspidal autom...

conjecture | open | MathDB 页面 | 原文线索

V2-061597 | MATHDB-324052

Delaunay's moment conjecture for torsion in quadratic class groups

摘录 / EXCERPT: Class-group torsion moment conjecture. As d varies over fundamental negative discriminants, the average of

conjecture | open | MathDB 页面 | 原文线索

V2-061598 | MATHDB-324053

Delaunay's moment conjecture for Tate-Shafarevich groups of elliptic curves

摘录 / EXCERPT: Tate-Shafarevich moment conjecture. The average of

conjecture | open | MathDB 页面 | 原文线索

V2-061599 | MATHDB-324054

Conjecture on moments of higher Selmer groups

摘录 / EXCERPT: Higher Selmer moment conjecture. The average of

conjecture | open | MathDB 页面 | 原文线索

V2-061600 | MATHDB-324063

Linear bound conjecture for non-list-colorable planar subgraphs

摘录 / EXCERPT: Linear bound conjecture. If G is not L-colorable, then G contains a subgraph F with $O(|V(H)|)$ vertices such that F is not L-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-061601 | MATHDB-324064

A Ky Fan singular-value inequality for concave functions

摘录 / EXCERPT: Let $A,B\in\mathbb{R}^{m\times n}$ be given, and let $f:\mathbb{R}_+\rightarrow\mathbb{R}_+$ be a concave function satisfying $f(0)=0$. For $k\in\{1,\ldots,\min\{m,n\}\}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061602 | MATHDB-324089

Engström's connectivity conjecture for large fiber graphs

摘录 / EXCERPT: Engström's conjecture. For any lattice ideal with a Gröbner basis, there is an N such that the connectivity of every N-large fiber graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061603 | MATHDB-324090

Categorical weak crepant resolution conjecture for orbit closures

摘录 / EXCERPT: Let (G,V) be a prehomogeneous vector space, meaning that a reductive linear group G acts on the finite-dimensional vector space V with a dense orbit. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061604 | MATHDB-324091

Wonderful polar resolution conjecture for orbit closures

摘录 / EXCERPT: Let (G,V) be a prehomogeneous vector space, let $Z\subset V$ be the closure of an orbit of G, and write Z_{sing} for its singular locus. For an int...

conjecture | open | MathDB 页面 | 原文线索

V2-061605 | MATHDB-324131

The Bernstein-Sato ideal factorization conjecture

摘录 / EXCERPT: Let $F=(f_1,\ldots,f_r)$ be a collection of non-zero polynomials f_j in $\mathbb{C}[x_1,\ldots,x_n]$, and let $B_F\subset\mathbb{C}[s_1,\ldots,s_r]$ be its Bernstein-Sato ideal,...

conjecture | open | MathDB 页面 | 原文线索

V2-061606 | MATHDB-324132

The Bernstein-Sato zero-locus and Sabbah specialization conjecture

摘录 / EXCERPT: Let $F=(f_1, \ellots,f_r)$ be a collection of non-zero polynomials on $X=\mathbb{C}^n$, let $D=\bigcup_{j=1}^rV(f_j)$, and let $B_{F,x}$ denote the local Bernstein-Sato ideal at...

conjecture | open | MathDB 页面 | 原文线索

V2-061607 | MATHDB-324133

The indecomposable central arrangement hyperplane conjecture

摘录 / EXCERPT: Let $F=(f_1, \ellots,f_r)$, where each f_j is a not necessarily reduced central hyperplane arrangement in $\mathbb{C}^n$ of degree d_j, and suppose that $\prod_{j=1}^rf_j$ is a...

conjecture | open | MathDB 页面 | 原文线索

V2-061608 | MATHDB-324156

Bridgeless-core conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a cubic graph, and let a core be a core associated with three 1-factors of G.
Bridgeless-core conjecture. Every bridgeless cubic graph has a bridgeless core. The claim...

conjecture | open | MathDB 页面 | 原文线索

V2-061609 | MATHDB-324163

Injection conjecture for flat virtual pure braid groups

摘录 / EXCERPT: Let $v\mathcal{B}_n$ be the virtual pure braid group on n strands, with generators σ_{ij} and relations ... The flat virtual pure braid injection conjecture. The quotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-061610 | MATHDB-324164

Darmon's conjecture on the rationality of Darmon points

摘录 / EXCERPT: Darmon's conjecture. The local point $P_\tau=\Phi_{\operatorname{Tate}}(J_\tau)$ belongs to $E(H_\tau)$, where H_τ is the ring class field associated with the order determined...

conjecture | open | MathDB 页面 | 原文线索

V2-061611 | MATHDB-324165

A minor characterization of signed graphs with signed Colin de Verdière parameter at most three

摘录 / EXCERPT: Minor characterization conjecture. A signed graph (G,Σ) has $\nu(G,\Sigma)\leq 3$ if and only if (G,Σ) has no minor isomorphic to K_5^o, $K_4^=$, or any signed grap...

conjecture | open | MathDB 页面 | 原文线索

V2-061612 | MATHDB-324172

The sharp Fischer-type determinantal inequality for accretive-dissipative matrices

摘录 / EXCERPT: Let $A\in \mathbb{M}_{n}(\bold C)$ be an accretive-dissipative matrix, partitioned as ... where A_{11} is an $m\times m$ principal block and A_{22} is the complementary block. Sh...

conjecture | open | MathDB 页面 | 原文线索

V2-061613 | MATHDB-324173

Shapiro's integral admissibility conjecture for orthogonal designs

摘录 / EXCERPT: Shapiro's conjecture. An OD over the field $\mathbb{C}$ with size $[p,n,k]$ is admissible if and only if the same size is admissible over the ring $\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-061614 | MATHDB-324174

Integral admissibility conjecture for complex orthogonal designs

摘录 / EXCERPT: COD integral admissibility conjecture. A COD over $\mathbb{C}$ with size $[p,n,k]$ exists if and only if it is admissible for COD over $\mathbb{Z}$, that is, COD without linear com...

conjecture | open | MathDB 页面 | 原文线索

V2-061615 | MATHDB-324189

The real groups construction conjecture for the tame local Langlands correspondence for PGSp(4)

摘录 / EXCERPT: Let $G(F)$ be the group under consideration, let $T(F)$ be the relevant torus, and let ${}^w\Theta_{\chi}$ be the character formula associated with a character χ and an elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-061616 | MATHDB-324200

Existence of self-similar embedded fractals

摘录 / EXCERPT: Existence of self-similar embedded fractals. A fractal of the form $z \leftarrow f(z)+c$ has quasi-self-similar embedded fractals under magnification if and only if it is invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-061617 | MATHDB-324203

Critical surface fugacity conjecture for the rotated honeycomb $O(n)$ loop model

摘录 / EXCERPT: Let the $O(n)$ loop model be defined on the semi-infinite honeycomb lattice with the boundary oriented as in Figure(b). For $n\in[-2,2]$, assign fugacity ... to occupied vertices and...

conjecture | open | MathDB 页面 | 原文线索

V2-061618 | MATHDB-324229

The weak local Langlands interpolation conjecture for framed deformation rings

摘录 / EXCERPT: Weak local Langlands interpolation conjecture. For any $ \overline{\rho}$, there is a map

conjecture | open | MathDB 页面 | 原文线索

V2-061619 | MATHDB-324230

The strong local Langlands interpolation conjecture for semisimple residual representations

摘录 / EXCERPT: Strong local Langlands interpolation conjecture. If $ \overline{\rho}$ is semisimple, there is a unique isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-061620 | MATHDB-324245

The antichain-sequence conjecture for naturally labeled posets

摘录 / EXCERPT: Let (P,ω) and (Q,τ) be naturally labeled posets, meaning that the labeling is compatible with the poset order. Write $(P,\omega)\sim(Q,\tau)$ when their P-partition g...

conjecture | open | MathDB 页面 | 原文线索

V2-061621 | MATHDB-324259

Dimca-Papadima conjecture on homaloidal hypersurfaces with isolated singularities

摘录 / EXCERPT: Let $V(h)\subseteq\mathbb{P}^n$ be a homaloidal hypersurface defined by a homogeneous polynomial h of degree $d\geq 3$, and suppose that $V(h)$ has only isolated singular points....

conjecture | open | MathDB 页面 | 原文线索

V2-061622 | MATHDB-324260

Classification conjecture for polar-degree-two hypersurfaces with isolated singularities

摘录 / EXCERPT: Let $V(h)\subseteq\mathbb{P}^n$ be a projective hypersurface defined by a homogeneous polynomial h of degree d, with only isolated singular points, and let its polar degree be...

conjecture | open | MathDB 页面 | 原文线索

V2-061623 | MATHDB-324261

The Haglund-Morse-Zabrocki parking-function bijection conjecture

摘录 / EXCERPT: Let PF be a parking function, let $\operatorname{area}(PF)$, $\operatorname{dinv}(PF)$, and $\operatorname{ides}(PF)$ denote its area, diagonal inversion number, and inverse desc...

conjecture | open | MathDB 页面 | 原文线索

V2-061624 | MATHDB-324262

The stronger parking-function bijection conjecture

摘录 / EXCERPT: Let $\mathcal{A}_c$ denote the set of parking functions with composition c, let $\operatorname{dinv}(PF)$ and $\operatorname{ides}(PF)$ be the diagonal inversion number and inver...

conjecture | open | MathDB 页面 | 原文线索

V2-061625 | MATHDB-324263

The diagonal-word generating-function corollary for parking functions

摘录 / EXCERPT: The diagonal-word generating-function conjecture. For $b\leq a-1$ and every permutation τ, one has

conjecture | open | MathDB 页面 | 原文线索

V2-061626 | MATHDB-324264

The Haglund-Morse-Zabrocki nabla conjecture for compositions

摘录 / EXCERPT: Let c be a composition, let ∇ be the nabla operator on symmetric functions, let $C_c=C_{c_1}\cdots C_{c_k}$ for $c=[c_1,\ldots,c_k]$, and let $\mathcal{A}_c$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-061627 | MATHDB-324272

Non-isomorphic Rees quotients of distinct symmetric numerical semigroups

摘录 / EXCERPT: Non-isomorphism conjecture. Different symmetric numerical semigroups correspond to non-isomorphic quotient numerical semigroups.

conjecture | open | MathDB 页面 | 原文线索

V2-061628 | MATHDB-324274

Spectral-measure decay conjecture for uniformly discrete invariant random subgroups

摘录 / EXCERPT: Let G be the Lie group under consideration, let k be a degree, and let an IRS μ of G be ε-discrete in the sense used in the source. Let ν_μ^k be the spe...

conjecture | open | MathDB 页面 | 原文线索

V2-061629 | MATHDB-324276

Existence of an irreducible SFT not conjugate to a renewal system

摘录 / EXCERPT: An irreducible shift of finite type (SFT) is a shift of finite type whose defining transition structure is irreducible. A renewal system is a shift generated by a finite collection...

conjecture | open | MathDB 页面 | 原文线索

V2-061630 | MATHDB-324281

Betti-table symmetry conjecture for graphs formed from glued four-cycles

摘录 / EXCERPT: Let G be a graph formed by gluing k copies of the four-cycle C_{4} together along one edge. Write $b_{i,j}(\overline{G})$ for the graded Betti numbers of its graph curve. Bet...

conjecture | open | MathDB 页面 | 原文线索

V2-061631 | MATHDB-324297

Rapinchuk's bounded-generation conjecture for higher-rank arithmetic groups

摘录 / EXCERPT: Let Γ be a noncocompact irreducible arithmetic group, and let $\operatorname{rank}_{\mathbb{R}}\Gamma$ denote its real rank. A group is boundedly generated by unipotent subg...

conjecture | open | MathDB 页面 | 原文线索

V2-061632 | MATHDB-324303

Simplicity, sign, and ratio conjecture for the Gauss transfer operator eigenvalues

摘录 / EXCERPT: Let λ_n be the eigenvalues of the transfer operator for the Gauss continued fraction map, arranged so that $|\lambda_1|\geq |\lambda_2|\geq\cdots$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061633 | MATHDB-324304

Monotonicity conjecture for Gaussian noise stability at fixed volume

摘录 / EXCERPT: For $a\in(0,1)$, let $J(a,a;\rho)$ denote the Gaussian noise-stability quantity for two sets of Gaussian measure a, and let $\mathbb{E}_\rho J(1_A(X),1_A(Y);\rho)$ denote the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-061634 | MATHDB-324306

Phase coexistence for q-colorings of high-dimensional lattices

摘录 / EXCERPT: Let ${\mathbb Z}^d$ be the nearest-neighbor graph on the d-dimensional integer lattice. For an integer $q\geq 3$, consider the q-coloring model, whose Gibbs measures are infini...

conjecture | open | MathDB 页面 | 原文线索

V2-061635 | MATHDB-324308

The Pandharipande-Thomas stable-pairs/Gromov-Witten correspondence

摘录 / EXCERPT: Let X be a Calabi-Yau threefold. For stable pairs with $\beta=\textrm{ch}_2(F)$ and $n=\chi(F)$, let $P_{n,\beta}$ be the degree of the virtual fundamental class of the moduli sp...

conjecture | open | MathDB 页面 | 原文线索

V2-061636 | MATHDB-324314

The CM supercongruence for higher truncations of the elliptic hypergeometric family

摘录 / EXCERPT: The CM supercongruence. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-061637 | MATHDB-324334

Quadratic relation conjecture for Askey-Wilson polynomials

摘录 / EXCERPT: Let n be a positive integer, let x be a variable, and let a, b, c, d, and q be parameters. Let $p_n(x;a,b,c,d;q)$ denote the Askey-Wilson polynomial. Quadratic relati...

conjecture | open | MathDB 页面 | 原文线索

V2-061638 | MATHDB-324338

Kollár's finiteness conjecture for local fundamental groups of klt singularities

摘录 / EXCERPT: Let $0\in (X,\Delta)$ be a klt singularity. The local fundamental group is ... where $L(0\in X)$ is the link of the singularity. Kollár's finiteness conjecture. The local fundamental...

conjecture | open | MathDB 页面 | 原文线索

V2-061639 | MATHDB-324343

Equality conjecture for the codes ${\mathbb C}(m,j)$ and ${\mathbb C}_{2^j}^{{\cal H}_m}$

摘录 / EXCERPT: Let ${\mathbb C}(m,j)$ be the code defined earlier in the source, and let ${\mathbb C}_{2^j}^{{\cal H}_m}$ be the code associated with the Hadamard matrix ${\cal H}_m$ and modulus...

conjecture | open | MathDB 页面 | 原文线索

V2-061640 | MATHDB-324347

GRR density conjecture for fixed-valency Cayley graphs

摘录 / EXCERPT: For fixed $d\geq 3$, let $CAY_d(n)$ be the number of isomorphism classes of d-valent Cayley graphs of order at most n, and let $GRR_d(n)$ count those that are graphical regular...

conjecture | open | MathDB 页面 | 原文线索

V2-061641 | MATHDB-324357

Characterization of representable polynomials by factorization modulo square ideals

摘录 / EXCERPT: Factorization characterization conjecture. The polynomial P is representable if and only if, for some (equivalently any) tuple of squares $\ell\in{\mathbb F}^m$,...

open_problem | open | MathDB 页面 | 原文线索

V2-061642 | MATHDB-324363

The Steiner subratio conjecture for the Euclidean plane

摘录 / EXCERPT: Let $\operatorname{ssr}(\mathbb R^2)$ denote the Steiner subratio of the Euclidean plane, defined as the infimum of the ratios of minimal-filling weight to Steiner minimal-tree len...

conjecture | open | MathDB 页面 | 原文线索

V2-061643 | MATHDB-324364

Generalized Poonen conjecture for unicritical polynomials

摘录 / EXCERPT: For an integer $d\geq 2$ and $c\in\mathbb{Q}$, let $f_{d,c}(z)=z^d+c$. Generalized Poonen conjecture. For $n>3$, there is no $f_{d,c}$ defined over $\mathbb{Q}$ with a $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-061644 | MATHDB-324365

Uniform boundedness conjecture for rational maps of the form $z^d+g(z)$

摘录 / EXCERPT: Let $g(z)$ be any rational map, and for an integer d define ... For a number field K, let $\operatorname{Preper}(f_{d,g},K)$ denote the set of K-rational preperiodic points of...

conjecture | open | MathDB 页面 | 原文线索

V2-061645 | MATHDB-324382

Cusa-Huygens-type inequality for generalized hyperbolic functions

摘录 / EXCERPT: Cusa-Huygens-type conjecture. The displayed inequality holds for every $p\in(2,\infty)$ and every $x\in(0,\pi_p/2)$. This is part of a sequence of Cusa-Huygens-type inequalities fo...

conjecture | open | MathDB 页面 | 原文线索

V2-061646 | MATHDB-324383

Integrality conjecture for the Painlevé I series coefficients

摘录 / EXCERPT: Let the series ... has ... The conjecture is suggested by known integrality results for the coefficients $a_{m,n}$ in the expansion of the sigma function and analogous results for sigm...

conjecture | open | MathDB 页面 | 原文线索

V2-061647 | MATHDB-324386

Modified higher-order Szegő conjecture for orthogonal polynomials on the unit circle

摘录 / EXCERPT: Modified higher-order Szegő conjecture. The weighted logarithmic integrability condition

conjecture | open | MathDB 页面 | 原文线索

V2-061648 | MATHDB-324403

Hoffmann's maximal splitting conjecture for quadratic forms

摘录 / EXCERPT: Hoffmann's maximal splitting conjecture. If q has maximal splitting, then q is a Pfister neighbour.

conjecture | open | MathDB 页面 | 原文线索

V2-061649 | MATHDB-324405

Extension of the Equal Ripple Theorem to the first Zolotarev problem on two complex intervals

摘录 / EXCERPT: Extended Equal Ripple conjecture. There are infinitely many arrays $bbda_1,bbda_2,bbda_3$ for which there exists l such that

conjecture | open | MathDB 页面 | 原文线索

V2-061650 | MATHDB-324416

General correspondence between real and complex solutions of the matrix equation

摘录 / EXCERPT: Let equation be the complex matrix equation considered in the paper, and let equation be its associated real matrix equation. If $\boldsymbol{Y}$ is the general real solution of eq...

conjecture | open | MathDB 页面 | 原文线索

V2-061651 | MATHDB-324417

Uniqueness correspondence for complex and real matrix equations

摘录 / EXCERPT: Let equation be the complex matrix equation and equation its associated real matrix equation. Denote the real representation of a solution X by X_{σ}. Uniqueness-correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-061652 | MATHDB-324421

Bartnicki-Grytczuk-Niwczyk list 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a graph with no component isomorphic to K_2. An edge k-list-weighting assigns to each edge a weight from an independently assigned list of k real numbers. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061653 | MATHDB-324422

Sequence list 1-2-3 Conjecture for arbitrary edge orderings

摘录 / EXCERPT: Let G be a graph with no component isomorphic to K_2. Fix an ordering of $E(G)$. A sequence edge k-list-weighting assigns to each edge a weight from its independently assigne...

conjecture | open | MathDB 页面 | 原文线索

V2-061654 | MATHDB-324424

General sequence list-irregularity-strength conjecture

摘录 / EXCERPT: Let G be a graph with no component isomorphic to K_2. For each degree i, let n_i be the number of vertices of degree i, let $\Delta(G)$ be the maximum degree, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-061655 | MATHDB-324428

Extension of the special SRD and LRD mixed-case limit theorem

摘录 / EXCERPT: Extension conjecture. The special SRD and LRD mixed-case limit theorem holds without the restriction that $k_{j,L}$ be 1 or 2; in particular,

conjecture | open | MathDB 页面 | 原文线索

V2-061656 | MATHDB-324446

Square-root conjecture for geometric Whittaker complexes

摘录 / EXCERPT: Square-root conjecture. For any $d\geq 0$, there is a complex ${\cal S}_E^d\in\operatorname{D}(\operatorname{RCov}^d)$ equipped with an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-061657 | MATHDB-324450

The Kotzig graph subdivision conjecture for 3-connected cubic graphs

摘录 / EXCERPT: Kotzig graph subdivision conjecture. Every 3-connected cubic graph has a spanning subgraph which is a subdivision of a Kotzig graph.

conjecture | open | MathDB 页面 | 原文线索

V2-061658 | MATHDB-324453

Welzl's mixing-time conjecture for random lattice triangulations

摘录 / EXCERPT: Let ... denote the ... T{rm mix}=O(mn(m+n)). ... T{rm mix}=exp(Omega(mn(m+n))). ... , while the stated extensions to all fixed values in the two regimes remain open.

conjecture | open | MathDB 页面 | 原文线索

V2-061659 | MATHDB-324454

Uniform distribution of van der Corput-type LS-sequences in the square

摘录 / EXCERPT: An LS-successione "alla van der Corput" nel quadrato is a sequence of points in the square generated by the van der Corput-type LS construction. A sequence of points in the squ...

conjecture | open | MathDB 页面 | 原文线索

V2-061660 | MATHDB-324456

Extension of acausal surfaces to Cauchy surfaces for cone-bundle embeddings

摘录 / EXCERPT: Let $C^\first$ and $C^{\first}$ be globally hyperbolic spacelike cone bundles over M and M', respectively, and let $\chi\colon M\to M'$ be an embedding inducing a chronological...

conjecture | open | MathDB 页面 | 原文线索

V2-061661 | MATHDB-324459

The criterion for relative quasiconvexity of tamely generated subgroups

摘录 / EXCERPT: Relative quasiconvexity criterion. Then H is relatively quasiconvex in G.

conjecture | open | MathDB 页面 | 原文线索

V2-061662 | MATHDB-324472

Wright q-log-concavity conjecture for the family f(μ;x)

摘录 / EXCERPT: Let $f(bc;x)$ be the family of functions defined in the paper, indexed by $bc\geq 0$. A family $c6(\mu)$ is Wright q-log-concave if its pairwise products satisfy the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-061663 | MATHDB-324473

Wright q-log-concavity conjecture for the family g_{a,c}(μ;x)

摘录 / EXCERPT: Let $g_{a,c}(\mu;x)$ be the family of functions defined in the paper, indexed by $\mu\geq 0$, with parameters a and c. A family is Wright q-log-concave when it satisfies the...

conjecture | open | MathDB 页面 | 原文线索

V2-061664 | MATHDB-324481

The fermionic-degree-one Macdonald superpolynomial conjugation conjecture

摘录 / EXCERPT: Let Λ be a superpartition of fermionic degree $m=1$, let H_Λ be the modified Macdonald superpolynomial, and let ψ be the linear map defined on Schur superpolyn...

conjecture | open | MathDB 页面 | 原文线索

V2-061665 | MATHDB-324482

The generalized Macdonald superpolynomial conjugation conjecture

摘录 / EXCERPT: Generalized conjugation conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-061666 | MATHDB-324483

The Macdonald superpolynomial norm conjecture

摘录 / EXCERPT: Let Λ be a superpartition of fermionic degree m, and let $P_\Lambda(x,\theta;q,t)$ be the Macdonald superpolynomial. Let $\mathcal B\Lambda$ be the set of bosonic boxes,...

conjecture | open | MathDB 页面 | 原文线索

V2-061667 | MATHDB-324484

The positivity conjecture for Macdonald superpolynomial Kostka coefficients

摘录 / EXCERPT: Kostka-positivity conjecture. Every $K_{\Omega\Lambda}(q,t)$ is a polynomial in q and t with nonnegative integer coefficients. The paper proves this conjecture in the stable se...

conjecture | open | MathDB 页面 | 原文线索

V2-061668 | MATHDB-324485

The evaluation formula conjecture for Macdonald superpolynomials

摘录 / EXCERPT: Let $F(x,\theta)$ be a polynomial in superspace of fermionic degree m, let $N\ge m$, and define ... where $u_i=t^{i-1}/q^{\max(m-i,0)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061669 | MATHDB-324498

Normal-defect inequality for cyclic matrices of consecutive sizes

摘录 / EXCERPT: Let $A \in \mathbb{C}^{n\times n}$ and $\tilde{A}\in \mathbb{C}^{(n+1)\times (n+1)}$ with $n \geq 4$ be the matrices defined by ... Here nd denotes the normal defect...

conjecture | open | MathDB 页面 | 原文线索

V2-061670 | MATHDB-324499

Propagation of excess normal defect for cyclic matrices

摘录 / EXCERPT: Let $A \in \mathbb{C}^{n\times n}$ and $\tilde{A}\in \mathbb{C}^{(n+1)\times (n+1)}$ with $n \geq 4$ be the matrices defined by ... Here nd is the normal defect and...

conjecture | open | MathDB 页面 | 原文线索

V2-061671 | MATHDB-324514

The Manna model relation between stationary, transition and threshold densities

摘录 / EXCERPT: Consider the Manna model on $\mathbb Z$. Let ρ_s be its stationary density, let ρ_{tr} be the transition density identified by the change in the long-time toppling behavi...

conjecture | open | MathDB 页面 | 原文线索

V2-061672 | MATHDB-324516

Wilson's conjecture on critical XOR-Ising loops

摘录 / EXCERPT: Wilson's conjecture. The scaling limit of the family of loops of the critical XOR Ising model are the level lines of the Gaussian free field corresponding to levels that are odd mu...

conjecture | open | MathDB 页面 | 原文线索

V2-061673 | MATHDB-324533

The total 1-2 Conjecture

摘录 / EXCERPT: Total 1-2 Conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-061674 | MATHDB-324534

Skowronek-Kaziów's total product-colouring conjecture

摘录 / EXCERPT: Skowronek-Kaziów's conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-061675 | MATHDB-324537

Bača-Jendrol'-Miller-Ryan's total edge irregularity strength conjecture

摘录 / EXCERPT: Bača-Jendrol'-Miller-Ryan's conjecture. If $G\not\cong K_5$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061676 | MATHDB-324538

The polynomiality conjecture for double ramification cycles

摘录 / EXCERPT: Let ${\rm DR}_g(a_1,\dots,a_n)$ be the double ramification cycle in $\overline{\mathcal M}_{g,n}$ associated with labels $a_1,\dots,a_n$. The class has codimension g, so its Poin...

conjecture | open | MathDB 页面 | 原文线索

V2-061677 | MATHDB-324539

The stable-map decomposition formula for powers of the cotangent class

摘录 / EXCERPT: Let G,d,m be as in the moduli problem, let x be the marked point, and let $\overline{\mathcal M}^0_{G,1,d}(\mathbb{CP}^1)$ denote the relevant moduli space of stable maps. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061678 | MATHDB-324541

Semi-primality conjecture for relative Chow motives

摘录 / EXCERPT: Semi-primality conjecture for relative Chow motives. The category $\operatorname{CHM}(X)_F$ is semi-primary.

conjecture | open | MathDB 页面 | 原文线索

V2-061679 | MATHDB-324543

KE deformations conjecture for Del Pezzo orbifolds

摘录 / EXCERPT: Let (X_0,ω_0) be a $\mathbb{Q}$-Gorenstein smoothable Kähler-Einstein orbifold Del Pezzo surface, and let ... be its versal $\mathbb{Q}$-Gorenstein deformation family, where...

conjecture | open | MathDB 页面 | 原文线索

V2-061680 | MATHDB-324544

Tian's conjecture on singularities of GH limits of KE Del Pezzo surfaces

摘录 / EXCERPT: Let (X_i,ω_i) be a sequence of smooth Kähler-Einstein Del Pezzo surfaces converging in the Gromov-Hausdorff sense to a limiting metric space X. Tian's conjecture. The limi...

conjecture | open | MathDB 页面 | 原文线索

V2-061681 | MATHDB-324548

The GIT-polystability conjecture for deformations of KE Del Pezzo orbifolds

摘录 / EXCERPT: Let (X_0,ω_0) be a Kähler-Einstein orbifold with T-singularities. Let $Kur(X_0)\subseteq TDef(X_0)$ be the versal Kuranishi space of $\mathbb{Q}$-Gorenstein deformations o...

conjecture | open | MathDB 页面 | 原文线索

V2-061682 | MATHDB-324551

Hansen-Li-Liu conjecture on the revised Szeged-Wiener difference for nonbipartite graphs

摘录 / EXCERPT: Let G be a connected graph with $n \geq 4$ vertices, $m \geq n$ edges, and an odd cycle. The Hansen-Li-Liu conjecture. asserts that ... Moreover, the bound should be best possible,...

conjecture | open | MathDB 页面 | 原文线索

V2-061683 | MATHDB-324554

Lattice and normal-form conjecture for the asymmetric Artin monoid A_n

摘录 / EXCERPT: Lattice and normal-form conjecture. 1. The ordered set $(A_n,\leq)$ is a lattice. 2. Let $x\in A_n$. Then $x\leq\Delta_n$ if and only if there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-061684 | MATHDB-324561

Mullen's conjecture on value-set bounds for even-degree polynomials

摘录 / EXCERPT: Mullen's conjecture. The value set satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061685 | MATHDB-324565

The inverse power index lower-bound conjecture for the distribution $(0.75,0.25,0,\dots)$

摘录 / EXCERPT: Inverse power index lower-bound conjecture. Every such simple game satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-061686 | MATHDB-324574

Conjecture on polynomial-degree covers detecting nonsolvable Torelli images

摘录 / EXCERPT: Let S be a surface and let H be a nonelementary subgroup of its mapping class group. A cover $\widetilde{S}$ to which H lifts is a cover on which every element of H admits...

conjecture | open | MathDB 页面 | 原文线索

V2-061687 | MATHDB-324590

The Lie-spray blockability conjecture for homogeneous spaces

摘录 / EXCERPT: Let G be a connected Lie group and let $\Gamma\subset G$ be a lattice. Set $M=G/\Gamma$, and call M blockable when every pair of points can be blocked by a finite set avoiding...

conjecture | open | MathDB 页面 | 原文线索

V2-061688 | MATHDB-324591

The midpoint blockability conjecture for simple Lie-group quotients

摘录 / EXCERPT: Let G be a simple, connected, noncompact Lie group and let $\Gamma\subset G$ be a lattice. For a pair of points in G/Γ, midpoint blockable means that the midpoints of the...

conjecture | open | MathDB 页面 | 原文线索

V2-061689 | MATHDB-324603

Generalized Kac conjecture for semistable quiver representations

摘录 / EXCERPT: Let Q be a quiver with a stability function Z and let $\alpha\in\Gamma_+(Q)$ be a dimension vector. Given a field k, let $M^{\operatorname{ind}}_Z(Q,\alpha)(k)$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-061690 | MATHDB-324604

Semistable Tutte polynomial positivity conjecture

摘录 / EXCERPT: Semistable Tutte polynomial positivity conjecture. One has $T_Z(Q;t,q)\in\mathbb{N}[t,q]$. The conjecture is motivated by the positivity theorem for $a_Z(Q)$, and the source sugges...

conjecture | open | MathDB 页面 | 原文线索

V2-061691 | MATHDB-324638

Amenable/non-amenable quasi-isometry conjecture for hyperbolic locally compact groups

摘录 / EXCERPT: Let G be a hyperbolic locally compact group, and let X be a metric space. A group is quasi-isometric to another group or space when their large-scale geometries are equivalent....

conjecture | open | MathDB 页面 | 原文线索

V2-061692 | MATHDB-324649

Chapoton's H-triangle identity for noncrystallographic root systems

摘录 / EXCERPT: Let Φ be a noncrystallographic root system of rank n. Define the antichain generating function ... Let $N\!C(\Phi)$ be the noncrossing partition lattice and let ... be its Möbiu...

conjecture | open | MathDB 页面 | 原文线索

V2-061693 | MATHDB-324650

Diagonal Catalan specialization conjecture for root posets

摘录 / EXCERPT: Let Φ be a root system, let $\mathcal{M}(\Phi)$ be its diagonal coinvariant ring, and define its bigraded Hilbert series by ... Let $\mathcal{I}(\Phi^+)$ denote the order ideals...

conjecture | open | MathDB 页面 | 原文线索

V2-061694 | MATHDB-324651

The unboundedness conjecture for the convexity region

摘录 / EXCERPT: Let $\mathcal C$ be the set of complex parameters λ for which R_λ is convex. The set $\mathcal C$ is known to be open and to contain substantial regions of the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-061695 | MATHDB-324652

The transcendental-parameter conjecture for the convexity region

摘录 / EXCERPT: Let $\mathcal C$ be the set of complex parameters λ for which R_λ is convex. The transcendental-parameter conjecture. Every transcendental complex number belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-061696 | MATHDB-324653

The discreteness conjecture for the complement of the convexity region

摘录 / EXCERPT: Let $\mathcal C$ be the set of complex parameters λ for which R_λ is convex. The complement-discreteness conjecture. The complement of $\mathcal C$ in the complex p...

conjecture | open | MathDB 页面 | 原文线索

V2-061697 | MATHDB-324654

McNulty's fundamental-triangle conjecture for R_λ

摘录 / EXCERPT: For a parameter $\lambda\in\mathbb C$, call the triangle with vertices 0, 1, and λ the fundamental triangle. Let R_λ be the associated closure. McNulty's fundam...

conjecture | open | MathDB 页面 | 原文线索

V2-061698 | MATHDB-324655

The affine-symmetry rigidity conjecture for $R_{2+\sqrt2}(O)$

摘录 / EXCERPT: Let O be a regular octagon and let $R_{2+\sqrt2}(O)$ be its associated closure. An $\mathbb R$-affine transformation of $\mathbb C$ is a symmetry when it preserves this set. The...

conjecture | open | MathDB 页面 | 原文线索

V2-061699 | MATHDB-324656

The auto-clonvex characterization conjecture

摘录 / EXCERPT: Let $S\subseteq\mathbb C$. Call S auto-clonvex when it is S-clonvex, meaning that it is closed and λ-convex for every $\lambda\in S$. For each parameter λ, the...

conjecture | open | MathDB 页面 | 原文线索

V2-061700 | MATHDB-324660

Edwards-King edge-local conjecture for graph coloring

摘录 / EXCERPT: Edwards-King conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-061701 | MATHDB-324665

Square-root upper-bound conjecture for Weyl remainder fluctuations

摘录 / EXCERPT: Let $X=\operatorname{SL}(2,\mathbb{Z})\backslash\mathcal{H}$ be the modular surface, and let $S(t)$ denote the Weyl remainder for its Maass spectrum. Square-root upper-bound conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-061702 | MATHDB-324666

Central limit conjecture for Weyl remainder fluctuations

摘录 / EXCERPT: Let $X=\operatorname{SL}(2,\mathbb{Z})\backslash\mathcal{H}$ be the modular surface, let $S(t)$ denote its Weyl remainder, and let $\sigma\approx0.140$. Central limit conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-061703 | MATHDB-324669

Right-cell description conjecture for the sets C_i

摘录 / EXCERPT: Right-cell description conjecture. The subsets

conjecture | open | MathDB 页面 | 原文线索

V2-061704 | MATHDB-324681

Amdeberhan's generating-function conjecture for the maximum number of 1-hooks

摘录 / EXCERPT: Amdeberhan's conjecture. The generating function for the sequence b_n is

conjecture | open | MathDB 页面 | 原文线索

V2-061705 | MATHDB-324683

Athanasiadis' shift relation for extended Catalan and Shi characteristic polynomials

摘录 / EXCERPT: Athanasiadis' shift-relation conjecture. If a,b are integers satisfying $-1\leq a\leq b$, except for $(a,b)=(-1,0)$ and $(-1,-1)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061706 | MATHDB-324684

Symmetry conjecture for minimizers of the circulant analytic functional

摘录 / EXCERPT: Let $N=4n$, and let ... be minimized over vectors $q\in\mathbb T^N$ satisfying $\bar q_i=q_{-i}$. In the parametrization described immediately before the conjecture, write the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-061707 | MATHDB-324685

Parity conjecture for critical points of the circulant analytic functional

摘录 / EXCERPT: For $N=4$, decompose the functional into components $\Phi=\Phi_0+\Phi_1+\Phi_2+\Phi_3$, where the Φ_i are the components defined in the preceding formulas. Parity conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-061708 | MATHDB-324686

Constructive reducibility conjecture for nonminimal trellises

摘录 / EXCERPT: A trellis represents a code, and a trellis is nonminimal when it is not minimal for that code. A reduction procedure consists of the reduction steps from the proof of Theorem; stat...

conjecture | open | MathDB 页面 | 原文线索

V2-061709 | MATHDB-324701

Exceptional-graph conjecture for fractional powers

摘录 / EXCERPT: For positive integers m,n with $1<m<n$, let $G^{\frac{m}{n}}$ be the mth power of the n-subdivision of a graph G, and let χ and ω denote its chromatic and cliqu...

conjecture | open | MathDB 页面 | 原文线索

V2-061710 | MATHDB-324718

Kühn, Lapinskas and Osthus's conjecture on Hamilton cycles and even factors

摘录 / EXCERPT: Kühn--Lapinskas--Osthus conjecture. G contains at least

conjecture | solved | MathDB 页面 | 原文线索

V2-061711 | MATHDB-324726

Koszul-duality conjecture for identities of corresponding operations

摘录 / EXCERPT: Let O and $\mathsf{O}^!$ be a dual pair of binary quadratic non-Sigma operads. Let $\circ\in\mathsf{O}(2)$ be a bilinear operation and let $\circ'$ be its corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-061712 | MATHDB-324736

Aganagic--Vafa and Aganagic--Klemm--Vafa mirror conjecture for disk amplitudes

摘录 / EXCERPT: Let W_H be the B-model prepotential obtained by integrating $\log y/q_0$ with respect to q_0, and write ... where $W_{H,\mathrm{inst}}$ is the instanton part. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061713 | MATHDB-324748

The density-one conjecture for Newton approximation over global fields

摘录 / EXCERPT: Let K be a finite extension of the rational field $\mathbb{Q}$ or of the rational function field $\mathbb{F}_p(T)$. Let (x_n) be the Newton approximation sequence associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-061714 | MATHDB-324749

Gopakumar-Vafa invariance and the GV-GW identity

摘录 / EXCERPT: Gopakumar-Vafa conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-061715 | MATHDB-324752

Finiteness conjecture for compactified torsors of Lagrangian fibrations

摘录 / EXCERPT: Finiteness conjecture for compactified torsors. Up to deformation, there are finitely many compactified torsors $\pi:X\rightarrow\mathbb{P}^n$ associated to the fixed Lagrangian fi...

conjecture | open | MathDB 页面 | 原文线索

V2-061716 | MATHDB-324754

The main conjecture on endoscopy correspondences

摘录 / EXCERPT: Let τ be the self-dual cuspidal automorphic datum and let the global Arthur parameters ψ_1, ψ_2, and $\psi=\psi_1\boxplus\psi_2$ determine the relevant classical gro...

conjecture | open | MathDB 页面 | 原文线索

V2-061717 | MATHDB-324755

The upper-bound conjecture for maximal Fourier coefficients

摘录 / EXCERPT: Let G be an F-quasisplit classical group, let $\psi\in\widetilde\Psi_2(G)$ be a global Arthur parameter, and let $\widetilde\Pi_\psi(\epsilon_\psi)$ be its automorphic L^2-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-061718 | MATHDB-324759

An integral inequality for Gaussian sequential decision making

摘录 / EXCERPT: Let $N=2$ and suppose the conditional densities of the observation Y given hypotheses $H=0$ and $H=1$ are ... ... where σ and h_1 are arbitrary positive numbers, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061719 | MATHDB-324776

Breaker’s conjecture for doubly-biased component games

摘录 / EXCERPT: Let G be a d-regular graph on n vertices, where $d\ge3$, and let m be a positive integer. In the $(m:(d-2)m)$ game on G, Breaker’s conjecture. Breaker can force Maker to...

conjecture | open | MathDB 页面 | 原文线索

V2-061720 | MATHDB-324784

Uniqueness conjecture for integrable viscous conservation laws

摘录 / EXCERPT: Let $a(u)$ be a non-vanishing functional parameter, and consider integrable viscous conservation laws obtained from the deformation procedure extended to arbitrary order in...

conjecture | open | MathDB 页面 | 原文线索

V2-061721 | MATHDB-324786

Uniform filling conjecture for finite quotients of Bruhat-Tits buildings

摘录 / EXCERPT: Let $\mathcal{B}$ be the Bruhat-Tits building associated with $\mathrm{PGL}_{d}(F)$, where F is a local field and $d\geq3$. For a finite quotient X of $\mathcal{B}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061722 | MATHDB-324792

Gao-Wang's conjecture on the maximal cross number of UFIMs

摘录 / EXCERPT: Gao-Wang's conjecture. For every finite abelian group G,

conjecture | open | MathDB 页面 | 原文线索

V2-061723 | MATHDB-324793

Narkiewicz's conjecture on the first Narkiewicz constant

摘录 / EXCERPT: Let G be an abelian group written as a sum of invariant factors ... where $n_i\mid n_j$ if $i\leq j$ and $n_1>1$. Let $N_1(G)$ denote the maximum length of a unique factorization i...

conjecture | open | MathDB 页面 | 原文线索

V2-061724 | MATHDB-324794

Sylow decomposition conjecture for maximal-cross-number UFIMs

摘录 / EXCERPT: Let G be a finite abelian group with Sylow decomposition ... where $p_1,\ldots,p_n$ are distinct primes and G_{p_i} is the Sylow p_i-group of G. Let S be a unique factoriza...

conjecture | open | MathDB 页面 | 原文线索

V2-061725 | MATHDB-324795

Baptista's adiabatic metric conjecture for Abelian vortices

摘录 / EXCERPT: Let Σ be the underlying compact Riemann surface, let $\nu_{k,0}(s)$ be the relevant open subset of the vortex moduli space, and let ... be the diffeomorphic correspondence. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-061726 | MATHDB-324796

The characterization of pure-tone simplicial arrangements

摘录 / EXCERPT: Let $c\mathcal{A}$ be the cone over a real line arrangement $\mathcal{A}$, and suppose that $c\mathcal{A}$ is a simplicial arrangement. Let $F_{\mathcal{A}}$ denote the Milnor fibe...

conjecture | open | MathDB 页面 | 原文线索

V2-061727 | MATHDB-324800

Conjecture on prolate spheroidal quadrature error for exponential functions

摘录 / EXCERPT: Let $c>0$ and $a\geq 0$ be real numbers, and let $n>2c/\pi$ be an integer. Let $\delta_n(e^{icax})$ denote the quadrature error associated with the quadrature nodes t_j and weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-061728 | MATHDB-324802

Conjecture on PSWF quadrature errors being bounded by the next eigenvalue

摘录 / EXCERPT: Let $c>0$, let $n>2c/\pi$ be an integer, and let m be an integer with $0\leq m<n$. Let $\delta_n(\psi_m)$ be the quadrature error for the prolate spheroidal wave function...

conjecture | open | MathDB 页面 | 原文线索

V2-061729 | MATHDB-324804

The denominator conjecture for corolla coefficients

摘录 / EXCERPT: Let $\mathtt{Crl}_n$ denote the corolla with n leaves, let $\sympawn_{\mathtt{Crl}_n}$ be its coefficient, and let Φ_d denote the dth cyclotomic polynomial. The denominato...

conjecture | open | MathDB 页面 | 原文线索

V2-061730 | MATHDB-324805

The Newton polygon conjecture for tree coefficients

摘录 / EXCERPT: Let T be a rooted tree, let $\sympawn_T$ be its coefficient, and let $\square_T$ be the Newton polygon of the numerator of $\sympawn_T$. Write $\#T$ for the number of vertices of...

conjecture | open | MathDB 页面 | 原文线索

V2-061731 | MATHDB-324806

The cyclotomic divisibility conjecture for odd tree powers

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\ldots,\lambda_m)$ be a partition, let T_λ be the rooted tree with $n+1$ vertices defined by ... let $k\geq 3$ be an odd integer, and let Φ_d den...

conjecture | open | MathDB 页面 | 原文线索

V2-061732 | MATHDB-324807

Conjecture on the Newton polytope of the numerator of the q-Ehrhart polynomial

摘录 / EXCERPT: Newton-polytope shape conjecture. The Newton polytope of the numerator of the q-Ehrhart polynomial L_P has an horizontal top edge, an horizontal bottom edge and a diagonal righ...

conjecture | open | MathDB 页面 | 原文线索

V2-061733 | MATHDB-324811

Connectivity conjecture for fixed-point components of real Higgs-bundle moduli spaces

摘录 / EXCERPT: Connectivity conjecture. Every connected component of the fixed-point set of Θ_{G} intersects the generic fibre of this Hitchin fibration.

conjecture | open | MathDB 页面 | 原文线索

V2-061734 | MATHDB-324820

Clique-Stable Set separation conjecture for H-free graphs

摘录 / EXCERPT: Let H be a fixed graph, and let an H-free graph be a graph with no induced subgraph isomorphic to H. A CS-separator is a family of cuts separating every disjoint clique from...

conjecture | open | MathDB 页面 | 原文线索

V2-061735 | MATHDB-324822

Oriented Alon-Saks-Seymour conjecture

摘录 / EXCERPT: Let G be a graph. Its oriented bipartite packing $\mathbf{bp}_{\mathrm{or}}(G)$ is the minimum number of oriented complete bipartite graphs such that every edge is covered by an...

conjecture | open | MathDB 页面 | 原文线索

V2-061736 | MATHDB-324837

Central-material conjecture for optimal one-dimensional refractive indices

摘录 / EXCERPT: Let $n_\star(x)$ be a locally optimal one-dimensional refractive index on the interval $[0,L]$, taking the values n_+ and $n_-=1$ on its material intervals. The point $x=L/2$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-061737 | MATHDB-324840

Weak conjugacy conjecture for stable cellular automata limit sets

摘录 / EXCERPT: Weak conjugacy conjecture. The limit set of any stable cellular automaton is weakly conjugate to an SFT.

conjecture | open | MathDB 页面 | 原文线索

V2-061738 | MATHDB-324841

Subdivided graphs preserve interval colorability

摘录 / EXCERPT: Subdivision conjecture. If $G\in \mathfrak{N}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061739 | MATHDB-324843

The second signed path conjecture for pairs of binary trees

摘录 / EXCERPT: Second signed path conjecture. For every pair (D,R) of finite, binary trees with the same number of leaves, there is a sign consistent path from D to R.

conjecture | open | MathDB 页面 | 原文线索

V2-061740 | MATHDB-324847

The weakly pseudo-effective pushforward conjecture

摘录 / EXCERPT: Weakly pseudo-effective pushforward conjecture. Any weakly pseudo-effective class $\alpha\in H^{2l}(X,\mathbf{R})$ such that $\varphi_*\alpha=0$ belongs to the real vector space sp...

conjecture | open | MathDB 页面 | 原文线索

V2-061741 | MATHDB-324858

Rényi information ordering conjecture for ranked set samples

摘录 / EXCERPT: Let $\alpha>1$ and let $n\in\mathbb{N}$. Write ${\rm H}_{\alpha}(\textbf{X}_{RSS})$, ${\rm H}_{\alpha}(\textbf{X}^*_{RSS})$, and ${\rm H}_{\alpha}(\textbf{X}_{SRS})$ for the Rényi...

conjecture | open | MathDB 页面 | 原文线索

V2-061742 | MATHDB-324862

Mixed-characteristic density conjecture for semiampleness

摘录 / EXCERPT: Let R be a finitely generated integral domain over $\mathbb Z$, let $S=\operatorname{Spec}R$, and let $\pi:\mathcal X\to S$ be a smooth proper morphism. Let $\mathcal L$ be an in...

conjecture | open | MathDB 页面 | 原文线索

V2-061743 | MATHDB-324864

Density conjecture for étale-trivializable reductions

摘录 / EXCERPT: Keep the mixed-characteristic model $\pi:\mathcal X\to S$ and the vector bundle E from the variation setup, and assume that E is numerically flat. A bundle is étale trivializab...

conjecture | open | MathDB 页面 | 原文线索

V2-061744 | MATHDB-324868

Faber-Zagier relations conjecture for the tautological ring of the moduli space of curves

摘录 / EXCERPT: Faber-Zagier relations conjecture. For all $g\geq 2$, the kernel of q is the Faber-Zagier ideal FZ_g.

conjecture | open | MathDB 页面 | 原文线索

V2-061745 | MATHDB-324869

Equivalence conjecture for Faber's and Faber-Zagier relations

摘录 / EXCERPT: Equivalence conjecture for Faber's and Faber-Zagier relations. For all $g\geq 2$, the relations of Theorem are equivalent to the Faber-Zagier relations.

conjecture | open | MathDB 页面 | 原文线索

V2-061746 | MATHDB-324870

Sarwate's minimum-distance conjecture for binary cyclic codes

摘录 / EXCERPT: Let $m=2t$, let ${\mathbb F}$ be the finite field of size 2^m, and let ${\mathcal C}_d$ be the binary cyclic code of length 2^m-1 and dimension $2m$ with two nonzeros...

conjecture | open | MathDB 页面 | 原文线索

V2-061747 | MATHDB-324874

The conjectural classification of Markoff-Rosenberger triples in arithmetic progression over quadratic fields

摘录 / EXCERPT: Classification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061748 | MATHDB-324877

Nakajima's singular-support characterization conjecture for \mathscr B(w)

摘录 / EXCERPT: Let $\mathscr B(w)$ be the subset of canonical-basis labels associated with w, and let $\mathscr B'(w)$ be the subset consisting of labels b for which the singular support...

conjecture | open | MathDB 页面 | 原文线索

V2-061749 | MATHDB-324879

Maffray's bipartisan even-pair conjecture

摘录 / EXCERPT: A graph is bipartisan if it is Berge, contains no $L(K_{3,3}\setminus e)$ and no double diamond, and neither G nor $\overline{G}$ contains a long prism. An even pair is a pair of...

conjecture | open | MathDB 页面 | 原文线索

V2-061750 | MATHDB-324880

Thomas's balanced-skew-partition even-pair conjecture

摘录 / EXCERPT: A graph is Berge if it has no odd hole and no odd antihole. A balanced skew partition is the balanced form of a skew partition used in the decomposition theory of Berge graphs. A g...

conjecture | open | MathDB 页面 | 原文线索

V2-061751 | MATHDB-324882

Pair-correlation conjecture for zeros of Dirichlet L-functions

摘录 / EXCERPT: For each Dirichlet character χ modulo q, let γ_χ and γ'_χ denote ordinates of nontrivial zeros of $L(s,\chi)$, counted with multiplicity. Fix $C>0$. Pair...

conjecture | open | MathDB 页面 | 原文线索

V2-061752 | MATHDB-324883

The one-point Brill-Noether codimension conjecture

摘录 / EXCERPT: Let g, r, and d be positive integers, and let $\bar{\alpha}$ be a Schubert index of type (r,d) with ... For a smooth curve C, write $G^r_d(C)$ for its linear series of degr...

conjecture | open | MathDB 页面 | 原文线索

V2-061753 | MATHDB-324886

Boyer-Gordon-Watson's surgery dichotomy conjecture for non-cable knots

摘录 / EXCERPT: Boyer-Gordon-Watson's surgery dichotomy conjecture. If K is not a cable of a nontrivial knot, then

conjecture | open | MathDB 页面 | 原文线索

V2-061754 | MATHDB-324891

The evenness conjecture for Dynkin quivers

摘录 / EXCERPT: Let Γ be a quiver and let $\mathbf{k}$ be a field. Call Γ $\mathbf{k}$-even if the complexes $\pi_{y*}\underline{\mathbf{k}}_{\widetilde{F}_{y}}$ are even for every...

conjecture | open | MathDB 页面 | 原文线索

V2-061755 | MATHDB-324892

Uniform distortion conjecture for simplicial power maps

摘录 / EXCERPT: Let $T=\triangle(,0,)$ be a triangle and denote its angle at 0 by $\theta=\measuredangle(\xi,0,\eta)$. Let T^q be the qth level of the regular 1-4 subdivision of T. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061756 | MATHDB-324893

The measure-of-non-orientability conjecture for Jack characters

摘录 / EXCERPT: Let λ be a Young diagram, let π be a partition, let $\alpha>0$, and let M range over non-oriented maps with face-type π. Write $V_\bullet(G)$ and $V_\circ(G)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-061757 | MATHDB-324894

The two-rectangle conjecture for Jack characters

摘录 / EXCERPT: Let $\lambda=(p_1,p_2)\times(q_1,q_2)$ be a multirectangular Young diagram consisting of at most two rectangles, and let $\operatorname{Ch}_\pi^{(\alpha)}(\lambda)$ and the orienta...

conjecture | open | MathDB 页面 | 原文线索

V2-061758 | MATHDB-324898

SASE equivalence conjecture for decreasing-increasing patterns

摘录 / EXCERPT: Let $M(p)$ denote the permutation matrix of a pattern p, and set ... Here $F_k\underset{1-\text{SASE}}{\sim}J_k$ denotes equivalence under 1-SASE. SASE equivalence conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-061759 | MATHDB-324899

Decreasing pattern minimizes avoidance among alternating permutations

摘录 / EXCERPT: Let $A_n(q)$ denote the set of alternating permutations of length n avoiding the pattern q, and let S_k be the symmetric group on k letters. Alternating-avoidance minimizat...

conjecture | open | MathDB 页面 | 原文线索

V2-061760 | MATHDB-324900

Non-equivalence conjecture for four length-four patterns

摘录 / EXCERPT: Let $D_n^k(p)$ denote the set of permutations of length n with the specified descent-type parameter k that avoid the pattern p. Non-equivalence conjecture. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-061761 | MATHDB-324902

Francisco-Herzog-Hibi replication conjecture for critical graphs

摘录 / EXCERPT: Francisco-Herzog-Hibi replication conjecture. For any positive integer k and any k-critical graph G, there is a set $W\subseteq V(G)$ such that G^W is $(k+1)$-critical.

conjecture | open | MathDB 页面 | 原文线索

V2-061762 | MATHDB-324907

Conjecture on Demazure crystal expansions of generalized minors

摘录 / EXCERPT: Demazure-crystal expansion conjecture. There exists a reduced expression of the longest Weyl-group element $\mathbf i=(i_1,\ldots,i_N)$ and a sign $p=(p_{i,j})_{i\ne j}$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-061763 | MATHDB-324908

The invariant-graph characterization of the random graph for permutation groups

摘录 / EXCERPT: Let G be a permutation group on a countable set. A graph is G-invariant if its edge relation is preserved by every element of G, and let a random G-invariant graph be chose...

conjecture | open | MathDB 页面 | 原文线索

V2-061764 | MATHDB-324917

Kohnert's conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let $w\in S_n$, let $\operatorname{Rothe}(w)$ be the Rothe diagram of w, and generate diagrams D from it using the K-Kohnert rule, in which each $+$ either moves according to...

conjecture | open | MathDB 页面 | 原文线索

V2-061765 | MATHDB-324936

Hall subgroups of finite simple groups are strongly pronormal

摘录 / EXCERPT: Simple-group strong pronormality conjecture. Every Hall subgroup of a finite simple group is strongly pronormal.

conjecture | open | MathDB 页面 | 原文线索

V2-061766 | MATHDB-324949

Newton non-degeneracy characterization for normal surface singularities

摘录 / EXCERPT: A normal surface singularity is a surface singularity that is normal, and a Newton non-degenerate singularity is one for which, for every $\mathbf{u}\in\mathbb{R}_{\geq 0}^n$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-061767 | MATHDB-324953

Non-negative least-squares column-exclusion conjecture

摘录 / EXCERPT: Non-negative least-squares column-exclusion conjecture. There exists at least one column ${\bf g}_*$ of G such that $\hat{\bf x}$ has non-negative entries.

conjecture | open | MathDB 页面 | 原文线索

V2-061768 | MATHDB-324955

Guo's distinction transfer conjecture for general linear groups

摘录 / EXCERPT: Let k be a number field with adele ring ${\mathbb A}$, let ${\mathbf G}={\operatorname{GL}}_{2n}$ and ${\mathbf H}={\operatorname{GL}}_n\times{\operatorname{GL}}_n$ embedded diag...

conjecture | open | MathDB 页面 | 原文线索

V2-061769 | MATHDB-324959

The uniqueness conjecture for non-degenerate potentials on twice-punctured positive-genus surfaces

摘录 / EXCERPT: Let $(\Sigma,\mathbb{M})$ be a positive-genus surface with empty boundary and exactly two punctures, and let τ be a tagged triangulation of $(\Sigma,\mathbb{M})$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-061770 | MATHDB-324960

Okada's (q,t)-hook formula conjecture for connected d-complete posets

摘录 / EXCERPT: Let P be a connected d-complete poset with maximum element v_0, rank function $r:P\to\mathbb{N}$, top tree T, and d-complete coloring $c:P\to T$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061771 | MATHDB-324962

The asymptotic classification conjecture for simplicial arrangements

摘录 / EXCERPT: Asymptotic classification conjecture. There exists an $N\in\mathbb{N}$ such that the incidence of every simplicial arrangement in $\mathbb{C}^3$ with more than N hyperplanes is i...

conjecture | open | MathDB 页面 | 原文线索

V2-061772 | MATHDB-324963

Non-realizability conjecture for the finite-field arrangement with $2q$ hyperplanes

摘录 / EXCERPT: Non-realizability conjecture. There is no arrangement in $\mathbb{C}^3$ with the same incidence as $\mathcal{A}$.

conjecture | open | MathDB 页面 | 原文线索

V2-061773 | MATHDB-324966

Two-cycle list-coloring conjecture for planar graphs of girth at least five

摘录 / EXCERPT: Let G be a planar graph of girth at least five, let C_1,C_2 be two cycles in G, and for every $v\in V(G)$ let $L(v)$ be a non-empty set such that $|L(v)|\ge 3$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-061774 | MATHDB-324974

Gornay's dimension conjecture for amenable groups

摘录 / EXCERPT: Let Γ be an amenable group and let $Y\subseteq l^{p}(\Gamma)^{\oplus n}$ for some $n\in\mathbb N$. Let $\dim_{l^{p}}^{G}(Y,\Gamma)$ denote the l^{p}-dimension defined by...

conjecture | open | MathDB 页面 | 原文线索

conjecture | open | MathDB 页面 | 原文线索

V2-061775 | MATHDB-324975

Equality of upper and lower sofic lp-dimensions

摘录 / EXCERPT: Let $2<p<\infty$, let Γ be a countable discrete sofic group, and let Σ be a sofic approximation of Γ. Equality conjecture. For every $n\in\mathbb N$, ... This co...

conjecture | open | MathDB 页面 | 原文线索

V2-061776 | MATHDB-324976

Sofic-approximation independence of lp-dimension for amenable groups

摘录 / EXCERPT: Let Γ be an amenable group, let Σ and Σ' be two sofic approximations of Γ, and let X be a uniformly bounded representation of Γ. Approximation-...

conjecture | open | MathDB 页面 | 原文线索

V2-061777 | MATHDB-324979

Sharpness of the maximal-planar-subgraph rigidity criterion

摘录 / EXCERPT: Let $\mathcal{C}$ be a disk configuration locally finite in $\mathbb{G}$, where $\mathbb{G}$ is one of $\hat{\mathbb{C}}$, $\mathbb{C}$, or $\mathbb{H}^2$, and let (G,Θ) be...

conjecture | open | MathDB 页面 | 原文线索

V2-061778 | MATHDB-324990

Two-sided Calogero-Moser and Kazhdan-Lusztig cells conjecture

摘录 / EXCERPT: Two-sided cell conjecture. There exists a choice of ${\bar{\mathfrak r}}_c$ such that the partition of W into two-sided Calogero-Moser c-cells coincides with the partition into...

conjecture | open | MathDB 页面 | 原文线索

V2-061779 | MATHDB-324994

The normality-K_5-minor conjecture for cut polytopes

摘录 / EXCERPT: Normality-K_5-minor conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-061780 | MATHDB-325005

Eu's lattice-path enumeration conjecture for standard Young tableaux with at most 2d rows

摘录 / EXCERPT: Let d be a positive integer, and let the admissible unit steps be ... where $\{\epsilon_1,\dots,\epsilon_{d+1}\}$ is the standard basis of $\mathbb{R}^{d+1}$. Consider paths from t...

conjecture | open | MathDB 页面 | 原文线索

V2-061781 | MATHDB-325009

Type B Schubert-constant rule for clan actions

摘录 / EXCERPT: Type B Schubert-constant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061782 | MATHDB-325011

Type D Schubert-constant rule for clan actions

摘录 / EXCERPT: Type D Schubert-constant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061783 | MATHDB-325012

Common-level-set conjecture for H-monotone elliptic systems

摘录 / EXCERPT: Common-level-set conjecture. Every H-monotone solution of this system has common level sets. This is presented as complementary to the conjecture of Fazly and Ghoussoub that, at...

conjecture | open | MathDB 页面 | 原文线索

V2-061784 | MATHDB-325019

Structure conjecture for matrices with weak multivariate orthogonality

摘录 / EXCERPT: Structure conjecture. Given $n>0$, a banded matrix ${\mathcal{A}}$ has a weak n-variate orthogonality property if and only if it has the displayed form

conjecture | open | MathDB 页面 | 原文线索

V2-061785 | MATHDB-325025

Rational defect conjecture for Butson matrices

摘录 / EXCERPT: Rational defect conjecture. For the Butson matrices we have $d_{\mathbb Q}(H)=d(H)$.

conjecture | open | MathDB 页面 | 原文线索

V2-061786 | MATHDB-325026

Defect bound by equivalent matrices conjecture

摘录 / EXCERPT: Defect bound conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-061787 | MATHDB-325029

The local-field non-embedding conjecture for completed positive Kac-Moody groups

摘录 / EXCERPT: Let $\mathcal{D}$ be a Kac-Moody root datum with indecomposable generalized Cartan matrix A, let $\mathbf{F}_q$ be the finite field of characteristic p with q elements, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-061788 | MATHDB-325039

The bounded domination conjecture for transitive colorings

摘录 / EXCERPT: Bounded domination conjecture. For each positive integer k there is a least integer $p(k)$ such that every k-transitive tournament T satisfies $dom(T)\le p(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-061789 | MATHDB-325040

The Erdős-Sands-Sauer-Woodrow path-domination conjecture

摘录 / EXCERPT: Erdős-Sands-Sauer-Woodrow conjecture. For each positive integer k there is a least integer $f(k)$ such that every k-colored tournament contains a set S of $f(k)$ vertices wit...

conjecture | open | MathDB 页面 | 原文线索

V2-061790 | MATHDB-325043

Baldwin's categoricity conjecture for atomic classes

摘录 / EXCERPT: Let T be a countable first-order theory and let K_T be its class of atomic models. Baldwin's conjecture. If K_T is categorical in $\beth_1$, then K_T is $\beth_0$-stable, e...

conjecture | open | MathDB 页面 | 原文线索

V2-061791 | MATHDB-325050

Superlimit-or-maximal-nonstructure conjecture for abstract elementary classes

摘录 / EXCERPT: Let ${\mathfrak k}$ be an abstract elementary class and put $\kappa={\rm LST}_{\mathfrak k}+|\tau_{\mathfrak k}|$. Superlimit-or-nonstructure conjecture. At least one of the follow...

conjecture | open | MathDB 页面 | 原文线索

V2-061792 | MATHDB-325051

Bar 1-visibility conjecture for 1-planar graphs

摘录 / EXCERPT: Bar 1-visibility conjecture. Every 1-planar graph is a bar 1-visible graph.

conjecture | open | MathDB 页面 | 原文线索

V2-061793 | MATHDB-325061

DLT extension conjecture

摘录 / EXCERPT: Let (X,Δ) be a projective divisorial log terminal pair, with Δ a $\mathbb{Q}$-divisor, and write $\lfloor\Delta\rfloor=S$. Suppose that $K_X+\Delta$ is nef and ... wher...

conjecture | open | MathDB 页面 | 原文线索

V2-061794 | MATHDB-325062

Nevo's gamma2 equality conjecture for flag spheres

摘录 / EXCERPT: Let $d\geq 4$ be an integer and let Δ be a flag simplicial $(d-1)$-sphere. Define $\gamma_2(\Delta)=f_1-(2d-3)f_0+2d(d-2)$. An edge contraction replaces an edge e by the c...

conjecture | open | MathDB 页面 | 原文线索

V2-061795 | MATHDB-325070

Fakhari’s conjecture on Stanley depth of integrally closed monomial ideals

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $S=\mathbb{K}[x_1,\ldots,x_n]$, and let $I\subset S$ be an integrally closed monomial ideal. Let $\ell(I)$ denote the analytic spread of I. Fakha...

conjecture | open | MathDB 页面 | 原文线索

V2-061796 | MATHDB-325071

Aluffi-Mihalcea conjecture on the nonnegativity of Schubert coefficients

摘录 / EXCERPT: Let $\underline{\alpha}$ and $\underline{\beta}$ be the index sequences used to define the coefficient $\gamma_{\underline{\alpha},\underline{\beta}}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-061797 | MATHDB-325082

The single-valuedness conjecture for two-dimensional graphical functions

摘录 / EXCERPT: Single-valued graphical-function conjecture. For any graph Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-061798 | MATHDB-325083

The single-valuedness conjecture for two-dimensional completed primitive periods

摘录 / EXCERPT: Single-valued period conjecture. The period of any two-dimensional completed primitive graph belongs to

conjecture | open | MathDB 页面 | 原文线索

V2-061799 | MATHDB-325084

The maximum-weight cell-zeta conjecture for two-dimensional periods

摘录 / EXCERPT: Maximum-weight cell-zeta conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061800 | MATHDB-325087

Dimension lower-bound conjecture for sums of projected planar sets and sets on a line

摘录 / EXCERPT: Projected-sum dimension conjecture. For almost every $\theta\in(0,2\pi)$,

conjecture | open | MathDB 页面 | 原文线索

V2-061801 | MATHDB-325092

Drużkowski's Structural Conjecture

摘录 / EXCERPT: Drużkowski's Structural Conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-061802 | MATHDB-325096

Fox-Pach separator conjecture for string graphs

摘录 / EXCERPT: A string graph is the intersection graph of a collection of curves in the plane. A separator in a graph $G=(V,E)$ is a subset $S\subseteq V$ such that no connected component of...

conjecture | open | MathDB 页面 | 原文线索

V2-061803 | MATHDB-325106

Nilpotent obstruction conjecture for small tripling in linear groups

摘录 / EXCERPT: Let K be a field, and let A be a finite subset of $\operatorname{GL}_n(K)$ satisfying $A=A^{-1}$ and $e\in A$. For every $R\geq 1$, consider the alternatives involving the trip...

conjecture | open | MathDB 页面 | 原文线索

V2-061804 | MATHDB-325118

Stevens's tautness and quasi-homogeneity conjecture for simple surface singularities

摘录 / EXCERPT: Stevens's conjecture. A normal surface singularity is simple if and only if it is taut and quasi-homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-061805 | MATHDB-325121

Open crepant resolution transformation formula for disk functions

摘录 / EXCERPT: Let $\mathcal{X}\rightarrow X\leftarrow Y$ be toric Calabi--Yau threefolds satisfying the closed crepant resolution conjecture, let...

conjecture | open | MathDB 页面 | 原文线索

V2-061806 | MATHDB-325125

The weak orthogonality relation for Fourier coefficients of Maass forms

摘录 / EXCERPT: The weak orthogonality relation. For positive integers $m_1,\ldots,m_{N-1}$,

conjecture | open | MathDB 页面 | 原文线索

V2-061807 | MATHDB-325129

Stasinski-Voll's signed L-generating-function conjecture for hyperoctahedral descent classes

摘录 / EXCERPT: Stasinski-Voll's conjecture. For every $n\in\mathbb{N}$ and every $I=\{i_1,\dots,i_l\}_{<}\subseteq[n-1]_0$,

conjecture | open | MathDB 页面 | 原文线索

V2-061808 | MATHDB-325130

The Laurent-series Pythagoras-number conjecture

摘录 / EXCERPT: Let k be a field and let $F_n=k(\!(t_1,\dotsc,t_n)\!)$ be the Laurent series field in n variables over k, with $n\geq 2$. For a finite field extension ℓ/k, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061809 | MATHDB-325134

Mocanu's conjecture on valence of harmonic functions

摘录 / EXCERPT: Mocanu's conjecture. Under these hypotheses, $f(z)$ is p-valent in $\mathbb{U}$.

conjecture | open | MathDB 页面 | 原文线索

V2-061810 | MATHDB-325135

Berestycki-Lin-Wei-Zhao's Gibbons-type conjecture for an elliptic system

摘录 / EXCERPT: Consider the elliptic system ... Let $N\geq 2$, and let (u,v) be a solution satisfying ... with the limits uniform in $x'\in\mathbb{R}^{N-1}$. Berestycki-Lin-Wei-Zhao's Gibbons-type...

conjecture | open | MathDB 页面 | 原文线索

V2-061811 | MATHDB-325158

Holonomic rank conjecture for tautological systems on homogeneous spaces

摘录 / EXCERPT: Holonomic rank conjecture. The solution rank of τ at the point $a\in V^\vee$ is given by

conjecture | solved | MathDB 页面 | 原文线索

V2-061812 | MATHDB-325162

Non-explosion conjecture for the interacting particle process

摘录 / EXCERPT: Let H_r be the state space of height configurations and let $\widetilde X^{(z,z')}_r$ be the Markov process on H_r whose deterministic component evolves each height according t...

conjecture | open | MathDB 页面 | 原文线索

V2-061813 | MATHDB-325165

Connectedness conjecture for the category of categories with adjunctions

摘录 / EXCERPT: Let $\operatorname{Cat}^{\triangleq}$ be the category whose objects are categories and whose morphisms are the adjunction-based morphisms described in the source. A category is con...

conjecture | open | MathDB 页面 | 原文线索

V2-061814 | MATHDB-325166

Existence and multiplicity of flows on the infinite sphere of categories

摘录 / EXCERPT: Let $\mathbb{S}^{\infty}[\operatorname{Cat}^{\triangleq}]$ denote the structure formed from $\operatorname{Cat}^{\triangleq}$ by the closed, compatible sequences of morphism compos...

conjecture | open | MathDB 页面 | 原文线索

V2-061815 | MATHDB-325167

Non-total-disconnectedness conjecture for the fractal infinity-category of categories

摘录 / EXCERPT: Let $\mathcal{F}_{\geq 0}\operatorname{Cat}_{\infty}$ be the fractal infinity-category constructed from the ranked categories and morphisms described in the source. It is called to...

conjecture | open | MathDB 页面 | 原文线索

V2-061816 | MATHDB-325169

The total-order conjecture for trees under the independence-polynomial order

摘录 / EXCERPT: Let $\succeq$ be the order on graphs defined by comparing their independence polynomials, and let $\mathcal{T}_n$ be the set of trees with n vertices. Total-order conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061817 | MATHDB-325170

The chain-length conjecture for the independence-polynomial order on trees

摘录 / EXCERPT: Let $\mathcal{T}_n$ be the set of all trees with n vertices, ordered by $\succeq$. The length of a chain is the number of elements in a sequence $T_1\succ T_2\succ\cdots$ under t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-061818 | MATHDB-325171

The total-order conjecture for fixed-degree starlike trees

摘录 / EXCERPT: Let $n_1\geq\cdots\geq n_k\geq 2$ and $m_1\geq\cdots\geq m_k\geq 2$. Let $T_1=T(n_1,\ldots,n_k)$ and $T_2=T(m_1,\ldots,m_k)$ be starlike trees of order n and maximum degree k....

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-061819 | MATHDB-325172

Chromatic-polynomial separation conjecture for signed Petersen graphs

摘录 / EXCERPT: Chromatic separation conjecture. (a) Two signed Petersen graphs that are not switching isomorphic have different zero-free chromatic polynomials; in particular, they have different...

conjecture | open | MathDB 页面 | 原文线索

V2-061820 | MATHDB-325173

Conjecture on the 2-ranks of submatrices of finite quadratic-space incidence matrices

摘录 / EXCERPT: The 2-rank conjecture. The 2-ranks satisfy the following:

conjecture | open | MathDB 页面 | 原文线索

V2-061821 | MATHDB-325180

Borel summability conjecture for transseries solutions of Painlevé III'

摘录 / EXCERPT: Let $R_{\rm odd}$ be the odd part of the formal WKB solution, and let $\lambda(t,{\bf c},\eta;\alpha)$ be the corresponding transseries solution of $(P_{\rm III'})_{D_{6}}$, with f...

conjecture | open | MathDB 页面 | 原文线索

V2-061822 | MATHDB-325183

Positive correlation of paths in the annealed random graph model for all parameters

摘录 / EXCERPT: Let G be the annealed randomly oriented Erdős-Rényi graph $\vec G(n,p)$, with distinct vertices a,s,b,t, and let $\{s\to a\}$ and $\{t\to b\}$ be the events that directed paths...

conjecture | open | MathDB 页面 | 原文线索

V2-061823 | MATHDB-325191

The linear connectivity conjecture for linked tournaments

摘录 / EXCERPT: Linear connectivity conjecture for linked tournaments. There exists a constant $C>0$ such that, for every $k\in\mathbb{N}$, every strongly Ck-connected tournament is k-linked.

conjecture | open | MathDB 页面 | 原文线索

V2-061824 | MATHDB-325192

The quadratic linkedness conjecture for edge-disjoint Hamilton cycles

摘录 / EXCERPT: Quadratic linkedness conjecture. There exists $C'>0$ such that, for every $k\in\mathbb{N}$, every $C'k^2$-linked tournament contains k edge-disjoint Hamilton cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-061825 | MATHDB-325193

The quadratic connectivity conjecture for edge-disjoint Hamilton cycles

摘录 / EXCERPT: Quadratic connectivity conjecture. There exists $C”>0$ such that, for every $k\in\mathbb{N}$, every strongly $C”k^2$-connected tournament contains k edge-disjoint Hamilton cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-061826 | MATHDB-325202

The bounded-order constant cycle curve density conjecture

摘录 / EXCERPT: Let X be a K3 surface. A constant cycle curve is a curve on X whose points all have the same class in the appropriate Chow group; its order is the integer used to measure the a...

conjecture | open | MathDB 页面 | 原文线索

V2-061827 | MATHDB-325203

The constant cycle curve conjecture through the Beauville-Voisin class

摘录 / EXCERPT: Let X be a complex K3 surface, and let $c_X\in\operatorname{CH}^2(X)$ be the distinguished Beauville-Voisin class. A constant cycle curve is a curve on X whose points have a co...

conjecture | open | MathDB 页面 | 原文线索

V2-061828 | MATHDB-325206

Goresky-Kottwitz-Macpherson purity conjecture for fundamental domains

摘录 / EXCERPT: Let F_{γ} be the fundamental domain for the action of the free abelian group generated by $\chi(\epsilon)$, with $\chi\in X_{*}(T)$, on the affine Springer fiber...

conjecture | open | MathDB 页面 | 原文线索

V2-061829 | MATHDB-325214

Kostochka-Woodall conjecture on the choice number of graph squares

摘录 / EXCERPT: Kostochka-Woodall conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061830 | MATHDB-325237

Aldous-Lyons conjecture for graphings

摘录 / EXCERPT: A graphing is a bounded-degree Borel graph satisfying the Intrinsic Mass Transport Principle. A bounded-degree graph sequence has a Benjamini-Schramm limit when the distribution of...

conjecture | open | MathDB 页面 | 原文线索

V2-061831 | MATHDB-325242

Third-order diagonal rational-function radius conjecture

摘录 / EXCERPT: Let $s\geq 2$, and let $\widehat{R}_{s/s,3}$ denote the maximal radius of absolute monotonicity in the restricted rational-function class used by the paper. Third-order diagonal ra...

conjecture | open | MathDB 页面 | 原文线索

V2-061832 | MATHDB-325243

Positivity rigidity conjecture for a binomial finite-difference sum

摘录 / EXCERPT: Positivity rigidity conjecture. Then $p_0\geq 0$, and for $m=1,2,\ldots,s+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-061833 | MATHDB-325244

Minimax risk prediction of phase transitions in convex programs

摘录 / EXCERPT: Let $\vct{x}_0\in\mathbb{R}^d$ be the target vector, let $R_{\mathrm{mm}}(\vct{x}_0)$ denote the minimax mean-square-error risk, and consider recovery by the convex program with me...

conjecture | open | MathDB 页面 | 原文线索

V2-061834 | MATHDB-325245

The sign conjecture for Conway polynomial coefficients of positive links

摘录 / EXCERPT: Conway coefficient sign conjecture. The conclusion of the two-component obstruction holds whenever $L\in\mathcal{P}_0$: in general, the sign $(-1)^k$ is replaced by $(-1)^{s(k,m)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-061835 | MATHDB-325250

Delaunay-Jouhet moment conjecture for class groups of quadratic fields

摘录 / EXCERPT: Delaunay-Jouhet moment conjecture. For every positive integer ℓ and partition $\lambda=1^{m_1}2^{m_2}\cdots \ell^{m_\ell}$, the average over fundamental negative discriminants...

conjecture | open | MathDB 页面 | 原文线索

V2-061836 | MATHDB-325251

Delaunay's moment conjecture for Tate-Shafarevich groups

摘录 / EXCERPT: Delaunay's moment conjecture. For every positive integer ℓ, partition $\lambda=1^{m_1}2^{m_2}\cdots \ell^{m_\ell}$, and nonnegative integer u, the average over...

conjecture | open | MathDB 页面 | 原文线索

V2-061837 | MATHDB-325252

Delaunay-Jouhet moment conjecture for elliptic-curve Selmer groups

摘录 / EXCERPT: Delaunay-Jouhet Selmer moment conjecture. For every positive integer ℓ and partition $\lambda=1^{m_1}2^{m_2}\cdots \ell^{m_\ell}$, the average over all elliptic curves...

conjecture | open | MathDB 页面 | 原文线索

V2-061838 | MATHDB-325254

Tverberg's theorem for max-T convexity

摘录 / EXCERPT: Let ${\mathcal B}^d=[0,1]^d$ be the d-dimensional box equipped with max-T convexity, and let X be a set of $(d+1)(r-1)+1$ points in ${\mathcal B}^d$. Tverberg's theorem for max...

conjecture | open | MathDB 页面 | 原文线索

V2-061839 | MATHDB-325255

Symmetry inequality conjecture for the entropy-difference penalty function

摘录 / EXCERPT: Let $d\geq 2$ and let $M(\Delta,d)$ denote the penalty function defined by the minimization over pairs of distributions with entropy difference Δ. For $\Delta\in[0,\log d]$,...

conjecture | open | MathDB 页面 | 原文线索

V2-061840 | MATHDB-325256

Tatami covering generating-function coefficient conjecture

摘录 / EXCERPT: Let $V\!H_n(z)$ be the generating polynomial for maximal tatami coverings of the square grid with the relevant number of vertical or horizontal dominoes, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-061841 | MATHDB-325272

Random character values on staircase principal hooks

摘录 / EXCERPT: Let P_n be the set of partitions of n, and let $\widehat\rho_k$ denote the principal-hook partition of the staircase partition $\rho_k=(k,k-1,\ldots,1)$, where $n=k(k+1)/2$. Ch...

conjecture | open | MathDB 页面 | 原文线索

V2-061842 | MATHDB-325273

Random character values on self-conjugate principal hooks

摘录 / EXCERPT: Let $\lambda\vdash n$ be a uniformly random partition of n, and let μ be a uniformly random self-conjugate partition of n. Write $\widehat\mu$ for the principal-hook partit...

conjecture | open | MathDB 页面 | 原文线索

V2-061843 | MATHDB-325274

Nonvanishing random entries in the character table of the symmetric group

摘录 / EXCERPT: Let $\lambda,\mu\vdash n$ be independently chosen random partitions of n, and let $\chi^\lambda[\mu]$ denote the value of the irreducible character indexed by λ on the co...

conjecture | open | MathDB 页面 | 原文线索

V2-061844 | MATHDB-325300

Hall-Littlewood polynomial identity for the generalized Rogers-Ramanujan series

摘录 / EXCERPT: Let $m,n\geq 1$. Let $h_{\lambda}^{(m)}(w,z;q)$ denote the modified Rogers-Szegő factor defined earlier, let P'_{λ} be the modified Hall-Littlewood polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061845 | MATHDB-325304

Connectedness conjecture for the graph of simply connected Calabi-Yau threefolds

摘录 / EXCERPT: Consider the graph whose nodes are deformation classes of simply connected Calabi-Yau threefolds and whose edges are given by geometric transitions. Call a node primitive if its de...

conjecture | open | MathDB 页面 | 原文线索

V2-061846 | MATHDB-325308

Abhyankar-Sathaye conjecture for polynomial coordinates

摘录 / EXCERPT: Let A be a ring of characteristic zero, and let $f \in A^{[n]}$. Abhyankar-Sathaye conjecture. If ... then f is an A-coordinate. This is a fundamental cancellation-type questio...

conjecture | open | MathDB 页面 | 原文线索

V2-061847 | MATHDB-325309

Stable coordinate conjecture over rings of characteristic zero

摘录 / EXCERPT: Let A be a ring of characteristic zero, and let $f \in A^{[n]}$. Stable coordinate conjecture. If there is some $m>0$ such that f is a coordinate in $A^{[n+m]}$, then f is a...

conjecture | open | MathDB 页面 | 原文线索

V2-061848 | MATHDB-325310

Dolgachev-Weisfeiler conjecture for partial residual coordinates

摘录 / EXCERPT: Let R be a ring and let $f_1,\ldots,f_m\in R^{[n]}$. A partial system of residual coordinates is a tuple for which the inclusion ... is an affine fibration. A partial system of coo...

conjecture | open | MathDB 页面 | 原文线索

V2-061849 | MATHDB-325311

Partial strongly residual coordinate conjecture

摘录 / EXCERPT: Let R be a ring with an element x, and let $f_1,\ldots,f_m\in R^{[n]}$. A partial system of strongly x-residual coordinates is a tuple that is a partial system of coordinates...

conjecture | open | MathDB 页面 | 原文线索

V2-061850 | MATHDB-325317

The order conjecture for automorphism groups of the norm-trace tower

摘录 / EXCERPT: Let q be the parameter of the norm-trace tower, and let T_m/K be its level of height $m\geq 1$. Order conjecture for the norm-trace tower. The automorphism group has order ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-061851 | MATHDB-325319

The u-invariant bound for function fields over mixed-characteristic complete discretely valued fields

摘录 / EXCERPT: Let K be a complete discretely valued field of characteristic 0 with residue field κ of characteristic 2, and let F be the function field of a curve over K. The u-...

conjecture | open | MathDB 页面 | 原文线索

V2-061852 | MATHDB-325331

Johansson-Kahn-Vu threshold conjecture for factors in random graphs

摘录 / EXCERPT: Johansson-Kahn-Vu threshold conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061853 | MATHDB-325332

Godbole-Goyt-Herdan-Pudwell recurrence conjecture for ordered partitions with blocks of size 2

摘录 / EXCERPT: An ordered partition of $[2k]$ with k blocks of size 2 is counted by $op_{[2^k]}(123)$ when it avoids the permutation pattern 123. Godbole-Goyt-Herdan-Pudwell's recurrence co...

conjecture | open | MathDB 页面 | 原文线索

V2-061854 | MATHDB-325341

Rational shuffle Frobenius-character conjecture

摘录 / EXCERPT: Let $\operatorname{Fr}_{n,m}$ be the symmetric function obtained by summing over m/n parking functions with the statistics and Gessel quasisymmetric functions defined in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-061855 | MATHDB-325349

Furter's Rigidity Conjecture for polynomial inverses

摘录 / EXCERPT: Let $m\in\mathbb N_+$ and let $a(X)\in{\mathbb C}[X]$ be a polynomial of degree at most $m+1$ such that $a(X)\equiv X\pmod{X^2}$. Let $a^{-1}(X)$ denote its formal inverse for comp...

conjecture | open | MathDB 页面 | 原文线索

V2-061856 | MATHDB-325350

The partial Rigidity Conjecture for consecutive inverse coefficients

摘录 / EXCERPT: Let $m,n\in\mathbb N_+$ and let $a(X)\in{\mathbb C}[X]$ be a polynomial of degree at most $m+1$ such that $a(X)\equiv X\pmod{X^2}$. Let $a^{-1}(X)$ be its formal inverse for compos...

conjecture | open | MathDB 页面 | 原文线索

V2-061857 | MATHDB-325352

The Vanishing Conjecture for homogeneous quartics

摘录 / EXCERPT: Let $m\in\mathbb N_+$, let ${\mathbb C}^{[m]}={\mathbb C}[X_1,\ldots,X_m]$, and let ... be the Laplace operator. Let $f\in{\mathbb C}^{[m]}$ be homogeneous of degree 4. Vanishing C...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-061858 | MATHDB-325353

The Generalized Vanishing Conjecture

摘录 / EXCERPT: Let $m\in\mathbb N_+$, let ${\mathbb C}^{[m]}={\mathbb C}[X_1,\ldots,X_m]$, and let $\Delta\in{\mathbb C}[\partial_1,\ldots,\partial_m]$ be a differential operator. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061859 | MATHDB-325354

The Special Image Conjecture

摘录 / EXCERPT: Let $A={\mathbb C}[\xi,z]$, where $\xi=(\xi_1,\ldots,\xi_m)$ and $z=(z_1,\ldots,z_m)$ are two sets of commuting indeterminates. For each i, define...

conjecture | open | MathDB 页面 | 原文线索

V2-061860 | MATHDB-325356

Furter's two-polynomial Rigidity Conjecture

摘录 / EXCERPT: Rigidity Conjecture. If $a\circ b(X)\equiv X\pmod{X^{m+n+2}}$, then $a(X)=b(X)=X$. The paper introduces this conjecture after recording partial results on the length 2 polydegree...

conjecture | open | MathDB 页面 | 原文线索

V2-061861 | MATHDB-325357

The global distribution conjecture for Selmer sequences

摘录 / EXCERPT: Fix a global field k. Let ${\mathscr E}$ be the set of elliptic curves under consideration, and for $E \in {\mathscr E}$ write ... Let ${\mathscr Q}$ be the limiting discrete proba...

conjecture | open | MathDB 页面 | 原文线索

V2-061862 | MATHDB-325358

Delaunay distribution conjecture for Shafarevich-Tate groups in fixed rank

摘录 / EXCERPT: Fix a global field k and $r \in {\mathbb Z}_{\ge 0}$ such that ${\mathscr E}_r=\{E\in{\mathscr E}:\operatorname{rk}E(k)=r\}$ is infinite. For each finite abelian p-group G, c...

conjecture | open | MathDB 页面 | 原文线索

V2-061863 | MATHDB-325364

The weak equivalence conjecture for the nerve of unbiased bicategories

摘录 / EXCERPT: Weak equivalence conjecture. The inclusion of simplicial sets $N(ubi\mathcal{C}tg)\longrightarrow i^*(w\mathcal{C}at^2)$ is a weak equivalence in the Joyal model structure on...

conjecture | open | MathDB 页面 | 原文线索

V2-061864 | MATHDB-325365

Cox ring quotient conjecture for the del Pezzo fibration

摘录 / EXCERPT: Let $\mathcal{F}$ be the projective bundle used to construct the threefold Mori dream space X, and let f and g be the defining equations of the two hypersurfaces whose comple...

conjecture | open | MathDB 页面 | 原文线索

V2-061865 | MATHDB-325374

Lower bounds for the eigenvalue growth rate when p equals 2

摘录 / EXCERPT: Let $f(x)$ be a polynomial and let $\lambda=\lambda(f)$ be the associated growth rate. Two polynomials are similar when they are related by the transformations described in the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-061866 | MATHDB-325382

The occupation measure conjecture for radial SLE(2)

摘录 / EXCERPT: Let $\tilde{\Omega}$ be the space of equivalence classes of SLE(2) curves in $\mathbb{D}$, let $\mathcal{M}$ be the relevant space of measures, and let $G(z)=G_{\mathbb{D}}(z)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-061867 | MATHDB-325401

Nevo's Colin de Verdière stress-freeness conjecture

摘录 / EXCERPT: Let G be a graph, let k be a positive integer, and let $\mu(G)$ denote the Colin de Verdière parameter of G. A graph is generically k-stress free if it has no nonzero gener...

conjecture | open | MathDB 页面 | 原文线索

V2-061868 | MATHDB-325404

Weak K_9-minor-free graph structural conjecture

摘录 / EXCERPT: Let $\delta(G)$ denote the minimum degree of G, and let a graph be K_9-minor free if it has no K_9 minor. Weak K_9 structural conjecture. Any K_9-minor-free graph G wit...

conjecture | open | MathDB 页面 | 原文线索

V2-061869 | MATHDB-325410

Expected second-degree distribution in the Buckley-Osthus model with multiple edges

摘录 / EXCERPT: Expected second-degree distribution conjecture. For

conjecture | open | MathDB 页面 | 原文线索

V2-061870 | MATHDB-325417

Discriminant formula conjecture for Chebyshev radical extensions

摘录 / EXCERPT: Let K be the number field generated by a root of the defining polynomial Φ, let ℓ and n be the parameters of the extension, and let $\operatorname{ind}_\ell(\Phi)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-061871 | MATHDB-325431

The infinite motion conjecture for permutation groups

摘录 / EXCERPT: Infinite motion conjecture for permutation groups. If A is closed and subdegree-finite with infinite motion, then

conjecture | open | MathDB 页面 | 原文线索

V2-061872 | MATHDB-325437

t-intersecting partition conjecture

摘录 / EXCERPT: Let P_n be the set of all integer partitions of n. For partitions ${\bf a}$ and ${\bf b}$, say that they t-intersect if they have t common parts, with multiplicities allowe...

conjecture | open | MathDB 页面 | 原文线索

V2-061873 | MATHDB-325438

Fixed-length t-intersecting partition conjecture

摘录 / EXCERPT: Let $P_{n,k}$ be the set of partitions of n of length k, and let $P_{n,k}\langle t\rangle$ be the set of partitions whose first t entries are 1. Two partitions t-intersec...

conjecture | open | MathDB 页面 | 原文线索

V2-061874 | MATHDB-325439

Properly t-intersecting partition conjecture

摘录 / EXCERPT: Let P_n be the set of all integer partitions of n. Two partitions t-intersect properly if they have t distinct common parts, and a subset is properly t-intersecting if ev...

conjecture | open | MathDB 页面 | 原文线索

V2-061875 | MATHDB-325440

Eremenko-Lyubich conjecture on detection of asymptotic values

摘录 / EXCERPT: Eremenko-Lyubich's conjecture. Then f has a finite asymptotic value. This conjecture connects the cylindrical area of large preimage sets with the existence of finite asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-061876 | MATHDB-325442

Almost-sure distinguishing conjecture for random colourings of locally finite graphs

摘录 / EXCERPT: Random distinguishing-colouring conjecture. The random colouring of G is almost surely distinguishing.

conjecture | open | MathDB 页面 | 原文线索

V2-061877 | MATHDB-325458

The Shuffle conjecture

摘录 / EXCERPT: Let n be a positive integer, let $\mu=(\mu_1,\mu_2,\ldots,\mu_k)\vdash n$, and let $E_1,E_2,\ldots,E_k$ be successive segments of the word $123\cdots n$ of respective lengths...

conjecture | open | MathDB 页面 | 原文线索

V2-061878 | MATHDB-325459

Genus-zero quasimap S-operator wall-crossing conjecture

摘录 / EXCERPT: Let $W//{\bf G}$ be a GIT quotient, let Λ be the Novikov ring, and let $\gamma\in H^*(W//{\bf G},\Lambda)$ be of the form $\gamma={\mathbb 1}+O(q)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-061879 | MATHDB-325465

The lattice Cayley polytope conjecture

摘录 / EXCERPT: Lattice Cayley polytope conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-061880 | MATHDB-325478

The k-sum embedding conjecture for graph families

摘录 / EXCERPT: k-sum embedding conjecture. For any family of graphs $\mathcal F$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-061881 | MATHDB-325479

Hypersimplex Ehrhart-root strip conjecture

摘录 / EXCERPT: Let d and n be integers with $1\leq d<n$, and let $\Delta(d,n)$ be the convex hull in $\mathbb{R}^n$ of the vectors $\mathbf e_{i_1}+\cdots+\mathbf e_{i_d}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-061882 | MATHDB-325485

Finite-basis conjecture for k-multiplicity-bounded permutations

摘录 / EXCERPT: For an integer $k\geq1$, a permutation w is k-multiplicity bounded when every coefficient $a_{w\lambda}$ in the Schur expansion ... satisfies $a_{w\lambda}\leq k$. Finite-basis c...

conjecture | open | MathDB 页面 | 原文线索

V2-061883 | MATHDB-325487

Hypergeometric inequality conjecture for parameter x

摘录 / EXCERPT: For an integer m in the range intended by the paper, let ${}_2F_1(a,b;c\mid z)$ denote the Gauss hypergeometric function. Hypergeometric inequality conjecture. The inequality ... h...

conjecture | open | MathDB 页面 | 原文线索

V2-061884 | MATHDB-325491

Universality conjecture for the edge spectrum of truncated unitary matrices

摘录 / EXCERPT: Let ${\mathcal{Q}}(\bm \phi)$ be the ensemble of matrices considered in the paper, with eigenvalues $\lambda_i(\bm \phi)$, and let $\rho(x)$ denote their local eigenvalue density....

conjecture | open | MathDB 页面 | 原文线索

V2-061885 | MATHDB-325499

Strong Bogomolov-Gieseker inequality for stable sheaves on quintic threefolds

摘录 / EXCERPT: Strong Bogomolov-Gieseker conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-061886 | MATHDB-325500

Clifford-type bound for stable coherent systems on quintic surfaces

摘录 / EXCERPT: Clifford-type bound conjecture. For every μ'-stable coherent system $(\mathcal{O}_S^{\oplus R}\to F)$ with

conjecture | open | MathDB 页面 | 原文线索

V2-061887 | MATHDB-325507

Signed mean value trace inequalities for Hermitian matrices

摘录 / EXCERPT: Let $\mathsfsl{A},\mathsfsl{B},\mathsfsl{C}\in\mathbb{H}^{d}$ be Hermitian matrices, let q be a positive integer, and let $s>0$. For a Hermitian matrix $\mathsfsl{X}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-061888 | MATHDB-325508

The LCW Enumeration Conjecture for permutation classes

摘录 / EXCERPT: A permutation class $\mathcal{C}$ is broadly rational if every finitely based subclass of $\mathcal{C}$ has a rational generating function, and it is strongly rational if it and al...

conjecture | open | MathDB 页面 | 原文线索

V2-061889 | MATHDB-325512

Concavity conjecture for the generalized inverse hyperbolic sine

摘录 / EXCERPT: For $x\in(0,1)$ fixed, let $p\mapsto {\rm arcsinh}_p(x)$ denote the generalized inverse hyperbolic sine as a function of p. Concavity conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-061890 | MATHDB-325514

Jinzenji-Shimizu categorical mirror formula for CP^2

摘录 / EXCERPT: Let h be the hyperplane class of CP^2, and let $w({\cal O}_{h^a}{\cal O}_{h^b}\mid(x^0,x^1,x^2))_0$ denote the generating function of multi-point virtual structure constants. D...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-061891 | MATHDB-325521

Finite-order strict total positivity conjecture for the Cauchy kernel

摘录 / EXCERPT: Let K_α denote the Cauchy kernel considered in the paper, and let ${\rm STP}_n$ and ${\rm SR}_n$ denote strict total positivity and sign-regularity of order n, respectivel...

conjecture | open | MathDB 页面 | 原文线索

V2-061892 | MATHDB-325522

Positivity conjecture for coefficients of the Chebyshev matrix

摘录 / EXCERPT: Let $B_{\alpha,n}$ be the $n\times(2n-1)$ band matrix whose coefficients are ... when $i\le k\le n-1+i$, and are zero otherwise. Let $B_\sigma(1)$ denote the associated coefficient i...

conjecture | open | MathDB 页面 | 原文线索

V2-061893 | MATHDB-325523

Total positivity conjecture for the Chebyshev matrix

摘录 / EXCERPT: Let $B_{\alpha,n}$ be the rectangular band matrix defined by ... for $i\le k\le n-1+i$, with zero entries otherwise. A matrix is totally positive if all of its minors are non-negativ...

conjecture | open | MathDB 页面 | 原文线索

V2-061894 | MATHDB-325526

Brualdi's conjecture in rainbow-matching form

摘录 / EXCERPT: Brualdi's conjecture. Such a partition has a rainbow matching of size $n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-061895 | MATHDB-325536

Asymptotic equality of Carmichael and weak Carmichael numbers

摘录 / EXCERPT: Let $C(n)$ and $W(n)$ denote the numbers of Carmichael numbers and weak Carmichael numbers in the interval $[1,n]$, respectively. Asymptotic equality conjecture. ... The conjecture p...

conjecture | open | MathDB 页面 | 原文线索

V2-061896 | MATHDB-325541

Super Carmichael numbers are Carmichael numbers

摘录 / EXCERPT: A super Carmichael number is a weak Carmichael number satisfying ... Super-Carmichael containment conjecture. Every super Carmichael number is a Carmichael number. The source present...

conjecture | open | MathDB 页面 | 原文线索

V2-061897 | MATHDB-325552

Cyclotomic defining-field conjecture for modular categories

摘录 / EXCERPT: A modular category is a braided fusion category over $\mathbb C$ equipped with a compatible ribbon structure whose braiding is nondegenerate; a defining number field is a number fi...

conjecture | open | MathDB 页面 | 原文线索

V2-061898 | MATHDB-325556

The branching-path conjecture for k-Schur functions

摘录 / EXCERPT: The branching-path conjecture. For all $\lambda\in\mathcal C^{k+1}$ and $\mu\in\mathcal C^k$,

conjecture | open | MathDB 页面 | 原文线索

V2-061899 | MATHDB-325566

Huang-Wong-Zhu on-line Ohba conjecture

摘录 / EXCERPT: Let G be a graph, and let $|V(G)|$ and $\chi(G)$ denote its number of vertices and chromatic number, respectively. A graph is on-line chromatic-choosable if $\chi_p(G)=\chi(G)$....

conjecture | open | MathDB 页面 | 原文线索

V2-061900 | MATHDB-325567

Weak on-line Ohba's conjecture

摘录 / EXCERPT: Let G be a graph, let $|V(G)|$ be its number of vertices, and let $\chi(G)$ and $\chi_p(G)$ denote its chromatic number and on-line choice number, respectively. Weak on-line Ohba...

conjecture | open | MathDB 页面 | 原文线索

V2-061901 | MATHDB-325569

The outer-bound conjecture for the $(0,R_L,R_M)$-setup when $L\leq M/2$

摘录 / EXCERPT: Outer-bound conjecture. For the $(0,R_L,R_M)$-setup with $L\leq \frac{M}{2}$, the inequality

conjecture | open | MathDB 页面 | 原文线索

V2-061902 | MATHDB-325582

Identity conjecture for tropical matrix monoids

摘录 / EXCERPT: For each positive integer n, let $M_n(\mathbb T)$ be the monoid of $n\times n$ tropical matrices. A nontrivial semigroup identity is an identity between distinct semigroup words...

conjecture | open | MathDB 页面 | 原文线索

V2-061903 | MATHDB-325584

The monotonicity conjecture for the gamma logarithmic ratio

摘录 / EXCERPT: The monotonicity conjecture. For every $\tau\geq0$, the function g_τ is strictly increasing on $(0,\infty)$. This conjecture is one of the monotonicity problems arising from r...

conjecture | solved | MathDB 页面 | 原文线索

V2-061904 | MATHDB-325585

The complete monotonicity conjecture for a logarithmic ratio

摘录 / EXCERPT: Complete monotonicity conjecture. The function h_1 is completely monotonic on $(0,\infty)$. This is a stronger regularity assertion about a logarithmic ratio related to the gamma...

conjecture | open | MathDB 页面 | 原文线索

V2-061905 | MATHDB-325587

Rimányi's positivity conjecture for Thom polynomials

摘录 / EXCERPT: Rimányi's positivity conjecture. The coefficients of the Thom polynomial are nonnegative, equivalently

conjecture | open | MathDB 页面 | 原文线索

V2-061906 | MATHDB-325593

Structural conjecture for the strict endoscopic part of modular Siegel threefolds

摘录 / EXCERPT: Structural conjecture for the strict endoscopic part. The strict endoscopic part contributes as a finite part of theta-lifts of cuspidal automorphic representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-061907 | MATHDB-325613

Brower's fixed-point conjecture for definable maps of the square

摘录 / EXCERPT: Let $\mathbb{K}$ be a definably complete expansion of an ordered field, and let $f:[0,1]^2\to[0,1]^2$ be a definable continuous function. Then Brower's fixed-point conjecture. Ther...

open_problem | open | MathDB 页面 | 原文线索

V2-061908 | MATHDB-325614

Mertens's asymptotic value conjecture for zero-sum repeated games

摘录 / EXCERPT: Let a finite two-person zero-sum repeated game have values v_n for its n-stage games and v_{λ} for its λ-discounted games, where $n\in\mathbb{N}^*$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061909 | MATHDB-325615

Mertens's maxmin conjecture for more-informed repeated games

摘录 / EXCERPT: Consider a finite two-person zero-sum repeated game with values v_n for its n-stage games and v_{λ} for its λ-discounted games. Player 1 is more informed than P...

conjecture | open | MathDB 页面 | 原文线索

V2-061910 | MATHDB-325616

Bernstein-von Mises conjecture for the kernel variance in normal location mixtures

摘录 / EXCERPT: Let $X_1,X_2,\ldots$ be an independent and identically distributed sample from $P_0=P_{\sigma_0,F_0}$ in the semiparametric normal location mixture model parametrized by the model...

conjecture | open | MathDB 页面 | 原文线索

V2-061911 | MATHDB-325617

Jakobson-Naud's essential spectral gap conjecture

摘录 / EXCERPT: Jakobson-Naud's essential spectral gap conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061912 | MATHDB-325629

Non-negativity conjecture for orbit-counting polynomials of pairs in abelian groups

摘录 / EXCERPT: Let R be a discrete valuation ring with residue field of order q, let M be the finite abelian group associated with a partition λ, and let G act on $M\times M$ as i...

conjecture | open | MathDB 页面 | 原文线索

V2-061913 | MATHDB-325630

Local K-theory connecting homomorphism isomorphism conjecture

摘录 / EXCERPT: Local K-theory connecting homomorphism conjecture. In local K-theory, $\partial^{K, loc}$ is an isomorphism rationally.

conjecture | open | MathDB 页面 | 原文线索

V2-061914 | MATHDB-325631

Local K-theory index theorem conjecture

摘录 / EXCERPT: Local K-theory index theorem conjecture. For any elliptic operator A,

conjecture | open | MathDB 页面 | 原文线索

V2-061915 | MATHDB-325635

Energy-independence conjecture for limit points of rescaled eigenvalue measures

摘录 / EXCERPT: Let $d\geq 4$ and let $\mu_{L,E}^0$ be the measures associated with the finite-volume model at energy $E\in(-2d,2d)$. Let n_d denote the density of states of the d-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-061916 | MATHDB-325640

Arithmetic-progression conjecture for rank-generating functions

摘录 / EXCERPT: Let $F_{\lambda}(q)$ be the rank-generating function for partitions contained inside a partition λ, and let ... be a partition whose parts form an arithmetic progression. A p...

conjecture | open | MathDB 页面 | 原文线索

V2-061917 | MATHDB-325641

Arbitrary-peak conjecture for rank-generating functions

摘录 / EXCERPT: Let $F_{\lambda}(q)$ be the rank-generating function for partitions contained inside a partition λ. A polynomial is unimodal if its coefficients weakly increase and then we...

conjecture | open | MathDB 页面 | 原文线索

V2-061918 | MATHDB-325642

Flawlessness conjecture for rank-generating functions

摘录 / EXCERPT: Let $F_{\lambda}(q)=\sum_{i=0}^{d}c_iq^i$ be the rank-generating function for partitions contained inside a partition λ. The polynomial is flawless if $c_i\le c_{d-i}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-061919 | MATHDB-325643

Partial rainbow matching conjecture for matchings of size k

摘录 / EXCERPT: Let k matchings of size k be given in a bipartite graph. A partial rainbow matching is a matching containing at most one edge from each given matching. Partial rainbow matching...

conjecture | open | MathDB 页面 | 原文线索

V2-061920 | MATHDB-325646

Hilton's Latin rectangle decomposition conjecture

摘录 / EXCERPT: An $n\times 2n$ Latin rectangle is an array with n rows and $2n$ columns in which each symbol occurs at most once in each row and column. A transversal is a set of entries meetin...

conjecture | open | MathDB 页面 | 原文线索

V2-061921 | MATHDB-325647

Disjoint rainbow matching conjecture

摘录 / EXCERPT: Let $k+1$ pairwise disjoint matchings of size k be given in a bipartite graph. A partial rainbow matching is a matching containing at most one edge from each given matching. Disj...

conjecture | open | MathDB 页面 | 原文线索

V2-061922 | MATHDB-325664

Two-moment criterion for normal convergence in a fixed eigenspace

摘录 / EXCERPT: Let $k,l\geq 2$ be distinct positive integers, and let $\{X_n\}_{n\geq 1}$ be a sequence of eigenfunctions lying in the same eigenspace of a Markov generator $\mathrm{\mathbf L}$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-061923 | MATHDB-325666

The KH property excludes pseudoalternating pseudodiagrams

摘录 / EXCERPT: KH-pseudoalternating conjecture. A pseudodiagram with the KH property cannot be pseudoalternating.

conjecture | open | MathDB 页面 | 原文线索

V2-061924 | MATHDB-325680

Congruence for 9-regular partitions modulo 12

摘录 / EXCERPT: Let $b_9(n)$ denote the number of 9-regular partitions of n, that is, partitions in which no part divisible by 9 appears. The higher-power congruence. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-061925 | MATHDB-325684

Isotropic realizability conjecture for periodic gradients

摘录 / EXCERPT: Isotropic realizability conjecture. Then ∇u is isotropically realizable in the torus with a positive conductivity σ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-061926 | MATHDB-325687

Sink-independence conjecture for recurrent-configuration class sums

摘录 / EXCERPT: Sink-independence conjecture. The sequence $\left(\textbf{\texttt{sum}}\,_{G,s}(B)\right)_{B\in\mathcal{C}/\sim}$ is independent of the choice of s, up to a permutation of its en...

conjecture | open | MathDB 页面 | 原文线索

V2-061927 | MATHDB-325693

Consistency of fixed-point partitions for branching matrices

摘录 / EXCERPT: Let $\mathbf{M}$ be a branching matrix, and for a specific value $\lambda<\operatorname{WSM}(T_{\mathbf{M}})$ let V_λ be the fixed points of the recurrences of the marginal...

conjecture | open | MathDB 页面 | 原文线索

V2-061928 | MATHDB-325699

The maximal and typical rank conjecture for real $2\times2\times2\times2$ tensors

摘录 / EXCERPT: Let $\mathbb{R}^{2\times2\times2\times2}$ denote the space of real tensors with four modes, each of size 2, and let tensor rank be the least number of rank-one tensors whose sum...

conjecture | open | MathDB 页面 | 原文线索

V2-061929 | MATHDB-325716

Young's shifted convolution conjecture for Maass forms

摘录 / EXCERPT: Let $U(z)$ be a Hecke-Maass cusp form with Laplace eigenvalue $1/4+T^2$, Fourier coefficients $\lambda(n)$, and first Fourier coefficient $\rho(1)$. For fixed $0<a\le y\le b$, nonz...

conjecture | open | MathDB 页面 | 原文线索

V2-061930 | MATHDB-325717

Young's square-root shifted convolution conjecture for Maass forms

摘录 / EXCERPT: Let $U(z)$ be a Hecke-Maass cusp form with Laplace eigenvalue $1/4+T^2$, Fourier coefficients $\lambda(n)$ and first coefficient $\rho(1)$. For fixed $0<a\le y\le b$,...

conjecture | open | MathDB 页面 | 原文线索

V2-061931 | MATHDB-325722

The converse supersolvability conjecture for ψ-graphical arrangements

摘录 / EXCERPT: Let (G,ψ) be a graph together with the data defining its ψ-graphical arrangement $\mathcal{A}_{G,\psi}$. Suppose the vertices of G can be ordered as $v_1,\dots,v_p$ so...

conjecture | open | MathDB 页面 | 原文线索

V2-061932 | MATHDB-325723

Freeness implies supersolvability for ψ-graphical arrangements

摘录 / EXCERPT: Let $\mathcal{A}_{G,\psi}$ be a ψ-graphical arrangement, and call it free when its cone is a free arrangement. Recall that an arrangement is supersolvable when the intersectio...

conjecture | open | MathDB 页面 | 原文线索

V2-061933 | MATHDB-325728

Sufficient conditions for generic global rigidity on the cylinder and cone

摘录 / EXCERPT: Let (G,p) be a generic framework on ${\mathcal{M}}$, where ${\mathcal{M}}\in\{{\mathcal{Y}},{\mathcal{C}}\}$. A framework is globally rigid on ${\mathcal{M}}$ when every framewor...

conjecture | open | MathDB 页面 | 原文线索

V2-061934 | MATHDB-325735

Kazhdan-Lusztig monotonicity property B

摘录 / EXCERPT: Let (W,S) be a Coxeter system, and let $P_{y,w}$ denote its Kazhdan-Lusztig polynomials for $y,w\in W$, with $\leq$ the Bruhat order. Property B. The polynomials $P_{y,w}$ are de...

conjecture | open | MathDB 页面 | 原文线索

V2-061935 | MATHDB-325736

Classical Kazhdan-Lusztig positivity package

摘录 / EXCERPT: The classical positivity package. Properties A, B, C, and D hold for all Coxeter systems (W,S). Properties A and C are known in this generality, and Property B is known for finit...

conjecture | open | MathDB 页面 | 原文线索

V2-061936 | MATHDB-325741

Logarithmic excess conjecture for snarks

摘录 / EXCERPT: Let G be a snark with n vertices, let m be the number of edges, and let $scc(G)$ denote the length of a shortest cycle cover. Logarithmic excess conjecture. There exist snark...

conjecture | open | MathDB 页面 | 原文线索

V2-061937 | MATHDB-325742

Bounded-oddness shortest cycle cover conjecture

摘录 / EXCERPT: Let G be a snark, let m be the number of edges of G, and let $o(G)$ denote its oddness. Write $scc(G)$ for the length of a shortest cycle cover. Bounded-oddness conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-061938 | MATHDB-325744

Everywhere convergence of Eisenstein series under a weakened Godement criterion

摘录 / EXCERPT: Everywhere-convergence conjecture. $E_\nu(g)$ converges absolutely for $g\in KA'N$ and ν satisfying $\mathrm{Re}~\nu(h_{\alpha_i})<-1$, $i\in I$. The proved result gives almost...

conjecture | open | MathDB 页面 | 原文线索

V2-061939 | MATHDB-325745

Everywhere convergence of one-parameter Eisenstein series beyond the proved range

摘录 / EXCERPT: Let $E_s(g)$ be the Eisenstein series induced by the parameter s, and let A' be the cone used in the paper. Let K be the unitary form and N the unipotent subgroup in the ra...

conjecture | open | MathDB 页面 | 原文线索

V2-061940 | MATHDB-325751

Virtual-4 automorphism group conjecture for primitive aperiodic substitution subshifts

摘录 / EXCERPT: Let σ be a primitive aperiodic substitution, and let X_σ be its associated subshift. The automorphism group $\operatorname{Aut}(X_\sigma)$ contains the subgroup of sh...

conjecture | open | MathDB 页面 | 原文线索

V2-061941 | MATHDB-325752

Positivity and uniqueness conjecture for the recurrence coefficients

摘录 / EXCERPT: Positivity and uniqueness conjecture. There is a unique positive solution, and it corresponds to the initial condition

conjecture | open | MathDB 页面 | 原文线索

V2-061942 | MATHDB-325764

Stevens's conjecture on simple curve singularities

摘录 / EXCERPT: Stevens's conjecture. The simple reduced curve singularities are exactly those with simple parametrisation.

conjecture | open | MathDB 页面 | 原文线索

V2-061943 | MATHDB-325765

Akbulut's conjecture on winding-one satellite operators

摘录 / EXCERPT: Let $*$ be either ex or top, denoting the relevant concordance category, and let $P:\mathcal{C}_*\to\mathcal{C}_*$ be the map induced by a winding...

conjecture | open | MathDB 页面 | 原文线索

V2-061944 | MATHDB-325769

Costa's corner-point conjecture for two-user Gaussian interference channels

摘录 / EXCERPT: Costa's corner-point conjecture. For reliable communication at both receivers, the following should hold: if $R_2\geq C_2-\varepsilon$ for an arbitrary $\varepsilon>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-061945 | MATHDB-325770

Coates-Perrin-Riou's interpolation conjecture for ordinary motives

摘录 / EXCERPT: Let $V=\{V_l\}$ be an ordinary compatible system of finite-dimensional l-adic representations of $G_{\mathbb{Q}}$, with coefficient field K, valuation ring $\mathcal{O}$, and D...

conjecture | open | MathDB 页面 | 原文线索

V2-061946 | MATHDB-325771

Greenberg's exceptional-zero conjecture for ordinary motives

摘录 / EXCERPT: Let $G(T,V)$ and $\mathcal{E}(V)$ be as above, and let g be the number of vanishing factors of $\mathcal{E}(V)$. Let $\mathcal{E}^*(V)$ denote the product of the nonzero factors,...

conjecture | open | MathDB 页面 | 原文线索

V2-061947 | MATHDB-325773

Eventual linearity of ultradiscrete Painlevé solutions

摘录 / EXCERPT: Let Y_m and Z_m be solutions of ultradiscrete Painlevé VI without parity variables, with fixed parity variables $\frak{y}_m=\frak{z}_m=-1$ for all m, and let Q,A_i,B_i be t...

conjecture | open | MathDB 页面 | 原文线索

V2-061948 | MATHDB-325778

Simplicity conjecture for the zeros of generalized Hermite polynomials

摘录 / EXCERPT: Let H_λ be the generalized Hermite polynomial associated with a partition λ. A zero of H_λ is simple if it has multiplicity one. Simplicity conjecture. All...

conjecture | open | MathDB 页面 | 原文线索

V2-061949 | MATHDB-325780

Generalized diffusion limit for queue-length processes

摘录 / EXCERPT: Generalized diffusion-limit conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-061950 | MATHDB-325784

The converse between weak K-Knuth and K-Knuth equivalence

摘录 / EXCERPT: The converse between weak K-Knuth and K-Knuth equivalence. For words ${\mathbf u}$ and ${\mathbf v}$ of integers,

conjecture | open | MathDB 页面 | 原文线索

V2-061951 | MATHDB-325790

ABC-type height conjecture for motives

摘录 / EXCERPT: For a motive M over a number field K, define ... where v ranges over all finite places of K at which M has bad reduction. Let D_K denote the discriminant of K, and fix...

conjecture | open | MathDB 页面 | 原文线索

V2-061952 | MATHDB-325793

Orthogonal symmetry conjecture for Maass cusp forms

摘录 / EXCERPT: Let $\mathcal{B}_N$ be the family of Maass cusp forms of level N, let h_T be the weight function defined in the source, let ϕ be an even Schwartz function with compactly s...

conjecture | open | MathDB 页面 | 原文线索

V2-061953 | MATHDB-325794

The Möbius function formula conjecture for intervals $[M_m,\pi]$

摘录 / EXCERPT: Möbius function formula conjecture. If π begins with 12, ends with $(n-1)n$, or contains a triple adjacency, then

conjecture | open | MathDB 页面 | 原文线索

V2-061954 | MATHDB-325804

The r-ELSV conjecture for r-spin Hurwitz numbers

摘录 / EXCERPT: Let r be a positive integer. For integers $k_1,\dots,k_n$, write ... where p_i is the quotient and $r-1-a_i$ is the remainder on division by r, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-061955 | MATHDB-325813

Kulikov's nondecomposability conjecture for transcendental Hodge structures of surfaces

摘录 / EXCERPT: Let S be a smooth projective surface over the complex numbers. Write T_S for the integral polarized Hodge structure on the transcendental part of $H^2(S,\mathbb Z)$. Nondecompo...

conjecture | open | MathDB 页面 | 原文线索

V2-061956 | MATHDB-325821

The Carleson-measure characterization of multi-nicely connected domains

摘录 / EXCERPT: Let G be a domain conformally equivalent to a circular domain W, and let $\phi:G\to W$ be a conformal map. A positive measure on G is a Carleson measure if it satisfies the C...

conjecture | open | MathDB 页面 | 原文线索

V2-061957 | MATHDB-325822

Mutual singularity of harmonic measures from componentwise Carleson measures

摘录 / EXCERPT: Let G be an open subset, with components whose harmonic measures are considered. A positive measure μ on G is a Carleson measure when it satisfies the relevant Carleson emb...

conjecture | open | MathDB 页面 | 原文线索

V2-061958 | MATHDB-325826

Uniqueness of rectifiers up to equivalence

摘录 / EXCERPT: Uniqueness of rectifiers. For T as above, T admits a unique rectifier up to equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-061959 | MATHDB-325829

Critical-mass blow-up and radiation dichotomy for quartic gKdV

摘录 / EXCERPT: Let Q be the soliton of the quartic generalized Korteweg-de Vries equation, and let $u(x,t)$ solve that equation for $n=4$ with smooth initial data $u_0\in L_2(\mathbb{R})$ of su...

conjecture | open | MathDB 页面 | 原文线索

V2-061960 | MATHDB-325830

Koch-profile blow-up conjecture for quintic gKdV

摘录 / EXCERPT: Let P be the solution of the ODE defining the proposed blow-up profile, and let $u(x,t)$ solve the generalized Korteweg-de Vries equation for $n=5$ with smooth initial data...

conjecture | open | MathDB 页面 | 原文线索

V2-061961 | MATHDB-325831

Dispersive-shock and exponential blow-up-time conjecture for small-dispersion gKdV

摘录 / EXCERPT: Let $u_0\in L_2(\mathbb R)$ be smooth initial data with a single maximum, let $\epsilon\ll1$ be the dispersion parameter, and let t_c be the break-up time of the corresponding ge...

conjecture | open | MathDB 页面 | 原文线索

V2-061962 | MATHDB-325835

Conjecture on the tensor-tubal-norm

摘录 / EXCERPT: Let a third-order tensor have a t-SVD with f-diagonal tensor $\boldsymbol{\mathscr{S}}$, and define its tensor-tubal-rank as the number of nonzero diagonal tubes...

conjecture | open | MathDB 页面 | 原文线索

V2-061963 | MATHDB-325836

Lin's conjecture on ternary ideal two-level autocorrelation sequences

摘录 / EXCERPT: Lin's conjecture. The sequence S has ideal two-level autocorrelation; that is, if its period is N, then

conjecture | open | MathDB 页面 | 原文线索

V2-061964 | MATHDB-325837

Berglund-Hübsch-Krawitz mirror symmetry for Landau-Ginzburg models

摘录 / EXCERPT: Let W be a nondegenerate invertible polynomial, let G be a group of diagonal symmetries of W, and let $(W^{\vee},G^{\vee})$ be the Berglund-Hübsch-Krawitz dual pair. Berglund...

conjecture | open | MathDB 页面 | 原文线索

V2-061965 | MATHDB-325840

Aut(W)-invariant Landau-Ginzburg/Calabi-Yau correspondence

摘录 / EXCERPT: Assume W satisfies the Calabi-Yau condition and G is a subgroup with $\langle j_W\rangle\subseteq G\subseteq\operatorname{Aut}(W)$. Consider the $\operatorname{Aut}(W)$-invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-061966 | MATHDB-325841

Large-complex-structure mirror map for Calabi-Yau orbifolds

摘录 / EXCERPT: Let W be a Calabi-Yau polynomial with group G, let $[X_W/\widetilde G]$ be the associated Calabi-Yau orbifold, and let ∞ denote the large complex structure point in the...

conjecture | open | MathDB 页面 | 原文线索

V2-061967 | MATHDB-325842

Higher-genus Calabi-Yau/Saito-Givental mirror symmetry

摘录 / EXCERPT: Let W be a Calabi-Yau polynomial, let G_W be the group appearing in the quotient $[X_W/\widetilde{G}_W]$, and let $W^{\vee}_{\infty}=\prod_i x_i$ be the large complex structure...

conjecture | open | MathDB 页面 | 原文线索

V2-061968 | MATHDB-325843

Higher-genus Landau-Ginzburg/Saito-Givental mirror symmetry

摘录 / EXCERPT: Let W be a nondegenerate polynomial and set $G=\operatorname{Aut}(W)$. Let $W^{\vee}_0=W^{\vee}$ be the Gepner point, let $\mathcal F^g_{\operatorname{FJRW}}$ be the genus-g FJ...

conjecture | open | MathDB 页面 | 原文线索

V2-061969 | MATHDB-325844

Analytic continuation of Saito-Givental theories between Gepner and large-complex-structure points

摘录 / EXCERPT: Let $W^{\vee}_0=W^{\vee}$ be the Gepner point and $W^{\vee}_{\infty}=\prod_i x_i$ the large complex structure point in the deformation space of $W^{\vee}$. Let the corresponding Sa...

conjecture | open | MathDB 页面 | 原文线索

V2-061970 | MATHDB-325871

Uniform Mordell-Lang conjecture for rational points on curves

摘录 / EXCERPT: Let $d\ge 1$, $g\ge 2$, and $r\ge 0$. Let K be a number field of degree d, and let C be a curve over K of genus g with Jacobian J. Uniform Mordell-Lang conjecture for r...

conjecture | open | MathDB 页面 | 原文线索

V2-061971 | MATHDB-325879

The bounded row and column sum discrepancy conjecture

摘录 / EXCERPT: Let V be an $n \times m$ real matrix with $|V_{i,j}| \leq 1$, $\|v^i\|_1 \leq R$, and $\|v_j\|_1 \leq \Delta$ for all $i \in [n]$ and $j \in [m]$. Assume $R \geq \Delta$. The bou...

conjecture | open | MathDB 页面 | 原文线索

V2-061972 | MATHDB-325880

The matrix paving conjecture

摘录 / EXCERPT: Let $A_1,\ldots,A_m$ be Hermitian, positive semi-definite matrices of the same size satisfying ... and $\operatorname{tr} A_i \leq \delta$ for all i. The matrix paving conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-061973 | MATHDB-325892

The perturbed-metric conjecture for Gaussian noise stability deficit

摘录 / EXCERPT: The perturbed-metric conjecture. For every $0<s<1$, there exist constants $C_s,c_s>0$ such that, whenever $\varepsilon(A)<e^{-1/\rho}$,

conjecture | open | MathDB 页面 | 原文线索

V2-061974 | MATHDB-325893

Strict concavity conjecture for the minimal Riesz energy at exponent minus one

摘录 / EXCERPT: Minus-one concavity conjecture. The map $N\mapsto v_{-1}(N)$ is locally strictly concave for all $N>2$, equivalently $\ddot v_{-1}(N)<0$ for all $N>2$. Numerical data suggest this...

conjecture | open | MathDB 页面 | 原文线索

V2-061975 | MATHDB-325894

Concavity-defect monotonicity conjecture for Riesz energies

摘录 / EXCERPT: Let $v_s(N)$ be the minimal average standardized Riesz pair-energy on $\mathbb{S}^2$, and let ${\cal C}_+(s)$ be the set of integers N at which strict local concavity fails, so t...

conjecture | open | MathDB 页面 | 原文线索

V2-061976 | MATHDB-325895

Five-point discrete-concavity conjecture under candidate optimizer assumptions

摘录 / EXCERPT: Let $v_s(N)$ be the minimal average standardized Riesz pair-energy on $\mathbb{S}^2$, and let $\ddot v_s(N)=v_s(N-1)-2v_s(N)+v_s(N+1)$. Let $s^\dagger=15.04807\ldots$, and assume t...

conjecture | open | MathDB 页面 | 原文线索

V2-061977 | MATHDB-325896

Fundamental asymptotic conjecture for subcritical Riesz energy

摘录 / EXCERPT: For $-2<s<2$ with $s\neq0$, let $u_s(N)$ be the minimal average unstandardized Riesz pair-energy on $\mathbb{S}^2$, let W_s denote the continuum energy, and let C_s be a consta...

conjecture | open | MathDB 页面 | 原文线索

V2-061978 | MATHDB-325897

Logarithmic energy asymptotic conjecture on the sphere

摘录 / EXCERPT: Let $u_0(N)$ be the minimal average logarithmic pair-energy of N points on $\mathbb{S}^2$, let W_{log} be its continuum contribution, and let C_{log} be the...

conjecture | open | MathDB 页面 | 原文线索

V2-061979 | MATHDB-325898

Hypersingular Riesz energy asymptotic conjecture for exponents between two and four

摘录 / EXCERPT: For $s\in(2,4)$, let ${\mathpzc E}_s(\mathbb{S}^2,N)$ be the minimal total Riesz s-energy, let W_s be the analytically continued continuum coefficient, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061980 | MATHDB-325912

Ih-type conjecture for preperiodic points of rational functions

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k containing all its archimedean places, and let $\mathfrak o_{k,S}$ be the subring of S-integers contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-061981 | MATHDB-325920

Polynomiality conjecture for lexicographic Gelfand-Tsetlin pattern counts

摘录 / EXCERPT: Fix partitions ${\lambda}/{\mu}$ and ${\nu}$, let G be an element of $GT_{{\lambda}/{\mu},{\nu}}$, and define ... Here $GT_{{\lambda}/{\mu},{\nu}}$ denotes the set of Gelfand-Tsetl...

conjecture | open | MathDB 页面 | 原文线索

V2-061982 | MATHDB-325921

Polynomiality conjecture for fibres of the Gelfand-Tsetlin pattern map

摘录 / EXCERPT: Fix partitions ${\lambda}/{\mu}$ and ${\nu}$. Let $\psi_k:GT_{k{\lambda}/k{\mu},k{\nu}}\to GT_{{\lambda}/{\mu},{\nu}}$ be the map sending a pattern to the unique lexicographically...

conjecture | open | MathDB 页面 | 原文线索

V2-061983 | MATHDB-325936

Forest embedding conjecture for tournaments

摘录 / EXCERPT: Let T be a tournament on n vertices, and let F be a forest on at most n vertices with c non-trivial components. Forest embedding conjecture. If F has at most ... edges, t...

conjecture | open | MathDB 页面 | 原文线索

V2-061984 | MATHDB-325941

Complex-parameter hypergeometric lemniscate conjecture

摘录 / EXCERPT: Complex-parameter clustering conjecture. The zeros of p_n asymptotically cluster on the loop of the lemniscate

conjecture | open | MathDB 页面 | 原文线索

V2-061985 | MATHDB-325942

Domain and logarithmic-potential description of asymptotic zero measures

摘录 / EXCERPT: Let $varPsi(z)=\operatorname{Max}\{\widetilde H_i(z)\}$, let K be the support of $triangle\varPsi$, and let mu be an asymptotic measure associated with the hypergeometric polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-061986 | MATHDB-325973

The AKSZ-formality conjecture for quasilocal shifted-Frobenius actions

摘录 / EXCERPT: For $d\geq 2$, let $\operatorname{Chains}_{\bullet}(\mathbb R^d)$ be the chains on $\mathbb R^d$, and let $\mathrm h\operatorname{invFrob}_{d}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-061987 | MATHDB-325974

Conjecture B on curvewise bounded ramification and logarithmic conductors

摘录 / EXCERPT: Keep the hypotheses of Conjecture A: k is perfect, X/k is normal, $j:U\hookrightarrow X$ is a k-smooth open immersion with complement supporting an effective Cartier divisor,...

conjecture | open | MathDB 页面 | 原文线索

V2-061988 | MATHDB-325982

Rényi entropy power inequality for generalized Gaussians

摘录 / EXCERPT: Rényi entropy power conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061989 | MATHDB-325983

Sharp two-summand Rényi entropy power inequality constant

摘录 / EXCERPT: Sharp two-summand Rényi entropy power conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-061990 | MATHDB-325988

The converse conjecture for F-support containment and row-overlap dominance

摘录 / EXCERPT: Let A and B be skew shapes of the same size. For each positive integer k, let $\mathrm{rows}_{k}(A)$ and $\mathrm{rows}_{k}(B)$ denote their row-overlap partitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-061991 | MATHDB-325989

Conjecture on elliptic-unit generators of ray class fields

摘录 / EXCERPT: Let K be an imaginary quadratic field, let $\mathfrak f$ be an ideal of $\mathcal O_K$, and let $K_{\mathfrak f}$ be the corresponding ray class field. Under the conditions of th...

conjecture | open | MathDB 页面 | 原文线索

V2-061992 | MATHDB-325995

Milnor-Tjurina conjecture for Q-Gorenstein smoothings

摘录 / EXCERPT: Let $(V,0)$ be a non-Gorenstein surface singularity admitting a $\mathbb Q$-Gorenstein smoothing, meaning a smoothing obtained as an r-cyclic quotient of a smoothing of the index...

conjecture | solved | MathDB 页面 | 原文线索

V2-061993 | MATHDB-325996

Milnor-Tjurina conjecture for non-complete-intersection surface singularities in C4

摘录 / EXCERPT: Let $(V,0)$ be a normal surface singularity in $\mathbb C^4$ that is not a complete intersection. Milnor-Tjurina conjecture in C4. ... with equality if and only if $(V,0)$ is quasi-h...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-061994 | MATHDB-326006

Necessary conditions for linear promotion eigenvalues of non-down-forest posets

摘录 / EXCERPT: Let P be a poset of size n that is not a down forest, and let M be its promotion transition matrix. Write the parameters as $x_1,\dots,x_n$. Necessary-conditions conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-061995 | MATHDB-326012

The proposed sharp threshold for rainbow connection number at least three

摘录 / EXCERPT: Proposed threshold conjecture. The function $p(n)$ is a sharp threshold for the graph property $\mathcal{R}_r$.

conjecture | open | MathDB 页面 | 原文线索

V2-061996 | MATHDB-326030

Characterization of local minima of the 1-Wasserstein population functional

摘录 / EXCERPT: Let $X_1,\ldots,X_m$ be persistence diagrams with only finitely many off-diagonal points. For a persistence diagram Y, define ... For each optimal pairing from $W=\{w_j\}$ to X_i...

conjecture | open | MathDB 页面 | 原文线索

V2-061997 | MATHDB-326034

Hé non-Lane-Emden conjecture for the polyharmonic system

摘录 / EXCERPT: Hé non-Lane-Emden conjecture. If (u,v) is a nonnegative solution of the system and (p,q) is under the critical hyperbola, then $u=v=0$. This is a Liouville-type conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-061998 | MATHDB-326035

Uniform boundedness conjecture for Sierpinski gasket sampling functions

摘录 / EXCERPT: Let ψ_m denote a sampling function on level m of the Sierpinski gasket. Uniform boundedness conjecture. The sampling functions are uniformly bounded by some constant, numeri...

conjecture | open | MathDB 页面 | 原文线索

V2-061999 | MATHDB-326036

L2 decay conjecture for Sierpinski gasket sampling functions

摘录 / EXCERPT: Let ψ_m be a sampling function on level m of the Sierpinski gasket SG. L2 decay conjecture. There is a constant c, numerically around 1.1, such that ... Unlike the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-062000 | MATHDB-326056

Coordinate-free rank-level duality map conjecture for conformal blocks

摘录 / EXCERPT: Let $\mathfrak{g}_1\oplus \mathfrak{g}_2\rightarrow\mathfrak{g}$ be a conformal embedding of Dynkin multi-index (ℓ_1,ℓ_2). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062001 | MATHDB-326066

The irreducible-lattice conjecture for Property FW

摘录 / EXCERPT: Let S be a semisimple connected Lie group with at least two simple factors and no compact factor. An irreducible lattice in S is a lattice satisfying the usual irreducibility c...

conjecture | open | MathDB 页面 | 原文线索

V2-062002 | MATHDB-326067

Hurwitz transitivity conjecture for Singer-cycle factorizations

摘录 / EXCERPT: Let c be a Singer cycle in $GL_n({\mathbb{F}}_q)$, and consider ordered factorizations $c=t_1t_2\cdots t_\ell$ into reflections. The multiset $\{\det(t_i)\}_{i=1}^\ell$ is an inv...

conjecture | open | MathDB 页面 | 原文线索

V2-062003 | MATHDB-326068

Uniform factorization-count hypothesis for regular elliptic elements

摘录 / EXCERPT: Let g be a regular elliptic element of $GL_n({\mathbb{F}}_q)$, and let $t_q(n,\ell)$ denote the number of ordered reflection factorizations of a Singer cycle of length ℓ. Un...

conjecture | open | MathDB 页面 | 原文线索

V2-062004 | MATHDB-326074

Completeness of compressed plane waves

摘录 / EXCERPT: Completeness of CPWs. There exists a constant μ_0 such that the set $\{\mathrm{b}^n_{\mathbf{j}}\}_{n,\mathbf{j}}$ generated from the recursive variational definitions is compl...

conjecture | open | MathDB 页面 | 原文线索

V2-062005 | MATHDB-326082

Nazarov-Shapiro conjecture on weak asymptotic Toeplitzness of composition operators

摘录 / EXCERPT: Let H^2 be the Hardy space, let $\boldsymbol{D}$ be the unit disk, and let C_{φ} denote the composition operator induced by a holomorphic self-map φ of...

conjecture | open | MathDB 页面 | 原文线索

V2-062006 | MATHDB-326085

Uniformity conjecture for period transfer when n is even

摘录 / EXCERPT: Let E/F be the quadratic extension, let $G'=\operatorname{GL}_{2n}$, and let π' be a cuspidal representation of $G'(\mathbf A)$. For each $D\in X(E:F:\pi')$, let π_D be t...

conjecture | open | MathDB 页面 | 原文线索

V2-062007 | MATHDB-326089

Cover-graph incidence-poset doubling conjecture

摘录 / EXCERPT: Let P be a poset, let G be the cover graph of P, and let Q be the incidence poset of G. Cover-graph incidence-poset doubling conjecture. For every $t\ge1$, there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-062008 | MATHDB-326092

Uniformization conjecture for strongly asymptotically log del Pezzo pairs

摘录 / EXCERPT: Uniformization conjecture. The surface S admits Kähler-Einstein edge metrics with angle β along C for all sufficiently small β if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062009 | MATHDB-326093

Small-angle metric degeneration conjecture for classes $(\aleph)$ and $(\beth)$

摘录 / EXCERPT: Small-angle metric degeneration conjecture. As β tends to zero, (S,C,ω_β) converges in an appropriate sense to a generalized Kähler-Einstein metric ω_∞...

conjecture | open | MathDB 页面 | 原文线索

V2-062010 | MATHDB-326099

The non-Gorenstein-locus conjecture for K-orbit closures

摘录 / EXCERPT: Let Y_{γ} be a K-orbit closure, and suppose the maximal components of its singular locus are $\mathcal O_{\alpha_1},\ldots,\mathcal O_{\alpha_k}$. Non-Gorenstein-locus con...

conjecture | open | MathDB 页面 | 原文线索

V2-062011 | MATHDB-326101

Apéry congruence conjecture for the second-kind numbers modulo 9

摘录 / EXCERPT: Second-kind Apéry congruence conjecture. The numbers $A^{(3)}_n$ satisfy the following congruences modulo 9:

conjecture | open | MathDB 页面 | 原文线索

V2-062012 | MATHDB-326103

The genus-zero GV invariants of the mirror to the strange Calabi-Yau threefold

摘录 / EXCERPT: Let $\widetilde{X}$ be the Calabi-Yau threefold considered in the paper, and let n_d denote its genus-zero Gopakumar-Vafa invariants on the mirror side. The preceding mirror-symm...

conjecture | open | MathDB 页面 | 原文线索

V2-062013 | MATHDB-326131

The conjecture that primitive groups synchronize every non-uniform map

摘录 / EXCERPT: Synchronization conjecture. Every primitive permutation group synchronizes every non-uniform map.

conjecture | open | MathDB 页面 | 原文线索

V2-062014 | MATHDB-326150

Wang and Lih's planar graph square-coloring conjecture

摘录 / EXCERPT: Let G be a planar graph, let $g(G)$ denote its girth, let $\operatorname{\Delta(G)}$ denote its maximum degree, and let $\chi^2(G)$ denote the chromatic number of its square. Wan...

conjecture | open | MathDB 页面 | 原文线索

V2-062015 | MATHDB-326155

Reduction conjecture for interval colorings of complete tripartite graphs

摘录 / EXCERPT: Let $K_{k,m,n}$ be the complete tripartite graph with part sizes k,m,n, where $k\leq m\leq n$ and $n>k+m$. Reduction conjecture. ... This is proposed as a generalization of the...

conjecture | open | MathDB 页面 | 原文线索

V2-062016 | MATHDB-326162

Fermat-hypersurface superpotential conjecture

摘录 / EXCERPT: Let V be the subspace spanned by the immersed-point generators $X_1,\ldots,X_{n+2}$, and write a deformation as $b=\sum_{i=1}^{n+2}x_iX_i$. Let W be the mirror superpotential obt...

conjecture | open | MathDB 页面 | 原文线索

V2-062017 | MATHDB-326164

Properties of parametric Presburger families

摘录 / EXCERPT: Let S_t be a parametric Presburger family, meaning a family of subsets of $\mathbb{N}^d$ definable over the natural numbers using quantifiers, boolean operations, and inequalitie...

conjecture | open | MathDB 页面 | 原文线索

V2-062018 | MATHDB-326167

Exponential decay conjecture for the robustness parameter of minimally redundant frames

摘录 / EXCERPT: Let $\mathcal{F}=\{f_j\}_{j=1}^{2n-1}$ be a frame with $\|f_j\|\leq L$ for all $f_j\in\mathcal{F}$, and let ω denote the robustness parameter discussed above. Exponential de...

conjecture | open | MathDB 页面 | 原文线索

V2-062019 | MATHDB-326171

The F-conjecture for the nef cone of the moduli space of stable pointed curves

摘录 / EXCERPT: Let $\overline{\operatorname{M}}_{0,\operatorname{n}}$ be the moduli space of stable n-pointed genus-zero curves. An F-conjecture. A divisor D on...

conjecture | open | MathDB 页面 | 原文线索

V2-062020 | MATHDB-326184

Characterization of inner-product norms by deformation

摘录 / EXCERPT: Let $(X,\|\cdot\|)$ be a complex normed vector space. Define its deformation by ... where $\langle\vec a\mid\vec b\rangle$ is the product induced by the norm through the polarization...

conjecture | open | MathDB 页面 | 原文线索

V2-062021 | MATHDB-326185

Characterization of inner products by phase covariance

摘录 / EXCERPT: Phase-covariance characterization. This identity holds for all $\vec x,\vec y\in X$ if and only if the product $\langle\cdot\mid\cdot\rangle$ is an inner product, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-062022 | MATHDB-326204

The enriched icon construction for symmetric monoidal higher categories

摘录 / EXCERPT: Enriched icon conjecture. For every symmetric monoidal $(n+1)$-category ${\mathcal {E}}$, there is a symmetric monoidal $(n+1)$-category ${\mathcal {E}}\text{-Icon}$ of categories...

conjecture | open | MathDB 页面 | 原文线索

V2-062023 | MATHDB-326205

The iterated icon construction for slightly strict higher categories

摘录 / EXCERPT: Iterated icon conjecture. We define a symmetric monoidal $(n+1)$-category ${\mathcal {E}}_{n,k}$ for each $k \geq 0$ by

conjecture | open | MathDB 页面 | 原文线索

V2-062024 | MATHDB-326206

The iterated weak-monoid construction in a symmetric monoidal higher category

摘录 / EXCERPT: Iterated weak-monoid conjecture. For every symmetric monoidal $(n+1)$-category ${\mathcal {E}}$, there is a symmetric monoidal $(n+1)$-category $\operatorname{Mon}({\mathcal {E}})$...

conjecture | open | MathDB 页面 | 原文线索

V2-062025 | MATHDB-326207

The equivalence between degenerate higher categories and iterated weak monoids

摘录 / EXCERPT: Degenerate-category conjecture. There is an equivalence of $(n+1)$-categories

conjecture | open | MathDB 页面 | 原文线索

V2-062026 | MATHDB-326209

Lyubashenko's decomposition conjecture for the automorphic sheaf

摘录 / EXCERPT: Let X be a smooth projective curve and let E be an irreducible SL_2-local system on X. Assume Lyubashenko's Conjecture 3 holds for the point...

conjecture | open | MathDB 页面 | 原文线索

V2-062027 | MATHDB-326210

Lichtman's conjecture on free subgroups in division rings

摘录 / EXCERPT: Let D be a division ring, and write $D^\times$ for its multiplicative group. A group is noncyclic free if it is a free group that is not cyclic. Lichtman's conjecture. The group...

conjecture | open | MathDB 页面 | 原文线索

V2-062028 | MATHDB-326218

The enemy-clique conjecture for minimally imperfect graphs

摘录 / EXCERPT: Let G be a minimally imperfect graph, and let K_1,K_2 be maximal cliques of G. A pair of enemy cliques is the relation defined in the source. The enemy-clique conjecture. A m...

conjecture | open | MathDB 页面 | 原文线索

V2-062029 | MATHDB-326222

Refined maximum choice number conjecture for graphs of prescribed order and chromatic number

摘录 / EXCERPT: Let $n\geq k\geq 2$. For a graph G, write $\alpha(G)$ for its independence number, $\chi(G)$ for its chromatic number, and $\operatorname{ch}(G)$ for its choice number. Refined m...

conjecture | open | MathDB 页面 | 原文线索

V2-062030 | MATHDB-326223

On-line Ohba conjecture

摘录 / EXCERPT: For a graph G, write $\operatorname{ch}^{\mathrm{OL}}(G)$ for its on-line choice number and $\chi(G)$ for its chromatic number. On-line Ohba conjecture. If ... then ... This is prese...

conjecture | open | MathDB 页面 | 原文线索

V2-062031 | MATHDB-326224

The Delta-squared conjecture for weighted graph labeling

摘录 / EXCERPT: Let (G,H) be a pair of graphs with $H\subset G$. Let $\operatorname{Span}(G,H)$ denote the minimum span of a (G,H)-labeling, and write $\Delta=\Delta(H)$ and $\Delta(G)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-062032 | MATHDB-326228

Positive-measure discontinuity conjecture for the lower spectral radius

摘录 / EXCERPT: Positive-measure discontinuity conjecture. The set of discontinuities of $\check{\varrho}$ has positive Lebesgue measure. This is presented as a weaker version of the resist-impuri...

conjecture | open | MathDB 页面 | 原文线索

V2-062033 | MATHDB-326239

Integral generators conjecture for square-free moonshine groups

摘录 / EXCERPT: Let N be square-free, let $g>0$ be the genus of the moonshine modular curve associated to the group $\Gamma_0(N)^+$, and consider its function field. Integral generators conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-062034 | MATHDB-326240

Universal Segre-SM classes for multi-singularity constructible functions

摘录 / EXCERPT: For a stable map $f:M\to N$, let $M(\underline{\eta})(f)$ be the closure of the locus of tuples of distinct points mapping to a common point, with the map-germ at each point having...

conjecture | open | MathDB 页面 | 原文线索

V2-062035 | MATHDB-326246

Exchange Laurent polynomial conjecture for generated period 1 seeds

摘录 / EXCERPT: Let $P=P(x_1,cdots,x_{n-1})$ be an irreducible polynomial in the class used to generate the seed $\mathbf{(x,P)}$, and let P_0 be the initial exchange polynomial of that seed. A...

conjecture | open | MathDB 页面 | 原文线索

V2-062036 | MATHDB-326247

Reversal-symmetry conjecture for multilinear period 1 polynomials

摘录 / EXCERPT: Let $P=P(x_1,\ldots,x_n)$ be a multilinear polynomial with positive coefficients that generates a period 1 seed. Reversal-symmetry conjecture. ... This predicts reversal symmetry f...

conjecture | open | MathDB 页面 | 原文线索

V2-062037 | MATHDB-326248

Classification conjecture for symmetric positive period 1 polynomials

摘录 / EXCERPT: Let $P=P(x_1,\ldots,x_{n-1})$ be a symmetric polynomial with positive coefficients that generates a period 1 seed. Symmetric-polynomial classification conjecture. The only possib...

conjecture | open | MathDB 页面 | 原文线索

V2-062038 | MATHDB-326256

The ratios conjecture for the second family of elliptic curves

摘录 / EXCERPT: Let ${\mathcal F}(X)$ be the second family of elliptic curves under consideration, with conductor parameter $C(t)$. Define ... and let $A(\alpha,\gamma)$ be the analytic Euler produc...

conjecture | open | MathDB 页面 | 原文线索

V2-062039 | MATHDB-326258

Lück-Sauer-Wegner conjecture on measure-equivalent groups and L2-torsion

摘录 / EXCERPT: Lück-Sauer-Wegner conjecture. Their L^2-torsions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-062040 | MATHDB-326260

Linear type for simplicial complexes with cyclic line graphs and compatible patches

摘录 / EXCERPT: Let Δ be a simplicial complex whose line graph is a cycle of length l. Let $\mathcal{P}=\{p_i^{j_i,k_i}\}_{i=1}^q$ be a system of compatible patches, meaning that each...

conjecture | open | MathDB 页面 | 原文线索

V2-062041 | MATHDB-326276

Linear-growth conjecture for maximal eigenvalues of cryo-EM covariance blocks

摘录 / EXCERPT: For frequency indices k_1 and k_2, let $\hat L^{k_1,k_2}$ denote the corresponding block of the matrix $\hat L$, and let $\lambda_{\max}(\hat L^{k_1,k_2})$ be its maximal eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-062042 | MATHDB-326277

Condition-number conjecture for cryo-EM covariance matrix blocks

摘录 / EXCERPT: For frequency indices k_1 and k_2, let $\hat L^{k_1,k_2}$ be the corresponding block of $\hat L$, and write $\kappa(\hat L^{k_1,k_2})$ for its condition number. Condition-numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-062043 | MATHDB-326282

Four-color conjecture for ISK4-free graphs

摘录 / EXCERPT: A graph is ISK4-free if it has no induced subdivision of K_4. ISK4-free four-color conjecture. Every ISK4-free graph is 4-colorable. The preceding theorem gives a constant bound...

conjecture | open | MathDB 页面 | 原文线索

V2-062044 | MATHDB-326296

The Eulerian conjecture for the augmented wiring-diagram poset

摘录 / EXCERPT: Let W_n be the poset of equivalence classes of lensless wiring diagrams, ordered by breaking crossings, and let $\widehat{0}$ be a new minimal element adjoined to W_n. Eulerian...

conjecture | open | MathDB 页面 | 原文线索

V2-062045 | MATHDB-326305

Log canonicity of the Proj of a finitely generated log canonical ring

摘录 / EXCERPT: The log canonical Proj conjecture. There is an effective $\mathbb Q$-divisor Δ_Y on Y such that (Y,Δ_Y) is log canonical. The result is known when K_Y is...

conjecture | open | MathDB 页面 | 原文线索

V2-062046 | MATHDB-326306

Log canonical base conjecture for lc-trivial fibrations

摘录 / EXCERPT: Log canonical base conjecture. There is an effective $\mathbb R$-divisor Δ_Y on Y such that (Y,Δ_Y) is log canonical and

conjecture | open | MathDB 页面 | 原文线索

V2-062047 | MATHDB-326309

Flip conjecture II

摘录 / EXCERPT: Flip conjecture II. The sequence terminates after finitely many steps; equivalently, there is no infinite sequence of flips. The source explains that the minimal model program can...

conjecture | open | MathDB 页面 | 原文线索

V2-062048 | MATHDB-326310

Dokos-Dwyer-Jelinek-Sagan-Savage conjecture on inversion-Wilf equivalence

摘录 / EXCERPT: Dokos et al.'s conjecture. The sets Π and Π' are inversion-Wilf equivalent if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062049 | MATHDB-326313

Arthur-parameter conjecture for Whittaker-Fourier coefficients

摘录 / EXCERPT: Arthur-parameter conjecture. For every elliptic Arthur parameter ρ of Ramanujan type,

conjecture | open | MathDB 页面 | 原文线索

V2-062050 | MATHDB-326314

Classical-group descent conjecture for Whittaker-Fourier coefficients

摘录 / EXCERPT: Classical-group descent conjecture. If π is an irreducible constituent of $\sigma^{\psi_N}(\{\pi_1,\dots,\pi_k\})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-062051 | MATHDB-326315

Arthur-component-group conjecture for generic cuspidal representations

摘录 / EXCERPT: Arthur-component-group conjecture. For every $\phi\in\Psi^t(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-062052 | MATHDB-326316

Descent multiplicity conjecture for classical and metaplectic groups

摘录 / EXCERPT: Descent multiplicity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062053 | MATHDB-326318

Frankl's lattice conjecture

摘录 / EXCERPT: Frankl's lattice conjecture. There is a join-irreducible element a with

conjecture | open | MathDB 页面 | 原文线索

V2-062054 | MATHDB-326330

The p-harmonic boundary extension conjecture

摘录 / EXCERPT: Let $1 < p \in \mathbb{R}$ and let X be a metric measure space. Let $\Delta_p(X)$ denote the p-harmonic boundary of X, and let $BD^p(X)$ and $HBD^p(X)$ denote the bounded Roy...

conjecture | open | MathDB 页面 | 原文线索

V2-062055 | MATHDB-326331

Idempotence conjecture for strongly productive ultrafilters

摘录 / EXCERPT: Idempotence conjecture. The ultrafilter p is idempotent; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-062056 | MATHDB-326336

Conjecture on the number of linear relations in length two

摘录 / EXCERPT: Let $k\geq 4$. Consider the linear relations among the generators of the weight-k, length-two component...

conjecture | open | MathDB 页面 | 原文线索

V2-062057 | MATHDB-326338

Rigidity conjecture for parabolic type semigroups

摘录 / EXCERPT: Let S and S^* be semigroups generated by functions of the form $a(1-z)^{1+\alpha}+R(z)$. Suppose that, for some $\varepsilon>0$, their remainders satisfy ... Here condition ... hol...

conjecture | open | MathDB 页面 | 原文线索

V2-062058 | MATHDB-326340

Jacobi-Stirling transformation conjecture on preserving log-convexity

摘录 / EXCERPT: A sequence transformation maps a sequence $\{z_n\}_n$ to a sequence $\{w_n\}_n$. For the Jacobi-Stirling numbers of the second kind $\operatorname{JS}(n,k;z)$ and the Jacobi-Stirli...

conjecture | open | MathDB 页面 | 原文线索

V2-062059 | MATHDB-326343

Polynomial-form conjecture for the hypergeometric Laplace transform

摘录 / EXCERPT: Polynomial-form conjecture. This integral is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-062060 | MATHDB-326367

The Erdős-Ginzburg-Ziv bound for finite commutative semigroups

摘录 / EXCERPT: Erdős-Ginzburg-Ziv conjecture for semigroups. For any finite commutative semigroup $\mathcal{S}$,

conjecture | open | MathDB 页面 | 原文线索

V2-062061 | MATHDB-326368

Existence of nontrivial infinite F-sets over finite fields

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field. An F-set is a set $\mathcal{S}\subset\mathbb{F}_q[x]$ such that, whenever $P\in\mathcal{S}$ is monic and irreducible with constant term...

conjecture | open | MathDB 页面 | 原文线索

V2-062062 | MATHDB-326371

The gcd-dependence conjecture for Gold exponent exponential sums

摘录 / EXCERPT: Gcd-dependence conjecture. For any positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-062063 | MATHDB-326372

The Kloosterman-sum identity conjecture for the auxiliary sum K_m'

摘录 / EXCERPT: Kloosterman-sum identity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062064 | MATHDB-326390

Nonexistence of continuous compensation functions for factor maps of subshifts

摘录 / EXCERPT: Let X and Y be subshifts, and let $pi:X \to Y$ be a factor map. A continuous function $f:X\to\mathbb{R}$ is a continuous compensation function for (X,Y,π) if it has the com...

conjecture | open | MathDB 页面 | 原文线索

V2-062065 | MATHDB-326398

Conjectural Wilf-equivalences for vincular patterns of length four

摘录 / EXCERPT: Wilf-equivalence conjecture. The following Wilf-equivalences hold:

conjecture | open | MathDB 页面 | 原文线索

V2-062066 | MATHDB-326399

The wheels-part characterization of the multivariable Alexander polynomial

摘录 / EXCERPT: Wheels-part Alexander conjecture. What remains of the wheels part of Z is precisely the multivariable Alexander polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-062067 | MATHDB-326401

The hypertree generation conjecture for the effective cone of \(\overline{M}_{0,n}\)

摘录 / EXCERPT: Let $\overline{M}_{0,n}$ be the moduli space of stable rational curves with n marked points. A boundary divisor δ_I is associated to a partition of the markings into I a...

conjecture | open | MathDB 页面 | 原文线索

V2-062068 | MATHDB-326405

Free elementary abelian action conjecture for products of odd-dimensional spheres

摘录 / EXCERPT: Free-rank conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062069 | MATHDB-326410

Chen-Wang conjecture on independence polynomials of clique-corona graphs

摘录 / EXCERPT: Let $K_{t,n}$ be the complete bipartite graph with parts of sizes t and n, and let $K_{t,n}\circ K_1$ denote its corona with K_1. Write $I(G;x)$ for the independence polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-062070 | MATHDB-326411

The minimal excluded minors conjecture for graphs with $g(H)\leq 2$

摘录 / EXCERPT: Let $g(H)$ denote the least integer k such that every k-connected H-minor-free graph has bounded pathwidth. Let $\mathcal{H}_2$ be the class of graphs H with $g(H)\leq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062071 | MATHDB-326413

Sarwate et al.'s lower-bound conjecture for binary m-sequence cross correlation

摘录 / EXCERPT: Let $n=2t$ and $p=2$. Define the cross-correlation quantity $\text{S}_d(z)=\text{C}_d(z)+1$. Sarwate et al.'s conjecture. One has ... The paper states that this conjecture is confirm...

conjecture | open | MathDB 页面 | 原文线索

V2-062072 | MATHDB-326415

Density conjecture for weak primary pseudoperfect numbers

摘录 / EXCERPT: Density conjecture. If $\mathfrak{N}_Q$ and $\mathfrak{N}_{Q'}$ are non-empty and $\mathfrak{n}_Q<\mathfrak{n}_{Q'}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-062073 | MATHDB-326419

Wang's 2-factor embedding conjecture for bipartite graphs

摘录 / EXCERPT: Let $G[U,V]$ and H be balanced bipartite graphs on $2n$ vertices. Write δ_U and δ_V for the minimum degrees of vertices in U and V in G, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062074 | MATHDB-326424

Zariski decomposition conjecture for the class gamma on \(\overline{M}_{0,7}^{S_7}\)

摘录 / EXCERPT: Let $\overline{M}_{0,7}^{S_7}$ be the symmetric moduli space appearing in the source, and let S_1,S_2,S_3,D_1,D_2 be the classes used there. Define ... A Zariski decomposition of...

conjecture | open | MathDB 页面 | 原文线索

V2-062075 | MATHDB-326427

Conjecture on the bounded signal property for uncountable SFTs

摘录 / EXCERPT: Let X be a shift of finite type over a multidimensional lattice. The bounded signal property is the property defined in the paper, and a direction means a direction in the lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-062076 | MATHDB-326463

A modified Donagi-Morrison conjecture for linear series on K3 surfaces

摘录 / EXCERPT: Let S be a $K3$ surface, let C be a smooth irreducible curve of genus $g\geq 2$ on S, and let A be a complete, base point free linear series g^r_d on C, with...

conjecture | open | MathDB 页面 | 原文线索

V2-062077 | MATHDB-326466

Forstnerič-Wold conjecture for compact bordered Riemann surfaces

摘录 / EXCERPT: Let F be a compact bordered Riemann surface. Forstnerič-Wold conjecture. The surface F embeds holomorphically in the plane $\mathbb C^2$. The source describes this finitary pro...

conjecture | open | MathDB 页面 | 原文线索

V2-062078 | MATHDB-326467

Proper embedding conjecture for open Riemann surfaces in \(\mathbb C^2\)

摘录 / EXCERPT: Let X be an open Riemann surface. Proper embedding conjecture. The surface X embeds properly in $\mathbb C^2$, equivalently its image is a closed set. The source calls this one...

open_problem | open | MathDB 页面 | 原文线索

V2-062079 | MATHDB-326476

Rigidity conjecture for Lyapunov-exponent maximizers on the two-torus

摘录 / EXCERPT: Let f be a C^2 volume-preserving Anosov diffeomorphism of $\mathbb{T}^2$, and let $\Lambda:\operatorname{Diff}^2_m(\mathbb{T}^2)\to\mathbb{R}$ be the functional defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-062080 | MATHDB-326482

Li-Zinger formula for reduced Gromov-Witten invariants of quintics

摘录 / EXCERPT: Let $Q\subset\mathbb{P}^4$ be a smooth quintic, and let $N_{g,d}(Q)$ denote its genus g, degree d Gromov-Witten invariant. Let $N'_{h,d}(Q)$ be the reduced invariant defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-062081 | MATHDB-326483

Relative derived resolution conjecture for stable-map moduli

摘录 / EXCERPT: Relative derived resolution conjecture. After pullback to $\widetilde{M}_g^k(\mathbb{P}^n,d)$, the derived object $\mathbf{R}\pi_*\mathfrak{f}^*\mathcal{O}_{\mathbb{P}^n}(k)$ becom...

conjecture | open | MathDB 页面 | 原文线索

V2-062082 | MATHDB-326484

Relative resolution conjecture for stable-map moduli

摘录 / EXCERPT: Relative resolution conjecture. The pullback of $\mathbf{R}\pi_*\mathfrak{f}^*\mathcal{O}_{\mathbb{P}^n}(1)$ to the resulting space becomes locally diagonalizable. If $d>2g-2$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-062083 | MATHDB-326488

AP abnormality for Cantor series normal numbers

摘录 / EXCERPT: Let $Q=(q_n)_{n\geq 1}$ be a basic sequence that is infinite in limit and fully divergent of type I. A real number is Q-normal if it belongs to $\mathscr{N}(Q)$, and is AP Q-ab...

conjecture | open | MathDB 页面 | 原文线索

V2-062084 | MATHDB-326489

Small-perturbation solvability conjecture for the Cantor series system

摘录 / EXCERPT: Small-perturbation solvability conjecture. Whenever $\lVert\vec{\epsilon}\rVert$ is sufficiently small, this system has a solution

conjecture | open | MathDB 页面 | 原文线索

V2-062085 | MATHDB-326490

Finite-order separation conjecture for Cantor series

摘录 / EXCERPT: Let $t\geq2$ be an integer and suppose that $\{1,\ldots,t\}=A\mathbin{\dot\cup}B$. Let Q be a basic sequence, and say that it is t-divergent when the relevant divergence condit...

conjecture | open | MathDB 页面 | 原文线索

V2-062086 | MATHDB-326491

Infinite-order separation conjecture for Cantor series

摘录 / EXCERPT: Suppose that $\mathbb{N}=A\mathbin{\dot\cup}B$ is a partition of the natural numbers. Let Q be a basic sequence, and say that it is fully divergent when the relevant divergence c...

conjecture | open | MathDB 页面 | 原文线索

V2-062087 | MATHDB-326492

Hausdorff-dimension separation conjecture for Cantor series

摘录 / EXCERPT: Let Q be a basic sequence. Hausdorff-dimension separation conjecture. If Q is infinite in limit and t-divergent, and $\{1,\ldots,t\}=A\mathbin{\dot\cup}B$, then the set ... has...

conjecture | open | MathDB 页面 | 原文线索

V2-062088 | MATHDB-326500

Main conjecture on Schur-positivity for complementary partition products

摘录 / EXCERPT: Main conjecture. The partition $\bmu$ covers $ u$ if and only if (C1) $ u$ or $ u'$ has the form $\beta+(s^s)+\alpha$, where β and $[?]$ are partitions, if $\beta_1=i$ then...

conjecture | open | MathDB 页面 | 原文线索

V2-062089 | MATHDB-326502

Bressan's mixing cost conjecture for incompressible flows

摘录 / EXCERPT: Let H be the left half of the torus, and let Ψ be the flow generated by an incompressible vector field u. The image of H under the flow Ψ after time T is called...

conjecture | open | MathDB 页面 | 原文线索

V2-062090 | MATHDB-326508

Anand's conjecture on minimizers of the nonlocal strain-gradient plasticity energy

摘录 / EXCERPT: Let ℓ and h be the material length-scale and specimen height, respectively, and let $\mathcal{T}(f)$ be the energy functional ... Let the space of admissible functions, the yi...

conjecture | open | MathDB 页面 | 原文线索

V2-062091 | MATHDB-326512

Rimányi-Szenes conjecture on iterated-residue formulas for double stable Grothendieck polynomials

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\ldots,\lambda_r)$ be an integer sequence, and let $\curly A$ and $\curly B$ be vector bundles. If $g_\lambda(\curly A-\curly B)$ is defined by the iterated...

conjecture | open | MathDB 页面 | 原文线索

V2-062092 | MATHDB-326514

Equality of the minimal and maximal travelling-wave speeds

摘录 / EXCERPT: Let $\underline{c}\leq\overline{c}$ be the speed bounds defined by Theorem. Speed-threshold conjecture. ... The conjecture asserts that the interval of admissible speed thresholds co...

conjecture | open | MathDB 页面 | 原文线索

V2-062093 | MATHDB-326519

Unbounded separation between orbital chromatic and chromatic roots

摘录 / EXCERPT: Unbounded separation conjecture. For any $N>0$, there exists a graph Γ and automorphism group G of Γ for which $OP_{\Gamma,G}(x)$ has a root at least N larger tha...

conjecture | open | MathDB 页面 | 原文线索

V2-062094 | MATHDB-326522

Submultiplicativity conjecture for the flag f-vector coefficients of CW-posets

摘录 / EXCERPT: Let P be a Cohen-Macaulay CW-poset or a Gorenstein poset of rank n, let $\mathcal A_n$ be the indexing family for its flag f-vector coefficients, and let $\alpha_S(P)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062095 | MATHDB-326523

Finiteness of the expected number of segments of Vervaat bridges

摘录 / EXCERPT: Expected-segment finiteness conjecture. The expected number of segments of the Vervaat bridges is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-062096 | MATHDB-326526

Relative rigidity conjecture for post-critically finite regular polynomial endomorphisms

摘录 / EXCERPT: Relative rigidity conjecture. Every such family of post-critically finite maps is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-062097 | MATHDB-326527

Beukers' 5-adic valuation conjecture for Apéry numbers

摘录 / EXCERPT: Beukers' conjecture. The integer 5^α divides $A_1(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062098 | MATHDB-326528

Beukers' valuation conjecture for the second Apéry sequence

摘录 / EXCERPT: Beukers' conjecture. The integer p^α divides $A_2(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062099 | MATHDB-326531

Weak Ichino-Ikeda conjecture for non-tempered Arthur packets

摘录 / EXCERPT: Weak Ichino-Ikeda conjecture. The period $P_{V',U'}$ on $\Pi\boxtimes \overline{\pi}$ is nonzero if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062100 | MATHDB-326532

Weak Ichino-Ikeda conjecture with local Gross-Prasad hypotheses

摘录 / EXCERPT: Let V' and U' be the quadratic spaces defining $SO(V')$ and $SO(U')$, and let $P_{V',U'}$ be the automorphic period. Let $\Pi\boxtimes \overline{\pi}$ be a cuspidal automorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-062101 | MATHDB-326533

Conjectured recursion for the ML degree of Grassmannians

摘录 / EXCERPT: For $n\geq4$, let $\operatorname{Gr}_{2,n}\subseteq\mathbb{P}^{\binom{n}{2}-1}$ denote the Grassmannian of lines in $\mathbb{P}^{n-1}$, with homogeneous coordinates p_{ij}, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-062102 | MATHDB-326536

Ervedoza-Zuazua small-time heat observability cost conjecture

摘录 / EXCERPT: Ervedoza-Zuazua conjecture. For every $L>0$ and every $K>L^2/4$, there exists some $C(K)>0$ such that, for every T sufficiently small,

conjecture | open | MathDB 页面 | 原文线索

V2-062103 | MATHDB-326537

Ervedoza-Zuazua finite-time observability constant conjecture

摘录 / EXCERPT: Let $L>0$ and $T>0$, and let $C_{int}(T,L)$ be a constant for which the finite-time weighted observability estimate ... holds for the relevant solutions of the one-dimensional heat e...

conjecture | open | MathDB 页面 | 原文线索

V2-062104 | MATHDB-326538

Coron-Guerrero uniform controllability conjecture for convection-diffusion equations

摘录 / EXCERPT: Let $M\ne0$, $\varepsilon\in(0,1)$, and consider the transport-diffusion equation on $(0,T)\times(0,L)$, ... with boundary control at $x=0$, homogeneous boundary condition at $x=L$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062105 | MATHDB-326544

Loray's analytic continuation conjecture for holonomy germs

摘录 / EXCERPT: Let $\mathcal{F}$ be a singular holomorphic foliation in $\mathbb{C}\mathbb{P}^2$. Let L_1 and L_2 be two non-invariant projective lines, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062106 | MATHDB-326549

Gaitsgory's dimension-torsor Whittaker equivalence conjecture

摘录 / EXCERPT: Let $\G$ be a loop group, let $\bN$ be its loop maximal unipotent subgroup, and let $\C$ be a category equipped with a $\bG$-action. Let χ be the character of $\bN$ defined fr...

conjecture | open | MathDB 页面 | 原文线索

V2-062107 | MATHDB-326551

Generic G-simplicity conjecture for Neumann eigenvalues in symmetric regions

摘录 / EXCERPT: Let G be a compact subgroup of $O(n)$. A bounded, regular region $\Omega\subset\mathbb{R}^n$ is G-symmetric when it is invariant under the natural action of G. For an eigenva...

conjecture | open | MathDB 页面 | 原文线索

V2-062108 | MATHDB-326556

Injectivity conjecture for Schrödinger-type elliptic growth

摘录 / EXCERPT: Injectivity conjecture. If $D(t)$ is described by both processes, then $u_1=u_2$; equivalently, under the Green-function conditions above, $u_1=u_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-062109 | MATHDB-326557

Two-point determination conjecture for planar permeability

摘录 / EXCERPT: Two-point determination conjecture. If both

conjecture | open | MathDB 页面 | 原文线索

V2-062110 | MATHDB-326562

Generalized Cranston-Kim conjecture for powers of graphs

摘录 / EXCERPT: Let G be a simple connected graph with maximum degree $\Delta\geq 3$. For $k\in\mathbb{N}^*$, let G^k be the graph obtained by joining vertices at distance at most k, and def...

conjecture | open | MathDB 页面 | 原文线索

V2-062111 | MATHDB-326563

Stronger finite-exception conjecture for choosability of graph powers

摘录 / EXCERPT: Stronger finite-exception conjecture. For any $k\in\mathbb{N}^*$, except for a finite number of graphs, the kth power G^k is $(D(k,\Delta)+1-k)$-choosable. This is posed as a s...

conjecture | open | MathDB 页面 | 原文线索

V2-062112 | MATHDB-326564

Quasi-projectivity conjecture for the moduli space of semistable pairs

摘录 / EXCERPT: Let ${\mathbf V}$ and ${\mathbf W}$ be representations of a group ${\mathbf G}$. For a semistable pair...

conjecture | open | MathDB 页面 | 原文线索

V2-062113 | MATHDB-326569

Nonabelian mixed Hodge structure conjecture for Brill-Noether stacks

摘录 / EXCERPT: Nonabelian mixed Hodge structure conjecture. There is a non-abelian mixed Hodge structure on

conjecture | open | MathDB 页面 | 原文线索

V2-062114 | MATHDB-326575

Macroscopic relaxation conjecture for lozenge Glauber dynamics

摘录 / EXCERPT: Let U, φ and their discretizations $(U_L,\varphi_L)_{L\ge1}$ be as above, and let h_t^η denote the height function under Glauber dynamics started from an arbitrary i...

conjecture | open | MathDB 页面 | 原文线索

V2-062115 | MATHDB-326582

Hindry-Pacheco conjecture for elliptic curves over function fields

摘录 / EXCERPT: Let K be a fixed function field and let E_i be a family of pairwise non-isomorphic elliptic curves over K. For each E_i, let $c_{E_i/K}$ be defined by the leading term of i...

conjecture | open | MathDB 页面 | 原文线索

V2-062116 | MATHDB-326583

Kunyavskii-Tsfasman conjecture for elliptic curves under base change

摘录 / EXCERPT: Let X be the curve with function field K, and let $\{E_i/K_i\}$ be a family of elliptic curves obtained by base change. Write g_{K_i} for the genus of the function field...

conjecture | open | MathDB 页面 | 原文线索

V2-062117 | MATHDB-326584

Unified Brauer-Siegel conjecture for elliptic curves over function fields

摘录 / EXCERPT: Let $\{E_i/K_i\}$ be an asymptotically very exact family of elliptic curves over function fields. Let c_{E_i/K_i} be the leading coefficient of the L-function at $s=1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062118 | MATHDB-326585

Monotonicity conjecture for the relative Banzhaf power index

摘录 / EXCERPT: Let n_1,n_2 be the numbers of players of the two types, and let $Bz_2(a,b,n_1,n_2)$ denote the relative Banzhaf power index of a player of type 2 in the corresponding complete...

conjecture | open | MathDB 页面 | 原文线索

V2-062119 | MATHDB-326586

Caputo's shuffle spectral gap conjecture

摘录 / EXCERPT: Let n be a positive integer with $n\ge 3$. For each $A\subset\{1,\ldots,n\}$, let $S_{n,A}$ be the set of permutations in S_n fixing every element outside A, and define ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-062120 | MATHDB-326587

Loughran's conjecture for counting points orthogonal to a Brauer subgroup

摘录 / EXCERPT: Loughran's conjecture. There exist an open subset $U\subset X$ with $\mathscr{B}\subset\operatorname{Br}U$ and a constant $c_{U,\mathscr{B},H}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062121 | MATHDB-326591

Strong base-change conjecture for unitary groups

摘录 / EXCERPT: Let E be a CM extension of a totally real field F, and let U_n be the unitary group specified in the source. Let BC denote local base change. Strong base-cha...

conjecture | open | MathDB 页面 | 原文线索

V2-062122 | MATHDB-326620

Kahn-Kalai large-subcube conjecture for monotone cube sets

摘录 / EXCERPT: Kahn-Kalai conjecture. For every $L>0$, there exist $L'>0$ and $\delta>0$ such that if

conjecture | open | MathDB 页面 | 原文线索

V2-062123 | MATHDB-326622

Derived automorphism group conjecture for K3 surfaces

摘录 / EXCERPT: Derived automorphism group conjecture. The group $\operatorname{Aut} D(X)$ preserves the connected component $\operatorname{Stab}^\dagger(X)$. Moreover,...

conjecture | open | MathDB 页面 | 原文线索

V2-062124 | MATHDB-326623

Positivity conjecture for the Saxl polynomials

摘录 / EXCERPT: Let d be a positive integer, let $\ol\nu\vdash d$, and let $s_{\ol\nu}$ be the polynomial map associated with the staircase Kronecker-square coefficients. Let $c(\ol\nu)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062125 | MATHDB-326624

Alternating-sign conjecture for the Saxl polynomials

摘录 / EXCERPT: Let d be a positive integer, let $\ol\nu\vdash d$, and let $s_{\ol\nu}(x)$ be the polynomial map associated with the staircase Kronecker-square coefficients. Let $c(\ol\nu)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-062126 | MATHDB-326630

Fixed-type Beauville structure conjecture for classical groups

摘录 / EXCERPT: Let $c\tau_1=(r_1,s_1,t_1)$ and $c\tau_2=(r_2,s_2,t_2)$ be two hyperbolic types. Let G be a finite simple classical group of Lie type, with Lie rank sufficiently large. Fixed-typ...

conjecture | open | MathDB 页面 | 原文线索

V2-062127 | MATHDB-326631

Motzkin numbers nonvanishing modulo the squares of 31, 37, and 61

摘录 / EXCERPT: Nonvanishing conjecture. For all $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062128 | MATHDB-326633

The LGL stretching property conjecture

摘录 / EXCERPT: Let $[a,b]$ be an interval, let $L=L_{[a,b]}:[a,(a+b)/2]\to[a,b]$ be given by $L(x)=2x-a$, and let $\{{\cal G}^{\mathrm{LGL}}_N\}_{N\in\mathbb{N}}$ be the family of Legendre-Gauss-...

conjecture | open | MathDB 页面 | 原文线索

V2-062129 | MATHDB-326635

The monotonicity conjecture for quotients of squared Bessel zeros

摘录 / EXCERPT: The Bessel-zero quotient monotonicity conjecture. The quotients $\hat q_k$ decrease monotonically as $k\in\mathbb{N}$ increases. The conjecture is supported by numerical experiment...

conjecture | open | MathDB 页面 | 原文线索

V2-062130 | MATHDB-326636

M/G/1 linear-asymptote conjecture under work-conserving policies

摘录 / EXCERPT: M/G/1 linear-asymptote conjecture. For any finite-third-moment service-time distribution and any work-conserving non-pre-emptive policy, there exists a finite $\overline{b}_\theta$...

conjecture | open | MathDB 页面 | 原文线索

V2-062131 | MATHDB-326641

Conjectured 9-bound for 5-circuits in 2-factors of 3-edge-connected cubic graphs

摘录 / EXCERPT: Let G be a 3-edge-connected cubic graph on n vertices, with G not isomorphic to the Petersen graph. A 2-factor is a spanning subgraph whose components are circuits. 9-bou...

conjecture | open | MathDB 页面 | 原文线索

V2-062132 | MATHDB-326646

Equidistribution criterion for polynomial sequences over function fields

摘录 / EXCERPT: Let p be the characteristic of the finite field ${\mathbb F_q}$, let ${\mathbb K}_\infty$ be the associated completion, and let ${\mathbb T}$ be the relevant torus. Let ... be a po...

conjecture | open | MathDB 页面 | 原文线索

V2-062133 | MATHDB-326647

Polynomial van der Corput conjecture over function fields

摘录 / EXCERPT: Let ${\mathbb F_q}[t]$ be the polynomial ring over the finite field ${\mathbb F_q}$. For a set ${\mathcal H}\subset {\mathbb F_q}[t]\setminus\{0\}$, being van der Corput means that...

conjecture | open | MathDB 页面 | 原文线索

V2-062134 | MATHDB-326662

Cranston-Kim conjecture on list coloring squares of graphs

摘录 / EXCERPT: Cranston-Kim conjecture. If G is not a Moore graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-062135 | MATHDB-326663

Finite-order injectivity conjecture for inverse semigroups

摘录 / EXCERPT: Finite-order injectivity conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-062136 | MATHDB-326664

Inversion conjecture for uniquely 2-divisible completely regular semigroups

摘录 / EXCERPT: Inversion conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-062137 | MATHDB-326672

Nonconcordance conjecture for iterated Whitehead doubles of the trefoil

摘录 / EXCERPT: Let $T_{2,3}$ be the $(2,3)$ torus knot, and let $D(K)$ denote its positively-clasped untwisted Whitehead double. Two knots are smoothly concordant when their connected sum with on...

conjecture | open | MathDB 页面 | 原文线索

V2-062138 | MATHDB-326685

Necessity of the polarization-preservation condition for $I[S](P)$

摘录 / EXCERPT: Necessity conjecture for preservation of IS. If IS is preserved by the polarization process, then such a subspace V_S must exist. The condition is presented as necessary for...

conjecture | open | MathDB 页面 | 原文线索

V2-062139 | MATHDB-326692

The universal dynamics monoid conjecture for finite directed acyclic graphs

摘录 / EXCERPT: Let Γ be a finite directed acyclic graph. Its universal dynamics monoid $D(\Gamma)$ is the smallest quotient of $\operatorname{\mathbf{HK}}_\Gamma$ through which all evaluat...

conjecture | open | MathDB 页面 | 原文线索

V2-062140 | MATHDB-326693

The initial-object conjecture for pointed update systems on path graphs

摘录 / EXCERPT: Let Γ_n be the graph considered in the paper, and let $\mathcal{S}_n^\star$ be the distinguished update system supported on Γ_n, with preferred state $\star$. A point...

conjecture | open | MathDB 页面 | 原文线索

V2-062141 | MATHDB-326696

Nonnegativity conjecture for the spt-crank of overpartitions

摘录 / EXCERPT: Let $N_{\text{S2}}(m,n)$ denote the coefficient associated with the spt-crank generating function $\text{S2}(z,q)$, where m is the crank value and n is the coefficient index. N...

conjecture | open | MathDB 页面 | 原文线索

V2-062142 | MATHDB-326704

Loebl's equivalence conjecture for the q-dichromate and U-polynomial

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let q be a non-negative-integer variable, and let x,y be real variables. Define the q-dichromate by ... where $\operatorname{\mathcal{C}}(A)$ is the s...

conjecture | open | MathDB 页面 | 原文线索

V2-062143 | MATHDB-326709

The isolation-or-local-identity conjecture for periodic points in positive characteristic

摘录 / EXCERPT: Isolation-or-local-identity conjecture. Every periodic point whose multiplier is a root of unity is either isolated as a periodic point, or has a neighborhood on which an iterate o...

conjecture | open | MathDB 页面 | 原文线索

V2-062144 | MATHDB-326723

Rigid curves conjecture

摘录 / EXCERPT: Let g,d,r be integers with $g-d+r\geq 2$, and let $\mathcal{W}^r_{d,g}$ denote the moduli space of smooth genus g curves with a degree d line bundle having at least $r+1$ ind...

conjecture | open | MathDB 页面 | 原文线索

V2-062145 | MATHDB-326739

The real-part bound conjecture for non-real eigenvalues of ${\cal A}_{m;c}$

摘录 / EXCERPT: Let m be a positive integer and let ${\cal A}_{m;c}$ denote the matrix pencil defined in the paper, with $c>0$. An eigenvalue is non-real when $\lambda\notin\mathbb{R}$. Real-par...

conjecture | open | MathDB 页面 | 原文线索

V2-062146 | MATHDB-326740

Compatibility of Ray-Singer metrics under the spectral-sequence identification

摘录 / EXCERPT: Compatibility conjecture. Under this identification of determinant lines, the Ray-Singer metric of the Gauss-Manin connection should map to the Ray-Singer metric of the original co...

conjecture | open | MathDB 页面 | 原文线索

V2-062147 | MATHDB-326741

Kahn and Kalai's dense-subcube conjecture

摘录 / EXCERPT: Let $A\subset\{0,1\}^n$ be monotone increasing, with measure $\alpha=|A|/2^n\leq 1/2$. Its edge-boundary is denoted by ∂A. Kahn and Kalai's conjecture. For every $L>0$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-062148 | MATHDB-326742

The ell^1-grid junta-size conjecture independent of the codomain size

摘录 / EXCERPT: Let $f:\mathbb{Z}_k^n\to\mathbb{Z}_l$ satisfy ... Here $|\cdot|'$ is the source's normalized difference notation. Junta-size conjecture. The function f is ϵ-close to an...

conjecture | open | MathDB 页面 | 原文线索

V2-062149 | MATHDB-326743

The coordinatewise junta approximation conjecture for Lipschitz maps between discrete tori

摘录 / EXCERPT: Let $f=(f_1,\ldots,f_m):\mathbb{Z}_k^n\to\mathbb{Z}_l^m$ be α-Lipschitz with respect to the L^1 norm. Coordinatewise junta conjecture. For every $\delta,\epsilon>0$, at le...

conjecture | open | MathDB 页面 | 原文线索

V2-062150 | MATHDB-326750

Universal building conjecture for spectral covers

摘录 / EXCERPT: Let X be a Riemann surface and let ϕ be a smooth spectral cover of X. A universal harmonic ϕ-map is a harmonic ϕ-map $h^{\phi}:\widetilde X\to\mathcal B^{\phi}$...

conjecture | open | MathDB 页面 | 原文线索

V2-062151 | MATHDB-326752

Universal-building category conjecture for semisimple Hitchin systems

摘录 / EXCERPT: Let G be a semisimple complex Lie group, let X be a Riemann surface, and let ... be a point in the corresponding Hitchin base. Let h^{ϕ} be the harmonic map to the universal...

conjecture | open | MathDB 页面 | 原文线索

V2-062152 | MATHDB-326753

Stability-condition component conjecture for Hitchin categories

摘录 / EXCERPT: Let G be a semisimple complex Lie group, let X be a Riemann surface, let ... be the 3-dimensional Calabi--Yau category associated to the singularities of the harmonic map to th...

conjecture | open | MathDB 页面 | 原文线索

V2-062153 | MATHDB-326754

Stokes wall-crossing conjecture for Hitchin stability conditions

摘录 / EXCERPT: Let G be a semisimple complex Lie group, let X be a Riemann surface, let $\mathcal C$ be the associated 3-dimensional Calabi--Yau category, and consider the component of its...

conjecture | open | MathDB 页面 | 原文线索

V2-062154 | MATHDB-326756

Inversion of adjunction for rational pairs

摘录 / EXCERPT: Let (X,D) be a pair with D a reduced Weil divisor. Let H be a Cartier divisor on X with no components in common with D. Assume that $(H,D\cap H)$ is Du Bois and that...

conjecture | open | MathDB 页面 | 原文线索

V2-062155 | MATHDB-326757

The positive-coefficient numerator conjecture for Carlitz-Riordan polynomial interpolation

摘录 / EXCERPT: Positive-coefficient numerator conjecture. One can write

conjecture | open | MathDB 页面 | 原文线索

V2-062156 | MATHDB-326759

The affinoid-analytic equality for multivariate Robba rings

摘录 / EXCERPT: Let G act continuously on $\tilde{\mathbf{A}}^{\dagger,r}_{L_\infty}$, and let $\tilde{\mathbf{A}}^{\dagger,r,\operatorname{aff}}_{K_\infty}$ be the subring of elements admitting...

conjecture | open | MathDB 页面 | 原文线索

V2-062157 | MATHDB-326762

TASEP-limit conjecture for the multi-state ASEP

摘录 / EXCERPT: Let $M_{i,i+1}^{(\ell)}$ be the multi-state ASEP update matrix formed from the fused generators $M_{i,i+1}^{(\ell;r)}$. TASEP-limit conjecture. The reparametrization ... gives a prop...

conjecture | open | MathDB 页面 | 原文线索

V2-062158 | MATHDB-326763

Heuristic least-nonresidue bound from a quadratic twist

摘录 / EXCERPT: Let ξ be an odd primitive Dirichlet character modulo k with $3\nmid k$, and define ... Let n_ξ denote the least positive integer n with $\xi(n)\neq 1$, and let $M(\chi)$...

conjecture | open | MathDB 页面 | 原文线索

V2-062159 | MATHDB-326781

Sharpness of rigidity estimates for pairs of quasiregular mappings

摘录 / EXCERPT: Sharpness conjecture. There exists a sequence of positive numbers $\epsilon_k\rightarrow 0$ and a sequence of pairs of Q-quasiregular maps $u_k:B_1\rightarrow \mathbb{R}^2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062160 | MATHDB-326783

Weak-tiling conjecture for SRS tiles

摘录 / EXCERPT: SRS weak-tiling conjecture. The collection

conjecture | open | MathDB 页面 | 原文线索

V2-062161 | MATHDB-326784

Zero-boundary-measure conjecture for SRS tiles

摘录 / EXCERPT: SRS tile boundary conjecture. For every $\mathbf{x}\in\mathbb{Z}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-062162 | MATHDB-326795

The conjecture on unicentroidal trees with edge-reconstruction number 3

摘录 / EXCERPT: The unicentroidal edge-reconstruction conjecture. The only infinite classes of unicentroidal trees which have $\text{ern}=3$ are the paths on an odd number of vertices, the caterpi...

conjecture | open | MathDB 页面 | 原文线索

V2-062163 | MATHDB-326803

Gain-tightness characterization for symmetric sphere frameworks with horizontal or improper rotational symmetry

摘录 / EXCERPT: Let $\mathcal{S}$ denote the sphere. For a S-symmetric framework, let S be either the group $\mathcal{C}_{mh}$ with m even or the group $\mathcal{S}_{2m}$ with m odd. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062164 | MATHDB-326804

Gain-tightness characterization for half-turn symmetric frameworks on the cylinder

摘录 / EXCERPT: Let $\mathcal{Y}$ denote the cylinder, and let S be the cyclic group $\mathcal{C}_2$ representing a two-fold rotation about an axis perpendicular to the z-axis. Let (G,p) be...

conjecture | open | MathDB 页面 | 原文线索

V2-062165 | MATHDB-326805

Gain-tightness characterization for rotational and reflectional cylinder symmetries

摘录 / EXCERPT: Let $\mathcal{Y}$ denote the cylinder. Let S be one of the groups $\mathcal{C}_{mh}$, $\mathcal{C}_{mv}$ or $\mathcal{S}_{2m}$, with the rotational axis equal to the cylinder axi...

conjecture | open | MathDB 页面 | 原文线索

V2-062166 | MATHDB-326806

Gain-tightness characterization for symmetric frameworks on the cone

摘录 / EXCERPT: Let $\mathcal{C}$ denote the cone. Let S be either the group $\mathcal{C}_2$ representing a two-fold rotation about an axis perpendicular to the z-axis, or the group...

conjecture | open | MathDB 页面 | 原文线索

V2-062167 | MATHDB-326807

Generic and symmetry-forced isostaticity on the cylinder

摘录 / EXCERPT: Let $\mathcal{Y}$ denote the cylinder. Let S be either the group $\mathcal{C}_s$ or the group $\mathcal{C}_i$, and let (G,p) be an S-generic framework on $\mathcal{Y}$. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-062168 | MATHDB-326817

Symmetric nonnegative matrix rank-one perturbation D-stability conjecture

摘录 / EXCERPT: Let $H\foral R^{n\times n}$ be symmetric and entrywise nonnegative, with geometrically simple spectral-radius eigenvalue $\rho(H)$. Let $v,w\in\mathbb R^n$ satisfy $v>0$ and $w>0$....

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-062169 | MATHDB-326818

Asymptotic positive-stability conjecture for the matrix family Gamma(t)

摘录 / EXCERPT: Let $\Gamma(t)$ be the matrix defined in the paper's Lemma. Asymptotic positive-stability conjecture. There exists $t_1>0$ such that, for every $t>t_1$, the matrix $\Gamma(t)$ is s...

conjecture | open | MathDB 页面 | 原文线索

V2-062170 | MATHDB-326822

KPZ line ensemble convergence conjecture to the Airy line ensemble

摘录 / EXCERPT: KPZ line ensemble convergence conjecture. As $t\to\infty$, the line ensemble $\{\mathcal{L}^{t}_n(x)\}$ converges, in the sense of line ensembles, to the Airy line ensemble...

conjecture | open | MathDB 页面 | 原文线索

V2-062171 | MATHDB-326830

GGC conjecture for negative powers of Beta variables

摘录 / EXCERPT: Let $\operatorname{Beta}(\alpha_1,\alpha_2)$ denote a Beta random variable with parameters $\alpha_1,\alpha_2>0$, and let $s\ge1$. A positive random variable is generalized gamma c...

conjecture | open | MathDB 页面 | 原文线索

V2-062172 | MATHDB-326831

GGC conjecture for large powers of positive stable variables

摘录 / EXCERPT: Let Z_α be a positive α-stable random variable with $\alpha\in(0,1)$, and let $\gamma\in\mathbb{R}$. A positive random variable is generalized gamma convolution (GGC)...

conjecture | open | MathDB 页面 | 原文线索

V2-062173 | MATHDB-326846

Universal weak-disorder asymptotics for directed-polymer free energy

摘录 / EXCERPT: Let $\beta>0$ denote the disorder strength, let $F(\beta)$ be the directed-polymer free energy, and let $\boldsymbol F(1)$ be the corresponding continuum free energy at unit coupli...

conjecture | open | MathDB 页面 | 原文线索

V2-062174 | MATHDB-326847

Universal critical-curve asymptotics for disordered pinning models

摘录 / EXCERPT: Let $h_c(\beta)$ be the critical curve of a disordered pinning model, and let $\boldsymbol h_c^{(\alpha)}(1)$ be the critical curve of the continuum model at unit coupling. For a s...

conjecture | open | MathDB 页面 | 原文线索

V2-062175 | MATHDB-326848

Fractional Hadwiger-number colouring conjecture

摘录 / EXCERPT: Fractional Hadwiger-number colouring conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-062176 | MATHDB-326849

Fractional chromatic versus fractional Hadwiger-number conjecture

摘录 / EXCERPT: Fractional chromatic versus fractional Hadwiger-number conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-062177 | MATHDB-326854

The congruence-kernel identification for free metabelian groups

摘录 / EXCERPT: Let Φ_r be the free metabelian group on two generators, and let $\Phi_{r,n}$ denote the corresponding finite quotient modulo the relevant power-n congruence subgroup. Using...

conjecture | open | MathDB 页面 | 原文线索

V2-062178 | MATHDB-326856

The Nase conjecture on triple cumulative edge bounds in simple graph drawings

摘录 / EXCERPT: Let D be a simple drawing of a graph, and let $E_{{\le}{\le}{\le}k}(D)$ denote the number of its $\le\!\le\!\le k$-edges, defined by ... Here $E_i(D)$ denotes the number of i-edg...

conjecture | open | MathDB 页面 | 原文线索

V2-062179 | MATHDB-326862

The tilting-bundle rationality conjecture for smooth projective surfaces

摘录 / EXCERPT: Let X be a smooth projective surface. A tilting bundle on X is a vector bundle $\cE$ such that $\cExt^i(\cE,\cE)=0$ for $i>0$, and the zero sheaf is the only sheaf $\cF$ satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-062180 | MATHDB-326868

Takahashi's logarithmic mirror symmetry conjecture for plane curves

摘录 / EXCERPT: Let $E\subset\mathbb P^{2}$ be an elliptic curve with a chosen group structure whose zero element is a flex point, and let $P\in E$ be a point of order $3d$. Let m_{d} be the num...

conjecture | open | MathDB 页面 | 原文线索

V2-062181 | MATHDB-326874

$(a+1)$-positivity conjecture for quarter-plane loop series

摘录 / EXCERPT: Let $Q(a,u)$ be the generating series for quarter-plane loops. $(a+1)$-positivity conjecture. The series $Q(a,u)$ can be expanded as ... where $P_n(x)\in\mathbb{N}[x]$. Thus, the coe...

conjecture | open | MathDB 页面 | 原文线索

V2-062182 | MATHDB-326875

Conjectured radius of convergence for the quarter-plane loop series

摘录 / EXCERPT: Let $Q(a,u)$ be the generating series for quarter-plane loops, and write $Q(a,\mathord\cdot)$ for the resulting univariate series in u. For $a\ge -1$, let $\rho_Q(a)$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-062183 | MATHDB-326876

Refined $(a+1)$-positivity conjecture for quarter-plane loop series

摘录 / EXCERPT: Let $Q(a,s,u)$ be the generating function for quarter-plane loops, refined by the variable s, and let $Q^{\bullet}(a,s,u)$ be the corresponding generating function counting primi...

conjecture | open | MathDB 页面 | 原文线索

V2-062184 | MATHDB-326877

The totally ramified extension conjecture for potentially good reduction

摘录 / EXCERPT: Totally ramified extension conjecture. If ϕ has potentially good reduction, then there is a totally ramified extension L/K with $[L:K]\leq B$ such that ϕ is conjugate ove...

conjecture | open | MathDB 页面 | 原文线索

V2-062185 | MATHDB-326878

The totally ramified point conjecture for disks

摘录 / EXCERPT: Totally ramified point conjecture. There is a point α in the disk V such that $K(\alpha)/K$ is totally ramified and

conjecture | open | MathDB 页面 | 原文线索

V2-062186 | MATHDB-326885

Conjecture on Dickson-polynomial auxiliary functions for CPP exponents

摘录 / EXCERPT: Dickson-polynomial conjecture. There exist $a\in\mathbb{F}_{p^{rk}}^*$ such that $h_a(x)$ are Dickson polynomials of degree $r+1$ over $\mathbb{F}_{p^k}$.

conjecture | open | MathDB 页面 | 原文线索

V2-062187 | MATHDB-326887

Tuple restrictions for zeta-like multizeta values

摘录 / EXCERPT: Let $(s_1,\dotsc,s_r)$ be a tuple of positive integers, and call it zeta-like when its associated multizeta value is zeta-like; call a tuple eulerian when its associated multizeta...

conjecture | open | MathDB 页面 | 原文线索

V2-062188 | MATHDB-326888

Splicing conjecture for zeta-like tuples when q=2

摘录 / EXCERPT: Splicing conjecture. If this total weight is a power of 2 or one less than a power of 2, then $(s_1,\dotsc,s_r)$ is zeta-like, except when the two tuples being spliced are...

conjecture | open | MathDB 页面 | 原文线索

V2-062189 | MATHDB-326889

Weight restrictions for eulerian and zeta-like multizeta values

摘录 / EXCERPT: Let p be the characteristic, let q be the size of the finite field, and let a tuple have depth r. Its weight is the sum of its components; call a tuple primitive when it is n...

conjecture | open | MathDB 页面 | 原文线索

V2-062190 | MATHDB-326891

Classification of primitive depth-two zeta-like tuples for q=2

摘录 / EXCERPT: Let $q=2$, and consider primitive tuples of depth 2. Depth-two classification conjecture for q=2. The zeta-like, equivalently eulerian, primitive tuples are exactly ... The source...

conjecture | open | MathDB 页面 | 原文线索

V2-062191 | MATHDB-326892

Existence and uniqueness of components for p-local compact groups

摘录 / EXCERPT: Component uniqueness conjecture. Every p-local compact group determines a unique irreducible component and a unique transporter system of components.

conjecture | open | MathDB 页面 | 原文线索

V2-062192 | MATHDB-326899

Exponential analyticity-shrinkage conjecture for small perturbations of AdS3

摘录 / EXCERPT: Let AdS_3 denote three-dimensional Anti-de Sitter space, and consider small smooth perturbations whose spatial dependence admits a radius of analyticity $\rho(t)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062193 | MATHDB-326901

Conjecture on the curves computing very general quasimonomial valuations

摘录 / EXCERPT: Let $t\in\mathbb{R}$, and let $\widehat\mu(t)$ be the invariant of a very general quasimonomial valuation; let $\mu_C(t)$ denote the quantity associated with a curve C, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062194 | MATHDB-326903

The virtually abelian image conjecture for unitary braided vector spaces

摘录 / EXCERPT: Virtually abelian image conjecture. For every n, the image $\rho^c(\mathcal{B}_n)$ is virtually abelian; moreover, if c has finite order, then $\rho^c(\mathcal{B}_n)$ is finite...

conjecture | open | MathDB 页面 | 原文线索

V2-062195 | MATHDB-326912

Rational minimum-rank conjecture for sign patterns with at most three zeros per column

摘录 / EXCERPT: Let ${\cal A}$ be a sign pattern, and let $mr({\cal A})$ denote its minimum rank over the reals while $mr_{\mathbb Q}({\cal A})$ denotes its minimum rank over the rationals. Ration...

conjecture | open | MathDB 页面 | 原文线索

V2-062196 | MATHDB-326913

Rigid special presentation conjecture for higher Brill-Noether loci

摘录 / EXCERPT: Rigid special presentation conjecture. There exists an irreducible, regular component $\mathcal B\subseteq\widetilde{B_C^{k_i}}(d)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062197 | MATHDB-326919

The unrestricted validity conjecture for Theorem a-f

摘录 / EXCERPT: Let n be a positive integer, let a,b,k be arbitrary nonnegative integers, and let $m,n_0\leq n\leq m+n_0$. Theorem a-f asserts a specific constant-term product formula for...

conjecture | open | MathDB 页面 | 原文线索

V2-062198 | MATHDB-326925

The explicit formula for the lowest spherical function in the second family

摘录 / EXCERPT: Explicit lowest-spherical-function conjecture. For all $n\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-062199 | MATHDB-326932

The cyclic-type conjecture for finite two-point homogeneous quandles

摘录 / EXCERPT: Cyclic-type conjecture. X is of cyclic type.

conjecture | open | MathDB 页面 | 原文线索

V2-062200 | MATHDB-326957

Amdeberhan's divisibility conjecture for binomial coefficients

摘录 / EXCERPT: Let a,b and n be positive integers with $a>b$. Amdeberhan's divisibility conjecture. ... This is a generalization of a theorem in the paper, found by T. Amdeberhan in personal co...

conjecture | open | MathDB 页面 | 原文线索

V2-062201 | MATHDB-326958

Unimodality conjecture for the q-factorial quotient

摘录 / EXCERPT: For $n\geqslant 2$, define ... where $[m]!=(1-q)(1-q^2)\cdots(1-q^m)$. Unimodality conjecture. The polynomial $P_n(q)$ is unimodal. The polynomial is known to be reciprocal, and the...

conjecture | open | MathDB 页面 | 原文线索

V2-062202 | MATHDB-326972

Conjecture on Voronoi polytopes of root lattices for polyhedral norms

摘录 / EXCERPT: Let A_n and D_n be the root lattices in their natural embeddings, and consider both the L^1 and L^{∞} polyhedral norms. Write $V_{<}(N_{\mathcal L})$...

conjecture | open | MathDB 页面 | 原文线索

V2-062203 | MATHDB-326974

The strengthened local epsilon-factor conjecture for metaplectic representations

摘录 / EXCERPT: Let F be a local field of characteristic 0, let $\mathbb{M}$ denote the relevant metaplectic group, and let c_π be the proportionality constant between the two local functi...

conjecture | open | MathDB 页面 | 原文线索

V2-062204 | MATHDB-326975

The entropy lower-bound conjecture for graphs with a given degree sequence

摘录 / EXCERPT: Let D be a strict graphic sequence of type ε, and let $H_{1}(x)$ be the entropy function whose maximum is attained at $\widetilde{\mathbf{p}}$. Write $\mathbb{G}^{D}$...

conjecture | open | MathDB 页面 | 原文线索

V2-062205 | MATHDB-326976

The weighted tree-indicator approximation conjecture for uniform graphs with a given degree sequence

摘录 / EXCERPT: Let D be a strict graphic sequence of type ε. Let $\mathbb{G}^{D}$ be the set of graphs with degree sequence D, and let $\mathcal{G}_{\mathbf{g}}(D)$ be uniformly d...

conjecture | open | MathDB 页面 | 原文线索

V2-062206 | MATHDB-326977

The logarithmic bound conjecture for the degree-sequence parameters

摘录 / EXCERPT: Let D be a strict graphic sequence, and for each $1\leq i\leq n$ let r_i be the variable defined in the source. Logarithmic bound conjecture. There is a constant $c>0$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-062207 | MATHDB-326979

The lower-density conjecture for complex zero strip decreasing operators

摘录 / EXCERPT: Lower-density conjecture. There exists a positive constant c_{ϕ} such that, for any $f(z) \in \operatorname{\mathcal{LP}}(r)$ with $r>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062208 | MATHDB-326980

Classification conjecture for differential operators that are complex zero strip decreasing

摘录 / EXCERPT: Classification conjecture. The operator $\phi(D)$ is a complex zero strip decreasing operator if and only if at least one of the following conditions holds: (i) $\beta>0$; (ii)...

conjecture | open | MathDB 页面 | 原文线索

V2-062209 | MATHDB-326982

Sandpile avalanche-size critical exponents

摘录 / EXCERPT: Let ν be the infinite-volume sandpile measure. For an avalanche started by adding a particle at the origin, let $S_o=\sum_{x\in\mathbb{Z}^d}n(o,x;\cdot)$ be the total number of...

conjecture | open | MathDB 页面 | 原文线索

V2-062210 | MATHDB-326983

Sandpile avalanche-radius exponent

摘录 / EXCERPT: Let ν be the infinite-volume sandpile measure and let $\mathrm{rad}(\mathrm{Av}_o)$ be the radius of the avalanche set generated by adding a particle at the origin. Avalanche-r...

conjecture | open | MathDB 页面 | 原文线索

V2-062211 | MATHDB-326999

The simultaneous positive-rank conjecture for quadratic, m-twist and 2m-twist Jacobians

摘录 / EXCERPT: Simultaneous twist-rank conjecture. The set of $d\in\mathbb{Q}$ such that the Jacobians of the quadratic, m-twist and $2m$-twist of C by d all have positive rank is infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-062212 | MATHDB-327000

The simultaneous positive-rank conjecture for three m-twists

摘录 / EXCERPT: Let $a,b,c\in\mathbb{Z}\setminus\{0\}$, and let C_1,C_2,C_3 be the hyperelliptic curves defined by the source's displayed family ${\rm}$. Three m-twist rank conjecture. The set o...

conjecture | open | MathDB 页面 | 原文线索

V2-062213 | MATHDB-327006

The quasilinear-part conjecture for viscous conservation laws

摘录 / EXCERPT: Quasilinear-part conjecture. The quasilinear part of a viscous conservation law is uniquely determined by $a(u)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062214 | MATHDB-327007

The viscous central-invariant conjecture for integrable viscous conservation laws

摘录 / EXCERPT: Viscous central-invariant conjecture. Two integrable viscous conservation laws admitting the same viscous central invariant $a(u)$ are Miura equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-062215 | MATHDB-327008

The even-power and independence conjecture for integrable dispersive conservation laws

摘录 / EXCERPT: Even-power and independence conjecture. The quasilinear part of an integrable dispersive conservation law contains only even powers of $epsilon$, and all coefficients of the quasil...

conjecture | open | MathDB 页面 | 原文线索

V2-062216 | MATHDB-327011

The sharp logarithmic estimate conjecture for the constant C_1

摘录 / EXCERPT: Constant-C_1 conjecture. In both cases considered in the cited works, the constant C_1 can be taken equal to 1.

conjecture | open | MathDB 页面 | 原文线索

V2-062217 | MATHDB-327012

Numerical global-regularity and blow-up conjecture for fractional KdV equations

摘录 / EXCERPT: Let $u_0 in L_2(\mathbb{R})$ be smooth initial data with a single hump. The fKdV equation is conjectured to have the following behavior. For $\alpha>0.5$, solutions remain smooth f...

conjecture | open | MathDB 页面 | 原文线索

V2-062218 | MATHDB-327013

Numerical cusp-formation conjecture for Whitham and fractional KdV equations

摘录 / EXCERPT: Let $u_0\in L_2(\mathbb{R})$ be smooth initial data with a single negative hump. For $-1<\alpha<0$, solutions to the Whitham equation and to the fKdV equations with sufficiently sm...

conjecture | open | MathDB 页面 | 原文线索

V2-062219 | MATHDB-327028

Mirkovic-Vilonen cycle and affine-paving conjecture for truncated affine Grassmannians

摘录 / EXCERPT: Mirkovic-Vilonen cycle and affine-paving conjecture. For every $x\in\mathscr{X}$, there exists $w\in W$ such that $w\cdot\mathscr{X}(\mathrm{Ec}(x))$ is a Mirkovic-Vilonen cycle. M...

conjecture | open | MathDB 页面 | 原文线索

V2-062220 | MATHDB-327041

Almost-sure welding and triangular-factorization conjecture

摘录 / EXCERPT: Let μ_0 be Werner's measure on self-avoiding loops, let ν_0 be its induced measure on welding homeomorphisms σ, and let W be the welding map. A quasicircle is a Jo...

conjecture | open | MathDB 页面 | 原文线索

V2-062221 | MATHDB-327042

Kontsevich-Suhov inverse-gamma conjecture for the diagonal distribution

摘录 / EXCERPT: Let μ_c be the conjectural family of measures with values in Det^c, let ν_c be its induced measure on welding homeomorphisms, and let a be the diagonal p...

conjecture | open | MathDB 页面 | 原文线索

V2-062222 | MATHDB-327048

Sills-Zeilberger conjecture on the polynomials P_{01r}(N)

摘录 / EXCERPT: Sills-Zeilberger's conjecture. For each $r\geqslant1$, $P_{01r}(N)$ is a monic, alternating, convex polynomial in N of degree $2r$ whose only real roots are 0 and 1. The sour...

conjecture | open | MathDB 页面 | 原文线索

V2-062223 | MATHDB-327053

The first-two-layers conjecture for spherical designs of Euclidean lattices

摘录 / EXCERPT: Let Λ be a lattice, and let $M_1(\Lambda)$ and $M_2(\Lambda)$ denote its first and second layers. A finite subset of a sphere is a spherical 2-design when it satisfies th...

conjecture | open | MathDB 页面 | 原文线索

V2-062224 | MATHDB-327060

Parameter identification conjecture for high-genus triangulation limits

摘录 / EXCERPT: Let $\mathcal{T}_{n,g}$ be the set of rooted triangulations of genus g with n vertices, and let $T_{n,g}$ be uniformly distributed in $\mathcal{T}_{n,g}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-062225 | MATHDB-327061

Laver's well-order conjecture for Hurwitz-action domains

摘录 / EXCERPT: Let $(S, \mathbin{*})$ be a left-cancellative LD-system, let B_n be the braid group on n strands, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062226 | MATHDB-327062

D., Fromentin and Gebhardt's formula for the function μ_3

摘录 / EXCERPT: Let B_3^+ be the positive braid monoid on three strands, let Δ_3 denote its fundamental braid, and let μ_3 be the function on B_3^+ introduced in the paper. D., From...

conjecture | open | MathDB 页面 | 原文线索

V2-062227 | MATHDB-327069

Gasarch-Hirst graph-colouring conjecture

摘录 / EXCERPT: Gasarch-Hirst conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062228 | MATHDB-327071

The polynomial Jacobian conjecture in dimension two

摘录 / EXCERPT: Let $f: \mathbb{C}^2\to\mathbb{C}^2$ be a polynomial map with maximal rank at every point. The polynomial Jacobian conjecture. The map f is injective. The source mentions this on...

conjecture | open | MathDB 页面 | 原文线索

V2-062229 | MATHDB-327074

Herzog's polarization conjecture for Stanley depth

摘录 / EXCERPT: Let $R=\mathbb{K}[X_1,\ldots,X_n]$ be a polynomial ring, let $I\subset R$ be a monomial ideal, and let $I^p\subset R^p$ be its polarization, where R^p is the polynomial ring in w...

conjecture | open | MathDB 页面 | 原文线索

V2-062230 | MATHDB-327080

Strong regularity conjecture for the graph of integral distances

摘录 / EXCERPT: Strong regularity conjecture. For even dimension m, the graph of integral distances $\mathfrak{G}_{m,q}$ is a strongly regular graph. The conjecture predicts that the common-neig...

conjecture | open | MathDB 页面 | 原文线索

V2-062231 | MATHDB-327081

The interrupted convergence conjecture for truth seekers

摘录 / EXCERPT: Let K be the set of truth seekers and let $\overline{K}$ be the set of ignorants in a generalized weighted asynchronous bounded-confidence opinion system (WASBOCOS) with structur...

conjecture | open | MathDB 页面 | 原文线索

V2-062232 | MATHDB-327082

The multidimensional generalized Hegselmann-Krause conjecture

摘录 / EXCERPT: Let $m\geq 1$, and consider opinions in $\mathbb{R}^m$ with a norm replacing the absolute value in the update formula. Let Ω be an arbitrary generalized weighted asynchronou...

conjecture | open | MathDB 页面 | 原文线索

V2-062233 | MATHDB-327085

The simple connectedness conjecture for stability conditions of the cuspidal category

摘录 / EXCERPT: Simple connectedness conjecture. The natural surjective homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-062234 | MATHDB-327092

Branch comparison conjecture for entropy and biaccessibility dimension

摘录 / EXCERPT: Let a be a branch point, and for each branch behind a let m be the β-type Misiurewicz point of lowest preperiod in that branch. Comparing branches conjecture. Among the...

conjecture | open | MathDB 页面 | 原文线索

V2-062235 | MATHDB-327096

Necessity of the singleton-partition condition for omnivocal communication

摘录 / EXCERPT: Omnivocality conjecture. Omnivocal communication is necessary for achieving SK capacity if and only if the singleton partition is the unique minimizer for $\mathbf{I}(X_{[m]})$.

conjecture | open | MathDB 页面 | 原文线索

V2-062236 | MATHDB-327099

Sato-Tate conjecture for automorphic families

摘录 / EXCERPT: Let $\mathfrak{F}=(W,F)$ be an infinite family of automorphic representations of GL_n, ordered by conductor $C(F(w))$, and let $p_0=p_0(\mathfrak{F})>0$. Put ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062237 | MATHDB-327102

Interleaving-tailored joint log-likelihood decoder is universally capacity-achieving

摘录 / EXCERPT: Universal joint-decoder conjecture. The joint log-likelihood decoder against the interleaving attack, using this score function, works against arbitrary attacks and asymptotically...

conjecture | open | MathDB 页面 | 原文线索

V2-062238 | MATHDB-327105

The generalized RR2T conjecture for equivariant Gromov-Witten potentials

摘录 / EXCERPT: Generalized RR2T conjecture. For $\mu_1=m\mu_2$, the full descendent all-genus Gromov-Witten potential of $(Y_{a,b},T)$ is the logarithm of a tau-function of the m-generalized RR...

conjecture | open | MathDB 页面 | 原文线索

V2-062239 | MATHDB-327119

Dilks-Petersen-Stembridge conjecture on affine Eulerian polynomials

摘录 / EXCERPT: Dilks-Petersen-Stembridge conjecture. For any irreducible finite Weyl group W, the affine Eulerian polynomial $\widetilde{W}(x)$ has only real zeros. The conjecture is an affine...

conjecture | open | MathDB 页面 | 原文线索

V2-062240 | MATHDB-327122

Existence of water waves evolving from a graph to a splash singularity

摘录 / EXCERPT: Graph-to-splash existence conjecture. There exist initial data $z_0(\alpha),\omega_0(\alpha)$ of solutions of the water wave equations such that at time 0 the curve $z_0(\alpha)$...

conjecture | open | MathDB 页面 | 原文线索

V2-062241 | MATHDB-327136

Speciality conjecture for linear systems on the blow-up of $(\mathbb{P}^1)^3$

摘录 / EXCERPT: Speciality conjecture. Let $\mathcal{L}:=\mathcal{L}_{(d_1,d_2,d_3)}(m_1,\dots,m_r)$ be a linear system in standard form and let D be a divisor in its strict transform. If...

conjecture | open | MathDB 页面 | 原文线索

V2-062242 | MATHDB-327143

The generalized tau-cell conjecture for finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group. The generalized $\tilde{\tau}$-cells are the equivalence classes obtained by iterating the generalized $\tilde{\tau}$-invariant defined using strin...

conjecture | open | MathDB 页面 | 原文线索

V2-062243 | MATHDB-327144

Lusztig's left-connectedness conjecture for left cells

摘录 / EXCERPT: Let W be a finite Coxeter group with generating set S, and let Γ be a left cell of W. Lusztig's left-connectedness conjecture. For any two elements $x,y\in\Gamma$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-062244 | MATHDB-327145

Kottwitz's involution-intersection conjecture for left cells

摘录 / EXCERPT: Let W be a finite Coxeter group, let C be a conjugacy class of involutions in W, and let V_C be the real vector space with basis $\{a_w\mid w\in C\}$ equipped with the W-...

conjecture | open | MathDB 页面 | 原文线索

V2-062245 | MATHDB-327156

The zero-density conjecture for square systems over free abelian groups

摘录 / EXCERPT: Let $F_{1,n,n}(r)$ denote the sum of the reciprocals of the absolute determinants of all full-rank $n\times n$ integer matrices whose entries have Euclidean norm at most r, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-062246 | MATHDB-327167

The real summation function randomness conjecture

摘录 / EXCERPT: Let $m\geq 3$ and let $X_1,\ldots,X_m$ be the inputs of an m-party computation, with $\rho(\emptyset,\ldots,\emptyset,X_1+\cdots+X_m)$ denoting the relevant randomness measure wh...

conjecture | open | MathDB 页面 | 原文线索

V2-062247 | MATHDB-327169

The quasimap-Gromov-Witten S-operator comparison conjecture

摘录 / EXCERPT: Quasimap-Gromov-Witten comparison conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062248 | MATHDB-327180

Convexity conjecture for the Bernstein basis square-sum

摘录 / EXCERPT: Convexity conjecture. The function φ_n is convex on $[0,1]$. This conjecture concerns the shape of the sum of squares of the Bernstein basis and would sharpen the understan...

conjecture | open | MathDB 页面 | 原文线索

V2-062249 | MATHDB-327181

Unimodality conjecture for the Bernstein basis square-sum

摘录 / EXCERPT: Unimodality conjecture. The function φ_n is decreasing on $[0,\frac12]$ and increasing on $[\frac12,1]$. This predicts that the coefficient governing the Bernstein Chebyshe...

conjecture | open | MathDB 页面 | 原文线索

V2-062250 | MATHDB-327182

Midpoint-minimum conjecture for the Bernstein basis square-sum

摘录 / EXCERPT: Midpoint-minimum conjecture. For every $x\in[0,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-062251 | MATHDB-327183

Path-pairable Cartesian products of two non-path-pairable graphs

摘录 / EXCERPT: Cartesian-product conjecture. There exist non-path-pairable graphs G and H such that

conjecture | open | MathDB 页面 | 原文线索

V2-062252 | MATHDB-327188

Monomial equivalence of two six-dimensional toric codes over the field with eight elements

摘录 / EXCERPT: Let $P_6^{(4)}$ and $P_6^{(5)}$ be the polytopes indexing the corresponding six-dimensional toric codes $C_{P_6^{(4)}}$ and $C_{P_6^{(5)}}$ over $\mathbb{F}_8$. The conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-062253 | MATHDB-327191

Gracefulness of shells of three-distant trees

摘录 / EXCERPT: Let T be a three-distant tree with a longest path P. Define the shell of a three-distant tree to be the graph obtained by contracting all branches not on P with size less tha...

conjecture | open | MathDB 页面 | 原文线索

V2-062254 | MATHDB-327192

Gracefulness of three-distant trees with perfect matchings

摘录 / EXCERPT: A three-distant tree is a tree whose vertices all have distance at most 3 from a longest path. Gracefulness conjecture for three-distant trees with perfect matchings. Every three...

conjecture | open | MathDB 页面 | 原文线索

V2-062255 | MATHDB-327215

Flip-map conjectures for Bethe solutions and rigged configurations

摘录 / EXCERPT: Let $\mathrm{BA}(\ell)$ be the set of regular or physical singular Bethe solutions, and let $\iota:\mathrm{BA}(\ell)\to\mathrm{BA}(\ell)$ be defined by ... For the generalized Heisen...

conjecture | open | MathDB 页面 | 原文线索

V2-062256 | MATHDB-327216

Bethe ansatz solution-counting conjecture

摘录 / EXCERPT: For the spin-$1/2$ Heisenberg model of length N, let ${\cal N}(N,\ell)$ denote the number of solutions with pairwise distinct roots, and let ${\cal N}_{sp}(N,\ell)$ denote the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-062257 | MATHDB-327221

HCM conjecture for square roots of products of gamma variables

摘录 / EXCERPT: Let γ_t and γ_s be independent gamma random variables with parameters $t,s>0$. The gamma-product conjecture. For all $s,t>0$, the independent product...

conjecture | open | MathDB 页面 | 原文线索

V2-062258 | MATHDB-327227

The inner-face inclusion conjecture for generalized simplices

摘录 / EXCERPT: Inner-face inclusion conjecture. For every $A\in\mathord{\trianglerighteqslant}$, the inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-062259 | MATHDB-327228

The weak cat equivalence conjecture for inner-face inclusions

摘录 / EXCERPT: Weak cat equivalence conjecture. For every $A\in\mathord{\trianglerighteqslant}$, the inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-062260 | MATHDB-327229

The fibrant-object conjecture for the model structure of n-cats

摘录 / EXCERPT: Fibrant-object conjecture for n-cats. The fibrant objects of the model structure for n-cats are precisely the n-cats.

conjecture | open | MathDB 页面 | 原文线索

V2-062261 | MATHDB-327230

The horn-generation conjecture for the model structure of groupoids

摘录 / EXCERPT: Groupoid horn-generation conjecture. The model structure for groupoids is equal to the Cisinski model structure generated by the set of horns. In particular, if an object...

conjecture | open | MathDB 页面 | 原文线索

V2-062262 | MATHDB-327244

Hall--Littlewood law of large numbers with a positive Plancherel parameter

摘录 / EXCERPT: Positive-Plancherel-parameter conjecture. The technical assumption $\gamma=0$ in the law of large numbers can be dropped: the same convergence should hold for every triplet...

conjecture | open | MathDB 页面 | 原文线索

V2-062263 | MATHDB-327245

Hall--Littlewood central limit conjecture for row and column lengths

摘录 / EXCERPT: Central limit conjecture. The infinite vector

conjecture | open | MathDB 页面 | 原文线索

V2-062264 | MATHDB-327246

Macdonald law of large numbers for row and column lengths

摘录 / EXCERPT: Macdonald law-of-large-numbers conjecture. With almost sure convergence,

conjecture | open | MathDB 页面 | 原文线索

V2-062265 | MATHDB-327256

Absence of nontrivial cycles for the $3x+1$ map

摘录 / EXCERPT: Absence of nontrivial cycles. There do not exist integers $n>2$ and $k>0$ such that $T^k(n)=n$.

conjecture | open | MathDB 页面 | 原文线索

V2-062266 | MATHDB-327257

Stable-set conjecture for the real $3x+1$ extension

摘录 / EXCERPT: Stable-set conjecture. The function f has no attracting cycle in the interval $[\mu_3,+\infty)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062267 | MATHDB-327258

No-wandering-intervals conjecture for the real $3x+1$ extension

摘录 / EXCERPT: No-wandering-intervals conjecture. The function f has no wandering intervals in $\mathbb{R}^{+}$.

conjecture | open | MathDB 页面 | 原文线索

V2-062268 | MATHDB-327260

Higher-order Stirling-number monotonicity and log-concavity conjectures

摘录 / EXCERPT: Higher-order Stirling-number conjecture. The following claims are valid: (1) for fixed $ell\in\mathbb{N}$ and $n\geq \ell+3$, the finite sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-062269 | MATHDB-327269

The magic-number conjecture for strong cyclotomic Delone sets

摘录 / EXCERPT: Let $n\ge 3$, let $N=[n,2]$, and let m_n denote the minimum number of pairwise nonparallel $\varLambda$-directions whose X-rays determine every convex subset of a strong n-cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-062270 | MATHDB-327270

The seven-direction conjecture for convex subsets of three-dimensional Delone sets

摘录 / EXCERPT: Let $\varLambda\subset\mathbb R^3$ be a Delone set. A set of $\varLambda$-directions is in general position when any three directions from it span the whole 3-space. Seven-direct...

conjecture | open | MathDB 页面 | 原文线索

V2-062271 | MATHDB-327275

Conjecture on the asymptotic second eigenvalue of the Schoenberg operator

摘录 / EXCERPT: Schoenberg eigenvalue conjecture. For fixed $k>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062272 | MATHDB-327282

The Fourier-transform conjecture for real transcendental entire functions in the Laguerre-Pólya class

摘录 / EXCERPT: Fourier-transform conjecture. The function F belongs to the Laguerre-Pólya class:

conjecture | open | MathDB 页面 | 原文线索

V2-062273 | MATHDB-327292

Monotonicity conjecture for the longest monotone subsequence in Latin squares

摘录 / EXCERPT: Let Λ_n denote the maximum, over all Latin squares of order n, of the length of a longest monotone subsequence. Monotonicity conjecture. The sequence Λ_n is nonde...

conjecture | open | MathDB 页面 | 原文线索

V2-062274 | MATHDB-327300

Satake parameters recover the Frobenius element of the local Langlands parameter

摘录 / EXCERPT: Let $W'_F:= W_F \rtimes \mathbb C$ be the Weil-Deligne group. For a representation $\pi \in \Pi(G/F,J)$ with local Langlands parameter $\varphi_\pi: W'_F \rightarrow {}^LG$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062275 | MATHDB-327337

The girth-five fractional-coloring gap conjecture for planar graphs

摘录 / EXCERPT: Girth-five fractional-coloring conjecture. There exists $\delta>0$ such that every planar graph of girth at least five has fractional chromatic number at most

conjecture | open | MathDB 页面 | 原文线索

V2-062276 | MATHDB-327352

Existence of nonlinear velocity constraints outside the Newtonian model

摘录 / EXCERPT: A mechanical system is described by constraints that may be nonlinear in the velocity. Existence conjecture. The existence of mechanical systems with nonlinear constraints in the v...

conjecture | open | MathDB 页面 | 原文线索

V2-062277 | MATHDB-327364

The order conjecture for birational rowmotion on northeast triangular posets

摘录 / EXCERPT: Let $\operatorname*{NEtri}(p)$ be the northeast triangular poset of the triangular poset construction, and let $R_{\operatorname*{NEtri}(p)}$ denote birational rowmotion on it. For...

conjecture | open | MathDB 页面 | 原文线索

V2-062278 | MATHDB-327367

The dimension-reduction conjecture for generalized Fibonacci cubes

摘录 / EXCERPT: Dimension-reduction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-062279 | MATHDB-327368

The index-invariance conjecture for generalized Fibonacci cubes

摘录 / EXCERPT: Index-invariance conjecture. If f,f' are a non-trivial pair such that

conjecture | open | MathDB 页面 | 原文线索

V2-062280 | MATHDB-327383

The full-rank coding conjecture for in-order streaming

摘录 / EXCERPT: Let $s_1,s_2,\ldots$ be the source packets, let $V[n]$ be the transmit index in slot n, and suppose that a transmission with index $V[n]$ is a linear combination of source packet...

conjecture | open | MathDB 页面 | 原文线索

V2-062281 | MATHDB-327387

Pattern-avoidance characterization of lci K-orbit closures

摘录 / EXCERPT: Let $G=GL(p+q,\mathbb{C})$, $K=GL(p,\mathbb{C})\times GL(q,\mathbb{C})$, and consider K-orbit closures in G/B, indexed by clans. A clan avoids a pattern when it contains no occ...

conjecture | open | MathDB 页面 | 原文线索

V2-062282 | MATHDB-327400

Morrey's rank-one convexity conjecture in two dimensions

摘录 / EXCERPT: Morrey's conjecture. Every rank-one convex function $\mathbf E:\mathbb R^{2\times2}\rightarrow\mathbb R$ is quasiconvex.

conjecture | open | MathDB 页面 | 原文线索

V2-062283 | MATHDB-327401

The quasiconcavity conjecture for Burkholder functions

摘录 / EXCERPT: For $p\geqslant n/2$ and $\lambda\geqslant\left|1-n/p\right|$, define the Burkholder matrix function ... The source also defines the planar Burkholder function...

conjecture | open | MathDB 页面 | 原文线索

V2-062284 | MATHDB-327402

The quasiconcavity conjecture for Burkholder functions at the zero matrix

摘录 / EXCERPT: Zero-matrix quasiconcavity conjecture. Burkholder functions are quasiconcave at the zero matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-062285 | MATHDB-327404

Conjectural convergent long-distance expansion for the sine-Gordon/Painlevé III3 tau function

摘录 / EXCERPT: Long-distance expansion conjecture. The sine-Gordon/Painlevé III_3 tau function is given by the convergent series

conjecture | open | MathDB 页面 | 原文线索

V2-062286 | MATHDB-327408

Conjecture on nonsingularity of polynomially bisected directions

摘录 / EXCERPT: Fix n, and let $I^n\subseteq\mathbb{R}^n$ be the unit cube. Given $\varepsilon,H>0$, a positive integer N, and a polynomial Q, subdivide I^n into cubes of edge length...

conjecture | open | MathDB 页面 | 原文线索

V2-062287 | MATHDB-327414

The $2M$ phase-transition conjecture for almost-injective complex phase retrieval

摘录 / EXCERPT: Let Φ be the measurement matrix for signals in $\mathbb{C}^M$, with N intensity measurements, and write $\mathrm{AlmInj}[\Phi,\mathbb{C}^{M\times N}]$ for almost injectivity...

conjecture | open | MathDB 页面 | 原文线索

V2-062288 | MATHDB-327417

Characterization of essentially lexicographic valuations by separated completions

摘录 / EXCERPT: Let $\nu:F^*\rightarrow\mathbb{Z}^n$ be a maximal valuation, and let $\mathcal{O}=\{f\in\mathscr{F}:\nu(f)\geq 0\}$ with maximal ideal $\mathcal{m}=\{f\in\mathscr{F}:\nu(f)>0\}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-062289 | MATHDB-327421

Loehr-Warrington rational q-Catalan partition formula conjecture

摘录 / EXCERPT: Fix $a,b\in\mathbb{N}$ with $\gcd(a,b)=1$. Let $\mathrm{D}(a,b)$ be the set of partitions whose diagrams fit in the triangle with vertices $(0,0)$, $(0,a)$, and (b,a). For a cell...

conjecture | open | MathDB 页面 | 原文线索

V2-062290 | MATHDB-327422

Loehr-Warrington nonstandard q-binomial formula conjecture

摘录 / EXCERPT: Fix $a,b\in\mathbb{N}$. Let $\mathrm{R}(a,b)$ be the set of partitions fitting in the $b\times a$ rectangle, let $\mathsf{ml}_{b,a}(\mu)$ be the minimum level on the frontier of...

conjecture | open | MathDB 页面 | 原文线索

V2-062291 | MATHDB-327423

Haglund-Haiman-Loehr diagonal-coinvariant Hilbert-series conjecture

摘录 / EXCERPT: Let PF_n be the set of labeled classical parking functions. For $P\in\mathsf{PF}_n$, let $\mathsf{area}(P)$ be its area statistic and let $\mathsf{dinv}(P)$ be its diago...

conjecture | open | MathDB 页面 | 原文线索

V2-062292 | MATHDB-327424

Haglund-Haiman-Loehr-Remmel-Ulyanov Shuffle Conjecture

摘录 / EXCERPT: Let DR_n be the doubly graded diagonal coinvariant ring, and let PF_n be the labeled classical parking functions. For each P, let $\mathsf{area}(P)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062293 | MATHDB-327425

Loehr-Warrington rational q,t-Catalan symmetry conjecture

摘录 / EXCERPT: Rational q,t-Catalan symmetry conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062294 | MATHDB-327426

Loehr-Warrington rational q,t-Catalan specialization conjecture

摘录 / EXCERPT: Let $\mathsf{Cat}_{a,b}(q,t)$ be the rational q,t-Catalan number and let $[m]_q$ and $\genfrac{[}{]}{0pt}{}{a+b}{a,b}_q$ denote the q-integer and q-binomial coefficient. Rati...

conjecture | open | MathDB 页面 | 原文线索

V2-062295 | MATHDB-327427

Loehr-Warrington rational parking-function Frobenius conjectures

摘录 / EXCERPT: Let $\mathsf{PF}_{a,b}(q,t)$ be the symmetric function defined from rational parking functions, and let ${\sf Sch}_{a,b;k}(q,t)$ denote the coefficient of $s_{(k+1,1^{a-k-1})}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-062296 | MATHDB-327449

Typical string graph convergence conjecture

摘录 / EXCERPT: Let $\mathcal{SG}_n$ and $\mathcal{SG}^L_n$ denote, respectively, the classes of unlabeled and labeled string graphs on n vertices, and let $W^4_{1/2}$ be the corresponding graph...

conjecture | open | MathDB 页面 | 原文线索

V2-062297 | MATHDB-327459

Existence of operators with the conjectured properties on Grassmannians

摘录 / EXCERPT: $S_{m}^{(b_{-})}$ operator conjecture. For any dimension b_{-} and any weight m, there exists an operator $S_{m}^{(b_{-})}$ satisfying properties (i)--(vii) stated in the sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-062298 | MATHDB-327461

Pseudo-dimension matching limit for dense graph sequences

摘录 / EXCERPT: Let G_n be a sequence of graphs admitting a graphon limit $\mathcal{G}$, with edge weights following a distribution of pseudo-dimension q. Suppose that there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-062299 | MATHDB-327463

The TKT a.1 defect-zero exclusion conjecture for quadratic fields

摘录 / EXCERPT: Let K be a quadratic field, and let $G=\operatorname{Gal}(\mathrm{F}_3^2(K)\mid K)$ be its second 3-class group. The TKT a.1 defect-zero exclusion conjecture. The transfer kern...

conjecture | open | MathDB 页面 | 原文线索

V2-062300 | MATHDB-327464

The weak leaf conjecture for principalization types G.19 and H.4

摘录 / EXCERPT: Let K be a quadratic base field, and let its second 3-class group have invariants m, n, and defect k. The weak leaf conjecture. The principalization types $\mathrm{G.19}$...

conjecture | open | MathDB 页面 | 原文线索

V2-062301 | MATHDB-327465

The exponentiated Hencky energy's failure of the E$^+$-TSS inequalities

摘录 / EXCERPT: Let $W_{\rm eH}$ denote the exponentiated Hencky-logarithmic strain energy, and let E$^+$-TSS denote the strict tension-tension stress inequalities. The exponentiated Hencky energy...

conjecture | open | MathDB 页面 | 原文线索

V2-062302 | MATHDB-327466

Planar polyconvexity of exponentiated Hencky energies

摘录 / EXCERPT: For $F\in\mathbb{R}^{n\times n}$, define the exponentiated Hencky energy by ... where U is the right stretch tensor and $\mu,\kappa,k,\widehat{k}$ are parameters. Planar polyconvex...

conjecture | open | MathDB 页面 | 原文线索

V2-062303 | MATHDB-327467

Rank-one convexity region for the isochoric exponentiated Hencky energy

摘录 / EXCERPT: Rank-one convexity region conjecture. For $n=2,3$, the energy $F\mapsto\frac{\mu}{k}e^{k\|\operatorname{dev}_n\log U\|^2}$ with $k>\frac{3}{16}$ is rank-one convex on a set contain...

conjecture | open | MathDB 页面 | 原文线索

V2-062304 | MATHDB-327468

Nonexistence of globally TSTS-M$^+$-monotone isochoric energies

摘录 / EXCERPT: Let U be the right stretch tensor and let $K_2^2=\|\operatorname{dev}_n\log U\|^2$. Global TSTS-M$^+$ nonexistence conjecture. For $n=2,3$, there is no elastic energy of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-062305 | MATHDB-327469

Wahl's conjecture for quasihomogeneous normal isolated complete intersections

摘录 / EXCERPT: A quasihomogeneous normal isolated complete intersection singularity (ICIS) is a normal isolated complete intersection singularity equipped with a positive grading. A derivation ha...

conjecture | open | MathDB 页面 | 原文线索

V2-062306 | MATHDB-327475

Conjecture on the global integral involving metaplectic theta functions

摘录 / EXCERPT: The global-integral conjecture. The integral $I(\varphi_\pi,s,f_s)$ has a meromorphic continuation to the full complex plane, and: (1) if $n<r+1$, it is identically zero for all...

conjecture | open | MathDB 页面 | 原文线索

V2-062307 | MATHDB-327476

Conjecture on the unipotent Fourier coefficient of the higher theta representation

摘录 / EXCERPT: Fourier-coefficient conjecture. The automorphic representation of $GL_2^{(r+1)}(\mathbb{A})$ generated by all the functions $f(g)$ is $\Theta_2^{(r+1)}$. The conjecture predicts th...

conjecture | open | MathDB 页面 | 原文线索

V2-062308 | MATHDB-327477

Friedberg-Ginzburg conjecture on descents and theta representations

摘录 / EXCERPT: For each group and covering degree treated in the paper, let $\sigma_G^{(m)}$ denote the representation generated by the corresponding descent, and let $\Theta_G^{(m)}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-062309 | MATHDB-327478

Scattering conjecture for global solutions below the mass-energy threshold

摘录 / EXCERPT: Scattering conjecture. Every global solution satisfying the conditions above has finite $S(\dot{H}^{1/2})$-norm, and it scatters in H^1.

conjecture | open | MathDB 页面 | 原文线索

V2-062310 | MATHDB-327488

Klein-Lewis-Morales coefficientwise inequality for permutation diagrams

摘录 / EXCERPT: Let w be a permutation in S_n, let $D(w)$ be its permutation diagram, and let $mat_q(n,D(w),n)$ denote the number of $n\times n$ matrices over $\mathbb{F}_q$ with rank n whos...

conjecture | open | MathDB 页面 | 原文线索

V2-062311 | MATHDB-327491

The cycle conjecture for 3-free Fibonacci sequences

摘录 / EXCERPT: A 3-free Fibonacci sequence is a Fibonacci-type sequence in which each newly appended term is obtained by dividing the sum of the preceding two terms by a power of 3 whenever the...

conjecture | open | MathDB 页面 | 原文线索

V2-062312 | MATHDB-327501

Thanatipanonda's generalized binomial-power expansion conjecture

摘录 / EXCERPT: Fix integers d, $k\geq 0$, and $r\geq 1$. The coefficients $a^{(r)}_{d,k,j}$ are intended to expand powers of $\binom{n+dk}{k}$ in the family $\binom{n+dj}{j}$. Generalized expan...

conjecture | open | MathDB 页面 | 原文线索

V2-062313 | MATHDB-327508

Hong's odd-set LCM nonsingularity conjecture

摘录 / EXCERPT: Let S be a GCD-closed set of odd positive integers, meaning that every element of S is odd and the greatest common divisor of any two elements of S belongs to S. Let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-062314 | MATHDB-327509

Hong's odd GCD-closed-set conjecture for real powers

摘录 / EXCERPT: Let $S=\{x_1,\ldots,x_n\}$ be an odd-GCD-closed set, meaning that S consists of odd positive integers and is closed under greatest common divisors. For a real exponent α,...

conjecture | open | MathDB 页面 | 原文线索

V2-062315 | MATHDB-327510

Hong's multiplicative-function conjecture for odd GCD-closed sets

摘录 / EXCERPT: Let $S=\{x_1,\ldots,x_n\}$ be an odd-GCD-closed set, and let f be a completely multiplicative function, meaning $f(ab)=f(a)f(b)$ for positive integers a,b. If f is strictly m...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-062316 | MATHDB-327513

Clunie and Sheil-Small's coefficient conjecture for normalized harmonic mappings

摘录 / EXCERPT: Clunie-Sheil-Small coefficient conjecture. For every $n\geq2$,

conjecture | open | MathDB 页面 | 原文线索

V2-062317 | MATHDB-327514

Equality of harmonic and analytic shear classes for normalized univalent mappings

摘录 / EXCERPT: Let $\mathcal{S}^0_H$ be the class of normalized sense-preserving univalent harmonic mappings $f=h+\overline{g}$ in the unit disk with $f_{\overline z}(0)=0$. Let $\mathcal{S}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-062318 | MATHDB-327515

Convex shear-class equality conjecture for harmonic mappings

摘录 / EXCERPT: Let $\mathcal{C}^0_H$ be the class of functions in $\mathcal{S}^0_H$ whose image of the unit disk is convex, and let $\mathcal{C}$ be the analytic class of normalized univalent fun...

conjecture | open | MathDB 页面 | 原文线索

V2-062319 | MATHDB-327516

Generalized Lang-Trotter conjecture for coinciding Frobenius fields

摘录 / EXCERPT: Generalized Lang-Trotter conjecture. The elliptic curves E_1 and E_2 satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-062320 | MATHDB-327517

The trace-forest content formula conjecture for skew tableaux of hook shape

摘录 / EXCERPT: Let λ and μ be partitions with $\lambda\vdash n$, and suppose that μ is a hook. Write $f^{\lambda/\mu}$ for the number of standard Young tableaux of skew shape...

conjecture | open | MathDB 页面 | 原文线索

V2-062321 | MATHDB-327522

Conjecture on the stability threshold and coincidence threshold of hypersurfaces

摘录 / EXCERPT: Let $V:f=0$ be a projective hypersurface of degree d in $\mathbb{P}^n$ with only isolated singularities. Writing $st(V)$ for its stability threshold, $ct(V)$ for its coincidence...

conjecture | open | MathDB 页面 | 原文线索

V2-062322 | MATHDB-327523

Conjecture on minimal degree relations and Arnold exponents of hypersurfaces

摘录 / EXCERPT: Let $V:f=0$ be a projective hypersurface of degree d in $\mathbb{P}^n$ with only weighted homogeneous isolated singularities. Let $mdr(V)$ denote the minimal degree of a nontrivi...

conjecture | open | MathDB 页面 | 原文线索

V2-062323 | MATHDB-327533

The additive coloring conjecture

摘录 / EXCERPT: Additive Coloring Conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-062324 | MATHDB-327537

Lewis-Reiner-Stanton conjecture on the cofixed Hilbert series

摘录 / EXCERPT: Let $G=GL_n({\mathbb F}_q)$ act on $S={\mathbb F}_q[x_1,\ldots,x_n]$, and let N be the ${\mathbb F}_q$-linear span of all $g(f)-f$ with $f\in S$ and $g\in G$. Define the G-cofi...

conjecture | open | MathDB 页面 | 原文线索

V2-062325 | MATHDB-327538

Three-variable Stanley decomposition conjecture

摘录 / EXCERPT: Let $f_3(t),f_2(t),f_1(t)$ be the Hilbert-series numerator data arising in the proposed Stanley-type decomposition discussed in the source, and let M_i be the corresponding sets...

conjecture | open | MathDB 页面 | 原文线索

V2-062326 | MATHDB-327542

The conjectured exact value of the Möbius triangular-ratio Lipschitz constant

摘录 / EXCERPT: The conjectured value of $L(a)$. For $n=2$, numerical experiments suggest that

conjecture | open | MathDB 页面 | 原文线索

V2-062327 | MATHDB-327544

Equality of the bifurcation and asymptotic sets at infinity

摘录 / EXCERPT: Let $f\colon \mathbb{R}^n\to\mathbb{R}$ be a real polynomial. Denote by $\mathcal{B}_{\infty}(f)$ the bifurcation set at infinity and by $\mathcal{S}_{\infty}(f)$ the set of asympt...

conjecture | open | MathDB 页面 | 原文线索

V2-062328 | MATHDB-327547

The non-centric K-normalizer conjecture for minimal characteristic bisets

摘录 / EXCERPT: Let $\mathcal F$ be a saturated fusion system on a finite p-group S, let $\Omega=\Lambda_{\mathcal F}$ be its minimal characteristic biset, and let $P\leq S$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062329 | MATHDB-327554

Unimodality and log-concavity of involution length profiles in classical Coxeter groups

摘录 / EXCERPT: Unimodality and log-concavity conjecture. The even and odd involution length profiles of $W(A_n)$ are log-concave, while the even and odd involution length profiles of $W(B_n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062330 | MATHDB-327555

Unimodality of length profiles for conjugacy classes of involutions

摘录 / EXCERPT: Unimodality conjecture. (i) If X is a conjugacy class of involutions in $W(A_n)$ or $W(B_n)$, then the even/odd length profile of X is unimodal. (ii) If X is the set of invol...

conjecture | open | MathDB 页面 | 原文线索

V2-062331 | MATHDB-327558

The integrality conjecture for the Manin constant of Cartan non-split parametrizations

摘录 / EXCERPT: Let E be the elliptic curve and let ... be the modular parametrization defined over $\mathbb{Q}$, with Manin constant c. Integrality conjecture. The Manin constant belongs to ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-062332 | MATHDB-327574

Conjecture on Theorem 2 for Artinian lex-ideals

摘录 / EXCERPT: Let L be an Artinian lex-ideal in the polynomial ring $\mathbf{k}[x,y,z]$. The Boij-Söderberg decomposition of L is the decomposition into pure diagrams described by Theorem 2....

conjecture | open | MathDB 页面 | 原文线索

V2-062333 | MATHDB-327578

Conjectured traveling-wave limit for branching reflected Brownian motion

摘录 / EXCERPT: Let $u^R(t,x,y):=\mathbb{P}_y(M^R_t\leq x)$, where $\mathbb{P}_y$ is the law of branching reflected Brownian motion starting at $y\geq 0$. Let $q^R_{\delta}(y,t)$ be defined as in...

conjecture | open | MathDB 页面 | 原文线索

V2-062334 | MATHDB-327583

Determinant formula for sinh-Gordon one-point functions in the fermionic basis

摘录 / EXCERPT: Determinant conjecture. As in the sine-Gordon model, the sinh-Gordon one-point functions in the fermionic basis satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-062335 | MATHDB-327585

Equivalence between weak recalages and simplicial triangulations

摘录 / EXCERPT: Let $\mathrm C$ be a sesquicategory. A weak recalage-simplicial triangulation equivalence conjecture. The category of weak recalages of $\operatorname{\bf LN}_+(\mathrm C)$ is equi...

conjecture | open | MathDB 页面 | 原文线索

V2-062336 | MATHDB-327588

Strengthened Ringel packing conjecture for bounded tree orders

摘录 / EXCERPT: Let n be a positive integer. Consider any family of trees in which every individual tree has order at most $n+1$ and the total number of edges is at most $\binom{2n+1}{2}$. Stren...

conjecture | open | MathDB 页面 | 原文线索

V2-062337 | MATHDB-327601

Jordan-domain conjecture for bounded immediate basins of transcendental entire functions

摘录 / EXCERPT: Jordan-domain conjecture. Every bounded immediate attracting or parabolic basin of a transcendental entire function is a Jordan domain.

conjecture | open | MathDB 页面 | 原文线索

V2-062338 | MATHDB-327606

The injective Yamanouchi-word conjecture for dual equivalence components

摘录 / EXCERPT: Let $b4$ be a diagram with n cells. Write $a3am(n)$ for the set of standard Yamanouchi words of length n, $a3am_b4(n)$ for the subset associated with $b4$, and let $c1$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062339 | MATHDB-327607

The Hall-Littlewood Schur-expansion conjecture via an injective Yamanouchi map

摘录 / EXCERPT: Let $b4$ be a diagram with n cells, let f_b4 be an injective function as in the injective Yamanouchi-word conjecture, and let $c0$ range over standard Yamanouchi words of parti...

conjecture | open | MathDB 页面 | 原文线索

V2-062340 | MATHDB-327618

The logarithmic mean conjecture for the multivariate osculating mean

摘录 / EXCERPT: Logarithmic mean conjecture. If $x_1(t)=t$, $x_2(t)=t^2$, $\ldots$, $x_{n-1}(t)=t^{n-1}$, and $x_n(t)=\log t$, then

conjecture | open | MathDB 页面 | 原文线索

V2-062341 | MATHDB-327619

Conjunctive-binding first-order logic positive-properties conjecture

摘录 / EXCERPT: Conjunctive-binding positive-properties conjecture. The fragment cb enjoys the finite-model property and has a decidable satisfiability problem.

conjecture | open | MathDB 页面 | 原文线索

V2-062342 | MATHDB-327620

Disjunctive-binding first-order logic decidability conjecture

摘录 / EXCERPT: Disjunctive-binding decidability conjecture. The fragment db enjoys a decidable satisfiability problem.

conjecture | open | MathDB 页面 | 原文线索

V2-062343 | MATHDB-327632

Conjecture on gradient catastrophe and dispersive scaling for two-dimensional Toda equations

摘录 / EXCERPT: Consider initial data that are the x-derivative of a rapidly decreasing smooth function in $L_{2}(\mathbb{R}^{2})$ with a single maximum. Write the two-dimensional dispersionless...

conjecture | open | MathDB 页面 | 原文线索

V2-062344 | MATHDB-327633

The $K(\pi,1)$ conjecture for the orbit space of an Artin-Coxeter arrangement

摘录 / EXCERPT: Let $\mathbf W$ be the Artin or Coxeter group considered in the paper, and let $\mathbf Y$ be the associated orbit space. A space is a $k(\pi,1)$ space if its universal cover is co...

conjecture | open | MathDB 页面 | 原文线索

V2-062345 | MATHDB-327635

Rank recovery conjecture for generalized Procrustes semidefinite relaxations

摘录 / EXCERPT: Let $d>1$, $n>2$, $m\geq d+1$, $A\in\mathbb{R}^{d\times m}$ represent m random points in $\mathbb{R}^d$ (i.i.d. uniform random coordinates in $[0,1]^d$), and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062346 | MATHDB-327639

Asymptotic speeds and profiles for all Fisher-KPP transition fronts

摘录 / EXCERPT: Asymptotic-speed and profile conjecture. Any such transition front has asymptotic past and future speeds $c_\pm$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-062347 | MATHDB-327640

Full classification of Fisher-KPP solutions by exponential decay

摘录 / EXCERPT: Full decay-characterization conjecture. For any solution $0<u<1$, the limit λ exists in $[0,\sqrt{f'(0)}]$, and u is a transition front connecting 0 and 1 if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-062348 | MATHDB-327647

Lapid-Mao's Ichino-Ikeda-type conjecture for Whittaker coefficients on the metaplectic group

摘录 / EXCERPT: Let F be a number field, let S be a sufficiently large finite set of its places, and let $\tilde\pi$ be the $\psi_{\tilde N}$-descent of an automorphic representation...

conjecture | open | MathDB 页面 | 原文线索

V2-062349 | MATHDB-327648

Geometric decomposition conjecture for the K3 elliptic genus

摘录 / EXCERPT: Let X be a K3 surface with holomorphic tangent bundle $T:=T^{1,0}X$, and let $\mathbb E_{q,-y}$ be the formal power series of holomorphic vector bundles used to define the geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-062350 | MATHDB-327651

The maximal-length exceptional collection conjecture

摘录 / EXCERPT: Let $\mathcal{T} = \langle E_1,\dots,E_n \rangle$ be a triangulated category generated by an exceptional collection. An exceptional collection in $\mathcal{T}$ is full if it genera...

conjecture | open | MathDB 页面 | 原文线索

V2-062351 | MATHDB-327654

The K3-type category conjecture for Küchle's fivefold hyperplane section

摘录 / EXCERPT: Let W be a 7-dimensional vector space, let $X\subset\operatorname{Gr}(3,W)$ be the 5-dimensional zero locus of a section of...

conjecture | open | MathDB 页面 | 原文线索

V2-062352 | MATHDB-327671

Finite multiple harmonic sums lie in the Bernoulli subspace

摘录 / EXCERPT: For positive integers $n>mgeq 1$ and primes $p>n$, define ... Let $\mathcal{P}$ be the set of rational primes, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062353 | MATHDB-327674

MNOP rationality and DT/GW correspondence conjecture

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau threefold. For a curve class $\beta\beta$ and the rank-one Donaldson-Thomas series $I_\beta(X)$, let $I_0(X)$ denote the degree-zero serie...

conjecture | open | MathDB 页面 | 原文线索

V2-062354 | MATHDB-327676

Pandharipande-Thomas multiple-cover formula conjecture

摘录 / EXCERPT: Let $N_{n,\beta}$ denote the invariants appearing in the Gopakumar-Vafa-type product expansion of the stable-pair generating series, and let (n,β) denote the common divisibil...

conjecture | open | MathDB 页面 | 原文线索

V2-062355 | MATHDB-327704

The shuffle conjecture for parking functions

摘录 / EXCERPT: Let ∇ be the linear operator on modified Macdonald polynomials defined by ... Let e_n be the elementary symmetric function, and let F_S be the Gessel quasisymmetric functi...

conjecture | open | MathDB 页面 | 原文线索

V2-062356 | MATHDB-327707

Klagsbrun-Mazur-Rubin average-rank conjecture for quadratic twists

摘录 / EXCERPT: Let K be a number field, let E be an elliptic curve over K, and let $C(K,X)$ denote the set of quadratic characters of K of norm less than X. Write E^χ for the quadr...

conjecture | open | MathDB 页面 | 原文线索

V2-062357 | MATHDB-327735

Complete transitivity conjecture for the codes $C^{(i)}$

摘录 / EXCERPT: Let $C^{(i)}$ be the nested binary codes defined in the paper, with parameter u and $i ein\{0,1,\ldots,u\}$. Complete transitivity conjecture. The code $C^{(i)}$ is completely tr...

conjecture | open | MathDB 页面 | 原文线索

V2-062358 | MATHDB-327737

Conrey-Farmer-Keating-Rubinstein-Snaith second-moment conjecture for holomorphic cusp-form L-functions

摘录 / EXCERPT: Second-moment conjecture. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062359 | MATHDB-327741

The BSD rank criterion over ring class fields

摘录 / EXCERPT: Birch and Swinnerton-Dyer conjecture. The predicted criterion is

conjecture | open | MathDB 页面 | 原文线索

V2-062360 | MATHDB-327743

The globality and reciprocity conjecture for mixed-signature Darmon points

摘录 / EXCERPT: Darmon point globality conjecture. The isogeny β can be chosen so that P_ψ is the image of a global point in $E(H_{\mathcal O}^+)$, and, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-062361 | MATHDB-327744

Oda-Yoshida's archimedean isogeny conjecture for Darmon points

摘录 / EXCERPT: Oda-Yoshida isogeny conjecture. There exists an isogeny

conjecture | open | MathDB 页面 | 原文线索

V2-062362 | MATHDB-327745

The archimedean globality and reciprocity conjecture for Darmon points

摘录 / EXCERPT: Archimedean Darmon point conjecture. The isogeny β can be chosen so that

conjecture | open | MathDB 页面 | 原文线索

V2-062363 | MATHDB-327752

The Katz-Klemm-Vafa product formula for BPS counts of K3 surfaces

摘录 / EXCERPT: Let $r_{g,h}$ denote the BPS count associated to a class β on a K3 surface whose norm square satisfies ... Let q and y be formal variables. Katz-Klemm-Vafa product formula....

conjecture | open | MathDB 页面 | 原文线索

V2-062364 | MATHDB-327755

Geometric Langlands compatibility with Eisenstein series

摘录 / EXCERPT: Let G be a reductive group, let $P\subset G$ be a standard parabolic with Levi subgroup M, and let $\bP\subset \bG$ be the corresponding parabolic in the Langlands dual group w...

conjecture | open | MathDB 页面 | 原文线索

V2-062365 | MATHDB-327765

Dependently rigid characterization of the maximum likelihood threshold

摘录 / EXCERPT: Dependently rigidity conjecture. The maximum likelihood threshold of G is greater than n if and only if G is generically n-dependently rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-062366 | MATHDB-327769

Subgraph-closed classes with strongly sublinear separators have subexponential expansion

摘录 / EXCERPT: Let ${\cal G}$ be a subgraph-closed class of graphs. Suppose that ${\cal G}$ has strongly sublinear separators. Subexponential expansion conjecture. The expansion of ${\cal G}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-062367 | MATHDB-327774

The eventual interval conjecture for unions of sets of lengths

摘录 / EXCERPT: Let H be a Krull monoid as in Theorem 5.3, with associated set G_P, and suppose in addition that ... For $k\in\mathbb N$, let $\mathcal U_k(H)$ denote the union of the sets of le...

conjecture | open | MathDB 页面 | 原文线索

V2-062368 | MATHDB-327778

Local Torelli conjecture for osculating cones to Brill-Noether loci

摘录 / EXCERPT: Local Torelli conjecture. The following hold: (a) away from points of C', the fibers of α are smooth or empty; and (b) for a smooth point p' of C', the corresponding p...

conjecture | open | MathDB 页面 | 原文线索

V2-062369 | MATHDB-327779

Equivalence of US, SUS and SDP under the regularity assumption

摘录 / EXCERPT: Let ξ be a point process satisfying Assumption (regularity-1), with the associated constant c_{ξ}. The properties (US), (SUS) and (SDP) are the conditions defined in the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-062370 | MATHDB-327780

The Selmer-complex main conjecture for pure motives

摘录 / EXCERPT: Let $M/\mathbf Q$ be a pure motive of weight $\leqslant -1$ which does not contain submotives of the form $\mathbf Q(m)$. Let V be its p-adic realisation satisfying conditions...

conjecture | open | MathDB 页面 | 原文线索

V2-062371 | MATHDB-327781

Greenberg's weak Leopoldt conjecture

摘录 / EXCERPT: Let V be a p-adic representation of $G_{\mathbf Q,S}$ that is potentially semistable at p, let T be a stable lattice in V, and let F_∞ be the cyclotomic extension...

conjecture | open | MathDB 页面 | 原文线索

V2-062372 | MATHDB-327782

The extra-zero conjecture for p-adic L-functions

摘录 / EXCERPT: Let V be a p-adic representation satisfying conditions C1-4a, let D be a regular subspace of $\mathbf D_{\mathrm{st}}(V)$, and let $e=\mathrm{rank}(\mathbf W_0)$. In the two...

conjecture | open | MathDB 页面 | 原文线索

V2-062373 | MATHDB-327786

Asymptotic logarithmic Kähler-Einstein conjecture for log del Pezzo surfaces

摘录 / EXCERPT: Asymptotic logarithmic Kähler-Einstein conjecture. The surface S admits conical Kähler-Einstein metrics with angle β along C for all sufficiently small β if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-062374 | MATHDB-327787

Zero-angle limit conjecture for Kähler-Einstein pairs on log del Pezzo surfaces

摘录 / EXCERPT: Zero-angle limit conjecture. As β tends to zero, (S,C,ω_β) converges in an appropriate sense to a generalized Kähler-Einstein metric ω_∞ on...

conjecture | open | MathDB 页面 | 原文线索

V2-062375 | MATHDB-327788

Conjecture on generating functions and structural formulas for generalized multivariate orthogonal polynomials

摘录 / EXCERPT: Structural-formula conjecture. Generating functions, orthogonality relations, difference equations, and recurrence formulas for the generalized multivariate Meixner, Charlier, and...

conjecture | open | MathDB 页面 | 原文线索

V2-062376 | MATHDB-327794

Conjectured sequence of generic Milnor numbers for homogeneous singularities

摘录 / EXCERPT: Generic Milnor number sequence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062377 | MATHDB-327795

Hooley-Sarnak conjecture on fundamental units

摘录 / EXCERPT: Let $D\geq1$ be a fundamental discriminant, let $\mathbb{Q}(\sqrt D)$ be the associated real quadratic field, and let ε_D denote its fundamental unit. Hooley-Sarnak con...

conjecture | open | MathDB 页面 | 原文线索

V2-062378 | MATHDB-327796

Existence of large cyclic subspace codes in every dimension

摘录 / EXCERPT: Let $\mathcal{G}_{q}(n,k)$ denote the Grassmannian of k-dimensional subspaces of $\mathbb{F}_{q}^{n}$, and let a cyclic code be a subspace code invariant under multiplication by...

conjecture | open | MathDB 页面 | 原文线索

V2-062379 | MATHDB-327798

The elementary abelian existence conjecture for skew Hadamard difference sets

摘录 / EXCERPT: The elementary abelian SHDS conjecture. If G admits an SHDS, then

conjecture | open | MathDB 页面 | 原文线索

V2-062380 | MATHDB-327799

The elementary abelian existence conjecture for generalized skew Hadamard difference sets

摘录 / EXCERPT: The elementary abelian GSHDS conjecture. If G affords a GSHDS, then

conjecture | open | MathDB 页面 | 原文线索

V2-062381 | MATHDB-327800

Fixed-core generating function conjecture for hook-difference statistics

摘录 / EXCERPT: Fixed-core generating function conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062382 | MATHDB-327806

Conjecture on the central bundle of aperiodic classes

摘录 / EXCERPT: Let Λ be an aperiodic class for a C^1-generic diffeomorphism f, and let ... be a dominated splitting such that E^s and E^u are respectively the maximal uniformly cont...

conjecture | open | MathDB 页面 | 原文线索

V2-062383 | MATHDB-327820

Negative explicit-bound conjecture for the Public Good Index

摘录 / EXCERPT: Let P be a power index, let $[q;w_1,\dots,w_n]$ denote a weighted majority game with $q\in(0,1)$, normalized nonnegative weights $w\in\mathbb{R}_{\ge0}^n$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-062384 | MATHDB-327821

Sharp nucleolus norm bound for normalized weights

摘录 / EXCERPT: Let $[q;w_1,\dots,w_n]$ be a weighted majority game with $q\in(0,1)$ and $w\in\mathbb{R}_{\ge0}^n$ normalized by $\Vert w\Vert_1=1$. Let $\Delta=\max_{1\le i\le n}w_i$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062385 | MATHDB-327822

Positive Alon-Edelman conjecture for several power indices

摘录 / EXCERPT: Let an Alon-Edelman type result mean a bound showing that a simple game's power vector is close to the power vector of a game with only the concentrated k voters and the remainin...

conjecture | open | MathDB 页面 | 原文线索

V2-062386 | MATHDB-327823

Uniform Banzhaf and Shapley-Shubik approximation lower bounds

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let $\sigma^n=(0.75,0.25,0,\dots,0)\in\mathbb{R}_{\ge0}^n$, and let v be a simple game on n voters. Uniform approximation lower-bound conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-062387 | MATHDB-327824

Shapley-Shubik one-third approximation conjecture

摘录 / EXCERPT: Let $n\ge3$, let $\sigma\in\mathbb{R}_{\ge0}^n$ satisfy $\Vert\sigma\Vert_1=1$, and write σ^n for the desired power distribution in dimension n. Shapley-Shubik approximati...

conjecture | open | MathDB 页面 | 原文线索

V2-062388 | MATHDB-327825

Banzhaf lower bound for the two-level target distribution

摘录 / EXCERPT: For each $n\ge2$, define ... A weighted majority game is a game represented by a quota and nonnegative voter weights. Two-level Banzhaf lower-bound conjecture. There is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-062389 | MATHDB-327826

Exact Banzhaf realization of the two-level target distribution

摘录 / EXCERPT: For each $n\ge6$, define ... A simple game is a monotone voting game on n voters, and $\operatorname{Bz}(v)$ denotes its normalized Banzhaf power vector. Exact two-level Banzhaf co...

conjecture | open | MathDB 页面 | 原文线索

V2-062390 | MATHDB-327829

The new-condition flatness conjecture for ramified unitary local models

摘录 / EXCERPT: Let F/F_0 be a ramified quadratic extension of discretely valued non-Archimedean fields with common residue field of characteristic not 2, let $n\geq 2$, let (r,s) be any sig...

conjecture | open | MathDB 页面 | 原文线索

V2-062391 | MATHDB-327840

Pseudo-filling symmetry conjecture for Gasharov-Reiner permutations

摘录 / EXCERPT: Let w be a permutation in the class avoiding 4231, 35142, 42513, and 351624. Let $PF^{{\rm L}}_w(q)$ and $PF^{\rotatebox[origin=c]{180}{\scriptsize {\rm L}}}_w(q)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-062392 | MATHDB-327842

Bessel polynomial Turán stability conjecture

摘录 / EXCERPT: Let $\mathcal{P} = \{Y_k(x)\}_{k=0}^\infty$, where Y_k is the k-th Bessel polynomial, with ... The extended Turán expression $\mathscr{T}_k^{(n)}(\mathcal{P};x)$ is defined for...

conjecture | open | MathDB 页面 | 原文线索

V2-062393 | MATHDB-327846

The forbidden ordering conjecture for omega-values in 3-generated numerical monoids

摘录 / EXCERPT: Let $\Gamma = \langle n_1,n_2,n_3\rangle$ be a numerical monoid minimally generated by $n_1<n_2<n_3$, and let $\omega(n_i)$ denote the omega-value of n_i. The forbidden ordering...

conjecture | open | MathDB 页面 | 原文线索

V2-062394 | MATHDB-327847

The exact-period conjecture for eventual quasilinearity of the omega-function

摘录 / EXCERPT: Let Γ be a numerical monoid with least generator n_1, and let $\omega(n)$ denote its omega-function. Exact-period conjecture. For all sufficiently large n, the function...

conjecture | open | MathDB 页面 | 原文线索

V2-062395 | MATHDB-327851

The backward orbit integrality conjecture for Lattés maps

摘录 / EXCERPT: Backward orbit integrality conjecture. If $Q\in\mathbb P^1(\overline K)$ is not ϕ-preperiodic, then $\phi^-(P)$ contains at most finitely many points in...

conjecture | open | MathDB 页面 | 原文线索

V2-062396 | MATHDB-327853

S-adelic Mordell-Lang conjecture for hyperbolic affine curves in tori

摘录 / EXCERPT: Let F be a global field, let X_F be a hyperbolic affine curve contained in $\mathbf P^1_F$, and let $j:X_F\rightarrow T_F$ be a closed immersion into a torus T_F. For a finit...

conjecture | open | MathDB 页面 | 原文线索

V2-062397 | MATHDB-327863

Shellability conjecture for P_1 intervals avoiding two patterns

摘录 / EXCERPT: Let P_1 be the permutation class denoted by P_1 in the source, and let π be a permutation in P_1. Write $[\textbf{1},\pi]$ for the interval from the one-element permutati...

conjecture | open | MathDB 页面 | 原文线索

V2-062398 | MATHDB-327866

The Markov degree conjecture for the incidence configuration of the loopless complete graph K5

摘录 / EXCERPT: Let A be the incidence matrix of the complete graph K_5 without self-loops, namely ... Let ${\mathbb N}A$ denote the semigroup generated by the columns of A, and for each...

conjecture | open | MathDB 页面 | 原文线索

V2-062399 | MATHDB-327869

Exceptional-speed solution-space conjecture for multiple-SLE null-state PDEs

摘录 / EXCERPT: Let $\mathcal{S}_N$ be the SLE partition-function solution space, let $\mathcal{B}_N$ be the distinguished family of solutions, and let $\mathcal{R}_N\subset\mathcal{S}_N$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-062400 | MATHDB-327870

Characterization of positivity by completely monotone Dirichlet-series products

摘录 / EXCERPT: Let $\zeta(s)$ be a generalized Dirichlet series absolutely convergent on the half-plane $\{z: \operatorname{Re} z\ge 1\}$, and suppose that it has a simple zero at $s=1$. A genera...

conjecture | open | MathDB 页面 | 原文线索

V2-062401 | MATHDB-327875

Simple connectedness of the stability space for the quiver Q_1

摘录 / EXCERPT: Let Q_1 be the quiver considered in the source, let $D^b(Q_1)$ denote its bounded derived category, and let $\mathrm{Stab}(D^b(Q_1))$ be the space of stability conditions on it....

conjecture | open | MathDB 页面 | 原文线索

V2-062402 | MATHDB-327893

The twistor-family conjecture for Fukaya-Seidel stability conditions

摘录 / EXCERPT: Twistor-family conjecture. The moduli space of stability conditions of a Fukaya-Seidel category can be included in a one-parameter twistor family.

conjecture | open | MathDB 页面 | 原文线索

V2-062403 | MATHDB-327898

The derived-category conjecture for Kuechle fourfolds

摘录 / EXCERPT: Derived-category conjecture for Kuechle fourfolds.

conjecture | open | MathDB 页面 | 原文线索

V2-062404 | MATHDB-327901

The Landau-Ginzburg monodromy conjecture for Orlov-spectrum gaps

摘录 / EXCERPT: Landau-Ginzburg monodromy conjecture. The monodromy of the Landau-Ginzburg models for X determines the gap in the spectrum of $D^b(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062405 | MATHDB-327902

The nonrationality conjecture for cubic, X_{10}, and Kuechle fourfolds

摘录 / EXCERPT: Nonrationality conjecture. These fourfolds are not rational if their semiorthogonal decompositions do not contain the derived category of a commutative K3 surface.

conjecture | open | MathDB 页面 | 原文线索

V2-062406 | MATHDB-327903

The phantom-category conjecture for Sarkisov conic bundles

摘录 / EXCERPT: Sarkisov phantom-category conjecture. The semiorthogonal decomposition of $D^b(X)$ contains a phantom category.

conjecture | open | MathDB 页面 | 原文线索

V2-062407 | MATHDB-327904

The phantom-category conjecture for degenerations of quadric bundles

摘录 / EXCERPT: Quadric-bundle phantom-category conjecture. The semiorthogonal decomposition of the derived category of such a degeneration contains a phantom category.

conjecture | open | MathDB 页面 | 原文线索

V2-062408 | MATHDB-327906

The uniform domination conjecture for Steiner triple systems

摘录 / EXCERPT: Uniform domination conjecture. There exists a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that, if $D\in STS(v)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-062409 | MATHDB-327915

Concavity conjecture for the fractional Dirichlet problem in dimensions at least three

摘录 / EXCERPT: Let $d\geq 3$, and let $D\subset\mathbb{R}^d$ be an arbitrary bounded convex set. Let φ denote the solution of the fractional Dirichlet problem ... with the boundary conditio...

conjecture | open | MathDB 页面 | 原文线索

V2-062410 | MATHDB-327916

Fractional torsion-function concavity conjectures

摘录 / EXCERPT: Let $d\geq 2$, let $D\subset\mathbb{R}^d$ be an arbitrary bounded convex set, and let φ be the solution of the fractional Dirichlet problem ... with the boundary condition fr...

conjecture | open | MathDB 页面 | 原文线索

V2-062411 | MATHDB-327919

Bradford-Ionascu conjecture for unit fractions in Gaussian integers

摘录 / EXCERPT: Bradford-Ionascu conjecture. Such a decomposition exists with $a,b,c\in\mathbb{Z}[i]$ such that the real and imaginary parts of each of a, b, and c are either both nonnegativ...

conjecture | open | MathDB 页面 | 原文线索

V2-062412 | MATHDB-327928

LG-LG mirror conjecture for FJRW and Saito-Givental theories

摘录 / EXCERPT: Let W be an invertible polynomial with maximal diagonal symmetry group G_W, and let $(W^T,\{1\})$ be its Berglund-Hübsch-Krawitz mirror pair. Write $(H_W,\bullet)$ for the FJRW...

conjecture | open | MathDB 页面 | 原文线索

V2-062413 | MATHDB-327934

Interlacing conjecture for Neumann and Dirichlet eigenvalues of self-similar measures

摘录 / EXCERPT: Let μ be a self-similar measure with contraction parameters r_1,r_2 and weights m_1,m_2 satisfying ... Write $\lambda_{N,k}$ and $\lambda_{D,k}$ for the Neumann and Dirichlet...

conjecture | open | MathDB 页面 | 原文线索

V2-062414 | MATHDB-327947

Zero-one law for k-connectivity in the intersection graph of a pairwise key graph and an unreliable-link graph

摘录 / EXCERPT: Let k be a fixed positive integer. Consider scalings $K:\mathbb{N}_0\rightarrow\mathbb{N}_0$ and $p:\mathbb{N}_0\rightarrow[0,1]$, with $K_n=K(n)$ and $p_n=p(n)$, such that ... a...

conjecture | open | MathDB 页面 | 原文线索

V2-062415 | MATHDB-327953

Zaidenberg's finiteness conjecture for Eisenbud-Neumann diagrams

摘录 / EXCERPT: Let $\bar E\subseteq\mathbb{P}^2$ be a rational cuspidal curve, and let $(X,D)\to(\mathbb{P}^2,\bar E)$ be the minimal log resolution of singularities. Zaidenberg's finiteness conj...

conjecture | open | MathDB 页面 | 原文线索

V2-062416 | MATHDB-327964

Finiteness conjecture for Thompson groups of loop braid groups

摘录 / EXCERPT: Let $\Vloop=\Thomp{\mathit{LB}_*}$ and $\Floop=\Thomp{\mathit{PLB}_*}$ be the generalized Thompson groups containing the loop braid groups and pure loop braid groups, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-062417 | MATHDB-327967

The plane extension distinguishing number conjecture

摘录 / EXCERPT: Plane extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062418 | MATHDB-327968

The 2-sphere extension distinguishing number conjecture

摘录 / EXCERPT: 2-sphere extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062419 | MATHDB-327969

The 3-dimensional Euclidean extension distinguishing number conjecture

摘录 / EXCERPT: 3-dimensional Euclidean extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062420 | MATHDB-327972

Akbari et al.'s planar edge Roman domination conjecture

摘录 / EXCERPT: Let G be a planar graph with n vertices, and let $\gamma'_R(G)$ denote its edge Roman domination number: the minimum of $\sum_{e\in E(G)}f(e)$ over all functions...

conjecture | open | MathDB 页面 | 原文线索

V2-062421 | MATHDB-327973

The large-girth planar edge Roman domination conjecture

摘录 / EXCERPT: Let G be a planar graph of girth at least $3k+2$ with n vertices, and let $\gamma'_R(G)$ denote its edge Roman domination number: the minimum of $\sum_{e\in E(G)}f(e)$ over all...

conjecture | open | MathDB 页面 | 原文线索

V2-062422 | MATHDB-327976

Douglas-Reinbacher-Yau conjecture on stable reflexive sheaves

摘录 / EXCERPT: Douglas-Reinbacher-Yau conjecture. There exists a rank-r stable reflexive sheaf $\mathcal E$ on X with respect to some polarization such that $c_1(\mathcal E)=0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062423 | MATHDB-327977

General gluing conjecture for real analytic torsion forms

摘录 / EXCERPT: General gluing conjecture. With the assumption of product structures, the following identity holds in $Q^{S}/Q^{S,0}$:

conjecture | open | MathDB 页面 | 原文线索

V2-062424 | MATHDB-327995

Breuil-Herzig's ordinary parts conjecture for principal series

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$, let G be a connected reductive group over F, and let $P\subset G$ be a standard parabolic subgroup with standard Levi subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-062425 | MATHDB-327996

The extension conjecture for parabolically induced supersingular representations

摘录 / EXCERPT: Let $P,P'\subset G$ be standard parabolic subgroups with standard Levi subgroups $L\subset P$ and $L'\subset P'$, and let π and π' be supersingular representations of...

conjecture | open | MathDB 页面 | 原文线索

V2-062426 | MATHDB-327997

McNamara-Sagan's infinite log-concavity conjecture for Pascal-triangle columns

摘录 / EXCERPT: McNamara-Sagan's conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-062427 | MATHDB-327998

McNamara-Sagan's infinite log-concavity conjecture for Pascal-triangle diagonals

摘录 / EXCERPT: McNamara-Sagan's conjecture. The sequence $\{\binom{n+mu}{mv}\}_{m\geq 0}$ is infinitely log-concave for all $n\geq 0$ if and only if $u<v$ or $v=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-062428 | MATHDB-328002

Conjecture on coincidence of nonlinear and constructed heat flow semigroups

摘录 / EXCERPT: Heat-flow coincidence conjecture. The heat flow semigroup (T_t) defined by the paper's formula for $F:=P_{(D)}$ coincides with the heat flow constructed by Sturm.

conjecture | open | MathDB 页面 | 原文线索

V2-062429 | MATHDB-328005

Characterization of balanced three-variable integral kernels

摘录 / EXCERPT: Three-variable kernel conjecture. If $\alpha\notin\{1/3,2/3\}$, then $f=0$ almost everywhere. If $\alpha=1/3$, then, almost everywhere,

conjecture | open | MathDB 页面 | 原文线索

V2-062430 | MATHDB-328010

Baker and Rumely's tau lower bound conjecture for metrized graphs

摘录 / EXCERPT: Let Γ be a metrized graph, let $\ell(\Gamma)=\int_{\Gamma} dx$ be its total length, and consider metrized graphs with $\ell(\Gamma)\neq 0$. Baker and Rumely's tau lower boun...

conjecture | open | MathDB 页面 | 原文线索

V2-062431 | MATHDB-328011

Cinkir's explicit tau lower bound conjecture for metrized graphs

摘录 / EXCERPT: Let Γ be a metrized graph, let $\tau(\Gamma)$ denote its tau constant, and let $\ell(\Gamma)$ denote its total length. Cinkir's explicit tau lower bound conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-062432 | MATHDB-328013

The u-deformed Macdonald Littlewood identity

摘录 / EXCERPT: Throughout this subsection, let $N=2n$, and let $P_\lambda(x_1,\dots,x_N;q,t)$ denote Macdonald polynomials, with $b^{\rm el}_\lambda(q,t)$ the even-columns Littlewood coefficient....

conjecture | open | MathDB 页面 | 原文线索

V2-062433 | MATHDB-328014

The u-refined Cauchy identity for lifted Koornwinder polynomials

摘录 / EXCERPT: Let $m_0(\lambda)$ be the multiplicity of zero parts of the partition λ, let $P_\lambda(x_1,\dots,x_n;t)$ be the Hall-Littlewood polynomial, and let $b_\lambda(t)$ be its s...

conjecture | open | MathDB 页面 | 原文线索

V2-062434 | MATHDB-328031

The face-number conjecture for perfect prismatoids

摘录 / EXCERPT: Face-number conjecture. For every d-dimensional perfect prismatoid P, there exists a d-dimensional lattice Delaunay polytope Q such that

conjecture | open | MathDB 页面 | 原文线索

V2-062435 | MATHDB-328040

Breadth-determinant conjecture for quasi-alternating links

摘录 / EXCERPT: Breadth-determinant conjecture. If L is a quasi-alternating link, then

conjecture | open | MathDB 页面 | 原文线索

V2-062436 | MATHDB-328047

The diagonal residue product conjecture for affine Weyl groups

摘录 / EXCERPT: Diagonal residue product conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-062437 | MATHDB-328052

Second-moment conjecture for matchings in random biregular graphs

摘录 / EXCERPT: Let G be the random (a,b)-biregular bipartite graph considered in the paper, and let $m_k(G)$ denote the number of matchings of size k. Second-moment conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-062438 | MATHDB-328056

The orthogonal-complement conjecture for transitive subspaces of an operator and its adjoint

摘录 / EXCERPT: Let T be an operator, let T^* be its adjoint, and let $d4dc_1$ and $d4dc_2$ be subspaces. Assume that T is $d4dc_1$-transitive and T^* is $d4dc_2$-transitive. Orthogonal-co...

conjecture | open | MathDB 页面 | 原文线索

V2-062439 | MATHDB-328057

Sufficiency conjecture for reflection-symmetric frameworks in quadrilateral normed planes

摘录 / EXCERPT: Let $\|\cdot\|_{\mathcal{P}}$ be a polyhedral norm on $\mathbb{R}^2$ whose unit ball $\mathcal{P}$ is a quadrilateral. Let G be a finite simple graph, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062440 | MATHDB-328059

Walsh's conjecture on weakly regular bent functions and weighted partial difference sets

摘录 / EXCERPT: Let $p>2$ and let $f:GF(p)^n\to GF(p)$ be a weakly regular bent function corresponding to a weighted strongly regular graph via the stated analogy. Let μ_{ii} denote the diagon...

conjecture | open | MathDB 页面 | 原文线索

V2-062441 | MATHDB-328060

Conjecture on level curves of even bent functions

摘录 / EXCERPT: Let $f:GF(p)^n\to GF(p)$ be an even bent function with $p>2$ and $f(0)=0$. For each value of f, its level curve is the corresponding level set; assume these level curves give ris...

conjecture | open | MathDB 页面 | 原文线索

V2-062442 | MATHDB-328062

Joint symmetry conjecture for generalized q,t-Catalan polynomials

摘录 / EXCERPT: Joint symmetry conjecture. For all $r,s\in\mathbb{Z}$ and all $a,b\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062443 | MATHDB-328063

Joint symmetry conjecture for q,t-rectangle numbers

摘录 / EXCERPT: For $a,b\geq 0$, define the q,t-rectangle numbers ... Here $\mathsf{area}^*_{b,-a}$ is the adjusted area statistic defined using the minimum of the $(b,-a)$-levels. Joint symmetry co...

conjecture | open | MathDB 页面 | 原文线索

V2-062444 | MATHDB-328064

Higher-power nabla conjecture for q,t-rectangle numbers

摘录 / EXCERPT: For $a,b\geq 0$, let $S_{a,b}(q,t)$ be the q,t-rectangle numbers defined from words with a letters N and b letters E, and let p_n and $s_{(1^n)}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062445 | MATHDB-328065

Generalized specialized q,t-rectangle formula

摘录 / EXCERPT: For $a,b\geq 0$, write $b=b'k$ and $a=a'k$ for integers $a',b',k\geq 0$ with $\gcd(a',b')=1$, and let $S_{a,b}(q,t)$ be the q,t-rectangle number. Let $[j]_{q^k}$ denote the q-integ...

conjecture | open | MathDB 页面 | 原文线索

V2-062446 | MATHDB-328068

The B_2 eigenfunction conjecture for the series f^{B_2}

摘录 / EXCERPT: Let s_1,s_2 be generic parameters, and let $lambda_1,lambda_2$ satisfy $s_1=tT^{1/2}q^{lambda_1}$ and $s_2=T^{1/2}q^{lambda_2}$. Define the series...

conjecture | open | MathDB 页面 | 原文线索

V2-062447 | MATHDB-328069

Metric conjecture for affine preferential attachment seed graphs

摘录 / EXCERPT: Affine preferential attachment metric conjecture. The function d_δ is a metric on trees with at least 3 vertices.

conjecture | open | MathDB 页面 | 原文线索

V2-062448 | MATHDB-328071

Ihara's lemma for unitary groups

摘录 / EXCERPT: Let G be the unitary group and retain the paper's notation for the coefficient system $S_{a,\{\rho_x\},\varnothing}$, the set of places T, and the level subgroup U. A place...

conjecture | open | MathDB 页面 | 原文线索

V2-062449 | MATHDB-328072

Stronger Ihara's lemma for unitary groups

摘录 / EXCERPT: Let G, F^+, F, T, S_{ℓ}, $S(B)$, U^v, $S_{a,\{\rho_x\},\varnothing}$, and $\bar{k}$ have the meanings used in the paper. Let a maximal ideal be non-Eisenstein when i...

conjecture | open | MathDB 页面 | 原文线索

V2-062450 | MATHDB-328081

Conjecture on regular, physical singular, and strange solutions for the spin-5 Heisenberg chain

摘录 / EXCERPT: Let s be the spin, N the system size, and ell the number of Bethe roots. A solution is called regular if it is a regular solution of the Bethe ansatz equations, physical sing...

conjecture | open | MathDB 页面 | 原文线索

V2-062451 | MATHDB-328090

The boundary-intersection generation conjecture for effective cycles on \(\overline{M}_{0,n}\)

摘录 / EXCERPT: Let $\overline{\operatorname{M}}_{0,n}$ be the moduli space of stable n-pointed rational curves. For an integer k with $1\leq k\leq \dim \overline{\operatorname{M}}_{0,n}-1$, c...

conjecture | open | MathDB 页面 | 原文线索

V2-062452 | MATHDB-328096

Cazenave-Kenig-Merle scattering conjecture for the energy-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Cazenave-Kenig-Merle conjecture. When $\mu=1$, solutions exist globally and scatter: for every $u_0\in\dot{H}^1(\mathbb{R}^d)$, there is a unique global solution...

conjecture | open | MathDB 页面 | 原文线索

V2-062453 | MATHDB-328109

Generalized abundance conjecture for dlt pairs

摘录 / EXCERPT: Let (X,Δ) be a $\mathbb Q$-factorial projective dlt pair. The numerical Kodaira dimension $\kappa_\sigma(X,K_X+\Delta)$ and the invariant Kodaira dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-062454 | MATHDB-328110

Good minimal model conjecture for dlt pairs

摘录 / EXCERPT: Let (X,Δ) be a $\mathbb Q$-factorial projective dlt pair, and assume that $K_X+\Delta$ is pseudo-effective. Good minimal model conjecture. The pair (X,Δ) has a good m...

conjecture | open | MathDB 页面 | 原文线索

V2-062455 | MATHDB-328111

Sign conjecture for Hardy inequalities in globally twisted waveguides

摘录 / EXCERPT: Let Ω be a globally twisted waveguide with twist derivative $\dot\theta=\beta+\varepsilon$, where β is the constant twisting rate and ε is a nonzero pertur...

conjecture | open | MathDB 页面 | 原文线索

V2-062456 | MATHDB-328120

Berndt-Chan-Chan-Liaw monotonicity conjecture for dissected crank products

摘录 / EXCERPT: Consider the five infinite products discussed in the source, namely ... Berndt-Chan-Chan-Liaw monotonicity conjecture. Each component in each of the dissections for these five produc...

conjecture | open | MathDB 页面 | 原文线索

V2-062457 | MATHDB-328121

General dissection monotonicity conjecture for products of Ramanujan theta functions

摘录 / EXCERPT: For positive integers α and β, consider the product ... An $(\alpha+\beta+1)$-dissection decomposes this power series according to exponents modulo $\alpha+\beta+1$. Gen...

conjecture | open | MathDB 页面 | 原文线索

V2-062458 | MATHDB-328122

The centrohermitian common-zero reality conjecture

摘录 / EXCERPT: Let $A\in\mathcal{M}(m,m+n)$ be a Toeplitz and centrohermitian matrix, and let P_I denote the associated generalized characteristic polynomials for index sets I with $|I|=m$. A...

conjecture | open | MathDB 页面 | 原文线索

V2-062459 | MATHDB-328123

The real common-zero reformulation for centrohermitian Toeplitz matrices

摘录 / EXCERPT: Let $A\in\mathcal{M}(m,m+n)$ be a Toeplitz and centrohermitian matrix, and let P_I be the associated generalized characteristic polynomials with $|I|=m$. Restrict these polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-062460 | MATHDB-328131

The relaxed locally identifying coloring conjecture for free-twin split graphs

摘录 / EXCERPT: Let G be a free-twin split graph, meaning that G is a split graph with no pair of free twins. Write $\chi_{rlid}(G)$ for its relaxed locally identifying chromatic number and...

conjecture | open | MathDB 页面 | 原文线索

V2-062461 | MATHDB-328147

Higher norm descent conjecture for Rubin-Stark units

摘录 / EXCERPT: Let $(L,S,V),(L',S',V')\in\Omega$ with $L\subset L'$, $S\subset S'$, and $V\supset V'$. Put $r=|V|$, $r'=|V'|$, and $d=r-r'$, and let i be the canonical injection from the d-th...

conjecture | open | MathDB 页面 | 原文线索

V2-062462 | MATHDB-328148

Refined higher norm formula for Rubin-Stark units

摘录 / EXCERPT: Let $(L,S,V),(L',S',V')\in\Omega$ with $L\subset L'$, $S\subset S'$, and $V\supset V'$. Put $r=|V|$, $r'=|V'|$, and $d=r-r'$. Let $\varphi_v:L^\times\to Q_{L'/L}^1$ be the local-re...

conjecture | open | MathDB 页面 | 原文线索

V2-062463 | MATHDB-328149

Burns-style congruence conjecture for Rubin-Stark units

摘录 / EXCERPT: Burns-style congruence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062464 | MATHDB-328155

All-coupling frequency independence of Sturmian spectral dimensions

摘录 / EXCERPT: Let $d706>0$ and let $d6fc\in(0,1)\setminus\mathbb{Q}$ be an irrational frequency. Denote by $d6a9_{\lambda,\alpha}$ the phase-independent spectrum of the corresponding Sturmian Sc...

conjecture | open | MathDB 页面 | 原文线索

V2-062465 | MATHDB-328157

Finiteness conjecture for permutation binomials over finite fields

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many pairs (q,a) for which f is a permutation polynomial of $\mathbb{F}_{q^2}$.

conjecture | open | MathDB 页面 | 原文线索

V2-062466 | MATHDB-328161

Yau's conjecture on Griffiths and Hironaka number inequalities

摘录 / EXCERPT: Yau's conjecture. The inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-062467 | MATHDB-328170

The k-ary rank conjecture for monomials

摘录 / EXCERPT: The k-ary rank conjecture. The upper bound is an equality:

conjecture | open | MathDB 页面 | 原文线索

V2-062468 | MATHDB-328175

The multiplicity-growth limit conjecture for automata-group transition operators

摘录 / EXCERPT: Let $M_{k=j}^{(n)}$ be the transition operator indexed by the non-negative integer j, let $\operatorname{Sp}(M_{k=j}^{(n)})$ denote its spectrum, and let $m^{(n;j)}(\lambda)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-062469 | MATHDB-328182

Lexicographic shellability conjecture for the uncrossing matching poset

摘录 / EXCERPT: Let P_n be the uncrossing partial order on matchings, and let ${\hat P}_n$ denote its augmented poset. Lexicographic shellability conjecture. The poset ${\hat P}_n$ is lexicograp...

conjecture | open | MathDB 页面 | 原文线索

V2-062470 | MATHDB-328185

Maximum-rank stress conjecture for global rigidity on surfaces

摘录 / EXCERPT: Let S be a surface, let (G,p) be a framework generic on S, and let (ω,λ) be an equilibrium stress for (G,p). Write $\Omega_S(\omega,\lambda)$ for its stress ma...

conjecture | open | MathDB 页面 | 原文线索

V2-062471 | MATHDB-328215

Fisk's Legendre zero-preservation conjecture

摘录 / EXCERPT: Let $f(x)=\sum_{k=0}^n a_kx^k\in\mathbb{R}[x]$ have all its zeros in $(-1,1)$, and define the linear transformation T by ... where $P_k(x)$ is the k-th Legendre polynomial. Fisk'...

conjecture | open | MathDB 页面 | 原文线索

V2-062472 | MATHDB-328216

Jacobi polynomial zero-preservation conjecture

摘录 / EXCERPT: Jacobi zero-preservation conjecture. If α and β are non-negative integers, then $T_{\alpha,\beta}$ preserves the set of polynomials whose zeros lie only in $(-1,1)$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-062473 | MATHDB-328223

Unique minimal diagram conjecture for free knots of girth at least four

摘录 / EXCERPT: Let K be a framed four-valent graph, viewed as a free knot diagram, and let its girth be the length of its shortest cycle. If ... then the unique minimal diagram conjecture. The di...

conjecture | open | MathDB 页面 | 原文线索

V2-062474 | MATHDB-328232

Kühn, Lapinskas, Osthus, and Patel's linear connectivity conjecture for tournaments

摘录 / EXCERPT: A tournament is a directed graph with exactly one directed edge between any two distinct vertices. A tournament is strongly r-connected if deleting fewer than r vertices leaves...

conjecture | open | MathDB 页面 | 原文线索

V2-062475 | MATHDB-328233

Kühn-Osthus-Townend partition conjecture for highly connected tournaments

摘录 / EXCERPT: Kühn-Osthus-Townend conjecture. There is a constant C such that the vertices of every strongly Ctk-connected tournament can be partitioned into t strongly k-connected subto...

conjecture | open | MathDB 页面 | 原文线索

V2-062476 | MATHDB-328241

The spanning conjecture for complete cd-indices of Bruhat intervals

摘录 / EXCERPT: Spanning conjecture. For all $n>0$, the complete cd-indices of all Bruhat intervals of rank $n+1$ span the whole space of cd-polynomials of degree bounded by n whose nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-062477 | MATHDB-328242

The nonexistence conjecture for universal relations among Bruhat-interval coefficients

摘录 / EXCERPT: Nonexistence conjecture. Let $n>0$. Then there are no nontrivial relations of the form

conjecture | open | MathDB 页面 | 原文线索

V2-062478 | MATHDB-328244

Growth-bound reconstruction conjecture for relational structures

摘录 / EXCERPT: Let $\mathcal{M}$ be a relational structure and let $G=\operatorname{Aut}(\mathcal{M})$. Let $F_n(G)$ denote the relevant orbit-counting sequence, and suppose that $F_n(G)$ is know...

open_problem | open | MathDB 页面 | 原文线索

V2-062479 | MATHDB-328251

Weak hypercyclicity criterion for bilateral weighted shifts

摘录 / EXCERPT: Let $1\leq p<\infty$, and let T be the bilateral weighted shift on $\ell^p(\mathbb{Z})$ defined by ... for a bounded positive weight sequence $(w_j)_{j\in\mathbb{Z}}$. Weak hypercy...

conjecture | refuted | MathDB 页面 | 原文线索

V2-062480 | MATHDB-328255

Non-realizability of the spun Kishino knot

摘录 / EXCERPT: Let KI be the Kishino virtual knot, and let $S_T(KI)$ denote its spun virtual 2-torus obtained by the construction $S_T(F\times [0,1],K)=(F\times S^1\times [0,1],K\times S^1)$....

conjecture | open | MathDB 页面 | 原文线索

V2-062481 | MATHDB-328256

Completeness of virtual broken surface diagrams with graph changes

摘录 / EXCERPT: A virtual broken surface diagram (VBSD) is a broken surface diagram for a virtual 2-link; a graph change is a modification that preserves the Gauss code of the double-point curve...

conjecture | open | MathDB 页面 | 原文线索

V2-062482 | MATHDB-328257

Independence of graph changes from Roseman moves

摘录 / EXCERPT: A graph change of a virtual broken surface diagram is a modification preserving the Gauss code of the double-point curves that either passes a double-point curve through another cu...

conjecture | open | MathDB 页面 | 原文线索

V2-062483 | MATHDB-328266

The twelve-neighbor common-neighbor conjecture for close-packed configurations

摘录 / EXCERPT: Common-neighbor conjecture. If $z,z'\in Z$ satisfy $\\#N(z)=\\#N(z')=12$ and $z\in N(z')$, then

conjecture | open | MathDB 页面 | 原文线索

V2-062484 | MATHDB-328286

Period-set characterization conjecture for degree-one circle maps

摘录 / EXCERPT: Period-set characterization conjecture. There exist sets $E_c,E_d\subset\mathbb N$, each a finite union of tails of the orderings $\leso{2}$ and $\leso{3}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-062485 | MATHDB-328288

The Tarski alternative for Boolean inverse monoids

摘录 / EXCERPT: The Tarski alternative. Exactly one of the following is true:

conjecture | open | MathDB 页面 | 原文线索

V2-062486 | MATHDB-328293

The stable-derived/Fukaya-Seidel category equivalence for exceptional unimodal singularities

摘录 / EXCERPT: Stable-derived/Fukaya-Seidel equivalence. There is an equivalence of triangulated categories

conjecture | open | MathDB 页面 | 原文线索

V2-062487 | MATHDB-328294

Dolgachev's lattice mirror-symmetry conjecture for K3 hypersurfaces

摘录 / EXCERPT: Dolgachev's conjecture. There exist a lattice $\check{M}_\Delta$ and an orthogonal decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-062488 | MATHDB-328301

Function-field higher additive divisor conjecture

摘录 / EXCERPT: Function-field higher additive divisor conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-062489 | MATHDB-328302

Non-universality conjecture for intersection graphs of subdivisions of a fixed graph

摘录 / EXCERPT: Let Y be a fixed graph, and let $\mathcal{G}(Y)$ be the class of intersection graphs of connected subgraphs of subdivisions of Y. Non-universality conjecture. For every fixed g...

conjecture | open | MathDB 页面 | 原文线索

V2-062490 | MATHDB-328310

The hard Lefschetz quantized crepant resolution conjecture

摘录 / EXCERPT: Let $\mathcal{X}\to X\leftarrow Y$ be a Hard Lefschetz crepant resolution diagram for which the Coates--Iritani--Tseng crepant resolution conjecture holds. For...

conjecture | open | MathDB 页面 | 原文线索

V2-062491 | MATHDB-328312

Kirillov's Conjecture 7(A) on positivity for reduced forms of complete graphs

摘录 / EXCERPT: Kirillov's Conjecture 7(A). For every k, the coefficient polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-062492 | MATHDB-328315

Embeddedness conjecture for the $(m,n,m+n-1)$ family of harmonic ends

摘录 / EXCERPT: Embeddedness conjecture. This end is a complete, properly embedded end of type $(m,n,m+n-1)$. The surrounding discussion says that numerical evidence supports embeddedness, but tha...

conjecture | open | MathDB 页面 | 原文线索

V2-062493 | MATHDB-328320

Refined invariant generating-function conjecture for weighted projective planes

摘录 / EXCERPT: Let $N_\delta(d,m;y)$ be the polynomial from the weighted-projective refined node polynomial conjecture, let $\widetilde{N}^{(\Sigma_m,dH),\delta}(y)$ be the refined invariant on th...

conjecture | open | MathDB 页面 | 原文线索

V2-062494 | MATHDB-328322

Coefficient polynomiality threshold conjecture for refined Severi degrees

摘录 / EXCERPT: Fix $\delta\ge1$, and write the refined plane Severi degree as ... Let $N_\delta(d;y)$ be the refined node polynomial, with coefficients $p^\delta_i(d)$ defined analogously. Coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-062495 | MATHDB-328326

Mason-Shorten-Angeli conjecture on stability of positive linear switched systems

摘录 / EXCERPT: Mason-Shorten-Angeli conjecture. If the system is a positive linear switched system, then this Hurwitz condition is sufficient for global uniform asymptotic stability.

conjecture | open | MathDB 页面 | 原文线索

V2-062496 | MATHDB-328327

Schatten-class characterization for truncated Hankel operators of one-component inner functions

摘录 / EXCERPT: Let θ be a one-component inner function, let $\mathcal K_\theta=H^2\ominus\theta H^2$, and let $\Gamma_{\phi}:\mathcal K_\theta\to\overline{z\mathcal K_\theta}$ be a truncat...

conjecture | open | MathDB 页面 | 原文线索

V2-062497 | MATHDB-328328

Decomposability conjecture for multiplicative groups of infinite fields

摘录 / EXCERPT: Let F be an infinite field, and write $F^{\times}$ for its multiplicative group. A group is decomposable if it is isomorphic to a direct sum of two nontrivial subgroups. Decompos...

conjecture | open | MathDB 页面 | 原文线索

V2-062498 | MATHDB-328333

Extension of the discretization bounds to general linear switching systems

摘录 / EXCERPT: Let ${\cal A}$ be a set of $d\times d$ matrices, let r be the maximal spectral radius of the matrices, let ε be the prescribed accuracy, and let $K(r,\varepsilon)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-062499 | MATHDB-328335

Steuding's joint universality conjecture for primitive Selberg-class functions

摘录 / EXCERPT: Let $\varphi_1,\ldots,\varphi_r$ be functions in the Selberg class, and suppose they are distinct and primitive. Steuding's joint universality conjecture. Any finite collection...

conjecture | open | MathDB 页面 | 原文线索

V2-062500 | MATHDB-328344

Genus-zero conjecture for reduced spectral curves of trivial flat connections

摘录 / EXCERPT: Genus-zero conjecture. The curve described by $R(X,y)=0$ has genus 0, although it may be singular and have several components.

conjecture | open | MathDB 页面 | 原文线索

V2-062501 | MATHDB-328352

Bellaterra graphs are two-sided expanders

摘录 / EXCERPT: Bellaterra expander conjecture. The Bellaterra graphs $\{B_i\}_{i=1}^\infty$ are a family of two-sided expanders. The authors' computations suggest an eigenvalue gap of roughly...

conjecture | open | MathDB 页面 | 原文线索

V2-062502 | MATHDB-328353

Stability conjecture for minimal relations of canonical rings on the projective line

摘录 / EXCERPT: Let D be a ... , supported at points $P_1,,P_n$, and let S_D be its canonical ring. A minimal relation means a relation in a minimal presentation of S_D by generators. The poin...

conjecture | open | MathDB 页面 | 原文线索

V2-062503 | MATHDB-328362

Finite-dimensionality criterion for local Picard groups of S2 surfaces

摘录 / EXCERPT: Let (s,S) be an S_2 surface. Finite-dimensionality conjecture. There is a finite-dimensional local Picard group $\operatorname{\mathbf{Pic}}^{\rm loc}(s,S)$ if and only if S...

conjecture | open | MathDB 页面 | 原文线索

V2-062504 | MATHDB-328363

Weight-two injectivity conjecture for links of complex analytic spaces

摘录 / EXCERPT: Let X be a complex analytic space that is S_2 and has pure dimension ≥ 3, let $x\in X$ be a point, and let $\pi:\bar X\to X$ be the normalization with corresponding point...

conjecture | open | MathDB 页面 | 原文线索

V2-062505 | MATHDB-328365

The Bordeaux Conjecture of Borodin and Raspaud

摘录 / EXCERPT: For a graph G, let $d^{\bigtriangledown}$ denote the minimum distance between distinct triangles of G. Borodin and Raspaud's Bordeaux conjecture. Every planar graph with...

conjecture | open | MathDB 页面 | 原文线索

V2-062506 | MATHDB-328368

The connected-graph bound for eternal connected domination

摘录 / EXCERPT: Let G be a connected graph on n vertices, let $\Delta(G)$ denote its maximum degree, let $\theta(G)$ denote its clique covering number, and let $\gamma_c^{\infty}(G)$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-062507 | MATHDB-328369

The upper bound for m-eternal eviction domination

摘录 / EXCERPT: Let $G=(V,E)$ be a graph with n vertices and no isolated vertices, and let $\rho_{\mathrm{m}}^{\infty}(G)$ denote its m-eternal eviction domination number. The m-eternal eviction...

conjecture | open | MathDB 页面 | 原文线索

V2-062508 | MATHDB-328383

Rigid deformation conjecture for monomial ideals on regular CW-complexes

摘录 / EXCERPT: Let I be a monomial ideal with a minimal resolution supported on a regular CW-complex having the intersection property, meaning that the intersection of any two cells is also a c...

conjecture | open | MathDB 页面 | 原文线索

V2-062509 | MATHDB-328386

Cocycle-invariant support conjecture for the Rig quandle Q(12,5)

摘录 / EXCERPT: Let $X=Q(12,5)$, let $A=\mathbb{Z}_4$, and let $\phi:X\times X\rightarrow A$ be the generating 2-cocycle chosen in the source's example. For a knot K, let $\Phi_\phi(K)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062510 | MATHDB-328387

Cocycle-invariant support conjecture for the Rig quandle Q(12,6)

摘录 / EXCERPT: Let $X=Q(12,6)$, let $A=\mathbb{Z}_4$, and let $\phi:X\times X\rightarrow A$ be the generating 2-cocycle chosen in the source's example. For a knot K, let $\Phi_\phi(K)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062511 | MATHDB-328388

Cocycle-invariant support conjecture for the Rig quandle Q(12,10)

摘录 / EXCERPT: Let $X=Q(12,10)$, let $A=\mathbb{Z}_6$, and let $\phi:X\times X\rightarrow A$ be the generating 2-cocycle chosen in the source's example. For each knot K, write its quandle coc...

conjecture | open | MathDB 页面 | 原文线索

V2-062512 | MATHDB-328397

Edge self-amalgamation genus-distribution conjecture

摘录 / EXCERPT: Let (H,e,f) be a doubly edge-rooted graph with 2-valent roots e and f. Consider the two graphs that can be formed by amalgamating the edge-roots e and f. Edge self-amalga...

conjecture | open | MathDB 页面 | 原文线索

V2-062513 | MATHDB-328403

The 3-colorability conjecture for triangle- and ISK4-free graphs

摘录 / EXCERPT: Here graphs are simple and finite. A graph is triangle-free if it contains no triangle and ISK4-free if it contains no induced subdivision of K_4, where K_4 is the complete gra...

conjecture | open | MathDB 页面 | 原文线索

V2-062514 | MATHDB-328404

Constant local Hölder exponent conjecture for refinable functions

摘录 / EXCERPT: Let φ be a refinable function, and suppose that its local Hölder exponent is constant and equal to α. Constant local Hölder exponent conjecture. Then ... The paper stu...

conjecture | open | MathDB 页面 | 原文线索

V2-062515 | MATHDB-328415

The GDN critical exponent and the index of primitivity

摘录 / EXCERPT: Let CE_n denote the critical exponent for n-by-n generalized doubly nonnegative matrices, and let MIP_n be the maximum index of primitivity among primitive n-by-n gener...

conjecture | open | MathDB 页面 | 原文线索

V2-062516 | MATHDB-328427

Forbidden-subgraph characterization of distributive Cambrian lattices

摘录 / EXCERPT: Forbidden-subgraph conjecture. The list in Proposition 1.6 is exhaustive: if $Gamma_{gamma}(W)$ contains none of the listed induced subgraphs, then $\mathcal{B}_{gamma}$ is distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-062517 | MATHDB-328428

The b-conjecture for Jack symmetric functions and nonorientable hypermaps

摘录 / EXCERPT: The b-conjecture. The series $\Psi(\mathbf{x},\mathbf{y},\mathbf{z};t,1+b)$ has coefficients that are polynomials in b with non-negative integer coefficients. The constant term...

conjecture | open | MathDB 页面 | 原文线索

V2-062518 | MATHDB-328438

Anosov's conjecture on identical cycles of generic polynomial foliations

摘录 / EXCERPT: Anosov's conjecture. A generic polynomial foliation has no identical cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-062519 | MATHDB-328440

Amdeberhan-Chen-Moll-Sagan valuation conjecture for generalized Fibonacci sequences

摘录 / EXCERPT: Let $f_n(s,t)$ denote the generalized Fibonacci sequence, and let $ u_d$ be the d-adic valuation. For an integer d', write $\delta_{d'\mathbb{Z}}(n)=1$ if d' divides n and...

conjecture | open | MathDB 页面 | 原文线索

V2-062520 | MATHDB-328441

ACMS reciprocal-sum formulas for negative generalized Fibonacci parameters

摘录 / EXCERPT: Let $f_n(s,t)$ be the generalized Fibonacci sequence, let $\mathbb{Z}^+$ denote the positive integers, and let $\lfloor\cdot\rfloor$ be the floor function. ACMS conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-062521 | MATHDB-328450

The conjecture that Catalan numbers are not tile counting functions

摘录 / EXCERPT: Catalan non-representability conjecture. The Catalan numbers C_n are not a tile counting function.

conjecture | open | MathDB 页面 | 原文线索

V2-062522 | MATHDB-328451

The cohort characterization of Wilf-equivalence for Catalan structures

摘录 / EXCERPT: Let arch systems be Catalan structures, and let $A\sim B$ denote the equivalence relation generated by the four intrinsic rules defined in the paper. For an arch system A, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062523 | MATHDB-328452

The maximal-size conjecture for the main cohort

摘录 / EXCERPT: For each positive integer n, let ${\mathcal M}_n$ denote the main cohort of arch systems with n arches, and let a cohort be an equivalence class under $\sim$. Main-cohort max...

conjecture | open | MathDB 页面 | 原文线索

V2-062524 | MATHDB-328453

The finite-size separation conjecture for distinct arch-system cohorts

摘录 / EXCERPT: Let A and B be arch systems, each with n arches. Let $\operatorname{Av}(A)$ and $\operatorname{Av}(B)$ be their avoidance classes, and let their enumeration sequences count o...

conjecture | open | MathDB 页面 | 原文线索

V2-062525 | MATHDB-328457

DoF conjecture for the general 3-user $(p,p+1)$ MIMO interference channel

摘录 / EXCERPT: Consider a 3-user $(p,p+1)$ MIMO interference channel with constant channel coefficients, where each transmitter has p antennas and each receiver has $p+1$ antennas. The degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-062526 | MATHDB-328458

The giant component size conjecture in the Waiter-Client game

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $\mathcal L(n,q)$ denote the largest component size that Waiter can force in Client's graph when playing a $(q:1)$ Waiter-C...

conjecture | open | MathDB 页面 | 原文线索

V2-062527 | MATHDB-328460

Achievability of symmetric capacity via equal-amplitude diagonal entries

摘录 / EXCERPT: Achievability conjecture. There exists a positive number $P^*(K)$ such that, for every $P\geq P^*(K)$, one can choose real numbers $\beta_2,\ldots,\beta_K$ with $|\beta_k|\geq 1$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-062528 | MATHDB-328466

Deterministic cyclic limit conjecture for polling systems with renewal arrivals

摘录 / EXCERPT: Let a cyclic polling system have general renewal arrival processes and deterministic switch-over times. The system is considered in the regime where the deterministic switch-over t...

conjecture | open | MathDB 页面 | 原文线索

V2-062529 | MATHDB-328481

A q-Clausen-type product identity for terminating basic hypergeometric sums

摘录 / EXCERPT: Let n and r be nonnegative integers with $r\leqslant n$, and let q,x be parameters for which the displayed expressions are defined. The q-Clausen-type conjecture. ... The text...

conjecture | open | MathDB 页面 | 原文线索

V2-062530 | MATHDB-328483

Strengthening of the q-supercongruences modulo [p]^2

摘录 / EXCERPT: Let the congruences denoted by (new-RV2)--(new-RV4) be the three earlier q-supercongruences referenced by the source. Strengthening conjecture. Those congruences also hold modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-062531 | MATHDB-328484

A further q-supercongruence for the 2k central-factorial sum

摘录 / EXCERPT: Let p,m,r satisfy the hypotheses of the source's Theorem 2k2k2k-1. Further q-supercongruence conjecture. ... The supplied text contains only the cross-reference to the theorem's hy...

conjecture | open | MathDB 页面 | 原文线索

V2-062532 | MATHDB-328488

Parabolic characterization conjecture for transitive quasiparabolic sets

摘录 / EXCERPT: Parabolic characterization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-062533 | MATHDB-328489

QP1-QP2 conjecture for conjugacy classes of twisted involutions

摘录 / EXCERPT: QP1-QP2 conjecture. Every conjugacy class in I^+ satisfying (QP1) also satisfies (QP2), and hence is quasiparabolic.

conjecture | open | MathDB 页面 | 原文线索

V2-062534 | MATHDB-328490

Strong exchange conjecture for twisted involution conjugacy classes

摘录 / EXCERPT: Strong exchange conjecture. Under this condition,

conjecture | open | MathDB 页面 | 原文线索

V2-062535 | MATHDB-328495

Hitting clause-sets minimize variable-degree in the generalized stable class

摘录 / EXCERPT: For $k,d\in\mathbb{N}$, let $\mathcal{U}\mathcal{HIT}_{\delta=k}^d$ be the class of generalized unsatisfiable hitting clause-sets of deficiency k with domain sizes at most d, a...

conjecture | open | MathDB 页面 | 原文线索

V2-062536 | MATHDB-328496

Finite-pattern classification of minimally unsatisfiable clause-sets by deficiency

摘录 / EXCERPT: For each $k\in\mathbb{N}$, let $\mathcal{M}\mathcal{U}_{\delta=k}$ be the class of minimally unsatisfiable clause-sets of deficiency k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062537 | MATHDB-328510

Felder-Hemery-Veselov conjecture on zeros of Wronskians of Hermite polynomials

摘录 / EXCERPT: Felder-Hemery-Veselov conjecture. For every doubled partition $\lambda=(\mu_1^2,\dots,\mu_n^2)$, H_λ has no real roots and has as many imaginary roots as there are odd numb...

conjecture | open | MathDB 页面 | 原文线索

V2-062538 | MATHDB-328513

The Hölder 1/2 a priori estimate conjecture for the Hilbert transport equation

摘录 / EXCERPT: Let θ be a bounded solution on $[0,T]$ of ... where H is the Hilbert transform. The $C^{1/2}$ Hölder seminorm measures the quotient $|\theta(t,x)-\theta(t,y)|/|x-y|^{1/2}$....

conjecture | open | MathDB 页面 | 原文线索

V2-062539 | MATHDB-328514

The smoothness conjecture for supercritically dissipative Hilbert models

摘录 / EXCERPT: Let θ_0 be initial data satisfying the hypotheses of the Hölder $1/2$ a priori estimate conjecture, and consider the supercritically dissipative Hilbert model ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-062540 | MATHDB-328515

The one-sided fractional Laplacian bound conjecture for vanishing-viscosity solutions

摘录 / EXCERPT: Let θ_0 satisfy the hypotheses of the Hölder $1/2$ a priori estimate conjecture, and let θ be a vanishing-viscosity limit of solutions of ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-062541 | MATHDB-328517

Anderson's paving conjecture

摘录 / EXCERPT: Let $\varepsilon>0$. For every $n\times n$ Hermitian matrix T with zero diagonal, a diagonal projection is an orthogonal projection represented by a diagonal matrix. Anderson's p...

conjecture | open | MathDB 页面 | 原文线索

V2-062542 | MATHDB-328534

Adversarial binary interactive capacity conjecture without shared randomness

摘录 / EXCERPT: Consider fully adversarial binary error channels without shared randomness, with noise rate $epsilon\to0$. Let R denote the interactive channel capacity, namely the maximal asymp...

conjecture | open | MathDB 页面 | 原文线索

V2-062543 | MATHDB-328535

The rigid Lyapunov-Harder-Narasimhan conjecture

摘录 / EXCERPT: Let C be a Teichmüller curve of genus g, with Lyapunov exponents $\lambda_1\geq\cdots\geq\lambda_g$ and normalized Harder-Narasimhan numbers $w_1\geq\cdots\geq w_g$. Let k be...

conjecture | open | MathDB 页面 | 原文线索

V2-062544 | MATHDB-328536

The Kontsevich-Zorich hyperelliptic Lyapunov exponent conjecture

摘录 / EXCERPT: For each genus g, consider the hyperelliptic components $\overline{\mathcal{H}}^{\mathrm{hyp}}_g(2g-2)$ and $\overline{\mathcal{H}}^{\mathrm{hyp}}_g(g-1,g-1)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062545 | MATHDB-328552

Bishop's conjecture for the minimal 3-partition of the sphere

摘录 / EXCERPT: Let Γ be the sphere and consider the problem of minimizing the sum of the first Dirichlet eigenvalues over partitions of Γ into three parts. A Y-partition is the part...

conjecture | open | MathDB 页面 | 原文线索

V2-062546 | MATHDB-328555

Image maximality conjecture for the DH-matrix

摘录 / EXCERPT: Image maximality conjecture. The system ${\bf D}$ is image maximal.

conjecture | open | MathDB 页面 | 原文线索

V2-062547 | MATHDB-328565

The Pieri formulas conjecture for Schur superpolynomials

摘录 / EXCERPT: Let s_Λ be a Schur superpolynomial indexed by a superpartition, and let $(;r)$, $(r;)$, $(;1^r)$ and $(0;1^r)$ denote the corresponding bosonic and fermionic row and column...

conjecture | open | MathDB 页面 | 原文线索

V2-062548 | MATHDB-328566

The bilinear identities conjecture for rectangular Schur superpolynomials

摘录 / EXCERPT: Let k,n be integers with $k>1$ and $n>1$. Let $r',r\in\{k,k-1,0\}$ satisfy $r'>r$, and set $\epsilon=\delta_{r,0}$. Here k^n denotes the ordinary partition with n parts equal...

conjecture | open | MathDB 页面 | 原文线索

V2-062549 | MATHDB-328567

The bilinear identities conjecture for almost rectangular Schur superpolynomials

摘录 / EXCERPT: Let k,n be integers with $k>1$ and $n>1$, and use the semicolon to separate the fermionic and bosonic parts of a superpartition. Almost rectangular bilinear identities conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-062550 | MATHDB-328582

The irreducibility characterization of primitive nonnegative tensors

摘录 / EXCERPT: Irreducibility characterization conjecture. A nonnegative tensor $\mathbb{A}$ is primitive if and only if $\mathbb{A}$ is irreducible and there exists some $j\in[n]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-062551 | MATHDB-328593

The circle rigidity conjecture for n-fine curves

摘录 / EXCERPT: Let γ be an n-fine curve, and consider its respective equitangent n-gons. Circle rigidity conjecture. If the vertices of the respective equitangent n-gons lie on a cir...

conjecture | open | MathDB 页面 | 原文线索

V2-062552 | MATHDB-328594

Near-unimodal coefficient conjecture for partition generating functions

摘录 / EXCERPT: Let $F_\lambda(q)$ and $G_\lambda(q)$ be the generating functions associated with a partition λ, and write their coefficients in increasing powers of q. A coefficient seq...

conjecture | open | MathDB 页面 | 原文线索

V2-062553 | MATHDB-328595

Near-rectangular unimodality conjecture for partition generating functions

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\ldots,\lambda_b)$ be a partition with b distinct parts. For positive quantities depending on b, write $\lambda_1\gg b$ for the source's asymptotic larg...

conjecture | open | MathDB 页面 | 原文线索

V2-062554 | MATHDB-328597

Primitive-point asymptotic for quotients of products of projective spaces

摘录 / EXCERPT: Let $G\subset S_n$ be a transitive subgroup, let $X=(\mathbb{P}^m)^n/G$, and assume $m\cdot\operatorname{ind}(G)\ge 2$. Let $X(K)^{\operatorname{prim}}$ denote the primitive K-po...

conjecture | open | MathDB 页面 | 原文线索

V2-062555 | MATHDB-328599

Douady-Hubbard mateability conjecture for quadratic polynomials

摘录 / EXCERPT: Douady-Hubbard's conjecture. The points c_1 and c_2 do not lie in conjugate limbs of the Mandelbrot set if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062556 | MATHDB-328600

Tan Lei's parameter-space conjecture for the cubic Newton family

摘录 / EXCERPT: Tan Lei's parameter-space conjecture. The fundamental part of the cubic Newton family is homeomorphic to the quotient of the union of those two subsets by the equivalence relation...

conjecture | open | MathDB 页面 | 原文线索

V2-062557 | MATHDB-328606

Congruences for self-conjugate partition counts at multiples of 36

摘录 / EXCERPT: Congruences for $[...]$.

conjecture | open | MathDB 页面 | 原文线索

V2-062558 | MATHDB-328608

Correction-factor conjecture for the 2-truncated hook-length formula

摘录 / EXCERPT: 2-truncation correction conjecture. For every n, its constant, b, and b^2 coefficients agree with those of

conjecture | open | MathDB 页面 | 原文线索

V2-062559 | MATHDB-328611

Shifted Lubell bound conjecture for induced diamond-free families

摘录 / EXCERPT: Shifted Lubell bound conjecture. Every induced diamond-free family $\mathcal{F}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-062560 | MATHDB-328616

Lattice Stanley projective dimension inequalities conjecture

摘录 / EXCERPT: Lattice Stanley projective dimension conjecture. For all finite lattices L, the following inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-062561 | MATHDB-328617

The sum of squared logarithms inequality in n dimensions

摘录 / EXCERPT: For $n\in\mathbb{N}$, let $\lambda_1,\lambda_2,\ldots,\lambda_n,\mu_1,\mu_2,\ldots,\mu_n\in\mathbb{R}_+$, and let ... be the elementary symmetric polynomials. The sum of squared loga...

conjecture | open | MathDB 页面 | 原文线索

V2-062562 | MATHDB-328620

Class-group pairing conjecture for cycles of reduced quadratic forms

摘录 / EXCERPT: Class-group pairing conjecture. The classes $\mathfrak{c}_1$ and $\mathfrak{c}_2$ have the same sum n, with $n\leq a$, and

conjecture | open | MathDB 页面 | 原文线索

V2-062563 | MATHDB-328622

Unimodality conjecture for r-stable second hypersimplices

摘录 / EXCERPT: Let $\Delta_{n,2}^{\operatorname{stab}(r)}$ denote the r-stable second hypersimplex, and let its Ehrhart $h^\ast$-polynomial be the polynomial whose coefficient sequence is the...

conjecture | open | MathDB 页面 | 原文线索

V2-062564 | MATHDB-328623

Shelling conjecture for the proposed order on $\max\nabla_{n,k}$

摘录 / EXCERPT: Let $\mathfrak{S}_{n,k}^r$ be the set of r-stable permutations, let $\Phi(\mathfrak{S}_{n,k}^r)$ be the associated set on which the graded colexicographic order is imposed, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-062565 | MATHDB-328627

Rational attainment conjecture for metric Mahler measure

摘录 / EXCERPT: Rational attainment conjecture. If α is rational and $t\in(0,\infty]$, then there exist rational points $\alpha_1,\ldots,\alpha_N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062566 | MATHDB-328635

The l2 sufficient condition for factorization limits to be homeomorphisms

摘录 / EXCERPT: Fix a permutation p of the natural numbers, and let σ_N be the associated finite factorization maps with limit $\sigma(p,w)$, where $w=(w_n)$ belongs to the product of uni...

conjecture | open | MathDB 页面 | 原文线索

V2-062567 | MATHDB-328636

The absolute continuity dichotomy for factorization measures

摘录 / EXCERPT: Fix a permutation p and write $\sigma(p,w)(e^{i\theta})=e^{i\Sigma(\theta)}$, where Σ is the associated nondecreasing lift. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062568 | MATHDB-328641

The covering-radius conjecture for small linear codes

摘录 / EXCERPT: Let $Q\subset \mathbb F_2^n$ be an $\mathbb F_2$-linear code of size n^t, where $t=\Theta(1)$. Its covering radius is the maximum, over $u\in\mathbb F_2^n$, of the minimum Hammin...

conjecture | open | MathDB 页面 | 原文线索

V2-062569 | MATHDB-328642

The square-root cancellation conjecture for the hyperbolic spectral exponential sum

摘录 / EXCERPT: Square-root cancellation conjecture. For $X>2$,

conjecture | open | MathDB 页面 | 原文线索

V2-062570 | MATHDB-328644

Sharp distortion conjecture for L_q integer grids in L_p

摘录 / EXCERPT: For $2<q<p$ and $m,n\in\mathbb N$, let $[m]_q^n$ denote the n-dimensional integer grid with its ℓ_q metric, and let $c_p([m]_q^n)$ be its least bi-Lipschitz distortion in...

conjecture | open | MathDB 页面 | 原文线索

V2-062571 | MATHDB-328646

The positivity conjecture for hyperbolic formal root coefficients

摘录 / EXCERPT: Let $v\le w$, let ... and set $z_i=y_{-\beta_i}$. Let $l_{vw\circ}$ be a reduced word as above, and let $\ell(v)$ denote the length of v. The positivity conjecture. (1) If $t=0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062572 | MATHDB-328649

Genericity conjecture for disclination-line energy concentration

摘录 / EXCERPT: In the Landau-de Gennes model, let point defects and disclination lines be the two defect forms, with excessive energy concentrated near the defects. Energy-concentration genericit...

conjecture | open | MathDB 页面 | 原文线索

V2-062573 | MATHDB-328650

Profile conjecture for local defect configurations

摘录 / EXCERPT: Consider defects in the Landau-de Gennes model, including k-disclination lines whose profiles are obtained from the corresponding profile equations. Profile conjecture. The local...

conjecture | open | MathDB 页面 | 原文线索

V2-062574 | MATHDB-328651

Stability conjecture for half-integer disclination lines

摘录 / EXCERPT: Consider disclination lines in the Landau-de Gennes theory, including $7k/2$-disclination lines and the $71/2$-disclination line. Half-integer disclination-line stability conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-062575 | MATHDB-328652

Uniaxial oblate-core conjecture for disclination lines

摘录 / EXCERPT: Let c be the Landau-de Gennes tensor order parameter at a point on a disclination line, and let s denote its scalar order parameter and $c{n}$ its director. Disclination-line c...

conjecture | open | MathDB 页面 | 原文线索

V2-062576 | MATHDB-328679

Stasinski and Voll's hyperoctahedral group factorization conjecture

摘录 / EXCERPT: Stasinski and Voll's conjecture. The polynomial $xt+1$ divides

conjecture | open | MathDB 页面 | 原文线索

V2-062577 | MATHDB-328682

Etingof's conjecture on pure cactus actions and crystals

摘录 / EXCERPT: Let $\CB_{\ul{\l}}$ be the set of highest elements in the tensor product $\CB_{\l_1}\otimes\cdots\otimes\CB_{\l_n}$ of $\fg$-crystals, and let $B_{\ul{\l}}(\ul{z})$ be the set of B...

conjecture | open | MathDB 页面 | 原文线索

V2-062578 | MATHDB-328684

Oper category conjecture for crystals

摘录 / EXCERPT: Let $M_{\l,\mu,\nu}$ be the set of monodromy-free ${}^L G$-opers on $\mathbb{C}\mathbb{P}^1$ with regular singularities of residues $\l,\mu,\nu$ at $0,1,\infty$, respectively. Defi...

conjecture | open | MathDB 页面 | 原文线索

V2-062579 | MATHDB-328687

Speyer covering conjecture for Gaudin monodromy

摘录 / EXCERPT: Let $\overline{M_{0,n+1}}(\mathbb{R})$ carry Speyer's covering and the covering constructed in the source, denoted the source's covering from Corollary. Speyer covering conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-062580 | MATHDB-328688

Berenstein-Kirillov quotient conjecture for the pure cactus group

摘录 / EXCERPT: Let $\fg=\mathfrak{gl}_N$, let the $V_{\l_i}$ be symmetric powers of the standard representation, and let PJ_n act on the associated Gelfand-Tsetlin polytope through Bethe-vector...

conjecture | open | MathDB 页面 | 原文线索

V2-062581 | MATHDB-328689

The voter-model interface straight-line convergence conjecture

摘录 / EXCERPT: Let B_L be the rhombic region of scale L, and let Z_L denote the voter-model interface curve in B_L. Straight-line convergence conjecture. As $L\rightarrow\infty$, the inte...

conjecture | open | MathDB 页面 | 原文线索

V2-062582 | MATHDB-328711

Complex Modular Ax-Schanuel conjecture for the j-function

摘录 / EXCERPT: Let A be an irreducible complex analytic subspace in a domain of $\mathbb{H}^n$, with coordinate functions $z_1,\ldots,z_n$ and their j-values meromorphic on A. The coordinat...

conjecture | open | MathDB 页面 | 原文线索

V2-062583 | MATHDB-328743

Balogh-Barát-Gerbner-Gyárfás-Sárközy cycle-partition conjecture

摘录 / EXCERPT: Balogh-Barát-Gerbner-Gyárfás-Sárközy conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-062584 | MATHDB-328744

Exact minimum-degree conjecture for two-colored cycle partitions

摘录 / EXCERPT: Exact cycle-partition conjecture. There exists an integer n_0 such that, whenever $n\geq n_0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-062585 | MATHDB-328746

Scattering below the ground-state threshold for the focusing energy-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Let $d\geq 3$, let I be the lifespan interval, and let $u:I\times\mathbf{R}^{d}\to\mathbf{C}$ solve the focusing energy-critical nonlinear Schrödinger equation with...

conjecture | open | MathDB 页面 | 原文线索

V2-062586 | MATHDB-328755

Open Virasoro conjecture for the full open partition function

摘录 / EXCERPT: Open Virasoro conjecture. The operators $\mathcal{L}_n$ annihilate the full partition function:

conjecture | open | MathDB 页面 | 原文线索

V2-062587 | MATHDB-328758

Loci equations conjecture for the first column of Zhao's table

摘录 / EXCERPT: Fix $Q=(u,u-r)$ with $u>r\ge2$, put $B=J_Q$, and use coordinates $a_1,\ldots,a_{u-1},b_1,\ldots,b_{u-r-1}$ on $\mathcal U_B$. For $\mu\in\{0,1,\ldots,u-1\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062588 | MATHDB-328760

Oblak's recursive conjecture for generic commuting nilpotent Jordan types

摘录 / EXCERPT: Let P be a partition, let $\mathcal D_P$ be the associated poset, and let C_a be a maximum-length U-chain in $\mathcal D_P$. Removing C_a gives a partition $P'(P,a)$; recursi...

conjecture | open | MathDB 页面 | 原文线索

V2-062589 | MATHDB-328763

Fabrici-Göring's unique maximal face-colour conjecture

摘录 / EXCERPT: Let G be a plane graph. A proper colouring of G assigns integers to its vertices so that adjacent vertices receive different integers. Fabrici-Göring's conjecture. Every plane...

conjecture | open | MathDB 页面 | 原文线索

V2-062590 | MATHDB-328771

Uniform essential spectral gap conjecture for congruence subgroups

摘录 / EXCERPT: Let $\mathcal{R}_q$ be the resonance set associated with the congruence subgroup $\Gamma(q)$, and let δ denote the Hausdorff dimension of the limit set. Uniform essential sp...

conjecture | open | MathDB 页面 | 原文线索

V2-062591 | MATHDB-328773

Filling-map factorization conjecture for tensor products of KR crystals

摘录 / EXCERPT: Let ... and let ... be organized into $r_i\times s_i$ rectangles as in T^{r_i,s_i}. Let $\operatorname{fill}\colon B\to T$ be the filling map. Filling-map factorization conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-062592 | MATHDB-328774

Affine realization conjecture for the spin KR crystal in type D

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type $D_{n+1}^{(1)}$, and let $\widetilde{B}^{n,s}$ be the affine crystal constructed by the stated classical decomposition and affine operators. Spin-crys...

conjecture | open | MathDB 页面 | 原文线索

V2-062593 | MATHDB-328781

The fine-category conjecture for the category of mixed motives

摘录 / EXCERPT: Let $\mathsf{DM}^\otimes$ be the ${\mathbb{Q}}$-linear symmetric monoidal stable presentable ∞-category of mixed motives. A symmetric monoidal stable presentable ∞-ca...

conjecture | open | MathDB 页面 | 原文线索

V2-062594 | MATHDB-328782

The length-set invariance conjecture for block monoids

摘录 / EXCERPT: Length-set invariance conjecture. If $|G|,|G'|>3$ and

conjecture | open | MathDB 页面 | 原文线索

V2-062595 | MATHDB-328802

Borodin-Raspaud Bordeaux conjecture on 3-coloring planar graphs

摘录 / EXCERPT: Borodin-Raspaud Bordeaux conjecture. Every planar graph without intersecting triangles and without 5-cycles is properly 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-062596 | MATHDB-328817

The Golod characterization of co-H moment-angle complexes

摘录 / EXCERPT: Golod characterization. The moment-angle complex $(D^2,S^1)^K$ is a co-H-space if and only if K is Golod.

conjecture | open | MathDB 页面 | 原文线索

V2-062597 | MATHDB-328830

White's nestedness conjecture for extreme rays of the $(N,2)$-Schur cone

摘录 / EXCERPT: White's conjecture. A is extreme in $\mathcal{SP}_N^2$ if and only if A is nested.

conjecture | open | MathDB 页面 | 原文线索

V2-062598 | MATHDB-328831

Extreme-ray induction conjecture for products in Schur cones

摘录 / EXCERPT: Extreme-ray induction conjecture. The product $s_A\otimes s_B$ is extreme in $\mathcal{C}_{N+M}^k$.

conjecture | open | MathDB 页面 | 原文线索

V2-062599 | MATHDB-328840

Atkinson's conjecture on the density of simple permutations

摘录 / EXCERPT: Atkinson's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062600 | MATHDB-328841

The finite-basis conjecture for monotone grid classes

摘录 / EXCERPT: Finite-basis conjecture for monotone grid classes. Every monotone grid class has a finite basis. The source notes that only limited cases are known, including skew-merged permutati...

conjecture | open | MathDB 页面 | 原文线索

V2-062601 | MATHDB-328843

The cyclicity conjecture for Abelian planar difference sets

摘录 / EXCERPT: Cyclicity conjecture for Abelian planar difference sets. The period of $\mathcal L$ in A_n along $\mathbf f_{i,j}$ is equal to n^2+n+1 for at least one vector $\mathbf f_{i,j}$...

conjecture | open | MathDB 页面 | 原文线索

V2-062602 | MATHDB-328846

Deflatability conjecture for parallel alternations

摘录 / EXCERPT: Deflatability conjecture. The class $\operatorname{Av}(\pi)$ is deflatable.

conjecture | open | MathDB 页面 | 原文线索

V2-062603 | MATHDB-328859

Facet-connectivity reduction conjecture for simplicial flows

摘录 / EXCERPT: Facet-connectivity reduction conjecture. If every k-facet-connected complex Δ has a q-flow for some $q<\kappa$, then every $(k-1)$-facet-connected complex Δ also...

conjecture | open | MathDB 页面 | 原文线索

V2-062604 | MATHDB-328862

The alpha-invariant conjecture for hypersurfaces without star points

摘录 / EXCERPT: Alpha-invariant conjecture. If X_d does not contain star points, then

conjecture | open | MathDB 页面 | 原文线索

V2-062605 | MATHDB-328867

Conjecture on the monotonicity threshold and complete monotonicity of G_a

摘录 / EXCERPT: Let R be the function defined in the source by equation (R(x)), and let G_a be the parameterized function defined earlier in the paper. Monotonicity and complete-monotonicity c...

conjecture | open | MathDB 页面 | 原文线索

V2-062606 | MATHDB-328868

Viehweg-Zuo conjecture on non-compact Jacobians of high-genus families

摘录 / EXCERPT: Viehweg-Zuo conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062607 | MATHDB-328869

The intermediate-word length conjecture for K-Knuth equivalent tableaux

摘录 / EXCERPT: Let T and T' be tableaux with $T\equiv T'$. For a tableau T, let its size be the number of boxes in its shape, and let $w\overset{k}{\equiv}w'$ mean that w and w' can be...

conjecture | open | MathDB 页面 | 原文线索

V2-062608 | MATHDB-328870

The asymptotic scarcity conjecture for unique rectification tableaux

摘录 / EXCERPT: Let I_n denote the number of K-Knuth equivalence classes of initial tableaux on $[n]$, and let U_n denote the number of unique rectification tableaux (URTs) on $[n]$. Asympto...

conjecture | open | MathDB 页面 | 原文线索

V2-062609 | MATHDB-328871

The interval conjecture for shapes in K-Knuth classes

摘录 / EXCERPT: Let $T_1,T_2,\dots,T_k$ be a K-Knuth class of straight tableaux, and let T_i have shape λ_i. Write ... Order Young diagrams by inclusion, and let $[\lambda_i,\lambda_j]$...

conjecture | open | MathDB 页面 | 原文线索

V2-062610 | MATHDB-328886

Bryan-Cadman-Young's Donaldson-Thomas crepant resolution conjecture

摘录 / EXCERPT: Let $\mathcal{Z}$ be a hard-Lefschetz 3-orbifold, and let ... be a crepant resolution. Let $DT_{mr}(-)$ denote the reduced, multi-regular Donaldson-Thomas potential, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062611 | MATHDB-328898

Pandharipande-Solomon-Tessler open Virasoro constraint conjecture

摘录 / EXCERPT: Pandharipande-Solomon-Tessler's open Virasoro constraint conjecture. The partition function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-062612 | MATHDB-328906

Autoequivalence conjecture for exceptional objects on the Hirzebruch surface \(\mathbb{F}_2\)

摘录 / EXCERPT: Autoequivalence conjecture. For any exceptional object $\mathcal{E}\in D(\mathbb{F}_2)$, there exists an autoequivalence $\Phi\in\operatorname{Auteq}(D(\mathbb{F}_2))$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-062613 | MATHDB-328913

The two-dimensional localisation and higher-dimensional delocalisation conjecture for the trimmed Anderson model

摘录 / EXCERPT: Let $H(g)$ be the trimmed Anderson random Schrödinger operator in dimension d, let H_Γ be the associated periodic operator, and let I be an interval contained in the abs...

conjecture | open | MathDB 页面 | 原文线索

V2-062614 | MATHDB-328915

The p-vanishing partition classification conjecture for primes at least 5

摘录 / EXCERPT: The p-vanishing partition classification conjecture. If $p\geq 5$, then a partition is p-vanishing if and only if it is of p-adic type.

conjecture | open | MathDB 页面 | 原文线索

V2-062615 | MATHDB-328916

The small-part bound conjecture for p-vanishing partitions

摘录 / EXCERPT: Small-part bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062616 | MATHDB-328919

The degree conjecture for multivariate Prouhet-Thue-Morse polynomials

摘录 / EXCERPT: Degree conjecture. One has $\operatorname{deg}_{t}H_{m,n}(X,t)=m-1$. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-062617 | MATHDB-328925

Stanley's conjecture on symmetric P-partition enumerators

摘录 / EXCERPT: A labelled poset P has a P-partition enumerator $F_P({\mathbf{x}})$, which is always quasisymmetric; a labelling is column-strict when it has the column-strict property used fo...

conjecture | open | MathDB 页面 | 原文线索

V2-062618 | MATHDB-328929

Rational-point formulation of Helleseth's conjecture for the curves D_k

摘录 / EXCERPT: Rational-point formulation. When $\gcd(k,m)=1$, the curves D_1 and D_k have the same number of rational points over ${\mathbb F}_{2^m}$:

conjecture | open | MathDB 页面 | 原文线索

V2-062619 | MATHDB-328933

Cyclic Young graph conjecture for the family in Z_{2n-1}

摘录 / EXCERPT: Let $Y(g,k)$ be a Young graph, and let Z_m denote the equivalence class of cyclic Young graphs on m nodes. Cyclic Young graph conjecture. For $t\ge 3$ and $n\ge 2$, the Young g...

conjecture | open | MathDB 页面 | 原文线索

V2-062620 | MATHDB-328934

Young graph isomorphism conjecture for the $Y(11,7)$ family

摘录 / EXCERPT: Let $Y(g,k)$ denote the Young graph associated with the base g and multiplier k. $Y(11,7)$ family conjecture. For all $n\ge 0$, the Young graphs ... ... are all isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-062621 | MATHDB-328935

Young graph isomorphism conjecture for the $Y(14,3)$ family

摘录 / EXCERPT: Let $Y(g,k)$ denote the Young graph associated with the base g and multiplier k. $Y(14,3)$ family conjecture. For all $t\ge 2$ and $n\ge 2$, ... This is an experimentally suggest...

conjecture | open | MathDB 页面 | 原文线索

V2-062622 | MATHDB-328936

Grimm-Ballew procedure conjecture on Young graph isomorphism

摘录 / EXCERPT: Let $d_1,d_2\in\mathbb{Z}^+$ and let m satisfy the Grimm-Ballew procedure's conditions; let g_1 and g_2 be different bases obtained by that procedure for the same d_1,...

conjecture | open | MathDB 页面 | 原文线索

V2-062623 | MATHDB-328937

Quadratic growth conjecture for Young graph equivalence classes

摘录 / EXCERPT: Let $Y(g,k)$ denote a Young graph, and let $\operatorname{Classes}(x)$ denote the number of equivalence classes occurring among all $Y(g,k)$ with $g\le x$. Quadratic growth conject...

conjecture | open | MathDB 页面 | 原文线索

V2-062624 | MATHDB-328939

Orbifold-resolution conjecture for BCOV invariants

摘录 / EXCERPT: Let Y be a Calabi-Yau orbifold and let $\widetilde{Y}\to Y$ be a crepant resolution. Orbifold-resolution conjecture. There is a constant $C(\widetilde{Y}/Y)$, depending only on t...

conjecture | open | MathDB 页面 | 原文线索

V2-062625 | MATHDB-328942

The Brauer-group conjecture for stable quiver moduli

摘录 / EXCERPT: Let Q be a quiver, let ${\bf d}$ be a dimension type for which stable representations exist, and let $M_{\bf d}^{\rm sst}(Q)$ denote the corresponding moduli space of stable repr...

conjecture | open | MathDB 页面 | 原文线索

V2-062626 | MATHDB-328945

The rectilinear crossing-number formula for complete tripartite graphs

摘录 / EXCERPT: Rectilinear crossing-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062627 | MATHDB-328946

The crossing-number equality conjecture for complete tripartite graphs

摘录 / EXCERPT: Crossing-number equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062628 | MATHDB-328949

Stability conjecture for normal bundles of general canonical curves

摘录 / EXCERPT: Let C be a general canonical curve of genus $g\geq 7$, embedded by its canonical linear system in ${\mathbf P}^{g-1}$, and let $N_{C/{\mathbf P}^{g-1}}$ denote its normal bundle....

conjecture | open | MathDB 页面 | 原文线索

V2-062629 | MATHDB-328962

The generalized Bogomolov-Gieseker conjecture for tilt-stable objects on threefolds

摘录 / EXCERPT: Let X be a smooth projective threefold. Let $ u_{\omega,B}$ be the tilt slope on the tilted heart $\operatorname{Coh}^{\omega,B}(X)$, where ω is ample and B is a real di...

conjecture | open | MathDB 页面 | 原文线索

V2-062630 | MATHDB-328964

The strengthened Bogomolov-Gieseker conjecture for tilt-semistable objects

摘录 / EXCERPT: Let X be a smooth projective threefold and let $H\in\operatorname{NS}(X)$ be an ample class. For a real number β, write $\operatorname{ch}_i^\beta$ for the βH-twiste...

conjecture | open | MathDB 页面 | 原文线索

V2-062631 | MATHDB-328965

The limiting Bogomolov-Gieseker conjecture for tilt-stable objects

摘录 / EXCERPT: Let X be a smooth projective threefold. For an object $E\in D^b(X)$, let $\overline{\beta}(E)$ be the limiting value of the parameter β associated with the kernel of the re...

conjecture | open | MathDB 页面 | 原文线索

V2-062632 | MATHDB-328975

Finiteness conjecture for monic cubics with depth-one emergent reducibility

摘录 / EXCERPT: A monic cubic with depth-one emergent reducibility is a monic polynomial $f(x)\in\mathbb{Z}[x]$ of degree 3 that is irreducible, while its first self-composition $f\circ f$ is re...

conjecture | open | MathDB 页面 | 原文线索

V2-062633 | MATHDB-328976

Conjecture on rainbow perfect matching packings in random bipartite graphs

摘录 / EXCERPT: Let n be a positive integer, let p satisfy ... and let $\mathcal{G}_c(K_{n,n},p)$ denote the random graph on the complete bipartite graph $K_{n,n}$ whose present edges receive a...

conjecture | open | MathDB 页面 | 原文线索

V2-062634 | MATHDB-328992

The converse bounded-cell conjecture for strongly β-mixing max-stable random fields

摘录 / EXCERPT: Let η be a stationary max-stable random field, and let $C(0)$ denote its cell containing the origin. The field is strongly β-mixing if for every compact set...

conjecture | open | MathDB 页面 | 原文线索

V2-062635 | MATHDB-328994

Stability from weak stability of all quotients

摘录 / EXCERPT: Let G be a group, let N range over the normal subgroups of G, and let stability and weak stability refer to the notions for systems of relator words introduced in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-062636 | MATHDB-328995

The sum of squared logarithms inequality

摘录 / EXCERPT: Sum of squared logarithms inequality. If

conjecture | open | MathDB 页面 | 原文线索

V2-062637 | MATHDB-328996

The dominance-path conjecture for elementary symmetric polynomials

摘录 / EXCERPT: Let Δ_n be the domain defined in the source, let S be a nonempty subset of $\{0,1,\ldots,n-1\}$, and let $a,b\in\Delta_n$ satisfy ... A point b dominates a, written...

conjecture | open | MathDB 页面 | 原文线索

V2-062638 | MATHDB-328997

The local root-splitting conjecture for perturbed real-rooted polynomials

摘录 / EXCERPT: Let $n\geq2$, let $a\in\Delta_n$, and let $S\subseteq\{1,2,\ldots,n-1\}$ be nonempty. Suppose there are numbers $A_k>0$ for $k\in S$ such that all roots of ... are real, and therefor...

conjecture | open | MathDB 页面 | 原文线索

V2-062639 | MATHDB-329002

Class-number-one conjecture for the cyclotomic $\mathbb{Z}_5$-extension

摘录 / EXCERPT: Let $\mathbb{B}_{5,n}$ denote the nth layer of the cyclotomic $\mathbb{Z}_5$-extension of $\mathbb{Q}$. $\mathbb{Z}_5$-extension class-number conjecture. For every positive integ...

conjecture | open | MathDB 页面 | 原文线索

V2-062640 | MATHDB-329004

Dokos et al.'s major-index distribution conjecture for three length-four patterns

摘录 / EXCERPT: Let $\mathcal{S}_n$ be the set of permutations of $[n]$, and let $\mathcal{S}_n(\tau)$ denote the permutations avoiding the pattern τ. For a permutation π, its major index...

conjecture | open | MathDB 页面 | 原文线索

V2-062641 | MATHDB-329013

Fernández de Bobadilla's smooth critical-locus conjecture

摘录 / EXCERPT: Let f be a holomorphic function germ with critical locus $\boldsymbol{\Sigma} f$, let C be its irreducible component, and let $F_{f,\mathbf p}$ denote the Milnor fiber of f a...

conjecture | open | MathDB 页面 | 原文线索

V2-062642 | MATHDB-329014

Generalized Fernández de Bobadilla conjecture on critical loci

摘录 / EXCERPT: Let f be a holomorphic function germ with critical locus $\boldsymbol{\Sigma} f$, let $F_{f,\mathbf 0}$ denote its Milnor fiber at the origin, and let $\stackrel{\circ}{\mu}_C$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-062643 | MATHDB-329034

Oscillation criterion for zeros and generalized Li coefficients

摘录 / EXCERPT: Let $F\in\mathcal{S}^{\sharp \flat}(\sigma_0,\sigma_1)$, let τ be the parameter in the generalized Li coefficients $\lambda_F(n,\tau)$, and consider the half-plane...

conjecture | open | MathDB 页面 | 原文线索

V2-062644 | MATHDB-329035

Cylindric loop Schur function conjecture for rigged-configuration shapes

摘录 / EXCERPT: Let $B=B^{1,s_1}\otimes\cdots\otimes B^{1,s_m}$ be a product of one-row crystals, let $p=b_1\otimes\cdots\otimes b_m\in B$, and define $x_j^{(i+j-1)}$ to be the number of occurrenc...

conjecture | open | MathDB 页面 | 原文线索

V2-062645 | MATHDB-329057

Non-realizability of powers of the exceptional 2-pairing

摘录 / EXCERPT: Let $\mathcal{E}_{2^r}$ denote the exceptional pairing on the cyclic group of order 2^r, as used in the paper. For an integer $k\geq 1$, consider the direct product of k copies...

conjecture | open | MathDB 页面 | 原文线索

V2-062646 | MATHDB-329061

Existence of a subforest preserving game chromatic number

摘录 / EXCERPT: Subforest preservation conjecture. There exists a subforest $F'\subseteq F$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062647 | MATHDB-329065

The crepant transformation conjecture for the Landau-Ginzburg and Calabi-Yau models

摘录 / EXCERPT: Let $I\mathcal{X}$ and $I\mathcal{Y}$ be the inertia orbifolds of the local targets $\mathcal{X}$ and $\mathcal{Y}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062648 | MATHDB-329066

The refined crepant transformation conjecture

摘录 / EXCERPT: Let $\mathscr{L}^{\mathcal{X}}\subset\mathscr{V}^{\mathcal{X}}$ and $\mathscr{L}^{\mathcal{Y}}\subset\mathscr{V}^{\mathcal{Y}}$ be the equivariant genus-zero Gromov-Witten Lagrangi...

conjecture | open | MathDB 页面 | 原文线索

V2-062649 | MATHDB-329073

The maximal-rank conjecture for regular polytopes with automorphism group \(\mathrm{A}_n\)

摘录 / EXCERPT: Let $n\geq 12$. An abstract regular polytope with automorphism group A_n has rank d, where rank means the number of involutory generators in its string C-group represe...

conjecture | open | MathDB 页面 | 原文线索

V2-062650 | MATHDB-329074

The strong divergence conjecture for arbitrary complete interpolating sequences

摘录 / EXCERPT: Strong divergence conjecture. There exists an $f\in\mathcal{PW}^{1}_{\pi}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062651 | MATHDB-329075

The pointwise operator-norm divergence conjecture for stable LTI systems

摘录 / EXCERPT: Pointwise operator-norm divergence conjecture. For every $t\in\mathbb{R}$, there exists a stable LTI system $\mathrm{H}:\mathcal{PW}^{1}_{\pi}\to\mathcal{PW}^{1}_{\pi}$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-062652 | MATHDB-329076

The strong divergence conjecture for LTI system approximations

摘录 / EXCERPT: Let $\Lambda = \{\lambda_n\}_{n\in\mathbb{Z}}$ be a complete interpolating sequence for $\mathcal{PW}^{2}_{\pi}$. Let H_N be the sampling-based approximation operators o...

conjecture | open | MathDB 页面 | 原文线索

V2-062653 | MATHDB-329089

Scalar-flat Kähler metrics on deformations of minimal resolutions of quotient singularities

摘录 / EXCERPT: Let $\Gamma \subset {\rm{U}}(2)$ be a finite subgroup containing no complex reflections, and let $\widetilde{X}$ be the minimal resolution of $\mathbb{C}^2/\Gamma$. A small deforma...

conjecture | open | MathDB 页面 | 原文线索

V2-062654 | MATHDB-329096

Equivalence of Volterra equations and Grunwald-Letnikov equations for fractional order below one

摘录 / EXCERPT: Volterra-Grunwald-Letnikov equivalence conjecture. Theorem 5, which asserts equivalence of these equations under the stated continuity and zero-initial-condition hypotheses, is val...

conjecture | open | MathDB 页面 | 原文线索

V2-062655 | MATHDB-329107

Full L^2-Alexander torsion conjecture for monodromy dilatation

摘录 / EXCERPT: Let K be a fibered oriented knot with monodromy f, and normalize its full L^2-Alexander torsion so that $\tau^{(2)}(K)(1)=1$. Full L^2-Alexander dilatation conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-062656 | MATHDB-329111

Penner's conjecture on powers of pseudo-Anosov mapping classes

摘录 / EXCERPT: Let $S_{g,n}$ be an orientable surface of genus g with n punctures, and let $f\in\mathrm{Mod}(S_{g,n})$ be a pseudo-Anosov mapping class, meaning that it has a representative p...

conjecture | open | MathDB 页面 | 原文线索

V2-062657 | MATHDB-329115

Conjecture on the exceptional Legendre torsion set and the four-element bound

摘录 / EXCERPT: For $\alpha,\beta\in\mathbb{C}\setminus\{0,1\}$ with $\alpha\ne\beta$, let $T(\alpha,\beta)$ be the set of Legendre parameters for which the points associated with α and...

conjecture | open | MathDB 页面 | 原文线索

V2-062658 | MATHDB-329117

Absolute height-bound conjecture for pairs with simultaneous torsion

摘录 / EXCERPT: Let $T(\alpha,\beta)$ be the set of Legendre parameters for which the points associated with α and β are both torsion, where...

conjecture | open | MathDB 页面 | 原文线索

V2-062659 | MATHDB-329127

The gamma-delta conjecture for polynomial endomorphisms

摘录 / EXCERPT: Let K be the underlying field, let $K[x,y]$ be the polynomial ring in two variables, and let f be an endomorphism of $K[x,y]$. Write K^* for the group of nonzero elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-062660 | MATHDB-329143

Billey-Burdzy-Sagan positivity conjecture for peak polynomials

摘录 / EXCERPT: Billey-Burdzy-Sagan's positivity conjecture. Each coefficient $(\Delta^j p_S)(m)$ is a positive integer for $1\le j\le m-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-062661 | MATHDB-329144

Bounded-roots conjecture for peak polynomials

摘录 / EXCERPT: Bounded-roots conjecture. Every complex root of $p_S(z)$ lies in

conjecture | open | MathDB 页面 | 原文线索

V2-062662 | MATHDB-329145

Integral and classified real-root conjecture for peak polynomials

摘录 / EXCERPT: Integral and classified real-root conjecture. If all roots of $p_S(x)$ are real, then all of them are integral. Moreover, $p_S(x)$ has only real roots if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062663 | MATHDB-329146

Real-root location conjecture for peak polynomials beyond the final gap

摘录 / EXCERPT: Real-root location conjecture. The inequality $x_0>\max(S_1)$ holds if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062664 | MATHDB-329148

Shrinking-hole density conjecture for the boundary of the Mandelbrot set

摘录 / EXCERPT: Let ${\mathcal M}$ be the parameter set under consideration, and let $\partial {\mathcal M}$ denote its boundary. A hole is a connected component of the complement of...

conjecture | open | MathDB 页面 | 原文线索

V2-062665 | MATHDB-329152

Goulden-Jackson-Vakil parity conjecture for double Hurwitz polynomials

摘录 / EXCERPT: Let $\mathbf{x}=(x_1,\ldots,x_n)\in\mathbb Z^n$ satisfy $\sum_i x_i=0$, and let $H_g^r(\mathbf{x})$ be the double Hurwitz number, viewed piecewise polynomially on the chambers of t...

conjecture | open | MathDB 页面 | 原文线索

V2-062666 | MATHDB-329153

Structure morphism conjecture for the double Hurwitz moduli spaces

摘录 / EXCERPT: Let $\overline{P}(\mathbf{x})$ be the moduli space appearing in the proposed ELSV formula for double Hurwitz numbers, and let Λ_{2g} be its tautological Chow class of degre...

conjecture | open | MathDB 页面 | 原文线索

V2-062667 | MATHDB-329154

Polynomiality conjecture for the double ramification cycle

摘录 / EXCERPT: Let $\mathbf{x}=(x_1,\ldots,x_n)$ satisfy the balancing condition $\sum_i x_i=0$, and let $DRC(\mathbf{x})$ denote the double ramification cycle in...

conjecture | open | MathDB 页面 | 原文线索

V2-062668 | MATHDB-329158

Conjecture on the limiting Evans function for regular internal solitary waves

摘录 / EXCERPT: Let $U^c(\xi,y)$ be a regular internal solitary wave (ISW), and let D_N denote its N-th order truncated Evans function. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-062669 | MATHDB-329159

Hoffmann's conjecture for the first higher isotropy index

摘录 / EXCERPT: Hoffmann's conjecture. The integer $\mathfrak{i}_{1}(\phi)-1$ is the remainder of $\dim\phi-1$ modulo 2^s for some $s<\log_2(\dim\phi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062670 | MATHDB-329160

Quasi-Pfister divisibility conjecture for quasilinear forms

摘录 / EXCERPT: Quasi-Pfister divisibility conjecture. The form ϕ is divisible by an $(n-1)$-fold quasi-Pfister form.

conjecture | open | MathDB 页面 | 原文线索

V2-062671 | MATHDB-329161

Expected classification of even-dimensional quadratic forms of height two

摘录 / EXCERPT: Expected height-two classification. Exactly one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-062672 | MATHDB-329162

Characteristic-two analogue of the height-two bilinear-form classification

摘录 / EXCERPT: Characteristic-two height-two conjecture. There exist a $(d-1)$-fold Pfister form π and a 4-dimensional form $\mathfrak{c}$ of nontrivial determinant such that

conjecture | open | MathDB 页面 | 原文线索

V2-062673 | MATHDB-329163

Vishik-type isotropy gap conjecture for quasilinear quadratic forms

摘录 / EXCERPT: Vishik-type isotropy gap conjecture. For every field extension L/F, either

conjecture | open | MathDB 页面 | 原文线索

V2-062674 | MATHDB-329166

Mutual absolute continuity of the Slepian and shifted Brownian zero sets

摘录 / EXCERPT: Zero-set absolute-continuity conjecture. For every $t\geq 0$, the Slepian zero set on $[0,t]$ is mutually absolutely continuous with respect to the zero set of Brownian motion star...

conjecture | open | MathDB 页面 | 原文线索

V2-062675 | MATHDB-329167

Berndt's generalized Hlawka inequality for positive definite matrices

摘录 / EXCERPT: Berndt's conjecture. The following generalization of the Hlawka inequality holds:

conjecture | open | MathDB 页面 | 原文线索

V2-062676 | MATHDB-329180

Conjecture on the Stanley depth of cyclic graph edge ideals

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ be a polynomial ring, and let J_n denote the ideal associated with the cyclic graph on n vertices as in the paper. The Stanley depth conjecture. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-062677 | MATHDB-329185

Exact cost of local monotonicity for the Banzhaf and proportional power indices

摘录 / EXCERPT: Let $n\ge 2$, let $\mathfrak{W}$ denote the class of weighted games, and let $c_{\mathcal{P}}(n,\mathfrak{W})$ be the cost of local monotonicity for the power-index collection...

conjecture | open | MathDB 页面 | 原文线索

V2-062678 | MATHDB-329188

The characterization of Hyperbrot sets

摘录 / EXCERPT: Let $\mathcal{H}^p$ denote the Hyperbrot set for the hyperbolic polynomial z^p+c, with $c=a+b\mathbf{j}$. Hyperbrot characterization conjecture. The Hyperbrot sets are squares an...

conjecture | open | MathDB 页面 | 原文线索

V2-062679 | MATHDB-329190

Kollár-Némethi lattice conjecture related to Elkies's theorem

摘录 / EXCERPT: Let L be the singularity lattice, let L' be its dual lattice, let $Z_{min}\in L$ be the minimal cycle, let Z_K be the canonical cycle, and let s denote the relevant lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-062680 | MATHDB-329192

Equidistribution of orthogonal grids on integer spheres

摘录 / EXCERPT: Equidistribution conjecture for orthogonal grids. The convergence

conjecture | open | MathDB 页面 | 原文线索

V2-062681 | MATHDB-329198

Thomassen's connected partition conjecture with cross-degree condition

摘录 / EXCERPT: Thomassen's partition conjecture. For every $k\in\mathbb{N}$ there exists $f(k)\in\mathbb{N}$ such that if G is an $f(k)$-connected graph and $M\subseteq V(G)$ consists of k ve...

conjecture | open | MathDB 页面 | 原文线索

V2-062682 | MATHDB-329205

Generic isolation conjecture for Hodge classes on high-dimensional hypersurfaces

摘录 / EXCERPT: Let V be the parameter space of smooth hypersurfaces of degree $d\geq 2\frac{{\sf m}+2}{{\sf m}-2}$ in $\mathbb{P}^{{\sf m}+1}$, with ${\sf m}\geq 4$. A Hodge class of Y_t is c...

conjecture | open | MathDB 页面 | 原文线索

V2-062683 | MATHDB-329210

The full-support facet conjecture for ${\operatorname{LS}}_+$-perfect graphs

摘录 / EXCERPT: Let ${\operatorname{LS}}_+(G)$ denote the Lovász-Schrijver positive semidefinite relaxation of the stable set polytope. A graph is ${\operatorname{LS}}_+$-perfect when...

conjecture | open | MathDB 页面 | 原文线索

V2-062684 | MATHDB-329214

Andrade-Chang-Miller conjecture for fixed genus

摘录 / EXCERPT: Fix $g\in\mathbb N$. For each good pair (q,D) with $\deg D=2g+1$, let Λ_D be the associated De Bruijn-Newman constant. Andrade-Chang-Miller conjecture. One has ... This con...

conjecture | open | MathDB 页面 | 原文线索

V2-062685 | MATHDB-329217

The Fuchsian length domination conjecture for Hitchin representations

摘录 / EXCERPT: Let S be the surface and ρ a representation in $Hit_n(S)$. A representation ρ' is in the Fuchsian locus of $Hit_n(S)$ when it is a Fuchsian representation viewed in...

conjecture | open | MathDB 页面 | 原文线索

V2-062686 | MATHDB-329257

Tate-Gross conjecture on Gross-Stark units

摘录 / EXCERPT: Tate-Gross conjecture. For every $\sigma\in G$, $u_{S,T,\sigma}\in{\mathscr E}_{\mathfrak p}$; for every $\sigma\in G$ and every ${\mathfrak q}\in T$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062687 | MATHDB-329277

Energy-critical NLS without potential global well-posedness and scattering conjecture

摘录 / EXCERPT: Let $d\geq 3$, let $\mu\in\{1,-1\}$, and consider the energy-critical nonlinear Schrödinger equation without potential ... For $\mu=-1$, define ... where W solves...

conjecture | open | MathDB 页面 | 原文线索

V2-062688 | MATHDB-329278

Energy-critical NLS with quadratic-growth potentials global well-posedness conjecture

摘录 / EXCERPT: Global well-posedness conjecture. When $\mu=1$, the equation is globally well posed: for each $u_0\in Q(H)$ there is a unique global solution...

conjecture | open | MathDB 页面 | 原文线索

V2-062689 | MATHDB-329283

Capacity characterization for group leaders satisfying the receiver side-information condition

摘录 / EXCERPT: Consider a private-message AWGN broadcast channel with $Q>3$ receivers, ordered by noise levels as $N_1\leq N_2\leq\cdots\leq N_Q$. A group leader is the common subgraph obtained f...

conjecture | open | MathDB 页面 | 原文线索

V2-062690 | MATHDB-329284

The generic K-semistability equivalence for fibrations over curves

摘录 / EXCERPT: Let $\pi\colon \mathcal{X}\to B$ be a fibration with polarized total space $(\mathcal{X},\mathcal{L})$, and let $(F,\mathcal{L}|_F)$ denote its generic fiber. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-062691 | MATHDB-329285

Petrosyan's binary-weight conjecture for interval colorings of complete graphs

摘录 / EXCERPT: Let K_{2n} be the complete graph on $2n$ vertices, and let $W(K_{2n})$ be the greatest number of colors in an interval coloring of K_{2n}. Write $\left\lfloor n_2\right\rfloor$ for the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-062692 | MATHDB-329287

Chen-Fröhlich-Walcher metastability conjecture for noncommutative Klein-Gordon solitons

摘录 / EXCERPT: The discrete nonlinear Klein-Gordon equation is ... Its quasi-stationary solutions have the form $u(t)=\cos(\mu t+\nu)\alpha_\mu$, where μ and ν are parameters and...

conjecture | open | MathDB 页面 | 原文线索

V2-062693 | MATHDB-329290

Lower bound for the dimension function of intersected factor posets

摘录 / EXCERPT: Let P_1 and P_2 be factor posets, and let $calD(P)$ denote the smallest dimension in which a frame with factor poset P has an inverse. Lower-bound conjecture. ... The preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-062694 | MATHDB-329292

Strict scaling from the scalability poset

摘录 / EXCERPT: Let $F=\{f_i\}_{i\in I}$ and $G=\{g_i\}_{i\in I}$ be unit-norm frames in $\mathbb{R}^n$. Suppose that F and G have the same scalability poset, and that G is a tight frame. St...

conjecture | open | MathDB 页面 | 原文线索

V2-062695 | MATHDB-329302

Oscillating walks in a Weyl chamber and bounded-height standard Young tableaux

摘录 / EXCERPT: Oscillating-walk conjecture. The set of W_k-oscillating lattice walks of length n ending at the boundary

conjecture | open | MathDB 页面 | 原文线索

V2-062696 | MATHDB-329304

Decidability of locally testable model checking for countable quasiminimal subshifts

摘录 / EXCERPT: Let X be a countable quasiminimal recursive subshift. The model-checking problem asks whether the language of X intersects a given formal language. Locally testable decidabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-062697 | MATHDB-329305

The density conjecture for 5/2-critical graphs

摘录 / EXCERPT: The density conjecture for $5/2$-critical graphs. If G is a $5/2$-critical graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-062698 | MATHDB-329315

The compact building strong transitivity characterization conjecture

摘录 / EXCERPT: Let Δ be an infinite thick irreducible compact spherical building of rank at least 2. Strong transitivity characterization conjecture. The building Δ is the spheric...

conjecture | open | MathDB 页面 | 原文线索

V2-062699 | MATHDB-329316

The Moufang property conjecture for strongly transitive compact polygons

摘录 / EXCERPT: Let Δ be an infinite thick irreducible compact totally disconnected spherical building of rank 2. Moufang property conjecture. If Δ is strongly transitive, then...

conjecture | open | MathDB 页面 | 原文线索

V2-062700 | MATHDB-329317

Mottram's path-selection conjecture for conditioned Brownian motion

摘录 / EXCERPT: Let $U\subseteq\mathbb{R}\times[0,\infty)$ be open with $(0,0)\in U$, and let $\operatorname{cost}(P)$ be the path cost. Assume that there is a unique path P from $(0,0)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-062701 | MATHDB-329318

Mottram's critical-speed conjecture for Brownian entropic repulsion

摘录 / EXCERPT: Let γ° and $\gamma^{\bullet}$ be the critical speeds with $1<\gamma^{\circ}<\gamma^{\bullet}$, and let $\mathcal{E}_a^{U_v}$ be the conditioning events associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-062702 | MATHDB-329319

Mottram's local-time exponent conjecture for ballistic limiting measures

摘录 / EXCERPT: Let $v>\gamma^{\circ}$ and let $\mathbb{Q}^{U_v}=\lim_{a\to\infty}\mathbb{W}(\mathord{\cdot}\mid\mathcal{E}_a^{U_v})$ be any measure supplied by the critical-speed framework. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062703 | MATHDB-329324

Conjecture on the unique singular point at infinity of GHV models

摘录 / EXCERPT: Let the GHV model be associated with a smooth hypersurface satisfying the assumptions of theorem, and let n, d, and w be the parameters appearing there, with $n+d-w>0$. A com...

conjecture | open | MathDB 页面 | 原文线索

V2-062704 | MATHDB-329326

Perfectness criterion for flop-functor equivalence

摘录 / EXCERPT: Let $X\to X_{\mathrm{con}}$ be a flopping contraction of 3-folds, with X having at worst Cohen-Macaulay rational singularities. Let the universal sheaf of the noncommutative de...

conjecture | open | MathDB 页面 | 原文线索

V2-062705 | MATHDB-329327

Mutation theorem for crepant curves with perfect universal sheaf

摘录 / EXCERPT: Let C_i be a crepant curve on a 3-fold X, and suppose that its universal sheaf is perfect and that X has at worst Cohen-Macaulay rational singularities. Mutation theorem fo...

conjecture | open | MathDB 页面 | 原文线索

V2-062706 | MATHDB-329331

Self-Kronecker-product lower-bound conjecture for nonnegative rank

摘录 / EXCERPT: Let A be a nonnegative matrix, let $A\otimes A$ denote its Kronecker square, and write $\operatorname{rank}(A)$ for its ordinary rank. Self-Kronecker-product conjecture. ... The so...

conjecture | open | MathDB 页面 | 原文线索

V2-062707 | MATHDB-329334

Emerton-Gee support conjecture for crystalline deformation spaces

摘录 / EXCERPT: Let $\mathfrak{X}^p$ be an irreducible component of $\mathfrak{X}_{\bar{\rho}^p}^\square$, and let ... be strictly dominant. Let $R_{\bar{\rho}_p}^{\square,{\bf k}-{\rm cr}}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-062708 | MATHDB-329337

The planar graph absolute clique number conjecture for (m,n)-colored mixed graphs

摘录 / EXCERPT: Planar absolute clique number conjecture. For the family $\mathcal{P}$ of planar graphs,

conjecture | open | MathDB 页面 | 原文线索

V2-062709 | MATHDB-329341

Weight-grading conjecture for multiple zeta values

摘录 / EXCERPT: For a convergent multiple zeta value $zeta_{m_1,\ldots,m_k}$, its weight is $m_1+\cdots+m_k$. A polynomial relation among MZVs is an equation $P(Z_1,\ldots,Z_n)=0$ with P having...

conjecture | open | MathDB 页面 | 原文线索

V2-062710 | MATHDB-329344

Maximum-transcendentality conjecture for loop amplitudes

摘录 / EXCERPT: Consider an L-loop amplitude in dimensional regularisation with $D=4-2\epsilon$, and write its Laurent expansion in ϵ. Maximum-transcendentality conjecture. The Laurent c...

conjecture | open | MathDB 页面 | 原文线索

V2-062711 | MATHDB-329346

Goncharov's weight-four classical-polylogarithm criterion

摘录 / EXCERPT: Let F be a combination of multiple polylogarithms of weight four, let $\mathcal{S}(F)$ denote its symbol, and define ... where $a\wedge b=a\otimes b-b\otimes a$. Goncharov's criter...

conjecture | open | MathDB 页面 | 原文线索

V2-062712 | MATHDB-329347

Asymptotic strength of random regular Tanner graphs

摘录 / EXCERPT: Let d_v and d_c be integers with $d_c>d_v\geq 3$. A Tanner graph is asymptotically strong if, for every constant $\beta>0$, there is a constant $\alpha>0$ such that its LP deco...

conjecture | open | MathDB 页面 | 原文线索

V2-062713 | MATHDB-329348

Nondegeneracy of random regular Tanner graphs with logarithmic girth

摘录 / EXCERPT: Let $d_c>d_v\geq 3$ be integers satisfying $d_v<\beta_{d_c}d_c$, where β_{d_c} is Calkin's threshold. For a positive constant λ, let Γ_{λ} be the ensembl...

conjecture | open | MathDB 页面 | 原文线索

V2-062714 | MATHDB-329349

Henkin-style second-order expressibility conjecture for NDL

摘录 / EXCERPT: Let NDL be the generalized dynamic logic discussed in the paper, and let Henkin-style second-order logic (HSOL) be the corresponding Henkin semantics for second-order logic. A pr...

conjecture | open | MathDB 页面 | 原文线索

V2-062715 | MATHDB-329361

Raşa's logarithmic-convexity conjecture for squared Baskakov functions

摘录 / EXCERPT: Raşa's conjecture. The function $\psi_{n,c}$ is logarithmically convex. The paper proves complete monotonicity of $\psi_{n,c}$, which implies logarithmic convexity; thus this conje...

conjecture | open | MathDB 页面 | 原文线索

V2-062716 | MATHDB-329362

Complete monotonicity conjecture for even powers of Baskakov functions

摘录 / EXCERPT: Even-power complete-monotonicity conjecture. The function $f_{\alpha}^{[r]}$ is completely monotonic on $[0,\infty)$. In particular, $\psi_{n,c}^{[r]}$ is completely monotonic on...

conjecture | open | MathDB 页面 | 原文线索

V2-062717 | MATHDB-329364

Kuznetsov's reconstruction conjecture for instantons on index-2 Fano threefolds

摘录 / EXCERPT: Kuznetsov's reconstruction conjecture. The generic line is not jumping, and E can be reconstructed from its theta-sheaf.

conjecture | solved | MathDB 页面 | 原文线索

V2-062718 | MATHDB-329365

Generic non-jumping-line conjecture for instantons on the Fano threefold Y_5

摘录 / EXCERPT: Generic non-jumping-line conjecture. The generic line is not jumping.

conjecture | open | MathDB 页面 | 原文线索

V2-062719 | MATHDB-329367

Bachmann's double stuffle completeness conjecture for bi-brackets

摘录 / EXCERPT: Let bi-brackets be the q-multiple zeta values defined in the paper, equipped with their stuffle and dual stuffle products and the duality relation. Bachmann's double stuffle conj...

conjecture | open | MathDB 页面 | 原文线索

V2-062720 | MATHDB-329369

Ehrhart equivalence conjecture for discrete equidecomposability

摘录 / EXCERPT: Let Ehrhart-equivalent rational polytopes be rational polytopes whose Ehrhart functions agree. Discrete equidecomposability is allowed to use an arbitrary number of simplices, and...

conjecture | open | MathDB 页面 | 原文线索

V2-062721 | MATHDB-329370

Ehrhart-function criterion for arbitrary equidecomposability

摘录 / EXCERPT: Ehrhart-function conjecture. There exists an equidecomposability relation $\mathcal{F}:(S,\mathcal{T})\to(S',\mathcal{T}')$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-062722 | MATHDB-329371

Marcus-Spielman-Srivastava optimal-bound conjecture for mixed characteristic polynomials

摘录 / EXCERPT: Let h be a degree-d polynomial hyperbolic with respect to $\boldsymbol{e}$, let $\boldsymbol{v}_1,\textellipsis,\boldsymbol{v}_m\boldsymbol{obreakspace}\text{?}$ be vectors in...

conjecture | open | MathDB 页面 | 原文线索

V2-062723 | MATHDB-329381

Mabuchi-Tian three-way Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Mabuchi-Tian three-way conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-062724 | MATHDB-329388

Tucker's conjecture for closed subdegree-finite permutation groups

摘录 / EXCERPT: Let $Gamma$ be a closed, subdegree finite permutation group acting on a set S. A distinguishing 2-colouring of S is a colouring with two colours that is preserved by no non-i...

conjecture | open | MathDB 页面 | 原文线索

V2-062725 | MATHDB-329389

The alternating-surgery tangle-replacement conjecture

摘录 / EXCERPT: Alternating-surgery tangle-replacement conjecture. Every alternating surgery arises as tangle replacement on an almost-alternating diagram of the unknot; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-062726 | MATHDB-329410

Maximum-efficiency conjecture for practical multi-user OFDM systems

摘录 / EXCERPT: Maximum-efficiency conjecture. The maximum efficiency is $d_{max}=1.4998$, achieved when there are $K=3$ users using $N=2{,}001$ subcarriers.

conjecture | open | MathDB 页面 | 原文线索

V2-062727 | MATHDB-329417

Kosaka's conjecture on virtual bridge number and primary data

摘录 / EXCERPT: Kosaka's conjecture. Virtual knots have the same virtual bridge number if they have the same primary data.

conjecture | open | MathDB 页面 | 原文线索

V2-062728 | MATHDB-329419

Unicity conjecture for archetypes of supercuspidal representations of $\mathbf{SL}_N(F)$

摘录 / EXCERPT: Let F be a non-Archimedean local field, let N be a positive integer, and let $\bar{\pi}$ be a supercuspidal representation of $\mathbf{SL}_N(F)$. Choose a supercuspidal represe...

conjecture | open | MathDB 页面 | 原文线索

V2-062729 | MATHDB-329434

Exponential discrete return-time conjecture for random unitary powers

摘录 / EXCERPT: Let $n_0(N,\epsilon)$ be the smallest integer $n\geq1$ for which all eigenvalues of U_N^n lie in $[e^{-i\pi\epsilon},e^{i\pi\epsilon}]$. Let $X_1,\dots,X_N$ be independent unifor...

conjecture | open | MathDB 页面 | 原文线索

V2-062730 | MATHDB-329454

The Gluing Conjecture for the global nilpotent cone

摘录 / EXCERPT: Gluing Conjecture. This functor is fully faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-062731 | MATHDB-329455

The global-nilpotent gluing conjecture for local systems

摘录 / EXCERPT: Global-nilpotent gluing conjecture. This functor is fully faithful.

conjecture | open | MathDB 页面 | 原文线索

V2-062732 | MATHDB-329464

The alternating-group bismash-product Schur indicator conjecture

摘录 / EXCERPT: Alternating-group indicator conjecture. The indicator satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-062733 | MATHDB-329468

Whittaker-Fourier coefficient identity for generic cuspidal representations of unitary groups

摘录 / EXCERPT: Whittaker-Fourier coefficient identity. Assume that σ weakly lifts to π as above. Then, for every $\varphi\in\sigma$, $\varphi^{\vee}\in\sigma^{\vee}$, and every suffici...

conjecture | open | MathDB 页面 | 原文线索

V2-062734 | MATHDB-329476

The optimal rearrangement-constant conjecture for BMO

摘录 / EXCERPT: Let Q be a d-dimensional cube and let f^* denote the monotone rearrangement of f on $[0,1]$. Let $c(d)$ be the constant appearing in the dimension-reduction conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-062735 | MATHDB-329481

Eigenvalue-one conjecture for bounded-degree simplicial complexes

摘录 / EXCERPT: Let X be a d-complex, and let $\mathscr{A}_{0}$ denote its operator on 0-faces. For each $(d-1)$-face $\sigma\in X^{d-1}$, write $\deg(\sigma)$ for its degree. Eigenvalue-one...

conjecture | open | MathDB 页面 | 原文线索

V2-062736 | MATHDB-329484

The folklore sup-norm conjecture for newforms with mildly ramified central character

摘录 / EXCERPT: Let f be a newform on $\operatorname{GL}_2(\mathbb{A}_{\mathbb{Q}})$ with level M and conductor N, and let N_1 be the smallest integer such that $N\mid N_1^2$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-062737 | MATHDB-329485

The global sup-norm conjecture for newforms with mildly ramified central characters

摘录 / EXCERPT: Let F be a number field, let $G=\operatorname{GL}_2$, and let π be an automorphic representation with finite conductor N and local conductor exponents n_v and central-cha...

conjecture | open | MathDB 页面 | 原文线索

V2-062738 | MATHDB-329489

Optimality of the tensor-power inversification bound

摘录 / EXCERPT: Optimality conjecture. If $K<L-1$, then there is always a channel $W\in\mathcal W([L],[L'])$ satisfying $W(\delta_i)\neq W(\delta_j)$ for all distinct $i,j\in[L]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-062739 | MATHDB-329497

Nonnegativity conjecture for Riesz kernels

摘录 / EXCERPT: Nonnegativity conjecture for Riesz kernels. For $\alpha\gg 0$, the Riesz kernel $q_\alpha(\sigma)$ takes nonnegative values on K.

conjecture | open | MathDB 页面 | 原文线索

V2-062740 | MATHDB-329498

The local-density kernel conjecture for Jordan components of quadratic lattices

摘录 / EXCERPT: Local-density kernel conjecture. The map φ_{κ} is surjective, and its kernel is isomorphic as a set to

conjecture | open | MathDB 页面 | 原文线索

V2-062741 | MATHDB-329499

The four-colour conjecture for signed planar graphs

摘录 / EXCERPT: Let G be a simple signed planar graph, and let $\chi(G)$ denote its chromatic number, defined as the smallest n for which G has a proper colouring using colours from M_n, w...

conjecture | open | MathDB 页面 | 原文线索

V2-062742 | MATHDB-329500

Veselov's simplicity conjecture for generalized Hermite polynomial zeros

摘录 / EXCERPT: Let $H_{\lambda}(z)$ be the generalized Hermite polynomial associated with a partition λ. Veselov's conjecture. For any partition λ, the zeros of $H_{\lambda}(z)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-062743 | MATHDB-329502

Renormalization conjecture for bounded nearly incompressible BV vector fields

摘录 / EXCERPT: Let b be a bounded, nearly incompressible vector field in $\operatorname{BV}_{\rm loc}(\mathbb{R} \times \mathbb{R}^{d})$. A bounded, nearly incompressible vector field...

conjecture | open | MathDB 页面 | 原文线索

V2-062744 | MATHDB-329503

The asymptotic lattice-counting conjecture for differential-operator classes

摘录 / EXCERPT: Asymptotic lattice-counting conjecture. There exist integers α, β, and ν such that $0<\alpha{\leqslant}\beta$, $0{\leqslant}\nu<d$, and, for all sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-062745 | MATHDB-329511

The conjecture on infinitely many rational curves on K3 surfaces

摘录 / EXCERPT: The rational-curve conjecture. The surface X contains infinitely many rational curves. This concerns the existence of special curves on K3 surfaces; the source presents the asser...

conjecture | open | MathDB 页面 | 原文线索

V2-062746 | MATHDB-329513

Talagrand's orthogonal structures conjecture for mixed p-spin glasses

摘录 / EXCERPT: Let G_N be the Gibbs measures of a mixed p-spin glass model on the hypercube with zero external field. The model admits an Orthogonal Structure if there is a sequence...

conjecture | solved | MathDB 页面 | 原文线索

V2-062747 | MATHDB-329530

Uniqueness of the maximal Fourier-coefficient partition

摘录 / EXCERPT: Uniqueness conjecture. The set $\frak{p}^m(\pi)$ contains only one partition in both the local and automorphic cases.

conjecture | open | MathDB 页面 | 原文线索

V2-062748 | MATHDB-329531

Arthur-parameter bound for maximal Fourier coefficients

摘录 / EXCERPT: Arthur-parameter conjecture. The following hold: (1) if $\underline{p}>\eta_{{\frak{g}^\vee,\frak{g}}}(\underline{p}(\psi))$, then $\underline{p}$ does not belong to...

conjecture | open | MathDB 页面 | 原文线索

V2-062749 | MATHDB-329548

Okounkov's dimension conjecture for q-multiple zeta values

摘录 / EXCERPT: Let ${\mathbf Z}_w^{\rm O}$ be the $\Q$-vector space generated by Okounkov's type O q-multiple zeta values $\frakz_q^{\rm O}[\bfs]$ with $|\bfs|\leq w$. Okounkov's dimension conj...

conjecture | open | MathDB 页面 | 原文线索

V2-062750 | MATHDB-329562

The differential multiplicity conjecture for nice exponents

摘录 / EXCERPT: Let L be a finite field of odd characteristic, and let s be an exponent. Write $s\not\approx 1$ for the stated exponent equivalence relation, and call s nice when it has the...

conjecture | open | MathDB 页面 | 原文线索

V2-062751 | MATHDB-329565

Absolute convergence of the geometric conjugacy-class expansion

摘录 / EXCERPT: Absolute convergence conjecture. For every $f\in C_c^\infty(G(\mathbb A)^1)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-062752 | MATHDB-329583

Realizability conjecture for sign patterns with positive pairs

摘录 / EXCERPT: Realizability conjecture. For an arbitrary sign pattern ${\bar \sigma}$, the only type of pairs (pos,neg) which can be non-realizable has either pos or neg vanishing. Equival...

conjecture | open | MathDB 页面 | 原文线索

V2-062753 | MATHDB-329609

Distinct asymptotic constants for non-isomorphic graph factors

摘录 / EXCERPT: Let H be a fixed graph on k vertices, and let β_H^d be the asymptotic constant for the minimum-length H-factor on random points in $[0,1]^d$. Given graphs H_1,H_2 and...

conjecture | open | MathDB 页面 | 原文线索

V2-062754 | MATHDB-329624

Exceptional coordinatizable MOLs without orthogonal large partial frames

摘录 / EXCERPT: Exceptional-frame conjecture. There are subdirectly irreducible coordinatizable MOLs of height at least 3 not containing an orthogonal large partial n-frame.

conjecture | open | MathDB 页面 | 原文线索

V2-062755 | MATHDB-329625

Representable $*$-regular rings and ultraproducts of artinian rings

摘录 / EXCERPT: Representable-ring ultraproduct conjecture. Every subdirectly irreducible representable $*$-regular ring can be embedded into a homomorphic image of a regular $*$-subring of an ult...

conjecture | open | MathDB 页面 | 原文线索

V2-062756 | MATHDB-329626

Unit regularity conjecture for proatomic $*$-regular rings

摘录 / EXCERPT: Unit-regularity conjecture. If R is $*$-regular and $\overline{L}(R)$ is proatomic with an orthogonal large partial n-frame, then R is unit regular. Every representable ring...

conjecture | open | MathDB 页面 | 原文线索

V2-062757 | MATHDB-329627

Positive-definite structured-tensor conjecture for uniqueness of nonlinear complementarity solutions

摘录 / EXCERPT: Let m and n be positive integers, let $\boldsymbol{q}in\boldsymbol{R}^{n}$, and let $\mathcal{A}in ST_{m,n}$ be a structured tensor. The tensor $\mathcal{A}$ is positive defini...

conjecture | open | MathDB 页面 | 原文线索

V2-062758 | MATHDB-329630

Finite-time turnover conjecture for a Muskat solution with initial amplitude 50

摘录 / EXCERPT: Let $f(x,t)$ be a solution of the Muskat interface graph equation ... and suppose that its initial profile satisfies $\|f(\cdot,0)\|_{L^{\infty}}=50$. Finite-time turnover conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-062759 | MATHDB-329631

Mahler's conjecture on prescribed rational powers of Liouville numbers

摘录 / EXCERPT: Let I and $f_{a,b}$ be as in Theorem. Let $A\subseteq\mathbb{Q}\setminus\{0\}$ satisfy $1\in A$ and the property that every non-zero integral multiple of each element of A also...

conjecture | open | MathDB 页面 | 原文线索

V2-062760 | MATHDB-329636

A squared Pólya-Szegö and Diaz-Metcalf inequality for positive linear maps

摘录 / EXCERPT: Let Φ be a unital positive linear map, and let A and B be positive operators satisfying ... and ... for positive real numbers $m_{1}\leq M_{1}$ and $m_{2}\leq M_{2}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-062761 | MATHDB-329647

Cyclotomic and modified cyclotomic difference-set classification conjecture

摘录 / EXCERPT: Cyclotomic difference-set classification conjecture. $H_{q,m}$ is a nontrivial difference set in $\mathbb{F}_q$ if and only if one of the following holds: (a) $m=2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062762 | MATHDB-329651

Chihara's conjecture on the essential spectrum of Jacobi matrices

摘录 / EXCERPT: Chihara's conjecture. The essential spectrum satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-062763 | MATHDB-329652

Roehner-Valent conjecture on eigenvalues of birth-and-death operators

摘录 / EXCERPT: Roehner-Valent conjecture. The operator has no eigenvalues:

conjecture | open | MathDB 页面 | 原文线索

V2-062764 | MATHDB-329666

The Schatten trace isoperimetric conjecture for the double layer potential

摘录 / EXCERPT: Let K be the double layer potential on the boundary of a smooth region in $\boldsymbol{R}^3$, and let tr denote the trace. For $p>1$, consider the Schatten trace...

conjecture | open | MathDB 页面 | 原文线索

V2-062765 | MATHDB-329675

Strong Fröberg-Iarrobino conjecture for homogeneous linear systems

摘录 / EXCERPT: Let ${\mathcal L}={\mathcal L}_{n,d}(m^s)$ be a homogeneous linear system of degree-d hypersurfaces in ${\mathbb P}^n$ through s general points, each with multiplicity m, and...

conjecture | open | MathDB 页面 | 原文线索

V2-062766 | MATHDB-329680

Singular-spectrum conjecture for Anderson models on antitrees

摘录 / EXCERPT: Let ${\mathbf s}$ be a sequence and let $H_\lambda={\mathcal A}_{\mathbf s}+\lambda {\mathcal V}$ be the Anderson model on the antitree ${\mathbb A}_{\mathbf s}$, with disorder par...

conjecture | open | MathDB 页面 | 原文线索

V2-062767 | MATHDB-329693

Bedford's basin conjecture for contracting automorphisms

摘录 / EXCERPT: Bedford's conjecture. The basin of attraction

conjecture | open | MathDB 页面 | 原文线索

V2-062768 | MATHDB-329694

Equality of the Loewner and univalent-function classes

摘录 / EXCERPT: Let $\mathcal D_n$ denote the unit polydisc, and let $E(\mathcal D_n)$ and $F(\mathcal D_n)$ be the two classes of normalized mappings defined in the paper, with tilded versions al...

conjecture | open | MathDB 页面 | 原文线索

V2-062769 | MATHDB-329695

The bounded-support-point conjecture for normalized Loewner mappings

摘录 / EXCERPT: Let $S^0(\mathcal D_n)$ be the compact class of normalized mappings generated by Loewner chains, and call $f\in S^0(\mathcal D_n)$ a support point if a nonconstant real part of a c...

conjecture | open | MathDB 页面 | 原文线索

V2-062770 | MATHDB-329696

The bounded-composition conjecture for normalized Loewner mappings

摘录 / EXCERPT: Let $S^0(\mathcal D_n)$ be the normalized Loewner class. Suppose that $f\in S^0(\mathcal D_n)$ is bounded by $M>0$, and let $G\in S^0(\mathcal D_n)$. Bounded-composition conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-062771 | MATHDB-329697

The bounded-Loewner-representation conjecture

摘录 / EXCERPT: Let $S^0(\mathcal D_n)$ be the normalized Loewner class, and let $\varphi_{s,t}$ be an evolution family associated with an $\mathcal M(\mathcal D_n)$-Herglotz vector field. Bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-062772 | MATHDB-329698

The Bieberbach coefficient conjecture for the class $S^0(\mathbb B_n)$

摘录 / EXCERPT: Let $\mathbb B_n$ be the unit ball in $\mathbb C^n$, and let $S^0(\mathbb B_n)$ be the normalized Loewner class. For $f\in S^0(\mathbb B_n)$, $D^kf(0)$ denotes its kth derivative...

conjecture | open | MathDB 页面 | 原文线索

V2-062773 | MATHDB-329703

Lubin's formal-group conjecture for commuting p-adic power series

摘录 / EXCERPT: Let $\mathfrak{f}$ and $\mathfrak{u}$ be a non-invertible and a non-torsion invertible series, respectively, defined over the ring of integers $\mathcal{O}$ of a finite extension o...

conjecture | open | MathDB 页面 | 原文线索

V2-062774 | MATHDB-329709

Postnikov-Stanley Riemann hypothesis for characteristic polynomials of truncated affine Weyl arrangements

摘录 / EXCERPT: Let Φ be a root system of rank ℓ with Coxeter number h. For integers a,b with $a\leq 1\leq b$, let $\mathcal{A}_{\Phi}^{[a,b]}$ be the truncated affine Weyl arrangeme...

conjecture | open | MathDB 页面 | 原文线索

V2-062775 | MATHDB-329712

The planar signed projective-cube homomorphism conjecture

摘录 / EXCERPT: Let $d\geq 2$. A signed graph is a graph together with a distinguished set of negative edges; its unbalanced-girth is the length of its shortest cycle containing an odd number of n...

conjecture | open | MathDB 页面 | 原文线索

V2-062776 | MATHDB-329713

The degree and ramification conjecture for Picard-Fuchs operators of maximally mutable Laurent polynomials

摘录 / EXCERPT: Degree and ramification conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-062777 | MATHDB-329715

Aldous's mixing-time conjecture for the constrained Ising process

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let p be the parameter of the constrained Ising process on G, and let $\tau_{\mathrm{mix}}^{\mathrm{RW}}$ denote the mixing time of the $\frac{1}{2}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-062778 | MATHDB-329718

The mutation-qG-deformation correspondence for class TG del Pezzo surfaces

摘录 / EXCERPT: Conjecture A. There is a one-to-one correspondence between $\mathfrak{P}$ and $\mathfrak{F}$. The correspondence sends P to a generic qG-deformation of the toric surface X_P.

conjecture | open | MathDB 页面 | 原文线索

V2-062779 | MATHDB-329720

The period-determines-mutation conjecture for Fano polygons

摘录 / EXCERPT: Period-determines-mutation conjecture. If P_1 and P_2 have the same singularity content and there is an affine-linear isomorphism $\varphi\colon L_{P_1}^T\to L_{P_2}^T$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-062780 | MATHDB-329722

Cummings-Feng boundary Korn inequality for elastic Helmholtz solutions

摘录 / EXCERPT: Boundary Korn inequality. There exists a positive constant $\tilde{K}$ such that, for every $\mathbf{u}\in\mathbf{H}^2(\Omega)$,

conjecture | open | MathDB 页面 | 原文线索

V2-062781 | MATHDB-329723

Zero-count conjecture for Meijer's G-function

摘录 / EXCERPT: Let $a_1,\ldots,a_p$ and $b_1,\ldots,b_p$ be real parameters, and define ... Assume $a_i\ge0$ for $i=1,\ldots,p$ and $\sum_{i=1}^{p}(b_i-a_i)>0$. Zero-count conjecture. The number of...

conjecture | open | MathDB 页面 | 原文线索

V2-062782 | MATHDB-329724

Boundary-measure conjecture for Fox's H-function

摘录 / EXCERPT: Let $W(x)=\dfrac{\prod_{i=1}^{p}\Gamma(A_i x+a_i)}{\prod_{j=1}^{q}\Gamma(B_j x+b_j)}$, let ρ be the endpoint appearing in its Fox H-function representation, and let $\mu=0$....

conjecture | open | MathDB 页面 | 原文线索

V2-062783 | MATHDB-329725

Zero-count conjecture for Fox's H-function

摘录 / EXCERPT: Zero-count conjecture. The number of zeros of

conjecture | open | MathDB 页面 | 原文线索

V2-062784 | MATHDB-329732

Hyatt's interlacing conjectures for half Eulerian polynomials

摘录 / EXCERPT: Hyatt's conjectures. For $n\geq 1$, $B_n^+(x)$ interlaces $x^nB_n^+(1/x)$, and hence $B_n(x)=B_n^+(x)+x^nB_n^+(1/x)$ has only real zeros. For $n\geq 2$, $D_n^+(x)$ interlaces...

conjecture | open | MathDB 页面 | 原文线索

V2-062785 | MATHDB-329733

Euler-zigzag stability threshold conjecture for Eulerian polynomials

摘录 / EXCERPT: Euler-zigzag stability threshold conjecture. For any $n\geq 3$, the polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-062786 | MATHDB-329744

Distribution of point classes in Picard-group cosets for odd-degree hyperelliptic curves

摘录 / EXCERPT: Let K) be the function field of a hyperelliptic curve X of odd degree. Write ... For a point $\mathfrak{p}\in X$, let $\deg\mathfrak{p}$ denote its degree and let its class in th...

conjecture | open | MathDB 页面 | 原文线索

V2-062787 | MATHDB-329746

Strong Gross-Zagier conjecture for Heegner points

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbf{Q}$, let K be an imaginary quadratic field satisfying the Heegner hypothesis, let $P_K\in E(K)$ be the Heegner point, let C be the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-062788 | MATHDB-329749

The exact-sequence conjecture for autoequivalences of elliptic surfaces

摘录 / EXCERPT: The exact-sequence conjecture. There should be a short exact sequence

conjecture | open | MathDB 页面 | 原文线索

V2-062789 | MATHDB-329753

Characterization of products with compressed fixed subgroups of automorphisms

摘录 / EXCERPT: Compressed fixed-subgroup characterization. The subgroup $\mathrm{Fix}\,\phi$ is compressed in G for every $\phi\in\operatorname{Aut}(G)$ if and only if G is of one of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-062790 | MATHDB-329758

Finite-index subgroup conjecture for bounded nonsolvable definable subgroups of SL_2

摘录 / EXCERPT: Let K be a p-adically closed field and let H be a definable subgroup of $SL_2(K)$. Assume that H is bounded and nonsolvable, and that w does not normalize H. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062791 | MATHDB-329759

Determinant identities for cubic categories at degrees seven and one and two

摘录 / EXCERPT: Determinant conjecture. In any cubic category,

conjecture | open | MathDB 页面 | 原文线索

V2-062792 | MATHDB-329767

Conjecture on intersections of Nash sets and maximal divisorial sets

摘录 / EXCERPT: Intersection conjecture. For some q, the intersection

conjecture | open | MathDB 页面 | 原文线索

V2-062793 | MATHDB-329768

Conjecture relating Nash adjacency to the thresholds k_E and k_F

摘录 / EXCERPT: Threshold conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062794 | MATHDB-329775

Conjecture on the L(1,1)-labeling number of direct products of cycles

摘录 / EXCERPT: Let C_m and C_n be cycles with $m,n\geq 12$, and suppose that neither m nor n is congruent to 0 modulo 5. The conjecture. ... The paper establishes this value in many cas...

conjecture | open | MathDB 页面 | 原文线索

V2-062795 | MATHDB-329776

Positivity conjectures for generalized Grothendieck polynomials

摘录 / EXCERPT: Let $w\in\mathbb{S}_n$, let ${\cal{KN}}_{w}^{(\beta,\alpha,\gamma)}(X)$ be the generalized Grothendieck polynomial, and let $T_w^{\beta=0,\alpha,\gamma}(x)$ denote the polynomial o...

conjecture | open | MathDB 页面 | 原文线索

V2-062796 | MATHDB-329778

Central-rank divisibility conjecture for monotonic-triangle rank polynomials

摘录 / EXCERPT: Let $r_{{\cal{MN}}_n}(t)$ be the rank-generating polynomial of the set ${\cal{MN}}_n$ of monotonic triangles, and let $\operatorname{ASM}(n)$ denote the number of alternating sign...

conjecture | open | MathDB 页面 | 原文线索

V2-062797 | MATHDB-329779

Super-activation conjecture for mean random correlated secrecy capacity

摘录 / EXCERPT: Super-activation conjecture. There exist pairs $(\mathfrak W_1,\mathfrak V_1)$ and $(\mathfrak W_2,\mathfrak V_2)$ of finite AVWCs such that

conjecture | open | MathDB 页面 | 原文线索

V2-062798 | MATHDB-329780

Kernel-growth conjecture for degree-preserving polynomial derivations

摘录 / EXCERPT: Kernel-growth conjecture. The dimension $\dim K_m$ grows polynomially in m of degree at most $N-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-062799 | MATHDB-329793

The open analog of Witten's conjecture

摘录 / EXCERPT: Open analog of Witten's conjecture. The open KdV system

conjecture | open | MathDB 页面 | 原文线索

V2-062800 | MATHDB-329803

Conjectural half-degree principle for finite reflection groups

摘录 / EXCERPT: Let V be the real reflection representation of a finite reflection group $\mathcal{W}$ of rank n, with root system Φ and basic invariants $\eta_1,\dots,\eta_n$ of degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-062801 | MATHDB-329811

Surjectivity conjecture for homomorphisms between projective cubes

摘录 / EXCERPT: Let r and k be integers with $r\geq k$, and let $\mathcal{PC}_{2r}$ and $\mathcal{PC}_{2k}$ denote projective cubes. A graph homomorphism is a map preserving adjacency. Surject...

conjecture | open | MathDB 页面 | 原文线索

V2-062802 | MATHDB-329818

Generalization of the Smith normal form theorem for the action of k partial p_k

摘录 / EXCERPT: The map $k\frac{\partial}{\partial p_k}p_k$ acts on homogeneous symmetric functions of degree n. With respect to an integral basis, let $M_k^{(n)}$ denote its matrix, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062803 | MATHDB-329820

Conjecture on linear independence of L-space knots

摘录 / EXCERPT: L-space-knot linear-independence conjecture. The set of L-space knots in S^3 is linearly independent in the knot concordance group. In particular, if L-space knots K_0 and...

conjecture | open | MathDB 页面 | 原文线索

V2-062804 | MATHDB-329824

Pairwise structure conjecture for equal smallest minors

摘录 / EXCERPT: Pairwise smallest-minors conjecture. There exists an arrangement of smallest minors $S\subset\binom{[n]}{k}$ containing I and J if and only if either the pair is 1-interlaced,...

conjecture | open | MathDB 页面 | 原文线索

V2-062805 | MATHDB-329825

Maesumi's measure-zero conjecture for reducible and defective matrix families

摘录 / EXCERPT: Let $\mathcal B$ be the space of bounded families of matrices, $\mathcal R$ the subset of reducible families, and $\mathcal D\subsetneq\mathcal R$ the subset of defective families....

conjecture | open | MathDB 页面 | 原文线索

V2-062806 | MATHDB-329826

Blondel-Maesumi generic finiteness conjecture for complex matrix families

摘录 / EXCERPT: Let the ambient space consist of finite families of complex square matrices, equipped with its natural measure, and let the finiteness property mean that some finite product attain...

conjecture | open | MathDB 页面 | 原文线索

V2-062807 | MATHDB-329828

Blondel-Jungers conjecture for pairs of binary matrices

摘录 / EXCERPT: A binary matrix is a matrix whose entries belong to $\{0,1\}$. For a finite matrix family, the finiteness property means that some finite product attains the joint spectral radius....

conjecture | open | MathDB 页面 | 原文线索

V2-062808 | MATHDB-329831

The idempotency conjecture for finitely generated parasemifields

摘录 / EXCERPT: Idempotency conjecture. Every parasemifield which is finitely generated as a semiring is additively idempotent.

conjecture | open | MathDB 页面 | 原文线索

V2-062809 | MATHDB-329833

The center of an indecomposable unitary multi-fusion category is closed

摘录 / EXCERPT: The center-closure conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062810 | MATHDB-329839

Smaller refuge with overcrowding conjecture

摘录 / EXCERPT: Smaller-refuge-with-overcrowding conjecture. There exist a patch Ω_2 and a coefficient d_4 such that every single refuge patch of measure at least $|\Omega_2|$, with the s...

conjecture | open | MathDB 页面 | 原文线索

V2-062811 | MATHDB-329853

Schrödinger subshift conjecture

摘录 / EXCERPT: Let $\mathcal{A}\subset \mathbb{R}$ be finite, and let $\Omega\subset \mathcal{A}^{\mathbb{Z}}$ be a minimal subshift that is not periodic. By minimality, the Schrödinger operators...

conjecture | open | MathDB 页面 | 原文线索

V2-062812 | MATHDB-329856

Freeness criterion for members of a pencil containing a unique non-reduced curve

摘录 / EXCERPT: Let $hh_1^{r_1}\cdots h_s^{r_s}=0$ be the equation of the unique non-reduced curve, where ${h=0}$ is reduced, $\operatorname{deg}(h_i)=m_i\geq 1$, and $r_i\geq 2$. Let D_k be a m...

conjecture | open | MathDB 页面 | 原文线索

V2-062813 | MATHDB-329864

Yui's modularity conjecture for rigid Calabi-Yau threefolds

摘录 / EXCERPT: Yui's conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-062814 | MATHDB-329869

Etingof-Nikshych-Ostrik's pivotal structure conjecture for fusion categories

摘录 / EXCERPT: Let $\mathbf{C}$ be a fusion category. A pivotal structure on $\mathbf{C}$ is a monoidal natural isomorphism ... from the identity functor to the double dual functor. Etingof-Nikshyk...

conjecture | open | MathDB 页面 | 原文线索

V2-062815 | MATHDB-329876

Melham's conjecture on odd power sums of Fibonacci numbers

摘录 / EXCERPT: Melham's conjecture. For any positive integers n,m, there is a polynomial $P_{2m-1}(x)$ of degree $2m-1$ with integer coefficients such that

conjecture | open | MathDB 页面 | 原文线索

V2-062816 | MATHDB-329877

Sign-count conjecture for the global quantity on rectangular tori

摘录 / EXCERPT: Sign-count conjecture. There are n branch points on the associated hyperelliptic curve for which $D(S)<0$ and $n+1$ branch points for which $D(S)>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-062817 | MATHDB-329880

Unbounded validity of the three-bifurcation results

摘录 / EXCERPT: Let ℓ be a positive integer. Theorem, Corollary and Corollary are the results established earlier in the paper for the corresponding construction, with ℓ bounded by 8....

conjecture | open | MathDB 页面 | 原文线索

V2-062818 | MATHDB-329907

Relative Donaldson-Thomas 4 virtual cycle product conjecture

摘录 / EXCERPT: Virtual cycle product conjecture. If orientable D-orbifolds associated with these stacks exist, then the virtual cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-062819 | MATHDB-329910

The minimizer-set conjecture for the PNS-free segment

摘录 / EXCERPT: The minimizer-set conjecture. The segment L is the minimizer set of both M_0 and N_0.

conjecture | open | MathDB 页面 | 原文线索

V2-062820 | MATHDB-329911

Cuttler-Greene-Skandera monotonicity conjecture for normalized Schur functions

摘录 / EXCERPT: Let λ and μ be partitions, and let $\bm{x}=(x_1,\ldots,x_n)$ have nonnegative entries. For a partition λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062821 | MATHDB-329916

Global epsilon-conjecture

摘录 / EXCERPT: Let S be a finite set of places containing p, let T be an R-representation of $G_{\mathbb{Q},S}$, and let T^* denote its relevant dual. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062822 | MATHDB-329957

Spectral multiplicity conjecture for Maaß newforms

摘录 / EXCERPT: Let q be an odd positive integer, and let $s(q)$ denote the number of distinct odd primes p for which p^2 divides q. Consider the new part of the discrete spectrum of the L...

conjecture | open | MathDB 页面 | 原文线索

V2-062823 | MATHDB-329959

Stability conjecture for Maaß newforms

摘录 / EXCERPT: Let $n_0(q,T)$ be the least integer such that two nonmonomial Maaß newforms of weight 0, level q, and principal character, with Laplacian eigenvalues in $(1/4,T]$ and identical...

conjecture | open | MathDB 页面 | 原文线索

V2-062824 | MATHDB-329960

Dyer's complete ortholattice conjecture for biclosed sets

摘录 / EXCERPT: Let (W,S) be a Coxeter system with positive root system $\Phi^{{\scriptscriptstyle +}}$, and let $\mathcal B$ be the poset of biclosed subsets of $\Phi^{{\scriptscriptstyle +}}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062825 | MATHDB-329961

Characterization of inversion sets of connected infinite reduced words

摘录 / EXCERPT: Let (W,S) be a Coxeter system with positive root system $\Phi^{{\scriptscriptstyle +}}$, let $\mathcal B$ denote the biclosed subsets of $\Phi^{{\scriptscriptstyle +}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062826 | MATHDB-329978

Systematic codes violating the Griesmer bound at distance 2^r+2

摘录 / EXCERPT: Systematic-code Griesmer conjecture. For every integer $r\geq 3$, there exists a systematic code with distance 2^r+2 that does not satisfy the Griesmer bound.

conjecture | open | MathDB 页面 | 原文线索

V2-062827 | MATHDB-329983

Strict separation of recursively, divisionally and multiplicatively free arrangements

摘录 / EXCERPT: Let $\mathcal{RF}^*$ denote the class of recursively free arrangements satisfying the additional condition denoted by a star in the source, let $\mathcal{DF}$ denote the class of d...

conjecture | open | MathDB 页面 | 原文线索

V2-062828 | MATHDB-329984

Divisionally free ideal-subarrangements and ideal-Shi arrangements

摘录 / EXCERPT: The ideal-arrangement conjecture. All ideal-subarrangements ${\mathcal{A}}_I$ and ideal-Shi arrangements $\operatorname{Shi}^k_{\pm I}$ are divisionally free.

conjecture | open | MathDB 页面 | 原文线索

V2-062829 | MATHDB-329987

The cycle-membership conjecture for exact skew-symmetric plane permutations

摘录 / EXCERPT: Cycle-membership conjecture. The elements n and s_n belong to the same cycle of π.

conjecture | open | MathDB 页面 | 原文线索

V2-062830 | MATHDB-330005

Saari's homographic conjecture for homogeneous potentials

摘录 / EXCERPT: Let N point particles have masses $m_k>0$ and positions $\bi{q}_k$, with homogeneous potential ... for $\alpha\ne0$. Define the moment of inertia and configurational measure by ... A...

conjecture | open | MathDB 页面 | 原文线索

V2-062831 | MATHDB-330006

Saari's original conjecture for the Newtonian N-body problem

摘录 / EXCERPT: Consider the Newtonian N-body problem, with potential corresponding to $\alpha=1$, and let I be the moment of inertia. A motion has constant moment of inertia when I is const...

conjecture | open | MathDB 页面 | 原文线索

V2-062832 | MATHDB-330007

Generalised Saari's conjecture for homogeneous potentials

摘录 / EXCERPT: Consider the homogeneous potential with exponent α as above, and the moment of inertia I. The exceptional exponents are $\alpha=-2$ and $\alpha=2$; the equal-mass rectilin...

conjecture | open | MathDB 页面 | 原文线索

V2-062833 | MATHDB-330009

The sparse random-graph path decomposition conjecture

摘录 / EXCERPT: Let $G\sim G_{n,p}$, where $p=o(1)$ and $pn/\log n\to\infty$, and let $odd(G)$ denote the number of odd-degree vertices of G. Sparse random-graph path decomposition conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-062834 | MATHDB-330012

Transfer-operator correspondence for automorphic cusp forms with unitary representations

摘录 / EXCERPT: Let (V,χ) be any unitary finite-dimensional representation of Γ_λ, and let s satisfy $\operatorname{Re}s\in(0,1)$. Let (Γ_λ,χ)-automorphic cusp...

conjecture | open | MathDB 页面 | 原文线索

V2-062835 | MATHDB-330013

Transfer-operator eigenfunctions determining resonance residues

摘录 / EXCERPT: Let Γ_λ be the Hecke triangle group, let χ be a unitary finite-dimensional representation, and let $\mathcal L_{s,\chi}$ be the associated transfer operator. At a...

conjecture | open | MathDB 页面 | 原文线索

V2-062836 | MATHDB-330019

Local convergence under conditioning of large maximal degree

摘录 / EXCERPT: Let X be the canonical process under the excursion measure $\mathbf{N}$ of the strong Markov process $X-I$ at zero. Write H for the height process, ζ for its lifetime, an...

conjecture | open | MathDB 页面 | 原文线索

V2-062837 | MATHDB-330021

Akbari-Khaghanpoor-Moazzeni colorful-path conjecture

摘录 / EXCERPT: Let G be a connected graph, let $\chi(G)$ denote its chromatic number, and call a path colorful under a proper $\chi(G)$-coloring when its $\chi(G)$ vertices have pairwise distin...

conjecture | open | MathDB 页面 | 原文线索

V2-062838 | MATHDB-330022

Forward-or-backward certifying-path conjecture

摘录 / EXCERPT: Let G be a connected k-chromatic graph. In a k-coloring, a certifying path is a path whose vertices meet the required distinct color classes, and call such a path forward or...

conjecture | open | MathDB 页面 | 原文线索

V2-062839 | MATHDB-330025

The versality conjecture for folding-in

摘录 / EXCERPT: Folding-in versality conjecture. Every such map $F\rightarrow B$ extends to a map $G\rightarrow B$, although the extension need not be unique.

conjecture | open | MathDB 页面 | 原文线索

V2-062840 | MATHDB-330030

The Schober reconstruction conjecture from versal harmonic maps

摘录 / EXCERPT: Schober reconstruction conjecture. The Schober $\mathfrak{S}$, together with a stability structure on it, can be recovered from the family

conjecture | open | MathDB 页面 | 原文线索

V2-062841 | MATHDB-330032

Ignat-Merlet conjecture on the viscosity solution as a line-energy minimizer

摘录 / EXCERPT: Ignat-Merlet conjecture. Then $(\nabla\varphi_0)^\perp$ is a global minimizer of $\mathcal{I}_f$.

conjecture | solved | MathDB 页面 | 原文线索

V2-062842 | MATHDB-330033

Unique gear-mapping parameter for prescribed prevertices

摘录 / EXCERPT: Let t_1,t_2 satisfy ... For a real parameter λ, let f be the solution of ... normalized by $J_f(0)=(0,1,0)$, and call f a gear mapping when its image is a gear-shaped dom...

conjecture | open | MathDB 页面 | 原文线索

V2-062843 | MATHDB-330034

Asciak's edge-reconstruction number conjecture for regular graphs

摘录 / EXCERPT: Asciak's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062844 | MATHDB-330036

The Main Conjecture for the Bernstein-Sato function of the Vandermonde determinant

摘录 / EXCERPT: Let $\xi_n=\prod_{1\leq i<j\leq n}(x_i-x_j)$ be the Vandermonde determinant, and for an integer partition $\lambda\vdash n$, let ξ_λ be the product of the Vandermonde det...

conjecture | open | MathDB 页面 | 原文线索

V2-062845 | MATHDB-330042

The universal Verlinde category conjecture for pre-Tannakian categories

摘录 / EXCERPT: Assume that $p>0$. A pre-Tannakian category is a $\boldsymbol{k}$-linear, rigid, symmetric monoidal abelian category with finite-dimensional Hom spaces and finite length objects. I...

conjecture | open | MathDB 页面 | 原文线索

V2-062846 | MATHDB-330053

LG mirror symmetry conjecture for BHK mirror pairs

摘录 / EXCERPT: LG mirror symmetry conjecture. Up to a change of variables, the generating function of the FJRW theory at all genera for (W,G_W) can be identified with the generating function of...

conjecture | open | MathDB 页面 | 原文线索

V2-062847 | MATHDB-330062

The inv-maj bijection for fillings and conjugate partition shapes

摘录 / EXCERPT: Let $\mathcal{F}$ be the set of all fillings of Young diagrams with positive integers. For a filling σ, let $\operatorname{inv}(\sigma)$ and $\operatorname{maj}(\sigma)$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-062848 | MATHDB-330063

Zhao's log-concavity conjecture for the Fennessey-Larcombe-French sequence

摘录 / EXCERPT: Let $\{V_n\}_{n\geq 0}$ be the Fennessey-Larcombe-French sequence, defined by $V_0=1$, $V_1=8$, and ... A sequence $\{a_n\}$ is log-concave when $a_n^2\geq a_{n-1}a_{n+1}$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-062849 | MATHDB-330070

Quasi-isometry conjecture for the electric curve-complex model

摘录 / EXCERPT: Electric-model quasi-isometry conjecture. The map φ is a quasi-isometry.

conjecture | open | MathDB 页面 | 原文线索

V2-062850 | MATHDB-330074

Weighted convolution monoidal structure from a braided strong promonoidal functor

摘录 / EXCERPT: Weighted convolution monoidal structure. If L is braided strong promonoidal then $\otimes^L$ defines a monoidal structure on $[\mathfrak{G}_q,\mathscr{V}]$.

conjecture | open | MathDB 页面 | 原文线索

V2-062851 | MATHDB-330078

Non-vanishing conjecture for homogeneous polynomial solutions of the p-Laplace equation

摘录 / EXCERPT: Non-vanishing conjecture. If $\Delta_pu=0$ for $p>1$ and $p\ne2$, then $u\equiv 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-062852 | MATHDB-330099

The rational Shuffle Conjecture

摘录 / EXCERPT: The rational Shuffle Conjecture. The following equation holds:

conjecture | open | MathDB 页面 | 原文线索

V2-062853 | MATHDB-330100

The permutation formulation of the rational Shuffle Conjecture

摘录 / EXCERPT: Permutation formulation of the rational Shuffle Conjecture. The following equation holds:

conjecture | open | MathDB 页面 | 原文线索

V2-062854 | MATHDB-330103

The SLP conjecture for nonequipointed linear spaces

摘录 / EXCERPT: Let (P,L) be a linear space with $d=\#P$, and let A be its associated ring. Set n and m to be the minimum and maximum, respectively, size of a line in L. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-062855 | MATHDB-330106

Minimal-energy equality conjecture for three, four and five partitions of the square torus

摘录 / EXCERPT: Minimal-energy equality conjecture. The three inequalities in Proposition 1 are equalities.

conjecture | open | MathDB 页面 | 原文线索

V2-062856 | MATHDB-330112

The differentiable-functions extension of the binomial product identity

摘录 / EXCERPT: Let n be a natural number, and let f and g be differentiable functions. Differentiable-functions identity. ... The preceding identities establish special cases involving linear...

conjecture | open | MathDB 页面 | 原文线索

V2-062857 | MATHDB-330118

Brändén-Krasikov-Shapiro mesh-preserving difference-operator conjecture

摘录 / EXCERPT: Brändén-Krasikov-Shapiro conjecture. The operator T preserves the set of real-rooted polynomials of degree at most m whose mesh is at least 1 if and only if $T((x)_m)$ is rea...

conjecture | open | MathDB 页面 | 原文线索

V2-062858 | MATHDB-330120

Forsgård-Shapiro coefficient-difference bound for real zeros

摘录 / EXCERPT: Forsgård-Shapiro coefficient-difference conjecture. The number of real zeros of f does not exceed $v(f)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062859 | MATHDB-330121

Shapiro's weighted Hawaiian conjecture for the first Pólya polynomial

摘录 / EXCERPT: Weighted Hawaiian conjecture.

conjecture | solved | MathDB 页面 | 原文线索

V2-062860 | MATHDB-330122

Shapiro's even-degree weighted Hawaiian inequality

摘录 / EXCERPT: Even-degree weighted Hawaiian conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062861 | MATHDB-330123

Shapiro's real-zero bound for Jensen polynomials

摘录 / EXCERPT: Jensen-polynomial zero-bound conjecture. For every such p and every relevant index i,

conjecture | open | MathDB 页面 | 原文线索

V2-062862 | MATHDB-330130

A volume bound for lattice polytopes with trinomial h^*-polynomials

摘录 / EXCERPT: Let Δ be a lattice polytope whose h^*-polynomial is ... where $b\ge 2$. Volume-bound conjecture. One has ... or equivalently, ... This conjecture extends the observed volume bou...

conjecture | open | MathDB 页面 | 原文线索

V2-062863 | MATHDB-330131

The volume lower bound for real reflexive polytopes

摘录 / EXCERPT: Real reflexive-polytope volume conjecture. If P is d-dimensional, then

conjecture | open | MathDB 页面 | 原文线索

V2-062864 | MATHDB-330132

The nonreal-root real-part bound in dimensions six and seven

摘录 / EXCERPT: Nonreal-root real-part conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-062865 | MATHDB-330133

Simon's phase-insensitivity conjecture for singular spectra

摘录 / EXCERPT: Simon's conjecture. For all $d,v\in\mathbb{N}$ and all $\theta\in\mathbb{T}^{\nu}$, $\sigma_{sing}(H_{\alpha,\theta})$ is constant in θ.

conjecture | open | MathDB 页面 | 原文线索

V2-062866 | MATHDB-330134

Connectedness characterisation of template Mandelbrot sets

摘录 / EXCERPT: Let ${\bf s} \in {\cal L}(\{0,1\})$ be a symbolic template, and let $(c_0,c_1) \in \mathbb{C}^2$. The fixed-template Mandelbrot set is ... and the fixed-map Mandelbrot set is ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-062867 | MATHDB-330135

Continuity of template Julia sets under common roots

摘录 / EXCERPT: Fix a parameter pair $(c_0,c_1)\in\mathbb{C}^2$ and let ${\bf s}$ be a template. For a template ${\bf t}=(t_k)$, its k-root is the finite sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-062868 | MATHDB-330136

Convergence of finite-root Julia sets to template Julia sets

摘录 / EXCERPT: Fix a parameter pair $(c_0,c_1)\in\mathbb{C}^2$ and let ${\bf s}$ be a template. For each n, let $\overline{\bf s}^{n}$ denote the finite n-root of ${\bf s}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062869 | MATHDB-330144

LOCC hierarchy conjecture for multipartite entanglement classes

摘录 / EXCERPT: LOCC hierarchy conjecture. For all $\underline{\boldsymbol{\alpha}},\underline{\boldsymbol{\beta}}\in P_\text{III}$,

conjecture | open | MathDB 页面 | 原文线索

V2-062870 | MATHDB-330149

Balanced generalized lower bound conjecture for simplicial polytopes

摘录 / EXCERPT: Let P be a balanced simplicial d-polytope, meaning that its underlying graph is d-colorable. For $-1\leq i\leq d-1$, let $f_i(P)$ be the number of i-dimensional faces, with...

conjecture | open | MathDB 页面 | 原文线索

V2-062871 | MATHDB-330153

Univariate Pythagorean 6-tuple factorization conjecture

摘录 / EXCERPT: Univariate factorization conjecture. Such polynomials satisfy the equation if and only if there exist $A,B\in\mathbb{H}[u]$ and $D\in\mathbb{R}[u]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-062872 | MATHDB-330157

Local freeness conjecture for pluricanonical direct images of weakly semistable morphisms

摘录 / EXCERPT: Local freeness conjecture for weakly semistable morphisms. The sheaf

conjecture | open | MathDB 页面 | 原文线索

V2-062873 | MATHDB-330158

Finite generation conjecture for log canonical rings of simple normal crossing pairs

摘录 / EXCERPT: Finite generation conjecture for log canonical rings. The log canonical ring

conjecture | open | MathDB 页面 | 原文线索

V2-062874 | MATHDB-330159

Glynn II five-weight conjecture for binary linear codes

摘录 / EXCERPT: Let m be odd with $m\geq 9$, and set ... Let ${\mathcal C}_{D(\Gamma_\rho)}$ denote the binary linear code in the Glynn II case. Glynn II code conjecture. The code...

conjecture | open | MathDB 页面 | 原文线索

V2-062875 | MATHDB-330160

Ding's conjectured difference sets from trinomials

摘录 / EXCERPT: Let $m\geq 5$ be odd. For $f\in\operatorname{GF}(2^m)[x]$, define ... The listed eleven trinomials f are the polynomials specified in the source statement. Ding's difference-set co...

conjecture | open | MathDB 页面 | 原文线索

V2-062876 | MATHDB-330161

Three-weight code conjecture for Ding's odd-m difference sets

摘录 / EXCERPT: Let $m\geq 5$ be odd, let D_f be defined from one of the eleven trinomials in Ding's difference-set conjecture, and let ${\mathcal C}_{D_f}$ be the associated binary linear code....

conjecture | open | MathDB 页面 | 原文线索

V2-062877 | MATHDB-330162

J170 three-weight code conjecture

摘录 / EXCERPT: J170 code conjecture. The binary code ${\mathcal C}_{D_f}$ has parameters

conjecture | open | MathDB 页面 | 原文线索

V2-062878 | MATHDB-330163

J171 four-weight code conjecture

摘录 / EXCERPT: J171 code conjecture. The binary code ${\mathcal C}_{D_f}$ has parameters

conjecture | open | MathDB 页面 | 原文线索

V2-062879 | MATHDB-330164

Ding's conjectured difference sets from x(x+1) polynomials

摘录 / EXCERPT: Ding's second difference-set conjecture. For the following three polynomials, with the indicated parity conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-062880 | MATHDB-330165

Takahashi's log mirror symmetry conjecture for the projective plane

摘录 / EXCERPT: Takahashi's conjecture. For all $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-062881 | MATHDB-330166

The inhomogeneous arithmetic transfer conjecture

摘录 / EXCERPT: Let F/F_0 be ramified and let $n\geq3$ be odd. Let $S(F_0)$ be the inhomogeneous symmetric space, let U_0 and U_1 be the relevant unitary groups, let $K_0\subset U_0(F_0)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-062882 | MATHDB-330170

Relative BPS integrality conjecture for del Pezzo surfaces

摘录 / EXCERPT: Relative BPS integrality conjecture. For every $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-062883 | MATHDB-330172

Uniqueness conjecture for the Angelesco equilibrium measure

摘录 / EXCERPT: Angelesco uniqueness conjecture. The measure $\mu=\lambda_1+\lambda_2$ is the only positive Borel measure in the plane satisfying these inequalities and $\mu(\mathbb C)=2$. This un...

conjecture | open | MathDB 页面 | 原文线索

V2-062884 | MATHDB-330189

Garaschuk's conjecture on fractional triangle decompositions

摘录 / EXCERPT: Let G be a graph with n vertices and minimum degree at least $\frac{3}{4}n$. Garaschuk's conjecture. If n is large enough, then G is fractionally K_3-decomposable. Fracti...

conjecture | open | MathDB 页面 | 原文线索

V2-062885 | MATHDB-330196

BCLMS conjecture on the first multiples in delta sets of three-generated semigroups

摘录 / EXCERPT: BCLMS conjectures. If $|\Delta(S)|>1$, then $2d\in\Delta(S)$. If $|\Delta(S)|>2$, then $3d\in\Delta(S)$. These conjectures refine the realization problem for delta sets by predicti...

conjecture | open | MathDB 页面 | 原文线索

V2-062886 | MATHDB-330199

Positive density conjecture for the nonzero coefficients of $\bar{B}(n)$

摘录 / EXCERPT: Let $\bar{B}(n)$ be the sequence defined in the paper, and let X tend to infinity. Positive density conjecture for $\bar{B}(n)$. There exists a constant $\sigma_2\in(0,1]$ such t...

conjecture | open | MathDB 页面 | 原文线索

V2-062887 | MATHDB-330202

Path-factor existence conjecture for graphs without small odd paths

摘录 / EXCERPT: Let $k\geq 3$, and let G be a graph. For a graph H, let $\mathcal{C}(H)$ be the set of components of H; for $i\geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062888 | MATHDB-330205

Conjecture on conjugacy of cuspidal types

摘录 / EXCERPT: Let G be an isometry group over a non-archimedean field, and let cuspidal types be representations of compact open subgroups used to construct irreducible cuspidal representation...

conjecture | open | MathDB 页面 | 原文线索

V2-062889 | MATHDB-330206

Natural matching conjecture for skew-semisimple characters

摘录 / EXCERPT: Let $[\Lambda,q,m\beta]$ and $[\Lambda',q,m\beta']$ be skew-semisimple strata, and let $\theta\in \mathcal C(\Lambda,m,\beta)^\sigma$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062890 | MATHDB-330211

The affine-irreducibility conjecture for dimensions of self-similar attractors

摘录 / EXCERPT: Let G be the relevant group of similarities of $\mathbb{R}^{d}$, and let $\Phi\subseteq G$ be an affinely irreducible iterated function system with attractor...

conjecture | open | MathDB 页面 | 原文线索

V2-062891 | MATHDB-330213

The frame-completeness conjecture for the class of all interpretability frames

摘录 / EXCERPT: Let $\mathfrak F$ be a class of ${\textup{\textbf{IL}}}$-frames, and define ... Let $\mathfrak{All}$ be the class of frames in which every S-relation implication that follows both...

conjecture | open | MathDB 页面 | 原文线索

V2-062892 | MATHDB-330218

Stability conjecture for optimal cross-intersecting families

摘录 / EXCERPT: Stability conjecture. There exists a constant $c=c(p_1,p_2)$ such that, for every pair of cross-intersecting families $U_1\subset\Omega_1$ and $U_2\subset\Omega_2$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-062893 | MATHDB-330225

Factorization conjecture for weighted Aztec-region matching generating functions

摘录 / EXCERPT: Let x,y,z be indeterminate edge weights, and let $A^{(i)}_{a,b,c}(x,y,z)$ and $F^{(i)}_{a,b,c}(x,y,z)$, for $i=1,2,3$, be the corresponding weighted versions of the graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-062894 | MATHDB-330231

Nonexistence conjecture for quadratic 5-periodic points

摘录 / EXCERPT: Nonexistence conjecture. There are no rational values c such that ϕ_c has a 5-periodic point in $\mathbb{Q}_{\mathrm{quad}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-062895 | MATHDB-330234

Bayer-Macrì-Stellari quadratic Bogomolov-Gieseker inequality conjecture

摘录 / EXCERPT: Bayer-Macrì-Stellari quadratic Bogomolov-Gieseker inequality conjecture. For every $\nu_{\omega,B}$-semistable object $E\in\mathcal{B}_{\omega,B}$,

conjecture | open | MathDB 页面 | 原文线索

V2-062896 | MATHDB-330235

Conjecture on limiting third twisted Chern character for tilt-stable objects

摘录 / EXCERPT: Limiting BG conjecture. Suppose there is an open neighborhood $U\subset\mathbb{R}^2$ containing $(0,\overline{\beta}(E))$ such that, for every $(\alpha,\beta)\in U$ with $\alpha>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-062897 | MATHDB-330238

Rationality conjecture for irreducible free plane curves

摘录 / EXCERPT: Let $C\subset\mathbb{P}^2$ be an irreducible plane curve of degree $d\geq 2$, and suppose that C is a free divisor, meaning that its logarithmic vector-field bundle splits as a d...

conjecture | open | MathDB 页面 | 原文线索

V2-062898 | MATHDB-330240

Freeness conjecture for the paper's rational cuspidal curve families

摘录 / EXCERPT: Let C be any of the rational cuspidal curves described in Proposition 2(i), Proposition 3, or Proposition 4, with degree d. Freeness conjecture. If $d\geq 5$, then C is a fre...

conjecture | open | MathDB 页面 | 原文线索

V2-062899 | MATHDB-330241

The harmonic-polynomial characterization of non-injectivity sets

摘录 / EXCERPT: Let $S\subset\mathbb R^d$. Call S an injectivity set for the spherical mean transform if vanishing of all spherical means centered on S forces the underlying function or distri...

conjecture | open | MathDB 页面 | 原文线索

V2-062900 | MATHDB-330250

Akiyama-Watanabe-Albertson-Haas conjecture on induced forests in bipartite planar graphs

摘录 / EXCERPT: Akiyama-Watanabe-Albertson-Haas conjecture. Every bipartite planar graph of order n admits an induced forest of order at least

conjecture | open | MathDB 页面 | 原文线索

V2-062901 | MATHDB-330251

Kowalik et al.'s conjecture on induced forests in planar graphs of girth at least five

摘录 / EXCERPT: Kowalik et al.'s conjecture. The graph G admits an induced forest of order at least

conjecture | open | MathDB 页面 | 原文线索

V2-062902 | MATHDB-330253

Kramer-Tunnell local root-number conjecture for hyperelliptic Jacobians

摘录 / EXCERPT: Let K be a local field of characteristic zero, let L/K be a quadratic extension, and let C/K be a hyperelliptic curve with Jacobian J. Write C^L and J^L for the quadrat...

conjecture | open | MathDB 页面 | 原文线索

V2-062903 | MATHDB-330265

Supercuspidal form of Jacquet's local converse conjecture

摘录 / EXCERPT: Let F be a locally compact non-archimedean local field, let ${\mathrm{G}}_n={\mathrm{GL}}_n(F)$, and let ψ be a nontrivial additive character of F. Assume that π_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-062904 | MATHDB-330313

Narkiewicz's conjecture on property (P) for $[200~\mathbb{Q}^{(d)}$[201~

摘录 / EXCERPT: For a field F, define property (P) by requiring that every polynomial $g(x)\in F[x]\setminus F$ for which there is an infinite set $X\subseteq F$ satisfying $g(X)=X$ has degree...

conjecture | open | MathDB 页面 | 原文线索

V2-062905 | MATHDB-330318

Complex BBM phase-diagram conjecture

摘录 / EXCERPT: Let $\beta=\sigma+i\tau\in\mathbb C$, let $\rho\in[-1,1]$, and let $p_t(\beta)$ be the log-partition function per unit time for the complex BBM energy model. Let B_1,B_2,B_3 deno...

conjecture | open | MathDB 页面 | 原文线索

V2-062906 | MATHDB-330322

Geelen-Gerards-Whittle conjecture for minor-closed classes of binary codes

摘录 / EXCERPT: Geelen-Gerards-Whittle conjecture. Either $\mathcal{G} \subseteq \mathcal{C}$ and

conjecture | open | MathDB 页面 | 原文线索

V2-062907 | MATHDB-330325

The pure braid group criterion for hyperplane arrangements

摘录 / EXCERPT: Let K be the subgroup of $\uppi_{1}(\mathbb{C}^{n}\backslash \mathcal{H}_{\mathbb{C}})$ generated by the J-twists, where J ranges over all subsets of the flopping curves, and...

conjecture | open | MathDB 页面 | 原文线索

V2-062908 | MATHDB-330326

The radial-limit identity for the universal mock theta function g_3 at cubic roots of unity

摘录 / EXCERPT: Radial-limit identity. The following identity should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-062909 | MATHDB-330329

Leader-Long shortest-path 1-switch conjecture for the hypercube

摘录 / EXCERPT: Leader-Long's conjecture. The following equivalent statements should hold: in every 2-coloring of $E(Q_n)$, there are antipodal vertices connected by a path of length n which c...

conjecture | open | MathDB 页面 | 原文线索

V2-062910 | MATHDB-330333

Reduced Gröbner basis conjecture for critical ideals of trees

摘录 / EXCERPT: Reduced Gröbner basis conjecture. If $1\leq j\leq n$, then $\mathcal{B}_j$ is a reduced Gröbner basis for the critical ideal $I_j(T,X)$ with respect to the degree lexicographic ord...

conjecture | open | MathDB 页面 | 原文线索

V2-062911 | MATHDB-330343

Field-extension conjecture for square roots of cyclotomic units

摘录 / EXCERPT: Let $\czeta_k$ and $\czeta_m$ be primitive roots of unity. Field-extension conjecture. ... Only the implication from left to right is open; the reverse implication is established in th...

conjecture | open | MathDB 页面 | 原文线索

V2-062912 | MATHDB-330345

Limit-point conjecture for the generalized Rogers-Ramanujan continued fraction

摘录 / EXCERPT: Let $a=\zeta_k$ and $x=\zeta_m$ be primitive roots of unity, with $m\geq3$. Limit-point conjecture. The continued fraction $K_a(x)$ converges exactly when ... and hence exactly when...

conjecture | open | MathDB 页面 | 原文线索

V2-062913 | MATHDB-330356

Bounded zero-dynamics conjecture for spline path following with redundancy resolution

摘录 / EXCERPT: Let x denote the system state, u the control input, and let x_c denote the configuration variables subject to joint limits. Let $r(x_c)$ be the redundancy-resolution objectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-062914 | MATHDB-330363

Deligne's conjecture for Shalika motive twists

摘录 / EXCERPT: Let $M(\Pi)$ be the conjectural motive over the totally real field F associated with the Shalika-type automorphic representation Π, and let n be the parameter for which...

conjecture | open | MathDB 页面 | 原文线索

V2-062915 | MATHDB-330368

The n/d-conjecture for central indecomposable hyperplane arrangements

摘录 / EXCERPT: Let $f_\mathscr{A}$ define a central reduced indecomposable arrangement of d hyperplanes in $\mathbb{C}^n$. The n/d-conjecture. The number $-n/d$ is a root of the Bernstein--Sa...

conjecture | open | MathDB 页面 | 原文线索

V2-062916 | MATHDB-330370

Nakamura's conjecture on consecutive Wilf-equivalence

摘录 / EXCERPT: Let τ and τ' be patterns. They are c-Wilf-equivalent if $P_{\tau}(0,z)=P_{\tau'}(0,z)$, and strongly c-Wilf-equivalent if $P_{\tau}(u,z)=P_{\tau'}(u,z)$. Nakamura's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-062917 | MATHDB-330371

Elizalde-Noy conjecture for the pattern 1423

摘录 / EXCERPT: Let $P_{1423}(0,z)$ be the generating function for permutations avoiding the consecutive pattern 1423, and write ... A formal power series is D-finite if it satisfies a linear diff...

conjecture | open | MathDB 页面 | 原文线索

V2-062918 | MATHDB-330378

The Bernstein-Sato root conjecture for indecomposable arrangements

摘录 / EXCERPT: Arrangement Bernstein-Sato conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-062919 | MATHDB-330382

The lattice identifiability threshold conjecture for uniform labels

摘录 / EXCERPT: Let ${\mathbf Z}_n^d$ be the d-dimensional lattice with periodic side length n, whose vertices receive independent uniformly distributed labels from q labels. Identifiability...

conjecture | open | MathDB 页面 | 原文线索

V2-062920 | MATHDB-330383

The critical-exponent conjecture for jigsaw puzzle identifiability

摘录 / EXCERPT: Consider the random jigsaw puzzle on the $n\times n$ lattice, where each puzzle piece has four edge labels chosen from q possible jigs, with the induced dependencies between adja...

conjecture | open | MathDB 页面 | 原文线索

V2-062921 | MATHDB-330384

Factorization-pattern conjecture for polynomials in short intervals

摘录 / EXCERPT: Let A be the polynomial ring over a finite field of size q. For a monic polynomial $f\in A$ of degree d and a positive integer $m<\deg(f)$, let ... For a partition λ of...

conjecture | open | MathDB 页面 | 原文线索

V2-062922 | MATHDB-330388

Counting diverging-root eigenstates near special temperatures

摘录 / EXCERPT: Fix the magnetic field in the XXZ chain and let $T_m=\frac{h}{2m\gamma}$ with $m\in\mathbb{N}$. Consider eigenstates in the sector with spin s, equivalently with...

conjecture | open | MathDB 页面 | 原文线索

V2-062923 | MATHDB-330395

Nešetřil-Ossona de Mendez limit-modeling conjecture for nowhere-dense classes

摘录 / EXCERPT: Let ${\cal C}$ be a nowhere-dense class of graphs. A sequence of graphs from ${\cal C}$ is first order convergent if the limiting frequency of every first-order property exists. A...

conjecture | open | MathDB 页面 | 原文线索

V2-062924 | MATHDB-330410

Granath's sign conjecture for truncation errors of the Landau constants expansion

摘录 / EXCERPT: Granath's conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-062925 | MATHDB-330414

Gopalan-Klivans-Zuckerman conjecture on Reed-Muller list-decoding radius

摘录 / EXCERPT: Gopalan-Klivans-Zuckerman conjecture. For fixed d and finite field $\mathbb{K}$,

conjecture | open | MathDB 页面 | 原文线索

V2-062926 | MATHDB-330415

The rational cuspidal curve freeness conjecture

摘录 / EXCERPT: Rational cuspidal freeness conjecture. (i) Any rational cuspidal curve C in the plane is either free or nearly free. (ii) An irreducible plane curve C which is either free or n...

conjecture | open | MathDB 页面 | 原文线索

V2-062927 | MATHDB-330416

The branch-number conjecture for free and nearly free plane curves

摘录 / EXCERPT: Branch-number conjecture. (i) Any free irreducible plane curve C has only singularities with at most two branches. (ii) Any nearly free irreducible plane curve C has only singu...

conjecture | open | MathDB 页面 | 原文线索

V2-062928 | MATHDB-330419

Rains's positivity conjecture for Koornwinder moments

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\lambda_2,\dots,\lambda_m)$ be a partition, and let $K_{\lambda}(\xi)$ denote the Koornwinder moment defined by ... where Z_N is the ASEP partition function...

conjecture | open | MathDB 页面 | 原文线索

V2-062929 | MATHDB-330421

Strong law of large numbers for loop lengths

摘录 / EXCERPT: Consider a random loop configuration in a box Λ of side length L, with loop lengths $\ell_j(\omega)$ ordered decreasingly. Let $m(\beta)$ denote the limiting density of p...

conjecture | open | MathDB 页面 | 原文线索

V2-062930 | MATHDB-330426

Stanley's P-partitions conjecture on symmetric P-partition enumerators

摘录 / EXCERPT: Let (P,ω) be a finite labeled poset, and let $\Gamma(P,\omega)$ be its P-partition enumerator, a quasisymmetric function. An isomorphism of labeled posets is a bijection p...

conjecture | open | MathDB 页面 | 原文线索

V2-062931 | MATHDB-330427

Angel-Gorin-Holroyd-Romik-Virág conjecture on random reduced words

摘录 / EXCERPT: Let $w_0=[n,n-1,\ldots,1]\in S_n$ be the longest permutation, let p be its inversion number, and choose a reduced word $(a_1,\ldots,a_p)$ uniformly at random. Form the initial pr...

conjecture | open | MathDB 页面 | 原文线索

V2-062932 | MATHDB-330450

The finite Garside-shadow conjecture for n-low elements

摘录 / EXCERPT: Let (W,S) be a Coxeter system. For $n\in\mathbb N$, define the dominance depth of a positive root as the number of positive roots it strictly dominates, let the n-small roots b...

conjecture | open | MathDB 页面 | 原文线索

V2-062933 | MATHDB-330451

The Sigma-map bijectivity conjecture for n-low elements

摘录 / EXCERPT: Let (W,S) be a Coxeter system. Let $L_n(W)$ denote the set of n-low elements and let $\Lambda_n(W)$ denote the corresponding set of admissible collections of n-small roots; f...

conjecture | open | MathDB 页面 | 原文线索

V2-062934 | MATHDB-330460

The winding-number-zero extinction-time conjecture for curve diffusion flow

摘录 / EXCERPT: Let a curve diffusion flow start from a smooth immersed curve with winding number zero, and let its initial length be L. Let T_{flow} denote its maximal existence time...

conjecture | open | MathDB 页面 | 原文线索

V2-062935 | MATHDB-330461

Stability of standard planar double bubbles under area-preserving gradient flows

摘录 / EXCERPT: Let a standard planar double bubble evolve by a sufficiently smooth area-preserving gradient flow. Stability conjecture. Standard planar double bubbles are stable under sufficientl...

conjecture | open | MathDB 页面 | 原文线索

V2-062936 | MATHDB-330462

Kuznetsov-Polishchuk conjecture on semi-orthogonal decompositions of Grassmannians

摘录 / EXCERPT: Let X be the Grassmannian underlying the bounded derived category $D^b(X)$, let ρ be a path in the rectangle $\bigl[0,n-k\bigr]\times\bigl[0,k\bigr]$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-062937 | MATHDB-330471

Dvořák's conjecture on coloring squares of girth-five planar graphs

摘录 / EXCERPT: Let G be a planar graph with girth 5, let Δ denote its maximum degree, and let M be a constant. The Dvořák conjecture. There exists some constant M such that every p...

conjecture | open | MathDB 页面 | 原文线索

V2-062938 | MATHDB-330479

Shareshian-Wachs generalized q-Eulerian q-unimodality conjecture

摘录 / EXCERPT: Shareshian and Wachs's conjecture. For all $r\in[n]$, the palindromic polynomial $A_n^{(r)}(q,t)$ is q-unimodal. This is presented as a consequence of the chromatic-quasisymmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-062939 | MATHDB-330494

Quadratic generation conjecture for toric ideals of twinned order polytopes

摘录 / EXCERPT: Let P and Q be finite posets with $|P|=|Q|=d$. For the toric ideal $I_{\Omega(P,-Q)}$ associated with the twinned order polytope $\Omega(P,-Q)$, quadratic generation conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-062940 | MATHDB-330509

Dehmer-Kraus-Schaumann's minimal dendrimer entropy conjecture

摘录 / EXCERPT: Dehmer-Kraus-Schaumann's conjecture. The star graph has the minimal value of $I_{f^3}(D)$.

conjecture | open | MathDB 页面 | 原文线索

V2-062941 | MATHDB-330513

Legendre-polynomial condition for increasing equilibrium density

摘录 / EXCERPT: Increasing-density conjecture. For any given $t>0$, $f_d(t)$ is an increasing function of d.

conjecture | open | MathDB 页面 | 原文线索

V2-062942 | MATHDB-330516

Feldman-Ilmanen-Knopf conjecture on blow-up limits of the Kähler-Ricci flow

摘录 / EXCERPT: Let $X=\mathbb{CP}^n$ blown up at one point, let E be the exceptional divisor, and let $g(t)$ be a $U(n)$-invariant solution of the Kähler-Ricci flow on X for $t\te[0,T)$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-062943 | MATHDB-330519

Incidence coloring conjecture for Cartesian products

摘录 / EXCERPT: Let a graph G be a $(\Delta+1)$-graph if it admits an incidence coloring with $\Delta(G)+1$ colors, and let $\mathbin{\square}$ denote the Cartesian product of graphs. Let H be...

conjecture | open | MathDB 页面 | 原文线索

V2-062944 | MATHDB-330527

Equilateral-set conjecture for finite-dimensional normed spaces

摘录 / EXCERPT: Let X be a normed vector space of dimension n. An equilateral set is a set of points whose pairwise distances are all equal. Equilateral-set conjecture. Every normed n-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-062945 | MATHDB-330531

Capacity conjecture for transitive codes with diverging primal and dual distances

摘录 / EXCERPT: Let $\{\mathcal{C}_n\}$ be a sequence of binary linear codes with blocklengths $N_n\to\infty$, rates $r_n\to r$ for some $r\in(0,1)$, and transitive permutation group for each code...

conjecture | open | MathDB 页面 | 原文线索

V2-062946 | MATHDB-330541

Jacobson's tree decomposition conjecture for regular bipartite graphs

摘录 / EXCERPT: Let T be a tree of size r, meaning a tree with r edges, and let a graph be T-decomposable when its edges can be partitioned into subgraphs isomorphic to T. Jacobson's con...

conjecture | open | MathDB 页面 | 原文线索

V2-062947 | MATHDB-330546

Virtualization conjecture for type $D_4^{(3)}$ KR crystals

摘录 / EXCERPT: Let $\mathfrak{g}$ be of type $D_4^{(3)}$, and let $\widehat{\mathfrak{g}}$ be the folded type $D_4^{(1)}$. Write $\widehat{B}^{r,s}$ for KR crystals of type $D_4^{(1)}$. KR virtua...

conjecture | open | MathDB 页面 | 原文线索

V2-062948 | MATHDB-330548

Tensor-product realization conjecture for the virtual type D4^(1) crystal

摘录 / EXCERPT: Let $\widetilde{B}^{3,s}$ be the crystal represented by tableaux in a $3\times s$ rectangle with classical decomposition ... Let $\widehat{B}^{3,s}$ and $\widehat{B}^{4,s}$ be the re...

conjecture | open | MathDB 页面 | 原文线索

V2-062949 | MATHDB-330549

X=M conjecture for type D4^(3) KR crystals

摘录 / EXCERPT: X=M conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-062950 | MATHDB-330550

Rigged-configuration bijection and energy conjecture for type D4^(3)

摘录 / EXCERPT: Let ... be a tensor product of type $D_4^{(3)}$ KR crystals. Let $\operatorname{RC}(B)$ be the corresponding rigged configurations, let Φ be the recursively defined map to B,...

conjecture | open | MathDB 页面 | 原文线索

V2-062951 | MATHDB-330551

Virtualization compatibility conjecture for the rigged-configuration bijection

摘录 / EXCERPT: Let ... be a tensor product of KR crystals, with virtualization map v to the corresponding virtual crystal, and let Φ be the rigged-configuration bijection. Virtualization com...

conjecture | open | MathDB 页面 | 原文线索

V2-062952 | MATHDB-330552

Affine crystal isomorphism conjecture for type D4^(3) rigged configurations

摘录 / EXCERPT: Let ... be a tensor product of type $D_4^{(3)}$ KR crystals, and let $\operatorname{RC}(B)$ be its rigged-configuration set equipped with the conjectural affine crystal operators. Af...

conjecture | open | MathDB 页面 | 原文线索

V2-062953 | MATHDB-330553

Uniform affine crystal structure conjecture for rigged configurations

摘录 / EXCERPT: Let $\mathfrak{g}$ be of affine type other than $A_n^{(1)}$. Let N_0 and κ be as in the source, and let $(c_a)_{a\in I}$ and $(t_a)_{a\in I}$ be the Kac-label data and ass...

conjecture | open | MathDB 页面 | 原文线索

V2-062954 | MATHDB-330554

Lowest-weight rigged-configuration formula conjecture

摘录 / EXCERPT: Let $\mathfrak{g}$ be of affine type other than $A_n^{(1)}$, with parameters N_0, κ, c_a, and t_a as defined in the source. For a classically lowest weight element...

conjecture | open | MathDB 页面 | 原文线索

V2-062955 | MATHDB-330556

The twisting conjecture for quadrangulation flip graphs and simplicial complexes

摘录 / EXCERPT: Let Q and Q' be quadrangulations related by twisting along one inner edge. Write γ_Q and $\gamma_{Q'}$ for their undirected flip graphs, and G_Q and $G_{Q'}$ for their...

conjecture | open | MathDB 页面 | 原文线索

V2-062956 | MATHDB-330564

The binary-operation identity for the discrete polynomials S and T

摘录 / EXCERPT: The binary-operation identity. The relations

conjecture | open | MathDB 页面 | 原文线索

V2-062957 | MATHDB-330570

The Dedekind-domain containment conjecture for idempotent plethories

摘录 / EXCERPT: Let D be a Dedekind domain of characteristic zero with quotient field K. The Dedekind-domain containment conjecture. Every idempotent D-plethory is contained in $K[e]$, or, e...

conjecture | open | MathDB 页面 | 原文线索

V2-062958 | MATHDB-330573

Quasi-polynomiality conjecture for monotone orbifold Hurwitz numbers

摘录 / EXCERPT: The quasi-polynomiality conjecture. The monotone orbifold Hurwitz numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-062959 | MATHDB-330576

Decidability of parity and Wilf-equivalence for single forbidden permutations

摘录 / EXCERPT: Let $\mathcal{F}$ and $\mathcal{F}'$ be forbidden sets of permutation patterns with $|\mathcal{F}|=|\mathcal{F}'|=1$. Single-permutation decidability conjecture. Both the parity pr...

conjecture | open | MathDB 页面 | 原文线索

V2-062960 | MATHDB-330585

The \(M5\)-brane conjecture for \((1,0)\) BCOV theory

摘录 / EXCERPT: Consider a single $M5$ brane on $\mathbb{R}^7\times(\mathbb{R}^4/\mathbb{Z}_N)$, located at the singular point of the A_N singularity and spanning six of the seven directions in...

conjecture | open | MathDB 页面 | 原文线索

V2-062961 | MATHDB-330586

The equivalence conjecture for the classical BCOV interactions \(I\) and \(\widehat I\)

摘录 / EXCERPT: On $\mathbb{C}^3$, let I be the previously defined classical BCOV interaction on the fields $\PV^{*,*}(\mathbb{C}^3)[[t]][2]$, and let $\widehat I$ be the classical interaction o...

conjecture | open | MathDB 页面 | 原文线索

V2-062962 | MATHDB-330587

The equivalence conjecture for the classical BCOV interactions \(J\) and \(I_{0,0,*}\)

摘录 / EXCERPT: Let J be the classical BCOV interaction defined using the coefficients obtained from integrals over $\overline{\mathcal{M}}_{0,n}$, and let $I_{0,0,*}$ be the purely closed-strin...

conjecture | open | MathDB 页面 | 原文线索

V2-062963 | MATHDB-330603

Eldar-Needell-Plan conjecture on the minimal measurement number for low-rank recovery

摘录 / EXCERPT: Let $\mathbb H$ be either $\mathbb R$ or $\mathbb C$, let $\mathcal L_r^{\mathbb H}$ denote the relevant class of matrices of rank at most r, and let ${\bf M}_{\mathcal A}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-062964 | MATHDB-330606

Braun-Solus stable hypersimplex h-star unimodality conjecture

摘录 / EXCERPT: For positive integers n and r, let $\Delta_{n,2}^{\operatorname{stab}(r)}$ denote the r-stable hypersimplex. Braun-Solus' conjecture. The h^*-vector of...

conjecture | open | MathDB 页面 | 原文线索

V2-062965 | MATHDB-330617

Recognition conjecture for generically highly transitive groups of finite Morley rank

摘录 / EXCERPT: Recognition conjecture. If (X,G) is transitive and generically $(n+2)$-transitive, then (X,G) is equivalent to

conjecture | open | MathDB 页面 | 原文线索

V2-062966 | MATHDB-330626

The Donaldson-number formula conjecture for moduli spaces of degree-d sheaves

摘录 / EXCERPT: Let $\pi:\mathrm{M}(d,1)\to |\mathcal{O}_{\mathbb{P}^{2}}(d)|$ be the Fitting map, and let ... Here g denotes the dimension of a general fiber of π. Donaldson-number formula co...

conjecture | open | MathDB 页面 | 原文线索

V2-062967 | MATHDB-330631

Gross-Zagier and Gross-Kohnen-Zagier conjecture for CM values of weight-4 automorphic Green's functions

摘录 / EXCERPT: Let $N\in\{1,2,3,4\}$, and let $z,z'\in\mathfrak H$ be quadratic irrational points satisfying ... Assume that $z\notin\varGamma_0(N)z'$, where $\varGamma_0(N)$ is the Hecke congrue...

conjecture | open | MathDB 页面 | 原文线索

V2-062968 | MATHDB-330635

Existence of maximal cells in Grassmann polytopes

摘录 / EXCERPT: Maximal-cell conjecture. Every nonempty Grassmann polytope has a maximal cell.

conjecture | open | MathDB 页面 | 原文线索

V2-062969 | MATHDB-330643

The multiple-cover formula for quotient invariants of abelian threefolds

摘录 / EXCERPT: Multiple-cover formula. For all $g\geq2$, $d_1,d_2>0$, and $d_3\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-062970 | MATHDB-330656

Dolgachev's Torelli conjecture for logarithmic bundles of hyperplane arrangements

摘录 / EXCERPT: Let $\mathcal{H}=\{H_{1},\ldots,H_{\ell}\}$ be an arrangement of $\ell\geq n+2$ hyperplanes on $\mathbf{P}^{n}$. Assume that the logarithmic bundle...

conjecture | open | MathDB 页面 | 原文线索

V2-062971 | MATHDB-330663

The cycle characterization conjecture for cyclically fully commutative elements in type A

摘录 / EXCERPT: Let $W(A_n)$ be the Coxeter group of type A_n, identified with the symmetric group S_{n+1}, and let $w\in W(A_n)$ correspond to a permutation with disjoint cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-062972 | MATHDB-330666

Conjecture on linear perfect codes in the l_2 metric

摘录 / EXCERPT: Conjecture. There are no linear perfect codes with parameters $(n,r,2)$ except when $r=1$ or

conjecture | open | MathDB 页面 | 原文线索

V2-062973 | MATHDB-330670

The UMTC realization conjecture for completely rational conformal nets

摘录 / EXCERPT: A unitary modular tensor category (UMTC) is a unitary braided fusion category with non-degenerate braiding. For a completely rational conformal net $\mathcal{A}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-062974 | MATHDB-330672

The reverse-braiding realization conjecture for completely rational conformal nets

摘录 / EXCERPT: For a completely rational conformal net $\mathcal{A}$, write $\operatorname{Rep}(\mathcal{A})^{\mathrm{rev}}$ for the unitary modular tensor category obtained by replacing its brai...

conjecture | open | MathDB 页面 | 原文线索

V2-062975 | MATHDB-330677

The longer-snake conjecture for Kendall snake-in-the-box codes

摘录 / EXCERPT: Longer-snake conjecture. There exists a $(2n+1,M_{2n+1},\mathcal{K})$-snake with

conjecture | open | MathDB 页面 | 原文线索

V2-062976 | MATHDB-330678

The super-long-snake conjecture for Kendall snake-in-the-box codes

摘录 / EXCERPT: Super-long-snake conjecture. There exists a $(2n+1,M_{2n+1},\mathcal{K})$-snake with

conjecture | open | MathDB 页面 | 原文线索

V2-062977 | MATHDB-330679

Conjecture on commutator rank and bidegrees of matrix-valued inner functions

摘录 / EXCERPT: Let Θ be a $d\times d$ matrix-valued inner function on the bidisk, and let S_{z_1} denote the associated compressed shift operator. Write $\deg_1\Theta$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-062978 | MATHDB-330686

Conjecture on generators for the palindromic IA-automorphism group of a free nilpotent group

摘录 / EXCERPT: Let ${\rm N}_{n,3}$ be the free nilpotent group of rank n and class 3, and let μ_{ij} denote the elementary palindromic automorphism that sends x_i to $x_jx_ix_j$ and fix...

conjecture | open | MathDB 页面 | 原文线索

V2-062979 | MATHDB-330689

Phase-alignment conjecture for the three-user semi-symmetric Gaussian interference channel

摘录 / EXCERPT: Consider the three-user semi-symmetric complex Gaussian interference channel ... ... where $P_k=P$ for $k=1,2,3$, and write $\phi(g_1)$ and $\phi(g_2)$ for the phases of g_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-062980 | MATHDB-330692

Signed-shift allowed-pattern inequality conjecture

摘录 / EXCERPT: Let $\mathcal A_n(\Sigma_k)$ and $\mathcal A_n(\Sigma^-_k)$ denote the sets of allowed patterns of length n for the shifts Σ_k and Σ^-_k, respectively. Allowed-patt...

conjecture | open | MathDB 页面 | 原文线索

V2-062981 | MATHDB-330697

The exhaustive classification of planar 4-critical P_7-free graphs

摘录 / EXCERPT: A graph is planar if it can be embedded in the plane without crossings. A graph is 4-critical if its chromatic number is 4 and deleting any edge lowers its chromatic number. A grap...

conjecture | open | MathDB 页面 | 原文线索

V2-062982 | MATHDB-330708

A multi-sequence formula for the Möbius function of the permutation poset

摘录 / EXCERPT: Let $\pi=\pi_1\oplus\cdots\oplus\pi_n$ be a decomposable permutation with t sequences of equal components ... of respective lengths α_i, for $1\le i\le t$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-062983 | MATHDB-330711

Finiteness conjecture for bounded-genus L-space knots in twist families

摘录 / EXCERPT: Let $\{K_n\}$ be a twist family of knots, obtained by twisting along a fixed unknot disjoint from the initial knot, and let $g(K_n)$ denote the genus of K_n. Twist-family finiten...

conjecture | open | MathDB 页面 | 原文线索

V2-062984 | MATHDB-330712

Finiteness conjecture for bounded-genus L-space surgeries in twist families

摘录 / EXCERPT: Let $\{(K_n,r_n)\}$ be a twist family of surgeries, where r_n is the corresponding surgery slope on K_n. Twist-family surgery finiteness conjecture. For every integer $N\geq 0$...

conjecture | open | MathDB 页面 | 原文线索

V2-062985 | MATHDB-330713

Lee-Zhang conjecture on length-spectrum domination by Fuchsian representations

摘录 / EXCERPT: Lee-Zhang conjecture. For any Hitchin representation $\rho:\Gamma\to\operatorname{PSL}(n,\mathbb{R})$, there is a Fuchsian representation j such that

conjecture | open | MathDB 页面 | 原文线索

V2-062986 | MATHDB-330716

The dimension formula for affine Springer fibers of regular symmetric-space elements

摘录 / EXCERPT: Let F be the Laurent series field with valuation v, let (G,G_0) be the symmetric pair considered in the paper, and let ... be regular semisimple. Regard XY as an element of...

conjecture | open | MathDB 页面 | 原文线索

V2-062987 | MATHDB-330718

The secant linear virtual dimension conjecture for hypersurfaces through general points

摘录 / EXCERPT: Let X be the blow-up of ${\mathbb P}^n$ at $n+3$ general points, and let D be a divisor on X as in the source. Let $\operatorname{\sigma ldim}(D)$ denote the secant linear vi...

conjecture | open | MathDB 页面 | 原文线索

V2-062988 | MATHDB-330720

The F-conjecture for divisors on \mathcal{X}_{n+2,(n-2)}

摘录 / EXCERPT: Let S be the indexing set used in the source, and let $\mathcal{X}_{n+2,(n-2)}$ be the corresponding space. For a divisor D of the form given in the source, define the intersec...

conjecture | open | MathDB 页面 | 原文线索

V2-062989 | MATHDB-330722

The rank distribution conjecture for quadratic twists ordered by canonical height

摘录 / EXCERPT: Rank distribution conjecture. For $\iota\in\{1,2\}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-062990 | MATHDB-330729

Escaping endpoints always explode

摘录 / EXCERPT: Escaping endpoints always explode. The set $\tilde{E}(f_a)$ is totally separated, and hence ∞ is an explosion point of $\tilde{E}(f_a)\cup\{\infty\}$.

conjecture | open | MathDB 页面 | 原文线索

V2-062991 | MATHDB-330741

Blondeau et al.'s differential 8-uniformity conjecture for two power functions

摘录 / EXCERPT: Blondeau et al.'s conjecture. For each of these two values of d, F_d is differentially 8-uniform, and all values $0,2,4,6,8$ occur in its differential spectrum. The paper pre...

conjecture | open | MathDB 页面 | 原文线索

V2-062992 | MATHDB-330742

Classification conjecture for Klein simple group Mori fiber spaces over the projective line

摘录 / EXCERPT: Let $G_K\cong \mathrm{PSL}_2(\mathbb{F}_7)$ be the Klein simple group, and let $X_n/\mathbb{P}^1$ be the G_K-Mori fiber spaces described above. A G_K-Mori fiber space is a thre...

conjecture | open | MathDB 页面 | 原文线索

V2-062993 | MATHDB-330744

Nonexistence conjecture for LR cubical and canonical cut-and-project sets with convex shapes

摘录 / EXCERPT: Let $d>1$, and let $\mathcal{C}_d'$ be the collection of convex sets in $\mathbb{R}^d$ with inradii at least $1/2$. A cut-and-project set is called cubical or canonical as in the p...

conjecture | open | MathDB 页面 | 原文线索

V2-062994 | MATHDB-330745

The Widom condition for equilibrium measures on regular compact sets

摘录 / EXCERPT: Let $K\subset\mathbb{R}$ be a compact set that is regular with respect to the Dirichlet problem, and let μ_K be its equilibrium measure. The Widom condition is the assertion th...

conjecture | open | MathDB 页面 | 原文线索

V2-062995 | MATHDB-330747

The endpoint Besov regularity conjecture for periodic symmetric-stable noises

摘录 / EXCERPT: Let w_{α} denote a periodic symmetric-α-stable noise on $\mathbb{T}^d$, with $0<p<\alpha<2$. The endpoint regularity conjecture. ... The paper obtains regularity result...

conjecture | open | MathDB 页面 | 原文线索

V2-062996 | MATHDB-330748

Criterion for nonaccumulation of complex eigenvalues near zero for Pauli operators

摘录 / EXCERPT: Nonaccumulation criterion. There is no accumulation of complex eigenvalues of $P_3(b,V)$ near zero if and only if $\Re(V)>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-062997 | MATHDB-330752

Maximal volume conjecture for the Kashaev invariant

摘录 / EXCERPT: Let K_n be a sequence of links that is both geometrically and diagrammatically maximal. For a link K, define its Kashaev invariant by ... Let $c(K_n)$ denote crossing number and...

conjecture | open | MathDB 页面 | 原文线索

V2-062998 | MATHDB-330758

Twined elliptic genus conjecture for K3 sigma models determined by positive planes

摘录 / EXCERPT: Let $\Pi=P_X\oplus P_Z$ be the positive plane determining a supersymmetric non-linear K3 sigma model, and let G_Π be the corresponding group of automorphisms. For $g\in G_\Pi$,...

conjecture | open | MathDB 页面 | 原文线索

V2-062999 | MATHDB-330759

Larsen-Lunts correction conjecture for piecewise isomorphism

摘录 / EXCERPT: Larsen-Lunts correction conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-063000 | MATHDB-330774

Dominant-singularity conjecture for adsorbing prudent loops

摘录 / EXCERPT: Let $W(z;1,0;a)$ be the generating function for prudent loops, with step variable z and surface fugacity $a>0$. Let $z_{1}^{\ell}\approx 0.412095$ denote a root of ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063001 | MATHDB-330775

Average-height criterion for the order of polymer adsorption transitions

摘录 / EXCERPT: Let $\mathcal{C}$ be an infinite subclass of self-avoiding walks in an upper half-plane, with starting point on the surface. For $\omega\in\mathcal{C}$, let $h(\omega)$ be the maxi...

conjecture | open | MathDB 页面 | 原文线索

V2-063002 | MATHDB-330776

Kuan-Lin-Yu's conjecture on zeta-like multizeta value families

摘录 / EXCERPT: Let $q=p^l>2$, let $1\leq p^m\leq q$, let $n>0$ and $r\geq 2$, and let $N_i\in \mathbb{N}\cup\{0\}$ for $0\leq i\leq n-1$ satisfy ... If ... then the displayed multizeta value below is...

conjecture | open | MathDB 页面 | 原文线索

V2-063003 | MATHDB-330778

Unbounded-order irreducible transformations for Parikh rewriting systems

摘录 / EXCERPT: Let $\mathfrak{P}$ be the Parikh rewriting system considered for the ternary alphabet, and let the order of an irreducible transformation denote its order in the system. An irreduc...

conjecture | open | MathDB 页面 | 原文线索

V2-063004 | MATHDB-330779

Uniqueness of irreducible-transformation length in Parikh rewriting

摘录 / EXCERPT: Let $\mathfrak{P}=(\Sigma,R,C)$ be a Parikh rewriting system, and suppose that $w\Rightarrow_{\mathfrak{P}}w'$ has a sequence of irreducible transformations as in Theorem, of lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-063005 | MATHDB-330780

Order-two decomposition conjecture for the ternary Parikh rewriting system

摘录 / EXCERPT: Let $\mathfrak{P}=(\Sigma,R,\{abc\})$, where ... For words $w,w'\in\Sigma^*$, let $d(w,w')$ denote their distance, and let an irreducible transformation of order at most two mean an...

conjecture | open | MathDB 页面 | 原文线索

V2-063006 | MATHDB-330782

Olson's conjecture on the Davenport constant of finite abelian groups

摘录 / EXCERPT: Let G be a finite commutative group. Write ... where $1<n_r\mid n_{r-1}\mid\cdots\mid n_1$, and define ... Let $D(G)$ denote the Davenport constant of G, the least length such that...

conjecture | open | MathDB 页面 | 原文线索

V2-063007 | MATHDB-330797

Rosen's fourth-power central limit conjecture for Brownian local time increments

摘录 / EXCERPT: Let $B=(B_t)_{t\geq 0}$ be Brownian motion, let L_t^x denote its local time at $x\in\mathbb{R}$, and define the increment ... Let $Z\sim\mathcal{N}(0,1)$ be independent of...

conjecture | open | MathDB 页面 | 原文线索

V2-063008 | MATHDB-330801

Sokal's distinct-moduli conjecture for the disturbed exponential function

摘录 / EXCERPT: Let ... where q is a complex number satisfying $0<|q|\leq 1$. Sokal's stronger conjecture. The function $\mathcal F(z;q)$ can have only simple zeros with distinct absolute values....

conjecture | open | MathDB 页面 | 原文线索

V2-063009 | MATHDB-330802

Sokal's simplicity problem for the partial theta function

摘录 / EXCERPT: Let ... where $\widetilde q\approx 0.3092493386$ is the positive threshold described in the source. Sokal's partial-theta problem. Is it true that all zeros of $\Theta_0(z;q)$ remain...

open_problem | open | MathDB 页面 | 原文线索

V2-063010 | MATHDB-330815

The eTNC for h^0(Spec L) over Z_p[G]

摘录 / EXCERPT: eTNC for $h^0(\operatorname{Spec}L)$ over $\mathbb{Z}_p[G]$.

conjecture | open | MathDB 页面 | 原文线索

V2-063011 | MATHDB-330816

Higher-rank Iwasawa main conjecture for zeta elements

摘录 / EXCERPT: Higher-rank Iwasawa main conjecture. There exists a Λ-basis $\mathcal{L}_{K_\infty/k,S,T}$ of $\det_\Lambda(C_{K_\infty,S,T})$ such that, for every $\chi\in\widehat\Delta$...

conjecture | open | MathDB 页面 | 原文线索

V2-063012 | MATHDB-330819

Iwasawa-theoretic Mazur-Rubin-Sano congruence conjecture

摘录 / EXCERPT: Iwasawa-theoretic Mazur-Rubin-Sano conjecture. There exists a unique

conjecture | open | MathDB 页面 | 原文线索

V2-063013 | MATHDB-330823

Teissier's equimultiplicity conjecture for μ-constant families

摘录 / EXCERPT: Teissier's conjecture. Any μ-constant family of isolated hypersurface singularities is equimultiple.

conjecture | open | MathDB 页面 | 原文线索

V2-063014 | MATHDB-330825

Strong asymptotics conjecture for Padé polynomials under Stahl's theorem

摘录 / EXCERPT: Strong asymptotics conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063015 | MATHDB-330829

The hypertree F-positivity conjecture

摘录 / EXCERPT: Let $H=(V,E)$ be a hypertree, and let X_H denote its chromatic symmetric function. The F-basis is the fundamental quasisymmetric-function basis. Hypertree F-positivity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-063016 | MATHDB-330834

Wang's universal Chern-class conjecture for tautological bundles on symmetric products of curves

摘录 / EXCERPT: Wang's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063017 | MATHDB-330839

Conjecture on the imaginary zeros and separation of Stirling permutation polynomials

摘录 / EXCERPT: Let $T_n(x)$ be the polynomial defined by the exponential generating function ... For polynomials with only real coefficients and only imaginary zeros, say that $f(x)$ separates...

conjecture | open | MathDB 页面 | 原文线索

V2-063018 | MATHDB-330840

Projectivity conjecture for Bridgeland moduli spaces on Del Pezzo surfaces

摘录 / EXCERPT: Let S be a Del Pezzo surface, and consider the program of constructing suitable hearts of Bridgeland stability conditions from exceptional collections and applying it to moduli s...

conjecture | open | MathDB 页面 | 原文线索

V2-063019 | MATHDB-330847

Shepherd-Sebő four-colour conjecture for t-perfect graphs

摘录 / EXCERPT: A graph is t-perfect if it satisfies the defining t-perfection property for graphs. Shepherd-Sebő's four-colour conjecture. Every t-perfect graph is 4-colourable. This is a...

conjecture | open | MathDB 页面 | 原文线索

V2-063020 | MATHDB-330848

The irreducibility conjecture for moduli of nodal curves on K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface, let L be a primitive polarization of genus g, and let $\mathcal{V}^n_{g,k}$ denote the moduli space of stable maps from smooth curves to X that are u...

conjecture | open | MathDB 页面 | 原文线索

V2-063021 | MATHDB-330862

Noether-Lefschetz and tautological relation conjecture for moduli of K3 surfaces

摘录 / EXCERPT: For each integer $\ell\geq 1$, let $\mathcal M_{2\ell}$ denote the moduli space of polarized $K3$ surfaces of degree 2ℓ. Let ${\mathsf{NL}}^\star(\mathcal M_{2\ell})$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-063022 | MATHDB-330872

The bounded-expansion characterization via poset dimension

摘录 / EXCERPT: Bounded-expansion characterization. The class $\mathcal{C}$ has bounded expansion if and only if, for each $h\geq 1$, the posets of height h whose cover graphs are in...

conjecture | open | MathDB 页面 | 原文线索

V2-063023 | MATHDB-330877

Wu-Hong revised conjecture on deep holes of primitive Reed-Solomon codes

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $D=\mathbb{F}_q^*$ be the evaluation set of a primitive Reed-Solomon code with code length $n=q-1$, dimension k, and codebook...

conjecture | open | MathDB 页面 | 原文线索

V2-063024 | MATHDB-330892

The 5-connectivity conjecture for block and hole graphs

摘录 / EXCERPT: Let $\hat{G}$ be a block and hole graph with one pentagonal block and two quadrilateral holes, or, two quadrilateral blocks and one pentagonal hole. The 5-connectivity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063025 | MATHDB-330895

The \P^{n-1}-functor conjecture for moduli spaces of sheaves on K3 surfaces

摘录 / EXCERPT: The $\mathbb P^{n-1}$-functor conjecture. This functor is a $\mathbb P^{n-1}$-functor.

conjecture | open | MathDB 页面 | 原文线索

V2-063026 | MATHDB-330896

Douady's conjecture on exotic Siegel disks for rational maps

摘录 / EXCERPT: Douady's conjecture. Polynomials and rational functions of degree $d\ge 2$ do not have exotic Siegel disks.

conjecture | open | MathDB 页面 | 原文线索

V2-063027 | MATHDB-330901

Closure of strong Keller maps under composition

摘录 / EXCERPT: Let k be a field of characteristic p. A strong Keller map is a polynomial endomorphism in $\operatorname{SKE}_n(k)$ satisfying the universal Keller equations. Composition conje...

conjecture | open | MathDB 页面 | 原文线索

V2-063028 | MATHDB-330903

Develin-Hartke conjecture on subcritical containment in square grids

摘录 / EXCERPT: Develin-Hartke conjecture. There exists an outbreak on $\mathbb{L}^d$ that cannot be contained by deploying $f(t)$ firefighters at time t.

conjecture | open | MathDB 页面 | 原文线索

V2-063029 | MATHDB-330905

Properness conjecture for the J-flow functional

摘录 / EXCERPT: Properness conjecture for the J-flow functional. Under these assumptions, the I_{μ_J} functional is proper. In particular, if $L_2=K_M$, then (X,L_1) has proper Mabuchi funct...

conjecture | open | MathDB 页面 | 原文线索

V2-063030 | MATHDB-330906

Equality of proper and right modality

摘录 / EXCERPT: Proper-modality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063031 | MATHDB-330910

The tempered L-packet conjecture for unitary groups

摘录 / EXCERPT: Let F be a local field and let ${\rm U}_N(F)$ be a unitary group. A Langlands parameter is a homomorphism ... A parameter is tempered when its image in ${\rm GL}_N(\mathbb{C})$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-063032 | MATHDB-330917

Covariant weak Lefschetz conjecture for zero-cycles

摘录 / EXCERPT: Covariant weak Lefschetz conjecture. Suppose

conjecture | open | MathDB 页面 | 原文线索

V2-063033 | MATHDB-330925

Classification conjecture for nonredundant jumping polynomials

摘录 / EXCERPT: Let $\{T_i\}_{i>0}$ be the sequence of jumping polynomials, and let $\{Q_i\}_{i\geq 0}$ and $\{R_i\}_{i\geq 0}$ be the polynomial sequences defined in the construction, with ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-063034 | MATHDB-330927

General recognition conjecture for $s \ell$-tannakian categories

摘录 / EXCERPT: General recognition conjecture. Under these hypotheses, if F is an $s \ell$-enriched open and faithful functor, then $\widetilde{F}$ is an equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-063035 | MATHDB-330934

Padovan dimension conjecture for finite multiple zeta values of superbity one

摘录 / EXCERPT: For each positive integer w, let $\FMZ_w$ be the $\Q$-vector space generated by finite multiple zeta values of weight w and superbity one. Define $d_0=1$ and $d_w=\dim\FMZ_w$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-063036 | MATHDB-330936

Padovan dimension conjecture for finite multiple zeta values of arbitrary superbity

摘录 / EXCERPT: For positive integers w and ℓ, let $\FMZ_{w,\ell}$ be the $\Q$-vector space generated by finite multiple zeta values of weight w and superbity ℓ. Arbitrary-superbity...

conjecture | open | MathDB 页面 | 原文线索

V2-063037 | MATHDB-330938

Broadhurst-type dimension conjecture for finite Euler sums of superbity one

摘录 / EXCERPT: Let $\FES_{w,1}$ denote the $\Q$-vector space of finite Euler sums of weight w and superbity one, and set $f_w=\dim_{\Q}\FES_{w,1}$ for $w\ge1$. Finite Euler-sum dimension conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-063038 | MATHDB-330939

The open KdV conjecture for open gravitational descendents

摘录 / EXCERPT: Let $F^o(s,t_0,t_1,ldots;u)$ be the open free energy, with double brackets denoting its derivatives with respect to the variables t_i and s, and let F^c denote the closed fre...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-063039 | MATHDB-330942

Blocki-Zwonek's convexity conjecture for Green-function sublevel sets

摘录 / EXCERPT: Let Ω be a pseudoconvex domain in $\mathbb C^n$, let $w\in\Omega$, let $G_{\Omega,w}$ denote the pluricomplex Green's function of Ω with pole at w, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063040 | MATHDB-330946

Ilmonen-Haukkanen-Merikoski conjecture for the constant c_n

摘录 / EXCERPT: Ilmonen-Haukkanen-Merikoski conjecture. The number c_n is equal to the smallest eigenvalue of $Y_0Y_0^T$.

conjecture | open | MathDB 页面 | 原文线索

V2-063041 | MATHDB-330947

Uniqueness conjecture for the minimising matrix of c_n

摘录 / EXCERPT: Uniqueness conjecture. There is a unique matrix $Y\in K_n$ such that the smallest eigenvalue of YY^T is c_n. Equivalently, if the smallest eigenvalue of YY^T equals c_n, th...

conjecture | open | MathDB 页面 | 原文线索

V2-063042 | MATHDB-330949

Gessel's conjectured gamma-coefficient inequality

摘录 / EXCERPT: Let $D4AE_n$ denote the symmetric group on n letters. Let $D4AE_n(231)$ be the set of permutations avoiding the pattern 231, and let $S_n(t)$ be its descent polynomial. Write .....

conjecture | open | MathDB 页面 | 原文线索

V2-063043 | MATHDB-330951

Asymptotic volume formula for strata of Abelian differentials

摘录 / EXCERPT: Asymptotic volume conjecture. For any $m\in\Pi_{2g-2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-063044 | MATHDB-330952

Uniform square-root error bound for Abelian differential stratum volumes

摘录 / EXCERPT: Uniform error-bound conjecture. There exists a universal constant C such that, for all g and all $m\in\Pi_g$,

conjecture | open | MathDB 页面 | 原文线索

V2-063045 | MATHDB-330953

Large-genus universality conjecture for the area Siegel-Veech constant

摘录 / EXCERPT: Siegel-Veech universality conjecture. For all such nonhyperelliptic connected components,

conjecture | open | MathDB 页面 | 原文线索

V2-063046 | MATHDB-330955

Toader-Qi mean versus the 3/2-order logarithmic mean conjecture

摘录 / EXCERPT: Toader-Qi/logarithmic-mean conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-063047 | MATHDB-330956

A chain inequality conjecture for the Toader-Qi mean and classical means

摘录 / EXCERPT: Let $a,b>0$ with $a\neq b$. Denote by $A(a,b)$, $G(a,b)$, $L(a,b)$, and $\mathcal{I}(a,b)$ the arithmetic, geometric, logarithmic, and identric means, respectively, and by...

conjecture | open | MathDB 页面 | 原文线索

V2-063048 | MATHDB-330957

Conjecture on log-canonical thresholds and a-invariants

摘录 / EXCERPT: Conjecture on log-canonical thresholds and a-invariants. 1. ...

conjecture | open | MathDB 页面 | 原文线索

V2-063049 | MATHDB-330968

Don's reachable-subset length conjecture for finite automata

摘录 / EXCERPT: Don's reachable-subset length conjecture. There is a word v such that $\delta(Q,v)=S$ and whose length is at most

conjecture | open | MathDB 页面 | 原文线索

V2-063050 | MATHDB-330973

Distance-regular graph extendability conjecture for diameter at least three

摘录 / EXCERPT: Distance-regular graph extendability conjecture. The extendability of G is at least

conjecture | open | MathDB 页面 | 原文线索

V2-063051 | MATHDB-330980

Conjecture on the matrix-factorization realization of the A-type defect category

摘录 / EXCERPT: For an integer $d\geqslant 3$, let $\mathbbm{Z}_d$ denote the integers modulo d, and let $P_{\{a,a+1,\dots,a+b\}}$ be the stated matrix factorizations indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-063052 | MATHDB-330994

The weak generic Arthur packet conjecture for GSpin groups

摘录 / EXCERPT: Weak generic Arthur packet conjecture. If $\Pi(\phi_{\psi})$ has a generic member, then the Langlands parameter ϕ_{ψ} is tempered.

conjecture | open | MathDB 页面 | 原文线索

V2-063053 | MATHDB-330995

The strong generic Arthur packet conjecture for GSpin groups

摘录 / EXCERPT: Strong generic Arthur packet conjecture. If $\Pi(\phi_{\psi})$ has a generic member, then $\Pi(\phi_{\psi})$ is a tempered L-packet; equivalently, all members of...

conjecture | open | MathDB 页面 | 原文线索

V2-063054 | MATHDB-330996

The period-ratio conjecture for CM Hilbert cuspforms

摘录 / EXCERPT: Let F be the CM field, let F^+ be its maximal totally real subfield, let $ u$ be the Hecke character giving rise to the Hilbert cuspform $ u(\theta)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063055 | MATHDB-331000

Han's iterated resultant conjecture for generic forms

摘录 / EXCERPT: Let $f(x_1,\ldots,x_n)$ be a generic form, and let $\Delta(f)$ denote its multivariate discriminant. Let ${\tt Hp}(f,[x_n,\ldots,x_1])$ be the polynomial associated with the iterat...

conjecture | open | MathDB 页面 | 原文线索

V2-063056 | MATHDB-331010

Minimum-degree refinement of the Barát-Thomassen conjecture

摘录 / EXCERPT: Let T be a fixed tree with maximum degree Δ_T, and let $|E(T)|$ denote its number of edges. A graph is T-decomposable if its edges can be partitioned into sets each indu...

conjecture | open | MathDB 页面 | 原文线索

V2-063057 | MATHDB-331017

Klaus-Teichner conjecture on signature multiplicativity modulo 8

摘录 / EXCERPT: Let $F^{2m} \to E \to B^{2n}$ be a fibration of oriented Poincaré complexes, and suppose that the action of $\pi_1(B)$ on $H^{m}(F;\mathbb Z_2)$ is trivial. Klaus-Teichner's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-063058 | MATHDB-331027

The set-of-length-sets conjecture for finite abelian groups

摘录 / EXCERPT: Set-of-length-sets conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-063059 | MATHDB-331039

Analogue of De Giorgi's conjecture for the fourth-order Allen-Cahn equation

摘录 / EXCERPT: Let $\beta\ge\sqrt{8}$ and let u be a bounded entire solution of ... Assume that $u_{x_N}(x)>0$ for every $x\in\mathbb{R}^N$. Analogue of De Giorgi's conjecture for the fourth-orde...

conjecture | open | MathDB 页面 | 原文线索

V2-063060 | MATHDB-331042

Triangular-row conjecture for optimal n-mean configurations

摘录 / EXCERPT: Triangular-row conjecture. For most n, there is an optimal configuration with N rows. The jth row has j points for $j\leq J$. If $n>N(N+1)/2$, the rows with $j>J$ each have...

conjecture | open | MathDB 页面 | 原文线索

V2-063061 | MATHDB-331047

Nonexistence conjecture for negacyclic Hadamard matrices

摘录 / EXCERPT: A negacyclic Hadamard matrix is a Hadamard matrix whose rows are generated from one another by negacyclic shifts. Negacyclic Hadamard nonexistence conjecture. There are no negacycl...

conjecture | open | MathDB 页面 | 原文线索

V2-063062 | MATHDB-331048

Existence conjecture for Hadamard matrices of 2N-type

摘录 / EXCERPT: A Hadamard matrix of 2N-type is a Hadamard matrix of order $2v$ of the block form ... where A and B are negacyclic blocks. 2N-type existence conjecture. For each even integer...

conjecture | open | MathDB 页面 | 原文线索

V2-063063 | MATHDB-331052

The constant-additive neighbour sum distinguishing total colouring conjecture

摘录 / EXCERPT: Let G be a graph with maximum degree Δ, and let $\chi”_{\sum}(G)$ denote the least number of colours in a proper total colouring whose weighted degrees distinguish adjacen...

conjecture | open | MathDB 页面 | 原文线索

V2-063064 | MATHDB-331053

The list neighbour sum distinguishing total colouring problem

摘录 / EXCERPT: Let G be a graph with maximum degree Δ. In the natural list version of neighbour sum distinguishing total colouring, colours are chosen from arbitrary lists of real number...

open_problem | open | MathDB 页面 | 原文线索

V2-063065 | MATHDB-331057

The three-colour decomposition conjecture for locally irregular graphs

摘录 / EXCERPT: Three-colour decomposition conjecture. Every connected graph G which does not belong to $\mathfrak{T}$ and is neither an odd-length path nor an odd-length cycle can be decomposed...

conjecture | open | MathDB 页面 | 原文线索

V2-063066 | MATHDB-331076

Haiman's Schur-P positivity conjecture for fixed-point-free involution Stanley symmetric functions

摘录 / EXCERPT: Let S_n be the symmetric group and let $\mathcal{I}(S_n)$ denote its involutions. For a fixed-point-free involution $y\in\mathcal{I}(S_n)$, let $\hat F^{\operatorname{FPF}}_y$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-063067 | MATHDB-331078

Nilpotency and inversion-step conjecture for cubic homogeneous perturbations

摘录 / EXCERPT: Let $F:K^d\rightarrow K^d$ be a polynomial map with $F_i=X_i+H_i$, where ... and ... for $1\leq i\leq d$. Write $H=(H_1,\dots,H_d)$ and let JH be its Jacobian matrix. Define polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-063068 | MATHDB-331080

Kohl's permutation product conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and let a,b,c be integers satisfying $1<a,b,c\leq n-2$. For permutations $x,y\in S_n$, say that they have prescribed orders a,b,c when x has order a,...

conjecture | open | MathDB 页面 | 原文线索

V2-063069 | MATHDB-331081

Marked chain-order polytope face-number monotonicity conjecture

摘录 / EXCERPT: Let $(P,A,\mathbf{\lambda})$ be a regular marked poset, and let $\operatorname*{St}(P)$ denote the set of star elements. An admissible decomposition of $P\setminus A$ is a pair...

conjecture | open | MathDB 页面 | 原文线索

V2-063070 | MATHDB-331082

Stanley-Hibi-Li conjecture for marked chain and order polytopes

摘录 / EXCERPT: Let $(P,A,\mathbf{\lambda})$ be a marked poset. For an N-dimensional polytope Q, let $f_i(Q)$ denote the number of its i-dimensional faces, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-063071 | MATHDB-331083

Refined Stanley-Hibi-Li conjecture for marked chain-order polytopes

摘录 / EXCERPT: Let $(P,A,\mathbf{\lambda})$ be a marked poset, and let (U_1,U_2) and (V_1,V_2) be different admissible decompositions of $P\setminus A$, meaning that each pair partitions...

conjecture | open | MathDB 页面 | 原文线索

V2-063072 | MATHDB-331084

Maximal strict scaling conjecture for scalable frames

摘录 / EXCERPT: Let F be a scalable frame with minimal scalings $v_j_{j\in I}$, and let $J\subset I$ be inclusion-minimal subject to ... Suppose that ... is the smallest orthogonal partition of...

conjecture | open | MathDB 页面 | 原文线索

V2-063073 | MATHDB-331086

The $(a+1)$-positivity conjecture for quarter-plane-loop series

摘录 / EXCERPT: $(a+1)$-positivity conjecture. Each polynomial P_n has positive coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-063074 | MATHDB-331087

Conjectural radius of convergence for the quarter-plane-loop series

摘录 / EXCERPT: Radius-of-convergence conjecture. The radius of convergence is

conjecture | open | MathDB 页面 | 原文线索

V2-063075 | MATHDB-331088

Convergence of the derivative at the conjectural quarter-plane-loop radius

摘录 / EXCERPT: Derivative-convergence conjecture. The series $Q_u(a,u)$ is convergent at $u=\rho_Q(a)$ for $a\geq -1/3$.

conjecture | open | MathDB 页面 | 原文线索

V2-063076 | MATHDB-331099

Finiteness criterion for non-real Dirac eigenvalues near the thresholds

摘录 / EXCERPT: Let $D_m(b,V)$ be the Dirac operator from Assumption 1.1, with $V=\Phi W$, where W is Hermitian of definite sign and...

conjecture | open | MathDB 页面 | 原文线索

V2-063077 | MATHDB-331100

The area-threshold conjecture for counter braids

摘录 / EXCERPT: For a counter-braid ensemble, let $\bar{\epsilon}$ denote the area threshold, namely the value at which the area under the relevant extended belief-propagation extrinsic informatio...

conjecture | open | MathDB 页面 | 原文线索

V2-063078 | MATHDB-331107

Log Donaldson-Tian-Yau conjecture

摘录 / EXCERPT: Log Donaldson-Tian-Yau conjecture. $((X,D);L)$ is log K-polystable with cone angle $2\pi\beta$ if and only if X admits a cscK metric in $c_1(L)$ with cone singularities along...

conjecture | open | MathDB 页面 | 原文线索

V2-063079 | MATHDB-331109

Linear free-boundary separation conjecture for optimal partial transport

摘录 / EXCERPT: Let the assumptions and notation be those of Theorem, including masses m_1,m_2, corresponding parameters λ_1,λ_2, free boundaries F_1,F_2, and neighborhoods...

conjecture | open | MathDB 页面 | 原文线索

V2-063080 | MATHDB-331110

Even-dimensional k-resultant modulus exponent conjecture

摘录 / EXCERPT: Let $\beta_k(d)$ denote the exponent governing the smallest size of a subset of $\mathbb{F}_q^d$ required to ensure that its k-resultant modulus set has the relevant full-size pr...

conjecture | open | MathDB 页面 | 原文线索

V2-063081 | MATHDB-331115

The equivariant K-theory conjecture for Bernstein components

摘录 / EXCERPT: Let G be a reductive p-adic group, let $\mathfrak{s}\in{{\mathfrak B}}(G)$ be a Bernstein component, let $T_{\mathfrak{s},\mathrm{un}}$ be its unitary torus, let…

conjecture | open | MathDB 页面 | 原文线索

V2-063082 | MATHDB-331119

Generic Bessel descent conjecture for cuspidal automorphic representations

摘录 / EXCERPT: Let τin{",

conjecture | open | MathDB 页面 | 原文线索

V2-063083 | MATHDB-331120

Local Gan-Gross-Prasad conjecture for Vogan packets

摘录 / EXCERPT: Let G_n^* and H_m^* be a relevant pair of F-quasisplit classical groups, let $R_{{\mathcal {O}}_\ell}(F)=H_m(F)\ltimes V_{\underline{p}_\ell}(F)$ be the Bessel subgroup attac…

conjecture | open | MathDB 页面 | 原文线索

V2-063084 | MATHDB-331122

Fourier-support conjecture for residual representations

摘录 / EXCERPT: Fourier-support conjecture. Every $\underline p\in\mathfrak p(\mathcal E_{\tau\otimes\sigma})$ satisfies $\underline p\leq\underline p_{\tau;\sigma,H_m^*}$; if $H_m=H_m^*$ is F-q…

conjecture | open | MathDB 页面 | 原文线索

V2-063085 | MATHDB-331123

Weak Fourier-support conjecture for twisted automorphic descent

摘录 / EXCERPT: Let τ be the datum in the descent diagram, let $m\leq n$, and define … Let $\mathfrak p(\mathcal E_{\tau\otimes\sigma})$ denote the relevant partitions for which the residual…

conjecture | open | MathDB 页面 | 原文线索

V2-063086 | MATHDB-331136

The nonexistence conjecture for gentile acute tetrahedra

摘录 / EXCERPT: Gentile acute tetrahedra conjecture. There are no gentile acute tetrahedra.

conjecture | open | MathDB 页面 | 原文线索

V2-063087 | MATHDB-331137

The nonexistence conjecture for gentile tetrahedra without right dihedral angles

摘录 / EXCERPT: Non-right-dihedral gentile tetrahedra conjecture. There are no gentile tetrahedra that do not have a dihedral angle of exactly $pi/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-063088 | MATHDB-331139

The Witten-Reshetikhin-Turaev correspondence for simplicial ribbon links

摘录 / EXCERPT: Let ${\mathfrak g}$ be as in Section 2, let $k\in\mathbb{N}$ satisfy $k>c_{\mathfrak g}$, and let L be a colored simplicial ribbon link in $S^2\times S^1$ as in Theorem 1 or Theo…

conjecture | open | MathDB 页面 | 原文线索

V2-063089 | MATHDB-331141

Density-one local-global conjecture for periodic points of quadratic polynomials

摘录 / EXCERPT: Let $f\in\mathbf Q[x]$ be a quadratic polynomial, and let $n>3$ be an integer. Let $\Phi_{n,f}$ denote the period-n dynatomic polynomial of f, and let $\mathcal S_{\Phi_{n,f}}$…

conjecture | open | MathDB 页面 | 原文线索

V2-063090 | MATHDB-331143

The chromatic-threshold conjecture for thundercloud-forest graphs

摘录 / EXCERPT: Thundercloud-forest conjecture. The chromatic threshold satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063091 | MATHDB-331151

Linear independence conjecture for the odd orthogonal-group construction

摘录 / EXCERPT: Conjecture 1 for $SO(2m+1)$. For every such configuration, the corresponding complex matrix M is nonsingular.

conjecture | open | MathDB 页面 | 原文线索

V2-063092 | MATHDB-331157

Oka's conjecture for irreducible sextic plane curves without toric decompositions

摘录 / EXCERPT: Let $C\subset\mathbf{P}^2$ be an irreducible sextic plane curve with no toric decomposition of type $(2,3)$. Let $\Delta_C(t)$ denote its Alexander polynomial, and let ℓ be a…

conjecture | open | MathDB 页面 | 原文线索

V2-063093 | MATHDB-331161

Local finiteness conjecture for fundamental groups of klt singularities

摘录 / EXCERPT: Let (X,D) be a klt pair and let $x\in X$ be a singular point, so that $\operatorname{Link}(x\in X)$ denotes the link of the singularity. Local finiteness conjecture. The local fu…

conjecture | solved | MathDB 页面 | 原文线索

V2-063094 | MATHDB-331163

The recursive term-order conjecture for lecture-hall Gröbner bases

摘录 / EXCERPT: Recursive term-order conjecture. For this choice of term order, the recursive Gröbner basis conjecture for I_n is true.

conjecture | open | MathDB 页面 | 原文线索

V2-063095 | MATHDB-331166

The global minimization conjecture for decay rates in layered resonators

摘录 / EXCERPT: Let b_1 and b_2 be constants satisfying $0\le b_1<b_2<+\infty$. For each admissible frequency parameter α, let $\beta(\alpha)$ denote the minimal decay rate among the co…

conjecture | open | MathDB 页面 | 原文线索

V2-063096 | MATHDB-331180

Monotonicity conjecture for the alternative discrete Painlevé I solutions

摘录 / EXCERPT: Let $b_n(t)$ and $a_n(t)$ denote the positive solutions and associated quantities introduced in the paper, with n a nonnegative integer. For fixed $t>0$, the expressions...

conjecture | open | MathDB 页面 | 原文线索

V2-063097 | MATHDB-331189

Uniform third-moment conjecture for affine Weyl group cores

摘录 / EXCERPT: Let Φ be a simply-laced root system of rank n, with Coxeter number h, let b satisfy $\gcd(h,b)=1$, and let $\mathcal{S}_\Phi(b)$ be the Sommers region. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063098 | MATHDB-331190

Higher-moment formulas for affine Weyl group cores

摘录 / EXCERPT: Higher-moment uniformity conjecture. In types $\widetilde{A}_n$ and $\widetilde{D}_n$, the displayed formulas for $\mathrm{top}_\Phi^b(i)$ for $i=4,5,6,7$ in type $\widetilde{A}_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-063099 | MATHDB-331191

Fuß-Catalan expected-size conjecture for type C affine cores

摘录 / EXCERPT: Let m,n be positive integers, let h be the Coxeter number in type $\widetilde{C}_n$, and consider the Fuß-Catalan case $b=mh+1$. Let ${\sf core}(\widetilde{C}_n,mh+1)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-063100 | MATHDB-331192

The Motzkin product conjecture for entangled threads

摘录 / EXCERPT: Let an even thread be a thread of even type and an odd thread be a thread of odd type. For an even thread α and an odd thread β, consider the threads entangled with ea...

conjecture | open | MathDB 页面 | 原文线索

V2-063101 | MATHDB-331193

The Main Conjecture for discrete Fourier restriction mean values

摘录 / EXCERPT: Main Conjecture. For every $\varepsilon>0$ and every complex sequence $({\mathfrak a}_n)$, the displayed bound holds.

conjecture | open | MathDB 页面 | 原文线索

V2-063102 | MATHDB-331198

Nonarithmetic rank-one sparsity conjecture for orbit closures

摘录 / EXCERPT: Nonarithmetic rank-one sparsity conjecture. Outside of genus two, a sequence of nonarithmetic rank-one orbit closures cannot equidistribute in a component of a stratum.

conjecture | open | MathDB 页面 | 原文线索

V2-063103 | MATHDB-331209

The minimum-cycle conjecture for non-bkd-colourable graphs

摘录 / EXCERPT: Minimum-cycle conjecture. If $\chi(G)>k$, then G contains at least

conjecture | open | MathDB 页面 | 原文线索

V2-063104 | MATHDB-331212

Conjecture on Siegel invariants generating fixed fields in ray class fields

摘录 / EXCERPT: Let K be a CM-field with reflex field K^*, let $\mathfrak{f}$ be a modulus of K, and let $\mathfrak{f}^*$ be the corresponding modulus of K^*. Write ... for the natural homom...

conjecture | open | MathDB 页面 | 原文线索

V2-063105 | MATHDB-331219

Batyrev-Popov conjecture on generators of del Pezzo Cox ring relations

摘录 / EXCERPT: Let S_r be a del Pezzo surface of degree $(-K_{S_r})^2=9-r$ with $4\leq r\leq 8$. Let G_r be a basis of the vector space ... for $r\leq 7$, and for $r=8$ let G_8 be a basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-063106 | MATHDB-331222

The minimum hyperplane-covering conjecture for finite grids

摘录 / EXCERPT: Let R be a ring, and let $A_1,\dots,A_n\subset R$ be nonempty (but possibly infinite) subsets. Let $\mathcal{H}=\{H_j\}_{j\in J}$ be a covering of the grid ... by hyperplanes. The...

conjecture | open | MathDB 页面 | 原文线索

V2-063107 | MATHDB-331228

Equidistribution of minimaj with inversion, major index, and diagonal inversion

摘录 / EXCERPT: Given an ordered set partition $\pi\in\mathcal{OP}_n$, define its shape $\operatorname{shape}(\pi)$ to be the composition whose ith part is the size of the ith block of π f...

conjecture | open | MathDB 页面 | 原文线索

V2-063108 | MATHDB-331238

The odd writhe formula for the zeroth Alexander polynomial of virtual knots

摘录 / EXCERPT: Let K be a virtual knot. Let $Delta_0(K)(u,v)$ denote the zeroth Alexander polynomial, and let $OW(K)$ denote the odd writhe of K. The odd writhe conjecture. ... The relationship...

conjecture | open | MathDB 页面 | 原文线索

V2-063109 | MATHDB-331250

Chao-Ping Chen's upper Wilker inequality for inverse trigonometric functions

摘录 / EXCERPT: For $0<x<1$, the functions $\frac{\arcsin x}{x}$ and $\frac{\arctan x}{x}$ are defined, and the associated inequality is considered. Chao-Ping Chen's conjecture. If $0<x<1$, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-063110 | MATHDB-331251

The 2-way infinite 2-walk conjecture for 3-connected planar graphs

摘录 / EXCERPT: The 2-way infinite 2-walk conjecture. The graph G has a 2-way infinite 2-walk.

conjecture | open | MathDB 页面 | 原文线索

V2-063111 | MATHDB-331254

Basis conjecture for the functions associated with two-step equivalence classes

摘录 / EXCERPT: For $k=0,1,2$, let $\{f^{(k)}\}$ be the set of functions realized as ... where $\mathcal{C}$ is a single equivalence class of $\equiv^k$. Basis conjecture. The set $\{f^{(2)}\}$ form...

conjecture | open | MathDB 页面 | 原文线索

V2-063112 | MATHDB-331280

The fine-set conjecture for vertically rotated one-column grid classes

摘录 / EXCERPT: Fine-set conjecture. For every one-column grid class ${\mathcal G}^{\mathbf{v}}$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-063113 | MATHDB-331281

The cycle and inverse descent-class fine-set conjecture

摘录 / EXCERPT: Cycle-inverse descent-class conjecture. For every $J\subseteq [n-2]$,

conjecture | open | MathDB 页面 | 原文线索

V2-063114 | MATHDB-331282

The inverse descent-class commutation conjecture for fine sets

摘录 / EXCERPT: Inverse descent-class commutation conjecture. For every fine set ${\mathcal B}\subset {\mathcal S}_n$ and every $J\subseteq [n-1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-063115 | MATHDB-331293

The facet lower-bound conjecture for k-neighborly polytopes

摘录 / EXCERPT: Let P be a d-dimensional convex polytope, and let $f_i(P)$ denote the number of its i-dimensional faces. A polytope is k-neighborly if every subset of k vertices is the v...

conjecture | open | MathDB 页面 | 原文线索

V2-063116 | MATHDB-331302

The tournament scheduling conjecture for on-line scheduling

摘录 / EXCERPT: Let G be a non-edgeless multigraph, let $t\in\mathbb{N}$, and let $\chi_{\textit{OL}}^t(G)$ denote the worst-case number of rounds required for on-line scheduling when each verte...

conjecture | open | MathDB 页面 | 原文线索

V2-063117 | MATHDB-331304

Associativity conjecture for finite Moufang p-loops with p not equal to 3

摘录 / EXCERPT: Let L be a finite Moufang p-loop, where $p\ne 3$. Suppose L is generated by a set $\{a_i\mid i\in I\}$ satisfying ... for all $i,j,k\in I$, where (x,y,z) denotes the associat...

conjecture | open | MathDB 页面 | 原文线索

V2-063118 | MATHDB-331310

Non-realizability of the spun Kishino torus

摘录 / EXCERPT: Let KI be the Kishino virtual knot, and let $S_T(F\times [0,1],K)=(F\times S^1\times [0,1],K\times S^1)$ be the generalized spinning construction. Non-realizability conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-063119 | MATHDB-331311

Graph-change conjecture for virtual broken surface diagrams

摘录 / EXCERPT: A virtual broken surface diagram (VBSD) is a broken surface diagram for a virtual 2-link. A graph change is a diagram change preserving the Gauss code of the double point curves...

conjecture | open | MathDB 页面 | 原文线索

V2-063120 | MATHDB-331315

The follower-set growth conjecture for subshifts

摘录 / EXCERPT: Let X be a subshift, meaning a closed, shift-invariant subset of $A^{\mathbb{Z}}$ for a finite set A. For each n, let $F_X(n)$ be the set of distinct follower sets of words o...

conjecture | open | MathDB 页面 | 原文线索

V2-063121 | MATHDB-331316

Gevrey smoothing conjecture for the non-cutoff homogeneous Boltzmann equation

摘录 / EXCERPT: Let $0<\nu<1$ be the order of the singular cross section kernel, and let f be a weak solution of the non-cutoff homogeneous Boltzmann equation in $\mathbb{R}^d$ with initial datu...

conjecture | open | MathDB 页面 | 原文线索

V2-063122 | MATHDB-331329

Beyond endoscopy pole detection conjecture for functorial transfers

摘录 / EXCERPT: Let F be a number field, let $\mathbb{A}_F$ be its ring of adeles, let G be a reductive group over F, and let ${}^{\lambda}G$ be the Zariski closure of the image of ${}^{L}G$...

conjecture | open | MathDB 页面 | 原文线索

V2-063123 | MATHDB-331330

Gabber-Kac simplicity conjecture for complete Kac-Moody groups

摘录 / EXCERPT: Gabber-Kac simplicity conjecture. The group $\mathfrak{G}_A^{pma}(k)$ is GK-simple.

conjecture | open | MathDB 页面 | 原文线索

V2-063124 | MATHDB-331335

Main bijection conjecture for physical Bethe solutions and rigged configurations

摘录 / EXCERPT: Let physical solutions of the Bethe ansatz equations be decomposed into strings, let ν be the partition formed by their string lengths, and let the riggings be assigned by the...

conjecture | open | MathDB 页面 | 原文线索

V2-063125 | MATHDB-331337

Ext-vanishing conjecture for standard Whittaker sheaves

摘录 / EXCERPT: Let x be a point of the curve, let $\Whit_x^\kappa$ be the metaplectic Whittaker category at x, and let $\mathcal F_{x,\mu}$ denote the Whittaker object indexed by a dominant c...

conjecture | open | MathDB 页面 | 原文线索

V2-063126 | MATHDB-331344

Classification conjecture for 2-ary order 2 Conolly recursions

摘录 / EXCERPT: Classification conjecture. For $\beta>0$, the only 2-ary order 2 (α,β)-Conolly recurrences are exactly the recurrences listed in the source for $(\alpha,\beta)=(-2,3)$...

conjecture | open | MathDB 页面 | 原文线索

V2-063127 | MATHDB-331345

Finite-versus-infinite conjecture for Conolly meta-Fibonacci recursions

摘录 / EXCERPT: Finite-versus-infinite conjecture. The number of order p (α,β)-Conolly meta-Fibonacci recursions is finite if $\beta>0$ and infinite if $\beta=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-063128 | MATHDB-331351

Adjacent triangular and small faces in 3-connected uniquely 3-colorable plane graphs

摘录 / EXCERPT: Adjacent-face conjecture. Every 3-connected uniquely 3-colorable plane graph G has adjacent $(3,k)$-faces for some $k\leq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-063129 | MATHDB-331361

Erdős's completely multiplicative discrepancy conjecture

摘录 / EXCERPT: Let $\epsilon_1,\epsilon_2,\epsilon_3,\dots$ be a completely multiplicative sequence taking values in the set $\{-1,1\}$. Erdős's completely multiplicative discrepancy conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063130 | MATHDB-331363

Kalai's modular Erdős discrepancy conjecture

摘录 / EXCERPT: Kalai's modular Erdős discrepancy conjecture. The assertion is a modular strengthening of the Erdős discrepancy problem, formulated by Gil Kalai; the source presents it as open.

conjecture | open | MathDB 页面 | 原文线索

V2-063131 | MATHDB-331365

Tao's completely multiplicative mean-square discrepancy conjecture

摘录 / EXCERPT: Let $\mathbb{N}$ and $\mathbb{R}$ denote the positive integers and real numbers, respectively. Tao's completely multiplicative mean-square discrepancy conjecture. There exists a fu...

conjecture | open | MathDB 页面 | 原文线索

V2-063132 | MATHDB-331366

Matrix strengthening of the Erdős discrepancy conjecture

摘录 / EXCERPT: Matrix strengthening of the Erdős discrepancy conjecture. This is presented as another strengthening of the Erdős discrepancy problem, motivated by an equivalence involving diagona...

conjecture | open | MathDB 页面 | 原文线索

V2-063133 | MATHDB-331367

Conjecture on the three-dimensional vectorial Gamma-limit for thin films with delamination

摘录 / EXCERPT: Three-dimensional vectorial Gamma-limit conjecture. If $\boldsymbol u\in L^2(\mathcal{O};\mathbb{R}^3)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-063134 | MATHDB-331371

Rational singularities for codimension-one orbit closures of arbitrary quivers

摘录 / EXCERPT: Let Q be a quiver. For an extended Dynkin quiver, let $T_1,\dots,T_m$ be pairwise non-isomorphic indecomposables in $\operatorname{Rep}(Q)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-063135 | MATHDB-331374

Van den Berg-Caffarelli-Lin hexagonal conjecture for minimal partitions

摘录 / EXCERPT: Let Ω be a bounded planar domain of area $A(\Omega)$, let $\mathfrak L_k(\Omega)$ denote the minimal energy of a k-partition of Ω, and let $Hexa_1$ be the regular h...

conjecture | open | MathDB 页面 | 原文线索

V2-063136 | MATHDB-331375

Hexagonal conjecture for Pleijel's constant

摘录 / EXCERPT: Hexagonal conjecture for Pleijel.

conjecture | open | MathDB 页面 | 原文线索

V2-063137 | MATHDB-331376

Polterovich's conjecture on the nodal-domain ratio

摘录 / EXCERPT: Let (ϕ_n) be a sequence of Laplace eigenfunctions, and let $\mu(\phi_n)$ denote the number of nodal domains of ϕ_n. Let Sq_1 be the square of area 1, with first eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-063138 | MATHDB-331382

Conjecture on main simple choreographies in the equal mass Newtonian N-body problem

摘录 / EXCERPT: Main choreography conjecture. For every $N\geq 3$, there is a main simple choreography solution different from the trivial circle solution. The number of such distinct main simple...

conjecture | open | MathDB 页面 | 原文线索

V2-063139 | MATHDB-331384

Coefficient-wise Wright log-convexity and log-concavity in the first parameter of the normalized incomplete beta function

摘录 / EXCERPT: For $a,b,\alpha,\beta>0$ and $x\tsim(0,1)$, write the relevant generalized Turán expression as a power series whose coefficients are denoted by ψ_k. Coefficient-wise first-par...

conjecture | open | MathDB 页面 | 原文线索

V2-063140 | MATHDB-331385

Recurrence relations for the remaining TRIP-Stern maximal terms

摘录 / EXCERPT: Conjectured recurrence relations. The numerical data in the table suggest recurrence relations for the maximal terms of the TRIP-Stern sequences corresponding to the remaining 96 m...

conjecture | open | MathDB 页面 | 原文线索

V2-063141 | MATHDB-331395

The inner mapping group conjecture for loops

摘录 / EXCERPT: Let Q be a loop, and let $\operatorname{Inn}(Q)$ denote its inner mapping group. A loop is centrally nilpotent if it has a central series; its nilpotence class is the length of t...

conjecture | open | MathDB 页面 | 原文线索

V2-063142 | MATHDB-331399

Conjecture on Lyapunov representations of observable all-pass functions

摘录 / EXCERPT: Lyapunov representation conjecture. There exists a symmetric matrix $P=P^{\top}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-063143 | MATHDB-331402

Concavity conjecture for entropy along normalized sums of log-concave variables

摘录 / EXCERPT: Concavity conjecture. The function f is concave on $[0,1]$. This conjecture proposes a stronger form of regularity for entropy under normalized sums of independent identically di...

conjecture | open | MathDB 页面 | 原文线索

V2-063144 | MATHDB-331411

Uniqueness of the unbiased filter delay

摘录 / EXCERPT: Uniqueness of the unbiased delay. The filter will be unbiased for only one value of the delay r; equivalently, there exists only one value of r satisfying the displayed conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-063145 | MATHDB-331412

Maximality of lcm lattices of maximal monomial ideals in Betti strata

摘录 / EXCERPT: Maximality conjecture. If $Q>P$ in $\mathcal{L}(n)$, then the minimal resolution of Q has total Betti numbers greater than those of P. Equivalently, P is maximal in its Betti...

conjecture | open | MathDB 页面 | 原文线索

V2-063146 | MATHDB-331417

Linear independence conjecture for observer-based maps

摘录 / EXCERPT: Linear independence conjecture for observer-based maps. For each configuration $\mathbf{x}\in C_n(\mathbb{R}^3)$, the corresponding polynomials p_I, as I varies in...

conjecture | open | MathDB 页面 | 原文线索

V2-063147 | MATHDB-331420

Phillips-Sarnak generic essential cuspidality conjectures

摘录 / EXCERPT: Let Γ vary in a Teichmüller space of finite-area hyperbolic surfaces. A surface is essentially cuspidal when its discrete spectrum satisfies the Weyl law. Phillips-Sarnak's...

conjecture | open | MathDB 页面 | 原文线索

V2-063148 | MATHDB-331424

Cheng-Huang-Li integral decay conjecture for the Lane-Emden system

摘录 / EXCERPT: Let (u,v) solve the Lane-Emden system in $\mathbb{R}^n$ with $p\geq q$, and let β denote the exponent used in the source's integral estimate. Cheng-Huang-Li conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-063149 | MATHDB-331426

Witten's conjecture on the KdV hierarchy for intersection numbers

摘录 / EXCERPT: Witten's conjecture. The formal power series $\exp(F(t_0,t_1,\ldots))$ is annihilated by every operator $L_{-1},L_0,L_1,\ldots$; equivalently, $L_k\exp(F)=0$ for all $k\geq-1$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-063150 | MATHDB-331427

The touchpoint 4-variable Catalan conjecture

摘录 / EXCERPT: Let $\operatorname{Cat}_{n,r}(q,t,z,w)$ and $\operatorname{Cat}'_{n,r}(q,t,z,w)$ be the refinements of the 4-variable Catalan polynomials obtained by summing over Dyck paths of ord...

conjecture | open | MathDB 页面 | 原文线索

V2-063151 | MATHDB-331428

The compositional 4-variable Catalan conjecture

摘录 / EXCERPT: Let n,k,ℓ be integers with $n>k$ and $\ell\geq0$, and let $\alpha\vDash n-\ell$ be a composition. Let $\operatorname{Cat}'_{n,\alpha}(q,t,z)$ be the decorated Dyck-path polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-063152 | MATHDB-331429

The partially labeled Dyck path formula for delta operators

摘录 / EXCERPT: Let n and ℓ be positive integers, and let $\mathcal{LD}^{\operatorname{Part}}_{n,\ell}$ be the set of Dyck paths of order $n+\ell$ with n positively labeled north steps an...

conjecture | open | MathDB 页面 | 原文线索

V2-063153 | MATHDB-331430

The touch refinement for partially labeled Dyck paths

摘录 / EXCERPT: Let n and ℓ be positive integers, let $r\geq0$, and let $\mathcal{LD}^{\operatorname{Part}}_{n,\ell}$ be the partially labeled Dyck paths defined above. For P in this set,...

conjecture | open | MathDB 页面 | 原文线索

V2-063154 | MATHDB-331431

Quadratic-field preperiodic graph conjecture

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a quadratic field and let $f\in K[z]$ be a quadratic polynomial. Let $\operatorname{PrePer}(f,K)$ be the set of K-rational preperiodic points of f, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-063155 | MATHDB-331433

Rank-symmetric nearly equal chain decomposition conjecture for the Boolean lattice

摘录 / EXCERPT: Rank-symmetric chain conjecture. The Boolean lattice $2^{[n]}$ can be partitioned into $\binom{n}{\lfloor n/2\rfloor}$ rank-symmetric chains such that the size of any two chains di...

conjecture | open | MathDB 页面 | 原文线索

V2-063156 | MATHDB-331434

Uniform rank-symmetric chain partition conjecture for the Boolean lattice

摘录 / EXCERPT: Uniform rank-symmetric chain conjecture. For every positive integer h there exists a positive integer $N(h)$ such that, whenever $n>N(h)$ and n is odd, the Boolean lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-063157 | MATHDB-331438

Local Lindelöf hypothesis for Whittaker newforms

摘录 / EXCERPT: Let π be the representation under consideration, with central character ω_π of conductor exponent $a(\omega_\pi)$, and let n_1 and m_1 be the parameters used in th...

conjecture | open | MathDB 页面 | 原文线索

V2-063158 | MATHDB-331467

The Faber-Zagier conjecture for tautological relations on moduli spaces of curves

摘录 / EXCERPT: Let τ_a denote the cotangent-line classes used to define intersection numbers on the moduli spaces of stable curves, and let $\boldsymbol{\beth}$ denote the tautological ring...

conjecture | open | MathDB 页面 | 原文线索

V2-063159 | MATHDB-331472

Ryser-Lovász conjecture for monochromatic tree covers

摘录 / EXCERPT: Ryser-Lovász conjecture. For every integer $r\geq 2$ and every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063160 | MATHDB-331473

Random-graph tree-cover threshold conjecture

摘录 / EXCERPT: Random-graph tree-cover threshold conjecture. For every $\epsilon>0$ and every $r\geq 1$, if

conjecture | open | MathDB 页面 | 原文线索

V2-063161 | MATHDB-331474

Distinct-color monochromatic partition and cover conjecture

摘录 / EXCERPT: Distinct-color conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-063162 | MATHDB-331479

Conjecture on the Hausdorff dimension of lower-spectral-radius discontinuities

摘录 / EXCERPT: Let $M_2(\mathbb{R})$ be the space of all 2×2 real matrices, and let ... be the lower spectral radius as a function on pairs of matrices. Discontinuity-dimension conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-063163 | MATHDB-331484

Genericity equivariance conjecture for L-packets

摘录 / EXCERPT: Let G be a quasi-split connected reductive group over a local field k of characteristic zero, with maximal torus T and adjoint torus $T_{\operatorname{ad}}=T/Z$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063164 | MATHDB-331485

Vogan L-packet genericity equivariance conjecture

摘录 / EXCERPT: Let G be a quasi-split connected reductive group over a local field k of characteristic zero, let G' be a pure inner form, and let Π'_{φ} be the standard L-packet...

conjecture | open | MathDB 页面 | 原文线索

V2-063165 | MATHDB-331489

Miller-Reiner conjecture on Smith normal forms of differential-poset operators

摘录 / EXCERPT: Miller-Reiner conjecture. For any differential poset P and any $n\geq 0$, the matrix

conjecture | open | MathDB 页面 | 原文线索

V2-063166 | MATHDB-331494

Noncommutative Schur positivity modulo the k-R relations

摘录 / EXCERPT: Noncommutative Schur positivity conjecture. For every integer partition λ and every positive integer k, $\mathfrak{J}_\lambda(\boldsymbol{u})$ is $\mathbb{Z}$-monomial po...

conjecture | open | MathDB 页面 | 原文线索

V2-063167 | MATHDB-331495

Lam's strengthened square-free positivity conjecture

摘录 / EXCERPT: Lam's strengthened square-free positivity conjecture. For every partition λ, $\mathfrak{J}_\lambda(\mathbf{u})$ is $\mathbb{Z}$-monomial positive modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-063168 | MATHDB-331502

Wong-Zhu's permanent-index conjecture for A_G

摘录 / EXCERPT: Let G be a graph, and let A_G be the matrix whose rows are indexed by the edges of G and whose columns are indexed by the vertices and edges of G, with entries determined b...

conjecture | open | MathDB 页面 | 原文线索

V2-063169 | MATHDB-331504

Generic structure conjecture for spectral edges of periodic operators

摘录 / EXCERPT: Generic spectral-edge conjecture. Generically, the extrema of band functions are attained by a single band, are isolated, and are non-degenerate, meaning that their Hessians are no...

conjecture | open | MathDB 页面 | 原文线索

V2-063170 | MATHDB-331505

Irreducibility conjecture for Fermi surfaces of periodic Schrödinger operators

摘录 / EXCERPT: Fermi-surface irreducibility conjecture. For $\lambda\in\mathbb{R}$, the Fermi surface $F_{\lambda,H}$ is irreducible modulo Γ^*, possibly except for a discrete set of value...

conjecture | open | MathDB 页面 | 原文线索

V2-063171 | MATHDB-331506

Higher-dimensional Borg uniqueness conjecture for periodic Schrödinger operators

摘录 / EXCERPT: Higher-dimensional Borg conjecture. The following claims are equivalent: 1. The potential V is constant. 2. There exists an entire function whose graph $\lambda=f(k)$ belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-063172 | MATHDB-331509

Generalized Yoshida conjecture for maximal CM subfields

摘录 / EXCERPT: Let K be a finite abelian extension of a totally real field F containing a CM-subfield, and let K_{CM} be its maximal CM-subfield. Put $G:=\operatorname{Gal}(K/F)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-063173 | MATHDB-331513

Ahmed's reverse Collatz convergence conjecture

摘录 / EXCERPT: Ahmed's reverse Collatz convergence conjecture. The reverse Collatz sequence converges to a multiple of 3 for every number greater than one.

conjecture | open | MathDB 页面 | 原文线索

V2-063174 | MATHDB-331514

Roune-Woods conjecture on the parametric Frobenius number

摘录 / EXCERPT: Roune-Woods conjecture. The function D is an eventual quasi-polynomial (EQP).

conjecture | open | MathDB 页面 | 原文线索

V2-063175 | MATHDB-331515

Parametric Gröbner-basis conjecture for the relation set

摘录 / EXCERPT: Parametric relation-set conjecture. There exists a PILP whose lattice point set equals $G(S(t))$ for $t\gg0$. The same is true for

conjecture | open | MathDB 页面 | 原文线索

V2-063176 | MATHDB-331522

Helffer-Nier's compact-resolvent conjecture for the Fokker-Planck operator

摘录 / EXCERPT: Helffer-Nier's conjecture. The operator P has a compact resolvent if and only if the Witten Laplacian $\Delta_{V/2}^{(0)}$ has a compact resolvent. This conjecture expresses a pr...

conjecture | open | MathDB 页面 | 原文线索

V2-063177 | MATHDB-331523

Universal Painlevé-I2 scaling conjecture for generalized KP wave breaking

摘录 / EXCERPT: Consider the generalized KP equation ... with ϵ-independent initial data $u(x,y,0;\epsilon)=u_0(x,y)$, and assume its solution is at least C^4 for $t>0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063178 | MATHDB-331524

Folklore conjecture on derived invariance of Hodge numbers

摘录 / EXCERPT: Hodge-number invariance conjecture. If X and Y are derived equivalent, then they have the same Hodge numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-063179 | MATHDB-331527

Minimality conjecture for the smallest Garside-shadow automaton

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let $\tilde S$ be its smallest Garside shadow, and let $\mathcal A_{\tilde S}(W,S)$ be the associated finite deterministic automaton. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-063180 | MATHDB-331528

Eriksson's minimality criterion for canonical automata

摘录 / EXCERPT: Let (W,S) be an irreducible Coxeter system. Let $\mathcal A_0(W,S)$ denote the canonical automaton recognizing the language of reduced words, let Σ denote the set of small...

conjecture | open | MathDB 页面 | 原文线索

V2-063181 | MATHDB-331557

Langlands' beyond endoscopy pole conjecture for detecting representations

摘录 / EXCERPT: Let F be a number field, let $\mathbb{A}_F$ be its ring of adeles, let H be a reductive group over F, and let ${}^{\lambda}H$ be the Zariski closure of the image of its L-g...

conjecture | open | MathDB 页面 | 原文线索

V2-063182 | MATHDB-331567

The structural decomposition conjecture for definable groups in dense pairs

摘录 / EXCERPT: Let $\langle G,*\rangle$ be a definable group. A group is small if it is a definable set of the corresponding smallness notion in the dense expansion, and a group is $\bigvee$-defi...

conjecture | open | MathDB 页面 | 原文线索

V2-063183 | MATHDB-331579

Bar-Lev-Bshouty-Enis conjecture for polynomial variance functions of natural exponential families

摘录 / EXCERPT: Bar-Lev-Bshouty-Enis conjecture. The pair $\left(v,\left(0,\infty\right)\right)$ is a variance function for all $n\in\mathbb{N}$ if and only if $\operatorname{Re}(u_1)\leq 0$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-063184 | MATHDB-331580

Breuillard's uniform exponential growth conjecture for linear groups

摘录 / EXCERPT: Given $d\in\mathbb{N}$, let S be a finite subset of $\operatorname{GL}_d(\mathbb{C})$ and define its exponential growth rate by ... where S^n is the set of products of n elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-063185 | MATHDB-331587

The alpha Jacobian conjecture

摘录 / EXCERPT: The α Jacobian conjecture. Then f is invertible. The paper subsequently states and proves this claim as a theorem, so this conjecture is solved.

conjecture | open | MathDB 页面 | 原文线索

V2-063186 | MATHDB-331588

The epsilon Jacobian conjecture

摘录 / EXCERPT: The ϵ Jacobian conjecture. Then f is invertible. The supplied passage only introduces this analogous claim and does not provide evidence that it has been proved or refut…

conjecture | open | MathDB 页面 | 原文线索

V2-063187 | MATHDB-331589

The weak symmetric-or-skew-symmetric reduction conjecture

摘录 / EXCERPT: The weak symmetric-or-skew-symmetric reduction conjecture. There exist such morphisms $g_1,\ldots,g_l$ such that $\prod g_iA$ is symmetric or skew-symmetric with respect to an invo…

conjecture | open | MathDB 页面 | 原文线索

V2-063188 | MATHDB-331593

Cohen-Lenstra-Soundararajan conjectural asymptotic for odd imaginary quadratic class numbers

摘录 / EXCERPT: The refined Soundararajan conjecture. As $h\to\infty$ through odd values,

conjecture | open | MathDB 页面 | 原文线索

V2-063189 | MATHDB-331594

Cyclicity-index conjecture for imaginary quadratic class groups

摘录 / EXCERPT: Fix $n\in\mathbb{N}$ and a partition $\lambda=(n_1,\dots,n_r)$ of n, with $n_1\geq\cdots\geq n_r\geq1$, and define … For a finite abelian group G, let $\mathcal{F}(G)$ denote t…

conjecture | open | MathDB 页面 | 原文线索

V2-063190 | MATHDB-331600

No long rational cycles for quadratic maps with a rational critical 2-cycle

摘录 / EXCERPT: No-long-cycles conjecture. The map phi has no $\mathbb{Q}$-periodic point of period greater than or equal to 3.

conjecture | open | MathDB 页面 | 原文线索

V2-063191 | MATHDB-331601

Optimality of full replication for log-convex service distributions

摘录 / EXCERPT: Let n be the number of servers, let $\bar{F}_X$ be the survival function of a task's service time, and let $\rX$ denote the service-time distribution. For a fork-join system, let…

conjecture | open | MathDB 页面 | 原文线索

V2-063192 | MATHDB-331602

Parametrised T-duality trivialisation conjecture for the bulk-boundary correspondence

摘录 / EXCERPT: Parametrised T-duality trivialisation conjecture. The diagram

conjecture | open | MathDB 页面 | 原文线索

V2-063193 | MATHDB-331614

The countable width-two lattice decomposition conjecture

摘录 / EXCERPT: The countable width-two lattice decomposition conjecture. The lattice L is semidistributive and

conjecture | open | MathDB 页面 | 原文线索

V2-063194 | MATHDB-331617

Entringer number conjecture for 000-avoiding inversion sequences

摘录 / EXCERPT: The Entringer numbers $d_{n,k}$ count down/up permutations of $[n+1]$ whose first entry is $k+1$, and satisfy … An Entringer number conjecture asserts that, for $n\geq 1$ and…

conjecture | open | MathDB 页面 | 原文线索

V2-063195 | MATHDB-331618

Ascent conjecture for Schröder paths and 021-avoiding inversion sequences

摘录 / EXCERPT: Ascent conjecture. The number of Schröder paths $p\in R_{n-1}$ with $k-1$ ascents is equal to the number of inversion sequences $e\in\mathbf{I}_n(021)$ with k distinct values.

conjecture | open | MathDB 页面 | 原文线索

V2-063196 | MATHDB-331627

Worst-case identifiability conjecture for general coalescents

摘录 / EXCERPT: Let $\Xi\in\mathcal{X}$ be the finite measure governing a general Ξ-coalescent, and let ψ^Ξ_n denote the quantity controlling the additional sample-size requirement for…

conjecture | open | MathDB 页面 | 原文线索

V2-063197 | MATHDB-331628

Evans-Gannon's modular-data conjecture for odd-order near-group categories

摘录 / EXCERPT: Let G be a finite abelian group of odd order, let q be a quadratic form on G, and let $\mathcal{NG}(G,q,b,c)$ be the corresponding near-group fusion category. Let E_j be th…

conjecture | open | MathDB 页面 | 原文线索

V2-063198 | MATHDB-331635

Zagier's inverse-matrix conjecture for odd-weight double zeta values

摘录 / EXCERPT: Let $k=2K+1\geq 5$ be odd, and let $\mathcal A=\mathcal A_K$ be the $(K-1)\times(K-1)$ matrix whose (r,s)-entry is … For $1\leq s,r\leq K-1$, let B_n denote the nth Bernoulli…

conjecture | open | MathDB 页面 | 原文线索

V2-063199 | MATHDB-331638

Debarre's ampleness conjecture for generic complete intersections

摘录 / EXCERPT: Debarre's ampleness conjecture. There is a positive lower bound $d\gg1$, depending on N and c, such that whenever $d_1,\dots,d_c\geqslant d$, the cotangent bundle Ω_X is…

conjecture | open | MathDB 页面 | 原文线索

V2-063200 | MATHDB-331642

Tracy-Widom crossover conjecture for rectangular Young tableaux

摘录 / EXCERPT: Let (a_n) be a nondecreasing sequence with $a_n\to\infty$ as $n\to\infty$, and let $X^{(n)}_{i,j}$ denote the relevant entry of the random rectangular Young tableau. Write TW f...

conjecture | open | MathDB 页面 | 原文线索

V2-063201 | MATHDB-331643

Quadratic diameter conjecture for Kempe reconfiguration graphs of degenerate graphs

摘录 / EXCERPT: Let $\mathcal{K}_k(G)$ denote the reconfiguration graph whose vertices are the k-colourings of G, with edges representing Kempe changes. Let d and k be positive integers wi...

conjecture | open | MathDB 页面 | 原文线索

V2-063202 | MATHDB-331648

The online power-of-choices geometric graph giant-component conjecture

摘录 / EXCERPT: Consider the online power-of-choices geometric graph process on n vertices, with radius ... Here $c>0$, f is a function from $(0,\infty)$ to $(0,1)$, and w.h.p. means with high p...

conjecture | open | MathDB 页面 | 原文线索

V2-063203 | MATHDB-331651

The normal-complex diameter conjecture

摘录 / EXCERPT: Let C be a pure normal simplicial complex of dimension $d-1$ with n vertices. Normal-complex diameter conjecture. The diameter of C is at most ... This conjecture strengthens t...

conjecture | open | MathDB 页面 | 原文线索

V2-063204 | MATHDB-331654

Wan's conjecture on limiting Newton polygons

摘录 / EXCERPT: Let f be a non-constant monic polynomial in $Q[x]$. A global permutation polynomial (GPP) over Q is a polynomial $P(x)\inQ[x]$ whose reduction modulo p is a permutation of...

conjecture | open | MathDB 页面 | 原文线索

V2-063205 | MATHDB-331664

Harmse-Remmel conjecture on c-Wilf and strong c-Wilf equivalence

摘录 / EXCERPT: Harmse-Remmel conjecture. P and Q are c-Wilf equivalent if and only if they are strongly c-Wilf equivalent. The conjecture extends Nakamura's $k=1$ formulation to $k\geq2$; the...

conjecture | open | MathDB 页面 | 原文线索

V2-063206 | MATHDB-331669

The left-closedness conjecture for semidirect products of monoidal categories

摘录 / EXCERPT: Let $\mathcal{X}$ and $\mathcal{C}$ be left closed monoidal categories, meaning that tensoring on the left has a right adjoint. Let there be a strong action of $\mathcal{X}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-063207 | MATHDB-331723

The real-slice conjecture for generalized Mandelbrot sets of even degree

摘录 / EXCERPT: Let ${\mathcal{M}^p}$ be the generalized Mandelbrot set for the polynomial $Q_{p,c}(z)=z^p+c$, where $z,c\in\mathbb{C}$ and $p\geq 2$ is an integer. The real-slice conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-063208 | MATHDB-331724

The square characterization conjecture for Hyperbrots of even degree

摘录 / EXCERPT: The square characterization conjecture. For every even integer $p\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-063209 | MATHDB-331727

The cuspidal Langlands parameter conjecture for supercuspidal representations

摘录 / EXCERPT: Cuspidal Langlands parameter conjecture. Le L-paquet $cPi_{cvarphi}(G)$ contient des représentations supercuspidales de G si et seulement si $cvarphi$ est un paramètre de Langl...

conjecture | open | MathDB 页面 | 原文线索

V2-063210 | MATHDB-331728

Vogan-Haines compatibility conjecture for parabolic induction

摘录 / EXCERPT: Vogan-Haines compatibility conjecture. La correspondance de Langlands devrait vérifier

conjecture | open | MathDB 页面 | 原文线索

V2-063211 | MATHDB-331729

Optimality conjecture for the harmonic kernel in Riesz energy

摘录 / EXCERPT: Let $cmathbb{S}^d$ be the unit sphere and let $n=cpi_L$. Let K be an isotropic kernel producing n points through its determinantal point process, and let $E_s(x)$ denote the Ri...

conjecture | open | MathDB 页面 | 原文线索

V2-063212 | MATHDB-331734

Piontkowski's three-cusp conjecture for rational cuspidal plane curves

摘录 / EXCERPT: Let C be a rational cuspidal plane curve of degree d. The source identifies three specific series of tricuspidal curves in a table. Piontkowski's conjecture. If $d\geq 6$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-063213 | MATHDB-331736

Piene's Hessian intersection-multiplicity conjecture

摘录 / EXCERPT: Let C be a rational cuspidal plane curve, let H_C be its Hessian curve, and let p be a cusp or an inflection point. Write $(C\cdot H_C)_p$ for their local intersection multip...

conjecture | open | MathDB 页面 | 原文线索

V2-063214 | MATHDB-331743

Local genus polynomials are log-concave

摘录 / EXCERPT: Let (G,v) be a rooted graph. For an almost complete rotation system at v, let the local genus polynomial be the genus polynomial of the embeddings consistent with that system....

conjecture | open | MathDB 页面 | 原文线索

V2-063215 | MATHDB-331747

The quaternion-brace counting conjecture

摘录 / EXCERPT: For $m\in\mathbb{N}$, let Q_{4m} be the generalized quaternion group, and call a brace a quaternion brace if its multiplicative group is isomorphic to a generalized quaternion gr...

conjecture | open | MathDB 页面 | 原文线索

V2-063216 | MATHDB-331748

The power-of-two quaternion-brace counting conjecture

摘录 / EXCERPT: Let a quaternion brace be a brace whose multiplicative group is isomorphic to a generalized quaternion group. Power-of-two quaternion-brace conjecture. There are seven isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-063217 | MATHDB-331749

The conjecture on non-synchronizing ranks of primitive groups

摘录 / EXCERPT: Let G be a primitive permutation group of degree n, and define ... to be its set of non-synchronizing ranks. Non-synchronizing-rank conjecture. ... This predicts that primitive gro...

conjecture | open | MathDB 页面 | 原文线索

V2-063218 | MATHDB-331753

Deligne-type conjecture for automorphic Rankin-Selberg L-values

摘录 / EXCERPT: Let F be a CM field, and let Π and Π' be cuspidal representations of $GL_n(\mathbb{A}_F)$ and $GL_{n'}(\mathbb{A}_F)$ with definable arithmetic automorphic periods. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063219 | MATHDB-331771

The 1/2-BPS black-hole index formula for the minimal representation of E8

摘录 / EXCERPT: 1/2-BPS index conjecture. The index is

conjecture | open | MathDB 页面 | 原文线索

V2-063220 | MATHDB-331772

The 1/4-BPS black-hole index formula from the next-to-minimal representation of E8

摘录 / EXCERPT: 1/4-BPS index conjecture. The index is

conjecture | open | MathDB 页面 | 原文线索

V2-063221 | MATHDB-331774

The quaternionic-discrete-series conjecture for DT invariants of CM-type rigid Calabi-Yau threefolds

摘录 / EXCERPT: Quaternionic-discrete-series conjecture. The invariants $\Omega(\gamma)$ are captured by the Fourier coefficients of an automorphic form attached to the quaternionic discrete serie...

conjecture | open | MathDB 页面 | 原文线索

V2-063222 | MATHDB-331775

The G2 quaternionic-discrete-series conjecture for Donaldson-Thomas invariants

摘录 / EXCERPT: G2 DT-invariant conjecture. There exist Calabi-Yau 3-folds X with $h_{1,1}(X)=1$ whose invariants $\Omega(\gamma)$ are captured by Fourier coefficients of automorphic forms attac...

conjecture | open | MathDB 页面 | 原文线索

V2-063223 | MATHDB-331807

Perron properties conjecture for irreducible nonaffine parabolic closures

摘录 / EXCERPT: Let (W,S) be a Coxeter system with $|S|>2$, let $cphi$ be a geometric representation, and let $w\redundant$ be an element of W. The parabolic closure of w is the smallest par...

conjecture | open | MathDB 页面 | 原文线索

V2-063224 | MATHDB-331808

Equivalence conjecture for Perron properties and irreducible parabolic closures

摘录 / EXCERPT: Let (W,S) be a Coxeter system with $|S|>2$, let $cphi$ be a geometric representation, and let $w\redundant$ be an element of W. Assume that the parabolic closure of w is irre...

conjecture | open | MathDB 页面 | 原文线索

V2-063225 | MATHDB-331818

ABPS conjecture for enhanced L-parameters

摘录 / EXCERPT: Let ${\mathcal H}$ be a connected reductive group over a local non-archimedean field. Let $\mathfrak{Lev}({\mathcal H})$ represent the conjugacy classes of Levi subgroups, let...

conjecture | open | MathDB 页面 | 原文线索

V2-063226 | MATHDB-331827

The multipartition equidistribution conjecture

摘录 / EXCERPT: Multipartition equidistribution conjecture. The density δ_t exists and equals $1/2$ for every odd positive integer t. Equivalently, if $t=2^k t_0$ with $t_0\geq 1$ odd, th...

conjecture | open | MathDB 页面 | 原文线索

V2-063227 | MATHDB-331832

HHPB correspondence for barbed-congruent reversible processes

摘录 / EXCERPT: HHPB correspondence conjecture. If $R\mathrel{\sim^{\tau}}S$, then there exists a HHPB $\mathrel{\mathcal{R}}$ between $\mathcal{C}_R$ and $\mathcal{C}_S$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063228 | MATHDB-331835

The recurrence classification conjecture for uniform shifts on the binary tree

摘录 / EXCERPT: Let $\mathbb{T}_{2}$ be the binary tree, and let ${\mathbf a}$ be an L-periodic rotor sequence. Define $\mathcal{A}_{\mathrm{rec}}^{\mathrm{conj}}$ to consist of all shifts of se...

conjecture | open | MathDB 页面 | 原文线索

V2-063229 | MATHDB-331836

The transience conjecture for uniform shifts on regular trees

摘录 / EXCERPT: Let $\mathbb{T}_{d}$ be the homogeneous d-ary tree with $d\geq 3$, and let ${\mathbf a}$ be an L-periodic rotor sequence. The regular-tree uniform-shift transience conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063230 | MATHDB-331845

Perrin-Riou's conjecture for twisted Kolyvagin systems

摘录 / EXCERPT: Let K be the relevant number field, let K_p denote its p-adic localization, let $\mathbb{T}$ be the associated big representation, and let $r_{\mathrm{an}}(\mathscr{M})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-063231 | MATHDB-331849

Sun's integrality conjectures for central-binomial sums

摘录 / EXCERPT: Let S_k and $S_k^{(2)}$ be the sequences defined by the corresponding central-binomial sums in the source. Sun's integrality conjectures. For every positive integer n, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-063232 | MATHDB-331850

Guo and Liu's congruences for the sums S_n^(r) and T_n^(r)

摘录 / EXCERPT: Guo and Liu's congruence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063233 | MATHDB-331851

Guo and Liu's existence conjecture for divisibility constants

摘录 / EXCERPT: Guo and Liu's existence conjecture. There exist integers a_{2r-1} and b_r, independent of n, such that

conjecture | open | MathDB 页面 | 原文线索

V2-063234 | MATHDB-331852

Guo and Liu's explicit constants conjecture for central-binomial congruences

摘录 / EXCERPT: Let a_{2r-1} and b_r be the integers occurring in the divisibility assertion for $S_n^{(r)}$. Guo and Liu's explicit constants conjecture. The listed values ... ... satisfy the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-063235 | MATHDB-331860

Bijectivity conjecture for the polar curve and polar discriminant

摘录 / EXCERPT: Let (X,x) be the germ under consideration, let f be the defining function, and let ℓ be a linear form general enough. Write $\phi_\ell=(\ell,f):(X,x)\to(\mathbb{C}^2,0)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-063236 | MATHDB-331868

Perrin-Riou's interpolation conjecture for analytic p-adic L-functions

摘录 / EXCERPT: Let $\mathcal{M}$ be the motive attached to the representation T, let $\mathfrak{I}_p$ index the subsets of basis vectors of size g_-, and let $\mathcal{H}_+$ be the relevant s...

conjecture | open | MathDB 页面 | 原文线索

V2-063237 | MATHDB-331870

The signed main conjecture for the $ \underline{I}$-Selmer group

摘录 / EXCERPT: Let $\underline{I}\in\mathfrak{I}_p$ and let η be an even Dirichlet character of conductor p. Let $n(\underline{I},\eta)$ be the integer specified by the cited lemma, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-063238 | MATHDB-331875

The authors' bump conjecture for two-weight bilinear forms

摘录 / EXCERPT: Bump conjecture. There is a constant $C>0$, independent of w and σ, such that

conjecture | open | MathDB 页面 | 原文线索

V2-063239 | MATHDB-331878

Kauffman's derivative-sliceness conjecture for slice knots

摘录 / EXCERPT: Let K be a knot with a slice disk $Delta$, and let J be a derivative of K associated to $Delta$. Kauffman's conjecture. If K is a slice knot via $Delta$, then J is a slic...

conjecture | open | MathDB 页面 | 原文线索

V2-063240 | MATHDB-331879

Solvable-filtration converse for associated derivatives

摘录 / EXCERPT: Let K be a knot, let J be a derivative associated to an $(n+1)$-solution for K, where n is a non-negative integer or half-integer, and let $Delta_K(t)$ denote the Alexander...

conjecture | open | MathDB 页面 | 原文线索

V2-063241 | MATHDB-331881

Type BC residue expansion identity

摘录 / EXCERPT: Fix $k\geq 1$ and $c\in(0,\infty)$. Let $\alpha_1,\ldots,\alpha_k$ be real numbers and let $F(z_1,\ldots,z_k)$ satisfy $\alpha_k=0$, $\alpha_j>\alpha_{j+1}+c$ for $1\leq j\leq k-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-063242 | MATHDB-331903

Uniqueness conjecture for minimizing normalized-volume valuations

摘录 / EXCERPT: Uniqueness conjecture. For any $\mathbb{Q}$-Gorenstein klt singularity (X,o), there exists a unique valuation minimizing $\widehat{\rm vol}$ on ${\rm Val}_{X,o}$.

conjecture | open | MathDB 页面 | 原文线索

V2-063243 | MATHDB-331906

Arithmetic Gan-Gross-Prasad conjecture for $\operatorname{SO}(3)\times\operatorname{SO}(4)$

摘录 / EXCERPT: Suppose that (E,A) is a pair of odd type, with associated Chow cycle $\Delta_{E,A}$ and motive $\mathsf{M}_{E,A}$. Arithmetic Gan-Gross-Prasad conjecture. The Chow cycle is nonze...

conjecture | open | MathDB 页面 | 原文线索

V2-063244 | MATHDB-331907

Bloch-Kato and Chow-rank conjecture for the motives $\mathsf{M}_{E,A}$

摘录 / EXCERPT: Let (E,A) be an arbitrary pair, with associated Chow motive $\mathsf{M}_{E,A}$, Chow group $\operatorname{CH}(\mathsf{M}_{E,A})$, Bloch-Kato Selmer groups...

conjecture | open | MathDB 页面 | 原文线索

V2-063245 | MATHDB-331910

Generic equidistribution conjecture for rays in positive-characteristic homogeneous dynamics

摘录 / EXCERPT: Generic equidistribution conjecture. For almost every $\xi\in\Omega$, the averages

conjecture | open | MathDB 页面 | 原文线索

V2-063246 | MATHDB-331911

Rational-ray escape-of-mass conjecture

摘录 / EXCERPT: Rational-ray escape-of-mass conjecture. For any rational $\xi\in\Omega$, the measures $\mu_{x_n^\xi}$ converge to the zero measure.

conjecture | open | MathDB 页面 | 原文线索

V2-063247 | MATHDB-331918

Ben Efraim's lexicographic edge-isoperimetric conjecture for the transposition graph

摘录 / EXCERPT: Let S_n be the symmetric group, and let T_n be its transposition graph. For $\mathcal{A} \subset S_n$, write $\partial \mathcal{A}$ for its edge-boundary in T_n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063248 | MATHDB-331920

Strauss's global existence and blowup conjecture for semilinear wave equations

摘录 / EXCERPT: Strauss's conjecture. If $p>p_1(n)$, then small-data solutions exist globally; if $1<p<p_1(n)$, then small-data solutions blow up in finite time.

conjecture | open | MathDB 页面 | 原文线索

V2-063249 | MATHDB-331926

The Manin conjecture for strong modular parametrizations

摘录 / EXCERPT: Manin conjecture. The Manin constant of a strong parametrization is ±1.

conjecture | open | MathDB 页面 | 原文线索

V2-063250 | MATHDB-331927

The generalized Manin conjecture for optimal parametrizations of quadratic curves

摘录 / EXCERPT: Generalized Manin conjecture. If $\pi\colon X_1(N)\to E$ is an optimal parametrization, then

conjecture | open | MathDB 页面 | 原文线索

V2-063251 | MATHDB-331929

The strong Birch and Swinnerton-Dyer conjecture for elliptic curves over number fields

摘录 / EXCERPT: Strong BSD conjecture. The weak BSD conjecture holds; moreover, $\operatorname{Sha}(E/K)$ is finite and

conjecture | open | MathDB 页面 | 原文线索

V2-063252 | MATHDB-331937

Garner's conjecture on corresponding Collatz stems

摘录 / EXCERPT: Garner's conjecture. Every pair of corresponding stems is of the form s_i and s_i' for some i; equivalently, any pair of consecutive integers of the same height has parity ve...

conjecture | open | MathDB 页面 | 原文线索

V2-063253 | MATHDB-331939

Dimca-Sticlaru conjectures on rational cuspidal plane curves

摘录 / EXCERPT: Dimca-Sticlaru's conjecture. (i) Any rational cuspidal curve C in the plane is either free or nearly free. (ii) An irreducible plane curve C which is either free or nearly free...

conjecture | open | MathDB 页面 | 原文线索

V2-063254 | MATHDB-331940

Dimca-Sticlaru conjecture on branches of singularities of free and nearly free curves

摘录 / EXCERPT: Dimca-Sticlaru's branch conjecture. (i) Any free irreducible plane curve C has only singularities with at most two branches. (ii) Any nearly free irreducible plane curve C has...

conjecture | open | MathDB 页面 | 原文线索

V2-063255 | MATHDB-331941

Weak convergence conjecture for rescaled chromatic measures of dense graph sequences

摘录 / EXCERPT: Let $(G_n)_n$ be a sequence of graphs convergent in the dense model. For a graph G of order n, let ν_G denote its rescaled chromatic measure, defined from the chromatic-roo...

conjecture | open | MathDB 页面 | 原文线索

V2-063256 | MATHDB-331950

Green-Lazarsfeld Secant Conjecture for line bundles on curves

摘录 / EXCERPT: Let C be a curve of genus g, and let L be a globally generated line bundle of degree d on C. Write $h^1(C,L)=\dim H^1(C,L)$, let $\operatorname{Cliff}(C)$ denote the Clif...

conjecture | open | MathDB 页面 | 原文线索

V2-063257 | MATHDB-331955

Cuspidal divisor group conjecture for Drinfeld modular Jacobians

摘录 / EXCERPT: Let A be the ring of functions regular away from a fixed place at infinity, and let $\mathfrak n\lhd A$ be a non-zero square-free ideal. Let $X_0(\mathfrak n)$ be the smooth proj...

conjecture | open | MathDB 页面 | 原文线索

V2-063258 | MATHDB-331957

The three-component skein invariant sum conjecture

摘录 / EXCERPT: Let L_1 and L_2 be links with three components. For each c, let $\Sigma^c(L)$ denote the sum of the elements of the set $\mathbf{F}^c_p$ associated with L, where p record...

conjecture | open | MathDB 页面 | 原文线索

V2-063259 | MATHDB-331959

No exponentially convergent spectral-ratio families in the P≤ cases

摘录 / EXCERPT: Let h be a pseudo-Anosov mapping class and let $\operatorname{\omega}(h)$ denote its spectral ratio. For a surface and an infinite family of pseudo-Anosov mapping classes, a P...

conjecture | open | MathDB 页面 | 原文线索

V2-063260 | MATHDB-331960

Ratio log-concavity conjecture for Sun's R-sequence

摘录 / EXCERPT: Ratio log-concavity conjecture. The sequence $\{r_n\}_{n\geq 4}$ is log-concave, equivalently, R_n is ratio log-concave for $n\geq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-063261 | MATHDB-331961

Infinite log-concavity conjecture for the R-sequence difference

摘录 / EXCERPT: Infinite log-concavity conjecture. The sequence $\{R_n^2-R_{n+1}R_{n-1}\}_{n\geq 6}$ is infinity-log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-063262 | MATHDB-331979

The rational odd-denominator limb conjecture for antipode-preserving cubic maps

摘录 / EXCERPT: Let ${\bf t}$ be a rotation number, and let ${\pazocal L}_{\bf t}$ denote the compact connected limb associated to ${\bf t}$, with distinguished point ${\boldsymbol\infty}_{\bf t}$...

conjecture | open | MathDB 页面 | 原文线索

V2-063263 | MATHDB-331981

The quadratic-binomial generation conjecture for isotonian toric ideals

摘录 / EXCERPT: Quadratic-binomial generation conjecture. If Q does not contain any induced poset cycle of length greater than 4, then $J_{P,Q}$ is generated by quadratic binomials.

conjecture | open | MathDB 页面 | 原文线索

V2-063264 | MATHDB-331991

Global asymptotic stability conjecture for locally stable steady states in competition models

摘录 / EXCERPT: Consider the competition model described in the paper, in the parameter regime where both semi-trivial steady states exist. A steady state is called locally stable if it is locally...

conjecture | open | MathDB 页面 | 原文线索

V2-063265 | MATHDB-331999

Similarity conjecture for cubic rational helical space curves

摘录 / EXCERPT: Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be cubic rational helical space curves with proportionality constants μ_1 and μ_2, respectively. The curves have proportionality con...

conjecture | open | MathDB 页面 | 原文线索

V2-063266 | MATHDB-332014

The weight-support system conjecture for perfect codes containing Preparata codes

摘录 / EXCERPT: Weight-support system conjecture. All codewords of weight 3 of a perfect code containing a Preparata code, and all codewords of weight 4 of an extended perfect code containing...

conjecture | open | MathDB 页面 | 原文线索

V2-063267 | MATHDB-332015

The linear-components conjecture for perfect codes containing Preparata codes

摘录 / EXCERPT: Linear-components conjecture. Every i-component of C, for every i, is linear; equivalently, it is equivalent to

conjecture | open | MathDB 页面 | 原文线索

V2-063268 | MATHDB-332017

Pivot-minor chi-boundedness conjecture

摘录 / EXCERPT: For a graph H, a graph class has no pivot-minor isomorphic to H if none of its members has such a pivot-minor. Pivot-minor conjecture. For every graph H, the class of graphs...

conjecture | open | MathDB 页面 | 原文线索

V2-063269 | MATHDB-332018

Weak convergence for rooted graphs excluding a finite set of minors

摘录 / EXCERPT: Rooted minor-exclusion convergence conjecture. The sequence R_n converges weakly.

conjecture | open | MathDB 页面 | 原文线索

V2-063270 | MATHDB-332019

Benjamini--Schramm convergence for graphs excluding a finite set of minors

摘录 / EXCERPT: Minor-exclusion Benjamini--Schramm conjecture. The sequence G_n Benjamini--Schramm converges.

conjecture | open | MathDB 页面 | 原文线索

V2-063271 | MATHDB-332020

Benjamini--Schramm convergence to the UIPT for maximal surface-embeddable graphs

摘录 / EXCERPT: Maximal surface-graph conjecture. The sequence G_n Benjamini--Schramm converges, and its limit coincides with the UIPT.

conjecture | open | MathDB 页面 | 原文线索

V2-063272 | MATHDB-332021

Compactified weak convergence for rooted minor-excluding graphs

摘录 / EXCERPT: Compactified rooted convergence conjecture. The sequence R_n converges weakly with respect to d.

conjecture | open | MathDB 页面 | 原文线索

V2-063273 | MATHDB-332022

Compactified local convergence for unrooted minor-excluding graphs

摘录 / EXCERPT: Compactified unrooted convergence conjecture. The sequence G_n converges weakly with respect to d.

conjecture | open | MathDB 页面 | 原文线索

V2-063274 | MATHDB-332023

The SCV-FCV comparison conjecture

摘录 / EXCERPT: Let $N\ge 3$. For each weight $w\ge 1$, let $\operatorname{CMZV}_{w,N}$, $\operatorname{FCV}_{w,N}$, and $\operatorname{SCV}_{w,N}$ be the $\mathbb{Q}(\mathcal{G}_N)$-spaces genera...

conjecture | open | MathDB 页面 | 原文线索

V2-063275 | MATHDB-332024

The basis conjecture for finite colored multiple zeta values at levels 3 and 4

摘录 / EXCERPT: The basis conjecture. When $N=3$ or $N=4$, these elements form a basis of the $\mathbb{Q}(\xi_N)$-vector space $\operatorname{FCV}_{w,N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-063276 | MATHDB-332027

Conjectural formula for the threshold at which two additional roots appear

摘录 / EXCERPT: Let $p\in\mathbb{Z}^+$, define $p^*=p\mod 9$, and let $n_0(p)$ denote the first value of n for which the two additional solutions in the paper's theorem occur. Threshold formula...

conjecture | open | MathDB 页面 | 原文线索

V2-063277 | MATHDB-332032

Weak Deligne conjecture up to F-factors

摘录 / EXCERPT: Let M be a critical motive with coefficients in a number field E, let $F\subset\mathbb{C}$ be a number field, let $L(M,s)$ be its L-function, and let $c^{+}(M)$ be its Delign...

conjecture | open | MathDB 页面 | 原文线索

V2-063278 | MATHDB-332036

Degree-two regulator conjecture for universal mixed elliptic motives

摘录 / EXCERPT: Let $\ast\in\{1,\vec{1}\}$, let $m\geq 0$ and $r\in\mathbb Z$, and let ${\mathsf{MEM}}_\ast$ be the corresponding category of universal mixed elliptic motives. The real and ℓ-...

conjecture | open | MathDB 页面 | 原文线索

V2-063279 | MATHDB-332044

Becker's categorical rationality conjecture for k-regular power series

摘录 / EXCERPT: Let $k\geq 2$, let $\mathpzc{k}$ be the coefficient field, and let $f\in\mathpzc{k}[[z]]$. A rational function $h(z)\in\mathpzc{k}(z)$ is k-regular if it is a k-regular rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-063280 | MATHDB-332045

Becker's naive precalmness conjecture

摘录 / EXCERPT: Let $k\geq 2$, let $\mathpzc{k}$ be the coefficient field, and let $f(z)\in\mathpzc{k}[[z]]$ satisfy the Mahler equation ... A sequence of coefficients is precalm in the sense define...

conjecture | open | MathDB 页面 | 原文线索

V2-063281 | MATHDB-332046

Unrestricted calm-coefficient criterion for regular Mahler power series

摘录 / EXCERPT: Let $k\geq 2$, let $\mathpzc{k}$ be the coefficient field, and let $f(z)\in\mathpzc{k}[[z]]$ be a Mahler power series. A sequence of rational functions is calm in the sense defined...

conjecture | open | MathDB 页面 | 原文线索

V2-063282 | MATHDB-332049

Replica-symmetry MP-structure conjecture for the independent set problem

摘录 / EXCERPT: Let (β,h,γ) be parameters of the independent set problem, and let an MP-structure consist of a hierarchy $(\zeta,\mathcal V_{\boldsymbol m})$ with K levels. For such a...

conjecture | open | MathDB 页面 | 原文线索

V2-063283 | MATHDB-332050

Broadhurst's generalized parity conjecture for sixth-root multiple polylogarithms

摘录 / EXCERPT: Generalized parity conjecture. If $\left\lvert\boldsymbol{n}\right\rvert+d$ is odd, then the real part $\operatorname{Re}\operatorname{Li}_{\boldsymbol{n}}(\boldsymbol{z})$ has dep...

conjecture | open | MathDB 页面 | 原文线索

V2-063284 | MATHDB-332054

The generating-series conjecture for perverse numbers of parabolic Hitchin systems

摘录 / EXCERPT: Let $\bm{\mu}$ be the multipartition encoding the parabolic data of $M_B(n)$, let k be as defined in the paper, and let $\mathbb{H}_{\bm{\mu}}$ be the function defined in Section...

conjecture | open | MathDB 页面 | 原文线索

V2-063285 | MATHDB-332055

Jang-Kim's maximal norm conjecture for indecomposable integers

摘录 / EXCERPT: Let D be a positive nonsquare integer, let $\mathbb Z[\sqrt D]$ be the quadratic order, and let N be the minimum of the absolute values of negative norms of elements of...

conjecture | open | MathDB 页面 | 原文线索

V2-063286 | MATHDB-332059

Conway's common-refinement conjecture for generalized power series

摘录 / EXCERPT: Let $K((\mathbb{R}^{\leq 0}))$ be the ring of generalized power series with exponents in $\mathbb{R}^{\leq 0}$. For a factorisation $a=b_1\cdot\ldots\cdot b_n$, a refinement is a f...

conjecture | open | MathDB 页面 | 原文线索

V2-063287 | MATHDB-332073

Optimality conjecture for the local-set-based piecewise-constant wavelet basis

摘录 / EXCERPT: Optimality conjecture. The local-set-based wavelet basis $\operatorname{W}_{\rm LSPC}$ minimizes, among all orthonormal bases F, the worst-case number of nonzero c...

conjecture | open | MathDB 页面 | 原文线索

V2-063288 | MATHDB-332078

Krichever's dispersive universal Whitham hierarchy conjecture

摘录 / EXCERPT: Let a punctured Riemann surface be a Riemann surface of arbitrary genus equipped with punctures, and let Krichever's construction associate to it a universal Whitham hierarchy. Kri...

conjecture | open | MathDB 页面 | 原文线索

V2-063289 | MATHDB-332082

Fermionic reformulation of dispersive universal Whitham hierarchies

摘录 / EXCERPT: Let C be any spectral curve, and let the dispersive universal Whitham integrable hierarchy associated with C be the corresponding hierarchy. Fermionic reformulation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063290 | MATHDB-332085

The h-measure exceptional-set conjecture for gap power series

摘录 / EXCERPT: Let $\Lambda=(\lambda_n)$ be the frequency sequence and let L and L^{-} denote the classes of auxiliary functions used in the paper. For $\varphi\in L$, define ... Let h be the...

conjecture | open | MathDB 页面 | 原文线索

V2-063291 | MATHDB-332087

The phase-portrait conjecture for generalized full symmetric Toda systems

摘录 / EXCERPT: Phase-portrait conjecture. The phase picture of the Toda system coincides with the Bruhat order on the corresponding Weyl group: stationary points correspond bijectively to Weyl-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-063292 | MATHDB-332090

Nonvanishing conjecture for Bernoulli-number finite zeta values

摘录 / EXCERPT: For every odd integer $k\geq 3$, define the element $(B_{p-k}\pmod p)_p\in\mathcal{A}_{\mathbb{Q}}$, where B_m denotes the m-th Bernoulli number. Bernoulli finite zeta nonvanis...

conjecture | open | MathDB 页面 | 原文线索

V2-063293 | MATHDB-332093

Korpelainen et al.'s conjecture on finite definition and labelled well-quasi-ordering

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary graph class, meaning that it is closed under taking induced subgraphs, and suppose that $\mathcal{C}$ is well-quasi-ordered under the induced subg...

conjecture | open | MathDB 页面 | 原文线索

V2-063294 | MATHDB-332094

Conjecture on generalised cycles and H-graphs

摘录 / EXCERPT: A hereditary graph class with finite distinguishing number is one whose distinguishing number, as defined in the source, is finite. The induced subgraph relation is the ordering by...

conjecture | open | MathDB 页面 | 原文线索

V2-063295 | MATHDB-332102

The orbit-by-orbit genomic tableau inequality

摘录 / EXCERPT: Let $\mathcal{O} \subseteq \mathrm{LR}(\alpha,\scalebox{.5}{\yng(1)},\beta, \gamma)$ be an orbit of ω. Let $K_1(\mathcal{O})$ and $K_2(\mathcal{O})$ denote the sets of genom...

conjecture | open | MathDB 页面 | 原文线索

V2-063296 | MATHDB-332105

Scale-free unique continuation for arbitrary compact energy intervals

摘录 / EXCERPT: Let $d\geq 1$, let $E_0,G>0$, let $\delta\in(0,G/2)$, and let (G,δ)-equidistributed sequences define the observation set S_δ. Let $V:\mathbb{R}^d\to\mathbb{R}$ be mea...

conjecture | open | MathDB 页面 | 原文线索

V2-063297 | MATHDB-332111

Widom-Rowlinson partition-function conjecture for distinct activities

摘录 / EXCERPT: Distinct-activity partition-function conjecture. For any $\lambda_1,\lambda_2>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-063298 | MATHDB-332117

Generator matrix conjecture for quasi-cyclic product codes in Pre-RGB/POT form

摘录 / EXCERPT: Generator matrix conjecture. A generator matrix in $\mathbb{F}_q[X]^{\ell\times\ell}$, in Pre-RGB/POT form, for the $[n=\ell m,k_Ak_B,d_Ad_B]_q$ ℓ-quasi-cyclic product code...

conjecture | open | MathDB 页面 | 原文线索

V2-063299 | MATHDB-332119

The alternating-group generation conjecture for Toffoli gates

摘录 / EXCERPT: Let A be a finite alphabet of even order, let $n\geq3$, let S_A be the symmetric group on A, and let $TG(i,\alpha,o)$ denote the i-ary Toffoli gate with permutation...

conjecture | open | MathDB 页面 | 原文线索

V2-063300 | MATHDB-332122

Iwasawa--Greenberg main conjecture for the anticyclotomic Selmer group

摘录 / EXCERPT: Let Σ be a finite set of finite places of $\mathcal K$ not dividing p, let $X_{\mathrm{ac}}^\Sigma(M)$ be the Pontryagin dual of the corresponding anticyclotomic Selmer gr...

conjecture | open | MathDB 页面 | 原文线索

V2-063301 | MATHDB-332124

Flag upper bound conjecture for Eulerian complexes

摘录 / EXCERPT: Flag Eulerian upper bound conjecture. For every $1\leq i\leq 2m-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-063302 | MATHDB-332136

The pair-approximation lower-bound conjecture for the evolving voter model

摘录 / EXCERPT: In the rewire-to-random evolving voter model, let $\nu_c(p)$ be the true critical value separating rapid disconnection from prolonged persistence for initial density p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063303 | MATHDB-332141

Integrality conjecture for generalized Macdonald integral-form coefficients

摘录 / EXCERPT: Let $|K_{\vec{\lambda}}\rangle$ and $\langle K_{\vec{\lambda}}|$ be the integral-form generalized Macdonald vectors, expanded in the ordered basis with coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-063304 | MATHDB-332142

Norm conjecture for generalized Macdonald integral forms

摘录 / EXCERPT: Let N be a positive integer, let $\vec{u}=(u_1,\ldots,u_N)$, and let $\vec{\lambda}=(\lambda^{(1)},\ldots,\lambda^{(N)})$ be an N-tuple of partitions. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-063305 | MATHDB-332149

Conjecture on the rightmost entries of the new Catalan triangle

摘录 / EXCERPT: Rightmost-entry conjecture. The rightmost numbers in rows numbered $2+3i$, for $i=0,1,\ldots$, are the Catalan numbers, and the rightmost numbers in rows numbered $1+3i$, for...

conjecture | open | MathDB 页面 | 原文线索

V2-063306 | MATHDB-332155

Existence conjecture for singular Kerr waves from compactly supported data

摘录 / EXCERPT: Let Σ_0 be a complete 2-ended asymptotically flat Cauchy hypersurface in a Kerr spacetime with $0<|a|<M$, and let ψ solve the linear wave equation ... The relevant assu...

conjecture | open | MathDB 页面 | 原文线索

V2-063307 | MATHDB-332156

Order-n oscillation conjecture for convex domains

摘录 / EXCERPT: Let $K\Subset\mathbb{C}$ be a compact convex domain with nonempty interior, let ${\mathcal P}_n(K)$ denote the polynomials of degree at most n on K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063308 | MATHDB-332158

The NP-hard criterion for canonical dual optimization

摘录 / EXCERPT: Let $({\cal P})$ be the primal nonconvex optimization problem and let $({\cal P}^d)$ be its canonical dual problem. The canonical dual problem is called solvable when it has a solu...

conjecture | open | MathDB 页面 | 原文线索

V2-063309 | MATHDB-332163

Triangulation conjecture for continued-fraction square-torus packings

摘录 / EXCERPT: Let N equal disks be packed in a square torus. A packing is said to correspond to a convergent of $1/\sqrt{3}$ or $2-\sqrt{3}$ when it is one of the Type I or Type II packings, r...

conjecture | open | MathDB 页面 | 原文线索

V2-063310 | MATHDB-332164

Uniform Markov-constant bound for square-torus disk packings

摘录 / EXCERPT: Square-torus Markov-bound conjecture. Every such packing satisfies $M\leq 6.25$. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-063311 | MATHDB-332166

Stability conjecture for high-Markov-constant square-torus packings

摘录 / EXCERPT: For each integer $N\geq 2$, let $\epsilon(N)$ be a positive quantity with ... as N tends to infinity. Let M be the Markov constant of a packing of N equal disks in a square tor...

conjecture | open | MathDB 页面 | 原文线索

V2-063312 | MATHDB-332167

Geometric boundary characterization by rank loci

摘录 / EXCERPT: Let M be a matrix represented as a slack matrix $M=S_{P,Q}$ of nested polytopes $P\subseteq Q\subseteq\mathbb R^{r-1}$. A spectrahedral shadow C of size k between P and Q...

conjecture | open | MathDB 页面 | 原文线索

V2-063313 | MATHDB-332169

Spectrahedron characterization of psd rank for normalized matrices

摘录 / EXCERPT: Let $M\in\mathbb R^{p\times q}_{\geq 0}$ have rank $k+1$ and satisfy ... Let P and Q be the two polytopes corresponding to M in the nested-polytope representation. Spectrahedro...

conjecture | open | MathDB 页面 | 原文线索

V2-063314 | MATHDB-332186

Kurihara-Pollack's greatest-common-divisor conjecture for supersingular p-adic L-functions

摘录 / EXCERPT: Kurihara-Pollack conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063315 | MATHDB-332188

Cyclotomic BSD conjecture for analytic rank jumps

摘录 / EXCERPT: Cyclotomic BSD conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063316 | MATHDB-332191

The conjecture that a gamma-function expression is logarithmically completely monotonic

摘录 / EXCERPT: Logarithmic complete monotonicity conjecture. The function q is logarithmically completely monotonic on $(0,\infty)$; that is,

conjecture | open | MathDB 页面 | 原文线索

V2-063317 | MATHDB-332197

The universal maximum conjecture for log-correlated fields

摘录 / EXCERPT: Universal maximum conjecture. The maximum satisfies, for some constant C and a random variable Z called the derivative martingale,

conjecture | open | MathDB 页面 | 原文线索

V2-063318 | MATHDB-332198

Ivanov's asphericity conjecture

摘录 / EXCERPT: Ivanov's asphericity conjecture. Then $\mathcal{Q}$ is aspherical.

conjecture | refuted | MathDB 页面 | 原文线索

V2-063319 | MATHDB-332204

Monotonicity of distinguishing total chromatic number for the (8)-condition

摘录 / EXCERPT: Let G be a simple graph and let H be a proper subgraph of G. Write $\chi \, ”_{(8)}(G)$ for the minimum number of colors in an (8)-distinguishing total coloring. Monotonici...

conjecture | open | MathDB 页面 | 原文线索

V2-063320 | MATHDB-332205

The supporting-functional inequality for the binary symmetric key-interaction conjecture

摘录 / EXCERPT: Supporting-functional inequality.

conjecture | open | MathDB 页面 | 原文线索

V2-063321 | MATHDB-332208

The multiple-harmonic-sum analogue of the Kontsevich-Zagier period conjecture

摘录 / EXCERPT: Fix $N>1$ and work on $\mathbb{P}^{1}-\{0,\mu_N,\infty\}$. Let p range over primes not dividing N, let $\alpha\in\mathbb{N}$, and let $\operatorname{har}_{p^{\alpha}}$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-063322 | MATHDB-332217

Universal-theory Herbrand consistency conjecture

摘录 / EXCERPT: Let $T\in\mathcal T$ be a theory axiomatized by a universal sentence, and let $HCS(T)$ denote its Herbrand Consistency Search problem. Universal-theory Herbrand consistency conject...

conjecture | open | MathDB 页面 | 原文线索

V2-063323 | MATHDB-332223

Goulden-Jackson's b-conjecture for hypermap coefficients

摘录 / EXCERPT: Let τ, μ, and ν be partitions of $n\geq 1$. The coefficient $h_{\mu,\nu}^{\tau}(\beta)$ is defined through the power-sum expansion of the connected Jack three-point se...

conjecture | open | MathDB 页面 | 原文线索

V2-063324 | MATHDB-332224

Macdonald strong factorization and cumulant conjecture

摘录 / EXCERPT: Let $\lambda^1,\dots,\lambda^r$ be partitions. For $I\subseteq [r]$, let λ^I denote the combined partition used in the Macdonald polynomial $J_{\lambda^I}^{(q,t)}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063325 | MATHDB-332227

The sharp lower-bound partition conjecture for split special orthogonal groups

摘录 / EXCERPT: Let ${\mathrm {G}}_n$ be an F-split ${\mathrm{SO}}_{2n+1}$ or ${\mathrm{SO}}_{2n}$, and let $\frak{p}(\pi)$ denote the partitions attached to a cuspidal automorphic representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-063326 | MATHDB-332228

Gutman-Mizoguchi real-rootedness conjecture for beta-polynomials

摘录 / EXCERPT: Let C be a circuit contained in a graph G. The matching polynomial is denoted by $\alpha(G,x)$, and the two circuit characteristic, or beta-polynomials, considered here are ... a...

conjecture | open | MathDB 页面 | 原文线索

V2-063327 | MATHDB-332232

Unique-limit conjecture for the smooth optimal transport gradient flow

摘录 / EXCERPT: Unique-limit conjecture. The flow has a unique limit in the set of strictly convex functions.

conjecture | open | MathDB 页面 | 原文线索

V2-063328 | MATHDB-332237

Schmidt's intersection conjecture for badly approximable sets

摘录 / EXCERPT: Schmidt's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063329 | MATHDB-332238

Winning criterion for inhomogeneous badly approximable sets on lines

摘录 / EXCERPT: Winning criterion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063330 | MATHDB-332241

Chen's complete-monotonicity remainder conjecture for Bernoulli polynomials

摘录 / EXCERPT: Chen's conjecture. For every $t>0$ and $m\in\mathbb{N}_0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-063331 | MATHDB-332247

Counting conjecture for S_{d+1}-fields with local conditions

摘录 / EXCERPT: Fix $d+1\geq3$ and a signature (r_1,r_2). Let $L(X)^{r_2}$ be the set of S_{d+1}-fields K with $X/2<|D_K|<X$, and let $L(X;\mathcal S)^{r_2}$ be the subset satisfying finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-063332 | MATHDB-332257

Optimality of the pre-coding for broadcast channels with receiver message side information

摘录 / EXCERPT: Consider a two-receiver memoryless broadcast channel with states and receiver message side information (RMSI), together with the corresponding channel without RMSI. Let a transmiss...

conjecture | open | MathDB 页面 | 原文线索

V2-063333 | MATHDB-332258

Pick-function conjecture for normalized multiple Gamma functions

摘录 / EXCERPT: Pick-function conjecture. For every $n\in\mathbb{N}$, f_n is a Pick function with Stieltjes representation

conjecture | open | MathDB 页面 | 原文线索

V2-063334 | MATHDB-332271

Strong equitable Brooks conjecture

摘录 / EXCERPT: Let G be a graph. A graph is k-equitable if $|G|=ks$, $\chi(G)\le k$, and every proper k-coloring of G has s vertices in each color class. Write H_1+H_2 for the disjoin...

conjecture | open | MathDB 页面 | 原文线索

V2-063335 | MATHDB-332276

The weighted higher-order discrepancy lower-bound conjecture

摘录 / EXCERPT: Let $D_r^w(m,d)_\infty$ denote the minimal weighted r-discrepancy in dimension d, where m is the number of nodes. Weighted higher-order discrepancy lower-bound conjecture. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-063336 | MATHDB-332280

Li-Magnant proper connection conjecture

摘录 / EXCERPT: Let G be a connected noncomplete graph with n vertices, where $n\geq 5$, and let $\delta(G)$ denote its minimum degree. Li-Magnant's proper connection conjecture. Then ... This c...

conjecture | open | MathDB 页面 | 原文线索

V2-063337 | MATHDB-332285

q-Delannoy positivity conjecture

摘录 / EXCERPT: For positive integers m,n,r, let $D_q(m,k)$ and $D_{q^{-1}}(m,k)$ be the natural q-Delannoy numbers defined by ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-063338 | MATHDB-332286

Integer-valued Delannoy-polynomial power conjecture

摘录 / EXCERPT: Let $d_k(x)$ and $s_k(x)$ be the two Delannoy-related polynomials considered in the paper. Integer-valued Delannoy-polynomial power conjecture. For all positive integers m and...

conjecture | open | MathDB 页面 | 原文线索

V2-063339 | MATHDB-332287

Higher-order Poisson-Charlier divergence lower-bound conjecture

摘录 / EXCERPT: Let X be a random variable with values in $\mathbb{N}_{0}$, let λ be the Poisson parameter, and let C_{k}^{λ} denote the Poisson-Charlier polynomial of order...

conjecture | open | MathDB 页面 | 原文线索

V2-063340 | MATHDB-332293

Bodirsky-Pinsker tractability conjecture for reducts of finitely bounded homogeneous structures

摘录 / EXCERPT: Bodirsky-Pinsker tractability conjecture. If the model-complete core of $fmathbb A$ has no such expansion $fmathbb C$ for which $\operatorname{Pol}(fmathbb C)$ has a continuous clo...

conjecture | open | MathDB 页面 | 原文线索

V2-063341 | MATHDB-332299

Transitive-code capacity conjecture for the binary erasure channel

摘录 / EXCERPT: Transitive-code capacity conjecture. Under these assumptions, the sequence achieves capacity on the binary erasure channel under bit-MAP decoding. This conjecture proposes that tra...

conjecture | open | MathDB 页面 | 原文线索

V2-063342 | MATHDB-332303

Conjectured properties of the Hankel determinants of the Prouhet-Thue-Morse sequence

摘录 / EXCERPT: Conjectured properties of the Hankel determinants. The following properties hold:

conjecture | open | MathDB 页面 | 原文线索

V2-063343 | MATHDB-332311

The low-degree automorphic quotient conjecture

摘录 / EXCERPT: Let π_1 and π_2 be non-isomorphic unitary cuspidal automorphic representations of $\operatorname{GL}_{d_1}(\mathbb{A}_{\mathbb{Q}})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-063344 | MATHDB-332312

The Scott map image growth conjecture for polygon tilings

摘录 / EXCERPT: Scott map image growth conjecture. The ratio of successive entries in the size of the image of the Scott map is smaller than the ratio of successive sizes of the corresponding perm...

conjecture | open | MathDB 页面 | 原文线索

V2-063345 | MATHDB-332314

The optimal-size clique transversal conjecture for 4-chordal graphs

摘录 / EXCERPT: Optimal-size clique transversal conjecture. Every 4-chordal graph with n vertices has a clique transversal of size at most $2n/7$.

conjecture | open | MathDB 页面 | 原文线索

V2-063346 | MATHDB-332320

Total positivity conjecture for the Narayana triangle

摘录 / EXCERPT: Total positivity conjecture. The Narayana triangle N is totally positive: every minor of N is nonnegative.

conjecture | open | MathDB 页面 | 原文线索

V2-063347 | MATHDB-332322

Lidman-Vafaee satellite L-space knot necessity conjecture

摘录 / EXCERPT: Let P be a pattern in the solid torus, let K be a knot, let U denote the unknot, and write $P(K)$ and $P(U)$ for the corresponding satellite knots. An L-space knot is a knot...

conjecture | open | MathDB 页面 | 原文线索

V2-063348 | MATHDB-332323

Precise satellite L-space knot characterization conjecture

摘录 / EXCERPT: Let P be a pattern in the solid torus, let K be a knot of genus g, let U denote the unknot, and write $P(U,-n)$ for the pattern knot after $-n$ full twists. Satellite L-spa...

conjecture | open | MathDB 页面 | 原文线索

V2-063349 | MATHDB-332324

Exponential propagation conjecture for fast doubly nonlinear diffusion

摘录 / EXCERPT: Fast-diffusion exponential propagation conjecture. For every $\sigma<\sigma_{\ast}$,

conjecture | open | MathDB 页面 | 原文线索

V2-063350 | MATHDB-332333

Loehr-Warrington Square Paths Conjecture

摘录 / EXCERPT: Loehr-Warrington's Square Paths Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063351 | MATHDB-332335

The ultimate Elekes-Rónyai theorem for rational functions

摘录 / EXCERPT: Let $f\in\mathbb C(x,y)$ be a rational function of degree d, where the degree is the maximum of the degrees of its numerator and denominator after removing common factors. Call...

conjecture | open | MathDB 页面 | 原文线索

V2-063352 | MATHDB-332340

Capacity characterization conjecture for general layered erasure channels

摘录 / EXCERPT: A general layered erasure channel has layers indexed by $l\in\{1,\dots,q\}$, and let Theorem 1 denote the theorem giving an outer bound on its capacity region. The moderate-inter...

conjecture | open | MathDB 页面 | 原文线索

V2-063353 | MATHDB-332348

The type-2 alignment-set characterization of rate-1/3 feasibility

摘录 / EXCERPT: An index coding problem consists of messages, receivers, and side-information constraints; the paper organizes its messages into alignment sets, with type-2 alignment sets subject...

conjecture | open | MathDB 页面 | 原文线索

V2-063354 | MATHDB-332352

The cubic graph upper-bound conjecture for rainbow-free P_4 colorings

摘录 / EXCERPT: Cubic graph upper-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063355 | MATHDB-332356

Amdeberhan's Fibonacci conjecture for core partitions into distinct parts

摘录 / EXCERPT: Let an (s,t)-core partition be a partition that is simultaneously s-core and t-core, and let F_n denote the nth Fibonacci number. Amdeberhan's conjecture. The number of...

conjecture | open | MathDB 页面 | 原文线索

V2-063356 | MATHDB-332364

The linking-transporter extension conjecture for almost simple fusion systems

摘录 / EXCERPT: Let $\calf$ be an almost simple fusion system over a p-group S. More generally, assume only that $O_p(\calf)=1$. Set $\cale=F^*(\calf)$; thus $\cale$ is simple, or a product of...

conjecture | open | MathDB 页面 | 原文线索

V2-063357 | MATHDB-332365

The automorphism-extension conjecture for models of normalizers in fusion systems

摘录 / EXCERPT: Let $\calf$ be a saturated fusion system over a p-group S such that $O_p(\calf)=1$. Set $\cale=F^*(\calf)$, a fusion subsystem over $T\nsg S$, and let $\call$ be a centric link...

conjecture | open | MathDB 页面 | 原文线索

V2-063358 | MATHDB-332366

Efficient linear-combination representation of generalized Boolean functions

摘录 / EXCERPT: Linear-combination conjecture. For $f\in\mathcal{GB}^q_n$, the function $\zeta^{f(x)}$ can be represented as

conjecture | open | MathDB 页面 | 原文线索

V2-063359 | MATHDB-332377

Chen-Cheung's generalized Wilker inequality conjecture

摘录 / EXCERPT: Chen-Cheung's generalized Wilker inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063360 | MATHDB-332378

Chen-Cheung's higher-order Wilker inequality conjecture

摘录 / EXCERPT: Chen-Cheung's higher-order Wilker inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063361 | MATHDB-332379

The recurrence conjecture for the q-Lommel polynomials

摘录 / EXCERPT: Let $r_n^{(\nu)}(x)$ be the polynomials introduced by the preceding recurrence, and let $s_n^{(\nu)}(x)$, $n\geqslant 0$, be the sequence uniquely defined by ... with initial values...

conjecture | open | MathDB 页面 | 原文线索

V2-063362 | MATHDB-332380

Conjecture on real-rootedness of the derived Boros-Moll polynomial Q_n

摘录 / EXCERPT: Real-rootedness conjecture for Q_n. For any $n\geq 1$, the polynomial $Q_n(x)$ has only real zeros. This conjecture is motivated by Brändén's theorem that the operator sending co...

conjecture | open | MathDB 页面 | 原文线索

V2-063363 | MATHDB-332382

Hard-edge universality conjecture for products of coupled non-Gaussian matrices

摘录 / EXCERPT: Non-Gaussian hard-edge universality conjecture. Under assumptions (C1)--(C4), the conclusion of Theorem (hardlimits) should still hold, with $\kappa=0$. This conjecture extends the...

conjecture | open | MathDB 页面 | 原文线索

V2-063364 | MATHDB-332389

The strong rationality characterization for permutation classes

摘录 / EXCERPT: Strong rationality conjecture. A permutation class is strongly rational if and only if it is well-quasi-ordered and does not contain the class of 312-avoiding permutations or any...

conjecture | open | MathDB 页面 | 原文线索

V2-063365 | MATHDB-332397

The minimum-length conjecture for odd tournaments and circulant tournaments

摘录 / EXCERPT: The minimum-length conjecture. For every odd $n\geq 3$, every $r\in[n-1]$, and every $T\in\mathrm{Tour}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-063366 | MATHDB-332398

The Bias Conjecture for second moments in one-parameter elliptic-curve families

摘录 / EXCERPT: Bias Conjecture. For any such family, the largest lower-order term in the second moment that does not average to 0 is on average negative; equivalently, the first $\beta_r(p)$ te...

conjecture | open | MathDB 页面 | 原文线索

V2-063367 | MATHDB-332401

The Beauville-Laszlo uniformization conjecture for G-bundles

摘录 / EXCERPT: Let E be the local field in the paper, let Gr be the corresponding affine Grassmannian, and let ... be the Beauville-Laszlo morphism. Beauville-Laszlo uniformizati...

conjecture | open | MathDB 页面 | 原文线索

V2-063368 | MATHDB-332403

The local geometric Langlands conjecture for discrete parameters

摘录 / EXCERPT: Let G be quasi-split over E, let $\Gamma=\operatorname{Gal}(\overline E/E)$, and let $\varphi:W_E\to{}^LG$ be a discrete Langlands parameter. Let ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063369 | MATHDB-332405

Conjecture on regular holonomy and Lie structures on infinitesimal holonomy groups

摘录 / EXCERPT: Let $\cal F$ be a Riemannian foliation, let H_p denote the holonomy group at p, and let $\mathcal V_p$ be the corresponding vertical space. The inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-063370 | MATHDB-332406

Conjecture on bounded holonomy and compact regular holonomy

摘录 / EXCERPT: Let $\cal F$ be a Riemannian foliation with bounded holonomy, and suppose that no leaf has infinite fundamental group. Let H_p denote the holonomy group at p. Bounded-holonomy...

conjecture | open | MathDB 页面 | 原文线索

V2-063371 | MATHDB-332413

Gross-Prasad conjecture for orthogonal groups

摘录 / EXCERPT: Let V_{m+1} be an orthogonal space of dimension $m+1$, and let V_m be a non-degenerate codimension-one subspace. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-063372 | MATHDB-332418

Hyponormality conjecture for the generalized Cesàro matrix of order two

摘录 / EXCERPT: Let $M\in\mathcal{B}(\ell^2)$ be the generalized Cesàro matrix of order two with parameter α. Hyponormality conjecture. The operator M is also hyponormal for...

conjecture | open | MathDB 页面 | 原文线索

V2-063373 | MATHDB-332423

Sufficiently general arcs have minimal valuation above the Nagata threshold

摘录 / EXCERPT: Minimal valuation conjecture. If C is sufficiently general and $s\geqslant 8+\frac{1}{36}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-063374 | MATHDB-332425

Segre's version of the SHGH conjecture for general fat points

摘录 / EXCERPT: Segre's version of the SHGH conjecture. For $X=m_1P_1+\cdots+m_rP_r$ with general points P_i, X fails to impose the expected number of conditions on V only if $[I_X]_j\ne0$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-063375 | MATHDB-332436

A Gaussian auxiliary-variable conjecture for the strengthened entropy power inequality

摘录 / EXCERPT: Auxiliary-variable conjecture. There exists a random variable V such that $Y_1\to Y_0\to V$ and

conjecture | open | MathDB 页面 | 原文线索

V2-063376 | MATHDB-332450

Abrikosov's triangular-lattice conjecture for beta-exponent planar zero packing

摘录 / EXCERPT: For a positive real β, let $\rho_\beta(\mathbb{C})$ denote the β-exponent minimal discrepancy density for planar zero packing, defined by ... where the infimum is over al...

conjecture | open | MathDB 页面 | 原文线索

V2-063377 | MATHDB-332452

Convergence of compact-surface beta-zero packing to the planar density

摘录 / EXCERPT: Let $\mathcal{S}$ be a compact Riemann surface, let $U(z,w)$ be its logarithmic monopole, and for a positive real β define ... where the infimum is over positive reals b and...

conjecture | open | MathDB 页面 | 原文线索

V2-063378 | MATHDB-332477

Generalized Hurwitz conjecture for zeros of Bessel function derivatives

摘录 / EXCERPT: Let $n\in\mathbb{N}_0$. For a real parameter ν, write $J_{\nu}^{(n)}$ for the nth derivative of the Bessel function of the first kind with respect to its argument. Generalize...

open_problem | open | MathDB 页面 | 原文线索

V2-063379 | MATHDB-332479

Monotonicity conjecture for positive zeros of Bessel function derivatives

摘录 / EXCERPT: For $n\in\mathbb{N}_0$ and $m\in\mathbb{N}$, let $j_{\nu,m}^{(n)}$ denote the mth positive zero of the nth derivative $J_{\nu}^{(n)}$ of the Bessel function of the first kind....

conjecture | open | MathDB 页面 | 原文线索

V2-063380 | MATHDB-332482

Infinite log-convexity of the Catalan-Larcombe-French transform sequence

摘录 / EXCERPT: Let $\{V_n\}_{n\geq 0}$ be the Fennessey-Larcombe-French sequence, and consider the sequence $\{V_n^2-V_{n-1}V_{n+1}\}_{n\geq 2}$. Infinite log-convexity conjecture. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-063381 | MATHDB-332483

Alternating log-concavity and log-convexity of iterated ratio sequences

摘录 / EXCERPT: Alternating ratio-sequence conjecture. For all integer $k\geq 1$, the sequence $\mathcal{R}^k(\{P_n\}_{n\geq 0})$, except for the first k terms at the beginning, is log-concave i...

conjecture | open | MathDB 页面 | 原文线索

V2-063382 | MATHDB-332484

Alternating log-convexity and log-concavity of iterated Fennessey-Larcombe-French ratios

摘录 / EXCERPT: Let $\mathcal{R}$ act on a sequence $\{S_n\}_{n\geq 0}$ by ... Let $\{V_n\}_{n\geq 1}$ be the Fennessey-Larcombe-French sequence. Alternating V_n ratio conjecture. For all integer...

conjecture | open | MathDB 页面 | 原文线索

V2-063383 | MATHDB-332497

The partition-function formula for the beta-gamma system on the cone of pure spinors

摘录 / EXCERPT: The partition-function formula. The partition function has the form

conjecture | open | MathDB 页面 | 原文线索

V2-063384 | MATHDB-332506

Extension principle for Floer field theories

摘录 / EXCERPT: Let $\mathcal{C}$ be a quilted 2-category whose underlying 1-category has Cerf decompositions. Let $\mathcal{F}$ be a Cerf-compatible partial functor from the objects and simpl...

conjecture | open | MathDB 页面 | 原文线索

V2-063385 | MATHDB-332507

Quilted Atiyah-Floer conjecture

摘录 / EXCERPT: Quilted Atiyah-Floer conjecture. The three extended Floer field theories induce isomorphic 2-functors

conjecture | open | MathDB 页面 | 原文线索

V2-063386 | MATHDB-332510

Critical energy conjecture for analytic quasi-periodic Jacobi operators

摘录 / EXCERPT: Critical energy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063387 | MATHDB-332512

Equidistribution of components on 5- and reversed 4-avoiding permutations

摘录 / EXCERPT: Equidistribution conjecture. The statistic “number of components” has the same distribution on Π_5-avoiders as on $\operatorname{Reverse}(\Pi_4)$-avoiders.

conjecture | open | MathDB 页面 | 原文线索

V2-063388 | MATHDB-332513

Nadjafi-Arani et al.'s Szeged-Wiener inequality for non-complete blocks

摘录 / EXCERPT: Let G be a connected graph and let $B_1,\ldots,B_k$ be all its non-complete blocks, of respective orders $n_1,\ldots,n_k$. Define the Szeged-Wiener difference by...

conjecture | open | MathDB 页面 | 原文线索

V2-063389 | MATHDB-332530

Bisztriczky's separation conjecture for totally sewn neighbourly 4-polytopes

摘录 / EXCERPT: Let $\mathbf{P}$ be a totally sewn neighbourly 4-polytope in $\mathbb{E}^4$, and let $\mathbf{o}$ be an interior point of $\mathbf{P}$. Bisztriczky's separation conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-063390 | MATHDB-332538

The large-parameter zero-measure conjecture for Fibonacci endomorphisms

摘录 / EXCERPT: Let f_c be the Fibonacci endomorphism of $\mathbb{C}^2$ with parameter c, and let $K^{+}(f_c)$ denote the set of points whose forward orbits under f_c remain bounded. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063391 | MATHDB-332539

Star-shapedness conjecture for the Hausdorff-dimension-greater-than-one locus

摘录 / EXCERPT: Let $\mathcal{M}$ be the Mandelbrot set, and write ... where J_c is the Julia set of the quadratic map with parameter c. Star-shapedness conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-063392 | MATHDB-332542

Equivalence conjecture for the two shellability conditions

摘录 / EXCERPT: Let W be a Weyl group, let $x\leq w$, and let $\mathfrak w$ be a reduced word for w. Let $\lambda_{x,\mathfrak w}$ be the deletion-index set, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063393 | MATHDB-332544

The Zindler-curve soliton conjecture

摘录 / EXCERPT: A Zindler curve is a closed planar curve that is in the bicycle correspondence with itself, meaning that ${\cal B}_{\ell}(\Gamma,\Gamma)$ holds for some fixed chord length ℓ....

conjecture | open | MathDB 页面 | 原文线索

V2-063394 | MATHDB-332545

OAMP state evolution conjecture

摘录 / EXCERPT: Let $\bm{A}$ be unitarily invariant, and let $\bm{W}_t$ be given by the specified linear estimator. Define ... with $v_0^2=\operatorname{E}\{X^2\}$, where Φ_t and Ψ_t are t...

conjecture | open | MathDB 页面 | 原文线索

V2-063395 | MATHDB-332546

Zhang's local spherical-character comparison conjecture

摘录 / EXCERPT: Let F be a number field, let E/F be a quadratic extension, and let v be a place of F. Let G and G' be the local groups in Zhang's comparison of relative trace formulas,...

conjecture | open | MathDB 页面 | 原文线索

V2-063396 | MATHDB-332548

Zhang's explicit local spherical-character comparison conjecture

摘录 / EXCERPT: Let E/F be a quadratic extension of local fields, let v be a place of F, and let $G=U(W)\times U(V)$ and G' be the corresponding local groups. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063397 | MATHDB-332560

Conjecture on logarithmic distances between Latin squares

摘录 / EXCERPT: Logarithmic distance conjecture. For each Latin square L of order n,

conjecture | open | MathDB 页面 | 原文线索

V2-063398 | MATHDB-332569

The q-Foulkes conjecture

摘录 / EXCERPT: For each positive integer n, let $H_n(\mathbf{x};q)$ be the Hall-Littlewood polynomial used in the paper, and for $0<a\leq b$ define ... Let $f_{a,b}=h_b[h_a]-h_a[h_b]$. The q-Foul...

conjecture | open | MathDB 页面 | 原文线索

V2-063399 | MATHDB-332570

The q-stability conjecture

摘录 / EXCERPT: For $0<a\leq b$, let $F_{a,b}(\mathbf{x};q)$ be the divided difference from the q-Foulkes conjecture, and let the overline operator remove the largest part from each Schur index....

conjecture | open | MathDB 页面 | 原文线索

V2-063400 | MATHDB-332571

The q-Doran conjecture

摘录 / EXCERPT: Let a and b be positive integers, set $n=ab$, let c be a divisor of n with $a\leq c\leq b$, and set $d=n/c$. Write $\preceq_s$ for Schur positivity. The q-Doran conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063401 | MATHDB-332572

The diagram q-Foulkes equivalence conjecture

摘录 / EXCERPT: Let $\alpha,\beta,\gamma,\delta$ be partitions such that ... Write $\prec_s$ for strict Schur order and let S_ρ denote the corresponding q-analogue used in the paper. The diag...

conjecture | open | MathDB 页面 | 原文线索

V2-063402 | MATHDB-332574

Failure of the one-half-Hölder maximal regularity implication

摘录 / EXCERPT: Let $\mathfrak{A}: [0,T] \to \mathcal{L}(V,V^*)$ be the operator associated with a non-autonomous form $\mathfrak{a}$, and suppose that ... The one-half-Hölder maximal regularity cou...

conjecture | open | MathDB 页面 | 原文线索

V2-063403 | MATHDB-332579

The average-degree conjecture for incomplete online k-list-critical graphs

摘录 / EXCERPT: An incomplete (online) k-list-critical graph is a graph satisfying the incomplete online k-list-criticality condition described in the source. Average-degree conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-063404 | MATHDB-332580

The inner-representation type conjecture for connected quandles

摘录 / EXCERPT: Inner-representation type conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063405 | MATHDB-332581

Conjecture on Bernstein bounds and dispersion estimates for the discrete Laguerre operator

摘录 / EXCERPT: Let ${\bf g}_n^{(\alpha,\beta)}$ be the function introduced earlier in the paper, and let H_α be the discrete Laguerre operator. The preceding estimates are stated for the r...

conjecture | open | MathDB 页面 | 原文线索

V2-063406 | MATHDB-332593

The affine stability-space conjecture for acyclic quivers

摘录 / EXCERPT: Let $0<\varepsilon<1$ and let Q be any acyclic quiver. The stability space $\operatorname{Stab}(D^b(Q))$ is affine if it is of the form $\mathbb{C}^n$. Affine stability-space con...

conjecture | open | MathDB 页面 | 原文线索

V2-063407 | MATHDB-332594

The intermediate norm conjecture for the quivers S_1, S_2 and S_3

摘录 / EXCERPT: Let S_1, S_2 and S_3 be the three quivers displayed in the source, and let $0<\varepsilon<1$. Intermediate norm conjecture. For each $i=1,2,3$, ... These quivers are examples f...

conjecture | open | MathDB 页面 | 原文线索

V2-063408 | MATHDB-332595

Rationality conjecture for residues of 2-flags of holomorphic foliations

摘录 / EXCERPT: Rationality conjecture. If the polynomials φ_i have rational coefficients, then

conjecture | open | MathDB 页面 | 原文线索

V2-063409 | MATHDB-332596

Beck-Stapledon conjecture on the real-rootedness of dilated Ehrhart h^*-polynomials

摘录 / EXCERPT: Let P be a d-dimensional lattice polytope, and let $h^*(rP)$ denote the h^*-polynomial of its dilation rP. Beck-Stapledon conjecture. The polynomial $h^*(rP)$ has only dist...

conjecture | open | MathDB 页面 | 原文线索

V2-063410 | MATHDB-332603

Infinite graph extension of the tree decomposition conjecture

摘录 / EXCERPT: Let T be a tree, and let a T-decomposition of a graph G mean a partition of the edge-set of G into subgraphs isomorphic to T. Canonical infinite tree decomposition conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-063411 | MATHDB-332611

Everywhere eventual stability conjecture

摘录 / EXCERPT: Let K be a field, let $phi\in K(z)$ have degree $d\geq 2$, and let $\alpha\in\mathbb{P}^1(K)$ be not periodic under ϕ. For a point $\alpha\ne\infty$, write (ϕ,α) a...

conjecture | open | MathDB 页面 | 原文线索

V2-063412 | MATHDB-332612

Eventual stability conjecture for reciprocal quadratic parameters

摘录 / EXCERPT: Let $a\in\mathbb{Z}$ with $a\notin\{0,-1\}$. The rational map $\phi(z)=z^2+1/a$ has degree two, and a pair (ϕ,α) is eventually stable when the number of irreducible facto...

conjecture | open | MathDB 页面 | 原文线索

V2-063413 | MATHDB-332616

O'Grady's conjecture on second Chern classes of sheaves on K3 surfaces

摘录 / EXCERPT: Let X be a projective K3 surface, let H be a polarization, and let $M_H(v)$ be the moduli space of sheaves with Mukai vector v. Let $M_H(v)^{ss}$ denote the moduli space of...

conjecture | open | MathDB 页面 | 原文线索

V2-063414 | MATHDB-332617

The sum-integral conjecture for the gamma-function logarithm

摘录 / EXCERPT: Let z_m be the sequence of complex numbers used in the preceding gamma-function expression, and let $A\in\mathbb{N}$. The logarithm is taken on the branch denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-063415 | MATHDB-332619

The K_{n+5} and $K_{3^2,1^n}$ families conjecture for n-apex obstructions

摘录 / EXCERPT: A graph is n-apex if deleting n or fewer vertices results in a planar graph. A graph is a minor of another graph if it can be obtained by contracting edges in a subgraph. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063416 | MATHDB-332630

Frankl's conjecture for reduced intersection-closed matrices

摘录 / EXCERPT: Let $F \in \mathbb{Q}^{m \times n}$ be a reduced intersection-closed matrix, and let e_k be the k-th unit coordinate vector. Write ΣX for the sum of all entries of a ma...

conjecture | open | MathDB 页面 | 原文线索

V2-063417 | MATHDB-332634

The relations conjecture for the generators of the pointed genus-one moduli ring

摘录 / EXCERPT: Let $n\geq 3$, and work over $\mathbb{Z}[1/6]$. For a fixed index i, let $\widetilde{{\cal U}}_{1,n}(i)$ be the corresponding moduli space, with coordinate-ring generators...

conjecture | open | MathDB 页面 | 原文线索

V2-063418 | MATHDB-332648

Effective determination of quintic FJRW invariants from localization relations

摘录 / EXCERPT: Let $\Theta_{g,k}$ denote the FJRW invariants for the quintic indexed by genus g and k, and let the relations ... . Effective determination conjecture. The equations ... , effectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-063419 | MATHDB-332655

Asymptotic almost periodicity conjecture for Widom-Hilbert factors

摘录 / EXCERPT: Let $K(\gamma)$ be the Cantor-type set associated with γ, let $\mu_{K(\gamma)}$ be its associated measure, and let $W_n^2(\mu_{K(\gamma)})$ be the corresponding Widom-Hilber...

conjecture | open | MathDB 页面 | 原文线索

V2-063420 | MATHDB-332670

Conjecture on cycles in the expression graph of a Coxeter group

摘录 / EXCERPT: Expression-graph conjecture. For every $w\in W$, the theorem of Bergeron, Ceballos and Labbé concerning cycles in $\mathcal{R}(w)$ remains valid when $\mathcal{R}(w)$ is replaced b...

conjecture | open | MathDB 页面 | 原文线索

V2-063421 | MATHDB-332673

Hurwitzness of A^{-} implies local stability in the on-demand agent system

摘录 / EXCERPT: Consider the fluid-limit dynamics of the on-demand agent service system, represented by switched linear systems with system matrices A^{-} and A^{+}. A matrix is Hurwitz when a...

conjecture | open | MathDB 页面 | 原文线索

V2-063422 | MATHDB-332674

Parameter-dependent local stability when A^{-} is not Hurwitz

摘录 / EXCERPT: Consider the fluid-limit dynamics of the on-demand agent service system with system matrix A^{-}. Non-Hurwitz parameter-dependence conjecture. If A^{-} is not Hurwitz, the syst...

conjecture | open | MathDB 页面 | 原文线索

V2-063423 | MATHDB-332687

Natural deformation conjecture for Jack-character structure coefficients

摘录 / EXCERPT: Let cS_n be the symmetric group and let $cdelta$ be the Jack deformation parameter. The structure coefficients $g^\mu_{\pi,\sigma}$ are defined by ... At $cdelta=0$, these coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-063424 | MATHDB-332688

Lassalle's Kerov-Lassalle positivity conjecture for single Jack characters

摘录 / EXCERPT: Let $cgamma=-\delta$, and let $cR_2,cR_3,cdots$ denote the free cumulants, viewed as functions on Young diagrams. For each partition cpi, the Jack character...

conjecture | open | MathDB 页面 | 原文线索

V2-063425 | MATHDB-332689

Lassalle's Kerov-Lassalle positivity conjecture for cumulants

摘录 / EXCERPT: Kerov-Lassalle cumulant-positivity conjecture. For all integers $k_1,\dots,k_\ell\geq 1$, the polynomial which expresses this signed cumulant in terms of...

conjecture | open | MathDB 页面 | 原文线索

V2-063426 | MATHDB-332690

Positivity conjecture for cumulant coefficients of Jack characters

摘录 / EXCERPT: Cumulant-coefficient positivity conjecture. For partitions $cpi_1,\dots,cpi_\ell$, the polynomial $(-1)^{\ell-1}d^\sigma_{\pi_1,\dots,\pi_\ell}(cdelta)$ has non-negative integer co...

conjecture | open | MathDB 页面 | 原文线索

V2-063427 | MATHDB-332691

The non-occurrence of $zeta(2)$ as a phi^4 period

摘录 / EXCERPT: Let $\mathcal{P}_{\phi^4}$ denote the space of ϕ^4 periods and let $\zeta(2)$ be the Riemann zeta value at 2. Non-occurrence conjecture. ... No ϕ^4 graph is known to eval...

conjecture | open | MathDB 页面 | 原文线索

V2-063428 | MATHDB-332692

The period-c_2 invariance conjecture for primitive log-divergent graphs

摘录 / EXCERPT: Let G be a connected graph with at least three vertices, let $P(G)$ denote its period, and let $c_2(G)$ denote its c_2-invariant. Period-c_2 invariance conjecture. For two pr...

conjecture | open | MathDB 页面 | 原文线索

V2-063429 | MATHDB-332693

The ancestor weight-drop conjecture for phi^4 periods

摘录 / EXCERPT: Let G be a primitive log-divergent ϕ^4 graph with loop number h_G and completion $\overline{G}$, and suppose $P(G)\in\mathcal{Z}_N[\xi_N]$ for some N. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063430 | MATHDB-332694

The $c_2=-1$ multiple-zeta-value conjecture

摘录 / EXCERPT: Let G be a primitive log-divergent graph with loop number h_G and c_2-invariant -1. $c_2=-1$ MZV conjecture. Its period is a multiple zeta value of weight $2h_G-3$: ... The p...

conjecture | open | MathDB 页面 | 原文线索

V2-063431 | MATHDB-332695

The refined weight-mixing conjecture for $c_2=0$ phi^4 graphs

摘录 / EXCERPT: Let G be a primitive log-divergent ϕ^4 graph with loop number h_G and $c_2(G)=0$. Suppose $P(G)$ is mixed Tate. Refined weight-mixing conjecture. Either $P(G)$ has pure we...

conjecture | open | MathDB 页面 | 原文线索

V2-063432 | MATHDB-332697

The $c_2=-z_3$ period conjecture

摘录 / EXCERPT: Let G be a primitive log-divergent graph with loop number h_G and $c_2(G)=-z_3$. $c_2=-z_3$ conjecture. ... The three known examples lie in the indicated space, and the source ob...

conjecture | open | MathDB 页面 | 原文线索

V2-063433 | MATHDB-332699

The weight decomposition conjecture for L_{10}

摘录 / EXCERPT: Let $L_{10}=P_{10,425}$ and let $\mathcal{W}_{12}$ denote the weight-12 component. L_{10} weight decomposition conjecture. The period L_{10} mixes weights 12 through 14, an...

conjecture | open | MathDB 页面 | 原文线索

V2-063434 | MATHDB-332704

Darboux classification conjecture for exceptional orthogonal polynomial systems

摘录 / EXCERPT: Gómez-Ullate-Kamran-Milson conjecture. Every exceptional orthogonal polynomial system of codimension m can be obtained by applying a sequence of at most m Darboux transformatio...

conjecture | open | MathDB 页面 | 原文线索

V2-063435 | MATHDB-332714

Cohn-Elkies sphere-packing conjecture in dimensions 2, 8, and 24

摘录 / EXCERPT: Let $n\in\{2,8,24\}$, and let f satisfy the hypotheses of the Cohn-Elkies linear-programming bound. Write r for the radius parameter in that bound. Cohn-Elkies sphere-packing c...

conjecture | solved | MathDB 页面 | 原文线索

V2-063436 | MATHDB-332717

E8 and Leech limiting-ratio conjecture

摘录 / EXCERPT: E8 and Leech limiting-ratio conjecture. In the E_8 case,

conjecture | open | MathDB 页面 | 原文线索

V2-063437 | MATHDB-332720

Gaussian energy Mellin-type identity conjecture in dimensions 8 and 24

摘录 / EXCERPT: Let $\varphi(x)=e^{-c|x|^2}$ in $\mathbb R^n$, where $n\in\{8,24\}$, and let h be the limiting auxiliary function from the sharp energy-bound convergence conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063438 | MATHDB-332727

Five-point triangular bipyramid or square-pyramid energy conjecture

摘录 / EXCERPT: Let S^2 be the unit sphere, and let a completely monotonic potential function be a potential function whose derivatives alternate in sign on the relevant domain. For five points...

conjecture | open | MathDB 页面 | 原文线索

V2-063439 | MATHDB-332731

The gradient-flow global-maximization conjecture for image-labeling assignments

摘录 / EXCERPT: Gradient-flow global-maximization conjecture. For any data and prior data, up to a subset of $\mathcal{W}$ of measure zero, the flow $W(t)$ approximates a global maximum in the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-063440 | MATHDB-332734

The orientifold Hodge equality conjecture

摘录 / EXCERPT: Orientifold Hodge equality conjecture. The canonical surjection

conjecture | open | MathDB 页面 | 原文线索

V2-063441 | MATHDB-332737

The constant-bias Client path conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let P_k be the path on k vertices, and write $CW(K_n,P_k,Cn)$ for the Client-Waiter game with bias Cn in which Client aims to...

conjecture | open | MathDB 页面 | 原文线索

V2-063442 | MATHDB-332746

Minkowski's measure regularity conjecture

摘录 / EXCERPT: Minkowski's measure regularity conjecture. Minkowski's measure μ is regular in the sense of Ullman--Stahl--Totik. This is a problem in logarithmic potential theory and orthogon...

conjecture | open | MathDB 页面 | 原文线索

V2-063443 | MATHDB-332747

Minkowski's measure Nevai-class conjecture

摘录 / EXCERPT: Minkowski's measure Nevai-class conjecture. The measure μ belongs to the Nevai class $N(\frac14,\frac12)$, equivalently its Jacobi coefficients have the limiting values...

conjecture | open | MathDB 页面 | 原文线索

V2-063444 | MATHDB-332753

Hurwitz-orbit conjecture for factorizations in well-generated complex reflection groups

摘录 / EXCERPT: Let W be a well-generated finite complex reflection group, let c be a Coxeter element of W, and consider factorizations of c into reflections. Two such reflection factoriza...

conjecture | open | MathDB 页面 | 原文线索

V2-063445 | MATHDB-332759

The nonvanishing conjecture for the FJRW recursion coefficient

摘录 / EXCERPT: Nonvanishing conjecture. The coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063446 | MATHDB-332768

Paraboloid L^2 extension range conjecture

摘录 / EXCERPT: Let $q=|\mathbb F_q|$, and let $P\subset \mathbb F_q^d$ be the paraboloid ... Write $R_P^*(p\to r)\lesssim 1$ when the finite-field extension operator for P is bounded from...

conjecture | open | MathDB 页面 | 原文线索

V2-063447 | MATHDB-332769

Critical endpoint conjecture for finite-field paraboloids

摘录 / EXCERPT: Let $q=|\mathbb F_q|$, and let $P\subset\mathbb F_q^d$ be the paraboloid ... Write $R_P^*(p\to r)\lesssim1$ for the uniform finite-field extension estimate. Critical endpoint conject...

conjecture | open | MathDB 页面 | 原文线索

V2-063448 | MATHDB-332771

Baraglia-Schaposnik's Langlands duality conjecture for the (A,B,A)-brane

摘录 / EXCERPT: Let G_c be a complex reductive group with Langlands dual group ^LG_c, let f be the anti-holomorphic involution of the Riemann surface defining i_2, and let $^L\rho$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-063449 | MATHDB-332772

Baraglia-Schaposnik's Nadler-group conjecture for Langlands dual branes

摘录 / EXCERPT: Let G_c be a complex reductive group with Langlands dual group ^LG_c. Let i_1 be the involution whose fixed points form a (B,A,A)-brane in $\mathcal{M}_{G_c}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063450 | MATHDB-332780

Kawarabayashi-Pedersen-Toft clique-minor conjecture for double-critical graphs

摘录 / EXCERPT: A finite simple graph G is double-critical if it is connected and, for every edge $uv\in E(G)$, $\chi(G-\{u,v\})=\chi(G)-2$. A graph H is a minor of G if it can be obtained f...

conjecture | open | MathDB 页面 | 原文线索

V2-063451 | MATHDB-332796

Boundedness equivalence for Widom factors and Chebyshev constants

摘录 / EXCERPT: Boundedness equivalence conjecture. The sequence $\left(W_n\left(\mu_K\right)\right)_{n=1}^\infty$ is bounded if and only if $\left(M_{n,K}\right)_{n=1}^\infty$ is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-063452 | MATHDB-332803

The Rational Conjecture for non-Gorenstein normal surface singularities

摘录 / EXCERPT: Let $(X,E)\rightarrow (V,0)$ be the minimal good resolution (MGR) of a complex normal surface singularity, and define $S_X=(\Omega^1_X(\operatorname{log} E))^*$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-063453 | MATHDB-332806

The generalised-triangle conjecture for chromatic roots

摘录 / EXCERPT: Let $\mathcal{G}$ be a minor-closed class of graphs. A generalised triangle is a graph in the family of graphs described in the paper as generalised triangles, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063454 | MATHDB-332807

Kirillov's unimodality conjecture for zig-zag poset chain polytopes

摘录 / EXCERPT: Kirillov's unimodality conjecture. For any $n\geq 1$, the Ehrhart δ-polynomial $\delta(\mathcal{P}_n;t)$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-063455 | MATHDB-332809

Uniqueness conjecture for holomorphic Lefschetz pencils on the four-torus

摘录 / EXCERPT: Consider holomorphic Lefschetz pencils on the four-torus, allowing arbitrary genera and divisibilities. Uniqueness conjecture. Two holomorphic Lefschetz pencils on the four-torus,...

conjecture | open | MathDB 页面 | 原文线索

V2-063456 | MATHDB-332813

Hirose-Watanabe-Yoshida's Gorenstein criterion for standard graded rings

摘录 / EXCERPT: Let R be a standard graded strongly F-regular ring, with R_0 an F-finite field of characteristic $p>0$. Let $\mathfrak{m}$ be the unique homogeneous maximal ideal of R. H...

conjecture | open | MathDB 页面 | 原文线索

V2-063457 | MATHDB-332814

Quasi-Gorenstein criterion via the F-pure threshold

摘录 / EXCERPT: Let $(R,\mathfrak{m})$ be an F-finite F-pure normal local ring of characteristic $p>0$. Quasi-Gorenstein threshold conjecture. The ring R is quasi-Gorenstein if and only if .....

conjecture | open | MathDB 页面 | 原文线索

V2-063458 | MATHDB-332816

The genomic-tableaux upper-bound conjecture for Lagrangian Grassmannian constants

摘录 / EXCERPT: Let $Y={\rm OG}(n,2n+1)$ and $Z={\rm LG}(n,2n)$, with structure constants $b_{\lambda,\mu}^{\nu}$ and $c_{\lambda,\mu}^{\nu}$, respectively, indexed by strict partitions λ,...

conjecture | open | MathDB 页面 | 原文线索

V2-063459 | MATHDB-332820

Cigler's moment conjecture for normalized q-Hermite polynomials

摘录 / EXCERPT: Cigler's conjecture. For the monic orthogonal polynomials $s_n(x)$ defined by the three-term recurrence

conjecture | open | MathDB 页面 | 原文线索

V2-063460 | MATHDB-332836

The True Comparison Theory conjecture

摘录 / EXCERPT: True Comparison Theory conjecture. Despite advances in the social sciences, an objective measuring stick for comparison-a so-called True Comparison Theory-has yet to be discovered...

conjecture | open | MathDB 页面 | 原文线索

V2-063461 | MATHDB-332847

Minuscule distributive-lattice mCDE conjecture

摘录 / EXCERPT: Minuscule mCDE conjecture. The distributive lattice $J(P)$ is mCDE.

conjecture | open | MathDB 页面 | 原文线索

V2-063462 | MATHDB-332848

Falling-by-twos shifted Young-lattice CDE conjecture

摘录 / EXCERPT: Falling-by-twos conjecture. The interval $[\varnothing,\lambda]_{\operatorname{shifted}}$ is CDE, and

conjecture | open | MathDB 页面 | 原文线索

V2-063463 | MATHDB-332856

The active spanning-tree Peano-curve conjecture

摘录 / EXCERPT: Let Λ be the triangular, hexagonal, or square lattice. Suppose either $y=0$ and $z>0$, or $y\in[0,1]$ and $z\in[1,\infty)$. Let T be a spanning tree on Λ sampled...

conjecture | open | MathDB 页面 | 原文线索

V2-063464 | MATHDB-332860

Baudon-Bensmail-Przybyło-Woźniak conjecture on the irregular chromatic index

摘录 / EXCERPT: Baudon-Bensmail-Przybyło-Woźniak conjecture. For every decomposable graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063465 | MATHDB-332861

Super-linear spreading conjecture for the nonlocal bistable cane toads model

摘录 / EXCERPT: Super-linear spreading conjecture. The model exhibits acceleration: the level sets of ρ move super-linearly in time.

conjecture | open | MathDB 页面 | 原文线索

V2-063466 | MATHDB-332866

The bounded fractional-chromatic-number-to-Hall-ratio conjecture

摘录 / EXCERPT: The bounded fractional-chromatic-number-to-Hall-ratio conjecture. There is an absolute constant such that, for every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063467 | MATHDB-332873

Bressan's conjecture on the geometric mixing scale

摘录 / EXCERPT: Let θ be a passive scalar transported on $\mathbf{R}^d$ by a time-dependent divergence-free velocity field u, and let $\theta(t,\cdot)$ denote the resulting scalar. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-063468 | MATHDB-332884

Pappas-Rapoport-Smithling conjecture on maximal elements in cocharacter classes

摘录 / EXCERPT: Pappas-Rapoport-Smithling conjecture. The set of maximal elements in $\{\bar{\mu}\}$ with respect to the Bruhat order is precisely the set $\Lambda_{\{\mu\}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-063469 | MATHDB-332891

Kumar-Courtade conjecture on Boolean functions and mutual information

摘录 / EXCERPT: Let $({\mathsf{x}},{\mathsf{y}})$ be dependent Rademacher random variables taking values in $\{-1,1\}$, and let ... consist of n independent, identically distributed copies. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-063470 | MATHDB-332893

Tight uniform stability conjecture for t-intersecting families

摘录 / EXCERPT: Tight uniform stability conjecture. There exists a t-umvirate $\mathcal{S}_B$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063471 | MATHDB-332894

Sharp stability conjecture for the Erdős matching theorem

摘录 / EXCERPT: Sharp stability conjecture. There exists $B\in[n]^{(s)}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063472 | MATHDB-332898

The arithmetic transfer conjecture in the unramified almost self-dual case

摘录 / EXCERPT: Assume that F/F_0 is unramified. Let $K'\subset S_n(F_0)$ and $K_1\subset\mathrm{U}(W_1)(F_0)$ be as in the almost self-dual transfer conjecture, let ω_S be the transfer f...

conjecture | open | MathDB 页面 | 原文线索

V2-063473 | MATHDB-332899

The ramified even-dimensional arithmetic transfer conjecture

摘录 / EXCERPT: Assume that F/F_0 is ramified and that $n\geq2$ is even. Let $K_0^{{{{{{{{{\flat}}}}}}}}}$ be the stabilizer of an almost π-modular lattice and let K_0^+ and K_0^- be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-063474 | MATHDB-332901

The almost self-dual fundamental lemma

摘录 / EXCERPT: Assume the relevant unramified hermitian-space setup and let $K_0(\varpi)$, K', $\mathfrak{k}'$, $K_1^{{{{{{{{{\flat}}}}}}}}}$, and K_1 be the compact subgroups and lattices defi...

conjecture | open | MathDB 页面 | 原文线索

V2-063475 | MATHDB-332902

Finite-jet conjecture for small-time local controllability

摘录 / EXCERPT: Finite-jet conjecture. There exists $N\in\mathbb{N}$ such that every such system $\mathcal{Y}$ is small-time locally controllable from x_0. The conjecture asks whether small-time...

conjecture | open | MathDB 页面 | 原文线索

V2-063476 | MATHDB-332904

Log-squared computation and communication cost for Pareto-optimal MP-AMP

摘录 / EXCERPT: MP-AMP cost conjecture. The total computation and communication cost scales as

conjecture | open | MathDB 页面 | 原文线索

V2-063477 | MATHDB-332921

The b-conjecture for unicellular bipartite maps

摘录 / EXCERPT: Let $n\geq 1$, and let τ, μ, and ν be partitions of n. For the coefficients $h_{\mu,\nu}^{\tau}(\beta)$ defined by the Jack-polynomial generating series, let the sum...

conjecture | open | MathDB 页面 | 原文线索

V2-063478 | MATHDB-332928

Pro-odd class tower moment conjecture for imaginary quadratic fields

摘录 / EXCERPT: Let D_X be the set of imaginary quadratic fields of absolute discriminant less than X. For each imaginary quadratic number field K, let $\mathcal{G}_K$ be the maximal pro-odd...

conjecture | open | MathDB 页面 | 原文线索

V2-063479 | MATHDB-332932

Affine Kac-Moody geometric Satake conjecture for Coulomb branches

摘录 / EXCERPT: Let Q be an affine quiver, let $\mathcal M_C(\lambda,\mu)$ be its Coulomb branch, and let $\mathfrak A_\chi(\lambda,\mu)$ be the attracting set for the chosen cocharacter χ....

conjecture | open | MathDB 页面 | 原文线索

V2-063480 | MATHDB-332938

Non-finite generation conjecture for non-zero CQP monoids

摘录 / EXCERPT: Non-finite generation conjecture. If M is a non-zero CQP monoid, then M is not finitely generated as a monoid.

conjecture | open | MathDB 页面 | 原文线索

V2-063481 | MATHDB-332939

Majorization conjecture for products of positive semidefinite matrices

摘录 / EXCERPT: Let A and B be $n\times n$ positive semidefinite matrices. Let $\lambda(X)$ denote the vector of eigenvalues of a Hermitian matrix X, arranged in decreasing order, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063482 | MATHDB-332942

Ball-Nayar-Tkocz planar Shannon reverse entropy power conjecture

摘录 / EXCERPT: Let (U,V) be the coordinates of a symmetric log-concave random vector in $\mathbb{R}^2$. Ball-Nayar-Tkocz's conjecture. ... This inequality is equivalent to the triangle inequality...

conjecture | open | MathDB 页面 | 原文线索

V2-063483 | MATHDB-332944

The Aubert-Baum-Plymen-Solleveld conjecture for Bernstein components

摘录 / EXCERPT: Let G be a split connected reductive p-adic group and let $\mathfrak{s} \in \mathcal{B}(G)$ be an inertial pair. Write $T_{\mathfrak{s}}$ for the associated complex torus,...

conjecture | open | MathDB 页面 | 原文线索

V2-063484 | MATHDB-332958

Fibonomial Lucas-type divisibility conjecture

摘录 / EXCERPT: Fibonomial Lucas-type divisibility conjecture. If $p^*\geq p$, then

conjecture | open | MathDB 页面 | 原文线索

V2-063485 | MATHDB-332965

Bhattacharya-Weitsman-Nadirashvili sharp quantitative Faber-Krahn conjecture

摘录 / EXCERPT: Bhattacharya-Weitsman-Nadirashvili conjecture. There exists a dimensional constant $C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063486 | MATHDB-332966

Strong Bordeaux Conjecture on 3-coloring planar graphs

摘录 / EXCERPT: Let G be a planar graph. Strong Bordeaux Conjecture. If no pair of cycles of length three in G shares an edge and G has no cycle of length five, then G is 3-colorable. This...

conjecture | open | MathDB 页面 | 原文线索

V2-063487 | MATHDB-332974

Circular Broersma conjecture for planar graphs and spanning trees

摘录 / EXCERPT: Let G be a planar graph and let T be a spanning tree of G. A circular backbone coloring uses colors in a cyclic color space; $CBC(G,T)$ denotes the minimum number of colors i...

conjecture | open | MathDB 页面 | 原文线索

V2-063488 | MATHDB-332975

Circular backbone bound for planar graphs without 4- or 5-cycles

摘录 / EXCERPT: Let G be a planar graph containing no C_4 or C_5 as a subgraph, and let $H\subseteq G$ be any backbone. Here $CBC_2(G,H)$ is the minimum k for which (G,H) has a circular...

conjecture | open | MathDB 页面 | 原文线索

V2-063489 | MATHDB-332981

Quantifier elimination with floor functions for parametric Presburger arithmetic

摘录 / EXCERPT: Let $R=\\Z[t]$ and $X=\\N$. Define ... where each $g_{\alpha(t)}$ is interpreted by ... with $g_{\alpha(t)}(x)=0$ when $\alpha(t)=0$. Quantifier-elimination conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-063490 | MATHDB-332985

Pattern Sturmian spectral conjecture

摘录 / EXCERPT: Pattern Sturmian spectral conjecture. Theorem 1 holds if $\bm{x}$ is a pattern Sturmian sequence: the spectrum $\sigma(H)$ is a zero Lebesgue measure Cantor set and all spectral me...

conjecture | open | MathDB 页面 | 原文线索

V2-063491 | MATHDB-332986

Cantorval spectrum conjecture for a two-dimensional quasicrystal Schrödinger operator

摘录 / EXCERPT: Cantorval spectrum conjecture. There exist values of λ_1 and λ_2 such that the spectrum $\sigma(H)$ is a Cantorval; that is, the spectrum is the closure of its inte...

conjecture | open | MathDB 页面 | 原文线索

V2-063492 | MATHDB-332990

Valette's conjecture on rapid decay for cocompact semisimple lattices

摘录 / EXCERPT: Let G) be a semisimple Lie group, either real or p-adic, and let Γ be a cocompact lattice in G. Valette's conjecture. The group Γ has the Rapid Decay property....

conjecture | open | MathDB 页面 | 原文线索

V2-063493 | MATHDB-332992

The generalized weight-monodromy and independence-of-ell conjecture

摘录 / EXCERPT: Weight-monodromy and independence-of-ell conjecture. The polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-063494 | MATHDB-332994

The Equivariant Tamagawa Number Conjecture

摘录 / EXCERPT: Equivariant Tamagawa Number Conjecture. There exists a basis $\mathbf z_\Sigma(M_K)$ of $\Delta_\Sigma(M_K)$ satisfying the complex period, p-adic lattice, and corestriction-comp...

conjecture | open | MathDB 页面 | 原文线索

V2-063495 | MATHDB-332995

The generalized Iwasawa main conjecture for zeta elements

摘录 / EXCERPT: Kato's general Iwasawa main conjecture. There exist equivariant fundamental lines and zeta elements for all such specializations, compatible with corestriction in K, specializati...

conjecture | open | MathDB 页面 | 原文线索

V2-063496 | MATHDB-333002

Sroysang's conjecture on the growth rate of Fibonacci functions with period k

摘录 / EXCERPT: Let f be a real-valued Fibonacci function with period $k\in\mathbb{N}$, meaning that ... for every $x\in\mathbb{R}$. Let ϕ denote the golden ratio. Sroysang's conjecture. ... Th...

conjecture | solved | MathDB 页面 | 原文线索

V2-063497 | MATHDB-333012

Wang's Hodge-theoretic criterion for finite Weil-Petersson distance

摘录 / EXCERPT: Wang's conjecture. X has finite Weil-Petersson distance if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-063498 | MATHDB-333013

Gross-Witten sum-rule conjecture for coupling below minus one

摘录 / EXCERPT: Let $\mu\in\mathcal M_1(\mathbb T)$ have Verblunsky coefficients $(\alpha_k)_{k\geq 0}\in\mathbb D^{\mathbb N}$. Let $\theta_{\tt g}$, N^-, N^+, λ_i^+, λ_i^-, a...

conjecture | open | MathDB 页面 | 原文线索

V2-063499 | MATHDB-333014

Matrix Gross-Witten sum-rule conjecture

摘录 / EXCERPT: Let $\Sigma\in\mathcal M_{p,1}(\mathbb T)$ have infinite support, and let $(\boldsymbol\alpha_k)_{k\geq0}\in(\mathbb B_p)^{\mathbb N_0}$ be its matrix Verblunsky coefficients. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063500 | MATHDB-333016

Non-hom-idempotence conjecture for stable Kneser graphs

摘录 / EXCERPT: Let $\operatorname{KG}(n,k)_{s-\operatorname{-stab}}$ denote the s-stable Kneser graph, and let n,k,s satisfy the stated inequalities. A graph is hom-idempotent when its endomo...

conjecture | open | MathDB 页面 | 原文线索

V2-063501 | MATHDB-333028

The embedding-dimension bound for rational isolated complete intersections

摘录 / EXCERPT: Let p be the dimension of a rational isolated complete intersection singularity equipped with a $\mathbb{C}^*$-action, and let its embedding dimension be the dimension of its Zar...

conjecture | open | MathDB 页面 | 原文线索

V2-063502 | MATHDB-333032

Winding-number conjecture for type II runaways under diagonal Toeplitz perturbations

摘录 / EXCERPT: Let T be the Toeplitz matrix and let A_{jj} denote the matrix with its (j,j) entry selected. Consider the rank-one deformation ... Let the first-order perturbation correction b...

conjecture | open | MathDB 页面 | 原文线索

V2-063503 | MATHDB-333033

Parity and defectivity conjecture for collisions of runaway eigenvalues

摘录 / EXCERPT: Let T be the Toeplitz matrix, let A_{jj} denote the indicated diagonal rank-one perturbation, and let V denote the perturbation appearing in the source. Consider the deformat...

conjecture | open | MathDB 页面 | 原文线索

V2-063504 | MATHDB-333038

Universality conjecture for critical one-dimensional random Schrödinger operators

摘录 / EXCERPT: Let E be distributed according to the arcsine law, U be uniform on $[0,1]$, and let $\mathcal{Z}$ be a standard two-sided Brownian motion started from 0, independent of E a...

conjecture | open | MathDB 页面 | 原文线索

V2-063505 | MATHDB-333046

Strassen's decomposition conjecture for disjoint-variable forms

摘录 / EXCERPT: Strassen's decomposition conjecture. If $F=\sum_{i=1}^sF_i\in S$ is a degree $d\geq3$ form such that $F_i\in S^{[i]}$ for all $i=1,\ldots,s$, then any minimal Waring decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-063506 | MATHDB-333053

The Telescope Conjecture for smashing subcategories

摘录 / EXCERPT: Telescope Conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-063507 | MATHDB-333079

Stein-Brualdi conjecture on near-transversals of Latin squares

摘录 / EXCERPT: Let a partial transversal in an $n\times n$ array be a set of entries in distinct rows, distinct columns, and with distinct symbols. Stein-Brualdi conjecture. Every $n\times n$ Lat...

conjecture | open | MathDB 页面 | 原文线索

V2-063508 | MATHDB-333080

Stein's equitable perfect matching conjecture

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with n vertices in each part, and suppose its edges are partitioned into sets $E_1,\ldots,E_m$, where $m\leqslant n$. A perfect matc...

conjecture | open | MathDB 页面 | 原文线索

V2-063509 | MATHDB-333081

ISR conjecture for line graphs of simple bipartite graphs

摘录 / EXCERPT: Let H be a simple bipartite graph, let $L(H)$ be its line graph, and let $V_1,\ldots,V_m$ be a partition of the edge set of $L(H)$. An independent system of representatives (ISR)...

conjecture | open | MathDB 页面 | 原文线索

V2-063510 | MATHDB-333082

Concavity conjecture for polar-code interpolation functions

摘录 / EXCERPT: Concavity conjecture. The function $\overline{g}_{n}(x)$ is concave on $[0,1]$ for every n.

conjecture | open | MathDB 页面 | 原文线索

V2-063511 | MATHDB-333083

Asymptotic mixing-rate conjecture for large-alphabet polarizing kernels

摘录 / EXCERPT: Asymptotic mixing-rate conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063512 | MATHDB-333089

Deep-hole classification conjecture for projective Reed-Solomon codes

摘录 / EXCERPT: Projective deep-hole conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-063513 | MATHDB-333090

The MDS conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, and let $[n,k]$ denote a linear code of length n and dimension k over $\mathbb{F}_q$. If $1<k<q$, then ... except when q is even and $k=3$...

conjecture | open | MathDB 页面 | 原文线索

V2-063514 | MATHDB-333091

General covering-radius conjecture for longest MDS codes

摘录 / EXCERPT: General covering-radius conjecture. For $2\le k\le q-2$,

conjecture | open | MathDB 页面 | 原文线索

V2-063515 | MATHDB-333107

The Delta Conjecture for the modified Macdonald delta operator

摘录 / EXCERPT: The Delta Conjecture. For any integers $n>k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-063516 | MATHDB-333114

Zhang's conjecture on l-torsion in class groups

摘录 / EXCERPT: Let K be a number field, let $n=[K:\mathbb{Q}]$, and let $\epsilon>0$. Write $Cl(K)[l]$ for the subgroup of elements of the class group $Cl(K)$ annihilated by l, and let D_K...

conjecture | open | MathDB 页面 | 原文线索

V2-063517 | MATHDB-333140

Conjecture identifying affine k-Schur functions with k-atoms

摘录 / EXCERPT: For $k>0$, let $\mathcal P^k$ be the set of partitions with first part at most k, let $s_\lambda^{(k)}[X;t]$ be the k-Schur function with parameter t, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063518 | MATHDB-333143

BFR-MSR and BFR-MBR point conjecture for $sigma>rho$

摘录 / EXCERPT: Let ${\cal M}$ be the file size, and let b, ρ, $sigma$, k, d, and k_c be the block-failure resilient coding parameters. Assume $d_r\geq k_c$ and $sigma>rho$. BFR-MSR a...

conjecture | open | MathDB 页面 | 原文线索

V2-063519 | MATHDB-333156

Large-load uniqueness conjecture for the canonical dual problem

摘录 / EXCERPT: Let $({\cal P})$ be a properly posed problem satisfying Assumption 1, let $\mathbf f$ be its load vector, and let $({\cal P}^d)$ be the associated canonical dual problem over...

conjecture | open | MathDB 页面 | 原文线索

V2-063520 | MATHDB-333163

Distinct tame root types determine distinct braid-group orbits

摘录 / EXCERPT: Root-type orbit conjecture. Let $E_1,E_2\in\mathcal{E}$. Assume that there are $g_1,g_2\in B_{n-1}$ such that g_1E_1 and g_2E_2 have tame type but different root types. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-063521 | MATHDB-333165

Mao's rank and M_2-rank inequalities modulo 6 and 10

摘录 / EXCERPT: Mao's conjectured inequalities. Computation evidence suggests that

conjecture | open | MathDB 页面 | 原文线索

V2-063522 | MATHDB-333167

Conjectured permutation trinomials over finite fields of characteristic three

摘录 / EXCERPT: Permutation trinomial conjectures. (1) If k is even and $\mathrm{gcd}(5+2l,q-1)=1$, then $f_1(x)$ is a permutation trinomial over $\mathbb{F}_{q^2}$. (2) If...

conjecture | open | MathDB 页面 | 原文线索

V2-063523 | MATHDB-333174

Degree-bounded bipartite rainbow matching conjecture

摘录 / EXCERPT: Degree-bounded bipartite rainbow matching conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-063524 | MATHDB-333175

Nested-size bipartite rainbow matching conjecture

摘录 / EXCERPT: Nested-size bipartite rainbow matching conjecture. If $F_i\subseteq K_{n,n}$ for $i\leq k$ and

conjecture | open | MathDB 页面 | 原文线索

V2-063525 | MATHDB-333176

The 1243 inversion generating-function conjecture for rectangular posets

摘录 / EXCERPT: Let $EN_{s,t}(\sigma)(q)$ denote the inversion-generating function for linear extensions of the rectangular poset with parameters s,t that avoid the permutation pattern σ,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-063526 | MATHDB-333177

The 2143 two-column inversion generating-function conjecture

摘录 / EXCERPT: Let $EN_{s,t}(\sigma)(q)$ denote the inversion-generating function for pattern-avoiding linear extensions of the rectangular poset with parameters s,t. The 2143 two-column invers...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-063527 | MATHDB-333178

The 2143 three-column Fibonacci-polynomial conjecture

摘录 / EXCERPT: Let $F_s(q)$ be defined by ... and, for $s\ge 2$, ... Let $EN_{s,t}(\sigma)(q)$ denote the inversion-generating function for pattern-avoiding linear extensions of the rectangular poset...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-063528 | MATHDB-333179

The coefficient-of-q^{2s-2} conjecture for the Fibonacci polynomials $F_s(q)$

摘录 / EXCERPT: The coefficient-of-q^{2s-2} conjecture. For all $s\ge2$, the coefficient of q^{2s-2} in $F_s(q)$ is the number of compositions of $s+9$ into s parts, none of which is 2 or...

conjecture | open | MathDB 页面 | 原文线索

V2-063529 | MATHDB-333180

The major-index generating-function conjectures for rectangular posets

摘录 / EXCERPT: Let $EN_{s,t}(\sigma)[q]$ and $NE_{s,t}(\sigma)[q]$ denote the major-index generating functions for the indicated pattern-avoiding linear extensions of rectangular posets. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063530 | MATHDB-333182

Bogomolov-Gieseker type conjecture for tilt-stable complexes

摘录 / EXCERPT: Bogomolov-Gieseker type conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-063531 | MATHDB-333188

Hypercover descent conjecture for locally free sheaves on soft ringed spaces

摘录 / EXCERPT: Let $(X,\mathcal{A})$ be a compact ringed space with soft structure sheaf $\mathcal{A}$. Let $U_{\bullet}$ be a hypercover of X, and let $\mathfrak{L}(X)$ denote the dg-category...

conjecture | open | MathDB 页面 | 原文线索

V2-063532 | MATHDB-333215

Badouel-Darondeau-Raoult recognizability conjecture for conflict event domains

摘录 / EXCERPT: Let $\mathcal E$ be an event structure, and let its conflict event domain be the domain of configurations associated with $\mathcal E$. A conflict event domain is recognizable when...

conjecture | open | MathDB 页面 | 原文线索

V2-063533 | MATHDB-333219

Undecidability of regular nice labelings for strongly regular event domains

摘录 / EXCERPT: Let $\mathcal D$ be a strongly regular event domain. A regular nice labeling is the labeling property used in Thiagarajan's conjecture. Undecidability conjecture. There does not ex...

conjecture | open | MathDB 页面 | 原文线索

V2-063534 | MATHDB-333222

Monotonicity of the optimal stopping region for the duration problem

摘录 / EXCERPT: Let $U_n(x)$ be the expected duration of owning the relatively best object whose rank remains within the two when the time to go is n, conditional on accepting the relatively bes...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-063535 | MATHDB-333223

Pivotality conjecture for fusion categories

摘录 / EXCERPT: Pivotality conjecture. Every fusion category admits a pivotal structure.

conjecture | open | MathDB 页面 | 原文线索

V2-063536 | MATHDB-333236

Critical-distance conjecture for anomalous localized resonance

摘录 / EXCERPT: Let $0<\gamma<\gamma_*$. Consider sources $f\in L^2(\mathcal{M})$ with compact support or distributional sources such as dipoles, and let $E_{\delta}(\xi)$ and $V(x,y)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-063537 | MATHDB-333243

The q-Kreweras cyclic sieving conjecture for principal-in-a-Levi nilpotent orbits

摘录 / EXCERPT: Let W be a Weyl group with Coxeter number h, let $m=sh+1$, and let $NC^{(s)}(W)$ denote the set of multichains $w_1\leq\cdots\leq w_s$ in the s-divisible noncrossing partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-063538 | MATHDB-333245

The partial divided-difference equation conjecture for multivariate Wilson polynomials

摘录 / EXCERPT: Let p be a positive integer, let ${\pmb{n}}=(n_1,\ldots,n_p)$ and ${\pmb{x}}=(x_1,\ldots,x_p)$, and let ${\pmb{e}}_p=(e_2,\ldots,e_p)$. The multivariate Wilson polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-063539 | MATHDB-333263

Twisted type IIB supergravity as BCOV theory

摘录 / EXCERPT: Let X be a Calabi-Yau 5-fold, with type IIB supergravity placed in the Calabi-Yau background and its bosonic ghost field set to the covariantly constant Killing spinor $(Q,0)$....

conjecture | open | MathDB 页面 | 原文线索

V2-063540 | MATHDB-333264

AdS type IIB twist as BCOV theory with a sourced five-form

摘录 / EXCERPT: Let $\mathbb{C}^2=\{z_1=z_2=z_3=0\}\subset\mathbb{C}^5$ and let $r=\sqrt{|z_1|^2+|z_2|^2+|z_3|^2}$ be the radius normal to $\mathbb{C}^2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063541 | MATHDB-333266

Boundary regularity conjecture for the Liouville density of the discrete Gaussian free field

摘录 / EXCERPT: Let D be a domain, let $t\in\mathbb R$, and let $\rho^D(t,\mathord\cdot)$ denote the density from Theorem 2.7. Boundary regularity conjecture. For each $t\in\mathbb R$, the funct...

conjecture | open | MathDB 页面 | 原文线索

V2-063542 | MATHDB-333267

Critical inverse-temperature scaling conjecture for the discrete Gaussian free field Gibbs measure

摘录 / EXCERPT: Let $\mathfrak D$ be the class of domains in the source, let D_N be the corresponding lattice approximation, let h_z be the discrete Gaussian free field, let m_N be its cente...

conjecture | open | MathDB 页面 | 原文线索

V2-063543 | MATHDB-333269

Migliore, Miró-Roig and Nagel's weak Lefschetz conjecture for an odd-dimensional complete intersection

摘录 / EXCERPT: Migliore, Miró-Roig and Nagel's conjecture. The ring R/I fails the Weak Lefschetz property if and only if $d>1$. Furthermore, if $n=3$, then R/I fails the Weak Lefschetz proper...

conjecture | open | MathDB 页面 | 原文线索

V2-063544 | MATHDB-333270

Inductive-freeness conjecture for arrangements of ideal type

摘录 / EXCERPT: Let Φ be a reduced root system with Weyl arrangement $\mathscr A=\mathscr A(\Phi)$, and let $\mathcal I$ be an ideal in Φ^+. Write $\mathscr A_{\mathcal I}$ for the subar...

conjecture | open | MathDB 页面 | 原文线索

V2-063545 | MATHDB-333276

Gross-Reeder conjecture for adjoint Swan conductors

摘录 / EXCERPT: Let k be a p-adic field with residue field $\mathfrak{f}$, and let $\mathcal{W}=\langle \operatorname{Fr}\rangle\ltimes\mathcal{I}$ be its Weil group, with wild inertia subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-063546 | MATHDB-333281

Spontaneous emergence of solitons in affine dKdV

摘录 / EXCERPT: Let n be chosen, let u and v be glides with fixed reduced words $u=s_{i_1}\cdots s_{i_l}$ and $v=s_{j_1}\cdots s_{j_m}$, and assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-063547 | MATHDB-333314

Finitely generated geometric divisor subgroup conjecture for complete intersection rings

摘录 / EXCERPT: Let R be the completion of a normal geometric complete intersection ring, and let $G\subset \operatorname{Cl} R$ be a finitely generated subgroup. A local ring is called geometri...

conjecture | open | MathDB 页面 | 原文线索

V2-063548 | MATHDB-333316

The colored strong Lefschetz property conjecture for balanced simplicial spheres

摘录 / EXCERPT: Colored SLP conjecture. Any balanced simplicial sphere, or at least any balanced simplicial polytope, has the colored SLP over a field of characteristic 0.

conjecture | open | MathDB 页面 | 原文线索

V2-063549 | MATHDB-333320

The retract characterization conjecture for hypercellular graphs

摘录 / EXCERPT: The retract characterization conjecture. A partial cube G is hypercellular if and only if G is a retract of a Cartesian product of bipartite cellular graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-063550 | MATHDB-333322

F-conjecture for \(\overline{\mathrm{M}}_{0,n}\)

摘录 / EXCERPT: F-conjecture. A divisor on $\overline{\mathrm{M}}_{0,n}$ is nef if and only if it is F-nef.

conjecture | open | MathDB 页面 | 原文线索

V2-063551 | MATHDB-333323

The \(S_n\)-invariant F-conjecture for \(\overline{\mathrm{M}}_{0,n}\)

摘录 / EXCERPT: The S_n-invariant F-conjecture. An S_n-invariant divisor on $\overline{\mathrm{M}}_{0,n}$ is nef if and only if it is F-nef.

conjecture | open | MathDB 页面 | 原文线索

V2-063552 | MATHDB-333339

Generalized closure relation conjecture for multi-indexed orthogonal polynomials

摘录 / EXCERPT: Let $Y(\eta)$ be any polynomial, let $X(\eta)$ be the polynomial specified in equation, and let A be the matrix specified in equations. Write $K=2L$. Generalized closure relation...

conjecture | open | MathDB 页面 | 原文线索

V2-063553 | MATHDB-333340

Explicit eigenvalue formulas conjecture for generalized closure relations

摘录 / EXCERPT: Let A be the matrix specified in equations, and let $K=2L$. The four systems under consideration are the Laguerre, Jacobi, Wilson, and Askey-Wilson cases, denoted by L, J,...

conjecture | open | MathDB 页面 | 原文线索

V2-063554 | MATHDB-333350

Gaussian local limit and large-deviation asymptotics for Lévy measures with a nontrivial absolutely continuous part

摘录 / EXCERPT: Let μ be a measure on $(0,1]$ such that ... Allow μ to have atoms and a continuous singular part, and suppose its absolutely continuous part is $g(x)\,dx$, with some...

conjecture | open | MathDB 页面 | 原文线索

V2-063555 | MATHDB-333352

Yoshinaga's sharp-pair conjecture on monodromy of real arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ be a real affine line arrangement and let $\overline{{\mathcal A}}$ be its projective arrangement. A sharp pair of lines is a pair of projective lines satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-063556 | MATHDB-333358

The triangle-free graph homomorphism-density conjecture

摘录 / EXCERPT: Triangle-free extension conjecture. For every graph H,

conjecture | open | MathDB 页面 | 原文线索

V2-063557 | MATHDB-333379

Hausel-Rodriguez-Villegas mixed Hodge polynomial conjecture

摘录 / EXCERPT: Let $\mathcal{M}_n$ be the moduli space under consideration, let d_n denote its dimension, and let $\mathcal{H}_{lambda}(z,w)$, $\overline{H}_n(z,w)$, and $H(\mathcal{M}_n;q,t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-063558 | MATHDB-333381

Willmore minimization conjecture for the bipolar Lawson Klein bottle

摘录 / EXCERPT: Let $\tilde\tau_{3,1}$ be Lawson's bipolar surface, and let $\varphi:\mathbb{S}^4\to\mathbb{R}^4$ be a stereographic projection. For $n\geq 4$, regard...

conjecture | open | MathDB 页面 | 原文线索

V2-063559 | MATHDB-333385

Positive-real-roots conjecture for quiver recurrence polynomials

摘录 / EXCERPT: Let $J_0=1,J_1,\dots,J_N,J_{N+1}=1$ be the coefficients of the minimal recurrence, and define its recurrence polynomial by ... The positive-real-roots conjecture. All roots of $J(z)$...

conjecture | open | MathDB 页面 | 原文线索

V2-063560 | MATHDB-333386

Reciprocity conjecture for recurrence polynomials under Dynkin involution

摘录 / EXCERPT: Assume the T-system has tensor-product form $\Lambda'\otimes\widehat{\Lambda}$, where Λ' is of type A_m, D_{2m+1}, or E_6, and let η be the canonical involutio...

conjecture | open | MathDB 页面 | 原文线索

V2-063561 | MATHDB-333392

Characterization conjecture for arithmetic functions satisfying a quadratic-form equation

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $f:\mathbb{N}\to\mathbb{C}$ satisfy ... for all $u,v\in\mathbb{N}$. Characterization conjecture. Exactly one of the following possibilities holds: (1)...

conjecture | open | MathDB 页面 | 原文线索

V2-063562 | MATHDB-333393

The systems-of-sets-of-lengths characterization of finite abelian groups

摘录 / EXCERPT: Systems-of-sets-of-lengths conjecture. If G' is an abelian group satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-063563 | MATHDB-333394

Ancient Ricci iteration implies ancient Ricci flow

摘录 / EXCERPT: An ancient Ricci iteration is a sequence of Riemannian metrics obtained by iterating the inverse Ricci operator for indefinitely many backward steps; an ancient Ricci flow is a Ric...

conjecture | open | MathDB 页面 | 原文线索

V2-063564 | MATHDB-333395

Convergence conjecture for Ricci iterations on homogeneous spaces

摘录 / EXCERPT: Let $M=G/H$ be a homogeneous space, let H be maximal in G, and let $\mathcal{M}$ denote the space of Riemannian metrics under consideration. Let $g_1\in\mathcal{M}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063565 | MATHDB-333407

Transitivity of the unipotent subgroup on opposite-ended apartments

摘录 / EXCERPT: Let $A^7=g\cdot A^7$ for some $g\in B_{i,\infty}^7$, where A^7 is the standard apartment and U_i is the corresponding unipotent subgroup. Transitivity conjecture. For each such...

conjecture | open | MathDB 页面 | 原文线索

V2-063566 | MATHDB-333408

Structure decomposition conjecture for the end stabilisers

摘录 / EXCERPT: Let $B_{i,\infty}^\pm$ be the stabiliser of the corresponding end, with subgroups $B_W^\pm$, U_i, and W^{even} as defined in the paper. Structure decomposition conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-063567 | MATHDB-333409

Spherical Iwasawa decomposition conjecture for the hyperbolic Kac--Moody group

摘录 / EXCERPT: Let G be the hyperbolic Kac--Moody group, let K be its relevant compact subgroup, let $T^{ad}_{\mathbb R}$ be the real adjoint torus, and define $U_{i,\infty}^\pm=U_W^\pm U_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-063568 | MATHDB-333410

The ideal-orbit conservation law for root-poset antichains

摘录 / EXCERPT: Ideal-orbit conservation law. For any $p\in\Delta(1)$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-063569 | MATHDB-333412

Poisson variance conjecture for shrinking spherical sets

摘录 / EXCERPT: Let $\mathcal E(n)$ be the projected lattice points on S^2, let N_n be their number, and let σ denote normalized spherical area. For a spherical cap or annulus...

conjecture | open | MathDB 页面 | 原文线索

V2-063570 | MATHDB-333413

Linnik's covering-radius conjecture for projected lattice points

摘录 / EXCERPT: Let $\mathcal E(n)$ be the projected lattice points on S^2, and let N_n denote their number. The covering radius $M(P_1,\dots,P_N)$ of a configuration on S^2 is the least...

conjecture | open | MathDB 页面 | 原文线索

V2-063571 | MATHDB-333416

Reiner-Tenner-Yong's first shifted Young lattice CDE conjecture

摘录 / EXCERPT: Let $\delta_r=(r,r-1,\ldots,1)$, and for partitions let $\nu+\lambda$ denote the partition obtained by adding corresponding parts. For $a,m,n\geq 1$ with $n>am$, set ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-063572 | MATHDB-333417

Reiner-Tenner-Yong's trapezoidal shifted Young lattice CDE conjecture

摘录 / EXCERPT: For $n\geq1$ and $0\leq k<n/2$, let ... The notation $[\varnothing,\lambda]_{\mathrm{shift}}$ denotes the corresponding initial interval in the shifted Young's lattice,...

conjecture | open | MathDB 页面 | 原文线索

V2-063573 | MATHDB-333429

Odd-dimensional hypercube path-pairability conjecture

摘录 / EXCERPT: For $k\in\mathbb{N}$, let Q_{2k+1} denote the $(2k+1)$-dimensional hypercube. A graph is path-pairable if every pairing of its vertices can be joined by pairwise edge-disjoint pa...

conjecture | open | MathDB 页面 | 原文线索

V2-063574 | MATHDB-333430

Existence of odd-order CASTs with minimal inflation multiplier and dihedral symmetry

摘录 / EXCERPT: A cyclotomic aperiodic substitution tiling (CAST) is a substitution tiling with the cyclotomic structure considered in the paper. For odd n, let D_n denote the dihedral symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-063575 | MATHDB-333431

Generalized Penrose CASTs for odd n

摘录 / EXCERPT: Let n be odd with $n\geq5$, and consider the CASTs described in Theorem. Let $\mu_{n,\lfloor n/2\rfloor}$ be the diagonal-based inflation multiplier used there, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063576 | MATHDB-333432

Dihedral-symmetry realization for generalized Penrose CASTs

摘录 / EXCERPT: Let n be odd with $n\geq5$, and consider a CAST as described in the generalized Penrose CAST conjecture, with its prototiles and substitution rules. A CAST has individual dihedra...

conjecture | open | MathDB 页面 | 原文线索

V2-063577 | MATHDB-333433

Rhombic CAST existence with prescribed edge configurations and dihedral symmetry

摘录 / EXCERPT: Let n be the polygon-order parameter, let the cases be those listed in Table, and let $\eta_{\min}$ and the edge configurations be the quantities specified in Tables--. A rhombic...

conjecture | open | MathDB 页面 | 原文线索

V2-063578 | MATHDB-333439

Density conjecture for zeros of generated polynomials

摘录 / EXCERPT: Density conjecture. For every integer m, all zeros of $H_m(z)$ lie in (a,b), and $\mathcal{Z}$ is dense in (a,b).

conjecture | open | MathDB 页面 | 原文线索

V2-063579 | MATHDB-333450

Yang's conjecture on coincident Einstein-Bogomol'nyi strings

摘录 / EXCERPT: Yang's conjecture. There is no solution of the Einstein--Bogomol'nyi equations when

conjecture | open | MathDB 页面 | 原文线索

V2-063580 | MATHDB-333457

Veronese-surface Betti-table conjecture

摘录 / EXCERPT: Veronese-surface Betti-table conjecture. If $d\geq2$, then the last non-zero entry on the linear strand is

conjecture | open | MathDB 页面 | 原文线索

V2-063581 | MATHDB-333462

Donagi-Pantev's Dolbeault geometric Langlands conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, let $G^{\vee}$ be its Langlands dual group, and let $T^*Bun_G(X)$ and $T^*Bun_{G^{\vee}}(X)$ be...

conjecture | solved | MathDB 页面 | 原文线索

V2-063582 | MATHDB-333464

Betti geometric Langlands for nilpotent sheaves

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, and let $G^{\vee}$ be its Langlands dual group. Let $\mathcal Shv_{\mathcal N}(Bun_G(X))$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-063583 | MATHDB-333465

Betti Hecke-functor compatibility conjecture

摘录 / EXCERPT: Let G be a complex reductive group, let X be a smooth complex projective curve, and let $Gr_G=G(\mathcal K)/G(\mathcal O)$ be the affine Grassmannian, where...

conjecture | open | MathDB 页面 | 原文线索

V2-063584 | MATHDB-333467

Automorphic gluing for nodal degenerations

摘录 / EXCERPT: Let G be a complex reductive group and let X be a smooth complex projective curve. Let $\mathcal Shv_{\mathcal N}(Bun_G(X))$ be the Betti category of nilpotent sheaves. Automor...

conjecture | open | MathDB 页面 | 原文线索

V2-063585 | MATHDB-333488

Existence of arbitrarily high-order periods for the complex rational delay recurrence

摘录 / EXCERPT: Let $z_{n+1}=\frac{\alpha+\beta z_{n-k}}{\gamma-z_n}$ be the complex rational delay difference equation, with parameters α, β, γ and delay k. A period p is...

conjecture | open | MathDB 页面 | 原文线索

V2-063586 | MATHDB-333489

The period formula for equal parameters in the complex rational delay recurrence

摘录 / EXCERPT: Let $z_{n+1}=\frac{\alpha+\beta z_{n-k}}{\gamma-z_n}$ be the complex rational delay difference equation, where k is the delay and p denotes the period of a periodic solution. E...

conjecture | open | MathDB 页面 | 原文线索

V2-063587 | MATHDB-333493

Existence of a Bartnik minimal mass extension via intrinsic-flat compactness

摘录 / EXCERPT: Let Ω_0 be a region and let $\mathcal{PM}(\Omega_0)$ denote the class of asymptotically flat extensions containing an isometric image of Ω_0. Bartnik extension compac...

conjecture | open | MathDB 页面 | 原文线索

V2-063588 | MATHDB-333496

Conjecture on negabent monomials of Niho type

摘录 / EXCERPT: Negabent Niho-exponent conjecture. The function ${{\rm Tr}}_1^n(\alpha x^d)$ is negabent if and only if one of the following conditions holds:

conjecture | open | MathDB 页面 | 原文线索

V2-063589 | MATHDB-333497

Cross-confluence of reduction for Artin-Tits monoids

摘录 / EXCERPT: Let M be an Artin-Tits monoid and let $\mathcal{R}_{M}$ be its reduction system on multifractions. Two multifractions are right reducts of the same multifraction if they are obta...

conjecture | open | MathDB 页面 | 原文线索

V2-063590 | MATHDB-333498

Uniform cross-confluence of reduction for Artin-Tits monoids

摘录 / EXCERPT: Let M be an Artin-Tits monoid and let $\mathcal{R}_{M}$ be its reduction system on multifractions. The system is uniformly cross-confluent if there is a map ∇ from multifr...

conjecture | open | MathDB 页面 | 原文线索

V2-063591 | MATHDB-333499

Tame reduction to the trivial multifraction for Artin-Tits monoids

摘录 / EXCERPT: Let M be an Artin-Tits monoid, let $\mathcal{F}_{M}$ be the set of multifractions over M, and let $\operatorname{red}_{\mathrm{t}}$ denote tame reduction. A multifraction is un...

conjecture | open | MathDB 页面 | 原文线索

V2-063592 | MATHDB-333500

The HL-clan-reversal conjecture for skew-symmetric matrices

摘录 / EXCERPT: Let A and B be $n\times n$ skew-symmetric real matrices. Two such matrices have equal corresponding principal minors of all orders if, for every subset $X\subseteq [n]$, the pr...

conjecture | open | MathDB 页面 | 原文线索

V2-063593 | MATHDB-333501

Positive gap above two-thirds for modularity of random cubic graphs

摘录 / EXCERPT: Let $G_{n,3}$ be a random cubic graph on n vertices, and let $q^*(G_{n,3})$ denote its maximum modularity. Positive-gap conjecture for random cubic graphs. There exists a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-063594 | MATHDB-333505

Stability conjecture for polarized chiral rings and SCFTs

摘录 / EXCERPT: Let $\mathcal{R}$ be a chiral ring equipped with a polarization ζ, so that $({\mathcal{R}},\zeta)$ is a polarized chiral ring. A polarized chiral ring is stable if it has no...

conjecture | open | MathDB 页面 | 原文线索

V2-063595 | MATHDB-333507

The global unramified geometric Langlands equivalence

摘录 / EXCERPT: Geometric Langlands conjecture. There is a canonically defined equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-063596 | MATHDB-333514

Generalized Gessel's conjecture for two-sided Eulerian polynomials

摘录 / EXCERPT: Let W be a finite Coxeter group of rank n, and let ... be its two-sided Eulerian polynomial. Generalized Gessel's conjecture. There exist nonnegative integers $\gamma_{a,b}^W$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-063597 | MATHDB-333515

Blažek-Koman conjecture for the k-page book crossing number of K_n

摘录 / EXCERPT: Blažek-Koman conjecture. For any positive integers k and n,

conjecture | open | MathDB 页面 | 原文线索

V2-063598 | MATHDB-333525

Modified cluster-expansion limit conjecture for proportional permanents

摘录 / EXCERPT: Modified cluster-expansion limit conjecture. Using E_1, E_2, or E, the following limit exists:

conjecture | open | MathDB 页面 | 原文线索

V2-063599 | MATHDB-333529

Holomorphic differential locus conjecture

摘录 / EXCERPT: Let $(a_1,\ldots,a_n)$ be a partition of $2g-2$ with non-negative parts. For sufficiently large r, let $\mathcal{W}_{g; a_1, \dots, a_n}(r)=r^{g-1}W^r_{g,n}(a_1, \dots, a_n)$ be...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-063600 | MATHDB-333542

Kouider-Lonc's balanced path-decomposition conjecture

摘录 / EXCERPT: Kouider-Lonc's conjecture. If G is 2ℓ-regular, then G admits a balanced P_ℓ-decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-063601 | MATHDB-333544

Oguiso's rational-curve conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Oguiso's conjecture. If X has a Cartier divisor D with nef-boundary class, then X contains a rational curve.

conjecture | open | MathDB 页面 | 原文线索

V2-063602 | MATHDB-333555

Large-positive-energy asymptotic behavior conjecture for the Novikov-Veselov equation

摘录 / EXCERPT: Let $E\gg 1$. A large-positive-energy behavior conjecture asserts that the KdV soliton for small a is stable under NV dynamics, whereas the KdV soliton for large a is unstable...

conjecture | open | MathDB 页面 | 原文线索

V2-063603 | MATHDB-333556

Large-negative-energy asymptotic behavior conjecture for the Novikov-Veselov equation

摘录 / EXCERPT: Let $E\ll -1$. A large-negative-energy behavior conjecture asserts that the KdV soliton is stable under NV dynamics and that localized initial data will be radiated away to infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-063604 | MATHDB-333557

Small-energy stability and self-similar blow-up conjecture for the Novikov-Veselov equation

摘录 / EXCERPT: Let E satisfy $|E|\ll 10$. A small-energy blow-up conjecture asserts that the KdV soliton for small a is stable under NV dynamics, while the KdV soliton for large a is unstab...

conjecture | open | MathDB 页面 | 原文线索

V2-063605 | MATHDB-333558

Jeker-Herzig-Breuil conjecture on extensions of parabolically induced representations

摘录 / EXCERPT: Let $F/\mathbb{Q}_p$ and $k/\mathbb{F}_p$ be finite extensions. Let $\mathbf{G}$ be a split connected reductive F-group with connected centre and simply connected derived subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-063606 | MATHDB-333559

Conjecture on derived ordinary parts of a Bruhat-cell compact induction

摘录 / EXCERPT: Let W be the Weyl group of $(\mathbf{G},\mathbf{S})$, let $I,J\subseteq\Delta$, let σ be a locally admissible smooth representation of L_I over A, let...

conjecture | open | MathDB 页面 | 原文线索

V2-063607 | MATHDB-333560

Conjecture on the natural morphism for derived ordinary parts

摘录 / EXCERPT: Let $I,J\subseteq\Delta$, let σ be a locally admissible smooth representation of L_I over A, let ${}^Iw^J\in{}^IW^J$, and let $n\in\mathbb{N}$. Let $d_{{}^Iw^J}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-063608 | MATHDB-333573

Diaconis-Perlman unique crossing conjecture for weighted gamma sums

摘录 / EXCERPT: Diaconis-Perlman unique crossing conjecture. If $\eta\prec\theta$, but θ is not a permutation of η, then $F_\eta(x)-F_\theta(x)$ changes signs exactly once, from $-$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-063609 | MATHDB-333575

Existence of exceptional polynomial bases with equidistributed random zeros

摘录 / EXCERPT: Exceptional-basis conjecture. It is possible to construct such sets E and sequences of random variables such that, for each of these polynomial bases, some subsequence of zero co...

conjecture | open | MathDB 页面 | 原文线索

V2-063610 | MATHDB-333581

Compatibility of characteristic cycles with proper push-forward

摘录 / EXCERPT: Characteristic-cycle push-forward conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-063611 | MATHDB-333590

Conjectured construction of solutions to the constrained Hamilton-Jacobi problem

摘录 / EXCERPT: Constrained-problem construction conjecture. First solve the unconstrained problem using the formula above. Then, whenever $\min \tilde u\leq 0$, replace $\tilde u$ by $v^2/2$ on t...

conjecture | open | MathDB 页面 | 原文线索

V2-063612 | MATHDB-333596

Equality conjecture for local Petrovskii lacuna counts

摘录 / EXCERPT: Equality conjecture. In every cell of Table 1, except possibly the cell corresponding to X_9^2, each inequality can be replaced by an equality.

conjecture | open | MathDB 页面 | 原文线索

V2-063613 | MATHDB-333599

Approximate projective homogeneity conjecture for the pseudo-circle

摘录 / EXCERPT: Pseudo-circle projective-homogeneity conjecture. For every $\epsilon>0$, there exists a local homeomorphism φ of C onto itself such that, for every $x\in C$,

conjecture | open | MathDB 页面 | 原文线索

V2-063614 | MATHDB-333600

Approximate projective homogeneity conjecture for pseudo-solenoids

摘录 / EXCERPT: Pseudo-solenoid projective-homogeneity conjecture. For every $\epsilon>0$, there exists a homeomorphism φ of C onto itself such that, for every $x\in C$,

conjecture | open | MathDB 页面 | 原文线索

V2-063615 | MATHDB-333605

Equidistribution and separation characterization of logarithmic-energy optimality

摘录 / EXCERPT: Let $D(\omega_N)$ denote the spherical cap discrepancy of a sequence of configurations $\{\omega_N\}_{N=2}^{\infty}\subset\mathbb{S}^2$. Call the sequence well-separated when its p...

conjecture | open | MathDB 页面 | 原文线索

V2-063616 | MATHDB-333606

Triangular-lattice expansion conjecture for hypersingular Riesz energy

摘录 / EXCERPT: Hypersingular triangular-lattice expansion conjecture. For $2<s<4$,

conjecture | open | MathDB 页面 | 原文线索

V2-063617 | MATHDB-333607

Logarithmic and Coulomb point-set optimality conjecture

摘录 / EXCERPT: Let $E_s(\omega_N)$ denote the Riesz s-energy of a configuration $\omega_N\subset\mathbb{S}^2$, let $E_{\log}(\omega_N)$ denote its logarithmic energy, and let $\omega_N^{\log}$...

conjecture | open | MathDB 页面 | 原文线索

V2-063618 | MATHDB-333612

McKay-Wanless conjecture on intercalates in random Latin squares

摘录 / EXCERPT: McKay-Wanless conjecture. The displayed concentration holds for every fixed $\varepsilon>0$. This predicts that a uniformly random Latin square has approximately $n^{2}/4$ intercal...

conjecture | open | MathDB 页面 | 原文线索

V2-063619 | MATHDB-333613

Linial-Luria discrepancy conjecture for Latin squares

摘录 / EXCERPT: Linial-Luria's conjecture. There exist arbitrarily large Latin squares L such that, for every box $T=I\times X\times Q$,

conjecture | open | MathDB 页面 | 原文线索

V2-063620 | MATHDB-333618

Shimura-Taniyama conjecture for elliptic curves over totally real fields

摘录 / EXCERPT: Shimura-Taniyama conjecture. Any elliptic curve over K is modular.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-063621 | MATHDB-333631

Enumerative interpretation of the KMK transform coefficients

摘录 / EXCERPT: Enumerative interpretation conjecture. The functions $\Phi_g(c)$ are generating functions for some family of branched coverings of the sphere, and the coefficients $\theta_g(k)$ en...

conjecture | open | MathDB 页面 | 原文线索

V2-063622 | MATHDB-333632

Beatson-zu Castell-Xu positivity conjecture for Gegenbauer integrals

摘录 / EXCERPT: Let $[? Let $[?

conjecture | open | MathDB 页面 | 原文线索

V2-063623 | MATHDB-333642

Loop-cluster metric conjecture for the continuum loop ensemble

摘录 / EXCERPT: Loop-cluster metric conjecture. There is a metric on the clusters of $\mathcal{L}_{g,1/2}$ inside Σ, measurable with respect to $\mathcal{L}_{g,1/2}$, obtained as the limit...

conjecture | open | MathDB 页面 | 原文线索

V2-063624 | MATHDB-333647

The effective-divisor spanning conjecture for moduli spaces on a quadric surface

摘录 / EXCERPT: Effective-divisor spanning conjecture. The method laid out in this paper produces a set of effective divisors spanning the effective cone of $M(\xi)$ for every ξ above Rudakov'...

conjecture | open | MathDB 页面 | 原文线索

V2-063625 | MATHDB-333651

Local non-rigidity conjecture for Clifford-Klein forms of nilpotent Lie groups

摘录 / EXCERPT: Let G be a 1-connected nilpotent Lie group, $H \subset G$ a closed subgroup, and $\Gamma \subset G$ a non-trivial discrete subgroup. Let $\mathcal{R}(\Gamma,G,H)$ denote the spac...

conjecture | open | MathDB 页面 | 原文线索

V2-063626 | MATHDB-333652

The vanishing Chern class conjecture for curve-semistable Higgs bundles

摘录 / EXCERPT: Vanishing Chern class conjecture. If, for every morphism $f\colon C\longrightarrow X$ from a smooth irreducible projective curve C, the pulled-back Higgs bundle $f^*\mathfrak E$...

conjecture | open | MathDB 页面 | 原文线索

V2-063627 | MATHDB-333656

The conjecture relating models of path and Sigma type theory to finitely complete infinity-categories

摘录 / EXCERPT: Models of path and Σ type theory conjecture. The $(\infty,1)$-category presented by the model category $T_\Sigma\text{-}\mathbf{Mod}$ is equivalent to the $(\infty,1)$-categ...

conjecture | open | MathDB 页面 | 原文线索

V2-063628 | MATHDB-333659

Bethe Ansatz in a generic composition sector

摘录 / EXCERPT: Let $\boldsymbol{m}=(m_1,\ldots,m_N)$ be a composition of M with length N, split $\boldsymbol{v}=(v_1,\ldots,v_M)$ into consecutive vectors $\boldsymbol{w}_j^{\boldsymbol{m}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-063629 | MATHDB-333660

Factorisation property for Bethe Ansatz polynomials

摘录 / EXCERPT: Let $\boldsymbol{m}=(m_1,\ldots,m_N)$ be a composition of M, let $\boldsymbol{w}_k^{\boldsymbol{m}}$ be its consecutive momentum blocks, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063630 | MATHDB-333661

Explicit solution for the one-block Bethe Ansatz polynomials

摘录 / EXCERPT: Let $Q^{(M)}(\boldsymbol{v})$ be the polynomial associated with the one-block composition (M), and let β be the coupling appearing in the equations of the source. Explicit...

conjecture | open | MathDB 页面 | 原文线索

V2-063631 | MATHDB-333662

3-reducibility of the explicit Bethe Ansatz equations

摘录 / EXCERPT: Assume $M\geq3$, and let $Q^{(M)}(\boldsymbol{v})$ be the explicit polynomials defined in the preceding conjecture. A polynomial is 3-reducible when it can be written as ... with eac...

conjecture | open | MathDB 页面 | 原文线索

V2-063632 | MATHDB-333673

Stability conjecture for regular polling policies

摘录 / EXCERPT: Let ${\cal P}$ be a regular polling policy, and let λ_1 and λ_2 be the intensities of the arrival processes in lanes 1 and 2, respectively. Let s be the servi...

conjecture | open | MathDB 页面 | 原文线索

V2-063633 | MATHDB-333674

Bayer-Macrì-Toda generalized Bogomolov inequality for tilt-stable objects

摘录 / EXCERPT: Bayer-Macrì-Toda conjecture. For any $\nu_{\alpha,\beta}$-stable object $E\in\operatorname{Coh}^{\beta}(X)$ with $\nu_{\alpha,\beta}(E)=0$, the inequality

conjecture | open | MathDB 页面 | 原文线索

V2-063634 | MATHDB-333678

Residue mirror symmetry for Grassmannian quasimaps and gauged linear sigma models

摘录 / EXCERPT: Let G act on a G-space V, and let $W\subset V$ be the zero locus of G-semi-invariant polynomials f_i, for $i=1,\ldots,r$. For $P\in\mathbb{C}[\mathfrak{t}]^{\mathcal{W}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-063635 | MATHDB-333680

Quasimap I-function conjecture for the Gromov-Witten Lagrangian cone

摘录 / EXCERPT: Let $E\sslash \textbf{G}$ be the relevant GIT quotient, let $I(\mathbf{t},z)$ be the quasimap I-function, let $P^Q(\mathbf{t},z)$ be the corresponding quasimap input, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063636 | MATHDB-333681

Fulton's conjecture on F-nef divisors on the moduli space of pointed rational curves

摘录 / EXCERPT: Let $\overline{M}_{0,n}$ be the moduli space of stable n-pointed rational curves, and let δ_S denote the class of the boundary divisor indexed by S, with...

conjecture | open | MathDB 页面 | 原文线索

V2-063637 | MATHDB-333683

Conjecture on minimizers of the body-centred-orthorhombic theta energy

摘录 / EXCERPT: Let $\alpha>0$ and $t>0$, and let $\tilde{E}_t(y;\alpha)$ be the reduced theta energy for the body-centred-orthorhombic family, with $t_0(\alpha)$ defined by the preceding lemma. C...

conjecture | open | MathDB 页面 | 原文线索

V2-063638 | MATHDB-333684

Tautologicality of reduced stable-map classes for lattice-polarized K3 surfaces

摘录 / EXCERPT: Let $\pi_\Lambda:\mathcal{X}_\Lambda\to\mathcal{M}_\Lambda$ be the universal lattice-polarized K3 surface, let $L\in\Lambda$ be an admissible class, and let ... be the evaluation map...

conjecture | solved | MathDB 页面 | 原文线索

V2-063639 | MATHDB-333685

The forest and bipartite-complement boundary conjecture for bipartite graph classes

摘录 / EXCERPT: Forest boundary conjecture. For any tree T, the class of $\{T,\overline{T^b}\}$-free bipartite graphs is at most factorial.

conjecture | open | MathDB 页面 | 原文线索

V2-063640 | MATHDB-333690

Landau-Ginzburg Hodge-number equality for Fano-type models

摘录 / EXCERPT: Let (Y,w) be a Landau-Ginzburg model of Fano type, with Landau-Ginzburg Hodge numbers $h^{p,q}(Y,w)$, $f^{p,q}(Y,w)$, and $i^{p,q}(Y,w)$ as defined in the paper. Landau-Ginzburg...

conjecture | open | MathDB 页面 | 原文线索

V2-063641 | MATHDB-333695

Classification conjecture for planar graphs with exactly two disparate faces

摘录 / EXCERPT: Let the families listed above be the fifteen families of planar graphs constructed in the paper with exactly two disparate faces. Classification conjecture. These families are the...

conjecture | open | MathDB 页面 | 原文线索

V2-063642 | MATHDB-333699

Nowhere-dense characterization of modeling limits

摘录 / EXCERPT: Let $\mathcal C$ be a monotone class of finite graphs, meaning that it is closed under taking subgraphs. A class is nowhere dense in the standard sparse-graph sense. The class...

conjecture | open | MathDB 页面 | 原文线索

V2-063643 | MATHDB-333713

Mixture classification conjecture for stationary gap distributions

摘录 / EXCERPT: Mixture classification conjecture. Under these conditions, every stationary gap distribution of an infinite system of competing Brownian particles is equal to π_ρ for some p...

conjecture | open | MathDB 页面 | 原文线索

V2-063644 | MATHDB-333714

Asymptotic velocity conjecture for rank particles in competing Brownian systems

摘录 / EXCERPT: Fix drift parameters $(g_n)_{n\geq 1}$ and a parameter a as in Theorem. Consider an infinite system of competing Brownian particles started from ... and with the initial gaps distr...

conjecture | open | MathDB 页面 | 原文线索

V2-063645 | MATHDB-333717

Bouchard-Hertz-Desaulniers deficiency bound for graphs

摘录 / EXCERPT: Bouchard-Hertz-Desaulniers conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063646 | MATHDB-333718

Deficiency bounds for complete graphs of power-of-two order

摘录 / EXCERPT: Complete-graph deficiency conjecture. For any $q\in\mathbb{Z}_+$,

conjecture | open | MathDB 页面 | 原文线索

V2-063647 | MATHDB-333719

The degree-associated edge-reconstruction conjecture for trees

摘录 / EXCERPT: Let T be a tree, and let $dern(T)$ denote the minimum number of degree-associated edge-cards needed to determine T up to isomorphism. Degree-associated edge-reconstruction conj...

conjecture | open | MathDB 页面 | 原文线索

V2-063648 | MATHDB-333720

The component-star conjecture for disconnected graphs

摘录 / EXCERPT: Let G be a disconnected graph whose components are all isomorphic to a graph H, and let $ern(G)$ be the minimum number of edge-cards needed to determine G up to isomorphism....

conjecture | open | MathDB 页面 | 原文线索

V2-063649 | MATHDB-333721

The infinite-class conjecture for trees with edge-reconstruction number three

摘录 / EXCERPT: Let $ern(T)$ denote the minimum number of edge-cards needed to determine a tree T up to isomorphism. The relevant infinite families are the paths on an odd number of vertices, th...

conjecture | open | MathDB 页面 | 原文线索

V2-063650 | MATHDB-333722

The degree-associated edge-reconstruction conjecture for disconnected equal-component graphs

摘录 / EXCERPT: Let G be as in Theorem 4.2: a disconnected graph whose non-trivial components are all isomorphic to a graph H. Let $dern(G)$ denote the minimum number of degree-associated edge...

conjecture | open | MathDB 页面 | 原文线索

V2-063651 | MATHDB-333726

The period-extended graph permanent conjecture for 4-point phi^4 graphs

摘录 / EXCERPT: Let a 4-point ϕ^4 graph be a graph of the type considered in the paper, with period and extended graph permanent as defined there. Period-extended graph permanent conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063652 | MATHDB-333727

The period-extended graph permanent conjecture for phi^4 graphs

摘录 / EXCERPT: Let a ϕ^4 graph be a graph in the class considered in the paper, and let its period and extended graph permanent be the associated graph invariants. Period-extended graph perm...

conjecture | open | MathDB 页面 | 原文线索

V2-063653 | MATHDB-333728

Kotzig's cyclic tree decomposition conjecture

摘录 / EXCERPT: Let T be a tree with $n+1$ vertices, and identify the vertices of K_{2n+1} with the integers modulo $2n+1$. A cyclic decomposition into copies of T consists of the cyclic shi...

conjecture | open | MathDB 页面 | 原文线索

V2-063654 | MATHDB-333731

Hibi's unimodality conjecture for Gorenstein lattice polytopes

摘录 / EXCERPT: Let $P\subseteq\mathbb{R}^n$ be a full-dimensional lattice polytope. It is Gorenstein when its $h^\ast$-polynomial is symmetric, and it is IDP when every positive integer m and e...

conjecture | refuted | MathDB 页面 | 原文线索

V2-063655 | MATHDB-333732

Unimodality conjecture for reflexive IDP simplices supported on three integers

摘录 / EXCERPT: Let $\Delta_{(1,q)}$ be a reflexive IDP lattice simplex of the family considered in the source, and suppose that its defining data q are supported on three integers. Three-suppor...

conjecture | open | MathDB 页面 | 原文线索

V2-063656 | MATHDB-333749

Positivity conjecture for Jack shifted functions and generalized Jack characters

摘录 / EXCERPT: Let $\lambda=\mathbf{r}^{\mathbf{p}}$ be a Young diagram in multirectangular coordinates. For a partition μ, let $J^{\star,(\alpha)}_\mu$ be the shifted Jack function and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063657 | MATHDB-333750

Isomorphism conjecture for the KSBA and semitoric compactifications of $D_{1,6}$-polarized Enriques surfaces

摘录 / EXCERPT: Let $\overline{M}_{D_{1,6}}^\nu$ be the KSBA compactification of the moduli space of $D_{1,6}$-polarized Enriques surfaces, and let $\overline{\mathcal{D}/\Gamma}^\Sigma$ be the se...

conjecture | open | MathDB 页面 | 原文线索

V2-063658 | MATHDB-333751

Bilinear relations for q-deformed conformal blocks

摘录 / EXCERPT: Bilinear relations. The q-deformed conformal blocks satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-063659 | MATHDB-333752

Bilinear equation for the q-deformed Painleve tau function

摘录 / EXCERPT: Bilinear equation. The function $\mathcal{T}(u,s;q|Z)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063660 | MATHDB-333754

Bilinear relation for generic q-deformed conformal blocks

摘录 / EXCERPT: Generic bilinear relation.

conjecture | open | MathDB 页面 | 原文线索

V2-063661 | MATHDB-333755

The conjectural formula for the Painlevé III(D_8) Whittaker coefficient

摘录 / EXCERPT: Coefficient formula. The coefficient satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063662 | MATHDB-333760

True minimum distance conjecture for the even-like LCD BCH code

摘录 / EXCERPT: Let q be odd and $m \geq 2$. For an integer t with $1 \leq t \leq \bar m$, let $\delta=(q^t-1)/2$ and let ${\mathcal{C}}_{(q,n,2\delta,\frac{n}{2}-\delta+1)}$ be the code from...

conjecture | open | MathDB 页面 | 原文线索

V2-063663 | MATHDB-333768

Rudakov's conjecture on rigid sheaves and weak exceptional collections on del Pezzo surfaces

摘录 / EXCERPT: Rudakov's conjecture. Any rigid sheaf on X is the direct sum of weak exceptional sheaves, and any weak exceptional sheaf can be included into a full weak exceptional collection.

conjecture | open | MathDB 页面 | 原文线索

V2-063664 | MATHDB-333786

The IDP conjecture for lecture hall polytopes

摘录 / EXCERPT: Let $\boldsymbol{s}=(s_1,\ldots,s_d)$ be an integer sequence, and let $P_d^{(\boldsymbol{s})}$ denote the lecture hall polytope associated to $\boldsymbol{s}$. A lattice polytope i...

conjecture | open | MathDB 页面 | 原文线索

V2-063665 | MATHDB-333787

The unimodular triangulation conjecture for lecture hall polytopes

摘录 / EXCERPT: Let $\boldsymbol{s}=(s_1,\ldots,s_d)$ be an integer sequence, and let $P_d^{(\boldsymbol{s})}$ denote the lecture hall polytope associated to $\boldsymbol{s}$. A triangulation is u...

conjecture | open | MathDB 页面 | 原文线索

V2-063666 | MATHDB-333789

Yakoubov's generating-function conjecture for 231-avoiding linear extensions of alpha-labeled combs

摘录 / EXCERPT: Yakoubov's conjecture. The generating function of these numbers is

conjecture | open | MathDB 页面 | 原文线索

V2-063667 | MATHDB-333790

Yakoubov's conjecture on doubly pattern-avoiding linear extensions of alpha-labeled combs

摘录 / EXCERPT: Yakoubov's conjecture. For every integer $s\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-063668 | MATHDB-333792

Planarity conjecture for golden tilings with nondegenerate weak alternation

摘录 / EXCERPT: A golden tiling is a tiling by the rhombohedra associated with the icosahedral construction, and nondegenerate weak alternation means weak alternation with no infinite run of rhomb...

conjecture | open | MathDB 页面 | 原文线索

V2-063669 | MATHDB-333814

Bentness criterion for binomial Boolean functions in degree four times an odd number

摘录 / EXCERPT: Bentness conjecture. The function $f_{a,b}$ is bent if and only if $K_{n/2}(a)=4$.

conjecture | open | MathDB 页面 | 原文线索

V2-063670 | MATHDB-333815

Conjectural Gauss-sum formula for the Walsh transform of binomial Boolean functions

摘录 / EXCERPT: Conjectural Gauss-sum formula. There exists a Boolean function $h_{a,b}(omega)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063671 | MATHDB-333820

A strengthening of the large-exponent zero-sum subsequence bound

摘录 / EXCERPT: Let G be a finite abelian p-group with $\exp(G)=n$ and ${\mathsf{D}}(G)\leq 2n-1$. Let $\ell\in [1,{\mathsf{D}}(G)+1-n]$, and let S be a sequence over G of length at least...

conjecture | open | MathDB 页面 | 原文线索

V2-063672 | MATHDB-333826

Bounded diameter arboricity plus-one conjecture

摘录 / EXCERPT: Bounded diameter arboricity plus-one conjecture. The class $\mathcal{A}_k$ has bounded diameter arboricity $k+1$, that is,

conjecture | open | MathDB 页面 | 原文线索

V2-063673 | MATHDB-333827

Forest-and-bounded-diameter-forest decomposition conjecture

摘录 / EXCERPT: Forest-and-bounded-diameter-forest decomposition conjecture. There exists a natural number $f(d)$ such that G can be partitioned into two forests, each of whose components has di...

conjecture | open | MathDB 页面 | 原文线索

V2-063674 | MATHDB-333828

Fractional arboricity bounded-diameter conjecture

摘录 / EXCERPT: Fractional arboricity bounded-diameter conjecture. There exists $d(k,\varepsilon)$ such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-063675 | MATHDB-333836

Monotonicity conjecture for infinite-order hyperbolicity-preserving operators

摘录 / EXCERPT: Monotonicity conjecture. If T is not monotone, then T is not hyperbolicity preserving.

conjecture | open | MathDB 页面 | 原文线索

V2-063676 | MATHDB-333837

All-level torsion growth conjecture for arithmetic groups

摘录 / EXCERPT: Let Γ be an arithmetic group, let Γ_k range over congruence subgroups of Γ of increasing level, and let δ denote the deficiency of the associated real s...

conjecture | open | MathDB 页面 | 原文线索

V2-063677 | MATHDB-333840

Icosahedral optimality conjecture for six unit vectors in three-dimensional real space

摘录 / EXCERPT: Let $v_1,dots,v_6$ be unit vectors in $\mathbf{R}^3$, and suppose one of the vectors lies at the north pole. An optimal configuration maximizes the minimum absolute determinant amo...

conjecture | open | MathDB 页面 | 原文线索

V2-063678 | MATHDB-333864

Nonattainment conjecture for Néel-wall energies at winding numbers d and d plus alpha over pi

摘录 / EXCERPT: Let $h\in[0,1)$, let $d\in\mathbb{N}=\{1,2,\dots\}$, and let $\mathcal{E}_h(w)$ denote the infimum of the energy among profiles with winding number w. Nonattainment conjecture. N...

conjecture | open | MathDB 页面 | 原文线索

V2-063679 | MATHDB-333865

Existence conjecture for Néel-wall energies at winding number d minus alpha over pi near h equals 1

摘录 / EXCERPT: Let $d\in\mathbb{N}$, and let $\mathcal{E}_h(w)$ denote the infimum of the energy among profiles with winding number w. Existence conjecture. There exists $H\in(0,1)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-063680 | MATHDB-333866

Low-h nonattainment conjecture for Néel-wall energies at winding number d minus alpha over pi

摘录 / EXCERPT: Let $d\in\mathbb{N}\setminus\{1\}$, and let $\mathcal{E}_h(w)$ denote the infimum of the energy among profiles with winding number w. Low-h nonattainment conjecture. There exis...

conjecture | open | MathDB 页面 | 原文线索

V2-063681 | MATHDB-333878

Stability conjecture for intersecting families in $PSL(2,q)$

摘录 / EXCERPT: Stability conjecture. There exists $\delta>0$ such that, whenever S is an intersecting family in $PSL(2,q)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-063682 | MATHDB-333890

The broken-line conjecture for holomorphic cylinder counts

摘录 / EXCERPT: Let B be the base of the log Calabi-Yau surface, and let $\beta\in\operatorname{NE}(Y)$. Let γ be a broken line in B for the canonical scattering diagram, and choose a n...

conjecture | open | MathDB 页面 | 原文线索

V2-063683 | MATHDB-333894

Bj\orner--Swartz conjecture on dual WLP for doubly Cohen--Macaulay complexes

摘录 / EXCERPT: Bjorner--Swartz conjecture. Every doubly Cohen--Macaulay simplicial complex has the dual WLP.

conjecture | open | MathDB 页面 | 原文线索

V2-063684 | MATHDB-333899

Higher-dimensional framed operadic formality conjecture

摘录 / EXCERPT: Higher-dimensional framed compatibility conjecture. If the framed little n-disks operad $\mathtt{fe}_{n}$ is formal, then the pair...

conjecture | open | MathDB 页面 | 原文线索

V2-063685 | MATHDB-333909

Ruas's conjecture on Whitney equisingularity of surface families

摘录 / EXCERPT: Let $f:(\mathbb{C}^2,0)\rightarrow(\mathbb{C}^3,0)$ be a finitely determined map germ, and let $F:(\mathbb{C}^2\times\mathbb{C},0)\rightarrow(\mathbb{C}^3\times\mathbb{C},0)$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-063686 | MATHDB-333915

Asymptotic polynomial conjecture for numerical semigroups generated by consecutive powers

摘录 / EXCERPT: Let R_n^k, with $n=T_km+j$, be a numerical semigroup given by its minimal relations on the residue class of n modulo T_k: ... If $k=2q$, the polynomials $E_{ij}^{(k)}(m)$ have...

conjecture | open | MathDB 页面 | 原文线索

V2-063687 | MATHDB-333918

Jackson-Sokal's density conjecture for Tutte-polynomial zeros

摘录 / EXCERPT: Jackson-Sokal's density conjecture. The zeros of the Tutte polynomials of graphs are dense in the following regions: (a) $q<0$ and $v<-2$; (b) $q<0$ and $0<v<-q/2$; (c)...

conjecture | open | MathDB 页面 | 原文线索

V2-063688 | MATHDB-333926

The non-vanishing conjecture for projective log canonical pairs

摘录 / EXCERPT: Non-vanishing conjecture. The stated effective representative exists.

conjecture | open | MathDB 页面 | 原文线索

V2-063689 | MATHDB-333938

Simplicity conjecture for the zeros of the adjacency function

摘录 / EXCERPT: For every $\omega>0$ and every integer $l\geq0$, let ... where ξ_l is the functional equation associated with entire Heun solutions and adjacencies. Simplicity conjecture. The ze...

conjecture | open | MathDB 页面 | 原文线索

V2-063690 | MATHDB-333947

The bilinear-equation conjecture for the Boussinesq polynomials

摘录 / EXCERPT: Let $\Theta_n^{\pm}(x,t)$ be the polynomials defined by ... where $P_n(x,t)$ and $Q_n(x,t)$ are the polynomials in the generalised rational solution of the Boussinesq equation. Bilin...

conjecture | open | MathDB 页面 | 原文线索

V2-063691 | MATHDB-333961

The Lang-Trotter primitive-point density conjecture for elliptic curves

摘录 / EXCERPT: Lang-Trotter conjecture. The limit exists, is positive, and P is a fixed primitive point for E in the indicated sense. This concerns primitive reductions of rational points in...

conjecture | open | MathDB 页面 | 原文线索

V2-063692 | MATHDB-333963

Nonvanishing conjecture for the mirror-quintic Wronskian pullback

摘录 / EXCERPT: Let ... and ... Define the Wronskian ... and let $F=W\circ\lambda$, where ... for q in the punctured unit disc $\Delta^*=\{q\in\mathbb C:0<|q|<1\}$. Wronskian nonvanishing conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-063693 | MATHDB-333971

Lin-Zeng's Jacobi-Stirling transformations conjecture

摘录 / EXCERPT: Lin-Zeng's conjecture. The Jacobi-Stirling transformations of the two kinds preserve log-convexity for $z=0,1$. This conjecture concerns preservation of log-convexity by the transf...

conjecture | open | MathDB 页面 | 原文线索

V2-063694 | MATHDB-333982

The integer decomposition property for nonempty $B_n(\Pi)$

摘录 / EXCERPT: Let $\mathfrak{S}_3$ denote the symmetric group on three letters, let $\Pi\subseteq\mathfrak{S}_3$, and let $B_n(\Pi)$ be the associated polytope. A conjecture on the integer decom...

conjecture | open | MathDB 页面 | 原文线索

V2-063695 | MATHDB-333989

Optimal dimension reduction bound for Euclidean point sets

摘录 / EXCERPT: Let $f(n,d,\varepsilon)$ denote the smallest m such that every n-point subset of ℓ_2^d can be embedded into ℓ_2^m with distortion at most $1+\varepsilon$. Dimension-r...

conjecture | open | MathDB 页面 | 原文线索

V2-063696 | MATHDB-334001

Kapovich-Kleiner conjecture for Sierpiński carpet boundaries

摘录 / EXCERPT: Let G be a hyperbolic group, and let S be the Sierpiński carpet. Let ∂G denote the boundary of G, and let $\mathbb{H}^3$ denote hyperbolic 3-space. Kapovich-Kleiner...

conjecture | open | MathDB 页面 | 原文线索

V2-063697 | MATHDB-334007

Bertolini-Darmon anticyclotomic formula for the theta element and Heegner class

摘录 / EXCERPT: Consider the setup of the anticyclotomic family in the paper and let $\widetilde r=\dim A^{\dagger}(K)_{\mathbf Q}$. If $\widetilde\varepsilon=+1$, then $\widetilde r$ is even and...

conjecture | open | MathDB 页面 | 原文线索

V2-063698 | MATHDB-334010

Regular unimodular triangulation conjecture for lecture hall polytopes

摘录 / EXCERPT: Let P be a naturally labeled ranked poset, let ρ be its rank function, and set $s=\rho+1$. The associated lecture hall polytope is $O(P,s)$. Regular unimodular triangulation...

conjecture | open | MathDB 页面 | 原文线索

V2-063699 | MATHDB-334013

The odd-plex separation conjecture for even-order Latin squares

摘录 / EXCERPT: Let k be an odd positive integer. A latin square of order n is an $n\times n$ array in which each of n symbols occurs exactly once in every row and column. A k-plex is a se...

conjecture | open | MathDB 页面 | 原文线索

V2-063700 | MATHDB-334016

Random-initialization success conjecture for alternating projections in phase retrieval

摘录 / EXCERPT: Let $x_0\in\mathbb{C}^n$ be the target vector, let $A\in\mathbb{C}^{m\times n}$ be the sensing matrix, and let alternating projections reconstruct x_0 from the phaseless measurem...

conjecture | open | MathDB 页面 | 原文线索

V2-063701 | MATHDB-334021

Unavoidability of sufficiently deep outbranching balanced binary trees

摘录 / EXCERPT: Balanced binary tree unavoidability conjecture. B_d is unavoidable for sufficiently large d, possibly with $d>1$ sufficient.

conjecture | open | MathDB 页面 | 原文线索

V2-063702 | MATHDB-334026

The optimal resonance lower bound for bounded compactly supported potentials

摘录 / EXCERPT: Let $V\in L^\infty_{\operatorname{comp}}(\mathbb R^n)$ be real-valued and non-zero, with n odd. Let $N_V(r)$ denote the number of resonances in $\{|\lambda|\le r\}$. Optimal reso...

conjecture | open | MathDB 页面 | 原文线索

V2-063703 | MATHDB-334027

The stability conjecture for Kerr-de Sitter quasinormal modes

摘录 / EXCERPT: For Kerr--de Sitter parameters a, Λ, and M satisfying ... and lying in the shaded region specified by the source, quasinormal modes are complex frequencies λ for...

conjecture | open | MathDB 页面 | 原文线索

V2-063704 | MATHDB-334028

The fractal Weyl law for resonance lower bounds

摘录 / EXCERPT: Let $N(\Omega):=N_{P(h)}(\Omega)$ be the resonance counting function, and let δ_E be the trapped-set dimension parameter defined in the source. Fractal Weyl law conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063705 | MATHDB-334035

Pseudo-isomorphism conjecture for the residual Selmer group

摘录 / EXCERPT: Let $epsilon$ be an odd character, let $S_{epsilon}=(\operatorname{Sel}(K,T^{\vee})^{\vee})^{epsilon}$, and define ... A pseudo-isomorphism is an isomorphism up to finite kernel and...

conjecture | open | MathDB 页面 | 原文线索

V2-063706 | MATHDB-334039

The length characterization of covers in the double-affine Bruhat order

摘录 / EXCERPT: Let $W_{\mathcal{T}}$ be the double-affine Weyl monoid, equipped with its Bruhat order $<$ and length function ℓ. Write $x\lhd y$ when y covers x, meaning that $x<y$ and t...

conjecture | open | MathDB 页面 | 原文线索

V2-063707 | MATHDB-334040

The polynomial-count conjecture for double-affine transverse slices

摘录 / EXCERPT: The transverse-slice counting conjecture. There exists a polynomial $R_{x,y}\in\mathbb{Z}[v]$, independent of k, of degree $\ell(y)-\ell(x)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063708 | MATHDB-334041

Deodhar's inequality for the double-affine Bruhat order

摘录 / EXCERPT: Let $W_{\mathcal{T}}$ be the double-affine Weyl monoid, let $\widetilde{\Delta}^+$ denote the positive double-affine real roots, and let ℓ be the length function. Deodhar's in...

conjecture | open | MathDB 页面 | 原文线索

V2-063709 | MATHDB-334044

Star-shaped connectedness of the root hybrid Mandelbrot set

摘录 / EXCERPT: Star-shaped connectedness conjecture. The set of infinitely well-behaved pairs $(0,c_1)$ is star-shaped connected.

conjecture | open | MathDB 页面 | 原文线索

V2-063710 | MATHDB-334045

Connectedness conjecture for the multi-Mandelbrot set

摘录 / EXCERPT: Multi-Mandelbrot connectedness conjecture. The set ${\cal MM}$ is a connected set in $\mathbb{C}^4$.

conjecture | open | MathDB 页面 | 原文线索

V2-063711 | MATHDB-334048

Superadditivity conjecture for propagation speeds

摘录 / EXCERPT: Let b_1 and b_2 be birth rates satisfying the assumptions of the paper, and let $s(b)$ denote the corresponding speed of propagation. Superadditivity conjecture. One has ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-063712 | MATHDB-334049

General maximum-root bound for the mixed characteristic polynomial

摘录 / EXCERPT: Let $M_n(\mathbb{C})^{+}$ denote the positive semidefinite $n\times n$ complex matrices, and let a positive contraction be a positive semidefinite matrix bounded above by the ident...

conjecture | open | MathDB 页面 | 原文线索

V2-063713 | MATHDB-334050

General r-paving bound via mixed characteristic polynomials

摘录 / EXCERPT: Let $M_n(\mathbb{C})^{+}_{1}$ denote the positive semidefinite $n\times n$ complex matrices bounded above by the identity, and let $\chi_r[A]$ be the r-characteristic polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-063714 | MATHDB-334052

The antipodal isoperimetric conjecture for the sphere

摘录 / EXCERPT: Let $n \in \mathbb{N}$ with $n \geq 2$, and let μ denote the n-dimensional Hausdorff measure on $\mathbb{S}^n$. Let $\mathcal{A} \subseteq \mathbb{S}^n$ be open and antipodal...

conjecture | open | MathDB 页面 | 原文线索

V2-063715 | MATHDB-334062

The Bogomolov-type lower bound for essential minima

摘录 / EXCERPT: Let $X\subset G$ be irreducible and not a translate, and let H be the irreducible translate of minimal dimension containing X. Write $\operatorname{codim}_H X$ for the codimens...

conjecture | open | MathDB 页面 | 原文线索

V2-063716 | MATHDB-334068

Polynomiality conjecture for indecomposable parabolic-bundle counts

摘录 / EXCERPT: Polynomiality conjecture. There exists a polynomial $A_{\boldsymbol{\mu};\mathbf{d}}^{\mathbf{m}}(t)\in\mathbb{Z}[t]$ such that for every finite field $\mathbb{F}_q$ and every divi...

conjecture | open | MathDB 页面 | 原文线索

V2-063717 | MATHDB-334069

Deligne's character-valued polynomiality conjecture for parabolic-bundle counts

摘录 / EXCERPT: Deligne's conjecture. There exists a polynomial $A^{{\bf b};\mathbf{m}}_{\boldsymbol{\mu}}(t)\in\mathcal{C}h(S_{\boldsymbol{\mu}})[t]$, with coefficients in the positive cone...

conjecture | open | MathDB 页面 | 原文线索

V2-063718 | MATHDB-334070

Degree-independence conjecture for sums of indecomposable parabolic structures

摘录 / EXCERPT: Degree-independence conjecture. For every integer d, the sum

conjecture | open | MathDB 页面 | 原文线索

V2-063719 | MATHDB-334082

The affine Springer fiber orbital-integral boundedness conjecture

摘录 / EXCERPT: Orbital-integral boundedness conjecture. For any $\gamma\in\mathfrak{g}$,

conjecture | open | MathDB 页面 | 原文线索

V2-063720 | MATHDB-334084

Accumulation conjecture for the wild monodromy of Painlevé V

摘录 / EXCERPT: Accumulation conjecture. The monodromy of P_{VI} in the confluent family possesses limits along these sequences, and the monodromy accumulates toward the family of such limits, d...

conjecture | open | MathDB 页面 | 原文线索

V2-063721 | MATHDB-334085

Greuel-Lossen-Shustin conjecture for general weighted plane configurations

摘录 / EXCERPT: Let $({\mathcal C},\boldsymbol{r})$ be a consistent finite weighted configuration of infinitely near points of $\mathbb{P}^2$, where r_p is the multiplicity assigned to p, and...

conjecture | open | MathDB 页面 | 原文线索

V2-063722 | MATHDB-334094

Splitting conjecture for Euler-Hadamard moment factorization

摘录 / EXCERPT: Let $\mathcal{H}_{2g+1}$ be the set of monic square-free polynomials of degree $2g+1$ over $\mathbb{F}_q[x]$, and define ... Let $P_X(\chi_D)=P_X(\tfrac12,\chi_D)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-063723 | MATHDB-334098

Vanishing conjecture for Whittaker inner integrals on GL_n

摘录 / EXCERPT: Whittaker vanishing conjecture. If J satisfies this dimension equation, then J is zero for every choice of data.

conjecture | open | MathDB 页面 | 原文线索

V2-063724 | MATHDB-334099

Magee-Avila-Gouëzel conjecture for moduli spaces of abelian differentials

摘录 / EXCERPT: Let $\mathcal{M}$ be a connected component of a stratum of abelian differentials, let $\mathcal{M}(q)$ be its level-q lift for $q\geq 2$, and let $\nu_{\mathcal{M}(q)}$ be the co...

conjecture | open | MathDB 页面 | 原文线索

V2-063725 | MATHDB-334101

Zimmer's finite-extension conjecture in the critical dimension

摘录 / EXCERPT: Let G be a higher-rank semisimple Lie group, let Γ be a higher-rank lattice in G, and let $r(G)$ be the root-theoretic dimension defined in the source. Let M be a mani...

conjecture | open | MathDB 页面 | 原文线索

V2-063726 | MATHDB-334104

The dynamic coloring conjecture for planar graphs

摘录 / EXCERPT: Dynamic coloring conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063727 | MATHDB-334106

The converse characterization of central weightings

摘录 / EXCERPT: The converse characterization of central weightings. If there exist constants $\alpha_1,\dots,\alpha_d,\beta$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063728 | MATHDB-334107

Universality classes for centrally weighted small-step models

摘录 / EXCERPT: Let ${\mathcal{S}}$ be one of the 19 small-step lattice-path models with finite non-zero orbit sum, and consider its centrally weighted walk model. Let $Q(1,1;t)$ be the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-063729 | MATHDB-334116

Conjecture on coincidences of ribbon stable Grothendieck polynomials

摘录 / EXCERPT: A ribbon is a connected skew Young diagram containing no 2×2 block. For a ribbon α, let G_α denote its stable Grothendieck polynomial, and let α^* denote...

conjecture | open | MathDB 页面 | 原文线索

V2-063730 | MATHDB-334117

Staircase conjugation conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let $\delta_n=\langle n-1,n-2,\ldots,1\rangle$ be the staircase partition, and let μ be a diagram contained in δ_n. For a skew shape A, write A^T for its transpose....

conjecture | open | MathDB 页面 | 原文线索

V2-063731 | MATHDB-334118

Greatest-element conjecture for ribbons with fixed dimensions

摘录 / EXCERPT: Consider the partial order on G-equivalence classes of skew shapes defined by $A\leq B$ when $a_{A,\nu}\leq a_{B,\nu}$ for every partition ν, equivalently when G_B-G_A is...

conjecture | open | MathDB 页面 | 原文线索

V2-063732 | MATHDB-334119

Greatest-element conjecture for permutations of a ribbon partition

摘录 / EXCERPT: Fix a partition λ, and consider the ribbons whose row lengths are a permutation of the parts of λ, ordered by G positivity: $A\leq B$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-063733 | MATHDB-334120

Conjugation automorphism conjecture for the G-positivity order

摘录 / EXCERPT: For skew shapes A and B, let $A\leq B$ mean that $a_{A,\nu}\leq a_{B,\nu}$ for every partition ν, equivalently that G_B-G_A is G positive. Transposition sends a skew sh...

conjecture | open | MathDB 页面 | 原文线索

V2-063734 | MATHDB-334126

The Planar Conjecture on embedding planar graph metrics into \ell_1

摘录 / EXCERPT: A planar graph metric is a shortest-path metric on a finite planar graph whose edges have arbitrary weights. The Planar Conjecture. Every metric supported on a finite planar graph...

conjecture | open | MathDB 页面 | 原文线索

V2-063735 | MATHDB-334127

The forbidden-minor embedding conjecture for graph metrics

摘录 / EXCERPT: Let L be a finite set of graphs. A graph excludes L as a minor if it contains no member of L as a minor, where minors are obtained by edge contractions, edge deletions, and v...

conjecture | open | MathDB 页面 | 原文线索

V2-063736 | MATHDB-334129

Steinberg's conjecture on 3-coloring planar graphs without 4- or 5-cycles

摘录 / EXCERPT: A planar graph is a graph that can be embedded in the plane. Steinberg's conjecture. Every planar graph with no cycles of length four or five is 3-colorable. Steinberg's conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-063737 | MATHDB-334131

Degeneration at the third page for plane-curve spectral sequences

摘录 / EXCERPT: Degeneration conjecture. The spectral sequences degenerate at the E_3-term for any reduced plane curve $C:f=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-063738 | MATHDB-334132

Third-page stabilization conjecture for plane-curve spectral sequences

摘录 / EXCERPT: Stabilization conjecture. In the spectral sequences of C, one has

conjecture | open | MathDB 页面 | 原文线索

V2-063739 | MATHDB-334133

Bounded stabilization conjecture for plane-curve spectral sequences

摘录 / EXCERPT: Bounded stabilization conjecture. There is a positive integer $q_0\leq 3d+1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063740 | MATHDB-334137

The local-global compatibility conjecture at Taylor-Wiles level

摘录 / EXCERPT: Local-global compatibility conjecture. The action of R_p^{loc} on

conjecture | open | MathDB 页面 | 原文线索

V2-063741 | MATHDB-334141

Scully's refined splitting-pattern conjecture for quadratic forms

摘录 / EXCERPT: Let F be a field of characteristic different from 2, and let p and q be anisotropic quadratic forms over F of dimension at least 2. Let $h(q)$ be the height of the spli...

conjecture | open | MathDB 页面 | 原文线索

V2-063742 | MATHDB-334143

Existence of rank 3 Ulrich bundles on Veronese surfaces of odd degree

摘录 / EXCERPT: Let $(\mathbb{P}^2,dH)$ denote the projective plane polarized by dH, where d is an odd integer and $d\geq 3$. Rank 3 Ulrich-bundle conjecture. There exists a rank 3 Ulrich bu...

conjecture | open | MathDB 页面 | 原文线索

V2-063743 | MATHDB-334144

The Q-graded Kontsevich-Penner matrix-model conjecture for open intersection numbers

摘录 / EXCERPT: Let $\mathcal{M}_{(g,b),k,l}$ be the moduli space of Riemann surfaces with boundary of genus g, with b boundary components, k boundary marked points, and l interior marked...

conjecture | open | MathDB 页面 | 原文线索

V2-063744 | MATHDB-334145

Verlinde-type generating-function conjecture for rational surfaces

摘录 / EXCERPT: The rational-surface conjecture. There is a polynomial $P^X_{c_1,L}(\Lambda)\in \Lambda^{-c_1^2}{\mathbb Z}_{\geq0}[\Lambda^{\pm4}]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063745 | MATHDB-334154

Polynomial functional-graph cycle-count conjecture

摘录 / EXCERPT: Polynomial cycle-count conjecture. For any $k\in\mathbb{Z}^{>0}$ and any $d\in\mathbb{Z}^{\geq 0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-063746 | MATHDB-334159

Ikeda-Yamana multiplicity conjecture for CAP representations of the metaplectic group

摘录 / EXCERPT: Let F be a totally real number field, let n be odd, and let $\pi\simeq\otimes'_v\pi_v$ be an irreducible cuspidal automorphic representation of $\mathrm{PGL}_2(\mathbb A)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-063747 | MATHDB-334160

Saito-Kurokawa multiplicity formula for even orthogonal CAP representations

摘录 / EXCERPT: Let F be a totally real number field, let n be even, and let $\pi\simeq\otimes'_v\pi_v$ be an irreducible cuspidal automorphic representation of $\mathrm{PGL}_2(\mathbb A)$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-063748 | MATHDB-334162

Looijenga's smoothability conjecture for cusp singularities

摘录 / EXCERPT: Looijenga's conjecture. The cusp (X,x) is smoothable if and only if the minimal resolution of the dual cusp sits on a rational surface.

conjecture | open | MathDB 页面 | 原文线索

V2-063749 | MATHDB-334163

Friedman's classification conjecture for Type III anticanonical-pair degenerations

摘录 / EXCERPT: Friedman's conjecture. There exists an admissible isometry ψ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063750 | MATHDB-334177

The conjecture on additive idempotence of finitely generated semifields

摘录 / EXCERPT: A semifield is a commutative semiring whose multiplicative structure is a group. A semiring is finitely generated as a semiring when it is generated by finitely many elements under...

conjecture | open | MathDB 页面 | 原文线索

V2-063751 | MATHDB-334182

Schepers-Van Langenhoven alternatingly increasing conjecture for IDP polytopes

摘录 / EXCERPT: Let P be an r-dimensional lattice polytope with the Integer Decomposition Property (IDP), meaning that for every integer $n\geq1$ and every $p\in nP\cap\mathbb{Z}^d$ there are...

conjecture | open | MathDB 页面 | 原文线索

V2-063752 | MATHDB-334183

De Joannis de Verclos-Kang-Pastor conjecture for squares of claw-free graphs

摘录 / EXCERPT: De Joannis de Verclos-Kang-Pastor conjecture. For any claw-free graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063753 | MATHDB-334184

SHGH-type conjecture for negative curves on the Klein blowup

摘录 / EXCERPT: Let $X_\mathcal K$ be the blowup associated with the Klein line configuration, let H be the pullback of a line, and let E_4 and E_3 denote the exceptional divisors over the q...

conjecture | open | MathDB 页面 | 原文线索

V2-063754 | MATHDB-334185

Conjecture on the nef divisor and Waldschmidt constant of the Klein configuration

摘录 / EXCERPT: Let $X_\mathcal K$ be the blowup associated with the Klein line configuration, with H the pullback of a line and E_4,E_3 the exceptional divisors over its quadruple and triple...

conjecture | open | MathDB 页面 | 原文线索

V2-063755 | MATHDB-334204

Flatness implies smoothness for solutions of the porous medium equation

摘录 / EXCERPT: Let ρ be a solution to the porous medium equation on $[-1,0] \times B_2(0)$. Assume that there is a nonempty open cone C such that $\rho(t,x) \leq \rho(t,y)$ whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-063756 | MATHDB-334205

Optimality of the B-E Construction for PIR array codes

摘录 / EXCERPT: Let $s>2$, and consider the PIR array codes produced by the B-E Construction. B-E optimality conjecture. These PIR array codes have optimal rate. The preceding discussion is based...

conjecture | open | MathDB 页面 | 原文线索

V2-063757 | MATHDB-334218

Planar triangulations have total domatic number at least two

摘录 / EXCERPT: Planar triangulation conjecture. Every planar triangulation G of order at least 4 satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063758 | MATHDB-334219

Planar triangulations admit paired total dominating color classes

摘录 / EXCERPT: Four-color extension conjecture. Every planar triangulation with at least four vertices has a proper 4-coloring (C_1,C_2,C_3,C_4) such that both $C_1\cup C_2$ and $C_3\cup C_4$...

conjecture | open | MathDB 页面 | 原文线索

V2-063759 | MATHDB-334220

Planar triangulations of minimum degree four have total domatic number at least three

摘录 / EXCERPT: Minimum-degree-three-set conjecture. If G is a planar triangulation with minimum degree at least 4, then

conjecture | open | MathDB 页面 | 原文线索

V2-063760 | MATHDB-334221

Connected cubic graphs have fractional total domatic number at least two

摘录 / EXCERPT: Cubic fractional domatic conjecture. If G is a connected cubic graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-063761 | MATHDB-334225

Cohen-Lenstra conjecture for class-group p-parts of number fields

摘录 / EXCERPT: CohenLenstra conjecture. For every $A\in S$,

conjecture | open | MathDB 页面 | 原文线索

V2-063762 | MATHDB-334227

Bounded-norm strengthening of the bipartite inner-product compression theorem

摘录 / EXCERPT: Let $a_1,a_2,\ldots,a_n,b_1,b_2,\ldots,b_n\in\mathbb{R}^n$ have Euclidean norm at most 1, let $0<\varepsilon<1$, and set ... where C is the absolute constant from Theorem ... , the...

conjecture | open | MathDB 页面 | 原文线索

V2-063763 | MATHDB-334231

Quillen equivalence conjecture for the properadic nerve

摘录 / EXCERPT: Let $\operatorname{sProperad}$ be the category of simplicially enriched properads, and let ... be the properadic nerve functor, where $\varGamma$ is the graphical category. Properadi...

conjecture | open | MathDB 页面 | 原文线索

V2-063764 | MATHDB-334235

The semilinear reducibility conjecture for zigzag graph completions

摘录 / EXCERPT: Semilinear reducibility conjecture. For every $h\geq 3$, the graph $\widehat{ZZ}_h$ is semilinearly reducible and, for every q with no primes of bad reduction,

conjecture | open | MathDB 页面 | 原文线索

V2-063765 | MATHDB-334249

Clique-index conjecture for decompositions of complete geometric graphs

摘录 / EXCERPT: Let K_n be a complete geometric graph of order n, and let $[\mathsf{K}_n,P]$ be a decomposition. The clique index $\omega'([\mathsf{K}_n,P])$ is the maximum size of a...

conjecture | open | MathDB 页面 | 原文线索

V2-063766 | MATHDB-334251

Conjecture on the weight of maximal $(s,ms-1,ms+1)$-core partitions

摘录 / EXCERPT: Let s and m be positive integers, and let t be determined by either $s=2t-1$ or $s=2t-2$. A maximal $(s,ms-1,ms+1)$-core partition is a core partition maximal under inclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-063767 | MATHDB-334253

Lattès-map bifurcation accumulation conjecture

摘录 / EXCERPT: Let f be a Lattès map on ${\mathbb P^2}$. Lattès-map bifurcation accumulation conjecture. The map f belongs to the closure of the interior of the bifurcation locus. This is pre...

conjecture | open | MathDB 页面 | 原文线索

V2-063768 | MATHDB-334264

The ranked-poset conjecture for dense Borel orbits in BB-cells

摘录 / EXCERPT: Fix a BB-cell decomposition of $\mathcal{X}_{n}$ and consider all the Borel orbits in $\mathcal{X}_{n}$ which are dense in some BB-cell. Denote this set by...

conjecture | open | MathDB 页面 | 原文线索

V2-063769 | MATHDB-334265

Intermediate-growth conjecture for dendrite Julia-set quadratic polynomials

摘录 / EXCERPT: Let P be a postcritically-finite non-renormalizable quadratic polynomial whose Julia set $\mathcal{J}_P$ is a dendrite. Intermediate-growth conjecture. The iterated monodromy gro...

conjecture | open | MathDB 页面 | 原文线索

V2-063770 | MATHDB-334267

Eventual monotonicity of recursively defined sequence growth

摘录 / EXCERPT: Let $f\in\mathbb Q(d_1,d_2)$ be a rational function defined on $\mathbb Z^+\times\mathbb Z^+$ and suppose ... and ... for some $k\in\mathbb R^+$. If n_d is a sequence recursively def...

conjecture | open | MathDB 页面 | 原文线索

V2-063771 | MATHDB-334274

Curtis's strong conjecture on the Bernstein center and invariant deformation rings

摘录 / EXCERPT: Let F be a non-archimedean local field, let $G_n=\operatorname{GL}_n(F)$, let Z_n be the Bernstein center, and let e be the primitive idempotent associated with an inertial t...

conjecture | open | MathDB 页面 | 原文线索

V2-063772 | MATHDB-334282

The elimination ideal conjecture for rank-five subgroups containing an \(\mathbb{F}_{p^2}\)-subspace

摘录 / EXCERPT: Elimination ideal conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063773 | MATHDB-334283

The elimination ideal conjecture for a dilation of \(\mathbb{F}_{p^2}\)

摘录 / EXCERPT: Dilation elimination conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063774 | MATHDB-334289

The automatic-generalised polynomial intersection conjecture

摘录 / EXCERPT: A sequence is generalised polynomial if it is obtained from polynomials by applying addition, multiplication, and the floor function, and it is ultimately periodic if it agrees wit...

conjecture | open | MathDB 页面 | 原文线索

V2-063775 | MATHDB-334291

Edwards-King fractional chromatic conjecture

摘录 / EXCERPT: For a graph Q and $r\in\mathbb{N}\cup\{\infty\}$, let $\mathcal{C}_r(Q)$ consist of the maximal cliques of size less than r together with the cliques of size exactly r, and d...

conjecture | open | MathDB 页面 | 原文线索

V2-063776 | MATHDB-334292

Edwards-King replacement conjecture for line graphs

摘录 / EXCERPT: For $r\in\mathbb{N}\cup\{\infty\}$ and a graph Q, define ... Let $\chi(Q)$ denote the chromatic number of Q. Edwards-King replacement conjecture. If Q is a line graph, then ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-063777 | MATHDB-334294

Naserasr's projective-hypercube conjecture for planar graphs

摘录 / EXCERPT: Let $PC(2k)$ be the projective hypercube of dimension $2k$, obtained from the $(2k+1)$-dimensional hypercube by identifying antipodal vertices. A graph H bounds the class of plan...

conjecture | open | MathDB 页面 | 原文线索

V2-063778 | MATHDB-334299

Denis's Schanuel conjecture for the Carlitz exponential

摘录 / EXCERPT: Denis's conjecture. The displayed transcendence-degree inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-063779 | MATHDB-334300

Denis's strengthened Carlitz-Schanuel conjecture

摘录 / EXCERPT: Denis's strengthened conjecture. The displayed transcendence-degree inequality holds.

conjecture | open | MathDB 页面 | 原文线索

V2-063780 | MATHDB-334309

The polylogarithmic susceptibility conjecture for bounded-degree graphs

摘录 / EXCERPT: Fix a maximum degree d. Let $(G_n)_{n\geq1}$ be a sequence of finite connected graphs, with $G_n=(V_n,E_n)$, $|V_n|\to\infty$, and maximum degree at most d. The polylogarithmic...

conjecture | open | MathDB 页面 | 原文线索

V2-063781 | MATHDB-334316

Li-Qu-Li-Fu's permutation trinomial conjecture over finite fields

摘录 / EXCERPT: Let k be a positive integer, set $q=3^k$, and let ${\mathbb F}_{q^2}$ be the finite field with q^2 elements. A polynomial is a permutation trinomial over ${\mathbb F}_{q^2}$ if...

conjecture | open | MathDB 页面 | 原文线索

V2-063782 | MATHDB-334317

Li-Qu-Li-Fu's rational-function permutation conjecture on roots of unity

摘录 / EXCERPT: Let k be a positive integer, set $q=3^k$, and define ... the set of $(q+1)$-th roots of unity in ${\mathbb F}_{q^2}$. A rational function permutes μ_{q+1} if it is defined on t...

conjecture | open | MathDB 页面 | 原文线索

V2-063783 | MATHDB-334320

Open dense full-measure conjecture for finite-type double rotations

摘录 / EXCERPT: Let $\mathcal{V}$ be the parameter space of pairs ${v_0,v_1}$, with a fixed partition, for which the double-rotation map f_{v_0,v_1} on a two-dimensional torus is of finite type....

conjecture | open | MathDB 页面 | 原文线索

V2-063784 | MATHDB-334321

Semi-continuity conjecture for double-rotation attractors

摘录 / EXCERPT: Let $\mathcal{V}$ be the parameter space of finite-type double rotations with a fixed partition, and let the associated map have an attractor. The attractor is semi-continuous with...

conjecture | open | MathDB 页面 | 原文线索

V2-063785 | MATHDB-334327

Generalized Lascoux-Pragacz determinant identity for factorial Schur functions

摘录 / EXCERPT: Let $(\theta_1,\ldots,\theta_k)$ be a Lascoux-Pragacz decomposition of $\lambda/\mu\subset d\times(n-d)$, and let $\lambda\setminus\theta_i\#\theta_j$ denote the partition obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-063786 | MATHDB-334328

Determinantal q-Eulerian conjecture for reverse plane partitions of thickened zigzag strips

摘录 / EXCERPT: Let δ_n denote the staircase shape, let $\operatorname{RPP}(\delta_{n+2k}/\delta_n)$ be the reverse plane partitions of the skew shape δ_{n+2k}/δ_n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063787 | MATHDB-334330

Universal sharpness of the four-point bound for five particles on the sphere

摘录 / EXCERPT: Let E_2 denote the second relaxation, or four-point bound, in the semidefinite hierarchy for energy minimization. For five particles on S^2, consider all completely monotonic p...

conjecture | open | MathDB 页面 | 原文线索

V2-063788 | MATHDB-334335

The Schumaker formula equality conjecture for planar splines

摘录 / EXCERPT: For a simplicial complex $\Delta\subseteq\mathbb{R}^2$, let $S^r_d(\Delta)$ be the real vector space of splines whose polynomial pieces have degree at most d and whose smoothness...

conjecture | open | MathDB 页面 | 原文线索

V2-063789 | MATHDB-334336

The polyhedral-complex degree-bound conjecture for spline dimensions

摘录 / EXCERPT: Let P be a planar polyhedral complex, and let $F=\max\{n\mid\text{\rm {there is an n-gon in }}P\}$. Let k and r be the parameters in the spline-dimension formula of Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-063790 | MATHDB-334349

Fargues-Rapoport conjecture on admissibility and weak admissibility

摘录 / EXCERPT: Assume that F has characteristic 0 and that μ is minuscule. Let G be the reductive group, let b_0 be the basic element associated with $(G,\{\mu\})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063791 | MATHDB-334355

Brown-Zagier relation for regularized multiple zeta values

摘录 / EXCERPT: Brown-Zagier relation. For any nonnegative integers m,n, one has

conjecture | open | MathDB 页面 | 原文线索

V2-063792 | MATHDB-334357

Runge-Gross density-potential mapping conjecture

摘录 / EXCERPT: Let ψ_0 be the initial wave function of an N-particle system, with initial density n_0 on a domain Ω. Assume that the external potential is time-analytic, that...

conjecture | open | MathDB 页面 | 原文线索

V2-063793 | MATHDB-334361

Gamburd-Jakobson-Sarnak spectral-gap conjecture for generic compact-group generators

摘录 / EXCERPT: Let G be a compact simple Lie group, let $(g_1,\ldots,g_k)\in G^k$ be a random k-tuple, and let $\Gamma(g_1,\ldots,g_k)<G$ be the subgroup it generates. Gamburd-Jakobson-Sarnak...

conjecture | open | MathDB 页面 | 原文线索

V2-063794 | MATHDB-334366

Adams' conjecture for tempered theta lifts

摘录 / EXCERPT: Let $G_{2,v}$ be the small member of a local Howe dual pair and let $G_{1,v}$ be the other member. Let ρ be a tempered representation of $G_{2,v}$ with Langlands parameter ... L...

conjecture | open | MathDB 页面 | 原文线索

V2-063795 | MATHDB-334367

Unitary Adams conjecture for the Plancherel decomposition of Howe duality

摘录 / EXCERPT: Unitary Adams conjecture. There is a direct integral decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-063796 | MATHDB-334368

Euler factorization conjecture for automorphic relative characters

摘录 / EXCERPT: Let $G=G_1\times G_2$ be the global Howe dual pair, let ${\boldsymbol{\omega}}$ be its global Weil representation, and let $\pi=\pi_1\otimes\pi_2$ be an automorphic representation...

conjecture | open | MathDB 页面 | 原文线索

V2-063797 | MATHDB-334371

Finkelberg-Mirković conjecture for the extended principal block

摘录 / EXCERPT: Let $G_{\Bbbk}$ be the semisimple group in characteristic p, let $\Repe$ be its extended principal block, let ${\EuScript Gr^\vee}$ be the affine Grassmannian of the Langlands du...

conjecture | open | MathDB 页面 | 原文线索

V2-063798 | MATHDB-334381

Critical-point shift criterion for pinning models with stable disorder

摘录 / EXCERPT: Let τ be the renewal process with inter-arrival law K, and let $h_c(\beta)$ denote the critical point at disorder intensity β. Assume that the environment satisfies th...

conjecture | open | MathDB 页面 | 原文线索

V2-063799 | MATHDB-334382

Supersmoothing conjecture for pinning models with stable disorder

摘录 / EXCERPT: Let $F(\beta,h)$ be the quenched free energy, let $h_c(\beta)$ be the critical point, and suppose the renewal inter-arrival law satisfies the regular-variation assumption and the e...

conjecture | open | MathDB 页面 | 原文线索

V2-063800 | MATHDB-334384

de Boor's maximal-hyperplane conjecture for geometrically characterized sets

摘录 / EXCERPT: Let X be a $GC_{d,n}$ set in ${\mathbb R}^d$: it has ${n+d\choose d}$ points, and for every $p\in X$ there is a product of n linear forms that equals 1 at p and 0 at ever...

conjecture | open | MathDB 页面 | 原文线索

V2-063801 | MATHDB-334385

de Boor's multiple-maximal-hyperplane conjecture for geometrically characterized sets

摘录 / EXCERPT: Let X be a $GC_{d,n}$ set in ${\mathbb R}^d$, and call a hyperplane maximal when it contains ${d-1+n\choose n}$ points of X. de Boor's multiple-maximal-hyperplane conjecture. E...

conjecture | open | MathDB 页面 | 原文线索

V2-063802 | MATHDB-334408

Local PT/GV correspondence for irreducible one-cycles

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, let β be an effective one-cycle class, and let $ gamma \in \operatorname{Chow}_{\beta}(X)$ be an irreducible one-cycle. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-063803 | MATHDB-334409

Local Pandharipande-Thomas invariants equal local Gopakumar-Vafa invariants

摘录 / EXCERPT: Let X be as in the definition of local Gopakumar-Vafa invariants, let β be an effective one-cycle class, and let $\gamma\in\operatorname{Chow}_{\beta}(X)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-063804 | MATHDB-334410

Gromov-Witten/Gopakumar-Vafa correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Suppose that X is a smooth projective Calabi-Yau threefold. For an effective curve class $\beta>0$ and $g\in\mathbb Z_{\geq 0}$, let $n_{g,\beta}^{GW}$ be the rational numbers de...

conjecture | open | MathDB 页面 | 原文线索

V2-063805 | MATHDB-334411

Independence of Gopakumar-Vafa invariants from Euler characteristic and stability

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, let $\beta>0$ and $k\in\mathbb Z$, and suppose that the moduli space $\operatorname{Sh}_{\beta,k}(X)$ is fine. Let π_k be its Hilbert-Chow ma...

conjecture | open | MathDB 页面 | 原文线索

V2-063806 | MATHDB-334414

The squarefree values conjecture for multivariable polynomials

摘录 / EXCERPT: Let $h\in\{1,2\}$, let $f\in\mathbb{Z}[x_1,\ldots,x_h]$ be primitive and squarefree, and let $\mathbb{Z}^h(X)=\{{\rm v}\in\mathbb{Z}^h:|{\rm v}|\leq X\}$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-063807 | MATHDB-334416

Chowla's conjecture for polynomial values

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x_1,\ldots,x_h]$ be a primitive squarefree polynomial, let $\mathcal{A}$ be an arithmetic progression, and let $\mathcal{A}(X)$ denote its points of norm at mos...

conjecture | open | MathDB 页面 | 原文线索

V2-063808 | MATHDB-334431

Large-deformation limit conjecture for the ASEP ground state and current generating function

摘录 / EXCERPT: The partially asymmetric simple exclusion process is described using a deformed Markov matrix, with deformation parameter ξ and system variables $\mathbf{x}$. For each positive...

conjecture | open | MathDB 页面 | 原文线索

V2-063809 | MATHDB-334435

q,t-positivity conjecture for threshold-graph flow polytopes

摘录 / EXCERPT: Let G be a threshold graph with $n+1$ vertices. The weighted Ehrhart series $\operatorname{Ehr}_{q,t}({\mathcal{F}}_G(-n,1,\ldots,1))$ is a polynomial with nonnegative coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-063810 | MATHDB-334436

Positivity conjecture for differences along the threshold-graph poset

摘录 / EXCERPT: Let $\mathcal{P}_n$ be the poset of connected threshold graphs with vertices $0,1,\ldots,n$, ordered by subgraph inclusion: $H\preceq G$ when H is a subgraph of G. For threshol...

conjecture | open | MathDB 页面 | 原文线索

V2-063811 | MATHDB-334439

Tightness of the partition-constrained minimization LP for Potts internal energy

摘录 / EXCERPT: Partition-LP tightness conjecture. The minimization LP is tight and yields, for every d-regular graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-063812 | MATHDB-334440

Critical phase-transition conjecture for parking on a random tree

摘录 / EXCERPT: Critical phase-transition conjecture. The expected number of cars that can leave the root, X, satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063813 | MATHDB-334448

The color-homomorphism tensor-square conjecture

摘录 / EXCERPT: Let G be a graph, let K_2 be the complete graph on two vertices, and let $G\times K_2$ denote their tensor product. Write $\hom(G,H)$ for the number of graph homomorphisms from...

conjecture | open | MathDB 页面 | 原文线索

V2-063814 | MATHDB-334449

The triangle-free regular graph homomorphism conjecture

摘录 / EXCERPT: Let G be a triangle-free d-regular graph and let H be a loop-graph. Write $v(G)$ for the number of vertices of G and $\hom(G,H)$ for the number of graph homomorphisms from...

conjecture | open | MathDB 页面 | 原文线索

V2-063815 | MATHDB-334452

Galvin's biregular homomorphism conjecture

摘录 / EXCERPT: Let G be a bipartite graph, let d_u denote the degree of $u\in V(G)$, and let H be a loop-graph. Write $\hom(G,H)$ for the number of graph homomorphisms from G to H. Galv...

conjecture | open | MathDB 页面 | 原文线索

V2-063816 | MATHDB-334453

Local constraints imply bounded extrema

摘录 / EXCERPT: Let $d\ge 3$ be an integer and let $\mathcal{F}$ be a finite list of graphs. Let $\mathcal{G}_d(\mathcal{F})$ be the set of finite d-regular graphs that do not contain any elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-063817 | MATHDB-334465

Faber's Gorenstein quotient conjecture from relations on the universal curve

摘录 / EXCERPT: Let C_g^{2g-1} be the $(2g-1)$-fold fiber product of the universal curve over M_g, let $\mathbb{E}$ be the Hodge bundle, and let K_i and D_{ij} denote the tautological clas...

conjecture | open | MathDB 页面 | 原文线索

V2-063818 | MATHDB-334468

Uniform bounded-points conjecture for tautological relations on universal curve powers

摘录 / EXCERPT: Let $\mathcal{C}_g^n$ be the n-fold fiber product of the universal curve over $\mathcal{M}_g$. Uniform bounded-points conjecture. For each genus g, there is a natural number...

conjecture | open | MathDB 页面 | 原文线索

V2-063819 | MATHDB-334472

Polynomial representation conjecture for the odd-power binomial sums

摘录 / EXCERPT: Let a,b,c be non-negative integers and, for $r\in\mathbb{Z}^+$, define ... Let $\mathcal{L}$ act on functions of a,b,c by ... where $E\cdot f(a,b,c)=f(a-1,b,c)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-063820 | MATHDB-334473

Even-dimensional Dwork hypersurface hypergeometric point count conjecture

摘录 / EXCERPT: Even-dimensional Dwork point count conjecture. The number of points over $\mathbb F_q$ on the Dwork hypersurface can be expressed as

conjecture | open | MathDB 页面 | 原文线索

V2-063821 | MATHDB-334475

Homogeneous boolean rank conjecture for polygon slack matrices

摘录 / EXCERPT: Homogeneous boolean rank conjecture. For any n,

conjecture | open | MathDB 页面 | 原文线索

V2-063822 | MATHDB-334477

The two-variable Iwasawa main conjecture for elliptic curves over an imaginary quadratic field

摘录 / EXCERPT: Let K be the imaginary quadratic field and K_∞ the relevant cyclotomic and anticyclotomic extension. For $\star,\circ\in\{\sharp,\flat\}$, let $L^{\star\circ}(X,Y)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-063823 | MATHDB-334484

Set-wise independence conjecture for odd and even staircase Schur functions

摘录 / EXCERPT: For each $k\in{\mathbb N}$, let δ_k denote the staircase partition used in the paper, and let s_{δ_k} be its Schur function under the random specialization over...

conjecture | open | MathDB 页面 | 原文线索

V2-063824 | MATHDB-334508

Polynomial ramification criterion for b-ramified power series

摘录 / EXCERPT: Polynomial ramification criterion. For every positive integer b, there exists a polynomial $P_b\in\mathbb{Z}[x_1,\dots,x_{b+1}]$ such that f is b-ramified if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-063825 | MATHDB-334515

Hong and Won's conjecture on the alpha-invariant of a del Pezzo surface

摘录 / EXCERPT: Let S be a smooth del Pezzo surface and let A be an ample divisor on S of one of the specified $\mathbb{P}^2$-, $\mathbb{F}_1$-, or $\mathbb{P}^1\times\mathbb{P}^1$-types. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-063826 | MATHDB-334518

Tutte-Coxeter graph occupancy-fraction conjecture

摘录 / EXCERPT: Let $\alpha_G(\lambda)$ denote the occupancy fraction of a graph G in the hard-core model, and let $H_{3,8}$ be the $(3,8)$-cage, also known as the Levi graph or Tutte-Coxeter gr...

conjecture | open | MathDB 页面 | 原文线索

V2-063827 | MATHDB-334519

(4,6)-cage occupancy-fraction conjecture

摘录 / EXCERPT: Let $\alpha_G(\lambda)$ denote the occupancy fraction of a graph G in the hard-core model, and let $H_{4,6}$ be the $(4,6)$-cage. (4,6)-cage occupancy-fraction conjecture. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-063828 | MATHDB-334535

Leader-Tan edge-decomposition conjecture for hypercubes

摘录 / EXCERPT: Let H be a non-empty subgraph of the hypercube Q_k for some k. Leader-Tan edge-decomposition conjecture. There exists a positive integer n such that the edges of Q_n can...

conjecture | open | MathDB 页面 | 原文线索

V2-063829 | MATHDB-334541

The point-count upper bound conjecture for nonsingular hypersurfaces in even-dimensional projective spaces

摘录 / EXCERPT: Let $m\geq 4$ be an even integer, and let $\mathcal{X}$ be a nonsingular hypersurface of degree d in $\mathbb{P}^m$ over $\mathbb{F}_q$. Write $N_q(\mathcal{X})$ for the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-063830 | MATHDB-334545

Heavy-traffic Gamma limit conjecture for total workload

摘录 / EXCERPT: Let V be the total amount of work in the system at the beginning of a cycle, and let r, ρ, $\tilde{b}_i$, $\tilde{B}_i$, $\tilde{A}_i$, τ, and β have the meanin...

conjecture | open | MathDB 页面 | 原文线索

V2-063831 | MATHDB-334546

Heavy-traffic Gamma limit conjecture for length-biased cycle time

摘录 / EXCERPT: Let C_i denote the cycle time seen by an arbitrary customer or arrival at queue i, and let $\bm{C_i}$ denote its scaled length-biased version. Let α and μ be as defi...

conjecture | open | MathDB 页面 | 原文线索

V2-063832 | MATHDB-334550

Conjecture on non-proper free boundary min-max hypersurfaces

摘录 / EXCERPT: Let n be the dimension parameter and consider compact smooth domains in $\mathbb{R}^{n+1}$. A free boundary min-max minimal hypersurface is a free boundary minimal hypersurface p...

conjecture | open | MathDB 页面 | 原文线索

V2-063833 | MATHDB-334551

Generic properness conjecture for free boundary min-max hypersurfaces

摘录 / EXCERPT: Let Ω be a compact domain in $\mathbb{R}^{n+1}$ with analytic boundary $\partial\Omega$. A free boundary min-max minimal hypersurface is a free boundary minimal hypersurface...

conjecture | open | MathDB 页面 | 原文线索

V2-063834 | MATHDB-334552

Weak Brill-Noether conjecture for general blowups of the projective plane

摘录 / EXCERPT: Weak Brill-Noether conjecture. The moduli space $M_{X,H}({\bf v})$ satisfies weak Brill-Noether.

conjecture | open | MathDB 页面 | 原文线索

V2-063835 | MATHDB-334555

The restricted k-broad l^p-decoupling conjecture

摘录 / EXCERPT: Let $2\leq k\leq n+1$. For a function F on $\mathbb{R}^{n+1}$ with Fourier support in $N_{1/D^2}\Sigma$, let $F_{D,V}$ denote its contribution associated with a k-dimensional s...

conjecture | open | MathDB 页面 | 原文线索

V2-063836 | MATHDB-334556

Conjecture on monotonicity of worst-case error for spherical t-designs

摘录 / EXCERPT: Worst-case error monotonicity conjecture. For fixed t, if $N'>N$, then

conjecture | open | MathDB 页面 | 原文线索

V2-063837 | MATHDB-334559

Exact two-Schur correlator formula for the r-fold q-deformed logarithmic matrix model

摘录 / EXCERPT: The exact two-Schur correlator conjecture. The average of the two Schur-polynomial insertions satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063838 | MATHDB-334560

Fair perfect matchings in partitions of K_{n,n}

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with edge set partitioned into $E_1,\ldots,E_m$. For a perfect matching F, write $|F\cap E_i|$ for the number of its edges in E_i....

conjecture | open | MathDB 页面 | 原文线索

V2-063839 | MATHDB-334566

Uniqueness conjecture for pulsating fronts of the competition system

摘录 / EXCERPT: Let $k>k^{\star}$. Let $\left(\hat{\varphi}_{1},\hat{\varphi}_{2}\right)$ and $\hat{c}$ be respectively the profile and the speed of a pulsating front solution...

conjecture | open | MathDB 页面 | 原文线索

V2-063840 | MATHDB-334571

Parabolic subgroup invariance conjecture for reduced reflection expressions

摘录 / EXCERPT: Let W be a Coxeter group with reflection set T, and let ${\mathrm{Red}_T}(w)$ denote the set of minimal-length expressions of w as a product of reflections. Suppose that ... an...

conjecture | open | MathDB 页面 | 原文线索

V2-063841 | MATHDB-334583

Captain-cycle conjecture for minimal TEQ-retentive sets

摘录 / EXCERPT: Let T be a tournament, and let $\mathfrak{Dom}(T)$ denote its directed domination graph. A set is TEQ-retentive when it satisfies the tournament-equilibrium-set retentiveness con...

conjecture | open | MathDB 页面 | 原文线索

V2-063842 | MATHDB-334590

Non-trivial identities in finite-rank plactic monoids

摘录 / EXCERPT: Finite-rank plactic identity conjecture. For each $n \geq 4$, there is a non-trivial identity satisfied by ${\mathsf{plac}}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-063843 | MATHDB-334599

The expressivity hierarchy conjecture for finitely iterated supervenience languages

摘录 / EXCERPT: Expressivity hierarchy conjecture. For all $n\in\mathbb{N}$, $\mathcal{L}_{\Rrightarrow^{n+1}}$ is more expressive than $\mathcal{L}_{\Rrightarrow^n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-063844 | MATHDB-334610

Liu's refined global Gross-Prasad conjecture for special Bessel periods

摘录 / EXCERPT: Liu's refined global Gross-Prasad conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063845 | MATHDB-334611

Wood's conjecture on unramified extensions of quadratic fields

摘录 / EXCERPT: Wood's conjecture. If there is a unique conjugacy class c of order 2 elements of G' not contained in G, then

conjecture | open | MathDB 页面 | 原文线索

V2-063846 | MATHDB-334612

Wood's moment conjecture for admissible pairs of 2-groups

摘录 / EXCERPT: Wood's moment conjecture. There exists a number $0<a<1$ such that, for every positive integer k,

conjecture | open | MathDB 页面 | 原文线索

V2-063847 | MATHDB-334630

The quadratic-differential characterization of the support

摘录 / EXCERPT: For a parameter b, let Θ be the variable in the quadratic differential ... where Λ is a complex parameter. Quadratic-differential support conjecture. The support of...

conjecture | open | MathDB 页面 | 原文线索

V2-063848 | MATHDB-334634

Sandpile group distribution conjecture for random directed graphs

摘录 / EXCERPT: Sandpile group distribution conjecture. For every finite abelian group G,

conjecture | open | MathDB 页面 | 原文线索

V2-063849 | MATHDB-334637

Wei Zhang's Arithmetic Fundamental Lemma for artinian unitary group elements

摘录 / EXCERPT: Let $S(E_0)$ be the relevant symmetric space, let $U(J_1)$ be the unitary group attached to the odd hermitian form, and let $Ω(γ)$, $\partial O_γ(1_{S(\mathcal{O}_{E_0})})$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-063850 | MATHDB-334641

Fundamental Lemma for regular semi-simple adjoint-stable pairs

摘录 / EXCERPT: Let V be a hermitian space over E, and let $I(x,j)$ be the number of self-dual lattices satisfying the stated stability conditions. Let $O(x,j)$ be the corresponding signed lat...

conjecture | open | MathDB 页面 | 原文线索

V2-063851 | MATHDB-334642

Arithmetic Fundamental Lemma for regular semi-simple self-adjoint artinian pairs

摘录 / EXCERPT: Let (x,j) be a pair in the odd hermitian setting. It is regular semi-simple and self-adjoint as defined in the source, and it is artinian when the schematic intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-063852 | MATHDB-334650

Căldăraru's conjecture on twisted K3 Fourier-Mukai partners

摘录 / EXCERPT: Let (X,α) be a twisted K3 surface, where α is a Brauer class, and let X' be an untwisted Fourier-Mukai partner of X. For a Hodge isometry ... consider the pullback...

conjecture | open | MathDB 页面 | 原文线索

V2-063853 | MATHDB-334651

Circular Brennan's conjecture

摘录 / EXCERPT: Let $f\colon\mathbb{D}\to\Omega$ be a bounded conformal map from the unit disk, and define the universal integral means spectrum by ... where $\beta_f(t)$ is the integral means spect...

conjecture | open | MathDB 页面 | 原文线索

V2-063854 | MATHDB-334656

Differentiability conjecture for Karpelevič arcs

摘录 / EXCERPT: Let a K-arc be an arc in the Karpelevič region, and let n denote its order. Differentiability conjecture. All K-arcs of order n are differentiable for every n. The paper note...

conjecture | open | MathDB 页面 | 原文线索

V2-063855 | MATHDB-334657

Strict-convexity conjecture for Karpelevič arcs

摘录 / EXCERPT: Let K be a K-arc, and let $d_K:[0,1]\longrightarrow\mathbb{R}_0^+$ be defined by ... where $\lambda(\alpha)$ is the point on K corresponding to $\alpha\in[0,1]$. Strict-convexity...

conjecture | open | MathDB 页面 | 原文线索

V2-063856 | MATHDB-334658

Petrosyan-Mkhitaryan conjecture on cyclic interval colorings

摘录 / EXCERPT: Let $\mathfrak{N}_c$ be the class of multigraphs admitting a cyclic interval coloring, and let $W_c(G)$ denote the maximum number of colors in a cyclic interval coloring of G. Pe...

conjecture | open | MathDB 页面 | 原文线索

V2-063857 | MATHDB-334682

Grivaux-Hubert conjecture on Lyapunov exponents with many poles

摘录 / EXCERPT: A translation surface is a surface equipped with a flat structure whose singularities may include poles, and its positive Lyapunov exponents are the exponents of the Kontsevich-Zor...

conjecture | open | MathDB 页面 | 原文线索

V2-063858 | MATHDB-334683

Conjecture on the first Lyapunov exponent for strata with many poles

摘录 / EXCERPT: Fix a genus g. Let Q_n be a sequence of connected components of strata, let p_n be the number of poles in the stratum containing Q_n, and suppose that p_n tends to infini...

conjecture | open | MathDB 页面 | 原文线索

V2-063859 | MATHDB-334692

Power-logarithmic asymptotic conjecture for Tate-Shafarevich group orders in the Neumann-Setzer family

摘录 / EXCERPT: For $i\in\{1,2\}$, let $f_k(i,X)$ count integers $u\equiv1\pmod 4$ with $|u|\leq X$ satisfying the paper's condition $(**)$, such that $L(E(u),1)\neq0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-063860 | MATHDB-334700

Failure of Bartnik mass minimization for nonembedded locally flat balls

摘录 / EXCERPT: Let $F:\bar B\rightarrow\mathbb R^3$ be a smooth immersion of a 3-ball that is not an embedding, and equip the ball with the pulled-back Euclidean metric $F^*(g_{\mathrm{Eucl}})$....

conjecture | open | MathDB 页面 | 原文线索

V2-063861 | MATHDB-334701

Nonexistence of static extensions for nonembedded locally flat boundary data

摘录 / EXCERPT: Let (γ,H) with $H>0$ be the Bartnik boundary data of a locally flat 3-ball that is not an embedded ball in $\mathbb R^3$. Let ${\mathcal M}$ denote the space of Bartnik boun...

conjecture | open | MathDB 页面 | 原文线索

V2-063862 | MATHDB-334706

Polynomial-dimensional embedding conjecture for finite subsets of \(\ell_1\)

摘录 / EXCERPT: Let $\mathcal{M}\subseteq \ell_1$ be a subset with $|\mathcal{M}|=n$. An embedding of $\mathcal{M}$ into a normed space has distortion D if distances are preserved up to a multip...

conjecture | open | MathDB 页面 | 原文线索

V2-063863 | MATHDB-334707

The alternating-restraint conjecture for bipartite graphs

摘录 / EXCERPT: Alternating-restraint conjecture. If r is any standard simple restraint on G and x is sufficiently large, then

conjecture | open | MathDB 页面 | 原文线索

V2-063864 | MATHDB-334717

The arithmetic correlations conjecture for convolutions of divisor sums

摘录 / EXCERPT: Arithmetic correlations conjecture. There are numbers $\phi<1$ and $\psi>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-063865 | MATHDB-334719

A Bloch-Kato-type formula for critical Asai motives

摘录 / EXCERPT: Let $\mathcal{M}=\mathcal{M}(\pi,r^{\pm}\otimes\Psi)$ be critical, with Betti and de Rham realizations $H_{\rm B}(\mathcal{M})$ and $H_{\rm dR}(\mathcal{M})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063866 | MATHDB-334726

The finiteness conjecture for constructible sheaves on Alexandrov spaces

摘录 / EXCERPT: Fix a finite field $\mathbb{F}$. Let X be a compact Alexandrov space, and let $D^b_c(\operatorname{Sh}_{\mathbb{F}}(X))$ denote the bounded derived category of constructible shea...

conjecture | open | MathDB 页面 | 原文线索

V2-063867 | MATHDB-334727

The compactification conjecture for constructible sheaf data on Alexandrov spaces

摘录 / EXCERPT: Fix a finite field $\mathbb{F}$ and let $\mathcal{L}_{\mathbb{F}}(n,D,\kappa)$ consist of pairs $(X,[\mathcal{F}])$, where X is an Alexandrov space of Hausdorff dimension at most...

conjecture | open | MathDB 页面 | 原文线索

V2-063868 | MATHDB-334728

Uniqueness of tournament-kernel representatives up to measure-preserving transformations

摘录 / EXCERPT: Let W and W' be equivalent tournament kernels, meaning that they represent the same tournament limit. A measure-preserving transformation is a map from $[0,1]$ to $[0,1]$ prese...

conjecture | open | MathDB 页面 | 原文线索

V2-063869 | MATHDB-334730

Asymptotic optimal pebbling conjecture for eight-wide staircase graphs

摘录 / EXCERPT: Eight-wide staircase conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-063870 | MATHDB-334731

Asymptotic optimal pebbling conjecture for multiples of seven-wide staircase graphs

摘录 / EXCERPT: Multiples-of-seven staircase conjecture. For all $k\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-063871 | MATHDB-334732

Stapledon decomposition conjecture for graph polynomials

摘录 / EXCERPT: Stapledon decomposition conjecture. The polynomial $h_G(z)$ can be decomposed as

conjecture | open | MathDB 页面 | 原文线索

V2-063872 | MATHDB-334733

Stapledon decomposition conjecture for open order polytopes

摘录 / EXCERPT: Stapledon decomposition conjecture. The polynomial $p_{\Pi}(z)$ can be decomposed as

conjecture | open | MathDB 页面 | 原文线索

V2-063873 | MATHDB-334738

Automatic-sequence uniformity conjecture

摘录 / EXCERPT: Let $a(n)$ be a 2-automatic sequence such that ... for every $q\in\mathbb{N}$ and $r\in\mathbb{N}_0$. Then the automatic-sequence uniformity conjecture. ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-063874 | MATHDB-334748

The maximal-radius distinctness conjecture of Robinson

摘录 / EXCERPT: For $N\geq 4$, let $r_{\max}(N)$ be the maximal radius of N equal spheres touching a central unit sphere. Robinson's conjecture. For all $N\geq 4$, one has ... except possibly for...

conjecture | open | MathDB 页面 | 原文线索

V2-063875 | MATHDB-334761

Carnevale-Voll's unitary-factor conjecture for Euler-Mahonian distributions

摘录 / EXCERPT: Carnevale-Voll's conjecture. The polynomial $C_\rho(x,q)$ has a unitary factor if and only if $\rho=(\rho_1,\ldots,\rho_m)$ has all parts equal to some odd integer r and m is e...

conjecture | open | MathDB 页面 | 原文线索

V2-063876 | MATHDB-334767

Generation conjecture for divisor multiplication operators on the Fock space

摘录 / EXCERPT: Let $\mathcal{F}_{\mathcal{A}_n}\cong \mathcal{F}_{[\mathbb{C}^2/\mathbb{Z}_{n+1}]}$ be the common Fock space, and let $D_0,\ldots,D_n$ be a basis of divisors. For each divisor, le...

conjecture | open | MathDB 页面 | 原文线索

V2-063877 | MATHDB-334768

Optimal sparse blow-up conjecture for bounded-bandwidth graphs

摘录 / EXCERPT: For each $\gamma>0$, $\Delta\ge2$, and $k\ge1$, there should exist constants $\beta^\ast>0$ and $C^\ast>0$ such that, asymptotically almost surely for $\Gamma=G(n,p)$ whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-063878 | MATHDB-334800

Galkin's fully faithful Fourier-Mukai conjecture for moduli spaces of rank-two bundles

摘录 / EXCERPT: Let C be a smooth curve of genus $g>1$, let $M=M_C(2,L)$ be the moduli space of stable rank-two vector bundles on C with fixed determinant L of odd degree, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063879 | MATHDB-334820

Faber-Voloch conjecture on Newton approximation

摘录 / EXCERPT: Let g be a polynomial of degree $d\geq 2$ with coefficients in a number field K, and let $y_0\in K$. Define the Newton map ... and, for each $n\geq 0$, set $y_{n+1}=N(y_n)$. Assu...

conjecture | open | MathDB 页面 | 原文线索

V2-063880 | MATHDB-334821

Cascade-channel capacity conjecture for finite-state channels

摘录 / EXCERPT: Cascade-channel capacity conjecture. If $|\mathcal S|<\infty$, so that the channel is finite-state, then the Shannon capacity of the m-fold cascade channel converges to...

conjecture | open | MathDB 页面 | 原文线索

V2-063881 | MATHDB-334828

Strict separation of Minimax and Maximin lower bounds for vector Gaussian rate-distortion with trace distortion

摘录 / EXCERPT: The vector Gaussian rate-distortion problem with variable side information under trace distortion, denoted ${\textsc{RDTr}}$, is considered together with its Minimax and Maximin lo...

conjecture | open | MathDB 页面 | 原文线索

V2-063882 | MATHDB-334830

Uniform condition-number conjecture for the trace Schur complement

摘录 / EXCERPT: Uniform condition-number conjecture. There exist $s<0$ and a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-063883 | MATHDB-334835

Chen-Xia conjecture on infinite log-convexity of the Apéry sequences

摘录 / EXCERPT: The Apéry numbers are the sequences ... A sequence is infinitely log-convex when every sequence obtained by iterating the log-convexity operator remains log-convex. Chen-Xia's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-063884 | MATHDB-334841

Completeness conjecture for Conway and umbral K3 twining genera

摘录 / EXCERPT: Let $\mathcal{T}(\Pi)$ be a $K3$ nonlinear sigma model with symmetry group G, let Γ_G be its co-invariant lattice, and let $\Phi(N)$ denote the corresponding set of Conway...

conjecture | open | MathDB 页面 | 原文线索

V2-063885 | MATHDB-334842

Realization conjecture for all Conway and umbral twining functions

摘录 / EXCERPT: Let N range over the 24 Niemeier lattices and let $\Phi(N)$ be the associated set of Conway or umbral twining functions. Realization conjecture. For every element...

conjecture | open | MathDB 页面 | 原文线索

V2-063886 | MATHDB-334843

Conjecture characterizing truncated Toeplitz operators by conjugation symmetries

摘录 / EXCERPT: Characterization conjecture. $A\in\mathcal{T}(\theta)$ if and only if A_α is C_α-symmetric for every nonconstant inner function α dividing θ.

conjecture | open | MathDB 页面 | 原文线索

V2-063887 | MATHDB-334847

The FPL expansion of geometric pushforwards

摘录 / EXCERPT: Geometric FPL expansion conjecture. The pushforward can be decomposed as

conjecture | open | MathDB 页面 | 原文线索

V2-063888 | MATHDB-334851

Gruslys-Leader-Tomon generalized bounded-remainder poset-packing conjecture

摘录 / EXCERPT: Gruslys-Leader-Tomon conjecture. The Boolean lattice $2^{[n]}$ can be partitioned into copies of P and a remainder of at most $c=c(P)$ elements.

conjecture | open | MathDB 页面 | 原文线索

V2-063889 | MATHDB-334854

Polynomiality conjecture for Sylvester sums of compound sequence semigroup complements

摘录 / EXCERPT: Let (A,B) be a suitable pair of k-tuples, let $S_m(A,B)$ denote the mth Sylvester sum of the complement of the associated compound sequence numerical semigroup, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-063890 | MATHDB-334858

Bump-Friedberg generating-function identity for covering groups

摘录 / EXCERPT: Let $(\widetilde{\operatorname{GL}}_r^{(n)},\widetilde{\operatorname{GL}}_R^{(n)})$ be a fundamental pair with $n\mid(\mathbf{p}\cdot n_\alpha)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063891 | MATHDB-334860

The local Langlands correspondence conjecture for Mp(2n)

摘录 / EXCERPT: The local Langlands correspondence conjecture. There should be a local Langlands correspondence

conjecture | open | MathDB 页面 | 原文线索

V2-063892 | MATHDB-334861

The refined character formula conjecture for Mp(2n)

摘录 / EXCERPT: The refined character formula conjecture. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-063893 | MATHDB-334862

The local intertwining relation conjecture for Mp(2n)

摘录 / EXCERPT: The local intertwining relation conjecture. Given ϕ_M and u as above, one expects

conjecture | open | MathDB 页面 | 原文线索

V2-063894 | MATHDB-334876

Drew-Johnson-Olesky-van den Driessche superpattern conjecture for sign patterns

摘录 / EXCERPT: A sign pattern is a matrix whose entries are 0, $+$, or $-$. It is spectrally arbitrary if every monic real polynomial of degree n is the characteristic polynomial of an...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-063895 | MATHDB-334880

Threshold conjecture for universality in bounded-degree graphs

摘录 / EXCERPT: For an integer $\Delta\geq 3$, let $\mathcal{H}(n,\Delta)$ denote the class of graphs on n vertices with maximum degree at most Δ, and say that a graph is...

conjecture | open | MathDB 页面 | 原文线索

V2-063896 | MATHDB-334888

The exceptional Hurwitz structure set conjecture for primitive NET maps

摘录 / EXCERPT: Exceptional Hurwitz structure set conjecture. The Hurwitz structure set of every primitive NET map with constant pullback map is either an MC2 or MC4 Hurwitz structure set, or it i...

conjecture | open | MathDB 页面 | 原文线索

V2-063897 | MATHDB-334914

Constant-energy minimal spectral partitions for symmetric planar domains

摘录 / EXCERPT: Let Ω be a disk, a square, or an equilateral triangle, let k be the number of cells, and let $\mathfrak L_{k,p}(\Omega)$ denote the optimal p-norm energy of a k-partit...

conjecture | open | MathDB 页面 | 原文线索

V2-063898 | MATHDB-334915

Endpoint behavior of optimal p-norm spectral-partition energies

摘录 / EXCERPT: Let Ω be a domain, let k be a positive integer, let $\mathfrak L_{k,p}(\Omega)$ be the optimal p-norm energy of a k-partition, and let $p_{\infty}(\Omega,k)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-063899 | MATHDB-334917

Nonexistence conjecture for Ehrhart-polynomial orthogonal polynomial sequences

摘录 / EXCERPT: Let $(P_j)_{j=0}^{\infty}$ be a family of reflexive polytopes whose Ehrhart polynomials $f_j(x)=H_{P_j}(x)$ form an orthogonal polynomial sequence and satisfy ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-063900 | MATHDB-334918

Interlacing conjecture for Ehrhart polynomials of complete multipartite graphs

摘录 / EXCERPT: Let G be a complete k-partite graph of type $(a_1,\dots,a_k)$, and let $H_{a_1,\dots,a_k}$ denote its Ehrhart polynomial. Complete multipartite Ehrhart conjecture. The polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-063901 | MATHDB-334928

Faber's conjecture on free divisors and normal crossing singularities

摘录 / EXCERPT: Faber's conjecture. A free divisor has only normal crossing singularities if and only if it is normal crossing in codimension 1.

conjecture | open | MathDB 页面 | 原文线索

V2-063902 | MATHDB-334938

The global-existence-versus-blow-up conjecture for radial fair-competition solutions

摘录 / EXCERPT: Let $\chi>0$, $k\in(-N,0)$, and $m=2N/(2N+k)$ in dimension $N\geq1$. Assume that the initial datum $\rho_0\in{\mathcal Y}$ is radially symmetric. Global-existence-versus-blow-up co...

conjecture | open | MathDB 页面 | 原文线索

V2-063903 | MATHDB-334939

The instability conjecture for the critical HLS stationary state

摘录 / EXCERPT: Let $\chi>0$, and let $\rho_{HLS,\lambda^*,c^*}\in{\mathcal Y}$ be the density whose parameters (λ^*,c^*) are given by the stated parameter relation. Instability conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-063904 | MATHDB-334943

Monotonicity conjecture for the degree-coloring index

摘录 / EXCERPT: Degree-coloring index monotonicity conjecture. The degree-coloring index $\tau(G)$ is a monotone function on multigraphs; equivalently, for every multigraph G and every submultig...

conjecture | open | MathDB 页面 | 原文线索

V2-063905 | MATHDB-334946

Converse to the parking-function criterion for strongly common radical solutions

摘录 / EXCERPT: Let $\mu=(\mu_1\geq\mu_2\geq\dots\geq\mu_r)$ be a partition. It admits a strongly common radical solution if, for some integer m, a common radical solution exists for any choice...

conjecture | open | MathDB 页面 | 原文线索

V2-063906 | MATHDB-334947

Refinement conjecture for the secant degeneracy length of binary-form strata

摘录 / EXCERPT: Let $\mu\vdash d$ be a partition, let $\bar\mu$ denote the associated standard stratum, and let $\ell_{\bar\mu}$ and $\ell_{\mu'}$ denote the corresponding secant degeneracy length...

conjecture | open | MathDB 页面 | 原文线索

V2-063907 | MATHDB-334952

Boundary strict-maximum conjecture for second derivatives of the Fokker-Planck semigroup

摘录 / EXCERPT: Let $h>0$, let $S_N(t)$ be the semigroup in the paper, and let u_0 satisfy the conditions of Theorem. Write N_0 for the threshold appearing there. Boundary strict-maximum conje...

conjecture | open | MathDB 页面 | 原文线索

V2-063908 | MATHDB-334953

Boundary-layer decay conjecture for second derivatives of the Fokker-Planck semigroup

摘录 / EXCERPT: Let $h>0$, let $S_N(t)$ be the semigroup in the paper, and let u_0 satisfy the conditions of Theorem. Let N_0 be the threshold appearing there. Boundary-layer decay conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-063909 | MATHDB-334961

Mirror-brane conjecture for Aganagic-Vafa branes

摘录 / EXCERPT: Let $X=X_\Delta^3$ be a Calabi-Yau 3-fold and let $L=L_{AV}$ be an Aganagic-Vafa A-brane. Let $X^\vee\subset(\mathbb{C}^\times)^2\times\mathbb{C}^2$ be its mirror, and let W be t...

conjecture | open | MathDB 页面 | 原文线索

V2-063910 | MATHDB-334963

Conjecture on the circular separation dimension of planar graphs

摘录 / EXCERPT: Planar circular separation dimension conjecture. The circular separation dimension of G is at most two:

conjecture | open | MathDB 页面 | 原文线索

V2-063911 | MATHDB-334966

The weak Hilbert-Smith conjecture for free proper actions

摘录 / EXCERPT: Weak Hilbert-Smith conjecture. G is a Lie group.

conjecture | open | MathDB 页面 | 原文线索

V2-063912 | MATHDB-334972

The small Fuss-Schröder path-sparse noncrossing partition bijection conjecture

摘录 / EXCERPT: Let k and n be positive integers, and let $\lambda=\lambda_1\lambda_2\ldots\lambda_\ell$ be a type with $\lvert\lambda\rvert$ denoting its size. A small (k,k)-Fuss-Schröder p...

conjecture | open | MathDB 页面 | 原文线索

V2-063913 | MATHDB-334973

Hyponormality conjecture for Cesàro matrices of positive integer order

摘录 / EXCERPT: Let $N>4$ be a positive integer. The Cesàro matrix $C(N)$ has entries ... and define the diagonal operator ... Hyponormality conjecture for Cesàro matrices. The operator $C(N)$ is boun...

conjecture | open | MathDB 页面 | 原文线索

V2-063914 | MATHDB-334982

The positivity conjecture for integrals of Gegenbauer polynomials

摘录 / EXCERPT: Positivity conjecture. $F_n^{\lambda,\delta}(t)>0$ for every $t\in(0,\pi]$ if $\delta\geq\lambda+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-063915 | MATHDB-334983

The positivity conjecture for Jacobi polynomial integrals

摘录 / EXCERPT: Let $\alpha,\beta\in(0,\infty)\setminus\mathbb{N}$, and let $F_n^{(\alpha,\beta),\delta}(t)$ denote the Jacobi-polynomial integral defined earlier in the paper. For $t\in(0,\pi]$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-063916 | MATHDB-334994

The length-64 bound for perfect simple-unit-quaternion sequences with the AOP

摘录 / EXCERPT: Length-64 conjecture. The longest perfect sequence over the simple unit quaternions with the AOP is of length 64. This conjecture is motivated by an exhaustive random search that...

conjecture | open | MathDB 页面 | 原文线索

V2-063917 | MATHDB-335008

Nonvanishing conjecture for the base-change trace

摘录 / EXCERPT: Let F and E be number fields as in the paper, let n be a positive integer, and let $_{E/F}(\Sigma_{\phi}^{S_0}(X))$ denote the associated base-change test function. For a cus...

conjecture | open | MathDB 页面 | 原文线索

V2-063918 | MATHDB-335009

Conjecture on transfers of test functions

摘录 / EXCERPT: Let $E \geq F' \geq F$ be a tower of number fields, let S be a finite set of places of F containing the infinite places, and let S' and S_0 be the places of F' and E ab...

conjecture | open | MathDB 页面 | 原文线索

V2-063919 | MATHDB-335011

Solvable base-change and descent conjecture

摘录 / EXCERPT: Let E/F' be a solvable extension of number fields, and let Π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_E)$ that is $\mathrm{Gal}(E/F')$-invariant....

conjecture | open | MathDB 页面 | 原文线索

V2-063920 | MATHDB-335025

Quartic Legendre multiplier sequence region conjecture

摘录 / EXCERPT: Consider the sequence ... A pair (b,c) lies in the multiplier sequence region when this sequence is a multiplier sequence for the Legendre basis. Quartic Legendre multiplier sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-063921 | MATHDB-335026

Conjecture on hyperbolicity preservation by powers of the Legendre differential operator

摘录 / EXCERPT: Hyperbolicity-preservation conjecture. The operator

conjecture | open | MathDB 页面 | 原文线索

V2-063922 | MATHDB-335036

Maximal non-Gorenstein locus conjecture for symmetric orbit closures

摘录 / EXCERPT: Let Y_{γ} be an orbit closure, and let $\operatorname{Maxsing}(Y_{\gamma})$ be the set of Bruhat-maximal orbits along which Y_{γ} is singular. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063923 | MATHDB-335037

Gorenstein pattern-avoidance conjecture for $(p,2)$-clans

摘录 / EXCERPT: Let γ be a $(p,2)$-clan with $p\geq 2$, and let Y_{γ} be its orbit closure. Gorenstein pattern-avoidance conjecture. Y_{γ} is Gorenstein if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-063924 | MATHDB-335038

lci and Gorenstein equivalence conjecture for $(p,2)$-clans

摘录 / EXCERPT: Let γ be any $(p,2)$-clan with $p\geq 2$, and let Y_{γ} be its orbit closure. lci-Gorenstein equivalence conjecture. Y_{γ} is a local complete intersection if...

conjecture | open | MathDB 页面 | 原文线索

V2-063925 | MATHDB-335041

Lower-bound conjecture for diamond-saturated families avoiding the extremes

摘录 / EXCERPT: Let n be a positive integer, let $\mathcal{B}_n$ be the Boolean lattice, and let $\mathcal{F}\subseteq\mathcal{B}_n$ be an induced-$\mathcal{D}_2$-saturated family that avoids bo...

conjecture | open | MathDB 页面 | 原文线索

V2-063926 | MATHDB-335047

Moment-graph formula conjecture for the Poincaré polynomial of a fundamental domain

摘录 / EXCERPT: Moment-graph formula conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-063927 | MATHDB-335050

Thomassé's 8-edge-connected antisymmetric 5-flow conjecture

摘录 / EXCERPT: Let $\vec{G}$ be a directed graph. An antisymmetric B-flow is a B-flow in which no two arcs receive inverse elements of B; in particular, it is nowhere-zero when $0=-0$. Thom...

conjecture | open | MathDB 页面 | 原文线索

V2-063928 | MATHDB-335056

Bipartite determinant inequality for Gaussian Markov random fields

摘录 / EXCERPT: Let $G=(V,E)$ be a finite bipartite graph. For $x\in(-1,1)$, let $\tau(G,x)=\det(\Sigma(G,x))$, where $\Sigma(G,x)$ is the unique positive-definite matrix with diagonal entries 1...

conjecture | open | MathDB 页面 | 原文线索

V2-063929 | MATHDB-335085

Sundaram's conjecture on tableaux with major index divisible by n

摘录 / EXCERPT: Let $\lambda\vdash n$ with $n\geq 1$, and let $a_{\lambda,0}$ denote the number of standard Young tableaux of shape λ whose major index is congruent to 0 modulo n. Sund...

conjecture | open | MathDB 页面 | 原文线索

V2-063930 | MATHDB-335092

Scaling limit conjecture for free Boltzmann triangulations with simple boundary

摘录 / EXCERPT: Scaling limit conjecture for free Boltzmann triangulations. For $i\in \\{\mathrm{I},\mathrm{II}\\}$ and every sequence satisfying the stated boundary-length scaling, the spaces...

conjecture | open | MathDB 页面 | 原文线索

V2-063931 | MATHDB-335112

The bounded-index gradient conjecture for sliced inverse regression

摘录 / EXCERPT: Let d be bounded, let $0<\lambda\leq 1$, and let $\mathcal{F}_{d}(\lambda,\kappa)$ be the function class of d-index models. A function f_{λ} is d-variate when it is d...

conjecture | open | MathDB 页面 | 原文线索

V2-063932 | MATHDB-335122

Steenrod-generation conjecture for the invariants $a_{m,2,k'}$

摘录 / EXCERPT: Fix m and let ... and ... Here the $a_{m,2,k'}$ are the invariants introduced for the $n=2$ case, and Steenrod operations act on them. Steenrod-generation conjecture. Every element o...

conjecture | open | MathDB 页面 | 原文线索

V2-063933 | MATHDB-335126

Additive-basis conjecture for parabolic invariants when $m=2$

摘录 / EXCERPT: Let P_α be the parabolic subgroup of $Gl_n(\mathbb{F}_q)$ associated with a composition α of n, and let ... Let a_n, $b_{r,k}$, $c_{r,s,k}$, and d_r be the famili...

conjecture | open | MathDB 页面 | 原文线索

V2-063934 | MATHDB-335127

Conjecture A on semi-convergence for Artin-Tits monoids

摘录 / EXCERPT: Let M be an Artin-Tits monoid, and let $\mathcal{R}$-reduction be the reduction system on multifractions associated with M. Conjecture A. $\mathcal{R}$-reduction is semi-conver...

conjecture | open | MathDB 页面 | 原文线索

V2-063935 | MATHDB-335128

Conjecture on padded semi-convergence for Artin-Tits monoids

摘录 / EXCERPT: Let M be an Artin-Tits monoid. A padding of a multifraction is obtained by inserting an even number of trivial components at its beginning; semi-convergence up to f-padding mea...

conjecture | open | MathDB 页面 | 原文线索

V2-063936 | MATHDB-335131

The partial-decoupling conjecture for critical points of the DPP likelihood

摘录 / EXCERPT: Partial-decoupling conjecture. The kernels of the partial decouplings of Z are the only critical points of Φ_{L^*}.

conjecture | open | MathDB 页面 | 原文线索

V2-063937 | MATHDB-335146

The Dirichlet-series Hardy-space duality conjecture

摘录 / EXCERPT: Let $p\leq2$, let $\mathcal{H}^p$ denote the Hardy space of Dirichlet series, and let $\varphi_{1/p}$ be the Dirichlet series defined in the source by equation. Dirichlet-series Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-063938 | MATHDB-335152

A q-Narayana positivity conjecture

摘录 / EXCERPT: Let n and r be positive integers, and let a be a non-negative integer. Define the q-Narayana numbers by ... Let $E_{n,r,a,j}(q)$ denote the Laurent polynomial expression in t...

conjecture | open | MathDB 页面 | 原文线索

V2-063939 | MATHDB-335153

A cyclic q-Narayana-type positivity conjecture

摘录 / EXCERPT: Let m be a positive integer, let $n_1,\ldots,n_m$ be positive integers, and set $n_{m+1}=n_1$. Let j be an integer and let r be a non-negative integer. The cyclic q-Narayana-...

conjecture | open | MathDB 页面 | 原文线索

V2-063940 | MATHDB-335155

Direction-separated tubes satisfy polynomial Wolff axioms

摘录 / EXCERPT: Direction-separated polynomial Wolff conjecture. Every set of tubes pointing in δ-separated directions satisfies the polynomial Wolff axioms; more precisely, it satisfies th...

conjecture | open | MathDB 页面 | 原文线索

V2-063941 | MATHDB-335158

Polynomiality conjecture for multispecies zero-range stationary states

摘录 / EXCERPT: Let $|\bar{P}\rangle$ be the stationary state in a sector k, with configurations $(\sigma_1,\ldots,\sigma_L)$ belonging to $\mathcal{B}(k)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063942 | MATHDB-335161

Super-exponential growth of non-isomorphic locally biased functions

摘录 / EXCERPT: Let $n>0$ be even. Let $p=b/2^k$, where $1\leq b\leq 2^k$, $k\geq 1$ and 2^k divides n. Let B_p^n be the set of non-isomorphic locally p-biased functions. Growth conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-063943 | MATHDB-335164

Farkas-Pandharipande conjecture for the class ${\rm H}_g(Z-P)$

摘录 / EXCERPT: Let $m\geq 1$. For a smooth curve $(C,x_1,\ldots,x_{n+m})$ with markings, let r be a positive integer and let L satisfy ... Let $\overline{\mathcal{M}}_{g,Z-P}^{1/r}$ be the comp...

conjecture | open | MathDB 页面 | 原文线索

V2-063944 | MATHDB-335165

Prellberg-Stanton overpartition coefficient nonnegativity conjecture

摘录 / EXCERPT: Let n be a positive integer, and write the resulting formal power series in q as a polynomial or power series with coefficients in the relevant coefficient ring. Prellberg-Stan...

conjecture | open | MathDB 页面 | 原文线索

V2-063945 | MATHDB-335166

Double unimodality conjecture for over-(q,t)-binomial coefficients

摘录 / EXCERPT: For positive integers m and n, let $\overline{{m+n\brack n}}_{q,t}$ denote the over-(q,t)-binomial coefficient. A polynomial $P(q,t)=\sum_{k=0}^r\sum_{j=0}^s a_{k,j}t^kq^j$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-063946 | MATHDB-335167

Unimodality conjecture for specialized over-(q,t)-binomial coefficients

摘录 / EXCERPT: For positive integers m and n, let $\overline{{m+n\brack n}}_{q,1}$ denote the specialization at $t=1$ of the over-(q,t)-binomial coefficient. Specialized over-(q,t)-binomi...

conjecture | open | MathDB 页面 | 原文线索

V2-063947 | MATHDB-335172

The union-of-balls simplex conjecture

摘录 / EXCERPT: For $N=n+1$, let $v_1,\ldots,v_N\in\mathbb R^n$ satisfy $|v_1|=\dots=|v_N|=r$, and let $B_{v_i}(R)$ denote the ball of radius R centered at v_i. The union-of-balls simplex conj...

conjecture | open | MathDB 页面 | 原文线索

V2-063948 | MATHDB-335183

The Prague-dimension upper-bound conjecture for random graphs

摘录 / EXCERPT: Let G be sampled from the binomial random graph model ${\mathcal G}(n,p)$, where p is a constant with $0<p<1$. Let $f_{\rm Pra}(G)$ denote the Prague dimension of G, namely t...

conjecture | open | MathDB 页面 | 原文线索

V2-063949 | MATHDB-335184

The clique-cover chromatic-index upper-bound conjecture for random graphs

摘录 / EXCERPT: Let G be sampled from the binomial random graph model ${\mathcal G}(n,p)$, where p is a constant with $0<p<1$. Let $\theta'_0(G)$ be the minimum, over all clique covers of G,...

conjecture | open | MathDB 页面 | 原文线索

V2-063950 | MATHDB-335188

Cusick-Li-Stănică-type conjecture for perturbations of symmetric Boolean functions

摘录 / EXCERPT: Cusick-Li-Stănică-type conjecture. No perturbation of the form $\sigma_{n,k}+X_1$ is balanced except for the trivial cases above.

conjecture | open | MathDB 页面 | 原文线索

V2-063951 | MATHDB-335189

Redheffer-type bounds for normalized Bessel functions

摘录 / EXCERPT: Redheffer-type bounds conjecture. For $|x|<j_{ u,1}$,

conjecture | open | MathDB 页面 | 原文线索

V2-063952 | MATHDB-335196

Canonical coloured-antichain conjecture for minimal unbounded clique-width classes

摘录 / EXCERPT: Let X be a minimal hereditary class of graphs of unbounded clique-width. A canonical infinite coloured antichain is the canonical infinite coloured antichain associated with such...

conjecture | open | MathDB 页面 | 原文线索

V2-063953 | MATHDB-335203

The distribution of the narrow class group rank difference for odd-degree S_n-fields

摘录 / EXCERPT: The rank-difference conjecture. As K varies over such S_n-fields, ordered by absolute discriminant, the density of fields satisfying $\rho^+-\rho=k$, for...

conjecture | open | MathDB 页面 | 原文线索

V2-063954 | MATHDB-335208

Maximum-gap sampling conjecture for shift-invariant spline spaces

摘录 / EXCERPT: Let Q_m denote the spline generator and let $V(Q_m)$ be the shift-invariant spline space it generates. A sequence $\\{x_i:i\in\mathbb{Z}\\}$ is separated if its distinct points h...

conjecture | open | MathDB 页面 | 原文线索

V2-063955 | MATHDB-335209

Maximum-gap derivative sampling conjecture for shift-invariant spaces

摘录 / EXCERPT: Let $V(\phi)$ be a shift-invariant space, let k and ν be positive integers, and let $\\{x_i:i\in\mathbb{Z}\\}$ be a separated set. For a function f, consider the nonuniform...

conjecture | open | MathDB 页面 | 原文线索

V2-063956 | MATHDB-335212

Existence of nice nef partitions and toric Landau-Ginzburg models for weighted Fano complete intersections

摘录 / EXCERPT: Let X be a smooth well formed Fano weighted complete intersection. A nice nef partition conjecture asserts that there exists a nice nef partition for X, and that X has a tori...

conjecture | open | MathDB 页面 | 原文线索

V2-063957 | MATHDB-335213

Gaiotto's scaling-limit conjecture for non-Abelian Hodge correspondence

摘录 / EXCERPT: Gaiotto's conjecture. The scaling limit

conjecture | open | MathDB 页面 | 原文线索

V2-063958 | MATHDB-335215

Trinomial family conjecture for finding polynomials satisfying Property condition

摘录 / EXCERPT: Trinomial family conjecture. The family $\mathcal{F}$ contains at least one polynomial satisfying Property condition for $r=2$.

conjecture | open | MathDB 页面 | 原文线索

V2-063959 | MATHDB-335220

Faithfulness conjecture for the Kontsevich integral of bottom tangles in handlebodies

摘录 / EXCERPT: Let $\bfF$ denote the category of finitely generated free groups. Consider the functor ... that assigns to each bottom tangle $T:m\to n$ the homomorphism $(i_T)_*:F_n\to F_m$. The fu...

conjecture | open | MathDB 页面 | 原文线索

V2-063960 | MATHDB-335225

KLP's compression symmetry characterization conjecture for truncated Toeplitz operators

摘录 / EXCERPT: Let u be an inner function on the unit disc, let $K_u=H^2\ominus uH^2$, and let $\mathcal{T}_u$ be the space of truncated Toeplitz operators on K_u. For an inner divisor v of...

conjecture | open | MathDB 页面 | 原文线索

V2-063961 | MATHDB-335226

Periodicity conjecture for increasing tableaux of shape $3 \times n$

摘录 / EXCERPT: Periodicity conjecture. For every $T \in \operatorname{Inc}^q(3 \times n)$,

conjecture | open | MathDB 页面 | 原文线索

V2-063962 | MATHDB-335228

Charleston and Steel's recurrence conjecture for ancestral sequence reconstruction

摘录 / EXCERPT: Charleston and Steel's conjecture. For T_k and $r\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-063963 | MATHDB-335230

The complete multipartite fractional-defect conjecture

摘录 / EXCERPT: For positive integers m, a, and k, let $K^{(m)}_a$ be the complete m-partite graph with a vertices in each partite set. Let $D(G,k)$ denote the minimum defect in a fracti...

conjecture | open | MathDB 页面 | 原文线索

V2-063964 | MATHDB-335232

Bounded chromatic number for locally bounded tournaments

摘录 / EXCERPT: Local-to-global tournament coloring conjecture. There is a function f such that every t-local tournament T satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063965 | MATHDB-335234

The GM-MDS conjecture for constrained generator matrices

摘录 / EXCERPT: Let M be an $m\times n$ binary support matrix. For a nonempty set $I\subseteq\{1,\dots,m\}$, write M_i for the ith row and $\operatorname{supp}(M_i)$ for its support. The mat...

conjecture | open | MathDB 页面 | 原文线索

V2-063966 | MATHDB-335235

The TM-MDS conjecture on generic nonsingularity

摘录 / EXCERPT: Let M be an $m\times n$ binary matrix satisfying the MDS condition. Let $N=\{\alpha_1,\dots,\alpha_n\}$ be independent indeterminates, let $V=V(N)$ be a generic $m\times n$ Vande...

conjecture | open | MathDB 页面 | 原文线索

V2-063967 | MATHDB-335239

Irreducibility conjecture for the class polynomial of χ^*

摘录 / EXCERPT: Let χ^* be the level 1 almost holomorphic modular function defined by ... where E_4 and E_6 are the usual Eisenstein series and...

conjecture | open | MathDB 页面 | 原文线索

V2-063968 | MATHDB-335254

Aleksandrov-Buryak-Tessler conjecture for the extended refined partition function

摘录 / EXCERPT: Extended refined Kontsevich-Penner conjecture. For any N we have

conjecture | open | MathDB 页面 | 原文线索

V2-063969 | MATHDB-335257

Kahn's bounded-degree spanning tree conjecture

摘录 / EXCERPT: Kahn's conjecture. There is a constant C such that, if $p\ge C\log n/n$, then $G(n,p)$ almost surely contains T.

conjecture | open | MathDB 页面 | 原文线索

V2-063970 | MATHDB-335258

The bounded-degree spanning graph threshold conjecture

摘录 / EXCERPT: Bounded-degree spanning graph threshold conjecture. The random graph $G(n,p)$ almost surely contains H whenever

conjecture | open | MathDB 页面 | 原文线索

V2-063971 | MATHDB-335259

Resilience conjecture for universality of bounded-degree spanning trees

摘录 / EXCERPT: Bounded-degree tree universality resilience conjecture. The conclusion of the cited resilience theorem-that the local resilience of $G(n,p)$ with respect to being universal for...

conjecture | open | MathDB 页面 | 原文线索

V2-063972 | MATHDB-335267

The Dubrovin-Yang conjecture on the special cubic Hodge hierarchy

摘录 / EXCERPT: Let $p,q,r e 0$ satisfy the local Calabi-Yau condition $1/p+1/q+1/r=0$, with $r=-pq/(p+q)$. Set $s_j=-(2j-2)!(p^{2j-1}+q^{2j-1}+r^{2j-1})$, $alpha=p/\sqrt{p+q}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-063973 | MATHDB-335268

Parity-subsequence extinction conjecture for the inter-stage Ricker model

摘录 / EXCERPT: Let $\{x_n\}$ be a positive solution of equation (4ab), and assume that condition (fxp0) holds. Consider its even subsequence $\{x_{2n}\}$ and odd subsequence $\{x_{2n-1}\}$. Parit...

conjecture | open | MathDB 页面 | 原文线索

V2-063974 | MATHDB-335277

Ciocan-Fontanine-Kim higher-genus quasimap wall-crossing conjecture

摘录 / EXCERPT: Let Z be a complete intersection in projective space, and fix $g,n\geq 0$. For each effective degree β, let q^β be the corresponding Novikov monomial, let...

conjecture | open | MathDB 页面 | 原文线索

V2-063975 | MATHDB-335285

Extension of the gamma-denominator inequalities to all positive alpha

摘录 / EXCERPT: Let $g(\mu;x)$ be the function defined in Theorem, and let $R_g>0$ be its radius of convergence in equation. Assume the conditions on the parameters in that theorem other than...

conjecture | open | MathDB 页面 | 原文线索

V2-063976 | MATHDB-335303

The spherical-twist conjecture for generic monodromy around discriminants

摘录 / EXCERPT: Generic monodromy conjecture. The generic monodromy around the component of the discriminant is given by a spherical twist, where A is the category of D-branes in the as...

conjecture | open | MathDB 页面 | 原文线索

V2-063977 | MATHDB-335308

Conjecture on the number of quadruple points in the example

摘录 / EXCERPT: Let f be the map-germ considered in the paper, and let a stable perturbation of f have quadruple points. Quadruple-point count conjecture. The number of quadruple points is...

conjecture | open | MathDB 页面 | 原文线索

V2-063978 | MATHDB-335317

Generic Strichartz conjecture for the Schrödinger equation on irrational tori

摘录 / EXCERPT: Generic Strichartz conjecture. For arbitrarily small $\varepsilon>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-063979 | MATHDB-335318

Polynomiality and degree of pre-moments in the parameter d

摘录 / EXCERPT: Polynomial pre-moment conjecture. For each n, the kth pre-moment $m_k(d,n)$ of $X_{d,n}$ is a polynomial in d. Further, the degree of this polynomial is...

conjecture | open | MathDB 页面 | 原文线索

V2-063980 | MATHDB-335320

Generalized-Fibonacci form of pre-moments for fixed d

摘录 / EXCERPT: Generalized-Fibonacci pre-moment conjecture. For each d, the kth pre-moment $m_k(d,n)$ of $X_{d,n}$ is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-063981 | MATHDB-335322

Joint generalized-Fibonacci form of pre-moments

摘录 / EXCERPT: Joint pre-moment conjecture. The kth pre-moment $m_k(d,n)$ of $X_{d,n}$ is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-063982 | MATHDB-335326

Trivial automorphism conjecture for component graphs of the uniform spanning forest

摘录 / EXCERPT: Let $\mathbb G$ be a d-dimensional transitive graph for some $d>8$, let $\mathfrak F$ be its uniform spanning forest, and let $r\geq 1$. Write $\mathcal C_r(\mathfrak F)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-063983 | MATHDB-335342

Point-vortex limit for a massive or massless particle in a two-dimensional perfect fluid

摘录 / EXCERPT: Point-vortex limit conjecture. A massive particle moves according to

conjecture | open | MathDB 页面 | 原文线索

V2-063984 | MATHDB-335352

Permutation conjecture for a rational function on roots of unity

摘录 / EXCERPT: Let k be an even integer and set $q=5^k$. Let … be the group of $(q+1)$-st roots of unity in ${\mathbb F}_{q^2}$. Permutation conjecture. The rational function … permutes...

conjecture | open | MathDB 页面 | 原文线索

V2-063985 | MATHDB-335362

The $(2k-1)$-edge-connected bounded-out-degree orientation conjecture

摘录 / EXCERPT: Let G be a graph, let k be an integer with $k\ge 3$, and let $p:V(G)\rightarrow Z_k$ satisfy ... A p-orientation is an orientation satisfying the prescribed modulo-k out-degr...

conjecture | open | MathDB 页面 | 原文线索

V2-063986 | MATHDB-335363

The bipartite tree-connected factor conjecture

摘录 / EXCERPT: Let G be a bipartite graph, let $k\ge 3$ and m be positive integers, and let $f:V(G)\rightarrow Z_k$ be a compatible mapping. The bipartite tree-connected factor conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-063987 | MATHDB-335364

The bounded-degree decomposition conjecture for 4-edge-connected graphs

摘录 / EXCERPT: Let G be a 4-edge-connected graph with maximum degree at most 5. The bounded-degree decomposition conjecture. The graph G can be decomposed into two connected factors G_1...

conjecture | open | MathDB 页面 | 原文线索

V2-063988 | MATHDB-335381

The strict support-dimension conjecture for Serre relations

摘录 / EXCERPT: Let $K=\mathbb{C}$ and let $\Pi=\Pi(C,D)$. Define the support of a constructible function $f\colon \operatorname{nil}_E(\Pi,d)\to\mathbb{C}$ by ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-063989 | MATHDB-335391

Sparse-signal extension of the one-sparse lower bound

摘录 / EXCERPT: Sparse-signal lower-bound conjecture. When $p\geq 8/m$, if the risk of an adaptive sensing and test procedure is less or equal to $b5$, then necessarily

conjecture | open | MathDB 页面 | 原文线索

V2-063990 | MATHDB-335399

McKay-Wormald's asymptotic enumeration conjecture for graphs with given degree sequence

摘录 / EXCERPT: Let ${\bf d}= (d_1,\ldots,d_n)$ be a degree sequence, let $g({\bf d})$ denote the number of graphs with degree sequence ${\bf d}$, let $d=\frac{1}{n}\sum_{i=1}^n d_i$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-063991 | MATHDB-335404

Coefficient-reflection conjecture for finite multiple harmonic sums

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, and let $\gb_a$ denote the $\mathcal{A}$-Bernoulli numbers. Write $a_1+\dots+a_l\vdash n$ for the summation condition used in the source, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-063992 | MATHDB-335407

Martingale-polar equivalence conjecture on the graph of convex components

摘录 / EXCERPT: Martingale-polar equivalence conjecture. In the setting of the main convex-component theorem, for every $4N\subseteq\operatorname{gr}(C)4$,

conjecture | open | MathDB 页面 | 原文线索

V2-063993 | MATHDB-335408

Isostatic conjecture for saturated triangulated disk packings

摘录 / EXCERPT: Isostatic conjecture. The density ρ of every such packing satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-063994 | MATHDB-335409

Universal formula conjecture for semistable Vafa-Witten invariants

摘录 / EXCERPT: Let S be a smooth projective surface, let $P^\perp_{\alpha}(n)$ be the Joyce-Song pair invariant for charge $\alpha=(r,c_1(L),c_2)$, and let $\chi(\alpha(n))$ denote the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-063995 | MATHDB-335418

Labourie's uniqueness conjecture for Hitchin representations

摘录 / EXCERPT: Labourie's conjecture. There is a unique complex structure $J\in\Teich(\Sigma)$ on Σ such that the ρ-equivariant harmonic map from $(\widetilde{\Sigma},J)$ to G/K is...

conjecture | open | MathDB 页面 | 原文线索

V2-063996 | MATHDB-335420

Generalized alternating binomial-sum divisibility conjecture

摘录 / EXCERPT: Generalized divisibility conjecture. is congruent to zero modulo

conjecture | open | MathDB 页面 | 原文线索

V2-063997 | MATHDB-335421

Positivity conjecture for alternating sums of q-Narayana numbers

摘录 / EXCERPT: Let n and r be positive integers, let $0\leqslant j\leqslant 2r-1$, and let $C_n(q)$ and $N_q(2n+1,n+k+1)$ denote the q-Catalan and q-Narayana quantities used in the paper. Pos...

conjecture | open | MathDB 页面 | 原文线索

V2-063998 | MATHDB-335422

Multivariate positivity conjecture for alternating q-binomial sums

摘录 / EXCERPT: Multivariate positivity conjecture. is a polynomial in q with nonnegative integer coefficients. This is stated as a generalization of the q-Narayana positivity conjecture; no res...

conjecture | open | MathDB 页面 | 原文线索

V2-063999 | MATHDB-335423

Polynomial Bézout identity for consecutive q-powers

摘录 / EXCERPT: Let n and r be positive integers. Polynomial Bézout conjecture. There exist polynomials $A_{n,r}$ and $B_{n,r}$ in q with integer coefficients such that ... This conjecture con...

conjecture | open | MathDB 页面 | 原文线索

V2-064000 | MATHDB-335425

Fatou's conjecture for quadratic rational maps

摘录 / EXCERPT: Consider complex rational maps of the sphere, in particular quadratic maps. Such a map satisfies axiom A when its non-wandering set is a finite union of basic pieces, and its criti...

conjecture | open | MathDB 页面 | 原文线索

V2-064001 | MATHDB-335431

Generalized Uniformness Conjecture for Kolakoski sequences

摘录 / EXCERPT: Generalized Uniformness Conjecture. The asymptotic density of S equals 2^{-k-1}.

conjecture | open | MathDB 页面 | 原文线索

V2-064002 | MATHDB-335433

Weight-ten basis conjecture for non-alternating Euler sums

摘录 / EXCERPT: Weight-ten basis conjecture. All non-alternating Euler sums of weight ten can be expressed as a rational linear combination of these seven quantities.

conjecture | open | MathDB 页面 | 原文线索

V2-064003 | MATHDB-335434

Alternating double-zeta reduction conjecture

摘录 / EXCERPT: Let m be a positive integer, and let $\zeta(\overline{r},s)$ denote an alternating double zeta value, while a non-alternating double zeta value has no barred index. Alternating d...

conjecture | open | MathDB 页面 | 原文线索

V2-064004 | MATHDB-335443

Equality of abelian and Artin-Schreier ramification functions

摘录 / EXCERPT: Equality conjecture. The two generalized ramification functions satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-064005 | MATHDB-335453

Converse to the finite-core-point criterion up to normalizer equivalence

摘录 / EXCERPT: Normalizer-equivalence conjecture. If $\operatorname{Fix}(G)^{\perp}$ contains a rational G-invariant subspace other than $\{0\}$ and $\operatorname{Fix}(G)^{\perp}$ itself, then...

conjecture | open | MathDB 页面 | 原文线索

V2-064006 | MATHDB-335454

The KGB obstruction conjecture for local extensions with cyclic p-Sylow subgroup

摘录 / EXCERPT: KGB obstruction conjecture. The KGB obstruction is the only obstruction to the local lifting problem for local G-extensions where G has cyclic p-Sylow subgroup. In particular...

conjecture | open | MathDB 页面 | 原文线索

V2-064007 | MATHDB-335455

The dihedral weak local Oort conjecture in characteristic two

摘录 / EXCERPT: Dihedral weak local Oort conjecture. The group D_{2^n} is a weak local Oort group for 2.

conjecture | open | MathDB 页面 | 原文线索

V2-064008 | MATHDB-335456

Saidi's Oort conjecture revisited on lifting cyclic extensions in towers

摘录 / EXCERPT: Oort conjecture revisited. Local cyclic extensions are liftable in towers: given the local G-extension $k[[z]]/k[[t]]$ and a characteristic-zero lift $R[[S]]/R[[T]]$ of a subexte...

conjecture | open | MathDB 页面 | 原文线索

V2-064009 | MATHDB-335457

Upper semicontinuity conjecture for the mother hedgehog

摘录 / EXCERPT: Let $c\in\partial\boldsymbol{\Delta}$ be a non-parabolic parameter, let f_c be the corresponding quadratic polynomial, and let H_c denote its mother hedgehog: if the closed Sie...

conjecture | open | MathDB 页面 | 原文线索

V2-064010 | MATHDB-335465

The mod-p Artin-Tate integrality conjecture

摘录 / EXCERPT: Let F be a totally real number field, and let $\rho: \operatorname{Gal}(\bar{\mathbb Q}/F) \rightarrow \operatorname{GL}_n(\bar{\mathbb Q}_p)$ be an irreducible representation cu...

conjecture | open | MathDB 页面 | 原文线索

V2-064011 | MATHDB-335466

The mod-p Artin L-value reduction conjecture

摘录 / EXCERPT: Let F be a totally real number field, and let $\bar\rho: \operatorname{Gal}(\bar{\mathbb Q}/F)\rightarrow\operatorname{GL}_n(\bar{\mathbb F}_p)$ be a semisimple modular represent...

conjecture | open | MathDB 页面 | 原文线索

V2-064012 | MATHDB-335468

The Deligne-Ribet integrality conjecture for CM characters

摘录 / EXCERPT: Let E be a totally real number field, and let $\chi:\mathbb A_E^\times/E^\times\rightarrow\bar{\mathbb Z}_p^\times$ be a finite-order character cutting out a non-real CM extensio...

conjecture | open | MathDB 页面 | 原文线索

V2-064013 | MATHDB-335469

Bergelson-Hindman-Leader partition conjecture for the nonzero rationals

摘录 / EXCERPT: Let $\mathbb{Q}\setminus\{0\}$ be partitioned into finitely many sets, ... For a sequence $\langle x_n\rangle_{n=1}^{\infty}$, let $FS(\langle x_n\rangle_{n=1}^{\infty})$ denote its...

conjecture | open | MathDB 页面 | 原文线索

V2-064014 | MATHDB-335473

Pole-constraint conjecture for stable-pairs descendent partition functions

摘录 / EXCERPT: Let X be a nonsingular projective 3-fold and let $\beta\in H_2(X,\mathbb Z)$ be a nonzero effective class. Let $d=\operatorname{div}(\beta)$ be the divisibility of the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-064015 | MATHDB-335474

Stable-pairs descendent correspondence for absolute 3-folds

摘录 / EXCERPT: Let X be a nonsingular projective 3-fold, let $\beta\in H_2(X,\mathbb Z)$, let $d_\beta=\int_\beta c_1(X)$, and let $\gamma_i\in H^*(X)$. The barred descendent insertion...

conjecture | open | MathDB 页面 | 原文线索

V2-064016 | MATHDB-335487

Unboundedness conjecture for the 2-adic valuations of $b_m(n)$

摘录 / EXCERPT: Let $b_m(n)$ be the sequence considered in the paper and let $m\in\mathbb{N}_{\geq2}$ be such that m is not of the form 2^k-1 for any $k\in\mathbb{N}_{+}$. Unbounded valuation...

conjecture | open | MathDB 页面 | 原文线索

V2-064017 | MATHDB-335488

Sign-exception density conjecture for $t_m(n)$

摘录 / EXCERPT: Sign-exception density conjecture. The sets $\mathcal{A}_{m,j}$ are infinite and have asymptotic density zero:

conjecture | open | MathDB 页面 | 原文线索

V2-064018 | MATHDB-335489

Kählerness conjecture for moduli spaces of stable sheaves on K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $v=(r,\xi,a)\in H^{2*}(S,\mathbb{Z})$ be a Mukai vector, and let ω be a v-generic Kähler class satisfying the hypotheses under which the modul...

conjecture | open | MathDB 页面 | 原文线索

V2-064019 | MATHDB-335506

Generic SNP conjecture for nonsymmetric Macdonald polynomials

摘录 / EXCERPT: Let $\alpha\in\mathbb Z_{\geq0}^n$, and let $E_\alpha(X;q,t)$ be the nonsymmetric Macdonald polynomial. Write ... where $<_S$ is the ordering generated by the stated transposition an...

conjecture | open | MathDB 页面 | 原文线索

V2-064020 | MATHDB-335507

SNP conjecture for key polynomials

摘录 / EXCERPT: Let $\alpha=(\alpha_1,\alpha_2,\ldots)\in\mathbb Z_{\geq0}^{\infty}$ have finite size, and let κ_α be the key polynomial defined using Demazure operators. A polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-064021 | MATHDB-335510

SNP conjecture for Demazure atoms

摘录 / EXCERPT: Let α be a weak composition, and let A_α be the Demazure atom obtained from x^α by applying the operators $\widehat\pi_i=\pi_i-\operatorname{id}$ according to...

conjecture | open | MathDB 页面 | 原文线索

V2-064022 | MATHDB-335511

SNP conjecture for Schubert polynomials

摘录 / EXCERPT: For $w\in S_\infty$, let ${\mathfrak S}_w$ be the Schubert polynomial defined recursively from the longest permutation using divided-difference operators. A polynomial has SNP when...

conjecture | open | MathDB 页面 | 原文线索

V2-064023 | MATHDB-335512

SNP conjecture for double Schubert polynomials

摘录 / EXCERPT: Let $X=\{x_1,x_2,\ldots\}$ and $Y=\{y_1,y_2,\ldots\}$, and let ${\mathfrak S}_w(X;Y)$ be the double Schubert polynomial defined recursively from the product for the longest permuta...

conjecture | open | MathDB 页面 | 原文线索

V2-064024 | MATHDB-335513

SNP conjecture for Grothendieck polynomials

摘录 / EXCERPT: For $w\in S_\infty$, let ${\mathfrak G}_w$ be the Grothendieck polynomial obtained recursively using $\overline\pi_i(f)=\partial_i((1-x_{i+1})f)$. A polynomial has SNP when every l...

conjecture | open | MathDB 页面 | 原文线索

V2-064025 | MATHDB-335514

SNP conjecture for Lascoux's Omega polynomials

摘录 / EXCERPT: Let $\alpha\in\mathbb Z_{\geq0}^{\infty}$, and let Ω_α be the inhomogeneous analogue of the key polynomial obtained by replacing the Demazure operators with...

conjecture | open | MathDB 页面 | 原文线索

V2-064026 | MATHDB-335515

SNP conjecture for Lascoux atoms

摘录 / EXCERPT: Let $\mathcal L_\alpha$ be the Lascoux atom obtained by replacing the Demazure operator in the definition of the key polynomial with $\widehat\tau_i=\tau_i-\operatorname{id}$, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-064027 | MATHDB-335516

Positive Ehrhart coefficients for normalized Schubitopes

摘录 / EXCERPT: Positive Ehrhart-coefficient conjecture. Every coefficient satisfies $c_i>0$ for $i=0,\ldots,d$. The claim was checked exhaustively for $n=4$ and in many random cases for $n=5$, bu...

conjecture | open | MathDB 页面 | 原文线索

V2-064028 | MATHDB-335518

Characteristic pure mapping class group and automorphism conjecture

摘录 / EXCERPT: Let S be a surface whose genus is finite and at least 4. The groups $\operatorname{PMap}(S)$, $\operatorname{Map}(S)$, and $\operatorname{Map}^\pm(S)$ denote the pure mapping c...

conjecture | open | MathDB 页面 | 原文线索

V2-064029 | MATHDB-335522

Asymptotic upper-bound conjecture for distant irregularity strength

摘录 / EXCERPT: Let $r\geq 2$ be an integer, and let G be a graph with maximum degree Δ and without an isolated edge. Write $s_r(G)$ for the least number of colours needed in an edge-colo...

conjecture | open | MathDB 页面 | 原文线索

V2-064030 | MATHDB-335523

A-F-M conjecture on sums of reproducing kernels

摘录 / EXCERPT: A-F-M conjecture. The multiplication operator M_z on $\mathcal H(K_1+K_2)$ is subnormal.

conjecture | open | MathDB 页面 | 原文线索

V2-064031 | MATHDB-335529

Positive key expansion conjecture for nonsymmetric Macdonald polynomials

摘录 / EXCERPT: Positive key expansion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064032 | MATHDB-335530

Refined Hall-Littlewood filling identity conjecture

摘录 / EXCERPT: Refined Hall-Littlewood filling identity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064033 | MATHDB-335538

Wu-Li's permutation-polynomial conjecture for odd extensions of 5

摘录 / EXCERPT: Let k be a positive odd integer and let $f(x)=x\big(\frac{x^2-x+2}{x^2+x+2}\big)^2$ over the finite field $\mathbb{F}_{5^k}$. Wu-Li's conjecture. The polynomial f is a permutat...

conjecture | open | MathDB 页面 | 原文线索

V2-064034 | MATHDB-335539

Wu-Li's permutation conjecture on 5-power roots of unity

摘录 / EXCERPT: Let k be an even integer, set $q=5^k$, and let ... Define $g(x)=-x\big(\frac{x^2-2}{x^2+2}\big)^2$. Wu-Li's conjecture. The map g permutates μ_{q+1}. This is the second of tw...

conjecture | open | MathDB 页面 | 原文线索

V2-064035 | MATHDB-335543

The necessity of all generalized Reidemeister moves for virtual-knot diagrams

摘录 / EXCERPT: Let K be a virtual knot, and let D and D' be diagrams of K. The generalized Reidemeister-move conjecture. There exist two diagrams D and D' such that every sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-064036 | MATHDB-335545

BBBL cyclic insertion conjecture for multiple zeta values

摘录 / EXCERPT: For non-negative integers $b_0,b_1,\ldots,b_{2n}$, let $C_{2n+1}=\langle(0\,1\,\cdots\,2n)\rangle$ act cyclically on the indices, and write $\{s\}^{\ell}$ for ℓ repetitions of...

conjecture | open | MathDB 页面 | 原文线索

V2-064037 | MATHDB-335546

Generalised cyclic insertion conjecture for 123-multiple zeta values

摘录 / EXCERPT: Let $\zeta(s_1,\ldots,s_d)$ be a 123-multiple zeta value of depth d, with n blocks in its alternating block decomposition, and let wt be its weight. Generalised cycl...

conjecture | open | MathDB 页面 | 原文线索

V2-064038 | MATHDB-335547

Alternating odd-position conjecture for block decompositions

摘录 / EXCERPT: Let $B=(0;\ell_1,\ldots,\ell_n)$ be any non-trivial block decomposition of even weight N. Alternate odd positions conjecture. ... The conjecture extends the analogous alternating i...

conjecture | open | MathDB 页面 | 原文线索

V2-064039 | MATHDB-335548

Odd-weight alternating odd-position conjecture for block decompositions

摘录 / EXCERPT: Odd-weight alternate odd-position conjecture. Subject to further restrictions on the block lengths and x,

conjecture | open | MathDB 页面 | 原文线索

V2-064040 | MATHDB-335556

The completed finite period conjecture

摘录 / EXCERPT: Period conjecture. For every integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-064041 | MATHDB-335562

The canonical rational-function push-forward conjecture for Amplituhedra

摘录 / EXCERPT: Let ${\mathcal A}(k,n,m;l^L)$ be an Amplituhedron, let Δ^D have positive coordinates $X_1,\ldots,X_D$, and let ... be a morphism from positive coordinates to the Amplituhedron...

conjecture | open | MathDB 页面 | 原文线索

V2-064042 | MATHDB-335566

MOP tautological conjecture for moduli of polarized K3 surfaces

摘录 / EXCERPT: Let $\mathcal{K}_g$ be the moduli space of primitively polarized K3 surfaces of genus g, and let ${\rm NL}^\ast(\mathcal{K}_g)$ be the subring of its Chow ring generated by Noeth...

conjecture | open | MathDB 页面 | 原文线索

V2-064043 | MATHDB-335579

The conditional Brauer-Manin conjecture for Kummer surfaces

摘录 / EXCERPT: Let k be a number field and let X be a Kummer surface over k. A 2-covering associated to X means one of the 2-coverings of the abelian surfaces associated to X. Kummer-su...

conjecture | open | MathDB 页面 | 原文线索

V2-064044 | MATHDB-335582

Mutation-qG-deformation correspondence for Fano polygons and TG del Pezzo surfaces

摘录 / EXCERPT: Conjecture A. There exists a bijective correspondence between the set of mutation-equivalence classes of Fano polygons and the set of qG-deformation equivalence classes of locally...

conjecture | open | MathDB 页面 | 原文线索

V2-064045 | MATHDB-335583

The real flat ETF Hadamard conjecture

摘录 / EXCERPT: Real flat ETF Hadamard conjecture. Every real flat ETF is Hadamard.

conjecture | open | MathDB 页面 | 原文线索

V2-064046 | MATHDB-335592

Digne-Gobet's conjecture on dual simple braids

摘录 / EXCERPT: Let W be a finite Coxeter group with braid group $\mathbb{B}_W$, standard Garside monoid with Garside element Δ, and dual positive monoid $\mathcal{B}e^+$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-064047 | MATHDB-335596

Bóna-Lackner-Sagan's log-concavity conjecture for involutions

摘录 / EXCERPT: Let n be a positive integer. For $1\leq k\leq n$, let $I_{n,k}$ be the set of involutions in the symmetric group $\mathfrak{S}_n$ whose longest increasing subsequence has length...

conjecture | open | MathDB 页面 | 原文线索

V2-064048 | MATHDB-335597

Bóna-Lackner-Sagan's log-concavity conjecture for even-column tableaux

摘录 / EXCERPT: Fix a positive integer n, and let Θ be the set of partitions of n all of whose column lengths are even. Let $\ell_{n,k}^{\Theta}$ denote the corresponding quantity assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-064049 | MATHDB-335599

The largest-part conjecture for wave-front sets and local descents

摘录 / EXCERPT: Largest-part conjecture. The largest part of the partition is

conjecture | open | MathDB 页面 | 原文线索

V2-064050 | MATHDB-335602

All-energy stability of the largest mode in the convex case

摘录 / EXCERPT: All-energy stability conjecture. Under these assumptions, Θ_m is linearly stable with respect to Θ_n for every $E_{\Theta_m}>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-064051 | MATHDB-335603

Small-energy stability in the limiting convex case

摘录 / EXCERPT: Let m,n be the mode indices, let P be the parameter, and let Θ_m and Θ_n denote the corresponding modes with energy E_{Θ_m}. Assume ... Let E_2 be the thre...

conjecture | open | MathDB 页面 | 原文线索

V2-064052 | MATHDB-335604

Intermediate-energy instability for the smaller mode in the stable high-energy regime

摘录 / EXCERPT: Let m,n be the mode indices, let P be the parameter, and define ... Let Θ_m and Θ_n be the corresponding modes with energy E_{Θ_m}. Assume ... Let E_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-064053 | MATHDB-335605

Endpoint stability and instability for high-energy mode ratios

摘录 / EXCERPT: Endpoint behavior conjecture. There exists $E_k>0$ such that if

conjecture | open | MathDB 页面 | 原文线索

V2-064054 | MATHDB-335610

Hodge ideals and m-adic approximation conjecture

摘录 / EXCERPT: Hodge ideals and m-adic approximation conjecture. There exists a positive integer $q(r)$ such that, for every such E,

conjecture | open | MathDB 页面 | 原文线索

V2-064055 | MATHDB-335611

Orbifold Zvonkine ELSV formula for q-orbifold r-spin Hurwitz numbers

摘录 / EXCERPT: Orbifold Zvonkine formula. The q-orbifold r-spin Hurwitz numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-064056 | MATHDB-335615

Uniqueness of tangent cones under disk decomposition near line segments

摘录 / EXCERPT: Disk-decomposition uniqueness conjecture. Under these hypotheses, Σ has a unique tangent cone at infinity.

conjecture | open | MathDB 页面 | 原文线索

V2-064057 | MATHDB-335617

Poisson/Gaudin-Mehta conjecture for random symmetric banded matrices

摘录 / EXCERPT: Let A be a random real symmetric banded matrix of size $N\times N$ with independent entries and bandwidth $b\asymp N^\alpha$. Poisson/Gaudin-Mehta conjecture. The limiting local...

conjecture | open | MathDB 页面 | 原文线索

V2-064058 | MATHDB-335618

Localization-length conjecture for eigenvectors of random banded matrices

摘录 / EXCERPT: Let A be a random banded matrix with independent entries and bandwidth b, and let l denote its eigenvector localisation length. Localization-length conjecture. The eigenvecto...

conjecture | open | MathDB 页面 | 原文线索

V2-064059 | MATHDB-335624

E2-degeneration conjecture for arrangements and free locally quasi-homogeneous hypersurfaces

摘录 / EXCERPT: Let f define a projective hypersurface $V:f=0$ in $\mathbb{P}^n$, and let $E_*(f)$ be the spectral sequence whose limit describes the monodromy action on the graded pieces of the...

conjecture | open | MathDB 页面 | 原文线索

V2-064060 | MATHDB-335634

Monin-Rana conjecture on the equations and Gröbner bases of \(\overline{M}_{0,n}\)

摘录 / EXCERPT: Monin-Rana's conjecture. The ideal I_n is the unique B-saturation of J_n; it is minimally generated by $\binom{n-1}{d+1}$ polynomials of degree d for $d=3,4,5,\ldots,n-2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-064061 | MATHDB-335636

Generalized dual equivalence conjecture for Macdonald polynomials

摘录 / EXCERPT: Generalized dual equivalence conjecture. For a partition μ, the quasisymmetric generating function of each generalized dual equivalence class under D_i^{μ} is symmetric and...

conjecture | open | MathDB 页面 | 原文线索

V2-064062 | MATHDB-335637

Partition-folding bijection conjecture for Macdonald polynomials

摘录 / EXCERPT: Partition-folding bijection conjecture. There exists a bijection φ on permutations that preserves the inverse descent set such that, if u and v are μ-equivalent, th...

conjecture | open | MathDB 页面 | 原文线索

V2-064063 | MATHDB-335638

Scaling-critical Strichartz estimates for rational metrics on compact Lie groups

摘录 / EXCERPT: Let G be a compact Lie group equipped with a rational metric g. Let d be the dimension of G and r the rank of G. Let $I\subset\mathbb{R}$ be a finite time interval. Com...

conjecture | open | MathDB 页面 | 原文线索

V2-064064 | MATHDB-335639

Scaling-critical Strichartz estimates on compact-type symmetric spaces

摘录 / EXCERPT: Let G/K be a Riemannian globally symmetric space of compact type, equipped with the negative of the Cartan--Killing form as the canonical Riemannian metric g. Let d and r b...

conjecture | open | MathDB 页面 | 原文线索

V2-064065 | MATHDB-335640

Improved Strichartz range on compact-type symmetric spaces

摘录 / EXCERPT: Improved symmetric-space Strichartz conjecture. Under these assumptions, this estimate holds for all $p>2+\frac{4}{r}$. This strengthens the preceding conjectured range...

conjecture | open | MathDB 页面 | 原文线索

V2-064066 | MATHDB-335642

The Majority is Least Stable conjecture for linear threshold functions

摘录 / EXCERPT: Let $f\colon\{-1,1\}^{n}\rightarrow\{-1,1\}$ be a linear threshold function, with n odd. For $\rho\in[0,1]$, let $\operatorname{Stab}_{\rho}[f]$ denote the noise stability of f...

conjecture | open | MathDB 页面 | 原文线索

V2-064067 | MATHDB-335647

Chromatic boundedness for graphs with spaghetti tree- and path-decompositions

摘录 / EXCERPT: Chromatic boundedness conjecture. There exists a function $f:\mathbb{N}\to\mathbb{N}$ such that $\chi(G)\leq f(k)$ for every $k\geq 1$ and every graph G admitting a spaghetti tre...

open_problem | open | MathDB 页面 | 原文线索

V2-064068 | MATHDB-335648

Černý's conjecture for D3-directing complete nondeterministic finite automata

摘录 / EXCERPT: Černý's conjecture for D3-directing CNFAs. Every D3-directing CNFA with n states admits a D3-directing word of length at most

conjecture | open | MathDB 页面 | 原文线索

V2-064069 | MATHDB-335650

Naturality conjecture for general paracanonical curves

摘录 / EXCERPT: Naturality conjecture. The resolution of $C\subseteq\mathbf{P}^{g-2}$ is natural.

conjecture | open | MathDB 页面 | 原文线索

V2-064070 | MATHDB-335654

Lescure-Meynial and Abu-Khzam-Langston chromatic immersion conjecture

摘录 / EXCERPT: Let t be a positive integer, and let G be a graph. An immersion of K_t in G consists of distinct branch vertices corresponding to the vertices of K_t, together with paths...

conjecture | refuted | MathDB 页面 | 原文线索

V2-064071 | MATHDB-335659

Girth-five conjecture for feasible partitions of multigraphs

摘录 / EXCERPT: Let G be a graph, and let $w_G(v)$ denote the number of edges parallel to an edge incident with v beyond the first. For functions $a,b:V(G)\to\mathbb{Z}_{\geq 2}$, an (a,b)-f...

conjecture | open | MathDB 页面 | 原文线索

V2-064072 | MATHDB-335677

The converse Haagerup-property conjecture for locally compact wreath products

摘录 / EXCERPT: Converse Haagerup-property conjecture. If $B\wr^A_X H$ has the Haagerup Property, then for every point stabilizer L of X, the relative profinite completion...

conjecture | open | MathDB 页面 | 原文线索

V2-064073 | MATHDB-335679

The placement delivery array additive bound conjecture

摘录 / EXCERPT: Additive bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064074 | MATHDB-335683

The X = M conjecture for tensor products of Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Let B be a tensor product of Kirillov--Reshetikhin crystals of type $\mathfrak{g}$. For a classical weight λ, let $X(B,\lambda;q)$ be the one-dimensional sum over classic...

conjecture | open | MathDB 页面 | 原文线索

V2-064075 | MATHDB-335694

Arithmetic completeness of the reflection calculus with nabla

摘录 / EXCERPT: Let T be a theory, and let ${\mathrm{RC}^\nabla\hspace{-0.7pt}}$ be the reflection calculus with modalities $\Diamond_n$ and ∇_n. An arithmetical interpretation $*$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-064076 | MATHDB-335702

Born winner/silver spoon regime conjecture for the multiplicative coalescent

摘录 / EXCERPT: Let $(\boldsymbol{x}^{\scriptscriptstyle(n)})_{n\geqslant 1}$ satisfy Condition IIτ for some $\tau\in(3,4)$, and let M_n be as in the leader theorem. Write $\mathcal{W}_n$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-064077 | MATHDB-335706

Kahn-Kalai conjecture on DNF approximation of low-influence biased Boolean functions

摘录 / EXCERPT: Let $f\colon\left\{0,1\right\}^{n}\to\left\{0,1\right\}$ be a monotone Boolean function with expectation $\mathbb{E}[f]=\mu$, and let $I[f]$ denote its total influence. A DNF of wi...

conjecture | open | MathDB 页面 | 原文线索

V2-064078 | MATHDB-335707

Conjectured improved Junta approximation for low-influence Boolean functions

摘录 / EXCERPT: Let $f\colon\left\{0,1\right\}^{n}\to\left\{0,1\right\}$ be a Boolean function, write $\mu=\mathbb{E}[f]$, and let $I(f)$ denote its total influence. A function g is a Junta if i...

conjecture | open | MathDB 页面 | 原文线索

V2-064079 | MATHDB-335711

Weighted dispersive estimates for point interactions

摘录 / EXCERPT: Let $H_{\alpha,y}$ be the Schrödinger operator with a point interaction of strength α at $y\in\mathbb{R}^3$, let P_{ac} denote the projection onto its absolutely continuou...

conjecture | open | MathDB 页面 | 原文线索

V2-064080 | MATHDB-335712

Weighted Lorentz dispersive estimates for point interactions

摘录 / EXCERPT: Weighted Lorentz dispersive estimates conjecture. For every $\alpha\ne0$ and $y\in\mathbb{R}^3$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-064081 | MATHDB-335715

The equality of regulator formulas for Gauss fibrations when a equals b

摘录 / EXCERPT: Gauss regulator equality conjecture. The first equality in Theorem is valid even when $a=b$.

conjecture | open | MathDB 页面 | 原文线索

V2-064082 | MATHDB-335727

Finite-time singularity conjecture for curve diffusion with free boundary

摘录 / EXCERPT: Suppose $|e|>0$. Let $\eta_1,\eta_2:\mathbb{R}\to\mathbb{R}^2$ be parallel straight lines defining the exterior problem, and let $\gamma_0:(-1,1)\to\mathbb{R}^2$ be an immersed cur...

conjecture | open | MathDB 页面 | 原文线索

V2-064083 | MATHDB-335731

The conjecture that iterated divisor-preimage minima are highly composite

摘录 / EXCERPT: Highly composite minima conjecture. All the integers n_k produced by Theorem 1 are highly composite numbers.

conjecture | open | MathDB 页面 | 原文线索

V2-064084 | MATHDB-335739

Conjecture on the perfection and linear syzygies of the principal-curve ideal

摘录 / EXCERPT: Principal-curve ideal conjecture. The ideal (f,g,q) is perfect, that is, unmixed. Moreover, it admits a linear syzygy if and only if either $s=3$ or $s=5$, or else $s>5$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064085 | MATHDB-335745

Spin variant of Saxl's conjecture for symmetric groups

摘录 / EXCERPT: Let $P(n)$ denote the set of partitions of n, and let $D^+(n)$ denote the relevant set of strict partitions indexing the spin characters under consideration. For...

conjecture | open | MathDB 页面 | 原文线索

V2-064086 | MATHDB-335746

Spin character-square conjecture for double covers of alternating groups

摘录 / EXCERPT: Let $\widetilde{A}_n$ be the double cover of the alternating group A_n, and let $\operatorname{Irr}(\widetilde{A}_n)$ denote its irreducible characters. A character is called spi...

conjecture | open | MathDB 页面 | 原文线索

V2-064087 | MATHDB-335747

Conjecture that a flattened-noise channel is at least as informative for every Boolean function

摘录 / EXCERPT: Flattened-noise informativeness conjecture. For every such b,

conjecture | open | MathDB 页面 | 原文线索

V2-064088 | MATHDB-335749

Determinant formula for the weak-order raising operator

摘录 / EXCERPT: Let W_n be the weak order on the symmetric group S_n, ranked by inversion number, and let U be the order-raising operator defined by ... For $0\leq k<\frac12\binom n2$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064089 | MATHDB-335754

The edge-injective neighbour-sum-distinguishing weighting conjecture

摘录 / EXCERPT: Let G be a nice graph, meaning a graph with no connected component isomorphic to K_2. An edge-weighting is edge-injective if it assigns distinct weights to all edges, and...

conjecture | open | MathDB 页面 | 原文线索

V2-064090 | MATHDB-335765

The congruence-kernel conjecture for super-modular categories

摘录 / EXCERPT: Let $\mathcal{B}$ be a super-modular category of rank $2k$, and let $\hat{S}$ and $\hat{T}^2$ be the corresponding matrices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064091 | MATHDB-335772

Ordered binary matrix removal lemma

摘录 / EXCERPT: Ordered binary matrix removal lemma. For any finite family $\mathcal{F}$ of ordered binary matrices and any $\epsilon > 0$ there exists $\delta = \delta(\mathcal{F}, \epsilon)$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-064092 | MATHDB-335774

Kalman conjecture for slope-restricted Lur'e systems

摘录 / EXCERPT: Kalman conjecture. Then the negative-feedback interconnection of the continuous-time (or discrete-time) LTI system $G\sim[A,B,C,0]$ and $c6$ is globally asymptotically stable if...

conjecture | refuted | MathDB 页面 | 原文线索

V2-064093 | MATHDB-335775

Discrete-time off-axis circle criterion conjecture

摘录 / EXCERPT: Consider a feedback system with $G\in\mathbf{RH}_\infty$, and let $c6$ be slope-restricted in $S[0,k]$ and monotonically increasing. The Nyquist plot of the linear part is the plot...

conjecture | open | MathDB 页面 | 原文线索

V2-064094 | MATHDB-335782

Convexity conjecture for Rényi entropy under symmetric \(\kappa\)-concavity

摘录 / EXCERPT: Let (X,Y) be a symmetric κ-concave random vector in $\mathbb{R}^2$, and suppose that X and Y have equal p-Rényi entropy. Convexity conjecture for Rényi entropy. The...

conjecture | open | MathDB 页面 | 原文线索

V2-064095 | MATHDB-335788

Morales-Mészáros conjecture for type C Chan-Robbins-Yuen polytopes

摘录 / EXCERPT: Let K^{C}_{n+1} be the complete signed graph with $n+1$ vertices, containing all edges $(i,j,\pm)$ for $1\leq i<j\leq n+1$ and $(i,i,+)$ for $1\leq i\leq n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064096 | MATHDB-335789

Global asymptotic stability conjecture for locally stable steady states in spatially heterogeneous competition models

摘录 / EXCERPT: Consider the reaction-diffusion competition model ... with Neumann boundary conditions on a habitat Ω, where $m(x)$ is nonconstant and $0<b,c<1$. Global asymptotic stability c...

conjecture | open | MathDB 页面 | 原文线索

V2-064097 | MATHDB-335795

Pointwise Christoffel function estimate for two-dimensional $[?]$-balls

摘录 / EXCERPT: Let $1<\alpha<2$ and let ${\boldsymbol{x}}\in B^2_\alpha$. Define ... where ${\boldsymbol{u}}$ satisfies ${\boldsymbol{x}}+\delta{\boldsymbol{u}}\in\partial D$ and ${\boldsymbol{v}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-064098 | MATHDB-335811

The Kakeya projection conjecture

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a Kakeya set, and let $G(n,k)$ denote the Grassmannian of k-dimensional subspaces of $\mathbb{R}^n$. For $\gamma\in G(n,k)$, write π_γ for...

conjecture | open | MathDB 页面 | 原文线索

V2-064099 | MATHDB-335816

The conjecture that K_6 is not reducible for the first Symanzik polynomial

摘录 / EXCERPT: A graph is reducible with respect to the first Symanzik polynomial when it has the corresponding reducibility property for that polynomial. The K_6 irreducibility conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-064100 | MATHDB-335827

Time-dependent conserved-quantity conjecture for affine-affine ADE T-systems

摘录 / EXCERPT: Let G be an affine-affine ADE bigraph in the classification, and let $A(t)$ be a function of the system parameters at time t. It is a time-dependent conserved quantity if the...

conjecture | open | MathDB 页面 | 原文线索

V2-064101 | MATHDB-335838

Chapoton's conjecture on Newton polygons of q-Ehrhart numerators

摘录 / EXCERPT: Let P be a poset on $\\{1,2,\dots,m\\}$, and let $\mathcal{O}(P)$ be its order polytope. For each $x\in P$, let $\overline{\operatorname{mc}}(x)$ be the maximum size of a chain s...

conjecture | open | MathDB 页面 | 原文线索

V2-064102 | MATHDB-335842

The finite D3-minus-C conjecture for short intervals

摘录 / EXCERPT: A finite poset is D3-minus-C if it satisfies the D3-minus-C axiom, and it has Short intervals are Small when that property holds. Finite D3-minus-C conjecture. If a finite poset is...

conjecture | open | MathDB 页面 | 原文线索

V2-064103 | MATHDB-335843

Infinite counterexamples to the finite-poset theorem

摘录 / EXCERPT: A locally finite poset is a poset in which every interval is finite. The preceding finite-poset theorem has several parts whose conclusions depend on finiteness. Infinite-counterex...

conjecture | open | MathDB 页面 | 原文线索

V2-064104 | MATHDB-335851

The period-determines-b2 invariant conjecture for 4-point c6^4 graphs

摘录 / EXCERPT: Period-determines-c_2 conjecture. If G and H have equal periods, then they have equal c_2 invariants.

conjecture | open | MathDB 页面 | 原文线索

V2-064105 | MATHDB-335853

Period-determines-extended-graph-permanent conjecture for 4-point c6^4 graphs

摘录 / EXCERPT: Period-determines-extended-graph-permanent conjecture. If G and H have equal periods, then they have equal extended graph permanents.

conjecture | open | MathDB 页面 | 原文线索

V2-064106 | MATHDB-335863

Vanishing of the L^2-torsion polytope for amenable groups

摘录 / EXCERPT: Let $G\neq \mathbb{Z}$ be an amenable group satisfying the Atiyah Conjecture. Suppose that G is of type F and that $\operatorname{Wh}(G)=0$. The L^2-torsion polytope $P(G)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-064107 | MATHDB-335865

Dvořák-Giannopoulou-Thilikos conjecture on the order of critical graphs

摘录 / EXCERPT: Dvořák-Giannopoulou-Thilikos conjecture. G has at most

conjecture | open | MathDB 页面 | 原文线索

V2-064108 | MATHDB-335866

Barrus-Sinković conjecture on the maximum degree of critical graphs

摘录 / EXCERPT: Barrus-Sinković conjecture. G has maximum degree at most

conjecture | open | MathDB 页面 | 原文线索

V2-064109 | MATHDB-335883

Nonnegativity conjecture for factorization coefficients of K-k-Schur functions

摘录 / EXCERPT: Nonnegativity conjecture. For every $4\lambda\in\mathcal{P}_k4$ and every μ, one has $a_{P,\lambda,\mu}\ge 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-064110 | MATHDB-335884

Interval conjecture for multiplication by a rectangular K-k-Schur function

摘录 / EXCERPT: Interval conjecture. For every $4\lambda\in\mathcal{P}_k4$ and $1\le t\le k$, there exists $4\mu\in\mathcal{P}_k4$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064111 | MATHDB-335916

Entropy-entropy production inequality for the non-cutoff linear Boltzmann equation

摘录 / EXCERPT: Entropy-entropy production conjecture. There exists $\lambda_{\gamma,b}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064112 | MATHDB-335920

Empty-interior conjecture for embedded Shannon-equivalence classes of channels

摘录 / EXCERPT: Let $\operatorname*{DMC}_{[n],[n]}^{(s)}$ denote the space of Shannon-equivalence classes of discrete memoryless channels with input and output alphabets $[n]=\{1,\ldots,n\}$, embe...

conjecture | open | MathDB 页面 | 原文线索

V2-064113 | MATHDB-335921

Factorization conjecture for arithmetic automorphic periods of unitary groups

摘录 / EXCERPT: Let F be a CM field, let Σ be a CM type, and let F^+ be the maximal totally real subfield of F. For each signature $I:\Sigma\to\{0,1,\ldots,n\}$, let U_I be the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-064114 | MATHDB-335928

Linear-combination decision-tree conjecture for point location

摘录 / EXCERPT: Let $H\subset\mathbb{R}^n$. Let $\mathcal{A}_H$ denote the associated point-location problem. A linear decision tree is a decision tree whose queries are linear forms. Linear-combi...

conjecture | open | MathDB 页面 | 原文线索

V2-064115 | MATHDB-335929

Optimal comparison decision-tree bounds for 3-SUM and related problems

摘录 / EXCERPT: Let $H\subset\{-1,0,1\}^n$, and let $\mathcal{A}_H$ denote the associated problem. A comparison decision tree uses comparison queries, and a linear decision tree is called k-spar...

conjecture | open | MathDB 页面 | 原文线索

V2-064116 | MATHDB-335930

The non-genericity classification conjecture for unramified metaplectic representations

摘录 / EXCERPT: Non-genericity classification conjecture. Ignoring the special cases mentioned in the paper, π is not generic if and only if it is a subrepresentation of

conjecture | open | MathDB 页面 | 原文线索

V2-064117 | MATHDB-335939

Observed slope and eigenvalue relations for $N=7$

摘录 / EXCERPT: For $p=2$ and $N=7$, let $\alpha_{\mathcal{L}}^\pm(k,2,7)$ denote the multisets of slopes on the ±1 Atkin-Lehner eigenspaces, let $\alpha_{\mathcal{L}}(k,2,7)$ be their combine...

conjecture | open | MathDB 页面 | 原文线索

V2-064118 | MATHDB-335944

Equality of perfectoid test ideals and multiplier ideals after inverting p

摘录 / EXCERPT: Let $(A,\fra^t)$ be a pair, where A is the regular ring and $\fra$ is the ideal appearing in the construction of the perfectoid test ideal ${\mathrlap{\mJ}\,\,\uptau}(A,\fra^t)$....

conjecture | open | MathDB 页面 | 原文线索

V2-064119 | MATHDB-335945

Classification conjecture for free multi-braid arrangements

摘录 / EXCERPT: Classification conjecture. The multi-braid arrangement $(A_\ell,\mathbf{m})$ is free if and only if it is one of the multi-braid arrangements constructed in Corollary. Equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-064120 | MATHDB-335955

The incomparability-graph shallow-minor star conjecture

摘录 / EXCERPT: Shallow-minor star conjecture. If G does not have an induced star on s leaves, then, for every $t\ge 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-064121 | MATHDB-335956

The incomparability-graph clique-separator conjecture

摘录 / EXCERPT: Clique-separator conjecture. There is a clique cover C in G such that removing

conjecture | open | MathDB 页面 | 原文线索

V2-064122 | MATHDB-335960

Regularity conjecture for Burau images of pure braids in B_4

摘录 / EXCERPT: Let B_4 be the braid group on four strands, let ρ denote the representation used in the paper, and let T_4 be the object on which the braid acts. For a Laurent polynomial,...

conjecture | open | MathDB 页面 | 原文线索

V2-064123 | MATHDB-335971

The conjecture that uniformly fully inert subgroups are commensurable with fully invariant subgroups

摘录 / EXCERPT: The conjecture. The class $\mathfrak I_*$ coincides with the class of all abelian groups. Equivalently, every uniformly fully inert subgroup H of an abelian group A is commensu...

conjecture | open | MathDB 页面 | 原文线索

V2-064124 | MATHDB-335979

The 2-lobe conjecture for rectangular constrained Willmore tori

摘录 / EXCERPT: For $b\geq 1$, consider tori in $\mathbb{R}^3$ with rectangular conformal type ... The relevant candidates are the homogeneous tori $S^1\times S^1(b)$ in the 3-sphere and the 2-l...

conjecture | open | MathDB 页面 | 原文线索

V2-064125 | MATHDB-335980

The constrained Willmore minimizer conjecture for tori

摘录 / EXCERPT: Let $\mathcal W$ denote the Willmore functional, and fix a conformal class of tori. The $(1,2)$-equivariant constrained Willmore tori are those constrained Willmore tori of orbit t...

conjecture | open | MathDB 页面 | 原文线索

V2-064126 | MATHDB-335981

The constrained Willmore Lawson conjecture

摘录 / EXCERPT: A constrained Willmore torus is a torus that is critical for the Willmore functional under conformal-class-preserving variations, and it is embedded when its immersion is embedded....

conjecture | open | MathDB 页面 | 原文线索

V2-064127 | MATHDB-335982

The stability conjecture for Willmore tori

摘录 / EXCERPT: A Willmore torus is called stable when the second variation of the Willmore functional is nonnegative, equivalently when the Jacobi operator satisfies ... The Clifford torus is the s...

conjecture | open | MathDB 页面 | 原文线索

V2-064128 | MATHDB-335983

The weak stability conjecture for Willmore tori

摘录 / EXCERPT: A Willmore torus is stable when its Willmore second variation is nonnegative, and it is isothermic when it admits the corresponding isothermic structure in conformal surface geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-064129 | MATHDB-335984

The higher-genus constrained Willmore conjecture for rectangular conformal types

摘录 / EXCERPT: Let g be a genus and let a surface of genus g have a conformal type called rectangular in the sense used for the quotient by the cyclic $(g+1)$-fold symmetry of Lawson's minima...

conjecture | open | MathDB 页面 | 原文线索

V2-064130 | MATHDB-335997

The extension conjecture for complex shifted powers

摘录 / EXCERPT: Let d be a positive integer, and let ... be a linearly independent family with $a_i\in\mathbb{C}$, $e_i\leq d$ for all i, and $s\leq ad+b$. Extension conjecture. There are absolu...

conjecture | open | MathDB 页面 | 原文线索

V2-064131 | MATHDB-336005

The modified Bogomolov-Gieseker inequality for families of stability parameters

摘录 / EXCERPT: Let X be a smooth projective threefold, let $H\in\operatorname{NS}(X)$ be ample, and let $B\in\operatorname{NS}_{\mathbb{R}}(X)$. For $\beta\in\mathbb{R}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064132 | MATHDB-336006

The zero-correction-function conjecture for the modified Bogomolov-Gieseker inequality

摘录 / EXCERPT: Let X be a smooth projective threefold, let $H\in\operatorname{NS}(X)$ be ample, and let $B\in\operatorname{NS}_{\mathbb{R}}(X)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064133 | MATHDB-336017

Cyclic subspace code existence conjecture for parameters n and k

摘录 / EXCERPT: Cyclic subspace code existence conjecture. There exists a cyclic subspace code

conjecture | open | MathDB 页面 | 原文线索

V2-064134 | MATHDB-336028

The DP-coloring analogue of Shannon's bound for line graphs

摘录 / EXCERPT: Let G be a loopless multigraph, let $\Delta(G)$ denote its maximum degree, and let $\mathsf{Line}(G)$ be the simple graph whose vertices are the edges of G, with adjacency when...

conjecture | open | MathDB 页面 | 原文线索

V2-064135 | MATHDB-336035

The construction conjecture for Gorenstein simplices with prescribed δ-polynomial

摘录 / EXCERPT: Let $v\geq 1$ and $k\geq 0$ be integers, and let $\Delta\subset\mathbb{R}^d$ be a Gorenstein simplex of dimension d with δ-polynomial ... Assume that Δ is not a latti...

conjecture | open | MathDB 页面 | 原文线索

V2-064136 | MATHDB-336036

The minimal modular extension conjecture for super-modular categories

摘录 / EXCERPT: A super-modular category is a premodular category whose Müger center is equivalent to $\sVec$. A spin modular category is a modular category containing a fermion. For a fusion cate...

conjecture | open | MathDB 页面 | 原文线索

V2-064137 | MATHDB-336037

Finiteness conjecture for premodular categories of fixed rank

摘录 / EXCERPT: A premodular category is a braided fusion category equipped with a compatible spherical, equivalently ribbon, structure. Its rank r is the number of isomorphism classes of simple...

conjecture | open | MathDB 页面 | 原文线索

V2-064138 | MATHDB-336038

Splitness conjecture for super-modular categories with modular quotient S-matrix

摘录 / EXCERPT: Splitness conjecture. Then $\mathcal{B}$ is split super-modular.

conjecture | open | MathDB 页面 | 原文线索

V2-064139 | MATHDB-336039

Completeness of the Mahonian 3-function equidistribution table for length-three pattern avoidance

摘录 / EXCERPT: Completeness conjecture. The table is the complete table of Mahonian 3-function equidistributions over permutations avoiding a single classical pattern of length three.

conjecture | open | MathDB 页面 | 原文线索

V2-064140 | MATHDB-336040

The L4-norm problem for Hecke-Maaß eigenforms

摘录 / EXCERPT: Let $\Gamma=\operatorname{SL}_2(\mathbb{Z})$, let $\mathbb{H}$ be the upper half-plane, and let $g\in\mathcal{B}_0(\Gamma)$ be a Hecke-Maaß eigenform normalised by...

conjecture | open | MathDB 页面 | 原文线索

V2-064141 | MATHDB-336045

Absolute-monotonicity conjecture for the second derivative of an elliptic-integral difference

摘录 / EXCERPT: Let $\mathcal{K}(r)$ denote the complete elliptic integral of the first kind and define ... A function is absolutely monotonic on an interval if all of its derivatives are nonnegativ...

conjecture | open | MathDB 页面 | 原文线索

V2-064142 | MATHDB-336063

The modular-data determination conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a modular category, and let (S,T) be its modular data. Modular-data determination conjecture. The S and T matrices of $\mathcal{C}$ determine it up to ri...

conjecture | open | MathDB 页面 | 原文线索

V2-064143 | MATHDB-336064

The Property F conjecture for Frobenius-Perron dimensions

摘录 / EXCERPT: Let $\mathcal{C}$ be a braided fusion category, and let $X,Y\in\mathcal{C}$ be simple objects. For each n, let ... be the associated braid group representation. Property F conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-064144 | MATHDB-336070

Universal limiting pair correlation conjecture for Apollonian gaskets

摘录 / EXCERPT: Let $\mathcal{P}$ be an Apollonian gasket and let $\mathcal R\subset\mathbb C$ satisfy $\mu(\mathcal R)>0$. For $T>0$, let $F_{T,\mathcal R}(s)$ denote the pair correlation functio...

conjecture | open | MathDB 页面 | 原文线索

V2-064145 | MATHDB-336071

Conjectured convergence of electrostatic energy for Apollonian gaskets

摘录 / EXCERPT: Electrostatic energy convergence conjecture. There exists a constant $b>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064146 | MATHDB-336072

The four-factor q-congruence conjecture

摘录 / EXCERPT: The four-factor q-congruence conjecture. The following four congruences hold:

conjecture | open | MathDB 页面 | 原文线索

V2-064147 | MATHDB-336073

The general q-congruence conjecture for products of $A_{n+i-1,k}(q)$

摘录 / EXCERPT: General q-congruence conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-064148 | MATHDB-336074

The all-j conjecture for the two alternating q-congruence theorems

摘录 / EXCERPT: Let the two preceding results, Theorems $\mathrm{thm:fatorodd\text{-}first}$ and $\mathrm{thm:fatorodd\text{-}second}$, be the stated congruence theorems involving the parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-064149 | MATHDB-336080

Goppa's conjecture on the binary Gilbert-Varshamov bound

摘录 / EXCERPT: Goppa's conjecture. The Gilbert-Varshamov bound is tight in the binary case; equivalently, the asymptotic achievable rate equals the Gilbert-Varshamov bound.

conjecture | open | MathDB 页面 | 原文线索

V2-064150 | MATHDB-336082

Lapid's conjecture on parabolic Kazhdan--Lusztig polynomials

摘录 / EXCERPT: Let $P_{u,w}$ be the Kazhdan--Lusztig polynomials for symmetric groups, and let $H\simeq S_m\times\dots\times S_m$ be the parabolic subgroup of S_{mn} of type $(m,\dots,m)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-064151 | MATHDB-336083

Kobayashi's rank conjecture on Clifford-Klein forms

摘录 / EXCERPT: A Clifford-Klein form of a homogeneous space G/H is a quotient space $\Gamma \backslash G/H$, where Γ is a discrete subgroup of G acting properly and freely on G/H. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-064152 | MATHDB-336097

Conjecture on linear stability and stability of Lazarsfeld-Mukai bundles

摘录 / EXCERPT: Let C be a curve, let L be a globally generated line bundle on C, and let M_L be the kernel bundle in the evaluation sequence ... The pair $(H^0(C,L),L)$ is linearly (semi)st...

conjecture | open | MathDB 页面 | 原文线索

V2-064153 | MATHDB-336099

Wormald's isomorphic linear-forest partition conjecture for cubic graphs

摘录 / EXCERPT: Wormald's conjecture. There exists a linear partition of G into two isomorphic linear forests.

conjecture | open | MathDB 页面 | 原文线索

V2-064154 | MATHDB-336100

Ando's bisection conjecture for cubic graphs

摘录 / EXCERPT: Ando's conjecture. Every cubic graph admits a bisection such that the two induced monochromatic subgraphs are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-064155 | MATHDB-336101

Strong Ando conjecture for cubic graphs

摘录 / EXCERPT: Strong Ando conjecture. Every cubic graph admits a bisection such that the two induced subgraphs are isomorphic linear forests.

conjecture | open | MathDB 页面 | 原文线索

V2-064156 | MATHDB-336103

Fekete-Szegő conjecture for the class of q-starlike functions of order alpha

摘录 / EXCERPT: Let q satisfy $0<q<1$, let $0\leq\alpha<1$, and let $f\in\mathcal{S}_q^*(\alpha)$ have the normalized expansion ... Let μ be any complex number. Fekete-Szegő conjecture. ... Equa...

conjecture | open | MathDB 页面 | 原文线索

V2-064157 | MATHDB-336107

Off-critical mean conjecture for the FK heat-bath coupling

摘录 / EXCERPT: Off-critical mean conjecture. There exists $C(p,q,d)\geq1$ such that, as $L\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-064158 | MATHDB-336108

Off-critical variance conjecture for the FK heat-bath coupling

摘录 / EXCERPT: Off-critical variance conjecture. There exists $C(p,q,d)\geq1$ such that, as $L\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-064159 | MATHDB-336114

The average-degree characterization of property M(k)

摘录 / EXCERPT: Average-degree characterization of property $M(k)$. Every graph with average degree $2k-2$ has property $M(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-064160 | MATHDB-336124

Nonvanishing of central Rankin-Selberg L-values after twisting

摘录 / EXCERPT: Let Π and Π' be the automorphic representations specified in the surrounding hypotheses, and let Π_χ and $\Pi_{\chi'}$ be the associated twists by conjugate self-dua...

conjecture | open | MathDB 页面 | 原文线索

V2-064161 | MATHDB-336125

Refined Gan-Gross-Prasad conjecture for unitary and orthogonal groups

摘录 / EXCERPT: Let $\mathbb F$ be a number field, let $\mathbb E/\mathbb F$ be either trivial or quadratic, and let $\mathscr G(\mathcal V)$ and $\mathscr G(\mathcal W)$ be the groups attached to...

conjecture | open | MathDB 页面 | 原文线索

V2-064162 | MATHDB-336126

Monotonicity conjecture for the asymptotic code rate

摘录 / EXCERPT: Let $q>2$ and let $\theta=1-1/q$. For x in the relevant relative-distance range, write $\alpha(x)=\lim_{n\to\infty}n^{-1}\log_q A_q(n,xn)$, where $A_q(n,d)$ is the maximum size o...

conjecture | open | MathDB 页面 | 原文线索

V2-064163 | MATHDB-336137

The permanent-non-singular $(1,1)$-matrix conjecture

摘录 / EXCERPT: Let G be a graph, and orient its edges arbitrarily. Define the graph matrix A_G with rows indexed by $E(G)$ and columns indexed by $V(G)\cup E(G)$ by assigning to an oriented e...

conjecture | open | MathDB 页面 | 原文线索

V2-064164 | MATHDB-336140

The harmonic-function characterization of polynomial growth

摘录 / EXCERPT: Harmonic-function characterization. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-064165 | MATHDB-336141

The nilpotent-by-finite characterization via polynomial harmonic functions

摘录 / EXCERPT: Nilpotent-by-finite characterization. The group G has a finite-index nilpotent subgroup if and only if $\dim HF_k(G,\mu)<\infty$ and there exists a finite-index subgroup H of...

conjecture | open | MathDB 页面 | 原文线索

V2-064166 | MATHDB-336147

The third-to-fourth iterate irreducibility conjecture for quadratic polynomials

摘录 / EXCERPT: Let $\phi_c(x)=x^2+c$ for $c\in\mathbb{Z}$, and let $b\in\mathbb{Z}$ be any basepoint. The polynomial $\phi_c^n(x)$ denotes the nth iterate of ϕ_c. Third-to-fourth iterate i...

conjecture | open | MathDB 页面 | 原文线索

V2-064167 | MATHDB-336148

Balanced normal bundle conjecture for rational normal curves on Fano intersections of quadrics

摘录 / EXCERPT: Let $R_n\subset\mathbb{P}^n$ be the rational normal curve, and let $X\subset\mathbb{P}^n$ be a general Fano complete intersection of k quadric hypersurfaces containing R_n. The...

conjecture | open | MathDB 页面 | 原文线索

V2-064168 | MATHDB-336151

Symmetric adjoining conjecture for tensor rank

摘录 / EXCERPT: Let $T\in\mathbb{C}^{I\times I\times I}$ be a symmetric tensor, and let $\mathcal{M}\subset\mathbb{C}^{I\times I}$ be a set of symmetric rank-one matrices. Let $\mathcal{T}$ be obt...

conjecture | open | MathDB 页面 | 原文线索

V2-064169 | MATHDB-336152

Waring-rank inequality for adjoining a variable

摘录 / EXCERPT: Let $f,g\in\mathbb{C}[x_1,\ldots,x_n]$ be homogeneous polynomials of degrees d and $d-1$, respectively. Let u be a new variable, and let $\ell[x_1,\ldots,x_n]$ be the set of al...

conjecture | open | MathDB 页面 | 原文线索

V2-064170 | MATHDB-336154

Field-extension conjecture for spectrally arbitrary zero-nonzero patterns

摘录 / EXCERPT: Let $\mathbb{F}\subset\mathbb{K}$ be an extension of fields, and let S be a zero-nonzero pattern that is spectrally arbitrary over $\mathbb{F}$. Field-extension conjecture. Then...

conjecture | refuted | MathDB 页面 | 原文线索

V2-064171 | MATHDB-336155

The $2n$-conjecture over the rationals

摘录 / EXCERPT: Let S be an $n\times n$ zero-nonzero pattern that is spectrally arbitrary with respect to $\mathbb{Q}$. The $2n$-conjecture over the rationals. Then S has at least $2n$ nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-064172 | MATHDB-336163

The nonwetting conjecture for the modified Steiner-like surface

摘录 / EXCERPT: Nonwetting conjecture. The surface $M_{h,s}$ satisfies condition (NW).

conjecture | open | MathDB 页面 | 原文线索

V2-064173 | MATHDB-336167

Unified weighted-condition conjecture for the multilinear maximal operator

摘录 / EXCERPT: Unified weighted-condition conjecture. There is a constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-064174 | MATHDB-336169

The Faltings height conjecture for elliptic curves

摘录 / EXCERPT: For an elliptic curve E over $\mathbb{Q}$, let $h(E)$ denote its Faltings height and let N_E be its conductor. Height conjecture. There is a constant κ such that for all...

conjecture | open | MathDB 页面 | 原文线索

V2-064175 | MATHDB-336170

The modular degree conjecture for elliptic curves over the rational numbers

摘录 / EXCERPT: For an elliptic curve E over $\mathbb{Q}$, let N_E be its conductor and let $\delta_{1,N}(E)$ be the degree of the standard modular parametrization associated with the optimal...

conjecture | open | MathDB 页面 | 原文线索

V2-064176 | MATHDB-336171

The modular degree conjecture over totally real fields

摘录 / EXCERPT: Let F be a totally real number field of degree n over $\mathbb{Q}$, and let E/F be an elliptic curve modular of Shimura level 1. Let δ_E be the relevant degree of a...

conjecture | open | MathDB 页面 | 原文线索

V2-064177 | MATHDB-336177

The Eisenstein-cocycle formula for Gross's p-adic regulator

摘录 / EXCERPT: The Eisenstein-cocycle regulator conjecture. For every subset $J\subset R$,

conjecture | open | MathDB 页面 | 原文线索

V2-064178 | MATHDB-336178

Gross's characteristic-polynomial conjecture for the p-adic regulator matrix

摘录 / EXCERPT: Let $R=\{\mathfrak p_1,\ldots,\mathfrak p_r\}$, let E_R^* be the group of totally positive R-units, let $\vartheta\in H_{n+r-1}(E_R^*,\mathbf Z)\cong\mathbf Z$ be a generator,...

conjecture | open | MathDB 页面 | 原文线索

V2-064179 | MATHDB-336179

Füredi's equi-sized chain partition conjecture for the Boolean lattice

摘录 / EXCERPT: Füredi's conjecture. For every positive integer n, the Boolean lattice $2^{[n]}$ can be partitioned into $\binom{n}{\lfloor n/2\rfloor}$ chains such that the size of each chain i...

conjecture | open | MathDB 页面 | 原文线索

V2-064180 | MATHDB-336180

Hsu-Logan-Shahriari chain partition conjecture for normalized matching posets

摘录 / EXCERPT: Hsu-Logan-Shahriari conjecture. If Q is a rank-symmetric, unimodal normalized matching poset of width w, then Q can be partitioned into w chains, each chain in the partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-064181 | MATHDB-336187

Popescu's higher-rank abelian Stark conjecture

摘录 / EXCERPT: Popescu's conjecture. Assuming Hypothesis StarkHhigher, one has

conjecture | open | MathDB 页面 | 原文线索

V2-064182 | MATHDB-336188

Burns's conjecture on higher-rank Stickelberger annihilators

摘录 / EXCERPT: Burns's conjecture. One has $\Theta(\phi)\in\mathbb{Z}[G]$, moreover $\Theta(\phi)\in\operatorname{Ann}_{\mathbb{Z}[G]}(Cl_S(K))$, and if S' satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-064183 | MATHDB-336195

SHGH conjecture for special divisors on blow-ups of the plane

摘录 / EXCERPT: Let S be the blow-up of s general points of $\mathbb P^2$, let F be a divisor on S, and let an exceptional curve mean a smooth rational curve E with $E^2=-1$. SHGH conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-064184 | MATHDB-336202

The unimodality conjecture for k-colored partition crank distributions

摘录 / EXCERPT: Unimodality conjecture. For $n\geq 0$ and $k\geq 2$, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-064185 | MATHDB-336206

The twisted conjugacy separability domination conjecture

摘录 / EXCERPT: Twisted conjugacy separability domination conjecture. For every automorphism $\varphi\in\operatorname{Aut}(N)$,

conjecture | open | MathDB 页面 | 原文线索

V2-064186 | MATHDB-336209

LLT e-positivity conjecture for circular Dyck diagrams

摘录 / EXCERPT: Let ${\boldsymbol\nu} = (\mathbf{a},\mathbf{s})$ be a circular Dyck diagram with some strict corner edges. Circular LLT e-positivity conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-064187 | MATHDB-336210

Circular chromatic e-positivity conjecture

摘录 / EXCERPT: Let $\mathbf{a}$ be a circular area sequence, and let $\mathrm{X}_{\mathbf{a}}(\mathbf{x};q)$ be the corresponding chromatic symmetric function. Circular chromatic e-positivity con...

conjecture | open | MathDB 页面 | 原文线索

V2-064188 | MATHDB-336211

Circular chromatic coefficient palindromicity and unimodality conjecture

摘录 / EXCERPT: Circular chromatic coefficient conjecture. The coefficients $c_\lambda(q)$ are palindromic and unimodal polynomials with non-negative integer coefficients. This extends and refines...

conjecture | open | MathDB 页面 | 原文线索

V2-064189 | MATHDB-336214

Circular LLT chromatic Schur-positivity difference conjecture

摘录 / EXCERPT: Let $\mathbf{a}$ be a circular area sequence, and let $\mathrm{G}_{\mathbf{a}}(\mathbf{x};q+1)$ and $\mathrm{X}_{\mathbf{a}}(\mathbf{x};q)$ denote the corresponding LLT polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-064190 | MATHDB-336217

Polynomial boundedness conjecture for elliptic-curve torsion

摘录 / EXCERPT: Let $T(d)$ be the supremum of $\#E(F)[{\rm tors}]$ as F ranges over number fields of degree d and E ranges over elliptic curves defined over F. Polynomial boundedness conje...

conjecture | open | MathDB 页面 | 原文线索

V2-064191 | MATHDB-336218

Polynomial boundedness conjecture for torsion with fixed j-invariant degree

摘录 / EXCERPT: Let $d_0\in\mathbb{Z}^+$, and for each positive integer d divisible by d_0, let $T_{d_0}(d)$ be the supremum of $\#E(F)[{\rm tors}]$ as E ranges over elliptic curves defined...

conjecture | open | MathDB 页面 | 原文线索

V2-064192 | MATHDB-336225

Zvonkine's qr-ELSV formula

摘录 / EXCERPT: Let $g\geq 0$, let $q,r\geq 1$, and let $\mu=(\mu_1,\ldots,\mu_n)$ be a partition with $n=\ell(\mu)$ and $|\mu|=\sum_{i=1}^n\mu_i$ divisible by q. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-064193 | MATHDB-336235

Net reproduction threshold conjecture for local stability of the trivial steady state

摘录 / EXCERPT: Let $u_*\equiv 0$ denote the trivial steady state of the nonlinear problem $\$. Assume that its linearisation exists, and that $E(0)={\bf 0}$. Net reproduction threshold conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-064194 | MATHDB-336236

Uniform subcritical reproduction conjecture for global stability

摘录 / EXCERPT: Let $u_*\equiv 0$ denote the trivial steady state of the nonlinear problem $\$, let $\mathcal{Y}$ be the parameter space, and let $\mathcal{R}:\mathcal{Y}\to\mathbb{R}$ be the net...

conjecture | open | MathDB 页面 | 原文线索

V2-064195 | MATHDB-336240

Costa-Morini-Pasotti-Pellegrini partial-sums conjecture for abelian groups

摘录 / EXCERPT: Let $(G,+)$ be an abelian group, and let A be a finite subset of $G\setminus\{0\}$. Assume that no two-element subset $\{x,-x\}$ is contained in A, and that ... For an ordering...

conjecture | open | MathDB 页面 | 原文线索

V2-064196 | MATHDB-336242

Common-denominator conjecture for multivariate characteristic functions

摘录 / EXCERPT: Common-denominator conjecture. For all $n,m,t_1,\ldots,t_{n+m}$, the functions $\chi_{n,m}^{(t_1,\ldots,t_{n+m})}(x_1,\ldots,x_{n+m}\mid\rho,q)$ are rational functions having the c...

conjecture | open | MathDB 页面 | 原文线索

V2-064197 | MATHDB-336243

Expansion conjecture for the bivariate derivative polynomials

摘录 / EXCERPT: Expansion conjecture for $d_n^{(2)}$. There exist functions $\beta_{n,j}(q)$ depending only on q, with $\beta_{n,0}(q)=1$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-064198 | MATHDB-336246

The first-column inverse-matrix partition-function conjecture

摘录 / EXCERPT: Let $p(n)$ denote the partition function, let $\mu(n)$ denote the Möbius function, and let $s_{n,1}^{(-1)}$ denote the entries in the first column of the inverse matrix under consi...

conjecture | open | MathDB 页面 | 原文线索

V2-064199 | MATHDB-336251

Contractive Riesz projection conjecture and its endpoint logarithmic inequality

摘录 / EXCERPT: Riesz projection contractivity conjecture. The operator P is a contraction from $L^r(\mathbb{T})$ to H^q. In addition, for every $f\in L^1(\mathbb{T})$,

conjecture | open | MathDB 页面 | 原文线索

V2-064200 | MATHDB-336253

Every shuffle-compatible permutation statistic is a descent statistic

摘录 / EXCERPT: Let a permutation statistic be a statistic on permutations, and call it shuffle-compatible if the multiset of its values on shuffles of two permutations depends only on the two inp...

conjecture | open | MathDB 页面 | 原文线索

V2-064201 | MATHDB-336257

Isolation conjecture for partial Hadamard matrices

摘录 / EXCERPT: Let $C_{M,N}$ be the space of partial complex Hadamard matrices, let $D_{M,N}$ be its dephased quotient, and for $H\in C_{M,N}$ write $d(H)=\dim(\widetilde{T}_H C_{M,N})$ for its un...

conjecture | open | MathDB 页面 | 原文线索

V2-064202 | MATHDB-336259

Regularity conjecture for Butson matrices

摘录 / EXCERPT: A partial complex Hadamard matrix $H\in M_{M\times N}(\mathbb T)$ is regular when the scalar products between its rows decompose as sums of cycles. A Butson matrix is a complex Had...

conjecture | open | MathDB 页面 | 原文线索

V2-064203 | MATHDB-336260

Deformation conjecture for regular Hadamard matrices

摘录 / EXCERPT: A regular matrix $H\in M_N(\mathbb C)$ is a complex Hadamard matrix whose row scalar products decompose as sums of cycles, and an affine deformation is the deformation notion used...

conjecture | open | MathDB 页面 | 原文线索

V2-064204 | MATHDB-336261

Weak isolation conjecture for regular partial Hadamard matrices

摘录 / EXCERPT: Let $H\in M_{M\times N}(\mathbb C)$ be a dephased partial Hadamard matrix. It is regular when its row scalar products decompose as sums of cycles, isolated when it has only trivial...

conjecture | open | MathDB 页面 | 原文线索

V2-064205 | MATHDB-336262

Regular complex Hadamard matrices of order seven

摘录 / EXCERPT: Let F_7 be the Fourier matrix of order 7, and let P_7^q denote the displayed Petrescu family, with $q\in\mathbb T$. A regular complex Hadamard matrix is one whose row scalar...

conjecture | open | MathDB 页面 | 原文线索

V2-064206 | MATHDB-336263

Master Hadamard conjecture for Diua deformations

摘录 / EXCERPT: In the square case, a master Hadamard matrix is a complex Hadamard matrix arising from the master-function construction, and F_N denotes the Fourier matrix of order N. A Dic tu...

conjecture | open | MathDB 页面 | 原文线索

V2-064207 | MATHDB-336266

Almost Hadamard conjecture in the complex setting

摘录 / EXCERPT: An almost Hadamard matrix in the complex sense is the notion introduced in the cited work, while a complex Hadamard matrix is a unitary matrix whose entries have equal modulus afte...

conjecture | open | MathDB 页面 | 原文线索

V2-064208 | MATHDB-336272

Bailey-Borwein-Broadhurst-Glasser sum rule for Bessel moments

摘录 / EXCERPT: B^{3}G sum rule. For every such pair (n,k), the sum vanishes identically:

conjecture | open | MathDB 页面 | 原文线索

V2-064209 | MATHDB-336284

Normal distribution of nonvanishing Betti rows for random flag complexes

摘录 / EXCERPT: Let k be a row index, and consider a Betti table for which Theorem yields nonvanishing Betti numbers in row k. The kth row is viewed as a sequence of Betti numbers indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-064210 | MATHDB-336285

Conjecture on Legendre-function GKN conditions and bases for left-definite domains

摘录 / EXCERPT: Legendre-function conjecture. These Legendre polynomials can be used as GKN conditions to define the $n/2$ left-definite domain, and together the $2n$ functions constitute a basis...

conjecture | open | MathDB 页面 | 原文线索

V2-064211 | MATHDB-336286

Conjecture on equality of the natural domains for left-definite powers

摘录 / EXCERPT: Left-definite domain equality conjecture. For every $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-064212 | MATHDB-336287

The two-variable Hida deformation Iwasawa main conjecture

摘录 / EXCERPT: Let $\mathbf{f}$ be a Hida family of p-ordinary p-stabilized cusp newforms of tame level N, and let $\mathcal{T}$ be the nearly ordinary Hida deformation associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-064213 | MATHDB-336288

The finite-surjectivity conjecture for four torsion images

摘录 / EXCERPT: Let $j:\mathcal E\rightarrow {\mathbb P}^1$ be the universal elliptic curve, let $\mathcal P$ be the quotient by the involution $x\mapsto -x$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-064214 | MATHDB-336289

The rational-embedding conjecture for four torsion images

摘录 / EXCERPT: Let $j:\mathcal E\rightarrow {\mathbb P}^1$ be the universal elliptic curve, let $\mathcal P$ be its quotient by $x\mapsto -x$, and let $Y_{4,\omega}\subset {\rm Sym}^4(Y)$ be an i...

conjecture | open | MathDB 页面 | 原文线索

V2-064215 | MATHDB-336290

The rational-embedding and distinct-image conjecture for five torsion images

摘录 / EXCERPT: Let $Y_{5,\omega}\subset {\rm Sym}^5(Y)$ be the irreducible component associated with a ${\rm PGL}_2({\mathbb Z})$-orbit ω, and let μ_5 be the torsion-image map to...

conjecture | open | MathDB 页面 | 原文线索

V2-064216 | MATHDB-336291

The rational-embedding and finite-intersection conjecture for six torsion images

摘录 / EXCERPT: Let $Y_{6,\omega}\subset {\rm Sym}^6(Y)$ be the irreducible component associated with a ${\rm PGL}_2({\mathbb Z})$-orbit ω, and let μ_6 be the torsion-image map to...

conjecture | open | MathDB 页面 | 原文线索

V2-064217 | MATHDB-336292

Bogomolov-Fu-Tschinkel boundedness conjecture for projectively related elliptic curves

摘录 / EXCERPT: Let E_r be an elliptic curve over a number field, let π_r be its quotient map to ${\mathbb P}^1$, and let $\gamma\in {\rm PGL}_2(\overline{{\mathbb Q}})$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-064218 | MATHDB-336293

The dense family conjecture for large torsion intersections

摘录 / EXCERPT: Let E_a and E_{a_2} be the elliptic curves in the family parametrized by ${\mathbb P}^1$, with quotient maps π_a and π_{a_2}, and let $E_a[n]$ denote the n-torsion su...

conjecture | open | MathDB 页面 | 原文线索

V2-064219 | MATHDB-336294

The generic-finiteness conjecture for torsion configurations

摘录 / EXCERPT: Let $Y(N)$ be the quotient modular curve of level N, let ${\rm Sym}^n(Y(N))$ be its n-fold symmetric product, and let $Z_{n,N}\subset\overline{\mathcal M}_{0,n}$ be the image o...

conjecture | open | MathDB 页面 | 原文线索

V2-064220 | MATHDB-336295

The density conjecture for torsion-configuration loci

摘录 / EXCERPT: For integers $n,N\geq5$, let $Z_{n,N}\subset\overline{\mathcal M}_{0,n}$ be the locus arising as the image of ${\rm Sym}^n(Y(N))$ under the torsion-image map. Torsion-locus density...

conjecture | open | MathDB 页面 | 原文线索

V2-064221 | MATHDB-336298

Anthony-Brightwell-Shawe-Taylor conjecture on minimum specification threshold functions

摘录 / EXCERPT: Anthony-Brightwell-Shawe-Taylor conjecture. If f has specification number $n+1$, then f is nested:

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-064222 | MATHDB-336299

The BSD conjecture on the sets $\mathfrak X_j$

摘录 / EXCERPT: $\mathfrak X_j$-BSD conjecture. If $\tau\in\mathfrak X_j$ and E_τ is defined over $\mathbb{Q}$, then the BSD conjecture for E_τ holds.

conjecture | open | MathDB 页面 | 原文线索

V2-064223 | MATHDB-336300

Chen's conjectures on inequalities for the Andrews spt-function

摘录 / EXCERPT: Let $p(n)$ denote the partition function and let ${\mathrm{spt}}(n)$ denote the smallest-parts function, which counts the smallest parts among the partitions of n. Chen's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-064224 | MATHDB-336310

Lužar-Škrekovski conjecture on injective coloring of planar graphs

摘录 / EXCERPT: Lužar-Škrekovski conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064225 | MATHDB-336314

The rigged-configuration characterization of solitons

摘录 / EXCERPT: Soliton characterization conjecture. The state p corresponds to a soliton of length ℓ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-064226 | MATHDB-336321

The equivalence between type-theoretic and Grothendieck-Maltsiniotis c9-categories

摘录 / EXCERPT: Let $\mathcal{S}_{\mathrm{cat}}$ be the category of type-theoretic models described in the paper, and let $\mathcal{T}$ be the canonical coherator for Grothendieck-Maltsiniotis...

conjecture | open | MathDB 页面 | 原文线索

V2-064227 | MATHDB-336322

The correspondence between type-theoretic and Grothendieck-Maltsiniotis c9-categories

摘录 / EXCERPT: A type-theoretic ω-category is a model of the type theory defined in the paper, while a Grothendieck-Maltsiniotis ω-category is a functor...

conjecture | open | MathDB 页面 | 原文线索

V2-064228 | MATHDB-336323

Sharpness conjecture for the Serre-type lower bound in weighted projective spaces

摘录 / EXCERPT: Let $\mathbb{P}(a_0,a_1,\ldots,a_m)$ be a weighted projective space over $\mathbb{F}_q$, let $e_q(d;1,a_1,a_2,\ldots,a_m)$ be the minimum number of $\mathbb{F}_q$-rational zeros of...

conjecture | open | MathDB 页面 | 原文线索

V2-064229 | MATHDB-336326

Rank-one abelian Stark conjecture

摘录 / EXCERPT: Let K/F be an abelian extension of number fields, let S be a finite set of places containing all infinite places of F and all places ramified in K/F, and suppose that a dis...

conjecture | open | MathDB 页面 | 原文线索

V2-064230 | MATHDB-336329

Spectrum invariance under toral and spherical Aluthge transforms of commuting 2-variable weighted shifts

摘录 / EXCERPT: Let $W_{(\alpha,\beta)}$ be a commuting 2-variable weighted shift, and suppose that its toral and spherical Aluthge transforms, denoted by $\widetilde{W}_{(\alpha,\beta)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064231 | MATHDB-336332

McMullen--Walkup generalized lower bound conjecture

摘录 / EXCERPT: Let Δ be the boundary of a simplicial convex d-polytope. For $1\leq r\leq\lfloor d/2\rfloor$, let $g_r(\Delta)$ denote the rth entry of its g-vector, and call Δ...

conjecture | open | MathDB 页面 | 原文线索

V2-064232 | MATHDB-336342

The higher cs lower bound equality conjecture for simplicial polytopes

摘录 / EXCERPT: Let P be a centrally symmetric simplicial d-polytope. The higher cs lower bound conjecture. If, for some $3\leq r<\left\lfloor\frac{d}{2}\right\rfloor$, ... then ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-064233 | MATHDB-336343

The polytopal cellulation conjecture for centrally symmetric polytopes with minimal higher g-number

摘录 / EXCERPT: The polytopal cellulation conjecture. There exists a unique polytopal complex $\mathcal C$ in $\mathbb R^d$ such that: (i) one face of $\mathcal C$ is the cross-polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-064234 | MATHDB-336344

Conjecture on achievement sets of a Cantor set and two Cantorvals

摘录 / EXCERPT: Let X be a Cantor set and let Y and Z be Cantorvals satisfying ... A set is an achievement set if it is the set of all subsums of an absolutely summable sequence. Achievement-s...

conjecture | open | MathDB 页面 | 原文线索

V2-064235 | MATHDB-336347

The Golomb--Welch weak conjecture for perfect Lee codes

摘录 / EXCERPT: For integers n,e,q, let a $PL(n,e,q)$-code mean a perfect e-error-correcting code in the Lee metric on $\mathbb{Z}_q^n$. Golomb--Welch weak conjecture. There is no $PL(n,e,q)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-064236 | MATHDB-336348

The Golomb--Welch strong conjecture for perfect Lee codes

摘录 / EXCERPT: For integers n,e, let a $PL(n,e)$-code mean a perfect e-error-correcting code in the Lee metric on $\mathbb{Z}^n$. Golomb--Welch strong conjecture. There is no $PL(n,e)$-code f...

conjecture | open | MathDB 页面 | 原文线索

V2-064237 | MATHDB-336353

Comon's conjecture for Hankel tensors

摘录 / EXCERPT: Let $\mathcal{H}\in\operatorname{H}^m(\mathbb{C}^n)$ be any Hankel tensor, where $\operatorname{rank}(\mathcal{H})$ denotes its tensor rank and $\operatorname{rank}_S(\mathcal{H})$...

conjecture | open | MathDB 页面 | 原文线索

V2-064238 | MATHDB-336361

Poisson-embedding convergence conjecture for critical Hawkes processes

摘录 / EXCERPT: Let F be an absolutely continuous displacement distribution, let $\lambda>0$, and let $B\in\mathcal{B}_b(\operatorname{\mathbb{R}})$. Let $N^{(g)}$ and $\lambda^{(g)}$ be defined...

conjecture | open | MathDB 页面 | 原文线索

V2-064239 | MATHDB-336364

Conjecture on the position of lumps emerging from the soliton front

摘录 / EXCERPT: Let ϵ be the small dispersion parameter, and let a, b, and c_0 be constants. Let (x_0,y_0) be the position where a singularity of the Whitham system is expected to...

conjecture | open | MathDB 页面 | 原文线索

V2-064240 | MATHDB-336366

Conjecture on the interpretation of the substitution rule in ParamDTT

摘录 / EXCERPT: Substitution-interpretation conjecture. The interpretation of the substitution rule is given by the substitution above. The source assumes this claim without proof as part of the s...

conjecture | open | MathDB 页面 | 原文线索

V2-064241 | MATHDB-336367

Conjecture on irrelevant pathhood of interpreted ParamDTT types

摘录 / EXCERPT: Pathhood-irrelevance conjecture. The interpretation of any type that can be constructed in ParamDTT has irrelevant pathhood. This property is introduced to ensure injectivity of th...

conjecture | open | MathDB 页面 | 原文线索

V2-064242 | MATHDB-336373

The one-node tree proportion conjecture for quadratic functional graphs

摘录 / EXCERPT: The one-node tree conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064243 | MATHDB-336375

Cohen-Martinet density conjecture for class numbers of cyclic cubic fields

摘录 / EXCERPT: Cohen-Martinet conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064244 | MATHDB-336386

The $1/3$-$2/3$ Conjecture for finite partially ordered sets

摘录 / EXCERPT: The $1/3$-$2/3$ Conjecture. Every finite partially ordered set that is not a chain has a $1/3$-balanced pair.

conjecture | open | MathDB 页面 | 原文线索

V2-064245 | MATHDB-336387

Ciocan-Fontanine-Kim wall-crossing conjecture for stable quasimaps

摘录 / EXCERPT: Ciocan-Fontanine-Kim wall-crossing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064246 | MATHDB-336388

Broad vanishing conjecture for cosection-localized GLSM virtual classes

摘录 / EXCERPT: Broad vanishing conjecture. For every $i\in\{1,\ldots,n\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-064247 | MATHDB-336390

The symmetric rank conjecture for the skew matrix multiplication polynomial of size three

摘录 / EXCERPT: Let $sM_{\langle 3\rangle}$ denote the skew matrix multiplication polynomial of size 3, and let ${\bold R}_s$ denote its symmetric rank, namely the least number of cubes of linea...

conjecture | open | MathDB 页面 | 原文线索

V2-064248 | MATHDB-336413

Bukh-Matoušek-Nivasch conjecture on simplicial depth

摘录 / EXCERPT: For a finite set P of n points in $\mathbb{R}^d$, define its simplicial depth as the maximum number of d-simplices spanned by P that contain a common point. Bukh-Matoušek-N...

conjecture | open | MathDB 页面 | 原文线索

V2-064249 | MATHDB-336416

Reay's relaxed Tverberg conjecture

摘录 / EXCERPT: Let $T(d,r,k)$ be the least positive integer n such that every set of n points in $\mathbb{R}^d$ admits a partition into r disjoint parts whose convex hulls have nonempty int...

conjecture | open | MathDB 页面 | 原文线索

V2-064250 | MATHDB-336417

Stahl's Kneser graph homomorphism conjecture

摘录 / EXCERPT: Let n,k,k' be integers, and write $k'=qk-r$ with $0\leq r<k$. Let $\operatorname{KG}(n,k)$ denote the Kneser graph on the k-subsets of $[n]$. Stahl's Kneser graph homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-064251 | MATHDB-336418

Brualdi-Stein colorful matching conjecture

摘录 / EXCERPT: Let $K_{n,n}$ be the complete bipartite graph with n vertices in each part, and partition its edge set into n classes $E_1,\ldots,E_n$, each of size n. Brualdi-Stein colorful...

conjecture | open | MathDB 页面 | 原文线索

V2-064252 | MATHDB-336421

The epsilon-union-closed sets conjecture

摘录 / EXCERPT: Let $\mathcal{A}$ be a finite union-closed collection of sets containing at least one non-empty set, and let $|\mathcal{A}|$ denote its number of sets. The ε-union-clos...

conjecture | open | MathDB 页面 | 原文线索

V2-064253 | MATHDB-336423

The des-Wilf equivalences among four-letter non-classical patterns

摘录 / EXCERPT: For a permutation pattern τ, let $F_n^{\operatorname{des}}(\tau;q)$ denote the descent generating polynomial over permutations in $\operatorname{Av}_n(\tau)$, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-064254 | MATHDB-336427

Mean-zero Sato-Tate law for conjugation-stable sets of analytic L-functions

摘录 / EXCERPT: Let $\mathcal{S}$ be a finite set of entire analytic L-functions that is stable under conjugation, so that $\overline{\mathcal{S}}=\mathcal{S}$. Let $(a_f)_{f\in\mathcal{S}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-064255 | MATHDB-336435

Buratti-Sajna conjecture on resolvable directed cycle decompositions

摘录 / EXCERPT: Buratti-Sajna conjecture. K_{2m}^* admits a resolvable directed m-cycle decomposition if and only if $m\geq 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-064256 | MATHDB-336439

The intrinsic-capacity decomposition conjecture

摘录 / EXCERPT: Let W be a channel and let $\operatorname{dec}(W)$ denote its convex decompositions. For $\lambda\in\operatorname{dec}(W)$, write $\operatorname{C}_{11}(\lambda)$ for the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-064257 | MATHDB-336440

The causal state-information uselessness conjecture for binary-input channels

摘录 / EXCERPT: Let W be a channel from $[\![1,2]\!]$ to $[\![1,n]\!]$, and let $\mathcal{S}$ be the state space. Suppose ... where S denotes the channel state. Causal state-information uselessn...

conjecture | open | MathDB 页面 | 原文线索

V2-064258 | MATHDB-336441

Conjecture on the maximal-elasticity difference invariant

摘录 / EXCERPT: Let H be a transfer Krull monoid over a finite abelian group G with $|G|>4$. Let $\Delta_{\rho}(H)$ denote the set of all positive integers d such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-064259 | MATHDB-336448

The pattern-pair characterization of restricted inversion-sequence interpretations

摘录 / EXCERPT: Let $S_{n,k}$ be the quantity defined earlier in the paper, let ρ and τ be patterns of length 4, and let $d53e_n(\rho,\tau)$ denote the permutations in $d53e_n$ avoidin...

conjecture | open | MathDB 页面 | 原文线索

V2-064260 | MATHDB-336449

Yan and Martinez-Savage's enumeration conjecture for restricted inversion sequences

摘录 / EXCERPT: For each $n\geq1$, let $\operatorname{{\bf I}}_n$ be the set of inversion sequences ... Let $\operatorname{{\bf I}}_n(\geq,\geq,-)$ be the subset of inversion sequences for which the...

conjecture | open | MathDB 页面 | 原文线索

V2-064261 | MATHDB-336454

The Colin de Verdière graph complement conjecture

摘录 / EXCERPT: Let G be a finite simple graph and let $\overline{G}$ denote its complement. Let $\mu(G)$ denote the Colin de Verdière parameter. Colin de Verdière's graph complement conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-064262 | MATHDB-336457

Triviality conjecture for Legendrian unknot racks

摘录 / EXCERPT: Let L be a Legendrian unknot, let $tb(L)$ denote its Thurston-Bennequin number, and let n-Legendrian racks be the racks associated to L by the n-Legendrian rack constructio...

conjecture | open | MathDB 页面 | 原文线索

V2-064263 | MATHDB-336458

Scaling conjecture for Legendrian unknot racks

摘录 / EXCERPT: Let L_1 and L_2 be Legendrian unknots, let $tb(L_i)$ denote their Thurston-Bennequin numbers, and let $\mathcal{LR}(L_i)$ denote their Legendrian racks. Scaling conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-064264 | MATHDB-336460

The rank-plus-one conjecture for second-order optimality under MFCQ

摘录 / EXCERPT: The rank-plus-one conjecture for second-order optimality. If x^* is a local minimizer, MFCQ holds at x^*, and the rank of...

conjecture | open | MathDB 页面 | 原文线索

V2-064265 | MATHDB-336466

The odd-degree peculiar linear mean relationship

摘录 / EXCERPT: Let f be an odd-degree polynomial of degree D, and let $\varphi_{D,0,\rho}$ denote the quantities used in the source for indices D, 0, and ρ. Odd-degree peculiar linea...

conjecture | open | MathDB 页面 | 原文线索

V2-064266 | MATHDB-336481

The limit-set conjecture for the mapping class group of a nonorientable surface

摘录 / EXCERPT: Limit-set conjecture. The limit set of $\operatorname{Map}(S)$ in $\mathcal{PML}(S)$ is $\mathcal{PML}^+(S)$.

conjecture | open | MathDB 页面 | 原文线索

V2-064267 | MATHDB-336484

The density conjecture for rational projective measured laminations

摘录 / EXCERPT: Rational-density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064268 | MATHDB-336487

Norm-one conjecture for dominant eigenvectors of non-negative premagic matrices

摘录 / EXCERPT: Let $\mathbf{M}$ be a non-negative premagic matrix, and let $\mathbf{v}$ be an eigenvector corresponding to its dominant eigenvalue. Norm-one conjecture. Both the 2-norm and the...

conjecture | open | MathDB 页面 | 原文线索

V2-064269 | MATHDB-336489

Scaling conjecture for dominant eigenpairs of non-negative premagic matrices

摘录 / EXCERPT: Scaling conjecture. The dominant eigenvalue of $\mathbf{M}_2$ is $\lambda_2=k\lambda_1$, and the corresponding eigenvector is $\mathbf{v}$. This is presented among computational co...

conjecture | open | MathDB 页面 | 原文线索

V2-064270 | MATHDB-336502

Exact-recovery phase-transition conjecture for logarithmic-degree Euclidean random graphs

摘录 / EXCERPT: For any $\lambda>0$, $d\in\mathbb{N}$, and $0\leq b<a\leq 1$, let ν_d denote the volume of the unit Euclidean ball in d dimensions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064271 | MATHDB-336505

Non-embeddability conjecture for right-angled Artin groups of tripod trees

摘录 / EXCERPT: Non-embeddability conjecture. If p,q,r are large enough, then $G(T_{p,q,r})$ does not embed into $G(P_n)$ for any n.

conjecture | open | MathDB 页面 | 原文线索

V2-064272 | MATHDB-336512

Higher-order Turán inequality for the partition function

摘录 / EXCERPT: Higher-order partition-function inequality. For $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-064273 | MATHDB-336513

Higher-order Jensen-polynomial real-rootedness conjecture for the partition function

摘录 / EXCERPT: Partition-function Jensen-polynomial conjecture. For every positive integer $m\geq 4$, there exists a positive integer $N(m)$ such that, for every $n\geq N(m)$, the polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-064274 | MATHDB-336517

The deep transfer conjecture for 3-class tower groups of totally real dihedral fields

摘录 / EXCERPT: Let $F=\mathbb{Q}(\sqrt{d})$ be a real quadratic field with fundamental discriminant $d>1$, 3-class group $\operatorname{Cl}_3(F)\simeq C_3\times C_3$, shallow transfer kernel ty...

conjecture | open | MathDB 页面 | 原文线索

V2-064275 | MATHDB-336521

Homogeneous equidistribution conjecture for divergent diagonal orbits

摘录 / EXCERPT: Let X_n be the space considered in the paper, let A be the diagonal subgroup acting on it, and for each $x\in\mathcal{H}_q(n)$ let μ_{xA} be the locally finite measure supp...

conjecture | open | MathDB 页面 | 原文线索

V2-064276 | MATHDB-336522

Adelic homogeneous equidistribution conjecture without the real-component restriction

摘录 / EXCERPT: Let $G_{\mathbb{A}}=\operatorname{PGL}_2(\mathbb{A})$, let $A_{\mathbb{A}}<G_{\mathbb{A}}$ be the diagonal subgroup, and let $\mu_{\tilde{x}_0A_{\mathbb{A}}}$ be the locally finite...

conjecture | open | MathDB 页面 | 原文线索

V2-064277 | MATHDB-336530

Schramm's conformal invariance conjecture for planar random-cluster interfaces

摘录 / EXCERPT: Fix $q\le 4$ and $p=p_c$. Let $(\Omega_\delta,a_\delta,b_\delta)$ be Dobrushin domains approximating a simply connected domain ${\bf \Omega}$ with marked boundary points ${\bf a}$...

conjecture | open | MathDB 页面 | 原文线索

V2-064278 | MATHDB-336532

Three-dimensional Ising correlation exponent bounds

摘录 / EXCERPT: Consider the critical three-dimensional Ising model, with spin variables $\mathit{\sigma_x}$ and critical Gibbs measure $\mathit{\mu_{\beta_c}}$. Three-dimensional Ising exponent c...

conjecture | open | MathDB 页面 | 原文线索

V2-064279 | MATHDB-336535

Delta-conjecture for the nef cone of blown-up planes

摘录 / EXCERPT: Let X be the blowup of $\mathbb{P}^2$ at n sufficiently general points. Let $\mathcal{Q}_n$ be the nonnegative cone and let $\Delta_n^{\preccurlyeq}$ be the cone defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-064280 | MATHDB-336536

Strong Delta-conjecture for rational boundary rays

摘录 / EXCERPT: Let X be the blowup of $\mathbb{P}^2$ at n sufficiently general points, and let $\mathcal{Q}_n$ and $\Delta_n^{\preccurlyeq}$ be as defined in the source. Strong Delta-conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-064281 | MATHDB-336538

Quadratic mating conjecture

摘录 / EXCERPT: Let P and Q be quadratic polynomials with locally connected Julia sets, and let p and q denote their corresponding parameters in the Mandelbrot set. The quadratic mating co...

conjecture | open | MathDB 页面 | 原文线索

V2-064282 | MATHDB-336544

Shift conjecture for signed odd-inversion generating functions on descent classes

摘录 / EXCERPT: Shift conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-064283 | MATHDB-336545

Varchenko-Stasinski's conjecture on odd-length generating functions for type B parabolic quotients

摘录 / EXCERPT: Varchenko-Stasinski's conjecture. The generating function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-064284 | MATHDB-336546

Sagan-Savage major-index unimodality conjecture for 321-avoiding permutations

摘录 / EXCERPT: Sagan-Savage conjecture. For all n and i, the polynomial $A_{n,i}(q)$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-064285 | MATHDB-336547

Unimodality conjecture for major index on two-row standard Young tableaux

摘录 / EXCERPT: Two-row tableau unimodality conjecture. For all n, k, and i, the polynomial $f_{(n-k,k),i}(q)$ is symmetric and unimodal, with central term $ni/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-064286 | MATHDB-336556

Univalent analytic shear conjecture for sense-preserving harmonic mappings

摘录 / EXCERPT: Let $f=h+\overline{g}$ be a sense-preserving and univalent harmonic mapping in a simply connected domain $D\subset\mathbb{C}$. Univalent analytic shear conjecture. There exists a c...

conjecture | open | MathDB 页面 | 原文线索

V2-064287 | MATHDB-336557

Ponnusamy-Sheil-Small analytic shear conjecture for normalized harmonic mappings

摘录 / EXCERPT: Let $\mathcal{S}_H^0$ be the subclass of normalized sense-preserving univalent harmonic mappings $f=h+\overline{g}$ with $g'(0)=0$, and let $\mathcal{S}$ be the class of normalized...

conjecture | open | MathDB 页面 | 原文线索

V2-064288 | MATHDB-336559

Asymptotic threshold conjecture for binary separating hash families

摘录 / EXCERPT: Asymptotic threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064289 | MATHDB-336569

Functoriality of similarity monoids for fractals

摘录 / EXCERPT: Let C be the category whose objects are fractals satisfying the conditions of Theorem, with suitable homeomorphisms as arrows. Let D be the category whose objects are Rees mono...

conjecture | open | MathDB 页面 | 原文线索

V2-064290 | MATHDB-336571

Heavy-traffic stable-limit conjecture for Hawkes/M/∞ queues with regularly varying service times

摘录 / EXCERPT: Let N denote the number of customers in a Hawkes/M/∞ queue, with ${\mathscr J}(u)=e^{-\mu u}$ and $h(u)=e^{-u}$ for $u\geqslant 0$. Assume that the service-time distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-064291 | MATHDB-336573

Doyle-Faber-Krumm conjecture on quadratic-field preperiodic graphs

摘录 / EXCERPT: Let K be a quadratic number field, let $f_c(x)=x^2+c$ with $c\in K$, and let $G(f_c,K)$ be the directed graph of K-rational preperiodic points, with an edge...

conjecture | open | MathDB 页面 | 原文线索

V2-064292 | MATHDB-336576

Martinez-Savage's inversion-sequence conjecture for 3-nonnesting set partitions

摘录 / EXCERPT: An inversion sequence of length n is a sequence $x=x_1x_2\cdots x_n$ satisfying $0\leq x_i<i$ for all $1\leq i\leq n$. A set partition of $[n]$ is enhanced 3-nonnesting (respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-064293 | MATHDB-336580

Higher-order Macdonald positivity conjecture

摘录 / EXCERPT: Higher-order Macdonald positivity conjecture. For every partition μ, the multivariate q,t-Kostka number $\tilde{K}^{(q,t)}_{\mu;\lambda^1,\dots,\lambda^r}$ is a polynomial in...

conjecture | open | MathDB 页面 | 原文线索

V2-064294 | MATHDB-336581

Conjecture on syndetic conjugates of a Cartan subgroup

摘录 / EXCERPT: Let G be a definably simple group definable in an o-minimal structure $\mathcal{M}$. A subset $X\subseteq G$ is syndetic if there is a finite set $E\subseteq G$ such that $XE=G$....

conjecture | open | MathDB 页面 | 原文线索

V2-064295 | MATHDB-336584

Crepant resolution conjecture for local projective four-space and $[\mathbb{C}^5/\mathbb{Z}_5]$

摘录 / EXCERPT: Crepant resolution conjecture. The transformation $\mathds{T}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-064296 | MATHDB-336612

The binary diagonal X-ray and tournament score-vector conjecture

摘录 / EXCERPT: A binary diagonal X-ray of an $n \times n$ permutation matrix is its diagonal X-ray with entries in $\{0,1\}$. A tournament is a loopless directed graph in which exactly one of...

conjecture | open | MathDB 页面 | 原文线索

V2-064297 | MATHDB-336614

Componentwise minimality conjecture for type B and C Waldspurger matrices

摘录 / EXCERPT: Componentwise minimality conjecture. Each element in the base for types B and C corresponds to a type B and C Waldspurger matrix which is componentwise least given a single fixed e...

conjecture | open | MathDB 页面 | 原文线索

V2-064298 | MATHDB-336621

Transitivity conjecture for strong unstable foliations near the toral automorphism A

摘录 / EXCERPT: Let $A\colon \mathbb T^3\to\mathbb T^3$ be the hyperbolic automorphism induced by the displayed integral matrix, and let W^{uu}_f denote the one-dimensional strong unstable folia...

conjecture | open | MathDB 页面 | 原文线索

V2-064299 | MATHDB-336622

Uniqueness conjecture for Gibbs u-measures near the toral automorphism A

摘录 / EXCERPT: Let $A\colon \mathbb T^3\to\mathbb T^3$ be the hyperbolic automorphism induced by the displayed integral matrix, and let f be an analytic diffeomorphism in a sufficiently small n...

conjecture | open | MathDB 页面 | 原文线索

V2-064300 | MATHDB-336624

Characterization of nonreducible associative idempotent monotone functions

摘录 / EXCERPT: Let X be a totally ordered Abelian group with respect to addition, and let $F:X^n\to X$ be an associative, idempotent, monotone function. The function F is reducible if there e...

conjecture | open | MathDB 页面 | 原文线索

V2-064301 | MATHDB-336631

Gross's first-part conjecture on orders of vanishing

摘录 / EXCERPT: Let $\psi\in\operatorname{Irr}_{\mathbb C_p}(\mathcal G)$ and let $r_S(\psi)$ be the order of vanishing at $s=0$ of the corresponding complex S-truncated Artin L-function. Gros...

conjecture | open | MathDB 页面 | 原文线索

V2-064302 | MATHDB-336632

Gross's second-part conjecture on leading coefficients

摘录 / EXCERPT: Let $\psi\in\operatorname{Irr}_{\mathbb C_p}(\mathcal G)$, choose $\iota:\mathbb C\simeq\mathbb C_p$, and let $r_S(\psi)$ be the corresponding complex order of vanishing. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064303 | MATHDB-336643

Non-realizability of the triple (3,5,6) for graph parameters

摘录 / EXCERPT: Let G be a graph, and write $\alpha(G)$ for its clique number, $\chi(G)$ for its chromatic number, and $\chi_{\rho}(G)$ for its packing chromatic number. Non-realizabili...

conjecture | open | MathDB 页面 | 原文线索

V2-064304 | MATHDB-336648

Conjecture on products and base change of l-i-modular extensions

摘录 / EXCERPT: Product and ascending base-change conjecture. (1) The compositum of two l-i-modular extensions is l-i-modular: if K/k and L/k are l-i-modular, then $K(L)/k$ is l-...

conjecture | open | MathDB 页面 | 原文线索

V2-064305 | MATHDB-336650

Singularity-theoretic determination conjecture for the invariant ν of plane curves

摘录 / EXCERPT: Singularity-theoretic determination conjecture. The invariant $\nu(C)$ is determined by the degree of C and the list of analytic types of the isolated singularities of C.

conjecture | open | MathDB 页面 | 原文线索

V2-064306 | MATHDB-336651

Colorability conjecture for cyclic pseudo-Loupekine graphs with parity conditions

摘录 / EXCERPT: Let S be a Loupekine 5-pole, let $x\in\{\alpha,\beta\}$, and let $S_x(m;a,a,c)$ denote the associated cyclic pseudo-Loupekine graph. Assume that $m/\gcd(m,a)$ is even and...

conjecture | open | MathDB 页面 | 原文线索

V2-064307 | MATHDB-336652

Cyclic 5-connectivity conjecture for the T families

摘录 / EXCERPT: Let $T^1_\alpha(m;a,a,c)$, $T^1_\beta(m;a,a,c)$, and $T^2_\alpha(m;a,a,c)$ be the snark families defined from the two triangle-destroying deletions of the Petersen-derived 5-pole....

conjecture | open | MathDB 页面 | 原文线索

V2-064308 | MATHDB-336653

Non-snark conjecture for the beta family $G_\beta(m;a,b,c)$

摘录 / EXCERPT: Let $G_\beta(m;a,b,c)$ be the beta family of cyclic graphs constructed in the paper. Beta-family non-snark conjecture. The graph $G_\beta(m;a,b,c)$ is never a snark. The displayed...

conjecture | open | MathDB 页面 | 原文线索

V2-064309 | MATHDB-336658

Li-Xu conjecture on quasi-monomial normalized-volume minimizers

摘录 / EXCERPT: Let $x\in (X=\operatorname{Spec}(R),D)$ be a klt singularity, and let v be a valuation minimizing the normalized volume $\widehat{\operatorname{vol}}_{(X,D)}$. A valuation is qua...

conjecture | open | MathDB 页面 | 原文线索

V2-064310 | MATHDB-336672

Choksi-Peletier conjecture on balls and dichotomy for the liquid-drop problem

摘录 / EXCERPT: For $n=3$ and $\alpha=2$, consider the liquid-drop energy $P(E)+\mathcal{V}_\alpha(E)$ among sets $E\subset\mathbb{R}^n$ of prescribed volume. A minimizer is a set attaining the in...

conjecture | open | MathDB 页面 | 原文线索

V2-064311 | MATHDB-336673

Two-dimensional ball-cluster conjecture for the liquid-drop problem

摘录 / EXCERPT: Let $n=2$, let $\alpha\in(0,2)$, and define $f(m)=P(B[m])+\mathcal{V}_\alpha(B[m])$, where $B[m]$ is a ball of volume m. Set $m_0=0$, and for $k\geq1$ let m_k be the critical m...

conjecture | open | MathDB 页面 | 原文线索

V2-064312 | MATHDB-336688

Tohăneanu's low-exponent supersolvability conjecture for free arrangements

摘录 / EXCERPT: Tohăneanu's low-exponent supersolvability conjecture. Then $\mathcal{A}$ is supersolvable. The conjecture concerns free arrangements with exactly one exponent equal to 3 and is v...

conjecture | open | MathDB 页面 | 原文线索

V2-064313 | MATHDB-336690

Amini's mak-stat equidistribution conjecture for pattern-avoiding permutations

摘录 / EXCERPT: For a pattern τ, let $S_n(\tau)$ denote the set of permutations in S_n avoiding τ. The statistics mak and $stat$ are generalized-pattern statistics, and equidistribut...

conjecture | open | MathDB 页面 | 原文线索

V2-064314 | MATHDB-336692

Thomas's half-integral Erdős-Pósa conjecture for graph minors

摘录 / EXCERPT: Thomas's conjecture. For every graph H, there exists a function f such that for every graph G and every positive integer k, either G half-integrally packs k graphs each...

conjecture | open | MathDB 页面 | 原文线索

V2-064315 | MATHDB-336695

The inclusion-property conjecture for finite point sets in multiprojective spaces

摘录 / EXCERPT: Inclusion-property conjecture. X is arithmetically Cohen-Macaulay (ACM).

conjecture | open | MathDB 页面 | 原文线索

V2-064316 | MATHDB-336696

The converse conjecture for ACM point sets and the condition $N_1(X) \in \mathcal D(X)$

摘录 / EXCERPT: Converse conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-064317 | MATHDB-336698

Monotonicity conjecture for a gamma-function quotient

摘录 / EXCERPT: Gamma-quotient monotonicity conjecture. The quotient decreases as c increases.

conjecture | open | MathDB 页面 | 原文线索

V2-064318 | MATHDB-336701

H. Cohn's finite-field correlation conjecture

摘录 / EXCERPT: H. Cohn's conjecture. Then f is a multiplicative character on $\mathbb{F}_p$. This problem is still open in the finite-field setting; partial results are known, but the conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-064319 | MATHDB-336704

The weak simplification number inequality for branched covering surface-knots

摘录 / EXCERPT: Let (F,Γ) be a branched covering surface-knot. The quantities $u_w(F,\Gamma)$ and $u(F,\Gamma)$ denote, respectively, the weak simplification number and simplification numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-064320 | MATHDB-336705

Conjectured bounds for the simplification number of branched covering surface-knots

摘录 / EXCERPT: Let (F,Γ) be a branched covering surface-knot of degree N. Let $u_w(F,\Gamma)$ and $u(F,\Gamma)$ denote the weak simplification number and simplification number, respectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-064321 | MATHDB-336710

The logarithmic psd-rank conjecture for regular polygon slack matrices

摘录 / EXCERPT: Let S_n be the slack matrix of the regular n-gon, and let $\operatorname{rank}_{\operatorname{psd}}(S_n)$ denote its positive semidefinite rank. Logarithmic psd-rank conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-064322 | MATHDB-336718

Positivity conjecture for matrix counts of a complementary diagonal permutation

摘录 / EXCERPT: For a positive integer n, let $v=(2n-1)(2n)(2n-3)(2n-2)\cdots3412$ be the indicated permutation, and let I_v be its diagram. Write $M_{2n}(\overline{I_v},q)$ for the matrix cou...

conjecture | open | MathDB 页面 | 原文线索

V2-064323 | MATHDB-336719

Positivity conjecture for matrix counts of 123-avoiding permutations

摘录 / EXCERPT: Let $w\in\mathfrak{S}_n$ be a 123-avoiding permutation, meaning that it contains no subsequence order-isomorphic to 123, and let I_w be its diagram. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-064324 | MATHDB-336730

Demailly-Kollár ascending chain conjecture for log-canonical thresholds

摘录 / EXCERPT: Demailly-Kollár's conjecture. The set $\mathcal{C}(n)$ satisfies the ascending chain condition: every ascending sequence in $\mathcal{C}(n)$ eventually stabilizes.

conjecture | open | MathDB 页面 | 原文线索

V2-064325 | MATHDB-336732

Huybrechts' K3 surfaces Hodge-motive-L-equivalence conjecture

摘录 / EXCERPT: Let S and S' be complex projective K3 surfaces. Write $H^2(S,\mathbb{Q})\cong H^2(S',\mathbb{Q})$ for an isomorphism of rational Hodge structures,...

conjecture | open | MathDB 页面 | 原文线索

V2-064326 | MATHDB-336733

Sharp-constant conjecture for the unbalanced cut problem

摘录 / EXCERPT: Let $G(N,d/N)$ be the Erdős-Rényi random graph, let $\mathbb{R}_d$ denote the infinite d-regular tree, and let $\operatorname{MCUT}_{\alpha}$ be the maximum unbalanced cut value...

conjecture | open | MathDB 页面 | 原文线索

V2-064327 | MATHDB-336739

The dimension-dependent importance of SC performance conjecture for polar codes

摘录 / EXCERPT: Consider polar codes decoded by list decoding with a fixed list size L, and let k denote the code dimension. Dimension-dependent SC-performance conjecture. The SC-performance o...

conjecture | open | MathDB 页面 | 原文线索

V2-064328 | MATHDB-336744

Singularity-model conjecture for non-collapsed steady Ricci solitons

摘录 / EXCERPT: Let the non-collapsed steady solitons be the complete steady solitons described in the paper's main theorem, and let a singularity model mean a blow-up limit arising from Ricci flo...

conjecture | open | MathDB 页面 | 原文线索

V2-064329 | MATHDB-336760

Periodic Gaussian surface-area conjecture for periodic half spaces

摘录 / EXCERPT: Let $\Omega\subseteq\mathbb{R}^{n}$ be a periodic set and let $B\subseteq\mathbb{R}^{n}$ be a periodic half space. Periodic Gaussian surface-area conjecture. The Gaussian surface a...

conjecture | open | MathDB 页面 | 原文线索

V2-064330 | MATHDB-336762

Global real-eigenvalue density conjecture for products of truncated orthogonal matrices

摘录 / EXCERPT: Let N and L tend to infinity with $\alpha=N/(N+L)\in(0,1)$, and let $m\geqslant1$ be fixed. Write $\rho_{(1)}^r(x)$ for the global spectral density of the real eigenvalues and...

conjecture | open | MathDB 页面 | 原文线索

V2-064331 | MATHDB-336763

Asymptotic real-eigenvalue count conjecture for products of truncated orthogonal matrices

摘录 / EXCERPT: Let N and L tend to infinity with $\alpha=N/(N+L)\in(0,1)$, and let $m>0$ be fixed. Let $\mathbb{E}(\#\operatorname{reals})$ denote the expected number of real eigenvalues of t...

conjecture | open | MathDB 页面 | 原文线索

V2-064332 | MATHDB-336784

The K_{6} forbidden-minor conjecture for first Symanzik reducibility

摘录 / EXCERPT: K_{6} forbidden-minor conjecture. The complete graph K_{6} is a forbidden minor for reducibility with respect to the first Symanzik polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-064333 | MATHDB-336788

Maximal differential uniformity distribution conjecture for polynomials over finite fields

摘录 / EXCERPT: Maximal differential uniformity distribution conjecture. There exists $\varepsilon_m>0$ such that, for all sufficiently large n, at least ε_m2^{2n} pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-064334 | MATHDB-336795

Asymptotic classification and uniqueness of one-sided minimisers

摘录 / EXCERPT: Assume the hypotheses of the one-force-one-solution conjecture, and work almost surely. A one-sided minimiser is a curve defined on $(-\infty,t]$ such that compact perturbations fi...

conjecture | open | MathDB 页面 | 原文线索

V2-064335 | MATHDB-336796

Brownian large-scale limit for the one-dimensional global solution

摘录 / EXCERPT: In one spatial dimension, let Φ be the global solution with $a=b=0$, and regard it modulo additive constants so that it has stationary spatial increments. Brownian limit conje...

conjecture | open | MathDB 页面 | 原文线索

V2-064336 | MATHDB-336798

Universal stable fixed points for the renormalisation transformation

摘录 / EXCERPT: Let R be the renormalisation transformation on point fields of concentration points and shocks, and suppose the density decays as $n^{-\alpha}$ for a critical exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-064337 | MATHDB-336805

Ossona de Mendez-Oum-Wood defective colouring conjecture

摘录 / EXCERPT: For a graph class $\mathcal{G}$, let $\chi_{\Delta}(\mathcal{G})$ be the minimum integer k such that, for some integer d, every graph in $\mathcal{G}$ is k-colourable with de...

conjecture | open | MathDB 页面 | 原文线索

V2-064338 | MATHDB-336806

Cluster-critical class characterization of clustered chromatic number

摘录 / EXCERPT: Let $\mathcal{M}$ be a minor-closed class of graphs, and let $k\geqslant 0$ be an integer. A graph class is k-cluster critical if it is obtained as a wedge-product of v copies...

conjecture | open | MathDB 页面 | 原文线索

V2-064339 | MATHDB-336808

Finite-codimension criterion for finite determinacy in positive characteristic

摘录 / EXCERPT: Let K be the ground field, let $R=K[[x_1,\ldots,x_s]]$ with maximal ideal $\mathfrak{m}$, and let $A\in\mathfrak{m}\cdot M_{m,n}$ be a matrix. Let G act on $M_{m,n}$ by the equ...

conjecture | open | MathDB 页面 | 原文线索

V2-064340 | MATHDB-336823

Existential generator conjecture for Erdős-Rényi graph Jacobians

摘录 / EXCERPT: Existential generator conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064341 | MATHDB-336824

Simplicity conjecture for generalized Laguerre polynomial zeros

摘录 / EXCERPT: Let $b1>-1$, let bb be an even partition, and let bc be the partition parameter occurring in the generalized Laguerre polynomial $a9_{bb,bc}^{(b1)}$. The zeros of...

conjecture | open | MathDB 页面 | 原文线索

V2-064342 | MATHDB-336829

Hamiltonicity conjecture for two-connected Cayley graphs generated by three involutions

摘录 / EXCERPT: Let G be a group generated by three involutions a,b,c such that $ab=ba$, and let $\Gamma(G,\{a,b,c\})$ be its Cayley graph. Three-involution conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-064343 | MATHDB-336830

Georgakopoulos-Mohar conjecture on Hamilton circles in planar Cayley graphs

摘录 / EXCERPT: Let Γ be a finitely generated 3-connected planar Cayley graph. A Hamilton circle is a homeomorphic image of S^1 in the Freudenthal compactification of the graph that conta...

conjecture | open | MathDB 页面 | 原文线索

V2-064344 | MATHDB-336833

Ayyer-Klee-Schilling conjecture on linear eigenvalues of promotion matrices

摘录 / EXCERPT: Let P be a poset of size n that is not a down forest, and let M^{P} be its promotion transition matrix. For each element x of P, call an element y a successor of x wh...

conjecture | open | MathDB 页面 | 原文线索

V2-064345 | MATHDB-336834

Conjecture on linear factorization for posets contained in $A_k\oplus A_2$

摘录 / EXCERPT: Let A_i denote an antichain of size i, and let P be a poset whose Hasse diagram is contained in the Hasse diagram of $A_k\oplus A_2$. Let M^{P} be the promotion matrix of...

conjecture | open | MathDB 页面 | 原文线索

V2-064346 | MATHDB-336841

Blue-red decomposition conjecture for 3-connected cubic graphs

摘录 / EXCERPT: Blue-red decomposition conjecture. The vertices of G can be colored blue and red such that the blue subgraph has maximum degree 1 (that is, it consists of a matching and some i...

conjecture | open | MathDB 页面 | 原文线索

V2-064347 | MATHDB-336847

The super edge-magic odd constellation conjecture

摘录 / EXCERPT: Let a constellation be a forest whose components are stars, and call it odd when it consists of an odd number of stars. A labeling of a graph is super edge-magic if it is an edge-m...

conjecture | open | MathDB 页面 | 原文线索

V2-064348 | MATHDB-336848

The duality between geometric thickness and geometric antithickness

摘录 / EXCERPT: For a graph, geometric thickness is the minimum number of plane geometric layers in a straight-line drawing, and geometric antithickness is the minimum number of geometric thrackle...

conjecture | open | MathDB 页面 | 原文线索

V2-064349 | MATHDB-336849

The duality between book thickness and convex antithickness

摘录 / EXCERPT: For a graph, book thickness is the minimum number of pages in a book embedding, and convex antithickness is the minimum number of convex geometric thrackles in a straight-line draw...

conjecture | open | MathDB 页面 | 原文线索

V2-064350 | MATHDB-336851

Confluence from connected dominant weights in type B

摘录 / EXCERPT: Let $Phi=B_n$. Let P be the weight lattice, and let $lambda\in P$ be a connected, dominant weight. Type B connected-weight confluence conjecture. Central-firing is confluent from...

conjecture | open | MathDB 页面 | 原文线索

V2-064351 | MATHDB-336857

Rainbow-partition conjecture for meandric partners

摘录 / EXCERPT: Let n be a positive integer. A meandric partner of a partition $\pi\in\mathrm{NC}(n)$ is a partition $\rho\in\mathrm{NC}(n)$ such that the associated meandric system...

conjecture | open | MathDB 页面 | 原文线索

V2-064352 | MATHDB-336859

Derived-equivalence conjecture for arbitrary weighted projective lines

摘录 / EXCERPT: Derived-equivalence conjecture. Given an arbitrary weighted projective line cX, Assertion holds for $cD=\mathcal{D}^b(cX)$.

conjecture | open | MathDB 页面 | 原文线索

V2-064353 | MATHDB-336860

HRS-tilt characterization conjecture for weighted projective lines

摘录 / EXCERPT: HRS-tilt characterization conjecture. The aisle $cD^{\leq 0}$ contains no nonzero Ext-projective if and only if it is a shift of the HRS-tilt associated with some torsion pair...

conjecture | open | MathDB 页面 | 原文线索

V2-064354 | MATHDB-336863

The bounded-expansion characterization by poset dimension

摘录 / EXCERPT: Let $\mathcal{C}$ be a monotone class of graphs. For each fixed $h\geq 1$, consider posets of height at most h whose cover graphs belong to $\mathcal{C}$. Bounded-expansion chara...

conjecture | open | MathDB 页面 | 原文线索

V2-064355 | MATHDB-336864

Conjecture on the recursive coefficients in Boij-Söderberg decompositions

摘录 / EXCERPT: Recursive-coefficient conjecture. For $\varepsilon<s<\varepsilon+c$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-064356 | MATHDB-336867

Restricted-type endpoint conjecture for orthonormal Schrödinger initial data

摘录 / EXCERPT: Restricted-type endpoint conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-064357 | MATHDB-336868

The E8 character-degree conjecture for the Sylow 5-subgroup

摘录 / EXCERPT: Let $q=5^f$, let $\mathrm E_8(q)$ denote the finite Chevalley group of type E_8 over $\boldsymbol{F}_q$, and let $U(\mathrm E_8(q))$ be its Sylow 5-subgroup. Consider the famil...

conjecture | open | MathDB 页面 | 原文线索

V2-064358 | MATHDB-336877

Andrews et al.'s twin chromatic index conjecture

摘录 / EXCERPT: Andrews et al.'s conjecture. If G is a connected graph of order at least 3 that is not a 5-cycle, then

conjecture | open | MathDB 页面 | 原文线索

V2-064359 | MATHDB-336879

Vanishing of the virtual Alexander polynomial for slice virtual knots

摘录 / EXCERPT: Vanishing conjecture. If K is slice, then

conjecture | open | MathDB 页面 | 原文线索

V2-064360 | MATHDB-336890

Supercuspidality criterion for unitary-group L-packets

摘录 / EXCERPT: Unitary-group supercuspidality conjecture. The packet Π_{ϕ} consists of supercuspidal representations if and only if ϕ is discrete and its restriction to...

conjecture | open | MathDB 页面 | 原文线索

V2-064361 | MATHDB-336902

Existence of Dickson polynomials among the associated polynomials

摘录 / EXCERPT: Existence conjecture. There exists $a\in\mathbb{F}_{p^{nk}}^*$ such that $h_a(x)$ is a Dickson polynomial of degree $n+1$ over $\mathbb{F}_{p^k}$. This conjecture was subsequently...

conjecture | solved | MathDB 页面 | 原文线索

V2-064362 | MATHDB-336915

B3 conjecture on expressing brackets by 123-brackets

摘录 / EXCERPT: Let brackets be the bracket q-series in the source, and let 123-brackets be those whose indices belong to $\{1,2,3\}$. B3 conjecture. Every bracket is a linear combination of...

conjecture | open | MathDB 页面 | 原文线索

V2-064363 | MATHDB-336918

Conjecture on the order of the non-giant part in the second phase transition

摘录 / EXCERPT: Let $S_g(n,m)$ be the random graph with n vertices and m edges embedded on an orientable surface of genus g, and let H_1 denote its largest component. Write ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-064364 | MATHDB-336921

Davis-Taussky conjecture on signatures of units in real cyclotomic fields

摘录 / EXCERPT: Let p and $q=2p+1$ be primes, let $L=\mathbb Q(\zeta_q+\zeta_q^{-1})$, and let C_L^+ denote the narrow class group of L, with C_L^2 its subgroup of squares. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-064365 | MATHDB-336924

Dye-Kaestner-Kauffman conjecture on slice genera of classical knots as virtual knots

摘录 / EXCERPT: Dye-Kaestner-Kauffman conjecture. For every classical knot K,

conjecture | open | MathDB 页面 | 原文线索

V2-064366 | MATHDB-336932

Boundedness conjecture for singular integrals on intrinsic Lipschitz graphs in the Heisenberg group

摘录 / EXCERPT: Let $\mathbb{W} \subset \mathbb{H}$ be a vertical subgroup with complementary subgroup $\mathbb{V}$, and let $\phi \colon \mathbb{W} \to \mathbb{V}$ be an intrinsic Lipschitz funct...

conjecture | open | MathDB 页面 | 原文线索

V2-064367 | MATHDB-336941

Hernandez-Sethuraman theta-quotient conjecture

摘录 / EXCERPT: Let $a,b\in\mathbb R_+$, and define the Jacobi theta function ... for $y\in\mathbb R_+$. Hernandez-Sethuraman conjecture. The function ... attains its unique maximum at $1/\sqrt{ab}$....

conjecture | open | MathDB 页面 | 原文线索

V2-064368 | MATHDB-336948

Arkani-Hamed-Trnka triangulation conjecture for the amplituhedron

摘录 / EXCERPT: Let $Z\in\operatorname{Mat}_{k+4,n}^{>0}$, and let $\mathcal{C}_{n,k,4}$ be the collection of BCFW cells in $\operatorname{Gr}_{k,n}^{\geq 0}$. Arkani-Hamed-Trnka triangulation con...

conjecture | open | MathDB 页面 | 原文线索

V2-064369 | MATHDB-336949

Continuum conjecture for unitarily non-equivalent MHDRs of bivariate polynomials

摘录 / EXCERPT: Continuum conjecture. There is a continuum of unitarily non-equivalent MHDRs for a bivariate polynomial of degree d.

conjecture | open | MathDB 页面 | 原文线索

V2-064370 | MATHDB-336954

The Day convolution compatibility conjecture for admissible double -categories

摘录 / EXCERPT: Let $\mathcal{X} \to \Delta^{\operatorname{op}}$ be an admissible double ∞-category and let $\mathcal{V}$ be a presentably monoidal ∞-category. Corollary 3.?? equips...

conjecture | open | MathDB 页面 | 原文线索

V2-064371 | MATHDB-336963

Fixed-point conjecture for Calogero-Moser spaces

摘录 / EXCERPT: Let W act as a complex reflection group on V, let τ be a finite-order element of the normalizer of W in $\operatorname{GL}_{\mathbb C}(V)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064372 | MATHDB-336965

Large graphs with bounded independence number have swap sets

摘录 / EXCERPT: Let G be a graph, let $\alpha(G)$ denote its independence number, and let a swap set mean the graph structure defined in the paper that supports the relevant disjoint dominating...

conjecture | open | MathDB 页面 | 原文线索

V2-064373 | MATHDB-336966

Drew-Johnson-Loewy conjecture on the CP-rank of completely positive matrices

摘录 / EXCERPT: Let A be a real completely positive matrix of order n, meaning that $A=BB^{T}$ for some nonnegative real matrix B. The CP-rank of A, denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-064374 | MATHDB-336968

Total nonnegativity of the flipped excedance matrix

摘录 / EXCERPT: Let $M(m,n)$ be the excedance matrix, and let $N(m,n)$ be obtained by reversing the order of its rows: its ith row is the $(m-i+1)$st row of $M(m,n)$. A matrix is totally nonnega...

conjecture | open | MathDB 页面 | 原文线索

V2-064375 | MATHDB-336988

Mirror conjecture for lattice-polarized weak del Pezzo pairs

摘录 / EXCERPT: Mirror conjecture. The mirror to (X,C) is the open set $Y\setminus D$ in an $\check{L}$-polarized rational elliptic surface (Y,D) of type d.

conjecture | open | MathDB 页面 | 原文线索

V2-064376 | MATHDB-336989

Period-map conjecture for lattice-polarized weak del Pezzo pairs

摘录 / EXCERPT: Period-map conjecture. There is a period map from the moduli space of such marked pairs to $\operatorname{Hom}(f_d^{\perp}/R_L,C)$, and this period map is an isomorphism over a den...

conjecture | open | MathDB 页面 | 原文线索

V2-064377 | MATHDB-336990

Analytic mirror conjecture for type II K3 degenerations

摘录 / EXCERPT: Analytic mirror conjecture. There exists an analytic open set in W isomorphic to an analytic open set $Y\setminus Z_D$.

conjecture | open | MathDB 页面 | 原文线索

V2-064378 | MATHDB-336991

Mirror-family conjecture for nef-cone faces

摘录 / EXCERPT: Mirror-family conjecture. For each such face F_i, there exists a codimension-one subfamily $\mathcal Y_i$ of $\mathcal Y$ whose members are $\langle\alpha_i\rangle$-polarized rat...

conjecture | open | MathDB 页面 | 原文线索

V2-064379 | MATHDB-336998

Underlying-field security conjecture for homomorphic encryption of classical matrix groups

摘录 / EXCERPT: Let G be a classical matrix group over a finite field, and consider homomorphic encryption of G and of its underlying field. Underlying-field security conjecture. Homomorphic e...

conjecture | open | MathDB 页面 | 原文线索

V2-064380 | MATHDB-337003

Oberdieck-Pixton quasi-Jacobi form conjecture for relative elliptic fibrations

摘录 / EXCERPT: Let $\pi:X\to B$ be an elliptic fibration with section and integral fibers, let $D\subset X$ be a nonsingular divisor restricting to an elliptic fibration $\pi_D:D\to A$ over a non...

conjecture | open | MathDB 页面 | 原文线索

V2-064381 | MATHDB-337004

Oberdieck-Pixton holomorphic anomaly equation for elliptic fibrations

摘录 / EXCERPT: Let $\pi:X\to B$ be an elliptic fibration, let $\mathsf{k}\in H_2(B,\mathbb{Z})$, and let ${\mathcal C}^{\pi}_{g,\mathsf{k}}(\gamma_1,\ldots,\gamma_n)$ be the associated relative G...

conjecture | open | MathDB 页面 | 原文线索

V2-064382 | MATHDB-337006

The multiplication-by-p conjecture for the Higgs-de Rham self-map

摘录 / EXCERPT: Multiplication-by-p conjecture. The diagram commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-064383 | MATHDB-337022

Family D conjecture for $D_{-1,2r}(2n)$

摘录 / EXCERPT: Let μ be an indeterminate, let $n\geqslant1$ and $r\geqslant0$ be integers, and define the rising factorial $(a)_k=a(a+1)\cdots(a+k-1)$. Let $D_{-1,2r}(2n)$ be the determinant...

conjecture | open | MathDB 页面 | 原文线索

V2-064384 | MATHDB-337031

Integrability conjecture for torsion fibers of the spherical elliptic DAHA

摘录 / EXCERPT: Torsion-fiber integrability conjecture. This fiber has $n+1$ global sections, all of which commute. The preceding proposition supplies a global section with dominant weight...

conjecture | open | MathDB 页面 | 原文线索

V2-064385 | MATHDB-337032

Full equivalence class criterion for ribbons

摘录 / EXCERPT: Let $\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_m)$ be a ribbon with $\alpha_1\geq\alpha_2\geq\cdots\geq\alpha_m$, where each $\alpha_i\geq2$ and $m\geq3$. Let N_j denote the quanti...

conjecture | open | MathDB 页面 | 原文线索

V2-064386 | MATHDB-337033

Exactness conjecture for the continuous-time zero-delay rate bound

摘录 / EXCERPT: Let $R^*(D)$ denote the infimum mutual-information rate over generalized causal reproductions GCR whose long-term distortion is at most D, and let $R(D)$ denote the ra...

conjecture | open | MathDB 页面 | 原文线索

V2-064387 | MATHDB-337034

The dimension quotient conjecture for dihedral quandle rings

摘录 / EXCERPT: Let R_n be the dihedral quandle and let R be a ring. For the quandle ring $R[\operatorname{R}_n]$, let $\Delta(\operatorname{R}_n)$ denote its augmentation ideal...

conjecture | open | MathDB 页面 | 原文线索

V2-064388 | MATHDB-337035

Rodney's 2-plex conjecture for latin squares

摘录 / EXCERPT: Let a latin square be a latin hypercube of dimension 2 and order n. A 2-plex is a selection of $2n$ different indices in which each row, column, and symbol is represented pre...

conjecture | open | MathDB 页面 | 原文线索

V2-064389 | MATHDB-337038

The Mallows-Sloane-type bounds for invariant polynomials and formal weight enumerators at q=3/2

摘录 / EXCERPT: Mallows-Sloane-type conjecture at q=3/2. (i) Every $\sigma_{3/2}$-invariant polynomial of this form in $R_{3/2}$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-064390 | MATHDB-337047

The generalized oscillating promotion conjecture for sl(n) web rotation

摘录 / EXCERPT: Assume there is a bijection between generalized oscillating tableaux of length k with n parts, whose final component is $(m,\ldots,m)$ for some $m\in\mathbb{Z}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-064391 | MATHDB-337048

Conjecture that the Sommerville and Goldberg lists classify face-to-face tetrahedral tiles

摘录 / EXCERPT: Sommerville-Goldberg classification conjecture. The Sommerville and Goldberg lists of face-to-face tetrahedral tiles are complete. In particular, every tetrahedral tile has at leas...

conjecture | open | MathDB 页面 | 原文线索

V2-064392 | MATHDB-337049

Conjecture that area-minimizing n-hedral tiles are convex

摘录 / EXCERPT: Convexity conjecture. For any n, the least-area n-hedral tile is convex.

conjecture | open | MathDB 页面 | 原文线索

V2-064393 | MATHDB-337050

Exact-solution-count conjecture for singular Liouville equations on rectangular tori

摘录 / EXCERPT: Exact-solution-count conjecture. The equation possesses exactly n solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-064394 | MATHDB-337051

Nonexistence conjecture for four-source Liouville equations on rectangular tori

摘录 / EXCERPT: Four-source nonexistence conjecture. The equation has no solution for any $\tau\in i\mathbb{R}_{>0}$ if and only if (n_0,n_1,n_2,n_3) satisfies neither of these two conditions.

conjecture | open | MathDB 页面 | 原文线索

V2-064395 | MATHDB-337056

The SLIN conjecture for special polynomial automorphism groups

摘录 / EXCERPT: Let ${\rm SA}_n(\mathbb{K})$ denote the group of special polynomial automorphisms, and let ${\rm SLIN}_n(\mathbb{K})$ denote the corresponding subgroup satisfying the special linea...

conjecture | open | MathDB 页面 | 原文线索

V2-064396 | MATHDB-337057

Finite excluded-minor characterization for 3-connected constraint graphs with boundedly many islands

摘录 / EXCERPT: Finite excluded-minor conjecture. The class of 3-connected constraint graphs with at most k islands is characterised by a finite list of excluded constraint minors.

conjecture | open | MathDB 页面 | 原文线索

V2-064397 | MATHDB-337065

Nefness criterion for divisors on the iterated blow-up $\mathcal{X}_{n+2,(n-2)}$

摘录 / EXCERPT: Nefness criterion. The divisor D is nef if and only if these F-nef inequalities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-064398 | MATHDB-337070

Exactness conjecture for Hom-complex connectivity of $K_2\times K_n$

摘录 / EXCERPT: For a bipartite graph of the form $K_2\times K_n$, let d denote its maximal degree and let K_m be a complete graph. The lower bound for the connectivity of...

conjecture | open | MathDB 页面 | 原文线索

V2-064399 | MATHDB-337090

The simple ordering conjecture for zero-sum subsets of abelian groups

摘录 / EXCERPT: Let $(G,+)$ be an abelian group. Let A be a finite subset of $G\setminus\{0\}$ such that no 2-subset $\{x,-x\}$ is contained in A, and suppose that ... A simple ordering of A...

conjecture | open | MathDB 页面 | 原文线索

V2-064400 | MATHDB-337093

Localization-number bound for unit disk graphs

摘录 / EXCERPT: Let G be a unit disk graph, let $\zeta(G)$ denote its localization number, and let $\omega(G)$ denote its clique number. Unit-disk localization conjecture. There is a function...

conjecture | open | MathDB 页面 | 原文线索

V2-064401 | MATHDB-337098

Bergeron's positivity and unimodality conjecture for q-binomial differences

摘录 / EXCERPT: Let a,b,c,d be positive integers such that a is the smallest and $ad=bc$. The q-binomial coefficient is ... Thus $\binom{m+n}{m}_q$ is a symmetric polynomial in q. Bergeron's...

conjecture | open | MathDB 页面 | 原文线索

V2-064402 | MATHDB-337100

Kawamata's conjecture on global generation of direct images of canonical bundles

摘录 / EXCERPT: Kawamata's conjecture. The locally free sheaf

conjecture | open | MathDB 页面 | 原文线索

V2-064403 | MATHDB-337104

Reconstruction of noncommutative Grothendieck-Witt motives without replacing objects

摘录 / EXCERPT: Let NcS be the category whose objects are dg-categories without additional structures, and let $\widetilde{NcS}(S)$ and $DM^{\mathcal{GW}}_{nc}(k)$ be the categories constructed...

open_problem | open | MathDB 页面 | 原文线索

V2-064404 | MATHDB-337105

Equivalence with the motive category built from paired dg-functors

摘录 / EXCERPT: Let $NcS^{\cdot\cdot}$ be the category whose objects are dg-categories and whose morphisms are pairs of dg-functors F,G equipped with a natural equivalence $F\simeq G$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064405 | MATHDB-337116

The Darmon-point rationality, reciprocity and Gross-Zagier conjecture

摘录 / EXCERPT: Let K_S^{ab} be the maximal abelian extension of K unramified and totally split at the primes in S, let $\mathcal G=\operatorname{Gal}(K_S^{\mathrm{ab}}/K)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064406 | MATHDB-337122

Yamaguchi-Yau genus-two mirror conjecture for the quintic threefold

摘录 / EXCERPT: Yamaguchi-Yau genus-two mirror conjecture. The function $F^{GW}_2(Q)$ is given by

conjecture | open | MathDB 页面 | 原文线索

V2-064407 | MATHDB-337130

The chart simplification conjecture for branched covering surface-knots

摘录 / EXCERPT: Chart simplification conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064408 | MATHDB-337134

Bruhat covering property for twisted involutions in Coxeter systems

摘录 / EXCERPT: Bruhat covering property. If $y\in I_*$ is a twisted involution in an arbitrary Coxeter group and $t\in T$, then there exists at most one $z\in I_*$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064409 | MATHDB-337143

Stability conjecture for independence polynomials of trees

摘录 / EXCERPT: Let G be a tree, and let $i(G,x)$ denote its independence polynomial. A polynomial is stable here when all of its roots lie in the left half-plane. Tree stability conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-064410 | MATHDB-337145

Major-index Wilf classes are generated by inversion

摘录 / EXCERPT: Let $M\mathcal I_n(\pi;q)$ denote the major-index generating function over involutions in $\mathfrak S_n$ avoiding a pattern π, and let $[\pi]_{M\mathcal I}$ denote the corresp...

conjecture | solved | MathDB 页面 | 原文线索

V2-064411 | MATHDB-337146

An involution-restricted major-index symmetry for layered pattern pairs

摘录 / EXCERPT: Let $\mathfrak i_j$ denote the increasing permutation of length j, let $\mathfrak d_j$ denote the decreasing permutation of length j, and let $12[\alpha,\beta]$ denote the dire...

conjecture | open | MathDB 页面 | 原文线索

V2-064412 | MATHDB-337151

The new square conjecture for decorated Dyck paths

摘录 / EXCERPT: New square conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-064413 | MATHDB-337169

Conjectural logarithmic identities for supersingular lambda invariants

摘录 / EXCERPT: Assume $p\geq5$. Let L be the set of supersingular lambda invariants in $\mathbf{F}_{N^2}$, let H' be the derivative of the associated polynomial H, and extend $\log$ to a su...

conjecture | open | MathDB 页面 | 原文线索

V2-064414 | MATHDB-337170

The analogous logarithmic identities for supersingular lambda invariants at \(p=3\)

摘录 / EXCERPT: Assume $p=3$. Let L be the set of supersingular lambda invariants in $\mathbf{F}_{N^2}$, let H' be the derivative of the associated polynomial H, and extend and lift $\log$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-064415 | MATHDB-337182

Ressayre's geometric fundamental-class conjecture for the Belkale-Kumar product

摘录 / EXCERPT: Ressayre's conjecture. If $v^\vee\lessdot u$, then

conjecture | open | MathDB 页面 | 原文线索

V2-064416 | MATHDB-337184

Ressayre's root-support conjecture for the Belkale-Kumar product

摘录 / EXCERPT: Ressayre's root-support conjecture. The product

conjecture | open | MathDB 页面 | 原文线索

V2-064417 | MATHDB-337185

Ressayre's translated fundamental-class conjecture for the Belkale-Kumar product

摘录 / EXCERPT: Ressayre's translated fundamental-class conjecture. If $u,v\in W^P$ satisfy $v^\vee\lessdot u$, then

conjecture | open | MathDB 页面 | 原文线索

V2-064418 | MATHDB-337191

Duke-Ellenberg-Venkatesh discriminant multiplicity conjecture

摘录 / EXCERPT: Fix $n\geq 2$. Property $\mathbf{D}_n(\varpi)$ means that, for every fixed integer $D>1$ and every $\epsilon>0$, the number of degree-n fields $K/\mathbb{Q}$ with absolute discri...

conjecture | open | MathDB 页面 | 原文线索

V2-064419 | MATHDB-337192

The \ell-torsion conjecture for number-field class groups

摘录 / EXCERPT: ell-torsion conjecture. For every integer $\ell\geq 1$ and every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-064420 | MATHDB-337200

The one-dimensional reduction conjecture for a magnetic barrier

摘录 / EXCERPT: One-dimensional reduction conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-064421 | MATHDB-337201

The density of hyperbolicity conjecture

摘录 / EXCERPT: Let ${\cal F}$ be one of the following families: unicritical polynomials of degree d, rational maps of degree d, the exponential family, or a family of entire maps with finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-064422 | MATHDB-337203

The quasiconformal rigidity conjecture

摘录 / EXCERPT: Let ${\cal F}$ be a family of maps, and let $f\in{\cal F}$ be non-hyperbolic. The map f is quasiconformally rigid when it is not quasiconformally conjugate to any other map in a...

conjecture | open | MathDB 页面 | 原文线索

V2-064423 | MATHDB-337204

The no invariant line fields conjecture

摘录 / EXCERPT: Let f be a polynomial, an entire transcendental function with finitely many singular values, or a rational map that is not double covered by an integral torus endomorphism. An in...

conjecture | open | MathDB 页面 | 原文线索

V2-064424 | MATHDB-337207

The triviality of fibers conjecture for unicritical and exponential families

摘录 / EXCERPT: Consider the family of unicritical polynomials of degree $d\geq2$ or the exponential family. A parameter fiber is trivial when it contains only the parameter itself; fibers are def...

conjecture | open | MathDB 页面 | 原文线索

V2-064425 | MATHDB-337208

Toric degeneration conjecture for Mori dream spaces

摘录 / EXCERPT: Let X be a Mori dream space, and let $Y_\bullet$ be a flag for which the global semigroup $\Gamma_{Y_\bullet}(X)$ is defined. The Cox ring of X is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-064426 | MATHDB-337209

Tilting-type equivalence for the remaining stratified Atiyah flops

摘录 / EXCERPT: Let $\operatorname{Gr}(r,N)$ be the Grassmannian appearing in the stratified Atiyah flop, and suppose that $2r=N-1$ or $2r=N$. Expected tilting-type equivalence. There exists a til...

conjecture | open | MathDB 页面 | 原文线索

V2-064427 | MATHDB-337211

Tree-model multiplicity conjecture

摘录 / EXCERPT: Let $\mathcal{M}=(G,In,Out,Leak)$ be a linear compartment model with input, output, and leak in compartment 1 only, so that $In=Out=Leak=\{1\}$, and suppose that G is a bidirec...

conjecture | open | MathDB 页面 | 原文线索

V2-064428 | MATHDB-337212

Collins-Trenk conjecture on the distinguishing chromatic number of graphs

摘录 / EXCERPT: Let G be a connected graph with girth at least 5, let $Delta$ denote its maximum degree, and let $chi_D(G)$ denote its distinguishing chromatic number. Collins-Trenk conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-064429 | MATHDB-337220

Grünbaum's interpolation conjecture for polytope skeleta

摘录 / EXCERPT: Let $\mathcal C$ be a k-dimensional complex that is the k-skeleton of both a d-polytope and a $d”$-polytope, where $d\le d”$. Grünbaum's conjecture. For every d' with...

conjecture | open | MathDB 页面 | 原文线索

V2-064430 | MATHDB-337234

Moment and asymptotic conjecture for admissible finite 2-groups

摘录 / EXCERPT: Let G be a finite 2-group and let $H\leq G$ have index 2, with (G,H) admissible. Let $f(K)$ be the associated unramified-extension counting function, let c be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-064431 | MATHDB-337235

Refined moment and asymptotic conjecture for admissible 2-group tuples

摘录 / EXCERPT: Let (G_i,H_i,T_i), for $1\leq i\leq k$, be admissible tuples of finite 2-groups and generating sets. Let $f_{T_i}(K)$ be the associated refined counting functions, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064432 | MATHDB-337240

The Legendrian Whitehead double conjecture

摘录 / EXCERPT: Let Λ be a Legendrian knot, and let W denote the Legendrian Whitehead double pattern. Write ${\Sigma(\Lambda,W+\operatorname{tw}_2)}$ for the untwisted Whitehead double o...

conjecture | open | MathDB 页面 | 原文线索

V2-064433 | MATHDB-337244

Bhargava's local-independence conjecture for quadratic class groups

摘录 / EXCERPT: Bhargava's local-independence conjecture. In any of these families, the distribution of the respective class groups is unchanged when restricting to those K having specified comp...

conjecture | open | MathDB 页面 | 原文线索

V2-064434 | MATHDB-337245

Conjecture on uniform distribution of split-place differences in class groups

摘录 / EXCERPT: Uniform-distribution conjecture. As such quadratic extensions are ordered by discriminant, v_1-v_2 is distributed uniformly in Cl_K. More precisely, in the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-064435 | MATHDB-337246

Beveridge-Bohman-Frieze-Pikhurko upper-bound conjecture for the game chromatic index

摘录 / EXCERPT: Let G be a graph, let $Delta(G)$ denote its maximum degree, and let $\chi'_g(G)$ denote its game chromatic index, the smallest number of colors for which Maker has a winning stra...

conjecture | open | MathDB 页面 | 原文线索

V2-064436 | MATHDB-337249

The conjectural two-dimensional Gaussian free field fluctuation theorem for random height functions

摘录 / EXCERPT: Conjectural two-dimensional fluctuation theorem. The centered random height function, multiplied by $\sqrt{\pi}$, converges in distribution to the pullback under Ω of the Ga...

conjecture | open | MathDB 页面 | 原文线索

V2-064437 | MATHDB-337254

Independence conjecture for generalized Gopakumar-Vafa invariants

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau threefold, let $v=(\beta,m)$ with $\beta eq 0$, let $\sigma\in U(X)$, and let $\gamma\in\operatorname{Chow}_X(\beta)$. Suppose that the mo...

conjecture | open | MathDB 页面 | 原文线索

V2-064438 | MATHDB-337259

The pointwise base-point-freeness conjecture for adjoint bundles

摘录 / EXCERPT: The pointwise base-point-freeness conjecture. The adjoint line bundle K_Y+L is base point free at y. The source presents this as a known generalization of Fujita's base point f...

conjecture | open | MathDB 页面 | 原文线索

V2-064439 | MATHDB-337262

Voevodsky's interpretation lifting conjecture for C-systems

摘录 / EXCERPT: Let ${\cal C}$ be a category, let CC be a C-system, and let $M:CC\rightarrow {\cal C}$ be a functor such that $M(pt_{CC})$ is a final object of ${\cal C}$ and M maps distinguis...

conjecture | open | MathDB 页面 | 原文线索

V2-064440 | MATHDB-337263

EL-labeling criterion for factorization posets

摘录 / EXCERPT: Let G be a group generated as a monoid by a conjugation-closed subset A, let $\mathfrak{c}\in G$, and let $A_{\mathfrak{c}}$ be the generators appearing in reduced A-factoriz...

conjecture | open | MathDB 页面 | 原文线索

V2-064441 | MATHDB-337267

The embedding conjecture for normalized univalent maps

摘录 / EXCERPT: Embedding conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064442 | MATHDB-337269

The extension conjecture for normalized univalent maps

摘录 / EXCERPT: Extension conjecture. Then $f\in{\mathcal{S}}^1$, that is, f embeds into a normalized Loewner chain. The source states that this is equivalent to the generalized Andérsen-Lempert...

conjecture | open | MathDB 页面 | 原文线索

V2-064443 | MATHDB-337274

Lang-Trotter fixed-Frobenius-field conjecture for elliptic curves

摘录 / EXCERPT: Lang-Trotter fixed-Frobenius-field conjecture. There exists a constant $C(E,F)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064444 | MATHDB-337282

Ayoub's Hodge realisation conjecture for the limit mixed motive

摘录 / EXCERPT: Let $\mathcal{Z}^{\vee}\in\textbf{DM}_{\text{gm}}(\mathbb{Q},\mathbb{Q})$ be the dual limit mixed motive, and let $\mathbf{Z}^{\bullet}$ be the complex in...

conjecture | open | MathDB 页面 | 原文线索

V2-064445 | MATHDB-337284

Kalai's intersecting triangulations conjecture for polygons

摘录 / EXCERPT: Kalai's conjecture. Every intersecting family $\mathcal F$ has at most as many elements as there are triangulations of the $(n-1)$-gon.

conjecture | open | MathDB 页面 | 原文线索

V2-064446 | MATHDB-337285

The pure-EKR conjecture for flag simplicial complexes without boundary

摘录 / EXCERPT: Pure-EKR conjecture. Every pure, flag simplicial complex without boundary is pure-EKR.

conjecture | open | MathDB 页面 | 原文线索

V2-064447 | MATHDB-337293

Odd-degree tree set-sequentiality conjecture

摘录 / EXCERPT: A graph is set-sequential if its vertices can be labeled by distinct nonzero vectors in a vector space over $\a0\mathbb{F}_2$ so that labeling each edge by the sum of its endpoint...

conjecture | open | MathDB 页面 | 原文线索

V2-064448 | MATHDB-337294

Balister-Győri-Schelp pairing conjecture for set-sequential trees

摘录 / EXCERPT: Let $n\geq 2$, and let $v_1,\dots,v_{2^{n-1}}$ be nonzero vectors in $\mathbb{F}_2^n$ satisfying ... A pairing partition is a partition of $\mathbb{F}_2^n$ into pairs (p_i,q_i) ind...

conjecture | open | MathDB 页面 | 原文线索

V2-064449 | MATHDB-337301

Simple-reflection descent conjecture for inverse Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let W be the Weyl group of a simply-laced root system Φ, with Bruhat order $\leqslant$, and let $Q_{u,v}$ denote the inverse Kazhdan-Lusztig polynomial. Assume $u<v$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064450 | MATHDB-337310

The density of hyperbolicity conjecture for rational maps

摘录 / EXCERPT: Let Rat_d be the space of rational maps of degree d, and call a rational map hyperbolic when all its critical points are attracted to attracting cycles. Density...

conjecture | open | MathDB 页面 | 原文线索

V2-064451 | MATHDB-337311

The J-stability conjecture for rational maps

摘录 / EXCERPT: Let f be a rational map of degree d on the Riemann sphere. It is J-stable if its Julia-set dynamics is stable under perturbation, as in structural stability on a neighborhood o...

conjecture | open | MathDB 页面 | 原文线索

V2-064452 | MATHDB-337312

The Mandelbrot homeomorphism conjecture for modular matings

摘录 / EXCERPT: Mandelbrot homeomorphism conjecture. The Mandelbrot set M is homeomorphic to M_{Γ}. The family $\mathcal{F}_a$ has been shown to mate parabolic quadratic maps with...

conjecture | open | MathDB 页面 | 原文线索

V2-064453 | MATHDB-337316

Hertling's strong-marking monodromy conjecture

摘录 / EXCERPT: Let f_0 be an isolated hypersurface singularity, and let $G^{smar}(f_0)$ be the subgroup of the Milnor-lattice automorphism group associated with the strong-marked moduli space....

conjecture | open | MathDB 页面 | 原文线索

V2-064454 | MATHDB-337317

Global Torelli conjecture for marked singularities

摘录 / EXCERPT: Fix a reference isolated hypersurface singularity f_0. Let M^{mar}_μ be the moduli space of marked singularities, D_{BL} the classifying space of marked Brieskorn lattices,...

conjecture | open | MathDB 页面 | 原文线索

V2-064455 | MATHDB-337319

Hoffman's coefficient formula conjecture for Louchard's asymptotic series

摘录 / EXCERPT: Hoffman's coefficient formula conjecture. This sequence gives every coefficient I_n in terms of alternating multiple zeta values. The formula is motivated by the explicitly compu...

conjecture | open | MathDB 页面 | 原文线索

V2-064456 | MATHDB-337320

Hoffman's rational zeta-value conjecture for Louchard's coefficients

摘录 / EXCERPT: Let I_n denote the coefficients in Louchard's asymptotic expansion, and let $\zeta(i)$ be the ordinary Riemann zeta values. Hoffman's rational zeta-value conjecture. For each n...

conjecture | open | MathDB 页面 | 原文线索

V2-064457 | MATHDB-337334

321-avoidance conjecture for shortest universal permutations

摘录 / EXCERPT: 321-avoidance conjecture. For every n, at least one shortest permutation that is n-universal for the class of 321-avoiding permutations avoids 321 itself.

conjecture | open | MathDB 页面 | 原文线索

V2-064458 | MATHDB-337337

Alkauskas's divisibility conjecture for binary partitions

摘录 / EXCERPT: Let $b_{2}(n)$ denote the number of binary partitions of n. Alkauskas's conjecture. For every $h\in\mathbb{N}$ with $8\nmid h$, there exist infinitely many n such that h divi...

conjecture | open | MathDB 页面 | 原文线索

V2-064459 | MATHDB-337338

Divisibility conjecture for the generalized recurrence sequence b_T

摘录 / EXCERPT: Let $L=(1,-1)$ and $R=(1)$, and let K be the parameter appearing in the definition of the generalized sequence b_T. Write $\operatorname{\underline{deg}}K$ for its indicated d...

conjecture | open | MathDB 页面 | 原文线索

V2-064460 | MATHDB-337339

Infinitely many divisibility instances for the generalized recurrence sequence b_T

摘录 / EXCERPT: Let $L=(1,-1)$, let K be the parameter appearing in the definition of the generalized sequence b_T, and let $h\in\mathbb{N}$. Infinitely-many-instances conjecture for b_T. If...

conjecture | open | MathDB 页面 | 原文线索

V2-064461 | MATHDB-337341

The standard-form negativity conjecture for blow-ups of the plane

摘录 / EXCERPT: Standard-form negativity conjecture. If $s\geq1$ and $D^{2}<0$, then D is not effective. This is presented as a consequence expected from the SHGH conjecture and concerns the eff...

conjecture | open | MathDB 页面 | 原文线索

V2-064462 | MATHDB-337347

Michel-Venkatesh mixing conjecture for joint CM packets

摘录 / EXCERPT: Mixing conjecture. If $\mathfrak{N}_i\to_{i\to\infty}\infty$, then μ_i^joint converges weak-$*$ to $\mathrm{m}_Y\times\mathrm{m}_Y$.

conjecture | open | MathDB 页面 | 原文线索

V2-064463 | MATHDB-337361

The closed extended r-spin potential conjecture for the Gelfand-Dickey wave function

摘录 / EXCERPT: Closed extended r-spin potential conjecture. For any $g\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-064464 | MATHDB-337363

The basic rank conjecture for Euler systems

摘录 / EXCERPT: Let T be a p-adic representation in the paper's setting, let $r(T)$ be defined from $d_-(T)$ and the de Rham filtration, and call T r-critical when $r(T)=r$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064465 | MATHDB-337364

The Panchishkin condition conjecture for Euler systems in the p-direction

摘录 / EXCERPT: Let T be r-critical. For every place $v\mid p$, a Panchishkin subrepresentation is a G_{K_v}-stable subspace $V_v^+\subseteq V$ whose Hodge-Tate weights are all at least 1,...

conjecture | open | MathDB 页面 | 原文线索

V2-064466 | MATHDB-337365

The twisting-interpolation conjecture for Euler systems

摘录 / EXCERPT: Let T be a representation, let $T_v^+\subseteq T|_{G_{K_v}}$ be chosen local subrepresentations for all $v\mid p$, and define ... Let $\Sigma(\mathcal{P})$ be the characters χ...

conjecture | open | MathDB 页面 | 原文线索

V2-064467 | MATHDB-337369

Asymptotic independence of descent sets and being an n-cycle

摘录 / EXCERPT: For each n, let $[n-1]=\{1,\ldots,n-1\}$, and let $\beta_n(I)$ denote the number of permutations in S_n with descent set $I\subseteq[n-1]$, while $\beta^{\mathrm{cyc}}_n(I)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-064468 | MATHDB-337373

The chord-diagram summation identity for Dodgson polynomials

摘录 / EXCERPT: Let $D_0\in \mathcal{D}^n_0$ with $n\in \mathbb{N}^{\ell}$ and $N=\sum n_i$. For the chord diagrams D in $\mathcal{D}^n_N$, let $s(D)$ be the number of chords and let E_D^0 den...

conjecture | open | MathDB 页面 | 原文线索

V2-064469 | MATHDB-337378

Okawa-Uehara conjecture on exceptional sheaves on weak del Pezzo surfaces

摘录 / EXCERPT: Okawa-Uehara conjecture. There exists an auto-equivalence $\Phi aut D(X)$ such that $\Phi(\mathcal{E})$ is an exceptional vector bundle or a line bundle on a (-1)-curve on X.

conjecture | open | MathDB 页面 | 原文线索

V2-064470 | MATHDB-337403

The planar-minor Erdős-Pósa conjecture with a tight logarithmic bound

摘录 / EXCERPT: The planar-minor Erdős-Pósa conjecture. There is a constant depending on H such that

conjecture | open | MathDB 页面 | 原文线索

V2-064471 | MATHDB-337404

The treewidth packing conjecture with a logarithmic bound

摘录 / EXCERPT: Treewidth packing conjecture. There is a function $f:\mathbb{N}\to\mathbb{N}$ such that every graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-064472 | MATHDB-337405

The uniform Erdős-Pósa conjecture for wheel models

摘录 / EXCERPT: Uniform wheel-model Erdős-Pósa conjecture. There are a constant $c>0$ and a function $g:\mathbb{N}\to\mathbb{N}$ such that, for every integer $t\geq 3$, wheel models in G have th...

conjecture | open | MathDB 页面 | 原文线索

V2-064473 | MATHDB-337411

Boundary universality conjecture for planar orthogonal polynomial density

摘录 / EXCERPT: Let Q be the planar potential defining the exponentially varying measure, let $\mathcal{S}_1$ be its droplet, and let $z_0\in\partial\mathcal{S}_1$ be a point where the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-064474 | MATHDB-337414

Stability criterion for exceptional post-critically finite quadratic polynomials

摘录 / EXCERPT: Let S be the set given in Proposition 3.8. For $m\in S$, define ... Stability conjecture. For every $m\in S$, the polynomial f is stable, meaning that every iterate f^n is irre...

conjecture | open | MathDB 页面 | 原文线索

V2-064475 | MATHDB-337415

Stability criterion for post-critically finite quadratic polynomials

摘录 / EXCERPT: Let K be a field of characteristic different from 2, and let $f(x)\in K[x]$ be a monic, post-critically finite quadratic polynomial. Let o_f denote the period of the post-cri...

conjecture | open | MathDB 页面 | 原文线索

V2-064476 | MATHDB-337417

Weyl-model asymptotic conjecture for zeros of random harmonic polynomials

摘录 / EXCERPT: Let $H_{n,m}(z)=p(z)+q(\overline{z})$ be a random harmonic polynomial in the Weyl model, with p and q analytic complex polynomials of degrees n and m, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064477 | MATHDB-337433

The KD hypothesis for quotient-type pairs

摘录 / EXCERPT: KD hypothesis. If $(X,B)\sim_K(Y,C)$, then there is an equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-064478 | MATHDB-337450

The monodromy group conjecture for toric surfaces

摘录 / EXCERPT: Let X be a smooth toric surface and let $\mathcal L$ be an ample line bundle on X whose generic fiber is not hyperelliptic. Assume $r>1$ or $g(\mathcal L)\geq 5$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064479 | MATHDB-337462

Weighted-resolution conjecture for the Jablan polynomial

摘录 / EXCERPT: A pseudoknot K has a WeRe set ... where the K_j are classical resolutions with weights α_j, and let Δ_J denote the Jablan polynomial. Weighted-resolution conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-064480 | MATHDB-337464

The conjectural dichotomy for normal subgroups of mapping class groups

摘录 / EXCERPT: Let N be a normal subgroup of either the orientation-preserving mapping class group $\operatorname{Mod}(S_g)$ or the extended mapping class group $\operatorname{MCG}(S_g)$. Norma...

conjecture | open | MathDB 页面 | 原文线索

V2-064481 | MATHDB-337465

The no-genus-bound automorphism conjecture for complexes of regions

摘录 / EXCERPT: Let S_g be a closed surface of genus g, and let $\mathcal C_A(S_g)$ be a complex of regions that is connected and has no holes or corks. Complex-of-regions conjecture. The natu...

conjecture | open | MathDB 页面 | 原文线索

V2-064482 | MATHDB-337478

The winding-number conjecture for dilated simplex Ehrhart h*-vectors

摘录 / EXCERPT: Let $\Delta_r^n=\{x\in[0,r]^n:\sum x_i=r\}$ be the $(n-1)$-dimensional simplex, and call a decorated ordered set partition $((L_1)_{l_1},\ldots,(L_k)_{l_k})$ simplicial when...

conjecture | open | MathDB 页面 | 原文线索

V2-064483 | MATHDB-337479

The winding-number conjecture for generic cube cross-section h*-vectors

摘录 / EXCERPT: Fix integers r,n and s with $1\le s\le rn-1$, and let ... Consider decorated ordered set partitions $((L_1)_{l_1},\ldots,(L_k)_{l_k})$ with $1\in L_1$, satisfying ... Let w_P be...

conjecture | open | MathDB 页面 | 原文线索

V2-064484 | MATHDB-337481

The non-full Reidemeister spectrum conjecture for free nilpotent groups

摘录 / EXCERPT: Let $N_{r,c}$ be a free nilpotent group with rank $r\geq 2$ and nilpotency class $c>2$. The Reidemeister spectrum $\operatorname{Spec}_R(N_{r,c})$ is the set of finite Reidemeister...

conjecture | open | MathDB 页面 | 原文线索

V2-064485 | MATHDB-337495

Judge-Keith-Zanello multipartition-density conjecture

摘录 / EXCERPT: Judge-Keith-Zanello conjecture. The density δ_t exists and equals $1/2$ for every positive odd integer t. Equivalently, if $t=2^k t_0$ with $t_0\geq1$ odd, then δ_t...

conjecture | open | MathDB 页面 | 原文线索

V2-064486 | MATHDB-337501

Caprace's polynomial bound conjecture for arithmetic lattices on products of two trees

摘录 / EXCERPT: Let q denote the parameter in the family of arithmetic lattices under consideration, and consider arithmetic lattices acting simply transitively on a product of two trees. Caprac...

conjecture | open | MathDB 页面 | 原文线索

V2-064487 | MATHDB-337504

The k-linear H"ormander conjecture

摘录 / EXCERPT: Let $1\leq k\leq n$, and let $\mathbf{T}^{\lambda}=(T_1^{\lambda},\dots,T_k^{\lambda})$ be a ν-transverse k-tuple of H"ormander-type operators with positive-definite phase fu...

conjecture | open | MathDB 页面 | 原文线索

V2-064488 | MATHDB-337522

Conjecture on connected sums of torus knots concordant to L-space knots

摘录 / EXCERPT: Concordance conjecture for sums of torus knots. If a connected sum of possibly several torus knots is concordant to an L-space knot, then it is concordant to a positive torus kno...

conjecture | open | MathDB 页面 | 原文线索

V2-064489 | MATHDB-337525

Secondary-term product conjecture for permutation groups

摘录 / EXCERPT: Secondary-term product conjecture. Then the asymptotic distribution of $G_1\times G_2$-extensions has a secondary term of order X^{c_1/n_2}. The conjecture proposes a general mec...

conjecture | open | MathDB 页面 | 原文线索

V2-064490 | MATHDB-337526

The Gorenstein characterization conjecture for s-lecture hall polytopes

摘录 / EXCERPT: Let ${\boldsymbol s}=(s_1,s_2,\ldots,s_n)$ be any sequence, and let $\overleftarrow{\boldsymbol s}$ denote its reverse. Let $\mathbf{P}_n^({\boldsymbol s})}$ be the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-064491 | MATHDB-337538

The symmetric monoidal functor conjecture for black-boxing open Markov processes

摘录 / EXCERPT: Let $\bold{Mark}$ and $\bold{LinRel}$ be the symmetric monoidal bicategories whose objects are finite sets, whose morphisms are open Markov processes and linear relations respectiv...

conjecture | open | MathDB 页面 | 原文线索

V2-064492 | MATHDB-337539

Schinzel's factorization conjecture for sparse Laurent polynomials

摘录 / EXCERPT: Schinzel's factorization conjecture. There is a finite set of matrices $\Omega\subset\mathbb{Z}^{n\times n}$ such that, for each ${\boldsymbol{a}}\in\mathbb{Z}^n$, there are...

conjecture | open | MathDB 页面 | 原文线索

V2-064493 | MATHDB-337541

Schinzel's non-cyclotomic factorization conjecture

摘录 / EXCERPT: Schinzel's non-cyclotomic factorization conjecture. There are finite sets $\Omega^0\subset\mathbb{Z}^{n\times n}$ of nonsingular matrices and $\Gamma\subset\mathbb{Z}^n$ of nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-064494 | MATHDB-337545

The optimal-order discrepancy conjecture for finite point sets

摘录 / EXCERPT: Optimal-order discrepancy conjecture. For every $d\geq 3$, there is a constant $C(d)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064495 | MATHDB-337546

Relative Lehmer conjecture for heights relative to divisible subgroups

摘录 / EXCERPT: Let $g_1,\dots,g_n\in\overline{\mathbb{Q}}^\times$ and let $\Gamma=\langle g_1,\dots,g_n\rangle^{\mathrm{div}}$ be a finite-rank divisible subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-064496 | MATHDB-337550

Planar induced outerplane subgraph conjecture

摘录 / EXCERPT: Let G be a planar graph on n vertices, and let an induced outerplane graph be an induced subgraph whose embedding inherited from G is an outerplanar embedding. Planar outerpl...

conjecture | open | MathDB 页面 | 原文线索

V2-064497 | MATHDB-337552

Aharoni-Barat-Wanless rainbow matching conjecture

摘录 / EXCERPT: Aharoni-Barat-Wanless conjecture. The graph G has a rainbow matching using every colour.

conjecture | refuted | MathDB 页面 | 原文线索

V2-064498 | MATHDB-337565

Soliton resolution conjecture for bounded energy-critical wave solutions

摘录 / EXCERPT: Consider a bounded solution u of the focusing energy-critical nonlinear wave equation (NLW). A soliton resolution conjecture predicts that, as time approaches the maximal time of...

conjecture | open | MathDB 页面 | 原文线索

V2-064499 | MATHDB-337566

Low-regularity scattering conjecture for defocusing nonlinear wave equations

摘录 / EXCERPT: Let $s_c\geq0$, $mu=-1$, and let $(u_0,u_1)\in\dot{H}^{s_c}\times\dot{H}^{s_c-1}$. Low regularity conjecture. The corresponding solution u is global and scatters: there exist uni...

conjecture | open | MathDB 页面 | 原文线索

V2-064500 | MATHDB-337571

Moreno-Socias's almost-degrevlex conjecture for generic ideals

摘录 / EXCERPT: Moreno-Socias's conjecture. The initial monomial ideal in degrevlex order of an ideal generated by generic forms is almost degrevlex.

conjecture | open | MathDB 页面 | 原文线索

V2-064501 | MATHDB-337592

A Wagner conjecture for transitive directed acyclic graphs

摘录 / EXCERPT: A Wagner conjecture for TDAGs. Every d-minor-closed class $\mathcal{G}$ of TDAGs has a finite set of excluded TDAGs $R=\{G_1,\ldots,G_r\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064502 | MATHDB-337607

Whirling homomesy conjecture for m-surjections

摘录 / EXCERPT: Let $[n]=\{1,2,\dots,n\}$ and $[k]=\{1,2,\dots,k\}$. Let $\operatorname{Sur}_m(n,k)$ denote the family of functions from $[n]$ to $[k]$ in which every element of $[k]$ occurs exact...

conjecture | open | MathDB 页面 | 原文线索

V2-064503 | MATHDB-337611

The free-or-nearly-free conjecture for rational cuspidal plane curves

摘录 / EXCERPT: Free-or-nearly-free conjecture. A reduced plane curve $C:f=0$ which is rational cuspidal is either free, or nearly free.

conjecture | open | MathDB 页面 | 原文线索

V2-064504 | MATHDB-337615

The signed planar graph 4-colourability conjecture

摘录 / EXCERPT: Signed planar graph 4-colourability conjecture. Every signed planar graph is 4-colourable.

conjecture | open | MathDB 页面 | 原文线索

V2-064505 | MATHDB-337618

Yui-Zagier's factorization conjecture for Weber CM values

摘录 / EXCERPT: Let $E_i=\mathbb{Q}(\sqrt{d_i})$ be imaginary quadratic fields of fundamental discriminants d_i with $(d_1,d_2)=1$, let $F=\mathbb{Q}(\sqrt D)$ with $D=d_1d_2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064506 | MATHDB-337630

Integrality conjecture distinguishing pseudo-Anosov mapping tori

摘录 / EXCERPT: Let Σ be a compact oriented surface and let $f\in\mathrm{Mod}(\Sigma)$ be pseudo-Anosov. Let ... be the mapping torus of f, and let $TV_r(M_f)$ denote its $SO(3)$ Turaev-Vir...

conjecture | open | MathDB 页面 | 原文线索

V2-064507 | MATHDB-337634

Normality conjecture for Boij-Söderberg coefficients of Veronese surfaces

摘录 / EXCERPT: For the d-uple Veronese embedding of $\mathbb P^2$, let $\beta(\mathbb P^2,b;d)$ denote its Betti table with twist parameter b, and let its Boij-Söderberg coefficients be the c...

conjecture | open | MathDB 页面 | 原文线索

V2-064508 | MATHDB-337642

The partially labelled Dyck path Delta conjecture

摘录 / EXCERPT: Partially labelled Dyck path Delta conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-064509 | MATHDB-337653

Optimality conjecture for geometric polynomial approximation of circular arcs

摘录 / EXCERPT: Let $\bfm{c}$ be a circular arc and let $\bfm{p}_n$ be a G^k geometric approximant of degree n. Choose a polynomial $p_{2n,k}^*$ minimizing ... where $q_{n,k}$ has degree...

conjecture | open | MathDB 页面 | 原文线索

V2-064510 | MATHDB-337661

Finite-monodromy conjecture for compatible rank 2 local systems

摘录 / EXCERPT: Finite-monodromy conjecture. The compatible system $\{\mathscr{L}_{l}\}_{l\ne p}$ should have finite monodromy.

conjecture | open | MathDB 页面 | 原文线索

V2-064511 | MATHDB-337669

Conjecture on maximal k-rank of binary forms

摘录 / EXCERPT: Let a binary form of degree kd be expressed as a sum of k-th powers of binary forms of degree d, and let $\mathrm{rk}^{\max}_k(2,kd)$ denote the maximum of the resulting k-...

conjecture | open | MathDB 页面 | 原文线索

V2-064512 | MATHDB-337678

Exponential stability bound for the heterogeneous Helmholtz equation

摘录 / EXCERPT: Let $ Omega y$ be a bounded Lipschitz domain in $ R^d$, let $a=1$, and let $c\in L^{\infty}(\Omega)$ satisfy ... For $ omega\geq omega_0>0$, consider the heterogeneous Helmholtz solu...

conjecture | open | MathDB 页面 | 原文线索

V2-064513 | MATHDB-337679

The characteristic-system-of-sets-of-lengths conjecture for finite abelian groups

摘录 / EXCERPT: Let H be a transfer Krull monoid over a finite abelian group G, and let $\mathcal B(G)$ be the monoid of zero-sum sequences over G. Write $\mathsf L(a)$ for the set of factor...

conjecture | open | MathDB 页面 | 原文线索

V2-064514 | MATHDB-337682

The AKPZ universality-class conjecture for two-dimensional growth models

摘录 / EXCERPT: Let $v(\cdot)$ be the speed function of a reasonable $(2+1)$-dimensional growth model, and let H_ρ be the Hessian of $v(\cdot)$ computed at the slope ρ. Write α fo...

conjecture | open | MathDB 页面 | 原文线索

V2-064515 | MATHDB-337683

Quadratic repulsion conjecture for determinants of independent Gaussian analytic functions

摘录 / EXCERPT: Let z_0 be a point in the bulk of the pseudospectrum, and let $\tilde G_{z_0}$ be the determinant of the $J\times J$ matrix of independent Gaussian analytic functions arising as...

conjecture | open | MathDB 页面 | 原文线索

V2-064516 | MATHDB-337685

Toda prepotential and geometric engineering

摘录 / EXCERPT: Toda prepotential conjecture. Locally around $a_i=\infty=t_i$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-064517 | MATHDB-337688

Bryan-Oberdieck-Pandharipande-Yin genericity conjecture for hyperelliptic curves on Abelian surfaces

摘录 / EXCERPT: Let B be a polarized Abelian surface, let C be a hyperelliptic curve, and let $f:C\to B$ be a degree-1 morphism with image $\bar{C}$ such that the polarization $[\bar{C}]$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-064518 | MATHDB-337689

Guo-Zeng's gamma-positivity conjecture for involution polynomials

摘录 / EXCERPT: For each positive integer n, let $I_n(x)$ denote the polynomial defined by the paper through the descent enumerator of standard Young tableaux with n squares. Guo-Zeng's conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-064519 | MATHDB-337691

Athanasiadis's gamma-positivity conjecture for flag simplex triangulations

摘录 / EXCERPT: Let V be an n-element set, let 2^V denote the simplex on V, and let $Gamma$ be a flag triangulation of 2^V. Let $ell_V(\Gamma,x)$ denote its local h-polynomial. Athanas...

conjecture | open | MathDB 页面 | 原文线索

V2-064520 | MATHDB-337697

Bounded rank-preservation conjecture for elliptic curves over number-field extensions

摘录 / EXCERPT: Let $K\subset L$ be an extension of number fields. Let $\alpha\in\mathcal O_K$ and nonzero $d\in\mathcal O_K$ satisfy the hypotheses of Theorem--in particular, the hypotheses refer...

conjecture | open | MathDB 页面 | 原文线索

V2-064521 | MATHDB-337700

Orthogonality conjecture for points on the cubic elliptic-curve family

摘录 / EXCERPT: Let $a\geq 1$ be an integer and $0\neq d\in\mathbb Q$ such that $f(X)+d^2$ has distinct roots, where f and its roots ρ,ρ_2,ρ_3, the splitting field L_f, and the elli...

conjecture | open | MathDB 页面 | 原文线索

V2-064522 | MATHDB-337706

Breuil-Schneider conjecture for unramified principal series

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$, let C be a sufficiently large finite extension, let $G=GL_2(F)$, and let S be the set of embeddings of F into C. For...

conjecture | open | MathDB 页面 | 原文线索

V2-064523 | MATHDB-337709

Period bound conjecture for quadratic rational maps with a rational critical 2-cycle

摘录 / EXCERPT: Let ϕ be a quadratic rational function defined over $\mathbb{Q}$ having a $\mathbb{Q}$-rational periodic critical point of period 2. Period bound conjecture for a rational c...

conjecture | open | MathDB 页面 | 原文线索

V2-064524 | MATHDB-337710

Period bound conjecture for quadratic rational maps with a rational critical 3-cycle

摘录 / EXCERPT: Let ϕ be a quadratic rational function defined over $\mathbb{Q}$, and let a critical 3-cycle be the set of iterates of a periodic critical point of period 3. Period bound...

conjecture | open | MathDB 页面 | 原文线索

V2-064525 | MATHDB-337718

Asymptotic bounds for the Nakamura number of weighted games

摘录 / EXCERPT: Let $\operatorname{Nak}^{\mathcal{T}}(n,t)$ denote the maximum possible Nakamura number among weighted games on n players with t types of players, where $t\in\mathbb{N}_{>0}$....

conjecture | open | MathDB 页面 | 原文线索

V2-064526 | MATHDB-337727

The accelero-summation conjecture for a composition-preserving Hardy field

摘录 / EXCERPT: Let $\mathbb{T}^{\operatorname{da}}$ denote the field of accelero-summable transseries, and let H be a Hardy field. In real accelero-summation, use the organic average whenever s...

conjecture | open | MathDB 页面 | 原文线索

V2-064527 | MATHDB-337729

The natural isomorphism conjecture between hyperseries and surreal numbers

摘录 / EXCERPT: Let $\textbf{Hy}$ be the field of hyperseries and $\textbf{No}$ the field of surreal numbers. For each hyperseries $f(x)\in\textbf{Hy}$, consider its value...

conjecture | open | MathDB 页面 | 原文线索

V2-064528 | MATHDB-337733

The characterization conjecture for Krull monoids by systems of sets of lengths

摘录 / EXCERPT: Characterization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-064529 | MATHDB-337735

The great open discrepancy conjecture in dimension at least three

摘录 / EXCERPT: Let $D(m,d)_\infty$ denote the minimal L_∞ discrepancy of an m-point set in dimension d, and let $C(d)>0$ depend only on d. Discrepancy lower-bound conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-064530 | MATHDB-337736

The conjecture on optimal lower bounds for weighted r-discrepancy

摘录 / EXCERPT: Let $D_q^r(\xi,\Lambda)$ be the r-discrepancy of a point set ξ with weights Λ, and define ... Here m is the number of points, d is the dimension, and $C(r,d)>0$ dep...

conjecture | open | MathDB 页面 | 原文线索

V2-064531 | MATHDB-337737

Instability conjecture for cosmological horizons in static solutions

摘录 / EXCERPT: Cosmological-horizon instability conjecture. Every horizon of cosmological type is necessarily unstable. In particular, every static solution to problem has at most one horizon of...

conjecture | open | MathDB 页面 | 原文线索

V2-064532 | MATHDB-337738

Dilcher's conjecture on the higher rows of the coefficient array

摘录 / EXCERPT: Let $b_n^{(p)}$ denote the coefficients appearing in the recurrence referenced in the source, with n,p positive integers. The first two rows are $b_1^{(p)}=p/12$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064533 | MATHDB-337740

De Giorgi's reverse approximation conjecture for gradient flows

摘录 / EXCERPT: Let $\mathbb H$ be a finite-dimensional Euclidean space, and let $\phi:\mathbb H\to\mathbb R$ be a continuously differentiable Lipschitz function. A map...

conjecture | open | MathDB 页面 | 原文线索

V2-064534 | MATHDB-337748

Sullivan's parabolicity conjecture for properly immersed minimal surfaces

摘录 / EXCERPT: A minimal surface in $\mathbb{R}^3$ is properly immersed if its immersion is proper, and an open Riemann surface is parabolic when it admits no nonconstant positive superharmonic f...

conjecture | open | MathDB 页面 | 原文线索

V2-064535 | MATHDB-337749

Schoen-Yau's harmonic-map parabolicity conjecture

摘录 / EXCERPT: Let M be a hyperbolic open Riemann surface, meaning an open Riemann surface that is not parabolic. A proper harmonic map is a harmonic map whose inverse image of every compact se...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-064536 | MATHDB-337752

The strong chromatic number bound conjecture

摘录 / EXCERPT: Strong chromatic number conjecture. For every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-064537 | MATHDB-337769

Klee-Nevo-Novik-Zheng's higher symmetric stackedness conjecture

摘录 / EXCERPT: Klee-Nevo-Novik-Zheng's conjecture. There exists a unique polytopal complex ${\mathcal C}$ in ${\mathbb R}^d$ such that: (i) one face of ${\mathcal C}$ is the cross-polytope...

conjecture | open | MathDB 页面 | 原文线索

V2-064538 | MATHDB-337787

Oka's cyclic fundamental-group conjecture for mixed projective curves

摘录 / EXCERPT: Oka's conjecture. The Milnor fiber F is simply connected. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-064539 | MATHDB-337794

The threshold conjecture for Latin squares in random three-dimensional arrays

摘录 / EXCERPT: Latin-square threshold conjecture. The threshold for $M\sim\mathcal{M}(n,n,n;p)$ to contain a Latin square is $p=\log(n)/n$. The scale is suggested by the obstruction consisting of...

conjecture | open | MathDB 页面 | 原文线索

V2-064540 | MATHDB-337801

Solus's Ehrhart positivity conjecture for base-r simplices

摘录 / EXCERPT: For a positive integer r and a positive integer dimension d, define $\boldsymbol{q}_r=(r-1,(r-1)r,(r-1)r^2,\dots,(r-1)r^{d-1})$ and let the base-r d-simplex be ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-064541 | MATHDB-337802

Stanley's Ehrhart positivity conjecture for Birkhoff polytopes

摘录 / EXCERPT: The Birkhoff polytope B_n is the convex hull of all $n\times n$ permutation matrices, equivalently the set of real nonnegative $n\times n$ matrices whose row and column sums are...

conjecture | open | MathDB 页面 | 原文线索

V2-064542 | MATHDB-337803

Morales's Ehrhart positivity conjecture for CRY and Tesler matrix polytopes

摘录 / EXCERPT: For $\boldsymbol{a}\in\mathbb{N}^n$, let $\operatorname{Tes}_n(\boldsymbol{a})$ be the polytope of upper triangular $n\times n$ matrices with nonnegative entries and hook sum...

conjecture | open | MathDB 页面 | 原文线索

V2-064543 | MATHDB-337804

McNamara-Pylyavskyy maximal-elements conjecture for connected skew shapes

摘录 / EXCERPT: Let $\mathcal{P}_N$ be the poset of skew shapes with N cells, ordered by Schur-positivity, and let $P_{R,S}$ be the box diagonal diagram with R rows, S columns, and $N=R+S-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-064544 | MATHDB-337805

Unique-maximal-element conjecture for ribbons with two adjacent row lengths

摘录 / EXCERPT: Let $\mathcal{R}((a+1)^n a^m)$ be the poset of ribbons whose row lengths consist of n copies of $a+1$ and m copies of a, ordered by Schur-positivity. Let $P_{R,S}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-064545 | MATHDB-337806

Unique-minimal-element conjecture for ribbons with two adjacent row lengths

摘录 / EXCERPT: Let $\mathcal{R}((a+1)^n a^m)$ be the poset of ribbons whose row lengths consist of n copies of $a+1$ and m copies of a, ordered by Schur-positivity. Unique-minimal-element c...

conjecture | open | MathDB 页面 | 原文线索

V2-064546 | MATHDB-337808

Fujino's Du Bois singularities conjecture for qlc pairs

摘录 / EXCERPT: Du Bois singularities conjecture. X has only Du Bois singularities.

conjecture | open | MathDB 页面 | 原文线索

V2-064547 | MATHDB-337809

Effective log canonical boundary conjecture for normal quasi-projective qlc pairs

摘录 / EXCERPT: Effective log canonical boundary conjecture. There exists an effective $mathbb Q$-divisor $Delta$ on X such that (X,Δ) is log canonical.

conjecture | open | MathDB 页面 | 原文线索

V2-064548 | MATHDB-337812

Average Selmer-size conjecture

摘录 / EXCERPT: Let $\mathscr{E}$ be the set of elliptic curves over $\mathbb{Q}$, let m be a positive integer, and let $\sigma(m)$ denote the sum of the positive divisors of m. The average ov...

conjecture | open | MathDB 页面 | 原文线索

V2-064549 | MATHDB-337818

The morsification-mutation conjecture for plane curve singularities

摘录 / EXCERPT: Let (C,z) and (C',z') be real isolated plane curve singularities, and choose real morsifications of them. Each morsification has an associated quiver; two quivers are mutation...

conjecture | open | MathDB 页面 | 原文线索

V2-064550 | MATHDB-337819

M. Shapiro's plabic-graph mutation conjecture

摘录 / EXCERPT: Let P and P' be plabic graphs, and associate to each its quiver. Two plabic graphs are move equivalent when they are related by local moves and changes of the colors of boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-064551 | MATHDB-337826

The stability criterion for McCammond expansions

摘录 / EXCERPT: Let (S,A) be a finite semigroup with generating set A, and let $\mathsf{KR}(S,A)$ and Mc denote the Karnofsky-Rhodes and McCammond expansions, respectively. The expa...

conjecture | open | MathDB 页面 | 原文线索

V2-064552 | MATHDB-337827

The Burnside semigroup stability conjecture

摘录 / EXCERPT: Let ... be a Burnside semigroup, and for $s\in\mathcal{B}(m,n)$ let $S^{\mathscr{J}}_s$ be the finite semigroup formed by the elements $\mathscr{J}$-above s, together with zero...

conjecture | open | MathDB 页面 | 原文线索

V2-064553 | MATHDB-337831

Maximal-chain conjecture for the subgroups of Thompson's group F

摘录 / EXCERPT: Let $\mathfrak{F}$ be the partially ordered family of relevant subgroups under embeddability, let ε_0 be the ordinal used to index the chain, and extend the chain by se...

conjecture | open | MathDB 页面 | 原文线索

V2-064554 | MATHDB-337839

Ek's recurrence conjecture for restricted-partition polynomials

摘录 / EXCERPT: Let $G_s(n,k)$ be the polynomial defined by summing over partitions $(d_1,\ldots,d_k)$ satisfying $\sum_{i=1}^k i d_i=k$ and $\sum_{i=1}^k d_i\leq s$, with the recurrence described...

conjecture | open | MathDB 页面 | 原文线索

V2-064555 | MATHDB-337841

Dominance of the complete-structure sector in the Turaev-Viro sum

摘录 / EXCERPT: Complete-structure dominance conjecture. The colored Jones terms with $s\in U$ dominate those with $s\notin U$:

conjecture | open | MathDB 页面 | 原文线索

V2-064556 | MATHDB-337845

Finiteness conjecture for conformal nets with fixed modular tensor category

摘录 / EXCERPT: Fix a modular tensor category $\mathcal C$ and a central charge c. Consider local conformal nets whose representation category is $\mathcal C$ and whose central charge is c. Fi...

conjecture | open | MathDB 页面 | 原文线索

V2-064557 | MATHDB-337855

Conjecture on spectral numbers and spectral pairs for triangular matrices

摘录 / EXCERPT: Let $T(n,\mathbb R)$ be the space of upper triangular matrices under consideration. A Seifert form stratum is a union of components of a fiber of the map to isomorphism classes of...

conjecture | open | MathDB 页面 | 原文线索

V2-064558 | MATHDB-337856

Conjecture on lifting spectral numbers to spectral pairs

摘录 / EXCERPT: Let S be a matrix in a Seifert form stratum of $T(n,\mathbb R)$ that is not met by $\bigcup_{k=1,2}T_{{\rm HOR}k}(n,\mathbb R)$. Let $\operatorname{Sp}(S)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064559 | MATHDB-337862

Stable degeneration conjecture for klt singularities

摘录 / EXCERPT: Stable degeneration conjecture. There is a unique minimiser v up to rescaling. Furthermore, v is quasi-monomial, R_0 is finitely generated, and the induced degeneration is a...

conjecture | open | MathDB 页面 | 原文线索

V2-064560 | MATHDB-337863

Finite-degree multiplication conjecture for local volumes

摘录 / EXCERPT: Finite-degree multiplication conjecture. The local volumes satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-064561 | MATHDB-337867

The 37-hyperplane conjecture for irreducible simplicial arrangements

摘录 / EXCERPT: 37-hyperplane conjecture. If $\lvert\mathcal{A}\rvert>37$, then

conjecture | open | MathDB 页面 | 原文线索

V2-064562 | MATHDB-337868

The qr^2 Eisenstein-kernel dimension conjecture

摘录 / EXCERPT: The qr^2 Eisenstein-kernel dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064563 | MATHDB-337881

Pieri rules for Macdonald polynomials in superspace

摘录 / EXCERPT: Macdonald superspace Pieri-rule conjecture. The Pieri rules are

conjecture | open | MathDB 页面 | 原文线索

V2-064564 | MATHDB-337883

Equal-mass conjecture for central configurations of homothetic polygons

摘录 / EXCERPT: Let a central configuration consist of L homothetic regular polygons, with the masses assigned to the vertices of each polygon. Equal-mass conjecture. Such a central configuratio...

conjecture | open | MathDB 页面 | 原文线索

V2-064565 | MATHDB-337888

GIT semistability conjecture for syzygy points of generic canonical curves

摘录 / EXCERPT: Let $C\subset \mathbb{P}^{g-1}$ be a generic canonical curve. For integers p with $p\geq \lfloor (g-3)/2\rfloor$, consider its $(p,2)$-syzygy point...

conjecture | open | MathDB 页面 | 原文线索

V2-064566 | MATHDB-337889

Syzygy semistability conjecture for balanced canonical ribbons

摘录 / EXCERPT: Let k be an integer and let R_{2k+1} be the balanced canonical ribbon of genus $2k+1$. For integers p satisfying $2\leq p\leq k-1$, consider its $(p,2)$-syzygy point...

conjecture | open | MathDB 页面 | 原文线索

V2-064567 | MATHDB-337893

Class-group p-rank polynomial conjecture for \p-adic towers of number fields

摘录 / EXCERPT: Class-group p-rank conjecture. There is a polynomial $R(x)\in\mathbb{Q}[x]$ of degree at most d, depending on the tower, such that for all sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-064568 | MATHDB-337894

Limiting solutions with 90-degree corners for rotating vortex patches

摘录 / EXCERPT: Limiting-solutions conjecture. The singular solutions with 90-degree corners seen in numerical computations exist as the weak limits of patches along $\mathscr C$ as $s\to\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-064569 | MATHDB-337896

Chigogidze's unstable Nöbeling space universality conjecture

摘录 / EXCERPT: Let N^n_m be the subset of $\mathbb{R}^n$ consisting of points with at most m rational coordinates. Chigogidze's conjecture. Every m-dimensional subset of $\mathbb{R}^n$ can...

conjecture | open | MathDB 页面 | 原文线索

V2-064570 | MATHDB-337908

Non-vanishing conjecture for global Miyawaki liftings

摘录 / EXCERPT: Let F be a totally real number field, let ψ be a non-trivial unitary character of $\mathbb{A}/F$, and fix a totally positive $\xi\in F^\times$. Let π be an irreducible r...

conjecture | open | MathDB 页面 | 原文线索

V2-064571 | MATHDB-337909

Refined Gan-Gross-Prasad conjecture for Fourier-Jacobi periods

摘录 / EXCERPT: Let F be a totally real number field, let $n'=n$ or $n'=n-1$, and let $\pi=\otimes'_v\pi_v$ and $\pi'=\otimes'_v\pi'_v$ be irreducible cuspidal automorphic representations with t...

conjecture | open | MathDB 页面 | 原文线索

V2-064572 | MATHDB-337925

The at-most-two non-real zeros conjecture for normalised polynomial sequences

摘录 / EXCERPT: At-most-two non-real zeros conjecture. Every polynomial $W_n(z)$ has at most two non-real zeros.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-064573 | MATHDB-337946

Minimal size conjecture for k-positivity tests

摘录 / EXCERPT: Let a k-positivity test be a collection of matrix minors whose positivity guarantees that every minor of size at most k is positive. For an $n\times n$ matrix, the size of a te...

conjecture | open | MathDB 页面 | 原文线索

V2-064574 | MATHDB-337947

Solid k-minor essentiality conjecture

摘录 / EXCERPT: Let a solid k-minor be a $k\times k$ minor using consecutive rows and consecutive columns, and call a minor k-essential if it is necessary in a k-positivity test in the sense...

conjecture | open | MathDB 页面 | 原文线索

V2-064575 | MATHDB-337948

Conjecture on sums of torus knots concordant to alternating knots

摘录 / EXCERPT: Torus-knot alternating-concordance conjecture. A sum of torus knots is concordant to an alternating knot if and only if it is a sum of $(2n+1,2)$ torus knots.

conjecture | open | MathDB 页面 | 原文线索

V2-064576 | MATHDB-337952

Monov's conjecture on moments of critical points of realizable lists

摘录 / EXCERPT: Monov's conjecture. If Λ is realizable, then

conjecture | open | MathDB 页面 | 原文线索

V2-064577 | MATHDB-337953

Johnson's conjecture on realizability of critical points

摘录 / EXCERPT: Let $\Lambda=\{\lambda_1,\dots,\lambda_n\}\subset\mathbb{C}$ with $n\ge 2$ be a list, let ... and let $\Lambda'=\{\mu_1,\dots,\mu_{n-1}\}$ be the list of zeros of p'. A list is rea...

conjecture | open | MathDB 页面 | 原文线索

V2-064578 | MATHDB-337954

Lando's congruence for the universal sl_2 weight system

摘录 / EXCERPT: For each $ngeq 1$, let $\mathcal{D}_n$ be the n-chord diagram in which every chord intersects every other chord, and let $D_n=varphi(\mathcal{D}_n)inmathbb{Z}[x]$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-064579 | MATHDB-337955

Lando's continued-fraction conjecture for the universal sl_2 weight system

摘录 / EXCERPT: Let $D_n(x)$ denote the polynomial associated with the n-chord diagram, and define ... For $k\geq 0$, set $b_k(x)=x-k(k+1)$, and for $k\geq 1$ set...

conjecture | open | MathDB 页面 | 原文线索

V2-064580 | MATHDB-337961

The superspecial hyperelliptic curve level conjecture

摘录 / EXCERPT: Superspecial level conjecture. If $\mathcal{C}$ is a superspecial hyperelliptic curve of genus g over k, then its level is strictly bigger than 2.

conjecture | open | MathDB 页面 | 原文线索

V2-064581 | MATHDB-337968

Hartshorne's weak Lefschetz conjecture for Chow groups

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a smooth hypersurface of dimension $n\ge 4$. Hartshorne's weak Lefschetz conjecture. The cycle class map ... is injective. This is a lon...

conjecture | open | MathDB 页面 | 原文线索

V2-064582 | MATHDB-337977

Tensor-ideal classification for the oriented Brauer-Clifford category

摘录 / EXCERPT: The oriented Brauer-Clifford category is the strict monoidal supercategory $\mathcal{OBC}$ equipped with full monoidal superfunctors ... for $n\in\mathbb{N}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064583 | MATHDB-337987

The second and third largest coset leader conjecture for characteristic-two BCH codes

摘录 / EXCERPT: Let q be a power of 2 and let $n=q^m+1$, with the quantities δ_2 and δ_3 defined as in the preceding cases for even m. Second and third largest coset leader con...

conjecture | open | MathDB 页面 | 原文线索

V2-064584 | MATHDB-337993

Lexi-bounds conjecture for short linear codes and saturating sets

摘录 / EXCERPT: Let $\ell_q(r,3)$ be the length function, let $\ell_q(r,3,5)$ be the d-length function, let $s_q(r-1,2)$ be the smallest size of a 2-saturating set in $\mathrm{PG}(r-1,q)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-064585 | MATHDB-337996

The optimal filtration bound for rational curves on cubic fourfolds

摘录 / EXCERPT: Optimal filtration bound conjecture. For every nonsingular connected rational curve $C\subset Y$ of degree $e\geq5$,

conjecture | open | MathDB 页面 | 原文线索

V2-064586 | MATHDB-338012

Lehmann-Tanimoto's geometric consistency conjecture for Manin's conjecture

摘录 / EXCERPT: Lehmann-Tanimoto's conjecture. Considering all such morphisms f for which f^*L is not big or

conjecture | open | MathDB 页面 | 原文线索

V2-064587 | MATHDB-338017

Frobenius semisimplicity and Heegner-Drinfeld classes conjecture

摘录 / EXCERPT: Frobenius semisimplicity conjecture. For the eigenvalue q, the generalized eigenspace of the Fr-action on W'_π coincides with the eigenspace, and...

conjecture | open | MathDB 页面 | 原文线索

V2-064588 | MATHDB-338019

Bounded front-width conjecture for branching Brownian motion with mass decay

摘录 / EXCERPT: Bounded front-width conjecture. For $b7<1$, almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-064589 | MATHDB-338021

Conjecture on escaping sets in the Eremenko-Lyubich class

摘录 / EXCERPT: Escaping-set conjecture. If $f\in\mathcal{B}$ is a transcendental entire function, then $I(f)$ is not a spider's web.

conjecture | open | MathDB 页面 | 原文线索

V2-064590 | MATHDB-338026

Characterization of closed upper sets in the MacNeille completion of the free monoid

摘录 / EXCERPT: Closed-upper-set characterization. An upper set Z of $\Lambda^{\ast}$ is closed if and only if it satisfies the four rules.

conjecture | open | MathDB 页面 | 原文线索

V2-064591 | MATHDB-338029

The pmaj formula for labelled parallelogram polyominoes

摘录 / EXCERPT: The pmaj formula. For $m\geq 1$ and $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-064592 | MATHDB-338030

Haglund-Remmel-Wilson's partially labelled Dyck path Delta conjecture

摘录 / EXCERPT: Haglund-Remmel-Wilson's conjecture. For $m\geq 0$, $n\geq 0$, and $k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-064593 | MATHDB-338031

Ichino-Ikeda conjecture for Gan-Gross-Prasad periods

摘录 / EXCERPT: Ichino-Ikeda conjecture. For every pure tensor $\phi=\bigotimes_v\phi_v\in\pi$,

conjecture | open | MathDB 页面 | 原文线索

V2-064594 | MATHDB-338034

The positive-soliton multi-soliton conjecture for a6(n,k)

摘录 / EXCERPT: Positive solitons are configurations of the cellular automaton $\Phi(n,k)$ in their minimal forms, and a multi-soliton solution is a configuration consisting of several such positi...

conjecture | open | MathDB 页面 | 原文线索

V2-064595 | MATHDB-338035

The finite-carrier conjecture for positive solitons of a6(n,k)

摘录 / EXCERPT: Finite-carrier conjecture. When only positive solitons occur in $\Phi(n,k)$, the carriers $\mathbf{Z}_i^t$ take values in the finite set M. This is intended to express the observ...

conjecture | open | MathDB 页面 | 原文线索

V2-064596 | MATHDB-338036

The positive-soliton duality conjecture for a6(n,k) and the \mathfrak{sl}_n box-ball system

摘录 / EXCERPT: Let $\Phi(n,k)$ have initial configuration $(\mathbf{Y}_i^0)_i$ consisting only of positive solitons. Interchange space and time so that $\mathbf{Y}_i^t$ and $\mathbf{Z}_i^t$ are r...

conjecture | open | MathDB 页面 | 原文线索

V2-064597 | MATHDB-338048

The countable-union conjecture for well-quasi-orders

摘录 / EXCERPT: Countable-union conjecture. Every well-quasi-ordered poset is a countable union of better quasi-orders.

conjecture | open | MathDB 页面 | 原文线索

V2-064598 | MATHDB-338054

The independence-of-realizations conjecture for log motives

摘录 / EXCERPT: Independence-of-realizations conjecture. Restriction of realizations induces equivalences of categories

conjecture | open | MathDB 页面 | 原文线索

V2-064599 | MATHDB-338055

The second Tate conjecture for log mixed motives

摘录 / EXCERPT: Second Tate conjecture. The set $\operatorname{Hom}(M,N)$ is in bijection with the set of pairs (a,b) where $a:M_{\ell}\to N_{\ell}$ is a morphism and $b:M_B\to N_B$ is a homomor...

conjecture | open | MathDB 页面 | 原文线索

V2-064600 | MATHDB-338058

Kenyon-Okounkov Gaussian Free Field conjecture for lozenge tilings

摘录 / EXCERPT: Fix an arbitrary tilable polygonal domain $Upsilon$ on the triangular grid. Let $H_L(x,y)$ be the random height function of a uniformly random lozenge tiling of the dilated domain...

conjecture | open | MathDB 页面 | 原文线索

V2-064601 | MATHDB-338061

Li-Wu's lower-bound conjecture for conflict-free connection numbers of trees

摘录 / EXCERPT: Let T be a tree of order n, and let P_n be the path on n vertices. Write $cfc(G)$ for the conflict-free connection number of a graph G. Li-Wu's conjecture. For every tree...

conjecture | open | MathDB 页面 | 原文线索

V2-064602 | MATHDB-338063

Tu-Zeng-Li-Helleseth conjecture on permutation trinomials over quadratic finite fields

摘录 / EXCERPT: Tu-Zeng-Li-Helleseth conjecture. If $f_{\alpha,\beta}(x)$ permutes $\mathbb{F}_{q^2}$, then condition (1) or condition (2) holds.

conjecture | open | MathDB 页面 | 原文线索

V2-064603 | MATHDB-338070

Non-moonshine-likeness conjecture for the completely replicable function 9a

摘录 / EXCERPT: Let ... be the indicated completely replicable function. Non-moonshine-likeness conjecture for f_{9a}. The function f_{9a} is non-moonshine-like. Establishing this would remove t...

conjecture | open | MathDB 页面 | 原文线索

V2-064604 | MATHDB-338071

Degree-three generic list recovery for MRA with projection

摘录 / EXCERPT: Let $p\geq 3$ be odd, let $V=\mathbb{R}^p$ with the cyclic-shift action of $G=\mathbb{Z}/p\mathbb{Z}$, and let the projection identify opposite entries as in MRA with projection. M...

conjecture | open | MathDB 页面 | 原文线索

V2-064605 | MATHDB-338074

Degree-two and degree-three recovery for symmetric cryo-EM

摘录 / EXCERPT: Let $H=\mathbb{Z}/L\mathbb{Z}\subset\mathrm{SO}(3)$ be generated by rotation through $2\pi/L$ about the z-axis, and consider homogeneous projected cryo-EM with S shells and F...

conjecture | open | MathDB 页面 | 原文线索

V2-064606 | MATHDB-338075

Conjecture on infinitely many asymptotic maximizers for lattice counts and Riesz means

摘录 / EXCERPT: Let $N_{\sigma,\tau}(\beta,\lambda)$ denote the shifted lattice-point counting function and let $R^\gamma_{-1/2}(\beta,\lambda)$ denote the corresponding Riesz mean for the harmoni...

conjecture | open | MathDB 页面 | 原文线索

V2-064607 | MATHDB-338076

The diagonal concavity conjecture for sums of elementary symmetric polynomials

摘录 / EXCERPT: Diagonal concavity conjecture. The function $f_{\overrightarrow{a}}$ is $\frac1p$-concave on Γ_n if and only if it is $\frac1p$-concave on Δ, if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-064608 | MATHDB-338077

The real-rootedness criterion for concavity of symmetric-polynomial sums

摘录 / EXCERPT: Real-rootedness concavity conjecture. If $\bar f_{\overrightarrow{a}}$ is real-rooted, then $f_{\overrightarrow{a}}$ is $\frac1p$-concave on Γ_n; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-064609 | MATHDB-338078

The hyperbolicity conjecture for semi-symmetric polynomials

摘录 / EXCERPT: For $n,p\in\mathbb{N}$, define the p-homogeneous semi-symmetric polynomial ... for $x\in\mathbb{R}^n$ and $\lambda\in\mathbb{R}^p$. Let $\mathbb{I}$ denote the all-ones direction,...

conjecture | open | MathDB 页面 | 原文线索

V2-064610 | MATHDB-338079

The derivative-convolution real-rootedness conjecture

摘录 / EXCERPT: Let $p\in\mathbb{N}^*$ and let $P,Q\in\mathbb{R}[X]$ have degree p. Derivative-convolution conjecture. If P and Q are real-rooted, then ... is real-rooted as well. This is pres...

conjecture | open | MathDB 页面 | 原文线索

V2-064611 | MATHDB-338080

The complex Gauss-Lucas-type conjecture for derivative convolutions

摘录 / EXCERPT: Let $p\in\mathbb{N}^*$ and let $P,Q\in\mathbb{C}[X]$ have degree p. Derivative-convolution Gauss-Lucas conjecture. The roots of ... lie in the convex hull of the roots of P and...

conjecture | open | MathDB 页面 | 原文线索

V2-064612 | MATHDB-338086

Babson-Billera-Chan cubical generalized lower bound conjecture

摘录 / EXCERPT: Let d be a positive integer. For a cubical d-polytope Q, let $g^c(Q)$ be its cubical g-vector, and let $\mathcal{C}_d$ be the minimal closed cone containing all such vector...

conjecture | open | MathDB 页面 | 原文线索

V2-064613 | MATHDB-338091

Three-dimensional maximum permanent conjecture for prescribed matrix sum

摘录 / EXCERPT: Three-dimensional maximum permanent conjecture. With $\alpha=\frac{-3+\sqrt{57}}{2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-064614 | MATHDB-338099

The golden inequality characterization of planarity for cubic graphs

摘录 / EXCERPT: Let G be a cubic bridgeless graph with E edges, let $F_G(Q)$ denote its flow polynomial, and let $\phi=(1+\sqrt{5})/2$ be the golden ratio. Golden inequality conjecture. ... More...

conjecture | open | MathDB 页面 | 原文线索

V2-064615 | MATHDB-338120

The square-free approximation conjecture for integer polynomials

摘录 / EXCERPT: Square-free approximation conjecture. For any $f\in\mathbb{Z}[x]$ of degree d, there is a square-free polynomial $g\in\mathbb{Z}[x]$ of degree at most d satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-064616 | MATHDB-338121

The palette-index conjecture for simple graphs

摘录 / EXCERPT: Let G be a simple graph, and let Δ denote its maximum degree. The palette-index conjecture asserts that ... Previous results and sporadic examples suggest that the palette in...

conjecture | open | MathDB 页面 | 原文线索

V2-064617 | MATHDB-338126

The maximal k-rank conjecture for binary forms

摘录 / EXCERPT: Let $\operatorname{rk}^{\max}_k(kd,2)$ be the least number of k-th powers of degree-d forms needed to represent every binary form of degree kd. Let ℓ_1 and ℓ_2 be n...

conjecture | open | MathDB 页面 | 原文线索

V2-064618 | MATHDB-338132

Eskin-Zorich large-genus volume conjecture for strata of differentials

摘录 / EXCERPT: Let g tend to infinity, let $\boldsymbol{\boldsymbol{k}}=(k_1,\ldots,k_n)$ be a partition of $2g-2$, and write ${\rm Vol}(k_1,\ldots,k_n)$ for the volume of the corresponding str...

conjecture | open | MathDB 页面 | 原文线索

V2-064619 | MATHDB-338133

Eskin-Zorich large-genus Siegel-Veech constant conjecture

摘录 / EXCERPT: Let g tend to infinity, let $\mu\vdash 2g-2$ be a partition indexing a stratum of Abelian differentials, and let $c_{\rm area}(\mu)$ denote its area Siegel-Veech constant. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-064620 | MATHDB-338134

Refined Eskin-Zorich asymptotics for volumes and Siegel-Veech constants

摘录 / EXCERPT: Let μ index a stratum of Abelian differentials, let ${\rm dim}({\mathcal H}(\mu))$ denote its dimension, and let $\epsilon_1(\mu)$ and $\epsilon_2(\mu)$ be the error terms in t...

conjecture | open | MathDB 页面 | 原文线索

V2-064621 | MATHDB-338137

The Shokurov-Kollár connectedness conjecture for non-klt loci

摘录 / EXCERPT: Shokurov-Kollár connectedness conjecture. For every point $s\in S$, the fiber of the non-klt locus has at most two connected components:

conjecture | open | MathDB 页面 | 原文线索

V2-064622 | MATHDB-338139

Götsche-Kool virtual elliptic genus conjecture for rank 2 sheaf moduli spaces

摘录 / EXCERPT: Let S be a smooth projective surface with $b_1(S)=0$, $p_g(S)>0$, $K_S\ne0$, and with only Seiberg-Witten basic classes 0 and K_S. Let H,c_1,c_2 be chosen so that there are...

conjecture | open | MathDB 页面 | 原文线索

V2-064623 | MATHDB-338140

Götsche-Kool universal cobordism power-series conjecture

摘录 / EXCERPT: Let S be a smooth projective surface with $b_1(S)=0$, $p_g(S)>0$, $K_S\ne0$, and only Seiberg-Witten basic classes 0 and K_S. Let H,c_1,c_2 be chosen so that there are no r...

conjecture | open | MathDB 页面 | 原文线索

V2-064624 | MATHDB-338141

Götsche-Kool numerical conjecture for rank 2 sheaf invariants

摘录 / EXCERPT: Let $a\in\{\operatorname{cob},\operatorname{ell}\}$, let $\underline\beta\in\mathbb Z^4$ satisfy $\beta_1\equiv\beta_2\pmod 2$ and $\beta_3\ge\beta_4-3$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064625 | MATHDB-338142

Götsche-Kool conjecture for arbitrary surfaces with holomorphic 2-forms

摘录 / EXCERPT: Let $a\in\{\operatorname{cob},\operatorname{ell}\}$, let S be a smooth projective surface with $b_1(S)=0$ and $p_g(S)>0$, and choose H,c_1,c_2 so that there are no rank 2 str...

conjecture | open | MathDB 页面 | 原文线索

V2-064626 | MATHDB-338144

Optimality conjecture for Levenshtein-bound refinements

摘录 / EXCERPT: Optimality conjecture. There exists a constant s_q such that whenever $s\in[-1,s_q)\cap T_n$, the refinements described above are the best bounds that can be obtained from Theore...

conjecture | open | MathDB 页面 | 原文线索

V2-064627 | MATHDB-338149

Tasaka's conjecture on multiple period-polynomial relations

摘录 / EXCERPT: Let $S_{N,r}$ be the index set used to define the coefficient space $\mathbb{P}_{N,r}$, let $\mathbf{W}_{N,r}\subseteq\mathbb{P}_{N,r}$ be the space of polynomials satisfying the s...

conjecture | open | MathDB 页面 | 原文线索

V2-064628 | MATHDB-338150

Brown's matrix conjecture for multiple zeta values

摘录 / EXCERPT: Let $C_{N,r}$ be the matrices defined from the matrices $E^{(i)}_{N,r}$, and let $\mathbb{O}(x)$ and $\mathbb{S}(x)$ denote the generating series appearing in the source. Brown's m...

conjecture | open | MathDB 页面 | 原文线索

V2-064629 | MATHDB-338152

Elementary-factorization generation conjecture for matrix factorizations over Bézout domains

摘录 / EXCERPT: Elementary-factorization generation conjecture. The inclusion functor ι is an equivalence of R-linear categories. This is established in the paper when R is an elementary...

conjecture | open | MathDB 页面 | 原文线索

V2-064630 | MATHDB-338153

Krull-Schmidt conjecture for matrix factorization categories over Bézout domains

摘录 / EXCERPT: Let R be a Bézout domain and let W be a critically-finite element of R. Krull-Schmidt conjecture. The category $\mathrm{hmf}(R,W)$ is a Krull-Schmidt category. The claim foll...

conjecture | open | MathDB 页面 | 原文线索

V2-064631 | MATHDB-338158

The conjectural evaluation formula for Jack superpolynomials

摘录 / EXCERPT: The evaluation conjecture. The evaluation of the Jack superpolynomial is

conjecture | open | MathDB 页面 | 原文线索

V2-064632 | MATHDB-338160

Cao-Leung-Borisov-Joyce genus-zero GW/DT4 conjecture

摘录 / EXCERPT: Genus-zero GW/DT4 conjecture. For a suitable choice of orientation,

conjecture | open | MathDB 页面 | 原文线索

V2-064633 | MATHDB-338162

Katz genus-zero GV/DT3 correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Katz's genus-zero GV/DT conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064634 | MATHDB-338163

Strong GW/PT correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Let Y be a smooth projective Calabi-Yau 3-fold. Let $n^{\mathrm{GW}}_{g,\beta}$ be the Gromov-Witten expansion coefficients and let $n^P_{g,\beta}$ be the coefficients in the l...

conjecture | open | MathDB 页面 | 原文线索

V2-064635 | MATHDB-338180

Fourier restriction conjecture for the paraboloid over \(\mathbb Z/N\mathbb Z\)

摘录 / EXCERPT: Let $n\geq 2$, let N be an odd positive integer, and let $\mathcal{E}$ denote the extension operator for the paraboloid over $\mathbb Z/N\mathbb Z$, acting on functions H on...

conjecture | open | MathDB 页面 | 原文线索

V2-064636 | MATHDB-338181

Sufficient conditions for discrete restriction estimates over \(\mathbb Z/N\mathbb Z\)

摘录 / EXCERPT: Let $n\geq 2$, let $1\leq r,s\leq\infty$, and write r' for the Hölder conjugate of r. Let the discrete restriction estimate refer to the estimate defined earlier in the source....

conjecture | open | MathDB 页面 | 原文线索

V2-064637 | MATHDB-338184

De Pierro's least-squares convergence conjecture for under-relaxed cyclic projections

摘录 / EXCERPT: De Pierro's conjecture. The least-squares solution set S is nonempty if and only if, for every starting point u_0, the limits

conjecture | solved | MathDB 页面 | 原文线索

V2-064638 | MATHDB-338190

The tau-function as conformal block conjecture for W(sl_d)-symmetric Toda theories

摘录 / EXCERPT: Let $\mathcal M_{\mathcal W({\mathfrak{sl}_d})}^{\mathbb P^1}$ be the moduli space of marked points, weights, and times, and let $\mathfrak T_{\bold z,\bold \alpha,\bold t}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-064639 | MATHDB-338197

Classification conjecture for symmetric universally self-predicting Boolean functions

摘录 / EXCERPT: Symmetric USP-function conjecture. The only symmetric USP functions are Majority and the largest character. In particular, Majority is the only monotone and symmetric USP function.

conjecture | open | MathDB 页面 | 原文线索

V2-064640 | MATHDB-338198

Acyclicity conjecture for the optimal-predictor graph

摘录 / EXCERPT: Optimal-predictor acyclicity conjecture. The graph $\mathcal{G}_{\rho,n}$ contains no cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-064641 | MATHDB-338200

Divisibility conjecture for the L-polynomials of the curves E_k

摘录 / EXCERPT: For a positive integer k, let E_k be the smooth projective model over $\mathbb{F}_2$ of ... Its L-polynomial is denoted by $\mathrm{L}_{E_k}(t)$. The E_k divisibility conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-064642 | MATHDB-338203

Equivalence of the semisimplification with the Verlinde quotient

摘录 / EXCERPT: Equivalence conjecture. For sufficiently large p, the functor F is an equivalence.

conjecture | open | MathDB 页面 | 原文线索

V2-064643 | MATHDB-338209

Rank-ordering conjecture for overpartition ranks modulo 6

摘录 / EXCERPT: Rank-ordering conjecture. For every integer $n\geq 11$, these inequalities and equalities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-064644 | MATHDB-338214

The characterization of MDS-PIR capacity-achieving codes

摘录 / EXCERPT: Let $\mathcal{C}$ be an $[n,k]$ code, and let $d_s^{\mathcal{C}}$ denote the quantity appearing in the necessary condition ... An MDS-PIR capacity-achieving matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-064645 | MATHDB-338232

The conjecture that very general degree-n hypersurfaces are not covered by rational surfaces

摘录 / EXCERPT: Let X be a very general hypersurface of degree n in ${\bf P}^n$, with $n\geq 5$. A hypersurface X is covered by rational surfaces if a family of rational surfaces in X has...

conjecture | open | MathDB 页面 | 原文线索

V2-064646 | MATHDB-338233

Non-trivial GPCGI conjecture for almost subnormal subgroups of general linear groups

摘录 / EXCERPT: Let D be a division ring with center F, and let $\operatorname{GL}_n(D)$ be the general linear group of degree n over D. If $n\geq 2$, assume that D is not a locally fini...

conjecture | open | MathDB 页面 | 原文线索

V2-064647 | MATHDB-338241

Oblomkov-Rozansky conjecture for the framed matrix factorization category

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathcal{H}(S_n)$ denote the corresponding monoidal category, and let $\mathbf{MF}^{\mathrm{fr}}_{n,1}$ be the framed matrix factorization ca...

conjecture | open | MathDB 页面 | 原文线索

V2-064648 | MATHDB-338243

The BCOV slope formula for semi-stable Calabi-Yau degenerations

摘录 / EXCERPT: Let $f\colon \mathcal{X}\to S$ be a relatively minimal family, let $s_i\in S$ be a point such that the singular fiber $f^{-1}(s_i)$ is semi-stable, let a_i denote its BCOV invari...

conjecture | open | MathDB 页面 | 原文线索

V2-064649 | MATHDB-338244

Converse of the logarithmic formula for hypergeometric functions ${}_3F_2$

摘录 / EXCERPT: Converse of the logarithmic formula. The condition

conjecture | open | MathDB 页面 | 原文线索

V2-064650 | MATHDB-338252

Beck's gap-free partition sum conjecture

摘录 / EXCERPT: A partition $lambda=(lambda_1,lambda_2,lambda_\ell)$ is gap-free, or compact, when $0\leqlambda_i-lambda_{i+1}\leq 1$ for all $1\leq i\leq lambda-1$. Let $a_2(n)$ denote the number...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-064651 | MATHDB-338253

Sharp large-time behavior conjecture for degree evolution

摘录 / EXCERPT: Sharp large-time behavior conjecture. For τ fixed,

conjecture | open | MathDB 页面 | 原文线索

V2-064652 | MATHDB-338263

Equivalence between SYLTDMOR1 and moment matching

摘录 / EXCERPT: Consider an LTI system with piecewise constant input and zero initial state. Choose Laguerre polynomials; Legendre polynomials with odd reduced order; Chebychev polynomials of the...

conjecture | open | MathDB 页面 | 原文线索

V2-064653 | MATHDB-338264

The test function conjecture for parahoric local models

摘录 / EXCERPT: Test function conjecture. As elements of $\mathcal Z(G(E_0),\mathcal G(\mathcal O_{E_0}))$,

conjecture | open | MathDB 页面 | 原文线索

V2-064654 | MATHDB-338266

The asphericity conjecture for arrangements of ideal type

摘录 / EXCERPT: Let Φ be a reduced root system with Weyl arrangement $\mathscr A(\Phi)$. An arrangement of ideal type $\mathscr A_{\mathcal I}$ is a subarrangement of $\mathscr A(\Phi)$ deter...

conjecture | open | MathDB 页面 | 原文线索

V2-064655 | MATHDB-338271

Davies-Jenssen-Perkins-Roberts minimisation conjecture for the Potts-model program

摘录 / EXCERPT: Let G be a d-regular graph, let $q\ge d+1$, let $\beta>0$, and let $U^q_G(\beta)$ denote the objective associated with the Potts-model linear-programming relaxation. Davies-Jen...

conjecture | open | MathDB 页面 | 原文线索

V2-064656 | MATHDB-338273

Relations among rational coefficients in reciprocal hyperbolic series

摘录 / EXCERPT: For each positive integer p, let $c_p,d_p,e_p,f_p,g_p,h_p,i_p,j_p,k_p,l_p$ be the rational coefficients defined by the evaluations of the reciprocal hyperbolic series at π. C...

conjecture | open | MathDB 页面 | 原文线索

V2-064657 | MATHDB-338277

Integer correlation conjecture for Kloosterman sums and cusp-form eigenvalues

摘录 / EXCERPT: Let f be a fixed primitive cusp form, holomorphic or Maass, and let $\lambda_f(n)$ denote its Hecke eigenvalue at n. Write $\mathcal{L}=\log X$. Integer correlation conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-064658 | MATHDB-338281

Conjectured scaling limit of critical random-cluster interfaces

摘录 / EXCERPT: Critical random-cluster interface conjecture. As $\delta\to 0$, the interface in $(\Omega^{\delta};x^{\delta},y^{\delta})$ converges weakly to the chordal...

conjecture | open | MathDB 页面 | 原文线索

V2-064659 | MATHDB-338285

A bipartite-subgraph-free bound for pebbling numbers of direct graph products

摘录 / EXCERPT: Direct-product pebbling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064660 | MATHDB-338289

Integer-coefficient conjecture for Wronskian Hermite polynomials

摘录 / EXCERPT: Let λ be a partition, and let $\operatorname{He}_{\lambda}(x)$ denote the monic Wronskian Hermite polynomial associated with λ. Integer-coefficient conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-064661 | MATHDB-338290

The relative Banzhaf index upper-bound conjecture

摘录 / EXCERPT: Let $w_1,w_2,w_3, \ldots, w_m$ be positive weights, let w be their maximum, and set ... For a voting game with these weights, let β_i denote the relative Banzhaf index of pla...

conjecture | open | MathDB 页面 | 原文线索

V2-064662 | MATHDB-338302

The Schubert flop derived-equivalence conjecture

摘录 / EXCERPT: Let $Y_{-C}\xleftarrow{\,l_{0,-C}\,}Y_0\xrightarrow{\,l_{0,C}\,}Y_C$ be the Schubert flop diagram for a one-dimensional face C in the remaining case described in the paper. The m...

conjecture | open | MathDB 页面 | 原文线索

V2-064663 | MATHDB-338306

Weak Lefschetz conjecture for Chow groups

摘录 / EXCERPT: Weak Lefschetz conjecture. This map is an isomorphism when $2p<\dim(L)$ and is injective modulo torsion when $2p=\dim(L)$. This is the Chow-group analogue of the classical Weak Lef...

conjecture | open | MathDB 页面 | 原文线索

V2-064664 | MATHDB-338307

Delta-only distance conjecture for optimal mixed-integer program solutions

摘录 / EXCERPT: Let A be the constraint matrix, let Δ be the largest absolute value of any determinant of a square submatrix of A, and let $I,J\subseteq[n]$ index the variables required...

conjecture | open | MathDB 页面 | 原文线索

V2-064665 | MATHDB-338308

Three-subgraph regular-irregular decomposition conjecture

摘录 / EXCERPT: A subgraph is called regular or locally irregular according as it is regular or locally irregular. The regular-irregular number $\operatorname{reg\text{-}irr}(G)$ is the minimum nu...

conjecture | open | MathDB 页面 | 原文线索

V2-064666 | MATHDB-338310

The gaunt-colimit characterization of flagged higher categories

摘录 / EXCERPT: Gaunt-colimit characterization. The restricted Yoneda functor

conjecture | open | MathDB 页面 | 原文线索

V2-064667 | MATHDB-338311

The maximum-deviation limit conjecture for the Sine-beta counting process

摘录 / EXCERPT: Maximum-deviation limit conjecture. There is a random variable ξ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064668 | MATHDB-338315

Optimal mixed-norm inequality for unimodular multilinear forms

摘录 / EXCERPT: Let m,n be positive integers, let $p\in[1,\infty]$, and let $T:\ell_p^n\times\cdots\times\ell_p^n\to\mathbb{K}$ be a unimodular m-linear form. The norm $\|T\|$ is its operator no...

conjecture | open | MathDB 页面 | 原文线索

V2-064669 | MATHDB-338316

Optimal dimension exponents for coefficient inequalities of unimodular multilinear forms

摘录 / EXCERPT: Let m be a positive integer, let $(r,p)\in(0,\infty)\times(1,\infty]$, and let $T:\ell_p^n\times\cdots\times\ell_p^n\to\mathbb{K}$ be a unimodular m-linear form, where n is a...

conjecture | open | MathDB 页面 | 原文线索

V2-064670 | MATHDB-338325

Normal-operator extension of the normalized-frame nonexistence theorem

摘录 / EXCERPT: Normalized-frame conjecture. The system above is not a frame for $\mathcal{H}$.

conjecture | refuted | MathDB 页面 | 原文线索

V2-064671 | MATHDB-338326

Continuous-frame interval invariance for normal reductive operators

摘录 / EXCERPT: Normal-reductive extension conjecture. Theorem remains true if A is a normal reductive operator.

conjecture | open | MathDB 页面 | 原文线索

V2-064672 | MATHDB-338327

Sun's conjecture on transversals in 3-dimensional latin hypercubes

摘录 / EXCERPT: A latin hypercube of dimension d and order n is the d-dimensional analogue of a latin square, with every symbol occurring exactly once on each axis-parallel line. Sun's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-064673 | MATHDB-338328

The one-two-three conjecture for genus-three minimal surfaces in flat three-tori

摘录 / EXCERPT: Let M be a compact oriented minimal surface in a flat three-torus, and suppose that M has genus three. Write $index_A$ for its Morse index with respect to the area functional....

conjecture | open | MathDB 页面 | 原文线索

V2-064674 | MATHDB-338333

Bloch-Quillen formula for the special fiber

摘录 / EXCERPT: Bloch-Quillen formula. The cycle class map cyc_X^M is an isomorphism. Together with the known surjectivity, this would identify the Levine-Weibel Chow group of zer...

conjecture | open | MathDB 页面 | 原文线索

V2-064675 | MATHDB-338337

Solvable Kauffman conjecture for derivatives

摘录 / EXCERPT: Let K be a knot, let $n\in \frac{1}{2}\mathbb{N}_0$, and let an (n)-solvable derivative mean a derivative of K that is (n)-solvable. Solvable Kauffman conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-064676 | MATHDB-338346

A truncated q-analogue of the Zudilin supercongruence

摘录 / EXCERPT: The truncated Zudilin q-congruence. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-064677 | MATHDB-338347

A q-analogue of Sun's second binomial supercongruence

摘录 / EXCERPT: The q-Sun binomial conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-064678 | MATHDB-338348

A refinement of the q-Zudilin congruence

摘录 / EXCERPT: The q-Zudilin refinement. Numerically, one expects

conjecture | open | MathDB 页面 | 原文线索

V2-064679 | MATHDB-338349

A q-analogue of Sun's supercongruence for n congruent to 1 modulo 4

摘录 / EXCERPT: The q-Sun congruence for $n\equiv1\pmod4$. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-064680 | MATHDB-338355

Ratios conjecture for cubic Dirichlet L-functions

摘录 / EXCERPT: Let f be an even Schwartz function, let w be the weight function, and define the weighted ratio average ... For $\operatorname{Re}(2\nu'+\nu)>-1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-064681 | MATHDB-338368

Component-size conjecture for random graphs locally modelled on lattices

摘录 / EXCERPT: Let $\mathbb{L}^d$ denote the d-dimensional lattice. Define ... Let $d\in\mathbb{N}$ with $d\geq 2$, and let $r\in\mathbb{N}$ with $r\geq r_0(d)$. Component-size conjecture. With h...

conjecture | open | MathDB 页面 | 原文线索

V2-064682 | MATHDB-338382

Unbounded minimal presentations for concatenation-generated numerical semigroups

摘录 / EXCERPT: Unbounded-presentation conjecture. The set $\{\mathfrak{h}_0,\ldots,\mathfrak{h}_{n+1}\}$ is part of a minimal generating set for $\mathcal{Q}_{(n,e,e-4)}$, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-064683 | MATHDB-338383

Uniqueness conjecture for the Schrödinger potential from the Dirichlet-to-Neumann map

摘录 / EXCERPT: Uniqueness conjecture for the Schrödinger potential. The knowledge of $\Lambda_{g,q}$ uniquely determines q. This is the potential-recovery version of the Calderón problem in a k...

conjecture | open | MathDB 页面 | 原文线索

V2-064684 | MATHDB-338388

Ichino-Ikeda-Neil Harris conjecture for unitary groups

摘录 / EXCERPT: Ichino-Ikeda-Neil Harris conjecture. There is an integer β, depending on the L-packets containing π and π', such that

conjecture | open | MathDB 页面 | 原文线索

V2-064685 | MATHDB-338390

Nonexistence of globally hyperbolic developments for quasilinear wave equations

摘录 / EXCERPT: Nonexistence conjecture. There are quasilinear wave equations of the form $$ for which there exists initial data such that there does not exist any UGHD.

conjecture | open | MathDB 页面 | 原文线索

V2-064686 | MATHDB-338397

Critical-orbit convergence conjecture for non-linearizable indifferent fixed points

摘录 / EXCERPT: Critical-orbit convergence conjecture. There exists a critical point c_0 of f such that

conjecture | open | MathDB 页面 | 原文线索

V2-064687 | MATHDB-338399

The boundary-support conjecture for the local time of super-Brownian motion

摘录 / EXCERPT: Let $S(X_t)=\overline{\{x:X(t,x)>0\}}$ be the closed support of X_t, let $\partial S(X_t)$ denote its boundary, and let L_t be the local-time measure associated with the bounda...

conjecture | open | MathDB 页面 | 原文线索

V2-064688 | MATHDB-338403

The arbitrary modal-combination derivability conjecture

摘录 / EXCERPT: Let Δ, ∇, $\bullet$, and $\circ$ denote the contingency, accident, and their dual modal operators in the bimodal logic under consideration. For $m\in\mathbb{N}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064689 | MATHDB-338407

Logarithmic separation choosability for bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph with maximum degree Δ, and let $\operatorname{ch}_{\operatorname{sep}}(G)$ denote its separation choosability. Alon-Krivelevich separation-choos...

conjecture | open | MathDB 页面 | 原文线索

V2-064690 | MATHDB-338419

The unimodular conjecture for Keller maps over local domains

摘录 / EXCERPT: Unimodular conjecture. The induced map $\overline{F}$ is a non-zero map.

conjecture | open | MathDB 页面 | 原文线索

V2-064691 | MATHDB-338424

The rank-two Euler-system conjecture for Beilinson-Flach classes

摘录 / EXCERPT: Let T be the rank-four representation, let $\mathcal{N}(\mathcal{P})$ be the specified set of auxiliary integers, and let $\mathcal{L}_{m,V}$ be the equivariant Perrin-Riou regul...

conjecture | open | MathDB 页面 | 原文线索

V2-064692 | MATHDB-338425

The signed decomposition conjecture for Beilinson-Flach classes

摘录 / EXCERPT: Let $\mathcal{N}(\mathcal{P})$ be the specified set of auxiliary integers, let $\mathbb{T}=T\otimes\Lambda^\iota$, and let $M_{\log}$ and Q be the matrices appearing in the signe...

conjecture | open | MathDB 页面 | 原文线索

V2-064693 | MATHDB-338434

Abelian Cayley graph bound implied by the Golomb-Welch conjecture

摘录 / EXCERPT: Abelian Cayley graph bound. For $k\ge2$ and $d\ge3$,

conjecture | open | MathDB 页面 | 原文线索

V2-064694 | MATHDB-338435

The two-dimensional Centralizer Conjecture over an integral domain

摘录 / EXCERPT: The two-dimensional Centralizer Conjecture over D. Suppose

conjecture | open | MathDB 页面 | 原文线索

V2-064695 | MATHDB-338442

Odd-order conjecture for nontrivially unstable circulants

摘录 / EXCERPT: A circulant is a Cayley graph of a cyclic group; write an order-n circulant as $\operatorname{Cay}(\mathbb{Z}_n,S)$. A circulant is nontrivially unstable when it is unstable but...

conjecture | open | MathDB 页面 | 原文线索

V2-064696 | MATHDB-338443

Marušič-Scapellato-Zagaglia Salvi compatibility conjecture

摘录 / EXCERPT: Let Γ be a graph. Its canonical double cover has adjacency matrix $\begin{pmatrix}0&A\\A&0\end{pmatrix}$ when A is the adjacency matrix of Γ. The matrix A is comp...

conjecture | open | MathDB 页面 | 原文线索

V2-064697 | MATHDB-338456

Pointed e-positivity conjecture for unit interval graphs

摘录 / EXCERPT: Let $m=m_1\cdots m_n$ satisfy $1\leq m_1\leq\cdots\leq m_n\leq n$ and $m_i\geq i$ for each i, and let $G(m)$ be the graph on $[n]$ with an edge (i,j) exactly when $i<j\leq m_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-064698 | MATHDB-338458

The claimed classification of orders for Hadamard matrices with one circulant core

摘录 / EXCERPT: One-circulant-core order claim. These are the only possible orders for a Hadamard matrix with one circulant core.

conjecture | open | MathDB 页面 | 原文线索

V2-064699 | MATHDB-338476

The representation-number doubling conjecture for maximal lattices

摘录 / EXCERPT: Representation-number doubling conjecture. Under these hypotheses, such a matrix T' exists and the representation number satisfies the displayed doubling identity. The claim conc...

conjecture | open | MathDB 页面 | 原文线索

V2-064700 | MATHDB-338479

Two-variable Iwasawa main conjecture for a Coleman family

摘录 / EXCERPT: Let $\mathbb{F}$ be a Coleman family over $\Lambda_{(k_0;r)}$ with slope $s\in\mathbb{Q}_{\geq 0}$, let $h\in\mathbb{Z}$ satisfy $h\geq s$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064701 | MATHDB-338480

BFGS for the unit ball

摘录 / EXCERPT: BFGS for the unit ball. The trial step s converges to zero.

conjecture | open | MathDB 页面 | 原文线索

V2-064702 | MATHDB-338485

Babai's thickness conjecture for primitive coherent configurations

摘录 / EXCERPT: Let $\mathfrak{X}$ be a primitive coherent configuration on n vertices. For a group G, its thickness $\theta(G)$ is the greatest t such that the alternating group A_t is in...

conjecture | open | MathDB 页面 | 原文线索

V2-064703 | MATHDB-338487

Sublinear-rank motion conjecture for primitive coherent configurations

摘录 / EXCERPT: Fix $0<\delta<1$. Let $\mathfrak{X}$ be a primitive coherent configuration on n vertices, and suppose every constituent has degree at most δn. Sublinear-rank motion conje...

conjecture | open | MathDB 页面 | 原文线索

V2-064704 | MATHDB-338492

The author's conjecture on heights of rational points on quadratic twists

摘录 / EXCERPT: Let $(A,B)\in\mathbb{Z}^2$ satisfy ... and for each squarefree integer $d\geq 1$ let E_d be the elliptic curve over $\mathbb{Q}$ defined by ... Let $\eta_d(A,B)$ be defined by ... when...

conjecture | open | MathDB 页面 | 原文线索

V2-064705 | MATHDB-338495

Worst concentrated load conjecture for the free rectangular plate

摘录 / EXCERPT: Worst concentrated load conjecture. When $D=\emptyset$, $\overline T_{\pi/2}$ and $-\overline T_{\pi/2}$ are the unique maximizers of the worst case problem.

conjecture | open | MathDB 页面 | 原文线索

V2-064706 | MATHDB-338496

The L^∞ maximizer conjecture for the plate load problem

摘录 / EXCERPT: L^∞ maximizer conjecture. For every $D\subset\Omega$, the unique maximizers of the problem are the odd function

conjecture | open | MathDB 页面 | 原文线索

V2-064707 | MATHDB-338497

The (f^1,D^0) minimaxmax optimum conjecture

摘录 / EXCERPT: Minimaxmax optimum conjecture. The optimum of the minimaxmax problem is the couple (f^1,D^0).

conjecture | open | MathDB 页面 | 原文线索

V2-064708 | MATHDB-338498

The strip minimaxmax optimum conjecture

摘录 / EXCERPT: Strip minimaxmax optimum conjecture. The optimum of the minimaxmax problem is the couple $(\overline e_1,\operatorname{Strips})$.

conjecture | open | MathDB 页面 | 原文线索

V2-064709 | MATHDB-338500

Gaussian likelihood-ratio fluctuation conjecture in the paramagnetic phase

摘录 / EXCERPT: Let $Gamma$ be the set of parameter pairs (α,β) for which the replica-symmetric formula vanishes, and let $Gamma'$ be its interior. For the likelihood ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-064710 | MATHDB-338503

Estimate for symmetric functions in the range of the spectral projection complement

摘录 / EXCERPT: Let Λ be the Dirichlet-to-Neumann operator, let P_τ denote the spectral projection associated with the threshold τ, and let J be the operator mapping v to...

conjecture | open | MathDB 页面 | 原文线索

V2-064711 | MATHDB-338516

Kuznetsov-Smirnov length conjecture for Grassmannian residual categories

摘录 / EXCERPT: Let $\operatorname{G}(k,n)$ be the Grassmannian, and let $\mathcal R_{k,n}$ be the residual category defined by the residual semiorthogonal decomposition associated with Fonarev's...

conjecture | open | MathDB 页面 | 原文线索

V2-064712 | MATHDB-338533

Exact relation between the idempotent-sum-free constant and the Erdős-Burgess constant for cyclic semigroups

摘录 / EXCERPT: Let ${\rm C}_{k;n}$ be a cyclic commutative semigroup with parameters k and n, and let ${\hat{\sl I}}({\rm C}_{k;n})$ denote the threshold associated with idempotent-sum-free s...

conjecture | open | MathDB 页面 | 原文线索

V2-064713 | MATHDB-338566

Lattice conjecture for inverse atom sets of involutions in the symmetric group

摘录 / EXCERPT: Let S_n be the symmetric group, let $\mathcal{I}(S_n)$ denote its set of involutions, and for $z\in\mathcal{I}(S_n)$ let $\mathcal{A}(z)$ be the corresponding set of atoms, equip...

conjecture | open | MathDB 页面 | 原文线索

V2-064714 | MATHDB-338571

Critical-selection conjecture for interacting Wright-Fisher diffusions

摘录 / EXCERPT: Consider the system of interacting Wright-Fisher diffusions on $\mathbb Z^d$ with $N=1$ and parameter $\mu>1$, where s is the selection parameter and coexistence refers to long-t...

conjecture | open | MathDB 页面 | 原文线索

V2-064715 | MATHDB-338597

Boundary-cell characterization conjecture for Wilson loop diagrams

摘录 / EXCERPT: Let $\mathcal{W}(k,n)$ be the cell complex associated to admissible Wilson loop diagrams with k propagators on n vertices. Let B be a cell of dimension $3k-1$, let W and...

conjecture | open | MathDB 页面 | 原文线索

V2-064716 | MATHDB-338599

Dai-He's large deviations exponent conjecture for the Halfin-Whitt diffusion

摘录 / EXCERPT: Let $W^{\infty}(\infty)$ denote the stationary workload of the diffusion limit arising from a $G/PH/n+G$ queue in the Halfin-Whitt regime. Assume the expected service time is 1;...

conjecture | open | MathDB 页面 | 原文线索

V2-064717 | MATHDB-338606

Recurrence conjecture for $(s,s+r)$-core partitions with d-distinct parts

摘录 / EXCERPT: For integers s, r, and d with $1\le r\le d$, let $N_{d,r}(s)$ denote the number of $(s,s+r)$-core partitions with d-distinct parts. Recurrence conjecture. The values of...

conjecture | open | MathDB 页面 | 原文线索

V2-064718 | MATHDB-338608

Positive-capacity Julia set conjecture for quasiregular maps

摘录 / EXCERPT: Positive-capacity Julia set conjecture. For every such map,

conjecture | open | MathDB 页面 | 原文线索

V2-064719 | MATHDB-338617

Dau et al.'s GM-MDS conjecture for zero patterns

摘录 / EXCERPT: GM-MDS conjecture. Let $S_1,\ldots,S_k\subset[n]$ be a set system satisfying the MDS condition. Then, for every field $\mathbb{F}$ with $|\mathbb{F}|\ge n+k-1$, there exists a...

conjecture | open | MathDB 页面 | 原文线索

V2-064720 | MATHDB-338619

The vector GM-MDS conjecture

摘录 / EXCERPT: Vector GM-MDS conjecture. Let $\mathcal{V}\subset\mathbb{N}^n$ and $k\ge1$. If $\mathcal{V}$ satisfies $V(k)$, then the polynomials in $P(k,\mathcal{V})$ are linearly independent o...

conjecture | open | MathDB 页面 | 原文线索

V2-064721 | MATHDB-338621

ACC conjecture for a-lc thresholds on smooth threefolds

摘录 / EXCERPT: Let $P\in X$ be the germ of a smooth threefold, and let $\mathfrak{a}$ and $\mathfrak{b}$ be $\mathbf{R}$-ideals on X. For a product of $\mathbf{R}$-ideals, membership in I mea...

conjecture | open | MathDB 页面 | 原文线索

V2-064722 | MATHDB-338622

Mustaţă's uniform m-adic semicontinuity conjecture for minimal log discrepancies

摘录 / EXCERPT: Let $P\in X$ be the germ of a smooth threefold, let $\mathfrak{m}$ be the maximal ideal in $\mathscr{O}_X$ defining P, and let ... be $\mathbf{R}$-ideals on X. Fix a finite subse...

conjecture | open | MathDB 页面 | 原文线索

V2-064723 | MATHDB-338634

Dyer's braid-graph connectivity conjecture

摘录 / EXCERPT: Let (W,S) be a Coxeter system with positive root system Φ^+, and let R be a finite subset of Φ^+. Form the braid graph whose vertices are the total orders on R obta...

conjecture | open | MathDB 页面 | 原文线索

V2-064724 | MATHDB-338645

Existence of a representing Borel measure for multiply connected cavities

摘录 / EXCERPT: Let $\omega=\cup_{k=1}^N\omega_k$ be a multiply connected set, where the sets ω_k, for $k=1,\dots,N$, are nonintersecting simply connected open domains with $C^{1,1}$ bounda...

conjecture | open | MathDB 页面 | 原文线索

V2-064725 | MATHDB-338653

Gibbons' one-dimensional symmetry conjecture for uniformly convergent solutions

摘录 / EXCERPT: Let $u:\mathbb{R}^n\to[-1,1]$ be a solution of the Allen-Cahn equation in the whole of $\mathbb{R}^n$ such that ... uniformly with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-064726 | MATHDB-338662

The line-finding conjecture for cubic surfaces

摘录 / EXCERPT: Let $\mathcal{H}_{3,3}$ denote the moduli space of smooth cubic surfaces, and let $\mathcal{H}_{3,3}(r)$ denote the moduli space of smooth cubic surfaces together with r specifie...

conjecture | open | MathDB 页面 | 原文线索

V2-064727 | MATHDB-338663

Brauer's conjecture on the resolvent degree of the general polynomial

摘录 / EXCERPT: For each n, let $\mathcal{P}_n$ be the parameter space of degree-n polynomials and let $\widetilde{\mathcal{P}}_n\to\mathcal{P}_n$ be the associated cover parametrizing a root....

conjecture | open | MathDB 页面 | 原文线索

V2-064728 | MATHDB-338673

Refined generic level-curve contact conjecture

摘录 / EXCERPT: Let ϕ satisfy condition (A1). For $\lambda\in\mathbb{T}$, let $\mathcal{L}_{\lambda}(\phi)$ have L_{λ} components near $(1,1)$, and let $\mathcal{Z}_{\tilde{p}}$ have...

conjecture | open | MathDB 页面 | 原文线索

V2-064729 | MATHDB-338684

Walther-bound equality conjecture for essential line arrangements

摘录 / EXCERPT: Walther-bound equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064730 | MATHDB-338689

Radial minimizer conjecture for the total combined energy

摘录 / EXCERPT: Let $\mathbb{A}$ and $\mathbb{B}$ be the annular domains appearing in the class $\mathcal{F}(\mathbb{A},\mathbb{B})$ of admissible mappings, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064731 | MATHDB-338690

The equivariant local Iwasawa main conjecture for unramified extensions

摘录 / EXCERPT: The equivariant local Iwasawa main conjecture. There exists $\zeta^{(j)}_{L_\infty/K}\in K_1(\mathcal{Q}^c(\mathcal{G}))$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064732 | MATHDB-338691

The logarithmic Riemann-Hilbert correspondence conjecture for geometric local systems

摘录 / EXCERPT: Logarithmic Riemann-Hilbert correspondence conjecture. The two tensor functors from the category of geometric p-adic étale local systems on X to the category of regular integra...

conjecture | open | MathDB 页面 | 原文线索

V2-064733 | MATHDB-338705

Row-count symmetry in the tableau correspondence $Z\leftrightarrow M$

摘录 / EXCERPT: Let Z be a tableau, let $Z^\ast$ denote its Schützenberger dual tableau, and let $i^\ast$ denote the opposite number of i. Let M and R be the tableaux in the correspondence...

conjecture | open | MathDB 页面 | 原文线索

V2-064734 | MATHDB-338707

Fukaya-category embedding conjecture for the open Calabi-Yau threefold Y_Φ

摘录 / EXCERPT: Let S be a complex curve with marked-point divisor D, let $m\geq 1$, let Φ be an admissible generic point of the Hitchin base, and let Y_Φ be the associated open Cala...

conjecture | open | MathDB 页面 | 原文线索

V2-064735 | MATHDB-338715

The connection-blocking conjecture for homogeneous spaces

摘录 / EXCERPT: Connection-blocking conjecture. The space M is blockable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-064736 | MATHDB-338728

Labourie's vector-bundle conjecture for Hitchin components

摘录 / EXCERPT: Labourie's conjecture. The quotient of $\mathcal{T}_H(S,\operatorname{PSL}(n,\mathbf{R}))$ by the mapping class group is a holomorphic vector bundle over $\mathcal{M}(S)$, with fib...

conjecture | open | MathDB 页面 | 原文线索

V2-064737 | MATHDB-338740

The symmetric-power bigness conjecture for direct images of pluricanonical bundles

摘录 / EXCERPT: Symmetric-power bigness conjecture. The symmetric power

conjecture | open | MathDB 页面 | 原文线索

V2-064738 | MATHDB-338744

Borodin-Kostochka-Woodall's List Total Coloring Conjecture

摘录 / EXCERPT: Let G be a finite multigraph, and let $T(G)$ be its total graph. Write $\chi(T(G))$ for the chromatic number and $\chi_l(T(G))$ for the list chromatic number. Borodin-Kostochka-W...

conjecture | open | MathDB 页面 | 原文线索

V2-064739 | MATHDB-338745

The List Equitable Total Coloring Conjecture

摘录 / EXCERPT: Let G be a finite simple graph, let $T(G)$ be its total graph, and let $\chi_l(T(G))$ denote the list chromatic number of $T(G)$. A graph is equitably k-choosable if every list...

conjecture | open | MathDB 页面 | 原文线索

V2-064740 | MATHDB-338748

Real ETF positivity-or-negativity conjecture

摘录 / EXCERPT: Let D and N be positive integers with $N>2D>2$, and let a real $\operatorname{ETF}(D,N)$ denote a real equiangular tight frame of N vectors in dimension D. Call (D,N) pos...

conjecture | open | MathDB 页面 | 原文线索

V2-064741 | MATHDB-338751

Projected gradient descent convergence conjecture for matrix completion

摘录 / EXCERPT: Let $M^*\in\mathcal{S}^{d\times d}$ be symmetric positive semidefinite, and let Ω be generated under the model $\operatorname{SMC}(M^*,p)$: each (i,j) with $i\geq j$ is ob...

conjecture | open | MathDB 页面 | 原文线索

V2-064742 | MATHDB-338752

The Kato-surface characterization conjecture for class VII surfaces

摘录 / EXCERPT: Kato-surface characterization conjecture. Any minimal class VII surface with $b_2>0$ containing a cycle of rational curves is a Kato surface.

conjecture | open | MathDB 页面 | 原文线索

V2-064743 | MATHDB-338754

Clustered chromatic conjecture for Colin de Verdière classes

摘录 / EXCERPT: Let $\mathcal{V}_k:=\{G:\mu(G)\leqslant k\}$, where $\mu(G)$ is the Colin de Verdière parameter, and let $\chi_{\star}(\mathcal{V}_k)$ be the clustered chromatic number of this cla...

conjecture | open | MathDB 页面 | 原文线索

V2-064744 | MATHDB-338755

Ouyang-Ossona de Mendez-Wood connected tree-depth conjecture

摘录 / EXCERPT: For a graph H, let $\operatorname{\overline{td}}(H)$ be its connected tree-depth, and let $\mathcal{M}_H$ be the class of graphs containing no H minor. Let...

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V2-064745 | MATHDB-338756

NSSW excluded-configuration conjecture for minor-closed classes

摘录 / EXCERPT: Let $\mathcal{X}_{k,c}$ be the graph family defined in the source, and let $\mathcal{M}$ be a minor-closed class. NSSW excluded-configuration conjecture. Every minor-closed class t...

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V2-064746 | MATHDB-338759

The EDT0L nonmembership conjecture for rank-one free inverse monoid word problems

摘录 / EXCERPT: Let FIM_1 be the free inverse monoid of rank 1. Write $\operatorname{WP}(\operatorname{FIM}_1)$ and $\iota(\operatorname{FIM}_1)$ for its two word problem langua...

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V2-064747 | MATHDB-338763

Explicit formula conjecture for truncated Ehrhart-like polynomials

摘录 / EXCERPT: Truncated Ehrhart-like polynomial conjecture. For every $\lambda\in P$ and every good $\mathbf{k}\in\mathbb{N}[\Phi]^W$,

conjecture | open | MathDB 页面 | 原文线索

V2-064748 | MATHDB-338775

Conjecture on dyadic local time for fractional Brownian motion

摘录 / EXCERPT: The dyadic local-time conjecture. Almost surely, $\widetilde{L}^{\pi_n}_t(x)$ converges uniformly for $(t,x)\in[0,T]\times\mathbb{R}$ to

conjecture | open | MathDB 页面 | 原文线索

V2-064749 | MATHDB-338783

The representation-to-TD-Delaunay conjecture

摘录 / EXCERPT: Let $H_d\simeq \mathbb{R}^{d-1}$ be the hyperplane used for the d-dimensional construction, let R be a d-representation, and let $\Sigma(R)$ be the abstract simplicial comple...

conjecture | open | MathDB 页面 | 原文线索

V2-064750 | MATHDB-338794

Higher-dimensional Morse-Smale conjecture for dilation surfaces and affine interval exchanges

摘录 / EXCERPT: Let g and n be fixed such that the moduli space of dilation surfaces $\mathcal{MD}_{g,n}$ is non-empty. For $k\geq 3$, consider k-affine interval exchange transformations and...

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V2-064751 | MATHDB-338807

Conjecture on Tracy-Widom fluctuations in the i.i.d. model

摘录 / EXCERPT: Let G_n be the hexagonal-lattice rhombus with $2n-1$ levels, and let H_n be its weighted adjacency operator. In the i.i.d. model, all edge weights are independent copies of a n...

conjecture | open | MathDB 页面 | 原文线索

V2-064752 | MATHDB-338808

Conjecture on Airy point-process fluctuations of higher eigenvalues

摘录 / EXCERPT: Higher-eigenvalue Airy-process conjecture. For any fixed k, as $n\to\infty$, the tuple $(\alpha_{n,1},\ldots,\alpha_{n,k})$ converges in distribution to the top k points of the...

conjecture | open | MathDB 页面 | 原文线索

V2-064753 | MATHDB-338819

Pollack's signed strong main conjecture

摘录 / EXCERPT: Let E be an elliptic curve with $a_p(E)=0$, and assume the conditions of Kurihara's strong main conjecture. Let ω_n and $\widetilde{\omega}_n^{\pm}$ be the signed cyclotom...

conjecture | open | MathDB 页面 | 原文线索

V2-064754 | MATHDB-338820

Mazur-Tate's vanishing-order conjecture

摘录 / EXCERPT: Let $\mathbb{Q}_n$ be a finite cyclotomic layer, let χ be a character of $\operatorname{Gal}(\mathbb{Q}_n/\mathbb{Q})$, and let I_χ be the kernel of the induced map from...

conjecture | open | MathDB 页面 | 原文线索

V2-064755 | MATHDB-338822

The quadratic-harmonic versus Seiffert mean conjecture

摘录 / EXCERPT: Quadratic-harmonic versus Seiffert mean conjecture. For all positive real numbers x and y,

conjecture | open | MathDB 页面 | 原文线索

V2-064756 | MATHDB-338823

The exponent-range conjecture for the four classical means

摘录 / EXCERPT: Exponent-range conjecture for the four classical means. The inequality holds for every negative real number n and every real number $n\geq\frac12$. Moreover, for every real numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-064757 | MATHDB-338826

Divisibility and palindromicity conjecture for local del Pezzo moduli spaces

摘录 / EXCERPT: Divisibility and palindromicity conjecture. The Poincaré polynomial $P_t(M_\beta)$ has a factor $P_t(\mathbb{P}^{w-1})$, and the quotient

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V2-064758 | MATHDB-338841

Erde's path-decomposition conjecture for hypercubes

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, with edge set of size $n2^{n-1}$. A path decomposition of Q_n is a decomposition of its edge set into paths of a common length. Erde's...

conjecture | open | MathDB 页面 | 原文线索

V2-064759 | MATHDB-338842

Ramras's fundamental-set conjecture for paths in hypercubes

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube and let $\operatorname{Aut}(Q_n)$ denote its automorphism group. A subset F of $E(Q_n)$ is a fundamental set with respect to a subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-064760 | MATHDB-338843

Conjecture of simple zeros for generalized Jacobi polynomials

摘录 / EXCERPT: Simple-zero conjecture. For any partitions λ and μ, take $\alpha>-1$ and $\beta>m_1$ such that the stated conditions are satisfied. If λ is an even partition, t...

conjecture | open | MathDB 页面 | 原文线索

V2-064761 | MATHDB-338847

Affine Gaudin eigenvector conjecture for Schechtman-Varchenko vectors

摘录 / EXCERPT: Affine Gaudin eigenvector conjecture. For every exponent i and every permitted contour $\boldsymbol{\beta}$,

conjecture | open | MathDB 页面 | 原文线索

V2-064762 | MATHDB-338856

Bapat-Lal's monotonicity conjecture for the μ-permanent

摘录 / EXCERPT: Let A be an $n\times n$ Hermitian positive definite nondiagonal matrix, and let $P_\mu(A)$ denote its μ-permanent. Bapat-Lal's conjecture. $P_\mu(A)$ is a strictly increasing fun...

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V2-064763 | MATHDB-338857

de Falco's extension of the monotonicity conjecture for the μ-permanent

摘录 / EXCERPT: Let A be a positive definite matrix, and let $P_\mu(A)$ denote its μ-permanent. de Falco's conjecture. There exists $\epsilon<-1$ such that $P_\mu(A)$ is a strictly increasing fu...

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V2-064764 | MATHDB-338858

Bapat-Lal's Soules-type conjecture for the μ-permanent

摘录 / EXCERPT: Let A be an $n\times n$ Hermitian positive semidefinite matrix, let $\Pi_\mu(A)$ be its μ-Schur power matrix, and let $\mu\in[0,1]$. Bapat-Lal's conjecture. The largest eigenvalu...

conjecture | open | MathDB 页面 | 原文线索

V2-064765 | MATHDB-338862

Charged-representation anomaly formula for elliptic Calabi-Yau threefolds

摘录 / EXCERPT: Let $X\to B$ be an elliptic Calabi-Yau threefold with $\mathbb{Q}$-factorial terminal singularities $\{P\}$, the relative minimal model of a Weierstrass model $W\to B$. Assume that...

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V2-064766 | MATHDB-338873

Parabolic aligned elements form a lattice quotient of parabolic weak order

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let $J\subseteq S$, and let $c\in W$ be a Coxeter element. An element of W^J is (W^J,c)-aligned if, whenever...

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V2-064767 | MATHDB-338874

Parabolic aligned elements and flips of the parabolic cluster complex

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let $J\subseteq S$, and let $c\in W$ be a Coxeter element. The (W^J,c)-cluster complex is the subword complex … and its facets carry the f...

conjecture | open | MathDB 页面 | 原文线索

V2-064768 | MATHDB-338881

Reduction of maximal non-lc ideals to non-F-pure ideals

摘录 / EXCERPT: Maximal non-lc reduction conjecture. The reduction of the maximal non-lc ideal filtration coincides with the non-F-pure ideal filtration on a dense set of closed fibers, as asser...

conjecture | open | MathDB 页面 | 原文线索

V2-064769 | MATHDB-338888

Ultradistribution characterisation conjecture for the sub-Laplacian on the Heisenberg group

摘录 / EXCERPT: Let $\mathbb H_n$ be the Heisenberg group, let $\gamma^{s}_{\mathbf{X},L^2}(\mathbb H_n)$ and $\gamma^{(s)}_{\mathbf{X},L^2}(\mathbb H_n)$ be the corresponding sub-Laplacian Gevrey...

conjecture | open | MathDB 页面 | 原文线索

V2-064770 | MATHDB-338889

Lang's real analogue of the Tsen-Lang conjecture

摘录 / EXCERPT: Lang's conjecture. The function field $\mathbb R(X)$ is C_i.

conjecture | open | MathDB 页面 | 原文线索

V2-064771 | MATHDB-338891

Commutativity conjecture for Heisenberg functors on moduli spaces of sheaves

摘录 / EXCERPT: Let S be the smooth projective surface and let $\CM$ denote the relevant smooth moduli space of sheaves. For each $n\in\BN$, write $e_{(0,...,0)}$ for the functor associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-064772 | MATHDB-338892

The centrosymmetric growth-rate inequality for permutation classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a permutation class. Write $\overline{\operatorname{gr}}(\mathcal{C})$ for its upper growth rate and $\overline{\operatorname{gr}}^{rc}(\mathcal{C})$ for the u...

conjecture | open | MathDB 页面 | 原文线索

V2-064773 | MATHDB-338893

The sum-closed centrosymmetric growth-rate equality conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a sum-closed and rc-invariant permutation class. Let $\operatorname{gr}(\mathcal{C})$ denote its growth rate and $\operatorname{gr}^{rc}(\mathcal{C})$ the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-064774 | MATHDB-338894

The first threshold conjecture for unbounded indecomposable permutations

摘录 / EXCERPT: Let $\mathcal{C}$ be a sum-closed permutation class, and let $\mathcal{C}^{\not\oplus}$ denote its indecomposable permutations. Define τ to be the unique positive root of ... so...

conjecture | open | MathDB 页面 | 原文线索

V2-064775 | MATHDB-338895

The linear lower-bound conjecture for indecomposable permutations

摘录 / EXCERPT: Let $\mathcal{C}$ be a sum-closed permutation class, and let $\mathcal{C}^{\not\oplus}$ denote its indecomposable permutations. Linear indecomposable-count conjecture. If the seque...

conjecture | open | MathDB 页面 | 原文线索

V2-064776 | MATHDB-338916

Hod-Piran asymptotic behaviour conjecture for weakly charged scalar fields

摘录 / EXCERPT: Let (ϕ,F) be a solution of the Maxwell-Charged-Scalar field system with no particular symmetry assumption on a Reissner--Nordström space-time. Define the Maxwell charge Q by...

conjecture | open | MathDB 页面 | 原文线索

V2-064777 | MATHDB-338918

The alternating Turán-type inequality for normalized Euler numbers

摘录 / EXCERPT: Alternating Turán-type inequality. For all $n\geq 0$ and $j,k\geq 1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-064778 | MATHDB-338923

Plancherel-shape conjecture for large Kronecker coefficients

摘录 / EXCERPT: Let $\{\lambda^{(n)}\vdash n\}$, $\{\mu^{(n)}\vdash n\}$, and $\{\nu^{(n)}\vdash n\}$ be three Plancherel partition sequences. The Plancherel Kronecker conjecture. Their Kronecker...

conjecture | open | MathDB 页面 | 原文线索

V2-064779 | MATHDB-338924

Vanishing conjecture for Littlewood-Richardson coefficients of Plancherel partitions

摘录 / EXCERPT: Let $\{\lambda^{(n)}\vdash n\}$, $\{\mu^{(n)}\vdash n/2\}$, and $\{\nu^{(n)}\vdash n/2\}$ be Plancherel partition sequences. The Plancherel LR vanishing conjecture. There exist suc...

conjecture | open | MathDB 页面 | 原文线索

V2-064780 | MATHDB-338925

Flag conjecture for largest Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $\lambda\vdash n$, and let μ and ν be partitions with $c^\lambda_{\mu,\nu}={\rm C}(n)$, where ${\rm C}(n)$ denotes the largest Littlewood-Richardson coefficient at size...

conjecture | open | MathDB 页面 | 原文线索

V2-064781 | MATHDB-338935

Painlevé II tau-function expansion from ramified irregular conformal blocks

摘录 / EXCERPT: Let G be the Barnes G-function, and let a be a complex number satisfying $a^{3/2}=-\sqrt{-2}$. Let $\Phi^{\Delta,\beta}_{\Lambda',\Lambda}$ denote the indicated ramified irre...

conjecture | open | MathDB 页面 | 原文线索

V2-064782 | MATHDB-338956

Eisenstein congruence conjecture for the Klingen parabolic of U(2,2)

摘录 / EXCERPT: Klingen Eisenstein congruence conjecture. There exists a Hecke eigenform $v\in M(V_{a,b},K_L)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064783 | MATHDB-338961

The generalized restricted-partition recurrence for unimodal polynomials

摘录 / EXCERPT: Let $G_s(n,k)$ be the recurrence obtained by summing over partitions of k having size at most s, with q an indeterminate and n,k,s in the domain where this recurrence is de...

conjecture | open | MathDB 页面 | 原文线索

V2-064784 | MATHDB-338964

Quadratic-threshold conjecture for poset partitions of the Boolean lattice

摘录 / EXCERPT: Quadratic-threshold conjecture. There exists a constant c such that, whenever

conjecture | open | MathDB 页面 | 原文线索

V2-064785 | MATHDB-338965

Quadratic-threshold conjecture for near-perfect poset packings

摘录 / EXCERPT: Quadratic near-perfect packing conjecture. There exists a constant c such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-064786 | MATHDB-338967

The freeness conjecture for rational cuspidal plane curves

摘录 / EXCERPT: Let $\mathcal C$ be a rational cuspidal curve in the plane. Here, rational means that $\mathcal C$ has genus zero, and cuspidal means that all its singularities are cusps. Freeness...

conjecture | open | MathDB 页面 | 原文线索

V2-064787 | MATHDB-338968

The conjecture on primitive formally dual pairs in cyclic groups

摘录 / EXCERPT: Let G be a finite cyclic group, and let $S,T\subseteq G$ be a primitive formally dual pair, meaning a formally dual pair that is not obtained from a formally dual pair in a prope...

conjecture | open | MathDB 页面 | 原文线索

V2-064788 | MATHDB-338974

Snake-decomposition characterization of c-Cambrian order element posets

摘录 / EXCERPT: Snake-decomposition characterization. A Φ-poset $\mathrel{\mathop{\rel}}\in\mathcal{P}(\Phi)$ is in $\mathsf{COEP}(c)$ if and only if it is in $\mathsf{COIP}(c)$ and every roo...

conjecture | open | MathDB 页面 | 原文线索

V2-064789 | MATHDB-338975

Sublattice conjecture for c-Cambrian order element posets

摘录 / EXCERPT: Sublattice conjecture for c-Cambrian order element posets. For every Coxeter element c, the set $\mathsf{COEP}(c)$ induces a sublattice of the weak order on $\mathcal{P}(\Phi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-064790 | MATHDB-338976

Sublattice conjecture for c-Cambrian order interval posets

摘录 / EXCERPT: Sublattice conjecture for c-Cambrian order interval posets. For every Coxeter element c, the set $\mathsf{COIP}(c)$ induces a sublattice of the weak order on $\mathcal{P}(\Phi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-064791 | MATHDB-338979

Generalized Kaplansky conjectures for cellular-automaton near rings

摘录 / EXCERPT: Let K be a field and G a group. Consider the near ring $(R(K,G),+,\star)$. For $g\in G$, write X_g for the corresponding basis element, let $K^*=K\setminus\{0\}$, and call el...

conjecture | open | MathDB 页面 | 原文线索

V2-064792 | MATHDB-338981

The closed formula for the refined tree index

摘录 / EXCERPT: Let γ_i be charge vectors, let $F_{{\rm tr},n}(\{\gamma_i\},z)$ be the partial tree index, let κ be the two-centered index factor, and define ... Let ${\rm Sym}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-064793 | MATHDB-338982

The flow tree formula for quiver moduli

摘录 / EXCERPT: Let a K-node quiver have adjacency matrix α_{ab}, dimension vector $\gamma=(N_1,\dots,N_K)$, stability parameters $\zeta=(\zeta_1,\dots,\zeta_K)$, and generic superpotenti...

conjecture | open | MathDB 页面 | 原文线索

V2-064794 | MATHDB-338983

The Coulomb branch formula for rational BPS indices

摘录 / EXCERPT: The Coulomb branch formula. The rational index is conjectured to be

conjecture | open | MathDB 页面 | 原文线索

V2-064795 | MATHDB-338984

Full rectangular Lefschetz collection conjecture for odd isotropic Grassmannians

摘录 / EXCERPT: Let $\IGr(n,2n+1)$ denote the odd isotropic Grassmannian parametrizing n-dimensional isotropic subspaces of a $(2n+1)$-dimensional vector space equipped with a nondegenerate skew...

conjecture | open | MathDB 页面 | 原文线索

V2-064796 | MATHDB-338985

Exotic edge-operator conjecture for analytic beta ensembles

摘录 / EXCERPT: Exotic edge-operator conjecture. After the appropriate scaling, T_n converges to the random operator $\mathcal{S}_{\beta,k}$. The source presents this as a conjecture and refers...

conjecture | open | MathDB 页面 | 原文线索

V2-064797 | MATHDB-338995

Ginzburg-Rallis conjecture on the exterior cube L-function and period

摘录 / EXCERPT: Ginzburg-Rallis conjecture. The following are equivalent:

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V2-064798 | MATHDB-339000

Strict complete monotonicity conjecture for F_3

摘录 / EXCERPT: Let F_3 be the function defined in Theorem 3 of the source paper. A function is strictly completely monotonic on an interval if all of its derivatives have alternating signs ther...

conjecture | open | MathDB 页面 | 原文线索

V2-064799 | MATHDB-339001

The Bernardi-process conjecture for determining ribbon-graph genus

摘录 / EXCERPT: Let (G,ρ) be a ribbon graph. Write $V(G)$ for its vertices, $E(G)$ for its edges, $\operatorname{Pic}^0(G)$ for its degree-zero Picard group, and $\mathcal{T}(G)\subset E(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-064800 | MATHDB-339005

Characteristic-zero purity conjecture for K3 unions of scrolls

摘录 / EXCERPT: Characteristic-zero purity conjecture. The union of scrolls $X_e(a,a)$ has a pure resolution unless $t_2/t_1\neq 1$ is a k-th root of unity for some $k\leq (a+1)/2$.

conjecture | open | MathDB 页面 | 原文线索

V2-064801 | MATHDB-339007

Co-Cartesian ramification-filtration diagrams for purely inseparable extensions

摘录 / EXCERPT: Let K'/K be a geometric, purely inseparable, untwisted extension of geometric discrete valuation fields with finite exponent. Write e for its ramification index, f for its du...

conjecture | open | MathDB 页面 | 原文线索

V2-064802 | MATHDB-339011

Savage's Ehrhart-positivity conjecture for base-r simplices

摘录 / EXCERPT: Savage's conjecture. For $r\geq 2$ and $d\geq 1$, the base-r d-simplex $\mathcal{B}_{(r,d)}$ is Ehrhart positive.

conjecture | open | MathDB 页面 | 原文线索

V2-064803 | MATHDB-339019

Conjecture on the binary two-user adder channel with complete feedback

摘录 / EXCERPT: Consider the two-user binary adder channel with complete feedback, and let $R({\mathcal O}_f)$ denote its average zero-error capacity. Binary adder-channel conjecture. The average...

conjecture | open | MathDB 页面 | 原文线索

V2-064804 | MATHDB-339020

Conjecture that feedback improves the average zero-error capacity of the two-user union channel

摘录 / EXCERPT: For $q\geq 2$, let $R({\mathcal E})$ be the average channel capacity of the two-user union channel without feedback, and let $R({\mathcal O}_f)$ be the average zero-error capacity...

conjecture | open | MathDB 页面 | 原文线索

V2-064805 | MATHDB-339024

Wang-Lih conjecture on 4-choosability of planar graphs without adjacent triangles

摘录 / EXCERPT: Wang-Lih conjecture. Every planar graph without adjacent triangles is 4-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-064806 | MATHDB-339033

Continuous evolution conjecture for discrete eigenvalues of Schrödinger operators

摘录 / EXCERPT: Let V_0 and V_1 be potentials supported on $[0,a]$, and let $H(t)=-\frac{d^2}{dx^2}+V_0+tV_1$, with $H(0)$ self-adjoint. Suppose that $V_1>0$ on $[0,a]$ or $V_1<0$ on $[0,a]$....

conjecture | open | MathDB 页面 | 原文线索

V2-064807 | MATHDB-339039

Characterization conjecture for right-angled Artin groups with MCF word problem

摘录 / EXCERPT: Let Γ be a finite graph, and let $A(\Gamma)$ be the associated right-angled Artin group. Say that a group has MCF word problem when its word problem is a multiple context-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-064808 | MATHDB-339049

Edelman-Greene maximality and height conjecture

摘录 / EXCERPT: Edelman-Greene height conjecture. The word $\mathrm{EG}^{-1}(Q)$ is a maximal element of $\mathcal{P}_Q$, and

conjecture | open | MathDB 页面 | 原文线索

V2-064809 | MATHDB-339059

Conjecture on possible rational roots of alternating sign triangle weight functions

摘录 / EXCERPT: Possible-roots conjecture. For M, the rational roots of $w_I(M)$ lie in

conjecture | open | MathDB 页面 | 原文线索

V2-064810 | MATHDB-339060

Divisibility characterization for centred Catalan sets

摘录 / EXCERPT: Divisibility conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-064811 | MATHDB-339061

Divisibility patterns for terminal segments of Motzkin paths

摘录 / EXCERPT: Terminal-pattern divisibility conjecture. The following implications hold:

conjecture | open | MathDB 页面 | 原文线索

V2-064812 | MATHDB-339064

Diagonal-positivity criterion for three-variable symmetric rational functions

摘录 / EXCERPT: Diagonal-positivity conjecture. The series diag_F is nonnegative if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-064813 | MATHDB-339067

The b-semi-ampleness conjecture for moduli b-divisors

摘录 / EXCERPT: Let $f:(X, B)\to Y$ be a basic slc-trivial fibration, and let $\mathbf M$ denote its moduli part, a moduli $\mathbb Q$-b-divisor on Y. B-semi-ampleness conjecture. The moduli par...

conjecture | open | MathDB 页面 | 原文线索

V2-064814 | MATHDB-339069

The geometric griddability and finite-letter-graph conjecture

摘录 / EXCERPT: Geometric griddability and finite-letter-graph conjecture. X is geometrically griddable if and only if ${\cal G}_X$ is a class of k-letter graphs for some finite value of k.

conjecture | open | MathDB 页面 | 原文线索

V2-064815 | MATHDB-339081

Lagarias' Periodicity Conjecture for the 3x+1 conjugacy map

摘录 / EXCERPT: Let $\mathbb Z_2$ denote the ring of 2-adic integers, and let $Q:\mathbb Z_2\to\mathbb Z_2$ be the 3x+1 conjugacy map. A 2-adic integer is rational when it belongs to $\mathbb Q$....

conjecture | open | MathDB 页面 | 原文线索

V2-064816 | MATHDB-339082

The 3x+1 problem in terms of the conjugacy map

摘录 / EXCERPT: Let $\mathbb Z^+$ denote the positive integers, let $\mathbb Z_2$ be the 2-adic integers, and let Q be the 3x+1 conjugacy map. The 3x+1 problem. One should have ... or equivalently...

open_problem | open | MathDB 页面 | 原文线索

V2-064817 | MATHDB-339086

Promotion conjecture for coherators with systems of inverses

摘录 / EXCERPT: Let $\mathfrak{D}$ be a coherator for $infty$-categories, and suppose it can be endowed with a system of inverses. A coherator is contractible when it is a coherator for $infty$-gr...

conjecture | open | MathDB 页面 | 原文线索

V2-064818 | MATHDB-339092

Casselgren-Toft's cyclic interval coloring conjecture for biregular graphs

摘录 / EXCERPT: Casselgren-Toft's conjecture. Every (a,b)-biregular graph admits a cyclic interval $\max\{a,b\}$-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-064819 | MATHDB-339093

The palette-index bound for biregular graphs

摘录 / EXCERPT: Biregular palette-index conjecture. For any (a,b)-biregular graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-064820 | MATHDB-339095

Closure relation conjecture for dual polynomials of multi-indexed $(q$-)Racah systems

摘录 / EXCERPT: Let $R_i(z)$, for $i=0,1,-1$, be the polynomials defined by the determinant formula in equation (Ri2), together with the preceding definitions of their coefficients, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064821 | MATHDB-339096

The barcode resolution conjecture for compactly supported piecewise linear sheaves

摘录 / EXCERPT: Barcode resolution conjecture. There exists a bounded complex

conjecture | open | MathDB 页面 | 原文线索

V2-064822 | MATHDB-339104

Jack Schur-expansion coefficient conjecture with real-rootedness

摘录 / EXCERPT: Let n be a positive integer, and let $\tilde{J}_\mu^{(\alpha)}(X)$ denote the Jack polynomial indexed by a partition μ of n. For partitions μ and λ of n, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-064823 | MATHDB-339105

Jack fundamental quasisymmetric expansion conjecture

摘录 / EXCERPT: Let n be a positive integer, let μ be a partition of size n, let S_n be the symmetric group, and let $Q_A(x)$ denote the fundamental quasisymmetric function indexed by a...

conjecture | open | MathDB 页面 | 原文线索

V2-064824 | MATHDB-339107

The plane-partition cell-count conjecture for amplituhedron triangulations

摘录 / EXCERPT: Let $M(a,b,c)$ denote the number of plane partitions fitting in an $a\times b\times c$ box, namely ... Let $Z\in\operatorname{Gr}_{>0}(k+m,n)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064825 | MATHDB-339109

The amplituhedron form triangulation-independence conjecture

摘录 / EXCERPT: Amplituhedron form conjecture. The form $\omega^{(\mathcal{T})}_{\mathcal{A}_{n,k,m}(Z)}$ does not depend on the choice of triangulation $\mathcal{T}$. The form is intended to be a...

conjecture | open | MathDB 页面 | 原文线索

V2-064826 | MATHDB-339113

Bistellar flip conjecture for amplituhedron triangulations

摘录 / EXCERPT: Let $g\in\tilde{S_n}(-k,n-k)$ be nearly (n,k,m)-admissible, meaning that $\operatorname{inv}(g)=k\ell-1$, $\operatorname{dim}(Z(\Pi^{>0}_{k+g}))=km$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-064827 | MATHDB-339114

Flip-connectivity conjecture for amplituhedron triangulations

摘录 / EXCERPT: A flip replaces one of the two triangulations associated with a nearly (n,k,m)-admissible affine permutation by the other. Flip-connectivity conjecture. Any two triangulations of...

conjecture | open | MathDB 页面 | 原文线索

V2-064828 | MATHDB-339115

BCFW triangulation conjecture for amplituhedra in four dimensions

摘录 / EXCERPT: For $m=4$, let $\mathcal{T}_{k,\ell}=\{f_1,\dots,f_N\}$ be the BCFW collection of positroid cells, with $\mathcal{T}_{\ell,k}=\{f_1^{-1},\dots,f_N^{-1}\}$ up to cyclic shift. BCFW...

conjecture | open | MathDB 页面 | 原文线索

V2-064829 | MATHDB-339117

Weyl-Schanuel law for cuspidal automorphic representations of general linear groups

摘录 / EXCERPT: Weyl-Schanuel law. As $Q\rightarrow\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-064830 | MATHDB-339130

Restricted Yano conjecture for generic plane curve singularities with simple monodromy

摘录 / EXCERPT: Restricted Yano conjecture. The b-exponents $\{\tilde{\beta}_1,\ldots,\tilde{\beta}_\mu\}$ of a generic element of Σ_μ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-064831 | MATHDB-339140

The integrality and zero-block conjecture for quadratic lattice extensions

摘录 / EXCERPT: Let $L=M\oplus N$ be a quadratic lattice, where M is spanned by $(e_1,\ldots,e_{n-d})$ and N by $(e_{n-d+1},\ldots,e_n)$. Let $L'=M'\oplus N$ be a lattice containing L in...

conjecture | open | MathDB 页面 | 原文线索

V2-064832 | MATHDB-339149

The admissible-lift conjecture for local-global convergence

摘录 / EXCERPT: Admissible-lift conjecture. Every local-global convergent sequence of graphs in a nowhere dense class has a modeling limit.

conjecture | open | MathDB 页面 | 原文线索

V2-064833 | MATHDB-339156

Unique rectification targets in d-complete posets

摘录 / EXCERPT: Let $\mathcal{P}$ be a d-complete poset and let $\lambda \subseteq \mathcal{P}$ be an order ideal. A rectification target is an increasing $\mathcal{P}$-tableau obtained as the r...

conjecture | open | MathDB 页面 | 原文线索

V2-064834 | MATHDB-339158

Bottom-tree invariance of rectification in d-complete posets

摘录 / EXCERPT: Let $\mathcal{P}$ be a d-complete poset with bottom tree $\mathcal{B}$. Let T be a skew increasing $\mathcal{P}$-tableau, and let R and S be rectifications of T. Bottom-t...

conjecture | open | MathDB 页面 | 原文线索

V2-064835 | MATHDB-339171

The anti-van der Waerden bound for Cartesian products of connected graphs

摘录 / EXCERPT: Cartesian-product anti-van der Waerden conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064836 | MATHDB-339175

M. Stein's dimension conjecture for linear relations among recursive-sequence Dirichlet series

摘录 / EXCERPT: M. Stein's conjecture. One has $\dim_{\mathbb{Q}}V_m=m$ for every positive integer m. Moreover, for every $\left(t_1,\ldots,t_{4m}\right)\in V_m$,

conjecture | open | MathDB 页面 | 原文线索

V2-064837 | MATHDB-339176

Polynomial growth relations are ET0L-uniform

摘录 / EXCERPT: Let $p:\mathbb{N}\rightarrow\mathbb{N}$ be a polynomial function. The relation associated with p is denoted by ρ_p. Polynomial ET0L conjecture. ... The claim extends the exhib...

conjecture | open | MathDB 页面 | 原文线索

V2-064838 | MATHDB-339177

Sorting relations eventually escape ET0L and linear languages

摘录 / EXCERPT: Sorting-language hierarchy conjecture. For sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-064839 | MATHDB-339178

Uniform indexed relations properly lie below transduced indexed relations

摘录 / EXCERPT: Let X be an alphabet, and let $U(\mathcal{I})$ and $T(\mathcal{I})$ denote the corresponding uniform and transduced classes of binary relations generated from indexed languages....

conjecture | open | MathDB 页面 | 原文线索

V2-064840 | MATHDB-339180

Transduced linear and uniform indexed relations are incomparable

摘录 / EXCERPT: Let $T(\mathcal{LIN})$ and $U(\mathcal{I})$ denote the transduced linear and uniform indexed classes of binary relations. Linear-indexed incomparability conjecture. ... The claim con...

conjecture | open | MathDB 页面 | 原文线索

V2-064841 | MATHDB-339181

Brownian-bridge perturbative expansion conjecture for TASEP cumulants

摘录 / EXCERPT: Let n be a non-negative integer and let h_0 be an arbitrary sufficiently regular continuous function satisfying ... For independent standard Brownian bridges $b_1,\ldots,b_n$, de...

conjecture | open | MathDB 页面 | 原文线索

V2-064842 | MATHDB-339182

Flat-initial-condition Brownian-bridge expansion conjecture

摘录 / EXCERPT: Let n be a non-negative integer. For independent standard Brownian bridges $b_0,\ldots,b_{2n}$, define ... Here $\max$ denotes the maximum over the bridge parameter. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064843 | MATHDB-339183

Stationary-initial-condition Brownian-bridge expansion conjecture

摘录 / EXCERPT: Let n be a non-negative integer. For independent standard Brownian bridges $b_0,\ldots,b_{2n}$, define ... Here b is a standard Brownian bridge whose average is represented by...

conjecture | open | MathDB 页面 | 原文线索

V2-064844 | MATHDB-339184

Gruslys-Leader-Tan bounded-dimension conjecture for one-dimensional tiles

摘录 / EXCERPT: A tile is a finite non-empty subset of $\mathbb{Z}$, and a tile T tiles $\mathbb{Z}^d$ if $\mathbb{Z}^d$ can be partitioned into translates, rotations, and reflections of T. Gr...

conjecture | open | MathDB 页面 | 原文线索

V2-064845 | MATHDB-339206

Gogolev-Maimon-Kolgomorov conjecture on transitivity of the strong unstable foliation

摘录 / EXCERPT: Gogolev-Maimon-Kolgomorov conjecture. For all analytic diffeomorphisms f in a sufficiently small neighborhood of A, the strong unstable foliation ${\mathcal W}^{uu}$ is transit...

conjecture | open | MathDB 页面 | 原文线索

V2-064846 | MATHDB-339219

Red-path realization conjecture for maximal green sequences

摘录 / EXCERPT: Red-path realization conjecture. Every maximal green sequence in $\mathcal{A}$ is induced by a red path in the wall and chamber structure of $\mathcal{A}$.

conjecture | open | MathDB 页面 | 原文线索

V2-064847 | MATHDB-339225

The GW/DT4 conjecture for genus-zero invariants of Calabi-Yau fourfolds

摘录 / EXCERPT: Let X be a sextic 4-fold, let β be a curve class, and let $\rho\in H^{4}(X,\mathbb{Z})$. Denote by $\mathrm{GW}_{0,\beta}(\gamma)$ the genus-zero Gromov-Witten invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-064848 | MATHDB-339231

The local antimagic chromatic number of generalized book graphs

摘录 / EXCERPT: Let $GB(a_1,a_2,\ldots,a_r)$ denote the generalized book graph obtained by edge-gluing cycles of orders $a_1,a_2,\ldots,a_r$, where $a_1\ge a_2\ge \cdots\ge a_r\ge 3$ and $r\ge 2$....

conjecture | open | MathDB 页面 | 原文线索

V2-064849 | MATHDB-339233

The join-graph characterization of local antimagic chromatic number

摘录 / EXCERPT: Join characterization. For $n\ge 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-064850 | MATHDB-339234

The complete-join characterization of local antimagic chromatic number

摘录 / EXCERPT: Complete-join characterization. For $n\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-064851 | MATHDB-339235

The even-order M{"o}bius ladder local antimagic chromatic number conjecture

摘录 / EXCERPT: M{"o}bius ladder conjecture. For even $n\ge 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-064852 | MATHDB-339241

Existence conjecture for weak solutions of the coupled p-Laplacian system

摘录 / EXCERPT: Existence conjecture. There exists $1<\underline{m}<(p^*)'$ such that, if $f\in L^m(\Omega)$ with $m\geq\underline{m}$, then system $$ admits a weak solution...

conjecture | open | MathDB 页面 | 原文线索

V2-064853 | MATHDB-339263

Duchêne-Kheddouci-Nowakowski-Tahraoui labeled packing conjecture for cycles

摘录 / EXCERPT: Let C_n be a cycle of order $n=2k+x$, with $k\geq2$ and $1\leq x\leq2k-1$. The parameter $\lambda^k(C_n)$ is the largest number of labels in a labeling of the vertices that admit...

conjecture | open | MathDB 页面 | 原文线索

V2-064854 | MATHDB-339266

Fan-Raspaud edge multiplicity conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Edge multiplicity conjecture. There is an FR-triple C of G such that $\nu_C(e)=i$.

conjecture | open | MathDB 页面 | 原文线索

V2-064855 | MATHDB-339267

Fan-Raspaud adjacent-edge multiplicity conjecture

摘录 / EXCERPT: Adjacent-edge multiplicity conjecture. The graph G contains an FR-triple C such that

conjecture | open | MathDB 页面 | 原文线索

V2-064856 | MATHDB-339282

The arbitrary-Newton-polygon conjecture for generic specializations

摘录 / EXCERPT: The arbitrary-Newton-polygon conjecture. For an arbitrary Newton polygon ξ, let $w\in{}^J W$ be a generic specialization of w_ξ. Then there exists a Newton polygon ζ...

conjecture | open | MathDB 页面 | 原文线索

V2-064857 | MATHDB-339289

Biran-Cornea identification of the discrete s-dot operads

摘录 / EXCERPT: Biran-Cornea identification conjecture. The categories T^s of Biran and Cornea are the discrete colored planar operads associated to the s-dot constructions of...

conjecture | open | MathDB 页面 | 原文线索

V2-064858 | MATHDB-339290

Cheung's conjecture on divergent trajectories in homogeneous spaces

摘录 / EXCERPT: Let G be a connected Lie group, let Γ be a lattice in G, and let $F=\{f_t:t\in\mathbb{R}\}$ be a one-parameter subgroup of G. Write $F^+=\{f_t:t\geq 0\}$ and let ... Her...

conjecture | open | MathDB 页面 | 原文线索

V2-064859 | MATHDB-339291

Akbari-Alipour conjecture on generalized Latin square transversals

摘录 / EXCERPT: Let S be an $n\times n$ generalized Latin square, meaning an array with an arbitrary number of symbols in which no symbol appears twice in any row or column. A transversal is a s...

conjecture | open | MathDB 页面 | 原文线索

V2-064860 | MATHDB-339293

Rainbow forest packing conjecture for globally bounded graphs

摘录 / EXCERPT: Let $\varepsilon>0$. A graph is globally m-bounded if no colour is used on more than m edges, and a forest of order k has k edges. Rainbow forest packing conjecture. There...

open_problem | open | MathDB 页面 | 原文线索

V2-064861 | MATHDB-339304

The Metropolis-partition conjecture for vertices of the integer-partition polytope

摘录 / EXCERPT: Let $p(r)$ be the number of integer partitions of r. For even $n=2r$, let $m_2(n)$ denote the number of partitions obtained as sums of two partitions of r, disregarding the req...

conjecture | open | MathDB 页面 | 原文线索

V2-064862 | MATHDB-339305

The three-partition conjecture for knapsack partitions

摘录 / EXCERPT: Let $K(n)$ be the set of knapsack partitions of n, and let P_n be the convex hull of the integer partitions of n. The three-partition conjecture. The majority of partitions i...

conjecture | open | MathDB 页面 | 原文线索

V2-064863 | MATHDB-339308

Huh's polar-degree-two classification conjecture for projective hypersurfaces

摘录 / EXCERPT: Let $V\subset \mathbb{P}^{n}$ be a projective hypersurface with only isolated singularities, defined by a homogeneous polynomial f of degree d, and let the polar degree of V...

conjecture | open | MathDB 页面 | 原文线索

V2-064864 | MATHDB-339309

Huh's finiteness conjecture for fixed polar degree

摘录 / EXCERPT: Let $k>2$, let $V\subset\mathbb{P}^{n}$ be a projective hypersurface with only isolated singular points, and let $d=\deg V$. Huh's finiteness conjecture. For sufficiently large n...

conjecture | open | MathDB 页面 | 原文线索

V2-064865 | MATHDB-339312

The exact 3- and 4-connectivity conjecture for balanced hypercubes

摘录 / EXCERPT: Exact connectivity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064866 | MATHDB-339330

C^1-rigidity conjecture at critical points

摘录 / EXCERPT: Let f and $\tilde f$ be maps in the setting of the main theorems, and let c be a critical point. Write $\omega(c)$ for the omega-limit set of c, and call it of bounded geomet...

conjecture | open | MathDB 页面 | 原文线索

V2-064867 | MATHDB-339338

Doss's Wilks phenomenon conjecture for concave regression

摘录 / EXCERPT: Assume the regression model holds with $Ee^{t\epsilon_{n,i}^2}<\infty$ for some $t>0$. Let r_0 be concave, satisfy $r_0(x_0)=y_0$, be twice continuously differentiable in a neigh...

conjecture | open | MathDB 页面 | 原文线索

V2-064868 | MATHDB-339345

Anthony-Brightwell-Shawe-Taylor characterization conjecture for the minimum specification number

摘录 / EXCERPT: Let $f535^n$ denote the Boolean cube, let $f539_n$ be the class of threshold Boolean functions on n variables, and let $f539_{f539_n}(f)$ denote the specification number of f w...

conjecture | open | MathDB 页面 | 原文线索

V2-064869 | MATHDB-339349

Katz-Langevin-Lee-Sapozhnikov conjecture on the maximum of $V_{F,d}$

摘录 / EXCERPT: Katz-Langevin-Lee-Sapozhnikov conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064870 | MATHDB-339350

Symmetry conjecture for three-valued Weil sums

摘录 / EXCERPT: Symmetry conjecture. $W_{F,d}$ is symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-064871 | MATHDB-339351

Helleseth's three-valued conjecture

摘录 / EXCERPT: Helleseth's three-valued conjecture. If $[F:{\mathbb F}_p]$ is a power of 2, then there is no invertible exponent d over F such that $W_{F,d}$ is three-valued.

conjecture | open | MathDB 页面 | 原文线索

V2-064872 | MATHDB-339352

Helleseth's vanishing conjecture

摘录 / EXCERPT: Helleseth's vanishing conjecture. There is some $a\in F^*$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064873 | MATHDB-339364

Uniform boundedness conjecture for ranks of quadratic twists

摘录 / EXCERPT: Uniform boundedness conjecture for ranks of quadratic twists. There is a positive integer $R=R(E,k)$ such that every elliptic curve E' over k that is a quadratic twist of E s...

conjecture | open | MathDB 页面 | 原文线索

V2-064874 | MATHDB-339380

Law of large numbers for the range of rotor walks in the null recurrent case

摘录 / EXCERPT: Let G be a finite graph on N vertices, let D be its adjacency matrix, and let T_i be a periodic tree directed cover of G with root of...

conjecture | open | MathDB 页面 | 原文线索

V2-064875 | MATHDB-339383

Local character of finite and partial function categories

摘录 / EXCERPT: Let Ξ be the universe of categories enriched in pointed DCPOs and Scott-continuous functors between them, respecting finite coproducts and initial objects. Let Θ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-064876 | MATHDB-339387

The base-changeability conjecture for mgs automorphic representations

摘录 / EXCERPT: Let F be the function field of a smooth projective curve over a finite field, let G be a split semisimple group over F, and let $d70$ be an automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-064877 | MATHDB-339394

Genus 9 and 11 minimal stretch factor conjecture for nonorientable surfaces

摘录 / EXCERPT: Genus 9 and 11 minimal stretch factor conjecture. The minimal stretch factors and minimal polynomials are predicted to be

conjecture | open | MathDB 页面 | 原文线索

V2-064878 | MATHDB-339395

Minimal polynomial conjecture for orientation-reversing stretch factors on odd-genus orientable surfaces

摘录 / EXCERPT: Odd-genus orientation-reversing stretch factor conjecture. For all $k\geq 2$, $\delta^+_{\mathrm{rev}}(S_{2k-1})$ is the largest root of

conjecture | open | MathDB 页面 | 原文线索

V2-064879 | MATHDB-339396

Even-odd monotonicity conjecture for orientation-reversing stretch factors

摘录 / EXCERPT: Even-odd monotonicity conjecture. For all $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-064880 | MATHDB-339397

Odd-genus normalized limit conjecture for orientation-reversing stretch factors

摘录 / EXCERPT: Odd-genus normalized limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064881 | MATHDB-339398

Odd-genus nonorientable normalized liminf conjecture

摘录 / EXCERPT: Odd-genus normalized liminf conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064882 | MATHDB-339406

Author's conjecture on improving a modified Bessel-function integral inequality

摘录 / EXCERPT: Let $n<1$, let $ u\geq 1-n+\sqrt{2(1-n)}$, and define ... For the modified Bessel function of the second kind K_μ, the author conjectures that, for every $x>0$, ... The exponent sh...

conjecture | open | MathDB 页面 | 原文线索

V2-064883 | MATHDB-339411

Disk conjecture for the roots of the descent polynomial

摘录 / EXCERPT: Disk conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-064884 | MATHDB-339414

GW descendent series polynomiality in the asymptotic function F3

摘录 / EXCERPT: GW descendent polynomiality conjecture. The series

conjecture | open | MathDB 页面 | 原文线索

V2-064885 | MATHDB-339426

Uniqueness of section-free equations for finite-orbit models

摘录 / EXCERPT: Let $d=2$ and consider a model whose orbit is finite. A section-free equation is a linear combination of the orbit equations in which the boundary sections do not occur. If the mod...

conjecture | open | MathDB 页面 | 原文线索

V2-064886 | MATHDB-339427

Hypergeometric closed form for the excursions of the second Kreweras-like model

摘录 / EXCERPT: Consider the model ... Let $Q(0,0;t^{1/2})$ be its excursion generating function after extracting the even subsequence. The hypergeometric closed-form conjecture. ... The formula is in...

conjecture | open | MathDB 页面 | 原文线索

V2-064887 | MATHDB-339430

Chen-Lai-Wang equitable chromatic threshold conjecture

摘录 / EXCERPT: For a graph G, let $\Delta(G)$ denote its maximum degree and let $\chi_e^*(G)$ be the smallest integer k such that G is equitably l-colorable for every $l\geq k$. Chen-Lai-...

conjecture | solved | MathDB 页面 | 原文线索

V2-064888 | MATHDB-339435

Cone theorem for generalized log canonical pairs

摘录 / EXCERPT: Cone theorem for g-lc pairs. There are countably many curves $C_j\subset X$ such that $0<-(K_X+B+M)\cdot C_j\leq 2\dim X$ and

conjecture | open | MathDB 页面 | 原文线索

V2-064889 | MATHDB-339436

Existence of flips for generalized log canonical pairs

摘录 / EXCERPT: Existence of flips for g-lc pairs. The flip of every such flipping contraction exists.

conjecture | open | MathDB 页面 | 原文线索

V2-064890 | MATHDB-339438

Proper containment of interval filament graphs in co-strongly pseudo transitive graphs

摘录 / EXCERPT: Containment conjecture. The class of interval filament graphs is properly contained in the class of co-strongly pseudo transitive graphs of the first type.

conjecture | open | MathDB 页面 | 原文线索

V2-064891 | MATHDB-339439

Morales-Pak-Panova product formula for shifted skew tableaux

摘录 / EXCERPT: Let $n,a,b\geq 0$ and $m\geq 1$. Define the shifted skew shape $\pi=\mathbf{V}(n,a,b,m)$ by ... where ν^* denotes the shifted Young diagram of a strict partition ν. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064892 | MATHDB-339454

Chen-Lewis-Sheng conjecture on rationally equivalent points on complete intersections

摘录 / EXCERPT: Let $X \subset \mathbb{P}^n$ be a very general complete intersection of multidegree $(d_1, \dots, d_k)$. For $p \in X$, let $R_{X,p}$ denote the space of points of X rationally e...

conjecture | open | MathDB 页面 | 原文线索

V2-064893 | MATHDB-339458

Thomas's virtual localisation conjecture for sheaf-counting invariants

摘录 / EXCERPT: Thomas's virtual localisation conjecture. The invariants should furthermore satisfy $\mathsf{VW}_\alpha=\mathsf{vw}_\alpha$ whenever $\deg K_S\leq0$. The preceding formulas conject...

conjecture | open | MathDB 页面 | 原文线索

V2-064894 | MATHDB-339460

Miana-Ohtsuka-Romero's Catalan triangle odd-power congruence

摘录 / EXCERPT: Let $C_{n,k}:=\frac{n-2k}{n}{n\choose k}={n-1\choose k}-{n-1\choose k-1}$, with the Catalan triangle relation $B_{n,k}=C_{2n,n-k}$. Let n and a be positive integers with $n>a$,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-064895 | MATHDB-339461

Cyclic q-binomial Laurent-polynomial positivity conjecture

摘录 / EXCERPT: Cyclic q-binomial conjecture. For every integer j, this expression is a Laurent polynomial in q, and it has non-negative integer coefficients when $0\leqslant j\leqslant m$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-064896 | MATHDB-339467

Embedded Calabi-Yau conjecture for finite genus

摘录 / EXCERPT: Let $M\subset\mathbb{R}^3$ be a connected, complete embedded minimal surface of finite genus with compact, possibly empty, boundary. Embedded Calabi-Yau conjecture for finite genus...

conjecture | open | MathDB 页面 | 原文线索

V2-064897 | MATHDB-339468

Massive end, middle end flux and quadratic area growth conjectures

摘录 / EXCERPT: Let $M\subset\mathbb{R}^3$ be a properly embedded minimal surface with more than one end and horizontal limit tangent plane at infinity. An end is massive if every open, proper sub...

conjecture | open | MathDB 页面 | 原文线索

V2-064898 | MATHDB-339493

An upper bound conjecture for mixing times in ballistic random environments

摘录 / EXCERPT: Let N denote the size of the segment, and consider the simple exclusion process in a ballistic random environment. Its mixing time is the time needed for the process to approach...

conjecture | open | MathDB 页面 | 原文线索

V2-064899 | MATHDB-339496

Andreae's ubiquity conjecture for locally finite connected graphs

摘录 / EXCERPT: Let G be a graph. For a graph relation $\vartriangleleft$, say that G is $\vartriangleleft$-ubiquitous if, whenever a graph Γ satisfies ... for every $n\in\mathbb{N}$, it...

conjecture | open | MathDB 页面 | 原文线索

V2-064900 | MATHDB-339498

The strong path localization conjecture for directed polymers

摘录 / EXCERPT: Let $ u$ be the law of the random environment, and let β denote the inverse temperature. For a polymer measure $\mu_{\beta,n}$ on paths of length n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064901 | MATHDB-339502

Uniformity conjecture for q-quasimultiplicative sequences

摘录 / EXCERPT: Uniformity conjecture. Then, for every $s\geq 1$, there exists $c_s>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-064902 | MATHDB-339507

Subregular Fourier-coefficient conjecture for relevant unitary Arthur packets

摘录 / EXCERPT: Let $G_n={\mathrm {U}}_{n+1,n-1}$ be defined over F with F-rank $n-1$, and let $G_n^*={\mathrm {U}}_{n,n}$. Let ϕ be a generic global Arthur parameter of G_n^* that is...

conjecture | open | MathDB 页面 | 原文线索

V2-064903 | MATHDB-339509

Subregular Fourier-coefficient conjecture for generic representations of ${\mathrm {U}}_{n,n}$

摘录 / EXCERPT: Let F be a number field, let ${\mathrm {U}}_{n,n}$ be the F-quasisplit unitary group, and let π be an irreducible cuspidal automorphic representation of...

conjecture | open | MathDB 页面 | 原文线索

V2-064904 | MATHDB-339510

Tilting-object conjecture for equivariant matrix factorizations of invertible polynomials

摘录 / EXCERPT: Let ${\bf w}\in {\mathbb C}[x_1,\ldots,x_n]$ be an invertible polynomial, let $\Gamma_{\bf w}$ be its group of diagonal symmetries, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064905 | MATHDB-339512

The $(3,n)$ Schur-coefficient symmetry conjecture

摘录 / EXCERPT: For nonnegative integers a,b,n, let $[s_{2^a1^b}]_{3,n}$ denote the coefficient of the Schur function $s_{2^a1^b}$ in the relevant Schur expansion for the $(3,n)$ case. The...

conjecture | open | MathDB 页面 | 原文线索

V2-064906 | MATHDB-339513

The recursive formula for $(3,n)$ Schur coefficients

摘录 / EXCERPT: Let a,b be nonnegative integers with $a<b$, and let $[s_{2^a1^b}]_{3,n}$ denote the coefficient of $s_{2^a1^b}$ in the Schur expansion for the $(3,n)$ case. Write $[r]_{q,t}$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-064907 | MATHDB-339515

The switch-map symmetry for Hall scalar products

摘录 / EXCERPT: Let $m,n>0$, let $\mu=\mu_1\cdots\mu_k\vdash n$ be a composition as used in the paper, and define ... Let $\mathrm{Q}_{m,n}$ be the rational shuffle operator, 1 the unit symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-064908 | MATHDB-339519

Heath-Pemmaraju height conjecture for planar posets

摘录 / EXCERPT: Let P be a planar poset, meaning that it admits a crossing-free diagram, and let h be its height, the maximum size of a chain. Let ${\operatorname{qn}}(P)$ denote its queue-num...

conjecture | open | MathDB 页面 | 原文线索

V2-064909 | MATHDB-339520

Heath-Pemmaraju square-root conjecture for planar posets

摘录 / EXCERPT: Let P be a planar poset on n elements, and let ${\operatorname{qn}}(P)$ denote its queue-number. Heath-Pemmaraju's square-root conjecture. Every planar poset on n elements ha...

conjecture | open | MathDB 页面 | 原文线索

V2-064910 | MATHDB-339526

Converse conjecture for finite-field gamma factors and Gauss sums

摘录 / EXCERPT: Let $n>m$. Let π and τ be irreducible cuspidal representations of $\operatorname{GL}_n(\mathbb F_q)$ and $\operatorname{GL}_m(\mathbb F_q)$, corresponding to regular chara...

conjecture | open | MathDB 页面 | 原文线索

V2-064911 | MATHDB-339533

Non-degeneracy conjecture for Riemannian Kottler metrics

摘录 / EXCERPT: Let P_L be the linearization, at a Riemannian Kottler metric with negative cosmological constant, of the TT-gauge-fixed Einstein operator. Here K is the constant sectional cu...

conjecture | open | MathDB 页面 | 原文线索

V2-064912 | MATHDB-339535

Conjecture on stable fluctuations of the minimum in branching Brownian motion

摘录 / EXCERPT: Conjecture on the minimum. As $t\to\infty$, for some constant $C>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-064913 | MATHDB-339543

Kojima-Mizushima-Tan uniqueness conjecture for circle packings

摘录 / EXCERPT: Let S be an oriented closed surface of genus $g\geq 2$, let τ be a polygonal cell decomposition of S, and let c be a complex structure on S. A complex projective struc...

conjecture | open | MathDB 页面 | 原文线索

V2-064914 | MATHDB-339544

Large monochromatic component conjecture for dense bipartite graphs

摘录 / EXCERPT: Dense bipartite component conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-064915 | MATHDB-339545

Positroid stratification regular CW-complex conjecture

摘录 / EXCERPT: Positroid stratification conjecture. The positroid stratification of $Gr^{\geq 0}(k,n)$ is a regular CW-complex. In particular, the closure $\overline{\Pi}_{\mathcal{M}}$ of each p...

conjecture | open | MathDB 页面 | 原文线索

V2-064916 | MATHDB-339546

Fibonacci multiplier conjecture for permuted van der Corput sequences

摘录 / EXCERPT: Let $p_n=\mathfrak{F}_n$ be the nth Fibonacci number and let $\mu_n=\sigma_{p_{n-1},0}$. Define $z(1)=z(2)=z(3)=1$, $z(4)=2$, and, for subsequent indices,...

conjecture | open | MathDB 页面 | 原文线索

V2-064917 | MATHDB-339547

Extension conjecture for iterated fractional linear permutations

摘录 / EXCERPT: Let τ_{A_0,A_1,A_2,A_3} be the permutation of a finite field defined by ... where $A_0,A_2\in\mathbb{F}_p^{\ast}$ and $A_1,A_3\in\mathbb{F}_p$. Let Theorem and the fractional af...

conjecture | open | MathDB 页面 | 原文线索

V2-064918 | MATHDB-339550

Volume-preserving singular adapted metric conjecture

摘录 / EXCERPT: Let X be a conservative C^1 vector field and let Γ be a singular hyperbolic set for X. A singular adapted metric is a Riemannian metric satisfying the contraction, dom...

conjecture | open | MathDB 页面 | 原文线索

V2-064919 | MATHDB-339551

Adapted metric conjecture for p-sectional hyperbolic sets

摘录 / EXCERPT: Let X be a C^1 vector field with flow X_t, and let Γ be a p-sectional hyperbolic set with invariant splitting $T_\Gamma M=E\oplus F$. A metric is required to provide...

conjecture | open | MathDB 页面 | 原文线索

V2-064920 | MATHDB-339555

The large-shift conjecture for graphs without isolated P_2 components

摘录 / EXCERPT: Let G be a graph, and let k be an integer. A k-shifted-antimagic labeling of G is an injective edge labeling by the consecutive integers $k+1,k+2,\ldots,k+|E(G)|$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-064921 | MATHDB-339563

The local-global compatibility conjecture for spectral p-adic Jacquet-Langlands

摘录 / EXCERPT: Let $E\subset\overline{\mathbb{Q}}_p$ be a finite extension of $\mathbb{Q}_p$, and let $\rho:G_{\mathbb{Q}}\to\operatorname{GL}_2(E)$ be an odd continuous absolutely irreducible re...

conjecture | open | MathDB 页面 | 原文线索

V2-064922 | MATHDB-339566

Monotonicity conjecture for energy density along the Higgs-bundle b1-action

摘录 / EXCERPT: Let $\mathcal{M}_{Higgs}$ be the moduli space of $SL(n,\mathbb{C})$-Higgs bundles, with the action ... of $\mathbb{C}^*$. For a corresponding harmonic map, write $e(f)$ for its energ...

conjecture | open | MathDB 页面 | 原文线索

V2-064923 | MATHDB-339567

Hitchin-section maximality conjecture in a Hitchin fiber

摘录 / EXCERPT: Let $(\tilde{E},\tilde{\phi})$ be a Higgs bundle in the Hitchin component, and let (E,ϕ) be a distinct polystable $SL(n,\mathbb{C})$-Higgs bundle in the same Hitchin fiber. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-064924 | MATHDB-339569

The $Delta$-Equitable Coloring Conjecture

摘录 / EXCERPT: The Δ-ECC. G is equitably $\Delta(G)$-colorable if it is different from K_m, C_{2m+1}, and $K_{2m+1,2m+1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-064925 | MATHDB-339570

The list Hajnál-Szemerédi conjecture

摘录 / EXCERPT: List Hajnál-Szemerédi conjecture. Every graph G is equitably k-choosable when $k\geq \Delta(G)+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-064926 | MATHDB-339571

The list $Delta$-Equitable Coloring Conjecture

摘录 / EXCERPT: List Δ-ECC. G is equitably k-choosable for each $k\geq\Delta(G)$ if it is different from K_m, C_{2m+1}, and $K_{2m+1,2m+1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-064927 | MATHDB-339578

Finite-generation conjecture for normalized-volume minimizers

摘录 / EXCERPT: Let $x\in (X=\operatorname{Spec}(R),D)$ be a klt singularity, and let v be a minimizer of the normalized-volume function at x. Finite-generation conjecture. The associated grad...

conjecture | open | MathDB 页面 | 原文线索

V2-064928 | MATHDB-339580

Finite degree formula for normalized volumes

摘录 / EXCERPT: Finite degree formula. The normalized volumes satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-064929 | MATHDB-339581

Constructibility conjecture for normalized volumes in families

摘录 / EXCERPT: Let ... be a $\mathbb{Q}$-Gorenstein flat family of klt singularities with a section $t\in T\mapsto x_t\in\mathcal{X}_t$. Constructibility conjecture. The function ... is constructible...

conjecture | open | MathDB 页面 | 原文线索

V2-064930 | MATHDB-339582

Normalized volume comparison for non-closed points

摘录 / EXCERPT: Let (X,D) be a klt pair, let η be a non-closed point, and let $Z=\overline{\{\eta\}}$ have dimension d. Choose a general closed point $x\in Z$, and let $n=\dim X$. Non-clo...

conjecture | open | MathDB 页面 | 原文线索

V2-064931 | MATHDB-339583

Normalized volume and F-volume reduction conjecture

摘录 / EXCERPT: Let $x\in (X,\Delta)$ be a klt singularity over characteristic 0, and let $(x_p\in (X_p,\Delta_p))$ be its reduction modulo $p\gg 0$. Reduction conjecture. ... This conjecture prop...

conjecture | open | MathDB 页面 | 原文线索

V2-064932 | MATHDB-339584

Borbon-Spotti conjecture for local orbifold Euler numbers

摘录 / EXCERPT: Let (X,D,x) be a germ of log canonical surface singularity with $\mathbb{Q}$-boundary. Borbon-Spotti conjecture. ... The conjecture relates Langer's local orbifold Euler number to...

conjecture | open | MathDB 页面 | 原文线索

V2-064933 | MATHDB-339588

Alternate evacuation Green-polynomial conjecture for tabloids

摘录 / EXCERPT: Suppose that n is even, and let $e' = e \circ \operatorname{pr}$ be one of the alternate evacuation maps described above. For a partition $\lambda \vdash n$, let $t'(\lambda)$ de...

conjecture | refuted | MathDB 页面 | 原文线索

V2-064934 | MATHDB-339590

The acyclic high-degree induced-subgraph conjecture for facial unique-maximum colorings

摘录 / EXCERPT: Acyclic high-degree conjecture. If the induced subgraph $G[X]$ is acyclic, then

conjecture | open | MathDB 页面 | 原文线索

V2-064935 | MATHDB-339591

The maximum-degree-two high-degree conjecture for facial unique-maximum colorings

摘录 / EXCERPT: Maximum-degree-two conjecture. If $G[X]$ has maximum degree at most 2, then

conjecture | open | MathDB 页面 | 原文线索

V2-064936 | MATHDB-339592

3-colorability conjecture for planar graphs without 4-cycles and close triangles

摘录 / EXCERPT: Let G be a planar graph, and let $d_{\triangle}(G)$ denote the minimum distance between distinct triangles of G. Close-triangle 3-colorability conjecture. If G has no 4-cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-064937 | MATHDB-339593

Gao's Property D conjecture

摘录 / EXCERPT: A positive integer m has Property D if every sequence S over C_m^2 of length ... and containing no zero-sum subsequence of length m has the form $S=T^{m-1}$ for some sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-064938 | MATHDB-339599

Mazur's conjecture on ranks in elliptic K3 surface families

摘录 / EXCERPT: Let $\pi:\mathcal{E}\to\mathbf{P}^1_{\mathbf{Q}}$ be an elliptic K3 surface, and for $t\in\mathbf{P}^1_{\mathbf{Q}}$ let $\mathcal{E}_t=\pi^{-1}(t)$ be the fibre. Mazur's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-064939 | MATHDB-339609

Uniform limiting level-crossing distribution for real Gaussian ensembles

摘录 / EXCERPT: Let A and B be independently sampled from the real Gaussian ensemble $GE_n^\mathbb R$ of real $n\times n$ matrices. The level crossings are the points of $\mathbb{CP}^1$ associ...

conjecture | open | MathDB 页面 | 原文线索

V2-064940 | MATHDB-339611

Brendle-Margalit large-support metaconjecture for normal subgroups

摘录 / EXCERPT: Let S_g be a closed orientable surface, let N be a normal subgroup of $\operatorname{Mod}(S_g)$, and call a subsurface not large if it is contained in a nonseparating subsurfac...

conjecture | open | MathDB 页面 | 原文线索

V2-064941 | MATHDB-339612

Automorphism conjecture for connected complexes of regions

摘录 / EXCERPT: Let S_g be a closed orientable surface and let $\mathcal C_A(S_g)$ be a complex of regions associated to S_g. Assume that $\mathcal C_A(S_g)$ is connected and admits no exchang...

conjecture | open | MathDB 页面 | 原文线索

V2-064942 | MATHDB-339613

Normal right-angled Artin subgroup classification conjecture

摘录 / EXCERPT: Let S_g be a closed orientable surface and let N be a normal subgroup of $\operatorname{Mod}(S_g)$ that is isomorphic to a right-angled Artin group. Normal right-angled Artin s...

conjecture | open | MathDB 页面 | 原文线索

V2-064943 | MATHDB-339614

Normal-subgroup automorphism dichotomy conjecture

摘录 / EXCERPT: Let S_g be a closed orientable surface and let N be a normal subgroup of $\operatorname{Mod}(S_g)$. Normal-subgroup automorphism dichotomy conjecture. Either ... or N is isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-064944 | MATHDB-339615

Farb-Leininger-Margalit symmetry conjecture for small stretch factors

摘录 / EXCERPT: Let S_g be a closed orientable surface, let $f\in\operatorname{Mod}(S_g)$ be pseudo-Anosov with stretch factor λ, and let C be the constant appearing in the small-stret...

conjecture | open | MathDB 页面 | 原文线索

V2-064945 | MATHDB-339622

The constancy conjecture for the sieve functions Ω_k and Ω_k^{ext}

摘录 / EXCERPT: Constancy conjecture for Ω_k and Ω_k^{ext}. The function $\Omega_k(\vartheta,\vartheta_0)$ is constant with respect to ϑ_0. The same applies to...

conjecture | open | MathDB 页面 | 原文线索

V2-064946 | MATHDB-339641

Counterparts of effective-topos local operators in realizability toposes

摘录 / EXCERPT: Counterpart conjecture. Some of these local operators have counterparts in the toposes

conjecture | open | MathDB 页面 | 原文线索

V2-064947 | MATHDB-339642

Propositional canonicity for natural numbers under intuitionistic LLPO

摘录 / EXCERPT: Propositional-canonicity conjecture. The context $(x:\mathbf{LLPO}_\vee)$ has propositional canonicity for $\mathbb{N}$ over type theory with bracket types, as studied by Awodey an...

conjecture | open | MathDB 页面 | 原文线索

V2-064948 | MATHDB-339643

Imatomi-Tanaka-Tasaka-Wakabayashi conjectures for multiple zeta star values

摘录 / EXCERPT: For integers $a_0,\ldots,a_{2n}\geq 0$, let $\eta^\star(s_1,\ldots,s_r)$ denote the multiple zeta star value, let wt denote the total weight of the arguments, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-064949 | MATHDB-339644

Borwein's cyclic insertion conjecture for multiple zeta values

摘录 / EXCERPT: For integers $a_0,\bfldots,a_{2n}\bfgeq 0$, let $eta(s_1,\bfldots,s_r)$ denote a multiple zeta value, let wt denote the total weight of the arguments, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064950 | MATHDB-339660

Monoidality conjecture for the shape functor of constructible polygraphs

摘录 / EXCERPT: Let $\mathbf{CPol}$ be the category of constructible polygraphs, with monoidal structures given by the lax Gray product $\scaleobj{0.75}{\boxtimes}$ and the join $\star$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064951 | MATHDB-339661

Free-generation conjecture for constructible directed complexes

摘录 / EXCERPT: Let P be a directed complex. Say that P is freely generating if $(\mathcal{M}o\ell_{}{P})^*$ admits the structure of a polygraph whose n-dimensional generators are the n-di...

conjecture | open | MathDB 页面 | 原文线索

V2-064952 | MATHDB-339662

Weak higher-category structure conjecture for representable constructible polygraphs

摘录 / EXCERPT: Let $\mathbf{CPol}$ be the category of constructible polygraphs and let $\mathbf{RCPol}$ be the category of representable constructible polygraphs and strong maps. For a constructi...

conjecture | open | MathDB 页面 | 原文线索

V2-064953 | MATHDB-339663

Automata-theoretic conjecture for game-semantic PCF-computations

摘录 / EXCERPT: Automata-theoretic conjecture. There is some class of automata that is strictly weaker than TMs yet powerful enough to implement all the game-semantic PCF-computations.

conjecture | open | MathDB 页面 | 原文线索

V2-064954 | MATHDB-339665

Anisotropic KPZ conjecture for two-dimensional interface growth

摘录 / EXCERPT: Anisotropic KPZ conjecture. If the Hessian matrix $D^2v(\rho)$ has two eigenvalues of the same sign, equivalently if $\det(D^2v(\rho))>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-064955 | MATHDB-339672

Lower-bound conjecture for adaptive linear-quadratic regulator regret

摘录 / EXCERPT: Let $\widehat{\pi}$ be an arbitrary adaptive policy, and let $\mathcal{R}_n(\widehat{\pi})$ denote its regret after n time points. Lower-bound conjecture. For an arbitrary adapti...

conjecture | open | MathDB 页面 | 原文线索

V2-064956 | MATHDB-339673

Restricted-isometry conjecture for truncated sparse approximation properties

摘录 / EXCERPT: Restricted-isometry conjecture. If A has restricted 2-isometry property of order tk with

conjecture | open | MathDB 页面 | 原文线索

V2-064957 | MATHDB-339679

Convexity of the coupling-operator set

摘录 / EXCERPT: Let $(\bar{E},\bar{\tau})$ be the product state space, let $(\mathcal{L}_i,\mathcal{D}(\mathcal{L}_i))$ be the generators of the marginal Feller semigroups, and let $\mathcal{G}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-064958 | MATHDB-339680

Convexity of the invariant coupling-operator set

摘录 / EXCERPT: Let $\mathcal{G}$ be the set of coupling operators and let $\mathcal{G}^0$ be the subset for which the corresponding Feller coupling has the required invariant measure. Convexity c...

open_problem | open | MathDB 页面 | 原文线索

V2-064959 | MATHDB-339682

Nonpositive asymptotic-variance change for the observable-gradient coupling

摘录 / EXCERPT: Let π be the invariant measure, let $f\in L_0^2(\pi)$, and let $\phi\in L_0^2(\pi)$ be the corresponding solution of the overdamped Poisson equation. Let α_f be the matr...

conjecture | open | MathDB 页面 | 原文线索

V2-064960 | MATHDB-339684

Involution word tableau enumeration conjecture for type D

摘录 / EXCERPT: Involution word tableau enumeration conjecture. If $n\geq 3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-064961 | MATHDB-339685

Irreducibility conjecture for quadratic rational-map curves

摘录 / EXCERPT: Let ${\mathcal M}_2$ be the moduli space of Möbius conjugacy classes of quadratic rational maps. For $k\geq 0$ with $k\neq 1$ and $n\geq 1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064962 | MATHDB-339687

Primary full-system conjecture for finite-rank submonoids

摘录 / EXCERPT: Primary full-system conjecture. For every dimension $d\geq 3$, there exists a primary monoid in $\mathcal{C}_d$ having full system of sets of lengths.

conjecture | open | MathDB 页面 | 原文线索

V2-064963 | MATHDB-339688

Sharp extension estimate conjecture for zero-radius spheres over finite fields

摘录 / EXCERPT: Sharp zero-radius sphere extension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-064964 | MATHDB-339690

The realizability conjecture for locating and neighbor-locating chromatic numbers

摘录 / EXCERPT: Realizability conjecture. For each pair h, k of integers with $3\le h\le k$, there exists a connected graph G such that

conjecture | open | MathDB 页面 | 原文线索

V2-064965 | MATHDB-339691

The product inequality conjecture for neighbor-locating chromatic number

摘录 / EXCERPT: Product inequality conjecture. If G and H are connected graphs, then

conjecture | open | MathDB 页面 | 原文线索

V2-064966 | MATHDB-339692

Permutation-polynomial conjecture for the polynomials $f_{a,q}$

摘录 / EXCERPT: Let $\mathbb F_q$ be the finite field with q elements, let $p=\operatorname{char}\mathbb F_q$, and for $a\geq 1$ define ... A polynomial over $\mathbb F_{q^e}$ is a permutation pol...

conjecture | open | MathDB 页面 | 原文线索

V2-064967 | MATHDB-339694

Refined sheaf-counting conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let S be a del Pezzo surface with a smooth anticanonical divisor, and let β be a curve class. The higher genus relative Gromov-Witten invariants of S relative to the divi...

conjecture | open | MathDB 页面 | 原文线索

V2-064968 | MATHDB-339695

Rationality conjecture for higher genus relative invariants of log Calabi-Yau surfaces

摘录 / EXCERPT: Let $(Y,\partial Y)$ be a log Calabi-Yau pair with respect to a class $\beta\in H_2(Y,\mathbb Z)$ such that $\beta\mathbin{.}(\partial Y)\ne 0$. Let $N_{g,\beta}^{Y/\partial Y}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-064969 | MATHDB-339698

Minimal global dimension conjecture for stability conditions on projective space

摘录 / EXCERPT: Let $n\geq 1$, and let $\operatorname{D}^b(\operatorname{Coh}\mathbb{P}^n)$ be the bounded derived category of coherent sheaves on projective n-space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-064970 | MATHDB-339699

Global-dimension contractibility conjecture for stability conditions on the projective plane

摘录 / EXCERPT: Let $\operatorname{D}^b(\operatorname{Coh}\mathbb{P}^2)$ be the bounded derived category of coherent sheaves on the projective plane, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-064971 | MATHDB-339703

Fibonacci recurrence conjecture for the auxiliary function u

摘录 / EXCERPT: Let $a_0=1$, $a_1=2$, and define $a_n=3a_{n-1}-a_{n-2}$. For ${\boldsymbol{i}}=(i_1,i_2)\in\langle a_n\rangle\times\langle a_{n+1}\rangle$, let $u(\alpha({\boldsymbol{i}}))$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-064972 | MATHDB-339704

Conjecture on the equivalence of two-impulse and one-impulse space interception problems

摘录 / EXCERPT: Consider the two-impulse space interception problem for a free-flight ballistic missile and the corresponding one-impulse problem, with the same numerical initial data and constrai...

conjecture | open | MathDB 页面 | 原文线索

V2-064973 | MATHDB-339706

The open and closed q-stability coincidence conjecture

摘录 / EXCERPT: Let $\operatorname{QStab}^{\oplus}_s\operatorname{\mathcal{D}}_\mathbb{X}$ denote the space of open q-stability conditions with parameter s, let...

conjecture | open | MathDB 页面 | 原文线索

V2-064974 | MATHDB-339707

The Br_Q-equivariant q-stability and twisted-period isomorphism conjecture

摘录 / EXCERPT: Let Q be a quiver of type ADE, let $\operatorname{QStab}_s\operatorname{\mathcal{D}}_\mathbb{X}(Q)$ be its space of q-stability conditions, and assume...

conjecture | open | MathDB 页面 | 原文线索

V2-064975 | MATHDB-339713

Acylindrical hyperbolicity conjecture for automorphism groups of virtually cocompact special groups

摘录 / EXCERPT: Let G be a virtually cocompact special group. An acylindrically hyperbolic automorphism-group conjecture. If G is acylindrically hyperbolic, then its automorphism group ... is ac...

conjecture | open | MathDB 页面 | 原文线索

V2-064976 | MATHDB-339722

Boros-Moll infinite q-log-convexity conjecture

摘录 / EXCERPT: Let $P_n(x)$ be the Boros-Moll polynomials, whose coefficients form a polynomial sequence in n. A polynomial sequence is infinitely q-log-convex if every iterate under the oper...

conjecture | open | MathDB 页面 | 原文线索

V2-064977 | MATHDB-339723

Infinite q-log-convexity of the polynomial sequences in Example basic-qSM

摘录 / EXCERPT: Let $\{f_i(q)\}_{i\geq 0}$ be any of the polynomial sequences defined in Example $\mathrm{basic\text{-}qSM}$. A sequence is infinitely q-log-convex if every iterate under ... has c...

conjecture | open | MathDB 页面 | 原文线索

V2-064978 | MATHDB-339729

The consecutive-edge-label conjecture for super graceful matchings

摘录 / EXCERPT: Let nK_2 be the disjoint union of n edges, and let a k-super graceful labeling be a bijective labeling of its vertices and edges by a consecutive integer interval beginning a...

conjecture | open | MathDB 页面 | 原文线索

V2-064979 | MATHDB-339730

The complete-bipartite super-graceful classification conjecture

摘录 / EXCERPT: Let $K(m,n)$ be the complete bipartite graph with part sizes m and n, and let k-super graceful have the labeling meaning used in the paper. Complete-bipartite classification...

conjecture | open | MathDB 页面 | 原文线索

V2-064980 | MATHDB-339738

Polynomial bound conjecture for sliding circuit sets of rigid braids

摘录 / EXCERPT: Polynomial bound conjecture. There exists a constant C such that, for every rigid braid x with N strands and Garside-length L,

conjecture | open | MathDB 页面 | 原文线索

V2-064981 | MATHDB-339739

Internal commutativity conjecture for sliding circuit sets

摘录 / EXCERPT: Internal commutativity conjecture. For every $N\geqslant 5$, there exists an integer L_N such that, whenever $L\geqslant L_N$ and $x\in B_N$ is maximising among rigid pseudo-Anos...

conjecture | open | MathDB 页面 | 原文线索

V2-064982 | MATHDB-339741

The classification conjecture for classical Bochner-Krall polynomial sequences

摘录 / EXCERPT: Let P_n be a sequence of monic polynomials obtained as eigenpolynomials of a linear differential operator solving the classical Bochner-Krall problem. Let T and T^{-1} denote...

conjecture | open | MathDB 页面 | 原文线索

V2-064983 | MATHDB-339742

Order-preservation conjecture for generalized Bochner-Krall operators

摘录 / EXCERPT: Let L be an irreducible differential operator of order k solving the generalized Bochner-Krall problem, and let Λ be its corresponding difference operator. Order-preser...

conjecture | open | MathDB 页面 | 原文线索

V2-064984 | MATHDB-339743

Classification conjecture for irreducible generalized Bochner-Krall operators

摘录 / EXCERPT: Let k be a positive integer, let ∂ denote differentiation, and let $a_j,a_m\in\mathbb C$. For a divisor ℓ of k, define ... with $a_{\ell-1}\neq0$, and let $q(t)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-064985 | MATHDB-339748

Menezes-Teske-Weng's finite-field quadratic conjecture

摘录 / EXCERPT: Let $l\in\mathbb{N}$, set $q=2^l$, and let $\beta\in\mathbb{F}_{q^6}^*$. Suppose that the quadratic equation ... has two solutions $u_1,u_2\in\mathbb{F}_{q^6}$. Menezes-Teske-Weng's...

conjecture | open | MathDB 页面 | 原文线索

V2-064986 | MATHDB-339751

Conjecture on Gaussian fluctuations of zero-critical-point distances

摘录 / EXCERPT: For each zero $Z_{k,n}$ of a random polynomial p_n, let $\zeta_{k,n}$ be a nearest critical point, and define ... ... Here $\mathcal N_{\mathbb C}(0,1)$ denotes the standard normal d...

conjecture | open | MathDB 页面 | 原文线索

V2-064987 | MATHDB-339759

Alexandersson-Panova unimodality conjecture for LLT polynomial coefficients

摘录 / EXCERPT: Let G be an oriented graph with no loops or multiple edges. Write the relevant LLT polynomial expansion as ... where Ψ_{α} is the quasisymmetric power sum indexed by the...

conjecture | open | MathDB 页面 | 原文线索

V2-064988 | MATHDB-339774

Bergeron's e-positivity conjecture for multivariate diagonal harmonics

摘录 / EXCERPT: Let $\mathrm{DH}_{n,r}$ denote the multivariate diagonal harmonics space, and let $\Phi_{n,r}(q)$ be the Frobenius characteristic of its character, with parameter q_r. A symmetri...

conjecture | open | MathDB 页面 | 原文线索

V2-064989 | MATHDB-339775

The refined coefficient conjecture for multivariate diagonal harmonics

摘录 / EXCERPT: Fix a partition $\lambda\vdash n$, and let $c_\lambda(q,r)$ be the coefficient of e_λ in the Frobenius characteristic $\Phi_{n,r}(q)$. Write $\lambda_i=0$ for i larger th...

conjecture | open | MathDB 页面 | 原文线索

V2-064990 | MATHDB-339776

Branched continued fraction for extended multivariate Eulerian polynomials of negative type

摘录 / EXCERPT: Let $m\geq 1$, let ${\mathbf{x}}=(x_0,\ldots,x_{m-1})$ and ${\mathbf{c}}=(c_L)_{L\geq 0}$ be indeterminates, and let ${\mathcal{P}}^{(m)}_n({\mathbf{x}},{\mathbf{c}})$ be the exte...

conjecture | open | MathDB 页面 | 原文线索

V2-064991 | MATHDB-339778

Lusztig duality conjecture for Hecke insertion crystals

摘录 / EXCERPT: A decreasing factorization is mapped under Hecke insertion to a semistandard set-valued tableau with entries in $\{1,\ldots,n\}$. The Lusztig dual crystal conjecture. The Lusztig d...

conjecture | open | MathDB 页面 | 原文线索

V2-064992 | MATHDB-339779

K-crystals on rectangular shapes conjecture

摘录 / EXCERPT: Let $\operatorname{SV}^n(k\Lambda_i)$ denote the set of semistandard set-valued tableaux of rectangular shape $k\Lambda_i$, where the rectangle has height or width determined by...

conjecture | open | MathDB 页面 | 原文线索

V2-064993 | MATHDB-339792

Haiman-Hugland k-Schur positivity conjecture for LLT polynomials

摘录 / EXCERPT: Haiman-Hugland's conjecture. The symmetric function

conjecture | open | MathDB 页面 | 原文线索

V2-064994 | MATHDB-339803

Conjecture on cofinite-index stable basins in transcendental semigroup dynamics

摘录 / EXCERPT: Stable-basin conjecture. Then a stable basin U has cofinite index in S.

conjecture | open | MathDB 页面 | 原文线索

V2-064995 | MATHDB-339804

Conjecture on stable basins in forward orbits of transcendental semigroups

摘录 / EXCERPT: Forward-orbit stable-basin conjecture. For each component U of $F(S)$, there is a stable basin V for S lying in the forward orbit of U that has cofinite index in S.

conjecture | open | MathDB 页面 | 原文线索

V2-064996 | MATHDB-339805

Zerbini's single-valued multiple zeta value conjecture for modular graph functions

摘录 / EXCERPT: Let Γ be a graph, and let ... be its modular graph function, with l edges, $q=\exp(2\pi i\tau)$, and $y=\pi\,\operatorname{Im}(\tau)$. Denote by $\mathcal{Z}^{\rm sv}$ the a...

conjecture | open | MathDB 页面 | 原文线索

V2-064997 | MATHDB-339806

The esv conjecture for B-cycle and modular graph functions

摘录 / EXCERPT: Brödel-Schlotterer-Zerbini's esv conjecture. For all graphs Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-064998 | MATHDB-339819

Converse characterization of sign patterns with unique sepr-sequences

摘录 / EXCERPT: Converse characterization conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-064999 | MATHDB-339821

Bopp-Hoff refined balancedness conjecture for relative canonical resolutions

摘录 / EXCERPT: Let $C\subset\mathbb{P}^{g-1}$ be a general canonical curve of genus g, let k be a positive integer with Brill-Noether number $\rho(g,k,1)\geq0$, and let g^1_k be a general p...

conjecture | open | MathDB 页面 | 原文线索

V2-065000 | MATHDB-339822

Bopp-Hoff balancedness conjecture for the first syzygy bundle

摘录 / EXCERPT: Let $\mathscr{H}_{g,k}$ denote the Hurwitz space of degree-k covers $C\to\mathbb{P}^1$, and let $\rho(g,k,1)$ be the Brill-Noether number. For a general cover, let N_1 be the f...

conjecture | open | MathDB 页面 | 原文线索

V2-065001 | MATHDB-339837

Formal quintic polynomiality conjecture for Birkhoff-factorization coefficients

摘录 / EXCERPT: Let $\widetilde{\mathcal{F}}^{\mathsf{B}}_g$ be the genus-g series associated to the formal quintic, let $T=I^{\mathsf{Q}}_1(q)/I^{\mathsf{Q}}_0(q)$ be the mirror coordinate, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065002 | MATHDB-339839

Birkhoff-factorization coefficient ring conjecture

摘录 / EXCERPT: Let $n\geq0$, let I_n be the equivariant I-function, and write its restriction to $H=\lambda_i$ as ... where $R_{k,i}$ are formal series and $\mathds{G}_n$ is the ring defined fr...

conjecture | open | MathDB 页面 | 原文线索

V2-065003 | MATHDB-339840

Ponnusamy-Sairam conjecture on analytic rotations of harmonic mappings

摘录 / EXCERPT: Ponnusamy-Sairam's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065004 | MATHDB-339843

Wasserstein-p local-dependence bound for normal approximation

摘录 / EXCERPT: Let p be a positive integer and let $W=\sum_{i\in I}X_i$, where ${\mathbbm{E}}X_i=0$, ${\mathbbm{E}}W^2=1$, and the variables satisfy (LD1)--(LD$(p+1)$): for each $i_1\in I$,...

conjecture | open | MathDB 页面 | 原文线索

V2-065005 | MATHDB-339845

Almkvist's strong Lefschetz conjecture for relative coinvariant rings

摘录 / EXCERPT: Fix positive integers m,n, let $W=G(m,1,n)$ and $K=\mathfrak{S}_n\subset W$, and define ... Here e_i is the i-th elementary symmetric function in $x_1,\ldots,x_n$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065006 | MATHDB-339855

The column-equivalence partition realization conjecture for permutations

摘录 / EXCERPT: Let n be a positive integer, let $\mathcal{F}$ be a partition of $\mathopen{[\![} 1,n\mathclose{]\!]}$, and let $\mathbb{F}$ be an infinite field. For a mapping...

conjecture | open | MathDB 页面 | 原文线索

V2-065007 | MATHDB-339857

Teissier's log-convexity conjecture for sectional Milnor numbers

摘录 / EXCERPT: Let (X,x) be a germ of an isolated hypersurface singularity, and let $\mu^{(i)}$ denote its ith sectional Milnor number. Teissier's log-convexity conjecture. Is it true that .....

open_problem | open | MathDB 页面 | 原文线索

V2-065008 | MATHDB-339875

Weak uniform discreteness conjecture for positive-characteristic simple groups

摘录 / EXCERPT: Weak uniform discreteness conjecture. The space $\operatorname{DIRS}(G)$ is weakly uniformly discrete. The statement extends weak uniform discreteness beyond the no-discrete-small-...

conjecture | open | MathDB 页面 | 原文线索

V2-065009 | MATHDB-339881

The modular Erdős-Burgess constant conjecture

摘录 / EXCERPT: Let $n>1$ be an integer, and let $R=\mathbb{Z}_n$ be the ring of integers modulo n. The modular Erdős-Burgess constant conjecture. ... This conjecture gives an exact formula for th...

conjecture | open | MathDB 页面 | 原文线索

V2-065010 | MATHDB-339886

Existence conjecture for cocompact lattices in the local pro-p completion

摘录 / EXCERPT: Let A be the rank-2 generalized Cartan matrix considered in the source, with $\max(a_{21},a_{12})\leq -2$ and without assuming $a_{21}=a_{12}$. Let $\widehat{G}$ denote the loc...

conjecture | open | MathDB 页面 | 原文线索

V2-065011 | MATHDB-339888

Affine-principal-minor conjecture for cocompact lattices

摘录 / EXCERPT: Let A be a generalized Cartan matrix and let $\widehat{G}$ be the associated local pro-p completion. An irreducible principal minor of A is a principal submatrix that is irre...

conjecture | open | MathDB 页面 | 原文线索

V2-065012 | MATHDB-339892

Mkrtchyan's Sylvester coloring conjecture for cubic graphs

摘录 / EXCERPT: Mkrtchyan's Sylvester coloring conjecture. For every cubic graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-065013 | MATHDB-339893

The perfect-matching covering conjecture for claw-free bridgeless cubic graphs

摘录 / EXCERPT: Claw-free Berge conjecture. For every claw-free bridgeless cubic graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-065014 | MATHDB-339894

The S_{12}-coloring conjecture for cubic graphs with perfect matchings

摘录 / EXCERPT: S_{12}-coloring conjecture. Every cubic graph with a perfect matching admits an S_{12}-coloring:

conjecture | open | MathDB 页面 | 原文线索

V2-065015 | MATHDB-339895

The P_{12}-coloring conjecture for cubic graphs covered by four perfect matchings

摘录 / EXCERPT: P_{12}-coloring conjecture. Every cubic graph G with $k(G)\leq 4$ admits a P_{12}-coloring:

conjecture | open | MathDB 页面 | 原文线索

V2-065016 | MATHDB-339900

Yan et al.'s Ohba-like conjecture for improper colorings

摘录 / EXCERPT: Let G be a graph and let d be a nonnegative integer. Let $\chi^d(G)$ and $\chi_l^d(G)$ denote respectively the d-improper chromatic number and the d-improper list chromatic...

conjecture | open | MathDB 页面 | 原文线索

V2-065017 | MATHDB-339914

Non-D-finiteness conjecture for Models 1, 3, 4, 17, 18 and 22

摘录 / EXCERPT: Let $G(t;a,b)$ denote the generating function for the corresponding model, with parameters t, a, and b. A formal power series is D-finite if it satisfies a linear differentia...

conjecture | open | MathDB 页面 | 原文线索

V2-065018 | MATHDB-339919

Sahin's extended recurrence conjecture for simultaneous core partitions

摘录 / EXCERPT: Sahin's extended conjecture. For any positive integers $r\leq d$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-065019 | MATHDB-339920

The odd-genus Penner dilatation limit conjecture

摘录 / EXCERPT: Odd-genus Penner dilatation limit conjecture. The limit is the largest real root of

conjecture | open | MathDB 页面 | 原文线索

V2-065020 | MATHDB-339931

Sharpness conjecture for Deodhar component closure containments

摘录 / EXCERPT: Sharpness conjecture. Then D is obtained from D' by one of the two procedures described in Theorem. This is intended to assert that the theorem gives a complete characterizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-065021 | MATHDB-339935

Complementation conjecture for Wilf equivalence of pattern pairs

摘录 / EXCERPT: Let $\pi_1 otin\Pi_k$ and $\pi_2 otin\Pi_l$ with $k otin l$. For a pattern π, write π^c for its complement, and let $\Pi_l(\pi_1)$ denote the set of length-l patterns avo...

conjecture | open | MathDB 页面 | 原文线索

V2-065022 | MATHDB-339937

Mouhot-Strain spectral gap conjecture for linearized Boltzmann operators

摘录 / EXCERPT: Let B be a Boltzmann collision kernel with parameters $\gamma\in(-d,\infty)$ and $\alpha\in[0,2)$. A spectral gap means that the associated linearized Boltzmann collision operato...

conjecture | open | MathDB 页面 | 原文线索

V2-065023 | MATHDB-339947

The symmetric-product semi-orthogonal decomposition conjecture

摘录 / EXCERPT: Let C be a projective smooth curve of genus $g\geq 2$, and let C_d denote its d-th symmetric product. Let $D(C_d)$ be its derived category, and call a semi-orthogonal decompo...

conjecture | open | MathDB 页面 | 原文线索

V2-065024 | MATHDB-339949

The one-circulant-core Hadamard matrix conjecture

摘录 / EXCERPT: Let p be a positive integer. A Hadamard matrix with one circulant core of order $p+1$ has block form $\left(\begin{smallmatrix}1&e\\e^{t}&X_C\end{smallmatrix}\right)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-065025 | MATHDB-339952

The nonexistence conjecture for perfect binary sequences with parameter $d=1$

摘录 / EXCERPT: For a binary sequence X of length n, write $PComS(n,q,c)$ for the family used in the paper, and let a perfect binary sequence have two-level nontrivial autocorrelation with val...

conjecture | open | MathDB 页面 | 原文线索

V2-065026 | MATHDB-339957

Nonuniqueness of the limit problem under strong confinement

摘录 / EXCERPT: Nonuniqueness conjecture. The sequence u_k does not converge, as $k\to\infty$, to the solution of a unique limit problem. More precisely, the mass outside Ω, namely

conjecture | open | MathDB 页面 | 原文线索

V2-065027 | MATHDB-339959

Connectivity conjecture for edge slide graphs of hypercube signatures

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, let $\mathcal{S}=(a_1,\dots,a_n)$ be a signature of Q_n, and let ${\mathcal{E}(\mathcal{S})}$ be the edge slide graph induced by the s...

conjecture | open | MathDB 页面 | 原文线索

V2-065028 | MATHDB-339964

Doubly-signed Iwasawa main conjecture for the symmetric square representation

摘录 / EXCERPT: Let $T:=\operatorname{Sym}^2R_f^*(1+\chi)$, let $\mathbb{T}=T\otimes\Lambda^{\iota}$, and let $\mathcal{S}=\{(+, -),(+, \bullet),(-, \bullet)\}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-065029 | MATHDB-339965

Benois-Pottharst analytic Iwasawa main conjecture

摘录 / EXCERPT: Let $V:=\operatorname{Sym}^2W_f^*(1+\chi)$, with $\lambda\in\{\pm\alpha\}$. Let $\delta_\chi:\mathbb{Q}_p^\times\to E^\times$ satisfy $\delta_\chi(p)=p\chi^{-1}(p)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-065030 | MATHDB-339968

Planar omega-decomposition conjecture for graphs of treewidth three

摘录 / EXCERPT: Planar ω-decomposition conjecture. Every graph G with treewidth 3 has a planar ω-decomposition of width $??$ and order $??\cdot|G|$, in which the spread of each...

open_problem | open | MathDB 页面 | 原文线索

V2-065031 | MATHDB-339970

Equivalence of the two-curve de la Vallée-Poussin procedure and the dual simplex method

摘录 / EXCERPT: Consider the two-curve Chebyshev approximation problem and its corresponding linear programming formulation. The de la Vallée-Poussin procedure is an iterative procedure for findin...

conjecture | open | MathDB 页面 | 原文线索

V2-065032 | MATHDB-339973

Maximum permanent conjecture for nonnegative matrices of bounded rank

摘录 / EXCERPT: Let $A=(a_{ij})$ be an $n\times n$ nonnegative matrix of rank at most k, with row sums $r_1,\ldots,r_n$ and column sums $c_1,\ldots,c_n$. For $k,n\in\mathbb{N}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-065033 | MATHDB-339980

The weighted sum vanishing conjecture for the index sequence $(1,1,2,3,\dots,r)$

摘录 / EXCERPT: Define the weighted sums of finite multiple zeta values by ... and define $W_{k}^{\star}(n_{1},\dots,n_{d})$ similarly with $\zeta_{\mathcal{A}}^{\star}$ in place of...

conjecture | open | MathDB 页面 | 原文线索

V2-065034 | MATHDB-339981

The weighted sum vanishing conjecture for odd repeated-terminal indices

摘录 / EXCERPT: Let $W_{k}(n_{1},\dots,n_{d})$ and $W_{k}^{\star}(n_{1},\dots,n_{d})$ denote the weighted sums of finite multiple zeta values and finite star multiple zeta values, respectively, de...

conjecture | open | MathDB 页面 | 原文线索

V2-065035 | MATHDB-339982

The arithmetic-progression symmetry conjecture for weighted finite multiple zeta sums

摘录 / EXCERPT: For $n_{1},\dots,n_{d}\in\mathbb{Q}$, define ... and let $W_{k}^{\star}(n_{1},\dots,n_{d})$ be the analogous sum with $\zeta_{\mathcal{A}}^{\star}$. For $k,r\in\mathbb{Z}_{\geq1}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-065036 | MATHDB-339997

Uniform density bound conjecture for proper subgroups of symmetric groups

摘录 / EXCERPT: Let $G\le S_n$ be a subgroup, and let its density mean its proportion of elements in S_n. The uniform density bound conjecture. There exists a constant c independent of n suc...

conjecture | open | MathDB 页面 | 原文线索

V2-065037 | MATHDB-340007

The consecutive-pair conjecture for order-three reflection groups

摘录 / EXCERPT: Let W be a reflection group whose generators all have order three, let $c\in W$ be a Coxeter element, and let T be a longer reflection factorization of c. A consecutive...

conjecture | open | MathDB 页面 | 原文线索

V2-065038 | MATHDB-340008

Equivalence of the Hurwitz transitivity and consecutive-pair conjectures

摘录 / EXCERPT: Let W be a reflection group whose generators all have order three. Let $c\in W$ be a Coxeter element. Let T be a longer reflection factorization of c. The Hurwitz transitivit...

conjecture | open | MathDB 页面 | 原文线索

V2-065039 | MATHDB-340009

The inversion-major-index equidistribution conjecture for reduced words of the longest permutation

摘录 / EXCERPT: Inversion-major-index equidistribution conjecture. For $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-065040 | MATHDB-340011

Sagan et al.'s unimodality conjecture for 321-avoiding permutations

摘录 / EXCERPT: Sagan et al.'s conjecture. The polynomials $A_{n,k}(q)$ are unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-065041 | MATHDB-340012

Conjectured formulas for three-rowed tableau major-index distributions

摘录 / EXCERPT: Conjectured three-rowed formulas. The two displayed identities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-065042 | MATHDB-340013

Unimodality conjecture for skew standard Young tableau distributions

摘录 / EXCERPT: Let λ and μ be partitions with $\mu\subseteq\lambda$, and let $f_{\lambda\setminus\mu,i}(q)$ denote the major-index generating polynomial for standard Young tableaux of...

conjecture | open | MathDB 页面 | 原文线索

V2-065043 | MATHDB-340017

Su's Schur-positivity conjecture for reverse-lexicographic initial segments

摘录 / EXCERPT: Let n be a positive integer, and let L_n denote the reverse lexicographic ordering on the set of partitions of n, whose minimal element is (1^n). For a partition $ u$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-065044 | MATHDB-340018

Schur positivity for parity-restricted initial segments

摘录 / EXCERPT: Let n be a positive integer, let $\mu\vdash n$, and use the reverse lexicographic order on partitions of n. For a partition λ, let $\ell(\lambda)$ denote its number of...

conjecture | open | MathDB 页面 | 原文线索

V2-065045 | MATHDB-340019

Generalized powers-of-two positivity conjecture for conjugacy-action functions

摘录 / EXCERPT: Let $k,m\geq1$, and let ψ_{2^r}, h_2, and p_1 denote the symmetric functions used in the paper. Generalized powers-of-two positivity conjecture. The symmetric function ... i...

conjecture | open | MathDB 页面 | 原文线索

V2-065046 | MATHDB-340023

Fractional Erdős-Hajnal conjecture for subgraphs of prescribed girth

摘录 / EXCERPT: Let $x\ge1$ be real and let $g\ge3$ be an integer. For a graph G, let $\chi_f(G)$ denote its fractional chromatic number, and let the girth of a graph be the length of its shorte...

conjecture | open | MathDB 页面 | 原文线索

V2-065047 | MATHDB-340036

Endoscopic classification conjecture for unitary similitude groups

摘录 / EXCERPT: Endoscopic classification conjecture. The set $\Pi_{irr,temp}({\mathrm{GU}}_{2n})$ is a disjoint union of finite local tempered Vogan L-packets:

conjecture | open | MathDB 页面 | 原文线索

V2-065048 | MATHDB-340046

Local linear matching conjecture for field extensions

摘录 / EXCERPT: Let $K\subset L$ be a field extension. The linear matching conjecture. If $K\subset L$ possesses the local linear matching property, then it possesses the linear matching property....

conjecture | open | MathDB 页面 | 原文线索

V2-065049 | MATHDB-340047

Conjecture on fractional chromatic number of triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph on n vertices, and let $\chi_f(G)$ denote its fractional chromatic number. Fractional chromatic-number conjecture. As $n\to\infty$, every such gr...

conjecture | open | MathDB 页面 | 原文线索

V2-065050 | MATHDB-340048

Conjecture on fractional chromatic number versus edges in triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph with m edges, and let $\chi_f(G)$ denote its fractional chromatic number. Edge-based fractional chromatic-number conjecture. As $m\to\infty$, eve...

conjecture | open | MathDB 页面 | 原文线索

V2-065051 | MATHDB-340049

Conjecture on list chromatic number of triangle-free graphs

摘录 / EXCERPT: Let G be a triangle-free graph on n vertices with m edges, and let $\chi_\ell(G)$ denote its list chromatic number. List-colouring conjecture. There are constants $C_1,C_2>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-065052 | MATHDB-340055

Borisov-Oblomkov-Pandharipande-Yin conjecture for Donaldson-Thomas invariants of abelian threefolds

摘录 / EXCERPT: Let (A,H) be a principally polarized abelian threefold with Picard rank $\rho(A)=1$. For nonzero $(\beta,n)\in\mathbb{Z}^2$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-065053 | MATHDB-340056

Fourier-Mukai wall-avoidance conjecture for rank-one Donaldson-Thomas classes

摘录 / EXCERPT: Let $(\beta,n)\in\mathbb{Z}^2$ satisfy ... Assume $\beta\neq 0$ or $n>0$. Fourier-Mukai wall-avoidance conjecture. For every integer solution (c,d) of ... one has ... This is a constra...

conjecture | open | MathDB 页面 | 原文线索

V2-065054 | MATHDB-340060

The Fukaya-categorical irregular Riemann--Hilbert correspondence conjecture

摘录 / EXCERPT: Fukaya-categorical irregular Riemann--Hilbert conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-065055 | MATHDB-340066

Strongly's superregular Toeplitz matrix conjecture and MDP convolutional code bound

摘录 / EXCERPT: Let $cgamma$ be an integer with $cgamma dagger 5$. A matrix is superregular Toeplitz if it is both superregular and Toeplitz. Let n, k, and $cdelta$ be convolutional-code param...

conjecture | open | MathDB 页面 | 原文线索

V2-065056 | MATHDB-340068

Monopole-branch formula for refined rank-three Vafa-Witten invariants

摘录 / EXCERPT: Let S be a smooth projective surface satisfying $H_1(S,\mathbb Z)=0$ and $p_g>0$. For any polarization H and first Chern class c_1, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065057 | MATHDB-340072

Higher-dimensional coboundary expansion for infinitely many cubical Ramanujan complexes

摘录 / EXCERPT: Let Γ_S be one of the lattices of finite characteristic not equal to 2 constructed in the paper, let $\Gamma \subseteq \Gamma_S$ be a congruence subgroup, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065058 | MATHDB-340075

Converse mean-speed conjecture for transition fronts in domains with cylindrical branches

摘录 / EXCERPT: Converse mean-speed conjecture. The complete propagation of all the front-like solutions u_i is equivalent to the property that all transition fronts connecting 0 and 1 have...

conjecture | open | MathDB 页面 | 原文线索

V2-065059 | MATHDB-340077

Convergence of cascade tree decomposition KL divergence to zero

摘录 / EXCERPT: Let $\underline{X}$ be the input random vector and let $\underline{X}_{{\mathcal M}_i}$ denote the model random vector obtained after i cascade-tree stages. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-065060 | MATHDB-340085

Primality conjecture for elliptic Fermat numbers

摘录 / EXCERPT: Let E be a fixed elliptic curve over $\mathbb{Q}$, let $P\in E(\mathbb{Q})$ be a rational point, and let $F=(F_n)_{n\in\mathbb{N}}$ be the associated sequence of elliptic Fermat...

conjecture | open | MathDB 页面 | 原文线索

V2-065061 | MATHDB-340086

Primality conjecture for generalized elliptic Fermat numbers

摘录 / EXCERPT: Let E be a fixed elliptic curve over $\mathbb{Q}$, let $P\in E(\mathbb{Q})$ be a rational point, and let m be the fixed integer indexing the generalized elliptic Fermat numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-065062 | MATHDB-340087

Cancellativity characterization for monoids of finite non-degenerate braided sets

摘录 / EXCERPT: Cancellativity conjecture. The monoid $S(X,r)$ is cancellative if and only if r is involutive.

conjecture | open | MathDB 页面 | 原文线索

V2-065063 | MATHDB-340090

Refined powered-Catalan conjecture for 23-1-4-avoiding permutations

摘录 / EXCERPT: For $n\geq 1$, let $AV_n(23\text{-}1\text{-}4)$ be the permutations of length n avoiding the generalized pattern $23\text{-}1\text{-}4$, and let k be the number of right-to-lef...

conjecture | open | MathDB 页面 | 原文线索

V2-065064 | MATHDB-340091

The alternatingly increasing conjecture for IDP lattice polytopes

摘录 / EXCERPT: Let P be a lattice polytope that is integrally closed (IDP), meaning that for every $m\in\mathbb{Z}_{>0}$ and every $x\in mP\cap\mathbb{Z}^n$ there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-065065 | MATHDB-340100

Chan and Mao's monotonicity conjecture for overpartition ranks

摘录 / EXCERPT: Let $\overline{N}(m,n)$ denote the number of overpartitions of n with D-rank m, and let $\overline{N2}(m,n)$ denote the number of overpartitions of n with M_2-rank m. C...

conjecture | open | MathDB 页面 | 原文线索

V2-065066 | MATHDB-340112

Kawarabayashi-Ozeki conjecture on disjoint paths preserving connectivity

摘录 / EXCERPT: Kawarabayashi-Ozeki conjecture. There exists a function $f(k,l)$ such that, for every $f(k,l)$-connected graph G and two distinct vertices s and t in G, there are k inter...

conjecture | open | MathDB 页面 | 原文线索

V2-065067 | MATHDB-340113

Saddle-point avoidance conjecture for backtracking gradient descent

摘录 / EXCERPT: Let $f:\mathbb{R}^m\rightarrow\mathbb{R}$ be a C^1 function that is C^2 near its critical points, and let $\{z_n\}$ be the sequence generated by the Backtracking GD method from...

conjecture | open | MathDB 页面 | 原文线索

V2-065068 | MATHDB-340122

The freeness conjecture for the Flenner-Zaidenberg first series

摘录 / EXCERPT: Freeness conjecture for ${\mathcal F}{\mathcal Z}_1$. The curves $C_{d,a}\in {\mathcal F}{\mathcal Z}_1(d,a)$ are free divisors and $mdr(f)=2$ for all possible values of (d,a) wh...

conjecture | open | MathDB 页面 | 原文线索

V2-065069 | MATHDB-340123

The freeness conjecture for the Flenner-Zaidenberg second series

摘录 / EXCERPT: Freeness conjecture for ${\mathcal F}{\mathcal Z}_2$. The curves $C_k\in {\mathcal F}{\mathcal Z}_2(k)$ are free divisors and

conjecture | open | MathDB 页面 | 原文线索

V2-065070 | MATHDB-340124

The classification conjecture for rational cuspidal plane curves with at least three cusps

摘录 / EXCERPT: Main classification conjecture. The curve C is free, and its exponents determine the infinite series to which it belongs as follows:

conjecture | open | MathDB 页面 | 原文线索

V2-065071 | MATHDB-340125

Flenner-Zaidenberg's rigidity conjecture for rational cuspidal curves

摘录 / EXCERPT: Rigidity conjecture. If C has at least three singularities, then C is FZ-projectively rigid and FZ-unobstructed. In particular, C is es-rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-065072 | MATHDB-340126

Flenner-Zaidenberg's weak rigidity conjecture

摘录 / EXCERPT: Weak rigidity conjecture. If C has at least three singularities, then

conjecture | open | MathDB 页面 | 原文线索

V2-065073 | MATHDB-340127

The small trace-norm perturbation conjecture for Hermitian matrix sequences

摘录 / EXCERPT: Small trace-norm perturbation conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065074 | MATHDB-340128

The real-part spectral-symbol conjecture for matrix sequences

摘录 / EXCERPT: Real-part spectral-symbol conjecture. Is it true that

open_problem | open | MathDB 页面 | 原文线索

V2-065075 | MATHDB-340132

Artin--Shioda--Rudakov--Safarevic conjecture for K3 surfaces

摘录 / EXCERPT: Artin--Shioda--Rudakov--Safarevic conjecture. A K3 surface is supersingular if and only if it is unirational.

conjecture | open | MathDB 页面 | 原文线索

V2-065076 | MATHDB-340139

Hao-Wang-Zhang equality conjecture for the Erdős-Burgess constant

摘录 / EXCERPT: For an integer $n>1$, let $l_r(\mathbb{Z}/n\mathbb{Z})$ denote the Erdős-Burgess constant of the multiplicative semigroup $\mathbb{Z}/n\mathbb{Z}$, let $\mathsf{D}(G)$ be the Daven...

conjecture | open | MathDB 页面 | 原文线索

V2-065077 | MATHDB-340142

Hibi-Tsuchiya-Olsen regular unimodular triangulation conjecture for lecture hall simplices

摘录 / EXCERPT: Let ${\boldsymbol s}\in\mathbb{Z}_{\geq 1}^n$ be given, and let $\mathbf{P}_n^{({\boldsymbol s})}$ denote the corresponding ${\boldsymbol s}$-lecture hall simplex. Hibi-Tsuchiya-Ol...

conjecture | open | MathDB 页面 | 原文线索

V2-065078 | MATHDB-340144

Brändén-Leander gamma-positivity conjecture for lecture hall order polytopes

摘录 / EXCERPT: Let P be a naturally labeled, graded poset with nonnegative rank function ρ, and set ${\boldsymbol s}=\rho+1$. Let $O(P,{\boldsymbol s})$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-065079 | MATHDB-340158

The Erdős-Faber-Lovász conjecture for segment colorings

摘录 / EXCERPT: Let M be a set of m segments drawn in the plane. An EFL coloring of M with k colors is a function ... such that if two intersection points p_1,p_2 belong to the same segmen...

conjecture | open | MathDB 页面 | 原文线索

V2-065080 | MATHDB-340169

Liu-et al. coefficient conjectures for univalent log-harmonic mappings

摘录 / EXCERPT: Liu-et al.'s coefficient conjectures. For every integer $n\geq 1$, the coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-065081 | MATHDB-340173

Conjecture on optimal allocation under fixed-size access

摘录 / EXCERPT: Let r denote the number of accessed nodes, let α be the allocation parameter, and let $\mu_s(\alpha)$ denote the average service rate under fixed-size access. Fixed-size a...

conjecture | open | MathDB 页面 | 原文线索

V2-065082 | MATHDB-340175

Conjecture on the effect of coded file size on optimal allocation

摘录 / EXCERPT: Let m be the coded file size ratio, and let α be the allocation parameter maximizing the average service rate $\mu_s(\alpha)$. The claim applies to both the fixed-size acc...

conjecture | open | MathDB 页面 | 原文线索

V2-065083 | MATHDB-340183

General-term conjecture for the short-time expansion of the KPZ generating function

摘录 / EXCERPT: Let g be the function used to define $Q_{t,\beta}(\sigma)$, let $\mathfrak{L}_i$ denote the corresponding coefficients, and let $\mathbf{n}=(n_1,\ldots,n_{r+2-2q})$. If the disti...

conjecture | open | MathDB 页面 | 原文线索

V2-065084 | MATHDB-340186

Classification conjecture for fault-line tilings in the Jeandel-Rao shift

摘录 / EXCERPT: Let Ω_4 be the Wang shift under consideration, let $X_4\subset\Omega_4$ be the associated subshift, and let Y_i^+ and Y_i^- be the horizontal-shift orbit closures of the...

conjecture | open | MathDB 页面 | 原文线索

V2-065085 | MATHDB-340197

The flow-tree polynomial ansatz for the unrefined index

摘录 / EXCERPT: Let $\check\gamma_i$ and c_i be the charge and stability data entering $g^{(0)}_n$, let S_j be the partial stability sums, and let $\rm Sym$ denote symmetrization over the cons...

conjecture | open | MathDB 页面 | 原文线索

V2-065086 | MATHDB-340198

The unrefined-limit identity for the refined index

摘录 / EXCERPT: Let $g^{\rm (ref)}_n(\{\check\gamma_i,c_i\},y)$ be defined by the refined generating-function construction and the reference solution $F^{\rm (ref)}_n$, with coefficients b_n fix...

conjecture | open | MathDB 页面 | 原文线索

V2-065087 | MATHDB-340199

The modified Erdős-Ginzburg-Ziv constant for powers of two

摘录 / EXCERPT: Modified Erdős-Ginzburg-Ziv conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-065088 | MATHDB-340200

Exponential growth of inequivalent codes from bent vectorial functions

摘录 / EXCERPT: Exponential-growth conjecture. For any given ℓ in the range $1\leq \ell\leq m$, the number of inequivalent codes with these parameters obtained from $(2m,\ell)$ bent vectorial...

conjecture | open | MathDB 页面 | 原文线索

V2-065089 | MATHDB-340219

Polynomial bias-rank conjecture

摘录 / EXCERPT: Let k be a finite field of characteristic p, let V be a finite-dimensional k-vector space, and let $P:V\to k$ be a polynomial of degree d and rank greater than r. The b...

conjecture | open | MathDB 页面 | 原文线索

V2-065090 | MATHDB-340223

Global convergence conjecture for pluriclosed flow on surfaces of general type

摘录 / EXCERPT: Let (M,J) be a complex surface of general type, and let ω_0 be a pluriclosed metric. General type convergence conjecture. The solution to pluriclosed flow with initial con...

conjecture | open | MathDB 页面 | 原文线索

V2-065091 | MATHDB-340224

Kähler-Ricci soliton convergence conjecture for pluriclosed flow on Fano surfaces

摘录 / EXCERPT: Let (M^4,J) be a compact Fano surface admitting a Kähler-Ricci soliton (ω_{KRS},X). Let $\omega_0\in c_1$ be a Kähler metric invariant under the one-parameter subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-065092 | MATHDB-340225

Geometrization conjecture for pluriclosed flow on properly elliptic surfaces

摘录 / EXCERPT: Let (M^4,J) be a properly elliptic surface with $\operatorname{kod}(M)=1$ and odd first Betti number, and let ω be a pluriclosed metric. Properly elliptic geometrization c...

conjecture | open | MathDB 页面 | 原文线索

V2-065093 | MATHDB-340226

Geometrization conjecture for pluriclosed flow on Inoue surfaces

摘录 / EXCERPT: Let (M^4,J) be an Inoue surface, and let ω_0 be a pluriclosed metric on M. Inoue surface geometrization conjecture. The solution ω_t to pluriclosed flow with this...

conjecture | open | MathDB 页面 | 原文线索

V2-065094 | MATHDB-340227

Steady-soliton limit conjecture for pluriclosed flow on Class VII+ surfaces

摘录 / EXCERPT: Let (M^4,J) be a compact Class VII_+ surface, let ω_0 be a pluriclosed metric on M, and let $p\in M$ be a generic point as described in the surrounding discus...

conjecture | open | MathDB 页面 | 原文线索

V2-065095 | MATHDB-340231

Rank conjecture for the lattice of orbifold contribution vectors

摘录 / EXCERPT: Rank conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065096 | MATHDB-340232

Uniform local-index conjecture for minimal cancelling tuples

摘录 / EXCERPT: Uniform local-index conjecture. Then $cell_{csigma_i}=cell_{csigma_j}$ for all i,j.

conjecture | open | MathDB 页面 | 原文线索

V2-065097 | MATHDB-340252

Linear upper-bound conjecture for group edge irregularity strength

摘录 / EXCERPT: Let G be a graph of size m, and let $es_g(G)$ denote its group edge irregularity strength, the least order of an Abelian group for which G admits an edge-irregular labeling b...

conjecture | open | MathDB 页面 | 原文线索

V2-065098 | MATHDB-340257

Conjecture on computing Whittaker integrals with boundary polynomials

摘录 / EXCERPT: Boundary-polynomial conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065099 | MATHDB-340258

Critical-resonance conjecture for failure of McNamara's vanishing

摘录 / EXCERPT: Critical-resonance conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-065100 | MATHDB-340260

Weak monotonicity of exclusion-process mixing time in the number of particles

摘录 / EXCERPT: Let G be an arbitrary graph with arbitrary transition rates, let n be its number of vertices, and let $\mathrm{EX}(k)$ denote the exclusion process with k particles. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-065101 | MATHDB-340266

The solvable maximal subgroup conjecture for non-central subnormal linear groups

摘录 / EXCERPT: Solvable maximal subgroup conjecture. If $n\geq 2$, then G contains no solvable maximal subgroups.

conjecture | open | MathDB 页面 | 原文线索

V2-065102 | MATHDB-340267

The GLSM-Gromov-Witten correspondence conjecture

摘录 / EXCERPT: GLSM-Gromov-Witten correspondence conjecture. There is an element $\mathfrak c\in H_{\it CR}^*(\bar X_W;\Lambda_+)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065103 | MATHDB-340269

Perles' conjecture for Perles pieces in dual polytopes

摘录 / EXCERPT: Let P be a simple polytope, and let P^* be its polar dual. A Perles piece is a connected simplicial complex such that every facet contains exactly one ridge in the boundary of...

conjecture | open | MathDB 页面 | 原文线索

V2-065104 | MATHDB-340270

The 2-categorical universal property of the optic construction

摘录 / EXCERPT: Optic biadjunction conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-065105 | MATHDB-340271

The optic interpretation of c-lenses

摘录 / EXCERPT: C-lens optic conjecture. C-lenses are the lawful, possibly mixed optics for some action on $\mathbf{Cat}$.

conjecture | open | MathDB 页面 | 原文线索

V2-065106 | MATHDB-340272

Conjecture on the connective constant of universal covers of punctured hexagonal lattices

摘录 / EXCERPT: Universal-cover connective-constant conjecture. There exists $k\in\mathbb N\cup\{0\}$ for which

conjecture | open | MathDB 页面 | 原文线索

V2-065107 | MATHDB-340275

Optimal regularity conjecture for two-dimensional KPZ scaling limits

摘录 / EXCERPT: For any $\delta>0$, let $\mathcal{C}_{\mathfrak{s};\mathrm{loc}}^{-\delta}(\mathbf{R}_{>0}\times\mathbf{T}^{2})$ and $\mathcal{C}^{-\delta}(\mathbf{T}^{2})$ denote the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-065108 | MATHDB-340277

The characterization of pure consequence relations by regular connectives

摘录 / EXCERPT: Let a consequence relation be a relation between sets of formulas that admits a collection of regular connectives, and let classical logic provide the reference collection of regul...

conjecture | refuted | MathDB 页面 | 原文线索

V2-065109 | MATHDB-340286

Polynomiality conjecture for Lecture Hall polynomials

摘录 / EXCERPT: Let ℓ_i denote the i-th Lecture Hall polynomial, constructed in the paper as a Laurent polynomial depending on $y_1,\dots,y_i$. Polynomiality conjecture for Lecture Hall pol...

conjecture | open | MathDB 页面 | 原文线索

V2-065110 | MATHDB-340287

Chapoton-type identity for parabolic Tamari lattices

摘录 / EXCERPT: Let $n>0$ and let $\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_r)$ be a composition of n into r parts. Let $H_{\alpha}(x,y)$ be the polynomial defined on order ideals of the releva...

conjecture | open | MathDB 页面 | 原文线索

V2-065111 | MATHDB-340288

Rankedness of the core label order of parabolic Tamari lattices

摘录 / EXCERPT: Let $n>0$ and let α be a composition of n. Let $\mathcal{T}_{\alpha}$ denote the parabolic Tamari lattice, and let $\operatorname{CLO}(\mathcal{T}_{\alpha})$ denote its co...

conjecture | open | MathDB 页面 | 原文线索

V2-065112 | MATHDB-340292

Sun's quadratic practical-values conjecture

摘录 / EXCERPT: Let a,b,c be positive integers with $2 \mid ab$ and $2 divides c$. A practical number is a positive integer each of whose integers from 1 through it can be written as a sum of d...

conjecture | open | MathDB 页面 | 原文线索

V2-065113 | MATHDB-340293

The expected contribution from isolated zeros of the vector field

摘录 / EXCERPT: Let f_l be the random spherical harmonic and let V be the vector field on S^2 considered above. Writing DF_l for the derivative of the map $F_l=(f_l,Vf_l)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-065114 | MATHDB-340303

Localized test-function conjecture for adaptive transport discretizations

摘录 / EXCERPT: Let $\breve{\boldsymbol{U}}_{\tilde{\mathcal{T}}}^{\delta}$ be the discrete trial space on the refined partition...

conjecture | open | MathDB 页面 | 原文线索

V2-065115 | MATHDB-340309

Möller's trajectory conjecture for the 3x + 1 problem

摘录 / EXCERPT: For any odd positive integer x_0, define its trajectory $(x_n)_{n\geqslant 0}$ by choosing positive integers a_n such that ... with every x_n odd. A trajectory is pure periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-065116 | MATHDB-340310

The Ω-divergence conjecture for non-periodic E-sequences

摘录 / EXCERPT: Let $(a_n)_{n\geqslant 1}$ be an E-sequence, meaning an infinite sequence of positive integers, and define ... An E-sequence is Ω-convergent to an odd positive integer x if it is t...

conjecture | open | MathDB 页面 | 原文线索

V2-065117 | MATHDB-340316

Resultant multiplicity conjecture for supersingular polynomials at levels 2, 3, 5 and 7

摘录 / EXCERPT: Resultant multiplicity conjecture. The following congruences hold modulo p:

conjecture | open | MathDB 页面 | 原文线索

V2-065118 | MATHDB-340321

Apéry-like coefficient formula for class 3C supersingular polynomials

摘录 / EXCERPT: Class 3C Apéry-like conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065119 | MATHDB-340322

Kadrani-Dussault-Benchakroun regularization convergence conjecture for MPSCs

摘录 / EXCERPT: Let X be the feasible set of the mathematical program with switching constraints (MPSC), and let $X_{\textup{KDB}}(t)$ denote the feasible set of its Kadrani-Dussault-Benchakroun...

conjecture | open | MathDB 页面 | 原文线索

V2-065120 | MATHDB-340323

Constraint qualification conjecture for Kadrani-Dussault-Benchakroun regularized MPSCs

摘录 / EXCERPT: Let X be the feasible set of a mathematical program with switching constraints (MPSC), and let $X_{\textup{KDB}}(t)$ be the feasible set of its Kadrani-Dussault-Benchakroun regul...

conjecture | open | MathDB 页面 | 原文线索

V2-065121 | MATHDB-340327

Witten's conjecture for open r-spin intersection numbers in higher genus

摘录 / EXCERPT: Let $r\ge 2$. For $g\ge 1$, denote by $\langle\tau^{\alpha_1}_{d_1}\cdots\tau^{\alpha_l}_{d_l}\sigma^k\rangle^{\frac{1}{r},o}_g$ the proposed open r-spin intersection numbers, an...

conjecture | open | MathDB 页面 | 原文线索

V2-065122 | MATHDB-340334

Permutation conjecture for a trinomial over fields of characteristic two

摘录 / EXCERPT: Let $n=2m$ be a positive integer, and let $f(x)$ be the trinomial defined by equation (1) in the source over ${\mathbb F}_{2^{2m}}$. A pair (s,t) specifies the exponents in this...

conjecture | open | MathDB 页面 | 原文线索

V2-065123 | MATHDB-340337

Q-system Demazure decomposition conjecture

摘录 / EXCERPT: Fix $a\in I_0$, set $c=\min\{c_b\mid A_{ba}\neq 0\}$, and let $\ell\geq\lceil m/c\rceil$. Write $L(k)=\left\lfloor\frac{mA_{ba}-k}{A_{ab}}\right\rfloor$. Q-system Demazure decompos...

conjecture | open | MathDB 页面 | 原文线索

V2-065124 | MATHDB-340367

Criticality conjecture for isometric embeddings in arbitrary codimension

摘录 / EXCERPT: Let Σ_a be the polar cap and let $\mathscr{I}^\alpha_1(\Sigma_a)$ denote the class of $C^{1,\alpha}$ isometric embeddings of Σ_a into the relevant codimension-one Euc...

conjecture | open | MathDB 页面 | 原文线索

V2-065125 | MATHDB-340371

Infinite-rank conjecture for winding-number-zero satellite operators

摘录 / EXCERPT: Let P be a non-constant satellite operator of winding number zero, and let K be a knot. Write nK for the n-fold connected sum when $n\geq 0$ and for the corresponding inver...

conjecture | open | MathDB 页面 | 原文线索

V2-065126 | MATHDB-340372

Critical norm conjecture for defocusing inter-critical nonlinear Schrödinger equations

摘录 / EXCERPT: Consider the defocusing Schrödinger initial value problem for $u:I\times\mathbb{R}^{d}\to\mathbb{C}$, ... where $s_c\geq 0$ and $p=\frac{4}{d-2s_c}$. The critical norm conjecture ass...

conjecture | open | MathDB 页面 | 原文线索

V2-065127 | MATHDB-340382

A signed Edelman-Greene expansion for doubled Stanley symmetric functions

摘录 / EXCERPT: Signed Edelman-Greene expansion. If $\omega\in C_{n+1}$ is unknotted, then

conjecture | open | MathDB 页面 | 原文线索

V2-065128 | MATHDB-340383

A barred signed Edelman-Greene expansion at opposite alphabets

摘录 / EXCERPT: Barred signed Edelman-Greene expansion. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065129 | MATHDB-340388

The Paulsen problem for nearly equal norm Parseval frames

摘录 / EXCERPT: Paulsen problem. For every ϵ-nearly equal norm Parseval frame V, is

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065130 | MATHDB-340406

Extension conjecture for contact systems of boxes

摘录 / EXCERPT: Extension conjecture. A set $\mathcal{C}$ of d-dimensional boxes in $[-1,+1]^d$ is a subset of a proper d-tiling $\mathcal{T}$ if and only if every set...

conjecture | open | MathDB 页面 | 原文线索

V2-065131 | MATHDB-340407

Characterization conjecture for simplicial complexes of proper box tilings

摘录 / EXCERPT: Characterization conjecture. A simplicial complex $\mathcal{S}$ is such that

conjecture | open | MathDB 页面 | 原文线索

V2-065132 | MATHDB-340408

Dvořák-Mnich conjecture on fractional coloring of planar graphs of girth five

摘录 / EXCERPT: Let G be a planar graph of girth at least five, and let $\chi_f(G)$ denote its fractional chromatic number. Dvořák-Mnich conjecture. There exists a real number $c<3$ such that ev...

conjecture | open | MathDB 页面 | 原文线索

V2-065133 | MATHDB-340412

He-Schramm's Cantor-set removability conjecture for circle-domain boundaries

摘录 / EXCERPT: He-Schramm's removability conjecture. If every Cantor set contained in $\partial\Omega$ is conformally removable, then $\partial\Omega$ is conformally removable.

conjecture | open | MathDB 页面 | 原文线索

V2-065134 | MATHDB-340421

Beilinson's Hodge-$\mathcal D$-Conjecture

摘录 / EXCERPT: Hodge-$\mathcal D$-Conjecture. The map $r_{\mathcal{D},\mathbb{R}}^{p,n}$ is surjective. This is a higher-dimensional and higher-cycle analogue of the Hodge conjecture, formulated...

conjecture | open | MathDB 页面 | 原文线索

V2-065135 | MATHDB-340423

Absolute-bound incoherence conjecture for equiangular lines

摘录 / EXCERPT: Let Ω be a set of $d(d+1)/2$ equiangular lines in $\mathbf{R}^d$. Here $\operatorname{Inc}(\Omega)$ denotes the maximum size of an incoherent subset of Ω. Absolute-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-065136 | MATHDB-340424

Labourie's conjecture on unique equivariant minimal surfaces

摘录 / EXCERPT: Labourie's conjecture. Does there exist a unique ρ-equivariant minimal surface in G/K?

open_problem | open | MathDB 页面 | 原文线索

V2-065137 | MATHDB-340426

The non-decoupling conjecture for Hitchin and maximal Higgs bundles

摘录 / EXCERPT: Non-decoupling conjecture. The Hitchin equation never decouples:

conjecture | open | MathDB 页面 | 原文线索

V2-065138 | MATHDB-340427

Li's pointwise monotonicity conjecture along the Higgs-bundle flow

摘录 / EXCERPT: Li's monotonicity conjecture. Along the $\mathbb C^*$-flow, the energy density decreases pointwise as $|t|$ decreases.

conjecture | open | MathDB 页面 | 原文线索

V2-065139 | MATHDB-340428

The nilpotent-cone domination conjecture for Higgs-bundle energy density

摘录 / EXCERPT: Nilpotent-cone domination conjecture. The energy density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065140 | MATHDB-340429

Dai-Li's Hitchin-fiber maximality conjecture

摘录 / EXCERPT: Dai-Li's conjecture. The energy densities satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-065141 | MATHDB-340430

Exponential-growth conjecture for Möbius values of the permutations κ_n

摘录 / EXCERPT: For each positive integer n, let $\kappa_n\in {\mathcal S}_{4n}$ be the permutation ... It is 321-free and can be split into four quadrants, each an increasing subsequence of len...

conjecture | open | MathDB 页面 | 原文线索

V2-065142 | MATHDB-340444

Kanade-Russell product identities

摘录 / EXCERPT: Let q be a formal variable, and for $1\leq \ell\leq 11$ let $H_\ell(x;q)$ be the series defined above from the sums over $i,j,k\geq 0$ and the specified polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-065143 | MATHDB-340447

Aubry-McGuire-Rodier exceptional APN polynomial conjecture

摘录 / EXCERPT: Let q be a power of two, and let $f\in\mathbb{F}_q[X]$ be an exceptional APN polynomial, meaning that f induces an APN function on $\mathbb{F}_{q^r}$ for infinitely many positi...

conjecture | open | MathDB 页面 | 原文线索

V2-065144 | MATHDB-340448

The admissible cone metric count conjecture for cyclically branched coverings

摘录 / EXCERPT: Admissible cone metric count conjecture. There are at least g admissible cone metrics of the form $a(d_1,\ldots,d_n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-065145 | MATHDB-340450

Noncentral concentration conjecture for empirical relative entropy

摘录 / EXCERPT: Let P be a distribution on a finite alphabet of size k, let $\hat{P}_{n,k}$ be its empirical distribution from n independent samples, and let $D(\hat{P}_{n,k} \\| P)$ denote...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065146 | MATHDB-340451

Centralized concentration conjecture for empirical relative entropy

摘录 / EXCERPT: Let P be a distribution on a finite alphabet of size k, let $\hat{P}_{n,k}$ be its empirical distribution from n independent samples, and let $D(\hat{P}_{n,k} \| P)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-065147 | MATHDB-340452

Kubertin's conjecture on generalized Erdős-Ginzburg-Ziv constants of C_n^r

摘录 / EXCERPT: Kubertin's conjecture. For any positive integers k,n with $k\ge r$,

conjecture | open | MathDB 页面 | 原文线索

V2-065148 | MATHDB-340453

Lattice signature and mirror isomorphism conjecture for invertible polynomials

摘录 / EXCERPT: Let $f(x_1,x_2,x_3)$ be an invertible polynomial, let $G_0\subset G$, and let $\Lambda^{(B)}_{f,G}$ and $\Lambda^{(A)}_{\widetilde{f},\widetilde{G}}$ be the associated lattices. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-065149 | MATHDB-340460

Kalai's conjecture for f-vectors of flag Cohen-Macaulay complexes

摘录 / EXCERPT: A simplicial complex is balanced if its graph is properly d-colorable when it has dimension $d-1$. Kalai's flag Cohen-Macaulay conjecture. The f-vector of any flag Cohen-Macaul...

conjecture | open | MathDB 页面 | 原文线索

V2-065150 | MATHDB-340461

The h-vector characterization conjecture for flag Cohen-Macaulay complexes

摘录 / EXCERPT: Let $h(\Delta)$ denote the h-vector of a simplicial complex and $f(\Delta)$ its f-vector. Flag Cohen-Macaulay h-vector conjecture. The set of h-vectors of flag Cohen-Macaulay...

conjecture | open | MathDB 页面 | 原文线索

V2-065151 | MATHDB-340462

Nevo-Lutz's gamma-two rigidity conjecture for flag spheres

摘录 / EXCERPT: Let $d\geq 4$ be an integer and let Δ be a flag simplicial $(d-1)$-sphere. Nevo-Lutz's gamma-two conjecture. The following are equivalent: 1. $\gamma_2(\Delta)=0$. 2. There...

conjecture | open | MathDB 页面 | 原文线索

V2-065152 | MATHDB-340464

The flag upper bound conjecture for odd-dimensional spheres

摘录 / EXCERPT: Let $J_m(n)$ be the simplicial $(2m-1)$-sphere on n vertices obtained as the join of m cycles, each having either $\lfloor n/m\rfloor$ or $\lceil n/m\rceil$ vertices. Flag uppe...

conjecture | open | MathDB 页面 | 原文线索

V2-065153 | MATHDB-340465

The flag upper bound conjecture for even-dimensional spheres

摘录 / EXCERPT: Let $J_m^*(n)$ be the suspension of a join of m balanced cycles as defined in the source, and let $\mathcal{J}_m^*(n)$ be the family of joins $S*C_2*\cdots*C_m$ with S a flag 2...

conjecture | open | MathDB 页面 | 原文线索

V2-065154 | MATHDB-340467

Gun-Ram Murty-Rath conjecture on logarithms of gamma values

摘录 / EXCERPT: Gun-Ram Murty-Rath conjecture. The dimension of $\overline{V_{\Gamma}(q)}$ is

conjecture | open | MathDB 页面 | 原文线索

V2-065155 | MATHDB-340483

The Bauer-Golinelli formula for the atomic mass at zero

摘录 / EXCERPT: Bauer-Golinelli conjecture. For any $c\in(0,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-065156 | MATHDB-340484

The Bauer-Golinelli conjecture on emergence of extended states at zero

摘录 / EXCERPT: Bauer-Golinelli conjecture. The following phase transition occurs: if $c<e$, then μ_c has no extended states at 0; if $c>e$, then μ_c has extended states at 0.

conjecture | open | MathDB 页面 | 原文线索

V2-065157 | MATHDB-340492

Separation conjecture for list arborability and equitable list arborability

摘录 / EXCERPT: Separation conjecture. There is a graph G and $k\in\mathbb{N}$ satisfying both properties above, but G is not equitably k-list arborable.

conjecture | open | MathDB 页面 | 原文线索

V2-065158 | MATHDB-340493

Equivalence of Milnor-Hamm tube and sphere fibrations

摘录 / EXCERPT: Let $G:({\mathbb R}^{m},0)\to({\mathbb R}^{p},0)$ be a nice analytic map germ with $m>p\geq2$ and radial discriminant, and suppose that Ψ_G is ρ-regular. Assume that the...

conjecture | open | MathDB 页面 | 原文线索

V2-065159 | MATHDB-340494

The density conjecture for G-irregular primes

摘录 / EXCERPT: Let ${\mathcal P}_G$ be the set of G-irregular primes, and write ${\mathcal P}_G(x)$ for the number of its elements not exceeding x. Let A be the Artin constant, ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065160 | MATHDB-340503

Andrews' generating-function conjecture for linked partition ideals

摘录 / EXCERPT: A linked partition ideal is a set of partitions closed under the linked-partition-ideal structure introduced by Andrews. Let $G_{\mathscr{I}}(x,q)$ denote its bivariate generating...

conjecture | open | MathDB 页面 | 原文线索

V2-065161 | MATHDB-340504

Kanade--Russell Conjecture I_1 on partition identities

摘录 / EXCERPT: For a nonnegative integer n, consider partitions into parts congruent to 1, 3, 6, or 8 modulo 9. Also consider partitions satisfying the difference at least 3 at dist...

conjecture | open | MathDB 页面 | 原文线索

V2-065162 | MATHDB-340505

The sharp L^2 extension conjecture for paraboloids when $d=4k-1$

摘录 / EXCERPT: Let $P\subset\mathbb F_q^d$ be the paraboloid. Suppose that $d=4k-1$ for some $k\in\mathbb N$ and that $-1\in\mathbb F_q$ is not a square. Sharp paraboloid extension conjecture. On...

conjecture | open | MathDB 页面 | 原文线索

V2-065163 | MATHDB-340506

The sharp $L^p\to L^4$ extension conjecture for odd-dimensional spheres

摘录 / EXCERPT: Let $S_j\subset\mathbb F_q^d$ be a sphere of nonzero radius j. Odd-dimensional sphere extension conjecture. The following estimates should hold: 1. If $d=4k+1$, $k\in\mathbb N$,...

conjecture | open | MathDB 页面 | 原文线索

V2-065164 | MATHDB-340507

The sharp even-dimensional sphere extension conjecture

摘录 / EXCERPT: Let $d\ge 2$ be even, and let $S_j\subset\mathbb F_q^d$ be the sphere of radius $j\ne 0$ centered at the origin. Even-dimensional sphere extension conjecture. One has ... This estima...

conjecture | open | MathDB 页面 | 原文线索

V2-065165 | MATHDB-340526

Odd-part partition excess conjecture by residue class

摘录 / EXCERPT: Let $\operatorname{part}(n,S_{\mathrm{odd}})$ be the set of partitions of n into odd parts, and let l denote the number of parts. Odd-part excess conjecture. For each residue c...

conjecture | open | MathDB 页面 | 原文线索

V2-065166 | MATHDB-340528

3-colorability of plane graphs with two-colored faces and an even face class

摘录 / EXCERPT: The face-coloring conjecture. Every simple plane graph whose faces can be properly colored with two colors such that one color class contains only even faces is 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-065167 | MATHDB-340541

Converse to the ferromagnetic clique-maximizing theorem

摘录 / EXCERPT: Let $H\colon\Omega\times\Omega\to\mathbb{R}_{\geq 0}$ be a symmetric measurable weighted graph, or model. Call H clique-maximizing if cliques maximize the relevant homomorphism q...

conjecture | open | MathDB 页面 | 原文线索

V2-065168 | MATHDB-340542

Antiferromagnetic biclique-maximizing conjecture

摘录 / EXCERPT: Let $H\colon\Omega\times\Omega\to\mathbb{R}_{\geq 0}$ be a symmetric measurable weighted graph, or model. Call H antiferromagnetic when all eigenvalues other than the top one are...

conjecture | open | MathDB 页面 | 原文线索

V2-065169 | MATHDB-340545

DeMarco-Kahn upper tail conjecture for strictly balanced graphs and dense regimes

摘录 / EXCERPT: Let H be a graph, let m_H denote its appearance-threshold parameter, and let the DeMarco-Kahn upper tail conjecture be the assertion that the upper-tail estimate stated above h...

conjecture | open | MathDB 页面 | 原文线索

V2-065170 | MATHDB-340553

Weighted Dirichlet-minimization conjecture for homeomorphisms between spherical rings

摘录 / EXCERPT: Let $\mathbb{A}(r,R)$ and $\mathbb{A}(r_\ast,R_\ast)$ be spherical rings in $\mathbf{R}^n$, and let $\mathcal{F}$ be the family of homeomorphisms between them belonging to...

conjecture | open | MathDB 页面 | 原文线索

V2-065171 | MATHDB-340556

Orlik-Randell exceptional-collection conjecture for invertible polynomials

摘录 / EXCERPT: Let $f\in S$ be an invertible polynomial, let L_f be its maximal grading, and let $\widetilde{f}$ be its Berglund-Hübsch transpose with Milnor number $\mu(\widetilde{f})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065172 | MATHDB-340568

Majorization inequality for Ferrers-good bipartite graphs

摘录 / EXCERPT: Let d be a partition with $\ell(d)=n$, and let λ be a non-increasing sequence of positive real numbers with $\ell(\lambda)=n-1$. Suppose that $d=a\oplus b$ for sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-065173 | MATHDB-340573

Comparison of precomplicial spaces with Rezk's Θ_n-spaces

摘录 / EXCERPT: Let $n\geq 0$. Write $\mathit{s}\mathcal{S}\!\mathit{et}^{t\Delta^{\operatorname{op}}}$ for the category of prestratified simplicial spaces, equipped with the cartesian model struc...

conjecture | open | MathDB 页面 | 原文线索

V2-065174 | MATHDB-340585

The Dyck-path-tree correspondence for modern intervals and dexter upper ideals

摘录 / EXCERPT: Let w be a Dyck path and let $t=\kappa(w)$ be the corresponding planar rooted binary tree under the bijection κ from Dyck paths to planar rooted binary trees. A correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-065175 | MATHDB-340588

Refined Joyce-Song pair conjecture for Vafa-Witten invariants

摘录 / EXCERPT: Let $(S,\mathcal O_S(1))$ be a polarised surface, let $X=K_S$, and let $\alpha=(r,c_1,c_2)\in H^{\operatorname{ev}}(S)$ be a charge. Let $P^\perp_\alpha(n,t)$ be the refined invari...

conjecture | open | MathDB 页面 | 原文线索

V2-065176 | MATHDB-340589

Refined K3 multiple-cover conjecture for Vafa-Witten invariants

摘录 / EXCERPT: Let S be a K3 surface, let α be a charge, and let $\chi_S(\alpha,\alpha)$ denote its Mukai pairing. Write $[r]_t=(t^{r/2}-t^{-r/2})/(t^{1/2}-t^{-1/2})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065177 | MATHDB-340596

BCOV's polynomiality conjecture for quintic threefolds

摘录 / EXCERPT: Let G_g^{BCOV} be the set of genus-g stable graphs for the quintic threefold, with solid, half-dotted-half-solid, and dotted edges. Place the propagators...

conjecture | open | MathDB 页面 | 原文线索

V2-065178 | MATHDB-340597

BCOV's polynomiality conjecture with legs

摘录 / EXCERPT: Let $G_{g,n}^{\mathrm{BCOV}}$ be the set of genus-g, n-leg stable graphs with the BCOV edge types and solid or dotted legs. Define $f_{g,n}^{\mathrm{BCOV}}$ as the sum over the...

conjecture | open | MathDB 页面 | 原文线索

V2-065179 | MATHDB-340599

Conjectural classification of near-equality for ribbon Schur functions

摘录 / EXCERPT: Near-equality classification conjecture. Up to reversal of α and β, the compositions α, β and partition ν are one of the five cases of Theorem 1.1, on...

conjecture | open | MathDB 页面 | 原文线索

V2-065180 | MATHDB-340602

Hedetniemi's conjecture on the chromatic number of categorical graph products

摘录 / EXCERPT: For graphs G and H, let $G\times H$ denote their categorical product and let $\chi(G)$ denote the chromatic number of a graph. Hedetniemi's conjecture. ... This is a famous open...

conjecture | open | MathDB 页面 | 原文线索

V2-065181 | MATHDB-340608

Bachoc-Robins conjecture on optimal density for tiling norms

摘录 / EXCERPT: Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ whose unit ball tiles $\mathbb{R}^n$ by translation. Let $m_1(\mathbb{R}^n,\|\cdot\|)$ denote the supremum of the upper densities of mea...

conjecture | open | MathDB 页面 | 原文线索

V2-065182 | MATHDB-340609

The large-error lower-bound conjecture for uniform attachment seed detection

摘录 / EXCERPT: Large-error lower-bound conjecture. The required confidence-set size is at least a constant multiple of $k\ell$ for sufficiently large seeds and a fixed positive error level. This...

conjecture | open | MathDB 页面 | 原文线索

V2-065183 | MATHDB-340615

The exponential deficit conjecture for near-optimal Kneser graph colorings

摘录 / EXCERPT: Let $KG_{n,k}$ be the Kneser graph on the k-element subsets of $[n]$, with edges joining disjoint sets. Consider a subset of its vertices that is properly colorable with $n-2k+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-065184 | MATHDB-340623

The k-invariant bound for fusion systems

摘录 / EXCERPT: Let S be a finite p-group, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. Let $\mathbf{k}(\mathcal{F},\alpha)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-065185 | MATHDB-340624

The k-m equality for fusion systems

摘录 / EXCERPT: Let S be a finite p-group, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. Let $\mathbf{k}(\mathcal{F},\alpha)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-065186 | MATHDB-340625

The defect distribution conjecture for nonabelian fusion systems

摘录 / EXCERPT: Let S be a nonabelian finite p-group of order p^d, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. The defect d...

conjecture | open | MathDB 页面 | 原文线索

V2-065187 | MATHDB-340626

The Eaton-Moreto defect minimum conjecture for fusion systems

摘录 / EXCERPT: Let S be a nonabelian finite p-group of order p^d, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. Let $r>0$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-065188 | MATHDB-340627

The commutator and conjugacy-class bounds for fusion-system invariants

摘录 / EXCERPT: Let S be a finite p-group of order p^d, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065189 | MATHDB-340628

The Knörr-Robinson reformulation for fusion systems

摘录 / EXCERPT: Let S be a nontrivial finite p-group, let $\mathcal{F}$ be a saturated fusion system on S, and let α be an $\mathcal{F}$-compatible family. For each fully normalized n...

conjecture | open | MathDB 页面 | 原文线索

V2-065190 | MATHDB-340635

The conjecture that non-SAGE certificates fail beyond the corollary's hypotheses

摘录 / EXCERPT: Non-SAGE separation conjecture. If every $\bm{a}_i$ lies in either $\operatorname{ext}\mathcal{P}(\bm{A})$ or $\operatorname{int}\mathcal{P}(\bm{A})$, but $\bm{A}$ does not satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-065191 | MATHDB-340640

Amdeberhan's refined Nekrasov-Okounkov hook length conjecture

摘录 / EXCERPT: Let λ be a partition of n, and let $\mathcal{H}(\lambda)$ be its multiset of hook lengths. Let $\mathcal{H}(\lambda)^{\diamond}$ be the multiset of hook lengths with triv...

conjecture | open | MathDB 页面 | 原文线索

V2-065192 | MATHDB-340641

Amdeberhan's root conjecture for Nekrasov-Okounkov polynomials

摘录 / EXCERPT: Amdeberhan's root conjecture. The polynomial $Q_n(z)$ has (i) only simple roots, (ii) only real roots, and (iii) only negative roots. The polynomial is considered in connection wit...

conjecture | open | MathDB 页面 | 原文线索

V2-065193 | MATHDB-340642

Amdeberhan's unimodality conjecture for Nekrasov-Okounkov polynomials

摘录 / EXCERPT: Amdeberhan's unimodality conjecture. For every positive integer n, the polynomial $Q_n(z)$ is unimodal. This concerns the coefficient sequence of the hook-length polynomial assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-065194 | MATHDB-340647

Finite-orbit conjecture for sufficiently large sets of quadratic polynomials

摘录 / EXCERPT: Let $S=\{x^2+c_1,x^2+c_2,\dots,x^2+c_s\}$ be a set of distinct quadratic polynomials with $c_i\in\mathbb{Q}$. A point $P\in\mathbb{Q}$ has finite orbit for S if its forward orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-065195 | MATHDB-340648

Uniform boundedness conjecture for finite orbits of rational maps

摘录 / EXCERPT: Let K be a number field, let d be an integer, and let $S=\{\phi_1,\phi_2,\dots,\phi_s\}$ be a set of rational maps defined over K such that $2\leq\deg(\phi)\leq d$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-065196 | MATHDB-340649

Point-independence conjecture for log BPS numbers

摘录 / EXCERPT: Let (S,E) be a rational log Calabi-Yau surface with smooth divisor and let $\beta\in\operatorname{H}_2(S,\mathbb{Z})$. For $P\in E(\beta)$, let m^P_β denote the log BPS num...

conjecture | open | MathDB 页面 | 原文线索

V2-065197 | MATHDB-340650

Log-local principle for BPS invariants

摘录 / EXCERPT: Assume that S is a del Pezzo surface and E is an anticanonical divisor. Let $P\in E(\beta)$ be β-primitive, where m^P_β is the log BPS number at P and n_β i...

conjecture | open | MathDB 页面 | 原文线索

V2-065198 | MATHDB-340652

Point-independence conjecture at primitive points

摘录 / EXCERPT: Let β be a curve class, let (S,E) be general with respect to β, and let $P,P'\in E(\beta)$ be β-primitive points. Primitive-point independence conjecture. ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-065199 | MATHDB-340653

Strict convexity characterization by linear extension of sphere embeddings

摘录 / EXCERPT: Let X be a normed space with unit sphere S_X, and let Y range over normed spaces with unit spheres S_Y. An isometric embedding $S_Y\to S_X$ is a distance-preserving map bet...

conjecture | open | MathDB 页面 | 原文线索

V2-065200 | MATHDB-340655

The horizontal-section diameter conjecture for subgroups of product groups

摘录 / EXCERPT: Horizontal-section diameter conjecture. There are absolute constants $C_{1},C_{2}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065201 | MATHDB-340672

Specialization map for strictly geometric triangulated categories

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field K and residue field k. For a field L, let $K_0(\mathrm{sGT}_L)$ be the Grothendieck group generated by bounded derive...

conjecture | open | MathDB 页面 | 原文线索

V2-065202 | MATHDB-340673

Specialization maps for Grothendieck groups of geometric triangulated categories

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field K and residue field k. For a field L, let $K_0(\mathrm{GT}_L)$ and $K_0(\mathrm{sGT}_L)$ denote the Grothendieck grou...

conjecture | open | MathDB 页面 | 原文线索

V2-065203 | MATHDB-340674

Factorization conjecture for the derived specialization map

摘录 / EXCERPT: Let R be a discrete valuation ring with fraction field K and residue field k. Let M_K be the abelian group freely generated by isomorphism classes of smooth proper...

conjecture | open | MathDB 页面 | 原文线索

V2-065204 | MATHDB-340680

Thiagarajan's finite 1-safe Petri-net characterization conjecture

摘录 / EXCERPT: Let ${\mathcal E}$ be an event structure, and let ${\mathcal E}_N$ denote the event-structure unfolding arising from a finite 1-safe Petri net N. Thiagarajan's finite 1-safe Petr...

conjecture | open | MathDB 页面 | 原文线索

V2-065205 | MATHDB-340681

The minimal-alphabet interior conjecture for irreducible unit Pisot matrices

摘录 / EXCERPT: Minimal-alphabet interior conjecture. For all irreducible unit Pisot matrices M with spectral radius β, we have

conjecture | open | MathDB 页面 | 原文线索

V2-065206 | MATHDB-340682

The natural-alphabet interior conjecture for irreducible unit Pisot substitutions

摘录 / EXCERPT: Natural-alphabet interior conjecture. For all irreducible unit Pisot substitutions s, we have

conjecture | open | MathDB 页面 | 原文线索

V2-065207 | MATHDB-340684

Frequency and bounded runs in generalized Thue-Morse sequences

摘录 / EXCERPT: Run-length conjecture. The frequency of 0's in $(c_n^{(p)})_{n\in\mathbb{N}}$ is 1, and the sequence contains arbitrarily long strings of consecutive 0's. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-065208 | MATHDB-340698

The instantonicity conjecture for rank 2 bundles on the Fano threefold V_5

摘录 / EXCERPT: Instantonicity conjecture. Every such bundle satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065209 | MATHDB-340708

Bounded maximum under positivity conditioning for the membrane model

摘录 / EXCERPT: Let $n=2$ or $n=3$, let V_N denote the finite domain of the membrane model, let $\psi=(\psi_x)_{x\in V_N}$ be its field, let E_N denote expectation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065210 | MATHDB-340709

Lower-bound conjecture for the Cheeger constant of random Cayley graphs

摘录 / EXCERPT: Cheeger lower-bound conjecture. There exists an absolute constant c such that, for all $\varepsilon\in(0,1)$, there are constants $N(\varepsilon)$ and $M(\varepsilon)$ such that,...

conjecture | open | MathDB 页面 | 原文线索

V2-065211 | MATHDB-340721

Berry-type conjecture for eigenfunctions on expanding locally symmetric spaces

摘录 / EXCERPT: Let X be the universal symmetric space and let $(M_n=\Gamma_n\backslash X)$ be a uniformly discrete expander family that BS-converges toward X. Fix an eigenvalue λ_0 of...

conjecture | open | MathDB 页面 | 原文线索

V2-065212 | MATHDB-340731

The variance asymptotic conjecture for sums of two squares

摘录 / EXCERPT: Let $b(n)$ be the indicator of integers representable as sums of two squares. Let $B(x)=\sum_{n\leq x}b(n)$, let $I(x;H)$ be the paper's smooth approximation to $B(x+H)-B(x)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065213 | MATHDB-340732

The divisor-sum variance asymptotic conjecture

摘录 / EXCERPT: For $z>0$, let $d_z(n)$ be the generalized divisor function, define $D_z(x)=\sum_{n\leq x}d_z(n)$, and let $\overline D_z(x)$ be the contour approximation defined in the paper. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-065214 | MATHDB-340733

The arithmetic-progression variance conjecture for generalized divisor functions

摘录 / EXCERPT: Let $d_z(n)$ be the generalized divisor function and define the arithmetic-progression variance by ... Let a_z and $\alpha_1^{(z)}$ be as in the paper. The arithmetic-progression d...

conjecture | open | MathDB 页面 | 原文线索

V2-065215 | MATHDB-340743

Phantom category conjecture for Barlow surfaces

摘录 / EXCERPT: Let X be a Barlow surface, and let $\mathrm{D}(X)$ denote its derived category. Barlow surface phantom conjecture. The category $\mathrm{D}(X)$ has an exceptional collection of l...

conjecture | open | MathDB 页面 | 原文线索

V2-065216 | MATHDB-340744

Orlov's rationality conjecture for surfaces with full exceptional collections

摘录 / EXCERPT: Let X be a smooth projective surface. A full exceptional collection in $\mathrm{D}(X)$ is an exceptional collection that generates the derived category; in the line-bundle case,...

open_problem | open | MathDB 页面 | 原文线索

V2-065217 | MATHDB-340745

Macpherson's conjecture on the growth constant for primitive oligomorphic groups

摘录 / EXCERPT: Let G be a primitive oligomorphic permutation group acting on a countable infinite set ... of size n. Macpherson's theorem states that there is a constant $c>1$ such that either...

conjecture | open | MathDB 页面 | 原文线索

V2-065218 | MATHDB-340746

Equality criterion for Hodge ideals and microlocal Hodge ideals

摘录 / EXCERPT: Let $D=\alpha Z$, where $0<\alpha\le 1$ and $Z=(f=0)$ is an integral and reduced effective divisor defined by a weighted homogeneous polynomial f with an isolated singularity at...

conjecture | open | MathDB 页面 | 原文线索

V2-065219 | MATHDB-340758

Anti-labeling conjecture for trees

摘录 / EXCERPT: Let T_n be a tree with bipartition (X_1,X_2), where $|X_i|=q_i$ for $i=1,2$. Write $q=\min\{q_1,q_2\}$. Anti-labeling conjecture for trees. One has ... This conjecture asserts th...

conjecture | open | MathDB 页面 | 原文线索

V2-065220 | MATHDB-340759

Anti-labeling conjecture for two-dimensional grids

摘录 / EXCERPT: Let $G=P_m\times P_n$ be a two-dimensional grid, and let $m=\min\{m,n\}$. Grid anti-labeling conjecture. One has ... The paper proves the displayed quantity as a lower bound by expli...

conjecture | open | MathDB 页面 | 原文线索

V2-065221 | MATHDB-340760

Anti-labeling conjecture for hypercubes

摘录 / EXCERPT: For each integer $n\geq 2$, let Q_n denote the n-dimensional cube, with 2^n vertices. Hypercube anti-labeling conjecture. One has ... The bound is stated after being shown to b...

conjecture | open | MathDB 页面 | 原文线索

V2-065222 | MATHDB-340765

Noncoincidence conjecture for covariance-operator eigenvalues

摘录 / EXCERPT: Let ρ_k, $k\in\mathbb{N}_0$, denote the quantities appearing in the spectral decomposition of the covariance operator $\mathcal{S}$, and let δ be an eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-065223 | MATHDB-340779

Global stability conjecture for locally stable steady states in competition models

摘录 / EXCERPT: Consider the nonlocal competition model described above, with two population densities and a locally stable steady state. Global stability conjecture. The locally stable steady sta...

conjecture | open | MathDB 页面 | 原文线索

V2-065224 | MATHDB-340782

The interior balanced-submatrix conjecture

摘录 / EXCERPT: Let $A\text{ }\in\mathcal{B}_{n}(\mathbb{R})$ be a square balanced matrix. An interior balanced-submatrix conjecture asserts that some interior of A, meaning a submatrix of A,...

conjecture | open | MathDB 页面 | 原文线索

V2-065225 | MATHDB-340783

The fair-discrepancy propagation conjecture for fully balanced matrices

摘录 / EXCERPT: Fair-discrepancy propagation conjecture. If one fixed row has discrepancy less than ϵ, then every row has discrepancy less than ϵ.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-065226 | MATHDB-340785

Sofic groups and finite Moufang loops

摘录 / EXCERPT: Sofic-Moufang conjecture. A group G is sofic if and only if it is locally embeddable into finite Moufang loops.

conjecture | open | MathDB 页面 | 原文线索

V2-065227 | MATHDB-340791

The dichotomy conjecture for the growth of irreducible-factor degrees

摘录 / EXCERPT: Throughout, let (f,g) be a pair of polynomials over $\mathbb{F}_q$ that are neither critical nor p-critical, with f irreducible and $deg g greater than or equal to 2$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065228 | MATHDB-340792

Exponential-growth conjecture for irreducible factors of iterated polynomials

摘录 / EXCERPT: Throughout, let (f,g) be a pair of polynomials over $\mathbb{F}_q$ that are neither critical nor p-critical, with f irreducible and $\deg g\geq 2$. Let $N_{f,g}(n)$ be the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-065229 | MATHDB-340795

The spider's-web conjecture for the quite fast escaping set

摘录 / EXCERPT: Quite fast escaping set conjecture. The set $Q(f)$ contains a spider's web. Consequently, $I(f)$ is a spider's web and is connected.

conjecture | open | MathDB 页面 | 原文线索

V2-065230 | MATHDB-340800

The P1 optimality conjecture for dynamic priority queues

摘录 / EXCERPT: Let $P0$, $P1$, and $P2$ denote the optimization problems described in the paper, and let S_p be the primary-class service level. Define ... and let $I^-\equiv(\hat{S}_p,I_l)$, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-065231 | MATHDB-340801

The Negativity Conjecture for smooth \(\mathbb{Q}\)-acyclic surfaces

摘录 / EXCERPT: Let S be a smooth $\mathbb{Q}$-acyclic surface, and let (X,D) be a log smooth completion of S. Here, K_X is the canonical divisor of X, D is the boundary divisor, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065232 | MATHDB-340802

Flenner-Zaidenberg's Strong Rigidity Conjecture

摘录 / EXCERPT: Let S be a $\mathbb{Q}$-acyclic surface of log general type, and let (X,D) be a minimal log smooth completion of S. Here, $\mathcal{T}_{X}(-\log D)$ denotes the logarithmic t...

conjecture | open | MathDB 页面 | 原文线索

V2-065233 | MATHDB-340806

Quasi-polarized isogeny realization conjecture for K3 surfaces

摘录 / EXCERPT: Let (X,ξ) be a quasi-polarized K3 surface over k. Let (Λ_p,λ_p) and (Λ^p,λ^p) be pointed lattices over $W(k)$ and $\hat{\mathbb{Z}}^p$, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-065234 | MATHDB-340815

Factorization conjecture for Kazhdan-Lusztig R-polynomials of 321-avoiding 2-repeating permutations

摘录 / EXCERPT: Let $w\in \mathfrak{S}_n$ be a 321-avoiding and 2-repeating permutation. Let $u,v\in\mathfrak{S}_n$ satisfy $u\leq v\leq w$, and let $\tilde{R}_{u,v}(t)$ denote the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-065235 | MATHDB-340819

Neftin's simultaneous conjugation conjecture for permutations

摘录 / EXCERPT: Let α and β be permutations in S_n. Write $[\alpha,\beta]=\alpha\beta\alpha^{-1}\beta^{-1}$ for their commutator, and let $\alpha^\gamma=\gamma^{-1}\alpha\gamma$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-065236 | MATHDB-340828

The Bruhat trajectory conjecture for non-split real forms

摘录 / EXCERPT: Bruhat trajectory conjecture. There exist trajectories connecting w and w' if and only if the corresponding elements are comparable in Bruhat order. In that case, the space of...

conjecture | open | MathDB 页面 | 原文线索

V2-065237 | MATHDB-340829

The Curve Histogram Conjecture for fully regular plane curves

摘录 / EXCERPT: Curve Histogram Conjecture. The distance histogram $H_{\mathcal{X}}$ determines a fully regular plane curve up to isometry.

conjecture | open | MathDB 页面 | 原文线索

V2-065238 | MATHDB-340834

Chromatic-choosability conjecture for signed graphs with few vertices

摘录 / EXCERPT: Chromatic-choosability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-065239 | MATHDB-340836

Convexity and endpoint maximum conjecture for spiderweb central configurations

摘录 / EXCERPT: Convexity and endpoint maximum conjecture. The sequence $(a_i)_{1\leq i\leq n-1}$ is convex. It is strictly increasing when $\ell=2$, and only when $\ell=2$. Moreover, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-065240 | MATHDB-340848

Convexity conjecture for Whitham's highest cusped wave

摘录 / EXCERPT: Let φ be the profile of a highest cusped periodic traveling-wave solution of Whitham's equation, with speed $\mu>0$, satisfying ... where L has Fourier symbol...

conjecture | open | MathDB 页面 | 原文线索

V2-065241 | MATHDB-340861

Dlt log Calabi--Yau compactification conjecture for the Betti moduli space

摘录 / EXCERPT: Let M_B be the Betti moduli space, and let $\mathcal{D}(\partial M_B)$ denote the dual complex associated with the boundary of a dlt compactification. Dlt log Calabi--Yau compact...

conjecture | open | MathDB 页面 | 原文线索

V2-065242 | MATHDB-340862

Deligne's conjecture on Frobenius traces for moduli of connections

摘录 / EXCERPT: Let C_0 be a smooth projective curve over the finite field $\mathbb F_q$ of characteristic $p>0$, let C be its base change to $\overline{\mathbb F}_q$, and let M_K and...

conjecture | open | MathDB 页面 | 原文线索

V2-065243 | MATHDB-340877

Lower bound conjecture for the top Lyapunov exponent on Hitchin components

摘录 / EXCERPT: Let C be a compact Riemann surface, and let the n-th Hitchin component be the connected component of ... containing symmetric powers of Fuchsian representations. For a representa...

conjecture | open | MathDB 页面 | 原文线索

V2-065244 | MATHDB-340881

The Weyl embedding conjecture for polar Grassmannians

摘录 / EXCERPT: The Weyl embedding conjecture. The Weyl embedding $\widehat{e}^W_n$ of $\widehat{\mathcal{Q}}_n$ induces on $\widetilde{\mathcal{Q}}_n$ its veronesean embedding $\widetilde{e}^W_n$...

conjecture | open | MathDB 页面 | 原文线索

V2-065245 | MATHDB-340884

Crystallisation implies fixed-height border-fixing

摘录 / EXCERPT: Crystallisation conjecture. If a shape X tiles the plane and has the crystallisation property, then it also has the fixed-height border-fixing property.

conjecture | open | MathDB 页面 | 原文线索

V2-065246 | MATHDB-340886

Morse-index transition conjecture for regular polygons

摘录 / EXCERPT: Let $N>4$, let A be the exponent of the homogeneous potential, and let f be the function whose Morse index is considered on the quotient configuration space $\mathcal{C}_N$. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065247 | MATHDB-340887

Four-body Morse-polynomial conjecture

摘录 / EXCERPT: Let f be the function whose critical points represent central configurations in the four-body problem, and let $\mathcal{C}_4$ be the quotient configuration space with Poincaré p...

conjecture | open | MathDB 页面 | 原文线索

V2-065248 | MATHDB-340888

Five-body Morse-structure conjecture

摘录 / EXCERPT: Let f be the function for the equal-mass planar five-body problem, let $\mathcal{C}_5$ be the quotient configuration space, and let $P(t)$ be its Poincaré polynomial. Five-body M...

conjecture | open | MathDB 页面 | 原文线索

V2-065249 | MATHDB-340889

Six-body Morse-polynomial conjecture

摘录 / EXCERPT: Consider the equal-mass planar six-body problem, with exponent A, function f, and Morse polynomial $M(t)$; let $P(t)$ be the Poincaré polynomial of the quotient configuration s...

conjecture | open | MathDB 页面 | 原文线索

V2-065250 | MATHDB-340890

Newtonian Morse-polynomial conjecture for seven-, eight-, and nine-body problems

摘录 / EXCERPT: Let f be the function for the equal-mass planar N-body problem, and let $M(t)$ denote its Morse polynomial. Newtonian Morse-polynomial conjecture. For $A=3$, the Morse polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-065251 | MATHDB-340891

Monotonicity conjecture for equal-mass central configurations

摘录 / EXCERPT: Consider the planar equal-mass N-body problem with homogeneous potential exponent A, and let $C_N(A)$ denote the number of equal-mass central configurations. Monotonicity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-065252 | MATHDB-340894

Cyclotomic composition irreducibility conjecture

摘录 / EXCERPT: Let p and q be distinct odd primes, and let $\Phi_p(x)$ and $\Phi_q(x)$ be the pth and qth cyclotomic polynomials. Cyclotomic composition irreducibility conjecture. At leas...

conjecture | open | MathDB 页面 | 原文线索

V2-065253 | MATHDB-340895

Density-zero conjecture for reducible cyclotomic compositions

摘录 / EXCERPT: For positive integers m,n, call the ordered pair (m,n) good if $\Phi_m(\Phi_n(x))$ factors over the integers, where Φ_m and Φ_n are the mth and nth cyclotomic pol...

conjecture | open | MathDB 页面 | 原文线索

V2-065254 | MATHDB-340896

Generalized density-zero conjecture for reducible cyclotomic compositions

摘录 / EXCERPT: Let $f(t)$ and $g(t)$ be strictly increasing functions tending to infinity as t tends to infinity. Let $D_{f,g}(t)$ count good pairs (m,n), where goodness means that...

conjecture | open | MathDB 页面 | 原文线索

V2-065255 | MATHDB-340909

Conjecture on parity-dependent convergence of periodic spline projections

摘录 / EXCERPT: Let ${\boldsymbol \tau}$ be a uniform knot sequence such that the dimension of $\mathcal{S}_{p,{\boldsymbol \tau},\operatorname{per}}$ is $2m$. For any $q\geq 0$, let Q_p^q be th...

conjecture | open | MathDB 页面 | 原文线索

V2-065256 | MATHDB-340912

The Gap Conjecture for normalised volumes of klt singularities

摘录 / EXCERPT: Gap Conjecture. The normalised volume satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065257 | MATHDB-340913

The conjecture on tangent cones and metric density of log Kähler-Einstein limits

摘录 / EXCERPT: Tangent-cone conjecture. The following properties hold:

conjecture | open | MathDB 页面 | 原文线索

V2-065258 | MATHDB-340916

Cigler's odd-index Hankel determinant formulas for Catalan convolution powers

摘录 / EXCERPT: Let $C(x)=\sum_{n\geq 0}C_nx^n$ be the Catalan generating function, let $F(x,r)=C(x)^r$, and write $H_n(F(x,r))$ for the $n\times n$ Hankel determinant of the coefficients of...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-065259 | MATHDB-340918

Polynomiality conjecture for Hankel determinants of generalized Catalan powers

摘录 / EXCERPT: Let $G(x,r)$ be the generalized generating function used in the paper, and write $H_n(G(x,r))$ for the Hankel determinant of its coefficients. For a positive integer t, consider...

conjecture | open | MathDB 页面 | 原文线索

V2-065260 | MATHDB-340923

Divisibility conjecture for the index of self-dual ternary codes

摘录 / EXCERPT: Let C be a self-dual ternary code of length c. Its index is the number of even maximal codewords minus the number of odd maximal codewords. Index divisibility conjecture. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-065261 | MATHDB-340931

Bhargava-Shankar and Poonen-Rains conjecture on average sizes of Selmer groups

摘录 / EXCERPT: Let n be a positive integer, and order elliptic curves by height. Bhargava-Shankar and Poonen-Rains conjecture. The average size of the n-Selmer groups is the sum of the diviso...

conjecture | open | MathDB 页面 | 原文线索

V2-065262 | MATHDB-340933

The elementary-symmetric-function formula for the two-species ASEP partition function

摘录 / EXCERPT: Let k,r,ell be nonnegative integers with $k+r+\ell=n$, and let $\mathcal{Z}_{(k,r,\ell)}(x_1,\ldots,x_n)$ be the partition function defined by summing the polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-065263 | MATHDB-340934

Superlogarithmic-in-k discrepancy lower-bound conjecture for permutation families

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$ and let a k-permutation family be a multiset union of k maximal chains arising from permutations of $[n]$. Its discrepancy is the minimum, over maps...

conjecture | open | MathDB 页面 | 原文线索

V2-065264 | MATHDB-340947

The optimal balance constant for width-two posets

摘录 / EXCERPT: Width-two optimality conjecture. If P cannot be formed from $\mathbf{1}$ and $\mathcal{E}$ using direct sums, then

conjecture | open | MathDB 页面 | 原文线索

V2-065265 | MATHDB-340948

A positive gap above one-third for non-decomposable posets

摘录 / EXCERPT: Let P be a finite poset, let $\mathbf{1}$ be the singleton poset, let $\mathcal{E}$ be the three-element poset with one relation, and let $\delta(P)$ be its balance constant. Pos...

conjecture | open | MathDB 页面 | 原文线索

V2-065266 | MATHDB-340949

Strict increase of infimal balance constants by poset width

摘录 / EXCERPT: For each positive integer w, let ... where $\delta(P)$ is the balance constant of P. Width-strictness conjecture. The infimal balance constants satisfy ... The equality...

conjecture | open | MathDB 页面 | 原文线索

V2-065267 | MATHDB-340953

Broustet-Höring's log canonical model conjecture for polarized endomorphisms

摘录 / EXCERPT: Broustet-Höring's conjecture. The morphism μ is an isomorphism in codimension one.

conjecture | open | MathDB 页面 | 原文线索

V2-065268 | MATHDB-340955

Conjecture on the most stably singular initial condition for Dyson Brownian motion

摘录 / EXCERPT: Most stably singular initial-condition conjecture. Among all initial conditions for Dyson Brownian motion, those maximizing $R^{(1)}_\beta(0)$ are precisely

conjecture | open | MathDB 页面 | 原文线索

V2-065269 | MATHDB-340968

Strict separation across one-round and eternal game coloring variants

摘录 / EXCERPT: Let G be a finite graph. For the variants indexed by 3, 2, and $*$, let $\chi_g^3(G)$, $\chi_g^2(G)$, and $\chi_g^*(G)$ denote the corresponding one-round game chromatic numb...

conjecture | open | MathDB 页面 | 原文线索

V2-065270 | MATHDB-340969

Reiner-Tenner-Yong's CDE conjecture for vexillary permutations

摘录 / EXCERPT: Let λ be a rectangular staircase of the form ... Let $w\in\mathfrak{S}_n$ be a vexillary permutation of shape λ, meaning that its Rothe diagram can be transformed to...

conjecture | open | MathDB 页面 | 原文线索

V2-065271 | MATHDB-340970

Chen-Haiman's Schur positivity conjecture for Catalan functions

摘录 / EXCERPT: Let Psi be a root ideal and let mu be a partition. The Catalan function $H(Psi;mu)$ is a symmetric function associated with this root ideal and partition. Chen-Haiman's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-065272 | MATHDB-340983

Complete Lagrangian subspace characterization of self-adjoint boundary conditions

摘录 / EXCERPT: Complete Lagrangian boundary-condition conjecture. The boundary condition

conjecture | open | MathDB 页面 | 原文线索

V2-065273 | MATHDB-340986

Nonexistence of an SVML for directed stars

摘录 / EXCERPT: Let S_n be a directed star (distar) with center v_0 and leaves $v_1,v_2,\ldots,v_n$. An SVML nonexistence claim. Distar S_n does not have an SVML for any n. The preceding r...

conjecture | open | MathDB 页面 | 原文线索

V2-065274 | MATHDB-340991

Piyaratne-Toda refined Bogomolov-Gieseker inequality conjecture

摘录 / EXCERPT: Fix real numbers $\alpha_0>0$ and β_0, and let $E\in\operatorname{Coh}^{\alpha_0\omega,B+\beta_0\omega}(X)$ be a $\nu_{\alpha_0\omega,B+\beta_0\omega}$-semistable object. Def...

conjecture | open | MathDB 页面 | 原文线索

V2-065275 | MATHDB-340992

Bilinear tau-function conjecture for q-Painlevé V

摘录 / EXCERPT: Let τ_i $(i=1,\dots,6)$ be the tau functions associated with the q-Painlevé V equation, with parameters θ_*, θ_t, and θ_0. Write $\overline{f}(t)=f(qt)$...

conjecture | open | MathDB 页面 | 原文线索

V2-065276 | MATHDB-340994

Sharp inequalities for h*-vectors of three-dimensional lattice polytopes

摘录 / EXCERPT: Let P be a three-dimensional lattice polytope having at least one interior lattice point, and write its h^*-vector as $(1,h^*_1,h^*_2,h^*_3)$. Conjecture on sharp h^*-vector...

conjecture | open | MathDB 页面 | 原文线索

V2-065277 | MATHDB-340999

Rogers-Zudilin type conjecture for the Gauss elliptic family

摘录 / EXCERPT: Let $X\to\mathbb P^1$ be the elliptic fibration defined by ... Let X_a be the fiber at $t=a\in\mathbb Q\setminus\{0,1\}$, and suppose that ... is integral in the sense of Scholl. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065278 | MATHDB-341001

Fermat and related curve Rogers-Zudilin type conjectures

摘录 / EXCERPT: Let $F_{N,M}$ be the Fermat curve defined by ... and let $F^*_{2,4}$ be the curve $z^2=w^4+1$. Let $\sigma=\sigma_1$, so $\sigma(t)=t^p$. Fermat and related curve conjectures. There...

conjecture | open | MathDB 页面 | 原文线索

V2-065279 | MATHDB-341002

Addition-deletion conjecture for free hyperplane arrangements

摘录 / EXCERPT: Addition-deletion conjecture. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-065280 | MATHDB-341004

Guenin's projective-cube conjecture for K_5-minor-free signed bipartite graphs

摘录 / EXCERPT: Let (G,Σ) be a signed bipartite graph, with unbalanced girth at least $2k$, that has no $(K_5,E(K_5))$-minor. The signed projective cube $SPC(2k-1)$ is the signed graph on v...

conjecture | open | MathDB 页面 | 原文线索

V2-065281 | MATHDB-341005

The conjecture on 0-completeness of finite abelian groups

摘录 / EXCERPT: Let G be a finite abelian group with exponent n, and let $A\subseteq\mathbb{Z}\backslash\left\{kn:k\in\mathbb{Z}\right\}$. 0-completeness conjecture. Every such group G is...

conjecture | open | MathDB 页面 | 原文线索

V2-065282 | MATHDB-341008

Kuperberg's packing-covering density conjecture for convex bodies

摘录 / EXCERPT: Let $d\geqslant 2$ be fixed. For a d-dimensional convex body C, let $\delta(C)$ and $\theta(C)$ denote its packing and covering densities by congruent copies. Kuperberg's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-065283 | MATHDB-341010

JNS conjecture on eight tau functions for q-Painlevé VI

摘录 / EXCERPT: Let $\tau(\boldsymbol{\theta};\sigma,s|t)$ be the tau function defined from the five-dimensional Nekrasov partition function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065284 | MATHDB-341011

The c=1 and c=-2 tau-function correspondence conjecture

摘录 / EXCERPT: Let $u=q^{2\sigma}$ and let τ and $\uptau$ denote the $c=1$ and $c=-2$ Painlevé VI tau functions, respectively. Let $\td{u}=\td{u}(u)$ and $\td{s}=\td{s}(s,\sigma)$ be functio...

conjecture | open | MathDB 页面 | 原文线索

V2-065285 | MATHDB-341012

The pure 5d Nekrasov partition-function relation at special parameters

摘录 / EXCERPT: Let $\mathcal{Z}_{\mathrm{inst}}(i,i,i,iq^{\pm1/2},u;q^{-1},q|z^{1/2})$ denote the five-dimensional Nekrasov instanton partition function with the displayed special values of v_i...

conjecture | open | MathDB 页面 | 原文线索

V2-065286 | MATHDB-341013

Pure O-sequence conjecture for PS ear-decomposable simplicial complexes

摘录 / EXCERPT: Let Δ be a PS ear-decomposable simplicial complex, and let $h(\Delta)$ denote its h-vector. The PS ear-decomposability conjecture. The vector $h(\Delta)$ is a pure...

conjecture | open | MathDB 页面 | 原文线索

V2-065287 | MATHDB-341016

Lê's conjecture on equisingular deformations of plane curve singularities

摘录 / EXCERPT: Let $(V(f),\0) \subseteq ({\mathbb C}^3,\0)$ be a reduced hypersurface with $\dim_\0 \Sigma f=1$, and suppose that the normalization of $V(f)$ is a bijection. Lê's conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-065288 | MATHDB-341018

Weyl-equivariance of equivariant multiplicities for MV cycles

摘录 / EXCERPT: Let $\mathbold{G}$ be a complex reductive group, let λ be a dominant coweight lying in the coroot lattice, and let W be the Weyl group of $\mathbold{G}$, canonically iden...

conjecture | open | MathDB 页面 | 原文线索

V2-065289 | MATHDB-341021

Term-order characterization of inductively pierced neural codes

摘录 / EXCERPT: Let $\mathcal C$ be a neural code on n neurons, and let $T_\mathcal C$ be its toric ideal. A code is 0- or 1-inductively pierced when it belongs to the corresponding inductively...

conjecture | open | MathDB 页面 | 原文线索

V2-065290 | MATHDB-341022

Quadratic Gröbner basis for toric ideals of inductively pierced codes

摘录 / EXCERPT: Let $\mathcal C$ be a neural code labeled so that neuron i is added as a piercing at the ith step, and let $T_\mathcal C$ be its toric ideal. Let $\prec$ be the monomial order...

conjecture | open | MathDB 页面 | 原文线索

V2-065291 | MATHDB-341030

Double Hurwitz coefficient-expansion conjecture

摘录 / EXCERPT: Let $DH_{g,n}(\mu_1,\ldots,\mu_n)$ be the double Hurwitz numbers, and let d and A_μ^i be defined by ... where λ is an integer partition with $\ell(\lambda)$ parts and...

conjecture | open | MathDB 页面 | 原文线索

V2-065292 | MATHDB-341034

Linear quantitative CFI conjecture for C_q weights

摘录 / EXCERPT: Let $1<p<q$, let $w\in C_q$, let T be a Calderón--Zygmund operator, and let $f\in C_c^\infty(\mathbb{R}^n)$. Linear quantitative CFI conjecture. The estimate ... should hold. The p...

conjecture | open | MathDB 页面 | 原文线索

V2-065293 | MATHDB-341047

Quadratic-growth extension of the convex-operator Beurling theorem

摘录 / EXCERPT: Let $A\in\mathcal{B}(\mathcal{H})$ be an expansive, convex analytic operator whose moment sequence has linear growth, meaning that there exist constants c_1,c_2 such that ... for e...

conjecture | open | MathDB 页面 | 原文线索

V2-065294 | MATHDB-341053

Almost maximally rigid non-closedness conjecture

摘录 / EXCERPT: Let $\operatorname{LS}_n(r,s)$ denote the set of matrices that can be written as the sum of an $n\times n$ matrix of rank at most r and an $n\times n$ matrix with at most s non...

conjecture | open | MathDB 页面 | 原文线索

V2-065295 | MATHDB-341055

Primitive homogeneous structures distinguishing-number conjecture

摘录 / EXCERPT: Primitive homogeneous structures conjecture. The distinguishing number of every primitive homogeneous countably infinite structure is two or infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-065296 | MATHDB-341056

The global velocity-properties conjecture for the massless XXZ chain

摘录 / EXCERPT: Global velocity-properties conjecture. The conclusions of Proposition 1 hold for all

conjecture | open | MathDB 页面 | 原文线索

V2-065297 | MATHDB-341059

Rubber PT invariance under Mukai-lattice isometries

摘录 / EXCERPT: Let S be a K3 surface, let $\gamma\in H_2(S,\mathbb{Z})$ be nonzero, and let $\varphi:H^*(S,\mathbb{R})\to H^*(S,\mathbb{R})$ be an orthogonal map for the Mukai lattice satisfyin...

conjecture | open | MathDB 页面 | 原文线索

V2-065298 | MATHDB-341062

DGGH's characterization conjecture for cop-win oriented graphs

摘录 / EXCERPT: An oriented graph D is a directed graph obtained by orienting the edges of a simple undirected graph; a directed cycle is cop-dominated if the robber cannot avoid capture by ente...

conjecture | open | MathDB 页面 | 原文线索

V2-065299 | MATHDB-341066

Andreatta-Barbieri-Viale-Bertapelle fullness conjecture

摘录 / EXCERPT: Fullness conjecture. The functor

conjecture | open | MathDB 页面 | 原文线索

V2-065300 | MATHDB-341071

Hohlweg-Nadeau-Williams conjecture on minimality of the Brink-Howlett automaton

摘录 / EXCERPT: Let (W,S) be a finitely generated Coxeter system with associated root system Φ and set of elementary roots $\mathscr{E}$. Let $\mathcal{A}_{BH}$ be the Brink-Howlett automat...

conjecture | open | MathDB 页面 | 原文线索

V2-065301 | MATHDB-341084

Subharmonic variation conjecture for Azukawa pseudometrics of balanced domains

摘录 / EXCERPT: Let $\widetilde{\Omega} \subset \mathbb{C}^n \times \Delta$ be a pseudoconvex domain. For each $t \in \Delta$, define the fiber ... Fix $w\in\mathbb{C}^n$, and assume that every...

conjecture | open | MathDB 页面 | 原文线索

V2-065302 | MATHDB-341085

The small-core convergence conjecture for the outer problem

摘录 / EXCERPT: Small-core convergence conjecture. For fixed V_0, the solution Ω of the outer problem converges to a ball as the diameter of D tends to 0.

conjecture | open | MathDB 页面 | 原文线索

V2-065303 | MATHDB-341086

The conductivity convergence conjecture for the outer problem

摘录 / EXCERPT: Conductivity convergence conjecture. For fixed V_0 and D, the solution Ω of the outer problem converges to a ball as $\sigma_c\to 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-065304 | MATHDB-341087

The large-volume uniqueness conjecture for the outer problem with high-conductivity core

摘录 / EXCERPT: Large-volume uniqueness conjecture. If $\sigma_c>1$, then there exists a threshold $V^*\geq |D|$ such that for all $V_0>V^*$, the outer problem has a unique solution Ω. In p...

conjecture | open | MathDB 页面 | 原文线索

V2-065305 | MATHDB-341088

The elliptic ordering conjecture for solutions of the outer problem

摘录 / EXCERPT: Elliptic ordering conjecture. The solutions of the outer problem form an elliptically ordered family: if $|\Omega_1|<|\Omega_2|$, then

conjecture | open | MathDB 页面 | 原文线索

V2-065306 | MATHDB-341089

The convexity conjecture for the outer problem

摘录 / EXCERPT: Convexity conjecture. If D is convex, then any solution Ω of the outer problem is convex.

conjecture | open | MathDB 页面 | 原文线索

V2-065307 | MATHDB-341093

Chan and Mao's ospt upper-bound conjecture

摘录 / EXCERPT: Chan and Mao's conjecture. For $n\geq 10$,

conjecture | open | MathDB 页面 | 原文线索

V2-065308 | MATHDB-341114

Gromov's surface subgroup conjecture for one-ended hyperbolic groups

摘录 / EXCERPT: Let G be a one-ended word-hyperbolic group, meaning a word-hyperbolic group with one end. Let the surface group of genus 2 be the fundamental group of a closed orientable surfa...

conjecture | open | MathDB 页面 | 原文线索

V2-065309 | MATHDB-341116

The List Colouring Conjecture for line graphs

摘录 / EXCERPT: Let a graph G be edge k-colourable if its line graph is k-colourable, and edge k-choosable if its line graph is k-choosable. The List Colouring Conjecture. Every edge k...

conjecture | open | MathDB 页面 | 原文线索

V2-065310 | MATHDB-341127

Kézdy-Snevily conjecture for the covering function of permutation space

摘录 / EXCERPT: Let $\mathcal{S}_n$ be the set of permutations of $[n]=\{1,2,\dots,n\}$, with Hamming distance d_H. Define $f(n,s)$ to be the minimum size of a subset of $\mathcal{S}_n$ having c...

conjecture | open | MathDB 页面 | 原文线索

V2-065311 | MATHDB-341131

The length conjecture for optimal locally recoverable codes

摘录 / EXCERPT: Let q be the alphabet size, and let an optimal locally recoverable code have minimum distance d and locality r. Length conjecture for optimal locally recoverable codes. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-065312 | MATHDB-341135

Jackson inequality for polynomials on a circular arc

摘录 / EXCERPT: For $T>1$, let A be the open arc of the complex unit circle ... and define ... For $F\in C^1(A)$, write $F'(z)$ for its derivative along the arc. Jackson inequality for polynomials o...

conjecture | open | MathDB 页面 | 原文线索

V2-065313 | MATHDB-341137

Arithmeticity conjecture for isotropic transvections in mapping class group representations

摘录 / EXCERPT: Arithmeticity conjecture. The group generated by the isotropic transvections $\{T_A[\chi]\}_{A\in G\backslash\operatorname{\mathscr{X}}(S,G)}$ maps onto an arithmetic subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-065314 | MATHDB-341144

The NILF conjecture for Julia sets of rational maps

摘录 / EXCERPT: Let f be a rational map, and let $J(f)$ denote its Julia set. An invariant line field on $J(f)$ is an invariant measurable field of unoriented lines defined on $J(f)$. NILF conje...

conjecture | open | MathDB 页面 | 原文线索

V2-065315 | MATHDB-341151

Fu and Tang's unimodality conjecture for the k-crank

摘录 / EXCERPT: Fu and Tang's unimodality conjecture. For $n\geq 0$ and $k\geq 2$, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-065316 | MATHDB-341158

Hodge-FVH correspondence for special cubic Hodge integrals

摘录 / EXCERPT: Let $\mathcal{H}({\bf t};p,q,r;\epsilon)$ be the cubic Hodge free energy, let $w=\epsilon^2\partial_{t_0}^2\mathcal{H}({\bf t};p,q,r;\epsilon)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065317 | MATHDB-341188

Integral period relations conjecture for Hilbert base change

摘录 / EXCERPT: Let F be the totally real field of degree d, let f be a normalized newform and f_F its normalized holomorphic base change. For $\epsilon\in\widehat{W_F}$, let d^+ and...

conjecture | open | MathDB 页面 | 原文线索

V2-065318 | MATHDB-341189

Congruence-ideal conjecture for non-base-change Hilbert cuspforms

摘录 / EXCERPT: Let f be a normalized newform and f_F its base change to F. Let $\eta_f^\sharp$ be the complementary factor in the factorization of the base-change congruence ideal, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065319 | MATHDB-341203

Multiplicative fractional list-coloring conjecture

摘录 / EXCERPT: Multiplicative fractional list-coloring conjecture. If G is (f,N)-list-colorable, then G is (f,mN)-list-colorable for every positive integer m.

conjecture | open | MathDB 页面 | 原文线索

V2-065320 | MATHDB-341205

Local injectivity conjecture for generalized Pak-Stanley labelings

摘录 / EXCERPT: Let $\mathcal{A}$ be a multigraphical arrangement. The generalized Pak-Stanley labeling assigns to each region a $\mathcal{G}_{\mathcal{A}}$-parking function. Local injectivity con...

conjecture | open | MathDB 页面 | 原文线索

V2-065321 | MATHDB-341206

Connectedness conjecture for regions with a repeated generalized Pak-Stanley label

摘录 / EXCERPT: Let $\mathcal{A}$ be a multigraphical arrangement, and let the generalized Pak-Stanley map assign labels to its regions. Assume that this map is not injective. For a fixed label...

conjecture | open | MathDB 页面 | 原文线索

V2-065322 | MATHDB-341207

The constant-redundancy FER conjecture for convolutional code-CRC pairs

摘录 / EXCERPT: Consider a minimal convolutional code with m memory elements and a degree-v distance-spectrum-optimal CRC code, used together under serial list Viterbi decoding at a fixed SNR....

conjecture | open | MathDB 页面 | 原文线索

V2-065323 | MATHDB-341208

Maximal third Chern character conjecture for stable sheaf moduli spaces

摘录 / EXCERPT: Let $M(r,c,d,e)$ denote the moduli space of Gieseker-semistable sheaves on ${\mathbb{P}}^3$ with Chern character (r,c,d,e). Let $r \in {\mathbb{Z}}_{\geq 0}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-065324 | MATHDB-341214

Transcendence conjecture for the continued fraction F(x,y)

摘录 / EXCERPT: Transcendence conjecture. If x and y are positive integers with $x\neq y$, then $F(x,y)$ is transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-065325 | MATHDB-341217

Quantized non-negative reduced partition function conjecture

摘录 / EXCERPT: Let $G=G(a,b,c)$ and let $Z_{{\rm red}}^{G(a,b,c)}(\mathbf{t},q^{1/2})$ denote the q-deformed reduced plane partition function. For...

conjecture | open | MathDB 页面 | 原文线索

V2-065326 | MATHDB-341228

Barna-type plainness conjecture for real Newton maps

摘录 / EXCERPT: Plainness conjecture. Every p satisfying the hypotheses of Barna's theorem is a plain map.

conjecture | open | MathDB 页面 | 原文线索

V2-065327 | MATHDB-341229

Weak plainness conjecture for real Newton maps on the plane

摘录 / EXCERPT: Weak plainness conjecture. Then:

conjecture | open | MathDB 页面 | 原文线索

V2-065328 | MATHDB-341234

The facet characterization conjecture for the cone of totally balanced games

摘录 / EXCERPT: Facet characterization conjecture. If $|N|\geq 3$, then the facet-defining inequalities for the cone $E(N)$ are exactly the inequalities corresponding to non-trivial irreducible mi...

conjecture | open | MathDB 页面 | 原文线索

V2-065329 | MATHDB-341240

The maximal-length conjecture for maximal green sequences of the n-cycle quiver

摘录 / EXCERPT: Let Γ_n be the n-cycle quiver, and let Φ^1 denote the corresponding root set, with $|\Phi^1|=\binom{n+1}{2}-1$. A maximal green sequence for Γ_n has a well-def...

conjecture | open | MathDB 页面 | 原文线索

V2-065330 | MATHDB-341261

Enhancement uniqueness conjecture for complicial Grothendieck prestable categories

摘录 / EXCERPT: Enhancement uniqueness conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065331 | MATHDB-341263

Enhancement uniqueness conjecture for bounded complicial triangulated categories

摘录 / EXCERPT: Enhancement uniqueness conjecture. Then T admits a unique ∞-categorical enhancement.

conjecture | open | MathDB 页面 | 原文线索

V2-065332 | MATHDB-341272

McDiarmid-Reed's conjecture for hexagonal graphs

摘录 / EXCERPT: McDiarmid-Reed's conjecture. Every hexagonal graph is $(9,4)$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-065333 | MATHDB-341273

Reducibility reformulation of McDiarmid-Reed's conjecture

摘录 / EXCERPT: McDiarmid-Reed reducibility conjecture. For every hexagonal graph G, $\operatorname{core}_{\mathcal{F}_{9,4}\cup\mathcal{F}'_{9,4}}(G)$ is $(9,4)$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-065334 | MATHDB-341274

Odd-girth multicoloring conjecture for hexagonal graphs

摘录 / EXCERPT: Odd-girth multicoloring conjecture. If G has odd-girth at least $2k+1$, then G is $(2k+1,k)$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-065335 | MATHDB-341275

Gamma-positivity conjectures for two-sided Eulerian polynomials of classical Weyl groups

摘录 / EXCERPT: For $n\equiv 0,1\pmod 4$, let $\mathcal{A}_n$ be the alternating subgroup of $\mathfrak{S}_n$, and define ... For even n, define the type-B polynomials $\mathrm{TSB}_n^+(s,t)$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-065336 | MATHDB-341287

Song-Cai-Yue rate conjecture for sequentially recoverable locally recoverable codes

摘录 / EXCERPT: Song-Cai-Yue conjecture. An achievable upper bound on the rate k/n is

conjecture | open | MathDB 页面 | 原文线索

V2-065337 | MATHDB-341288

The final-matrix gcd conjecture for multigenic Laver tables

摘录 / EXCERPT: Let $FM_n^{-}(x,y)$ denote the negative final-matrix value associated with the multigenic Laver table at level n, and let $*_{n}$ denote its operation on A_n. For a in this s...

conjecture | open | MathDB 页面 | 原文线索

V2-065338 | MATHDB-341289

The multigenic Laver-table sharpness conjecture

摘录 / EXCERPT: Let M be a multigenic Laver table, let t be an n-ary term in the language of LD-systems, and let $\mathbf{x}_i=a_{i,1}\ldots a_{i,n_i}$ for $1\leq i\leq n$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-065339 | MATHDB-341300

Genus-growth conjecture for geometric p-adic Lie towers

摘录 / EXCERPT: Let U be a smooth curve over k with compactification C. Let M be an irreducible overconvergent F-isocrystal on U whose fiber Newton polygon is constant as x varies in...

conjecture | open | MathDB 页面 | 原文线索

V2-065340 | MATHDB-341301

Local ramification-growth conjecture for unit-root F-isocrystals

摘录 / EXCERPT: Let M be an irreducible overconvergent F-isocrystal on $\operatorname{Spec}(F)$ with nonnegative slopes. Let M^{u-r} be its unit-root subcrystal, and let η be its first...

conjecture | open | MathDB 页面 | 原文线索

V2-065341 | MATHDB-341307

Information-distillation lower-bound tightness conjecture

摘录 / EXCERPT: Let $\mathcal{X}$ be a finite alphabet with $|\mathcal{X}|>2$. For an information-distillation function $\mathrm{ID}_M(|\mathcal{X}|,\beta)$, where M is the number of quantizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-065342 | MATHDB-341311

Staircase equality for tableau and dual Grothendieck functions

摘录 / EXCERPT: Let $\rho_k=(k,k-1,\ldots,2,1)$ be the staircase partition, and let $gp_{\rho_k}({\bf y})$ and $g_{\rho_k}({\bf y})$ denote the tableau-defined and stable dual Grothendieck functio...

conjecture | open | MathDB 页面 | 原文线索

V2-065343 | MATHDB-341317

e-positivity conjecture for a three-parameter natural unit interval order

摘录 / EXCERPT: Let $P(\mathbf{m})$ be the natural unit interval order associated with a sequence $\mathbf{m}=(2,m_2,m_3,n,\ldots,n)$, and let G be its incomparability graph. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065344 | MATHDB-341318

e-positivity conjecture for generalized natural unit interval orders

摘录 / EXCERPT: Let $P(\mathbf{m})$ be the natural unit interval order associated with a sequence $\mathbf{m}=(m_1,m_2,m_3,n,\ldots,n)$, and let G be its incomparability graph. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065345 | MATHDB-341321

Sharpness conjecture for products of commutators in associative rings

摘录 / EXCERPT: Sharpness conjecture. If $(m,n)\ne(3,3)$, then there is a unital associative ring A and elements $a_i,b_j\in A$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065346 | MATHDB-341328

Sakellaridis-Venkatesh conjecture on parameters of relative discrete series

摘录 / EXCERPT: Sakellaridis-Venkatesh conjecture. A relative discrete series representation π in $L^2(X)$ is contained in an Arthur packet corresponding to an X-distinguished A-parameter

conjecture | open | MathDB 页面 | 原文线索

V2-065347 | MATHDB-341342

The star-discrepancy lower-bound conjecture

摘录 / EXCERPT: Let m points be chosen in $[0,1)^d$, and let $D(m,d)_\infty$ denote the infimum of their L_∞ discrepancy. Star-discrepancy lower-bound conjecture. For $d\ge 3$, ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-065348 | MATHDB-341343

The smooth discrepancy lower-bound conjecture

摘录 / EXCERPT: For a point set ξ with m points in the d-dimensional domain and arbitrary weights, let $D^{r,o}_\infty(m,d)$ be the infimum of the r-discrepancy over all such point-weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-065349 | MATHDB-341344

The smooth $L_{\mathbf p}$-discrepancy lower-bound conjecture

摘录 / EXCERPT: For a point set ξ with m points in the d-dimensional domain and arbitrary weights, let $\widetilde D^{r,o}_{\mathbf p}(m,d)$ denote the infimum of the smooth...

conjecture | open | MathDB 页面 | 原文线索

V2-065350 | MATHDB-341355

Asymptotic stability conjecture for wave equations with positive-real impedance

摘录 / EXCERPT: Let $\hat{z}$ be the impedance introduced earlier, and let the Cauchy problem be the multidimensional wave equation with the associated impedance boundary condition. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-065351 | MATHDB-341356

Left-right symmetry conjecture for affine involution Stanley symmetric functions

摘录 / EXCERPT: Let n be a positive integer, let $\tilde I_n$ denote the set of affine involutions, let $z\in\tilde I_n$, let $\hat{F}_z$ be the affine involution Stanley symmetric function, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065352 | MATHDB-341357

Generalized 2413-balloon Möbius function conjecture

摘录 / EXCERPT: Let β be a permutation, let i,j be indexes, and let $2413 \circledcirc_{i,j} \beta$ denote the permutation formed by inserting β into 2413 between its i-th and...

conjecture | open | MathDB 页面 | 原文线索

V2-065353 | MATHDB-341361

The symmetric-power automorphy conjecture

摘录 / EXCERPT: Let k be a global field and let Π be a cuspidal automorphic representation of ${\rm GL}_n(\mathbb{A}_k)$. Symmetric-power automorphy conjecture. For every n, the symmetric-...

conjecture | open | MathDB 页面 | 原文线索

V2-065354 | MATHDB-341369

Folding compatibility for classical multiplicative Langlands duality

摘录 / EXCERPT: Folding conjecture for classical multiplicative Langlands duality. There is an equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-065355 | MATHDB-341370

Compact-fiber conjecture for the reduced multiplicative Hitchin system

摘录 / EXCERPT: Let G be a reductive group, D a divisor on $\mathbb{CP}^1$, and $\omega^\vee$ the prescribed coweight data. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065356 | MATHDB-341372

Conjecture on the asymptotic balance of the optimal continuation function

摘录 / EXCERPT: Asymptotic balance conjecture. As $\epsilon\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-065357 | MATHDB-341374

Vignesh et al.'s total coloring conjecture for line graphs of complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let $L(K_n)$ be its line graph, and let $\chi”$ denote total chromatic number. Vignesh et al.'s conjecture. For every complete grap...

conjecture | open | MathDB 页面 | 原文线索

V2-065358 | MATHDB-341378

Critical-line conjecture for the disordered two-community noisy Kuramoto model

摘录 / EXCERPT: Let μ be the common distribution of natural frequencies in the two communities, let $\psi=\psi_{2}-\psi_{1}\in\{0,\pi\}$, and let r_{1},r_{2} be the two synchronization order...

conjecture | open | MathDB 页面 | 原文线索

V2-065359 | MATHDB-341392

Periodic sign conjecture for coefficients of Ramanujan theta-function q-series

摘录 / EXCERPT: Periodic sign conjecture. For any integer $n\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-065360 | MATHDB-341393

The skewed projection inequality for weak antichains

摘录 / EXCERPT: Skewed projection inequality. If A is a weak antichain, then

conjecture | open | MathDB 页面 | 原文线索

V2-065361 | MATHDB-341405

Uniqueness conjecture for the no-spin Hartree-Fock minimisation problem

摘录 / EXCERPT: Let w be a positive radial non-increasing function. For density $\rho\geq 0$ and temperature $T\geq 0$, let $E_{\textnormal{no-spin}}^{\rm HF}(\rho,T)$ denote the no-spin Hartree...

conjecture | open | MathDB 页面 | 原文线索

V2-065362 | MATHDB-341417

Bounds conjecture for dimer-monomer entropy on generalized Tower of Hanoi graphs

摘录 / EXCERPT: Bounds conjecture. For every positive integer k, the entropy per site is bounded by

conjecture | open | MathDB 页面 | 原文线索

V2-065363 | MATHDB-341422

Exposed-circuit reformulation of Simon's conjecture

摘录 / EXCERPT: Exposed-circuit reformulation of Simon's conjecture. Then ${\mathcal C}$ contains an exposed circuit.

conjecture | open | MathDB 页面 | 原文线索

V2-065364 | MATHDB-341423

Linear quotients imply ridge-chordality for clutters

摘录 / EXCERPT: Linear-quotients ridge-chordality conjecture. If $I_{\overline{\mathcal C}}$ has linear quotients, then ${\mathcal C}$ is ridge-chordal.

open_problem | open | MathDB 页面 | 原文线索

V2-065365 | MATHDB-341431

Conjecture on the transformed function in the class $\mathcal{U}$

摘录 / EXCERPT: Conjecture. The function g belongs to the class $\mathcal{U}$ in the disc

conjecture | open | MathDB 页面 | 原文线索

V2-065366 | MATHDB-341435

The theta block characterization conjecture for Jacobi forms

摘录 / EXCERPT: Let L be the lattice indexing the Jacobi-form space $J_{k,L,1}$, and let $\phi\in J_{k,L,1}$. Let $T_{-}(2)$ denote the index-raising Hecke operator, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065367 | MATHDB-341437

The rainbow antichain-union conjecture

摘录 / EXCERPT: Let A_k be an antichain of size k, and let $\bigvee_s$ and $\bigwedge_s$ denote the posets consisting respectively of a common minimal or maximal element together with s inco...

conjecture | open | MathDB 页面 | 原文线索

V2-065368 | MATHDB-341438

The sharp exponent conjecture for rainbow antichain colorings

摘录 / EXCERPT: Let A_k be an antichain of size k, and let $f(n,c,A_k)$ be the maximum size of a color class in a c-coloring of $B_n=2^{[n]}$ avoiding a rainbow copy of A_k. Sharp exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-065369 | MATHDB-341441

Small-parameter stability conjecture for Courant-sharp Robin eigenvalues on the square

摘录 / EXCERPT: Small-parameter stability conjecture. There exists $h_0>0$ such that for $0<h\leq h_0$, the Courant-sharp cases for the Robin problem are the same, except the fifth one, as those f...

conjecture | open | MathDB 页面 | 原文线索

V2-065370 | MATHDB-341448

Conjectured upper bound for the maximum modulus of tree independence roots

摘录 / EXCERPT: Let T be a tree on n vertices, with $n\ge 6$ even, and let $\textnormal{maxmod}_{T}(n)$ denote the maximum modulus of an independence root among trees on n vertices. The tree...

conjecture | open | MathDB 页面 | 原文线索

V2-065371 | MATHDB-341453

The self-similarity conjecture for open subgroups of norm-one quaternion groups

摘录 / EXCERPT: Self-similarity conjecture. For all primes $p\geqslant 5$, no open subgroup of $SL_1^1(\Delta_p)$ is self-similar. For all primes $p\geqslant 3$, no open subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-065372 | MATHDB-341456

Silverman-Morton strong uniform boundedness conjecture

摘录 / EXCERPT: Let $D\ge1$, $N\ge1$, and $d\ge2$. For a number field $K/\mathbb Q$, write $[K:\mathbb Q]$ for its degree, and let $\operatorname{End}_d^N(K)$ be the degree-d endomorphisms of...

conjecture | open | MathDB 页面 | 原文线索

V2-065373 | MATHDB-341465

Approximation of statistical dimension by distance to the scaled subdifferential

摘录 / EXCERPT: Approximation conjecture. For these choices of f, the statistical dimension is well approximated by the squared Euclidean distance to the scaled subdifferential:

conjecture | open | MathDB 页面 | 原文线索

V2-065374 | MATHDB-341474

The generic-set conjecture for definably generated abelian groups

摘录 / EXCERPT: Let $\mathcal U$ be an abelian, connected, definably generated locally definable group. A definable set $X\subseteq\mathcal U$ is generic if $AX=\mathcal U$ for some countable subs...

conjecture | open | MathDB 页面 | 原文线索

V2-065375 | MATHDB-341480

Canonical isomorphism between geometric Hitchin space and Hitchin's component

摘录 / EXCERPT: Let Σ be a closed surface of genus $g\geq 2$, and let $\hat{\mathcal{T}}^n_{\Sigma}$ denote geometric Hitchin space and $\mathcal{T}^n_{\Sigma}$ Hitchin's component. Canonic...

conjecture | open | MathDB 页面 | 原文线索

V2-065376 | MATHDB-341484

A double-strength q-congruence for truncated basic hypergeometric sums

摘录 / EXCERPT: Let n be a positive integer with $n\equiv 3\pmod{4}$, let r be any positive integer, and let $\Phi_n(q)$ denote the nth cyclotomic polynomial. Then the conjectured q-congruen...

conjecture | open | MathDB 页面 | 原文线索

V2-065377 | MATHDB-341498

Lozin-Razgon-Zamaraev's finite-definition conjecture for induced-subgraph well-quasi-ordering

摘录 / EXCERPT: Lozin-Razgon-Zamaraev's labelled induced-subgraph conjecture. If $\mathcal{G}$ is defined by a finite set of forbidden induced subgraphs, then $\mathcal{G}$ is well-quasi-ordered b...

conjecture | open | MathDB 页面 | 原文线索

V2-065378 | MATHDB-341505

The determinant conjecture

摘录 / EXCERPT: Determinant conjecture. For every group G and every $A\in M_{r,s}({\mathbb Z}G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-065379 | MATHDB-341506

The approximation conjecture for Fuglede-Kadison determinants

摘录 / EXCERPT: Approximation conjecture for Fuglede-Kadison determinants. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065380 | MATHDB-341516

Simplex-code conjecture for functional batch codes

摘录 / EXCERPT: Let r be a positive integer, and let the $[2^r-1,r]$ simplex code be the binary simplex code of length 2^r-1 and dimension r. Let $FB(r,k)$ denote the minimum number of serve...

conjecture | open | MathDB 页面 | 原文线索

V2-065381 | MATHDB-341517

Gross-Prasad's global conjecture for special orthogonal groups

摘录 / EXCERPT: Let F be a number field and $\mathbb{A}$ its ring of adeles. Let $(V_n,q_n)\subset (V_{n+1},q_{n+1})$ be an inclusion of quadratic spaces over F of dimensions n and $n+1$, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-065382 | MATHDB-341518

Ichino-Ikeda refinement of the global Gross-Prasad conjecture

摘录 / EXCERPT: Let F, $\mathbb{A}$, V_n, V_{n+1}, SO_n, and SO_{n+1} be as above, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065383 | MATHDB-341519

Component-group formula for the Gan-Gross-Prasad constant

摘录 / EXCERPT: Let $G_n=\operatorname{GSpin}_n$ and $G_{n+1}=\operatorname{GSpin}_{n+1}$, and let π_n and π_{n+1} be cuspidal automorphic representations with conjectural global Langlands...

conjecture | open | MathDB 页面 | 原文线索

V2-065384 | MATHDB-341528

Boundedness conjecture for the ϕ_i parameters of positive representations

摘录 / EXCERPT: Boundedness conjecture. The collection

conjecture | open | MathDB 页面 | 原文线索

V2-065385 | MATHDB-341530

Lusztig's conjecture on connectivity of Kazhdan-Lusztig right cells

摘录 / EXCERPT: Let W be a Coxeter group, with Kazhdan-Lusztig right cells defined by the Kazhdan-Lusztig right preorder. A subset $C\subseteq W$ is right-connected if for every $x,y\in C$ there...

conjecture | open | MathDB 页面 | 原文线索

V2-065386 | MATHDB-341531

Geck-Halls' conjecture on left cells and generalized tilde-tau invariants

摘录 / EXCERPT: Let W_f be a finite Coxeter group, and let $x,y\in W_f$. Say that x and y have the same generalized $\widetilde{\tau}$-invariant when they are equivalent under all the relati...

conjecture | open | MathDB 页面 | 原文线索

V2-065387 | MATHDB-341532

Rationality conjecture for fully supported Bradley-Terry optimal-design weights

摘录 / EXCERPT: Let m be the number of alternatives, let $\binom{m}{2}$ pairwise-comparison intensities be denoted by $\boldsymbol{\lambda}=(\lambda_{ij})$, and let w_{ij} be the weights of a...

conjecture | open | MathDB 页面 | 原文线索

V2-065388 | MATHDB-341535

Conjecture that holomorphic isometric submersions of Teichmüller spaces are geometric

摘录 / EXCERPT: Geometric-origin conjecture. Any holomorphic and isometric submersion between finite-dimensional Teichmüller spaces of complex dimension at least 2 is a composition of forgetful...

conjecture | open | MathDB 页面 | 原文线索

V2-065389 | MATHDB-341537

The constant-term formula for the $(1,2)$ double-torus curve

摘录 / EXCERPT: Let N be a positive integer, let $q_u=q^{\frac{1}{2}}$, and let $S_{N-1}(\mathcal{A}_{(1,2)})$ be the indicated Chebyshev-type expression in the DAHA operator...

conjecture | open | MathDB 页面 | 原文线索

V2-065390 | MATHDB-341540

Oblomkov-Okounkov-Pandharipande stationary DT rationality conjecture

摘录 / EXCERPT: Oblomkov-Okounkov-Pandharipande conjecture. If all insertions are stationary, meaning $\alpha_i\in H^{>0}(X)$, then the renormalized invariant is a rational function of q regular...

conjecture | open | MathDB 页面 | 原文线索

V2-065391 | MATHDB-341541

DT/PT correspondence conjecture with descendents and Euler-class corrections

摘录 / EXCERPT: DT/PT correspondence conjecture. For any such data,

conjecture | open | MathDB 页面 | 原文线索

V2-065392 | MATHDB-341544

The conjecture on Camacho-Sad indices of homogeneous convex foliations

摘录 / EXCERPT: Let $d\geq 2$. For a homogeneous convex foliation $\mathcal{H}$ of degree d on $\mathbb{P}^{2}_{\mathbb{C}}$, let $\mathcal{CS}_d$ denote the set of Camacho-Sad indices at non-ra...

conjecture | open | MathDB 页面 | 原文线索

V2-065393 | MATHDB-341545

The lower-bound conjecture for Camacho-Sad indices of homogeneous convex foliations

摘录 / EXCERPT: Let $\mathcal{H}$ be a homogeneous convex foliation of degree $d\geq2$ on $\mathbb{P}^{2}_{\mathbb{C}}$, and let $s\in\ell_\infty$ be a non-radial singularity of $\mathcal{H}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-065394 | MATHDB-341550

One-dimensional symmetry conjecture for the generalized fractional Allen-Cahn equation

摘录 / EXCERPT: Let $a\in(-1,1)$ and let u be a bounded, smooth solution of the generalized fractional Allen-Cahn equation ... Assume that $\partial_{x_n}u>0$. Generalized fractional De Giorgi con...

conjecture | open | MathDB 页面 | 原文线索

V2-065395 | MATHDB-341563

Uniqueness conjecture for pitch-one tetragonal and rhombohedral TPMSg3s

摘录 / EXCERPT: Uniqueness conjecture. For every $r\in(-1,1)$, there is a unique τ with $\operatorname{Re}\tau=r$ that solves the period condition for $\mathcal{T}$ with pitch 1. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-065396 | MATHDB-341564

Classification conjecture for tetragonal and rhombohedral TPMSg3 families

摘录 / EXCERPT: Classification conjecture. The tP, tD, CLP, and tG surfaces are the only members of $\mathcal{T}$. The H, rPD, and rGL surfaces are the only members of $\mathcal{R}$.

conjecture | open | MathDB 页面 | 原文线索

V2-065397 | MATHDB-341565

Classification conjecture for TPMSg3s with order-3 and order-4 screw symmetry

摘录 / EXCERPT: Screw-symmetry classification conjecture. The tP, tD, CLP, and tG surfaces are the only TPMSg3s with an order-4 screw symmetry. The H, rPD, and rGL surfaces are the only TPMSg3s wi...

conjecture | open | MathDB 页面 | 原文线索

V2-065398 | MATHDB-341570

Morton-Silverman Uniform Boundedness Conjecture for rational maps

摘录 / EXCERPT: Let $D,N\ge 1$ and $d\ge 2$ be integers. For a number field K with $[K:\mathbb{Q}]=D$ and a morphism $\phi:\mathbb{P}^N\to\mathbb{P}^N$ of degree d defined over K, a point of...

conjecture | open | MathDB 页面 | 原文线索

V2-065399 | MATHDB-341572

Genericity conjecture for the automorphic sheaf constructed by theta lifting

摘录 / EXCERPT: In the situation of Theorem, let $F_G(K_{E,\chi, \mathbb H})$ be the complex on $\operatorname{Bun}_{\mathbb G}$ constructed there, and let $\Whit$ denote the first Whittaker coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-065400 | MATHDB-341577

The completeness conjecture for realizable join and intersection ranks

摘录 / EXCERPT: Let F be a free group, and let h,k,v,c be integers satisfying ... For subgroups $H,K\le F$, the quantities $\operatorname{rk}(H)$, $\operatorname{rk}(K)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-065401 | MATHDB-341586

Reading's maximum cd-index conjecture for Bruhat intervals

摘录 / EXCERPT: Let (W,S) be a Coxeter system, and let $[u,v]$ be an interval in the Bruhat order of W. Write k for the length of u and suppose that the length of v is $n+k+1$. The cd-...

conjecture | open | MathDB 页面 | 原文线索

V2-065402 | MATHDB-341587

The classification conjecture for number-conserving binary cellular automata

摘录 / EXCERPT: Let the dimension $dgeq 1$ be given, and consider binary cellular automata with the von Neumann neighborhood. A cellular automaton is number-conserving when it preserves the total...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065403 | MATHDB-341589

Generalized intersection-exponent cascade conjecture for SLE restriction measures

摘录 / EXCERPT: Fix $\kappa\in(0,4]$, let $n\geq 1$, and let $\alpha_1,\ldots,\alpha_n$ denote the parameters of the restriction measures. Define ... The generalized intersection exponents are the e...

conjecture | open | MathDB 页面 | 原文线索

V2-065404 | MATHDB-341594

Conjecture on first excited states in the general-mass N-body problem

摘录 / EXCERPT: Consider the N-body problem with general positive masses and its relative equilibria, namely solutions generated by central configurations. The first excited state is the relative...

conjecture | open | MathDB 页面 | 原文线索

V2-065405 | MATHDB-341597

Repeated orthonormal sequence threshold conjecture for p-frame energy

摘录 / EXCERPT: Let $N=m+kd$ points be given in $\mathbb{S}^{d-1}$, where $1\leq m<d$ and $d\geq 2$, and let $\mathbf{A}\in\mathbb{R}^{N\times N}$ be their Gram matrix. The repeated orthonormal se...

conjecture | open | MathDB 页面 | 原文线索

V2-065406 | MATHDB-341606

The self-interlacing conjecture for the geometric-sum polynomial

摘录 / EXCERPT: Self-interlacing conjecture. The polynomial $\Delta_{\theta,h}(p)(x)$ is self-interlacing, or is a self-interlacing polynomial multiplied by x.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065407 | MATHDB-341607

The simple-roots conjecture for finite difference operators

摘录 / EXCERPT: Simple-roots conjecture. For almost all polynomials p, the polynomial $\Delta_{\theta,h}(p)$ has only simple roots, apart from the stated exclusion.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-065408 | MATHDB-341610

Logarithmic Chowla conjecture

摘录 / EXCERPT: Let λ be the Liouville function. For every $t\in\mathbb{N}$ and every collection of distinct natural numbers $h_1,\ldots,h_t$, logarithmic Chowla conjecture. One has ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-065409 | MATHDB-341622

Long-time energy stabilization conjecture for hyperbolic obstacle problems

摘录 / EXCERPT: Let $s=1$ and consider an obstacle problem of the form of the wave equation with obstacle, with obstacle g, and let u be the weak solution obtained through the convex minimizat...

conjecture | open | MathDB 页面 | 原文线索

V2-065410 | MATHDB-341623

The asymptotic optimal learning-rate conjecture for a binary symmetric cascade

摘录 / EXCERPT: Huleihel et al.'s conjecture. In the regime $p\to 1/2$, the optimal learning rate of the student is

conjecture | open | MathDB 页面 | 原文线索

V2-065411 | MATHDB-341624

The strict causal penalty conjecture for optimal teaching and learning

摘录 / EXCERPT: Causal penalty conjecture. The optimal learning rate of the student is strictly less than

conjecture | open | MathDB 页面 | 原文线索

V2-065412 | MATHDB-341641

Nonexistence of cyclotomic norm representations for two candidate multipliers

摘录 / EXCERPT: Let n be an even integer with $n\geq 16$ and $n\equiv 0\pmod{4}$. Let ζ_{2n} be a primitive $2n$th root of unity, let $\overline{\eta}$ denote complex conjugation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065413 | MATHDB-341649

A q-congruence for a second truncated ${}_3\phi_2$ series modulo a cyclotomic square

摘录 / EXCERPT: The conjecture. The truncated series should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-065414 | MATHDB-341650

A family of q-congruences for truncated ${}_3\phi_2$ series when n is 2 modulo 3

摘录 / EXCERPT: The conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065415 | MATHDB-341654

Unique information lower-bound conjecture for the two-way secret key rate

摘录 / EXCERPT: Unique information lower-bound conjecture. The unique information lower bounds the two-way secret key rate:

conjecture | open | MathDB 页面 | 原文线索

V2-065416 | MATHDB-341665

Zero-sum partition conjecture for finite Abelian groups with multiple involutions

摘录 / EXCERPT: Let Γ be a finite Abelian group of order n with more than one involution. Let ... be a partition of $n-1$ with $r_i\geq 3$ for $1\leq i\leq t$, where t is any positive int...

conjecture | open | MathDB 页面 | 原文线索

V2-065417 | MATHDB-341667

Asymptotic ratio conjecture for the first Sierpinski gasket monomial boundary values

摘录 / EXCERPT: The asymptotic ratio conjecture. For $r\neq 1$ and $r\neq \frac{\sqrt{17}-3}{4}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065418 | MATHDB-341668

Asymptotic ratio conjecture for the second Sierpinski gasket monomial boundary values

摘录 / EXCERPT: The asymptotic ratio conjecture. For $r\neq \frac{\sqrt{17}-3}{4}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065419 | MATHDB-341669

Existence conjecture for ratios of the third Sierpinski gasket monomial boundary values

摘录 / EXCERPT: The ratio-existence conjecture. For all but at most finitely many values of r,

conjecture | open | MathDB 页面 | 原文线索

V2-065420 | MATHDB-341670

Boundary-value decay conjecture for Sierpinski gasket monomials

摘录 / EXCERPT: Boundary-value decay conjecture. For $r\neq 1$ and $r\neq\frac{\sqrt{17}-3}{4}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065421 | MATHDB-341673

Global minimizer conjecture for the Morse potential lattice energy

摘录 / EXCERPT: Let $r_0>0$. For a Bravais lattice L in dimension 3, let $E_{\alpha,r_0}[L]$ denote the Morse lattice energy, let $\mathcal{L}_3$ be the class of Bravais lattices, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065422 | MATHDB-341694

Bleher-Chinburg-Greenberg-Kakde-Pappas-Sharifi-Taylor pseudo-nullity conjecture for a finite character deformation

摘录 / EXCERPT: Let K be an imaginary quadratic field in which p splits, let $\chi:\operatorname{Gal}(\overline{\mathbb{Q}}/K)\to\mathbb{Z}_p^\times$ be a finite continuous character, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065423 | MATHDB-341709

The lexicographic characterization of the sequences S_k

摘录 / EXCERPT: Let $\mathcal{S}_k$ be the sequence written in base $3/2$, and let S_k denote the result of interpreting that representation in base 3. The lexicographic characterization conje...

conjecture | open | MathDB 页面 | 原文线索

V2-065424 | MATHDB-341710

Bogomolov-Tschinkel uniform common-torsion conjecture for elliptic curves

摘录 / EXCERPT: Bogomolov-Tschinkel's uniform common-torsion conjecture. There exists a uniform constant B such that

conjecture | open | MathDB 页面 | 原文线索

V2-065425 | MATHDB-341712

Compactness conjecture for minimizing sequences of subdivision surfaces

摘录 / EXCERPT: Compactness conjecture for minimizing sequences. There is a sequence of Möbius transformations G_j of $\mathbb{R}^3$ such that the surfaces $G_j\circ\mathbf{x}_j(K)$ can be parti...

conjecture | open | MathDB 页面 | 原文线索

V2-065426 | MATHDB-341730

Peter Borwein's first Borwein conjecture

摘录 / EXCERPT: Peter Borwein's conjecture. These polynomials have non-negative coefficients. The conjecture concerns the repeating sign pattern in the coefficients of the product...

conjecture | open | MathDB 页面 | 原文线索

V2-065427 | MATHDB-341731

Peter Borwein's second Borwein conjecture

摘录 / EXCERPT: Second Borwein conjecture. These polynomials have non-negative coefficients. This is one of two further Borwein conjectures mentioned in the discussion; the supplied text gives no...

conjecture | open | MathDB 页面 | 原文线索

V2-065428 | MATHDB-341732

Peter Borwein's third Borwein conjecture

摘录 / EXCERPT: Third Borwein conjecture. These polynomials have non-negative coefficients. This conjecture concerns the analogous sign decomposition for modulus 5; the supplied text gives no re...

conjecture | open | MathDB 页面 | 原文线索

V2-065429 | MATHDB-341733

Bressoud's generalized Borwein conjecture

摘录 / EXCERPT: Suppose that $m,n\in\mathbb{Z}^+$, α and β are positive rational numbers, and K is a positive integer such that αK and βK are integers. If ... with stri...

conjecture | open | MathDB 页面 | 原文线索

V2-065430 | MATHDB-341739

Minimality conjecture for products of two independent non-centered normal random variables

摘录 / EXCERPT: Let X and Y be independent non-centered normal random variables, and let $W=XY$. A second-order Stein operator for W is a differential operator acting on smooth functions wit...

conjecture | open | MathDB 页面 | 原文线索

V2-065431 | MATHDB-341740

Klemm-Pandharipande genus-zero integrality conjecture for Calabi-Yau fourfolds

摘录 / EXCERPT: Klemm-Pandharipande's genus-zero integrality conjecture. The invariants $n_{0,\beta}(\gamma_1,\ldots,\gamma_n)$ are integers. This is the genus-zero integrality prediction for Cala...

conjecture | open | MathDB 页面 | 原文线索

V2-065432 | MATHDB-341741

Stable-pair genus-one correspondence conjecture for Calabi-Yau fourfolds

摘录 / EXCERPT: Stable-pair genus-one correspondence conjecture. For a suitable choice of orientation,

conjecture | open | MathDB 页面 | 原文线索

V2-065433 | MATHDB-341742

Equivariant genus-zero correspondence conjecture for the local projective line

摘录 / EXCERPT: Let $X=\mathcal{O}_{\mathbb{P}^{1}}(l_1,l_2,l_3)$ with $l_1+l_2+l_3=-2$, and let $\mathrm{GW}_{0,2}(X)$ and $P_{1,d[\mathbb{P}^{1}]}(X)$ denote the equivariant invariants defined i...

conjecture | open | MathDB 页面 | 原文线索

V2-065434 | MATHDB-341757

Upper-bound conjecture for MaxEJS expected blocklength via a genie-aided decoder

摘录 / EXCERPT: MaxEJS blocklength upper-bound conjecture. The expected blocklength satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065435 | MATHDB-341760

The comparison conjecture for weighted cones and comma quasi-categories

摘录 / EXCERPT: Let Q be a quasi-category, let $d\colon J\to Q$ be a diagram, and let $p\colon\tilde J\to J$ be a weight. Write $st(p)^{bifib}\colon J\to\mathfrak N[\mathcal{K}\!\mathit{an}]$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065436 | MATHDB-341761

The formal adjunction conjecture for weighted cone models

摘录 / EXCERPT: Let Q be a quasi-category tensored and cotensored over $\mathfrak N[\mathcal{K}\!\mathit{an}]$, let $d\colon J\to Q$ be a diagram, and let $p\colon\tilde J\to J$ be a weight. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065437 | MATHDB-341764

The local Shimura conjecture for standard intertwining operators

摘录 / EXCERPT: Let F be the relevant number field, let ν be a place of F, and let τ_ν be the local representation used to form the induced representations. For $l\geq 2$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-065438 | MATHDB-341765

The pole conjecture for the partial Rankin-Selberg L-function

摘录 / EXCERPT: Pole conjecture. $L^S_{\vartheta,\vartheta^{-1}}(s,\tau\times\tau^{\vee})$ has a simple pole at $s=1$ and is holomorphic and nonzero for $\operatorname{Re}(s)>1$. This assertion co...

conjecture | open | MathDB 页面 | 原文线索

V2-065439 | MATHDB-341766

Duality compatibility for Fukaya-category cobordism cone decompositions

摘录 / EXCERPT: Duality compatibility conjecture. For $j=1,\ldots,s-1$, the quasi-isomorphisms

conjecture | open | MathDB 页面 | 原文线索

V2-065440 | MATHDB-341769

Alexandersson's minimal-polynomial conjecture for skew Schur functions

摘录 / EXCERPT: Let $\pmb{\kappa},\pmb{\lambda},\pmb{\mu},\pmb{\nu}$ be as in Theorem. For every vector $\mathbf{w}$, let $\overline{\mathbf{w}}$ denote the vector obtained by rearranging its coor...

conjecture | open | MathDB 页面 | 原文线索

V2-065441 | MATHDB-341771

Uniform erasure robustness conjecture for random Gabor frames

摘录 / EXCERPT: Uniform erasure-robustness conjecture. Suppose that $|\Lambda|=O(M\log^\alpha M)$, where $\alpha\geq0$ is to be specified. Then, for every $p\in(0,1)$, $\Delta(p)\geq C$ with high...

conjecture | open | MathDB 页面 | 原文线索

V2-065442 | MATHDB-341775

Universality conjecture for the random graph of a good polygon configuration

摘录 / EXCERPT: Let $(\mathcal{P}_{n})_{n \geqslant 1}$ be a good sequence of configurations. For each n, let $\mathbf{G}_{\mathcal{P}_{n}}$ be the associated random graph, let $\mathbb{G}_{n}$...

open_problem | open | MathDB 页面 | 原文线索

V2-065443 | MATHDB-341782

Rigidity conjecture for the Hausdorff dimension of symmetric limit sets

摘录 / EXCERPT: Let $\Gamma\subset\operatorname{SL}(n,\mathbb{R})$ be a projectively convex cocompact subgroup. Here $\delta_{1,n}$ is the critical exponent associated to the gap between the larg...

conjecture | open | MathDB 页面 | 原文线索

V2-065444 | MATHDB-341785

The discriminant multiplicity conjecture

摘录 / EXCERPT: For each integer $n\geq 2$, let D_K denote the absolute discriminant of a degree-n number field $K/\mathbb{Q}$. The discriminant multiplicity conjecture. For every $\epsilon>0$...

conjecture | open | MathDB 页面 | 原文线索

V2-065445 | MATHDB-341788

Dicks-Ventura retract inertia conjecture for free groups

摘录 / EXCERPT: Dicks-Ventura conjecture. Every retract of F_n is inert.

conjecture | open | MathDB 页面 | 原文线索

V2-065446 | MATHDB-341789

Test-element conjecture for finitely generated subgroups of free groups

摘录 / EXCERPT: Test-element conjecture. Every test element of H is a test element of F_n.

conjecture | open | MathDB 页面 | 原文线索

V2-065447 | MATHDB-341792

Subadditivity of mutual information for spin synchronization models

摘录 / EXCERPT: Mutual-information subadditivity conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-065448 | MATHDB-341799

Komori-Yamashita conjecture on connectedness thresholds of linear slices

摘录 / EXCERPT: Let C_0 and C_1 be positive real constants such that the linear slice $\mathcal L_c(\Sigma_{1,1})$ is connected for every $0<c<C_0$, while it is not connected for every $c>C_1$...

conjecture | open | MathDB 页面 | 原文线索

V2-065449 | MATHDB-341814

Compatibility of the two constructions of the extension $\pi^\alpha(\rho_p)$

摘录 / EXCERPT: Let $\pi^\alpha(\rho_p)$ be the nonsplit extension attached to ρ_p in the cited work, and let $\pi^\alpha(\operatorname{Fil}^{\rm max}_\alpha(\rho_p))$ be the extension constr...

conjecture | open | MathDB 页面 | 原文线索

V2-065450 | MATHDB-341815

The $n=3$ local identification conjecture for $\pi^\alpha(\rho_p)$

摘录 / EXCERPT: Let $\rho_p:{\rm Gal}(\overline{\mathbb Q}_p/{\mathbb Q}_p)\to{\rm GL}_3(E)$ be semistable as in the preceding setup, with $\dim_E W=3$ and $N^2\ne0$. The $n=3$ local identificatio...

conjecture | open | MathDB 页面 | 原文线索

V2-065451 | MATHDB-341819

Universal optimality conjecture for lifted ETFs in the p-frame potential

摘录 / EXCERPT: Universal optimality conjecture for lifted ETFs. The following configurations minimize the p-frame potential $\operatorname{FP}_{p,d+1,d}$: when $p\in(p_{k-1},p_k]$, the L_k^d...

conjecture | open | MathDB 页面 | 原文线索

V2-065452 | MATHDB-341826

Chen's fluctuation conjecture for internal DLA on the Sierpinski gasket

摘录 / EXCERPT: Let SG be the Sierpinski gasket graph, let $\mathcal{B}_n$ be the graph-metric ball of radius n, and let $\mathcal{I}_{|\mathcal{B}_n|}$ be the internal DLA cluster fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065453 | MATHDB-341828

Concatenation conjecture for standard Young tableaux and vacillating tableaux

摘录 / EXCERPT: Let standard Young tableaux be concatenated when their row lengths all have the same parity, and let vacillating tableaux of shape $\emptyset$ be concatenated in the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-065454 | MATHDB-341829

Evacuation conjecture for standard Young tableaux and vacillating tableaux

摘录 / EXCERPT: Let Q be a standard Young tableau and let V be its corresponding vacillating tableau. Let evacuation, also called the Schützenberger involution, act on Q, and let reversal ac...

conjecture | open | MathDB 页面 | 原文线索

V2-065455 | MATHDB-341832

Goncharov's five-term generation conjecture for the Bloch relations

摘录 / EXCERPT: Let F be a field and let $R_2(F)$ be the subgroup of the relevant free abelian group generated by functional relations in weight two. For $x,y\in P^1_F$, write $FT[x,y]$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-065456 | MATHDB-341833

The inverted five-term generation conjecture for weight-two relations

摘录 / EXCERPT: Let F be a field and let $R_2(F)$ be the weight-two relation subgroup in the modified definition using $F\setminus\{0,1\}$. An inverted five term relation is a relation obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-065457 | MATHDB-341835

Infinite-point conjecture for pointed nonsingular curves over finitely generated fields

摘录 / EXCERPT: Let K be a field finitely generated over $\mathbb{Q}$, let X be a pointed nonsingular curve over K, and let $\bar K(\boldsymbol{\sigma})$ denote a field obtained by fixing a...

conjecture | open | MathDB 页面 | 原文线索

V2-065458 | MATHDB-341841

Upper semicontinuity conjecture for Gevrey slopes of A-hypergeometric systems

摘录 / EXCERPT: Gevrey-slope conjecture. The map $\beta\mapsto {\operatorname{d}_s}(A,\beta)$ is upper-semicontinuous. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-065459 | MATHDB-341842

The first partial theta Borwein conjecture

摘录 / EXCERPT: The first partial theta Borwein conjecture. For each $m,k\geq0$, there is a non-negative integer $N_{m,k}$ such that, if $n\geq N_{m,k}$, then $A_{m,n,k}(q)$, $B_{m,n,k}(q)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065460 | MATHDB-341843

The generalized partial theta Borwein conjecture

摘录 / EXCERPT: The generalized partial theta Borwein conjecture. For fixed k, if $m_1,m_2\geq1$ and n_1,n_2,n_3 are sufficiently large, then $A(q)$, $B(q)$, and $C(q)$ have non-negative coeff...

conjecture | open | MathDB 页面 | 原文线索

V2-065461 | MATHDB-341844

The generalized odd-modulus partial theta Borwein conjecture

摘录 / EXCERPT: The generalized odd-modulus partial theta Borwein conjecture. For fixed k and K, if $m_1,m_2\geq1$ and n_1,n_2,n_3 are sufficiently large, then

conjecture | open | MathDB 页面 | 原文线索

V2-065462 | MATHDB-341858

The factorization conjecture for the two-variable Bernoulli determinant

摘录 / EXCERPT: Let $r\geq 1$, and let $\bar F_{r,2}(x_1,x_2)$ be the determinant polynomial defined in the paper for $D=2$. Factorization conjecture. One has ... The formula was supported by comput...

conjecture | open | MathDB 页面 | 原文线索

V2-065463 | MATHDB-341860

Conjecture on optimal linear programming functions for Gaussian energy

摘录 / EXCERPT: Fix a dimension $d\geq 3$, density $\rho=1$, and $\alpha>0$. For the potential $p(r)=e^{-\alpha r^2}$, consider the optimal linear programming bound $\widehat{f}(0)-f(0)$ from the...

conjecture | open | MathDB 页面 | 原文线索

V2-065464 | MATHDB-341863

The natural nerve model-structure conjecture for strict n-categories

摘录 / EXCERPT: Let n be a nonnegative integer, let $n\mathcal{C}\!\mathit{at}$ denote the category of strict n-categories, and let ... be the natural n-nerve, with...

conjecture | open | MathDB 页面 | 原文线索

V2-065465 | MATHDB-341878

Voros connection formula conjecture for the Painlevé I WKB solution

摘录 / EXCERPT: Fix $(t,\nu)\in D_{t_\ast}\times D_{\nu_\ast}$ so that the Stokes graph has no Stokes segment. Let C be a Stokes curve connecting a branch point e and ∞, forming a commo...

conjecture | open | MathDB 页面 | 原文线索

V2-065466 | MATHDB-341880

Initiality conjecture for the cubical cwf term model

摘录 / EXCERPT: Let $\mathcal{C}$, $\II$, and $\FF$ be chosen parameters, and let $\Syn$ be the cubical category-with-families term model whose contexts are judgmental-equality classes of syntacti...

conjecture | open | MathDB 页面 | 原文线索

V2-065467 | MATHDB-341885

De Klerk-Dobre conjecture on the subtour LP for circulant TSP

摘录 / EXCERPT: De Klerk-Dobre conjecture. For every circulant TSP instance,

conjecture | open | MathDB 页面 | 原文线索

V2-065468 | MATHDB-341887

Equivariant Tamagawa number conjecture for motives

摘录 / EXCERPT: Let K' be a finite extension of K, let ε be an idempotent of $A[\mathcal{G}_{K'}]$, and let $\mathcal{R}$ be a $\mathbb{Z}_p$-order in...

conjecture | open | MathDB 页面 | 原文线索

V2-065469 | MATHDB-341888

Bloch-Kato-Kato equality for elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be the elliptic curve and let $F/\mathbb{Q}$ be the fixed totally real extension of odd degree. Assume that $\Sha(E/F)[p^\infty]$ is finite. Let T be the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-065470 | MATHDB-341896

Minuscule P-partition rowmotion orbit conjecture

摘录 / EXCERPT: Minuscule P-partition rowmotion orbit conjecture. For every $\ell\geq 1$, there is a bijection φ between the row-orbits of $\mathrm{PP}^{\ell}(P)$ and those of...

conjecture | open | MathDB 页面 | 原文线索

V2-065471 | MATHDB-341897

Minuscule P-partition cyclic sieving conjecture

摘录 / EXCERPT: Minuscule P-partition cyclic sieving conjecture. For every $\ell\geq 1$, the triple

conjecture | open | MathDB 页面 | 原文线索

V2-065472 | MATHDB-341905

Hu-Zhang determinant conjecture for a Sylvester-Kac type matrix

摘录 / EXCERPT: Hu-Zhang determinant conjecture. Its determinant is

conjecture | open | MathDB 页面 | 原文线索

V2-065473 | MATHDB-341909

The definability conjecture for strongly first order dependencies

摘录 / EXCERPT: Let $\mathbf D(R)$ be a strongly first order dependency, meaning that every sentence of $\texttt{FO}(\mathbf D)$ is equivalent to a first order sentence. Let $\mathcal D^\uparrow$...

conjecture | open | MathDB 页面 | 原文线索

V2-065474 | MATHDB-341910

The global-disjunction definability conjecture for strongly first order dependencies

摘录 / EXCERPT: Let $\mathbf D(R)$ be a strongly first order dependency, and let global disjunctions be added to the language of team semantics alongside upwards closed dependencies and constancy...

conjecture | open | MathDB 页面 | 原文线索

V2-065475 | MATHDB-341911

Relativizability conjecture for strongly first order generalized dependencies

摘录 / EXCERPT: Let $\mathbf D$ be a generalized dependency. It is relativizable if every sentence of $\texttt{FO}(\mathbf D^{(P)})$ is equivalent to some sentence of $\texttt{FO}(\mathbf D)$. Rel...

conjecture | open | MathDB 页面 | 原文线索

V2-065476 | MATHDB-341912

Decomposition conjecture for strongly first order dependencies

摘录 / EXCERPT: Let $\mathbf D(R)$ be a strongly first order dependency. A dependency is non-jumping when it has the non-jumping property considered in the paper. Decomposition conjecture. Every s...

conjecture | open | MathDB 页面 | 原文线索

V2-065477 | MATHDB-341913

Kraus's conjecture on elliptic curves over totally real fields

摘录 / EXCERPT: Kraus's conjecture. There are no elliptic curves over K with full 2-torsion and conductor $d513$.

conjecture | open | MathDB 页面 | 原文线索

V2-065478 | MATHDB-341914

Pisier's conjecture on bounded and completely bounded Schur multipliers

摘录 / EXCERPT: Let $d4d8_p$ be the collection of symbols that are bounded Schur multipliers of the Schatten-von Neumann class $d4d4_p$, and let $d4d8_p^{\mathrm{cb}}$ be the collection of symbols...

conjecture | open | MathDB 页面 | 原文线索

V2-065479 | MATHDB-341917

Full monodromy conjecture for rational curves on smooth complete toric surfaces

摘录 / EXCERPT: Let X be a smooth complete toric surface, let $\mathcal{L}$ be a line bundle in the setting of the monodromy map, and let C be a curve whose nodes form the set on which monodro...

conjecture | open | MathDB 页面 | 原文线索

V2-065480 | MATHDB-341922

Conjecture on the average size of the BSD factor for elliptic curves

摘录 / EXCERPT: BSD-factor growth conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065481 | MATHDB-341923

The generization conjecture for pure sheaves on ribbons

摘录 / EXCERPT: Let X be a ribbon and let ${\mathscr F}$ be a pure sheaf on X that is not a generalized vector bundle. A sheaf generizes to another sheaf if it occurs as a special fiber of a f...

conjecture | open | MathDB 页面 | 原文线索

V2-065482 | MATHDB-341924

Irreducibility conjecture for generalized vector bundles on ribbons

摘录 / EXCERPT: Let X be a ribbon, let r be a positive integer, and let $\delta=-\deg({\mathscr N})$, where ${\mathscr N}$ is the conormal sheaf of X_{red} in X. Fix the generaliz...

conjecture | open | MathDB 页面 | 原文线索

V2-065483 | MATHDB-341925

Large-nilradical component conjecture for moduli of sheaves on ribbons

摘录 / EXCERPT: Let X be a ribbon, let $\delta=-\deg({\mathscr N})$, and let $\bar{g}$ be the genus of X_{red}. The large-nilradical component conjecture. If ... then the only irreducib...

conjecture | open | MathDB 页面 | 原文线索

V2-065484 | MATHDB-341928

Hitchin's conjecture on stabilizers of invariant quadruples of torsion points

摘录 / EXCERPT: Let $S=\{(r_i,\theta_i)\}_{i=1}^{4}\in\mathcal{R}_4$ have common order n. Let $\Gamma_S=\{\gamma^{-T}\in\operatorname{SL}_2(\mathbb{Z}):\gamma\cdot S=S\}$ be the stabilizer of...

conjecture | open | MathDB 页面 | 原文线索

V2-065485 | MATHDB-341931

Havet and Yu's $(p,1)$-total labelling conjecture

摘录 / EXCERPT: Let G be a finite, simple, undirected graph, let p be a positive integer, and let $\lambda^T_p(G)$ denote the minimum k for which G has a $(p,1)$-total k-labelling. Havet...

conjecture | open | MathDB 页面 | 原文线索

V2-065486 | MATHDB-341937

Persistence conjecture for critical sets with one fast and two slow variables

摘录 / EXCERPT: Persistence conjecture. For one fast and two slow variables, the critical set $\mathcal{C}[g]$ is persistent under perturbations for $g\in V_f$ if all folds are non-degenerate, tha...

conjecture | open | MathDB 页面 | 原文线索

V2-065487 | MATHDB-341938

Codimension-one bifurcation classification conjecture for one-fast, two-slow critical sets

摘录 / EXCERPT: Codimension-one bifurcation conjecture. The codimension-one bifurcations of critical sets $\mathcal{C}[g]$ are characterised by the stated classification: precisely one set...

conjecture | open | MathDB 页面 | 原文线索

V2-065488 | MATHDB-341941

Conjecture on the irreducible components of the moduli space of generalized line bundles on primitive triple curves

摘录 / EXCERPT: Let X be a primitive multiple curve of multiplicity 3, with reduced subcurve C of genus $g_1\ge 2$, and let $\delta=-\operatorname{deg}(\mathcal{C})>0$. Fix an integer D, a...

conjecture | open | MathDB 页面 | 原文线索

V2-065489 | MATHDB-341942

Kalman's convergence-rate conjecture for Lur'e systems

摘录 / EXCERPT: Kalman's convergence-rate conjecture. For any stable G and $\Delta\in S[0,K]$ with $K<K_N$,

conjecture | open | MathDB 页面 | 原文线索

V2-065490 | MATHDB-341956

The primitive-to-coprimitive comparison conjecture

摘录 / EXCERPT: Let G be a nontrivial finite abelian group and $n\ge 1$. Let $H_{n,\mathrm{prim}}^{BM}(\mathbb X_n,\mathcal L_{n,G})$ be the common kernel of the nontrivial co-multiplication map...

conjecture | open | MathDB 页面 | 原文线索

V2-065491 | MATHDB-341958

Specific mirror conjecture for Schoen Calabi-Yau threefolds from rational elliptic surfaces

摘录 / EXCERPT: Specific mirror conjecture. The threefolds $\widetilde{X}_0^{\ast}$, $\widetilde{X}_1^{\ast}$, $\widetilde{X}_2^{\ast}$, and $\widetilde{X}_3^{\ast}$ are, respectively, mirror Cala...

conjecture | open | MathDB 页面 | 原文线索

V2-065492 | MATHDB-341965

Zagier-Gangl conjecture on the symmetric-square value of an elliptic curve

摘录 / EXCERPT: Let E) be an elliptic curve over $\mathbb{Q}$. For any divisor $\xi=\sum n_i(P_i)\in \mathbb{Z}[E(\overline{\mathbb{Q}})]$ and homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-065493 | MATHDB-341979

Superlinear area conjecture for subtree-separated orthogonal drawings of complete ternary trees

摘录 / EXCERPT: Superlinear area conjecture. There exists a constant $\varepsilon>0$ such that every n-node complete ternary tree requires

conjecture | open | MathDB 页面 | 原文线索

V2-065494 | MATHDB-341984

The equality conjecture for Christoffel functions and birth-death processes

摘录 / EXCERPT: Equality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065495 | MATHDB-341991

Finiteness conjecture for curve components of derived categories of threefolds

摘录 / EXCERPT: Let X be a threefold satisfying the stated properties, including a decomposition of its intermediate Jacobian ... where the S_i are smooth projective curves of positive genus and...

conjecture | open | MathDB 页面 | 原文线索

V2-065496 | MATHDB-341998

Extension of the word-function observations for Hecke groups

摘录 / EXCERPT: Let G_k be the Hecke group with parameter k, let W denote the words under consideration, and let $\nu_1,\ldots,\nu_8$ be the functions defined in Observation 1, using the Fib...

conjecture | open | MathDB 页面 | 原文线索

V2-065497 | MATHDB-341999

Parity formula for the Kepler limit of p_λ

摘录 / EXCERPT: Let $p=p_\lambda$ be the polynomial family considered in the paper, and let $\gamma(p)$ denote the quantity defined in the preceding observation. Parity formula for p_λ. Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065498 | MATHDB-342000

Generalization of the recurrence and Kepler-limit observations

摘录 / EXCERPT: Let $k\geq 5$, let W be the relevant word set, let $\nu^{(k)}$ be the coefficient function, let $L_p\in\Lambda_p$ be the linear recurrence associated with the coefficients, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-065499 | MATHDB-342003

Monotone upper-bound DNF compression conjecture

摘录 / EXCERPT: A monotone DNF is a DNF containing no negated literals. In the improved upper-bound DNF compression conjecture, f_{upper} is an upper bound of f when...

conjecture | open | MathDB 页面 | 原文线索

V2-065500 | MATHDB-342004

Polylogarithmic bound for intersecting regular set systems

摘录 / EXCERPT: Let $\beta(w)$ denote the largest κ for which there exists an intersecting κ-regular w-set system. Here a w-set system is a family of sets of size at most w, an...

conjecture | open | MathDB 页面 | 原文线索

V2-065501 | MATHDB-342007

Purity conjecture for refined Gopakumar-Vafa invariants of flopping curves

摘录 / EXCERPT: Let C be a flopping curve, let r be a positive integer, and let $\operatorname{GV}_{C,r,0}$ be the associated monodromic mixed Hodge structure. Write $n_{C,r}(q^{1/2})$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-065502 | MATHDB-342016

The higher-rank generalization of Lehmer's conjecture for units

摘录 / EXCERPT: Let F be a number field, let L_1 be the logarithmic image of $\mathcal{O}_F^*$, and for each k define $\mu_{1,k}(F)$ as the minimum of the L^1-norms of wedges...

conjecture | open | MathDB 页面 | 原文线索

V2-065503 | MATHDB-342022

Weighted flexibility conjecture for planar graphs without diamonds

摘录 / EXCERPT: Weighted flexibility conjecture. There exists $varepsilon>0$ such that every planar graph G without diamonds and assignment L of lists of size five is weighted $varepsilon$-fle...

conjecture | open | MathDB 页面 | 原文线索

V2-065504 | MATHDB-342030

Normalized-flow conjecture for the absolute order of type D

摘录 / EXCERPT: Let W be the Coxeter group of type D_n. A normalized flow on a ranked poset is a flow with parameter function $ u$ identically equal to 1; the absolute order...

conjecture | open | MathDB 页面 | 原文线索

V2-065505 | MATHDB-342031

The maximal embedding dimension criterion for good subsemigroups of \(\mathbb{N}^2\)

摘录 / EXCERPT: Let S be a good subsemigroup of $\mathbb{N}^2$. Its multiplicity vector is denoted by $\boldsymbol{e}$, and let $M=S\setminus\{\boldsymbol{0}\}$. A good semigroup is said to have...

conjecture | open | MathDB 页面 | 原文线索

V2-065506 | MATHDB-342035

The higher-weight Elliptic Stark conjecture

摘录 / EXCERPT: Let (f,g,h) have weights (k,ℓ,m) with $k\geq\ell+m$ and $k,\ell,m\geq2$, and let $M(f\otimes g\otimes h)$ be the associated triple-product motive. Put $c=(k+\ell+m-2)/2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-065507 | MATHDB-342053

The solid Pfaffian graph matching-colouring conjecture

摘录 / EXCERPT: Solid Pfaffian conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065508 | MATHDB-342054

The matching-minor analogue of Hadwiger's conjecture

摘录 / EXCERPT: Matching-minor Hadwiger conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-065509 | MATHDB-342056

The generalized Delta conjecture for labelled decorated Dyck paths

摘录 / EXCERPT: Generalized Delta conjecture. For any integers $n>k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-065510 | MATHDB-342067

Oura's analogy conjecture for Eisenstein polynomials

摘录 / EXCERPT: Oura's conjecture. The following analogous properties hold for $Th(\widetilde{\varphi_\ell^{G_{\rm{II}}}})$: (1) all zeros lie on a segment of the circle...

conjecture | open | MathDB 页面 | 原文线索

V2-065511 | MATHDB-342070

Boundary trace uniqueness conjecture for the CGO-Faddeev Dirac system

摘录 / EXCERPT: Let $\mathcal O$ be the domain, let $S_{\lambda,w}:H^{1/2}(\partial\mathcal O)\to H^{1/2}(\partial\mathcal O)$ be the boundary integral operator defined by ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065512 | MATHDB-342074

The blow-up conjecture for the rank-three logarithmic-connection moduli space

摘录 / EXCERPT: Let Σ be the Riemann sphere with n parabolic points, fix generic eigenvalues $\left\{\rho_{i,k}\right\}_{i=0,k=1}^{n,3}$, and let ${\mathcal M}$ be the moduli space of log...

conjecture | open | MathDB 页面 | 原文线索

V2-065513 | MATHDB-342077

The conjecture on rigid singularities

摘录 / EXCERPT: Rigidity conjecture. There exist no rigid fat points, no rigid reduced curve singularities, and no rigid normal surface singularities.

conjecture | open | MathDB 页面 | 原文线索

V2-065514 | MATHDB-342078

The Tjurina-torsion conjecture for isolated complete intersection singularities

摘录 / EXCERPT: Tjurina-torsion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065515 | MATHDB-342081

The triple multiple T-value sum formula

摘录 / EXCERPT: Let $T(a,b,c)$ denote the multiple T-value of indices (a,b,c), and let $T(k)$ denote the single T-value of weight k. For $k\geq 4$, the indices satisfy $a+b+c=k$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-065516 | MATHDB-342082

The containment conjecture for shuffle and stuffle multiple T-value spaces

摘录 / EXCERPT: Let $\mathcal{Z}$ be the space of ordinary multiple zeta values, and let $\mathcal{T}^{\sh}$ and $\mathcal{T}^*$ be the shuffle and stuffle spaces of multiple T-values. The conta...

conjecture | open | MathDB 页面 | 原文线索

V2-065517 | MATHDB-342083

The classification conjecture for multiple T-values in ordinary multiple zeta values

摘录 / EXCERPT: Let $T(p,q,r)$ denote a triple multiple T-value, let $T(k)$ denote a single T-value, and let $\mathcal{Z}$ be the space of ordinary multiple zeta values. By duality, restrict t...

conjecture | open | MathDB 页面 | 原文线索

V2-065518 | MATHDB-342084

The parity-weighted double T-value conjecture

摘录 / EXCERPT: Let $T(p+i,q+j)$ denote a double multiple T-value, with $m\geq 1$, $p\geq 1$, and $q\geq 2$. Assume that $p+q+m$ is even. The parity-weighted double T-value conjecture. ... The c...

conjecture | open | MathDB 页面 | 原文线索

V2-065519 | MATHDB-342085

The Lyndon basis conjecture for multiple zeta values via multiple t-values

摘录 / EXCERPT: Let $\mathcal{Z}_k$ be the weight-k space of ordinary multiple zeta values. For a sequence $(k_1,\ldots,k_r)$, call it Lyndon when every proper right subsequence is greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-065520 | MATHDB-342089

Alexandrov immersion obstruction conjecture for the Bartnik mass

摘录 / EXCERPT: Let $\operatorname{Imm}_{A}(S^2,\mathbb R^3)$ be the space of smooth Alexandrov immersed 2-spheres, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065521 | MATHDB-342090

Maximal static vacuum immersion conjecture

摘录 / EXCERPT: Let (γ,H) be Bartnik boundary data on S^2, normalized by $\operatorname{area}_{\gamma}(S^2)=1$. A maximal static vacuum solution is an asymptotically flat static vacuum tr...

conjecture | open | MathDB 页面 | 原文线索

V2-065522 | MATHDB-342095

Power-sum equality for orientation posets

摘录 / EXCERPT: Let $\mathbf{a}$ be an area sequence of length n, let $O(\mathbf{a})$ be the set of orientations associated with $\mathbf{a}$, let $P(\theta)$ be the poset obtained from the asce...

conjecture | open | MathDB 页面 | 原文线索

V2-065523 | MATHDB-342099

Exterior-cubic pole criterion for tempered cuspidal representations of GU(6)

摘录 / EXCERPT: Let F be the global field and let E/F be the quadratic extension underlying the unitary similitude group. Let $\ba$ denote the ring of adeles of F, let π be an irreducibl...

conjecture | open | MathDB 页面 | 原文线索

V2-065524 | MATHDB-342105

Khavinson's gradient-direction conjecture for bounded harmonic functions

摘录 / EXCERPT: Let h^{∞} denote the space of bounded harmonic functions on the unit ball $\mathbb{B}^n:=\{x\in\mathbb{R}^n:|x|<1\}$. For $x\in\mathbb{B}^n$ and $l\in\mathbb{S}^{n-1}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065525 | MATHDB-342107

Local finiteness of intermediate logics of finite powers of the natural numbers

摘录 / EXCERPT: Let n be finite, and let ${\operatorname{ILog}}(\omega^n,\preceq)$ denote the intermediate logic of the frame $(\omega^n,\preceq)$. Local finiteness claim. For all finite n, ...

conjecture | open | MathDB 页面 | 原文线索

V2-065526 | MATHDB-342114

Mean Lindelöf hypothesis for symmetric-square L-functions

摘录 / EXCERPT: Let $\Gamma = \mathrm{PSL}_{2}(\mathcal{O})$, let u_j be a Hecke-Maass cusp form for Γ, and let t_j be its spectral parameter. Write $w=1/2+i\tau$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-065527 | MATHDB-342122

Conjecture on the tempered positive-energy interpretation of the crossed-product K-group

摘录 / EXCERPT: Tempered positive-energy conjecture. The K-group $K_\bullet(\Gamma_0(\tau_G^G)\rtimes G)$ reflects the tempered, positive energy representations of LG in a suitable sense, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065528 | MATHDB-342124

Finite expected angular condition number for identifiable tensors

摘录 / EXCERPT: Finite expected angular condition number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065529 | MATHDB-342127

Deterministic optimal channel proportions for moderate parameters

摘录 / EXCERPT: Deterministic-maximizer conjecture. For all $\alpha,\lambda\in\mathbb{R}_+$, the vectors $\mathbf{a}_\alpha^*$ and $\mathbf{b}_\alpha^*$ that maximize...

conjecture | open | MathDB 页面 | 原文线索

V2-065530 | MATHDB-342137

The Markov-partition conjecture for the Jeandel-Rao toral rotation

摘录 / EXCERPT: Let $\mathcal{P}_0$ be the toroidal partition associated with the Jeandel-Rao Wang shift, let Γ_0 be the corresponding lattice, and let R_0 denote the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-065531 | MATHDB-342145

The KTV conjecture on rapid mixing of switch Markov chains

摘录 / EXCERPT: Let a switch Markov chain be the chain on realizations of a degree sequence that performs a switch, for an unconstrained, bipartite, or directed degree sequence. A degree sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-065532 | MATHDB-342153

Walk dimension conjecture for fractional Brownian motion graphs

摘录 / EXCERPT: Let B^H be a fractional Brownian motion with Hurst index $H\in(0,1)$, and let $\varphi(W)$ be the Hunt process on $\operatorname{gr}(B^H)$ induced by a Wiener process through the...

conjecture | open | MathDB 页面 | 原文线索

V2-065533 | MATHDB-342181

Integral-form conjecture for Jack superpolynomials

摘录 / EXCERPT: Integral-form conjecture. The coefficients $v_{{\Lambda}\Om}(\alpha)$ are polynomials in α with integer coefficients, so that $v_{{\Lambda}\Om}(\alpha)\in\mathbb Z[\alpha]$....

conjecture | open | MathDB 页面 | 原文线索

V2-065534 | MATHDB-342182

Clustering-basis conjecture for Jack superpolynomials

摘录 / EXCERPT: Clustering-basis conjecture. The Jack superpolynomials $P_{\Lambda}^{(\alpha_{k,2})}(x_1,\ldots,x_N,\ta_1,\ldots,\ta_N)$, with Λ $(k,2,N)$-admissible, form a basis of...

conjecture | open | MathDB 页面 | 原文线索

V2-065535 | MATHDB-342183

Macdonald superpolynomial positivity conjecture

摘录 / EXCERPT: Macdonald superpolynomial positivity conjecture. For all superpartitions Λ and $\Om$, the coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-065536 | MATHDB-342187

Converse characterisation of separately exchangeable random matrices by empirical distributions

摘录 / EXCERPT: Converse characterisation conjecture. If the empirical distributions $(\eta_{n,m})$ form a reverse measure-valued martingale, then X is separately exchangeable, meaning that for...

conjecture | open | MathDB 页面 | 原文线索

V2-065537 | MATHDB-342193

Maximal area conjecture for filled Julia sets

摘录 / EXCERPT: Maximal area conjecture. The maximal area of K_c is

conjecture | open | MathDB 页面 | 原文线索

V2-065538 | MATHDB-342194

Nonexistence of box-counting dimension for typical connected Julia sets

摘录 / EXCERPT: Nonexistence conjecture. The box-counting dimension of a typical connected Julia set does not exist.

conjecture | open | MathDB 页面 | 原文线索

V2-065539 | MATHDB-342201

The phase-transition conjecture for the marginal of a random contingency table

摘录 / EXCERPT: Let $B,C>0$, let $0<\delta<1$, and let $X=(X_{ij})$ be sampled uniformly from $\mathcal M_{n,\delta}(B,C)$. Let B_c denote the critical value. Phase-transition conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-065540 | MATHDB-342212

Integral period basis conjecture for the six-line K3 family

摘录 / EXCERPT: Let $d=2$, and let ω_{0}, $\omega^{(2)}$, and $\omega_{i}^{(1)}$ be the period solutions used to form ... Let (M_{ij}) be the symmetric form introduced for the period relati...

conjecture | open | MathDB 页面 | 原文线索

V2-065541 | MATHDB-342213

Mirror conjecture for the double-cover K3 surface from six lines

摘录 / EXCERPT: Mirror conjecture for the double cover. The mirror of the singular double-cover K3 surface $\overline{X}$ is the singular K3 surface $\overline{S}_{6}$ defined above. This conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-065542 | MATHDB-342217

The classification conjecture for perfect colorings of infinite circulant graphs with odd distances

摘录 / EXCERPT: Classification conjecture. The set of perfect k-colorings of $\mathrm{Ci}_\infty(D_n)$ consists of perfect colorings induced from perfect colorings of the infinite path graph and...

conjecture | open | MathDB 页面 | 原文线索

V2-065543 | MATHDB-342228

The analytic continuation conjecture for complete adjoint L-functions

摘录 / EXCERPT: Adjoint L-function analytic continuation conjecture. The complete adjoint L-function $\Lambda(s,\pi,\operatorname{Ad})$ admits an analytic continuation to the whole complex pla...

conjecture | open | MathDB 页面 | 原文线索

V2-065544 | MATHDB-342237

Černý-Starke conjecture on synchronizing automata

摘录 / EXCERPT: A synchronizing automaton is a finite automaton whose transition semigroup contains a word sending every state to the same state; its reset threshold $rt(\mathcal{A})$ is the lengt...

conjecture | open | MathDB 页面 | 原文线索

V2-065545 | MATHDB-342238

Conjectured eventual coincidence of F_k(n) and B_k(n)

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and define ... Let $F_k(n)$ and $B_k(n)$ be the bounds defined in the paper. Eventual coincidence conjecture. If $2\leq k\leq 6$, then for every $n>k^2$, ... If...

conjecture | open | MathDB 页面 | 原文线索

V2-065546 | MATHDB-342239

Reiher-Schacht conjecture on constant rooted clique densities

摘录 / EXCERPT: Let $r\ge 3$ and let W be a graphon. For $x,y\in[0,1]$, write $t_{x,y}(K_r^{\bullet\bullet},W)$ for the rooted homomorphism density of the clique K_r with two distinguished roo...

conjecture | open | MathDB 页面 | 原文线索

V2-065547 | MATHDB-342240

Polynomial-time lower bound for regression risk

摘录 / EXCERPT: Fix $\pi\in\mathcal P_2(\mathbb R)$, $\delta\in(0,\infty)$, and $\sigma>0$. Consider sequences of dimensions, Gaussian design matrices, coefficient vectors, and noise vectors satis...

conjecture | open | MathDB 页面 | 原文线索

V2-065548 | MATHDB-342241

Symmetry conjecture for vectors working with the all-ones-complement vector

摘录 / EXCERPT: Symmetry conjecture. If v and $\mathantt{e}$ work together, then v is symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-065549 | MATHDB-342244

The non-relativistic rapidity substitution conjecture for massless AdS/CFT R-matrices

摘录 / EXCERPT: Let p be the momentum of a massless mode and define the non-relativistic rapidity ... For relativistic massless scattering, let θ denote the rapidity difference, and let rig...

conjecture | open | MathDB 页面 | 原文线索

V2-065550 | MATHDB-342251

Hoffmann's conjecture on the first Witt index of quadratic forms

摘录 / EXCERPT: Let φ be an anisotropic quadratic form over a field F, let $\mathrm{dim}\varphi\geq 2$, let $\frak{i}_{1}(\varphi)$ denote its first Witt index, and for a nonzero integer...

conjecture | open | MathDB 页面 | 原文线索

V2-065551 | MATHDB-342252

The Burau representation's surjectivity conjecture

摘录 / EXCERPT: Let B_n be the braid group, let $\beta:B_n\to\Gamma$ be the unreduced Burau representation, and let $\beta':B_n\to\Gamma'$ be its reduced representation, where Γ is the un...

conjecture | open | MathDB 页面 | 原文线索

V2-065552 | MATHDB-342256

The monotonicity conjecture for the Potts boundary functions $\alpha_\pm$

摘录 / EXCERPT: Monotonicity conjecture. The function $\alpha_-(\theta)$ is monotone decreasing for all $\theta\geq\theta_{\rm c}$, while $\alpha_+(\theta)$ is decreasing for...

conjecture | open | MathDB 页面 | 原文线索

V2-065553 | MATHDB-342257

The monotonicity conjecture for the Potts functions α_m

摘录 / EXCERPT: Monotonicity conjecture. For each $m=1,\dots,q-2$, the function $\theta\mapsto\alpha_m(\theta)$ is monotone increasing. This would give the monotonicity observed in the plotted exa...

conjecture | open | MathDB 页面 | 原文线索

V2-065554 | MATHDB-342270

The local-curve stable-pair generating-series conjecture

摘录 / EXCERPT: Let $X=\operatorname{Tot}_{\mathbb P^1}(\mathcal O\oplus\mathcal O(-1)\oplus\mathcal O(-1))$ and let $P_{n,d}(X;T)$ be the equivariant stable-pair invariant, where d is the degre...

conjecture | open | MathDB 页面 | 原文线索

V2-065555 | MATHDB-342274

List decodability of Reed-Solomon codes up to capacity

摘录 / EXCERPT: List decodability of Reed-Solomon codes up to capacity. For every $\rho>0$, there is a constant C_ρ such that every Reed-Solomon code of length n and rate ρ is list-dec...

conjecture | open | MathDB 页面 | 原文线索

V2-065556 | MATHDB-342276

Fractional matching gap conjecture for critical graphs without 1-factors

摘录 / EXCERPT: Let $k\geq 3$ and let G be a k-critical graph. Let $\mu(G)$ denote the matching number and $\mu_f(G)$ the fractional matching number. Fractional matching gap conjecture. If G...

conjecture | open | MathDB 页面 | 原文线索

V2-065557 | MATHDB-342283

Asymptotic capacity conjecture for PIR from MDS-coded colluding and adversarial servers

摘录 / EXCERPT: Asymptotic PIR capacity conjecture. The asymptotic capacity as the number of files m tends to infinity is

conjecture | open | MathDB 页面 | 原文线索

V2-065558 | MATHDB-342284

Symmetric PIR capacity conjecture for MDS-coded colluding and adversarial servers

摘录 / EXCERPT: Symmetric PIR capacity conjecture. The capacity of SPIR is

conjecture | open | MathDB 页面 | 原文线索

V2-065559 | MATHDB-342291

Full elasticity conjecture for cyclic rational semirings

摘录 / EXCERPT: Let $r\in\mathbb{Q}_{>1}\setminus\mathbb{N}$, and write $r=\mathsf{n}(r)/\mathsf{d}(r)$ in lowest terms. Let S_r be the cyclic rational semiring generated by the geometric sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-065560 | MATHDB-342292

The nonnegative hook-content formula conjecture for skew partitions

摘录 / EXCERPT: Let $\overline{H}_{\lambda/\mu}(n)$ be the normalized hook-content quantity associated with partitions $\mu\subseteq\lambda$, and let ℓ denote the length of λ. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-065561 | MATHDB-342293

The positive rationality conjecture for skew q-hook-length polynomials

摘录 / EXCERPT: Let $\mu\subseteq\lambda$ be partitions, and let $f^{\lambda/\mu}_q$ be the skew q-hook-length quantity. Assume it can be written as a rational function ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-065562 | MATHDB-342294

The positive rationality conjecture for normalized skew q-hook-content formulas

摘录 / EXCERPT: Let $\mu\subseteq\lambda$ be partitions, let ℓ be the length of λ, and fix an integer $n\geq\ell$. Let $H_{\lambda/\mu}(n;q)$ be the skew q-hook-content quantity and...

conjecture | open | MathDB 页面 | 原文线索

V2-065563 | MATHDB-342304

A q-supercongruence associated with a truncated basic hypergeometric series

摘录 / EXCERPT: Let n be an integer greater than 1. Define the q-integer $[m]=(1-q^m)/(1-q)$ and let $(a;q)_k$ denote the q-shifted factorial. The q-supercongruence. ... This is presented as...

conjecture | open | MathDB 页面 | 原文线索

V2-065564 | MATHDB-342307

A parametric q-congruence for truncated basic hypergeometric series

摘录 / EXCERPT: Let $d\geqslant4$ and let $n>1$ satisfy $n\equiv-1\pmod d$. Let $(a;q)_k$ denote the q-shifted factorial and let $\Phi_n(q)$ be the nth cyclotomic polynomial. The q-congruence....

conjecture | open | MathDB 页面 | 原文线索

V2-065565 | MATHDB-342322

Reiner-Shimozono's positivity conjecture for products of key polynomials

摘录 / EXCERPT: Let a and b be weak compositions, and let ${\mathfrak{D}}_a$ and ${\mathfrak{D}}_b$ denote the corresponding key polynomials. Let ${\mathfrak{A}}_c$ denote the Demazure atom in...

conjecture | open | MathDB 页面 | 原文线索

V2-065566 | MATHDB-342328

Explicit reciprocity conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let $S:=S_\infty(k)\cup S_p(k)\cup S_{\rm ram}(L/k)$, assume the Rubin-Stark conjecture for all finite subextensions of $Lk(p^\infty)/k$, and suppose that p splits completely in...

conjecture | open | MathDB 页面 | 原文线索

V2-065567 | MATHDB-342330

Hida-Tilouine Iwasawa Main Conjecture for CM fields

摘录 / EXCERPT: Let M_Σ be the maximal abelian pro-p extension of $Lk(p^\infty)$ that is Σ-ramified, let $X_\Sigma:=\operatorname{Gal}(M_\Sigma/Lk(p^\infty))$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065568 | MATHDB-342339

Ando-type unitarily invariant norm inequalities for operator functions

摘录 / EXCERPT: Let A and B be positive matrices, let f be a function on $[0,infty)$ with $f(0)=0$, and let $|||\cdot|||$ be an arbitrary unitarily invariant norm. Ando-type conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-065569 | MATHDB-342344

Hyperbolicity of the parabolic generating set for spherical Artin-Tits groups

摘录 / EXCERPT: Let $A=A_S$ be a non-cyclic irreducible Artin-Tits group of spherical type. Let X_P^A be the union of all proper irreducible standard parabolic subgroups of A and the cyclic su...

conjecture | open | MathDB 页面 | 原文线索

V2-065570 | MATHDB-342348

Langlands correspondence for weight two eigenforms over number fields

摘录 / EXCERPT: Let K be a number field and let ${\mathfrak N}$ be an ideal of ${\mathcal O}_K$. Let ${\mathfrak f}$ be a non-trivial, new weight 2 complex eigenform over K of level...

conjecture | open | MathDB 页面 | 原文线索

V2-065571 | MATHDB-342360

Small phase theorem with necessity for MIMO LTI systems

摘录 / EXCERPT: Let $G\in\mathcal{RH}^{m\times m}_\infty$ and let $\mathfrak{C}[\alpha]$ be the set of phase-bounded systems, where $\alpha\in[0,\pi)$. The feedback system $G\#H$ is stable for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-065572 | MATHDB-342361

Linear decay conjecture for the transformed gradient of p-harmonic maps

摘录 / EXCERPT: Let $d,n\in\mathbb{N}$ and $1<p<\infty$. Let $h\colon\Omega\to\mathbb{R}^n$ be p-harmonic on an open set $\Omega\subset\mathbb{R}^d$, and let V denote the standard nonlinear tr...

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V2-065573 | MATHDB-342364

The 2-adic valuation conjecture for even-colored 2-ary partition functions

摘录 / EXCERPT: Let $c_m(n)$ denote the m-colored binary partition function, and let $ u_2$ be the 2-adic valuation. For positive integers k,n and nonnegative integers k as specified below,...

conjecture | open | MathDB 页面 | 原文线索

V2-065574 | MATHDB-342365

The p-adic valuation conjecture for differences of m-colored p-ary partition functions

摘录 / EXCERPT: Let $d_m(n)$ denote the m-colored p-ary partition function, let $p\in\mathbb{P}_{\geq 3}$, and let ν_p denote the p-adic valuation. Write $m=pk+i$ with $k\in\mathbb{N}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-065575 | MATHDB-342366

The conjectures on 4-adic valuations of colored 4-ary partition functions

摘录 / EXCERPT: Let $S_{k,m}(n)$ be the nth coefficient in the power-series expansion defining the m-colored k-ary partition function, and define ... 4-adic valuation conjectures. The followin...

conjecture | open | MathDB 页面 | 原文线索

V2-065576 | MATHDB-342369

Collinear triples conjecture for pairs of almost holomorphic modular values

摘录 / EXCERPT: Collinear triples conjecture. There are only finitely many triples with the P_i pairwise distinct and lying on a straight line which is neither horizontal, vertical nor the diago...

conjecture | open | MathDB 页面 | 原文线索

V2-065577 | MATHDB-342370

Effective André-Oort-type conjecture for pairs of almost holomorphic modular values

摘录 / EXCERPT: Effective André-Oort-type conjecture. There are effectively computable constants $c_i=c_i(V)$ such that whenever

conjecture | open | MathDB 页面 | 原文线索

V2-065578 | MATHDB-342391

Braidedness conjecture for annular Khovanov-Rozansky functors

摘录 / EXCERPT: Let $\mathrm{A}\boldsymbol{\mathrm{Link}}^+$ be the category of consistently oriented annular links and cobordisms, and let $K^b(\mathrm{A}\boldsymbol{\mathrm{Foam}}^+)$ be the hom...

conjecture | open | MathDB 页面 | 原文线索

V2-065579 | MATHDB-342398

The graded-ring generalization of Kaplansky's conjectures

摘录 / EXCERPT: Graded-ring generalization of Kaplansky's conjectures. Every unit in R is homogeneous, R is a domain, and every idempotent in R is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-065580 | MATHDB-342399

Ghorbani's determinant conjecture for Seidel matrices

摘录 / EXCERPT: Let G_n be a simple graph on $V_n=\{v_1,\dots,v_n\}$, and let $S(G_n)$ be its Seidel matrix. Let $\det(S(G_n))$ denote its determinant. For a uniformly chosen graph on V_n, con...

conjecture | open | MathDB 页面 | 原文线索

V2-065581 | MATHDB-342400

Conjectured optimal trajectory for imbalance with idle servers

摘录 / EXCERPT: Let $T>0$, let $j\geq 2$, and let $c\in(0,1)$. The event that some servers have queues of length j while the remaining servers are idle is ... This corresponds to nc queues of le...

conjecture | open | MathDB 页面 | 原文线索

V2-065582 | MATHDB-342401

Piecewise-linear optimal trajectory for partially long queues

摘录 / EXCERPT: Let $T>0$, let $j\geq 2$, and let $c\in(0,1)$. The event that, at some time $t\in[0,T]$, some servers have queues of length j while the remaining servers have queues of length...

conjecture | open | MathDB 页面 | 原文线索

V2-065583 | MATHDB-342402

The index-weighted vanishing conjecture for semi-ample growth

摘录 / EXCERPT: Let S be the bihomogeneous polynomial ring and let $\overline{S}$ denote the quotient defined in the paper. Fix $\mathbf n=(n_1,n_2)\in \mathbb{Z}^2_{\geq 1}$, set...

conjecture | open | MathDB 页面 | 原文线索

V2-065584 | MATHDB-342404

Hildebrand's conjecture on patterns in stable sets

摘录 / EXCERPT: Hildebrand's stable sets conjecture. If $k\geq 2$ and $A\subset\mathbb{N}$ is stable with $d_-(A)>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-065585 | MATHDB-342408

Fourth moment conjecture for real-valued Maaß newforms

摘录 / EXCERPT: Let q and χ be fixed, and let $g\in\mathcal{B}_0^{\ast}(q,\chi)$ be a real-valued newform with spectral parameter t_g. Equip $\Gamma_0(q)\backslash\mathbb{H}$ with hyperbo...

conjecture | open | MathDB 页面 | 原文线索

V2-065586 | MATHDB-342411

Nonexistence of infinite uniquely-Wilf classes containing all permutations of length three

摘录 / EXCERPT: Let ${\mathcal{C}}$ be a class of permutations, and let ${\mathcal{C}}_3$ denote its permutations of length three. Let ${\mathcal{S}}_3$ be the set of all permutations of length th...

conjecture | open | MathDB 页面 | 原文线索

V2-065587 | MATHDB-342413

The girth-shifting conjecture for Hall ratios

摘录 / EXCERPT: Let $\rho(d,g)$ denote the maximum Hall ratio among graphs of maximum degree at most d and girth at least g. Girth-shifting conjecture. The values presented in the table are up...

conjecture | open | MathDB 页面 | 原文线索

V2-065588 | MATHDB-342429

Gupta's bounded transcendence-degree conjecture for polynomial sets

摘录 / EXCERPT: Gupta's bounded transcendence-degree conjecture. There is a function λ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065589 | MATHDB-342431

Minimum dominator chromatic number of orientations of generalized stars

摘录 / EXCERPT: Let GS_{m}^{k} be a generalized star, and let $\chi_{d}(GS_{m}^{k})$ denote its dominator chromatic number. The generalized-star minimum conjecture. The minimum dominator chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-065590 | MATHDB-342432

Maximum dominator chromatic number of orientations of generalized stars

摘录 / EXCERPT: Let GS_{m}^{k} be a generalized star, and let $\chi_{d}(GS_{m}^{k})$ denote its dominator chromatic number. An (anti-)arborescence is an orientation of the generalized star that...

conjecture | open | MathDB 页面 | 原文线索

V2-065591 | MATHDB-342438

The list coloring conjecture for multigraphs

摘录 / EXCERPT: List coloring conjecture. For every multigraph G,

conjecture | open | MathDB 页面 | 原文线索

V2-065592 | MATHDB-342443

Bifurcation-set equality conjecture for beta transformations with a hole

摘录 / EXCERPT: Let $\beta\in(1,2]$. The bifurcation-set equality conjecture. ... if and only if $\mathcal{E}_\beta$ has no isolated points. Here $\mathcal{E}_\beta$ is the set-valued bifurcation se...

conjecture | open | MathDB 页面 | 原文线索

V2-065593 | MATHDB-342444

Ji-Zhang-Zhao inequalities for overpartition ranks

摘录 / EXCERPT: Ji-Zhang-Zhao inequalities. For $n\geq 0$ and $1\leq i\leq 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-065594 | MATHDB-342445

Wei-Zhang inequalities for overpartition ranks modulo 6

摘录 / EXCERPT: Wei-Zhang inequalities. For $n\geq 11$,

conjecture | open | MathDB 页面 | 原文线索

V2-065595 | MATHDB-342447

The torsion-growth classification conjecture for degrees at most 23

摘录 / EXCERPT: Torsion-growth classification conjecture. The set $\Phi_{\mathbb{Q}}(d)$ consists of the union of all $\Theta(d')$ such that $d'\mid d$:

conjecture | open | MathDB 页面 | 原文线索

V2-065596 | MATHDB-342448

The primitive torsion-growth classification conjecture for degrees at most 23

摘录 / EXCERPT: Primitive torsion-growth classification conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065597 | MATHDB-342449

Conjecture on the landscape of planted dense subgraphs and overlap gaps

摘录 / EXCERPT: Landscape conjecture. Suppose ... If $k=o(\sqrt n)$, then:

conjecture | open | MathDB 页面 | 原文线索

V2-065598 | MATHDB-342454

Nonlocal stable De Giorgi conjecture

摘录 / EXCERPT: Nonlocal stable De Giorgi conjecture. If $d\le 7$, then u is a 1-D solution. Stable solutions include local minimizers and monotone stationary solutions, and the conjecture con...

conjecture | open | MathDB 页面 | 原文线索

V2-065599 | MATHDB-342455

Segre-Harbourne-Gimigliano-Hirschowitz conjecture

摘录 / EXCERPT: SHGH Conjecture. If $r\geq 3$ and $d>m_1+m_2+m_3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-065600 | MATHDB-342459

Eigenvalue convergence for kernels under total-variation convergence

摘录 / EXCERPT: Let (X,d,μ) be a bounded metric measure space with kernel K and associated operator $T_K\colon L^2(X,\mu)\to L^2(X,\mu)$. Let (X,d,μ_n) have analogous kernel K_n and op...

conjecture | open | MathDB 页面 | 原文线索

V2-065601 | MATHDB-342468

Meunier's chromatic number conjecture for stable Kneser graphs

摘录 / EXCERPT: Let $KG(n,k)_{s-\mathrm{stab}}$ be the induced subgraph of the Kneser graph $KG(n,k)$ on the s-stable k-subsets of $[n]$, where a subset A is s-stable when...

conjecture | open | MathDB 页面 | 原文线索

V2-065602 | MATHDB-342472

Veech's conjecture on non-negative r-median functions

摘录 / EXCERPT: Veech's conjecture. Every non-negative, r-median function on D is harmonic.

conjecture | open | MathDB 页面 | 原文线索

V2-065603 | MATHDB-342480

Conjectural formula for the Tate-Shafarevich group of a Gross-curve twist

摘录 / EXCERPT: Conjectural Tate-Shafarevich formula. The order of the Tate-Shafarevich group satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065604 | MATHDB-342486

Purity conjecture for zero-sumfree complexes at half the modulus

摘录 / EXCERPT: For an odd integer n, consider the complexes $\Delta_{n,(n-1)/2}$ and $\Delta_{n,(n+1)/2}$ of ℓ-zero-sumfree subsets. A simplicial complex is pure when all of its facets hav...

conjecture | open | MathDB 页面 | 原文线索

V2-065605 | MATHDB-342487

h-vector criterion for purity of zero-sumfree complexes

摘录 / EXCERPT: Let $\Delta_{n,\ell}$ be the simplicial complex of ℓ-zero-sumfree subsets, and let $h_1,\ldots,h_d$ be the relevant entries of its h-vector. A simplicial complex is pure whe...

conjecture | open | MathDB 页面 | 原文线索

V2-065606 | MATHDB-342488

Connectivity conjecture for zero-sumfree complexes

摘录 / EXCERPT: Let $\Delta_{n,\ell}$ be the simplicial complex of ℓ-zero-sumfree subsets. Regard it as connected when for any two vertices there is a sequence of vertices joining them throug...

conjecture | open | MathDB 页面 | 原文线索

V2-065607 | MATHDB-342496

Vishik's conjecture on local Chow groups and numerical equivalence

摘录 / EXCERPT: Let k be a flexible field. The local Chow groups are the groups obtained from the local Chow theory over k, while $\operatorname{Ch}^*_{Num}$ denotes Chow groups with...

conjecture | open | MathDB 页面 | 原文线索

V2-065608 | MATHDB-342500

Outer-function characterization in the upper half-plane

摘录 / EXCERPT: Outer-function characterization. The function h above is outer if and only if F has no zeros in the open upper half-plane.

conjecture | open | MathDB 页面 | 原文线索

V2-065609 | MATHDB-342503

The divisibility conjecture for generalized near-group categories

摘录 / EXCERPT: Divisibility conjecture. The order $|G_\rho|$ divides K. This conjecture proposes an arithmetic constraint on the multiplicities in generalized near-group fusion categories; the...

conjecture | open | MathDB 页面 | 原文线索

V2-065610 | MATHDB-342505

Kohnert-diagram formula conjecture for Lascoux polynomials

摘录 / EXCERPT: Let $a=(a_1,\dots,a_n)\in\boldsymbol{Z}_{\geq 0}^n$. A K-Kohnert diagram is a subset D of $\boldsymbol{Z}_{>0}^n$ together with a subset of marked boxes, obtained from the skyline di...

conjecture | open | MathDB 页面 | 原文线索

V2-065611 | MATHDB-342507

Flagged Grothendieck characterization conjecture for Lascoux polynomials

摘录 / EXCERPT: Let λ be any partition and let $w\in S_n^{\lambda}$. A flagged Grothendieck polynomial is understood in the sense used in the paper, and w is 312-avoiding when it has no s...

conjecture | open | MathDB 页面 | 原文线索

V2-065612 | MATHDB-342508

Lusztig-involution conjecture for general Lascoux polynomials and atoms

摘录 / EXCERPT: Lusztig-involution conjecture. For every partition $lambda$,

conjecture | open | MathDB 页面 | 原文线索

V2-065613 | MATHDB-342509

Splitting conjecture for two-dimensional torsion-free definable abelian groups

摘录 / EXCERPT: Let $\mathcal{M}$ be an o-minimal structure, and let G be an abelian, two-dimensional, torsion-free group definable in $\mathcal{M}$. Splitting conjecture. The group G is the p...

conjecture | open | MathDB 页面 | 原文线索

V2-065614 | MATHDB-342517

The Gröbner basis characterization of 0- and 1-inductively pierced codes

摘录 / EXCERPT: The Gröbner basis characterization. For each n, there exists a monomial order such that $\mathcal{C}$ is 0- or 1-inductively pierced if and only if the reduced Gröbner basis...

conjecture | open | MathDB 页面 | 原文线索

V2-065615 | MATHDB-342520

The discrete-excess conjecture for the contradictory component in random 2-SAT

摘录 / EXCERPT: Let F be chosen uniformly at random from the set $\mathcal F(n,m)$ of 2-CNF formulas, and let ξ denote the total excess of the strongly connected contradictory component in t...

conjecture | open | MathDB 页面 | 原文线索

V2-065616 | MATHDB-342521

The critical-point equations for degree-constrained random 2-SAT

摘录 / EXCERPT: Critical-point conjecture. The phase-transition point r can be computed from the system

conjecture | open | MathDB 页面 | 原文线索

V2-065617 | MATHDB-342525

Finite-generation conjecture for quasimap potentials of K3 fibrations

摘录 / EXCERPT: K3-fibration finite-generation conjecture. For $\gamma\in H_2(B,\mathbb Z)$,

conjecture | open | MathDB 页面 | 原文线索

V2-065618 | MATHDB-342526

Finite-generation conjecture for twisted quasimap series of ${\mathbb P}^2 \times {\mathbb P}^1$

摘录 / EXCERPT: Let ... be the $((3;2),(0;0))$-twisted genus-g quasimap series of ${\mathbb P}^2\times{\mathbb P}^1$, and let ... with $X(q_1)=q_1(\partial B_1'/\partial q_1)/(1+B_1')$. Twisted elli...

conjecture | open | MathDB 页面 | 原文线索

V2-065619 | MATHDB-342527

The grand discrepancy conjecture for infinite sequences

摘录 / EXCERPT: Let $s\in\mathbb{N}$ and let $\mathcal{S}$ be a sequence in $[0,1)^s$. Write $D_N^{\ast}(\mathcal{S})$ for its star discrepancy at N points. Grand discrepancy conjecture. For eve...

conjecture | open | MathDB 页面 | 原文线索

V2-065620 | MATHDB-342532

The generating-function conjecture for valid hook configurations avoiding length-3 and length-4 patterns

摘录 / EXCERPT: Let $\operatorname{Av}_n(\sigma_1,\ldots,\sigma_r)$ denote the permutations of length n avoiding the listed patterns, and let $\operatorname{\mathsf{VHC}}(\mathcal{S})$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-065621 | MATHDB-342535

The auxiliary-character p-adic Stark conjecture

摘录 / EXCERPT: Let $\eta,\psi\in\mathcal{W}(\mathbb{C}_p)$ be finite-order characters of orders p^m and p^n, respectively. Let M_m and M_n be the fixed fields of the kernels of...

conjecture | open | MathDB 页面 | 原文线索

V2-065622 | MATHDB-342536

The p-adic Stark conjecture in the cyclotomic tower

摘录 / EXCERPT: For each $n\ge 0$, let $\mathbb{Q}_n$ be the nth layer of the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$, let $M_n=M\mathbb{Q}_n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065623 | MATHDB-342537

Existence of a non-trivial intertwining from the infinite Ehrenfest process to the Ornstein-Uhlenbeck process

摘录 / EXCERPT: Let D_{∞} denote the infinite Ehrenfest process and let L be the Ornstein-Uhlenbeck operator. An intertwining existence conjecture. There exists a non-trivial intertwining...

conjecture | open | MathDB 页面 | 原文线索

V2-065624 | MATHDB-342538

The p-adic Brauer-Siegel conjecture for p-ramification torsion

摘录 / EXCERPT: The p-adic Brauer-Siegel conjecture. There exists a constant ${\mathcal C}_p({\mathcal K})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065625 | MATHDB-342552

Morita adjoints conjecture for higher Morita categories

摘录 / EXCERPT: Let $\mathcal{V}$ be a symmetric monoidal $(\infty,1)$-category compatible with $\mathsf{simp}^{n,\operatorname{op}}$-colimits. Let $\mathfrak{alg}_{n}(\mathcal{V})$ denote the und...

conjecture | open | MathDB 页面 | 原文线索

V2-065626 | MATHDB-342572

The list coloring conjecture for edge and total coloring

摘录 / EXCERPT: Let G be a simple graph. Write $\chi'(G)$ and $\chi”(G)$ for its edge-chromatic and total-chromatic numbers, and $\chi'_{\ell}(G)$ and $\chi”_{\ell}(G)$ for the corresponding lis...

conjecture | open | MathDB 页面 | 原文线索

V2-065627 | MATHDB-342586

Gimigliano-Harbourne-Hirschowitz conjecture for effective divisors on blown-up planes

摘录 / EXCERPT: Let $X=X_{2,s}$ be the blow-up of the projective plane at s general points, and let $D\text{ in }\operatorname{Pic}(X)$ be an effective divisor of the specified form. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-065628 | MATHDB-342587

Nagata's conjecture for homogeneous divisors

摘录 / EXCERPT: Let $X=X_{n,s}$ be the blow-up of projective n-space at s general points, and let $D=dH-mE_1-\cdots-mE_s$ be a homogeneous divisor, with degree d and common multiplicity m....

conjecture | open | MathDB 页面 | 原文线索

V2-065629 | MATHDB-342591

Bounded-carrier Brownian box-ball invariance conjecture

摘录 / EXCERPT: Bounded-carrier Brownian invariance conjecture. If S^K is the Brownian motion with drift $c>0$ conditioned to stay within K of its past maximum, then

conjecture | open | MathDB 页面 | 原文线索

V2-065630 | MATHDB-342601

Cho-Huh-Sohn conjecture for self-conjugate consecutive-core partitions

摘录 / EXCERPT: For positive integers s and k, a partition is a self-conjugate $(s,s+1,\ldots,s+k)$-core partition if it is equal to its conjugate and is simultaneously an s-core, an $(s+1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-065631 | MATHDB-342604

Square-root cancellation conjecture for spectral exponential sums on arithmetic hyperbolic surfaces

摘录 / EXCERPT: Let $X>2$, let Γ be an arithmetic Fuchsian group defining a hyperbolic surface, and let $\mathcal{S}(T,X)$ denote the spectral exponential sum associated with that surface....

conjecture | open | MathDB 页面 | 原文线索

V2-065632 | MATHDB-342608

Tautologicality conjecture for curves on Enriques surfaces

摘录 / EXCERPT: Let E be an Enriques surface, and let $C\subset E$ be an irreducible nonsingular curve of genus $g\geq 2$. Its moduli point in $\overline{\mathcal{M}}_g$ determines a 0-cycle....

conjecture | open | MathDB 页面 | 原文线索

V2-065633 | MATHDB-342615

Del Zotto-Lockhart's conjecture for rank-one instanton strings

摘录 / EXCERPT: Let G be the global symmetry group of a rank-one H_G theory, let v and m_G denote its fugacities, and let Q_{τ} be the elliptic-parameter fugacity. Write ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065634 | MATHDB-342617

Near-unimodality conjecture for major-index generating polynomials

摘录 / EXCERPT: Near-unimodality conjecture. The polynomials $\SYT(\lambda)^{\maj}(q)$ are nearly unimodal but not unimodal for partitions λ or λ' in the following cases: any parti...

conjecture | open | MathDB 页面 | 原文线索

V2-065635 | MATHDB-342618

Almost-everywhere log-concavity conjecture for major-index generating polynomials

摘录 / EXCERPT: Log-concavity conjecture. The polynomials $\SYT(\lambda)^{\maj}(q)$ are almost always log-concave for partitions $\lambda\vdash n$ for large n.

conjecture | open | MathDB 页面 | 原文线索

V2-065636 | MATHDB-342619

Asymptotic normality conjecture for major indices of general skew tableaux

摘录 / EXCERPT: General skew-shape asymptotic normality conjecture. Asymptotic normality for general skew shapes, rather than only block diagonal skew shapes, holds if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-065637 | MATHDB-342633

The squared coefficient-of-variation conjecture for irreducible MMPPs

摘录 / EXCERPT: An MMPP is a Markovian modulated Poisson process; write c^2 for the squared coefficient of variation of its interarrival time. Squared coefficient-of-variation conjecture. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-065638 | MATHDB-342642

The Bessel-period obstruction to first occurrence

摘录 / EXCERPT: Bessel-period obstruction conjecture. The integer ℓ, with $0\leq\ell\leq {\mathfrak {r}}_n$, is not the first occurrence index of π.

conjecture | open | MathDB 页面 | 原文线索

V2-065639 | MATHDB-342645

Finiteness conjecture for Fuchsian equations with unitarizable monodromy

摘录 / EXCERPT: Let X be a compact Riemann surface, and fix prescribed singularities and exponents for Fuchsian equations on X. Finiteness conjecture. The set of such Fuchsian equations with u...

conjecture | open | MathDB 页面 | 原文线索

V2-065640 | MATHDB-342648

The height-function conjecture for bounded lecture hall tableaux

摘录 / EXCERPT: For a sequence of bounded lecture hall tableaux with limiting height function $h(x,y)=\lim h_n(x,y)/n$, let $u(x,y)$ be the solution of the Burgers equation introduced for the asso...

conjecture | open | MathDB 页面 | 原文线索

V2-065641 | MATHDB-342652

Alekseev-Mamedov conjecture on homogeneous braid monoids

摘录 / EXCERPT: For $n\geq 2$, let B_n be the braid group with standard generators $\sigma_1,\ldots,\sigma_{n-1}$. For $e_1,\ldots,e_{n-1}\in\{1,-1\}$, let $\mathcal{HBM}(e_1,\ldots,e_{n-1})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-065642 | MATHDB-342653

Homogeneous block count conjecture for braid words

摘录 / EXCERPT: Let w_1,w_2 be homogeneous braid words representing the same braid. For each word, partition the strand indices into its homogeneous blocks: indices $i,j\in\{1,2,\ldots,n-1\}$ li...

conjecture | open | MathDB 页面 | 原文线索

V2-065643 | MATHDB-342654

Conjecture on mixing time with a fixed number of random hubs

摘录 / EXCERPT: Consider a cycle of n vertices and choose K hubs randomly, where $K\in\mathbb{Z}_+$ and >0 are fixed. Equip the hubs with an appropriate interconnection structure, and consid...

conjecture | open | MathDB 页面 | 原文线索

V2-065644 | MATHDB-342656

Classification conjecture for Lagrangian fibrations by Jacobians

摘录 / EXCERPT: Let $\mathcal{C}\rightarrow\mathbb{P}^n$ be a family of reduced and irreducible curves of arithmetic genus n. Suppose that the relative compactified Jacobian ... is an irreducible h...

conjecture | open | MathDB 页面 | 原文线索

V2-065645 | MATHDB-342659

Nonlinear instability conjecture for traveling waves with decreasing myosin mass

摘录 / EXCERPT: Nonlinear instability conjecture. In the range of parameters such that the total myosin mass M of traveling wave solutions decreases as a function of V for small V, these sol...

conjecture | open | MathDB 页面 | 原文线索

V2-065646 | MATHDB-342670

Surface bracket state characterization of virtual closure images

摘录 / EXCERPT: Let K be a virtual knot of genus one, and let a torus representation be a representation of K by a knot diagram in the thickened torus. The surface bracket states of such a rep...

conjecture | open | MathDB 页面 | 原文线索

V2-065647 | MATHDB-342671

A colorability bound for the hat chromatic number

摘录 / EXCERPT: Let G be a graph. Write $\operatorname{col}(G)$ for its coloring number and let $\mu(G)$ denote its hat chromatic number. Coloring-number conjecture. There is a function F such...

conjecture | open | MathDB 页面 | 原文线索

V2-065648 | MATHDB-342672

The degree-sequence conjecture for hat guessing

摘录 / EXCERPT: Let G be a graph with degree sequence $(d_1,d_2,\dots,d_n)$, and let a sequence be winning for G when it gives the corresponding hat-guessing game a winning strategy. Degree-se...

conjecture | open | MathDB 页面 | 原文线索

V2-065649 | MATHDB-342673

The maximum-degree bound for the hat chromatic number

摘录 / EXCERPT: Let G be a graph with maximum degree Δ, and let $\mu(G)$ denote its hat chromatic number. Maximum-degree conjecture. Every graph G with maximum degree Δ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-065650 | MATHDB-342676

Fractional Calabi-Yau conjecture for the category of holomorphic triples on an elliptic curve

摘录 / EXCERPT: Fractional Calabi-Yau conjecture. The Serre functor satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065651 | MATHDB-342678

Arithmetic BSD invariant conjecture for Artin twists of elliptic curves

摘录 / EXCERPT: Let $E/{\mathbb Q}$ be an elliptic curve. For every Artin representation ρ over ${\mathbb Q}$, suppose there is an invariant $\BSD(E,\rho)\in{\mathbb C}^{\times}$. If ρ a...

conjecture | open | MathDB 页面 | 原文线索

V2-065652 | MATHDB-342703

Cichacz-Froncek conjecture on distance magic graphs

摘录 / EXCERPT: Cichacz-Froncek conjecture. If G is a distance magic graph, then G is group distance magic.

conjecture | open | MathDB 页面 | 原文线索

V2-065653 | MATHDB-342704

Conjecture on group distance magic labelings of Cartesian products of cycles

摘录 / EXCERPT: Conjecture on Cartesian products of cycles. The Cartesian product $C_m\square C_n$ admits a Γ-distance magic labeling if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-065654 | MATHDB-342722

Vanishing leading asymptotic coefficient for fixed-occurrence word enumeration

摘录 / EXCERPT: vanishing-coefficient conjecture. There exist $k,r\in\mathbb N$ and a pattern $v\in[k]^*$ such that $d>1$ and $K_r=0$. In that case, there exist $L_r\in\mathbb N$, $L_r<M_r$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-065655 | MATHDB-342723

Linear growth and repeated values of the automaton exponent M_r

摘录 / EXCERPT: Fix $n,k\in\mathbb N$ and a pattern $v\in[k]^*$ using $d>1$ distinct letters. Let M_r be the maximal number of states with $d-1$ loops in a simple path in the automaton...

conjecture | open | MathDB 页面 | 原文线索

V2-065656 | MATHDB-342724

Finiteness of the exponential generating limit for pattern-occurrence polynomials

摘录 / EXCERPT: For a pattern $\xi\in S_k$, let $c_n^\xi(x)$ denote the pattern-occurrence generating polynomial over S_n, so that its coefficients enumerate permutations according to the number...

conjecture | open | MathDB 页面 | 原文线索

V2-065657 | MATHDB-342725

Distinguished $\a0\a0\mathbb{P}^n$-structure with prescribed initial data

摘录 / EXCERPT: Let $F\colon D({\mathcal{A}})\to D({\mathcal{B}})$ be an enhanceable functor, and suppose that (H,Q_n,γ) is a $\mathbb{P}^n$-structure on F. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065658 | MATHDB-342726

Kusner's conjecture on the first Willmore min-max width

摘录 / EXCERPT: Let β_0 be the min-max width associated with a nontrivial loop in the space of immersions of the round sphere S^2 into $\mathbb{R}^3$, equivalently the infimum of the maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-065659 | MATHDB-342735

The characterization of abelian periods for the slope $[0;\overline{2}]$

摘录 / EXCERPT: Abelian-period conjecture. The abelian period set of a Sturmian word of slope α is

conjecture | open | MathDB 页面 | 原文线索

V2-065660 | MATHDB-342740

Palindromicity and real-rootedness conjecture for two-by-n-by-(n+1) grids

摘录 / EXCERPT: For every positive integer n, let $[r]$ denote a chain with r elements and let $\mathcal{N}_{[2]\times[n]\times[n+1]}(x)$ be the antichain generating polynomial of the product...

conjecture | open | MathDB 页面 | 原文线索

V2-065661 | MATHDB-342749

The GW/DT correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau threefold. Write $Z'_{\mathrm{GW}}(X;\lambda,Q)$ for the reduced Gromov-Witten partition function and $Z'_{\mathrm{DT}}(X;q,Q)$ for the re...

conjecture | open | MathDB 页面 | 原文线索

V2-065662 | MATHDB-342750

Generalized Grothendieck period conjecture for 1-motives

摘录 / EXCERPT: Let M be a 1-motive defined over a sub-field K of $\mathbb{C}$. Its periods are the coefficients of the comparison isomorphism between its De Rham and Hodge realizations, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-065663 | MATHDB-342762

Robin spectral-ratio conjecture for the ball

摘录 / EXCERPT: Ball spectral-ratio conjecture. For $\alpha\geq-1/R$ with $\alpha\neq0$, the ratio

conjecture | open | MathDB 页面 | 原文线索

V2-065664 | MATHDB-342763

Length-scaled Robin spectral-ratio conjecture for convex planar domains

摘录 / EXCERPT: Length-scaled spectral-ratio conjecture. For every $\alpha\geq-2\pi$, the quantity

conjecture | open | MathDB 页面 | 原文线索

V2-065665 | MATHDB-342767

The two-factor conjecture for iterated Q_c-transforms

摘录 / EXCERPT: Let f be an irreducible polynomial over $\mathbb F_q$ of degree $n\ge 3$, and let Q_c be a polynomial of degree D. Suppose that n is odd and $D\equiv 2\pmod 4$, and that th...

conjecture | open | MathDB 页面 | 原文线索

V2-065666 | MATHDB-342777

Bufetov's equivalence conjecture for determinantal point process kernels

摘录 / EXCERPT: Let X be the underlying space, let $\mathbb{F}$ be the scalar field, and let $K,Q:X^2\to\mathbb{F}$ be kernels. They are equivalent when, for every $r\geq 1$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-065667 | MATHDB-342794

The parity conjecture for elliptic-cover loci in genus two

摘录 / EXCERPT: Let $d>1$ and $n\geq 1$ be integers. Let $W_{d^2}[n]\subset\mathcal M_2$ be the locus of genus-two Riemann surfaces admitting a primitive degree-d elliptic cover with two critica...

conjecture | open | MathDB 页面 | 原文线索

V2-065668 | MATHDB-342795

The illumination conjecture for modular curves

摘录 / EXCERPT: Let $X(d)$ be the modular curve equipped with its square-tiling and the metric induced by the quadratic differential q. Say that a point $A\in X(d)$ illuminates a point...

conjecture | open | MathDB 页面 | 原文线索

V2-065669 | MATHDB-342800

The higher Morita conjecture for finite tensor categories

摘录 / EXCERPT: Let $\mathbf{Vec}$ denote the category of finite-dimensional vector spaces over a field $\mathbb K$. The object $\Sigma^2\mathbf{Vec}$ is the twofold delooping associated to vector...

conjecture | open | MathDB 页面 | 原文线索

V2-065670 | MATHDB-342802

Equality of the Borisov-Nuer and Enriques-image loci

摘录 / EXCERPT: Let $g=H_Y^2+1\geq 5$ be an odd integer, let $\mathcal F_g$ be the moduli space of polarized $K3$ surfaces of genus g, and let $\Sigma_g\subset\mathcal F_g$ be the naturally defi...

conjecture | open | MathDB 页面 | 原文线索

V2-065671 | MATHDB-342804

Kuroda's formula conjecture for the density of totally real quadratic class fields

摘录 / EXCERPT: Kuroda's density conjecture. For every $m\in\{0,1,\ldots,t-1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065672 | MATHDB-342806

Boundedness conjecture for semistable torsion-free sheaves on Grassmannians

摘录 / EXCERPT: Boundedness conjecture. The class $\mathfrak{F}^{ss}_G(r,c_1,c_2)$ forms a bounded family.

conjecture | open | MathDB 页面 | 原文线索

V2-065673 | MATHDB-342808

Polynomial formula conjecture for the fixed-degree coefficients $H_n[d]$

摘录 / EXCERPT: Let $H_n[d]$ be the fixed-degree coefficients considered in the generating series, and for each d define ... For each $d\geq 0$, there exists a polynomial P_d of degree d with...

conjecture | open | MathDB 页面 | 原文线索

V2-065674 | MATHDB-342809

Polynomial structure conjecture for Masur-Veech volumes

摘录 / EXCERPT: Let $MV_{g,n}$ denote the Masur-Veech volume, and set ... For each $g\geq 0$, consider polynomials $p_g,q_g\in\mathbb{Q}[n]$. Polynomial structure conjecture. For any $g\geq 0$, ther...

conjecture | open | MathDB 页面 | 原文线索

V2-065675 | MATHDB-342810

Ratios-model formula for the averaged one-level density of Hecke L-functions

摘录 / EXCERPT: Let $\mathcal{F}(K)=\{L_k(s):1\le k\le K\}$, let $M=\log K$, and let $R_K(\alpha,\gamma)$ be the ratios-conjecture prediction built from the Euler product $G(\alpha,\gamma)$. For s...

conjecture | open | MathDB 页面 | 原文线索

V2-065676 | MATHDB-342813

Poonen's separating intersection-closed sets conjecture

摘录 / EXCERPT: Poonen's conjecture. Either S is a power set or there is some $\alpha\in X_S$ with

conjecture | open | MathDB 页面 | 原文线索

V2-065677 | MATHDB-342814

Poonen's unique low-frequency element conjecture

摘录 / EXCERPT: Poonen's conjecture. If there is exactly one $\alpha\in X_S$ with

conjecture | open | MathDB 页面 | 原文线索

V2-065678 | MATHDB-342816

The maximal covering ideal conjecture for extended submonoids of \mathbb{F}_2^n

摘录 / EXCERPT: Maximal covering ideal conjecture. The set M has a maximal covering ideal with an associated submonoid that is properly contained in M.

conjecture | open | MathDB 页面 | 原文线索

V2-065679 | MATHDB-342817

The unique maximal covering ideal conjecture for extended submonoids

摘录 / EXCERPT: Unique maximal covering ideal conjecture. If M has a unique maximal covering ideal, then $\mathbb{F}_2$ is the only covered submonoid contained in M.

conjecture | open | MathDB 页面 | 原文线索

V2-065680 | MATHDB-342828

The Schwarz symmetrization conjecture for residue Monge-Ampère mass

摘录 / EXCERPT: Schwarz symmetrization conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065681 | MATHDB-342840

Weak containment and conjugate subgroups of the Bohr compactification

摘录 / EXCERPT: Let G be a finitely generated group. Let $G \curvearrowright^\pi \mathcal{H}$ be an orthogonal representation, and let $G \curvearrowright^{\pi'} \mathcal{H}'$ be an irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-065682 | MATHDB-342843

Reineke's stable-weight-system conjecture for Dynkin quivers

摘录 / EXCERPT: Let Q be a quiver of Dynkin type. A weight system Θ on Q is understood in the sense of stability for representations of Q, and a representation is stable with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-065683 | MATHDB-342844

Modified Reineke conjecture on maximal stable categories of Dynkin quivers

摘录 / EXCERPT: Let Q be a Dynkin quiver, and let k be the field over which its representations are taken. Write $\mathrm{\mathbf{Rep}}_k(Q)$ for the abelian category of finite-dimensional k...

conjecture | open | MathDB 页面 | 原文线索

V2-065684 | MATHDB-342857

Optimal anti-concentration conjecture for subgraph counts in random graphs

摘录 / EXCERPT: Fix $p\in(0,1)$ and a graph H with h non-isolated vertices. Let $G\in\mathbb{G}(n,p)$ be a binomial random graph, and let X_H denote the number of copies of H in G. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-065685 | MATHDB-342860

Berger-Coburn conjecture on bounded Toeplitz operators on Fock spaces

摘录 / EXCERPT: Let $t>0$ and let f be a symbol for which the Toeplitz operator T_f^t and its Berezin transform $\tilde{f}^{(t/2)}$ are defined on the t-Fock space. Berger-Coburn conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-065686 | MATHDB-342863

Conjecture on the limiting genetic composition of an exponentially growing cell population

摘录 / EXCERPT: Limiting composition conjecture. Suppose

conjecture | open | MathDB 页面 | 原文线索

V2-065687 | MATHDB-342867

Orlik-Randell conjecture on Seifert matrices of chain-type singularities

摘录 / EXCERPT: Let $f:({\mathbb C}^{n+1},0)\to({\mathbb C},0)$ be the weighted homogeneous polynomial ... Let $r_k=a_0a_1\cdots a_k$ for $k=0,1,\ldots,n$, let $r_{-1}=1$, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-065688 | MATHDB-342868

Il'yuta's monotone-cycle conjecture for Coxeter-Dynkin diagrams

摘录 / EXCERPT: Assume $n\equiv2\pmod 4$. Let D be a Coxeter-Dynkin diagram, with vertices given by a distinguished basis and edge weights $\langle\delta_i,\delta_j\rangle$. A monotone cycle is...

conjecture | open | MathDB 页面 | 原文线索

V2-065689 | MATHDB-342871

The relative hyperbolicity classification conjecture for hierarchical twist-free graphs of multicurves

摘录 / EXCERPT: Relative hyperbolicity classification conjecture. $d4a2$ is relatively hyperbolic if and only if this HHS structure has isolated orthogonality.

conjecture | open | MathDB 页面 | 原文线索

V2-065690 | MATHDB-342872

Rational strong Novikov conjecture

摘录 / EXCERPT: Strong Novikov conjecture. The map e induces an injection

conjecture | open | MathDB 页面 | 原文线索

V2-065691 | MATHDB-342876

Anticoncentration conjecture for subgraph counts in random graphs

摘录 / EXCERPT: Anticoncentration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065692 | MATHDB-342878

The strengthened point-modification conjecture for APN functions

摘录 / EXCERPT: Let $F:{\mathbb F}_{2^n}\to{\mathbb F}_{2^n}$ be an APN function. Fix $x_0\in{\mathbb F}_{2^n}$ and let F' be its (x_0,ϵ)-modification, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-065693 | MATHDB-342885

Asymptotic Stanley-depth conjecture for symbolic powers of cover ideals

摘录 / EXCERPT: Let G be a graph with n vertices, let $J(G)$ be its cover ideal, and let $J(G)^{(k)}$ denote its k-th symbolic power. Let $\nu_o(G)$ denote the parameter used in the source f...

conjecture | open | MathDB 页面 | 原文线索

V2-065694 | MATHDB-342899

The one-dimensional point-interaction ground-state optimization conjecture

摘录 / EXCERPT: Ground-state optimization conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-065695 | MATHDB-342906

Sharp bounds for successive minimum spanning tree weights

摘录 / EXCERPT: For each $k\geqslant1$, let γ_k be the limiting constant for the weight of the kth successive minimum spanning tree. Sharp weight bounds conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-065696 | MATHDB-342907

Asymptotic affine form of successive giant-component thresholds

摘录 / EXCERPT: For each $k\geqslant1$, let σ_k be the threshold at which the limiting giant-component function $\rho_k(t)$ becomes positive. Threshold asymptotics conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-065697 | MATHDB-342914

Classification of isomorphisms among k-signaletic operads

摘录 / EXCERPT: For any $k\geq 1$, let $\operatorname{op}(l)$, $\odot$, and $\operatorname{op}(r)$ be the three symbols used to form constraints in the tidy parallel k-signaletic operads. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-065698 | MATHDB-342922

Exponential synchronization conjecture for circular automata of composite order

摘录 / EXCERPT: Let $\mathcal{A}_n(\mathbf{a},\mathbf{b})$ be a circular automaton of order n, where $\mathbf{a}:\mathbb{Z}_n\to\mathbb{Z}_n$ is fixed and $\mathbf{b}\in\mathcal{M}_n$ is chosen...

conjecture | open | MathDB 页面 | 原文线索

V2-065699 | MATHDB-342928

Monotone coupling conjecture for random regular graphs

摘录 / EXCERPT: Let n be the number of vertices, and let $0\le d_1\le d_2\le n-1$ be integers such that d_1n and d_2n are even. Write ${\mathcal G}(n,d_i)$ for the uniformly random d_i-reg...

conjecture | open | MathDB 页面 | 原文线索

V2-065700 | MATHDB-342929

The interval conjecture for second-level universal lower bounds

摘录 / EXCERPT: Interval conjecture. For every dimension n, the values $N\in [D(n,\tau),D(n,\tau+1)]$ for which second-level bounds exist form an interval.

conjecture | open | MathDB 页面 | 原文线索

V2-065701 | MATHDB-342931

Positivity conjecture for the massless time-like XX-chain correlation

摘录 / EXCERPT: Consider the massless time-like regime of the XX chain and the complex correlation function $(-1)^m\langle\sigma^-_1(0)\sigma^+_{m+1}(t)\rangle$. Positivity conjecture. Both its re...

conjecture | open | MathDB 页面 | 原文线索

V2-065702 | MATHDB-342936

Unimodality conjecture for the Hilbert series of $R_n({\mathbf a})$

摘录 / EXCERPT: Unimodality conjecture. The matrix of coefficients of ${\mathrm {Hilb}}(R_n({\mathbf a});q,z)$ has unimodal rows and columns. This would extend the observed symmetry and unimodalit...

conjecture | open | MathDB 页面 | 原文线索

V2-065703 | MATHDB-342959

Gamma-positivity conjecture for locally anti-blocking reflexive polytopes

摘录 / EXCERPT: Let ${\mathcal P}\subset {\mathbb R}^d$ be a locally anti-blocking reflexive polytope, and let $h^*({\mathcal P},x)$ denote its h^*-polynomial. Locally anti-blocking reflexive po...

conjecture | open | MathDB 页面 | 原文线索

V2-065704 | MATHDB-342960

Gamma-positivity conjecture for symmetric edge polytopes of type A

摘录 / EXCERPT: Let ${\mathcal A}_G$ be a symmetric edge polytope of type A, and let $h^*({\mathcal A}_G,x)$ denote its h^*-polynomial. Type A symmetric edge-polytope conjecture. The h^*-polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-065705 | MATHDB-342966

Dimension conjecture for dual cells

摘录 / EXCERPT: Dimension conjecture. For every dual k-cell, the dimension of its affine span is equal to k.

conjecture | open | MathDB 页面 | 原文线索

V2-065706 | MATHDB-342968

Shilov-boundary accumulation conjecture for stretching rays

摘录 / EXCERPT: Let $Y_d(r)$ be the object introduced in the source, let d denote the degree, and let the connected locus be the relevant connectedness locus. Shilov-boundary conjecture. All str...

conjecture | open | MathDB 页面 | 原文线索

V2-065707 | MATHDB-342969

Green's conjecture for canonical curves

摘录 / EXCERPT: Let $X\subseteq \mathbf{P}^{g-1}$ be a non-hyperelliptic canonical curve of genus g. Its Clifford index is ... Here $r(\varphi)$ is the dimension of the linear span of the image an...

conjecture | open | MathDB 页面 | 原文线索

V2-065708 | MATHDB-342970

Folk conjecture for tangent developable surfaces

摘录 / EXCERPT: Folk conjecture. The tangent developable surface T satisfies Property (N_p) for

conjecture | open | MathDB 页面 | 原文线索

V2-065709 | MATHDB-342972

Existence of weak signed edge-labelings in 3-connected planar graphs

摘录 / EXCERPT: Let H be a 3-connected planar graph with an even number of negative vertices. A $\{0,a,b\}$-edge-labeling c is a weak signed edge-labeling if the parity conditions on...

conjecture | open | MathDB 页面 | 原文线索

V2-065710 | MATHDB-342973

Existence of strong signed edge-labelings in 3-connected planar graphs

摘录 / EXCERPT: Let H be a 3-connected planar graph with an even number of negative vertices. A $\{0,a,b\}$-edge-labeling c is a strong signed edge-labeling if it is a weak signed edge-labelin...

conjecture | open | MathDB 页面 | 原文线索

V2-065711 | MATHDB-342976

Hliněný-Kwon-Obdržálek-Ordyniak conjecture on rank-depth

摘录 / EXCERPT: Hliněný-Kwon-Obdržálek-Ordyniak conjecture. The class $\mathcal{C}$ has bounded rank-depth if and only if there exists an integer t such that no graph $G\in\mathcal{C}$ contains...

conjecture | open | MathDB 页面 | 原文线索

V2-065712 | MATHDB-342978

McMullen's g-theorem for simplicial spheres

摘录 / EXCERPT: Let Σ be a simplicial sphere of dimension $d-1$, with f-vector $(f_{-1},f_0,\ldots,f_{d-1})$ and associated g-vector defined by $g_k(\Sigma)=h_k-h_{k-1}$ for $k\leq d/2$...

conjecture | open | MathDB 页面 | 原文线索

V2-065713 | MATHDB-342982

Width-two conjecture for the first missing linear extension number

摘录 / EXCERPT: For each positive integer n, let $M_2(n)$ be the smallest positive integer that is not the linear extension number of a width-2 poset. Width-two conjecture. ... The motivation is...

conjecture | open | MathDB 页面 | 原文线索

V2-065714 | MATHDB-342995

The no-common-part conjecture for distributed source simulation

摘录 / EXCERPT: Let U and V be source variables, and let X and Y be target variables. Write $C_{\mathsf{GK}}(U;V)$ for the Gács-Körner common information of U and V. A target distribut...

conjecture | open | MathDB 页面 | 原文线索

V2-065715 | MATHDB-342996

Hyperbolicity and infinite diameter conjecture for the complex of parabolic subgroups

摘录 / EXCERPT: Let A_S be an irreducible non-cyclic FC type Artin group, and let ${\mathcal{CP}_{A_S}}$ denote its complex of parabolic subgroups. Hyperbolicity and infinite-diameter conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-065716 | MATHDB-343001

Evans-Gannon-type near-group modular-data conjecture for arbitrary abelian groups

摘录 / EXCERPT: Let A be a finite abelian group, let $G=A\times A$, and let $\langle\langle\cdot,\cdot\rangle\rangle$ be a non-degenerate symmetric bicharacter on A. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-065717 | MATHDB-343002

Near-group realization conjecture for the inverse-involution family

摘录 / EXCERPT: Let A be an odd finite abelian group and let $G=\mathbb{Z}_{2^n}\times A$ with $n\geq1$. Inverse-involution near-group conjecture. The constructed modular data are the modular da...

conjecture | open | MathDB 页面 | 原文线索

V2-065718 | MATHDB-343003

Quadratic-category modular data conjecture for odd abelian groups

摘录 / EXCERPT: Let A be a finite abelian group of odd order, and let $\mathcal{C}$ be a quadratic category of type $(\mathbb{Z}_2,A,1)$ whose $\operatorname{Vec}_{\mathbb{Z}_2}$ subcategory has...

conjecture | open | MathDB 页面 | 原文线索

V2-065719 | MATHDB-343004

De-equivariantized generalized Haagerup modular data for cyclic even parts

摘录 / EXCERPT: Let A be an abelian group with $A_e\cong\mathbb{Z}_{2^{n+1}}$. Cyclic-even-part de-equivariantization conjecture. The modular data of the Drinfeld center of the $\mathbb{Z}_2$-de...

conjecture | open | MathDB 页面 | 原文线索

V2-065720 | MATHDB-343005

De-equivariantized generalized Haagerup modular data for mixed even parts

摘录 / EXCERPT: Let A be an abelian group with $A_e\cong\mathbb{Z}_{2^n}\times\mathbb{Z}_2$ for $n\geq2$. Mixed-even-part conjecture. The modular data of the Drinfeld center of the...

conjecture | open | MathDB 页面 | 原文线索

V2-065721 | MATHDB-343007

Descent conjecture for the first steps of the Shimura curve tower at 3

摘录 / EXCERPT: Let $X_0(\mathfrak{p}_3)$ and $X_0(\mathfrak{p}_3^2)$ be the indicated Shimura curves with models over $\mathbb{F}_{2^3}$. Descent conjecture. The canonical branched cover ... and th...

conjecture | open | MathDB 页面 | 原文线索

V2-065722 | MATHDB-343016

Kontsevich-Soibelman metatree conjecture for the Fulton-MacPherson operad

摘录 / EXCERPT: Let $FM'_k=F_k({\mathbb C})/({\mathbb C}\rtimes{\mathbb R}^+)$ be the moduli space of configurations of k points in the complex plane modulo translations and positive rescalings,...

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V2-065723 | MATHDB-343020

The curve analogue of Nekrasov's conjecture for the resolved conifold

摘录 / EXCERPT: Let $X=\operatorname{Tot}_{\mathbb P^1}(\mathcal O(-1)\oplus\mathcal O(-1)\oplus\mathcal O)$, and let $\mathcal Z_X(y,q,Q)=\sum_{n,d}P_{n,d[\mathbb P^1]}(\mathcal O,y)q^nQ^d$ be it...

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V2-065724 | MATHDB-343022

Turán's distance-two conjecture for irreducible integer polynomials

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ have degree $n\geq 1$. Define $L(f-g)=\sum_{j=0}^{n}|b_j-a_j|$ when $f(x)=\sum_{j=0}^{n}a_jx^j$ and $g(x)=\sum_{j=0}^{n}b_jx^j$. Turán's distance-two con...

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V2-065725 | MATHDB-343023

Dubickas-Sha squarefree distance-two conjecture for integer polynomials

摘录 / EXCERPT: Let $f(x)\in\mathbb{Z}[x]$ have degree $n\geq 0$. A polynomial is squarefree if it is not divisible by the square of an irreducible polynomial over $\mathbb{Q}$. Define...

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V2-065726 | MATHDB-343024

The planar-poset standard-example conjecture

摘录 / EXCERPT: Let P be a planar poset, let $d\ge2$, and let S_d denote the standard example of order d. The dimension of P is denoted by $\dim(P)$. The planar-poset standard-example conj...

conjecture | open | MathDB 页面 | 原文线索

V2-065727 | MATHDB-343025

The standard-example or clique-minor conjecture for posets

摘录 / EXCERPT: Let P be a poset, let $\dim(P)$ denote its dimension, let S_d be the standard example of order d, and let the cover graph of P be the graph whose vertices are the elements...

conjecture | open | MathDB 页面 | 原文线索

V2-065728 | MATHDB-343026

The bounded Boolean-dimension conjecture for planar posets

摘录 / EXCERPT: Let P be a planar poset, and let the Boolean dimension of P denote its Boolean dimension. The bounded Boolean-dimension conjecture. There is a constant d_0 such that if P i...

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V2-065729 | MATHDB-343031

Baruchel's conjecture on the zeroth digit in base-phi representations

摘录 / EXCERPT: Baruchel's conjecture. The digit $d_0(N)=1$ if and only if

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V2-065730 | MATHDB-343041

Generalized Nagao conjecture for hyperelliptic Jacobians

摘录 / EXCERPT: Generalized Nagao conjecture.

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V2-065731 | MATHDB-343049

The Lorentzian conjecture for normalized Schur P-polynomials

摘录 / EXCERPT: Let λ be a strict partition, meaning a decreasing sequence of positive integers, and let $P_\lambda(x_1,\ldots,x_m)$ be its Schur P-polynomial. Schur-P conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-065732 | MATHDB-343050

The Lorentzian conjecture for normalized homogeneous Grothendieck polynomials

摘录 / EXCERPT: Homogeneous-Grothendieck conjecture. The normalized polynomial

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V2-065733 | MATHDB-343051

The Lorentzian conjecture for normalized key polynomials

摘录 / EXCERPT: Let $\alpha\in\mathbb{N}^n$ be a composition, and let $\kappa_\alpha(x_1,\ldots,x_n)$ be its key polynomial. Key-polynomial conjecture. The normalized polynomial ... is Lorentzian fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065734 | MATHDB-343052

The Lorentzian conjecture for normalized skew Schur polynomials in partition notation

摘录 / EXCERPT: Let $\lambda,\mu\in\mathbb{Z}_{\geq0}^n$ be partitions satisfying $\lambda_i\geq\mu_i$ for every i, and let $s_{\lambda/\mu}$ be the skew Schur polynomial. Skew-Schur conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-065735 | MATHDB-343053

The Lorentzian conjecture for normalized Schur P-functions

摘录 / EXCERPT: Let λ be a partition whose nonzero parts are distinct, and let P_λ be its Schur P-function. Schur-P conjecture. The normalized function ... is Lorentzian for ever...

conjecture | open | MathDB 页面 | 原文线索

V2-065736 | MATHDB-343063

Weak modularity of the 1-skeleta of A_n buildings

摘录 / EXCERPT: Let a building of type $\widetilde{A}_n$ be a simplicial complex built from Coxeter complexes of type $\widetilde{A}_n$, and let its 1-skeleton be the graph whose vertices and edge...

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V2-065737 | MATHDB-343064

Odd-path local antimagic total chromatic number conjecture

摘录 / EXCERPT: Let P_n be the path of order $n\geq 2$, and let $\chi_{lat}(P_n)$ denote its local antimagic total chromatic number. Odd-path conjecture. For odd $n\geq 3$, ... The source verifies...

conjecture | open | MathDB 页面 | 原文线索

V2-065738 | MATHDB-343070

Non-generation conjecture for higher-dimensional polar Grassmannians

摘录 / EXCERPT: Let $\mathbb{F}_0$ be a proper subfield of a field $\mathbb{F}$, and let $Q^+_{n-1}(2n-1,\mathbb{F})$ denote the polar Grassmannian of maximal singular subspaces of the hyperbolic...

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V2-065739 | MATHDB-343111

Haglund-Remmel-Wilson Delta Conjecture

摘录 / EXCERPT: Let $X=\{x_1,x_2,\ldots,x_n\}$ and $Y=\{y_1,y_2,\ldots,y_n\}$ be sets of commuting variables, and let $\mathcal{WPF}_n$ denote the set of word parking functions of size n. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-065740 | MATHDB-343112

Rise version of the Extended Delta Conjecture

摘录 / EXCERPT: Let $\mathcal{WPF}_{n;r}$ denote the set of extended word parking functions with blank valleys having parameters n and r. For such an object PF, use the statistics...

conjecture | open | MathDB 页面 | 原文线索

V2-065741 | MATHDB-343116

The standard-example conjecture for posets with planar cover graphs

摘录 / EXCERPT: Let P be a poset, let its dimension be denoted by $\dim(P)$, and let S_n denote the standard example of size n. A cover graph of P is planar when it can be embedded in the...

conjecture | open | MathDB 页面 | 原文线索

V2-065742 | MATHDB-343121

The coefficient formula for Leibniz graphs in the factorizing operator

摘录 / EXCERPT: Let c_n be the coefficients in the factorizing operator ... where w_Γ is the weight of a Leibniz graph Γ and $\#!\operatorname{Aut}(\Gamma)$ denotes the order of its...

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V2-065743 | MATHDB-343122

Lin's conjecture on a cubic partition-pair congruence modulo 81

摘录 / EXCERPT: Let $b(n)$ denote the cubic partition-pair function defined by ... where $f_k=(q^k;q^k)_\infty$. Lin's conjecture. For each $n\geq 0$, ... This conjecture strengthens the known congrue...

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V2-065744 | MATHDB-343130

The linked-position conjecture for Gröbner bases of finite point sets

摘录 / EXCERPT: Linked-position conjecture. If q is in a linked position with respect to S and the convex hull of the points in T has no holes, then

conjecture | open | MathDB 页面 | 原文线索

V2-065745 | MATHDB-343135

The infinite rational curves conjecture for projective K3 surfaces

摘录 / EXCERPT: Infinite rational curves conjecture. X contains infinitely many rational curves.

conjecture | open | MathDB 页面 | 原文线索

V2-065746 | MATHDB-343138

Convergence of multidimensional scaling embeddings under Gromov-Wasserstein convergence

摘录 / EXCERPT: Let (X_n,d_n,μ_n) for $n\in\mathbb{N}$ be a sequence of metric measure spaces that converges to (X,d,μ) in the Gromov-Wasserstein distance. The associated multidimensional...

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V2-065747 | MATHDB-343140

Kernel nondegeneracy conjecture for periodic fractional Korteweg-de Vries waves

摘录 / EXCERPT: Let $\psi_0 \in H^{\alpha}_{\rm per}(\mathbb{T})$ be the solution to the boundary-value problem with $c=c_0$ obtained from the main existence theorem, and let X_0 and...

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V2-065748 | MATHDB-343154

Propagation of half-derivative regularity for the Boltzmann equation

摘录 / EXCERPT: Regularity propagation conjecture. The local solution $f(t)$ carries the $\frac{1}{2}+$ regularity of the initial data up to time T_0; in particular, for every $T\in(0,T_0)$,

conjecture | open | MathDB 页面 | 原文线索

V2-065749 | MATHDB-343166

Difference-set graceful-labeling conjecture for regular connected graphs

摘录 / EXCERPT: Let D be a (v,k,μ) difference set in a group G, and define ... for $1\leq i\leq\gcd(k-1,\mu)$. A graph is D-graceful when its vertices can be labeled by the elements of D...

conjecture | open | MathDB 页面 | 原文线索

V2-065750 | MATHDB-343182

Product decomposition conjecture for characteristic functions of multiplicity-free cnu contractive associators

摘录 / EXCERPT: Product decomposition conjecture. The characteristic function of any multiplicity-free cnu contractive associator in $B_n^*(\mathbb D)$ is the pointwise product of the characterist...

conjecture | open | MathDB 页面 | 原文线索

V2-065751 | MATHDB-343190

Conjecture on the radius of convergence of Fox-Wright expansions

摘录 / EXCERPT: Let ${}_{p}\Psi_q$ denote the Fox-Wright function, and consider the series in equations,, and, each expanded about the relevant point in the variable $1-z$. Radius-of-convergence c...

conjecture | open | MathDB 页面 | 原文线索

V2-065752 | MATHDB-343195

The stable character formula for supercuspidal Langlands parameters

摘录 / EXCERPT: Let F be a non-archimedean local field, let G be a reductive group that splits over a tame extension of F, and assume that the residue characteristic is sufficiently large fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065753 | MATHDB-343198

Genus-two Hankel-determinant sigma-function conjecture

摘录 / EXCERPT: Let Δ_n be the Hankel determinants whose moments are defined by the recurrence in the source. Let $\tilde{\mathcal C}$ be the quintic curve ... which is isomorphic to the rele...

conjecture | open | MathDB 页面 | 原文线索

V2-065754 | MATHDB-343206

Conjectural extension of Bass' theorem for locally presentable abelian categories

摘录 / EXCERPT: Let $\mathsf B$ be a locally presentable abelian category with a projective generator P. Let $\mathsf B_{\operatorname{proj}}$ denote the class of projective objects in...

conjecture | open | MathDB 页面 | 原文线索

V2-065755 | MATHDB-343207

The distributive-law conjecture for Hopf operads generated by primitive nilpotent elements

摘录 / EXCERPT: Let $\mathcal{P}$ be a Hopf operad generated by primitive nilpotent elements, and let $\mathcal{Q}$ be the suboperad they generate, assumed to be Koszul. Write Comm for...

conjecture | open | MathDB 页面 | 原文线索

V2-065756 | MATHDB-343215

Metric thickening sphere-spectrum conjecture for the circle

摘录 / EXCERPT: Let S^1 be the unit circle, let $r>0$, and let $\mathrm{VR}^m(S^1;r)$ denote its Vietoris-Rips metric thickening. For each integer $k\geq 1$, let $\mathcal{B}_{2k}$ be the Barvin...

conjecture | open | MathDB 页面 | 原文线索

V2-065757 | MATHDB-343216

Diameter-preservation conjecture for the Vietoris-Rips thickening retraction

摘录 / EXCERPT: Let $k\geq 1$, let r satisfy ... let $\mu\in\mathrm{VR}^m(S^1;r)$, and let ι map the boundary of the Barvinok-Novik orbitope $\mathcal{B}_{2k}$ into the Vietoris-Rips metric...

conjecture | open | MathDB 页面 | 原文线索

V2-065758 | MATHDB-343217

The conjecture on intersection theory for simplicial-complex-stable curves

摘录 / EXCERPT: Let Γ be a finite simplicial complex. A Γ-stable curve is a nodal marked curve satisfying the stability conditions determined by Γ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065759 | MATHDB-343222

The planar graph positive-valued cover degeneracy conjecture

摘录 / EXCERPT: Let G be a planar graph and let (H,f) be a positive-valued cover, meaning that the cover function f is positive-valued. Let $\kappa\geq 2$. A strictly f-degenerate transver...

conjecture | open | MathDB 页面 | 原文线索

V2-065760 | MATHDB-343224

Small-polarization stability conjecture for exceptional bundles on Hirzebruch surfaces

摘录 / EXCERPT: Let $e\geq 2$ and let $\mathcal{V}$ be an exceptional bundle on the Hirzebruch surface $\mathbb{F}_e$. Small-polarization stability conjecture. The bundle $\mathcal{V}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-065761 | MATHDB-343228

Planarity conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Planarity conjecture. If $\partial_B(G,\mathcal{P})$ is planar and has no cut points, then G is virtually isomorphic to a Kleinian group.

conjecture | open | MathDB 页面 | 原文线索

V2-065762 | MATHDB-343229

Dual Grothendieck positivity classification conjecture

摘录 / EXCERPT: Let Λ be the ring of symmetric functions, let $g_{\lambda/\mu}$ be skew dual Grothendieck polynomials, and let a homomorphism $\rho:\Lambda\to\mathbb R$ be g-positive whe...

conjecture | open | MathDB 页面 | 原文线索

V2-065763 | MATHDB-343244

Optimal strong approximation conjecture for the four-square form

摘录 / EXCERPT: Let $N,m,\lambda_1,\dots,\lambda_4\in\mathbb{Z}$ satisfy ... Four-square strong approximation conjecture. If $N\gg m^{4+\varepsilon}$, then there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-065764 | MATHDB-343255

Interior L^p estimate conjecture for fractional-Laplacian eigenfunctions

摘录 / EXCERPT: Interior L^p estimate conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-065765 | MATHDB-343265

Sharp linear critical-point bound for quadrature domains

摘录 / EXCERPT: Quadrature-domain critical-point conjecture. The number of critical points satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065766 | MATHDB-343266

Circular-domain critical-point conjecture

摘录 / EXCERPT: Circular-domain critical-point conjecture. For $k\geq 3$,

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V2-065767 | MATHDB-343267

Magnanini's critical-point conjecture for the landscape function

摘录 / EXCERPT: Magnanini's conjecture. The number of critical points satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-065768 | MATHDB-343269

Outerplanar graph polygon-chain determination conjecture

摘录 / EXCERPT: Let G be a bi-connected outerplanar graph. Let G^* be its inner dual, and let $P_1\cup\cdots\cup P_t$ be the polygon chain decomposition of G corresponding to the maximal pat...

conjecture | open | MathDB 页面 | 原文线索

V2-065769 | MATHDB-343287

The gradedness conjecture for Q-Bruhat posets

摘录 / EXCERPT: Let $Q i\operatorname{POS}(n)$ and $k\in[n]$. For each $I\in Q^k_B(\mathbb{C})$, let $[Q]_I(\mathbb{C})$ denote the associated incidence stratum, and let $\dim([Q]_I(\mathbb{C}))$...

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V2-065770 | MATHDB-343289

Categorical DT wall-crossing conjecture for MNOP and stable-pair theories

摘录 / EXCERPT: Fix (β,n) and choose stability parameters $t_1>t_2>\cdots>t_N>0$ away from the finitely many walls. Let $I_n(X,\beta)$ be the MNOP moduli space, $P_n(X,\beta)$ the stable-pai...

conjecture | open | MathDB 页面 | 原文线索

V2-065771 | MATHDB-343291

Categorical DT MNOP/PT comparison conjecture

摘录 / EXCERPT: Let $I_n(X,\beta)$ and $P_n(X,\beta)$ be the MNOP and stable-pair moduli spaces, respectively, and let $\mathcal{DT}^{\mathbb{C}^{\ast}}(-)$ denote their $\mathbb{C}^{\ast}$-equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-065772 | MATHDB-343292

Categorical DT wall-crossing conjecture for stable-pair moduli

摘录 / EXCERPT: Fix $\sigma=iH$ for an ample divisor H, $v=(\beta,n)\in N_{\leq 1}(S)$, and a positive rational wall parameter $t_0\in W\cap\mathbb{Q}_{>0}$. For $t_{\pm}=t_0\pm\varepsilon$ with...

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V2-065773 | MATHDB-343293

Categorical DT flop invariance at the zero wall

摘录 / EXCERPT: Let $P_n(X,\beta)_{\varepsilon}$ and $P_n(X,\beta)_{-\varepsilon}$ be the stable-pair moduli spaces for parameters on the two sides of the wall at $t_0=0$, with $0<\varepsilon\ll1$...

conjecture | open | MathDB 页面 | 原文线索

V2-065774 | MATHDB-343296

Trace-zero form conjecture for totally real octic fields

摘录 / EXCERPT: Trace-zero form conjecture for totally real octic fields. A totally real octic number field with fundamental discriminant is determined, among tamely ramified number fields, by the...

conjecture | open | MathDB 页面 | 原文线索

V2-065775 | MATHDB-343301

Transpose-and-rowmotion cyclic sieving conjecture for even-height plane partitions

摘录 / EXCERPT: Transpose-and-rowmotion cyclic sieving conjecture. For every $k\in\mathbb Z$,

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V2-065776 | MATHDB-343305

Asymptotic equivalence conjecture for powerful 132-avoidance

摘录 / EXCERPT: Powerful 132-avoidance asymptotic conjecture. As n tends to infinity,

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V2-065777 | MATHDB-343311

Componentwise p-positivity conjecture for ribbon Schur Q-functions

摘录 / EXCERPT: Let $D=D_1\cup\cdots\cup D_r$ be a ribbon with connected components D_i, for $i=1,\dots,r$, and let $\mathfrak{r}_D$ denote its ribbon Schur Q-function. Componentwise p-positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-065778 | MATHDB-343318

Nadler-LPS conjecture for the dual brane of a real Higgs-bundle brane

摘录 / EXCERPT: Nadler-LPS conjecture. The support of the dual brane to the fixed-point set of i_1 is the moduli space

conjecture | open | MathDB 页面 | 原文线索

V2-065779 | MATHDB-343329

DeWitt's generalized identity for shifted stable Grothendieck polynomials

摘录 / EXCERPT: Let $m,k\in\mathbb{Z}_{>0}$, let $\mu=(m^k)$, and let $\nu=(m+k-1,m+k-3,\dots,|m-k|+1)$. For partitions λ and μ with $\mu\subseteq\lambda$, let...

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V2-065780 | MATHDB-343330

Finite vertical-strip expansion formula for shifted stable Grothendieck polynomials

摘录 / EXCERPT: Let μ be a strict partition. For strict partitions $\lambda\supseteq\mu$ with $\ell(\lambda)=\ell(\mu)$, let $\textsf{SD}_{\lambda/\mu}$ be the shifted skew diagram. A vertical...

conjecture | open | MathDB 页面 | 原文线索

V2-065781 | MATHDB-343332

Minimum-weight MIPPR word conjecture for minimal PPRIC codes

摘录 / EXCERPT: Let r, s, and L be nonnegative integers with $L\geqslant 2s+r+1$, and let $C\subseteq {\cal W}_s$ be a minimal (L,s,r) PPRIC code. An MIPPR word is a word associated with...

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V2-065782 | MATHDB-343334

Chen-Gonzales-Goodman-Kang-Okoudjou conjecture on minimizers of the p-frame potential

摘录 / EXCERPT: Chen-Gonzales-Goodman-Kang-Okoudjou's conjecture. If $p\in(p_{k-1},p_k]$ for $k=1,2,\ldots,d$, then L_k^d minimizes the p-frame potential when $N=d+1$.

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V2-065783 | MATHDB-343337

Charpentier's coloring conjecture for squares of graphs with maximum average degree below 4

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ denote its maximum degree, let $\operatorname{mad}(G)$ denote its maximum average degree, and let G^2 be its square, with chromatic number...

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V2-065784 | MATHDB-343354

Nearly valid higher-degree pseudomoment extension of the Montanari-Sen witness

摘录 / EXCERPT: Let $\delta \in (0,1)$, and let $\bm{M}^{(\delta)}$ be the Montanari-Sen witness. Fix $d \in 2\mathbb{N}$, and choose constants σ_k^2 depending only on δ. Let...

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V2-065785 | MATHDB-343355

Billera-Brenti's strong conjecture on complete cd-indices of Bruhat intervals

摘录 / EXCERPT: Billera-Brenti's strong conjecture. One has the coefficientwise inequality

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V2-065786 | MATHDB-343385

de Weger's conjecture on large Tate-Shafarevich groups over function fields

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$, and for an elliptic curve E/K let $H(E)$ denote its exponential differential height, $N(E)$ its numerical conductor, and $\@math@cyr{\cyracc SH}(E)$ its...

conjecture | open | MathDB 页面 | 原文线索

V2-065787 | MATHDB-343393

Freeness conjecture for generic twisted quasi-invariants

摘录 / EXCERPT: Let m be a nonnegative integer, let $f_1,f_2,\dots,f_n$ be one-variable meromorphic functions, and let $Q_m(f_1,\dots,f_n)$ be the space of polynomials...

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V2-065788 | MATHDB-343407

Coherence-band gradient similarity conjecture for one-bit millimetre-wave channel estimation

摘录 / EXCERPT: Coherence-band gradient similarity conjecture. For sufficiently large η, if $j\in B_{\eta}(i)$ and $x_i=x_j$, then

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V2-065789 | MATHDB-343408

Blidia-Maffray-Zemir conjecture on the b-chromatic number of regular graphs

摘录 / EXCERPT: Blidia-Maffray-Zemir conjecture. If G is not the Petersen graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-065790 | MATHDB-343414

Compactified Gaussian free field conjecture for Temperleyan forest height forms

摘录 / EXCERPT: Compactified Gaussian free field conjecture. The limiting height form is given by the compactified Gaussian Free Field in the appropriately punctured surface.

conjecture | open | MathDB 页面 | 原文线索

V2-065791 | MATHDB-343416

Universal optimality of the 85-point code in complex dimension five

摘录 / EXCERPT: Let $\mathcal C$ be the code constructed in the paper consisting of 85 points in $\mathbb C^5$, viewed projectively. A code is universally optimal if it minimizes every admissibl...

conjecture | open | MathDB 页面 | 原文线索

V2-065792 | MATHDB-343422

Fu's Equitable Total Coloring Conjecture

摘录 / EXCERPT: Let G be a graph and let $T(G)$ be its total graph, whose vertices represent the vertices and edges of G. An equitable k-coloring is a proper k-coloring whose color classes...

conjecture | open | MathDB 页面 | 原文线索

V2-065793 | MATHDB-343430

Conjecture on cubic-root values of the quadratic-residue product $S_p(x)$

摘录 / EXCERPT: The cubic-root product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065794 | MATHDB-343442

The harmonic-oscillator lower-bound and equality conjecture for maximal multiplicities

摘录 / EXCERPT: For $N\in\mathbb N^*$, let $\mathfrak A_N^{har}$ be the isotropic harmonic-oscillator array, let λ_k^{har} denote its k-th eigenvalue, and let $m(k,\mathfrak A)$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-065795 | MATHDB-343443

The sparse square-function criterion for Calderón-Zygmund operators

摘录 / EXCERPT: Sparse square-function criterion. Suppose that for every $0<\eta<1$ and every η-sparse family $\mathcal{S}$ of cubes in $\mathbb{R}^d$,

conjecture | open | MathDB 页面 | 原文线索

V2-065796 | MATHDB-343447

Higher-order Compton scattering smoothing conjecture

摘录 / EXCERPT: Higher-order Compton scattering smoothing conjecture. For each scattering order i, g_i is approximated by a Fourier integral operator $\mathcal{L}_i(\cdot,n_e)$ of order

conjecture | open | MathDB 页面 | 原文线索

V2-065797 | MATHDB-343458

Convexity conjecture for the Gamma-type non-negativity domains

摘录 / EXCERPT: Convexity conjecture. For every $a,b>0$, the set ${\mathcal{D}}_{a,b}$ is convex.

open_problem | open | MathDB 页面 | 原文线索

V2-065798 | MATHDB-343460

Isett-Oh optimal energy regularity conjecture for Hölder Euler solutions

摘录 / EXCERPT: Isett-Oh conjecture. For every $\theta<\frac{1}{3}$, there exists a weak Euler solution $v\in C^\theta(\mathbb R\times\mathbb T^n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-065799 | MATHDB-343464

Quadratic-order conjecture for 1-subdivisions of transitive tournaments

摘录 / EXCERPT: Quadratic-order conjecture. There is a constant $C>0$ such that any tournament with at least Ck^2 vertices contains a 1-subdivision of T_k.

conjecture | open | MathDB 页面 | 原文线索

V2-065800 | MATHDB-343476

The Riesz-kernel conjecture for complete hyperbolic polynomials

摘录 / EXCERPT: Let $p\in \mathbb{R}[x_1,\dots,x_n]$ be a complete hyperbolic polynomial with hyperbolicity cone C. For a real $\alpha>n$, let q_{α} be the function defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-065801 | MATHDB-343477

Scott-Sokal conjecture for elementary symmetric polynomials

摘录 / EXCERPT: For integers m,n with $2\leq m\leq n$, let $E_{m,n}(x)=\sum_{1\leq i_1<\dots<i_m\leq n}x_{i_1}\cdots x_{i_m}$ be the mth elementary symmetric polynomial, and consider its negat...

conjecture | open | MathDB 页面 | 原文线索

V2-065802 | MATHDB-343482

Clancy-Leake-Payne conjecture on cyclic critical groups of random graphs

摘录 / EXCERPT: Clancy-Leake-Payne conjecture. As n tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-065803 | MATHDB-343485

Local-time convergence conjecture for conditioned Galton-Watson trees

摘录 / EXCERPT: Let ξ, T_{ξ}, and $|T_{\xi}|$ be as in the conditioned Galton-Watson tree theorem, and write $\sigma^2=\operatorname{Var}\xi$. Let H_n^{ξ} and C_n^{ξ} denote respec...

conjecture | open | MathDB 页面 | 原文线索

V2-065804 | MATHDB-343501

Every line segment intersects the escaping set of the exponential map

摘录 / EXCERPT: Let $E(e^z)=\{z\in\mathbb{C}: e^{z},e^{e^{z}},\ldots,e^{\circ n}(z)\to\infty\}$ be the escaping set of the map $z\mapsto e^z$. Escaping-set intersection conjecture. Every line segm...

conjecture | open | MathDB 页面 | 原文线索

V2-065805 | MATHDB-343502

Real-analyticity conjecture for the hairs of the exponential map

摘录 / EXCERPT: Let $f:\mathbb{C}\to\mathbb{C}$ be the exponential map $f(z)=e^z$, and let its hairs denote the corresponding invariant curves in the escaping set. Real-analyticity conjecture for...

open_problem | open | MathDB 页面 | 原文线索

V2-065806 | MATHDB-343503

The P=W conjecture for Higgs and Betti moduli spaces

摘录 / EXCERPT: Let G) be a reductive group, and let M_H and M_B be the corresponding Hitchin and Betti moduli spaces for a curve C. Let $h:M_\mathrm{H}\to\mathbb{A}^d$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-065807 | MATHDB-343505

Medvedoff-Morrison conjecture on generalised shuffle groups

摘录 / EXCERPT: Let $k\geqslant 3$ and let n be a positive integer that is not a power of k; when $k=4$, also assume that $n\neq 2^f$ for every positive integer f. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-065808 | MATHDB-343522

The sign conjecture for the quadratic Gauss sum $G(Q)$

摘录 / EXCERPT: Sign conjecture. With the above notations, $G(Q)$ is ±1.

conjecture | open | MathDB 页面 | 原文线索

V2-065809 | MATHDB-343546

The low-degree polynomial test conjecture

摘录 / EXCERPT: Consider a broad class of hypothesis-testing problems H_0 versus H_1. A polynomial f is a D-simple statistic if it has degree at most D and satisfies ... while...

conjecture | open | MathDB 页面 | 原文线索

V2-065810 | MATHDB-343550

Lu-Wenzel fundamental conjecture for real matrices

摘录 / EXCERPT: Let $M(n,\mathbb{R})$ be the space of real $n\times n$ matrices. Let $B,B_2,\ldots,B_m\in M(n,\mathbb{R})$, with B_α^* denoting transpose and with the Frobenius inner produc...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065811 | MATHDB-343551

Lu-Wenzel majorization conjecture for real matrices

摘录 / EXCERPT: Let $M(n,\mathbb{R})$ be the space of real $n\times n$ matrices. For $X\in M(n,\mathbb{R})$ with Frobenius norm $\|X\|=1$, let $T_X(Y)=[X^*,[X,Y]]$ and let $\lambda(T_X)$ denote it...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-065812 | MATHDB-343552

Complex Lu-Wenzel even partial-sum conjecture

摘录 / EXCERPT: Let $M(n,\mathbb{C})$ be the space of complex $n\times n$ matrices, with Frobenius norm $\|X\|=\sqrt{\operatorname{Tr}(XX^*)}$. For $X\in M(n,\mathbb{C})$ with $\|X\|=1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065813 | MATHDB-343556

Optimal strong approximation conjecture for $\operatorname{SL}_{N}(\mathbb{Z})$

摘录 / EXCERPT: Optimal strong approximation conjecture. For every $\epsilon>0$, as $q\to\infty$, there exists a set $Y\subset G_q$ with

conjecture | open | MathDB 页面 | 原文线索

V2-065814 | MATHDB-343558

Barát's good-coloring conjecture for subcubic graphs

摘录 / EXCERPT: A good coloring of a graph is a 2-coloring of its vertices in blue and red such that the subgraph induced by the blue vertices has maximum degree at most 1, the subgraph induced...

conjecture | open | MathDB 页面 | 原文线索

V2-065815 | MATHDB-343559

Terao's minimal-degree conjecture for free line arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ and ${\mathcal B}$ be line arrangements with isomorphic intersection lattices ... For an arrangement ${\mathcal A}$, let $r({\mathcal A})$ denote the minimal degre...

conjecture | open | MathDB 页面 | 原文线索

V2-065816 | MATHDB-343560

The stronger Terao conjecture for minimal degrees of line arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ and ${\mathcal B}$ be arrangements of d lines with isomorphic intersection lattices ... Let $r({\mathcal A})$ denote the minimal degree of a Jacobian relation. S...

conjecture | open | MathDB 页面 | 原文线索

V2-065817 | MATHDB-343561

Generic splitting-type conjecture for logarithmic bundles of line arrangements

摘录 / EXCERPT: Let ${\mathcal A}$ and ${\mathcal B}$ be line arrangements with isomorphic intersection lattices ... Let $E({\mathcal A})$ and $E({\mathcal B})$ be the associated rank 2 vector bun...

conjecture | open | MathDB 页面 | 原文线索

V2-065818 | MATHDB-343562

Non-birationality conjecture for hyperplane restrictions of maps from very general hypersurfaces

摘录 / EXCERPT: Let $X_d\bsubseteq \mathbb{P}^{n+1}$ be a very general hypersurface of degree $d\geq n+3$, and let Y be a smooth projective n-fold that is non-uniruled. Let $H_d:=X_d\cap H$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-065819 | MATHDB-343564

Universal decoupling conjecture for real analytic surfaces

摘录 / EXCERPT: Let Ssubset${\mathbb R}^3$ be the graph of a nonconstant real analytic function $g:[-1,1]^2\to{\mathbb R}$. For $0<\delta\leq 1$, let ${\mathcal N}_\delta(S)$ be its δ-nei...

conjecture | open | MathDB 页面 | 原文线索

V2-065820 | MATHDB-343567

The geometric Breuil-Mézard conjecture for crystalline special fibres

摘录 / EXCERPT: Let $\boldsymbol{\rho}$ range over residual representations and let $\boldsymbol{\lambda}$ range over regular tuples of labeled Hodge-Tate weights. For each Serre weight...

conjecture | open | MathDB 页面 | 原文线索

V2-065821 | MATHDB-343569

Bloch-Kato conjecture for motives

摘录 / EXCERPT: Let M be a motive with L-function $L(M,s)$ and p-adic Selmer group $\mathrm{Sel}_{p^\infty}(M)$. Suppose its functional equation is ... with $s=\frac{1}{2}$ a critical value. B...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-065822 | MATHDB-343571

Set-valued tableau formulas for Lascoux polynomials and atoms

摘录 / EXCERPT: Set-valued tableau conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-065823 | MATHDB-343572

Skyline-tableau formula for Lascoux atoms

摘录 / EXCERPT: Skyline-tableau conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-065824 | MATHDB-343573

Transpose-invariance and positivity conjectures for stable Schur expansions of degeneracy loci

摘录 / EXCERPT: Let $\Sigma^\wedge_{\infty,i}$ and $\Sigma^S_{\infty,i}$ denote the stable skew-symmetric and symmetric degeneracy loci, and write their Segre-Schwartz-MacPherson classes as formal...

conjecture | open | MathDB 页面 | 原文线索

V2-065825 | MATHDB-343575

Suzuki's conjecture on automorphic representations for metaplectic covers

摘录 / EXCERPT: Let ${\bf A}$ be the ring of adeles, let τ be an irreducible cuspidal representation of $GL_m({\bf A})$, and let $GL_{nm}^{(n)}({\bf A})$ denote the n-fold metaplectic cover...

conjecture | open | MathDB 页面 | 原文线索

V2-065826 | MATHDB-343576

Wang's 5-power congruence conjecture for the spt$_\omega$ function

摘录 / EXCERPT: Let $\textup{spt}_\omega(n)$ denote the partition statistic used in the paper. For integers $k\geq 1$ and $n\geq 0$, Wang's conjecture. ... ... The conjecture concerns an infinite fami...

conjecture | open | MathDB 页面 | 原文线索

V2-065827 | MATHDB-343585

The flag-complex Turán lower-bound conjecture

摘录 / EXCERPT: Turán lower-bound conjecture. The face vector of Δ is componentwise at least that of T:

conjecture | open | MathDB 页面 | 原文线索

V2-065828 | MATHDB-343586

Disjoint-interval conjecture for two-type queue equilibria

摘录 / EXCERPT: Let (F_a,F_b) be a Nash equilibrium arrival profile of the two-type queueing game, with $c3(F_a)$ and $c3(F_b)$ denoting the arrival times used by types a and b, respectively...

conjecture | open | MathDB 页面 | 原文线索

V2-065829 | MATHDB-343587

Gaiotto's conformal limit conjecture for the Hitchin component

摘录 / EXCERPT: Let a Higgs bundle lie in the Hitchin component, and let its conformal limit be the flat connection obtained from the conformal-limit construction. An oper is a flat bundle satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-065830 | MATHDB-343588

Simpson's foliation conjecture for de Rham moduli spaces

摘录 / EXCERPT: Let $M_{\rm dR}(X,r)$ be the de Rham moduli space, and for each index α let G_α^1 be a stratum with projection ... Assume that the fibers of these projections are Lagra...

conjecture | open | MathDB 页面 | 原文线索

V2-065831 | MATHDB-343590

Simpson's oper stratum conjecture

摘录 / EXCERPT: Let $M_{\rm dR}(X,r)$ be the de Rham moduli space with its stratification, and let G_u^1 denote the oper stratum. The source identifies G_u^1 as the highest stratum. Simpson's...

conjecture | open | MathDB 页面 | 原文线索

V2-065832 | MATHDB-343591

Simpson's VHS characterization conjecture for de Rham strata

摘录 / EXCERPT: For each index α, let G_α^1 be a de Rham stratum and let $\big(G_\alpha^1\big)^{\rm VHS}\subset G_\alpha^1$ be the subset of polarized complex variations of Hodge str...

conjecture | open | MathDB 页面 | 原文线索

V2-065833 | MATHDB-343592

Characterization of posets embeddable in the nonnegative reals

摘录 / EXCERPT: Poset extension conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-065834 | MATHDB-343594

Conjecture on variance lower bounds for white-noise Schrödinger operators

摘录 / EXCERPT: White-noise semigroup variance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065835 | MATHDB-343603

Radiative Uniqueness Conjecture for bubbling wave maps

摘录 / EXCERPT: Let $\boldsymbol{u}_c(t)$ denote the constructed blow-up solution from Part (a) of Theorem, and let $\boldsymbol{u}(t) \in \mathcal{H}_1$ be any other solution that blows up backwa...

conjecture | open | MathDB 页面 | 原文线索

V2-065836 | MATHDB-343610

Revised LMOV integrality conjecture for framed knots

摘录 / EXCERPT: Revised framed LMOV conjecture. For every partition μ,

conjecture | open | MathDB 页面 | 原文线索

V2-065837 | MATHDB-343611

Finite rank and crank moment inequality conjecture

摘录 / EXCERPT: Let N be a natural number. For $k\geq 1$, define the finite rank and crank moments for vector partitions by ... where $N_{S_1}(m,n)$ and $M_{S_2}(m,n)$ are the corresponding finite...

conjecture | open | MathDB 页面 | 原文线索

V2-065838 | MATHDB-343624

The level-size characterization of monoid posets for bi-infinite weak orders

摘录 / EXCERPT: Level-size characterization. A bi-infinite weak order is a monoid poset if and only if all its levels are of the same size.

conjecture | open | MathDB 页面 | 原文线索

V2-065839 | MATHDB-343625

Existence of uniform posets that are not monoid posets

摘录 / EXCERPT: Uniform-poset nonexistence conjecture. There are uniform posets that are not monoid posets.

conjecture | open | MathDB 页面 | 原文线索

V2-065840 | MATHDB-343629

The second analytic Langlands conjecture on discreteness of the joint spectrum

摘录 / EXCERPT: Second analytic Langlands conjecture. The spectrum $\operatorname{Spec}_{\mathcal A}(\mathcal H)$ is discrete; equivalently, $\mathcal H$ is the completed direct sum of finite-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-065841 | MATHDB-343632

The Hepp-bound faithfulness conjecture for primitive phi-four graphs

摘录 / EXCERPT: Let G be a graph that is p-logarithmic in $d=4$ dimensions and whose vertices all have degree at most 4; call such a graph a ϕ^4 graph. Let $\mathcal{P}(G)$ denote its per...

conjecture | open | MathDB 页面 | 原文线索

V2-065842 | MATHDB-343633

FUSF heaviness and branching conjecture for nonunimodular transitive pairs

摘录 / EXCERPT: FUSF heaviness and branching conjecture. Each tree in the FUSF is Γ-heavy and has branching number bigger than one almost surely.

conjecture | open | MathDB 页面 | 原文线索

V2-065843 | MATHDB-343651

Direct-product immersion conjecture

摘录 / EXCERPT: Let G and H be graphs, and write $\operatorname{im}(G)=t$ and $\operatorname{im}(H)=r$. Direct-product immersion conjecture. Then ... This conjecture asserts that the direct prod...

conjecture | open | MathDB 页面 | 原文线索

V2-065844 | MATHDB-343652

Strong-product immersion conjecture

摘录 / EXCERPT: Let G and H be graphs, and write $\operatorname{im}(G)=t$ and $\operatorname{im}(H)=r$. Strong-product immersion conjecture. Then ... This conjecture is motivated by the identity...

conjecture | open | MathDB 页面 | 原文线索

V2-065845 | MATHDB-343653

Chen--Lih--Wu's equitable Brooks conjecture

摘录 / EXCERPT: Chen--Lih--Wu's conjecture. The graph G has an equitable Δ-coloring unless one of the following holds: $\Delta=2$ and G is an odd cycle; $G \cong K_{\Delta+1}$; or...

conjecture | open | MathDB 页面 | 原文线索

V2-065846 | MATHDB-343654

The aperiodic Borel equitable Brooks conjecture

摘录 / EXCERPT: Aperiodic Borel equitable Brooks conjecture. The graph G has a μ-equitable Δ-coloring.

conjecture | open | MathDB 页面 | 原文线索

V2-065847 | MATHDB-343656

Generic-level variance conjecture for excursion and level sets

摘录 / EXCERPT: Let f satisfy Assumptions,, and, and let $D\subset\mathbb{R}^2$ be an open rectangle centred at the origin. Write D_R for its dilation by R, and let $N_\mathrm{ES}(D_R,\ell)$...

conjecture | open | MathDB 页面 | 原文线索

V2-065848 | MATHDB-343657

Anomalous-level variance conjecture for fields with spectral singularity

摘录 / EXCERPT: Let f satisfy Assumptions and, and suppose there exist $\alpha\in(0,2)$ and $r_0>0$ such that ... for all $|x|<r_0$. Let $D\subset\mathbb{R}^2$ be an open rectangle centred at the...

conjecture | open | MathDB 页面 | 原文线索

V2-065849 | MATHDB-343660

Minh's symbolic-power regularity conjecture for edge ideals

摘录 / EXCERPT: Let G be a graph, let $I(G)$ be its edge ideal, and let $I(G)^{(s)}$ denote its s-th symbolic power. The invariant ${\rm reg}(-)$ denotes Castelnuovo-Mumford regularity. For ev...

conjecture | open | MathDB 页面 | 原文线索

V2-065850 | MATHDB-343661

Unbounded partial sums for squarefree-supported multiplicative functions

摘录 / EXCERPT: Let $g:\mathds{N}\to\{-1,1\}$ be multiplicative, let μ denote the Möbius function, and define $f=\mu^2g$. Squarefree multiplicative discrepancy conjecture. The partial sums of...

conjecture | open | MathDB 页面 | 原文线索

V2-065851 | MATHDB-343664

Graceful difference labeling conjecture for disjoint unions of circuits

摘录 / EXCERPT: Graceful difference labeling conjecture. If G is the disjoint union of circuits, then G has a gdl, unless

conjecture | open | MathDB 页面 | 原文线索

V2-065852 | MATHDB-343672

Cusick-Li-Stănică conjecture on balanced elementary symmetric Boolean functions

摘录 / EXCERPT: Let $\boldsymbol{e}_{n,k}$ denote the elementary symmetric Boolean function of degree k in n variables. A Boolean function is balanced when it takes each value equally often, a...

conjecture | open | MathDB 页面 | 原文线索

V2-065853 | MATHDB-343697

The global annulus conjecture for the first normalized Steklov eigenvalue

摘录 / EXCERPT: Let Λ_{ϵ} be a bounded connected planar annulus with two boundary components, and let $\lambda_{1}(\Lambda_{\epsilon})$ be its first nontrivial Steklov eigenvalue, g...

conjecture | open | MathDB 页面 | 原文线索

V2-065854 | MATHDB-343700

Counting admissible polynomial Bethe-root solutions for diagonal and generic parameters

摘录 / EXCERPT: Let $M=2s$, and let $P_a^M(U_i,\bar U_i)=0$ be the polynomial Bethe-root equations for parameter choice $a=d$ or $a=g$. A solution is admissible if its coordinates are pairwise dis...

conjecture | open | MathDB 页面 | 原文线索

V2-065855 | MATHDB-343710

Uehara's autoequivalence-group conjecture for surfaces

摘录 / EXCERPT: Let S be a smooth projective complex surface with $K_S\not\equiv 0$ that admits no minimal elliptic fibration. Let Z be the union of all (-2)-curves, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065856 | MATHDB-343711

Generalized Auslander conjecture for nil-affine crystallographic groups

摘录 / EXCERPT: Let N be a connected and simply connected nilpotent Lie group, and let $\Gamma\subseteq\operatorname{Aff}(N)$ act properly discontinuously and cocompactly on N; such an action...

conjecture | open | MathDB 页面 | 原文线索

V2-065857 | MATHDB-343714

Classification conjecture for simple groups with a generically 4-transitive action

摘录 / EXCERPT: Let G be a simple group of finite Morley rank admitting a generically 4-transitive action. Generically 4-transitive classification conjecture. Either...

conjecture | open | MathDB 页面 | 原文线索

V2-065858 | MATHDB-343723

Takahashi's higher-genus correspondence for relative invariants of $(\mathbb P^2,E)$

摘录 / EXCERPT: For every $d\in\mathbb Z_{>0}$, let $k\in\mathbb Z_{>0}$ divide d, and let $N_{g,d}^{\mathbb P^2/E,k}$ denote the higher-genus relative Gromov-Witten invariant associated with de...

conjecture | open | MathDB 页面 | 原文线索

V2-065859 | MATHDB-343724

Takahashi's higher-genus BPS invariance conjecture for $(\mathbb P^2,E)$

摘录 / EXCERPT: For every $d\in\mathbb Z_{>0}$ and every positive divisor k of d, let $\Omega_{d,k}^{\mathbb P^2/E}(\hbar)\in\mathbb Q[\![\hbar]\!]$ be the higher-genus relative BPS invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-065860 | MATHDB-343726

Real sheaves/Gromov-Witten correspondence for $K_{\mathbb P^2}$

摘录 / EXCERPT: Let $K_{\mathbb P^2}$ carry its natural real structure, with real locus $K_{\mathbb P^2}^{\mathbb R}$ restricting to $\mathbb R\mathbb P^2$. For each positive integer d, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065861 | MATHDB-343729

Uniqueness of the Gibbs measure away from the critical point

摘录 / EXCERPT: Uniqueness conjecture. For $p\neq p_c$ there is a unique Gibbs measure.

conjecture | open | MathDB 页面 | 原文线索

V2-065862 | MATHDB-343730

Critical-point threshold conjecture for Gibbs-measure uniqueness

摘录 / EXCERPT: Critical-point threshold conjecture. There is a $q_0=q_0(d)$ such that for $q>q_0$ there is non-uniqueness of Gibbs measures

conjecture | open | MathDB 页面 | 原文线索

V2-065863 | MATHDB-343732

Conjecture on the asymptotic k-path partition function

摘录 / EXCERPT: For c, β, and $\frac{1}{N^2}\frac{1}{2}$ as in the preceding definitions, let $R^-_c(]alpha,\beta)$ and $R^+_c(]alpha,\beta)$ denote the lower and upper limits of the scal...

conjecture | open | MathDB 页面 | 原文线索

V2-065864 | MATHDB-343736

The gamma2 conjecture for flag simplicial spheres

摘录 / EXCERPT: The gamma2 conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065865 | MATHDB-343738

The reflectional multi-tiling conjecture for concrete integer polytopes

摘录 / EXCERPT: Reflectional multi-tiling conjecture. If P is a concrete integer polytope, then P multi-tiles $\mathbb{R}^{d}$ by translations together with a finite set of reflections.

conjecture | open | MathDB 页面 | 原文线索

V2-065866 | MATHDB-343744

Trivariate shuffle conjecture for selected hook shapes

摘录 / EXCERPT: Let $\rho=(a\,|\,b)$ be a hook, let Γ_a be the Dyck path defined in the source, and let $\mathbb{L}_\beta(t;\bm{z})$ be the LLT polynomial associated with a Dyck path...

conjecture | open | MathDB 页面 | 原文线索

V2-065867 | MATHDB-343748

Lovász's planar graph homomorphism isomorphism question

摘录 / EXCERPT: Lovász's planar homomorphism question. If

open_problem | open | MathDB 页面 | 原文线索

V2-065868 | MATHDB-343761

Local polynomial convexity conjecture for the surfaces M_t

摘录 / EXCERPT: Let M_t be the real surface under consideration, parametrized by the parameter t. A surface is locally polynomially convex at a point if sufficiently small compact neighbourhoo...

conjecture | open | MathDB 页面 | 原文线索

V2-065869 | MATHDB-343765

Artin-Rudakov-Shafarevich-Shioda conjecture on supersingular K3 surfaces

摘录 / EXCERPT: Artin-Rudakov-Shafarevich-Shioda conjecture. A K3 surface over k is supersingular if and only if it is unirational.

conjecture | open | MathDB 页面 | 原文线索

V2-065870 | MATHDB-343767

S-duality conjecture for SU(r) Vafa-Witten partition functions

摘录 / EXCERPT: Let S be a smooth projective surface and let $r\in{\mathbb Z}_{>0}$. For $L\in\operatorname{Pic}(S)$, define the $\operatorname{SU}(r)$ Vafa-Witten partition function by ... Define...

conjecture | open | MathDB 页面 | 原文线索

V2-065871 | MATHDB-343768

Joyce-Song wall-crossing formula for twisted Vafa-Witten invariants

摘录 / EXCERPT: Let S be a smooth projective surface, let ${\mathfrak S}\to S$ be a μ_r-gerbe, and let $\alpha=(\operatorname{rk},L,c_2)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065872 | MATHDB-343769

The fixed-error hybrid isoperimetric conjecture for the Hamming cube

摘录 / EXCERPT: Fixed-error hybrid isoperimetric conjecture. There is a fixed constant K such that, whenever $\mu(A)=1/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-065873 | MATHDB-343770

The maximal hybrid isoperimetric conjecture for the Hamming cube

摘录 / EXCERPT: Maximal hybrid isoperimetric conjecture. If $|A|=a$, then

conjecture | open | MathDB 页面 | 原文线索

V2-065874 | MATHDB-343771

The all-codimension stability conjecture for Hamming-cube subcubes

摘录 / EXCERPT: All-codimension stability conjecture. The statement in Theorem $$ holds for all $k\in\mathbb P$, even with n^β replaced by $2^n/\partial(|A|)$.

conjecture | open | MathDB 页面 | 原文线索

V2-065875 | MATHDB-343776

Hadamard-product convexity conjecture for the class Ω

摘录 / EXCERPT: Hadamard-product convexity conjecture. For any $f_1,f_2\in\Omega$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-065876 | MATHDB-343781

Lai et al.'s conjecture on dynamic coloring of planar graphs

摘录 / EXCERPT: Lai et al.'s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065877 | MATHDB-343784

Equality of discriminant Milnor and Tjurina numbers for matrix families

摘录 / EXCERPT: Equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065878 | MATHDB-343785

Maulik-Pandharipande-Thomas stable-pair/GV correspondence for Calabi-Yau fourfolds

摘录 / EXCERPT: Let (X,ω) be a smooth projective Calabi-Yau 4-fold, let $P_{n,\beta}(\gamma)$ be its stable-pair invariants with $\gamma\in H^4(X,\mathbb Z)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065879 | MATHDB-343787

Cao-Maulik-Toda wall-crossing conjecture for stable pairs on Calabi-Yau fourfolds

摘录 / EXCERPT: Let (X,ω) be a smooth projective Calabi-Yau 4-fold, let $\beta\in H_2(X,\mathbb Z)$ and $n\in\mathbb Z_{\geqslant0}$, and choose a generic $t\in\mathbb R_{>0}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065880 | MATHDB-343788

Joyce-Song/GV special-case conjecture for stable pairs on Calabi-Yau fourfolds

摘录 / EXCERPT: In the setting of the wall-crossing conjecture, let $P^{\mathrm{JS}}_{n,\beta}(\gamma)$ denote the Joyce-Song chamber stable-pair invariant, and let $n_{0,\beta}(\gamma)$ be the ge...

conjecture | open | MathDB 页面 | 原文线索

V2-065881 | MATHDB-343789

Equivariant Joyce-Song stable-pair formula for the local curve \(\mathcal{O}_{\mathbb{P}^{1}}(-1,-1,0)\)

摘录 / EXCERPT: Let $X=\mathcal O_{\mathbb P^1}(-1,-1,0)$ and let T be the Calabi-Yau torus, with equivariant parameter λ_3. Let $P^{\mathrm{JS}}_{n,d}$ denote the T-equivariant Joyce-...

conjecture | open | MathDB 页面 | 原文线索

V2-065882 | MATHDB-343793

Directed abelian Cayley graph cop-number conjecture

摘录 / EXCERPT: Let G be a finite abelian group, let S be a generating set defining a directed Cayley graph, and let $n=|G|$. Write $c(\operatorname{Cay}(G,S))$ for its cop number. Directed ab...

conjecture | open | MathDB 页面 | 原文线索

V2-065883 | MATHDB-343794

Farber's diameter conjecture for higher associahedra

摘录 / EXCERPT: Let k and n be positive integers with $n=2k$. For a bipartite $(k,2k)$-plabic graph G, let G^{op} be the graph obtained from G by a 180° rotation fo...

conjecture | open | MathDB 页面 | 原文线索

V2-065884 | MATHDB-343796

Generic polygon invariance conjecture for higher associahedra

摘录 / EXCERPT: Let $\mathcal{A}$ be the set of vertices of a generic convex n-gon, and let $\Upsigma_{\mathcal{A},k}$ be the associated higher associahedron. The generic polygon invariance conj...

conjecture | open | MathDB 页面 | 原文线索

V2-065885 | MATHDB-343797

Braverman-Kazhdan's kernel conjecture for rho-Bessel sheaves

摘录 / EXCERPT: Let G be a reductive group over $\mathbb F_q$, let ρ be a representation as in the Braverman-Kazhdan construction, and let $\Phi_{G,\rho,\psi}$ be the corresponding ρ-B...

conjecture | open | MathDB 页面 | 原文线索

V2-065886 | MATHDB-343804

The Julia-set inclusion for the magnetic Laplacian spectrum on the infinite Sierpinski gasket

摘录 / EXCERPT: Let G_∞ be the infinite Sierpinski gasket lattice, let $\mathcal{U}$ be the spectral-decimation map, and let $\mathcal{J}(\mathcal{U})$ denote its Julia set. For parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-065887 | MATHDB-343810

The signed projective cube conjecture for planar signed graphs

摘录 / EXCERPT: Let (G,σ) be a signed graph, and let $g_{ij}(G,\sigma)$ denote the shortest length of a closed walk of type ij, where $ij\in\{00,01,10,11\}$. Let $\operatorname{SPC}(k)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-065888 | MATHDB-343811

The planar signed-graph characterization by four girth conditions

摘录 / EXCERPT: Let (G,σ) be a planar signed graph. For $ij\in\{00,01,10,11\}$, let $g_{ij}(G,\sigma)$ be the shortest length of a closed walk of type ij, and let $\operatorname{SPC}(k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-065889 | MATHDB-343816

Split-and-merge scaling-limit conjecture for random stirring cycles

摘录 / EXCERPT: Let $\widetilde\eta^N(t)$ be the random stirring process, let $\mathbf p(\widetilde\eta^N(t))$ be its vector of macroscopically scaled cycle lengths, let $m(\tau)$ be the macroscop...

conjecture | open | MathDB 页面 | 原文线索

V2-065890 | MATHDB-343826

The telescope conjecture without smash product

摘录 / EXCERPT: Let $\mathcal{T}$ be a rigidly compactly generated tensor triangulated category. A localization $L:\mathcal{T}\to\mathcal{T}$ is smashing if it preserves coproducts, and it is fini...

conjecture | open | MathDB 页面 | 原文线索

V2-065891 | MATHDB-343837

The twisted Linnik-Selberg conjecture over function fields

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$, $K_{\infty}=\mathbb{F}_q(!(1/t)!)$, and let ψ be the nontrivial additive character on K_{∞} that is trivial on $\mathbb{F}_q[t]$. For nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-065892 | MATHDB-343844

Regular directed complexes are freely generating

摘录 / EXCERPT: Let P be a directed complex. It is freely generating when $(\mathcal{M}o\ell_{}(P))^*$ admits the structure of a polygraph whose n-dimensional generators are the n-dimensiona...

conjecture | open | MathDB 页面 | 原文线索

V2-065893 | MATHDB-343846

The Frobenius structure conjecture for theta-function products

摘录 / EXCERPT: Assume that X is in the absolute log Calabi-Yau case. Fix $p_1,\ldots,p_n\in B(\mathbb{Z})$ and a curve class A in X. Let β be the type of genus-0 punctured map with...

conjecture | open | MathDB 页面 | 原文线索

V2-065894 | MATHDB-343868

Andrade-Jung-Shamesaldeen moment conjecture for quadratic Dirichlet L-functions

摘录 / EXCERPT: Andrade-Jung-Shamesaldeen moment conjecture. For any $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065895 | MATHDB-343869

Bui-Florea partial Hadamard product moment conjecture

摘录 / EXCERPT: Bui-Florea partial Hadamard product moment conjecture. For any $k\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065896 | MATHDB-343870

Splitting conjecture for hybrid Euler-Hadamard product moments

摘录 / EXCERPT: Splitting conjecture. For any $k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-065897 | MATHDB-343872

Positivity conjecture for the Jacobian of the multigraph moment map

摘录 / EXCERPT: Let $h:\mathbb{R}^n\setminus\{\mathbf{0}\}\to\mathbb{R}^n$ be defined componentwise by ... Let $\mathbf{J}$ be the Jacobian of h, and let ... For $\boldsymbol{\beta}\in\mathcal{B}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-065898 | MATHDB-343873

Uniform perturbation conjecture for motif-support size

摘录 / EXCERPT: Let $\degree$ be a degree sequence, and let $\psi(\degree)$ denote the number of nonzero entries of the associated multigraph representation. Define $v(\degree)$ as the smallest re...

conjecture | open | MathDB 页面 | 原文线索

V2-065899 | MATHDB-343879

Triangle-free signed graph 3-coloring conjecture

摘录 / EXCERPT: A signed graph is a graph equipped with a signature assigning signs to its edges; it is signed 3-colorable if it admits a signed coloring with three colors. Triangle-free signed gr...

conjecture | open | MathDB 页面 | 原文线索

V2-065900 | MATHDB-343882

The Goldberg snark circular-flow-number conjecture

摘录 / EXCERPT: For every positive integer k, let G_{2k+1} be the Goldberg snark on $8(2k+1)$ vertices, and let $\Phi_c(G_{2k+1})$ denote its circular flow number. The Goldberg snark circular-...

conjecture | open | MathDB 页面 | 原文线索

V2-065901 | MATHDB-343886

The short-range scattering exponent conjecture for time-dependent density functional theory

摘录 / EXCERPT: Consider the time-dependent density-matrix equation with self-interaction satisfying condition (SI2), and let d denote the spatial dimension and β the exponent in that cond...

conjecture | open | MathDB 页面 | 原文线索

V2-065902 | MATHDB-343887

The modified scattering conjecture for critical and long-range density-functional interactions

摘录 / EXCERPT: Let d be the spatial dimension, let $v(x)=|x|^{-\alpha}$ with $0<\alpha\leq 1$, and consider ... with $0<\beta\leq 1/d$. Let I^1 denote the trace-class space used for the density...

conjecture | open | MathDB 页面 | 原文线索

V2-065903 | MATHDB-343891

Guo and Schlosser's generalized q-congruence for d-adic parameters

摘录 / EXCERPT: Let $d\geqslant 3$ and let n be a positive integer with $n\equiv -1\pmod{d}$. Let $(a;q)_k=(1-a)(1-aq)\cdots(1-aq^{k-1})$ denote the q-shifted factorial, let...

conjecture | open | MathDB 页面 | 原文线索

V2-065904 | MATHDB-343892

Guo and Schlosser's companion q-congruence for d-adic parameters

摘录 / EXCERPT: Let $d\geqslant 3$ and let $n>1$ be an integer with $n\equiv 1\pmod{d}$. With $(a;q)_k=(1-a)(1-aq)\cdots(1-aq^{k-1})$, $[m]=1+q+\cdots+q^{m-1}$, and $\Phi_n(q)$ the n-th cyclotom...

conjecture | open | MathDB 页面 | 原文线索

V2-065905 | MATHDB-343903

Conjectures on real roots and reducibility of the polynomials B_{p_{k,n}}

摘录 / EXCERPT: Conjectures for $B_{p_{k,n}}$. (1) If $k\equiv0\pmod 2$, there is $c_k\in\mathbb{N}_{+}$ such that $B_{p_{k,n}}(t)$ has exactly one real root for $n\in\mathbb{N}_{\geq c_k}$; under...

conjecture | open | MathDB 页面 | 原文线索

V2-065906 | MATHDB-343904

Conjectures on roots, irreducibility and monotonicity of the polynomials B_{s_{i,n}}

摘录 / EXCERPT: Let $s_{i,n}$, for $i\in\{0,1,2,3\}$, be the sequences defined earlier in the paper, and let $B_{s_{i,n}}(t)$ be their associated Stern polynomials. Conjectures for $B_{s_{i,n}}$....

conjecture | open | MathDB 页面 | 原文线索

V2-065907 | MATHDB-343906

Random free-product extension conjecture for stable subgroups of right-angled Artin groups

摘录 / EXCERPT: Let $A(\Gamma)$ be a right-angled Artin group whose defining graph Γ is connected and not a join, and let H be a stable subgroup of $A(\Gamma)$. Let (μ_i) be a sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-065908 | MATHDB-343919

Novak's positive-sem definiteness conjecture for trigonometric integration matrices

摘录 / EXCERPT: Novak's conjecture. The matrix

conjecture | open | MathDB 页面 | 原文线索

V2-065909 | MATHDB-343922

Equivalent version of the P=W conjecture with tautological and multiplicative splitting

摘录 / EXCERPT: Let C be an irreducible nonsingular projective curve of genus $g\geq 2$, let ${\mathcal M}_{\mathrm{Dol}}$ be its twisted Dolbeault moduli space with Hitchin fibration h, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-065910 | MATHDB-343924

The immunity-weight conjecture for Hurwitz orbits with simply intersecting cycles

摘录 / EXCERPT: Let Σ be a homogeneous B_3-space covering a finite homogeneous $PSL(2,\mathbb{Z})$-space $\overline{\Sigma}$. Suppose that the cycles of $\overline{\Sigma}$ are simply int...

conjecture | open | MathDB 页面 | 原文线索

V2-065911 | MATHDB-343929

The first-increase inequalities for connected unit interval graphs

摘录 / EXCERPT: First-increase inequalities. One has

conjecture | open | MathDB 页面 | 原文线索

V2-065912 | MATHDB-343930

The Markov convexity threshold conjecture for Carnot groups

摘录 / EXCERPT: Markov convexity threshold conjecture. G is not Markov p-convex for every

conjecture | open | MathDB 页面 | 原文线索

V2-065913 | MATHDB-343931

The largest Carnot subgroup conjecture for Markov convexity

摘录 / EXCERPT: Largest Carnot subgroup conjecture. The set of p for which G is Markov p-convex is the same as the set of p for which H is Markov p-convex.

conjecture | open | MathDB 页面 | 原文线索

V2-065914 | MATHDB-343936

The p-resolution obstruction conjecture for compacta

摘录 / EXCERPT: The p-resolution obstruction conjecture. No compactum Y with

conjecture | open | MathDB 页面 | 原文线索

V2-065915 | MATHDB-343945

The generalized Wilf conjecture for Frobenius generalized numerical semigroups

摘录 / EXCERPT: Let $S\subseteq\mathbb{N}^{d}$ be a Frobenius generalized numerical semigroup with Frobenius element $\mathbf{f}=(f^{(1)},\ldots,f^{(d)})$. With $e(S)$ the number of minimal genera...

conjecture | open | MathDB 页面 | 原文线索

V2-065916 | MATHDB-343947

The bipartite rainbow matching conjecture for an almost-full matching

摘录 / EXCERPT: Let $f_{\mathcal B}(n,m)$ be the least number of n-element matchings in a bipartite graph that guarantees a rainbow matching of size m. Bipartite rainbow matching conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-065917 | MATHDB-343948

The cycle rainbow matching conjecture

摘录 / EXCERPT: Let C_t be the cycle on t vertices, and let $f_{C_t}(s,s)$ denote the least number of independent sets of size s in C_t that guarantees a rainbow independent set of size...

conjecture | open | MathDB 页面 | 原文线索

V2-065918 | MATHDB-343949

The stronger bipartite rainbow matching conjecture

摘录 / EXCERPT: Let $f_{\mathcal B}(n,m)$ be the least number of n-element matchings in a bipartite graph that guarantees a rainbow matching of size m. Strong bipartite rainbow matching conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-065919 | MATHDB-343950

The exceptional-line-graph rainbow matching conjecture

摘录 / EXCERPT: Let $f_G(n,m)$ denote the least number of independent n-sets in a graph G that guarantees a rainbow independent set of size m. Exceptional-line-graph conjecture. If G is th...

conjecture | open | MathDB 页面 | 原文线索

V2-065920 | MATHDB-343951

The maximum-degree-two rainbow matching conjecture

摘录 / EXCERPT: Let $f_G(n,m)$ denote the least number of independent n-sets in a graph G that guarantees a rainbow independent set of size m. Maximum-degree-two conjecture. If G has maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-065921 | MATHDB-343953

The refined bounded-degree conjecture for full rainbow sets

摘录 / EXCERPT: Let $f_{\mathcal D(k)}(n)$ denote the full-rainbow parameter for the class $\mathcal D(k)$ of graphs with maximum degree at most k, and let q be the quantity used in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-065922 | MATHDB-343970

Conjecture A on adjoining prescribed matrix blocks to division-ring embeddings

摘录 / EXCERPT: Let $n\in\mathbb{N}$ with $n>1$, and let $\varphi:F\to M_n(G)$ be a ring embedding, where G and F are division rings. Suppose that a and b are positive integers satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-065923 | MATHDB-343971

The depth-one realization conjecture for the weight-five Grassmannian polylogarithm

摘录 / EXCERPT: Weight-five depth-one conjecture. (i) There exists a formal linear combination $\mathcal{Q}_5(v_1,\dots,v_{10})$ of rational functions on $\operatorname{Conf}_{10}(5)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-065924 | MATHDB-343985

The Malle conjecture for 2-ranks of cyclic cubic and quintic class groups

摘录 / EXCERPT: Malle's class-group rank conjecture. As K ranges over cyclic number fields of degree ℓ, for every $r\in\mathbb{Z}_{\geq0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-065925 | MATHDB-343988

Reznick's dimension-two classification conjecture for simplicial sets

摘录 / EXCERPT: Reznick's conjecture. Δ is either an M-simplex or an H-simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-065926 | MATHDB-343989

The numerical-range conjecture for periodic tridiagonal operators

摘录 / EXCERPT: The numerical-range conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-065927 | MATHDB-343992

Somos 4 conjecture for generalized Schröder Hankel transforms

摘录 / EXCERPT: Let $a_n(p,s,t)$ be the generalized Schröder numbers and let $h_n(p,s,t)$ be their Hankel transform. A (α,β) Somos 4 sequence satisfies ... Generalized Schröder Somos 4...

conjecture | open | MathDB 页面 | 原文线索

V2-065928 | MATHDB-343993

Explicit Somos 4 conjecture for the five-parameter generalized Catalan sequence

摘录 / EXCERPT: Let $a_n(p,q,r,s,t)$ be the five-parameter generalized Catalan sequence and let h_n denote the Hankel transform of $\frac{1}{p}a_{n+1}(p,q,r,s,t)$. A (α,β) Somos 4 s...

conjecture | open | MathDB 页面 | 原文线索

V2-065929 | MATHDB-343994

Specialized Somos 4 conjecture for the five-parameter generalized Catalan sequence

摘录 / EXCERPT: Let $a_n(p,q,r,s,t)$ be the five-parameter generalized Catalan sequence. Assume ... Specialized generalized Catalan Somos 4 conjecture. The Hankel transform of $a_n(p,q,1,q-p+1,1)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-065930 | MATHDB-343995

Recurrence formulation of the alpha-family Hankel-transform conjecture

摘录 / EXCERPT: Let $a_n(\beta)$ be defined by ... with $a_0=1$, $a_1=0$, and $a_2=\beta$, and let $h_n(\beta)$ be its Hankel transform. Recurrence-form Hankel-transform conjecture. The Hankel trans...

conjecture | open | MathDB 页面 | 原文线索

V2-065931 | MATHDB-343996

Explicit Hankel transform conjecture for the cubic Catalan family

摘录 / EXCERPT: Let $c(x)$ denote the Catalan generating function and define a sequence by ... Define ... ... Cubic-family Hankel-transform conjecture. The Hankel transform of the sequence with generati...

conjecture | open | MathDB 页面 | 原文线索

V2-065932 | MATHDB-343997

Somos 4 conjecture for the elliptic-curve family sequence

摘录 / EXCERPT: Let b_n be the sequence whose generating function is ... arising from the elliptic-curve family $E_t:y^2+4xy+y=x^3+(t-1)x^2+tx$. A (α,β) Somos 4 sequence satisfies ... E...

conjecture | open | MathDB 页面 | 原文线索

V2-065933 | MATHDB-343998

Somos 4 conjecture for the transformed elliptic-curve sequence

摘录 / EXCERPT: Let t be a parameter and let a_n be the sequence whose generating function is ... Transformed elliptic-curve Somos 4 conjecture. The Hankel transform of a_n is a $(1,t^2+3t+1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-065934 | MATHDB-343999

Atia's orthogonal-polynomial recurrence conjecture

摘录 / EXCERPT: Atia's conjecture. There is a sequence of positive numbers $(\gamma_n)_{n\geq 1}$ satisfying these conditions such that $(P_n)_{n\geq 0}$ is orthogonal with respect to the weight...

conjecture | open | MathDB 页面 | 原文线索

V2-065935 | MATHDB-344011

Cyclotomic maximum-or-minimum coefficient conjecture

摘录 / EXCERPT: For a cyclotomic polynomial, call a coefficient non-zero when it is not equal to zero. Cyclotomic maximum-or-minimum coefficient conjecture. Every non-zero integer occurs either as...

conjecture | open | MathDB 页面 | 原文线索

V2-065936 | MATHDB-344034

Full-dimensionality conjecture for feasible regions of consecutive permutation patterns

摘录 / EXCERPT: Let clP_k denote the feasible region of density vectors of consecutive permutation patterns of size at most k, and let $\mathcal{L}_k$ be the set of Lyndon permutations of size...

conjecture | open | MathDB 页面 | 原文线索

V2-065937 | MATHDB-344042

Conjecture on infinite-dimensional random attractors under infinite energy mean

摘录 / EXCERPT: Infinite-dimensional attractor conjecture. There are a nonlinearity f and right-hand side g satisfying those assumptions such that $$ mathcal A(...

conjecture | open | MathDB 页面 | 原文线索

V2-065938 | MATHDB-344044

Energy measure singularity dichotomy for MMD spaces

摘录 / EXCERPT: Energy measure singularity dichotomy. Then $\Gamma(f,f) \perp m$ for all $f \in \mathcal{F}$. The conjecture predicts singularity of every energy measure under the stated structura...

conjecture | open | MathDB 页面 | 原文线索

V2-065939 | MATHDB-344048

Sun's conjecture on reciprocal sums over adjacent permutation values

摘录 / EXCERPT: Let n be a positive integer and let $\pi\in\mathfrak{S}_n$ be a permutation of $\{1,\ldots,n\}$. Sun's reciprocal-sum conjecture. If $n>6$, there is a permutation π such that...

conjecture | open | MathDB 页面 | 原文线索

V2-065940 | MATHDB-344049

Sun's conjecture on reciprocal squared differences in a permutation

摘录 / EXCERPT: Let n be a positive integer and let $\pi\in\mathfrak{S}_n$ be a permutation of $\{1,\ldots,n\}$. Sun's squared-difference conjecture. If $n>7$, there is a permutation π such...

conjecture | open | MathDB 页面 | 原文线索

V2-065941 | MATHDB-344050

Cyclic reciprocal squared-difference conjecture for permutations

摘录 / EXCERPT: Let n be a positive integer and let $\pi\in\mathfrak{S}_n$ be a permutation of $\{1,\ldots,n\}$. Cyclic squared-difference conjecture. If $n>11$, there is a permutation π suc...

conjecture | open | MathDB 页面 | 原文线索

V2-065942 | MATHDB-344058

Sharifi's Eisenstein-ideal annihilation conjecture

摘录 / EXCERPT: Let $\widetilde{\varpi}:H_1(X_1(N),C_{\infty},\mathbf Z_p)_+\to\mathcal K$ be the cyclotomic map, and let $\mathfrak I_{\infty}$ be the Eisenstein ideal acting on the source. Shari...

conjecture | open | MathDB 页面 | 原文线索

V2-065943 | MATHDB-344067

The lattice equivalence conjecture for facial weak order

摘录 / EXCERPT: Lattice equivalence conjecture. The poset of regions PR is a lattice if and only if the facial weak order FW is a lattice.

conjecture | open | MathDB 页面 | 原文线索

V2-065944 | MATHDB-344082

Relative gerbe duality conjecture for Gromov-Witten theory

摘录 / EXCERPT: Let $\mathcal{Y}$ be a gerbe with dual $\widehat{\mathcal{Y}}$ and associated $\mathbb{C}^*$-gerbe c. Let $D\subset \mathcal{B}$ be a smooth irreducible divisor, and define its i...

conjecture | open | MathDB 页面 | 原文线索

V2-065945 | MATHDB-344092

Simon's conjecture on primitive groups with near-2^n growth

摘录 / EXCERPT: Simon's near-2^n conjecture. Then G is either $\operatorname{Aut}(S(2))$, or the group of automorphisms and anti-automorphisms of $S(2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-065946 | MATHDB-344105

Finiteness conjecture for automorphism groups of derivations with zero eigenvalue degree

摘录 / EXCERPT: Let D be a nonzero derivation of ${\rm K}[x,y]$, where ${\rm K}$ is the underlying field, and let ${\mathbb e}(D)$ denote the eigenvalue-degree invariant used in the paper. Finit...

conjecture | open | MathDB 页面 | 原文线索

V2-065947 | MATHDB-344109

Noncommutative Iwasawa main conjecture for the Bloch-Kato Selmer group

摘录 / EXCERPT: Let $V\cong\mathcal K^{2d}$ be an even-dimensional p-adic representation of $G_{\mathbb Q}$ satisfying condition $\mathrm{(Geom)}$, with no Artin representation as a subquotient....

conjecture | open | MathDB 页面 | 原文线索

V2-065948 | MATHDB-344116

The super-approximation conjecture for finitely generated linear groups

摘录 / EXCERPT: Super-approximation conjecture. The group Γ has super-approximation, without restriction on the moduli, if and only if the Zariski closure $\mathbb G$ of Γ has perfec...

conjecture | open | MathDB 页面 | 原文线索

V2-065949 | MATHDB-344118

Abdesselam's conjecture on operator product expansions and correlation bounds in conformal field theory

摘录 / EXCERPT: Let I be an index set, let $D_\iota\in[0,d/2)$ for every $\iota\in I$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-065950 | MATHDB-344121

Henselianity conjecture for dp-finite valued fields

摘录 / EXCERPT: Henselianity conjecture. The valuation v is henselian.

conjecture | open | MathDB 页面 | 原文线索

V2-065951 | MATHDB-344126

Vanishing conjecture for squares of ample generators on fake projective planes

摘录 / EXCERPT: Let S be a fake projective plane, let G be an ample generator of $\operatorname{Pic}(S)$, and let σ be a torsion line bundle on S. Vanishing conjecture. One has ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-065952 | MATHDB-344127

Hawk-Dove steady-state convergence conjecture

摘录 / EXCERPT: Hawk-Dove steady-state convergence conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-065953 | MATHDB-344137

Structure conjecture for sRC quotients of smooth klt pairs

摘录 / EXCERPT: Structure conjecture for sRC quotients. There exists an orbifold morphism

conjecture | open | MathDB 页面 | 原文线索

V2-065954 | MATHDB-344138

The spectral edge conjecture for periodic Schrödinger operators

摘录 / EXCERPT: Let $-elta+V$ be a Schrödinger operator on $\mathbb{R}^n$ with periodic potential, and let $\lambda_j(k)$ denote its band functions, with k the quasimomentum. Spectral edges are...

conjecture | open | MathDB 页面 | 原文线索

V2-065955 | MATHDB-344139

Persistence of generic non-degeneracy under parameter extension

摘录 / EXCERPT: Consider a family of periodic operators whose parameters include the periodic potential, and possibly additional parameters such as a metric. Generic non-degeneracy means that dege...

conjecture | open | MathDB 页面 | 原文线索

V2-065956 | MATHDB-344152

Refinement conjecture for Levenshtein-type polynomial nodes

摘录 / EXCERPT: Let $q\geq 3$, let n and ℓ be fixed, and let T_n denote the discrete set of admissible points used in the relaxed linear programs over $[\ell,s]\cap T_n$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-065957 | MATHDB-344155

Perrin-Riou's conjecture on Kato's zeta element and the analytic rank

摘录 / EXCERPT: Let $E/\mathbb Q$ be the elliptic curve, let $r_{\rm an}=\operatorname{ord}_{s=1}L(E,s)$ and $r_{\rm alg}=\operatorname{rank}E(\mathbb Q)$, and let ${}_{c,d}z_{\mathbb Q}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-065958 | MATHDB-344156

The generalized Perrin-Riou conjecture for finite-level Darmon norms

摘录 / EXCERPT: Assume the paper's Hypothesis and set $r=r_{\rm alg}$. Let $F/\mathbb Q$ be the fixed finite real abelian extension, let I_F be the augmentation ideal, let ${}_{c,d}z_F$ be the m...

conjecture | open | MathDB 页面 | 原文线索

V2-065959 | MATHDB-344157

The finite-level Fitting-ideal conjecture for Kato's zeta elements

摘录 / EXCERPT: Let $F/\mathbb Q$ be the fixed finite abelian extension, let $G=\operatorname{Gal}(F/\mathbb Q)$, let I_F be its augmentation ideal, let T be the relevant p-adic representati...

conjecture | open | MathDB 页面 | 原文线索

V2-065960 | MATHDB-344158

The Iwasawa-theoretic generalized Perrin-Riou conjecture

摘录 / EXCERPT: Assume the paper's Hypothesis, set $r=r_{\rm alg}$, and let $\bm{x}$ be a $\mathbb Z_p$-basis of $\bigwedge_{\mathbb Z_p}^{r-1}H^2(\mathbb Z_S,T)_{\rm tf}$. Let κ_∞ be...

conjecture | open | MathDB 页面 | 原文线索

V2-065961 | MATHDB-344160

Double integral relation for a fifth Painlevé transcendent

摘录 / EXCERPT: Let $0<\tilde{\omega}<1/2$ and let $\sigma(s)$ be the solution of the fifth Painlevé transcendent satisfying the equation and boundary conditions specified in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-065962 | MATHDB-344163

Guo-Zudilin q-analogue with half truncation

摘录 / EXCERPT: Let $(a;q)_k=\prod_{j=0}^{k-1}(1-aq^j)$ be the q-shifted factorial, let $\Phi_n(q)$ be the nth cyclotomic polynomial, and let $n>1$ be odd and $r\geqslant 1$. Guo-Zudilin's hal...

conjecture | open | MathDB 页面 | 原文线索

V2-065963 | MATHDB-344164

Partial q-analogues of the Long-Tauraso-Zudilin supercongruences

摘录 / EXCERPT: Let $(a;q)_k=\prod_{j=0}^{k-1}(1-aq^j)$ be the q-shifted factorial, let $\Phi_n(q)$ be the nth cyclotomic polynomial, and let $n>1$ be odd and $r\geqslant 1$. Partial q-analogu...

conjecture | open | MathDB 页面 | 原文线索

V2-065964 | MATHDB-344168

The even-odd-drop and D-cycle correspondence conjecture

摘录 / EXCERPT: Even-odd-drop and D-cycle correspondence conjecture. For every finite subset $A\subset\mathbb{Z}_{>0}$, the number of cycles on A with only even-odd drops is equal to the number...

conjecture | open | MathDB 页面 | 原文线索

V2-065965 | MATHDB-344169

Nevanlinna's generalized volume-capacity conjecture

摘录 / EXCERPT: Let h be a continuous, increasing, concave function defined and positive in some right neighborhood of 0. For a set $E\subset\mathbb{R}$, define its h-volume by ... where ... Ass...

conjecture | open | MathDB 页面 | 原文线索

V2-065966 | MATHDB-344172

p-adic Artin integrality conjecture

摘录 / EXCERPT: Let k be a totally real number field and let $chi:G_k\to\overline{\mathbb Q}_p$ be an even p-adic Artin character. Let $G_chi(T)$ be its Iwasawa power series. p-adic Artin conj...

conjecture | open | MathDB 页面 | 原文线索

V2-065967 | MATHDB-344173

Conjectural Ohno-sum relation for the indices $(4,2)$ and $(2,1,1,2)$

摘录 / EXCERPT: Let $\mathcal{O}$ denote the Ohno sum and let $\widetilde{\mathcyr{sh}}$ denote the interleaving operation on indices. For an integer $s\ge2$, Conjectural Ohno-sum relation. ... This...

conjecture | open | MathDB 页面 | 原文线索

V2-065968 | MATHDB-344174

Generalized dual conjecture for quasiperiodic bobbin lace patterns

摘录 / EXCERPT: Generalized dual conjecture. For every such T, there exists a set of edge directions of T' that satisfies $C0$--$C4$ and thus forms a workable bobbin lace pattern.

conjecture | open | MathDB 页面 | 原文线索

V2-065969 | MATHDB-344175

Ammann lines conjecture for bobbin lace patterns

摘录 / EXCERPT: Ammann lines conjecture. For every set of Ammann lines in general position, there exists a set of edge directions for the associated graph drawing that satisfies $C0$--$C4$. Theref...

conjecture | open | MathDB 页面 | 原文线索

V2-065970 | MATHDB-344180

Bressoud's partition identity conjecture

摘录 / EXCERPT: Let λ, k, r and $j=0$ or 1 be integers such that ... Let $A_j(\alpha_1,\ldots,\alpha_\lambda;\eta,k,r;n)$ count the partitions defined by the stated congruence and repe...

conjecture | open | MathDB 页面 | 原文线索

V2-065971 | MATHDB-344194

The Hitchin-Kostant-Rallis section conjecture for real moduli spaces

摘录 / EXCERPT: Let $\mathfrak{g}_{\mathbb{R}}$ be a real form, let $\hat{\mathcal{T}}_{\mathfrak{g}_{\mathbb{R}}}$ be the moduli space of $\mathfrak{g}_{\mathbb{R}}$-complex structures, and let t...

conjecture | open | MathDB 页面 | 原文线索

V2-065972 | MATHDB-344203

Interval-rank conjecture for finite simple groups

摘录 / EXCERPT: Interval-rank conjecture. The group A_{11} is the only finite simple group whose set of ranks of string C-group representations is not an interval in the set of integers.

conjecture | open | MathDB 页面 | 原文线索

V2-065973 | MATHDB-344206

Three-dimensional Catalan formula for reduced valid hook configurations

摘录 / EXCERPT: Three-dimensional Catalan conjecture. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-065974 | MATHDB-344207

Alternating-sum Catalan conjecture for reduced valid hook configurations

摘录 / EXCERPT: Alternating-sum Catalan conjecture. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-065975 | MATHDB-344208

Weak Bruhat monotonicity conjecture for valid hook configuration counts

摘录 / EXCERPT: Weak Bruhat monotonicity conjecture. If $\sigma,\tau\in S_r$ satisfy $\sigma\leq_B\tau$, then

conjecture | open | MathDB 页面 | 原文线索

V2-065976 | MATHDB-344217

Strong differentiation conjecture for axis-parallel rectangles

摘录 / EXCERPT: Strong differentiation conjecture. If $\widetilde{M}_{\mathcal{B}}f(x)$ is finite almost everywhere, then f is strongly differentiable.

conjecture | open | MathDB 页面 | 原文线索

V2-065977 | MATHDB-344218

Density-basis differentiation conjecture under maximal-operator finiteness

摘录 / EXCERPT: Density-basis differentiation conjecture. If $M_{\mathcal{B}}f(x)<\infty$ almost everywhere, then

conjecture | open | MathDB 页面 | 原文线索

V2-065978 | MATHDB-344219

List-coloring version of Linear Hadwiger's conjecture

摘录 / EXCERPT: Let G be a finite simple graph, and let K_t denote the complete graph on t vertices. A graph is k-list colorable if every assignment of lists of at least k colors to its...

conjecture | open | MathDB 页面 | 原文线索

V2-065979 | MATHDB-344220

Cheltsov's conjecture on Kawamata decompositions of nodal Fano threefolds

摘录 / EXCERPT: Cheltsov's conjecture. The only nodal Fano threefolds of Picard rank one with Kawamata type decompositions are the quadric, V_5, and V_{22}.

conjecture | open | MathDB 页面 | 原文线索

V2-065980 | MATHDB-344222

Smooth transfer conjecture for relative endoscopy of unitary symmetric spaces

摘录 / EXCERPT: Smooth transfer conjecture. For any relative endoscopic datum (ξ,α,β) and any $f\in C_c^\infty(\mathfrak g_1)$, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-065981 | MATHDB-344245

Strict log-concavity conjecture for the rank of strongly unimodal sequences

摘录 / EXCERPT: Let $u(m,n)$ denote the number of strongly unimodal sequences of size n and rank m. Strict log-concavity conjecture. For $n\ge37$ and $|m|\le n-23$, we have ... The conjecture is...

conjecture | open | MathDB 页面 | 原文线索

V2-065982 | MATHDB-344246

Strict log-concavity conjecture for the rank and crank of ordinary partitions

摘录 / EXCERPT: Let $N(m,n)$ and $M(m,n)$ denote, respectively, the rank and crank counting functions for ordinary partitions of n with statistic m. Strict log-concavity conjecture. The follow...

conjecture | open | MathDB 页面 | 原文线索

V2-065983 | MATHDB-344261

Variational characterization conjecture for additive difference equations

摘录 / EXCERPT: Let k be a positive integer, and consider an additive $2k$th-order difference equation ... Here $\mathbf{x}_{n}^{(m,l)}=(x_{n+m},\dots,x_{n+k})$ as defined in the source, and f,...

conjecture | open | MathDB 页面 | 原文线索

V2-065984 | MATHDB-344269

Jørgensen's apex conjecture for 6-connected graphs without K_6 minors

摘录 / EXCERPT: Jørgensen's conjecture. If G is 6-connected and does not have a K_6 minor, then G is apex.

conjecture | open | MathDB 页面 | 原文线索

V2-065985 | MATHDB-344278

Conjecture on logarithmic integrals in \mathcal{I}_{2n+2}

摘录 / EXCERPT: Logarithmic-integral space conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-065986 | MATHDB-344279

Interior-point conjecture for wedges generated by horizontal half-spaces

摘录 / EXCERPT: Let ${\mathbb G}$ be a Carnot group and let W be a horizontal half-space. Denote by S_W the semigroup generated by W, by $\bar S_W$ its closure, by $\mathfrak w_{\bar S_W}$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-065987 | MATHDB-344284

Linear independence conjecture for finite wavelet systems generated by Schwartz functions

摘录 / EXCERPT: Let $\phi\in\mathcal{S}(\mathbb{R})\setminus\{0\}$, and let a finite wavelet system generated by ϕ be a finite collection of wavelet dilates and translates of ϕ. Finite-w...

conjecture | open | MathDB 页面 | 原文线索

V2-065988 | MATHDB-344288

Bipartite sufficient-set conjecture for triangle-free planar graphs

摘录 / EXCERPT: Let G be a triangle-free planar graph, and let X be a subset of $V(G)$ such that the induced subgraph $G[X]$ is bipartite. A subset X is 3-sufficient when G is $f_{X,3}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-065989 | MATHDB-344299

David-Fearnley-Kisilevsky weak conjecture on cubic and quintic twists

摘录 / EXCERPT: Let E be the elliptic curve under consideration, let $\mathcal{C}_p(X)$ denote the even primitive Dirichlet characters of order p and conductor at most X, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-065990 | MATHDB-344300

Finite generation conjecture for elliptic curves over infinite real abelian extensions

摘录 / EXCERPT: Let E be the elliptic curve under consideration, and let $F/\mathbb{Q}$ be an infinite real abelian extension containing only finitely many subfields of degree 2, 3, or 5 o...

conjecture | open | MathDB 页面 | 原文线索

V2-065991 | MATHDB-344301

The asymptotic equidistribution conjecture for singleton double descent sets

摘录 / EXCERPT: Let $dd(I;n)$ denote the number of permutations in the symmetric group on n letters whose double descent set is I. For fixed $0<\alpha<\beta<1$, the asymptotic equidistribution...

conjecture | open | MathDB 页面 | 原文线索

V2-065992 | MATHDB-344302

The down-up-down-up conjecture for singleton double descent counts

摘录 / EXCERPT: Let $dd(I;n)$ denote the number of permutations in the symmetric group on n letters whose double descent set is I. Given a fixed $n\in\mathbb{N}$, the down-up-down-up conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-065993 | MATHDB-344303

The alternating ratio conjecture for singleton double descent counts

摘录 / EXCERPT: Let $dd(I;n)$ denote the number of permutations in the symmetric group on n letters whose double descent set is I. For $n,i\in\mathbb{N}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-065994 | MATHDB-344304

The positive ratio-limit conjecture for singleton double descent counts

摘录 / EXCERPT: Let $dd(I;n)$ denote the number of permutations in the symmetric group on n letters whose double descent set is I. For fixed $i,j\in\mathbb{Z}_{\geq2}$, the positive ratio-limi...

conjecture | open | MathDB 页面 | 原文线索

V2-065995 | MATHDB-344306

The proposed optimal bound for the Milnor-Tjurina quotient in dimension two

摘录 / EXCERPT: Let $(X,0)\subset(\mathbb{C}^3,0)$ be an isolated hypersurface singularity, with Milnor number μ and Tjurina number τ. The conjecture. For every such singularity, the boun...

conjecture | open | MathDB 页面 | 原文线索

V2-065996 | MATHDB-344320

Zhou's orthogonality relation for Maass forms on GL(n)

摘录 / EXCERPT: Fix $n\geq 2$. Let $\{\phi_j\}_{j=1,2,\ldots}$ be an orthogonal basis of Maass cusp forms for $\operatorname{SL}(n,\mathbb Z)$, with Langlands parameters $\alpha^{(j)}$ and normali...

conjecture | open | MathDB 页面 | 原文线索

V2-065997 | MATHDB-344321

Property H conjecture for subdegree-finite primitive permutation groups

摘录 / EXCERPT: Let G be a subdegree-finite primitive permutation group with more than one end. Recall that G has property H if, in its faithful action on its structure tree T, the pointwise...

conjecture | open | MathDB 页面 | 原文线索

V2-065998 | MATHDB-344323

The extended-spectrum characterization of interweaving relations for finite Markov generators

摘录 / EXCERPT: Extended-spectrum interweaving conjecture. There exists an interweaving relation from $(\exp(tL))_{t\geq 0}$ to $(\exp(t\widetilde L))_{t\geq 0}$ if and only if the extended spectr...

conjecture | open | MathDB 页面 | 原文线索

V2-065999 | MATHDB-344332

Minimality conjecture for reachable and observable linear structured systems

摘录 / EXCERPT: Minimality conjecture. If $(\cC(s), \cK(s))$ is $\C^n$--observable and $(\cK(s), \cB(s))$ is $\C^n$--reachable, then there is no lower-order realization, constructible via Petrov--...

conjecture | open | MathDB 页面 | 原文线索

V2-066000 | MATHDB-344333

The contact-process threshold conjecture for random graphs

摘录 / EXCERPT: Let cmu be a degree distribution satisfying the giant component condition, and let $ctilde{\mu}$ be its size-biased distribution. Write $clambda_c^-(\mu)$ and $clambda_c^+(\mu)$...

conjecture | open | MathDB 页面 | 原文线索

V2-066001 | MATHDB-344335

Central limit conjecture for zeros of random Newman and Littlewood polynomials

摘录 / EXCERPT: Let $\mathcal N_n$ and $\mathcal L_n$ denote the relevant families of Newman and Littlewood polynomials of degree n, and let N and $\widetilde N$ count zeros in the open unit d...

conjecture | open | MathDB 页面 | 原文线索

V2-066002 | MATHDB-344339

SHGH conjecture for effective line bundles on blowups of the plane

摘录 / EXCERPT: Let X_m be the blowup of $\operatorname{\mathbb{P}}^2$ at m general points. A line bundle $\operatorname{\mathcal{O}}(D)$ is (-1)-special if D is effective and...

conjecture | open | MathDB 页面 | 原文线索

V2-066003 | MATHDB-344346

Conjecture on the sign of the odd-part index difference

摘录 / EXCERPT: Let $o_{\mathcal{O}}(n)$ and $e_{\mathcal{O}}(n)$ denote the relevant counts of partitions associated with $\mathcal{O}$ having odd and even index, respectively. Sign conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-066004 | MATHDB-344347

Generalized sign conjecture for powers of the odd-part set

摘录 / EXCERPT: Let $o_{\mathcal{O}^d}(n)$ and $e_{\mathcal{O}^d}(n)$ denote the relevant odd- and even-index counts associated with $\mathcal{O}^d$. Generalized sign conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-066005 | MATHDB-344348

Nonnegativity conjecture for a q-Pochhammer reciprocal

摘录 / EXCERPT: For $m\ge 4$, let $(a;q)_\infty=\prod_{j\ge 0}(1-aq^j)$ and $(a,b;q)_\infty=(a;q)_\infty(b;q)_\infty$. Coefficient nonnegativity conjecture. Every coefficient of ... is nonnegative....

conjecture | open | MathDB 页面 | 原文线索

V2-066006 | MATHDB-344349

Eventual coefficient monotonicity conjecture for a q-Pochhammer reciprocal

摘录 / EXCERPT: For $m\ge 4$, define $[q^n]F(q)$ to be the coefficient of q^n in the series $F(q)$. Increasing-sequences-by-m conjecture. There exists a positive integer $N(m)$ such that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-066007 | MATHDB-344354

Gross-Zagier non-vanishing criterion for Stark-Heegner cycles

摘录 / EXCERPT: Let $\chi:\operatorname{Gal}(H_{\mathcal C}/K)\to\mathbb C^\times$ be a character, let $\mathfrak s_\chi$ be the associated Stark-Heegner cycle class, and let $L(f/K,\chi,s)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-066008 | MATHDB-344382

The Standard (2,2)-Conjecture on decomposition into two {1,2}-weight colourable subgraphs

摘录 / EXCERPT: Let a graph be $\{1,2\}$-weight colourable if its edges can be assigned weights from $\{1,2\}$ so that adjacent vertices receive distinct sums of the weights on their incident edge...

conjecture | open | MathDB 页面 | 原文线索

V2-066009 | MATHDB-344389

The no-conjugate-points conjecture for the Zermelo navigation problem with a vortex

摘录 / EXCERPT: No-conjugate-points conjecture. For any $(x_0,\mu)\in M\times\mathbb{R}^{*}$, the conjugate locus is empty.

conjecture | open | MathDB 页面 | 原文线索

V2-066010 | MATHDB-344394

List-size conjecture for Reed-Solomon codes beyond the Johnson radius

摘录 / EXCERPT: For a code of rate R over a finite field, list-decodable from relative radius r with list size L means that every Hamming ball of radius $r n$ contains at most L codewords.…

conjecture | open | MathDB 页面 | 原文线索

V2-066011 | MATHDB-344395

Kosyak et al.'s height conjecture for cyclotomic polynomials

摘录 / EXCERPT: Let m be a natural number. For a polynomial f, its height $h(f)$ is the maximum of the absolute values of its coefficients. Kosyak et al.'s height conjecture. For every natural…

conjecture | open | MathDB 页面 | 原文线索

V2-066012 | MATHDB-344400

Ichino-Ikeda-N. Harris conjecture for unitary groups

摘录 / EXCERPT: Let $\f$ and $\f'$ be factorizable vectors in representations $\c0$ and $\c0'$, respectively. Let $\a$ be the period pairing, ${\overline{\frak b}}$ the global arithmetic factor,…

conjecture | open | MathDB 页面 | 原文线索

V2-066013 | MATHDB-344405

Near-perfect matching conjecture for 5-connected 1-planar graphs

摘录 / EXCERPT: Near-perfect matching conjecture. Every 5-connected 1-planar graph has a matching of size

conjecture | open | MathDB 页面 | 原文线索

V2-066014 | MATHDB-344407

Uniform boundedness conjecture for common preperiodic points of rational maps

摘录 / EXCERPT: Uniform boundedness conjecture. For each degree $d\geq 2$, there exists a constant $B=B(d)$ such that either

conjecture | open | MathDB 页面 | 原文线索

V2-066015 | MATHDB-344409

Joyal's cylinder conjecture

摘录 / EXCERPT: For simplicial sets A and B, let $\mathbf{Cyl}(A,B)$ be the category of cylinders from A to B. Thus an object is a simplicial set X equipped with a map $X\to\Delta[1]$ wh…

conjecture | open | MathDB 页面 | 原文线索

V2-066016 | MATHDB-344414

Reed's list-coloring conjecture

摘录 / EXCERPT: Let G be a graph, and let L be a list-assignment for G. The list-chromatic number $\chi_\ell(G)$ is the smallest integer k such that G is k-list-colorable. Write…

conjecture | open | MathDB 页面 | 原文线索

V2-066017 | MATHDB-344417

Global regularized-period conjecture for Arthur packets

摘录 / EXCERPT: Let $G=G(V)\times G(W)$ over a number field, let $\pi\otimes\pi'$ be an irreducible automorphic representation occurring in $\mathcal{A}_{\rm aut}(G)$, and let $F(\nu)$ denote the…

conjecture | open | MathDB 页面 | 原文线索

V2-066018 | MATHDB-344419

Vector-bundle-injection conjecture for the infinitesimal period map

摘录 / EXCERPT: Let $(f:V\to U,\mathcal{L})\in\mathcal{M}_h(U)$ be the polarized family referred to in Theorem, and let S and T be the loci appearing there. Vector-bundle-injection conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-066019 | MATHDB-344426

Bressan's conjecture for anomalous dissipation with rough velocity fields

摘录 / EXCERPT: Let θ^κ be a solution of the viscous transport equation on $\mathbb{T}^d$ with velocity field $u\in L^1([0,\infty),W^{1,1}(\mathbb{T}^d))$ and smooth initial data…

conjecture | open | MathDB 页面 | 原文线索

V2-066020 | MATHDB-344428

The location conjecture for unique maximum absolute coefficients of q-factorials

摘录 / EXCERPT: Let $(q;q)_n=(1-q)(1-q^2)\cdots(1-q^n)$, and let i denote an exponent at which the maximum absolute coefficient of $(q;q)_n$ occurs. Location conjecture. For $n\geq 35$, if…

conjecture | open | MathDB 页面 | 原文线索

V2-066021 | MATHDB-344429

The bounded-difference conjecture for maximum-coefficient locations

摘录 / EXCERPT: For odd $n\geq 35$, let D_n be the half-difference between the two locations of the maximum absolute coefficients of $(q;q)_n$, and define $E_n=D_n-D_{n-4}$. Bounded-difference c…

conjecture | open | MathDB 页面 | 原文线索

V2-066022 | MATHDB-344431

The explicit periodic formulas conjecture for maximum-coefficient locations

摘录 / EXCERPT: Let $L(n)$ denote the lower exponent at which a maximum absolute coefficient of $(q;q)_n$ occurs. Explicit periodic formulas conjecture. For every $k\geq 0$, the stated formulas ho…

conjecture | open | MathDB 页面 | 原文线索

V2-066023 | MATHDB-344432

The unified difference conjecture for maximum-coefficient locations

摘录 / EXCERPT: For odd $n\geq 35$, define ... where D_n is the half-difference between the two locations of the maximum absolute coefficients of $(q;q)_n$. Unified difference conjecture. If n i...

conjecture | open | MathDB 页面 | 原文线索

V2-066024 | MATHDB-344433

Goldschmidt and Przykucki's conjecture on the mean number of cars

摘录 / EXCERPT: Goldschmidt and Przykucki's conjecture. The expected number of cars arriving at the root is

conjecture | open | MathDB 页面 | 原文线索

V2-066025 | MATHDB-344434

Brešar, Klavžar, Rall, and Wash's packing-coloring conjecture for subdivided subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, meaning that its maximum degree is at most 3. Let $D(G)$ be the graph obtained by subdividing every edge of G, and let $\chi_p(G)$ denote the packi...

conjecture | open | MathDB 页面 | 原文线索

V2-066026 | MATHDB-344437

Russell's conjecture for level 2 modular equations

摘录 / EXCERPT: Russell's conjecture for level 2. There exists a polynomial $P(u,v)$ of degree $(p+1)/2$ in u and v such that

conjecture | open | MathDB 页面 | 原文线索

V2-066027 | MATHDB-344439

Hexagon patch wavenumber-selection conjecture for the planar Swift-Hohenberg equation

摘录 / EXCERPT: Hexagon patch wavenumber-selection conjecture. Every stationary and invading stable hexagon 2D patch solution selects a unique hexagon cellular pattern. Its wavenumbers are determi...

conjecture | open | MathDB 页面 | 原文线索

V2-066028 | MATHDB-344445

Conjectured characterizations and reductions of intuitionistic provability logics of arithmetic

摘录 / EXCERPT: Characterizations and reductions conjecture. The following characterizations and reductions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-066029 | MATHDB-344457

Local root-number formula for C2D4 curves

摘录 / EXCERPT: Local root-number formula. The local root number of the Jacobian satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-066030 | MATHDB-344477

Characterization of the tableaux-graph order by row entries

摘录 / EXCERPT: Let $S,T\in\mathcal{T}(n,n)$ be standard Young tableaux of shape (n,n), and let a_i and b_i denote the entries in the first row of S and T, respectively. Characterization...

conjecture | open | MathDB 页面 | 原文线索

V2-066031 | MATHDB-344479

Eventual hyperbolicity conjecture for Jensen polynomials of k-regular partitions

摘录 / EXCERPT: k-regular Jensen polynomial conjecture. Let $d\geq1$ and $k\geq2$ be integers. There exists a positive integer $N(d)$ such that $J_{p_k}^{d,n}(X)$ is hyperbolic for all...

conjecture | open | MathDB 页面 | 原文线索

V2-066032 | MATHDB-344481

The group-isomorphism conjecture for superelliptic Jacobians over finite fields

摘录 / EXCERPT: Superelliptic Jacobian group-isomorphism conjecture. There exists an isomorphism of abstract groups

conjecture | open | MathDB 页面 | 原文线索

V2-066033 | MATHDB-344484

Conjecture on the unbounded uMAT-MAT DoF gap for any number of cells

摘录 / EXCERPT: uMAT-MAT gap conjecture. For the same scenario, the DoF gap defined above holds for any number of cells C.

conjecture | open | MathDB 页面 | 原文线索

V2-066034 | MATHDB-344485

Existence conjecture for D-optimal rhombic designs

摘录 / EXCERPT: Let K be the dimension, let D be the covariance matrix of the random coefficients, and let $(d_0,d_1,d_2)^T\in\mathcal{C}_K$ denote its parameter vector. A rhombic design is an...

conjecture | open | MathDB 页面 | 原文线索

V2-066035 | MATHDB-344486

Flexible 2-periodic placement characterization for connected gain graphs

摘录 / EXCERPT: Flexible 2-periodic placement characterization. There exists a full placement-lattice (p,L) of G in $\mathbb{R}^2$ such that (G,p,L) is a flexible full 2-periodic framework...

conjecture | open | MathDB 页面 | 原文线索

V2-066036 | MATHDB-344487

Realization conjecture for type 3 flexible 2-lattice colourings

摘录 / EXCERPT: Type 3 realization conjecture. If G has a type 3 flexible 2-lattice NBAC-colouring, then there exists a full placement-lattice (p,L) of G in $\mathbb{R}^2$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-066037 | MATHDB-344488

Weak flexible 2-periodic placement characterization without type 3 colourings

摘录 / EXCERPT: Weak flexible 2-periodic placement characterization. There exists a full placement-lattice (p,L) of G in $\mathbb{R}^2$ such that (G,p,L) is a flexible full 2-periodic fram...

conjecture | open | MathDB 页面 | 原文线索

V2-066038 | MATHDB-344490

Conjecture on Higgs branch Schur operators under KK reduction

摘录 / EXCERPT: Higgs-branch survival conjecture. The vector space of Higgs branch Schur operators is not affected by the KK reduction.

conjecture | open | MathDB 页面 | 原文线索

V2-066039 | MATHDB-344504

Noncontainment and strict growth of affine involution symmetric-function spans

摘录 / EXCERPT: Let n be a positive integer, and let ... For $n=2$, set $\tilde{\Omega}_1=\mathbb{Z}$. The span-growth conjecture. If n is even, then ... This predicts both strict growth at the ad...

conjecture | open | MathDB 页面 | 原文线索

V2-066040 | MATHDB-344505

Noncontainment of affine fixed-point-free involution symmetric-function spans

摘录 / EXCERPT: The fixed-point-free span conjecture. If n is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-066041 | MATHDB-344510

Fourth-moment conjecture for character twists over number fields

摘录 / EXCERPT: Let F be a number field, let chi be a unitary character of $\mathbb{A}^{\times}$, and let $B=(B_\mathfrak{p})_{\mathfrak{p}}$ be real numbers indexed by the places of F, with...

conjecture | open | MathDB 页面 | 原文线索

V2-066042 | MATHDB-344526

The Hilbert-cube Overlap Gap Property conjecture for even p-spin tensors

摘录 / EXCERPT: Hilbert-cube OGP conjecture. For every even $p\geq4$, there exist $\mu>0$ and $0<\nu_1<\nu_2<1$ such that $\mathcal{A}$ satisfies the OGP with domain $\mathcal{S}_N=\mathcal{H}_N$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066043 | MATHDB-344527

Near-optimal Hilbert-cube points are close to binary points

摘录 / EXCERPT: Let $A\in(\mathbb{R}^N)^{\otimes p}$ have independent entries distributed as $\mathcal{N}(0,N^{-(p-1)})$, and let $\mathcal{B}_N\subset\mathcal{H}_N$ denote the binary and Hilbert-...

open_problem | open | MathDB 页面 | 原文线索

V2-066044 | MATHDB-344537

The inner-product characterization of equivalent frames

摘录 / EXCERPT: Equivalent-frame inner-product conjecture. F and G are equivalent if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-066045 | MATHDB-344544

Stability characterization for bounded-linear-rankwidth graph classes

摘录 / EXCERPT: Let $\mathscr{C}$ be a class of graphs with bounded linear rankwidth. A first-order transduction is a graph interpretation obtained using first-order formulas. The class...

conjecture | open | MathDB 页面 | 原文线索

V2-066046 | MATHDB-344547

Getz-Wambach conjecture for unitary Friedberg-Jacquet periods

摘录 / EXCERPT: Let E/F be a quadratic extension of number fields, let W_1 and W_2 be n-dimensional Hermitian spaces over E, and suppose that $U(W_1\oplus W_2)(\mathbb A_F)$ is quasi-spl...

conjecture | open | MathDB 页面 | 原文线索

V2-066047 | MATHDB-344570

The critical-density conjecture for zero-temperature Glauber dynamics on integer lattices

摘录 / EXCERPT: Let $p_c(G)$ be the infimum of the densities $p\in[0,1]$ for which zero-temperature Glauber dynamics on G, started from i.i.d. $\operatorname{Bernoulli}(p)$ opinions, converges a...

conjecture | open | MathDB 页面 | 原文线索

V2-066048 | MATHDB-344571

The marginal monotonicity conjecture for median dynamics

摘录 / EXCERPT: Let G be a graph, let $x\in G$, and let $(\eta_t)_{t\geq0}$ denote median dynamics with i.i.d. uniform initial opinions on $[0,1]$. For $\alpha\in[0,1]$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-066049 | MATHDB-344573

The fixation conjecture for noncritical majority dynamics

摘录 / EXCERPT: Let $(\eta_t)_{t\geq0}$ be median dynamics and, for an initial density $p\in[0,1]$, let $(\xi_t^p)_{t\geq0}$ be the coupled majority-dynamics process. Say that majority dynamics fi...

conjecture | open | MathDB 页面 | 原文线索

V2-066050 | MATHDB-344574

The stable-structure tail conjecture for median dynamics on the square lattice

摘录 / EXCERPT: Consider median dynamics on $\mathbb{Z}^2$, with $\eta_\infty(0)$ denoting the limiting opinion at the origin. Stable-structure tail conjecture. ... The exponent 4 comes from the s...

conjecture | open | MathDB 页面 | 原文线索

V2-066051 | MATHDB-344575

The almost-sure coin-median convergence conjecture for majority dynamics with coins

摘录 / EXCERPT: Let $\operatorname{MD}_{\mathrm{coins}}$ denote the coin-valued median dynamics process on $\mathbb{Z}^d$, and let $\eta_t(\underline{0})$ be the opinion at the origin. Coin-median...

conjecture | open | MathDB 页面 | 原文线索

V2-066052 | MATHDB-344578

The abelian-closure conjecture for Property (N) of elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve defined over a number field K. The field K^{ab} is the compositum of all abelian extensions of K, and Property $\mathrm{(N)}$ is the pro...

conjecture | open | MathDB 页面 | 原文线索

V2-066053 | MATHDB-344588

Hasse principle conjecture for hypersurfaces in products of projective spaces

摘录 / EXCERPT: Let d_i,n_i be positive integers for $i\in\{1,\ldots,k\}$, and consider multidegree $(d_1,\ldots,d_k)$ hypersurfaces in $\mathbb{P}^{n_1}\times\cdots\times\mathbb{P}^{n_k}$ over...

conjecture | open | MathDB 页面 | 原文线索

V2-066054 | MATHDB-344589

Conjecture on eigenvalues of the reduced spectral covariance matrix

摘录 / EXCERPT: Let $\lGamma_{yy}$ and $\bGamma_{yy|x}$ be the matrices arising from the single-regression Granger-causality estimator, and let λ_i denote the eigenvalues of...

conjecture | open | MathDB 页面 | 原文线索

V2-066055 | MATHDB-344592

The clustering relation for minority voter populations

摘录 / EXCERPT: Let a voter distribution have a minority voter population, let ClusP denote its clustering measure, and let $\mathbb{E}(\mathsf{Rep}_\Delta)$ denote the expected represe...

conjecture | open | MathDB 页面 | 原文线索

V2-066056 | MATHDB-344593

The triangle-free Hedetniemi conjecture

摘录 / EXCERPT: Let G and H be finite undirected simple triangle-free graphs, meaning that neither graph contains a copy of K_3. Let $G\times H$ denote their tensor product and let $\chi(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-066057 | MATHDB-344594

Orlov's block conjecture for categories of singularities of invertible polynomials

摘录 / EXCERPT: Orlov's block conjecture. This category admits a full strongly exceptional collection consisting of at most $n+1$ blocks, where each block consists of pairwise orthogonal objects....

conjecture | open | MathDB 页面 | 原文线索

V2-066058 | MATHDB-344595

Exceptional-collection conjecture for coordinate subspaces of invertible hypersurfaces

摘录 / EXCERPT: Let $w\in \mathbf{k}[x_1,\dots,x_n]$ be an invertible polynomial, and consider the hypersurface $\{w=0\}$ in $\mathbb{A}_{\mathbf{k}}^n$. Let L_w be the maximal grading group, an...

conjecture | open | MathDB 页面 | 原文线索

V2-066059 | MATHDB-344622

The distinguishing-index conjecture for connected finite regular graphs

摘录 / EXCERPT: Let G be a connected, finite, regular graph, and let $D'(G)$ denote the least number of colours needed to colour the edges of G so that the only colour-preserving automorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-066060 | MATHDB-344640

Generalized extinction scenario for positive symmetric matrices

摘录 / EXCERPT: Generalized extinction conjecture. With a suitable modification of condition $$, the conclusions of Theorem should hold for a general positive symmetric matrix ${\bf A}$.

conjecture | open | MathDB 页面 | 原文线索

V2-066061 | MATHDB-344641

Quasi-stationary concentration and extinction-rate conjecture

摘录 / EXCERPT: Quasi-stationary distribution conjecture. Then:

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V2-066062 | MATHDB-344642

Quasi-stationary extinction-event conjecture

摘录 / EXCERPT: Quasi-stationary extinction-event conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066063 | MATHDB-344643

Logarithmic quasi-stationary extinction-rate conjecture

摘录 / EXCERPT: Logarithmic extinction-rate conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066064 | MATHDB-344659

Finiteness conjecture for eigenvalues of the Neumann-Poincaré operator on planar corners

摘录 / EXCERPT: Let Γ be a piecewise C^3 Jordan curve with finitely many corners, whose angles lie strictly between 0 and π. Let $K \colon H^{1/2}(\Gamma) \to H^{1/2}(\Gamma)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-066065 | MATHDB-344661

Degree-distribution convergence to BB(m) in regime R2

摘录 / EXCERPT: Let $m\in\mathbb{N}$, let G_t be a graph in Regime $R2$, and let $\mathbf{deg}_t$ be its empirical degree distribution. Let $\mathbf{deg}_t^{BB(m)}$ be the empirical degree distr...

conjecture | open | MathDB 页面 | 原文线索

V2-066066 | MATHDB-344662

Linear-time threshold for condensation in regime R3

摘录 / EXCERPT: Let $m(t)$ denote the time-dependent memory parameter in Regime $R3$, let μ_ϵ be the mean of the fitness increments, and let the cumulative fitness distribution be the d...

conjecture | open | MathDB 页面 | 原文线索

V2-066067 | MATHDB-344664

Fraughnaugh and Locke's 2-connectivity conjecture for triangle-free subcubic graphs

摘录 / EXCERPT: Let G be a 2-connected triangle-free subcubic graph on n vertices. Fraughnaugh and Locke's conjecture. ... This conjecture strengthens the known lower bounds for connected tria...

conjecture | open | MathDB 页面 | 原文线索

V2-066068 | MATHDB-344665

Bajnok and Brinkmann's six-exception conjecture for triangle-free subcubic graphs

摘录 / EXCERPT: Let F_{11}, $F_{14}^{(1)}$, $F_{14}^{(2)}$, $F_{19}^{(1)}$, $F_{19}^{(2)}$, and F_{22} denote the six specified forbidden graphs. Bajnok and Brinkmann's conjecture. There are e...

conjecture | open | MathDB 页面 | 原文线索

V2-066069 | MATHDB-344666

Fraughnaugh and Locke's six-forbidden-graphs conjecture

摘录 / EXCERPT: Let G be a triangle-free subcubic graph, and let F_{11}, $F_{14}^{(1)}$, $F_{14}^{(2)}$, $F_{19}^{(1)}$, $F_{19}^{(2)}$, and F_{22} be the six specified graphs. Fraughnaugh a...

conjecture | solved | MathDB 页面 | 原文线索

V2-066070 | MATHDB-344667

The six-forbidden-graphs fractional chromatic conjecture

摘录 / EXCERPT: Let G be a triangle-free subcubic graph, and let F_{11}, $F_{14}^{(1)}$, $F_{14}^{(2)}$, $F_{19}^{(1)}$, $F_{19}^{(2)}$, and F_{22} be the six specified graphs. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066071 | MATHDB-344668

Kadell's orthogonality conjecture

摘录 / EXCERPT: Let $a=(a_0,\dots,a_n)$ be a sequence of non-negative integers, let $x^{(a)}$ be the alphabet defined by ... and write $|a|=a_0+\cdots+a_n$. For a composition $v=(v_0,\dots,v_n)$, wr...

conjecture | open | MathDB 页面 | 原文线索

V2-066072 | MATHDB-344671

The general Grassmannian reconstruction conjecture

摘录 / EXCERPT: General Grassmannian reconstruction conjecture. For any $1\leq r<n$, the morphism

conjecture | open | MathDB 页面 | 原文线索

V2-066073 | MATHDB-344675

The quasi-projective moduli conjecture for polystable fibrations

摘录 / EXCERPT: Consider fibrations over a fixed base, allowing the total space or fibres to be possibly singular, and impose the paper’s notion of polystability. Quasi-projective moduli conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-066074 | MATHDB-344689

Root conjecture for 2-twist spins of torus knots

摘录 / EXCERPT: Let $T(2,p)$ denote the $(2,p)$-torus knot, and let K_p be its 2-twist spin. A knot is a root of the ribbon concordance graph if no distinct knot ribbon-concords to it from below...

conjecture | open | MathDB 页面 | 原文线索

V2-066075 | MATHDB-344690

Successive-maxima conjecture for non-genus class numbers of abelian fields

摘录 / EXCERPT: Successive-maxima conjecture. (i) As f increases from its minimum, the successive maxima of $C_{\varepsilon}({\mathcal F}_{G,f}^s)$, $C'_{\varepsilon}({\mathcal F}'{}_{G,f}^s)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066076 | MATHDB-344692

Generalized correlation-immunity bound for perfect 2-colorings of Hamming graphs

摘录 / EXCERPT: Generalized correlation-immunity bound. If there exists a (b,c)-coloring in $H(n,q)$ with $b'+c'>q$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066077 | MATHDB-344693

Generalized face-uniformity bound for two-valued functions on Hamming graphs

摘录 / EXCERPT: Let $f: \{0,\ldots,q-1\}^n \rightarrow \{0,1\}$ be a function with B zeros and C ones, where $q\geq 3$. Suppose that for some k, every k-face has the same number ... of ones...

conjecture | open | MathDB 页面 | 原文线索

V2-066078 | MATHDB-344694

Admissibility criterion for perfect 2-colorings of Hamming graphs

摘录 / EXCERPT: For fixed parameters q, b, and c, call (q,b,c) admissible if there is an integer $n_0=n_0(b,c;q)$ such that a (b,c)-coloring of $H(n,q)$ exists if and only if $n\geq n_0$...

conjecture | open | MathDB 页面 | 原文线索

V2-066079 | MATHDB-344699

The edge-contraction lower-bound conjecture for dominated chromatic number

摘录 / EXCERPT: Let G be a graph and let e be an edge of G. Write G/e for the graph obtained by contracting e, and let $\chi_{dom}(G)$ denote the dominated chromatic number of G. Edge-...

conjecture | open | MathDB 页面 | 原文线索

V2-066080 | MATHDB-344700

Siegel's conjecture on intersections of Hirsch polytopes with cubes

摘录 / EXCERPT: Siegel's Conjecture. If H satisfies the Hirsch Conjecture and C is a cube, then $H\cap C$ satisfies the Hirsch Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066081 | MATHDB-344703

Non-idealness conjecture for sufficiently wise-intersecting clutters

摘录 / EXCERPT: Non-idealness conjecture. There exists an integer $k\geq 4$ such that every k-wise intersecting clutter is non-ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-066082 | MATHDB-344704

Seymour's quarter-integral packing conjecture for ideal clutters

摘录 / EXCERPT: Seymour's quarter-integral packing conjecture. Every ideal clutter $\mathcal{C}$ has a $\frac14$-integral packing of value $\tau(\mathcal{C})$.

conjecture | open | MathDB 页面 | 原文线索

V2-066083 | MATHDB-344705

Cube-ideal set conjecture for bounded non-agreement certificates

摘录 / EXCERPT: Cube-ideal set conjecture. There exists an integer $k\geq4$ such that for every cube-ideal set, either all the points agree on a coordinate, or there is a subset of at most k poi...

conjecture | open | MathDB 页面 | 原文线索

V2-066084 | MATHDB-344708

Sufficient stability condition for redundancy-d systems with processor sharing

摘录 / EXCERPT: Let a redundancy-d system use processor sharing (PS), and let job sizes have general distributions. Write $\tilde{\rho}$ for the corresponding system load parameter. Sufficient s...

conjecture | open | MathDB 页面 | 原文线索

V2-066085 | MATHDB-344716

The non-closed-convexity conjecture of Goldrup and Phillipson

摘录 / EXCERPT: Let $\mathcal{C}$ be a max-intersection incomplete open convex code, where $\Delta(\mathcal{C})$ has at least two non-mandatory codewords not contained in $\mathcal C$. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-066086 | MATHDB-344717

The seven-neuron conjecture for locally good codes

摘录 / EXCERPT: A code is locally good if every link of every codeword is contractible. Seven-neuron conjecture. Every locally good code on at most 7 neurons is open convex or closed convex. The...

conjecture | open | MathDB 页面 | 原文线索

V2-066087 | MATHDB-344723

Somos 4 conjecture for a quadratic A-matrix

摘录 / EXCERPT: Somos 4 conjecture. The Hankel transform of the expansion of $f(x)/x$ is a

conjecture | open | MathDB 页面 | 原文线索

V2-066088 | MATHDB-344742

Coefficient-positivity reformulation for rational q-Catalan parity unimodality

摘录 / EXCERPT: For positive integers m,n, define ... and, for a polynomial $P(q)$ of degree d, define its normalization by ... Let PT_q extract the terms with positive powers of...

conjecture | open | MathDB 页面 | 原文线索

V2-066089 | MATHDB-344743

Generating-function reformulation of rational q-Catalan parity unimodality

摘录 / EXCERPT: For integers $m>r\geq 0$, define the coefficient-extraction operator ... and ... Let PT_q extract positive powers of q. Generating-function reformulation. If $m>r$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-066090 | MATHDB-344744

Generalized generating-function positivity conjecture for rational q-Catalan polynomials

摘录 / EXCERPT: For integers $m>r\geq0$, let $d=\gcd(m,r)$, define $X_{m,r}$ by ... and let ... Let PT_q extract positive powers of q. Generalized generating-function positivity con...

conjecture | open | MathDB 页面 | 原文线索

V2-066091 | MATHDB-344749

The intersection-theoretic formulas for Siegel-Veech constants and Lyapunov exponents

摘录 / EXCERPT: The Siegel-Veech and Lyapunov intersection formulas. The area Siegel-Veech constant and the sum of Lyapunov exponents are

conjecture | open | MathDB 页面 | 原文线索

V2-066092 | MATHDB-344787

Quantitative Hausdorff measure conjecture for singular strata of Alexandrov spaces

摘录 / EXCERPT: Let $n\in\mathds{N}$, let k be a nonnegative integer, and let $(X,p)\in\operatorname{Alex}^n(-1)$, where $\mathcal{S}^k_\epsilon$ denotes the ϵ-singular stratum. Hausdor...

conjecture | open | MathDB 页面 | 原文线索

V2-066093 | MATHDB-344788

Summable Hausdorff measure conjecture for singular strata of Alexandrov spaces

摘录 / EXCERPT: Let $n\in\mathds{N}$, let k be a nonnegative integer, let $(X,p)\in\operatorname{Alex}^n(-1)$, and set ... Here $\mathcal{S}^k_{\epsilon_i}$ denotes the ϵ_i-singular strat...

conjecture | open | MathDB 页面 | 原文线索

V2-066094 | MATHDB-344789

Martinez and Savage's Wilf-equivalence conjecture for inversion sequences

摘录 / EXCERPT: Let $\mathbf I_n(\rho_1,\rho_2,\rho_3)$ denote the set of inversion sequences of length n avoiding the indicated patterns or relations. Martinez and Savage's conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066095 | MATHDB-344790

Unbalanced Wilf-equivalence conjecture for inversion-sequence pattern pairs

摘录 / EXCERPT: For a tuple of patterns, write $(\rho_1,\ldots,\rho_k)\thicksim(\sigma_1,\ldots,\sigma_\ell)$ when the corresponding inversion-sequence avoidance classes are Wilf-equivalent, meani...

conjecture | open | MathDB 页面 | 原文线索

V2-066096 | MATHDB-344791

Jun Ma and Lin's enumeration conjecture for inversion sequences avoiding 0012

摘录 / EXCERPT: Let $\mathbf I_n(0012)$ denote the set of inversion sequences of length n avoiding the pattern 0012. Jun Ma and Lin's conjecture. For $n\geq 1$, we have ... In other words, the u...

conjecture | open | MathDB 页面 | 原文线索

V2-066097 | MATHDB-344792

Log-concavity conjecture for sextet polynomials

摘录 / EXCERPT: Let H be a hexagonal system, and let its sextet polynomial be the polynomial whose coefficients count sextet-related structures of H. Log-concavity conjecture for sextet polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-066098 | MATHDB-344793

Zhang-Zhang's unimodality conjecture for Clar covering polynomials

摘录 / EXCERPT: Let H be a hexagonal system, and let its Clar covering polynomial be ... where $c(H,k)$ is the number of Clar covers of H having precisely k hexagons. Zhang-Zhang's unimodality...

conjecture | open | MathDB 页面 | 原文线索

V2-066099 | MATHDB-344794

Log-concavity conjecture for Clar covering polynomials

摘录 / EXCERPT: Let H be a hexagonal system, and let its Clar covering polynomial be ... where $c(H,k)$ is the number of Clar covers of H having precisely k hexagons. Log-concavity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-066100 | MATHDB-344799

Criticality conjecture for interlacing edges of large Schrijver graphs

摘录 / EXCERPT: Let d be any integer, let k_d be a threshold depending on d, and let ${\rm SG}(2k+d,k)$ be the Schrijver graph. An edge is interlacing according to the standard interlacing-e...

conjecture | open | MathDB 页面 | 原文线索

V2-066101 | MATHDB-344808

The three-weight sum-set characterization for odd sums

摘录 / EXCERPT: The three-weight sum-set conjecture. Ω is an s-sum set for some $s>3$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-066102 | MATHDB-344809

The middle-weight conjecture for three-weight codes

摘录 / EXCERPT: Middle-weight conjecture. Then $w_2=n$.

conjecture | open | MathDB 页面 | 原文线索

V2-066103 | MATHDB-344824

Ingalls-Kuznetsov equivalence conjecture for Artin-Mumford quartic double solids

摘录 / EXCERPT: Let Y be a general Artin-Mumford quartic double solid, let Y' be its blow-up at its 10 singular points, and let X be the Enriques surface associated to Y. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066104 | MATHDB-344825

Movable Bend-and-Break conjecture for sections

摘录 / EXCERPT: Movable Bend-and-Break conjecture. There is a constant $Q=Q(\mathcal{X})$ such that every movable section C satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-066105 | MATHDB-344826

Dominant-families conjecture for sections of del Pezzo fibrations

摘录 / EXCERPT: Dominant-families conjecture. There is some translate $\mathcal{T}$ of Nef_{λ} in N_{λ} such that every numerical class in $\mathcal{T}_{\mathbb{Z}}$ is re...

conjecture | open | MathDB 页面 | 原文线索

V2-066106 | MATHDB-344841

A strengthened truncated pentagonal number series positivity conjecture

摘录 / EXCERPT: The strengthened truncated pentagonal positivity conjecture. Its coefficients are zero for $0\leq n<k(3k+1)/2$ and for $n=k(3k+1)/2+1$, while the coefficients for $n=k(3k+1)/2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066107 | MATHDB-344842

Linear inequalities for the partition function from truncated pentagonal series

摘录 / EXCERPT: Let $A(n)$ denote the partition function introduced in the paper, and let $\delta(n)$ denote the quantity appearing in the source. For an integer $k>0$, define the finite alternati...

conjecture | open | MathDB 页面 | 原文线索

V2-066108 | MATHDB-344843

Mercer's two-product non-negativity conjecture

摘录 / EXCERPT: Mercer's non-negativity conjecture. The coefficients of both series are non-negative. The source recalls this conjecture because it has a similar appearance to the preceding trunca...

conjecture | open | MathDB 页面 | 原文线索

V2-066109 | MATHDB-344844

The D-regular graphs largest-component large-deviation conjecture

摘录 / EXCERPT: Let $D\in\mathbb{N}$ with $D\geq 3$, and suppose that $p_k=0$ and $n_k=0$ for $k\ne D$, while $p_D=1$ and $n_D=n$. For each $x\in[0,1]$, let $q_D^{(x)}=x$, let $q_k^{(x)}=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-066110 | MATHDB-344850

A generalized Hawaii conjecture for real zeros of differential rational functions

摘录 / EXCERPT: Generalized Hawaii conjecture. If $\varkappa>\dfrac{n-1}{n}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066111 | MATHDB-344855

Factorial-basis positivity conjecture for shifted Jack polynomials

摘录 / EXCERPT: Let μ be a partition, let $P_\mu^\#(x_1,\dots,x_n)$ be the shifted Jack polynomial, let H_μ be the usual Jack hook product, and let $\ell(\mu)$ denote the length of μ....

conjecture | open | MathDB 页面 | 原文线索

V2-066112 | MATHDB-344857

Characterization of rational multiples of powers of pi among even-weight relations

摘录 / EXCERPT: Let k be an even integer with $k\geqslant 2$. Let $\mathcal{A}_k^{\rm even}$ be the set of even-weight admissible compositions, let ${\bf Z}^{\mathcal{A}_k^{\rm even}}$ be the co...

conjecture | open | MathDB 页面 | 原文线索

V2-066113 | MATHDB-344875

Asymptotic locating chromatic number conjecture for regular palms

摘录 / EXCERPT: Let $S_n(k)$ be the regular palm with n arms, each having length k, and let $\chi_L(S_n(k))$ denote its locating chromatic number. Asymptotic locating chromatic number conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-066114 | MATHDB-344888

The 2m diagrams attain the lower trefoil and upper unknot bounds

摘录 / EXCERPT: Let a free $2m$ diagram denote the free closure of the $2m$-tangle referred to in the source. A free diagram with m crossings is considered sufficiently complicated as in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-066115 | MATHDB-344890

Non-foil free knot diagrams realize the figure-eight knot

摘录 / EXCERPT: Let K be a minimal free knot diagram with at least four crossings that is not a foil or a connected sum of solely foil knots. An assignment of crossings resolves every free cross...

conjecture | open | MathDB 页面 | 原文线索

V2-066116 | MATHDB-344894

The 21n diagrams have the proposed resultant classification

摘录 / EXCERPT: A free $21n$ knot is the free knot diagram formed from the 21-tangle and a free n-tangle, as used in the source. Resultants are knots obtained by assigning all free crossings....

conjecture | open | MathDB 页面 | 原文线索

V2-066117 | MATHDB-344897

Conjecture equating Calogero-Moser cellular and Leclerc-Miyachi constructible characters

摘录 / EXCERPT: Cellular-constructible character conjecture. There exists an explicit choice of parameter cc such that the set of Calogero-Moser cc-cellular characters and the set of Leclerc-M...

conjecture | open | MathDB 页面 | 原文线索

V2-066118 | MATHDB-344901

Grassmann-necklace containment conjecture for covers of positroids

摘录 / EXCERPT: Let P_n be the poset of positroids on $[n]$, ordered by the quotient relation. For $\tau_1,\tau_2 otin P_n$? Let $\tau_1,\tau_2 otin P_n$ be such that...

conjecture | open | MathDB 页面 | 原文线索

V2-066119 | MATHDB-344906

Eigenvalue-determination conjecture for sandpile groups

摘录 / EXCERPT: Let M be a generating multiset for a Cayley graph, and suppose the greatest common divisor of all generator multiplicities in M is 1. Eigenvalue-determination conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-066120 | MATHDB-344908

Doubling conjecture for Sylow-2 sandpile groups of consecutive hypercubes

摘录 / EXCERPT: For $n\geq 1$, let Q_m denote the m-dimensional hypercube and let $\operatorname{Syl}_2(K(Q_m))$ denote the Sylow-2 component of its sandpile group. Doubling conjecture. ... Eq...

conjecture | open | MathDB 页面 | 原文线索

V2-066121 | MATHDB-344912

Linear independence of shifted log-sine integrals

摘录 / EXCERPT: Let $\operatorname{SLs}(k_{1},\dots,k_{n})$ denote a shifted log-sine integral, with $m,n\geq 0$ and $k_i\geq 2$. Linear-independence conjecture. The elements of ... are $\mathbb{Q}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066122 | MATHDB-344913

Multiple zeta values generated by shifted log-sine integrals

摘录 / EXCERPT: For $k\in\mathbb{Z}_{\geq 0}$ and $d\in\mathbb{Z}_{\geq 0}$, let ... and let a multiple zeta value have weight k and depth d. Generation conjecture. Every multiple zeta value wit...

conjecture | open | MathDB 页面 | 原文线索

V2-066123 | MATHDB-344914

Shifted log-sine basis conjecture for multiple zeta values

摘录 / EXCERPT: Shifted log-sine basis conjecture. The set of real numbers $S'_{k,k}$ is a basis of $\mathcal{Z}_{k}$. If true, this would imply Zagier's dimension conjecture for multiple zeta val...

conjecture | open | MathDB 页面 | 原文线索

V2-066124 | MATHDB-344915

Multiple Clausen and Glaisher values generated by shifted log-sine integrals

摘录 / EXCERPT: For $k,d\in\mathbb{Z}_{\geq 0}$, define ... ... Let $\mathcal{C}_k$ and $\mathcal{G}_k$ be the spaces spanned by multiple Clausen and multiple Glaisher values of weight k, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-066125 | MATHDB-344916

Iterated log-sine basis conjecture at pi over 3

摘录 / EXCERPT: Let $\mathcal{L}^{o}_{k}$ and $\mathcal{L}^{e}_{k}$ denote the spaces of iterated log-sine integrals at $\pi/3$ of weight k with odd and even...

conjecture | open | MathDB 页面 | 原文线索

V2-066126 | MATHDB-344917

The parity conjecture for disconnected 4-coloring complexes of planar triangulations

摘录 / EXCERPT: Parity conjecture. If $B(T)$ has at least two components, then $B(T)$ has a component of even parity and a component of odd parity.

conjecture | open | MathDB 页面 | 原文线索

V2-066127 | MATHDB-344921

The existence conjecture for p-adic Asai L-functions

摘录 / EXCERPT: The existence conjecture for p-adic Asai L-functions. There exists an element

conjecture | open | MathDB 页面 | 原文线索

V2-066128 | MATHDB-344929

Dunfield's orderable Dehn filling conjecture for two-bridge knots

摘录 / EXCERPT: Dunfield's conjecture. For every rational slope $r\in I$, Dehn filling K along slope r is orderable.

conjecture | open | MathDB 页面 | 原文线索

V2-066129 | MATHDB-344938

Conjecture on monotone catenary degree for generalized arithmetic monoids with gcd one

摘录 / EXCERPT: Let M be the generalized arithmetic monoid under consideration, with parameters h and d, and let $c(M)$, $c_{mon}(M)$, and $c_{eq}(M)$ denote its regular, monotone, and equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-066130 | MATHDB-344939

Conjecture on adjacent factorizations in generalized arithmetic monoids with gcd one

摘录 / EXCERPT: Let M be the generalized arithmetic monoid under consideration, let $m\in M$, and let $\mathcal Z(m)$ be its set of factorizations. For factorizations $z,z'\in\mathcal Z(m)$, wri...

conjecture | open | MathDB 页面 | 原文线索

V2-066131 | MATHDB-344943

Hiraga-Ichino-Ikeda conjecture for unipotent representations

摘录 / EXCERPT: Let F be a nonarchimedean local field, let G be the group of F-points of a reductive group, and let $P=MU$ be a parabolic subgroup with Levi factor M. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066132 | MATHDB-344954

Modified Grinenko conjecture on birational rigidity of del Pezzo fibrations

摘录 / EXCERPT: Let d be an integer with $1\leq d\leq 3$. Let $X\to\mathbb P^1$ be a $\mathfrak D$-semistable del Pezzo fibration of degree d, and assume that it satisfies the K-condition, m...

conjecture | open | MathDB 页面 | 原文线索

V2-066133 | MATHDB-344955

The cyclic version of Matsumoto's theorem for infinite parabolic closure

摘录 / EXCERPT: Cyclic Matsumoto conjecture. The CVMT holds for a W-conjugacy class as long as the parabolic closure $W_{\operatorname{supp}(w)}$ of its minimal elements contains only infinite i...

conjecture | open | MathDB 页面 | 原文线索

V2-066134 | MATHDB-344966

Bourgain-Sarnak-Rudnick-type conjecture for shrinking almost-round regions

摘录 / EXCERPT: Bourgain-Sarnak-Rudnick-type conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066135 | MATHDB-344971

The generating-level conjecture for the linear free divisors of type D_n

摘录 / EXCERPT: Let $D\subset X=\mathbb{C}^{4n-10}$ be the linear free divisor D_n. The Hodge filtration on $\mathcal{O}_X(*D)$ is generated at level k when ... for every $l\geq 0$. Generating-l...

conjecture | open | MathDB 页面 | 原文线索

V2-066136 | MATHDB-344977

Seo-Yee nonnegativity conjecture for a nonmodular infinite product

摘录 / EXCERPT: Throughout, use the q-series notation ... and $(A,B,\ldots,C;q)_\infty:=(A;q)_\infty(B;q)_\infty\cdots(C;q)_\infty$. Seo-Yee's nonnegativity conjecture. The series expansion of ... h...

conjecture | open | MathDB 页面 | 原文线索

V2-066137 | MATHDB-344978

Zero distribution conjecture for v-adic multiple zeta values over $\mathbb{F}_q(t)$

摘录 / EXCERPT: Zero distribution conjecture. The v-adic multiple zeta value satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-066138 | MATHDB-344981

Thakur's linear independence conjecture for function-field multiple zeta values

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$ be a rational function field, and let $\zeta(\mathbf{s})$ denote the multiple zeta values at positive-integer tuples $\mathbf{s}$. Thakur's linear independe...

conjecture | open | MathDB 页面 | 原文线索

V2-066139 | MATHDB-344982

The characteristic-two spanning conjecture for v-adic multiple zeta relations

摘录 / EXCERPT: Let $K=\mathbb{F}_q(t)$, and let $W(K)$ be the product over monic primes v of the vector spaces $W_v(K)$ spanned over the relevant coefficient field by the v-adic multiple zeta...

conjecture | open | MathDB 页面 | 原文线索

V2-066140 | MATHDB-344986

Murakami-Murakami simplicial-volume conjecture for links

摘录 / EXCERPT: Murakami-Murakami simplicial-volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066141 | MATHDB-344988

Detcherry-Kalfagianni-Yang generalized Turaev-Viro volume conjecture for links

摘录 / EXCERPT: Detcherry-Kalfagianni-Yang generalized Turaev-Viro volume conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066142 | MATHDB-344990

Resolvent construction conjecture for asymptotically stationary abstract Klein-Gordon operators

摘录 / EXCERPT: Let K be an abstract Klein-Gordon operator on $L^2(\mathbb{R},\mathcal K)$, and let $R_z(t,s)$ and $\mathcal Z_{\pm,z}^{(\pm)}(t)$ be the evolution and asymptotic subspaces defin...

conjecture | open | MathDB 页面 | 原文线索

V2-066143 | MATHDB-344996

Hod-Piran conjecture on decay of charged scalar fields

摘录 / EXCERPT: Hod-Piran conjecture. Among all admissible, sufficiently regular and decaying data, there exists a generic subclass such that, whenever the maximal future development satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-066144 | MATHDB-344997

Conjecture on the space-like character of the singularity

摘录 / EXCERPT: Let $\mathcal{Q}^+$ be the quotient spacetime, $\mathcal{I}^+$ future null infinity, and $\mathcal{S}$ the relevant singular boundary component. Space-like singularity conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-066145 | MATHDB-344999

Kommemi's nonzero asymptotic charge conjecture

摘录 / EXCERPT: Let $\mathcal{Q}^+$ be the quotient spacetime, $\mathcal{I}^+$ future null infinity, and let the black-hole asymptotic charge mean the limiting charge appearing in the model. Nonze...

conjecture | open | MathDB 页面 | 原文线索

V2-066146 | MATHDB-345000

Dimension-one conjecture for non-golden-ratio Pisot reciprocal conjugates

摘录 / EXCERPT: Dimension-one conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-066147 | MATHDB-345004

Bonamy-Bousquet's conjecture on color savings for graph powers

摘录 / EXCERPT: Bonamy-Bousquet's conjecture. For every $k\geqslant 2$, only finitely many graphs G satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-066148 | MATHDB-345005

Eventual periodicity conjecture for signs of q-series coefficients

摘录 / EXCERPT: Eventual sign-periodicity conjecture. For given r, s, and t, the signs of $a_{r,s,t}(n)$'s and $b_{r,s,t}(n)$'s are periodic with period t or $2t$ from sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-066149 | MATHDB-345008

Parity conjecture for transversal types in even-order Latin squares

摘录 / EXCERPT: Let L be a Latin square of even order n. Let w, x, y, and z be the numbers of transversals in L of types T^{000}, T^{011}, T^{101}, and T^{110}, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-066150 | MATHDB-345009

Congruence relations among diagonal counts of Latin squares

摘录 / EXCERPT: Let L be a Latin square of order n. For $1\leqslant k\leqslant n$, let E_k be the number of diagonals of L containing exactly k distinct symbols. Let t_{ij} be the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-066151 | MATHDB-345014

Completeness conjecture for zeroth-order conservation-law extensions of two-dimensional shallow water equations

摘录 / EXCERPT: Consider the class of two-dimensional shallow water equations with variable bottom topography, denoted by $\mathcal L$, and its spaces of zeroth-order conservation laws. Two cases...

conjecture | open | MathDB 页面 | 原文线索

V2-066152 | MATHDB-345020

Large-genus asymptotics of area Siegel-Veech constants of quadratic-differential strata

摘录 / EXCERPT: Let $\boldsymbol{d}\in\hat\Pi_{4g-4}$, and let ${\mathcal Q}(\boldsymbol{d})$ be a non-hyperelliptic connected component of a stratum of meromorphic quadratic differentials with at...

conjecture | open | MathDB 页面 | 原文线索

V2-066153 | MATHDB-345024

The new-root volume conjecture for hyperbolic links

摘录 / EXCERPT: Let L be a hyperbolic link in $\mathbb S^3$, and let $J_{\vec N}(L,t)$ denote its colored Jones polynomial with all components colored by N. New-root hyperbolic-link volume con...

open_problem | open | MathDB 页面 | 原文线索

V2-066154 | MATHDB-345026

Asymptotic counting conjecture for rational points on quartic del Pezzo surfaces

摘录 / EXCERPT: Let $\phi_{1}(x_{1},\ldots,x_{5})$ and $\phi_{2}(x_{1},\ldots,x_{5})$ be quadratic forms over $\mathbb{Z}$ in five variables, and let $\mathbf V\subset\mathbb{P}^{4}$ be the quarti...

conjecture | open | MathDB 页面 | 原文线索

V2-066155 | MATHDB-345030

Surjectivity conjecture for the weakly integral Witt group

摘录 / EXCERPT: Let $\mathcal{W}$ be the Witt group of non-degenerate braided fusion categories, let $s\mathcal{W}$ be the corresponding group of slightly degenerate braided fusion categories, and...

conjecture | open | MathDB 页面 | 原文线索

V2-066156 | MATHDB-345034

Constant-coefficient recurrence conjecture for arbitrary-parameter multi-indexed polynomials

摘录 / EXCERPT: Let $P_{\mathcal{D},n}(\eta;\boldsymbol{\lambda})$ be the multi-indexed Meixner-Pollaczek or continuous Hahn polynomial, with parameters $\boldsymbol{\lambda}$ and index set...

conjecture | open | MathDB 页面 | 原文线索

V2-066157 | MATHDB-345035

Rational-form conjecture for the central recurrence coefficient

摘录 / EXCERPT: Let $X(\eta)=I\Xi_{\mathcal{D}}Y$ have degree L, and let $r_{n,k}^{X,\mathcal{D}}$ be the coefficients in the $1+2L$-term constant-coefficient recurrence for the multi-indexed Me...

conjecture | open | MathDB 页面 | 原文线索

V2-066158 | MATHDB-345043

Density of Misiurewicz-model parameters on the common boundary

摘录 / EXCERPT: Let $\mathcal{M}$ and $\mathcal{M}_0$ be the connectivity locus and Thurston set, respectively, for the iterated function system generated by $\mathfrak{s}_-$ and $\mathfrak{s}_+$....

conjecture | open | MathDB 页面 | 原文线索

V2-066159 | MATHDB-345044

Common-boundary parameters lie on the principal Thurston-set component boundary

摘录 / EXCERPT: Let $\mathcal{M}$ be the connectivity locus and let $\mathcal{M}_0$ be the Thurston set, both contained in the unit disk $\mathbb{D}$. Let U be the connected component of...

conjecture | open | MathDB 页面 | 原文线索

V2-066160 | MATHDB-345051

The converse subsequence quantile conjecture for point-stationary circle packings

摘录 / EXCERPT: Converse quantile conjecture. If $[\boldsymbol{G},\boldsymbol{o}]$ has a point-stationary CP whose CP-cocycle is $\boldsymbol{c}$, then the quantile condition $$ holds along some s...

conjecture | open | MathDB 页面 | 原文线索

V2-066161 | MATHDB-345052

Conjecture on burying pairs of binary lattices in rank 3

摘录 / EXCERPT: Let ℓ_1 and ℓ_2 be binary $\mathbb{Z}$-lattices with equal discriminants ... Assume that the pair (ℓ_1,ℓ_2) is buried in a genus of rank 3. Burying conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-066162 | MATHDB-345053

Rationality of the generating function for Frobenius dimensions

摘录 / EXCERPT: Let $\mathcal{C}$ be the tensor category under consideration and let X be the object used in the source to define the generating function $f_X(z)$. The rationality conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-066163 | MATHDB-345059

Rainbow induced-path conjecture for K_r-free graphs

摘录 / EXCERPT: Rainbow induced-path conjecture. For each $r\geq 3$, every properly coloured K_r-free graph of chromatic number χ contains a rainbow induced path of length $\chi^{1/(r-2)}$.

conjecture | open | MathDB 页面 | 原文线索

V2-066164 | MATHDB-345060

The linear-logarithmic induced-path conjecture for triangle-free graphs

摘录 / EXCERPT: Linear-logarithmic induced-path conjecture. There is some constant $c>0$ such that every triangle-free graph of chromatic number χ contains an induced path of length at least...

conjecture | open | MathDB 页面 | 原文线索

V2-066165 | MATHDB-345061

The linear-logarithmic induced-cycle conjecture for triangle-free graphs

摘录 / EXCERPT: Linear-logarithmic induced-cycle conjecture. There is some constant $c>0$ such that every triangle-free graph of chromatic number $\chi>2$ contains an induced cycle of length at le...

conjecture | open | MathDB 页面 | 原文线索

V2-066166 | MATHDB-345062

The induced-forest chromatic bound conjecture

摘录 / EXCERPT: Induced-forest chromatic bound conjecture. There is some constant $c>0$ such that every such graph G has chromatic number at most

conjecture | open | MathDB 页面 | 原文线索

V2-066167 | MATHDB-345064

Tautological vanishing conjecture for moduli spaces of sheaves on K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface and ${\mathsf M}$ a moduli space of stable sheaves on X with virtual universal sheaf $\overline{\mathcal F}$. Let $R_{\star}({\mathsf M})$ denote the taut...

conjecture | open | MathDB 页面 | 原文线索

V2-066168 | MATHDB-345065

Tautological injectivity conjecture for products of moduli spaces and K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface, ${\mathsf M}$ a moduli space of stable sheaves on X, and $R_{\star}({\mathsf M}\times X^{\ell})$ the tautological subring. Let ... be the restriction of th...

conjecture | open | MathDB 页面 | 原文线索

V2-066169 | MATHDB-345067

Extended tautological vanishing conjecture for products of K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface and ${\mathsf M}$ a moduli space of stable sheaves on X. For integers $k\geq 0$ and $\ell>0$, consider ${\mathsf M}\times X^k\times X^{\ell}$, with factor...

conjecture | open | MathDB 页面 | 原文线索

V2-066170 | MATHDB-345069

The sine-process hereditary completeness conjecture

摘录 / EXCERPT: Let X be a configuration sampled from the sine process, with law denoted by ${{\mathbb P}_{\mathscr S}}$, and let $p\in X$ be a particle. A set is hereditarily complete if every...

conjecture | open | MathDB 页面 | 原文线索

V2-066171 | MATHDB-345070

The sine-process maximum-growth conjecture at zero shift

摘录 / EXCERPT: The sine-process maximum-growth conjecture at zero shift. The same asymptotic formula holds for $A=0$:

conjecture | open | MathDB 页面 | 原文线索

V2-066172 | MATHDB-345087

Descent equidistribution for k-arrangements

摘录 / EXCERPT: Let ${\mathsf{A_n^k}}$ denote the set of k-arrangements of size n, represented in permutation and derangement forms, and let $\mathop{\sf des}\nolimits$ denote the descent stat...

conjecture | open | MathDB 页面 | 原文线索

V2-066173 | MATHDB-345088

Descent distribution for pattern-avoiding 2-arrangements

摘录 / EXCERPT: Let $[n]=\{1,2,\ldots,n\}$, let a 2-arrangement of $[n-1]$ be represented in permutation form, and let $\mathop{\sf des}\nolimits$ denote the number of descents. Descent-distributi...

conjecture | open | MathDB 页面 | 原文线索

V2-066174 | MATHDB-345089

Mahonian equidistribution in color encodings of k-arrangements

摘录 / EXCERPT: Let ${\mathsf{A_n^k}}$ denote the set of k-arrangements of size n. For a k-arrangement represented by a permutation π, color each fixed point using a map from...

conjecture | open | MathDB 页面 | 原文线索

V2-066175 | MATHDB-345099

The non-crossing ELP-cut conjecture for matching-covered graphs

摘录 / EXCERPT: Non-crossing ELP-cut conjecture. If C is a nontrivial tight cut of G, then G has an ELP-cut that does not cross C.

conjecture | open | MathDB 页面 | 原文线索

V2-066176 | MATHDB-345102

Polynomial comparability conjecture for exhaustion subsets of convex structures

摘录 / EXCERPT: Polynomial comparability conjecture. The following subsets of $\mathcal{H}(S_{g,m})(\mathbf{L})$ are polynomially comparable to each other:

conjecture | open | MathDB 页面 | 原文线索

V2-066177 | MATHDB-345110

The X-ray transform analogue of the endpoint restriction conjecture

摘录 / EXCERPT: X-ray endpoint conjecture. For every $\varepsilon>0$, there is a constant $C_\varepsilon<\infty$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066178 | MATHDB-345123

Liu's exceptional-set conjecture for radial projections

摘录 / EXCERPT: Let $E\subset\mathbb{R}^d$ be a Borel set with $\dim_{\mathcal{H}} E\in(k-1,k]$, where $k=0,1,\dots,d-1$. For $y\in\mathbb{R}^d$, let $\pi^y(x)=(x-y)/|x-y|$ be the radial projectio...

conjecture | open | MathDB 页面 | 原文线索

V2-066179 | MATHDB-345130

The \ell-torsion conjecture for class groups of number field extensions

摘录 / EXCERPT: Let $\ell>1$ be an integer, let k be a number field, and let d be a positive integer. For any degree-d extension F/k, write $\operatorname{Cl}_{F}[\ell]$ for the subgroup o...

conjecture | open | MathDB 页面 | 原文线索

V2-066180 | MATHDB-345142

The asymptotic gap conjecture for PIR codes with block length $k=2^{s-2}$

摘录 / EXCERPT: Let $P(s,k)$ denote the optimal block length of a PIR code with parameters s and k, and let $N(s,k)$ denote the coding-theoretic lower bound. Set $k=2^{s-2}$. Asymptotic gap co...

conjecture | open | MathDB 页面 | 原文线索

V2-066181 | MATHDB-345144

Complete base-point-free linear-series stability conjecture

摘录 / EXCERPT: Let C be a smooth projective curve and let L be a line bundle such that $V=H^0(C,L)$ gives a complete base point free linear series. Write $|L|$ for this complete linear series...

conjecture | open | MathDB 页面 | 原文线索

V2-066182 | MATHDB-345147

Greenberg's nonvanishing conjecture for Hida-family p-adic L-functions

摘录 / EXCERPT: Assume that $\mathbb{I}=\mathfrak{O}_L[[T]]$, and for each $k\in\mathbf{Z}_{\geqslant 2}$ let $\boldsymbol{f}_k$ be the p-stabilized newform obtained by setting $T=(1+p)^{k-2}-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-066183 | MATHDB-345151

Conjecture on complements of Schubert polynomials

摘录 / EXCERPT: Complement partition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-066184 | MATHDB-345152

1432-pattern obstruction conjecture for complements of Schubert polynomials

摘录 / EXCERPT: 1432-pattern obstruction conjecture. If $w\in S_n$ contains a 1432 pattern, then

conjecture | open | MathDB 页面 | 原文线索

V2-066185 | MATHDB-345156

Speck-Strain conjecture on the relativistic thermodynamic coordinate map

摘录 / EXCERPT: Let $n>0$ and $gamma>0$ denote the particle number density and inverse temperature, respectively, and let k_B,e,m,c,h be the Boltzmann constant, elementary charge, particle mass,...

conjecture | open | MathDB 页面 | 原文线索

V2-066186 | MATHDB-345157

Speck-Strain conjecture on the relativistic equation of state and sound speed

摘录 / EXCERPT: Under the relations referred to in the source as equations ... , let p be the pressure, e the energy density, and S the entropy variable. The conjecture asserts that p is a s...

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V2-066187 | MATHDB-345166

Braided n-ary coherence conjecture for polyadic tensor categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a braided semigroupal polyadic category with an n-ary tensor product, an n-ary associator $\scalebox{1.15}{\textsf{\upshape A}}^{\left(2n-1\right)\otimes}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066188 | MATHDB-345167

The non-radial extension conjecture for planar mappings of finite distortion

摘录 / EXCERPT: Let f be a planar homeomorphic mapping of finite distortion satisfying the hypotheses of Theorem $\mathrm{}$, except that radiality is not assumed. The conclusions of Theorem...

conjecture | open | MathDB 页面 | 原文线索

V2-066189 | MATHDB-345173

Finite Laplace transform representation for a Fox H-function kernel

摘录 / EXCERPT: Finite Laplace transform question. Can the finite Laplace transform of this function be expressed in terms of the Fox H-function?

open_problem | open | MathDB 页面 | 原文线索

V2-066190 | MATHDB-345177

Strong Coloring Conjecture for the strong chromatic number

摘录 / EXCERPT: Let H be a graph, let $Delta(H)$ denote its maximum degree, and let $schi(H)$ denote its strong chromatic number, the minimum number of colors needed so that every graph formed b...

conjecture | open | MathDB 页面 | 原文线索

V2-066191 | MATHDB-345186

The classification conjecture for graphs attaining local antimagic chromatic number equal to order

摘录 / EXCERPT: Let G be a graph of order n, and let K_n, $K_{1,n-1}$, $K(m;n_1,n_2,\ldots,n_m)$, and $Ct(3;n_1,0,n_3)$ denote the graph families used in the paper. Order-equality classifica...

conjecture | open | MathDB 页面 | 原文线索

V2-066192 | MATHDB-345187

The almost-all-trees conjecture for local antimagic chromatic number

摘录 / EXCERPT: Let T be a tree with k pendant vertices. The families $K(2;a,b)$ and $Ct(3;a,0,b)$ are the exceptional tree families described in the supplied statement. Almost-all-trees conje...

conjecture | open | MathDB 页面 | 原文线索

V2-066193 | MATHDB-345197

Random higher-dimensional permutation universality conjecture

摘录 / EXCERPT: For $d\geqslant 2$, a d-permutation of order n is a $(d+1)$-array over $\{0,1\}$ of order n in which every line contains a unique 1. A d-pattern of order k is a sequenc...

conjecture | open | MathDB 页面 | 原文线索

V2-066194 | MATHDB-345205

Capacity characterization for the Poisson channel with increasing power

摘录 / EXCERPT: Consider the Poisson channel ... with power constraint $\tfrac{1}{n}\sum_{t=1}^n x[t] \leq n^\alpha$. This is the Poisson channel with increasing power (PCIP). A normalized rate...

conjecture | open | MathDB 页面 | 原文线索

V2-066195 | MATHDB-345220

Disjoint base-block conjecture for cyclic $(v,k,1)$-designs

摘录 / EXCERPT: Let $(V,\mathcal{B})$ be a cyclic $(v,k,1)$-design: the blocks are partitioned into orbits under a cyclic automorphism, and a base block is one chosen block from each orbit. Disjoi...

conjecture | open | MathDB 页面 | 原文线索

V2-066196 | MATHDB-345221

Conjecture on the block max-min estimator's pivotal limit distribution

摘录 / EXCERPT: Let f_0 be the monotone regression function, let x_0 be the point of inference, and let $\widehat{f}_n^{-}(x_0)$ be the block max-min estimator with associated block count...

conjecture | open | MathDB 页面 | 原文线索

V2-066197 | MATHDB-345223

The non-emptiness criterion for coherent systems on the projective line

摘录 / EXCERPT: Non-emptiness conjecture. The space $G(\alpha;n,an-t,k)$ is non-empty for some $\alpha>0$ if and only if one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-066198 | MATHDB-345224

Monotonicity of non-emptiness for coherent systems on the projective line

摘录 / EXCERPT: Let $G(\alpha;n,d,k)$ denote the moduli space of α-stable coherent systems of type (n,d,k) on the projective line. Suppose that $a\ge2$, $1\le t\le n-1$, and $k\ge n$. Mon...

conjecture | open | MathDB 页面 | 原文线索

V2-066199 | MATHDB-345225

Spiro's ballot-permutation and odd-order-permutation equidistribution conjecture

摘录 / EXCERPT: Let $n\ge 1$. A ballot permutation is a permutation whose initial segments contain at least as many ascents as descents, and an odd order permutation is a permutation all of whose...

conjecture | open | MathDB 页面 | 原文线索

V2-066200 | MATHDB-345226

Refinement conjecture for ballot and odd-order permutation counts

摘录 / EXCERPT: Let $b_{n,d}(i,j)$ denote the number of ballot permutations of length n with descent d and the relevant endpoint data (i,j), and let $p_{n,d}(i,j)$ denote the corresponding n...

conjecture | open | MathDB 页面 | 原文线索

V2-066201 | MATHDB-345242

Existence for general nonlinearities in the critical case

摘录 / EXCERPT: Let f satisfy (F1) with $q\geq 1$, suppose that $f'(u)F(u)\leq q$ for all sufficiently large $u>0$, and let $\phi\geq 0$. Set ... If ... then (S1E1) has a local-in-time solution. Exi...

conjecture | open | MathDB 页面 | 原文线索

V2-066202 | MATHDB-345243

Universal Gaussian off-diagonal decay conjecture for Toeplitz operators

摘录 / EXCERPT: Let d and N be integers, and let U,V be open subsets of $(\mathbb{S}^2)^d$. Write $\operatorname{dist}(U,V)$ for their distance and let S_N denote the Toeplitz operator app...

conjecture | open | MathDB 页面 | 原文线索

V2-066203 | MATHDB-345248

Guo and Zudilin's second q-analogue of a supercongruence

摘录 / EXCERPT: Define the q-shifted factorial by $(a;q)_0=1$ and $(a;q)_k=(1-a)(1-aq)\cdots(1-aq^{k-1})$ for $k\geq 1$, and let $\Phi_n(q)$ denote the n-th cyclotomic polynomial. Let m and...

conjecture | open | MathDB 页面 | 原文线索

V2-066204 | MATHDB-345249

A second q-analogue of the Swisher-type supercongruence

摘录 / EXCERPT: Define the q-shifted factorial by $(a;q)_0=1$ and $(a;q)_k=(1-a)(1-aq)\cdots(1-aq^{k-1})$ for $k\geq 1$, let $\Phi_n(q)$ be the n-th cyclotomic polynomial, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-066205 | MATHDB-345250

The continuous-limit correspondence between differential and q-difference spectral types

摘录 / EXCERPT: Continuous-limit correspondence. If, as $q\to1$, the zeros α^k_j of $\det A(x)$ corresponding to m^k_j satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-066206 | MATHDB-345259

Dousse-Kim unimodality conjecture for overpartitions in a box

摘录 / EXCERPT: Dousse-Kim's unimodality conjecture. Any linear cut through this array is a unimodal sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-066207 | MATHDB-345265

The weak Boolean lattice local dimension conjecture

摘录 / EXCERPT: Weak Boolean lattice local dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066208 | MATHDB-345266

The asymptotic local dimension conjecture for powers of chains

摘录 / EXCERPT: Asymptotic local dimension conjecture for powers of chains. There exists a universal constant c such that, for any $t\in\operatorname{\mathbb{N}}$, as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-066209 | MATHDB-345269

Brockett's condition implies exponential composition-operator stabilization

摘录 / EXCERPT: Let $ \overset{\raisebox{0ex}{\scalebox{0.45}{$bullet$}}}{x} =f(x,u)$ be the control system under consideration. Brockett's condition means that the vector field f is open at the...

conjecture | open | MathDB 页面 | 原文线索

V2-066210 | MATHDB-345276

Uniqueness conjecture for maximizing Bregman divergence on one side of a codimension-one family

摘录 / EXCERPT: Uniqueness conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-066211 | MATHDB-345278

Three-dimensional mixing conjecture for the auxiliary complex of a zig-zag construction

摘录 / EXCERPT: Let G be a d-regular triangle-free graph with n vertices, where ... Under the notion in Chapman (2018), let ... Here $\operatorname{Aux}(\Gamma)$ denotes the auxiliary complex as...

conjecture | open | MathDB 页面 | 原文线索

V2-066212 | MATHDB-345280

Graph-product structure conjecture for k-nearest-neighbour graphs

摘录 / EXCERPT: A finite set $P\subset\mathbb{R}^d$ defines a k-nearest-neighbour graph by joining two points whenever one is among the k points closest to the other. Here,...

conjecture | open | MathDB 页面 | 原文线索

V2-066213 | MATHDB-345286

Approximate ERM with constant conditional mutual information

摘录 / EXCERPT: Let $\mathcal{X}$ be a domain, let $\mathcal{W}$ be a class of functions $f:\mathcal{X}\to\{0,1\}$ with VC dimension $d\geq 1$, and let $n\geq 1$. For a dataset...

conjecture | open | MathDB 页面 | 原文线索

V2-066214 | MATHDB-345291

Signed Selmer cotorsion conjecture over the cyclotomic extension

摘录 / EXCERPT: Let E be an elliptic curve over the number field F', let F be a finite extension satisfying the hypotheses of the paper, and let F^{cyc} be the cyclotomic...

conjecture | open | MathDB 页面 | 原文线索

V2-066215 | MATHDB-345292

Signed Selmer cotorsion conjecture over the noncommutative extension

摘录 / EXCERPT: Let F_{∞}/F be the $\mathbb{Z}_p^d$-extension considered in the paper, let $G=\operatorname{Gal}(F_{\infty}/F)$, and let Σ_{ss} be the set of primes where...

conjecture | open | MathDB 页面 | 原文线索

V2-066216 | MATHDB-345293

Coates-Fukaya-Kato-Sujatha-Venjakob $M_H(G)$-conjecture for signed Selmer groups

摘录 / EXCERPT: Let F_{∞}/F be the extension in the paper, let $G=\operatorname{Gal}(F_{\infty}/F)$, and let H be the subgroup specified by the extension, with completed group ring...

conjecture | open | MathDB 页面 | 原文线索

V2-066217 | MATHDB-345304

Shin-Templier's average conductor growth conjecture for functorial L-functions

摘录 / EXCERPT: Let ρ be an L-homomorphism and let $C_{\rho,X}$ denote the average analytic conductor associated with the family indexed by representations with analytic conductor bounded b...

conjecture | open | MathDB 页面 | 原文线索

V2-066218 | MATHDB-345305

Shin-Templier's negligible-poles conjecture for functorial L-functions

摘录 / EXCERPT: Let ρ be an L-homomorphism, and let $\mathcal{P}$ be the characteristic function on $\hat{\mathbb{X}}_{\mathrm{gen}}$ of the representations π for which...

conjecture | open | MathDB 页面 | 原文线索

V2-066219 | MATHDB-345306

The comparison conjecture for the SDP bound and the generalized Schrijver bound

摘录 / EXCERPT: Comparison conjecture. The upper bound $\theta^3_k(G)$ is at least as good as the upper bound $\vartheta'_k(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-066220 | MATHDB-345323

Generation conjecture for multiple zeta values by symmetric Mordell-Tornheim values

摘录 / EXCERPT: Let $\Omega(\mathbf{k})$ be the real symmetric Mordell-Tornheim value associated with an index $\mathbf{k}\in\mathbb{N}^r$, defined by the limiting construction in the paper. Symme...

conjecture | open | MathDB 页面 | 原文线索

V2-066221 | MATHDB-345334

Hod-Piran charged massless scalar decay conjecture

摘录 / EXCERPT: Charged massless scalar decay conjecture. For smooth, regular, generic admissible data with a non-empty black hole and $m^2=0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066222 | MATHDB-345335

Massive scalar-field $v^{-5/6}$ tail conjecture

摘录 / EXCERPT: Let $\mathcal{H}^+$ be the event horizon and let v be the standard advanced time null coordinate. Massive scalar-field tail conjecture. For smooth, regular, generic admissible da...

conjecture | open | MathDB 页面 | 原文线索

V2-066223 | MATHDB-345337

Spherical trapped-surface conjecture for charged gravitational collapse

摘录 / EXCERPT: Let $\mathcal{Q}^+$ be the quotient of the maximal future development, $J^-(\mathcal{I}^+)$ the causal past of future null infinity, $\mathcal{A}$ the apparent horizon, and...

conjecture | open | MathDB 页面 | 原文线索

V2-066224 | MATHDB-345346

Lindahl's linearizability conjecture for positive-characteristic polynomials

摘录 / EXCERPT: Lindahl's conjecture. The polynomial f is linearizable if and only if $a_i=0$ for every $i\geq 2$ such that $p\nmid i$. This conjecture refines Lindahl's known examples of non-li...

conjecture | open | MathDB 页面 | 原文线索

V2-066225 | MATHDB-345347

Finite-type representing measure conjecture for moment matrices with an involutive relation

摘录 / EXCERPT: Finite-type representing measure conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-066226 | MATHDB-345358

Xu's conditional edge-connectivity conjecture

摘录 / EXCERPT: Xu's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066227 | MATHDB-345360

A shift-extension conjecture for Fibonacci divisibility sums

摘录 / EXCERPT: Let F_i denote the ith Fibonacci number, and for a positive integer m write $\sum_{i=1}^m F_i$ for the sum of the first m Fibonacci numbers. Shift-extension conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066228 | MATHDB-345361

Infinitely many consecutive Fibonacci divisibility sums

摘录 / EXCERPT: Let F_i denote the ith Fibonacci number. Consecutive-pair conjecture. There are infinitely many pairs of positive integers $(n,n+1)$ such that ... This conjecture asks whether th...

conjecture | open | MathDB 页面 | 原文线索

V2-066229 | MATHDB-345362

Conjecture on the limiting fraction of non-positive Hessian eigenvalues

摘录 / EXCERPT: Limiting Hessian-spectrum conjecture. For $a,b,c>0$, as $L\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066230 | MATHDB-345364

Kudla-Rapoport arithmetic intersection conjecture for special cycles

摘录 / EXCERPT: Let $x_1,\dots,x_n\in {\mathbb V}$, where ${\mathbb V}$ is the hermitian space of special homomorphisms, and let h be its hermitian form. Put ... The special cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-066231 | MATHDB-345365

Conjectural Whittaker formula for mixed special-cycle intersections

摘录 / EXCERPT: Let $0\leq h\leq 2n$, and let $x_1,\dots,x_{2n-h},y_1,\dots,y_h\in {\mathbb V}$ be the special homomorphisms defining cycles ${\mathcal Z}(x_i)$ and ${\mathcal Y}(y_j)$. Let B be...

conjecture | open | MathDB 页面 | 原文线索

V2-066232 | MATHDB-345366

Whittaker-function formula for the top mixed intersection number

摘录 / EXCERPT: Let $x_1,\dots,x_n,y_1,\dots,y_n$ be a basis of the hermitian space ${\mathbb V}$, and let $\mathbf{x}=(x_1,\dots,x_n)$ and $\mathbf{y}=(y_1,\dots,y_n)$. Define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066233 | MATHDB-345367

Bertoncello-Levcovitz conjecture on Shamsuddin derivations and isotropy groups

摘录 / EXCERPT: Let $K[x_1,x_2,\ldots,x_n]$ be a polynomial ring, and let D be a Shamsuddin derivation, meaning a derivation of the form ... where $a_i,b_{i,j}\in K[x]$ and $a_i\neq a_l$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-066234 | MATHDB-345371

Characterization conjecture for unavoidable graphs in poset cover graphs

摘录 / EXCERPT: Let H be a graph. Call H unavoidable if the cover graph of every poset of sufficiently large dimension contains H as a minor. Kelly's construction refers to the family of pla...

conjecture | open | MathDB 页面 | 原文线索

V2-066235 | MATHDB-345376

Heuristic for counting elliptic curves with prescribed level structures

摘录 / EXCERPT: Let $\mathcal{Z}_{1,\mathbb{Q}}^{[\Gamma]}(\mathcal{B})$ count generalized elliptic curves with $[\Gamma]$-structures over $\mathbb{Z}$ ordered by...

conjecture | open | MathDB 页面 | 原文线索

V2-066236 | MATHDB-345377

Heuristic for counting multiply marked stable genus-one curves

摘录 / EXCERPT: For $2\leq m\leq5$, let $\mathcal{Z}_{1,m,\mathbb{Q}}(\mathcal{B})$ count m-marked $(m-1)$-stable genus-one curves over $\mathbb{Z}$ ordered by...

conjecture | open | MathDB 页面 | 原文线索

V2-066237 | MATHDB-345379

Arthur's existence and uniqueness conjecture for distinguished unipotent parameters

摘录 / EXCERPT: Let G be the reductive group under consideration, let X be the underlying curve, and let $\check{G}$ be its Langlands dual group. Let ... be a distinguished unipotent Arthur para...

conjecture | open | MathDB 页面 | 原文线索

V2-066238 | MATHDB-345380

Conjecture on the subregular unipotent Arthur sheaf for G2

摘录 / EXCERPT: The subregular Arthur-sheaf conjecture. If $\operatorname{char}(k)\neq 2$ or 3, then there exists a unique sheaf $\mathcal F_{\mathrm{sr}}$ on Bun_G such that, o...

conjecture | open | MathDB 页面 | 原文线索

V2-066239 | MATHDB-345390

Unique hypotenuse multiplicity conjecture for congruent number triangles

摘录 / EXCERPT: Unique hypotenuse multiplicity conjecture. There do not exist two dissimilar primitive right triangles with the same hypotenuse and whose squarefree parts of the areas are equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-066240 | MATHDB-345405

Embedding conjecture for optimal dimension-four binary codes

摘录 / EXCERPT: Let $[n,4]$ codes be binary linear codes of length n and dimension 4. A code is optimal if its minimum distance is $d(n,4)$, and it is self-orthogonal if it is orthogonal to it...

conjecture | open | MathDB 页面 | 原文线索

V2-066241 | MATHDB-345406

Dimension-five self-orthogonal distance conjecture

摘录 / EXCERPT: Let $d(n,5)$ denote the largest minimum distance of a binary linear $[n,5]$ code, and let $\dso(n,5)$ denote the largest minimum distance of a binary self-orthogonal $[n,5]$ code....

conjecture | open | MathDB 页面 | 原文线索

V2-066242 | MATHDB-345413

The sharp upper-bound conjecture for the Yokota invariant

摘录 / EXCERPT: Let Γ be a planar 3-connected graph, let $\overline{\Gamma}$ denote the associated maximal-volume proper generalized hyperbolic polyhedron, and let col be any r-admiss...

conjecture | open | MathDB 页面 | 原文线索

V2-066243 | MATHDB-345414

Fiorini-Wilson-Fisk conjecture on uniquely 4-colorable planar graphs

摘录 / EXCERPT: A uniquely 4-colorable simple planar graph is a simple planar graph with exactly one proper 4-coloring up to permutation of the colors. Fiorini-Wilson-Fisk conjecture. Every unique...

conjecture | open | MathDB 页面 | 原文线索

V2-066244 | MATHDB-345415

Constructive conjecture for uniquely 4-colorable planar graphs

摘录 / EXCERPT: Let G be a uniquely 4-colorable simple planar graph, meaning that G has exactly one proper 4-coloring up to permutation of the colors. Constructive conjecture. Every such graph...

conjecture | open | MathDB 页面 | 原文线索

V2-066245 | MATHDB-345416

The classification conjecture for strongly hereditarily self-similar pro-p groups

摘录 / EXCERPT: Classification conjecture. G is strongly hereditarily self-similar of index p if and only if G is isomorphic to

conjecture | open | MathDB 页面 | 原文线索

V2-066246 | MATHDB-345421

Branching coefficient formula for the type B_n Toda eigenfunction

摘录 / EXCERPT: The branching coefficient formula. The coefficients are

conjecture | open | MathDB 页面 | 原文线索

V2-066247 | MATHDB-345427

Leaving principle for Bloch eigenvalues of an optical Schrödinger operator

摘录 / EXCERPT: Let $H(c)$ be the operator with pure imaginary potential $2ic\cos 2x$, and let $L(V)$ be the corresponding operator whose Bloch eigenvalues depend on V. Leaving principle. If c...

conjecture | solved | MathDB 页面 | 原文线索

V2-066248 | MATHDB-345428

Irreversibility principle for Bloch eigenvalues

摘录 / EXCERPT: Let $L(V)$ be the operator considered in the paper, with Bloch eigenvalues depending on V. A Bloch eigenvalue is said to leave the real line when it becomes nonreal as V varies...

conjecture | open | MathDB 页面 | 原文线索

V2-066249 | MATHDB-345429

Simplicity principle for nonreal periodic and antiperiodic eigenvalues

摘录 / EXCERPT: Let $L(V)$ be the operator in the paper, and let periodic and antiperiodic eigenvalues mean Bloch eigenvalues with the corresponding periodic or antiperiodic boundary condition. Si...

conjecture | open | MathDB 页面 | 原文线索

V2-066250 | MATHDB-345436

Mocanu's conjecture on a harmonic mapping class

摘录 / EXCERPT: Mocanu's conjecture. Every mapping in $\mathcal{M}$ belongs to $\mathcal{S}_{\mathcal{H}}^0$; equivalently, $\mathcal{M}\subset\mathcal{S}_{\mathcal{H}}^0$. The conjecture concerns...

conjecture | open | MathDB 页面 | 原文线索

V2-066251 | MATHDB-345439

The twin-Reed-Muller code equals the Reed-Muller code

摘录 / EXCERPT: Twin-Reed-Muller conjecture. The twin-Reed-Muller code is the Reed-Muller code; equivalently, for $i,j\in[n]$,

conjecture | open | MathDB 页面 | 原文线索

V2-066252 | MATHDB-345440

Layer-crossing conditional-entropy conjecture for Reed-Muller codes

摘录 / EXCERPT: Layer-crossing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066253 | MATHDB-345441

Equivalence of the twin-Reed-Muller codes for the BEC and BSC

摘录 / EXCERPT: BEC-BSC twin-Reed-Muller conjecture. The twin-Reed-Muller codes for the BEC and BSC are equivalent; that is, for some $\epsilon=o(1/n)$,

conjecture | open | MathDB 页面 | 原文线索

V2-066254 | MATHDB-345455

Stability threshold conjecture for ground states of the trapped nonlinear Schrödinger equation

摘录 / EXCERPT: Stability threshold conjecture. For every p satisfying $+\frac{4}{n}<p<p_n^*$, there exists $\omega_{p,n}$ such that ϕ_ω is stable whenever $-n<\omega\leq\omega_{p,n}$...

open_problem | open | MathDB 页面 | 原文线索

V2-066255 | MATHDB-345464

The radial SLE scaling-limit conjecture for multiple self-avoiding walks

摘录 / EXCERPT: Radial SLE scaling-limit conjecture. There exist b, $\tilde b_n$, a critical value $\beta=\beta_{\cent}$, and a partition function $\Psi^*(D;{\bf z},0)$ such that, as...

conjecture | open | MathDB 页面 | 原文线索

V2-066256 | MATHDB-345484

Approximation conjecture for analytic torsion along shrinking congruence subgroups

摘录 / EXCERPT: Let G be the reductive group defining the symmetric space $\widetilde{X}$, let $\tau\in\operatorname{Rep}(G(\mathbb{R}))$ satisfy $\tau\not\cong\tau_\theta$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066257 | MATHDB-345491

Down-degree homomesy conjecture for trapezoid posets

摘录 / EXCERPT: Down-degree homomesy conjecture. For every toggle on antichains-symmetric distribution μ on $J(\mathscr T(a,b))$,

conjecture | open | MathDB 页面 | 原文线索

V2-066258 | MATHDB-345500

Lewis-Reiner's Hurwitz orbit conjecture for reflection factorizations

摘录 / EXCERPT: Let W be a well-generated finite complex reflection group, and let T and T' be reflection factorizations of a Coxeter element of W. Two factorizations are said to have the...

conjecture | open | MathDB 页面 | 原文线索

V2-066259 | MATHDB-345502

Conjecture on total internal branch length in dust-free coalescents

摘录 / EXCERPT: Let $\hat \ell_n:=\ell_n-\bar \ell_n$ be the total internal branch length of an n-coalescent, where ℓ_n is its total length and $\bar \ell_n$ is its total external branch le...

conjecture | open | MathDB 页面 | 原文线索

V2-066260 | MATHDB-345503

Conjectured stable fluctuations of total length for regularly varying Lambda-coalescents

摘录 / EXCERPT: Total-length fluctuation conjecture. As $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-066261 | MATHDB-345508

Gyárfás's conjecture on chromatic boundedness of triangle-free graphs with bounded hole length

摘录 / EXCERPT: A hole is an induced cycle of length at least 4. For every integer ℓ, consider triangle-free graphs having no hole of length greater than ℓ. Gyárfás's conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-066262 | MATHDB-345509

Tree-chromatic weakening of Hadwiger's conjecture

摘录 / EXCERPT: Let G be a graph and let t be an integer. The tree-chromatic number $\mathsf{tree}\textnormal{-}\chi(G)$ is the minimum, over tree-decompositions of G, of the maximum chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-066263 | MATHDB-345510

Local Hadwiger conjecture for closed-neighborhood chromatic number

摘录 / EXCERPT: Let G be a graph, let t be an integer, and for $v\in V(G)$ let $N_G[v]$ denote the closed neighborhood of v. Local Hadwiger conjecture. If G is a graph without a K_{t+1}-...

conjecture | open | MathDB 页面 | 原文线索

V2-066264 | MATHDB-345511

Universal local Hadwiger conjecture

摘录 / EXCERPT: Let G be a graph, let t be an integer, and for $v\in V(G)$ let $N_G[v]$ denote the closed neighborhood of v. Universal local Hadwiger conjecture. If G is a graph without a...

conjecture | open | MathDB 页面 | 原文线索

V2-066265 | MATHDB-345513

The mass monotonicity conjecture for cubic-quintic solitary waves

摘录 / EXCERPT: In dimension $d=3$, let ϕ_ω denote the ground state solution of the cubic-quintic Schrödinger equation at frequency ω, with $0<\omega<\tfrac{3}{16}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066266 | MATHDB-345515

The conjectural boundary two-point function formula in boundary Liouville theory

摘录 / EXCERPT: Let γ and Q be the Liouville parameters, with $\beta\in(\frac{\gamma}{2},Q)$ and $\mu,\mu_1,\mu_2>0$. For $i=1,2$, define σ_i by ... Let $R(\beta,\sigma_1,\sigma_2)$...

conjecture | open | MathDB 页面 | 原文线索

V2-066267 | MATHDB-345520

The stable blow-up conjecture for the L^2-critical generalized Hartree equation

摘录 / EXCERPT: Let u be a stable blow-up solution of the L^2-critical generalized Hartree equation, with blow-up time T, spatial dimension d, scale $L(t)$, center $x(t)$, phase parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-066268 | MATHDB-345526

Cell-boundary conjecture for T-duality

摘录 / EXCERPT: Let S_{π} be a loopless $(n-c)$-dimensional cell of $Gr^+_{k+1,n}$ with c connected components, where c is a positive integer. Then $S_{\hat{\pi}}$ is a coloopless...

conjecture | open | MathDB 页面 | 原文线索

V2-066269 | MATHDB-345527

T-duality conjecture for hypersimplex and amplituhedron dissections

摘录 / EXCERPT: Let $\{S_{\pi}\}$ be a collection of cells of $Gr^+_{k+1,n}$, and let $\{S_{\hat{\pi}}\}$ be its T-dual collection in $Gr^+_{k,n}$. T-duality conjecture. The collection...

conjecture | open | MathDB 页面 | 原文线索

V2-066270 | MATHDB-345528

Amplituhedron facet-colooplessness conjecture

摘录 / EXCERPT: Let $S_{\pi^{(1)}}$ and $S_{\pi^{(2)}}$ be positroid cells in $Gr^+_{k,n}$ corresponding to coloopless permutations $\pi^{(1)}$ and $\pi^{(2)}$, with ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-066271 | MATHDB-345529

Good T-duality correspondence for hypersimplex and amplituhedron

摘录 / EXCERPT: Let $\{S_{\pi}\}$ be a collection of cells of $Gr^+_{k+1,n}$, and let $\{S_{\hat{\pi}}\}$ be its T-dual collection. Good T-duality correspondence conjecture. The collection...

conjecture | open | MathDB 页面 | 原文线索

V2-066272 | MATHDB-345530

Regular positroidal subdivision conjecture for the m=2 amplituhedron

摘录 / EXCERPT: A regular positroidal subdivision of $\mathcal{A}_{n,k,2}$ is a positroidal dissection of the form $\{S_{\hat{\pi}}\}$, where $\{S_{\pi}\}$ is a regular positroidal subdivision of...

conjecture | open | MathDB 页面 | 原文线索

V2-066273 | MATHDB-345532

Refinement-order compatibility under T-duality

摘录 / EXCERPT: Consider regular positroidal subdivisions $\{\Gamma_{\pi}\}$ and $\{\Gamma_{\pi'}\}$ of $\Delta_{k+1,n}$ and corresponding subdivisions $\{Z_{\hat{\pi}}\}$ and $\{Z_{\hat{\pi'}}\}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066274 | MATHDB-345533

T-duality conjecture for the momentum and ordinary amplituhedra

摘录 / EXCERPT: Let m be even and let $\{S_{\pi}\}$ be a collection of cells of $Gr^+_{k+m/2,n}$. Generalized T-duality conjecture. The collection gives a triangulation, respectively dissection,...

conjecture | open | MathDB 页面 | 原文线索

V2-066275 | MATHDB-345535

Maximal F-signature conjecture for non-regular local rings

摘录 / EXCERPT: Let $(A,\mathfrak{m})$ be a non-regular local ring of dimension $d \geq 1$, and let $A_{p,d}$ be the simple singularity defined above. Maximal F-signature conjecture. ... Since...

conjecture | open | MathDB 页面 | 原文线索

V2-066276 | MATHDB-345536

Gaussian comparison conjecture for maxima under variance and increment domination

摘录 / EXCERPT: Let $\mathbf{X}=(X_1,\ldots,X_d)$ and $\mathbf{Y}=(Y_1,\ldots,Y_d)$ be d-dimensional Gaussian vectors with common mean $\bm{\mu}=(\mu_1,\ldots,\mu_d)$ and covariance matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-066277 | MATHDB-345543

The asymptotic degree of the bubble-sort map on partitions

摘录 / EXCERPT: Bubble-sort degree conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066278 | MATHDB-345551

Conjectured sharp upper bounds for induced degenerate subgraphs of planar graphs

摘录 / EXCERPT: Let $\bar\alpha_d$ denote the minimum, over planar graphs, of the maximum order of a d-degenerate induced subgraph divided by the number of vertices. The octahedron and icosahedr...

conjecture | open | MathDB 页面 | 原文线索

V2-066279 | MATHDB-345565

Stahl's conjecture on multichromatic numbers of Kneser graphs

摘录 / EXCERPT: Stahl's conjecture. For all integers k and $d\geq 2s$,

conjecture | open | MathDB 页面 | 原文线索

V2-066280 | MATHDB-345572

Differentiability and weighted Laplacian formula on RCD spaces

摘录 / EXCERPT: Let $(X,\mathsf{d},\mathfrak{m})$ be the RCD space under discussion, let $k=\dim(X,\mathsf{d},\mathfrak{m})$, and let $\mathcal{R}_k^*$ be the full-measure regular set. Let ϕ...

conjecture | open | MathDB 页面 | 原文线索

V2-066281 | MATHDB-345575

Global rigidity conjecture for graphs containing triangulations of surfaces

摘录 / EXCERPT: Let G be a graph with a triangulation T of a surface S as a spanning subgraph. A graph is globally rigid when its generic realization is uniquely determined, up to Euclidean...

conjecture | open | MathDB 页面 | 原文线索

V2-066282 | MATHDB-345580

Positivity conjecture for the top-defect matrix coefficient

摘录 / EXCERPT: Positivity conjecture. In the redundancy range,

conjecture | open | MathDB 页面 | 原文线索

V2-066283 | MATHDB-345593

The shifted moments conjecture for Dirichlet L-functions

摘录 / EXCERPT: Let q tend to infinity through primes, let Y be the shift-height parameter, and let R be a piecewise smooth positively oriented path encircling 0 and $-s$. Dirichlet shifte...

conjecture | open | MathDB 页面 | 原文线索

V2-066284 | MATHDB-345598

Soundness and strong completeness of ${\bf K^W}$ for filters and (mc)-frames

摘录 / EXCERPT: Let ${\bf K^W}$ be the logic considered in the paper, let a filter be a neighborhood frame whose neighborhood collections satisfy the filter conditions defined in the surrounding d...

conjecture | open | MathDB 页面 | 原文线索

V2-066285 | MATHDB-345603

Semiprimitive local action finiteness conjecture for tree groups

摘录 / EXCERPT: Let T_d be the d-regular tree, let Ω be a finite set, and let $F\leq\operatorname{Sym}(\Omega)$ be semiprimitive, meaning that every normal subgroup of F is either tra...

conjecture | open | MathDB 页面 | 原文线索

V2-066286 | MATHDB-345609

Measure-zero conjecture for non-unique sparse phase-retrieval solutions

摘录 / EXCERPT: Let K be the sparsity level and let a_x denote the periodic auto-correlation of a K-sparse signal x. Measure-zero conjecture. If $K/N<1/2$, then the set of K-sparse signa...

conjecture | open | MathDB 页面 | 原文线索

V2-066287 | MATHDB-345610

Generic signal-recovery conjecture with known support

摘录 / EXCERPT: Let $S\subset[0,N-1]$ and let L_S be the subspace of signals supported in S. Let D_S be the subgroup of intrinsic symmetries D that preserves L_S. Generic signal-recovery...

conjecture | open | MathDB 页面 | 原文线索

V2-066288 | MATHDB-345613

Steuding's universality conjecture for Selberg-class functions

摘录 / EXCERPT: Let S be the Selberg class, let $\mathcal L\in \widetilde{\mathcal S}\cap S$, let K be a compact subset of the strip $\mathfrak S=\{\sigma_m<\sigma<1\}$ with connected compleme...

conjecture | open | MathDB 页面 | 原文线索

V2-066289 | MATHDB-345616

Vizing-Goldberg type gap-one conjecture for equitable coloring of block graphs

摘录 / EXCERPT: Let G be a block graph. Write $\omega(G)$ for its clique number, $\alpha_{\min}(G)$ for the minimum size of a maximal independent set, and $\chi_{=}(G)$ for its equitable chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-066290 | MATHDB-345620

The divisoriality conjecture for the Gieseker-Petri locus

摘录 / EXCERPT: Let M_g be the moduli space of smooth irreducible curves of genus g, and let $GP_g\subset M_g$ be the locus of curves possessing a line bundle A such that the Petri map ... is...

conjecture | open | MathDB 页面 | 原文线索

V2-066291 | MATHDB-345636

Three-dimensional Brownian loop-soup cluster dimension conjecture

摘录 / EXCERPT: Let $d=3$, and let the cable-graph loop-soup clusters be formed on finer and finer meshes. Denote by Δ the Hausdorff dimension of the Brownian loop-soup clusters. Three-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-066292 | MATHDB-345637

Four- and five-dimensional cable-graph cluster scaling-limit conjecture

摘录 / EXCERPT: Let $d\in\{4,5\}$, and consider the cable-graph clusters in $(0,1)^d\cap\delta\mathbb Z^d$ as $\delta\to0$. Four- and five-dimensional scaling-limit conjecture. The limit in distri...

conjecture | open | MathDB 页面 | 原文线索

V2-066293 | MATHDB-345653

Burstein-Jones Wilf-equivalence conjecture for Dumont-1 permutations

摘录 / EXCERPT: Let $\mathfrak{D}^1_{2n}$ denote the set of Dumont-1 permutations of length $2n$, and let $\mathfrak{D}^1_{2n}(\tau)$ denote those avoiding the pattern τ. Burstein-Jones's Wil...

conjecture | open | MathDB 页面 | 原文线索

V2-066294 | MATHDB-345654

Burstein-Jones dominance conjecture for vincular-pattern statistics

摘录 / EXCERPT: Burstein-Jones's dominance conjecture. For all $n\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066295 | MATHDB-345656

Perelman's conjecture for noncompact ancient three-dimensional Ricci flows

摘录 / EXCERPT: Perelman's conjecture. Then $(M,g(t))$ is the Bryant soliton.

conjecture | solved | MathDB 页面 | 原文线索

V2-066296 | MATHDB-345658

Lower-bound conjecture for the Gaussian free field reconstruction threshold

摘录 / EXCERPT: Let T_{rec}^- denote the lower reconstruction threshold for the statistical reconstruction problem considered in the paper. Lower-threshold conjecture. ... The claim is pr...

conjecture | open | MathDB 页面 | 原文线索

V2-066297 | MATHDB-345659

Discrete Gaussian free field level-line convergence to SLE4

摘录 / EXCERPT: Let ϕ_n be the discrete Gaussian free field, let u_n be the auxiliary function appearing in the construction, and let $\eta^{(n)}$ be the original level line. For sufficient...

conjecture | open | MathDB 页面 | 原文线索

V2-066298 | MATHDB-345660

Villain-model level lines with piecewise boundary values converge to SLE4

摘录 / EXCERPT: Let Λ_n be the finite domain, let T be sufficiently small, and let ψ_n be a Villain model in Λ_n at temperature T. Give it boundary values...

conjecture | open | MathDB 页面 | 原文线索

V2-066299 | MATHDB-345661

Villain-model level lines with general boundary angles converge to SLE(4,ρ)

摘录 / EXCERPT: Let Λ_n be the finite domain, let T be sufficiently small, and let ψ_n be a Villain model in Λ_n at temperature T with boundary values $\exp(-ia)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-066300 | MATHDB-345662

Discrete Gaussian free field ALE convergence under exponential observation

摘录 / EXCERPT: Let ϕ_n be a zero-boundary discrete Gaussian free field converging to a continuum Gaussian free field Φ, and let $\mathbb{A}_{-\lambda,\lambda}$ denote the continuum ALE....

conjecture | open | MathDB 页面 | 原文线索

V2-066301 | MATHDB-345663

Villain-model ALE convergence

摘录 / EXCERPT: Let Λ_n be the finite domain and let ψ_n be a Villain model in Λ_n at sufficiently small temperature T. Let the discrete ALE be the union of the interfaces a...

conjecture | open | MathDB 页面 | 原文线索

V2-066302 | MATHDB-345664

Upper-bound conjecture for the Gaussian free field reconstruction threshold

摘录 / EXCERPT: Let T_{rec}^+ denote the upper reconstruction threshold for the statistical reconstruction problem considered in the paper. Upper-threshold conjecture. ... The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-066303 | MATHDB-345668

Square-placement Wilf-equivalence conjecture for three length-four patterns

摘录 / EXCERPT: For a distant pattern, write a square between two consecutive letters to indicate the constrained gap at that position. Consider the classical patterns 1234, 1243, and 2143,...

conjecture | open | MathDB 页面 | 原文线索

V2-066304 | MATHDB-345678

Existence of marked $(\infty,2)$-colimits

摘录 / EXCERPT: Existence conjecture. There is a theory of marked $(\infty,2)$-colimits in every $(\infty,2)$-category, with input and output as above.

conjecture | open | MathDB 页面 | 原文线索

V2-066305 | MATHDB-345679

Marked colimits in the infinity-bicategory of infinity-categories

摘录 / EXCERPT: Special-cases conjecture. In $\mathfrak{C}\!\operatorname{at}_{\infty}$, the marked $(\infty,2)$-colimit of $F:X^\flat\to\mathfrak{C}\!\operatorname{at}_{\infty}$ coincides with th...

conjecture | open | MathDB 页面 | 原文线索

V2-066306 | MATHDB-345680

The cofinality conjecture for marked infinity-2-functors

摘录 / EXCERPT: The cofinality conjecture. Precomposition with F preserves marked $(\infty,2)$-colimits if and only if, for every object $d\in\mathbfcal{D}$: (1) there are $c\in\mathbfcal{C}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-066307 | MATHDB-345682

The hook-restricted Schur expansion conjecture for the nabla operator

摘录 / EXCERPT: Let n be a positive integer, let $ abla$ be the nabla operator on symmetric functions, and let e_n and s_{μ} denote the elementary and Schur symmetric functions, respective...

conjecture | open | MathDB 页面 | 原文线索

V2-066308 | MATHDB-345686

The affine Stanley-Wilf growth-rate conjecture

摘录 / EXCERPT: Affine Stanley-Wilf growth-rate conjecture. The proper growth rate

conjecture | open | MathDB 页面 | 原文线索

V2-066309 | MATHDB-345689

Noncriticality conjecture for the Hausdorff dimension on the Mandelbrot main cardioid

摘录 / EXCERPT: Let $\mathcal{H}$ be the main cardioid of the Mandelbrot set, and let $\delta: \mathcal{H} \to [0,2)$ be the Hausdorff dimension function. Noncriticality conjecture. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-066310 | MATHDB-345704

The descendent DT4 formula for genus-one Gopakumar-Vafa type invariants

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau 4-fold, let $\beta\in H_2(X,\mathbb{Z})$, and let $M_1(X,\beta)$ be the moduli space of one-dimensional stable sheaves F on X with...

conjecture | open | MathDB 页面 | 原文线索

V2-066311 | MATHDB-345729

The p-adic Beilinson conjecture for totally real fields

摘录 / EXCERPT: The p-adic Beilinson conjecture. For each odd integer $j>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-066312 | MATHDB-345736

Morris's infinite-fluctuation conjecture for critical two-dimensional Ising dynamics

摘录 / EXCERPT: Let $\mathcal U$ be a critical two-dimensional family, meaning that its stable directions intersect every open semicircle and have finite intersection with some semicircle. Start t...

conjecture | open | MathDB 页面 | 原文线索

V2-066313 | MATHDB-345741

Partial List Coloring Conjecture

摘录 / EXCERPT: Let G be a graph, let L be a list assignment, and let $\alpha_L(G)$ be the maximum number of vertices that can be properly L-colored. For a positive integer t, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066314 | MATHDB-345748

The list 1-2-3 Conjecture

摘录 / EXCERPT: Let G be a connected graph with at least two edges, and assign to each edge e a list $L(e)$ of allowable real numbers. A decoration is cool if the sums at adjacent vertices are...

conjecture | open | MathDB 页面 | 原文线索

V2-066315 | MATHDB-345750

The list cascade coloring conjecture

摘录 / EXCERPT: For a positive integer k and a positive integer r, a cascade of length k rooted at r is the set $\{r,2r,\dots,kr\}$. The list cascade coloring conjecture. Every graph G i...

conjecture | open | MathDB 页面 | 原文线索

V2-066316 | MATHDB-345760

Spatzier's rank-rigidity lattice conjecture

摘录 / EXCERPT: Let G be a simple Lie group with real rank at least 2. For a countable group Γ, let A_i consist of elements whose centralizer contains a free abelian subgroup of rank...

conjecture | open | MathDB 页面 | 原文线索

V2-066317 | MATHDB-345770

The τ-twisted Dedekind conjecture for extensions of global fields

摘录 / EXCERPT: Let E/F be an extension of global fields, and let τ be as in the preceding notation. Write $N_{E/F}$ for the relative norm and let $\Lambda_E(s,\tau\circ N_{E/F})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066318 | MATHDB-345771

Li, Wu, Meng and Ma's k-tree connectivity conjecture for line graphs

摘录 / EXCERPT: Let G be a connected graph with at least k vertices and at least k edges, and let $L(G)$ be its line graph. For a set $S\subseteq V(G)$ with $|S|\geq 2$, let $\lambda_G(S)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-066319 | MATHDB-345775

Cutoff conjecture for symmetric simple exclusion with open boundaries

摘录 / EXCERPT: Symmetric cutoff conjecture. The lower bound in the symmetric pre-cutoff estimate is sharp, and cutoff occurs.

conjecture | open | MathDB 页面 | 原文线索

V2-066320 | MATHDB-345776

High-density mixing-time constant conjecture for asymmetric open exclusion

摘录 / EXCERPT: Consider the asymmetric simple exclusion process with open boundaries, with parameters a,b,p satisfying the assumptions of the paper's two-sided asymmetric regime. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066321 | MATHDB-345777

N^{3/2}-scale mixing conjecture in the maximal-current phase

摘录 / EXCERPT: Consider the simple exclusion process with open boundaries, with phase parameters a and b, and let $t^N_{\mathrm{mix}}(\varepsilon)$ denote its ε-mixing time. Maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-066322 | MATHDB-345780

The all-charge multicenter Dirac lower-bound conjecture

摘录 / EXCERPT: Let $M\geqslant2$, let $R_1,\ldots,R_M\in\mathbb{R}^3$, and let $\nu_m\geqslant0$ satisfy $\sum_{m=1}^M\nu_m\leqslant1$. Let D_0 be the free Dirac operator and let λ_1 de...

conjecture | solved | MathDB 页面 | 原文线索

V2-066323 | MATHDB-345782

Optimality of a delta distribution for the lowest Dirac-Coulomb eigenvalue

摘录 / EXCERPT: Optimality for a delta. The first eigenvalue is minimal when the charge distribution is concentrated at the origin, namely when $\mu=\mu(\mathbb{R}^3)\delta_0$. Equivalently, for a...

conjecture | open | MathDB 页面 | 原文线索

V2-066324 | MATHDB-345799

The dichotomy conjecture for induced poset saturation

摘录 / EXCERPT: Induced saturation dichotomy conjecture. Either there exists a constant K_P such that

conjecture | open | MathDB 页面 | 原文线索

V2-066325 | MATHDB-345800

The boundedness conjecture for induced saturation under adjoining a maximum

摘录 / EXCERPT: Adjoining-a-maximum conjecture. The quantity ${\rm sat}^*(n,P)$ is bounded as a function of n if and only if ${\rm sat}^*(n,\dot P)$ is bounded as a function of n.

conjecture | open | MathDB 页面 | 原文线索

V2-066326 | MATHDB-345808

Conjecture on Weil-Petersson metric completion for stable families

摘录 / EXCERPT: Weil-Petersson extension and completion conjecture. The metric ω_{WP} extends uniquely to a non-negative closed $(1,1)$-current on $O{S}$ with continuous local potentials, a...

conjecture | open | MathDB 页面 | 原文线索

V2-066327 | MATHDB-345816

Unimodality conjecture for $\operatorname{LS}_{n,1}(112)$

摘录 / EXCERPT: Let $\operatorname{LS}_{n,1}(112)$ be the generating polynomial under consideration for restricted growth functions of length n containing the pattern 112 exactly once; its coe...

conjecture | open | MathDB 页面 | 原文线索

V2-066328 | MATHDB-345817

Linear independence conjecture for regularized double zeta values with first argument 1

摘录 / EXCERPT: Let $J_{\mathfrak{D}}(1;{\rm odd},{\rm odd})$ denote the regularized double zeta values whose first argument is 1 and whose remaining arguments are odd. Linear independence conje...

conjecture | open | MathDB 页面 | 原文线索

V2-066329 | MATHDB-345829

Lang's conjecture on integral points of elliptic curves

摘录 / EXCERPT: Let L be a number field, let S be a finite set of places of L containing the archimedean places, let R_S be the ring of S-integers of L, and let E be an elliptic curv...

conjecture | open | MathDB 页面 | 原文线索

V2-066330 | MATHDB-345842

Bonsma-Cereceda diameter conjecture for colouring graphs

摘录 / EXCERPT: Bonsma-Cereceda's conjecture. For $k\geq\operatorname{col}(G)+1$,

conjecture | open | MathDB 页面 | 原文线索

V2-066331 | MATHDB-345843

Mynhardt-Roux path and cycle non-realisation conjecture for irredundance graphs

摘录 / EXCERPT: Mynhardt-Roux's conjecture. For every $n\geq3$, P_n is not an IR-graph, and for every $n\geq5$, C_n is not an IR-graph.

conjecture | open | MathDB 页面 | 原文线索

V2-066332 | MATHDB-345846

Mestrano-Simpson conjecture for stable bundles on a very general sextic surface

摘录 / EXCERPT: Mestrano-Simpson conjecture. The 13-dimensional family in Theorem 2 constitutes a full irreducible component of $\mathcal{M}(H,11)$; this component is nonreduced and obstructed....

conjecture | open | MathDB 页面 | 原文线索

V2-066333 | MATHDB-345859

Valuation formula for Kac polynomials of multi-arrowed quivers

摘录 / EXCERPT: Valuation conjecture. Provided $A_{Q_{\underline{n}},\operatorname{\mathbf{d}}}(q)\neq 0$, its valuation is

conjecture | open | MathDB 页面 | 原文线索

V2-066334 | MATHDB-345860

Direction-independent stabilization conjecture for normalized Kac polynomials

摘录 / EXCERPT: Stabilization conjecture. The sequence

conjecture | open | MathDB 页面 | 原文线索

V2-066335 | MATHDB-345866

Positive Lyapunov exponent conjecture for projective K3 automorphisms

摘录 / EXCERPT: Positive Lyapunov exponent conjecture. One has $\Lambda>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-066336 | MATHDB-345869

Stable line shelling conjecture for cubical polytopes

摘录 / EXCERPT: Let $\mathcal{C}$ be the boundary complex of a cubical polytope. A stable line shelling is a line shelling satisfying the stability condition defined through the realization and in...

conjecture | open | MathDB 页面 | 原文线索

V2-066337 | MATHDB-345873

Faraday's charge-cleaning conjecture for inverse electrostatic potentials

摘录 / EXCERPT: Let $B=\{x\in\mathbb{R}^d:|x|<1\}$ with $d>2$, and let μ be a signed charge distribution supported on $\overline{B}$ such that ... Thus, outside $\overline{B}$, μ produces th...

conjecture | open | MathDB 页面 | 原文线索

V2-066338 | MATHDB-345874

Rationality conjecture for Gushel fourfolds with associated K3 surfaces

摘录 / EXCERPT: Rationality conjecture. A fourfold $[X]\in\mathcal M_4$ is rational if and only if it has an associated K3 surface, equivalently if and only if $[X]$ belongs to

conjecture | open | MathDB 页面 | 原文线索

V2-066339 | MATHDB-345885

Goulden-Jackson-Vakil conjecture for one-part double Hurwitz numbers

摘录 / EXCERPT: Let $h_{g;\mu_1,\ldots,\mu_n}^{\textnormal{one-part}}$ denote the double Hurwitz number with one ramification profile equal to the one-part partition of size...

conjecture | open | MathDB 页面 | 原文线索

V2-066340 | MATHDB-345896

Carvalho-Lucchesi-Murty laminar ELP cut conjecture

摘录 / EXCERPT: Carvalho-Lucchesi-Murty conjecture. The graph G has a nontrivial ELP cut that is laminar with C.

conjecture | open | MathDB 页面 | 原文线索

V2-066341 | MATHDB-345899

The LG mirror model proposal for hyperelliptic curves

摘录 / EXCERPT: Let Y be a general hyperelliptic curve of genus $g\geq2$, with Kähler parameters t_1,t_2, and let φ,ψ satisfy ... The proposed LG mirror model has torus...

conjecture | open | MathDB 页面 | 原文线索

V2-066342 | MATHDB-345913

Scaling-limit conjecture for the interface length in critical Ising triangulations

摘录 / EXCERPT: Let $(\mathfrak{t},\sigma)$ be sampled under $\mathbb{P}_{p,q}$, and let $\eta_{p,q}$ be the length of its leftmost interface. Let E be the expectation under $\mathbb{P}_\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-066343 | MATHDB-345915

Jiang's multiple-cover conjecture for twisted Vafa-Witten invariants

摘录 / EXCERPT: Jiang's twisted multiple-cover conjecture. If $H^{0,1}(S)=H^{0,2}(S)=0$, there exist rational numbers $\operatorname{VW}^{\operatorname{tw}}_{\alpha_i}({\mathfrak S})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066344 | MATHDB-345918

The conjecture on asymptotic normality and log-normality of hidden-word counts

摘录 / EXCERPT: Let Z be the number of occurrences of a pattern w as a subsequence in a random text of length n, and let $\binom{n}{m}$ be the number of possible positions for a subsequence...

conjecture | open | MathDB 页面 | 原文线索

V2-066345 | MATHDB-345920

Unimodality conjecture for the convexity-deficit distribution of benzenoids

摘录 / EXCERPT: Let $h\geq 0$, and let $F(h,k)$ denote the number of benzenoids on h hexagons having convexity deficit equal to k. Let $\operatorname{mcd}(h)$ be the maximum convexity deficit...

conjecture | open | MathDB 页面 | 原文线索

V2-066346 | MATHDB-345925

The cycle-to-path local metamorphosis conjecture

摘录 / EXCERPT: Let k be a positive integer, and consider parallel classes whose blocks are copies of the cycle C_k or the path P_k. Cycle-to-path local metamorphosis conjecture. The union o...

conjecture | open | MathDB 页面 | 原文线索

V2-066347 | MATHDB-345926

Conjecture on monomials with the intersection distribution of x^3

摘录 / EXCERPT: Monomial intersection-distribution conjecture. The following two families of monomials x^d over $\mathbb{F}_q=\mathbb{F}_{3^m}$ have the same intersection distribution as x^3:

conjecture | open | MathDB 页面 | 原文线索

V2-066348 | MATHDB-345927

The 4/4 conjecture for permutation rational functions

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, and let $N_{v,u}(q)$ denote the number of permutation rational functions $V(x)/U(x)$ over $\mathbb{F}_q$ with numerator degree v and denomin...

conjecture | open | MathDB 页面 | 原文线索

V2-066349 | MATHDB-345930

Existence of a reversible physically universal one-dimensional Turing machine

摘录 / EXCERPT: A one-dimensional reversible Turing machine is a reversible Turing machine whose tape is one-dimensional. Reversible physical universality means physical universality in the specif...

conjecture | open | MathDB 页面 | 原文线索

V2-066350 | MATHDB-345942

The critical-diagonal conjecture for phase transition in the area-interaction model

摘录 / EXCERPT: Let $z\geq 0$ be the activity and let $\beta\geq 0$ be the inverse temperature of the area-interaction model. Phase transition means non-uniqueness of the area-interaction measures...

conjecture | open | MathDB 页面 | 原文线索

V2-066351 | MATHDB-345947

The discrete negative sign uncertainty square-size conjecture

摘录 / EXCERPT: Discrete square-size conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066352 | MATHDB-345948

The discrete Hankel-transform feasibility conjecture

摘录 / EXCERPT: Discrete Hankel-transform feasibility conjecture. If (k,q) is (s,ν)-feasible, then $(k+1,q)$ and $(k,q-1)$ are (s,ν)-feasible. The function...

conjecture | open | MathDB 页面 | 原文线索

V2-066353 | MATHDB-345949

The structured discrete Hankel-transform feasibility conjecture

摘录 / EXCERPT: Structured discrete Hankel-transform conjecture. The following pairs are feasible:

conjecture | open | MathDB 页面 | 原文线索

V2-066354 | MATHDB-345950

The equivalence between excellence and constancy of the image Milnor number

摘录 / EXCERPT: Let $f\colon(\mathbb{C}^{n},S)\rightarrow(\mathbb{C}^{n+1},0)$ be an $\mathscr{A}$-finite germ, and let $F(x,t)=(f_t(x),t)$ be an origin-preserving one-parameter unfolding. The exc...

conjecture | open | MathDB 页面 | 原文线索

V2-066355 | MATHDB-345957

Courant-sharpness conjecture for negative Robin eigenvalues of the square

摘录 / EXCERPT: Courant-sharpness conjecture. If $\lambda_{0,q,h}(S)$ is negative, then no associated eigenfunction is Courant-sharp.

conjecture | open | MathDB 页面 | 原文线索

V2-066356 | MATHDB-345959

Conjecture on kinetic equations for smooth decaying potentials

摘录 / EXCERPT: Let Φ be a smooth decaying potential and consider the scaling ... with $\varepsilon\to0$, where N is the average number of particles per macroscopic unit volume. Smooth decayi...

conjecture | open | MathDB 页面 | 原文线索

V2-066357 | MATHDB-345960

Conjecture on kinetic equations for scale-invariant potentials

摘录 / EXCERPT: Let Φ be a scale-invariant potential of the form ... and consider the scaling ... with $\gamma>0$. Scale-invariant potentials conjecture. If $s\in(\frac12,1)$, then the one-partic...

conjecture | open | MathDB 页面 | 原文线索

V2-066358 | MATHDB-345964

Lin and Yan's zero-refinement conjecture for 120-avoiding inversion sequences

摘录 / EXCERPT: Let $\mathbf{I}_n$ be the set of inversion sequences of length n, and let $\mathcal{PA}_n$ be the set of primitive ascent sequences of length n. For an inversion sequence e,...

conjecture | open | MathDB 页面 | 原文线索

V2-066359 | MATHDB-345965

Beaton-Bouvel-Guerrini-Rinaldi's right-to-left-minimum refinement for 2314-avoiding permutations

摘录 / EXCERPT: Let $\mathfrak{S}_n(\underline{23}14)$ be the permutations avoiding the vincular pattern $\underline{23}14$, and let $c_{n,k}$ be the powered-Catalan refinement defined recursively...

conjecture | open | MathDB 页面 | 原文线索

V2-066360 | MATHDB-345966

Fu and Lin's four-statistic equidistribution conjecture for 2314-avoiding permutations

摘录 / EXCERPT: Let $\mathfrak{S}_n(\underline{23}14)$ be the permutations avoiding the vincular pattern $\underline{23}14$, and let $\mathbf{I}_n(110)$ be the inversion sequences avoiding 110....

conjecture | open | MathDB 页面 | 原文线索

V2-066361 | MATHDB-345967

Last-entry and right-to-left-maximum equidistribution for 2314-avoiding permutations

摘录 / EXCERPT: Let $\mathfrak{S}_n(\underline{23}14)$ be the permutations avoiding the vincular pattern $\underline{23}14$, and let $\mathbf{I}_n(\underline{12}0)$ be the inversion sequences avoi...

conjecture | open | MathDB 页面 | 原文线索

V2-066362 | MATHDB-345968

Non-trivial automorphism conjecture for APN functions and permutations

摘录 / EXCERPT: Let $F\colon\mathbb{F}_2^n\rightarrow\mathbb{F}_2^n$ be an APN function, and let $\mathsf{Aut}(F)$ denote its automorphism group. If F is an APN permutation, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066363 | MATHDB-345978

The ell-torsion conjecture for class groups of number fields

摘录 / EXCERPT: ell-torsion conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-066364 | MATHDB-345979

The discriminant multiplicity conjecture for number-field extensions

摘录 / EXCERPT: Discriminant multiplicity conjecture. For every $D\in\mathbb{N}$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-066365 | MATHDB-345982

Dirichlet disk eigenfunction growth conjecture for $1\leq\gamma<3$

摘录 / EXCERPT: Dirichlet disk growth conjecture. If $1\leq\gamma<3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066366 | MATHDB-345983

Neumann disk eigenfunction growth conjecture for $\gamma\geq1$

摘录 / EXCERPT: Let $F^N_{n,m}$ be the L^2-normalized eigenfunctions of the Neumann Laplace operator on the unit disk, with eigenvalues $\lambda^N_{n,m}$. For a subsequence with...

conjecture | open | MathDB 页面 | 原文线索

V2-066367 | MATHDB-345984

Connected Julia set dichotomy conjecture for rational functions

摘录 / EXCERPT: Let $f:\widehat{\mathbb C}\to\widehat{\mathbb C}$ be a rational function of degree at least 2. Assume that the Julia set $J(f)$ is connected. Connected Julia set dichotomy conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-066368 | MATHDB-345987

Dichotomy conjecture for free products of automaton semigroups

摘录 / EXCERPT: Let $\mathcal{A}$ and $\mathcal{B}$ be automata, and let $\Sigma(\mathcal{A})$ and $\Sigma(\mathcal{B})$ be the automaton semigroups generated by their states. Their free product i...

conjecture | open | MathDB 页面 | 原文线索

V2-066369 | MATHDB-345992

Rank-width conjecture for bipartite circle graph pivot-minors

摘录 / EXCERPT: A graph H is a bipartite circle graph if it is both bipartite and a circle graph. Oum's rank-width conjecture. For every bipartite circle graph H, there is an integer $r(H)$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-066370 | MATHDB-345995

Bartoli-Timpanella conjecture on a power function with planar-like differential uniformity

摘录 / EXCERPT: Bartoli-Timpanella conjecture. For $c=-1$, the power function F is PcN over ${\mathrm{GF}}(p^n)$. This conjecture concerns the existence of power functions with perfect c-non...

conjecture | open | MathDB 页面 | 原文线索

V2-066371 | MATHDB-345997

Grand uniqueness conjecture for the genus-zero and genus-one Canham problems

摘录 / EXCERPT: Grand uniqueness conjecture. The genus-$g=0$ or 1 Canham problem with any isoperimetric ratio constraint $v_0\in(0,1)$ has a unique solution up to homothety. Moreover: (i) for...

conjecture | open | MathDB 页面 | 原文线索

V2-066372 | MATHDB-346021

Integrality conjecture for Seifert-fibered surgeries on hyperbolic knots

摘录 / EXCERPT: Integrality conjecture. Any Seifert-fibered surgery on a hyperbolic knot in S^3 is integral.

conjecture | open | MathDB 页面 | 原文线索

V2-066373 | MATHDB-346022

Primitive/Seifert position conjecture for tunnel-number-one knots

摘录 / EXCERPT: Primitive/Seifert position conjecture. Any Seifert-fibered surgery on a hyperbolic tunnel-number-one knot arises from a primitive/Seifert position whose surface slope corresponds t...

conjecture | open | MathDB 页面 | 原文线索

V2-066374 | MATHDB-346026

Conjecture on transitory canard and saddle-node canard cycles

摘录 / EXCERPT: Transitory canard cycles conjecture. There exist values of $\varepsilon>0$ and $k\approx 1$ such that the corresponding system exhibits one, two, or three canard limit cycles in a...

conjecture | open | MathDB 页面 | 原文线索

V2-066375 | MATHDB-346028

Spin q-Whittaker polynomial orthogonality conjecture

摘录 / EXCERPT: Spin q-Whittaker orthogonality conjecture. For all signatures $lambda,mu$,

conjecture | open | MathDB 页面 | 原文线索

V2-066376 | MATHDB-346049

Eldredge-Gordina-Saloff-Coste eigenvalue-diameter conjecture

摘录 / EXCERPT: Let G be a compact connected Lie group, and let $\mathcal M^G$ denote the space of left-invariant Riemannian metrics on G. For $g\in\mathcal M^G$, write $\lambda_1(G,g)$ for th...

conjecture | open | MathDB 页面 | 原文线索

V2-066377 | MATHDB-346050

Sarnak's Density Conjecture

摘录 / EXCERPT: Let G be a real, semisimple, almost-simple and simply connected Lie group, let Γ_1 be a cocompact arithmetic lattice of G, and let (Γ_N) be a sequence of finite-i...

conjecture | open | MathDB 页面 | 原文线索

V2-066378 | MATHDB-346051

Weak Injective Radius Property implies the General Density and Spherical Density Hypotheses

摘录 / EXCERPT: Let a sequence of congruence subgroups be equipped with the Weak Injective Radius Property, the General Density Hypothesis, and the Spherical Density Hypothesis as defined in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-066379 | MATHDB-346052

Optimal Weak Injective Radius and General Density conjecture for Archimedean congruence subgroups

摘录 / EXCERPT: Let G be an Archimedean, semisimple, almost-simple and simply connected group, let Γ_1 be an arithmetic lattice, and let (Γ_N) be a sequence of congruence subgroups...

conjecture | open | MathDB 页面 | 原文线索

V2-066380 | MATHDB-346054

Kollár's local fundamental group conjecture for klt singularities

摘录 / EXCERPT: Kollár's local fundamental group conjecture. The group $\pi_1^{\rm loc}(X,x)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-066381 | MATHDB-346055

Finiteness conjecture for the smooth locus of weakly Fano pairs

摘录 / EXCERPT: Finiteness conjecture for weakly Fano pairs. The fundamental group

conjecture | open | MathDB 页面 | 原文线索

V2-066382 | MATHDB-346056

Rank-zero Bloch-Kato conjecture for triple product motives

摘录 / EXCERPT: Rank-zero Bloch-Kato conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-066383 | MATHDB-346067

Alexandersson-Amini's cyclic sieving conjecture for stretched skew tableaux

摘录 / EXCERPT: Let λ/μ be a skew shape, let m be a positive integer, and let k be a positive integer. Write $m\lambda/m\mu$ for the skew shape obtained by multiplying every part of...

conjecture | open | MathDB 页面 | 原文线索

V2-066384 | MATHDB-346074

The lax- and pseudo-slice characterization of 2-limits

摘录 / EXCERPT: Lax- and pseudo-slice characterization. The pair (L,λ) is a 2-limit of the 2-functor F precisely when it is a 2-terminal object in the lax-slice (or pseudo-slice)...

conjecture | open | MathDB 页面 | 原文线索

V2-066385 | MATHDB-346076

Esperet's conjecture on polynomially chi-bounded graph classes

摘录 / EXCERPT: Let $\mathcal G$ be a class of graphs. A class is χ-bounded if there is a function $f:\mathbb{Z}^+\to\mathbb{Z}^+$ such that every graph $G\in\mathcal G$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-066386 | MATHDB-346078

The Fargues-Rapoport conjecture for weakly admissible Newton strata

摘录 / EXCERPT: Assume that G is quasi-split and that $[b]\in A(G,\mu)$ is basic. Let $[b']$ be an element of ... Suppose that $(G,\nu_b(w_0\mu^{-1})^{\diamond},b')$ is HN-indecomposable. Fargues-...

conjecture | open | MathDB 页面 | 原文线索

V2-066387 | MATHDB-346081

Border symmetric rank conjecture for monomials

摘录 / EXCERPT: Let $x_0,\ldots,x_n$ be variables and let $0\leq\alpha_0\leq\cdots\leq\alpha_n$. Write bsrk for border symmetric rank. Border symmetric rank conjecture. For these...

conjecture | open | MathDB 页面 | 原文线索

V2-066388 | MATHDB-346085

The primitive derangement subgroup index conjecture

摘录 / EXCERPT: Let G be a primitive permutation group of degree n, and let $D(G)$ be the subgroup generated by the derangements in G. A Frobenius group is a transitive permutation group who...

conjecture | open | MathDB 页面 | 原文线索

V2-066389 | MATHDB-346088

The GM series large-color expansion conjecture

摘录 / EXCERPT: Let K be a knot, let $J_K(n;q)$ be its reduced colored Jones polynomial, and let $\Delta_K(x)$ be its Alexander polynomial. Write the large-color expansion near $\hbar=0$ while k...

conjecture | open | MathDB 页面 | 原文线索

V2-066390 | MATHDB-346090

The GM series conjecture for links

摘录 / EXCERPT: Let L be an l-component link, let $J_L(n_1,\ldots,n_l;q)$ be its colored Jones polynomials, and let $\nabla_L(x_1,\ldots,x_l)$ be its Alexander-Conway function. The link GM-ser...

conjecture | open | MathDB 页面 | 原文线索

V2-066391 | MATHDB-346092

The large-color R-matrix conjecture for fibered strongly quasi-positive braid knots

摘录 / EXCERPT: The fibered strongly quasi-positive braid-knot conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-066392 | MATHDB-346095

Alon-Tarsi bound for Earth-Moon graphs

摘录 / EXCERPT: Let $G=G_1\cup G_2$ be an Earth-Moon graph, meaning that G_1 and G_2 are planar subgraphs. The Alon-Tarsi conjecture. Every Earth-Moon graph G satisfies ... This would follow i...

conjecture | open | MathDB 页面 | 原文线索

V2-066393 | MATHDB-346106

Le Boudec's conjecture on small points in quadratic twist families

摘录 / EXCERPT: Le Boudec's conjecture. For every $\alpha>0$, there exists a constant $c_{A,B}(\alpha)>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066394 | MATHDB-346107

Universal edge fluctuation conjecture for dimer models

摘录 / EXCERPT: Universal Edge Fluctuation Conjecture. Under suitable assumptions on the convergence of G_n, all scaling limits of the determinantal point processes at the frozen boundary are gi...

conjecture | open | MathDB 页面 | 原文线索

V2-066395 | MATHDB-346112

Conjecture on bounded-degree induced trees in random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the binomial random graph. For a graph G, write $T(G)$ for the size of a largest induced subtree, and set $q=(1-p)^{-1}$. Let $\Delta\geq 2$ be an integer and le...

conjecture | open | MathDB 页面 | 原文线索

V2-066396 | MATHDB-346127

Minimality of the rock-salt structure for inverse-power and Gaussian interactions

摘录 / EXCERPT: Minimality of the rock-salt structure. There exist $delta_0,delta_1>0$, depending only on d, such that the global minimizer of E_{f_1,f_2} is of the form

conjecture | open | MathDB 页面 | 原文线索

V2-066397 | MATHDB-346135

High-weight generator polynomial conjecture for shortened second-order Reed-Muller codes

摘录 / EXCERPT: Let $m\ge 5$ be an integer, let α be a primitive element of ${\mathrm{GF}}(2^m)$, and let $g^*_{2,\alpha}(X)$ denote the generator polynomial of the shortened second-order R...

conjecture | open | MathDB 页面 | 原文线索

V2-066398 | MATHDB-346136

The antimagic labeling conjecture for lobsters

摘录 / EXCERPT: Let T be a lobster, that is, a tree whose vertices are at distance at most two from a fixed path. An antimagic labeling of T is a bijection $\tau:E(T)\rightarrow [|E(T)|]$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-066399 | MATHDB-346138

The tower ground state conjecture for selected imaginary quadratic fields

摘录 / EXCERPT: Tower ground state conjecture. The fields with discriminants $d\in\{-225\,299,-343\,380,-423\,476,-486\,264\}$ of type $\mathrm{F}.7$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066400 | MATHDB-346161

Parity-density conjecture for partitions with odd multiplicities

摘录 / EXCERPT: Let $a(n)$ denote the number of partitions of n in which every part occurs with odd multiplicity. For a sequence, say that its odd values have density d when the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-066401 | MATHDB-346183

The universal framed KZB connection conjecture

摘录 / EXCERPT: Let C be a smooth curve, let $\operatorname{Conf}(C,n)$ be the configuration space of n distinct points on C, and let $\operatorname{Conf}^f(C,n)$ denote its framed version....

conjecture | open | MathDB 页面 | 原文线索

V2-066402 | MATHDB-346191

Krivelevich-Alon conjecture on the list chromatic number of bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph with maximum degree at most Δ, and let k be a positive integer. The graph is k-choosable if every assignment of lists of k colours to its...

conjecture | open | MathDB 页面 | 原文线索

V2-066403 | MATHDB-346192

Asymmetric list-colouring conjecture for bipartite graphs

摘录 / EXCERPT: Let $G=(V=A\cup B,E)$ be a bipartite graph with maximum degrees on A and B at most Δ_A and Δ_B, respectively. For positive integers k_A and k_B, say that G...

conjecture | open | MathDB 页面 | 原文线索

V2-066404 | MATHDB-346198

Ignjatović's asymptotic Christoffel-Darboux kernel conjecture

摘录 / EXCERPT: Ignjatović's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-066405 | MATHDB-346207

Optimal-diffusion conjecture for the zero-Dirichlet river competition model

摘录 / EXCERPT: Consider the two-species river competition system with diffusion rates $d,D>0$, common advection rate $\alpha>0$, growth rate $r(x)$, upstream flux boundary conditions, and zero Di...

conjecture | open | MathDB 页面 | 原文线索

V2-066406 | MATHDB-346208

Critical-sinking-velocity conjecture for phytoplankton diffusion competition

摘录 / EXCERPT: Consider the two-species phytoplankton competition model with equal advection rates $\alpha_1=\alpha_2=\alpha$, equal growth functions $g_1=g_2$, equal additional diffusion paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-066407 | MATHDB-346211

Giga's conjecture on global existence of embedded curve diffusion flows

摘录 / EXCERPT: Giga's conjecture. Then $T=\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-066408 | MATHDB-346213

The Pixton formula conjecture for weighted fundamental classes of twisted canonical divisors

摘录 / EXCERPT: Let g,n,k be as above, let $A=(a_1,\ldots,a_n)$ satisfy $\sum_{i=1}^n a_i=k(2g-2)$, and let $\widetilde A=(a_1+k,\ldots,a_n+k)$. For $k\geq 1$, assume that A is not of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-066409 | MATHDB-346214

The genus-2 mutation invariance conjecture for tau invariants

摘录 / EXCERPT: Genus-2 mutation invariance conjecture. The values of $\tau^\sharp$ and τ are preserved by genus-2 mutation.

conjecture | open | MathDB 页面 | 原文线索

V2-066410 | MATHDB-346220

Schwarz-Pick inequality for bounded harmonic functions on the unit ball

摘录 / EXCERPT: Let $n=2$ or $n\geq 4$, let u satisfy the hypotheses of Theorem (i), and let $x\in\mathbb{B}_n$. Schwarz-Pick conjecture. One has ... This proposed refinement combines the sharp gr...

conjecture | open | MathDB 页面 | 原文线索

V2-066411 | MATHDB-346221

Liu-Ponnusamy conjectures on Bohr radii for harmonic quasi-subordination

摘录 / EXCERPT: Liu-Ponnusamy's conjectures. (a) If φ is univalent and convex in $\mathbb D$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066412 | MATHDB-346228

Vanishing Chern classes conjecture for numerically flat Higgs bundles

摘录 / EXCERPT: Vanishing Chern classes conjecture. If $\mathfrak{E}=(E,\varphi)$ is an H-nflat Higgs bundle, then

conjecture | open | MathDB 页面 | 原文线索

V2-066413 | MATHDB-346230

Regularity conjecture for quasiconformal homeomorphisms of rigid Carnot groups

摘录 / EXCERPT: Let G be a Carnot group. A group is rigid if the space of smooth contact embeddings $G\supset U\to G$ is finite-dimensional for every connected open subset $U\subset G$. Regulari...

conjecture | open | MathDB 页面 | 原文线索

V2-066414 | MATHDB-346231

Xie's bilipschitz conjecture for quasiconformal mappings of Carnot groups

摘录 / EXCERPT: Let G be a Carnot group, and let $U,U'\subset G$ be open sets. Xie's conjecture. If G is neither $\mathbb{R}^n$ nor a Heisenberg group $\mathbb{H}_n$, then every quasiconformal...

conjecture | open | MathDB 页面 | 原文线索

V2-066415 | MATHDB-346237

Functional L_p-Gardner-Zvavitch conjecture

摘录 / EXCERPT: Functional L_p-Gardner-Zvavitch conjecture. There exists an absolute constant $C\geq1$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-066416 | MATHDB-346239

The conjecture on real periodic points of Hermite polynomials

摘录 / EXCERPT: For each $d\geq 1$, define the Hermite polynomial by ... Let $\mathscr{R}_d$ denote the set of real polynomials of degree d having only real periodic points. Hermite polynomial con...

conjecture | open | MathDB 页面 | 原文线索

V2-066417 | MATHDB-346243

An upper-bound conjecture for weighted Hermite polynomials

摘录 / EXCERPT: Hermite-polynomial upper-bound conjecture. For all $n\in\mathbb{N}^+$ and $x>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066418 | MATHDB-346248

Unique continuation conjecture for harmonic functions in Lipschitz domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a Lipschitz domain and let $\Sigma\subset\partial\Omega$ be relatively open with respect to $\partial\Omega$. Let u be harmonic in Ω and...

conjecture | open | MathDB 页面 | 原文线索

V2-066419 | MATHDB-346258

Per Alexandersson's MNability conjecture for Petrie functions times p_2

摘录 / EXCERPT: Let $G(k,m)\in\Lambda$ be the Petrie symmetric function, let p_2 be the second power-sum symmetric function, let Par denote the set of partitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066420 | MATHDB-346259

François Bergeron's nabla-positivity conjecture for Petrie functions

摘录 / EXCERPT: Let $G(k,m)\in\Lambda$ be the Petrie symmetric function, let ∇ be the nabla operator on symmetric functions, let Par denote the set of partitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066421 | MATHDB-346262

Oddness conjecture for valid hook configurations

摘录 / EXCERPT: For $n\geq 3$, let $\operatorname{VHC}(S_{n-1})$ denote the set of valid hook configurations of permutations in S_{n-1}. Oddness conjecture. If $n\geq 3$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-066422 | MATHDB-346266

The maximal-order containment bound for Bass orders

摘录 / EXCERPT: Maximal-order containment conjecture. For every $\epsilon>0$, the number of maximal orders containing Λ is

conjecture | open | MathDB 页面 | 原文线索

V2-066423 | MATHDB-346275

The lc-conjecture for very basic slc-trivial fibrations

摘录 / EXCERPT: Let $f\colon (X, B_X) \to (Y,D)$ be a very basic slc-trivial fibration with structure decomposition ... Here B_Y is the discriminant part and M_Y is the moduli part. The lc-conje...

conjecture | open | MathDB 页面 | 原文线索

V2-066424 | MATHDB-346276

The abundance conjecture for very basic slc-trivial fibrations

摘录 / EXCERPT: Let $f\colon (X, B_X) \to (Y,D)$ be a very basic slc-trivial fibration, where D is the divisor in its structure decomposition. The abundance conjecture. If D is nef, then D i...

conjecture | open | MathDB 页面 | 原文线索

V2-066425 | MATHDB-346285

Gross-Mansour-Tucker interpolation conjecture for non-orientable ribbon graphs

摘录 / EXCERPT: Let G be a non-orientable ribbon graph, and define its partial-dual Euler-genus polynomial by ... A polynomial is interpolating when its non-zero coefficients are all equal to 1....

conjecture | refuted | MathDB 页面 | 原文线索

V2-066426 | MATHDB-346295

Conjectured two-point correlations for the type B MultiTASEP

摘录 / EXCERPT: Let n be the rank parameter of the type B MultiTASEP, and write $\langle a,b\rangle$ for the two-point correlation of entries in the last two positions, with barred symbols den...

conjecture | open | MathDB 页面 | 原文线索

V2-066427 | MATHDB-346316

Conjecture on convexity and concavity of the Arimoto information-combining bound

摘录 / EXCERPT: Let k_α^A denote the Arimoto conditional-entropy function and let $\ast$ denote binary convolution. There exists a value $1<\hat\alpha<1.5783$ such that ... is convex for...

conjecture | open | MathDB 页面 | 原文线索

V2-066428 | MATHDB-346317

Conjecture on convexity and concavity of the conditional Rényi information-combining bound

摘录 / EXCERPT: Let h_α denote the binary conditional Rényi-entropy function and let $\ast$ denote binary convolution. There exists a value $1<\hat\alpha<1.3863$ such that ... is convex for...

conjecture | open | MathDB 页面 | 原文线索

V2-066429 | MATHDB-346333

Bilinear Strichartz estimate conjecture for compact symmetric spaces

摘录 / EXCERPT: Let M be a symmetric space of compact type of dimension d and rank r, and let M_0 range over its irreducible components, with dimension d_0 and rank r_0. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066430 | MATHDB-346334

Geometric interpretation of the differential on antisymmetrized moduli-space forms

摘录 / EXCERPT: Geometric interpretation conjecture. Under this identification, the operation Δ on HGK_0 corresponds to pullback under π, and this pullback induces a well-def...

conjecture | open | MathDB 页面 | 原文线索

V2-066431 | MATHDB-346335

Căldăraru's conjecture for the ribbon graph complex in genera at least two

摘录 / EXCERPT: Căldăraru's conjecture. For every genus $g\geq 2$, there is a natural identification

conjecture | open | MathDB 页面 | 原文线索

V2-066432 | MATHDB-346338

The antichain obstruction conjecture for projectively unique order polytopes

摘录 / EXCERPT: Antichain obstruction conjecture. If $\operatorname{\mathcal{O}}(\operatorname{\mathscr{P}})$ is projectively unique, then $\operatorname{\mathscr{P}}$ has no antichain of size 3...

conjecture | open | MathDB 页面 | 原文线索

V2-066433 | MATHDB-346339

Fukaya-Kato's conjectural interpolation formula for a p-adic L-function

摘录 / EXCERPT: Fukaya-Kato's conjectural interpolation formula. There is a p-adic L-function, an element of the quotient ring of $\widetilde{\Lambda}$, such that for every appropriate pair...

conjecture | open | MathDB 页面 | 原文线索

V2-066434 | MATHDB-346342

Monotonicity conjecture for the second partition eigenvalue

摘录 / EXCERPT: Let $\mathcal{G}$ be a finite, compact, connected metric graph. An exhaustive rigid 2-partition of $\mathcal{G}$ is a partition ${\mathcal P}_2=\{\mathcal{G}_1,\mathcal{G}_2\}$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-066435 | MATHDB-346343

Parametric optimal-partition conjecture for metric graphs

摘录 / EXCERPT: Let $\mathcal{G}$ be a metric graph and let $k\geq1$. A rigid k-partition is a partition of $\mathcal{G}$ of the type considered in the paper, and $\mathcal{P}$ is called interna...

conjecture | open | MathDB 页面 | 原文线索

V2-066436 | MATHDB-346344

Loose-versus-non-exhaustive partition conjecture for metric graphs

摘录 / EXCERPT: Loose-versus-non-exhaustive partition conjecture. The infimum over non-exhaustive rigid k-partitions satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-066437 | MATHDB-346348

Character-sum bound for primitive normal values of rational functions

摘录 / EXCERPT: Rational-function primitive normal conjecture. There exists an element $\alpha\in\mathbb{F}_{q^m}$ such that α and $f(\alpha)$ are simultaneously primitive normal over...

conjecture | open | MathDB 页面 | 原文线索

V2-066438 | MATHDB-346350

Rudnick-Sarnak Poissonian local statistics conjecture for quadratic sequences

摘录 / EXCERPT: Let Y be uniformly distributed on $[0,1)$, and for $N\in\mathbb{N}$ and $0<L\leq N$ define the random variable ... Here $\operatorname{Po}(L)$ denotes a Poisson-distributed random...

conjecture | open | MathDB 页面 | 原文线索

V2-066439 | MATHDB-346359

Non-power-of-two lifting conjecture for $Lie_n^{(2)}$

摘录 / EXCERPT: Let $Lie_n^{(2)}$ denote the S_n-representation considered in the paper, and let $p_1Lie_{n-1}^{(2)}-Lie_n^{(2)}$ be the corresponding symmetric function; equivalently, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066440 | MATHDB-346366

Asymptotic Betti-number formula for ordered configuration spaces on an elliptic curve

摘录 / EXCERPT: Asymptotic Betti-number conjecture. We claim that

conjecture | open | MathDB 页面 | 原文线索

V2-066441 | MATHDB-346370

Universal degree distribution formula for recursive trees with fitnesses

摘录 / EXCERPT: Let $\mathcal{T}$ be a (μ,f,ℓ)-RIF tree, and define ... For $\alpha<\infty$, let $N_k(t,B)$ denote the number of degree-k vertices present by time t whose fitness lies in...

conjecture | open | MathDB 页面 | 原文线索

V2-066442 | MATHDB-346371

Degenerate degree distribution for infinite generalized Malthusian parameter

摘录 / EXCERPT: Let $\mathcal{T}$ be a (μ,f,ℓ)-RIF tree, and define ... Assume that $\alpha=\infty$, and let $N_0(t,B)$ denote the number of degree-zero vertices present by time t whose fit...

conjecture | open | MathDB 页面 | 原文线索

V2-066443 | MATHDB-346392

The determinant magnitude conjecture for random sign matrices

摘录 / EXCERPT: Determinant magnitude conjecture. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-066444 | MATHDB-346396

The symmetric spectrum simplicity asymptotic

摘录 / EXCERPT: Symmetric spectrum simplicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066445 | MATHDB-346401

The spectral fingerprint conjecture for random sign matrices

摘录 / EXCERPT: Spectral fingerprint conjecture. Almost all ±1 matrices are determined by their spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-066446 | MATHDB-346421

Merca's inequality conjecture for the distinct-part partition function

摘录 / EXCERPT: Let $b(n)$ denote the number of partitions of n into distinct parts, and let $s(m)$ denote the number of partitions of m into distinct parts. For integers n and k, define t...

conjecture | open | MathDB 页面 | 原文线索

V2-066447 | MATHDB-346424

The fiber-base duality conjecture for Vafa-Witten generating functions

摘录 / EXCERPT: Let Λ_S be the lattice of a del Pezzo surface. Let $v_0,v'_0\in\Lambda_S$ be null vectors satisfying ... Let $\{E_I\}_{I=1}^{b_2-2}$ be an orthonormal basis of the orthogonal...

conjecture | open | MathDB 页面 | 原文线索

V2-066448 | MATHDB-346427

The tiling characterization of 4-colorable maximal planar graphs

摘录 / EXCERPT: A maximal planar graph is a planar graph to which no edge can be added while preserving planarity. The every-zero graphic group $\{F_{\textrm{inner}\bigtriangleup};\oplus\}$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-066449 | MATHDB-346434

Strong sausage conjecture for finite parametric sphere packings

摘录 / EXCERPT: Let B^d be the unit ball in d-dimensional Euclidean space, let $\delta_\rho(B^d,n)$ be the optimal parametric density for a finite packing of n balls with parameter $\rho>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066450 | MATHDB-346443

Brešar, Klavžar, Rall, and Wash's packing chromatic number conjecture for subdivisions of subcubic graphs

摘录 / EXCERPT: Let G be a subcubic graph, and let $D(G)$ denote its subdivision, obtained by replacing each edge of G by a path of length two. The packing chromatic number $\chi_p(G)$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-066451 | MATHDB-346452

Rational realization conjecture for the Thue-Morse iterated monodromy covering

摘录 / EXCERPT: Let H_q be the Thue-Morse group, realized as the iterated monodromy group of the degree-q branched covering described using the post-critical set...

conjecture | open | MathDB 页面 | 原文线索

V2-066452 | MATHDB-346460

The semisimplification principle for Rozansky-Witten invariants

摘录 / EXCERPT: Let X be a non-compact space whose Rozansky-Witten invariant $Z_{\mathrm{RW}[X]}(M_3)$ computes invariants associated with a non-semisimple modular tensor category, denoted...

conjecture | open | MathDB 页面 | 原文线索

V2-066453 | MATHDB-346463

k-repeating higher-order Arf invariant conjecture

摘录 / EXCERPT: k-repeating higher-order Arf conjecture. The map Arf^k_j is an isomorphism for all k and j. This is the k-repeating analogue of the higher-order Arf invariant co...

conjecture | open | MathDB 页面 | 原文线索

V2-066454 | MATHDB-346471

The pointed Shafarevich conjecture for moduli spaces

摘录 / EXCERPT: Pointed Shafarevich conjecture. The set of morphisms $f:C\to U$ satisfying $f(c)=u$ is finite. This predicts finiteness of families over a curve after fixing one fibre over one mar...

conjecture | open | MathDB 页面 | 原文线索

V2-066455 | MATHDB-346491

Folklore conjecture on ranks with prescribed torsion

摘录 / EXCERPT: Let G be a torsion group, and let $\mathcal{E}_G$ denote the family of elliptic curves over $\mathbb{Q}$ with torsion subgroup prescribed by G. Folklore conjecture on ranks wit...

conjecture | open | MathDB 页面 | 原文线索

V2-066456 | MATHDB-346507

Quantitative stability conjecture for optimal potentials on general domains

摘录 / EXCERPT: Quantitative stability conjecture. There exists $\eta>0$ such that, for every $V\in\mathcal M(\Omega)$,

conjecture | open | MathDB 页面 | 原文线索

V2-066457 | MATHDB-346509

The cubical-complicial Quillen equivalence conjecture

摘录 / EXCERPT: Let cSet^+ denote the category of marked cubical sets and let $\mathsf{PreComp}$ denote the category of marked simplicial sets (pre-complicial sets). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066458 | MATHDB-346512

Exponential bounds conjecture for the tempered fractional kernel equivalence factor

摘录 / EXCERPT: Let n and s be the dimension and fractional-order parameters, let $\boldsymbol{x},\boldsymbol{y}$ be points in the underlying Euclidean space, and let λ be the temperin...

conjecture | open | MathDB 页面 | 原文线索

V2-066459 | MATHDB-346517

Conjecture on mutual information near the broadcasting threshold

摘录 / EXCERPT: Let $I(delta)$ denote the mutual information in the binary-tree broadcasting model, let δ_c be the critical channel parameter, and write $\delta=\delta_c-\tau$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-066460 | MATHDB-346519

Poisson distribution conjecture for cokernels of random matrices over p-adic integers

摘录 / EXCERPT: Let $\mathbb{Z}_{p}$ be the ring of p-adic integers. Fix distinct $d_{1},\dots,d_{r}\in\mathbb{Z}_{\geq 1}$ and not necessarily distinct $k_{1},\dots,k_{r}\in\mathbb{Z}_{\geq 0}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066461 | MATHDB-346523

The stronger secret-leakage planted clique conjecture

摘录 / EXCERPT: Let ρ be sufficiently symmetric, and let $p_\rho(s)$ denote the intersection-size distribution for two independent sets sampled from ρ. Stronger Secret Leakage Planted Cl...

conjecture | open | MathDB 页面 | 原文线索

V2-066462 | MATHDB-346526

Marchesi-Miró-Roig classification conjecture for uniform Steiner bundles

摘录 / EXCERPT: Let k be a positive integer, and let ${\mathcal E}$ be a k-type uniform vector bundle on ${\mathbb P}^n$. Consider integers α_j and β satisfying $\alpha_j\geq 0$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-066463 | MATHDB-346536

Special Lagrangian torus fibre conjecture for toric Fano surfaces

摘录 / EXCERPT: Let Y be a toric Fano surface, and let $W(u)$ denote the superpotential of a special Lagrangian torus fibre L_u; let W^{HV} be the Hori-Vafa superpotential. Special Lagrangia...

conjecture | open | MathDB 页面 | 原文线索

V2-066464 | MATHDB-346537

Maximal-tangency correspondence conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let (Y,D) be the del Pezzo surface with anticanonical divisor considered in the paper, and let $\tilde{\Omega}(\gamma;I)$ denote the limiting open Gromov-Witten invariant associa...

conjecture | open | MathDB 页面 | 原文线索

V2-066465 | MATHDB-346538

Stability conjecture for special Lagrangians in Fukaya categories

摘录 / EXCERPT: Let a complex structure determine a Fukaya category, and let special Lagrangians be the relevant geometric objects in that category. Stability conjecture. Complex structures are st...

conjecture | open | MathDB 页面 | 原文线索

V2-066466 | MATHDB-346544

Forest-Wilf equivalences for pairs of patterns

摘录 / EXCERPT: Forest-Wilf equivalence conjecture. The following three equivalences hold:

conjecture | open | MathDB 页面 | 原文线索

V2-066467 | MATHDB-346545

Forest-Wilf equivalence implies Wilf equivalence

摘录 / EXCERPT: Forest-Wilf implication conjecture. If S and S' are forest-Wilf equivalent, then S and S' are Wilf equivalent with respect to pattern avoidance in permutations.

conjecture | open | MathDB 页面 | 原文线索

V2-066468 | MATHDB-346549

Stronger Newell-Littlewood saturation conjecture

摘录 / EXCERPT: Stronger Newell-Littlewood saturation conjecture. If $N_{k\mu,k\nu,k\lambda}>0$, then there exist partitions α,β,γ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066469 | MATHDB-346556

Gómez-Reñasco-López-Gómez multiplicity conjecture for indefinite boundary-value problems

摘录 / EXCERPT: Gómez-Reñasco-López-Gómez conjecture. There exists $lambda_c<pi^2$ such that, for every $lambda<lambda_c$, the problem admits at least $2^{n+1}-1$ positive solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-066470 | MATHDB-346557

Replica-symmetric self-overlap identification conjecture for deep restricted Boltzmann machines

摘录 / EXCERPT: Replica-symmetric self-overlap identification conjecture. Then, with respect to the unique stationary point $\bar q$ from the minimax theorem, $\hat q=\bar q$. This asserts uniquen...

conjecture | open | MathDB 页面 | 原文线索

V2-066471 | MATHDB-346558

The global spherical shell conjecture for class VII_0 surfaces

摘录 / EXCERPT: Global spherical shell conjecture. Every class VII_0 surface X with

conjecture | open | MathDB 页面 | 原文线索

V2-066472 | MATHDB-346564

Ohsugi-Hibi conjecture on unimodality of Gorenstein IDP polytopes

摘录 / EXCERPT: Let $\mathcal{P}$ be a lattice polytope. It is Gorenstein if some positive integer dilate of $\mathcal{P}$ is reflexive, and it has the integer decomposition property if every latt...

conjecture | open | MathDB 页面 | 原文线索

V2-066473 | MATHDB-346567

The 1/3-2/3 Conjecture for finite posets

摘录 / EXCERPT: The 1/3-2/3 Conjecture. For any finite poset P which is not a total order, $b(P)\geq\frac{1}{3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-066474 | MATHDB-346583

Parity-unimodality conjecture for fake-degree polynomials

摘录 / EXCERPT: Parity-unimodality conjecture. The fake-degree polynomials $f^{clambda}(q)$ are parity-unimodal for all partitions $clambda$.

conjecture | open | MathDB 页面 | 原文线索

V2-066475 | MATHDB-346584

The analytic stability data conjecture for arbitrary central charges

摘录 / EXCERPT: Analytic stability data conjecture. The proposition asserting exponential bounds on $|a(\gamma)|$ for $\gamma\in C(I)$ holds for any central charge Z, not necessarily a rational...

conjecture | open | MathDB 页面 | 原文线索

V2-066476 | MATHDB-346585

The resurgence conjecture for sections of the wall-crossing bundle

摘录 / EXCERPT: Resurgence conjecture. In the canonical formal trivialization, the Taylor series of $\chi_\gamma(s(t))$ is resurgent. The same is true for the Taylor series of...

conjecture | open | MathDB 页面 | 原文线索

V2-066477 | MATHDB-346586

The analytic stability data and nonlinear connection correspondence conjecture

摘录 / EXCERPT: Analytic stability data correspondence conjecture. The data and axioms described in 1)-8) are in bijection with the set of continuous families of analytic stability data parametriz...

conjecture | open | MathDB 页面 | 原文线索

V2-066478 | MATHDB-346591

The degree conjecture for the associated Boolean function of odd-variable APN functions

摘录 / EXCERPT: Degree conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-066479 | MATHDB-346592

The component-degree conjecture for the associated map of quadratic APN functions

摘录 / EXCERPT: Component-degree conjecture. For every nonzero $v\in\mathbb{F}_2^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-066480 | MATHDB-346603

Conjectures on bistable and endemic behavior in the simplicial SIS model

摘录 / EXCERPT: Behaviors in the bistable and endemic domains. For the simplicial SIS model, in the bistable domain and at fixed β_2, the domain of attraction of the disease-free equilibrium...

conjecture | open | MathDB 页面 | 原文线索

V2-066481 | MATHDB-346612

The Cohen-Lenstra-Martinet heuristic for Arakelov class groups

摘录 / EXCERPT: Cohen-Lenstra-Martinet conjecture. If P is finite, then

conjecture | open | MathDB 页面 | 原文线索

V2-066482 | MATHDB-346622

Thue-Morse Stieltjes continued-fraction minimal-polynomial conjecture

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let a,b be two distinct elements of $\mathbb{F}_{2^k}^{\times}$, and let $\mathbf{u}$ be the (a,b)-Thue-Morse sequence. Thue-Morse Stieltjes minima...

conjecture | open | MathDB 页面 | 原文线索

V2-066483 | MATHDB-346623

Period-doubling Stieltjes continued-fraction transcendence conjecture

摘录 / EXCERPT: Let $J\in\mathbf{F}_4\backslash\{0,1\}$, and let $\mathbf{u}$ be the $(1,J)$-period-doubling sequence. Period-doubling Stieltjes transcendence conjecture. The Stieltjes continued f...

conjecture | open | MathDB 页面 | 原文线索

V2-066484 | MATHDB-346657

Sun-Yang-Zuo conjecture on periodic Higgs bundles and torsion points

摘录 / EXCERPT: Sun-Yang-Zuo conjecture. A Higgs bundle in $\mathcal M_\lambda$ over a finite unramified extension is twisted f-periodic if and only if $\pi^*(\theta_0)$ is a $(p^f\pm1)$-torsion...

conjecture | open | MathDB 页面 | 原文线索

V2-066485 | MATHDB-346658

Sun-Yang-Zuo conjecture on finiteness of crystalline local systems

摘录 / EXCERPT: Sun-Yang-Zuo conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066486 | MATHDB-346660

Conjecture that the Bernstein-Sato function of a free hyperplane arrangement is lattice-determined

摘录 / EXCERPT: Lattice-determined b-function conjecture. The b-function of a free hyperplane arrangement is determined from its intersection lattice.

conjecture | open | MathDB 页面 | 原文线索

V2-066487 | MATHDB-346661

Budur's multivariable monodromy conjecture

摘录 / EXCERPT: Budur's multivariable monodromy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066488 | MATHDB-346666

Conjectured optimal strategy for nested group testing

摘录 / EXCERPT: Let $p\binom{}{}$ be the prevalence parameter, let $D_k(m,p)$ denote the cost of a nested strategy (k,m), and let m_{33} and m_{34} be the strategies defined by ... and ... A str...

conjecture | open | MathDB 页面 | 原文线索

V2-066489 | MATHDB-346667

Finite generation of lower central series quotients for graph braid groups

摘录 / EXCERPT: Finite generation of LCS quotients. Every successive quotient of the lower central series,

conjecture | open | MathDB 页面 | 原文线索

V2-066490 | MATHDB-346668

Finite generation of dual K-theory under graph deletion and contraction

摘录 / EXCERPT: Finite generation of dual K-theory. For every fixed n, the dual of the Grothendieck group of vector bundles over configuration spaces of graphs

conjecture | open | MathDB 页面 | 原文线索

V2-066491 | MATHDB-346669

The Collatz orbit conjecture

摘录 / EXCERPT: Let ... Let ${\cal O}_1=\{1,2\}$. An eventual finite orbit is a finite orbit in which the iterates of an initial integer eventually remain. Collatz orbit conjecture. ${\cal O}_1$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-066492 | MATHDB-346670

Transitive formal-codegree conjecture for fusion categories

摘录 / EXCERPT: Formal-codegree conjecture. If $\mathcal{C}$ has a formal codegree f such that for all $m\in\mathbb{Z}_{\geq2}$, m does not divide f, then $\mathcal{C}$ is equivalent to a fu...

conjecture | open | MathDB 页面 | 原文线索

V2-066493 | MATHDB-346678

Dai-Li's energy-maximizing Hitchin section conjecture

摘录 / EXCERPT: Let Σ be a Riemann surface and let $\mathcal M_{\text{Higgs}}(\Sigma)$ be the moduli space of Higgs bundles, equipped with its Hitchin fibration. A Hitchin section is the se...

conjecture | open | MathDB 页面 | 原文线索

V2-066494 | MATHDB-346682

Characterization of cluster complexes by root independence and full support

摘录 / EXCERPT: Cluster-complex characterization conjecture. The following statements are equivalent: (i) up to commutations of consecutive commuting letters, $Q=c\,\mathrm{w_\circ}(c)$ for some C...

conjecture | open | MathDB 页面 | 原文线索

V2-066495 | MATHDB-346683

Wei's fourth-power q-supercongruence for n congruent to 3 modulo 4

摘录 / EXCERPT: Let n be a positive integer with $n\equiv3\pmod4$, let $[n]=(1-q^n)/(1-q)$, let $(a;q)_k$ be the q-shifted factorial, and let $(a_1,\ldots,a_r;q)_k=\prod_{i=1}^r(a_i;q)_k$. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-066496 | MATHDB-346695

Local existence and uniqueness conjecture for higher-complex-structure connections

摘录 / EXCERPT: Let U be a neighborhood of the zero-section in $T^*\bm\hat{\mathcal{T}}^n$, and let $[(\mu_k,t_k)]\in U$. A connection ... is considered up to unitary gauge. Local existence and un...

conjecture | open | MathDB 页面 | 原文线索

V2-066497 | MATHDB-346698

The zero-evaluation criterion for Bernstein's second theorem

摘录 / EXCERPT: Let F be the system of sphere equations, and let $\widetilde{F}(\mathbf{t})$ be the system obtained by zero evaluations of the δ-variables. Zero-evaluation criterion. The...

conjecture | open | MathDB 页面 | 原文线索

V2-066498 | MATHDB-346700

Sufficiency of the minimal proper-interval completion characterization

摘录 / EXCERPT: Minimal proper-interval completion conjecture. The only-if condition of that theorem is also sufficient. Moreover, if this condition is sufficient, then the problem of finding a mi...

conjecture | open | MathDB 页面 | 原文线索

V2-066499 | MATHDB-346705

The centered single-well and convex potential spectral-gap lower-bound conjecture

摘录 / EXCERPT: Let V be a single-well potential with centered transition point $\tau=0$ or a convex potential, and let $\Lambda(V,\alpha)$ denote the fundamental spectral gap of the one-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-066500 | MATHDB-346709

The flatness conjecture for affine Weyl group S-cells

摘录 / EXCERPT: Let W be the affine Weyl group, let $x\in W$, and let $[w,\mathbf{r}]\in\frak{P}$. Let $\call(x)^+_{w,\mathbf{r}}$ be the corresponding space and let $\frak{c}_{w,\mathbf{r}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-066501 | MATHDB-346716

The higher-dimensional GPEVP ground-state L^∞ bound

摘录 / EXCERPT: Let $d>1$, and let $u_{\text{\tiny\rm gs}}$ denote the ground state of the Gross-Pitaevskii eigenvalue problem (GPEVP). Higher-dimensional ground-state bound. The ground state sati...

conjecture | open | MathDB 页面 | 原文线索

V2-066502 | MATHDB-346717

The higher-dimensional localization conjecture for disordered Bose-Einstein condensates

摘录 / EXCERPT: Let $d>1$, let $u_{\text{\tiny\rm gs}}$ be the ground state of the Gross-Pitaevskii eigenvalue problem (GPEVP), and let V_ε be a disorder potential satisfying assumptio...

conjecture | open | MathDB 页面 | 原文线索

V2-066503 | MATHDB-346723

Boyle-Handelman spectral conjecture for primitive matrices over subrings of the reals

摘录 / EXCERPT: Let $\mathcal R$ be a subring of $\mathbb R$, and let $\Lambda=(\lambda_1,\dots,\lambda_k)$ be a k-tuple of complex numbers. The necessary conditions are the Perron condition, th...

conjecture | open | MathDB 页面 | 原文线索

V2-066504 | MATHDB-346733

The k-linear oscillatory integral estimate for Hrmnder-type operators

摘录 / EXCERPT: Let $1\leq k\leq n$, and let $(T_1,\dots,T_k)$ be a ν-transverse k-tuple of H"ormander-type operators of the same signature σ. Thus, if G_j denotes the generalised G...

conjecture | open | MathDB 页面 | 原文线索

V2-066505 | MATHDB-346737

Low shrubdepth coloring conjecture for powers of nowhere dense classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a nowhere dense class of graphs, let d be a positive integer, and let ε be a positive real. For a graph G, write G^d for its dth power, and...

conjecture | open | MathDB 页面 | 原文线索

V2-066506 | MATHDB-346740

Recht-Ré conjecture for noncommutative arithmetic-geometric means

摘录 / EXCERPT: Let n be a positive integer, $A_1,\dots,A_n$ be symmetric positive semidefinite matrices, and let $\lVert\,\cdot\,\rVert$ be the spectral norm. For $m\leq n$, compare the average...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-066507 | MATHDB-346741

Recht-Ré conjecture with norms inside the sums

摘录 / EXCERPT: Let $A_1,\dots,A_n$ be positive semidefinite matrices, and let $\lVert\,\cdot\,\rVert$ denote the matrix norm used in the conjecture. For ordered products of length m, distinguis...

conjecture | open | MathDB 页面 | 原文线索

V2-066508 | MATHDB-346742

Hopkins-Rubey cyclic sieving conjecture for linear extensions of V(n)

摘录 / EXCERPT: Let $V(n)$ be the poset in the source, let $\mathcal{L}(V(n))$ denote its set of linear extensions, and let Pro denote promotion. Hopkins-Rubey's conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-066509 | MATHDB-346743

Rowmotion-equivariant bijection for minuscule doppelgänger pairs

摘录 / EXCERPT: Let (P,Q) be a minuscule doppelgänger pair, and let $\mathcal{PP}^{m}(P)$ and $\mathcal{PP}^{m}(Q)$ denote their bounded plane-partition sets. Minuscule doppelgänger rowmotion co...

conjecture | open | MathDB 页面 | 原文线索

V2-066510 | MATHDB-346744

Rowmotion order conjecture for V(n)

摘录 / EXCERPT: Let $V(n)$ be the poset in the source, let $r(P)$ denote the rank of a poset P, let Row denote rowmotion, and let δ denote the involution used in the sour...

conjecture | open | MathDB 页面 | 原文线索

V2-066511 | MATHDB-346746

Exit-path realization conjecture for quasicategories with invertible endomorphisms

摘录 / EXCERPT: Let K be a quasicategory in which all endomorphisms are equivalences. Write $||K||$ for its stratified realization, and let C_*K denote the chain complex associated to K. Exi...

conjecture | open | MathDB 页面 | 原文线索

V2-066512 | MATHDB-346751

Coefficient monotonicity conjecture for the classes \(\mathcal{S}_{l,n}^*\)

摘录 / EXCERPT: Coefficient monotonicity conjecture. If $f_n\in\mathcal{S}_{l,n}^*$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066513 | MATHDB-346764

Uniqueness conjecture for fixed-mass energy minimizers

摘录 / EXCERPT: Let $d\geqslant2$ and $p>q>1$, and let $I(\lambda)$ be the fixed-mass variational problem ... Let λ_c denote the critical mass appearing in the variational theory. Uniqueness...

conjecture | open | MathDB 页面 | 原文线索

V2-066514 | MATHDB-346766

The one-half Minimizer-start Enclaveless Game Conjecture

摘录 / EXCERPT: Let G be a graph of order n, and let $\delta(G)$ denote its minimum degree. In the Minimizer-start competition-enclaveless game, Minimizer starts, and $\Psi_g^-(G)$ is the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-066515 | MATHDB-346773

Third Borwein conjecture on coefficients of quintic Borwein polynomials

摘录 / EXCERPT: Third Borwein conjecture. Each of $v_n(q)$, $\phi_n(q)$, $\chi_n(q)$, $\psi_n(q)$, and $\omega_n(q)$ has non-negative coefficients. This is one of the sign-pattern conjectures for...

conjecture | open | MathDB 页面 | 原文线索

V2-066516 | MATHDB-346783

Linearity of the Ketonen order implies the Ultrapower Axiom

摘录 / EXCERPT: Let $\mathscr B(\lambda)$ denote the class of countably complete incompressible ultrafilters relevant to the Ketonen order. Ketonen-order conjecture. If the Ketonen order is linear...

conjecture | open | MathDB 页面 | 原文线索

V2-066517 | MATHDB-346799

Ryser's conjecture on transversals in latin squares

摘录 / EXCERPT: A latin square of order n is an $n\times n$ array filled with n symbols so that each row and column contains every symbol exactly once. A transversal is a set of n entries co...

conjecture | open | MathDB 页面 | 原文线索

V2-066518 | MATHDB-346800

Brualdi's conjecture for near transversals in latin hypercubes

摘录 / EXCERPT: A latin hypercube of dimension d and order n is a d-dimensional array filled with n symbols so that every line contains all symbols exactly once. A near transversal is a co...

conjecture | open | MathDB 页面 | 原文线索

V2-066519 | MATHDB-346802

Kobayashi's standardness conjecture for compact Clifford-Klein forms

摘录 / EXCERPT: Kobayashi's conjecture. If G/H has a compact Clifford-Klein form, then there is a structure of a Clifford-Klein form on G/H which is standard.

conjecture | open | MathDB 页面 | 原文线索

V2-066520 | MATHDB-346807

Positivity conjecture for canonical Monsky polynomials

摘录 / EXCERPT: Let an honest triangulation be a triangulation in the sense used in the paper, and let its canonical Monsky polynomials be the associated pair of polynomials. A polynomial is posit...

conjecture | open | MathDB 页面 | 原文线索

V2-066521 | MATHDB-346816

Conjectured dynamic convergence for the Hellan-Herrmann-Johnson Kirchhoff plate method

摘录 / EXCERPT: Kirchhoff plate convergence conjecture. Assuming a smooth solution of the Kirchhoff plate problem, the error estimates

conjecture | open | MathDB 页面 | 原文线索

V2-066522 | MATHDB-346820

Generic Morse-extrema conjecture for band functions

摘录 / EXCERPT: Let V be a real-valued periodic potential, let $H_0=-\Delta+V$ be the associated self-adjoint operator on $\ell^2(\mathbb{Z}^d)$, and let $\lambda_m(k)$, $k\in\mathbb{R}^d$, be i...

conjecture | solved | MathDB 页面 | 原文线索

V2-066523 | MATHDB-346836

Krejčiřík and Zuazua's heat-semigroup decay conjecture

摘录 / EXCERPT: Let Ω be an open connected subset of $\mathbb{R}^d$. Let H_0 and H_+ be two self-adjoint operators in $L^2(\Omega)$ such that ... Assume that there is a positive function...

conjecture | open | MathDB 页面 | 原文线索

V2-066524 | MATHDB-346853

Main asymptotic conjecture for spherical two-distance sets

摘录 / EXCERPT: Fix $-1\le\beta<0\le\alpha<1$. For a spherical two-distance set in $\mathbb R^d$ with inner products in $\{\alpha,\beta\}$, let $N_{\alpha,\beta}(d)$ be the maximum possible size....

conjecture | open | MathDB 页面 | 原文线索

V2-066525 | MATHDB-346868

Recurrence of tangent-vector graphs for leaf-constructed measures

摘录 / EXCERPT: Let T be a triangulation, let x be a tangent vector on T, and let G^x denote the corresponding graph of discrete tangent vectors. Suppose that G^x is recurrent and that t...

conjecture | open | MathDB 页面 | 原文线索

V2-066526 | MATHDB-346872

Defect-resonance conjecture for truncated Schrödinger operators

摘录 / EXCERPT: In $\mathbb{R}^d$, where d is an arbitrary positive integer, let $H=-\Delta+V$ be a Schrödinger operator with $V(\boldsymbol{x})$ of unbounded support. Suppose that H has a bou...

conjecture | open | MathDB 页面 | 原文线索

V2-066527 | MATHDB-346879

Finite Kishore conjecture for ratios of Lommel polynomials

摘录 / EXCERPT: Finite Kishore conjecture. $N^{(m)}_{n,\nu}$ is a polynomial in ν with nonnegative integer coefficients. This finite version would imply Kishore's theorem, as stated in the pap...

conjecture | open | MathDB 页面 | 原文线索

V2-066528 | MATHDB-346880

Coefficient-denominator conjecture for a second basic hypergeometric ratio

摘录 / EXCERPT: Second coefficient-denominator conjecture. For every n, there is a polynomial $P'_n(a,b,c)$ in a,b,c,q with integer coefficients such that

conjecture | open | MathDB 页面 | 原文线索

V2-066529 | MATHDB-346881

Necessary and sufficient condition for the principle of optimality in deterministic MSOPs

摘录 / EXCERPT: Consider sets $X_t subseteq\mathbb{R}^{n\times T}$ for $0\leq t\leq T$, a control set $U\subseteq\mathbb{R}^m$, and cost functions ... Let $\mathcal{F}$ denote the set of pairs...

conjecture | open | MathDB 页面 | 原文线索

V2-066530 | MATHDB-346883

Monotonicity of Heckman-Opdam hypergeometric functions under W-majorization

摘录 / EXCERPT: Let V be the real vector space associated with a root system, let W be its Weyl group, and let $F_{k,u}$ denote the Heckman-Opdam hypergeometric function with nonnegative real...

conjecture | open | MathDB 页面 | 原文线索

V2-066531 | MATHDB-346890

Large-root stabilization and polynomiality conjecture for orbifold Gromov-Witten invariants

摘录 / EXCERPT: Large-root stabilization and polynomiality conjecture. Genus-zero orbifold Gromov-Witten invariants of $X_{D,\vec r}$, after multiplication by suitable powers of the r_i, stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-066532 | MATHDB-346898

The new version of Gordon's identities

摘录 / EXCERPT: Let $r\geq 2$ and $1\leq i\leq r$. For a partition $\Lambda=(\lambda_1,\ldots,\lambda_m)$, define its (i,ℓ)-new parts recursively by ... where $\lambda_j=0$ for $j\leq 0$, and i...

conjecture | open | MathDB 页面 | 原文线索

V2-066533 | MATHDB-346899

The six-color conjecture for recoloring planar graphs of girth at least five

摘录 / EXCERPT: Six-color conjecture. $R_6(G)$ has diameter $O(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-066534 | MATHDB-346900

The list-recoloring conjecture for planar graphs

摘录 / EXCERPT: List-recoloring conjecture. If either

conjecture | open | MathDB 页面 | 原文线索

V2-066535 | MATHDB-346906

Linear circumference conjecture for random cubic planar multigraphs

摘录 / EXCERPT: Linear circumference conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066536 | MATHDB-346907

Weakly supercritical circumference conjecture for random planar graphs

摘录 / EXCERPT: Weakly supercritical circumference conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066537 | MATHDB-346914

The small conjecture for hereditary graph classes

摘录 / EXCERPT: Small conjecture. Every small hereditary class has bounded twin-width.

conjecture | open | MathDB 页面 | 原文线索

V2-066538 | MATHDB-346919

The rational-cycle conjecture for non-Kählerian surfaces

摘录 / EXCERPT: Rational-cycle conjecture. If every such T belongs to $L^2_{-1}(X)$, but not every such T satisfies $I(T)=0$, then X admits a cycle of rational curves.

conjecture | open | MathDB 页面 | 原文线索

V2-066539 | MATHDB-346928

Apollonius-boundary conjecture for active pursuers in the MPSE problem

摘录 / EXCERPT: Let an MPSE problem have pursuers and evaders at positions at time t, with $0\leq t<t_c$. Let $\mathcal{B}^t$ be the Apollonius boundary and let $\mathcal{A}_i$ denote the Apollo...

conjecture | open | MathDB 页面 | 原文线索

V2-066540 | MATHDB-346940

TanakaThomas conjecture on JoyceSong pair invariants

摘录 / EXCERPT: Tanaka--Thomas conjecture. The invariant $\mathrm{VW}_S^H(r,c_1,n)$ is independent of the choice of $\nu\gg0$.

conjecture | open | MathDB 页面 | 原文线索

V2-066541 | MATHDB-346941

The ILM.4 characterization conjecture for the frame $\widetilde{\mathfrak{G}_1^{\bullet}}$

摘录 / EXCERPT: Let $\widetilde{\mathfrak{G}_1^{\bullet}}$ be the frame whose validity on the closed fragment $\mathcal{F}_1$ characterizes provability in ${\mathrm{PRA}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066542 | MATHDB-346944

The geometric Langlands conjecture for irreducible local systems

摘录 / EXCERPT: Let k be a field, let X be a smooth projective curve over k, and let G be a reductive group over $k(X)$ with Langlands dual group $\LG$. Let $S\subset X$ be a finite set. F...

conjecture | open | MathDB 页面 | 原文线索

V2-066543 | MATHDB-346962

The $\mathfrak M_H(G)$-conjecture for supersingular elliptic curves

摘录 / EXCERPT: Let E be the elliptic curve and let $X_{\infty,f}^{\pm}$, $\Lambda(H)$, and the groups and fields denoted by the notation of Section 1 in the source be as defined there....

conjecture | open | MathDB 页面 | 原文线索

V2-066544 | MATHDB-346964

Finite abelian descent VSA conjecture for the thrice-punctured projective line

摘录 / EXCERPT: Let k be a number field and let $\poneminusthreepoints_k$ denote the thrice-punctured projective line over k. VSA for $(\fab,f,k)$ is the finite abelian descent property used i...

conjecture | open | MathDB 页面 | 原文线索

V2-066545 | MATHDB-346966

Generalized Bloom characterization conjecture for iterated commutators

摘录 / EXCERPT: Let T be a non-degenerate Calderón-Zygmund operator with Dini-continuous kernel, let $m\in\mathbb N$ and $p>1$, and let λ, μ, and η be arbitrary weights. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-066546 | MATHDB-346967

The rank-sensitive matrix discrepancy conjecture

摘录 / EXCERPT: The rank-sensitive matrix discrepancy conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-066547 | MATHDB-346968

Niho's five-value conjecture for crosscorrelation spectra

摘录 / EXCERPT: Let F) be a finite field of order 4^m, where m is even, and set ... Let $W_{F,d}(a)$ denote the Weil sum associated with this decimation. Niho's five-value conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-066548 | MATHDB-346972

Conjecture on the weight distribution of a binary linear code from a two-to-one polynomial

摘录 / EXCERPT: The weight-distribution conjecture. The polynomial $f(x)$ is two-to-one over ${\mathbb F}_{2^n}$, the code $\mathcal{C}_{D(f)}$ has parameters

conjecture | open | MathDB 页面 | 原文线索

V2-066549 | MATHDB-346973

Conjecture on the parameters and weight distribution of codes from almost bent exponents

摘录 / EXCERPT: Parameter and weight-distribution conjecture. The parameters of $\mathcal{C}_f$ should be the same as those of the code in the cited theorem, namely the theorem's parameters for th...

conjecture | open | MathDB 页面 | 原文线索

V2-066550 | MATHDB-346975

Automorphy conjecture for isolated representations of split classical groups

摘录 / EXCERPT: Let G be a split classical group, and let π be an isolated representation in the unitary dual of G. Automorphy conjecture. The representation π is automorphic. The conj...

conjecture | open | MathDB 页面 | 原文线索

V2-066551 | MATHDB-346976

Critical-type conjecture for isolated representations

摘录 / EXCERPT: Let G be a split classical group, and let π be an isolated representation in the unitary dual of G. Critical-type conjecture. The representation π is a subquotient of a...

conjecture | open | MathDB 页面 | 原文线索

V2-066552 | MATHDB-346983

Microlocal characterization conjecture for Lusztig sheaves

摘录 / EXCERPT: Microlocal characterization conjecture. Any irreducible $G_{\boldsymbol{d}}$-equivariant perverse sheaf on $E_{\boldsymbol{d}}$ whose singular support is contained in...

conjecture | open | MathDB 页面 | 原文线索

V2-066553 | MATHDB-346992

Novak-Rhoades numerical Kronecker-coefficient conjecture

摘录 / EXCERPT: For $n\geq3$, let $\nu\vdash n$, let $g_{\lambda\mu}^{\nu}$ be the Kronecker coefficient in $V^\lambda\otimes V^\mu$, and let H_λ denote the hook product of the partition...

conjecture | open | MathDB 页面 | 原文线索

V2-066554 | MATHDB-346993

Novak-Rhoades Schur-positivity conjecture for increasing subsequence functions

摘录 / EXCERPT: Let Λ_n be the ring of homogeneous symmetric functions of degree n with Kronecker product $*$, and define ... where s_λ is the Schur function indexed by λ....

conjecture | open | MathDB 页面 | 原文线索

V2-066555 | MATHDB-347002

Path-space hydrodynamic limit for the boundary driven zero-range process

摘录 / EXCERPT: Let Q^N be the law on $\mathcal D([0,T],\mathcal M_+)$ of the empirical process $\{\pi_t^N:t\in[0,T]\}$. Let $\rho:[0,T]\times[0,1]\to\mathbb R_+$ be a weak solution of the hydro...

conjecture | open | MathDB 页面 | 原文线索

V2-066556 | MATHDB-347006

The eta-part of Kato's main conjecture for residual representations

摘录 / EXCERPT: Let $\overline{\rho}$ be an absolutely irreducible residual rank-two representation, let $\Delta=\operatorname{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066557 | MATHDB-347007

The eta-part of Kato's main conjecture for universal deformations

摘录 / EXCERPT: Let ρ_x be a rank-two p-adic representation in the universal deformation family, let Σ be a finite set of primes containing the ramified primes and p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066558 | MATHDB-347010

Path-cordiality characterization for finite abelian groups

摘录 / EXCERPT: Let A be a finite abelian group, and let $\mathbb{P}$ be the class of path graphs. A group is $\mathbb{P}$-cordial if every path graph is A-cordial, where A-cordiality means...

conjecture | open | MathDB 页面 | 原文线索

V2-066559 | MATHDB-347011

Weak path-cordiality conjecture for finite abelian groups

摘录 / EXCERPT: Let A be a finite abelian group, and let $\mathbb{P}$ be the class of path graphs. A group is weakly $\mathbb{P}$-cordial if all but finitely many path graphs are A-cordial. We...

conjecture | open | MathDB 页面 | 原文线索

V2-066560 | MATHDB-347017

Optimality of the ascending-descending order for odd sequential juries

摘录 / EXCERPT: ADO optimality conjecture. For any fixed set of an odd number of distinct abilities for a jury, the ordering that produces the highest reliability is the ADO.

conjecture | open | MathDB 页面 | 原文线索

V2-066561 | MATHDB-347032

Berkovich-Uncu's eventual partition inequality for $C^{*}_{L,s}$ and $D_{L,s}$

摘录 / EXCERPT: For positive integers L and s with $L\geq s+1$, let $C^{*}_{L,s}$ be the set of partitions whose smallest part is s, whose parts are at most $L+s$, and in which L does not...

conjecture | open | MathDB 页面 | 原文线索

V2-066562 | MATHDB-347033

Berkovich-Uncu's eventual positivity conjecture for $H_{L,s,k}(q)$

摘录 / EXCERPT: For positive integers $n\geq 1$, define the q-Pochhammer symbol by ... with $(a;q)_0=1$. For positive integers L, s, and k, define ... A series $\sum_{n\geq 0}a_nq^n$ is eventu...

conjecture | open | MathDB 页面 | 原文线索

V2-066563 | MATHDB-347034

Berkovich-Uncu's corrected nonnegativity conjecture for $G_{L,2}(q)$

摘录 / EXCERPT: For a positive integer L, let $G_{L,2}(q)$ be the difference of the generating series for partitions with smallest part 2 and largest-minus-smallest part at most L, and for p...

conjecture | open | MathDB 页面 | 原文线索

V2-066564 | MATHDB-347039

Orbit identifiability from mixed low-order invariants

摘录 / EXCERPT: Let K signals be represented through their averaged mean, power spectrum, and bispectrum, and let $G_{\Pi,L}x$ denote the orbit of the underlying signal under the parameterized s...

conjecture | open | MathDB 页面 | 原文线索

V2-066565 | MATHDB-347040

Higher-order autocorrelation conjecture for super-resolution

摘录 / EXCERPT: Let N observations of a signal be collected from the model, with each observation sampled at L equally spaced locations; let M denote the number of grid points and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066566 | MATHDB-347041

The relative critical height conjecture for regular polynomial endomorphisms

摘录 / EXCERPT: Let M_d^N be the moduli space of endomorphisms of $\mathbb{P}^N$, let $\mathsf{P}_d^N\subseteq\mathsf{M}_d^N$ be the moduli space of regular polynomial endomorphisms, an...

conjecture | open | MathDB 页面 | 原文线索

V2-066567 | MATHDB-347044

Keating-Snaith moment conjecture for cubic Dirichlet L-functions

摘录 / EXCERPT: Let $q\equiv 2\,(\mathrm{mod}\,3)$, and let $\mathcal{C}(g)$ be the set of primitive cubic Dirichlet characters of genus g over $\mathbb{F}_q[T]$. For a positive integer k, def...

conjecture | open | MathDB 页面 | 原文线索

V2-066568 | MATHDB-347047

Five-valued conjecture for Weil sums with Niho exponents

摘录 / EXCERPT: Five-valued conjecture. If either $d_1+d_2\geq 5$, or $d_1+d_2=3$ and $p^n\equiv 11\pmod{12}$, then the Weil spectrum over L is at least five-valued. In the first case its five v...

conjecture | open | MathDB 页面 | 原文线索

V2-066569 | MATHDB-347053

Prasad-Takloo-Bighash conjecture for distinguished representations of inner forms of general linear groups

摘录 / EXCERPT: Prasad-Takloo-Bighash conjecture. Let π be an irreducible admissible representation of G with trivial central character. If π is H-distinguished, then:

conjecture | open | MathDB 页面 | 原文线索

V2-066570 | MATHDB-347054

Demailly's strong continuity conjecture for Monge-Ampère masses

摘录 / EXCERPT: Demailly's conjecture. The n-th Lelong numbers converge:

conjecture | open | MathDB 页面 | 原文线索

V2-066571 | MATHDB-347055

Valuative invariance conjecture for Monge-Ampère masses

摘录 / EXCERPT: Valuative invariance conjecture. If φ and ψ are valuatively equivalent, then their n-th Lelong numbers at 0 are equal:

conjecture | open | MathDB 页面 | 原文线索

V2-066572 | MATHDB-347081

Veselov's conjecture on simple zeros of Wronskians of Hermite polynomials

摘录 / EXCERPT: Veselov's conjecture. This Wronskian has simple zeros, except possibly at $x=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-066573 | MATHDB-347082

The circular chromatic conjecture for planar graphs

摘录 / EXCERPT: Let t be a positive integer and let G be a planar graph of girth at least $2t$. The circular chromatic conjecture. ... The cases $t=1$ and $t=2$ follow from the Four Color Theore...

conjecture | open | MathDB 页面 | 原文线索

V2-066574 | MATHDB-347084

Conjecture on Frobenius norms of projected bidiagonal blocks

摘录 / EXCERPT: Let ${\bf B}_{m+\ell}$ be partitioned as ... and define ... Here $\widetilde{{\bf B}}_\ell$ is the corresponding projected bidiagonal block produced by the recycled hybrid projection m...

conjecture | open | MathDB 页面 | 原文线索

V2-066575 | MATHDB-347085

Lescure-Meyniel and Abu-Khzam-Langston immersion coloring conjecture

摘录 / EXCERPT: Let t be a positive integer. For a graph H, an H-immersion is a graph isomorphic to a graph obtained from a subgraph by repeatedly splitting off pairs of edges with a common...

conjecture | open | MathDB 页面 | 原文线索

V2-066576 | MATHDB-347086

Bounded-gap conjecture for strong-immersion-free clustered coloring

摘录 / EXCERPT: For a graph H, let $\chi_*(H)$ denote the strong-immersion coloring parameter used in the paper, and let the clustered chromatic number of a graph class be the least number of co...

conjecture | open | MathDB 页面 | 原文线索

V2-066577 | MATHDB-347090

The generic parameterization conjecture for flat parabolic connections

摘录 / EXCERPT: Let Σ be a surface, and let $\mathcal{A}(\lambda)$ be a family of $\operatorname{PSL}_n(\mathbb{C})$-connections satisfying ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-066578 | MATHDB-347091

The period-coordinates conjecture for the spectral curve

摘录 / EXCERPT: Let $\tilde{\Sigma}$ be the spectral curve associated with a point of $T^*\bm\bm\bm\bm\bm\bm\bm\bm\hat{\mathcal{T}}^n$, and let α be the Liouville form on...

conjecture | open | MathDB 页面 | 原文线索

V2-066579 | MATHDB-347095

Equivalence preservation under properadification

摘录 / EXCERPT: Let $\mathbf{G}^{\mathcal{V}}_{{\text{sc}}}$ be the subcategory of graphs corresponding to simplicial-enriched dioperads, let $\mathbf{G}^{\mathcal{V}}$ be the graph category for s...

conjecture | open | MathDB 页面 | 原文线索

V2-066580 | MATHDB-347103

Minkowski indecomposability of type B shard polytopes

摘录 / EXCERPT: Let a B-arc be an arc in the type B model, and let $\shardPolytope[\beta]$ denote its shard polytope. Minkowski indecomposability conjecture. For any B-arc β, the shard...

conjecture | open | MathDB 页面 | 原文线索

V2-066581 | MATHDB-347104

Polytopality of quotient fans of congruence-uniform posets of regions

摘录 / EXCERPT: Let $\mathcal{H}$ be a hyperplane arrangement defining a fan fan, and let B be a base region of fan. Assume that the poset of regions...

conjecture | open | MathDB 页面 | 原文线索

V2-066582 | MATHDB-347105

Existence of shard polytopes for congruence-uniform posets of regions

摘录 / EXCERPT: Let $\mathcal{H}$ be a hyperplane arrangement and let B be a base region such that the poset of regions $\mathsf{PR}(\mathcal{H},\mathsf{B})$ is a congruence-uniform lat...

conjecture | open | MathDB 页面 | 原文线索

V2-066583 | MATHDB-347108

Bethe-Sommerfeld conjecture

摘录 / EXCERPT: Let $d\geq 2$. Let $V:\mathbb{R}^d\to\mathbb{R}$ be a periodic potential such that the Schrödinger operator H_V is self-adjoint. Bethe-Sommerfeld conjecture. The spectrum...

conjecture | open | MathDB 页面 | 原文线索

V2-066584 | MATHDB-347111

The non-perfectness conjecture for odd-dimensional modular tensor categories

摘录 / EXCERPT: Non-perfectness conjecture. The category $\mathcal{C}$ has at least one non-trivial invertible object; equivalently, it is not perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-066585 | MATHDB-347112

The solvability conjecture for odd-dimensional fusion categories

摘录 / EXCERPT: Solvability conjecture. Every odd-dimensional fusion category is solvable.

conjecture | open | MathDB 页面 | 原文线索

V2-066586 | MATHDB-347121

Bannai et al.'s classification conjecture for spherical 3-distance 5-designs

摘录 / EXCERPT: Bannai et al.'s classification conjecture. One of the following holds: (1) $n=2$ and C is a regular hexagon or a regular heptagon; (2) C is a tight spherical 5-design; or (3)...

conjecture | open | MathDB 页面 | 原文线索

V2-066587 | MATHDB-347122

Real spectral-radius eigenvalue conjecture for ALS on canonical tensor decomposition

摘录 / EXCERPT: Let q be the ALS fixed-point map for canonical tensor decomposition, and let x^* be a fixed point. Write q' for the Jacobian evaluated at x^*, and let $\rho_{q'}$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-066588 | MATHDB-347123

Local linear convergence conjecture for AA and NGMRES with infinite window

摘录 / EXCERPT: Let q be a nonlinear fixed-point map with fixed point x^*, and consider the linearized fixed-point problem with matrix $I-q'(x^*)$. Suppose GMRES(∞) applied to this line...

conjecture | open | MathDB 页面 | 原文线索

V2-066589 | MATHDB-347124

Failure of weak mixed-norm inclusion into a sum space

摘录 / EXCERPT: Let k and m be parameters satisfying $1\leq k<+\infty$ and $1<m<+\infty$, and let p,q,p',q' satisfy ... ... with $m\leq q'<q<+\infty$. The weak mixed-norm non-inclusion conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-066590 | MATHDB-347127

Hille-Perling's standard augmentation conjecture for strong exceptional collections

摘录 / EXCERPT: Let X be a rational surface. A standard augmentation is a toric system obtained from a toric system on a Hirzebruch surface by successive blow-ups and the corresponding augmentat...

conjecture | open | MathDB 页面 | 原文线索

V2-066591 | MATHDB-347147

Special-case strong Bogomolov-Gieseker conjecture for the classes used in curve counting

摘录 / EXCERPT: Let X be the smooth projective threefold under consideration, let H be its ample divisor, and let β and m be the numerical data defining v_n. Set ... Let the strong Bog...

conjecture | open | MathDB 页面 | 原文线索

V2-066592 | MATHDB-347151

The $(\mathrm{C}_n,\mathrm{B}_n)$ branching rule

摘录 / EXCERPT: The $(\mathrm{C}_n,\mathrm{B}_n)$ branching-rule conjecture. The two displayed identities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-066593 | MATHDB-347152

The vanishing integral conjecture

摘录 / EXCERPT: Vanishing integral conjecture. The integral $I_{\lambda}(a,b;q,t,t^n)$ vanishes unless $\operatorname{2\text{-}core}(\lambda)=0$. Moreover, the two specializations satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-066594 | MATHDB-347153

The bounded Littlewood identities

摘录 / EXCERPT: Bounded Littlewood identities. Both displayed identities hold.

conjecture | open | MathDB 页面 | 原文线索

V2-066595 | MATHDB-347154

The unbounded Littlewood identities

摘录 / EXCERPT: Let $x=(x_1,x_2,\ldots)$ be an infinite set of variables, and let $\kappa^{(1)}_{\lambda}(q,t)$ and $\kappa^{(2)}_{\lambda}(q,t)$ be the limiting coefficients defined in the source...

conjecture | open | MathDB 页面 | 原文线索

V2-066596 | MATHDB-347155

Hoshino-Shiraishi formulas for the branching coefficients in ranks two and three

摘录 / EXCERPT: Let $s=(s_1,\ldots,s_n)$, let $\theta=\{\theta_{ij}\}_{1\leqslant i<j\leqslant n}$ be nonnegative integers, and define ... The branching coefficients $e_n(s;\theta;q,t)$ are defined...

conjecture | open | MathDB 页面 | 原文线索

V2-066597 | MATHDB-347160

Sato-Tate conjecture for non-CM elliptic curves

摘录 / EXCERPT: Sato-Tate conjecture. The angles φ_p are equidistributed in $[0,\pi]$ with respect to the Sato-Tate density measure

conjecture | open | MathDB 页面 | 原文线索

V2-066598 | MATHDB-347164

Double ramification cycle conjecture for Gromov-Witten invariants of K3 surfaces

摘录 / EXCERPT: Double ramification cycle conjecture. There exist quasi-Jacobi forms $\varphi_m(z,\tau)$ and $\varphi_{m,n}(z,\tau)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066599 | MATHDB-347170

Mock theta functions' parity-type conjecture

摘录 / EXCERPT: Let $g(q)$ be a mock theta function with integer coefficients, and let $c(g;n)$ denote the coefficient of q^n in its series expansion. Say that g is of parity type...

conjecture | open | MathDB 页面 | 原文线索

V2-066600 | MATHDB-347173

Finiteness conjecture for N-admissible rational functions

摘录 / EXCERPT: Finiteness conjecture. For every positive integer N, the number of N-admissible rational functions is finite. The conjecture is stated as plausible without a rigorous proof in...

conjecture | open | MathDB 页面 | 原文线索

V2-066601 | MATHDB-347174

CMZV membership conjecture for polylogarithms at admissible values

摘录 / EXCERPT: Let N be a positive integer and let ... For a word $w\in\{x_0,x_1\}^\ast$ of weight n, let $\operatorname{Li}_w(x)$ denote the corresponding multiple polylogarithm, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066602 | MATHDB-347187

BPF-psef duality conjecture for nef b-divisor classes

摘录 / EXCERPT: BPF-psef duality conjecture. A b-numerical class $\alpha\in\operatorname{w-N}^k(\mathcal{X})$ is psef if and only if, for every b-numerical class...

conjecture | open | MathDB 页面 | 原文线索

V2-066603 | MATHDB-347189

Uniform mod-Poisson convergence for Abelian square-tiled surfaces

摘录 / EXCERPT: Abelian mod-Poisson conjecture. For every $x>0$, uniformly for $k\in\{0,1,\ldots,\lfloor x\log(g)\rfloor\}$, as $g\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-066604 | MATHDB-347191

Uniform Abelian mod-Poisson convergence in strata

摘录 / EXCERPT: Abelian-stratum mod-Poisson conjecture. For every $x>0$, uniformly for $k\in\{0,1,\ldots,\lfloor x\log(g)\rfloor\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-066605 | MATHDB-347192

Boundedness criterion for compact differences of composition operators

摘录 / EXCERPT: Let $0<p,q<\infty$, let $\omega\in\widehat{\mathcal{D}}$, and let ν be a positive Borel measure on $\mathbb{D}$. Let φ and ψ be analytic self-maps of $\mathbb{D}$....

conjecture | open | MathDB 页面 | 原文线索

V2-066606 | MATHDB-347194

Chen-Haiman Schur-positivity conjecture for Catalan functions

摘录 / EXCERPT: Chen-Haiman's conjecture. The Catalan functions $H(\Psi;\mu)$ are Schur positive:

conjecture | solved | MathDB 页面 | 原文线索

V2-066607 | MATHDB-347195

Popov criterion conjecture for the scalar heat equation

摘录 / EXCERPT: Let $\beta(\theta)$ be defined for $\theta\in(0,1)$ by ... For a given θ, let $q(\theta)>0$ be the quantity specified by the slope-direction formula. Popov criterion conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-066608 | MATHDB-347203

The versal Breuil-Mézard conjecture

摘录 / EXCERPT: Let $K/\mathbb{Q}_p$ be a finite extension, let $\overline{\rho}:G_K\rightarrow\operatorname{GL}_n(\mathbb{F})$ be continuous, and regard $\overline{\rho}$ as an $\mathbb{F}$-point...

conjecture | open | MathDB 页面 | 原文线索

V2-066609 | MATHDB-347207

The unbounded range conjecture for pseudopivot points

摘录 / EXCERPT: Let $\{a_n,b_n,c_n\}$ denote the set of pseudopivot points at iteration n for the initial set under consideration. Unbounded range conjecture. The range of pseudopivot points wil...

conjecture | open | MathDB 页面 | 原文线索

V2-066610 | MATHDB-347209

Rank-three semiorthogonal decomposition conjecture for moduli spaces of vector bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus $g\geq 2$, let L be a line bundle of degree 1 on C, and let $M(3,L)$ be the moduli space of rank-three stable vector bundles on...

conjecture | open | MathDB 页面 | 原文线索

V2-066611 | MATHDB-347210

The lemniscate starlike class Bohr-radius conjecture

摘录 / EXCERPT: Let $\mathcal{SL}^{*}$ denote the class of lemniscate starlike functions, associated with $\psi(z)=\sqrt{1+z}$. The Bohr-radius conjecture. The Bohr radius of $\mathcal{SL}^{*}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-066612 | MATHDB-347226

The omitted-value conjecture for Herman rings of meromorphic functions

摘录 / EXCERPT: Let f be a general meromorphic function. An omitted point is a point $z_0\in\mathbb{C}$ such that $f(z)\neq z_0$ for every $z\in\mathbb{C}$. A Herman ring is a periodic doubly co...

conjecture | open | MathDB 页面 | 原文线索

V2-066613 | MATHDB-347230

The kernel conjecture for the second-kind interval transform

摘录 / EXCERPT: Kernel conjecture. For each $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-066614 | MATHDB-347231

The eigenvector generation conjecture for the second-kind interval transform

摘录 / EXCERPT: Eigenvector generation conjecture. For each $n\geq 1$, a generating set of $S_{\mathbb Q}\langle a,b\rangle_n$, consisting only of eigenvectors, may be obtained by applying all pos...

conjecture | open | MathDB 页面 | 原文线索

V2-066615 | MATHDB-347233

Leading term conjectures for Rubin-Stark elements

摘录 / EXCERPT: Let L_n/K, S, and Σ be as in the source, let r_T be the specified rank, and let $\varepsilon_{L_n\mid K,S,\Sigma}$ be the r_T-th Rubin-Stark element. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066616 | MATHDB-347236

The capacity formula conjecture for strictly positive channels

摘录 / EXCERPT: Let $P_{Z|X}$ be a strictly positive discrete memoryless channel (DMC), let $\hat{P}_{X_1^n}$ and $\hat{P}_{Y_1^n}$ denote the empirical distributions of the input and output seque...

conjecture | open | MathDB 页面 | 原文线索

V2-066617 | MATHDB-347240

The asymptotic-bias dichotomy for maximum likelihood estimation of structured anomalies

摘录 / EXCERPT: Asymptotic-bias dichotomy. The technical condition in the unbiasedness theorem should be replaceable by $\mu \geq \mu_{\textnormal{detect}}$. Under this condition, the MLE is asymp...

conjecture | open | MathDB 页面 | 原文线索

V2-066618 | MATHDB-347243

Bound on zero-block lengths in the 2-adic expansion of zeta

摘录 / EXCERPT: Let $\zeta\in\mathbb{Z}_2$ be the 2-adic number defined in Equation, and let $\ell_\zeta(a)$ denote the length of the zero block beginning at the relevant digit position indexed...

conjecture | open | MathDB 页面 | 原文线索

V2-066619 | MATHDB-347245

The kernel conjecture for v-adic multiple zeta values

摘录 / EXCERPT: Let k be the rational function field underlying the paper, let v be a finite place of k, and let $\overline{\mathcal{Z}}$ and $\overline{\mathcal{Z}}_v$ denote the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-066620 | MATHDB-347247

CGDG finite-element error estimates for the Mindlin plate

摘录 / EXCERPT: Let Ω be the spatial domain, h the mesh size, and k the polynomial order in the CGDG discretization. Let e_w, $\bm{e}_\theta$, $\bm{E}_\kappa$, and $\bm{e}_\gamma$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-066621 | MATHDB-347248

Finite CRG representation conjecture for edit distance functions

摘录 / EXCERPT: Finite CRG representation conjecture. For every nontrivial hereditary property $\mathcal{H}$ and every $\varepsilon>0$, the edit distance function...

conjecture | open | MathDB 页面 | 原文线索

V2-066622 | MATHDB-347272

Normal distribution and flip connectivity conjecture for random tilings of large boxes

摘录 / EXCERPT: Consider the cubiculated box ... Let $\mathbf T,\mathbf T_0,\mathbf T_1$ be independent uniformly distributed random tilings of $\mathcal R$, and let Tw denote the t...

conjecture | open | MathDB 页面 | 原文线索

V2-066623 | MATHDB-347273

Monotonicity and convergence of discretized quartic minimization problems

摘录 / EXCERPT: Consider the discretized problem with discretization parameter N, its smallest eigenvalue, and its optimal value, together with the corresponding original problem. Monotonicity a...

conjecture | open | MathDB 页面 | 原文线索

V2-066624 | MATHDB-347277

A mod 9 generating-function congruence for the Gordon-McIntosh mock theta coefficients

摘录 / EXCERPT: Let $p_{\xi}(n)$ denote the coefficients of the Gordon-McIntosh mock theta function $\xi(q)$, so that $\xi(q)=\sum_{n\geq 0}p_{\xi}(n)q^n$. Generating-function congruence. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-066625 | MATHDB-347283

The maximal-length minimal zero-sum sequence conjecture

摘录 / EXCERPT: Let G be a finite abelian group, and call a sequence over G a minimal zero-sum sequence if its sum is zero and no nonempty proper subsum is zero. Write $\mathsf D(G)$ for the D...

conjecture | open | MathDB 页面 | 原文线索

V2-066626 | MATHDB-347284

The conjecture on maximal elasticity differences in transfer Krull monoids

摘录 / EXCERPT: Let G be a finite abelian group, and let H be a transfer Krull monoid over G, meaning that there is a transfer homomorphism $H\to\mathcal B(G)$ to the monoid of zero-sum sequ...

conjecture | open | MathDB 页面 | 原文线索

V2-066627 | MATHDB-347285

Infinitesimal Torelli conjecture for elliptic surfaces with d=1

摘录 / EXCERPT: Infinitesimal Torelli conjecture. X does not satisfy infinitesimal Torelli.

conjecture | open | MathDB 页面 | 原文线索

V2-066628 | MATHDB-347295

Cancellativity conjecture for the monoids \mathcal{H}_n^+ and \Sigma_n

摘录 / EXCERPT: Let $n\geq 3$. The monoid $\mathcal{H}_n^+$ is defined by the presentation ... Let $\mathcal{B}_{n+1}$ be the braid group on $n+1$ strands, with standard generators...

conjecture | open | MathDB 页面 | 原文线索

V2-066629 | MATHDB-347298

The differentiable-map conjecture on closed special alpha-limit sets

摘录 / EXCERPT: Let $f:[0,1]\to[0,1]$ be continuously differentiable, and let $s\alpha(x)$ denote a special α-limit set. Differentiable-map conjecture. Every special α-limit set of...

conjecture | open | MathDB 页面 | 原文线索

V2-066630 | MATHDB-347306

Even-characteristic lower-bound conjecture for complete symmetric hypersurfaces

摘录 / EXCERPT: Let $h(x_1,\ldots,x_k)$ be a complete symmetric polynomial in $k\geq 4$ variables over the finite field $\mathbb{F}_q$ of degree m, with $1\leq m\leq q-4$. Even-characteristic lo...

conjecture | open | MathDB 页面 | 原文线索

V2-066631 | MATHDB-347307

Zero-of-complete-symmetric-polynomial conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $3\leq k\leq q-2$ (or $4\leq k\leq q-3$ if q is even), and let $h_e(x_1,\ldots,x_k)$ denote the complete symmetric polynomial of degree...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-066632 | MATHDB-347308

Low-degree distinct-coordinate zero conjecture for complete symmetric polynomials

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $3\leq k\leq q-2$ (or $4\leq k\leq q-3$ if q is even), and let h be a complete symmetric polynomial of positive degree m in k vari...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-066633 | MATHDB-347309

Whole-field generalized Vandermonde zero conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field. For a polynomial $f\in\mathbb{F}_q[x]$ and pairwise distinct elements $\alpha_1,\ldots,\alpha_k\in\mathbb{F}_q$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066634 | MATHDB-347314

Berman-Rantzer conjecture on large rank-one perturbations of singular M-matrices

摘录 / EXCERPT: Berman-Rantzer conjecture. There is a positive t_1 such that, for every $t>t_1$, all eigenvalues of $B(t)$ lie in the open right half plane.

conjecture | open | MathDB 页面 | 原文线索

V2-066635 | MATHDB-347321

Picard-number conjecture for moduli spaces of sheaves on a quadric surface

摘录 / EXCERPT: Let $X=D53F^1\timesD53F^1$ and let $M_{H_m}({\bf v})$ be the moduli space associated with a character ${\bf v}=(r,\nu,\Delta)\in K(X)$, where $r\geq 2$ and $\Delta\geq\frac{1}{2}$....

conjecture | open | MathDB 页面 | 原文线索

V2-066636 | MATHDB-347323

Cheap balanced separator conjecture for graphs with polynomial expansion

摘录 / EXCERPT: Let G be a graph and let p be a polynomial such that the expansion of G is bounded by p. A cost assignment is a function ... For integers $t\geq 1$ and $q\geq 0$, a set...

conjecture | open | MathDB 页面 | 原文线索

V2-066637 | MATHDB-347334

Decreasing-density conjecture for quantile mixtures

摘录 / EXCERPT: Decreasing-density conjecture. For $\Lambda\in\mathcal Q_n$ and $\boldsymbol{F}\in\mathcal M_D^n$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-066638 | MATHDB-347354

Kitaev and Zhang's Stieltjes continued fraction conjecture for mesh pattern 3

摘录 / EXCERPT: For a pattern p and a permutation π, let $p(\pi)$ denote the number of occurrences of p in π. Let S_n be the set of permutations of $\{1,\ldots,n\}$, and let y and...

conjecture | open | MathDB 页面 | 原文线索

V2-066639 | MATHDB-347355

Height finiteness conjecture for stable infinitely semiadditive presentable categories

摘录 / EXCERPT: Height Finiteness Conjecture. For every $\mathcal{C}\in\Pr_{\mathrm{st}}^{\mathrm{\infty\text{-}semiadditive}}$, the full subcategory $\mathcal{C}_{\infty}$ is trivial.

conjecture | open | MathDB 页面 | 原文线索

V2-066640 | MATHDB-347367

Recurrence conjecture for symmetrized poly-Bernoulli polynomials

摘录 / EXCERPT: Recurrence conjecture. The polynomials $t_n^k(x,y)$ coincide with $T_n^k(x,y)$.

conjecture | open | MathDB 页面 | 原文线索

V2-066641 | MATHDB-347386

Castravet's conjecture on the rational quotient of elliptic quintics

摘录 / EXCERPT: Castravet's conjecture. The maximally rationally connected quotient of ${\mathcal C}$ is birationally equivalent to the twisted intermediate Jacobian of Y.

conjecture | open | MathDB 页面 | 原文线索

V2-066642 | MATHDB-347388

Unit-scaled determinant-count conjecture for circulant matrices over finite chain rings

摘录 / EXCERPT: Let R be a commutative finite chain ring (CFCR) of nilpotency index e, let n be a positive integer, and suppose that the maximal ideal of R is generated by $gamma$. Write...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-066643 | MATHDB-347389

Unit determinant conjecture for circulant matrices over finite chain rings

摘录 / EXCERPT: Let R be a commutative finite chain ring (CFCR) of nilpotency index e, and let n be a positive integer. Let $C_n(R)$ denote the ring of $n\times n$ circulant matrices over...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-066644 | MATHDB-347395

The non-cylindricity conjecture for very general Fano threefolds of genus 9 or 10

摘录 / EXCERPT: Let X be a very general smooth Fano threefold such that $\rho(X)=1$, $\iota(X)=1$, and $\operatorname{g}(X)=9$ or $\operatorname{g}(X)=10$. The non-cylindricity conjecture. Then...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-066645 | MATHDB-347396

The non-cylindricity conjecture for Fano threefolds of genus 7

摘录 / EXCERPT: Let X be a smooth Fano threefold such that $\rho(X)=1$, $\iota(X)=1$, and $\operatorname{g}(X)=7$. The non-cylindricity conjecture. Then X is not cylindrical. The source states...

conjecture | open | MathDB 页面 | 原文线索

V2-066646 | MATHDB-347398

The automorphism conjecture for complements of del Pezzo surfaces

摘录 / EXCERPT: Let S be the del Pezzo surface under consideration, and let $\mathbb P$ denote the ambient projective threefold in which it is embedded. The automorphism conjecture for del Pezzo...

conjecture | open | MathDB 页面 | 原文线索

V2-066647 | MATHDB-347429

Transcendence conjecture for e+π

摘录 / EXCERPT: Let e denote Euler's number and let π denote the circle constant. Transcendence conjecture for $e+\pi$. The real number $e+\pi$ is transcendental. The preceding result establ...

conjecture | open | MathDB 页面 | 原文线索

V2-066648 | MATHDB-347433

Existence of smooth minimisers for the ${\cal A}$-mean distortion energy

摘录 / EXCERPT: Let ${\Bbb D}$ be the unit disk, and let $f_0:\overline{{\Bbb D}}\to\overline{{\Bbb D}}$ be a homeomorphism of finite distortion with ... Here ${\cal A}:[1,\infty)\to[1,\infty)$ is c...

conjecture | open | MathDB 页面 | 原文线索

V2-066649 | MATHDB-347434

The sloshing eigenvalue counting asymptotic conjecture for a triangular prism

摘录 / EXCERPT: Let $N(c3)$ be the eigenvalue counting function of the sloshing problem, and let $N^e(c3)$ and $N^s(c3)$ denote respectively the counting functions for edge-wave and surface-wave q...

conjecture | open | MathDB 页面 | 原文线索

V2-066650 | MATHDB-347435

Asymptotic completeness conjecture for edge-wave and surface-wave quasi-eigenvalues

摘录 / EXCERPT: Let $\{\tilde{\sigma}_j\}_{j\in\mathbb N_0}$ be the edge-wave and surface-wave quasi-eigenvalues arranged in ascending order. The sequence is asymptotically complete if the indexin...

conjecture | open | MathDB 页面 | 原文线索

V2-066651 | MATHDB-347438

The main temperedness conjecture for quiver A-polynomials

摘录 / EXCERPT: Temperedness conjecture. Quiver A-polynomials are tempered for every choice of the adjacency matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-066652 | MATHDB-347440

Dvořák-Postle edge conjecture for C_5-critical graphs

摘录 / EXCERPT: Dvořák-Postle conjecture. If G is C_5-critical, then

conjecture | open | MathDB 页面 | 原文线索

V2-066653 | MATHDB-347441

Circular-colouring density conjecture for 4-critical graphs

摘录 / EXCERPT: Circular-colouring density conjecture. If G has no (p,q)-colouring, then there exist positive rational numbers $\varepsilon_{p,q}$ and $c_{p,q}$, depending on p and q, such...

conjecture | open | MathDB 页面 | 原文线索

V2-066654 | MATHDB-347446

Universal optimality of the hexagonal, E_8, and Leech lattices for charged periodic configurations

摘录 / EXCERPT: Universal optimality conjecture. For all $N\in2\mathbb{N}$, A_2, E_8, and the Leech lattice are the unique maximizers of $E_f^\pm$ on $\mathcal{A}_N^1$ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-066655 | MATHDB-347447

The sparsity conjecture for graph independence in non-Euclidean p_q^d

摘录 / EXCERPT: Sparsity conjecture. For $q\in(1,\infty)$ with $q\not=2$, and $d\geq 1$, a graph G is independent in ℓ_q^d if and only if G is (d,d)-sparse.

conjecture | open | MathDB 页面 | 原文线索

V2-066656 | MATHDB-347455

Exponential-field-size conjecture for lower triangular Toeplitz superregular matrices

摘录 / EXCERPT: Let $\gamma\geq 2$, and let $\mathbb{F}$ be a finite field. A lower triangular Toeplitz superregular matrix is a lower triangular Toeplitz matrix whose relevant minors are all nonz...

conjecture | open | MathDB 页面 | 原文线索

V2-066657 | MATHDB-347459

The contractibility conjecture for sublevel spaces of global dimension

摘录 / EXCERPT: Contractibility conjecture. The differential of gldim provides a flow such that $\operatorname{Stab}_{<x}\mathcal{D}$ contracts to...

conjecture | open | MathDB 页面 | 原文线索

V2-066658 | MATHDB-347468

Nine-weight conjecture for the subfield code associated with x^6

摘录 / EXCERPT: Let $q=2^m$, and let ${\mathcal{C}}_{(f,q)}^{(2)}$ be the binary subfield code defined by the trace representation for an oval polynomial f. Nine-weight conjecture. For odd...

conjecture | open | MathDB 页面 | 原文线索

V2-066659 | MATHDB-347469

Nine-weight conjecture for the subfield code associated with the second oval polynomial

摘录 / EXCERPT: Let $q=2^m$, with $m\equiv 3\pmod 4$ and $m\geq 5$, and define ... Let ${\mathcal{C}}_{(f,q)}^{(2)}$ be the associated binary subfield code. Nine-weight conjecture. The code...

conjecture | open | MathDB 页面 | 原文线索

V2-066660 | MATHDB-347471

Pathwidth-treedepth conjecture for graphs without long paths

摘录 / EXCERPT: Let G be a graph, let p be its pathwidth, and let ℓ be a positive integer. A path of order 2^ℓ is a path with 2^ℓ vertices. Pathwidth-treedepth conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-066661 | MATHDB-347472

The conjecture on formal periods of moduli spaces and regularized double shuffle relations

摘录 / EXCERPT: Let $\mathcal{M}^\delta_{0,*}$ denote the family of partial compactifications of moduli spaces of genus-zero curves with marked points, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066662 | MATHDB-347473

The chromatic-number conjecture for preferential attachment graphs

摘录 / EXCERPT: Let $PA_t(m,\delta)$ be the preferential attachment graph at time t, with parameters m and δ, and let $(PA_t(m,\delta))_{t\ge 1}$ denote the resulting graph sequence. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-066663 | MATHDB-347476

Minimum-degree Oberwolfach conjecture

摘录 / EXCERPT: Minimum-degree Oberwolfach conjecture. For every $\varepsilon>0$, the following holds for sufficiently large n: if

conjecture | open | MathDB 页面 | 原文线索

V2-066664 | MATHDB-347478

Exact high-girth triangle-decomposition conjecture

摘录 / EXCERPT: High-girth triangle-decomposition conjecture. For every fixed g, every sufficiently large K_3-divisible complete graph K_n has a K_3-decomposition with girth at least g.

conjecture | open | MathDB 页面 | 原文线索

V2-066665 | MATHDB-347479

High-girth Nash-Williams conjecture for triangle decompositions

摘录 / EXCERPT: High-girth Nash-Williams conjecture. For every fixed g, every sufficiently large K_3-divisible graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-066666 | MATHDB-347486

Non-emptiness conjecture for fields with persistently irreducible polynomial iterates

摘录 / EXCERPT: For a field K, let $K\in\mathcal{N}_{d,n}^{\infty}$ mean that there exists a degree-d polynomial over K whose first n iterates, and all further iterates, remain irreducible...

conjecture | open | MathDB 页面 | 原文线索

V2-066667 | MATHDB-347493

Infinite log-concavity conjecture for the sparse-polynomial sequence

摘录 / EXCERPT: Let $f_n(z)=f_{2,n}(z)$, and for a sequence (a_n) define the operator ... A sequence is called k-log-concave when $\mathcal L^j(a_n)\geq 0$ for $j=0,1,\ldots,k$, and infinite log...

conjecture | open | MathDB 页面 | 原文线索

V2-066668 | MATHDB-347495

One uniqueness to rule them all

摘录 / EXCERPT: Let A be the function defining the Zakharov-Shabat reflection coefficient, let ... and let A_s be the associated operator, with kernel $K_s=A_s^2$. Consider the Riemann-Hilbert p...

conjecture | open | MathDB 页面 | 原文线索

V2-066669 | MATHDB-347510

Injectivity after restriction to an Iwahori subgroup in semisimple rank zero

摘录 / EXCERPT: Iwahori-restriction conjecture. If $\pi\in\mathfrak{Rep}^{\textnormal{ladm}}(G)$ is injective, then

conjecture | open | MathDB 页面 | 原文线索

V2-066670 | MATHDB-347525

Langer's asymptotic conjecture for cusps on plane curves

摘录 / EXCERPT: Let $k_{\max}(d)$ be the maximal number of ordinary cusps of a plane curve of degree d. Langer's conjecture. ... This conjecture asserts that the coefficient of d^2 in the known...

conjecture | open | MathDB 页面 | 原文线索

V2-066671 | MATHDB-347526

The critical-parameter conjecture for weak competition equilibria

摘录 / EXCERPT: Consider the competition model given by equations, with $p=q=1$, and assume ... The competitors coexist. Critical-window conjecture. There is a critical window...

conjecture | open | MathDB 页面 | 原文线索

V2-066672 | MATHDB-347531

Scaling-limit conjecture for the unlabelled (α,γ)-chain

摘录 / EXCERPT: Let $0 < \gamma \leq \alpha < 1$. Consider the unlabelled (α,γ)-chain on rooted multifurcating trees with n leaves, obtained by selecting and deleting a uniform leaf...

conjecture | open | MathDB 页面 | 原文线索

V2-066673 | MATHDB-347537

Splitter theorem for locally 3-connected graphs

摘录 / EXCERPT: Let r be a positive integer and let G be an r-locally 3-connected graph, meaning that every separation of G of order at most r has the prescribed local 3-connectivity...

conjecture | open | MathDB 页面 | 原文线索

V2-066674 | MATHDB-347538

Infinite-graph extension of the local 2-separator theorem

摘录 / EXCERPT: Infinite-graph extension conjecture. The local 2-separator theorem is true for infinite graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-066675 | MATHDB-347543

The conjecture that every quintic graph has at least two Kempe equivalence classes

摘录 / EXCERPT: Let a quintic graph be a 5-regular graph, and let two 1-factorisations be Kempe equivalent if one can be obtained from the other by a sequence of Kempe switches. Kempe-class conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-066676 | MATHDB-347546

The conjecture on spectral separation for rank-one quasinilpotent perturbations

摘录 / EXCERPT: Let T be a quasinilpotent operator. For a positive scalar m chosen so that $4\widetilde{T}=mT4$ has norm a, write $4\widetilde{T}=mT4$. Let F be a rank-1 quasinilpotent o...

conjecture | open | MathDB 页面 | 原文线索

V2-066677 | MATHDB-347547

Fixed-point homomesy conjecture for the complement-inversion Foatic action

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, and let $\overline{\gamma}$ denote the complement-inversion Foatic action on $\mathfrak{S}_n$. For a permutation w, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066678 | MATHDB-347548

Fixed-point homomesy and orbit-structure conjectures for the complement-rotation Foatic action

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, and let $\overline{\rho}$ denote the complement-rotation Foatic action on $\mathfrak{S}_n$. For $w\in\mathfrak{S}_n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066679 | MATHDB-347549

Fixed-point homomesy conjecture for the reversal-rotation Foatic action

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, and let τ denote the reversal-rotation Foatic action on $\mathfrak{S}_n$. For a permutation w, let $\operatorname{Fix}(w)$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-066680 | MATHDB-347550

NWU speed-variation load conjecture for redundancy systems

摘录 / EXCERPT: NWU load conjecture. For NWU distributed speed variations, the loads satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-066681 | MATHDB-347559

Strict paired domination inequality for direct products of trees

摘录 / EXCERPT: Let T_1 and T_2 be trees of order at least 2, and let $\gamma_{\mathrm{pr}}(T)$ denote the paired domination number of a graph T. Strict paired domination conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-066682 | MATHDB-347560

Bourgain-Rudnick conjecture on nodal intersections with curves

摘录 / EXCERPT: Bourgain-Rudnick conjecture. The lower bound for the number of nodal intersections is of order λ; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-066683 | MATHDB-347562

Kingston et al.'s perimeter surveillance convergence conjecture

摘录 / EXCERPT: A configuration consists of a time t together with the positions, directions, and estimates of all drones at that time. A drone is left synchronized at time t if thereafter it...

conjecture | open | MathDB 页面 | 原文线索

V2-066684 | MATHDB-347565

Conjecture on the ground state and principal eigenvalue of the Gierer-Meinhardt core problem

摘录 / EXCERPT: Let $S>0$ and consider the core problem eqref{eq:core}, whose ground state solution is denoted by $(U(\rho;S),V(\rho;S))$, with $\mu(S)$ the associated eigenvalue function. A soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-066685 | MATHDB-347574

Aliniaeifard-van Willigenburg-Wang conjecture for a generalized family of spiders

摘录 / EXCERPT: Aliniaeifard-van Willigenburg-Wang conjecture. The family of spiders

conjecture | open | MathDB 页面 | 原文线索

V2-066686 | MATHDB-347575

Conjecture that combining spider legs preserves e-positivity

摘录 / EXCERPT: Leg-combination conjecture. If S is e-positive, then S' is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-066687 | MATHDB-347576

Conjecture that line graphs of e-positive spiders are e-positive

摘录 / EXCERPT: Line-graph conjecture. If a spider S is e-positive, then its line graph S_L is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-066688 | MATHDB-347590

Conjecture on the asymptotic threshold for asymmetric choosability

摘录 / EXCERPT: Asymptotic threshold conjecture. There exist positive constants $\xi_g(k_A)$ such that every bipartite graph G satisfying $\xi<\xi_g(k_A)$ is (k_A,k_B)-choosable. Moreover, if...

conjecture | open | MathDB 页面 | 原文线索

V2-066689 | MATHDB-347591

Conjecture that the two-uniform threshold equals $\log 2$

摘录 / EXCERPT: Two-uniform threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066690 | MATHDB-347592

Conjecture equating the asymptotic thresholds ξ_m and ξ'_m

摘录 / EXCERPT: Threshold-limit conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-066691 | MATHDB-347601

Dense resolvent conjecture for the Bloch-Torrey operator

摘录 / EXCERPT: Let $\mathcal B^D$ be the Bloch-Torrey operator on the periodically perforated domain, and let $\rho(\mathcal B^D)$ denote its resolvent set. Dense resolvent conjecture. ... The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-066692 | MATHDB-347602

Exponential spectral formula conjecture for the Bloch-Torrey fiber semigroup

摘录 / EXCERPT: Fix $q\in\mathbb R$ and let $\mathcal B^D_q$ be the corresponding fiber operator. Set ... where g is the parameter in the Bloch-Torrey operator. Let $\lambda_k(q)$ denote the eigen...

conjecture | open | MathDB 页面 | 原文线索

V2-066693 | MATHDB-347607

Type-I irreducible ancient Ricci flow solutions are Ricci shrinkers

摘录 / EXCERPT: Type-I ancient-solution conjecture. $(M^n,g(t))$ is isometric to a Ricci shrinker, up to scaling.

conjecture | open | MathDB 页面 | 原文线索

V2-066694 | MATHDB-347610

Convergence of generalized Ricci flow on compact semisimple Lie groups

摘录 / EXCERPT: Let G be a compact semisimple Lie group with bi-invariant metric g_{∞}. Given a left-invariant metric g_0 and ... consider generalized Ricci flow with initial condition...

conjecture | open | MathDB 页面 | 原文线索

V2-066695 | MATHDB-347619

Hildebrand's regularity conjecture for conformally invariant Lagrangians

摘录 / EXCERPT: In the setting of variational problems with a coercive conformally invariant Lagrangian of quadratic growth, consider its critical points. Hildebrand's regularity conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-066696 | MATHDB-347624

Parabola conjecture for the inverse V-question-mark function

摘录 / EXCERPT: Let $y\in{\mathbb R}_1$. Choose $k\in{\mathbb N}_0$ such that ... Let $\overline{?}_V^{-1}$ denote the inverse-function graph quantity used in the source. Parabola conjecture. With t...

conjecture | open | MathDB 页面 | 原文线索

V2-066697 | MATHDB-347635

Hou-Jagadeesan convexity conjecture for partition ranks modulo 2

摘录 / EXCERPT: Let $N(r,2;n)$ denote the number of partitions of n whose rank is congruent to r modulo 2, for $r\in\{0,1\}$. Hou-Jagadeesan's conjecture. If $r=0$ (respectively, $r=1$), the...

conjecture | open | MathDB 页面 | 原文线索

V2-066698 | MATHDB-347652

Zhang's equitable list arboricity conjecture

摘录 / EXCERPT: Zhang's conjecture. Any graph G is equitably k-list arborable for each k satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-066699 | MATHDB-347654

Improved equitable list arboricity conjecture for connected graphs

摘录 / EXCERPT: Improved equitable list arboricity conjecture. Any connected graph G is equitably $\lceil\Delta(G)/2\rceil$-list arborable provided G is neither a cycle nor a complete graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-066700 | MATHDB-347655

Equitable list arboricity conjecture for powers of paths

摘录 / EXCERPT: Path-power equitable list arboricity conjecture. The graph G is equitably k-list arborable if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-066701 | MATHDB-347661

Shortest origami edge-paths minimize successive endpoint intersections

摘录 / EXCERPT: Let α and β be the endpoints of a shortest origami edge-path, and let α_i and α_{i+1} be consecutive curves on it, with α_{i+1} intersecting...

conjecture | open | MathDB 页面 | 原文线索

V2-066702 | MATHDB-347662

Uniqueness conjecture for submaximal curves on principally polarized abelian surfaces

摘录 / EXCERPT: Uniqueness conjecture. For every ample $\mathbb Q$-line bundle L on X, there exists at most one irreducible curve C that is submaximal for L if and only if either

conjecture | open | MathDB 页面 | 原文线索

V2-066703 | MATHDB-347664

Bounds relating the difference index and sum index

摘录 / EXCERPT: Let G be a nonempty simple graph. Let $s(G)$ be its sum index and $d(G)$ its difference index. Difference-sum index bounds conjecture. ... These bounds would place the difference i...

conjecture | open | MathDB 页面 | 原文线索

V2-066704 | MATHDB-347668

The positivity-induced triangle inequality for even-power forms

摘录 / EXCERPT: Let $q\geq 1$, and let $s_t en\mathbb{R}$ and $v_t\in\mathbb{R}^m$ for $t\in [w]$ satisfy ... for every $x\in\mathbb{R}^m$. Positivity-induced triangle inequality. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-066705 | MATHDB-347670

The local antimagic chromatic number conjecture for spiders

摘录 / EXCERPT: The local antimagic spider conjecture. $\chi_{la}(Sp(y_1,y_2,\ldots,y_d))=d+1$, except for $Sp(2^{[n]},3^{[m]})$ when (n,m) belongs to the listed exceptional set:

conjecture | open | MathDB 页面 | 原文线索

V2-066706 | MATHDB-347672

The two-actions conjecture for degenerate projective hypersurfaces

摘录 / EXCERPT: Let $X\subseteq\mathbb{P}^{n+1}$ be a degenerate hypersurface admitting an induced additive action, where an induced additive action is an additive action obtained from the ambient...

conjecture | open | MathDB 页面 | 原文线索

V2-066707 | MATHDB-347683

Realizability of prescribed numbers of non-atomic molecules in numerical monoids

摘录 / EXCERPT: Realizability conjecture. For every $n \in \mathbb{N}_{\ge 2}$ there exists a numerical monoid N such that

conjecture | open | MathDB 页面 | 原文线索

V2-066708 | MATHDB-347684

Molecularity of the class $\mathcal{C}_3$

摘录 / EXCERPT: Molecularity conjecture for $\mathcal{C}_3$. The class $\mathcal{C}_3$ is molecular.

conjecture | open | MathDB 页面 | 原文线索

V2-066709 | MATHDB-347687

The representability bound for simplicial complexes with bounded Leray number

摘录 / EXCERPT: Representability bound conjecture. If $h(X)\leq d$ for $d\geq 1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066710 | MATHDB-347688

The representability problem for the affine-plane Steiner system on nine vertices

摘录 / EXCERPT: Affine-plane representability problem. Determine whether

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-066711 | MATHDB-347696

Equilibrium measure conjecture for TASEP on Galton-Watson trees

摘录 / EXCERPT: Consider TASEP with a reservoir of rate $\lambda=\rho q$ for some $\rho \in (0,\infty)$, and suppose that a flow rule holds for a flow of strength q. Let π_{λ} be the e...

conjecture | open | MathDB 页面 | 原文线索

V2-066712 | MATHDB-347700

Metric Kaplansky conjecture for bounded metric groups

摘录 / EXCERPT: Let K be a finite field, let $\varepsilon>0$ and $N\in\mathbb{N}$. For a metric group $(G,\|\cdot\|)$ with $\|\cdot\|\leq 1$, let $K[G]$ be its group ring, let $l(a)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-066713 | MATHDB-347708

The enriched presheaf conjecture for presentable monoidal infinity-categories

摘录 / EXCERPT: Let $\mathcal{V}$ be a presentable, closed monoidal ∞-category. Write $\operatorname{Cat}^{\mathcal{V}}$ for the ∞-category of $\mathcal{V}$-enriched categories and...

conjecture | open | MathDB 页面 | 原文线索

V2-066714 | MATHDB-347718

Spectral condition conjecture for attractive delta NLS solitary waves

摘录 / EXCERPT: Fix $q<0$ and $p>4$. Let $\sigma=-1$ denote the focusing case and $\sigma=1$ the defocusing case, and let $\Omega=\Omega(q,p)$ be the minimal unstable frequency. The spectral condi...

conjecture | open | MathDB 页面 | 原文线索

V2-066715 | MATHDB-347719

Qualitative spectrum conjecture for the linearized focusing delta NLS

摘录 / EXCERPT: Fix $q<0$ and $p>4$, and consider the linearized operator $\mathcal{L}(\omega)$ for $\omega\in(\tfrac12q^2,\infty)$. Let $\csigma_c(\mathcal{L}(\omega))$ denote its continuous spect...

conjecture | open | MathDB 页面 | 原文线索

V2-066716 | MATHDB-347720

The extreme-coefficient conjecture for ED polynomials of Segre products

摘录 / EXCERPT: Extreme-coefficient conjecture. The extreme coefficients are

conjecture | open | MathDB 页面 | 原文线索

V2-066717 | MATHDB-347721

The ED-degree stabilization conjecture for hypersurface Segre products

摘录 / EXCERPT: ED-degree stabilization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066718 | MATHDB-347723

Finite-time wave breaking criterion for the generalized 0-Holm-Staley equation

摘录 / EXCERPT: Let $k=1$, let $u_0\in H^s(\mathbb{R})$ with $s>3/2$, and let u be the corresponding solution of equation (8.0.1), with lifespan T. Finite-time wave breaking conjecture. The so...

conjecture | open | MathDB 页面 | 原文线索

V2-066719 | MATHDB-347725

Descent along atlases is equivalent to hyperdescent

摘录 / EXCERPT: Atlas descent conjecture. The following conditions on F are equivalent: (1) F satisfies descent along atlases; (2) F satisfies hyperdescent. In particular, the full subcatego...

conjecture | open | MathDB 页面 | 原文线索

V2-066720 | MATHDB-347727

Kalman conjecture for absolute stability of feedback interconnections

摘录 / EXCERPT: Let G be a stable transfer function, let k_N be its Nyquist value, and let ϕ be a memoryless nonlinearity in $S[0,k]$. Kalman conjecture. The feedback interconnection of...

conjecture | open | MathDB 页面 | 原文线索

V2-066721 | MATHDB-347728

Zames-Falb multiplier equality conjecture for absolute stability

摘录 / EXCERPT: Let G be an LTI system, let k_N denote its Nyquist gain, and let $0<k<k_N$. Suppose there is no suitable LTI Zames-Falb multiplier for $G+1/k$. Zames-Falb multiplier equality c...

conjecture | open | MathDB 页面 | 原文线索

V2-066722 | MATHDB-347729

Numerical improvement conjecture for four-cell stratified sampling

摘录 / EXCERPT: Let $\mathbf{v}=(v_1,v_2,v_3)$ satisfy $v_1<v_2<v_3$ with each $v_i\in[0,\sqrt{2}]$. Let $\mathbf{\Omega}_{\mathbf{v}}^{(4)}$ be the associated four-cell partition, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066723 | MATHDB-347740

Convex recovery conjecture for well-structured sensing matrices

摘录 / EXCERPT: Let K be the number of scatters, let $N_{\mathsf c}$ and $\bar{\mathsf X},\bar{\mathsf Y},\bar{\mathsf Z}$ be the sensing and array parameters, and let $r_{\ell_1}^{\circ}$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-066724 | MATHDB-347756

Sofic and hyperlinear profile comparison

摘录 / EXCERPT: Let G be a sofic group. The sofic profile $\mathcal{D}_{G}^{sof}(n)$ and hyperlinear profile $\mathcal{D}_{G}^{hyp}(n)$ are compared using the relation $\preccurlyeq$. The profile comp...

conjecture | open | MathDB 页面 | 原文线索

V2-066725 | MATHDB-347757

Arzhantseva's residual finiteness conjecture for Gromov hyperbolic groups

摘录 / EXCERPT: A Gromov hyperbolic group is a group satisfying the usual metric hyperbolicity condition; a group is residually finite if its nontrivial elements survive in finite quotients, and i...

conjecture | open | MathDB 页面 | 原文线索

V2-066726 | MATHDB-347758

Virtually abelian groups' linear profile conjecture

摘录 / EXCERPT: Let G be a finitely generated virtually abelian group, and let $\operatorname{rank} G$ denote its rank. Write $\mathcal{D}_{G,S}^{lin}(n)$ for the linear sofic profile of G rel...

conjecture | open | MathDB 页面 | 原文线索

V2-066727 | MATHDB-347759

Heisenberg groups' linear and weakly sofic profile conjecture

摘录 / EXCERPT: Let H_{2l+1} denote the $(2l+1)$-dimensional Heisenberg group, and let $\mathcal{D}_{H_{2l+1},S}^{fin}(n)$ and $\mathcal{D}_{H_{2l+1},S}^{lin}(n)$ denote its weakly sofic and lin...

conjecture | open | MathDB 页面 | 原文线索

V2-066728 | MATHDB-347760

Virtually nilpotent groups' profile equivalence conjecture

摘录 / EXCERPT: Let G be a finitely generated virtually nilpotent group, let S be a finite generating set, and let $\mathcal{D}_{G,S}^{sof}$, $\mathcal{D}_{G,S}^{hyp}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066729 | MATHDB-347761

Linear sofic free-product profile conjecture

摘录 / EXCERPT: Let G and H be linear sofic groups, let $G\ast H$ be their free product, and let F_2 denote the free group of rank 2. Write $\mathcal{D}^{lin}_G(n)$ for the linear sofic pr...

conjecture | open | MathDB 页面 | 原文线索

V2-066730 | MATHDB-347781

Vandiver's conjecture

摘录 / EXCERPT: Vandiver's conjecture. $X_\infty^{+}=0$. Equivalentemente, p no divide el número de clases del subcampo $\mathbb{Q}(\mu_p)^{+}$ de $\mathbb{Q}(\mu_p)$ fijado por la conjugación c...

conjecture | open | MathDB 页面 | 原文线索

V2-066731 | MATHDB-347782

Mazur's main conjecture for elliptic curves

摘录 / EXCERPT: Mazur's main conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-066732 | MATHDB-347785

The Panchishkin p-adic L-function conjecture for motives

摘录 / EXCERPT: Panchishkin's conjecture. Existe un único elemento $\mu_M\in\widetilde{\Phi}(G)$ tal que, para cada ψ y n como arriba,

conjecture | open | MathDB 页面 | 原文线索

V2-066733 | MATHDB-347786

Greenberg's main conjecture for motives

摘录 / EXCERPT: Greenberg's main conjecture for motives. Existe $h\in\Lambda(G)$ tal que

conjecture | open | MathDB 页面 | 原文线索

V2-066734 | MATHDB-347787

The Generalized Lax Conjecture for hyperbolicity cones

摘录 / EXCERPT: A homogeneous polynomial $h\in\mathbb{R}[x_1,\ldots,x_n]$ is hyperbolic with respect to $e\in\mathbb{R}^n$ if $h(e)>0$ and, for every $a\in\mathbb{R}^n$, the polynomial $h(te-a)$ h...

conjecture | open | MathDB 页面 | 原文线索

V2-066735 | MATHDB-347793

Vanishing conjecture for stable-pairs descendents of holomorphic $(p,0)$-classes

摘录 / EXCERPT: Let X be a simply-connected smooth projective 3-fold, let $\gamma\in H^{p,0}(X)$ with $p=2$ or $p=3$, and let $D\in\mathbb D_{\textup{PT}}^{X}$. For $k\geq 0$, the stable-pairs d...

conjecture | open | MathDB 页面 | 原文线索

V2-066736 | MATHDB-347808

Equality of admissible and numerically trivial Chow cycles

摘录 / EXCERPT: Let X be as above, and let $CH^i(X)^{(0)}$ be the subgroup defined in Proposition 7. Let $CH^i_{\operatorname{num}}(X)$ denote the subgroup of cycles numerically equivalent to...

conjecture | open | MathDB 页面 | 原文线索

V2-066737 | MATHDB-347811

Sensitivity conjecture for Cayley graphs of Coxeter groups

摘录 / EXCERPT: Let G be the Cayley graph of a Coxeter group, and let Q_d be the largest subgraph of G isomorphic to a cube. Write $\boldsymbol{\kappa}(G)=d$ for its dimension, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066738 | MATHDB-347812

Generalized GLT spectral-distribution conjecture for ALIF iteration matrices

摘录 / EXCERPT: Let k be an even, non-negative, bounded, compactly supported measurable function with $\\|k\\|_1=1$, supported on $[-1,1]$, and let $L(x)$ be strictly positive on $[0,1]$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-066739 | MATHDB-347813

Discrete ALIF inner-loop convergence conjecture under the spectral condition

摘录 / EXCERPT: The Discrete ALIF inner-loop convergence conjecture asserts that if

conjecture | open | MathDB 页面 | 原文线索

V2-066740 | MATHDB-347817

Uniqueness conjecture for positive solutions of indefinite ϕ-Laplacian boundary value problems

摘录 / EXCERPT: Let $a\in L^{\infty}(0,T)$ satisfy the assumptions in (hp-a), and let $\gamma\in\mathbb{R}\setminus\{-1,0,1\}$. Consider the Neumann and periodic boundary value problems associated...

conjecture | open | MathDB 页面 | 原文线索

V2-066741 | MATHDB-347819

Primitivity rank bound in terms of stable commutator length

摘录 / EXCERPT: Let $w\in\mathbb{F}$ be a word in a free group, let $\pi(w)$ be its primitivity rank, and let $\mathrm{scl}(w)$ be its stable commutator length. Primitivity-rank inequality. For an...

conjecture | open | MathDB 页面 | 原文线索

V2-066742 | MATHDB-347822

Maximum Möbius value conjecture for rank-3 intervals in the matching pattern poset

摘录 / EXCERPT: Rank-3 maximum conjecture. The maximum absolute value of the Möbius function on intervals of rank 3 is

conjecture | open | MathDB 页面 | 原文线索

V2-066743 | MATHDB-347823

Alternating Möbius function conjecture for the matching pattern poset

摘录 / EXCERPT: Alternating Möbius function conjecture. The Möbius function is alternating, meaning that it is

conjecture | open | MathDB 页面 | 原文线索

V2-066744 | MATHDB-347825

Strictly nef canonical bundle conjecture

摘录 / EXCERPT: Strictly nef canonical bundle conjecture. If K_X is strictly nef, then K_X is ample.

conjecture | open | MathDB 页面 | 原文线索

V2-066745 | MATHDB-347827

Exterior-power tangent and cotangent ampleness conjecture

摘录 / EXCERPT: Exterior-power ampleness conjecture. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-066746 | MATHDB-347840

The KP hierarchy characterization of Calabi-Yau triple Hodge integrals

摘录 / EXCERPT: Let $Z_{q,p}({\bf T})$ be the generating function of triple Hodge integrals ... with ... The parameters satisfy the Calabi-Yau condition, and $\mathbf T$ denotes the Hodge-integral var...

conjecture | open | MathDB 页面 | 原文线索

V2-066747 | MATHDB-347846

The Planted Affine Planes SoS conjecture for nearly quadratic sample size

摘录 / EXCERPT: Planted Affine Planes SoS conjecture. Theorem $$ holds with the bound on the number of sampled vectors loosened to

conjecture | open | MathDB 页面 | 原文线索

V2-066748 | MATHDB-347847

The Planted Boolean Vector SoS conjecture for large random-subspace dimension

摘录 / EXCERPT: Planted Boolean Vector SoS conjecture. Theorem $$ holds with the bound on the dimension p of a random subspace loosened to

conjecture | open | MathDB 页面 | 原文线索

V2-066749 | MATHDB-347850

Conjectured growth series of the three-strand braid group with dual generators

摘录 / EXCERPT: Let B_3 be the three-strand braid group, let $\Sigma_3^{\scriptstyle *}$ denote its dual generating set, and let $\mathcal{S}(B_3,\Sigma_3^{\scriptstyle *})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066750 | MATHDB-347853

Resonance-pole instability conjecture for inhomogeneous coupling

摘录 / EXCERPT: Consider the modified system ... where the coupling term βv has been replaced by $\beta\sigma(x)v$, and suppose that the Fourier transform of $\sigma(x)$ has full support. Res...

conjecture | open | MathDB 页面 | 原文线索

V2-066751 | MATHDB-347865

The discrete-time Kalman conjecture for Lurye systems

摘录 / EXCERPT: Let G be a discrete-time, linear time-invariant plant in negative feedback with a nonlinearity $\phi\in S_{0,k}$. Let k_{AS} be the supremum of the values of k for which the...

conjecture | open | MathDB 页面 | 原文线索

V2-066752 | MATHDB-347871

Initial-distribution scaling conjecture for one-dimensional multi-particle DLA

摘录 / EXCERPT: Let $Z_1,Z_2,\dots$ be independent identically distributed random variables with ... and initialize the one-dimensional multi-particle Diffusion Limited Aggregation model with Z_i...

conjecture | open | MathDB 页面 | 原文线索

V2-066753 | MATHDB-347876

Karthick's star-chi-boundedness conjecture for claw-free graphs

摘录 / EXCERPT: For a graph G, the star chromatic number $\chi_s(G)$ is the minimum number of colors in a proper coloring such that any two color classes induce a star forest. A graph is...

conjecture | open | MathDB 页面 | 原文线索

V2-066754 | MATHDB-347877

Karthick's star-chi-boundedness conjecture for (T,C_4)-free graphs

摘录 / EXCERPT: For a graph F, an F-free graph has no induced subgraph isomorphic to F; a graph is (T,C_4)-free if it is both T-free and C_4-free. Here χ_s denotes the star chroma...

conjecture | open | MathDB 页面 | 原文线索

V2-066755 | MATHDB-347878

Barrett-Butler-Hall conjecture on equimodular inverses of symmetric 0-1 matrices

摘录 / EXCERPT: Let $n\geq 3$ and let $A\in\{0,1\}^{n\times n}$ be a nonsingular symmetric 0-1 matrix. The enhanced principal rank characteristic sequence of a symmetric matrix records, for each o...

conjecture | open | MathDB 页面 | 原文线索

V2-066756 | MATHDB-347885

Mohar's conjecture on the crossing number of antipodal multipartite graphs

摘录 / EXCERPT: Let $M_{n,t}$ be the graph considered in the construction above, and write $k=\lfloor n/2\rfloor$. If a set $P'\subseteq P$ has strength 0 and $t=|P\setminus P'|$, then the const...

conjecture | open | MathDB 页面 | 原文线索

V2-066757 | MATHDB-347886

Symmetric-group controllability conjecture for semisimple Lie groups

摘录 / EXCERPT: Symmetric-group controllability conjecture. The system is controllable on G if and only if there exists $\Sigma\subseteq\Gamma$ such that $\iota(\Sigma)$ is a cycle of maximal le...

conjecture | open | MathDB 页面 | 原文线索

V2-066758 | MATHDB-347895

Carlen-Lieb lower rearrangement inequality conjecture for Schatten norms

摘录 / EXCERPT: Let $A,B\in M_{n\times n}(\mathbb{C})$, and let $\sigma_\downarrow(A)$ and $\sigma_\downarrow(B)$ denote their singular-value vectors arranged in decreasing order. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066759 | MATHDB-347896

Carlen-Lieb upper rearrangement inequality conjecture for Schatten norms

摘录 / EXCERPT: Let $A,B\in M_{n\times n}(\mathbb{C})$, and let $\sigma_\uparrow(A)$ denote the singular-value vector of A arranged in increasing order, while $\sigma_\downarrow(B)$ is arranged...

conjecture | open | MathDB 页面 | 原文线索

V2-066760 | MATHDB-347903

Submodularity conjecture for noncooperative ACC H_2 performance

摘录 / EXCERPT: Let S be the set of autonomous vehicles in the mixed traffic system, let $J_1(S)=-\lVert \mathbf{G}_1(S)\rVert_2^2$ be the negative squared $\mathcal{H}_2$ norm of the transfer f...

conjecture | open | MathDB 页面 | 原文线索

V2-066761 | MATHDB-347904

The LMAX extension of the set-valued Wilf-equivalence problem

摘录 / EXCERPT: Let P be a set of indecomposable patterns. Let LMAX denote the set-valued statistic recording left-to-right maxima, and let iar and comp denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-066762 | MATHDB-347908

Persistence of remote-fold singular cycles in three-timescale systems

摘录 / EXCERPT: Let the assumptions concerning the parameters, reduced flow, and slow drift hold, and suppose that the singularities of the system are remote. Let θ denote the system parame...

conjecture | open | MathDB 页面 | 原文线索

V2-066763 | MATHDB-347909

Persistence of aligned or connected singular-cycle families

摘录 / EXCERPT: Let $varepsilon$ and $delta$ be positive perturbation parameters, and suppose that the assumptions concerning the parameters and reduced flow hold. Assume that the singularities of...

conjecture | open | MathDB 页面 | 原文线索

V2-066764 | MATHDB-347923

Regulator and torsion growth conjecture for totally real fields

摘录 / EXCERPT: Let ${\mathcal K}$ be the set of totally real number fields. For $K\in{\mathcal K}$, let D_K be its discriminant, let ${\mathcal T}_K$ be the relevant p-torsion group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066765 | MATHDB-347924

The K-theory-class group conjecture for local cDV singularities

摘录 / EXCERPT: K-theory-class group conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-066766 | MATHDB-347926

The vanishing conjecture for singularity-category K-theory of minimal models

摘录 / EXCERPT: Vanishing conjecture. The Grothendieck group of the singularity category vanishes:

conjecture | open | MathDB 页面 | 原文线索

V2-066767 | MATHDB-347929

Lefschetz decomposition conjecture for moduli spaces of rank-two bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus g, let ${\mathcal L}$ be the fixed determinant, and let ${\moduli_C(2,\mathcal L)}$ be the moduli space of rank-two bundles with det...

conjecture | open | MathDB 页面 | 原文线索

V2-066768 | MATHDB-347930

Ringel-Samokhin decomposition conjecture for moduli spaces of rank-two bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus $g\geq 2$, let ${\mathcal L}$ be the fixed determinant, and let ${\moduli_C(2,\mathcal L)}$ be the moduli space of rank-two bundles wi...

conjecture | open | MathDB 页面 | 原文线索

V2-066769 | MATHDB-347931

Komlós' discrepancy conjecture

摘录 / EXCERPT: Let n be a positive integer, and let $\bm{v}_1,\dots,\bm{v}_n\in\mathbb{R}^n$ satisfy $\|\bm{v}_i\|_2\leq 1$ for $i\in[n]$. Komlós' conjecture. There exists a constant C such t...

conjecture | open | MathDB 页面 | 原文线索

V2-066770 | MATHDB-347932

The sharp Hermite-ratio upper-bound conjecture

摘录 / EXCERPT: Let $h_n(x)$ and $F_n(x)=h_n(x)/h_{n+1}(x)$ be the ratios considered above, and let $g_{\alpha,\beta}(x)$ denote the comparison function introduced in the paper. Hermite-ratio uppe...

conjecture | open | MathDB 页面 | 原文线索

V2-066771 | MATHDB-347934

Gross-Mansour-Tucker's interpolating conjecture for partial-dual Euler-genus polynomials

摘录 / EXCERPT: Let G be a non-orientable ribbon graph. Its partial-dual Euler-genus polynomial is the generating function ... where G^A is the partial dual of G with respect to...

conjecture | refuted | MathDB 页面 | 原文线索

V2-066772 | MATHDB-347949

ADE classification of Lagrangian fillings

摘录 / EXCERPT: ADE classification of Lagrangian fillings. Exactly one of the following possibilities occurs:

conjecture | open | MathDB 页面 | 原文线索

V2-066773 | MATHDB-347958

Local well-posedness and blow-up conjecture for the Schrödinger equation with concentrated nonlinearity

摘录 / EXCERPT: Well-posedness and blow-up conjecture. If $0<\sigma<\frac{2s}{n}-1$, the solutions are global, whereas if $\sigma\geq\frac{2s}{n}-1$, finite-time blow-up is possible for some initi...

conjecture | open | MathDB 页面 | 原文线索

V2-066774 | MATHDB-347960

The bounded-color monochromatic-cover conjecture

摘录 / EXCERPT: Bounded-color cover conjecture. Every such coloring has a monochromatic cover of order at most $r-1$ in which the subgraphs use at most $\lceil r/2\rceil$ distinct colors.

conjecture | open | MathDB 页面 | 原文线索

V2-066775 | MATHDB-347961

The bounded-diameter bipartite monochromatic-cover conjecture

摘录 / EXCERPT: Bipartite bounded-diameter conjecture. There is an integer d such that, for every $r\geq 2$, every r-coloring of every $K\in\mathcal{K}_2$ has a monochromatic cover of order at...

conjecture | open | MathDB 页面 | 原文线索

V2-066776 | MATHDB-347964

Relative Scott adjunction conjecture for Grothendieck topoi

摘录 / EXCERPT: Let $\mathcal{E}$ be a Grothendieck topos, and let $\omega \leq \lambda \leq \kappa$. Relative Scott adjunction conjecture. There is a 2-adjunction ... Moreover, if $\mathcal{E}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-066777 | MATHDB-347968

Subgroup projection-distance conjecture for closed orthogonal subgroups

摘录 / EXCERPT: Let $\mathcal{G}$ be a closed subgroup of the orthogonal group $\mathcal{O}(d)$, and let $\operatorname{conv}(\mathcal{G})$ denote its convex hull. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066778 | MATHDB-347976

Bipartite random friends-and-strangers graph component conjecture

摘录 / EXCERPT: Let r be a positive integer, let $p\in[0,1]$, and let X and Y be independently chosen random graphs from $\mathcal G(K_{r,r},p)$, where $K_{r,r}$ is the complete bipartite gr...

conjecture | open | MathDB 页面 | 原文线索

V2-066779 | MATHDB-347984

Li-Horacek-Luo-Miao's planar girth-five list star chromatic index conjecture

摘录 / EXCERPT: Li-Horacek-Luo-Miao's conjecture. There exists a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066780 | MATHDB-347994

Conjecture on Betti numbers of powers of closed binomial edge ideals

摘录 / EXCERPT: Let G be a closed graph, let J_G be its binomial edge ideal, and let $\ini_<(J_G)$ denote its initial ideal with respect to the specified term order. For every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-066781 | MATHDB-347999

Average rank zero for elliptic surfaces ordered by coefficient size

摘录 / EXCERPT: For positive integers d and M, let ... where $\mu(p)$ is the Mahler measure, and define ... Here $\mathcal{S}_{m,n}=\mathcal{S}_{m,n}(M)$ is the family of elliptic surfaces with fi...

conjecture | open | MathDB 页面 | 原文线索

V2-066782 | MATHDB-348000

Half positive-rank density for elliptic surfaces over finite fields

摘录 / EXCERPT: For positive integers m and n, let $\mathcal{S}_{\ell;m,n}$ be the finite set of elliptic surfaces over $\mathbb{F}_\ell$ of the form $y^2=x^3+A(T)x+B(T)$ with $\deg(A)=m$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066783 | MATHDB-348003

Converse compatibility conjecture for connected tensor products of signed graphs

摘录 / EXCERPT: Converse compatibility conjecture. If Σ_1 and Σ_2 are two compatible signed graphs, then the connected tensor product $\Sigma_1\otimes\Sigma_2$ is compatible.

conjecture | open | MathDB 页面 | 原文线索

V2-066784 | MATHDB-348005

Rank-symmetry conjecture for odd diagram classes

摘录 / EXCERPT: Let D be an odd diagram, and let $\operatorname{Perm}_n(D)$ denote its associated odd diagram class in Bruhat order. A subset of a graded poset is rank-symmetric when the number...

conjecture | open | MathDB 页面 | 原文线索

V2-066785 | MATHDB-348007

The optimal quasi-simply-ordered polychromatic coloring conjecture

摘录 / EXCERPT: Optimal quasi-simply-ordered coloring conjecture. If $n\geq z$ and $n-q$ is even when r is odd, then there exist optimal quasi-simply-ordered $\mathscr{R}(n,r,q)$- and...

conjecture | open | MathDB 页面 | 原文线索

V2-066786 | MATHDB-348008

The optimal simply-ordered polychromatic coloring conjecture in the large-q regime

摘录 / EXCERPT: Large-q optimality conjecture. Let $r\geq 1$ and $q\geq 0$ be integers with $q\geq 2r-2$, and let $n\geq q+r+1$, with r and $n-q$ not both odd. Then there exists an optimal sim...

conjecture | open | MathDB 页面 | 原文线索

V2-066787 | MATHDB-348013

Nonexistence of canonical small resolutions for symmetric rank loci

摘录 / EXCERPT: Nonexistence conjecture. There do not exist canonical small resolutions for symmetric rank loci.

conjecture | open | MathDB 页面 | 原文线索

V2-066788 | MATHDB-348015

Polynomial asymptotic conjecture for multi-party auction vulnerability counts

摘录 / EXCERPT: Let n be a positive integer and let m be the input size. Define ... where f is the auction-output function. Asymptotic form conjecture for c_n. There exists a positive ration...

conjecture | open | MathDB 页面 | 原文线索

V2-066789 | MATHDB-348016

The generalized Khavinson conjecture for hyperbolic harmonic mappings

摘录 / EXCERPT: Let $p\in[1,\infty]$ and let q be its conjugate. For $\Omega=\mathbb{B}^{n}$ or $\mathbb{H}^{n}$, suppose that $u=\mathcal{P}_{\Omega}[\phi]$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-066790 | MATHDB-348034

Borodin's cyclic coloring conjecture

摘录 / EXCERPT: Borodin's cyclic coloring conjecture. Every connected plane graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-066791 | MATHDB-348035

Borodin-Broersma-Glebov-van den Heuvel conjecture on cyclic coloring

摘录 / EXCERPT: Borodin-Broersma-Glebov-van den Heuvel conjecture. Every plane graph G with $\Delta^*(G)$ and k^* sufficiently large has a cyclic coloring with

conjecture | open | MathDB 页面 | 原文线索

V2-066792 | MATHDB-348036

Cyclic coloring conjecture for subdivisions of simple 3-connected plane graphs

摘录 / EXCERPT: Subdivision cyclic coloring conjecture. Every such graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-066793 | MATHDB-348037

The variational principle and asymptotic enumeration for skew Young tableaux

摘录 / EXCERPT: The variational principle for skew Young tableaux. The function $g(t,x)$ maximizes

conjecture | open | MathDB 页面 | 原文线索

V2-066794 | MATHDB-348038

The pathwise variational principle for skew Young tableaux

摘录 / EXCERPT: Pathwise asymptotic enumeration conjecture. As $n_N\to\infty$ and $\varepsilon\to0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066795 | MATHDB-348040

Strong synchronisation of alternating-group descent enumerators

摘录 / EXCERPT: Strong-synchronisation conjecture. For every positive integer n, the sequences $(A_{n,k}^+)_{k=0}^{n-1}$ and $(A_{n,k}^-)_{k=0}^{n-1}$ are strongly synchronised.

conjecture | open | MathDB 页面 | 原文线索

V2-066796 | MATHDB-348041

Strong synchronisation of alternating-group descent and excedance enumerators

摘录 / EXCERPT: Descent-excedance synchronisation conjecture. For every positive integer n, the sequences $(A_{n,k}^+)_{k=0}^{n-1}$ and $(P_{n,k})_{k=0}^{n-1}$ are strongly synchronised. Similar...

conjecture | open | MathDB 页面 | 原文线索

V2-066797 | MATHDB-348049

Wall-crossing conjecture for perverse coherent systems on the local resolved conifold

摘录 / EXCERPT: Wall-crossing conjecture. If $\Theta\in L_-^(k)$ or $L_+^(k)$ with $k\geqslant1$, then there exist choices of signs such that

conjecture | open | MathDB 页面 | 原文线索

V2-066798 | MATHDB-348052

Deformed GKM conjecture for r-spin enumerative invariants

摘录 / EXCERPT: Consider the deformed generalized Kontsevich matrix model (GKM) with potential given by equation, and allow a shift of its variables. Deformed GKM conjecture. The shifted deformed...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-066799 | MATHDB-348062

Theta-function coefficients in products of Jacobi forms

摘录 / EXCERPT: Let L, K, v, w, and ι be as in Theorem $\operatorname{MFprodJac}$. Write D_L and D_K for the discriminant groups of L and K, and let $D_{L+_{\iota}K}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-066800 | MATHDB-348066

Polynomiality of the number of star operations for length four

摘录 / EXCERPT: Let $\mathbf{s}$ be a sequence of pairs of positive integers with $n(\mathbf{s})=4$, and let $\Lambda(\mathbf{s},q)$ denote the number of star operations associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-066801 | MATHDB-348068

Monotonicity of the p-cell a-function under the two-sided p-preorder

摘录 / EXCERPT: Let W be the Weyl group under consideration, and let ${}^{p} \!\mkern 1mu {\mathbf a}_{\mathbf c}:W\to\mathbb{R}_{\ge 0}$ be the function obtained by extending the cell invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-066802 | MATHDB-348071

Quantized multiplication conjecture for truncated shifted Yangians

摘录 / EXCERPT: Let λ be a dominant coweight and let $\mu\leq\lambda$ be a coweight such that $\mu+\alpha_i^\vee\leq\lambda$. Let $Y_\mu^\lambda(\mathbf{R})$ be the truncated shifted Yangi...

conjecture | open | MathDB 页面 | 原文线索

V2-066803 | MATHDB-348073

Independence and positivity conjecture for monodromy heights

摘录 / EXCERPT: For each $d\geq 2$, let $N_{v,d}$ be the canonical monodromy element associated with the ℓ-adic realization M_ℓ of a mixed motive M, and let $\langle\,\ \rangle_{N_v}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066804 | MATHDB-348078

Infinite-length conjecture for blocked wild labyrinth fractals

摘录 / EXCERPT: Let ${\cal A}$ be a wild labyrinth pattern: its graph is connected, it has at least one vertical and one horizontal exit pair, and it satisfies the stated labyrinth-pattern connect...

conjecture | open | MathDB 页面 | 原文线索

V2-066805 | MATHDB-348080

Improved homogenization bound for Steklov eigenvalues with rapidly oscillating weights

摘录 / EXCERPT: Let $\lambda_{k,\varepsilon}$ and $\lambda_{k,0}$ denote the k-th Steklov eigenvalues, respectively, for the rapidly oscillating boundary weight ρ_ε and its homoge...

conjecture | open | MathDB 页面 | 原文线索

V2-066806 | MATHDB-348081

Kusner's equilateral-set conjecture in finite-dimensional lp spaces

摘录 / EXCERPT: Kusner's conjecture.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-066807 | MATHDB-348085

Tightness of the strengthened SDP relaxation in dimension two

摘录 / EXCERPT: Tightness conjecture. The intersection

conjecture | open | MathDB 页面 | 原文线索

V2-066808 | MATHDB-348086

Tightness of locally valid cuts for the nonnegative two-dimensional feasible space

摘录 / EXCERPT: Locally valid-cut conjecture. The intersection of $\mathcal{R}_{\mathrm{shor}} \cap \mathcal{R}_{\mathrm{ksoc}}$ with the locally valid cuts captures $\mathcal{G}$.

conjecture | open | MathDB 页面 | 原文线索

V2-066809 | MATHDB-348091

Optimality of the round-robin policy for more than two sources

摘录 / EXCERPT: Let $M>2$ and let N be the number of channels. For a problem instance $(M,N,\boldsymbol\mu)$, consider all possible permutations of arms in the set $\mathcal{S}(M)$ and the round...

conjecture | open | MathDB 页面 | 原文线索

V2-066810 | MATHDB-348093

Asymptotic translation lengths for mapping class groups of punctured surfaces

摘录 / EXCERPT: Let $S_{g,n}$ be a genus-g surface with n punctures, let $\Mod(S_{g,n})$ be its mapping class group, let $\PMod(S_{g,n})$ be the pure mapping class group, and let $L_{\C}(G)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-066811 | MATHDB-348094

Uniform lower bound for proper normal subgroups of mapping class groups

摘录 / EXCERPT: Let $S_{g,n}$ be a genus-g surface with n punctures, let $\Mod(S_{g,n})$ be its mapping class group, and let $L_C(H)$ denote the least asymptotic translation length on the curv...

conjecture | open | MathDB 页面 | 原文线索

V2-066812 | MATHDB-348099

Uniform entropy lower bound conjecture for proper subgraphs of metric graphs

摘录 / EXCERPT: Let $r\geq 3$. A metric graph of rank r has unit entropy when its entropy is normalized to equal 1, and a subgraph is proper when it is a strict subgraph of the graph. Uniform...

conjecture | open | MathDB 页面 | 原文线索

V2-066813 | MATHDB-348106

The structure and substructure connectivity conjecture for hypercubes

摘录 / EXCERPT: Let Q_n be the n-dimensional hypercube, let $K_{1,r}$ be the star with r leaves, and write $\kappa(Q_n;K_{1,r})$ and $\kappa^{s}(Q_n;K_{1,r})$ for the $K_{1,r}$-structure con...

conjecture | open | MathDB 页面 | 原文线索

V2-066814 | MATHDB-348107

The structure and substructure connectivity conjecture for folded hypercubes

摘录 / EXCERPT: Let FQ_n be the n-dimensional folded hypercube, let $K_{1,r}$ be the star with r leaves, and write $\kappa(FQ_n;K_{1,r})$ and $\kappa^{s}(FQ_n;K_{1,r})$ for the $K_{1,r}$-str...

conjecture | open | MathDB 页面 | 原文线索

V2-066815 | MATHDB-348108

The limiting energy conjecture for Barabási-Albert trees

摘录 / EXCERPT: Limiting energy conjecture. For every α, there exists $g(\alpha)$ such that, almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-066816 | MATHDB-348111

Odaka's non-compact Yau-Tian-Donaldson conjecture

摘录 / EXCERPT: Let $((X,D),L)$ be a polarized dlt log Calabi-Yau pair with $\lfloor D\rfloor=\sum_iD_i$, $K_X+D\equiv0$, and L ample on $X^{o}:=X\setminus\operatorname{Supp}(\lfloor D\rfloor)$....

conjecture | open | MathDB 页面 | 原文线索

V2-066817 | MATHDB-348114

Odaka's product behavior conjecture for minimally non-collapsing limits

摘录 / EXCERPT: Let $(\mathcal X^*,\mathcal L^*)\to\Delta^*$ be a degeneration satisfying the paper's special maximal-degeneration expectation, and let (S,M) be a fixed pointed polarized m-dim...

conjecture | open | MathDB 页面 | 原文线索

V2-066818 | MATHDB-348119

Random-phase triangular factorization conjecture for loops in $SU(2)$

摘录 / EXCERPT: Let k_2 be the loop associated with a sequence $\zeta=(\zeta_n)$ satisfying $\zeta\in l^2$, and suppose the phases of the ζ_k are uniform and independent random variables....

conjecture | open | MathDB 页面 | 原文线索

V2-066819 | MATHDB-348127

The converse periodicity conjecture for Beauville's list

摘录 / EXCERPT: Let $ obreak\lambda\neq 0,1$ and let $(E_{\mathrm{unif}},\theta_{\mathrm{unif}})$ denote the uniformizing Higgs bundle on $(\mathbb{P}^1,0+1+\lambda+\infty)$. The converse periodic...

conjecture | open | MathDB 页面 | 原文线索

V2-066820 | MATHDB-348129

F-property conjecture for generalized Hermite polynomials

摘录 / EXCERPT: F-property conjecture. Every partition ν has the F-property; equivalently, no nonzero root of $P^{[\nu]}$ has multiplicity greater than one.

conjecture | open | MathDB 页面 | 原文线索

V2-066821 | MATHDB-348130

Spectral formula conjecture for the matrix associated with a partition

摘录 / EXCERPT: Spectral formula conjecture. The eigenvalues of $J^{[\nu]}$ are

conjecture | open | MathDB 页面 | 原文线索

V2-066822 | MATHDB-348131

The finite-difference characterization of summability for completely convex functions

摘录 / EXCERPT: Let $\mathbb{R}_+=(0,\infty)$, let $\mathcal{K}^{\infty}$ denote the class of functions that are completely convex, and let $\mathcal{D}^{\infty}_{\mathbb{N}}$ denote the class of...

conjecture | open | MathDB 页面 | 原文线索

V2-066823 | MATHDB-348132

The Hyperplane Conjecture for periods of anticanonical hypersurfaces

摘录 / EXCERPT: Let $X_{\underline{t}}$ be the family of anticanonical hypersurfaces associated with the polytope Δ, let ∇ be its Gauss-Manin connection, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066824 | MATHDB-348133

The Beilinson-Hodge conjecture for geometric higher normal functions

摘录 / EXCERPT: Let $\mathcal{X}_{\mathcal{U}}$ be the total space of the family over $\mathcal{U}$, let $\mathcal{H}^{2p-r}_v(p)$ be the varying part of the relevant polarized variation of Hodge...

conjecture | open | MathDB 页面 | 原文线索

V2-066825 | MATHDB-348146

Kyureghyan-Li-Pott conjecture on intersection distributions of ternary monomials

摘录 / EXCERPT: Let $\mathbb{F}_{q}=\mathbb{F}_{3^m}$, with m odd, and let $f(x)=x^d$ be a monomial over $\mathbb{F}_{q}$. Consider the two families of exponents ... and ... For a polynomial f ove...

conjecture | open | MathDB 页面 | 原文线索

V2-066826 | MATHDB-348147

The common monodromy-invariant cycle conjecture for glued K3 fibrations

摘录 / EXCERPT: Common cycle conjecture. The class $e'\cap e”$ is primitive in both $H_{1}(e',\mathbb{Z})$ and $H_{1}(e”,\mathbb{Z})$, and is monodromy invariant with respect to both π' and...

conjecture | open | MathDB 页面 | 原文线索

V2-066827 | MATHDB-348160

Conjecture on disconnected decomposition graphs and the Ritt ideal

摘录 / EXCERPT: Disconnected-graph conjecture. If $\Gamma_B\backslash\{B\}$ is disconnected, then B belongs to the Ritt ideal.

conjecture | open | MathDB 页面 | 原文线索

V2-066828 | MATHDB-348172

Symmetric equidistribution conjecture for inversion-sequence statistics

摘录 / EXCERPT: Let $\mathcal{I}_n$ be the set of inversion sequences of length n, let $\mathfrak{S}_n$ be the symmetric group, and let $b:\mathfrak{S}_n\to\mathcal{I}_n$ be the bijection due to...

conjecture | open | MathDB 页面 | 原文线索

V2-066829 | MATHDB-348173

Conjecture on dimension of Bernoulli convolutions from Pisot conjugates

摘录 / EXCERPT: Let ν_{β_1} denote the Bernoulli convolution associated with β_1. Suppose that $\beta_1\in(1,2)$, $\beta_1\neq(1+\sqrt{5})/2$, and that β_1 has a real conjugate...

conjecture | open | MathDB 页面 | 原文线索

V2-066830 | MATHDB-348182

The chord-word formula for singular height of planar knotoids

摘录 / EXCERPT: Let D be a linear chord diagram with one chord for a planar knotoid, and write its corresponding word as $c=a^{n_1}b^{m_1}\dots a^{n_k}b^{m_k}$, where $k<\infty$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066831 | MATHDB-348188

Equidistribution conjecture for non-BPS attractors

摘录 / EXCERPT: Non-BPS attractor equidistribution conjecture. The non-BPS attractors equidistribute according to ρ; more precisely,

conjecture | open | MathDB 页面 | 原文线索

V2-066832 | MATHDB-348191

Coding or splitting beats replication under additive scaling

摘录 / EXCERPT: Let n be the number of workers, let $Y_{k:n}$ denote the job completion time when the diversity/parallelism parameter is k, and consider a general CU service time under additiv...

conjecture | open | MathDB 页面 | 原文线索

V2-066833 | MATHDB-348192

Replication is optimal only without a deterministic service-time component

摘录 / EXCERPT: Let $X=\delta+Z$ be a CU service time, where $\delta\ge 0$ is constant and the support of the random variable Z includes 0. Under server-dependent scaling, consider the expecte...

conjecture | open | MathDB 页面 | 原文线索

V2-066834 | MATHDB-348193

Gonçalves-Ramos equality conjecture for the linear-programming bound

摘录 / EXCERPT: Let $\boldsymbol v=(v_1,\ldots,v_{n-1})$ be a vector of increasing positive integers, and let $\operatorname{gap}(\boldsymbol v)$ and $\lambda_+(\boldsymbol v)$ be the quantities d...

conjecture | open | MathDB 页面 | 原文线索

V2-066835 | MATHDB-348199

Multiplicity and archimedean packet conjecture for G_2 lifts from GL_2

摘录 / EXCERPT: Multiplicity and packet conjecture. For every $v\in S$, there is a representation Π_v^- of $\mathrm{G}_2(\mathbb{Q}_v)$, different from $\mathcal{L}_\alpha(\pi_v,1/10)$, such t...

conjecture | open | MathDB 页面 | 原文线索

V2-066836 | MATHDB-348200

The minimum-test conjecture for the MSRD non-existence criterion

摘录 / EXCERPT: Let $(n_1,\ldots,n_t)\in\mathbb{N}_0^t$ with $n_1\geq\cdots\geq n_t$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-066837 | MATHDB-348202

The Lieb-Thirring constant phase-alternative conjecture

摘录 / EXCERPT: Let $d\geq1$ and let γ satisfy the constraint defining the admissible Lieb-Thirring regime. Let $L_{\gamma,d}$ be the optimal Lieb-Thirring constant, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066838 | MATHDB-348210

Brumer-Stark conjecture for ray class groups

摘录 / EXCERPT: Let L/K be an abelian CM extension of number fields, with K totally real, and let $G=\operatorname{Gal}(L/K)$. For finite sets of places S,T of K, assume S contains all r...

conjecture | open | MathDB 页面 | 原文线索

V2-066839 | MATHDB-348214

Sign-change symmetry conjecture for the polynomials T_r and f_r

摘录 / EXCERPT: Let $T_r({\bf x}^m)$ and $f_r({\bf x}^m)$ be the symmetric polynomials defined in the source's formulas (j13) and (j17), and let $E_1,\ldots,E_r$ denote their corresponding element...

conjecture | open | MathDB 页面 | 原文线索

V2-066840 | MATHDB-348215

Binomial recurrence conjecture for the symmetric polynomials T_r

摘录 / EXCERPT: Let $T_r({\bf x}^m)$ be the symmetric polynomials defined in the source's formula (j13), and let $T_1({\bf x}^m)$ be the first polynomial in this family. Binomial recurrence conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-066841 | MATHDB-348216

The soficity conjecture for finitely generated groups

摘录 / EXCERPT: A group is sofic when every finitely generated group can be homomorphically embedded into a metric ultraproduct of permutation groups. Let $\mathcal{S}=(S_n,\|\cdot\|_H)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-066842 | MATHDB-348217

Metric uniform simplicity of simple metric ultraproducts

摘录 / EXCERPT: Let $({\mathcal G}_{\mathrm{met}},\|\cdot\|)$ be a metric ultraproduct of metric groups. A metric group is metrically uniformly simple if, for every $r>0$, there is...

conjecture | open | MathDB 页面 | 原文线索

V2-066843 | MATHDB-348221

The KP-ABT partition-function equality

摘录 / EXCERPT: Let $Z_{\rm KP}$ and $Z_{\rm ABT}$ denote the partition functions associated with the Kontsevich-Penner and Alexandrov-Buryak-Tessler constructions. KP-ABT equality conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-066844 | MATHDB-348222

The Bertola-Yang weak open r-spin comparison conjecture

摘录 / EXCERPT: Fix $r\geq 2$, let $\overline g$ denote the doubled genus, and suppose $b,m>0$. Let the superscript $r\mathrm{BY}$ denote the Bertola-Yang intersection numbers and $r\star$ the num...

conjecture | open | MathDB 页面 | 原文线索

V2-066845 | MATHDB-348225

Uniform positivity conjecture for correlation numbers along distinct cyclic Higgs rays

摘录 / EXCERPT: Let X_0 be a hyperbolic structure, and let $(\rho_t)_{t\geq 0}$ and $(\eta_t)_{t\geq 0}$ be rays associated to two different d-th order holomorphic differentials q_1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-066846 | MATHDB-348231

The Bottman-Wehrheim correspondence conjecture for the earring tangle

摘录 / EXCERPT: Let P^* be the pillowcase, let u_s and v_δ be the immersed Lagrangian correspondences associated to the earring tangle and its model, respectively, and let A be a...

conjecture | open | MathDB 页面 | 原文线索

V2-066847 | MATHDB-348234

The naturality conjecture for the earring endofunctor

摘录 / EXCERPT: Let P^* be the pillowcase, let $\mathcal{W}^\star(P^*)\simeq\operatorname{Tw}\mathcal{B}$ be the indicated subcategory, and let $\mathcal{N}$ denote the curve-to-twisted-complex...

conjecture | open | MathDB 页面 | 原文线索

V2-066848 | MATHDB-348235

The Bottman-Wehrheim immersed composition conjecture

摘录 / EXCERPT: Bottman-Wehrheim conjecture. There is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-066849 | MATHDB-348252

Focusing modified fractional Korteweg-de Vries dynamics conjecture

摘录 / EXCERPT: Let $u_0\in L^2(\mathbb{R})$ be smooth initial data with a single hump for the focusing modified fractional Korteweg-de Vries equation, with dispersion parameter α. Let Q...

conjecture | open | MathDB 页面 | 原文线索

V2-066850 | MATHDB-348253

Defocusing modified fractional Korteweg-de Vries dispersion and shock conjecture

摘录 / EXCERPT: Let $u_0\in L^2(\mathbb{R})$ be smooth initial data with a single hump for the defocusing modified fractional Korteweg-de Vries equation, with dispersion parameter α. Defocu...

conjecture | open | MathDB 页面 | 原文线索

V2-066851 | MATHDB-348254

Fractional Benjamin-Bona-Mahony global behavior and cusp conjecture

摘录 / EXCERPT: Let $u_0\in L^2(\mathbb{R})$ be smooth initial data with a single hump for the fractional Benjamin-Bona-Mahony equation, with parameter $0<\alpha$. Fractional Benjamin-Bona-Mahony...

conjecture | open | MathDB 页面 | 原文线索

V2-066852 | MATHDB-348262

Hong-Zhang's log-concavity conjecture for power coefficients of the divisor-sum series

摘录 / EXCERPT: Let ... and, for $m\in\mathbb{N}$, let $b_{m,n}$ denote the coefficient of q^n in $f(q)^m$, with $b_{m,n}=0$ otherwise. Hong-Zhang's conjecture. There exists a constant $C>1$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-066853 | MATHDB-348273

Dimension conjecture for feasible regions of pattern-avoiding permutations

摘录 / EXCERPT: Fix $k\in\mathbb{Z}_{\geq 1}$. For a pattern $\tau\in\mathcal{S}$, let $P^{\operatorname{Av}(\tau)}_k$ denote the feasible region associated with permutations avoiding τ, and...

conjecture | open | MathDB 页面 | 原文线索

V2-066854 | MATHDB-348282

Eventual uniqueness of good components on smooth Fano threefolds

摘录 / EXCERPT: Let X be a smooth Fano threefold, and let $\mathrm{Nef}_1(X)_{\mathbb Z}$ denote the integral points of the nef cone of curve classes. A good component is a component of the spac...

conjecture | open | MathDB 页面 | 原文线索

V2-066855 | MATHDB-348287

Asymptotics conjecture for quarter-plane spiral walks

摘录 / EXCERPT: Let $P_m^{{\mathrel{\tikz{\draw (0ex,1ex) -- (0ex,0ex) -- (1ex,0ex) (-0.2ex,0ex)}}}}$ be the number of quarter-plane spiral walks of length m. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066856 | MATHDB-348289

Enumeration conjecture for symmetry-group orders of two-step rules

摘录 / EXCERPT: Let $\mathcal Q$ be the class of two-step rules, and let $\mathcal N_\ell$ denote the set of rules in $\mathcal Q$ whose symmetry group has order ℓ, with $\ell=\infty$ allowed...

conjecture | open | MathDB 页面 | 原文线索

V2-066857 | MATHDB-348291

The dual canonical basis conjecture for the base affine space

摘录 / EXCERPT: The dual canonical basis conjecture. For every $T\in {\rm SSYT}(k-1,[k])$,

conjecture | open | MathDB 页面 | 原文线索

V2-066858 | MATHDB-348292

The conjecture that 3-colorability implies both 4-colorability and Klein-four colorability

摘录 / EXCERPT: The $\mathbb Z_3$-coloring conjecture. If G is $\mathbb Z_3$-colorable, then G is both $\mathbb Z_4$-colorable and $\mathbb Z_2\times\mathbb Z_2$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-066859 | MATHDB-348293

The noncrossing matching ensemble realization conjecture

摘录 / EXCERPT: Noncrossing matching ensemble realization conjecture. Every noncrossing matching ensemble appears as ${\mathcal E}(P)$ for some generic simple polypositroid P.

conjecture | open | MathDB 页面 | 原文线索

V2-066860 | MATHDB-348294

The surjectivity conjecture for generalized W-permutohedra

摘录 / EXCERPT: Surjectivity conjecture for generalized W-permutohedra. The maps in Theorem are surjective.

conjecture | open | MathDB 页面 | 原文线索

V2-066861 | MATHDB-348295

The alternating and c-noncrossing face conjecture for prepolypositroids

摘录 / EXCERPT: Alternating and c-noncrossing face conjecture. If $(\widetilde\beta,\widetilde\gamma)\in\widetilde R$ span a two-dimensional face of the normal fan of P, then (β,γ) mu...

conjecture | open | MathDB 页面 | 原文线索

V2-066862 | MATHDB-348302

Scaling reparametrization conjecture for identifiable path/cycle models

摘录 / EXCERPT: Let $\mathcal{M}=(G,In,Out,V)$ be a linear compartmental model. Suppose either that G is strongly input-output connected and $|Out|=1$, or that G is strongly connected and...

conjecture | open | MathDB 页面 | 原文线索

V2-066863 | MATHDB-348304

Second shuffle conjecture for symmetric elliptic Hodge correlators

摘录 / EXCERPT: Let E be a CM elliptic curve, let s be a base point, and let D denote the depth filtration on the space of symmetric elliptic Hodge correlators. The second shuffle relations...

conjecture | open | MathDB 页面 | 原文线索

V2-066864 | MATHDB-348308

The positivity conjecture for cyclic-intersection functions

摘录 / EXCERPT: Let $J=\{j_1,\ldots,j_k\}$ be a k-subset of $[n]$, and let I_{j_α} be the j_αth cyclic k-interval. Let ξ_J be the quantity defined in the source's geometric c...

conjecture | open | MathDB 页面 | 原文线索

V2-066865 | MATHDB-348312

Arithmetic biquadratic fundamental lemma

摘录 / EXCERPT: Let F be the base local field, let K/K_3 be unramified, and let ... be embeddings such that (Φ_1,Φ_2) is a regular semisimple pair and (Φ_0,Φ_3) is regular semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-066866 | MATHDB-348316

Polynomial-degree conjecture for $\mathbb{Z}_2$ Hurwitz-Hodge integrals

摘录 / EXCERPT: Let $\vec{i}$ be a tuple of nonnegative integers, with $|\vec{i}|$ denoting the sum of its entries. Let $D_{(\vec{i}),2g+2}$ and $d_{(\vec{i}),2g+2}$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-066867 | MATHDB-348317

Nonnegativity and log-concavity conjecture for the binomial coefficients of $\mathbb{Z}_2$ Hurwitz-Hodge integrals

摘录 / EXCERPT: Nonnegativity and log-concavity conjecture. The integers $c_{\vec{i},k}$ and $\widetilde{c}_{\vec{i},k}$ are nonnegative, and, for fixed $\vec{i}$, the sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-066868 | MATHDB-348318

Vanishing and log-concavity conjecture for $\mathbb{Z}_2$ Hurwitz-Hodge integrals

摘录 / EXCERPT: Let $g>0$ and let $\vec{\ell}\in\mathbb{Z}_{\geq0}^n$ satisfy $|\vec{\ell}|\leq g-1$, where $|\vec{\ell}|$ is the sum of the entries of $\vec{\ell}$. Let $D_{(\vec{\ell},i),2g+2}$...

conjecture | open | MathDB 页面 | 原文线索

V2-066869 | MATHDB-348319

Odd-Hadwiger conjecture for signed graphs

摘录 / EXCERPT: Let $(G,-)$ be a signed graph whose underlying graph is G, with every edge negative, and let $(K_{k+1},-)$ denote the complete graph on $k+1$ vertices with every edge negative. A...

conjecture | open | MathDB 页面 | 原文线索

V2-066870 | MATHDB-348321

Multi-process maximum-age threshold probing policy for age-of-information minimization

摘录 / EXCERPT: Let $N>1$, let E denote the energy state, let $T_1,\ldots,T_N$ denote the ages of information of the N processes, and define ... Let E_p and E_s be the probing and sampling e...

conjecture | open | MathDB 页面 | 原文线索

V2-066871 | MATHDB-348345

Maximality conjecture for the triangular and face-centred-cubic lattices in the grid cells model

摘录 / EXCERPT: Let B_R be the radius-R ball, let σ_R be the uniform measure on B_R, and set $\alpha=\frac{10}{\pi}$. Define ... Let $\mathcal{F}_{\sigma_R}^{\alpha}$ denote the grid-ce...

conjecture | open | MathDB 页面 | 原文线索

V2-066872 | MATHDB-348349

The generalized double-exclusion Alder conjecture

摘录 / EXCERPT: Generalized double-exclusion Alder conjecture. Let a,d be positive integers with $1\leq a\leq d+2$. Then for all $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-066873 | MATHDB-348360

Alternative characterizations of asymptotic dimension for fuzzy metric spaces

摘录 / EXCERPT: Let X be a fuzzy metric space, and let n be a nonnegative integer. Conditions,, and in Theorem are the three conditions stated there for X and n. Alternative-characterizati...

conjecture | open | MathDB 页面 | 原文线索

V2-066874 | MATHDB-348375

Infinite-family conjecture for self-similar congruences of the 19-regular partition function

摘录 / EXCERPT: Infinite-family conjecture for b_{19}. For a positive proportion of primes p, one has

conjecture | open | MathDB 页面 | 原文线索

V2-066875 | MATHDB-348376

Odd density conjecture for residue classes of 21-regular partitions

摘录 / EXCERPT: Odd density conjecture for b_{21}. The series

conjecture | open | MathDB 页面 | 原文线索

V2-066876 | MATHDB-348377

The Flynn-Poonen conjecture on rational cycles of quadratic polynomials

摘录 / EXCERPT: Flynn-Poonen conjecture. There is no quadratic polynomial $f(x)\in{\mathbb Q}[x]$ with a rational point of exact period N.

conjecture | open | MathDB 页面 | 原文线索

V2-066877 | MATHDB-348383

Sharpness of the logarithmic inverse-temperature threshold for uniform separation

摘录 / EXCERPT: Let (β_n) be a sequence of inverse temperatures, and consider the Coulomb gas configurations at these temperatures. Sharpness conjecture. If almost sure uniform separation ho...

conjecture | open | MathDB 页面 | 原文线索

V2-066878 | MATHDB-348402

Uniform conjugacy invariance of shifts of finite type

摘录 / EXCERPT: Uniform conjugacy invariance conjecture. Any shift which is uniformly conjugate to an SFT is itself an SFT.

conjecture | open | MathDB 页面 | 原文线索

V2-066879 | MATHDB-348403

Conjugacy invariance of finitely defined shifts

摘录 / EXCERPT: Conjugacy invariance conjecture for FDSs. A shift space which is conjugated to an FDS is itself an FDS.

conjecture | open | MathDB 页面 | 原文线索

V2-066880 | MATHDB-348404

Conjecture on the stationary support of a quadratic Markov chain over finite fields

摘录 / EXCERPT: Stationary-support conjecture. Conjecturally, the chain has a unique stationary distribution π, and

conjecture | open | MathDB 页面 | 原文线索

V2-066881 | MATHDB-348406

Exact maximization conjecture for minimal codewords in binary linear codes of codimension three

摘录 / EXCERPT: Let C be a binary linear $[k+3,k]_2$ code, and let $M_2(k+3,k)$ denote the maximum possible number of minimal codewords among such codes. For $k\geq 4$, write ... where the entries...

conjecture | open | MathDB 页面 | 原文线索

V2-066882 | MATHDB-348410

The normal-spanning-tree chromatic bound conjecture

摘录 / EXCERPT: A normal spanning tree of a graph G is a spanning tree T with the normality property used in the source, and an edge of G is secant with respect to T in the sense defined t...

conjecture | open | MathDB 页面 | 原文线索

V2-066883 | MATHDB-348414

Universality conjecture for the microscopic roots of random polynomials

摘录 / EXCERPT: Let $f_n(z)=\sum_{j=0}^n\varepsilon_j z^j$, where $\{\varepsilon_j\}_j$ are iid real random variables satisfying $\mathbb{E}\,\varepsilon_1=0$ and $\mathbb{E}\,\varepsilon_1^2=1$....

conjecture | open | MathDB 页面 | 原文线索

V2-066884 | MATHDB-348422

Classification of $(2,0;2)$-bridge trisections

摘录 / EXCERPT: Classification conjecture. Every $(2,0;2)$-bridge trisection is diffeomorphic to one of those described in Example and Figure.

conjecture | open | MathDB 页面 | 原文线索

V2-066885 | MATHDB-348424

Conjecture on centroidal critical points under curve-shortening flow

摘录 / EXCERPT: Let Γ be a generic plane curve evolving under the curve-shortening flow. Denote by $n_U(t)$ the number of critical points of its radial function relative to the fixed point...

conjecture | open | MathDB 页面 | 原文线索

V2-066886 | MATHDB-348429

Sankar's three-dimensional Catalan formula for reduced valid hook configurations

摘录 / EXCERPT: Let $\operatorname{RedVHC}_k(\operatorname{Av}_{3k-i}(312))$ denote the set of 312-avoiding reduced valid hook configurations with k hooks on permutations with $2k+i$ points. D...

conjecture | open | MathDB 页面 | 原文线索

V2-066887 | MATHDB-348432

Sankar's real-rootedness conjecture for valid hook configuration polynomials

摘录 / EXCERPT: For each positive integer k, let $h_k(x)$ and $f_k(x)$ be the polynomials associated with 312-avoiding reduced valid hook configurations, with $h_k(x)=f_k(x-1)$. Sankar's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-066888 | MATHDB-348439

Free-energy asymptotics conjecture for directed polymers on fractal graphs

摘录 / EXCERPT: Let $F_q(\beta)$ and $F_a(\beta)$ denote the quenched and annealed free energies of the directed polymer, respectively, let d_s be the spectral dimension of the underlying fracta...

conjecture | open | MathDB 页面 | 原文线索

V2-066889 | MATHDB-348442

Universality conjecture for stacks of fiber functors

摘录 / EXCERPT: Let $\mathcal{X}$ be a quasi-compact fibered category over R, and let $\mathcal{C}$ be as in the resolution-property hypothesis of the paper. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-066890 | MATHDB-348448

Elbert-Laforgia conjecture on McMahon bounds for Bessel zeros

摘录 / EXCERPT: Let $j_{\nu,\kappa}$ denote the κth positive zero of the Bessel function J_ν, and let McMahon's expansion be the asymptotic expansion for these zeros. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066891 | MATHDB-348449

Fabijonas-Olver conjecture on error terms in Airy-zero expansions

摘录 / EXCERPT: Let a_k and b_k be the quantities whose asymptotic expansions are under consideration, and let the Nth error term mean the error obtained by stopping the corresponding expans...

conjecture | open | MathDB 页面 | 原文线索

V2-066892 | MATHDB-348456

Higher-order asymptotic conjecture for parabolic double cosets

摘录 / EXCERPT: Higher-order asymptotic conjecture. There exists a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-066893 | MATHDB-348458

A q-analogue of Sun's supercongruence relating two truncated sums

摘录 / EXCERPT: Let n be a positive odd integer, let q be an indeterminate, let $\Phi_n(q)$ be the nth cyclotomic polynomial, and use $(x;q)_k=\prod_{j=0}^{k-1}(1-xq^j)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066894 | MATHDB-348459

Deformability in towers for cyclic extensions

摘录 / EXCERPT: Deformability in towers conjecture. If G is cyclic, then the natural morphism

open_problem | open | MathDB 页面 | 原文线索

V2-066895 | MATHDB-348460

Shallow minimal reduction type conjecture

摘录 / EXCERPT: Shallow minimal reduction type conjecture. For every $[w]\in\underline W$,

conjecture | open | MathDB 页面 | 原文线索

V2-066896 | MATHDB-348461

Transitivity conjecture for centralizers on reduction-type fibers

摘录 / EXCERPT: Centralizer transitivity conjecture. The group $(LG)_{\mathfrak g}$ acts transitively on $\operatorname{Gr}_{\mathfrak g,\mathcal O}$. The paper notes that this fiber is nonempty a...

conjecture | open | MathDB 页面 | 原文线索

V2-066897 | MATHDB-348463

Arjevani et al.'s lower-bound conjecture for polynomial iterations

摘录 / EXCERPT: Arjevani et al.'s lower-bound conjecture. For every such q, r, and μ, some $\nu\in[\mu,\ell]$ satisfies the stated spectral-radius lower bound. This conjecture concerns tig...

conjecture | open | MathDB 页面 | 原文线索

V2-066898 | MATHDB-348466

Alspach-Mason-Pullman consistency conjecture for even-order tournaments

摘录 / EXCERPT: Let T be a tournament, let $d_T^+(v)$ and $d_T^-(v)$ denote the outdegree and indegree of v, and define its excess by ... A path decomposition of T is a decomposition of its ed...

conjecture | open | MathDB 页面 | 原文线索

V2-066899 | MATHDB-348467

Alspach-Pullman-Mason path-number range conjecture for odd-order tournaments

摘录 / EXCERPT: Let T be a tournament on n vertices, and let $\operatorname{pn}(T)$ be its path number. Alspach-Pullman-Mason's conjecture. If n is odd, then $\operatorname{pn}(T)=k$ for som...

conjecture | open | MathDB 页面 | 原文线索

V2-066900 | MATHDB-348470

The unlabelled Diestel-Leader graph seeded tiling conjecture

摘录 / EXCERPT: For integers $p,q\geq 2$, let ${\mathsf{DL}(p,q)}$ be the Diestel-Leader graph, and let the unlabelled graph mean this graph considered without its standard edge labelling. A seede...

conjecture | open | MathDB 页面 | 原文线索

V2-066901 | MATHDB-348471

Bryson-Ono-Pitman-Rhoades mod 4 conjectures for strongly unimodal sequences

摘录 / EXCERPT: Bryson-Ono-Pitman-Rhoades conjecture. For all n,

conjecture | open | MathDB 页面 | 原文线索

V2-066902 | MATHDB-348475

Limit law for repeated derivatives of polynomials with i.i.d. roots

摘录 / EXCERPT: Let $P_n(z)=\prod_{k=1}^n(z-Z_k)$, where $Z_1,\ldots,Z_n$ are i.i.d. random variables with rotationally invariant law μ_0 on $\mathbb{C}$ having no atom at 0. Let μ_t be...

conjecture | open | MathDB 页面 | 原文线索

V2-066903 | MATHDB-348479

Minimal-class conjecture for shrub-depth and rank-depth

摘录 / EXCERPT: Minimal-class conjecture for shrub-depth and rank-depth. Shrub-depth and rank-depth are unbounded in $\mathcal X$ if and only if $\mathcal X$ contains a minimal hereditary class of...

conjecture | open | MathDB 页面 | 原文线索

V2-066904 | MATHDB-348490

Critical-point unitarizability and Arthur-packet conjecture

摘录 / EXCERPT: Let π be an irreducible subquotient at a critical point for a classical group, with no restriction on the rank. Critical-point Arthur-packet conjecture. The following are equiv...

conjecture | open | MathDB 页面 | 原文线索

V2-066905 | MATHDB-348491

Isolated representations are of critical type

摘录 / EXCERPT: Let π be an isolated representation in the unitary dual of a classical group. A representation is of critical type if it is an irreducible subquotient of...

conjecture | open | MathDB 页面 | 原文线索

V2-066906 | MATHDB-348492

Critical-type unitarizability and Arthur-packet conjecture

摘录 / EXCERPT: Let π be an irreducible representation of a split classical group and suppose that π is of critical type, meaning that it is an irreducible subquotient of a representation...

conjecture | open | MathDB 页面 | 原文线索

V2-066907 | MATHDB-348493

Isolated representations belong to Arthur packets

摘录 / EXCERPT: Let π be an isolated representation in the unitary dual of a classical group. Isolated-representation Arthur-packet conjecture. The representation π belongs to an Arthur pa...

conjecture | open | MathDB 页面 | 原文线索

V2-066908 | MATHDB-348499

Finite generation of the graded ring of a normalized-volume minimizer

摘录 / EXCERPT: Finite-generation conjecture. The associated graded ring $\operatorname{gr}_v R$ is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-066909 | MATHDB-348507

Fukaya-category matrix-factorization conjecture for a one-critical-value Landau-Ginzburg model

摘录 / EXCERPT: Let $(X,f:X\to\mathbb{C})$ be a Landau-Ginzburg Weinstein sector such that f has no critical values outside 0, and let F be a general fiber of f. Let $\mathcal{W}(X,f)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-066910 | MATHDB-348509

Affine relation for critical points with an admissible column subset

摘录 / EXCERPT: Assume that $\mathcal{L}_r^S$ is irreducible. Let $I\subset[n]$ satisfy $|I|=m$, and let $X_{[m],I}$ and $U_{[m],I}$ denote the corresponding $m\times m$ submatrices; write...

conjecture | open | MathDB 页面 | 原文线索

V2-066911 | MATHDB-348517

The geometric-moduli conjecture for integral PCF rational maps

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k including all the archimedean places, and let $d\geq 2$. Let $\overline{\mathcal{M}}_d^1[\mathcal{P}]$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-066912 | MATHDB-348518

The integrality conjecture for pairs of PCF unicritical parameters

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k including all the archimedean places, and let $d\geq 2$ be an integer. Write $f_{d,c}(z):=z^d+c$. Let D be a n...

conjecture | open | MathDB 页面 | 原文线索

V2-066913 | MATHDB-348527

Nonnegative polynomial and symmetry conjecture for Grassmannian K-theoretic structure constants

摘录 / EXCERPT: Let $\mathcal{I}$ be the indexing set for Schubert cells of $\operatorname{Gr}(d,n)$, and let $I=(i_1,\ldots,i_d)$, $I'=(i'_1,\ldots,i'_d)$, $J=(j_1,\ldots,j_d)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066914 | MATHDB-348539

Strict unimodality conjecture for Reiner-Stanton polynomials

摘录 / EXCERPT: Let ... where $k\geq 2$ and n is even. A polynomial is strictly unimodal, except for the two terms at both ends when its coefficients strictly increase to the middle, apart from th...

conjecture | open | MathDB 页面 | 原文线索

V2-066915 | MATHDB-348549

Bruhn-Rautenbach's Fano-plane determinant conjecture

摘录 / EXCERPT: Let $A\in\{0,1\}^{n\times n}$ have at most $3n$ non-zero entries. Bruhn-Rautenbach's conjecture. The determinant of A is at most ... The bound is motivated by the incidence matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-066916 | MATHDB-348552

Strict containment conjecture for weakly locally finite and Stewart's division rings

摘录 / EXCERPT: Strict containment conjecture. The class of weakly locally finite division rings strictly contains the class of Stewart's division rings.

conjecture | open | MathDB 页面 | 原文线索

V2-066917 | MATHDB-348559

Conjecture on valid orientations with four exceptional boundary vertices

摘录 / EXCERPT: Conjecture on valid orientations with four exceptional boundary vertices. Then G has a valid orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-066918 | MATHDB-348560

Conjecture on valid orientations for two specified faces

摘录 / EXCERPT: Conjecture on valid orientations for two specified faces. Then G has a valid orientation.

conjecture | open | MathDB 页面 | 原文线索

V2-066919 | MATHDB-348561

The point pair function quasi-metric constant conjecture

摘录 / EXCERPT: Let $G e\rrbracket\mathbb{R}^n$ be a domain, and let p_G denote the point pair function. A function d is a quasi-metric if there is a constant $c\geq 1$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-066920 | MATHDB-348575

q-gamma positivity conjecture for alternating Eulerian polynomials

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, and let $\operatorname{altdes}(\pi)$ and $\operatorname{altmaj}(\pi)$ denote the alternating descent number and alternating major index...

conjecture | open | MathDB 页面 | 原文线索

V2-066921 | MATHDB-348576

Two-sided alternating gamma-expansion conjecture

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the symmetric group, and define the two-sided alternating Eulerian polynomial by ... Two-sided alternating gamma-expansion conjecture. For every $n\geq 1$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-066922 | MATHDB-348580

The decidability equivalence for Schreier graphs of contracting self-similar groups

摘录 / EXCERPT: Decidability equivalence. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-066923 | MATHDB-348583

Near-critical massive Gaussian free field conjecture for the high-dimensional Ising model

摘录 / EXCERPT: Near-critical scaling-limit conjecture. As $a\downarrow 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066924 | MATHDB-348593

Small-limit conjecture for presentable n-categories

摘录 / EXCERPT: Small-limit conjecture. The category $n\operatorname{Pr}^L$ has all small limits for all $n\geq 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-066925 | MATHDB-348598

Direct formula conjecture for the multitype derivative-martingale limit

摘录 / EXCERPT: Let $\mathcal{N}_t^2$ be the set of type-2 particles at time t, with positions $X_u(t)$, and let $\overline Z_\infty$ denote the previously defined limit of the relevant multit...

conjecture | open | MathDB 页面 | 原文线索

V2-066926 | MATHDB-348607

Polystability conjecture for periodic Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective complex curve, and let a periodic Higgs bundle on C mean a Higgs bundle satisfying the source's periodicity condition. Polystability conjecture. An...

conjecture | open | MathDB 页面 | 原文线索

V2-066927 | MATHDB-348611

Apéry-limit conjecture for sums of powers of binomial coefficients

摘录 / EXCERPT: Apéry-limit conjecture. For $d\geq 3$, there is a unique such solution $B^{(d)}(n)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-066928 | MATHDB-348612

Higher Apéry-limit conjecture for sums of powers of binomial coefficients

摘录 / EXCERPT: Higher Apéry-limit conjecture. For $d\geq 5$, there is a unique such solution $C^{(d)}(n)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-066929 | MATHDB-348624

Corti-Prokhorov-Süß conjecture for index-two del Pezzo hypersurfaces

摘录 / EXCERPT: Let S_d be a quasi-smooth and well-formed hypersurface of degree d in $\mathbb{P}(a_0,a_1,a_2,a_3)$, where $a_0\leq a_1\leq a_2\leq a_3$ are positive integers. Define the index...

conjecture | open | MathDB 页面 | 原文线索

V2-066930 | MATHDB-348625

K-stability conjecture for the remaining index-two del Pezzo hypersurfaces

摘录 / EXCERPT: K-stability conjecture. Then S_d is K-stable, and hence admits an orbifold Kähler-Einstein metric.

conjecture | open | MathDB 页面 | 原文线索

V2-066931 | MATHDB-348626

Delta-invariant conjecture for the exceptional index-two family

摘录 / EXCERPT: Delta-invariant conjecture. For every n, one has

conjecture | open | MathDB 页面 | 原文线索

V2-066932 | MATHDB-348631

Slackened Reed-Solomon list-decoding conjecture

摘录 / EXCERPT: Let L be a positive integer, let $C\geq1$ be a positive real, and let $R=\frac{k}{n}$ be the rate of an $[n,k]$ Reed-Solomon code. A code is list-decodable from radius ρ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-066933 | MATHDB-348636

Constancy conjecture for the integrated density of states at inverse-beta energies

摘录 / EXCERPT: Let p be the Bernoulli parameter, let $\beta^{-1}(n)$ denote the decreasing sequence appearing in the model, and let $I_{p,\zeta}(x)$ be the integrated density of states at disor...

conjecture | open | MathDB 页面 | 原文线索

V2-066934 | MATHDB-348638

Coefficient bound conjecture for the class of analytic functions associated with sine hyperbolic functions

摘录 / EXCERPT: Let $f\in\mathcal{G}_{sh}$ be an analytic function of the form given by equation (1), with Taylor coefficients a_n. Coefficient bound conjecture. One should have ... The preceding...

conjecture | open | MathDB 页面 | 原文线索

V2-066935 | MATHDB-348642

Seidl's conjecture for deterministic solutions of the three-marginal radial Coulomb problem

摘录 / EXCERPT: Seidl's conjecture. The map T is optimal for the radial problem.

conjecture | open | MathDB 页面 | 原文线索

V2-066936 | MATHDB-348647

Singular-component conjecture for A-L measures on Dirichlet-regular Widom sets

摘录 / EXCERPT: Conjecture on singular A-L measures. There exists a Dirichlet-regular Widom set ${\mathsf{E}}$ with DCT such that A-L holds and μ^α has a nontrivial singular component wit...

conjecture | open | MathDB 页面 | 原文线索

V2-066937 | MATHDB-348648

Large-genus asymptotics conjecture for Masur-Veech volumes

摘录 / EXCERPT: Let $\mathcal{Q}_{g,0}$ denote the moduli space of holomorphic quadratic differentials in genus g without marked points. Large-genus asymptotics conjecture. As $g\to+\infty$, its...

conjecture | open | MathDB 页面 | 原文线索

V2-066938 | MATHDB-348665

Existence of plt blow-ups from positive local volume

摘录 / EXCERPT: Let $n\geq 2$ be a positive integer and let η,ϵ be positive real numbers. Let $x\in (X,\Delta)$ be an n-dimensional klt singularity, with ... where each...

conjecture | open | MathDB 页面 | 原文线索

V2-066939 | MATHDB-348666

Boundedness up to special degeneration from positive local volume

摘录 / EXCERPT: Let n be a positive integer, and let ϵ,η be positive numbers. A special degeneration of a klt singularity is the central fiber of a special test configuration as defi...

conjecture | open | MathDB 页面 | 原文线索

V2-066940 | MATHDB-348672

Minimal-interface characterization conjecture for the relaxed functional

摘录 / EXCERPT: Let $\rho(z,\phi)$ be the relaxed functional for an admissible pair (z,phi), and define ... If $\lambda^n(C\Phi)>0$, then z is not uniquely identified and may attain every value...

conjecture | open | MathDB 页面 | 原文线索

V2-066941 | MATHDB-348673

Modularity conjecture for elliptic curves over function fields

摘录 / EXCERPT: Let X be a smooth, projective, geometrically irreducible curve with function field K, and let E be an elliptic curve over K with conductor N and irreducible associated Ga...

conjecture | open | MathDB 页面 | 原文线索

V2-066942 | MATHDB-348686

The GL corner energy conjecture

摘录 / EXCERPT: Let $E_{\mathrm{corner},\beta}$ denote the effective Ginzburg-Landau energy associated with a boundary corner of opening angle β, and let E_{corr} be the correctio...

conjecture | open | MathDB 页面 | 原文线索

V2-066943 | MATHDB-348692

Derivative-generation conjecture for invariant subspaces of the weighted Bergman space

摘录 / EXCERPT: Let A^2_1 be the weighted Bergman space of holomorphic functions on the unit disk with norm ... For $H\in A^2_1$, write ... the smallest closed subspace of A^2_1 containing H and...

conjecture | open | MathDB 页面 | 原文线索

V2-066944 | MATHDB-348700

Nakamura's epsilon-isomorphism conjecture for relative Robba rings

摘录 / EXCERPT: Nakamura's conjecture. For every such M, there exists an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-066945 | MATHDB-348704

The Strong Diamond Conjecture for full loci of covers

摘录 / EXCERPT: The Strong Diamond Conjecture. Under these assumptions, $\mathcal{M}$ is a full locus of covers. This would extend the full-locus conclusion beyond the cases established by the pre...

conjecture | open | MathDB 页面 | 原文线索

V2-066946 | MATHDB-348706

The stationary log/local correspondence for maximal log Calabi-Yau pairs

摘录 / EXCERPT: Let $(Y,D=D_1+\dots+D_l)$ be a log-smooth log Calabi-Yau pair of maximal boundary. For $\mathsf{m}=1,\dots,l+1$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-066947 | MATHDB-348707

The log-open correspondence for maximal-boundary log Calabi-Yau surfaces

摘录 / EXCERPT: Let $Y(D)$ be a log Calabi-Yau surface with maximal boundary, let d be a D-convex curve class, and let $Y^{\rm op}(D)=(Y^{(l)}\setminus D^{(l)},L)$ be the associated open geome...

conjecture | open | MathDB 页面 | 原文线索

V2-066948 | MATHDB-348708

The all-genus log-open correspondence for maximal-boundary log Calabi-Yau surfaces

摘录 / EXCERPT: Let $Y(D)$ be a log Calabi-Yau surface with maximal boundary, let d be a D-convex curve class, and use the all-genus generating functions...

conjecture | open | MathDB 页面 | 原文线索

V2-066949 | MATHDB-348710

Conjectural tail estimates for discriminant-divisible elements

摘录 / EXCERPT: Conjectural tail estimates. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-066950 | MATHDB-348720

The cyclicity conjecture for optimal covering maps

摘录 / EXCERPT: Let $\mathcal{M}$ be a geminal orbit closure, and let the optimal covering map associated with $\mathcal{M}$ be the covering map of minimal degree arising in the classification of...

conjecture | open | MathDB 页面 | 原文线索

V2-066951 | MATHDB-348722

The deterministic context-free word problem conjecture for special monoids

摘录 / EXCERPT: Let $M=\operatorname{Mon}\langle A\mid w_i=1\ (1\leq i\leq p)\rangle$ be a finitely presented special monoid. A monoid has deterministic context-free word problem when its word pro...

conjecture | open | MathDB 页面 | 原文线索

V2-066952 | MATHDB-348723

Haglund-Remmel-Wilson generalized Delta conjecture

摘录 / EXCERPT: Let m,n,k be in the range for which the partially labelled Dyck-path set $\mathsf{PLD}(m,n)^{\ast k}$ is defined. Let Δ_{h_m} and $\Delta'_{e_{n-k-1}}$ be Delta operators,...

conjecture | open | MathDB 页面 | 原文线索

V2-066953 | MATHDB-348726

The square form of the Delta conjecture

摘录 / EXCERPT: Let $n\geq1$, let $\mathsf{PSQ^E}(n)$ be the relevant set of square paths, and let $\delta(P)$ be the dinv reading word of P. Write $\mathsf{Des}(\delta(P))$ for its descent set...

conjecture | open | MathDB 页面 | 原文线索

V2-066954 | MATHDB-348727

Roger-Ramanujan categorification conjecture for modified Euler characteristics

摘录 / EXCERPT: Let N be a positive integer, let $P(N)$ be a partition of N into two parts, and let ... be the modified Euler characteristic, where $f_{P(N)}(i)$ is an integer-valued function de...

conjecture | open | MathDB 页面 | 原文线索

V2-066955 | MATHDB-348740

Li et al.'s neighbor product distinguishing total coloring conjecture

摘录 / EXCERPT: Let G be a graph with at least two vertices. The neighbor product distinguishing total chromatic number $\chi”_{\prod}(G)$ is the smallest number of colors in a neighbor product...

conjecture | open | MathDB 页面 | 原文线索

V2-066956 | MATHDB-348744

K-semistability of smooth Fano threefolds with Picard rank one

摘录 / EXCERPT: K-semistability conjecture. Every smooth Fano threefold with $\rho(X)=1$ is K-semistable.

conjecture | open | MathDB 页面 | 原文线索

V2-066957 | MATHDB-348752

The degree-sum conjecture for feasible disjoint cycles in bipartite graphs

摘录 / EXCERPT: Let $G[X,Y]$ be a balanced bipartite graph of order $2n$, let S be a subset of X with $|S|\geq 2k+1$, and let $\sigma_{1,1}(S)$ denote the degree-sum parameter used in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-066958 | MATHDB-348753

The prescribed partition conjecture for disjoint cycles in bipartite graphs

摘录 / EXCERPT: Let $G[X,Y]$ be a balanced bipartite graph of order $2n$, let S be a subset of X with $|S|\geq 2k+2$, and let S_{C_i} denote the vertices of S on a cycle C_i. Prescribed...

conjecture | open | MathDB 页面 | 原文线索

V2-066959 | MATHDB-348754

The unimodality conjecture for q-rational numerator and denominator polynomials

摘录 / EXCERPT: Let $\left[\frac{r}{s}\right]_{q}=\frac{\mathcal{R}(q)}{\mathcal{S}(q)}$ be a q-rational, where $\mathcal{R}(q)$ and $\mathcal{S}(q)$ are its numerator and denominator polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-066960 | MATHDB-348779

The inscribed square conjecture for conformal angle spaces

摘录 / EXCERPT: Let $(\mathcal X,\mathcal B,\angle)$ be an angle space conformal to a Jordan curve, meaning that it arises from the angle structure induced by a Jordan curve under conformal equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-066961 | MATHDB-348790

The non-negative coefficient conjecture for cylindric partition generating functions

摘录 / EXCERPT: Non-negative coefficient conjecture. For any profile $c=(c_1,c_2,c_3)$, the polynomial $P_{n,c}(q)$ has non-negative coefficients. Moreover, if $\ell\not\equiv 0\pmod{3}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066962 | MATHDB-348792

Phase-transition conjecture for connectedness of the weighted free uniform spanning forest

摘录 / EXCERPT: Let H) be a finite connected graph and let $\mathbb{T}^d$ be the d-regular tree. For $w>0$, define the edge-weight function $\hat w$ on $G=\mathbb{T}^d\square H$ by assigning w...

conjecture | open | MathDB 页面 | 原文线索

V2-066963 | MATHDB-348797

Nonconstancy conjecture for the spectral asymptotic function

摘录 / EXCERPT: Let s be the continuous function appearing in the eigenvalue-counting asymptotics ... for a Sturm-Liouville problem with a non-constant weight whose primitive is arithmetically sel...

conjecture | open | MathDB 页面 | 原文线索

V2-066964 | MATHDB-348807

Cohen-Lenstra-Martinet-Malle conjecture for even-degree class groups

摘录 / EXCERPT: Let K be a nowhere-evenly-ramified number field of even degree n with real signature (r_1,r_2). Enumerate such fields by discriminant. Cohen-Lenstra-Martinet-Malle conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-066965 | MATHDB-348813

Converse prisoner-set inclusion conjecture for quadratic mutations

摘录 / EXCERPT: Converse prisoner-set inclusion conjecture. If the radius R is such that $\mathcal{P}(f)\subseteq\mathcal{P}(f_{c_1})$, then

conjecture | open | MathDB 页面 | 原文线索

V2-066966 | MATHDB-348819

The sign-stability estimate for pA mutation loops

摘录 / EXCERPT: Let ϕ be a pA mutation loop on a marked surface Σ. Write $\mathcal{E}_\phi^a$ for its associated asymptotic entropy and $\lambda_{\pi(\phi)}$ for the spectral radius of...

conjecture | open | MathDB 页面 | 原文线索

V2-066967 | MATHDB-348820

The asymptotic consecutive-pattern distribution conjecture

摘录 / EXCERPT: The asymptotic consecutive-pattern distribution conjecture. For any $k\geq 3$, any $\pi\in S_k$, and any finite $I\subseteq\mathbb{Z}^+$, there are constants...

conjecture | open | MathDB 页面 | 原文线索

V2-066968 | MATHDB-348821

The equidistribution conjecture for separated consecutive-pattern occurrences

摘录 / EXCERPT: Fix $k\geq 3$, let $\pi\in S_k$, and let $a\in\mathbb{Z}^+$. For $n\in\mathbb{Z}^+$ and $I=\{i_1,i_2,\ldots,i_a\}\subseteq[n]$, suppose that ... Here $p_{\pi}(I,n)$ counts permutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-066969 | MATHDB-348826

The Lazy Caterer's sequence conjecture for restricted fractional partitions

摘录 / EXCERPT: Let $j>2$ and let $k,j\in\mathbb{N}$. For a sequence of numerators $\{n_j\}$, define ... A sequence enables solutions when every positive integer $k<\lfloor G\rfloor$ can be represen...

conjecture | open | MathDB 页面 | 原文线索

V2-066970 | MATHDB-348827

The unimodality-or-bimodality conjecture for restricted fractional-partition generating functions

摘录 / EXCERPT: Let $r(x)$ be the generating function whose coefficients $f_{O_j}(k)$ count the relevant partitions into fractions with fixed denominator j, with k ranging over the integer val...

conjecture | open | MathDB 页面 | 原文线索

V2-066971 | MATHDB-348855

The global Gross-Prasad conjecture for GSpin(4) × GSpin(5)

摘录 / EXCERPT: Global Gross-Prasad conjecture. The period is non-zero if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-066972 | MATHDB-348861

Novodvorsky compatibility conjecture for GSp(4) × GL(2) local L-factors

摘录 / EXCERPT: Novodvorsky compatibility conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-066973 | MATHDB-348862

Exceptional-pole conjecture for GSp(4) × GL(2) L-factors

摘录 / EXCERPT: Exceptional-pole conjecture. The point s_0 is an exceptional pole of $L(\pi\times\sigma,s)$ if and only if it is a pole of

conjecture | open | MathDB 页面 | 原文线索

V2-066974 | MATHDB-348866

Borie's bijection conjecture for 1234-avoiding up-down permutations

摘录 / EXCERPT: Let n be a positive integer. An up-down permutation of size $2n$ is a permutation of $\{1,\ldots,2n\}$ whose successive entries alternately rise and fall, and a permutation avoid...

conjecture | open | MathDB 页面 | 原文线索

V2-066975 | MATHDB-348881

Below-ground-state scattering conjecture for the focusing mass-critical generalized KdV equation

摘录 / EXCERPT: Let u solve the focusing, mass-critical generalized KdV equation (1.1) with initial data $u(0,x)=u_0\in L^2(\mathbb{R})$, and let ... be the ground state, which satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-066976 | MATHDB-348882

Nonexistence of below-ground-state almost periodic solutions for focusing generalized KdV

摘录 / EXCERPT: Below-ground-state almost-periodic-solution conjecture. There does not exist a nonzero almost periodic solution u to (1.1) satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-066977 | MATHDB-348883

Left-definite domain equality conjecture for Sturm-Liouville operators

摘录 / EXCERPT: Let ${\bf L}^n$ be a self-adjoint operator defined by left-definite theory on $L^2[(a,b),w]$ with domain $\mathcal{D}_{\bf L}^n$ that includes a complete system of orthogonal polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-066978 | MATHDB-348884

Friedrichs extension and powers conjecture

摘录 / EXCERPT: Let A be a closed semi-bounded symmetric operator, let ${\bf A}$ denote the associated operator, and let ${\bf A}_F$ be its Friedrichs self-adjoint extension. For...

conjecture | open | MathDB 页面 | 原文线索

V2-066979 | MATHDB-348886

Finite-rank perturbation parametrization for powers of Sturm-Liouville operators

摘录 / EXCERPT: Let ${\bf L}^n_0$, with $n\in\mathbb{N}$, be a self-adjoint extension of the minimal operator $L^n_{\scriptstyle \text{min}}$ on $L^2[(a,b),w]$, with deficiency indices (m,m) whe...

conjecture | open | MathDB 页面 | 原文线索

V2-066980 | MATHDB-348893

Frögoh"amosa conjecture on equality in Wilf's inequality

摘录 / EXCERPT: Frögoh"amosa conjecture. The equality

conjecture | open | MathDB 页面 | 原文线索

V2-066981 | MATHDB-348895

Bound conjecture for the minimum Wilf number of a gap

摘录 / EXCERPT: Bound conjecture. For any semigroup Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-066982 | MATHDB-348897

Conjecture on vortex corrections to the effective inverse temperature and free energies

摘录 / EXCERPT: Let $\hat\beta_{\mathrm{eff}}$ be the effective inverse temperature defined in the source, and let $f^{\operatorname{Coul}}(\beta)$, $f^{\operatorname{IV}}(\beta^{-1})$,...

conjecture | open | MathDB 页面 | 原文线索

V2-066983 | MATHDB-348898

Higher-order Arf invariant conjecture

摘录 / EXCERPT: Higher-order Arf invariant conjecture. The maps Arf_{j} are isomorphisms for all j. This is known for $j=1$, where it is given by the classical Arf invariants of...

conjecture | open | MathDB 页面 | 原文线索

V2-066984 | MATHDB-348899

Finite-type concordance invariant from the framed Arf trees

摘录 / EXCERPT: Consider the sum of the trees depicted in Figure 8 of the source, associated to 2-component string links, with coefficients in $\mathbb Z/2\mathbb Z$. Framed Arf finite-type conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-066985 | MATHDB-348900

Bing doubles do not bound framed Whitney towers of order six

摘录 / EXCERPT: Let K be a knot with non-trivial classical Arf invariant, and let its Bing double be the associated 2-component link. Framed Bing-double conjecture. The Bing double of any such k...

conjecture | open | MathDB 页面 | 原文线索

V2-066986 | MATHDB-348903

Liu and Postle's density conjecture for triangle-free 4-critical graphs

摘录 / EXCERPT: A graph G is 4-critical if it is 4-colourable but every proper subgraph is 3-colourable; it is triangle-free if it contains no subgraph isomorphic to K_3. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-066987 | MATHDB-348920

The descending conjecture for geometric subrepresentations

摘录 / EXCERPT: Descending conjecture. The geometric sublocal system in Theorem 1.2 descends to a ${\mathbb Z}_p$-étale subrepresentation

conjecture | open | MathDB 页面 | 原文线索

V2-066988 | MATHDB-348932

A product formula for a family of Kaneko-Yamamoto multiple zeta values

摘录 / EXCERPT: For positive integers r, m, and p, let $\{1\}_r$ denote the index consisting of r successive entries equal to 1, let $\zeta^\star$ denote a multiple zeta-star value, and...

conjecture | open | MathDB 页面 | 原文线索

V2-066989 | MATHDB-348933

Conjecture on expressing multiple zeta-binomial-star values by alternating multiple zeta values

摘录 / EXCERPT: For ${\bf k}=(k_1,\ldots,k_r)\in\mathbb{N}^r$, let $\zeta_B^\star({\bf k})$ denote the multiple zeta-binomial-star value and let alternating multiple zeta values be multiple zeta v...

conjecture | open | MathDB 页面 | 原文线索

V2-066990 | MATHDB-348945

Generic Kerr bounded-phase-space diffusion conjecture

摘录 / EXCERPT: Define the bounded-motion region ... where R is the Kerr radial potential, r^* is the local maximum of $-R$, and r_* is the local minimum of $-R$ to its right. Generic Kerr bou...

conjecture | open | MathDB 页面 | 原文线索

V2-066991 | MATHDB-348954

Strict tessellation characterization of trigonometric Laplace eigenfunctions

摘录 / EXCERPT: Strict tessellation conjecture. Ω has a complete set of trigonometric eigenfunctions for the Laplace eigenvalue equation with the Dirichlet boundary condition if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-066992 | MATHDB-348956

The L^2-critical mBO global-existence and blow-up dichotomy

摘录 / EXCERPT: Let $u_0\in H^{1/2}(\mathbb R)$ be sufficiently localized, let Q be the ground state solution of the equation defining Q with $m=3$, and let $u(x,t)$ be the mBO time evolution...

conjecture | open | MathDB 页面 | 原文线索

V2-066993 | MATHDB-348957

The L^2-supercritical gBO blow-up and global-existence dichotomy

摘录 / EXCERPT: Let $m>3$, let $u_0\in X\cap H^{1/2}(\mathbb R)$ be sufficiently localized, where X is the corresponding local well-posedness space for the given m, and let Q be the ground s...

conjecture | open | MathDB 页面 | 原文线索

V2-066994 | MATHDB-348963

The heat-flow conjecture for average lozenge orientations in dimer systems

摘录 / EXCERPT: Let $\bar\Omega$ be the limiting region with red boundary segments corresponding to constrained boundary conditions and blue segments corresponding to free boundary conditions. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-066995 | MATHDB-348993

The sublinear conjugacy-class conjecture for primitive wreath-product subgroups

摘录 / EXCERPT: Let A be an almost simple primitive group on m points appearing in the paper's table of final exceptions, and assume that $k(A)<m$. For each primitive subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-066996 | MATHDB-348994

NSSW19's clustered colouring conjecture for minor-free graphs

摘录 / EXCERPT: For a graph H, let $\mathcal{M}_H$ be the class of graphs with no H minor. Write $\raisebox{1.5 5pt}{\scalebox{1.25}{\chi}}_{\star}(\mathcal{M}_H)$ for its clustered chromatic n...

conjecture | open | MathDB 页面 | 原文线索

V2-066997 | MATHDB-348996

Postnikov-Stanley conjecture on the roots of truncated affine Weyl arrangement polynomials

摘录 / EXCERPT: Let Φ be an irreducible root system with Coxeter number h, and let V be its ambient vector space. For integers $a\leq b$, define the truncated affine Weyl arrangement ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-066998 | MATHDB-348997

A hyperbolic lower bound for the Kilbas-Saigo function at the boundary parameter

摘录 / EXCERPT: Let $\alpha\in(0,1]$, $m>0$, and $x\geq 0$. Define the Kilbas-Saigo function $E_{\alpha,m,l}$ by its usual power series, and let G denote the Barnes-type function used in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-066999 | MATHDB-348998

Hyperbolic two-sided bounds for the Kilbas-Saigo function

摘录 / EXCERPT: Let $\alpha\in(0,1]$, $m>0$, $l>m-1/\alpha$, and $x\geq 0$. Let $E_{\alpha,m,l}$ be the Kilbas-Saigo function. The conjectured two-sided bound. One has ... The bounds are natural bec...

conjecture | open | MathDB 页面 | 原文线索

V2-067000 | MATHDB-349004

Asymptotic enumeration conjecture for bipartite maps with prescribed face degrees

摘录 / EXCERPT: Let $(\mathbf{f}^{n})_{n \geq 1}$ be a sequence of face degree sequences, and let (g_n) be a sequence such that $v(\mathbf{f}^{n},g_n)\geq 2$ for all $n\geq 1$. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-067001 | MATHDB-349009

Diehl-Reizenstein conjecture on polynomial invariants of path orbits

摘录 / EXCERPT: Let G be a subgroup of $ ame{SL}_d(\mathbb{R})$, and let $Z,Z':[0,T]\to\mathbb{R}^d$ be curves. Suppose that ... for every $\varphi\in T(\mathbb{R}^d)$ fixed by...

conjecture | open | MathDB 页面 | 原文线索

V2-067002 | MATHDB-349010

Almost-polynomial moving-frame conjecture for orthogonal invariants

摘录 / EXCERPT: For $d\geq 2$, let $\mathcal U_{d;\leq n}$ be the domain of the moving frame and let ... be the moving frame defined in the preceding section. Almost-polynomial moving-frame conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-067003 | MATHDB-349015

The middle-weight conjecture for binary projective three-weight codes

摘录 / EXCERPT: Middle-weight conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067004 | MATHDB-349016

The q-ary middle-weight and symmetric-gap conjecture for projective three-weight codes

摘录 / EXCERPT: q-ary three-weight-code conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067005 | MATHDB-349020

Gu's conjecture on reducing subspaces of truncated shifts with one-dimensional defects

摘录 / EXCERPT: Gu's conjecture. The following are equivalent: A_B^θ has a reducing subspace such that the restriction has one-dimensional defects; and $\theta(z)=b(z)u(z^N)$ for some inner...

conjecture | open | MathDB 页面 | 原文线索

V2-067006 | MATHDB-349029

Logarithmic diameter conjecture for uniform high-genus quadrangulations

摘录 / EXCERPT: Let $q^{(n)}$ be the random uniform quadrangulation in the high-genus regime, with genus parameter determined by θ, and let $\operatorname{diam}(q^{(n)})$ denote its graph d...

conjecture | open | MathDB 页面 | 原文线索

V2-067007 | MATHDB-349031

Tightness conjecture for the systole of high-genus quadrangulations

摘录 / EXCERPT: Let $q^{(n)}$ be the random uniform high-genus quadrangulation, and let $\operatorname{syst}(q^{(n)})$ denote the size of its shortest non-contractible cycle. Systole tightness con...

conjecture | open | MathDB 页面 | 原文线索

V2-067008 | MATHDB-349040

Polytopality conjecture for the oriented graph of a positive braid

摘录 / EXCERPT: Let β be an arbitrary positive braid word, and let D_β be the oriented graph obtained from β by including edges for $ss\to s$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067009 | MATHDB-349044

Huang's conjecture on local obstructions in Zaremba theory

摘录 / EXCERPT: Let $A\subset\mathbb N$ be a finite alphabet and let X_A be its limit set. For each $m\in\mathbb N$, define reduction modulo m by ... The alphabet A has no local obstructions i...

conjecture | open | MathDB 页面 | 原文线索

V2-067010 | MATHDB-349053

Brilleslyper and Schaubroeck's unit-circle root-count conjecture for trinomials

摘录 / EXCERPT: Brilleslyper and Schaubroeck's conjecture. The number of roots of x^n+x^m-1 in the interior of the unit circle is

conjecture | open | MathDB 页面 | 原文线索

V2-067011 | MATHDB-349054

Boyd et al.'s density conjecture for square-free trinomial discriminants

摘录 / EXCERPT: Boyd et al.'s conjecture. The set S has density

conjecture | open | MathDB 页面 | 原文线索

V2-067012 | MATHDB-349055

Mironov-Morozov expansion formula for the BGW tau-function

摘录 / EXCERPT: Let DP denote the set of strict partitions, let $Q_\lambda({\bf t})$ be the normalized Schur Q-function, and let $\delta_{k,j}$ denote the specialization with valu...

conjecture | open | MathDB 页面 | 原文线索

V2-067013 | MATHDB-349056

Hypergeometric BKP conjecture for the generalized BGW tau-function

摘录 / EXCERPT: Let DP be the set of strict partitions. A hypergeometric BKP tau-function has the form ... where $r_\lambda=\prod_{j=1}^{\ell(\lambda)}\prod_{m=1}^{\lambda_j}r(m)$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-067014 | MATHDB-349057

Hypergeometric BKP conjecture for the Kontsevich-Witten tau-function

摘录 / EXCERPT: Let DP be the set of strict partitions, and let a hypergeometric BKP tau-function be expressed as ... with $r_\lambda=e^{\sum_{j=1}^{\ell(\lambda)}\xi(\lambda_j)}$....

conjecture | open | MathDB 页面 | 原文线索

V2-067015 | MATHDB-349071

Diskoid representation and correspondence conjecture for abstract key polynomials

摘录 / EXCERPT: Let $b[?]$ be a valuation on $K[X]$ and let $Qb[?]K[X]$ be an abstract key polynomial for $b[?]$, with $b?b[?]b[?]b?$. A diskoid is a subset $b?b[?]b[?]$ whose polynomial-value min...

conjecture | open | MathDB 页面 | 原文线索

V2-067016 | MATHDB-349072

Valuation conjectures for diskoids associated with abstract key polynomials

摘录 / EXCERPT: Let $b[?]$ be a valuation and Q an abstract key polynomial (ABKP) for $b[?]$. For an associated diskoid $b?)$, let $b[?]_{b?)}$ denote the induced function on $b[?][X]$. An RT ex...

conjecture | open | MathDB 页面 | 原文线索

V2-067017 | MATHDB-349089

The two-base automaticity conjecture

摘录 / EXCERPT: Let p and q be two multiplicatively independent natural numbers, and let automatic in base b mean that the real number's expansion in base b can be generated by a finite au...

conjecture | open | MathDB 页面 | 原文线索

V2-067018 | MATHDB-349091

Finite generation conjecture for invariant jet polynomials under reparametrization

摘录 / EXCERPT: Finite generation conjecture. The ring of invariant polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-067019 | MATHDB-349110

The 6-color conjecture for signed triangular grids

摘录 / EXCERPT: Signed triangular-grid chromatic-number conjecture. The chromatic number of every signed triangular grid is 6.

conjecture | open | MathDB 页面 | 原文线索

V2-067020 | MATHDB-349113

Knörrer periodicity for singular complete intersections

摘录 / EXCERPT: Singular complete-intersection Knörrer periodicity conjecture. Under appropriate hypotheses on $f_1,\dots,f_k$, there is a quasiequivalence of A_∞-categories

conjecture | open | MathDB 页面 | 原文线索

V2-067021 | MATHDB-349114

Injectivity conjecture for forgetful charts of prestable curves

摘录 / EXCERPT: Injectivity conjecture. There exists $m_0\geq 0$ such that, for every $m\geq m_0$, the pullback

conjecture | open | MathDB 页面 | 原文线索

V2-067022 | MATHDB-349121

Thomas-Yau-type conjecture for stability conditions on the Fukaya category

摘录 / EXCERPT: Fukaya-category stability-equation conjecture. There is a class of stability conditions on the Fukaya category such that for each stability condition in this class, there is an ass...

conjecture | open | MathDB 页面 | 原文线索

V2-067023 | MATHDB-349132

Generalized Marchal's conjecture for odd-body polygon and figure-eight choreographies

摘录 / EXCERPT: Generalized Marchal's conjecture. For any odd number of bodies, the n-gon choreography and the n-body figure eight are in the same continuation class.

conjecture | open | MathDB 页面 | 原文线索

V2-067024 | MATHDB-349134

Quenched Brownian scaling conjecture for IID horizontal-vertical rotor walks

摘录 / EXCERPT: Quenched scaling conjecture. For almost every such ρ,

conjecture | open | MathDB 页面 | 原文线索

V2-067025 | MATHDB-349140

Complete-monotonicity criterion for the generalized Mittag-Leffler function

摘录 / EXCERPT: For $\alpha,\beta,\gamma>0$, let $F_{\alpha,\beta}^{(\gamma)}$ denote the generalized Mittag-Leffler function, and let $x>0$. Complete-monotonicity criterion conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-067026 | MATHDB-349146

Existence of (2,3)-cordial oriented hypercubes in dimensions congruent to 1 modulo 3

摘录 / EXCERPT: An oriented hypercube C_d is an orientation of the d-dimensional hypercube, and it is (2,3)-cordial if its vertices admit a friendly $(0,1)$-labeling such that the induced arc...

conjecture | open | MathDB 页面 | 原文线索

V2-067027 | MATHDB-349152

Confocality conjecture for incenter and excenter loci of Poncelet 3-periodics

摘录 / EXCERPT: Let a pair of conics admit a family of Poncelet 3-periodics, and consider the locus traced by the incenter X_1 and by the excenters of the triangles in that family. Confocality c...

conjecture | open | MathDB 页面 | 原文线索

V2-067028 | MATHDB-349153

The bias conjecture for second moments of elliptic-curve families

摘录 / EXCERPT: Let $\mathcal{F}_k$ be a one-parameter family of elliptic curves over $\mathbf{Q}(T)$. Write the second moment as $M_{2,p}(\mathcal{F}_k)=\sum_{i=0}^{4}f_i(p)$, where the lower ord...

conjecture | open | MathDB 页面 | 原文线索

V2-067029 | MATHDB-349157

Orbital stability of cubic-quintic ground states in two dimensions

摘录 / EXCERPT: For $d=2$, let Q_ω denote a cubic-quintic ground state solution at frequency $\omega\in(0,\tfrac{3}{16})$. Orbital stability conjecture. All cubic-quintic ground state solut...

conjecture | open | MathDB 页面 | 原文线索

V2-067030 | MATHDB-349159

Conjectures on the roots and interlacing of Lambert W-function polynomials

摘录 / EXCERPT: Let $L_n(X)$ and $L_{n,k}(X)$ be the polynomial families defined earlier in the paper, and let the roots of a polynomial be ordered decreasingly when indexed by j. For $k\leq 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067031 | MATHDB-349161

The global spherical-function estimate for complex root systems

摘录 / EXCERPT: The global spherical-function estimate. On flat Riemannian symmetric spaces with complex group G, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067032 | MATHDB-349162

The sharp estimate for the Weyl-invariant heat kernel in the complex case

摘录 / EXCERPT: The Weyl-invariant heat-kernel estimate. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067033 | MATHDB-349163

The sharp estimate for the Weyl-invariant Dunkl heat kernel

摘录 / EXCERPT: The Weyl-invariant Dunkl-kernel estimate. The heat kernel satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067034 | MATHDB-349170

Irreducibility conjecture for the bivariate Hahn polynomial numerator

摘录 / EXCERPT: Irreducibility conjecture. The polynomial $R_n(x;\ell_1,\ell_2,y)$ is irreducible unless $\ell_1=\ell_2$ and n is odd. In that exceptional case, $R_n(x;\ell_1,\ell_2,y)$ is the p...

conjecture | open | MathDB 页面 | 原文线索

V2-067035 | MATHDB-349172

Rank-alternating homomesy conjecture for bounded labelings of the type A positive root poset

摘录 / EXCERPT: Rank-alternating homomesy conjecture. The triple $(\mathcal{A}^{\ell}(\widetriangle_n),\operatorname*{TogPro},\mathcal{R})$ is 0-mesic when n is even and $\frac{\ell}{2}$-mesic...

conjecture | open | MathDB 页面 | 原文线索

V2-067036 | MATHDB-349173

Cyclic sieving conjecture for staircase flagged tableaux

摘录 / EXCERPT: Staircase flagged-tableaux cyclic sieving conjecture. The triple $(\mathrm{FT}(sc_n,b),\langle\operatorname*{Pro}\rangle,Cat_{\ell}(x))$ exhibits the cyclic sieving phenomenon.

conjecture | open | MathDB 页面 | 原文线索

V2-067037 | MATHDB-349181

The circular limit-shape conjecture for totally symmetric self-complementary plane partitions

摘录 / EXCERPT: Circular limit-shape conjecture. As $n\to\infty$, the limit-shape curve is

conjecture | open | MathDB 页面 | 原文线索

V2-067038 | MATHDB-349200

The determinant-positive extension of the non-Eguchi-Hanson obstruction theorem

摘录 / EXCERPT: Let L_∞ be the limiting linear map appearing in the desingularization and obstruction setup, where the theorem was originally stated under the assumption...

conjecture | open | MathDB 页面 | 原文线索

V2-067039 | MATHDB-349205

Replica symmetry breaking condition for indefinite multi-species Sherrington-Kirkpatrick models

摘录 / EXCERPT: Let the multi-species Sherrington-Kirkpatrick model have an indefinite interaction matrix $\boldsymbol{D}^2$, and let $\boldsymbol{C}$, $\boldsymbol{C}'$, $\boldsymbol{L}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067040 | MATHDB-349214

Generalized Kasteleyn theorem for planar non-bipartite graphs in the disk

摘录 / EXCERPT: Let $\mathcal G$ be a graph in the broader class considered by the cited work, with edge set $\mathcal E$, and let $\sigma:\mathcal E\mapsto\{\pm1\}$ be a signature. For every fini...

conjecture | open | MathDB 页面 | 原文线索

V2-067041 | MATHDB-349234

Conjecture on vanishing of Picard-Lefschetz pairings modulo the identity component

摘录 / EXCERPT: In the situation of the Picard-Lefschetz formula, let $c_1=w_1c_0$ and $c_2=w_2c_0$ with $w_1,w_2\in W$, and let W_χ^0 be the specified subgroup of W. Assume that the cosets...

conjecture | open | MathDB 页面 | 原文线索

V2-067042 | MATHDB-349239

The realization conjecture for finite 2-special groups as narrow tower groups

摘录 / EXCERPT: Narrow realization conjecture. Every finite 2-special 2-group arises as the narrow tower group of some cyclic cubic field.

conjecture | open | MathDB 页面 | 原文线索

V2-067043 | MATHDB-349240

The realization conjecture for finite 2-select groups as wide tower groups

摘录 / EXCERPT: Wide realization conjecture. Every finite 2-select 2-group arises as the wide tower group of some cyclic cubic field.

conjecture | open | MathDB 页面 | 原文线索

V2-067044 | MATHDB-349242

The narrow/wide 2-class group frequency conjecture for cyclic cubic fields

摘录 / EXCERPT: Narrow/wide frequency conjecture. For each relevant n, the proportion of fields satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-067045 | MATHDB-349243

The extension-frequency conjecture for unramified [2,2]-extensions

摘录 / EXCERPT: Extension-frequency conjecture. Precisely one half of unramified $[2,2]$-extensions of cyclic cubic fields extend to unramified $\langle8,4\rangle$-extensions.

conjecture | open | MathDB 页面 | 原文线索

V2-067046 | MATHDB-349247

Soliton stability and resolution conjectures for the 3D Zakharov-Kuznetsov equation

摘录 / EXCERPT: Let the 3D Zakharov-Kuznetsov equation be given by, and let the soliton solutions be those defined by --. A perturbation is understood to be of soliton-like initial data, and suffi...

conjecture | open | MathDB 页面 | 原文线索

V2-067047 | MATHDB-349249

Wilf-class and palindromic-prefix enumeration conjecture for Hertzsprung patterns

摘录 / EXCERPT: Let $\mathcal{S}_k$ denote the set of Hertzsprung patterns of length k, and let $\Omega(\sigma,\sigma)$ be the autocorrelation polynomial of $\sigma\in\mathcal{S}_k$. For a palin...

conjecture | open | MathDB 页面 | 原文线索

V2-067048 | MATHDB-349250

Bóna's maximal-avoidance conjecture for Hertzsprung patterns

摘录 / EXCERPT: Let $\mathcal{S}_k$ be the set of Hertzsprung patterns of length k, let $\tau\in\mathcal{S}_k$, and let $|\mathcal{S}_n(\tau)|$ denote the number of length-n objects avoiding...

conjecture | open | MathDB 页面 | 原文线索

V2-067049 | MATHDB-349251

Rational generating-function conjecture for a non-Hertzsprung mesh pattern

摘录 / EXCERPT: Let p be the mesh pattern represented by the diagram encoded in the source, and let $|\mathcal{S}_n(p)|$ denote the number of length-n objects avoiding p. Rational generating...

conjecture | open | MathDB 页面 | 原文线索

V2-067050 | MATHDB-349261

Demailly's convergence conjecture for Monge-Ampère masses

摘录 / EXCERPT: Let φ be plurisubharmonic in a bounded pseudoconvex neighborhood of $0\in\mathbb{C}^n$, locally bounded away from 0. For $m\in\mathbb{Z}_{>0}$, let φ_m be the ana...

conjecture | open | MathDB 页面 | 原文线索

V2-067051 | MATHDB-349262

Kim-Rashkovskii's valuative invariance conjecture for Monge-Ampère masses

摘录 / EXCERPT: Let φ and ψ be plurisubharmonic germs with isolated singularities at $0\in\mathbb{C}^n$. For every divisorial valuation v centered at 0, write $v(\varphi)$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-067052 | MATHDB-349264

Rowvacuation homomesy for the type B piecewise-linear statistic

摘录 / EXCERPT: The type B piecewise-linear homomesy conjecture. The statistic $h^{\mathrm{PL}}_*\colon \mathcal{A}^{\mathrm{PL}}_{\kappa}(\mathsf{B}^n)\to \mathbb{R}$ is κ-mesic for...

conjecture | open | MathDB 页面 | 原文线索

V2-067053 | MATHDB-349265

Birational rowvacuation homomesy for type B quotient posets

摘录 / EXCERPT: Birational type B homomesy conjecture. The statistics $h^{\mathrm{B}}_i\colon \mathcal{A}^{\mathrm{B}}_{\kappa}(\mathsf{B}^n)\to \mathbb{R}_{>0}$, for $1\leq i\leq 2n-1$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067054 | MATHDB-349266

Birational OY-invariant conjecture for the Lalanne-Kreweras involution

摘录 / EXCERPT: Birational OY-invariant conjecture. The function $\EuScript{Y}^{\mathrm{B}}\colon\mathcal{A}^{\mathrm{B}}(\mathsf{A}^n)\to\mathbb{R}_{>0}$ is invariant under both...

conjecture | open | MathDB 页面 | 原文线索

V2-067055 | MATHDB-349274

Conjecture on the quadratic conductor formula for quasi-homogeneous hypersurface singularities

摘录 / EXCERPT: Conjecture on the quadratic conductor formula. Then

conjecture | open | MathDB 页面 | 原文线索

V2-067056 | MATHDB-349277

Lions' maximal regularity conjecture for viscous Hamilton-Jacobi equations

摘录 / EXCERPT: Consider the viscous Hamilton-Jacobi equation ... where Ω is a domain, $\gamma>1$, and $f\in L^q(\Omega)$ with $q>1$. Write $\gamma'=\frac{\gamma}{\gamma-1}$ for the conjugate...

conjecture | open | MathDB 页面 | 原文线索

V2-067057 | MATHDB-349279

Cauchy subsequences and hulls of sequences in labeled trees

摘录 / EXCERPT: Cauchy-hull conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-067058 | MATHDB-349287

Nonexistence of three-dimensional transformation cloaking for holes of arbitrary shape

摘录 / EXCERPT: Let the virtual body be an isotropic, non-centrosymmetric gradient solid in dimension three, and let the cloaked region contain a hole of arbitrary shape. Nonexistence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-067059 | MATHDB-349293

Stability under adding a common first-row parameter to Newell-Littlewood coefficients

摘录 / EXCERPT: Stability conjecture. For an integer a, there exists $N\boldsymbol{ u}=(a,\boldsymbol{ u}_0)$, $\boldsymbol{ u}'=(a,\boldsymbol{ u}'_0)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067060 | MATHDB-349303

The growth-rate order conjecture for standard PCP zig-zags

摘录 / EXCERPT: Fix $m\geq 2$, and let $f_1,f_2\in\operatorname{PA}(m)$ be m-modal standard PCP zig-zags of pseudo-Anosov type. Let λ_i be the growth rate of f_i, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067061 | MATHDB-349305

Dimension greater than one for the attractors of a self-affine IFS

摘录 / EXCERPT: Dimension conjecture. For all $0 < \lambda < \mu < 1$ with $\lambda + \mu > 1$, the set $A_{\lambda,\mu}$ has dimension strictly greater than 1.

conjecture | open | MathDB 页面 | 原文线索

V2-067062 | MATHDB-349307

Equidistribution of the descent and cycle bistatistics with the excedance bistatistic

摘录 / EXCERPT: Let S_n be the set of permutations of $\\{1,2,\ldots,n\\}$, and let des_2, cyc, pex, des, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067063 | MATHDB-349308

Cotwist conjecture for functor categories on spheres

摘录 / EXCERPT: Cotwist conjecture. There is a commutative diagram in LinCat_R identifying $T_{\operatorname{Fun}(S^n,\operatorname{RMod}_R)}$ under the equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-067064 | MATHDB-349313

Off-diagonal bump conjecture for sparse operators

摘录 / EXCERPT: Let T be a sparse operator, let $1<p<q<\infty$, and let $\mathcal{D}_{p,p}$ and $\mathcal{D}^{\ast}_{p,p}$ denote the corresponding direct comparison bump quantities for the case...

conjecture | open | MathDB 页面 | 原文线索

V2-067065 | MATHDB-349314

Generalized Fibonacci analogue of the coefficient-statistics rationality conjecture

摘录 / EXCERPT: Generalized Fibonacci analogue conjecture. The generating function

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067066 | MATHDB-349315

Explicit formula conjecture for the third coefficient moment

摘录 / EXCERPT: Fix $k\geq 2$ and let $v_3^{(k)}(n,t)$ be the sum of the cubes of the coefficients of ... Define ... Third-moment generating-function conjecture. One has ... where ... This conjecture give...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067067 | MATHDB-349316

Denominator-shape conjecture for generalized Fibonacci coefficient moments

摘录 / EXCERPT: Fix $r\geq 2$ and $k\geq 2$, and let $v_r^{(k)}(n,t)$ be the sum of the rth powers of the coefficients of ... with ... Denominator-shape conjecture. There is an integer $m\geq 1$, de...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-067068 | MATHDB-349317

Rationality conjecture for generalized Fibonacci coefficient congruence counts

摘录 / EXCERPT: Fix $k\geq 2$, $m\geq 1$, and $0\leq a<m$. Set ... Let $h_{m,a}^{(k)}(n)$ be the number of coefficients of $I_n^{(k)}(x)$ congruent to a modulo m, and define ... Congruence-count r...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067069 | MATHDB-349318

Explicit parity-count generating function conjecture for generalized Fibonacci products

摘录 / EXCERPT: Fix $k\geq 2$ and define $H_{m,a}^{(k)}(x)$ as the generating function counting coefficients of $\prod_{i=1}^n(1+x^{F_{i+k-1}^{(k)}})$ that are congruent to a modulo m. Parity-...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-067070 | MATHDB-349320

Conjecture on the periodic fractional KdV Green's function

摘录 / EXCERPT: Periodic Green's function conjecture. For each $\alpha\in(2,4]$, there exists $c_0>0$ such that, for $c\in(0,c_0)$, $G_{\mathbb{T}}$ is even, strictly positive on $\mathbb{T}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067071 | MATHDB-349321

Conjecture on oscillations of the real-line fractional KdV Green's function

摘录 / EXCERPT: Real-line Green's function conjecture. For every $c>0$ and every $\alpha\in(2,4]$, $G_{\mathbb{R}}$ is not strictly positive on $\mathbb{R}$ and is not monotonically decreasing on...

conjecture | open | MathDB 页面 | 原文线索

V2-067072 | MATHDB-349324

Panyushev's nonnesting partition duality conjecture

摘录 / EXCERPT: Let W be a finite Weyl group with root system Φ, positive roots Φ^+, rank r, Coxeter number h, and simple roots $\\{\alpha_1,\ldots,\alpha_r\\}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-067073 | MATHDB-349331

The generalised valley version of the Delta conjecture

摘录 / EXCERPT: Let $m,n,k\in\mathbb{N}$ with $k<n$, and let $\mathsf{LD}(m,n)^{\bullet k}$ denote the partially labelled rectangular Dyck paths with k decorated valleys. For such a path P, le...

conjecture | open | MathDB 页面 | 原文线索

V2-067074 | MATHDB-349332

The touching valley version of the Delta conjecture

摘录 / EXCERPT: Let $n,k,r\in\mathbb{N}$ with $k<n$, and let $\mathsf{LD}(n)^{\bullet k}$ be the partially labelled Dyck paths with k decorated valleys. For P in this set, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067075 | MATHDB-349333

The modified valley version of the Delta square conjecture

摘录 / EXCERPT: Let $n,k\in\mathbb{N}$ with $k<n$, and let $\mathsf{LSQ}'(n)^{\bullet k}$ denote the partially labelled square paths with k decorated valleys. For such a path P, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067076 | MATHDB-349334

Localization Dichotomy Conjecture for non-periodic systems

摘录 / EXCERPT: Localization Dichotomy Conjecture. The following statements are equivalent: P admits a generalized Wannier basis that is exponentially localized; P admits a generalized Wannier...

conjecture | open | MathDB 页面 | 原文线索

V2-067077 | MATHDB-349343

Log-gamma polymer convergence to the Airy process

摘录 / EXCERPT: Fix $\theta,r>0$ and let $\tilde{f}_N^{LG}$ be the random continuous function from Definition 2. Then, as $N\to\infty$, the random functions $\tilde{f}_N^{LG}$ converge weakly in...

conjecture | open | MathDB 页面 | 原文线索

V2-067078 | MATHDB-349350

Pate's conjecture on the largest eigenvalue of the complementary-permanent matrix

摘录 / EXCERPT: Pate's conjecture. The largest eigenvalue of $\mathscr{C}_k(A)$ is $\operatorname{per}(A)$. This conjecture is refuted; the paper states that its $k=1$ case has an 8×8 count...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067079 | MATHDB-349351

Loxton and Van der Poorten's conjecture for Mahler functions

摘录 / EXCERPT: Let k and l be multiplicatively independent positive integers. Let $F\in K[[x]]$ be a power series over a number field K that is both k-Mahler and l-Mahler. Loxton and Va...

conjecture | open | MathDB 页面 | 原文线索

V2-067080 | MATHDB-349352

The irrational multiplicatively independent Mahler Hahn-series conjecture

摘录 / EXCERPT: Let α and β be irrational and multiplicatively independent. Let $F(x)$ be a Hahn series over $\mathbb{R}$ that is both α-Mahler and β-Mahler. Irrational M...

open_problem | open | MathDB 页面 | 原文线索

V2-067081 | MATHDB-349354

Cyclic-sum conjecture for finite multiple harmonic q-series

摘录 / EXCERPT: Cyclic-sum conjecture. For every integer $n>r$ and every primitive root of unity ζ_n, the following hold. With $r=\sum_{j=0}^{t}d_j+2t$,

conjecture | open | MathDB 页面 | 原文线索

V2-067082 | MATHDB-349358

Chudnovsky bound for monomial ideals

摘录 / EXCERPT: Let I be a monomial ideal. Define its initial degree by $[?]$ and its big-height by ... The Chudnovsky bound. The Waldschmidt constant satisfies ... This conjecture strengthens the S...

conjecture | open | MathDB 页面 | 原文线索

V2-067083 | MATHDB-349359

Naive Chudnovsky bound for monomial ideals

摘录 / EXCERPT: Let I be a monomial ideal. Define its initial degree by $\alpha(I)$, its naive Waldschmidt constant by $\widetilde{\alpha}(I)$, and its big-height by ... The naive Chudnovsky bound...

conjecture | open | MathDB 页面 | 原文线索

V2-067084 | MATHDB-349364

Strong form of the variance estimate conjecture for stationary Gaussian processes

摘录 / EXCERPT: Let $N(T)$ denote the number of zeroes in an interval of length T of a non-degenerate stationary Gaussian process (SGP), with covariance function r and parameter σ as in...

conjecture | open | MathDB 页面 | 原文线索

V2-067085 | MATHDB-349368

Hausdorff-dimension conjecture for exceptional triples in the directed landscape

摘录 / EXCERPT: Let X be the set of triples (y,s,t) with $y\in\mathbb{R}$ and $s<t$. Equip X with the metric ... For $k\geq2$, let E_k be the subset of triples with $|s-t|\geq1$ for which th...

conjecture | open | MathDB 页面 | 原文线索

V2-067086 | MATHDB-349390

The Poisson spacing conjecture for unfolded Sato-Tate angles

摘录 / EXCERPT: Let E be an elliptic curve for which the Sato-Tate conjecture holds. If $\theta_p\in[0,\pi]$ are the Sato-Tate angles and $\rho(\theta)$ is their density, define the unfolded var...

conjecture | open | MathDB 页面 | 原文线索

V2-067087 | MATHDB-349412

Bohr-Sommerfeld conjecture for the type G classical confluent conformal block

摘录 / EXCERPT: Bohr-Sommerfeld conjecture for type $\mathcal G$.

conjecture | open | MathDB 页面 | 原文线索

V2-067088 | MATHDB-349426

Cyclicity conjecture for elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a non-CM elliptic curve. For each positive integer k, let $E[k]$ be the k-torsion subgroup and let $\mathbb{Q}(E[k])$ be the field obtained by adjoining t...

conjecture | open | MathDB 页面 | 原文线索

V2-067089 | MATHDB-349433

Central-value weighting conjecture for low-lying zeros

摘录 / EXCERPT: Central-value weighting conjecture. The density $W_w(\mathcal F)$ should differ from $W(\mathcal F)$ only when w_Π essentially contains the central value $L(1/2,\Pi)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067090 | MATHDB-349436

Double critical mass conjecture for radial Keller-Segel steady states

摘录 / EXCERPT: Let $\Omega=B_R(0)\subset\mathbb{R}^2$ with $R>0$ and $k>0$. A steady state is a solution (u,v) of the stationary problem considered in the paper, with prescribed mass...

conjecture | open | MathDB 页面 | 原文线索

V2-067091 | MATHDB-349448

The degree-nine conjecture for rational plane curves with triple-point singularities

摘录 / EXCERPT: A reduced plane curve has multiplicity three singularities when all its singularities have multiplicity 3. Degree-nine conjecture. There is no reduced plane curve in any characte...

conjecture | open | MathDB 页面 | 原文线索

V2-067092 | MATHDB-349449

The reduced plane-curve sigma-four conjecture

摘录 / EXCERPT: Reduced sigma-four conjecture. For any reduced singular complex curve $C\subset\mathbb{P}^2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067093 | MATHDB-349450

The node-existence conjecture for high-degree plane curves with nodes and triple points

摘录 / EXCERPT: Let $C\subset\mathbb{P}^2$ be a reduced plane curve whose singularities are all nodes or triple points, and let n_2 denote the number of nodes. Node-existence conjecture. If C...

conjecture | open | MathDB 页面 | 原文线索

V2-067094 | MATHDB-349460

The phase-shift formula for multi-block box-ball systems

摘录 / EXCERPT: Let a box-ball system have n blocks, with Q_k balls in the k-th block, and suppose the blocks are separated sufficiently so that the asymptotic soliton structure can be ident...

conjecture | open | MathDB 页面 | 原文线索

V2-067095 | MATHDB-349467

Matsuzawa-Meng-Zhang-Sano conjecture on bounded-degree points of maximal arithmetic degree

摘录 / EXCERPT: Matsuzawa-Meng-Zhang-Sano conjecture. The set $Z_f(d)$ is not Zariski dense in X. This conjecture predicts that, among points of bounded degree, points whose arithmetic degree is...

conjecture | open | MathDB 页面 | 原文线索

V2-067096 | MATHDB-349469

Gross-inspired mirror conjecture for cominuscule fixed points

摘录 / EXCERPT: Let $\alpha_{i_1},\dots,\alpha_{i_k}$ be distinct cominuscule roots. Suppose the corresponding sections b_{i_j} of $L_{\alpha_{i_j}}K$ have m distinct zeros $x_1,\dots,x_m$, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-067097 | MATHDB-349470

Mirror of the canonical Hecke-cycle Lagrangian

摘录 / EXCERPT: Let C be the underlying curve, let $c\in C$, and let W^+_0 and W^+_c be the upward flows from the canonical fixed point and from the fixed point indexed by c, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-067098 | MATHDB-349471

Conjectural Bernoulli congruences for multiple harmonic sums of depths 8, 9, and 10

摘录 / EXCERPT: Bernoulli congruences for $R_8^{(m)}(p)$, $R_9^{(m)}(p)$, and $R_{10}^{(m)}(p)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067099 | MATHDB-349472

Conjecture on uniformly signed 6-cycles in bipartite planar graphs

摘录 / EXCERPT: Let G be a bipartite planar graph of girth at least 6, and let σ be a signature on G, assigning a sign to every edge. Say that the 6-cycles have the same sign when eve...

conjecture | open | MathDB 页面 | 原文线索

V2-067100 | MATHDB-349473

The multiples-of-factorial conjecture for Ehrhart-equivalent integral polytopes

摘录 / EXCERPT: The multiples-of-factorial conjecture. For any $n\in\mathbb{Z}_{>3}$, if $P,Q\subset\mathbb{R}^n$ are two arbitrary Ehrhart-equivalent integral n-polytopes, then

conjecture | open | MathDB 页面 | 原文线索

V2-067101 | MATHDB-349475

Ciolan-García-Sánchez-Moree conjecture on cyclotomic numerical semigroups

摘录 / EXCERPT: Ciolan-García-Sánchez-Moree conjecture. A numerical semigroup S is cyclotomic if and only if it is complete intersection.

conjecture | open | MathDB 页面 | 原文线索

V2-067102 | MATHDB-349476

Cyclotomic exponent sequence conjecture for minimal generators

摘录 / EXCERPT: Cyclotomic exponent sequence conjecture. For $n\in\mathbb{N}$, n is a minimal generator of S if and only if $e_n<0$.

conjecture | open | MathDB 页面 | 原文线索

V2-067103 | MATHDB-349477

Positive-exponent bound for cyclotomic numerical semigroup polynomials

摘录 / EXCERPT: Positive-exponent bound conjecture. The positive exponent indices satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067104 | MATHDB-349478

The complete intersection characterization of cyclotomic numerical semigroups

摘录 / EXCERPT: Cyclotomic complete intersection conjecture. A numerical semigroup is a complete intersection if and only if it is cyclotomic.

conjecture | open | MathDB 页面 | 原文线索

V2-067105 | MATHDB-349479

The minimal-generator conjecture for cyclotomic numerical semigroups

摘录 / EXCERPT: Minimal-generator conjecture. Then $n \in \mathbb{N}$ is a minimal generator of S if and only if $e_n < 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-067106 | MATHDB-349480

Conjecture on dessins corresponding to families of conformal blocks

摘录 / EXCERPT: General dessin conformal-block conjecture. For a dessin satisfying the proposition's conditions, it corresponds to a family of 4-point CBs whose states follow the stated table.

conjecture | open | MathDB 页面 | 原文线索

V2-067107 | MATHDB-349482

Robin-to-Neumann and Dirichlet convergence conjecture for the porous medium equation

摘录 / EXCERPT: Let $T>0$, let m be the porous-medium exponent, and let $g:{\mathbb T}\to[0,1]$ be measurable. For each $\kappa>0$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067108 | MATHDB-349492

Cayley correspondence conjecture for higher Teichmüller components

摘录 / EXCERPT: Let X be a compact Riemann surface, let $ \Sigma$ be its underlying surface, and let $\mathrm{G}^\mathbb{R}$ be a simple Lie group. Write ${\mathcal M}(X,\mathrm{G}^\mathbb{R})$...

conjecture | open | MathDB 页面 | 原文线索

V2-067109 | MATHDB-349493

Openness and closedness conjecture for Θ-positive Anosov representations

摘录 / EXCERPT: Let Σ be a closed oriented surface and let $(\mathrm{G},\mathrm{P}^\mathbb{R}_\Theta)$ admit a Θ-positive structure. A representation...

conjecture | open | MathDB 页面 | 原文线索

V2-067110 | MATHDB-349510

The SHGH conjecture for uniform points in the projective plane

摘录 / EXCERPT: Let $R={\mathbb C}[x,y,z]$, let $Y=\{P_1,\ldots,P_r\}$ be a general set of points in $\mathbb{P}^2$, and let I_Y be its defining ideal. An irreducible homogeneous polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-067111 | MATHDB-349513

The weighted-capacity optimization conjecture for the binary adder channel

摘录 / EXCERPT: Weighted-capacity optimization conjecture. For every $(c_1,c_2)\in\Delta^1$, $R(\mathcal{O}_f;c_1,c_2)$ equals the optimum of

open_problem | open | MathDB 页面 | 原文线索

V2-067112 | MATHDB-349528

The modulo 5 decomposition conjecture for the crank generating function

摘录 / EXCERPT: Let q be a formal variable, and define ... Here $(a;q)_{\infty}=\prod_{j\geq 0}(1-aq^j)$ denotes the infinite q-Pochhammer symbol. The modulo 5 decomposition conjecture. ... This i...

conjecture | open | MathDB 页面 | 原文线索

V2-067113 | MATHDB-349529

The modulo 7 decomposition conjecture for the crank generating function

摘录 / EXCERPT: Let q be a formal variable, and let A, B, C, D, and E denote the quantities used in the paper's modulo 7 decomposition. The modulo 7 decomposition conjecture. ... The i...

conjecture | open | MathDB 页面 | 原文线索

V2-067114 | MATHDB-349533

Deformation invariance of plurigenera under a globally generated pluricanonical bundle

摘录 / EXCERPT: Deformation invariance under global generation. The plurigenera are locally constant at $0\in S$.

conjecture | open | MathDB 页面 | 原文线索

V2-067115 | MATHDB-349536

The bounded-discrepancy conjecture for spanning forests in zero-sum complete graphs

摘录 / EXCERPT: Let K_n be a complete graph, let $c:E(K_n)\to\{-1,1\}$ be a zero-sum labeling satisfying ... and let F be a spanning forest of K_n. Write $\Delta(F)$ for the maximum degree of...

conjecture | open | MathDB 页面 | 原文线索

V2-067116 | MATHDB-349538

Stability and existence of selected monotonic nontrivial traveling waves

摘录 / EXCERPT: The systems under consideration admit traveling waves, including trivial traveling waves and, in some parameter regimes, nontrivial traveling waves with profiles N and P. The n...

conjecture | open | MathDB 页面 | 原文线索

V2-067117 | MATHDB-349539

Marchioro-Pulvirenti conjecture on the measure of point-vortex collapses

摘录 / EXCERPT: Marchioro-Pulvirenti conjecture. The set of collapses ${\mathfrak C}$ has Lebesgue measure 0.

conjecture | open | MathDB 页面 | 原文线索

V2-067118 | MATHDB-349543

Gepner-type stability condition conjecture for chain-type invertible polynomials

摘录 / EXCERPT: Let f_n be the chain-type invertible polynomial, let S_n and L_{f_n} be the associated ring and grading group, and let $\omega_1,\dots,\omega_n$ be positive rational numbers...

conjecture | open | MathDB 页面 | 原文线索

V2-067119 | MATHDB-349547

Conjecture on oscillations and coexistence in the two-sex predator-prey branching process

摘录 / EXCERPT: Let $\{(Z_n,\widetilde{Z}_n)\}_{n\in\mathbb{N}_0}$ be a PP-2SDDBPO, with total population variables T_n and $\widetilde{T}_n$, parameter μ, and positive initial values...

conjecture | open | MathDB 页面 | 原文线索

V2-067120 | MATHDB-349548

Bias Conjecture for second moments of one-parameter elliptic-curve families

摘录 / EXCERPT: Bias Conjecture. If i is the largest index for which $\mu_i\neq 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067121 | MATHDB-349552

The coefficient-sign conjecture for cable knots of the figure-eight knot

摘录 / EXCERPT: Let $\bm{4_1}$ denote the figure-eight knot, and consider the cable knots ... where $r>8$ is odd and the Alexander polynomial is monic. Write the invariant as ... Let $\{f_m(q)\}$ be p...

conjecture | open | MathDB 页面 | 原文线索

V2-067122 | MATHDB-349553

The coefficient-pattern conjecture for cable knots of the figure-eight knot

摘录 / EXCERPT: Let $K_r=C_{(r,2)}(\bm{4_1})\subseteq S^3$ be the family of cable knots from the preceding conjecture, with $r>8$ odd and monic Alexander polynomial. Let $\{f_m(q)\}$ be the coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-067123 | MATHDB-349556

Gradient descent optimality conjecture for gradient-norm minimization

摘录 / EXCERPT: Let $K>0$. Consider a method whose iterates have the form ... where $\boldsymbol{x}_0\in\mathbb{R}^d$ is the initial point, f is a convex function accessed through a gradient oracl...

conjecture | open | MathDB 页面 | 原文线索

V2-067124 | MATHDB-349577

Translation-tiling conjecture for hunger-game basins of attraction

摘录 / EXCERPT: Let a finite rational irreducible Markov chain have n states, transition matrix with hunger-game matrix H, and basin of attraction in the hyperplane Z. The vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-067125 | MATHDB-349580

Prasad's conjecture on distinction and base change for quasi-split classical groups

摘录 / EXCERPT: Let ${\bf Y}$ be an F-quasi-split group, let E/F be the quadratic extension implicit in the notation, and let $\mathbf{BC}_F^E$ denote the base-change map from L-parameters o...

conjecture | open | MathDB 页面 | 原文线索

V2-067126 | MATHDB-349581

Global transfer conjecture for distinction of classical groups

摘录 / EXCERPT: Let K/k be a quadratic extension of number fields, let $\mathbb{H}^\circ$ be a k-quasisplit classical group, and let T be the functorial lift to...

conjecture | open | MathDB 页面 | 原文线索

V2-067127 | MATHDB-349606

Gross-Mansour-Tucker conjecture on orientable ribbon graph partial-dual genus polynomials

摘录 / EXCERPT: A ribbon graph is orientable if its underlying surface is orientable, and its partial-dual genus polynomial records the genera of all partial duals. A polynomial is non-constant if...

conjecture | open | MathDB 页面 | 原文线索

V2-067128 | MATHDB-349610

Wormhole conjecture for KSBA moduli of surfaces

摘录 / EXCERPT: Wormhole conjecture. The MMP requires only flips and gives a minimal model. The KSBA replacement of $(X'\subset\mathcal{X}')\to(0\in\mathbb{D})$ lives on the same moduli space as t...

conjecture | open | MathDB 页面 | 原文线索

V2-067129 | MATHDB-349611

Wormhole conjecture for punctured W-surfaces

摘录 / EXCERPT: Wormhole conjecture. The MMP on the new W-surface finishes in a minimal model and requires only flips; that is, both punctured W-surfaces live in the same moduli space.

conjecture | open | MathDB 页面 | 原文线索

V2-067130 | MATHDB-349613

Kundu-Ray density conjecture for Iwasawa invariants of elliptic curves

摘录 / EXCERPT: Let $E_{/\mathbb{Q}}$ be an elliptic curve of rank r_E. For primes p of good ordinary reduction, let μ and λ be the ordinary Iwasawa invariants; for primes of good...

conjecture | open | MathDB 页面 | 原文线索

V2-067131 | MATHDB-349615

Density-one vanishing conjecture for Iwasawa invariants at varying primes

摘录 / EXCERPT: Let $E_{/\mathbb{Q}}$ be an elliptic curve of rank r_E with good ordinary or supersingular reduction at p. Let $\mu_p^{\ddag}(E)$ and $\lambda_p^{\ddag}(E)$ denote the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-067132 | MATHDB-349626

The generalized Collatz orbit criterion

摘录 / EXCERPT: Let $m\in\mathbb{N}$, and let Col be the Collatz function defined by ... Consider the sequence $\operatorname{Col}^n(m)$ for $n\geq1$. Generalized Collatz conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-067133 | MATHDB-349628

The partial reconstruction threshold conjecture for sparse graph alignment

摘录 / EXCERPT: Partial reconstruction threshold conjecture. If $\lambda s \leq 1$, then partial reconstruction is impossible: for every $\alpha>0$ and every estimator $\hat{\pi}$,

conjecture | open | MathDB 页面 | 原文线索

V2-067134 | MATHDB-349629

Miller's negative bias conjecture for second moments

摘录 / EXCERPT: Let $E: y^2=x^3+a(T)x+b(T)$ be a one-parameter family of elliptic curves over $\mathbb{Q}(T)$, with $a(T),b(T)\in\mathbb{Z}[T]$, and let ... be its second moment of Dirichlet coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-067135 | MATHDB-349630

Odd-moment rank conjecture for the third moment of elliptic-curve coefficients

摘录 / EXCERPT: Third-moment rank conjecture. The average value of the main term of the third moment is $-2\operatorname{rank}(E(\mathbb{Q}))p^2$. The conjecture is based on numerical data relatin...

conjecture | open | MathDB 页面 | 原文线索

V2-067136 | MATHDB-349637

Finite-length augmentation of minimal Dyck rationals to represent real numbers

摘录 / EXCERPT: Let $\mathcal{D}_{r_{\text{min}}}$ be the language of minimal Dyck-word representations, and consider augmenting its words with additional notation or productions. Finite-length re...

conjecture | open | MathDB 页面 | 原文线索

V2-067137 | MATHDB-349638

Context-free grammar approximation conjecture for irrational numbers

摘录 / EXCERPT: Let x be an irrational real number. Let G_x be a context-free grammar that generates an infinite sequence of Dyck rationals $(s_i)_{i\in\mathbb{N}_{+}}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067138 | MATHDB-349645

The 6-coupled-choosability conjecture for planar partial 3-trees

摘录 / EXCERPT: Planar partial 3-tree conjecture. Every planar partial 3-tree is 6-coupled-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-067139 | MATHDB-349646

The 5-coupled-choosability conjecture for IO-graphs

摘录 / EXCERPT: IO-graph coupled-choosability conjecture. Every IO-graph is 5-coupled-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-067140 | MATHDB-349661

The Schmidt rank versus analytic rank conjecture for multilinear polynomials

摘录 / EXCERPT: Let $d\geq 2$, let $\mathbb F_q$ be a finite field, and let P be a multilinear polynomial of degree d. Denote by $r_{\mathbb F_q}(P)$ its Schmidt rank and by...

conjecture | open | MathDB 页面 | 原文线索

V2-067141 | MATHDB-349674

The endpoint-maximization conjecture for the Lebesgue function of modified Chebyshev points

摘录 / EXCERPT: Endpoint-maximization conjecture. The maximum of the Lebesgue function is attained at $x=-1$, namely

conjecture | open | MathDB 页面 | 原文线索

V2-067142 | MATHDB-349679

Monotonicity and threshold conjecture for optimal spreading under fixed-size access

摘录 / EXCERPT: In the fixed-size access model, let N be the number of nodes, m the redundancy level, r the number of accessed nodes, and $b1$ the spreading parameter, with minimal spreading...

conjecture | open | MathDB 页面 | 原文线索

V2-067143 | MATHDB-349688

Gross-Harrington-Meshkat-Shiu leak-removal conjecture for linear compartmental models

摘录 / EXCERPT: Let $\widetilde{\mathcal{M}}$ be a strongly connected linear compartmental model with at least one input and exactly one leak. A model is generically locally identifiable if its pa...

conjecture | open | MathDB 页面 | 原文线索

V2-067144 | MATHDB-349690

Leak-addition conjecture for unidentifiable linear compartmental models

摘录 / EXCERPT: Let $\mathcal{M}$ be an unidentifiable linear compartmental model. Leak-addition conjecture. If one leak is added, the resulting model $\widetilde{\mathcal{M}}$ is also unidentifia...

conjecture | open | MathDB 页面 | 原文线索

V2-067145 | MATHDB-349693

Woźniakowski's conjecture on the exponent of tractability of star discrepancy

摘录 / EXCERPT: Woźniakowski's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067146 | MATHDB-349695

Yeo's complementary cycle-factor conjecture for regular multipartite tournaments

摘录 / EXCERPT: Let D be a regular c-partite tournament, with $c\geq 5$. A $(p,|V(D)|-p)$-cycle-factor is a cycle-factor consisting of cycles of lengths p and $|V(D)|-p$. Yeo's conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-067147 | MATHDB-349697

Monotonicity conjecture for the average energy of minimax spherical designs

摘录 / EXCERPT: Monotonicity conjecture. For each fixed positive integer p, the sequence $R_p(n)$ is non-increasing in n.

conjecture | open | MathDB 页面 | 原文线索

V2-067148 | MATHDB-349702

The two-disjoint-triangles conjecture for maximum subtour-LP integrality ratio instances

摘录 / EXCERPT: Let an instance have a fixed number of vertices, and let x^* be an optimal fractional solution of its subtour LP. The two-disjoint-triangles conjecture. The instances maximizing...

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V2-067149 | MATHDB-349706

The contact-star formula for up to four general fat points

摘录 / EXCERPT: Let $s\le 4$ and let m_i be the multiplicities of s general fat points. For each i, take a contact star configuration formed by m_i+1 lines tangent to the same conic. The c...

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V2-067150 | MATHDB-349716

Equidistribution of d-th power partitions modulo every integer

摘录 / EXCERPT: Equidistribution conjecture. For every $m\in\mathbb{N}_{\geq 2}$ and every $r\in\\{0,\ldots,m-1\\}$, there are infinitely many $n\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067151 | MATHDB-349725

Sharpness conjecture for the singular-time dimension of wild Euler solutions

摘录 / EXCERPT: Sharpness conjecture. For every $\beta<\frac{1}{3}$, there exists a non-conservative weak solution

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V2-067152 | MATHDB-349727

The local-global compatibility conjecture for mod p representations of GL_n

摘录 / EXCERPT: Let $\overline{r}:{\operatorname{Gal}}(\overline F/F)\rightarrow {\operatorname{GL}}_n(\mathbb{F})$ be a continuous homomorphism satisfying conditions (i)-(iii) in the paper, and l...

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V2-067153 | MATHDB-349732

Almost High Rank Conjecture for orbit closures of translation surfaces

摘录 / EXCERPT: Let $\mathcal{M}$ be an orbit closure of surfaces of genus g. Its rank is denoted by $\operatorname{rank}(\mathcal{M})$. Almost High Rank Conjecture. If ... then $\mathcal{M}$ is a...

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V2-067154 | MATHDB-349733

The type A arrangement pizza-sharing conjecture

摘录 / EXCERPT: Let $\mathcal{H}$ be a hyperplane arrangement of type A_n in the n-dimensional space $\{(x_1,x_2,\ldots,x_{n+1}): x_1+x_2+\cdots+x_{n+1}=0\}$, and let $a\in V$ satisfy...

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V2-067155 | MATHDB-349737

Nonexistence conjecture for girth-regular graphs with signatures 32^21 and 4^33

摘录 / EXCERPT: Let Γ be a girth-regular graph with signature 32^21 or 4^33, and let egc mean an edge-girth coloring. Nonexistence conjecture. Graphs Γ with signatures 32^21 an...

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V2-067156 | MATHDB-349738

Klein-bottle prism conjecture for hexagonal tessellations

摘录 / EXCERPT: Let Γ be a prism of a hexagonal tessellation of the Klein bottle, and suppose its cutout contains an $m\times n$ array of tiles, with m and n even. Assume that both type...

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V2-067157 | MATHDB-349739

The toroidal choice-chromatic gap conjecture for 3-chromatic graphs

摘录 / EXCERPT: Let $\jump(g,r)$ denote the maximum of the difference between list-chromatic number and chromatic number over graphs embeddable on the orientable surface of genus g with chromati...

conjecture | open | MathDB 页面 | 原文线索

V2-067158 | MATHDB-349740

Unimodality conjecture for the choice-chromatic gap

摘录 / EXCERPT: Let $\jump(g,r)$ be the maximum choice-chromatic gap among graphs embeddable on the orientable surface of genus g with chromatic number r, and let $H(g)$ be the maximum chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-067159 | MATHDB-349742

Stronger variable reduction conjecture for monomial ideals with linear quotients

摘录 / EXCERPT: Let K be a field, let $S=K[x_1,x_2,\ldots,x_n]$, let $I\subset S$ be a proper monomial ideal with linear quotients, and write $[n]=\\{1,\ldots,n\\}$. The stronger variable reduct...

conjecture | open | MathDB 页面 | 原文线索

V2-067160 | MATHDB-349750

Large-degree Pisot Bernoulli convolution dimension conjecture

摘录 / EXCERPT: Given a number $\beta\in(1,2)$, let ν_{β} denote the associated Bernoulli convolution and let $\dim_H$ denote Hausdorff dimension. Large-degree Pisot dimension conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-067161 | MATHDB-349761

Quantitative rigidity conjecture for global minimal-surface-type equations

摘录 / EXCERPT: Let u be a global solution of an equation of minimal-surface type on $\mathbb{R}^n$, corresponding to a functional A_{Ψ}. Here B_r denotes the radius-r ball, ∇u...

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V2-067162 | MATHDB-349763

p-adic L-function conjecture for the elliptic K3 families

摘录 / EXCERPT: Elliptic-family p-adic L-function conjecture. There is a constant $C_{n,a}\in {\mathbb Q}^{\times}$ independent of p such that

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V2-067163 | MATHDB-349766

The list multiplicative 1-2-3 bound

摘录 / EXCERPT: Let G be a nice graph, and let $\operatorname{ch}_\Pi^*(G)$ denote the list parameter for the multiplicative labelling problem defined in the paper. The list multiplicative 1-2-3...

open_problem | open | MathDB 页面 | 原文线索

V2-067164 | MATHDB-349768

The nonnegative Laguerre-integral conjecture

摘录 / EXCERPT: Let $n\ge1$, $\alpha=(\alpha_1,\dots,\alpha_n)\in\mathbb N^n$, and $a=(a_1,\dots,a_n)\in(0,+\infty)^n$ be as in the Laguerre-integral conjecture, and let $K_\alpha(a)$ denote the c...

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V2-067165 | MATHDB-349779

DF linearization conjecture for Goursat bundles not ESFT-equivalent to Brunovský normal form

摘录 / EXCERPT: Let a control system with at least two controls be a Goursat bundle, and suppose it cannot be transformed to Brunovský normal form via an ESFT. DF linearization conjecture. The con...

conjecture | open | MathDB 页面 | 原文线索

V2-067166 | MATHDB-349780

Characterization conjecture for ESFL reductions by codimension 1 partial contact curves

摘录 / EXCERPT: Let $p(z)$ denote the function associated with an ESFL reduction by a codimension 1 partial contact curve, and let the truncated Euler operator and the higher-order Euler operato...

conjecture | open | MathDB 页面 | 原文线索

V2-067167 | MATHDB-349781

Codimension-one reduction conjecture for ESFL contact sub-connections

摘录 / EXCERPT: Let a contact sub-connection admit an ESFL reduction by a partial contact curve of codimension $s>1$. Codimension-one reduction conjecture. The contact sub-connection must also adm...

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V2-067168 | MATHDB-349782

ESFL reduction conjecture for one-dimensional symmetry groups

摘录 / EXCERPT: Let a contact sub-connection have a control admissible symmetry group G with $\dim G=1$. One-dimensional symmetry reduction conjecture. If the contact sub-connection is EDFL by p...

conjecture | open | MathDB 页面 | 原文线索

V2-067169 | MATHDB-349783

EDFL implies CFL conjecture for systems with control admissible symmetries

摘录 / EXCERPT: Let a control system possess a control admissible symmetry group. EDFL-to-CFL conjecture. If the control system is EDFL, then it must be CFL. The conjecture concerns the relation b...

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V2-067170 | MATHDB-349795

Weyl-equivariant splitting conjecture for saturated covers

摘录 / EXCERPT: Let $\overline{G}_0$ be a saturated cover, let $\mathscr{X}_{Q,n}$ be the relevant lattice, and let $\mathscr{X}_{0,Q,n}$ be the Weyl-stable sublattice. Let W act by the usual We...

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V2-067171 | MATHDB-349796

Canonical splitting and unramifiedness compatibility conjecture

摘录 / EXCERPT: Let $\mathcal{K}$ be the set of relevant subgroups, let H_φ be the image defined in the preceding diagrams, and let $\phi_\overline{\overline{\overline{\overline{\chi}}}}$...

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V2-067172 | MATHDB-349797

Whittaker multiplicity invariance under torus twisting

摘录 / EXCERPT: Let $\tilde{\varphi}_0:X_{Q,n}/X_{Q,n}^{sc}\to X/X_G^{sc}$ be well-defined, let $\mathfrak{w}=(B,U,\psi)$ be Whittaker data, and let...

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V2-067173 | MATHDB-349799

The simple current conjecture for modular tensor categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a modular tensor category, let $\mathcal{C}^{\operatorname{ad}}$ be its adjoint subcategory, and let $\mathcal{F}$ be an auto-equivalence of $\mathcal{C}$ rest...

conjecture | open | MathDB 页面 | 原文线索

V2-067174 | MATHDB-349839

The constant-factor equivalence conjecture for partition and analytic rank

摘录 / EXCERPT: Constant-factor equivalence conjecture. For every $k ge 2$ there is a constant $C=C(k)$ such that, for every finite field F and every k-tensor T over F,

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V2-067175 | MATHDB-349851

Inclusion-exclusion positivity conjecture for Schubert polynomial specializations

摘录 / EXCERPT: Let $w\in S_n$ and let u be a subword of w. For words $u\leq v\leq w$, write $\lvert w\rvert$ for word length, and let $\mathfrak{S}_{\operatorname{perm}(v)}({\bf 1})$ denote t...

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V2-067176 | MATHDB-349853

Regularity conjecture for the flip graph of generalized snake order polytopes

摘录 / EXCERPT: Let $\mathbf{w} in \mathcal{V}$, and consider the flip graph whose vertices are the regular triangulations of the order polytope $\mathcal{O}(Q_{\mathbf{w}})$. Let k denote the d...

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V2-067177 | MATHDB-349854

Noncanonical dual-graph conjecture for generalized snake order polytopes

摘录 / EXCERPT: Let $Q_{\mathbf{w}}$ be a poset with Jordan--Hölder poset $J(Q_{\mathbf{w}})=\widehat{P}(\mathbf{w})$, and let a turn mean the specified local turn in the generalized snake poset....

conjecture | open | MathDB 页面 | 原文线索

V2-067178 | MATHDB-349855

Nearly-ladder dual-graph counting conjecture for order polytopes

摘录 / EXCERPT: Assume $J(Q_{\mathbf{w}})=\widehat{P}(\mathbf{w})$, where $\mathbf{w}=varepsilon L R^{n-2}$ and $ngeq 3$. Consider triangulations of $\mathcal{O}(Q_{\mathbf{w}})$ and their dual gr...

conjecture | open | MathDB 页面 | 原文线索

V2-067179 | MATHDB-349856

Regularity conjecture for all triangulations of generalized snake order polytopes

摘录 / EXCERPT: Let $\mathbf{w} in \mathcal{V}$, and let $\mathcal{O}(Q_{\mathbf{w}})$ be the associated order polytope. A triangulation is regular if it arises from a lifting function, equivalent...

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V2-067180 | MATHDB-349857

Odd-Catalan enumeration conjecture for regular triangulations of the permutohedron order polytope

摘录 / EXCERPT: Let S_n denote the poset used in the paper, and let $\mathcal{O}(S_n)$ be its order polytope. Let $\operatorname{Cat}(m)$ denote the mth Catalan number. Odd-Catalan enumeration...

conjecture | open | MathDB 页面 | 原文线索

V2-067181 | MATHDB-349859

Bern-Carrasco-Johansson colour-kinematics duality conjecture

摘录 / EXCERPT: Bern-Carrasco-Johansson colour-kinematics duality conjecture. There exists a choice of kinematic numerators for the trivalent diagrams entering $\scA_{n,L}$ such that whenever a tr...

conjecture | open | MathDB 页面 | 原文线索

V2-067182 | MATHDB-349864

The generic bigraded Hilbert-series conjecture for ideals in a product of projective lines

摘录 / EXCERPT: Let $S=k[x_0,x_1,y_0,y_1]$ with its standard bigrading, and let $I=(f_1,\ldots,f_r)\subset S$, where the f_i are generic bihomogeneous forms of degree (d_i,e_i). For a formal p...

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V2-067183 | MATHDB-349866

Relative fundamental lemma for U(n) × U(n)

摘录 / EXCERPT: Relative fundamental lemma. For every regular semisimple orbit (ζ,y), the normalized orbital integral with the characteristic functions of $\mathrm{S}_n(O_F)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067184 | MATHDB-349868

Regular truncated-octahedron conjecture for unit-volume parallelohedra

摘录 / EXCERPT: Let P be a three-dimensional parallelohedron of unit volume, and let $\operatorname{surf}(P)$ denote its surface area. Regular truncated-octahedron conjecture. Among three-dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-067185 | MATHDB-349871

Abelian-surface multiple-cover conjecture

摘录 / EXCERPT: Let A be an abelian surface, let $\beta\in H_2(A,\mathbb{Z})$ be effective, and for each divisor $k\mid\beta$ choose an abelian surface A_k and a parallel-transport morphism...

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V2-067186 | MATHDB-349872

Large expander conjecture for high-genus triangulations

摘录 / EXCERPT: Let g_n satisfy ... and let $\mathbf{T_{n,g_n}}$ be a uniform triangulation of genus g_n with $3n$ edges. An induced subgraph is a κ-expander if every set of vertices has...

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V2-067187 | MATHDB-349873

Generalized recurrent integral formula for indefinite integrals

摘录 / EXCERPT: Let $f_1,\ldots,f_n$ be functions of x. For $n\in\mathbb{N^*}$, let S_n be the permutation group on $\{1,\ldots,n\}$, and let $C_{n,k}$ be the set of k-element subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-067188 | MATHDB-349874

Generalized recurrent integral formula for definite integrals

摘录 / EXCERPT: Let $f_1,\ldots,f_n$ be functions defined on the interval $[a,b]$. Let S_n be the permutation group on $\{1,\ldots,n\}$, and for $\sigma\in S_n$ write $\sigma(i)$ for its i-th...

conjecture | open | MathDB 页面 | 原文线索

V2-067189 | MATHDB-349876

Folklore conjecture on nonvanishing of quadratic twists

摘录 / EXCERPT: Let E be any elliptic curve over ${\mathbb Q}$, let r be a positive integer, and let $E^{(M)}$ denote the quadratic twist of E by M. A square-free integer has exactly r p...

conjecture | open | MathDB 页面 | 原文线索

V2-067190 | MATHDB-349887

Unramified multiplet-frequency conjecture for totally real cubic fields

摘录 / EXCERPT: Let $K=\mathbb{Q}(\sqrt{d})$ be a quadratic field with 3-class rank ϱ, and let the associated unramified nilets, singlets, and quartets be classified by their types. The...

conjecture | open | MathDB 页面 | 原文线索

V2-067191 | MATHDB-349896

Finite global attractor conjecture for pullbacks of curves under rational Thurston maps

摘录 / EXCERPT: Let f be a rational Thurston map, and consider the pullback relation on nontrivial curves associated to f. A finite global attractor conjecture asserts that, if f is not a fl...

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V2-067192 | MATHDB-349902

Agashe's divisibility conjecture for quadratic twists of optimal elliptic curves

摘录 / EXCERPT: Agashe's conjecture. Up to a power of 2, the square of the order of $E^{-D}(\mathbb{Q})$ divides

conjecture | open | MathDB 页面 | 原文线索

V2-067193 | MATHDB-349906

Massive CLE2 boundary conjecture for the canonical evaporated UST component

摘录 / EXCERPT: Condition on the event $|{\bf t}(r)|=n$, and let ${\sf Canonical}({\bf t}^{<n}(r))$ be the canonical embedding of the root component obtained by edge evaporation. The massive...

conjecture | open | MathDB 页面 | 原文线索

V2-067194 | MATHDB-349907

Finite-torus approximation conjecture for the DLA tree

摘录 / EXCERPT: Let ${\sf TDLA}_0(n,{ \sf Torus}(n^c))$ denote the finite-graph DLA tree with n vertices rooted at 0, and let ${\bf dla}_n$ denote the standard DLA tree with n vertices. The...

conjecture | open | MathDB 页面 | 原文线索

V2-067195 | MATHDB-349909

Square-root width exponent conjecture for uniform spanning trees

摘录 / EXCERPT: Let t_n be a uniformly sampled tree of size approximately n in the square-torus uniform spanning-tree model, and let $w(t_n)$ be its Euclidean width. The UST width exponent con...

conjecture | open | MathDB 页面 | 原文线索

V2-067196 | MATHDB-349910

Infinite Burr-Erdős conjecture for positive-density monochromatic copies

摘录 / EXCERPT: A graph G is d-degenerate if it has an ordering of its vertices $v_1,v_2,\ldots$ such that ... for every $i\geq 1$. The upper density of a set $V\subseteq\mathbb N$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-067197 | MATHDB-349911

Kelly-Kühn-Osthus conjecture on directed cycle semidegree thresholds

摘录 / EXCERPT: An oriented graph is a directed graph without loops or multiple edges. For an oriented graph G, let $\delta^\pm(G)$ be its minimum semidegree, the smaller of its minimum indegree...

conjecture | open | MathDB 页面 | 原文线索

V2-067198 | MATHDB-349912

Outdegree threshold conjecture for directed cycles

摘录 / EXCERPT: An oriented graph is a directed graph without loops or multiple edges. For an oriented graph G, let $\delta^+(G)$ denote its minimum outdegree, and let C_ℓ denote a directed...

conjecture | open | MathDB 页面 | 原文线索

V2-067199 | MATHDB-349913

ALE scaling-limit conjecture for boundary-touching curves in free boundary dimers

摘录 / EXCERPT: For any $z>0$, superimpose two independent free boundary dimer models in the upper half-plane, and consider the resulting collection of boundary-touching curves. Free-boundary dime...

conjecture | open | MathDB 页面 | 原文线索

V2-067200 | MATHDB-349914

ALE scaling-limit conjecture for the folded dimer model on \(\hat{\mathcal G}\)

摘录 / EXCERPT: Let $\hat{\mathcal G}$ be obtained by adjoining to a graph $\mathcal G$ its reflection across the real axis while retaining one copy of vertices on the real axis. Let M be an inf...

conjecture | open | MathDB 页面 | 原文线索

V2-067201 | MATHDB-349916

Conjectured universally tight lower bound for personalized federated bandits

摘录 / EXCERPT: Let M clients play a multi-armed bandit for T time slots. For client m, let $k'_{*,m}$ denote its optimal arm, let $X'_{k,m}$ denote the reward distribution of arm k for cl...

conjecture | open | MathDB 页面 | 原文线索

V2-067202 | MATHDB-349921

Categorical local Langlands conjecture for the Fargues-Fontaine curve

摘录 / EXCERPT: Let G be a quasisplit reductive group over E, let $B\subset G$ be a Borel subgroup with unipotent radical U, and let $\psi:U(E)\to\mathcal O_L^\times$ be a generic character,...

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V2-067203 | MATHDB-349923

Categorical local Langlands conjecture over $\overline{\mathbb Q}_\ell$

摘录 / EXCERPT: Let G be a quasisplit reductive group over E, choose Whittaker data consisting of a Borel $B\subset G$ with unipotent radical U and a generic character...

conjecture | open | MathDB 页面 | 原文线索

V2-067204 | MATHDB-349924

Categorical local Langlands conjecture for elliptic parameters

摘录 / EXCERPT: Let G be quasisplit over E, fix a Whittaker datum, and assume that the connected split center of G is trivial. Let φ be an elliptic L-parameter, let S_φ be...

conjecture | open | MathDB 页面 | 原文线索

V2-067205 | MATHDB-349931

The strong-generation characterization of abelian envelopes

摘录 / EXCERPT: Let $\mathbf{D}\subset\mathbf{T}$ be a pseudo-tensor subcategory of a tensor category. It is strongly generating if every object in $\mathbf{T}$ is a quotient of an object in...

conjecture | open | MathDB 页面 | 原文线索

V2-067206 | MATHDB-349936

A conjectured upper bound for the iterated Mycielski L(2,1)-labeling number

摘录 / EXCERPT: Iterated Mycielski labeling conjecture. For all $t\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-067207 | MATHDB-349940

The conjecture that smeariness begets finite sample smeariness

摘录 / EXCERPT: Let X be a random variable on a metric space. A distribution is smeary when its Fréchet mean has a limiting convergence rate slower than the classical $n^{-1/2}$ rate, and it has...

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V2-067208 | MATHDB-349942

Type I finite sample smeariness for random variables on spheres

摘录 / EXCERPT: Let X be a random variable supported on a set $A \subset \mathbb{S}^m$ whose convex closure has nonzero volume and which has a unique mean μ. A point μ is Type I finite s...

conjecture | open | MathDB 页面 | 原文线索

V2-067209 | MATHDB-349943

Non-negativity conjecture for absolutely indecomposable representation polynomials

摘录 / EXCERPT: Let Γ be a quiver, let $\mathbb{F}_q$ be a finite field, and suppose that for every $\alpha\in\mathbb{N}^n$ there is a polynomial $r(\alpha,q)\in\mathbb{Z}[q]$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-067210 | MATHDB-349954

Second-derivative conjecture for the triple-product p-adic L-function

摘录 / EXCERPT: Let E be the elliptic curve and H the number field in the running assumptions, let $V_{gh}^{\vee}$ be the contragredient representation of V_{gh}, and let P,Q be generators...

conjecture | open | MathDB 页面 | 原文线索

V2-067211 | MATHDB-349961

Baker-Wang conjecture on Bernardi and rotor-routing torsors

摘录 / EXCERPT: Baker-Wang conjecture. The torsors b_v and r_v agree for all vertices v if and only if G is planar.

conjecture | open | MathDB 页面 | 原文线索

V2-067212 | MATHDB-349966

Eigenvalue formula conjecture for the first Hecke operator of SL_n

摘录 / EXCERPT: Let χ be an SL_n-oper with real monodromy. Let $(\mathcal V_{\omega_1},\nabla_\chi)$ be its associated flat vector bundle, with determinant identified with the trivial flat...

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V2-067213 | MATHDB-349975

Kim et al.'s vacuum expectation value conjecture for principal-series operators

摘录 / EXCERPT: Let the theory satisfy the assumptions of Proposition, and let $\mathcal{O}_{h_\star}$ be the operator with scaling dimension $h_\star$. In the true vacuum of the theory, Kim et al...

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V2-067214 | MATHDB-349983

Gross's conjecture on the Brumer-Stark reciprocity law

摘录 / EXCERPT: Let F, H, S, T, and $\mathfrak{p}$ be as above, and let L/F be a finite abelian CM extension containing H and unramified outside...

conjecture | open | MathDB 页面 | 原文线索

V2-067215 | MATHDB-349992

Majda's global regularity conjecture for totally linearly degenerate systems

摘录 / EXCERPT: Let $d\geq 2$ and consider a d-dimensional nonlinear symmetric system that is totally linearly degenerate. Suppose the initial data belong to $H^s(\mathbb{R}^d)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-067216 | MATHDB-349993

Klein-Saut conjecture on blow-up and global existence for dispersive Burgers equations

摘录 / EXCERPT: Klein-Saut conjecture. The following assertions hold:

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V2-067217 | MATHDB-349995

Nonnegativity conjecture for flag Hilbert-Poincaré numerators of hyperplane arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a hyperplane arrangement, and let $\mathcal{N}_{\mathcal{A}}(Y,T)\in\mathbb{Z}[Y,T]$ denote its flag Hilbert-Poincaré numerator. Nonnegativity conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-067218 | MATHDB-349997

Fifth-moment rank bias conjecture for elliptic-curve families

摘录 / EXCERPT: Fifth-moment rank bias conjecture. The average value of the main term of the fifth moment is $-5rp^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-067219 | MATHDB-349998

Catalan-number conjecture for odd moments of elliptic-curve families

摘录 / EXCERPT: Catalan-number odd-moment conjecture. The average value of the main term of the $(2k+1)$st moment is

conjecture | open | MathDB 页面 | 原文线索

V2-067220 | MATHDB-350002

Alzer-Qiu's monotonicity and convexity conjecture for the elliptic-integral ratio

摘录 / EXCERPT: Alzer-Qiu's conjecture. The function G is strictly increasing and convex from $(0,1)$ onto $(3/4,1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067221 | MATHDB-350003

Monotonicity conjecture for the normalized elliptic-integral approximation error

摘录 / EXCERPT: Monotonicity conjecture for f. The function f is strictly increasing from $(0,1)$ onto $(1/320,1/4)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067222 | MATHDB-350004

Conjectures on convexity and absolute monotonicity of auxiliary elliptic-integral functions

摘录 / EXCERPT: Auxiliary-function conjectures. (1) The function f is convex on $(0,1)$. (2) The functions h_{11} and h_{12} are both strictly increasing and convex on $(0,1)$, with ranges...

conjecture | open | MathDB 页面 | 原文线索

V2-067223 | MATHDB-350026

Pixton-class conjecture for the noncommutative KdV hierarchy

摘录 / EXCERPT: Pixton-class conjecture. The function u^P satisfies the full noncommutative KdV hierarchy.

conjecture | open | MathDB 页面 | 原文线索

V2-067224 | MATHDB-350027

The smoothing-scale conjecture for zeta maxima

摘录 / EXCERPT: Let $T>1$, let τ be chosen uniformly from $[T,2T]$, and let $0<\alpha<1$ with $\theta\sim(\log\log T)^{-\alpha}$. Consider an interval of size ... The smoothing-scale conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-067225 | MATHDB-350038

Joint equidistribution conjecture for rational subspaces and their shapes

摘录 / EXCERPT: Joint equidistribution conjecture. These sets equidistribute in $\mathrm{Gr}_{n,k}(\mathbb{R})\times\mathcal{S}_k\times\mathcal{S}_{n-k}$ with respect to the product of the invaria...

conjecture | open | MathDB 页面 | 原文线索

V2-067226 | MATHDB-350039

Equidistribution conjecture for moduli of basis extensions

摘录 / EXCERPT: Let $L\subset\mathbb{R}^n$ be a k-dimensional subspace and let Λ be a full-rank lattice such that $L\cap\Lambda$ is a lattice in L. Let $[L,\Lambda]$ denote the equival...

conjecture | open | MathDB 页面 | 原文线索

V2-067227 | MATHDB-350040

Joint equidistribution conjecture for primitive restriction forms

摘录 / EXCERPT: Let Q be a positive definite integral quadratic form, let L be a rational k-dimensional subspace, and let q_L and $q_{L^\perp}$ be the associated integral quadratic forms o...

conjecture | open | MathDB 页面 | 原文线索

V2-067228 | MATHDB-350051

The integer-finite local-system conjecture in intersection K-theory

摘录 / EXCERPT: Integer-finite local-system conjecture. Then L is an integer finite local system, and there is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-067229 | MATHDB-350058

A q-supercongruence from Rahman's summation formula

摘录 / EXCERPT: Let $n \equiv 1\pmod{4}$ be a positive integer. Define the q-integer $[m]=(1-q^m)/(1-q)$, the q-shifted factorial $(a;q)_k=\prod_{j=0}^{k-1}(1-aq^j)$, and let $\Phi_n(q)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-067230 | MATHDB-350067

Non-D-finiteness conjecture for the generating function of $\mathcal{A}v(A_{5,1})$

摘录 / EXCERPT: Non-D-finiteness conjecture. The generating function for $\mathcal{A}v(A_{5,1})$ is not D-finite.

conjecture | open | MathDB 页面 | 原文线索

V2-067231 | MATHDB-350078

Conjecture on the left adjoint morphisms in Ord-enriched partial-map categories

摘录 / EXCERPT: Let $\mathcal{U}$ be the class of morphisms satisfying $\operatorname{id}_{A}\leq f^*\cdot f$ for some f^*, and let $\mathcal{U}^{\mathrel{\ooalign{{\wedge}\cr\hfil$\mid$\hfil}}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-067232 | MATHDB-350079

Conjecture on the oplax weak orthogonal complement of split epimorphisms

摘录 / EXCERPT: Let LI denote the class of split epimorphisms, and let $\overline{\mathcal{S}}$ denote the class of morphisms whose associated comparison cell is an isomorphism. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-067233 | MATHDB-350086

Multiplicity equality conjecture for elliptic A-parameters

摘录 / EXCERPT: Let G be a unitary group or an even orthogonal group, and let ... be an elliptic A-parameter for G. Let π be an irreducible representation of $G(\mathbb A)$ such that the...

conjecture | open | MathDB 页面 | 原文线索

V2-067234 | MATHDB-350089

Gross-Pavanelli mirror-symmetry conjecture for quotients of V_{8,w}^{1}

摘录 / EXCERPT: Let $V_{8,w}^{1}$ be the Calabi-Yau threefold fibered by $(1,8)$-polarized abelian surfaces, with a free action of $\mathbb{Z}_{8}\times\mathbb{Z}_{8}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067235 | MATHDB-350091

Pouzet's 2-wqo conjecture for permutation classes

摘录 / EXCERPT: For a permutation class $\mathcal{C}$ and an integer $n\geq 1$, say that $\mathcal{C}$ is n-well-quasi-ordered (or n-wqo) if the set of permutations in $\mathcal{C}$ labeled by...

conjecture | open | MathDB 页面 | 原文线索

V2-067236 | MATHDB-350092

The lwqo consequence of wqo for one-point extensions of permutation classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a permutation class, and let $\mathcal{C}^{+1}$ denote its one-point extension. A class is lwqo when it is well-quasi-ordered under labeling by arbitrary wqo l...

conjecture | open | MathDB 页面 | 原文线索

V2-067237 | MATHDB-350093

The 2-wqo one-point-extension conjecture for permutation classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a permutation class, and let $\mathcal{C}^{+1}$ denote its one-point extension. The 2-wqo one-point-extension conjecture. If $\mathcal{C}$ is 2-wqo, then...

conjecture | open | MathDB 页面 | 原文线索

V2-067238 | MATHDB-350094

The substitution-closure lwqo conjecture for permutation classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a permutation class, and let $\langle\mathcal{C}\rangle$ denote its substitution closure. The substitution-closure conjecture. If $\langle\mathcal{C}\rangle$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-067239 | MATHDB-350095

Huczynska-Vatter conjecture on finite bases of monotone grid classes

摘录 / EXCERPT: A monotone grid class is a permutation class defined by a finite grid whose nonempty cells contain monotone permutations. Huczynska-Vatter's conjecture. Every monotone grid class i...

conjecture | open | MathDB 页面 | 原文线索

V2-067240 | MATHDB-350104

Positive-relator quotient conjecture for reductions to amenable groups

摘录 / EXCERPT: Positive-relator quotient conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067241 | MATHDB-350110

Conjectural minimality criterion for Veech extensions with $m=\bar m n+1$

摘录 / EXCERPT: Let $0<\alpha<1$ be irrational, let T be the rotation by angle α, viewed as a two-interval exchange with discontinuity $1-\alpha$, and let T_f be the Veech extension fro...

conjecture | open | MathDB 页面 | 原文线索

V2-067242 | MATHDB-350125

Orthogonality conjecture for Macdonald complexes

摘录 / EXCERPT: Let $\sEM_{\lambda}$ denote the conjectured Macdonald complex associated with an integral weight λ, and for an integral dominant weight λ let...

conjecture | open | MathDB 页面 | 原文线索

V2-067243 | MATHDB-350129

Nešetřil-Ossona de Mendez strong mass-transport conjecture for nowhere-dense classes

摘录 / EXCERPT: Nešetřil-Ossona de Mendez's strong mass-transport conjecture. Every first-order convergent sequence of graphs from ${\cal C}$ has a modeling limit that satisfies the strong finitar...

conjecture | open | MathDB 页面 | 原文线索

V2-067244 | MATHDB-350133

VC-dimension sunflower conjecture

摘录 / EXCERPT: Let $f^d_r(k)$ be the least positive integer m such that every family $\mathcal F$ of k-sets with $|\mathcal F|\geq m$ and $\text{VC-dim}(\mathcal F)\leq d$ contains an r-sun...

conjecture | open | MathDB 页面 | 原文线索

V2-067245 | MATHDB-350138

Boundary-sum conjecture for the coefficients of generalized hyperharmonic numbers

摘录 / EXCERPT: Let $a(r,m,\ell)$ be the coefficients appearing in the expansion under consideration. Boundary-sum conjecture. For $r\in\mathbb N$, ... and ... This is one of four conjectures suggeste...

conjecture | open | MathDB 页面 | 原文线索

V2-067246 | MATHDB-350139

Sign conjecture for the coefficients of generalized hyperharmonic numbers

摘录 / EXCERPT: Let $a(r,m,\ell)$ be the coefficients appearing in the expansion under consideration, and let $\operatorname{sgn}(x)$ be the signum function: ... For $r,m\in\mathbb N$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-067247 | MATHDB-350140

Recursive Gittins-index conjecture for sequential multistage jobs

摘录 / EXCERPT: Let S_j be the service time of stage j in a sequential multistage job with J stages, let $G(j,a)$ denote the Gittins index when the job is at stage j with attained service...

conjecture | open | MathDB 页面 | 原文线索

V2-067248 | MATHDB-350142

Las Vergnas-Meyniel conjecture on Kempe equivalence and graph minors

摘录 / EXCERPT: For a graph G, a t-colouring is a proper colouring using t colours, and two colourings are Kempe equivalent when they are connected by a sequence of Kempe changes, each swapp...

conjecture | open | MathDB 页面 | 原文线索

V2-067249 | MATHDB-350144

Total domatic number conjecture for Cartesian products

摘录 / EXCERPT: Total domatic product conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067250 | MATHDB-350145

Domatic number conjecture for Cartesian products

摘录 / EXCERPT: Domatic product conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067251 | MATHDB-350146

Equitable chromatic number conjecture for Pancake graphs

摘录 / EXCERPT: Pancake graph equitable coloring conjecture. For every $n\geqslant 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-067252 | MATHDB-350154

Generalized Perrin-Riou conjecture for an elliptic curve over the cyclotomic extension

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve, let $r={\rm rank}(E(\mathbb{Q}))$, let I_∞ be the augmentation ideal associated with the cyclotomic $\mathbb{Z}_p$-extension...

conjecture | open | MathDB 页面 | 原文线索

V2-067253 | MATHDB-350162

Adaptive optimality conjecture for LCB in offline Markov decision processes

摘录 / EXCERPT: Adaptive optimality conjecture. The LCB approach, together with value iteration, is adaptively optimal for solving offline MDPs for all ranges of $C^\star$.

conjecture | open | MathDB 页面 | 原文线索

V2-067254 | MATHDB-350163

Smooth-transition conjecture for episodic VI-LCB

摘录 / EXCERPT: Smooth-transition conjecture. If $1 \leq C^\pi \leq 2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067255 | MATHDB-350164

Kostochka-Nahvi-West-Zirlin tree reconstructibility conjecture

摘录 / EXCERPT: Tree reconstructibility conjecture. For $\ell\ne2$, every tree with at least $2\ell+1$ vertices is ℓ-reconstructible.

conjecture | open | MathDB 页面 | 原文线索

V2-067256 | MATHDB-350173

Rips's tame-generation conjecture in positive characteristic

摘录 / EXCERPT: Rips's conjecture. Linear automorphisms, together with ψ, generate the entire tame automorphism group of $K[x_1,\ldots,x_n]$. The characteristic-zero analogue is known, while...

conjecture | open | MathDB 页面 | 原文线索

V2-067257 | MATHDB-350176

GKS logarithm-removal conjecture for approximation numbers of Schatten embeddings

摘录 / EXCERPT: GKS's conjecture. The approximation numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067258 | MATHDB-350177

Conjectured Gelfand and approximation number bound for Schatten embeddings

摘录 / EXCERPT: The Gelfand-approximation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067259 | MATHDB-350180

Baril and Kirgizov's equidistribution conjecture for descent and excedance bistatistics

摘录 / EXCERPT: Baril and Kirgizov's conjecture. The two bistatistics $(\operatorname{des}_2,\operatorname{cyc})$ and $(\operatorname{pex},\operatorname{cyc})$ are equidistributed on...

conjecture | open | MathDB 页面 | 原文线索

V2-067260 | MATHDB-350181

Vajnovszki's equidistribution conjecture for descent and excedance bistatistics

摘录 / EXCERPT: Vajnovszki's conjecture. The two bistatistics $(\operatorname{des}_2,\operatorname{des})$ and $(\operatorname{pex},\operatorname{exc})$ are equidistributed on $\mathfrak{S}_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-067261 | MATHDB-350182

The symmetry conjecture for descent and ear statistics

摘录 / EXCERPT: Symmetry conjecture. The distribution of $(\operatorname{des}_2,\operatorname{ear})$ over permutations is symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-067262 | MATHDB-350187

Characterization of bi-UFS positive semirings

摘录 / EXCERPT: Let S be a positive semiring. Bi-UFS characterization conjecture. The semiring S is a bi-UFS if and only if ... The paper notes that $\mathbb{N}_0$ is a bi-UFS and presents no ot...

conjecture | open | MathDB 页面 | 原文线索

V2-067263 | MATHDB-350188

Nýdl's sharp threshold conjecture for tree reconstruction

摘录 / EXCERPT: For a tree T, let $\mathcal{D}_\ell(T)$ be the multiset of all induced subgraphs of T on ℓ vertices. Nýdl's sharp threshold conjecture. For any $n\geq 4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067264 | MATHDB-350192

Boundedness conjecture for canonical models with fixed Iitaka volume

摘录 / EXCERPT: Let n be a positive integer, let v be a non-negative rational number, and let $\mathcal{I}\subset [0,1]\cap \mathbb{Q}$ be a DCC set of rational numbers. For a klt pair...

conjecture | open | MathDB 页面 | 原文线索

V2-067265 | MATHDB-350195

Recurrence classification conjecture for dual multiple-orthogonal-polynomial Markov chains

摘录 / EXCERPT: Let $\boldsymbol{\kappa}_\lambda$ be the left eigenvector associated with the dual Markov chains, and let λ be the parameter used to construct them. The chains are called t...

conjecture | open | MathDB 页面 | 原文线索

V2-067266 | MATHDB-350199

Worst-case variance conjecture for the chromatic number of random graphs

摘录 / EXCERPT: Worst-case variance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067267 | MATHDB-350200

Best-case variance conjecture for the chromatic number of random graphs

摘录 / EXCERPT: Best-case variance conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067268 | MATHDB-350201

Asymptotic normality conjecture for the chromatic number of random graphs

摘录 / EXCERPT: Asymptotic normality conjecture. There are functions $f(n)$ and $g(n)$ such that, at least for good n,

conjecture | open | MathDB 页面 | 原文线索

V2-067269 | MATHDB-350202

Independent-set explanation conjecture for chromatic-number fluctuations

摘录 / EXCERPT: Independent-set explanation conjecture. There are functions $f(n)$ and $g(n)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067270 | MATHDB-350203

Four-regime variance conjecture for the chromatic number of random graphs

摘录 / EXCERPT: Four-regime variance conjecture. The following asymptotic estimates hold: (i) if $x\to0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067271 | MATHDB-350206

Lex-plus-powers conjecture for socle dimensions

摘录 / EXCERPT: Let $R=K[x_1,\ldots,x_n]$, and let I be a homogeneous ideal containing a regular sequence of degrees $2\leq a_1\leq\ldots\leq a_n$. Let L be a lex-plus-powers ideal associated...

conjecture | open | MathDB 页面 | 原文线索

V2-067272 | MATHDB-350211

KdV approximation conjecture for generalized moving pulse solutions in the FPU model

摘录 / EXCERPT: Let $m\in\mathbb{N}$ and $\tilde{c}>0$. For sufficiently small $\varepsilon>0$, consider the FPU system and the speed $c=1-\varepsilon^2\tilde{c}$. A generalized moving pulse solut...

conjecture | open | MathDB 页面 | 原文线索

V2-067273 | MATHDB-350213

McMullen's exponential-growth conjecture for bounded continued fractions

摘录 / EXCERPT: Consider the periodic continued fractions $[\overline{a_1,\ldots,a_k}]$ with entries in $\mathbb{Q}(\sqrt{5})$ satisfying $a_i\leq 2$. McMullen's exponential-growth conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-067274 | MATHDB-350214

The local-global conjecture for traces in the commutator subgroup of \(\Gamma(2)\)

摘录 / EXCERPT: Let $\Gamma(2)'$ be the commutator subgroup of the level-2 principal congruence subgroup $\Gamma(2)\subset\operatorname{SL}_2(\mathbb{Z})$, and let an integer t be admissible if,...

conjecture | open | MathDB 页面 | 原文线索

V2-067275 | MATHDB-350216

The commutator-width conjecture for traces in \(\Gamma(2)'\)

摘录 / EXCERPT: Let $\Gamma(2)'$ be the commutator subgroup of $\Gamma(2)$, and let $\operatorname{Tr}(\Gamma(2)')$ denote its trace set. The commutator width of an element is the minimum number o...

conjecture | open | MathDB 页面 | 原文线索

V2-067276 | MATHDB-350217

The local-global multiplicity conjecture for traces in \(\Gamma(2)'\)

摘录 / EXCERPT: The local-global multiplicity conjecture. If $t\asymp X$ is admissible, then

conjecture | open | MathDB 页面 | 原文线索

V2-067277 | MATHDB-350221

The conditional Poincaré inequality inheritance conjecture for filters

摘录 / EXCERPT: Conditional Poincaré inequality inheritance conjecture. Suppose the standard form of the Poincaré inequality holds for X with constant c_0, and suppose ${\cal Z}$ is obtained u...

conjecture | open | MathDB 页面 | 原文线索

V2-067278 | MATHDB-350228

The finitely generated subgroup embedding conjecture for Thompson's group F

摘录 / EXCERPT: Let $G\leq \mathrm{PL}_+ I$ be a finitely generated subgroup. An F-obstruction is a pair of elements of G forming an F-obstruction in the sense of Theorem. Embedding conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-067279 | MATHDB-350263

Tilting-bundle equivalence for the heart of the exotic t-structure on Grassmannians

摘录 / EXCERPT: Tilting-bundle equivalence conjecture. Under this derived equivalence, the restriction of the functor to the heart of the exotic t-structure gives an equivalence of abelian categor...

conjecture | open | MathDB 页面 | 原文线索

V2-067280 | MATHDB-350265

Nonexistence conjecture for even-order deeply critical oriented cliques

摘录 / EXCERPT: Let n be the order of a deeply critical oriented clique, meaning an oriented graph $\overrightarrow{G}$ that is an absolute oriented clique and satisfies ... for every arc xy of...

conjecture | open | MathDB 页面 | 原文线索

V2-067281 | MATHDB-350269

Instability conjecture for traveling waves with decreasing myosin mass

摘录 / EXCERPT: Let M denote the total myosin mass of a traveling wave solution and let V denote its velocity. Consider parameter ranges in which M decreases as a function of V for small...

conjecture | open | MathDB 页面 | 原文线索

V2-067282 | MATHDB-350285

Rigidity conjecture for Eikonal solutions with integrable entropy productions

摘录 / EXCERPT: Let $[...].$

conjecture | open | MathDB 页面 | 原文线索

V2-067283 | MATHDB-350297

General weighted higher-order Hardy inequality with a boundary weight

摘录 / EXCERPT: Let B_R be the ball of radius R centred at the origin in $\mathbb{R}^N$, let $k\geq 2$, let $1<p<N/k$, let $0<\gamma\leq (N-kp)/(kp-1)$, and let $u\in C_c^k(B_R)$. Weighted hig...

conjecture | open | MathDB 页面 | 原文线索

V2-067284 | MATHDB-350298

Sharp higher-order Hardy inequality with distance to the boundary

摘录 / EXCERPT: Let B_R be the ball of radius R in $\mathbb{R}^N$, let $1<p<\infty$, let $k\geq 2$, and let $u\in C_c^k(B_R)$. Write $\operatorname{dist}(x,\partial B_R)$ for the distance from...

conjecture | open | MathDB 页面 | 原文线索

V2-067285 | MATHDB-350299

Critical logarithmic higher-order Hardy inequality

摘录 / EXCERPT: Let B_R be the ball of radius R in $\mathbb{R}^N$, let $N>k\geq 2$, and let $u\in C_c^k(B_R)$. Critical logarithmic Hardy conjecture. The inequality ... holds. Moreover, the cons...

conjecture | open | MathDB 页面 | 原文线索

V2-067286 | MATHDB-350300

Presentability of limit- and filtered-colimit-closed full subcategories

摘录 / EXCERPT: Let $\mathcal{C}$ be a presentable ∞-category and let $\mathcal{C}_0\subseteq\mathcal{C}$ be a full subcategory. Suppose that $\mathcal{C}_0$ is closed under small limits an...

conjecture | open | MathDB 页面 | 原文线索

V2-067287 | MATHDB-350314

Weak variance conjecture for the spectral norm of Gaussian matrices

摘录 / EXCERPT: Weak variance conjecture. The expected spectral norm satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067288 | MATHDB-350319

Dyer-Hohlweg bijection conjecture for low elements and small inversion sets

摘录 / EXCERPT: Let (W,S) be a Coxeter system with root system $cPhi$, positive roots $cPhi^+$, and inversion set $N(w)$ for $w eqthinspacecin W$. A positive root $a eqthinspacecincPhi^+$ is sma...

conjecture | open | MathDB 页面 | 原文线索

V2-067289 | MATHDB-350340

The modularity conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over the rational numbers. The modularity conjecture. Every elliptic curve is modular. This is a central theorem in the arithmetic of elliptic curves a...

conjecture | open | MathDB 页面 | 原文线索

V2-067290 | MATHDB-350353

The characterization conjecture for infinite trees with finite locating chromatic number

摘录 / EXCERPT: Let T be an infinite tree with bounded degree. Define $f(T)$ by removing all end-paths of T except the end-branches, and for $n\geq 2$ set $f^n=f^{n-1}\circ f$. A path may be i...

conjecture | open | MathDB 页面 | 原文线索

V2-067291 | MATHDB-350358

The non-associative Moufang loop conjecture for cubic surfaces over number fields

摘录 / EXCERPT: Let $V \subset \mathbb{P}^3_L$ be a smooth cubic surface defined over a number field L. Let ν_1 and ν_2 be two different places of L. Suppose that there exists a non-as...

conjecture | open | MathDB 页面 | 原文线索

V2-067292 | MATHDB-350362

The spanning-tree bound conjecture for weighted triangle-free graphs

摘录 / EXCERPT: Let G be a weighted triangle-free graph, and let T be a spanning tree of G. Spanning-tree bound conjecture. One should have ... The conjecture would determine the optimal value...

conjecture | open | MathDB 页面 | 原文线索

V2-067293 | MATHDB-350363

The weighted subcubic triangle-free maximum-cut conjecture

摘录 / EXCERPT: Let G be a weighted triangle-free graph with maximum degree at most 3; equivalently, G is a weighted triangle-free subcubic graph. Weighted subcubic conjecture. One should ha...

conjecture | open | MathDB 页面 | 原文线索

V2-067294 | MATHDB-350378

Biclique version of the Scott-Seymour conjecture

摘录 / EXCERPT: For positive integers k and ℓ, let $K_{\ell,\ell}$ denote the complete bipartite graph with parts of size ℓ. A k-wheel is a graph consisting of an induced cycle of le...

conjecture | open | MathDB 页面 | 原文线索

V2-067295 | MATHDB-350379

The Kottwitz conjecture for unitary PEL-type Rapoport-Zink spaces

摘录 / EXCERPT: Let (G,b,μ) be a PEL-type Rapoport-Zink datum, with G a connected reductive group over $\mathbb{Q}_p$, μ a minuscule cocharacter, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067296 | MATHDB-350393

The permanent anti-concentration conjecture

摘录 / EXCERPT: Let ${\mathcal G^{k \times k}}$ be the distribution of random $k\times k$ matrices whose entries are independent complex Gaussian random variables with mean 0 and variance 1. T...

conjecture | open | MathDB 页面 | 原文线索

V2-067297 | MATHDB-350395

The permanent moment growth conjecture with fixed matrix size

摘录 / EXCERPT: Let M be a random $k\times k$ matrix with independent standard complex Gaussian entries, where $k\geq 3$ is an integer, and let $t\geq 3$ be real. The non-integer moment growth c...

conjecture | open | MathDB 页面 | 原文线索

V2-067298 | MATHDB-350396

Nick Hunter-Jones's moment conjecture for permanents of Haar unitary matrices

摘录 / EXCERPT: Let $U(d)$ denote a Haar-random $d\times d$ unitary matrix, and let t be a positive integer. Hunter-Jones's conjecture. The permanent moment satisfies ... The source attributes thi...

conjecture | open | MathDB 页面 | 原文线索

V2-067299 | MATHDB-350397

GUE Tracy-Widom fluctuation conjecture for individual sticky Brownian atoms

摘录 / EXCERPT: GUE Tracy-Widom fluctuation conjecture. There are functions $J:\mathbb{R}\to\mathbb{R}$ and $\sigma:\mathbb{R}\to\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067300 | MATHDB-350401

Tunnell's congruent number conjecture criterion

摘录 / EXCERPT: Tunnell's conjecture. The integer n is a congruent number if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-067301 | MATHDB-350402

Goldfeld's conjecture for quadratic twists of automorphic representations

摘录 / EXCERPT: Automorphic Goldfeld conjecture. Among quadratic characters χ' satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-067302 | MATHDB-350403

The Poonen-Rains rank conjecture for elliptic curves

摘录 / EXCERPT: Poonen-Rains rank conjecture. For each $r\in\{0,1\}$, 50% of elliptic curves over F have Mordell-Weil rank r.

conjecture | open | MathDB 页面 | 原文线索

V2-067303 | MATHDB-350404

The average-size conjecture for Selmer groups

摘录 / EXCERPT: Selmer average-size conjecture. The average of $\#\operatorname{Sel}_n(A/F)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-067304 | MATHDB-350409

Harrington-Henninger-Voss-Karhadkar-Robinson-Wong sum-difference index equality conjecture

摘录 / EXCERPT: Let G be a finite simple graph. Its sum index, denoted by $\operatorname{sum}(G)$, is the minimum number of distinct values of $f(u)+f(v)$ over the edges uv among injective lab...

conjecture | open | MathDB 页面 | 原文线索

V2-067305 | MATHDB-350411

Harrington et al.'s unbounded exclusive-sum-number gap conjecture

摘录 / EXCERPT: Let $\epsilon(G)$ denote the exclusive sum number of a graph: the least r such that G together with r isolated vertices has an exclusive sum labelling, meaning that the only...

conjecture | open | MathDB 页面 | 原文线索

V2-067306 | MATHDB-350412

Conjecture on the relative difference-index bound

摘录 / EXCERPT: Let G be a graph with sum index $\operatorname{sum}(G)$ and difference index $\operatorname{dif}(G)$. Relative difference-index bound conjecture. Every graph G satisfies ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-067307 | MATHDB-350414

Interior regularity conjecture for non-degenerate nonlocal mean field games

摘录 / EXCERPT: Let $2\sigma\in(0,2)$ and let $\mathcal{L}$ be a non-degenerate nonlocal operator satisfying condition (LC), let F satisfy the non-degeneracy and structural assumptions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067308 | MATHDB-350415

Schreieder-Kotschick equivalence conjecture for holomorphic 1-forms

摘录 / EXCERPT: Schreieder-Kotschick conjecture. The following three statements are equivalent: (1) X admits a global holomorphic 1-form without zeros; (2) X admits a smooth real closed 1-form...

conjecture | open | MathDB 页面 | 原文线索

V2-067309 | MATHDB-350416

Grid-shell optimality conjecture for vaults under finitely many point loads

摘录 / EXCERPT: Let f be a finite collection of point forces, written as ... A truss-like solution of $(\mathcal{P})$ is a measure of the form ... where x^-_k,x^+_k belong to...

conjecture | open | MathDB 页面 | 原文线索

V2-067310 | MATHDB-350420

Conjectured relative clique number of planar graphs of girth at least 3

摘录 / EXCERPT: Planar girth-three conjecture. For the family $\mathcal{P}_3$ of planar graphs,

conjecture | open | MathDB 页面 | 原文线索

V2-067311 | MATHDB-350421

Conjectured relative clique number of triangle-free planar graphs

摘录 / EXCERPT: Triangle-free planar conjecture. For the family $\mathcal{P}_4$ of triangle-free planar graphs,

conjecture | open | MathDB 页面 | 原文线索

V2-067312 | MATHDB-350424

Nadeau-Tewari sign conjecture for staircase-basis coefficients

摘录 / EXCERPT: Nadeau-Tewari sign conjecture. For fixed j, $a_j^{m,n}(\lambda,q)$ is a Laurent polynomial in q whose coefficients are integers of the same sign.

conjecture | open | MathDB 页面 | 原文线索

V2-067313 | MATHDB-350425

Nadeau-Tewari deletion-contraction conjecture for q-hit numbers

摘录 / EXCERPT: Nadeau-Tewari deletion-contraction conjecture. The q-hit numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067314 | MATHDB-350434

Non-APN and non-permutation conjecture for the family C_u

摘录 / EXCERPT: Non-APN and non-permutation conjecture. The function C_u is not APN. Further, if $u\neq 0$, then C_u is not a permutation.

conjecture | open | MathDB 页面 | 原文线索

V2-067315 | MATHDB-350442

Shattering-range conjecture for spherical pure p-spin glasses

摘录 / EXCERPT: Let T_s and T_{sh} denote the static and shattering transition temperatures, respectively. For a temperature T, say that the free energy landscape is (E,q,r)-shattered when...

conjecture | open | MathDB 页面 | 原文线索

V2-067316 | MATHDB-350446

Knop-Sahi positivity conjecture for normalized interpolation polynomials

摘录 / EXCERPT: Knop-Sahi positivity conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067317 | MATHDB-350453

The one-third-two-thirds conjecture for finite posets

摘录 / EXCERPT: Let $P=(X,\prec)$ be a finite poset, let $\operatorname{\mathcal{E}}(P)$ be its set of linear extensions, and write $e(P)=|\operatorname{\mathcal{E}}(P)|$. The one-third-two-thirds...

conjecture | open | MathDB 页面 | 原文线索

V2-067318 | MATHDB-350454

Brightwell-Felsner-Trotter's cross-product conjecture for posets

摘录 / EXCERPT: Let $P=(X,\prec)$ be a finite poset, let $x,y,z\in X$ be distinct, and for integers k,ℓ let $F(k,\ell)$ denote the number of linear extensions...

conjecture | open | MathDB 页面 | 原文线索

V2-067319 | MATHDB-350456

The generalized cross-product inequality for posets

摘录 / EXCERPT: Assume the conditions of the generalized cross-product conjecture, so that $F(i,j)$ counts the relevant linear extensions for integer indices. Generalized cross-product inequality....

conjecture | open | MathDB 页面 | 原文线索

V2-067320 | MATHDB-350463

The bottom-dissection reformulation of the Andrews-Gordon companion conjecture

摘录 / EXCERPT: Let $r\geq 2$ and $1\leq i\leq r$. Let $\mathcal{B}_{r,i}$ be the set of partitions whose $(i-1)$-bottom dissection consists of $i-1$ successive bottom squares followed by bottom r...

conjecture | open | MathDB 页面 | 原文线索

V2-067321 | MATHDB-350464

Martinec's annulus bootstrap conjecture for boundary Liouville CFT

摘录 / EXCERPT: Consider an annulus represented as a cylinder of length $\pi\tau$ and radius 1. Set $q=e^{-2\pi\tau}$, and let $\mathcal{Z}_{\mathrm{Annulus}}$ denote the no-insertion-point part...

conjecture | open | MathDB 页面 | 原文线索

V2-067322 | MATHDB-350466

Dihedral symmetry conjecture for critical points of the coupled dipolar/Maier-Saupe potential

摘录 / EXCERPT: Consider the coupled dipolar/Maier-Saupe potential with parameters (σ,κ) and its critical points on the space of admissible orientational densities. A critical point ha...

conjecture | open | MathDB 页面 | 原文线索

V2-067323 | MATHDB-350467

Uniqueness conjecture for non-axisymmetric critical points of the coupled dipolar/Maier-Saupe potential

摘录 / EXCERPT: Let (σ_0,κ_0) be a parameter pair for the coupled dipolar/Maier-Saupe potential, and identify critical points that differ only by a rotation. A critical point is non-ax...

conjecture | open | MathDB 页面 | 原文线索

V2-067324 | MATHDB-350468

Monotonicity conjecture for non-axisymmetric states in the coupled dipolar/Maier-Saupe potential

摘录 / EXCERPT: Fix σ_0 and suppose that the coupled dipolar/Maier-Saupe potential has a non-axisymmetric critical point at parameter (σ_0,κ_0). Monotonicity conjecture. If a no...

conjecture | open | MathDB 页面 | 原文线索

V2-067325 | MATHDB-350469

Axisymmetric-minimizer conjecture for the Onsager potential

摘录 / EXCERPT: Let μ be the Onsager interaction parameter, and consider the local and global minimizers of the Onsager free-energy functional. A state is axisymmetric if its density is invari...

conjecture | open | MathDB 页面 | 原文线索

V2-067326 | MATHDB-350470

Dihedral-family conjecture for non-axisymmetric critical points of the Onsager potential

摘录 / EXCERPT: Consider the Onsager potential with interaction parameter $\mu>0$. For a positive integer n, let $D^{2n\mathrm{h}}$ denote the relevant horizontal dihedral symmetry group, and ca...

conjecture | open | MathDB 页面 | 原文线索

V2-067327 | MATHDB-350471

The triangular lattice minimizer conjecture for logarithmic lattice energies

摘录 / EXCERPT: Triangular lattice minimizer conjecture. For every $s>2$, the triangular lattice A_2 is the unique minimizer of $\mathcal{F}_s$ among all two-dimensional lattices of uni...

conjecture | open | MathDB 页面 | 原文线索

V2-067328 | MATHDB-350487

Tree-symmetrization identity for the Joyce potential

摘录 / EXCERPT: Let $\mathcal{T}$ be an unrooted, unlabeled tree with n vertices, and let $K_\mathcal{T}(\{\gamma_i,t_i\})$ be the tree kernel defined by the product over its oriented edges. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-067329 | MATHDB-350492

Almost-everywhere uniqueness conjecture for complex phase retrieval least squares

摘录 / EXCERPT: Almost-everywhere uniqueness conjecture. For all vectors ${\bm b}\in\mathbb{R}^m$ except for a measure zero set, this program has a unique solution. The real analogue is establishe...

conjecture | open | MathDB 页面 | 原文线索

V2-067330 | MATHDB-350507

Jordan-curve conjecture for eigenvalue contour lines of Kac-Murdock-Szegő matrices

摘录 / EXCERPT: Let $L_{n,N}^{(1)}$ and $L_{n,N}^{(2)}$ denote the two types of closed eigenvalue contour lines associated with the Kac-Murdock-Szegő matrix of size n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067331 | MATHDB-350508

Connectedness conjecture for the complement of the zero-locus

摘录 / EXCERPT: For each integer $\Delta\geq 3$, let $\mathcal{G}_\Delta$ be the class of graphs of maximum degree at most Δ, and let ... where $Z_G(\lambda)$ is the independence polynomial o...

conjecture | open | MathDB 页面 | 原文线索

V2-067332 | MATHDB-350510

Higher-dimensional Payne-type inequality for buckling eigenvalues

摘录 / EXCERPT: Let Ω be a bounded domain in $\mathbb R^d$ with smooth boundary. Let λ_j, Λ_j, and σ_j denote the eigenvalues used in the paper for the Dirichlet Lapl...

conjecture | open | MathDB 页面 | 原文线索

V2-067333 | MATHDB-350512

Normality conjecture for spanning subgraph counts in random regular graphs

摘录 / EXCERPT: Normality conjecture. The numbers of perfect matchings, Hamilton cycles, spanning trees, and k-factors where $k\le d-1$, are all asymptotically normally distributed in...

conjecture | open | MathDB 页面 | 原文线索

V2-067334 | MATHDB-350513

Normality conjecture for cycle and matching counts in random regular graphs

摘录 / EXCERPT: Normality conjecture. Suppose

conjecture | open | MathDB 页面 | 原文线索

V2-067335 | MATHDB-350518

Robust connectivity conjecture for planar triangulations

摘录 / EXCERPT: Let G be a planar triangulation, and let $\kappa_\rho(G)$ denote its robust connectivity. Planar triangulation robust connectivity conjecture. ... Planar triangulations on at least...

conjecture | open | MathDB 页面 | 原文线索

V2-067336 | MATHDB-350521

Isometric dimension conjecture for braid graphs of reduced expressions

摘录 / EXCERPT: Reduced-expression isometric-dimension conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067337 | MATHDB-350523

Theta-packet induction irreducibility conjecture

摘录 / EXCERPT: Theta-packet induction conjecture. The representation $I(\pi_0)$ is irreducible for every $\pi_0\in\Pi_{\psi_0}^\theta(G_0)$. Moreover, if $I(\pi_0)\simeq I(\pi'_0)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-067338 | MATHDB-350524

Theta-packet Arthur multiplicity formula conjecture

摘录 / EXCERPT: Theta-packet Arthur multiplicity conjecture. There is a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-067339 | MATHDB-350529

GNS mixing-time conjecture for asymmetric exclusion in the maximal current phase

摘录 / EXCERPT: Consider the asymmetric simple exclusion process on a segment of size N with open boundaries, in which particles have a drift, may move in both directions within the segment, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067340 | MATHDB-350530

Jara's conjecture on KPZ mixing for particle-conserving systems

摘录 / EXCERPT: Consider a particle-conserving system on a ring of size N with only nice local interactions, and let the average current be a function of the particle density. Assume that the se...

conjecture | open | MathDB 页面 | 原文线索

V2-067341 | MATHDB-350533

Erman-Yang normal-distribution conjecture for rows of random edge-ideal Betti tables

摘录 / EXCERPT: Let $G\sim G(n,c/n)$ be an Erdős-Rényi random graph, let Δ be its random flag complex, and consider the Betti table of R/I_G. Suppose the cited density theorem yields nonv...

conjecture | open | MathDB 页面 | 原文线索

V2-067342 | MATHDB-350534

Kontsevich formula conjecture for thin-monodromy variations of Hodge structures

摘录 / EXCERPT: Let $\mathcal{H}^n$ be a weight n variation of Hodge structures with thin monodromy over a hyperbolic Riemann surface C with singular locus S. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067343 | MATHDB-350535

Necessity and sufficiency for PcN power functions over finite fields

摘录 / EXCERPT: PcN power-function conjecture. $F(x)=x^d$ is a PcN function if and only if one of the following conditions holds:

conjecture | open | MathDB 页面 | 原文线索

V2-067344 | MATHDB-350541

Goulden-Rattan conjecture on coefficients of Goulden-Rattan polynomials

摘录 / EXCERPT: Let $C_0=1$, $C_1=0$, and for $k\geq 2$ define C_k on Young diagrams by ... Let L_k be the polynomial with rational coefficients determined by $K_k-R_{k+1}=L_k(C_2,C_3,\ldots)$;...

conjecture | open | MathDB 页面 | 原文线索

V2-067345 | MATHDB-350546

Dwork's p-adic hypergeometric inversion conjecture

摘录 / EXCERPT: Dwork's p-adic hypergeometric inversion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067346 | MATHDB-350551

The recursive-construction conjecture for permanents of 312-avoiding matrices

摘录 / EXCERPT: Let $f_{312}(n)$ be the maximum permanent of an $n\times n$ 0-1 matrix avoiding the permutation pattern 312. Let $\mathcal{M}_n(\ast,\circ)$ be the set of $n\times n$ matrices obta...

conjecture | open | MathDB 页面 | 原文线索

V2-067347 | MATHDB-350559

PSTV-A conjecture for Campana points on Fano orbifolds

摘录 / EXCERPT: Let (X,D) be a klt Campana orbifold over K with a good integral model $(\mathcal{X},\mathcal{D})$ over $\mathcal{O}_{K,S}$. Let $H_{\mathcal{L}}$ be the height associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-067348 | MATHDB-350561

Hibi-Ohsugi conjecture on unimodality of Gorenstein IDP polytopes

摘录 / EXCERPT: Let P be a lattice polytope. It is Gorenstein if its associated Ehrhart ring is Gorenstein, and it has the integer decomposition property (IDP) if every lattice point in mP is...

conjecture | open | MathDB 页面 | 原文线索

V2-067349 | MATHDB-350566

Mori-Matano Wulff-shape conjecture for spreading fronts

摘录 / EXCERPT: Let $K(\theta)$ be the directional spreading speed and define the Wulff shape by ... Let $\hat{\mathscr{F}}$ be the convex hull of the Frank diagram, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067350 | MATHDB-350586

Monotonicity conjecture for the Möbius metric under midpoint rotation in ring domains

摘录 / EXCERPT: Let $R(r,1)$ be the annular ring with $0<r<1$, let k satisfy $r<k<1$, and let q satisfy ... For $\mu\in[0,\pi/2]$, consider the two points $qe^{\mu i}+k$ and $qe^{(\pi+\mu)i}+k$...

conjecture | open | MathDB 页面 | 原文线索

V2-067351 | MATHDB-350590

Growth-rate conjecture for cyclically reduced automorphic images

摘录 / EXCERPT: Let F_r be a free group and let $w\in F_r$ be a non-primitive element. Let $CA(w)$ denote the set of cyclically reduced automorphic images of w. Cyclically reduced orbit growth...

conjecture | open | MathDB 页面 | 原文线索

V2-067352 | MATHDB-350592

Minimal extension conjecture for slightly degenerate fusion categories

摘录 / EXCERPT: Minimal extension conjecture. Any slightly degenerate fusion category admits a minimal extension. This is the special case of Müger's conjecture that remains open after counterexam...

conjecture | open | MathDB 页面 | 原文线索

V2-067353 | MATHDB-350608

Formal asymptotic conjecture for front propagation with evolving dispersion

摘录 / EXCERPT: Formal front-propagation conjecture. There exists an interval of trait values J_0 centered at 1 such that, for every open interval $J\subset J_0$ centered at 1, the density...

conjecture | open | MathDB 页面 | 原文线索

V2-067354 | MATHDB-350612

Hitting-time conjecture for the Erdős-Ko-Rado property in random Kneser graphs

摘录 / EXCERPT: For integers n and k, let $K_i(n,k)$ denote the random subgraph process of the Kneser graph, and define … … … … Here an EKR graph is one whose maximum independent sets are prec…

conjecture | open | MathDB 页面 | 原文线索

V2-067355 | MATHDB-350618

Uniform Boundedness Conjecture for quadratic polynomials

摘录 / EXCERPT: Let $f_c(x)=x^2+c\in\mathbb{Q}[x]$ be a quadratic polynomial, and let a rational point have exact period n if its period is n and no smaller positive period. Uniform Boundednes…

conjecture | open | MathDB 页面 | 原文线索

V2-067356 | MATHDB-350621

The recursive Fukaya-Seidel category claim for chain polynomials

摘录 / EXCERPT: For a tuple $a=(a_1,\ldots,a_n)$, let $-a=(a_2,\ldots,a_n)$, and let $\mathcal{A}_a$ be the directed Fukaya-Seidel A_∞-category associated with the chain polynomial indexed…

conjecture | open | MathDB 页面 | 原文线索

V2-067357 | MATHDB-350622

Generation by distinguished collections in Fukaya-Seidel categories

摘录 / EXCERPT: Let f be the Lefschetz fibration under consideration, let $\mathcal{F}(f)$ be its associated Fukaya-Seidel category, and let $Tw(\mathcal{F}(f))$ be its category of twisted compl…

conjecture | open | MathDB 页面 | 原文线索

V2-067358 | MATHDB-350634

The logarithmic cycle-packing bound for inversion number

摘录 / EXCERPT: Logarithmic cycle-packing conjecture. For all k, $g(k)=O(k\log k)$; equivalently, there is an absolute constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-067359 | MATHDB-350635

Belkhechine et al.'s lower-bound conjecture for maximum inversion number

摘录 / EXCERPT: Belkhechine et al.'s lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067360 | MATHDB-350660

Near-linear mixing-time conjecture for uniform quadrangulations

摘录 / EXCERPT: Let Q_n be a uniform quadrangulation with n vertices, let $Q_n^\dagger$ be its dual, and let $\tau_{Q_n}(\varepsilon)$, $\tau_{Q_n^\dagger}(\varepsilon)$ denote uniform mixing…

conjecture | open | MathDB 页面 | 原文线索

V2-067361 | MATHDB-350661

Even-dimensional single-valuedness conjecture for graphical functions

摘录 / EXCERPT: Let G be a weighted graph whose graphical function $f_G(z)$ exists in dimension $D=2\lambda+2\geq3$. Let $\mathcal{S}\mathcal{V}_{\{0,1,\infty\}}$ denote the space of functions o…

conjecture | open | MathDB 页面 | 原文线索

V2-067362 | MATHDB-350662

Weight bound for constructible graphical functions and periods

摘录 / EXCERPT: Let G be a constructible graph in an even dimension $D\geq4$, and let P_G be its associated period. The preceding reduction argument concerns the weight of graphical functions…

conjecture | open | MathDB 页面 | 原文线索

V2-067363 | MATHDB-350663

Basso-Dixon fishnet determinant conjecture

摘录 / EXCERPT: Basso-Dixon fishnet determinant conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067364 | MATHDB-350664

Universal Gegenbauer expansion conjecture for graphical functions

摘录 / EXCERPT: Let G be a graphical function for which the radial representation and Gegenbauer expansion are defined. Universal Gegenbauer expansion conjecture. The Gegenbauer expansion (gfr)…

conjecture | open | MathDB 页面 | 原文线索

V2-067365 | MATHDB-350683

Immersion number conjecture for generalized Mycielski graphs of complete graphs

摘录 / EXCERPT: For integers $m\geq 3$, let K_{m+1} be the complete graph on $m+1$ vertices, let $\mu_m(K_{m+1})$ be its m-Mycielskian, and let $\operatorname{im}(G)$ denote the largest t su…

conjecture | open | MathDB 页面 | 原文线索

V2-067366 | MATHDB-350700

The global curve attractor problem for rational Thurston maps

摘录 / EXCERPT: Let $f\colon S^2\to S^2$ be a Thurston map with a hyperbolic orbifold that is realized by a rational map. Let P_f denote its postcritical set, and let pullbacks be considered up…

open_problem | open | MathDB 页面 | 原文线索

V2-067367 | MATHDB-350702

Logarithmic version of Serrano's conjecture

摘录 / EXCERPT: Log version of Serrano's conjecture. The $\mathbb Q$-Cartier divisor $K_X+\Delta+tL$ is ample for $t>2n$.

conjecture | open | MathDB 页面 | 原文线索

V2-067368 | MATHDB-350703

Bigness conjecture for almost strictly nef divisors on log canonical pairs

摘录 / EXCERPT: Bigness conjecture. The $\mathbb Q$-Cartier divisor $K_X+\Delta+tL$ is big for $t>2n$.

conjecture | open | MathDB 页面 | 原文线索

V2-067369 | MATHDB-350705

The free-Boson formula for the CKP Darboux transformation

摘录 / EXCERPT: Let $T(q)$ be the CKP Darboux transformation associated with a solution q, let $g\in Sp_{\infty}$, and write the corresponding tau function as ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067370 | MATHDB-350714

The cubical or simplicial-set construction of free cocompletions and finite n-groupoids

摘录 / EXCERPT: The paper considers an absolute free cocompletion of a point constructed from cubical sets, simplicial sets, or semisimplicial sets, together with three reflective localizations fo...

conjecture | open | MathDB 页面 | 原文线索

V2-067371 | MATHDB-350715

Lin-Kim's descent-statistic equivalence conjecture for six (4,4)-avoidance classes

摘录 / EXCERPT: Let (ν,μ) be a pair of patterns of length four. For a permutation π, let $\operatorname{asc}(\pi)$ denote its number of ascents and let $\operatorname{last}(\pi)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-067372 | MATHDB-350723

Weighted Langlands-Shelstad transfer conjecture

摘录 / EXCERPT: Weighted Langlands-Shelstad transfer conjecture. For each G and M, there is a function $\hat{S}^{{\tilde G}'}_{{\tilde M}'}(\delta',f')$ on ... such that

conjecture | open | MathDB 页面 | 原文线索

V2-067373 | MATHDB-350727

Conjecture on Pythagoras numbers of maximal orders in biquadratic fields

摘录 / EXCERPT: Let K be a totally real biquadratic field, and let $\mathcal{O}_K$ be its ring of integers. The Pythagoras number $\mathsf{P}(\mathcal{O}_K)$ is the least number of squares neede...

conjecture | open | MathDB 页面 | 原文线索

V2-067374 | MATHDB-350728

Conjecture that Wilf equivalence implies enumerating equivalence for binary-tree patterns

摘录 / EXCERPT: Let s and t be patterns on binary trees. Two patterns are Wilf equivalent when their avoidance functions are equal, and they are enumerating equivalent when their enumerating g...

conjecture | open | MathDB 页面 | 原文线索

V2-067375 | MATHDB-350729

Conjecture on the computed Wilf and enumeration classes of one-relator operads

摘录 / EXCERPT: For $n=8,9,10,11,12$, let $A(n)$ be the number of Wilf classes and let $E(n)$ be the number of enumeration classes of one-relator binary operads with a relation of degree n. The...

conjecture | open | MathDB 页面 | 原文线索

V2-067376 | MATHDB-350740

Conjecture on a spectral gap for the collisionless kinetic semigroup

摘录 / EXCERPT: Let T_{H} be the generator of the C_{0}-semigroup $\left(U_{\mathsf{H}}(t)\right)_{t\geqslant0}$, let $\mathbf{P}_{0}$ be the spectral projection associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-067377 | MATHDB-350746

Denominator conjecture for a basic hypergeometric coefficient ratio

摘录 / EXCERPT: Denominator conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-067378 | MATHDB-350761

Conjecture on the cost of the type H2 service-station walk

摘录 / EXCERPT: Let n be the number of targets, and suppose that $\mathcal{R}^*(n+1) > \mathcal{RD}^*(n+2)$. Let $\mathcal{W}^1_{\operatorname{STID}}$ be the walk used in the type $H2$ solution...

conjecture | open | MathDB 页面 | 原文线索

V2-067379 | MATHDB-350764

Average-rank conjecture for elliptic surface specializations

摘录 / EXCERPT: Average-rank conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067380 | MATHDB-350765

The Zero Density Conjecture for quadratic Dirichlet L-functions over function fields

摘录 / EXCERPT: Let $\varphi(x)$ be a Schwartz class function on $\mathbb{R}$ with Fourier transform having compact support. Define the density ... and let $\rho=\frac12+i\gamma$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-067381 | MATHDB-350770

Soulé-Stark conjecture for rank-one motives

摘录 / EXCERPT: Soulé-Stark conjecture. The Soulé-Stark element coincides with the Bloch-Kato element:

conjecture | open | MathDB 页面 | 原文线索

V2-067382 | MATHDB-350771

Kurihara-et al. Iwasawa main conjecture for Rubin-Stark elements

摘录 / EXCERPT: Let L_∞/K be the relevant p-adic extension, let $\Lambda=\mathds{Z}_p\llbracket\operatorname{Gal}(L_\infty/K)\rrbracket$, let ε be the specified idempotent, an...

conjecture | open | MathDB 页面 | 原文线索

V2-067383 | MATHDB-350773

The five-dimensional Bessel-process conjecture for the concave majorant of Brownian motion

摘录 / EXCERPT: Let $B=(B_t)_{t\geq 0}$ be a standard one-dimensional Brownian motion, and let $K=(K_t)_{t\geq 0}$ be its almost surely unique concave majorant on $[0,\infty)$. Write R_5 for a f...

conjecture | open | MathDB 页面 | 原文线索

V2-067384 | MATHDB-350777

Conjecture on the zeroes of the Lah generating polynomial

摘录 / EXCERPT: Let $P_{n,k}(t)$ be the generating polynomial associated with the Lah distribution. Negative-real-part zeroes conjecture. All complex zeroes of $P_{n,k}$ have negative real parts....

conjecture | open | MathDB 页面 | 原文线索

V2-067385 | MATHDB-350781

The $\mathcal{O}(k\log k)$ Erdős-Pósa conjecture for long holes in C_4-free graphs

摘录 / EXCERPT: For each $\ell\geq 3$, let C_ℓ be the cycle of length ℓ. A hole is an induced cycle of length at least 4, and a graph is C_4-free if it contains no C_4. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067386 | MATHDB-350786

The E3-structure conjecture for the bubbling spherical Hecke category

摘录 / EXCERPT: Let G be the reductive group underlying the spherical Hecke category, and let ... be the bubbling spherical Hecke category, equipped with its monoidal structure. The E3-structure c...

conjecture | open | MathDB 页面 | 原文线索

V2-067387 | MATHDB-350787

The automorphic gluing equivalence for the bubbling degeneration

摘录 / EXCERPT: Let α be the functor ... constructed from the bubbling family, where $\mathcal{H}^{\mathit{bub},\otimes R}$ is the relevant bubbling Hecke category over R. The automorphic g...

conjecture | open | MathDB 页面 | 原文线索

V2-067388 | MATHDB-350789

The universal affine Hecke category Langlands duality conjecture

摘录 / EXCERPT: Let G be the reductive group, let $G^\vee$ be its Langlands dual, and let $\widetilde{G}^\vee$ denote the space appearing in the universal affine Hecke construction. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067389 | MATHDB-350790

The genus-one Betti geometric Langlands conjecture

摘录 / EXCERPT: Let E be a genus-one curve, let $\mathcal{H}$ be the universal affine Hecke category, and let ... be the canonical map of Corollary. Let $\operatorname{Loc}_{G^\vee}(E)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-067390 | MATHDB-350791

Campos-de Mello total coloring conjecture for powers of cycles

摘录 / EXCERPT: Campos-de Mello's conjecture. If $G=C_n^k$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067391 | MATHDB-350795

Isotypic modularity conjecture for CM cycles

摘录 / EXCERPT: Let $\mathcal A$ be the set of cuspidal automorphic representations defined above. For $\pi\in\mathcal A$, let ... where S ranges over the relevant finite sets and...

conjecture | open | MathDB 页面 | 原文线索

V2-067392 | MATHDB-350814

Safe gluing preserves equality of the vanishing and shortest trek ideals

摘录 / EXCERPT: Let G_1 and G_2 be directed acyclic graphs (DAGs), and let I_{G_i} be their vanishing ideals and ST_{G_i} their shortest trek ideals. Suppose that $I_{G_1}=ST_{G_1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067393 | MATHDB-350815

Toricity characterized by the shortest trek ideal

摘录 / EXCERPT: Let G be a directed acyclic graph (DAG), with vanishing ideal I_G and shortest trek ideal ST_G. Shortest-trek characterization conjecture. The vanishing ideal I_G is toric...

conjecture | open | MathDB 页面 | 原文线索

V2-067394 | MATHDB-350816

Recursive classification of DAGs with toric vanishing ideals by gluing operations

摘录 / EXCERPT: Let G be a directed acyclic graph (DAG) with toric vanishing ideal I_G. The relevant constructions are safe gluing, gluing at sinks, and adding a sink. Recursive classification...

conjecture | open | MathDB 页面 | 原文线索

V2-067395 | MATHDB-350817

Higher d-separation obstructs toric vanishing ideals

摘录 / EXCERPT: Let G be a directed acyclic graph (DAG), and let i,j be vertices of G. A d-separating set for i and j is a set of vertices that d-separates them; consider the minimum car...

conjecture | open | MathDB 页面 | 原文线索

V2-067396 | MATHDB-350818

Safe-gluing decomposition for chordal DAGs with toric vanishing ideals

摘录 / EXCERPT: Assume the higher d-separation conjecture: whenever two vertices of a DAG have minimum d-separating-set size at least 2, the vanishing ideal is not toric. Let G be a chordal di...

conjecture | open | MathDB 页面 | 原文线索

V2-067397 | MATHDB-350823

Smallest singular value bound for general deterministic shifts of real Ginibre matrices

摘录 / EXCERPT: Let X be an $N\times N$ real Ginibre matrix, let A be a deterministic matrix, let $\gamma>0$, and define the Hermitian imaginary part by ... Write $\sigma_1(\gamma X+A)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-067398 | MATHDB-350829

Phase-transition conjecture for support vector proliferation in ℓ_1-SVMs

摘录 / EXCERPT: Let $(\bX,y) \in \mathbb{R}^{n \times d} \times \mathbb{R}^n$ be an isotropic Gaussian sample, and let SVP denote support vector proliferation, namely coincidence between the solut...

conjecture | open | MathDB 页面 | 原文线索

V2-067399 | MATHDB-350854

Gross-Manturov-Tucker's interpolating partial-duality polynomial conjecture

摘录 / EXCERPT: Let G be a ribbon graph, and let $~^{\partial}{\mathscr{E}}^{*}_{G}(z)$ denote its partial-$*$ polynomial. A polynomial is odd or even when all terms with nonzero coefficients ha...

conjecture | open | MathDB 页面 | 原文线索

V2-067400 | MATHDB-350860

The shifted generalized forest-matrix Hankel-total positivity conjecture

摘录 / EXCERPT: Let $\tilde{\mathsf{F}}(q,y,z)$ be the generalized forest matrix, and define $\bar{\mathsf{F}}(q,y,z)=\tilde{\mathsf{F}}(q,y,z)\operatorname{diag}((q^{k(k-1)/2})_{k\geq0})$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067401 | MATHDB-350866

Confocality criterion for elliptic incenter loci

摘录 / EXCERPT: Let a pair of ellipses admit Poncelet 3-periodics, and let X_1 denote the incenter of a member of the family. The pair is confocal when its two ellipses share the same foci. Ince...

conjecture | open | MathDB 页面 | 原文线索

V2-067402 | MATHDB-350867

Elliptic excenter loci for confocal or poristic ellipse pairs

摘录 / EXCERPT: Let a pair of ellipses admit Poncelet 3-periodics, and consider the loci of the excenters of the triangles in the family. The pair is poristic when the associated Poncelet family h...

conjecture | open | MathDB 页面 | 原文线索

V2-067403 | MATHDB-350870

Max-min compact-set conjecture for Tschebyshev-Padé approximants

摘录 / EXCERPT: Let $F\in\mathcal F_f$ be an admissible compact set for f, and let $\{\Omega_n\}$ be an interpolation table disjoint from F. Assume that the limiting density ω in the in...

conjecture | open | MathDB 页面 | 原文线索

V2-067404 | MATHDB-350879

Kamatchi-Arumugam characterization conjecture for distance-antimagic graphs

摘录 / EXCERPT: Kamatchi-Arumugam conjecture. G is distance antimagic if and only if $N(u)\ne N(v)$ for any two distinct vertices $u,v\in V(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067405 | MATHDB-350884

Rational-point conjecture for the surfaces $\widetilde{Z}(17,1)$ and $\widetilde{Z}(17,3)$

摘录 / EXCERPT: Rational-point conjecture. The only ${\mathbb Q}$-points on $\widetilde{Z}(17,1)$ lie above one of the listed curves in (x,y), the curve $T=0$, or one of the points in the indica...

conjecture | open | MathDB 页面 | 原文线索

V2-067406 | MATHDB-350900

Equivalence between 2-simplicial and admissible 2-paracyclic stable infinity-categories

摘录 / EXCERPT: Let $({\operatorname{Cat}}_\infty)_{\mathbb{D}\Delta}$ and $({\operatorname{Cat}}_\infty)_{\mathbb{L}\Lambda_\infty}$ denote the ∞-categories of 2-simplicial and 2-parac...

conjecture | open | MathDB 页面 | 原文线索

V2-067407 | MATHDB-350901

Categorical characterization of spherical adjunctions by relative S-constructions

摘录 / EXCERPT: Let Sph be the ∞-category formed by spherical adjunctions, and let ${\mathcal{S}}t_{\Lambda_\infty}^{\operatorname{adm},\,2\text{-Segal}}$ be the ∞-c...

conjecture | open | MathDB 页面 | 原文线索

V2-067408 | MATHDB-350908

Minimal unbounded-bb classes in hereditary graph classes

摘录 / EXCERPT: A graph class is minimal of unbounded λ if it has unbounded λ, while every proper hereditary subclass has bounded λ. Minimal-class conjecture for λ....

conjecture | open | MathDB 页面 | 原文线索

V2-067409 | MATHDB-350909

Chains-in-a-cycle conjecture for bounded lettericity

摘录 / EXCERPT: Let $\mathcal X$ be a graph class with bounded λ and unbounded lettericity, and let $\mathcal C_k$ denote the chains-in-a-cycle classes defined earlier in the paper. Chains...

conjecture | open | MathDB 页面 | 原文线索

V2-067410 | MATHDB-350910

Bounded-σ criterion for hereditary closures of linked chain graphs

摘录 / EXCERPT: Let $(G_n)_{n \geq 1}$ be a family of linked chain graphs with linking permutations $\pi_1,\pi_2,\dots$, and let $\mathcal X$ be the hereditary closure of this family. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-067411 | MATHDB-350911

Twisted-cycle obstruction to PMM-griddability

摘录 / EXCERPT: Let M be a partial multiplication matrix, meaning a matrix whose nonzero entries are monotone classes, and suppose its cell graph is cyclic. Let $\mathcal X$ be a permutation cla...

conjecture | open | MathDB 页面 | 原文线索

V2-067412 | MATHDB-350912

Minimality of twisted cycles in a chain for unbounded lettericity

摘录 / EXCERPT: Let $\mathcal T$ denote the graph class of twisted cycles in a chain, namely the class represented by the constructions discussed around Figure. A graph class is minimal of unbound...

conjecture | open | MathDB 页面 | 原文线索

V2-067413 | MATHDB-350913

The intertwining conjecture for pivot-minors

摘录 / EXCERPT: A pivot-minor H of a graph G is proper if $|V(H)|<|V(G)|$. A graph G is an intertwine of graphs H_{1} and H_{2} for pivot-minors if it contains both H_{1} and H_{2} a...

conjecture | open | MathDB 页面 | 原文线索

V2-067414 | MATHDB-350915

Greenberg-Ochiai's two-variable Iwasawa main conjecture for Hida deformations

摘录 / EXCERPT: Let $d53c$ be the coefficient ring of a residually reducible Hida deformation, let $d54b$ be a stable free lattice, and put ... Write $\operatorname{Sel}_{\mathcal{A}}$ for the assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-067415 | MATHDB-350918

Characterization of Chebyshev-like maps by affine Weyl iterated monodromy groups

摘录 / EXCERPT: Let $f: \mathbb{C}^n \to \mathbb{C}^n$ be post-critically finite. An affine Weyl iterated-monodromy characterization conjecture asserts that if the iterated monodromy group of f...

conjecture | open | MathDB 页面 | 原文线索

V2-067416 | MATHDB-350923

The core conjecture for right-angled Artin groups

摘录 / EXCERPT: Let $G=\Gamma(\Gamma,\{\mathbb Z\})$ and $G'=\Gamma(\Gamma',\{\mathbb Z\})$ be right-angled Artin groups. Core conjecture. Then ... if and only if the cores of G and G' are isomo...

conjecture | open | MathDB 页面 | 原文线索

V2-067417 | MATHDB-350937

Mutual asymptotic Fekete sequence existence conjecture

摘录 / EXCERPT: Let $(L_1,\phi_1),\ldots,(L_m,\phi_m)$ be a collection of pairs as above, with normalized volumes as defined in the surrounding text. For each pair, let $\mu_{(L_j,\phi_j)}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-067418 | MATHDB-350938

Low-rank perturbation conjecture for Toeplitz spectral measures

摘录 / EXCERPT: Let A satisfy the conditions of Theorem 3.1, and let A' be the matrix constructed in Lemma 3.4. For the Toeplitz coefficients a_j, define the limiting measure μ as the di...

conjecture | open | MathDB 页面 | 原文线索

V2-067419 | MATHDB-350943

The optimal point-concentration bound for definable sets

摘录 / EXCERPT: Optimal point-concentration conjecture. In the setting of Theorem baby-forwards, one has $\Pr(X\in S)\le C_S/\sqrt n$ for some constant C_S depending only on S.

conjecture | open | MathDB 页面 | 原文线索

V2-067420 | MATHDB-350945

Bohr-radius conjecture for cardioid starlike functions

摘录 / EXCERPT: Let $\mathcal{S}^{*}_{\wp}$ denote the class of cardioid starlike functions associated with $\psi(z)=1+ze^z$, and let r_0 denote the Bohr radius for this class. Bohr-radius conje...

conjecture | open | MathDB 页面 | 原文线索

V2-067421 | MATHDB-350952

Categorical Donaldson-Thomas wall-crossing conjecture for PT stable-pair moduli

摘录 / EXCERPT: Let S be a smooth projective surface over $\mathbb{C}$, let $X=\operatorname{Tot}_S(\omega_S)$, and let $P_n^t(X,\beta)$ denote the moduli space of $\mu_t^{\dagger}$-semistable D...

conjecture | open | MathDB 页面 | 原文线索

V2-067422 | MATHDB-350957

Patrias-Pechenik's weak path-cordiality conjecture

摘录 / EXCERPT: Let A be a finite Abelian group, and let $\mathbb{P}$ denote the class of path graphs. A group is weakly $\mathbb{P}$-cordial if all but finitely many graphs in $\mathbb{P}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-067423 | MATHDB-350967

Mercat's conjecture on higher-rank Clifford indices

摘录 / EXCERPT: Let C be a curve of genus g, and let E be a semistable vector bundle on C of rank n and degree d. Write $\mu(E)=d/n$, and let γ_1 denote the Clifford index of...

conjecture | open | MathDB 页面 | 原文线索

V2-067424 | MATHDB-350975

Polynomial upper bound for genus-increment growth ratios

摘录 / EXCERPT: Let $\mathcal A^h$ be one of the graph classes considered in the paper, indexed by graphs on $[n]$ embeddable in surfaces of Euler genus at most h. Genus-increment growth-ratio c...

conjecture | open | MathDB 页面 | 原文线索

V2-067425 | MATHDB-350976

Growth-constant threshold for order-dependent surface classes

摘录 / EXCERPT: Let g be a genus function, and let $\mathcal A^g$ denote one of the graph classes considered in the paper, such as $\mathcal E^g$, $\mathcal O\mathcal E^g$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067426 | MATHDB-350977

Growth constant at the logarithmic genus threshold

摘录 / EXCERPT: Let g be a genus function and let $\mathcal A^g$ be one of the labelled graph classes considered in the paper. Logarithmic-threshold growth-constant conjecture. If $c>0$ is a con...

conjecture | open | MathDB 页面 | 原文线索

V2-067427 | MATHDB-350989

Resolved noncrossing formula for the biadjoint scalar amplitude

摘录 / EXCERPT: Resolved amplitude conjecture. There exists a set of linear functions $\hat{\eta}_J(s)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067428 | MATHDB-350992

Spectral-gap conjecture for the Berezin transform at fixed weight deficit

摘录 / EXCERPT: Let j be a positive parameter, let $d\in\mathbb{N}$, and set $m=j-d$. Denote the eigenvalues of the Berezin transform $\mathcal{B}_{j,j-d}$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-067429 | MATHDB-350993

Oscillation conjecture for Berezin-transform eigenvalues at fixed weight deficit

摘录 / EXCERPT: Let j be a positive parameter, let $d\in\mathbb{N}$, and set $m=j-d$. Let $\lambda^{(0)},\lambda^{(1)},\ldots,\lambda^{(2j)}$ be the eigenvalues of the Berezin transform...

conjecture | open | MathDB 页面 | 原文线索

V2-067430 | MATHDB-351000

Soliton resolution conjecture for FPUT lattices

摘录 / EXCERPT: Soliton resolution conjecture. Asymptotically, the (n,t)-space splits into regions separated by straight lines through the origin. Some regions are periodic, with period equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-067431 | MATHDB-351006

The extension conjecture for parahoric torsors

摘录 / EXCERPT: Let E be a p-adic field with perfect residue field, let C be a perfect non-archimedean field of characteristic p with an open and bounded valuation subring C^+, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-067432 | MATHDB-351007

Antimagic labeling conjecture for subdivided caterpillars

摘录 / EXCERPT: Let T be a caterpillar, and let G be a subdivided caterpillar, meaning a subdivision of T in which all edges of T that are not on the central path of T are subdivided the...

conjecture | open | MathDB 页面 | 原文线索

V2-067433 | MATHDB-351010

Conjecture on transversal representatives in Schubert structure-constant classes

摘录 / EXCERPT: Transversal-representative conjecture. For any $[(\{u,v\},\{w\})]\in\hat{\mathcal{U}}$, there exists $(u_1',\ldots,u_{m+1}',\{w'\})\in[(\{u,v\},\{w\})]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067434 | MATHDB-351012

Yano's generic b-exponent conjecture for irreducible plane curves

摘录 / EXCERPT: Yano's conjecture. For generic curves in some Γ-constant deformation of f, the b-exponents are

conjecture | open | MathDB 页面 | 原文线索

V2-067435 | MATHDB-351023

Sign-control conjecture for dual canonical basis immanants

摘录 / EXCERPT: Let $\operatorname{Imm}_v X$ be a Kazhdan-Lusztig immanant that is k-positive, and let R,C be index sets for which the dual canonical basis element...

conjecture | open | MathDB 页面 | 原文线索

V2-067436 | MATHDB-351024

Strong pattern-avoidance conjecture for dual canonical basis immanants

摘录 / EXCERPT: Let $0<k<n\leq m$ be integers, let $v\in S_n$ avoid the pattern $12\cdots(k+1)$, and let $R,C\in\binom{[m]}{n}$, where repetitions are allowed. Strong pattern-avoidance conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-067437 | MATHDB-351029

Sign-vector reconstruction conjecture for simplicial polytopes

摘录 / EXCERPT: Let $P\subseteq \mathbb R^d$ be a simplicial d-polytope. For an affine k-stress on P, record the signs of the coefficients in its squarefree part on the $(k-1)$-faces; let...

conjecture | open | MathDB 页面 | 原文线索

V2-067438 | MATHDB-351030

Stress-sign conjecture for missing faces of simplicial polytopes

摘录 / EXCERPT: Stress-sign conjecture. There exists a $(k-2)$-face $F\subset M$ and an affine k-stress λ on P such that, for every $(k-1)$-face $G=F\cup v_{F,G}$ of ∂P, one...

conjecture | open | MathDB 页面 | 原文线索

V2-067439 | MATHDB-351031

Missing-face extension conjecture for affine stresses

摘录 / EXCERPT: Let $P\subseteq \mathbb R^d$ be a simplicial polytope of dimension $d\geq 2k-1$, with $k\geq 2$. Let G be a missing $(k-1)$-face of ∂P, and let F be a $(k-1)$-subset...

conjecture | open | MathDB 页面 | 原文线索

V2-067440 | MATHDB-351032

Strong-coupling effective mass conjecture for regular polaron models

摘录 / EXCERPT: Strong-coupling effective mass conjecture. The effective mass satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067441 | MATHDB-351040

Okounkov's shift-operator R-matrix conjecture

摘录 / EXCERPT: Okounkov's shift-operator R-matrix conjecture. In the stable basis, the operator S_w is conjugate to the R-matrix for

conjecture | open | MathDB 页面 | 原文线索

V2-067442 | MATHDB-351051

Volume-normalized p-capacity conjecture for linear images

摘录 / EXCERPT: Let $n\geq 2$ and $0<p<n$. Let $K\subset\mathbb{R}^n$ be a compact set with an irreducible group of isometries, let M be a volume-preserving invertible linear map, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-067443 | MATHDB-351069

Anosov density conjecture for compact hyperbolic triangle reflection groups

摘录 / EXCERPT: Anosov density conjecture. The representation ρ is discrete and faithful if and only if it lies in the closure of the space of Anosov representations in...

conjecture | open | MathDB 页面 | 原文线索

V2-067444 | MATHDB-351070

The dimension bound conjecture for torsion motives

摘录 / EXCERPT: Dimension bound conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-067445 | MATHDB-351077

Tree packing conjecture for families of trees

摘录 / EXCERPT: Let $n\in\mathbb{N}$, and let $(T_s)_{s\in[n]}$ be a family of trees such that $v(T_s)=s$ for every $s\in[n]$. Tree packing conjecture. The family $(T_s)_{s\in[n]}$ packs into the...

conjecture | open | MathDB 页面 | 原文线索

V2-067446 | MATHDB-351078

Conjecture on packing families of bounded-degree trees with non-spanning members

摘录 / EXCERPT: Let a tree family be a family $(T_s)_{s\in[N]}$ of trees. Bounded-degree tree-family packing conjecture. There exist $\delta>0$ and $n_0\in\mathbb{N}$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-067447 | MATHDB-351080

Ringel-type conjecture for degenerate graphs

摘录 / EXCERPT: Fix an integer D. Degenerate-graph Ringel-type conjecture. There exists $n_0\in\mathbb{N}$ such that, for every $n\ge n_0$, if G is a D-degenerate graph on n vertices with...

conjecture | open | MathDB 页面 | 原文线索

V2-067448 | MATHDB-351084

Woronowicz-type integral representations for the triple sine function

摘录 / EXCERPT: Woronowicz-type integral representations.

conjecture | open | MathDB 页面 | 原文线索

V2-067449 | MATHDB-351085

The correspondence conjecture for trinomial-associated polycyclic codes

摘录 / EXCERPT: Let $t_1(x)=x^n-ax^i-b$ and $t_2(x)=x^n-a'x^{n-i}-b'$ be polynomials in $\bbF_q[x]$ with equal order, and let C_1 and C_2 denote the sets of length-n polycyclic codes over...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067450 | MATHDB-351088

Bhargava-Shankar's average n-Selmer size conjecture

摘录 / EXCERPT: Let n be any positive integer. For elliptic curves E over $\mathbb{Q}$ ordered by height, let $\operatorname{Sel}_n(E/\mathbb{Q})$ denote the n-Selmer group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067451 | MATHDB-351091

Tsai's newform conjecture for generic representations of split odd special orthogonal groups

摘录 / EXCERPT: Let $G_n={\rm SO}_{2n+1}$ be the split odd special orthogonal group of rank $n\geq 1$ over F, let $U_n\subset G_n$ be its upper triangular maximal unipotent subgroup, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067452 | MATHDB-351114

Reiner-Yong conjecture on Grothendieck-to-Lascoux expansions

摘录 / EXCERPT: Reiner-Yong conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067453 | MATHDB-351116

Elliptic Fourier-transform compatibility conjecture

摘录 / EXCERPT: Let G be a simple F-split group. Let ${\mathcal R}^p_{{\mathrm{un}},{\mathrm{ell}}}(G)$ be the elliptic unipotent representation space, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067454 | MATHDB-351143

Nonexistence conjecture for an unbiased photon transfer conversion gain estimator

摘录 / EXCERPT: Nonexistence conjecture. There is no estimator $\mathscr T$ satisfying this equation when both κ_1 and κ_2 are unknown.

conjecture | open | MathDB 页面 | 原文线索

V2-067455 | MATHDB-351145

Sharpness conjecture for the radio-k-number of even cycles

摘录 / EXCERPT: Let n be even and k be odd with ... Set ... Write $\langle h\rangle$ for the subgroup generated by h in the relevant cyclic group. Sharpness conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-067456 | MATHDB-351146

Generalized positivity conjecture for matching and constellation coefficients

摘录 / EXCERPT: Let $k\geq 1$ and let $\lambda,\mu^0,\ldots,\mu^k$ be partitions of size $n\geq 1$. Define $c^\lambda_{\mu^0,\ldots,\mu^k}(b)$ and $h^\lambda_{\mu^0,\ldots,\mu^k}(b)$ by the coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-067457 | MATHDB-351147

Generalized b-conjecture for non-oriented constellations

摘录 / EXCERPT: Let $k\geq 1$ and let $\lambda,\mu^0,\ldots,\mu^k$ be partitions of size $n\geq 1$. A connected rooted k-constellation with profile $(\lambda,\mu^0,\ldots,\mu^k)$ is denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-067458 | MATHDB-351148

Lassalle's positivity conjecture for Jack character polynomials

摘录 / EXCERPT: Let λ be a partition and let $\theta^{(\alpha)}_\mu(\lambda)$ be the Jack-character coefficient defined by the paper, with $b:=\alpha-1$. Let $m_1(\mu)$ denote the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-067459 | MATHDB-351161

Stable anisotropic equilibria conjecture

摘录 / EXCERPT: Let $n\geq 2$, let ρ^n denote the function associated with equilibria whose largest eigenspace has dimension $d=1$, and let $\rho^*>0$ be the threshold density. For...

conjecture | open | MathDB 页面 | 原文线索

V2-067460 | MATHDB-351165

Sharper Randić index-diameter ratio conjecture for cactus graphs

摘录 / EXCERPT: Let G be a cactus with k cycles and b bridges; write n for its number of vertices, R for its Randić index, and d for its diameter. A BC-tree is the block-cut tree of...

conjecture | open | MathDB 页面 | 原文线索

V2-067461 | MATHDB-351171

The periodic Hankel transform of $1-x/r(x^2)$

摘录 / EXCERPT: Let $r(x)$ be the generating function of the Rueppel sequence, and let the Hankel transform of a sequence be the sequence of determinants of its successive Hankel matrices. Periodi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067462 | MATHDB-351172

The modulo-2 Hankel transform of $1-x+x^2/r(x^2)$

摘录 / EXCERPT: Let $r(x)$ be the generating function of the Rueppel sequence, and let the Hankel transform be reduced termwise modulo 2. Modulo-2 periodicity conjecture. The Hankel transform of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067463 | MATHDB-351173

The signed A005811 Hankel transform for $x+1/r(x^2)$

摘录 / EXCERPT: Let $r(x)$ be the generating function of the Rueppel sequence, and let the Hankel transform be the sequence of determinants of successive Hankel matrices. A005811 Hankel-transform...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-067464 | MATHDB-351175

The Stieltjes-parameter formula for generalized Rueppel sequences

摘录 / EXCERPT: Let ... and write its Stieltjes continued fraction as ... Let $s_{b,c}(n)$ denote the corresponding Stieltjes parameters. Generalized Rueppel parameter conjecture. Then ... where $P(k)$...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-067465 | MATHDB-351177

The A268411 product formula for Rueppel Hankel transforms

摘录 / EXCERPT: Let h_n be the Hankel transform of the Rueppel sequence r_n, and let H_n be the Hankel transform of the once-shifted Rueppel sequence r_{n+1}. A268411 product conjecture. T...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067466 | MATHDB-351178

Factorizability conjecture for Stanley Jack g-polynomials

摘录 / EXCERPT: Factorizability conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067467 | MATHDB-351179

Factorization conjecture for two-component skew diagrams

摘录 / EXCERPT: Two-component factorization conjecture. The ratio

conjecture | open | MathDB 页面 | 原文线索

V2-067468 | MATHDB-351204

Bui-Keating moment conjecture for Dirichlet L-functions over function fields

摘录 / EXCERPT: Bui-Keating moment conjecture. For all non-negative integers k,

conjecture | open | MathDB 页面 | 原文线索

V2-067469 | MATHDB-351234

Dissipation-minimization conjecture for periodic porous-medium solutions

摘录 / EXCERPT: Dissipation-minimization conjecture. The parameters (V,A_s,A_f) are chosen in such a way that the dissipation R is minimal. The claim concerns the selection of parameters for a...

conjecture | open | MathDB 页面 | 原文线索

V2-067470 | MATHDB-351240

Three regular polytope conjecture for multicomplex Mandelbrot sets

摘录 / EXCERPT: Let $n\geq3$. The multicomplex Mandelbrot set ${\mathcal{M}_{n}}$ is considered through its principal and idempotent n-dimensional slices, including slices associated with regula...

conjecture | open | MathDB 页面 | 原文线索

V2-067471 | MATHDB-351244

Definable Baire conjecture for closed definable sets

摘录 / EXCERPT: Let $\mathbb{K}$ be a structure in the setting of the paper, and let $X \subseteq \mathbb{K}^n$ be definable and closed. Definable Baire conjecture. The set X is definably Baire....

conjecture | open | MathDB 页面 | 原文线索

V2-067472 | MATHDB-351247

Uniqueness conjecture for relatively free sections of del Pezzo fibrations

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a smooth del Pezzo fibration. Fix an intersection profile λ, let N_λ be the affine subset of $N_1(\mathcal{X})$ consisting of curve cl...

conjecture | open | MathDB 页面 | 原文线索

V2-067473 | MATHDB-351248

Ohno-Nakagawa reflection conjecture for odd-dimensional symmetric-matrix boxes

摘录 / EXCERPT: Let n be a positive odd integer. Let f be a polynomial of degree n with no multiple roots and only one real root. A $2\times n\times n$ box is a pair (A,B) of $n\times n$ i...

conjecture | solved | MathDB 页面 | 原文线索

V2-067474 | MATHDB-351250

Local Ohno-Nakagawa reflection conjecture for symmetric-matrix boxes

摘录 / EXCERPT: Let K be a local field with $\operatorname{char}K\neq2$, let n be odd, and let τ be a nonzero parameter. Let $\mathcal V_{\tau,\tau^{n-1}f}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067475 | MATHDB-351264

Uniqueness conjecture for the pure-type UBCMPG

摘录 / EXCERPT: Let G be the 17-order pure-type UBCMPG shown in Figure 4, where $C_4^0(G)$ contains eight pairs of complement UBC-colorings. Uniqueness conjecture. The graph G is the uniquely...

conjecture | open | MathDB 页面 | 原文线索

V2-067476 | MATHDB-351266

Conjecture on Brownian tracking horizon for geometric matchings

摘录 / EXCERPT: Let n particles move in d dimensions, sampled at intervals of length δ, and let $\widehat{\pi}_k$ be the estimator at time step k. Let $T_{\max}$ denote the maximum tr...

conjecture | open | MathDB 页面 | 原文线索

V2-067477 | MATHDB-351267

Alon-Krivelevich-Sudakov conjecture on coloring graphs with forbidden subgraphs

摘录 / EXCERPT: Alon-Krivelevich-Sudakov conjecture. For every graph F, there is a constant $c_F>0$ such that, whenever G is an F-free graph of maximum degree $Delta\geqslant 2$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067478 | MATHDB-351268

DP-coloring extension of the Alon-Krivelevich-Sudakov conjecture

摘录 / EXCERPT: DP-coloring extension of the Alon-Krivelevich-Sudakov conjecture. For every graph F, there is a constant $c_F>0$ such that, if H is F-free, has maximum degree $d\geqslant 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067479 | MATHDB-351277

The necessary-and-sufficient condition conjecture for non-strict local minimizers of homogeneous QCQPs

摘录 / EXCERPT: Let ... hold; and (ii) there exists a nonzero vector $\bar v$ satisfying $-$. This conjecture seeks a complete characterization of non-strict local non-global minimizers; the supplie...

conjecture | open | MathDB 页面 | 原文线索

V2-067480 | MATHDB-351290

Finiteness conjecture for accumulation points of edit distance functions

摘录 / EXCERPT: Let $\mathcal{H}$ be a non-trivial hereditary graph property, and call a point an accumulation point if it is a limit point of the edit distance function of $\mathcal{H}$. Finitene...

conjecture | open | MathDB 页面 | 原文线索

V2-067481 | MATHDB-351291

Half-interval conjecture for accumulation points of edit distance functions

摘录 / EXCERPT: Let $\mathcal{H}$ be a non-trivial hereditary graph property, and let accumulation points mean limit points of its edit distance function. Half-interval conjecture. For any non-tri...

conjecture | open | MathDB 页面 | 原文线索

V2-067482 | MATHDB-351302

Hard-phase conjecture for partial graph alignment

摘录 / EXCERPT: Fix parameters (λ,s) in the correlated Erdős-Rényi model $\mathcal{G}(n,\lambda/n,s)$. One-sided correlation detection in trees is said to fail when none of the equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-067483 | MATHDB-351311

The predicted virtual chi_y-genus blowup formula

摘录 / EXCERPT: Let Φ be the multiplicative class specified in Examples 3 and 3, and let Z_a be the power series defined by the virtual χ_y-genus blowup relation. For the generating se...

conjecture | open | MathDB 页面 | 原文线索

V2-067484 | MATHDB-351314

Moment criterion conjecture for ultra-small distances

摘录 / EXCERPT: Let $\mathbb{G}_n$ and the uniformly chosen vertices $U_{n,1},U_{n,2}$ be as in Theorem ... , there is a constant $\varepsilon_{\ast}\in(0,1)$ depending on the model parameters such...

conjecture | open | MathDB 页面 | 原文线索

V2-067485 | MATHDB-351315

Buttcane's conjecture on the interchange of integral

摘录 / EXCERPT: Buttcane's conjecture. Let g be holomorphic with rapid decay on an open tube domain in $\mathfrak{a}^*_{\mathbb{C}}$ containing $\mathfrak{a}^*$, let $t\in G(\mathbb{R})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067486 | MATHDB-351322

Spum conjecture for paths

摘录 / EXCERPT: Spum conjecture for paths. For $n\geq 8$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-067487 | MATHDB-351323

Sum-diameter conjecture for paths

摘录 / EXCERPT: Sum-diameter conjecture for paths. For $n\geq 3$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-067488 | MATHDB-351327

The one-component conjecture for bounded truncated Toeplitz operator symbols

摘录 / EXCERPT: Let I be an inner function, let K_I^2 denote the corresponding model space, and let a bounded symbol of a truncated Toeplitz operator mean a symbol in $L^\infty(\mathbb{T})$ re...

conjecture | open | MathDB 页面 | 原文线索

V2-067489 | MATHDB-351330

David-Fearnley-Kisilevsky conjecture on rank jumps in degree-p extensions

摘录 / EXCERPT: David-Fearnley-Kisilevsky conjecture. As $X\to\infty$: if $p=3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067490 | MATHDB-351341

Milnor's homeomorphism conjecture for the parabolic Mandelbrot set

摘录 / EXCERPT: Let $\mathbf M$ be the Mandelbrot set of quadratic polynomials, and let $\mathbf M_1$ be the connectedness locus in the family of rational maps with multiplier 1 at a fixed point...

conjecture | open | MathDB 页面 | 原文线索

V2-067491 | MATHDB-351348

Conjecture on decomposable derived categories of elliptic fibrations

摘录 / EXCERPT: Let X be a minimal surface of Kodaira dimension 1 with an elliptic fibration ... Suppose that $P_g(X)=0$ and $q(X)=1$. Elliptic-fibration decomposability conjecture. Then $D(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-067492 | MATHDB-351353

The order-two non-classicality conjecture for almost classical knot 6.85091

摘录 / EXCERPT: Let $\mathscr{VC}$ be the concordance group of long virtual knots. The almost classical knot 6.85091 is the closure of a long virtual knot υ. The order-two non-classical...

conjecture | open | MathDB 页面 | 原文线索

V2-067493 | MATHDB-351355

The corner permuton limit conjecture for almost square permutations

摘录 / EXCERPT: Corner permuton limit conjecture. If $k,n\to\infty$ with $k/n\to\alpha\in\mathbb{R}_{>0}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067494 | MATHDB-351360

The semi-Baxter and strong-Baxter universality conjecture

摘录 / EXCERPT: Semi-Baxter and strong-Baxter universality conjecture. The permuton limit of uniform semi-Baxter or strong-Baxter permutations is the skew Brownian permuton. In particular, for str...

conjecture | open | MathDB 页面 | 原文线索

V2-067495 | MATHDB-351368

The gate set is a Garside shadow

摘录 / EXCERPT: Let (W,S) be a Coxeter system, let Γ be the set of gates of W, and let joins be taken in the weak order on W. A Garside shadow conjecture. The set Γ is closed u...

conjecture | open | MathDB 页面 | 原文线索

V2-067496 | MATHDB-351369

The boundary-root characterization of gates

摘录 / EXCERPT: Let (W,S) be a Coxeter system. For $x\in W$, let $T(x^{-1})$ be its cone type, let $\partial T(x^{-1})$ be the boundary roots of that cone type, and let $\Phi^1(x)$ be the base o...

conjecture | open | MathDB 页面 | 原文线索

V2-067497 | MATHDB-351389

The dual-partition refinement conjecture for poset-weight partitions

摘录 / EXCERPT: Dual-partition refinement conjecture. If $|H_i|\geqslant 2$ for all $i\in\Omega$, then the dual partition of $\mathcal{Q}(\boldsymbol{H},\boldsymbol{P})$ is finer than...

conjecture | open | MathDB 页面 | 原文线索

V2-067498 | MATHDB-351393

Immanant positivity conjecture for Hankel matrices

摘录 / EXCERPT: Let $H=(h_{i,j})_{i,j\geq 0}$ be the Hankel matrix defined in the paper, and for $k\geq 1$ let $I=(i_1,\ldots,i_k)$ and $J=(j_1,\ldots,j_k)$ be sequences satisfying ... Let $H_{I,J}$...

conjecture | open | MathDB 页面 | 原文线索

V2-067499 | MATHDB-351397

Quadratic-field-size conjecture for MDS(3) codes

摘录 / EXCERPT: An (n,k)-$\operatorname{MDS}(\ell)$ code is a code with the higher-order MDS property for parameters n,k,ℓ. Quadratic-field-size conjecture. There exist $(n,3)$-...

conjecture | open | MathDB 页面 | 原文线索

V2-067500 | MATHDB-351404

Finiteness-property conjecture for commutator subgroups of thick buildings

摘录 / EXCERPT: Let $d \in \mathbb{N}$ and let $(\Delta_n)_{n \in \mathbb{N}}$ be a sequence of finite d-dimensional buildings whose thickness satisfies ... as $n \rightarrow \infty$. For each bui...

conjecture | open | MathDB 页面 | 原文线索

V2-067501 | MATHDB-351407

Unique travelling-wave shooting-parameter conjecture for low ECM density

摘录 / EXCERPT: Fix $\kappa>0$ and define ... Let $\bar{m}\in(0,m^*(\kappa)]$ and $c\geq2\sqrt{1-\bar{m}}$. For the travelling-wave system in the source, let $\alpha_0(c)$ and $\alpha_1(c)$ be the s...

conjecture | open | MathDB 页面 | 原文线索

V2-067502 | MATHDB-351414

Induced zesting conjecture for Drinfeld centers

摘录 / EXCERPT: Let $\mathcal{C}$ be a fusion category, and let an associative zesting of $\mathcal{C}$ induce a zesting on its center $\mathcal{Z}(\mathcal{C})$. Induced zesting conjecture. The i...

conjecture | open | MathDB 页面 | 原文线索

V2-067503 | MATHDB-351444

Inoue-surface Chern-Ricci flow convergence conjecture

摘录 / EXCERPT: Let M be any Inoue surface. Let $\omega(t)$ be the Chern-Ricci flow and let ω_∞ be the closed semipositive $(1,1)$-form representing a multiple of $c_1(K_M)$ describe...

conjecture | open | MathDB 页面 | 原文线索

V2-067504 | MATHDB-351455

Nonexistence of lattice t-PTMCs in the intermediate range

摘录 / EXCERPT: Let $\Lambda_n=\mathbb{Z}^n$ and let a lattice t-PTMC be a lattice perfect total-multipacking code with parameter t. Nonexistence conjecture. There is no lattice t-PTMC in...

conjecture | open | MathDB 页面 | 原文线索

V2-067505 | MATHDB-351464

Shen's conjecture on gradient conditions for magnetic Schrödinger operators

摘录 / EXCERPT: Shen's conjecture. Some condition on $\nabla {\bf B}$ is necessary to prove Theorem 0.11 of Shen.

conjecture | open | MathDB 页面 | 原文线索

V2-067506 | MATHDB-351465

Gaussian upper-bound conjecture for walk counts on a high-dimensional torus

摘录 / EXCERPT: Let $c_n^{\mathbb{T}}$ be the n-step weakly self-avoiding-walk count on the torus, let μ be the growth constant on $\mathbb{Z}^d$, and let V be the number of vertices in th...

conjecture | open | MathDB 页面 | 原文线索

V2-067507 | MATHDB-351466

Critical-window susceptibility conjecture for weakly self-avoiding walk on a torus

摘录 / EXCERPT: Let $\chi^{\mathbb{T}}(z)$ be the torus susceptibility, let z_c be the critical point, and let V be the number of vertices in the torus. Critical-window susceptibility conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-067508 | MATHDB-351467

Distinctness conjecture for frayed ribbon Schur Q-functions

摘录 / EXCERPT: Let D and E be shifted skew shapes. A frayed ribbon is a shifted near-ribbon containing two squares on the staircase, and let D^a denote the antipodal reflection of D acros...

conjecture | open | MathDB 页面 | 原文线索

V2-067509 | MATHDB-351484

The leading-monomial stability conjecture for symmetric-group invariant ideals

摘录 / EXCERPT: Order the variables as ... Let $G_{k,n}$ be the Gröbner basis for the ideal $cmathcal{I}_{k,n}$. Let $LG_{k,n}$ be the set of leading monomials of $G_{k,n}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067510 | MATHDB-351487

Chern-Fu-Tang conjecture on log-concavity of k-colored partitions

摘录 / EXCERPT: Chern-Fu-Tang conjecture. Let $n>m\geq 1$ and $k\geq 2$. Except for $(k,n,m)=(2,6,4)$,

conjecture | open | MathDB 页面 | 原文线索

V2-067511 | MATHDB-351488

Heim-Neuhauser conjecture for D’Arcais polynomials

摘录 / EXCERPT: Heim-Neuhauser conjecture. For all $x\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-067512 | MATHDB-351503

The bounded-degree wall or line-graph induced-subgraph conjecture

摘录 / EXCERPT: Let $W_{k\times k}$ be the $(k\times k)$-wall, let $L(H)$ denote the line graph of H, and let $\Delta(G)$ and $\operatorname{tw}(G)$ denote the maximum degree and treewidth of a...

conjecture | open | MathDB 页面 | 原文线索

V2-067513 | MATHDB-351507

The abundance conjecture for klt log pairs

摘录 / EXCERPT: Work over $\mathbb{C}$. Let (X,Δ) be a projective klt log pair of dimension $n\geq 2$, and let $\kappa(X,\Delta)$ denote the Kodaira dimension of $K_X+\Delta$. A divisor is...

conjecture | open | MathDB 页面 | 原文线索

V2-067514 | MATHDB-351508

The numerical-dimension-zero abundance conjecture

摘录 / EXCERPT: Work over $\mathbb{C}$. Let (X,Δ) be a projective klt log pair of dimension $n\geq 2$. Let $\kappa(X,\Delta)$ denote the Kodaira dimension of $K_X+\Delta$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067515 | MATHDB-351515

The converse Laguerre-operator characterization for shifted Laguerre-Pólya functions

摘录 / EXCERPT: Let $\phi(x)$ be a real entire function in the shifted Laguerre-Pólya class $\mathcal{SL}^{-}P$. For each $d\in\mathbb{N}$, suppose there exists $N_3(d)$ such that ... for all...

conjecture | open | MathDB 页面 | 原文线索

V2-067516 | MATHDB-351516

Eventual positivity of iterated Turán inequalities for shifted multiplier sequences

摘录 / EXCERPT: Let $\{\gamma_i\}_{i\geq 0}$ be a shifted multiplier sequence that is eventually increasing. For $j\in\mathbb{N}$, let T_j denote the order-j Turán operator, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-067517 | MATHDB-351517

Turán transforms preserve eventually increasing shifted multiplier sequences

摘录 / EXCERPT: Let $\{\gamma_i\}_{i\geq 0}$ be a shifted multiplier sequence that is eventually increasing, and define ... The Turán-transform preservation conjecture. For every $j\in\mathbb{N}$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-067518 | MATHDB-351518

Laguerre operators preserve the positive shifted Laguerre-Pólya class

摘录 / EXCERPT: Let $\phi(x)$ belong to the positive shifted Laguerre-Pólya class $\mathcal{SL\text{-}P}^{+}$, and let L_j be the Laguerre operator defined by the coefficient expansion ... The Lag...

conjecture | open | MathDB 页面 | 原文线索

V2-067519 | MATHDB-351524

Buhl cycle-condition converse for the weak maximum likelihood threshold

摘录 / EXCERPT: Let G be a graph with at least one edge. An orientation of G is acyclic if it has no directed cycles, and a cycle is stretched when it has the form...

conjecture | open | MathDB 页面 | 原文线索

V2-067520 | MATHDB-351534

Orlik's conjecture on the decomposition of the Milnor lattice

摘录 / EXCERPT: Let f be an isolated quasihomogeneous singularity. Let $(M_1,\ldots,M_{l_\psi})$ be the standard covering associated with its weight-system data, and let (H_{Mil},h_{Mil}) deno...

conjecture | open | MathDB 页面 | 原文线索

V2-067521 | MATHDB-351536

The degree-four planar line-graph conjecture

摘录 / EXCERPT: Let G be a graph, let $L(G)$ denote its line graph, and let $\Delta(G)$ denote its maximum degree. A graph is word-representable if it admits a word representation, and an orient...

conjecture | open | MathDB 页面 | 原文线索

V2-067522 | MATHDB-351537

The planar maximum-degree-four characterization of semi-transitive line graphs

摘录 / EXCERPT: Let G be a graph, let $L(G)$ denote its line graph, and let $\Delta(G)$ denote its maximum degree. An orientation is semi-transitive when it is acyclic and has no shortcuts. Plan...

conjecture | open | MathDB 页面 | 原文线索

V2-067523 | MATHDB-351538

Numerical exceptional collection conjecture for fake projective spaces

摘录 / EXCERPT: Numerical exceptional collection conjecture. For a suitable choice of the line bundles E_i, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-067524 | MATHDB-351547

Global minimality conjecture for the magnetic Ginzburg-Landau solutions

摘录 / EXCERPT: Global minimality conjecture. The solutions of the magnetic Ginzburg-Landau equations provided in Theorem 1.1 and Theorem 1.2 are energy minimizers of the energy functional.

conjecture | open | MathDB 页面 | 原文线索

V2-067525 | MATHDB-351548

Bayer-Macrì-Toda Bogomolov-Gieseker inequality

摘录 / EXCERPT: Bayer-Macrì-Toda Bogomolov-Gieseker inequality. For any $(b,w)\in U$ and $\nu_{b,w}$-semistable $E\in\mathcal D(X)$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067526 | MATHDB-351556

Irreducibility conjecture for trinomials with large middle coefficient

摘录 / EXCERPT: Irreducibility conjecture. If $|a|\geq 5$, then $x^{n}+ax^{m}\pm1$ is irreducible over $\mathbb{Q}$. Furthermore, there are only finitely many reducible polynomials of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-067527 | MATHDB-351563

Hilton-Zhao's Core Conjecture for class 2 graphs

摘录 / EXCERPT: Let G be a simple connected graph with maximum degree $\Delta\ge 3$. The core G_Δ is the subgraph induced by the vertices of degree Δ, and $\Delta(G_\Delta)\le 2$....

conjecture | open | MathDB 页面 | 原文线索

V2-067528 | MATHDB-351564

The (σ,ρ)-Poisson summation formula

摘录 / EXCERPT: Let k be a number field, let G be a k-split reductive group, and let $\rho\colon G^\vee(\mathbb{C})\to\operatorname{GL}_n(\mathbb{C})$ be a finite-dimensional representation....

conjecture | open | MathDB 页面 | 原文线索

V2-067529 | MATHDB-351565

The binary polycyclic-code isoduality characterization

摘录 / EXCERPT: Let C be a polycyclic code over $\mathbb{F}_2$ generated by $g(x)$ dividing x^n-ax^i-b. Binary polycyclic-code isoduality conjecture. The code C is iso-dual if and only if...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-067530 | MATHDB-351566

The binary nonexistence conjecture for self-dual polycyclic codes

摘录 / EXCERPT: A polycyclic code over the binary field $\mathbb{F}_2$ may be self-dual, self-orthogonal, or dual-containing. Binary nonexistence conjecture. There is no self-dual, self-orthogonal...

conjecture | open | MathDB 页面 | 原文线索

V2-067531 | MATHDB-351567

The semiorder quotient-isomorphism conjecture

摘录 / EXCERPT: Let $f(x)$ be a polynomial whose semiorder is e. Semiorder quotient-isomorphism conjecture. Then there is a ring isomorphism ... This proposes a general quotient-ring isomorphism g...

conjecture | open | MathDB 页面 | 原文线索

V2-067532 | MATHDB-351572

The UPS sign-period conjecture for powers of the infinite Borwein product

摘录 / EXCERPT: Let $c_t^{(m)}(n)$ denote the coefficient of q^n in $(q;q)_\infty^m(q^t;q^t)_\infty^{-m}$, and let a sequence be a UPS sequence when its signs are periodic with a least sign peri...

conjecture | open | MathDB 页面 | 原文线索

V2-067533 | MATHDB-351574

Conjecture that wreath products preserve intersection density

摘录 / EXCERPT: Wreath-product intersection-density conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067534 | MATHDB-351588

The N-th Hankel relaxation Riemann hypothesis

摘录 / EXCERPT: N-th Hankel relaxation Riemann hypothesis. The function Λ is in the N-th order Laguerre-Pólya class. These are proposed relaxations of the Riemann hypothesis: membershi...

conjecture | open | MathDB 页面 | 原文线索

V2-067535 | MATHDB-351589

The first Hankel relaxation Riemann hypothesis

摘录 / EXCERPT: First Hankel relaxation Riemann hypothesis.

conjecture | open | MathDB 页面 | 原文线索

V2-067536 | MATHDB-351593

Dahlberg's (e)-positivity conjecture for labelled unit interval graphs

摘录 / EXCERPT: Dahlberg's conjecture. All labelled unit interval graphs are (e)-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-067537 | MATHDB-351594

Appendable (e)-positivity conjecture for labelled unit interval graphs

摘录 / EXCERPT: Appendable (e)-positivity conjecture. Every labelled unit interval graph is appendable (e)-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-067538 | MATHDB-351599

The Bernstein-Sato formula for the singularity level of local complete intersections

摘录 / EXCERPT: Let Z be a reduced local complete intersection of codimension r. Let $p(Z)$ denote its singularity level, and let $\widetilde{\alpha}(Z)$ be the negative of the largest root of...

conjecture | open | MathDB 页面 | 原文线索

V2-067539 | MATHDB-351601

Mirror Shioda-Inose conjecture for Kähler rigid structures

摘录 / EXCERPT: Mirror Shioda-Inose conjecture. There is a bijective correspondence between the isomorphism classes of Kähler rigid structures on M and $\mathfrak{L}^{ev}_{(2,0)}$, given by

conjecture | open | MathDB 页面 | 原文线索

V2-067540 | MATHDB-351611

Polynomial χ-boundedness of grounded segment graphs

摘录 / EXCERPT: A grounded segment graph is the intersection graph of a collection of line segments grounded on a common line. Grounded segment-graph conjecture. The class of grounded segment grap...

conjecture | open | MathDB 页面 | 原文线索

V2-067541 | MATHDB-351635

The smaller-ring refinement of the perverse stable-pair functional equation

摘录 / EXCERPT: Smaller-ring refinement. The preceding proposition still holds with this smaller ring R in place of the ring allowing all factors $(1-Q^j)^{-1}$; in particular,

conjecture | open | MathDB 页面 | 原文线索

V2-067542 | MATHDB-351648

The sign and congruence conjecture for tangent permanents

摘录 / EXCERPT: Let t_n and t_n' be the quantities defined earlier in the paper, and let $\left(\frac{\cdot}{p}\right)$ denote the Legendre symbol. The tangent permanent conjecture. (i) For ev...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067543 | MATHDB-351655

Asymptotic resistance increment conjecture for straight linear K-trees

摘录 / EXCERPT: Let G be the straight linear K-tree with n vertices and parameter $k\geq 1$, and let H be the straight linear K-tree with $n+1$ vertices. Write $r_G(u,v)$ for the resista...

conjecture | open | MathDB 页面 | 原文线索

V2-067544 | MATHDB-351656

Asymptotic resistance increment conjecture for block tower graphs

摘录 / EXCERPT: Let G be the block tower graph $C_4\square P_n$, the Cartesian product of the 4-cycle and the path of length n, with $4n$ vertices, and let H be the block tower graph...

conjecture | open | MathDB 页面 | 原文线索

V2-067545 | MATHDB-351657

Exponential resistance growth conjecture for square grid graphs

摘录 / EXCERPT: Let $G=P_n\square P_n$ be the $n\times n$ grid graph with n^2 vertices, and let $H=P_{n+1}\square P_{n+1}$ be the $(n+1)\times(n+1)$ grid graph with $(n+1)^2$ vertices. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067546 | MATHDB-351658

Exponential resistance growth conjecture for triangular grid graphs

摘录 / EXCERPT: Let T_n be the triangular grid graph shown in Figure, with n rows and $m=n^2$ cells. Let a and b be distinct vertices of degree 2, and let $r_n(a,b)$ be the resistance di...

conjecture | open | MathDB 页面 | 原文线索

V2-067547 | MATHDB-351659

Second-smallest-value conjecture for CDT subproblems and quartically constrained quadratic optimization

摘录 / EXCERPT: A CDT subproblem is a cubic-regularized optimization subproblem, and a nonconvex quadratic optimization problem with a quartic constraint is an optimization problem whose objective...

conjecture | open | MathDB 页面 | 原文线索

V2-067548 | MATHDB-351676

The integral asymptotic expansion conjecture for irreducible Calabi-Yau Picard-Fuchs systems

摘录 / EXCERPT: Let $\mathcal{P}$ be a Picard-Fuchs operator of a family of Calabi-Yau threefolds with a MUM point at 0, and suppose that its associated local system is irreducible. For a loop...

conjecture | open | MathDB 页面 | 原文线索

V2-067549 | MATHDB-351678

Converse to derived isogeny implying quasi-liftable isogeny for abelian surfaces

摘录 / EXCERPT: Let X and Y be abelian surfaces over $\bar{k}$ with finite height and $\operatorname{char}(k)\neq 2$. They are quasi-liftably isogenous if they are isogenous through suitable l...

conjecture | open | MathDB 页面 | 原文线索

V2-067550 | MATHDB-351681

The generalized Bogomolov-Gieseker conjecture for the quadratic-quartic threefold

摘录 / EXCERPT: Let X be a smooth projective threefold that is the complete intersection of a quadratic and a quartic hypersurface in $\mathbb{P}^5$, and let H be an ample class. For...

conjecture | open | MathDB 页面 | 原文线索

V2-067551 | MATHDB-351684

Cohen-Lenstra heuristics for 3-torsion in quadratic fields

摘录 / EXCERPT: Cohen-Lenstra conjecture for $p=3$. $\operatorname{Conj}^{+}(3,\alpha)$ and $\operatorname{Conj}^{-}(3,\alpha)$ are true for every positive integer α.

conjecture | open | MathDB 页面 | 原文线索

V2-067552 | MATHDB-351688

Complete monotonicity of squared derivative ratios of Mill's ratio

摘录 / EXCERPT: Let r denote Mill's ratio, and write $r^{(n)}$ for its nth derivative. A function is completely monotone (CM) if all derivatives exist on $(0,\infty)$ and satisfy the alternati...

conjecture | open | MathDB 页面 | 原文线索

V2-067553 | MATHDB-351702

The connectivity conjecture for the crossing graph of random right-angled Coxeter groups

摘录 / EXCERPT: Connectivity conjecture. For every $\epsilon>0$, if

conjecture | open | MathDB 页面 | 原文线索

V2-067554 | MATHDB-351703

Ding's tri-weight code conjecture for trinomial value-sets

摘录 / EXCERPT: Let $m\ge 5$ be an odd integer. Let f be any trinomial specified in the paper, let $D(f)^*$ be its punctured value-set, and let $\mathcal{C}_{D(f)^*}$ denote the associated binar...

conjecture | open | MathDB 页面 | 原文线索

V2-067555 | MATHDB-351707

Finiteness conjecture for back stable double Grothendieck polynomial products

摘录 / EXCERPT: Back stable double Grothendieck polynomials have structure constants $\overleftarrow{c}^{w}_{uv}(a)$ defined by their products, with indices $u,v,w\in S_\mathbb{Z}$. Equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-067556 | MATHDB-351708

Eremenko's conjecture on unbounded components of the escaping set

摘录 / EXCERPT: Let f be a transcendental entire function, and let ... be its escaping set. Eremenko's conjecture. Every connected component of $I(f)$ is unbounded. The conjecture has been confirm...

conjecture | open | MathDB 页面 | 原文线索

V2-067557 | MATHDB-351710

Completeness conjecture for the simplicial model axiom system of $KB4_n$

摘录 / EXCERPT: Let A be the set of agents, and let $\KBfour+\mathbf{NE}+(\mathbf{SA_a})_{a \in A}$ be the axiom system consisting of the $KB4_n$ axioms together with the non-emptiness axiom...

conjecture | open | MathDB 页面 | 原文线索

V2-067558 | MATHDB-351711

The Seagull problem for graphs with independence number two

摘录 / EXCERPT: Seagull problem. For every graph G with $\alpha(G)=2$, we have

open_problem | open | MathDB 页面 | 原文线索

V2-067559 | MATHDB-351725

The 1-2-3 Conjecture for sum-proper edge-labellings

摘录 / EXCERPT: The 1-2-3 Conjecture. If G is a nice graph, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067560 | MATHDB-351726

The multiplicative 1-2-3 Conjecture for multiset-proper edge-labellings

摘录 / EXCERPT: The multiplicative 1-2-3 Conjecture for multisets. If G is a nice graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-067561 | MATHDB-351729

Time monotonicity conjecture for Bayesian sequential composite hypothesis testing

摘录 / EXCERPT: Let $V(n,\pi)$ be the value function defined in, where n denotes the time index, π is the relevant state variable, μ is a prior distribution, and ν is the parameter a...

conjecture | open | MathDB 页面 | 原文线索

V2-067562 | MATHDB-351733

Le Boudec's height lower-bound conjecture for quadratic twists

摘录 / EXCERPT: For integers A,B with $4A^3+27B^2\neq 0$, consider the quadratic twists ... for positive square-free integers d, and let $\eta_d(A,B)$ be defined by ... with $\eta_d(A,B)=+\infty$...

conjecture | open | MathDB 页面 | 原文线索

V2-067563 | MATHDB-351734

Height lower-bound conjecture for sextic twists of Mordell curves

摘录 / EXCERPT: The sextic-twist height conjecture. For fixed $\epsilon>0$, the set of $d\in S_6(X)\setminus\{0\}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067564 | MATHDB-351747

Poisson point-process and Gumbel conjecture for the low-lying eigenvalues

摘录 / EXCERPT: Poisson point-process and Gumbel conjecture. The point process

conjecture | open | MathDB 页面 | 原文线索

V2-067565 | MATHDB-351748

Eigenfunction and noise-profile localization conjecture

摘录 / EXCERPT: Let $d\in\{1,2,3\}$. For each $n\ge 1$, let $\varphi_{n,L}$ be the corresponding eigenfunction, let $U_{n,L}\in[-L/2,L/2]^d$ be a point where $|\varphi_{n,L}|$ reaches its global m...

conjecture | open | MathDB 页面 | 原文线索

V2-067566 | MATHDB-351752

Coxeter-element invariance conjecture for antichain toggles on fences

摘录 / EXCERPT: Coxeter-element invariance conjecture. Any linear combination of the indicator functions χ_y for $y\in S$ is c-mesic or orbomesic under the action of W if and only if it i...

conjecture | open | MathDB 页面 | 原文线索

V2-067567 | MATHDB-351754

Uniform lower-bound conjecture for the Schwarz preconditioner

摘录 / EXCERPT: Let A be the discretized integral operator in the referenced two- or three-dimensional models, and let T^{-1}_{Schwarz} be the Schwarz preconditioner associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-067568 | MATHDB-351757

Existence of optimal solutions for the convex control problem

摘录 / EXCERPT: Let $A\in\mathbb{R}^{n \times n}$, $B\in\mathbb{R}^{n \times m}$, $T>0$, $\beta\in\{1,2,\dots,m-1\}$, and $\alpha_j\in(0,T]$ for $j=1,2,\dots,m$, as in Problem ... exist. The exist...

conjecture | open | MathDB 页面 | 原文线索

V2-067569 | MATHDB-351764

The spectral fractional-Laplacian sign-transform inequality conjecture

摘录 / EXCERPT: Spectral sign-transform inequality conjecture. For $s\in(1,\frac32)$, the analogous inequalities should hold for spectral quadratic forms. The corresponding inequality has been pro...

conjecture | open | MathDB 页面 | 原文线索

V2-067570 | MATHDB-351765

Eventual unimodality criterion for colored-partition crank functions

摘录 / EXCERPT: Eventual unimodality conjecture. The function $\mathcal{D}(z;\tau)$ is eventually unimodal if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-067571 | MATHDB-351767

Locality conjecture for dimension-distinguishing graph properties

摘录 / EXCERPT: Locality conjecture. A graph property distinguishes the dimension if and only if it is local.

conjecture | open | MathDB 页面 | 原文线索

V2-067572 | MATHDB-351770

The existence conjecture for recursive APN functions

摘录 / EXCERPT: Recursive APN function conjecture. There exists a recursive APN function in every dimension $n\in\mathbb{N}$ with $n\geq 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-067573 | MATHDB-351774

Edge-only orbit coloring symmetry-fafulness conjecture

摘录 / EXCERPT: Edge-only orbit coloring conjecture. The edge-only orbit coloring is symmetry faithful for both the linear symmetry group and the orthogonal symmetry group:

conjecture | open | MathDB 页面 | 原文线索

V2-067574 | MATHDB-351779

Graph isomorphism conjecture for Milnor K-theory and higher Chow groups

摘录 / EXCERPT: Graph isomorphism conjecture. For every such A, the graph homomorphism is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-067575 | MATHDB-351780

Pseudo-localization conjecture for completed higher Chow cycle complexes

摘录 / EXCERPT: Pseudo-localization conjecture. The respective cokernels $\mathcal{C}_{\bullet}$ and $\mathcal{C}'_{\bullet}$ of these two restriction morphisms are acyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-067576 | MATHDB-351789

Monotonicity of Mahler measure along the Mahler flow

摘录 / EXCERPT: Let $P(z,w)=k-p(z,w)$ be a Newton polynomial undergoing the Mahler flow, with Mahler measure $m(P)$ and flow parameter $k>0$. Mahler-flow monotonicity conjecture. The Mahler measur...

conjecture | open | MathDB 页面 | 原文线索

V2-067577 | MATHDB-351790

Monotonicity of the Ronkin function along the Mahler flow

摘录 / EXCERPT: Let $P(z,w)=k-p(z,w)$ be a Newton polynomial, and let $R_k(x,y)$ denote its Ronkin function at flow parameter $k>0$. For $k_2>k_1$, distinguish the non-linear, bounded linear, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067578 | MATHDB-351791

Large-parameter scaling of bounded amoeba-complement volumes

摘录 / EXCERPT: Let $P(\boldsymbol{z})=k-p(\boldsymbol{z})$ be a Laurent polynomial in $\boldsymbol{z}\in\mathbb{C}^n$, and let $\mathcal{A}(P)$ be its amoeba. Write $V_\textup{bdd}$ for the volum...

conjecture | open | MathDB 页面 | 原文线索

V2-067579 | MATHDB-351796

Small-energy majorization conjecture on the bi-tree

摘录 / EXCERPT: Let $f,g:T^2\to\mathbb R_+$, where g is superadditive in each variable and ... Let $\lambda\geq 10\delta$. Small-energy majorization conjecture. There exists...

conjecture | refuted | MathDB 页面 | 原文线索

V2-067580 | MATHDB-351802

Berkovich's conjecture on non-inner automorphisms of finite p-groups

摘录 / EXCERPT: Berkovich's conjecture. Every finite nonabelian p-group admits a non-inner automorphism of order p.

conjecture | open | MathDB 页面 | 原文线索

V2-067581 | MATHDB-351805

Existence of arbitrarily long bicrucial permutations of even length

摘录 / EXCERPT: A bicrucial permutation is a square-free permutation such that prepending or appending any single element creates a square. The existence conjecture. There exist arbitrarily long b...

conjecture | open | MathDB 页面 | 原文线索

V2-067582 | MATHDB-351813

Nagao's conjecture for the rank of an elliptic curve over $\mathbb{Q}(T)$

摘录 / EXCERPT: Nagao's conjecture. The rank r of $\mathcal{E}(\mathbb{Q}(T))$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067583 | MATHDB-351815

RDC+ lower-density conjecture for rank-zero elliptic curves

摘录 / EXCERPT: Let $\mathscr{E}$ be the set of isomorphism classes of elliptic curves over ${\mathbb Q}$, ordered by height, and let $\mathscr{E}'$ be the subset consisting of rank-zero elliptic...

conjecture | open | MathDB 页面 | 原文线索

V2-067584 | MATHDB-351818

The zero-entry conjecture for satisfying Loosened Collatz tuples

摘录 / EXCERPT: Let a tuple $(y_1,\dots,y_n)$ be satisfying when it satisfies the Loosened Collatz Function, and let $(2,2,\ldots)$ denote the constant tuple with every entry equal to 2. Zero-en...

conjecture | open | MathDB 页面 | 原文线索

V2-067585 | MATHDB-351824

The logarithmic treewidth conjecture for (theta, triangle)-free graphs

摘录 / EXCERPT: Logarithmic treewidth conjecture. There exists a constant c such that every (theta, triangle)-free graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067586 | MATHDB-351825

The bounded-degree induced-wall conjecture

摘录 / EXCERPT: Bounded-degree induced-wall conjecture. For all $k,\Delta>0$, there exists a constant $c=c(k,\Delta)$ such that every graph G with maximum degree at most Δ and treewidth m...

conjecture | open | MathDB 页面 | 原文线索

V2-067587 | MATHDB-351826

The logarithmic treewidth conjecture excluding cliques, bicliques and induced walls

摘录 / EXCERPT: Induced-wall exclusion conjecture. For all $t\geq 0$, there exists a constant $c=c(t)$ such that every graph G with no induced subgraph isomorphic to K_t, $K_{t,t}$, a subdivis...

conjecture | open | MathDB 页面 | 原文线索

V2-067588 | MATHDB-351847

Cao-Maulik-Toda genus-zero DT$_4$-Gopakumar-Vafa correspondence

摘录 / EXCERPT: Cao-Maulik-Toda's genus-zero correspondence conjecture. Via some suitable choice of orientation on the moduli space, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067589 | MATHDB-351848

Cao-Toda genus-one DT$_4$ descendent correspondence conjecture

摘录 / EXCERPT: Cao-Toda's genus-one descendent correspondence conjecture. Via some suitable choice of orientation on the moduli space, one has, for all $\gamma\in H^2(X,\mathbb Z)$,

conjecture | open | MathDB 页面 | 原文线索

V2-067590 | MATHDB-351849

Genus-zero DT$_4$-Gromov-Witten correspondence for local Fano threefolds

摘录 / EXCERPT: Local Fano threefold genus-zero correspondence. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067591 | MATHDB-351857

Tolerance complexes preserve Lerayness

摘录 / EXCERPT: Tolerance-Leray conjecture. Then $\mathcal{T}_{t}(K)$ is $(\eta(d+1,t+1)-1)$-Leray.

conjecture | open | MathDB 页面 | 原文线索

V2-067592 | MATHDB-351858

Tolerance complexes of collapsible complexes are Leray

摘录 / EXCERPT: Collapsible-to-Leray tolerance conjecture. Then $\mathcal{T}_{t}(K)$ is $(\eta(d+1,t+1)-1)$-Leray.

conjecture | open | MathDB 页面 | 原文线索

V2-067593 | MATHDB-351869

The differential operator conjecture for pseudo-reflection groups

摘录 / EXCERPT: Differential operator conjecture. If $G=G(m,1,n)$, then this identity holds. The conjecture extends the proved cases for rank at most two and the real pseudo-reflection groups; com...

conjecture | open | MathDB 页面 | 原文线索

V2-067594 | MATHDB-351871

Zabrocki's Hilbert-series conjecture for type A super-diagonal coinvariants

摘录 / EXCERPT: Zabrocki's conjecture. For $0\leq k\leq n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-067595 | MATHDB-351872

The refined type A Hilbert-series conjecture

摘录 / EXCERPT: Refined type A Hilbert-series conjecture. For $1\leq j\leq n-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-067596 | MATHDB-351874

Conjecture F for fibrations with split fibres away from a finite locus

摘录 / EXCERPT: Conjecture F. The subset

conjecture | open | MathDB 页面 | 原文线索

V2-067597 | MATHDB-351875

Conjecture F+ for fibrations with controlled Brauer classes

摘录 / EXCERPT: Conjecture F+. The subset

conjecture | open | MathDB 页面 | 原文线索

V2-067598 | MATHDB-351876

Conjecture F_const for constant Brauer classes

摘录 / EXCERPT: Conjecture Fconst. The subset

conjecture | open | MathDB 页面 | 原文线索

V2-067599 | MATHDB-351877

Conjecture on the primitivity of the over-exceptional lattice for K3 surfaces

摘录 / EXCERPT: Primitivity conjecture for the over-exceptional lattice. If X has infinite automorphism group, then the embedding

conjecture | open | MathDB 页面 | 原文线索

V2-067600 | MATHDB-351879

Gross's conjectural formula for the Tate-Shafarevich group of a Gross curve

摘录 / EXCERPT: Gross's conjectural formula. The order of the Tate-Shafarevich group of the Gross curve over H is

conjecture | open | MathDB 页面 | 原文线索

V2-067601 | MATHDB-351884

The $mathfrak{M}_H(G)$-conjecture for Selmer groups over p-adic Lie extensions

摘录 / EXCERPT: Let F_∞/F be a strongly admissible p-adic Lie extension, set $G=\operatorname{Gal}(F_\infty/F)$ and $H=\operatorname{Gal}(F_\infty/F^{\mathrm{cyc}})$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067602 | MATHDB-351893

Pilehrood-Pilehrood-Tauraso conjecture for cyclic sums on 1-2 indices

摘录 / EXCERPT: Pilehrood-Pilehrood-Tauraso's conjecture. For any positive integer n with $n>k$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067603 | MATHDB-351895

Sierksma's Dutch cheese conjecture over valued fields

摘录 / EXCERPT: Sierksma's Dutch cheese conjecture. There are at least $((r-1)!)^d$ partitions of X into parts whose convex hulls intersect. The conjecture extends the Dutch cheese conjecture fr...

conjecture | open | MathDB 页面 | 原文线索

V2-067604 | MATHDB-351896

Second selection lemma over valued fields

摘录 / EXCERPT: For each $d\in\mathbb{N}_{\geq 1}$, let $c,s>0$ be constants. Second selection lemma over valued fields. For every valued field K, every $\alpha\in(0,1]$, every $n\in\mathbb{N}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067605 | MATHDB-351898

Gross-Reeder root number conjecture for adjoint gamma-factors

摘录 / EXCERPT: Let $T\subset G$ be a maximal torus split over F, with root datum $(X(T),\Phi(T),X^{\vee}(T),\Phi^{\vee}(T))$, and let $G^\sphat$ be the dual group. Define ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067606 | MATHDB-351906

Whittaker nonvanishing conjecture for the theta representation of $Sp_{r-1}^{(r)}$

摘录 / EXCERPT: Let $r>1$ be odd, let F be a global field, and let $\Theta_{r-1}^{(r)}$ be the theta representation of the r-fold metaplectic cover of $Sp_{r-1}({\mathbb A})$. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-067607 | MATHDB-351907

The Descent Conjecture for the metaplectic theta representation

摘录 / EXCERPT: Let r be odd, and let ρ be the representation of $Sp_{2n+r-2k+1}^{(2r)}({\mathbb A})$ generated by the descent functions defined by the integral construction in the paper. D...

conjecture | open | MathDB 页面 | 原文线索

V2-067608 | MATHDB-351909

Conjecture on regularity of symbolic and ordinary powers of edge ideals

摘录 / EXCERPT: Let G be a simple graph, and let $I(G)$ be its edge ideal. For each integer $s\geq 1$, the symbolic power $I(G)^{(s)}$ and ordinary power $I(G)^s$ are ideals in the associated po...

conjecture | open | MathDB 页面 | 原文线索

V2-067609 | MATHDB-351913

EKR property conjecture for the remaining primitive groups of degree a product of two odd primes

摘录 / EXCERPT: Let G range over the groups listed in Table 2 of the source, namely the remaining socles of primitive groups of degree pq that do not admit imprimitive subgroups, where p and...

conjecture | open | MathDB 页面 | 原文线索

V2-067610 | MATHDB-351919

Conjecture on complete fractional Yamabe metrics with high-dimensional singularities

摘录 / EXCERPT: High-dimensional fractional Yamabe singularity conjecture. There exists a complete Yamabe metric that is singular on Σ^k.

conjecture | open | MathDB 页面 | 原文线索

V2-067611 | MATHDB-351921

Pun's Demazure-character atom-positivity conjecture

摘录 / EXCERPT: Let $\alpha,\beta\in\mathbb{N}^m$, let κ_α and κ_β be Demazure characters, and let $\mathcal{A}_\gamma$ denote the Demazure atom indexed by γ. Pun's c...

conjecture | open | MathDB 页面 | 原文线索

V2-067612 | MATHDB-351922

Characterization of fixed points by the weak regularity inequality

摘录 / EXCERPT: Fixed-point characterization. The following conditions are equivalent: (i) F has a fixed point in every nonempty spherically complete subspace of M; and (ii) the displayed ineq...

conjecture | open | MathDB 页面 | 原文线索

V2-067613 | MATHDB-351923

Extension of Theorem 1 to arbitrary spherically complete sets

摘录 / EXCERPT: Let A and B be nonempty spherically complete subsets of an ultrametric space, and let Statements (i)--(iv) denote the four statements from Theorem 1. Extension conjecture. Stat...

conjecture | open | MathDB 页面 | 原文线索

V2-067614 | MATHDB-351930

Ismail's characterization conjecture for continuous q-Jacobi and Al-Salam-Chihara polynomials

摘录 / EXCERPT: Let $(P_n)_{n\geq0}$ be a monic orthogonal polynomial sequence (OPS), and let π be a polynomial of degree at most 2 that does not depend on n. For $0<q<1$, define the Askey...

conjecture | open | MathDB 页面 | 原文线索

V2-067615 | MATHDB-351931

The Abelian Cayley graph independent-set counting conjecture

摘录 / EXCERPT: Let $\epsilon>0$, and let Γ be a connected Abelian Cayley graph on $2n$ vertices with degree $(2+\epsilon)\log n$. Let $i(\Gamma)$ denote the number of independent sets of...

conjecture | open | MathDB 页面 | 原文线索

V2-067616 | MATHDB-351941

Conjecture on rank separations for polynomials in two variables

摘录 / EXCERPT: Rank-separation conjecture. There exist no functions $f,g,h\colon\mathbb{N}\to\mathbb{N}$ such that, for every $p\in\mathbb{R}[x,y]$, independently of the degree of p,

conjecture | open | MathDB 页面 | 原文线索

V2-067617 | MATHDB-351943

Vitushkin's quasiadditivity conjecture for continuous analytic capacity

摘录 / EXCERPT: Let K_1 and K_2 be compact subsets of the plane, and let α denote continuous analytic capacity. Vitushkin's quasiadditivity conjecture. There exists a constant C such...

conjecture | open | MathDB 页面 | 原文线索

V2-067618 | MATHDB-351957

Hachimori-Lim rank-growth conjecture for elliptic curves over imaginary quadratic fields

摘录 / EXCERPT: Let E be an elliptic curve over an imaginary quadratic field F satisfying the conditions stated in the source, and let $F^{(n)}$ and F_{cyc} denote the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-067619 | MATHDB-351959

Goldberg's conjecture on total chromatic numbers

摘录 / EXCERPT: Goldberg's conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-067620 | MATHDB-351972

Komlós's conjecture for vector balancing

摘录 / EXCERPT: Komlós's conjecture. There exists a constant $K>0$ such that, for every such matrix A,

conjecture | open | MathDB 页面 | 原文线索

V2-067621 | MATHDB-351993

Twin conjecture for periodic branches of groups of maximal class

摘录 / EXCERPT: Twin conjecture. There exists $n_0=n_0(p)$ such that for all $n\geqslant n_0$ and each group G at depth $n-c$ in $\mathcal B_p(n+d)$, there exists a twin H at depth $n-c-d$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-067622 | MATHDB-351996

Grinshpun-Sárközy conjecture on tiling bounded-degree graph sequences

摘录 / EXCERPT: Grinshpun-Sárközy conjecture. For every positive integer r there exists a constant C_r such that, for every $\Delta\geq 2$ and every Δ-bounded graph sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-067623 | MATHDB-351999

Vallentin's conjecture for the least Euclidean distortion of distance-regular graphs

摘录 / EXCERPT: Let G) be a distance-regular graph with diameter d, valency k, eigenvalues $\theta_0>\theta_1>ots>\theta_d$, and standard sequences $(w_0(\theta_j),...,w_d(\theta_j))$ corres...

conjecture | open | MathDB 页面 | 原文线索

V2-067624 | MATHDB-352000

Revised Vallentin conjecture for distance-regular graphs

摘录 / EXCERPT: Let G be a distance-regular graph with diameter d, with eigenvalues $\theta_0>\theta_1>\cdots>\theta_d$ and cosine sequences $(w_0(\theta_j),\ldots,w_d(\theta_j))$ correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-067625 | MATHDB-352012

Conjectured asymptotics for Hermite-Padé polynomials with a nonreal symmetric condenser plate

摘录 / EXCERPT: Let $f\in\mathbb C(z,w)$ and $f_\infty\in\mathscr H(\infty)$. Assume the setting of the subsection in which $E\subset\mathbb R$, $E\cap F=\varnothing$, and F is symmetric with re...

conjecture | open | MathDB 页面 | 原文线索

V2-067626 | MATHDB-352016

An integral test for the stationary pinned string

摘录 / EXCERPT: Integral-test conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067627 | MATHDB-352022

Ratios conjecture for quadratic Dirichlet L-functions over function fields

摘录 / EXCERPT: Ratios Conjecture. Under these constraints,

conjecture | open | MathDB 页面 | 原文线索

V2-067628 | MATHDB-352023

Symmetric cube period formula for Maass cusp forms

摘录 / EXCERPT: Let θ be the cubic theta series and let $cphi$ be a Maass cusp form with ground field K containing the cube roots of unity. Write $L^*(s,cphi,cmathrm{sym}^3)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-067629 | MATHDB-352029

Singular-weight dimension conjecture for E8-invariant holomorphic Jacobi forms

摘录 / EXCERPT: Let $H(t)$ be the dimension of the space of $W(E_8)$-invariant holomorphic Jacobi forms of weight 4 and positive index t. Let $N(t)$ be the number of distinct Weyl orbits of ve...

conjecture | open | MathDB 页面 | 原文线索

V2-067630 | MATHDB-352030

Finite-generation conjecture for Weyl-invariant E8 weak Jacobi forms

摘录 / EXCERPT: Let $W(E_8)$ be the Weyl group of E_8, and consider all $W(E_8)$-invariant weak Jacobi forms of integral weight and integral index. Let $M_*(\operatorname{SL}_2(\mathbb Z))$ deno...

conjecture | open | MathDB 页面 | 原文线索

V2-067631 | MATHDB-352038

Conjecture on the ordered continued-fraction coefficients of quadratic integers

摘录 / EXCERPT: Let ... be a purely periodic quadratic integer, and let $\mathcal{C}(\Delta)=\{a_0,a_1,\dots,a_{l-1}\}$. For each positive integer n, let w_n be the n-th largest element of...

conjecture | open | MathDB 页面 | 原文线索

V2-067632 | MATHDB-352040

Vu's list-coloring conjecture for graphs with few common neighbors

摘录 / EXCERPT: Vu's conjecture. For every $czeta,ceps>0$, the following holds for all sufficiently large D: if G has maximum degree at most D and every two distinct vertices have at most...

conjecture | open | MathDB 页面 | 原文线索

V2-067633 | MATHDB-352045

Invariant means on non-exceptional left amenable rational-map semigroups

摘录 / EXCERPT: Invariant-means conjecture. The set of left invariant means on S coincides with the set of right invariant means on S.

conjecture | open | MathDB 页面 | 原文线索

V2-067634 | MATHDB-352050

The non-degeneracy conjecture for solutions of the non-critical $SU(3)$ Toda system

摘录 / EXCERPT: Non-degeneracy conjecture. Except for finitely many τ modulo the $SL(2,\mathbb{Z})$ action, every solution of the Toda system is non-degenerate. If true, this would support th...

conjecture | open | MathDB 页面 | 原文线索

V2-067635 | MATHDB-352052

Uniqueness conjecture for minimal embeddings of the three-dimensional torus in the four-sphere

摘录 / EXCERPT: Let $T^3=S^1\times S^1\times S^1$ be the three-dimensional torus, embedded minimally in the round four-dimensional sphere $\mathbb{S}^4$. Two embeddings are considered equivalent i...

conjecture | open | MathDB 页面 | 原文线索

V2-067636 | MATHDB-352054

Multidimensional moment-bound conjecture for PDMP approximations

摘录 / EXCERPT: Let ψ satisfy the growth bounds in $\$, and let $H(x,v)$ be the function in inequality $\$. The mesh satisfies $0=t_0\leq t_1\leq\cdots$ with $\delta_k=t_k-t_{k-1}<\delta_0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067637 | MATHDB-352066

Jaeger-Zhang conjecture for signed planar graphs

摘录 / EXCERPT: Let p be a positive integer, and let $\mathcal{P}^*_{4p+1}$ denote the class of signed planar graphs with the indicated girth condition. Jaeger-Zhang conjecture. ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-067638 | MATHDB-352067

Bipartite analogue of the Jaeger-Zhang conjecture

摘录 / EXCERPT: Let p be a positive integer, and let $\mathcal{P}^*_{4p}$ denote the class of signed bipartite planar graphs with the indicated girth condition. Bipartite analogue of the Jaeger-...

conjecture | open | MathDB 页面 | 原文线索

V2-067639 | MATHDB-352068

The $S(K_3)$-free signed bipartite planar graph conjecture

摘录 / EXCERPT: Let (G,σ) be a signed bipartite planar graph, let $S(K_3)$ be the signed subdivision of K_3, and write $(G,\sigma)\to(K_{3,3},M)$ for the corresponding signed homomorphism...

open_problem | open | MathDB 页面 | 原文线索

V2-067640 | MATHDB-352074

Kolmogorov's converse exponential inequality for lower capacities

摘录 / EXCERPT: Let $\{X_{n,i};i=1,\ldots,k_n\}$ be an array of independent random variables in the sub-linear expectation space $(\Omega,\mathscr{H},\widehat{\mathbb E})$ with ... Let x_n be a se...

conjecture | open | MathDB 页面 | 原文线索

V2-067641 | MATHDB-352076

Polynomial-rank symmetrization conjecture for multilinear forms

摘录 / EXCERPT: Symmetrization conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067642 | MATHDB-352094

Weakly periodic tiling conjecture for exact clusters

摘录 / EXCERPT: Weakly periodic tiling conjecture. There exists a weakly periodic F-tiling of $\mathbb Z^d$; more precisely, there exists an F-tiling $T\subseteq\mathbb Z^d$ admitting such a f...

conjecture | open | MathDB 页面 | 原文线索

V2-067643 | MATHDB-352095

Nonexistence of absorbing points for coupling resonance functions

摘录 / EXCERPT: Let H_0 and V be the operators specified in the preceding setup, and let an absorbing point mean a point of the resolvent set at which some coupling resonance function tends to...

conjecture | open | MathDB 页面 | 原文线索

V2-067644 | MATHDB-352097

Existence of a second-order rigid but locally and continuously flexible framework

摘录 / EXCERPT: Existence conjecture. There exists a normed space with a second-order rigid framework that is locally and continuously flexible.

conjecture | open | MathDB 页面 | 原文线索

V2-067645 | MATHDB-352098

Random-information decay threshold for polynomially decaying ellipsoids

摘录 / EXCERPT: Let $1\le p\le\infty$ and let the semiaxes be $\sigma_j=j^{-\lambda}$ for $j\in\mathbb{N}$, with $\lambda>\frac{1}{s}=\left(\frac{1}{2}-\frac{1}{p}\right)_+$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-067646 | MATHDB-352104

Hausdorff-measure lower-bound conjecture for the truncated Laplacian

摘录 / EXCERPT: Hausdorff-measure lower-bound conjecture. There exists a constant $C>0$, depending on n, k, hR, and an upper bound for $\mathscr{H}^k(E)$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-067647 | MATHDB-352106

The large-characteristic Newton polygon conjecture for twisted binomial L-functions

摘录 / EXCERPT: Let χ be a multiplicative character of order c over a finite field, let $f(x)=x^d+\lambda x^e$ with $(d,e)=1$, and let p be the characteristic. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-067648 | MATHDB-352107

Turán inequality conjecture for the plane-partition polynomials

摘录 / EXCERPT: Let $P_n(x)$ denote the polynomials associated with the generalized plane partitions, and let a be an integer with $a\geq 12$. Turán inequality conjecture. For every real number...

conjecture | open | MathDB 页面 | 原文线索

V2-067649 | MATHDB-352108

Generalized Turán inequality conjecture for plane-partition polynomials

摘录 / EXCERPT: Let $P_n(x)$ be the polynomials associated with generalized plane partitions, and define ... Let a and b be integers satisfying $a-1>b\geq 0$ and $(a,b)\notin\{(4,2),(6,4)\}$. Ge...

conjecture | open | MathDB 页面 | 原文线索

V2-067650 | MATHDB-352111

Gendulphe's density conjecture for projective measured foliations

摘录 / EXCERPT: Let $\no_g$ be a non-orientable surface with $g\geq 4$. Let $\pmf^+(\no_g)$ be the space of projective measured foliations on $\no_g$ containing no one-sided leaves, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067651 | MATHDB-352112

Low-lying zeros conjecture for cubic twists of an elliptic curve over a function field

摘录 / EXCERPT: Low-lying zeros conjecture. For every $n\in\mathbb{Z}_{\geq1}$,

conjecture | open | MathDB 页面 | 原文线索

V2-067652 | MATHDB-352116

Bhargava's local specification density conjecture for S_n-number fields

摘录 / EXCERPT: Let n be a positive integer, and let $\Sigma=(\Sigma_2,\Sigma_3,\Sigma_5,\ldots)$ be an acceptable collection of local specifications, where each Σ_p is a set of...

conjecture | open | MathDB 页面 | 原文线索

V2-067653 | MATHDB-352119

Kitaev's determinant conjecture for Fredholm commutators

摘录 / EXCERPT: Kitaev's determinant conjecture. Under these assumptions, the Fredholm determinant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067654 | MATHDB-352142

Eremenko's disconjugacy conjecture for Wronskians

摘录 / EXCERPT: Let $V\in\operatorname{Gr}_{k,n}(\mathbb{R})$. Suppose that all complex zeros of the Wronskian $\operatorname{Wr}(V)$ are real, and let $I\subseteq\mathbb{R}$ be any interval on wh...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-067655 | MATHDB-352143

Totally positive secant conjecture for Schubert problems

摘录 / EXCERPT: Let $0\le k\le n$, and let $I_1,\dots,I_{k(n-k)}$ be pairwise disjoint intervals of $\mathbb{P}^1(\mathbb{R})$. For each I, let X_I be a multiset of k points in I_I, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-067656 | MATHDB-352144

Wronskian positivity conjecture for totally nonnegative and totally positive spaces

摘录 / EXCERPT: Let $V\in\operatorname{Gr}_{k,n}(\mathbb{R})$. Regard the Wronskian $\operatorname{Wr}(V)$ as having a zero at $-\infty$ when its degree is less than $k(n-k)$. Wronskian positivity...

conjecture | open | MathDB 页面 | 原文线索

V2-067657 | MATHDB-352155

The modularity conjecture for elliptic curves over totally real fields

摘录 / EXCERPT: Modularity conjecture. Every elliptic curve over a totally real field F is modular.

conjecture | open | MathDB 页面 | 原文线索

V2-067658 | MATHDB-352162

Conjecture for the average logarithmic derivative of quadratic Dirichlet L-functions

摘录 / EXCERPT: The logarithmic-derivative conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-067659 | MATHDB-352164

The folklore conjecture on spherical 4-distance 7-designs

摘录 / EXCERPT: Folklore conjecture. Let $C\subset\mathbb{S}^{n-1}$, with $n\geq 2$, be a spherical 4-distance 7-design. Then C is a tight spherical 7-design. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-067660 | MATHDB-352167

Tightness conjecture for the fat-shattering dimension of bounded affine maxima

摘录 / EXCERPT: Let $B ewcommand{\mathbb{R}}{\mathbb{R}}\subset\mathbb{R}^d$ be the Euclidean unit ball. For $i\in[k]$, let ... be collections of bounded affine functions on B, and let $F_{\max}$...

conjecture | open | MathDB 页面 | 原文线索

V2-067661 | MATHDB-352168

Logarithmic aggregation conjecture for fat-shattering dimension

摘录 / EXCERPT: Let Ω be a domain, let $F_i\subseteq\mathbb{R}^{\Omega}$ for $i\in[k]$, and let $F_{\max}$ be their k-fold maximum. Logarithmic aggregation conjecture. For some universal...

conjecture | open | MathDB 页面 | 原文线索

V2-067662 | MATHDB-352169

Nonsingularity of superelliptic Legendre inflectionary curves away from the singular fibers

摘录 / EXCERPT: Let F be the base field, let $n\geq 2$, and let a,b,c be positive integers. Let $\mathcal{C}^{\ell}_m$ be the inflectionary curve derived from the pencil...

conjecture | open | MathDB 页面 | 原文线索

V2-067663 | MATHDB-352170

Coefficient factorization conjecture for Weierstrass atomic inflection polynomials

摘录 / EXCERPT: Let $n=2\ell$, assume that the characteristic of F is zero or sufficiently positive, and let $P^{\ell,*}_m$ be the atomic inflection polynomial of the Weierstrass pencil...

conjecture | open | MathDB 页面 | 原文线索

V2-067664 | MATHDB-352171

Newton non-degeneracy conjecture for Weierstrass inflectionary curves

摘录 / EXCERPT: Let $n=2\ell$, assume that $\operatorname{char}(F)$ is zero or sufficiently positive, and let $P^{\ell,*}_m$ be the atomic inflection polynomial of $y^n=x^3+\lambda x+2$. For even...

conjecture | open | MathDB 页面 | 原文线索

V2-067665 | MATHDB-352172

Nonsingularity away from the singular fibers for Weierstrass inflectionary curves

摘录 / EXCERPT: Let $n=2\ell$, let F have characteristic zero or sufficiently positive characteristic different from three, and let $\mathcal{C}^{\ell}_m\subset\mathbb{P}(1,2,1)$ be the inflecti...

conjecture | open | MathDB 页面 | 原文线索

V2-067666 | MATHDB-352174

Singularity conjecture for D4 inflectionary curves

摘录 / EXCERPT: Let $n=2\ell$, assume that $\operatorname{char}(F)$ is zero or sufficiently positive, and let $\mathcal{C}_m=\mathcal{C}^{\ell}_m$ be the m-th inflectionary curve from the D_4...

conjecture | open | MathDB 页面 | 原文线索

V2-067667 | MATHDB-352175

Local Newton polygon conjecture for D4 inflectionary curves

摘录 / EXCERPT: Let $n=2\ell$, assume that $\operatorname{char}(F)$ is zero or sufficiently positive, and let $\mathcal{C}_m$ be the m-th inflectionary curve from $y^n=x^5+x^3+sx$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-067668 | MATHDB-352178

Oprea-Pandharipande conjecture on tautological rings of lattice-polarized K3 moduli spaces

摘录 / EXCERPT: Let $\mathcal{F}_{\Lambda}$ be the moduli space of lattice-polarized K3 surfaces, let $R^*(\mathcal{F}_{\Lambda})\subset A^*(\mathcal{F}_{\Lambda})$ be its tautological ring, and s...

conjecture | open | MathDB 页面 | 原文线索

V2-067669 | MATHDB-352185

Correspondence packing number for triangle-free graphs

摘录 / EXCERPT: Triangle-free correspondence packing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067670 | MATHDB-352186

Asymptotic independent-transversal packing bounds for list and correspondence covers

摘录 / EXCERPT: Independent-transversal packing conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067671 | MATHDB-352187

K_r-free correspondence-cover packing conjecture

摘录 / EXCERPT: Kr-free correspondence-cover packing conjecture. For every $r\ge3$, there is some $C_r>0$ such that the following holds for

conjecture | open | MathDB 页面 | 原文线索

V2-067672 | MATHDB-352196

General critical-value rationality conjecture for degree-six L-functions

摘录 / EXCERPT: Let F be the totally real field with set of archimedean places $Sigma_{\infty}$, and let π and τ be the automorphic representations considered in the paper, with balance...

conjecture | open | MathDB 页面 | 原文线索

V2-067673 | MATHDB-352198

The generalized New Way integral conjecture for \(\mathrm{Sp}_{2n}\times\mathrm{GL}_n\)

摘录 / EXCERPT: Generalized New Way conjecture. This integral is Eulerian in the sense of the New Way method and represents the tensor-product partial L-function

conjecture | open | MathDB 页面 | 原文线索

V2-067674 | MATHDB-352199

Optimal Gegenbauer measure for logarithmic energy

摘录 / EXCERPT: Optimal-measure claim. For any fixed $n\geq 1$, the optimal measure is $d\sigma=w^\lambda dx$ for some $\lambda\in(-1/2,\infty)$. The value of λ may depend on n.

conjecture | open | MathDB 页面 | 原文线索

V2-067675 | MATHDB-352217

Super Markoff-number conjecture for the once-punctured torus

摘录 / EXCERPT: Begin with an oriented triangulation of the once-punctured torus described in the source, and retain e as a formal variable rather than setting it equal to 1. Iterated flips in...

conjecture | open | MathDB 页面 | 原文线索

V2-067676 | MATHDB-352222

The local root-number formula for Sturm-polynomial error terms

摘录 / EXCERPT: Local conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-067677 | MATHDB-352223

The conjecture that Paley type II Hadamard matrix subfactors have Temperley-Lieb standard invariants

摘录 / EXCERPT: Paley type II Temperley-Lieb conjecture. Paley type II Hadamard matrix subfactors have Temperley-Lieb standard invariants.

conjecture | open | MathDB 页面 | 原文线索

V2-067678 | MATHDB-352226

Delecroix-Goujard-Zograf-Zorich polynomiality conjecture for large-genus psi-class intersections

摘录 / EXCERPT: For nonnegative integers $g,n,d_1, oops$ satisfying $2g-2+n>0$ and $d_1+cdots+d_n=3g-3+n$, let $\mathcal{G}_{d_1,\dots,d_n}(g)$ denote the normalized psi-class intersection numbers...

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V2-067679 | MATHDB-352230

The modularity conjecture for higher theta series of unitary groups

摘录 / EXCERPT: Let X'/X be the fixed double cover, let $U(2m)$ be the associated quasi-split unitary group, and let $\widetilde{Z}^{r}_{m}$ be the generating series taking values in...

conjecture | open | MathDB 页面 | 原文线索

V2-067680 | MATHDB-352231

The modularity conjecture in the split Lagrangian case

摘录 / EXCERPT: Let $\mathcal{E}$ be a rank m vector bundle on X', let $\mathcal{E}'=\sigma^{*}\underline{\operatorname{Hom}}(\mathcal{E},\nu^{*}\mathfrak{L})$, and let...

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V2-067681 | MATHDB-352232

The large-slope special case of the modularity conjecture

摘录 / EXCERPT: Let $\mathcal{E}$ be a rank m vector bundle on X', let $\mathcal{E}'=\sigma^{*}\underline{\operatorname{Hom}}(\mathcal{E},\nu^{*}\mathfrak{L})$, and suppose the maximal slope o...

conjecture | open | MathDB 页面 | 原文线索

V2-067682 | MATHDB-352233

The genus-square-root conjecture for proper chromatic number

摘录 / EXCERPT: Genus-square-root conjecture. If G is a graph of genus g, then

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V2-067683 | MATHDB-352239

Canonical-subspace conjecture for maximum cliques in semi-primitive pseudo-Paley graphs

摘录 / EXCERPT: Let $X=PP(q,2d,l)$ be a semi-primitive pseudo-Paley graph with $q=p^{2rt}$, where r is even. A maximum clique is a clique of size $\omega(X)$. Canonical-subspace conjecture. If ……

conjecture | open | MathDB 页面 | 原文线索

V2-067684 | MATHDB-352240

Two-clique conjecture for square-root maximum cliques

摘录 / EXCERPT: Let $X=PP(q,2d,l)$ be a semi-primitive pseudo-Paley graph with $q=p^{2rt}$, where r is even, and assume … A maximum clique is a clique of size $\omega(X)$. Two-clique conjecture.…

conjecture | open | MathDB 页面 | 原文线索

V2-067685 | MATHDB-352242

Classification conjecture for two-dimensional cliques in pseudo-Paley graphs

摘录 / EXCERPT: Let V be a 2-dimensional F_p-subspace of $\mathbb{F}_{p^4}$ containing 1, and let g be the primitive element used to define the graph $PP(p^4,p+1,l)$. Two-dimensional cli…

conjecture | open | MathDB 页面 | 原文线索

V2-067686 | MATHDB-352247

Bergeron-Garsia-Haiman-Tesler nabla positivity conjecture

摘录 / EXCERPT: Let λ and μ be partitions of n, let $k\geq 1$, let ∇ be the nabla operator on modified Macdonald polynomials, and let $(\,, \,)$ denote the Hall inner product.…

conjecture | open | MathDB 页面 | 原文线索

V2-067687 | MATHDB-352263

Effective asymmetric coloring conjecture for inverse limits of finite permutation groups

摘录 / EXCERPT: Effective asymmetric coloring conjecture. There exists a function $g:\mathscr{Gr}\to\mathbb N$ such that, whenever $k\geq g(G_0)$, the inverse limit of this system admits a zero-ne…

conjecture | open | MathDB 页面 | 原文线索

V2-067688 | MATHDB-352264

Polylogarithmic bound for effective asymmetric coloring of inverse limits

摘录 / EXCERPT: Polylogarithmic bound conjecture. There exists a polynomial p such that the effective asymmetric coloring conjecture holds with

conjecture | open | MathDB 页面 | 原文线索

V2-067689 | MATHDB-352270

Rigging conjecture for semi-regular numbers

摘录 / EXCERPT: Rigging conjecture. Every real number that is not an eigenvalue of infinite multiplicity is semi-regular for some compatible choice of rigging.

conjecture | open | MathDB 页面 | 原文线索

V2-067690 | MATHDB-352271

The (I)-to-(L) singularity conjecture

摘录 / EXCERPT: (I)-to-(L) conjecture. Every (I)-singular number is (L)-singular.

conjecture | open | MathDB 页面 | 原文线索

V2-067691 | MATHDB-352272

Equality conjecture for the (F)-singular subspace

摘录 / EXCERPT: Equality conjecture. For every semi-regular λ and every $H\in\mathcal A$, the inclusion in the preceding result can be replaced by equality:

conjecture | open | MathDB 页面 | 原文线索

V2-067692 | MATHDB-352273

Scalar-resolvent singularity conjecture

摘录 / EXCERPT: Scalar-resolvent singularity conjecture. Under these assumptions, λ is essentially singular.

conjecture | open | MathDB 页面 | 原文线索

V2-067693 | MATHDB-352297

Conjecture on great-circle thrackleable trees and augmented caterpillars

摘录 / EXCERPT: A great-circle thrackleable tree is a tree admitting a thrackle drawing on the sphere in which the edges are great-circle arcs. An augmented caterpillar is the graph class defined…

conjecture | open | MathDB 页面 | 原文线索

V2-067694 | MATHDB-352302

Guo-Schlosser-Zudilin's divisibility conjecture for sums of even powers of q-binomial coefficients

摘录 / EXCERPT: Let n be a positive integer and let r be an arbitrary integer. Write $[n]=(1-q^n)/(1-q)$ for the q-integer and let $\Phi_n(q)$ denote the nth cyclotomic polynomial. For int…

conjecture | open | MathDB 页面 | 原文线索

V2-067695 | MATHDB-352310

Uniform limiting distribution in the profit-driven regime

摘录 / EXCERPT: Profit-driven regime conjecture. The scaled steady-state variable should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067696 | MATHDB-352318

Rich 2-to-1 Games conjecture

摘录 / EXCERPT: Let $\Psi=(L\cup R,E,\Sigma_L,\Sigma_R,\Phi)$ be a Rich 2-to-1 Games instance with $|\Sigma_L|=n$ and $|\Sigma_R|=n/2$. For each $u\in L$, the distribution of partitions of…

conjecture | open | MathDB 页面 | 原文线索

V2-067697 | MATHDB-352320

Conjecture on inclusion divergence for sparse random geometric graphs

摘录 / EXCERPT: Inclusion-divergence conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-067698 | MATHDB-352323

Kottwitz's multiplicity conjecture for global rigid inner forms

摘录 / EXCERPT: Kottwitz's multiplicity conjecture. The multiplicity of π in the discrete spectrum of G is given by

conjecture | open | MathDB 页面 | 原文线索

V2-067699 | MATHDB-352327

Nesbitt-Shapiro inequality at the remaining endpoint values

摘录 / EXCERPT: Nesbitt-Shapiro endpoint conjecture. If $p=12$ or $p=23$, then

conjecture | open | MathDB 页面 | 原文线索

V2-067700 | MATHDB-352329

The $\mathcal{A}^\star$-optimality conjecture for OBL-G$_\flat$

摘录 / EXCERPT: OBL-G$_\flat$'s $\mathcal{A}^\star$-optimality conjecture. OBL-G$_\flat$ is $\mathcal{A}^\star$-optimal in the sense that

conjecture | open | MathDB 页面 | 原文线索

V2-067701 | MATHDB-352332

Unimodality conjecture for Garvan k-rank functions

摘录 / EXCERPT: Garvan k-rank unimodality conjecture. For each integer $k\ge2$ and integer $n\ge n_u(k)$, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-067702 | MATHDB-352333

Rank and crank log-concavity conjectures

摘录 / EXCERPT: Let $M(m,n)$ and $N(m,n)$ denote respectively the number of partitions of n with crank m and rank m. Rank and crank log-concavity conjecture. The inequalities ... hold for...

conjecture | open | MathDB 页面 | 原文线索

V2-067703 | MATHDB-352350

Unimodality conjecture for minimal obstruction counts of uniform clutters

摘录 / EXCERPT: For a d-uniform clutter $\mathcal{C}$, let $I(\mathcal{C})$ be its edge ideal, and let $\Omega_{d,k}$ consist of the clutters satisfying ... ... and having no proper induced sub-clut...

conjecture | open | MathDB 页面 | 原文线索

V2-067704 | MATHDB-352357

Theorem 7.7 bound conjecture for projective hypersurface singularities

摘录 / EXCERPT: Let $Z\subseteq\mathbb{P}^n$ be a hypersurface of degree d with only isolated singular points. Theorem 7.7 gives a bound on the number of these singularities, while Theorem 3.1 g...

conjecture | open | MathDB 页面 | 原文线索

V2-067705 | MATHDB-352358

Kikuta-Takahashi's categorical Gromov-Yomdin conjecture

摘录 / EXCERPT: Kikuta-Takahashi's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067706 | MATHDB-352369

Mkrtchyan's S_{12}-Conjecture

摘录 / EXCERPT: S_{12}-Conjecture. For each cubic graph G admitting a perfect matching, $S_{12}\prec G$.

conjecture | open | MathDB 页面 | 原文线索

V2-067707 | MATHDB-352371

The bounded-constituent-degree motion conjecture for coherent configurations

摘录 / EXCERPT: Motion conjecture under bounded constituent degrees. If every constituent has degree at most δn, then

conjecture | open | MathDB 页面 | 原文线索

V2-067708 | MATHDB-352373

The conformal-dimension characterization of Sierpiński carpet energy exponents

摘录 / EXCERPT: Conformal-dimension characterization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067709 | MATHDB-352376

da Silva-Sellers congruences for three-colour 3-regular partitions

摘录 / EXCERPT: da Silva-Sellers conjecture. For all $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-067710 | MATHDB-352377

Extended Double Cover conjecture for $\mathcal{C}_{11}\cup\mathcal{C}_{10}$

摘录 / EXCERPT: Let (B,π) be a signed graph in $\mathcal{C}_{11}\cup\mathcal{C}_{10}$. A signed graph is $\mathcal{P}$-complete if every signed planar graph satisfying the relevant girth inequ...

conjecture | open | MathDB 页面 | 原文线索

V2-067711 | MATHDB-352379

The $(d+1)$-rigidity conjecture for flag spheres

摘录 / EXCERPT: Let G be the graph of a flag triangulation of the $(d-1)$-dimensional sphere. A graph is $(d+1)$-rigid if it has the corresponding generic rigidity property. The $(d+1)$-rigidity...

conjecture | open | MathDB 页面 | 原文线索

V2-067712 | MATHDB-352380

The exact maximum-stable-set conjecture for flag spheres

摘录 / EXCERPT: Let G be the graph of a flag triangulation of the $(d-1)$-dimensional sphere on n vertices, and let $\alpha_M(d,n)$ denote the maximum possible size of a stable set in such a g...

conjecture | open | MathDB 页面 | 原文线索

V2-067713 | MATHDB-352385

Positivity conjecture for Sylow branching coefficients of powers of two

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $\lambda\vdash 2^k$. Write $l(\lambda)$ for the number of parts of λ, let $\lambda_{l(\lambda)}$ be its smallest part, and let Z^λ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-067714 | MATHDB-352386

The BS/PT wall-crossing formula with insertions

摘录 / EXCERPT: Let X be a projective threefold with Gorenstein and rational singularities, let $f:Y\to X$ be a resolution of singularities of relative dimension 1, and let Y be a projective...

conjecture | open | MathDB 页面 | 原文线索

V2-067715 | MATHDB-352397

Gordon's conjecture on Seifert fibered surgeries on hyperbolic knots

摘录 / EXCERPT: Gordon's conjecture. If $S^3_{p/q}(K)$ is a Seifert fibered space, then

conjecture | open | MathDB 页面 | 原文线索

V2-067716 | MATHDB-352398

The Montesinos rational unknotting conjecture

摘录 / EXCERPT: Montesinos rational unknotting conjecture. These three conditions are equivalent for all Montesinos knots.

conjecture | open | MathDB 页面 | 原文线索

V2-067717 | MATHDB-352400

The characterization of destabilization sets by stability characters

摘录 / EXCERPT: Characterization conjecture. Every element of the destabilization set of U comes from vanishing conditions on the map M. Moreover, knowing the destabilization set of U is equ...

conjecture | open | MathDB 页面 | 原文线索

V2-067718 | MATHDB-352409

Schlosser's quadratic-residue q-product conjecture

摘录 / EXCERPT: Schlosser's conjecture. For every $n\geq0$, the four families of coefficient identities stated in the source hold for $p=13,17,29,53$, with the displayed residue sets and shifts.

conjecture | open | MathDB 页面 | 原文线索

V2-067719 | MATHDB-352410

Matching-coefficient conjecture for the 21-, 28-, and 77-modulus products

摘录 / EXCERPT: Matching-coefficient conjecture. For every $n\geq0$, the following identities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-067720 | MATHDB-352411

Branching-number factorization conjecture for Cohen-Lenstra series of curve singularities

摘录 / EXCERPT: Let p be an $\mathbb F_q$-point of a reduced curve X. Write $\widehat Z_{X,p}(x)$ for the associated Cohen-Lenstra series, and let $(xq^{-1};q^{-1})_\infty$ denote the q-Poch...

conjecture | open | MathDB 页面 | 原文线索

V2-067721 | MATHDB-352412

Smooth-coefficient conjecture for the Cohen-Lenstra factor $H(x;t)$

摘录 / EXCERPT: Smooth-coefficient conjecture. The function $H(x;t)$ satisfies the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-067722 | MATHDB-352420

Bachmann's conjecture that brackets and bi-brackets span the same space

摘录 / EXCERPT: Bachmann's conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-067723 | MATHDB-352422

Bachmann's polynomial reformulation of the bracket conjecture

摘录 / EXCERPT: Refinement of Bachmann's conjecture. There exist polynomials $Q_j\in\mathbb{Q}[X_1,\dots,X_j]$, for $j\geq 1$, with $Q_j\equiv 0$ for all but finitely many j, such that for every...

conjecture | open | MathDB 页面 | 原文线索

V2-067724 | MATHDB-352423

Conjecture on the Neveu CSBP limit in the pulled regime

摘录 / EXCERPT: Let $\rho_1=1+\frac{\pi^2}{4}$, and let $(\overline N_t)_{t\geq 0}$ denote the rescaled particle-number process. Pulled-regime CSBP conjecture. If $\rho<\rho_1$, under suitable ass...

conjecture | open | MathDB 页面 | 原文线索

V2-067725 | MATHDB-352424

Conjecture on the Feller diffusion limit in the fully pushed regime

摘录 / EXCERPT: Let ρ_2 be the threshold separating the weakly pushed and fully pushed regimes, and let $(\overline N_t)_{t\geq 0}$ denote the rescaled particle-number process. Fully pushed F...

conjecture | open | MathDB 页面 | 原文线索

V2-067726 | MATHDB-352437

Kuznetsov-Smirnov Grassmannian residual-category conjecture

摘录 / EXCERPT: Let $X=\operatorname{Gr}(k,n)$ be the Grassmannian, and let $\mu(d)$ be the Möbius function. Define ... Kuznetsov-Smirnov Grassmannian residual-category conjecture. The category...

conjecture | open | MathDB 页面 | 原文线索

V2-067727 | MATHDB-352444

Eguchi-Hanson limit conjecture for Kähler-Einstein edge metrics on the second Hirzebruch surface

摘录 / EXCERPT: Let ${\mathbb F}_2$ be the second Hirzebruch surface, equipped with the family of Kähler-Einstein edge metrics ω_{β_1,β_2} described in the source, with...

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V2-067728 | MATHDB-352445

Addition-chain defect truncation order-type conjecture

摘录 / EXCERPT: Addition-chain defect order-type conjecture. For each whole number k, the set $\mathscr{D}^{\ell}\cap[0,k]$ has order type ω^k. This conjecture concerns the fine structure...

conjecture | open | MathDB 页面 | 原文线索

V2-067729 | MATHDB-352452

Circular-locus pencil conjecture for triangle centers of polar families

摘录 / EXCERPT: Let X_k' denote a triangle center of the polar family, and suppose its locus is a circle with nonzero radius. Circular-locus pencil conjecture. That circle belongs to the parabol...

conjecture | open | MathDB 页面 | 原文线索

V2-067730 | MATHDB-352453

Pentagon centroid-locus conjecture for parabola-inscribed Poncelet families

摘录 / EXCERPT: Let $\cal{P}$ be a parabola of focal distance f, and let $\cal{C}$ be a focus-centered circle of radius r such that they admit a Poncelet family of inscribed pentagons. Pentago...

conjecture | open | MathDB 页面 | 原文线索

V2-067731 | MATHDB-352454

Polar-pentagon centroid-locus conjecture

摘录 / EXCERPT: Let $\cal{P}$ be a parabola and consider the family of polar polygons associated with parabola-inscribed Poncelet pentagons. Polar-pentagon centroid-locus conjecture. The loci of t...

conjecture | open | MathDB 页面 | 原文线索

V2-067732 | MATHDB-352455

General parabola centroid-locus conjecture for Poncelet polygons

摘录 / EXCERPT: Let $\cal{P}$ be a parabola and $\cal{C}$ a focus-centered circle, and let $\cal{R}$ be a Poncelet family of N-gons inscribed in $\cal{P}$ and circumscribed about $\cal{C}$. Gene...

conjecture | open | MathDB 页面 | 原文线索

V2-067733 | MATHDB-352456

Polar-parabola centroid-locus conjecture for arbitrary polygon size

摘录 / EXCERPT: Let $\cal{P}$ be a parabola, let $\cal{C}$ be a focus-centered circle, and let $\cal{R}$ be a Poncelet family of N-gons inscribed in $\cal{P}$ and circumscribed about $\cal{C}$....

conjecture | open | MathDB 页面 | 原文线索

V2-067734 | MATHDB-352457

Perimeter-centroid conic conjecture for polar images of bicentric families

摘录 / EXCERPT: Let $\cal{B}'$ be the conic-inscribed polar image of a generic bicentric family of N-gons with respect to its bicentric circumcircle. Perimeter-centroid conic conjecture. Over...

conjecture | open | MathDB 页面 | 原文线索

V2-067735 | MATHDB-352458

Nonconic perimeter-centroid conjecture for polar polygons of bicentric images

摘录 / EXCERPT: Let $\cal{B}'$ be the conic-inscribed polar image of a generic bicentric family of N-gons, and let $\cal{P}'$ be the conic to which $\cal{B}'$ is inscribed. Nonconic perimeter-ce...

conjecture | open | MathDB 页面 | 原文线索

V2-067736 | MATHDB-352459

Conjecture on the minimizers of the exceptional-point construction

摘录 / EXCERPT: Minimizer conjecture. For every $1\leq i\leq n-1$, $\mathbf y_n(i,i+1)$ is the unique element of $\mathcal F_n^+$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067737 | MATHDB-352465

General partition-function conjecture for ${\mathbb C}^2/{\mathbb Z}_n\times{\mathbb C}$

摘录 / EXCERPT: General partition-function conjecture. With $\kappa=t_1t_2t_3$, $t_4=q/\sqrt{\kappa}$, $t_5=1/(q\sqrt{\kappa})$, and, after symmetrization, $q=-p_0p_1\cdots p_{n-1}$,

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V2-067738 | MATHDB-352481

The non-all-net conjecture for the 24-cell, 120-cell, and 600-cell

摘录 / EXCERPT: All-net conjecture. The 24-cell, the 120-cell, and the 600-cell fail to be all-net.

conjecture | open | MathDB 页面 | 原文线索

V2-067739 | MATHDB-352482

Conjecture on the secrecy gain of formally unimodular lattices

摘录 / EXCERPT: Let Λ be a formally unimodular lattice, with secrecy function $\Xi_{\Lambda}(\tau)$, and denote its secrecy gain by ξ_{Λ}. Formally unimodular secrecy-gain conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-067740 | MATHDB-352483

Equality of the K- and $ 0\bb{C}_p$-Julia sets for rational maps

摘录 / EXCERPT: Julia-set equality conjecture. For every rational map $\phi\in K(z)$,

conjecture | open | MathDB 页面 | 原文线索

V2-067741 | MATHDB-352508

The converse binomial inequality for alpha>2

摘录 / EXCERPT: Let $k\in\mathbb N$, $\alpha>2$, and $x,y\geq 0$. Converse binomial inequality. One has ... This is proposed as an appropriate inequality comparing the generalized binomial sum with...

conjecture | open | MathDB 页面 | 原文线索

V2-067742 | MATHDB-352514

General resolution-proof dual-certificate optimality conjecture

摘录 / EXCERPT: General resolution-proof certificate conjecture. For any unsatisfiable formula F and resolution proof T, the resulting vector $\bm\mu$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-067743 | MATHDB-352525

CPS rank decomposition conjecture for real nonsymmetric fourth-order tensors

摘录 / EXCERPT: Let $\mathcal{A}\in\mathbb{R}_{cps}^{n^4}\backslash\mathbb{R}_s^{n^4}$ be a real fourth-order conjugate partial-symmetric tensor that is not symmetric. A CPS rank decomposition is...

conjecture | open | MathDB 页面 | 原文线索

V2-067744 | MATHDB-352542

Chen, Wang and Wang's unimodality conjecture for q-Catalan polynomials

摘录 / EXCERPT: Let $C_n(q)$ be the q-Catalan polynomial defined by ... Chen, Wang and Wang's conjecture. For $n\ge 16$, the q-Catalan polynomial $C_n(q)$ is unimodal. This conjecture concerns t...

conjecture | open | MathDB 页面 | 原文线索

V2-067745 | MATHDB-352543

Xin and Zhong's unimodality conjecture for q-Catalan polynomials

摘录 / EXCERPT: Let $C_n(q)$ be the q-Catalan polynomial defined by ... Xin and Zhong's conjecture. For $n\ge 1$, the polynomial $(1+q)C_n(q)$ is unimodal. This is a conjecture about the unimodali...

conjecture | open | MathDB 页面 | 原文线索

V2-067746 | MATHDB-352547

The consistency and perfect-matching conjecture for brane tilings

摘录 / EXCERPT: Consistency and perfect-matching conjecture. A brane tiling is consistent if and only if the corresponding toric diagram has maximally tempered coefficients for its boundary points...

conjecture | open | MathDB 页面 | 原文线索

V2-067747 | MATHDB-352548

The Hauptmodul and monodromy-group conjecture for Mahler flows

摘录 / EXCERPT: Hauptmodul and monodromy-group conjecture. Then k^n is a Hauptmodul for Γ^b,

conjecture | open | MathDB 页面 | 原文线索

V2-067748 | MATHDB-352549

The stronger congruence-subgroup relation conjecture

摘录 / EXCERPT: Stronger congruence-subgroup conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-067749 | MATHDB-352550

The canonical-weight criterion for the $\tau_B=\tau_R$ condition

摘录 / EXCERPT: Canonical-weight criterion conjecture. Up to $\operatorname{SL}(2,\mathbb{Z})$, the condition

conjecture | open | MathDB 页面 | 原文线索

V2-067750 | MATHDB-352552

Multiplicity of genus 2 helicoids

摘录 / EXCERPT: Multiplicity conjecture. There exist multiple translation-invariant genus 2 helicoids and multiple genus 2 helicoids.

conjecture | open | MathDB 页面 | 原文线索

V2-067751 | MATHDB-352553

Existence of minimally screw-symmetric minimal surfaces

摘录 / EXCERPT: Screw-only symmetry conjecture. There exist SMIMS which are only symmetric with respect to screw motions.

conjecture | open | MathDB 页面 | 原文线索

V2-067752 | MATHDB-352568

Rationality conjecture for critical L-values of RACARs on GL(3)

摘录 / EXCERPT: Let Π be a RACAR of GL_3, let E be its coefficient field, and let η be a Dirichlet character with Gauss sum $G(\eta)$. For $j\in\mathbb{Z}_{\geq 0}$, set ... As...

conjecture | open | MathDB 页面 | 原文线索

V2-067753 | MATHDB-352571

Acharya-Kureethara conjecture on the rna number of parity signed graphs

摘录 / EXCERPT: Acharya-Kureethara conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067754 | MATHDB-352572

Conjecture on uniqueness of the rna number for graphs

摘录 / EXCERPT: Uniqueness conjecture for the rna number.

conjecture | open | MathDB 页面 | 原文线索

V2-067755 | MATHDB-352574

Takemura's factorization conjecture for Fuchsian operators

摘录 / EXCERPT: Let H be the Fuchsian operator ... where $m_1,\ldots,m_M$ are positive integers, $\alpha+\beta-\gamma-\delta+\sum_{k=1}^M m_k+1=0$, and $a_1,\ldots,a_M$ are apparent singularities...

conjecture | open | MathDB 页面 | 原文线索

V2-067756 | MATHDB-352582

Rational connectedness conjecture for klt pairs with strictly nef anti-log canonical divisor

摘录 / EXCERPT: Rational connectedness conjecture. If $-(K_X+\Delta)$ is strictly nef, then X is rationally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-067757 | MATHDB-352585

The 3-zero-sum partition property conjecture

摘录 / EXCERPT: Let Γ be a finite Abelian group, and let $I(\Gamma)$ denote its set of involutions. The 3-zero-sum partition property conjecture. If $|I(\Gamma)|>1$, then Γ has the 3...

conjecture | open | MathDB 页面 | 原文线索

V2-067758 | MATHDB-352586

The complete multipartite $\mathbb{Z}_{2^n}$-distance-magic conjecture

摘录 / EXCERPT: Let $G=K_{n_1,n_2,\ldots,n_t}$ be a complete t-partite graph such that ... and $n_i\geq 2$ for all i. A $\mathbb{Z}_{2^n}$-distance-magic labeling is a bijection from the vertice...

conjecture | open | MathDB 页面 | 原文线索

V2-067759 | MATHDB-352587

The equal-sum partition conjecture for cyclic groups of order 2^n

摘录 / EXCERPT: Let $\mathbb{Z}_m$ be a cyclic group of order $m=2^n$. Let t and $m_1,\ldots,m_t$ be positive integers satisfying ... with $m_1\geq 1$ and $m_i\geq 2$ for $2\leq i\leq t$. Equal-su...

conjecture | open | MathDB 页面 | 原文线索

V2-067760 | MATHDB-352599

Seymour's TDD conjecture for integral set covering systems

摘录 / EXCERPT: Seymour's TDD conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067761 | MATHDB-352601

Geometric-signature conjecture for Kasteleyn matrices of PBDTP graphs

摘录 / EXCERPT: Let G be a planar bicoloured PBDTP graph, and let its edges carry the geometric signature. Form the associated sign matrix by assigning the corresponding geometric-signature sign...

conjecture | open | MathDB 页面 | 原文线索

V2-067762 | MATHDB-352609

Conjecture on nilpotent conjugacy growth and lower-central ranks

摘录 / EXCERPT: Let G be a finitely generated nilpotent group, and for each i let r_i be the torsion-free rank of the quotient $G^{(i+1)}/G^{(i)}$. Nilpotent conjugacy-growth rank conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-067763 | MATHDB-352612

The special-horn generation conjecture for special-horn trivial cofibrations

摘录 / EXCERPT: Let SpHorn be the set of special horn inclusions. A map of simplicial sets is bijective on 0-simplices when it induces a bijection on its sets of 0-simplices....

conjecture | open | MathDB 页面 | 原文线索

V2-067764 | MATHDB-352618

Stability of eigenvectors of equiangular tight frame decomposable tensors

摘录 / EXCERPT: Let $\mathcal{T}$ be an equiangular tight frame decomposable tensor, meaning a symmetric tensor admitting a decomposition into powers of vectors from an equiangular tight frame. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-067765 | MATHDB-352621

Zagier-Hoffman-Saha dimension and basis conjectures for multiple zeta values and multiple t-values

摘录 / EXCERPT: Let $(d_n)_{n\geqslant1}$ be the Padovan sequence defined by $d_1=0$, $d_2=1$, $d_3=1$ and $d_n=d_{n-2}+d_{n-3}$ for $n\geqslant4$, and let $(f_n)_{n\geqslant1}$ be the Fibonacci s...

conjecture | open | MathDB 页面 | 原文线索

V2-067766 | MATHDB-352622

Identity for the multiple t-value $t(\{2\}^N,1)$

摘录 / EXCERPT: Let $N\in\mathbb{N}$, and write $t(\{2\}^N,1)$ for the multiple t-value with N consecutive entries equal to 2, followed by 1. Let $I(2N)$ and $\eta(s)$ denote the quantitie...

open_problem | open | MathDB 页面 | 原文线索

V2-067767 | MATHDB-352624

Andrews-Paule congruence conjecture for 2-elongated plane partitions

摘录 / EXCERPT: For $k\formul{ obreakspace}= obreakspace2$, let $d_2(n)$ denote the number of 2-elongated plane partitions of n, equivalently the coefficient defined by ... where n is a nonnegat...

conjecture | open | MathDB 页面 | 原文线索

V2-067768 | MATHDB-352632

Corteel--Dousse--Uncu positivity conjecture for cylindric partitions

摘录 / EXCERPT: For a nonnegative integer n and a profile $c=(c_0,c_1,c_2)$, define $d:=c_0+c_1+c_2$, $\ell:=\gcd(d,3)$, and ... Assume $d\not\equiv0\pmod 3$. Corteel--Dousse--Uncu's conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-067769 | MATHDB-352633

Corteel--Dousse--Uncu positivity conjecture for arbitrary-rank cylindric partitions

摘录 / EXCERPT: For integers $d,r\geqslant1$, let $c=(c_0,\dots,c_{r-1})$ be a profile of level d, meaning ... Let $Q_{n,c}(q)$ be the formal power series defined from the cylindric-partition gene...

conjecture | open | MathDB 页面 | 原文线索

V2-067770 | MATHDB-352643

The converse from completely uniquely paired to semidistrim lattices

摘录 / EXCERPT: Let L be a lattice. It is completely uniquely paired if there is a unique bijection $\lambda:L\to L$ such that, for every $x\in L$, $\lambda(x)$ is maximal among the elements...

conjecture | open | MathDB 页面 | 原文线索

V2-067771 | MATHDB-352645

Uniqueness conjecture for exterior quadrilaterals of prescribed angles and conformal modulus

摘录 / EXCERPT: Let $M>0$ be a positive real number. Consider exterior quadrilaterals with prescribed angles and vertices $A_1=1$, $A_2=0$, A_3, and A_4, and let their conformal modulus be M...

conjecture | open | MathDB 页面 | 原文线索

V2-067772 | MATHDB-352659

Cofiber Morava K-theory dimension formula conjecture for real Grassmannians

摘录 / EXCERPT: Let $C_d({\mathbb R}^m)$ be the cofiber of the inclusion $\operatorname{Gr}_d({\mathbb R}^{m-1})\hookrightarrow\operatorname{Gr}_d({\mathbb R}^m)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067773 | MATHDB-352665

Generalized fine Selmer torsion conjecture over p-adic Lie extensions

摘录 / EXCERPT: Let $\bar{A}$ be one of the coefficient objects considered in the source, with coefficient ring $\bar{R}$, and let F_∞/F be a p-adic Lie extension such that...

conjecture | open | MathDB 页面 | 原文线索

V2-067774 | MATHDB-352668

Nakagawa-Naruse shifted reverse plane partition formulas for connective Schur functions

摘录 / EXCERPT: Let β be a parameter, let β-dependent skew functions $g\hspace{-0.2mm}p_{ u}$ and $g\hspace{-0.2mm}q_{ u}$ be indexed by strict partitions, and let $ u$ be a strict par...

conjecture | open | MathDB 页面 | 原文线索

V2-067775 | MATHDB-352669

Shifted bar tableau formulas for connective Schur functions

摘录 / EXCERPT: Let β be a parameter, let β-dependent skew functions $j\hspace{-0.2mm}p_{\nu}$ and $j\hspace{-0.2mm}q_{\nu}$ be indexed by strict partitions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067776 | MATHDB-352674

The Abel-map linearization conjecture for reflectionless KdV potentials

摘录 / EXCERPT: Let E be the spectral set, let $\mathcal{R}(\mathsf{E})$ denote the reflectionless potentials with spectrum E, and let ... be the divisor map. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067777 | MATHDB-352675

The comb-domain generic-position conjecture for KdV frequencies

摘录 / EXCERPT: Let $\eta=\{\eta_k\}_{k\geq1}$ and $\eta^{(1)}=\{\eta_k^{(1)}\}_{k\geq1}$ be the frequency collections associated with the comb domains describing the KdV flows. Comb-domain generi...

conjecture | open | MathDB 页面 | 原文线索

V2-067778 | MATHDB-352687

Periodic ABC arrangement conjecture for the ternary long-range Ising model

摘录 / EXCERPT: Consider a one-dimensional periodic cell containing n balls of each type A, B, and C, with positive radii parameters ω_i satisfying $\sum_i\omega_i=1$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067779 | MATHDB-352697

The alternating inequality conjecture for 4-regular partitions counted by $Q_2(n)$

摘录 / EXCERPT: Alternating inequality conjecture. For $n,k\geqslant 0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-067780 | MATHDB-352706

Arithmetic autocorrelation value distribution conjecture for binary m-sequences

摘录 / EXCERPT: Let $\mathcal{M}$ be a binary m-sequence of period 2^n-1, where $n\in\mathbb{N}$. For an integer shift τ with $1\leq\tau<2^n-1$, let $\mathcal{A}^{A}_{\mathcal{M}}(\tau)$...

conjecture | open | MathDB 页面 | 原文线索

V2-067781 | MATHDB-352716

Gromov--Hausdorff coupling conjecture for unicellular maps and hyperbolic surfaces

摘录 / EXCERPT: Gromov--Hausdorff coupling conjecture. The variables $\mathbf U_{n,g}$ and $\mathbf S_g$ can be coupled so that

conjecture | open | MathDB 页面 | 原文线索

V2-067782 | MATHDB-352720

Cisinski-style conjecture on cellularly generated localisers being sketchable

摘录 / EXCERPT: Let C be an $infty$-category, and let $W\subseteq (\operatorname{Cat}\!\slice C)^{\Delta^1}$ be a set of maps satisfying three conditions: it is stable under 2-out-of-3 and retra...

conjecture | open | MathDB 页面 | 原文线索

V2-067783 | MATHDB-352727

The categorical p-half-twist conjecture

摘录 / EXCERPT: Categorical p-half-twist conjecture. For any w in the p-cell λ,

conjecture | open | MathDB 页面 | 原文线索

V2-067784 | MATHDB-352728

The p-cell eigenmap conjecture

摘录 / EXCERPT: The p-cell eigenmap conjecture. For every p-cell λ, there exists a chain map

conjecture | open | MathDB 页面 | 原文线索

V2-067785 | MATHDB-352729

Modified nonexistence conjecture for optimal self-orthogonal codes of dimension 5

摘录 / EXCERPT: Let $d(n,5)$ denote the maximum minimum distance of a binary linear $[n,5]$ code, and let $d_{\mathrm{so}}(n,5)$ denote the maximum minimum distance of a binary self-orthogonal...

conjecture | open | MathDB 页面 | 原文线索

V2-067786 | MATHDB-352747

Commutation of deformation and scaling limits for the Laplace-Beltrami operator

摘录 / EXCERPT: Commutation-of-limits conjecture. For sufficiently small ϵ_1 and ϵ_2, the scaling limit exists uniformly with respect to T_{cr}; the deformation limits...

conjecture | open | MathDB 页面 | 原文线索

V2-067787 | MATHDB-352751

A gamma-ratio continued fraction formula with four gamma-function pairs

摘录 / EXCERPT: Let $x,l,n,\eta\in\mathbb{C}$, and define P by ... Suppose that x is complex with $\operatorname{Re}(x)>0$, or that either n or η is an odd integer, or that l is an even...

conjecture | open | MathDB 页面 | 原文线索

V2-067788 | MATHDB-352761

Existence of a finite-term recurrence relation for eigenpolynomials

摘录 / EXCERPT: Let a differential operator be given as in $\$, and let $\{P_n\}\$ be its sequence of eigenpolynomials. The coefficients $\{\alpha_{n,k}\}\$ are those occurring in the recurrence r...

conjecture | open | MathDB 页面 | 原文线索

V2-067789 | MATHDB-352763

The biequivalence and infinity-properad conjecture for labelled cospan categories

摘录 / EXCERPT: A labelled cospan category is a symmetric monoidal category equipped with a symmetric monoidal functor to the category of cospans satisfying the pullback condition specified in the...

conjecture | open | MathDB 页面 | 原文线索

V2-067790 | MATHDB-352770

The second overpartition identity of Andrews and Uncu

摘录 / EXCERPT: Andrews-Uncu's conjectural overpartition identity. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-067791 | MATHDB-352772

Hypercube decomposition formula for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let $X=[x,y]$ be the Bruhat graph of an interval in a symmetric group. Let $J\subset X$ be a hypercube decomposition, and let $I_{x,y,J}$ and $Q_{x,y,J}$ denote the inductive and h...

conjecture | open | MathDB 页面 | 原文线索

V2-067792 | MATHDB-352777

The planar positive-valued cover conjecture

摘录 / EXCERPT: Planar positive-valued cover conjecture. Under these assumptions, H has a strictly f-degenerate transversal.

conjecture | open | MathDB 页面 | 原文线索

V2-067793 | MATHDB-352789

Full pattern support conjecture for skew Brownian permutons

摘录 / EXCERPT: Let $\bm \mu_{\rho,q}$ be a skew Brownian permuton with parameters $(\rho,q)\in(-1,1)\times(0,1)$. For $k\in\mathbb{Z}_{>0}$, let π be a permutation pattern of size k, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-067794 | MATHDB-352802

Decay conjecture for the wave equation in flat FLRW universes

摘录 / EXCERPT: Decay conjecture in the flat case.

conjecture | open | MathDB 页面 | 原文线索

V2-067795 | MATHDB-352803

Decay conjecture for the wave equation in hyperbolic FLRW universes

摘录 / EXCERPT: Decay conjecture in the hyperbolic case.

conjecture | open | MathDB 页面 | 原文线索

V2-067796 | MATHDB-352819

Hadamard-type inequalities for hyperbolic polynomials

摘录 / EXCERPT: Hadamard-type inequality for hyperbolic polynomials. The claim extends the diagonal-versus-eigenvalue inequality from elementary symmetric polynomials to homogeneous, real, symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-067797 | MATHDB-352822

Infinite-family conjecture for doubly exotic superintegrable systems

摘录 / EXCERPT: Let N be the order of a superintegrable system, let L_z denote the angular momentum, and let p_1,p_2 be the Cartesian momenta. Let V_1 and V_2 be functions of the Cartesi...

conjecture | open | MathDB 页面 | 原文线索

V2-067798 | MATHDB-352824

Ottazzi-Warhurst regularity conjecture for rigid Carnot groups

摘录 / EXCERPT: Let Gsupset U$ and $U' ormal?" Wait. Cannot introduce invalid notation. Let G be a rigid Carnot group, and let $U,U' e othing$ be open subsets of G. A quasiconformal homeomor...

conjecture | open | MathDB 页面 | 原文线索

V2-067799 | MATHDB-352825

Local bilipschitz conjecture for quasiconformal maps of Carnot groups

摘录 / EXCERPT: Let G be a Carnot group other than $\mathbb{R}^n$ or the Heisenberg group $\mathbb{H}_n$ for any n, and let U,U' be open subsets of G. Let $f:U\to U'$ be a quasiconformal h...

conjecture | open | MathDB 页面 | 原文线索

V2-067800 | MATHDB-352829

Regular CW decomposition conjecture for the momentum amplituhedron

摘录 / EXCERPT: Let $\mathcal{M}_{n,k}$ be the momentum amplituhedron, and let $\mathcal{S}_{n,k}$ consist of the decorated permutations whose strata are boundary strata, together with the decorat...

conjecture | open | MathDB 页面 | 原文线索

V2-067801 | MATHDB-352830

Grassmannian forest characterization of momentum-amplituhedron boundaries

摘录 / EXCERPT: Let $\mathcal{M}_{n,k}$ be the momentum amplituhedron. For a Grassmannian graph G, let Φ°_G be the associated stratum, and let $\dim_{\mathcal{M}}(F)$ denote the dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-067802 | MATHDB-352833

Invariant reciprocal area sum for lateral harmonic polygons

摘录 / EXCERPT: Let $\mathcal{H}$ be a Poncelet family of N-gons interscribed between two homothetic, concentric ellipses, and let A_1 and A_2 be the areas of the two lateral harmonic polygo...

conjecture | open | MathDB 页面 | 原文线索

V2-067803 | MATHDB-352834

Invariant sine-angle ratio for harmonic polygons

摘录 / EXCERPT: Let P be a harmonic polygon with area A, sidelengths s_i, and angles θ_i as used in the paper. Sine-angle invariant conjecture. The quantity ... is invariant. Equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-067804 | MATHDB-352835

Vanishing global quasiconformal Assouad spectrum dimension below one

摘录 / EXCERPT: Let $0\le \theta<1$ and let $E\subset\mathbb{R}^n$. Write $\dim_A^\theta(E)$ for the Assouad spectrum with parameter θ, and let the global quasiconformal Assouad spectrum di...

conjecture | open | MathDB 页面 | 原文线索

V2-067805 | MATHDB-352836

Arkani-Hamed and Trnka's BCFW triangulation conjecture for the amplituhedron

摘录 / EXCERPT: Let $k\geq 1$ and $n\geq k+4$. Let $\mathcal{A}_{n,k,4}$ be the amplituhedron, and let $\mathcal{BCFW}_{n,k}$ denote the collection of $4k$-dimensional BCFW positroid cells in the...

conjecture | open | MathDB 页面 | 原文线索

V2-067806 | MATHDB-352845

Central binomial sum evaluation involving zeta values and powers of pi

摘录 / EXCERPT: Central binomial sum conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067807 | MATHDB-352849

The rational cuspidal groups conjecture for modular Jacobians

摘录 / EXCERPT: Let N be a positive integer. Let $C_N(\mathbb{Q})$ be the subgroup of $J_0(N)(\mathbb{Q})$ represented by degree-zero cuspidal divisors, and let $C(N)$ be the subgroup formed by...

conjecture | open | MathDB 页面 | 原文线索

V2-067808 | MATHDB-352851

The same-recording-tableau conjecture for steady-state time

摘录 / EXCERPT: For permutations $v,w\in S_n$, let $\operatorname{Q}(v)$ and $\operatorname{Q}(w)$ denote their recording tableaux under the Robinson-Schensted correspondence. The steady-state tim...

conjecture | solved | MathDB 页面 | 原文线索

V2-067809 | MATHDB-352860

Asymptotic conjecture for Euler characteristics of minimal strata of differentials

摘录 / EXCERPT: Let ${\mathcal{M}}_g(2g-1)$ denote the moduli space of minimal holomorphic strata of differentials in genus g, and let χ denote its orbifold Euler characteristic. Asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-067810 | MATHDB-352863

Extension of the Pfaffian asymptotics to Simon-Griffiths spin systems

摘录 / EXCERPT: Let a similar finite-range, ferromagnetic, translation-invariant and reflection-invariant system be obtained by replacing the Ising spins in Theorem 1 with spin variables in the Si...

conjecture | open | MathDB 页面 | 原文线索

V2-067811 | MATHDB-352870

The eTNC minus component in Fitting-ideal form for CM-extensions

摘录 / EXCERPT: Let H/F be a finite abelian CM-extension, and write $G=\operatorname{Gal}(H/F)$. Let Σ and Σ' be finite sets of places of F satisfying (H1), (H2), (H3), and (H4)....

conjecture | open | MathDB 页面 | 原文线索

V2-067812 | MATHDB-352882

The inversion-number conjecture for the expected first letter

摘录 / EXCERPT: Inversion-number conjecture. The expected value is the rational function

conjecture | open | MathDB 页面 | 原文线索

V2-067813 | MATHDB-352883

The major-index polynomial-degree conjecture

摘录 / EXCERPT: Major-index conjecture. This expected value is a rational function in n that is $1/(k+1)$ times the quotient of a monic degree-$(k+1)$ polynomial by a monic degree-k polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-067814 | MATHDB-352902

Unassigned-outflow truncation error conjecture

摘录 / EXCERPT: Let N_i^* be the truncation levels, let $\bar{t}$ be the time parameter, and let $\relax\epsilon_1(N_1^*,N_2^*,\bar{t})$ and $\relax\epsilon_2(N_1^*,N_2^*,\bar{t})$ denote the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-067815 | MATHDB-352905

e-positivity conjecture for the sporadic spiders $S(b^2+b,b,1)$

摘录 / EXCERPT: Spider e-positivity conjecture. If b is even and $b\not\equiv2\pmod3$, then the spider $S(b^2+b,\,b,\,1)$ is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-067816 | MATHDB-352907

Classification conjecture for positive-mass planar equilateral 5-body central configurations

摘录 / EXCERPT: Consider planar equilateral 5-body central configurations with all masses positive in the Newtonian case. The regular pentagon and the star are the configurations with equal masses...

conjecture | open | MathDB 页面 | 原文线索

V2-067817 | MATHDB-352915

The APT Collatz convergence claim

摘录 / EXCERPT: For $n\in\mathbb{N}_{>0}$, let $A(n)$ be the alternating parity Terras-modified Collatz sequence defined by ... where r is the ruler function. APT Collatz claim. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-067818 | MATHDB-352917

Coincidence of asymptotic coefficients with the coefficients of the mean L_c

摘录 / EXCERPT: Let M be a symmetric homogeneous mean that has an asymptotic power series expansion, and suppose that it fulfills the requirements of the open question from Farhi. Let L_c deno...

conjecture | open | MathDB 页面 | 原文线索

V2-067819 | MATHDB-352928

Conjecture on optimal stability constants for 3-wise intersecting families

摘录 / EXCERPT: Conjecture on optimal stability constants. There exists a constant C_p' such that the inequality in case (ii) of the theorem can be replaced by

conjecture | open | MathDB 页面 | 原文线索

V2-067820 | MATHDB-352933

The compact self-adjointness conjecture for local Hecke operators

摘录 / EXCERPT: Let $\mathcal C$ be a smooth projective curve over a local field F of genus at least 2, let G be a split connected semi-simple group, and let Bun be the modu...

conjecture | open | MathDB 页面 | 原文线索

V2-067821 | MATHDB-352934

The unitary Eisenstein embedding conjecture for cuspidal functions

摘录 / EXCERPT: Let F be a non-archimedean local field with residue field k, let $\mathcal C$ be a curve over the ring of integers of F, and let $E_{\operatorname{Bun},1/2}$ be the Eisenstei...

conjecture | open | MathDB 页面 | 原文线索

V2-067822 | MATHDB-352936

The real-oper spectral support conjecture

摘录 / EXCERPT: Let $\operatorname{Oper}_{G^{\vee}}(\mathcal C)^{\mathbb R}$ denote the real locus of $G^{\vee}$-opers on a compact complex curve $\mathcal C$, and let $L^2(\operatorname{Bun})_o$...

conjecture | open | MathDB 页面 | 原文线索

V2-067823 | MATHDB-352937

The real-oper Hecke eigenvalue conjecture

摘录 / EXCERPT: Let $\mathcal C$ be a compact complex curve, let $G^{\vee}$ be the Langlands dual group, and let o be a real $G^{\vee}$-oper. For a dominant coweight λ, let V_λ b...

conjecture | open | MathDB 页面 | 原文线索

V2-067824 | MATHDB-352939

The period conjecture for Laurent polynomials

摘录 / EXCERPT: Let $f,g\typically$ be Laurent polynomials in two variables, and let ... and similarly for g denote their classical periods. Two Laurent polynomials are mutation equivalent if a co...

conjecture | open | MathDB 页面 | 原文线索

V2-067825 | MATHDB-352945

Optimal bound on the a-number of curves in positive characteristic

摘录 / EXCERPT: Let C be a curve of genus g over a field of characteristic $p>0$. Write $a(C)$ for the a-number of its Jacobian, and let r be the rank of the Cartier operator on...

conjecture | open | MathDB 页面 | 原文线索

V2-067826 | MATHDB-352952

The strict-decrease conjecture for asymptotic conformal graph energies

摘录 / EXCERPT: Let (π,ϕ) be a recurrent virtual graph endomorphism, and for $q\in[1,\infty]$ let $\overline{E}{}^q[\pi,\phi]$ denote its asymptotic q-conformal graph energy. Strict-decre...

conjecture | open | MathDB 页面 | 原文线索

V2-067827 | MATHDB-352956

Lalín-Rogers-type conjecture for Ramanujan-type polynomial zeros

摘录 / EXCERPT: Ramanujan-type polynomial zero conjecture. The polynomial $\mathfrak{R}_{2k+1,p}(z)$ has only the real zero $z=0$ of multiplicity 2; all remaining zeros are non-real, lie on the...

conjecture | open | MathDB 页面 | 原文线索

V2-067828 | MATHDB-352967

ACC conjecture for generalized pairs

摘录 / EXCERPT: Let $d\in\mathbb{Z}_{>0}$ and let $I\subset[0,+\infty)$ be a subset satisfying the DCC. Define ... where $M\in I$ has the meaning specified in the source's definition of generalized...

conjecture | open | MathDB 页面 | 原文线索

V2-067829 | MATHDB-352976

Brevig-Ortega-Cerdà-Seip-Zhao contractive inequality for weighted Bergman spaces

摘录 / EXCERPT: Let H^2 be the Hardy space on the unit disk, let A^p_β denote the standard weighted Bergman space with exponent p and weight parameter β, and let the Szegő kernel b...

conjecture | open | MathDB 页面 | 原文线索

V2-067830 | MATHDB-352982

The Sen-operator conjecture for rational Hodge-Tate crystals

摘录 / EXCERPT: Let K be a complete discretely valued field of mixed characteristic $(0,p)$ with perfect residue field. Let ${\mathbb M}$ be a rational Hodge-Tate crystal in ... with associated pa...

conjecture | open | MathDB 页面 | 原文线索

V2-067831 | MATHDB-352997

Iwaniec's quasiconvexity conjecture for the planar Burkholder integrand

摘录 / EXCERPT: Let $B_p\colon \mathbb{R}^{2\times 2}\to\mathbb{R}$ be the Burkholder integrand, defined by ... where $\lvert A\rvert$ denotes the operator norm. Iwaniec's conjecture. The integrand...

conjecture | open | MathDB 页面 | 原文线索

V2-067832 | MATHDB-353001

Conjectured Brownian-sphere limit for planar sparse random maps

摘录 / EXCERPT: Brownian-sphere conjecture. The rescaled maps

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V2-067833 | MATHDB-353012

Polynomial vanishing relations for kappa classes on the moduli space of stable curves

摘录 / EXCERPT: Kappa-class vanishing conjecture. The polynomial relations

conjecture | open | MathDB 页面 | 原文线索

V2-067834 | MATHDB-353013

Hodge-class formula for the top J-polynomial on the moduli space of curves

摘录 / EXCERPT: Hodge formula conjecture. For $n=0$ or $n=1$, the top polynomial is

conjecture | open | MathDB 页面 | 原文线索

V2-067835 | MATHDB-353014

Characterisation of the top K-polynomial on the moduli space of curves

摘录 / EXCERPT: Top K-polynomial conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-067836 | MATHDB-353015

Theta-class formula for the top K-polynomial on the moduli space of curves

摘录 / EXCERPT: For $g,n\in\mathbb{N}$ with $2g-2+n>0$, let K_m be the homogeneous polynomials in the kappa classes defined by ... where ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067837 | MATHDB-353026

The GW/PT correspondence for absolute threefolds

摘录 / EXCERPT: The GW/PT correspondence. $Z^{X}_{\mathsf{PT},\beta}(\boldsymbol{\tau}_{\alpha-1}(\boldsymbol{\gamma}))$ is the Fourier expansion of a rational function in p, and under $p=e^z$,

conjecture | open | MathDB 页面 | 原文线索

V2-067838 | MATHDB-353027

The relative GW/PT correspondence

摘录 / EXCERPT: The relative GW/PT correspondence. $Z^{(X,D)}_{\mathsf{PT},\beta}(\lambda\mid\tau_{\alpha_1-1}(\gamma_1)\cdots\tau_{\alpha_r-1}(\gamma_r))$ is the Fourier expansion of a rational f...

conjecture | open | MathDB 页面 | 原文线索

V2-067839 | MATHDB-353028

The marked relative GW/PT correspondence

摘录 / EXCERPT: The marked relative GW/PT correspondence. For all $\gamma\in H^{\ast}((X,D)^r)$, $Z^{(X,D)}_{\mathsf{PT},\beta}(\lambda\mid\tau_{\alpha_1-1}\cdots\tau_{\alpha_r-1}(\gamma))$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-067840 | MATHDB-353029

The K3 multiple cover formula for reduced Gromov-Witten invariants

摘录 / EXCERPT: Multiple Cover Formula.

conjecture | open | MathDB 页面 | 原文线索

V2-067841 | MATHDB-353030

A supercongruence relating two cubic binomial sums

摘录 / EXCERPT: The supercongruence.

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V2-067842 | MATHDB-353031

A strengthened parametric q-congruence for Theorem 3

摘录 / EXCERPT: Let n be a positive odd integer, and let a be an indeterminate. Write $[n]$ for the q-integer and let $\Phi_n(q)$ denote the nth cyclotomic polynomial. Let the congruences...

conjecture | open | MathDB 页面 | 原文线索

V2-067843 | MATHDB-353032

A fifth-power q-congruence for the convolution in Theorem 4

摘录 / EXCERPT: Let n be a positive odd integer, and let $c_q(k)$ be the sequence defined in Theorem 4. Let $[n]$ denote the q-integer and let $\Phi_n(q)$ denote the nth cyclotomic polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-067844 | MATHDB-353034

Stronger convolution q-congruences extending Theorems 7 and 8

摘录 / EXCERPT: Let n be a positive odd integer, and let $c_q(k)$ be the sequence defined in Theorem 7. Let $[n]$ denote the q-integer and let $\Phi_n(q)$ denote the nth cyclotomic polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-067845 | MATHDB-353035

A strengthened convolution q-congruence for Theorem 8

摘录 / EXCERPT: Let n be a positive odd integer, and let $c_q(k)$ be the sequence defined in Theorem 8 by ... Let $[n]$ denote the q-integer and let $\Phi_n(q)$ denote the nth cyclotomic polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-067846 | MATHDB-353044

Triple LLT product tableau expansion conjecture

摘录 / EXCERPT: Let $\bm{\nu}=(\lambda^1/\mu^1,\lambda^2/\mu^2,\lambda^3/\mu^3)$ be a triple of skew diagrams. For every triple $\{i,j,k\}=\{1,2,3\}$ with $i<j$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-067847 | MATHDB-353053

Density of stable accessibility for partially hyperbolic diffeomorphisms

摘录 / EXCERPT: Let $r\in[2,+\infty]$, and consider C^{r} partially hyperbolic diffeomorphisms, whether volume preserving or not. A diffeomorphism is stably accessible if it has the accessibilit...

conjecture | open | MathDB 页面 | 原文线索

V2-067848 | MATHDB-353069

The spectral containment conjecture for homogeneous multiaffine stable polynomials

摘录 / EXCERPT: A homogeneous multiaffine stable polynomial $p\in\mathbb{R}[x_1,\ldots,x_n]$ has the spectral containment property if, for every symmetric matrix $A\in H(P)$, some eigenvalue vecto...

conjecture | open | MathDB 页面 | 原文线索

V2-067849 | MATHDB-353070

The quadratic spectral containment conjecture for multiaffine stable polynomials

摘录 / EXCERPT: A homogeneous multiaffine stable polynomial is a stable polynomial that is homogeneous and multiaffine. Quadratic spectral containment conjecture. Every quadratic homogeneous multi...

conjecture | open | MathDB 页面 | 原文线索

V2-067850 | MATHDB-353071

The diagonal-rescaled elementary symmetric spectral containment conjecture

摘录 / EXCERPT: Let D be a positive definite diagonal matrix, and let $p(x)=e_k(Dx)$. Diagonal-rescaled elementary symmetric conjecture. The polynomial $p(x)$ has the spectral containment proper...

conjecture | open | MathDB 页面 | 原文线索

V2-067851 | MATHDB-353072

The Schur-Horn conjecture for homogeneous multiaffine polynomials

摘录 / EXCERPT: A homogeneous multiaffine polynomial is a polynomial in which every variable has degree at most one and all monomials have the same total degree. Multiaffine Schur-Horn conjecture....

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V2-067852 | MATHDB-353081

Gordon-Martino conjecture for Coxeter groups

摘录 / EXCERPT: Gordon-Martino conjecture. If W is a Coxeter group, then

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V2-067853 | MATHDB-353082

Lusztig-Calogero-Moser cuspidal-family conjecture

摘录 / EXCERPT: Cuspidal-family conjecture. The τ-cuspidal Lusztig k-families coincide with the τ-cuspidal Calogero-Moser k-families.

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V2-067854 | MATHDB-353084

Gordon-Martino conjecture on unipotent families and Calogero-Moser fixed points

摘录 / EXCERPT: Unipotent-family conjecture. There exists a unique bijection

conjecture | open | MathDB 页面 | 原文线索

V2-067855 | MATHDB-353088

The character-sum refinement for fixed Calogero-Moser components

摘录 / EXCERPT: Character-sum refinement. Assume that τ is regular and that the fixed-component Calogero-Moser isomorphism conjecture holds. If...

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V2-067856 | MATHDB-353089

The regular-number character criterion for fixed Calogero-Moser spaces

摘录 / EXCERPT: Regular-number character criterion. Recall that d is a regular number. For every $p\in{\mathcal Z}_k^{{\mathbb C}^\times}$, p belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-067857 | MATHDB-353090

The five-color conjecture for the odd chromatic number of planar graphs

摘录 / EXCERPT: Five-color conjecture. For every planar graph G,

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V2-067858 | MATHDB-353091

The four-color conjecture for odd colorings of planar graphs of girth at least six

摘录 / EXCERPT: Girth-six odd-coloring conjecture. Every planar graph G of girth at least 6 has

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V2-067859 | MATHDB-353103

Robust nullspace property conjecture for sparse Bernoulli matrices

摘录 / EXCERPT: Let $A\in\{0,1\}^{m\times n}$ be a matrix whose entries are independent and identically distributed Bernoulli p random variables. The robust nullspace property conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-067860 | MATHDB-353109

Conjecture on singular regularisation parameters for multiple t-values

摘录 / EXCERPT: Conjecture on singular regularisation parameters. The sequence is increasing and bounded by 3:

conjecture | open | MathDB 页面 | 原文线索

V2-067861 | MATHDB-353114

Hydrodynamic limit conjecture for roots of repeatedly differentiated random polynomials

摘录 / EXCERPT: Let $(\xi_j)_{j\geq 1}$ be independent complex-valued random variables with distribution μ, let $P_n(z)=\prod_{j=1}^n(z-\xi_j)$, and let $\mu\lsem P_n^{(k)}\rsem$ denote the em...

conjecture | open | MathDB 页面 | 原文线索

V2-067862 | MATHDB-353125

Silverman's transcendence conjecture for canonical heights

摘录 / EXCERPT: Silverman's conjecture. For every $a\in\mathbb{P}^1(\overline{\mathbb{Q}})$, $\hat{h}_f(a)$ is either 0 or transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-067863 | MATHDB-353138

The equality of Landau-Ginzburg Hodge invariants

摘录 / EXCERPT: Let (Y,w) be a Landau-Ginzburg model admitting a tame compactification. Let $f^{p,q}(Y,w)$ be the Hodge-theoretic invariants defined from the Hodge filtration on relative cohomol...

conjecture | open | MathDB 页面 | 原文线索

V2-067864 | MATHDB-353139

Conjecture on the singular regularisation parameters for alternating multiple t-values

摘录 / EXCERPT: Singular-parameter conjecture. Discounting the $\lambda_1=0$ term, the sequence satisfies: the sign of λ_i is $(-1)^i$ for all $i>1$; the odd-indexed and even-indexed subse...

conjecture | open | MathDB 页面 | 原文线索

V2-067865 | MATHDB-353141

The coefficient conjecture for the class of univalent functions with real coefficients

摘录 / EXCERPT: Let $0<\lambda\leq 1$, and let ${\mathcal U}^+(\lambda)$ denote the class of functions with real coefficients considered in the paper. Write $f(z)=z+a_2z^2+a_3z^3+\cdots$. Coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-067866 | MATHDB-353151

Sufficiency conjecture for multiplication profiles of hyperelliptic Jacobian point classes

摘录 / EXCERPT: Let B) be a hyperelliptic curve of genus $\ga$, with a marked Weierstrass point q, and let $\Theta_1^0(B)\subset \operatorname{Jac}(B)$ denote the relevant locus of point class...

conjecture | open | MathDB 页面 | 原文线索

V2-067867 | MATHDB-353160

The two-dimensional threshold-generation conjecture for the Molchanov-Vainberg operator

摘录 / EXCERPT: Let D be the discrete Molchanov-Vainberg Laplacian, let $\boldsymbol{\Theta}_{m,\kappa}(D)$ denote the threshold sets defined in the paper, and let $\mathbb{N}_e$ denote the even...

conjecture | open | MathDB 页面 | 原文线索

V2-067868 | MATHDB-353162

The Mourre-estimate conjecture for successive bands in the second gap

摘录 / EXCERPT: Fix $\kappa\geq4$ with $\kappa\in\mathbb{N}_e$, and let $\{\mathcal{E}_n\}$ be the sequence from the stated theorem. Let $A_{j\kappa}=\sum_{i=1}^{2}A_i(j,\kappa)$ be the conjugate...

conjecture | open | MathDB 页面 | 原文线索

V2-067869 | MATHDB-353163

The third-band absolutely-continuous-spectrum conjecture

摘录 / EXCERPT: For even $\kappa\geq6$, define $J_3(\kappa)=(\cos(2\pi/\kappa),\cos^2(\pi/\kappa))$, and let $\boldsymbol{\mu}_{\kappa}(D)$ denote the absolutely-continuous-spectrum set defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-067870 | MATHDB-353165

The finite-combination Mourre conjecture between consecutive thresholds

摘录 / EXCERPT: Let D be the discrete Molchanov-Vainberg Laplacian in dimension 2, let κ be even, and let $\mathcal{E}_{i_1},\mathcal{E}_{i_2}\in\boldsymbol{\Theta}_{\kappa}(D)$ be cons...

conjecture | open | MathDB 页面 | 原文线索

V2-067871 | MATHDB-353166

The three-dimensional second-gap Mourre conjecture

摘录 / EXCERPT: Three-dimensional Mourre conjecture. On every interval $(\mathcal{E}_n,\mathcal{E}_{n-1})$, the Mourre estimate holds with

conjecture | open | MathDB 页面 | 原文线索

V2-067872 | MATHDB-353168

The Chebyshev interlacing inequality conjecture

摘录 / EXCERPT: Let $\kappa\geq2$ be even and fix $1\leq j\leq\lfloor\kappa/2\rfloor$. Let T_κ be the Chebyshev polynomial used in the paper, and choose real numbers satisfying ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-067873 | MATHDB-353180

Straight-condition conjecture for Hörmander-type operators

摘录 / EXCERPT: Let $n\geq 3$ and let T^{λ} be a Hörmander-type operator satisfying the straight condition: for each ξ, the associated quantity $G(x,\xi)$ is invariant under changes in...

conjecture | open | MathDB 页面 | 原文线索

V2-067874 | MATHDB-353190

Heim-Neuhauser-Tröger's log-concavity conjecture for the plane partition function

摘录 / EXCERPT: Heim-Neuhauser-Tröger's conjecture. The function $\mathrm{PL}(n)$ is log-concave for every integer $n\geq 12$.

conjecture | open | MathDB 页面 | 原文线索

V2-067875 | MATHDB-353193

Sharpness conjecture for odd colorings of sparse graphs

摘录 / EXCERPT: Let $\mathrm{mad}(G)$ denote the maximum average degree of G, and let $\chi_o(G)$ be the smallest number of colors admitting an odd coloring of G. Fix ϵ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-067876 | MATHDB-353194

Kozlov's non-evasiveness conjecture for order complexes

摘录 / EXCERPT: Kozlov's conjecture. The order complex of P is non-evasive.

conjecture | open | MathDB 页面 | 原文线索

V2-067877 | MATHDB-353196

The genus zero log/local/open correspondence for quasi-tame Looijenga pairs

摘录 / EXCERPT: The genus zero log/local/open correspondence. The invariants satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-067878 | MATHDB-353198

The level-4 multiple-polylogarithm conjecture

摘录 / EXCERPT: Let $\operatorname{Li}_n(z)$ denote the polylogarithm of positive integer order n, and let $i=\sqrt{-1}$. Let $\textsf{CMZV}^{2}_{n}$ denote the space of cyclotomic multiple zeta...

conjecture | open | MathDB 页面 | 原文线索

V2-067879 | MATHDB-353199

The level-5 multiple-polylogarithm conjecture at the golden-ratio argument

摘录 / EXCERPT: Let $\operatorname{Li}_n(z)$ denote the polylogarithm of positive integer order n, let ... and let $\textsf{CMZV}^{5}_{n}$ denote the space of cyclotomic multiple zeta values of we...

conjecture | open | MathDB 页面 | 原文线索

V2-067880 | MATHDB-353202

The analogue of Coates-Sujatha's fine Selmer conjecture over function fields

摘录 / EXCERPT: Let K be the function field considered in the paper, let L be a finite extension of K, and let L_cyc be its cyclotomic extension. For an elliptic curve E/L, let...

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V2-067881 | MATHDB-353206

The c-pathwidth lower-bound conjecture

摘录 / EXCERPT: Let c be a positive integer. For a graph G, write c-$pw(G)$ for its c-pathwidth, $tw(G)$ for its treewidth, and n for the number of vertices of G. The c-pathwidth lower...

conjecture | open | MathDB 页面 | 原文线索

V2-067882 | MATHDB-353218

Improved magnetic Hardy inequality with a critical remainder

摘录 / EXCERPT: Let $d\geq 2$, let ∇_A denote the magnetic gradient, let $h_{A,p}$ be the corresponding magnetic p-Laplacian form domain, and let $\mu_{p,d}$ be the Hardy constant. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067883 | MATHDB-353219

Aharonov-Bohm magnetic Hardy inequality for nonintegral flux

摘录 / EXCERPT: Aharonov-Bohm Hardy conjecture. If $\beta\notin\mathbb{Z}$, then there exists a constant $\lambda(p)>\left(\frac{2-p}{p}\right)^p$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067884 | MATHDB-353227

Morier-Genoud-Ovsienko unimodality conjecture for fence rank sequences

摘录 / EXCERPT: Let $F(\beta)$ be a fence and let $L(\beta)$ be the distributive lattice of lower order ideals of $F(\beta)$. Its rank sequence is $r(\beta):r_0,r_1,\ldots,r_n$, where r_k is the...

conjecture | open | MathDB 页面 | 原文线索

V2-067885 | MATHDB-353228

Oğuz-Ravichandran unimodality conjecture for restricted fence rank sequences

摘录 / EXCERPT: Let $F(\beta)$ be a fence with an even number of parts, and let $\overline{r}(\beta)$ denote the restricted rank sequence defined in the source. Oğuz-Ravichandran's conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-067886 | MATHDB-353229

McConville-Sagan-Smyth chain decomposition conjecture for fence lattices

摘录 / EXCERPT: Let $F(\beta)$ be a fence and let $L(\beta)$ be its distributive lattice of lower order ideals, with rank sequence governed by the alternatives in Theorem heavy. A saturated chain...

conjecture | open | MathDB 页面 | 原文线索

V2-067887 | MATHDB-353230

Lexicographic chain decomposition conjecture for fence lattices

摘录 / EXCERPT: Let $F(\beta)$ be a fence and let $L(\beta)$ be its distributive lattice of lower order ideals. For a linear extension of $F(\beta)$, form the corresponding lexicographic chain dec...

conjecture | open | MathDB 页面 | 原文线索

V2-067888 | MATHDB-353232

The independence conjecture for von Neumann Sylvester matrix rank

摘录 / EXCERPT: Let G be a group, let K be a field, and let $\phi_1,\phi_2:K\to\mathbb{C}$ be embeddings of K into $\mathbb{C}$. For a matrix $A\in\operatorname{Mat}_{n\times m}(K[G])$, writ...

conjecture | open | MathDB 页面 | 原文线索

V2-067889 | MATHDB-353233

Lück's twisted conjecture for von Neumann Sylvester matrix rank

摘录 / EXCERPT: Let G be a group and let $\sigma:G\to\operatorname{GL}_k(\mathbb{C})$ be a homomorphism. Define $\widetilde{\sigma}:\mathbb{C}[G]\to\operatorname{Mat}_k(\mathbb{C}[G])$ by sendin...

conjecture | open | MathDB 页面 | 原文线索

V2-067890 | MATHDB-353234

Nonvanishing of quadratic twists at the central value

摘录 / EXCERPT: Let K be a local field of characteristic zero and let L_K be its Weil group when K is archimedean and its Weil-Deligne group when K is non-archimedean. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-067891 | MATHDB-353239

The maximum-degree bound for odd chromatic number

摘录 / EXCERPT: Let G be a connected graph with maximum degree Δ. Maximum-degree odd chromatic number conjecture. If $\Delta\geq3$, then ... This would extend the preceding bounds for subcu...

conjecture | open | MathDB 页面 | 原文线索

V2-067892 | MATHDB-353246

Bayer-Macrì-Toda modified Bogomolov inequality conjecture

摘录 / EXCERPT: Bayer-Macrì-Toda's modified Bogomolov inequality conjecture. There exists such a class γ and an upper semicontinuous function $f^\gamma\colon\mathbb{R}\to\mathbb{R}$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-067893 | MATHDB-353249

The finite representational witnesses conjecture for divergent trajectories

摘录 / EXCERPT: Let G be a Lie group, Γ a lattice, A a subgroup acting on G/Γ, and let $K\subseteq K_1\subset G/\Gamma$ be compact sets. For a trajectory Ak with $k\in K$, say...

conjecture | open | MathDB 页面 | 原文线索

V2-067894 | MATHDB-353256

The conjecture on the ternary sum of triangular numbers $\Delta_{1,4,5}$

摘录 / EXCERPT: For positive integers $a_1,\a_2,\a_3$, write Δ_{a_1,a_2,a_3} for the corresponding sum of triangular numbers; in particular, $\Delta_{1,4,5}$ is the ternary sum with coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-067895 | MATHDB-353259

Exoo's conjecture on the chromatic number of interval distance graphs

摘录 / EXCERPT: Exoo's conjecture. For $b>1$ sufficiently close to 1, it holds

conjecture | open | MathDB 页面 | 原文线索

V2-067896 | MATHDB-353260

Open problem on realizing every chromatic number by interval distance graphs

摘录 / EXCERPT: Realization problem. For any integer $k\ge 7$, there exists $b>1$ such that

open_problem | open | MathDB 页面 | 原文线索

V2-067897 | MATHDB-353262

The rank-jump density conjecture for elliptic curves over rational function fields

摘录 / EXCERPT: Let $\mathcal{E}$ be an elliptic curve over the function field $\Q(T)$ with nonconstant j-invariant. For $t\in\Q$, let $\mathcal{E}_t$ denote its specialization when it is an ell...

conjecture | open | MathDB 页面 | 原文线索

V2-067898 | MATHDB-353275

The Euler product conjecture for parabolic rigid Gross G-motives

摘录 / EXCERPT: Let X be a smooth proper curve over $\mathbb{F}_p$, let $G\to X$ satisfy the stated hypotheses, fix a parabolic subgroup P, and let $\mathcal{E}$ be an augmented commutative...

conjecture | open | MathDB 页面 | 原文线索

V2-067899 | MATHDB-353279

The w_k-equivalence characterization for welded string links

摘录 / EXCERPT: The w_k-equivalence conjecture. Two welded string links are w_k-equivalent if and only if they cannot be distinguished by finite type invariants of degree $<k$.

conjecture | open | MathDB 页面 | 原文线索

V2-067900 | MATHDB-353280

The RC_k-equivalence characterization for ribbon tubes

摘录 / EXCERPT: The RC_k-equivalence conjecture. Two ribbon tubes are RC_k-equivalent if and only if they cannot be distinguished by finite type invariants of degree $<k$.

conjecture | open | MathDB 页面 | 原文线索

V2-067901 | MATHDB-353281

The degree-three relation $A\cdot D\stackrel{4}{\sim}B\cdot C$

摘录 / EXCERPT: The degree-three w_4-equivalence relation.

conjecture | open | MathDB 页面 | 原文线索

V2-067902 | MATHDB-353287

Ohba's conjecture on list chromatic number

摘录 / EXCERPT: Ohba's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-067903 | MATHDB-353288

Necessity conjecture for Hilbert-series deviation of q-deformed quasi-invariants

摘录 / EXCERPT: Let m and n be positive integers, let q be a nonzero complex number, and let $Q_{m,q}(n)$ and $Q_m(n,\mathbb{C})$ denote the q-deformed and ordinary m-quasi-invariant pol...

conjecture | open | MathDB 页面 | 原文线索

V2-067904 | MATHDB-353294

Scrollar invariants from standard Young tableaux on Hirzebruch surfaces

摘录 / EXCERPT: Scrollar-invariant tableau conjecture. The multiset of scrollar invariants of the partition λ with respect to φ is

conjecture | open | MathDB 页面 | 原文线索

V2-067905 | MATHDB-353295

Nonexistence conjecture for superlinear elliptic inequalities on weighted graphs

摘录 / EXCERPT: Let $(V,\b5)$ be a weighted graph, let $o\in V$, and let $\sigma>1$. A nonnegative solution u of the elliptic inequality ... is a function on V satisfying this inequality, where...

conjecture | open | MathDB 页面 | 原文线索

V2-067906 | MATHDB-353297

Threshold-location conjecture for the exponential kernel

摘录 / EXCERPT: Assume that $\mathcal{K}(x)=\mathrm{e}^{-|x|}/2$ and that $f_a(u)=u(1-u)(u-a)$ with $0<a<1/2$. Let $(a,d)\in \mathcal{R}_2\cup\mathcal{R}_3$ such that $0<\ell_1^*<\infty$ is well d...

conjecture | open | MathDB 页面 | 原文线索

V2-067907 | MATHDB-353298

Universality conjecture for threshold regions in nonlocal bistable equations

摘录 / EXCERPT: Assume that $f_a(u)=u(1-u)(u-a)$ with $0<a<1/2$. Let $\mathcal{K}$ satisfy the stated hypotheses with normalized second moment ... Let $\mathcal{R}_j$, $j=1,\ldots,5$, be the regions...

conjecture | open | MathDB 页面 | 原文线索

V2-067908 | MATHDB-353299

The low-element characterization of Shi regions in Coxeter systems

摘录 / EXCERPT: Shi-region low-element conjecture. One has $L_{\operatorname{Shi}}(W,S)=L(W,S)$. Moreover, every Shi region contains a unique low element, which is its unique element of minimal le...

conjecture | open | MathDB 页面 | 原文线索

V2-067909 | MATHDB-353314

Lorentzianity conjecture for chromatic symmetric functions of Dyck paths

摘录 / EXCERPT: Let d be a Dyck path, let $G(d)$ be its indifference graph, and let $X_{G(d)}$ be its chromatic symmetric function. For any finite number of variables, restrict $X_{G(d)}$ to tho...

conjecture | open | MathDB 页面 | 原文线索

V2-067910 | MATHDB-353316

The zeta-map correspondence for unit interval orders

摘录 / EXCERPT: Let P be a unit interval order corresponding to an incomparability graph $G(d)$ and to a lex-maximal part listing encoded by ${\bf a}=(a_1,\ldots,a_n)$. Let d' be the Dyck path...

conjecture | open | MathDB 页面 | 原文线索

V2-067911 | MATHDB-353318

The densely sparse cluster characterization for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. Write K_G for its maximal conflicting subgraph, $r(G)$ for its resistance, and call G a densely sparse cluster when it has the correspondin...

conjecture | open | MathDB 页面 | 原文线索

V2-067912 | MATHDB-353319

Fiol et al.'s oddness-resistance conjecture for snarks with empty buffer subgraph

摘录 / EXCERPT: Let G be a snark, let $\omega(G)$ denote its oddness, let $r(G)$ denote its resistance, and suppose that the buffer subgraph of G is empty. Oddness-resistance conjecture with e...

conjecture | refuted | MathDB 页面 | 原文线索

V2-067913 | MATHDB-353320

Bonamy-Bousquet-Feghali-Johnson quadratic diameter conjecture for coloring graphs

摘录 / EXCERPT: Bonamy-Bousquet-Feghali-Johnson conjecture. If $G=(V,E)$ is d-degenerate and $k>d$ is an integer, then $\mathcal{K}_k(G)$ has diameter

conjecture | open | MathDB 页面 | 原文线索

V2-067914 | MATHDB-353326

Tightness of the lower bound under multiple LP solutions

摘录 / EXCERPT: Let the standing assumptions of the paper hold. Let ${\cal S}_{\text{LP}}$ denote the set of solutions to the linear program, and suppose that this set contains multiple solutions.…

conjecture | open | MathDB 页面 | 原文线索

V2-067915 | MATHDB-353327

Irreducibility conjecture for Misiurewicz polynomials

摘录 / EXCERPT: Fix an integer $d\geq 2$, let $m\geq 2$ and $n\geq 1$, and let $\zeta\neq 1$ be a d-th root of unity. For the Misiurewicz polynomial $G_{d,m,n}^{\zeta}(c)$ defined by the paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-067916 | MATHDB-353329

Concavity conjecture for mutual information in a two-target vector Poisson channel

摘录 / EXCERPT: Mutual-information concavity conjecture. The mutual information $I(X_1,X_2;Y_1,Y_2,Y_3)$ is concave in (T_1,T_2,T_3). The conjecture concerns the sensing-time allocation in the t...

conjecture | open | MathDB 页面 | 原文线索

V2-067917 | MATHDB-353330

Fang-Xia conjecture on cubic graphical regular representations

摘录 / EXCERPT: Fang-Xia conjecture. Except for a finite number of cases, every finite non-abelian simple group has a cubic GRR.

conjecture | open | MathDB 页面 | 原文线索

V2-067918 | MATHDB-353331

The hypergeometric description conjecture for classical-group de Rham local systems

摘录 / EXCERPT: Let G) be one of the classical groups under consideration, with standard representation Std, and let $(\delta=0,\mu=\rho\times\phi)$ be rigid hypergeometric automorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-067919 | MATHDB-353340

Tensor-product geometric Satake conjecture for Coulomb branches

摘录 / EXCERPT: Let $\mathcal M^\nu(\lambda,\mu)$ be the deformation associated with a decomposition $W=W^1\oplus W^2$, let λ^1 and λ^2 be the corresponding dominant coweights, and...

conjecture | open | MathDB 页面 | 原文线索

V2-067920 | MATHDB-353341

Bounded unit-hypercube dimension implies vanishing information rate

摘录 / EXCERPT: Bounded-hypercube conjecture. If the sequence $\{M(P_i)\}$ is bounded, then

conjecture | open | MathDB 页面 | 原文线索

V2-067921 | MATHDB-353342

Bounded Minkowski length implies vanishing information rate

摘录 / EXCERPT: Bounded-Minkowski-length conjecture. If the sequence $\{L(P_i)\}$ is bounded, then

conjecture | open | MathDB 页面 | 原文线索

V2-067922 | MATHDB-353343

Ellipse conjecture for the roots of the Fibonacci-level characteristic polynomial

摘录 / EXCERPT: Ellipse conjecture. The roots of $f_n(\lambda)=F_{n+1}$ lie on an ellipse.

conjecture | open | MathDB 页面 | 原文线索

V2-067923 | MATHDB-353354

The order conjecture for entire quasiregular Yosida mappings

摘录 / EXCERPT: Let $n\geq 2$, and let $f:\mathbb{R}^n\to\mathbb{R}^n$ be an entire quasiregular Yosida mapping, meaning that the family of translates ... is normal. Order conjecture. The order of g...

conjecture | open | MathDB 页面 | 原文线索

V2-067924 | MATHDB-353363

Regulated extensions independence-of-ell conjecture for A-motives

摘录 / EXCERPT: Regulated independence-of-ℓ conjecture. The inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-067925 | MATHDB-353370

The analytic torsion formula for Shannon products with complete graphs

摘录 / EXCERPT: Let C_n be a circle and K_m a complete graph, with $C_n * K_m$ their Shannon product. The complete-graph product conjecture. ... This is presented as an experimentally observed f...

conjecture | open | MathDB 页面 | 原文线索

V2-067926 | MATHDB-353371

The analytic torsion formula for Shannon products with wheel graphs

摘录 / EXCERPT: Let C_n be a circle and W_m the wheel graph with m vertices, with $C_n * W_m$ their Shannon product. The wheel-product conjecture. ... The formula is suggested by computations...

conjecture | open | MathDB 页面 | 原文线索

V2-067927 | MATHDB-353372

The analytic torsion formula for Shannon products with linear graphs

摘录 / EXCERPT: Let C_n be a circle and L_n a linear graph with n vertices, with $C_n * L_n$ their Shannon product. The linear-graph product conjecture. ... The notation is inconsistent: the l...

conjecture | open | MathDB 页面 | 原文线索

V2-067928 | MATHDB-353373

The interpolation conjecture for Brill-Noether curves

摘录 / EXCERPT: Let d, g, r, and n be nonnegative integers with ... A BN-curve is a stable map from a general curve of genus g to $\mathbb{P}^r$ of degree d corresponding to the unique c...

conjecture | open | MathDB 页面 | 原文线索

V2-067929 | MATHDB-353374

Grothendieck support poset conjecture

摘录 / EXCERPT: Let $w\in S_n$, and let $\operatorname{supp}(\mathfrak{G}_w)$ be the set of exponent vectors of the nonzero terms of the Grothendieck polynomial $\mathfrak{G}_w$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-067930 | MATHDB-353375

Ranked Grothendieck support conjecture

摘录 / EXCERPT: Let $w\in S_n$, and let $\operatorname{supp}(\mathfrak{G}_w)$ be the set of exponent vectors of the nonzero terms of $\mathfrak{G}_w$. For $\alpha,\beta\in\mathbb{Z}^n$, compare ve...

conjecture | open | MathDB 页面 | 原文线索

V2-067931 | MATHDB-353376

Interval-closed support conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let $w\in S_n$, and let $\operatorname{supp}(\mathfrak{G}_w)$ be the set of exponent vectors of the nonzero terms of $\mathfrak{G}_w$. Order $\mathbb{Z}^n$ by componentwise compari...

conjecture | open | MathDB 页面 | 原文线索

V2-067932 | MATHDB-353378

Top-support coefficient-sum conjecture for Grothendieck polynomials

摘录 / EXCERPT: Let $w\in S_n$ and write ... Let $\mathfrak{G}_w^{\mathrm{top}}$ denote the top-degree part of $\mathfrak{G}_w$, and order exponent vectors componentwise. Top-support coefficient-sum...

conjecture | open | MathDB 页面 | 原文线索

V2-067933 | MATHDB-353379

Conjectured conjugate operators for Mourre estimates on discrete Schrödinger bands

摘录 / EXCERPT: Conjugate-operator conjecture. There exists such a conjugate operator $\mathbb{A}(n)$ for which the Mourre estimate for Δ holds with respect to $\mathbb{A}(n)$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-067934 | MATHDB-353380

Liftability conjecture for non-split metacyclic actions on surfaces

摘录 / EXCERPT: Liftability conjecture. Every non-split metacyclic action on S_g lifts under a suitably chosen finite regular cyclic cover to a split metacyclic action.

conjecture | open | MathDB 页面 | 原文线索

V2-067935 | MATHDB-353383

Ohsugi-Tsuchiya gamma-nonnegativity conjecture for symmetric edge polytopes

摘录 / EXCERPT: Ohsugi-Tsuchiya conjecture. For every $i\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-067936 | MATHDB-353386

Undecidability of testing whether the joint spectral radius is at least one

摘录 / EXCERPT: Undecidability conjecture. It is undecidable to check whether $\rho\geq 1$ for the joint spectral radius ρ.

conjecture | open | MathDB 页面 | 原文线索

V2-067937 | MATHDB-353389

Extension of the limiting free-energy theorem beyond symmetric disorder

摘录 / EXCERPT: Let $\mathcal{D}$ be the distribution of the disorder random variable, and suppose that it has finite second moment. Throughout the paper the disorder is assumed to be symmetric, w...

conjecture | open | MathDB 页面 | 原文线索

V2-067938 | MATHDB-353391

All-or-nothing conjecture for higher-order Gaussian fields

摘录 / EXCERPT: Consider a higher-order Gaussian generative field of order bb, with load R, channel capacity $\mathcal{C}(\sigma)$, overlap $m^\star$, and information rate...

conjecture | open | MathDB 页面 | 原文线索

V2-067939 | MATHDB-353392

Narasimhan's semiorthogonal decomposition conjecture for the rank-two moduli space

摘录 / EXCERPT: Let X be a smooth projective curve of genus $g\geq 2$, let $L\in\operatorname{Pic}^{d}(X)$ with $(2,d)=1$ and $0<d<2$, and let $\mathrm{M}(2,L)$ be the moduli space of stable ran...

conjecture | open | MathDB 页面 | 原文线索

V2-067940 | MATHDB-353393

Generalized semiorthogonal decomposition conjecture for moduli spaces of vector bundles

摘录 / EXCERPT: Let X be a smooth projective curve, let $\mathrm{M}(r,L)$ be the moduli space of stable rank-r vector bundles on X with fixed determinant L, and let $\operatorname{Jac}(X)$...

conjecture | open | MathDB 页面 | 原文线索

V2-067941 | MATHDB-353399

Degree conjecture relating interior and gamma polynomials of bipartite graphs

摘录 / EXCERPT: Let ${\mathcal{G}}$ be a bipartite graph. Denote by $I_{\mathcal{G}}$ its interior polynomial and by $\gamma_{\mathcal{G}}$ its gamma-polynomial. Degree conjecture. The degrees of...

conjecture | open | MathDB 页面 | 原文线索

V2-067942 | MATHDB-353400

Coefficientwise minimality conjecture for standard orientations of bipartite graphs

摘录 / EXCERPT: Let ${\mathcal{G}}$ be a bipartite graph, and consider its facet graphs and the two standard orientations of ${\mathcal{G}}$. For each facet graph, take its interior polynomial. St...

conjecture | open | MathDB 页面 | 原文线索

V2-067943 | MATHDB-353405

The standard derived equivalence characterization for colax dg-category diagrams

摘录 / EXCERPT: Standard derived equivalence characterization. The following conditions should be equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-067944 | MATHDB-353415

Conjecture on inverse pairs of primitive 1-normal elements

摘录 / EXCERPT: Let t be a positive integer. A pair (α,α^{-1}) is a pair of primitive 1-normal elements in $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ when both elements are primitive a...

conjecture | open | MathDB 页面 | 原文线索

V2-067945 | MATHDB-353429

The strict P-partition enumerator conjecture for tree posets

摘录 / EXCERPT: Let P and Q be finite posets whose Hasse diagrams are trees, and let $\overline{K}_{P}(\mathbf{x})$ denote the strict P-partition enumerator. The strict P-partition enumerato...

conjecture | open | MathDB 页面 | 原文线索

V2-067946 | MATHDB-353431

The principal specialization conjecture for tree posets

摘录 / EXCERPT: Let P and Q be posets with n elements whose Hasse diagrams are trees. Let $\overline{K}_{P}(\mathbf{x})$ denote the strict P-partition enumerator, and specialize...

conjecture | open | MathDB 页面 | 原文线索

V2-067947 | MATHDB-353432

Generalized Terao conjecture for d-arrangements

摘录 / EXCERPT: Let $\mathcal{C}_{1},\mathcal{C}_{2} \subset \mathbb{P}^{2}_{\mathbb{C}}$ be two d-arrangements for a fixed $d\geqslant 1$. Let G_{1} and G_{2} be their associated Levi graph...

conjecture | open | MathDB 页面 | 原文线索

V2-067948 | MATHDB-353433

Graded extension conjecture for generalized Asaeda-Haagerup categories

摘录 / EXCERPT: Let generalized Asaeda-Haagerup categories be de-equivariantizations of generalized Haagerup categories for the groups $\mathbb{Z}_{4m}\times\mathbb{Z}_2$ with $\epsilon_{(0,1)}$ t...

conjecture | open | MathDB 页面 | 原文线索

V2-067949 | MATHDB-353441

Lin's conjecture on boundary nodal sets of harmonic functions

摘录 / EXCERPT: Let $\Omega \subset \mathbb R^d$ be a Lipschitz domain and let $\Sigma=B\cap\partial\Omega$ for a ball B centered at a point of $\partial\Omega$. Let u be harmonic in Ω,...

conjecture | open | MathDB 页面 | 原文线索

V2-067950 | MATHDB-353445

The Third Borwein conjecture at modulus five

摘录 / EXCERPT: Third Borwein conjecture. The sign pattern of the coefficients in the expansion of $S_n(q)$ is $+----+----+----\cdots$, with the convention that zero coefficients may be regarded a...

conjecture | open | MathDB 页面 | 原文线索

V2-067951 | MATHDB-353446

Ismail-Kim-Stanton's generalized Borwein sign conjecture

摘录 / EXCERPT: Ismail-Kim-Stanton conjecture. If $m\equiv \pm(2l+1)a\pmod K$ for some l with $0\le l<K/2$, then $b_m\le0$; otherwise $b_m\ge0$. Thus the sign of b_m is determined by m modul...

conjecture | open | MathDB 页面 | 原文线索

V2-067952 | MATHDB-353447

The modulus-seven Borwein sign-pattern conjecture

摘录 / EXCERPT: Modulus-seven Borwein conjecture. For all m and n, $d_{7m}(n)\ge0$ and $d_{7m+1}(n),d_{7m+3}(n),d_{7m+4}(n),d_{7m+6}(n)\le0$. Further, $d_{7m+5}(n)\ge0$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-067953 | MATHDB-353448

Bressoud's coefficient-positivity conjecture

摘录 / EXCERPT: For nonnegative integers A,B, define the q-binomial coefficient by ... Suppose that $M,N\in\mathbb Z^+$, that α and β are positive rational numbers, and that K is...

conjecture | open | MathDB 页面 | 原文线索

V2-067954 | MATHDB-353452

The planar Bowditch boundary conjecture for relatively hyperbolic groups

摘录 / EXCERPT: Let $(G,\mathcal{P})$ be a relatively hyperbolic group pair, with Bowditch boundary $\partial(G,\mathcal{P})$. A boundary is planar if it embeds in the plane, and it has no cut poi...

conjecture | open | MathDB 页面 | 原文线索

V2-067955 | MATHDB-353456

López-Bracho et al.'s robust coloring conjecture for paths

摘录 / EXCERPT: López-Bracho et al.'s robust coloring conjecture. There exists a 3-coloring of G such that the number of monochromatic edges of H is at most

conjecture | open | MathDB 页面 | 原文线索

V2-067956 | MATHDB-353460

The braid-class ratio conjecture for reduced words

摘录 / EXCERPT: Let $d53e$ be the symmetric group on n letters. For $d70e i d53e$, let $d4a1(d70e)$ be the set of reduced words of $d70e$ and let $d49e(d70e)$ be the set of braid classes of redu...

conjecture | open | MathDB 页面 | 原文线索

V2-067957 | MATHDB-353464

The $O(G)$ conjecture on minimal synchronizing factors

摘录 / EXCERPT: $O(G)$ conjecture. The set of graphs H with $H\fold_S G$ has a unique $\fold_S$-minimal element $O(G)$.

conjecture | open | MathDB 页面 | 原文线索

V2-067958 | MATHDB-353466

The lcH-slenderness conjecture for Artin groups

摘录 / EXCERPT: A discrete group G is lcH-slender if every group homomorphism from a locally compact Hausdorff group into G is continuous. The lcH-slenderness conjecture. All Artin groups ar...

conjecture | open | MathDB 页面 | 原文线索

V2-067959 | MATHDB-353469

The fixed-point conjecture for elliptic subgroups of Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group, let C_Γ be its associated clique-cube complex, and let ... be the action by left multiplication. For a subgroup $H\subset A_\Gamma$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-067960 | MATHDB-353471

Neumann-Lara's planar graph dichromatic conjecture

摘录 / EXCERPT: Neumann-Lara's conjecture. Every orientation of a planar graph has dichromatic number at most 2.

conjecture | open | MathDB 页面 | 原文线索

V2-067961 | MATHDB-353480

Qiu's Morse-function conjecture for the global dimension on Dynkin-quiver stability spaces

摘录 / EXCERPT: Qiu's Morse-function conjecture. The function gldim is a piecewise Morse function on

conjecture | open | MathDB 页面 | 原文线索

V2-067962 | MATHDB-353481

The conjugacy-to-triangularity conjecture for tame exponential automorphisms

摘录 / EXCERPT: Conjugacy-to-triangularity conjecture. For every $\phi\in\mathcal{E}_3(k)\cap\operatorname{T}_3(k)$, there exists $\sigma\in\operatorname{T}_3(k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067963 | MATHDB-353482

The invariant-coordinate conjecture for positive-characteristic tame automorphisms

摘录 / EXCERPT: Invariant-coordinate conjecture. For every such ϕ, there exists $\sigma\in\operatorname{T}_3(k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067964 | MATHDB-353483

The principal-plinth-ideal conjecture for exponential automorphisms

摘录 / EXCERPT: Principal-plinth-ideal conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067965 | MATHDB-353494

The Casson invariant conjecture

摘录 / EXCERPT: Casson invariant conjecture. The Casson invariant of $a3$ equals one-eighth the signature of the Milnor fiber.

conjecture | open | MathDB 页面 | 原文线索

V2-067966 | MATHDB-353495

The splice-diagram decomposition conjecture for Milnor fibers

摘录 / EXCERPT: Splice-diagram decomposition conjecture. The Milnor fiber F is homeomorphic to the result $\overline F$ of pasting

conjecture | open | MathDB 页面 | 原文线索

V2-067967 | MATHDB-353496

Linear-size total-variation approximation for the uniform infinite cubic planar graph

摘录 / EXCERPT: Linear-size approximation conjecture. As $n\in 2\mathbb{N}$ tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-067968 | MATHDB-353508

The C_{θ}-real matrix representation conjecture for truncated Toeplitz operators

摘录 / EXCERPT: The C_{θ}-real matrix representation conjecture. If M is a complex symmetric matrix and is unitarily equivalent to a TTO, then there exist an inner function u and a...

open_problem | open | MathDB 页面 | 原文线索

V2-067969 | MATHDB-353512

Marcus's block permanent conjecture

摘录 / EXCERPT: Marcus's block permanent conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-067970 | MATHDB-353513

Soules's maximum-eigenvalue conjecture for Schur power matrices

摘录 / EXCERPT: Soules's Schur power matrix conjecture. The permanent $\mathop{\rm per} A$ is the maximum eigenvalue of $\pi(A)$. The conjecture is refuted: the source gives a positive-semidefinit...

conjecture | open | MathDB 页面 | 原文线索

V2-067971 | MATHDB-353514

Chollet's permanental Oppenheim inequality conjecture

摘录 / EXCERPT: Let ${\cal H}_n$ be the cone of positive-semidefinite Hermitian matrices, and let $A\circ B$ denote the Hadamard (entrywise) product. Chollet's conjecture. For $A,B\in{\cal H}_n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-067972 | MATHDB-353516

Pate's two-index permanent inequality challenge

摘录 / EXCERPT: Let $A\in{\cal H}_n$, and let $A(i|j)$ denote the matrix obtained by deleting row i and column j. Pate's inequality challenge. ... The source presents this as a consequence of th...

open_problem | refuted | MathDB 页面 | 原文线索

V2-067973 | MATHDB-353532

Presentation conjecture for Dehn quandles of Artin groups

摘录 / EXCERPT: Let S be a generating set and let ... be an Artin group. For quandle elements, write $(s*t)_m$ for the alternating product of s and t under the quandle operation, beginning wit...

conjecture | open | MathDB 页面 | 原文线索

V2-067974 | MATHDB-353535

Generalized Catalan derivative formula for unit-step solutions

摘录 / EXCERPT: Let $p=1,2,\dots$, let $j=1,\dots,p+1$, and let $n=0,1,\dots$. For the functions f_j corresponding to initial data in the form of the unit step, let $C^p_{j,n}$ denote the genera...

conjecture | open | MathDB 页面 | 原文线索

V2-067975 | MATHDB-353536

Improper conflict-free and unique-maximum coloring conjecture for planar and outerplanar graphs

摘录 / EXCERPT: Improper coloring conjecture. For the classes $\mathcal{P}$ and $\mathcal{O}$, the following equalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-067976 | MATHDB-353537

Proper conflict-free and unique-maximum coloring conjecture for planar graphs

摘录 / EXCERPT: Proper coloring conjecture. For the class of planar graphs $\mathcal{P}$,

conjecture | open | MathDB 页面 | 原文线索

V2-067977 | MATHDB-353539

Collapsibility-number conjecture for Vietoris-Rips complexes of hypercube graphs

摘录 / EXCERPT: Let $\mathbb{I}_n$ denote the n-dimensional hypercube graph, and let $\mathcal{VR}(\mathbb{I}_n;r)$ be its Vietoris-Rips complex at scale r. Let the collapsibility number of a...

conjecture | open | MathDB 页面 | 原文线索

V2-067978 | MATHDB-353541

Signature-independence conjecture for Adinkra invariant factors

摘录 / EXCERPT: Let an Adinkra be a signed graph with signed Laplacian L, and let its invariant factors mean the diagonal entries in the Smith normal form of L. A signature is the choice of si...

conjecture | open | MathDB 页面 | 原文线索

V2-067979 | MATHDB-353550

Primitive quasi-Jacobi form property and holomorphic anomaly equation

摘录 / EXCERPT: Primitive conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067980 | MATHDB-353552

The asymptotic threshold conjecture for complete bipartite graphs

摘录 / EXCERPT: Asymptotic threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-067981 | MATHDB-353554

Cuspidal occurrence conjecture for the long-root Langlands quotient of G_2

摘录 / EXCERPT: Cuspidal occurrence conjecture. The $\Pi_{F,f}^v$-isotypic component of the cuspidal spectrum is

conjecture | open | MathDB 页面 | 原文线索

V2-067982 | MATHDB-353555

Functorial lifting conjecture from G_2 to GL_7

摘录 / EXCERPT: Functorial lifting conjecture. There is a self-dual automorphic representation $\widetilde\Pi$ of $GL_7(\mathbb A)$ satisfying all five listed properties: matching unramified Satak...

conjecture | open | MathDB 页面 | 原文线索

V2-067983 | MATHDB-353559

Sokal's reversed second-order Eulerian polynomial conjecture

摘录 / EXCERPT: Sokal's reversed second-order Eulerian conjecture. The sequence $(E_n^{[2]*}(x))_{n\geq0}$ is coefficientwise Hankel-totally positive in x.

conjecture | open | MathDB 页面 | 原文线索

V2-067984 | MATHDB-353560

Conjecture on coincidence of the optimal and asymptotic vaccine allocations

摘录 / EXCERPT: Let R_0 be the reproductive number, let $\vec{v}_{\rm optimal}$ denote an optimal vaccine allocation solving the optimization problem in, and let $\vec{v}^*$ denote the asymptoti...

conjecture | open | MathDB 页面 | 原文线索

V2-067985 | MATHDB-353561

The exceptional-pair conjecture for quadratic forms over finite fields

摘录 / EXCERPT: Exceptional-pair conjecture. For the fields $\mathbb{F}_{q^m}$ with $m=3$ or $m=4$, the only possible exceptional pairs (q,m) are

conjecture | open | MathDB 页面 | 原文线索

V2-067986 | MATHDB-353562

Constant-depth decision-tree conjecture for Boolean functions on Grassmann graphs

摘录 / EXCERPT: Let $J_q(n,k)$ be the Grassmann graph on the k-dimensional subspaces of an n-dimensional vector space over the finite field of order q. A Boolean degree d function is a Boo...

conjecture | open | MathDB 页面 | 原文线索

V2-067987 | MATHDB-353563

Ringel-Kotzig decomposition conjecture

摘录 / EXCERPT: Let $n\geq 1$, and let K_{2n+1} be the complete graph on $2n+1$ vertices. Ringel-Kotzig decomposition conjecture. The graph K_{2n+1} can be decomposed into $2n+1$ subgraphs, al...

conjecture | open | MathDB 页面 | 原文线索

V2-067988 | MATHDB-353592

The CSP conjecture for lattices in the simple Lie group $F_{4(-20)}$

摘录 / EXCERPT: Let $F_{4(-20)}$ be the simple real Lie group of the indicated type, and let Γ be a lattice in $F_{4(-20)}$. CSP conjecture for $F_{4(-20)}$. Every lattice Γ in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-067989 | MATHDB-353594

Uniform stable approximation conjecture for the parametric approximation sets

摘录 / EXCERPT: Uniform stable approximation conjecture. The sequence $\boldsymbol{\Psi}_{P}$ is a stable approximation for $\mathcal{A}_{P}$: there exist $s=s(P)\geq 0$ and $C=C(P)\geq 0$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-067990 | MATHDB-353595

The Theta Conjecture for fully tiered rooted trees

摘录 / EXCERPT: Let $ u$ be a composition, let $ u(u)$ denote the partition obtained by rearranging its parts, and let $ \mathsf{RTT}(\nu)$ be the set of fully tiered rooted trees associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-067991 | MATHDB-353596

The symmetric Theta Conjecture for rooted trees with a prescribed root label

摘录 / EXCERPT: Let n be a positive integer, let $ \mathsf{RTT}_j(1^{n+1})$ denote the set of standard fully tiered rooted trees on $n+1$ vertices whose root is labelled j, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-067992 | MATHDB-353597

Conjectural asymptotics for numerical-semigroup sequences

摘录 / EXCERPT: Let s_g, $\hat n_g$, and $\mathsf{r}_{1.51}$ be the sequences and constant defined in the source, where g tends to infinity. Conjectural asymptotics. The paper gives three succ...

conjecture | open | MathDB 页面 | 原文线索

V2-067993 | MATHDB-353604

The hook-tableau conjecture for recording tableaux of uncrowding

摘录 / EXCERPT: Let λ be a partition, let $\mu\subseteq\lambda$, and let $\mathcal{H}_{\lambda/\mu}$ be the set of recording tableaux arising from the uncrowding map on edge labeled tablea...

conjecture | open | MathDB 页面 | 原文线索

V2-067994 | MATHDB-353617

Boundedness of minimal log discrepancies of Kollár components with positive local volume

摘录 / EXCERPT: Boundedness conjecture. There exists a constant $A=A(n,\varepsilon,I)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-067995 | MATHDB-353618

Boundedness of minimal log discrepancies of Kollár components

摘录 / EXCERPT: Boundedness conjecture. There exists a constant $A=A(n,I)$, depending only on n and I, such that

conjecture | open | MathDB 页面 | 原文线索

V2-067996 | MATHDB-353624

The monoid Cayley characterization conjecture for generalized Petersen graphs

摘录 / EXCERPT: Let $G(n,k)$ be a generalized Petersen graph, and let ${\rm Cay}(M,C)$ denote a monoid Cayley graph with connection set C of size 2. The monoid Cayley characterization conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-067997 | MATHDB-353625

The retract-image conjecture for non-bipartite generalized Petersen graphs

摘录 / EXCERPT: Let $G(n,k)$ be a non-bipartite generalized Petersen graph, and let f be an endomorphism of $G(n,k)$. The retract-image conjecture. The image of f is a retract of $G(n,k)$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-067998 | MATHDB-353626

The semigroup representation conjecture for generalized Petersen graphs

摘录 / EXCERPT: Let $G(n,k)$ be a generalized Petersen graph. A semigroup graph is a graph admitting a Cayley representation by a semigroup. The semigroup representation conjecture. Every generali...

conjecture | open | MathDB 页面 | 原文线索

V2-067999 | MATHDB-353630

The conjectural sharp bound for the separating degree of a diagonalizable group

摘录 / EXCERPT: Let G be the diagonalizable group considered in Theorem, let $\mathcal{B}$ be the associated object, let $\tau(\mathcal{B})$ denote its separating degree, let $\dim(G)$ be the di...

conjecture | open | MathDB 页面 | 原文线索

V2-068000 | MATHDB-353631

Ghosh-Sarnak strong approximation conjecture for the Markoff surface

摘录 / EXCERPT: For an integer m, let U_m be the affine Markoff surface ... For $m=0$, let $U_0^*(\mathbb{Z}/p\mathbb{Z})=U_0(\mathbb{Z}/p\mathbb{Z})\setminus\{(0,0,0)\}$. Let Γ be the gr...

conjecture | open | MathDB 页面 | 原文线索

V2-068001 | MATHDB-353635

The Strong Four Exponentials Conjecture

摘录 / EXCERPT: Let $\widetilde{\mathcal L}\subset\overline{\mathbf Q}_p$ be the $\overline{\mathbf Q}$-vector space generated by 1 and the elements of $\log_p(\overline{\mathbf Q}^{\times})$. L...

conjecture | open | MathDB 页面 | 原文线索

V2-068002 | MATHDB-353640

Mass-map conjecture for 2-Calabi-Yau categories

摘录 / EXCERPT: Let $\mathcal T$ be the 2-Calabi-Yau category associated to a connected quiver, let $\mathbb{S}$ be the set of spherical objects of $\mathcal T$, and let m_q be the mass map for...

conjecture | open | MathDB 页面 | 原文线索

V2-068003 | MATHDB-353649

Strictification conjecture for reflexive objects in compact closed categories

摘录 / EXCERPT: Strictification conjecture. The restrictions that the compact closed category arise from the $\bf Int$ or $\bf Gol$ construction and have trivial scalars are not essential: reflexi...

conjecture | open | MathDB 页面 | 原文线索

V2-068004 | MATHDB-353658

Metricity threshold for the point pair function outside the unit ball

摘录 / EXCERPT: Metricity conjecture. For a constant $\alpha>0$, this function, defined for $x,y\in\mathbb{R}^n\backslash\overline{\mathbb{B}}^n$ and $n\geq2$, is a metric if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-068005 | MATHDB-353659

Metricity of the point pair function off the z-axis

摘录 / EXCERPT: Let Z be the z-axis in $\mathbb{R}^3$, and let $G=\mathbb{R}^3\backslash Z$. The point pair function p_G is defined on pairs of points in G. Metricity conjecture. The point...

conjecture | open | MathDB 页面 | 原文线索

V2-068006 | MATHDB-353664

Alternating bi-3-positivity conjecture for BorosMoll polynomials

摘录 / EXCERPT: For $m1$, define the BorosMoll polynomial $M_m(x)$ and its transform ... A polynomial is symmetric if its coefficients read the same forwards and backwards, unimodal if its coefficie...

conjecture | open | MathDB 页面 | 原文线索

V2-068007 | MATHDB-353665

The uniform-distribution conjecture for bases with four zeros in their Pisano period

摘录 / EXCERPT: Uniform-distribution conjecture. For every base β with $\omega(\beta)=4$, there exists a K such that, for all $k\geq K$, the digits in the sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-068008 | MATHDB-353675

The non-quasiconvex subgroup conjecture for Cannon-Thurston maps

摘录 / EXCERPT: Let $H<G$ be a hyperbolic subgroup of a hyperbolic group G. Suppose the Cannon-Thurston map ... exists, and suppose that H is not quasiconvex in G. Non-quasiconvex subgroup con...

conjecture | open | MathDB 页面 | 原文线索

V2-068009 | MATHDB-353682

Conjecture on the giant d-rigid component and its double core

摘录 / EXCERPT: Giant d-rigid component conjecture. Then a.a.s.:

conjecture | open | MathDB 页面 | 原文线索

V2-068010 | MATHDB-353688

Dervan-Sektnan's automorphism-free fiber conjecture

摘录 / EXCERPT: Let $f:(X,H)\to(B,L)$ be a polarized smooth fibration such that every fiber (X_b,H_b) is K-stable. Assume that $\operatorname{Aut}(X_b,H_b)=0$ for every $b\in B$. Dervan-Sektnan'...

conjecture | open | MathDB 页面 | 原文线索

V2-068011 | MATHDB-353689

The semicompleteness characterization for graph products of abelian groups

摘录 / EXCERPT: Semicompleteness characterization. The graph product G_Γ is semicomplete if and only if Γ has no separating star.

conjecture | open | MathDB 页面 | 原文线索

V2-068012 | MATHDB-353692

Champanerkar-Kofman-Lalín's bipyramid volume conjecture for periodic alternating links

摘录 / EXCERPT: Let $\mathcal{L}$ be a $\mathbb{Z}^2$-periodic alternating link, with toroidally alternating quotient link L. Let $P(z,w)$ be the characteristic polynomial of the toroidal dimer...

conjecture | open | MathDB 页面 | 原文线索

V2-068013 | MATHDB-353696

The four-group structure conjecture for optimal sensing-time proportions

摘录 / EXCERPT: Four-group structure conjecture. The optimal time-proportions for both information-theoretic and detection-theoretic metrics have the form

conjecture | open | MathDB 页面 | 原文线索

V2-068014 | MATHDB-353698

Binomial approximation conjecture for the Poisson bootstrap quantile index distribution

摘录 / EXCERPT: Let the sample $\mathbf{y}$ consist of elements that can be uniquely ordered, and let the quantile $q\in(0,1)$ be fixed. Let ψ denote the index of the original order statistic...

conjecture | open | MathDB 页面 | 原文线索

V2-068015 | MATHDB-353704

Multiplicity-free inverse Grassmannian Schubert product conjecture

摘录 / EXCERPT: Multiplicity-free inverse Grassmannian conjecture. If u and v are inverse Grassmannian, then $\mathfrak S_u\cdot\mathfrak S_v$ is a multiplicity-free sum of Schubert polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-068016 | MATHDB-353706

Cranston's maximum-average-degree conjecture for odd colorings

摘录 / EXCERPT: Let G be a finite simple graph, let $\operatorname{mad}(G)$ be the maximum of $\frac{2|E(H)|}{|V(H)|}$ over all non-empty subgraphs H of G, and let $\chi_o(G)$ be the least p...

conjecture | open | MathDB 页面 | 原文线索

V2-068017 | MATHDB-353728

The unit-circle-rootedness conjecture for prism-slice generating polynomials

摘录 / EXCERPT: Unit-circle-rootedness conjecture. Every complex root of $p_{n,m,\mathbf{c}}(x)$ lies on the unit circle $|z|=1$. This conjecture concerns the generating polynomials arising from l...

conjecture | open | MathDB 页面 | 原文线索

V2-068018 | MATHDB-353729

The real-rootedness and interlacing conjecture for prism-slice h*-polynomials

摘录 / EXCERPT: Real-rootedness and interlacing conjecture. The polynomial $h^*(\mathscr{R}_{k,\mathbf{c}},x)$ is real-rooted. Moreover, when c_ngeq2, the roots of...

conjecture | open | MathDB 页面 | 原文线索

V2-068019 | MATHDB-353731

Weighted Erdős-Burgess equality conjecture for finite principal ideal rings

摘录 / EXCERPT: Weighted Erdős-Burgess equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068020 | MATHDB-353746

Conjecture on definiteness of separable symmetric tensors

摘录 / EXCERPT: Definiteness conjecture. Then $\mathcal{A}$ is neither a positive semidefinite tensor nor a negative semidefinite tensor.

conjecture | open | MathDB 页面 | 原文线索

V2-068021 | MATHDB-353748

Dimension-independent Welch-bound conjecture for positive definite functions

摘录 / EXCERPT: Let $U(d)$ be the unitary group, let $H(d)$ be the set of complex Hadamard matrices in $U(d)$, and let $B(d)$ be the optimal positive-definite-functions upper bound for mutually un...

conjecture | open | MathDB 页面 | 原文线索

V2-068022 | MATHDB-353752

Mirror invariance conjecture for the Thue-Morse two-block language

摘录 / EXCERPT: Let $x^{(00)}$ be the fixed point of the Thue-Morse two-block substitution, and let $L_{\rm\sc TM}$ be the language of all finite words occurring in $x^{(00)}$. A set of binary wor...

conjecture | open | MathDB 页面 | 原文线索

V2-068023 | MATHDB-353754

Uniform recurrence conjecture for the Thue-Morse two-block fixed point

摘录 / EXCERPT: Let $x^{(00)}$ be the fixed point of the Thue-Morse two-block substitution. A word is uniformly recurrent when every finite factor occurring in $x^{(00)}$ occurs there with bounded...

conjecture | open | MathDB 页面 | 原文线索

V2-068024 | MATHDB-353755

Conjecture on the frequency of 1 in the Thue-Morse two-block fixed point

摘录 / EXCERPT: Let $x^{(00)}$ be the Thue-Morse two-block fixed point, and let the frequency of 1 mean the limiting proportion of positions occupied by 1 in its prefixes, if that limit exists...

conjecture | open | MathDB 页面 | 原文线索

V2-068025 | MATHDB-353758

Odd rational palindrome quotient conjecture

摘录 / EXCERPT: Let $p>1$ be odd. A rational number p/q has a palindromic quotient representation if there are palindromic numbers A and B such that $p/q=A/B$. Odd rational palindrome quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-068026 | MATHDB-353759

Zero-density conjecture for quotients of antipalindromic numbers

摘录 / EXCERPT: Let $Q_{\rm anti}$ be the set of integers representable as quotients of binary antipalindromic numbers. Zero-density conjecture. The set $Q_{\rm anti}$ has natural density zero. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-068027 | MATHDB-353761

Rational antipalindromic quotient conjecture

摘录 / EXCERPT: A rational number p/q has an antipalindromic quotient representation if there are antipalindromic numbers A and B such that $p/q=A/B$. Rational antipalindromic quotient conje...

conjecture | open | MathDB 页面 | 原文线索

V2-068028 | MATHDB-353766

Bogoliubov excitation-spectrum conjecture for the Gross-Pitaevskii three-body Bose gas

摘录 / EXCERPT: Excitation-spectrum conjecture. The low-lying spectrum of $H_N^{\rm GP}$ is given by

conjecture | open | MathDB 页面 | 原文线索

V2-068029 | MATHDB-353768

Second-order low-density expansion conjecture for the dilute three-body Bose gas

摘录 / EXCERPT: Let $e_{\rm 3B}$ denote the ground-state energy per unit volume in the thermodynamic limit, let ρ be the particle density, and let $b_{\mathcal M}(V)$ be the three-body scatte...

conjecture | open | MathDB 页面 | 原文线索

V2-068030 | MATHDB-353777

Bétermin's conjecture on phase transitions for one-well potentials

摘录 / EXCERPT: Let L be a two-dimensional lattice of area A, and let $E_f(L)$ denote its lattice energy for a potential f. A potential is completely monotone if it satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-068031 | MATHDB-353778

Square-lattice minimization conjecture for a difference of exponentials

摘录 / EXCERPT: Let $E_f(L)$ denote the lattice energy of a two-dimensional lattice L, and consider the potential $f(r)=e^{-\beta\pi r}-e^{-\alpha\pi r}$, where α and β are paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-068032 | MATHDB-353787

Concentration conjecture for empirical relative entropy

摘录 / EXCERPT: Concentration conjecture. There are positive constants c_1 and c_2 such that for every n, k, P, and $\varepsilon\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068033 | MATHDB-353788

Centered gamma moment-generating-function conjecture for empirical relative entropy

摘录 / EXCERPT: Centered gamma moment-generating-function conjecture. For every n, k, and P,

conjecture | open | MathDB 页面 | 原文线索

V2-068034 | MATHDB-353790

Tameness conjecture for automorphisms of free Bol loops of exponent two

摘录 / EXCERPT: Let $B(X)$ be a free Bol loop of exponent two. An automorphism of $B(X)$ is tame if it belongs to the group generated by the elementary automorphisms that, for a free generating se...

conjecture | open | MathDB 页面 | 原文线索

V2-068035 | MATHDB-353794

The odd chromatic number conjecture for connected graphs

摘录 / EXCERPT: Odd chromatic number conjecture. If G is a connected graph of maximum degree $\Delta\geq3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068036 | MATHDB-353795

The PCF chromatic number conjecture for bounded-degree connected graphs

摘录 / EXCERPT: PCF chromatic number conjecture. If G is a connected graph of maximum degree $\Delta\geq3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068037 | MATHDB-353796

Two spanning temporal arborescences under half-connectivity

摘录 / EXCERPT: Two-arborescence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-068038 | MATHDB-353801

The unilateral frame number conjecture for rigid matrix model operators

摘录 / EXCERPT: The unilateral frame number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068039 | MATHDB-353806

Thomas-Yau's equivalent variational existence conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be the class of quantitatively almost calibrated exact unobstructed Lagrangian objects under consideration, and let $L_0\in\mathcal{L}$ be exact, quantitatively a...

conjecture | open | MathDB 页面 | 原文线索

V2-068040 | MATHDB-353807

Thomas-Yau's minimizer conjecture for the Solomon functional

摘录 / EXCERPT: Let $\mathcal{L}$ be the class of admissible Lagrangian objects and let $\mathcal{S}$ be the Solomon functional. A minimizer is an element $L\in\mathcal{L}$ attaining the infimum o...

conjecture | open | MathDB 页面 | 原文线索

V2-068041 | MATHDB-353808

Thomas-Yau's minimizer-to-special-Lagrangian conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be the admissible class of Lagrangian currents and let $\mathcal{S}$ be the Solomon functional. Let $\hat{\theta}$ be the phase determined by the relevant Fukaya-...

conjecture | open | MathDB 页面 | 原文线索

V2-068042 | MATHDB-353810

Thomas-Yau's special-Lagrangian minimizer conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be the admissible class of Lagrangian objects and let $\mathcal{S}$ be the Solomon functional. A special Lagrangian is a Lagrangian on which the relevant holomorp...

conjecture | open | MathDB 页面 | 原文线索

V2-068043 | MATHDB-353818

The conflict-free chromatic index conjecture for bipartite graphs

摘录 / EXCERPT: Let G be a bipartite graph without isolated vertices. The bipartite conflict-free chromatic index conjecture. ... The preceding results show that the optimal general upper bound fo...

conjecture | open | MathDB 页面 | 原文线索

V2-068044 | MATHDB-353821

Strong-coupling quasi-particle scaling conjecture for the polaron

摘录 / EXCERPT: Strong-coupling quasi-particle scaling conjecture. There exists a continuous function $f:[0,\infty)\to[0,\infty)$ such that $f(s)\to1$ as $s\to\infty$ and

conjecture | open | MathDB 页面 | 原文线索

V2-068045 | MATHDB-353835

Uniqueness and classification conjecture for the heated Riemann problem

摘录 / EXCERPT: Uniqueness and classification conjecture. The solution of the Riemann problem consisting of the governing equations and initial condition is unique for any given γ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068046 | MATHDB-353836

Positivity conjecture for the distribution in the Bôcher-type characterization

摘录 / EXCERPT: Let u and v be as in Theorem, with v the distribution associated to u and supported on the singular set Γ. For a test function $\phi\in C_c^\infty(B_2)$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-068047 | MATHDB-353838

Simon’s genus monotonicity conjecture for knot-group epimorphisms

摘录 / EXCERPT: Let J and K be knots with knot groups π_J and π_K, and let $\varphi:\pi_J\twoheadrightarrow\pi_K$ be a surjective homomorphism. Simon’s genus monotonicity conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-068048 | MATHDB-353843

The girth analogue for hereditary graph classes

摘录 / EXCERPT: Girth analogue conjecture. For every graph F with at least one cycle, there exists a constant b_F and graphs G of arbitrarily large chromatic number and the same girth as F...

conjecture | open | MathDB 页面 | 原文线索

V2-068049 | MATHDB-353855

Existence and asymptotics of a shrinking soliton on the blowup of \mathbb{C}\times\mathbb{P}^1

摘录 / EXCERPT: Let $\operatorname{Bl}_{p}(\mathbb{C}\times \mathbb{P}^{1})$ be the blowup of $\mathbb{C}\times\mathbb{P}^{1}$ at a fixed point p of the standard torus action, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068050 | MATHDB-353857

The atom bound conjecture for inhibiting sets

摘录 / EXCERPT: Inhibiting-set atom bound conjecture. Every inhibiting set of a non-completable partial matrix with variability k contains at most $k+1$ atoms.

conjecture | open | MathDB 页面 | 原文线索

V2-068051 | MATHDB-353858

The growth conjecture for TP obstructions

摘录 / EXCERPT: TP-obstruction growth conjecture. There are infinitely many TP obstructions, and the number of TP obstructions with a given number of rows grows quickly, perhaps exponentially, wit...

conjecture | open | MathDB 页面 | 原文线索

V2-068052 | MATHDB-353859

The finiteness conjecture for k-variable TN obstructions

摘录 / EXCERPT: TN-obstruction finiteness conjecture. For every k, there are finitely many k-variable TN obstructions.

conjecture | open | MathDB 页面 | 原文线索

V2-068053 | MATHDB-353860

The obstruction-containment conjecture relating TP and TN completion

摘录 / EXCERPT: TP-to-TN obstruction conjecture. All TP obstructions contain a TN obstruction.

conjecture | open | MathDB 页面 | 原文线索

V2-068054 | MATHDB-353861

Batyrev-Juny's reducibility conjecture for high-dimensional Gorenstein polytopes

摘录 / EXCERPT: Let P be a Gorenstein polytope of dimension d and degree s. A Gorenstein polytope is reducible with factors F and G if it is a Cayley join of F and G and ... equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-068055 | MATHDB-353868

Mazur's non-vanishing conjecture for Rankin-Selberg L-values

摘录 / EXCERPT: Mazur's conjecture. If $e(\chi_0)=+1$ (the definite case), then

conjecture | open | MathDB 页面 | 原文线索

V2-068056 | MATHDB-353879

Modular relation for H_(k,k,k-3)

摘录 / EXCERPT: Let k be in the finite-level setting of the paper, with $H_{(c_0,c_1,c_2)}(z,q)$ denoting the associated generating function. The modular relation conjecture. ... This is presented...

conjecture | open | MathDB 页面 | 原文线索

V2-068057 | MATHDB-353880

Sufficiency of the R1--R4 relations for the recurrence proofs

摘录 / EXCERPT: Let $S_m(\rho\mid\sigma)$ denote the finite-level (z,q) sums, and let R_1--R_4 be the relations listed for the relevant moduli. The R1--R4 sufficiency conjecture. In each mod...

conjecture | open | MathDB 页面 | 原文线索

V2-068058 | MATHDB-353883

The non-Mathieu-Zhao image conjecture for simple derivations

摘录 / EXCERPT: Let K be a field, let $n\geq 2$, and let D be a simple derivation of the polynomial ring $K[x_1,\ldots,x_n]$. A K-subspace M of a ring is a Mathieu-Zhao space if, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-068059 | MATHDB-353884

Lieb-Thirring conjecture for optimal spectral constants

摘录 / EXCERPT: Let $L_{\gamma,d}$ be the optimal constant in the Lieb-Thirring inequality, let $L_{\gamma,d}^{\mathrm{cl}}$ be the semiclassical constant, and let $L_{\gamma,d}^{\mathrm{one}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-068060 | MATHDB-353886

Dual Lieb-Thirring conjecture for the kinetic-energy constant

摘录 / EXCERPT: Let K_d be the optimal constant in the kinetic-energy inequality referred to as equation (kinetic), and let K_d^{one} and K_d^{cl} be the corresponding one-...

conjecture | open | MathDB 页面 | 原文线索

V2-068061 | MATHDB-353889

The conjecture that half-isomorphisms of automorphic loops are special

摘录 / EXCERPT: Special half-isomorphism conjecture. Every half-isomorphism between automorphic loops is special.

conjecture | open | MathDB 页面 | 原文线索

V2-068062 | MATHDB-353896

The forest-path partition conjecture for planar graphs without 4-cycles and 6-cycles

摘录 / EXCERPT: Forest-path partition conjecture. Every planar graph without 4-cycles and 6-cycles has an $(\mathcal{F}_1,\mathcal{F})$-partition.

conjecture | open | MathDB 页面 | 原文线索

V2-068063 | MATHDB-353903

Local List Linear Hadwiger's Conjecture

摘录 / EXCERPT: Let G be a graph, let $v(G)$ denote its number of vertices, and let $B_c[v]$ be the subgraph induced by the vertices at distance at most c from v. For a positive integer c,...

conjecture | open | MathDB 页面 | 原文线索

V2-068064 | MATHDB-353904

Deletable-subgraph size conjecture for minor-free graphs

摘录 / EXCERPT: Let G be a graph, let $X\subseteq V(G)$, and let an r-deletable subgraph mean an induced subgraph that is colourable from every list assignment whose list sizes satisfy the deg...

conjecture | open | MathDB 页面 | 原文线索

V2-068065 | MATHDB-353907

The bounded-degree wall-or-line-wall induced-subgraph conjecture

摘录 / EXCERPT: Let G be a graph, let k and Δ be positive integers, and let $W_{k \times k}$ denote the $(k \times k)$-wall. A subdivision of a graph is obtained by replacing its edges b...

conjecture | open | MathDB 页面 | 原文线索

V2-068066 | MATHDB-353915

Uniform Alon-Krivelevich-Sudakov conjecture

摘录 / EXCERPT: Uniform Alon-Krivelevich-Sudakov conjecture. There is a constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-068067 | MATHDB-353916

DP-version of the Alon-Krivelevich-Sudakov conjecture

摘录 / EXCERPT: DP-version of the Alon-Krivelevich-Sudakov conjecture. For every graph F, there exist constants $c,d_0>0$ such that, whenever H is F-free, has maximum degree $d\geqslant d_0$...

conjecture | open | MathDB 页面 | 原文线索

V2-068068 | MATHDB-353917

Cambie-Kang conjecture for triangle-free DP-coloring

摘录 / EXCERPT: Cambie-Kang conjecture. For every $\varepsilon>0$, there is $d_0\in\mathbb{N}$ such that, if H has maximum degree $d\geqslant d_0$ and

conjecture | open | MathDB 页面 | 原文线索

V2-068069 | MATHDB-353921

Characterization of the edge-universality regime for dense random regular graphs

摘录 / EXCERPT: Edge-universality regime conjecture. The assertion in (1.2) holds if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-068070 | MATHDB-353927

Automorphic Goldfeld conjecture for quadratic twists

摘录 / EXCERPT: Let X be the set of quadratic fields, and let π' be an irreducible cuspidal automorphic representation of $\operatorname{PGL}_2(\mathbb{A})$. For $E\in X$, let η_E be th...

conjecture | open | MathDB 页面 | 原文线索

V2-068071 | MATHDB-353930

Exhaustion of polynomial cases for alternating $(1,3)$ indices in t-adic symmetric multiple zeta values

摘录 / EXCERPT: Let $\boldsymbol{k}$ be an index in which 1 and 3 appear alternately, let m be a positive integer, and let $\zeta_{\mathcal{S}_m}(\boldsymbol{k})$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-068072 | MATHDB-353931

The limiting nonunimodular-root ratio for $P_{m,1}(x)$

摘录 / EXCERPT: Limiting-ratio conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068073 | MATHDB-353932

A Lehmer-type gap conjecture for bounded-coefficient polynomial sequences

摘录 / EXCERPT: Bounded-coefficient limit-ratio conjecture. If the limit ratio of such a sequence is strictly below some constant $c(N)>0$, then its limit ratio is equal to 0.

conjecture | open | MathDB 页面 | 原文线索

V2-068074 | MATHDB-353933

Bijection between constrained Delannoy paths and 102-avoiding inversion sequences

摘录 / EXCERPT: Let n be a positive integer. Let $D\mathcal{P}^{(2)}_{n,2n}(NE,EN)$ be the set of Delannoy paths from $(0,0)$ to $(n,2n)$ consisting of north-steps $N=(0,1)$, east-steps...

conjecture | open | MathDB 页面 | 原文线索

V2-068075 | MATHDB-353934

Correspondence between Carter diagrams and Dynkin-Minchenko completion nodes

摘录 / EXCERPT: Carter diagrams with cycles and extra nodes in the Dynkin-Minchenko completion procedure are the objects under consideration. Correspondence conjecture. There is a correspondence b...

conjecture | open | MathDB 页面 | 原文线索

V2-068076 | MATHDB-353935

Homogeneous Carter diagrams have similar signed enhanced Dynkin diagrams

摘录 / EXCERPT: Let $\{\widetilde\Gamma,\Gamma\}$ be a homogeneous pair of Carter diagrams. Associate signed enhanced Dynkin diagrams $\Delta(\widetilde\Gamma)$ and $\Delta(\Gamma)$ to them. Homog...

conjecture | open | MathDB 页面 | 原文线索

V2-068077 | MATHDB-353940

Categorical Torelli conjecture for general ordinary Gushel-Mukai threefolds

摘录 / EXCERPT: Categorical Torelli conjecture. The intermediate Jacobian $J(X)$ determines the Kuznetsov component $\mathcal{K}u(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-068078 | MATHDB-353957

Model-independent spanning conjecture for q-bracket spaces

摘录 / EXCERPT: Model-independent q-bracket conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068079 | MATHDB-353975

Generalized double-flip decomposition conjecture for friends-and-strangers graphs

摘录 / EXCERPT: Let Y be a graph, let $\overline{Y}$ be its complement, and let $\operatorname{Acyc}(\overline{Y})$ denote the acyclic orientations of $\overline{Y}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068080 | MATHDB-353977

The q-to-e positivity conjecture for Delta and Xi operators

摘录 / EXCERPT: Let F be a symmetric function with a positive monomial-basis expansion, let G be a symmetric function with a positive Schur-basis expansion, and let Δ_F and Ξ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-068081 | MATHDB-353978

The t-statistic conjecture for gamma-parking functions

摘录 / EXCERPT: Let γ and λ be the parameters used to define the set $\operatorname{PF}^{\gamma}_{\lambda}$ of γ-parking functions, let S be a subset of the area cells of...

conjecture | open | MathDB 页面 | 原文线索

V2-068082 | MATHDB-353980

The inverse mean value property for panharmonic functions

摘录 / EXCERPT: Let $D \subset \mathbb{R}^m$, with $m \geqslant 2$, be a bounded domain with sufficiently smooth boundary, and let $r>0$ satisfy $|B_r|=|D|$. A function u is panharmonic if it be...

conjecture | open | MathDB 页面 | 原文线索

V2-068083 | MATHDB-353995

Grünbaum-Cuntz conjecture for rank-3 simplicial arrangements

摘录 / EXCERPT: Grünbaum-Cuntz conjecture. Every simplicial arrangement of rank 3 is isomorphic to an arrangement in the Grünbaum-Cuntz catalog.

conjecture | open | MathDB 页面 | 原文线索

V2-068084 | MATHDB-353999

Transversal restriction conjecture for the order bound

摘录 / EXCERPT: Let $\mathcal{F}$ be a codimension-1 foliation, let V range over all truly transversal hypersurfaces, and write $\mathcal{F}|_V$ for the restricted foliation. If $g'=g|_V$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068085 | MATHDB-354004

Asymptotic change-of-variable conjecture for preconditioned Toeplitz eigenvalues

摘录 / EXCERPT: Let l,g be two even functions with $g>0$ on $(0,\pi)$, and suppose that $f\equiv l/g$ is monotone increasing over $(0,\pi)$. Set $X_n\equiv T_n^{-1}(g)T_n(l)$ for all n. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068086 | MATHDB-354006

Miranda's conjecture on Chow stability of rational Weierstrass fibrations

摘录 / EXCERPT: Let W be a rational Weierstrass fibration, let X be its minimal resolution, and let H be an arbitrary ample line bundle. Miranda's conjecture. The pair (W,H) is asymptotica...

conjecture | open | MathDB 页面 | 原文线索

V2-068087 | MATHDB-354015

Annulus Liouville CFT boundary three-point bootstrap conjecture

摘录 / EXCERPT: Let Q, β, μ, μ_∂, P, q, G, $U_{\rm FZZ}$, η, and H be the quantities in the annulus Liouville CFT bootstrap formula. For $\mu\;\geqslant\; 0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-068088 | MATHDB-354017

Lacoin's mixing-time conjecture for asymmetric exclusion on the circle

摘录 / EXCERPT: Lacoin's conjecture. The mixing time is of order

conjecture | open | MathDB 页面 | 原文线索

V2-068089 | MATHDB-354018

Jara's mixing-time conjecture for particle-conserving systems on the circle

摘录 / EXCERPT: Jara's conjecture. The mixing time is of order

conjecture | open | MathDB 页面 | 原文线索

V2-068090 | MATHDB-354028

Deep Riemann hypothesis for Artin L-functions

摘录 / EXCERPT: Deep Riemann hypothesis. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-068091 | MATHDB-354041

The negativity-locus characterization of compact torus orbit spaces

摘录 / EXCERPT: Let $\mathbb{T}^n$ be a compact torus, let $\mathcal{G}$ act on it, and let $\mathcal{T}$ denote the image of $\mathbb{T}^n$ under the orbit-map parametrization...

conjecture | open | MathDB 页面 | 原文线索

V2-068092 | MATHDB-354051

Conjecture on singular points outside the upper level set for a unit-circle obstacle

摘录 / EXCERPT: Let Ω be a unit circle, and let the upper level set (ULS) be the set where an optimal body's height attains its maximum. Assume that the ULS has nonempty interior. A singula...

conjecture | open | MathDB 页面 | 原文线索

V2-068093 | MATHDB-354053

2-speciality conjecture for quadratic multiplier polynomials

摘录 / EXCERPT: For $m\geq 2$ and $n\geq 1$, let $P_{m,n}$ denote the multiplier polynomial in the quadratic case $d=2$. A polynomial is called 2-special when it has the special coefficient and di...

conjecture | open | MathDB 页面 | 原文线索

V2-068094 | MATHDB-354063

Exact threshold conjecture for adding three-paths

摘录 / EXCERPT: Let G be a graph and let t be a positive integer. Write $G\cup tP_3$ for the disjoint union of G and t copies of P_3, and let $\tau(G)$ be the maximum integer such that...

conjecture | open | MathDB 页面 | 原文线索

V2-068095 | MATHDB-354068

Klivans's uniqueness conjecture for sandpile torsor structures on plane graphs

摘录 / EXCERPT: Let (G,χ) be a plane graph, meaning a planar graph equipped with a plane ribbon structure. A sandpile torsor structure assigns a free transitive action of the sandpile group...

conjecture | open | MathDB 页面 | 原文线索

V2-068096 | MATHDB-354071

The hydrodynamic-limit conjecture for ASEP with time-dependent reservoirs

摘录 / EXCERPT: Let u solve the hydrodynamic equation ... with boundary parameters determined by the reservoir rates $(\alpha,\beta,\gamma,\delta)(t)$ through the relations defining $\nu_\pm(t)$....

conjecture | open | MathDB 页面 | 原文线索

V2-068097 | MATHDB-354073

The Andrews-Uncu conjecture for an asymmetric Rogers-Ramanujan identity

摘录 / EXCERPT: Let $(a;q)_n$ denote the q-Pochhammer symbol, and let $(a_1,a_2;q)_\infty=(a_1;q)_\infty(a_2;q)_\infty$. The variables m,n range over nonnegative integers. The Andrews-Uncu con...

conjecture | open | MathDB 页面 | 原文线索

V2-068098 | MATHDB-354075

Characterisation of minimal hereditary graph classes of unbounded clique-width

摘录 / EXCERPT: Characterisation conjecture. The hereditary graph class $\mathcal{G}^\delta$ is minimal of unbounded clique-width if and only if $\delta \in \Delta_{min}$.

conjecture | open | MathDB 页面 | 原文线索

V2-068099 | MATHDB-354084

The quarter-gap conjecture for random hyperbolic surfaces with cusps

摘录 / EXCERPT: Quarter-gap conjecture. For every $\epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068100 | MATHDB-354088

Balanced max-part conjecture for K_4-free graphs

摘录 / EXCERPT: Let n be even and let G be a K_4-free graph on n vertices. A balanced 2-partition is a partition $V(G)=A\cup B$ with $|A|=|B|=n/2$. Balanced max-part conjecture for K4-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-068101 | MATHDB-354089

Balanced 3-partition conjecture for K_4-free graphs

摘录 / EXCERPT: Let n be divisible by 3 and let G be a K_4-free graph on n vertices. A balanced 3-partition divides $V(G)$ into three classes of size $n/3$; class-edges are edges whose...

conjecture | open | MathDB 页面 | 原文线索

V2-068102 | MATHDB-354095

The full support eigenfunctions conjecture for metric graphs

摘录 / EXCERPT: Full support eigenfunctions conjecture. For any metric graph $(\Gamma,\boldsymbol{\ell})$, and any choice of a complete orthonormal sequence of eigenfunctions, there are infinitely...

conjecture | open | MathDB 页面 | 原文线索

V2-068103 | MATHDB-354098

Colliot-Thélène-Parimala-Suresh Hasse principle conjecture for projective homogeneous spaces

摘录 / EXCERPT: Let k be a p-adic field, and let F be the function field of a smooth projective geometrically integral curve defined over k. Let Ω denote the set of discrete (rank 1...

conjecture | open | MathDB 页面 | 原文线索

V2-068104 | MATHDB-354100

Asymptotic shape conjecture for the contact process in an evolving random environment

摘录 / EXCERPT: Asymptotic shape conjecture. There exists a bounded convex set $U\subset\mathds{R}^d$ such that, for every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068105 | MATHDB-354102

The $(3,1)$-choosability conjecture for planar graphs

摘录 / EXCERPT: A graph is $(3,1)$-choosable if every list assignment in which each list has size at least 3 and the lists of adjacent vertices have intersection of size at most 1 admits a pro...

conjecture | open | MathDB 页面 | 原文线索

V2-068106 | MATHDB-354105

The Hodge-theoretic Gross-Zagier conjecture for Hecke characters

摘录 / EXCERPT: Let F be a quadratic imaginary field, let ϕ be a Hecke character of odd weight w, and let H_{ϕ} be the pure Hodge structure associated with the motive attached to...

conjecture | open | MathDB 页面 | 原文线索

V2-068107 | MATHDB-354106

The classification conjecture for non-orientably fillable torus knots

摘录 / EXCERPT: Classification conjecture. The only non-orientably fillable torus knots are of the form $T(p,2)$ with $p<0$.

conjecture | open | MathDB 页面 | 原文线索

V2-068108 | MATHDB-354109

Rotational symmetry conjecture for entire critical k-Hessian solutions

摘录 / EXCERPT: Rotational symmetry conjecture. Every such solution u on $\mathbb{R}^n$ is rotationally symmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-068109 | MATHDB-354112

Anti-diagonal equivariant Gromov-Witten partition-function correspondence for the resolved conifold

摘录 / EXCERPT: Anti-diagonal equivariant GW correspondence. The identity

conjecture | open | MathDB 页面 | 原文线索

V2-068110 | MATHDB-354113

Shapiro-Tater scaling conjecture for discriminant roots and Painlevé II poles

摘录 / EXCERPT: Let $C_n(t,\lambda)$ be the secular polynomial of the quartic anharmonic oscillator, and let ... be its discriminant with respect to λ. Let $Y_n(t)$ denote the Vorob'ev-Yablo...

conjecture | open | MathDB 页面 | 原文线索

V2-068111 | MATHDB-354119

Šverák's conjecture on non-precompact vorticity orbits

摘录 / EXCERPT: Let $M\subset\mathbb{R}^2$ be a bounded planar domain, let $\omega_0\in L^{\infty}(M)$ be initial vorticity, and let $\{\omega(t)\}_{t\in\mathbb{R}}$ be its inviscid incompressible...

conjecture | open | MathDB 页面 | 原文线索

V2-068112 | MATHDB-354121

The parity conjecture for one-dimensional vorticity models

摘录 / EXCERPT: Parity conjecture. In general, H^s solutions should develop singularities when $s<3/2$; if m is odd, there exist analytic solutions that become singular in finite time; and if...

conjecture | open | MathDB 页面 | 原文线索

V2-068113 | MATHDB-354123

The subcritical Hölder blow-up conjecture for axisymmetric Euler flow

摘录 / EXCERPT: Let $\alpha<1/3$. Subcritical Hölder blow-up conjecture. There exists a compactly supported, axisymmetric, no-swirl local Euler solution with initial regularity C^α_c that b...

conjecture | open | MathDB 页面 | 原文线索

V2-068114 | MATHDB-354124

Radial smoothness conjecture for axisymmetric Euler blow-up

摘录 / EXCERPT: Radial smoothness conjecture. The initial vorticity ω_0 can be C^∞ as a function of ρ. This sharpens the preceding conjecture by requiring the lack of regularity...

conjecture | open | MathDB 页面 | 原文线索

V2-068115 | MATHDB-354125

Artificial-boundary conjecture for axisymmetric Euler blow-up

摘录 / EXCERPT: Consider smooth axisymmetric Euler solutions in the axisymmetric coordinate formulation, with vorticity non-vanishing on the axis of symmetry. Artificial-boundary conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-068116 | MATHDB-354126

Expanding-ring conjecture for the post-singularity set

摘录 / EXCERPT: Expanding-ring conjecture. The singular set of the solutions of the cited self-similar Euler construction propagates within the natural symmetry class as the expanding ring...

conjecture | open | MathDB 页面 | 原文线索

V2-068117 | MATHDB-354132

Error-order conjecture for sample optimal portfolio weights

摘录 / EXCERPT: Error-order conjecture. The expected coordinatewise error satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068118 | MATHDB-354134

Coherence conjecture for affine Springer fiber sheaves

摘录 / EXCERPT: The construction associates to each $gamma\bin g(cK)$ a ${cC}^*$-equivariant quasi-coherent sheaf cF_gamma on $cmathfrak C_{ccheck G}$. Coherence conjecture. For any regular semi...

conjecture | open | MathDB 页面 | 原文线索

V2-068119 | MATHDB-354142

The e-local character-counting conjecture for finite reductive groups

摘录 / EXCERPT: The e-local counting conjecture. For every ℓ-block B of ${\mathbf{G}}^F$ and $d\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068120 | MATHDB-354143

The block-isomorphism character-triple conjecture for finite reductive groups

摘录 / EXCERPT: The block-isomorphism character-triple conjecture. For every ℓ-block B of ${\mathbf{G}}^F$ and $d\geq 0$, there is an ${\rm Aut}_{\mathbb F}({\mathbf{G}}^F)_B$-equivariant b...

conjecture | open | MathDB 页面 | 原文线索

V2-068121 | MATHDB-354145

The weak e-local character-counting conjecture for finite reductive groups

摘录 / EXCERPT: Weak e-local counting conjecture. For every ℓ-block B of ${\mathbf{G}}^F$,

conjecture | open | MathDB 页面 | 原文线索

V2-068122 | MATHDB-354149

Symmetry-breaking conjecture for the two-point variational problem

摘录 / EXCERPT: Let $q_{\mathrm{ic}}=\delta_{-\zeta}+\delta_{\zeta}$ for $\zeta>0$, set $\mathfrak{m}=1$ and $\xi_1=0$, and consider the variational problem in which T and these data are fixed....

conjecture | open | MathDB 页面 | 原文线索

V2-068123 | MATHDB-354157

Padé continued-fraction coefficient matching conjecture

摘录 / EXCERPT: Coefficient matching conjecture. In the limit as m tends to infinity, the continued fraction $g_m(n)$ agrees with $B(n)$; equivalently, their numerator sequences satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-068124 | MATHDB-354158

The unique minimal-b-invariant term conjecture for canonical basis vectors

摘录 / EXCERPT: Fix a multicharge $\mathbf{s}=(s_1,\ldots,s_l)\in\mathbb{Z}^l$ and let λ be a cylindric l-partition. Expand the canonical basis vector $G(\lambda,\mathbf{s})$ in the stan...

conjecture | open | MathDB 页面 | 原文线索

V2-068125 | MATHDB-354159

The minimal-content conjecture for canonical basis vectors in Calogero-Moser families

摘录 / EXCERPT: Let $\mathbf{s}$ be a multicharge of the form ... Let λ be a cylindric l-partition, and let $m(\lambda)$ denote the unique l-partition with minimal b-invariant occurrin...

conjecture | open | MathDB 页面 | 原文线索

V2-068126 | MATHDB-354161

The canonical-basis characterization of Calogero-Moser cellular characters

摘录 / EXCERPT: Let $G(l,1,n)$ be the imprimitive complex reflection group, let c be a parameter satisfying the paper's standing hypothesis, and let $V_{\mathbf{s}}$ be the corresponding Fock-sp...

conjecture | open | MathDB 页面 | 原文线索

V2-068127 | MATHDB-354162

Kalmykov-Karp's Laguerre-Pólya conjecture for hypergeometric functions

摘录 / EXCERPT: Let p and q be integers with $p<q$, let ${\mathbf b}=(b_1,\ldots,b_q)$ satisfy $b_i>0$, and let ${\mathbf a}=(a_1,\ldots,a_p)$ satisfy $a_i\geq b_i$ for $1\leq i\leq p$. The ge...

conjecture | open | MathDB 页面 | 原文线索

V2-068128 | MATHDB-354167

Arveson-Douglas Schatten-class conjecture

摘录 / EXCERPT: Let $I \subset \mathbb{C}[z_1,\ldots,z_d]$ be a homogeneous ideal of infinite co-dimension, let ... and set $I^\bot=H^2_d\ominus I$. Define $S^I=(S_1,\ldots,S_d)$ on $I^\bot$ by ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-068129 | MATHDB-354169

Residue reduction conjecture for generalized amplitudes

摘录 / EXCERPT: Residue reduction conjecture. The $(k-2)(n-k-2)$-dimensional residue of $m^{(k)}_n$ on the subspace where $\eta_J=0$ for every $\eta_J\in\text{base}_{k,n}\cup\text{comb}_{k,n}$ can...

conjecture | open | MathDB 页面 | 原文线索

V2-068130 | MATHDB-354170

Restriction conjecture for truncated-semistable pure sheaves

摘录 / EXCERPT: Restriction conjecture. The restriction $E|_D$ remains ℓ-(semi)stable for $a\gg0$.

conjecture | open | MathDB 页面 | 原文线索

V2-068131 | MATHDB-354175

BHHT levelwise orbifold Euler characteristic mirror symmetry

摘录 / EXCERPT: Let f be an invertible polynomial in n variables, let $G\subset G_f$ be a group of diagonal symmetries, and let $S\subset S_n$ preserve f and G. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068132 | MATHDB-354178

Fixed-weight-matrix conjecture for spurious local minima

摘录 / EXCERPT: Fixed-weight-matrix conjecture. There exists a large constant $\Gamma(\epsilon)>0$ such that every instance $\mathcal{MC}(C,u^*)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-068133 | MATHDB-354183

Connectivity conjecture for rigidity of abstract simplicial complexes

摘录 / EXCERPT: Let d be a positive integer, and let G be the graph of an abstract simplicial $(d+1)$-complex. Connectivity conjecture. If G is $(2d-1)$-connected, then G is rigid in...

conjecture | open | MathDB 页面 | 原文线索

V2-068134 | MATHDB-354187

Existence of finite-field solutions to the Hopf equations

摘录 / EXCERPT: Finite-field Hopf-solution conjecture. The equations have real solutions $\varphi\neq0$.

conjecture | open | MathDB 页面 | 原文线索

V2-068135 | MATHDB-354188

Moss-Ward conjecture on Dold congruences for quadratic recurrence subsequences

摘录 / EXCERPT: Let $P,Q\in\thinspace\mathbb Z$ and let (u_n) be defined by ... with initial conditions $u_0=0$ and $u_1=1$. A sequence satisfies condition (D) when its Dold congruences hold. Moss-...

conjecture | open | MathDB 页面 | 原文线索

V2-068136 | MATHDB-354189

Nonlinear stability conjecture for N-spot ring patterns

摘录 / EXCERPT: Nonlinear stability conjecture. All N-spot ring patterns are nonlinearly stable unless the spot-spot distance is the smallest binding distance.

conjecture | open | MathDB 页面 | 原文线索

V2-068137 | MATHDB-354191

Random-DFA synchronization-time mean-field conjecture

摘录 / EXCERPT: Let $\tau_{\rm sync}$ denote the synchronization time of the random DFA, and retain the notation in which $\mathbf Q$ is the uniform distribution on words of the relevant length. R...

conjecture | open | MathDB 页面 | 原文线索

V2-068138 | MATHDB-354204

Solitons in the defocusing case

摘录 / EXCERPT: Let $sigma=-1$. Consider the stationary nonlinear Schrödinger equation ... where H has lowest eigenvalue $e_0<0$, and let S denote the set of solitons, while M is the mass,...

conjecture | open | MathDB 页面 | 原文线索

V2-068139 | MATHDB-354209

Zabrocki's superspace diagonal coinvariant conjecture

摘录 / EXCERPT: Let $\Omega_n[y_1,\dots,y_n]=\mathbb{C}[x_1,\dots,x_n]\otimes\mathbb{C}[y_1,\dots,y_n]\otimes\bigwedge\{\theta_1,\dots,\theta_n\}$, with S_n acting triply diagonally, and let ... b...

conjecture | open | MathDB 页面 | 原文线索

V2-068140 | MATHDB-354211

Rhoades-Wilson double superspace Vandermonde conjecture

摘录 / EXCERPT: For $1\leq k\leq n$, let $\mathbb{V}_{n,k}$ be the smallest subspace of $\Omega_n[y_1,\dots,y_n]$ containing the superspace Vandermonde $\delta_{n,k}$, closed under...

conjecture | open | MathDB 页面 | 原文线索

V2-068141 | MATHDB-354216

Milnor's conjecture on the rational independence of Hurwitz zeta values

摘录 / EXCERPT: Let q and k be positive integers greater than one. Consider the Hurwitz zeta values $\zeta(k,a/q)$ for integers a satisfying $1\le a\le q$ and $(a,q)=1$. Milnor's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-068142 | MATHDB-354225

Budaghyan et al.'s solution-count conjecture for a finite-field equation

摘录 / EXCERPT: Budaghyan et al.'s conjecture. The equation has q^2 solutions for one value of b, q^2-q solutions for 2^n values of b, and at most 2 solutions for all remaining points...

conjecture | open | MathDB 页面 | 原文线索

V2-068143 | MATHDB-354226

Conjecture on edge-disjoint 1-factors in realizations of degree sequences

摘录 / EXCERPT: Edge-disjoint 1-factor conjecture. Some realization of $\mathrm{\pi}$ has k edge-disjoint 1-factors if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-068144 | MATHDB-354228

Even-arity complete full tree local antimagic chromatic number conjecture

摘录 / EXCERPT: Let T be a complete full t-ary tree, meaning a rooted tree in which every nonleaf has exactly t children and all leaves have the same depth. Let l be the number of leaves a...

conjecture | open | MathDB 页面 | 原文线索

V2-068145 | MATHDB-354239

Unipotent representation and ideal conjecture for real reductive groups

摘录 / EXCERPT: Let $G_{\mathbb R}$ be a real reductive group, let $\widetilde{\mathcal O}$ be a birationally rigid nilpotent cover for its complexification, and let $\mathcal O_{\theta}$ be a K...

conjecture | open | MathDB 页面 | 原文线索

V2-068146 | MATHDB-354243

Convex upper bounds for robustness of positive systems

摘录 / EXCERPT: Convex upper-bound conjecture. For positive systems, there exists a convex upper bound for a robustness measure against such uncertainties.

conjecture | open | MathDB 页面 | 原文线索

V2-068147 | MATHDB-354244

Exactness of the upper robustness bound for some norm-bounded disturbances

摘录 / EXCERPT: Exactness conjecture. $\overline{\nu}$ is exact for some class of norm-bounded disturbances.

conjecture | open | MathDB 页面 | 原文线索

V2-068148 | MATHDB-354247

Equal-size equal-distance minimizer conjecture for the generalized Ohta-Kawasaki model

摘录 / EXCERPT: Let $\omega\ll 1$ and $\gamma\gg 1$, and let $\mathcal{A}_N^{\omega}$ be the class of periodic step functions considered in the generalized Ohta-Kawasaki model. Let U_N^{ω}...

conjecture | open | MathDB 页面 | 原文线索

V2-068149 | MATHDB-354248

The conjectural formula for the top-degree double ramification cycle coefficient

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,2}$ be the moduli space of stable curves with two marked points, let λ_g denote the top Chern class of the Hodge bundle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068150 | MATHDB-354261

Conjecture on the least-degree relations among tautological generators of the moduli space of one-dimensional sheaves

摘录 / EXCERPT: Let $M_{d,\chi}$ be the moduli space under consideration, and let the tautological generators in $A^*(M_{d,\chi})$ be those defined in. A relation among these generators has degree...

conjecture | open | MathDB 页面 | 原文线索

V2-068151 | MATHDB-354266

The conjecture on embedding-independent temperedness

摘录 / EXCERPT: Let G be a reductive group over the global function field K, let $[\Pi]$ be a cuspidal automorphic representation of $G([\mathbf A_K])$ with coefficients in $\bar{\mathbb Q}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-068152 | MATHDB-354274

Three-colorability conjecture for graphs with prescribed girth and no long odd holes

摘录 / EXCERPT: Three-colorability conjecture. Every graph G in any of these families satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068153 | MATHDB-354276

The Landau-Coulomb L^∞ growth conjecture

摘录 / EXCERPT: Let $f:[0,T]\times\mathbb R^3\to[0,\infty)$ be a classical solution of the space-homogeneous Landau equation with the Landau-Coulomb operator Q, and let $f(0,v)=f_0(v)$. The Land...

conjecture | open | MathDB 页面 | 原文线索

V2-068154 | MATHDB-354283

Conjecture on piecewise linear Seshadri functions for non-simple abelian surfaces

摘录 / EXCERPT: Piecewise linearity conjecture. The Seshadri function of A is piecewise linear precisely in these cases.

conjecture | open | MathDB 页面 | 原文线索

V2-068155 | MATHDB-354290

Conjecture on minimum-degree construction time in the semi-random graph process

摘录 / EXCERPT: Minimum-degree construction-time conjecture. The conclusion of Theorem holds for $k=o(n)$.

conjecture | open | MathDB 页面 | 原文线索

V2-068156 | MATHDB-354291

Linear lower-bound conjecture for factors of non-central graphs

摘录 / EXCERPT: Non-central-factor conjecture. Let H be a fixed non-central graph. Then, for every strategy $\sigma\in\operatorname{\mathcal{S}}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-068157 | MATHDB-354300

The weak epsilon dichotomy conjecture for the GL4 times GL2 model

摘录 / EXCERPT: Let $G=\operatorname{GL}_4\times\operatorname{GL}_2$, let $H=\operatorname{GL}_2\times\operatorname{GL}_2$ be embedded by $(a,b)\mapsto(\operatorname{diag}(a,b),b)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068158 | MATHDB-354301

The geometric multiplicity formula for the GL6 model

摘录 / EXCERPT: Let $G=\operatorname{GL}_6$ with the model $H=\operatorname{GL}_2\ltimes U$, let $\rho_X=\wedge^3$, and let π be an irreducible tempered representation with central character t...

conjecture | open | MathDB 页面 | 原文线索

V2-068159 | MATHDB-354302

The generalized epsilon dichotomy conjecture for the GL6 model

摘录 / EXCERPT: Let $G=\operatorname{GL}_6$, $H=\operatorname{GL}_2\ltimes U$, and let χ' be defined by $\chi'(\operatorname{diag}(h,h,h)u)=\alpha(\det h)\xi(u)$. Let π be irreducible tem...

conjecture | open | MathDB 页面 | 原文线索

V2-068160 | MATHDB-354303

The weak epsilon dichotomy conjecture for the GSp4 times GL2 times GL2 model

摘录 / EXCERPT: Let $G=\operatorname{GSp}_4\times\operatorname{GL}_2\times\operatorname{GL}_2$ with $H=(\operatorname{GL}_2\times\operatorname{GL}_2)^0$, let (G_D,H_D) be its quaternionic inner...

conjecture | open | MathDB 页面 | 原文线索

V2-068161 | MATHDB-354304

The weak epsilon dichotomy conjecture for smaller GSpin and GHSpin models

摘录 / EXCERPT: Let (G,H) be one of the smaller models $(\operatorname{GSpin}_{10}\times\operatorname{GSpin}_3,\operatorname{GSpin}_3\ltimes U)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-068162 | MATHDB-354310

Classical pure local cabled braid groups conjecture

摘录 / EXCERPT: Classical pure local cabled braid groups conjecture. There exists such an operad $\ms P$ whose evaluation on every generalised fission tree recovers the corresponding group:

conjecture | open | MathDB 页面 | 原文线索

V2-068163 | MATHDB-354316

One-weight Griesmer code conjecture for repeated simplicial defining sets

摘录 / EXCERPT: Let $\mathbb{F}_{2^n}=\mathbb{F}_{2}(\eta)$, let $m\in\mathbb{N}$, and let $\emptyset\neq M\subsetneq[m]$. Set ... Let $C_{D^{\ast}}$ be the associated punctured simplicial code, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068164 | MATHDB-354317

Complement-code parameters conjecture over fields of characteristic 2

摘录 / EXCERPT: Let $\mathbb{F}_{2^n}=\mathbb{F}_{2}(\eta)$ with $\eta\in\mathbb{F}_{2^n}$, and let ... where $m\in\mathbb{N}$, $\emptyset\neq M_i\subsetneq[m]$ for $1\leq i\leq n$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068165 | MATHDB-354318

Parameters of subfield complement codes over fields of characteristic 2

摘录 / EXCERPT: Let $D=\Delta_{M_1}+\eta\Delta_{M_2}+\cdots+\eta^{n-1}\Delta_{M_n}\subset\mathbb{F}_{2^n}^m$ be as above, with the hypotheses stated in the source, and let $C^{(2)}_{D^c}$ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-068166 | MATHDB-354319

Asymptotic optimality of the alternative payoff for parallel fingerprinting channels

摘录 / EXCERPT: Let L be the number of TV channels, let k be the coalition size, and let $\tilde{L}=1$. Write $J_{k,L}$ for the original payoff and define the alternative payoff ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-068167 | MATHDB-354320

Asymptotic binary fingerprinting capacity for multiple TV channels

摘录 / EXCERPT: Let k be the coalition size, let L be the number of parallel TV channels, and let $C_{\rm fp}^{\rm binary}(k,L)$ denote the binary fingerprinting capacity in this setting. Assu...

conjecture | open | MathDB 页面 | 原文线索

V2-068168 | MATHDB-354325

The $SL(n,\mathbb R)/SL(m,\mathbb R)$ cocompact-discontinuous-group conjecture

摘录 / EXCERPT: Kobayashi's special-case conjecture. This homogeneous space does not admit a cocompact discontinuous group. This is presented as a special case of the compact-standard-quotient con...

conjecture | open | MathDB 页面 | 原文线索

V2-068169 | MATHDB-354334

Gross's explicit reciprocity conjecture for Brumer-Stark units

摘录 / EXCERPT: Let H/F be as in the Brumer-Stark setting, let L/F be a finite abelian CM extension containing H, and write $\mathfrak{g}=\operatorname{Gal}(L/F)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068170 | MATHDB-354335

Atsuta-Kataoka's shifted Fitting ideal conjecture for class groups

摘录 / EXCERPT: Let $R=\mathbf{Z}[1/2][G]^-$, let $\operatorname{Cl}^T(H)^-$ be the minus part of the T-smoothed class group, and for each ramified place v let $A_v=\mathbf{Z}[G/I_v]/(g_v)$. W...

conjecture | open | MathDB 页面 | 原文线索

V2-068171 | MATHDB-354340

The orbit-interval conjecture for automorphisms of Bruhat graphs

摘录 / EXCERPT: Let $w \in \mathfrak{S}_n$ and let $\mathcal{O}=\\{\varphi(e) \mid \varphi \in \operatorname{Aut}(\Gamma(e,w))\\}$ be the orbit of the identity under graph automorphisms of...

conjecture | open | MathDB 页面 | 原文线索

V2-068172 | MATHDB-354341

The general Coxeter-group conjecture for special matchings and Bruhat graph automorphisms

摘录 / EXCERPT: Let W be a Coxeter group and let $u \leq v$ be elements of W. A perfect matching of the Hasse diagram of $[u,v]$ is a special matching if and only if it is an automorphism of t...

conjecture | open | MathDB 页面 | 原文线索

V2-068173 | MATHDB-354342

The strengthened orbit-interval conjecture for Bruhat-irreducible permutations

摘录 / EXCERPT: Strengthened orbit-interval conjecture. Then $\mathcal{O}=[e,v(w)]$. This is presented as a strengthening of the orbit-interval conjecture; its status is not resolved in the suppli...

conjecture | open | MathDB 页面 | 原文线索

V2-068174 | MATHDB-354343

The butterfly weakening conjecture for Coxeter-group Bruhat intervals

摘录 / EXCERPT: Let W be any Coxeter group, let $u \leq v \in W$, and suppose that $x_1,x_2,y_1,y_2 \in [u,v]$ form a butterfly, meaning $x_1 \lessdot y_1,y_2$ and $x_2 \lessdot y_1,y_2$. Butter...

conjecture | open | MathDB 页面 | 原文线索

V2-068175 | MATHDB-354349

Zariski Dense Orbit conjecture for rational self-maps

摘录 / EXCERPT: Zariski Dense Orbit conjecture. If there is no non-constant rational function g on X satisfying $g\circ f=g$, then $\mathcal{Z}(f)\neq\emptyset$.

conjecture | open | MathDB 页面 | 原文线索

V2-068176 | MATHDB-354353

Fulkerson's six-perfect-matching conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A Fulkerson cover is a list of six perfect matchings of G in which every edge is contained in exactly two matchings. Fulkerson's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-068177 | MATHDB-354354

Berge's five-perfect-matching cover conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. A collection of perfect matchings is an edge-covering collection when every edge of G belongs to at least one matching in the collection. Ber...

conjecture | open | MathDB 页面 | 原文线索

V2-068178 | MATHDB-354355

The punctured-disk zero-free conjecture for series-parallel graphs

摘录 / EXCERPT: Let $B_{5/27}(1)$ denote the disk of radius $5/27$ centered at 1, and let $Z(G;q)$ be the chromatic polynomial of a series-parallel graph G. The punctured-disk zero-free conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-068179 | MATHDB-354356

The degree-three threshold conjecture for leaf joined trees

摘录 / EXCERPT: Let Δ_0 be the threshold parameter in Theorem, concerning the accumulation of chromatic zeros of leaf joined trees relative to the degree bound Δ. The degree-three th...

conjecture | open | MathDB 页面 | 原文线索

V2-068180 | MATHDB-354364

Nonexistence conjecture for infinite fields with indecomposable multiplicative groups

摘录 / EXCERPT: Let F be a field, and write $F^\times$ for its multiplicative group. A group is indecomposable if it cannot be expressed as a nontrivial direct product of groups. Nonexistence co...

conjecture | open | MathDB 页面 | 原文线索

V2-068181 | MATHDB-354369

The positivity and real-rootedness conjecture for increasing-subsequence character polynomials

摘录 / EXCERPT: Positivity and real-rootedness conjecture. For all $k,n\in\mathbb{N}$, $M_{\textnormal{id}_k,n}$ is a nonnegative linear combination of irreducible symmetric group characters. Furt...

conjecture | open | MathDB 页面 | 原文线索

V2-068182 | MATHDB-354370

The single-row root-at-minus-one conjecture for permutation pattern character polynomials

摘录 / EXCERPT: Single-row root-at-minus-one conjecture. When λ has a single row, the polynomials $a_\sigma^\lambda(n)$ have $n=-1$ as a root.

conjecture | open | MathDB 页面 | 原文线索

V2-068183 | MATHDB-354372

Classification conjecture for invariant lattices of pairs of ternary quadratic forms

摘录 / EXCERPT: Invariant-lattice classification conjecture. Every primitive, $\operatorname{GL}_3(\mathcal{O}_K)$-invariant lattice L in $\mathcal{V}(\mathcal{O}_K)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-068184 | MATHDB-354374

Ohno-Nakagawa duality conjecture for symmetric $2\times n\times n$ boxes

摘录 / EXCERPT: Local Ohno-Nakagawa box conjecture. The lattices $\mathcal V_{\tau,\tau^{n-1}f}$ and $\mathcal V_{2\tau^{-1},(2\tau^{-1})^{n-1}f}$ are naturally dual with duality constant...

conjecture | open | MathDB 页面 | 原文线索

V2-068185 | MATHDB-354375

Hall's conjecture on integral points of Mordell curves

摘录 / EXCERPT: Hall's conjecture. Given $\epsilon>0$, there exists a positive constant C_ϵ such that, for every integral point P on E_B,

conjecture | open | MathDB 页面 | 原文线索

V2-068186 | MATHDB-354379

Continuity and Radon-Nikodym conjecture for Nevanlinna counting functions on the Shilov boundary

摘录 / EXCERPT: Continuity and Radon-Nikodym conjecture. The values $\tau_\varphi(z_\alpha)$ converge for every such net, so this limit defines τ_φ on $\mathfrak{M}_{L^{\infty}}$. Moreo...

conjecture | open | MathDB 页面 | 原文线索

V2-068187 | MATHDB-354381

Trivial-idempotent conjecture for integral quandle rings of semi-latin quandles

摘录 / EXCERPT: Let X be a quandle and let $\mathbb{Z}[X]$ denote its integral quandle ring. A quandle is semi-latin when every right multiplication restricted to each orbit is injective; a fini...

conjecture | open | MathDB 页面 | 原文线索

V2-068188 | MATHDB-354384

Finiteness conjecture for two-generator arithmetic subgroups of the hyperbolic isometry group

摘录 / EXCERPT: Let $\operatorname{PSL}(2,\mathbb{C})$ be the group under consideration, and call a subgroup arithmetic when it is commensurable with an arithmetic subgroup. Finiteness conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-068189 | MATHDB-354390

Fallat-Gekhtman-Johnson conjecture on factorization of bounded ratios of minors

摘录 / EXCERPT: Let A be a totally positive $n\times n$ matrix. For index sequences α and β defining products of minors, write α/β for the corresponding ratio. A ratio is...

conjecture | open | MathDB 页面 | 原文线索

V2-068190 | MATHDB-354396

Cowan's conjecture on the average Mordell-Weil rank of elliptic curves over the rational function field

摘录 / EXCERPT: Cowan's conjecture. For every pair of positive integers m,n,

conjecture | open | MathDB 页面 | 原文线索

V2-068191 | MATHDB-354397

Inclusion conjecture for NN and BD generalized doubly nonnegative cones

摘录 / EXCERPT: Inclusion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068192 | MATHDB-354403

Super-expander conjecture for congruence quotients of \(\mathrm{SL}_3(\mathbf{Z})\)

摘录 / EXCERPT: Let $\mathcal{G}_n$ be the Cayley graph of $\mathrm{SL}_3(\mathbf{Z}/n\mathbf{Z})$ with respect to the images of the elementary matrices. A sequence of bounded-degree graphs is a s...

conjecture | open | MathDB 页面 | 原文线索

V2-068193 | MATHDB-354405

The equal-order correspondence conjecture for polycyclic codes associated with trinomials

摘录 / EXCERPT: Let $t_1(x)=x^n-ax^i-b$ and $t_2(x)=x^n-a'x^{n-i}-b'$ satisfy ... Let S_1 be the set of all polycyclic codes of length n over $\mathbb{F}_q$ associated with $t_1(x)$, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068194 | MATHDB-354406

The binary characterization conjecture for isodual polycyclic codes

摘录 / EXCERPT: Binary isoduality characterization conjecture. The code C is isodual if and only if

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068195 | MATHDB-354409

Heckman-Thomas fractional colorability conjecture for planar subcubic triangle-free graphs

摘录 / EXCERPT: Heckman-Thomas planar conjecture. Every subcubic triangle-free planar graph is fractionally $8/3$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-068196 | MATHDB-354416

Isomorphism and equivariance of the Whittaker-space embedding

摘录 / EXCERPT: Whittaker-space isomorphism conjecture. For every W-orbit $\mathcal{O}\subset\mathscr{X}_{Q,n}$, the embedding is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-068197 | MATHDB-354423

The McKinsey axiom is not d-persistent

摘录 / EXCERPT: Let p be a proposition letter, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068198 | MATHDB-354437

Odd-weight evaluation conjecture for multiple t-values

摘录 / EXCERPT: Let $t^{\sfrac{1\!}{2}}(2,\{1\}^{2n+1},2)$ denote the multiple t-value with the indicated indices, and let $t(k)$ denote the ordinary t-value of weight k. Odd-weight evaluati...

conjecture | open | MathDB 页面 | 原文线索

V2-068199 | MATHDB-354441

Luo, Tian and Wu's connectivity-preserving tree conjecture for bipartite graphs

摘录 / EXCERPT: Let T be a tree with bipartition X and Y, and set ... Let G be a k-connected bipartite graph with minimum degree $\delta(G)\geq k+t$. Luo, Tian and Wu's conjecture. The gra...

conjecture | open | MathDB 页面 | 原文线索

V2-068200 | MATHDB-354459

Fan-Tringali's sets-of-lengths conjecture for power monoids of numerical monoids

摘录 / EXCERPT: Let S be a numerical monoid, and let $\mathcal P_{\mathrm{fin}}(S)$ denote its power monoid of finite nonempty subsets. For $A\in\mathcal P_{\mathrm{fin}}(S)$, let $\mathsf L(A)$...

conjecture | open | MathDB 页面 | 原文线索

V2-068201 | MATHDB-354460

The isomorphism conjecture for power monoids of numerical monoids

摘录 / EXCERPT: Let S and S' be numerical monoids. For a numerical monoid T, let $\mathcal P_{\mathrm{fin},0}(T)$ denote the power monoid of finite subsets of T containing 0. Isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-068202 | MATHDB-354473

Hibi-Tsuchiya's conjecture on the h-vector of cycle graph stable set Ehrhart rings

摘录 / EXCERPT: Let G be a cycle graph with $\ell\geq 3$, let $R=E_{\mathbb{K}}[\operatorname{STAB}(G)]$ be its Ehrhart ring, and write its h-vector as $(h_0,h_1,\ldots,h_s)$ with $h_s\neq 0$. S...

conjecture | open | MathDB 页面 | 原文线索

V2-068203 | MATHDB-354477

The iterated Weak Lefschetz Property conjecture for quadratic complete intersections

摘录 / EXCERPT: Iterated Weak Lefschetz Property conjecture. For any linear forms $\ell_1,\dots,\ell_i$ in R, the quotient R_i enjoys the Weak Lefschetz Property.

conjecture | open | MathDB 页面 | 原文线索

V2-068204 | MATHDB-354482

Frontal version of Mond's conjecture

摘录 / EXCERPT: Let $f\colon (\mathbb{C}^n,S) \to (\mathbb{C}^{n+1},0)$ be an $\mathcal{F}$-finite frontal map, and let $\mu_{\mathcal{F}}(f)$ denote its frontal Milnor number and...

conjecture | open | MathDB 页面 | 原文线索

V2-068205 | MATHDB-354483

The integrability converse for Lie algebroid foliations

摘录 / EXCERPT: Integrability converse. The Lie algebroid $\mathfrak{g}\to X$ is integrable if and only if it is equivalent, as an LA-groupoid, to a simple foliation.

conjecture | open | MathDB 页面 | 原文线索

V2-068206 | MATHDB-354490

The sums-of-squares conjecture for Gaussian product polynomials

摘录 / EXCERPT: Let $n\ge 3$, let $m_1,\dots,m_n\in\mathbb{N}$, and let $U_1,\dots,U_n$ be independent standard Gaussian random variables. Define the polynomial H on $\mathbb{R}^{n(n-1)/2}$ by .....

conjecture | open | MathDB 页面 | 原文线索

V2-068207 | MATHDB-354497

Fuss-Catalan enumeration and construction of regions for \(A^{\phi^p}_n\)

摘录 / EXCERPT: Place $n-2$ points labeled $0,1,\ldots,n-3$ in increasing order on the real line, and let $H(x)$ be the function in the global Schwinger formula for $A^{\phi^p}_n$. For a non-cross...

conjecture | open | MathDB 页面 | 原文线索

V2-068208 | MATHDB-354499

Graham's log-concavity conjecture for order-preserving maps

摘录 / EXCERPT: Let $P=(X,\prec)$ be a finite poset, let $x\in X$, and let $a,t\in\mathbb N_{\geq 1}$ with $1<a<t$. Write $\Omega(P,t;x,a)$ for the number of order-preserving maps $g:X\to[t]$ sati...

conjecture | open | MathDB 页面 | 原文线索

V2-068209 | MATHDB-354500

Matching conjecture for the Stanley inequality graph

摘录 / EXCERPT: Let $\mathcal G=(V\sqcup W,E)$ be the bipartite graph defined above, with the parts and edges constructed from the relevant sets of linear extensions. Matching conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-068210 | MATHDB-354510

Gross-Mansour-Tucker conjecture on partial duality polynomials of orientable ribbon graphs

摘录 / EXCERPT: A ribbon graph is orientable if its associated ribbon surface is orientable, and its partial duality polynomial is the generating function that enumerates its partial duals by Eule...

conjecture | open | MathDB 页面 | 原文线索

V2-068211 | MATHDB-354514

Periodic eigenvalue conjecture for post-critically finite endomorphisms of projective space

摘录 / EXCERPT: Let $f \colon {\mathbb{CP}}^n \to {\mathbb{CP}}^n$, with $n \ge 2$, be a post-critically finite endomorphism of degree $d \ge 2$, and let λ be an eigenvalue of f along a...

conjecture | open | MathDB 页面 | 原文线索

V2-068212 | MATHDB-354515

Analytic-boundary alternative for nice harmonic-function quotients

摘录 / EXCERPT: Let Ω be a one-sided NTA domain, and let u and v be positive harmonic functions as above, vanishing on the relevant portion of the boundary, such that their quotient agr...

conjecture | open | MathDB 页面 | 原文线索

V2-068213 | MATHDB-354522

Real-rootedness and Eulerian interlacing conjecture for geometric lattices

摘录 / EXCERPT: Real-rootedness and Eulerian interlacing conjecture. The polynomial $h_{\mathcal L}(x)$ has only real roots and is interlaced by the Eulerian polynomial $A_n(x)$ for every geometri...

conjecture | open | MathDB 页面 | 原文线索

V2-068214 | MATHDB-354525

Compatibility of cyclic base change with semisimple local Langlands parameters

摘录 / EXCERPT: Let F' / F be an extension, let $F”/F'$ be cyclic, and let $BC_{F”/F'}$ be the conjectural cyclic base-change map on representations of $G(F')$. Let ${\mathcal{L}}^{ss}_{F'}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068215 | MATHDB-354526

The conjecture on pure incorrigible supercuspidal representations

摘录 / EXCERPT: Let F be a local field, let G be a reductive group over F, and call a supercuspidal representation pure incorrigible when it is pure and remains supercuspidal in the base-cha...

conjecture | open | MathDB 页面 | 原文线索

V2-068216 | MATHDB-354527

The conjecture on cyclic base change eliminating supercuspidal representations

摘录 / EXCERPT: Let F_0 be a local field, let G be a reductive group over F_0, let $\pi_0\in\Pi(G/F_0)$ be any supercuspidal representation, and let $\Pi_0\subset\Pi(G/F_0)$ be the packet co...

conjecture | open | MathDB 页面 | 原文线索

V2-068217 | MATHDB-354533

The spectral transition conjecture for the extended Harper model

摘录 / EXCERPT: Spectral transition conjecture. For $\lambda\in\mathcal{R}_1\cap\mathcal{R}_s$, the operator $H^{\mathrm{EHM}}_{\lambda,\alpha,\theta}$ has purely singular continuous spectrum if

conjecture | open | MathDB 页面 | 原文线索

V2-068218 | MATHDB-354540

Seven-colour conjecture for odd colouring of toroidal graphs

摘录 / EXCERPT: Seven-colour conjecture for toroidal graphs. Every toroidal graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068219 | MATHDB-354542

Kopczyński's finite-union conjecture for half-positional winning conditions

摘录 / EXCERPT: Kopczyński's conjecture. Prefix-independent half-positional winning conditions are closed under finite unions.

conjecture | open | MathDB 页面 | 原文线索

V2-068220 | MATHDB-354543

ETOG representation conjecture for bi-positional winning conditions

摘录 / EXCERPT: ETOG representation conjecture. Every bi-positional prefix-independent winning condition coincides on periodic sequences with some energy condition over a totally ordered group.

conjecture | open | MathDB 页面 | 原文线索

V2-068221 | MATHDB-354544

Invariant-subsemigroup separation conjecture for free groups

摘录 / EXCERPT: Invariant-subsemigroup separation conjecture. For every $P\subseteq C^+$ such that P and $C^+\setminus P$ are closed under concatenations and cyclic shifts, there exists an invar...

conjecture | open | MathDB 页面 | 原文线索

V2-068222 | MATHDB-354552

Strong Laplace Hopf-lemma conjecture for ordered graph functions

摘录 / EXCERPT: Strong Laplace Hopf-lemma conjecture. If, in addition, $u_t(0,0)=0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068223 | MATHDB-354553

Finite-order vanishing Laplace Hopf-lemma conjecture

摘录 / EXCERPT: Let u and v satisfy the hypotheses of the preceding strong Laplace Hopf-lemma conjecture, including $u_t(0,0)=0$. Finite-order vanishing conjecture. If, in addition, $u(t,0)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-068224 | MATHDB-354554

The converse higher Du Bois-rational singularities conjecture

摘录 / EXCERPT: Higher Du Bois implies lower rational conjecture. If X has lci singularities and X is k-Du Bois, then X is $(k-1)$-rational.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068225 | MATHDB-354556

Kapouleas-Li conjecture for antiprismatic free boundary minimal surfaces

摘录 / EXCERPT: Let $g\geq2$ be an integer. A free boundary minimal surface in $\mathbb{B}^3$ is a minimal surface whose boundary lies on and meets $\partial\mathbb{B}^3$ orthogonally. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068226 | MATHDB-354558

Folha-Pacard-Zolotareva conjecture for prismatic free boundary minimal surfaces

摘录 / EXCERPT: Let $n\geq2$ be an integer. A free boundary minimal surface in $\mathbb{B}^3$ is a minimal surface whose boundary lies on and meets $\partial\mathbb{B}^3$ orthogonally. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068227 | MATHDB-354559

Maximal-area conjecture for genus-zero free boundary minimal surfaces

摘录 / EXCERPT: Let $b\geq3$ be an integer. A free boundary minimal surface in $\mathbb{B}^3$ is a minimal surface whose boundary lies on and meets $\partial\mathbb{B}^3$ orthogonally. Maximal-are...

conjecture | open | MathDB 页面 | 原文线索

V2-068228 | MATHDB-354561

Antipodal rowmotion homomesy conjecture for $[a] \times [2] \times [2]$

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let $\mathcal{A}^2(P)$ denote the set of P-partitions of height at most 2, and let χ_S be the indicator statistic of S. For antipodal elements...

conjecture | open | MathDB 页面 | 原文线索

V2-068229 | MATHDB-354562

Translated promotion homomesy conjecture for $[a] \times [2] \times [2]$

摘录 / EXCERPT: Let $\mathcal{L}_{P \times [\ell]}(R^q)$ denote the set of labelings subject to the indicated restriction function, let $\xi(f,(x_1,x_2),b)$ be the statistic defined by ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068230 | MATHDB-354563

Translated triple promotion homomesy conjecture for $[2] \times [2] \times [2]$

摘录 / EXCERPT: Let $\mathcal{L}_{P\times[\ell]}(R^q)$ denote the set of labelings with the stated restriction function, and let ξ be the statistic defined in the preceding formulation. Transl...

conjecture | open | MathDB 页面 | 原文线索

V2-068231 | MATHDB-354564

Hopkins's rowmotion order conjecture for the V poset

摘录 / EXCERPT: Let $V=\{a,b,c\}$ be the three-element poset with $a\lessdot b$ and $a\lessdot c$, let $\mathcal{A}^{\ell}(P)$ denote the set of P-partitions of height ℓ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068232 | MATHDB-354565

Translated V-strict promotion order conjecture

摘录 / EXCERPT: Let $V=\{a,b,c\}$ be the three-element poset with $a\lessdot b$ and $a\lessdot c$, let $\mathcal{L}_{V\times[\ell]}(R^q)$ denote the set of V-strict labelings with restriction fu...

conjecture | open | MathDB 页面 | 原文线索

V2-068233 | MATHDB-354566

Translated promotion order conjecture for the dual V poset

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$, let β be the restriction function on $[2]\times[b]$ defined by ... and let $\mathcal{L}_{([2]\times[b])\times[\ell]}(R^{\beta})$ denote the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-068234 | MATHDB-354567

Existence of smooth Mallows processes with non-total graph structure

摘录 / EXCERPT: Let $n\geq 3$. A smooth Mallows process is a process ... for which the associated graph group $\langle{\mathcal{G}}_{\mathcal{M}}\rangle$ is defined, and let $\mathcal{T}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-068235 | MATHDB-354568

Non-Markovian jumping process conjecture for the birth Mallows process

摘录 / EXCERPT: Let $n\geq 4$, and let $({\mathcal{M}}_t)_{t\in[0,\infty)}$ be the birth Mallows process. If T_k are its jump times, its jumping process is...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068236 | MATHDB-354569

Markov Mallows processes with Markovian jumping chains

摘录 / EXCERPT: Let $n\geq 4$. A smooth Mallows process is a process ... , with jumping process ... Let $\mathcal{T}^a$ and $\mathcal{T}$ denote the graph groups appearing in the source. Markovian-jum...

conjecture | open | MathDB 页面 | 原文线索

V2-068237 | MATHDB-354570

Gaussian Markov-process limit for the birth Mallows jump count

摘录 / EXCERPT: For the birth Mallows process, let $J_t=|\{k\leq\binom{n}{2}:T_k\leq t\}|$ be the number of jumps by time t. Gaussian Markov-limit conjecture. As n tends to infinity, ... where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068238 | MATHDB-354572

Monotonicity Conjecture for orbit portraits of cubic polynomials

摘录 / EXCERPT: Monotonicity Conjecture. As we cross any primary ray of co-period q into the face which contains $H_\mathfrak{p}$, the period q orbit portrait is replaced by a strictly larger...

conjecture | open | MathDB 页面 | 原文线索

V2-068239 | MATHDB-354603

Cross Gramian coherence lower-bound conjecture for dual frames

摘录 / EXCERPT: Let F be a frame for $\mathbb{F}^n$, and let G be one of its dual frames. Write $Gr(F,G)$ for their cross Gramian and let $\mu(Gr(F,G))$ denote its maximum off-diagonal magnitu...

conjecture | open | MathDB 页面 | 原文线索

V2-068240 | MATHDB-354604

Exclusive Grassmannian pair canonical-dual conjecture

摘录 / EXCERPT: Let F be a frame for $\mathbb{F}^n$. A dual of F forms an exclusive Grassmannian pair with F if it is the only dual frame satisfying the paper's Grassmannian coherence minimi...

conjecture | open | MathDB 页面 | 原文线索

V2-068241 | MATHDB-354605

Co-equidistribution characterization of exclusive Grassmannian pairs

摘录 / EXCERPT: Let F be a frame for $\mathbb{F}^n$, let $\tilde{F}$ be a dual frame, and let $(F,\tilde{F})$ be the associated frame pair. Co-equidistribution characterization conjecture. The f...

conjecture | open | MathDB 页面 | 原文线索

V2-068242 | MATHDB-354607

The KPZ sheet-to-Airy-sheet conjecture

摘录 / EXCERPT: Let $\mathcal{Z}(s,x;t,y)$ denote the continuum directed random polymer partition function, and define ... Let $\mathcal{S}(x,y)$ be the parabolic Airy sheet. KPZ sheet-to-Airy-sheet...

conjecture | open | MathDB 页面 | 原文线索

V2-068243 | MATHDB-354614

The degree-four Fekete-set conjecture for real simplices

摘录 / EXCERPT: Let S_d be the real d-simplex, let F_4 be the set of $N_4=\binom{d+4}{4}$ points constructed in the paper, and let ℓ_i be its fundamental Lagrange polynomials. The degre...

conjecture | open | MathDB 页面 | 原文线索

V2-068244 | MATHDB-354615

The degree-four Lagrange-ℓ^4 bound conjecture for real simplices

摘录 / EXCERPT: Let S_d be the real d-simplex, let F_4 be the set of $N=\binom{d+4}{4}$ points constructed in the paper, and let ℓ_j be the associated fundamental Lagrange polynomials....

conjecture | open | MathDB 页面 | 原文线索

V2-068245 | MATHDB-354621

Twisted Gan-Gross-Prasad conjecture for biquadratic unitary groups

摘录 / EXCERPT: Let F be a non-Archimedean local field of characteristic 0, let $E\neq K$ be distinct quadratic extensions of F, and put $L=E\otimes_F K$. Let V be an n-dimensional skew-...

conjecture | open | MathDB 页面 | 原文线索

V2-068246 | MATHDB-354622

The unchanged-symbol access conjecture for stable MDS convertible codes

摘录 / EXCERPT: Let λ^{F}, k^{F}, and $k^{I}=\lambda^{F}k^{F}$ describe the dimensions of a stable MDS $(n^{I},k^{I};n^{F},k^{F})$ convertible code, and let β_1 and β_2 denot...

conjecture | open | MathDB 页面 | 原文线索

V2-068247 | MATHDB-354623

Eliahou-Villarreal conjecture on projective dimension of edge ideals

摘录 / EXCERPT: Eliahou-Villarreal conjecture. If the edge ideal of a connected graph G has a 2-linear resolution, then

conjecture | open | MathDB 页面 | 原文线索

V2-068248 | MATHDB-354624

The total coloring conjecture in fractional-power form

摘录 / EXCERPT: Let G be a simple graph. The graph $G^{\frac{2}{2}}$ is the square of the subdivision of G obtained by replacing every edge by a path of length two, and χ and ω den...

conjecture | open | MathDB 页面 | 原文线索

V2-068249 | MATHDB-354625

The comparison conjecture for the $2/2$- and $3/3$-powers

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta(G)\geq 2$. For positive integers m,n, $G^{\frac{m}{n}}$ denotes the m-power of the n-subdivision of G; in particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-068250 | MATHDB-354629

Conrey-Farmer-Keating-Rubinstein-Snaith shifted-moment conjecture for elliptic-curve twists

摘录 / EXCERPT: Let $k,\beta\in\mathbb{N}$ and let $\underline{\theta}=(\theta_1,\dots,\theta_k)\in\mathbb{R}^k$. For the even-sign quadratic twists of an elliptic curve $E/\mathbb{Q}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-068251 | MATHDB-354631

The critical-exponent asymptotic conjecture for self-avoiding walks

摘录 / EXCERPT: The critical-exponent asymptotic conjecture. For every $d\ge2$, such a critical exponent exists, and

conjecture | open | MathDB 页面 | 原文线索

V2-068252 | MATHDB-354656

The maximum-average-degree conjecture for 2-distance coloring

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta(G)=4$ and maximum average degree $\operatorname{mad}(G)<4$. The maximum-average-degree conjecture. asserts ... Here $\chi^2(G)$ is the 2-...

conjecture | open | MathDB 页面 | 原文线索

V2-068253 | MATHDB-354657

The exact square coloring conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph with maximum degree Δ. The exact square coloring conjecture. asserts ... Here $\chi^{\#2}(G)$ is the exact square chromatic number, where conflicts occ...

conjecture | open | MathDB 页面 | 原文线索

V2-068254 | MATHDB-354658

EGD statistical-radius conjecture in the middle SNR regime

摘录 / EXCERPT: EGD statistical-radius conjecture. There exist universal constants C_{3},C_{4},C_{5} such that, whenever $n\geq C_{3}(d\log(d/\delta))^{4}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068255 | MATHDB-354660

Felsner-Hurtado-Noy-Streinu conjecture for simple great-circle arrangements

摘录 / EXCERPT: Felsner-Hurtado-Noy-Streinu conjecture. The arrangement graph of every simple arrangement of great-circles on the sphere is 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-068256 | MATHDB-354661

Infinite-family conjecture for 4-edge-critical pseudocircle arrangement graphs

摘录 / EXCERPT: Infinite-family conjecture. There exists an infinite family of simple arrangement graphs of 4-edge-critical arrangements of pseudocircles.

conjecture | open | MathDB 页面 | 原文线索

V2-068257 | MATHDB-354662

Strengthened colorability and antipodal-coloring conjectures for pseudocircle arrangements

摘录 / EXCERPT: Strengthened pseudocircle-arrangement conjectures. The following three statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-068258 | MATHDB-354682

Scattering below the ground state for the Zakharov system

摘录 / EXCERPT: Let $(f,g)\in H^1\times L^2$ satisfy ... Here $\mathcal{E}_Z$ denotes the Zakharov energy and Q is the ground state. Scattering below the ground state conjecture. There exists a un...

conjecture | open | MathDB 页面 | 原文线索

V2-068259 | MATHDB-354685

Equality of the two arithmetic Higgs functors

摘录 / EXCERPT: Let F_1, F_2, and F_3 be the functors defined above from enhanced log Higgs bundles to arithmetic Higgs bundles, with F_3 the direct construction and $F_2\circ F_1$ the con...

conjecture | open | MathDB 页面 | 原文线索

V2-068260 | MATHDB-354698

Bostan's conjecture on Catalan numbers modulo 10

摘录 / EXCERPT: Bostan's conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-068261 | MATHDB-354700

Todd's dimension conjecture for positive-characteristic multiple zeta values

摘录 / EXCERPT: Todd's dimension conjecture. For every positive integer w,

conjecture | open | MathDB 页面 | 原文线索

V2-068262 | MATHDB-354701

Thakur's basis conjecture for positive-characteristic multiple zeta values

摘录 / EXCERPT: Thakur's basis conjecture. The set $\mathcal{B}^{\mathrm{T}}_w$ is a basis of the k-vector space $\mathcal{Z}_w$.

conjecture | open | MathDB 页面 | 原文线索

V2-068263 | MATHDB-354702

Gałęski's smooth rank-constrained approximation conjecture

摘录 / EXCERPT: Gałęski's conjecture. If $f\in C^1(\Omega,\mathbb R^n)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068264 | MATHDB-354703

Local smooth rank-constrained approximation conjecture

摘录 / EXCERPT: Local rank-constrained approximation conjecture. If $f\in C^1(\Omega,\mathbb R^n)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068265 | MATHDB-354709

Barát-Thomassen conjecture on tree edge-decompositions

摘录 / EXCERPT: Barát-Thomassen conjecture. For every tree T, there exists a positive integer k_T such that every k_T-edge-connected simple graph whose size is divisible by $|E(T)|$ admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-068266 | MATHDB-354710

Barát-Gerbner-Thomassé conjecture on tree decompositions of tree-connected graphs

摘录 / EXCERPT: Barát-Gerbner-Thomassé conjecture. If G is an $|E(T)|$-tree-connected simple graph whose size is divisible by $|E(T)|$, then G admits a T-edge-decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-068267 | MATHDB-354711

Polynomial edge-connectivity conjecture for tree edge-decompositions

摘录 / EXCERPT: Polynomial edge-connectivity conjecture. There are two positive integers a and n such that every $a|E(T)|^n$-edge-connected simple graph whose size is divisible by $|E(T)|$ adm...

conjecture | open | MathDB 页面 | 原文线索

V2-068268 | MATHDB-354713

Stevens's P-modification reformulation of Kollár's conjecture

摘录 / EXCERPT: Stevens's reformulation of Kollár's conjecture. There exists such a modification for which some multiple of $K_{\mathcal{U}}$ is Cartier, $K_{\mathcal{U}}$ is π'-ample, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068269 | MATHDB-354718

The tree-even-cycle graph-product conjecture

摘录 / EXCERPT: Let T be a tree and let n be an even integer. Let $T\square C_n$ be the Cartesian product of T and the cycle C_n, and let $\operatorname{diam}(T\square C_n)$ denote its dia...

conjecture | open | MathDB 页面 | 原文线索

V2-068270 | MATHDB-354725

The surface-Hadamard well-posedness conjecture for semiclassical electrodynamics

摘录 / EXCERPT: Let $\mathcal C$ be an initial constant-time surface, let $\mathbf E^{\rm in}$ and $\mathbf B^{\rm in}$ be initial electric and magnetic fields on $\mathcal C$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068271 | MATHDB-354729

Dual Haglund positivity conjecture

摘录 / EXCERPT: Let $k_{\lambda,\mu}(q,t)=\langle s_{\lambda},J_{\mu}(q,t)\rangle_{q,t}$, where $\langle\,\rangle_{q,t}$ is the (q,t)-deformation of the Hall scalar product, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068272 | MATHDB-354740

The variance asymptotic conjecture for squarefull numbers in short intervals

摘录 / EXCERPT: Variance asymptotic conjecture. The variance of the count of squarefull numbers in these intervals satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068273 | MATHDB-354751

The Q-invariant and shape-invariant generation conjecture for ${\rm Inv}_{R\overline{e}}$

摘录 / EXCERPT: Q-invariant and shape-invariant generation conjecture. The field ${\rm Inv}_{R\overline{e}}$ is generated by the Q-invariants and shape invariants. Moreover, every element of...

conjecture | open | MathDB 页面 | 原文线索

V2-068274 | MATHDB-354752

The Q-invariant and barred P-invariant generation conjecture for ${\rm Inv}_{R\overline{R}}$

摘录 / EXCERPT: Q-invariant and barred P-invariant generation conjecture. The field ${\rm Inv}_{R\overline{R}}$ is generated by the Q-invariants, barred P-invariants, and shape invariants. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-068275 | MATHDB-354753

The Weyl-invariant generation conjecture for the Gelfand-Tsetlin geometric crystal

摘录 / EXCERPT: Weyl-invariant generation conjecture. The subfield of $\mathbb{C}(z'_{j,i})$ fixed by this birational action is generated by the reduced Q-invariants and shape invariants.

conjecture | open | MathDB 页面 | 原文线索

V2-068276 | MATHDB-354756

The Newton-polygon divisibility criterion for the two-dimensional Jacobian conjecture

摘录 / EXCERPT: The Newton-polygon criterion. Then $[F,G]=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-068277 | MATHDB-354760

The Valqui-Guccione-Guccione criterion for the Jacobian conjecture

摘录 / EXCERPT: The Valqui-Guccione-Guccione criterion. Then

conjecture | open | MathDB 页面 | 原文线索

V2-068278 | MATHDB-354764

Neumann-Wahl Casson invariant conjecture

摘录 / EXCERPT: Casson invariant conjecture. The Casson invariant of $\partial(X,o)$ is equal to one-eighth of the signature of the Milnor fiber of (X,o).

conjecture | open | MathDB 页面 | 原文线索

V2-068279 | MATHDB-354765

Neumann-Wahl Milnor fiber conjecture

摘录 / EXCERPT: Milnor fiber conjecture. The Milnor fiber of (X,o) is obtained by a four-dimensional splicing operation from the Milnor fibers associated to Γ_a and Γ_b.

conjecture | open | MathDB 页面 | 原文线索

V2-068280 | MATHDB-354767

Kinetic Schauder estimate for time-irregular Landau coefficients

摘录 / EXCERPT: Let Q_1 and $Q_{1/2}$ denote the kinetic cylinders appearing in the equation, let a, c, f, g, and $\bar a$, $\bar c$ be the corresponding coefficients and functions, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068281 | MATHDB-354768

Weak-strong uniqueness for rough Landau initial data

摘录 / EXCERPT: Let $f_{\rm in}$ be the initial datum in the setting of the paper's Landau weak-strong uniqueness theorem, and let $L^{\infty,k}$ denote the weighted bounded space used there. The...

conjecture | open | MathDB 页面 | 原文线索

V2-068282 | MATHDB-354788

Bottom-heaviness conjecture for q-Stirling coefficients

摘录 / EXCERPT: For a finite coefficient sequence $(a_0,\ldots,a_n)$, call it bottom heavy when $a_k\geq a_{n-k}$ for $k<n/2$. Let $S[n,k]$, $c[n,k]$, $S_B[n,k]$, and $c_B[n,k]$ be the type A an...

conjecture | open | MathDB 页面 | 原文线索

V2-068283 | MATHDB-354798

Pairwise locality compatibility of bilinear relations

摘录 / EXCERPT: Let $(E,\top)$ be a locality vector space, and let V_1,V_2 be subspaces of E. Write $I_{\rm bil}(V_1,V_2)$ for the subspace of bilinear relations and...

conjecture | open | MathDB 页面 | 原文线索

V2-068284 | MATHDB-354805

Schur and shifted-e positivity conjecture for generalized LLT polynomials

摘录 / EXCERPT: Let $w\in S_n$ and define the generalized LLT polynomial by ... where C'_w is a Kazhdan-Lusztig element and ω is the standard involution on symmetric functions. A symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-068285 | MATHDB-354806

Log-concavity conjecture for Kazhdan-Lusztig Frobenius characters

摘录 / EXCERPT: Let $w\in S_n$, let C'_w be the Kazhdan-Lusztig basis element, and let $\ell(w)$ be its Coxeter length. Expand the Frobenius character $\operatorname{ch}(q^{\ell(w)/2}C'_w)$ in t...

conjecture | open | MathDB 页面 | 原文线索

V2-068286 | MATHDB-354814

Byott's conjecture on solubility of finite skew braces

摘录 / EXCERPT: Let A be a finite skew brace with multiplicative group $(A,\cdot)$ and circle group $(A,\circ)$. Byott's conjecture. If $(A,\cdot)$ is soluble, then $(A,\circ)$ is soluble. The p...

conjecture | open | MathDB 页面 | 原文线索

V2-068287 | MATHDB-354822

Blasius's conjecture for tensor products of automorphic representations

摘录 / EXCERPT: Blasius's conjecture. The tensor product L-function admits meromorphic continuation to $\mathbb C$ and is entire if it has critical points. For every critical point...

conjecture | open | MathDB 页面 | 原文线索

V2-068288 | MATHDB-354824

Coxeter polynomial conjecture for non-negative posets of Dynkin type

摘录 / EXCERPT: Coxeter polynomial conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-068289 | MATHDB-354835

The polynomial-positive codominant character decomposition conjecture

摘录 / EXCERPT: Polynomial-positive decomposition conjecture. For each permutation $w\in S_n$, there exist codominant permutations $w_1,\ldots,w_k\in S_n$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-068290 | MATHDB-354838

Local well-posedness conjecture for the static vacuum Bartnik boundary problem

摘录 / EXCERPT: Let $(M,\bar g,\bar u)$ be an asymptotically flat, static vacuum triple, and let the geometric boundary value problem be ... where the interior equations are...

conjecture | open | MathDB 页面 | 原文线索

V2-068291 | MATHDB-354839

The rectangular shuffle theorem for rectangular paths

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, let $d=\gcd(m,n)$, and let $p_{m,n}$ be the symmetric function associated with rectangular paths. For a rectangular path $\pi\in\mathsf{LRP}(m,n)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-068292 | MATHDB-354840

The rectangular Delta conjecture for decorated Dyck paths

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, let $e_{m,n}$ be the rectangular Dyck-path symmetric function, and let Θ_{e_k} be the Theta operator indexed by e_k. For...

conjecture | open | MathDB 页面 | 原文线索

V2-068293 | MATHDB-354841

The rectangular Delta conjecture for decorated rectangular paths

摘录 / EXCERPT: Let $m,n\in\mathbb{N}$, let $d=\gcd(m,n)$, and let $p_{m,n}$ be the symmetric function associated with rectangular paths. Let Θ_{e_k} be the Theta operator indexed by e_k....

conjecture | open | MathDB 页面 | 原文线索

V2-068294 | MATHDB-354866

Plummer-Zha's 3-colorability conjecture for pentagraphs

摘录 / EXCERPT: Plummer-Zha's conjecture. Every pentagraph is 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-068295 | MATHDB-354871

Small cap moment-curve exponential-sum conjecture

摘录 / EXCERPT: Let $N\geq 1$, $0\leq \sigma\leq 2$, and $s\geq 1$, and write $e(t)=e^{2\pi i t}$. Then the small cap decoupling conjecture. There is a constant C_ε such that ... This co...

conjecture | open | MathDB 页面 | 原文线索

V2-068296 | MATHDB-354882

Conjecture on sup-norms of diagonal pullbacks of Saito-Kurokawa lifts

摘录 / EXCERPT: Let $F\in\mathrm{SK}_k$ be a Hecke eigenform, let F° denote its pullback to the diagonal, and suppose that $F^\circ\ne0$ and $\lVert F^\circ\rVert_2=1$. The Saito-Kurokawa p...

conjecture | open | MathDB 页面 | 原文线索

V2-068297 | MATHDB-354892

Magnitude-intrinsic-volume finiteness conjecture for compact convex subsets of L_1

摘录 / EXCERPT: Let L_1 denote the space of integrable functions, and let $K\subseteq L_1$ be compact and convex. Write $\mu_1^{L_1}(K)$ for the supremum of the first Holmes-Thompson intrinsic v...

conjecture | open | MathDB 页面 | 原文线索

V2-068298 | MATHDB-354895

Convergence conjecture for Artin Euler products at the central point

摘录 / EXCERPT: Let K be a global field and let ρ be an Artin representation with Artin L-function $L(s,\rho)$. Set $m=\operatorname{ord}_{s=1/2}L(s,\rho)$. Convergence conjecture. There...

conjecture | open | MathDB 页面 | 原文线索

V2-068299 | MATHDB-354901

The characteristic-cycle pullback conjecture for log-clean ramification

摘录 / EXCERPT: Suppose that X is purely of dimension d and that D has simple normal crossings. Let $I'subset I$ be a subset containing $I_{\mathrm{T},\mathcal{F}}$, and let ... Assume that th...

conjecture | open | MathDB 页面 | 原文线索

V2-068300 | MATHDB-354906

Borwein-Bradley-Broadhurst Eulerian multiple zeta value conjecture

摘录 / EXCERPT: For a multi-index $\boldsymbol{k}=(k_1,\ldots,k_d)$, write $\zeta(\boldsymbol{k})$ for its multiple zeta value, and write $\{2\}^m$ for m consecutive entries equal to 2. For...

conjecture | open | MathDB 页面 | 原文线索

V2-068301 | MATHDB-354907

Cyclic insertion conjecture for multiple zeta values

摘录 / EXCERPT: Cyclic insertion conjecture. For $n\geq0$ and $m_0,\ldots,m_{2n}\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068302 | MATHDB-354908

Generalized cyclic insertion conjecture for Z-values

摘录 / EXCERPT: Generalized cyclic insertion conjecture. For $a_1,a_2,a_3,b_1,b_2\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068303 | MATHDB-354912

Manes's conjecture on rational periodic points of degree-2 maps with an involution

摘录 / EXCERPT: Let $\phi(z)\in\mathbb{Q}(z)$ be a degree-2 rational map, and suppose its automorphism group satisfies $\operatorname{Aut}(\phi)\cong C_2$. A rational point has exact period N...

conjecture | open | MathDB 页面 | 原文线索

V2-068304 | MATHDB-354915

Curran's standard bicentrally balanced labeling conjecture for projective grid graphs

摘录 / EXCERPT: Curran's standard bicentrally balanced labeling conjecture. If X is a standard bicentrally balanced C_4-face-magic projective labeling on $\mathcal{P}_{m,n}$, then either there...

conjecture | open | MathDB 页面 | 原文线索

V2-068305 | MATHDB-354917

Lei-Lim's signed Selmer torsion conjecture over a quadratic imaginary field

摘录 / EXCERPT: Let E/L be an elliptic curve satisfying the standing hypotheses, let L_∞ be the relevant extension with $G\cong\mathbb{Z}_p^2$, and let $\Lambda(G)$ denote its Iwasawa alg...

conjecture | open | MathDB 页面 | 原文线索

V2-068306 | MATHDB-354921

Sharpness of the conformal-radius and hyperbolic-distance bounds in rectangles

摘录 / EXCERPT: Rectangle inequality. For all points $u,v\in R$,

conjecture | open | MathDB 页面 | 原文线索

V2-068307 | MATHDB-354926

Fastest-convergence conjecture for the complementing flip process

摘录 / EXCERPT: Fastest-convergence conjecture. For every such $\mathcal{R}$ and every $\delta\in(0,\frac{1}{2})$,

conjecture | open | MathDB 页面 | 原文线索

V2-068308 | MATHDB-354939

Stability conjecture for irreducible polynomials of the form z^d+1/c

摘录 / EXCERPT: Stability conjecture. If $f(z)=z^d+\frac{1}{c}$ is irreducible over $\mathbb{Q}$, then $f(z)$ is stable over $\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-068309 | MATHDB-354941

The Donagi-Morrison conjecture for Brill-Noether series on K3 surfaces

摘录 / EXCERPT: Let (S,H) be a polarized K3 surface and let $C\in|H|$ be a smooth irreducible curve of genus $g\ge 2$. Suppose A is a complete basepoint-free g^r_d on C with $d\le g-1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068310 | MATHDB-354943

De Giorgi's phase-field approximation conjecture for the Willmore functional

摘录 / EXCERPT: Let $n\geq 2$ be an integer, let $E\subset \mathbb{R}^n$ have boundary $\Sigma:=\partial E$ of class C^2, and let $\Omega\subset \mathbb{R}^n$ be open. For $\lambda>0$, define .....

conjecture | open | MathDB 页面 | 原文线索

V2-068311 | MATHDB-354950

Chen's generalized partition inequality

摘录 / EXCERPT: Let $p(n)$ denote the classical partition function, and let $a>b$ be integers such that $an,bn\in\mathbb{N}$. Chen's conjecture. For all $a>b$, one has ... This is an analogue of ine...

conjecture | open | MathDB 页面 | 原文线索

V2-068312 | MATHDB-354959

Convergence of the graphical-model support approximation sequence

摘录 / EXCERPT: Support approximation conjecture. The sequence of sets converges to the MLE support:

conjecture | open | MathDB 页面 | 原文线索

V2-068313 | MATHDB-354962

Stanley's chromatic conjecture for trees

摘录 / EXCERPT: Stanley's chromatic conjecture for trees. Trees T and U are isomorphic as graphs if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-068314 | MATHDB-354963

Finite principal-specialization conjecture for chromatic symmetric functions of trees

摘录 / EXCERPT: Finite principal-specialization conjecture. For all such trees,

conjecture | open | MathDB 页面 | 原文线索

V2-068315 | MATHDB-354964

Rooted $(3+1)$-free e-positivity conjecture

摘录 / EXCERPT: Rooted $(3+1)$-free e-positivity conjecture. For every $r\in P$, if $P-r$ is $(\mathbf{3}+\mathbf{1})$-free, then

conjecture | open | MathDB 页面 | 原文线索

V2-068316 | MATHDB-354966

Leclere-Morier-Genoud trace unimodality conjecture

摘录 / EXCERPT: Let $c_1,c_2,\ldots,c_k$ be integers, and let $M_q(c_1,c_2,\ldots,c_k)$ be the associated matrix in $\mathrm{PSL}_q(2,\mathbb{Z})$. Leclere-Morier-Genoud trace unimodality conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-068317 | MATHDB-354967

Trace counterexample classification conjecture

摘录 / EXCERPT: Let $c_1,c_2,\ldots,c_k$ be integers with $c_i\geq 2$, and let $M_q(c_1,c_2,\ldots,c_k)$ be the associated matrix. Two sequences are considered equivalent when one is obtained from...

conjecture | refuted | MathDB 页面 | 原文线索

V2-068318 | MATHDB-354972

Conjecture on unions of minor-minimal separating projective planar graphs

摘录 / EXCERPT: Disjoint-union conjecture. If both G and H are not II3L, then $G\dot{\cup}H$ is minor-minimal IPPI3L.

conjecture | open | MathDB 页面 | 原文线索

V2-068319 | MATHDB-354973

Conjecture on closed nonseparation of the epsilon and zeta families

摘录 / EXCERPT: Epsilon-zeta conjecture. The graph $b5_1$ and every member of the $b6$ family except $b6_3$ are closed nonseparating.

conjecture | open | MathDB 页面 | 原文线索

V2-068320 | MATHDB-354974

Conjecture on minor-minimal IPPI3L graphs from two copies of epsilon_1

摘录 / EXCERPT: Two-copy epsilon conjecture. The resulting graph G is minor-minimal IPPI3L.

conjecture | open | MathDB 页面 | 原文线索

V2-068321 | MATHDB-354981

Subconvexity conjecture for the Rankin-Selberg convolution of Siegel cusp forms

摘录 / EXCERPT: Subconvexity conjecture. There exists $\delta>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-068322 | MATHDB-354986

GAS versus Lyapunov stability and global attractivity for time-delay systems

摘录 / EXCERPT: Let the autonomous time-delay system be given on the history space $\mathcal X^n$, with solution $x(t,x_0)$ from initial history x_0. GAS conjecture. The system is GAS if and onl...

conjecture | open | MathDB 页面 | 原文线索

V2-068323 | MATHDB-354987

Forward completeness implies robust forward completeness for time-delay systems

摘录 / EXCERPT: Let the autonomous time-delay system be given on the history space $\mathcal X^n$. It is forward complete if every solution exists for all positive times, and it is robustly forwar...

conjecture | open | MathDB 页面 | 原文线索

V2-068324 | MATHDB-354988

GES under point-wise dissipation for time-delay systems

摘录 / EXCERPT: GES under point-wise dissipation conjecture. Under these assumptions, the system is GES. The converse is known from the cited theorem, so the conjecture would characterize GES thro...

conjecture | open | MathDB 页面 | 原文线索

V2-068325 | MATHDB-354989

ISS under point-wise dissipation for time-delay systems

摘录 / EXCERPT: ISS under point-wise dissipation conjecture. These assumptions imply that the system is ISS. The converse is known, so this would provide an alternative characterization of ISS and...

conjecture | open | MathDB 页面 | 原文线索

V2-068326 | MATHDB-354990

0-GAS and point-wise ULIM imply ISS for time-delay systems

摘录 / EXCERPT: Consider the time-delay system on $\mathcal X^n$ with inputs in $\mathcal U^m$. 0-GAS plus point-wise ULIM conjecture. If the system is 0-GAS and satisfies the point-wise version o...

conjecture | open | MathDB 页面 | 原文线索

V2-068327 | MATHDB-354991

Point-wise integral-to-integral estimate implies ISS for time-delay systems

摘录 / EXCERPT: Let the time-delay system act on $\mathcal X^n$ with inputs in $\mathcal U^m$. Assume that there are $\alpha,\overline\alpha\in\mathcal K_\infty$ and $\gamma\in\mathcal N$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-068328 | MATHDB-354992

iISS implies an iISS Lyapunov-Krasovskii functional with point-wise dissipation

摘录 / EXCERPT: Consider the time-delay system on $\mathcal X^n$ with inputs in $\mathcal U^m$. Converse point-wise iISS-LKF conjecture. If the system is iISS, then it admits an iISS Lyapunov-Kras...

conjecture | open | MathDB 页面 | 原文线索

V2-068329 | MATHDB-354993

0-GAS and bounded input-frequently bounded state imply iISS

摘录 / EXCERPT: Consider the time-delay system on $\mathcal X^n$ with inputs in $\mathcal U^m$. 0-GAS plus BEFBS conjecture. Assume the system is 0-GAS and, for some $\zeta\in\mathcal K_\infty$, e...

conjecture | open | MathDB 页面 | 原文线索

V2-068330 | MATHDB-354995

The asymptotic Carathéodory-rank conjecture for normal polytopes

摘录 / EXCERPT: Asymptotic Carathéodory-rank conjecture. The ratio $\operatorname{CR}(d)/d$ is monotonically converging to 2 as $d\to\infty$. The source records a counterexample to the integral...

conjecture | open | MathDB 页面 | 原文线索

V2-068331 | MATHDB-354997

The pyramidal and quasi-pyramidal growth conjectures for polytopes

摘录 / EXCERPT: Let $\mathbf{POL}$ be the poset of polytopes, let $\underset{\Delta}{\le}$ denote pyramidal growth, and let $\underset{\mathbf q}{\le}$ denote quasi-pyramidal growth. For a quasi-p...

conjecture | open | MathDB 页面 | 原文线索

V2-068332 | MATHDB-354999

The optimal scalar-fluctuation upper-bound conjecture for random band matrices

摘录 / EXCERPT: Scalar-fluctuation upper-bound conjecture. If $\varepsilon>0$ is small, $|\lambda|<\varepsilon$, $M\geq C_\varepsilon$, and $1<k<N$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068333 | MATHDB-355001

The proposed classification of 4-chambars on an open subset of the complex line

摘录 / EXCERPT: 4-chambar classification conjecture. Up to affine conjugacy, every 4-chambar on an open subset of $\mathbb{C}$ is one of the following types:

conjecture | open | MathDB 页面 | 原文线索

V2-068334 | MATHDB-355002

The colinearity and singular-set conjecture for homogeneous chambars

摘录 / EXCERPT: Homogeneous-chambar singular-set conjecture. For every $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068335 | MATHDB-355004

Delanoy's zircon-to-lower-interval conjecture in types A, D and E

摘录 / EXCERPT: Let $[u,w]$ be a Bruhat interval in a Weyl group of type A, D, or E. A zircon is a Bruhat interval with the corresponding compression property, and a lower interval is an int...

conjecture | open | MathDB 页面 | 原文线索

V2-068336 | MATHDB-355005

Delanoy's descent-set criterion for zircons in types A, D and E

摘录 / EXCERPT: Let $[u,w]$ be a Bruhat interval in a Weyl group of type A, D, or E. Write $D_L(w)$ for the left descent set of w, namely the set of simple reflections s such that $sw<w$...

conjecture | open | MathDB 页面 | 原文线索

V2-068337 | MATHDB-355020

The Kertész-line critical-scaling conjecture for the planar Ising model

摘录 / EXCERPT: Let p_c be the critical edge parameter and let $h_c(p)$ denote the Kertész line in the planar Ising case. The planar Ising Kertész-line conjecture. As $p\to p_c$, ... for some cons...

conjecture | open | MathDB 页面 | 原文线索

V2-068338 | MATHDB-355021

The openness conjecture for discontinuity on the Kertész line

摘录 / EXCERPT: For parameters $q\geq1$ and $h\geq0$, let $\mathfrak{Z}$ denote the relevant pressure function, and consider the phase transition as a function of p. The Kertész-line discontinui...

conjecture | open | MathDB 页面 | 原文线索

V2-068339 | MATHDB-355031

Milnor's non-local-connectivity conjecture for hyperbolic component boundaries

摘录 / EXCERPT: Milnor's non-local-connectivity conjecture. If the maps in $\mathcal H$ have an attracting cycle with two distinct free critical points in its attracting basin, then...

conjecture | open | MathDB 页面 | 原文线索

V2-068340 | MATHDB-355034

The homomorphism-threshold conjecture for simply connected graphs

摘录 / EXCERPT: Homomorphism-threshold conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-068341 | MATHDB-355040

Equality of nonnegative and zero-one tensor spectral-to-Frobenius ratios

摘录 / EXCERPT: For a tensor dimension $n_1\times n_2\times\dots\times n_d$, let $\phi(\mathbb{R}^{n_1\times n_2\times\dots\times n_d}_+)$ be the infimum of the spectral-to-Frobenius norm ratio ov...

conjecture | open | MathDB 页面 | 原文线索

V2-068342 | MATHDB-355043

The antipodal-point logarithmic-energy asymptotic conjecture on the sphere

摘录 / EXCERPT: Let $\omega_N\subset\mathbb{S}^2$ be a collection of N antipodal points, meaning that $p\in\omega_N$ implies $-p\in\omega_N$. Let $\mathscr{E}_{\log}(\omega_N)$ denote its logari...

open_problem | open | MathDB 页面 | 原文线索

V2-068343 | MATHDB-355047

Sudoku number characterization of complete graphs

摘录 / EXCERPT: Let G be a graph of order n. The Sudoku number characterization. ... Here $sn(G)$ denotes the Sudoku number of G, and K_n is the complete graph on n vertices. The conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-068344 | MATHDB-355048

Existence of an infinite bounded non-lattice with a Dacey orthogonality space

摘录 / EXCERPT: Let P be a bounded poset, and let $(Q^+(P),\perp)$ denote the orthogonality space associated with P. A poset is Dacey when every maximal orthogonal subset of $Q^+(P)$ is a base...

conjecture | open | MathDB 页面 | 原文线索

V2-068345 | MATHDB-355057

Spectral inequality conjecture for power-growth Schrödinger operators with ball-containing sensor sets

摘录 / EXCERPT: Let $\tau>0$ and let $\function{P}_\tau$ denote the spectral projector associated with $-Delta+|x|^\tau$. Suppose that $omega\five subset\real^d$ is measurable and that, for some...

conjecture | open | MathDB 页面 | 原文线索

V2-068346 | MATHDB-355071

La conjecture du $K(\pi,1)$ pour les groupes d'Artin affines

摘录 / EXCERPT: Soit W un groupe de Coxeter affine agissant par réflexions sur $\mathbb{R}^n$. On note Y_W l'espace de configuration de type W, obtenu comme le complémentaire dans...

conjecture | open | MathDB 页面 | 原文线索

V2-068347 | MATHDB-355073

Equality of divisorial and K-stability thresholds

摘录 / EXCERPT: Let $(X,B;L)$ be a polarized pair, and let $\sigma_{\mathrm{div}}(X,B;L)$ and $\sigma_{\mathrm{K}}(X,B;L)$ denote the divisorial and K-stability thresholds, respectively. Threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-068348 | MATHDB-355076

de Cataldo-Hausel-Migliorini P=W conjecture

摘录 / EXCERPT: de Cataldo-Hausel-Migliorini P=W conjecture. Under this identification, the perverse filtration associated with π satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068349 | MATHDB-355087

Gao's beta-Grothendieck positivity conjecture

摘录 / EXCERPT: Gao's beta-Grothendieck conjecture. For all permutations w, $c^{(\beta)}_w\in\mathbb Z_{\geq0}[\beta]$.

conjecture | open | MathDB 页面 | 原文线索

V2-068350 | MATHDB-355090

Conjecture on generalized energy-distance bounds for bivariate spherical copulas

摘录 / EXCERPT: Generalized energy-distance copula conjecture. The solution is attained by the circular bivariate L_p-norm spherical copula with $p=3-\beta$ for arbitrary $\beta\in(0,2)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-068351 | MATHDB-355092

Ismail's structure conjecture for orthogonal polynomials and the Askey-Wilson operator

摘录 / EXCERPT: Ismail's structure conjecture. Then $(P_n)_{n\geq 0}$ are continuous q-Jacobi polynomials, Al-Salam-Chihara polynomials, or special or limiting cases of these families. The same...

conjecture | open | MathDB 页面 | 原文线索

V2-068352 | MATHDB-355093

Distinctness conjecture for minor- and union-closed graph families

摘录 / EXCERPT: Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two distinct minor- and union-closed families of graphs. For a graph family $\mathcal{F}$, write $G\cong_{\mathcal{F}}H$ when G and H...

conjecture | open | MathDB 页面 | 原文线索

V2-068353 | MATHDB-355094

Inclusion conjecture for homomorphism indistinguishability relations

摘录 / EXCERPT: Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be minor- and union-closed families of graphs. Write $\cong_{\mathcal{F}_1}\Rightarrow\cong_{\mathcal{F}_2}$ when the first homomorphism ind...

conjecture | open | MathDB 页面 | 原文线索

V2-068354 | MATHDB-355095

Minor-separating graph-pair conjecture

摘录 / EXCERPT: Let G be a connected graph, and let H and H' be graphs. Write $\hom(F,H)$ for the number of homomorphisms from F to H. Minor-separating graph-pair conjecture. For every c...

conjecture | open | MathDB 页面 | 原文线索

V2-068355 | MATHDB-355096

Homomorphism-distinguishing closure conjecture for minor- and union-closed families

摘录 / EXCERPT: Let $\mathcal{F}$ be a family of graphs. Call it homomorphism distinguishing closed if, for every graph $F\notin\mathcal{F}$, there exist graphs G and H such that...

conjecture | open | MathDB 页面 | 原文线索

V2-068356 | MATHDB-355097

Non-isomorphism conjecture for proper minor- and union-closed families

摘录 / EXCERPT: Let $\mathcal{F}$ be a minor- and union-closed family of graphs, and let $\cong_{\mathcal{F}}$ denote homomorphism indistinguishability over $\mathcal{F}$. Non-isomorphism conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-068357 | MATHDB-355098

Clique-minor separating-pair conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$. Clique-minor separating-pair conjecture. There exist non-isomorphic graphs H and H' such that ... Here K_n is the complete graph on n vertices. The pape...

conjecture | open | MathDB 页面 | 原文线索

V2-068358 | MATHDB-355114

Virtually nilpotent kernel conjecture for log canonical singularities

摘录 / EXCERPT: Virtually nilpotent kernel conjecture. The normal subgroup N is virtually nilpotent of rank at most $2n+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-068359 | MATHDB-355117

Mkrtchyan-Vardanyan edge-incidence conjecture for Fan-Raspaud triples

摘录 / EXCERPT: Let G be a bridgeless cubic graph, let $e\in E(G)$, and let $\nu_{\mathcal T}(e)$ denote the number of members of a multiset $\mathcal T$ of perfect matchings that contain e. A...

conjecture | open | MathDB 页面 | 原文线索

V2-068360 | MATHDB-355118

Unique Apéry expansions and Cohen-Macaulay tangent cones for the semigroups Γ_m

摘录 / EXCERPT: Unique-expansion conjecture. The elements of $\operatorname{Ap}(\Gamma_m,a)$ have unique expressions. Hence the tangent cone of $k[[\Gamma_m]]$ is Cohen-Macaulay. This would extend...

conjecture | open | MathDB 页面 | 原文线索

V2-068361 | MATHDB-355119

Explicit Apéry set formula for Γ_6

摘录 / EXCERPT: Explicit Apéry-set conjecture for Γ_6. The Apéry set of Γ_6 with respect to a is

conjecture | open | MathDB 页面 | 原文线索

V2-068362 | MATHDB-355120

Nonvanishing conjecture for the relevant p-adic L-functions

摘录 / EXCERPT: Nonvanishing conjecture for the relevant p-adic L-functions. The p-adic L-function attached to the π-power-torsion does not vanish at negative powers of χ, and the one...

conjecture | open | MathDB 页面 | 原文线索

V2-068363 | MATHDB-355121

The Artin-gluing conjecture for replacing an open subtopos

摘录 / EXCERPT: Artin-gluing conjecture. The theory $\mathbb{T}+\mathbb{E}/\phi$ is classified by the topos

conjecture | open | MathDB 页面 | 原文线索

V2-068364 | MATHDB-355123

The removable-edge conjecture for surjective polymorphisms of reflexive cycles

摘录 / EXCERPT: Let $n\geq 4$, let $\mathbb{G}$ be an n-cycle, and let f be a surjective homomorphism from a product of subpaths of $\mathbb{G}$ of length $n-2$ to $\mathbb{G}$. An edge e of...

conjecture | open | MathDB 页面 | 原文线索

V2-068365 | MATHDB-355127

Extended valley Delta square conjecture

摘录 / EXCERPT: Let m,n,k be nonnegative integers. Write ∇, Θ_{e_k}, and Δ_{h_m} for the nabla, Theta, and Delta operators on symmetric functions, let ω be the standa...

conjecture | open | MathDB 页面 | 原文线索

V2-068366 | MATHDB-355129

Boundedness conjecture for the derivative-Hilbert operator on Hardy spaces

摘录 / EXCERPT: Let H^p denote the Hardy space on the unit disk, let μ be a positive Borel measure on $[0,1)$, and let $\mathcal{DH}_\mu$ denote the derivative-Hilbert operator associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-068367 | MATHDB-355139

The multipartite semiregular factorisation conjecture for complete bipartite graphs

摘录 / EXCERPT: The multipartite semiregular factorisation conjecture. If $n\to\infty$ with $2\le m\le n$ and $1\le k=o(m)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068368 | MATHDB-355153

Conjecture de Greenberg étendue

摘录 / EXCERPT: Soit K un corps de nombres arbitraire ayant r_K places réelles et c_K places complexes. Soit Z sa pro-ℓ-extension abélienne canonique, avec ... Pour les sous-corps L d...

conjecture | open | MathDB 页面 | 原文线索

V2-068369 | MATHDB-355159

Conjecture on the critical moment blowup threshold for the parabolic Anderson model

摘录 / EXCERPT: Critical moment blowup threshold conjecture. For all $p>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068370 | MATHDB-355163

The Shokurov-polytope version of the relative cone conjecture

摘录 / EXCERPT: Let $f:(X,\Delta)\to S$ be a klt Calabi-Yau fiber space. Let $\operatorname{Eff}(X/S)$, $\operatorname{Mov}(X/S)$, and $\operatorname{Amp}(X/S)$ denote the effective, movable, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068371 | MATHDB-355168

Uniformity conjecture for elementary generation lengths over global fields

摘录 / EXCERPT: Uniformity conjecture. There is a function $f:\mathbb{N}\to\mathbb{N}$ such that, for any K, S, and R as above,

conjecture | open | MathDB 页面 | 原文线索

V2-068372 | MATHDB-355176

Belfiore-Solé secrecy function conjecture for formally unimodular lattices

摘录 / EXCERPT: Let Λ be a formally unimodular lattice, and let $\Xi_{\Lambda}(\tau)$ denote its secrecy function and ξ_{Λ} its maximum value. Belfiore-Solé secrecy function conj...

conjecture | open | MathDB 页面 | 原文线索

V2-068373 | MATHDB-355177

Secrecy function conjecture for formally unimodular packings

摘录 / EXCERPT: Let Γ be a formally unimodular packing, and let $\Xi_{\Gamma}(\tau)$ denote its secrecy function and ξ_{Γ} its maximum value. Formally unimodular packing secrecy fu...

conjecture | open | MathDB 页面 | 原文线索

V2-068374 | MATHDB-355178

Secrecy-optimality criterion for formally self-dual codes

摘录 / EXCERPT: Let $\code{C}^\ast$ be a formally self-dual code of length n. Let $f_{\code{C}}(t)$ be the polynomial associated with a formally self-dual code $\code{C}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068375 | MATHDB-355183

Random list threshold conjecture for Latin hypercubes

摘录 / EXCERPT: For $d\in\mathbb N$, a d-dimensional order-n Latin hypercube is a d-dimensional array indexed by $[n]^d$ with entries from $[n]$, such that every line contains each symbol ex...

conjecture | open | MathDB 页面 | 原文线索

V2-068376 | MATHDB-355194

The high-genus modular-curve conjecture for rational non-CM points

摘录 / EXCERPT: High-genus modular-curve conjecture. If G is not conjugate in $\operatorname{GL}_2(\widehat{\mathbb Z})$ to G_1 or G_2, then $X_G(\mathbb Q)$ has no non-CM points.

conjecture | open | MathDB 页面 | 原文线索

V2-068377 | MATHDB-355203

Exponential localization and rotational symmetry of INB density-of-states measures

摘录 / EXCERPT: Exponential localization and symmetry conjecture. The densities of states $\nu^{\gamma,\beta}_{\mathrm{INB},+,k}$ and $\nu^{\gamma,\beta}_{\mathrm{INB},\times,k}$ exist and satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-068378 | MATHDB-355205

Sharpness conjecture for the lattice three-point bound in dimension 4

摘录 / EXCERPT: A lattice packing in $\mathbb{R}^4$ has density bounded above by the lattice three-point bound, obtained from a semidefinite-programming formulation of the three-point bound. Sharp...

conjecture | open | MathDB 页面 | 原文线索

V2-068379 | MATHDB-355227

Periodic restriction conjecture for the Schrödinger evolution

摘录 / EXCERPT: Let ... where a_m is supported in ... For $\lambda>0$, define the superlevel set $U_\lambda(\nu)=\{(x,t):|\nu(x,t)|\geq\lambda\}$. Periodic restriction conjecture. If $|a_m|\leq 1$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-068380 | MATHDB-355259

Shafarevich-Tate finiteness conjecture for elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over a number field K, and let $Ш(E/K)$ denote its Shafarevich-Tate group. Shafarevich-Tate finiteness conjecture. The group $Ш(E/K)$ is finite...

conjecture | open | MathDB 页面 | 原文线索

V2-068381 | MATHDB-355265

Bednarska-Bzdęga-Hefetz-Łuczak conjecture for Waiter-Client subgraph counts

摘录 / EXCERPT: Let H be a graph with $e(H)\ge 1$, and let $m(H)$ denote its maximum density over nonempty subgraphs, namely ... Consider the b-Waiter-Client game on the edges of K_n, where Wa...

conjecture | open | MathDB 页面 | 原文线索

V2-068382 | MATHDB-355275

Herzog-Kaplan-Lev product-length conjecture for cycle classes

摘录 / EXCERPT: Let $n(k,l)$ be the largest integer n such that every element of the alternating group A_n is a product of k many l-cycles, where $k,l\geq 2$ and either l is odd or k i...

conjecture | open | MathDB 页面 | 原文线索

V2-068383 | MATHDB-355280

Minimal rigidity implies rigidity for conic frameworks

摘录 / EXCERPT: Minimal rigidity conjecture. If (Γ,p) is minimally rigid, then (Γ,p) is rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-068384 | MATHDB-355281

Equivalence of minimal rigidity and rigidity for generic conic frameworks

摘录 / EXCERPT: Generic rigidity equivalence conjecture. The framework (Γ,p) is minimally rigid if and only if it is rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-068385 | MATHDB-355282

Sun-Wang-Yao conjecture on transforming strongly graceful trees into paths

摘录 / EXCERPT: Let T be a tree with a perfect matching. An edge-mismatched transfer-operation removes an edge uv from T and adds a non-edge xy when the edges uv and xy have the same e...

conjecture | open | MathDB 页面 | 原文线索

V2-068386 | MATHDB-355297

The convex-radii cover conjecture for metric trees

摘录 / EXCERPT: Let $(r_1,\dots,r_k)$ be a sequence of non-negative real numbers satisfying ... so that $(r_1,\dots,r_k)$ is convex. Let T be a metric tree with total length...

conjecture | open | MathDB 页面 | 原文线索

V2-068387 | MATHDB-355303

Congruence-class bias conjecture for parts in k-regular partitions

摘录 / EXCERPT: Congruence-class bias conjecture. For $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068388 | MATHDB-355309

The F-rational structure conjecture for wavefront sets of enhanced L-parameters

摘录 / EXCERPT: Let F be a local field of characteristic zero, let G_n^* be an F-quasisplit classical group of the form $\operatorname{Isom}(V,q)^\circ$ for a non-degenerate ϵ-Hermi...

conjecture | open | MathDB 页面 | 原文线索

V2-068389 | MATHDB-355312

The list color function threshold conjecture for complete bipartite graphs

摘录 / EXCERPT: List color threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068390 | MATHDB-355313

A q-product identity for F_0^star(q) and G_0^star(q)

摘录 / EXCERPT: Let q be a formal variable, let ω be a primitive third root of unity, and define ... ... Here $(a;Q)_\infty=\prod_{j\geq0}(1-aQ^j)$, multiple parameters are interpreted multip...

conjecture | open | MathDB 页面 | 原文线索

V2-068391 | MATHDB-355316

Contractible stable perturbation conjecture for finitely determined germs

摘录 / EXCERPT: Contractible stable perturbation conjecture. There are such germs for which

conjecture | open | MathDB 页面 | 原文线索

V2-068392 | MATHDB-355317

No coalescence conjecture for image-contractible deformations

摘录 / EXCERPT: No coalescence conjecture. The coalescence problem does not occur: a one-parameter deformation f_t cannot have more than one instability and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-068393 | MATHDB-355321

Distinct-eigenvalue conjecture for subcritical covariance matrices

摘录 / EXCERPT: Let Σ_n be the covariance matrix of the length power functionals in the subcritical regime, for $n\geq 2$, with eigenvalues $\lambda_1\leq\ldots\leq\lambda_n$. Distinct-eige...

conjecture | open | MathDB 页面 | 原文线索

V2-068394 | MATHDB-355324

The diameter-c0 radial-or-one-dimensional conjecture for the Liouville equation

摘录 / EXCERPT: Let u be a solution of the Liouville equation ... in $\mathbb{R}^2$, and let $delta$ be the standard Euclidean metric. Assume that $\mathbb{R}^2$ has diameter pi under the metric...

conjecture | open | MathDB 页面 | 原文线索

V2-068395 | MATHDB-355325

The concave Liouville solution one-dimensionality conjecture

摘录 / EXCERPT: Let u be a solution of the Liouville equation ... in $\mathbb{R}^2$. The concave-solution conjecture. If u is concave, then u is one-dimensional. The paper proves the analogous...

conjecture | open | MathDB 页面 | 原文线索

V2-068396 | MATHDB-355332

Convergence to the minimal-speed front along level sets

摘录 / EXCERPT: Let u be the solution of the homogeneous reaction-diffusion equation with initial datum as in the paper, let $X_\lambda(t,x')$ be its λ-level set, and let φ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-068397 | MATHDB-355333

De Giorgi type conjecture for reaction-diffusion omega-limit sets

摘录 / EXCERPT: Assume the invasion property, Hypothesis ... d{mathcal{H}}(U,Urho)<+infty ... . Let $\Omega(u)$ be the omega-limit set of u under time translations. De Giorgi type conjecture. Every...

conjecture | open | MathDB 页面 | 原文线索

V2-068398 | MATHDB-355339

Andrews-Paule congruence conjectures for $d_2(n)$ modulo 81, 243, and 729

摘录 / EXCERPT: Andrews-Paule's congruences.

conjecture | open | MathDB 页面 | 原文线索

V2-068399 | MATHDB-355371

Refined conjecture on pluripolar hulls and finely analytic extensions

摘录 / EXCERPT: Let f be an analytic function, and consider its graph and pluripolar hull. A finely analytic extension of f may be possibly multi-valued; its maximal such extension is understo...

conjecture | open | MathDB 页面 | 原文线索

V2-068400 | MATHDB-355372

The fixed-point conjecture for the bijection $f_{4,n}\circ f_{1,n}^{-1}$

摘录 / EXCERPT: Fixed-point conjecture. There is only 1 fixed point of $f_{4,n}\circ f_{1,n}^{-1}:S_n\to S_n$, namely the identity permutation, corresponding to the word $1\dots1$.

conjecture | open | MathDB 页面 | 原文线索

V2-068401 | MATHDB-355384

Equal-density conjecture for p2mg, cm, and p4 packings

摘录 / EXCERPT: Let n-gons be regular polygons, and write the density of the densest packing with plane group symmetry G as the corresponding densest G-packing density. An n-gon has a k-...

conjecture | open | MathDB 页面 | 原文线索

V2-068402 | MATHDB-355404

Einsiedler-Lindenstrauss-Michel-Venkatesh individual equidistribution conjecture

摘录 / EXCERPT: Let $\rho>0$, and consider a torsor for $\operatorname{Pic}(\mathcal{O}_{-D})$, such as $\mathcal{H}_{-D}$ or $\mathbb{S}^2(D)$, together with an orbit of a coset of a subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-068403 | MATHDB-355405

Michel-Venkatesh mixing conjecture for shifted packets

摘录 / EXCERPT: In the setting of shifted packets P_D^{shifted} on $X_1\times\cdots\times X_r$, assume individual equidistribution, or assume $k_1=\dots=k_r=1$. For every pair $i\ne j$...

conjecture | open | MathDB 页面 | 原文线索

V2-068404 | MATHDB-355407

The operator norm conjecture for differentiation on discrete function spaces

摘录 / EXCERPT: Let T be a rooted tree, let $\mathcal{X}$ be a discrete function space on T, let D be the differentiation operator on $\mathcal{X}$, and let C_b be the composition operator...

conjecture | open | MathDB 页面 | 原文线索

V2-068405 | MATHDB-355408

Bessel-number formula for generalized Eulerian triangles

摘录 / EXCERPT: Let $p\in\mathbb{N}$, let $\zeta\in\{0,1\}$, and let c be arbitrary. Write $E_{n,k}(-1,2;c_0,c_\infty)$ for the generalized Eulerian triangle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068406 | MATHDB-355411

Lenstra's spectrum-polytope conjecture for tower types

摘录 / EXCERPT: Let $\mathfrak{T}$ be a tower type. Let $\mathcal{S}(\mathfrak{T})$ be the associated family of orders, let $\operatorname{Spectrum}(\mathcal{S}(\mathfrak{T}))$ be its spectrum of...

conjecture | open | MathDB 页面 | 原文线索

V2-068407 | MATHDB-355414

Redundancy of the detectability condition for the optimal matrix pair

摘录 / EXCERPT: Detectability redundancy conjecture. The detectability condition in Theorem is redundant: if $(\Gamma,\Pi,\hat{\Sigma})$ solves the optimization in, then...

conjecture | open | MathDB 页面 | 原文线索

V2-068408 | MATHDB-355428

Attainment conjecture for the optimal vertical perimeter exponent

摘录 / EXCERPT: Attainment conjecture. There exist finite exponents q that answer Questions ... positively, and the infimum q_G among such q is attained. This conjecture is motivated by known...

conjecture | open | MathDB 页面 | 原文线索

V2-068409 | MATHDB-355429

Explicit triangular limit shape for Young diagrams on the diamond

摘录 / EXCERPT: Let Ω be the open diamond ... and let Φ be the limiting variational function for a homogeneous Poisson point process on Ω; its derivative Φ' is the correspond...

conjecture | open | MathDB 页面 | 原文线索

V2-068410 | MATHDB-355430

Convexity conjecture for the maximal limit shape

摘录 / EXCERPT: Let $F_{\max}$ be the maximal value function whose derivative appears as the limit shape in the paper's limit-shape corollary, and let $F_{\max}'$ denote that limit shape. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-068411 | MATHDB-355453

The token-reconstruction-family conjecture

摘录 / EXCERPT: Let G be a graph, let $F_k(G)$ denote its k-token graph, and let a k-token reconstruction family of $F_k(G)$ be a family of subsets as defined in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068412 | MATHDB-355457

The distant 2-colored components extension conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph and let $W\subseteq V(G)$ be such that $G[W]$ is bipartite, with any two components of $G[W]$ at distance at least 100 from each other. A coloring of...

open_problem | open | MathDB 页面 | 原文线索

V2-068413 | MATHDB-355460

Equality of asymptotic norms in the free-group completion

摘录 / EXCERPT: Let G be a group with norm $\|\cdot\|$, and let $F(G)$ denote the free group on the underlying set of G, with generators written $\overline{g}$. For $g_1,\dots,g_n\in G$, compar...

conjecture | open | MathDB 页面 | 原文线索

V2-068414 | MATHDB-355467

Crude Siegel-Weil formula for theta correspondences

摘录 / EXCERPT: Let $1 leq \ell<n$. Let $\theta_{\Delta(\tau,\ell)}$ and $\theta_{\Delta(\tau,2n+\ell)}$ be given theta functions, and let $f'_{\Delta(\tau,2n),s}$ be a section of the relevant ind...

conjecture | open | MathDB 页面 | 原文线索

V2-068415 | MATHDB-355468

Refined Siegel-Weil conjecture for theta lifts

摘录 / EXCERPT: Let $\theta_{\Delta(\tau,\ell)}$ and $\theta_{\Delta(\tau,2n+\ell)}$ be theta functions, and let $\Phi(\theta_{\Delta(\tau,\ell)},\theta_{\Delta(\tau,2n+\ell)},s)$ be the proposed...

conjecture | open | MathDB 页面 | 原文线索

V2-068416 | MATHDB-355469

Generalized crude Siegel-Weil conjecture for Speh theta representations

摘录 / EXCERPT: Let τ be as in the surrounding discussion, and let $\theta_{\Delta(\tau,\ell)}\in\Theta_{\Delta(\tau,\ell)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068417 | MATHDB-355482

Invariants Heuristics for anti-cyclotomic extensions

摘录 / EXCERPT: Invariants Heuristics. Among such imaginary quadratic fields, one always has

conjecture | open | MathDB 页面 | 原文线索

V2-068418 | MATHDB-355490

The strengthened burning number conjecture for trees with a prescribed number of leaves

摘录 / EXCERPT: The strengthened burning number conjecture. The tree T is m-burnable.

conjecture | open | MathDB 页面 | 原文线索

V2-068419 | MATHDB-355509

The virtual Todd class formula for almost perfect obstruction theories

摘录 / EXCERPT: Suppose that $f\colon X\to Y$ is a morphism equipped with an almost perfect obstruction theory with virtual tangent class $[T_{X/Y}]-[\mathcal{O}b_X]\in K_0(X)$. Assume that this c...

conjecture | open | MathDB 页面 | 原文线索

V2-068420 | MATHDB-355513

The cycle complement stable-set polynomial conjecture

摘录 / EXCERPT: Let C_n be the cycle graph on n vertices, let $p(C_n;z)$ denote its perfectly matchable set polynomial, and let $\overline{h}^*(C_n;z)$ denote the normalized h^*-polynomial o...

conjecture | open | MathDB 页面 | 原文线索

V2-068421 | MATHDB-355515

Anti-concentration conjecture for the independence number of sparse random graphs with a fixed number of edges

摘录 / EXCERPT: Let $G_{n,m}$ be the random graph with n vertices and m edges, and let $falpha(G_{n,m})$ denote its independence number. Suppose ... where $falpha$ is a constant such that...

conjecture | open | MathDB 页面 | 原文线索

V2-068422 | MATHDB-355516

Uniform anti-concentration conjecture for the independence number of sparse random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the binomial random graph with n vertices, where $p=n^{\gamma}$ and $-1<\gamma<-2/3$, and let $\alpha(G_{n,p})$ denote its independence number. For an arbitrary...

conjecture | open | MathDB 页面 | 原文线索

V2-068423 | MATHDB-355519

dSKP diagonal singularity conjecture

摘录 / EXCERPT: Let $\mathcal{L}$ be the lattice on which the dSKP recurrence is defined, let $x:\mathcal{L}\to\hat{\mathbb{C}}$ be a solution, and set $a_{i,j}=x(i,j,[i+j]_2)$. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-068424 | MATHDB-355520

Polygon recutting singularity position conjecture

摘录 / EXCERPT: Let $m\in\mathbb{Z}$ and let $\tilde z$ be an m-closed polygon recutting lattice map, meaning $\tilde z_{i,j}=\tilde z_{i+m,j-m}$ for all i,j. Assume $\tilde z_{i,0}=0$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-068425 | MATHDB-355525

Nonexistence of binary quasi group codes with involutory permutation automorphism group

摘录 / EXCERPT: Let C be a binary quasi group code of length $n=2m>2$, and let $\operatorname{PAut}(C)$ denote its permutation automorphism group. Nonexistence conjecture. There is no binary qua...

conjecture | open | MathDB 页面 | 原文线索

V2-068426 | MATHDB-355530

Cho's local representation density formula for hermitian special cycles

摘录 / EXCERPT: Let E/F be the unramified quadratic extension, let ${\mathbb V}$ be the space of special homomorphisms with hermitian form h, and let $\lbrace x_1,\dots,x_{2n}\rbrace$ be a bas...

conjecture | open | MathDB 页面 | 原文线索

V2-068427 | MATHDB-355544

Boundedness conjecture for relative Shalika germs

摘录 / EXCERPT: Let $A_e^{w_i}$ and $A_e^{w_{G_n}}$ be the domains associated with the relative Shalika germs $K_e^{w_i}$ and $K_e^{w_{G_n}}$, respectively, and let Δ be the discriminant fu...

conjecture | open | MathDB 页面 | 原文线索

V2-068428 | MATHDB-355550

Top-degree HOMFLY conjecture for arbitrary plabic graphs

摘录 / EXCERPT: Let G be a plabic graph with $\operatorname{c}(G)$ connected components and n interior faces. Let $L^{\operatorname{plab}}_G$ be its associated plabic graph link, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068429 | MATHDB-355551

Full HOMFLY invariance under mutation for connected simple plabic graphs

摘录 / EXCERPT: Let G and G' be connected simple plabic graphs whose quivers Q_G and $Q_{G'}$ are mutation equivalent. Let $L^{\operatorname{plab}}_G$ and $L^{\operatorname{plab}}_{G'}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-068430 | MATHDB-355553

Affine plabic-fence move classification conjecture

摘录 / EXCERPT: Let $G(\mathbf w,f)$ and $G(\mathbf w',f')$ be affine plabic fences, with restricted move equivalence defined using the relevant affine plabic-fence moves. Affine plabic-fence clas...

conjecture | open | MathDB 页面 | 原文线索

V2-068431 | MATHDB-355555

Greenberg's functional equation conjecture for ordinary Selmer groups

摘录 / EXCERPT: Let V be a finite-dimensional $\d835\dd34_p$-vector space with a continuous...

conjecture | open | MathDB 页面 | 原文线索

V2-068432 | MATHDB-355556

Cerbai-Claesson-Ferrari bijection conjecture for 201-avoiding ascent sequences

摘录 / EXCERPT: Cerbai-Claesson-Ferrari conjecture. For every $n\geq 0$, there is a bijection between $A_n(201)$ and the permutations in S_n that are 312-machine sortable.

conjecture | open | MathDB 页面 | 原文线索

V2-068433 | MATHDB-355557

Gil-Weiner enumeration conjecture for three Fishburn pattern classes II

摘录 / EXCERPT: Gil-Weiner conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068434 | MATHDB-355558

Grassmannian enumeration conjecture for Fishburn pattern classes

摘录 / EXCERPT: Grassmannian enumeration conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068435 | MATHDB-355559

Four-pattern Fishburn enumeration conjecture

摘录 / EXCERPT: Four-pattern enumeration conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068436 | MATHDB-355560

Fishburn-to-classical pattern enumeration conjecture for 2143 and 3124

摘录 / EXCERPT: Fishburn-classical enumeration conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068437 | MATHDB-355561

Fishburn and classical permutation enumeration conjecture for 1243 and 2134

摘录 / EXCERPT: Enumeration conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068438 | MATHDB-355562

Fishburn and classical permutation enumeration conjecture for 1243 and 3124

摘录 / EXCERPT: Enumeration conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068439 | MATHDB-355563

Catalan-convolution conjecture for pairs of Fishburn pattern classes

摘录 / EXCERPT: Catalan-convolution conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068440 | MATHDB-355564

Fine-sequence conjecture for indecomposable Fishburn classes

摘录 / EXCERPT: Fine-sequence conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068441 | MATHDB-355565

Additional enumerations for 321-avoiding Fishburn permutations

摘录 / EXCERPT: Additional enumeration conjectures. The following identities are conjectured:

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068442 | MATHDB-355567

Lefschetz properties for codimension-four Artin Gorenstein rings with prescribed regularity

摘录 / EXCERPT: Let A be an Artin Gorenstein ring with codimension c and regularity r, and let a Betti table refer to the graded Betti table of a minimal free resolution of A. The codimens...

conjecture | open | MathDB 页面 | 原文线索

V2-068443 | MATHDB-355568

Higher Zagier formulas for multiple polylogarithms

摘录 / EXCERPT: Let $k\geq 2$, let $a_1,\dots,a_k$ be elements for which the displayed multiple polylogarithms are defined, and let $\mathcal{L}_{2k}$ be the weight-$2k$ component of the Lie coalg...

conjecture | open | MathDB 页面 | 原文线索

V2-068444 | MATHDB-355573

Serre's congruence subgroup property conjecture

摘录 / EXCERPT: Let K be a global field, let S be a finite set of places, and retain the paper's notation for the corresponding arithmetic group $({\mathbf{G}},S)$. The pair has the congruence...

conjecture | open | MathDB 页面 | 原文线索

V2-068445 | MATHDB-355575

Spiro's minimum Wiener index conjecture for signed trees

摘录 / EXCERPT: A signed graph is a graph whose edges are assigned positive or negative signs. For a signed graph (G,σ), let $W_\sigma(G)$ denote the sum of the signed distances over all un...

conjecture | refuted | MathDB 页面 | 原文线索

V2-068446 | MATHDB-355588

Kumar-Gangania conjecture on the third Hankel determinant for cardioid starlike functions

摘录 / EXCERPT: Kumar-Gangania conjecture. If $f\in\mathcal{S}_{\wp}^{*}$, then the sharp bound is

conjecture | open | MathDB 页面 | 原文线索

V2-068447 | MATHDB-355591

Lee's spin Gromov-Witten/Hurwitz correspondence

摘录 / EXCERPT: Let (B,ϑ) be a spin curve, let g,d be integers, and let $k_1,\ldots,k_n$ satisfy ... Write $\overline{c}_k=\sum_{\mu\ \textup{odd}}\kappa_{k,\mu}\cdot\mu$ for the spin co...

conjecture | open | MathDB 页面 | 原文线索

V2-068448 | MATHDB-355606

The twisted Fukaya equivalence for normal crossings surfaces

摘录 / EXCERPT: Let X be a normal crossings surface with graph-like singular locus and orientable dual intersection complex, arising as the zero locus of a section s of a line bundle L on a...

conjecture | open | MathDB 页面 | 原文线索

V2-068449 | MATHDB-355609

The twisted Fukaya-category conjecture for normal crossings surfaces

摘录 / EXCERPT: Let X be a proper normal crossings surface with graph-like singular locus described by a trivalent graph $G(X)$ and orientable dual intersection complex. For each edge e whose...

conjecture | open | MathDB 页面 | 原文线索

V2-068450 | MATHDB-355612

Frieze-Krivelevich-Michaeli conjecture on d-connectivity under restricted budget

摘录 / EXCERPT: Frieze-Krivelevich-Michaeli conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-068451 | MATHDB-355620

Finite-field sphere extension conjecture

摘录 / EXCERPT: Sphere extension conjecture. The $L^2\to L^{(2n+4)/n}$ extension estimate should hold uniformly in q:

conjecture | open | MathDB 页面 | 原文线索

V2-068452 | MATHDB-355629

The stability conjecture for the infinity-Laplacian and H-type deviation

摘录 / EXCERPT: Let $\mathbb{G}$ be a step two Carnot group with horizontal metric g_h. Denote by Σ the horizontal layer in exponential coordinates, ... Let u be the fundamental solution...

conjecture | open | MathDB 页面 | 原文线索

V2-068453 | MATHDB-355630

Balogh-Tyson conjecture on polarizable Carnot groups

摘录 / EXCERPT: Let $\mathbb{G}$ be a Carnot group of homogeneous dimension $Q\geq 3$, and let u be Folland's fundamental solution for the sub-Laplacian. The group is polarizable if...

conjecture | open | MathDB 页面 | 原文线索

V2-068454 | MATHDB-355634

Generalized uniqueness conjecture for compatible Calabi-Yau metrics

摘录 / EXCERPT: Fix a polystable negative valuation v on X. Let $\mathcal M_v$ be the space of compatible Calabi-Yau metrics on X with $v=v_\omega$, let $\mathcal C_v$ be the corresponding w...

conjecture | open | MathDB 页面 | 原文线索

V2-068455 | MATHDB-355635

Andrews-Merca inequality for truncated partition sums

摘录 / EXCERPT: Let $p(n)$ denote the number of partitions of n, and let $M_\nu(n)$ denote the number of partitions of n in which the first part greater than $2\nu-1$ is odd and appears exactl...

conjecture | open | MathDB 页面 | 原文线索

V2-068456 | MATHDB-355638

André-Grothendieck period conjecture for Nori motives

摘录 / EXCERPT: André-Grothendieck period conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068457 | MATHDB-355639

Functional André-Grothendieck period conjecture for Nori motives

摘录 / EXCERPT: André-Grothendieck period conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068458 | MATHDB-355642

Eisenstein isoperimetric inequality for Fibonacci colorings

摘录 / EXCERPT: Let $\beta=a+b\alpha$ and $\beta'=a'+b'\alpha$ be primitive Eisenstein integers, and write $\beta\preceq\beta'$ when $b\leq b'$. Let β be a Fibonacci Eisenstein integer, let...

conjecture | open | MathDB 页面 | 原文线索

V2-068459 | MATHDB-355671

A two-term relation conjecture for depth-two multizeta values

摘录 / EXCERPT: Let q be a power of 2, and let $0\leq i\leq2$ and $0\leq j\leq2-i$. Define ... For $a\in\{4q,q^2+q-1,q^2+q,q^2+2q-2,q^2+2q-1,q^2+2q\}$, the two-term relation conjecture asserts .....

conjecture | open | MathDB 页面 | 原文线索

V2-068460 | MATHDB-355674

Thin-tree conjecture for 5-edge-connected planar graphs

摘录 / EXCERPT: Thin-tree conjecture for 5-edge-connected planar graphs. There exists an $\varepsilon\in(0,1)$ such that every 5-edge-connected planar graph admits two edge-disjoint...

conjecture | open | MathDB 页面 | 原文线索

V2-068461 | MATHDB-355677

The generalized ideq-tilting equivalence conjecture

摘录 / EXCERPT: Let $\mathcal{E}$ be an exact category, and suppose that there exists an integer $n\geq 0$ for which $\mathcal{E}$ has at least one n-tilting subcategory. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068462 | MATHDB-355678

Least-denominator conjecture for the Poincaré series of generic matrix representations

摘录 / EXCERPT: Let $G(t)$ be the rational function ... whose extra numerator factor is $(1+t+\cdots+t^{n-1})^2$. Equivalently, if $G(t)=\prod_i(1-t^i)^{-\beta(i)}$, then ... Here $\bar{R}(n,k)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-068463 | MATHDB-355679

Courtade-Kumar conjecture on information-maximizing Boolean functions

摘录 / EXCERPT: Let $\mathcal{C}_n=\{-1,1\}^n$ be the Boolean hypercube, let X be uniformly distributed on $\mathcal{C}_n$, and let $\mathcal{N}_{X,\rho}$ denote the corresponding noisy version...

conjecture | open | MathDB 页面 | 原文线索

V2-068464 | MATHDB-355682

Cornulier's pointed-sphere conjecture for quasiisometries of non-special Heintze groups

摘录 / EXCERPT: Let S be a Heintze group, and let ϕ be a self-quasiisometry of S. The Gromov boundary of S is the boundary on which ϕ induces an extension. Assume that S is not o...

conjecture | open | MathDB 页面 | 原文线索

V2-068465 | MATHDB-355688

Integrality conjecture for a truncated hypergeometric series

摘录 / EXCERPT: Integrality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-068466 | MATHDB-355690

Kalai's conjecture on recovering lower affine-stress spaces

摘录 / EXCERPT: Kalai's lower-stress reconstruction conjecture. The space $\mathcal S^a_i(P)$ determines the space $\mathcal S^a_j(P)$.

conjecture | open | MathDB 页面 | 原文线索

V2-068467 | MATHDB-355691

Kalai's affine-stress support conjecture for simplicial spheres

摘录 / EXCERPT: Kalai's affine-stress support conjecture. Every $(i-1)$-face of Δ participates in some affine i-stress on (Δ,p). In particular, the space of affine i-stresses det...

conjecture | open | MathDB 页面 | 原文线索

V2-068468 | MATHDB-355694

The dihedral permutation-group size conjecture for indecomposable Yang-Baxter solutions

摘录 / EXCERPT: Let X be an indecomposable solution of the Yang-Baxter equation whose permutation group is isomorphic to the dihedral group D_{2n}. Dihedral size conjecture. Then ... For an inde...

conjecture | open | MathDB 页面 | 原文线索

V2-068469 | MATHDB-355703

Strong Martio's conjecture for quasiregular mappings

摘录 / EXCERPT: Let $U \subset \mathbb R^n$ be open with $n \ge 3$, let $f:U\to f(U)\subset\mathbb R^n$ be a non-constant quasiregular mapping, let $K_I(f)$ denote its inner dilatation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068470 | MATHDB-355714

Woodall's conjecture on complete bipartite graph minors

摘录 / EXCERPT: Let s and t be positive integers. A graph has no $K_{s,t}$ minor if it does not contain the complete bipartite graph $K_{s,t}$ as a minor. Woodall's conjecture. Every graph wit...

conjecture | open | MathDB 页面 | 原文线索

V2-068471 | MATHDB-355720

Quiver analogue of Higman's conjecture for commuting radical endomorphisms

摘录 / EXCERPT: Let Q be an acyclic quiver, let $\mathbb{k}=\mathbb{F}_q$, and let $P\in\operatorname{rep}Q$ be a projective representation over $\mathbb{k}$. Consider the radical...

conjecture | open | MathDB 页面 | 原文线索

V2-068472 | MATHDB-355732

Mixing-time and cutoff conjecture for open asymmetric simple exclusion processes

摘录 / EXCERPT: Mixing-time and cutoff conjecture. The following claims should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-068473 | MATHDB-355735

The Weber class-number conjecture for the fields \(\mathbb{B}_n\)

摘录 / EXCERPT: Let $\mathbb{B}_n$ be the number field introduced above, and let h_n denote its class number for $n\in\mathbb{Z}_{\geq 1}$. Weber's conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-068474 | MATHDB-355752

Conjecture on archimedean Rankin-Selberg zeta integrals for GL_n

摘录 / EXCERPT: Conjecture on archimedean zeta integrals. There exists a number field $E(\varepsilon^{(n+1)}_v,\varepsilon^{(n)}_v)$ such that, for every $m\in\mathrm{Crit}(\pi^{(n+1)},\pi^{(n)})$...

conjecture | open | MathDB 页面 | 原文线索

V2-068475 | MATHDB-355757

The four-color locally irregular decomposition conjecture

摘录 / EXCERPT: Let $\mathfrak{T'}$ be the family consisting of the recursively defined family $\mathfrak{T}$, all odd-length paths and all odd-length cycles. For a graph G, let ${\rm lir}(G)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-068476 | MATHDB-355758

The bow-tie exception conjecture for locally irregular decompositions

摘录 / EXCERPT: Let $\mathfrak{T'}$ be the family consisting of the recursively defined family $\mathfrak{T}$, all odd-length paths and all odd-length cycles, and let B denote the bow-tie graph....

conjecture | open | MathDB 页面 | 原文线索

V2-068477 | MATHDB-355764

Decomposability conjecture for D-optimal designs in the Bradley-Terry model

摘录 / EXCERPT: Decomposability conjecture. The graph G of a D-optimal design in the Bradley--Terry model is decomposable.

conjecture | open | MathDB 页面 | 原文线索

V2-068478 | MATHDB-355777

Tribonacci dimension conjecture for alternating multiple two-one values

摘录 / EXCERPT: For each weight n, let AMTV_n denote the rational vector space of alternating multiple two-one values of weight n. Tribonacci dimension conjecture. The generating fu...

conjecture | open | MathDB 页面 | 原文线索

V2-068479 | MATHDB-355778

Exponential dimension conjecture for alternating multiple t-values

摘录 / EXCERPT: For each $n\ge2$, let AMtV_n denote the rational vector space of alternating multiple t-values of weight n. Exponential dimension conjecture. ... The paper lists this...

conjecture | open | MathDB 页面 | 原文线索

V2-068480 | MATHDB-355779

Recurrence dimension conjecture for alternating multiple s-values

摘录 / EXCERPT: For each weight n, let AMSV_n denote the rational vector space of alternating multiple s-values. Recurrence dimension conjecture. ... and, for every $n\ge3$, ... The pap...

conjecture | open | MathDB 页面 | 原文线索

V2-068481 | MATHDB-355782

Refined h-positivity conjecture for equivalence classes of heaps

摘录 / EXCERPT: Let P be a natural unit interval order on $[n]$, let $\mu\in\mathbb{N}^n$, and let $\mathcal{H}(P,\mu)/\sim$ be the set of equivalence classes of heaps. For an equivalence class...

conjecture | open | MathDB 页面 | 原文线索

V2-068482 | MATHDB-355789

Mordell-Weil finite-generation conjecture over suitable real abelian fields

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, and let $\mathbf L\subset\mathbb{Q}^{\operatorname{ab}}$ be a real abelian field containing only finitely many extensions of...

conjecture | open | MathDB 页面 | 原文线索

V2-068483 | MATHDB-355795

The Guo-Jacquet fundamental lemma

摘录 / EXCERPT: Guo-Jacquet fundamental lemma. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068484 | MATHDB-355802

McKernan-Shokurov conjecture for generalized pairs

摘录 / EXCERPT: Let d be a positive integer and ϵ a positive real number. Let $(X,B+\mathbf{M})$ be a generalized pair of dimension d and let $\pi:X\to Z$ be a contraction between pro...

conjecture | open | MathDB 页面 | 原文线索

V2-068485 | MATHDB-355807

Thomas-Williams' cyclic sieving conjecture for rational Tamari lattices

摘录 / EXCERPT: Thomas-Williams' cyclic sieving conjecture. The triple

conjecture | open | MathDB 页面 | 原文线索

V2-068486 | MATHDB-355808

The rational Tamari down-degree homomesy conjecture

摘录 / EXCERPT: Rational Tamari down-degree homomesy conjecture. The statistic ddeg is homomesic for rowmotion, with average

conjecture | open | MathDB 页面 | 原文线索

V2-068487 | MATHDB-355809

Asymptotic homomesy for repeated lattice-path Tamari lattices

摘录 / EXCERPT: Asymptotic homomesy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068488 | MATHDB-355811

Star-like graph mode conjecture for domination polynomials

摘录 / EXCERPT: For a graph G, let $D(G,x)$ be its domination polynomial, and call the index of a largest coefficient its mode. A graph is star-like in the sense intended by the source. Star-lik...

conjecture | open | MathDB 页面 | 原文线索

V2-068489 | MATHDB-355816

Norin-Scott-Seymour-Wood clustered coloring conjecture for minor-free graphs

摘录 / EXCERPT: Let H be a graph, let ${\mathcal F}$ be the class of graphs with no H minor, and let $\chi_*({\mathcal F})$ denote the clustered chromatic number of ${\mathcal F}$. Norin-Scott...

conjecture | open | MathDB 页面 | 原文线索

V2-068490 | MATHDB-355836

Brown's uniqueness conjecture for the Calderón problem with $W^{1,n}$ conductivities

摘录 / EXCERPT: Let $n \geq 3$ and let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain. Suppose that $\gamma_1,\gamma_2 \in W^{1,n}(\Omega)$ are uniformly elliptic, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068491 | MATHDB-355843

The word-measure orbit conjecture

摘录 / EXCERPT: Let F_r be the free group on r generators. For $w,w'\in F_r$ and a compact group G, let $\mu_{w,G}$ be the push-forward of Haar measure on G^r under the word map determined...

conjecture | open | MathDB 页面 | 原文线索

V2-068492 | MATHDB-355858

Aubert-Moussaoui-Solleveld Bernstein-block LLC conjecture

摘录 / EXCERPT: Aubert-Moussaoui-Solleveld's Bernstein-block conjecture. The local Langlands correspondence for L induces a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-068493 | MATHDB-355874

Distinct positive zeros of the component polynomials in the general case

摘录 / EXCERPT: Distinct-zeros conjecture. The positive zeros of $\Sigma(R_1)$ and $\Sigma(R_2)$ are distinct.

conjecture | open | MathDB 页面 | 原文线索

V2-068494 | MATHDB-355875

Distinct positive zeros of the derivative-polynomial combinations

摘录 / EXCERPT: Polynomial distinct-zeros conjecture. The positive real zeros of the polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-068495 | MATHDB-355876

Boundedness of the zero set of a two-dimensional finite electrical field

摘录 / EXCERPT: Bounded-zero-set conjecture. The zero set $Z(F)$ is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-068496 | MATHDB-355877

Finiteness of the zero set of a two-dimensional finite electrical field

摘录 / EXCERPT: Finite-zero-set conjecture. The zero set of a two-dimensional finite electrical field consists of only finitely many points.

conjecture | open | MathDB 页面 | 原文线索

V2-068497 | MATHDB-355896

The special-orbit filtration conjecture

摘录 / EXCERPT: Special-orbit filtration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068498 | MATHDB-355897

The special-orbit geometric Langlands correspondence conjecture

摘录 / EXCERPT: Special-orbit geometric Langlands conjecture. For every ${\mathbb O}\in\mathcal X(\mathfrak{g})^{sp}$, there is a Hecke-compatible equivalence

conjecture | open | MathDB 页面 | 原文线索

V2-068499 | MATHDB-355898

The Eisenstein-series filtration conjecture

摘录 / EXCERPT: Eisenstein-series filtration conjecture. For every $K\in F_{\bar{\mathbb O}}(M)$,

conjecture | open | MathDB 页面 | 原文线索

V2-068500 | MATHDB-355901

The realization question for Rasmussen-invariant vectors

摘录 / EXCERPT: Let $\underline{s},\underline{\vartheta}\in\mathbb{Z}^{\infty}$ be infinite vectors such that, for each vector, all but finitely many entries equal the first entry, and suppose the...

open_problem | open | MathDB 页面 | 原文线索

V2-068501 | MATHDB-355902

The characteristic-two generalized crossing-change inequality conjecture

摘录 / EXCERPT: Let K_- and K_+ be knots such that K_+ is obtained from K_- by a negative generalized crossing change on $2n$ strands. Characteristic-two generalized crossing-change inequa...

conjecture | open | MathDB 页面 | 原文线索

V2-068502 | MATHDB-355908

The compact tame tangle approximation conjecture

摘录 / EXCERPT: Tame tangles approximate transversal tangles. Any compact framed transversal m-tangle can be approximated by compact tame m-tangles.

conjecture | open | MathDB 页面 | 原文线索

V2-068503 | MATHDB-355912

The classification conjecture for stable tangle singularities

摘录 / EXCERPT: Classifications of stable tangle singularities is tractable. The $\sF$-equivalence classes of stable m-tangle singularities in codimension p are constructively classifiable, an...

conjecture | open | MathDB 页面 | 原文线索

V2-068504 | MATHDB-355919

Fixed-denominator joint Gaussian distribution conjecture for additive twists

摘录 / EXCERPT: Let $(\phi_j)_{j=1}^r$ be a finite orthogonal family of Hecke-Maaß cusp forms for $\operatorname{SL}(2,\mathbb{Z})$. Let $\mathcal{V}(x)$ be the associated vector of additive-twist...

conjecture | open | MathDB 页面 | 原文线索

V2-068505 | MATHDB-355922

Compactification of ALG and ALG* gravitational instantons

摘录 / EXCERPT: An ALG or ALG^* gravitational instanton may, after a suitable hyperKähler rotation, admit a compactification obtained by adding a canonical cycle to a rational surface. Compact...

open_problem | open | MathDB 页面 | 原文线索

V2-068506 | MATHDB-355927

Infinite equality cases for the local antimagic chromatic number of lexicographic products

摘录 / EXCERPT: Let G and H be disjoint non-null graphs. Write $G[H]$ for their lexicographic product, and let $\chi_{la}(G)$ denote the local antimagic chromatic number and $\chi(G)$ the chro...

conjecture | open | MathDB 页面 | 原文线索

V2-068507 | MATHDB-355928

Characterization of equality for local antimagic chromatic numbers of lexicographic products

摘录 / EXCERPT: Let G and H be graphs, let $G[H]$ be their lexicographic product, and let χ_{la} and χ denote the local antimagic chromatic number and chromatic number, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-068508 | MATHDB-355930

Upper-bound conjecture for disjoint perfect matchings in edge-connected r-graphs

摘录 / EXCERPT: For $0\leq t\leq r$, let $m(t,r)$ be the maximum integer s such that every t-edge-connected r-graph has s pairwise disjoint perfect matchings. The upper-bound conjecture. F...

conjecture | open | MathDB 页面 | 原文线索

V2-068509 | MATHDB-355931

The Aboulker-Bousquet conjecture on chi-boundedness of graphs with no k-chord cycle

摘录 / EXCERPT: For an integer $k\geq 1$, let $\mathcal{C}_k$ be the class of graphs that contain no cycle with exactly k chords. A family of graphs is chi-bounded if there is a function f suc...

conjecture | open | MathDB 页面 | 原文线索

V2-068510 | MATHDB-355935

Amended Gaussian optimality conjecture for the Han-Kobayashi region

摘录 / EXCERPT: Let (N_1,N_2,u,q_1,q_2) denote positive channel and power parameters, and let the isotropic condition mean that the maximum in the stated theorem is achieved by isotropic Gaussia...

conjecture | open | MathDB 页面 | 原文线索

V2-068511 | MATHDB-355938

The tau-function formulation of the Ablowitz-Ladik flow conjecture

摘录 / EXCERPT: Tau-function Ablowitz-Ladik conjecture. The functions U and W satisfy the positive flows of the Ablowitz-Ladik hierarchy when $t^{2,k}$ is identified with t_k for $k\geq 0$....

conjecture | open | MathDB 页面 | 原文线索

V2-068512 | MATHDB-355941

Rectangular-flat-torus tessellation representation conjecture

摘录 / EXCERPT: Rectangular-flat-torus tessellation conjecture. Every toroidal graph without loops has a tessellation representation on the rectangular flat torus.

conjecture | open | MathDB 页面 | 原文线索

V2-068513 | MATHDB-355942

Rectangular-flat-torus grid contact representation conjecture

摘录 / EXCERPT: Rectangular-flat-torus grid contact conjecture. Every bipartite toroidal graph without loops has a grid contact representation on the rectangular flat torus.

conjecture | open | MathDB 页面 | 原文线索

V2-068514 | MATHDB-355946

Hafner's recursive leading-monomial conjecture for Grothendieck polynomials

摘录 / EXCERPT: Fix $w\in S_n$ and any term order with $x_1<x_2<\cdots<x_n$. Let $\mathfrak{G}_w^{(k)}$ denote the degree-k homogeneous component of $\mathfrak{G}_w$, and let $m_k(\bm{x})$ be it...

conjecture | open | MathDB 页面 | 原文线索

V2-068515 | MATHDB-355953

Lewis's Hurwitz-orbit classification conjecture for finite Coxeter groups

摘录 / EXCERPT: Lewis's conjecture. Two reflection factorizations of g belong to the same Hurwitz orbit if and only if they generate the same subgroup $H\leq W$ and have the same multiset of H...

conjecture | open | MathDB 页面 | 原文线索

V2-068516 | MATHDB-355960

Generic smoothness conjecture for the Signorini free boundary

摘录 / EXCERPT: Consider the Signorini problem, also known as the thin obstacle problem, and its free boundary. Generic smoothness conjecture. Generically, the free boundary in the Signorini probl...

conjecture | open | MathDB 页面 | 原文线索

V2-068517 | MATHDB-355961

Preserved-measure conjecture for homogeneous quadratic divergence-free systems

摘录 / EXCERPT: Preserved-measure conjecture. The Kahan map of f has a preserved measure with density

conjecture | open | MathDB 页面 | 原文线索

V2-068518 | MATHDB-355965

Hibi-Matsuda's h-polynomial bound for binomial edge ideals

摘录 / EXCERPT: Let G be a graph on n vertices, let J_G be its binomial edge ideal in a polynomial ring S, and let $h_{S/J_G}(t)$ be the numerator polynomial of the Hilbert series of...

conjecture | open | MathDB 页面 | 原文线索

V2-068519 | MATHDB-355969

The sharp two-step convergence-rate conjecture for gradient descent-ascent

摘录 / EXCERPT: Let $F\in\mathcal{F}(L_x,L_y,L_{xy},\mu_x,\mu_y)$, where L_x,L_y,L_{xy} are the smoothness parameters, μ_x,μ_y are the strong-convexity and strong-concavity parameters, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068520 | MATHDB-355985

2-torsion and 4-torsion conjecture for double coverings of arrangement complements

摘录 / EXCERPT: Let $\mathcal{A}$ be a complex hyperplane arrangement, let $M(\mathcal{A})$ be its complement, and let $\omega\in H^1(M(\mathcal{A}),\mathbb{Z}_2)$. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-068521 | MATHDB-355987

Positivity conjecture for fractional Gaussian noise predictor coefficients

摘录 / EXCERPT: Let $H>1/2$, and consider fractional Gaussian noise with Hurst parameter H. For any element of this process, project it onto any finite number of its subsequent elements, and den...

conjecture | open | MathDB 页面 | 原文线索

V2-068522 | MATHDB-355989

Connectedness-locus embedding conjecture for exponential and error functions

摘录 / EXCERPT: Connectedness-locus embedding conjecture. There exists a continuous map from $\mathcal{M}_{\exp}$ to $\mathcal{M}_{\mathrm{err}}$ that is a homeomorphism onto its image.

conjecture | open | MathDB 页面 | 原文线索

V2-068523 | MATHDB-355991

The K3 categorical enumerative invariant conjecture

摘录 / EXCERPT: Let X be a smooth projective $K3$ surface and let $\mathcal{C}=D^b(\operatorname{Coh}(X))$. Write ϕ for a weak Calabi-Yau structure, $s^{\sf BT}$ for the specified splitting...

conjecture | open | MathDB 页面 | 原文线索

V2-068524 | MATHDB-355993

Vertical asymptotics conjecture for the meromorphic 3D-index

摘录 / EXCERPT: Let $I^{\mathrm{mer}}(0,0)$ be the meromorphic 3D-index of a hyperbolic knot, let σ range over the relevant boundary-parabolic $\operatorname{PSL}_2(\mathbb C)$-representat...

conjecture | open | MathDB 页面 | 原文线索

V2-068525 | MATHDB-356001

Directed-cycle orientation-counting conjecture

摘录 / EXCERPT: Let $C_k^\circlearrowright$ denote the directed cycle of length k, and let $D(G(n,p),C_k^\circlearrowright)$ be the number of orientations of $G(n,p)$ containing no copy of...

conjecture | open | MathDB 页面 | 原文线索

V2-068526 | MATHDB-356002

Separation-number conjecture for complete graphs

摘录 / EXCERPT: Separation-number conjecture. Under these conditions,

conjecture | open | MathDB 页面 | 原文线索

V2-068527 | MATHDB-356003

Refined separation-number conjecture for the complete graph K4

摘录 / EXCERPT: K4 separation-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068528 | MATHDB-356005

Thomassé's sibling-number conjecture for countable relations

摘录 / EXCERPT: Let R be a relation, with domain a non-empty set and with any fixed finite arity. For a structure R, let $Sib(R)$ denote the number of isomorphism classes of structures equimor...

conjecture | open | MathDB 页面 | 原文线索

V2-068529 | MATHDB-356006

The three-part classification conjecture for countable direct sums of chains

摘录 / EXCERPT: Let a DSC be a direct sum of chains, and let $Sib(\mathcal{D})$ denote its sibling number. Suppose ... is a countable DSC, where $\mathcal{D}_1$ is bounded, $\mathcal{D}_2$ is a non-...

conjecture | open | MathDB 页面 | 原文线索

V2-068530 | MATHDB-356014

The Kerr-Griffiths-Green mixed Tate lifting conjecture for limit Hodge structures

摘录 / EXCERPT: Let $f:X\longrightarrow C$ be a proper semi-stable degeneration over a smooth curve C/k, with $k\subset\mathbb C$. Let X_t be a fiber at a complex point $t\neq s$, and write it...

conjecture | open | MathDB 页面 | 原文线索

V2-068531 | MATHDB-356017

The closure ordering conjecture for local L-parameters in Arthur packets

摘录 / EXCERPT: Closure ordering conjecture. For every $pi e Pi_psi$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-068532 | MATHDB-356018

The dual generic representation conjecture

摘录 / EXCERPT: Dual generic representation conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068533 | MATHDB-356019

The Gross-Prasad--Rallis genericity conjecture

摘录 / EXCERPT: Gross-Prasad--Rallis conjecture. The parameter phi is generic if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-068534 | MATHDB-356020

Factorization conjecture for the determinant of the Lyapunov Jacobian

摘录 / EXCERPT: Factorization conjecture. The determinant of $A(\Sigma)_{\cdot,E}$ factorizes as

conjecture | open | MathDB 页面 | 原文线索

V2-068535 | MATHDB-356023

The Gaussian vs. MZ derivatives conjecture

摘录 / EXCERPT: Let $D_{\mathcal{A}}$ be an n-th generalized Riemann differentiation at a point x. An MZ differentiation is one for which, for every function f that is $n-1$ times Peano diff...

conjecture | open | MathDB 页面 | 原文线索

V2-068536 | MATHDB-356024

The Riemann vs. MZ derivatives conjecture

摘录 / EXCERPT: Let D_n denote the n-th Riemann differentiation, and let an MZ differentiation mean an n-th generalized Riemann differentiation equivalent to the n-th Peano derivative for...

conjecture | open | MathDB 页面 | 原文线索

V2-068537 | MATHDB-356025

The symmetric Gaussian vs. MZ derivatives conjecture

摘录 / EXCERPT: For $n\geq2$, let $D_{\mathcal{A}}^s$ be an n-th symmetric generalized Riemann differentiation. A symmetric MZ differentiation is one for which, for every function f, both the...

conjecture | open | MathDB 页面 | 原文线索

V2-068538 | MATHDB-356026

The symmetric Riemann vs. MZ derivatives conjecture

摘录 / EXCERPT: Let D_n^s be the n-th symmetric Riemann differentiation, and let a symmetric MZ differentiation be an n-th symmetric generalized Riemann differentiation satisfying the symmet...

conjecture | open | MathDB 页面 | 原文线索

V2-068539 | MATHDB-356027

The first updated symmetric Gaussian vs. MZ derivatives conjecture

摘录 / EXCERPT: Let a symmetric MZ differentiation be an n-th symmetric generalized Riemann differentiation satisfying the symmetric Peano equivalence condition, and let a scale of a symmetric G...

conjecture | open | MathDB 页面 | 原文线索

V2-068540 | MATHDB-356028

The second updated symmetric Gaussian vs. MZ derivatives conjecture

摘录 / EXCERPT: Let a symmetric MZ differentiation be an n-th symmetric generalized Riemann differentiation satisfying the symmetric Peano equivalence condition. Two generalized Riemann differen...

conjecture | open | MathDB 页面 | 原文线索

V2-068541 | MATHDB-356039

Ismail's characterization conjecture for orthogonal polynomial sequences on the Askey-Wilson lattice

摘录 / EXCERPT: Ismail's characterization conjecture. Then $(P_n)_{n\geq 0}$ are continuous q-Jacobi polynomials, Al-Salam-Chihara polynomials, or special or limiting cases of these families. Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-068542 | MATHDB-356042

The exact 3-cube densities of W_7, W_8, W_9, W_10, and W_12

摘录 / EXCERPT: Let W_7,W_8,W_9,W_{10}, and W_{12} be the configurations in Q_3 listed in Table, and let their lower bounds there be the corresponding values of their 3-cube densities. The e...

conjecture | open | MathDB 页面 | 原文线索

V2-068543 | MATHDB-356043

The classification of d-cube densities above 5/8

摘录 / EXCERPT: Let H be a configuration in Q_d, and let $\lambda(H,d)$ denote its d-cube density. Classification conjecture above $5/8$. If ... then either H is layered and...

conjecture | open | MathDB 页面 | 原文线索

V2-068544 | MATHDB-356044

The exact density of E(d,i)

摘录 / EXCERPT: Let d and i be positive integers with $1\leq i<d$. Define ... Let $i(K_{i,d-i})$ denote the inducibility of the complete bipartite graph $K_{i,d-i}$. E(d,i) density conjecture. E...

conjecture | open | MathDB 页面 | 原文线索

V2-068545 | MATHDB-356045

Garg and Sharma's permutation trinomial conjecture over fields of characteristic two

摘录 / EXCERPT: Let m be a positive integer and define ... A polynomial over a finite field is a permutation polynomial if it induces a one-to-one map from the field to itself. Garg-Sharma's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-068546 | MATHDB-356048

Shokurov's index conjecture for singularities

摘录 / EXCERPT: Shokurov's index conjecture. There exists a positive integer $r(n,a)$ such that, whenever $\operatorname{mld}_p(X)=a$, the Cartier index of K_X at p is at most $r(n,a)$.

conjecture | open | MathDB 页面 | 原文线索

V2-068547 | MATHDB-356049

Shokurov's Gorensteinness conjecture for quotient singularities

摘录 / EXCERPT: Shokurov's Gorensteinness conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-068548 | MATHDB-356052

Extension of the main theorem to degrees at least the characteristic

摘录 / EXCERPT: Extension conjecture. Theorem still holds for $d\geq p$.

conjecture | open | MathDB 页面 | 原文线索

V2-068549 | MATHDB-356057

Perron-number conjecture for growth rates of Coxeter systems of dimension at most two

摘录 / EXCERPT: Perron-number conjecture. The growth rate of any Coxeter system of dimension at most 2 is a Perron number.

conjecture | open | MathDB 页面 | 原文线索

V2-068550 | MATHDB-356058

Seymour's H-Hadwiger conjecture

摘录 / EXCERPT: Let H be a graph on t vertices. A graph has no H minor if H cannot be obtained from any of its subgraphs by contracting edges. Seymour's H-Hadwiger conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-068551 | MATHDB-356061

Uniqueness of very free curve components in large nef classes

摘录 / EXCERPT: Let X be a smooth Fano threefold. Write $\operatorname{Nef}_1(X)_{\mathbb{Z}}$ for the integral nef curve classes, and let $\operatorname{Mor}(\mathbb{P}^1,X,\alpha)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-068552 | MATHDB-356063

Łaba-Wang conjecture on spectral self-similar measures

摘录 / EXCERPT: Łaba-Wang conjecture. If $\mu_{\rho,\mathcal{D},{\bf p}}$ is a spectral measure, then ${\bf p}$ is equal-weight, so $p_j=1/m$; $\rho=1/N$ for some integer N; and there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-068553 | MATHDB-356072

Chern's coefficient-positivity conjecture for a double q-series

摘录 / EXCERPT: Chern's conjecture. This double series has nonnegative coefficients in its expansion. This would provide a uniform q-series proof of the parity-bias inequalities previously estab...

conjecture | open | MathDB 页面 | 原文线索

V2-068554 | MATHDB-356073

Chern's inner-series coefficient-positivity conjecture

摘录 / EXCERPT: Chern's conjecture. This q-series has nonnegative coefficients in its expansion. The source presents this as a sufficient condition for the preceding double-series conjecture aft...

conjecture | open | MathDB 页面 | 原文线索

V2-068555 | MATHDB-356099

Linear Hadwiger's conjecture

摘录 / EXCERPT: Let K_t be the complete graph on t vertices, and call a graph K_t-minor-free if it has no K_t minor. A graph is k-colourable if it has a proper colouring using at most...

conjecture | open | MathDB 页面 | 原文线索

V2-068556 | MATHDB-356101

Refined list Hadwiger's conjecture

摘录 / EXCERPT: Let λ be a multiset of positive integers, let k_λ be the sum of its elements, and let $|\lambda|$ be its number of elements. For a graph G, define $h(\lambda)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-068557 | MATHDB-356103

Conjecture on zeroes of functions in the evaluation space

摘录 / EXCERPT: Zero-count conjecture. Any function $f\in V$ will have fewer than

conjecture | open | MathDB 页面 | 原文线索

V2-068558 | MATHDB-356138

The discrete periodic tiling conjecture

摘录 / EXCERPT: Let $G=(G,+)$ be a finitely generated discrete abelian group. For subsets $A,F\subseteq G$, write $A\oplus F=G$ when every element of G has a unique representation $a+f$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-068559 | MATHDB-356140

The multiple periodic tiling conjecture

摘录 / EXCERPT: Let $G=(G,+)$ be a finitely generated discrete abelian group, and let $F_1,\dots,F_M$ be finite non-empty subsets of G. For each m, a set $A\subseteq G$ solves the tiling equat...

conjecture | open | MathDB 页面 | 原文线索

V2-068560 | MATHDB-356146

Singularity-free conjecture for torus one-point conformal blocks

摘录 / EXCERPT: Let $H^{(N)}_{\Delta}$ denote the contribution at order N to the torus one-point conformal block, expressed in terms of the central charge c. Singularity-free conjecture. At ea...

conjecture | open | MathDB 页面 | 原文线索

V2-068561 | MATHDB-356165

The Motzkin enumeration conjecture for good tableaux

摘录 / EXCERPT: Let T be a standard tableau, and call T good when every permutation w satisfying $\operatorname{Q}(w)=T$ is BBS good, equivalently when T is the recording tableau of some g...

conjecture | open | MathDB 页面 | 原文线索

V2-068562 | MATHDB-356175

The reduced partition-function conjecture for compressed partitions

摘录 / EXCERPT: Reduced partition-function conjecture. The partition function of this mTAZRP chain is reduced.

conjecture | open | MathDB 页面 | 原文线索

V2-068563 | MATHDB-356176

The extension-weight conjecture for strict partitions

摘录 / EXCERPT: Let $\tilde{\boldsymbol{ői}}$ be a strict partition, let $n\in\mathbb{N}$, and let $\sigma\in\operatorname{Tab}(\tilde{\boldsymbol{ői}}^{\mathrm{c}},n)$ be a filling. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068564 | MATHDB-356192

Plurality is Stablest Conjecture

摘录 / EXCERPT: Let n be large, let $k\geq 3$ be the number of candidates, and let voters cast independent uniformly random votes. A voting method is balanced when each candidate has equal proba...

conjecture | open | MathDB 页面 | 原文线索

V2-068565 | MATHDB-356193

Borda count is Stablest Conjecture

摘录 / EXCERPT: Let S_k be the set of rankings of k candidates, and let $f\colon S_k^n\to\Delta_k$ be a ranked-choice voting method, where Δ_k is the simplex of randomized outcomes on t...

conjecture | open | MathDB 页面 | 原文线索

V2-068566 | MATHDB-356194

Standard Simplex Conjecture for ranked choice voting

摘录 / EXCERPT: Let $\rho\in(0,1)$ and $n\geq k!+k-1$. Consider partitions $\Omega_1,\ldots,\Omega_k$ of $\mathbb{R}^n$ with Gaussian measure $\gamma_n(\Omega_i)=1/k$ that maximize the ranked-choi...

conjecture | open | MathDB 页面 | 原文线索

V2-068567 | MATHDB-356195

Plurality is stablest conjecture, discrete version

摘录 / EXCERPT: Let $m\geq 2$, let $\rho\in[0,1]$, and let $\varepsilon>0$. For a function $f:\{1,\ldots,m\}^n\to\Delta_m$, write $\operatorname{Inf}_i(f_j)$ for the influence of coordinate i on...

conjecture | open | MathDB 页面 | 原文线索

V2-068568 | MATHDB-356213

Eulerian-statistics equidistribution conjecture for two pattern classes

摘录 / EXCERPT: For a permutation π, let $\mathsf{des}(\pi)$ and $\mathsf{asc}(\pi)$ denote its numbers of descents and ascents, and let $\mathsf{ides}(\pi)=\mathsf{des}(\pi^{-1})$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068569 | MATHDB-356218

Eventual rank unimodality conjecture

摘录 / EXCERPT: Let $N(m,n)$ denote the number of partitions of n with rank m. Eventual rank unimodality conjecture. For $n\geq39$ and $0\leq m\leq n-3$, one has ... By the symmetry...

conjecture | open | MathDB 页面 | 原文线索

V2-068570 | MATHDB-356220

Conjecture on the inverse matrix of the degenerate LGF $L_f(1,0,2,1;q)$

摘录 / EXCERPT: Inverse-matrix conjecture. For $1\leq k\leq n$, the entries satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-068571 | MATHDB-356221

The optimal OGF conjecture for cross-correlation with a kernel

摘录 / EXCERPT: Optimal-OGF conjecture. An optimal ordinary generating function maximizing those correlation coefficients is

conjecture | open | MathDB 页面 | 原文线索

V2-068572 | MATHDB-356222

The Euler-transform expansion conjecture for the optimal OGF

摘录 / EXCERPT: Euler-transform optimality conjecture. An optimal selection of $\mathcal{C}(q)$ has the expansions

conjecture | open | MathDB 页面 | 原文线索

V2-068573 | MATHDB-356223

The cross-correlation equivalence conjecture for convolution kernels

摘录 / EXCERPT: Equivalence conjecture. The statistic $\operatorname{Corr}_1(\mathcal{C},\mathcal{D})$ is maximized or minimized over all possible $\mathcal{C}(q)$ if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-068574 | MATHDB-356230

Nested CLE scaling-limit conjecture for the fuzzy Potts model

摘录 / EXCERPT: Let $q\in (0,4]$ and let ... Suppose that D is a Jordan domain, $\epsilon_n\to 0$ as $n\to\infty$, and that $\mathcal{D}_n$ is a discrete domain in $\epsilon_n\mathbb{Z}^2$ for eac...

conjecture | open | MathDB 页面 | 原文线索

V2-068575 | MATHDB-356237

Polarization independence of local BPS invariants

摘录 / EXCERPT: Let S be a Gorenstein surface with ample anticanonical sheaf, let D be a polarization of S, and let n^D_γ be the local BPS invariant associated with a curve class...

conjecture | open | MathDB 页面 | 原文线索

V2-068576 | MATHDB-356238

The genus-zero multiple-cover formula for local snc del Pezzo surfaces

摘录 / EXCERPT: Let S be a rank 2 local snc del Pezzo surface, let D be a polarization, and let N^0_γ and $n^{0,D}_\gamma$ denote its local genus-zero Gromov-Witten and local genus-ze...

conjecture | open | MathDB 页面 | 原文线索

V2-068577 | MATHDB-356245

Pilgrim-Thurston conjecture on strict decrease of conformal energy

摘录 / EXCERPT: Let $f\colon (S^2,A)\righttoleftarrow$ be a post-critically finite hyperbolic-type branched covering, and let $\overline{E}^p(f)$ denote its asymptotic p-conformal energy for...

conjecture | open | MathDB 页面 | 原文线索

V2-068578 | MATHDB-356246

Conjecture on conformal dimensions under decomposition by multicurves

摘录 / EXCERPT: Suppose $f\colon (S^2,A)\righttoleftarrow$ is a post-critically finite hyperbolic-type branched covering without a Levy cycle. Let Γ be a completely f-invariant multicurve...

conjecture | open | MathDB 页面 | 原文线索

V2-068579 | MATHDB-356250

The graduated-order conjecture for the order $H_{n,\lambda}$

摘录 / EXCERPT: Graduated-order conjecture. The order $H_{n,\lambda}$, represented as a matrix order using the standard basis of $S_\lambda(V)$, is a graduated order: there exists a matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-068580 | MATHDB-356253

Finite positivity conjecture for copartition generating functions

摘录 / EXCERPT: Let $a,b,m,N,M\in\mathbb{N}$, and let $(x;q)_k$ denote the finite q-Pochhammer symbol. Finite positivity conjecture. If $b\mid a$ and $a+b=m$, then the q-series ... has non-negat...

conjecture | open | MathDB 页面 | 原文线索

V2-068581 | MATHDB-356260

The Montgomery family quartic oscillator eigenvalue critical-point conjecture

摘录 / EXCERPT: Let $\lambda_j(\alpha)$ denote the j-th eigenvalue of the Montgomery family of quartic oscillators, for $j\geq 1$ and $\alpha\in\mathbb R$. A critical point $\alpha_{j,c}$ satisf...

conjecture | open | MathDB 页面 | 原文线索

V2-068582 | MATHDB-356268

The log canonical center realization conjecture for Du Bois subsets

摘录 / EXCERPT: Let (X, Δ) be a quasi-projective, log canonical pair of dimension n. Let $V\subset X$ be a closed subset such that ... and suppose that V contains all log canonical center...

open_problem | open | MathDB 页面 | 原文线索

V2-068583 | MATHDB-356273

The irregular Riemann-Hilbert isomorphism conjecture for homogeneous elements

摘录 / EXCERPT: Irregular Riemann-Hilbert isomorphism conjecture. For any reductive G over $\mathbb{C}$ and homogeneous element ψ in $\mathfrak{g}((t))$ of slope $\nu>0$, and any...

conjecture | open | MathDB 页面 | 原文线索

V2-068584 | MATHDB-356279

Sun-Wang-Yao transformation conjecture for strongly graceful trees

摘录 / EXCERPT: Let T be a tree with a perfect matching. An adding-edge-subtracting dual graph transformation replaces an edge $u_{k-1}v_{k-1}\in E(H_{k-1})$ by an edge $x_ky_k\notin E(H_{k-1})$...

conjecture | open | MathDB 页面 | 原文线索

V2-068585 | MATHDB-356282

Wang's equitable total coloring conjecture

摘录 / EXCERPT: For a graph G, let $\Delta(G)$ denote its maximum degree, and let $\chi\,"_e(G)$ denote its equitable total chromatic number, namely the least number of colors in an equitable to...

conjecture | open | MathDB 页面 | 原文线索

V2-068586 | MATHDB-356293

Jiang-Liu-Shahidi wavefront bound conjecture for Arthur packets

摘录 / EXCERPT: Let ψ be an Arthur parameter for a reductive group $\mathbf G$ over a nonarchimedean local field ${\mathsf k}$, and let $\mathcal O^{\vee}_{\psi}$ be the nilpotent $G^{\vee}$-...

conjecture | open | MathDB 页面 | 原文线索

V2-068587 | MATHDB-356311

Large-sample convergence conjecture for infinitesimal gradient boosting

摘录 / EXCERPT: Let $(\hat{F}_t)_{t\geq 0}$ be the infinitesimal gradient boosting trajectory, let $\beta\geq 0$ be the tree-randomization parameter, let $\mathcal{P}_d(F^*)$ denote the projection...

conjecture | open | MathDB 页面 | 原文线索

V2-068588 | MATHDB-356318

Six-aggregation conjecture for three quadratics under PDLC

摘录 / EXCERPT: Let S be a set described by three quadratic inequalities, and assume the inequalities satisfy PDLC. A six-aggregation conjecture asserts that there exists such an S for which i...

conjecture | open | MathDB 页面 | 原文线索

V2-068589 | MATHDB-356322

Classification of conic divisorial ideals of toric rings associated to posets

摘录 / EXCERPT: Let P be a poset, let $\widehat{P}$ be its augmented poset, let $\mathcal{Q}(\widehat{P})$ denote the set of maximal chains of $\widehat{P}$, and let $\mathfrak{C}_P$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068590 | MATHDB-356323

Directed KKL inequality

摘录 / EXCERPT: Directed KKL inequality. Given a Boolean function $f\colon\{0,1\}^n\to\{0,1\}$, there exists $i\in[n]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-068591 | MATHDB-356327

The character identity conjecture for disconnected groups

摘录 / EXCERPT: Let $\phi:L_F\to{}^LG$ be a tempered Langlands parameter, let $\tilde s\in\tilde S_\phi^{[z]}$ be semisimple, and let $\mathfrak e$ be the associated endoscopic datum. Choose a z...

conjecture | open | MathDB 页面 | 原文线索

V2-068592 | MATHDB-356329

Functoriality conjecture for the local Langlands correspondence under automorphisms

摘录 / EXCERPT: Let A act on the rigid inner twists of a connected reductive group G, on tempered refined Langlands parameters, and on the set Π_{temp} of tempered representations...

conjecture | open | MathDB 页面 | 原文线索

V2-068593 | MATHDB-356349

The conjecture on subgroup perfect codes in connected Cayley sum graphs of symmetric groups

摘录 / EXCERPT: Let $n\geq 3$, and let a Cayley sum graph of the symmetric group S_n be connected. A subgroup perfect code is a subgroup of S_n that is a perfect code in this graph. Subgroup p...

conjecture | open | MathDB 页面 | 原文线索

V2-068594 | MATHDB-356350

The conjecture on subgroup total perfect codes in connected Cayley sum graphs of symmetric groups

摘录 / EXCERPT: Let $n\geq 3$, and let a Cayley sum graph of the symmetric group S_n be connected. A subgroup total perfect code is a subgroup of S_n that is a total perfect code in this graph...

conjecture | open | MathDB 页面 | 原文线索

V2-068595 | MATHDB-356358

The infinite virtual connect-sum conjecture

摘录 / EXCERPT: Infinite virtual connect-sum conjecture. There are infinitely many distinct virtual links that decompose as a connect sum of V_1 and V_2.

conjecture | open | MathDB 页面 | 原文线索

V2-068596 | MATHDB-356360

The maximum VC-dimension conjecture for quadratic residues

摘录 / EXCERPT: Maximum VC-dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068597 | MATHDB-356361

The maximum VC-dimension conjecture for bounded-index power subgroups

摘录 / EXCERPT: Maximum VC-dimension conjecture for power subgroups. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068598 | MATHDB-356363

The sharp random-graph threshold conjecture for friends-and-strangers graphs

摘录 / EXCERPT: Let $X,Y\sim\mathcal{G}(n,p)$ denote independent Erdős-Rényi random graphs on n vertices, and let $X,Y\sim\mathcal{G}(K_{n,n},p)$ denote independent random bipartite graphs with...

conjecture | open | MathDB 页面 | 原文线索

V2-068599 | MATHDB-356364

The bridge characterization conjecture for double-multiplicity friends-and-strangers graphs

摘录 / EXCERPT: Let X be a connected simple graph, and let $c^{(X)}$ and $c^{(\mathsf{Star}_n)}$ be multiplicity lists, with equal total capacity. Let the center of Star_n have multip...

conjecture | open | MathDB 页面 | 原文线索

V2-068600 | MATHDB-356370

Inagaki-Tamura shift identity conjecture for partition inequalities

摘录 / EXCERPT: Inagaki-Tamura shift identity conjecture. If $d\geq 12$ and $n\geq d+2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068601 | MATHDB-356371

Higher-dimensional density conjecture for Hessian-Schatten variation

摘录 / EXCERPT: Let $n\geq 2$, let $\Omega=(0,1)^n$, and consider functions with bounded Hessian--Schatten variation, together with the class $\operatorname{CPWL}(\Omega)$ of continuous piecewise-...

conjecture | open | MathDB 页面 | 原文线索

V2-068602 | MATHDB-356374

Completeness and discreteness conjecture for the d-categorical nerve of a pasting shape

摘录 / EXCERPT: Nerve completeness conjecture. The d-categorical nerve I is a complete and discrete d-fold Segal space.

conjecture | open | MathDB 页面 | 原文线索

V2-068603 | MATHDB-356388

The formal scalelessness conjecture for monomials in Feynman integral families

摘录 / EXCERPT: Let $I(z_1,\ldots,z_n)$ be an L-loop integral with nonnegative indices, and let $D_1,\ldots,D_n$ be the propagator generators in the associated noncommutative rational double-shi...

conjecture | open | MathDB 页面 | 原文线索

V2-068604 | MATHDB-356400

Havet-van den Heuvel-McDiarmid-Reed planar square list-coloring conjecture

摘录 / EXCERPT: Let G be a planar graph. Havet-van den Heuvel-McDiarmid-Reed conjecture. ... The conjecture was disproved in 2022 by Hasanvand, although it remains open for graphs with maximum deg...

conjecture | open | MathDB 页面 | 原文线索

V2-068605 | MATHDB-356401

Havet-van den Heuvel-McDiarmid-Reed conjecture for nice graph classes

摘录 / EXCERPT: A graph class $\mathcal{G}$ is nice if it is minor-closed and does not contain $K_{3,t}$ for some positive integer t. Havet-van den Heuvel-McDiarmid-Reed conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-068606 | MATHDB-356402

Wang-Lih planar high-girth square-coloring conjecture

摘录 / EXCERPT: Let G be a planar graph of girth at least k, and let Δ be its maximum degree. Wang-Lih conjecture. For every $k\ge 5$, there exists Δ_k such that, if...

conjecture | open | MathDB 页面 | 原文线索

V2-068607 | MATHDB-356403

Dvořák-Kráľ-Nejedlý-Škrekovski girth-five planar square-coloring conjecture

摘录 / EXCERPT: Let G be a planar graph of girth at least 5, and let Δ be its maximum degree. Dvořák-Kráľ-Nejedlý-Škrekovski conjecture. There exists Δ_0 such that, if...

conjecture | open | MathDB 页面 | 原文线索

V2-068608 | MATHDB-356407

Griggs-Yeh L(2,1)-labeling conjecture

摘录 / EXCERPT: Let G be a graph with maximum degree Δ, and let $\lambda^{2,1}(G)$ be the minimum span of an $L(2,1)$-labeling, in which vertices at distance one receive labels differing...

conjecture | open | MathDB 页面 | 原文线索

V2-068609 | MATHDB-356408

Non differentiability at the energy wells

摘录 / EXCERPT: Let $X\subset[0,1]\cap\mathbb Q$ be such that ... for all z, and let $\theta^\ast\in X$ be an energy well satisfying ... for some z. Non differentiability at the energy wells. The...

conjecture | open | MathDB 页面 | 原文线索

V2-068610 | MATHDB-356410

Conjecture on largest minimum distances of binary self-orthogonal codes of dimension 5

摘录 / EXCERPT: Let $d(n,k)$ denote the largest minimum distance of a binary linear $[n,k]$ code, and let $d_{so}(n,k)$ denote the largest minimum distance of a binary self-orthogonal $[n,k]$ code...

conjecture | open | MathDB 页面 | 原文线索

V2-068611 | MATHDB-356418

The bi-LFS/bi-UFS equivalence conjecture for positive semidomains

摘录 / EXCERPT: Let S be a positive semidomain. It is a bi-LFS if both $(S,+)$ and $S^\bullet$ are length-factorial monoids (LFMs), and a bi-UFS if both $(S,+)$ and $S^\bullet$ are unique-factor...

conjecture | open | MathDB 页面 | 原文线索

V2-068612 | MATHDB-356419

Waldhausen S-construction conjecture for microlocalization

摘录 / EXCERPT: Waldhausen S-construction conjecture. The relative Waldhausen S-construction of m_Λ^l is the simplicial ∞-category

conjecture | open | MathDB 页面 | 原文线索

V2-068613 | MATHDB-356422

Cofinality conjecture for products of boundary Dehn twists

摘录 / EXCERPT: Let $g\geq 2$ and let $b_1,\ldots,b_n$ be curves isotopic to the boundary components of the surface Σ_g^n. For positive exponents $k_1,\ldots,k_n$, consider the product ... He...

conjecture | open | MathDB 页面 | 原文线索

V2-068614 | MATHDB-356427

Simple connectivity of limits of universal covers of RCD spaces

摘录 / EXCERPT: Let $(X_i,d_i,\mathfrak{m}_i)$ be a sequence of $RCD^{\ast}(K,N;D)$ spaces, and let $(\widetilde{X}_i,\widetilde{d}_i,\widetilde{\mathfrak{m}}_i)$ be their universal covers. Assume...

open_problem | open | MathDB 页面 | 原文线索

V2-068615 | MATHDB-356429

Mirror correspondence for class TG \Q-Fano threefolds

摘录 / EXCERPT: Mirror correspondence. There is a bijective correspondence between qG-deformation families of Q-Fano threefolds X of class TG and mutation-equivalence classes of rigid maximally...

conjecture | open | MathDB 页面 | 原文线索

V2-068616 | MATHDB-356443

Weak concentration conjecture for deletion-channel trace ambiguity

摘录 / EXCERPT: Concentration conjecture, weak form. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-068617 | MATHDB-356446

Asymptotic conjecture for s-distance sets in Johnson spaces

摘录 / EXCERPT: Let $\mathcal{J}^{n,w}$ be the Johnson space of w-subsets of an n-element set, and let $A(\mathcal{J}^{n,w},s)$ denote the maximum size of an s-distance set in this space. Jo...

conjecture | open | MathDB 页面 | 原文线索

V2-068618 | MATHDB-356447

The vanishing conjecture for the classes $B^m_{g,\overline{d}}$

摘录 / EXCERPT: Vanishing conjecture for $B^m_{g,\overline{d}}$. For any $m\ge 2$, $g\ge 0$, $n\ge 1$, and $\overline{d}$ such that $\sum d_i\ge 2g+m-1$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-068619 | MATHDB-356448

The identification conjecture for $B^1_{g,\overline{d}}$ and $A^1_{g,\overline{d}}$

摘录 / EXCERPT: Identification conjecture. For any such g, n, and $\overline{d}$,

conjecture | open | MathDB 页面 | 原文线索

V2-068620 | MATHDB-356449

The identification conjecture for $B^0_{g,\overline{d}}$ and $A^0_{g,\overline{d}}$

摘录 / EXCERPT: Identification conjecture. For any such g, n, and $\overline{d}$,

conjecture | open | MathDB 页面 | 原文线索

V2-068621 | MATHDB-356450

Shareshian-Wachs q-analogue of the Stanley-Stembridge conjecture

摘录 / EXCERPT: Shareshian-Wachs conjecture. If G is the incomparability graph of a natural unit interval order, then $X_G(\mathbf{x},q)$ is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-068622 | MATHDB-356457

Conjecture on shrinking stability domains for fractional-order Mandelbrot sets

摘录 / EXCERPT: Let $q<0.5$, let S^q be the stability domain of the fixed points of the fractional-order Mandelbrot set (FOM), and let the main body of the associated fractal set be understood i...

conjecture | open | MathDB 页面 | 原文线索

V2-068623 | MATHDB-356467

The transverse-distance conjecture for the E_n D3-brane probe

摘录 / EXCERPT: Transverse-distance conjecture. The transverse distance between the D3-brane probe and the E_n 7-branes is proportional to $\operatorname{vol}(V_2)$ in the local Calabi-Yau three...

conjecture | open | MathDB 页面 | 原文线索

V2-068624 | MATHDB-356475

Quintuple equidistribution conjecture for Euler-Stirling statistics on permutations

摘录 / EXCERPT: Quintuple equidistribution conjecture. There is a bijection $\Omega:\mathsf{I}_n\to\mathsf{I}_n$ such that, for every $s\in\mathsf{I}_n$,

conjecture | open | MathDB 页面 | 原文线索

V2-068625 | MATHDB-356476

Quadruple equidistribution conjecture for ascent-sequence statistics

摘录 / EXCERPT: Quadruple equidistribution conjecture. For every $n\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068626 | MATHDB-356484

Equivalence of Proca Hadamard-state definitions on globally hyperbolic spacetimes

摘录 / EXCERPT: Let a globally hyperbolic spacetime be given, and consider the two Proca Hadamard-state conditions referred to as Definition and Definition. Equivalence conjecture. The two definit...

conjecture | open | MathDB 页面 | 原文线索

V2-068627 | MATHDB-356490

The palindromic surjective-strategy conjecture for wreath products

摘录 / EXCERPT: Let $G\wr H$ be a wreath product, and call a surjective strategy palindromic when its sequence of moves $S'=\{k'_i\in K\}_{i=1}^N$ satisfies $k'_i=k'_{N-i+1}$ for every i. Palind...

conjecture | open | MathDB 页面 | 原文线索

V2-068628 | MATHDB-356495

Christol's conjecture on diagonals of rational functions

摘录 / EXCERPT: Let $A(t)=\sum_{n\ge 0} a_n t^n\in\mathbb Z[[t]]$, suppose that $|a_n|<c^n$ for all $n\in\mathbb N$ and some $c>0$, and suppose that A is D-finite. Christol's conjecture. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-068629 | MATHDB-356501

Poisson conjecture for the distribution of primes in short intervals

摘录 / EXCERPT: Poisson conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-068630 | MATHDB-356502

Extended exponential tail bound for primes in short intervals

摘录 / EXCERPT: Extended tail-bound conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-068631 | MATHDB-356527

Conjecture on uniform convergence under weak competition and fear

摘录 / EXCERPT: Let (u,v) solve the reaction diffusion system referenced as ... . Suppose ... and the parameters satisfy the restrictions in ... . Uniform-convergence conjecture. The solution (u,v)...

conjecture | open | MathDB 页面 | 原文线索

V2-068632 | MATHDB-356528

Linear independence conjecture for iterated Mazur-pattern knots

摘录 / EXCERPT: Let Q be the Mazur pattern, let 4_1 denote the figure eight knot, and let $Q^n(4_1)$ denote the knot obtained by applying Q iteratively n times to 4_1. Let $cmathcal{C}$...

conjecture | open | MathDB 页面 | 原文线索

V2-068633 | MATHDB-356551

Abreu-Goedgebeur-Labbate-Mazzuoccolo isomorphic linear-forest conjecture

摘录 / EXCERPT: Let G be a cubic graph. A linear forest is a forest whose components are paths, and a two-colouring of $V(G)$ induces one subgraph on each colour class. Abreu-Goedgebeur-Labbate-...

conjecture | open | MathDB 页面 | 原文线索

V2-068634 | MATHDB-356557

Bloom and Sagan's conjecture on large symmetrically avoided permutation sets

摘录 / EXCERPT: Bloom and Sagan's conjecture. For each $p\geq 3$, there exists a $K=K(p)$ such that, if $|\Pi|=p$ and $\Pi\subseteq S_k$ for some $k\geq K$, then Π is not symmetrically avoided...

conjecture | open | MathDB 页面 | 原文线索

V2-068635 | MATHDB-356561

Tightness of the quadratic Gromov-Wasserstein correspondence-plan theorem

摘录 / EXCERPT: Tightness conjecture. Theorem is tight: there exist μ and ν for which optimal correspondence plans for the quadratic Gromov-Wasserstein problem are not maps, but are instea...

conjecture | open | MathDB 页面 | 原文线索

V2-068636 | MATHDB-356563

Monotonicity conjecture for the number of rainbow trees

摘录 / EXCERPT: Let $k,c,c'\in\mathbb{N}$ with $c'\leq c$, and let G be a graph. For $r\in\{c,c'\}$, let $T(r,k)$ be the random variable counting the number of rainbow trees of order k when th...

conjecture | open | MathDB 页面 | 原文线索

V2-068637 | MATHDB-356565

Classification conjecture for finite-mass entire Keller-Segel solutions

摘录 / EXCERPT: Let (u,v) be an entire solution of the Keller-Segel system on $\mathbb{R}^2\times\mathbb{R}$, meaning that u and ∇v are smooth and satisfy the system, with the second e...

conjecture | open | MathDB 页面 | 原文线索

V2-068638 | MATHDB-356578

Bounded path covering with forests number for non-crossing shortest paths

摘录 / EXCERPT: Let G be a plane graph, and let P be a set of non-crossing single-touch shortest paths in G. The path covering with forests number of P, denoted by $\operatorname{PCFN}(P)$...

conjecture | open | MathDB 页面 | 原文线索

V2-068639 | MATHDB-356586

Bayer-Macrì-Toda generalized Bogomolov-Gieseker inequality for tilt-stable objects

摘录 / EXCERPT: Let (X,H) be a polarized smooth projective 3-fold. For $b\in\mathbb{R}$ and $a>0$, let $\sigma_{a,b}=(\mathcal{A}^b,Z_{a,b})$ be the tilt-stability condition, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068640 | MATHDB-356591

Sufficiency of the Hautus criterion for exact controllability of difference delay equations

摘录 / EXCERPT: Let $\Lambda_N>0$, $d\in\mathbb{N}$, $q\in[1,+\infty)$, and consider the linear difference delay system with delay parameters and matrices A_N and B described above. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068641 | MATHDB-356592

The list-coloring conjecture for squares of planar graphs of maximum degree four

摘录 / EXCERPT: List-coloring conjecture. For every planar graph G satisfying $\Delta(G) \le 4$,

conjecture | open | MathDB 页面 | 原文线索

V2-068642 | MATHDB-356599

Monotonicity conjecture for the point-evaluation constants

摘录 / EXCERPT: Let $\mathscr{C}_p$ denote the point-evaluation constant associated with the Paley-Wiener space PW^p. Monotonicity conjecture. The function ... is strictly decreasing on...

conjecture | open | MathDB 页面 | 原文线索

V2-068643 | MATHDB-356600

Korevaar-type conjecture for imaginary-axis point evaluation

摘录 / EXCERPT: Fix $0<p<\infty$, and let $\mathscr{K}_p$ be the best constant in Korevaar's inequality for point evaluation on the imaginary axis, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-068644 | MATHDB-356613

The non-isotrivial elliptic-surface rank conjecture

摘录 / EXCERPT: Non-isotrivial elliptic-surface rank conjecture. If π is non-isotrivial, then both $\mathscr{J}(X,\pi)$ and $\mathscr{N}(X,\pi)$ are infinite.

conjecture | open | MathDB 页面 | 原文线索

V2-068645 | MATHDB-356614

The rank-zero fibre conjecture for non-isotrivial elliptic surfaces

摘录 / EXCERPT: Rank-zero fibre conjecture. There are infinitely many $b\in\mathbb{P}^1(\mathbb{Q})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-068646 | MATHDB-356623

Clark and Ismail's positivity conjecture for derivatives of a quotient

摘录 / EXCERPT: Let $n\in\mathbb{N}$ and define the function $x\mapsto x^n/(1-e^{-x})$ on $(0,\infty)$. Clark and Ismail's positivity conjecture. For all $x\in(0,\infty)$ and $n\in\mathbb{N}$, ... T...

conjecture | refuted | MathDB 页面 | 原文线索

V2-068647 | MATHDB-356628

Independent sums conjecture for TMXY-UI

摘录 / EXCERPT: TMXY-UI independent sums conjecture. TMXY-UI has the independent sums property. The paper introduces TMXY-UI as a unique-information measure obtained by optimization under Markov-r...

conjecture | open | MathDB 页面 | 原文线索

V2-068648 | MATHDB-356629

Stability and quenching-profile conjectures for self-similar solutions of the MEMS wave equation

摘录 / EXCERPT: Stability and quenching-profile conjecture. The following statements are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-068649 | MATHDB-356630

Gessel's equidistribution conjecture for permutations with disjoint cycle types

摘录 / EXCERPT: Let $\mu\vdash m$ and $\nu\vdash n$ be integer partitions with no common part. Let π and σ be permutations of cycle types μ and ν, respectively, with disjoint su...

conjecture | open | MathDB 页面 | 原文线索

V2-068650 | MATHDB-356641

Numerical determination conjecture for partial Okounkov bodies of locally symmetric spaces

摘录 / EXCERPT: Partial Okounkov body conjecture. The collection of $\Delta_{\nu}(\hat{E})$ for linear valuations ν determines the numerical class of E.

conjecture | open | MathDB 页面 | 原文线索

V2-068651 | MATHDB-356644

The conjectured max-domain-of-attraction limit for clustered Archimax copulas

摘录 / EXCERPT: Let $C_{\mathcal{G},\bm{\psi},\bm{\ell},Q}$ be a clustered Archimax copula satisfying the extended max-domain-of-attraction assumption and the standard stable tail dependence funct...

conjecture | open | MathDB 页面 | 原文线索

V2-068652 | MATHDB-356647

Real regulator conjecture for products of very general elliptic curves

摘录 / EXCERPT: Let $E_1,E_2,\ldots,E_n$ be very general complex elliptic curves and let ... For $1\leq m\leq n$, define ... The real regulator is the map $r_{k,1}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-068653 | MATHDB-356653

Merkulov-Willwacher's properadic correspondence conjecture for framed curves

摘录 / EXCERPT: Let $\Omega(\overline{\textsf N}^{fr})$ be the cobar construction of the properad of framed curves, let $\Omega(\textsf{Frob}^*)$ be the cobar construction of the dual Frobenius pr...

conjecture | open | MathDB 页面 | 原文线索

V2-068654 | MATHDB-356659

Schechtman's vector-balancing conjecture for zonotopes

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^d$ be a zonotope. For symmetric convex bodies, the vector balancing constant $\operatorname{vb}(K,K)$ is the least $r\geq0$ such that every finite sequenc...

open_problem | open | MathDB 页面 | 原文线索

V2-068655 | MATHDB-356663

The minimum-eigenvalue conjecture for the two-way Gaussian channel objective

摘录 / EXCERPT: Let ${\bf F}_1$ and ${\bf F}_2$ be the linear encoding matrices, with ${\bf F}_2$ of the form specified in. Let α be the objective-weight parameter, let σ_1^2 and...

conjecture | open | MathDB 页面 | 原文线索

V2-068656 | MATHDB-356672

Directed Sidorenko conjecture for oriented graphs

摘录 / EXCERPT: Directed Sidorenko conjecture. If B is a bipartite oriented graph with a homomorphism $B\to\vec K_2$, then B has the directed Sidorenko property. This is a directed analogue of...

conjecture | open | MathDB 页面 | 原文线索

V2-068657 | MATHDB-356678

UV-to-IR line-operator categorification conjecture

摘录 / EXCERPT: Let $C(S,S):=\operatorname{Hom}_{\mathrm{Ft}(M,\Omega_I)}(S,S)$ and $C(S,F_b)$ be the corresponding Fueter Hom-categories, with the functor $RG_b(\zeta):C(S,S)\to C(S,F_b)$ induced...

conjecture | open | MathDB 页面 | 原文线索

V2-068658 | MATHDB-356680

Sharpness conjecture for generic focusing flows

摘录 / EXCERPT: Let a focusing flow satisfy the hypotheses of the paper, and let the result in Theorem provide its $epsilon^{-8}$ lifespan bound and accompanying estimates. Sharpness conjecture fo...

conjecture | open | MathDB 页面 | 原文线索

V2-068659 | MATHDB-356685

Generalized Gan-Gross-Prasad relevance characterizes quotient branching

摘录 / EXCERPT: Let $pi in \mathrm{Irr}(G_{n+1})$ and $pi' in \mathrm{Irr}(G_n)$. A pair (pi,pi') is relevant if there exist multisegments $mathfrak m,mathfrak n$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-068660 | MATHDB-356690

Zero-multiplicity conjecture for the k-generalized Fibonacci sequence with non-positive indices

摘录 / EXCERPT: Let $H^{(k)}$ be the k-generalized Fibonacci sequence with non-positive indices, and let $\mathcal{Z}(H^{(k)})=\{n\in\mathbb{Z}:H_n^{(k)}=0\}$ denote its zero set. The zero-multi...

conjecture | open | MathDB 页面 | 原文线索

V2-068661 | MATHDB-356694

Higher-rank non-commutative main conjecture for \mathbb{G}_m

摘录 / EXCERPT: Let $\mathcal{K}_\infty/K$ and S be as in the source, put $\Sigma=\Sigma_S(\mathcal{K}_\infty)$ and $r=|\Sigma|$, and let $\Lambda(\mathcal{G}_\infty)$ be the relevant Iwasawa al...

conjecture | open | MathDB 页面 | 原文线索

V2-068662 | MATHDB-356695

Generalized Gross-Stark conjecture

摘录 / EXCERPT: Fix a subset $\Sigma\subseteq\Sigma_S(E)$ satisfying the source's condition, set $r=|\Sigma_S(\mathcal{K}_\infty)|$, $r'=|\Sigma|$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068663 | MATHDB-356698

Hyperellipticity conjecture for the families J_g(\lambda,1)

摘录 / EXCERPT: For $g\in\{3,4,5\}$, let $J_g(\lambda,1)$ denote the corresponding family of translation surfaces of genus g. A compact Riemann surface is hyperelliptic if it admits a degree-two...

conjecture | open | MathDB 页面 | 原文线索

V2-068664 | MATHDB-356706

Borodin-Kostochka-Woodall List Total Coloring Conjecture

摘录 / EXCERPT: Let G be a graph, and let $T(G)$ be its total graph, whose vertices are the vertices and edges of G, with adjacency given by adjacency, incidence, or edge adjacency in G. The...

conjecture | open | MathDB 页面 | 原文线索

V2-068665 | MATHDB-356707

Vizing-Gupta-Albertson-Collins-Bollobás-Harris line-graph List Coloring Conjecture

摘录 / EXCERPT: Let G be a graph and let $L(G)$ be its line graph, whose vertices are the edges of G, with two vertices adjacent when the corresponding edges share an endpoint. Write χ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-068666 | MATHDB-356708

Revised square-graph chromatic-choosability conjecture

摘录 / EXCERPT: Let G be a square graph, meaning a graph of the form H^2 for some graph H, and let $\Delta(G)$ denote its maximum degree. Write χ for chromatic number and χ_ℓ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-068667 | MATHDB-356709

Bounded list-chromatic gap under a maximum-degree threshold

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ be its maximum degree, and let $\chi(G)$ and $\chi_\ell(G)$ denote its chromatic and list chromatic numbers. Bounded-gap conjecture. For every n...

conjecture | open | MathDB 页面 | 原文线索

V2-068668 | MATHDB-356710

Half-maximum-degree chromatic-choosability conjecture

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ be its maximum degree, and let $\chi(G)$ and $\chi_\ell(G)$ denote its chromatic and list chromatic numbers. Half-maximum-degree conjecture. Eve...

conjecture | open | MathDB 页面 | 原文线索

V2-068669 | MATHDB-356711

Low-chromatic-number choosability conjecture

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ be its maximum degree, and let $\chi(G)$ denote its chromatic number. A graph is k-choosable when every assignment of lists of size k admits...

conjecture | open | MathDB 页面 | 原文线索

V2-068670 | MATHDB-356713

Relaxed avoiding-coloring threshold conjecture

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ be its maximum degree, let $\chi(G)$ be its chromatic number, and let $\bar{\chi}_m(G)$ denote the m-avoiding chromatic number used in the sou...

conjecture | open | MathDB 页面 | 原文线索

V2-068671 | MATHDB-356726

Logarithmic face conjecture for Erdős-Rényi random graphs

摘录 / EXCERPT: Erdős-Rényi face conjecture. The expected number of faces is

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-068672 | MATHDB-356728

Shifted Grothendieck positivity conjecture for skew products

摘录 / EXCERPT: Shifted Grothendieck positivity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068673 | MATHDB-356729

Shifted J-Grothendieck expansion conjecture

摘录 / EXCERPT: Shifted J-Grothendieck expansion conjecture. If λ is a strict partition, then

conjecture | open | MathDB 页面 | 原文线索

V2-068674 | MATHDB-356750

Generic-parameter extension of the hypergeometrization change-of-variable formula

摘录 / EXCERPT: Generic-parameter extension conjecture. The formula holds for generic values of $a,c\in\mathbb{C}$ and all nonzero $m\in\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-068675 | MATHDB-356751

Coboundary expansion conjecture for random balanced Cayley complexes

摘录 / EXCERPT: Coboundary expansion conjecture for random balanced Cayley complexes. For any fixed $k\geq 1$ there exist constants $C(k)<\infty$ and $\epsilon(k)>0$ such that for any group G, t...

conjecture | open | MathDB 页面 | 原文线索

V2-068676 | MATHDB-356752

Coboundary expansion conjecture for Ramanujan balanced Cayley complexes

摘录 / EXCERPT: Let p and q be distinct odd primes with $q>2\sqrt p$ and Legendre symbol $\left(\frac pq\right)=1$, let $G_q=\operatorname{PSL}_2(\mathbb{F}_q)$, and let $S_{p,q}\subset G_q$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-068677 | MATHDB-356753

Arboreal Sen Conjecture for PCF maps

摘录 / EXCERPT: Arboreal Sen Conjecture for PCF maps. If $p \mid d$, then the iterated extension

conjecture | open | MathDB 页面 | 原文线索

V2-068678 | MATHDB-356757

Limiting FDP and TPP conjecture for TPoP and CPoP

摘录 / EXCERPT: Limiting FDP and TPP conjecture. The limiting false discovery proportion and true positive proportion of TPoP satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-068679 | MATHDB-356759

Bell-Ho-Strichartz conjecture on the asymptotic Kusuoka-measure distribution

摘录 / EXCERPT: Bell-Ho-Strichartz conjecture. As $m\to\infty$, the law of $r\circ c^{(w)}$ under λ_m converges to the Dirac measure at $\sqrt{8/75}$, while the law of...

conjecture | open | MathDB 页面 | 原文线索

V2-068680 | MATHDB-356766

Odaka's CM minimization conjecture for polarized families

摘录 / EXCERPT: Odaka's CM minimization conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-068681 | MATHDB-356791

Strength-super edge-magic deficiency conjecture for trees

摘录 / EXCERPT: Let T be a nontrivial tree. A numbering of T is a bijection $f:V(T)\to\{1,2,\ldots,|V(T)|\}$, and its strength is ... The strength of T is ... where the minimum is over all numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-068682 | MATHDB-356793

Three-class conjecture for phased block-gluing Hom shifts

摘录 / EXCERPT: Let G be an undirected, simple, connected graph, with self-loops allowed, and let a two-dimensional Hom shift be the set of graph morphisms from $\mathbb{Z}^2$ to G. A Hom shif...

conjecture | open | MathDB 页面 | 原文线索

V2-068683 | MATHDB-356794

Three-class conjecture for gap functions of Hom shifts

摘录 / EXCERPT: Let G be an undirected, simple, connected graph, with self-loops allowed, and let a Hom shift be the set of graph morphisms from $\mathbb{Z}^d$ to G. Consider the gap function...

conjecture | open | MathDB 页面 | 原文线索

V2-068684 | MATHDB-356811

Beilinson-Deligne rationality conjecture for the fundamental line

摘录 / EXCERPT: Let $\mathcal{M}$ be the modular motive, let $\Delta(\mathcal{M})$ be its one-dimensional fundamental line over F, and assume the non-degeneracy conjecture for the Gillet-Soulé h...

conjecture | open | MathDB 页面 | 原文线索

V2-068685 | MATHDB-356812

Completed-L-function variant of the rationality conjecture

摘录 / EXCERPT: Let $\Delta(\mathcal{M})$ be the fundamental line, let $\theta_\infty:\Delta(\mathcal{M})_\infty\overset\simeq\longrightarrow F_\infty$ be the comparison isomorphism, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068686 | MATHDB-356813

Finiteness conjecture for Bloch-Kato Shafarevich-Tate groups

摘录 / EXCERPT: Let K be a number field and let $\text{Ш}^{\operatorname{BK}}(K,\mathcal{M})=\bigoplus_p\text{Ш}_p^{\operatorname{BK}}(K,\mathcal{M})$ be the Bloch-Kato Shafarevich-Tate group of...

conjecture | open | MathDB 页面 | 原文线索

V2-068687 | MATHDB-356818

Categorical compactness conjecture for conically smooth stratified spaces

摘录 / EXCERPT: Categorical compactness conjecture. If (X,P) is of finite stratified type, then (X,P) is also categorically compact.

conjecture | open | MathDB 页面 | 原文线索

V2-068688 | MATHDB-356820

Density hypothesis for Hecke eigenvalues

摘录 / EXCERPT: Density hypothesis, Hecke eigenvalue version. For every $l\geq 0$, $T\geq 1$, and $\delta>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-068689 | MATHDB-356824

Inf-sup stability conjecture for Scott-Vogelius elements on Freudenthal meshes

摘录 / EXCERPT: Let V_h be constructed with continuous Lagrange elements of degree k, and choose $\Pi_h={\nabla\cdot}\,}V_h$. The inf-sup condition is the existence of a constant $\beta>0$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-068690 | MATHDB-356827

The bound on the number of sign vectors associated with an input

摘录 / EXCERPT: Let n be a positive integer, let P_x be the region associated with a sign vector $x\in\{-1,+1\}^n$, and let (C,c) be a quadratic-programming input. The preceding proposition...

conjecture | open | MathDB 页面 | 原文线索

V2-068691 | MATHDB-356860

The e-positivity conjecture for chromatic quasisymmetric functions of indifference graphs

摘录 / EXCERPT: The e-positivity conjecture. For each $\gamma \in \mathcal{I}\mathcal{G}$, $X_{\gamma}(\mathbf{x};t)$ is e-positive; equivalently, $a_{\lambda}(t)\in\mathbb{Z}_{\geq 0}[t]$ for e...

conjecture | open | MathDB 页面 | 原文线索

V2-068692 | MATHDB-356875

Uniform short-time Fisher-information bound for regularized Stein variational flows

摘录 / EXCERPT: Let $\rho_0\in\mathcal{P}_V$, let S_r be the stated class of flows, and for $u\in S_r$ define the pushforward measure $\rho_t=u(t,\cdot)_{\#}\rho_0$. For u-regularized Stein F...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068693 | MATHDB-356884

The bow-tie version of the Local Irregularity Conjecture

摘录 / EXCERPT: Let G be a connected graph, let $\mathfrak{T'}$ be the family consisting of the recursively defined family $\mathfrak{T}$ together with all odd-length paths and odd-length cycles...

conjecture | open | MathDB 页面 | 原文线索

V2-068694 | MATHDB-356885

Syzygy shifting for canonical bundles on symmetric powers

摘录 / EXCERPT: Let C be a smooth complex projective curve of genus $g\geq 2$, let C_k denote its k-th symmetric power, and let K_C be the canonical bundle. Say that a line bundle L on...

conjecture | open | MathDB 页面 | 原文线索

V2-068695 | MATHDB-356886

Unimodality conjecture for rank polynomials of circular fence posets

摘录 / EXCERPT: Let $\bar{a}$ be a tuple of positive integers, and let $\overline{F}(\bar{a})$ be the corresponding circular fence poset with rank polynomial $\overline{R}(\bar{a};q)$. A polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-068696 | MATHDB-356889

Inverse zero-sum conjecture for short zero-sum sequences in rank-two groups

摘录 / EXCERPT: Let $n\geq 2$, let $G=C_n \oplus C_n$, let $k\in [0,n-1]$, and let S be a sequence of terms from G with length $|S|=\operatorname{D}(G)+k-1=2n-2+k$ having no nonempty zero-sum...

conjecture | open | MathDB 页面 | 原文线索

V2-068697 | MATHDB-356897

Cambie-Cranston list-packing conjecture

摘录 / EXCERPT: For a graph G, let $\chi_{\ell}(G)$ denote its list-chromatic number and let $\chi_{\ell}^*(G)$ denote its list packing number, the least k such that every k-assignment admit...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-068698 | MATHDB-356898

The entropy-KL-divergence strengthening of the union-closed sets conjecture

摘录 / EXCERPT: Let A and B be independent identically distributed samples from a distribution supported on a family of subsets of $[n]$. Write $H(A)$ for the entropy of A, and...

conjecture | open | MathDB 页面 | 原文线索

V2-068699 | MATHDB-356901

The sRC quotient structure conjecture for klt pairs

摘录 / EXCERPT: sRC quotient structure conjecture. Such an orbifold morphism with all the stated properties exists.

conjecture | open | MathDB 页面 | 原文线索

V2-068700 | MATHDB-356902

The boundedness of the index conjecture for klt log Calabi-Yau pairs

摘录 / EXCERPT: Let d be a positive integer and let Λ be a set of rational numbers satisfying the descending chain condition. A projective d-dimensional klt log Calabi-Yau pair is a pr...

conjecture | open | MathDB 页面 | 原文线索

V2-068701 | MATHDB-356909

Modular Schanuel conjecture without derivatives

摘录 / EXCERPT: Let $z_1,\ldots,z_n\in\mathbb{H}^{+}$, write $\mathbf{z}=(z_1,\ldots,z_n)$, and let j be the modular function. Let Σ be the set of special points, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068702 | MATHDB-356921

The quadratic upper-bound conjecture for intersecting triangles

摘录 / EXCERPT: Let S be a point set with $3n$ elements in the plane in general position. An intersecting edge-disjoint triangle family is a family of edge-disjoint triangles such that every two...

conjecture | open | MathDB 页面 | 原文线索

V2-068703 | MATHDB-356922

Weak cordiality conjecture for trees and abelian groups

摘录 / EXCERPT: A family of graphs is weakly A-cordial if all but finitely many of its elements are A-cordial, where A is an abelian group. Weak cordiality conjecture. For any abelian group...

conjecture | open | MathDB 页面 | 原文线索

V2-068704 | MATHDB-356928

Chebotarev-Sato-Tate square-root error conjecture for certain abelian surfaces

摘录 / EXCERPT: Square-root error conjecture. For fixed t, m, and $I\subset[-1,1]$,

conjecture | open | MathDB 页面 | 原文线索

V2-068705 | MATHDB-356941

Higher modularity conjecture for nonisotrivial elliptic curves

摘录 / EXCERPT: Let $K/\mathbf{F}_q$ be a function field and let E/K be a nonisotrivial elliptic curve. For $r\geq 1$, E is called r-modular when it is (r,Σ_∞)-modular for every...

conjecture | open | MathDB 页面 | 原文线索

V2-068706 | MATHDB-356950

The nonvanishing linear-part conjecture for Hodge-locus ideals

摘录 / EXCERPT: Let V_δ be a Hodge locus of a continuous family of Hodge cycles on smooth hypersurfaces, and let $I(\delta)$ be the associated Artinian Gorenstein ideal of the Hodge locus....

conjecture | open | MathDB 页面 | 原文线索

V2-068707 | MATHDB-356951

The trivial-foliation conjecture for high-degree hypersurfaces

摘录 / EXCERPT: Let V be the parameter space of hypersurfaces referred to in the source, let ${\cal F}$ be the foliation defined by the displayed differential forms, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068708 | MATHDB-356962

The lamplighter exclusion conjecture for diagram groups

摘录 / EXCERPT: Let G be a diagram group, and let $\mathbb{Z} \wr \mathbb{Z}^2$ denote the wreath product of $\mathbb{Z}$ by $\mathbb{Z}^2$. Lamplighter exclusion conjecture. The group G does...

conjecture | open | MathDB 页面 | 原文线索

V2-068709 | MATHDB-356963

The free-product conjecture for relatively hyperbolic diagram groups

摘录 / EXCERPT: Let G be a relatively hyperbolic diagram group. Free-product conjecture. Relatively hyperbolic diagram groups are free products. The claim gives a more precise form of the preced...

conjecture | open | MathDB 页面 | 原文线索

V2-068710 | MATHDB-356970

Zhu's reciprocal conjecture for weighted Bergman spaces

摘录 / EXCERPT: Let $\mathbb{B}_n$ denote the unit ball, and let A^p_α be the weighted Bergman space on $\mathbb{B}_n$, where $0<p<\infty$ and α is real. Zhu's reciprocal conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-068711 | MATHDB-356986

The equal-time-scale conjecture for Mittag-Leffler renewal queues

摘录 / EXCERPT: Let $\alpha,\beta\in(0,1]$, and let X_{α} and X_{β} be independent Mittag-Leffler random variables with power indices α and β, respectively, and equal tim...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068712 | MATHDB-356989

The scale-free conjecture for infinity-PageRank

摘录 / EXCERPT: Scale-free infinity-PageRank conjecture. If $d_i\sim X$ where X is a regularly varying random variable, then $PR_\infty(i)$ will also be scale-free.

conjecture | open | MathDB 页面 | 原文线索

V2-068713 | MATHDB-357007

Cutoff at the symmetric boundary case for the SSEP with reservoirs

摘录 / EXCERPT: Consider the symmetric simple exclusion process with reservoirs on a segment of size N, with parameters p and q, and let t^* denote the characteristic mixing-time scale. In...

conjecture | open | MathDB 页面 | 原文线索

V2-068714 | MATHDB-357041

Spin k-twisted double ramification conjecture

摘录 / EXCERPT: Let $g,n\geq 0$ and let k be a positive odd integer. Let $\mu=(m_1,\ldots,m_n)\in\mathbb{Z}^n\setminus k\mathbb{Z}^n_{\geq 0}$ satisfy ... with all entries of μ even. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068715 | MATHDB-357042

Rationality conjecture for general Peskine sixfolds with trivial twisting

摘录 / EXCERPT: Rationality conjecture for Peskine sixfolds. If the twisting class β is trivial, then X_1^σ is rational.

conjecture | open | MathDB 页面 | 原文线索

V2-068716 | MATHDB-357047

The higher-genus open r-spin potential formula

摘录 / EXCERPT: Let $r\ge 2$, let L be the specified solution of the r-th Gelfand--Dickey hierarchy, and let Φ solve ... with $\Phi|_{T_{\ge 2}=0}=1$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-068717 | MATHDB-357054

Asymptotic linear bound conjecture for the second Lasserre level

摘录 / EXCERPT: Asymptotic linear bound conjecture. The optimal objective satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068718 | MATHDB-357062

Three-ary factorization conjecture for generalized biadjoint amplitudes

摘录 / EXCERPT: Three-ary factorization conjecture. To leading order in these three parameters,

conjecture | open | MathDB 页面 | 原文线索

V2-068719 | MATHDB-357065

The extended Stanley inequality for minimum positions of subsets

摘录 / EXCERPT: Let $P=(X,\prec)$ be a poset with $|X|=n$ elements. Fix a nonempty subset $A\subseteq X$, and let f be chosen uniformly from the set of linear extensions of P. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-068720 | MATHDB-357066

The 3-connected degree-swappability conjecture

摘录 / EXCERPT: Degree-swappability conjecture. Every 3-connected graph distinct from the complete graph and the triangular prism is degree-swappable.

conjecture | open | MathDB 页面 | 原文线索

V2-068721 | MATHDB-357079

The reflection-compatible twist conjecture for Coxeter systems

摘录 / EXCERPT: The reflection-compatible twist conjecture. The complete graph $'$ can be obtained from $$ by a sequence of elementary twists.

conjecture | open | MathDB 页面 | 原文线索

V2-068722 | MATHDB-357080

The expanded angle-compatible twist conjecture for Coxeter systems

摘录 / EXCERPT: The expanded angle-compatible twist conjecture. The complete graph $'$ can be obtained from $$ by a sequence of elementary twists.

conjecture | open | MathDB 页面 | 原文线索

V2-068723 | MATHDB-357081

The separating-set characterization of tame quiver representation type

摘录 / EXCERPT: Separating-set conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-068724 | MATHDB-357087

The adjoint-motive existence conjecture for automorphic representations

摘录 / EXCERPT: Let $\Pi=\Pi_f\otimes\Pi_\infty$ satisfy the paper's Assumption, and let E be the reflex field and F_Π the coefficient field. Adjoint-motive existence conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-068725 | MATHDB-357089

Top-bottom period comparison conjecture for archimedean L-packets

摘录 / EXCERPT: Let Π, V, $\Pi_{\min}$, $\Pi_{\max}$, $i_{\min}$ and $i_{\max}$ be as in the paper's main metric conjecture: V is the relevant automorphic vector bundle, and $\Pi_{\min}$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-068726 | MATHDB-357090

Lapid--Mao conjecture on adjoint L-values and Whittaker periods

摘录 / EXCERPT: Let Π be a globally generic representation satisfying the paper's newvector assumption, and let $f=\bigotimes_p f_p\in\Pi$ be a newform. Let W and W_∞ be the Whittaker...

conjecture | open | MathDB 页面 | 原文线索

V2-068727 | MATHDB-357093

Poisson tau-method solvability conjecture on hypercubes

摘录 / EXCERPT: Poisson tau-method solvability conjecture. If $f\in\Pi_N^d$ and the tau polynomials P_k span $\Pi_{N-2,N}^{(d)}$, then the tau-modified equation

conjecture | open | MathDB 页面 | 原文线索

V2-068728 | MATHDB-357096

Bhargava-Cremona-Fisher-Gajović symmetry conjecture for polynomial factorization densities

摘录 / EXCERPT: Let $h(z)$ be a degree-n polynomial over a local field with residue-field size q, and let $\rho(n,\sigma;q)$ denote the density of polynomials whose associated finite étale alg...

conjecture | open | MathDB 页面 | 原文线索

V2-068729 | MATHDB-357098

Chowla-Chowla nonvanishing conjecture for periodic functions

摘录 / EXCERPT: Chowla-Chowla conjecture. $L(2,f)\ne0$, except when

conjecture | open | MathDB 页面 | 原文线索

V2-068730 | MATHDB-357100

Largest-loop exponent conjecture for uniform meandric systems

摘录 / EXCERPT: Let $\mathfrak S_n$ be a uniform meandric system of size n, and order its loops by decreasing size. For each fixed $k\in\mathbb N$, let the kth largest loop be measured by its...

conjecture | open | MathDB 页面 | 原文线索

V2-068731 | MATHDB-357105

The asymptotic upper-bound conjecture for the 2-tone chromatic number of planar graphs

摘录 / EXCERPT: Let G be a planar graph, let $\tau_2(G)$ denote its 2-tone chromatic number, and let $\Delta(G)$ denote its maximum degree. Planar 2-tone bound conjecture. There exists a constan...

conjecture | open | MathDB 页面 | 原文线索

V2-068732 | MATHDB-357106

The sharp planar upper-bound conjecture for the 2-tone chromatic number

摘录 / EXCERPT: Let G be a planar graph, let $\tau_2(G)$ denote its 2-tone chromatic number, and let $\Delta(G)$ denote its maximum degree. Sharp planar 2-tone bound conjecture. Every planar gra...

conjecture | open | MathDB 页面 | 原文线索

V2-068733 | MATHDB-357107

The girth-five planar conjecture for the 2-tone chromatic number

摘录 / EXCERPT: Let G be a planar graph of girth at least 5, let $\tau_2(G)$ denote its 2-tone chromatic number, and let $\Delta(G)$ denote its maximum degree. Girth-five planar 2-tone bound c...

conjecture | open | MathDB 页面 | 原文线索

V2-068734 | MATHDB-357114

The equal odd-hole-length 3-colorability conjecture for triangle-free graphs

摘录 / EXCERPT: Let G be a finite simple undirected graph. An odd hole is an induced cycle of odd length at least five, and G is triangle-free if it has no cycle of length three. Equal odd-hol...

conjecture | open | MathDB 页面 | 原文线索

V2-068735 | MATHDB-357115

The odd-hole-length chromatic bound conjecture

摘录 / EXCERPT: Let $\ell\geq2$ be an integer and let G be a finite simple undirected graph with girth ... Let k be the number of distinct lengths occurring among the odd holes of G. Odd-hole-...

conjecture | open | MathDB 页面 | 原文线索

V2-068736 | MATHDB-357120

Conjectured exponential-square-root bound for the cop-pebbling number

摘录 / EXCERPT: Let G be a graph on n vertices, and let $\pi^{\sf c}(G)$ denote its cop-pebbling number. Cop-pebbling bound conjecture. Every graph G on n vertices satisfies ... This conject...

conjecture | open | MathDB 页面 | 原文线索

V2-068737 | MATHDB-357121

The KPZ sheet and Busemann functions of the KPZ line ensemble

摘录 / EXCERPT: For $T>0$, let $d4cd^T=d4cd^T(y)$ denote the KPZ line ensemble and let $d4d5^T(x,y)$ denote the KPZ sheet. For points on the line ensemble, write $d4cd^T[(x,k)\to(y,1)]$ for the po...

conjecture | open | MathDB 页面 | 原文线索

V2-068738 | MATHDB-357127

Existence conjecture for generalized reflectors with finitely many point targets

摘录 / EXCERPT: Generalized-reflector existence conjecture. There exists a generalized reflector $R\in \mathcal{R}_1^{\mathscr{C}_U^\delta(z')}(T)$ such that $G_1(\{x_i\})=f_i$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-068739 | MATHDB-357128

Existence conjecture for interpolated reflectors with finitely many point targets

摘录 / EXCERPT: Interpolated-reflector existence conjecture. There exists an interpolated reflector $R\in \mathcal{R}_2^{\mathscr{C}_U^\delta(z')}(T)$ such that $G_2(\{x_i\})=f_i$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-068740 | MATHDB-357138

Kanade-Russell's cylindric partition profile conjecture

摘录 / EXCERPT: Let $k\geq 3$, let $c=(c_1,c_2,c_3)$ be a cylindric partition profile with $|c|+3=m=3k+\{-1,0,1\}$, and, using cyclic symmetries, assume $c_1\geq c_2,c_3$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068741 | MATHDB-357139

Kanade-Russell's recurrence-generation conjecture for cylindric Kanade-Russell identities

摘录 / EXCERPT: Let $m\geq5$ be a modulus of the form $m=3k+\{-1,0,1\}$ with $k\geq3$, and let $S_m(\rho\mid\sigma)$ be the multiple-sum functions indexed by integer vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-068742 | MATHDB-357144

Geometric Manin's conjecture for non-nef sections

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a Fano fibration, and let $\operatorname{Sec}(\mathcal{X}/B)$ be its moduli space of sections. A component is accumulating when it is governed by the...

conjecture | open | MathDB 页面 | 原文线索

V2-068743 | MATHDB-357145

Polyhedrality conjecture for nef section classes

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a Fano fibration. Let A be the affine space of curve classes having intersection 1 with a general fiber, and let $\mathcal{P}$ be the convex hull...

conjecture | open | MathDB 页面 | 原文线索

V2-068744 | MATHDB-357160

Autoduality conjecture for motives

摘录 / EXCERPT: Let M be a motive with rational coefficients and suppose that M has weight n, meaning that it is a direct summand of a motive of the form $\mathfrak{h}^n(X)$. Autoduality con...

conjecture | open | MathDB 页面 | 原文线索

V2-068745 | MATHDB-357166

The Fukaya-Seidel and categorical intersection complex correspondence for hyperplane arrangements

摘录 / EXCERPT: Let $X=(\mathbb{C}^*)^n\subset\mathbb{C}^{n+1}$. A normal crossings divisor in $\mathbb{P}^n$ is the union of $n+1$ hyperplanes in general position. The categorical intersection co...

conjecture | open | MathDB 页面 | 原文线索

V2-068746 | MATHDB-357171

Dobbertin's conjecture on the linearity of monomial vectorial functions

摘录 / EXCERPT: Let m and n be the parameters of the vectorial function F under consideration, and let u satisfy ... with $u\not\equiv 2^k\pmod{2^m-1}$ for every $k\geq 0$. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-068747 | MATHDB-357184

The cyclic-polytope conjecture for unique rainbow simplices

摘录 / EXCERPT: Let K be a triangulation of the d-simplex Δ^d associated with a d-dimensional cyclic polytope, meaning that it is obtained from the boundary of the polytope by removin...

conjecture | open | MathDB 页面 | 原文线索

V2-068748 | MATHDB-357187

Rodriguez-Villegas's integrality conjecture for structure coefficients

摘录 / EXCERPT: For partitions $\boldsymbol{\mu}$ and λ, let $c_{\boldsymbol{\mu}}^{\lambda}(q,t)$ denote the structure coefficients introduced through the product of modified Hall-Littlew...

conjecture | open | MathDB 页面 | 原文线索

V2-068749 | MATHDB-357200

Bondal-Polishchuk transitivity conjecture for full exceptional sequences

摘录 / EXCERPT: Let $\mathcal{T}$ be a triangulated category admitting a full exceptional sequence of length n, and let $\mathfrak{B}_n$ denote the braid group on n strands. The group...

conjecture | solved | MathDB 页面 | 原文线索

V2-068750 | MATHDB-357216

Non-vanishing conjecture for lc pairs

摘录 / EXCERPT: Non-vanishing conjecture. The relative real linear system is nonempty:

conjecture | open | MathDB 页面 | 原文线索

V2-068751 | MATHDB-357224

Conjecture on the orbit stratification of the convolution Grassmannian

摘录 / EXCERPT: Orbit-stratification conjecture. There exists a finite partially ordered set $(X,<)$ describing the L^+G-orbits in $\widetilde{\operatorname{Gr}}_{\mu_{\bullet}}$. Equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-068752 | MATHDB-357232

Odd-moment equidistribution conjecture for quartic Gauss sums

摘录 / EXCERPT: For odd integers k, consider primitive quartic Dirichlet characters χ of odd conductor q, and let $\tau(\chi)$ denote their Gauss sums. Odd-moment equidistribution conject...

conjecture | open | MathDB 页面 | 原文线索

V2-068753 | MATHDB-357233

Equidistribution conjecture for squared sextic Gauss sums

摘录 / EXCERPT: Let $\Psi_6(X)$ be the primitive Dirichlet characters of order 6 with conductor at most X, and let $\tau(\chi)$ be the Gauss sum of χ. Sextic Gauss-sum equidistribution co...

conjecture | open | MathDB 页面 | 原文线索

V2-068754 | MATHDB-357234

Gross-Prasad conjecture on generic members and adjoint L-functions

摘录 / EXCERPT: Let G be a split connected reductive group over a nonarchimedean local field F, let ${}^LG$ be its L-group with connected component $\widehat{G}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068755 | MATHDB-357243

Fourier-Mukai/decomposition correspondence for integrable systems

摘录 / EXCERPT: Let $\pi_M:(M,\sigma)\to B$ be an integrable system, with partial Fourier-Mukai transform ϕ_{FM}, sheaf of Kähler differentials Ω_M^k, and decomposition-theore...

conjecture | open | MathDB 页面 | 原文线索

V2-068756 | MATHDB-357252

Helmke's numerical Fujita freeness conjecture

摘录 / EXCERPT: Helmke's numerical Fujita freeness conjecture. Under these conditions, the adjoint divisor

conjecture | open | MathDB 页面 | 原文线索

V2-068757 | MATHDB-357254

Large-population weak-limit conjecture for the finite-population Fisher-KPP equation

摘录 / EXCERPT: Consider the one-dimensional Dirichlet problem ... with the Dirichlet boundary conditions and initial data specified in the paper, and let u denote its solution. Large-population w...

conjecture | open | MathDB 页面 | 原文线索

V2-068758 | MATHDB-357255

Long-time convergence conjecture for the Neumann finite-population Fisher-KPP equation

摘录 / EXCERPT: Let $u(x,t)$ solve the Neumann problem ... with ... and initial data $u(x,0)=u_0(x)$, where $u_0(x)>0$ for some x. Long-time convergence conjecture. ... The claim describes the predict...

conjecture | open | MathDB 页面 | 原文线索

V2-068759 | MATHDB-357256

The graph-isomorphism conjecture for rational quadratic-form graphs

摘录 / EXCERPT: Let $q_1,q_2\boxtimes$ be quadratic forms on $\operatorname{\mathbb{Q}}^n$, and let $G_0(\operatorname{\mathbb{Q}}^n,q)$ denote the connected component containing...

conjecture | open | MathDB 页面 | 原文线索

V2-068760 | MATHDB-357259

Sparse curve singularity delta-invariant conjecture

摘录 / EXCERPT: Let $B_1,\ldots,B_k\subset\mathbb Z^1_{\geqslant 0}$ be support sets, let $d_i:=\min B_i>0$, and let ... Assume that $j_\infty=1$, equivalently that the associated map germ is inject...

conjecture | open | MathDB 页面 | 原文线索

V2-068761 | MATHDB-357262

Scaling-limit conjecture for uniformly random graphic sequences

摘录 / EXCERPT: Let $D_1\ge\dotsb\ge D_n$ be a uniformly random graphic sequence of length n. The scaling-limit conjecture. There exists a random continuous function $D:[0,1]\to\mathbb{R}$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-068762 | MATHDB-357271

Expected boundary vanishings for rank-zero Donaldson-Thomas invariants on the quintic

摘录 / EXCERPT: Let X_5 be the quintic threefold, let H be its hyperplane class, and set $\kappa=H^3=5$. Write $\overline{\Omega}_H$ for the rank-zero invariant with Chern-character arguments...

conjecture | open | MathDB 页面 | 原文线索

V2-068763 | MATHDB-357274

The Chet Conjecture for entry-wise nonnegative Chet matrices

摘录 / EXCERPT: Let C_n denote the $n\times n$ Chet matrix, and let $C=\{n:C_n\text{ is entry-wise nonnegative}\}$. Chet Conjecture. One has ... More strongly, ... The paper presents this as its mai...

conjecture | open | MathDB 页面 | 原文线索

V2-068764 | MATHDB-357290

Rigidity conjecture for the half-space Allen-Cahn linearized equation

摘录 / EXCERPT: Let $\phi: \mathbb{R}^{n-1} \times \mathbb{R}^+ \to \mathbb{R}$. Suppose that $\phi(s,0) \equiv 0$. Let $f: \mathbb{R}^{n-1} \to \mathbb{R}$, and suppose ... Here g is the one-dime...

conjecture | open | MathDB 页面 | 原文线索

V2-068765 | MATHDB-357291

The reducibility conjecture for homomorphisms between braid groups

摘录 / EXCERPT: Reducibility conjecture. For $n \ge 5$ and $m > n$, every homomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-068766 | MATHDB-357292

Elliptic Stark Conjecture

摘录 / EXCERPT: Let $(\kappa,\lambda,\mu)\in \mathcal{W}_\textup{\bf f}^{\rm cl}$ satisfy ${\rm w}(\kappa)=2$ and ${\rm w}(\lambda)={\rm w}(\mu)=1$. Assume that $\textup{\bf f}_\kappa$ is the p-...

conjecture | open | MathDB 页面 | 原文线索

V2-068767 | MATHDB-357294

The smooth monomial-Hessian conjecture

摘录 / EXCERPT: Let $S=\mathbb C[x_1,\ldots,x_n]$ and let $g\in S$ be homogeneous of degree $d\geq 3$, with $n\geq 2$. Assume that its Hessian $H(g)$ is a monomial. Smooth monomial-Hessian conject...

conjecture | open | MathDB 页面 | 原文线索

V2-068768 | MATHDB-357308

The Kesten-McKay condition removal conjecture for MANOVA spectral edges

摘录 / EXCERPT: Let $N=N(\iota)$ be an increasing sequence, let $\bm A,\bm B\in\mathbb C^{N\times N}$ be random orthogonal projection matrices, and suppose that ... Assume that the empirical spectra...

conjecture | open | MathDB 页面 | 原文线索

V2-068769 | MATHDB-357309

Newton series conjecture for the principal inverse gamma function

摘录 / EXCERPT: Let $\tilde{\Gamma}(x)$ denote the principal branch of the inverse of the gamma function, so that $\tilde{\Gamma}(x)=y$ when $ Gamma(y)=x$. Let $ alpha=1.4616\ldots$ be the unique...

conjecture | open | MathDB 页面 | 原文线索

V2-068770 | MATHDB-357322

The directed degree-f arboricity conjecture

摘录 / EXCERPT: Directed degree-f arboricity conjecture. For every directed multigraph D,

conjecture | open | MathDB 页面 | 原文线索

V2-068771 | MATHDB-357326

Small-independent-set conjecture for Hasse diagrams of bounded-dimensional posets

摘录 / EXCERPT: A finite poset has dimension d when d is the least number of linear orders whose intersection is the poset order. Its Hasse diagram is the graph whose vertices are the poset el...

conjecture | open | MathDB 页面 | 原文线索

V2-068772 | MATHDB-357327

Large-independent-set conjecture for bounded-dimensional Hasse diagrams

摘录 / EXCERPT: A finite poset has dimension d when its order is the intersection of d linear orders. Its Hasse diagram is the graph of the poset's cover relations, and the independence number...

open_problem | open | MathDB 页面 | 原文线索

V2-068773 | MATHDB-357335

The strongly proper connection conjecture for 3-connected graphs

摘录 / EXCERPT: Strongly proper connection conjecture. Every 3-connected graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068774 | MATHDB-357336

Bounded nonrepetitive connection conjecture for 2-connected graphs

摘录 / EXCERPT: Bounded nonrepetitive connection conjecture. There exists a constant t such that every 2-connected graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068775 | MATHDB-357337

Connectivity threshold conjecture for honest connected colorings

摘录 / EXCERPT: Connectivity threshold conjecture for honest connected colorings. For every honest property $\mathcal{P}$, there exists a constant $c(\mathcal{P})$ such that every $c(\mathcal{P})$...

conjecture | open | MathDB 页面 | 原文线索

V2-068776 | MATHDB-357348

The triviality conjecture for toroidal automorphic forms over the projective line

摘录 / EXCERPT: Triviality conjecture. For all $n\geqslant 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068777 | MATHDB-357350

Conservativity of the left Quillen strictification functor

摘录 / EXCERPT: For an integer m, let $\mathbf{sSet}^m$ denote the category of stratified simplicial sets and let $\infty\text{-}\mathbf{Cat}^m$ denote the category of marked strict ∞-cat...

conjecture | open | MathDB 页面 | 原文线索

V2-068778 | MATHDB-357351

The DT/PT crepant resolution conjecture for abelian quotient fourfolds

摘录 / EXCERPT: Let $G=\mathbb{Z}_r<\mathrm{SU}(2)<\mathrm{SU}(4)$ or $G=\mathbb{Z}_2\times\mathbb{Z}_2<\mathrm{SO}(3)<\mathrm{SU}(4)$. Let $X\to\mathbb{C}^4/G$ be the Nakamura G-Hilbert crepant...

conjecture | open | MathDB 页面 | 原文线索

V2-068779 | MATHDB-357353

Helson's zero-free conjecture for twisted Beurling Dirichlet series

摘录 / EXCERPT: Helson's conjecture. The function f_χ never has any zeros in its half-plane of convergence.

conjecture | open | MathDB 页面 | 原文线索

V2-068780 | MATHDB-357361

Quasi symmetric plane partitions enumeration conjecture

摘录 / EXCERPT: Let $\operatorname{qspp}(a,c)$ denote the number of quasi symmetric plane partitions inside an (a,a,c)-box. Let $p_a(c)$ be a polynomial in $\mathbb{Q}[c]$. Quasi symmetric plane...

conjecture | open | MathDB 页面 | 原文线索

V2-068781 | MATHDB-357362

QTC2 plane partitions enumeration conjecture

摘录 / EXCERPT: Let π be a plane partition inside an (a,a,c)-box. It is quasi transpose complementary of second kind (QTC2) if it is transpose complementary except along the diagonal, namely...

conjecture | open | MathDB 页面 | 原文线索

V2-068782 | MATHDB-357366

Au's conjecture on iterated integrals of unital functions

摘录 / EXCERPT: Au's conjecture. For all $U(x)\in\mathsf{U}_N$, the renormalized value

conjecture | open | MathDB 页面 | 原文线索

V2-068783 | MATHDB-357374

The common relative Jacobian conjecture for Type i KSS pairs

摘录 / EXCERPT: Let (A_i,B_i), for $i=1,\ldots,12$, be one of the twelve Type i KSS pairs of smooth elliptic Calabi--Yau 3-folds over $\mathbb{P}^2$, with elliptic fibrations...

conjecture | open | MathDB 页面 | 原文线索

V2-068784 | MATHDB-357375

Tate--Shafarevich group conjecture for Type i KSS Jacobians

摘录 / EXCERPT: Let $J_i\to\mathbb{P}^2$ be the common relative Jacobian of the Type i KSS pair. The Tate--Shafarevich group is the group of torsors of the Jacobian fibration that are locally tr...

conjecture | open | MathDB 页面 | 原文线索

V2-068785 | MATHDB-357377

The Sato-Tate independence conjecture for successive Frobenius angles

摘录 / EXCERPT: The Sato-Tate independence conjecture. For every integer $s\geq1$, the sequence $\{X_k\}_{k\geq1}$ is uniformly distributed with respect to $\mu_{ST}^{[s]}$; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-068786 | MATHDB-357378

Rank-one orbifold instanton conjecture on $[\mathbb{C}^3/(\mathbb{Z}_2\times\mathbb{Z}_2)]\times\mathbb{C}$

摘录 / EXCERPT: Let $\vec{\mathtt{q}}=(\mathtt{q}_0,\mathtt{q}_1,\mathtt{q}_2,\mathtt{q}_3)$ and $\mathtt{Q}=\mathtt{q}_0\mathtt{q}_1\mathtt{q}_2\mathtt{q}_3$. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-068787 | MATHDB-357392

The division conjecture for arithmetic Gevrey functions

摘录 / EXCERPT: Let $\mathfrak{f}(z)$ be an arithmetic Gevrey function and let $\theta\in(-\pi/2,\pi/2)$ satisfy $\mathfrak{f}_\theta(1)=0$. Division conjecture for arithmetic Gevrey functions. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-068788 | MATHDB-357393

The division conjecture for mixed arithmetic Gevrey functions

摘录 / EXCERPT: A mixed function is a formal series ... where F is an E-function and $\mathfrak{f}$ is an arithmetic Gevrey function. For a non-anti-Stokes direction θ, write...

conjecture | open | MathDB 页面 | 原文线索

V2-068789 | MATHDB-357400

Caro-Petruševski-Škrekovski's proper conflict-free coloring Brooks conjecture

摘录 / EXCERPT: Proper conflict-free coloring Brooks conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-068790 | MATHDB-357407

The infinite Koebe-Andreev-Thurston theorem for plane cellular decompositions

摘录 / EXCERPT: Let G and $\hat{G}$ be a pair of simple, 3-connected plane graphs corresponding to a cellular decomposition of $\mathbb{C}$ and its dual. A circle configuration is a configuratio...

conjecture | open | MathDB 页面 | 原文线索

V2-068791 | MATHDB-357422

The weak null conjecture for differentiated semilinear wave systems

摘录 / EXCERPT: Weak null conjecture. The original system of wave equations also admits global solutions for sufficiently small initial data.

conjecture | open | MathDB 页面 | 原文线索

V2-068792 | MATHDB-357425

Additional-term conjecture for moments of quadratic Dirichlet L-functions

摘录 / EXCERPT: Let $\EuScript F_{\EuScript D}$ be the family of real primitive quadratic Dirichlet characters χ_d, with d nonzero and squarefree, and let $r\geq4$. For integers $N\geq1$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-068793 | MATHDB-357426

Conjecture on the dual equivalence graph inside the skeleton of a crystal

摘录 / EXCERPT: Let λ be a partition, and let $Skeleton(\lambda)$ be the graph obtained from the crystal of tableaux $B(\lambda)$ by replacing each quasicrystal with its associated standar...

conjecture | open | MathDB 页面 | 原文线索

V2-068794 | MATHDB-357428

Colorful Tverberg conjecture for regression depth

摘录 / EXCERPT: Let r,d be positive integers, and let $A_1,\dots,A_{d+1}$ be sets of r hyperplanes each in $\mathbb{R}^d$. A partition of their union into r sets $B_1,\dots,B_r$ is colorful...

conjecture | open | MathDB 页面 | 原文线索

V2-068795 | MATHDB-357430

Conjecture on self-similarity of non-escaping loci

摘录 / EXCERPT: Self-similarity conjecture. For every stationary type irrational number $\theta=[0;N,N,N,\ldots]$, the non-escaping locus $\mathcal{M}_{d_0,d_\infty}$ is asymptotically self-simila...

conjecture | open | MathDB 页面 | 原文线索

V2-068796 | MATHDB-357436

The regulator-growth conjecture for real quadratic fields

摘录 / EXCERPT: Let K range over real quadratic fields, with regulator R_K, discriminant d_K, and parameter m_K as used in the source. Regulator-growth conjecture. There exists an infinite...

conjecture | open | MathDB 页面 | 原文线索

V2-068797 | MATHDB-357443

The Weyl-law conjecture for Love waves in layered elastic half-spaces

摘录 / EXCERPT: Let $n\geqslant3$, let $y\in[1/C_\infty,1/C_0)$, and let $\widetilde{C}_1\leqslant\cdots\leqslant\widetilde{C}_{n+1}$ be the nondecreasing reordering of the parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-068798 | MATHDB-357449

Geometric S-duality for Vafa-Witten invariants

摘录 / EXCERPT: Let $\hat{\mathsf{VW}}^{\mathsf m,\mathsf a}_{\mathsf d}(B,q)_t$ denote the Vafa-Witten generating series, with indices recording degree, monopole data, and flux. Geometric S-duali...

conjecture | open | MathDB 页面 | 原文线索

V2-068799 | MATHDB-357450

S-duality for zero-degree Vafa-Witten invariants

摘录 / EXCERPT: Let $\hat{\mathsf{VW}}^{\mathsf m,\mathsf a}_{\mathsf d,0}$ denote the zero-degree Vafa-Witten invariants and set $\omega=e^{2\pi i/\mathsf r}$. S-duality without μ-inserti...

conjecture | open | MathDB 页面 | 原文线索

V2-068800 | MATHDB-357455

One-dimensional worst-case conjecture for gradient descent on composed operator classes

摘录 / EXCERPT: Let $\mathcal{C}_{\mu_g}^{0}$ and $\mathcal{D}_{\mu_g}^{\mu_M}$ be the operator-composed function classes defined in the source, and let $\mathcal{C}_{\mu_g}^{0,(1)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-068801 | MATHDB-357456

Closed-form worst-case conjecture for gradient descent on $\mathcal{D}_{\mu_g}^{\mu_M}$

摘录 / EXCERPT: Closed-form worst-case conjecture. The worst-case performance satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068802 | MATHDB-357468

Grand Density Hypothesis for Selberg class L-functions

摘录 / EXCERPT: Let $\mathcal{L}$ belong to the subclass $\mathcal{S}'$ of the Selberg class, and let $N_\mathcal{L}(\sigma,T)$ denote the number of zeros...

conjecture | open | MathDB 页面 | 原文线索

V2-068803 | MATHDB-357476

Chenciner's regular polygon conjecture for equal-mass circular central configurations

摘录 / EXCERPT: Let n bodies have equal masses and lie on a common circle, with their center of mass coinciding with the center of the circle. A central configuration is one in which the acceler...

open_problem | open | MathDB 页面 | 原文线索

V2-068804 | MATHDB-357477

Dubrovin's conjecture for FJRW theory

摘录 / EXCERPT: Let (W,G) be an admissible Landau-Ginzburg pair, with W a nondegenerate quasihomogeneous polynomial and G an admissible subgroup of its maximal symmetry group containing the...

conjecture | open | MathDB 页面 | 原文线索

V2-068805 | MATHDB-357483

Extension of adversarial non-convergence to expander graphs

摘录 / EXCERPT: Expander-graph conjecture. There is a choice of corrupted set such that the conclusion to Theorem holds.

conjecture | open | MathDB 页面 | 原文线索

V2-068806 | MATHDB-357492

Higman-Kaplansky unit conjecture for torsion-free group rings

摘录 / EXCERPT: Let G) be a torsion-free group and let K be a field. A unit of the group ring $K[G]$ is an element with a multiplicative inverse, and the trivial units are the elements kg wi...

conjecture | open | MathDB 页面 | 原文线索

V2-068807 | MATHDB-357495

The tree-decomposition minor conjecture

摘录 / EXCERPT: The tree-decomposition minor conjecture. There is a function f such that every connected graph G has a tree decomposition $(T,\mathcal{B})$ of width at most...

conjecture | open | MathDB 页面 | 原文线索

V2-068808 | MATHDB-357496

The spanning-tree decomposition conjecture

摘录 / EXCERPT: The spanning-tree decomposition conjecture. There is a function f such that every connected graph G has a tree decomposition $(T,\mathcal{B})$ of width at most...

conjecture | open | MathDB 页面 | 原文线索

V2-068809 | MATHDB-357497

The strong spanning-tree decomposition conjecture

摘录 / EXCERPT: The strong spanning-tree decomposition conjecture. There is a function f such that every connected graph G has a tree decomposition $(T,\mathcal{B})$ of width at most...

conjecture | open | MathDB 页面 | 原文线索

V2-068810 | MATHDB-357502

The binary entropy inequality for f_k

摘录 / EXCERPT: Binary entropy inequality conjecture. The function $f_k(x)$ is nonnegative on $[0,1]$. The case $k=2$ was proved by Boppana and by AHS using differential calculus, with interval ar...

conjecture | solved | MathDB 页面 | 原文线索

V2-068811 | MATHDB-357506

Gross's integral Gross-Stark conjecture

摘录 / EXCERPT: Let $L/H/F$ be a tower in which F is totally real and H and L are finite abelian CM extensions of F, with L containing H. Write $\mathcal{G}=\operatorname{Gal}(L/F)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-068812 | MATHDB-357511

The even zigzag-poset LE-symmetry conjecture

摘录 / EXCERPT: For even n, let Z_n be the poset on n elements with relations ... A poset is LE-symmetric when its Bender-Knuth group acts as the full symmetric group on its set of linear exte...

conjecture | open | MathDB 页面 | 原文线索

V2-068813 | MATHDB-357512

The Ferrers-poset LE-symmetry conjecture for specified partitions

摘录 / EXCERPT: Let F_λ be the Ferrers poset associated with a partition λ, and let λ^t be the conjugate partition; the posets F_λ and F_{λ^t} are isomorphi...

conjecture | open | MathDB 页面 | 原文线索

V2-068814 | MATHDB-357525

Simó-Yoccoz uniqueness conjecture for convex planar four-body central configurations

摘录 / EXCERPT: Let four bodies with positive masses be arranged in a specified ordering along the boundary of their convex hull. A planar configuration is convex if no body lies inside or on the...

conjecture | open | MathDB 页面 | 原文线索

V2-068815 | MATHDB-357531

Two-point-function dichotomy below the critical temperature

摘录 / EXCERPT: Let $s\in\mathbb{S}^{d-1}$, let n tend to infinity, let $\beta_{\rm sat}^{*}(s)$ be the saturation threshold for the truncated two-point function, let $\beta_{\rm c}$ be the crit...

conjecture | open | MathDB 页面 | 原文线索

V2-068816 | MATHDB-357533

The finite-field exceptional-set estimate in the plane

摘录 / EXCERPT: Let $A\subset \mathbb{F}_p^2$ satisfy ... For $0<s<\max\{1,a\}$, let $E_s(A)$ be the exceptional set defined for projection to lines, with $n=1$ and $k=1$. The finite-field exception...

conjecture | open | MathDB 页面 | 原文线索

V2-068817 | MATHDB-357535

The finite-field sharp exceptional-set conjecture in three dimensions

摘录 / EXCERPT: Let $A\subset\mathbb{F}_p^3$ satisfy ... Recall the exceptional set $E_s(A)$ for projection to lines or planes. Let $t(a,s)$ be the lower bound for $T(a,s)$ given by Proposition 1 fo...

conjecture | open | MathDB 页面 | 原文线索

V2-068818 | MATHDB-357540

Bonamy et al.'s odd-degree planar edge-partition conjecture

摘录 / EXCERPT: Odd-degree planar edge-partition conjecture. The edges of G can be partitioned into $\tfrac{\Delta-1}{2}$ linear forests and one matching.

conjecture | open | MathDB 页面 | 原文线索

V2-068819 | MATHDB-357554

Monotone metric rigidity conjecture for polytopes

摘录 / EXCERPT: Let $P\subset\mathbb{R}^d$ and $Q\subset\mathbb{R}^e$ be polytopes with isomorphic edge-graphs. Assume that $0\in\operatorname{int}(Q)$, every edge of Q is at most as long as its...

conjecture | open | MathDB 页面 | 原文线索

V2-068820 | MATHDB-357556

Translation conjecture for nullspace embeddings

摘录 / EXCERPT: Let v be a nullspace embedding of an expansion matrix M, and let $c\in\operatorname{conv}\{v_1,\ldots,v_n\}$. Translation conjecture for nullspace embeddings. There is an expan...

conjecture | open | MathDB 页面 | 原文线索

V2-068821 | MATHDB-357563

Subcritical pitchfork conjecture for the Ren-Wei branch

摘录 / EXCERPT: Let χ be the fissility parameter and let χ_* be the bifurcation value. The Ren-Wei branch is the equilibrium branch emerging at this bifurcation. Subcritical pitchfork co...

conjecture | open | MathDB 页面 | 原文线索

V2-068822 | MATHDB-357568

Two-dimensional scission-branch conjecture for the liquid-drop model

摘录 / EXCERPT: Let χ be the fissility parameter and let $\chi_3\approx0.428$ be the two-dimensional analogue of the Businaro-Gallone point. A scission point is a transition state on the sepa...

conjecture | open | MathDB 页面 | 原文线索

V2-068823 | MATHDB-357576

Robust eigenvectors conjecture for regular simplex tensors

摘录 / EXCERPT: Robust eigenvectors conjecture. The robust eigenvectors of a regular simplex tensor are precisely the vectors in the frame.

conjecture | open | MathDB 页面 | 原文线索

V2-068824 | MATHDB-357579

Stable Schur-positivity conjecture for nabla applied to $[200~\operatorname{Xi}e_\mu$

摘录 / EXCERPT: Let $[200~\mu$ be a partition, let $[200~\boldsymbol{x}$ be a set of variables, and let $[200~\operatorname{\oldXi}$ be the operator appearing in the expression below. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068825 | MATHDB-357580

Schur-positivity conjecture for triangular-partition step sequences

摘录 / EXCERPT: Let $[200~\boldsymbol{v}$ be a vector that is the step sequence of a triangular partition, and let $[200~\mathcal{E}_{\boldsymbol{v}}(\boldsymbol{x})$ be the symmetric function def...

conjecture | open | MathDB 页面 | 原文线索

V2-068826 | MATHDB-357581

Contractibility conjecture for the bicategorical walking adjoint equivalence

摘录 / EXCERPT: Let $\mathrm{bi}\mathcal{E}^{\mathrm{adj}}$ be the candidate contractible 3-category with two objects obtained by adjoining bicategorical adjoint-equivalence data. An equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-068827 | MATHDB-357591

Fractional list and correspondence packing conjectures

摘录 / EXCERPT: Fractional packing conjectures. There exists a constant $C>0$ such that, for every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-068828 | MATHDB-357592

Packing version of Dinitz's problem

摘录 / EXCERPT: Packing version of Dinitz's problem. For $n\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-068829 | MATHDB-357593

Thread-set conjecture for iterated localizations of stratified tensor triangulated categories

摘录 / EXCERPT: Thread-set conjecture. Let ${\mathbb{A}}$ and ${\mathbb{B}}$ be tuples of subsets of $\operatorname{Spc}({\mathcal{T}}^c)$. If they have the same thread sets, then there is a canon...

conjecture | open | MathDB 页面 | 原文线索

V2-068830 | MATHDB-357594

Sarnak's bounded-clustering conjecture for cofinite Fuchsian groups

摘录 / EXCERPT: Sarnak's conjecture. If $\mathrm{Tr}(\Gamma)$ satisfies the B-C property, then Γ is arithmetic. If $\operatorname{Gap}(\mathrm{Tr}(\Gamma))>0$, then Γ is derived from...

conjecture | open | MathDB 页面 | 原文线索

V2-068831 | MATHDB-357599

Conjecture on active complementarity in the Beta relaxation

摘录 / EXCERPT: Let $\text{\sc Beta}$ denote the convex relaxation of the ball-constrained nonconvex quadratic program defined in the paper, with matrix variable W and vectors ℓ_1,ℓ_2 as...

conjecture | open | MathDB 页面 | 原文线索

V2-068832 | MATHDB-357601

The zero residual Monge-Ampère mass conjecture

摘录 / EXCERPT: Let $D=B_1$ be the unit ball in ${\mathbb C}^n$, and let u be a plurisubharmonic function on B_1 that is locally bounded outside the origin. The Monge-Ampère measure...

conjecture | open | MathDB 页面 | 原文线索

V2-068833 | MATHDB-357606

The diagonal-maximality conjecture for correspondence packing of complete bipartite graphs

摘录 / EXCERPT: For positive integers a and b with $a>b$, let $K_{a,b}$ denote the complete bipartite graph with parts of sizes a and b, and let $\chi_c^\star$ denote its correspondence pa...

conjecture | open | MathDB 页面 | 原文线索

V2-068834 | MATHDB-357614

The p-adic Structural Rank Conjecture

摘录 / EXCERPT: p-adic Structural Rank Conjecture. For every

conjecture | open | MathDB 页面 | 原文线索

V2-068835 | MATHDB-357619

Coefficient-Length conjecture for triangular diagonal harmonics

摘录 / EXCERPT: Coefficient-Length conjecture. For all triangular partitions τ,

conjecture | open | MathDB 页面 | 原文线索

V2-068836 | MATHDB-357621

Conjectural formula for the reproducing kernel on the positive cone

摘录 / EXCERPT: Let Ω be a bounded symmetric domain of rank r, with invariants p, b, d, and q_Ω, and let Γ_Ω denote the associated gamma function. Let m and...

conjecture | open | MathDB 页面 | 原文线索

V2-068837 | MATHDB-357625

The New Way Eulerianity conjecture for $\mathrm{Sp}_{2n}\times\mathrm{GL}_k$

摘录 / EXCERPT: Let n be a positive even integer, let π and τ be the automorphic representations occurring in the integral, and let $\mathcal{L}(\phi,f_{2n,k,s})$ be the integral in equ...

conjecture | open | MathDB 页面 | 原文线索

V2-068838 | MATHDB-357630

Regularity bound for squarefree parts of symbolic powers of edge ideals

摘录 / EXCERPT: Let G be a graph, let $I(G)$ denote its edge ideal, let $I(G)^{\{s\}}$ denote the squarefree part of its s-th symbolic power, and let ${\rm match}(G)$ denote the matching numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-068839 | MATHDB-357632

The Geer-Young TQFT identification for punctured surfaces

摘录 / EXCERPT: Let $F_{g,p}$ be the surface of genus g with $p\geq 1$ boundary components, viewed with all boundary components as punctures labeled by $V(0,0)_{\overline{0}}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068840 | MATHDB-357636

Uniqueness of binary phylogenetic trees from their A_k alignments

摘录 / EXCERPT: Let $k\in\mathbb{N}_{\geq 3}$, let X be a taxon set, and let T and $\widetilde{T}$ be binary phylogenetic trees on X. For a binary phylogenetic tree T, let $A_k(T)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-068841 | MATHDB-357649

Kapranov-Voevodsky conjecture on higher Bruhat-to-Stasheff-Tamari maps

摘录 / EXCERPT: Kapranov-Voevodsky conjecture. The map f is a quotient by a weak order congruence. This conjecture asks whether the higher-dimensional analogue of the map from weak Bruhat order...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-068842 | MATHDB-357651

The twisted K3 categorical rationality conjecture for cubic fourfolds

摘录 / EXCERPT: Let $X\subset\mathbb{P}^5$ be a smooth cubic fourfold, let $\mathcal{A}_X$ be its K3 category, and let (S,α) be a twisted K3 surface, with $\alpha\in\operatorname{Br}(S)$. T...

conjecture | open | MathDB 页面 | 原文线索

V2-068843 | MATHDB-357652

The derived Torelli conjecture for K3 categories of cubic fourfolds

摘录 / EXCERPT: Let $X,X'\subset\mathbb{P}^5$ be smooth cubic fourfolds, let $\mathcal{A}_X$ and $\mathcal{A}_{X'}$ be their K3 categories, and let $\widetilde H(X,\mathbb{Z})$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-068844 | MATHDB-357653

The $(T,\mathcal{V})$-proarrow conjecture for enriched approaches to automata

摘录 / EXCERPT: Let T be a monad on Set and let $\mathcal{V}$ be a quantale. Write $T\mathcal{V}\operatorname{Prof}$ for the locally thin bicategory associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-068845 | MATHDB-357654

The proarrow-equipment approach to nondeterministic bicategorical automata

摘录 / EXCERPT: Let $\bbB$ be a bicategory. Recall that automata in the Kleisli category of the powerset monad model nondeterministic automata in Set, and that the presheaf constr...

conjecture | open | MathDB 页面 | 原文线索

V2-068846 | MATHDB-357662

Bounded symmetric difference for P_7-free bipartite graphs

摘录 / EXCERPT: Let G be a graph in the class of P_7-free bipartite graphs. The bounded symmetric-difference conjecture. The symmetric difference is bounded in the class of P_7-free bipartit...

conjecture | open | MathDB 页面 | 原文线索

V2-068847 | MATHDB-357666

Generalized pebbling product conjecture for weighted stars

摘录 / EXCERPT: Generalized weighted-star pebbling conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068848 | MATHDB-357672

Subpolynomial interval colouring thickness conjecture

摘录 / EXCERPT: Let $\theta(n)$ be the maximum interval colouring thickness of a graph on n vertices. Subpolynomial thickness conjecture. ... The paper has bounds...

conjecture | open | MathDB 页面 | 原文线索

V2-068849 | MATHDB-357674

Bounded thickness conjecture for biregular graphs

摘录 / EXCERPT: Let G be a biregular graph, meaning a bipartite graph whose vertices in each part all have the same degree, and let $\theta(G)$ denote its interval colouring thickness. Bounded b...

conjecture | open | MathDB 页面 | 原文线索

V2-068850 | MATHDB-357689

Craig-Pun's log-concavity and higher-order Turán conjecture for distinct partitions

摘录 / EXCERPT: Craig-Pun's conjecture. The function $q(n)$ is log-concave for $n\geq33$ and satisfies the higher-order Turán inequalities for $n\geq121$.

conjecture | open | MathDB 页面 | 原文线索

V2-068851 | MATHDB-357690

Conjectured quartic-invariant inequalities for the distinct partition function

摘录 / EXCERPT: The quartic-invariant conjecture for $q(n)$. The following inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-068852 | MATHDB-357712

Existence and uniqueness conjecture for Linnell division rings

摘录 / EXCERPT: Let K be a field and let G be a torsion-free group. A Linnell division $K[G]$-ring is an epic division $K[G]$-ring that is (N,G)-free for every subgroup N of G. Linnell d...

conjecture | open | MathDB 页面 | 原文线索

V2-068853 | MATHDB-357728

The gluing characterization of universally free numerical semigroups

摘录 / EXCERPT: Let $\mathcal{S}$ be a numerical semigroup minimally generated by $A = \{a_1,\ldots,a_n\}$. A nontrivial partition of A is a partition $A = B \sqcup C$ with both parts nonempty....

conjecture | open | MathDB 页面 | 原文线索

V2-068854 | MATHDB-357736

Isometry classification conjecture for essential-rank-metric spaces

摘录 / EXCERPT: Let $d53d$ be a field, let $S_{n,d}(d53d)$ denote the space of homogeneous degree-d polynomials in n variables, and let $\operatorname{d}_{\operatorname{ess}}$ be the essential...

conjecture | open | MathDB 页面 | 原文线索

V2-068855 | MATHDB-357740

The conjectural Joyce structure from isomonodromic connections

摘录 / EXCERPT: Let M be the moduli space of pairs (C,Q_0), let $\mathcal O=\mathcal O(g,m)$ be the space of data (C,Q_0,Q_1,Q_2) described above, and let $X^\#$ be the space equipped with t...

conjecture | open | MathDB 页面 | 原文线索

V2-068856 | MATHDB-357741

Linear nodal-volume lower bound for harmonic functions with stable growth

摘录 / EXCERPT: Let $B\subset\mathbb{R}^n$ be a unit ball, with $n\geq 3$, and let u be harmonic in $4B$, vanish at the center of B, have stable growth in B, and satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-068857 | MATHDB-357743

Brenti-Carnevale's explicit evaluations for odd-length generating functions in type D

摘录 / EXCERPT: Let $n\geq 5$ and $3\leq i\leq n-1$. The polynomials $\mathcal{D}_n^{\{0,i\}}(x)$ and $\mathcal{D}_n^{\{0,1,i\}}(x)$ are the sign-twisted odd-length generating functions associated...

conjecture | open | MathDB 页面 | 原文线索

V2-068858 | MATHDB-357744

Stembridge's cyclotomicity conjecture for type-D parabolic quotients

摘录 / EXCERPT: Let $W=W(\Phi)$ be a Weyl group, where Φ is a crystallographic irreducible root system of type D with simple system Δ. For $I\subseteq\Delta$, let $W^I(x)$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-068859 | MATHDB-357750

The parity conjecture for Artin twists of elliptic curves

摘录 / EXCERPT: Parity conjecture for Artin twists.

conjecture | open | MathDB 页面 | 原文线索

V2-068860 | MATHDB-357751

The minimalist conjecture for elliptic-curve Artin twists

摘录 / EXCERPT: Minimalist conjecture for twists. For 100% of elliptic curves $E/\mathbb Q$,

conjecture | open | MathDB 页面 | 原文线索

V2-068861 | MATHDB-357753

The polynomial-factor conjecture for the number of tiles of \(\mathbb{Z}^d\)

摘录 / EXCERPT: For each d, let $t_{n,d}$ denote the number of tiles contained in $[n]^d$, and let v_n denote the volume parameter used in the paper. Polynomial-factor conjecture. There is a c...

open_problem | open | MathDB 页面 | 原文线索

V2-068862 | MATHDB-357759

Gaussian upper-bound conjecture for unsupervised binary signal reconstruction

摘录 / EXCERPT: Let $A_1,\dots,A_G\in\mathbb{R}^{m\times n}$ be measurement operators whose entries are independently and identically distributed according to a standard Gaussian distribution. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068863 | MATHDB-357760

Strong arboricity versus arboricity conjecture

摘录 / EXCERPT: Strong arboricity conjecture. Every graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068864 | MATHDB-357761

Large-girth strong arboricity conjecture

摘录 / EXCERPT: Large-girth strong arboricity conjecture. For every integer k there is an integer $g(k)$ such that every graph G with

conjecture | open | MathDB 页面 | 原文线索

V2-068865 | MATHDB-357768

Sun-Yang-Zuo's torsion-point conjecture for four-punctured Higgs bundles

摘录 / EXCERPT: Let $\lambda\in{\mathbb P}^1\setminus\{0,1,\infty\}$, and let C_λ be the elliptic curve that double-covers ${\mathbb P}^1$ and is ramified over $\{0,1,\lambda,\infty\}$. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-068866 | MATHDB-357770

Revised Kempe-class conjecture for almost bipartite graphs

摘录 / EXCERPT: Revised Kempe-class conjecture. If G is a $(k-1)$-colorable $B+E_\ell$ graph with $k\geq 4$ and

conjecture | open | MathDB 页面 | 原文线索

V2-068867 | MATHDB-357783

Equality of critical exponents for metric measure spaces with the same heat-kernel exponents

摘录 / EXCERPT: Let $(M_1,d_1,{\Greekmath 0116}_1)$ and $(M_2,d_2,{\Greekmath 0116}_2)$ be metric measure spaces admitting heat kernels satisfying the heat kernel estimate (hk), with the same expo...

open_problem | open | MathDB 页面 | 原文线索

V2-068868 | MATHDB-357784

Heat kernel estimate implies energy comparability

摘录 / EXCERPT: Let (KE) denote the energy-comparability property, and suppose a metric measure space admits a heat kernel satisfying the heat kernel estimate (hk). Heat-kernel implication questio...

open_problem | open | MathDB 页面 | 原文线索

V2-068869 | MATHDB-357796

The sorting-alternating quotient conjecture for subword complexes

摘录 / EXCERPT: Let Q be a word and let ω be an element of a Coxeter group. For the non-empty subword complex $\mathcal{SC}(Q,\omega)$, let $[e,\omega]$ be the corresponding weak-order in...

conjecture | open | MathDB 页面 | 原文线索

V2-068870 | MATHDB-357797

The brick-polytope quotient conjecture for the longest element

摘录 / EXCERPT: Let Q be a sorting and alternating word in a Coxeter group with longest element $\omega_\circ$, let $\mathsf{Brick}(Q,\omega_\circ)$ be the associated brick polytope, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068871 | MATHDB-357799

Uniqueness of bubble limits at a singular Einstein point

摘录 / EXCERPT: Let (X_j,g_j,p_j) be a sequence of Einstein metrics converging with non-collapsing volume to a singular Einstein metric $(X_\infty,g_\infty,p_\infty)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068872 | MATHDB-357805

Vilfred's non-distance-magic conjecture for cylindrical grid graphs

摘录 / EXCERPT: Let P_m be the path graph on m vertices and C_n the cycle graph on n vertices. A graph is Non-Distance Magic (NDM) if it admits no bijective labeling...

conjecture | open | MathDB 页面 | 原文线索

V2-068873 | MATHDB-357806

Odd-height cylindrical grid graphs are non-distance-magic

摘录 / EXCERPT: Let P_{2k+1} be the path graph on $2k+1$ vertices and C_n the cycle graph on n vertices. A graph is Non-Distance Magic (NDM) if it admits no bijective labeling...

conjecture | open | MathDB 页面 | 原文线索

V2-068874 | MATHDB-357811

Nonvanishing of the total face color polynomial for bridgeless graphs

摘录 / EXCERPT: Let G be a bridgeless connected graph and let Γ be a signed ribbon diagram of G. For n a positive integer, write $T(\Gamma,n,t)$ for the total face color polynomial. T...

conjecture | open | MathDB 页面 | 原文线索

V2-068875 | MATHDB-357819

Nielsen-Thurston classification conjecture for tame maps of infinite-type surfaces

摘录 / EXCERPT: Nielsen-Thurston classification conjecture. There exists such a collection Γ so that, for each component X of $S \smallsetminus \Gamma$, the first return of f to X is...

conjecture | open | MathDB 页面 | 原文线索

V2-068876 | MATHDB-357825

Frantzikinakis's logarithmic Chowla conjecture along Beatty sequences

摘录 / EXCERPT: Frantzikinakis's conjecture. For any such multiplicative functions, one has

conjecture | open | MathDB 页面 | 原文线索

V2-068877 | MATHDB-357831

No relaxation gap for peak estimation of time-delay systems

摘录 / EXCERPT: No-relaxation-gap conjecture. There is no relaxation gap between the trajectory formulation ... , so that $p^*=P^*$.

conjecture | open | MathDB 页面 | 原文线索

V2-068878 | MATHDB-357846

Conjecture on robust eigenvectors of regular simplex tensors

摘录 / EXCERPT: Let $\mathbf{w}_1,\ldots,\mathbf{w}_{n+1}\in\mathbb{R}^n$ be the vectors of a regular simplex frame, and let ... be the associated symmetric tensor of order m and dimension n, wh...

conjecture | open | MathDB 页面 | 原文线索

V2-068879 | MATHDB-357858

Zhang et al.'s unimodality conjecture for cube polynomials of median graphs

摘录 / EXCERPT: Zhang et al.'s conjecture. The cube polynomial $C(G,x)$ is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-068880 | MATHDB-357859

Zhang and Zhang's unimodality conjecture for Clar covering polynomials

摘录 / EXCERPT: Zhang and Zhang's conjecture. The polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-068881 | MATHDB-357860

Li et al.'s log-concavity conjecture for Clar covering polynomials

摘录 / EXCERPT: Li et al.'s conjecture. The polynomial $\zeta(H,x)$ is log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-068882 | MATHDB-357866

Arthur's theta-lift A-parameter conjecture for unitary groups

摘录 / EXCERPT: Let E/k be the relevant quadratic extension, let ${\rm U}(V)\times {\rm U}(W)$ be a dual pair in the stable range, and let $\pi\subset\mathcal{A}_{\rm cusp}({\rm U}(V))$ have A-p...

conjecture | open | MathDB 页面 | 原文线索

V2-068883 | MATHDB-357876

Weighted logarithmic mean characterisation of harmonic functions

摘录 / EXCERPT: Let D be a bounded domain in $\mathbb{R}^m$, with $m\geqslant 2$. Let $A_x(r_1,r_2)$ be an admissible annular domain centred at x, with $r_2>r_1>0$, and let $B_{r_2}(x)$ be adm...

conjecture | open | MathDB 页面 | 原文线索

V2-068884 | MATHDB-357878

Conjectural coefficientwise bound for modified Kazhdan-Lusztig R-polynomials

摘录 / EXCERPT: Let $[u,v]$ be a Bruhat interval, let $I\subset [u,v]$ be a strong hypercube decomposition, and let $\widetilde{H}_{u,v,I}(q)$ and $\widetilde{R}_{u,v}(q)$ be the polynomials defin...

conjecture | open | MathDB 页面 | 原文线索

V2-068885 | MATHDB-357879

Jonsson's polytopality conjecture for multiassociahedra

摘录 / EXCERPT: Jonsson's polytopality conjecture. For every $n\geq 2k+1$, the complex $\overline{\mathcal{A}\hskip-0pt\mathit{ss}}_{k}(n)$ is a polytopal sphere: there is a simplicial polytope of...

conjecture | open | MathDB 页面 | 原文线索

V2-068886 | MATHDB-357890

The rainbow triangle-packing conjecture for properly coloured complete graphs

摘录 / EXCERPT: Rainbow triangle-packing conjecture. The graph K_n contains a rainbow subgraph that is a disjoint union of triangles covering all but at most C vertices, for some absolute cons...

conjecture | open | MathDB 页面 | 原文线索

V2-068887 | MATHDB-357893

Density of typical Hodge loci at the expected dimension

摘录 / EXCERPT: Let $\mathbb V$ be as in Theorem mainqsimple, and let $(\mathbf M,D_M)\subsetneq(\mathbf G,D)$ be a strict Hodge sub-datum satisfying ... The expected-dimension typical-point conject...

conjecture | open | MathDB 页面 | 原文线索

V2-068888 | MATHDB-357904

Even-hook divisibility conjecture for self-conjugate partitions

摘录 / EXCERPT: Even-hook divisibility conjecture. For all integers $n\geq 0$ and $m\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-068889 | MATHDB-357918

The Abel-Jacobi injectivity conjecture for special cycles

摘录 / EXCERPT: Let $\rho^{?}\in\{\rho,\rho^*(1)\}$, let $K^p\subset \mathrm{H}(\mathbf{A}^{p\infty})$ be open compact, and let K_p° be the chosen hyperspecial subgroup. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-068890 | MATHDB-357929

Guo's extension of two q-congruences modulo the square of a cyclotomic polynomial

摘录 / EXCERPT: Let n be a positive odd integer, let $(a;q)_k$ denote the q-shifted factorial, and let $\Phi_n(q)$ be the nth cyclotomic polynomial. Guo's conjecture. Modulo $\Phi_n(q)^2$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-068891 | MATHDB-357933

The finite simple group k-valent GRR conjecture

摘录 / EXCERPT: A graphical regular representation (GRR) of a group G is a Cayley graph whose full automorphism group is isomorphic to G; a GRR is k-valent when the corresponding graph has v...

conjecture | open | MathDB 页面 | 原文线索

V2-068892 | MATHDB-357934

Time-average code-length conjecture for zero-delay LQG quantizer coding

摘录 / EXCERPT: Time-average code-length conjecture. The time-average expected codeword lengths should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-068893 | MATHDB-357944

The multi-phase concentric-balls conjecture

摘录 / EXCERPT: Let $m\in\mathbb{N}$ and let Ω_k, $k\in\{1,\dots,m\}$, be as in the introduction. Consider the multi-phase overdetermined problem described by problem ... admits a solution...

conjecture | open | MathDB 页面 | 原文线索

V2-068894 | MATHDB-357958

Kuznetsov's conjecture on generic non-jumping lines for instanton bundles

摘录 / EXCERPT: Let Y be a del Pezzo threefold, and let E be an instanton bundle on Y, meaning a stable vector bundle of rank two with $c_1(E)=0$ and $H^1(Y,E(-1))=0$. A line $L\subset Y$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-068895 | MATHDB-357962

Absolute continuity conjecture for periodic second-order elliptic operators

摘录 / EXCERPT: Let L be a periodic second-order elliptic operator with sufficiently nice coefficients. Absolute continuity conjecture. The spectrum of L is absolutely continuous, and hence...

conjecture | open | MathDB 页面 | 原文线索

V2-068896 | MATHDB-357964

Generic non-degeneracy conjecture for spectral gap edges

摘录 / EXCERPT: Consider an elliptic periodic operator and vary its parameters, for example its periodic potential. Generic gap-edge non-degeneracy conjecture. For almost all such parameters, the...

conjecture | open | MathDB 页面 | 原文线索

V2-068897 | MATHDB-357966

Closed-range conjecture for composition operators on the Dirichlet space

摘录 / EXCERPT: Let $\mathbb D$ be the unit disk, let φ be an analytic self-mapping of $\mathbb D$, and let $C_{\varphi}f=f\circ\varphi$ be bounded on the Dirichlet space...

conjecture | open | MathDB 页面 | 原文线索

V2-068898 | MATHDB-357971

The tree-coloring strong spatial mixing conjecture

摘录 / EXCERPT: Let T) be a tree of maximum degree Δ, and let q be the number of colors. The uniform distribution on proper q-colorings of T is considered with strong spatial mixing...

conjecture | open | MathDB 页面 | 原文线索

V2-068899 | MATHDB-357972

Jerrum's Glauber-dynamics conjecture for graph colorings

摘录 / EXCERPT: Let G be a graph of maximum degree Δ, and consider the Glauber dynamics for sampling proper q-colorings of G. Jerrum's conjecture. The Glauber dynamics for sampling...

conjecture | open | MathDB 页面 | 原文线索

V2-068900 | MATHDB-357973

The antiferromagnetic Potts-model spatial-mixing threshold conjecture

摘录 / EXCERPT: Let $q \geq 2$ be an integer, let $\beta \in [0,1]$, and consider the q-state antiferromagnetic Potts model on graphs of maximum degree Δ. Define ... Weak spatial mixing (WS...

conjecture | open | MathDB 页面 | 原文线索

V2-068901 | MATHDB-357986

Reed-Muller capacity conjecture for binary-input memoryless symmetric channels

摘录 / EXCERPT: Let $\mathcal{P}$ be a binary-input memoryless symmetric (BMS) channel with output alphabet $\mathcal{Y}$, and let ... be its capacity under the uniform input distribution. For a seq...

conjecture | open | MathDB 页面 | 原文线索

V2-068902 | MATHDB-357991

The admissible-function convex-hull conjecture for $SU(N)$

摘录 / EXCERPT: Let $N\geq 2$, set $k=\frac{N(N-1)}{2}$ and $l=\frac{N(N+1)}{2}-1$, and let S^* denote the relevant multiplicative parameter space. A $SU(N)$-admissible function is a function .....

conjecture | open | MathDB 页面 | 原文线索

V2-068903 | MATHDB-357996

Scaling-limit conjecture for the excursion decomposition of the two-dimensional DGFF

摘录 / EXCERPT: Excursion-decomposition scaling-limit conjecture. The discrete decompositions converge to a decomposition $(E_k,\theta_k,\mu_k)_{k\geq 1}$ such that the union of outermost positive...

conjecture | open | MathDB 页面 | 原文线索

V2-068904 | MATHDB-357998

Conjecture on simultaneous Waring identifiability for ternary forms of consecutive degrees

摘录 / EXCERPT: Simultaneous Waring identifiability conjecture. The polynomial vector admits a unique simultaneous Waring decomposition with

conjecture | open | MathDB 页面 | 原文线索

V2-068905 | MATHDB-358000

Local differential precision operator conjecture for Gaussian Markov random fields on metric graphs

摘录 / EXCERPT: Let $Gamma$ be a compact metric graph with edge set $\mathcal{E}$. A Gaussian Markov random field of order p is assumed to satisfy some mild regularity conditions. On each edge...

conjecture | open | MathDB 页面 | 原文线索

V2-068906 | MATHDB-358001

The standard-orientation conjecture for interior polynomials of bipartite graphs

摘录 / EXCERPT: Standard-orientation conjecture. Among all semi-balanced orientations of a bipartite graph, the standard orientations minimize every coefficient of the interior polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-068907 | MATHDB-358006

Conjecture on the joint cokernel distribution of shifted random p-adic matrices

摘录 / EXCERPT: Let $X_m:=\{x_1,\ldots,x_m\}$ be a finite ordered subset of $\mathbb{Z}_p$ whose elements have distinct reductions modulo p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-068908 | MATHDB-358007

The piecewise quasi-polynomial conjecture for lattice tetrahedra by multi-width

摘录 / EXCERPT: The multi-width tetrahedron counting conjecture. There is a piecewise quasi-polynomial with four components counting lattice tetrahedra of multi-width (w_1,w_2,w_3). There is one...

conjecture | open | MathDB 页面 | 原文线索

V2-068909 | MATHDB-358013

Heng's NMDS code conjecture for the extended trace code

摘录 / EXCERPT: Heng's conjecture. If $q>2$, then $\widetilde{\overline{C_D}}$ is a $[q+1,3,q-2]_q$ NMDS code. This conjecture concerns the remaining family of NMDS codes identified by Heng; provi...

conjecture | open | MathDB 页面 | 原文线索

V2-068910 | MATHDB-358016

Poissonian gap distribution conjecture for αn^θ

摘录 / EXCERPT: Let $\alpha\neq 0$ and let $\theta>0$ satisfy $\theta\neq\tfrac12$ and $\theta\notin\mathbb Z$. For each N, let $y_1^{(N)}<\cdots<y_N^{(N)}$ be an ordering of the points...

conjecture | open | MathDB 页面 | 原文线索

V2-068911 | MATHDB-358018

Rigidity conjecture for 4-connected symmetric simplicial 2-circuits

摘录 / EXCERPT: Rigidity conjecture. If $G({\cal S})$ is 4-connected and non-planar, then $G({\cal S})$ is $\Gamma_{1,3}$-rigid in $\mathbb R^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-068912 | MATHDB-358019

Redundant symmetry-rigidity conjecture for symmetric simplicial circuits

摘录 / EXCERPT: Redundant symmetry-rigidity conjecture. For every such Γ and e, $(G({\cal S})-e-e^*,*)$ is Γ-rigid in $\mathbb R^{k+1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-068913 | MATHDB-358024

Gessel's non-D-finiteness conjecture for the full generating function

摘录 / EXCERPT: Let $G(x,y,t)$ be the full generating function for Gessel walks with step set $\{\nearrow,\swarrow,\leftarrow,\rightarrow\}$. Gessel's conjecture. The generating function...

conjecture | open | MathDB 页面 | 原文线索

V2-068914 | MATHDB-358031

The perfect Martin invariant conjecture for cyclically 6-connected 4-regular graphs

摘录 / EXCERPT: Let G_1 and G_2 be cyclically 6-connected 4-regular graphs, and let v_1 and v_2 be vertices whose deletion produces graphs with periods $\mathcal{P}(G_1\setminus v_1)$...

conjecture | open | MathDB 页面 | 原文线索

V2-068915 | MATHDB-358033

The perfect Martin sequence conjecture for cyclically 4-connected 3-regular graphs

摘录 / EXCERPT: Perfect Martin sequence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068916 | MATHDB-358040

The root conjecture for frame lattices corresponding to lattices $T_{[abc]}$

摘录 / EXCERPT: Let $T_{[abc]}$ be a lattice in the notation of the paper, and let W be a frame lattice corresponding to $T_{[abc]}$. A root is a lattice vector of minimal norm as defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-068917 | MATHDB-358061

Resurgence and radial-limit conjecture for Habiro elements of knots

摘录 / EXCERPT: Let K be a knot. Let $F_K(x)=\Phi_K(e^{-1/x})$ be the formal expansion of the Habiro element Φ_K around a root of unity, let $G_K(p)$ denote the Borel transform of $F_K(x)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-068918 | MATHDB-358068

Fundamental decoupling identity for generalized color orderings

摘录 / EXCERPT: Fundamental decoupling identity. Fix a fundamental triple $\mathcal{T}=(\Sigma_0,\Sigma_1,\Sigma_2)$. As $\Sigma\in CO(k,n)$ ranges over all double $\mathcal{T}$-extensions...

conjecture | open | MathDB 页面 | 原文线索

V2-068919 | MATHDB-358077

The Deformation Conjecture on the emergence of internal symmetries

摘录 / EXCERPT: The Standard Model has internal symmetries describing elementary particles, while relativistic symmetries describe spacetime transformations. The Deformation Conjecture. Internal s...

conjecture | open | MathDB 页面 | 原文线索

V2-068920 | MATHDB-358080

Clustering conjecture for Cucker-Smale systems

摘录 / EXCERPT: Let N agents have initial data $({\bf x}_0,{\bf v}_0)\in\mathbb{T}^{nN}\times\mathbb{R}^{nN}$ and evolve according to the Cucker-Smale system. A cluster is a group of agents shar...

conjecture | open | MathDB 页面 | 原文线索

V2-068921 | MATHDB-358081

Generic two-agent alignment conjecture under confinement

摘录 / EXCERPT: Consider the confined Cucker-Smale system for N agents with initial data $({\bf x}_0,{\bf v}_0)\in\mathbb{R}^{nN}\times\mathbb{R}^{nN}\setminus\mathcal{H}$, where $\mathcal{H}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-068922 | MATHDB-358085

Dimension-boundedness conjecture for cover-planar posets

摘录 / EXCERPT: Dimension-boundedness conjecture. The class of cover-planar posets is $\dim$-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-068923 | MATHDB-358086

Multiplicative lower-bound conjecture for dimension of cover-planar posets

摘录 / EXCERPT: Multiplicative lower-bound conjecture. Among cover-planar posets,

conjecture | open | MathDB 页面 | 原文线索

V2-068924 | MATHDB-358102

Nonemptiness and infinitude conjecture for geometric-progression value sets

摘录 / EXCERPT: Let $f(x,y)=(y^2-x^3)/x$. For $Q\in\mathbb{N}_{\geq 2}$, let C_Q be the set defined in the paper for this rational function and quotient Q. Nonemptiness and infinitude conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-068925 | MATHDB-358105

Tightness conjecture for oriented chromatic number in bounded-degeneracy classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a class of graphs with bounded degeneracy. For a graph G, let $\chi_2(G)$ denote its 2-dipath chromatic number and let $\chi_o(G)$ denote its oriented chro...

conjecture | open | MathDB 页面 | 原文线索

V2-068926 | MATHDB-358111

Finite accumulation point conjecture for transmission eigenvalues

摘录 / EXCERPT: Assume $k_\epsilon>1/\sqrt{2}$ and define ... This number is a pole of the Drude-Lorentz term, and the transmission eigenvalues are discrete outside the compact region K described...

conjecture | open | MathDB 页面 | 原文线索

V2-068927 | MATHDB-358112

Conjecture on finiteness of Born transmission eigenvalues in bounded horizontal strips

摘录 / EXCERPT: Let $d=2$ or $d=3$, let B_2 be the ball of radius 2, and let ... A Born transmission-eigenvalue strip-finiteness conjecture. For any $C>0$, the strip ... contains at most finitely...

conjecture | open | MathDB 页面 | 原文线索

V2-068928 | MATHDB-358113

Global non-scattering solutions conjecture for the cubic wave equation

摘录 / EXCERPT: Global non-scattering solutions conjecture. Let u be a solution of the cubic wave equation with the prescribed initial-data problem, defined for $t\in[0,\infty)$ and not scatteri...

conjecture | open | MathDB 页面 | 原文线索

V2-068929 | MATHDB-358116

Logarithmic critical-norm criterion for global scattering of cubic-wave solutions

摘录 / EXCERPT: Let u be a solution of the cubic wave equation on its maximal interval $[0,t_+)$, with initial data in $\mathcal{H}^{1/2}$. Logarithmic critical-norm criterion. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-068930 | MATHDB-358133

Conjecture on Gaussian absolute continuity and Gibbs measures in Benjamin-Ono Birkhoff coordinates

摘录 / EXCERPT: Gaussian absolute-continuity conjecture. If 4μ is a suitable Gaussian measure on $h^{s+\frac{1}{2}}$, then the corresponding measure $4\rho_s$ is absolutely continuous with res...

conjecture | open | MathDB 页面 | 原文线索

V2-068931 | MATHDB-358134

Finiteness conjecture for non-decomposable critically frustrated signed graphs

摘录 / EXCERPT: Finiteness conjecture. For every positive integer k, the set $\mathcal{L}^*(k)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-068932 | MATHDB-358135

Finiteness conjecture for the subclass of critically frustrated signed graphs

摘录 / EXCERPT: Finiteness conjecture. For every positive integer k, the set $\mathcal{S}^*(k)$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-068933 | MATHDB-358136

Facial-cycle conjecture for non-decomposable critically frustrated signed plane graphs

摘录 / EXCERPT: Facial-cycle conjecture. Every non-decomposable critically k-frustrated signed plane graph has exactly $2k$ facial cycles, each of which is a negative cycle.

conjecture | open | MathDB 页面 | 原文线索

V2-068934 | MATHDB-358137

Maximum-degree conjecture for critically frustrated signed graphs

摘录 / EXCERPT: Maximum-degree conjecture. Every critically k-frustrated signed graph (G,σ) satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-068935 | MATHDB-358158

Optimality of TC-dominant generating sets for odd-length secondary structure avoidance codes

摘录 / EXCERPT: Optimality conjecture. When m is odd, the generating set S of all TC-dominant length-m sequences gives an m-secondary-structure-avoidance code with the largest rate among a...

conjecture | open | MathDB 页面 | 原文线索

V2-068936 | MATHDB-358171

Existence and uniqueness conjecture for two polynomial difference equations

摘录 / EXCERPT: Polynomial difference-equation conjecture. For each nonnegative integer u, the first equation has, up to multiplication by a nonzero constant, a unique polynomial solution P, a...

conjecture | open | MathDB 页面 | 原文线索

V2-068937 | MATHDB-358177

Maximum-period conjecture for the logistic map over $\mathbb{Z}_{3^n}$

摘录 / EXCERPT: Maximum-period conjecture. When $\mu\bmod 3=1$, the maximum period is

conjecture | open | MathDB 页面 | 原文线索

V2-068938 | MATHDB-358181

K-polystability conjecture for rank-2 Fano threefolds of degree 28

摘录 / EXCERPT: Let X be the smooth Fano threefold obtained by blowing up a smooth quadric threefold $Q\subset\mathbb{P}^4$ along the smooth twisted rational quartic curve C_4, so that...

conjecture | open | MathDB 页面 | 原文线索

V2-068939 | MATHDB-358185

Conti-Otto-Serfaty dimension conjecture for the irrigated measure

摘录 / EXCERPT: Consider the branched transport energy functional referred to as ... , then $\mu_{\pm T}$ is of dimension $8/5$. This conjecture predicts the Hausdorff dimension of the boundary trac...

conjecture | open | MathDB 页面 | 原文线索

V2-068940 | MATHDB-358187

The Easy Coefficients Conjecture for rotation-symmetric quadratic Boolean functions

摘录 / EXCERPT: Let I be an index set, let $Q_{I,n}$ be the associated rotation-symmetric quadratic Boolean function, and set ... Let $w_1,w_2,\ldots$ be the sequence obtained by extending the rec...

conjecture | open | MathDB 页面 | 原文线索

V2-068941 | MATHDB-358197

Flexible stability conjecture for higher-rank arithmetic lattices

摘录 / EXCERPT: Let Γ be the arithmetic group specified by the local-to-global conjecture above. A group G is flexibly stable if every sufficiently accurate approximate homomorphism from...

conjecture | open | MathDB 页面 | 原文线索

V2-068942 | MATHDB-358198

Ultraproduct normal-subgroup conjecture for higher-rank arithmetic groups

摘录 / EXCERPT: Let Γ be the arithmetic group specified by the local-to-global conjecture above, fix a non-principal ultrafilter $\mathcal{N}$ on the natural numbers, and let O^* and...

conjecture | open | MathDB 页面 | 原文线索

V2-068943 | MATHDB-358210

Versal Breuil--Mézard conjecture for \(\mathrm{GSp}_4\)

摘录 / EXCERPT: Let ${\mathcal S}$ be a set of types, and let $\mathrm{JH}(\overline{\sigma}({\mathcal S}))$ be the union of the Jordan--Hölder factors of the reductions...

conjecture | open | MathDB 页面 | 原文线索

V2-068944 | MATHDB-358219

Theorem 1.1 limit-shape conjecture for general rate vectors

摘录 / EXCERPT: Theorem 1.1 limit-shape conjecture. The results of Theorem 1.1 hold for all $\vec{\mathbf{r}}\in\mathscr{R}$ under $N\geq1$ and $N^2T\to\infty$.

conjecture | open | MathDB 页面 | 原文线索

V2-068945 | MATHDB-358220

Sokal-Zeng's J-fraction conjecture for record-and-cycle permutation polynomials

摘录 / EXCERPT: Let $\widehat{Q}_n(x_1,x_2,y_1,y_2,u_1,u_2,v_1,v_2,\mathbf{w},\lambda)$ be the permutation polynomials defined by weighting the record-and-cycle statistics, with...

conjecture | open | MathDB 页面 | 原文线索

V2-068946 | MATHDB-358227

Nakamuta's conjecture on c-Wilf equivalence of permutation patterns

摘录 / EXCERPT: Let v and w be permutation patterns. They are c-Wilf-equivalent in permutations if ... for every $n\in{\mathbb N}$, and they are strongly c-Wilf-equivalent in permutations if ... f...

conjecture | open | MathDB 页面 | 原文线索

V2-068947 | MATHDB-358232

The purported trace on the dynamic tracing category satisfies the traced monoidal axioms

摘录 / EXCERPT: Let t be a monad equipped with an exceptional structure, and let the purported trace on the category denoted by $\mathbf{DK}_{t}$ be the trace constructed above from the correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-068948 | MATHDB-358238

Larrión-Neumann-Lara-Pizaña conjecture on clique divergence of non-negative Euler characteristic surfaces

摘录 / EXCERPT: Let G be a locally cyclic graph of minimum degree $\delta\ge 4$ triangulating a closed surface of Euler characteristic $\chi\ge 0$; equivalently, the surface is a sphere, project...

conjecture | open | MathDB 页面 | 原文线索

V2-068949 | MATHDB-358239

Clique convergence conjecture for locally cyclic triangulations of a disc

摘录 / EXCERPT: Let G be a locally cyclic graph with boundary and minimum degree $\delta\ge 4$ triangulating a disc. Such a graph is clique null if its clique dynamics converges to the one-verte...

conjecture | open | MathDB 页面 | 原文线索

V2-068950 | MATHDB-358242

The conjecture on equality of F-pure and log canonical thresholds for ideals

摘录 / EXCERPT: Threshold-equality conjecture. There are infinitely many primes p for which

conjecture | open | MathDB 页面 | 原文线索

V2-068951 | MATHDB-358245

Passare's conjecture on solid amoebas of maximally sparse polynomials

摘录 / EXCERPT: Let $p(z)$) be a maximally sparse polynomial, meaning that the support of p is equal to the set of vertices of its Newton polytope. The Passare conjecture. The amoeba of $p(z)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-068952 | MATHDB-358268

The negative-part contraction conjecture for lifted flip MMPs

摘录 / EXCERPT: Let (X_1,B_1+M_1) be a projective $\mathbb{Q}$-factorial NQC dlt generalised pair such that $K_{X_1}+B_1+M_1$ is pseudoeffective. Consider a sequence of flips in a...

conjecture | open | MathDB 页面 | 原文线索

V2-068953 | MATHDB-358269

Arbitrarily deep failure of log-concavity for independence polynomials of trees

摘录 / EXCERPT: Let G be a tree, let $\alpha(G)$ denote its independence number, and write its independence polynomial as ... Here, log-concavity at the jth coefficient means...

conjecture | open | MathDB 页面 | 原文线索

V2-068954 | MATHDB-358284

The fractional chromatic conjecture for 4-cycle-free planar graphs

摘录 / EXCERPT: Let G be a 4-cycle-free planar graph, and let $\chi_f(G)$ denote its fractional chromatic number. Fractional chromatic conjecture. There exists some $\epsilon>0$ such that ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-068955 | MATHDB-358291

Compact-closedness conjecture for pointed bimodular profunctors

摘录 / EXCERPT: Let $\mathbb{M}$ be a monoidal category, and let $\mathbb{M}^{\mathrm{Rev}}$ denote its reverse monoidal category, with tensor product ... The objects, 1-cells, 2-cells and 3-cells of the tricategory...

conjecture | open | MathDB 页面 | 原文线索

V2-068956 | MATHDB-358295

Boyd and Rodriguez Villegas's Mahler measure identity for a conductor-15 elliptic curve

摘录 / EXCERPT: Let $m(P)$ denote the Mahler measure of a Laurent polynomial P. Let E be the elliptic curve ... which has conductor 15. Boyd and Rodriguez Villegas's conjecture. The Mahler mea...

conjecture | open | MathDB 页面 | 原文线索

V2-068957 | MATHDB-358299

Gluing conjecture for globally linked pairs

摘录 / EXCERPT: Gluing conjecture. If $\{u,v\}$ is linked in G_i in $\mathbb{R}^d$ for $i\in\{1,2\}$, then $\{u,v\}$ is globally linked in G in $\mathbb{R}^d$.

conjecture | open | MathDB 页面 | 原文线索

V2-068958 | MATHDB-358300

Gluing characterization conjecture for globally linked pairs

摘录 / EXCERPT: Gluing characterization conjecture. If $\{u,v\}$ is not linked in G_2 in $\mathbb{R}^d$, then $\{u,v\}$ is globally linked in G in $\mathbb{R}^d$ if and only if it is globally...

conjecture | open | MathDB 页面 | 原文线索

V2-068959 | MATHDB-358301

Oh, Harks, and Condorelli-Watson's breakpoint derivative conjecture for routing games

摘录 / EXCERPT: Let $({G},\boldsymbol{c},\mathcal H,\boldsymbol{\mu}(\,\cdot\,))$ be a routing game whose cost functions are continuous, differentiable, strictly increasing, and have positive seco...

conjecture | open | MathDB 页面 | 原文线索

V2-068960 | MATHDB-358303

Hitting time conjecture for connectivity in the W-edge-incremental process

摘录 / EXCERPT: Let W be a connected graphon and $K\in\mathbb{N}$. For each $n\in\mathbb{N}$, consider the W-edge-incremental process $\mathfrak{X}_n=(G_n^t)_{t=0}^{\binom{n}{2}}$ of order n...

conjecture | open | MathDB 页面 | 原文线索

V2-068961 | MATHDB-358304

Monotonicity conjecture for flame-front velocity and reaction width

摘录 / EXCERPT: Let $c^*(\theta,\Lambda,\alpha)$ denote the velocity of propagation and let $R^*(\theta,\Lambda,\alpha)$ denote the reaction width, where θ is the ignition temperature,...

conjecture | open | MathDB 页面 | 原文线索

V2-068962 | MATHDB-358305

Dimension-free Lipschitz conjecture for Brenier maps onto log-Lipschitz Gaussian perturbations

摘录 / EXCERPT: Let γ be the standard Gaussian measure on $\mathbb{R}^d$, and let μ be a log-Lipschitz perturbation of γ. Let $T=\nabla\varphi$ be the Brenier map transporting...

conjecture | open | MathDB 页面 | 原文线索

V2-068963 | MATHDB-358308

Léveque-Maffray-Trotignon conjecture on 4-coloring ISK4-free graphs

摘录 / EXCERPT: Léveque-Maffray-Trotignon conjecture. Every ISK_4-free graph is 4-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-068964 | MATHDB-358319

Conjecture on the motive isomorphism between a symmetric Kloosterman motive and a rank-three motive

摘录 / EXCERPT: Let M_2^6 and $\mathrm{M}_3^{(2,2)}$ be the motives introduced in the paper, and let (-3) denote the Tate twist. Motive-isomorphism conjecture. The two motives ... are i...

conjecture | open | MathDB 页面 | 原文线索

V2-068965 | MATHDB-358321

Blecher-Knopfmacher's fixed-point inequality for integer partitions

摘录 / EXCERPT: Let n be a positive integer, and let $p(n)$ denote the number of partitions of n. A partition $\lambda=(\lambda_1,\ldots,\lambda_j)$ of n, written $\lambda\vdash n$, has a fi...

conjecture | open | MathDB 页面 | 原文线索

V2-068966 | MATHDB-358322

Quartic lifespan conjecture for small-data ILW solutions

摘录 / EXCERPT: The Intermediate Long Wave equation has small initial data of size ϵ and dispersion relation ... which is KdV-like at low frequencies and Benjamin-Ono-like at high frequencie...

conjecture | open | MathDB 页面 | 原文线索

V2-068967 | MATHDB-358323

Zero-distribution conjecture for generated Taylor polynomials with nonpositive discriminant

摘录 / EXCERPT: Let $\{P_m(z)\}_{m\geq 0}$ be a sequence of functions of z generated by ... Here $a,b,c\in\mathbb{R}\backslash\{0\}$ and $b^2-4ac\leq 0$. Let t_1 be the zero of at^2+bt+c havin...

conjecture | open | MathDB 页面 | 原文线索

V2-068968 | MATHDB-358346

Poor-edge conjecture for normal colorings of bridgeless cubic graphs

摘录 / EXCERPT: Let G be a bridgeless cubic graph. Let $\mathrm{NC}(G)$ be the set of all normal 5-colorings of G, and, assuming this set is nonempty, let $\mathrm{poor}(G)$ be the maximum n...

conjecture | open | MathDB 页面 | 原文线索

V2-068969 | MATHDB-358348

The Prym-Petri expectation for double covers of curves on Nikulin surfaces

摘录 / EXCERPT: Prym-Petri expectation. If $-r\leq \rho(2g-1,r,2g-2)<r$, then

conjecture | open | MathDB 页面 | 原文线索

V2-068970 | MATHDB-358357

Almost-cover conjecture for zonotopes

摘录 / EXCERPT: Let Z be a zonotope, and let $\operatorname{rk}(Z)$ denote its rank, namely the number of summands in a representation of Z as the Minkowski sum of a nondegenerate collection o...

conjecture | open | MathDB 页面 | 原文线索

V2-068971 | MATHDB-358358

Linear-order conjecture for immersions of transitive tournaments in tournaments

摘录 / EXCERPT: Let k be a positive integer. A tournament is an orientation of a complete graph, and an immersion of a transitive tournament on k vertices consists of k branch vertices toget...

conjecture | open | MathDB 页面 | 原文线索

V2-068972 | MATHDB-358376

Foliated complement conjecture

摘录 / EXCERPT: Foliated complement conjecture. For every positive real number ϵ and positive integer d, there exists a positive real number n depending only on ϵ, d and...

conjecture | open | MathDB 页面 | 原文线索

V2-068973 | MATHDB-358377

Foliated global index conjecture

摘录 / EXCERPT: Foliated global index conjecture. For every positive integer d, there exists a positive integer I depending only on d and Γ such that, whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-068974 | MATHDB-358378

Uniform boundedness conjecture for minimal log discrepancies of foliations

摘录 / EXCERPT: Uniform boundedness conjecture for foliated minimal log discrepancies. For every positive integer d, there exists a positive real number l depending only on d and Γ su...

conjecture | open | MathDB 页面 | 原文线索

V2-068975 | MATHDB-358380

The full exceptional collection conjecture for semisimple homogeneous spaces

摘录 / EXCERPT: Full exceptional collection conjecture. There is a full exceptional collection in

conjecture | open | MathDB 页面 | 原文线索

V2-068976 | MATHDB-358385

Two-sided majorization conjecture for cyclotomic generating functions

摘录 / EXCERPT: Let ... where $a_1\leq\cdots\leq a_m$ and $b_1\leq\cdots\leq b_m$. Two-sided majorization conjecture. For every ℓ, one has ... Equivalently, the multiset of numerator parameters w...

conjecture | open | MathDB 页面 | 原文线索

V2-068977 | MATHDB-358387

Explicit reciprocity law for the GSp4 times GL2 Euler-system class

摘录 / EXCERPT: Let $\underline{\pi}$ and $\underline{\sigma}$ be the Coleman families and let $\mathbf{z}_1(\underline{\pi}\times\underline{\sigma},\gamma_S)$ be the associated Euler-system cohom...

conjecture | open | MathDB 页面 | 原文线索

V2-068978 | MATHDB-358388

Bessel coupling conjecture for the local behavior at the Brownian maximum

摘录 / EXCERPT: Bessel coupling conjecture. There exists a coupling of (R,Y) and $\widetilde{\mathbf{B}}$ such that almost surely there are $\delta_1>0$ and $n_1\in\mathbb{N}$ for which, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-068979 | MATHDB-358390

Churchley's totally odd immersion conjecture

摘录 / EXCERPT: Churchley's conjecture. Every graph G with $\chi(G)\ge t$ contains a totally odd immersion of K_t.

conjecture | open | MathDB 页面 | 原文线索

V2-068980 | MATHDB-358391

The totally odd strong immersion conjecture

摘录 / EXCERPT: Totally odd strong immersion conjecture. Every graph G with $\chi(G)\ge t$ contains a totally odd strong immersion of K_t.

conjecture | open | MathDB 页面 | 原文线索

V2-068981 | MATHDB-358399

Maximal Hilbert function conjecture for generic 2-square points

摘录 / EXCERPT: Maximal Hilbert function conjecture. For every s and t,

conjecture | open | MathDB 页面 | 原文线索

V2-068982 | MATHDB-358410

Multivariate big-step induction conjecture for right cancellation of list concatenation

摘录 / EXCERPT: Let T_{1} be the base theory for list concatenation. For a finite sequence of pairwise distinct list variables $\vec{X}=(X_{1},\dots,X_{m})$ and a sequence of non-zero natural nu...

conjecture | open | MathDB 页面 | 原文线索

V2-068983 | MATHDB-358416

Ekin's missing crank inequality modulo 11

摘录 / EXCERPT: Let $p(n)$ be the number of partitions of n, and let $M(r,11,n)$ denote the number of partitions of n whose crank is congruent to r modulo 11. Ekin's crank inequality. For...

conjecture | open | MathDB 页面 | 原文线索

V2-068984 | MATHDB-358417

Conjectural two-term rank-crank inequalities modulo 11

摘录 / EXCERPT: Let $N(r,11,n)$ and $M(r,11,n)$ denote the numbers of partitions of n with rank and crank, respectively, congruent to r modulo 11. For each residue m modulo 11, write...

conjecture | open | MathDB 页面 | 原文线索

V2-068985 | MATHDB-358430

The asymptotic Cobham conjecture for regular sequences

摘录 / EXCERPT: Let $k,\ell\in\mathbb{N}$ be multiplicatively independent, meaning that no positive powers of k and ℓ are equal. Let $\mathbf a=(a_n)_{n\in\mathbb{N}_0}$ be an integer seque...

conjecture | open | MathDB 页面 | 原文线索

V2-068986 | MATHDB-358433

Injectivity conjecture for actions of essentially trivial twisted K3 autoequivalences

摘录 / EXCERPT: Let X be a K3 surface. An essentially trivial twisted K3 surface $\mathscr{X}\to X$ is one whose associated Brauer class is zero. Let ... be the subgroup generated by the actions o...

conjecture | open | MathDB 页面 | 原文线索

V2-068987 | MATHDB-358435

The normalized arrival conjecture for the Collatz recurrence

摘录 / EXCERPT: Let $(a_{k,n})$ be the normalized arrival sequence defined by the recurrence described in the source, with $a_k:=1$ as its normalization, and let $a_{k,1}$ denote its value at 1....

conjecture | open | MathDB 页面 | 原文线索

V2-068988 | MATHDB-358436

The formal power series conjecture equivalent to Collatz

摘录 / EXCERPT: For each $k\geq 1$, let $A_k(x):=\sum_{n\geq 0}a_{k,n}x^n$ be the modified formal power series introduced from the arrival recurrence, and let $A_k'(0)$ denote its derivative at ze...

conjecture | open | MathDB 页面 | 原文线索

V2-068989 | MATHDB-358438

The Solomon-Terao palindromicity conjecture for free arrangements

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a hyperplane arrangement in a vector space of dimension ℓ, and let $\Psi({\mathcal{A}};x,t)$ denote its Solomon-Terao polynomial. A polynomial is palind...

conjecture | open | MathDB 页面 | 原文线索

V2-068990 | MATHDB-358446

The Klemm-Mariño formula for Gromov-Witten invariants of Enriques surfaces

摘录 / EXCERPT: Klemm-Mariño's formula. For all g and β,

conjecture | open | MathDB 页面 | 原文线索

V2-068991 | MATHDB-358447

Tanaka-Thomas's Vafa-Witten invariant formula for Joyce-Song pairs

摘录 / EXCERPT: Tanaka-Thomas's conjecture. There exist rational numbers $\mathsf{VW}(v_i)$ such that, for all $n\gg0$,

conjecture | open | MathDB 页面 | 原文线索

V2-068992 | MATHDB-358448

Relative Gromov-Witten quasi-Jacobi-form and holomorphic anomaly conjecture

摘录 / EXCERPT: Relative holomorphic anomaly conjecture. The series lies in

conjecture | open | MathDB 页面 | 原文线索

V2-068993 | MATHDB-358449

LDP under the weak Kato hypothesis

摘录 / EXCERPT: LDP under the weak Kato hypothesis. The statement of the main large-deviations theorem holds true if the strong Kato hypothesis is replaced with the weak Kato condition.

conjecture | open | MathDB 页面 | 原文线索

V2-068994 | MATHDB-358454

Unipotent polynomial monofibration conjecture

摘录 / EXCERPT: Let F_n be a free group of rank n, let $Gamma=F_n\rtimes_{alpha}\langle t\rangle$ be a unipotent polynomial suspension, and let F_m be a free group of rank m. Suppose that...

conjecture | open | MathDB 页面 | 原文线索

V2-068995 | MATHDB-358455

The LCT-strong Euler-homogeneity conjecture for free divisors

摘录 / EXCERPT: LCT-strong Euler-homogeneity conjecture. If D satisfies the Logarithmic Comparison Theorem, then D is strongly Euler-homogeneous.

conjecture | open | MathDB 页面 | 原文线索

V2-068996 | MATHDB-358460

Chowla-type correlation conjecture for omega sequences

摘录 / EXCERPT: Omega correlation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-068997 | MATHDB-358464

Growth problem for strong boundedness constants of special linear groups over number fields

摘录 / EXCERPT: For $n\in\mathbb{N}$, define ... Here $\mathscr{O}_K$ denotes the ring of integers of the number field K, and Δ_k is the strong boundedness invariant considered in the paper...

open_problem | open | MathDB 页面 | 原文线索

V2-068998 | MATHDB-358466

Gyárfás's conjecture on chi-boundedness of forest-free graphs

摘录 / EXCERPT: Let H be a fixed forest, and let $\mathcal{G}(H)$ denote the hereditary class of graphs with no induced subgraph isomorphic to H. A graph class is chi-bound if there is a funct...

conjecture | open | MathDB 页面 | 原文线索

V2-068999 | MATHDB-358467

Equality of the asymptotic clique bounds for bounded maximum average degree

摘录 / EXCERPT: For each positive integer k, let α_k be the minimum value such that there is a constant c_k with ... whenever G is k-degenerate and has maximum degree at most D. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069000 | MATHDB-358468

The Tits Centre Conjecture

摘录 / EXCERPT: Tits Centre Conjecture. If Σ is a convex non-completely-reducible subcomplex of Δ, then Σ has a simplicial Γ-centre.

conjecture | solved | MathDB 页面 | 原文线索

V2-069001 | MATHDB-358469

The strong Tits Centre Conjecture for spherical edifices

摘录 / EXCERPT: Strong Tits Centre Conjecture. If Σ is a closed convex non-completely-reducible subset of $\Delta_G(\mathbb{K})$, then Σ has an unopposed Γ-centre.

conjecture | open | MathDB 页面 | 原文线索

V2-069002 | MATHDB-358470

The strong Tits Centre Conjecture for vector edifices

摘录 / EXCERPT: Strong Tits Centre Conjecture for vector edifices. If C is a closed convex non-completely-reducible cone in $V_G(\mathbb{K})$, then C has an unopposed Γ-centre.

conjecture | solved | MathDB 页面 | 原文线索

V2-069003 | MATHDB-358481

Möbius-rigidity conjecture for homeomorphic critical points of conformal energy

摘录 / EXCERPT: Möbius-rigidity conjecture. There is an $a>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069004 | MATHDB-358484

Weak form of the Main Conjecture for circuit-array rows

摘录 / EXCERPT: Let $C^X_r(j)$ and $C^M_r(j)$ denote the entries in rows r of the corresponding circuit arrays, and let Num and Den denote their numerators and denominators. A sequence satis...

conjecture | open | MathDB 页面 | 原文线索

V2-069005 | MATHDB-358485

Strong form of the Main Conjecture for circuit-array rows

摘录 / EXCERPT: Let $C^X_{r,j}$ denote the entries of the X-circuit array. For $c=1,r=0$ and for each $c \ge 2$ with corresponding $r \in \{2(c-1),2(c-1)-1\}$, suppose there is a sequence of pol...

conjecture | open | MathDB 页面 | 原文线索

V2-069006 | MATHDB-358494

The local antimagic 3-coloring conjecture for balanced bipartite bridge graphs

摘录 / EXCERPT: Let θ be a bipartite bridge graph whose two partite sets have the same size. Balanced bridge-graph conjecture. If θ is a bipartite bridge graph with both partite sets...

conjecture | open | MathDB 页面 | 原文线索

V2-069007 | MATHDB-358495

Canonical Deligne-Riemann-Roch isomorphism

摘录 / EXCERPT: Canonical Deligne-Riemann-Roch isomorphism. There is a canonical Deligne-Riemann-Roch isomorphism, functorially in E.

conjecture | open | MathDB 页面 | 原文线索

V2-069008 | MATHDB-358508

Existence of optimal cyclic subspace codes

摘录 / EXCERPT: Existence conjecture. There exists a cyclic subspace code $\mathcal{C}\subseteq \mathcal{G}_q(n,k)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069009 | MATHDB-358512

Fundamental gap conjecture for stationary states of the dissipative rotational NLS equation

摘录 / EXCERPT: Let $m>0$, and let $\mathcal{S}_m$ denote the set of stationary solutions with total mass m. Let Q_{gs} be a ground state, and let $E[Q]$ denote the total energy of a stationar...

conjecture | open | MathDB 页面 | 原文线索

V2-069010 | MATHDB-358520

Batanin's conjecture on slices of globular operads

摘录 / EXCERPT: Let $\boldsymbol{G_n}$ be a contractible n-globular operad. Its k^{th} slice is a symmetric operad encoding the structure of the k-cells with lower-dimensional cells fixed as...

conjecture | open | MathDB 页面 | 原文线索

V2-069011 | MATHDB-358526

Scott and Seymour's tournament-local chromatic number conjecture

摘录 / EXCERPT: Scott and Seymour's conjecture. For every positive integer k there exists a positive integer χ such that, for every graph G with $\chi(G) \geqslant \chi$ and every tournam...

conjecture | open | MathDB 页面 | 原文线索

V2-069012 | MATHDB-358528

Sublogarithmic out-neighbourhood chromatic conjecture

摘录 / EXCERPT: Sublogarithmic out-neighbourhood conjecture. There exists a function $f(N)$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-069013 | MATHDB-358536

Collapsing limits conjecture for semi-flat ALH_d^*-gravitational instantons

摘录 / EXCERPT: Consider a sequence of pointed semi-flat ALH_d^*-gravitational instantons satisfying the assumption of Theorem, with parameters c_i and τ_i. Collapsing limits conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069014 | MATHDB-358547

Andrews' unboundedness conjecture for the coefficients of the mock theta function sigma

摘录 / EXCERPT: Let ... where $(a;q)_n=\prod_{j=0}^{n-1}(1-aq^j)$. Andrews' unboundedness conjecture. ... The coefficients $S(n)$ arise from the mock theta function studied by Andrews, Dyson, and Hick...

conjecture | open | MathDB 页面 | 原文线索

V2-069015 | MATHDB-358548

Kato's conjecture for positive Howland-Kato commutators

摘录 / EXCERPT: Let P and Q be the momentum and position operators, and let f and g be bounded, real, measurable functions. Define ... For a positive parameter α, let K_{α} den...

conjecture | open | MathDB 页面 | 原文线索

V2-069016 | MATHDB-358551

Coherence conjecture for the free cornering with choice and iteration

摘录 / EXCERPT: Let $\mathbb{A}$ be a distributive monoidal category. Write $\mathbf{V}\,{}^\ulcorner_\llcorner{\mathbb{A}}\!_\lrcorner^\urcorner^\oplus$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069017 | MATHDB-358556

Martio's conjecture on branching of low-distortion quasiregular maps

摘录 / EXCERPT: Martio's conjecture. If $n\geq3$ and $K_I(f)<2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069018 | MATHDB-358565

Grothendieck-Plancherel conjecture for the homogeneous limit shape

摘录 / EXCERPT: Let $x,y\searrow 0$ and let $\beta=\beta(x)\nearrow 0$ satisfy $-\beta(x)\ll x$. Let $\mathfrak{W}(u\mid x,y,\beta(x))$ denote the Grothendieck limit shape, and let $\Omega(u)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-069019 | MATHDB-358566

Grothendieck limit-shape conjecture in the large negative-beta regime

摘录 / EXCERPT: Let $x,y\searrow 0$ and let $\beta=\beta(y)=-K/y$, with fixed $K>0$. Let $\mathfrak{W}(u\mid x,y,\beta(y))$ be the Grothendieck limit shape, and let Ω be the Vershik-Kerov-L...

conjecture | open | MathDB 页面 | 原文线索

V2-069020 | MATHDB-358567

Grothendieck limit-shape conjecture for extended parameter range

摘录 / EXCERPT: Let $\mathfrak{W}(u)=\mathfrak{W}(u\mid x,y,\beta)$ be the curve obtained from the surface and implicit-curve procedure described immediately beforehand, and consider Grothendieck...

conjecture | open | MathDB 页面 | 原文线索

V2-069021 | MATHDB-358569

Laguerre inequality threshold conjecture for the broken k-diamond partition function

摘录 / EXCERPT: Laguerre threshold conjecture. For $k=1$ or 2 and $1\leq m\leq14$, the sequence $\Delta_k(n)$ satisfies the Laguerre inequality of order m, namely $L_m(\Delta_k(n))\geq0$, for...

conjecture | open | MathDB 页面 | 原文线索

V2-069022 | MATHDB-358570

Determinantal inequality threshold conjecture for the broken k-diamond partition function

摘录 / EXCERPT: Determinantal threshold conjecture. For $k=1$ or 2 and $1\leq m\leq14$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-069023 | MATHDB-358575

Fixed-element dual odd extension conjecture for polygon maps

摘录 / EXCERPT: Fixed-element dual odd extension conjecture. The map

conjecture | open | MathDB 页面 | 原文线索

V2-069024 | MATHDB-358580

Data-dependent equivalence conjecture for random features

摘录 / EXCERPT: Let the assumptions in Crefrange{asm:model}{asm:feat} hold. Define $\phi_n=p/n$. Let $k\le n$ be the subsample size and set $\bar{\psi}_n=p/k$. Suppose that φ satisfies cer...

conjecture | open | MathDB 页面 | 原文线索

V2-069025 | MATHDB-358603

The directional-modulation vulnerability conjecture

摘录 / EXCERPT: Let $c_{\mathcal{A}}(s,n)$ denote the relevant coefficient for symbol $s\in\mathcal{S}$ and index n. For a fixed n, compare the values $c_{\mathcal{A}}(s,n)$ as s ranges over...

conjecture | open | MathDB 页面 | 原文线索

V2-069026 | MATHDB-358612

Characterization of flip process rules with the same flip process distributions

摘录 / EXCERPT: Flip-process distribution conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069027 | MATHDB-358613

Milnor's irreducibility conjecture for preperiodic curves of cubic polynomials

摘录 / EXCERPT: Let $f_{a,b}(z)=z^{3}-3a^{2}z+2a^{3}+b$ be a monic, reduced cubic polynomial with marked critical point a, and let $\a7Sigma_{k,n}$ be the Zariski closure in the moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-069028 | MATHDB-358621

Involution Schubert polynomial expansion conjecture in P- and Q-key polynomials

摘录 / EXCERPT: Involution Schubert expansion conjecture. Each involution Schubert polynomial $\mathfrak S^{\mathsf{Sp}}_z$ (respectively, $\mathfrak S^{\mathsf{O}}_z$) is an $\mathbb{N}$-linear c...

conjecture | open | MathDB 页面 | 原文线索

V2-069029 | MATHDB-358624

Vexillary involution crystal isomorphism conjecture

摘录 / EXCERPT: Vexillary involution crystal conjecture. There is an isomorphism of $\mathfrak q^+_n$-crystals

conjecture | open | MathDB 页面 | 原文线索

V2-069030 | MATHDB-358631

Large-data extension conjecture for local well-posedness

摘录 / EXCERPT: Let the cubic problems ... be as in Theorem ... also holds in the large-data case. The paper considers only small data, so this extension is stated as expected rather than proved here.

conjecture | open | MathDB 页面 | 原文线索

V2-069031 | MATHDB-358640

The non-isolatedness conjecture for two-unitary complex Hadamard matrices

摘录 / EXCERPT: Non-isolatedness conjecture. Only non-isolated complex Hadamard matrices of order $N=d^2$ provide two-unitary structures.

conjecture | open | MathDB 页面 | 原文线索

V2-069032 | MATHDB-358647

Local weak convergence of random fullerenes to the planar hexagonal lattice

摘录 / EXCERPT: Let F_n be a random fullerene, meaning a finite, connected, 3-regular planar graph with faces of degrees both 5 and 6, chosen uniformly from the set of fullerenes with n vertic...

conjecture | open | MathDB 页面 | 原文线索

V2-069033 | MATHDB-358652

Torso-extension equivalence for parameters strictly above treewidth

摘录 / EXCERPT: Let ${\sf p}:{\mathcal{G}}_{{\text{\,\textsf{all}}}}\to\mathbb{N}$ be a minor-monotone parameter. For an annotated graph (G,X), let ${\sf p}(G,X)$ denote the annotated extension...

conjecture | open | MathDB 页面 | 原文线索

V2-069034 | MATHDB-358655

The geometric Breuil-Mézard conjecture for potentially crystalline deformation stacks

摘录 / EXCERPT: Let λ be a Hodge type, viewed as an element of the character lattice of G, and let τ be a tame inertial type. Let $V(\lambda-\eta_{\mathbb{Q}_p})$ denote the restric...

conjecture | open | MathDB 页面 | 原文线索

V2-069035 | MATHDB-358656

The refined geometric Breuil-Mézard conjecture for tame groups

摘录 / EXCERPT: Let σ range over parahoric Serre weights, and let $\mathcal{X}^{\operatorname{crys},\lambda,\tau}_{{^{L}\!G},\mathbb{F}_p}$ have cycle class in the Chow group of...

conjecture | open | MathDB 页面 | 原文线索

V2-069036 | MATHDB-358658

Sharp coefficient bounds for Gregory-starlike functions

摘录 / EXCERPT: Let $f(z)=z+a_{2}z^{2}+a_{3}z^{3}+\cdots$ belong to the class $\mathcal{S}_{G}^{\ast}$. The coefficients a_n are the Taylor coefficients of f. Coefficient-bound conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-069037 | MATHDB-358665

Nonemptiness conjecture for correspondences on very general polarized K3 surfaces

摘录 / EXCERPT: Nonemptiness conjecture. $Z_n(L)$ is not empty for any $g\leq 2n-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-069038 | MATHDB-358671

Quiver Hecke categorification conjecture for K-theoretic Coulomb branches

摘录 / EXCERPT: Let Γ be an ADE root system, let Q be an orientation of its Dynkin diagram, and let $\alpha=\sum_{i\in Q_0}n_i\alpha_i$ be a positive root. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-069039 | MATHDB-358672

The general formula for pattern-avoiding stabilized-interval-free permutations

摘录 / EXCERPT: Let $f_{n,d}^{(2)}(123)$ denote the number of permutations of size n that avoid the pattern 123 and have stabilized interval parameter d. Write $n=2m+r$ with $r\in\{0,1\}$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-069040 | MATHDB-358674

Transversality conjecture for sections of a general rational elliptic surface

摘录 / EXCERPT: Let $R\to\mathbb P^1$ be a general rational elliptic surface, and let sections mean section curves of this elliptic fibration. Transversality conjecture. Any two sections meet tran...

conjecture | open | MathDB 页面 | 原文线索

V2-069041 | MATHDB-358677

Plurality is Stablest conjecture for k candidates

摘录 / EXCERPT: Plurality is Stablest conjecture. If $\rho\geq0$ and $\mathbb{E}f=\frac1k\sum_{j=1}^{k}e_j$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069042 | MATHDB-358679

Li-Naber integral singularity conjecture for Alexandrov spaces

摘录 / EXCERPT: Let $X\in\text{Alex}^n(-1)$. For $0\le k\le n-1$, define the k-singularity function by setting $\theta_k(x)=2\pi$ for $x\in\mathcal{S}^{k-1}(X)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069043 | MATHDB-358685

Conjectured asymptotic expansion for expected longest monotone subsequence lengths

摘录 / EXCERPT: Let $L_n^{\varoast}$ denote the relevant longest monotone subsequence length for $\varoast\in\{\boxslash,\boxbslash,\boxdot\}$, let $\gamma(\varoast)$ and $\delta(\varoast)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-069044 | MATHDB-358686

Conjectured asymptotic expansion for variances of longest monotone subsequence lengths

摘录 / EXCERPT: Let $L_n^{\varoast}$ denote the relevant longest monotone subsequence length for $\varoast\in\{\boxslash,\boxbslash,\boxdot\}$, and let $\gamma(\varoast)$ be the corresponding scal...

conjecture | open | MathDB 页面 | 原文线索

V2-069045 | MATHDB-358698

Nontrivial units for Type 3 hyperelliptic curves over $\mathbb{F}_2$

摘录 / EXCERPT: Nontrivial-unit conjecture. If $\deg g>\deg h$ and $h\ne 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069046 | MATHDB-358701

Exact optimality of the simplex codebook

摘录 / EXCERPT: Let $\mathbb{S}^{d-1}$ be the unit sphere, and consider codebooks consisting of points on this sphere. A simplex codebook is a maximally separated codebook; the paper also consider...

conjecture | open | MathDB 页面 | 原文线索

V2-069047 | MATHDB-358702

Diwan's safe-colouring conjecture for cycles

摘录 / EXCERPT: Let $C=v_1v_2\cdots v_lv_1$ be a cycle, and let $f:V(C)\to\{c_1,c_2,\ldots,c_k\}$ be a proper k-colouring. Call f safe if, whenever C is an induced cycle in a planar graph...

conjecture | open | MathDB 页面 | 原文线索

V2-069048 | MATHDB-358703

Hankel-transform Somos-4 conjecture for elliptic-curve division polynomials

摘录 / EXCERPT: Let $E=E(a,b,c,d)$ be the elliptic curve ... with nonzero discriminant, and let $(0,0)$ be the reference point. Let h_n be the Hankel transform of the sequence generated by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-069049 | MATHDB-358704

Jacobi continued fraction conjecture from elliptic-curve multiples

摘录 / EXCERPT: Let $E=E(a,b,c,d)$ be the elliptic curve ... with nonzero discriminant, and let (x_n,y_n) be the coordinates of $n(0,0)$ on E. Let $g(x)$ be the generating function defined above...

conjecture | open | MathDB 页面 | 原文线索

V2-069050 | MATHDB-358706

Random-current phase transition conjecture on the hexagonal lattice

摘录 / EXCERPT: Hexagonal-lattice random-current conjecture. All phase transitions of the random current and random-cluster model on the hexagonal lattice coincide:

conjecture | open | MathDB 页面 | 原文线索

V2-069051 | MATHDB-358717

The Generalized Wilf Conjecture for generalized numerical monoids

摘录 / EXCERPT: Let $n\geq 2$, and let T be a generalized numerical monoid (GMS), meaning a complement-finite submonoid of $\mathbb N^{n-1}$. Let $H(T)=\mathbb N^{n-1}\setminus T$ be its hole se...

conjecture | open | MathDB 页面 | 原文线索

V2-069052 | MATHDB-358718

The conductor-sporadicity ratio conjecture for unipotent numerical monoids

摘录 / EXCERPT: Let $n\geq 2$, let $M=\mathbf P(n,\mathbb N)$, and let S be a unipotent numerical monoid in M. Let $\varphi(S)$ be its image in $\mathbb N^{n-1}$ under the monoid isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-069053 | MATHDB-358719

The Gopakumar-Vafa/Pandharipande-Thomas correspondence

摘录 / EXCERPT: The Gopakumar-Vafa/Pandharipande-Thomas correspondence. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-069054 | MATHDB-358720

The canonical oriented d-critical structure conjecture for stable pair moduli

摘录 / EXCERPT: Canonical oriented d-critical structure conjecture. There exists a canonical oriented d-critical locus structure on $P_n(X,\beta)$ inducing $\phi_{n,\beta}$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-069055 | MATHDB-358722

Signed Lovász induced negative cycle conjecture

摘录 / EXCERPT: Signed Lovász conjecture. For every natural number k, there exists a least natural number $f'(k)$ such that, for every $f'(k)$-connected, unbalanced signed graph G, there is an...

conjecture | open | MathDB 页面 | 原文线索

V2-069056 | MATHDB-358732

Nagata's ampleness conjecture for blowups of the projective plane

摘录 / EXCERPT: Nagata's conjecture. If $n\geq 10$ and $t>\sqrt{n}$, then A_t is ample. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-069057 | MATHDB-358733

Segre-Harbourne-Gimigliano-Hirschowitz conjecture for plane blowups

摘录 / EXCERPT: Let X be the blowup of $\mathbb{P}^2$ at n very general points, and let $D=dH-\sum_i m_iE_i$ be a divisor class with $d\geq 0$. Call D special when both $h^0(\mathcal{O}(D))$...

conjecture | open | MathDB 页面 | 原文线索

V2-069058 | MATHDB-358734

List distinguishing index conjecture for connected graphs

摘录 / EXCERPT: List distinguishing index conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069059 | MATHDB-358736

Bounded-arithmetic provability of Arrow's theorem

摘录 / EXCERPT: Bounded-arithmetic conjecture for Arrow's theorem. The first-order formalisation of Arrow's theorem is provable in

conjecture | open | MathDB 页面 | 原文线索

V2-069060 | MATHDB-358738

Characterization of non-varying strata of quadratic differentials

摘录 / EXCERPT: Non-varying-stratum characterization. The stratum $\mathcal{Q}_g(d_1,\ldots,d_n)$ is non-varying if and only if L^+ equals the sum of the g smallest numbers in S. This is pre...

conjecture | open | MathDB 页面 | 原文线索

V2-069061 | MATHDB-358757

Chakraborty-Das-Nandi-Roy-Sen's relative clique conjecture for triangle-free planar graphs

摘录 / EXCERPT: Chakraborty-Das-Nandi-Roy-Sen's triangle-free relative clique conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069062 | MATHDB-358758

The Desbot-Wilf classification conjecture for patterns in S_4

摘录 / EXCERPT: Let S_4 be the set of permutations of length 4. For a permutation τ, define its descent bottom set by ... Two patterns are Desbot-Wilf equivalent if their avoidance classes...

conjecture | open | MathDB 页面 | 原文线索

V2-069063 | MATHDB-358759

The joint Destop-Desbot Wilf-equivalence conjecture for 3142, 3241, and 4132

摘录 / EXCERPT: For a permutation τ, let $\operatorname{Destop}(\tau)$ and $\operatorname{Desbot}(\tau)$ denote its descent top and descent bottom sets. Patterns are...

conjecture | open | MathDB 页面 | 原文线索

V2-069064 | MATHDB-358760

The Ascbot shape-Wilf equivalence conjecture for 231⊕σ and 312⊕σ

摘录 / EXCERPT: Let σ be a nonempty permutation, let $\rho\oplus\sigma$ denote their direct sum, and let F be a traversable Ferrers board. For a traversal τ of F, define the F-as...

conjecture | open | MathDB 页面 | 原文线索

V2-069065 | MATHDB-358761

The planar high-girth circular chromatic number conjecture

摘录 / EXCERPT: Circular chromatic number conjecture. If $f(g) \geq 4g$, then the circular chromatic number of G is at most $2+\frac{1}{g}$.

conjecture | open | MathDB 页面 | 原文线索

V2-069066 | MATHDB-358765

Existence of nice nef-partitions for smooth well-formed Fano weighted complete intersections

摘录 / EXCERPT: Let $X \subset \mathbb{P}(a_0,\ldots,a_N)$ be a smooth well-formed Fano weighted complete intersection of multidegree $(d_1,\ldots,d_c)$. A nef-partition for X is a splitting ... I...

conjecture | open | MathDB 页面 | 原文线索

V2-069067 | MATHDB-358766

Dehornoy's class bound for indecomposable involutive solutions

摘录 / EXCERPT: Let (X,r) be an indecomposable involutive solution, and let the Dehornoy's class denote its associated class invariant. Dehornoy's class conjecture for indecomposable solutions....

conjecture | open | MathDB 页面 | 原文线索

V2-069068 | MATHDB-358767

Sharp universal bound for Dehornoy's class

摘录 / EXCERPT: Let (X,r) be an involutive solution of size n, and define ... Dehornoy's class bound conjecture. The Dehornoy's class is bounded by a_n, and the bound a_n is minimal. The cla...

conjecture | open | MathDB 页面 | 原文线索

V2-069069 | MATHDB-358768

Dehornoy's class bound for solutions with abelian permutation group

摘录 / EXCERPT: Let (X,r) be an involutive solution of size n, and let $g(n)$ denote the bound used in the surrounding text for solutions whose permutation group is abelian. Abelian permutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-069070 | MATHDB-358775

Wrapping-degree equality conjecture for solid torus links

摘录 / EXCERPT: Let L be a solid torus link with wrapping number wr, and let ∇_L denote its Kauffman polynomial. Wrapping-degree equality conjecture. ... Equality holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-069071 | MATHDB-358778

Antisymmetry conjecture for the top quasi-Betti numbers of monomial complete intersections

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ and let $I\subset S$ be a monomial complete intersection minimally generated by m monomials. Write $\beta^d_k(S/I)$ for its quasi-Betti numbers. Antisym...

conjecture | open | MathDB 页面 | 原文线索

V2-069072 | MATHDB-358793

The classification conjecture for incompressible moderate-growth categories

摘录 / EXCERPT: Classification conjecture. Every incompressible category of moderate growth over $\mathbf{k}$ is a tensor subcategory of Ver_{p^∞}.

conjecture | open | MathDB 页面 | 原文线索

V2-069073 | MATHDB-358794

The moderate-growth characterization conjecture for pretannakian categories

摘录 / EXCERPT: Moderate-growth characterization conjecture. The category $\mathcal C$ admits a tensor functor to an incompressible category if and only if it is of moderate growth. Equivalently, e...

conjecture | open | MathDB 页面 | 原文线索

V2-069074 | MATHDB-358795

The Grothendieck-ring growth-dimension conjecture

摘录 / EXCERPT: Grothendieck-ring growth-dimension conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-069075 | MATHDB-358798

McKean-type blowup criterion for radial EPDiff equations

摘录 / EXCERPT: McKean-type blowup conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-069076 | MATHDB-358799

Finite-time breakdown conjecture for the b-equations hierarchy

摘录 / EXCERPT: Finite-time breakdown conjecture for the b-equations. If, for some operator A, the Euler-Arnold equation has C^1 solutions that break down in finite time, then for every $b>1$...

conjecture | open | MathDB 页面 | 原文线索

V2-069077 | MATHDB-358802

Novik-Zheng affine-stress support conjecture for simplicial spheres

摘录 / EXCERPT: Let $2\leq i\leq d/2$. Let Δ be either the boundary complex of a simplicial d-polytope with its natural embedding p, or a simplicial $(d-1)$-sphere with a generic embedd...

conjecture | open | MathDB 页面 | 原文线索

V2-069078 | MATHDB-358805

The spectral conjecture for hyperbolic groups

摘录 / EXCERPT: Spectral conjecture. There exists such a function f and such representations and vectors for every compact $K\subset\Gamma$ and every $\varepsilon>0$. This would yield, by a limi...

conjecture | open | MathDB 页面 | 原文线索

V2-069079 | MATHDB-358813

Hironaka's golden ratio conjecture for minimum dilatations

摘录 / EXCERPT: For each genus g, let $\delta_{g,0}$ denote the minimum dilatation among pseudo-Anosov maps on the closed surface of genus g, and let μ be the golden ratio. Hironaka's gold...

conjecture | open | MathDB 页面 | 原文线索

V2-069080 | MATHDB-358815

Chourasiya-Jamal-Maji-Sarkar conjecture on zeros of Ramanujan-type polynomials

摘录 / EXCERPT: Conjecture on the zeros of $R_{2k+1,p}$. The only real root of $R_{2k+1,p}(z)$ is $z=0$, with multiplicity 2; moreover, its non-real roots are simple and lie on the circle...

conjecture | open | MathDB 页面 | 原文线索

V2-069081 | MATHDB-358816

Maji-Sarkar conjecture on zeros of higher-power Ramanujan polynomials

摘录 / EXCERPT: Let k and ℓ be fixed natural numbers, and define ... Here B_m denotes the mth Bernoulli number; the variable Z replaces $z^{2\ell}$ in the one-variable generalization of...

conjecture | open | MathDB 页面 | 原文线索

V2-069082 | MATHDB-358818

The discriminant formula for the invariant εΓ(X)

摘录 / EXCERPT: Let Γ be a Fuchsian group, let Σ be a double coset of Γ, and let X be a Γ-conjugacy class contained in Σ. For $M\in X$, define the discriminant...

conjecture | open | MathDB 页面 | 原文线索

V2-069083 | MATHDB-358826

Coskun-Woolf stabilization conjecture for Betti numbers of sheaf moduli spaces

摘录 / EXCERPT: Let X be a smooth projective surface and let H be an ample line bundle. Fix a rank $r>0$ and a first Chern class c, and let $M_{X,H}(r,c,\Delta)$ be the corresponding moduli...

conjecture | open | MathDB 页面 | 原文线索

V2-069084 | MATHDB-358827

Higher-rank SHGH conjecture for weak Brill-Noether on plane blowups

摘录 / EXCERPT: Let X be a blowup of $\mathbb{P}^2$ at m very general points, and let H be an ample class on X such that $H\cdot K_X<0$. Let ${\bf v}$ be a Chern character such that its no...

conjecture | open | MathDB 页面 | 原文线索

V2-069085 | MATHDB-358844

Non-conservatism of the graph hierarchy for robust feedback stabilization

摘录 / EXCERPT: Let $M\in\mathbb{Z}^{+}$ and let ... Suppose that the switched system under consideration is robust feedback stabilizable. Non-conservatism conjecture. There exists an...

conjecture | open | MathDB 页面 | 原文线索

V2-069086 | MATHDB-358847

Strong Donagi-Morrison conjecture for Lazarsfeld-Mukai bundles

摘录 / EXCERPT: Strong Donagi-Morrison conjecture. There is a nontrivial line bundle $N\hookrightarrow E_{C,A}$ with $h^0(S,N)\geq 2$ such that $E_{C,A}/N$ is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-069087 | MATHDB-358848

Bounded strong Donagi-Morrison conjecture for Lazarsfeld-Mukai bundles

摘录 / EXCERPT: Bounded strong Donagi-Morrison conjecture. There is a bound β depending on C and S such that, if $d<\beta$, then there is a line bundle $N\hookrightarrow E_{C,A}$ with...

conjecture | refuted | MathDB 页面 | 原文线索

V2-069088 | MATHDB-358850

Bounded additive group irregularity strength for graphs without small components

摘录 / EXCERPT: Let G be a graph of order n with no connected components of order less than 3. A Γ-irregular labeling is a labeling of the edges of G by elements of an abelian group...

conjecture | open | MathDB 页面 | 原文线索

V2-069089 | MATHDB-358852

Stronger colored-tree embedding conjecture for pseudorandom graph families

摘录 / EXCERPT: Let $\Delta\geq 2$. An (n,D,λ)-graph is a graph with the corresponding order, degree and spectral parameters, and a $[t]$-colored tree is a tree whose edges receive colors i...

conjecture | open | MathDB 页面 | 原文线索

V2-069090 | MATHDB-358868

The DT-PT virtual structure sheaf localization conjecture

摘录 / EXCERPT: The DT-PT virtual structure sheaf localization conjecture. There exists a canonical halving $\mathbb{E}_{1/2}$, given explicitly in the source's Definition, such that

conjecture | open | MathDB 页面 | 原文线索

V2-069091 | MATHDB-358869

Gezmis-Pellarin injectivity conjecture for the map \mathcal{G}_{\Sigma}

摘录 / EXCERPT: Let K be the field and let Σ be the index set used to define the spaces of multiple polylogarithms and multiple zeta values. Let ... be the map obtained by composing the map...

conjecture | open | MathDB 页面 | 原文线索

V2-069092 | MATHDB-358876

A mod 4 congruence for partitions with parts separated by parity

摘录 / EXCERPT: Mod 4 congruence conjecture. For every $n\geq0$ and every integer j with $0\leq j\leq\ell-2$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-069093 | MATHDB-358877

Density conjecture for even values of the partition function p_0

摘录 / EXCERPT: Density conjecture. The natural density of indices m for which $p_0(m)$ is even exists and equals

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-069094 | MATHDB-358878

Inequalities between parity-separated partition functions

摘录 / EXCERPT: Let $\overline{\mathcal{OE}}(n)$ and $\overline{\mathcal{EO}}(n)$ denote the paper's parity-separated partition-counting functions, and let $p_0(n)$ and $p_1(n)$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-069095 | MATHDB-358880

Poor-edge strengthening of the Petersen Coloring Conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let $\mathrm{NC}(G)$ be the set of all normal 5-colorings of G. Assuming the Petersen Coloring Conjecture, define $\mathrm{poor}(G)$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-069096 | MATHDB-358894

Unimodality conjecture for higher dimer-cover rank generating functions

摘录 / EXCERPT: Let $\mathcal{G}$ be the snake graph and let $P(\mathcal{G}^*)$ be the associated poset; write $U_{P(\mathcal{G}^*);m}(q)$ for the rank generating function of m-dimer covers. Hig...

conjecture | open | MathDB 页面 | 原文线索

V2-069097 | MATHDB-358895

Conjectured sharp orthonormal Strichartz estimate for the wave equation

摘录 / EXCERPT: Let $d \geq 2$, $q \in (2,\infty)$, and $r \in [2,\infty)$ satisfy ... Let $\beta_{d/2}(q,r)$ denote the exponent appearing in the conjectured estimate, and let the estimate ... holds...

conjecture | open | MathDB 页面 | 原文线索

V2-069098 | MATHDB-358897

Killing-field classification conjecture for non-Kähler ALF gravitational instantons

摘录 / EXCERPT: Killing-field classification conjecture. The conclusion of the Hermitian classification conjecture holds for $(\mathcal{M},g_{ab})$.

conjecture | open | MathDB 页面 | 原文线索

V2-069099 | MATHDB-358921

Herden's congruence conjecture for 2-regular partitions with designated summands

摘录 / EXCERPT: Let $PD_2(n)$ denote the number of 2-regular partitions of n with designated summands; equivalently, these are partitions into odd parts with exactly one occurrence of each dis...

conjecture | open | MathDB 页面 | 原文线索

V2-069100 | MATHDB-358922

Dyadic congruence conjecture for 2-regular partitions with designated summands

摘录 / EXCERPT: Let $PD_2(n)$ denote the number of 2-regular partitions of n with designated summands, that is, partitions into odd parts with exactly one occurrence of each distinct part desi...

conjecture | open | MathDB 页面 | 原文线索

V2-069101 | MATHDB-358931

Real-rootedness conjecture for Primitive Eulerian polynomials of simplicial arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a simplicial arrangement, and let $P_{\mathcal{A}}(z)$ denote its Primitive Eulerian polynomial. Simplicial-arrangement real-rootedness conjecture. The polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-069102 | MATHDB-358932

Guo-Wang's uniqueness conjecture for positive harmonic functions in the unit ball

摘录 / EXCERPT: Guo-Wang's conjecture. If $1<q<\frac{n}{n-2}$ and $0<\lambda\leq\frac{1}{q-1}$, then u is constant.

conjecture | open | MathDB 页面 | 原文线索

V2-069103 | MATHDB-358939

Gamma-positivity conjecture for peak-refined inverse descent polynomials

摘录 / EXCERPT: For $n\geq1$ and $0\leq k\leq\lfloor(n-1)/2\rfloor$, define ... A symmetric polynomial with center of symmetry $(n+1)/2$ is γ-positive when its expansion in the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-069104 | MATHDB-358940

Pinnacle-set refinement of gamma-positivity for inverse descent polynomials

摘录 / EXCERPT: For $n\geq1$ and $S\subseteq[n]$, let $\operatorname{Pin}(\pi)$ be the set of values at peak positions of $\pi\in\mathfrak{S}_n$, and define ... A polynomial is γ-positive wit...

conjecture | open | MathDB 页面 | 原文线索

V2-069105 | MATHDB-358955

A Rogers-Ramanujan type double-sum identity with index $(1,3)$

摘录 / EXCERPT: For $|q|<1$, let $(a;q)_n$ denote the q-Pochhammer symbol, defined by $(a;q)_n=\prod_{k=0}^{n-1}(1-aq^k)$ and $(a;q)_\infty=\prod_{k=0}^{\infty}(1-aq^k)$; for multiple parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-069106 | MATHDB-358959

Approximate minimum path-metric conjecture for SCL decoding with list size at least 8

摘录 / EXCERPT: Let $L=2^m$ with $m\ge 3$, and let ${\mathsf L}_{k-m}$ denote the log-likelihood ratio at bit position $k-m$. For each ${\bf v}_{k-m+1}^k\in\{0,1\}^m$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069107 | MATHDB-358963

Foulkes' plethysm inequality conjecture

摘录 / EXCERPT: Let $a_\lambda(d[n])$ denote the plethysm coefficient indexed by a partition $\lambda\vdash nd$. Foulkes' conjecture. If $d>n$, then ... for all $\lambda\vdash nd$. This is a fundame...

conjecture | open | MathDB 页面 | 原文线索

V2-069108 | MATHDB-358986

The permutation-representation conjecture for comparability graphs

摘录 / EXCERPT: Let G be a comparability graph, meaning the graph of a partial order. Write $\mathcal{R}_2$ for the class of graphs with representation number at most two, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069109 | MATHDB-358995

Expectation for the stable wavefront set of a theta representation

摘录 / EXCERPT: Stable wavefront-set conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069110 | MATHDB-358997

von Stengel's conjecture on estranged facet pairs of simplicial polytopes

摘录 / EXCERPT: Let P be a simplicial d-polytope with $2d$ vertices. Two facets are estranged if they are disjoint. Von Stengel's conjecture. The maximum number of pairs of estranged facets of...

conjecture | open | MathDB 页面 | 原文线索

V2-069111 | MATHDB-359002

The index-three conjecture for Kontsevich-Zorich monodromy of origamis in \(\mathcal{B}_d\)

摘录 / EXCERPT: Let $\mathcal{O}$ be an origami of degree d and genus 2, let $\Gamma\subseteq\operatorname{SL}_2(\mathbb{Z})$ be its Kontsevich-Zorich monodromy, and let $\mathcal{B}_d$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-069112 | MATHDB-359003

The interval inner Bernoulli solution-count conjecture for the fractional Laplacian

摘录 / EXCERPT: Interval solution-count conjecture. There exists a constant $bbda_{b1,D}>0$ such that the problem has exactly two solutions for $bbda>bbda_{b1,D}$, exactly one solution for...

conjecture | open | MathDB 页面 | 原文线索

V2-069113 | MATHDB-359004

The ball inner Bernoulli solution-count conjecture for the fractional Laplacian

摘录 / EXCERPT: Ball solution-count hypothesis. There exists a constant $bbda_{b1,D}>0$ such that the problem has exactly two solutions for $bbda>bbda_{b1,D}$, exactly one solution for...

conjecture | open | MathDB 页面 | 原文线索

V2-069114 | MATHDB-359005

The variational solution threshold conjecture for the fractional Bernoulli problem on an interval

摘录 / EXCERPT: Variational solution threshold conjecture. There exists a constant $bbda_{b1,D}$ with $bbda_{b1,D}>bbda_{b1,D}$ such that, for every $bbda a bbda_{b1,D}$, the variational problem h...

conjecture | open | MathDB 页面 | 原文线索

V2-069115 | MATHDB-359021

Bonnet's sparse twin-width 3 conjecture for bounded tree-width

摘录 / EXCERPT: Let $\mathcal{C}$ be a class of graphs of twin-width at most 3, and suppose there exists an integer t such that no graph $G\in\mathcal{C}$ contains $K_{t,t}$ as a subgraph. Spa...

conjecture | open | MathDB 页面 | 原文线索

V2-069116 | MATHDB-359022

The stable twin-width 2 conjecture for bounded clique-width

摘录 / EXCERPT: Let $\mathcal{C}$ be a stable class of graphs of twin-width at most 2. Stable twin-width 2 conjecture. Then $\mathcal{C}$ has bounded clique-width. This conjecture is the correct...

conjecture | open | MathDB 页面 | 原文线索

V2-069117 | MATHDB-359039

Expected Syzygy Conjecture for chopped ideals of general points

摘录 / EXCERPT: Let S be the homogeneous coordinate ring of $\mathbb{P}^n$, let Z be a general set of r points, and let $I(Z)$ be its vanishing ideal. For an integer d that is the smallest...

conjecture | open | MathDB 页面 | 原文线索

V2-069118 | MATHDB-359042

The current-density conjecture for lightcone operator accumulation

摘录 / EXCERPT: Let Z be a torus partition function of the form specified in the source, with at least one chiral current, and let $D(j)$ be the current density satisfying ... where $a>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-069119 | MATHDB-359043

The shifted twist-accumulation conjecture for infinitely many currents

摘录 / EXCERPT: Let Z be a partition function of the form specified in the source, with current degeneracy $D(j)$ given by the source's shifted ansatz and parameter $a\in(0,\sqrt A)$. Define the...

conjecture | open | MathDB 页面 | 原文线索

V2-069120 | MATHDB-359044

The Delta Conjecture skewing formula for the quadratic t coefficient

摘录 / EXCERPT: Let n and k be the parameters in the Delta Conjecture, let $s_\lambda^\perp$ denote the operator adjoint to multiplication by s_λ, let $H_\lambda(x;q)$ be the Hall-Litt...

conjecture | open | MathDB 页面 | 原文线索

V2-069121 | MATHDB-359050

Existence conjecture for smooth higher-degree plane-filling curves

摘录 / EXCERPT: Existence conjecture for $C_{k,r}$. There exists an integer $m=m(r)$ such that, for every finite field $\mathbb{F}_q$ with $q>m$ and characteristic not dividing r, there is some...

conjecture | open | MathDB 页面 | 原文线索

V2-069122 | MATHDB-359055

Asymptotic volume-determinant conjecture for knots

摘录 / EXCERPT: Let K be a knot, let $\det(K)$ denote its determinant, let $\operatorname{Vol}(K)$ denote its hyperbolic volume, and let c denote the crossing number. Asymptotic volume-determi...

conjecture | open | MathDB 页面 | 原文线索

V2-069123 | MATHDB-359067

Breuil's Ext conjecture for de Rham non-trianguline representations of \(\operatorname{GL}_2(\mathbb Q_p)\)

摘录 / EXCERPT: Breuil's conjecture. There is a natural E-linear bijection

conjecture | open | MathDB 页面 | 原文线索

V2-069124 | MATHDB-359069

The minimum-distance conjecture for extended codes over fields of characteristic five

摘录 / EXCERPT: Let $q=5^m$ with $m\geq3$ odd. Let $\mathcal C_{(q,q+1,4,1)}$ be the BCH code appearing in the source, and let $\overline{\mathcal C}_4$ be the extended classical MDS cyclic code....

conjecture | open | MathDB 页面 | 原文线索

V2-069125 | MATHDB-359077

Reid's canonical-surface inequality conjecture

摘录 / EXCERPT: Reid's conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-069126 | MATHDB-359078

Reduced-fiber-free slope equality conjecture for fibered surfaces

摘录 / EXCERPT: Reduced-fiber-free slope equality conjecture. The slope equality

conjecture | open | MathDB 页面 | 原文线索

V2-069127 | MATHDB-359079

Xiao-Reid's atomic-fiber Morsification conjecture

摘录 / EXCERPT: Xiao-Reid's Morsification conjecture. A fiber germ $f^{-1}(p)$ is atomic if and only if it is a stable curve with one node or a multiple of a smooth curve.

conjecture | open | MathDB 页面 | 原文线索

V2-069128 | MATHDB-359081

El-Zahar-Erdős conjecture on disconnected high-chromatic graphs

摘录 / EXCERPT: El-Zahar-Erdős conjecture. For every pair of integers $t,c\geq 1$, there exists $d\geq 1$ such that if $\chi(G)\geq d$ and $\omega(G)<t$, then there are subsets $A,B\subseteq V(G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-069129 | MATHDB-359082

Nguyen-Scott-Seymour conjecture on large directed pairs in tournaments

摘录 / EXCERPT: Nguyen-Scott-Seymour conjecture. For every integer $c\geq 0$, there exists $d\geq 0$ such that if $\vec\chi(T)\geq d$, then there are two sets $A,B\subseteq V(T)$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-069130 | MATHDB-359084

Lin's 3-adic congruence conjecture for the partition function $\text{PDO}_t(n)$

摘录 / EXCERPT: A partition with designated summands is obtained from an ordinary partition by tagging exactly one occurrence of each part size. Let $\text{PDO}_t(n)$ denote the total number of ta...

conjecture | open | MathDB 页面 | 原文线索

V2-069131 | MATHDB-359085

Conjectured 2-adic congruences for the partition function $\text{PDO}_t(n)$

摘录 / EXCERPT: Let $\text{PDO}_t(n)$ denote the total number of tagged parts over all partitions of n with designated summands in which every part is odd. For nonnegative integers k and n,...

conjecture | open | MathDB 页面 | 原文线索

V2-069132 | MATHDB-359087

Beauville's conjecture on stability of pushforwards of general vector bundles

摘录 / EXCERPT: Let $\alpha\colon X \to Y$ be a finite morphism between smooth irreducible projective curves, and let V be a general vector bundle on X. The pushforward α_*V is a vector...

conjecture | open | MathDB 页面 | 原文线索

V2-069133 | MATHDB-359089

Finiteness conjecture for indecomposable elements in real biquadratic fields

摘录 / EXCERPT: Finiteness conjecture. There are only finitely many real biquadratic fields K such that

conjecture | open | MathDB 页面 | 原文线索

V2-069134 | MATHDB-359093

Henning-Löwenstein-Rautenbach domination-packing conjecture for subcubic graphs

摘录 / EXCERPT: Henning-Löwenstein-Rautenbach conjecture. Every connected subcubic graph G except the three graphs H_1, H_2, and H_3 satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069135 | MATHDB-359094

Nested solutions for products of powers of paths and complete graphs

摘录 / EXCERPT: Let P_{n_1} denote the path on n_1 vertices and K_{n_2} the complete graph on n_2 vertices. For nonnegative integers d_1,d_2 and integers $n_1,n_2\geq 2$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-069136 | MATHDB-359095

Standard block lexicographic order for products involving powers of the Petersen graph

摘录 / EXCERPT: Let C_m be the cycle graph on m vertices, K_2 and C_3 the indicated factors, and let G be the Petersen graph. For nonnegative exponents, the displayed Cartesian products...

conjecture | solved | MathDB 页面 | 原文线索

V2-069137 | MATHDB-359106

Herzog-Stamate bound for generators of monomial curve ideals

摘录 / EXCERPT: Herzog-Stamate conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069138 | MATHDB-359110

Recurrence conjecture for the Maclaurin-series coefficients of Somos-related sequences

摘录 / EXCERPT: Recurrence conjecture. The sequence $(\tau_{n,r})$ satisfies the linear recurrence of order $2r+2$

conjecture | open | MathDB 页面 | 原文线索

V2-069139 | MATHDB-359116

The Deep Riemann Hypothesis for normalized partial Euler products

摘录 / EXCERPT: Let K be a global field. For all but finitely many primes $\mathfrak p$ of K, let $M(\mathfrak p)\in\operatorname{GL}_{r_{\mathfrak p}}(\mathbb C)$ be a unitary matrix, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-069140 | MATHDB-359123

The Collins-Dykema-Torres-Ayala positivity conjecture for symmetric matrices

摘录 / EXCERPT: Collins-Dykema-Torres-Ayala conjecture. This coefficient is non-negative.

conjecture | open | MathDB 页面 | 原文线索

V2-069141 | MATHDB-359124

The Lieb-Seiringer BMV positivity conjecture for positive semidefinite matrices

摘录 / EXCERPT: Lieb-Seiringer conjecture. This coefficient is non-negative.

conjecture | open | MathDB 页面 | 原文线索

V2-069142 | MATHDB-359129

Reversed Dickson second-kind permutation classification over the field of five elements

摘录 / EXCERPT: Let $E_n(1,x)$ denote the reversed Dickson polynomial of the second kind, and let a permutation polynomial (PP) be a polynomial that permutes the field. The $p=5$ reversed Dickson...

conjecture | open | MathDB 页面 | 原文线索

V2-069143 | MATHDB-359133

The random-graph conjecture for external and internal bisections

摘录 / EXCERPT: For a graph on n vertices, a bisection is a red-blue colouring in which the colour-class sizes are equal, or differ by one when n is odd. An external bisection has at least hal...

conjecture | open | MathDB 页面 | 原文线索

V2-069144 | MATHDB-359142

The complexification of the volume conjecture

摘录 / EXCERPT: Let $\mathscr{H}\subset S^3$ be a hyperbolic knot. Define its complex volume by ... where $\operatorname{CS}^{\rm{SO(3)}}(\mathscr{H})$ is the Chern-Simons invariant modulo π^2....

conjecture | open | MathDB 页面 | 原文线索

V2-069145 | MATHDB-359146

Lusztig's Fourier-matrix conjecture for non-crystallographic Coxeter groups

摘录 / EXCERPT: Let W be a non-crystallographic finite Coxeter group, let c be a two-sided cell in W, and let $\mathcal{U}_W^c=\mathcal{Z}(\mathcal{H}_W^c)$ be the Drinfeld center of the asy...

conjecture | open | MathDB 页面 | 原文线索

V2-069146 | MATHDB-359175

Ziegler's generic restriction conjecture for logarithmic forms

摘录 / EXCERPT: Let ${\mathcal{A}}'$ be an essential and free arrangement in $V={\mathbb{K}}^\ell$ with $\ell\geq 4$. Let X be a generic subspace with $\dim X>2$, and define the restriction ... He...

conjecture | open | MathDB 页面 | 原文线索

V2-069147 | MATHDB-359176

Ziegler's nonfreeness conjecture for generic cuts of free arrangements

摘录 / EXCERPT: Ziegler's nonfreeness conjecture. For ${\mathcal{A}}'$ and X as in Ziegler's generic restriction conjecture, the arrangement ${\mathcal{A}}'\cap X$ is never free; equivalently, a...

conjecture | open | MathDB 页面 | 原文线索

V2-069148 | MATHDB-359177

Ziegler's critical-arrangement restriction conjecture

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a hyperplane arrangement, and for an integer k call it k-critical when ... and ... for every $H\in{\mathcal{A}}$. Here $|{\mathcal{A}}|$ denotes the number o...

conjecture | open | MathDB 页面 | 原文线索

V2-069149 | MATHDB-359185

Asymptotic normality conjecture for sample PLS

摘录 / EXCERPT: Sample PLS asymptotic normality conjecture. The sample PLS is asymptotically normal with

conjecture | open | MathDB 页面 | 原文线索

V2-069150 | MATHDB-359187

Maximal-minus-minimal homomesy conjecture for products of two chains

摘录 / EXCERPT: Maximal-minus-minimal homomesy conjecture. The number of maximal elements minus the number of minimal elements is 0-mesic under rowmotion on $\mathcal{IC}(P)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-069151 | MATHDB-359210

Mao's identities for Beck's partition statistics modulo 8

摘录 / EXCERPT: Let $NT(r,m,n)$ denote the total number of parts in partitions of n whose rank is congruent to r modulo m, and let $M_{\omega}(r,m,n)$ denote the total number of ones in part...

conjecture | open | MathDB 页面 | 原文线索

V2-069152 | MATHDB-359211

Equidistribution conjecture for ascents and descents of perfect matchings

摘录 / EXCERPT: Equidistribution conjecture. For all $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-069153 | MATHDB-359219

First-moment lower-bound conjecture for average-case matrix discrepancy

摘录 / EXCERPT: Let $\bm A_1,\dots,\bm A_T\in\mathbb{R}^{n\times n}_{\operatorname{sym}}$ be i.i.d. random symmetric matrices, where $T=T(n)$, $\|\bm A_i\|\leq1$ almost surely, the distribution is...

conjecture | open | MathDB 页面 | 原文线索

V2-069154 | MATHDB-359224

Boundedness conjecture for K-semistable Fano cone singularities

摘录 / EXCERPT: Let n be a positive integer. A set of klt singularities is bounded if all its members occur as fibers of one $\mathbb{Q}$-Gorenstein family; for Fano cone singularities, the fami...

conjecture | open | MathDB 页面 | 原文线索

V2-069155 | MATHDB-359225

Multiplicity and embedding-dimension boundedness conjecture

摘录 / EXCERPT: Let n be a positive integer, let $\varepsilon>0$, and let mult_xX denote the multiplicity and the embedded dimension of $x\in X$ its embedding dimension. Multipl...

conjecture | open | MathDB 页面 | 原文线索

V2-069156 | MATHDB-359226

Uniform delta-plt blowup conjecture for klt singularities

摘录 / EXCERPT: Let n be a positive integer and let $\varepsilon>0$. A δ-plt blowup of a klt singularity $x\in X$ is a plt blowup $Y\to X$ of a Kollár component E such that (Y,E) is...

conjecture | open | MathDB 页面 | 原文线索

V2-069157 | MATHDB-359227

ACC conjecture for local volumes in fixed dimension

摘录 / EXCERPT: Let n be a positive integer. ACC conjecture. The set of all possible local volumes $\widehat{\rm vol}(x,X)$ of n-dimensional klt singularities is discrete away from zero. This...

conjecture | open | MathDB 页面 | 原文线索

V2-069158 | MATHDB-359228

Ordinary double point volume gap conjecture

摘录 / EXCERPT: Let n be a positive integer, and let an ordinary double point denote the corresponding n-dimensional klt quadratic singularity. Ordinary double point volume gap conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-069159 | MATHDB-359229

Local volume gap near the smooth point

摘录 / EXCERPT: Let n be a positive integer, and let a smooth point mean a nonsingular n-dimensional klt singularity. Local volume gap conjecture. There exists $\varepsilon>0$ such that the on...

conjecture | open | MathDB 页面 | 原文线索

V2-069160 | MATHDB-359230

Boundedness of K-minimal log discrepancies at fixed local volume

摘录 / EXCERPT: Let n be a positive integer, let $\varepsilon>0$, and let $\mathrm{mld}^{\mathrm{K}}(x,X)$ denote the K-minimal log discrepancy of an n-dimensional klt singularity $x\in X$. K-...

conjecture | open | MathDB 页面 | 原文线索

V2-069161 | MATHDB-359231

Local volume versus F-signature conjecture

摘录 / EXCERPT: Let $x\in X$ be an n-dimensional klt singularity in characteristic 0, and for primes $p\gg0$ let $x_p\in X_p$ be its reduction modulo p. Let $s(x_p,X_p)$ denote the F-signatu...

conjecture | open | MathDB 页面 | 原文线索

V2-069162 | MATHDB-359232

Finite degree formula for local volumes of klt pairs

摘录 / EXCERPT: Let $f\colon\big(y\in(Y,\Delta_Y)\big)\to\big(x\in(X,\Delta)\big)$ be a finite surjective morphism between klt pairs, and suppose it is crepant, meaning ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069163 | MATHDB-359235

Andrews-Merca-Guo-Zeng coefficient non-negativity conjecture for averaged Jacobi truncations

摘录 / EXCERPT: Andrews-Merca-Guo-Zeng conjecture. This coefficient is non-negative. This conjecture generalizes non-negativity results for averaged truncations of Euler's pentagonal number theore...

conjecture | open | MathDB 页面 | 原文线索

V2-069164 | MATHDB-359236

Unimodality conjecture for the truncated Jacobi refinement

摘录 / EXCERPT: For any $k\in\mathbb N_0$ and $n\in\mathbb N$, define the integers $J_k^{\textsc{t}}(m,n)$ by ... Recall that a finite sequence of real numbers is unimodal if it first increases and...

conjecture | open | MathDB 页面 | 原文线索

V2-069165 | MATHDB-359237

Stanley's conjecture on unimodality of IDP polytopes

摘录 / EXCERPT: Let P be a lattice polytope with the integer decomposition property (IDP), meaning that every lattice point in mP is a sum of m lattice points in P. Let $h^*_P(x)$ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-069166 | MATHDB-359272

Asymptotic equivalence of full and Gaussian maxT power

摘录 / EXCERPT: Let $\overline{H}$ be the matrix appearing in the maxT procedure, let ι denote the all-ones vector, and define ... Suppose that E has independent and identically distributed...

conjecture | open | MathDB 页面 | 原文线索

V2-069167 | MATHDB-359288

Density of hyperbolicity conjecture for the Mandelbrot set

摘录 / EXCERPT: Density of hyperbolicity conjecture. The set $\mathcal{H}$ is dense in $\mathcal{M}$.

conjecture | open | MathDB 页面 | 原文线索

V2-069168 | MATHDB-359289

Conjecture that distinct mass-label sets do not share a cluster

摘录 / EXCERPT: Mass-label separation conjecture. Mass labels from distinct $L(m)$'s never belong to the same cluster.

conjecture | open | MathDB 页面 | 原文线索

V2-069169 | MATHDB-359291

Sintiari and Trotignon's logarithmic-treewidth conjecture for even-hole-free graphs

摘录 / EXCERPT: For a graph $G=(V(G),E(G))$, let $\operatorname{tw}(G)$ denote its treewidth, defined as the minimum width of a tree decomposition, where the width is the maximum bag size minus on…

conjecture | open | MathDB 页面 | 原文线索

V2-069170 | MATHDB-359312

The cyclotomic-divisibility equidistribution conjecture for Fekete polynomials

摘录 / EXCERPT: Let n be a positive squarefree integer, let $d>1$ be an integer not dividing n, set $d_1=\gcd(d,n)$, and let … Let F_n be the Fekete polynomial and let…

conjecture | open | MathDB 页面 | 原文线索

V2-069171 | MATHDB-359317

The correspondence coloring conjecture for dense random graphs

摘录 / EXCERPT: Dense random graph correspondence coloring conjecture. For every fixed positive $p\leq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-069172 | MATHDB-359319

Twisted Gan-Gross-Prasad nonvanishing conjecture for unitary theta periods

摘录 / EXCERPT: Let E/F be a quadratic extension of number fields, let V be an n-dimensional skew-Hermitian space over E, and let $\U(V)$ be its unitary group. Let π be an irreducible…

conjecture | open | MathDB 页面 | 原文线索

V2-069173 | MATHDB-359320

Unramified local refined twisted Gan-Gross-Prasad conjecture

摘录 / EXCERPT: Let v be a nonarchimedean place of F. Assume that E_v/F_v is unramified, the Hermitian space V is split at v, the conductor of ψ'_v is $\mathcal O_{F_v}$, and…

conjecture | open | MathDB 页面 | 原文线索

V2-069174 | MATHDB-359321

Refined twisted Gan-Gross-Prasad period formula

摘录 / EXCERPT: Refined twisted Gan-Gross-Prasad formula. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069175 | MATHDB-359330

Dolgachev-Nikulin duality conjecture for compactified Landau-Ginzburg models

摘录 / EXCERPT: Dolgachev-Nikulin duality conjecture. There is an isomorphism of lattices

conjecture | open | MathDB 页面 | 原文线索

V2-069176 | MATHDB-359339

Regular polar spaces admit tight embeddings

摘录 / EXCERPT: Embedding conjecture. Every regular polar space admits a tight embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-069177 | MATHDB-359340

Non-regularity of polar spaces induced from proper dual subspaces

摘录 / EXCERPT: Non-regularity conjecture. The polar space associated to this induced form is non-regular.

conjecture | open | MathDB 页面 | 原文线索

V2-069178 | MATHDB-359350

The Arc-Floer conjecture for isolated hypersurface singularities

摘录 / EXCERPT: Let $f \in \mathbb{C}\{z_0,\dots,z_n\}$ be a convergent power series such that $f(\mathbf{0})=0$ that defines an isolated hypersurface singularity at the origin. For every integer…

conjecture | open | MathDB 页面 | 原文线索

V2-069179 | MATHDB-359355

Measure minimization conjecture for continuous frames

摘录 / EXCERPT: Measure minimization conjecture. Then f is the unique solution of

conjecture | open | MathDB 页面 | 原文线索

V2-069180 | MATHDB-359359

The B+E characterization conjecture for generalized Bassian groups

摘录 / EXCERPT: Let G be an abelian group. Say that G is B+E if it decomposes as $G=A\oplus E$, where A is Bassian and E is elementary. A group is B+E characterization conjecture. B+E if a…

conjecture | open | MathDB 页面 | 原文线索

V2-069181 | MATHDB-359362

Finite string system conjecture for finitely supported loads

摘录 / EXCERPT: Let Ω be a bounded convex domain, and let the load $f \in \mathcal{M}(\overline{\Omega};\mathbb{R})$ be finitely supported. An optimal string system consists of a pair…

conjecture | open | MathDB 页面 | 原文线索

V2-069182 | MATHDB-359365

The branched-cover conjecture for Garland posets

摘录 / EXCERPT: Branched-cover conjecture. For every $k>0$ there is d such that, whenever Q is a finite cubical poset of dimension $k+1$ and every k-face is contained in at least d facets,…

conjecture | open | MathDB 页面 | 原文线索

V2-069183 | MATHDB-359370

The repeated-degree conjecture for the family x^n+x+1

摘录 / EXCERPT: Let $A=x^n+x+1$ with $n\geq 2$, and let $A_1,A_3,\ldots$ be the odd subsequence in the binary-polynomial Collatz transformation. Let s be the greatest integer such that…

conjecture | open | MathDB 页面 | 原文线索

V2-069184 | MATHDB-359371

The length conjecture for the family x^n+x+1

摘录 / EXCERPT: Let $A=x^n+x+1$ with $n\geq 2$, and let ℓ_A denote the length of its binary-polynomial Collatz transformation sequence. Let s be the greatest integer such that...

conjecture | open | MathDB 页面 | 原文线索

V2-069185 | MATHDB-359372

Signed Hadwiger conjecture

摘录 / EXCERPT: A signed graph $\hat{G}$ may have loops and parallel edges, and its balanced chromatic number $\chi_b(\hat{G})$ is the minimum number of parts in a partition of its vertices such t...

conjecture | open | MathDB 页面 | 原文线索

V2-069186 | MATHDB-359374

Sharp estimate for the Dunkl kernel of type A_n via shortest Weyl-chamber realizations

摘录 / EXCERPT: Let A_n be a root system with positive roots Σ^+, Weyl group W, and positive Weyl chamber C^+. For a Weyl chamber C, let $\mathcal{Short}(C)\subset W$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-069187 | MATHDB-359383

Recursive asymptotic-variance conjecture for stratified particle filters

摘录 / EXCERPT: Recursive asymptotic-variance conjecture. For each $f\in\mathcal{B}_b(\mathbb{R}^d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-069188 | MATHDB-359384

Asymptotic independence conjecture for fractional weights in stratified resampling

摘录 / EXCERPT: Let $t\in\mathbb{N}$, let $\psi\in\mathcal{B}_b(\mathbb{R}^{t+2})$ be continuous, and let $h\in\mathcal{B}_b(\mathbb{R}^{d(t+1)})$. For the step-$n$ particles $X_n^{M,m}$, define .....

conjecture | open | MathDB 页面 | 原文线索

V2-069189 | MATHDB-359387

Borozenets' rank-crank inequalities modulo 11

摘录 / EXCERPT: Borozenets' conjecture. The following chains of inequalities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-069190 | MATHDB-359396

The p-refined Friedberg-Jacquet conjecture

摘录 / EXCERPT: Let $G=\mathrm{GL}_{2n}$, let $P\subsetneq G$ be a proper spin parabolic with associated parabolic $\cP\subset\mathrm{GSpin}_{2n+1}$, and let $\tilde\pi^P=(\pi,\alpha^P)$ be a P-...

conjecture | open | MathDB 页面 | 原文线索

V2-069191 | MATHDB-359403

Complex product expansion conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Complex product expansion conjecture. The element $(X,t^{[\cdot]})\in H^0_c(\mathcal Z^{\mathrm{semi\textrm{-}Fano}}(\operatorname{Cpx}_3))[[t^{H_2(X)}]]$ is an infinite product

conjecture | open | MathDB 页面 | 原文线索

V2-069192 | MATHDB-359405

MNOP correspondence for projective threefolds

摘录 / EXCERPT: MNOP correspondence. The invariants

conjecture | open | MathDB 页面 | 原文线索

V2-069193 | MATHDB-359420

The covolume conjecture for 48-dimensional lattices of minimal norm 6

摘录 / EXCERPT: Let $L\subset\mathbb{R}^{48}$ be a lattice with minimal norm $\sqrt{6}$. Covolume conjecture. If there is $x\in L$ with $\sqrt{8}<|x|<\sqrt{10}$, then L has covolume greater than...

conjecture | open | MathDB 页面 | 原文线索

V2-069194 | MATHDB-359421

Periodic optimality for one-dimensional compact forbidden-distance sets

摘录 / EXCERPT: Let $K\subset[1,\infty)$ be a compact set with $1\in K$. A sphere packing is K-admissible if all distances between distinct centers avoid K in the paper's normalization, and a...

conjecture | open | MathDB 页面 | 原文线索

V2-069195 | MATHDB-359434

Manvel's bounded-deletion reconstruction conjecture for graphs

摘录 / EXCERPT: Manvel's conjecture. For each $\ell\in\mathbb{N}$, there exists a threshold M_ℓ such that every graph with at least M_ℓ vertices is ℓ-reconstructible.

conjecture | open | MathDB 页面 | 原文线索

V2-069196 | MATHDB-359437

Period-preserving involution conjecture for elliptic surfaces with $p_g=q=1$

摘录 / EXCERPT: Period-preserving involution conjecture. The space $F_{1,1}$ admits a period-preserving birational involution $S\leftrightarrow S'$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069197 | MATHDB-359438

Toroidal compactification conjecture for elliptic-surface moduli

摘录 / EXCERPT: Toroidal compactification conjecture. There is a morphism

conjecture | open | MathDB 页面 | 原文线索

V2-069198 | MATHDB-359440

Real Abelian Main Conjecture for arithmetic class groups

摘录 / EXCERPT: Let $\chi = \chi_0 \chi_p$, with $\chi_0 \ne 1$, be an even irreducible rational character and let $K=K_\chi$. For every irreducible p-adic character φ dividing χ, s...

conjecture | open | MathDB 页面 | 原文线索

V2-069199 | MATHDB-359442

Terai's conjecture on sequential Cohen-Macaulay edge ideals of weighted graphs

摘录 / EXCERPT: Terai's conjecture. The ideal $I(G_{\mathbf w})$ is sequentially Cohen-Macaulay for every weight function $\mathbf w$. This conjecture extends questions about Cohen-Macaulayness an...

conjecture | open | MathDB 页面 | 原文线索

V2-069200 | MATHDB-359455

GinOE number variance scaling limits

摘录 / EXCERPT: GinOE number variance scaling limits. As $N\to\infty$, the following limits are expected. For fixed $a\in(0,1)$,

conjecture | open | MathDB 页面 | 原文线索

V2-069201 | MATHDB-359461

The Bernoulli-number formula for the orbifold Euler characteristic of moduli space

摘录 / EXCERPT: Bernoulli-number conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069202 | MATHDB-359462

Modulo 32 congruences for Schur-type overpartition numbers

摘录 / EXCERPT: Let $S(n)$ denote the Schur-type overpartition function considered in the paper, and let n be a nonnegative integer. Modulo 32 congruences. The following congruences are supporte...

conjecture | open | MathDB 页面 | 原文线索

V2-069203 | MATHDB-359496

Larsen-Lubotzky conjecture on representation growth of irreducible lattices

摘录 / EXCERPT: Let H be a higher-rank semisimple group of the form ... where each F_i is a local field, each $\mathbf{G}_i$ is an absolutely almost simple F_i-group, and ... For a group...

conjecture | open | MathDB 页面 | 原文线索

V2-069204 | MATHDB-359501

West-Wu's conjecture on packing T-connectors

摘录 / EXCERPT: West-Wu's conjecture. For every positive integer k, if T is $3k$-edge-connected in G, then G admits k pairwise edge-disjoint T-connectors.

conjecture | open | MathDB 页面 | 原文线索

V2-069205 | MATHDB-359502

The conjecture that d-complete posets are LE-cactus

摘录 / EXCERPT: A d-complete poset is a poset in the sense of the standard definition of d-completeness. A poset is LE-cactus when the Bender-Knuth involutions acting on its linear extensions sati...

conjecture | open | MathDB 页面 | 原文线索

V2-069206 | MATHDB-359515

Margulis-Zimmer conjecture on commensurated subgroups of S-arithmetic lattices

摘录 / EXCERPT: Margulis-Zimmer conjecture. Every commensurated subgroup of Λ is either finite or S'-arithmetic for some $S'\subseteq S$.

conjecture | open | MathDB 页面 | 原文线索

V2-069207 | MATHDB-359518

The minimal-volume conjecture for log canonical pairs with reduced boundary

摘录 / EXCERPT: Let (X,B) be a projective log canonical pair of dimension n, where B is a nonzero reduced divisor and K_X+B is ample. The volume of K_X+B is the top self-intersection num...

conjecture | open | MathDB 页面 | 原文线索

V2-069208 | MATHDB-359520

The lower-bound conjecture for the first gap of global log canonical thresholds

摘录 / EXCERPT: For every $n\geq 2$, let m_n be the integer defined by the construction in Theorem. Let $(X,(1-b)S)$ be a complex klt Calabi-Yau pair of dimension n, where S is a nonzero eff...

conjecture | open | MathDB 页面 | 原文线索

V2-069209 | MATHDB-359523

The convex-hull characterization conjecture in the integer case

摘录 / EXCERPT: Integer-case convex-hull conjecture. Every such vector should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069210 | MATHDB-359524

The non-integer optimality conjecture for duo-equidistant vectors

摘录 / EXCERPT: Non-integer optimality conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-069211 | MATHDB-359525

The componentwise characterization conjecture for non-integer optimal vectors

摘录 / EXCERPT: Non-integer characterization conjecture. A vector $\boldsymbol{v}\in\Lambda(v)$ is optimal if and only if it has this decomposition and satisfies, for the relevant indices I,

conjecture | open | MathDB 页面 | 原文线索

V2-069212 | MATHDB-359526

The convex-hull inclusion conjecture for non-integer optimal vectors

摘录 / EXCERPT: Non-integer convex-hull conjecture. Every vector satisfying that characterization should belong to

conjecture | open | MathDB 页面 | 原文线索

V2-069213 | MATHDB-359537

Stabilization conjecture for ramified Hecke graph fibers

摘录 / EXCERPT: Fix a degree-one place $x\in |P^1|$. For each $r\in\mathbb N$, let P_r be the canonical projection and let c_{nx} denote the corresponding class. Stabilization conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069214 | MATHDB-359538

Dimension conjecture for ramified Hecke eigenspaces

摘录 / EXCERPT: Fix a degree-one place $x\in |P^1|$, let $r\in\mathbb N$, and let P_r be the canonical projection. Assuming the stabilization conjecture, let m_r be the minimal integer with th...

conjecture | open | MathDB 页面 | 原文线索

V2-069215 | MATHDB-359540

Super approximation conjecture for finitely generated subgroups of \(\operatorname{SL}_n(\mathbb Z)\)

摘录 / EXCERPT: Super approximation conjecture. The group Γ has the super approximation property with respect to all positive integers if and only if $\mathbb G_0$ is perfect.

conjecture | open | MathDB 页面 | 原文线索

V2-069216 | MATHDB-359543

Derived p-adic leading term conjecture

摘录 / EXCERPT: Let $\mathcal O$ be the coefficient ring, let I be the relevant augmentation ideal, let $c\in\bigcap_{\Lambda_I}^r H^1_\Sigma(\mathcal O_{K,S},\mathbb T)_I$ be the special elemen...

conjecture | open | MathDB 页面 | 原文线索

V2-069217 | MATHDB-359544

A p-adic Birch and Swinnerton-Dyer formula for the anticyclotomic Heegner element

摘录 / EXCERPT: Let $z^{\rm Hg}$ be the Λ-adic Heegner element, let $\mathcal D$ be the derived descent map, let ${\rm Eul}_S$ be the product of Euler factors at the finite places in S,...

conjecture | open | MathDB 页面 | 原文线索

V2-069218 | MATHDB-359545

Laptev-Safronov conjecture for complex Schrödinger operators

摘录 / EXCERPT: Let $d\geq2$ and $q\in(d/2,d]$. Let V be a bounded complex-valued potential on $\mathbb{R}^d$, and let z be an eigenvalue of $-\Delta-V$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-069219 | MATHDB-359553

Row-column transformation dependence for two and three copies

摘录 / EXCERPT: Let $\upsilon,\upsilon_2,\upsilon_3,\ldots$ be the weighted-average ITV family defined in the source, and let the row-column-type transformations be the transformations on the tran...

conjecture | open | MathDB 页面 | 原文线索

V2-069220 | MATHDB-359554

The multiple cover conjecture for descendent Gromov-Witten invariants of K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $\beta\in H_2(S,\mathbb{Z})$ be an effective curve class, and let $\langle \tau_{k_1}(\gamma_1) \cdots \tau_{k_n}(\gamma_n) \rangle^S_{g,\beta}$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-069221 | MATHDB-359556

The polynomiality conjecture for descendent invariants of K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface, let $\beta\in H_2(S,\mathbb Z)$ be primitive and effective with $\beta^2\ne0$, and let $\delta_1,\ldots,\delta_t\in H^2(S)$ satisfy $\delta_i\cdot\beta=0$....

conjecture | open | MathDB 页面 | 原文线索

V2-069222 | MATHDB-359557

The upgraded polynomiality conjecture for descendent invariants of K3 surfaces

摘录 / EXCERPT: Let S, β, δ_i, and the normalized descendent invariant be as in the polynomiality conjecture, with $k_i,m_i\geq1$. For subsets $I_x\subset\{1,\ldots,x\}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-069223 | MATHDB-359558

The Virasoro constraints conjecture for K3 surfaces with fiber and point insertions

摘录 / EXCERPT: Let S be a K3 surface, let γ_a be a homogeneous basis of $H^*(S)$ with $\gamma_a\in H^{p_a,q_a}(S)$, and set $b_a=p_a-1/2$. Let $[\alpha]^p_q$ denote the coefficient of...

conjecture | open | MathDB 页面 | 原文线索

V2-069224 | MATHDB-359560

Compactness and multiple-strings degeneration conjecture for Einstein-Bogomol'nyi metrics

摘录 / EXCERPT: Compactness and degeneration conjecture. If $V_i\to V\in(\underline V,+\infty)$, then there exists a sequence $\sigma_i\in PSL(2,\mathbf{C})$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069225 | MATHDB-359567

Global fibrancy conjecture for Green complexes

摘录 / EXCERPT: Let ${\mathscr{Green}}$ denote the simplicial presheaf assigning to each object the simplicial set of Green complexes. A simplicial presheaf is a presheaf of Kan complexes when eac...

conjecture | open | MathDB 页面 | 原文线索

V2-069226 | MATHDB-359569

Thickened Green resolution morphism conjecture

摘录 / EXCERPT: Let ${\mathscr{Green}}'$ and ${\mathscr{sTwist}}'$ be the simplicial presheaves obtained by replacing the simplices used in the constructions of ${\mathscr{Green}}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069227 | MATHDB-359570

The weak local Arthur packets conjecture

摘录 / EXCERPT: The weak local Arthur packets conjecture. The set $\Pi_{\psi}^{\textrm{Weak}}$ is a union of local Arthur packets.

conjecture | open | MathDB 页面 | 原文线索

V2-069228 | MATHDB-359571

The Polya-Schur characterization for 0-sum hyperbolicity preservers

摘录 / EXCERPT: Let $T:\mathbb{R}[t]_{n,0}\to\mathbb{R}[t]_{d,0}$ be a diagonal linear map. A 0-sum hyperbolicity preserver is a map preserving the relevant 0-sum hyperbolicity property. Polya-Sch...

conjecture | open | MathDB 页面 | 原文线索

V2-069229 | MATHDB-359572

The SOS characterization of symmetric hyperbolicity

摘录 / EXCERPT: Let $\Gamma_{n,d}$ denote the class of symmetric degree-d polynomials in n variables, and let $\Delta_{\vec{1},\vec{1}}p$ be the indicated second directional derivative. A poly...

conjecture | open | MathDB 页面 | 原文线索

V2-069230 | MATHDB-359573

The extendability and weak SOS-hyperbolicity conjecture

摘录 / EXCERPT: Let $\Gamma_{n,d}$ be the class of symmetric degree-d polynomials in n variables. For a symmetric hyperbolic polynomial p, let its associated operator be the operator defined...

conjecture | open | MathDB 页面 | 原文线索

V2-069231 | MATHDB-359576

Salehi-Golsefidy-Varjú super approximation conjecture

摘录 / EXCERPT: Salehi-Golsefidy-Varjú conjecture. A finitely generated subgroup $G<\operatorname{SL}_n(\mathbb Z)$ has super approximation with respect to all positive integers exactly when the i...

conjecture | open | MathDB 页面 | 原文线索

V2-069232 | MATHDB-359581

The discrete tiling conjecture for self-replicating words

摘录 / EXCERPT: Discrete tiling conjecture. The translates satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069233 | MATHDB-359595

The anticyclotomic Iwasawa-Greenberg main conjecture for the BDP p-adic L-function

摘录 / EXCERPT: Iwasawa-Greenberg main conjecture. The group ${\rm Sel}_{\overline{\mathfrak p}}(K_\infty,W)$ should be Λ-cotorsion and satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069234 | MATHDB-359600

Existence conjecture for modular realizations of the representation

摘录 / EXCERPT: Let p and q be the parameters introduced earlier in the paper, and let ρ_1 and ρ_2 be the corresponding representations. Existence conjecture. There exists a modular...

conjecture | open | MathDB 页面 | 原文线索

V2-069235 | MATHDB-359601

The chordal Lipschitz constant conjecture for Möbius transformations

摘录 / EXCERPT: Let $\mathbb{B}^n$ be the unit ball, let $\overline{\mathbb{R}}^n$ be the one-point compactification of $\mathbb{R}^n$, and let q denote the chordal metric. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069236 | MATHDB-359605

Finite-fiber conjecture for characteristic polynomials of symmetric tensors

摘录 / EXCERPT: Finite-fiber conjecture for symmetric tensors. All fibers of Φ_{sym} are finite.

conjecture | open | MathDB 页面 | 原文线索

V2-069237 | MATHDB-359606

The dimensions conjecture for fixed subspaces of the Cartier operator

摘录 / EXCERPT: The dimensions conjecture. For every divisor m of $\operatorname{lcm}(\deg R_1,\dots,\deg R_k)$, this vector space has dimension

conjecture | open | MathDB 页面 | 原文线索

V2-069238 | MATHDB-359607

A basis conjecture for fixed points of the Cartier operator

摘录 / EXCERPT: The fixed-point basis conjecture. For every $i\in\{1,2,\dots,k\}$, the polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-069239 | MATHDB-359620

Syzygy resolution conjecture for a specified plane curve

摘录 / EXCERPT: Let $R=k[x,y,z]$ and let $f=(x^2-y^2)z^{d-1} -(x^{d-1}-y^{d-1})x^2 -y^{d+1}$, with $d\geq 2$, and let J_f be its gradient ideal. Syzygy-resolution conjecture. The maximal degree...

conjecture | open | MathDB 页面 | 原文线索

V2-069240 | MATHDB-359624

Faudree-Gyárfás-Schelp star-forest ascending decomposition conjecture

摘录 / EXCERPT: Let G be a graph with $\binom{m+1}{2}$ edges. An ascending subgraph decomposition is a decomposition $H_1,\dots,H_m$ in which H_i has i edges and is a subgraph of H_{i+1} f...

conjecture | open | MathDB 页面 | 原文线索

V2-069241 | MATHDB-359627

Injectivity of the dual winding quotient map for strong Weil curves of odd analytic rank

摘录 / EXCERPT: Let E be a strong Weil curve over $\mathbb{Q}$ of conductor $M\mid N$. Suppose that E has odd analytic rank, and let $\xi_{E,N}^\vee$ be the dual map associated with the windin...

conjecture | open | MathDB 页面 | 原文线索

V2-069242 | MATHDB-359628

Fourier-Laumon invariance conjecture for cuspidal perverse sheaves

摘录 / EXCERPT: Let G be a possibly disconnected reductive group, let $\mathcal N_G$ be its nilpotent cone, and let $\mathcal O$ be a nilpotent orbit. If...

conjecture | open | MathDB 页面 | 原文线索

V2-069243 | MATHDB-359629

Cuspidality criterion in the refined local Langlands correspondence

摘录 / EXCERPT: Fix a reductive group G, a rigid inner twist, a Whittaker datum $\mathfrak w$, and the refined local Langlands reciprocity map $\operatorname{rec}_{\mathfrak w}$. An enhanced L...

conjecture | open | MathDB 页面 | 原文线索

V2-069244 | MATHDB-359631

Levi-subgroup compatibility conjecture for local Langlands correspondences

摘录 / EXCERPT: Let G be a reductive group and let $\dot\pi$ be supercuspidal. Levi-subgroup compatibility conjecture. Every Levi subgroup of G satisfies the cuspidality criterion and satisfie...

conjecture | open | MathDB 页面 | 原文线索

V2-069245 | MATHDB-359632

ABPS twisted-extended-quotient compatibility conjecture

摘录 / EXCERPT: Let $(M',\xi_M,\mathcal T_M,\bar h_M)$ be a Levi subgroup of $(G',\xi,\mathcal T,\bar h)$, let $s=(M',\sigma)$ be a cuspidal datum for G', and let $\hat s=({}^LM,\psi,v,\varrho)$...

conjecture | open | MathDB 页面 | 原文线索

V2-069246 | MATHDB-359636

Triangle-free Cohen-Macaulay oriented graph conjecture

摘录 / EXCERPT: Let $\operatorname{\mathcal{D}}$ be a weighted oriented graph whose underlying graph G is triangle-free, meaning that G contains no cycles of length 3. The graph G is Cohen...

conjecture | open | MathDB 页面 | 原文线索

V2-069247 | MATHDB-359638

Wreath Macdonald P-polynomial pairing conjecture

摘录 / EXCERPT: Let r be the wreath parameter, let $w\in\hat{\mathfrak{S}}_I$, and let $P^w_{\mu^\bullet}$ denote the wreath Macdonald P polynomial. For a partition ... write $a(s)$, $l(s)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069248 | MATHDB-359639

Wreath Macdonald-Kostka leading-term conjecture

摘录 / EXCERPT: Let $w\in\hat{\mathfrak{S}}_I$ and let $\tilde{K}^w_{\mu^\bullet,\mu^\bullet}(q,t)$ be the diagonal wreath Macdonald-Kostka polynomial. Let $f^w_{\mu^\bullet}$ be the sum of the te...

conjecture | open | MathDB 页面 | 原文线索

V2-069249 | MATHDB-359647

The non-4-divisible Frobenius-Perron dimension solvability conjecture

摘录 / EXCERPT: Let $\mathcal C$ be a modular tensor category, and let $\operatorname{FPdim}(\mathcal C)$ denote its Frobenius-Perron dimension. Solvability conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-069250 | MATHDB-359663

The n-roach conjecture for Shehtman's first problem

摘录 / EXCERPT: For each n, let ${\sf L}_n$ denote the logic associated with $\beta(\omega^n)$, and let ${\sf L}(\mathcal R_n)$ denote the logic of the class of n-roaches. n-roach conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069251 | MATHDB-359664

The nine-frame Fine-Jankov conjecture for ${\sf L}_3$

摘录 / EXCERPT: Let ${\sf L}_3$ be the logic associated with $\beta(\omega^3)$, and let ${\sf S4.1}$ be the base logic. The Fine-Jankov formula of a finite frame is the characteristic formula used...

conjecture | open | MathDB 页面 | 原文线索

V2-069252 | MATHDB-359675

Existence of graphs of every prescribed minimum girth without the AOP property

摘录 / EXCERPT: Let $g\in\mathbb{N}$. A graph has the AOP property if it admits an acyclic orientation with at most one directed path between any pair of vertices, and its girth is the length of i...

conjecture | open | MathDB 页面 | 原文线索

V2-069253 | MATHDB-359683

Positive-proportion conjecture for irreducible polynomials in quadratic semigroups

摘录 / EXCERPT: Let $S=\{x^2+c_1,\dots,x^2+c_r\}$ for some distinct $c_1,\dots,c_r\in\mathbb{Q}$ and $r\geq2$. Let M_S denote the semigroup generated by S under composition, and call a subset...

conjecture | open | MathDB 页面 | 原文线索

V2-069254 | MATHDB-359684

Poonen's rational-period conjecture for quadratic polynomials

摘录 / EXCERPT: Let $\phi(x)=x^2+c$ for some $c\in\mathbb{Q}$, and let $a\in\mathbb{Q}$ be a periodic point for ϕ, meaning that $\phi^n(a)=a$ for some $n\geq1$; the least such n is the peri...

conjecture | open | MathDB 页面 | 原文线索

V2-069255 | MATHDB-359694

The Mumford-Shah conjecture on planar minimizer structure

摘录 / EXCERPT: Mumford-Shah conjecture. If (K,u) is a minimizer of E_λ, then K has the structure described above.

conjecture | open | MathDB 页面 | 原文线索

V2-069256 | MATHDB-359695

Uniqueness conjecture for global generalized Mumford-Shah minimizers with one unbounded component

摘录 / EXCERPT: Uniqueness conjecture. The pair (K,u) is either an elementary global minimizer or a cracktip.

conjecture | open | MathDB 页面 | 原文线索

V2-069257 | MATHDB-359696

The bipartite list-cover independent transversal conjecture

摘录 / EXCERPT: Let G and H be bipartite graphs, with bipartitions $V(G)=A_G\cup B_G$ and $V(H)=A_H\cup B_H$. Let H be a bipartite list-cover of G with respect to a mapping L, meaning th...

conjecture | open | MathDB 页面 | 原文线索

V2-069258 | MATHDB-359704

Swinnerton-Dyer's refined Manin conjecture for cubic surfaces

摘录 / EXCERPT: Let S' be a cubic surface over $\mathbb{Q}$, let U' be the relevant open subset, let H be its height function, and let ρ' denote the corresponding Picard rank. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069259 | MATHDB-359714

Analytic properties of the density matrix for the \(\mathfrak{sl}_{n+1}\)-invariant model

摘录 / EXCERPT: Analytic-properties conjecture. The function $D_{1,\dots,m}$ is meromorphic in $\lambda_1,\dots,\lambda_m$, with at most simple poles when

conjecture | open | MathDB 页面 | 原文线索

V2-069260 | MATHDB-359721

Conjecture on maximum divergence from trapezoid models

摘录 / EXCERPT: Let $\T_{a,b,d}$ be the trapezoid model in the simplex described above, with parameters a, b, and d, and let $D(\T_{a,b,d})$ denote its maximum divergence. Trapezoid-model di...

conjecture | open | MathDB 页面 | 原文线索

V2-069261 | MATHDB-359723

Linear oriented chromatic number for bounded-degeneracy graphs

摘录 / EXCERPT: Bounded-degeneracy linearity conjecture. For every integer $d\geq 1$, there exists a constant c depending on d such that

conjecture | open | MathDB 页面 | 原文线索

V2-069262 | MATHDB-359725

Odd-minor clustered and defective colouring conjecture

摘录 / EXCERPT: Let H be a graph, let $\mathcal{G}_H^{\text{odd}}$ be the class of graphs with no H odd-minor, let $\operatorname{\overline{td}}(H)$ denote the connected tree-depth of H, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069263 | MATHDB-359728

The threshold conjecture for star decompositions of random regular graphs

摘录 / EXCERPT: Let $d\geq 3$ be fixed, and let $\kindactual{d}$ denote the largest positive integer k such that, asymptotically almost surely, the random d-regular graph $\mathcal{G}_{n,d}$ c...

conjecture | open | MathDB 页面 | 原文线索

V2-069264 | MATHDB-359731

The transcendence-degree versus horizontal-codimension conjecture for Hodge structures

摘录 / EXCERPT: Transcendence-degree versus horizontal-codimension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-069265 | MATHDB-359732

Hodge classes are motivated

摘录 / EXCERPT: Hodge classes are motivated. The functor ω_{Hdg} is fully faithful. This is presented as a weaker form of the Hodge conjecture and is not known in the stated genera...

conjecture | open | MathDB 页面 | 原文线索

V2-069266 | MATHDB-359733

André's generalized Grothendieck period conjecture

摘录 / EXCERPT: André's generalized Grothendieck period conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069267 | MATHDB-359748

The shared-stress-kernel witness conjecture

摘录 / EXCERPT: Let G be a graph and let $u,v\in V(G)$ be a pair of vertices. Let $K(G,p)$ denote the shared stress kernel associated with a generic realization (G,p). Suppose that $\{u,v\}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-069268 | MATHDB-359753

The Gamma-bound dominance conjecture for integer D-optimality

摘录 / EXCERPT: Gamma-bound dominance conjecture. For $s-q\geq 1$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-069269 | MATHDB-359761

Sprinkling conjecture for random regular graphs

摘录 / EXCERPT: Let $d_1=d_1(n),d_2=d_2(n)\in\mathbb{N}:=\{0,1,\ldots\}$ satisfy ... Let $\mathcal{S}_n(d_1,d_2)$ be the set of pairs of edge-disjoint graphs on $[n]$ whose respective degrees are...

conjecture | open | MathDB 页面 | 原文线索

V2-069270 | MATHDB-359762

Asymptotic equivalence conjecture for sums of random regular graphs

摘录 / EXCERPT: For $d_1,d_2\in\mathbb{N}$ with $d_1+d_2\leq n-1$, let $\mathcal{G}(n,d)$ denote the uniform random d-regular graph. Let $\mathcal{G}(n,d_1)\oplus\mathcal{G}(n,d_2)$ denote the d...

conjecture | open | MathDB 页面 | 原文线索

V2-069271 | MATHDB-359764

Gao-Isaev-McKay embedding conjecture for random regular graphs

摘录 / EXCERPT: Let $\mathcal{G}(n,d)$ denote the uniform random labelled d-regular graph. Let $0\leq d_1\leq d\leq n-1$ be integers, excluding $(d_1,d)=(1,2)$ and $(d_1,d)=(n-3,n-2)$. Gao-Isaev...

conjecture | open | MathDB 页面 | 原文线索

V2-069272 | MATHDB-359767

The Fekete-Pommerenke energy conjecture for Coulomb gases on Jordan curves

摘录 / EXCERPT: Fekete-Pommerenke energy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069273 | MATHDB-359783

Other-bases conjecture for non-half residue stopping sets

摘录 / EXCERPT: Fix an even modulus $n\geq2$, a subset $S\subset\{0,\dots,n-1\}$ of residue classes, and the associated discrete stick-fragmentation process with stopping set determined by S. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-069274 | MATHDB-359793

Divisibility conjecture for the deformed polynomials of type B2

摘录 / EXCERPT: Divisibility conjecture. For any $m,i\geq 0$, the polynomial $\widetilde{f^m_i}(x,y)+\widetilde{f^m_i}(y,x)$ is divisible by

conjecture | open | MathDB 页面 | 原文线索

V2-069275 | MATHDB-359804

Büyükboduk-Lei's cyclotomic cotorsion conjecture for signed Selmer groups

摘录 / EXCERPT: Let F be the number field and let F_cyc be its cyclotomic Z_p-extension. Let $\mathcal{I}_p$ be the indexing set for the signed local conditions, let $\mathcal{M}$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-069276 | MATHDB-359806

Conjecture on zero-counting measures for weighted Chebyshev polynomials

摘录 / EXCERPT: For $s>0$, let $T_n^{w_s}$ be the weighted Chebyshev polynomial associated with the weight w_s, and let $\nu_{n,s}$ denote its normalised zero-counting measure on the unit circle...

conjecture | open | MathDB 页面 | 原文线索

V2-069277 | MATHDB-359808

Conjectured form of the M/M/1 processor-sharing queue's average age of information

摘录 / EXCERPT: M/M/1-PS AAoI conjecture. The average age of information is

conjecture | open | MathDB 页面 | 原文线索

V2-069278 | MATHDB-359810

Leading-order Gamma conjecture for del Pezzo mirror central charges

摘录 / EXCERPT: Let Y be a del Pezzo surface, let $(\mathcal X,W)$ be its Landau-Ginzburg mirror, let $\alpha:K(Y)\to H_2(\mathcal X,W^{-1}(+\infty);\mathbb Z)$ be the map from the preceding cen...

conjecture | open | MathDB 页面 | 原文线索

V2-069279 | MATHDB-359814

SLE trace minimality conjecture

摘录 / EXCERPT: For $\kappa\in[0,8]$, let SLE_κ denote Schramm-Loewner evolution with parameter κ, and let its trace be the associated random curve. SLE trace minimality con...

conjecture | open | MathDB 页面 | 原文线索

V2-069280 | MATHDB-359820

The integral-model intersection conjecture for singular support

摘录 / EXCERPT: Integral-model intersection conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069281 | MATHDB-359823

The generalized zero-divisor conjecture for graded rings

摘录 / EXCERPT: Let $R=\bigoplus_{n i G}R_n$ be a ring graded by a torsion-free group G, and let I be a left ideal of R. The ideal I is unfaithful when its annihilator is nonzero. The gene...

conjecture | open | MathDB 页面 | 原文线索

V2-069282 | MATHDB-359824

The generalized idempotent conjecture for torsion-free graded rings

摘录 / EXCERPT: Let $R=\bigoplus\limits_{n\in G}R_n$ be a ring graded by a torsion-free group G, where 0 denotes the identity element of G. An idempotent is an element e satisfying $e^2=e$...

conjecture | open | MathDB 页面 | 原文线索

V2-069283 | MATHDB-359825

The generalized idempotent conjecture for Abelian graded rings

摘录 / EXCERPT: Let $R=\bigoplus\limits_{n\in G}R_n$ be a ring graded by an Abelian group G, and let H be the torsion subgroup of G. An idempotent is an element e satisfying $e^2=e$. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-069284 | MATHDB-359826

The Jacobson-radical conjecture for torsion-free graded rings

摘录 / EXCERPT: Let $R=\bigoplus_{n\in G}R_n$ be a ring graded by a torsion-free group G, and let $J(R)$ denote its Jacobson radical. An ideal is graded when it is the direct sum of its homogene...

open_problem | open | MathDB 页面 | 原文线索

V2-069285 | MATHDB-359827

Gaetz-Gao's separable-element splitting conjecture for finite Weyl groups

摘录 / EXCERPT: Let W be a finite Weyl group, and let $U=[e,u]_R\subseteq W$, where $[e,u]_R$ denotes the interval from the identity to u in the right weak order. A pair (X,Y) of subsets of...

conjecture | open | MathDB 页面 | 原文线索

V2-069286 | MATHDB-359833

A divisibility conjecture for MacMahon's generalized divisor sums

摘录 / EXCERPT: Let $MO(t,n)$ denote the MacMahon-type arithmetic function used in the paper, and let t,n be integers. MacMahon divisibility conjecture. If $t=10$, then ... The paper presents this...

conjecture | open | MathDB 页面 | 原文线索

V2-069287 | MATHDB-359846

Gras's finiteness conjecture for exceptional primes of dihedral Artin representations

摘录 / EXCERPT: Let $\rho=\operatorname{Ind}_{\operatorname{G}_L}^{\mathbb{Q}}\zeta$ be an odd two-dimensional dihedral Artin representation, with L imaginary quadratic, and let $S(\rho)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-069288 | MATHDB-359859

Non-tensorial norm estimate for random matrices with Weibull entries

摘录 / EXCERPT: Non-tensorial Weibull norm conjecture. The expected operator norm is comparable, with constants depending on p and q, to $D_1+D_2+D_{3,r}$.

open_problem | open | MathDB 页面 | 原文线索

V2-069289 | MATHDB-359860

The q-rational radius-of-convergence conjecture

摘录 / EXCERPT: For a real number $x>0$, let $[x]_q$ denote the associated q-deformation and let $R(x)$ be the radius of convergence of its Taylor series. Write ... for the golden ratio, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069290 | MATHDB-359865

Sintiari and Trotignon's diamond conjecture for even-hole-free graphs

摘录 / EXCERPT: Sintiari and Trotignon's conjecture. Every (even-hole, K_4)-free graph of sufficiently large treewidth contains a diamond as an induced subgraph.

conjecture | open | MathDB 页面 | 原文线索

V2-069291 | MATHDB-359868

Sottile's general secant conjecture

摘录 / EXCERPT: Let $\nu\vdash n$ and let $\boldsymbol{\mu}\Vdash\kappa$, where $\kappa=(\kappa_1,\dots,\kappa_s)$ is a composition of n. Let $l_1,\dots,l_s\subseteq\mathbb{R}$ be pairwise disjo...

conjecture | open | MathDB 页面 | 原文线索

V2-069292 | MATHDB-359869

Boris Shapiro's total reality conjecture for convex curves

摘录 / EXCERPT: Let $\gamma:\mathbb{R}\to\mathbb{R}_{m-1}[u]$ be a convex curve, meaning that any m distinct points along γ are linearly independent. Let $z_1,\dots,z_{d(m-d)}$ be distinc...

conjecture | open | MathDB 页面 | 原文线索

V2-069293 | MATHDB-359873

Convergence of powers of a dual number matrix

摘录 / EXCERPT: Let $A=A_s+A_d\varepsilon$ be an $n\times n$ dual number matrix and let $k\in\mathbb{N}$. Here O denotes the zero matrix, and $\rho(A_s)$ denotes the spectral radius of the stand...

conjecture | open | MathDB 页面 | 原文线索

V2-069294 | MATHDB-359877

Odd-order nilpotent Malle-Bhargava principle

摘录 / EXCERPT: Let G be a nilpotent group, and let $c_{(H,\phi)}(G)$ be the leading constant in the constrained fair-counting asymptotic for $\operatorname{Epi}_{(H,\phi)}(G_{\mathbb{Q}},G)$. O...

conjecture | open | MathDB 页面 | 原文线索

V2-069295 | MATHDB-359878

Explicit leading constant for odd nilpotent groups

摘录 / EXCERPT: Let G be an odd-order nilpotent group, with $b_{(H,\phi)}(G)$ and the local factors $\alpha_1,\alpha_2,\alpha_3$ as defined in the surrounding discussion and in Proposition. Expl...

conjecture | open | MathDB 页面 | 原文线索

V2-069296 | MATHDB-359887

Finiteness conjecture for non-simplifiable rectangular diagrams

摘录 / EXCERPT: Let $\mathscr L$ be a link type in $\mathbb S^3$. A rectangular diagram representing $\mathscr L$ is non-simplifiable if it cannot be reduced by the simplifying moves considered fo...

conjecture | open | MathDB 页面 | 原文线索

V2-069297 | MATHDB-359888

Total-variation convergence for the multineighbor construction

摘录 / EXCERPT: Let $\beta=k-\frac{1}{n_m}$, $p=n^{-\frac{1}{\beta}}$, and $q=n^{-\frac{km}{\beta}}$, with $n=n_m$. Let $\Gamma_{n,m,p}$ and $Y_{n,k,q}$ denote the two distributions defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-069298 | MATHDB-359891

Kontsevich-Zorich conjecture on the asymptotic second Lyapunov exponent

摘录 / EXCERPT: Kontsevich-Zorich conjecture. For the hyperelliptic components $\mathcal{M}^{\mathcal{H}}_{2g-2}$ and $\mathcal{M}^{\mathcal{H}}_{(g-1,g-1)}$,

conjecture | open | MathDB 页面 | 原文线索

V2-069299 | MATHDB-359894

Monotonicity of the Poisson order-k median offset

摘录 / EXCERPT: Median offset monotonicity conjecture. The function $\alpha_{k,n}-n$ is decreasing in n under one of the following conditions: (i) $k=2$ and $n\ge3\kappa$; (ii) $k\in[3,6]$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069300 | MATHDB-359901

Conjecture on Bernstein interpolation and operator-theoretic properties of geometrically regular weighted shifts

摘录 / EXCERPT: The conjecture. In Sector II, except on its boundary with Sector I, no α is interpolated by a Bernstein function; in Sector III, W_α is subnormal, and except on its b...

conjecture | open | MathDB 页面 | 原文线索

V2-069301 | MATHDB-359914

The block-and-hole minimal-rigidity conjecture in ℓ_p^3

摘录 / EXCERPT: Block-and-hole minimal-rigidity conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069302 | MATHDB-359915

The one-discus global-rigidity conjecture for discus-and-hole graphs

摘录 / EXCERPT: One-discus global-rigidity conjecture. The graph $G^\dagger$ is generically globally rigid in $\mathbb R^3$ if and only if it is 4-connected and redundantly rigid in...

conjecture | open | MathDB 页面 | 原文线索

V2-069303 | MATHDB-359918

Hilbert-Burch matrix parametrization conjecture for Gröbner cells of zero-dimensional monomial ideals

摘录 / EXCERPT: Let $E\subset \mathbf{k}[[x,y]]$ be a zero-dimensional monomial ideal. Let $\mathcal{N}_{<\underline{d}}(E)$ be the set of matrices defined above, and let H be the Hilbert-Burch...

conjecture | open | MathDB 页面 | 原文线索

V2-069304 | MATHDB-359930

Givental-type mirror conjecture for general-type LG singularities

摘录 / EXCERPT: Let W be a nondegenerate quasihomogeneous singularity of general type. Let $I_{\rm FJRW}^{\rm sm}(t,z)$ be the small FJRW I-function, let $\tau(t)$ be determined by the coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-069305 | MATHDB-359931

Givental-type genus-zero mirror symmetry conjecture for the small FJRW I-function

摘录 / EXCERPT: Let $(W,\langle J\rangle)$ be an admissible LG pair, and let $I_{\rm FJRW}^{\rm sm}(t,-z)$ be its small FJRW I-function. Let $\tau(t)$ be defined by the coefficient of z^0 in t...

conjecture | open | MathDB 页面 | 原文线索

V2-069306 | MATHDB-359935

Finite forbidden-subdiagram characterization of ranked Kohnert posets

摘录 / EXCERPT: Finite forbidden-subdiagram conjecture. There exists a finite number of special families of subdiagrams $\mathcal{F}$ such that, for any diagram D_0, $\mathcal{P}(D_0)$ is ranked...

conjecture | open | MathDB 页面 | 原文线索

V2-069307 | MATHDB-359936

Spectrum-aware debiasing conjecture for non-separable convex penalties

摘录 / EXCERPT: Spectrum-aware debiasing conjecture. Under suitable conditions, there is a unique solution $\widehat{\mathsf{adj}}$ of this equation and

conjecture | open | MathDB 页面 | 原文线索

V2-069308 | MATHDB-359940

The Assmus-Mattson nonexistence conjecture for higher-weight codes

摘录 / EXCERPT: Assmus-Mattson nonexistence conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-069309 | MATHDB-359941

Serre-invariant locus conjecture for triangulated categories

摘录 / EXCERPT: Let $\mathcal T$ be a triangulated category with $X \in \operatorname{FM} \mathcal T$. The Fourier-Mukai locus $\operatorname{Spec}^{\mathsf{FM}} \mathcal T$ and the Serre-invarian...

conjecture | open | MathDB 页面 | 原文线索

V2-069310 | MATHDB-359942

Serre-geometric locus conjecture for proper Fourier-Mukai partners

摘录 / EXCERPT: Let $\mathcal T$ be a triangulated category with $\operatorname{FM} \mathcal T\operatorname{NE}\emptyset$. The proper Fourier-Mukai locus is...

conjecture | open | MathDB 页面 | 原文线索

V2-069311 | MATHDB-359953

K3-category equivalence conjecture for quasi-BPS categories at genus one

摘录 / EXCERPT: Let S be a K3 surface, let $v=dv_0$ with $d\geqslant 1$ and v_0 primitive satisfying $\langle v_0,v_0\rangle=0$, and let S' and α be the surface and Brauer class occur...

conjecture | open | MathDB 页面 | 原文线索

V2-069312 | MATHDB-359955

Odd-order off-diagonally symmetric alternating sign matrix product formula conjecture

摘录 / EXCERPT: Let an off-diagonally symmetric alternating sign matrix (OSASM) be an alternating sign matrix invariant under reflection in the off-diagonal axis, and let n be a nonnegative inte...

conjecture | open | MathDB 页面 | 原文线索

V2-069313 | MATHDB-359956

Even-order off-diagonally symmetric alternating sign matrix generating-function symmetry conjecture

摘录 / EXCERPT: Let $X^{\mathrm O}_{2n}(r,t)$ be the even-order OSASM generating function, and let $\bar t$ denote the parameter appearing in the source's generating-function notation. For an OSAS...

conjecture | open | MathDB 页面 | 原文线索

V2-069314 | MATHDB-359962

Monotonicity of asymptotic outbreak size for short-period infection rates

摘录 / EXCERPT: Short-period monotonicity conjecture. There exists $\bar{\omega}>0$ such that, for all $\omega<\bar{\omega}$, the asymptotic outbreak size $1-s^{(\infty)}(\infty)$ is decreasing in...

conjecture | open | MathDB 页面 | 原文线索

V2-069315 | MATHDB-359964

Kresch-Tamvakis conjecture for a terminating hypergeometric series

摘录 / EXCERPT: Kresch-Tamvakis conjecture. The following terminating hypergeometric series has absolute value at most 1:

conjecture | open | MathDB 页面 | 原文线索

V2-069316 | MATHDB-359971

The v-number conjecture for monomial ideals with linear powers

摘录 / EXCERPT: The v-number conjecture. If I has linear powers, then

conjecture | open | MathDB 页面 | 原文线索

V2-069317 | MATHDB-359973

Hoffmann-Ostenhof's 3-Decomposition Conjecture

摘录 / EXCERPT: Let G be a connected cubic graph. Hoffmann-Ostenhof's 3-Decomposition Conjecture. G can be decomposed into a spanning tree, a collection of cycles, and a possibly empty matchin...

conjecture | open | MathDB 页面 | 原文线索

V2-069318 | MATHDB-359976

Bloom's conjecture on type, contact and order of vanishing

摘录 / EXCERPT: Let $M\subset \mathbb{C}^n$ be a pseudoconvex real hypersurface with $n\geq 3$. For $1\leq s\leq n-2$ and $p\in M$, write $t^{(s)}(M,p)$, $c^{(s)}(M,p)$ and $a^{(s)}(M,p)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-069319 | MATHDB-359977

Generalized D'Angelo conjecture for complex vector subbundles

摘录 / EXCERPT: Let $M\subset \mathbb{C}^n$ be a pseudoconvex real hypersurface with $n\geq 3$, let $1\leq s\leq n-2$, let $p\in M$, and let B be a smooth complex vector subbundle of...

conjecture | open | MathDB 页面 | 原文线索

V2-069320 | MATHDB-359983

Wan-Xi's constant-equality conjecture for CM elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be a CM elliptic curve, let $r\neq 0$ be a fixed integer, let $C_{E,r}$ be the Lang-Trotter constant, and let $\overline{\omega}_{E,r}$ be the explicit constant...

conjecture | open | MathDB 页面 | 原文线索

V2-069321 | MATHDB-359984

Conjecture on the exponential decay rate of building-block straightforwardness parameters

摘录 / EXCERPT: Exponential-rate conjecture. There exists $c>0$ such that for all $k\geq 1$, the building block $\mathfrak{h}^{(k)}/2$ is straightforward with

conjecture | open | MathDB 页面 | 原文线索

V2-069322 | MATHDB-359986

Necessary and sufficient Kempe-chain condition for edge-critical non-4-colorable planar graphs

摘录 / EXCERPT: Necessary and sufficient condition. $M\in e\mathcal{MPGN}4$ if and only if there exists an edge $e=ab$ in M such that, for every $T_b\in\mathcal{BT}_{4b}$ and every extended...

conjecture | open | MathDB 页面 | 原文线索

V2-069323 | MATHDB-359990

Zuo's arithmetic Higgs bundle conjecture

摘录 / EXCERPT: Let K be a number field, let (Y,D_Y) be a log smooth pair over K, and let (E,θ) be a Higgs bundle over $(Y,D_Y)/K$. The bundle (E,θ) is arithmetic if there is a...

conjecture | open | MathDB 页面 | 原文线索

V2-069324 | MATHDB-359993

The $=5$ case of Vizing's conjecture for triangle-free graphs

摘录 / EXCERPT: The $=5$ case of Vizing's conjecture. Every 5-regular triangle-free graph has chromatic number at most 4.

conjecture | open | MathDB 页面 | 原文线索

V2-069325 | MATHDB-359994

The C_4-free analogue of Vizing's conjecture

摘录 / EXCERPT: The C_4-free analogue of Vizing's conjecture. It holds that

open_problem | open | MathDB 页面 | 原文线索

V2-069326 | MATHDB-360000

The type-\a0B transitive-partition enumeration conjecture

摘录 / EXCERPT: Let B_n be the Coxeter system of type B_n, and let $\Phi^+(B_n)$ denote its set of positive roots. A type-B_n transitive-partition enumeration conjecture. The number of maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-069327 | MATHDB-360005

Existence conjecture for timelike Liouville correlation functions

摘录 / EXCERPT: Let $c\geq 25$ and consider the inner product on conformal blocks ... where Z_{bc} is the ghost partition function and Z_{26-c} is the timelike Liouville correlation function wit...

conjecture | open | MathDB 页面 | 原文线索

V2-069328 | MATHDB-360006

Completeness conjecture for Liouville conformal blocks

摘录 / EXCERPT: Liouville completeness conjecture. Liouville conformal blocks form a complete basis with respect to the inner product above. This is the second mathematical gap in the rigorous def...

conjecture | open | MathDB 页面 | 原文线索

V2-069329 | MATHDB-360009

The conjectural critical mean-value bound for monomial Weyl sums

摘录 / EXCERPT: Critical mean-value conjecture. For every $p\geq d+1$ and every $\varepsilon>0$, there is a constant $C_{d,p,\varepsilon}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069330 | MATHDB-360022

Darmon's Stark-Heegner point conjecture

摘录 / EXCERPT: Let f be a weight-2 newform of level $\Gamma_0(p)$ with integer Fourier coefficients, let $E=E_f$ be the associated elliptic curve with multiplicative reduction at p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069331 | MATHDB-360025

Dinh-Sibony equidistribution conjecture for generic analytic subsets of projective space

摘录 / EXCERPT: Let $0\le s\le k$ be an integer. Let $f:\mathbb{P}^k\to\mathbb{P}^k$ be a holomorphic endomorphism of degree $d\geq 2$, and let T be its Green $(1,1)$-current. For an analytic su...

conjecture | open | MathDB 页面 | 原文线索

V2-069332 | MATHDB-360028

Torelli conjecture for polarized Hodge structures of opers

摘录 / EXCERPT: Torelli conjecture for opers. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069333 | MATHDB-360033

The KdV initial-condition formula for the non-perturbative descendent potential

摘录 / EXCERPT: Let $U({\bf p};\hbar)=\partial_{p_1}^2\log\bigl(\mathcal D^{\rm NP}_{\mu,\nu}(\hbar\cdot{\bf t}({\bf p});\hbar;w)\bigr)$ be the stated solution of the KdV equations. Let G_2,...

conjecture | open | MathDB 页面 | 原文线索

V2-069334 | MATHDB-360038

Density-Ramanujan conjecture for congruence complexes

摘录 / EXCERPT: Let G be a definite Gross inner form and let $\mathfrak{p}$ be a finite place. Let $\{X_{\mathfrak{p}}(q)\}$ denote its family of congruence complexes. Density-Ramanujan conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-069335 | MATHDB-360039

DSXDH-shape conjecture

摘录 / EXCERPT: Let G be a definite Gross inner form, let $\mathcal{M}(G)$ be the set of A-shapes, and let $h(G,\varsigma;q)$ be the counting function defined from discrete global A-paramete...

conjecture | open | MathDB 页面 | 原文线索

V2-069336 | MATHDB-360041

The universality conjecture for finite connected graph minor classes

摘录 / EXCERPT: Universality conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069337 | MATHDB-360052

Caprace-Kalantar-Monod conjecture on type I locally compact groups

摘录 / EXCERPT: Let G be a second countable locally compact group of type I, meaning that any two irreducible unitary representations of G that are weakly equivalent are unitarily equivalent....

conjecture | open | MathDB 页面 | 原文线索

V2-069338 | MATHDB-360056

Grünbaum's rank-two self-dual configuration conjecture

摘录 / EXCERPT: A self-dual geometric configuration is a geometric configuration equipped with a duality mapping its points and lines to one another; its rank is the order of the self-duality. Grü...

conjecture | open | MathDB 页面 | 原文线索

V2-069339 | MATHDB-360059

Shifted L_∞ master equation for spliced treed holomorphic discs

摘录 / EXCERPT: Let X^{00} be the relevant open space, let $\pi^*\mathcal{O}_{an}$ be the pulled-back analytic-function sheaf, and let $C^*(X^{00},\pi^*\mathcal{O}_{an})$ be its Morse complex. F...

conjecture | open | MathDB 页面 | 原文线索

V2-069340 | MATHDB-360060

The duplication lower-bound conjecture for Betti numbers of symbolic powers

摘录 / EXCERPT: Let G be a graph on m vertices and let $\alpha=(\alpha_1,\dots,\alpha_m)\in(\mathbb Z^+)^m$, where G^α is obtained by duplicating the vertices of G according to...

conjecture | open | MathDB 页面 | 原文线索

V2-069341 | MATHDB-360074

Equality and strict upper-bound conjecture for the estimates of direct powers of free distributive lattices

摘录 / EXCERPT: Let $g_3^{\ast}(n)$, $g_3^{\ast\ast}(n)$, and $g_r(n)$ be the functions defined earlier in the paper, and let $\mathbb{N}^{\geq k}=\{n\in\mathbb{N}:n\geq k\}$. Equality and strict...

conjecture | open | MathDB 页面 | 原文线索

V2-069342 | MATHDB-360081

The Cantor Julia set conjecture for dissipative hyperbolic automorphisms

摘录 / EXCERPT: Let $f$colon ${\mathbb C}^2\to{\mathbb C}^2$ be a dissipative and hyperbolic automorphism, meaning that $|\operatorname{Jac} f|<1$, and suppose that f has no attracting points. L...

conjecture | open | MathDB 页面 | 原文线索

V2-069343 | MATHDB-360082

The local product structure conjecture for the core of a quasi-solenoidal component

摘录 / EXCERPT: Core local product structure conjecture. The set $\operatorname{Core}(\Lambda)$ has local product structure near any regular point, and is locally the product of a Jordan arc by a...

conjecture | open | MathDB 页面 | 原文线索

V2-069344 | MATHDB-360083

The Almeida-Thouless conjecture for the replica-symmetric regime of the planted Sherrington-Kirkpatrick model

摘录 / EXCERPT: Let c denote the planted-clique proportion, let θ and θ_1 be the model parameters, and let q, μ, h, and z be the quantities appearing in the planted Sherr...

conjecture | open | MathDB 页面 | 原文线索

V2-069345 | MATHDB-360085

Polytopality conjecture for pure intervals of the s-weak order

摘录 / EXCERPT: Let s be a weak composition. For a pure interval $[T,T+A]$ of the s-weak order, consider the poset of pure intervals contained in it and the restriction of the s-weak order t...

conjecture | open | MathDB 页面 | 原文线索

V2-069346 | MATHDB-360086

Facet-removal conjecture for the s-permutahedron and s-associahedron

摘录 / EXCERPT: Let s be a weak composition with no zero entries except possibly $s(1)$. Removing facets of the s-permutahedron. There exists a geometric realization of the s-permutahedron s...

conjecture | open | MathDB 页面 | 原文线索

V2-069347 | MATHDB-360091

Open mirror conjecture for (W_5,mu_5)

摘录 / EXCERPT: Let $F^{\mathrm{LG}}_{0,1}(\tau)$ be the generating function of genus-zero open FJRW invariants of (W_5,μ_5), and let $\mathcal{T}^{\mathrm{LG}}(t)$ and $I^{\mathrm{LG}}_k(t)$...

conjecture | open | MathDB 页面 | 原文线索

V2-069348 | MATHDB-360092

Mirror conjecture for the extended GLSM in the positive phase

摘录 / EXCERPT: Consider the extended GLSM in the positive phase $\zeta>0$, with GLSM I-function $I^+_w(q,z)$ and GLSM small J-function $J^+_w(Q,z)$. Let $I^{\mathrm{CY}}_0(q,z)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069349 | MATHDB-360095

Semiampleness conjecture for the generalized Hodge line bundle on KSBA spaces

摘录 / EXCERPT: Semiampleness conjecture. The generalized Hodge line bundle λ is semiample.

conjecture | open | MathDB 页面 | 原文线索

V2-069350 | MATHDB-360099

Unbounded open embedding dimension for Steiner triple codes

摘录 / EXCERPT: Steiner triple code embedding-dimension conjecture. For every $d\ge 1$ there exists a Steiner triple code $\mathcal{C}$ with

conjecture | open | MathDB 页面 | 原文线索

V2-069351 | MATHDB-360100

Lubin's formal-group conjecture for commuting power series

摘录 / EXCERPT: Let K be a finite extension of $\mathbf{Q}_p$, with ring of integers $\mathcal{O}_K$ and maximal ideal $\mathfrak{m}_K$. Let $P,U\in T\cdot\mathcal{O}_K[[T]]$ satisfy ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-069352 | MATHDB-360104

Behrend's Gog-Magog pentagon conjecture

摘录 / EXCERPT: For $n\in\mathbb{N}$, let $(0,n,k,l)$-Magog and $(0,n,k,l)$-Gog pentagons denote the corresponding pentagonal objects. Behrend's conjecture. If $n\in\mathbb{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069353 | MATHDB-360105

Gog-GOGAm pentagon correspondence for the new Magog shape

摘录 / EXCERPT: For $n\in\mathbb{N}$, let $(0,n,k,l)$-Magog pentagons and $(0,n,k,l)$-GOGAm pentagons denote the corresponding objects. Gog-GOGAm pentagon conjecture. If $n\in\mathbb{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069354 | MATHDB-360107

Nonunital E-infinity structure conjecture for the first derivative spectrum

摘录 / EXCERPT: Let $F \colon \mathcal{J} \to \mathcal{S}$ be lax symmetric monoidal, and let Θ^1F denote its first derivative spectrum. First derivative multiplication conjecture. The spec...

conjecture | open | MathDB 页面 | 原文线索

V2-069355 | MATHDB-360115

The conjecture on obstructions to injectivity of the multiplier spectrum

摘录 / EXCERPT: Let $d\geq 2$, and let f,g be rational maps of degree d whose conjugacy classes are different. Write τ_d for the multiplier spectrum map. Two rational maps are equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-069356 | MATHDB-360116

Generic injectivity conjecture for the multiplier spectrum of periods one and two

摘录 / EXCERPT: For $d\geq 2$, let $\mathcal{M}_d(\mathbb{C})$ be the moduli space of rational maps of degree d, and let N_1 and N_2 denote the dimensions used by the multiplier-spectrum mor...

conjecture | open | MathDB 页面 | 原文线索

V2-069357 | MATHDB-360117

Length-spectrum versus multiplier-spectrum conjecture for non-Lattès maps

摘录 / EXCERPT: Let $d\geq 2$, and let f,g be non-Lattès rational maps of degree d. Let τ_d denote the multiplier spectrum map, and let $\overline{g}$ be the coefficientwise complex conju...

conjecture | open | MathDB 页面 | 原文线索

V2-069358 | MATHDB-360118

Generic finite convergence conjecture for generalized Moment-SOS hierarchies

摘录 / EXCERPT: Generic finite convergence conjecture. There exists a finite set of polynomials $\varphi_1,\ldots,\varphi_L$ in the coefficients of f, a, b, and c such that, if

conjecture | open | MathDB 页面 | 原文线索

V2-069359 | MATHDB-360121

Equitable 2-colourability conjecture for even cycle decompositions

摘录 / EXCERPT: Even-cycle equitable-colourability conjecture. There exists an equitably 2-colourable ℓ-cycle decomposition of K_v-I if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-069360 | MATHDB-360124

Riemannian gauge-completeness conjecture for anisotropic elasticity

摘录 / EXCERPT: Consider generic stiffness tensor fields c_i, density fields ρ_i, and Riemannian metrics g_i in the setting of the stated Riemannian anisotropic elasticity problem, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-069361 | MATHDB-360125

The gravitating coupled vortex conjecture for triples on the projective line

摘录 / EXCERPT: Gravitating coupled vortex conjecture. The pair $(\mathbb{P}^1,T)$ admits a solution to the gravitating coupled vortex equations if and only if T is σ-polystable and the p...

conjecture | open | MathDB 页面 | 原文线索

V2-069362 | MATHDB-360131

The equivalent-definition conjecture for dlt generalized pairs

摘录 / EXCERPT: Let $(X,B,\mathbf{M})/U$ be an lc g-pair. The terms log resolution, descent of the b-divisor, and lc center are understood in the usual sense. The dlt characterization conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069363 | MATHDB-360140

Mond conjecture for image Milnor and -codimension numbers

摘录 / EXCERPT: Mond conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069364 | MATHDB-360142

The positivity conjecture for asymptotic realisation constants

摘录 / EXCERPT: For each integer $d>1$, let α_d denote the corresponding asymptotic growth constant for d-realisation numbers. Positivity conjecture. For every $d>1$ we have ... The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-069365 | MATHDB-360143

The Bergman N-polynomial content conjecture for convex polygons

摘录 / EXCERPT: Let Ω be a planar region, let $\rho_N(\Omega)$ be the squared distance from $\bar z$ to the polynomials of degree at most N in $L^2(\Omega,dA)$, and call the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-069366 | MATHDB-360144

The no-maximizer conjecture for Bergman N-polynomial content

摘录 / EXCERPT: Let Ω be a simply connected Jordan region in the plane. For $N\in\mathbb N$, let $\rho_N(\Omega)$ denote the squared distance from $\bar z$ to the polynomials of degree at m...

conjecture | open | MathDB 页面 | 原文线索

V2-069367 | MATHDB-360145

The fixed-angle Bergman N-polynomial content conjecture for triangles

摘录 / EXCERPT: Let $N\in\mathbb N$ and let $\theta\in(0,\pi)$. Consider triangles of area 1 having a fixed interior angle θ, and let ρ_N denote their Bergman N-polynomial content...

conjecture | open | MathDB 页面 | 原文线索

V2-069368 | MATHDB-360154

Poncelet polygon density conjecture for pairs of conics

摘录 / EXCERPT: Poncelet polygon density conjecture. The density satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069369 | MATHDB-360159

Steinberg-map compatibility conjecture for symmetric-pair orbit embeddings

摘录 / EXCERPT: Assume an orbit embedding ... and Steinberg maps ... where $\iota^{\pm\theta}:\mathcal{N}_{\mathfrak{g}}^{\pm\theta}/K\to\mathcal{N}_{\mathfrak{G}}^{\pm\theta}/\mathbb{K}$ is the natur...

conjecture | open | MathDB 页面 | 原文线索

V2-069370 | MATHDB-360166

Langhede and Thomassen's exponential correspondence-colouring conjecture for planar graphs

摘录 / EXCERPT: Langhede and Thomassen's conjecture. Planar graphs have exponentially many 5-correspondence colourings.

conjecture | open | MathDB 页面 | 原文线索

V2-069371 | MATHDB-360167

Local girth correspondence-colourability conjecture for planar graphs

摘录 / EXCERPT: Local girth correspondence-colourability conjecture. If G is a planar graph and (L,M) is a correspondence assignment for G where L is a local girth list assignment, then...

conjecture | open | MathDB 页面 | 原文线索

V2-069372 | MATHDB-360174

Octad picture stable-reduction conjecture for plane quartics

摘录 / EXCERPT: Octad picture stable-reduction conjecture. The stable reduction type, including whether or not it is hyperelliptic, of any smooth plane quartic over any non-archimedean local field...

conjecture | open | MathDB 页面 | 原文线索

V2-069373 | MATHDB-360180

Arlandini-Loeffler twisted factorization conjecture for adjoint p-adic L-functions

摘录 / EXCERPT: Let ψ be a finite order Dirichlet character, let U be the weight domain of a Coleman family $\mathcal{F}$, let $\mathcal{W}$ be the cyclotomic weight space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069374 | MATHDB-360182

Validity conjecture for reciprocal-distance weight functions on augmented hypercubes

摘录 / EXCERPT: Validity conjecture. The weight function w_n is valid.

conjecture | open | MathDB 页面 | 原文线索

V2-069375 | MATHDB-360186

Silverman's height comparison conjecture for rational maps

摘录 / EXCERPT: For an individual rational function $f\in\operatorname{Rat}_d(k)$, let $h_{\operatorname{crit},\mathbb{Q}}$ be the critical height on the moduli space M_d, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069376 | MATHDB-360187

Orbifold Fano bound conjecture for locally soluble fibres

摘录 / EXCERPT: Orbifold Fano bound conjecture. If the $\mathbb{Q}$-divisor $-(K_{\mathbb{P}^1}+\partial_\pi)$ is ample, then, for every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-069377 | MATHDB-360188

Twisted-fibration asymptotic conjecture for locally soluble fibres

摘录 / EXCERPT: Let $\pi:X\to\mathbb{P}^1$ be a standard fibration, and let $\theta:\mathbb{P}^1\to(\mathbb{P}^1,\partial_\pi)$ be a finite étale orbifold morphism that is a G-torsor under a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-069378 | MATHDB-360198

Completeness conjectures for second and third Bernoulli polynomial derivatives

摘录 / EXCERPT: For $k\geq1$, define ... and let $\mathcal{S}_k$ be the corresponding computable subsets, with ... ... Second- and third-derivative completeness conjecture. One has ... The paper proves fi...

conjecture | open | MathDB 页面 | 原文线索

V2-069379 | MATHDB-360201

Weak Ax-Schanuel conjecture for the Gamma function

摘录 / EXCERPT: Let $f_1,\ldots,f_n$ be holomorphic functions satisfying, for every distinct $i,j\in\{1,\ldots,n\}$, $f_i\notin\mathbb{Q}f_j+\mathbb{C}$. Write $\mathbf{f}=(f_1,\ldots,f_n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069380 | MATHDB-360209

The ring-domain automatic concavity conjecture

摘录 / EXCERPT: Let $S^1=\{y\in\mathbb{R}^2\mid y_1^2+y_2^2=1\}$, and let $\mathfrak{B}^\circ_\delta[f]$ and $\mathfrak{B}^{\circ,*}_\delta[f]$ denote, respectively, the minimal locally concave an...

conjecture | open | MathDB 页面 | 原文线索

V2-069381 | MATHDB-360210

$C^{1,1}$ regularity conjecture for the minimal Bellman function

摘录 / EXCERPT: Let f be a smooth uniformly bounded function, and let $\mathbb{B}_\varepsilon$ denote the minimal function associated with f on the domain Ω_ε. $C^{1,1}$ regul...

conjecture | open | MathDB 页面 | 原文线索

V2-069382 | MATHDB-360211

$C^{1,1}$ regularity conjecture for the functions $\mathbb{U}$ and $\mathbb{V}$

摘录 / EXCERPT: Let f be a smooth uniformly bounded function, and let $\mathbb{U}$ and $\mathbb{V}$ be the two functions associated with f in the paper's Bellman-function formulation....

conjecture | open | MathDB 页面 | 原文线索

V2-069383 | MATHDB-360223

The conjecture that natural-language-trained large language models cannot reliably reason at professional mathematics

摘录 / EXCERPT: Large language models (LLMs) are systems trained to generate text from natural-language data. Natural-language reasoning conjecture. Large language models (LLMs) trained on natural...

conjecture | open | MathDB 页面 | 原文线索

V2-069384 | MATHDB-360224

The conjecture that AI-assisted mathematical research requires an adversarial theorem-proving process

摘录 / EXCERPT: An adversarial process is a process in which AI-generated mathematical arguments are checked by an independent formal mechanism, such as an interactive theorem prover. Interactive...

conjecture | open | MathDB 页面 | 原文线索

V2-069385 | MATHDB-360225

The conjecture that formalizing more professional mathematics is the rate-limiting step for AI proof assistance

摘录 / EXCERPT: AI proof assistance refers to AI systems that help mathematicians develop and verify formal proofs, potentially by interacting with an interactive theorem prover. Formalization bot...

conjecture | open | MathDB 页面 | 原文线索

V2-069386 | MATHDB-360226

Higher-depth positivity conjecture for Kac polynomials

摘录 / EXCERPT: Let Q be a quiver, let $\mathbf{r}\in\mathbb{Z}_{\geq0}^{Q_0}$ be a rank vector, and let $\alpha\geq1$. Higher-depth positivity conjecture. ... The conjecture extends positivity fr...

conjecture | open | MathDB 页面 | 原文线索

V2-069387 | MATHDB-360233

The signed-tiling conjecture for contributions of regions in the double dimer model

摘录 / EXCERPT: Signed-tiling conjecture. If $n=3$ or $n=6$, then the contribution of L is 0 unless there exists a signed tiling of L by stones, bones, and snakes; if such a signed tiling ex...

conjecture | open | MathDB 页面 | 原文线索

V2-069388 | MATHDB-360236

Parametrized minimal-resolution conjecture for Kunz faces

摘录 / EXCERPT: Let C_m be the Kunz cone, let F be a face of C_m, and let S range over the numerical semigroups whose corresponding points lie in the interior of F. For each such S, le...

conjecture | open | MathDB 页面 | 原文线索

V2-069389 | MATHDB-360237

Face-dependent artinian reduction conjecture for semigroup resolutions

摘录 / EXCERPT: Let C_m be the Kunz cone, let F be a face of C_m, and let S be a numerical semigroup whose corresponding point lies in F. Let J_S be the ideal in the Apéry-resolution c...

conjecture | open | MathDB 页面 | 原文线索

V2-069390 | MATHDB-360247

The bounded twin-width conjecture for tournaments

摘录 / EXCERPT: Let $k\geq 1$. A tournament has twin-width at most k if its twin-width is at most k. The bounded twin-width conjecture. The class of tournaments with twin-width at most k is...

conjecture | open | MathDB 页面 | 原文线索

V2-069391 | MATHDB-360248

The good ordering conjecture for tournament twin-width and clique number

摘录 / EXCERPT: For a tournament T and an ordering $\prec$ of $V(T)$, let $T^{\prec}$ be its backedge graph. A BST-ordering is the ordering associated with a binary search tree satisfying the to...

conjecture | open | MathDB 页面 | 原文线索

V2-069392 | MATHDB-360250

The BIG implies BIG conjecture for tournaments

摘录 / EXCERPT: A class of tournaments $\mathcal T$ has the $BIG\Rightarrow BIG$ property if there exists a function f such that, for every $T\in\mathcal T$, if...

conjecture | open | MathDB 页面 | 原文线索

V2-069393 | MATHDB-360251

The domination-to-clique cluster conjecture

摘录 / EXCERPT: For a tournament T, let $\operatorname{dom}(T)$ be its domination number and let $\operatorname{\overrightarrow{\omega}}(T)$ be its clique number. The domination-to-clique cluste...

conjecture | open | MathDB 页面 | 原文线索

V2-069394 | MATHDB-360252

The poset tournament rebel conjecture

摘录 / EXCERPT: A tournament R is a rebel if the class of tournaments not containing R has bounded domination number. A poset tournament is a tournament admitting an ordering whose backedge gr...

conjecture | open | MathDB 页面 | 原文线索

V2-069395 | MATHDB-360253

The local-to-global conjecture for tournament clique number

摘录 / EXCERPT: For a tournament T, let $N^+(v)$ denote the out-neighbourhood of v, and let $\operatorname{\overrightarrow{\omega}}(T)$ denote the clique number. The local-to-global clique-num...

conjecture | open | MathDB 页面 | 原文线索

V2-069396 | MATHDB-360254

The infinite critical tournaments conjecture

摘录 / EXCERPT: A tournament T is k-clique-critical if $\operatorname{\overrightarrow{\omega}}(T)=k$ and $\operatorname{\overrightarrow{\omega}}(T-v)=k-1$ for every $v\in V(T)$. The infinite c...

conjecture | open | MathDB 页面 | 原文线索

V2-069397 | MATHDB-360259

The restricted-product conjecture for quasi-endomorphisms of localized integers

摘录 / EXCERPT: Restricted-product conjecture. The ring of quasi-endomorphisms satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069398 | MATHDB-360265

Weak Lefschetz conjecture for balanced minimal cycle complexes

摘录 / EXCERPT: Let $\bm{a}=(a_1,\ldots,a_m)\in\mathbb{Z}_{>0}^m$ with $d=\sum_{i=1}^m a_i\geq 3$ and $\bm{a}\neq(d-1,1),(1,d-1)$. An $(d-1)$-dimensional simplicial complex Δ is $\bm{a}$-ba...

conjecture | open | MathDB 页面 | 原文线索

V2-069399 | MATHDB-360268

Within-class collapsing conjecture for neural features

摘录 / EXCERPT: Within-class collapsing. All features within each class are identical: for each $k=1,\ldots,K$, $\boldsymbol{h}_{k,i}^*\equiv\boldsymbol{h}_k^*$ for every i.

conjecture | open | MathDB 页面 | 原文线索

V2-069400 | MATHDB-360269

Self-duality conjecture for neural-collapse solutions

摘录 / EXCERPT: Self-duality conjecture. The class feature aggregate is proportional to its corresponding classifier weight: for each $k=1,\ldots,K$, $\boldsymbol{h}_k^*=\alpha\boldsymbol{w}_k^*$...

conjecture | open | MathDB 页面 | 原文线索

V2-069401 | MATHDB-360270

Weight-feature equality conjecture at full-rank stationary points

摘录 / EXCERPT: Weight-feature equality conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-069402 | MATHDB-360272

Gao-Han-Peng-Sun conjecture for Erdős-Ginzburg-Ziv constants

摘录 / EXCERPT: Let G be a finite abelian group, let $\beta=\exp(G)$ denote its exponent, and let $k\in\mathbb N$. The invariant $\mathsf{s}_{k\beta}(G)$ is the smallest length guaranteeing a ze...

conjecture | open | MathDB 页面 | 原文线索

V2-069403 | MATHDB-360278

The ALUT non-unknowability conjecture

摘录 / EXCERPT: ALUT non-unknowability conjecture. For all $\varphi\in\textbf{ALUT}$ and all $i\in\textbf{I}$,

conjecture | open | MathDB 页面 | 原文线索

V2-069404 | MATHDB-360302

Hu-Li diagonalization conjecture for reduced Gromov-Witten invariants of Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, let β be a curve class with degree ${\rm deg}(\beta)>2g-2$, and let N_β^g and r_β^h denote the genus-g standard and genus-h...

conjecture | open | MathDB 页面 | 原文线索

V2-069405 | MATHDB-360303

Zinger's comparison conjecture for Fano threefold Gromov-Witten invariants

摘录 / EXCERPT: Zinger's comparison conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069406 | MATHDB-360304

Lower-genus boundary contribution conjecture for desingularized moduli spaces

摘录 / EXCERPT: Let X be a threefold that is a complete intersection in projective space, and let $[\widetilde{\mathcal M}_{g,n}(\mathbb{P}^r,d)^p]_i^{\mathrm{vir}}$ denote the i-th component...

conjecture | open | MathDB 页面 | 原文线索

V2-069407 | MATHDB-360306

Transcendence conjecture for power towers of ultra-Liouville numbers

摘录 / EXCERPT: Let θ be the fixed point of $x\mapsto \xi^x$ and define the iterated power towers by $h_1(\xi)=\xi$ and $h_{k+1}(\xi)=\xi^{h_k(\xi)}$ for $k\geq 1$. An ultra-Liouville numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-069408 | MATHDB-360308

The consequence of the extended Lang-Vojta conjecture for pointed maps

摘录 / EXCERPT: Consequence of the extended Lang-Vojta conjecture. There is a proper closed subset $\Delta\subsetneq U$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-069409 | MATHDB-360314

The geometric Breuil-Mézard conjecture for general linear groups

摘录 / EXCERPT: Let $G=\operatorname{GL}_n$. For each Serre weight σ of $G(\mathbf{F}_p)$, let $\mathcal{Z}(\sigma)$ be a cycle in the d-dimensional cycle group of the reduced Emerton-Gee...

conjecture | open | MathDB 页面 | 原文线索

V2-069410 | MATHDB-360338

The average-size conjecture for Selmer groups of elliptic curves

摘录 / EXCERPT: Average-size conjecture. The average size of the n-Selmer groups is

conjecture | open | MathDB 页面 | 原文线索

V2-069411 | MATHDB-360340

Tetrahedron-instanton construction with D6-branes

摘录 / EXCERPT: Tetrahedron-instanton BPS/CFT conjecture. After suitable conditions, the deformation parameters can be reparametrized to at most three independent parameters. The partition functio...

conjecture | open | MathDB 页面 | 原文线索

V2-069412 | MATHDB-360356

Nonnegativity conjecture for the curve invariant epsilon

摘录 / EXCERPT: Let C_1 and C_2 be curves without common irreducible components, and let $q\in C_1\cap C_2$ be an intersection point. Nonnegativity conjecture for ϵ. One has ... This co...

conjecture | open | MathDB 页面 | 原文线索

V2-069413 | MATHDB-360357

Symmetric Tjurina-number inequality for unions of plane curves

摘录 / EXCERPT: Let C_1 and C_2 be curves without common irreducible components, and let $q\in C_1\cap C_2$ be an intersection point. Write $\tau(C_i,q)$ for the Tjurina number of (C_i,q),...

conjecture | open | MathDB 页面 | 原文线索

V2-069414 | MATHDB-360361

Embedding conjecture for weak double categories and double bicategories

摘录 / EXCERPT: Embedding conjecture. There is a fully faithful embedding of weak double categories into Verity's double bicategories with essential image given by double bicategories in which the...

conjecture | open | MathDB 页面 | 原文线索

V2-069415 | MATHDB-360362

Enriched profunctor example for univalent weak double categories

摘录 / EXCERPT: Enriched profunctor conjecture. (Enriched) categories, (enriched) functors and (enriched) profunctors assemble into a univalent weak double category, but not into a univalent doubl...

conjecture | open | MathDB 页面 | 原文线索

V2-069416 | MATHDB-360363

Univalence conjecture for the tricategory of univalent weak double categories

摘录 / EXCERPT: Tricategorical univalence conjecture. The tricategory of univalent weak double categories is univalent.

conjecture | open | MathDB 页面 | 原文线索

V2-069417 | MATHDB-360376

Distance-3 precoloring extension conjecture for graph products with an edge

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta(G)$, and let K_2 be the complete graph on two vertices. Consider the Cartesian product $G\square K_2$. A precoloring is extendable...

conjecture | open | MathDB 页面 | 原文线索

V2-069418 | MATHDB-360396

Llosa Isenrich-Pallier-Tessera conjecture on Dehn functions of central products

摘录 / EXCERPT: Let K and L be simply connected nilpotent Lie groups with one-dimensional centres, of nilpotency classes k and ℓ, respectively, and let their central product be ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-069419 | MATHDB-360398

The neutral-hypermultiplet conjecture for frozen F-theory compactifications

摘录 / EXCERPT: Consider a six-dimensional F-theory compactification with a frozen 7-brane localized along a divisor D and residual discriminant $\widetilde{\Delta}$. Neutral-hypermultiplet co...

conjecture | open | MathDB 页面 | 原文线索

V2-069420 | MATHDB-360401

Zeng's deflation conjecture for ultrasingular irreducible components

摘录 / EXCERPT: Zeng's deflation conjecture. The recursive deflation process terminates after finitely many steps, the deflation sequences of all points $\textbf{x}\in U$ are equal and regular, an...

conjecture | open | MathDB 页面 | 原文线索

V2-069421 | MATHDB-360405

The multiplicativity conjecture for reduced Weierstrass isomorphism classes

摘录 / EXCERPT: Let $n=p_1^{e_1}p_2^{e_2}\ldots p_l^{e_l}$ be an odd composite integer with $e_i\geq 1$, and let $C_R(\mathbb{Z}_n)$ denote the number of isomorphism classes of reduced Weierstrass...

conjecture | open | MathDB 页面 | 原文线索

V2-069422 | MATHDB-360406

The generalized Weierstrass curve-count conjecture over integer residue rings

摘录 / EXCERPT: Generalized Weierstrass counting conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069423 | MATHDB-360407

The classification problem for generalized Weierstrass elliptic curves over integer residue rings

摘录 / EXCERPT: Let n be a positive integer, and let $C_G(\mathbb{Z}_n)$ denote the number of isomorphism classes of generalized Weierstrass elliptic curves over $\mathbb{Z}_n$. Generalized Weie...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-069424 | MATHDB-360408

The Kuga--Satake Hodge conjecture

摘录 / EXCERPT: Kuga--Satake Hodge conjecture. The embedding ι_V is induced by a correspondence in $A^\ast(X\times KS(V)\times KS(V))$. This is a special case of the Hodge conjecture; the pa...

conjecture | open | MathDB 页面 | 原文线索

V2-069425 | MATHDB-360413

Resurgent Stokes-coefficient integrality conjecture

摘录 / EXCERPT: Let α and β be critical points, let $\tau(\alpha)$ and $\tau(\beta)$ denote their corresponding torsions, and let $\mathcal{S}_{\alpha}^{\beta}$ be the Stokes coeffici...

conjecture | open | MathDB 页面 | 原文线索

V2-069426 | MATHDB-360417

Le Mogne-Préville-Ratelle's symmetry and Schur-positivity conjecture for ν-Tamari intervals

摘录 / EXCERPT: Let λ be a triangular partition, and let its top-down tableau be the Young tableau associated with a maximal chain in the ν-Tamari lattice. Say that this tableau is sim...

conjecture | open | MathDB 页面 | 原文线索

V2-069427 | MATHDB-360418

Polytopal-subdivision realization conjecture for s-permutahedra

摘录 / EXCERPT: For each sequence s, let the s-permutahedron be the generalized permutahedral object defined in the source. A polytopal subdivision is a subdivision whose cells are polytopes....

conjecture | open | MathDB 页面 | 原文线索

V2-069428 | MATHDB-360419

Clark-Ismail positivity conjecture for the functions f_j and g_j

摘录 / EXCERPT: Clark-Ismail's positivity conjecture. The functions $f_j(v)$, or equivalently $g_j(v)$, are positive on $(0,\infty)$ for all $j>1$. This conjecture concerns the positivity needed t...

conjecture | open | MathDB 页面 | 原文线索

V2-069429 | MATHDB-360420

Clark-Ismail coefficient-positivity conjecture for the power series of g_j

摘录 / EXCERPT: Clark-Ismail's coefficient-positivity conjecture. Then $\gamma(j,k)\ge0$ for all $j>1$ and $k\ge0$. This is a coefficientwise strengthening of the positivity question for g_j and...

conjecture | open | MathDB 页面 | 原文线索

V2-069430 | MATHDB-360429

Spectral fourth moment conjecture for weight 1 Maass forms

摘录 / EXCERPT: Let N be the level, and let $\{\mu_j^1\}_j$ be an orthonormal basis of Maass forms of weight 1 on $\Gamma_0(N)$, with spectral parameters t_j and Fourier coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-069431 | MATHDB-360435

The asymptotic proportionality conjecture for Collatz counts

摘录 / EXCERPT: Let $T(n)$ denote the paper's count of Collatz integers in the interval $[2^n+1,2^{n+1}]$, and let $\delta(n)$ be the formula defined in the preceding discussion for the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-069432 | MATHDB-360449

The A-Philosophy dynamic Riemann Hypothesis

摘录 / EXCERPT: For $N\in\mathbb{N}$, let $\mathcal{Z}_N$ be the parameter space of functions ... with $\overline{a}\in\mathbb{R}^N$. A path $\gamma(r)$ in this space starts at the core function...

conjecture | open | MathDB 页面 | 原文线索

V2-069433 | MATHDB-360450

The corrected Gram-law conjecture for isolated bad Gram points

摘录 / EXCERPT: Let g_n be an isolated bad Gram point, meaning the middle point of a Gram block whose adjacent boundary Gram points g_{n-1} and g_{n+1} are good. In the two-dimensional param...

conjecture | open | MathDB 页面 | 原文线索

V2-069434 | MATHDB-360451

The RH energy-bound conjecture for isolated Gram points

摘录 / EXCERPT: Let g_n be an isolated Gram point, and let $\gamma^{\mathrm{descend}}$ denote the descending curve in the two-dimensional deformation associated with the shifting and descending...

conjecture | open | MathDB 页面 | 原文线索

V2-069435 | MATHDB-360452

The signed four-cycle detection conjecture for L_q random geometric graphs

摘录 / EXCERPT: Let n,d be positive parameters, let $q\in[1,\infty]$, and consider testing ... against ... Here the signed four-cycle test is the signed subgraph-count test based on the four-cycle...

conjecture | open | MathDB 页面 | 原文线索

V2-069436 | MATHDB-360453

The signed triangle detection conjecture for L_q random geometric graphs

摘录 / EXCERPT: Let n,d be positive parameters, let $q\in[1,\infty]$, and consider testing ... against ... Assume $dq^3=\omega(n^3)$. The signed triangle test is the signed subgraph-count test based...

conjecture | open | MathDB 页面 | 原文线索

V2-069437 | MATHDB-360458

The second q-analogue of Liu's Dwork congruence

摘录 / EXCERPT: Let m and n be positive integers with $n\equiv1\;\allowbreak(\operatorname{mod}4)$ and $n>1$, and let $(a;q)_k$ denote the q-shifted factorial. The q-analogue conjecture. ... T...

conjecture | open | MathDB 页面 | 原文线索

V2-069438 | MATHDB-360459

The strengthening of Lemma 3

摘录 / EXCERPT: Let n and s be the parameters occurring in Lemma 3, and let $\Phi_{n^s}(q)$ be the corresponding cyclotomic polynomial. The strengthening conjecture. The q-congruence in Lemma...

conjecture | open | MathDB 页面 | 原文线索

V2-069439 | MATHDB-360460

A factorial-ratio q-congruence for powers of an odd integer

摘录 / EXCERPT: Let $n\equiv1\;\allowbreak(\operatorname{mod}4)$ be an integer greater than 1, and let r,s be positive integers with $r>s$. For $(a;q)_k$ the q-shifted factorial, the factorial...

conjecture | open | MathDB 页面 | 原文线索

V2-069440 | MATHDB-360461

A fifth-order q-supercongruence

摘录 / EXCERPT: Let m and n be positive integers with $n\equiv1\;\allowbreak(\operatorname{mod}4)$, and let $(a;q)_k$ denote the q-shifted factorial. The fifth-order q-supercongruence conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-069441 | MATHDB-360462

A cubic q-supercongruence with shifted factors

摘录 / EXCERPT: Let m and n be positive integers with n odd, and let $(a;q)_k$ denote the q-shifted factorial. The cubic q-supercongruence conjecture. ... The $m=1$ case was proved by the auth...

conjecture | open | MathDB 页面 | 原文线索

V2-069442 | MATHDB-360463

A cubic q-supercongruence with $(q;q^2)_k$ factors

摘录 / EXCERPT: Let m and n be positive integers with n odd, and let $(a;q)_k$ denote the q-shifted factorial. The cubic q-supercongruence conjecture. ... The source records this as another si...

conjecture | open | MathDB 页面 | 原文线索

V2-069443 | MATHDB-360479

Classification conjecture for strongly-neutralizable threshold functions

摘录 / EXCERPT: A threshold function is a map $f:\Sigma^N\to\Sigma$, where Σ is the binary state space. A function is strongly neutralizable when it has the property that its associated Isi...

conjecture | open | MathDB 页面 | 原文线索

V2-069444 | MATHDB-360480

Engbers-Erey-Fox-He conjecture for chromatic polynomials of l-connected graphs

摘录 / EXCERPT: Let G be a k-chromatic l-connected graph on n vertices, where $k\geq 4$ and $l\geq 3$, and let $P(G,x)$ denote its chromatic polynomial. Engbers-Erey-Fox-He conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069445 | MATHDB-360484

The mock plectic invariant rank-distribution conjecture

摘录 / EXCERPT: Let K be a quadratic imaginary field, let $E/\mathbb Q$ be the elliptic curve in the construction, let E' be its quadratic twist attached to K, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069446 | MATHDB-360491

Strong multiplicity one conjecture for self-dual cuspidal automorphic representations

摘录 / EXCERPT: Let π and π' be self-dual unitary cuspidal automorphic representations of $\operatorname{GL}_m(\mathbb A)$ and $\operatorname{GL}_{m'}(\mathbb A)$, respectively. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069447 | MATHDB-360492

Selberg's orthogonality conjecture for self-dual cuspidal automorphic representations

摘录 / EXCERPT: Let π and π' be self-dual unitary cuspidal automorphic representations of $\operatorname{GL}_m(\mathbb A)$ and $\operatorname{GL}_{m'}(\mathbb A)$, respectively, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069448 | MATHDB-360500

The injective map expansion conjecture for Jack polynomials

摘录 / EXCERPT: Injective map expansion conjecture. There exists a statistic of non-orientability ϑ on $\operatorname{\mathcal{ID}}_n(\lambda)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069449 | MATHDB-360501

Hanlon's permutation-pair expansion conjecture for Jack polynomials

摘录 / EXCERPT: Hanlon's conjecture. There exists a statistic w on the orientable maps in $\operatorname{\mathcal{ID}^O}_n(\lambda)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069450 | MATHDB-360502

The degree bound for generators involving the second variable in Jacobian generic initial ideals

摘录 / EXCERPT: Let $\mathcal{A}$ be a central arrangement in $\mathbb{K}^{n+1}$. Write $\operatorname{rgin}(J(\mathcal{A}))$ for the generic initial ideal of its Jacobian ideal, and set ... A minim...

conjecture | open | MathDB 页面 | 原文线索

V2-069451 | MATHDB-360503

Choosability with weak diameter for bounded layered tree-width

摘录 / EXCERPT: Layered-tree-width weak-diameter conjecture. For every positive integer w, there exists a positive integer N such that every graph with layered tree-width at most w is 3-choo...

conjecture | open | MathDB 页面 | 原文线索

V2-069452 | MATHDB-360516

Relative descendent Gromov-Witten/Pandharipande-Thomas correspondence

摘录 / EXCERPT: Relative descendent GW/PT correspondence.

conjecture | open | MathDB 页面 | 原文线索

V2-069453 | MATHDB-360521

Havet-Thomassé conjecture for trees with few leaves

摘录 / EXCERPT: Let T be an oriented tree with k edges and ll leaves, and let a tournament be an orientation of a complete graph. Havet-Thomassé's conjecture. Every tournament on $k+\ell$ ve...

conjecture | open | MathDB 页面 | 原文线索

V2-069454 | MATHDB-360524

Stein's minimum-semidegree conjecture for oriented paths

摘录 / EXCERPT: Let $k\in\mathbb N$, and let $\delta^0(D)$ denote the minimum semidegree of an oriented graph D. Stein's conjecture. Every oriented graph with $\delta^0(D)>k/2$ contains every or...

conjecture | open | MathDB 页面 | 原文线索

V2-069455 | MATHDB-360525

Minimum-semidegree conjecture for balanced antidirected trees

摘录 / EXCERPT: Let D be an oriented graph on n vertices, let $\delta^0(D)$ denote its minimum semidegree, and let a balanced antidirected tree have equally many sources and sinks. The balance...

conjecture | open | MathDB 页面 | 原文线索

V2-069456 | MATHDB-360527

Minimum-pseudo-semidegree conjecture for balanced antidirected trees

摘录 / EXCERPT: Let D be an oriented graph on n vertices, let $\bar\delta^0(D)$ denote its minimum pseudo-semidegree, and let a balanced antidirected tree have equally many sources and sinks....

conjecture | open | MathDB 页面 | 原文线索

V2-069457 | MATHDB-360529

Gaetz-Pierson's real-rootedness conjecture for identity-pattern character polynomials

摘录 / EXCERPT: Let $k\geq 0$ and let λ be a partition. The polynomial $a_{\mathrm{id}_k}^\lambda(n)$ is the character coefficient polynomial associated with the identity pattern of length...

conjecture | open | MathDB 页面 | 原文线索

V2-069458 | MATHDB-360530

Nonnegative-coefficients conjecture for chain-type polynomials

摘录 / EXCERPT: Let τ be a chain type, and let $P^\tau\in\mathbb{Z}[z,w]$ be the polynomial associated with τ in the paper's recursive construction. Nonnegative-coefficients conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069459 | MATHDB-360549

The perverse-weight correspondence conjecture for compactified Picards

摘录 / EXCERPT: Let C be a complex integral projective curve whose normalization has genus zero and whose unique planar singularity is locally given by $f(x,y)=0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069460 | MATHDB-360568

Allcock's Bimonster covering conjecture

摘录 / EXCERPT: A 13-dimensional complex ball quotient is obtained from the Allcock lattice construction, with X denoting the corresponding ball quotient and $X^{\rm sg}$ its singular or special...

conjecture | open | MathDB 页面 | 原文线索

V2-069461 | MATHDB-360570

Anti-van der Waerden conjecture for products with peripheral trees

摘录 / EXCERPT: Peripheral-tree product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069462 | MATHDB-360573

Stanley's symmetry characterization conjecture for Schur labeled skew shape posets

摘录 / EXCERPT: Stanley's symmetry characterization conjecture. If K_P is symmetric, then $P\in\sfSP_n$.

conjecture | open | MathDB 页面 | 原文线索

V2-069463 | MATHDB-360578

Effective finiteness conjecture for good Atkin-Lehner quotients

摘录 / EXCERPT: Effective finiteness conjecture. The set of ideals ${\mathfrak N}$ for which $Q_0({\mathfrak N})_{\mathfrak a}$ is good is an effectively computable finite set.

conjecture | open | MathDB 页面 | 原文线索

V2-069464 | MATHDB-360597

Elliptic-reflection functor conjecture for orbifold surfaces

摘录 / EXCERPT: Let $[E/\Gamma]$ be the orbifold curve, let Ξ denote the deformation data, and let IW_R^{ell} act on $HH_0(E)$. Elliptic-reflection functor conjecture. There is a group G a...

conjecture | open | MathDB 页面 | 原文线索

V2-069465 | MATHDB-360600

Blokh-Oversteegen-Ptacek-Timorin conjecture on the cubic main hyperbolic component

摘录 / EXCERPT: Let $\theta\in\mathbb{R}/\mathbb{Z}$. Let $\mathbf{P}^3$ be the space of conformal conjugacy classes of cubic polynomials, let $\mathbf{H}^3\subset\mathbf{P}^3$ be the hyperbolic c...

conjecture | open | MathDB 页面 | 原文线索

V2-069466 | MATHDB-360605

Shokurov's conjecture on generalized singularities in adjunction for Fano-type fibrations

摘录 / EXCERPT: Shokurov's conjecture. Fix a positive integer r and a real number $0<\epsilon\leq 1$. There exists $\delta>0$ depending only on r,ϵ such that, if $\dim X-\dim Z=r$,...

conjecture | open | MathDB 页面 | 原文线索

V2-069467 | MATHDB-360624

Monotonicity of relative convexity among binary Rényi entropies

摘录 / EXCERPT: Rényi relative-convexity conjecture. If $\alpha\geq\beta\geq 0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069468 | MATHDB-360629

Linear resolution conjecture for higher-dimensional path complexes

摘录 / EXCERPT: Let L_n be the path graph on vertices $\{x_1,\ldots,x_n\}$, and let $\Delta_k^t(L_n)$ denote the associated complex. The preceding result establishes the claim when $n=2k-1$. Lin...

conjecture | open | MathDB 页面 | 原文线索

V2-069469 | MATHDB-360630

Cohen-Macaulay linear-resolution conjecture for complexes from squared paths

摘录 / EXCERPT: Let L_n^2 be the graph on $[1,n]$ with edges $\{i,i+1\}$ for $i=1,\ldots,n-1$ and $\{i,i+2\}$ for $i=1,\ldots,n-2$. Let $\Delta_k^t(L_n^2)$ be the associated complex, and let its...

conjecture | open | MathDB 页面 | 原文线索

V2-069470 | MATHDB-360632

The Monasson et al. mixed 2- and 3-SAT threshold conjecture

摘录 / EXCERPT: Monasson et al.'s mixed-SAT conjecture. For every $p\in[0,1]$ there exists a value $\alpha_\mathrm{sat}(2+p)>0$ such that, whenever $\alpha_2,\alpha_3>0$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-069471 | MATHDB-360644

Conjecture on flux-ratio behavior at bifurcation in ionic flows

摘录 / EXCERPT: Assume $k=1$ and consider the bifurcation moment where the flux ratio $\lambda_1=1$. Let $r_0>0$ be a fixed boundary condition, and let $(I^*,\mathcal{A}^*,V^*,\mathcal{I}^*)$ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-069472 | MATHDB-360645

Conjecture on boundary-concentration effects at bifurcation in ionic flows

摘录 / EXCERPT: Consider the bifurcation moment where the flux ratio $\lambda_1=1$ in the system under study. Let $r_0>0$ be fixed, let $(I^*,\mathcal{A}^*,V^*,\mathcal{I}^*)$ be a critical soluti...

conjecture | open | MathDB 页面 | 原文线索

V2-069473 | MATHDB-360649

Mock plectic invariant rank conjecture for elliptic curves over ring class fields

摘录 / EXCERPT: Let E be an elliptic curve, let K be an imaginary quadratic field, let H_c be the relevant ring class field, and let $\chi\colon\Delta_c\to\mathbb{C}^\times$ be a primitive c...

conjecture | open | MathDB 页面 | 原文线索

V2-069474 | MATHDB-360650

Regulator image conjecture for mock plectic invariants

摘录 / EXCERPT: Regulator image conjecture. If $L(E/K,\chi,1)=0$, then $\mathcal{Q}^\chi$ belongs to the image of this regulator. This predicts a direct relation between the mock plectic invariant...

conjecture | open | MathDB 页面 | 原文线索

V2-069475 | MATHDB-360661

Holomorphic extension conjecture for the stability-condition embedding

摘录 / EXCERPT: Let $\mathcal{MC}(m)^{\Gamma}_{\ast,s}$ be the open subspace of framed cubic differentials whose zeros are distinct. Let $\mathcal D$ be the associated CY_3 category, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069476 | MATHDB-360662

The local weight-monodromy conjecture for the mixed central functor

摘录 / EXCERPT: Let V be a representation of the Langlands dual group, and let $\scrZ^{\mathrm{mix}}(V)$ be the associated mixed perverse sheaf obtained from the mixed central functor. Its monod...

conjecture | open | MathDB 页面 | 原文线索

V2-069477 | MATHDB-360663

The generalized McKay-Wanless conjecture for Latin rectangles

摘录 / EXCERPT: For $k\leq n$, let $\mathcal{LR}_{k,n}$ be the set of $k\times n$ Latin rectangles with symbol set $[n]$, and let $\randLR$ be chosen uniformly from this set. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-069478 | MATHDB-360665

Almost-sure convergence conjecture for non-leaf-free multicyclic subgraph counts

摘录 / EXCERPT: Fix a connected multicyclic graph H with 2-core H', and let t be the number of vertices in $H\setminus H'$. For a fixed $m\geqslant m_H$, let W_n and W'_n denote the nu...

conjecture | open | MathDB 页面 | 原文线索

V2-069479 | MATHDB-360678

Doubly layered maximizer conjecture at the principal specialization q = -1

摘录 / EXCERPT: Doubly layered maximizer conjecture. For every n, all permutations reaching the maximum $\widetilde{u}(n)$ are doubly layered permutations. Thus, there is a limit

conjecture | open | MathDB 页面 | 原文线索

V2-069480 | MATHDB-360679

k-multi-layered maximizer conjecture at a root-of-unity specialization

摘录 / EXCERPT: Fix $k\in\mathbb{Z}_{>0}$, let ζ be a k-th root of unity, and let Φ_w^k denote the principal specialization $\mathfrak{S}_w(1,q,q^2,\ldots)$ at $q=\zeta$. Define ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-069481 | MATHDB-360683

Strict monotonicity conjecture for the growth rate under time-independent migration

摘录 / EXCERPT: Let $\Lambda(m,T)$ denote the growth rate for migration intensity m and period parameter T, and suppose that the migration matrix is time independent. Let $r_i(\tau)$ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-069482 | MATHDB-360697

The suitable-order conjecture for root systems

摘录 / EXCERPT: Let Φ be a root system. An ordering of its positive roots is called suitable when it has the structural properties required in the paper to obtain the extension and maximal-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-069483 | MATHDB-360698

Real-part sampling conjecture for Benjamin-Ono bulk eigenvalue branches

摘录 / EXCERPT: Let S_U and S_L denote the upper and lower branches of the bulk eigenvalues, with real parts indexed increasingly by $\mu_{U,k}(t)$ and $\mu_{L,k}(t)$, and let t_b be the bre...

conjecture | open | MathDB 页面 | 原文线索

V2-069484 | MATHDB-360699

Lower-branch imaginary-part sampling conjecture for Benjamin-Ono eigenvalues

摘录 / EXCERPT: Let $t>t_b$ and let $\nu_{L,k}(t)$ be the imaginary parts of the lower-branch eigenvalues. Lower-branch imaginary-part conjecture. There is an ϵ-independent function...

conjecture | open | MathDB 页面 | 原文线索

V2-069485 | MATHDB-360700

Benjamin-Ono soliton-ensemble connection with Whitham modulation theory

摘录 / EXCERPT: Let $\psi_U(x,t)$, $\psi_L(x,t)$, and $\phi_L(x,t)$ be the modulation functions defined from the scaling functions of the upper- and lower-branch eigenvalues, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069486 | MATHDB-360705

The chromatic-number conjecture for irredundance compelling colorings

摘录 / EXCERPT: Chromatic-number conjecture. Any IRC-colorable graph G admits an IRC-coloring using $\chi(G)$ colors.

conjecture | open | MathDB 页面 | 原文线索

V2-069487 | MATHDB-360720

Kuznetsov's Fano threefold correspondence conjecture

摘录 / EXCERPT: Kuznetsov's conjecture. There is a correspondence

conjecture | refuted | MathDB 页面 | 原文线索

V2-069488 | MATHDB-360721

Jonsson-Mustață's equivalent valuation-computing conjecture for subadditive sequences

摘录 / EXCERPT: Jonsson-Mustață's equivalent conjecture. There is a quasimonomial valuation ν on $\mathcal{O}_o$ that computes $\mathrm{lct}^{\mathfrak{q}}(\mathfrak{b}_{\bullet})$, namely

conjecture | open | MathDB 页面 | 原文线索

V2-069489 | MATHDB-360724

Cocycle expansion for quotients of Bruhat-Tits buildings with permutation coefficients

摘录 / EXCERPT: Cocycle-expansion conjecture. For every such lattice Γ,

conjecture | open | MathDB 页面 | 原文线索

V2-069490 | MATHDB-360725

Cocycle Cheeger constants for Steinberg and GKKL presentations

摘录 / EXCERPT: Cocycle-Cheeger conjecture. The cocycle Cheeger constants in permutations of the Steinberg presentations have a uniform positive lower bound, at least when n is fixed and...

conjecture | open | MathDB 页面 | 原文线索

V2-069491 | MATHDB-360729

Hereditary admissibility conjecture for hereditarily structurally complete intermediate logics

摘录 / EXCERPT: Let L be an intermediate logic which is hereditarily structurally complete. Hereditary admissibility conjecture. Hereditary admissibility is equivalent to admissibility over L....

conjecture | open | MathDB 页面 | 原文线索

V2-069492 | MATHDB-360732

Conjecture on MDYPL state evolutions with an intercept

摘录 / EXCERPT: Consider the logistic regression model with intercept ... where $\theta_0\in\mathbb{R}$, the covariates are independent realizations...

conjecture | open | MathDB 页面 | 原文线索

V2-069493 | MATHDB-360742

Classification conjecture for rational isolated j-invariants on X_1(N)

摘录 / EXCERPT: Classification conjecture. If $x\in X_1(N)$ is an isolated point with $j(x)\in\bQ$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069494 | MATHDB-360753

Feng-Yun-Zhang's modularity conjecture for special cycles on unitary shtuka spaces

摘录 / EXCERPT: Feng-Yun-Zhang's modularity conjecture. There is an automorphic form

conjecture | open | MathDB 页面 | 原文线索

V2-069495 | MATHDB-360755

Refined Gan-Gross-Prasad-type norm formula for Miyawaki lifts

摘录 / EXCERPT: Let $\sigma=\otimes_v\sigma_v$, $\Sigma=\otimes_v\Sigma_v$, and $\pi=\otimes_v\pi_v$ be the cuspidal representations corresponding to $\mathbf{f}$, $\mathbf{F}$, and $\mathbf{g}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-069496 | MATHDB-360756

The Lubin-Tate Iwasawa main conjecture

摘录 / EXCERPT: Let V be a representation for which the L-analytic p-adic L-function $L_p(V)$ is defined, and let $\widetilde{H}^2(V)$ be the Selmer group described above over the L-anal...

conjecture | open | MathDB 页面 | 原文线索

V2-069497 | MATHDB-360760

The Bayes-Unexpectedness correspondence conjecture

摘录 / EXCERPT: Bayes-Unexpectedness correspondence conjecture. Bayes’ rule is a specific instantiation of a more general template captured in ST by Unexpectedness. This conjecture proposes a theo...

conjecture | open | MathDB 页面 | 原文线索

V2-069498 | MATHDB-360765

Feigin-Wang-Yoshinaga conjecture for the extended Catalan arrangement of type B_2

摘录 / EXCERPT: Let $m,i\in\mathbb{Z}_{\geq 0}$, and let $\widetilde{f_i^m}(x,y)$ be the deformed polynomial defined from the coefficients of ... For a polynomial g, write $g\in((x+y+m)_{2m+1})$ w...

conjecture | open | MathDB 页面 | 原文线索

V2-069499 | MATHDB-360766

Transience conjecture for unions of WUSFs on transitive transient graphs

摘录 / EXCERPT: Let G be a transitive transient graph. For $\varepsilon>0$, let an ε-WUSF be obtained by independently retaining each edge of a wired uniform spannin...

conjecture | open | MathDB 页面 | 原文线索

V2-069500 | MATHDB-360768

Generalized tadpole conjecture for the Hodge locus

摘录 / EXCERPT: Let $E\rightarrow\mathcal{M}$ be a variation of polarized Hodge structure of weight $D=2k$. For a positive integer L, define ... Here $\mathcal{M}$ is the complex-structure moduli...

conjecture | open | MathDB 页面 | 原文线索

V2-069501 | MATHDB-360770

Psi-expansion conjecture for interval graphs

摘录 / EXCERPT: Let $G=([n],E)$ be an interval graph. For a composition $\alpha\vDash n$, let Ψ_α be the type 1 power sum quasisymmetric function, let $z_\alpha=z_{\lambda(\alpha)}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069502 | MATHDB-360771

Composition-partition invariance conjecture for Dyck graphs

摘录 / EXCERPT: Let $G=([n],E)$ be a Dyck graph. For a composition $\alpha\vDash n$, let $\lambda(\alpha)$ be the partition obtained by rearranging its parts, and let $\mathcal{N}_{G,\alpha}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-069503 | MATHDB-360772

Psi-expansion conjecture for interval-graph permutation functions

摘录 / EXCERPT: Let $G=([n],E)$ be an interval graph and let $\tau\in\mathfrak{S}_n$. Let $\Phi_G(\tau)$, $\mathcal{Q}_{\Phi_G(\tau)}^{(G)}$, CoInv_G, and Inv_G be the functi...

conjecture | open | MathDB 页面 | 原文线索

V2-069504 | MATHDB-360776

Fargues-Scholze categorical Langlands conjecture

摘录 / EXCERPT: Fargues-Scholze's categorical Langlands conjecture. There is a canonical isomorphism between these two ∞-categories.

conjecture | open | MathDB 页面 | 原文线索

V2-069505 | MATHDB-360780

Reid's fundamental-group conjecture for surfaces of general type

摘录 / EXCERPT: Reid's conjecture. The fundamental group $\pi_1(X)$ is either finite or commensurable with the fundamental group of a curve.

conjecture | open | MathDB 页面 | 原文线索

V2-069506 | MATHDB-360794

Strong Approximation for the Markoff surface modulo primes

摘录 / EXCERPT: Let $\mathcal{X}$ be the affine surface in $\mathbb{A}^3$ defined by ... and let Γ be the group of affine integral morphisms generated by coordinate permutations and the Vieta...

conjecture | open | MathDB 页面 | 原文线索

V2-069507 | MATHDB-360799

The inhomogeneous-shift classification conjecture for multiplicative functions

摘录 / EXCERPT: Inhomogeneous-shift classification conjecture. If $\delta(\mathcal{N}_{f,a,b})\neq 0$ for some nonzero $a,b\in\mathbb{Z}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069508 | MATHDB-360802

The variance conjecture for entropy-regularized optimal transport maps

摘录 / EXCERPT: Variance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069509 | MATHDB-360803

The squaring conjecture for subGaussian optimal transport map estimation

摘录 / EXCERPT: Squaring conjecture. In this setting,

conjecture | open | MathDB 页面 | 原文线索

V2-069510 | MATHDB-360804

The Gaussian n^{-1} convergence conjecture for entropy-regularized optimal transport maps

摘录 / EXCERPT: Gaussian n^{-1} convergence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069511 | MATHDB-360809

The Langlands correspondence conjecture for weight-two eigenforms over number fields

摘录 / EXCERPT: Langlands correspondence conjecture. The alternatives and point-count identities above hold for every such eigenform.

conjecture | open | MathDB 页面 | 原文线索

V2-069512 | MATHDB-360811

Hedden's Whitehead-doubling sliceness conjecture

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot, and let $Wh^\pm(K)$ denote its positive or negative clasped, untwisted Whitehead double. Hedden's Whitehead-doubling conjecture. The knot...

conjecture | open | MathDB 页面 | 原文线索

V2-069513 | MATHDB-360812

Gordon's order-two concordance conjecture

摘录 / EXCERPT: Let $K\subseteq S^3$ be a knot, and write $K\#K$ for its connected sum with itself. A knot is negative amphichiral if it is isotopic to $r\overline{K}$, where r denotes reversal...

conjecture | open | MathDB 页面 | 原文线索

V2-069514 | MATHDB-360826

Boundedness conjecture for klt stable minimal models

摘录 / EXCERPT: Boundedness conjecture. The family $\mathcal{S}_{klt}(d,\Phi,\leq u,v)$ is bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-069515 | MATHDB-360827

Boundedness conjecture with bounded volume for klt stable minimal models

摘录 / EXCERPT: Bounded-volume boundedness conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-069516 | MATHDB-360831

The L^2-Hall characterization for graphs of free groups with cyclic edge groups

摘录 / EXCERPT: L^2-Hall characterization conjecture. The following conditions should be equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069517 | MATHDB-360836

Blidia et al.'s conjecture on b-colorings of regular graphs

摘录 / EXCERPT: Blidia et al.'s conjecture. Every d-regular graph of girth at least 5, different than the Petersen graph, has a b-coloring with $d+1$ colors.

conjecture | open | MathDB 页面 | 原文线索

V2-069518 | MATHDB-360847

Torsion local-global compatibility at $\ell=p$

摘录 / EXCERPT: Assume Conjecture 5.4. Let $\mathfrak{m}\subset \mathbf{T}^{T,\lambda,\tau}(K^p)$ be a non-Eisenstein maximal ideal. For $v\in S_p(F)$, set...

conjecture | open | MathDB 页面 | 原文线索

V2-069519 | MATHDB-360850

Conjecture on special values of zpread polynomials

摘录 / EXCERPT: Special-value conjecture. The following equivalences hold:

conjecture | open | MathDB 页面 | 原文线索

V2-069520 | MATHDB-360862

Edge-count threshold conjecture for rigidity in binomial random graphs

摘录 / EXCERPT: Random-graph rigidity conjecture. For every $p=\omega(\log n/n)$, $G\sim G(n,p)$ is $\textbf{whp}$ d-rigid for

conjecture | open | MathDB 页面 | 原文线索

V2-069521 | MATHDB-360865

Meyer's equitable coloring conjecture

摘录 / EXCERPT: Let G be a connected graph, and let ... Delta(G)-coloring. This conjecture is an equitable analogue of Brooks' theorem and concerns the equitable chromatic number. The paper's ab...

conjecture | open | MathDB 页面 | 原文线索

V2-069522 | MATHDB-360866

The odd-order latin-cuboid transversal conjecture

摘录 / EXCERPT: A latin n-cuboid of order q is an n-dimensional array with one dimension of length less than q and the remaining dimensions of length q, whose lines contain different sym...

conjecture | open | MathDB 页面 | 原文线索

V2-069523 | MATHDB-360869

The divisibility conjecture for even hook numbers in self-conjugate partitions

摘录 / EXCERPT: Let $a_h^*(n)$ denote the number of hooks of length h among self-conjugate partitions of n. The even-hook divisibility conjecture. For every integer $n\geq0$ and every integer...

conjecture | open | MathDB 页面 | 原文线索

V2-069524 | MATHDB-360876

Arthur's discrete-spectrum decomposition conjecture

摘录 / EXCERPT: Let G be a connected reductive group over a global function field, let $U\subset G(\mathbf A_K)$ be open compact, and let $S=S(U)$ be the finite set of places where U does not...

conjecture | open | MathDB 页面 | 原文线索

V2-069525 | MATHDB-360884

Twisted cubic moment conjecture for $GL(3)\times GL(2)$ L-functions

摘录 / EXCERPT: Twisted cubic moment conjecture. The full set of main terms for

conjecture | open | MathDB 页面 | 原文线索

V2-069526 | MATHDB-360892

Numerical evaluation of a Berndt-type integral of order three

摘录 / EXCERPT: Numerical evaluation. We believe that

conjecture | open | MathDB 页面 | 原文线索

V2-069527 | MATHDB-360902

A lattice-counting conjecture for paired quaternionic matrices

摘录 / EXCERPT: Let $g\in \operatorname{SL}_{2}(\mathbb{R})$. For parameters ℓ, δ, L, and $\heartsuit$, consider pairs $\gamma_1,\gamma_2\in R(\ell;g)$ lying in...

conjecture | open | MathDB 页面 | 原文线索

V2-069528 | MATHDB-360919

The thick-disc configuration-space extension conjecture

摘录 / EXCERPT: Let n be the number of thick discs in the collinear configuration, and let $\mathrm{Conf}_{n,\frac{1}{n}+\varepsilon}$ denote the corresponding configuration space. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069529 | MATHDB-360931

Intersection-of-series block bijection conjecture

摘录 / EXCERPT: Let $\mathbb{G}$ be a generic group, let $e,m>0$, and let $(\mathbb{L},\lambda)$ and $(\mathbb{M},\mu)$ be respectively Φ_e- and Φ_m-cuspidal pairs. Let the intersection...

conjecture | open | MathDB 页面 | 原文线索

V2-069530 | MATHDB-360932

Level-rank Koszul duality conjecture for cyclotomic rational DAHAs

摘录 / EXCERPT: Let the matching blocks be as in the intersection-of-series block bijection conjecture, and let $\vec{\nu}_e$ and $\vec{\nu}_m$ be charges for the corresponding specializations. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-069531 | MATHDB-360938

The commutator inequality with optimal constant for matrix monotone functions

摘录 / EXCERPT: Let A, B, and X be operators in $B({\mathcal H})$. Assume that $A,B\geq 0$ are compact with finite unitarily invariant norm...

conjecture | open | MathDB 页面 | 原文线索

V2-069532 | MATHDB-360939

The normalized commutator inequality with optimal constant for matrix monotone functions

摘录 / EXCERPT: Let A, B, and X be operators in $B({\mathcal H})$. Assume that $A,B\geq 0$ and that X is compact with finite unitarily invariant norm...

conjecture | open | MathDB 页面 | 原文线索

V2-069533 | MATHDB-360940

The function-approximation conjecture for operator monotone functions

摘录 / EXCERPT: Let $\widehat{\mathcal M}_+$ be the space of Fourier transforms of finite positive Borel measures on $\mathbb R$, and define ... Let f be a non-negative function that is operator m...

conjecture | open | MathDB 页面 | 原文线索

V2-069534 | MATHDB-360941

The commutator function-approximation conjecture for operator monotone functions

摘录 / EXCERPT: Let $\operatorname{CL}_{\lVert\!\lVert\!\lVert\cdot\rVert\!\rVert\!\rVert}([0,\infty))$ denote the class of commutator-Lipschitz functions for the chosen unitarily invariant norm,...

conjecture | open | MathDB 页面 | 原文线索

V2-069535 | MATHDB-360948

The stable square-length asymptotic formula for unitary characters

摘录 / EXCERPT: Let $1\ne w\in\mathbb{\mathbf{F}}$, let $\mu=(\mu_{1},\ldots,\mu_{k})$ be an integer sequence with no zero entries, and let $d_{w,\mu}$ be the number of orientable stable-square-le...

conjecture | open | MathDB 页面 | 原文线索

V2-069536 | MATHDB-360953

Brenti's log-concavity conjecture for lower Bruhat intervals

摘录 / EXCERPT: Let W be a finite Weyl group, and let $u,v\in W$. For each integer i, define ... A sequence $a_0,a_1,\ldots,a_n$ is log-concave if $a_{i-1}a_{i+1}\leq a_i^2$ for every $0<i<n$. B...

conjecture | open | MathDB 页面 | 原文线索

V2-069537 | MATHDB-360954

Unimodality conjecture for dominant affine Bruhat intervals

摘录 / EXCERPT: Let t_λ be the affine Weyl group element associated with the dominant coweight λ, and let f denote the finite Weyl group element used to define the dominant lower...

conjecture | open | MathDB 页面 | 原文线索

V2-069538 | MATHDB-360975

The parametrized volume conjecture for hyperbolic knots

摘录 / EXCERPT: Let K be a hyperbolic knot. For a parameter u, let $\rho_u\colon\pi_1(S^3\setminus K)\to\operatorname{SL}(2;\mathbb{C})$ be the irreducible representation specified by the meri...

conjecture | open | MathDB 页面 | 原文线索

V2-069539 | MATHDB-360977

The List Coloring Conjecture for claw-free graphs

摘录 / EXCERPT: Let G be a graph. Write $\chi(G)$ for its chromatic number and $\chi_l(G)$ for its list chromatic number, the least integer k such that every assignment of lists of size k to t...

conjecture | open | MathDB 页面 | 原文线索

V2-069540 | MATHDB-360978

Bressoud's partition identity conjecture for even moduli

摘录 / EXCERPT: Bressoud's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069541 | MATHDB-360979

Waiter-Client tree-universality conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. A graph is tree-universal at degree bound D if it contains a copy of every tree T with n vertices and maximum degree...

conjecture | open | MathDB 页面 | 原文线索

V2-069542 | MATHDB-360980

Client-Waiter tree-universality conjecture

摘录 / EXCERPT: Let K_n be the complete graph on n vertices. In the $(1:1)$ Client-Waiter game, Waiter offers two board elements at each round and Client claims one; Client's claimed edges for...

conjecture | open | MathDB 页面 | 原文线索

V2-069543 | MATHDB-360984

The pseudo-Anosov prevalence conjecture for simple braids

摘录 / EXCERPT: Let $\Br_n$ be the braid group on n strands, identified with the mapping class group of a disk with n marked points relative to its boundary. A braid is pseudo-Anosov if its ma...

conjecture | open | MathDB 页面 | 原文线索

V2-069544 | MATHDB-360990

The threshold interpolation conjecture for consecutive patterns in random permutations

摘录 / EXCERPT: The threshold interpolation conjecture. If $n^{1-1/s}\ll m$ and $\binom{n}{2}-m\gg n^{1-1/s'}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069545 | MATHDB-360996

Optimal Steiner tree construction conjecture for regular simplices

摘录 / EXCERPT: Let d be the number of terminals in a regular simplex, and let $k\geq 1$. The construction described immediately before the claim recursively splits Steiner points and coordinate...

conjecture | open | MathDB 页面 | 原文线索

V2-069546 | MATHDB-360998

Simon's higher-order Szegő sum-rule conjecture

摘录 / EXCERPT: Let w be the density of the absolutely continuous part of a measure on the unit circle, let $\alpha=(\alpha_n)_{n\geq 0}$ be its Verblunsky-coefficient sequence, and let S be t...

conjecture | open | MathDB 页面 | 原文线索

V2-069547 | MATHDB-361000

Uniqueness conjecture for powers of one minus cosine

摘录 / EXCERPT: Uniqueness conjecture. $P_n(\theta)$ is the only trigonometric polynomial of order n satisfying (5.59). The claim proposes uniqueness within that class, beyond the explicitly ver...

conjecture | open | MathDB 页面 | 原文线索

V2-069548 | MATHDB-361009

Kim's conjecture on local constituents of residual Arthur packets

摘录 / EXCERPT: Kim's conjecture. The parameter ψ is associated with a distinguished orbit $\mathcal{O}$ of $G^{\vee}$, and

conjecture | open | MathDB 页面 | 原文线索

V2-069549 | MATHDB-361028

Generation conjecture for invariant differential operators on the Siegel-Jacobi space

摘录 / EXCERPT: Let ${\mathscr P}_{n,m}$ be the Siegel-Jacobi space. Let D_j for $1\leq j\leq n$ and $\Omega_{pq}^{(k)}$ for $0\leq k\leq n-1$ and $1\leq p\leq q\leq m$ be the invariant differen...

conjecture | open | MathDB 页面 | 原文线索

V2-069550 | MATHDB-361029

Relations conjecture for invariant differential operators on the Siegel-Jacobi space

摘录 / EXCERPT: Let ${\mathscr P}_{n,m}$ be the Siegel-Jacobi space, and let D_j for $1\leq j\leq n$ and $\Omega_{pq}^{(k)}$ for $0\leq k\leq n-1$ and $1\leq p\leq q\leq m$ be the invariant diff...

conjecture | open | MathDB 页面 | 原文线索

V2-069551 | MATHDB-361030

Ballantine-Burson-Craig-Folsom-Wen divisibility conjecture for self-conjugate partition hook lengths

摘录 / EXCERPT: Let $a_t^{\star}(n)$ denote the number of hooks of length t among all self-conjugate partitions of n. For integers $n\geq0$ and $m\geq1$, define divisibility in the usual sense...

conjecture | open | MathDB 页面 | 原文线索

V2-069552 | MATHDB-361031

Characterization of positivity for traces of 2-webs in the honeycomb graph

摘录 / EXCERPT: Positivity characterization conjecture. Traces of all 2-webs in H are positive if and only if, up to gauge equivalence, one of the following conditions holds:

conjecture | open | MathDB 页面 | 原文线索

V2-069553 | MATHDB-361037

Abrishami, Henning, and Rahbarnia's bipartite cubic planar domination conjecture

摘录 / EXCERPT: Let G be a bipartite cubic planar graph, and let $\gamma(G)$ denote its domination number. Abrishami, Henning, and Rahbarnia's conjecture. ... This is described in the paper as a w...

conjecture | open | MathDB 页面 | 原文线索

V2-069554 | MATHDB-361038

The asymptotic weighted domination bound for connected subcubic planar graphs

摘录 / EXCERPT: Let G be a connected subcubic planar graph, and for $i\in\{0,1,2,3\}$ let $n_i(G)$ denote the number of vertices of degree i in G. Let $\gamma(G)$ denote the domination numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-069555 | MATHDB-361040

Disconnectedness conjecture for the Hénon connectedness locus

摘录 / EXCERPT: Disconnectedness conjecture. The connectedness locus is disconnected:

conjecture | open | MathDB 页面 | 原文线索

V2-069556 | MATHDB-361041

Period classification conjecture for quadratic Hénon maps over the rationals

摘录 / EXCERPT: Period classification conjecture. Over $\mathbb{Q}$, the map has no point of period N except possibly when

conjecture | open | MathDB 页面 | 原文线索

V2-069557 | MATHDB-361042

Generic period-two conjecture for quadratic Hénon maps over the rationals

摘录 / EXCERPT: Generic period-two conjecture. For all but finitely many $\delta\in\mathbb{Q}$, the $\mathbb{Q}$-rational periodic points of every such map, with $c\in\mathbb{Q}$, have period divi...

conjecture | open | MathDB 页面 | 原文线索

V2-069558 | MATHDB-361046

Conjecture on strictly bounded objects in symmetric tensor categories

摘录 / EXCERPT: Strict boundedness conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-069559 | MATHDB-361047

Jordan conjecture for fundamental groups of log canonical Fano pairs

摘录 / EXCERPT: Let n be a positive integer, and let (X,Δ) be a log canonical Fano pair. Write $\pi_1^{\rm orb}(X,\Delta)$ for its orbifold fundamental group. Jordan conjecture for log ca...

conjecture | open | MathDB 页面 | 原文线索

V2-069560 | MATHDB-361048

Solvable-rank conjecture for fundamental groups of log canonical log Calabi-Yau pairs

摘录 / EXCERPT: Let n be a positive integer, and let (X,Δ) be an n-dimensional log canonical log Calabi-Yau pair. Write $\pi_1^{\rm orb}(X,\Delta)$ for its orbifold fundamental group. S...

conjecture | open | MathDB 页面 | 原文线索

V2-069561 | MATHDB-361055

Triviality of automorphisms of reduced power monoids of proper numerical monoids

摘录 / EXCERPT: A numerical monoid is a submonoid S of $(\mathbb N,+)$ such that $\mathbb N\smallsetminus S$ is finite. Let the reduced power monoid of S be the reduced power monoid associated...

conjecture | open | MathDB 页面 | 原文线索

V2-069562 | MATHDB-361078

Inversion of adjunction for minimal log discrepancies on quotient singularities

摘录 / EXCERPT: PIA conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069563 | MATHDB-361079

Finite basis conjecture for relational structures without finite monomorphic decomposition

摘录 / EXCERPT: Let μ be a finite relational signature, and let $\mathscr S_{\mu}$ be the class of all relational structures of signature μ without any finite monomorphic decomposition. A...

conjecture | open | MathDB 页面 | 原文线索

V2-069564 | MATHDB-361107

The supersolvable plane curve freeness conjecture

摘录 / EXCERPT: Let C be a reduced plane curve in $\mathbb{P}^2$. A point $p\in C$ is modular if the central projection from p induces a locally trivial fibration of the complement...

conjecture | open | MathDB 页面 | 原文线索

V2-069565 | MATHDB-361109

Symmetry characterization for zig-zag classes

摘录 / EXCERPT: Symmetry characterization for zig-zag classes. If two zig-zag classes are symmetrically equivalent, then one is formed by complementing the labels of the other or by reversing all...

conjecture | open | MathDB 页面 | 原文线索

V2-069566 | MATHDB-361110

Bipartite characterization of rational POP generating functions

摘录 / EXCERPT: Bipartite characterization conjecture. A POP class has a rational generating function if and only if it is bipartite.

conjecture | open | MathDB 页面 | 原文线索

V2-069567 | MATHDB-361127

Exhaustiveness conjecture for factorized A_2-Leonard pairs

摘录 / EXCERPT: A factorized A_2-Leonard pair is an object defined in the paper, and Sections and give lists of examples of such pairs. Exhaustiveness conjecture. The list of factorized A_2-Le...

conjecture | open | MathDB 页面 | 原文线索

V2-069568 | MATHDB-361134

The Petersen Coloring Conjecture in terms of normal chromatic index

摘录 / EXCERPT: Petersen Coloring Conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069569 | MATHDB-361139

The Gaudin no-monodromy correspondence conjecture

摘录 / EXCERPT: Let L_1 be the displayed scalar oper with distinct nonzero points s_i, integer r, and parameters k,l,n_1 as in the source. Gaudin no-monodromy conjecture. For generic k,l...

conjecture | open | MathDB 页面 | 原文线索

V2-069570 | MATHDB-361140

The BLZ-oper no-monodromy correspondence conjecture

摘录 / EXCERPT: Let L_0 be the displayed BLZ-type oper, with distinct points $t_j\notin\{0,1\}$, satisfying the condition $b_0+b_1+\sum_iq_i=0$ and having monodromy-free singularities at $x=1$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-069571 | MATHDB-361142

The exponential Gaudin existence conjecture outside the Virasoro highest-weight locus

摘录 / EXCERPT: Let the exponential Gaudin Bethe Ansatz system be as in the source, and let $d_0=r^2$, $d_1=r(r-1)$ characterize the Virasoro highest-weight locus. Exponential Gaudin nonexistence...

conjecture | open | MathDB 页面 | 原文线索

V2-069572 | MATHDB-361143

The no-common-roots conjecture for the polynomials W_r and U_r

摘录 / EXCERPT: Let $W_r(x)$ and $U_r(x)$ be the recursively defined polynomials associated with the unique exponential-Gaudin solutions indexed by $r\in\mathbb Z$. Polynomial-root conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069573 | MATHDB-361144

The uniqueness conjecture for solutions of the W_r monodromy system

摘录 / EXCERPT: Let W_r be the polynomial defined from the unique exponential-Gaudin solution indexed by r, and consider the monodromy system in the variables w_i displayed in the source....

conjecture | open | MathDB 页面 | 原文线索

V2-069574 | MATHDB-361158

The Weyl-monodromy realization conjecture for D and E singularities

摘录 / EXCERPT: Let W be the Weyl group associated with a singularity of type D_n or E_n. For some proper smooth surface X/K, consider an integral model of X whose special fiber has a si...

conjecture | open | MathDB 页面 | 原文线索

V2-069575 | MATHDB-361160

Functoriality of the rigid refined local Langlands correspondence

摘录 / EXCERPT: Functoriality conjecture. The representation

conjecture | open | MathDB 页面 | 原文线索

V2-069576 | MATHDB-361162

Critical maximum-cut conjecture for random triangle-free graphs

摘录 / EXCERPT: Critical maximum-cut conjecture. There exists $c^*>0$ and a continuous function $\delta:(c^*,\infty)\to(0,1/2]$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-069577 | MATHDB-361163

Anticyclotomic Iwasawa main conjecture for the BDP p-adic L-function

摘录 / EXCERPT: Let K be an imaginary quadratic field satisfying the Heegner, discriminant, and splitting hypotheses, let K_∞ be its anticyclotomic Z_p-extension, let Λ be the a...

conjecture | open | MathDB 页面 | 原文线索

V2-069578 | MATHDB-361164

Mistretta-Stoppino stability equivalence conjecture for generated linear series

摘录 / EXCERPT: Mistretta-Stoppino's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-069579 | MATHDB-361165

Mistretta-Stoppino gonality-bound conjecture for non-complete linear series

摘录 / EXCERPT: Let C be an irreducible projective smooth complex curve, let L be a line bundle on C, and let $V\subseteq H^0(C,L)$ define a base point free non-complete linear series. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069580 | MATHDB-361166

Choi et al.'s optimal-code characterization of the LCP lower bound

摘录 / EXCERPT: Choi et al.'s conjecture. There exists a unique binary optimal $[n,k,d_L(n,k)]$ code C which is even-like and contains ${\bf1}$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-069581 | MATHDB-361167

Guenda et al.'s conjecture on linear intersection pairs of codes

摘录 / EXCERPT: Guenda et al.'s conjecture. Provided that $[n,k_1,d_1]_q$ and $[n,k_2,d_2]_q$ codes exist, there exists a linear ℓ-intersection pair of such codes for every integer ℓ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-069582 | MATHDB-361170

El Sahili's universality conjecture for oriented paths

摘录 / EXCERPT: El Sahili's conjecture. Every oriented path of order n is n-universal.

conjecture | open | MathDB 页面 | 原文线索

V2-069583 | MATHDB-361171

Burr's universality conjecture for oriented trees

摘录 / EXCERPT: Burr's conjecture. Every oriented tree of order n is $(2n-2)$-universal.

conjecture | open | MathDB 页面 | 原文线索

V2-069584 | MATHDB-361187

Threshold conjecture for Laguerre inequalities and determinant positivity of partition-function differences

摘录 / EXCERPT: Threshold conjecture. For $1\leq m\leq 11$ and $1\leq k\leq 5$, $\Delta^k p(n)$, respectively $\Delta^k\bar{p}(n)$, satisfies the Laguerre inequality of order m whenever...

conjecture | open | MathDB 页面 | 原文线索

V2-069585 | MATHDB-361188

Threshold conjecture for third-order Turán inequalities and invariant positivity

摘录 / EXCERPT: Threshold conjecture. For $1\leq k\leq 5$, $\Delta^k p(n)$, respectively $\Delta^k\bar{p}(n)$, satisfies the third-order Turán inequality whenever $n\geq T_p(k)$, respectively...

conjecture | open | MathDB 页面 | 原文线索

V2-069586 | MATHDB-361189

The deficient path forest conjecture on impossibly burnable forests

摘录 / EXCERPT: Deficient path forest conjecture. Let $n\geq4$. If T is a deficient n-path forest with $l_1\geq L_{n-1}$, then T is impossibly burnable. The conjecture is motivated by comput...

conjecture | open | MathDB 页面 | 原文线索

V2-069587 | MATHDB-361190

The linear upper-bound conjecture for path-forest burning thresholds

摘录 / EXCERPT: For each $n\geq4$, let L_n denote the threshold parameter for n-path forests introduced in the paper. Linear upper-bound conjecture. ... If true, this would give a good asymptoti...

conjecture | open | MathDB 页面 | 原文线索

V2-069588 | MATHDB-361193

Generalized self-concordance of the log trace-inverse function

摘录 / EXCERPT: Log trace-inverse self-concordance conjecture. The function $k(X)$ is (ν,M)-generalized self-concordant for some $\nu\in[2,3]$ and $M>0$ on some subset of the positive-definite...

conjecture | open | MathDB 页面 | 原文线索

V2-069589 | MATHDB-361207

Complete quasi-equidistribution of the golden-ratio Hammersley point set

摘录 / EXCERPT: Complete quasi-equidistribution claim. The Hammersley point set in base $$ is completely quasi-equidistributed in base 2.

conjecture | open | MathDB 页面 | 原文线索

V2-069590 | MATHDB-361208

Discrepancy-order conjecture for golden-ratio Hammersley point sets

摘录 / EXCERPT: Discrepancy-order conjecture. For $m e 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-069591 | MATHDB-361210

Exponential DP-4-colorings of triangle-free planar graphs

摘录 / EXCERPT: Let G be a triangle-free planar graph with n vertices, and let $P_{DP}(G,4)$ denote its DP color function. The exponential DP-coloring conjecture. There exists a constant $c>1$...

conjecture | open | MathDB 页面 | 原文线索

V2-069592 | MATHDB-361217

The positivity conjecture for NV2 spectral determinants

摘录 / EXCERPT: Positivity conjecture. For spectral data $\{\lambda_k,\mathbf{b}_k\}_{k=1}^N\in\mathcal{S}(N)$, all determinants in these conditions are positive. These inequalities would show tha...

conjecture | open | MathDB 页面 | 原文线索

V2-069593 | MATHDB-361218

The total-positivity conjecture for the NV2 extended spectral data

摘录 / EXCERPT: Let $\mathcal{S}_{\mathrm{ext}}(N)$ be the extended spectral data for an N-peakon NV2 system, let $\mathcal{S}(N)$ be the actual spectral data set, and let $\Lambda_2(N)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-069594 | MATHDB-361227

Conjecture on the optimality of the tree-strategy bound for the fourth weak Bruhat graph

摘录 / EXCERPT: Tree-strategy optimality conjecture for B_4. The bound

conjecture | open | MathDB 页面 | 原文线索

V2-069595 | MATHDB-361229

Schmidt's conjecture on integral points on smooth strictly convex curves

摘录 / EXCERPT: Let $N>0$, let I be the interval under consideration, and let $f:[0,N]\to[0,N]$ be a strictly convex function such that $f\in C^3([0,N])$ and $f^{(3)}(x)\neq 0$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-069596 | MATHDB-361231

The Griffiths-group rank prediction for bielliptic Picard curves

摘录 / EXCERPT: Griffiths-group rank prediction. One expects

conjecture | open | MathDB 页面 | 原文线索

V2-069597 | MATHDB-361237

Precoloring extension for sufficiently generic near-Eulerian triangulations

摘录 / EXCERPT: Generic precoloring-extension conjecture. For every positive even integer ℓ, there exists an integer d such that, if no 4-cycle in $G-U$ separates a face of S from the out...

conjecture | open | MathDB 页面 | 原文线索

V2-069598 | MATHDB-361238

The relative GW/PT correspondence for elliptic-surface geometries

摘录 / EXCERPT: Let S be an elliptic surface, let C be a curve of genus h, and let S_z denote the specified relative divisors in $S\times C$. For curve data (β,n), insertions...

conjecture | open | MathDB 页面 | 原文线索

V2-069599 | MATHDB-361242

Anomaly cascade for fermionic symmetry actions

摘录 / EXCERPT: Let $\mathcal{C}$ be a super modular tensor category with fermionic symmetry data (G_b,s,ω), and let $\mathcal{B}$ be a unitary modular tensor category that is a minimal mod...

conjecture | open | MathDB 页面 | 原文线索

V2-069600 | MATHDB-361252

The exponent-realization conjecture for modular invariants and NIM-reps

摘录 / EXCERPT: Let $\cal{C}$ be a modular tensor category of rank k with modular data S and T. A modular invariant is a matrix M satisfying $MS=SM$, $MT=TM$, $M_{ab}\in\mathbb{N}$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-069601 | MATHDB-361261

Grad's conjecture for toroidally nested MHD equilibria

摘录 / EXCERPT: Grad's conjecture. If there exists an MHD equilibrium (X,p) on M whose pressure has toroidally nested level sets, then the equilibrium is axisymmetric.

conjecture | open | MathDB 页面 | 原文线索

V2-069602 | MATHDB-361263

Reduced systems conjecture for blow-ups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb{F}_{e,r}$ be the blow-up of a Hirzebruch surface $\mathbb{F}_e$ at very general points, and let $\widetilde{C}_e$ be the strict transform of its negative section. A no...

conjecture | open | MathDB 页面 | 原文线索

V2-069603 | MATHDB-361264

Negative curves conjecture for blow-ups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb{F}_{e,r}$ denote the blow-up of the Hirzebruch surface $\mathbb{F}_e$ at r very general points $p_1,\dots,p_r$. A curve is irreducible and reduced if it is an irredu...

conjecture | open | MathDB 页面 | 原文线索

V2-069604 | MATHDB-361265

The entropy inequality for real exponents

摘录 / EXCERPT: The entropy inequality for real exponents. For $0\leq x\leq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-069605 | MATHDB-361266

The entropy polynomial root conjecture

摘录 / EXCERPT: The entropy polynomial root conjecture. The polynomial $p_{k,r}(x)$ has exactly two real roots in $(0,1)$, counting multiplicity. The paper explains that this root statement would...

conjecture | open | MathDB 页面 | 原文线索

V2-069606 | MATHDB-361270

Casselgren-Petrosyan conjecture for outerplanar graphs

摘录 / EXCERPT: Let G be an outerplanar graph, and let $\mu_{\operatorname{int}}(G)$ denote its interval coloring impropriety. Casselgren-Petrosyan's outerplanar graph conjecture. ... This would i...

conjecture | open | MathDB 页面 | 原文线索

V2-069607 | MATHDB-361273

Genericity of non-degenerate policies in H-infinity control

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the space of full-order dynamic policies, and let $\mathcal{C}_{\mathrm{nd}}$ be the subset of non-degenerate policies, namely those admitting a certificate...

conjecture | open | MathDB 页面 | 原文线索

V2-069608 | MATHDB-361274

Minimal stationary policies are non-degenerate in H-infinity control

摘录 / EXCERPT: Let $\mathcal{C}_n$ be the space of full-order dynamic policies for $\mathcal{H}_\infty$ control. A policy $\mathsf{K}\in\mathcal{C}_n$ is Clarke stationary when...

conjecture | open | MathDB 页面 | 原文线索

V2-069609 | MATHDB-361284

Aharoni and Howard's rainbow matching conjecture

摘录 / EXCERPT: Let $\mathcal{G}=\{G_1,\ldots,G_t\}$ be a family of t bipartite graphs, and let Δ be a bound such that $\Delta(G_i)\leq \Delta$ for every i. Aharoni and Howard's rainbow...

conjecture | open | MathDB 页面 | 原文线索

V2-069610 | MATHDB-361287

The fundamental trade-off conjecture for expander-code decoding

摘录 / EXCERPT: Let G be a (c,d,α,δ)-bipartite expander and let $C_0\subseteq\mathbb{F}_2^d$ be an inner code with minimum Hamming distance $d(C_0)\ge d_0$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069611 | MATHDB-361307

Connectivity conjecture for the square-root queer crystal on set-valued decomposition tableaux

摘录 / EXCERPT: Connectivity conjecture. For each strict partition λ with $\ell(\lambda)\leq n$, the square-root queer crystal $\mathsf{SetDecTab}_n(\lambda)$ is connected and the tableau...

conjecture | open | MathDB 页面 | 原文线索

V2-069612 | MATHDB-361308

The Fine-number enumeration conjecture for valleyless-tieless parking functions

摘录 / EXCERPT: A valleyless-tieless parking function of length n is a parking function $(a_1,\ldots,a_n)$ with no index $1\leq i\leq n-1$ such that $a_i=a_{i+1}$ and no index $2\leq i\leq n-1$...

conjecture | open | MathDB 页面 | 原文线索

V2-069613 | MATHDB-361309

Bergeron-Garsia Schur-positivity conjecture for nabla applied to monomial symmetric functions

摘录 / EXCERPT: Let Λ be the ring of symmetric functions, let m_{μ} and s_{λ} denote the monomial and Schur symmetric functions indexed by partitions μ and λ, respe...

conjecture | open | MathDB 页面 | 原文线索

V2-069614 | MATHDB-361311

Conjecture on superspecial hyperelliptic curves of genus 4 with dihedral automorphisms

摘录 / EXCERPT: Superspecial genus-4 dihedral automorphism conjecture. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-069615 | MATHDB-361315

The Hankel determinant periodicity and recurrence conjecture for q-deformed metallic numbers

摘录 / EXCERPT: Hankel determinant periodicity and recurrence conjecture. The $k+2$ sequences $\Delta_n^{(0)},\Delta_n^{(1)},\dots,\Delta_n^{(k+1)}$ take only the values $-1,0,1$. They are...

conjecture | open | MathDB 页面 | 原文线索

V2-069616 | MATHDB-361321

Conjecture on fractional list packing for planar triangle-free graphs

摘录 / EXCERPT: Let G be a planar triangle-free graph. The fractional list-packing conjecture. ... This conjecture would extend the paper's list-coloring results to fractional packing numbers and...

conjecture | open | MathDB 页面 | 原文线索

V2-069617 | MATHDB-361329

Existence of a Hopfian finitely presented group with undecidable weak Whitehead problem

摘录 / EXCERPT: A Hopfian finitely presented group is a finitely presented group for which every surjective endomorphism is an automorphism; its word problem asks whether a given word represents t...

conjecture | open | MathDB 页面 | 原文线索

V2-069618 | MATHDB-361334

Individual Volterra-Lyapunov stability implies a non-square D-stability-like property

摘录 / EXCERPT: Let $A\in\mathbb{R}^{m\times n}$ be a real nonsquare matrix. Let $\mathcal{M}$ be the index set consisting of k-tuples of integers in the range $1,\ldots,m$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069619 | MATHDB-361336

Bollobás-Thomason threshold conjecture for binomial random multigraphs

摘录 / EXCERPT: Bollobás-Thomason threshold conjecture. The property $\mathcal{P}$ has a threshold in $\mathbb{G}(n,p\,; \,\binom{[n]}{k})$.

conjecture | open | MathDB 页面 | 原文线索

V2-069620 | MATHDB-361341

Quasi-geometricity conjecture for connections from braided fusion categories

摘录 / EXCERPT: Let $\mathcal C$ be a braided fusion category over $\Bbb C$. For objects $X_1,\ldots,X_n,Y\in\mathcal C$, the space $\operatorname{Hom}(Y,X_1\otimes\cdots\otimes X_n)$ carries an a...

conjecture | open | MathDB 页面 | 原文线索

V2-069621 | MATHDB-361350

The power-function integration formula for arbitrary real exponents

摘录 / EXCERPT: Let $a,b\in\mathbb R$ and let $p\in\mathbb R$ be an exponent for which the integral and the displayed powers are defined. Power-function integration conjecture. The integration for...

conjecture | open | MathDB 页面 | 原文线索

V2-069622 | MATHDB-361357

The even-hole-free graph and K4-free chordal graph conjecture

摘录 / EXCERPT: Let $t\geq 4$ be an integer, let H be a graph, and call H K_4-free chordal if it is chordal and contains no induced subgraph isomorphic to K_4. An even-hole-free graph has...

conjecture | open | MathDB 页面 | 原文线索

V2-069623 | MATHDB-361358

The 2-forest conjecture for even-hole-free graphs

摘录 / EXCERPT: For an integer $k\geq 1$, a k-forest is a K_{k+2}-free chordal graph; in particular, a 2-forest is a K_4-free chordal graph. An even-hole-free graph has no induced cycle of...

conjecture | open | MathDB 页面 | 原文线索

V2-069624 | MATHDB-361359

The bounded-treewidth conjecture for even-hole-free graphs excluding a 2-forest

摘录 / EXCERPT: Let $t\geq 4$ be an integer, let H be a 2-forest, and let an $(\mathcal{F})$-free graph mean a graph with no induced subgraph isomorphic to a member of $\mathcal{F}$. The bound...

conjecture | open | MathDB 页面 | 原文线索

V2-069625 | MATHDB-361360

The sparse induced-subgraph conjecture for even-hole-free graphs

摘录 / EXCERPT: Let $\operatorname{tw}(G)$ denote the treewidth of G. The sparse induced-subgraph conjecture for even-hole-free graphs. For every integer $t\geq 1$, every even-hole-free graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-069626 | MATHDB-361361

The general sparse induced-subgraph conjecture

摘录 / EXCERPT: Let $\operatorname{tw}(G)$ denote the treewidth of G. The general sparse induced-subgraph conjecture. For every $t\geq 1$, every graph of sufficiently large treewidth has an indu...

conjecture | open | MathDB 页面 | 原文线索

V2-069627 | MATHDB-361363

The bounded-index virtually solvable fundamental group conjecture

摘录 / EXCERPT: Let d and c be the dimension and absolute coregularity, respectively, and let (X,D) be a klt Calabi-Yau type pair of dimension d and absolute coregularity c. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069628 | MATHDB-361364

Numerical evaluation conjecture for a Berndt-type integral with denominator power six

摘录 / EXCERPT: Let $\Gamma=\Gamma(1/4)$. The Berndt-type integral ... should equal ... This is a numerically verified special case of the paper's broader arithmetic-pattern results, but no proof of t...

conjecture | open | MathDB 页面 | 原文线索

V2-069629 | MATHDB-361365

Arithmeticity conjecture for fifth-moment Berndt-type integrals

摘录 / EXCERPT: Let $\Gamma=\Gamma(1/4)$ and, for $n\geq1$, define ... Here $\mathbb{Q}$ denotes the rational numbers. Fifth-moment arithmeticity conjecture. For every integer $n\geq1$, ... and ... Thes...

conjecture | open | MathDB 页面 | 原文线索

V2-069630 | MATHDB-361369

Asymptotic formula for flip-graph vertices of repeated even chains

摘录 / EXCERPT: Let ℓ be a fixed even positive integer, and let $G(\underbrace{\ell,\dots,\ell}_{n\text{ copies}})$ denote the flip graph associated with the product of n chains of length...

conjecture | open | MathDB 页面 | 原文线索

V2-069631 | MATHDB-361371

No-hot-spots conjecture for simply connected planar mixed-boundary triples

摘录 / EXCERPT: Let (Ω,D,N) be a triple where Ω is simply connected, D is a Euclidean line segment, and ... Say that (Ω,D,N) has no hot spots when each first mixed eigenfuncti...

conjecture | open | MathDB 页面 | 原文线索

V2-069632 | MATHDB-361372

Stanley's strong conjecture for Jack Littlewood-Richardson coefficients when the Schur coefficient is one

摘录 / EXCERPT: Strong Stanley conjecture. If $c_{\mu\nu}^\lambda=1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069633 | MATHDB-361373

The rule formulation of the general structure conjecture

摘录 / EXCERPT: Rule formulation of the general structure conjecture. There exists an integer rule $\bm g_{\mu\nu}^{\lambda}\in\mathcal H_{\mu,\nu}^{\lambda}(\mathbb Z)$ of the form

conjecture | open | MathDB 页面 | 原文线索

V2-069634 | MATHDB-361374

The window positivity conjecture for Jack Littlewood-Richardson coefficients

摘录 / EXCERPT: Window positivity conjecture. The window factor accounts for all boxes outside R in the positivity expression, namely

conjecture | open | MathDB 页面 | 原文线索

V2-069635 | MATHDB-361375

The channel conjugation symmetry for the six-parameter solution

摘录 / EXCERPT: Channel conjugation symmetry. The transformed coefficients

conjecture | open | MathDB 页面 | 原文线索

V2-069636 | MATHDB-361376

The kernel-exhaustion conjecture for the six-parameter channel equation

摘录 / EXCERPT: Kernel-exhaustion conjecture. The relations of Lemma exhaust the kernel of the channel equation. The claim concerns the completeness of the listed linear relations among channel so...

conjecture | open | MathDB 页面 | 原文线索

V2-069637 | MATHDB-361377

The factorization conjecture for Jack Littlewood-Richardson coefficients

摘录 / EXCERPT: Factorization conjecture. The Jack Littlewood-Richardson coefficient has an expansion

conjecture | open | MathDB 页面 | 原文线索

V2-069638 | MATHDB-361378

The regular factorization conjecture for local Schur channel coefficients

摘录 / EXCERPT: Regular factorization conjecture. For certain factorizations κ, there exist factorization solutions for which

conjecture | open | MathDB 页面 | 原文线索

V2-069639 | MATHDB-361382

Threshold conjecture for monotone properties of random geometric graphs

摘录 / EXCERPT: Let $G_d(n,p)$ be the random geometric graph model, and let $\mathcal{F}$ be a non-trivial monotone graph property, where monotonicity means monotonicity under adding edges. Random...

conjecture | open | MathDB 页面 | 原文线索

V2-069640 | MATHDB-361386

Bourgain-Kontorovich local-global conjecture for continued-fraction semigroups

摘录 / EXCERPT: Let $\mathcal{A}\subseteq\mathbb{N}$ be finite, let v be the primitive vector from the associated orbit, and let L be a linear functional. Define ... and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069641 | MATHDB-361387

Local-global conjecture for the semigroup orbit of \(\Psi_1\)

摘录 / EXCERPT: Define ... The table of congruence and reciprocity obstructions specifies the stated restrictions on numerators and denominators in these orbits. Local-global conjecture for Ψ_1...

conjecture | open | MathDB 页面 | 原文线索

V2-069642 | MATHDB-361388

Explicit numerator conjecture for the orbit of \(\Psi_1\)

摘录 / EXCERPT: Let ... and consider its orbit of the indicated vector. Explicit Ψ_1 numerator conjecture. Every non-square integer $n>10569$ appears as a numerator in the orbit...

conjecture | open | MathDB 页面 | 原文线索

V2-069643 | MATHDB-361389

Large-denominator conjecture for continued fractions avoiding squares

摘录 / EXCERPT: Let v be a non-square integer with $v\geq10^{13}$. Consider rational numbers with continued fraction expansions ... where the a_i are positive integers. Large-denominator continu...

conjecture | open | MathDB 页面 | 原文线索

V2-069644 | MATHDB-361400

Domination bound for complementary self-centered graphs

摘录 / EXCERPT: Domination conjecture for $SCC(2)$ graphs. If $G\in SCC(2)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-069645 | MATHDB-361408

Linear-average stopping-time conjecture for the polynomial Collatz map

摘录 / EXCERPT: Let $T\colon\mathbb{F}_2[x]\to\mathbb{F}_2[x]$ be the polynomial Collatz map, with $T(f)=(1+x)f+1$ for odd f and $T(f)=f/x$ for even f. For polynomials of degree d, define th...

conjecture | open | MathDB 页面 | 原文线索

V2-069646 | MATHDB-361409

Dimension conjecture for alternating Mordell--Tornheim zeta values

摘录 / EXCERPT: The alternating Mordell--Tornheim dimension conjecture. For every $w\geq3$, the space $\AMT_w$ can be generated by C_w. The conjecture proposes an explicit family of alternating...

conjecture | open | MathDB 页面 | 原文线索

V2-069647 | MATHDB-361426

Non-negativity conjecture for the $R=2$, $S=1$ truncated theta series

摘录 / EXCERPT: Non-negativity conjecture. This series has non-negative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-069648 | MATHDB-361430

Arhancet's Hodge-Dirac functional-calculus conjecture

摘录 / EXCERPT: Let G be a Lie group of polynomial volume growth, generated by vector fields $X_1,\ldots,X_m$, and let $\slashed{D}_p$ be the L^p realization of the Hodge-Dirac operator on ... F...

conjecture | open | MathDB 页面 | 原文线索

V2-069649 | MATHDB-361437

Real-analyticity conjecture for center distributions

摘录 / EXCERPT: Let f be a holomorphic partially hyperbolic diffeomorphism, with center distribution E^c. Real-analyticity conjecture. The center distribution E^c is real analytic. The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-069650 | MATHDB-361444

Quantitative Riesz rearrangement conjecture with Fraenkel deficit

摘录 / EXCERPT: Let $0<\delta<1$. For a measurable set $\Omega\subset\mathbb{R}^d$ of finite measure, let Ω^* be the centered ball with $|\Omega^*|=|\Omega|$, and let $\alpha[\Omega]$ be it...

conjecture | open | MathDB 页面 | 原文线索

V2-069651 | MATHDB-361447

Span preserves lax-univalent discrete cocartesian fibrations

摘录 / EXCERPT: Span-preservation conjecture. Let $\boxed{}$ be a univalent discrete cocartesian fibration. Then

conjecture | open | MathDB 页面 | 原文线索

V2-069652 | MATHDB-361448

Span constructions provide enough lax-univalent fibrations

摘录 / EXCERPT: Sufficiency conjecture. Suppose $\boxed{}$ is a sufficient family of univalent fibrations for $\boxed{}$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-069653 | MATHDB-361449

Rank-conductor conjecture for elliptic curves over global fields

摘录 / EXCERPT: Rank-conductor conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069654 | MATHDB-361459

Enveloping expansion for series 33

摘录 / EXCERPT: Let series 33 in Table have $t=-1$, and let $\mathcal{R}_n$ and $\mathcal{F}_n$ be the quantities associated with this series. Let c_j be the coefficients defined by Theorem. Env...

conjecture | open | MathDB 页面 | 原文线索

V2-069655 | MATHDB-361461

Bourin-Lee's multivariable unitary-orbit inequality conjecture

摘录 / EXCERPT: Bourin-Lee's multivariable conjecture. The unitary matrices can be chosen so that

conjecture | open | MathDB 页面 | 原文线索

V2-069656 | MATHDB-361466

Periodic replication conjecture for zero-dipole configurations

摘录 / EXCERPT: Periodic replication conjecture. Without requiring $(\widetilde U,\widetilde V)$ to be a global minimizer or compactly supported in Ω, one has

conjecture | open | MathDB 页面 | 原文线索

V2-069657 | MATHDB-361482

The generalized Sato-Tate conjecture for Jacobians

摘录 / EXCERPT: Let C be a smooth projective curve over a number field k. Let X_k be the set of conjugacy classes of $\operatorname{ST}(\operatorname{Jac}(C)_k)$, and let $\{p_i\}_{i\geq 1}$...

conjecture | open | MathDB 页面 | 原文线索

V2-069658 | MATHDB-361493

Clemens's cofinite-image conjecture for primitive vanishing cycles

摘录 / EXCERPT: Let X be a smooth hypersurface and suppose that $n=\dim(X)$ is odd. Let T_v be the connected component of the analytic covering space of the smooth hyperplane-section parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-069659 | MATHDB-361495

Costate proportionality conjecture for the Douglas-Rachford projector

摘录 / EXCERPT: Costate proportionality conjecture. The costate variables satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069660 | MATHDB-361505

Aboulker-Aubian-Charbit-Lopes conjecture on forest backedge graphs

摘录 / EXCERPT: Let T be a tournament and suppose that one of its backedge graphs is a forest. For a tournament, let its clique number mean the minimum clique number of a backedge graph over all...

conjecture | open | MathDB 页面 | 原文线索

V2-069661 | MATHDB-361507

Null-density conjecture for finitely generated proper subgroupoids of the cyclic free groupoid

摘录 / EXCERPT: Let $\mathbb{M}$ be the cyclic free groupoid, and let N be a finitely generated proper subgroupoid of $\mathbb{M}$. Null-density conjecture. The subgroupoid N has null density....

conjecture | open | MathDB 页面 | 原文线索

V2-069662 | MATHDB-361508

Unimodality conjecture for colored-permutation Hilbert series

摘录 / EXCERPT: Unimodality conjecture. For any integers n,r, there exists an integer k such that

conjecture | open | MathDB 页面 | 原文线索

V2-069663 | MATHDB-361509

The integer-scaling criterion for C-realizability

摘录 / EXCERPT: Integer-scaling criterion. If $\{\frac{\lambda_1}{\alpha_1},\ldots,\frac{\lambda_n}{\alpha_n}\}$ is $C_{\mathbb{Q},3}$-realizable, then it is $C_{\mathbb{Q},3}$-realizable if and o...

conjecture | open | MathDB 页面 | 原文线索

V2-069664 | MATHDB-361513

The pseudo-automorphism group conjecture for the rational threefold X

摘录 / EXCERPT: Let X be the rational threefold, S its Jacobian Kummer surface, R the unique rational normal curve through the six points blown up in constructing X, and $PsAut(X)$ the gro...

conjecture | open | MathDB 页面 | 原文线索

V2-069665 | MATHDB-361514

Model 2 conjecture on eigenvalue and pseudospectrum structures

摘录 / EXCERPT: Let $S_{\delta J}^{(2)}(m)$ denote the model 2 matrix, with matrix size n, parameters δ and a, and let λ_0 be the distinguished eigenvalue of the original system...

conjecture | open | MathDB 页面 | 原文线索

V2-069666 | MATHDB-361515

Model 1 conjecture on the limiting pseudospectrum

摘录 / EXCERPT: Let n be the matrix size, let $m\geq 1$ be fixed, and let $S_{\delta J}^{(1)}(m)$ denote model 1. Let λ_0 be the distinguished point included in the pseudospectrum, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-069667 | MATHDB-361516

Scaling-limit conjecture for model 1 and model 2 spectra

摘录 / EXCERPT: Let n be the matrix size and let $m\leq n$ be the model parameter, with fixed parameters δ and a. Spectral scaling conjecture. For both model 1 and model 2, when n and...

conjecture | open | MathDB 页面 | 原文线索

V2-069668 | MATHDB-361521

Gamma-product evaluations for the E-determinant family

摘录 / EXCERPT: Gamma-product conjecture for the E-determinants. For all non-negative integers x and all $n\geq x$, the three displayed formulas in the source hold:

conjecture | open | MathDB 页面 | 原文线索

V2-069669 | MATHDB-361522

Recurrence conjecture for the polynomial factor in the F-determinants

摘录 / EXCERPT: Let $\operatorname{Pol}_n(x)$ be the monic polynomial of degree $2n-2$ occurring in the factorization of the determinant in Theorem MS1. Polynomial-factor recurrence conjecture. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-069670 | MATHDB-361523

Unshifted determinant evaluation with binomial indices divisible by four

摘录 / EXCERPT: Let n be a positive integer. The $4j$ determinant conjecture. ... ... This is one of the experimentally discovered determinant evaluations in the paper. The authors state that they...

conjecture | open | MathDB 页面 | 原文线索

V2-069671 | MATHDB-361534

Zero-freeness transfer to a neighborhood of zero for 2-spin systems

摘录 / EXCERPT: Fix parameters β and γ, not both zero, and let $Z^{\sigma_\Lambda}_{G}(\lambda)$ denote the partition function of the 2-spin system with boundary condition...

conjecture | open | MathDB 页面 | 原文线索

V2-069672 | MATHDB-361535

Zero-freeness transfer to a neighborhood of the Ising points

摘录 / EXCERPT: Fix a nonzero parameter λ, and let $Z^{\sigma_\Lambda}_{G}(\beta)$ denote the partition function of the Ising model with boundary condition σ_Λ. Suppose it is...

conjecture | open | MathDB 页面 | 原文线索

V2-069673 | MATHDB-361536

Plausibility-likelihood decomposition for binary product frames

摘录 / EXCERPT: Let ${\mathbb X}_1,\ldots,{\mathbb X}_n$ be binary frames with ${\mathbb X}_1=\cdots={\mathbb X}_n=\{T,F\}$, and let $(x_1,\ldots,x_n)$ be a singleton element of...

conjecture | open | MathDB 页面 | 原文线索

V2-069674 | MATHDB-361544

Conjecture on the trace description of the unit for closed symmetric semigroup categories

摘录 / EXCERPT: Trace-unit conjecture. The category $[\mathcal{S}^{\operatorname{op}},\mathbf{Vec}_{\Bbbk}]$ is symmetric monoidal, and its unit is equivalent to the trace of the underlying semigr...

conjecture | open | MathDB 页面 | 原文线索

V2-069675 | MATHDB-361545

Stevens's conjecture on the Manin constant for $X_1(N)$

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, and let $\mathrm{c}_1(E)$ be the Manin constant obtained from a modular parametrisation by the modular curve $\mathrm{X}_1(N)$ assoc...

conjecture | open | MathDB 页面 | 原文线索

V2-069676 | MATHDB-361550

Weighted triangle-free subcubic bisection conjecture

摘录 / EXCERPT: Triangle-free subcubic bisection conjecture. Every weighted triangle-free subcubic graph other than $K_{1,3}$ has a bisection of weight at least

conjecture | open | MathDB 页面 | 原文线索

V2-069677 | MATHDB-361551

Weighted Berge-Fulkerson matching conjecture for bridgeless cubic graphs

摘录 / EXCERPT: Weighted perfect-matching conjecture. There exists a perfect matching M such that

conjecture | open | MathDB 页面 | 原文线索

V2-069678 | MATHDB-361556

Self-affine tile digit-set stabilization conjecture

摘录 / EXCERPT: Let R be an $n\times n$ expanding integer matrix and let $0\in\mathcal D\subset\mathbb Z^n$ satisfy $\#\mathcal D=|\det(R)|$. Define $T(R,\mathcal D)$ to be the associated self-a...

conjecture | open | MathDB 页面 | 原文线索

V2-069679 | MATHDB-361560

The chromatic-number conjecture for clique immersions

摘录 / EXCERPT: Let G be a graph, let $\chi(G)$ be its chromatic number, and let K_k denote the complete graph on k vertices. Clique-immersion conjecture. If ... then G contains a K_k-imme...

conjecture | open | MathDB 页面 | 原文线索

V2-069680 | MATHDB-361570

Conjecture on the automorphism group of symmetric Reed-Muller codes

摘录 / EXCERPT: Automorphism-group conjecture. For $q>r>\frac{n(n-1)}{2}$, the automorphism group of $SRM_q[n,r]$ contains a subgroup isomorphic to M; equivalently, $SRM_q[n,r]$ is invariant und...

conjecture | open | MathDB 页面 | 原文线索

V2-069681 | MATHDB-361575

Congruence conjecture for initial RGB-tilings

摘录 / EXCERPT: Congruence conjecture. For every fixed $e\in E(EP)$, all possible initial RGB-tilings on $EP-\{e\}$ are congruent to each other.

conjecture | open | MathDB 页面 | 原文线索

V2-069682 | MATHDB-361589

Star-placement symmetry conjecture for knotoid potentials

摘录 / EXCERPT: Star-placement symmetry conjecture. The potentials satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069683 | MATHDB-361591

Kriesell's spanning-tree deletion conjecture

摘录 / EXCERPT: Kriesell's conjecture. There exists a smallest integer $g(k)$ such that every $g(k)$-connected graph G contains a spanning tree T for which

conjecture | open | MathDB 页面 | 原文线索

V2-069684 | MATHDB-361594

Hamilton-cycle conjecture for involutions of type D

摘录 / EXCERPT: Let I_n^D denote the involutions in the type-D signed permutation group, and let $G(\mathcal S}_n^B,X_1^D\cup X_2^D)$ be the Cayley graph with generating sets ... ... Consider the r...

conjecture | open | MathDB 页面 | 原文线索

V2-069685 | MATHDB-361595

The chromatic pairs polynomial refines the chromatic polynomial

摘录 / EXCERPT: Let G_1 and G_2 be graphs, and let $\pi_{G_i}^{(P_2)}(k)$ denote their chromatic pairs polynomials and $\pi_{G_i}(k)$ their chromatic polynomials. Chromatic-pairs refinement co...

conjecture | open | MathDB 页面 | 原文线索

V2-069686 | MATHDB-361597

The coloring-graph family is a complete graph invariant

摘录 / EXCERPT: For a finite graph G, let $\mathcal C_k(G)$ be its coloring graph for $k\geq 1$, and let $\pi_G^{(H)}$ count induced copies of a graph H in these coloring graphs. Coloring-grap...

conjecture | open | MathDB 页面 | 原文线索

V2-069687 | MATHDB-361598

Finitely many coloring graphs determine the graph

摘录 / EXCERPT: For a finite graph G, let $\mathcal C_k(G)$ be its coloring graph. Finite coloring-graph determination conjecture. There exists a function...

conjecture | open | MathDB 页面 | 原文线索

V2-069688 | MATHDB-361601

Locally analytic socle conjecture for ${\rm GL}_2(L)$

摘录 / EXCERPT: Let $S\subseteq\Sigma_L$, let $I^c_S(\alpha,\mathbf h)=C(1,s_J)$ be the corresponding locally analytic representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069689 | MATHDB-361607

The large-n higher Prym monodromy conjecture

摘录 / EXCERPT: Large-n higher Prym monodromy conjecture. There is a function $c(g,\dim\rho)$ such that, whenever $\Delta>c(g,\dim\rho)$, the image of $\operatorname{Mod}_\varphi$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-069690 | MATHDB-361610

The continuous Babai conjecture for compact simple groups

摘录 / EXCERPT: Continuous Babai conjecture. There is an integer m such that

conjecture | open | MathDB 页面 | 原文线索

V2-069691 | MATHDB-361618

Conjecture on monotonicity of the linearized population mean for grouped exponential data

摘录 / EXCERPT: Monotonicity conjecture. The function $g_{tT}(\theta)$ is strictly increasing.

conjecture | open | MathDB 页面 | 原文线索

V2-069692 | MATHDB-361619

M-convexity conjecture for k-Schur polynomials

摘录 / EXCERPT: Let k and d be positive integers, and let $\lambda\in \mathrm{Par}^k(d)$ be a partition. The k-Schur polynomial associated with λ is denoted by $s_{\lambda}^{(k)}$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-069693 | MATHDB-361620

Lorentzianity conjecture for normalized dual k-Schur polynomials

摘录 / EXCERPT: Let $f=\sum_{\alpha\in\mathbb{N}^n}c_{\alpha}x^{\alpha}$ be a polynomial, and define its normalization by ... For $\lambda\in \mathrm{Par}^k(d)$, let $\mathfrak{S}_{\lambda}^{(k)}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-069694 | MATHDB-361621

The local total antimagic bound for graphs with few pendant edges

摘录 / EXCERPT: Let G be a graph with k pendant edges, let $\Delta(G)$ be its maximum degree, and let $\chi_{lt}(G)$ denote its local total antimagic chromatic number. Local total antimagic bo...

conjecture | open | MathDB 页面 | 原文线索

V2-069695 | MATHDB-361622

The inactive sum-rate bound condition for sliding secure multilevel diversity coding

摘录 / EXCERPT: Let L and s be the parameters of the sliding secure symmetric multilevel diversity coding problem, let X_α denote the source associated with level α, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069696 | MATHDB-361625

Conjecture on the zeros of the Mittag-Leffler function with second parameter 2

摘录 / EXCERPT: Smallest-zero conjecture for $E_{\alpha,2}$. For each $\alpha\in[\alpha_0,2]$, the equation

conjecture | open | MathDB 页面 | 原文线索

V2-069697 | MATHDB-361631

Strictification conjecture for linear fibrational double categories

摘录 / EXCERPT: A linear fibrational double category (LFDC) is a categorical structure with the substructural type-theoretic features described in the source, including type-level weakening, funct...

conjecture | open | MathDB 页面 | 原文线索

V2-069698 | MATHDB-361634

Diameter conjecture for large subsets of classical compact groups

摘录 / EXCERPT: Let G be one of $\mathrm{SU}(n)$, $\mathrm{SO}(n)$, $\mathrm{Spin}(n)$, or $\mathrm{Sp}(n)$, and let $\mathcal{A}\subseteq G$ be measurable with measure ν. The diameter with...

conjecture | open | MathDB 页面 | 原文线索

V2-069699 | MATHDB-361643

Duality conjecture for recovery balanced codes

摘录 / EXCERPT: Duality conjecture. A code $\mathcal{C}$ is recovery balanced if and only if its dual code $\mathcal{C}^\perp$ is recovery balanced.

conjecture | open | MathDB 页面 | 原文线索

V2-069700 | MATHDB-361645

Characterization of ultrametric-preserving functions for unbounded proper ultrametric spaces

摘录 / EXCERPT: Let $f: [0,\infty) \to [0,\infty)$ be a function. Let $\mathbf{A}$ be the class of all unbounded boundedly compact (proper) ultrametric spaces, and let $\mathbf{P}_{\mathbf{A}}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-069701 | MATHDB-361647

The Non-Cancelling Intersections Conjecture, formulation II

摘录 / EXCERPT: Non-Cancelling Intersections Conjecture, formulation II. For every tight intersection lattice L,

conjecture | open | MathDB 页面 | 原文线索

V2-069702 | MATHDB-361669

Stability of blow-ADE singularities under μ*-constant deformations

摘录 / EXCERPT: Let f define a general blow-ADE singularity of blow-order $m\geq 1$ at $\mathbf{0}$. Let $\{f_s\}$, $0\leq s\leq 1$, be a μ^*-constant piecewise complex-analytic family sta...

conjecture | open | MathDB 页面 | 原文线索

V2-069703 | MATHDB-361691

Lu-Wenzel SOS conjecture for real Toeplitz matrices

摘录 / EXCERPT: Let X and Y be arbitrary real Toeplitz matrices, let $\langle X,Y\rangle$ denote their Frobenius inner product, and let $\|\cdot\|$ denote the Frobenius norm. Lu-Wenzel conject...

conjecture | open | MathDB 页面 | 原文线索

V2-069704 | MATHDB-361692

Lu-Wenzel constrained DDVV conjecture

摘录 / EXCERPT: Let $\mathbb{K}=\mathbb{R}$ and let $B_1,\ldots,B_m\in M(n,\mathbb{K})$, where $M(n,\mathbb{K})$ is the space of $n\times n$ matrices over $\mathbb{K}$. Let $[A,B]=AB-BA$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069705 | MATHDB-361693

Fundamental Lu-Wenzel conjecture

摘录 / EXCERPT: Let $\mathbb{K}=\mathbb{R}$ and let $B,B_2,\ldots,B_m\in M(n,\mathbb{K})$. Let $B^* $ denote the conjugate transpose, let $\bot$ denote orthogonality for the Frobenius inner produc...

conjecture | open | MathDB 页面 | 原文线索

V2-069706 | MATHDB-361694

Lu-Wenzel eigenvalue conjecture

摘录 / EXCERPT: Let $T_X:M(n,\mathbb{C})\to M(n,\mathbb{C})$ be defined by $T_X(Y)=[X^*,[X,Y]]$, and let $\lambda_1(T_X)\geq\cdots\geq\lambda_N(T_X)$ be its eigenvalues. Lu-Wenzel eigenvalue conje...

conjecture | open | MathDB 页面 | 原文线索

V2-069707 | MATHDB-361695

Ge-Li-Lu-Zhou even-eigenvalue-sum conjecture

摘录 / EXCERPT: Let $T_X:M(n,\mathbb{C})\to M(n,\mathbb{C})$ be defined by $T_X(Y)=[X^*,[X,Y]]$, and let $\lambda_1(T_X)\geq\cdots\geq\lambda_N(T_X)$ be its eigenvalues. Ge-Li-Lu-Zhou conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069708 | MATHDB-361696

Ge-Li-Lu-Zhou factor-two constrained commutator conjecture

摘录 / EXCERPT: Let $\mathbb{K}=\mathbb{R}$ or $\mathbb{K}=\mathbb{C}$, and let $B,B_2,\ldots,B_m\in M(n,\mathbb{K})$. Let B^* denote the conjugate transpose and let $[A,B]=AB-BA$. Assume ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-069709 | MATHDB-361699

Hull-dimension monotonicity hypothesis for binary and ternary linear codes

摘录 / EXCERPT: Let $B_{n,k,\ell,q}$ be the number of inequivalent linear codes of length n, dimension k, and hull dimension ℓ over $\mathbb{F}_q$. Hull-dimension monotonicity hypothesis....

conjecture | open | MathDB 页面 | 原文线索

V2-069710 | MATHDB-361723

Linear independence conjecture for higher-order blow-up scalar fields

摘录 / EXCERPT: Let V be a set with one Poisson particle source for each element, and let a degree-r arrival experiment receive particles in successive batches of r, silencing each source af...

conjecture | open | MathDB 页面 | 原文线索

V2-069711 | MATHDB-361724

Higher-order blow-up differential-form complex conjecture

摘录 / EXCERPT: Let T be a simplex, let $\widetilde{T}$ denote its blow-up, and let $b\mathcal P_1^-\Lambda^k$ be the shadow-form or blow-up Whitney-form complex and $b\mathcal P_r\Lambda^0$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-069712 | MATHDB-361729

Conjecture on poles of abelian extension counting Dirichlet series

摘录 / EXCERPT: Pole-order conjecture. The series $D_{\Q,G}(s)$ should have a meromorphic continuation to

conjecture | open | MathDB 页面 | 原文线索

V2-069713 | MATHDB-361734

Classical sufficiency conjecture for the exceptional diagram category

摘录 / EXCERPT: Let R be a ring and let $\boldsymbol{\lambda}$ be a parameter such that the category $\mathsf{Exc}_{R,\lambda}$ is defined; write $\mathsf{Exc}_{R,\lambda}(n)$ for its space of d...

conjecture | open | MathDB 页面 | 原文线索

V2-069714 | MATHDB-361735

Classical consistency conjecture for the exceptional diagram category

摘录 / EXCERPT: Let R be a ring and let $\mathsf{Exc}_{R,\lambda}$ be the classical exceptional diagram category, with $\mathsf{Exc}_{R,\lambda}(0)$ its closed-diagram space and the empty diagra...

conjecture | open | MathDB 页面 | 原文线索

V2-069715 | MATHDB-361736

Classical semisimplicity conjecture for the exceptional series

摘录 / EXCERPT: Let U be a subset of $\mathbb C$, and let the Cauchy completion mean the additive and idempotent completion of a category. Classical semisimplicity conjecture. There is a dense s...

conjecture | open | MathDB 页面 | 原文线索

V2-069716 | MATHDB-361745

The type-BC indifference graph equivalence conjecture for Kazhdan-Lusztig basis elements

摘录 / EXCERPT: Let $v,w\in\mathfrak{B}_n$ avoid the patterns 3412 and 4231. Write $\Gamma(v)$ and $\Gamma(w)$ for their type-BC indifference graphs, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069717 | MATHDB-361747

Asgarli-Krehbiel-Levinson-Russell finite colouring-graph determination conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ be the set of finite graphs. For a graph G and $k\in\mathbb{N}$, let $\mathcal{C}_k(G)$ be the graph whose vertices are the proper k-colourings of G, with e...

conjecture | open | MathDB 页面 | 原文线索

V2-069718 | MATHDB-361755

The continuity conjecture for the optimal negative coefficient bound

摘录 / EXCERPT: Continuity conjecture. The largest $t\in\mathbb{R}^+$ such that $p_t\in\mathcal{P}_n$ for all such m,s is a continuous function of m and s, bounded between 1 and 2. The c...

conjecture | open | MathDB 页面 | 原文线索

V2-069719 | MATHDB-361757

The finite quadratic-form characterization conjecture for matrix-preserving polynomials

摘录 / EXCERPT: Let $\mathcal{P}_{n,2n}$ be the cone of polynomials of degree at most $2n$ that preserve nonnegative matrices of order n, and let $\mathbf{v}$ denote the coefficient vector of su...

conjecture | open | MathDB 页面 | 原文线索

V2-069720 | MATHDB-361762

The satellite-operator dichotomy conjecture

摘录 / EXCERPT: Satellite-operator dichotomy conjecture. Every satellite operator $P\colon \mathcal{C}\longrightarrow \mathcal{C}$ is constant or has image generating an infinite-rank subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-069721 | MATHDB-361766

Large ensemble decoupling for the paraboloid

摘录 / EXCERPT: Let R be a large parameter, let B_R be a ball or cube of radius or side length R in ${\mathbb R}^3$, and let E_SF denote the Fourier extension operator associated to the tw...

conjecture | open | MathDB 页面 | 原文线索

V2-069722 | MATHDB-361767

Sharp small-ensemble decoupling for the paraboloid

摘录 / EXCERPT: Let R be a large parameter, let B_R be a ball or cube of radius or side length R in ${\mathbb R}^3$, and let E_SF be the extension operator for the two-dimensional parabolo...

conjecture | open | MathDB 页面 | 原文线索

V2-069723 | MATHDB-361768

Characterization of totally positive sequence preservers

摘录 / EXCERPT: The TP_∞-preserver conjecture. The sequence ${\mathbf A}$ is a TP_∞-preserver if and only if, for every $l\in\mathbb{N}\cup\{0\}$, the formal power series

conjecture | open | MathDB 页面 | 原文线索

V2-069724 | MATHDB-361771

Informal stable-CSBP and genealogy convergence conjecture for spatial branching processes

摘录 / EXCERPT: Let $\tilde h^\tinfty$ denote the invariant spatial distribution of particles, and suppose that, for some slowly varying function ℓ and $alpha\backslashin(1,2)$, condition ... h...

conjecture | open | MathDB 页面 | 原文线索

V2-069725 | MATHDB-361781

Asymptotic threshold conjecture for wide-sense frameproof codes

摘录 / EXCERPT: Wide-sense frameproof threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069726 | MATHDB-361782

Grünbaum's conjecture on spanning trees and co-trees of planar graphs

摘录 / EXCERPT: Grünbaum's conjecture. Every planar 3-connected graph G contains a 3-tree T whose co-tree $\neg T^*$ is also a 3-tree.

conjecture | open | MathDB 页面 | 原文线索

V2-069727 | MATHDB-361784

Polytopality conjecture for hyperbolic posets

摘录 / EXCERPT: Let F be a real-rooted polynomial and let $\mathcal{H}_s(F)$ be its hyperbolic slice, stratified by root multiplicity compositions; the associated hyperbolic poset records the in...

conjecture | open | MathDB 页面 | 原文线索

V2-069728 | MATHDB-361785

Polytope-complex conjecture for boundaries of hyperbolic posets

摘录 / EXCERPT: Let F be a real-rooted polynomial, let $\mathcal{L}_s(F)$ be the lattice associated with its hyperbolic slice, and let $\mathcal{L}^\Delta_s(F)$ denote its dual; write...

conjecture | open | MathDB 页面 | 原文线索

V2-069729 | MATHDB-361790

Wreath Macdonald norm conjecture

摘录 / EXCERPT: Wreath Macdonald norm conjecture. The wreath Macdonald polynomials satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-069730 | MATHDB-361792

The matching characterization of the proper Z4 x Z2-coloring conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, let F be a 2-factor of G, and let M be a matching in F. Put $H=G-M$. An F-matching is the matching notion defined in the paper, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069731 | MATHDB-361794

The chordality characterization conjecture for homaloidal spanning tree polynomials

摘录 / EXCERPT: Let G be an undirected graph, and let P_G denote the spanning tree generating function corresponding to G. A polynomial is homaloidal if its polar map is birational. Chordali...

conjecture | open | MathDB 页面 | 原文线索

V2-069732 | MATHDB-361795

Random Steiner quasigroup row-cycle and subsystem conjecture

摘录 / EXCERPT: Let S be a Steiner quasigroup of order n selected uniformly at random. Let $\ell(S)$ be the length of its longest row cycle, and let $m(S)$ be the size of its largest proper St...

conjecture | open | MathDB 页面 | 原文线索

V2-069733 | MATHDB-361796

Random 1-factorisation substructure conjecture

摘录 / EXCERPT: Let $\mathcal{U}$ be the Latin square representing a uniformly random 1-factorisation of K_{2n}. Let $\ell(\mathcal{U})$ be the length of its longest row cycle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069734 | MATHDB-361797

Longest row-cycle conjecture for random Latin squares

摘录 / EXCERPT: Let L be a Latin square of order n, and let $\ell(L)$ denote the length of its longest row cycle. Longest row-cycle conjecture. As $n\to\infty$, asymptotically almost all Latin...

conjecture | open | MathDB 页面 | 原文线索

V2-069735 | MATHDB-361824

Symmetry equality conjecture for König maps

摘录 / EXCERPT: Let p be a normalized polynomial, let $n\geq 2$, and let $K_{p,n}$ be the König root-finding map. Write Σp for the symmetry group associated with p and $\Sigma K_{p,n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-069736 | MATHDB-361825

Containment conjecture for spectra of universal covers and limiting eigenvalue supports

摘录 / EXCERPT: Containment conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069737 | MATHDB-361827

Jetchev-Stein modularity conjecture for classes in Tate-Shafarevich groups

摘录 / EXCERPT: Jetchev-Stein conjecture. Every class in $\Sh(E/\mathbf{Q})$ is modular.

conjecture | open | MathDB 页面 | 原文线索

V2-069738 | MATHDB-361828

The characterization of pure-type UBCMPGs

摘录 / EXCERPT: Pure-type UBCMPG conjecture. An MPG is a pure-type UBCMPG if and only if it is the graph shown in Figure.

conjecture | open | MathDB 页面 | 原文线索

V2-069739 | MATHDB-361829

Strong Ext relevance conjecture for non-tempered Arthur representations

摘录 / EXCERPT: Let F be a non-archimedean local field. Let π_1 and π_2 be Arthur type representations of $\operatorname{GL}_n(F)$ and $\operatorname{GL}_{n-1}(F)$, respectively. Say tha...

conjecture | open | MathDB 页面 | 原文线索

V2-069740 | MATHDB-361830

Prasad-Chan same-group Ext conjecture for Arthur type representations

摘录 / EXCERPT: Let F be a non-archimedean local field, let $\Delta\operatorname{SL}_2$ denote the diagonally embedded copy of SL_2 in...

conjecture | open | MathDB 页面 | 原文线索

V2-069741 | MATHDB-361843

Frobenius liftability conjecture for cusp singularities

摘录 / EXCERPT: Frobenius liftability conjecture. Every cusp singularity is F-liftable.

conjecture | open | MathDB 页面 | 原文线索

V2-069742 | MATHDB-361844

Finite Euler sums and symmetric Euler sums correspondence

摘录 / EXCERPT: Let $\bbN$ denote the positive integers. For each weight $w\in\mathbb{N}$, let FES_w be the $\mathbb{Q}$-vector space generated by finite Euler sums of weight w, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-069743 | MATHDB-361845

Conjectural relations among finite multiple-value subspaces

摘录 / EXCERPT: For each weight w, let FMZV_w, $\mathsf{FMZV}^{(2)}_w$, FMSV_w, FES_w, FMTV_w, FMMVe_w, FMMV_w,...

conjecture | open | MathDB 页面 | 原文线索

V2-069744 | MATHDB-361846

Conjectural dimensions of finite multiple mixed-value subspaces

摘录 / EXCERPT: Let F_w denote the w-th Fibonacci number, and let the symbols $\mathsf{FMZV}^{(2)}_w$, FMMVe_w, FMMV_w, FES_w, FMMVo_w,...

conjecture | open | MathDB 页面 | 原文线索

V2-069745 | MATHDB-361848

Anti-Sidorenko orientations conjecture for trees

摘录 / EXCERPT: Anti-Sidorenko orientations conjecture for trees. For every undirected tree, there exists some orientation that has the tournament anti-Sidorenko property. The paper says that this...

conjecture | open | MathDB 页面 | 原文线索

V2-069746 | MATHDB-361849

Gorodnik-Varjú's super approximation conjecture

摘录 / EXCERPT: Let G be a finitely generated subgroup of $\mathrm{SL}_n(\mathbb{Z})$. Write $\mathbb{G}$ for the Zariski closure of G, and let $\mathbb{G}_0$ be its identity component. The gr...

conjecture | open | MathDB 页面 | 原文线索

V2-069747 | MATHDB-361859

Complete classification conjecture for critical nonlocal quasi-linear equation

摘录 / EXCERPT: Complete classification conjecture. Either $u\equiv 0$, or there exist $\lambda:=u(0)^{\frac{p}{N-p}}>0$ and $x_{0}\in\mathbb{R}^{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069748 | MATHDB-361861

Bipartite monochromatic k-connection conjecture

摘录 / EXCERPT: Let G be a k-connected bipartite graph with partition classes of sizes s and t, where $t\ge s\ge k\ge 2$, and let $H\subset G$ be a minimum spanning k-connected subgraph....

conjecture | open | MathDB 页面 | 原文线索

V2-069749 | MATHDB-361862

Complete bipartite graph monochromatic k-connection conjecture

摘录 / EXCERPT: Let $K_{s,t}$ be the complete bipartite graph with parts of sizes s and t. Complete bipartite monochromatic k-connection conjecture. For $t\ge s\ge k\ge 2$, ... and, for...

conjecture | open | MathDB 页面 | 原文线索

V2-069750 | MATHDB-361863

Uniqueness conjecture for power integral basis generators in totally complex pure octic fields

摘录 / EXCERPT: Let $0\ne m\in\mathbb Z$ be a rational integer satisfying ()-that is, $m\ne 1$ is square-free and $m\not\equiv 1\pmod 4$-with $m<0$. Set $K=\mathbb Q(\sqrt[8]{m})$ and identify gen...

conjecture | open | MathDB 页面 | 原文线索

V2-069751 | MATHDB-361867

Undecidability conjecture for rapidly growing generalised polynomials

摘录 / EXCERPT: Undecidability conjecture. The first-order theory of $(\mathbb{Z};<,+,g)$ is undecidable.

conjecture | open | MathDB 页面 | 原文线索

V2-069752 | MATHDB-361877

Liouville theorem for the free boundary Monge-Ampère equation

摘录 / EXCERPT: Let l,m,k be parameters and let $\varphi:(\mathbb R_+)^l\times \mathbb R^{m}\to \mathbb R_{\geq 0}$ solve ... in $(\mathbb R_+)^l\times \mathbb R^{m}$, with ... on...

conjecture | open | MathDB 页面 | 原文线索

V2-069753 | MATHDB-361887

1-2-3 Conjecture on neighbor-sum-distinguishing edge-weightings

摘录 / EXCERPT: Let G be a graph with no isolated edge. A neighbor-sum-distinguishing (nsd) 3-edge-weighting is a mapping from $E(G)$ to $\{1,2,3\}$ such that the sums of the weights incident...

conjecture | open | MathDB 页面 | 原文线索

V2-069754 | MATHDB-361889

Alt-GDA convergence lower-bound conjecture for non-quadratic objectives

摘录 / EXCERPT: Let ${\mathcal F}(\mu_x,\mu_y,L_x,L_y,L_{xy})$ denote the function class considered for the alternating gradient descent-ascent method, and let κ_x, κ_y, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069755 | MATHDB-361890

Alex-GDA last-iterate convergence conjecture for convex-concave objectives

摘录 / EXCERPT: Let f be a convex-concave objective function with (L_x,L_y,L_{xy})-Lipschitz gradients, and let Alex-GDA denote the extrapolated alternating gradient descent-ascent method. Ale...

conjecture | open | MathDB 页面 | 原文线索

V2-069756 | MATHDB-361893

The sigma-weight conjecture for binary linear codes

摘录 / EXCERPT: Let C be a binary linear code, and let σ be a permutation in its permutation automorphism group $\operatorname{PAut}(C)$. Let $E_\sigma(C)$ denote the associated σ-...

conjecture | open | MathDB 页面 | 原文线索

V2-069757 | MATHDB-361903

Pach-Saghafian-Schnider lower-bound conjecture for plane star-forest decompositions

摘录 / EXCERPT: A complete geometric graph is a complete graph drawn with vertices in general position and straight-line edges. A plane k-star-forest is a star-forest with at most k connected...

conjecture | open | MathDB 页面 | 原文线索

V2-069758 | MATHDB-361904

Lower-bound conjecture for plane star-forest decompositions of complete geometric graphs

摘录 / EXCERPT: A complete geometric graph is a complete graph drawn with vertices in general position and straight-line edges. A plane star-forest is a forest whose connected components are stars...

conjecture | open | MathDB 页面 | 原文线索

V2-069759 | MATHDB-361905

Convex-hull-size conjecture for optimal star-forest decompositions

摘录 / EXCERPT: Let $n\ge3$ be odd, and let G be a complete geometric graph on n vertices that can be decomposed into $\left\lceil\frac{n}{2}\right\rceil+1$ plane star-forests. Convex-hull-siz...

conjecture | open | MathDB 页面 | 原文线索

V2-069760 | MATHDB-361909

Dallard et al.'s equivalence conjecture for tree-independence number

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary graph class. A graph class is (tw,ω)-bounded if its treewidth is bounded by a function of its clique number, and it has bounded tr...

conjecture | open | MathDB 页面 | 原文线索

V2-069761 | MATHDB-361911

The induced-grid conjecture for tree-independence number

摘录 / EXCERPT: The induced-grid conjecture. The class $\mathcal{C}$ has bounded tree-independence number if and only if $\widehat{\mathcal{C}}$ does not contain all planar graphs. This is propose...

conjecture | open | MathDB 页面 | 原文线索

V2-069762 | MATHDB-361912

The wall-subdivision formulation of the induced-grid conjecture

摘录 / EXCERPT: Let d be a positive integer and let $\mathcal{C}$ be a hereditary graph class excluding $K_{1,d}$. A subdivision of the elementary k-wall is obtained by subdividing edges of th...

conjecture | open | MathDB 页面 | 原文线索

V2-069763 | MATHDB-361930

Weakened Minimal Resolution Conjecture for points in \(\mathbb P^1\times\mathbb P^2\)

摘录 / EXCERPT: Weakened Minimal Resolution Conjecture. The set X has a generic Hilbert matrix as in the source definition, and, for every fixed $(i,j)>(0,0)$, one has $\beta_{1,(i,j)}>0$ if and...

conjecture | open | MathDB 页面 | 原文线索

V2-069764 | MATHDB-361934

Constant-additive coloring conjecture for pseudoline arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be an arrangement of pseudolines, and let $\operatorname{mx}(\mathcal{A})$ denote the maximum number of crossings on any pseudoline. A coloring of the crossings i...

conjecture | open | MathDB 页面 | 原文线索

V2-069765 | MATHDB-361936

Colliot-Thélène-Sansuc-Kato-Saito conjecture for zero-cycles

摘录 / EXCERPT: Colliot-Thélène-Sansuc-Kato-Saito conjecture. The complex

conjecture | open | MathDB 页面 | 原文线索

V2-069766 | MATHDB-361938

Hong-Kang-Yu induced-cycle conjecture

摘录 / EXCERPT: Let m and l be positive integers. A graph is r-connected if it remains connected after deletion of any set of fewer than r vertices. Let G be a graph, let e be an edge...

conjecture | open | MathDB 页面 | 原文线索

V2-069767 | MATHDB-361939

Bourgain-Rudnick-Sarnak variance conjecture for lattice points on the sphere

摘录 / EXCERPT: Let $\mathcal N=\{n\in\mathbb N:n\not\equiv 0,4,7\pmod{8}\}$, let $\mathcal E(n)=\{\mathbf{x}\in\mathbb Z^3:|\mathbf{x}|^2=n\}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069768 | MATHDB-361942

The largeness conjecture for gluings of complex analytic space germs

摘录 / EXCERPT: Largeness conjecture. Every gluing

conjecture | open | MathDB 页面 | 原文线索

V2-069769 | MATHDB-361945

Lawson surfaces minimize Morse index among minimal surfaces of fixed genus

摘录 / EXCERPT: Let M be a closed minimal surface in $\mathbb{S}^3$ of genus γ, and let $\operatorname{ind}(M)$ denote its Morse index. Let $\xi_{\gamma,1}$ be the Lawson surface, whose i...

conjecture | open | MathDB 页面 | 原文线索

V2-069770 | MATHDB-361946

Critical catenoid uniqueness for first-Steklov-eigenfunction embeddings

摘录 / EXCERPT: Let N be an orientable surface with two boundary components, and let $\Psi:(N,g)\to\mathbb{B}^3$ be a free boundary minimal embedding by $\sigma_1(N,g)$-eigenfunctions, with mult...

conjecture | open | MathDB 页面 | 原文线索

V2-069771 | MATHDB-361947

Linear growth of distinct free boundary minimal embeddings

摘录 / EXCERPT: Let $N_{\gamma,k}$ be the compact orientable surface of genus γ with k boundary components. Embedding-growth conjecture. The number of distinct embeddings of...

conjecture | open | MathDB 页面 | 原文线索

V2-069772 | MATHDB-361954

Quantitative Ryser conjecture for circulant matrices

摘录 / EXCERPT: Let $n>4$, let A be an $n\times n$ circulant matrix with entries in $\{-1,1\}$, and let $\|\cdot\|$ denote the Euclidean norm on $\mathbb R^n$. Quantitative Ryser conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-069773 | MATHDB-361962

Planchon's space-time conjecture for the Schrödinger equation

摘录 / EXCERPT: Planchon's space-time conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069774 | MATHDB-361963

Lee-Rogers-Seeger local space-time conjecture

摘录 / EXCERPT: Lee-Rogers-Seeger local space-time conjecture. The estimate

conjecture | open | MathDB 页面 | 原文线索

V2-069775 | MATHDB-361964

Bilinear space-time estimate conjecture

摘录 / EXCERPT: Bilinear space-time estimate conjecture. For $2\leq p,r<\infty$, the estimate

conjecture | open | MathDB 页面 | 原文线索

V2-069776 | MATHDB-361966

Uniform two-thirds decay conjecture for the discrete Klein-Gordon oscillatory integral

摘录 / EXCERPT: Uniform two-thirds decay conjecture. For any $\zeta\in\mathcal{C}_c^{\infty}(\mathbb{R}^d)$,

conjecture | open | MathDB 页面 | 原文线索

V2-069777 | MATHDB-361967

Strong semigroup version of Ih's finiteness conjecture

摘录 / EXCERPT: Let K be a number field and let S be a finite set of places of K containing all archimedean ones. Let $f_i:\mathbb{P}^1\to\mathbb{P}^1$, $i\in I$, be rational maps of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-069778 | MATHDB-361968

Ih's finiteness conjecture for preperiodic points of rational-function semigroups

摘录 / EXCERPT: Let K be a number field and let S be a finite set of places of K containing all archimedean ones. Let $f_1,\ldots,f_s:\mathbb{P}^1\to\mathbb{P}^1$ be rational maps of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-069779 | MATHDB-361970

Bouillot's multitangent function decomposition conjecture

摘录 / EXCERPT: For an index $\bm{k}=(k_{1},\dots,k_{r})$, its weight is $k_{1}+\cdots+k_{r}$. Let MTGF_{w} be the $\mathbb{Q}$-vector space generated by multitangent functions of weigh...

conjecture | open | MathDB 页面 | 原文线索

V2-069780 | MATHDB-361972

Full stress-rank conjecture for generic global (d,k)-rigidity

摘录 / EXCERPT: Full stress-rank conjecture. If (G,p) is globally (d,k)-rigid and G is not complete, then there exists $\sigma \in \ker DR_k(G,p)^T$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069781 | MATHDB-361974

Generalized Dumas-Eisenstein irreducibility conjecture

摘录 / EXCERPT: Generalized Dumas-Eisenstein irreducibility conjecture. The polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-069782 | MATHDB-361975

Sintiari-Trotignon conjecture on treewidth of diamond-free even-hole-free graphs

摘录 / EXCERPT: For a graph G, let $\operatorname{tw}(G)$ denote its treewidth. A graph is even-hole-free if it has no induced cycle of even length at least four, and the diamond is the graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-069783 | MATHDB-361980

Buryak-Shadrin tautological vanishing conjecture

摘录 / EXCERPT: Let $g,m\geq 0$, $n\geq 1$, and let $\textbf{d}=(d_1,\ldots,d_n)\in(\Bbb Z_{\geq 0})^n$, with $|\textbf{d}|:=d_1+\cdots+d_n$. For a nondegenerate balanced genus-g stable rooted t...

conjecture | open | MathDB 页面 | 原文线索

V2-069784 | MATHDB-361998

Exact minimal normalized stretch factor conjecture for orientation-reversing fully-punctured pseudo-Anosov maps

摘录 / EXCERPT: Exact minimal stretch factor conjecture. The normalized stretch factor of f is greater than or equal to the largest real root of

conjecture | open | MathDB 页面 | 原文线索

V2-069785 | MATHDB-361999

The CTI-ITC polynomial equality conjecture for complete split graphs

摘录 / EXCERPT: Let $S_{n,d}$ be the complete split graph, and let $\mathfrak{F}_{n,d}^{CTI}(q,t)$ and $\mathfrak{F}_{n,d}^{ITC}(q,t)$ denote the q,t-CTI and q,t-ITC polynomials, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-069786 | MATHDB-362000

The CTI-Schröder polynomial equality conjecture

摘录 / EXCERPT: Let $S_{n,d}$ be the complete split graph, let $\mathfrak{F}_{n,d}^{CTI}(q,t)$ be the q,t-CTI polynomial, and let $\mathsf{Schr\ddot{o}der}_{n,d}$ be the set of (n,d)-Schröder...

conjecture | open | MathDB 页面 | 原文线索

V2-069787 | MATHDB-362001

A statistic-preserving CTI-ITC bijection conjecture for sorted recurrent configurations

摘录 / EXCERPT: Let $S_{n,d}$ be the complete split graph, and let $\mathsf{SortedRec}(S_{n,d})$ be the set of sorted recurrent configurations on $S_{n,d}$. For a configuration c, write...

conjecture | open | MathDB 页面 | 原文线索

V2-069788 | MATHDB-362009

Chi-boundedness conjecture for graphs of bounded induced matching treewidth

摘录 / EXCERPT: Chi-boundedness conjecture. For any two integers $k,c\in\mathbb{N}$ there exists an integer r such that every graph with induced matching treewidth at most k and clique number...

conjecture | open | MathDB 页面 | 原文线索

V2-069789 | MATHDB-362028

Coxeter-length difference under the Simion-Schmidt map

摘录 / EXCERPT: Coxeter-length difference conjecture. The difference in Coxeter length of a permutation and its image under the Simion-Schmidt map (Statistic 123) exhibits the cyclic sieving phe...

conjecture | open | MathDB 页面 | 原文线索

V2-069790 | MATHDB-362029

Weak-excedance mid-points and cycle descents

摘录 / EXCERPT: Weak-excedance mid-point conjecture. The number of weak excedances that are also mid-points of a decreasing subsequence of length 3 (Statistic 373) is equidistributed with the...

conjecture | open | MathDB 页面 | 原文线索

V2-069791 | MATHDB-362030

Short-distance inversion conjecture for even permutations

摘录 / EXCERPT: Short-distance inversion conjecture. For $n\geq 2$ even, the number of inversions of distance at most 3 (Statistic 494) exhibits the cyclic sieving phenomenon under involutions...

conjecture | open | MathDB 页面 | 原文线索

V2-069792 | MATHDB-362037

Generalized conjecture on v-numbers of graded ideals with linear powers

摘录 / EXCERPT: Let S be a standard graded polynomial ring and let $I\subset S$ be a graded ideal with linear powers, meaning that every power I^k has a linear free resolution. Let $\alpha(I)$...

conjecture | open | MathDB 页面 | 原文线索

V2-069793 | MATHDB-362043

Active common randomness outside the no-signaling polytope

摘录 / EXCERPT: Active-common-randomness conjecture. If $P(X,Y\mid A,B)$ has the displayed active-common-randomness factorization and $P(X,Y\mid A,B)\notin\mathscr{L}$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-069794 | MATHDB-362045

The torus-generation conjecture for Fukaya categories of Grassmannians

摘录 / EXCERPT: Let $\mathcal{A}(k,n)$ be the maximal positroid stratum in $\operatorname{Gr}(k,n)$, and let $\mathbf{D}\operatorname{Fuk}_\lambda(\operatorname{Gr}(k,n))$ denote the Fukaya catego...

conjecture | open | MathDB 页面 | 原文线索

V2-069795 | MATHDB-362049

Uniqueness of the unitary group representation minimizer for HyperCube regularization

摘录 / EXCERPT: HyperCube regularizer minimizer conjecture. The unitary group representation $A_g=B_g=C_g^\intercal=\varrho(g)$ describes the unique minimizer of $\mathcal{H}$ up to unitary basis...

conjecture | open | MathDB 页面 | 原文线索

V2-069796 | MATHDB-362050

Looijenga's conjecture on the smooth Torelli group of Enriques surfaces

摘录 / EXCERPT: Looijenga's conjecture. The smooth Torelli group is trivial:

conjecture | open | MathDB 页面 | 原文线索

V2-069797 | MATHDB-362053

Sharp global-existence condition for the supercooled Stefan problem

摘录 / EXCERPT: Let η_0 be the initial data, with initial domain ${\eta_0>0}$. Condition (C) means that η_0 is strictly subharmonically ordered with ν with respect to ${\eta_0>0}$...

conjecture | open | MathDB 页面 | 原文线索

V2-069798 | MATHDB-362056

Gaiotto's positivity and filtered nondegeneracy conjecture

摘录 / EXCERPT: Let $V=\mathbb C[z_1,\dots,z_n]$ be filtered by polynomial degree, with $F_NV=\mathbb C[z_1,\dots,z_n]_{\le N}$. Let $W\in S^d\mathbb C^{n*}$ be a homogeneous polynomial of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-069799 | MATHDB-362057

Gaiotto's nonvanishing integral conjecture

摘录 / EXCERPT: Gaiotto's conjecture. The number $I(W)$ is strictly positive for all W, or equivalently, nonzero for all W. This is the $N=0$ case of part (ii) of the preceding conjecture, so...

conjecture | open | MathDB 页面 | 原文线索

V2-069800 | MATHDB-362061

Conjecture on global rigidity of 2-edge-apex graphs

摘录 / EXCERPT: Let $G=(V,E)$ be a graph. A framework is globally $\mathcal{R}_3$-rigid if every framework in $\mathbb{R}^3$ with the same edge lengths differs from it by a composition of isometri...

conjecture | open | MathDB 页面 | 原文线索

V2-069801 | MATHDB-362065

Pouzet's conjecture on totally ordering omega-well structured relational structures

摘录 / EXCERPT: Pouzet's conjecture. Every ω-well structured relational structure $\mathcal{A}$ is a reduct of an ω-well structured, totally ordered structure.

conjecture | open | MathDB 页面 | 原文线索

V2-069802 | MATHDB-362075

Yuster's triangle-packing conjecture for random graphs

摘录 / EXCERPT: Yuster's triangle-packing conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-069803 | MATHDB-362076

Yuster's triangle-decomposition conjecture for random regular graphs

摘录 / EXCERPT: Yuster's triangle-decomposition conjecture. If $d\gg\sqrt{n\log n}$ and d is even, then asymptotically almost surely $G_{n,d}$ has a K_3-decomposition provided $3\mid dn$.

conjecture | open | MathDB 页面 | 原文线索

V2-069804 | MATHDB-362079

Prescribed-parity perfect matching conjecture in random graphs

摘录 / EXCERPT: Prescribed-parity matching conjecture. If $p\geq n^{-1/2}$, then asymptotically almost surely $G(n,p)$ contains a perfect matching on

conjecture | open | MathDB 页面 | 原文线索

V2-069805 | MATHDB-362081

Johnson graph pathwidth conjecture

摘录 / EXCERPT: For integers n and k with $1\le k\le n-1$, let $F_k(K_n)$ be the k-token graph of the complete graph K_n, also known as the Johnson graph. The tree decomposition of this gr...

conjecture | open | MathDB 页面 | 原文线索

V2-069806 | MATHDB-362082

Optimal path-decomposition conjecture for Johnson graphs

摘录 / EXCERPT: Let $F_k(K_n)$ be the k-token graph of the complete graph K_n, and let pw and tw denote pathwidth and treewidth. Optimal path-decomposition co...

conjecture | open | MathDB 页面 | 原文线索

V2-069807 | MATHDB-362086

The global rigidity and coordinated self-stress conjecture in ℓ_p^d

摘录 / EXCERPT: Global rigidity and self-stress conjecture. The framework $(G,\bm p)$ is globally rigid in ℓ_p^d if and only if it has a k-th coordinated self-stress ω^k for some k...

conjecture | open | MathDB 页面 | 原文线索

V2-069808 | MATHDB-362087

The general-dimensional identifiability characterization of global rigidity

摘录 / EXCERPT: General-dimensional characterization conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-069809 | MATHDB-362097

The four locally irregular subgraphs conjecture

摘录 / EXCERPT: Four-subgraph conjecture. Each connected graph G which does not belong to $\mathfrak{T}'$ is decomposable to 4 locally irregular subgraphs.

conjecture | open | MathDB 页面 | 原文线索

V2-069810 | MATHDB-362104

Putman-Wieland conjecture on finite-index subgroups of surface groups

摘录 / EXCERPT: Putman-Wieland conjecture. For every $g\ge 2$, every $p\ge 0$, and every finite-index subgroup $K<\pi_1(\Sigma^p_g,\star)$, the subgroup K has the Putman-Wieland property.

conjecture | refuted | MathDB 页面 | 原文线索

V2-069811 | MATHDB-362107

The roots-of-unity and irrational-angle conjecture for complex Markov triples

摘录 / EXCERPT: Let (z_1,z_2,z_3) be a triple of complex roots of unity, and let n be the minimal positive integer such that $z_i^n=1$ for each i. Write the arguments as fractions of a full...

conjecture | open | MathDB 页面 | 原文线索

V2-069812 | MATHDB-362112

Heuristic conjecture for the limiting distribution of a-numbers of hyperelliptic curves in characteristic three

摘录 / EXCERPT: Continue with $p=3$, let $F={{\mathbb{F}_q}}(X)$, and for $\epsilon\in\{1,2\}$ let $\mu'_{\epsilon,g}(a)$ denote the proportion of genus-g hyperelliptic curves in the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-069813 | MATHDB-362113

Cobordism conjecture for Morin and de-suspended singular loci of twisted prim maps

摘录 / EXCERPT: Cobordism conjecture. This cobordism is oriented if M and N are oriented, the codimension k is odd, and r is even.

conjecture | open | MathDB 页面 | 原文线索

V2-069814 | MATHDB-362125

Maximum parsimony recovery conjecture for low-score binary characters

摘录 / EXCERPT: Let T be a binary phylogenetic X-tree with $|X|=n$. For a character alignment $A_k(T)$ consisting of the binary characters of parsimony score k on T, let $k<\frac{n}{4}$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-069815 | MATHDB-362126

Global maximum parsimony recovery conjecture for low-score characters

摘录 / EXCERPT: Let T be a binary phylogenetic X-tree with $|X|=n$. For a character alignment $A_k(T)$ consisting of the binary characters of parsimony score k on T, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069816 | MATHDB-362132

The PL action conjecture for dagger $(infty,n)$-categories

摘录 / EXCERPT: PL action conjecture. There is a natural map

conjecture | open | MathDB 页面 | 原文线索

V2-069817 | MATHDB-362135

McNamara-van Willigenburg's factorization conjecture for equivalent skew diagrams

摘录 / EXCERPT: McNamara-van Willigenburg's conjecture. The diagrams $\mathcal{D}$ and $\mathcal{E}$ satisfy $\mathcal{D}\sim\mathcal{E}$ if and only if the factors satisfy:...

conjecture | open | MathDB 页面 | 原文线索

V2-069818 | MATHDB-362141

Conjecture on the minimum acyclic number of oriented triangle-free graphs

摘录 / EXCERPT: For each positive integer n, let $\vec a(n)$ be the minimum of $\vec\alpha(D)$ over all oriented triangle-free graphs D of order n, where $\vec\alpha(D)$ denotes the maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-069819 | MATHDB-362142

Conjecture on the maximum dichromatic number of oriented triangle-free graphs

摘录 / EXCERPT: For each positive integer n, let $\vec t(n)$ be the maximum dichromatic number of an oriented triangle-free graph of order n. Maximum dichromatic number conjecture. ... This conj...

conjecture | open | MathDB 页面 | 原文线索

V2-069820 | MATHDB-362144

The partial saturation ratio bound for planar graphs with bounded twin multiplicities

摘录 / EXCERPT: Partial saturation ratio conjecture. Every such graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069821 | MATHDB-362145

The partial saturation ratio bound for planar graphs of minimum degree at least three

摘录 / EXCERPT: Planar partial saturation ratio conjecture. Every such graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069822 | MATHDB-362150

Integrality and positivity conjecture for integral binomial coefficients of interpolation polynomials

摘录 / EXCERPT: Let λ and μ be partitions with $\lambda\supseteq\mu$, and let $B_{\lambda\mu}$ denote the integral binomial coefficient for the interpolation-polynomial family...

conjecture | open | MathDB 页面 | 原文线索

V2-069823 | MATHDB-362151

Integrality and positivity conjecture for Macdonald-type integral binomial coefficients

摘录 / EXCERPT: Let λ and μ be partitions with $\lambda\supseteq\mu$, and let $B_{\lambda\mu}$ be the integral binomial coefficient for the interpolation-polynomial family $\mathcal F$...

conjecture | open | MathDB 页面 | 原文线索

V2-069824 | MATHDB-362152

Support conjecture for Littlewood-Richardson coefficients of interpolation polynomials

摘录 / EXCERPT: Fix partitions μ and ν. For each family $\mathcal F$, let ... and let $M_{\mu\nu}$ be the set of partitions $\lambda\supseteq\mu,\nu$ for which there exists a Molev tableau o...

conjecture | open | MathDB 页面 | 原文线索

V2-069825 | MATHDB-362158

Universal lower bound conjecture for binary search tree height of permuton samples

摘录 / EXCERPT: Universal height lower-bound conjecture. For any permuton cmu and $cvarepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-069826 | MATHDB-362167

Eventual gonality conjecture for king's graphs

摘录 / EXCERPT: Let $K_{m\times n}$ denote the king's graph on an $m\times n$ grid, with $4\leq m\leq n$. The gonality of a graph is denoted by $\operatorname{gon}(G)$. Eventual gonality conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-069827 | MATHDB-362176

The Andreadakis conjecture for automorphism groups of free groups

摘录 / EXCERPT: Let F_n be the free group of rank n, let $\mathrm{IA}_n=\mathcal{A}_n'(1)$ be the group of automorphisms of F_n acting trivially on its abelianization, and let ... be the lower...

conjecture | open | MathDB 页面 | 原文线索

V2-069828 | MATHDB-362188

Folsom-Homma-Ryu-Tong congruence conjecture for M-ary partitions

摘录 / EXCERPT: Folsom-Homma-Ryu-Tong conjecture. For all integers $n\geq1$ and $0\leq c\leq m_1-1$,

conjecture | open | MathDB 页面 | 原文线索

V2-069829 | MATHDB-362189

Factorial-partition congruence conjecture

摘录 / EXCERPT: Factorial-partition congruence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069830 | MATHDB-362190

Injectivity of the monodromy map for stratified braid groups

摘录 / EXCERPT: Let $n\geq 3$, let $\kappa=k_1\geq\dots\geq k_p$ be a partition of $n-1$ with $p\geq 3$ parts, and let ... be the stratified braid group. Let $\rho:\sB_n[\kappa]\to B_{n+p}$ be the m...

conjecture | open | MathDB 页面 | 原文线索

V2-069831 | MATHDB-362198

The conjecture relating cubic roots to ring class fields

摘录 / EXCERPT: Let d be as in the paper, let $\mathcal{O}(d)$ be the associated quadratic order, and let P_d and R_d be the subsets defined there. For $\beta\in\mathcal{O}(d)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069832 | MATHDB-362199

The discriminant norm conjecture for cubic roots

摘录 / EXCERPT: Let $u\in S_d$, write $\nu=u^{1/3}$, and let $c=c(u)$ be the product of the distinct rational primes dividing u but not d. Let c' be the odd part of c. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069833 | MATHDB-362200

The conjecture that cubic roots lie in the enlarged ring class field

摘录 / EXCERPT: Let $u\in S_d$, write $\nu=u^{1/3}$, and let $c=c(u)$ be the product of the distinct rational primes dividing u but not d. Let M_{3c} denote the ring class field associated w...

conjecture | open | MathDB 页面 | 原文线索

V2-069834 | MATHDB-362205

The planar forest-star forest decomposition conjecture

摘录 / EXCERPT: Let G be a planar graph of girth at least 5. A star forest is a forest whose connected components are stars. Planar forest-star forest decomposition conjecture. The edge set of...

conjecture | open | MathDB 页面 | 原文线索

V2-069835 | MATHDB-362206

The three-spanning-trees thin tree conjecture

摘录 / EXCERPT: Three-spanning-trees thin tree conjecture. There exists an $\varepsilon>0$ such that every graph with three edge-disjoint spanning trees contains an ε-thin tree.

open_problem | open | MathDB 页面 | 原文线索

V2-069836 | MATHDB-362207

Rank one elliptic Stark conjecture for balanced triple product p-adic L-functions

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}$, let V_g and V_h be two-dimensional L-linear Artin representations of $\operatorname{Gal}(H/\mathbb{Q})$, and set...

conjecture | open | MathDB 页面 | 原文线索

V2-069837 | MATHDB-362208

Vanishing conjecture for stable tree expressions with at least two frozen legs

摘录 / EXCERPT: Let $Q=\mathbb{Q}[a_1,\dots,a_n]$, and let $\Omega^m_{g,n}\in R^*({\overline{\mathcal{M}}}_{g,n+m})\otimes_{\mathbb{Q}}Q$ be the class obtained by summing the tree contributions de...

conjecture | open | MathDB 页面 | 原文线索

V2-069838 | MATHDB-362209

A-class conjecture for one frozen leg

摘录 / EXCERPT: Let $Q=\mathbb{Q}[a_1,\dots,a_n]$. For a stable rooted tree with one frozen leg, let $A^1_{g,n}\in R^*({\overline{\mathcal{M}}}_{g,n+1})\otimes_{\mathbb{Q}}Q$ be the sum of the cla...

conjecture | open | MathDB 页面 | 原文线索

V2-069839 | MATHDB-362210

A-class conjecture for no frozen legs

摘录 / EXCERPT: Let $Q=\mathbb{Q}[a_1,\dots,a_n]$, let $\pi\colon {\overline{\mathcal{M}}}_{g,n+1}\to {\overline{\mathcal{M}}}_{g,n}$ forget the last marked point, and define ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069840 | MATHDB-362227

Scott-Seymour conjecture on induced odd subdivisions

摘录 / EXCERPT: For a graph H, an odd subdivision is a subdivision in which every replacing path has an odd number of edges. Scott-Seymour's conjecture. For every graph H and integer t, the...

conjecture | open | MathDB 页面 | 原文线索

V2-069841 | MATHDB-362228

Logarithmic degree-boundedness for intersection graphs of proper minor-closed classes

摘录 / EXCERPT: A graph class $\mathcal{F}$ is proper if some graph is not isomorphic to any graph in $\mathcal{F}$. Write $int(\mathcal{F})$ for the class of intersection graphs of collections of...

conjecture | open | MathDB 页面 | 原文线索

V2-069842 | MATHDB-362233

Stable dual-surjunctivity for NUCA with uniformly bounded singularity

摘录 / EXCERPT: Let G be a finitely generated group and let A be a finite alphabet. Let $M\subset G$ be finite, let $S=A^{A^M}$, and let $s\in S^G$. Suppose that the NUCA map...

conjecture | open | MathDB 页面 | 原文线索

V2-069843 | MATHDB-362234

Andrews-Merca averaged truncation conjecture for Jacobi's triple product

摘录 / EXCERPT: Andrews-Merca's conjecture. The coefficient of q^m in this series is non-negative.

conjecture | open | MathDB 页面 | 原文线索

V2-069844 | MATHDB-362235

Merca's truncation conjecture for Jacobi's triple product

摘录 / EXCERPT: Merca's conjecture. This theta series has non-negative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-069845 | MATHDB-362236

Schlosser-Zeng conjecture on unimodality of truncated theta coefficients

摘录 / EXCERPT: Schlosser-Zeng's conjecture. For every $k\in\mathbb N_0$ and $n\in\mathbb N$, the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-069846 | MATHDB-362237

The invariant-subspace certificate conjecture for noncommutative polynomials

摘录 / EXCERPT: Invariant-subspace certificate conjecture. The following are equivalent: (i) for every $n\in\mathbb{N}$ and every $\underline X\in\operatorname{M}_n(\mathbb k)^d$, every joint inv...

conjecture | open | MathDB 页面 | 原文线索

V2-069847 | MATHDB-362238

Makar-Limanov's low-rank values conjecture for noncommutative polynomials

摘录 / EXCERPT: Makar-Limanov's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069848 | MATHDB-362239

The rank Nullstellensatz conjecture for noncommutative polynomials

摘录 / EXCERPT: Rank Nullstellensatz conjecture. The following are equivalent: (i) there is $K\in\mathbb{N}$ such that for every $n\in\mathbb{N}$ and every...

conjecture | open | MathDB 页面 | 原文线索

V2-069849 | MATHDB-362244

Non-negativity conjecture for q-Catalan germs

摘录 / EXCERPT: For each integer $a\geq3$, let $\operatorname{Cat}((a;c))_q$ denote the q-Catalan germ associated with an integer c, as defined in the paper. q-Catalan germ positivity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-069850 | MATHDB-362246

Ribbon-partition conjecture for q-Catalan germs

摘录 / EXCERPT: Let $\mathcal{L}=\mathbb{Z}^{a-1}$ be the weight lattice, let $\operatorname{Box}[c'+1,c]$ be the indicated slab of the fundamental box, and order lattice points by the tilted part...

conjecture | open | MathDB 页面 | 原文线索

V2-069851 | MATHDB-362247

Generalized monotonicity conjecture for q-Ramanujan sums

摘录 / EXCERPT: For positive integers a,b and k, define ... where $c_d(k)_q=\sum_{s\mid\gcd(d,k)}\mu(d/s)[s]_q$. Here $[n]_q$ is the q-integer, μ is the Möbius function, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069852 | MATHDB-362248

Converse characterization of generating skew Hall-Littlewood functions

摘录 / EXCERPT: Let $\{\lambda^n/\mu^n\}$ be a graded skew partition sequence, and for each $n\geq0$ set $u_n=P_{\lambda^n/\mu^n}$. A skew diagram λ^n/μ^n is column-connected if it canno...

conjecture | open | MathDB 页面 | 原文线索

V2-069853 | MATHDB-362254

Koyama's mean Lindelöf hypothesis for Rankin-Selberg L-functions

摘录 / EXCERPT: Koyama's mean Lindelöf hypothesis. The displayed bound should hold.

conjecture | open | MathDB 页面 | 原文线索

V2-069854 | MATHDB-362255

The square mean Lindelöf hypothesis over the Picard orbifold

摘录 / EXCERPT: Square mean Lindelöf hypothesis. The value $\eta=0$ is admissible.

conjecture | open | MathDB 页面 | 原文线索

V2-069855 | MATHDB-362265

Energy-length convergence for saddle-point interfaces through conical singularities

摘录 / EXCERPT: Let E_ε be the energy functional, and let I_0 be an interface that may pass through the conical singularity. Consider sequences of saddle points of E_ε wh...

conjecture | open | MathDB 页面 | 原文线索

V2-069856 | MATHDB-362266

Real-rootedness, log-concavity, and Frankl-Füredi-Kalai conjecture for zig-zag Eulerian polynomials

摘录 / EXCERPT: For $n\geq 1$, write $Z_n(t)=\sum_k z(n,k)t^k$, and let ... be the gamma-vector associated with $Z_n(t)$. A sequence is log-concave when $a_k^2\geq a_{k-1}a_{k+1}$ wherever the terms...

conjecture | open | MathDB 页面 | 原文线索

V2-069857 | MATHDB-362272

Simon's conjecture on shelling completable complexes

摘录 / EXCERPT: Simon's conjecture. Every shellable complex is shelling completable.

conjecture | open | MathDB 页面 | 原文线索

V2-069858 | MATHDB-362276

Ulas's conjecture on the distribution of roots of m-ary partition polynomials

摘录 / EXCERPT: Ulas's conjecture. For every $\varepsilon>0$ and every $m\geq 2$, there exists N, possibly depending on ε and m, such that

conjecture | open | MathDB 页面 | 原文线索

V2-069859 | MATHDB-362309

Set-valued characterization of test Morita equivalences

摘录 / EXCERPT: Let $F\colon\mathcal{S}\to\mathcal{T}$ be a morphism of limit sketches. A test Morita equivalence is a morphism of sketches that induces an equivalence ... for every test sketch. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-069860 | MATHDB-362312

The upper-bound conjecture for equiangular lines in dimensions 8 through 13

摘录 / EXCERPT: Let t'_n denote the largest number of lines in an equiangular family obtained from the minimal vectors of a lattice in dimension n. Dimension 8-13 conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-069861 | MATHDB-362313

The maximal-line conjecture for minimal vectors in dimensions 16, 18, 19 and 20

摘录 / EXCERPT: For a lattice in dimension n, consider the equiangular lines produced by its minimal vectors. Minimal-vector line-count conjecture. For $n=16$, 18, 19 and 20, the largest n...

conjecture | open | MathDB 页面 | 原文线索

V2-069862 | MATHDB-362327

Fischer operator bijectivity conjecture for polynomials

摘录 / EXCERPT: Fischer operator bijectivity conjecture. The operator F_P is a bijection. The source states that this conjecture remains open in general, although it records several special case...

conjecture | open | MathDB 页面 | 原文线索

V2-069863 | MATHDB-362339

Randomly shifted Gumbel convergence conjecture for the maximum of Ginzburg-Landau fields

摘录 / EXCERPT: Randomly shifted Gumbel convergence conjecture. There exists a random variable $Z>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-069864 | MATHDB-362340

Odd 5-coloring conjecture for planar graphs

摘录 / EXCERPT: Odd 5-coloring conjecture. Every planar graph is odd 5-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-069865 | MATHDB-362341

Odd chromatic number conjecture for biplanar graphs excluding complete graphs

摘录 / EXCERPT: Biplanar odd-coloring conjecture. If G is a biplanar graph without $K_{n+1}^\ast$ as a subgraph, then

conjecture | open | MathDB 页面 | 原文线索

V2-069866 | MATHDB-362342

Andrews-Merca and Guo-Zeng's truncated Jacobi triple product conjecture

摘录 / EXCERPT: Let ℓ, R and S be positive integers with $1\leq S<R/2$, and let $(a;q)_\infty$ denote the q-Pochhammer symbol. Andrews-Merca and Guo-Zeng's conjecture. The coefficient o...

conjecture | open | MathDB 页面 | 原文线索

V2-069867 | MATHDB-362343

Ballantine-Merca's 6-regular partition truncation conjecture

摘录 / EXCERPT: Ballantine-Merca's conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069868 | MATHDB-362349

Jiang's conjecture on wavefront sets in local Arthur packets

摘录 / EXCERPT: Let $\mathrm G$ be a connected reductive group over F and let $G=\mathrm G(F)$. Assume a local Arthur-packet theory for G. Let $\Psi^+(G)$ be the set of local Arthur parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-069869 | MATHDB-362350

Generalized Shahidi conjecture for local ABV-packets

摘录 / EXCERPT: Let $\mathrm G$ be a connected reductive group over F with a quasi-split pure inner form, and let $G=\mathrm G(F)$. Let $\Phi(G)$ be the set of local L-parameters, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069870 | MATHDB-362351

Generalized Shahidi conjecture for local L-packets

摘录 / EXCERPT: Let $\mathrm G$ be a connected reductive group and let $G=\mathrm G(F)$. Assume the local Langlands correspondence for G. For a local L-parameter ϕ, let Π_ϕ be its...

conjecture | open | MathDB 页面 | 原文线索

V2-069871 | MATHDB-362357

Shapiro's refined Newton-difference bound for real zeros

摘录 / EXCERPT: Let $P(x)=\sum_{k=0}^n a_kx^k$ be a polynomial with positive coefficients. Define ... where $a_{-1}=a_{n+1}=0$. Let $0=k_1<k_2<\cdots<k_m=n$ be the sequence of all indices k_i for...

conjecture | open | MathDB 页面 | 原文线索

V2-069872 | MATHDB-362368

Gyárfás-Lehel conjecture for monochromatic tree covers

摘录 / EXCERPT: Let $K_{n,m}$ be the complete bipartite graph with parts of sizes n and m, and let $\operatorname{tc}_r(K_{n,m})$ be the minimum number of monochromatic trees whose vertices co...

conjecture | open | MathDB 页面 | 原文线索

V2-069873 | MATHDB-362369

Three-colour random bipartite tree-cover threshold conjecture

摘录 / EXCERPT: Let $G\sim G(n,n,p)$ be a random bipartite graph with two parts of size n, and let $\operatorname{tc}_3(G)$ be the minimum number of monochromatic trees whose vertices cover G...

conjecture | open | MathDB 页面 | 原文线索

V2-069874 | MATHDB-362376

Moments of Riesz p-equilibrium measures are minimal for the ball

摘录 / EXCERPT: Moments of Riesz p-equilibrium measures are minimal for the ball. For every $q\in(0,\infty)$,

conjecture | open | MathDB 页面 | 原文线索

V2-069875 | MATHDB-362395

The polynomial decay conjecture for the linearised equation

摘录 / EXCERPT: Let ψ^{dat} be data as in the paper's scattering-data definition, with $d>0$ and $N>5$, for the linearised equation on the hypersurface Σ_1. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069876 | MATHDB-362396

The non-finite axiomatizability conjecture for inequality over the Weihrauch lattice

摘录 / EXCERPT: Let $\mathfrak{W}$ be the Weihrauch lattice equipped with meet $\sqcap$, product $\times$, and unit 1, and consider the partial axiomatization of inequalities over...

conjecture | open | MathDB 页面 | 原文线索

V2-069877 | MATHDB-362397

The finite axiomatizability conjecture for the pointed Weihrauch lattice

摘录 / EXCERPT: Let $\mathfrak{W}_\bullet$ be the pointed Weihrauch lattice equipped with meet $\sqcap$, product $\times$, and unit 1, and let the paper's partial axiomatization be given by the...

conjecture | open | MathDB 页面 | 原文线索

V2-069878 | MATHDB-362403

Supercuspidal wavefront-set conjecture

摘录 / EXCERPT: Supercuspidal wavefront-set conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069879 | MATHDB-362404

Arthur-packet equality conjecture for tempered L-packets

摘录 / EXCERPT: Let $\pi=\pi(\lambda,s,{{\mathcal O}}^\vee_{\lambda,s},{{\mathcal L}}^\vee)$ be an irreducible representation in the tempered L-packet...

conjecture | open | MathDB 页面 | 原文线索

V2-069880 | MATHDB-362447

Conjecture on congruences for partitions with distinct even parts

摘录 / EXCERPT: Let $ped(n)$ denote the number of partitions of n in which the even parts are distinct. Congruence conjecture. For each nonnegative integer n, ... These are proposed divisibili...

conjecture | open | MathDB 页面 | 原文线索

V2-069881 | MATHDB-362452

Kerdock spherical-code optimality conjecture

摘录 / EXCERPT: For each integer $k e 0$, let a Kerdock spherical code be the spherical code in $\mathbb{R}^{2^{2k}}$ constructed from the Kerdock binary code, with $2^{4k}+2^{2k+1}$ points and ma...

conjecture | open | MathDB 页面 | 原文线索

V2-069882 | MATHDB-362453

Snub-cube optimality conjecture for spherical codes

摘录 / EXCERPT: Let $\mathcal{C}$ be the set of 24 vertices of a snub cube, regarded as unit vectors in $\mathbb{R}^3$. Snub-cube optimality conjecture. Three-point bounds prove that...

conjecture | open | MathDB 页面 | 原文线索

V2-069883 | MATHDB-362454

Global minimum conjecture for the theta series ratio of unimodular lattices

摘录 / EXCERPT: Global-minimum conjecture. The theta series ratio achieves its global minimum at $\tau=1$:

conjecture | open | MathDB 页面 | 原文线索

V2-069884 | MATHDB-362459

Hilbert-depth comparison for squarefree monomial ideals in at most nine variables

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ and let $I\subset S$ be a squarefree monomial ideal. The nine-variable Hilbert-depth conjecture. If ... then ... The paper gives examples showing failure of t...

conjecture | open | MathDB 页面 | 原文线索

V2-069885 | MATHDB-362463

The Jacobi index-one sup-norm conjecture

摘录 / EXCERPT: Let $J^{cusp}_{k,1}(N)$ denote the space of Jacobi cusp forms of index 1 and level N, and let $\sup(J^{cusp}_{k,1}(N))$ denote its associated supremum quantity. Jacobi index-on...

conjecture | open | MathDB 页面 | 原文线索

V2-069886 | MATHDB-362464

The Jacobi newform sup-norm conjecture

摘录 / EXCERPT: Let $J^{cusp}_{k,1}(N)$ denote the space of Jacobi cusp forms of index 1 and level N, and let $\phi\in J^{cusp}_{k,1}(N)$ be an L^2-normalized newform. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069887 | MATHDB-362465

The Saito-Kurokawa lift space-size conjecture

摘录 / EXCERPT: Let $\mathrm{SK}_k(N)$ denote the space of Saito-Kurokawa lifts of weight k and level N, let $B_k^{SK}(N)$ be an orthonormal basis, and define ... with...

conjecture | open | MathDB 页面 | 原文线索

V2-069888 | MATHDB-362466

The Saito-Kurokawa newform sup-norm conjecture

摘录 / EXCERPT: Let $\mathrm{SK}_k(N)$ denote the space of Saito-Kurokawa lifts of weight k and level N, and let $F\in\mathrm{SK}_k(N)$ be an L^2-normalized newform. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069889 | MATHDB-362475

WKB asymptotic conjecture for Stokes matrices

摘录 / EXCERPT: Let A be a generic matrix, let u be the diagonal irregular-type parameter, and let $S_+(u,A,\varepsilon)$ be the upper triangular Stokes matrix with $S_-=S_+^\dagger$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-069890 | MATHDB-362476

Mneimneh-type binomial-sum conjecture for multiple harmonic-star sums

摘录 / EXCERPT: Let $r\in\mathbb{N}_0$ and $n\in\mathbb{N}$. For any reals x,y, let ${\bf p}_{r+1}:=(p_1,p_2,\ldots,p_{r+1})\in\mathbb{N}^{r+1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069891 | MATHDB-362480

Hovey's 3-cordiality conjecture

摘录 / EXCERPT: A graph G is 3-cordial if its vertices are labeled by 0, 1, and 2 with label-class sizes differing by at most one, and each edge receives the sum of its endpoint labels mod...

conjecture | open | MathDB 页面 | 原文线索

V2-069892 | MATHDB-362482

The finite Euler sum quotient-space isomorphism conjecture

摘录 / EXCERPT: For $w\in\mathbb{N}$, let FES_{w} and ES_w be the $\mathbb{Q}$-vector spaces generated by finite Euler sums and Euler sums of weight w, respectively. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069893 | MATHDB-362483

Zhao's Fibonacci basis conjecture for finite Euler sums

摘录 / EXCERPT: Let FES_w be the $\mathbb{Q}$-vector space generated by finite Euler sums of weight w, and let $F_0=F_1=1$ and $F_k=F_{k-1}+F_{k-2}$ for $k\geq2$. Zhao's finite Euler...

conjecture | open | MathDB 页面 | 原文线索

V2-069894 | MATHDB-362484

Zlobin's Fibonacci basis conjecture for ordinary Euler sums

摘录 / EXCERPT: Let ES_w be the $\mathbb{Q}$-vector space generated by Euler sums of weight w, and let $F_0=F_1=1$ and $F_k=F_{k-1}+F_{k-2}$ for $k\geq2$. Zlobin's Euler sum basis con...

conjecture | open | MathDB 页面 | 原文线索

V2-069895 | MATHDB-362485

The odd-height all-ones multiple T-value conjecture

摘录 / EXCERPT: Let $T_{\mathcal A}$ and $T_\shuffle^{\mathcal S}$ denote the finite and shuffle-regularized symmetric multiple T-values, respectively; let $S_{\mathcal A}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-069896 | MATHDB-362486

The even all-ones S-value basis conjecture

摘录 / EXCERPT: Let $S_{\mathcal A}$ and $S_\shuffle^{\mathcal S}$ denote the finite and shuffle-regularized symmetric S-values, respectively. Even all-ones S-value conjecture. For every even...

conjecture | open | MathDB 页面 | 原文线索

V2-069897 | MATHDB-362488

The finite multiple T-value dimension recurrence conjecture

摘录 / EXCERPT: Let FMT_w be the $\mathbb{Q}$-vector space generated by finite multiple T-values of weight w. Finite multiple T-value dimension conjecture. For all $k\geq1$, ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-069898 | MATHDB-362489

Hölder continuity conjecture for core entropy of quadratic polynomials

摘录 / EXCERPT: For a quadratic polynomial, let its core entropy be the entropy of the dynamics on its Hubbard tree, and let the polynomial vary in the space of quadratic polynomials equipped with...

conjecture | open | MathDB 页面 | 原文线索

V2-069899 | MATHDB-362493

Rank-one joint Gaussian distribution conjecture for elliptic-curve twists

摘录 / EXCERPT: Rank-one joint distribution conjecture. As d ranges over $\mathcal{F}$, the joint distribution of $\log|L'(\frac12,E_d)|$ and $\log(|\Sha(E_d)|R(E_d)/\sqrt{|d|})$ is asymptotical...

conjecture | open | MathDB 页面 | 原文线索

V2-069900 | MATHDB-362494

Strichartz conjecture for the nonelliptic Schrödinger equation on irrational tori

摘录 / EXCERPT: Let $\tilde{\Delta}:=\frac{1}{2\pi}(\partial_{11}+\alpha\partial_{22})$ on $\mathbb{T}^2$, where $\alpha<0$ is irrational, and let $e^{it\tilde{\Delta}}f$ denote the solution to...

conjecture | open | MathDB 页面 | 原文线索

V2-069901 | MATHDB-362498

Integral coformality conjecture for flag moment-angle complexes

摘录 / EXCERPT: Let $\mathcal{K}$ be a flag simplicial complex. The spaces $\mathrm{DJ}(\mathcal{K})$ and $\mathcal{Z}_\mathcal{K}$ are defined by the Davis-Januszkiewicz and moment-angle construc...

conjecture | open | MathDB 页面 | 原文线索

V2-069902 | MATHDB-362501

Regular polygon conjecture for the first Dirac eigenvalue

摘录 / EXCERPT: Regular polygon conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069903 | MATHDB-362508

The local smoothing conjecture for the Hermite wave equation

摘录 / EXCERPT: Let $d\geq 1$, let $2<p<\infty$, let $0<T<\infty$, and let u solve the Hermite wave equation ... with $u(0)=u_0$ and $\dot u(0)=0$. The local smoothing conjecture for the Hermite w...

conjecture | open | MathDB 页面 | 原文线索

V2-069904 | MATHDB-362509

Lapid-Mao's local identity for Whittaker-period constants

摘录 / EXCERPT: Lapid-Mao's local conjecture. For every $\eta\in E_v^\times$ such that $\bar\eta=-\eta$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-069905 | MATHDB-362524

Extended Mond conjecture for mappings on ICIS

摘录 / EXCERPT: Let $(X,S)\subset(\mathbb{C}^N,S)$ be an n-dimensional isolated complete intersection singularity, and let $f:(X,S)\rightarrow(\mathbb{C}^{n+1},0)$ be an $\mathscr{A}$-finite map...

conjecture | open | MathDB 页面 | 原文线索

V2-069906 | MATHDB-362528

Extension of 2-Morita equivalence to braided fusion categories

摘录 / EXCERPT: Braided 2-Morita conjecture. This definition of E_2-Morita equivalence extends to braided fusion categories without requiring non-degeneracy.

conjecture | open | MathDB 页面 | 原文线索

V2-069907 | MATHDB-362533

Associativity conjecture for fusion of one-dimensional domain walls

摘录 / EXCERPT: Wall-fusion associativity conjecture. Fusion through $\CC$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069908 | MATHDB-362538

Luo-Rao-Xiong's local connectedness conjecture for planar self-similar sets

摘录 / EXCERPT: Let Φ be a self-similar iterated function system on $\mathbb R^2$ and let K be its attractor. The Luo-Rao-Xiong local connectedness conjecture. If Φ satisfies the open...

conjecture | open | MathDB 页面 | 原文线索

V2-069909 | MATHDB-362546

Hibi-Li's f-vector conjecture for order and chain polytopes

摘录 / EXCERPT: Let P be a finite poset with $|P|=d>1$. Write $f_i(Q)$ for the number of i-dimensional faces of a polytope Q, and let ${\mathcal O}(P)$ and ${\mathcal C}(P)$ denote the order...

conjecture | open | MathDB 页面 | 原文线索

V2-069910 | MATHDB-362559

The unit trapezoid upper-bound conjecture for reptile trapezoids

摘录 / EXCERPT: Let T be a unit trapezoid, and let $\mathbf{ub}(T)$ denote the set of its relevant upper bounds. Unit trapezoid upper-bound conjecture. If T is a reptile, then ... This conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-069911 | MATHDB-362560

The rational-angle conjecture for reptile trapezoids

摘录 / EXCERPT: Let T be a unit trapezoid, and let $\theta(T)$ and $\psi(T)$ denote the angle parameters associated with T. For a right trapezoid, define $\psi(T)=\pi/2$. Rational-angle conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-069912 | MATHDB-362561

The $R(\pi/3,a)$ reptile-trapezoid conjecture

摘录 / EXCERPT: Let $R(\pi/3,a)$ denote the trapezoid in the paper's notation, with parameter a. $R(\pi/3,a)$ reptile conjecture. If ... is a reptile, then ... This is a specific classification conj...

conjecture | open | MathDB 页面 | 原文线索

V2-069913 | MATHDB-362571

Fulman-Kim-Lee-Petersen real-rootedness conjecture for even and odd excedance enumerators

摘录 / EXCERPT: Fulman-Kim-Lee-Petersen conjecture. The polynomials $\sum_{i=0}^n P_{n,k}t^k$ and $\sum_{i=0}^n Q_{n,k}t^k$ are all real-rooted for $n\ge 5$. This is an open problem proposed as pa...

conjecture | open | MathDB 页面 | 原文线索

V2-069914 | MATHDB-362579

Higashitani-Kummer-Michałek conjecture for complete multipartite graphs

摘录 / EXCERPT: Higashitani-Kummer-Michałek conjecture. (i) For any complete multipartite graph $K_{a_1,\dots,a_k}$, the roots of $E_{a_1,\dots,a_k}$ lie on CL. (ii) If...

conjecture | open | MathDB 页面 | 原文线索

V2-069915 | MATHDB-362580

Conjecture on cross-degree bounds and interlacing for multipartite symmetric edge polytopes

摘录 / EXCERPT: Cross-degree and interlacing conjecture. The inequalities

conjecture | open | MathDB 页面 | 原文线索

V2-069916 | MATHDB-362582

The arithmetic transfer conjecture for odd type $(r,0)$

摘录 / EXCERPT: Arithmetic transfer conjecture for odd type $(r,0)$. There exists $\varphi'\in C_c^\infty(G')$ transferring to $(c_r^2\\,\mathbf{1}_{K_n^{[r]}\times K_{n+1}},0)$ such that, for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-069917 | MATHDB-362584

The arithmetic transfer conjecture for type $(0,r)$

摘录 / EXCERPT: Let $0\leq r\leq n+1$, let ε be the parity of r, and let W_ε and $W_{\varepsilon+1}$ be the hermitian spaces of dimension $n+1$ with opposite invariants....

conjecture | open | MathDB 页面 | 原文线索

V2-069918 | MATHDB-362585

Absolute maximum principle for tight forms

摘录 / EXCERPT: Absolute maximum principle for tight forms. The restriction of $|F|$ to $\overline{U}$ attains its maximum on ∂U. This would extend the absolute-minimization theorem from...

conjecture | open | MathDB 页面 | 原文线索

V2-069919 | MATHDB-362590

Superiority conjecture for the cross-polytope bound

摘录 / EXCERPT: Superiority conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-069920 | MATHDB-362605

Intersection-lattice conjecture for the defect of line arrangements

摘录 / EXCERPT: For a reduced plane curve $C=\{f=0\}$ in $\mathbb{P}^{2}_{\mathbb{C}}$, let $\nu(C)$ denote its defect. A line arrangement is a union of finitely many lines in...

conjecture | open | MathDB 页面 | 原文线索

V2-069921 | MATHDB-362610

Total positivity of the barycentric subdivision operator

摘录 / EXCERPT: Let $\mathcal{F}$ be the barycentric subdivision, and let $H_{\mathcal{F}}$ denote its associated operator. A matrix is totally positive if every minor of every order is positive,...

conjecture | open | MathDB 页面 | 原文线索

V2-069922 | MATHDB-362616

The positivity conjecture for the Macdonald character operator

摘录 / EXCERPT: The positivity conjecture. The quantity

conjecture | open | MathDB 页面 | 原文线索

V2-069923 | MATHDB-362617

The Macdonald version of Stanley's positivity conjecture

摘录 / EXCERPT: Macdonald version of Stanley's conjecture. For any partitions λ,μ,ν, the quantity

conjecture | open | MathDB 页面 | 原文线索

V2-069924 | MATHDB-362618

The b-conjecture for Jack map coefficients

摘录 / EXCERPT: The b-conjecture. The coefficients $c^\pi_{\mu,\nu}$ are polynomials in b with non-negative integer coefficients. The paper states that the conjecture remains open, although its...

conjecture | open | MathDB 页面 | 原文线索

V2-069925 | MATHDB-362619

The b-conjecture for connected Jack map coefficients

摘录 / EXCERPT: The b-conjecture. The coefficients $h^\pi_{\mu,\nu}$ are polynomials in b with non-negative integer coefficients. The paper explicitly says both Jack conjectures remain open.

conjecture | open | MathDB 页面 | 原文线索

V2-069926 | MATHDB-362620

The Macdonald generalization of the Matchings-Jack conjecture

摘录 / EXCERPT: The Macdonald generalization of the Matchings-Jack conjecture. For any positive integer n and partitions π,μ,ν of n,

conjecture | open | MathDB 页面 | 原文线索

V2-069927 | MATHDB-362621

The positivity conjecture for normalized Macdonald structure coefficients

摘录 / EXCERPT: The positivity conjecture. Let π,μ,ν be three partitions. Then

conjecture | open | MathDB 页面 | 原文线索

V2-069928 | MATHDB-362622

Conjecture on short admissible profiles in the hypercube

摘录 / EXCERPT: Let Q^n be the n-dimensional hypercube, and let an admissible tuple be a tuple that occurs as the profile of a matching in Q^n. For $\bm{x}=(x_1,\ldots,x_n)\in\mathbb{N}^n$,...

conjecture | open | MathDB 页面 | 原文线索

V2-069929 | MATHDB-362625

Finiteness of even-dimensional thin simplices of lattice width at least two

摘录 / EXCERPT: A thin simplex is a lattice simplex of lattice width at most one. Finiteness conjecture. In each even dimension there are only finitely many thin simplices of lattice width...

conjecture | open | MathDB 页面 | 原文线索

V2-069930 | MATHDB-362626

Width-one conjecture for odd-dimensional thin simplices

摘录 / EXCERPT: A thin simplex is a lattice simplex of lattice width at most one, and its lattice width is the minimum width over all nonzero integral linear functionals. Width-one conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-069931 | MATHDB-362644

Pattern-conditioned symmetric chain decomposition conjecture for Young's lattice boxes

摘录 / EXCERPT: Let $L(w,h)$ be the poset of partitions in a box of height h and width w, ordered componentwise. A saturated chain in $L(w,h)$ has an edge labelling recording the columns in wh...

conjecture | open | MathDB 页面 | 原文线索

V2-069932 | MATHDB-362645

McMullen's Carathéodory hyperbolicity conjecture for rational-map moduli spaces

摘录 / EXCERPT: Let f be a rational map of degree $d\geq 2$ on the Riemann sphere, and let $\mathcal{M}_f$ be its moduli space of quasiconformal conjugacy classes. A rational map is flexible Lat...

conjecture | open | MathDB 页面 | 原文线索

V2-069933 | MATHDB-362648

The monomer-dimer spectral independence conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, let $\lambda>0$, and let μ be the Gibbs distribution of the monomer-dimer model on G with fugacity λ. The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-069934 | MATHDB-362649

The simple-graph monomer-dimer spectral independence conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a simple graph, let $\lambda>0$, and let μ be the Gibbs distribution of the monomer-dimer model on G with fugacity λ. The quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-069935 | MATHDB-362650

The hardcore-model tree spectral-gap conjecture

摘录 / EXCERPT: Let T be a tree with n vertices, and let λ be the fugacity of the hardcore model. The Glauber dynamics is the single-site Markov chain for this Gibbs distribution, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069936 | MATHDB-362660

Gaussoid axioms suffice for CI implication in singly constrained Gaussian DAG models

摘录 / EXCERPT: Gaussoid-axioms sufficiency conjecture. If $a \perp\!\!\!\perp b \mid C$ holds for every covariance matrix $\Sigma \in \mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j\mid K}$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-069937 | MATHDB-362665

Cowling's conjecture on weak amenability and the Haagerup property

摘录 / EXCERPT: Let G be a locally compact group. The Haagerup property equivalence conjecture. G has the Haagerup property if and only if G is weakly amenable with $\boldsymbol\Lambda(G)=1$...

conjecture | refuted | MathDB 页面 | 原文线索

V2-069938 | MATHDB-362666

Cowling's conjecture on weak amenability constant one and the Haagerup property

摘录 / EXCERPT: Let G be a locally compact group. The one-way Cowling conjecture. If G is weakly amenable with $\boldsymbol\Lambda(G)=1$, then it has the Haagerup property. This is the directi...

conjecture | open | MathDB 页面 | 原文线索

V2-069939 | MATHDB-362669

Strata-effectivity conjecture for Griffiths-Hodge Chern classes

摘录 / EXCERPT: Let G be a connected reductive group over $\boldsymbol{F}_p$, let μ be a cocharacter of G, and let r be a representation of G. The associated Griffiths-Hodge bundle is...

conjecture | open | MathDB 页面 | 原文线索

V2-069940 | MATHDB-362671

Dual enhanced Shahidi conjecture for unramified Arthur-type representations

摘录 / EXCERPT: Let G be a connected reductive group over a non-Archimedean local field k_v, and let $G=\mathrm{G}(k_v)$. Assume that a theory of local Arthur packets for G exists....

conjecture | open | MathDB 页面 | 原文线索

V2-069941 | MATHDB-362680

The LSC conjecture for families

摘录 / EXCERPT: Let $\varphi: Y \to T$ be a flat morphism, let $\mathfrak{a}$ be an $\mathbb{R}$-ideal sheaf on Y, and let $\tau: T \to Y$ be a section of φ. For a closed point $t \in T$...

conjecture | open | MathDB 页面 | 原文线索

V2-069942 | MATHDB-362683

The pseudoadjunction conjecture for bicategorical graphs and bicategories

摘录 / EXCERPT: Pseudoadjunction conjecture. There is a pseudoadjunction between the locally discrete 2-category of bicategorical graphs and the 2-category of bicategories, pseudofunctors and icon...

conjecture | open | MathDB 页面 | 原文线索

V2-069943 | MATHDB-362684

The physical-lax-tensor conjecture for wiring diagrams and monoidal lenses

摘录 / EXCERPT: Physical-lax-tensor conjecture. The multicategory of wiring diagrams of symmetric monoidal categories is the multicategory induced by the produoidal category of monoidal lenses.

conjecture | open | MathDB 页面 | 原文线索

V2-069944 | MATHDB-362712

Optimality conjecture for the equidistant vector set in even dimensions

摘录 / EXCERPT: Let $m\geq 4$ be an even integer, let $\boldsymbol{E}_m$ denote the vector set defined in the paper, and let $\beta_{\boldsymbol{A}}$ be the condition number associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-069945 | MATHDB-362713

Main genus-1 mirror-map conjecture for Fano hypersurfaces

摘录 / EXCERPT: Let M_N^k be a degree-k hypersurface in ${\mathbb P}^{N-1}$. Let $F_1^{N,k,A}(t^0,t^1,\ldots,t^{N-2})$ be the generating function of genus-1 Gromov-Witten invariants, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069946 | MATHDB-362714

Elliptic virtual-structure-constant conjecture for Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let M_k^k be a degree-k hypersurface in ${\mathbb P}^{k-1}$, let $F_{1,\mathrm{vir.}}^{k,k,B}(x)$ be the genus-1 virtual-structure-constant generating function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-069947 | MATHDB-362715

BCOV-Zinger loop-residue identities for Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let k be the degree of the Calabi-Yau hypersurface M_k^k, let $f_{\Gamma_d^{\mathrm{loop}}}$ denote the loop contribution indexed by degree d, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069948 | MATHDB-362719

Weight preservation for deletion-confusable Naisargik images of quaternary VT codes

摘录 / EXCERPT: Let $VT_{(a,b)}(n;4)$ be a quaternary Varshamov-Tenengolts code, let ϕ be a Naisargik map, and let $D_1(X)$ denote the set of binary sequences obtained from X by one deletio...

conjecture | open | MathDB 页面 | 原文线索

V2-069949 | MATHDB-362720

The Naisargik mapping between quaternary and binary Helberg codes

摘录 / EXCERPT: Let $H(n,q,s,a)$ denote a Helberg code with length n, alphabet size q, deletion parameter s, and residue a. Let a be the residue corresponding to the maximum number of co...

conjecture | open | MathDB 页面 | 原文线索

V2-069950 | MATHDB-362731

Behrend-Fischer-Koutschan symmetry conjecture for off-diagonally symmetric ASMs

摘录 / EXCERPT: Behrend-Fischer-Koutschan symmetry conjecture. Given n, for any ρ and τ, we have

conjecture | open | MathDB 页面 | 原文线索

V2-069951 | MATHDB-362732

Asymptotic growth conjecture for symmetric domino tilings of Aztec diamonds

摘录 / EXCERPT: Let $AD(n)$ be the Aztec diamond of order n. Let $O(2n,[2n])$ be the set of off-diagonally symmetric domino tilings of $AD(2n)$ with no boundary defect, and let $D(2n-1)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-069952 | MATHDB-362734

Mean Lindelöf hypothesis for the fourth spectral moment

摘录 / EXCERPT: Let $K,P\geqslant 1/2$ be real numbers, let $\upvarepsilon>0$, and let $\text{\raisebox{- 2 \depth}{\scalebox{1.1}{$ text{usefont{U}{BOONDOX-calo}{m}{n}M} hskip 0.5pt...

conjecture | open | MathDB 页面 | 原文线索

V2-069953 | MATHDB-362735

Chatterjee-Garg linear independence conjecture for generalized Stieltjes constants

摘录 / EXCERPT: Chatterjee-Garg conjecture. The numbers

conjecture | open | MathDB 页面 | 原文线索

V2-069954 | MATHDB-362736

Number-field linear independence conjecture for generalized Stieltjes constants

摘录 / EXCERPT: Number-field linear independence conjecture. The numbers

conjecture | open | MathDB 页面 | 原文线索

V2-069955 | MATHDB-362737

Full-dimension conjecture for the generalized Stieltjes constant span

摘录 / EXCERPT: Full-dimension conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-069956 | MATHDB-362738

Property B conjecture for expansions in completions of global function fields

摘录 / EXCERPT: Let K be a global function field with normalized valuation v, let K_v be its completion, and let $\pi\in K_v$ satisfy $v(\pi)>0$. Let A_v be the completion of the valuation...

conjecture | open | MathDB 页面 | 原文线索

V2-069957 | MATHDB-362739

Cancellation conjecture for nonsingular full Q-forms

摘录 / EXCERPT: Let Q be a form parameter, and let a nonsingular full Q-form be a nonsingular Q-form whose linearisation is surjective. Write $H_{b\epsilon_Q}(\mathbb{Z}^m)$ for the standard...

conjecture | open | MathDB 页面 | 原文线索

V2-069958 | MATHDB-362740

Conjecture on the map from W_0(Q_beta(q)) to twisted L-groups

摘录 / EXCERPT: Let (B_*,β) be a chain bundle, let $q\geq0$, and let $\eta_\beta\colon W_0(Q_\beta(q))\to L^{2q}(B_*,\beta)$ be the natural homomorphism. Conjecture on η_β. If...

conjecture | open | MathDB 页面 | 原文线索

V2-069959 | MATHDB-362743

Congruence kernel conjecture for Higman representations

摘录 / EXCERPT: Congruence kernel conjecture. The projective representation of $\operatorname{SL}(2,\mathbb Z)$ given by ρ has kernel that is a congruence subgroup of level...

conjecture | open | MathDB 页面 | 原文线索

V2-069960 | MATHDB-362749

Nonsplit Langlands conjecture for \(n=4\)

摘录 / EXCERPT: Let $G_4=\operatorname{Res}_{\mathbf C[[t^{1/4}]]/\mathbf C[[t]]}\mathrm{PGL}_2$. Let $\overline{\mathcal O}$ be the minimal nilpotent coadjoint orbit closure of $\mathfrak{so}_8$,...

conjecture | open | MathDB 页面 | 原文线索

V2-069961 | MATHDB-362750

Two-dimensional and higher-dimensional quasilinear global well-posedness conjecture

摘录 / EXCERPT: Consider quasilinear dispersive flows in two and higher spatial dimensions with cubic nonlinearities and sufficiently small initial data. Two-dimensional and higher-dimensional qua...

conjecture | open | MathDB 页面 | 原文线索

V2-069962 | MATHDB-362753

Critical non-standard central limit conjecture for the triangle count

摘录 / EXCERPT: Let T_n be the triangle count under the edge-triangle model with law ${\mathbb P}_{n;\alpha,h}$, and let (α_c,h_c) be the critical parameter pair. At this pair, let...

conjecture | open | MathDB 页面 | 原文线索

V2-069963 | MATHDB-362761

Aubian et al.'s universal dijoin strictness conjecture

摘录 / EXCERPT: Let $\ell\geq 3$. Aubian et al.'s conjecture. There exists an oriented graph D_1 with $\operatorname{inv}(D_1)=\ell$ such that, for every oriented graph D_2 with...

conjecture | open | MathDB 页面 | 原文线索

V2-069964 | MATHDB-362762

Tournament minimum-rank additivity for a directed triangle

摘录 / EXCERPT: Let D be a tournament, and let $\overrightarrow{C_3}$ denote the directed cycle on three vertices. The tournament minimum rank $\operatorname{tmr}(D)$ is the invariant used in th...

conjecture | open | MathDB 页面 | 原文线索

V2-069965 | MATHDB-362765

Wilkie's conjecture on integer-valued definable functions

摘录 / EXCERPT: Let $\mathbb{R}_{\mathrm{an},\exp}$ denote the real field expanded by restricted analytic functions and the exponential function. Let $f:\mathbb{R}\to\mathbb{R}$ be definable in...

conjecture | open | MathDB 页面 | 原文线索

V2-069966 | MATHDB-362770

The F-conjecture for F-curves on moduli spaces of stable curves

摘录 / EXCERPT: F-conjecture. The cone $\overline{\text{NE}}_1(\overline{\rm M}_{g,n})$ is generated by the cycle classes of F-curves. Equivalently, a line bundle on $\overline{\rm M}_{g,n}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-069967 | MATHDB-362771

Parallel LTC amplification conjecture

摘录 / EXCERPT: Parallel LTC amplification conjecture. There exists such a family for which the construction gives a parallel variant of the theorem on LTC amplification:

conjecture | open | MathDB 页面 | 原文线索

V2-069968 | MATHDB-362776

Acylindrical Hyperbolicity Conjecture for irreducible Artin groups

摘录 / EXCERPT: Let A_Γ be an irreducible Artin group, and let $Z(A_\Gamma)$ denote its centre. The central quotient is $A_\Gamma/Z(A_\Gamma)$. Acylindrical Hyperbolicity Conjecture. The gr...

conjecture | open | MathDB 页面 | 原文线索

V2-069969 | MATHDB-362777

Weak Malnormality Conjecture for standard parabolic subgroups of Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group. A standard parabolic subgroup is a subgroup generated by the standard generators associated to a subgraph of the presentation graph, and a subgrou...

conjecture | open | MathDB 页面 | 原文线索

V2-069970 | MATHDB-362778

Intersection Conjecture for parabolic subgroups of Artin groups

摘录 / EXCERPT: Let A_Γ be an Artin group, and let a parabolic subgroup mean a conjugate of a subgroup generated by a subgraph of the presentation graph. Intersection Conjecture. The inters...

conjecture | open | MathDB 页面 | 原文线索

V2-069971 | MATHDB-362785

The induced-path conjecture for the graphs $G_{q,p}$

摘录 / EXCERPT: Let $q\geq 3$ be odd, let $p\geq 1$, and let $G_{q,p}$ be the graph constructed in the paper. Here, P_t denotes the path on t vertices, and a graph is P_t-free if it has no i...

conjecture | open | MathDB 页面 | 原文线索

V2-069972 | MATHDB-362791

Segre's SHGH conjecture for plane linear systems

摘录 / EXCERPT: Let X_r be the blow-up of $[?]$?

conjecture | open | MathDB 页面 | 原文线索

V2-069973 | MATHDB-362800

Bertolini-Darmon-Prasanna main conjecture

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be an imaginary quadratic field satisfying the Heegner hypothesis and suppose that $(p)=v\overline{v}$ splits in K. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-069974 | MATHDB-362804

The amazing-hypercube R-element conjecture

摘录 / EXCERPT: Amazing-hypercube R-element conjecture. The element z is an R-element, meaning that

conjecture | open | MathDB 页面 | 原文线索

V2-069975 | MATHDB-362805

The triviality conjecture for double-shortcut equivalence classes

摘录 / EXCERPT: Triviality conjecture for double-shortcut equivalence classes. The above equivalence relation is trivial; that is, it has only one equivalence class.

conjecture | open | MathDB 页面 | 原文线索

V2-069976 | MATHDB-362806

The amazing R-element existence conjecture for double-shortcut classes

摘录 / EXCERPT: Amazing R-element existence conjecture. Every equivalence class of this relation contains an amazing R-element.

conjecture | open | MathDB 页面 | 原文线索

V2-069977 | MATHDB-362812

Strict crossing-number separation for balanced complete 4-partite graphs

摘录 / EXCERPT: Let K_n^4 denote the complete balanced 4-partite graph with n vertices, and let $\operatorname{cr}(G)$ and $\overline{\operatorname{cr}}(G)$ denote respectively its ordinary...

conjecture | open | MathDB 页面 | 原文线索

V2-069978 | MATHDB-362813

Realization conjecture for metric-preserving function monoids

摘录 / EXCERPT: Let $\mathbf{M}$ denote the class of metric spaces, let $\mathbf{P}_{\mathbf{X}}$ be the monoid associated with a metric space $\mathbf{X}$, and let $\mathbf{Am}$ be the monoid of...

conjecture | open | MathDB 页面 | 原文线索

V2-069979 | MATHDB-362814

Isomorphic realization conjecture for submonoids of metric-preserving functions

摘录 / EXCERPT: Let $\mathbf{M}$ denote the class of metric spaces, let $\mathbf{F}$ be the monoid of functions under composition, and let $\mathbf{P}_{\mathbf{X}}$ be the monoid associated with a...

conjecture | open | MathDB 页面 | 原文线索

V2-069980 | MATHDB-362828

Golomb-Welch conjecture on perfect Lee codes

摘录 / EXCERPT: Let $\mathcal A_q^n$ be the ambient Lee-metric space, and let a perfect $(n,M,d)_q$-Lee code be a Lee code with M codewords whose packing and covering radii coincide. Golomb-Welc...

conjecture | open | MathDB 页面 | 原文线索

V2-069981 | MATHDB-362832

Choe-Choi-Kim-Park independence-of-p conjecture for differences of composition operators

摘录 / EXCERPT: Let $\mathbb D$ be the unit disk, let $\mathcal{S}(\mathbb D)$ denote the holomorphic self-maps of $\mathbb D$, and for $\varphi\in\mathcal{S}(\mathbb D)$ let...

conjecture | open | MathDB 页面 | 原文线索

V2-069982 | MATHDB-362841

The stress-flex conjecture for coned polytope frameworks

摘录 / EXCERPT: Let P be a convex polytope with n vertices and full-dimensional span in $\mathbb R^d$. Let $(G^*,\hat{\bf p})$ be the tensegrity framework on $n+1$ vertices obtained by putting...

conjecture | open | MathDB 页面 | 原文线索

V2-069983 | MATHDB-362842

The strong stress-flex conjecture for coned polytope frameworks

摘录 / EXCERPT: Let P be any “polytope” with a full-dimensional affine span in $\mathbb R^d$. Let $\mathbf p_{n+1}$ be placed anywhere in $\mathbb R^d$. Let Ω be any stress for the bar fr...

conjecture | open | MathDB 页面 | 原文线索

V2-069984 | MATHDB-362848

The converse density conjecture for even-indexed G_t

摘录 / EXCERPT: Let $t=t_0 2^\alpha$ be even, with t_0 odd and ... Here $\delta^{(t)}$ denotes the odd density of the coefficients of G_t. Even-index density conjecture. Then ... The paper notes t...

conjecture | open | MathDB 页面 | 原文线索

V2-069985 | MATHDB-362849

No-even-progression conjecture for G_t when $t\equiv1\pmod4$

摘录 / EXCERPT: No-even-progression conjecture. There is no arithmetic progression $An+B$, with $A\neq0$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-069986 | MATHDB-362850

Parity-modulus conjecture for even progressions of odd-indexed G_t

摘录 / EXCERPT: Even-modulus conjecture. If an arithmetic progression satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-069987 | MATHDB-362866

The rank-n conformal-block conjecture for multi-leg PT vertices

摘录 / EXCERPT: Let n be the rank of the gauge-origami construction, and let the conformal blocks ... be the blocks defined in the source. The corresponding multi-leg Pandharipande-Thomas (PT) ver...

conjecture | open | MathDB 页面 | 原文线索

V2-069988 | MATHDB-362880

Finite forbidden-subdiagram conjecture for shellable Kohnert posets

摘录 / EXCERPT: Let D be a diagram, let $\mathcal{P}(D)$ be its Kohnert poset, and let $\mathcal{F}$ be a finite collection of families of subdiagrams. Finite forbidden-subdiagram conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-069989 | MATHDB-362881

Pattern characterization conjecture for shellable Kohnert posets of key diagrams

摘录 / EXCERPT: Let $\mathbf{a}=(a_1,\ldots,a_n)$ be a weak composition, and let $\mathcal{P}(\mathbb{D}(\mathbf{a}))$ be the Kohnert poset of its key diagram. Pattern characterization conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-069990 | MATHDB-362882

The KMT homeomorphism conjecture for Delaunay circle patterns

摘录 / EXCERPT: Let S_g be a closed surface of genus g, let E be the edge set of a triangulation of S_g, and let $\Theta:E\to[0,\pi)$ be a Delaunay angle structure. Let $P(\Theta)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-069991 | MATHDB-362890

Unimodality conjecture for statistics on separable permutations

摘录 / EXCERPT: Let S_n be the set of separable permutations of length n, and for $\pi\in S_n$ let $\operatorname{lmax}(\pi)$, $\operatorname{rmax}(\pi)$, $\operatorname{lmin}(\pi)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-069992 | MATHDB-362928

Conjectured lower bound for near-maximal clique packings

摘录 / EXCERPT: Let $G(n,p)$ be the Erdős-Rényi random graph, let $p\in(0,1)$ and $C\in\mathbb{N}_{\geq 3}$ be constants, and let k_0 be the least integer such that the expected number of k_0-...

conjecture | open | MathDB 页面 | 原文线索

V2-069993 | MATHDB-362931

Černý's conjecture for synchronizing DFAs

摘录 / EXCERPT: Černý's conjecture. If M is synchronizable, then M has a synchronizing word of length at most

conjecture | open | MathDB 页面 | 原文线索

V2-069994 | MATHDB-362932

Černý bound for difference DFAs in Euclidean space

摘录 / EXCERPT: Difference-DFA Černý conjecture. The DFA M has a synchronizing word of length at most

conjecture | open | MathDB 页面 | 原文线索

V2-069995 | MATHDB-362933

Synchronizability of products of at least three cornered DFAs

摘录 / EXCERPT: Product-corner conjecture. If $M_1,\ldots,M_k$ are distinct and each contains an f-corner z_i, then M is $(z_1,\ldots,z_k)$-synchronizable. The conjecture extends the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-069996 | MATHDB-362934

Progressive Feedforward Collapse conjecture for ResNet training

摘录 / EXCERPT: Progressive Feedforward Collapse conjecture. At the terminal phase of ResNet training, neural collapse emerges in the last-layer features. Before the effective depth, there exists...

conjecture | open | MathDB 页面 | 原文线索

V2-069997 | MATHDB-362935

Joint Gaussian value distribution conjecture for orthogonal Hecke-Maass cusp forms

摘录 / EXCERPT: Let $J\geq2$, and let f_j, $1\leq j\leq J$, be pairwise orthogonal real-valued Hecke-Maass cusp forms for $\operatorname{SL}_2(\mathbb{Z})$, with spectral parameters t_{f_j}. A...

conjecture | open | MathDB 页面 | 原文线索

V2-069998 | MATHDB-362942

Kaneko-Koyama-Kurokawa Deep Riemann Hypothesis

摘录 / EXCERPT: Let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb A_\mathbb Q)$, with entire L-function ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-069999 | MATHDB-362944

Young-regime well-posedness conjecture for locally monotone equations

摘录 / EXCERPT: Let $V\to H\to V^*$ be the compact Gelfand triple from the assumptions, let A satisfy the stated local monotonicity and coercivity hypotheses, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070000 | MATHDB-362960

Lipa's pseudo-monodromy marker conjecture for herds

摘录 / EXCERPT: Lipa's pseudo-monodromy marker conjecture. The monodromy action of $\rho(\gamma)$ is described by compositions of the markers

conjecture | open | MathDB 页面 | 原文线索

V2-070001 | MATHDB-362961

Weak Shanks Conjecture on zeros of bidisk optimal polynomial approximants

摘录 / EXCERPT: Let $H^2(\mathbb{D}^2)$ be the Hardy space on the bidisk, and let f be a polynomial in this space with no zeros in the closed bidisk. For each multidegree, an optimal polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-070002 | MATHDB-362969

Cyclicity threshold conjecture for polynomials in Dirichlet-type spaces of the unit ball

摘录 / EXCERPT: Let $\mathbb B_n$ be the unit ball in $\mathbb C^n$, let $\mathbb S_n$ be its boundary sphere, and let $\mathcal{Z}(p)$ denote the zero set of a polynomial p. A polynomial is cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-070003 | MATHDB-362984

The weak and strong spectral genus conjectures for isolated hypersurface singularities

摘录 / EXCERPT: Let $f\colon (\mathbb C^{n+1},0)\to(\mathbb C,0)$ define an n-dimensional isolated hypersurface singularity, with $n\geq 1$. Let μ be its Milnor number, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070004 | MATHDB-362987

Conjecture on equivalence of average-dwell-time stability notions

摘录 / EXCERPT: Average-dwell-time stability equivalence conjecture. For every $\tau>0$, System $$ is $\tau\text{-UGB}$ if and only if it is UGAS with respect to...

conjecture | open | MathDB 页面 | 原文线索

V2-070005 | MATHDB-363000

Inversion diameter conjecture for graphs

摘录 / EXCERPT: Let G be a graph, let $\mathcal{I}(G)$ be its inversion graph, and let $\Delta(G)$ denote its maximum degree. Inversion diameter conjecture. For every graph G, ... This would giv...

conjecture | open | MathDB 页面 | 原文线索

V2-070006 | MATHDB-363001

Blow-up conjecture for inversion diameter

摘录 / EXCERPT: Let G be a graph, let t be a positive integer, let $\overline{K}_t$ be the edgeless graph on t vertices, and let $G[\overline{K}_t]$ denote the blow-up of G in which each v...

conjecture | open | MathDB 页面 | 原文线索

V2-070007 | MATHDB-363002

Unlabelled inversion diameter conjecture for complete graphs

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, and let $\mathcal{U}(K_n)$ be its unlabelled inversion graph, obtained by identifying vertices of the inversion graph corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-070008 | MATHDB-363004

Stability conjecture for two-dimensional hyperbolic systems with directional viscosity

摘录 / EXCERPT: Stability conjecture. If the eigenvalues of all four stability operators are negative for every wavenumber k, then the system (1.2) is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-070009 | MATHDB-363007

Wooley's lower-bound conjecture for rational points on cubic hypersurfaces

摘录 / EXCERPT: Wooley's conjecture. There exists a constant $c>0$ depending only on X such that

conjecture | open | MathDB 页面 | 原文线索

V2-070010 | MATHDB-363008

Critical-moment growth conjecture for directed polymers

摘录 / EXCERPT: Critical-moment growth conjecture. The critical moment satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070011 | MATHDB-363009

Infinite-order criticality conjecture for the directed-polymer free energy

摘录 / EXCERPT: Infinite-order criticality conjecture. The free energy satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070012 | MATHDB-363015

Song-Tian's Gromov-Hausdorff convergence conjecture for the Kähler-Ricci flow

摘录 / EXCERPT: Song-Tian's Gromov-Hausdorff conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070013 | MATHDB-363017

Vanishing-Lelong-number and bounded-potential conjecture for the Kähler-Ricci flow limit

摘录 / EXCERPT: Vanishing-Lelong-number and boundedness conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070014 | MATHDB-363023

Aliniaeifard, van Willigenburg and Wang's generalized spider e-positivity conjecture

摘录 / EXCERPT: For positive integers m,n, let $S(a,b,c)$ denote the spider formed from three paths of lengths a,b,c with a common end. A graph is e-positive when its chromatic symmetric fun...

conjecture | open | MathDB 页面 | 原文线索

V2-070015 | MATHDB-363025

The rationality conjecture for cubic fourfolds with associated K3 surfaces

摘录 / EXCERPT: Rationality conjecture. A smooth cubic fourfold X is rational if and only if $X\in\mathcal{C}_d$ for some d satisfying ().

conjecture | open | MathDB 页面 | 原文线索

V2-070016 | MATHDB-363037

Figueroa-Centeno et al.'s product antimagic conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a connected graph of size m. A labeling of its edges by $\{1,2,\dots,m\}$ is product antimagic if the products of the labels incident with the vertices are pairw...

conjecture | open | MathDB 页面 | 原文线索

V2-070017 | MATHDB-363047

Optimal-mixing conjecture for Kawasaki dynamics above the analytic threshold

摘录 / EXCERPT: Let $\mathcal G_\Delta$ be the class of graphs under consideration, let $\hat{\mu}_{G,\beta,\eta}$ be the fixed-magnetization Ising measure on $G\in\mathcal G_\Delta$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070018 | MATHDB-363049

Existence conjecture for maximal Tjurina curves and line arrangements

摘录 / EXCERPT: Maximal Tjurina existence conjecture. For every integer $d\geq 3$ and every integer r with $d/2\leq r\leq d-1$, there are maximal Tjurina curves of type (d,r). Moreover, for...

conjecture | open | MathDB 页面 | 原文线索

V2-070019 | MATHDB-363050

Existence conjecture for curves with Jacobian syzygies of the same degree

摘录 / EXCERPT: Existence conjecture for curves of type (d,r,m). For any integers $d,m\geq 3$ and any integer r such that

conjecture | open | MathDB 页面 | 原文线索

V2-070020 | MATHDB-363062

Well-quasi-ordering conjecture for finite-tree-width graphs

摘录 / EXCERPT: Let a graph have finite tree-width if its tree-width is finite, and consider the minor relation on graphs. Finite-tree-width well-quasi-ordering conjecture. The graphs of finite tr...

conjecture | open | MathDB 页面 | 原文线索

V2-070021 | MATHDB-363067

Generic transitivity conjecture for primitive groups of finite Morley rank

摘录 / EXCERPT: Let (G,X) be a connected primitive permutation group of finite Morley rank. Generic transitivity conjecture. Almost all such groups have maximum degree of generic transitivity at...

conjecture | open | MathDB 页面 | 原文线索

V2-070022 | MATHDB-363070

Universality conjecture for generalized patterned Gaussian random matrices

摘录 / EXCERPT: Let X_n) be an $n\times n$ generalized random Toeplitz, circulant, or symmetric circulant matrix with standard Gaussian entries. Let $p\geq 2$ be a positive integer and set...

conjecture | open | MathDB 页面 | 原文线索

V2-070023 | MATHDB-363071

Sub-Gaussian universality conjecture for generalized patterned random matrices

摘录 / EXCERPT: Let X_n be an $n\times n$ generalized random Toeplitz, circulant, or symmetric circulant matrix with symmetric standard sub-Gaussian entries, meaning mean-zero, variance-one sub-...

conjecture | open | MathDB 页面 | 原文线索

V2-070024 | MATHDB-363075

Uniqueness conjecture for boundary Schwarzian data in the universal Teichmüller space

摘录 / EXCERPT: Uniqueness conjecture. Let $\phi\in\overline{\mathbf{S}}_q$. Then the function f_ϕ obtained in this way is unique, so that the IMS functional

conjecture | open | MathDB 页面 | 原文线索

V2-070025 | MATHDB-363087

The Catalan-polynomial structure conjecture for the moment generating function of random graphs

摘录 / EXCERPT: Catalan-polynomial structure conjecture. There exists a unique power series $P(x)$ with non-negative integer coefficients such that every $V_i(z)$ is the product of $C(z)$ and a po...

conjecture | open | MathDB 页面 | 原文线索

V2-070026 | MATHDB-363088

Finite-cover descent conjecture for similarity of matrices over discrete valuation rings

摘录 / EXCERPT: Let R be a discrete valuation ring (DVR) with fraction field K, and let A and B be $n\times n$ matrices over R. Suppose that L/K is a finite extension and S is the in...

conjecture | open | MathDB 页面 | 原文线索

V2-070027 | MATHDB-363091

Dejter's four-operation conjecture for efficient total colorings of cubic graphs

摘录 / EXCERPT: Dejter's four-operation conjecture. Every ETC of Γ is obtained via the iterative genus-increasing procedure of Corollary, departing from Q_3, through four constructive ope...

conjecture | open | MathDB 页面 | 原文线索

V2-070028 | MATHDB-363093

Parameter self-similarity conjecture for critical quasicircle maps

摘录 / EXCERPT: Let d_0 and d_∞ be the local degrees of the critical fixed points at 0 and ∞, respectively, and let $\{F_c\}_{c\in\mathbb{C}^*}$ be the one-parameter family of ra...

conjecture | open | MathDB 页面 | 原文线索

V2-070029 | MATHDB-363102

Ishii's moduli-space characterization conjecture for G-constellations

摘录 / EXCERPT: Let $G\subset\operatorname{GL}(2,\mathbb{C})$ be a finite subgroup, let $X=\mathbb{C}^2/G$ with boundary divisor B, and let $\mathcal{M}_{\theta}$ denote the moduli space of...

conjecture | open | MathDB 页面 | 原文线索

V2-070030 | MATHDB-363117

Nonexistence of minimizers for sufficiently large charge

摘录 / EXCERPT: Let $q>0$ be the charge parameter, and consider the minimization problem defined by equation (minmin1), whose minimizers are configurations attaining its minimum energy. Nonexisten...

conjecture | open | MathDB 页面 | 原文线索

V2-070031 | MATHDB-363122

May's strictification conjecture for symmetric bimonoidal functors

摘录 / EXCERPT: A bipermutative category has additive and multiplicative symmetric permutative structures, with the multiplicative structure distributing over the additive one. A symmetric bimonoi...

conjecture | open | MathDB 页面 | 原文线索

V2-070032 | MATHDB-363125

Log-concavity conjecture for genus distributions of surface-triangulating graphs

摘录 / EXCERPT: Let G be a graph that triangulates some surface, and let its genus distribution be the sequence of numbers of 2-cell embeddings of G in orientable surfaces of each genus. Sur...

conjecture | open | MathDB 页面 | 原文线索

V2-070033 | MATHDB-363130

Haettel's downward flag conjecture for the type D_n subdivision

摘录 / EXCERPT: Suppose Λ is of type D_n. Let Δ_Λ be its Artin complex, and let Δ' be the (δ_1,δ_2)-subdivision obtained by subdividing edges between type...

conjecture | open | MathDB 页面 | 原文线索

V2-070034 | MATHDB-363135

Mirror symmetry for lattice-polarised weak del Pezzo surfaces

摘录 / EXCERPT: Mirror symmetry for lattice-polarised weak del Pezzo surfaces. The mirror to (V,C) is given by an elliptic fibration $W\to\Delta$ with fibre F_p such that ϕ is a quasi del...

conjecture | open | MathDB 页面 | 原文线索

V2-070035 | MATHDB-363136

DHT conjecture for K3 surfaces

摘录 / EXCERPT: DHT conjecture. The surface Y admits an elliptic fibration $\pi\colon Y\to\mathbb{P}^1$ and an allowable loop $\gamma\subset\mathbb{P}^1$ compatible with the $\check{L}$-quasi-po...

conjecture | open | MathDB 页面 | 原文线索

V2-070036 | MATHDB-363140

Contractibility conjecture for cores of relative Artin complexes

摘录 / EXCERPT: Let A_S be an Artin group, and let $S'\subset S$ be almost spherical. Let $\Delta_{S,S'}$ denote the relative Artin complex spanned by vertices of types indexed by S'. Assume tha...

conjecture | open | MathDB 页面 | 原文线索

V2-070037 | MATHDB-363142

Six-cycle link criterion conjecture for relative Artin complexes

摘录 / EXCERPT: Let Λ and Λ' satisfy the assumptions of the source's six-cycle link lemma, with Λ' not necessarily admissible in Λ. Consider the links of vertices i...

conjecture | open | MathDB 页面 | 原文线索

V2-070038 | MATHDB-363143

Upward-flag conjecture for linear subdiagrams of spherical Artin diagrams

摘录 / EXCERPT: Let A_S be an irreducible spherical Artin group with Dynkin diagram Λ. Let Λ' be a linear subdiagram with consecutive vertices $\{s_i\}_{i=1}^n$, and assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-070039 | MATHDB-363144

Downward-flag conjecture for D_n subdivisions of relative Artin complexes

摘录 / EXCERPT: Let A_S be an irreducible spherical Artin group. Let $S'\subset S$ be such that $A_{S'}$ has Dynkin diagram isomorphic to type D_n for $n\geq 3$, without requiring the isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-070040 | MATHDB-363152

Generalized A=B relations

摘录 / EXCERPT: Let $A^1_{g,n}$ and $B^1_{g,n}$ be the polynomial-valued tautological classes associated with the rubber and corresponding constructions in the paper, and let $B^m_{g,n}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070041 | MATHDB-363153

Master relation for the classes $\Xi^m_{g,n}$

摘录 / EXCERPT: For $g\geq0$ and $m,n\geq1$ with $2g-2+n+m>0$, let $\Xi^m_{g,n}$ be the Laurent polynomial in u with coefficients in the tautological ring of $\overline{\mathcal{M}}_{g,n+m}$ obt...

conjecture | open | MathDB 页面 | 原文线索

V2-070042 | MATHDB-363157

Linear Euclidean realizability conjecture for median systems

摘录 / EXCERPT: Let a median system be a set system whose 1-inclusion graph is a median graph, and let d denote its dimension. Median-system realization conjecture. Every median system of dimens...

conjecture | open | MathDB 页面 | 原文线索

V2-070043 | MATHDB-363159

Existence of strictly polystable extensions for generic line bundles

摘录 / EXCERPT: Existence conjecture. There exists a strictly polystable extension of L^{-1} by L. This would extend the existence results for reducible cone spherical metrics to generic line...

conjecture | open | MathDB 页面 | 原文线索

V2-070044 | MATHDB-363160

Rank conjecture for the restricted ramification divisor map

摘录 / EXCERPT: Rank conjecture. The rank of the complex derivative of ρ at (i_1,i_2) satisfies:

conjecture | open | MathDB 页面 | 原文线索

V2-070045 | MATHDB-363163

Chen-Reissig's threshold conjecture for semilinear wave equations

摘录 / EXCERPT: Chen-Reissig's conjecture. If $C_{\mathrm{Str}}>0$, then the solution blows up at finite time for some initial data; if $C_{\mathrm{Str}}=0$, then small-data global existence holds...

conjecture | open | MathDB 页面 | 原文线索

V2-070046 | MATHDB-363164

Integral criterion conjecture for semilinear wave equations

摘录 / EXCERPT: Integral criterion conjecture. Small-data global existence holds if

conjecture | open | MathDB 页面 | 原文线索

V2-070047 | MATHDB-363173

Conjecture on the sign of the Stolarsky and power-mean difference

摘录 / EXCERPT: Sign conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-070048 | MATHDB-363174

Conjecture on the sign of the Stolarsky and power-mean difference at exceptional parameters

摘录 / EXCERPT: Sign conjecture. At $\lvert\alpha\rvert=\frac12\sqrt{2\pm\sqrt3}$,

conjecture | open | MathDB 页面 | 原文线索

V2-070049 | MATHDB-363175

Conjecture on the sign of the M_4 and power-mean difference

摘录 / EXCERPT: Sign conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070050 | MATHDB-363182

The conjecture that random generalized power series are irreducible

摘录 / EXCERPT: Let K be a field and let $b\in K((\mathbb{R}^{\leq 0}))$ be a random generalized power series, meaning that either $\operatorname{cl}(\operatorname{supp}(b))-\{0\}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-070051 | MATHDB-363184

The five-term case of Goncharov's Depth Conjecture

摘录 / EXCERPT: Let F be the field under consideration, let k be a positive integer, let $x_2,\ldots,x_k\in F^\times$, and let $w_0,\ldots,w_4\in\mathbb{P}^1(F)$. For each i, omit w_i from...

conjecture | open | MathDB 页面 | 原文线索

V2-070052 | MATHDB-363193

Wilf-equivalence conjecture for the 13 pattern sets in Corollary 3

摘录 / EXCERPT: Let the 13 pattern sets be the sets referred to in Corollary, and let $\{13452,23451\}$ be represented by the partially ordered pattern (POP) shown in the source. Wilf-equivalenc...

conjecture | open | MathDB 页面 | 原文线索

V2-070053 | MATHDB-363197

Morita invariance conjecture for the connected higher automorphism group

摘录 / EXCERPT: Let $\mathcal{C}$ be a graded $\Bbbk$-linear category such that ... for all $C,D\in\operatorname{Ob}(\mathcal{C})$ and all $i\geq 1$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070054 | MATHDB-363200

Automaticity of number walls

摘录 / EXCERPT: Let k and q be positive integers, and let a sequence over $\mathcal{F}_q$ be k-automatic. Its number wall is the associated two-dimensional array of Hankel determinants. Auto...

conjecture | open | MathDB 页面 | 原文线索

V2-070055 | MATHDB-363207

Local-Global Conjecture for heights of Kleinian group orbits

摘录 / EXCERPT: Let $\mathbb H^3$ be the upper half-space model of hyperbolic 3-space, and let Γ be a geometrically finite subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-070056 | MATHDB-363209

Positivity conjecture for Kac polynomials of quivers with multiplicities

摘录 / EXCERPT: Let Q be a quiver, let $\mathbf{r}\in\mathbb{Z}_{\geq0}^{Q_{0}}$ be a rank vector, and let $\alpha\geq1$. Let $A_{(Q,\alpha),\mathbf{r}}$ be the associated Kac polynomial for the...

conjecture | open | MathDB 页面 | 原文线索

V2-070057 | MATHDB-363213

The parabolic Airy process rate-function contraction conjecture

摘录 / EXCERPT: Let D be the space of admissible directed-landscape environments, let I be the associated rate function, and let Q be the Dirichlet energy of a function...

open_problem | open | MathDB 页面 | 原文线索

V2-070058 | MATHDB-363214

The one-branch-point conjecture for crossing-free Hubbard trees

摘录 / EXCERPT: Let a Hubbard tree be a forward-invariant tree associated with a quadratic polynomial, and call it crossing-free when the relevant tree dynamics satisfy the source's crossing-free...

conjecture | open | MathDB 页面 | 原文线索

V2-070059 | MATHDB-363234

Complete quasi-equidistribution conjecture for van der Corput sequences in irrational bases

摘录 / EXCERPT: Complete quasi-equidistribution conjecture. Let γ be the largest root of the polynomial x^2-px-q for $1\leq q\leq p$. The van der Corput sequence in base γ is c.q.e...

conjecture | open | MathDB 页面 | 原文线索

V2-070060 | MATHDB-363243

The Roos-category conjecture for dualizable locally presentable linear categories

摘录 / EXCERPT: Let $\mathcal{C}$ be a locally presentable linear category over a field. A Grothendieck category is a Roos category if it satisfies Ab6 and Ab4, where Ab6 is Grothendieck's distrib...

conjecture | open | MathDB 页面 | 原文线索

V2-070061 | MATHDB-363245

The deviating subgroup conjecture for the Burau representation of B_3

摘录 / EXCERPT: Let F be the deviating subgroup associated with the image of the Burau representation of B_3, and let a group be free of countably infinite rank when it has a countably infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-070062 | MATHDB-363248

The v-function formula for powers of ideals generated by regular sequences

摘录 / EXCERPT: The v-function formula. For all $k\ge1$,

conjecture | open | MathDB 页面 | 原文线索

V2-070063 | MATHDB-363250

The Noetherian-Pfaffian counting o-minimality conjecture

摘录 / EXCERPT: Let $\mathbb R_{\mathrm{LN},\mathrm{PF}}$ be the structure generated by the larger class of Noetherian and Pfaffian functions considered in the paper. Noetherian-Pfaffian counting...

conjecture | open | MathDB 页面 | 原文线索

V2-070064 | MATHDB-363256

Non-entropy shock comparison conjecture

摘录 / EXCERPT: Let $\mu\in\mathscr{N}_+$ and write $v:=\partial_x g:=\partial_x\mathsf{H}[\mu]$, where v is a tractable solution. Non-entropy shock comparison conjecture. The measure μ rest...

conjecture | open | MathDB 页面 | 原文线索

V2-070065 | MATHDB-363258

Formal uncertainty principle implies duality

摘录 / EXCERPT: Formal uncertainty-duality conjecture. If A and B satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-070066 | MATHDB-363264

The effective critical point conjecture for free-boundary high-dimensional Ising models

摘录 / EXCERPT: Free-boundary effective critical point conjecture. There exists such a $\beta_{c,r}$ for which all statements of the non-Gaussian limit and universal profile conjecture hold with...

conjecture | open | MathDB 页面 | 原文线索

V2-070067 | MATHDB-363271

String operators realizing the Drinfeld center in Levin-Wen models

摘录 / EXCERPT: String-operator realization conjecture. There exist string operators associated to the simple objects of $Z(\mathcal{C})$ such that, when they are used to define F-symbols and...

conjecture | open | MathDB 页面 | 原文线索

V2-070068 | MATHDB-363272

Arthur's characterization conjecture for local Arthur packets

摘录 / EXCERPT: Let $G=G(V)$ be the classical group associated to a quadratic space (V,q_V), and let ψ be a local Arthur parameter of G. For an endoscopic group G' of G and a local Ar...

conjecture | open | MathDB 页面 | 原文线索

V2-070069 | MATHDB-363274

Foley-Hoàng-Merkel's twinning conjecture for $(3+1)$-free posets

摘录 / EXCERPT: Foley-Hoàng-Merkel's twinning conjecture. If P is $(3+1)$-free and $X_{\operatorname{Inc}(P)}(\mathbf{x})$ is e-positive, then $X_{\operatorname{Inc}(P_v)}(\mathbf{x})$ is e-...

conjecture | open | MathDB 页面 | 原文线索

V2-070070 | MATHDB-363275

Twin-free reduction conjecture for $(3+1)$-free posets

摘录 / EXCERPT: Twin-free reduction conjecture. If P is $(3+1)$-free and twin-free, then $X_{\operatorname{Inc}(P)}(\mathbf{x})$ is e-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-070071 | MATHDB-363277

Generalized Langlands geometric objects for two-fold Weil coverings

摘录 / EXCERPT: Let K be a finite extension of $ \mathbb{Q}_p$ or $ \mathbb{F}_q((z))$ with $ \ell\neq p$ and $p>2$, let G/K be reductive, and let $W_{K,2}$ denote the indicated two-fold cover...

conjecture | open | MathDB 页面 | 原文线索

V2-070072 | MATHDB-363295

The injectivity conjecture for Tits homomorphisms

摘录 / EXCERPT: Let $M(\mathbf d)$ be a Coxeter monoid and let $\mathbf T_{\mathbf d}$ be a Tits homomorphism associated with it. Injectivity conjecture. All Tits homomorphisms...

conjecture | solved | MathDB 页面 | 原文线索

V2-070073 | MATHDB-363296

The submonoid conjecture for multiparabolic elements

摘录 / EXCERPT: Let M be a Coxeter matrix over I, let $W(M)$ be its Coxeter group with Hecke product $\star$, and let $\tilde{\mathsf P}(M)$ be the set of multiparabolic elements in $W(M)$. Su...

conjecture | open | MathDB 页面 | 原文线索

V2-070074 | MATHDB-363297

The classification conjecture for fully supported standard homomorphisms from A_2

摘录 / EXCERPT: Let $M\in\mathbf{Cox}(I)$ be irreducible of finite type, and let $\Phi\in\operatorname{Hom}_{\mathscr A}(A_2,M)$ be fully supported and standard with $\Phi\ne\Phi$. Classification...

conjecture | open | MathDB 页面 | 原文线索

V2-070075 | MATHDB-363298

The three-family conjecture for standard homomorphisms from $I_2(N)$ to A_n

摘录 / EXCERPT: Let $1\le b\le a<n-1$ and $a>1$. Let $w_\circ^{[r,s]}$ denote the longest element associated with the interval $[r,s]$. Three-family conjecture for $I_2(N)$ homomorphisms. The assi...

conjecture | open | MathDB 页面 | 原文线索

V2-070076 | MATHDB-363299

The $12/d$ family conjecture for standard homomorphisms to A_{n+1}

摘录 / EXCERPT: For $i\in[2,n-2]$, let $d=1+\delta_{2i,n}+\delta_{2i,n-1}+\delta_{2i,n+1}$. $12/d$ family conjecture. The assignments ... define a standard homomorphism ... This is one of the conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-070077 | MATHDB-363300

The conjectural families of standard homomorphisms to D_{n+1}

摘录 / EXCERPT: Use the displayed labeling of the Coxeter graph of type D_{n+1}, let $K_1,K_2\subset[0,n]$, and let $w_\circ^K$ denote the longest element associated with K. Conjectural...

conjecture | open | MathDB 页面 | 原文线索

V2-070078 | MATHDB-363303

Mazur-Tate-Teitelbaum p-adic BSD conjecture for elliptic curves

摘录 / EXCERPT: Mazur-Tate-Teitelbaum's p-adic BSD conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-070079 | MATHDB-363304

Bertolini-Darmon-Prasanna p-adic BSD conjecture over an imaginary quadratic field

摘录 / EXCERPT: Bertolini-Darmon-Prasanna p-adic BSD conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070080 | MATHDB-363312

The \Ver_{p^\infty}-characterization conjecture for moderate-growth tensor categories

摘录 / EXCERPT: $\operatorname{Ver}_{p^\infty}$-conjecture. The following conditions are equivalent on $\mathcal{C}$:

conjecture | open | MathDB 页面 | 原文线索

V2-070081 | MATHDB-363314

The \Ver_4-fibering characterization conjecture

摘录 / EXCERPT: Ver_4-fibering conjecture. The following conditions are equivalent on $\mathcal{C}$:

conjecture | open | MathDB 页面 | 原文线索

V2-070082 | MATHDB-363316

Babai's independence-ratio conjecture for minimal Cayley graphs

摘录 / EXCERPT: Babai's conjecture. For every $\varepsilon>0$, there exists a minimal Cayley graph $G=(V,E)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070083 | MATHDB-363318

Deift's almost-periodic dynamics conjecture for the KdV equation

摘录 / EXCERPT: Let $u(t,x)$ solve the KdV equation ... Assume that the initial data $u(0,x)$ is almost periodic in x. Deift's conjecture. The solution $u(t,x)$ is almost periodic in time t and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-070084 | MATHDB-363321

The sharp second-coefficient conjecture for harmonic K-quasiconformal mappings

摘录 / EXCERPT: Let $\mathcal{S}^0_\mathcal{H}(K)$ be the class of normalized harmonic K-quasiconformal mappings, and write $f=h+\overline{g}$. Second-coefficient conjecture. One has ... The bound...

conjecture | open | MathDB 页面 | 原文线索

V2-070085 | MATHDB-363322

The Hardy-order conjecture for harmonic K-quasiconformal mappings

摘录 / EXCERPT: Let $\mathcal{S}_\mathcal{H}(K)$ be the class of normalized harmonic K-quasiconformal mappings, and write $f=h+\overline{g}$. Harmonic K-quasiconformal Hardy-order conjecture. On...

conjecture | open | MathDB 页面 | 原文线索

V2-070086 | MATHDB-363323

The covering-radius conjecture for harmonic K-quasiconformal mappings

摘录 / EXCERPT: Covering-radius conjecture. The radius satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070087 | MATHDB-363328

Period-collapse conjecture for deleted Shi arrangements of root systems

摘录 / EXCERPT: Let $k\in\mathbb{Z}_{>0}$, let Φ be a root system, let Φ^+ be its set of positive roots, and let $\delta\in\Phi^+$. Write δ^c for the complement of δ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-070088 | MATHDB-363335

The large-q exponential-regime conjecture for sphere packing of proper colorings

摘录 / EXCERPT: Large-q exponential-regime conjecture. For all q sufficiently large, there exists an infinite sequence of d-regular graphs with $\lambda_2=\Theta(1/\sqrt{d})$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-070089 | MATHDB-363336

The Gorenstein contraction conjecture for multiscale differentials

摘录 / EXCERPT: Let (X,η) be a generalized multiscale differential with semistable modification $(\tilde{X},\tilde{\eta})$, dual graph $\tilde{\Gamma}$, levels indexed by i, and vanishing o...

conjecture | open | MathDB 页面 | 原文线索

V2-070090 | MATHDB-363338

Malkevitch's consecutive-cycle conjecture for 4-connected planar graphs

摘录 / EXCERPT: Let G be a 4-connected planar graph on n vertices. A cycle of length 4 is a cycle with four vertices. Malkevitch's conjecture. If G contains a cycle of length 4, then G...

conjecture | open | MathDB 页面 | 原文线索

V2-070091 | MATHDB-363349

Local closedness and finiteness conjectures for Kollár valuations

摘录 / EXCERPT: Local closedness and finiteness conjectures. The locus $\mathcal{D}^{\rm KV}(X,\Delta+D)$ satisfies all of the following:

conjecture | open | MathDB 页面 | 原文线索

V2-070092 | MATHDB-363350

Dolorfino et al.'s bounded-hypercube conjecture for toric-code families

摘录 / EXCERPT: Dolorfino et al.'s bounded-hypercube conjecture. If $\{M(P_i)\}_i$ is bounded, then

conjecture | open | MathDB 页面 | 原文线索

V2-070093 | MATHDB-363352

The logarithmic-form cokernel bound for an Euler restriction

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a hyperplane arrangement, let $H\in{\mathcal{A}}$, and write ${\mathcal{A}}'$ for the deletion and ${\mathcal{A}}^H$ for the restriction to H. Let ... be the...

conjecture | open | MathDB 页面 | 原文线索

V2-070094 | MATHDB-363353

The cokernel-dimension identity for restriction maps

摘录 / EXCERPT: Let ${\mathcal{A}}$ be a hyperplane arrangement and let $H\in{\mathcal{A}}$. Let ρ^H and res^H denote the maps defined in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070095 | MATHDB-363360

Total nonnegativity of the permanental analogue of the Sihong polynomial

摘录 / EXCERPT: Permanental Sihong polynomial conjecture. The polynomial above is totally nonnegative for all n. This would provide a permanental analogue of the known totally nonnegative determ...

conjecture | open | MathDB 页面 | 原文线索

V2-070096 | MATHDB-363362

Goussarov--Habiro conjecture for string links in the cylinder

摘录 / EXCERPT: Let $L(m)$ denote the string links on m strands. For $n\geq 1$, let V_n-equivalence be the equivalence relation induced by the Vassiliev filtration, and let C_n-equivalence b...

conjecture | open | MathDB 页面 | 原文线索

V2-070097 | MATHDB-363375

Conjecture on j-invariants of isogenous elliptic curves

摘录 / EXCERPT: j-invariant conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070098 | MATHDB-363378

Isogeny invariance conjecture for valuations of elliptic-curve j-invariants

摘录 / EXCERPT: Let $\mathcal{O}_K$ be a discrete valuation domain with field of fractions K and residue field k, and let v be the normalized valuation on K. Let E_1/K and E_2/K be ell...

conjecture | open | MathDB 页面 | 原文线索

V2-070099 | MATHDB-363380

Optimality of random linear codes for LCL properties

摘录 / EXCERPT: Let $\cP$ be a b-local, monotone-decreasing LCL property, and let $R_\cP$ denote its threshold rate. For every $\eps>0$ and $b\in\mathbb{N}$, there exist constants n_0 and...

conjecture | open | MathDB 页面 | 原文线索

V2-070100 | MATHDB-363387

Peak-valley antichain statistic conjecture for odd uniform self-dual fences

摘录 / EXCERPT: Let a be a positive integer, let t be odd, and let $F=\breve{F}(a^t)$ be the uniform self-dual fence with t segments. For each peak p and valley v of F, let χ_p an...

conjecture | open | MathDB 页面 | 原文线索

V2-070101 | MATHDB-363388

Antichain-indicator symmetry conjecture for odd uniform self-dual fences

摘录 / EXCERPT: Let a be a positive integer, let t be odd, and let $F=\breve{F}(a^t)$ be the uniform self-dual fence with t segments. For $i\in\{1,\ldots,t-1\}$, let s_i and s_{t-i} be t...

conjecture | open | MathDB 页面 | 原文线索

V2-070102 | MATHDB-363389

Full-range codimension conjecture for homomesic spaces of fences

摘录 / EXCERPT: Let F be a fence with t segments. Let $I_H(F)$ be its ideal-statistic homomesic space and $I_T(F)$ its ideal-toggleability space. Full-range codimension conjecture. For every i...

conjecture | open | MathDB 页面 | 原文线索

V2-070103 | MATHDB-363393

Characterization of lattices by unique quasi-suprema and quasi-infima

摘录 / EXCERPT: Let a partially ordered family be a family equipped with a partial order. For two elements, a quasi-supremum is a minimal upper bound and a quasi-minimum is a maximal lower bound....

conjecture | open | MathDB 页面 | 原文线索

V2-070104 | MATHDB-363394

Conjecture on the starlikeness order of non-linear resolvents

摘录 / EXCERPT: Resolvent starlikeness conjecture. Every mapping G_λ is starlike of order at least $\frac12$. Furthermore, the order of starlikeness of G_λ tends to 1 as...

conjecture | open | MathDB 页面 | 原文线索

V2-070105 | MATHDB-363413

The specific sequence irregularity strength conjecture

摘录 / EXCERPT: Let G be a graph without a component isomorphic to K_2. For each $1\le i\le\Delta(G)$, let n_i be the number of vertices of degree i, and define ... The specific sequence irr...

conjecture | open | MathDB 页面 | 原文线索

V2-070106 | MATHDB-363415

Construction 2 weight-spectrum conjecture for Reed-Muller codes

摘录 / EXCERPT: Construction 2 weight-spectrum conjecture. The set S contains all weights in the two families

conjecture | open | MathDB 页面 | 原文线索

V2-070107 | MATHDB-363419

Conjecture characterizing spanning sets in \mathbb{Z}_2\times\mathbb{Z}_{2k}

摘录 / EXCERPT: Let k and s be positive integers. An s-spanning set of size 2 in a finite abelian group is a two-element set whose signed sums of at most s terms cover the group. The...

conjecture | open | MathDB 页面 | 原文线索

V2-070108 | MATHDB-363427

The non-3-colorability conjecture for countable undirected multigraphs

摘录 / EXCERPT: Multigraph non-3-colorability conjecture. There exists an undirected countable multigraph that is not majority 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-070109 | MATHDB-363429

Large induced bounded-degree trees conjecture for random graphs

摘录 / EXCERPT: Large induced bounded-degree trees conjecture. For every $cDelta\textgreater=1$, the stated containment holds for all such p.

conjecture | open | MathDB 页面 | 原文线索

V2-070110 | MATHDB-363430

Escape conjecture for nonsmooth folds with higher-order terms

摘录 / EXCERPT: Let f be a map of the form ... where the higher-order terms satisfy ... as $(x;\mu)\to(0;0)$, and let $B_\eta=\{x\in\mathbb{R}^n\mid\|x\|\leq\eta\}$. Assume $\det(I-A_L)>0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-070111 | MATHDB-363434

Strong Nine Dragon Tree Conjecture

摘录 / EXCERPT: Strong Nine Dragon Tree Conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-070112 | MATHDB-363437

Akbari-Dalirrooyfard-Ehsani-Ozeki-Sherkati conjecture on forbidden out-degrees

摘录 / EXCERPT: Let G be a loopless d-regular graph, with parallel edges permitted, and let $F\subseteq\{0,1,\ldots,d\}$ be a list of forbidden out-degrees. An orientation is F-avoiding if n...

conjecture | open | MathDB 页面 | 原文线索

V2-070113 | MATHDB-363444

Simultaneous reflection-length reduction conjecture for Coxeter groups

摘录 / EXCERPT: Simultaneous reflection-length reduction conjecture. There exists a letter s in $u_1\cdots u_p$ such that, after omitting it,

conjecture | open | MathDB 页面 | 原文线索

V2-070114 | MATHDB-363449

The negative-energy Hill-boundary radius conjecture

摘录 / EXCERPT: Let M_E be the negative-energy configuration space equipped with the Jacobi-Maupertuis metric d_E, let ∂M_E denote its Hill boundary, and let 0 be the origin config...

conjecture | open | MathDB 页面 | 原文线索

V2-070115 | MATHDB-363464

Shi-Wang-Zuo's generalized Hertling conjecture for the Tjurina spectrum

摘录 / EXCERPT: Shi-Wang-Zuo's generalized Hertling conjecture. The original inequality

conjecture | open | MathDB 页面 | 原文线索

V2-070116 | MATHDB-363468

Monotonicity along increasing curves in the antiferromagnetic isotropic Ashkin-Teller model

摘录 / EXCERPT: Monotonicity conjecture. For $\beta_1,\beta_2\in(0,1)$ with $\beta_1<\beta_2$, there exists $\varepsilon>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070117 | MATHDB-363480

Klein-Roudenko-Stoilov blow-up rate conjecture for the critical generalized Zakharov-Kuznetsov equation

摘录 / EXCERPT: Klein-Roudenko-Stoilov conjecture. The solution blows up at a finite time T, and, as $t\to T$,

conjecture | open | MathDB 页面 | 原文线索

V2-070118 | MATHDB-363495

Generic double multiplicity conjecture for symmetric singular matrix pencils

摘录 / EXCERPT: Generic double multiplicity conjecture. There exists a generic set $\Omega\subseteq\mathbb R^{n,k}$ such that, for every $U\in\Omega$ and every $\tau\in\mathbb R\setminus\{0\}$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-070119 | MATHDB-363498

Functional strong slicing conjecture

摘录 / EXCERPT: Let L_f, $\widetilde{L}_f$, and $\widehat{L}_f$ be the three functional isotropic constants defined for integrable centered log-concave functions. Define ... Functional strong slic...

conjecture | open | MathDB 页面 | 原文线索

V2-070120 | MATHDB-363506

Yip's lower-bound conjecture for the Tesler flow polytope

摘录 / EXCERPT: Yip's conjecture. For $\bm{N}=(1,1,\ldots,1,-n)$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-070121 | MATHDB-363511

Nilpotency conjecture for the Collatz adjacency submatrices

摘录 / EXCERPT: Nilpotency conjecture for the Collatz adjacency submatrices. If $n\geqslant 3$, then the matrix C_n is nilpotent.

conjecture | open | MathDB 页面 | 原文线索

V2-070122 | MATHDB-363518

Gross-Prasad-Rallis genericity criterion

摘录 / EXCERPT: Let bbG be a connected reductive group over bbF, let $bbphi$ be an $bbSL_2$-parameter, and let $L(s,bboperatorname{Ad}\circbbphi)$ denote its adjoint L-factor. Gross-Prasad-R...

conjecture | open | MathDB 页面 | 原文线索

V2-070123 | MATHDB-363520

Extended packet conjecture

摘录 / EXCERPT: Suppose bbG is bbF-quasi-split and fix a Whittaker datum $(bbU,bbpsi)$. An extended L-packet is a fibre of the composite semisimplification map from representations to semisi...

conjecture | open | MathDB 页面 | 原文线索

V2-070124 | MATHDB-363521

Haines's invariance conjecture under isomorphisms

摘录 / EXCERPT: Let $bbalpha:bbG\tobbG'$ be an isomorphism of connected reductive groups, let $bbpi\inbbPi_{bbC}(bbG)$, and let ${}^{bbalpha}bbpi$ be the irreducible representation of bbG' obtai...

conjecture | open | MathDB 页面 | 原文线索

V2-070125 | MATHDB-363536

The P=C conjecture for the projective plane

摘录 / EXCERPT: Let $M_{d,\chi}$ be the moduli space associated with the curve class $\beta=dH$ and Euler characteristic $\chi\in\mathbb{Z}$ on $\mathbb{P}^2$, where H is the class of a line and...

conjecture | open | MathDB 页面 | 原文线索

V2-070126 | MATHDB-363538

Shellability conjecture for intervals in the second class of level Eulerian posets

摘录 / EXCERPT: Shellability conjecture. The intervals in the level Eulerian poset from Section are shellable, and hence their order complexes are homeomorphic to spheres.

conjecture | open | MathDB 页面 | 原文线索

V2-070127 | MATHDB-363545

The Gorenstein criterion conjecture for tautological rings of moduli spaces of curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable curves of genus g with n markings. Let $\mathsf{R}^*(\overline{\mathcal{M}}_{g,n})$ be its tautological Chow ri...

conjecture | open | MathDB 页面 | 原文线索

V2-070128 | MATHDB-363547

Besau-Werner's isoperimetric inequality for floating area

摘录 / EXCERPT: Let $\mathcal{K}(\mathbb{S}^n)$ denote the set of spherically convex bodies, and let $\Omega_s(K)$ be the floating area of $K\in\mathcal{K}(\mathbb{S}^n)$: ... For each K, let...

conjecture | solved | MathDB 页面 | 原文线索

V2-070129 | MATHDB-363550

The inductive signed projective cube bounding conjecture

摘录 / EXCERPT: Inductive bounding conjecture. If $\mathcal{SP}_{k}$ is bounded by a signed graph $\hat{B}$, then $\mathcal{SP}_{k+1}$ is bounded by $EDC(\hat{B})$.

conjecture | open | MathDB 页面 | 原文线索

V2-070130 | MATHDB-363551

The signed planar signature-packing conjecture

摘录 / EXCERPT: Signed planar signature-packing conjecture. Any signed planar graph in

conjecture | open | MathDB 页面 | 原文线索

V2-070131 | MATHDB-363553

Stabilization conjecture for log-concavity of k-regular partitions

摘录 / EXCERPT: Log-concavity stabilization conjecture. Let $k\geq 2$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070132 | MATHDB-363556

Liu-Lou-Zhu's Carleson embedding conjecture for ${\mathcal Q}_s$

摘录 / EXCERPT: Let $0<s<1$ and let μ be a positive Borel measure on the open unit disc $\mathbb D$. The spaces ${\mathcal Q}_s$ and ${\mathcal T}_{s,2}^2(\mu)$ are linked by the identity map...

conjecture | open | MathDB 页面 | 原文线索

V2-070133 | MATHDB-363557

Pau-Zhao's Carleson embedding conjecture for $F(p,p-2,s)$

摘录 / EXCERPT: Let $p>1$, let $0<s<1$, let μ be a positive Borel measure on the open unit disc $\mathbb D$, and let $F(p,p-2,s)$ and ${\mathcal T}_{s,p}^p(\mu)$ denote the function and tent s...

conjecture | open | MathDB 页面 | 原文线索

V2-070134 | MATHDB-363558

Pau-Zhao's generalized Carleson embedding conjecture for $F(p,p\alpha-2,s)$

摘录 / EXCERPT: Let $p>1$, let $\alpha>1$, let $0<s<1$, let μ be a positive Borel measure on the open unit disc $\mathbb D$, and let $F(p,p\alpha-2,s)$ and ${\mathfrak T}_{s,p}^{\infty}(\mu)$...

conjecture | open | MathDB 页面 | 原文线索

V2-070135 | MATHDB-363560

Conjecture on the detailed shape of the Sadovskii vortex patch

摘录 / EXCERPT: Let Ω be the bounded open set defining a Sadovskii vortex patch as described in Theorem 1. Detailed-shape conjecture. The set Ω is connected, convex, and makes a righ...

conjecture | open | MathDB 页面 | 原文线索

V2-070136 | MATHDB-363561

Uniqueness conjecture for small-impulse Sadovskii vortex patches

摘录 / EXCERPT: Let $S_{\mu,1,1}$ denote the set of maximizers associated with impulse μ, and let μ be the impulse parameter. Small-impulse uniqueness conjecture. There exists a universal...

conjecture | open | MathDB 页面 | 原文线索

V2-070137 | MATHDB-363567

P=C conjecture for one-dimensional sheaf moduli on del Pezzo surfaces

摘录 / EXCERPT: Let S be a del Pezzo surface, let H be the polarization, let β be an effective divisor, let $\chi\in\mathbb{Z}$, and let $M_{\beta,\chi}$ be the moduli space of one-dimen...

conjecture | open | MathDB 页面 | 原文线索

V2-070138 | MATHDB-363568

Stabilization conjecture for refined BPS numbers of one-dimensional sheaf moduli

摘录 / EXCERPT: Refined BPS stabilization conjecture. The refined BPS number $n_\beta^{i,j}$ stabilizes when β is sufficiently positive. This conjectures asymptotic stability of the refined...

conjecture | open | MathDB 页面 | 原文线索

V2-070139 | MATHDB-363569

Bobadilla's conjecture on the smoothness of an irreducible singular locus

摘录 / EXCERPT: Bobadilla's conjecture. Under these assumptions, Σf is smooth at $\mathbf 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-070140 | MATHDB-363573

The countable-index conjecture for automatic continuity in Polish groups

摘录 / EXCERPT: Let G be a Polish group. Say that G has the countable index property when every subgroup of index at most countable is open, and say that G has the automatic continuity prope...

conjecture | open | MathDB 页面 | 原文线索

V2-070141 | MATHDB-363575

Extension of the double-dimer result to CP hyperbolic triangulations

摘录 / EXCERPT: Let G be a CP hyperbolic triangulation, without necessarily assuming bounded degree or nonamenability. Extension conjecture. The statements of Theorems $\mathrm{thm:main0}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070142 | MATHDB-363580

Conjecture on pattern-avoiding ascent sequences under the maximum bijection

摘录 / EXCERPT: Pattern-avoidance conjecture. The map $\mathrm{hat}_{\max}$ restricts to the following bijections:

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-070143 | MATHDB-363582

Fast-speed comparison conjecture for contact-process critical values

摘录 / EXCERPT: Fast-speed comparison conjecture. Under these assumptions, $\lambda_1>0$ if and only if $\lambda_1^p>0$, and $\lambda_2>0$ if and only if $\lambda_2^p>0$.

conjecture | open | MathDB 页面 | 原文线索

V2-070144 | MATHDB-363583

Immunisation conjecture for fast degree penalisation

摘录 / EXCERPT: Immunisation conjecture. If $\alpha\geq 1$, then for every sufficiently small $\kappa>0$ there exists $\nu'=\nu'(\kappa,\eta)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070145 | MATHDB-363584

Penalty relaxation feasibility bound for convex constrained optimization

摘录 / EXCERPT: Let f and g define a convex constrained optimization problem of the form ... For a penalty parameter Λ, consider the penalty relaxation ... where $[g(x)]_+$ denotes the pos...

conjecture | open | MathDB 页面 | 原文线索

V2-070146 | MATHDB-363601

Hoffman basis conjecture for multiple zeta values

摘录 / EXCERPT: Hoffman basis conjecture. The Hoffman elements $\zeta(k_1,\ldots,k_d)$ with $k_i\in\{2,3\}$ form a basis for $\mathcal{Z}$; in particular,

conjecture | open | MathDB 页面 | 原文线索

V2-070147 | MATHDB-363611

Existence of symEF(n-1) allocations

摘录 / EXCERPT: Let an instance consist of n agents and m items. A symEF$(n-1)$ allocation is an allocation satisfying symmetric envy-freeness up to $n-1$ items. SymEF$(n-1)$ existence conject...

conjecture | open | MathDB 页面 | 原文线索

V2-070148 | MATHDB-363616

Anisotropic Calderón conjecture at zero frequency

摘录 / EXCERPT: Anisotropic Calderón conjecture at zero frequency. If

conjecture | open | MathDB 页面 | 原文线索

V2-070149 | MATHDB-363617

Modified anisotropic Calderón conjecture at nonzero frequency

摘录 / EXCERPT: Modified anisotropic Calderón conjecture at nonzero frequency. If

conjecture | open | MathDB 页面 | 原文线索

V2-070150 | MATHDB-363632

Play-once DG game Nash-solvability conjecture

摘录 / EXCERPT: Let an n-person deterministic graphical game be play-once if every player makes at most one move during a play. The game is Nash-solvable if it has a Nash equilibrium in pure sta...

conjecture | open | MathDB 页面 | 原文线索

V2-070151 | MATHDB-363634

Conjecture that Simpson datasets are pathological for logistic regression

摘录 / EXCERPT: A Simpson dataset is a $2\times 2\times 2$ dataset exhibiting Simpson's paradox; a pathological regularization regime is a set of regularization parameters for which the trend obta...

conjecture | open | MathDB 页面 | 原文线索

V2-070152 | MATHDB-363637

Large-system factorial-growth conjecture for explosive birth processes

摘录 / EXCERPT: Let $F_1=\ldots=F_A$ be the common birth-rate function for A agents, and let the initial state be ... For each $a\geq 2$, let $c(A,a)$ denote the tail-dependence constant appearing...

conjecture | open | MathDB 页面 | 原文线索

V2-070153 | MATHDB-363643

The logarithmic susceptibility asymptotic for the four-dimensional Ising model

摘录 / EXCERPT: Let $\chi(\beta)$ denote the susceptibility of the high-dimensional Ising model and let β_c be its critical inverse temperature. Logarithmic susceptibility conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070154 | MATHDB-363647

Rubin's range characterization conjecture for the spherical mean transform

摘录 / EXCERPT: Rubin's range characterization conjecture. A function $g\in C_c^\infty(\mathbb{S}^{n-1}\times(0,2))$ belongs to the range of the operator $f\mapsto\mathcal{M}f$ if and only if

conjecture | solved | MathDB 页面 | 原文线索

V2-070155 | MATHDB-363660

Convergence of the Laplace expansion process for suitable matrix families

摘录 / EXCERPT: Consider a family of matrices to which the Laplace expansion process described in the source is applied. Convergence conjecture. For matrix families satisfying specified criteria,...

conjecture | open | MathDB 页面 | 原文线索

V2-070156 | MATHDB-363661

A common annihilator for resistance numerator and denominator families

摘录 / EXCERPT: Let a family of graphs have Laplacian matrices, and apply the procedures LaplaceExpand and SystemReduce together with the mathematical theorems described in the source. The numerat...

conjecture | open | MathDB 页面 | 原文线索

V2-070157 | MATHDB-363665

The Erdős-Pósa parameter conjecture for minor-closed graph classes

摘录 / EXCERPT: Erdős-Pósa parameter conjecture. For every graph H, there exists a minor-monotone graph parameter $\text{\scriptsize\textsf{EP}}_H$ such that $\mathcal{M}_H$ has the Erdős-Pósa p...

conjecture | open | MathDB 页面 | 原文线索

V2-070158 | MATHDB-363670

Condensed generalized Langlands parametrization for ℓ-adic perverse sheaves

摘录 / EXCERPT: Let $n\geq 2$, let X be a finite extension of a local nonarchimedean field, let $G(X)$ be a reductive group, and let $\mathrm{Gal}_{X,n}$ and $\mathrm{Weil}_{X,n}$ denote the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-070159 | MATHDB-363671

Breuil-Schneider conjecture for generalized Robba-ring parametrizations

摘录 / EXCERPT: Let X be as in the paper, let $n\geq 2$, let $G(X)$ be a reductive group, and let $\Pi_X[\log(1+T)^{1/n}]$ be the generalized Robba ring in the imperfect setting. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070160 | MATHDB-363679

Lyapunov positivity conjecture for coded Kalman filtering

摘录 / EXCERPT: Let $B=\operatorname{diag}(b_1,\ldots,b_k)$ with $b_i<1$ for all i, let $H=\operatorname{diag}(h_1,\ldots,h_n)$, and let $\Gamma\in\mathbb{R}^{n\times k}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070161 | MATHDB-363682

Sharp geometric characterization of uniform stabilization for the damped wave equation

摘录 / EXCERPT: Uniform stabilization conjecture. Equation is uniformly stable if and only if b satisfies the Geometric Control Condition and the Turning Point Condition.

conjecture | open | MathDB 页面 | 原文线索

V2-070162 | MATHDB-363692

Closure of the circuit reduction property under gluing

摘录 / EXCERPT: Closure conjecture. The set $\mathcal A$ is closed under gluing: if $A,B\in\mathcal A$ admit a gluing, then

conjecture | open | MathDB 页面 | 原文线索

V2-070163 | MATHDB-363693

Distance-irreducible elements for three-variable complete intersections

摘录 / EXCERPT: Distance-irreducible basis conjecture. If M is not distance reducing, then

conjecture | open | MathDB 页面 | 原文线索

V2-070164 | MATHDB-363694

Inclusion of distance-reducing move sets

摘录 / EXCERPT: Distance-reducing inclusion conjecture. For every matrix A,

conjecture | open | MathDB 页面 | 原文线索

V2-070165 | MATHDB-363699

Ballantine--Burson--Craig--Folsom--Wen hook length bias conjecture for self-conjugate partitions

摘录 / EXCERPT: Ballantine--Burson--Craig--Folsom--Wen hook length bias conjecture. For every integer $t\geq 2$, both of the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-070166 | MATHDB-363718

Random cellular-semimetric graph Laplacian convergence on the Sierpiński gasket

摘录 / EXCERPT: Let SG be the Sierpiński gasket, let $\mathcal{L}$ be its Laplacian, and let $\mathcal{L}_\epsilon(\omega)$ be the random graph averaging Laplacian defined from random s...

conjecture | open | MathDB 页面 | 原文线索

V2-070167 | MATHDB-363724

The type B super-Artin basis conjecture

摘录 / EXCERPT: Let $R_{\mathfrak{B}_n}^{(1,1)}$ be the type B $(1,1)$ coinvariant ring, and let $B_{\mathfrak{B}_n}^{(1,1)}$ be the type B super-Artin set. Type B super-Artin conjecture. .....

conjecture | open | MathDB 页面 | 原文线索

V2-070168 | MATHDB-363733

The conjecture on the maximal order of vanishing of torus Laplacian eigenfunctions

摘录 / EXCERPT: Let $d\geq 2$. Let $\Gamma_d(\lambda)$ denote the maximal order of vanishing of a Laplacian eigenfunction on the d-dimensional torus with frequency parameter λ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070169 | MATHDB-363734

The conjectured sharp lower bound for observability constants on high-dimensional tori

摘录 / EXCERPT: Let $m_d(r)$ be the optimal observability constant for Laplacian eigenfunctions on the d-dimensional torus observed on a ball of radius r. For $d\geq 4$, let C_d and C'_d b...

conjecture | open | MathDB 页面 | 原文线索

V2-070170 | MATHDB-363736

The maximal Brill-Noether loci conjecture

摘录 / EXCERPT: Let $\mathcal{M}^{r}_{g,d}$ denote the Brill-Noether locus of smooth genus-g curves possessing a linear series g^r_d, and let $\rho(g,r,d)=g-(r+1)(g-d+r)$ be the Brill-Noether...

conjecture | open | MathDB 页面 | 原文线索

V2-070171 | MATHDB-363740

Shellability conjecture for the DKK-triangulation of the flow polytope

摘录 / EXCERPT: Shellability conjecture. The DKK-triangulation $ obreak\root_n^{\textup{LRL}}$ is shellable because both the modified lexicographic ordering and its reverse are shellings.

conjecture | open | MathDB 页面 | 原文线索

V2-070172 | MATHDB-363749

Syzygy behavior in the base locus decomposition program

摘录 / EXCERPT: Syzygy behavior conjecture. A sheaf U is F-admissible, where F is a Bridgeland destabilizing object that yields the wall W when U enters the stable base locus at the wall...

conjecture | open | MathDB 页面 | 原文线索

V2-070173 | MATHDB-363757

The co-chromaticity conjecture for signed complete graphs

摘录 / EXCERPT: A signed graph is a graph whose edges are assigned positive or negative signs; its bivariate chromatic polynomial is denoted by $\mathscr B(\Sigma,x,y)$, and its chromatic polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-070174 | MATHDB-363758

The signed-threshold conjecture for bivariate chromatic polynomials

摘录 / EXCERPT: For each integer $n\geqslant 0$ and each vector $\bm{a}\in\{-1,0,1\}^n$, let $\mathcal{T}_{\bm{a}}$ be the associated signed threshold graph, with bivariate chromatic polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-070175 | MATHDB-363759

The chromatic-polynomial characterization of signed complete graphs

摘录 / EXCERPT: Let Σ_1 and Σ_2 be signed complete graphs. Write $\Sigma_1\cong\Sigma_2$ when they are isomorphic, and let $\mathscr B(\Sigma,x,y)$ and $\mathsf E(\Sigma,x,y)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-070176 | MATHDB-363778

Thurston's genericity conjecture for pseudo-Anosov mapping classes

摘录 / EXCERPT: Let Σ be a surface, let $\operatorname{MCG}(\Sigma)$ denote its mapping class group, and let S be a finite generating set for $\operatorname{MCG}(\Sigma)$. Write d_S for...

conjecture | open | MathDB 页面 | 原文线索

V2-070177 | MATHDB-363788

Hyperuniformity crossover conjecture for the facilitated exclusion process

摘录 / EXCERPT: Let $d\ge 2$ and consider the facilitated exclusion process (FEP), with critical density ρ_c and stationary measure ν_ρ. Write $\delta=\rho_c-\rho>0$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070178 | MATHDB-363807

Conjectural formula for the next-to-leading coefficient of path distance matrices

摘录 / EXCERPT: Coefficient formula conjecture. The coefficient of x^{2n-5} is

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-070179 | MATHDB-363813

Two-row partition covering-number conjecture for symmetric-group characters

摘录 / EXCERPT: Let $n\geq 5$ and let $\lambda=(n-k,k)$, where $1\leq k\leq \lfloor\frac n2\rfloor$. Write χ_λ for the irreducible character of S_n indexed by λ, let...

conjecture | open | MathDB 页面 | 原文线索

V2-070180 | MATHDB-363819

Folklore coupling-independence conjecture for proper list-colorings

摘录 / EXCERPT: Folklore conjecture. Let $\delta>0$ be a constant. If

conjecture | open | MathDB 页面 | 原文线索

V2-070181 | MATHDB-363820

PŁ-like gradient-mapping relation for trust-region optimisation

摘录 / EXCERPT: Let E be the objective function, let $\boldsymbol{s}^{(k)}$ be the iterate at iteration k, let E^* denote the optimal objective value, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070182 | MATHDB-363830

The quasi-homogeneity criterion for 3-syzygy plane curves

摘录 / EXCERPT: Let $C=V(f)$ be a reduced 3-syzygy plane curve, meaning that it has a first syzygy matrix M_f that is a square matrix of size 3. Let $p\in\operatorname{Sing}(C)$ be an isolat...

conjecture | open | MathDB 页面 | 原文线索

V2-070183 | MATHDB-363836

Arthur-Langlands conjecture for semisimple groups

摘录 / EXCERPT: Let G be a semisimple $\mathbb{Q}$-group admitting a reductive $\mathbb{Z}$-model. Let $\pi\in\Pi_{\mathrm{disc}}(G)$ and let $r:\widehat{G}\to\mathrm{SL}_{n,\mathbb{C}}$ be an a...

conjecture | open | MathDB 页面 | 原文线索

V2-070184 | MATHDB-363840

The planar induced-minor Erdős-Pósa conjecture

摘录 / EXCERPT: For a graph H, kH denotes the disjoint union of k copies of H, and a graph is H-induced-minor-free if it does not contain H as an induced minor. The planar induced-mino...

conjecture | open | MathDB 页面 | 原文线索

V2-070185 | MATHDB-363844

e-log-concavity conjecture for chromatic quasisymmetric functions

摘录 / EXCERPT: e-log-concavity conjecture. Let G be a natural unit interval graph. Then $X_G(\bm x;q)$ is e-log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-070186 | MATHDB-363845

Conjectured enumeration of 132-avoiding permutations fixed by the fundamental bijection

摘录 / EXCERPT: Let $t_n^k(\tau)$ denote the number of permutations of length n avoiding the pattern τ that are fixed by the kth iterate of the fundamental bijection. For $ngeq2$, the con...

conjecture | open | MathDB 页面 | 原文线索

V2-070187 | MATHDB-363846

Rationality conjecture for generating functions of 231- and 312-avoiding permutations

摘录 / EXCERPT: Rationality conjecture. For $\sigma\in\{231,312\}$, the generating functions $F_\sigma^k(x)$ are rational. The paper gives the conjectured examples

conjecture | open | MathDB 页面 | 原文线索

V2-070188 | MATHDB-363847

Eventual constancy conjecture for 213- and 132-avoiding permutations

摘录 / EXCERPT: Eventual constancy conjecture. For each $\sigma\in\{213,132\}$ and $k\geq1$, $f_n^k(\sigma)$ is eventually constant as n becomes large. The paper conjectures that for $n\geq8$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-070189 | MATHDB-363848

Power-of-two periodicity conjecture for pattern-avoidance counts

摘录 / EXCERPT: Let $f_n^k(\sigma)$ denote the number of permutations of length n associated with a pattern $\sigma\in S_3$ and fixed by the kth iterate of the fundamental bijection. Power-of-...

conjecture | open | MathDB 页面 | 原文线索

V2-070190 | MATHDB-363850

The rationality and reciprocity conjecture for q-Ehrhart series

摘录 / EXCERPT: Let P be a d-dimensional lattice polytope in $\mathbb{R}^n$. Define ... and ... where $\mathrm{i}_P(m;q)=\operatorname{Hilb}(R(\mathbb{Z}^n\cap mP),q)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070191 | MATHDB-363859

Global EISL optimality conjecture for CP-free OFDM under sub-Gaussian constellations

摘录 / EXCERPT: Let N be the number of signaling dimensions, let $\mathbf{U}\in\mathbb{U}(N)$ denote the unitary signaling basis, and let $\mathbf{F}_N$ be the normalized N-point discrete Four...

conjecture | open | MathDB 页面 | 原文线索

V2-070192 | MATHDB-363864

The abelian Iwasawa main conjecture for CM fields

摘录 / EXCERPT: Iwasawa main conjecture for ψ. In the abelian case, for a finite Hecke character $\psi^{\mathrm{gal}}\colon G_F\to\operatorname{Aut}_{\mathcal{E}}V_\psi$ with F_ψ linearl...

conjecture | open | MathDB 页面 | 原文线索

V2-070193 | MATHDB-363867

The eventual structure of non-terminal residue classes for Stirling-number valuations

摘录 / EXCERPT: Let $S(n,k)$ denote the number of partitions of an n-element set into exactly k nonempty subsets, and let v_2 denote the 2-adic valuation. For fixed $k\in\mathbb{N}$, consi...

conjecture | open | MathDB 页面 | 原文线索

V2-070194 | MATHDB-363870

DCC conjecture for Iitaka volumes of log canonical pairs

摘录 / EXCERPT: DCC of Iitaka volumes. The set $\operatorname{Ivol}_{\mathrm{lc}}^{\Gamma(d,\kappa)}$ is a DCC set.

conjecture | open | MathDB 页面 | 原文线索

V2-070195 | MATHDB-363872

DCC conjecture for invariant Iitaka volumes

摘录 / EXCERPT: DCC of invariant Iitaka volumes. This set is a DCC set.

conjecture | open | MathDB 页面 | 原文线索

V2-070196 | MATHDB-363874

Fonarev's conjecture on the mutation Lefschetz basis for Grassmannians

摘录 / EXCERPT: Let ${\mathsf{Gr}}(k,V)$ be a Grassmannian, let $\mathrm{Y}^{\mathrm{mu}}_{k,n-k}$ denote the lexicographically minimal upper-triangular representatives of the cyclic orbits of You...

conjecture | open | MathDB 页面 | 原文线索

V2-070197 | MATHDB-363876

The fullness conjecture for exceptional collections on orthogonal Grassmannians

摘录 / EXCERPT: Let ${\mathsf{OGr}}(k,V)$ be an orthogonal Grassmannian, and let the collections in the source be the semiorthogonal collections constructed in Theorems,, and. The orthogonal Grass...

conjecture | open | MathDB 页面 | 原文线索

V2-070198 | MATHDB-363877

The unique minimal parabolic subgroup conjecture for Artin groups

摘录 / EXCERPT: Let B be an Artin group, and let ${\mathbf b}\in B$. A minimal parabolic subgroup containing ${\mathbf b}$ is a parabolic subgroup containing ${\mathbf b}$ that is minimal under...

conjecture | open | MathDB 页面 | 原文线索

V2-070199 | MATHDB-363878

The pure-centralizer conjecture for minimal-support elements in Artin groups

摘录 / EXCERPT: Let B be an Artin group with standard generating set ${\mathbf S}$, let ${\mathbf b}\in B$ have minimal length in its conjugacy class, and let $\supp({\mathbf b})$ denote its sup...

conjecture | open | MathDB 页面 | 原文线索

V2-070200 | MATHDB-363879

Upper-bound conjecture for the growth rate in the three-patch seasonal system

摘录 / EXCERPT: Assume $a>0>b$ and $a+2b<0$, and let $\Lambda(m,T)$ be the growth rate of the three-patch seasonal system under consideration, with migration rate m and period T. Upper-bound c...

conjecture | open | MathDB 页面 | 原文线索

V2-070201 | MATHDB-363880

Asymptotic growth conjecture for the first three-patch seasonal system

摘录 / EXCERPT: Assume $a>0>b$ and $a+2b<0$. For the seasonal system denoted by (47ab), let $\Lambda(m,T)$ be its growth rate, with migration rate m and period T. Asymptotic growth conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-070202 | MATHDB-363881

Asymptotic growth conjecture for the season-swapped three-patch system

摘录 / EXCERPT: Assume $a>0>b$ and $a+2b<0$. For the seasonal system denoted by (47ab1), let $\Lambda(m,T)$ be its growth rate, with migration rate m and period T. Asymptotic growth conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-070203 | MATHDB-363888

A theta-block crank conjecture for $(k+j)$-colored partitions

摘录 / EXCERPT: The theta-block crank conjecture. The function $C_{k,j}(\textbf{z},\tau)$ defines a crank for the desired $(k+j)$-colored partitions and explains most of the Ramanujan-like congrue...

conjecture | open | MathDB 页面 | 原文线索

V2-070204 | MATHDB-363889

Drużkowski's invertibility conjecture for cubic maps

摘录 / EXCERPT: Let $n\geq 1$ and let $A=\{a_{ij}\}_{1\leq i,j\leq n}$ be an $n\times n$ complex matrix. Define the map $T:\mathbb{C}^n\to\mathbb{C}^n$ by ... Such a map is called a Drużkowski map....

conjecture | open | MathDB 页面 | 原文线索

V2-070205 | MATHDB-363896

Appleby-Flammia-McConnell-Yard ray class field conjecture for SICs

摘录 / EXCERPT: Let $d\geq 4$, set $\Delta_d=(d+1)(d-3)$, and let $K_d=\Q(\sqrt{\Delta_d})$. For a Weyl-Heisenberg line-SIC S in dimension d, let $\fieldvec{S}$ denote its SIC field, and let...

conjecture | solved | MathDB 页面 | 原文线索

V2-070206 | MATHDB-363898

Strong Zauner conjecture for Weyl-Heisenberg SICs

摘录 / EXCERPT: For $d\geq1$, let a Weyl-Heisenberg SIC be a set of d^2 equiangular complex lines in $\C^d$ generated by a Weyl-Heisenberg fiducial vector. Strong Zauner Conjecture. For every di...

conjecture | open | MathDB 页面 | 原文线索

V2-070207 | MATHDB-363902

Geometric count conjecture for Weyl-Heisenberg SICs

摘录 / EXCERPT: Fix $d\in\mathbb Z_{>0}$, $d\neq3$, let $\Delta=(d+1)(d-3)$, and let $\mathcal O_\Delta$ be the quadratic order of discriminant Δ. Geometric Count Conjecture. The number of...

conjecture | open | MathDB 页面 | 原文线索

V2-070208 | MATHDB-363904

Extended order-to-multiplet conjecture for SIC fields

摘录 / EXCERPT: Fix $d\in\mathbb Z_{>0}$, $d\neq3$, write $\Delta=\Delta_d=(d+1)(d-3)=f^2\Delta_0$ with Δ_0 a fundamental discriminant, and let $\mathcal M(f')$ be the multiplet indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-070209 | MATHDB-363905

Extended ray class fields of orders conjecture for SICs

摘录 / EXCERPT: Fix $d\geq4$, write $\Delta=(d+1)(d-3)=f^2\Delta_0$ with Δ_0 a fundamental discriminant, and let $f'\mid f$. Let $\mathcal M$ satisfy the extended order-to-multiplet conject...

conjecture | open | MathDB 页面 | 原文线索

V2-070210 | MATHDB-363908

Measure-valued Hessian conjecture for Lipschitz subsolutions

摘录 / EXCERPT: Let w be a Lipschitz subsolution to the generalized k-convexity equation for $\beta=1$. A measure-valued Hessian conjecture asserts that there exist signed Borel measures...

conjecture | open | MathDB 页面 | 原文线索

V2-070211 | MATHDB-363911

The Elliott-type conjecture on correlations of aperiodic multiplicative functions

摘录 / EXCERPT: Elliott-type correlation conjecture. If P_1 and P_2 are irreducible binary quadratic forms that are not multiples of each other, then they are good for vanishing of correlation...

conjecture | open | MathDB 页面 | 原文线索

V2-070212 | MATHDB-363912

Syzygy conjecture for general projections of canonical curves

摘录 / EXCERPT: Let $C\hookrightarrow\mathbb{P}^{g-1}$ be a generic canonical curve of genus $g\geq 9$, and let $p\in\mathbb{P}^{g-1}$ be a generic point. Write C' for the projection of C away...

conjecture | open | MathDB 页面 | 原文线索

V2-070213 | MATHDB-363913

Syzygy conjecture for general projections of paracanonical curves

摘录 / EXCERPT: Assume that C is a generic paracanonical curve of genus $g\geq 11$, embedded by the paracanonical bundle $\omega_C\otimes\eta$, where η is a generic non-trivial torsion line...

conjecture | open | MathDB 页面 | 原文线索

V2-070214 | MATHDB-363915

Parity conjecture for the triple twist of slab tilings

摘录 / EXCERPT: Triple-twist parity conjecture. For every slab tiling ${\mathbf t}$ of ${\cal R}$, the integers

conjecture | open | MathDB 页面 | 原文线索

V2-070215 | MATHDB-363922

Unimodality conjecture for matching polytopes of wheel graphs

摘录 / EXCERPT: Unimodality conjecture. The h^*-vector of $P_M(W_n)$ is unimodal for every $n\geq 4$.

conjecture | open | MathDB 页面 | 原文线索

V2-070216 | MATHDB-363932

The derived-series analogue for one-relator groups

摘录 / EXCERPT: Let F be a free group, let $w\in F$, and let $G=F/\langle\!\langle w\rangle\!\rangle$ be the corresponding one-relator group. For a subset of F, write...

conjecture | open | MathDB 页面 | 原文线索

V2-070217 | MATHDB-363937

The confusion-number upper-bound conjecture for signed graphs

摘录 / EXCERPT: Let (G,σ) be a signed graph on n vertices. The confusion-number upper-bound conjecture. ... This conjecture proposes a general upper bound for the confusion number of a sign...

conjecture | open | MathDB 页面 | 原文线索

V2-070218 | MATHDB-363938

The relaxed confusion-number bound conjecture for signed graphs

摘录 / EXCERPT: Let G be a graph of order $n \geq 1$, let σ be a signing of G, and let $\ell(G,\sigma)$ denote the frustration index. Let $C_r(G,\sigma)$ denote the relaxed confusion nu...

conjecture | open | MathDB 页面 | 原文线索

V2-070219 | MATHDB-363939

Partition-function quiver description for equal-sign Hopf link conormal fillings

摘录 / EXCERPT: Let L_{st} be a Lagrangian filling of the Hopf link conormal, with $s,t\in\{l,m,d\}$ and $|s|=|t|$. Let ${\mathsf{Z}}_{L_{st}}$ denote its skein-valued partition function, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-070220 | MATHDB-363971

Structured low-rank certificate conjecture for minimax gradient descent

摘录 / EXCERPT: Fix N, a 1-smooth convex function f, an initial point x_0, a minimizer $x_\star$, and gradient-descent iterates $x_0,\ldots,x_N$ with constant stepsize α. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070221 | MATHDB-363973

Modified positivity conjecture for L-classes of smooth complete intersections

摘录 / EXCERPT: Modified positivity conjecture for complete intersections. For every integer j with $0\leq j\leq n/2$,

conjecture | open | MathDB 页面 | 原文线索

V2-070222 | MATHDB-363974

The single-point modification conjecture for APN functions

摘录 / EXCERPT: Single-point modification conjecture. The function G is not APN.

conjecture | open | MathDB 页面 | 原文线索

V2-070223 | MATHDB-363975

The uniform-exclude-distribution maximality conjecture for APN graphs

摘录 / EXCERPT: Uniform-exclude-distribution conjecture. If $d_{\mathcal{G}_F}$ is uniform on $\mathcal{Q}(\mathbb{F}_2^n,F)$, then $\mathcal{G}_F$ is maximal.

conjecture | open | MathDB 页面 | 原文线索

V2-070224 | MATHDB-363976

The axis-value conjecture for power APN functions

摘录 / EXCERPT: Axis-value conjecture. For every $a\in\mathbb{F}_2^n$ and every nonzero $b\in\mathbb{F}_2^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-070225 | MATHDB-363977

Iwasawa-Greenberg Main Conjecture for anticyclotomic Selmer groups

摘录 / EXCERPT: Iwasawa-Greenberg Main Conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-070226 | MATHDB-363993

Harmonic-map metastability conjecture for O(N) models

摘录 / EXCERPT: Harmonic-map metastability conjecture. The relaxation time of the corresponding Langevin dynamics grows exponentially fast with β if and only if $\mathcal E$ has non-trivial,...

conjecture | open | MathDB 页面 | 原文线索

V2-070227 | MATHDB-363997

The sharp random-graph threshold for relative hyperbolicity and thickness

摘录 / EXCERPT: Let $\mathcal{G}_{n,p}$ denote the Erdős-Rényi random graph, let $\Gamma\sim\mathcal{G}_{n,p}$, and let W_{Γ} be the right-angled Coxeter group with presentation graph...

conjecture | open | MathDB 页面 | 原文线索

V2-070228 | MATHDB-363998

The threshold sequence for polynomial divergence in random right-angled Coxeter groups

摘录 / EXCERPT: Let $\mathcal{G}_{n,p}$ denote the Erdős-Rényi random graph, and let $\Gamma\sim\mathcal{G}_{n,p}$. For a group or graph, “thick of order at most k” is used in the sense of the p...

conjecture | open | MathDB 页面 | 原文线索

V2-070229 | MATHDB-364009

Self-adjointness conjecture for operators with positive real part and self-adjoint powers

摘录 / EXCERPT: Self-adjointness conjecture. T is self-adjoint.

conjecture | open | MathDB 页面 | 原文线索

V2-070230 | MATHDB-364011

Booher-Cais conjecture for higher a-numbers in minimal-break-ratio towers

摘录 / EXCERPT: Booher-Cais conjecture. For each $r\geq 1$, there exist an integer m_r and functions $b_r,\nu_r,\lambda_r:\mathbf Z/m_r\mathbf Z\to\mathbf Q$ such that, for $n\gg0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070231 | MATHDB-364012

Periodicity conjecture for higher a-numbers under the transformation r to (r+1)p+1

摘录 / EXCERPT: Periodicity conjecture. The minimal periods of the sequences $a^{r_1}(X_n)$ and $a^{r_0}(X_n)$ are equal.

conjecture | open | MathDB 页面 | 原文线索

V2-070232 | MATHDB-364019

The Ratios Conjectures lemma for pair correlations of zeros

摘录 / EXCERPT: Let f_d be a quadratic twist in the family and define ... where $s=1/2+it$. For $-1/4<\operatorname{Re}(\alpha),\operatorname{Re}(\beta)<1/4$,...

conjecture | open | MathDB 页面 | 原文线索

V2-070233 | MATHDB-364023

Boavida-Weiss equivalence conjecture for the PL Stiefel space

摘录 / EXCERPT: Let $n-m\geq 3$. Write $\mathrm{V}^{pl}_{n,m}=\mathrm{PL}_n/\mathrm{PL}_{n,m}$ for the PL Stiefel space, and let $\mathrm{Operad}^h(E_m,E_n)$ be the derived mapping space of the li...

conjecture | open | MathDB 页面 | 原文线索

V2-070234 | MATHDB-364024

Pucci-Serrin conjecture for radial solutions of polyharmonic equations

摘录 / EXCERPT: Let B be the unit ball of $\mathbb{R}^n$, and let $k\in\mathbb{N}$ satisfy $n>2k\geq 2$. Consider $\lambda\in\mathbb{R}$ and $u\in C^{2k}(\overline{B})$ solving ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-070235 | MATHDB-364027

Universality of the empirical spectral distribution without sub-Gaussianity

摘录 / EXCERPT: Non-sub-Gaussian ESD conjecture. The conclusion of Theorem (ii) holds without assuming that the base measure μ is sub-Gaussian.

conjecture | open | MathDB 页面 | 原文线索

V2-070236 | MATHDB-364028

Circular-law universality for constant linear margins

摘录 / EXCERPT: Circular-law universality conjecture. For constant linear margins and every base measure μ satisfying the hypothesis of Corollary, the empirical eigenvalue distribution of

conjecture | open | MathDB 页面 | 原文线索

V2-070237 | MATHDB-364032

Winning-player conjecture for loopy cycles

摘录 / EXCERPT: Let $C(n,k)$ be a loopy cycle formed from a cycle C_n by adding k loops to consecutive vertices, with $n\geq 3$. Loopy-cycle winner conjecture. For $k=1$ and $k=2$, Player 1 wi...

conjecture | open | MathDB 页面 | 原文线索

V2-070238 | MATHDB-364033

Ferris Wheel score conjecture for Dots and Boxes

摘录 / EXCERPT: Let W_n be a wheel graph and let $C(n,n)$ be the corresponding Ferris Wheel graph, for $n\geq 3$. Suppose Player X wins on W_n with score (a,b). Ferris Wheel score conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-070239 | MATHDB-364037

Higher-dimensional extension of the X-ray theorem for modular correspondences

摘录 / EXCERPT: Higher-dimensional extension conjecture. Theorem on X-rays holds with respect to any fixed point $\star$ of F.

conjecture | open | MathDB 页面 | 原文线索

V2-070240 | MATHDB-364053

The Pollyanna conjecture for box intersection graphs

摘录 / EXCERPT: Fix a positive integer d, and let $\mathcal{B}_d$ be the class of intersection graphs of axis-aligned boxes in $\mathbb{R}^d$. A hereditary class is Pollyanna if every hereditary...

conjecture | open | MathDB 页面 | 原文线索

V2-070241 | MATHDB-364063

Conjecture on outlier traps under Λ-poisedness

摘录 / EXCERPT: Let $\mathcal{Y}$ be the set of experimental points, and let $y^1,y^2,y^3\in\mathcal{Y}$. A point is called bad under Λ-poisedness when it satisfies the paper's bad-point c...

conjecture | open | MathDB 页面 | 原文线索

V2-070242 | MATHDB-364064

Conjecture on directional extension of bad points under Λ-poisedness

摘录 / EXCERPT: Let y^1,y^2,y^3,y^4 be interpolated points, and write $\overrightarrow{y^i y^j}$ for the vector from y^i to y^j. A point is called bad under Λ-poisedness when it sati...

conjecture | open | MathDB 页面 | 原文线索

V2-070243 | MATHDB-364066

Watkins's conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve. Its modular degree m_E is the minimal degree of a surjective modular parametrization $X_0(N)\to E$ defined over $\mathbb{Q}$ and sending...

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V2-070244 | MATHDB-364067

Mean-Lindelöf bounds for twisted automorphic L-functions

摘录 / EXCERPT: Let T be a positive parameter, let $\Pi_{c}^{\natural}(T)$ be the family of cuspidal automorphic representations appearing in the paper, and for each...

conjecture | open | MathDB 页面 | 原文线索

V2-070245 | MATHDB-364069

Minimal forbidden-minor conjecture for the sign pattern s_*

摘录 / EXCERPT: Let $\Sigma \in \operatorname{PR}_5$ be a real symmetric matrix whose principal-minor sign pattern $\operatorname{signs}_\Sigma$ agrees with s_* on all subsets of sizes 1 and...

conjecture | open | MathDB 页面 | 原文线索

V2-070246 | MATHDB-364074

Rank-difference lower-bound conjecture for strongly unimodal sequences

摘录 / EXCERPT: Let $u(m,n)$ denote the number of strongly unimodal sequences of weight n and rank m, and let $N(0,n)$ denote the auxiliary counting function used in the paper. Rank-difference...

open_problem | open | MathDB 页面 | 原文线索

V2-070247 | MATHDB-364079

The Redei-Berge function conjecture for sums of minimal and maximal elements

摘录 / EXCERPT: Redei-Berge function conjecture. If two posets P and Q have the same Redei-Berge function, then

conjecture | open | MathDB 页面 | 原文线索

V2-070248 | MATHDB-364081

Self-intersection conjecture for curves on ruled surfaces

摘录 / EXCERPT: Self-intersection conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070249 | MATHDB-364089

The toroidal 5-choosability and 5-colorability conjecture

摘录 / EXCERPT: The toroidal 5-choosability conjecture. A toroidal graph is 5-choosable if and only if it is 5-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-070250 | MATHDB-364090

The edge-width four conjecture for 5-choosability on the torus

摘录 / EXCERPT: The edge-width four conjecture. Every graph drawn on the torus with edge-width at least four is 5-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-070251 | MATHDB-364091

The spacing-five conjecture for critical prism-canvases

摘录 / EXCERPT: The spacing-five conjecture. There is no critical prism-canvas of spacing at least five.

conjecture | open | MathDB 页面 | 原文线索

V2-070252 | MATHDB-364093

Almost-all non-congruence presentations conjecture for finite simple groups

摘录 / EXCERPT: Almost-all non-congruence presentations conjecture. As the order of S tends to infinity, the proportion of non-congruence presentations among all presentations tends to 1:

conjecture | open | MathDB 页面 | 原文线索

V2-070253 | MATHDB-364105

Computably simple copies for homogeneous completely decomposable groups

摘录 / EXCERPT: Let H be a subgroup of $\mathbb{Q}$, and consider a computable group of the form ... A computably simple-copy conjecture. Theorem holds for any computable group of this form. That...

conjecture | open | MathDB 页面 | 原文线索

V2-070254 | MATHDB-364111

The degree of irrationality of very general polarized K3 surfaces

摘录 / EXCERPT: Let $\{(S_d,L_d)\}_{d\in\mathbb N}$ be very general polarized $K3$ surfaces with $L_d^2=2d-2$. Bastianelli-De Poi-Ein-Lazarsfeld-Ullery conjecture. ... This predicts that the degrees...

conjecture | open | MathDB 页面 | 原文线索

V2-070255 | MATHDB-364118

The Cayley-equivalence maximum conjecture

摘录 / EXCERPT: Let p and q be Cayley permutations of the same length, and let $\max(p)$ denote the largest letter occurring in p. Write $p\sim_c q$ for Cayley-equivalence. The Cayley-equiva...

conjecture | open | MathDB 页面 | 原文线索

V2-070256 | MATHDB-364126

Quasi-isometric rigidity conjecture for large-type Artin groups

摘录 / EXCERPT: Quasi-isometric rigidity conjecture. 1. Every self quasi-isometry of A_Γ is at bounded distance from an automorphism of the Cayley complex C_Γ if and only if Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-070257 | MATHDB-364135

Optimal Castelnuovo bound conjecture for complete intersection Calabi-Yau 3-folds

摘录 / EXCERPT: Optimal Castelnuovo bound conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070258 | MATHDB-364148

Oshima's realizability conjecture for abstract spectral types

摘录 / EXCERPT: Let $\mathbf{S}=(S_i)_{i=1,\ldots,d}$ be a collection of abstract spectral types. A collection is irreducibly realizable if it is the spectral type of an irreducible meromorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-070259 | MATHDB-364155

Minimality conjecture for the BPD energy in the plane

摘录 / EXCERPT: Minimality for $\textnormal{E}_{\textnormal{BPD}}$ in general. For all $N\in\mathbb N\backslash\{0,1\}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-070260 | MATHDB-364156

Conjecture on norm-dependent optimality of triangular and square lattices

摘录 / EXCERPT: Optimality conjecture for triangular and square lattices. There exist $1<p_1<p_2$ and $1<p_3<p_4$ such that, up to rotation, the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-070261 | MATHDB-364166

The AGI finite-determination conjecture for language-model entropy

摘录 / EXCERPT: Let $\mathcal{S}$ be the set of sentences, let $\mathscr{L}$ be a causal language model, and let ... be its entropy function. The AGI finite-determination conjecture. If a causal lan...

conjecture | open | MathDB 页面 | 原文线索

V2-070262 | MATHDB-364168

Uncrossed number can differ arbitrarily from outerthickness

摘录 / EXCERPT: Uncrossed-number separation conjecture. The uncrossed number can be arbitrarily far apart from the outerthickness. This conjecture asks whether the difference between these two gra...

conjecture | open | MathDB 页面 | 原文线索

V2-070263 | MATHDB-364172

The refined Vafa-Witten multiple-cover conjecture for Enriques surfaces

摘录 / EXCERPT: The refined Vafa-Witten multiple-cover conjecture. The refined Vafa-Witten invariant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070264 | MATHDB-364173

Vanishing conjecture for the chi-genus of Enriques moduli spaces

摘录 / EXCERPT: Let Y be an Enriques surface and let $v=(r,\beta,n)\in H^{\ast}(Y,\mathbb{Z})$ be primitive, with r even and $2\nmid\beta$. Let $M^Y(v)$ be the automatically smooth moduli spac...

conjecture | open | MathDB 页面 | 原文线索

V2-070265 | MATHDB-364174

The bounded-degree path lower-bound conjecture for clustered colouring of graph products

摘录 / EXCERPT: Let $c\geqslant 2$ be a fixed integer. Let H be a graph of treewidth t, and let P be a path. The bounded-degree path lower-bound conjecture. There exists an integer t such...

conjecture | open | MathDB 页面 | 原文线索

V2-070266 | MATHDB-364175

The three-colour lower-bound conjecture for clustered colouring of graph products

摘录 / EXCERPT: Let H_1 and H_2 be graphs of treewidth t. The three-colour lower-bound conjecture. There exists a positive integer t such that for infinitely many graphs H_1,H_2, every...

conjecture | open | MathDB 页面 | 原文线索

V2-070267 | MATHDB-364176

The unrestricted clustered-colouring lower-bound conjecture for graph products

摘录 / EXCERPT: Let $c\geqslant 4$ be a fixed integer, and let H_1 and H_2 be graphs of treewidth k. The unrestricted lower-bound conjecture. There exists an integer k such that for infini...

conjecture | open | MathDB 页面 | 原文线索

V2-070268 | MATHDB-364178

Height-function limit for intermediate q-reduced pipe-dream permutations

摘录 / EXCERPT: For fixed $p,q\in(0,1)$, let $\boldsymbol{w}^{(q)}(n)\in S_n$ be the random permutation from the q-reduced pipe-dream model, and let $H^{(q)}$ be its height function. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070269 | MATHDB-364179

The spouse-loving Oberwolfach conjecture for four groups of odd cycle length

摘录 / EXCERPT: Let m be a positive odd integer. The spouse-loving variant of the Oberwolfach problem, denoted $\operatorname{OP}^+(n;m)$, asks whether the graph K_n+I admits a C_m-factoriza...

conjecture | open | MathDB 页面 | 原文线索

V2-070270 | MATHDB-364183

Bounded modified saturation for an isolated minimal edge

摘录 / EXCERPT: Let G) be an edge-ordered graph and let e_0 be an isolated minimal edge of G. Write $F=e_0+G$. The isolated-edge saturation conjecture. ... The authors present this as a candid...

conjecture | open | MathDB 页面 | 原文线索

V2-070271 | MATHDB-364184

Linear modified saturation under disjoint neighborhoods

摘录 / EXCERPT: Disjoint-neighborhood saturation conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-070272 | MATHDB-364185

Near-linear saturation conjecture for edge-ordered graphs

摘录 / EXCERPT: Let G be an edge-ordered graph, and let $sat_e(n,G)$ denote its edge-ordered saturation function. Near-linear saturation conjecture. For every edge-ordered graph G, ... The paper...

conjecture | open | MathDB 页面 | 原文线索

V2-070273 | MATHDB-364186

Almost-linear saturation conjecture for edge-ordered graphs

摘录 / EXCERPT: Let G be an edge-ordered graph, and let $sat_e(n,G)$ denote its edge-ordered saturation function. Almost-linear saturation conjecture. For every edge-ordered graph G, ... This is...

conjecture | open | MathDB 页面 | 原文线索

V2-070274 | MATHDB-364189

Genericity conjecture for pseudo-Anosov mapping classes in metric balls

摘录 / EXCERPT: Let Σ be a complete hyperbolic surface of finite type that is not a three-punctured sphere, and let $\operatorname{Mod}(\Sigma)$ be its mapping class group. Let S be a spe...

conjecture | open | MathDB 页面 | 原文线索

V2-070275 | MATHDB-364193

Ferroni-Matherne-Vecchi's real-rootedness conjecture for characteristic Chow polynomials

摘录 / EXCERPT: Let P be a finite graded bounded poset, and let P be Cohen-Macaulay. Denote by the characteristic Chow polynomial of P the polynomial associated with P in the Chow-polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-070276 | MATHDB-364199

The structured Newton-polygon conjecture for Jacobian pairs

摘录 / EXCERPT: The structured Newton-polygon conjecture. Suppose $(a,b,m,n)\in\mathcal{Q}$ and $F,G\in\mathcal{R}$ satisfy $[F,G]\in\mathbb{C}$, $F\in R_{m,n}$, and $G\in R_{bm/a,bn/a}$. Then...

conjecture | open | MathDB 页面 | 原文线索

V2-070277 | MATHDB-364200

The generator-composition conjecture for Jacobian pairs

摘录 / EXCERPT: Let $\mathcal{Q}$ and $R_{m,n}$ be as above. For a polynomial $F\in R_{m,n}$, let W_F be the F-generator constructed in the paper, and let $\mathbb{T}$ be the corresponding cla...

conjecture | open | MathDB 页面 | 原文线索

V2-070278 | MATHDB-364201

The vanishing innermost-polynomial conjecture

摘录 / EXCERPT: Let $\mathcal{Q}$ and $R_{m,n}$ be as above. For $F\in R_{m,n}$, let Z denote the innermost polynomial of F. The vanishing innermost-polynomial conjecture. If...

conjecture | open | MathDB 页面 | 原文线索

V2-070279 | MATHDB-364202

The inner-polynomial lattice-point conjecture

摘录 / EXCERPT: Let $\mathcal{Q}$ and $R_{m,n}$ be as above, and let $\zeta(F)$ denote the inner polynomial of F. The inner-polynomial lattice-point conjecture. If $(a,b,m,n)\in\mathcal{Q}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070280 | MATHDB-364204

The fractional-support exclusion conjecture

摘录 / EXCERPT: Let $\overline{\mathcal{R}}=\mathbb{C}[x^{1/\mathbf d},y]$ for a fixed positive integer $\mathbf d$, and let $\overline{T}_{m,n,a}$ be the class defined in the paper. Suppose...

conjecture | open | MathDB 页面 | 原文线索

V2-070281 | MATHDB-364214

O'Neill's Möbius-function bound for regular Tesler matrices

摘录 / EXCERPT: Let $A\in\\T\left(\mathbf{1}^n\right)$ and let μ be the Möbius function for the corresponding Tesler poset. O'Neill's conjecture. ... This conjecture concerns the growth of Möbiu...

conjecture | open | MathDB 页面 | 原文线索

V2-070282 | MATHDB-364215

Equivalence of the two-sided dictionary and merger orders on Lusztig data

摘录 / EXCERPT: Let $A(\varnu)$ be the set of Lusztig data equipped with the two-sided dictionary partial order from the cited definition and the merger partial order from the cited definition. Eq...

conjecture | open | MathDB 页面 | 原文线索

V2-070283 | MATHDB-364222

Critical height-moduli height conjecture for endomorphisms of projective space

摘录 / EXCERPT: Let $\mathcal{M}_d^N$ be the moduli space of degree-d endomorphisms of $\mathbb{P}^N$, let $h_{\mathcal{M}}$ be an ample moduli height, and let $\widehat h^{\mathrm{crit}}$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-070284 | MATHDB-364227

Conjectured exact minimum distances for D-sequence cyclic codes

摘录 / EXCERPT: Let $m=2^l e>2$, where $2\nmid e$ and $l\geq1$, and let $\mathcal{C}_{\mathcal{D},i}$ denote the cyclic code associated with the sequence family $\mathcal{D}$. Let h be the param...

conjecture | open | MathDB 页面 | 原文线索

V2-070285 | MATHDB-364228

Conjecture on the extended period map for I-surfaces

摘录 / EXCERPT: Let $\mathcal{M}$ be the moduli space of I-surfaces, let $\widetilde{\mathcal{M}}$ be its natural blowup along the boundary strata, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070286 | MATHDB-364229

Conjecture on the unblown extended period map for I-surfaces

摘录 / EXCERPT: Let $\mathcal{M}$ be the moduli space of I-surfaces, let $\widetilde{\mathcal{M}}$ be the natural blowup of the moduli space along the boundary, let...

conjecture | open | MathDB 页面 | 原文线索

V2-070287 | MATHDB-364230

Quasi-BPS categorical duality for Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus g, let L be a line bundle on C of degree l, and let $g^{\rm{sp}}$ be the genus of the spectral curve. For integers...

conjecture | open | MathDB 页面 | 原文线索

V2-070288 | MATHDB-364231

Reduced quasi-BPS categorical duality for canonical Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve, let Ω_C be its canonical bundle, and let B be the Hitchin base. For rank r, Euler characteristic χ, and weight w, write...

conjecture | open | MathDB 页面 | 原文线索

V2-070289 | MATHDB-364246

Embedded resolution with an snc boundary

摘录 / EXCERPT: Embedded resolution with boundary. The triple $(\bar{X},D,\bar{Z})$ admits a resolution by blowups in smooth centers

conjecture | open | MathDB 页面 | 原文线索

V2-070290 | MATHDB-364247

The perverse Hodge number generating-series conjecture for Enriques moduli spaces

摘录 / EXCERPT: The perverse Hodge number generating-series conjecture. The perverse Hodge numbers depend on β only through β^2, and, writing ${^ph}^{i,j}(M_{\beta})={^ph}^{i,j}_d$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-070291 | MATHDB-364248

Maulik-Toda conjecture for stable-pair and sheaf-theoretic invariants

摘录 / EXCERPT: Maulik-Toda conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070292 | MATHDB-364249

The refined Gopakumar-Vafa and perverse Hodge number conjecture

摘录 / EXCERPT: Refined Gopakumar-Vafa conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070293 | MATHDB-364251

The perverse Hodge generating-series conjecture for Enriques moduli spaces

摘录 / EXCERPT: Enriques perverse Hodge generating-series conjecture. The perverse Hodge numbers satisfy the displayed generating-series identity involving the theta functions, eta functions, and...

conjecture | open | MathDB 页面 | 原文线索

V2-070294 | MATHDB-364275

Conjecture on class numbers in the family Q(sqrt(p^{2r}+1))

摘录 / EXCERPT: Class-number infinitude conjecture. There are infinitely many $r\in\mathbb{Z}^+$ such that p does not divide the class number of K_r.

conjecture | open | MathDB 页面 | 原文线索

V2-070295 | MATHDB-364279

Grünbaum's orthogonal line equipartition conjecture

摘录 / EXCERPT: Grünbaum's conjecture. For every $t\in[0,1/4]$, there exists a pair of orthogonal lines that partitions P into four pieces whose areas, in clockwise order, are...

conjecture | open | MathDB 页面 | 原文线索

V2-070296 | MATHDB-364280

The identity-assignment planar measure partition conjecture

摘录 / EXCERPT: Identity-assignment partition conjecture. There exists a positive integer r such that the plane can be partitioned, using r lines, into $2k$ not necessarily connected regions...

conjecture | open | MathDB 页面 | 原文线索

V2-070297 | MATHDB-364283

Lüroth-type uniqueness conjecture for invariant fields in infinitely many variables

摘录 / EXCERPT: Lüroth-type conjecture. There is a unique field extension $L|k$ in F such that

conjecture | open | MathDB 页面 | 原文线索

V2-070298 | MATHDB-364287

Odd-indexed generalized Gauss polynomial conjecture

摘录 / EXCERPT: For integers $n\ge 0$ and $d\ge 1$, define … Here $\mathscr{U}_{q,z}^2$ denotes the class used in the paper for the relevant two-variable unimodality property. Odd-indexed generali...

conjecture | open | MathDB 页面 | 原文线索

V2-070299 | MATHDB-364288

Unimodality conjecture for refined plane-partition Betti numbers

摘录 / EXCERPT: For integers $n\ge 0$ and $\delta\in\frac12\mathbb Z$, let ... where PP is the set of plane partitions and w_-, w_0, and w_+ are the sums of the entries strictly bel...

conjecture | open | MathDB 页面 | 原文线索

V2-070300 | MATHDB-364289

Strict unimodality conjecture for the V-statistics

摘录 / EXCERPT: Let $V(m,n)$ and $V_d(m,n)$ be the coefficients of the bivariate formal series $\mathcal{V}(z,q)$ and $\mathcal{V}_d(z,q)$, respectively, and let $\mathscr{U}_{z,q}^1$ denote the p...

conjecture | open | MathDB 页面 | 原文线索

V2-070301 | MATHDB-364290

The counting conjecture for joins of critical graphs and cliques

摘录 / EXCERPT: Counting conjecture. For all sufficiently large t and every $(k+t)$-fold cover $\mathcal{H}$ of $G\vee K_t$, the number of proper $\mathcal{H}$-colorings of $G\vee K_t$ is at lea...

open_problem | open | MathDB 页面 | 原文线索

V2-070302 | MATHDB-364293

Cuspidality conjecture for the categorical local Langlands correspondence

摘录 / EXCERPT: Let G be a reductive group, let $B(G)_{\operatorname{bas}}$ denote the basic $sigma$-conjugacy classes, and let G_b be the extended pure inner form corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-070303 | MATHDB-364294

Han's conjecture on Hecke compatibility of geometric Eisenstein series

摘录 / EXCERPT: Let P be a parabolic subgroup of G with Levi subgroup M and unipotent radical U, let $V\in\operatorname{Rep}({}^LG)$, and restrict V to ${}^LP$. Choose a finite filtratio...

conjecture | open | MathDB 页面 | 原文线索

V2-070304 | MATHDB-364303

Abreu-Diwan-Jackson-Labbate-Sheehan constituent conjecture for pseudo 2-factors

摘录 / EXCERPT: Let G be an essentially 4-edge-connected pseudo 2-factor isomorphic cubic bipartite graph, where essentially 4-edge-connected means 3-edge-connected with no non-trivial 3-e...

conjecture | open | MathDB 页面 | 原文线索

V2-070305 | MATHDB-364304

Aldred-Labbate-Robertson-Seymour conjecture on cyclically 5-edge-connected odd 2-factored snarks

摘录 / EXCERPT: Let G be a cyclically 5-edge-connected odd 2-factored snark, where a snark is a bridgeless cubic graph of chromatic index four and odd 2-factored means that every cycle in ev...

conjecture | open | MathDB 页面 | 原文线索

V2-070306 | MATHDB-364305

Morton-Vivaldi irreducibility conjecture for quadratic delta factors

摘录 / EXCERPT: Let $f_c(z)=z^2+c$. For positive integers m,n with $m\mid n$, define the delta factors $\Delta_{n,m}$ by ... and, when $m=n$, by ... Here Φ^{cyc}_k is the k-th cyclot...

conjecture | open | MathDB 页面 | 原文线索

V2-070307 | MATHDB-364309

The reducible-tree conjecture for time-periodic solutions

摘录 / EXCERPT: Reducible-tree conjecture. The infinite reducible tree and its rescalings according to the relevant scaling provide an effective description for the structure of time-periodic solu...

conjecture | open | MathDB 页面 | 原文线索

V2-070308 | MATHDB-364317

Arithmetic kei coloring average-order conjecture

摘录 / EXCERPT: Arithmetic kei coloring average-order conjecture. Let $\mathscr{K}\in\mathscr{K}\operatorname{ei}$ be finite. Then there exists $a_{\mathscr{K}}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070309 | MATHDB-364318

Generic summatory-type conjecture for arithmetic kei colorings

摘录 / EXCERPT: Let $\mathbb{N}^*$ denote the positive integers. For squarefree s, write ... and for a function $f\colon\mathbb{N}^*\to\mathbb{C}$ define its average over the squarefree integers c...

conjecture | open | MathDB 页面 | 原文线索

V2-070310 | MATHDB-364320

Higher-order monotonicity conjecture for contiguous parabolic-cylinder-function ratios

摘录 / EXCERPT: Higher-order ratio monotonicity conjecture. The functions $R_n^{[k]}(x)$ are positive and increasing in x, satisfy $R_n^{[k+1]}(x)>R_n^{[k]}(x)$, and obey $R_n^{[k]}(x)<1$ when...

conjecture | open | MathDB 页面 | 原文线索

V2-070311 | MATHDB-364324

Weak Granville conjecture for smooth shifted primes in a quadratic setting

摘录 / EXCERPT: For $X,Y\to\infty$ with $X\geq Y$, define ... and let ... where (Δ/p) is the Legendre symbol and $\Psi(X,Y)$ counts the Y-smooth integers at most X. Weak Granville conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-070312 | MATHDB-364326

Francis-Gaitsgory's Koszul duality conjecture

摘录 / EXCERPT: Let $\mathcal{O}$ be an operad in an ambient ∞-category $\mathcal{C}$, and let $B\mathcal{O}$ denote its bar construction, a cooperad. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070313 | MATHDB-364333

The authors' Fujita-type jet-ampleness conjecture

摘录 / EXCERPT: Fujita-type jet-ampleness conjecture. There exist integer-valued polynomial functions $a_\ell(n)$ and $b_\ell(n)$ depending only on n such that

conjecture | open | MathDB 页面 | 原文线索

V2-070314 | MATHDB-364339

The positive-characteristic intersection-of-orbits conjecture for polynomials

摘录 / EXCERPT: Intersection-of-orbits conjecture. At least one of the following holds: (1) there exist $m,n\in{\mathbb N}$ such that $f^m=g^n$; or (2) there exist linear polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-070315 | MATHDB-364340

The same-degree positive-characteristic intersection-of-orbits conjecture

摘录 / EXCERPT: Same-degree intersection-of-orbits conjecture. At least one of the following holds: (1) there exists $m\in{\mathbb N}$ such that $f^m=g^m$; or (2) there exist linear polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-070316 | MATHDB-364358

Iwasawa's μ=0 conjecture for cyclotomic ℤp-extensions

摘录 / EXCERPT: Let K be a number field and let K^{cyc} be the cyclotomic $\mathbf{Z}_p$-extension of K. The Iwasawa μ-invariant is the coefficient governing the exponential term...

conjecture | open | MathDB 页面 | 原文线索

V2-070317 | MATHDB-364368

The 3MIN threshold conjecture for reliable noisy computation

摘录 / EXCERPT: Let $\delta^*_{\operatorname{3MIN}}$ denote the threshold for reliable computation with noisy circuits of 3MIN gates, and let $\delta^*(3)$ denote the threshold for reliable comput...

conjecture | open | MathDB 页面 | 原文线索

V2-070318 | MATHDB-364375

Jedwab-Pender conjecture on even-length quaternary Legendre pairs

摘录 / EXCERPT: Jedwab-Pender conjecture. There exists a quaternary Legendre pair of even length $2N$ for every $N\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-070319 | MATHDB-364378

Generalized Ramanujan master theorem for powers of the cosecant

摘录 / EXCERPT: Let $H(\delta)=\{z\in\mathbb{C}\mid\operatorname{Re}(z)\ge -\delta\}$ for some $0<\delta<1$. Define polynomials by $P_1(x)=1$, $P_2(x)=x$, and ... for $m>2$. Let $g(z)$ be analytic o...

conjecture | open | MathDB 页面 | 原文线索

V2-070320 | MATHDB-364380

Modulo-p^3 conjecture for products of truncated hypergeometric series

摘录 / EXCERPT: Product supercongruence conjecture. For primes $p\geq 5$,

conjecture | open | MathDB 页面 | 原文线索

V2-070321 | MATHDB-364384

Orbifold KKP conjecture for nef-anticanonical orbifolds

摘录 / EXCERPT: Orbifold KKP conjecture. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-070322 | MATHDB-364387

Chabert-Halberstadt's pre-Pólya group conjecture for D4-fields of class number two

摘录 / EXCERPT: Chabert-Halberstadt's conjecture. If $h_K=2$, then

conjecture | open | MathDB 页面 | 原文线索

V2-070323 | MATHDB-364392

Ultra log-concavity conjecture for dependence polynomials

摘录 / EXCERPT: Ultra log-concavity conjecture. For every graph G, $D(G,x)$ is ultra log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-070324 | MATHDB-364406

Rationality conjecture for symmetric Hilbert modular surfaces

摘录 / EXCERPT: Let $Z^{\mathrm{sym}}_{N,r}$ denote the symmetric Hilbert modular surface associated with an (N,r)-congruence, and suppose (N,r) is one of the pairs in part (i) of the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-070325 | MATHDB-364407

Elliptic-fibration conjecture for symmetric Hilbert modular surfaces

摘录 / EXCERPT: Let $Z^{\mathrm{sym}}_{N,r}$ be the symmetric Hilbert modular surface, and consider each (N,r) in parts (ii)-(iii) of the paper's classification theorem. A Jacobian elliptic fibr...

conjecture | open | MathDB 页面 | 原文线索

V2-070326 | MATHDB-364415

The conjecture on a stationary process without a good ARMA approximation

摘录 / EXCERPT: Conjecture on the absence of a good ARMA approximation. The process X_t does not have a good ARMA model of the displayed form that is better than the simple $M+N$-term truncation...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-070327 | MATHDB-364425

Strict monotonicity conjecture for growth rates under random environmental switching

摘录 / EXCERPT: Let L be an irreducible dispersal matrix, let $A_i=R_i+L$ be the matrices in the differential inclusion model, and let x_i be eigenvectors of A_i. Assume that the eigenvector...

conjecture | open | MathDB 页面 | 原文线索

V2-070328 | MATHDB-364426

Poonen's strict-majority conjecture for union-closed families

摘录 / EXCERPT: Let $\mathcal{F}$ be a union-closed family of sets. A power set is a family of the form $\mathcal{P}(X)$ for some set X. Poonen's conjecture. Unless $\mathcal{F}$ is a power set,...

conjecture | open | MathDB 页面 | 原文线索

V2-070329 | MATHDB-364427

The exact-half reformulation of Poonen's conjecture

摘录 / EXCERPT: Let $\mathcal{F}$ be a union-closed family of sets, and suppose that a most frequent element of $U(\mathcal{F})$ belongs to exactly half the sets in $\mathcal{F}$. Exact-half conje...

conjecture | open | MathDB 页面 | 原文线索

V2-070330 | MATHDB-364429

Jiang's higher-order union-closed families conjecture

摘录 / EXCERPT: Jiang's conjecture. For every positive integer $k\leq n$, there exists a set $S\subseteq U(\mathcal{F})$ with $|S|=k$ that is contained in at least

conjecture | open | MathDB 页面 | 原文线索

V2-070331 | MATHDB-364431

Critical-corner classification conjecture for skew Howe limit shapes

摘录 / EXCERPT: Let t_+ be the right endpoint of the support of the limit shape, let z_+ be the corresponding root, and let $ Omega$ denote the limit shape. Critical-corner classification conj...

conjecture | open | MathDB 页面 | 原文线索

V2-070332 | MATHDB-364432

Constant-q asymptotic limit-shape and discrete-sine-kernel conjecture

摘录 / EXCERPT: Consider the specializations ... Let $\mu_{n,k}^b$ be the corresponding measure, with $q>0$, $b\neq 0$, and let $\rho(t)$ and $\Omega(t)$ denote its limiting density and limit shape....

conjecture | open | MathDB 页面 | 原文线索

V2-070333 | MATHDB-364436

Unobstructedness conjecture for non-rigid Calabi-Yau families

摘录 / EXCERPT: Unobstructedness conjecture. The deformations of both restricted log maps are unobstructed for all $a\in H_1$ and $b\in H_2$.

conjecture | open | MathDB 页面 | 原文线索

V2-070334 | MATHDB-364444

The Gopakumar-Vafa/Gromov-Witten correspondence conjecture

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau 3-fold. For a curve class $\beta \in H_2(X, \mathbb{Z})$ and a non-negative integer $g \in \mathbb{Z}_{\geq 0}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-070335 | MATHDB-364450

ACC conjecture for log canonical thresholds of generalized pairs

摘录 / EXCERPT: ACC conjecture for log canonical thresholds. The set

conjecture | open | MathDB 页面 | 原文线索

V2-070336 | MATHDB-364451

Finite-coefficient conjecture for numerically trivial generalized pairs

摘录 / EXCERPT: Finite-coefficient conjecture. There exists a finite set $\Gamma_0\subset\Gamma$, depending only on d and Γ, such that $B\in\Gamma_0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070337 | MATHDB-364453

Invariant graph conjecture for marked rational maps

摘录 / EXCERPT: Let (f,P) be a marked rational map, with Julia set J_f, and let G be a graph in the sphere. An f-invariant graph satisfies $f(G)=G$; the post-critical set is denoted by P...

conjecture | open | MathDB 页面 | 原文线索

V2-070338 | MATHDB-364454

Invariant graph conjecture for marked expanding Thurston maps

摘录 / EXCERPT: Let (F,Q) be a marked expanding Thurston map, where F is an expanding Thurston map of the sphere and Q is the marked set. An F-invariant graph is a graph preserved by F....

conjecture | open | MathDB 页面 | 原文线索

V2-070339 | MATHDB-364455

Fischer-Kubitzke conjecture on eventual real-rootedness of Hadamard powers

摘录 / EXCERPT: Let p be a polynomial with nonnegative coefficients, and let $\mathscr{W}(p)$ denote the polynomial obtained by the operator $\mathscr{W}$. Fischer-Kubitzke conjecture. For all s...

conjecture | open | MathDB 页面 | 原文线索

V2-070340 | MATHDB-364460

Cranston-Rabern equivalent form of the Borodin-Kostochka conjecture

摘录 / EXCERPT: Let G be a simple graph. Write $\chi=\chi(G)$ for its chromatic number and $\Delta=\Delta(G)$ for its maximum degree. For graphs G_1 and G_2, let $G_1\lor G_2$ denote their j...

conjecture | open | MathDB 页面 | 原文线索

V2-070341 | MATHDB-364461

Casselgren-Häggkvist conjecture on random list coloring of balanced complete-bipartite line graphs

摘录 / EXCERPT: Let D be an even natural number and let $G= obreak\frac{}{}\mathcal{L}(K_{\frac{D}{2}+1,\frac{D}{2}+1})$ be the line graph of a complete bipartite graph. Thus G has...

conjecture | solved | MathDB 页面 | 原文线索

V2-070342 | MATHDB-364473

Schur non-positivity conjecture for products of two chains

摘录 / EXCERPT: For positive integers m and n, let $\mathbf m\times\mathbf n$ be the product of two chains, and let $\operatorname{inc}(\mathbf m\times\mathbf n)$ denote its incomparability gr...

conjecture | open | MathDB 页面 | 原文线索

V2-070343 | MATHDB-364480

Odd-part dominance for elementary symmetric products of partition parts

摘录 / EXCERPT: For a partition $\lambda=(\lambda_1,\ldots,\lambda_\ell)$, let $\operatorname{pre}_i(\lambda)$ be the partition whose parts are all products $\lambda_{i_1}\cdots\lambda_{i_i}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-070344 | MATHDB-364481

Injectivity of pairwise products of partition parts

摘录 / EXCERPT: For a partition $\lambda=(\lambda_1,\ldots,\lambda_\ell)$, let $\operatorname{pre}_2(\lambda)$ be the partition whose parts are the products $\lambda_i\lambda_k$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-070345 | MATHDB-364484

Conjecture on stable primes of postcritically infinite polynomials

摘录 / EXCERPT: Let K be a number field and let $f\in K[X]$ be a postcritically infinite polynomial, meaning that the forward orbit of at least one critical point of f is infinite. Let the sta...

conjecture | open | MathDB 页面 | 原文线索

V2-070346 | MATHDB-364498

Eventual residue-class bias for unrestricted partitions and related partitions

摘录 / EXCERPT: Eventual bias conjecture. There exists a constant $N_{a,b,m}>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070347 | MATHDB-364505

Fatou's density of hyperbolicity conjecture for rational maps

摘录 / EXCERPT: Let f be a rational map of the Riemann sphere, and fix its degree. A rational map is hyperbolic when every critical point is attracted to an attracting periodic cycle. Fatou's de...

conjecture | open | MathDB 页面 | 原文线索

V2-070348 | MATHDB-364506

The no invariant line field conjecture for rational maps

摘录 / EXCERPT: Let f be a rational map, let J be its Julia set, and let an invariant line field on J mean a line field fixed by the pushforward operator ... A rational map is double covered b...

conjecture | open | MathDB 页面 | 原文线索

V2-070349 | MATHDB-364509

ACC conjecture for local volumes

摘录 / EXCERPT: ACC conjecture for local volumes. If b is finite, then the only accumulation point of $\operatorname{Vol}_{d,b}^{\operatorname{loc}}$ is 0. If b satisfies the descending chai...

conjecture | open | MathDB 页面 | 原文线索

V2-070350 | MATHDB-364512

Functoriality of the lasso construction on schemata

摘录 / EXCERPT: A schema is a small category, and for a schema T, $\mathsf{Set}^{\mathsf{T}}$ denotes its associated copresheaf category. Let $\mathsf{Lasso}(\mathsf{Set}^{\mathsf{T}})$...

conjecture | open | MathDB 页面 | 原文线索

V2-070351 | MATHDB-364513

Composition conjecture for lasso contractions

摘录 / EXCERPT: Let $(\operatorname{\mathcal{L}},\eta)$ be a lasso on a category C. For monomorphisms $f_1\colon X_1\hookrightarrow Y$ and $f_2\colon X_2\to Y_{/f}$ arranged as in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-070352 | MATHDB-364519

Kuratowski-type coarse planarisation conjecture

摘录 / EXCERPT: Kuratowski-type coarse planarisation conjecture. There exists a function f such that every graph with no K-fat K_5 and $K_{3,3}$ minor is $f(K)$-quasi-isometric to a planar g...

conjecture | open | MathDB 页面 | 原文线索

V2-070353 | MATHDB-364529

Baricz's conjecture on parameter monotonicity of Gaussian hypergeometric functions

摘录 / EXCERPT: Let $F(a,b;c;r)={}_2F_1(a,b;c;r)$ denote the Gaussian hypergeometric function, defined for the parameters under consideration by ... For $a>0$, $b>0$, and $r\in(0,1)$, consider the r...

conjecture | open | MathDB 页面 | 原文线索

V2-070354 | MATHDB-364530

Conjectured parameter ranges for two Gaussian hypergeometric inequalities

摘录 / EXCERPT: Let $a,b>0$ and $r\in(0,1)$. Let $\Lambda(a,r)$ and $\Lambda_2(a,b_1,b_2,r)$ denote the quantities defined earlier in the source by the inequalities referenced as ... , respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-070355 | MATHDB-364536

The vanishing conjecture for $\kappa(2^a3^b)$

摘录 / EXCERPT: For $N\in\mathbb{Z}_{\geq1}$, let $\kappa(N)$ be the dimension of the cokernel of the weight-two map D_{2}^{iter}. Let $a,b\in\mathbb{Z}_{\geq0}$. Vanishing conjecture f...

conjecture | open | MathDB 页面 | 原文线索

V2-070356 | MATHDB-364539

Almost equitable factorization conjecture with controlled parity defects

摘录 / EXCERPT: Almost equitable factorization conjecture. The graph G can be edge-decomposed into factors $G_1,\ldots,G_k$ such that, for every $v\in V(G)$,

conjecture | open | MathDB 页面 | 原文线索

V2-070357 | MATHDB-364540

Briggs's inequality for polynomials with only negative zeros

摘录 / EXCERPT: Let $f(x)=\sum_{l=0}^{n}a_lx^l$ be a polynomial with nonnegative coefficients, and suppose that all zeros of $f(x)$ are negative. Briggs's conjecture. For $1\leq l\leq n-1$, one ha...

conjecture | open | MathDB 页面 | 原文线索

V2-070358 | MATHDB-364541

Asymptotic Briggs inequalities for regular partitions and overpartitions

摘录 / EXCERPT: Let $p_k(n)$ and $\overline{p}_k(n)$ denote the k-regular partition and k-regular overpartition functions, respectively, and let the inequalities referred to in the source as (...

conjecture | open | MathDB 页面 | 原文线索

V2-070359 | MATHDB-364544

Explicit basis conjecture for parabolic invariants of truncated polynomial rings

摘录 / EXCERPT: Explicit parabolic basis conjecture. The resulting elements

conjecture | open | MathDB 页面 | 原文线索

V2-070360 | MATHDB-364545

Explicit basis conjecture for general linear invariants in all ranks

摘录 / EXCERPT: Explicit general-linear basis conjecture. The set $\mathcal{B}_m(n)$ consisting of these elements forms an $\mathbb{F}_q$-basis of ${\mathcal{Q}}_m(n)^{G_n}$. The conjecture is the...

conjecture | open | MathDB 页面 | 原文线索

V2-070361 | MATHDB-364548

Blanchet-Sadri's four-letter alphabet conjecture for unbordered partial words

摘录 / EXCERPT: Let $\mathrm{HB}_k(n)$ be the maximum length of an unbordered partial word with parameter n over an alphabet of size k, and let $\mathrm{HB}_\infty(n)$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-070362 | MATHDB-364560

The eigenvalue-form conjecture for the substitution matrix

摘录 / EXCERPT: Eigenvalue-form conjecture. Every eigenvalue of M_n has the form $\pm\frac{1}{s_{n,j}}$ for all odd $n\geq 5$. If this holds and $g_n(x)$ is irreducible for odd $n\geq 11$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-070363 | MATHDB-364572

Critical-threshold conjecture for the zeros of partial sums

摘录 / EXCERPT: Critical-threshold conjecture. There exists a threshold $t_{\textrm{crit}}=t_{\textrm{crit}}(N)>0$ such that $x\mapsto q_N(x;t)$ has N real simple zeros for $t>t_{\textrm{crit}}$...

conjecture | open | MathDB 页面 | 原文线索

V2-070364 | MATHDB-364579

The local theta correspondence conjecture for quaternionic dual pairs

摘录 / EXCERPT: Local theta correspondence conjecture. If $s\in S_\phi^+$ and $s'\in S_{\phi'}^+$ are associated via ξ, and $\pi\in\Pi_\phi(G(V))$, then $\theta_\psi(\pi,W)$ has L-parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-070365 | MATHDB-364580

The theta-lift transfer conjecture for quaternionic dual pairs

摘录 / EXCERPT: Theta-lift transfer conjecture. If such a parameter ϕ' exists, then

conjecture | open | MathDB 页面 | 原文线索

V2-070366 | MATHDB-364593

The alternating-group derangement bound conjecture

摘录 / EXCERPT: Alternating-group derangement bound conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-070367 | MATHDB-364595

The converse-type conjecture for sparse Haar-random matrices

摘录 / EXCERPT: For each n, let $\Sigma_n=(\sigma_{n,1},\ldots,\sigma_{n,n})$, where each $\sigma_{n,j}$ is a subset of $[n]=\{1,2,\ldots,n\}$. Let $X_n\in\operatorname{M}_n(\mathbb{Z}_p)$ be Ha...

conjecture | open | MathDB 页面 | 原文线索

V2-070368 | MATHDB-364601

Arone's twisted Koszul self-duality conjecture for orthogonal epimorphisms

摘录 / EXCERPT: Arone's conjecture. The category formed by these sections is equivalent to $K(\mathsf{OEpi})$. Moreover, OEpi is a twisted Koszul self-dual category: $K(\mathsf{OEpi})$...

conjecture | open | MathDB 页面 | 原文线索

V2-070369 | MATHDB-364603

Equiconstitution conjecture for the Toda-Uehara tilting generator

摘录 / EXCERPT: Equiconstitution conjecture. The tilting generator $\mathcal{Z}$ is equiconstituted with any Toda-Uehara tilting generator.

conjecture | open | MathDB 页面 | 原文线索

V2-070370 | MATHDB-364613

The canonical-drawing characterization of product structure for regular polygon intersection graphs

摘录 / EXCERPT: The canonical-drawing characterization. The class of intersection graphs of α-free homothetic regular n-gons admits product structure if and only if their canonical drawin...

conjecture | open | MathDB 页面 | 原文线索

V2-070371 | MATHDB-364614

The no-product-structure conjecture below the independent-crossing threshold

摘录 / EXCERPT: The no-product-structure conjecture. For every $\alpha < s(n)$, the class of intersection graphs of α-free homothetic regular n-gons does not have product structure.

conjecture | open | MathDB 页面 | 原文线索

V2-070372 | MATHDB-364615

The product-structure conjecture for k-independent crossing graphs

摘录 / EXCERPT: Product-structure conjecture for k-independent crossing graphs. The class of k-independent crossing graphs admits product structure.

conjecture | open | MathDB 页面 | 原文线索

V2-070373 | MATHDB-364616

The product-structure conjecture above the independent-crossing threshold

摘录 / EXCERPT: The above-threshold product-structure conjecture. For every $\alpha \geq s(n)$, the class of intersection graphs of α-free homothetic regular n-gons has product structure.

conjecture | open | MathDB 页面 | 原文线索

V2-070374 | MATHDB-364618

Scaled Graham-Häggkvist conjecture for tree decompositions

摘录 / EXCERPT: Let T be a tree with n edges, and let x be a positive integer. A graph decomposes another graph if the latter's edges can be partitioned into edge-disjoint subgraphs isomorph...

conjecture | open | MathDB 页面 | 原文线索

V2-070375 | MATHDB-364619

Scully's generalized separation conjecture for quadratic forms

摘录 / EXCERPT: Scully's conjecture. If $q_{F(p)}$ is isotropic, then

conjecture | open | MathDB 页面 | 原文线索

V2-070376 | MATHDB-364621

Silverman's rank conjecture for elliptic curves in one-parameter families

摘录 / EXCERPT: Let E be an elliptic curve over $\mathbb{Q}(t)$, let $E(\mathbb{A}^1)$ denote the elliptic curve over the function field $\mathbb{Q}(t)$, and let $E(t)$ denote its specialization...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-070377 | MATHDB-364629

Partial root number murmurations for elliptic curves

摘录 / EXCERPT: Let $(\mathcal N,\mathcal Q)$ be an arithmetically compatible sequence of pairs (N,Q) of Type I. Let $\mathcal E$ be the set of rational newforms lying in $S_2(N)$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-070378 | MATHDB-364630

The SSOT energy formula for type C Lusztig q-weight multiplicities

摘录 / EXCERPT: Let $\tilde{\lambda}$ be the rectangular complement partition associated with λ in (g^n), and let $\tilde{\mu}$ be the corresponding complement for μ. Let T range o...

conjecture | open | MathDB 页面 | 原文线索

V2-070379 | MATHDB-364631

Nonnegativity conjecture for type C \mathfrak{gl}_n q-weight multiplicities

摘录 / EXCERPT: For a function L on the positive roots, define ${\mathrm{KL}}^{\mathfrak{g},L}_{\lambda,\mu}(q)$ by ... For $L=L_A$, where $L_A(\alpha)=1$ for positive roots of type A and...

conjecture | open | MathDB 页面 | 原文线索

V2-070380 | MATHDB-364651

Equality conjecture for the Riesz capacity decay of strictly self-similar fractals

摘录 / EXCERPT: Equality conjecture. If A is a strictly self-similar fractal with dimension d, then

conjecture | open | MathDB 页面 | 原文线索

V2-070381 | MATHDB-364653

Invariant bisolution basis conjecture for the universal cover of anti-de Sitter space

摘录 / EXCERPT: Let W_0 denote the universal cover of anti-de Sitter space, and let Z be the invariant anti-de Sitter quantity appearing in the Gegenbauer kernels. For the relevant parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-070382 | MATHDB-364656

Classification conjecture for cubic power functions with optimal second-order differential uniformity

摘录 / EXCERPT: Let $n\geq 4$, let $f:\mathbb{F}_{2^n}\to\mathbb{F}_{2^n}$ be the power function $f(x)=x^d$, and identify exponents under the affine-equivalence transformation $d\mapsto 2^i d$ mod...

conjecture | open | MathDB 页面 | 原文线索

V2-070383 | MATHDB-364681

Adiprasito-Schmidt stability conjecture for slice rank of subspaces

摘录 / EXCERPT: Let $d\geq 2$, let $\mathbb{K}$ be a field, and let $\mathsf{W}\subseteq\mathbb{K}^{n_1}\otimes\cdots\otimes\mathbb{K}^{n_d}$ be a linear subspace. Write $\overline{\mathbb{K}}$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-070384 | MATHDB-364682

Stability conjecture for higher slice rank of subspaces

摘录 / EXCERPT: Let $d\geq 2$, let $\mathbb{K}$ be a field, let $\mathsf{W}\subseteq\mathbb{K}^{n_1}\otimes\cdots\otimes\mathbb{K}^{n_d}$ be a linear subspace, and let k be a positive integer sa...

conjecture | open | MathDB 页面 | 原文线索

V2-070385 | MATHDB-364685

The canonical-quotient conjecture for Springer-fiber components

摘录 / EXCERPT: Canonical-quotient conjecture. The $\bar{A}(e)$-sets $\Irr(\spr)/K_e$ and Y are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-070386 | MATHDB-364697

Critical-point conjecture for multireference alignment loss landscapes

摘录 / EXCERPT: Let $x\in\mathcal{M}_{m_2}$ be an odd-size signal and let $\tau>0$. Write $\operatorname{supp}(\widehat{m}_2)$ for the support of the Fourier transform of the second moment, and id...

conjecture | open | MathDB 页面 | 原文线索

V2-070387 | MATHDB-364698

Babai spectrum conjecture for paths

摘录 / EXCERPT: Babai spectrum conjecture for paths. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-070388 | MATHDB-364701

Martel-Merle nonexistence conjecture for outgoing multi-solitons

摘录 / EXCERPT: Let ϕ solve the three-dimensional energy-critical wave equation and satisfy the no-outgoing-radiation condition, and consider multi-solitons whose ground-state components are...

conjecture | open | MathDB 页面 | 原文线索

V2-070389 | MATHDB-364703

Parabolic-orbit conjecture for supercritical wave solitons

摘录 / EXCERPT: Let $\mathcal{D}^{\mathrm{par}}$ be a compactification of Minkowski space, let W^{sup} be the supercritical ground-state soliton, and let $z\in\mathbb{R}^3$. Define ... A...

conjecture | open | MathDB 页面 | 原文线索

V2-070390 | MATHDB-364705

Necessary and sufficient condition for bounded distance equivalence of parallelotope model sets

摘录 / EXCERPT: Let $\Gamma \subset \mathbb{R}^m \times \mathbb{R}^n$ be a lattice, let p_2 denote projection onto the second factor, and let $W \subset \mathbb{R}^n$ be the half-open parallelot...

conjecture | open | MathDB 页面 | 原文线索

V2-070391 | MATHDB-364717

Bensmail et al.'s planar (n,m)-complete graph bound conjecture

摘录 / EXCERPT: Let n,m be nonnegative integers, and let G be a planar (n,m)-complete graph, meaning a planar graph in which every pair of distinct vertices is joined by a walk of length at...

conjecture | open | MathDB 页面 | 原文线索

V2-070392 | MATHDB-364719

Converse to the octahedral relation for Sato tau functions

摘录 / EXCERPT: Let $\tau(n_1,\ldots,n_s)=\langle g\mid n_1,\ldots,n_s\rangle$ be the discrete function defined for integer tuples with $\operatorname{deg}(\mathbf n)=\sum_i n_i=0$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-070393 | MATHDB-364721

Weak Hadwiger conjecture for graphs with independence number at most two

摘录 / EXCERPT: Weak Hadwiger conjecture. Every graph G with $\alpha(G)\leq 2$ contains

conjecture | open | MathDB 页面 | 原文线索

V2-070394 | MATHDB-364722

Cycle depth formula conjecture for square-free powers

摘录 / EXCERPT: Let C_n be a cycle of length n, let $I(C_n)$ be its edge ideal in the corresponding polynomial ring $R=\mathbb{K}[x\mid x\in V(C_n)]$, and let $\nu(C_n)$ denote its matching nu...

conjecture | open | MathDB 页面 | 原文线索

V2-070395 | MATHDB-364725

Eventual equality of list colorings and colorings for edgeless graphs

摘录 / EXCERPT: Eventual equality conjecture. For each $n \in \mathbb{N}$, there is an $N \in \mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070396 | MATHDB-364727

Sharpness conjecture for forbidden induced subgraphs and JEP decidability

摘录 / EXCERPT: Let H be a graph, let S be a finite set of graphs, and write $\operatorname{Forb}_\subseteq(S)$ for the graphs containing no member of S as an induced subgraph. One-forbidden...

conjecture | open | MathDB 页面 | 原文线索

V2-070397 | MATHDB-364728

Variational Yau-Tian-Donaldson conjecture for the Mabuchi functional

摘录 / EXCERPT: Variational Yau-Tian-Donaldson conjecture. The Mabuchi functional is coercive if, and only if, (X,α) is uniformly K-stable.

conjecture | open | MathDB 页面 | 原文线索

V2-070398 | MATHDB-364730

Cisto's Generalized Wilf conjecture for N^p-semigroups

摘录 / EXCERPT: Cisto's Generalized Wilf conjecture. The inequality

conjecture | open | MathDB 页面 | 原文线索

V2-070399 | MATHDB-364732

Boundedness-by-one conjecture for ratios of minors

摘录 / EXCERPT: Let A be a totally positive element of GL_n, and let ... where $I_k,I'_k,J_k,J'_k\subseteq\{1,2,\ldots,n\}$ satisfy $|I_k|=|I'_k|$ and $|J_k|=|J'_k|$. A ratio is subtraction free...

conjecture | open | MathDB 页面 | 原文线索

V2-070400 | MATHDB-364736

Compatibility of the enhanced spectral Eisenstein functors with the global self-duality

摘录 / EXCERPT: Compatibility conjecture. The following diagram of functors commutes:

conjecture | open | MathDB 页面 | 原文线索

V2-070401 | MATHDB-364741

Square-root critical behavior conjecture for the formal two-matrix model

摘录 / EXCERPT: Square-root critical behavior conjecture. The formal matrix model has the asymptotic expansion

conjecture | open | MathDB 页面 | 原文线索

V2-070402 | MATHDB-364742

Equality of the (g,g,g) and $(g,-g,g)$ two-matrix model solutions

摘录 / EXCERPT: Equality-of-solutions conjecture. The (g,g,g) and $(g,-g,g)$ two-matrix models have the same unique solution satisfying the positivity constraints.

conjecture | open | MathDB 页面 | 原文线索

V2-070403 | MATHDB-364743

One-dimensional search-space conjecture for the generalized three-color matrix model

摘录 / EXCERPT: One-dimensional search-space conjecture. The matrix model has search-space dimension one: all moments in the large-N limit can be written in terms of the coupling constant g an...

conjecture | open | MathDB 页面 | 原文线索

V2-070404 | MATHDB-364755

Generalized Laguerre inequality for Speyer's generating polynomial

摘录 / EXCERPT: Generalized Laguerre conjecture. Given $t>0$, the polynomial $h_n(x;t)$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070405 | MATHDB-364756

Polynomial torsion bound conjecture for elliptic curves

摘录 / EXCERPT: Let F be a number field of degree $d=[F:\mathbb{Q}]$, and let E/F be an elliptic curve. Write $E(F)[\operatorname{tors}]$ for the subgroup of F-rational points of finite orde...

conjecture | open | MathDB 页面 | 原文线索

V2-070406 | MATHDB-364759

Kalai's generalized upper bound conjecture for shellable CW spheres

摘录 / EXCERPT: Let the Generalized Upper Bound Theorem mean that, for a simplicial d-polytope P and the cyclic d-polytope $C(d,n)$, the inequality $f_{d-1}(P) \geq f_{d-1}(C(d,n))$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-070407 | MATHDB-364762

Generalized Gross--Kudla conjecture for Chow cycles

摘录 / EXCERPT: Let r be a positive integer and let $(\kappa,\lambda)\in\mathcal A_r^{(2)}$. Write $X_r=X_1(Np^r)$, $f=\mathbf f_\kappa$, and $g=\mathbf g_\lambda$. Assume that f and g are n...

conjecture | open | MathDB 页面 | 原文线索

V2-070408 | MATHDB-364777

Kaplan-Lev-Roditty's antimagic conjecture for trees

摘录 / EXCERPT: Let A be a finite Abelian group, and write $I(A)$ for the set of involutions of A. Set $A^*=A\setminus\{0\}$. An A^*-antimagic labeling of a graph with $|A|$ vertices is a bi...

conjecture | open | MathDB 页面 | 原文线索

V2-070409 | MATHDB-364779

The global extreme-zero product conjecture for transitional Laguerre polynomials

摘录 / EXCERPT: For $n=0,1,\dots$ and $\alpha>-1$, define the transitional Laguerre polynomials $\widehat{L}^{(\alpha)}_{n+1}(\cdot,t)$ by ... where ... Let the extreme zeros denote the smallest and l...

conjecture | open | MathDB 页面 | 原文线索

V2-070410 | MATHDB-364783

Existence conjecture for a column permutation with no zero principal minors

摘录 / EXCERPT: Column-permutation conjecture. For every natural number N, there exists a permutation σ such that the Fourier matrix with permuted columns has no zero principal minors.

conjecture | open | MathDB 页面 | 原文线索

V2-070411 | MATHDB-364785

Weak particle-system convergence conjecture

摘录 / EXCERPT: Let $\bm X^P$ be the continuous-time particle-system approximation and let $Z(t)$ be the true solution of the corresponding McKean-Vlasov equation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070412 | MATHDB-364786

Variance-decay conjecture for the particle estimator

摘录 / EXCERPT: Let $\mathcal{A}(M,P,N)$ be the single-level estimator, and let $\bm X_N^P$ denote the particle-system realization. For M samples and P particles, variance-decay conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-070413 | MATHDB-364787

Extrapolated multilevel variance-decay conjecture

摘录 / EXCERPT: Let $\prescript{\mathrm{Ext}}{}{\Phi}^{\ell}$ and $\prescript{\mathrm{Ext}}{}{\Psi}^{\ell}$ denote the extrapolated fine- and coarse-level quantities at level ℓ. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-070414 | MATHDB-364794

Asymptotic equivalence of affine and unimodular lattice-polytope classes

摘录 / EXCERPT: Let K_V^d be the set of equivalence classes of convex lattice polytopes of normalized volume V under affine equivalence, and let L_V^d be the corresponding set under unimodul...

conjecture | open | MathDB 页面 | 原文线索

V2-070415 | MATHDB-364805

The exact hexagon-count conjecture for strongly regular graphs with parameters $\lambda=1$ and $\mu=2$

摘录 / EXCERPT: Let G be a strongly regular graph with parameters $\lambda=1$ and $\mu=2$, order n, and valency k. Let p_6 denote the number of hexagons in G. Exact hexagon-count conject...

conjecture | open | MathDB 页面 | 原文线索

V2-070416 | MATHDB-364808

Pantev-Toën-type Fourier-Mukai symmetry conjecture for quasi-BPS categories of Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus g, let L be a line bundle on C with $l=\deg L>2g-2$, and let B^L be the Hitchin base. For rank r, numerical invariant hi,...

conjecture | open | MathDB 页面 | 原文线索

V2-070417 | MATHDB-364809

Pantev-Toën SL/PGL symmetry conjecture for quasi-BPS categories of Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve, let L be a line bundle on C, and let $B^L_{\geqslant 2}=\bigoplus_{i=2}^r H^0(C,L^{\otimes i})$ be the Hitchin base for trace-free Higgs f...

conjecture | open | MathDB 页面 | 原文线索

V2-070418 | MATHDB-364818

Dimca's conjecture on the Tjurina number of reducible plane curves

摘录 / EXCERPT: Let $C=C_J\cup C_K$ be a reducible plane curve with the notation used above, where C_J and C_K are the corresponding subcurves and $\textup{I}(C_J,C_K)$ denotes their intersect...

conjecture | open | MathDB 页面 | 原文线索

V2-070419 | MATHDB-364822

Finite-difference moment identities for elementary symmetric partition images

摘录 / EXCERPT: Let $\mathcal P(n)$ be the set of partitions of n, let $\mathcal P[m,n]=\biguplus_{i=m}^n\mathcal P(i)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070420 | MATHDB-364823

OEIS formulas for statistics of image-partition sets

摘录 / EXCERPT: For $n\ge0$, let $\chi(\mathcal S)$ be the number of distinct parts occurring in a set of partitions $\mathcal S$, and let $\tau(\mathcal S)$ count parts with multiplicity. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070421 | MATHDB-364824

Wild non-uniqueness conjecture for two-dimensional isentropic Euler Riemann problems

摘录 / EXCERPT: Wild non-uniqueness conjecture. For any such Riemann initial data, there exists a smooth pressure law p with $p'>0$ such that the corresponding system admits infinitely many boun...

conjecture | open | MathDB 页面 | 原文线索

V2-070422 | MATHDB-364831

Relation between continuous and discrete Rayleigh systems

摘录 / EXCERPT: Consider a continuous Rayleigh system $(L,\mathcal{R})$ on TQ, where L is natural, meaning that ... Here g is a Riemannian metric on Q and V is a function on Q. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070423 | MATHDB-364858

Minor-closed-family conjecture for oriented chromatic number

摘录 / EXCERPT: Minor-closed-family conjecture. There exists a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that $f(n)=n^{1+o(1)}$ and, for every minor-closed family of graphs $\mathcal{G}$,

conjecture | open | MathDB 页面 | 原文线索

V2-070424 | MATHDB-364862

The Milnor-number equality criterion including exceptional vertices

摘录 / EXCERPT: Let $f\in\mathbb{K}[[x,y]]$ be a reduced power series, and let $\mathcal{T}$ be a minimal tree of f. Let $\mathcal{V}$ be the set of vertices of $\mathcal{T}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-070425 | MATHDB-364876

Hartke's integrality gap conjecture for trees with at most 12 vertices

摘录 / EXCERPT: Let T be a tree, and consider the linear program obtained from the MacGillivray-Wang integer program for virus spread by relaxing $x(v)\in\{0,1\}$ to a linear constraint and addi...

conjecture | open | MathDB 页面 | 原文线索

V2-070426 | MATHDB-364877

One-vaccination conjecture for the Pentagon Graph

摘录 / EXCERPT: Let the Pentagon Graph be the infinite graph obtained from a pentagonal tiling of the plane, with vertices of degrees three and four. Consider the virus-containment process in whic...

conjecture | open | MathDB 页面 | 原文线索

V2-070427 | MATHDB-364878

Conjectural determinant factorization for the matrix C

摘录 / EXCERPT: The determinant factorization conjecture. Writing $n=2r$ or $n=2r-1$, respectively,

conjecture | open | MathDB 页面 | 原文线索

V2-070428 | MATHDB-364879

Conjecture on the real-rootedness and strong log-concavity of the Birkhoff-polytope h-star polynomial

摘录 / EXCERPT: The real-rootedness and log-concavity conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070429 | MATHDB-364889

Wishart Gaussian product inequality conjecture

摘录 / EXCERPT: Wishart Gaussian product inequality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070430 | MATHDB-364891

The UT-free monotone polytope conjecture

摘录 / EXCERPT: Let a UT-free monotone polytope be a monotone polytope with the UT-free property, and let a USFP be a unimodular smooth Fano polytope. UT-free monotone polytope conjecture. UT-free...

conjecture | open | MathDB 页面 | 原文线索

V2-070431 | MATHDB-364893

Quantitative rank distribution conjecture for elliptic curves over $\mathbb{F}_q(t)$

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{\geq 0}$ and suppose that $\operatorname{char}(\mathbb{F}_q)>3$. Let $\mathcal{N}^{r}_{T}(\mathbb{F}_q(t),B)$ count the number of $\mathbb{F}_q$-isomorphism cl...

conjecture | open | MathDB 页面 | 原文线索

V2-070432 | MATHDB-364905

The subdivision packing-coloring conjecture for subcubic graphs

摘录 / EXCERPT: Subdivision packing-coloring conjecture. The graph $S(G)$ admits a $(1,2,3,4,5)$-packing coloring. Equivalently, every subdivision of a subcubic graph has packing chromatic number...

conjecture | open | MathDB 页面 | 原文线索

V2-070433 | MATHDB-364939

Conjecture on global monodromy of the bi-degree arrangement

摘录 / EXCERPT: Let m,n be positive integers with $n\geq 2$, and let $\mathcal A_{m,n}$ be the arrangement whose linear functions are ... for $(j,\ell)\in[0;m-1]\times[0;n-1]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070434 | MATHDB-364958

Manin's conjecture for Châtelet surfaces

摘录 / EXCERPT: Let $\Delta\in\mathbb{Q}^\times$ satisfy $\sqrt{-\Delta}\notin\mathbb{Q}$, and let $f(z)\in\mathbb{Z}[z]$ be a separable polynomial of degree 3 or 4 that decomposes as a produc...

conjecture | open | MathDB 页面 | 原文线索

V2-070435 | MATHDB-364965

Conjecture on the chromatic-cochromatic gap in random graphs

摘录 / EXCERPT: Let G be sampled from the binomial random graph model $G_{n,1/2}$, and let $ u$ denote the chromatic number of G and $\bar\chi$ its cochromatic number, so that the gap is...

conjecture | open | MathDB 页面 | 原文线索

V2-070436 | MATHDB-364970

The conjecture on simple fibrations for canonical threefolds on the Noether line

摘录 / EXCERPT: A canonical threefold on the Noether line is a canonical threefold whose canonical volume and geometric genus satisfy ... A simple fibration in $(1,2)$-surfaces means the type of fib...

conjecture | open | MathDB 页面 | 原文线索

V2-070437 | MATHDB-364974

Perfectoid purity under split finite quasi-torsors

摘录 / EXCERPT: Suppose that $R\subseteq S$ is a finite μ_n-quasi-torsor: there exists an open subset $U\subseteq\operatorname{Spec}R$ whose complement has codimension at least 2, such that...

conjecture | open | MathDB 页面 | 原文线索

V2-070438 | MATHDB-364975

Log canonicity of normal Q-Gorenstein lim-perfectoid pure singularities

摘录 / EXCERPT: Let R be a normal $\mathbf{Q}$-Gorenstein ring. The lim-perfectoid log-canonicity conjecture. If R is lim-perfectoid pure, then R is log canonical. The paper proves the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-070439 | MATHDB-364976

Critical-mass conjecture for discrete coagulation-fragmentation equations

摘录 / EXCERPT: Consider the discrete coagulation-fragmentation equation with multiplicative coagulation and constant fragmentation kernels, and let ρ_0 be initial data satisfying the paper's...

conjecture | open | MathDB 页面 | 原文线索

V2-070440 | MATHDB-364977

Stationary-solution and classical-solution conjecture below unit mass

摘录 / EXCERPT: Consider the discrete coagulation-fragmentation equation with multiplicative coagulation and constant fragmentation kernels, with initial data ρ_0 satisfying the paper's initi...

conjecture | open | MathDB 页面 | 原文线索

V2-070441 | MATHDB-364978

Injectivity conjecture for the intersection complexes of rational singularities

摘录 / EXCERPT: Intersection-complex injectivity conjecture. The natural morphism

conjecture | open | MathDB 页面 | 原文线索

V2-070442 | MATHDB-364979

The tangent-measure conjecture for spectral measures of sparse decaying Jacobi matrices

摘录 / EXCERPT: Tangent-measure conjecture. For every ... μ has a unique tangent measure at $$ ξ, and this tangent measure is Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-070443 | MATHDB-364981

Interval colorability of planar Cartesian products

摘录 / EXCERPT: Let G and H be non-trivial graphs, and suppose that their Cartesian product $G\square H$ is planar. Planar Cartesian product conjecture. The graph $G\square H$ is interval colo...

conjecture | open | MathDB 页面 | 原文线索

V2-070444 | MATHDB-364982

The property R_∞ conjecture for the Artin group of type D_5

摘录 / EXCERPT: Let $A[D_5]$ be the Artin group of spherical type D_5, and let $Z(A[D_5])$ denote its center. The D_5 property R_∞ conjecture. The Artin group $A[D_5]$ and its central q...

conjecture | open | MathDB 页面 | 原文线索

V2-070445 | MATHDB-364987

Banks-Panzer-Pym's generation conjecture for normalized multiple zeta values

摘录 / EXCERPT: Let $\mzvswithhalf^n$ be the weight-n component of the filtered ring generated by normalized multiple zeta values, with $\mzvswithhalf^n=\mzvs^n+\frac{1}{2}\mzvs^{n-1}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-070446 | MATHDB-364989

3-colorability conjecture for string graphs of sufficiently large odd girth

摘录 / EXCERPT: Let k be an integer, and consider the class of string graphs whose odd girth is at least k. String-graph coloring conjecture. There is an integer k such that the class of str...

conjecture | open | MathDB 页面 | 原文线索

V2-070447 | MATHDB-365007

Equivalence of positive corotational rates and strong monotonicity conditions

摘录 / EXCERPT: Let $\mathcal{E}: \operatorname{Sym}^{++}(3) \to \operatorname{Sym}(3)$ be a spatial strain tensor with strongly monotone scale function $e: \mathbb{R}^+ \to \mathbb{R}$. For a cor...

conjecture | open | MathDB 页面 | 原文线索

V2-070448 | MATHDB-365034

Brenti's log-concavity conjecture for Eulerian polynomials of labeled posets

摘录 / EXCERPT: Brenti's conjecture. The polynomial $W_{P,\omega}$ is log-concave with no internal zeroes.

conjecture | open | MathDB 页面 | 原文线索

V2-070449 | MATHDB-365047

Unstable circulants of twice an odd order

摘录 / EXCERPT: The conjecture. The graph Γ is unstable if and only if either there exists a nonzero $h\in 2\mathbb{Z}_{2m}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070450 | MATHDB-365052

Baker's conjecture on S-integral preperiodic points

摘录 / EXCERPT: Let K be a number field, and let S be a finite set of places of K containing all archimedean places. Let $\varphi:\mathbb{P}_K^{1}\to\mathbb{P}_K^{1}$ be a nonconstant ration...

conjecture | open | MathDB 页面 | 原文线索

V2-070451 | MATHDB-365055

Local boundedness conjecture for critical-drift Stokes solutions

摘录 / EXCERPT: Local boundedness conjecture. Under these hypotheses, $u\in L^\infty(B_1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-070452 | MATHDB-365084

The finite-cycle character-value coincidence conjecture for symmetric groups

摘录 / EXCERPT: Finite-cycle character-value conjecture. For all sufficiently large n, there exist k distinct irreducible characters of S_n whose values on permutations $g\in S_n$ coincide f...

conjecture | open | MathDB 页面 | 原文线索

V2-070453 | MATHDB-365085

The restriction-coincidence conjecture for irreducible characters of symmetric groups

摘录 / EXCERPT: Restriction-coincidence conjecture. For all sufficiently large n, there are k distinct irreducible characters of S_n whose restrictions to S_m coincide.

conjecture | open | MathDB 页面 | 原文线索

V2-070454 | MATHDB-365103

Brauer p-dimension bounds for henselian discretely valued fields

摘录 / EXCERPT: Brauer p-dimension conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070455 | MATHDB-365104

Symbol-length conjecture for differential forms

摘录 / EXCERPT: Symbol-length conjecture for differential forms. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070456 | MATHDB-365114

Weak resolution of singularities

摘录 / EXCERPT: Let L_0 be a field and let L_1 be a nontrivial finitely generated extension of L_0. Suppose there exists a valuation v on L_1 with residue field L_0. Weak resolution of...

conjecture | open | MathDB 页面 | 原文线索

V2-070457 | MATHDB-365118

Multiplicity conjecture for Proudfoot-Speyer degenerations of scattering equations

摘录 / EXCERPT: Let L_m be the relevant linear space arrangement, let $\mathbb{L}_s$ be the linear space determined by the kinematic data s, and let $I_r(W)$ be a type (ii) flat, with associat...

conjecture | open | MathDB 页面 | 原文线索

V2-070458 | MATHDB-365122

Conjecture on continuity of the Clifford linear gap function

摘录 / EXCERPT: Continuity conjecture. The Clifford linear gap function is pointwise continuous.

conjecture | open | MathDB 页面 | 原文线索

V2-070459 | MATHDB-365123

Conjecture on approximate joint eigenstates from the Clifford linear pseudospectrum

摘录 / EXCERPT: Approximate joint-eigenstate conjecture. If $(\bm{\lambda},\nu)\in\bar{\Lambda}_{\epsilon}^{\mathrm{C}}(\bm{A},B)$, then there is a unit state $\bm{\psi}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070460 | MATHDB-365124

Classification conjecture for smooth affine cubic surfaces with infinite discrete automorphism group

摘录 / EXCERPT: Let S be a smooth, affine, cubic surface whose automorphism group is infinite and discrete. A classification conjecture asserts that S is either a Markov-type surface, or the c...

conjecture | open | MathDB 页面 | 原文线索

V2-070461 | MATHDB-365131

Šámal's normal chromatic-index conjecture

摘录 / EXCERPT: Let G be a bridgeless cubic graph, and let $\chi'_{N}(G)$ be its normal chromatic index. Šámal's conjecture. Every bridgeless cubic graph G satisfies ... This is presented as an...

conjecture | open | MathDB 页面 | 原文线索

V2-070462 | MATHDB-365133

Infinite cyclically 6-edge-connected counterexamples to non-conflicting flows

摘录 / EXCERPT: Let G be a cubic graph and let F be a perfect matching; write $\overline{F}$ for the complementary 2-factor. A non-conflicting nowhere-zero $Z_2\times Z_2$-flow with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-070463 | MATHDB-365140

Hua-Huang-Li conjecture on independent random waves for Hecke-Maass cusp forms

摘录 / EXCERPT: Hua-Huang-Li conjecture. The powers $\{f_j^{a_j}\}_{j=1}^J$ are statistically independent: for every $\psi\in\mathcal{C}_c^\infty(\mathbb{X})$,

conjecture | open | MathDB 页面 | 原文线索

V2-070464 | MATHDB-365142

Efficient cleaning sequence for directed graphs

摘录 / EXCERPT: Let G be a directed graph, and consider cleaning sequences in which multiple vertices that are ready to fire may fire simultaneously. Efficient-cleaning-sequence problem. Determi...

open_problem | open | MathDB 页面 | 原文线索

V2-070465 | MATHDB-365150

Amdeberhan's character table conjecture

摘录 / EXCERPT: Let $\lambda \vdash n$. Write $Ev(\lambda)$ for the set of even refinements of λ, let $\mathcal{R}_{2N+1}(2|\lambda|)$ and $\mathcal{R}^{c}_{2N}(2|\lambda|)$ be the partiti...

conjecture | open | MathDB 页面 | 原文线索

V2-070466 | MATHDB-365156

The nonlinear instability conjecture for Kerr-AdS

摘录 / EXCERPT: Let Kerr-AdS be a rotating asymptotically anti-de Sitter black-hole solution of the Einstein vacuum equations, and impose Dirichlet boundary conditions. A generic perturbation of i...

conjecture | open | MathDB 页面 | 原文线索

V2-070467 | MATHDB-365159

Existence conjecture for optimal extended irregular dominating labelings of cycles

摘录 / EXCERPT: Let C_{4n} be the cycle on $4n$ vertices, and let an optimal extended irregular dominating labeling mean a labeling satisfying the paper's extended irregular domination condition...

conjecture | open | MathDB 页面 | 原文线索

V2-070468 | MATHDB-365160

Conjecture for the extended irregular domination number of paths

摘录 / EXCERPT: Let P_n be the path on n vertices, and let $\gamma_e(P_n)$ denote its extended irregular domination number. Path-domination conjecture. For $n\geq 4$, ... The equality is known f...

conjecture | open | MathDB 页面 | 原文线索

V2-070469 | MATHDB-365161

Existence conjecture for polynomial flows with prescribed heteroclinic regions

摘录 / EXCERPT: Existence conjecture for polynomial flows with prescribed heteroclinic regions. For all $n,m\in\mathbb{N}$, there exists a complex polynomial $F_{n,m}$ of degree $n+m$ such that th...

conjecture | open | MathDB 页面 | 原文线索

V2-070470 | MATHDB-365167

Almost-sure global convergence of the subgradient method for \ell_1-norm rank-one symmetric matrix factorization

摘录 / EXCERPT: Consider the ℓ_1-norm rank-one symmetric matrix factorization problem and initialize the subgradient method randomly, with an initialization distribution absolutely continuous...

conjecture | open | MathDB 页面 | 原文线索

V2-070471 | MATHDB-365172

Sarnak's trace-set arithmeticity conjecture for cofinite Fuchsian groups

摘录 / EXCERPT: Sarnak's conjecture. If $\mathrm{Tr}(\Gamma)$ satisfies the bounded clustering property, then Γ is arithmetic. If $\operatorname{Gap}(\mathrm{Tr}(\Gamma))>0$, then Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-070472 | MATHDB-365173

Schmutz's linear-growth conjecture for trace sets of cofinite Fuchsian groups

摘录 / EXCERPT: Schmutz's conjecture. If $\mathrm{Tr}(\Gamma)$ has linear growth, then Γ is arithmetic.

conjecture | open | MathDB 页面 | 原文线索

V2-070473 | MATHDB-365174

Cocompact trace-set volume conjecture for large-genus hyperbolic surfaces

摘录 / EXCERPT: Let V_g^ϵ and $\overline{V_g^\epsilon}$ be the sets defined in Theorem, and let $\mathcal{M}_g$ be the corresponding moduli space of genus-g surfaces, with Weil--Peterss...

conjecture | open | MathDB 页面 | 原文线索

V2-070474 | MATHDB-365175

Countable branches of radial self-similar singular solutions for the supercritical CGL equation

摘录 / EXCERPT: Branch-counting conjecture. The supercritical CGL equation has a countable number of branches of nontrivial radial self-similar singular solutions. These branches can be labeled by...

conjecture | open | MathDB 页面 | 原文线索

V2-070475 | MATHDB-365176

Ogg's torsion conjecture for elliptic curves over the rationals

摘录 / EXCERPT: Ogg's torsion conjecture. If E is an elliptic curve over $\mathbb{Q}$, then its rational torsion subgroup has one of these forms; equivalently, $X_1(N)$ has no $\mathbb{Q}$-ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-070476 | MATHDB-365177

The uniform boundedness conjecture for rational torsion of elliptic curves

摘录 / EXCERPT: Let K be a number field and let E be an elliptic curve over K. Uniform boundedness conjecture. The order of the rational torsion subgroup $E(K)_\mathrm{tor}$ is bounded by a...

conjecture | open | MathDB 页面 | 原文线索

V2-070477 | MATHDB-365188

Splitting principle for objects of finite bosonic subdimension

摘录 / EXCERPT: Let $[?]$ be a 2-rig and let $r\triangleright [?]$ be an object of finite bosonic subdimension. Splitting principle conjecture. There exists a 2-rig $[?]$ and a map of 2-rigs...

conjecture | open | MathDB 页面 | 原文线索

V2-070478 | MATHDB-365192

Universality of the central extension of the annulus semigroup

摘录 / EXCERPT: Universality conjecture. This central extension is a universal central extension. The claim concerns universality in the category of central extensions of the annulus semigroup. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-070479 | MATHDB-365193

Universality of the annulus central extension for diffeological semigroups

摘录 / EXCERPT: Let Ann be the annulus semigroup and let ... be the stated central extension. Diffeological universality conjecture. This extension is a universal central extension...

conjecture | open | MathDB 页面 | 原文线索

V2-070480 | MATHDB-365196

Morrison-Totaro cone conjecture for klt log Calabi-Yau pairs

摘录 / EXCERPT: Let (X,Δ) be a klt log Calabi-Yau pair, meaning that X is smooth projective and $\Delta \in |-K_X|$ is linearly equivalent to $\sum_i a_iD_i$ for effective divisors D_i...

conjecture | open | MathDB 页面 | 原文线索

V2-070481 | MATHDB-365198

Mohr-Pardey-Rautenbach conjecture on almost colour-balanced spanning forests

摘录 / EXCERPT: Mohr-Pardey-Rautenbach conjecture. There exists a copy F' of F in K_n such that

conjecture | open | MathDB 页面 | 原文线索

V2-070482 | MATHDB-365205

The rational-curves density conjecture for terminal Calabi--Yau models

摘录 / EXCERPT: Rational-curves density conjecture. The union of rational curves in X' is Zariski dense in X'.

conjecture | open | MathDB 页面 | 原文线索

V2-070483 | MATHDB-365220

Small-coupling continuity conjecture for Sturmian spectrum dimension

摘录 / EXCERPT: For an irrational $\alpha\in(0,1)$, let $\sigma_{\lambda,\alpha}$ denote the corresponding Sturmian spectrum and let $\dim_H$ denote Hausdorff dimension. Small-coupling continuity...

conjecture | open | MathDB 页面 | 原文线索

V2-070484 | MATHDB-365228

Fractional moment asymptotics for characteristic polynomials of the real elliptic ensemble

摘录 / EXCERPT: Let $A_{N,\tau}$ be a matrix from the real elliptic ensemble with symmetry parameter $\tau\in(-1,1)$, and let μ lie in the real bulk interval $(-(1+\tau),1+\tau)$. Let G deno...

conjecture | open | MathDB 页面 | 原文线索

V2-070485 | MATHDB-365231

Kirillov's non-negativity conjecture for Hecke-Grothendieck and generalized key polynomials

摘录 / EXCERPT: Let n be a positive integer, let $w\in S_n$ be a permutation, let ζ be a weak composition, and let $\boldsymbol{x}=(x_1,\ldots,x_n)$. Write...

conjecture | refuted | MathDB 页面 | 原文线索

V2-070486 | MATHDB-365235

Positivity of products with Lascoux polynomials

摘录 / EXCERPT: Let $\mathbb N^n$ denote the set of compositions indexing the Lascoux polynomials $\mathfrak L_\alpha$, and let $i\in[n]$. Lascoux product-positivity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-070487 | MATHDB-365236

Crab bucket non-destabilization conjecture for Legendrian knots

摘录 / EXCERPT: Crab bucket non-destabilization conjecture. For all odd $n\geq 5$, β_n cannot be negatively destabilized without increasing its mosaic number.

conjecture | open | MathDB 页面 | 原文线索

V2-070488 | MATHDB-365237

Conjecture on the h-vector of the coarse Hall-Littlewood-Schubert series at $Y=-1$

摘录 / EXCERPT: The $Y=-1$ h-vector conjecture. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-070489 | MATHDB-365239

Openness conjecture for proper and cocompact actions

摘录 / EXCERPT: Let G/H be a homogeneous space of reductive type, and let Γ be a discrete subgroup of G acting properly discontinuously and cocompactly on G/H. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070490 | MATHDB-365241

Reductive projection conjecture for compact Clifford-Klein forms

摘录 / EXCERPT: Let G be a connected real linear reductive Lie group, let H be a reductive subgroup of G, and let Γ be a discrete subgroup of G. Let the Zariski closure of Γ...

conjecture | open | MathDB 页面 | 原文线索

V2-070491 | MATHDB-365259

Capuano-Veneziano-Zannier conjecture on Ruban continued fractions

摘录 / EXCERPT: Let $\alpha^{(p)}$ be a p-adic quadratic irrational, and consider its Ruban continued fraction. Capuano-Veneziano-Zannier conjecture. The Ruban continued fraction of a quadratic ir...

conjecture | open | MathDB 页面 | 原文线索

V2-070492 | MATHDB-365261

Simultaneous real and p-adic convergence implies periodicity

摘录 / EXCERPT: Let $[a_0,a_1,\ldots]$ be a continued fraction with $a_n\in\mathbb{Q}$ for all n. Suppose it converges simultaneously in $\mathbb{R}$ and $\mathbb{Q}_p$ to a real quadratic irrat...

conjecture | open | MathDB 页面 | 原文线索

V2-070493 | MATHDB-365263

Bi-SP Conjecture on common shortest paths in positive two-person games

摘录 / EXCERPT: Bi-SP Conjecture. If G has an (s,V_T)-path, then P_1 and P_2 have such a path in common, that is,

conjecture | open | MathDB 页面 | 原文线索

V2-070494 | MATHDB-365269

Ricci-flatness of Kähler-Ricci shrinkers on Fano cones

摘录 / EXCERPT: Fano-cone Ricci-flatness conjecture. On a Fano cone X, every Kähler-Ricci shrinker is a Ricci-flat Kähler cone metric.

conjecture | open | MathDB 页面 | 原文线索

V2-070495 | MATHDB-365270

Two-step weighted-volume degeneration conjecture for Fano fibration germs

摘录 / EXCERPT: Two-step degeneration conjecture for weighted volume. 1. There is a unique minimizer v_* of $\mathbb W$ on $\operatorname{Val}_{\pi,p}^*$, which is quasi-monomial. 2. The graded...

conjecture | open | MathDB 页面 | 原文线索

V2-070496 | MATHDB-365271

K-moduli conjecture for polarized Fano fibrations

摘录 / EXCERPT: K-moduli conjecture for polarized Fano fibrations. Given a positive integer n and $\epsilon>0$, there exists a projective moduli space parametrizing K-polystable polarized Fano f...

conjecture | open | MathDB 页面 | 原文线索

V2-070497 | MATHDB-365272

Two-step degeneration conjecture for Kähler-Ricci flows at finite-time singularities

摘录 / EXCERPT: Finite-time two-step degeneration conjecture. For any $p\in Y$ and $q\in\pi^{-1}(p)$: 1. The Kähler-Ricci flow $\omega(t)$ naturally induces a weighted-volume-minimizing, hence K-s...

conjecture | open | MathDB 页面 | 原文线索

V2-070498 | MATHDB-365273

Tangent-flow-at-infinity conjecture for ancient Kähler-Ricci flows

摘录 / EXCERPT: Tangent flows at infinity conjecture. Given a non-collapsed ancient Kähler-Ricci flow $(X,\omega(t))$, $t\in(-\infty,0]$: 1. X admits a non-collapsing weak Fano fibration structu...

conjecture | open | MathDB 页面 | 原文线索

V2-070499 | MATHDB-365281

The log-concavity conjecture for Coxeter R-polynomials

摘录 / EXCERPT: R-polynomial log-concavity conjecture. The polynomial $Q_{u,v}(t)$ is log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-070500 | MATHDB-365282

The rank log-concavity conjecture for Weyl-group Bruhat intervals

摘录 / EXCERPT: Rank log-concavity conjecture. The interval $[u,v]$ is rank log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-070501 | MATHDB-365283

The monotone total positivity conjecture for the Eulerian matrix

摘录 / EXCERPT: Monotone total positivity conjecture. For all $i,j,r\in\mathbb{N}$, the determinant of this submatrix is a monotonically increasing function of $i\in\mathbb{N}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070502 | MATHDB-365291

Symmetric q-stability conjecture

摘录 / EXCERPT: For $q\geq 1$, let $\ln_q:(0,+\infty)\to\mathbb{R}$ be the q-logarithm ... and define $\Phi_q^{\mathrm{sym}}(t)=t\ln_q(t)+(1-t)\ln_q(1-t)$ and the associated symmetric q-stabilit...

conjecture | open | MathDB 页面 | 原文线索

V2-070503 | MATHDB-365313

C_4-saturation conjecture for the complete four-partite graph

摘录 / EXCERPT: Let K_4^n be the complete 4-partite graph with n vertices in each part, and let C_4 be the cycle of length four. Complete four-partite C_4-saturation conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-070504 | MATHDB-365321

The irreducibility and degree conjecture for logarithmic discriminants of ${\mathcal M}_{0,m}$

摘录 / EXCERPT: Let ${\mathcal M}_{0,m}$ denote the moduli space of m marked points on the projective line, and let $\nabla_{\rm log}({\mathcal M}_{0,m})$ be its logarithmic discriminant in...

conjecture | open | MathDB 页面 | 原文线索

V2-070505 | MATHDB-365322

The unitary-dual exhaustion conjecture for closures of Arthur representations

摘录 / EXCERPT: Unitary-dual exhaustion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070506 | MATHDB-365323

The good-parity Arthur-type conjecture for classical groups

摘录 / EXCERPT: Good-parity Arthur-type conjecture. For a classical group G,

conjecture | open | MathDB 页面 | 原文线索

V2-070507 | MATHDB-365324

Tadić's critical-type conjecture for classical groups

摘录 / EXCERPT: Tadić's critical-type conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070508 | MATHDB-365325

Tadić's preservation of unitarizability conjecture

摘录 / EXCERPT: Tadić's preservation of unitarizability conjecture. If $\pi\in\Pi(G_n)$ is weakly real, then π is unitary if and only if $X_{\rho_i}(\pi)$ is unitary for every $i=1,\ldots,r$.

conjecture | open | MathDB 页面 | 原文线索

V2-070509 | MATHDB-365327

The parameter-boundary conjecture for the reverse Mittag-Leffler inequality

摘录 / EXCERPT: The parameter-boundary conjecture. There exists an increasing convex function $f:[2,\infty)\to[3,\infty)$ with $f(2)=3$ such that, for every $\alpha\ge2$,

conjecture | open | MathDB 页面 | 原文线索

V2-070510 | MATHDB-365338

Coincidence conjecture for optimal concatenation indices of primitive stepsize schedules

摘录 / EXCERPT: Let $h_{\circ}^{(n)}$ denote the primitive stepsize schedule generated by Definition, let $\Call{ConPP}$ denote the concatenation operation, and let $\mathbf 1$ be the all-ones vec...

conjecture | open | MathDB 页面 | 原文线索

V2-070511 | MATHDB-365344

Hall-Littlewood P-positivity of q-divided symmetrized Schubert polynomials

摘录 / EXCERPT: Let $\mathfrak{S}_{w}(x_1,\dots,x_n)$ be a Schubert polynomial, and let $\langle \mathfrak{S}_{w}\rangle_n^q$ denote its q-divided symmetrization. A symmetric polynomial is Hall-...

conjecture | open | MathDB 页面 | 原文线索

V2-070512 | MATHDB-365345

Hall-Littlewood positivity for q-divided symmetrized Schubert polynomials

摘录 / EXCERPT: Let $\mathfrak{S}_{w}\in \operatorname{Pol}_n$ be a Schubert polynomial, and let $\langle \mathfrak{S}_{w}\rangle_n^q$ denote its q-divided symmetrization. For a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-070513 | MATHDB-365346

Positive boundary dissipation conjecture for BIBO stability of hBCS

摘录 / EXCERPT: Positive boundary dissipation conjecture. If $\widetilde{W}_B R_0^{-1} \Sigma \left(\widetilde{W}_B R_0^{-1}\right)^* > 0$, then the hBCS is BIBO stable. In the setting where the d...

conjecture | open | MathDB 页面 | 原文线索

V2-070514 | MATHDB-365347

Higher-order boundedness conjecture for contractive hBCS matrices

摘录 / EXCERPT: Let M be the matrix associated with a one-dimensional hyperbolic boundary control system, let $\mathbf{U}(s)$ be the corresponding matrix-valued function, let n be the system d...

conjecture | open | MathDB 页面 | 原文线索

V2-070515 | MATHDB-365348

Exponential stability conjecture for BIBO stability of hBCS

摘录 / EXCERPT: Let an hBCS be a one-dimensional hyperbolic boundary control system satisfying Assumption ... is BIBO stable. The authors state that they have not encountered an exponentially stable...

conjecture | open | MathDB 页面 | 原文线索

V2-070516 | MATHDB-365358

Böttcher-Hladký-Piguet-Taraz tree-packing conjecture

摘录 / EXCERPT: Böttcher-Hladký-Piguet-Taraz conjecture. Each family of trees of individual orders at most $n+1$ and total number of edges at most $\binom{2n+1}{2}$ packs into K_{2n+1}.

conjecture | open | MathDB 页面 | 原文线索

V2-070517 | MATHDB-365359

Graham-Häggkvist conjecture for complete bipartite hosts

摘录 / EXCERPT: Graham-Häggkvist complete-bipartite conjecture. Any tree of order $n+1$ decomposes the complete bipartite graph $K_{n,n}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070518 | MATHDB-365360

Hobbs-Bourgeois-Kasiraj bipartite Tree Packing Conjecture

摘录 / EXCERPT: Hobbs-Bourgeois-Kasiraj conjecture. Any family of trees $T_1,\dots,T_n$ packs into $K_{\lceil n/2\rceil,n-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070519 | MATHDB-365361

Balanced tree decomposition conjecture for complete bipartite graphs

摘录 / EXCERPT: Balanced tree decomposition conjecture. Any tree $T_{n,n}$ decomposes $K_{2n-1,2n-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070520 | MATHDB-365383

Explicit perfect-matching threshold conjecture for randomly perturbed unbalanced complete bipartite graphs

摘录 / EXCERPT: Perfect-matching threshold conjecture. The sharp d-threshold for containing a perfect matching is C/n.

conjecture | open | MathDB 页面 | 原文线索

V2-070521 | MATHDB-365403

Geng et al.'s AMDS conjecture for the dual BCH code

摘录 / EXCERPT: Geng et al.'s AMDS conjecture. The code $\mathcal{C}^\perp_{(q,q+1,3,4)}$ is an AMDS code with parameters $[q+1,4,q-3]$.

conjecture | open | MathDB 页面 | 原文线索

V2-070522 | MATHDB-365404

Monotonicity conjecture for hook counts in t-regular partitions

摘录 / EXCERPT: Monotonicity conjecture. Let $t\geq3$ be an integer. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070523 | MATHDB-365415

The conjecture on immortal Ricci flows from quotients of Bryant's soliton

摘录 / EXCERPT: Bryant-quotient conjecture. There exist immortal Ricci flows approaching any quotient of Bryant's steady soliton by a subgroup of $\operatorname{SU}(2)$, bubbling off any hyperkähl...

conjecture | open | MathDB 页面 | 原文线索

V2-070524 | MATHDB-365424

Basic g-composable schedules capture minimax-optimal gradient-norm schedules

摘录 / EXCERPT: Basic g-composable schedule conjecture. For each n, every minimax-optimal stepsize schedule solving

conjecture | open | MathDB 页面 | 原文线索

V2-070525 | MATHDB-365428

Intermediate-regime diameter conjecture for random spanning trees in random environment

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let ω be the edge-disorder environment, and let ${{\mathcal T}}^\omega_{n,\beta_n}$ be the random spanning tree in random en...

conjecture | open | MathDB 页面 | 原文线索

V2-070526 | MATHDB-365435

Mohr-Pardey-Rautenbach's zero-sum tree-factor conjecture

摘录 / EXCERPT: Mohr-Pardey-Rautenbach's conjecture. Fix integers $k,n\geq 1$ such that $\binom{n}{2}$ and $(k-1)n/k$ are both even integers. If $\mathbf{c}:E(K_n)\to\{-1,1\}$ is a 0-sum labeling...

conjecture | open | MathDB 页面 | 原文线索

V2-070527 | MATHDB-365441

Conjecture on Ehrhart quasipolynomials of half-integral polygons

摘录 / EXCERPT: Let $i\in\mathbb{Z}_{\geq 2}$ be the number of interior lattice points of a half-integral polygon, and require the polygon to have at least 2 boundary lattice points. An Ehrhart...

conjecture | open | MathDB 页面 | 原文线索

V2-070528 | MATHDB-365442

The weak $d\times 2d$ equiangular tight frame conjecture

摘录 / EXCERPT: Weak $d\times 2d$ conjecture. For every d, there exists a $d\times 2d$ ETF.

conjecture | open | MathDB 页面 | 原文线索

V2-070529 | MATHDB-365446

Linear growth conjecture for genetic pseudo-Anosov braid closures

摘录 / EXCERPT: Linear growth conjecture. The entropy of braids in the sequence $\\{\beta^{mk+1}\\}_{m=M+1}^\infty$, a lower bound on the knot genus of...

conjecture | open | MathDB 页面 | 原文线索

V2-070530 | MATHDB-365453

The local antimagic chromatic number conjecture for joins of matching and null graphs

摘录 / EXCERPT: Local antimagic chromatic number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070531 | MATHDB-365471

The boundary conjecture for anomaly-free rational conformal field theories

摘录 / EXCERPT: An RCFT has vanishing gravitational anomaly when its left- and right-moving gravitational anomaly cancels. A boundary condition has finite g-function when its boundary entropy pa...

conjecture | open | MathDB 页面 | 原文线索

V2-070532 | MATHDB-365474

The coherent and incoherent Iwasawa main conjectures for weight interlacing strings

摘录 / EXCERPT: The coherent and incoherent Iwasawa main conjectures. There are $[?]$ main conjectures, indexed by weight interlacing strings. In the coherent case, the Selmer group is torsion ove...

conjecture | open | MathDB 页面 | 原文线索

V2-070533 | MATHDB-365475

The sign-dependent Iwasawa main conjecture for weight interlacing strings

摘录 / EXCERPT: Sign-dependent Iwasawa main conjecture. If $[?]$, then $[?]$ is torsion and

conjecture | open | MathDB 页面 | 原文线索

V2-070534 | MATHDB-365476

Existence of p-adic L-functions for weight interlacing strings

摘录 / EXCERPT: Existence of p-adic L-functions. For each of the $[?]$ weight interlacing strings $[?]$, there exists a p-adic L-function $[?]$ in $[?]$ interpolating $L(\frac{1}{2},\Pi)$...

conjecture | open | MathDB 页面 | 原文线索

V2-070535 | MATHDB-365489

The characterization of 1-trivial motions for size-k psd factorizations

摘录 / EXCERPT: The 1-trivial-motion conjecture. The set of 1-trivial motions of size-k psd factorizations of matrices in $\mathcal{M}_{\binom{k+1}{2},k}^{p \times q}$ is equal to...

conjecture | open | MathDB 页面 | 原文线索

V2-070536 | MATHDB-365495

The commuting gauging diagram for fermionic symmetry-enriched string-net models

摘录 / EXCERPT: Let G be a symmetry group, let $\mathcal{S}_G$ be a G-graded super fusion category with trivial ω_2, and let $\mathcal{S}_1$ be its trivial grading sector. The category...

conjecture | open | MathDB 页面 | 原文线索

V2-070537 | MATHDB-365499

Linear treewidth-Hadwiger bound for hereditary graph classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary graph class. It is $(\operatorname{tw},\operatorname{had})$-bounded if there is a function f such that...

conjecture | open | MathDB 页面 | 原文线索

V2-070538 | MATHDB-365500

The cut-partition conjecture for t-triangles in fullerenes

摘录 / EXCERPT: Let $C_{56,622}$ and $C_{80,31924}$ denote the indicated fullerene graphs. A cut-partition of a hexagonal subgraph is the set of components generated by the stated cut-partition co...

conjecture | open | MathDB 页面 | 原文线索

V2-070539 | MATHDB-365501

The cut-partition characterization of fullerenes without generalized Stone-Wales paths

摘录 / EXCERPT: Let T_n be a dual fullerene graph and let T_n^6 be its hexagonal subgraph, for feasible n. A cut-partition of T_n^6 is the set of components generated by the stated cut-par...

conjecture | open | MathDB 页面 | 原文线索

V2-070540 | MATHDB-365506

Feng-Huang-Rao affine embedding conjecture for self-similar sets

摘录 / EXCERPT: Feng-Huang-Rao's conjecture. If, for some $i\in I$, $\log\alpha_i\notin\operatorname{span}_{\mathbb{Q}}\{\log\beta_j\}_{j\in J}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-070541 | MATHDB-365514

Gimbert's cyclotomic irreducibility conjecture

摘录 / EXCERPT: Gimbert's cyclotomic irreducibility conjecture. If k is even, then $F_{i,k}(x)$ is reducible in $\mathbb Q[x]$ if and only if $i\mid(k+2)$, and in that case it has exactly two fa...

conjecture | open | MathDB 页面 | 原文线索

V2-070542 | MATHDB-365530

The itinerary-preservation conjecture for marked cycles in the basilica degeneration

摘录 / EXCERPT: Let g_a be the family described above, let $C_{p,2}$ denote the set of period-p itineraries, and let … be the indicated partitions of the Julia sets for $a<-1$ and $a=0$, respe...

conjecture | open | MathDB 页面 | 原文线索

V2-070543 | MATHDB-365531

Connectivity conjecture for marked cycle curves in the basilica limb

摘录 / EXCERPT: Let $\mathcal{F}_2$ denote the relevant parameter space and let $\operatorname{Cyc}_p(\mathcal{F}_2)$ be the marked cycle curve of period p. Connectivity conjecture. For $p\ne 3$...

conjecture | open | MathDB 页面 | 原文线索

V2-070544 | MATHDB-365532

Skewer conjecture for marked cycle curves

摘录 / EXCERPT: Let π be the projection from the marked cycle curve $\operatorname{Cyc}_p(\mathcal{F}_1)$ to its parameter space, and consider the lift of the real axis under this projection.…

conjecture | open | MathDB 页面 | 原文线索

V2-070545 | MATHDB-365542

The disjoint-path connectivity conjecture for friends-and-strangers graphs with a star

摘录 / EXCERPT: Disjoint-path connectivity conjecture. The maximum number of disjoint paths between any two permutations σ and ρ in G equals $\min\{\deg(\sigma),\deg(\rho)\}$. This w...

conjecture | open | MathDB 页面 | 原文线索

V2-070546 | MATHDB-365545

Overparameterization Improves Optimization for Neural Operators

摘录 / EXCERPT: Let a neural operator have adjustable capacity, for example through its number of layers or neurons, and consider the non-convex optimization landscape of its training loss. Overpa...

conjecture | open | MathDB 页面 | 原文线索

V2-070547 | MATHDB-365548

Numerical contractibility conjecture for curves in Calabi-Yau threefolds

摘录 / EXCERPT: Numerical contractibility conjecture. In a suitably local neighbourhood of C in X, the following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-070548 | MATHDB-365549

The doubly-presentable dualizable-object presentability conjecture

摘录 / EXCERPT: Let $\mathcal P$ be a “doubly-presentably” symmetric monoidal $(\infty,2)$-category. The $(\infty,1)$-category $\mathcal P^{\mathrm{dbl}}$ of dualizable objects and $\mathcal P$-in...

conjecture | open | MathDB 页面 | 原文线索

V2-070549 | MATHDB-365550

The internal-left-adjoint compact-generation conjecture for linear categories

摘录 / EXCERPT: Internal-left-adjoint compact-generation conjecture. If the coevaluation morphism is an internal left adjoint, then C is compactly generated. The paper states that no counterexam...

conjecture | open | MathDB 页面 | 原文线索

V2-070550 | MATHDB-365551

Classification conjecture for multiple radial SLE null-vector solutions

摘录 / EXCERPT: Let n growth points and m screening variables define the null-vector equations and Ward identities for multiple radial SLE, and let $u\in\mathbb{H}$ be the marked interior poin...

conjecture | open | MathDB 页面 | 原文线索

V2-070551 | MATHDB-365552

Radial quadratic-differential link-pattern conjecture

摘录 / EXCERPT: Let $\mathcal{QD}(\boldsymbol{z})$ be the specified space of quadratic differentials with m poles and n zeros, and let $\Gamma(Q)$ denote the horizontal trajectories of...

conjecture | open | MathDB 页面 | 原文线索

V2-070552 | MATHDB-365553

Involution-symmetry conjecture for master-function critical points

摘录 / EXCERPT: Let $\boldsymbol{z}$ be a generic configuration on the unit circle, and let $\Phi_{m,n}(\boldsymbol{z},\boldsymbol{\xi})$ be the master function. Involution-symmetry conjecture. Ev...

conjecture | open | MathDB 页面 | 原文线索

V2-070553 | MATHDB-365554

Enumeration conjecture for critical points of the radial master function

摘录 / EXCERPT: Let $\boldsymbol{z}$ be generic on the unit circle, let $\Phi_{m,n}(\boldsymbol{z},\boldsymbol{\xi})$ be the master function, and write $\#\operatorname{LP}(n,m)$ for the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-070554 | MATHDB-365556

Nonvanishing conjecture for the Benjamin-Ono critical tau function

摘录 / EXCERPT: Let Z denote the zero set of the function $\tau(T,X)$ associated with the critical Benjamin-Ono profile. Nonvanishing conjecture. ... i.e., $\tau(T,X)\neq 0$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-070555 | MATHDB-365557

Masoero-Raimondo string-equation conjecture for the Benjamin-Ono critical profile

摘录 / EXCERPT: Masoero-Raimondo string-equation conjecture. Under the nonvanishing assumption, $U(T,X)$ is a global solution of this equation.

conjecture | open | MathDB 页面 | 原文线索

V2-070556 | MATHDB-365563

Audenaert-Kittenah Schatten norm compression conjecture

摘录 / EXCERPT: Audenaert-Kittenah conjecture. For $1\le p\le 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-070557 | MATHDB-365569

Stable Chow groups conjecture for Hurwitz spaces

摘录 / EXCERPT: Stable Chow groups conjecture. The ith rational Chow groups of $\operatorname{CHur}^{G,c_1,\ldots,c_k}_{n_1,\ldots,n_k}$ stabilize, in the sense that they are independent of...

conjecture | open | MathDB 页面 | 原文线索

V2-070558 | MATHDB-365578

Criado-Quintero-Lundqvist-Nenashev dimension conjecture for two additional squares

摘录 / EXCERPT: Let $P_n=\mathbb{F}[x_1,\ldots,x_n]$, let ℓ_1,ℓ_2 be general linear forms in P_n, and let $a_{n,d}$ denote the number of lattice paths from $(0,0)$ to $(n+2-2s,n+2)$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-070559 | MATHDB-365579

The almost-regular orbit neighbourhood conjecture for generalised Saxl graphs

摘录 / EXCERPT: Let $G\leqslant\operatorname{Sym}(\Omega)$ be a finite primitive permutation group with base size $b(G)\geqslant 2$. For $\alpha\in\Omega$, call a G_α-orbit on Ω almo...

conjecture | open | MathDB 页面 | 原文线索

V2-070560 | MATHDB-365604

Non-permutation conjecture for the trivariate family C_u

摘录 / EXCERPT: Non-permutation conjecture. The function C_u is not a permutation and is not APN.

conjecture | open | MathDB 页面 | 原文线索

V2-070561 | MATHDB-365609

Sumner's conjecture for oriented trees

摘录 / EXCERPT: Sumner's conjecture. Every oriented tree of order $n>1$ is $(2n-2)$-unavoidable; equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-070562 | MATHDB-365615

Berestycki's total-variation mixing conjecture for the giant component of the random transposition walk

摘录 / EXCERPT: Let X_t be the random transposition walk on the symmetric group, let G_t be the graph formed by the transpositions selected up to time t, and let C_t be the largest connect...

conjecture | open | MathDB 页面 | 原文线索

V2-070563 | MATHDB-365635

List-coloring transfer meta-conjecture for graph classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a natural graph class, and let k be a positive integer. For a graph G with a k-assignment L, let $\mathcal{C}_L(G)$ be the graph of proper L-coloring...

conjecture | open | MathDB 页面 | 原文线索

V2-070564 | MATHDB-365636

Finite global attractor conjecture for Thurston pullback relations

摘录 / EXCERPT: Let $f\colon (\widehat{\mathbb{C}},P)\to(\widehat{\mathbb{C}},P)$ be a rational Thurston map with a hyperbolic orbifold. Let $\overline{\mathscr{C}_P}$ be the set of isotopy classe...

conjecture | open | MathDB 页面 | 原文线索

V2-070565 | MATHDB-365637

The n/d-conjecture for central essential indecomposable hyperplane arrangements

摘录 / EXCERPT: Let $f\in\mathbb C[z_1,\ldots,z_n]\setminus\mathbb C$ be homogeneous of degree d, and suppose that the hypersurface $\{f=0\}$ is a central essential indecomposable hyperplane arr...

conjecture | open | MathDB 页面 | 原文线索

V2-070566 | MATHDB-365648

Díaz-Park sharpness conjecture for fusion systems

摘录 / EXCERPT: Let S be a finite p-group, let $\mathcal{F}$ be a fusion system over S, and let $M=(M_{*},M^{*})$ be a Mackey functor over $\mathcal{F}$ on $\mathbb{F}_{p}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070567 | MATHDB-365653

Petersen-minor-free fractional perfect-matching conjectures for r-graphs

摘录 / EXCERPT: Let G be an r-graph and let a (t,r)-PM be a multiset of $t\cdot r$ perfect matchings of G such that every edge is contained in exactly t of them. Petersen-minor-free frac...

conjecture | open | MathDB 页面 | 原文线索

V2-070568 | MATHDB-365665

Stability-space uniformization conjecture for generalized root systems of affine ADE type

摘录 / EXCERPT: Let $A=(a_1,a_2,a_3)$ satisfy $\chi_A>0$. Let ${\mathcal X}_A$ be the associated parameter space, $\widehat{W}_A$ its extended Weyl group, ${\mathcal D}_A$ the associated triangula...

conjecture | open | MathDB 页面 | 原文线索

V2-070569 | MATHDB-365669

Fukaya category conjecture for compatible tangential structures

摘录 / EXCERPT: Let (Θ,Φ) be a tangential pair, meaning based spaces equipped with maps fitting into a commutative diagram over $BO \to BU$. Let X admit a Φ-structure, set...

conjecture | open | MathDB 页面 | 原文线索

V2-070570 | MATHDB-365676

Near-optimal lower-bound conjecture for one-sided estimation under shift-invariance

摘录 / EXCERPT: One-sided estimation lower-bound conjecture. There exist constants $c_0,c,r>0$ such that, for every $s,n\in\mathbb{N}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-070571 | MATHDB-365678

The refined graph-container bound at activity of order inverse degree

摘录 / EXCERPT: Let d_X and d_Y be sufficiently large integers, let $\delta\geq 1$ and $\delta',\delta”>0$, and let $\Sigma=X\sqcup Y$ be a δ-approximately (d_X,d_Y)-biregular graph s...

conjecture | open | MathDB 页面 | 原文线索

V2-070572 | MATHDB-365681

Tight Bayesian optimal uniform reserve approximation for quasi-regular buyers

摘录 / EXCERPT: Let asymmetric and symmetric buyers have quasi-regular valuation distributions, and let $\calC_{{\sf BOM}}^{\textsf{Bayesian Optimal Uniform Reserve}}$ denote the revenue approxima...

conjecture | open | MathDB 页面 | 原文线索

V2-070573 | MATHDB-365682

No separation in Bayesian uniform-pricing approximation for quasi-regular buyers

摘录 / EXCERPT: Let asymmetric regular and quasi-regular buyers be compared using the revenue approximation of Bayesian Optimal Uniform Pricing to Bayesian Optimal Mechanism, with tight ratio deno...

conjecture | open | MathDB 页面 | 原文线索

V2-070574 | MATHDB-365689

Coherence conjecture for n-fold monoidal categories

摘录 / EXCERPT: Coherence conjecture. All such maps defined as composites of associators, unitors, and interchange morphisms are equal.

conjecture | open | MathDB 页面 | 原文线索

V2-070575 | MATHDB-365690

Iterated lax monoid bicategory conjecture

摘录 / EXCERPT: Let $\mathcal{B}$ be a monoidal bicategory. For each monoid object in $\mathcal{B}$, consider lax monoidal 1-cells and monoidal 2-cells. Iterated lax monoid conjecture. It shou...

conjecture | open | MathDB 页面 | 原文线索

V2-070576 | MATHDB-365694

Mirror-family conjecture for affine partial minimal models

摘录 / EXCERPT: Let $U\subset Y$ be a smooth affine partial minimal model with smooth boundary E, let V be the mirror fibre, and let θ_E be the corresponding theta function. Let F be...

conjecture | open | MathDB 页面 | 原文线索

V2-070577 | MATHDB-365695

Homogeneous-coordinate-ring conjecture for the mirror filtration

摘录 / EXCERPT: Let $U\subset Y$ be an snc partial minimal model, let W be an effective ample divisor supported on the boundary, and let $\widetilde{A}_{\theta}$ be the graded subring obtained fr...

conjecture | open | MathDB 页面 | 原文线索

V2-070578 | MATHDB-365697

Gamma-positivity conjecture for Coxeter Chow polynomials

摘录 / EXCERPT: Let W be a Coxeter group and let $u<v$ be elements of W. The Coxeter Chow polynomial H_{uv} is the polynomial associated with the Bruhat interval $[u,v]$; it is...

conjecture | open | MathDB 页面 | 原文线索

V2-070579 | MATHDB-365701

Regularity of finitary semi-upho posets

摘录 / EXCERPT: Regularity conjecture. Every finitary semi-upho poset is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-070580 | MATHDB-365702

Greedy 0-monoid characterization of regular semi-upho functions

摘录 / EXCERPT: Greedy 0-monoid conjecture. The formal power series $g(x)$ is a regular semi-upho function if and only if it admits an infinite greedy 0-monoid series $\{M_k^0\}_{k\ge 1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070581 | MATHDB-365706

Integral representation of the invariant hull

摘录 / EXCERPT: Suppose the functional I has the integral form ... for a continuous integrand ... Let I_i be its invariant hull with respect to an invariant class $\mathcal D_0$, and define ... wher...

conjecture | open | MathDB 页面 | 原文线索

V2-070582 | MATHDB-365715

The bordification conjecture for the moduli space of grafted surfaces

摘录 / EXCERPT: The bordification conjecture. The natural inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-070583 | MATHDB-365719

Halin's ray graph conjecture

摘录 / EXCERPT: Halin's ray graph conjecture. Every graph admits a ray graph for each of its ends.

conjecture | open | MathDB 页面 | 原文线索

V2-070584 | MATHDB-365729

Border Comon's conjecture for minimal border rank

摘录 / EXCERPT: Border Comon's conjecture for minimal border rank. If

conjecture | open | MathDB 页面 | 原文线索

V2-070585 | MATHDB-365730

The 2-Tillmann conjecture on dualizable objects in 2Kar

摘录 / EXCERPT: 2-Tillmann conjecture. An object of the symmetric monoidal 3-category $\mathbf{2Kar}$ is 1-dualizable if and only if it is a finite semisimple 2-category.

conjecture | open | MathDB 页面 | 原文线索

V2-070586 | MATHDB-365732

The very stable locus conjecture for elliptic bundles of rank d+1 and degree d

摘录 / EXCERPT: Very stable locus conjecture. The set of L-very stable bundles is the open dense subset

conjecture | open | MathDB 页面 | 原文线索

V2-070587 | MATHDB-365735

Stark--Tate conjecture for rank-one abelian extensions

摘录 / EXCERPT: Let H/F be an abelian extension of number fields, let W be the number of roots of unity in H, and let S be a finite set of places of F containing the ramified and infinit...

conjecture | open | MathDB 页面 | 原文线索

V2-070588 | MATHDB-365738

The congruence subgroup property for mapping class groups of genus at least 3

摘录 / EXCERPT: Let Σ be a surface of genus at least 3. A finite-index subgroup of $\operatorname{Mcg}(\Sigma)$ is a subgroup of finite index containing the kernel of the action on a fini...

conjecture | open | MathDB 页面 | 原文线索

V2-070589 | MATHDB-365740

The orbit-containment criterion for congruence subgroups of mapping class groups

摘录 / EXCERPT: Let Γ be a finite-index subgroup of $\operatorname{Mcg}(\Sigma)$, and let α be a simple closed curve on Σ. A subgroup of $\operatorname{Mcg}(\Sigma)$ is congru...

conjecture | open | MathDB 页面 | 原文线索

V2-070590 | MATHDB-365742

The zonotopal shifted Lonely Runner Conjecture

摘录 / EXCERPT: Let $Z_{\mathbf v}$ be a Strong Lonely Runner Zonotope, meaning a Lonely Runner Zonotope associated with a velocity vector $\mathbf v=(v_1,\dotsc,v_n)\in\mathbb{Z}_{>0}^n$ whose en...

conjecture | open | MathDB 页面 | 原文线索

V2-070591 | MATHDB-365746

Pullman's conjecture on strongly consistent odd-regular graphs

摘录 / EXCERPT: Pullman's conjecture. For odd d, every d-regular graph is strongly consistent. This generalizes the even-order tournament conjecture from complete graphs to all odd-regular gra...

conjecture | open | MathDB 页面 | 原文线索

V2-070592 | MATHDB-365765

Lee's conjecture that the fractional Steiner triple system threshold is $2/3$

摘录 / EXCERPT: Let $\theta_{\operatorname{STS}}$ denote the minimum codegree threshold for Steiner triple systems, and let $\theta_{\operatorname{STS}}^*$ denote the corresponding fractional thre...

conjecture | open | MathDB 页面 | 原文线索

V2-070593 | MATHDB-365777

The Oda-Hamahata conjecture for real quadratic elliptic curves

摘录 / EXCERPT: Let K be a real quadratic field, let I be an ideal of ${\mathcal O}_K$, and let $H_{I,K}$ be the Baily-Borel compactification of the Hilbert modular surface of level I. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-070594 | MATHDB-365778

A conjectural Hilbert-modular construction for certain ${\mathbb Q}(\alpha)$-curves

摘录 / EXCERPT: Let $K=\mathbb Q(\sqrt D)$, let I be a level, and let f be a Hilbert cuspidal newform of level I over K. Suppose there exist a quadratic field $\mathbb Q(\sqrt n)$ and a qu...

conjecture | open | MathDB 页面 | 原文线索

V2-070595 | MATHDB-365784

Poset-type conjecture for quadratic multiplicities of key-polynomial generating series

摘录 / EXCERPT: Poset-type conjecture. The multiplicity $m^{k,l}_{\eta}(w)$ depends only on the poset type of the set of sums $\alpha+\beta=\eta$, with $\alpha\in A_l(w)$ and $\beta\in A_k(w)$. Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-070596 | MATHDB-365785

Uniform-speed conjecture for a non-escaping impurity in TASEP

摘录 / EXCERPT: Let an impurity in the totally asymmetric simple exclusion process have jumping rates $b1$ and $b2$, and initially be located at the interface of a 1-0 profile. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-070597 | MATHDB-365786

Non-vacuity of improved covariate-adjusted variance

摘录 / EXCERPT: Let ν^{rd}, $\nu^{\mathsf{rd}\dag}$, and $\mathsf{var}(\widehat{\tau}_{\mathsf{unadj}})$ denote the variance quantities defined in the paper for the covariate-adjusted...

conjecture | open | MathDB 页面 | 原文线索

V2-070598 | MATHDB-365805

Nelson's essential skew-adjointness conjecture

摘录 / EXCERPT: Let $p=2$ and let $\mathbf{v}\in L^2(\mathbb{R}^d)$ be a compactly supported, divergence-free vector field. Define ... as an operator on $L^2(\mathbb{R}^d)$. Nelson's conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-070599 | MATHDB-365807

The exponential valuation conjecture for real polytopes

摘录 / EXCERPT: Let n be a positive integer, let $\mathcal{P}^n$ denote the family of polytopes in $\mathbb{R}^n$, and let $\mathcal{M}(\mathbb{R}^n)$ denote the space of real-valued functions o...

conjecture | open | MathDB 页面 | 原文线索

V2-070600 | MATHDB-365816

Gromov-Witten conjecture for elliptic Noether-Lefschetz loci

摘录 / EXCERPT: Let $\overline{\mathcal M}_{g,1}(\pi,d)$ be the moduli space of log-stable maps of degree d to the fibers of the universal family of elliptic curves, with evaluation map...

conjecture | open | MathDB 页面 | 原文线索

V2-070601 | MATHDB-365817

Functoriality conjecture for the span construction

摘录 / EXCERPT: Span functoriality conjecture. The Span construction underlies a limit-preserving $(\infty,2)$-functor from the $(\infty,2)$-category of categories with pullbacks,...

conjecture | open | MathDB 页面 | 原文线索

V2-070602 | MATHDB-365821

Modularity conjecture for higher theta series of unitary shtukas

摘录 / EXCERPT: Let $m\le n$ be fixed. Let X'/X be the given double cover, let U_{2m} be the associated quasi-split unitary group, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070603 | MATHDB-365823

The Chevalley basis Khovanskii basis conjecture

摘录 / EXCERPT: Let $\X=\G/\P$ be a homogeneous space with a dominant weight ϖ yielding a projective embedding $\X\hookrightarrow\bP(V_\varpi)$, and let $\bs$ be a reduced expression for...

conjecture | open | MathDB 页面 | 原文线索

V2-070604 | MATHDB-365832

Bessel-integral asymptotic conjecture for the Euclidean ball

摘录 / EXCERPT: Bessel-integral asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070605 | MATHDB-365839

Equality of the Helly and Radon numbers for connected graphs under Δ-convexity

摘录 / EXCERPT: Helly-Radon equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070606 | MATHDB-365841

Bunde-Milans-West-Wu conjecture on parity colorings of bipartite graphs

摘录 / EXCERPT: Bunde-Milans-West-Wu conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070607 | MATHDB-365849

The componentwise linear symbolic power conjecture for linear-resolution edge ideals

摘录 / EXCERPT: Componentwise linear symbolic power conjecture. If $I(G)$ has a linear resolution, then $I(G)^{(k)}$ is componentwise linear for every $k\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-070608 | MATHDB-365850

The linear quotients conjecture for symbolic powers of linear-resolution edge ideals

摘录 / EXCERPT: Linear quotients conjecture for symbolic powers. If $I(G)$ has a linear resolution, then $I(G)^{(k)}$ has linear quotients for every $k\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-070609 | MATHDB-365853

The De Gennes upper-bound conjecture for the magnetic ground-state energy

摘录 / EXCERPT: Let $\beta>0$ denote the magnetic-field parameter, let $\lambda(\beta)$ be the lowest eigenvalue of the Neumann magnetic Laplacian in the unit disk, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-070610 | MATHDB-365854

The monotonicity conjecture for crossing eigenvalue parameters

摘录 / EXCERPT: For each $n\in\mathbb N$, let β_n be the unique parameter satisfying $\lambda(n,\beta_n)=\lambda(n+1,\beta_n)$, and define ... The crossing-sequence monotonicity conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-070611 | MATHDB-365855

The strong diamagnetism conjecture for the magnetic ground-state eigenvalue

摘录 / EXCERPT: Let $\lambda(\beta)$ denote the lowest eigenvalue of the Neumann magnetic Laplacian in the unit disk as a function of the magnetic-field parameter $\beta\geq0$. Strong diamagnetism...

conjecture | open | MathDB 页面 | 原文线索

V2-070612 | MATHDB-365877

Half-space conjecture for ranges of products of Toeplitz operators

摘录 / EXCERPT: Let E_1 and the relevant Hardy-space-valued symbols be as in Theorem 1, with T_{Φ} and T_{Θ} the associated Toeplitz operators. Define ... Here a half-space is a closed...

conjecture | open | MathDB 页面 | 原文线索

V2-070613 | MATHDB-365878

Partition conjecture for strengthening sets of a crossing family

摘录 / EXCERPT: Strengthening-set partition conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-070614 | MATHDB-365879

Equivariantization and delooping commute for multifusion categories

摘录 / EXCERPT: Let G be a group. For a multifusion category $\mathcal{C}$ with a G-action, write $\mathcal{C}^G$ for its equivariantization and $\mathbf{Mod}(\mathcal{C})$ for its delooping....

conjecture | open | MathDB 页面 | 原文线索

V2-070615 | MATHDB-365891

Generalized truncated Jacobi triple product conjecture

摘录 / EXCERPT: Let R and S be positive integers with $1\leq S<R$, let $k>\ell\geq1$, and assume $k\geq4$. There exists an integer N such that the theta series ... has nonnegative coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-070616 | MATHDB-365892

Zimmer's conjecture for higher-rank lattice actions

摘录 / EXCERPT: Let G be a connected semisimple Lie group with finite center, all of whose simple factors have real rank at least 2. Let $\Gamma\subset G$ be a lattice, M be a compact manifo...

conjecture | open | MathDB 页面 | 原文线索

V2-070617 | MATHDB-365903

Conjectures on secondary bifurcations for single-hump kernel perturbations

摘录 / EXCERPT: Let $\overline{\phi}\in \overline{K}(\mathbb{R})$ be an admissible kernel perturbation supported in $[-1/2,1/2]$, with $||\overline{\phi}||_1^m$ small. Consider parameters...

conjecture | open | MathDB 页面 | 原文线索

V2-070618 | MATHDB-365908

Ramirez's conjecture on the structure of the second Jacobian ideal

摘录 / EXCERPT: Ramirez's conjecture. The ideal of maximal minors of K is

conjecture | open | MathDB 页面 | 原文线索

V2-070619 | MATHDB-365909

Schofield's completion conjecture for universal bundles on quiver moduli

摘录 / EXCERPT: Let Q, ${\mathbf{d}}$, and θ satisfy the assumptions of the theorem on partial tilting bundles, and let $\mathcal{U}$ be the universal representation on...

conjecture | open | MathDB 页面 | 原文线索

V2-070620 | MATHDB-365915

General-domain square-root asymptotic conjecture for magnetic Dirichlet-to-Neumann ground states

摘录 / EXCERPT: Let Ω be a regular domain in $\mathbb{R}^2$, and let A_0 be a magnetic potential whose magnetic field is constant and has norm 1. Let Λ_{bA_0} be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-070621 | MATHDB-365917

Strong diamagnetism conjecture for magnetic Dirichlet-to-Neumann ground states

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be any bounded set, and let the magnetic field be constant. Denote by $\lambda^{\rm DN}(A,\Omega)$ the ground-state energy of the magnetic Dirichlet...

open_problem | open | MathDB 页面 | 原文线索

V2-070622 | MATHDB-365924

Uniqueness and stabilization conjecture for two-orthogonal tensor decompositions

摘录 / EXCERPT: Two-orthogonal uniqueness and stabilization conjecture. A generic two-orthogonal tensor in $\mathbb{R}^{n_1}\otimes\cdots\otimes\mathbb{R}^{n_d}$ has a unique two-orthogonal decomp...

conjecture | open | MathDB 页面 | 原文线索

V2-070623 | MATHDB-365926

Type-and-Jordan-type invariance conjecture for unitary stable orbital integral strata

摘录 / EXCERPT: Let L be a unimodular hermitian lattice over the ring of integers $\boldsymbol{o}_E$ of an unramified quadratic extension E/F, and let M,M' be sublattices of L. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070624 | MATHDB-365929

Hassett's weighted blow-up conjecture for unordered points on the projective line

摘录 / EXCERPT: Let n be a positive integer and let $\frac{1}{n}+\frac{}{}\epsilon$ be the common weight on $2n$ marked points, with the weighted moduli space...

conjecture | open | MathDB 页面 | 原文线索

V2-070625 | MATHDB-365941

Pattern-containment characterization of SCNP for dual Schubert polynomials

摘录 / EXCERPT: Let S_n be the symmetric group on n letters. For permutations $u,w\in S_n$, let D_u^w denote the dual Schubert polynomial, and let SCNP denote the property that its Newton po...

conjecture | open | MathDB 页面 | 原文线索

V2-070626 | MATHDB-365946

Mundy's Arthur multiplicity conjecture for certain G2 representations

摘录 / EXCERPT: Mundy's conjecture. The representation Π occurs in $L^2_{\rm disc}(G_2(\mathbb Q)\backslash G_2(\mathbb A))$ with multiplicity zero or one, and it occurs with multiplicity one...

conjecture | open | MathDB 页面 | 原文线索

V2-070627 | MATHDB-365949

Cusick's conjecture on sums of continued-fraction Cantor sets

摘录 / EXCERPT: Let $S(k)$ be the set of numbers in $[0,1]$ whose continued-fraction partial quotients satisfy $a_n(x)\geq k$ whenever they are defined, with 0 included. Cusick's conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-070628 | MATHDB-365951

Rank-index conjecture for projected rational normal curves

摘录 / EXCERPT: Let $\widetilde{\mathcal{C}}$ be the rational normal curve and let $p\in\operatorname{\mathbb{P}}^{r+1}\setminus\widetilde{\mathcal{C}}^3$ be a projection center, with...

conjecture | open | MathDB 页面 | 原文线索

V2-070629 | MATHDB-365952

Rank-index conjecture for a curve union its trisecant line

摘录 / EXCERPT: Let $\mathcal{C}\subset\operatorname{\mathbb{P}}^r$, let $\mathbb{L}$ be its unique trisecant line, and set ... Assume these objects are as in the paper's second main theorem. The ra...

conjecture | open | MathDB 页面 | 原文线索

V2-070630 | MATHDB-365961

The classical-pattern basis conjecture for the union of dihedral subgroups

摘录 / EXCERPT: Let D_n be the dihedral subgroup of the symmetric group on n letters, and let D denote the union of the groups D_n. For a set P of classical patterns, write...

conjecture | open | MathDB 页面 | 原文线索

V2-070631 | MATHDB-365962

The marked mesh-pattern characterization of permutations with RSK tableaux avoiding $(3,2)$

摘录 / EXCERPT: Under the Robinson-Schensted-Knuth correspondence, a permutation is mapped to a pair of Young tableaux of the same shape. Let A be the set of permutations whose tableaux avoid th...

conjecture | open | MathDB 页面 | 原文线索

V2-070632 | MATHDB-365977

Real-rootedness and root-location conjecture for generalized snake posets

摘录 / EXCERPT: Let $\mathbf{w}$ be a generalized snake word of length $n+1$. Let $h^*(\mathbf{w};z)$ denote its h^*-polynomial, and let $L(\mathcal{O}(P(\mathbf{w}));t)$ denote the Ehrhart poly...

conjecture | open | MathDB 页面 | 原文线索

V2-070633 | MATHDB-365978

Coefficientwise monotonicity conjecture for Ehrhart polynomials under the swap operator

摘录 / EXCERPT: Let $\mathbf{w}$ be a generalized snake word, let $\varphi_i(\mathbf{w})$ be the word obtained by the swap operator, and write $L(\mathbf{w};t)$ for the Ehrhart polynomial associat...

conjecture | open | MathDB 页面 | 原文线索

V2-070634 | MATHDB-365989

The phase limit set conjecture for discriminant coamoebas

摘录 / EXCERPT: Phase limit set conjecture. The closure of the discriminant coamoeba equals the phase limit set:

conjecture | open | MathDB 页面 | 原文线索

V2-070635 | MATHDB-365993

The negative-parameter extension conjecture for block-beta random polytopes

摘录 / EXCERPT: Let $\bbbeta=(\beta_1,\ldots,\beta_m)$ be the block-parameter vector for the block-beta random polytope $\mathcal{P}_{n,d}^{\bbbeta}$. Negative-parameter extension conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-070636 | MATHDB-365994

The Poissonization conjecture for block-beta random polytopes

摘录 / EXCERPT: For $n\in\mathbb{N}$, let N_n be a Poisson random variable with parameter n, and let $\mathcal{P}_{N_n,d}^{\bbbeta}$ and $\mathcal{P}_{n,d}^{\bbbeta}$ denote the corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-070637 | MATHDB-365998

Universal limiting law conjecture for the normalized Stieltjes-transform fluctuations

摘录 / EXCERPT: Universal fluctuation conjecture. The distribution of Z is independent of $\vb{M}$ and z, satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070638 | MATHDB-366022

Cell-count conjecture for face Schubert arrangements of polytopes

摘录 / EXCERPT: Let $P\subseteq \mathbb{P}^{n-1}$ be a full-dimensional polytope, and let $P^{[k]}$ denote its k-stabbing set. Consider the k-face Schubert arrangement of P inside $P^{[k]}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-070639 | MATHDB-366033

The product-of-projective-spaces conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Product-of-projective-spaces conjecture. The Kazhdan-Lusztig polynomial indexed by w_{l}^{k} is the Hilbert-Poincaré polynomial of $(\mathbb{C}\mathbb{P}^{l})^{\times(k-1)}$:

conjecture | open | MathDB 页面 | 原文线索

V2-070640 | MATHDB-366034

The asymptotic unimodality conjecture for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: For each n, let uni_{n} be the proportion of distinct first-row Kazhdan-Lusztig polynomials for S_{n} that are unimodal, and let uni_{n}^{mult} b...

conjecture | open | MathDB 页面 | 原文线索

V2-070641 | MATHDB-366036

Upper bound conjecture for the h-star polynomial of cosmological polytopes

摘录 / EXCERPT: Let $G=(V,E)$ be a graph, and let $h^*(\mathbb{C}_G;z)$ denote the h^*-polynomial of its cosmological polytope. For polynomials with nonnegative coefficients, write $f\preceq g$...

conjecture | open | MathDB 页面 | 原文线索

V2-070642 | MATHDB-366037

Theta-graph h-star polynomial conjecture

摘录 / EXCERPT: Let P_k, P_ℓ, and P_m be paths of lengths k, ℓ, and m, respectively, and let G be the graph obtained by identifying the three starting nodes and the three endin...

conjecture | open | MathDB 页面 | 原文线索

V2-070643 | MATHDB-366038

Squiggly-and-double-edge formula for cosmological polytopes

摘录 / EXCERPT: Let G be a graph, let $\mathbb{C}_G$ be its cosmological polytope, and let T be a triangulation of $\mathbb{C}_G$ arising from a good term order on R_G. For each simplex...

conjecture | open | MathDB 页面 | 原文线索

V2-070644 | MATHDB-366042

Eigenvalue formula conjecture for completely factorizable operators

摘录 / EXCERPT: Let T be a completely factorizable operator of degree d, meaning that ... Then the eigenvalue formula conjecture states that the first few eigenvalues...

conjecture | open | MathDB 页面 | 原文线索

V2-070645 | MATHDB-366043

Revised minimum-dilatation conjecture for 10-strand pseudo-Anosov braids

摘录 / EXCERPT: Let $\underline{\delta}_9$ be the stated candidate value for the minimum dilatation of pseudo-Anosov braids with 9 strands, and let x denote the variable in its defining polynomi...

conjecture | open | MathDB 页面 | 原文线索

V2-070646 | MATHDB-366044

Hironaka's asymptotic dilatation conjecture for closed surfaces

摘录 / EXCERPT: For each genus g, let $\delta_{g,0}$ be the minimum dilatation of pseudo-Anosov maps on the closed surface of genus g. Let $\mu=\frac{1+\sqrt{5}}{2}\approx1.618$ be the golden...

conjecture | open | MathDB 页面 | 原文线索

V2-070647 | MATHDB-366045

The Double Mirror conjecture for theta functions of Looijenga pairs

摘录 / EXCERPT: Let (X,E) be a smooth Looijenga pair with affine interior $U=X\setminus E$, let $u:\mathcal T\to X$ be the universal torsor, and set $G=\mathcal T|_U$. Let $U^{\vee}$ and V den...

conjecture | open | MathDB 页面 | 原文线索

V2-070648 | MATHDB-366050

Existence of a proper retract to an embedded tree

摘录 / EXCERPT: Let X be a noncompact generalized continuum. Let T be an infinite tree and let ... be a proper embedding. Assume that the induced map on end spaces ... is injective. Existence of a...

conjecture | open | MathDB 页面 | 原文线索

V2-070649 | MATHDB-366071

Graph-distance scaling limit conjecture for planar maps

摘录 / EXCERPT: Let $a\from (2,5/2)$, let $\bigl(f_i\bigr)_{i\geq 1}$ be the sampled marked vertices or faces, let $d^\dagger_\mathrm{gr}$ denote the graph distance, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070650 | MATHDB-366075

Folklore eventual monotonicity conjecture for alpha-invariants

摘录 / EXCERPT: Folklore monotonicity conjecture. There exists $k_0\in\mathbb{N}_+$ such that, whenever $k_0\leq k_1\leq k_2$,

conjecture | open | MathDB 页面 | 原文线索

V2-070651 | MATHDB-366079

The bijection between alcove covers and hypersimplicial decorated ordered set partitions

摘录 / EXCERPT: Let n be any positive integer and let A_0 be an alcove in the hypersimplex $Delta_{2,n}$. Let $\mathcal{P}_{A_0}=(V,E)$ be the breadth-first search order of $\Gamma_{2,n}$ root...

conjecture | open | MathDB 页面 | 原文线索

V2-070652 | MATHDB-366094

Merzon-Smirnov's layered-permutation conjecture for maximal Schubert specialization

摘录 / EXCERPT: For $n\geq 1$, define ... where $\nu_w=\mathfrak{S}_w(x)|_{x_i=1}$. A layered permutation $w(b_1,\ldots,b_k)$ is the concatenation of decreasing blocks on consecutive intervals, wher...

conjecture | open | MathDB 页面 | 原文线索

V2-070653 | MATHDB-366095

The layered-permutation conjecture for maximal Newton-polytope support

摘录 / EXCERPT: For $n\geq 1$, let θ_w denote the number of supports of the Schubert polynomial $\mathfrak{S}_w(x)$, equivalently the number of lattice points in its Newton polytope, and de...

conjecture | open | MathDB 页面 | 原文线索

V2-070654 | MATHDB-366096

Gao's positivity conjecture for permutation-pattern coefficients

摘录 / EXCERPT: For $w\in S_n$, write ... where $p_u(w)$ counts occurrences of the pattern u in w, and define the coefficients recursively by ... Here u ranges over permutations and c_u has th...

conjecture | open | MathDB 页面 | 原文线索

V2-070655 | MATHDB-366097

Positivity conjecture for Newton-polytope support coefficients

摘录 / EXCERPT: For $w\in S_n$, write ... where θ_w is the number of supports of $\mathfrak{S}_w(x)$ and $p_u(w)$ counts occurrences of the permutation pattern u in w. The coefficients ar...

conjecture | open | MathDB 页面 | 原文线索

V2-070656 | MATHDB-366101

Berger's invariant-elements conjecture for overconvergent substitutions

摘录 / EXCERPT: Berger's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070657 | MATHDB-366108

The determinant- p conjecture for twisted derivations of cyclotomic integer rings

摘录 / EXCERPT: The determinant-p conjecture. The matrix $A=[a_{ij}]$ is a $(p-1)\times(p-1)$ matrix with determinant p.

conjecture | open | MathDB 页面 | 原文线索

V2-070658 | MATHDB-366118

Unique post-collision jamiton conjecture

摘录 / EXCERPT: Let two compatible jamitons be given, with their parameters describing their densities, velocities, and lengths. Unique post-collision jamiton conjecture. The collision of two comp...

conjecture | open | MathDB 页面 | 原文线索

V2-070659 | MATHDB-366119

Relaxation-time invariance of post-collision sonic velocity

摘录 / EXCERPT: Let τ be the relaxation time in the second-order macroscopic traffic model, and consider the jamiton produced by a collision. Sonic-velocity invariance conjecture. The sonic v...

conjecture | open | MathDB 页面 | 原文线索

V2-070660 | MATHDB-366120

Relaxation-time invariance of post-collision amplitude and downstream density

摘录 / EXCERPT: Let τ be the relaxation time, and consider the jamiton produced by a collision. Denote its amplitude by A and its downstream density by ρ_+. Amplitude and downstream-de...

conjecture | open | MathDB 页面 | 原文线索

V2-070661 | MATHDB-366121

Relaxation-time dependence of post-collision jamiton length

摘录 / EXCERPT: Let τ be the relaxation time, and consider the jamiton produced by a collision. Denote its length by L. Length-dependence conjecture. The length of the jamiton resulting fro...

conjecture | open | MathDB 页面 | 原文线索

V2-070662 | MATHDB-366122

Additivity conjecture for colliding jamiton lengths

摘录 / EXCERPT: Let two jamitons collide, and denote their lengths before collision by L and L_{test}, and the length of the resulting jamiton by L_{col}. Jamiton-length ad...

conjecture | open | MathDB 页面 | 原文线索

V2-070663 | MATHDB-366126

Local Irregularity Conjecture for 2-multigraphs

摘录 / EXCERPT: Local Irregularity Conjecture for 2-multigraphs. For every connected graph G which is not isomorphic to K_2,

conjecture | refuted | MathDB 页面 | 原文线索

V2-070664 | MATHDB-366127

Three-color conjecture for connected locally irregular graphs

摘录 / EXCERPT: Three-color conjecture for connected locally irregular graphs. Every simple connected locally irregular colorable graph G different from the bow-tie graph satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070665 | MATHDB-366130

The mapping class group power-quotient conjecture for hierarchically hyperbolic groups

摘录 / EXCERPT: Let S be any surface of finite type, and let $g_1,\dots,g_l\in \mathcal{MCG}^\pm(S)$. For a subgroup H of a group G, write $\langle\langle H\rangle\rangle$ for its normal clo...

conjecture | open | MathDB 页面 | 原文线索

V2-070666 | MATHDB-366138

Mass-threshold global well-posedness and scattering for the mass-critical nonlinear Schrödinger equation

摘录 / EXCERPT: Global existence and scattering conjecture. All maximal-lifespan solutions are globally well-posed and scatter.

conjecture | open | MathDB 页面 | 原文线索

V2-070667 | MATHDB-366139

Conjecture on Airy limits for solvable random block tridiagonal matrix ensembles

摘录 / EXCERPT: Let $r\geq 2$ and let $\beta=1$ or 2. Consider the point process associated with the $r=2$, $\beta s=2$ joint density in Theorem 1, and the point process associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-070668 | MATHDB-366141

Conjecture on Bessel limits for solvable random block tridiagonal ensembles

摘录 / EXCERPT: Let $\operatorname{Bessel}_{\beta,a}$ be the hard-edge point process defined by the $r=1$, $\gamma=1$ limiting operator, and consider the solvable ensembles with the explicit joint...

conjecture | open | MathDB 页面 | 原文线索

V2-070669 | MATHDB-366143

The two-dimensional sample path central limit conjecture for voter-model occupation time

摘录 / EXCERPT: Two-dimensional occupation-time central limit conjecture. If $d=2$ and η_0 is distributed according to ν_p, then

conjecture | open | MathDB 页面 | 原文线索

V2-070670 | MATHDB-366153

The lattice-tree and lattice-animal finite-size scaling profile conjecture

摘录 / EXCERPT: Lattice-tree and lattice-animal profile conjecture. The same critical finite-size scaling profile I_0 occurs for both lattice trees and lattice animals on $\mathbb{T}_r^d$ above...

conjecture | open | MathDB 页面 | 原文线索

V2-070671 | MATHDB-366159

Werner's critical-cluster scaling-limit conjecture

摘录 / EXCERPT: For functions f and g depending on the dimension d, define ... Here a critical cluster is either a loop cluster of $\widetilde{\mathcal{L}}_{1/2}$ or, when non-empty, a positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-070672 | MATHDB-366163

Graph-cover characterization conjecture for Bethe partition functions of DE-NFGs

摘录 / EXCERPT: Graph-cover characterization conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-070673 | MATHDB-366164

Navin-type multiplicative inequality for degree-M Bethe permanents

摘录 / EXCERPT: Navin-type conjecture. It holds that

conjecture | open | MathDB 页面 | 原文线索

V2-070674 | MATHDB-366166

Asymptotic maximal plane-saturation ratio conjecture

摘录 / EXCERPT: For a maximal planar graph G on n vertices, let its plane-saturation ratio be the relevant ratio measuring the size of a largest plane-saturated subgraph relative to G. Plane...

conjecture | open | MathDB 页面 | 原文线索

V2-070675 | MATHDB-366170

Efficient-ordering conjecture for powers of edge ideals

摘录 / EXCERPT: Let G be a graph with edge ideal $I(G)$. Suppose that, for some integer $q\in\mathbb{N}$, $I(G)^q$ has linear quotients with an ordering $\mathcal N^{(q)}:u_1>\dots>u_r$ of its g...

conjecture | open | MathDB 页面 | 原文线索

V2-070676 | MATHDB-366178

Finiteness conjecture for martingale dimension under sub-Gaussian heat kernel estimates

摘录 / EXCERPT: Let $(X,d,m,\mathcal E,\mathcal F)$ be an MMD space, meaning a metric measure Dirichlet space equipped with the associated diffusion. Suppose it satisfies sub-Gaussian heat kernel...

conjecture | open | MathDB 页面 | 原文线索

V2-070677 | MATHDB-366179

Quantitative martingale-dimension conjecture under sub-Gaussian heat kernel estimates

摘录 / EXCERPT: Let $(X,d,m,\mathcal E,\mathcal F)$ be an MMD space. Suppose it satisfies sub-Gaussian heat kernel estimates with volume growth exponent α and walk dimension β, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-070678 | MATHDB-366180

Hausdorff-dimension bound for martingale dimension under sub-Gaussian estimates

摘录 / EXCERPT: Let $(X,d,m,\mathcal E,\mathcal F)$ be an MMD space, and let d_m denote the martingale dimension of its associated diffusion. Suppose that $(X,d,m,\mathcal E,\mathcal F)$ satisfi...

conjecture | open | MathDB 页面 | 原文线索

V2-070679 | MATHDB-366192

Goldfeld's asymptotic rank conjecture for quadratic twists

摘录 / EXCERPT: Let $E:y^2=f(x)$ be an elliptic curve over $\mathbb{Q}$. For a squarefree integer d, let $E^{(d)}$ be the quadratic twist of E by d, and let $n_{E,r}^{\ast}(X)$ count fundame...

conjecture | open | MathDB 页面 | 原文线索

V2-070680 | MATHDB-366203

Conjecture on the $\a$-function and $\m$-equivalence

摘录 / EXCERPT: Let x and x' be elements for which $x\rightarrow_\mathbf{m}x'$; that is, there is a directed relation from x to x' in the $\mathbf{m}$-graph. Let $\mathbf{a}$ be the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-070681 | MATHDB-366208

Common-eigenvalue conjecture for periodic tridiagonal matrices

摘录 / EXCERPT: Let N be a positive integer and let T^- and T^+ be the $(N-1)\times(N-1)$ tridiagonal matrices with ... for $j=1,\ldots,N-1$, and $T^\pm_{j,j+1}=T^\pm_{j+1,j}=1$. Assume $a>0$....

conjecture | open | MathDB 页面 | 原文线索

V2-070682 | MATHDB-366209

Gir2o-Snyder-Popielarz linear out-degree conjecture for linked tournaments

摘录 / EXCERPT: Gir2o-Snyder-Popielarz conjecture. There exists a constant $C>0$ such that every $(2k+1)$-connected tournament with minimum out-degree at least Ck is k-linked.

conjecture | open | MathDB 页面 | 原文线索

V2-070683 | MATHDB-366210

The q-deformed chromatic polynomial limit conjecture for graphical arrangements

摘录 / EXCERPT: Let G be a graph, let $\mathcal{A}_G^q$ be its q-deformed graphical arrangement, let $\chi(\mathcal{A}_G^q,t)$ denote the characteristic polynomial of this arrangement, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070684 | MATHDB-366213

Numerical Terao's conjecture for plane curves with ordinary quasi-homogeneous singularities

摘录 / EXCERPT: Let C_{1}, C_{2} be reduced curves in $\mathbb{P}^{2}_{\mathbb{C}}$ whose irreducible components are smooth and whose singularities are ordinary quasi-homogeneous singularities....

conjecture | refuted | MathDB 页面 | 原文线索

V2-070685 | MATHDB-366219

Table-profile formulas for universal finite-size scaling

摘录 / EXCERPT: Table-profile conjecture. The profile functions $f(s)$ are the functions listed in Table for both SR and LR models in every dimension $d\ge d_{\mathrm{c},\alpha}$, including the up...

conjecture | open | MathDB 页面 | 原文线索

V2-070686 | MATHDB-366220

Pseudocritical finite-size scaling for free boundary conditions

摘录 / EXCERPT: Pseudocritical-profile conjecture. For $d>d_{\mathrm{c},\alpha}$, there is a pseudocritical point

conjecture | open | MathDB 页面 | 原文线索

V2-070687 | MATHDB-366227

The sub-nine acyclic list-colouring conjecture for locally planar graphs

摘录 / EXCERPT: Let G be a graph embedded in a surface Σ, and let $g(\Sigma)$ denote the genus of Σ. A graph is ρ-locally planar if the relevant local planarity condition hold...

conjecture | open | MathDB 页面 | 原文线索

V2-070688 | MATHDB-366228

The logarithmic edge-width conjecture for acyclic list-colouring locally planar graphs

摘录 / EXCERPT: Let Σ be a surface and let $g(\Sigma)$ denote its genus. Fix an integer k for which there is a value $\rho_0=\rho_0(g(\Sigma))$ such that every ρ_0-locally planar gra...

conjecture | open | MathDB 页面 | 原文线索

V2-070689 | MATHDB-366235

BBDVW conjectural recurrence for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let $[u,v]\subset S_n$ be a Bruhat interval, let I be a hypercube decomposition of $[u,v]$, and let $\ell(u,v)$ denote the length of the interval. The polynomials $N_{u,v,I}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070690 | MATHDB-366252

Universal Airy-one particle-position scaling conjecture for exclusion processes

摘录 / EXCERPT: Universal Airy-one scaling conjecture. There exist universal constants c_1 and c_2 such that

conjecture | open | MathDB 页面 | 原文线索

V2-070691 | MATHDB-366268

Logarithmic evaluation conjecture for a family of generalized continued fractions

摘录 / EXCERPT: Logarithmic evaluation conjecture. For all $k\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-070692 | MATHDB-366274

Bilu's conjecture on unbounded n-rank of class groups

摘录 / EXCERPT: Let $n>1$ be an integer, let F be a number field, and let d be a positive integer. For a number field L, write $Cl(L)[n]$ for the subgroup of the ideal class group of L ann...

conjecture | open | MathDB 页面 | 原文线索

V2-070693 | MATHDB-366278

The projection-term conjecture for lax cones in CaTT

摘录 / EXCERPT: Projection-term conjecture. Under these hypotheses, there exists a term $K\vdash p_t:T_t$ with the recursively specified type T_t, and its free variables satisfy the displayed un...

conjecture | open | MathDB 页面 | 原文线索

V2-070694 | MATHDB-366282

The differential-value conjecture for Bernstein-Sato roots of cusps

摘录 / EXCERPT: The differential-value conjecture. For any element $\lambda\in\Lambda_C\setminus\Gamma_C$, the rational number $-\lambda/nm$ is a root of the Bernstein-Sato polynomial of C.

conjecture | open | MathDB 页面 | 原文线索

V2-070695 | MATHDB-366285

Ryser-Brualdi-Stein conjecture on rainbow matchings

摘录 / EXCERPT: Ryser-Brualdi-Stein conjecture. Every such coloring contains a rainbow matching of size $n-1$; moreover, if n is odd, it contains a perfect rainbow matching.

conjecture | open | MathDB 页面 | 原文线索

V2-070696 | MATHDB-366286

Ringel's tree-packing conjecture

摘录 / EXCERPT: Ringel's conjecture. Every tree with $n+1$ vertices packs $2n+1$ times into the complete graph K_{2n+1}.

conjecture | open | MathDB 页面 | 原文线索

V2-070697 | MATHDB-366287

Graham-Sloane harmonious labeling conjecture

摘录 / EXCERPT: Graham-Sloane conjecture. Every tree is harmonious.

conjecture | open | MathDB 页面 | 原文线索

V2-070698 | MATHDB-366292

Generalised Ogg's conjecture for rational cuspidal torsion

摘录 / EXCERPT: Generalised Ogg's conjecture. For every positive integer N,

conjecture | open | MathDB 页面 | 原文线索

V2-070699 | MATHDB-366293

Rational cuspidal divisor class conjecture for modular curves

摘录 / EXCERPT: Let N be a positive integer. Let $X_0(N)$ be the modular curve of level N, let $J_0(N)$ be its Jacobian, and let $C_N(\mathbb{Q})$ be its rational cuspidal subgroup. Let $C(N)$...

conjecture | open | MathDB 页面 | 原文线索

V2-070700 | MATHDB-366294

Drinfeld modular rational cuspidal torsion conjecture

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $\mathcal{A}=\mathbb{F}_q[T]$, and let $K=\mathbb{F}_q(T)$. For a monic $\mathfrak{n}\in\mathcal{A}$, let $X_0(\mathfrak{n})$ be the Drinf...

conjecture | open | MathDB 页面 | 原文线索

V2-070701 | MATHDB-366308

Eventual stability conjecture for associated radicals of ordinary powers

摘录 / EXCERPT: Let I be a square-free monomial ideal. Eventual stability conjecture. Then, $\operatorname{assrad} I^s$ is constant for $s\gg 0$. This is the ordinary-power analogue of the event...

conjecture | open | MathDB 页面 | 原文线索

V2-070702 | MATHDB-366311

The puncturing conjectures for specialness and the weak Hilbert property

摘录 / EXCERPT: The puncturing conjectures. The following statements hold:

conjecture | open | MathDB 页面 | 原文线索

V2-070703 | MATHDB-366313

Witt-space functor conjecture for extended symmetric multifusion categories

摘录 / EXCERPT: Let $\mathds{k}$ be a field, and let $\widehat{\mathbf{SMF1C}}^{\mathrm{dom,faith}}_{\mathds{k}}$ be the $(2,1)$-category of extended symmetric (multi)fusion 1-categories over...

conjecture | open | MathDB 页面 | 原文线索

V2-070704 | MATHDB-366316

Chen's log-concavity conjecture for longest increasing subsequence counts

摘录 / EXCERPT: Chen's conjecture. For any fixed n, the sequence $\ell_{n,1},\ell_{n,2},\ldots,\ell_{n,n}$ is log-concave. Equivalently, the distribution of $\ell_n(\sigma)$ for a uniformly chos...

conjecture | open | MathDB 页面 | 原文线索

V2-070705 | MATHDB-366322

The -1-curves conjecture for blowups of the projective plane

摘录 / EXCERPT: -1-curves conjecture. The Mori cone of S is generated by the light cone and the -1-curves. Dually,

conjecture | open | MathDB 页面 | 原文线索

V2-070706 | MATHDB-366326

Beale-Kato-Majda-type continuation criterion for free boundary MHD

摘录 / EXCERPT: Let d be the spatial dimension, let $\frac{d}{2}+1<\sigma<\infty$, and let $(v,B,\Gamma)\in C([0,T);\mathbf{H}^{\sigma})$ be a solution to the free boundary MHD equations. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070707 | MATHDB-366328

The twisting classification conjecture for three-functor formalisms

摘录 / EXCERPT: Twisting classification conjecture. There is a canonical isomorphism of anima

conjecture | open | MathDB 页面 | 原文线索

V2-070708 | MATHDB-366333

K(π,1) conjecture for the stability-condition and hyperplane-complement spaces

摘录 / EXCERPT: K(π,1) conjecture. The space $\operatorname{Stab}^\dagger_{\mathcal C}(\mathcal D)$ is contractible; consequently, both $\Upsilon_{\operatorname{reg}}/\mathbb W$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070709 | MATHDB-366352

Dual-DP Shameful inequality conjecture

摘录 / EXCERPT: Let G be a graph on n vertices, and let $P^{*}_{DP}(G,k)$ denote its dual DP color function, the maximum number of colorings over all full k-fold covers of G. Dual-DP Shame...

conjecture | open | MathDB 页面 | 原文线索

V2-070710 | MATHDB-366371

The majorization characterization conjecture for ordinal spaces embeddable in the line

摘录 / EXCERPT: Let (X,δ) be an ordinal space with $|X|=n\in\mathbb N^+$, and let $x_1,x_2,\ldots,x_n$ be an enumeration of its points satisfying the majorization property. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-070711 | MATHDB-366372

The finite Menger-type embedding conjecture for ordinal spaces

摘录 / EXCERPT: Let (X,δ) be a finite ordinal space and let $n\in\mathbb N^+$. For a subset $A\subseteq X$, write $A\hookrightarrow\mathbb R^n$ when the induced ordinal space on A embeds...

conjecture | open | MathDB 页面 | 原文线索

V2-070712 | MATHDB-366382

Mazur-Tate's refined class-group formula conjecture

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve, let $G_M=\left(\mathbb{Z}/M\mathbb{Z}\right)^*/\langle-1\rangle$, and let $\theta_M=\sum_{a\in G_M}\lambda(a,M)[a]$ be the Mazur-Tate eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-070713 | MATHDB-366391

The strict-partition coloring conjecture

摘录 / EXCERPT: Strict-partition coloring conjecture. For each $\lambda\in\mathcal{S}_\alpha$, there exists an admissible $Q\in\operatorname{SYT}(\lambda)$ such that $Q^\uparrow$ admits an...

conjecture | open | MathDB 页面 | 原文线索

V2-070714 | MATHDB-366392

The almost Pieri rule for Robinson-Schensted shapes

摘录 / EXCERPT: Almost Pieri rule.

conjecture | open | MathDB 页面 | 原文线索

V2-070715 | MATHDB-366393

The almost Pieri coloring conjecture

摘录 / EXCERPT: Almost Pieri coloring conjecture. There exists an admissible $\widetilde{Q}\in\operatorname{SYT}(\lambda)$ such that $\widetilde{Q}^\uparrow$ admits a $\widetilde{\alpha}$-coloring...

conjecture | open | MathDB 页面 | 原文线索

V2-070716 | MATHDB-366396

Turánability conjecture for oriented graphs

摘录 / EXCERPT: Let D_r be the tournament on $3r$ vertices obtained from the r-blow-up of the directed cycle C_3 by replacing each of its three independent sets with a transitive tournament....

conjecture | open | MathDB 页面 | 原文线索

V2-070717 | MATHDB-366399

Manturov's parity group projection conjecture for connected sums

摘录 / EXCERPT: Let K_1 and K_2 be representatives of virtual knots $[K_1]$ and $[K_2]$, and choose any connected sum $K_1\#K_2$. Let upr denote Manturov's parity group projection from virtu...

conjecture | open | MathDB 页面 | 原文线索

V2-070718 | MATHDB-366405

Congruence conjecture for overpartitions into nonmultiples of 4 and 9

摘录 / EXCERPT: Let $\overline{R_{4,9}}(m)$ denote the number of overpartitions of m into parts divisible by neither 4 nor 9. Congruence conjecture. For all $n\geq 0$, $\ell\geq 2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-070719 | MATHDB-366406

The support approximation conjecture for vexillary involutions

摘录 / EXCERPT: Let I^{vex}_∞ be the set of vexillary involutions, and let $\mathcal{B}_{\mathsf{inv}}(z)$, $\mathcal{B}_{\mathsf{inv}}^+(z)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-070720 | MATHDB-366419

The vanishing intersection conjecture for right-closure and conjugation invariants

摘录 / EXCERPT: Let rcl denote the right-closure operator, and let $\operatorname{im}\operatorname{rcl}$ be its image, the space of right-closure invariants. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070721 | MATHDB-366440

Vietoris-Rips ultrametric condition for Laplacian approximation

摘录 / EXCERPT: Vietoris-Rips ultrametric condition. The Vietoris-Rips clusters for some $\epsilon>0$ realise the sufficient ultrametric condition

conjecture | open | MathDB 页面 | 原文线索

V2-070722 | MATHDB-366446

Gallai's conjecture for chordal graphs with subdivided caterpillar representations

摘录 / EXCERPT: Let G be a connected chordal graph admitting a tree representation T, where T is a subdivided caterpillar. Subdivided-caterpillar Gallai conjecture. Then ... This asserts that...

conjecture | open | MathDB 页面 | 原文线索

V2-070723 | MATHDB-366462

Complexified volume conjecture for hyperbolic knots and links

摘录 / EXCERPT: Let K be a hyperbolic knot or link, so that $S^3\setminus K$ admits a hyperbolic structure. Let $J_{N-1}(K)$ be the normalized colored Jones polynomial at $q=\exp(\pi i/N)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-070724 | MATHDB-366467

Joint distribution equivalence of mesh patterns S21 and S22

摘录 / EXCERPT: Joint distribution equivalence of S21 and S22. The first pair of mesh patterns, with points at $(1,1),(2,2),(3,3)$ and at $(1,3),(2,2),(3,1)$, satisfies $\sim_{jd}$, and the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-070725 | MATHDB-366472

Cames van Batenburg-de Joannis de Verclos-Kang-Pirot fractional chromatic conjecture

摘录 / EXCERPT: Let n tend to infinity, and let G be a triangle-free graph on n vertices. Write $\chi_f(G)$ for its fractional chromatic number. Cames van Batenburg-de Joannis de Verclos-Kan...

conjecture | open | MathDB 页面 | 原文线索

V2-070726 | MATHDB-366487

Initial-condition universality of the KPZ upper tail field

摘录 / EXCERPT: Let the KPZ fixed point have a general initial condition that grows sufficiently more slowly than $f(x)=x^2$ as $|x|$ becomes large, so that the KPZ fixed point exists at $(0,1)$....

conjecture | open | MathDB 页面 | 原文线索

V2-070727 | MATHDB-366488

Landis's conjecture for exponentially decaying Schrödinger solutions

摘录 / EXCERPT: Let Ω be either $\mathbb{R}^N$ or an exterior domain in $\mathbb{R}^N$, and let $V\in L^{\infty}(\mathbb{R}^N)$ satisfy $\|V\|_{\infty}\leq 1$. Suppose that u solves ... and...

conjecture | open | MathDB 页面 | 原文线索

V2-070728 | MATHDB-366492

The Widom-Arinkin-Borodin determinant formula

摘录 / EXCERPT: Let J be a jump matrix with factorization $J=\Phi_-^{-1}\Phi_+$, let $\tau_W[J]$ be the Widom determinant, let $\tau_{AB}[J]$ be the corresponding Arinkin-Borodin tau function, a...

conjecture | open | MathDB 页面 | 原文线索

V2-070729 | MATHDB-366493

The q-lattice representation conjecture for the matrix Fredholm determinant

摘录 / EXCERPT: Let q be the deformation parameter, t the time parameter, and let $\mathcal{D}$ denote the discrete q-lattice ... Let $\mathcal{D}$ be the matrix Fredholm determinant defined e...

conjecture | open | MathDB 页面 | 原文线索

V2-070730 | MATHDB-366496

Weight-derivative factorization for the beta p-adic L-function

摘录 / EXCERPT: Let r be an integer and let τ and ψ be Dirichlet characters satisfying $(-1)^s=-\psi(-1)$. Let $L_p'(f_{\beta})(s)$ denote the weight derivative of the beta p-adic...

conjecture | open | MathDB 页面 | 原文线索

V2-070731 | MATHDB-366497

Weight-derivative factorization conjecture for an Eisenstein family

摘录 / EXCERPT: Let r be an integer, let τ and ψ be Dirichlet characters, let ${\mathbf{f}}$ denote the relevant Eisenstein family, and let ${\mathbf j}$ denote the cyclotomic-variable...

conjecture | open | MathDB 页面 | 原文线索

V2-070732 | MATHDB-366505

Altinsik-Yildiz-Keskin conjecture on gcd-closed sets and GCD-LCM divisibility

摘录 / EXCERPT: Altinsik-Yildiz-Keskin conjecture. If $\max_{x\in S}|G_S(x)|\geq 2$ and S does not satisfy condition $\mathcal M$, then (S) does not divide $[S]$.

conjecture | open | MathDB 页面 | 原文线索

V2-070733 | MATHDB-366506

Zhao-Chen-Hong conjecture on power GCD-LCM divisibility and condition C

摘录 / EXCERPT: Zhao-Chen-Hong conjecture. If $\max_{x\in S}|G_S(x)|\geq 4$, then $[S^e]$ is divisible by (S^e) in $M_n(\mathbb Z)$ if and only if S satisfies condition $\mathcal C$.

conjecture | open | MathDB 页面 | 原文线索

V2-070734 | MATHDB-366512

Guy's equality conjecture for complete graphs in specified genus ranges

摘录 / EXCERPT: Let K_n be the complete graph, let S_t be the orientable surface of genus t, and let $\delta_t(K_n)$ and $\mu_t(K_n)$ denote its Euler excess and skewness, respectively. Guy'...

conjecture | open | MathDB 页面 | 原文线索

V2-070735 | MATHDB-366529

Aharoni-Berger strong conjecture for bipartite graphs

摘录 / EXCERPT: Aharoni-Berger strong conjecture. Every collection of n matchings of size $n+1$ in a bipartite graph admits a rainbow matching of size n.

conjecture | open | MathDB 页面 | 原文线索

V2-070736 | MATHDB-366534

Hard-core Caro-Wei conjecture

摘录 / EXCERPT: Hard-core Caro-Wei conjecture. The expected size of X satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-070737 | MATHDB-366535

Davies-Jenssen-Perkins-Roberts conjecture on independent sets in triangle-free graphs

摘录 / EXCERPT: Davies-Jenssen-Perkins-Roberts conjecture. As $d\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-070738 | MATHDB-366547

Lattice-polytope conjecture for generic surface-map singularities

摘录 / EXCERPT: Let cA be the parameter space of maps associated with a triple $cbmA\inccIthth$ of integer polytopes, and let cA_1, cA_2, and cA_3 denote the indicated singularity types. L...

conjecture | open | MathDB 页面 | 原文线索

V2-070739 | MATHDB-366562

Congested shallow-minor obstruction for toroidal graphs

摘录 / EXCERPT: Toroidal shallow-minor conjecture. There is no constant $k\in\mathbb N$ with the following property: every toroidal graph is a congestion-k depth-k minor of a planar graph.

open_problem | open | MathDB 页面 | 原文线索

V2-070740 | MATHDB-366563

MSO obstruction characterization for bounded linear cliquewidth

摘录 / EXCERPT: Linear cliquewidth CMSO obstruction conjecture. A class of graphs $\mathscr{C}$ has bounded linear cliquewidth if and only if the class of trees cannot be CMSO-transduce...

conjecture | open | MathDB 页面 | 原文线索

V2-070741 | MATHDB-366564

The SNP conjecture for sums of Schur polynomials of symmetric polytopes

摘录 / EXCERPT: SNP conjecture. For $t\in\mathbb Z_{>0}$, we have

conjecture | open | MathDB 页面 | 原文线索

V2-070742 | MATHDB-366567

Alavi-Malde-Schwenk-Erdős unimodality conjecture for independence polynomials of trees

摘录 / EXCERPT: Alavi-Malde-Schwenk-Erdős conjecture. The independence polynomial of every tree is unimodal.

conjecture | open | MathDB 页面 | 原文线索

V2-070743 | MATHDB-366572

Traub's basin conjecture for polynomial root-finding maps

摘录 / EXCERPT: Let p be a polynomial of degree $d\geq 2$. Assume that $p(\alpha)=0$, and let $T_{p,1}$ be the corresponding Traub's map. Denote by $\mathcal A^{\star}(\alpha)$ the basin of attr...

conjecture | open | MathDB 页面 | 原文线索

V2-070744 | MATHDB-366581

Budur-Mustaţă-Teitler's n/d-conjecture for indecomposable hyperplane arrangements

摘录 / EXCERPT: Budur-Mustaţă-Teitler's n/d-conjecture. The number $-\frac{n}{d}$ is a root of $b_{f,0}(s)$.

conjecture | open | MathDB 页面 | 原文线索

V2-070745 | MATHDB-366585

Gabrielov-Novikov-Shapiro monotonicity conjecture for generalized electrostatic potentials

摘录 / EXCERPT: Gabrielov-Novikov-Shapiro conjecture. For n unit point charges in generic position, and all $p\geq 1$, the number of index-j equilibria of V_p is at most $\#_j$.

conjecture | open | MathDB 页面 | 原文线索

V2-070746 | MATHDB-366597

Bousseau's refined sheaves/Gromov-Witten correspondence

摘录 / EXCERPT: Let S be a del Pezzo surface, let E be a suitable anticanonical divisor, and let M_{β} be the moduli space of stable one-dimensional sheaves of class β and Euler ch...

conjecture | open | MathDB 页面 | 原文线索

V2-070747 | MATHDB-366598

Leading Betti-number expansion conjecture for moduli spaces on $\mathbb P^2$

摘录 / EXCERPT: Leading Betti-number expansion conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070748 | MATHDB-366599

Divisibility conjecture for shifted Poincaré polynomials on $\mathbb P^2$

摘录 / EXCERPT: Let d be a positive integer and let $\hat\Omega_d^{\mathbb P^2}$ be the shifted Poincaré polynomial associated with the moduli space of degree-d one-dimensional sheaves on...

conjecture | open | MathDB 页面 | 原文线索

V2-070749 | MATHDB-366602

The conjectural formulas for earliest Thue-Morse progressions in particular families

摘录 / EXCERPT: The conjectural formulas for earliest Thue-Morse progressions.

conjecture | open | MathDB 页面 | 原文线索

V2-070750 | MATHDB-366603

The Fibonacci-index formula conjecture for earliest progressions

摘录 / EXCERPT: The Fibonacci-index formula conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070751 | MATHDB-366610

Nakamura's GSS conjecture for class VII$_0^+$ surfaces

摘录 / EXCERPT: A compact complex surface in class VII$_0^+$ is a surface with first Betti number one and positive second Betti number. A Kato surface is a compact complex surface obtained by Kato...

conjecture | open | MathDB 页面 | 原文线索

V2-070752 | MATHDB-366616

Factorization conjecture for the $\mathbf{g}$-dominant triple product p-adic L-function

摘录 / EXCERPT: Suppose that $\varepsilon(\textup{\bf f})=-1$ and the stated hypothesis holds. Let ${\rm Log}_{\omega_\textup{\bf f}}({\rm BK}_\textup{\bf f}^\dagger)$ denote the logarithm of the...

conjecture | open | MathDB 页面 | 原文线索

V2-070753 | MATHDB-366617

Existence conjecture for adjoint triple-product p-adic L-functions

摘录 / EXCERPT: Let $y=(\kappa,\lambda)$ be an arithmetic, not necessarily crystalline, specialization, and let $?=\textup{\bf f},{\rm ad}$. The quantities...

conjecture | open | MathDB 页面 | 原文线索

V2-070754 | MATHDB-366618

Twist-equivalence conjecture for characters of complex reflection groups

摘录 / EXCERPT: Let $G(m,p,n)$ be a complex reflection group, let J_1 and $J_{-\mathbf i}$ be the character-twisting maps considered in the paper, and let χ be a character of $G(m,p,n)$. Tw...

conjecture | open | MathDB 页面 | 原文线索

V2-070755 | MATHDB-366622

Theta-order modularity conjecture for preferential attachment graphs

摘录 / EXCERPT: Let $G_n^h=(V,E)$ be a preferential attachment graph, where n is the number of vertices and h is the number of edges added per step. Write $\operatorname{mod}(G_n^h)$ for its m...

conjecture | open | MathDB 页面 | 原文线索

V2-070756 | MATHDB-366633

Sittitrai-Pimpasalee defective coloring conjecture for planar graphs without 3-, 4-, and 6-cycles

摘录 / EXCERPT: Sittitrai-Pimpasalee's conjecture. If $d_1+d_2\geq 6$, then G is (d_1,d_2)-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-070757 | MATHDB-366640

The one-third time-decay conjecture for the periodically forced Dirac propagator

摘录 / EXCERPT: Consider the periodically forced Dirac equation with mass $m=1$, and let M^n denote its n-period propagator. For $\beta>0$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-070758 | MATHDB-366643

Minimal models from nef positive parts of Nakayama-Zariski decompositions

摘录 / EXCERPT: Minimal-model conjecture. Then $(X,B+M)$ has a minimal model.

conjecture | open | MathDB 页面 | 原文线索

V2-070759 | MATHDB-366648

The Petite Conjecture on almost periodicity and weakly localized solutions

摘录 / EXCERPT: A localized solution is a spatially localized solution, and regularity in space refers to the spatial regularity assumed in the surrounding discussion. A weakly localized solution...

conjecture | open | MathDB 页面 | 原文线索

V2-070760 | MATHDB-366649

Weakly localized solutions for localized time-dependent potentials

摘录 / EXCERPT: Consider a time-dependent potential that is localized in space. A weakly localized solution (WLS) is a solution with a component exhibiting weak localization, while a localized sol...

conjecture | open | MathDB 页面 | 原文线索

V2-070761 | MATHDB-366671

Janson's asymptotic Gaussianity conjecture for the number of vertices

摘录 / EXCERPT: Janson's asymptotic Gaussianity conjecture. Under this assumption, the number of vertices $|V_n|$ is asymptotically Gaussian.

conjecture | open | MathDB 页面 | 原文线索

V2-070762 | MATHDB-366674

A q-analogue of the Van Hamme-Sun supercongruence equivalence

摘录 / EXCERPT: Let $n\equiv 1\pmod{4}$ be a positive integer. Define the q-integer $[m]=(1-q^m)/(1-q)$, the q-shifted factorial $(x;q)_k=(1-x)(1-xq)\cdots(1-xq^{k-1})$, and let $\Phi_n(q)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-070763 | MATHDB-366683

Effective cone conjecture for klt Calabi-Yau pairs

摘录 / EXCERPT: Let (X,Δ) be a klt Calabi-Yau pair. Let $\mathcal{B}^e(X)$ denote the effective cone appearing in the source, and let $PsAut(X,\Delta)$ act on it by pullback. Effective cone...

conjecture | open | MathDB 页面 | 原文线索

V2-070764 | MATHDB-366684

The binomial-factor conjecture for exponent discrepancies

摘录 / EXCERPT: Exponent-discrepancy factorization conjecture. For every $m\geqslant6$ there exists an irreducible polynomial $r_m(t)\in\mathbb{Q}[t]$ of degree at most $m-6$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-070765 | MATHDB-366686

Zeros-of-the-derivative conjecture for nonunique minimal sub-sums

摘录 / EXCERPT: Let ... be a polynomial potential on the unit circle, and suppose that minimizing over the stated class of equally distributed measures reduces to choosing a minimal sub-sum of its c...

conjecture | open | MathDB 页面 | 原文线索

V2-070766 | MATHDB-366694

Blomer-Khan-Young fourth-moment conjecture for holomorphic cusp forms

摘录 / EXCERPT: Blomer-Khan-Young's fourth-moment conjecture. Under the normalization

conjecture | open | MathDB 页面 | 原文线索

V2-070767 | MATHDB-366695

The optimal functional 2^{s-1}-batch code conjecture

摘录 / EXCERPT: Optimal functional 2^{s-1}-batch code conjecture. For any s,

conjecture | open | MathDB 页面 | 原文线索

V2-070768 | MATHDB-366704

Quasiconcavity conjecture for marginal likelihoods of ridge, lasso, and group lasso

摘录 / EXCERPT: Let $Z(\lambda)$ denote the marginal likelihood for a regularized linear regression model, with ridge, lasso, and group lasso corresponding to their respective regularizers. Quasic...

conjecture | open | MathDB 页面 | 原文线索

V2-070769 | MATHDB-366730

Liu's conjecture on gap-level excedance and Denert statistics

摘录 / EXCERPT: Let $g\geq 1$ and $\ell\geq 1$, and let $g\mathsf{exc}_\ell$ and $g\mathsf{den}_\ell$ denote the g-gap ℓ-level excedance number and g-gap ℓ-level Denert statistic, re...

conjecture | open | MathDB 页面 | 原文线索

V2-070770 | MATHDB-366731

The hierarchy conjecture for products of irreducible polynomials in PRACs

摘录 / EXCERPT: Let n_1,n_2,r_1,r_2,k,ℓ be positive integers with $n_1<r_1<2n_1$, and let $f_1(x),\ldots,f_\ell(x)$ be irreducible polynomials of degree n_1n_2 and exponent r_1r_2. Suppos...

conjecture | open | MathDB 页面 | 原文线索

V2-070771 | MATHDB-366735

Open problem on the point spectrum of multiplication operators on weighted Dirichlet spaces

摘录 / EXCERPT: Point-spectrum conjecture. The point spectrum of V_a is empty:

open_problem | open | MathDB 页面 | 原文线索

V2-070772 | MATHDB-366739

The CFKRS recipe prediction for elliptic-curve quadratic twists

摘录 / EXCERPT: CFKRS conjecture. If $|\operatorname{Re}(s)-\frac12|\ll1/\log X$ for all $s\in S$, then as $X\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-070773 | MATHDB-366740

The multiple-Dirichlet-series pole conjecture for unitary moments

摘录 / EXCERPT: Unitary multiple-Dirichlet-series pole conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-070774 | MATHDB-366741

The multiple-Dirichlet-series pole conjecture for elliptic-curve twists

摘录 / EXCERPT: Elliptic-curve multiple-Dirichlet-series pole conjecture. The function

conjecture | open | MathDB 页面 | 原文线索

V2-070775 | MATHDB-366762

Pizzato-Sano-Tasin Frobenius bound for regular pairs

摘录 / EXCERPT: Pizzato-Sano-Tasin conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070776 | MATHDB-366763

Generalized Frobenius bound for h-regular pairs

摘录 / EXCERPT: Generalized Frobenius conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-070777 | MATHDB-366764

Polarization monotonicity conjecture for the transfinite diameter

摘录 / EXCERPT: Let Esubset ${\mathbb R}^2$ be compact and let H be a polarizer. The polarization monotonicity conjecture. ... Moreover, equality holds only if $P_H(E)\cong E$ or...

conjecture | open | MathDB 页面 | 原文线索

V2-070778 | MATHDB-366775

Cohn-de Laat-Leijenhorst conjecture on maximal spherical codes from triangle-free strongly regular graphs

摘录 / EXCERPT: Cohn-de Laat-Leijenhorst conjecture. Three-point semidefinite programming bounds prove that C is a maximal spherical code.

conjecture | open | MathDB 页面 | 原文线索

V2-070779 | MATHDB-366779

The singleton-root conjecture for the Bernstein-Sato polynomial of a regular sequence

摘录 / EXCERPT: Let X be the ambient space, let $f_1,\dots,f_r\in \mathscr{O}_X(X)$ be a regular sequence, and let g and U be as in the preceding construction. Write $b_g(s)$ for the Bernste...

conjecture | open | MathDB 页面 | 原文线索

V2-070780 | MATHDB-366782

Approximate Lagrange multiplier convergence for secant-function-augmented SOPICs

摘录 / EXCERPT: Approximate multiplier convergence conjecture. Under the direct-adjoining and secant-function-augmentation formulations, $\tilde{\eta}_i$ approaches η_i as $\rho_i\to 0$:

conjecture | open | MathDB 页面 | 原文线索

V2-070781 | MATHDB-366783

The conjecture separating proper actions of free and surface groups

摘录 / EXCERPT: Let G/H be a homogeneous space of reductive type. Say that G/H satisfies (P-free) if it admits a proper action of a non-abelian free discrete subgroup of G, and satisfies (P-...

conjecture | open | MathDB 页面 | 原文线索

V2-070782 | MATHDB-366785

The cocompact hyperbolic-lattice proper-action conjecture

摘录 / EXCERPT: For a homogeneous space G/H of reductive type, write (P-cocH^n) if there exists a discrete subgroup of G isomorphic to a cocompact lattice of $\operatorname{O}(n,1)$ acting p...

conjecture | open | MathDB 页面 | 原文线索

V2-070783 | MATHDB-366787

Conjecture on the sharp threshold for embedding subdivisions in finite-field distance graphs

摘录 / EXCERPT: Let $E \subseteq \mathbb{F}_q^d$ and let $R \subseteq \mathbb{F}_q \setminus \{0\}$. Let C be the constant appearing in Theorem, whose conclusion guarantees the existence of the...

conjecture | open | MathDB 页面 | 原文线索

V2-070784 | MATHDB-366794

Conjectured Sprague-Grundy values for king moves on generalized staircases with odd equal parameters

摘录 / EXCERPT: Let $\mathbb{SG}_{\mathrm{King}}(\mathcal{S}_{r,r}^{k})$ denote the Sprague-Grundy value of the impartial chess game with the king move rule on the generalized staircase determined...

conjecture | open | MathDB 页面 | 原文线索

V2-070785 | MATHDB-366795

Eventual polynomiality of general rank sequences

摘录 / EXCERPT: Let $s=(s(n))_{n\geq 0}$ be a constant-recursive sequence. A general rank sequence is the rank sequence determined by a recurrence and its associated general exponential-polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-070786 | MATHDB-366796

Degree conjecture for general rank sequences

摘录 / EXCERPT: Let a recurrence have characteristic polynomial with k distinct roots, without multiplicity, and let $N\subseteq\mathbb{Z}^k$ be the subgroup of relations between these roots. Wr...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-070787 | MATHDB-366799

Modulo-two orbit conjecture for the Hecke group of the double heptagon

摘录 / EXCERPT: Modulo-two orbit conjecture. An element $s=\frac{x}{y}\in\mathbb{Q}[\lambda_7]$ belongs to $H_7\cdot\infty$ if and only if $[\overline{x}:\overline{y}]$ belongs to the orbit of...

conjecture | open | MathDB 页面 | 原文线索

V2-070788 | MATHDB-366800

Connection-point characterization for the double heptagon

摘录 / EXCERPT: Double-heptagon connection-point conjecture. The point P is a connection point if and only if N is even and $[\overline{x}:\overline{y}]$ belongs to the orbit of...

conjecture | open | MathDB 页面 | 原文线索

V2-070789 | MATHDB-366801

Gartland's breakability conjecture for wall-subdivision-free graphs

摘录 / EXCERPT: Gartland's conjecture. For every positive integer t, there is an integer $d=d(t)$ such that every $\mathcal{L}_t$-free graph G with no induced subgraph isomorphic to a subdivis...

conjecture | open | MathDB 页面 | 原文线索

V2-070790 | MATHDB-366819

Bogomolov-Gieseker inequality in the reduced parameter range

摘录 / EXCERPT: Let X be the canonical line bundle of $\mathbb{P}^3$, with projection $\pi:X\to\mathbb{P}^3$, and let H be the pullback of the hyperplane class. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070791 | MATHDB-366820

Exceptional-sheaf version of the Bogomolov-Gieseker inequality

摘录 / EXCERPT: Let X be the canonical line bundle of $\mathbb{P}^3$, with projection $\pi:X\to\mathbb{P}^3$. Let $\beta\in\mathbb{R}$ and let $\mathcal{E}$ be an exceptional locally free sheaf...

conjecture | open | MathDB 页面 | 原文线索

V2-070792 | MATHDB-366827

Permutation criterion for two families of pentanomials over finite fields

摘录 / EXCERPT: Permutation criterion. The polynomials f_1 and f_2 are permutation polynomials over $\mathbb{F}_{q^3}$ if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-070793 | MATHDB-366831

Spherical connectedness conjecture for square-tiled surfaces

摘录 / EXCERPT: Let (μ,k) be a spherical profile, meaning profile data for square-tiled surfaces on the sphere. Let $\operatorname{ST}(\mu,k)$ denote the corresponding set of square-tiled surf...

conjecture | open | MathDB 页面 | 原文线索

V2-070794 | MATHDB-366841

Lascoux positivity for characters of normal square root crystals

摘录 / EXCERPT: Let $m\in\mathbb Z_{>0}$, let $\mathcal C\subseteq\mathsf{RPD}_{m,n}$ be a $\sqrt{\mathfrak{gl}_n}$-crystal, and let $w\in S_n$. Write ... A polynomial is Lascoux positive if it is a...

conjecture | open | MathDB 页面 | 原文线索

V2-070795 | MATHDB-366856

The separation of increasing and decreasing Turing-degree chains

摘录 / EXCERPT: For $p\in\mathbb{N}^{\mathbb{N}}$, let $\mathsf{TDINC}(p)$ consist of sequences $\langle p_0,p_1,p_2,\ldots\rangle$ such that, for every i,...

conjecture | open | MathDB 页面 | 原文线索

V2-070796 | MATHDB-366866

Stability conjecture for expressions constructed from transductions

摘录 / EXCERPT: Expression-stability conjecture. There is a function $g:\mathbb{N}\to\mathbb{N}$ such that each $(\mathcal{Q}_r^\circ,\ell)$-expression constructed in the proof of the transduction lemma...

conjecture | open | MathDB 页面 | 原文线索

V2-070797 | MATHDB-366867

Transduction preservation conjecture for bounded expression-stable clique-width

摘录 / EXCERPT: Transduction preservation conjecture. The property of being a perturbation of a class of bounded expression-stable $\mathcal{P}^\circ$-clique-width is preserved under taking first-order...

conjecture | open | MathDB 页面 | 原文线索

V2-070798 | MATHDB-366868

Essential-image conjecture for the generalized Montréal functor

摘录 / EXCERPT: Let $\Delta=\{\alpha_1,\dots,\alpha_{n-1}\}$, and let V be an object of $\operatorname{Rep}^{\operatorname{ord}}_h(G_{\mathbb{Q}_p,\Delta}\times\mathbb{Q}_p^\times)$. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-070799 | MATHDB-366877

Positive Burau eigenvalue conjecture for order-preserving braids

摘录 / EXCERPT: Let B_n be the braid group on n strands, let $\mathbb{E}$ be the relevant ordered field with quadratic form Q, and let ... be the reduced Burau representation. A braid...

conjecture | open | MathDB 页面 | 原文线索

V2-070800 | MATHDB-366887

The perfect-graph characterization of quadratic stable-set toric ideals

摘录 / EXCERPT: Stable-set ideal conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-070801 | MATHDB-366889

The finite companion hierarchy for Andrews-Gordon identities

摘录 / EXCERPT: Finite companion Andrews-Gordon conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070802 | MATHDB-366890

The finite companion hierarchy for Bressoud identities

摘录 / EXCERPT: Finite companion Bressoud conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070803 | MATHDB-366891

The infinite cylindric-partition generating-function hierarchy

摘录 / EXCERPT: Cylindric-partition hierarchy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070804 | MATHDB-366892

The finite Andrews-Gordon Bailey hierarchy

摘录 / EXCERPT: Finite Andrews-Gordon Bailey hierarchy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070805 | MATHDB-366894

Conjectured existence of universal ternary lattices over thirteen real quadratic fields

摘录 / EXCERPT: Let D be a positive squarefree integer, let ... and let $\mathcal{O}_F$ denote the ring of integers of F. A universal ternary $\mathcal{O}_F$-lattice is a universal quadratic lat...

conjecture | open | MathDB 页面 | 原文线索

V2-070806 | MATHDB-366895

Universality conjecture for the 34 listed ternary lattices

摘录 / EXCERPT: Consider the 34 ternary lattices listed in the theorem immediately preceding the conjecture, over the relevant real quadratic fields. A universal ternary lattice is a ternary qua...

conjecture | open | MathDB 页面 | 原文线索

V2-070807 | MATHDB-366910

Sharp bandwidth exponent conjecture for surfaces of revolution

摘录 / EXCERPT: Sharp bandwidth conjecture. The main projector estimate should hold for

conjecture | open | MathDB 页面 | 原文线索

V2-070808 | MATHDB-366919

The Ancestor Conjecture for finite Coxeter groups

摘录 / EXCERPT: The Ancestor Conjecture. Every finite Coxeter group has the ancestor property.

conjecture | open | MathDB 页面 | 原文线索

V2-070809 | MATHDB-366920

Bound on involution length in finite Coxeter groups assuming the Ancestor Conjecture

摘录 / EXCERPT: Bound on involution length. For every $w\in W$,

conjecture | open | MathDB 页面 | 原文线索

V2-070810 | MATHDB-366937

Sparse relaxation order conjecture for finite convergence

摘录 / EXCERPT: Let a polynomial optimization problem have a sparse moment relaxation and the corresponding dense moment relaxation, with relaxation order measured by the parameter in the respecti...

conjecture | open | MathDB 页面 | 原文线索

V2-070811 | MATHDB-366945

Han and Ren's maximal-length conjecture for MDS elliptic codes

摘录 / EXCERPT: Let $\mathcal{E}$ be an elliptic curve over $\mathbb{F}_q$, and let $C_L(D,G)$ be an $[n,k]$ MDS code from $\mathcal{E}$. Assume that $3\leq k\leq n-3$. Han and Ren's conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-070812 | MATHDB-366960

The k-formality conjecture for hyperpolygonal arrangements

摘录 / EXCERPT: For $n\in{\mathbb N}$, let ${\mathscr H}_n$ denote the hyperpolygonal arrangement. An arrangement is k-formal when it has the stronger formality property indexed by k, with...

conjecture | open | MathDB 页面 | 原文线索

V2-070813 | MATHDB-366969

Thin code conjecture for binary linear codes

摘录 / EXCERPT: Let $\mathcal{C}\subseteq\mathbb{F}_2^n$ be a linear code. A hitting set for $\mathcal{C}$ is a set $S\subseteq[n]$ such that $c_S\neq0$ for every nonzero codeword...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-070814 | MATHDB-366975

Kantarcı Oğuz's unimodality conjecture for traces of q-deformed matrices

摘录 / EXCERPT: For a sequence of positive integers $a_1,\ldots,a_{2m}$, let $M(a_1,\ldots,a_{2m})$ be the corresponding matrix and call a polynomial $f(q)=\sum_{i=0}^n a_iq^i$ unimodal if...

conjecture | open | MathDB 页面 | 原文线索

V2-070815 | MATHDB-366979

Vanishing-mass limit for locally constrained IMD problems

摘录 / EXCERPT: Let $p\in(1,\infty)$ and $t_{\max}>0$. For each natural number n, let $(\hat{k}_n,\hat{\mu}_n)\in L^\infty(\Omega;\mathbb{R}_+)^2$ solve the (vp-IMD) problem, respectively the (s...

conjecture | open | MathDB 页面 | 原文线索

V2-070816 | MATHDB-366995

Hain's Hodge structure from a cotangent-fiber microsheaf

摘录 / EXCERPT: Cotangent-fiber Hodge microsheaf conjecture. Hain's Hodge structure is induced by a Hodge microsheaf whose underlying microsheaf corresponds to a cotangent fiber.

conjecture | open | MathDB 页面 | 原文线索

V2-070817 | MATHDB-367000

Türkelli's modification of Malle's conjecture

摘录 / EXCERPT: Let $G\subset S_n$ be a transitive permutation group and let Q be a global field with $(|G|,\operatorname{char}(Q))=1$. Let $b(G,Q)$ denote the relevant logarithmic exponent and...

conjecture | open | MathDB 页面 | 原文线索

V2-070818 | MATHDB-367001

The Malle-Türkelli bounds for number fields

摘录 / EXCERPT: Let k be a number field and let $G\subset S_n$ be a transitive permutation group. Let $b_M(G,k)$ be Malle's logarithmic constant, $b(G,k)$ the actual logarithmic constant, and...

conjecture | open | MathDB 页面 | 原文线索

V2-070819 | MATHDB-367002

Fast mixing conjecture for face flips on the triangle lattice

摘录 / EXCERPT: Fast mixing conjecture. The mixing time of the face-flip Markov chain on $T_{m,n}$ is $O(mn \log(mn))$.

conjecture | open | MathDB 页面 | 原文线索

V2-070820 | MATHDB-367008

The planar E1a doubling conjecture for minimally rigid graphs

摘录 / EXCERPT: Let G be a minimally rigid graph, and let G' be obtained from G by a 1-extension of type E1a. Let $\mathbf{c}_{2}(G)$ denote the planar realization count. The planar E1a doub...

conjecture | open | MathDB 页面 | 原文线索

V2-070821 | MATHDB-367009

The spherical E1a doubling conjecture for minimally rigid graphs

摘录 / EXCERPT: Let G be a minimally rigid graph, and let G' be obtained from G by a 1-extension of type E1a. Let $\mathbf{c}^{\circ}_{2}(G)$ denote the spherical realization count. The sphe...

conjecture | open | MathDB 页面 | 原文线索

V2-070822 | MATHDB-367010

The E1s63 doubling conjecture for minimally 3-rigid graphs

摘录 / EXCERPT: Let G be a minimally 3-rigid graph, and let G' be obtained from G by a 1-extension of type E1s63. Let $\mathbf{c}_{3}(G)$ denote the three-dimensional realization count. The...

conjecture | open | MathDB 页面 | 原文线索

V2-070823 | MATHDB-367011

The E2Xs511/E2Vs511 doubling conjecture for minimally 3-rigid graphs

摘录 / EXCERPT: Let G be a minimally 3-rigid graph, and let G' be obtained from G by a 1-extension of type E2Xs511 or E2Vs511. Let $\mathbf{c}_{3}(G)$ denote the three-dimensional realizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-070824 | MATHDB-367013

Conjecture on two-parameter contact inequivalence in characteristic p

摘录 / EXCERPT: Contact-inequivalence conjecture. For general $a,b,c,d a\in F$, these singularities are not contact equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-070825 | MATHDB-367022

Removal of the auxiliary boundedness assumption in the compactness theorem

摘录 / EXCERPT: Let v be a weight, let $\phi\colon\mathbb D\to\mathbb D$ be an analytic symbol, and let ψ be an analytic function. The boundedness-assumption conjecture. The assumption in T...

conjecture | open | MathDB 页面 | 原文线索

V2-070826 | MATHDB-367024

Comparison of local oscillation quantities for weighted BMOA spaces

摘录 / EXCERPT: Let v be an admissible weight, let X be either BMOA_v or VMOA_v, and assume $X\not\subset H^\infty$. Let δ_z denote evaluation at z,...

conjecture | open | MathDB 页面 | 原文线索

V2-070827 | MATHDB-367031

Ultra-log-concavity of weak-order Eulerian polynomials

摘录 / EXCERPT: Ultra-log-concavity conjecture. For every $\pi\in\mathfrak{S}_n$, Q_W^π is ultra-log-concave.

conjecture | open | MathDB 页面 | 原文线索

V2-070828 | MATHDB-367032

The periodicity and unboundedness conjecture for shifted Hankel determinant sequences

摘录 / EXCERPT: The case $\ell\geq n+2$. The sequence $\Delta_j^{(n+2)}$ is $2n(n+1)$-periodic when n is even and $2n(n+1)$-antiperiodic, hence $4n(n+1)$-periodic, when n is odd, and its terms...

conjecture | open | MathDB 页面 | 原文线索

V2-070829 | MATHDB-367037

Relative fundamental lemma in the semi-Lie case

摘录 / EXCERPT: Relative fundamental lemma in the semi-Lie case.

conjecture | open | MathDB 页面 | 原文线索

V2-070830 | MATHDB-367039

Stein's conjecture for weighted extension operators

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^d$ be a C^2 hypersurface with surface measure $\mathrm{d}\sigma$, let $f\in L^2(\Sigma;\mathrm{d}\sigma)$, and let w be a nonnegative weight. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070831 | MATHDB-367046

Zilber-Pink density conjecture for real multiplication in low genus

摘录 / EXCERPT: Low-genus density conjecture. If $g\leq 6$, then for each K, $E_{g,K}$ is dense in $\mathcal{M}_g(\mathbb{C})$. This is the complementary low-genus prediction to the high-genus f...

conjecture | open | MathDB 页面 | 原文线索

V2-070832 | MATHDB-367062

Gaussian free field and discrete Gaussian fluctuation conjecture for general periodic dimer models

摘录 / EXCERPT: Gaussian free field-discrete Gaussian conjecture. The fluctuations of the dimer-model height function in the liquid region of a graph of scale N are approximated in distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-070833 | MATHDB-367064

Equality of Segre-Stiefel-Whitney classes for Morin and corank-two singularities

摘录 / EXCERPT: Let l be an integer, and let $\overline{A_2}(l+1)$ and $\overline{\Sigma}^2(l)$ denote the corresponding singularity loci. Their Segre-Stiefel-Whitney classes are defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-070834 | MATHDB-367065

Fold-lift conjecture for Morin maps

摘录 / EXCERPT: Let $f:M\to N$ be a Morin map. A lift of f is a map $\tilde{f}:M\to N\times\mathbb{R}$ whose composition with the projection $N\times\mathbb{R}\to N$ is f. Fold-lift conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-070835 | MATHDB-367082

The Adler-zero conjecture for generic pure generalized NLSM 3-amplitudes

摘录 / EXCERPT: Let $\mathcal{A}^{(3),\sigma}_n$ be a generic, pure generalized nonlinear sigma model 3-amplitude, and consider any single soft limit obtained by sending one external kinematic d...

conjecture | open | MathDB 页面 | 原文线索

V2-070836 | MATHDB-367085

Tight fitting conjecture for Hardy and weighted Bergman spaces on the unit ball

摘录 / EXCERPT: Tight fitting conjecture. (a) The embedding $A^p_\alpha\subset A^q_\beta$ is a tight fitting if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-070837 | MATHDB-367090

Positive asymptotic fraction below the b-loading threshold

摘录 / EXCERPT: Let $\mathcal P(n)$ be the set of partitions of n, let $\mathbf t\in\mathcal P(n)^3$, and let $g(\mathbf t)$ denote the corresponding Kronecker coefficient. Define ... Let F_n be...

conjecture | open | MathDB 页面 | 原文线索

V2-070838 | MATHDB-367095

Rectangular multi-t Yang-Baxter trace conjecture

摘录 / EXCERPT: Rectangular multi-t Yang-Baxter trace conjecture. The equivariant trace of $Y_{c3_bc}(v_bc)$ is proportional to

conjecture | open | MathDB 页面 | 原文线索

V2-070839 | MATHDB-367096

Monomial-product property for multi-t Macdonald coefficients

摘录 / EXCERPT: Monomial-product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-070840 | MATHDB-367103

Finite-time round-point convergence conjecture for the torsion-driven flow

摘录 / EXCERPT: Let Ω be a bounded smooth convex domain in $\mathbb{R}^n$. Let $X(t,x)$ be the flow defined by ... where ν is the unit outer normal, $u(t)$ is the torsion function on the...

conjecture | open | MathDB 页面 | 原文线索

V2-070841 | MATHDB-367104

Alternating-group monodromy conjecture for minimal [1,1]-origamis

摘录 / EXCERPT: Alternating-group monodromy conjecture. All of the $[1,1]$-origamis in Theorem have monodromy group isomorphic to $\operatorname{Alt}_{2g-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-070842 | MATHDB-367105

SL(2,Z)-orbit conjecture for the constructed origamis

摘录 / EXCERPT: SL(2,Z)-orbit conjecture. For odd genus $g\geq 5$, the origamis constructed by Aougab-Menasco-Nieland lie in exactly two $\operatorname{SL}(2,\mathbb{Z})$-orbits inside the odd com...

conjecture | open | MathDB 页面 | 原文线索

V2-070843 | MATHDB-367116

The S-number conjecture for E- and M-function values

摘录 / EXCERPT: Let ${\bf E}$ and ${\bf M}$ denote the sets of values under consideration of E-functions and M-functions, respectively. A complex number is an S-number when its Mahler expone...

conjecture | open | MathDB 页面 | 原文线索

V2-070844 | MATHDB-367117

The most-informative Boolean function conjecture

摘录 / EXCERPT: Let $\mathbf{X}\sim\operatorname{Unif}(\mathbb{H}^n)$ be uniformly distributed on the Boolean hypercube, and let $\mathbf{Y}$ be obtained by passing each bit of $\mathbf{X}$ throug...

conjecture | open | MathDB 页面 | 原文线索

V2-070845 | MATHDB-367118

The Hellinger conjecture for Boolean functions

摘录 / EXCERPT: The Hellinger conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-070846 | MATHDB-367119

The candidate function conjecture for the finite-dimensional functional inequality

摘录 / EXCERPT: The candidate function conjecture. The function ϕ belongs to the class Ψ. If true, this would establish the most-informative Boolean function conjecture for balanced func...

conjecture | open | MathDB 页面 | 原文线索

V2-070847 | MATHDB-367121

The kappa inequalities for the finite-dimensional functional approach

摘录 / EXCERPT: For $(u,w)\in(0,1)^2$, define ... with $\kappa(u,u)=0$. The kappa inequalities. The following two assertions hold: for $u,w\in(0,1/2]$, $\kappa(u,w)\leq\kappa(1-u,w)$; and the functi...

conjecture | open | MathDB 页面 | 原文线索

V2-070848 | MATHDB-367122

The explicit lower-bound conjecture for zeta

摘录 / EXCERPT: Let $m_u,m_w,e_u,e_w\in[0,1]$ satisfy $H_2(m_u)\geq e_u$ and $H_2(m_w)\geq e_w$. Let ϕ and J be the functions defined in the paper. The explicit lower-bound conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-070849 | MATHDB-367125

Reed's recolouring conjecture

摘录 / EXCERPT: Reed's recolouring conjecture. Any graph G is k-recolourable for all

conjecture | open | MathDB 页面 | 原文线索

V2-070850 | MATHDB-367126

Recolouring conjecture for triangle-free graphs

摘录 / EXCERPT: Triangle-free recolouring conjecture. Any triangle-free graph G is k-recolourable for all

conjecture | open | MathDB 页面 | 原文线索

V2-070851 | MATHDB-367127

Recolouring conjecture for odd-hole-free graphs

摘录 / EXCERPT: Odd-hole-free recolouring conjecture. All odd-hole-free graphs of maximum degree Δ and clique number ω are k-recolourable for

conjecture | open | MathDB 页面 | 原文线索

V2-070852 | MATHDB-367128

Canonical MambaZero implementation conjecture for add-beta estimation of binary order-k Markov chains

摘录 / EXCERPT: Let $k>1$ and consider the binary order-k Markov setting. Let the canonical $\mathsf{MambaZero}$ model have state dimension d, number of state channels N, expansion dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-070853 | MATHDB-367131

Least-area conjecture for free boundary minimal surfaces in the 3-ball

摘录 / EXCERPT: Let $N_{\gamma,1}$ and $N_{\gamma,2}$ be the basic reflection surfaces and let Γ_{12} and Γ_{12}' be the reflection groups defined in the source. For a free boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-070854 | MATHDB-367132

Jiménez-Quiroz-Thraves Caro totally odd immersion conjecture

摘录 / EXCERPT: For a graph G, let $\operatorname{toi}(G)$ be the maximum integer t such that G contains a totally odd strong immersion of K_t, and let $\chi(G)$ denote its chromatic numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-070855 | MATHDB-367150

The double Hall path-cover conjecture

摘录 / EXCERPT: Double Hall path-cover conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-070856 | MATHDB-367158

Characterization conjecture for weighted projective spaces as Fano orbifolds

摘录 / EXCERPT: Weighted-projective-space characterization conjecture. If, for every such ℓ, g, and f,

conjecture | open | MathDB 页面 | 原文线索

V2-070857 | MATHDB-367165

Fixed-perimeter inequality for parts divisible by and repeated at least k times

摘录 / EXCERPT: Let $FO_{j,k}(n)$ be the number of partitions of n with exactly j parts divisible by k, and let $FD_{j,k}(n)$ be the number of partitions of n with exactly j parts appear...

conjecture | open | MathDB 页面 | 原文线索

V2-070858 | MATHDB-367170

Finite monoid catenary degree for conductor submonoids

摘录 / EXCERPT: Finite catenary-degree conjecture. If F_{red} is finitely generated, then

conjecture | open | MathDB 页面 | 原文线索

V2-070859 | MATHDB-367181

Adapted Masur-Veech volume asymptotics conjecture for odd strata

摘录 / EXCERPT: Let $M\in{\mathbb Z}_+$ be fixed. For every $0\leq p\leq M$ and every odd positive partition ${\underline{k}}$ of $4g-4+p$, write its parts as $\kappa_1,\dots,\kappa_n$. Adapted vo...

conjecture | open | MathDB 页面 | 原文线索

V2-070860 | MATHDB-367184

1-Rayleigh conjecture for the normalized Kahn-Saks polynomial

摘录 / EXCERPT: Let $x_1\leq\ldots\leq x_k$ be a chain, let P_{KS} be its Kahn-Saks polynomial, and let $N(P_{\mathrm{KS}})$ denote its normalization. For exponent vectors α and...

conjecture | open | MathDB 页面 | 原文线索

V2-070861 | MATHDB-367202

Coxeter code distance conjecture

摘录 / EXCERPT: Coxeter code distance conjecture. The distance is

conjecture | open | MathDB 页面 | 原文线索

V2-070862 | MATHDB-367203

Trotignon's string-graph or biclique-induced-minor conjecture

摘录 / EXCERPT: A hereditary class is a graph class closed under induced subgraphs. A string graph is an intersection graph of curves in the plane, and $K_{\ell,\ell}$ is the complete bipartite gr...

conjecture | open | MathDB 页面 | 原文线索

V2-070863 | MATHDB-367204

The weakly sparse large-treewidth subclass meta-conjecture

摘录 / EXCERPT: Let Π be a graph-class property. A hereditary weakly sparse class is a hereditary graph class excluding some biclique $K_{t,t}$ as a subgraph. Meta-conjecture-ws(Π). Every...

conjecture | open | MathDB 页面 | 原文线索

V2-070864 | MATHDB-367241

Twisted GGP conjecture for unitary periods and twisted Asai L-values

摘录 / EXCERPT: Let π be an irreducible cuspidal automorphic representation of $\mathrm U(V_{E_0})(\mathbb A_{E_0})$, let Π be a weak base change of π to $\mathrm {GL}_n(\mathbb A_E)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-070865 | MATHDB-367242

Arithmetic twisted GGP conjecture for Fourier-Jacobi periods

摘录 / EXCERPT: Arithmetic twisted GGP conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-070866 | MATHDB-367244

Conjecture on norm comparison for matrices with independent columns

摘录 / EXCERPT: Norm-comparison conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-070867 | MATHDB-367247

The signed supersingular Iwasawa main conjecture for elliptic curves

摘录 / EXCERPT: Let $E/\mathbb{Q}$ have supersingular reduction at p, let $L_p^\pm(E,T)$ be Pollack's signed p-adic L-functions, and let $f_E^\pm(T)$ be the characteristic series of the corr...

conjecture | open | MathDB 页面 | 原文线索

V2-070868 | MATHDB-367248

Density and rank-stability conjectures for generalized non-Wieferich primes

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve and let $P\in E^0(\mathbb{Q})$ be a non-torsion point. Define the sets of good primes ... and $W_{E,P}$ as the set of primes for which...

conjecture | open | MathDB 页面 | 原文线索

V2-070869 | MATHDB-367260

Unimodality conjecture for the coefficients $b(n_k,\ldots,n_1)$

摘录 / EXCERPT: Let $k\geq 2$ be an integer and let $n_k,\ldots,n_1$ be non-negative integers. Let $b(n_k,\ldots,n_1)$ denote the polynomial constructed in Theorem 3. Unimodality conjecture. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-070870 | MATHDB-367265

The reproducing-kernel asymptotics conjecture for weighted residual polynomials on Jordan arcs

摘录 / EXCERPT: Let K be a C^{1+} Jordan arc, let $z_0 \in \overline{\mathbb{C}}\backslash K$, and let $\rho:K\to[0,\infty)$ be upper-semicontinuous. Writing $t_n(\rho,z_0)$ for the weighted r...

conjecture | open | MathDB 页面 | 原文线索

V2-070871 | MATHDB-367272

Betti-element count distinguishes finite abelian groups via block monoids

摘录 / EXCERPT: Let G be a finite abelian group, and let $\mathcal{F}(G)$ be the free abelian monoid on G. Define ... and let ... be the block monoid on G. The Betti elements of an affine monoid...

conjecture | open | MathDB 页面 | 原文线索

V2-070872 | MATHDB-367274

The dense analogue characterization conjecture for merge-width and flip-width

摘录 / EXCERPT: Let $\mathcal C$ be a graph class. Following the paper, say that $\mathcal C$ is in the dense analogue of bounded expansion if, for every weakly sparse graph class $\mathcal D$ suc...

conjecture | open | MathDB 页面 | 原文线索

V2-070873 | MATHDB-367276

DP-4-colorability conjecture for planar graphs without chorded 6-cycles

摘录 / EXCERPT: DP-4-colorability conjecture. Every planar graph without chorded 6-cycles is DP-4-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-070874 | MATHDB-367278

Flag-accuracy conjecture for connected subgraph arrangements

摘录 / EXCERPT: Let G be a connected graph, and let ${\mathscr A}_G$ denote its connected subgraph arrangement. A free arrangement is flag-accurate if it is accurate and the tuple of flats witne...

conjecture | open | MathDB 页面 | 原文线索

V2-070875 | MATHDB-367279

Universal formality conjecture for connected subgraph arrangements

摘录 / EXCERPT: Let G be a connected graph, and let ${\mathscr A}_G$ denote its connected subgraph arrangement. For an arrangement ${\mathscr A}$, k-formality is the hierarchy of formality pro...

conjecture | open | MathDB 页面 | 原文线索

V2-070876 | MATHDB-367282

Compatibility conjecture for quantization and reduction via adapted formality

摘录 / EXCERPT: Compatibility conjecture. The diagram of curved L_∞-morphisms

conjecture | open | MathDB 页面 | 原文线索

V2-070877 | MATHDB-367314

Bean's informal high-dimensional asymptotics for the full conformal LASSO

摘录 / EXCERPT: Let $(X_i,Y_i)_{i=1}^{n+1}$ follow the data-generating process described in Section 2, let $\hat{\beta}$ be the LASSO estimator fit on all $n+1$ observations, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070878 | MATHDB-367315

High-dimensional coverage conjecture for full conformal quantile regression

摘录 / EXCERPT: Let $\ell_{\alpha}(u)=\alpha u$ for $u\geq 0$ and $(\alpha-1)u$ for $u<0$ be the pinball loss, let $\hat{\eta}_{n+1}$ be the dual quantile-regression score, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070879 | MATHDB-367323

Conjecture on Hankel determinants of Motzkin convolution powers when the exponent is divisible by 3

摘录 / EXCERPT: Let $M(x)=\sum_{n\geq 0}M_nx^n$ be the generating function of the Motzkin numbers, let $F(x,r)=M(x)^r$, and write $H_n(F(x,r))$ for the Hankel determinant of the coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-070880 | MATHDB-367325

Conjecture on shifted periodicity of Hankel determinants for Motzkin convolution powers

摘录 / EXCERPT: Let $F(x,r)=M(x)^r$, where $M(x)$ is the generating function of the Motzkin numbers, and let $H_n(F(x,r))$ be its Hankel determinants. The conjecture asserts that, for...

conjecture | open | MathDB 页面 | 原文线索

V2-070881 | MATHDB-367328

Möbius-value conjecture for staircase partition posets

摘录 / EXCERPT: Let λ be a partition with repeating elements, and let ${\mathbb D}_{\mathsf{S}}(\lambda)$ be the associated poset of staircase matrices. Denote its Möbius function by...

conjecture | open | MathDB 页面 | 原文线索

V2-070882 | MATHDB-367330

Asymptotic F-nefness conjecture for differences of Virasoro conformal-block divisors

摘录 / EXCERPT: Fix $r>0$. For integers k, n, and labels $a_1,\ldots,a_n$, consider the difference of conformal-block divisors on $\overline{\mathcal{M}}_{0,n}$ associated with the Virasoro ve...

conjecture | open | MathDB 页面 | 原文线索

V2-070883 | MATHDB-367331

The F-conjecture for moduli spaces of stable curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{g,n}$ be the moduli space of stable curves, and let an F-curve be a one-dimensional codimension-$3g-4+n$ boundary stratum. A line bundle is F-nef if it...

conjecture | open | MathDB 页面 | 原文线索

V2-070884 | MATHDB-367353

Conjecture on weak Lyapunov functionals for the inertial and first-order Kuramoto models

摘录 / EXCERPT: Weak-Lyapunov-functional conjecture. Either there is a constant $c\ge\frac12$ such that, whenever $\kappa>c\mathcal{D}(\mathcal{V})$, both models admit weak Lyapunov functionals, o...

conjecture | open | MathDB 页面 | 原文线索

V2-070885 | MATHDB-367354

Conjecture on generic-data asymptotic phase-locking in the inertial Kuramoto model

摘录 / EXCERPT: Generic-data phase-locking conjecture. (Weak form) For every $\varepsilon>0$, there exists $\delta=\delta(\varepsilon)$ such that, if $\mathcal{D}(\mathcal{V})/\kappa<\delta$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-070886 | MATHDB-367374

Veech-group commensurator conjecture for isogenous interval exchange systems

摘录 / EXCERPT: Let two interval exchange systems be isogenous, and suppose one is minimal and self-induced. Then the other is also minimal and self-induced. Associate to such systems their transl...

conjecture | open | MathDB 页面 | 原文线索

V2-070887 | MATHDB-367375

Eventual flow equivalence conjecture for Sturmian systems

摘录 / EXCERPT: Let $(X_{\alpha},\operatorname{\sigma}_{\alpha})$ and $(X_{\beta},\operatorname{\sigma}_{\beta})$ be Sturmian systems. Sturmian eventual-flow conjecture. Eventual flow equivalence...

conjecture | open | MathDB 页面 | 原文线索

V2-070888 | MATHDB-367377

Kapranov-Schechtman's total-positivity conjecture for finite Coxeter groups

摘录 / EXCERPT: Kapranov-Schechtman's conjecture. The metamatrix $M(W)$ is totally positive.

conjecture | open | MathDB 页面 | 原文线索

V2-070889 | MATHDB-367384

Jain's incidence-code parameter conjecture for unit graphs

摘录 / EXCERPT: Let n be a natural number, let $G(\mathbb{Z}_n)$ be a unit graph, and let H be a $|V|\times|E|$ incidence matrix of $G(\mathbb{Z}_n)$. Let $\phi(n)$ denote Euler's totient func...

conjecture | open | MathDB 页面 | 原文线索

V2-070890 | MATHDB-367385

The Picard rank conjecture for simply branched Hurwitz spaces

摘录 / EXCERPT: Let $d>0$ and let n be even with $n\geq 2d-2$. Let $c\subset S_d$ be the conjugacy class of transpositions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070891 | MATHDB-367393

Optimality of the isosceles trapezoid inscription theorem

摘录 / EXCERPT: Let $\gamma_1,\gamma_2\colon\mathbb{R}\to\mathbb{C}$ be continuous disjoint periodic embeddings of the real line into the plane. A quadrilateral Q admits an inscription in any pa...

conjecture | open | MathDB 页面 | 原文线索

V2-070892 | MATHDB-367398

Linear bound for the treewidth contraction function

摘录 / EXCERPT: Let $p,q\geqslant 1$ be integers. Define $\ell(p,q)$ to be the minimum integer such that every graph G with treewidth at least $\ell(p,q)$ contains pairwise disjoint connected su...

conjecture | open | MathDB 页面 | 原文线索

V2-070893 | MATHDB-367399

The arbor polytope h-vector realization conjecture

摘录 / EXCERPT: The arbor polytope h-vector realization conjecture. For every arbor t, there exists a simple polytope Φ_t such that

conjecture | open | MathDB 页面 | 原文线索

V2-070894 | MATHDB-367400

The real-rootedness conjecture for arbor Ehrhart polynomials

摘录 / EXCERPT: The real-rootedness conjecture for arbor Ehrhart polynomials. For any arbor t, all roots of $E_t(u)$ are negative real numbers in the interval $[-1,0]$, excluding 0.

conjecture | open | MathDB 页面 | 原文线索

V2-070895 | MATHDB-367401

The Ehrhart-Zeta duality conjecture for linear arbors

摘录 / EXCERPT: The Ehrhart-Zeta duality conjecture. For every linear arbor t,

conjecture | open | MathDB 页面 | 原文线索

V2-070896 | MATHDB-367403

The gentle arbor F-triangle conjecture

摘录 / EXCERPT: The gentle arbor F-triangle conjecture. In SC_t, there exists a facet C such that the F-triangle for (SC_t,C) coincides with the F-triangle associated with the transmuted...

conjecture | open | MathDB 页面 | 原文线索

V2-070897 | MATHDB-367404

The Hochschild polytope F-triangle conjecture

摘录 / EXCERPT: The Hochschild polytope F-triangle conjecture. In SC_n, there exists a facet C such that the F-triangle for (SC_n,C) coincides with the F-triangle associated with the tra...

conjecture | open | MathDB 页面 | 原文线索

V2-070898 | MATHDB-367405

The $C^{p'}$-regularity conjecture for elliptic p-Laplace equations

摘录 / EXCERPT: Let $p\in(2,\infty)$, let B_1 be the unit ball, and let $\Delta_p u\coloneqq\operatorname{div}(|Du|^{p-2}Du)$. A function is locally of class $C^{1,\alpha}$ if it has locally Höl...

conjecture | open | MathDB 页面 | 原文线索

V2-070899 | MATHDB-367435

Nath's modulo 192 congruences for partitions with distinct even parts

摘录 / EXCERPT: Nath's conjecture. For every $n\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070900 | MATHDB-367438

The tropical monomial-realization conjecture for crystal bases

摘录 / EXCERPT: Let $ciota$ be an adapted sequence and let $cPsi_{ciota}$ be the map in Theorem for $cmathfrak{g}$. For each $s\in\mathbb{Z}_{\geq1}$ and $k\in I$, let $cmathcal{M}_{s,k,ciota}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-070901 | MATHDB-367451

The sphericity conjecture for dichotomous ordinal complete graphs

摘录 / EXCERPT: Let $\mathrm{pdd}(n)$ denote the minimum dimension required to realize a dichotomous ordinal complete graph on n vertices, and let $\mathrm{sp}(G)$ be the sphericity of a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-070902 | MATHDB-367462

The p-adic Artin formalism factorization for triple product L-functions

摘录 / EXCERPT: The p-adic Artin formalism conjecture. In full generalities, there should be a factorization

conjecture | open | MathDB 页面 | 原文线索

V2-070903 | MATHDB-367464

Localization conjecture for differences under optimal ERGM couplings

摘录 / EXCERPT: Let two random graphs be sampled under a coupling of an ERGM and a corresponding Erdős-Rényi model. Localization conjecture. For certain ERGM specifications, under some coupling wi...

open_problem | open | MathDB 页面 | 原文线索

V2-070904 | MATHDB-367483

Conjectured period-five sign patterns for coefficients of Rogers-Ramanujan products

摘录 / EXCERPT: Conjectured sign pattern. For all integers $n\geq0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070905 | MATHDB-367486

Edge Monotonicity Conjecture for tessellations of cubic parameter spaces

摘录 / EXCERPT: Let ${\mathfrak p}$ be a parabolic endpoint of edges in ${\bf Tes}_q({\overline{\mathcal S}}_p)$, and let $H_{\mathfrak p}$ be its Type D hyperbolic component. Edge Monotonicity Co...

conjecture | open | MathDB 页面 | 原文线索

V2-070906 | MATHDB-367487

Across the Boundary Conjecture for Mandelbrot copies

摘录 / EXCERPT: Let a Mandelbrot copy in ${\mathcal S}_p$ cut across the boundary between two escape regions at a parabolic point ${\mathfrak p}$ of period q. The associated dynamic configuratio...

conjecture | open | MathDB 页面 | 原文线索

V2-070907 | MATHDB-367488

Tan Lei similarity for cubic polynomial parameter spaces

摘录 / EXCERPT: Tan Lei Similarity. Under iterated magnification, the connectedness locus around a Misiurewicz map $F\in{mathcal S}_p$ looks more and more like the filled Julia set of F around t...

conjecture | open | MathDB 页面 | 原文线索

V2-070908 | MATHDB-367493

General congruence transformation to a diagonal-noise adjacency representation

摘录 / EXCERPT: General diagonal-noise representation conjecture. There exists a general adjacency matrix

conjecture | open | MathDB 页面 | 原文线索

V2-070909 | MATHDB-367494

Star decomposition threshold conjecture for random regular graphs

摘录 / EXCERPT: Let $d\geq 3$, let $\mathcal{G}_{N,d}$ be a uniformly random d-regular graph on N vertices, and let $k_d^{\mathrm{ind}}=\lfloor\kappa_d(\alpha_d^\star)\rfloor$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-070910 | MATHDB-367500

Noetherianity conjecture for fixed subgroups of untwisted automorphisms

摘录 / EXCERPT: Noetherianity conjecture. The set

conjecture | open | MathDB 页面 | 原文线索

V2-070911 | MATHDB-367504

Alternating-sign conjecture for SSM-Thom polynomials at relative dimension zero

摘录 / EXCERPT: Let $T^{\mathrm{SSM}}(\tau,\ell)$ denote an SSM-Thom polynomial of a singularity τ and relative dimension ℓ. For $\ell=0$, the SSM-Thom polynomials have alternating signs...

conjecture | open | MathDB 页面 | 原文线索

V2-070912 | MATHDB-367513

Girã o, Snyder, and Popielarz's linear minimum out-degree conjecture for linked tournaments

摘录 / EXCERPT: Girã o, Snyder, and Popielarz's conjecture. There exists a constant $C>0$ such that every $(2k+1)$-connected tournament with minimum out-degree at least Ck is k-linked.

conjecture | open | MathDB 页面 | 原文线索

V2-070913 | MATHDB-367514

The asymptotic expansion conjecture for zeros of $d_2(B)$ in Heun equations

摘录 / EXCERPT: Let $d_2(B)$ be the function associated with the Heun equation, the confluent Heun equation, and the reduced confluent Heun equation, and let D_k and $D_k^{[j],(k+j+1)}$ denote t...

conjecture | open | MathDB 页面 | 原文线索

V2-070914 | MATHDB-367541

A family of modulo 128 congruences for overpartitions with 6-regular nonoverlined parts

摘录 / EXCERPT: Let $\overline{R_6^\ast}(n)$ denote the overpartition function in which the nonoverlined parts are 6-regular. Modulo 128 congruence family. For all $n \geq 0$ and $k \geq 1$, we...

conjecture | open | MathDB 页面 | 原文线索

V2-070915 | MATHDB-367546

The maximal-element invariance conjecture for induced poset saturation

摘录 / EXCERPT: Let $\mathcal P$ be a finite poset, and let $\dot{\mathcal P}$ be the poset obtained by adding a new element greater than every element of $\mathcal P$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070916 | MATHDB-367548

The unboundedness conjecture for glued posets

摘录 / EXCERPT: Let $\mathcal P_1$ and $\mathcal P_2$ be finite posets, and let $\mathcal P_2*\mathcal P_1$ denote their gluing, obtained by placing every element of $\mathcal P_1$ below every ele...

conjecture | open | MathDB 页面 | 原文线索

V2-070917 | MATHDB-367549

The boundedness conjecture for gluing posets

摘录 / EXCERPT: Let $\mathcal P_1$ and $\mathcal P_2$ be finite posets with bounded saturation number, and let $\mathcal P_2*\mathcal P_1$ denote their gluing, obtained by placing every element of...

conjecture | open | MathDB 页面 | 原文线索

V2-070918 | MATHDB-367561

Roverato's conjecture on G-Wishart normalising constants

摘录 / EXCERPT: Roverato's conjecture. For every real number $\delta>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070919 | MATHDB-367577

Conjecture that Szöllősi-family matrices cannot belong to MUB-quadruplets

摘录 / EXCERPT: Szöllősi-family non-extendability conjecture. Any matrix $S\in X(\alpha)$ in the Szöllősi family cannot be part of an MUB-quadruplet.

conjecture | open | MathDB 页面 | 原文线索

V2-070920 | MATHDB-367588

The root conjecture for jointly hyponormal spherically quasinormal tuples

摘录 / EXCERPT: Root conjecture. If $\mathbf{T}^n$ is spherically quasinormal for some $n\in\mathbb{N}$, then $\mathbf{T}$ is also spherically quasinormal.

conjecture | open | MathDB 页面 | 原文线索

V2-070921 | MATHDB-367589

The antimagic threshold characterization for unions with paths on three vertices

摘录 / EXCERPT: Let G be a graph, let P_3 be the path on three vertices, and let $\tau(G)$ be the maximum integer such that the disjoint union of G and c copies of P_3 is antimagic for e...

conjecture | open | MathDB 页面 | 原文线索

V2-070922 | MATHDB-367606

Cowling's conjecture for Hausdorff étale groupoids

摘录 / EXCERPT: Let $\mathcal{G}$ be a Hausdorff étale groupoid. Weak amenability means that $\Lambda_{\text{cb}}(\mathcal{G})$ is finite, and the condition $\Lambda_{\text{cb}}(\mathcal{G})=1$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-070923 | MATHDB-367612

Positive order polynomials for width-two and shifted or cylindric skew-shape posets

摘录 / EXCERPT: Let $\Omega(P;t)$ denote the order polynomial of a finite poset P. The classes under consideration are posets of width two, cell posets of cylindric skew shapes, and cell posets...

conjecture | open | MathDB 页面 | 原文线索

V2-070924 | MATHDB-367613

Nonnegative order polynomial coefficients for width-two posets

摘录 / EXCERPT: Let $\lambda/\mu\subset [m]\times[n]$ be a skew shape encoding a width-two poset $R_{\lambda/\mu}$, with $m+n$ elements. Width-two positivity conjecture. ... Thus the normalized orde...

conjecture | open | MathDB 页面 | 原文线索

V2-070925 | MATHDB-367614

Nonnegative order polynomial coefficients for cylindric skew shapes

摘录 / EXCERPT: Let $P_{\lambda/\mu/d}$ be the cell poset associated with a cylindric skew shape λ/μ of period parameter d, and let $\Omega(P_{\lambda/\mu/d};t)$ be its order polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-070926 | MATHDB-367615

Lewis-Marberg skew GQ positivity conjecture

摘录 / EXCERPT: Skew GQ positivity conjecture. For any shifted shape λ/μ,

conjecture | open | MathDB 页面 | 原文线索

V2-070927 | MATHDB-367628

Isolated-singularity version of Zariski's multiplicity conjecture

摘录 / EXCERPT: Let $f,g\colon(\mathbb{C}^n,0)\to(\mathbb{C},0)$ be holomorphic function-germs with isolated singularities at the origin. Suppose there is a homeomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-070928 | MATHDB-367636

Converse characterization of the essential approximate point spectrum for elliptic and double parabolic Blaschke products

摘录 / EXCERPT: Converse characterization conjecture. The set $\{\lvert\lambda\rvert: \lambda \mathds{T} \subset \sigma_{usf}(T)\}$ is at most countable and its only accumulation point, if any, is...

conjecture | open | MathDB 页面 | 原文线索

V2-070929 | MATHDB-367652

The composition-lift conjecture for path-complete Lyapunov functions

摘录 / EXCERPT: Let $d4a2,d4a2$ be path-complete Lyapunov functions in $d52f\mathrm{pc}(\Sigma)$ satisfying Assumption. Suppose that $d53d$ is a subset of linear invertible maps. The composition-l...

conjecture | open | MathDB 页面 | 原文线索

V2-070930 | MATHDB-367664

The packing-coloring conjecture for bounded-degree graphs

摘录 / EXCERPT: Let $k\geq 3$, and let a k-degree graph mean a graph of maximum degree at most k. A coloring is $(1^{k-1},2^{k-1},3)$-packing colorable if its vertices can be partitioned into...

conjecture | open | MathDB 页面 | 原文线索

V2-070931 | MATHDB-367665

Characterization of depth-one Artin-Schreier defect towers

摘录 / EXCERPT: Let (K,v) be a valued field, and let an Artin-Schreier (AS) tower mean a tower M/K formed through Artin-Schreier extensions. Assume that (K,v) is Henselian, perfect, and has...

conjecture | open | MathDB 页面 | 原文线索

V2-070932 | MATHDB-367666

Avdonin-Zhao's watershed conjecture for optimal graph control time

摘录 / EXCERPT: Avdonin-Zhao's watershed conjecture. The watershed description of Avdonin and Zhao provides the optimal time for the observation and control of the free wave equation on a graph.

conjecture | open | MathDB 页面 | 原文线索

V2-070933 | MATHDB-367667

Generic graph geometric control conjecture for the Schrödinger equation

摘录 / EXCERPT: Generic Schrödinger control conjecture. Generically with respect to the lengths $(\ell_i)_{i=1,\dots,N}$, (GGCC) is a necessary and sufficient condition for the internal control of...

conjecture | open | MathDB 页面 | 原文线索

V2-070934 | MATHDB-367668

Generic graph geometric control conjecture for the wave equation

摘录 / EXCERPT: Generic wave control conjecture. Generically with respect to the lengths $(\ell_i)_{i=1,\dots,N}$, (GGCC) is a necessary and sufficient condition for the internal control of the wa...

conjecture | open | MathDB 页面 | 原文线索

V2-070935 | MATHDB-367669

Generic small-time null-controllability conjecture for the heat equation on graphs

摘录 / EXCERPT: Generic heat-control conjecture. Generically with respect to the lengths $(\ell_i)_{i=1,\dots,N}$, the heat equation is small-time null-controllable.

conjecture | open | MathDB 页面 | 原文线索

V2-070936 | MATHDB-367673

Categorical hypercompleteness conjecture for finite and infinite categories

摘录 / EXCERPT: Let $n\text{-}\operatorname{Cat}$ denote the category of n-categories, let $\operatorname{Surj}_{\omega}$ denote the class of ω-surjective morphisms, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070937 | MATHDB-367677

The dimension growth conjecture for smooth projective hypersurfaces

摘录 / EXCERPT: Dimension growth conjecture. For every $\varepsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070938 | MATHDB-367690

Jossen's conjecture on division and common zeros of E-functions

摘录 / EXCERPT: Jossen's conjecture. (i) If f and g are two E-functions such that f/g is entire, then f/g is an E-function. (ii) If two E-functions f and g share at least one com...

conjecture | open | MathDB 页面 | 原文线索

V2-070939 | MATHDB-367691

Jossen's factorization conjecture for E-functions

摘录 / EXCERPT: Jossen's factorization conjecture. Every non-zero E-function f can be written as a finite product of powers of E-functions with simple zeros, with distinct factors having no...

conjecture | open | MathDB 页面 | 原文线索

V2-070940 | MATHDB-367692

Bessel common-zero conjecture for E-functions

摘录 / EXCERPT: Bessel common-zero conjecture. If f and J_α share a common zero in $\mathbb C^*$, then

conjecture | open | MathDB 页面 | 原文线索

V2-070941 | MATHDB-367693

Ritt-type factorization conjecture for E-functions

摘录 / EXCERPT: Ritt-type factorization conjecture. For every non-zero E-function f, there exist a unit $u\in\mathcal E^\times$, simple normalized E-functions $s_1,\ldots,s_p$ with pairwise...

conjecture | open | MathDB 页面 | 原文线索

V2-070942 | MATHDB-367694

Irreducible common-zero conjecture for E-functions

摘录 / EXCERPT: Irreducible common-zero conjecture. If h_1 and h_2 have at least one common zero in $\mathbb C$, then

conjecture | open | MathDB 页面 | 原文线索

V2-070943 | MATHDB-367711

The square-free palindrome conjecture in every base

摘录 / EXCERPT: Fix a base $b\geq 2$ and let $\overleftarrow{n}$ denote the integer obtained by reversing the base-b digits of n. A square-free palindrome is a positive integer that is square-...

conjecture | open | MathDB 页面 | 原文线索

V2-070944 | MATHDB-367714

Kolyvagin-inspired conjecture on all 2-Selmer ranks in quadratic twist families

摘录 / EXCERPT: Let ${\mathcal {E}}$ be a family of type (B) or (C). For a Σ-equivalence class ${\mathfrak{X}}$, let $r({\mathfrak{X}})$ denote its parity parameter and let...

conjecture | open | MathDB 页面 | 原文线索

V2-070945 | MATHDB-367716

Kuznetsov-Shinder conjecture that D-equivalence implies L-equivalence

摘录 / EXCERPT: Kuznetsov-Shinder conjecture. If X and Y are D-equivalent, then they are L-equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-070946 | MATHDB-367718

The local-global conjecture for the fourth level of number fields

摘录 / EXCERPT: Let K be a number field with finite fourth level, and let $s_4(K)$ denote the least number of fourth powers of elements of K whose sum is -1 when such a representation exists...

conjecture | open | MathDB 页面 | 原文线索

V2-070947 | MATHDB-367719

The leapfrogging conjecture for concentrated vortex rings

摘录 / EXCERPT: Let $N>1$ and $0<\varepsilon\ll 1$. Suppose that the initial vorticity ω_0^ε is decomposed into N blobs, each with circulation $\gamma>0$, sharply concentrated a...

conjecture | open | MathDB 页面 | 原文线索

V2-070948 | MATHDB-367720

Poisson summation conjecture for the triple product space

摘录 / EXCERPT: Poisson summation conjecture. One can choose these Schwartz spaces so that the properties (FT1), (FT2), (FT3), (basic:norm), (rapidlydecreasing), (Frechet), (nA:ratio), (Arch:ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-070949 | MATHDB-367734

Burstein-Han-Kitaev-Zhang shape-Wilf-equivalence conjecture

摘录 / EXCERPT: Burstein-Han-Kitaev-Zhang conjecture. The two displayed partially ordered patterns are shape-Wilf-equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-070950 | MATHDB-367736

Arras-Joos conjecture for independent sets in Cartesian products of triangles

摘录 / EXCERPT: Arras-Joos conjecture. As t tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-070951 | MATHDB-367742

Stein's conjecture on oriented paths in graphs of large semidegree

摘录 / EXCERPT: Let G be an oriented graph, let $\b7$ denote its minimum semidegree, and let k be a nonnegative integer. An orientation of the k-edge path is an oriented graph obtained by as...

conjecture | open | MathDB 页面 | 原文线索

V2-070952 | MATHDB-367743

Isoperimetric inequality for spacelike star-shaped hypersurfaces in De Sitter space

摘录 / EXCERPT: Let $M\subset\mathbb S^{n+1}_1$ be a spacelike, compact, star-shaped, and k-convex hypersurface. For $1\leq k\leq n$, let $\mathcal A_k$ denote the quermassintegral-type quantity...

conjecture | open | MathDB 页面 | 原文线索

V2-070953 | MATHDB-367761

Bayer-Macrì-Toda generalized Bogomolov-Gieseker conjecture

摘录 / EXCERPT: Bayer-Macrì-Toda generalized Bogomolov-Gieseker conjecture. For any $Z_{\alpha,\beta}^{2}$-semistable object $E\in\mathcal{B}_{H}^{\beta}$ satisfying

conjecture | open | MathDB 页面 | 原文线索

V2-070954 | MATHDB-367762

Conjecture on delayed blowup under dispersion or nonlinearity management

摘录 / EXCERPT: Delayed-blowup conjecture. The corresponding solution of either managed model does not blow up in $[0,T]$; thus, blowup can be delayed by dispersion or nonlinearity management.

conjecture | open | MathDB 页面 | 原文线索

V2-070955 | MATHDB-367763

Existence of dispersion-manageable managed NLS data

摘录 / EXCERPT: Dispersion-manageability conjecture. There exists a Schwartz function u_0 such that, with $u(0,x)=u_0(x)$, either $(NLS)_\gamma$ or $(NLS)^\gamma$ is dispersion-manageable.

conjecture | open | MathDB 页面 | 原文线索

V2-070956 | MATHDB-367765

The golden-ratio conjecture for radii of convergence of q-deformed real numbers

摘录 / EXCERPT: Let $[x]_q$ denote the q-deformation of a real number $x>0$, viewed as a power series in the complex variable q, and let $R(x)$ be its radius of convergence. For the golden rat...

conjecture | open | MathDB 页面 | 原文线索

V2-070957 | MATHDB-367768

Recursive decomposition coefficient inequality for real-rooted f-polynomials

摘录 / EXCERPT: Recursive decomposition inequality. For every relevant index i, one has $h_i\leq g_i$.

conjecture | open | MathDB 页面 | 原文线索

V2-070958 | MATHDB-367778

The period-integral commutator relation

摘录 / EXCERPT: Period-integral commutator relation. Under $\thref{commsetup}$,

conjecture | open | MathDB 页面 | 原文线索

V2-070959 | MATHDB-367779

The semi-local period-integral commutator relation

摘录 / EXCERPT: Semi-local period-integral commutator relation. Under $\thref{genassum}$,

conjecture | open | MathDB 页面 | 原文线索

V2-070960 | MATHDB-367782

Hydrodynamic speed convergence for the liquid bin model

摘录 / EXCERPT: Let $p_1+\dots+p_N=1$, let $a_1,\ldots,a_N>0$, and for $s\geq 1/a_1$ let $X^{(s)}$ be the infinite bin model with move distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-070961 | MATHDB-367784

Bangachev's minimum-degree conjecture for friends-and-strangers graphs

摘录 / EXCERPT: Let X and Y be connected graphs on n vertices. For a graph G, write $\delta(G)$ for its minimum degree, and let $\operatorname{FS}(X,Y)$ denote the friends-and-strangers gr...

conjecture | open | MathDB 页面 | 原文线索

V2-070962 | MATHDB-367791

Non-smoothability conjecture for Gorenstein Stöhr-Torres monomial curves

摘录 / EXCERPT: Let C be a Gorenstein monomial curve of Stöhr-Torres type. Stöhr-Torres non-smoothability conjecture. The curve C is not smoothable. The conjecture is motivated by the example...

conjecture | open | MathDB 页面 | 原文线索

V2-070963 | MATHDB-367792

Smooth-branch conjecture for generic curve singularities

摘录 / EXCERPT: Let C be a generic curve singularity, in the sense proposed in the surrounding discussion: it is parametrised by a generic point of a component of a base space of a versal deform...

conjecture | open | MathDB 页面 | 原文线索

V2-070964 | MATHDB-367795

The asymptotic-dimension conjecture for bounded sphere dimension

摘录 / EXCERPT: Let $d\geq 1$ be an integer, and consider the class of graphs whose sphere dimension is at most d. Asymptotic-dimension conjecture. This class has asymptotic dimension at most...

conjecture | open | MathDB 页面 | 原文线索

V2-070965 | MATHDB-367796

The quasi-isometry conjecture for bounded sphere dimension

摘录 / EXCERPT: Let $d\geq 2$ be an integer, and consider the class of graphs with sphere dimension at most d. Quasi-isometry conjecture. This class is quasi-isometric to the class of intersecti...

conjecture | open | MathDB 页面 | 原文线索

V2-070966 | MATHDB-367811

Guo-Kang-Zwaneveld conjecture on defective coloring of clique blowups

摘录 / EXCERPT: Guo-Kang-Zwaneveld conjecture. For every graph G and every integer $d\ge 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070967 | MATHDB-367813

Classification of minimal-mass blowup solutions

摘录 / EXCERPT: Classification of minimal-mass blowup solutions. Only the following two cases occur:

conjecture | open | MathDB 页面 | 原文线索

V2-070968 | MATHDB-367814

Shallow quasinormal-mode convergence conjecture

摘录 / EXCERPT: Shallow-QNM convergence conjecture. As $r_{\rm C} earrow r_{\rm e}$, the set

conjecture | open | MathDB 页面 | 原文线索

V2-070969 | MATHDB-367820

Arsie-Lorenzoni-Moro conjecture on normal forms of conservation-law deformations

摘录 / EXCERPT: Consider a deformation of the Riemann hierarchy ... where the flow with respect to t_1 is in normal form: ... with $c_\lambda(u)\in\mathbb C[[u]]$. Arsie-Lorenzoni-Moro conjecture. I...

conjecture | open | MathDB 页面 | 原文线索

V2-070970 | MATHDB-367833

Infinite ultrametric extension conjecture with four-point obstructions

摘录 / EXCERPT: Let (X,d) be an infinite ultrametric space. A metric space is weakly similar to another when there is a similarity between them, allowing a positive rescaling of distances. Infin...

conjecture | open | MathDB 页面 | 原文线索

V2-070971 | MATHDB-367834

Starlike rayless-tree characterization conjecture for compact generated ultrametric spaces

摘录 / EXCERPT: Let (X,d) be an infinite ultrametric space. A labeled ray is a ray equipped with a labeling that generates the indicated ultrametric space. A tree is starlike if it has exactly o...

conjecture | open | MathDB 页面 | 原文线索

V2-070972 | MATHDB-367843

Cohen-Danchin logarithmic-sharpness conjecture for acute-corner cusps

摘录 / EXCERPT: Consider a vortex patch whose boundary has a single corner of acute angle, and define the order of a cusp at $x=x_0$ by ... where f_1 and f_2 represent the graphs of the two side...

conjecture | open | MathDB 页面 | 原文线索

V2-070973 | MATHDB-367870

The type B hypersimplex reciprocity conjecture

摘录 / EXCERPT: For $n\geq 3$, let $\Psi_{B_n}(t)$ denote the Ehrhart h^*-polynomial of the type B_n hypersimplex, and let $E_{D_n}(t)$ denote the type D_n Eulerian polynomial. Type B recipr...

conjecture | open | MathDB 页面 | 原文线索

V2-070974 | MATHDB-367876

Gross-Prasad's genericity conjecture for L-packets

摘录 / EXCERPT: Let G be a reductive p-adic group and let ϕ be a Langlands parameter, with associated L-packet $\Pi_\phi(G)$. A representation is generic if it admits a Whittaker model. G...

conjecture | open | MathDB 页面 | 原文线索

V2-070975 | MATHDB-367879

Hausel-Hitchin mirror conjecture for upward flows

摘录 / EXCERPT: Let C be a smooth projective curve, let $M_{r,d}$ be the moduli space of semistable Higgs bundles on C, and let $\mathbb G_m$ act by $\lambda\cdot(E,\phi)=(E,\lambda\phi)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-070976 | MATHDB-367882

Paper Boat zone partition conjecture

摘录 / EXCERPT: Paper Boat zone conjecture. For every affine Weyl group there is a partition $\mathcal{Z}$ of $(\Lambda^\vee)^+$ with at most 3^n parts such that, for every $Z\in\mathcal{Z}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-070977 | MATHDB-367883

Geometric polynomial conjecture for lower Bruhat interval sizes

摘录 / EXCERPT: Geometric polynomial conjecture. For any affine Weyl group, the map q_a is a geometric polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-070978 | MATHDB-367885

Anderson-Vamanamurthy-Vuorinen elliptic-integral monotonicity conjecture

摘录 / EXCERPT: Let $\mathcal{K}(r)=(\pi/2)F(1/2,1/2;1;r^2)$ be the complete elliptic integral of the first kind. For $r\in(0,1)$, set $r'=\sqrt{1-r^2}$. Anderson-Vamanamurthy-Vuorinen's conjectur...

conjecture | solved | MathDB 页面 | 原文线索

V2-070979 | MATHDB-367890

Nikolov's non-negative Chebyshev expansion conjecture for snake polynomials

摘录 / EXCERPT: Let $\mu\in C[-1,1]$ be a nonnegative majorant, and let $\omega_\mu\in\mathcal{P}_n$ be the snake polynomial associated with μ, meaning that $|\omega_\mu(x)|\leq\mu(x)$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-070980 | MATHDB-367902

Integrality conjecture for the recursively defined polynomials u_n

摘录 / EXCERPT: Let $u_n\in\mathbb{Q}[b,\lambda]$ be a sequence defined by the recursion ... where $u_{-1}=0$ and $u_0=1$. Integrality conjecture. Then ... This conjecture formalizes the observed inte...

conjecture | open | MathDB 页面 | 原文线索

V2-070981 | MATHDB-367918

Affine invariance conjecture for the Poisson-series function class

摘录 / EXCERPT: Let $\mathscr C$ denote the class of functions for which the Poisson series under consideration has the required convergence properties, and let an affine transformation of the var...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-070982 | MATHDB-367926

Conjecture on extreme Dirichlet L-values in cosets

摘录 / EXCERPT: Let $q>1$, let $\mathcal{X}$ be the group of Dirichlet characters modulo q, and let $H\leq\mathcal{X}$ be a subgroup satisfying $\#H\gg q^\varepsilon$. For any coset representati...

conjecture | open | MathDB 页面 | 原文线索

V2-070983 | MATHDB-367930

Domination of prefix sums for surjective binary cellular automata

摘录 / EXCERPT: Let F be a surjective binary cellular automaton with radius r, and let $N^1_1(F,r,k)$ denote the corresponding symbol-frequency quantity; write id for the identity c...

conjecture | open | MathDB 页面 | 原文线索

V2-070984 | MATHDB-367931

The substantial unfolding characterization conjecture for quasi-homogeneous map-germs

摘录 / EXCERPT: Let $f\colon(\mathbb{C}^n,0)\to(\mathbb{C}^p,0)$ be an $\mathcal{A}$-finite map-germ whose stable unfolding with the minimal number of parameters lies in the nice dimensions. A sta...

conjecture | open | MathDB 页面 | 原文线索

V2-070985 | MATHDB-367932

Buckmaster-Vicol Onsager-type conjecture for Elsässer energies in ideal MHD

摘录 / EXCERPT: Let (u,b) be a weak solution of the ideal MHD system, and define the total energy and cross helicity by ... ... Here $C_{x,t}^\alpha$ denotes joint Hölder regularity and...

conjecture | open | MathDB 页面 | 原文线索

V2-070986 | MATHDB-367934

Non-localized data long-time well-posedness conjecture

摘录 / EXCERPT: Consider a one-dimensional dispersive flow on the real line with a cubic conservative nonlinearity and initial data of size $epsilon \ll 1$. Non-localized data long-time well-posed...

conjecture | open | MathDB 页面 | 原文线索

V2-070987 | MATHDB-367935

Higher-dimensional non-localized data global well-posedness conjecture

摘录 / EXCERPT: Consider a cubic quasilinear dispersive flow with small initial data in spatial dimension at least two. Higher-dimensional non-localized data global well-posedness conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-070988 | MATHDB-367937

Global well-posedness conjecture for non-trapping initial data

摘录 / EXCERPT: Let the initial data be non-trapping. Global well-posedness conjecture for non-trapping initial data. The result in Theorem holds for all non-trapping initial data. The source give...

conjecture | open | MathDB 页面 | 原文线索

V2-070989 | MATHDB-367941

The type I singularity conjecture for Kähler-Ricci flow on Kähler surfaces

摘录 / EXCERPT: Type I singularity conjecture. Every finite-time singularity of the Kähler-Ricci flow on a Kähler surface is type I.

conjecture | open | MathDB 页面 | 原文线索

V2-070990 | MATHDB-367942

Intersection bounds for affine half-intervals over finite fields

摘录 / EXCERPT: Intersection-bound conjecture. If $t_1^{-1}t_2\notin\{1,n-1\}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-070991 | MATHDB-367946

Hayman's conjecture on generalized Picard exceptional values of $f^n f'$

摘录 / EXCERPT: Hayman's conjecture. The differential polynomial $f^n f'$ cannot have any non-zero generalized Picard exceptional values.

conjecture | open | MathDB 页面 | 原文线索

V2-070992 | MATHDB-367948

Saikia and Sarma's congruence conjectures for overcubic partition triples

摘录 / EXCERPT: Saikia and Sarma's conjecture. For $n,\alpha\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-070993 | MATHDB-367951

Hegedüs' conjecture on non-exceptional q-ary equidistant codes

摘录 / EXCERPT: Let H_q^n be the Hamming space over an alphabet of size q, and let $C\subseteq H_q^n$ be an equidistant code with distance d. Hegedüs' conjecture. If ... then ... This conjecture...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-070994 | MATHDB-367956

The categorical arithmetic local Langlands conjecture

摘录 / EXCERPT: Assume that G is quasi-split over F, equipped with a pinning (B,T,e), and fix a non-trivial additive character $\psi:F\to {\mathbb Z}_\ell[\mu_{p^\infty}]^\times$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-070995 | MATHDB-367959

Clozel-Harris-Taylor's generalized Ihara lemma for definite unitary groups

摘录 / EXCERPT: Clozel-Harris-Taylor's generalized Ihara lemma. Then V is Whittaker-generic.

conjecture | open | MathDB 页面 | 原文线索

V2-070996 | MATHDB-367967

Fritzsche-Külshammer-Reiche character-theoretic formula for the base size of ${\rm S}_{n,k}$

摘录 / EXCERPT: Let $n\geq 2k$, let $G={\rm S}_{n,k}$, and let H be a point stabiliser of G. Write $b(G)$ for the base size of the permutation group G, let $\mathcal{K}(G,H)$ be the Külshamm...

conjecture | open | MathDB 页面 | 原文线索

V2-070997 | MATHDB-367968

Bekka-Valette conjecture on representations of lattices

摘录 / EXCERPT: Bekka-Valette conjecture. There exist unitary representations of Γ that are not weakly contained in any representation restricted from G.

conjecture | open | MathDB 页面 | 原文线索

V2-070998 | MATHDB-367976

Polynomial χ-boundedness conjecture for bounded merge-width classes

摘录 / EXCERPT: A graph class is χ-bounded if there is a function f such that every graph G in the class satisfies $\chi(G)\leq f(\omega(G))$, where $\chi(G)$ is the chromatic number and...

conjecture | open | MathDB 页面 | 原文线索

V2-070999 | MATHDB-367978

Faudree-Fronček-Ryjáček-Locke-Langley conjecture on 2-factors in claw-free graphs

摘录 / EXCERPT: Let G be a claw-free graph, meaning that G has no induced subgraph isomorphic to $K_{1,3}$. Let $\delta(G)$ denote its minimum degree and let $\alpha(G)$ denote its independenc...

conjecture | open | MathDB 页面 | 原文线索

V2-071000 | MATHDB-367979

Bounded-cycle 2-factor conjecture under degree-sum and independent-set conditions

摘录 / EXCERPT: Let k be a positive integer and let G be a graph of order n. For an independent set of order $k+1$, define ... when $\alpha(G)\geq k+1$, and set $\sigma_{k+1}(G)=\infty$ otherw...

conjecture | open | MathDB 页面 | 原文线索

V2-071001 | MATHDB-367980

Clunie-Sheil-Small coefficient conjecture for normalized harmonic univalent mappings

摘录 / EXCERPT: Clunie-Sheil-Small's coefficient conjecture. For every $n\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-071002 | MATHDB-367981

Conjectured covering radius for K-quasiconformal harmonic mappings

摘录 / EXCERPT: Covering-radius conjecture. Every function $f\in\mathcal{S}^0_H(K)$ contains the disk

conjecture | open | MathDB 页面 | 原文线索

V2-071003 | MATHDB-367982

Coefficient conjecture for K-quasiconformal harmonic mappings

摘录 / EXCERPT: Let $f=h+\overline{g}\in\mathcal{S}^0_H(K)$, with h and g having the series representations specified in the source. Let $A(n,k)$ and $B(n,k)$ be the quantities defined in the...

conjecture | open | MathDB 页面 | 原文线索

V2-071004 | MATHDB-367986

Uniqueness conjecture for the positive dP_I^{(2)} solution

摘录 / EXCERPT: Let $\kappa,\tau\in\mathbb{R}$, and consider the dP$_{\rm I}^{(2)}$ equation for a sequence $\{\beta_n(\kappa,\tau)\}_{n\geq 0}$, with initial values specified by the source. A sol...

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V2-071005 | MATHDB-367987

Higher-order cluster-expansion conjecture for the Ising model on even tori

摘录 / EXCERPT: Fix $k\in\mathbb{N}$. For the Ising partition function $Z_{\mathbb{Z}_m^t}(\lambda,\beta)$ on the even torus $\mathbb{Z}_m^t$, let L_j denote the sum of the weights of clusters o...

conjecture | open | MathDB 页面 | 原文线索

V2-071006 | MATHDB-367988

Strict monotonicity of the equilibrium parameters in non-symmetric Showcase Showdown

摘录 / EXCERPT: For each integer $n>1$, let (ϵ_n,δ_n) be the unique solution in $[0,1]\times[0,1]$ of ... with $\delta_n>\epsilon_n$. Strict-monotonicity conjecture. The sequences...

conjecture | open | MathDB 页面 | 原文线索

V2-071007 | MATHDB-367996

Malle's prediction for twist classes of genus two curves with dihedral automorphisms

摘录 / EXCERPT: Malle's prediction. If $\operatorname{Aut}_{\overline{K}}(C)\cong D_4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071008 | MATHDB-368002

Structural implications of the d_{An}-Cauchy condition

摘录 / EXCERPT: Structural implications of the d_{An}-Cauchy condition. The requirement that $\{X_i\}$ be d_{An}-Cauchy ensures that the limit flow θ inherits strong s...

conjecture | open | MathDB 页面 | 原文线索

V2-071009 | MATHDB-368007

No-local-minimum conjecture for the discrete signed-mass energy

摘录 / EXCERPT: Let $x=(x_1,\ldots,x_d)$ with $x_i\in{\mathbb R}^n$, signed masses $m(x_i)$ satisfying ... and define ... A configuration is stationary and stable when its first variation vanishes and...

conjecture | open | MathDB 页面 | 原文线索

V2-071010 | MATHDB-368041

Profinite rigidity of direct products of finitely generated free groups among residually free groups

摘录 / EXCERPT: Direct-product profinite rigidity conjecture. If a finitely presented residually free group Γ has the same finite quotients as $F\times F'$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071011 | MATHDB-368042

Chordal graph conjecture for square-free powers

摘录 / EXCERPT: Let G be a chordal graph, let $I(G)$ be its edge ideal, let $I(G)^{[k]}$ denote its k-th square-free power, let $\nu(G)$ be its matching number, and let $\mathrm{aim}(G,k)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-071012 | MATHDB-368049

Kalinin's polynomial congruence conjecture for Gaussian integers

摘录 / EXCERPT: Kalinin's conjecture. If $p\equiv1\pmod 4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071013 | MATHDB-368050

Mészáros's critical-distribution conjecture for band-matrix cokernels

摘录 / EXCERPT: Mészáros's critical-distribution conjecture. For every finite abelian p-group G,

conjecture | open | MathDB 页面 | 原文线索

V2-071014 | MATHDB-368053

Yu's folklore conjecture for Hamming-cube t-codes

摘录 / EXCERPT: Let $\mathcal{H}_n=\mathbb{F}_2^n$ with Hamming distance, and let $A(\mathcal{H}_n,t)$ denote the largest size of a t-code in $\mathcal{H}_n$, meaning a set with at most t perm...

conjecture | open | MathDB 页面 | 原文线索

V2-071015 | MATHDB-368056

Yoo's conjecture on the rational cuspidal divisor class group

摘录 / EXCERPT: Let N be a positive integer. Let $\mathscr{C}_N(\mathbf{Q})$ be the rational cuspidal subgroup of $J_0(N)$, and let $\mathscr{C}(N)$ be the rational cuspidal divisor class group...

conjecture | open | MathDB 页面 | 原文线索

V2-071016 | MATHDB-368058

The $8/5$ balanced-coloring conjecture for signed subcubic graphs

摘录 / EXCERPT: Let G be a signed subcubic graph, meaning a graph of maximum degree at most 3 whose edges have signs. Let $(K_4,-)$ denote the negatively signed complete graph on four vertices...

conjecture | open | MathDB 页面 | 原文线索

V2-071017 | MATHDB-368064

Grassmannian and bigrassmannian signed permutation avoidance conjecture

摘录 / EXCERPT: Grassmannian and bigrassmannian avoidance conjecture. A signed permutation is Grassmannian if and only if it globally avoids every pattern in P, and it is bigrassmannian if and o...

conjecture | open | MathDB 页面 | 原文线索

V2-071018 | MATHDB-368070

Bogomolov property conjecture for elliptic torsion fields

摘录 / EXCERPT: Let K be a number field and let E be an elliptic curve over K. Write E_{tor} for the torsion subgroup of $E(\overline{\mathbb{Q}})$, and let $K(E_{\text{tor}})$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-071019 | MATHDB-368080

The split semi-elliptic conjecture

摘录 / EXCERPT: Split semi-elliptic conjecture. The transcendence degree of K is at least $s+2n$, unless $2\pi\mathrm{i}\mathbb{Q}\subseteq\mathbb{Q}t_1+\cdots+\mathbb{Q}t_s$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071020 | MATHDB-368081

The elliptic Schanuel conjecture

摘录 / EXCERPT: Let Ω be a lattice in $\mathbb{C}$, let k be the associated elliptic curve's endomorphism field, and let g_2,g_3 be its invariants. Let $p_1,\dots,p_n$ be k-linearly i...

conjecture | open | MathDB 页面 | 原文线索

V2-071021 | MATHDB-368084

The elliptico--toric conjecture

摘录 / EXCERPT: Let $\mathcal{E}$ be an elliptic curve with period lattice Ω, invariants g_2,g_3, and endomorphism field k. Let $s,n\geq0$, let $\mathrm{e}^{t_1},\dots,\mathrm{e}^{t_s}$...

conjecture | open | MathDB 页面 | 原文线索

V2-071022 | MATHDB-368087

Genus-zero mirror-map conjecture for complete intersections in weighted projective space

摘录 / EXCERPT: Let $P(a_1,\dots,a_N\mid k_1,\dots,k_m)$ be a nonsingular complete intersection in weighted projective space. Let $t^p(x^*)$ be the mirror maps defined from the genus-zero virtual...

conjecture | open | MathDB 页面 | 原文线索

V2-071023 | MATHDB-368094

Conjectured Chernoff limit for the uniform deconvolution MLE

摘录 / EXCERPT: Let F_0 have a continuous positive density f_0 on $(0,M)$, where $M>0$, and let $t_0\to(a,b)$ be an interior point. Let $\hat F_n$ be the nonparametric maximum likelihood estim...

conjecture | open | MathDB 页面 | 原文线索

V2-071024 | MATHDB-368102

Grünbaum's equitable coloring conjecture

摘录 / EXCERPT: Let G be a graph with maximum degree $\Delta(G)$, and let a k-equitable coloring be a proper coloring with k color classes whose sizes differ by at most one. Grünbaum's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-071025 | MATHDB-368103

Czygrinow-DeBiasio-Kierstead-Molla universal-graph tiling conjecture

摘录 / EXCERPT: Let $\vec{G}$ be a directed graph, and call a directed graph H universal if every tournament T with $|T|=|H|$ is a subgraph of H. Let $n=km$. Czygrinow-DeBiasio-Kierstead-Mol...

conjecture | open | MathDB 页面 | 原文线索

V2-071026 | MATHDB-368105

Kostochka-Kierstead Ore-degree Chen-Lih-Wu conjecture

摘录 / EXCERPT: For a graph G, define its Ore-degree by ... A proper coloring is equitable when its color classes differ in size by at most one. Kostochka-Kierstead's Ore-degree conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071027 | MATHDB-368106

Strong Ore-degree Chen-Lih-Wu decomposition conjecture

摘录 / EXCERPT: Let $k\geq3$, and let a k-decomposable graph be one admitting the k-decomposition defined in the source. Strong Ore-degree Chen-Lih-Wu conjecture. If G is a k-colorable gra...

conjecture | open | MathDB 页面 | 原文线索

V2-071028 | MATHDB-368107

Kostochka-Yu equitable list-coloring Ore conjecture

摘录 / EXCERPT: For a graph G, define its Ore-degree by ... A graph is equitably k-choosable if every k-list assignment admits a proper coloring in which each color class has size at most...

conjecture | open | MathDB 页面 | 原文线索

V2-071029 | MATHDB-368108

Strongly equitable Ore-degree list-coloring conjecture

摘录 / EXCERPT: For a graph G, define its Ore-degree by ... A graph is SE k-choosable if every k-list assignment admits a strongly equitable list coloring. Strongly equitable Ore-degree conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071030 | MATHDB-368123

The surjectivity characterization conjecture for LR-separated cellular automata

摘录 / EXCERPT: Surjectivity characterization conjecture. F is surjective if and only if either

conjecture | open | MathDB 页面 | 原文线索

V2-071031 | MATHDB-368127

The Möbius-band Gram determinant conjecture for diagrams with one crosscap curve

摘录 / EXCERPT: Let $Mb_{n,1}$ be the set of crossingless-connection diagrams between $2n$ marked boundary points of the Möbius band, with exactly one curve intersecting the crosscap. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071032 | MATHDB-368128

The Gram determinant conjecture for Möbius-band diagrams with one crosscap curve

摘录 / EXCERPT: Let $R=\mathbb{Z}[A^{\pm1},w,x,y,z]$. For $n\geq 1$, let $D_n^{(Mb)_1}$ denote the Gram determinant of type $(Mb)_1$, where the Gram matrix is formed from crossingless-connection d...

conjecture | open | MathDB 页面 | 原文线索

V2-071033 | MATHDB-368129

Qi Chen's Gram determinant conjecture for the Möbius band

摘录 / EXCERPT: Let $R=\mathbb{Z}[A^{\pm1},w,x,y,z]$. For $n\geq1$, let $D_n^{\mathit{Mb}}(d,w,x,y,z)$ be the Gram determinant of type Mb. Let T_k and S_k denote the Chebyshev polyn...

conjecture | open | MathDB 页面 | 原文线索

V2-071034 | MATHDB-368132

Christol-André conjecture for globally bounded D-finite power series

摘录 / EXCERPT: Let $fin \mathbb{Q}[[z]]$ be a globally bounded and D-finite power series, and let L^{min}_f denote its minimal annihilating differential operator. Christol-André conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071035 | MATHDB-368135

Miller-Scheinerman zero-density conjecture for the character table of the symmetric group

摘录 / EXCERPT: Let $Z(N)$ be the number of zero entries in the character table of the symmetric group S_N, and let $p(N)$ be the number of partitions of N. Miller-Scheinerman's conjecture. As...

conjecture | open | MathDB 页面 | 原文线索

V2-071036 | MATHDB-368144

Objective-value conjecture for the naive semidefinite relaxation of the mKPC

摘录 / EXCERPT: Consider an instance of the multiple knapsack problem with compacity constraints, its linear relaxation $(\mathrm{mKPC})_{\mathrm{LP}}$, and its naive semidefinite relaxation. Writ...

conjecture | open | MathDB 页面 | 原文线索

V2-071037 | MATHDB-368145

Counterexample conjecture to the ROAD property for compact knapsack instances

摘录 / EXCERPT: For each integer $m\geq 2$, consider the integer program $(\mathrm{CE})_m$ with variables $x\in\{0,1\}^{2m}$ and its linear relaxation. Let x_m° denote an optimal solution o...

conjecture | refuted | MathDB 页面 | 原文线索

V2-071038 | MATHDB-368149

Rank-four tadpole Rogers-Ramanujan companion identity

摘录 / EXCERPT: Let q be a formal variable, and for $n\geq0$ let $(q^4;q^4)_n$ and $(q^m;q^m)_\infty$ denote the usual finite and infinite q-Pochhammer products; products with several argument...

open_problem | open | MathDB 页面 | 原文线索

V2-071039 | MATHDB-368164

The claw-free four-perfect-matching conjecture

摘录 / EXCERPT: A cubic graph is claw-free if it has no induced subgraph isomorphic to $K_{1,3}$. A set of perfect matchings covers a graph when every edge belongs to at least one matching. Claw-f...

conjecture | open | MathDB 页面 | 原文线索

V2-071040 | MATHDB-368173

Weak kappa conjecture for codes of defect s

摘录 / EXCERPT: Let $\kappa(s,q)$ be the invariant introduced in the paper for defect parameter s and field size q, with $s e 0$ allowed only where the paper's definition applies. Weak...

conjecture | open | MathDB 页面 | 原文线索

V2-071041 | MATHDB-368174

Strong kappa conjecture for codes of defect s

摘录 / EXCERPT: Let $\kappa(s,q)$ be the invariant introduced in the paper for defect parameter s and field size q. Strong $\kappa(s,q)$ conjecture. If $s\ge 0$, then ... This refines the weak c...

conjecture | open | MathDB 页面 | 原文线索

V2-071042 | MATHDB-368184

The Frobenius uniqueness conjecture for degree-9 Wahl chains

摘录 / EXCERPT: Frobenius uniqueness conjecture. A Wahl chain $\texttt{W}$ in $\mathbb{W}_9$ is determined by its index.

conjecture | open | MathDB 页面 | 原文线索

V2-071043 | MATHDB-368189

Invertible-summand conjecture for anisotropic quadrics

摘录 / EXCERPT: Let Q be an anisotropic quadric over k, with defining quadratic form q, and let $\widetilde{M}_{\operatorname{K}(n)}(Q)$ be its $\operatorname{K}(n)$-kernel motive. Invertibl...

conjecture | open | MathDB 页面 | 原文线索

V2-071044 | MATHDB-368190

The dilation asymptotic conjecture for compact sets

摘录 / EXCERPT: Let K be a non-empty compact subset of the complex plane, and let δ_n and C denote the quantities defined in the preceding results. Dilation asymptotic conjecture. ... Thi...

conjecture | open | MathDB 页面 | 原文线索

V2-071045 | MATHDB-368191

The polar-set characterization for strongly incomplete polynomial approximation

摘录 / EXCERPT: Let $K\subset {\mathbb{C}}\setminus \{0\}$ be a compact set. A polynomial of the indicated form is $\sum_{k=n/\tau_n}^{n}a_kz^k$, where $\tau_n\to 1$. Polar-set characterization co...

conjecture | open | MathDB 页面 | 原文线索

V2-071046 | MATHDB-368196

Repulsive periodic-point equidistribution conjecture for non-Archimedean rational maps

摘录 / EXCERPT: Fix a rational map R over K of degree $d\geq 2$. A fixed point x is called repulsive if either $x\in \operatorname{\mathcal{C}}_R\cap \mathbb{H}_K$, or $x\in \mathbb{P}^1_K$...

conjecture | open | MathDB 页面 | 原文线索

V2-071047 | MATHDB-368197

Maximal-entropy measure conjecture for non-Archimedean rational maps

摘录 / EXCERPT: Fix a rational map R over K of degree $d\geq 2$, and for $n\in\mathbb{N}^*$ let $\operatorname{RFix}(R^n)$ be the set of repulsive fixed points of R^n. Let μ_R be a measu...

conjecture | open | MathDB 页面 | 原文线索

V2-071048 | MATHDB-368198

Vanishing hyperbolic repulsive-point mass conjecture for non-Archimedean rational maps

摘录 / EXCERPT: Fix a rational map R over K of degree $d\geq 2$, let $\chi(R)$ be its Lyapunov exponent, let $\mathbb{H}_K$ denote the hyperbolic space, and let $\operatorname{RFix}(R^n)$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-071049 | MATHDB-368199

Uniform bounded-period conjecture for repelling periodic points of non-Archimedean rational maps

摘录 / EXCERPT: Uniform bounded-period conjecture. There exists an integer $N(d)\ge1$ such that every such map f admits a rigid repelling periodic point of period at most $N(d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-071050 | MATHDB-368200

The Hermite Wigner zero-set rigidity conjecture

摘录 / EXCERPT: Let $Wf(z)$ be the Wigner distribution of some function $f\in L^2(\mathbb{R})$, with bounded nodal set $\mathcal{N}(Wf)$, and suppose that Wf is centered at $z=0$. For a Hermite...

conjecture | open | MathDB 页面 | 原文线索

V2-071051 | MATHDB-368201

The Laguerre polynomial divisibility conjecture

摘录 / EXCERPT: Let $n,k\in\mathbb{N}$ and $m\in\mathbb{N}_0$. Laguerre polynomial divisibility conjecture. If the Laguerre polynomial L_k divides the generalized Laguerre polynomial $L_n^{(m)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-071052 | MATHDB-368203

Universal Liouville and noncommutative integrability conjecture for magnetic flows on spheres

摘录 / EXCERPT: Let m be the mass of a material point moving on the sphere $S^{n-1}\subset\mathbb{R}^n$, let the point be placed in a constant homogeneous magnetic field described by the paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-071053 | MATHDB-368205

The conjecture on globally synchronizing random 3-regular graphs

摘录 / EXCERPT: Let a graph be globally synchronizing when its adjacency matrix is globally synchronizing, meaning that the only local minima of the Kuramoto energy are the global synchronized min...

conjecture | open | MathDB 页面 | 原文线索

V2-071054 | MATHDB-368208

The Kikuchi spectral-threshold conjecture

摘录 / EXCERPT: Fix even r and ℓ with $n\gg\ell\geq r/2$, and let $Z_S\sim\mathcal N(0,1)$ independently for each r-subset $S\subset[n]$. Define the Kikuchi matrix $M(\lambda)$, indexed b...

conjecture | open | MathDB 页面 | 原文线索

V2-071055 | MATHDB-368209

The K51ml\f3s discrepancy conjecture

摘录 / EXCERPT: K51mlf3s conjecture. There exists a universal constant K such that every square matrix A whose columns have unit ℓ_2 norm satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071056 | MATHDB-368211

The odd-dimensional Sylvester Hadamard discrepancy conjecture

摘录 / EXCERPT: Let H_k be the $2^k\times2^k$ Hadamard matrix obtained by the Sylvester construction, so $H_0=1$ and $H_k=H_1\otimes H_{k-1}$. For $x\in\\{\pm1\\}^{2^k}$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-071057 | MATHDB-368212

Gross's Euler-Poincaré orbital-integral conjecture

摘录 / EXCERPT: Let F be a non-Archimedean local field, let G be the reductive group under consideration, and let f^{EP} be the Euler-Poincaré function with respect to a Haar measure dg on...

conjecture | open | MathDB 页面 | 原文线索

V2-071058 | MATHDB-368213

Lefschetz-type conjecture for the base-change sum of L-functions

摘录 / EXCERPT: Assume the setup in which G is simply connected and almost simple, S is a set of places with at least two elements, and $T=\emptyset$. For each positive integer m, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071059 | MATHDB-368223

The locally dense induced-subgraph conjecture for P_5-free graphs

摘录 / EXCERPT: Let G be a finite simple graph. For an induced subgraph F of G and $v\in V(F)$, let $N_F(v)$ be the neighbours of v in F. P_5 local-density conjecture. There exists...

conjecture | open | MathDB 页面 | 原文线索

V2-071060 | MATHDB-368232

Poonen-Rains conjecture on Selmer group distributions of elliptic curves

摘录 / EXCERPT: Poonen-Rains conjecture. As E varies over all elliptic curves over F ordered by height,

conjecture | open | MathDB 页面 | 原文线索

V2-071061 | MATHDB-368239

Ellipsoidal positivity-set conjecture for cubic thin-obstacle solutions

摘录 / EXCERPT: Let u be a global solution to the thin obstacle problem with cubic growth, and suppose that its positivity set on the thin space is bounded. Cubic positivity-set conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-071062 | MATHDB-368240

Separation of choice principles in elementary topoi

摘录 / EXCERPT: Choice-separation conjecture. 1. There exists an elementary topos in which $\mathrm{AC}_{\mathbb{N}\text{-bounded}}$ holds but $\mathrm{AC}_{\mathbb{N}\mathbb{N}}$ does not. 2. The...

conjecture | open | MathDB 页面 | 原文线索

V2-071063 | MATHDB-368241

Cauchy-completeness separation conjecture for elementary topoi

摘录 / EXCERPT: Cauchy-completeness separation conjecture. 1. There exists an elementary topos in which $\mathbb{R}_C\neq\mathbb{R}_E$; equivalently, there exists an elementary topos in which...

conjecture | open | MathDB 页面 | 原文线索

V2-071064 | MATHDB-368244

Axisymmetry conjecture for magnetic ground-state minimizers

摘录 / EXCERPT: Let $B\geq 0$ be the strength of a constant magnetic field, and let $\Omega^*(B)$ be a domain minimizing the first eigenvalue among the admissible domains of fixed volume. The magn...

conjecture | open | MathDB 页面 | 原文线索

V2-071065 | MATHDB-368254

Exponential average-availability conjecture for local girth in planar graphs

摘录 / EXCERPT: Exponential average-availability conjecture. There exists a constant $c>1$ such that, for every planar graph G and local girth function f for G, there is a legal sequence of...

conjecture | open | MathDB 页面 | 原文线索

V2-071066 | MATHDB-368261

Cartier-Manin matrix conjecture for the parameter a

摘录 / EXCERPT: Let H_a be a genus-4 hyperelliptic curve, let M_{H_a} be its Cartier-Manin matrix, and let a denote the parameter defining the curve. Cartier-Manin matrix conjecture. The gre...

conjecture | open | MathDB 页面 | 原文线索

V2-071067 | MATHDB-368270

Stable-part and stable-cover decomposition conjecture

摘录 / EXCERPT: Stable-part decomposition conjecture. Every dependent hereditary class of graphs has a modelization in a dependent hereditary class consisting of the expansion of a hereditary stab...

conjecture | open | MathDB 页面 | 原文线索

V2-071068 | MATHDB-368271

Elkies' conjecture on rational points of Atkin-Lehner quotients

摘录 / EXCERPT: Elkies' conjecture. For all integers $N\gg0$, the points in

conjecture | open | MathDB 页面 | 原文线索

V2-071069 | MATHDB-368294

Sun's 1/pi series conjecture for S_n^{(2)}(1/11) and S_n^{(2)}(1/18)

摘录 / EXCERPT: Let ... The two displayed series are conjectured to converge to the stated multiples of $1/\pi$. Sun's 1/pi series conjecture. ... and ... The source presents these as open conjectures m...

conjecture | open | MathDB 页面 | 原文线索

V2-071070 | MATHDB-368296

Linear-size totally odd immersion conjecture

摘录 / EXCERPT: Let G be a finite, undirected, loopless graph, and let $\chi(G)$ denote its chromatic number. A graph H is a totally odd immersion of G when the edges of H are represented...

conjecture | open | MathDB 页面 | 原文线索

V2-071071 | MATHDB-368304

The Pfaffian generator conjecture for truncated shifted twisted Yangians

摘录 / EXCERPT: Pfaffian generator conjecture. A Pfaffian generator exists if and only if either ${\mathscr{Y}}_{N,\ell}^{+}(\sigma)$ has N even and ℓ odd, or...

conjecture | open | MathDB 页面 | 原文线索

V2-071072 | MATHDB-368310

Woodall-Seymour bipartite minor conjecture

摘录 / EXCERPT: Let G be a finite simple graph, and let $\chi(G)$ denote its chromatic number. For positive integers ℓ with $\ell<\chi(G)$, let $K_{\ell,\chi(G)-\ell}$ be the complete bipar...

conjecture | open | MathDB 页面 | 原文线索

V2-071073 | MATHDB-368311

Odd minor analogue for graphs with independence number two

摘录 / EXCERPT: Let G be a finite simple graph with independence number $\alpha(G)$ and chromatic number $\chi(G)$. For positive integers ℓ with $2\ell\leq\chi(G)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071074 | MATHDB-368316

Max-minus-min homomesy conjecture for interval-closed sets of products of two chains

摘录 / EXCERPT: Let $P=[m]\times[n]$ be the product of two chains, and let $\mathcal{IC}(P)$ denote its interval-closed sets. Under rowmotion on $\mathcal{IC}(P)$, consider the statistic that assi...

conjecture | open | MathDB 页面 | 原文线索

V2-071075 | MATHDB-368320

Karthick et al.'s perfect divisibility conjecture for fork-free graphs

摘录 / EXCERPT: A graph G is fork-free if it has no induced subgraph isomorphic to the graph obtained from $K_{1,3}$ by subdividing one edge once. A graph is perfectly divisible if, for each ind...

conjecture | open | MathDB 页面 | 原文线索

V2-071076 | MATHDB-368323

General tame SL_k-frieze pattern counting conjecture

摘录 / EXCERPT: General counting conjecture. If $\gcd(k,w+k+1)=1$, then the number of tame SL_k-frieze patterns of width w over K is

conjecture | open | MathDB 页面 | 原文线索

V2-071077 | MATHDB-368326

The trivialization conjecture for the orthogonal action on rigid higher categories

摘录 / EXCERPT: Let ${\mathsf{RigidCat}}_n$ denote the category of rigid symmetric monoidal n-categories, let O_n act on it through the action supplied by the adjoint-category conjecture, and...

conjecture | open | MathDB 页面 | 原文线索

V2-071078 | MATHDB-368330

Politeness suffices for theory combination

摘录 / EXCERPT: Let Σ_1 and Σ_2 be disjoint signatures, and let ${\mathcal{T}}_1$ and ${\mathcal{T}}_2$ be decidable theories over them. Let ${\mathcal{T}}_1\oplus{\mathcal{T}}_2$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-071079 | MATHDB-368331

Norton's replicability conjecture for rational formal q-series

摘录 / EXCERPT: Let $f=q^{-1}+\sum_{n=1}^\infty a_nq^n$ be a formal q-series with rational coefficients that is replicable. Norton's conjecture. The series f is either a Hauptmodul for a genus...

conjecture | open | MathDB 页面 | 原文线索

V2-071080 | MATHDB-368338

Monotonicity conjecture for cyclic 321-avoidance growth

摘录 / EXCERPT: For each $n\geq 1$, let $c_n=|\mathcal C_n(321)|$, where $\mathcal C_n(321)$ is the set of cyclic permutations of size n avoiding 321. The monotonicity conjecture. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-071081 | MATHDB-368339

The join conjecture for enumerative chromatic-choosability

摘录 / EXCERPT: Let G be a graph, let K_p be the complete graph on p vertices, and let $G\vee K_p$ denote their join. A graph is enumeratively chromatic-choosable when its list color functio...

conjecture | open | MathDB 页面 | 原文线索

V2-071082 | MATHDB-368343

The Sturm-sequence conjecture for (q,t)-Hoggatt sums

摘录 / EXCERPT: For integers $q>0$ and $m=2,3$, let $H_n^{[m]}(q,t)$ denote the (q,t)-Hoggatt sums. The Sturm-sequence conjecture. The polynomial sequence ... forms a Sturm sequence. The authors v...

conjecture | open | MathDB 页面 | 原文线索

V2-071083 | MATHDB-368354

Gartland-Lokastov induced-minor separator conjecture

摘录 / EXCERPT: Gartland-Lokastov's conjecture. For every planar graph H, there exists $k\in\mathbb{N}$ such that every H-induced-minor-free graph G admits a $(k,1)$-balanced separator.

conjecture | open | MathDB 页面 | 原文线索

V2-071084 | MATHDB-368361

Pinned distinct distances for most norms

摘录 / EXCERPT: Pinned most-norms conjecture. For most d-norms $\|\cdot\|$, every finite point set $P\subseteq\mathbb R^d$ contains a point $x\in P$ determining $(1-o(1))|P|$ distances to the ot...

conjecture | open | MathDB 页面 | 原文线索

V2-071085 | MATHDB-368393

Linear-diameter conjectures for list-recolouring graphs

摘录 / EXCERPT: Let G be a connected graph with n vertices, and let L be a list-assignment of G. Write $\mathcal{C}(G,L)$ for the L-recolouring graph and $\widehat{\mathcal{C}}(G,L)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-071086 | MATHDB-368394

The uniqueness conjecture for good involutions of connected quandles

摘录 / EXCERPT: A quandle is a set equipped with a binary operation satisfying the quandle axioms; it is connected when its inner automorphism group acts transitively, involutory when each right t...

conjecture | open | MathDB 页面 | 原文线索

V2-071087 | MATHDB-368402

Isoperimetric inequality for the ground-state magnetic eigenvalue with localized field

摘录 / EXCERPT: Let $ u\text{?}$ and $ u$? No. Let β?

conjecture | open | MathDB 页面 | 原文线索

V2-071088 | MATHDB-368407

A 3-adic valuation formula for cubic binomial sums

摘录 / EXCERPT: For an integer $n\geq 0$, define ... Let $s_3(m)$ denote the sum of the base-3 digits of the nonnegative integer m. The cubic binomial-sum valuation conjecture. For every integer...

open_problem | open | MathDB 页面 | 原文线索

V2-071089 | MATHDB-368411

The L^p decoupling conjecture for the hyperbolic paraboloid

摘录 / EXCERPT: Hyperbolic-paraboloid decoupling conjecture. For $p>3$, for every $\varepsilon>0$ there exists $C_\varepsilon>0$ such that for all $R\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-071090 | MATHDB-368417

Hodge-theoretic adjacency conjecture for k-Du Bois degenerations

摘录 / EXCERPT: Let $\mathfrak{X}\to\Delta$ be a flat, projective degeneration with reduced special fiber X_0, and let $\pi\colon\mathfrak{X}^*\to\Delta^*$ be smooth, where...

conjecture | open | MathDB 页面 | 原文线索

V2-071091 | MATHDB-368430

Odd Hadwiger's conjecture

摘录 / EXCERPT: Let $(G,-)$ be the signed graph obtained from a graph G by assigning a negative sign to every edge, and let $(K_t,-)$ be the all-negative signed complete graph. A signed graph mi...

conjecture | open | MathDB 页面 | 原文线索

V2-071092 | MATHDB-368444

The Arthur-closure conjecture for the unitary dual

摘录 / EXCERPT: Let F) be a non-Archimedean local field of characteristic zero, let G be a connected reductive group over F, and set $G=\mathrm{G}(F)$. Let $\Pi_u(G)$ denote the uni...

conjecture | open | MathDB 页面 | 原文线索

V2-071093 | MATHDB-368471

The bounded tree-independence conjecture for $\{P_k,K_{t,t}\}$-free graphs

摘录 / EXCERPT: Bounded tree-independence conjecture. For any two positive integers k and t, the class of $\{P_k,K_{t,t}\}$-free graphs has bounded tree-independence number.

conjecture | open | MathDB 页面 | 原文线索

V2-071094 | MATHDB-368472

Gartland and Lokshtanov's induced grid minor conjecture

摘录 / EXCERPT: Induced Grid Minor Conjecture. There exists a function f such that for every planar graph H, every H-induced-minor-free graph G has a balanced separator dominated by $f(H)$...

conjecture | open | MathDB 页面 | 原文线索

V2-071095 | MATHDB-368482

Kuznetsov's rationality conjecture for cubic fourfolds

摘录 / EXCERPT: Let $\mathcal{C}$ be the moduli space of cubic fourfolds, and let $\mathcal{C}_d$ denote a special divisor, consisting of cubic fourfolds with discriminant d. A discriminant d...

conjecture | open | MathDB 页面 | 原文线索

V2-071096 | MATHDB-368491

Burness-Giudici Common Neighbour Conjecture for Saxl graphs

摘录 / EXCERPT: Let $G\leq \operatorname{Sym}(\Omega)$ be a primitive permutation group with base size $b(G)=2$, and let $\Sigma(G)$ be its Saxl graph, whose vertices are the points of Ω an...

conjecture | open | MathDB 页面 | 原文线索

V2-071097 | MATHDB-368507

Unique q-series decomposition conjecture for Mordell-Borel integrals

摘录 / EXCERPT: Let p be an odd integer and let $J^{(A,p)}_j(t)$, $J^{(B,p)}_j(t)$, $J^{(C,p)}_j(t)$, and $J^{(D,p)}_j(t)$ denote the vectors of Mordell-Borel integrals defined in the source. Wr...

conjecture | open | MathDB 页面 | 原文线索

V2-071098 | MATHDB-368508

The Elementary Type Conjecture for finite quaternionic structures

摘录 / EXCERPT: Elementary Type Conjecture. Every finite quaternionic structure is of elementary type.

conjecture | refuted | MathDB 页面 | 原文线索

V2-071099 | MATHDB-368509

The weak Elementary Type Conjecture for field-realized quaternionic structures

摘录 / EXCERPT: Weak Elementary Type Conjecture. Every finite quaternionic structure that is realized by a field is of elementary type.

conjecture | open | MathDB 页面 | 原文线索

V2-071100 | MATHDB-368512

Coulomb-gas representation of radial pure partition functions

摘录 / EXCERPT: Let $\mathrm{LP}(n,m)$ be the set of radial link patterns, let $\mathcal{Z}_{\alpha}(\boldsymbol{\theta})$ be the pure partition functions, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071101 | MATHDB-368513

Analytic continuation of radial pure partition functions

摘录 / EXCERPT: Let $\mathcal{Z}_{\alpha}(\boldsymbol{\theta})$ and $\mathcal{Z}_{\alpha}^{\eta}(\boldsymbol{\theta})$ denote the zero-spin and spin-η pure partition functions, respectively,...

conjecture | open | MathDB 页面 | 原文线索

V2-071102 | MATHDB-368524

Gaetz-Gao's strong hull conjecture for Coxeter groups

摘录 / EXCERPT: Gaetz-Gao's strong hull conjecture. Every Coxeter group W has the property that its Cayley graph $\mathrm{Cay}(W)$ satisfies the strong hull property.

conjecture | open | MathDB 页面 | 原文线索

V2-071103 | MATHDB-368525

Critical-point count conjecture for heterogeneous quadratics with two vectors

摘录 / EXCERPT: Consider the heterogeneous quadratics minimization problem ... where $A_i \in {\rm Sym}(\mathbb{R}^n)$ and Z_i is the ith column of Z. Critical-point count conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-071104 | MATHDB-368528

The enriched model structure conjecture for marked simplicial spaces

摘录 / EXCERPT: Let $s\mathscr{S}\mathrm{et}$ be the category of simplicial sets, $s\mathscr{S}\mathrm{et}^+$ the category of marked simplicial sets, and $s\mathscr{S}^+$ the category of simplicia...

conjecture | open | MathDB 页面 | 原文线索

V2-071105 | MATHDB-368535

Bonamy-Kardos-Kelly-Postle conjecture on fractional arboricity of planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let $a_f(G)$ denote its fractional arboricity: the minimum ratio p/q for which vertices can be assigned at least q colors from a total of at most...

conjecture | open | MathDB 页面 | 原文线索

V2-071106 | MATHDB-368543

Conjecture that nontrivial snarks of high perfect matching index are strong

摘录 / EXCERPT: Strong-snark conjecture. With the sole exception of the Petersen graph, every nontrivial snark with perfect matching index at least 5 is strong; equivalently, all its edges are s...

conjecture | open | MathDB 页面 | 原文线索

V2-071107 | MATHDB-368544

Conjecture on cyclically 4-edge-connected cubic graphs attaining cycle covering ratio 7/5

摘录 / EXCERPT: Cyclic 4-edge-connectivity conjecture. Up to isomorphism, the Petersen graph is the only cyclically 4-edge-connected cubic graph with cycle covering ratio $7/5$.

conjecture | open | MathDB 页面 | 原文线索

V2-071108 | MATHDB-368554

Sturmfels conjecture on Graver basis degrees and true circuit degrees

摘录 / EXCERPT: Let I_A be a toric ideal, and let the true degree of a circuit of A mean the degree computed without dividing by the common factor c in the circuit formula. Sturmfels's conje...

conjecture | open | MathDB 页面 | 原文线索

V2-071109 | MATHDB-368557

The unimodality conjecture for h^*-vectors of IDP polytopes

摘录 / EXCERPT: Let P be an IDP polytope, meaning a lattice polytope such that for every $k\in\mathbb{N}_{\ge 2}$, each lattice point in kP is a sum of k lattice points in P. If ... is the...

conjecture | open | MathDB 页面 | 原文线索

V2-071110 | MATHDB-368558

Basis conjecture for hybrid Grothendieck polynomials

摘录 / EXCERPT: Let $\Lambda=\bigoplus_{\lambda}\mathbb{Q}(\alpha,\beta)s_{\lambda}(\mathbf{x})$ be the ring of symmetric functions over the field of rational functions in α and β, an...

conjecture | open | MathDB 页面 | 原文线索

V2-071111 | MATHDB-368559

Stembridge-type coincidence conjecture for hybrid Grothendieck polynomials

摘录 / EXCERPT: Let $\delta_n=(n,n-1,\ldots,1)$ be the staircase partition, and let $\rho\subseteq\delta_n$. Write ρ' for the conjugate partition. Stembridge-type coincidence conjecture. One...

conjecture | open | MathDB 页面 | 原文线索

V2-071112 | MATHDB-368566

Mazur's growth number conjecture for Selmer groups over p-extensions

摘录 / EXCERPT: Mazur's growth number conjecture. For all sufficiently large n,

conjecture | open | MathDB 页面 | 原文线索

V2-071113 | MATHDB-368573

Orbifold classification conjecture for asymptotically cylindrical steady solitons

摘录 / EXCERPT: Let $\mathcal{S}$ be one of the spherical space forms appearing in Table 1, and consider asymptotically $\mathcal{S}$-cylindrical steady gradient Ricci solitons on smooth Riemannia...

conjecture | open | MathDB 页面 | 原文线索

V2-071114 | MATHDB-368576

Missing-value conjecture for the next-to-middle missing-face number of neighborly spheres

摘录 / EXCERPT: Let $k\geq 2$, and let Δ be a neighborly $2k$-sphere with n vertices. Write $m_{k+1}(\Delta)$ for its number of missing faces of dimension $k+1$. Missing-value conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071115 | MATHDB-368586

Alanazi-Munagi-Saikia's congruence conjecture for $(4,9)$-regular overpartitions

摘录 / EXCERPT: Let $\overline{R_{4,9}}(n)$ denote the number of $(4,9)$-regular overpartitions of n. For integers $n\geq0$, $\ell\geq2$, and $1\leq k\leq\ell$, Alanazi-Munagi-Saikia's conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-071116 | MATHDB-368594

Biquadratic linear AFL conjecture for GL(4)

摘录 / EXCERPT: Let $\alpha:(K_1, K_2) \to \operatorname{Mat}_{2h}(F)$ and $\beta:(K_0, K_3) \to \operatorname{Mat}_{2h}(F)$ be matching regular semisimple pairs. Let $\mathcal{H}_{2h}$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-071117 | MATHDB-368595

Gao-Isaev-McKay monotone coupling conjecture for random regular graphs

摘录 / EXCERPT: Gao-Isaev-McKay conjecture. There exists a coupling such that

conjecture | open | MathDB 页面 | 原文线索

V2-071118 | MATHDB-368596

Isaev-McKay-Southwell-Zhukovskii decomposition conjecture for random regular graphs

摘录 / EXCERPT: Isaev-McKay-Southwell-Zhukovskii conjecture. If $d_2=\omega(1)$, then there exists a coupling of $G_{\oplus}\sim G(n,d_1)\oplus G(n,d_2-d_1)$ and $G\sim G(n,d_2)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071119 | MATHDB-368598

The obstruction conjecture for C^2 Lusin approximation of convex functions

摘录 / EXCERPT: Let $n\geq 2$. A convex function is a function whose restriction to every line segment in its domain is convex. Write $B^n(0,1)$ for the open unit ball in $\mathbb R^n$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071120 | MATHDB-368600

RPCD worst-case conjecture for positive-definite quadratics

摘录 / EXCERPT: RPCD worst-case conjecture. The upper bound above, proved for the specified permutation-invariant class of quadratic Hessians, extends to all positive-definite quadratic functions....

conjecture | open | MathDB 页面 | 原文线索

V2-071121 | MATHDB-368608

Sagan-Swanson's type B Hilbert-series conjecture for the superspace coinvariant ring

摘录 / EXCERPT: Let $\Omega_n=\mathbb{C}[x_1,\ldots,x_n]\otimes\bigwedge\langle\theta_1,\ldots,\theta_n\rangle$ be the superspace, and let SR^B_n be its type B superspace coinvariant ring. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-071122 | MATHDB-368613

The periodic-point version of Morton-Silverman uniform boundedness

摘录 / EXCERPT: Let $n\geq 1$ and $d\geq 2$ be integers. Let $K/\mathbb{Q}$ be a number field of degree at most n, and let $\varphi:\mathbb{P}^1(K)\to\mathbb{P}^1(K)$ be a morphism of degree d...

conjecture | open | MathDB 页面 | 原文线索

V2-071123 | MATHDB-368615

The positive semidefinite Graph Complement Conjecture

摘录 / EXCERPT: Positive semidefinite Graph Complement Conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-071124 | MATHDB-368617

The Weak Graph Complement Conjectures

摘录 / EXCERPT: Weak Graph Complement Conjectures. There exist universal constants $b,b_+,b_\nu<2$ such that, for every graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-071125 | MATHDB-368619

The chromatic lower-bound conjecture for the Colin de Verdière parameter

摘录 / EXCERPT: Chromatic lower-bound conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-071126 | MATHDB-368620

The independence-number lower-bound conjecture for the Colin de Verdière parameter

摘录 / EXCERPT: Independence-number lower-bound conjecture. For any graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-071127 | MATHDB-368636

The neighborhood nefness conjecture for log canonical pairs

摘录 / EXCERPT: Neighborhood nefness conjecture. Then $K_X+\Delta$ is π-nef over some open neighborhood of W.

conjecture | open | MathDB 页面 | 原文线索

V2-071128 | MATHDB-368641

The generic Arthur packet conjecture

摘录 / EXCERPT: Let ψ be an Arthur parameter, let φ_{ψ} be its associated Langlands parameter, and let $\Pi(\varphi_{\psi})$ be the L-packet associated to φ_{ψ}. A rep...

conjecture | open | MathDB 页面 | 原文线索

V2-071129 | MATHDB-368642

The symmetry-group dimension conjecture for conformal field theories

摘录 / EXCERPT: Let $H=\bigoplus_{p,q}H^{p,q}$ be the state space of a conformal field theory, with $H^{1,0}$ its subspace of states of conformal weights $(1,0)$, and let the group of symmetries m...

conjecture | open | MathDB 页面 | 原文线索

V2-071130 | MATHDB-368643

The compactness and deformation-dimension conjecture for CFT moduli

摘录 / EXCERPT: Fix $c_0\geq 0$ and $E_{min}>0$, and let ${\cal M}_{c\leq c_0}^{E_{min}}$ be the moduli space of irreducible conformal field theories with central charge $c\leq c_0$ and ... For a gi...

conjecture | open | MathDB 页面 | 原文线索

V2-071131 | MATHDB-368644

Partition-function bound for collapsing conformal field theories

摘录 / EXCERPT: Consider a family of unitary conformal field theories with fixed central charge c, parameterized by $\varepsilon\to 0$, and closed surfaces Σ in the family described by lo...

conjecture | open | MathDB 页面 | 原文线索

V2-071132 | MATHDB-368649

Affine meander-matrix conjecture for pure partition functions

摘录 / EXCERPT: Fix irrational $\kappa\in(0,8)$ and let $\mathrm{LP}(n,m)$ be the set of link patterns. Let $\mathcal{Z}_{\alpha}$ be the pure partition function indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-071133 | MATHDB-368662

The conjectural mean value estimate for Weyl sums associated with (n^3,n)

摘录 / EXCERPT: The conjectural Weyl-sum estimate. For $d\geq2$, $0\leq u\leq d-1$, and even p, one has

conjecture | open | MathDB 页面 | 原文线索

V2-071134 | MATHDB-368670

Maximal extension conjecture for the paraboloid

摘录 / EXCERPT: Let S be the relevant two-dimensional paraboloid and let E_Sf denote its extension operator. For $p,q\in[1,\infty]$, use the mixed norm $L^q_{x_1}L^\infty_{x_2}$. Maximal exten...

conjecture | open | MathDB 页面 | 原文线索

V2-071135 | MATHDB-368671

Interval graphs are boundary-distance-matrix reconstructible

摘录 / EXCERPT: Let G be an interval graph. A graph is called BDM-constructible if it is uniquely determined by its boundary distance matrix (with the relevant order and boundary fixed). Interva...

conjecture | open | MathDB 页面 | 原文线索

V2-071136 | MATHDB-368678

The monoid realization conjecture for pre-upho colored lattices

摘录 / EXCERPT: Monoid realization conjecture. The monoid M is left-cancellative, every pair of elements of M has a greatest common left divisor, and therefore...

conjecture | open | MathDB 页面 | 原文线索

V2-071137 | MATHDB-368679

Grzesik-Győri-Salia-Tompkins bipartite Turán-goodness conjecture

摘录 / EXCERPT: A graph H is K_3-Turán-good when, for all sufficiently large n, the Turán graph with two parts attains ${\mathrm{ex}}(n,H,K_3)$. The Grzesik-Győri-Salia-Tompkins conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071138 | MATHDB-368682

The symmetric exchange generation conjecture for unicyclic graph toric ideals

摘录 / EXCERPT: Let G be a unicyclic graph on n vertices, and let $\mathfrak{c} \in \mathbb{Z}_{>0}^n$. Write $\operatorname{Ker}(\pi_G^{\mathfrak{c}})$ for the associated toric ideal, and cal...

conjecture | open | MathDB 页面 | 原文线索

V2-071139 | MATHDB-368686

The symmetric-function conjecture for b-deformed Jucys-Murphy operators

摘录 / EXCERPT: Let k be a positive integer, let $\mathcal{Z}\mathcal{V}_k$ be the subspace of vectors constant on coset-type, and let $\mathfrak{p}_\lambda,\mathfrak{w}_\lambda$ be the b-defo...

conjecture | open | MathDB 页面 | 原文线索

V2-071140 | MATHDB-368687

The Laplace-Beltrami correspondence for b-deformed Jucys-Murphy operators

摘录 / EXCERPT: Let k be a positive integer, let $\mathcal{Z}\mathcal{V}_k$ be the coset-type-constant subspace, let $\mathfrak{p}_\lambda$ be the b-deformed class vector for $\lambda\vdash k$...

conjecture | open | MathDB 页面 | 原文线索

V2-071141 | MATHDB-368714

Malle's moment predictions for the 2-class groups of cubic fields

摘录 / EXCERPT: Malle's conjecture. The average size of $\operatorname{Cl}_2(K)$ over totally real cubic fields K is $5/4$, and its second moment is $15/8$; over complex cubic fields K, the av...

conjecture | open | MathDB 页面 | 原文线索

V2-071142 | MATHDB-368723

Full-rank invariance conjecture for gradient-flow synaptic dynamics

摘录 / EXCERPT: Let $X \in \mathbb{R}^{n \times T}$ satisfy $C_X \succ 0$, and let $W(t)$ solve the continuous-time gradient-flow feedforward synaptic dynamics ... Write $\mathcal{R}_{m,n}$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-071143 | MATHDB-368724

Global-minimum and strict-saddle conjecture for the third-layer synaptic objective

摘录 / EXCERPT: Let $X \in \mathbb{R}^{n \times T}$ satisfy $C_X \succ 0$, and let $W(t)$ follow the continuous-time gradient-flow feedforward synaptic dynamics. Let $\mathcal{S}$ denote the stati...

conjecture | open | MathDB 页面 | 原文线索

V2-071144 | MATHDB-368728

Geometric irreducibility of the non-ordinary hyperelliptic locus

摘录 / EXCERPT: Let k be a field of characteristic $p\geq 3$, and let $\mathcal{H}_g$ be the moduli space of smooth hyperelliptic curves of genus g over k. Write $\mathcal{H}_g^{g-1}$ for it...

conjecture | open | MathDB 页面 | 原文线索

V2-071145 | MATHDB-368729

Geometric irreducibility of the next-to-non-ordinary hyperelliptic stratum

摘录 / EXCERPT: Let k be a field of characteristic $p\geq 3$, and let $\mathcal{H}_g$ be the moduli space of smooth hyperelliptic curves of genus g over k. Denote by $\mathcal{H}_g^{g-2}$ th...

conjecture | open | MathDB 页面 | 原文线索

V2-071146 | MATHDB-368730

Unboundedness of the geometric component count of the genus-three p-rank-zero locus

摘录 / EXCERPT: Let $\mathcal{H}_3^0$ denote the moduli space of smooth hyperelliptic curves of genus 3 and p-rank 0 in characteristic $p\geq 3$. Let $c_{3,0,p}$ be the number of geometric i...

conjecture | open | MathDB 页面 | 原文线索

V2-071147 | MATHDB-368731

Positive p-rank for hyperelliptic curves with rational branch points

摘录 / EXCERPT: Assume $p\equiv 1\pmod 4$. Let C be a hyperelliptic curve of genus at least 3 whose branch points are all defined over $\mathbb{F}_p$. Rational-branch-point p-rank conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071148 | MATHDB-368739

Zung's spectral-sequence conjecture relating $SFC(\phi,\mathcal{C})$ and $CA(\phi)$

摘录 / EXCERPT: Let ϕ be the flow under consideration, let G_+ be its associated graph, and let $SFC(\phi,\mathcal{C})$ and $CA(\phi)$ be the chain complexes described above, where...

conjecture | open | MathDB 页面 | 原文线索

V2-071149 | MATHDB-368747

Convergence of the population's average opinion toward the leaders' average opinion

摘录 / EXCERPT: Let $f(t,A,w)$ be the population distribution, let $f_\ell(z)$ be the leaders' opinion distribution, and define the population's average opinion and the leaders' average opinion by...

conjecture | open | MathDB 页面 | 原文线索

V2-071150 | MATHDB-368748

Ward's divisibility conjecture for Griesmer codes

摘录 / EXCERPT: Ward's divisibility conjecture. Suppose that $p^e\mid d$ and $p^e\geq q$. Then p^{e+1}/q is a divisor of C.

conjecture | open | MathDB 页面 | 原文线索

V2-071151 | MATHDB-368749

Diagonal conjugation conjecture for match Yang-Baxter representations

摘录 / EXCERPT: Let $N\in\mathbb{N}$, let $(1,R)\in YB^1(\mathsf{Match}^N)$, and let X be a diagonal matrix in $\mathsf{Mat}^N(2,2)$. Define ... For each n, let ρ_n^R and ρ_n^S denote...

conjecture | open | MathDB 页面 | 原文线索

V2-071152 | MATHDB-368755

Burillo's positive pure braid crossing-matrix conjecture

摘录 / EXCERPT: Let n be a positive integer, and let M be an $n\times n$ integer matrix. A zero-diagonal matrix is T0 if, whenever $1\leq i<j<k\leq n$, the conditions $M(i,j)=M(j,k)=0$ imply...

conjecture | open | MathDB 页面 | 原文线索

V2-071153 | MATHDB-368756

OU-matrix conjecture for pure braids

摘录 / EXCERPT: Let n be a positive integer. For an n-braid diagram, its OU matrix $U(b)$ is the non-negative integer zero-diagonal matrix whose (i,j) entry counts crossings between strands...

conjecture | open | MathDB 页面 | 原文线索

V2-071154 | MATHDB-368757

CN-matrix conjecture for pure braid projections

摘录 / EXCERPT: Let n be a positive integer. A braid projection is a braid diagram without over/under-crossing information. Its CN matrix $N(B)$ is the zero-diagonal matrix whose (i,j) entry c...

conjecture | open | MathDB 页面 | 原文线索

V2-071155 | MATHDB-368758

T-structure criterion for CN-realizable matrices

摘录 / EXCERPT: Let M be a strictly upper triangular $(0,2)$-matrix, meaning that its entries are 0 or 2, and suppose that for each $k\geq 3$ there is at most one pair (i,j) with $j-i=k$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-071156 | MATHDB-368765

Locally finite perturbation conjecture for non-transitive tournaments

摘录 / EXCERPT: Let T be a finite tournament. A locally finite perturbation of a tournament is obtained by changing the direction of arcs in a locally finite set. Tournament perturbation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-071157 | MATHDB-368767

Equivalent Redei-Berge formulation of the Stanley-Stembridge conjecture

摘录 / EXCERPT: Equivalent Stanley-Stembridge formulation. If P is $(3+1)$-free, then U_P is h-positive.

conjecture | open | MathDB 页面 | 原文线索

V2-071158 | MATHDB-368770

Standard atom positivity conjecture for flagged LLT polynomials

摘录 / EXCERPT: Let ${\boldsymbol\nu}$ be a tuple of ordinary, non-ragged-right skew diagrams, and let σ be the standard compatible permutation. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071159 | MATHDB-368771

Unimodal and anti-unimodal atom positivity conjecture for flagged LLT polynomials

摘录 / EXCERPT: Let ${\boldsymbol\nu},\sigma$ be any flagged LLT indexing data, and let $\mathcal G_{{\boldsymbol\nu},\sigma}[x_1,\ldots,x_l;t]$ be the associated flagged LLT polynomial. If ... then...

conjecture | open | MathDB 页面 | 原文线索

V2-071160 | MATHDB-368779

Graham positivity conjecture for triple Grothendieck polynomial coefficients

摘录 / EXCERPT: Let β be a formal variable, let $u,v,w\in S_\infty$, and let $\mathfrak{G}_w(\mathbf{x};\mathbf{y})$ denote the double Grothendieck polynomial. Define the coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-071161 | MATHDB-368781

Locality tensor-product conjecture

摘录 / EXCERPT: Let $(E,\top)$ be a locality vector space, and let V and W be locality vector subspaces of E. The locality tensor product $V\otimes_\top W$ is equipped with the quotient loca...

conjecture | open | MathDB 页面 | 原文线索

V2-071162 | MATHDB-368788

The critical connectivity and Sierpiński-carpet conjecture for SLE

摘录 / EXCERPT: For $\kappa \in (4,8)$, let η be an SLE$_\kappa$ curve from 0 to ∞ in $\mathbb{H}$. The adjacency graph of the connected components of $\mathbb{H} \setminus \eta$ has...

conjecture | open | MathDB 页面 | 原文线索

V2-071163 | MATHDB-368789

Phase-transition conjecture for SLE with a boundary force point

摘录 / EXCERPT: For each $\underline{\kappa}\in(0,4)$, let $\rho_c=\rho_c(\underline{\kappa})$ lie in $(-2-\underline{\kappa}/2,-2)\cap[\underline{\kappa}/2-4,-2)$, and let η be an SLE...

conjecture | open | MathDB 页面 | 原文线索

V2-071164 | MATHDB-368799

Borradaile-Le-Sherman-Bennett conjecture on induced outerplanar subgraphs

摘录 / EXCERPT: Borradaile-Le-Sherman-Bennett conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071165 | MATHDB-368800

The induced K_4-minor-free subgraph conjecture for planar graphs

摘录 / EXCERPT: Induced K_4-minor-free subgraph conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071166 | MATHDB-368801

Matheson-Tarjan domination conjecture for plane triangulations

摘录 / EXCERPT: Matheson-Tarjan conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071167 | MATHDB-368802

Total domination conjecture for plane triangulations

摘录 / EXCERPT: Total domination conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071168 | MATHDB-368803

Connected domination conjecture for plane triangulations

摘录 / EXCERPT: Connected domination conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071169 | MATHDB-368804

Induced maximal outerplane subgraph conjecture

摘录 / EXCERPT: Induced maximal outerplane subgraph conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071170 | MATHDB-368805

Expected dimension conjecture for maps to general hypersurfaces in Grassmannians

摘录 / EXCERPT: Let C be a smooth curve of genus g, let $G(r,n)$ be a Grassmannian, and let X_{ℓ} be a general hypersurface of degree ℓ in $G(r,n)$ with $\ell<n$. Denote by...

conjecture | open | MathDB 页面 | 原文线索

V2-071171 | MATHDB-368821

González-Alonso-Torelli maximal-rank conjecture for curves on projective surfaces

摘录 / EXCERPT: González-Alonso-Torelli conjecture. Such a deformation can have rank

conjecture | open | MathDB 页面 | 原文线索

V2-071172 | MATHDB-368832

The conjecture that matched upper- and lower-bounded Lüroth sets cover modulo 1

摘录 / EXCERPT: For $k\in\mathbb{N}\setminus\{1\}$, let $L_{\leq k}$ be the Lüroth set with digits at most k, and let $L_{\geq k}$ be the Lüroth set with digits at least k. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-071173 | MATHDB-368835

The conjecture that two lower-bounded Lüroth sets contain an interval

摘录 / EXCERPT: For $k\in\mathbb{N}\setminus\{1\}$, let $L_{\geq k}$ be the Lüroth set whose digits are all at least k. For subsets of $\mathbb{R}$, write $A+B=\{a+b:a\in A,b\in B\}$. The interv...

conjecture | open | MathDB 页面 | 原文线索

V2-071174 | MATHDB-368845

Generalised Payne shift inequality for buckling eigenvalues

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^d$ be a domain of finite measure, and let $\lambda_k^{(m,t)}(\Omega)$ denote its kth buckling eigenvalue. Generalised Payne buckling inequality. For e...

conjecture | open | MathDB 页面 | 原文线索

V2-071175 | MATHDB-368846

Pilatte's eventual periodicity conjecture for binary multiplicative functions

摘录 / EXCERPT: Pilatte's conjecture. These sums are bounded uniformly for all $x>0$ and all $\alpha\in E$ if and only if f is eventually periodic. The conjecture was communicated to the authors...

conjecture | open | MathDB 页面 | 原文线索

V2-071176 | MATHDB-368849

Equality of asymptotic redundancy factors for multiple bursts and substitutions

摘录 / EXCERPT: Multiple burst-substitution redundancy conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071177 | MATHDB-368856

Generalised Trinh-Xue conjecture for spetsial reflection groups

摘录 / EXCERPT: Generalised Trinh-Xue conjecture. The intersection of the Φ- and Φ'-Harish-Chandra series

conjecture | open | MathDB 页面 | 原文线索

V2-071178 | MATHDB-368859

Refined leading-term conjecture for regular supercuspidal connections

摘录 / EXCERPT: Let $G_1,\ldots,G_p$ be twisted Levi subgroups, let $\phi_{(i)}:\fa_i\to\bC$ be refined leading terms, let $X_{(i)}\in\check{\fa}_i$ be their duals, and let A' and the central su...

conjecture | open | MathDB 页面 | 原文线索

V2-071179 | MATHDB-368866

Partial homogenization conjecture for adaptive weight systems

摘录 / EXCERPT: Partial homogenization conjecture. If the corresponding solutions of are denoted by $(\bm{X}^{(k)}_i(t))_{i\in[N_k]}$ and $(\bm{w}^{(k)}_{i,j}(t))_{i,j\in[N_k]}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071180 | MATHDB-368879

Completeness of the Bethe Ansatz for the periodic XXZ spin chain

摘录 / EXCERPT: Completeness conjecture. The Bethe Ansatz completely determines the spectrum for all but finitely many values of Δ with $|\Delta|<\varepsilon$.

conjecture | open | MathDB 页面 | 原文线索

V2-071181 | MATHDB-368882

The higher-residue conjecture for the VC-dimension of Cayley graphs

摘录 / EXCERPT: Higher-residue conjecture. As $N\to\infty$ through the primes congruent to 1 modulo r,

conjecture | open | MathDB 页面 | 原文线索

V2-071182 | MATHDB-368883

Faudree-Lehel's irregularity-strength conjecture

摘录 / EXCERPT: Let G be a d-regular graph on n vertices with $d>1$. An irregular edge weighting assigns real weights to the edges so that all vertices have distinct weighted degrees, and...

conjecture | open | MathDB 页面 | 原文线索

V2-071183 | MATHDB-368884

Bartnicki-Grytczuk-Niwczyk edge-weight 3-choosability conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a nice graph, meaning a graph without isolated edges. A graph is edge-weight 3-choosable if, for every assignment L of a three-element set of real numbers to e...

conjecture | open | MathDB 页面 | 原文线索

V2-071184 | MATHDB-368895

Sun-Zhang's boundedness conjecture for equivariant Fano fibrations

摘录 / EXCERPT: Let T be a torus and let $c>0$ be a real number. Consider isomorphism classes of T-equivariant $(\mathbb{Q})$-Fano fibrations $T\curvearrowright(X\xrightarrow{\pi}Y)$, with wei...

conjecture | open | MathDB 页面 | 原文线索

V2-071185 | MATHDB-368896

Higher Theta-stratification conjecture for equivariant Fano fibrations

摘录 / EXCERPT: Consider T-equivariant Fano fibrations $\pi\colon X\to Y$, fixing the multi-Hilbert function of $T\curvearrowright Y$ and the Hilbert polynomial of $-K_X$ on the fibers of π....

conjecture | open | MathDB 页面 | 原文线索

V2-071186 | MATHDB-368900

Panyushev's order conjecture for the positive Panyushev map

摘录 / EXCERPT: Let W be a finite, crystallographic Coxeter group with Coxeter number h, let m be the parameter from the preceding theorem, and let ${\sf P}_+$ be the positive Panyushev map...

conjecture | open | MathDB 页面 | 原文线索

V2-071187 | MATHDB-368904

Conditional central limit conjecture for general subgraph counts in exponential random graphs

摘录 / EXCERPT: Let H) be a fixed graph containing t triangles and s two stars. Under the same conditions as in the two-star conditional central limit theorem, let F_n denote the number of...

conjecture | open | MathDB 页面 | 原文线索

V2-071188 | MATHDB-368910

Lipsman's conjecture on properness and the CI property for nilpotent homogeneous spaces

摘录 / EXCERPT: Lipsman's conjecture. The following two conditions are equivalent: (i) the L-action on G/H is proper; (ii) the L-action on G/H has the CI property.

conjecture | open | MathDB 页面 | 原文线索

V2-071189 | MATHDB-368913

The nonexistence conjecture for cocompact quotients of $SL(n,\mathbb{R})/SL(m,\mathbb{R})$

摘录 / EXCERPT: Nonexistence conjecture. The homogeneous space

conjecture | open | MathDB 页面 | 原文线索

V2-071190 | MATHDB-368915

Long-time asymptotic behaviour conjecture for unbounded crystalline elastic flows

摘录 / EXCERPT: Let $\Gamma^0:=\bigcup_{i=1}^nS_i^0$ be an unbounded admissible polygonal curve, and let $\{\Gamma(t)\}_{t\ge0}$ be the globally defined elastic flow with finitely many restarts. W...

conjecture | open | MathDB 页面 | 原文线索

V2-071191 | MATHDB-368918

Relative Witt quotient conjecture for faithfully flat enriched finite tensor categories

摘录 / EXCERPT: Let $\mathcal{E}$ be a finite symmetric tensor category. Let $\mathrm{Mor}_2^{\mathrm{fin}}(\mathbf{Pr}_{\mathcal{E}})$ be the symmetric monoidal sub-4-category whose objects are f...

conjecture | open | MathDB 页面 | 原文线索

V2-071192 | MATHDB-368919

Graph-cover characterization of the Bethe partition function for double-edge factor graphs

摘录 / EXCERPT: Let N be a double-edge normal factor graph (DE-NFG). For each integer $M\geq 1$, let $Z_{\mathrm{B},M}(\mathsf{N})$ denote the degree-M Bethe partition function, defin...

conjecture | open | MathDB 页面 | 原文线索

V2-071193 | MATHDB-368922

Regularity conjecture for antiunitary representations of simply connected Lie groups

摘录 / EXCERPT: Let G be a simply connected Lie group, let τ_h be the involution associated with an Euler element $h\in\mathfrak g$, and let $(U,\mathcal H)$ be an antiunitary representatio...

conjecture | open | MathDB 页面 | 原文线索

V2-071194 | MATHDB-368923

Ye-Lim's universal Toeplitz factorization conjecture

摘录 / EXCERPT: Ye-Lim's conjecture. For all $n\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-071195 | MATHDB-368924

Bounded invertible reduction conjecture for Toeplitz factorizations

摘录 / EXCERPT: Bounded invertible reduction conjecture. There exists a constant $C>0$ such that every matrix can be expressed as a product of an invertible matrix and C Toeplitz matrices. In pa...

conjecture | open | MathDB 页面 | 原文线索

V2-071196 | MATHDB-368925

Quasi-Toeplitz row-equivalence conjecture

摘录 / EXCERPT: Quasi-Toeplitz row-equivalence conjecture. Every $n\times n$ matrix is row equivalent to a quasi-Toeplitz matrix.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-071197 | MATHDB-368926

Ceiling-bound conjecture for the maximal Toeplitz number

摘录 / EXCERPT: Ceiling-bound conjecture. For $n>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-071198 | MATHDB-368927

Local Hölder regularity conjecture for homogeneous fractional p-Laplace systems

摘录 / EXCERPT: Local Hölder regularity conjecture. Then u is locally Hölder continuous in Ω.

conjecture | open | MathDB 页面 | 原文线索

V2-071199 | MATHDB-368928

Eventual evenness of standard Young tableau counts for generic partitions

摘录 / EXCERPT: Let c be a constant. A partition λ of n boxes is called generic if its Young diagram fits into a $c\sqrt n$ by $c\sqrt n$ grid. Write $\operatorname{SYT}(\lambda)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-071200 | MATHDB-368929

Eventual alternating-group containment for cactus-group images

摘录 / EXCERPT: Let C_n be the cactus group acting on the set of standard Young tableaux of a partition, and let A_N denote the alternating group on N tableaux, where N is the number of st...

conjecture | open | MathDB 页面 | 原文线索

V2-071201 | MATHDB-368939

Tightness of the intermediate regime for relatively inexact gradient descent

摘录 / EXCERPT: Intermediate-regime tightness conjecture. The intermediate regime in Theorem 1 is tight. Furthermore, the corresponding worst-case function is bivariate.

conjecture | open | MathDB 页面 | 原文线索

V2-071202 | MATHDB-368941

A spectral-norm generalization of the Böttcher-Wenzel inequality for rectangular matrices

摘录 / EXCERPT: Let $M_{m,n}(\mathbb{C})$ denote the space of complex $m\times n$ matrices. For a matrix, write $\|\cdot\|_F$ for the Frobenius norm, $\|\cdot\|_2$ for the spectral norm, and...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071203 | MATHDB-368945

The ODP conjecture on the second largest local volume of klt singularities

摘录 / EXCERPT: Let $x\in X$ be an n-dimensional klt singularity, and let $\widehat{\operatorname{vol}}(x,X)$ denote its local volume, defined as the infimum of the normalized volumes of valuati...

conjecture | open | MathDB 页面 | 原文线索

V2-071204 | MATHDB-368956

The multiple genus two Cauchy sum leading-term formula

摘录 / EXCERPT: For a positive integer m, let $\widetilde X_{12}^{(a)},\widetilde X_{13}^{(a)},\widetilde X_{23}^{(a)}$ for $1\leq a\leq m$ be the formal variables associated with the multiple g...

conjecture | open | MathDB 页面 | 原文线索

V2-071205 | MATHDB-368957

The root-order conjecture for decomposable inversion roots

摘录 / EXCERPT: Root-order conjecture. If $\check\alpha\in\check R^+(w)$ is decomposable, then there exists a decomposition

conjecture | open | MathDB 页面 | 原文线索

V2-071206 | MATHDB-368958

The reflection deletion conjecture for Coxeter groups

摘录 / EXCERPT: Reflection deletion conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071207 | MATHDB-368963

A coefficient conjecture for a family of double Lambert series

摘录 / EXCERPT: Let a be a positive integer, let n be a positive integer, and write $[q^k]F(q)$ for the coefficient of q^k in $F(q)$. Let ... be the sum of the divisors of n. Coefficient con...

conjecture | open | MathDB 页面 | 原文线索

V2-071208 | MATHDB-368964

A coefficient identity involving a double Lambert series

摘录 / EXCERPT: Let r be a positive integer, and let $[q^k]F(q)$ denote the coefficient of q^k in $F(q)$. Coefficient conjecture. Then ... This is a proposed coefficient relation between two Lam...

conjecture | open | MathDB 页面 | 原文线索

V2-071209 | MATHDB-368985

Uniform lower-bound and convergence conjecture for the Bers constants of metric graphs

摘录 / EXCERPT: For each rank $r\geq 3$, let C_r be the constant from the theorem asserting that every rank-r metric graph contains a proper subgraph whose entropy is at least C_r. Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-071210 | MATHDB-368992

Gyárfás' boundedness conjecture for graphs with odd-cycle chromatic number at most three

摘录 / EXCERPT: Let G be a graph, and let $r(G)$ denote the maximum chromatic number of a subgraph spanned by an odd cycle of G, while $\chi(G)$ denotes the chromatic number of G. Gyárfás' b...

conjecture | open | MathDB 页面 | 原文线索

V2-071211 | MATHDB-368993

Conjecture that P_5-free graphs are path-perfect

摘录 / EXCERPT: Let G be a graph. A graph is path-perfect if its chromatic number equals the maximum chromatic number of a subgraph spanned by an odd cycle, that is, $\chi(G)=r(G)$. P_5-free path...

conjecture | open | MathDB 页面 | 原文线索

V2-071212 | MATHDB-368994

Conjecture that chromatic number is not bounded by a constant multiple of r(G)

摘录 / EXCERPT: Let $r(G)$ be the maximum chromatic number of a subgraph spanned by an odd cycle of a graph G, and let $\chi(G)$ be the chromatic number of G. Linear separation conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-071213 | MATHDB-368995

Scaling conjecture for longest subsequences in uniform separable permutations and cographs

摘录 / EXCERPT: Uniform separable scaling conjecture. There is a deterministic constant $c>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071214 | MATHDB-368998

Allocca et al.'s three-reflection bound for dihedral groups

摘录 / EXCERPT: Let D_n be the dihedral group, with order-n cyclic subgroup generated by the usual rotation. Let S be a generating set composed of three involutions, none of which belongs to...

conjecture | open | MathDB 页面 | 原文线索

V2-071215 | MATHDB-369001

Weight-additivity conjecture for weakly separated tableaux

摘录 / EXCERPT: Let $T \in \operatorname{SSYT}(2,[n])$ have m columns. Let $(S_1,\ldots,S_m)$ be the unique unordered m-tuple of pairwise weakly separated one-column tableaux whose union is...

conjecture | open | MathDB 页面 | 原文线索

V2-071216 | MATHDB-369012

Stable-pair/Donaldson-Thomas partition-function correspondence for logarithmic Calabi-Yau 3-folds

摘录 / EXCERPT: Stable-pair/Donaldson-Thomas correspondence conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071217 | MATHDB-369013

Gromov-Witten/Donaldson-Thomas correspondence for logarithmic Calabi-Yau 3-folds

摘录 / EXCERPT: Gromov-Witten/Donaldson-Thomas correspondence conjecture. With this change of variables,

conjecture | refuted | MathDB 页面 | 原文线索

V2-071218 | MATHDB-369022

Krzakala-Qiu-TAP conjecture for spherical Bayesian linear regression

摘录 / EXCERPT: Consider the Gaussian linear regression model with responses $\mathbf{y}\in\mathbb{R}^N$, design matrix $\mathbf{x}\in\mathbb{R}^{N\times p}$, coefficient vector...

conjecture | open | MathDB 页面 | 原文线索

V2-071219 | MATHDB-369023

Totally asymmetric SIC existence conjecture over finite fields

摘录 / EXCERPT: Let a SIC be a system of d^2 equiangular lines, represented here by vectors over a finite field extension as in the paper. A SIC is totally asymmetric when its automorphism group...

conjecture | open | MathDB 页面 | 原文线索

V2-071220 | MATHDB-369028

Joint empirical density formula for left and right eigenvectors

摘录 / EXCERPT: Let X be an $N\times N$ matrix, let z be an eigenvalue, and let ${\bf v}$ and ${\bf u}$ be corresponding right and left eigenvectors. Normalize the right eigenvector by...

conjecture | open | MathDB 页面 | 原文线索

V2-071221 | MATHDB-369029

Rubey's lattice conjecture for generalized chute moves

摘录 / EXCERPT: For a permutation w, let $\rp(w)$ denote its set of reduced pipe dreams, and order it by generalized chute moves. Rubey's lattice conjecture. This poset is a lattice. Certain spe...

conjecture | open | MathDB 页面 | 原文线索

V2-071222 | MATHDB-369030

Billey-Pawlowski pattern-avoidance conjecture for multiplicity-free Stanley symmetric functions

摘录 / EXCERPT: Let G_w be the Stanley symmetric function of a permutation w. Call G_w multiplicity-free when, in its Schur expansion, every coefficient is either 0 or 1. Billey-Pawlowsk...

conjecture | open | MathDB 页面 | 原文线索

V2-071223 | MATHDB-369036

Conjecture on support convergence for paired Gaussian matrix products

摘录 / EXCERPT: Support convergence conjecture. The support of $\mu_{XY^*}^N$ converges to the support of the limiting μ_{XY^*}.

conjecture | open | MathDB 页面 | 原文线索

V2-071224 | MATHDB-369043

Zhao's conjecture on the short zero-sum threshold below the Davenport midpoint

摘录 / EXCERPT: Let G be a finite abelian group with rank $r(G)\geq 2$ and $\mathsf{D}(G)=\mathsf{D}^*(G)$, and let $\exp(G)$ denote its exponent. Let $\mathsf{s}_{\leq m}(G)$ be the smallest le...

conjecture | open | MathDB 页面 | 原文线索

V2-071225 | MATHDB-369044

Zhao's midpoint conjecture for bounded-length zero-sum constants

摘录 / EXCERPT: Let G be a finite abelian group with rank $r(G)\geq 2$ and $\mathsf{D}(G)=\mathsf{D}^*(G)$. Let k be a positive integer with $\mathsf{D}(G)-k\in[\exp(G),\mathsf{D}(G)]$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071226 | MATHDB-369045

Zhao's conjecture on the midpoint bound for exact-length zero-sum constants

摘录 / EXCERPT: Let G be a finite abelian group, let $\exp(G)$ denote its exponent, and let $\mathsf{D}(G)$ denote its Davenport constant. Let $\mathsf{s}_{k\exp(G)}(G)$ be the smallest length f...

conjecture | open | MathDB 页面 | 原文线索

V2-071227 | MATHDB-369059

Existence of planar pulsating traveling waves with whole-space periodicity

摘录 / EXCERPT: Let $e\in\mathbb{S}^{n-1}$ and let $c_e^\star$ denote the critical speed in direction $-e$. A planar pulsating traveling wave solution has speed c, propagates in direction $-e$,...

conjecture | open | MathDB 页面 | 原文线索

V2-071228 | MATHDB-369064

A strengthened equality conjecture for real zeros of logarithmic derivatives

摘录 / EXCERPT: Strengthened Hawaii conjecture. There exists $\sigma\in\mathbb{R}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071229 | MATHDB-369074

Conjectured joint equidistributions for pairs X^(4)_7 and Y^(4)_7

摘录 / EXCERPT: Let a mesh pattern be specified by a set of marked points in a 3×3 grid together with a set of shaded cells; let $\sim_{jd}$ denote equality of joint distributions for the c...

conjecture | open | MathDB 页面 | 原文线索

V2-071230 | MATHDB-369076

Kaplansky-type zero-divisor conjecture for non-inert semi-latin quandles

摘录 / EXCERPT: Let $\mathbb{k}$ be an integral domain with unity and let X be a semi-latin quandle, meaning a quandle whose left and right translations satisfy the relevant latin-type condition...

conjecture | open | MathDB 页面 | 原文线索

V2-071231 | MATHDB-369077

Idempotent conjecture for integral quandle rings of semi-latin quandles

摘录 / EXCERPT: For a quandle X, its integral quandle ring is $\mathbb{Z}[X]$, and an idempotent is an element e satisfying $e^2=e$. The idempotents 0 and the basis elements of X are calle...

conjecture | open | MathDB 页面 | 原文线索

V2-071232 | MATHDB-369078

Idempotent conjecture for quandle rings of free products

摘录 / EXCERPT: Let ${X_i}_{i in I}$ be a family of quandles, and suppose that each integral quandle ring $\mathbb{Z}[X_i]$ has only trivial idempotents. Let $\star_{i in I}X_i$ denote their free pr...

conjecture | open | MathDB 页面 | 原文线索

V2-071233 | MATHDB-369090

Conjecture on optimal independent spanning trees in Erdős-Rényi graphs

摘录 / EXCERPT: Let $G\sim G(n,p)$ be an Erdős-Rényi random graph, and write $\delta(G)$ for its minimum degree. A spanning tree rooted at r is an independent spanning tree (IST) family member w...

conjecture | open | MathDB 页面 | 原文线索

V2-071234 | MATHDB-369099

Quillen equivalence conjecture for the complicial and marked Kan model structures

摘录 / EXCERPT: Let $\mathfrak{K}_{n}$ and $\mathfrak{M}_{n}$ be the model categories of marked Kan complexes and marked complicial sets, respectively, and let $(i_{\Delta},-_{\Delta})$ be the adj...

conjecture | open | MathDB 页面 | 原文线索

V2-071235 | MATHDB-369115

The MichelangeRoll expected-toss lower-bound conjecture

摘录 / EXCERPT: Let N be the number of possible outcomes in a uniform sample, and consider the bulleted procedure at the beginning of the subsection that generates tosses from this sample. Condi...

conjecture | open | MathDB 页面 | 原文线索

V2-071236 | MATHDB-369119

Prokhorov-Shokurov's effective b-semiampleness conjecture

摘录 / EXCERPT: Effective b-semiampleness conjecture. There is an integer $I=I(d)$ such that one can choose the diagram with $K_{X'}+\Delta'=\phi^*(K_X+\Delta)$, $I\mathbb M_{Y'}$ Cartier and free...

conjecture | open | MathDB 页面 | 原文线索

V2-071237 | MATHDB-369153

The $K_7^{=}$-minor 6-colorability conjecture

摘录 / EXCERPT: $K_7^{=}$-minor 6-colorability conjecture. Every graph with no $K_7^{=}$-minor is 6-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-071238 | MATHDB-369160

Mock rational point divisor conjecture for rank-2 bielliptic modular curves

摘录 / EXCERPT: Let X be a bielliptic modular curve of genus 2 and level N, and write $Jac(X)$ for its Jacobian. A mock rational point over $\mathbb{Q}(\sqrt{D})$ is a point of the type cons...

conjecture | open | MathDB 页面 | 原文线索

V2-071239 | MATHDB-369164

The flat-act cover conjecture for right-reversible monoids

摘录 / EXCERPT: Let S be a monoid, and write Act-S for the category of right S-acts. The monoid S is right-reversible when any two principal right ideals intersect. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071240 | MATHDB-369176

Lv-Zhang joint equidistribution conjecture for mesh patterns P_{11} and P_{12}

摘录 / EXCERPT: Let $\mathfrak{S}_n$ be the set of permutations of $[n]$, and for a mesh pattern P let $\mathrm{P}(\sigma)$ denote the number of occurrences of P in σ....

conjecture | open | MathDB 页面 | 原文线索

V2-071241 | MATHDB-369179

The large-genus Brill-Noether containment threshold conjecture for $r<s$

摘录 / EXCERPT: Large-genus containment threshold conjecture. If $e<T$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071242 | MATHDB-369184

Beilinson's conjecture for the special value of an elliptic curve over a number field

摘录 / EXCERPT: Beilinson's conjecture. If $L(1,A)\neq0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071243 | MATHDB-369189

Reverse log-majorization for nonnegative matrix powers

摘录 / EXCERPT: Let $A,B\geq 0$, and let p,q,r be real parameters satisfying $0<p\leq q$ and $r\geq 0$. Here $\succ_{\log}$ denotes log-majorization. Reverse log-majorization conjecture. One has...

conjecture | open | MathDB 页面 | 原文线索

V2-071244 | MATHDB-369200

Trotignon's bounded-treewidth conjecture for graphs excluding induced minors

摘录 / EXCERPT: Trotignon's conjecture. For all $r,t\in\mathbb{N}$, there exists $c=c(r,t)\in\mathbb{N}$ such that for every graph G, if G does not contain $W_{t \times t}$ or $K_{t,t}$ as an...

conjecture | refuted | MathDB 页面 | 原文线索

V2-071245 | MATHDB-369205

The proposed Macdonald index formula for the (A_1,D_3) theory

摘录 / EXCERPT: Let $(q;q)_m=\prod_{j=1}^{m}(1-q^j)$ and let $\binom{2m}{k}_q$ be the q-binomial coefficient. The (A_1,D_3) Macdonald-index conjecture. The Macdonald index of the Argyres-Dougl...

conjecture | open | MathDB 页面 | 原文线索

V2-071246 | MATHDB-369206

The Macdonald index formula for (A_1,D_{2n+1}) theories

摘录 / EXCERPT: The (A_1,D_{2n+1}) Macdonald-index conjecture. The Macdonald index is

conjecture | open | MathDB 页面 | 原文线索

V2-071247 | MATHDB-369212

Stevenhagen's conjecture on Eisenstein discriminants

摘录 / EXCERPT: Stevenhagen's conjecture. As $x\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-071248 | MATHDB-369218

Non-categorifiability conjecture for iterated integral fusion-ring extensions

摘录 / EXCERPT: Let $\mathcal{R}=\mathcal{R}_0$ be an integral fusion ring, and let $(\mathcal{R}_n)$ be the sequence obtained by iterating the integral extension construction of Proposition, star...

conjecture | open | MathDB 页面 | 原文线索

V2-071249 | MATHDB-369224

Finite Mal'tsev embedding conjecture for finite-domain predicates

摘录 / EXCERPT: Finite Mal'tsev embedding conjecture. Any predicate over a finite domain with an infinite Mal'tsev embedding also has a finite Mal'tsev embedding. In particular, any predicate with...

conjecture | open | MathDB 页面 | 原文线索

V2-071250 | MATHDB-369228

The finiteness conjecture for large Fatou components of polynomials

摘录 / EXCERPT: Let $f:\mathbb{C}\to\mathbb{C}$ be a polynomial, and let a Fatou component mean a connected component of the Fatou set of f. For $\epsilon>0$, measure diameters in the usual Eucl...

conjecture | open | MathDB 页面 | 原文线索

V2-071251 | MATHDB-369230

Frobenius divisibility conjecture for partial character tables of spetses

摘录 / EXCERPT: Let $\mathbb{G}=(W,\Lambda)$ be a simply connected $\mathbb{Z}_\ell$-spets with $\ell>2$ and $(|W|,\ell)=1$. Let T be the Sylow ℓ-subgroup of $\mathbb{G}(q)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071252 | MATHDB-369231

Unipotent Frobenius divisibility conjecture for spetses

摘录 / EXCERPT: Let $\mathbb{G}=(W\phi,\Lambda)$ be a simply connected $\mathbb{Z}_\ell$-spets and let $q\in\mathbb{Z}_\ell^\times$. Let S be the Sylow ℓ-subgroup of $(\mathbb{G},q)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071253 | MATHDB-369232

Polynomiality conjecture for almost-character scalar products

摘录 / EXCERPT: Let $\mathbb{G}=(W,\Lambda)$ be a $\mathbb{Z}_\ell$-reflection group with $(|W|,\ell)=1$, and let $T_a=\Lambda/\ell^a\Lambda$. For $b\ge a$, regard each...

conjecture | open | MathDB 页面 | 原文线索

V2-071254 | MATHDB-369234

The K-semistable degeneration conjecture for Fano threefolds of Family 2.16

摘录 / EXCERPT: Let X be a K-semistable $\mathbb{Q}$-Gorenstein degeneration of a smooth Fano threefold of Family textnumero 2.16. Degeneration conjecture. X is isomorphic to the blow-up of a...

conjecture | open | MathDB 页面 | 原文线索

V2-071255 | MATHDB-369235

Periodic Morton-Silverman conjecture over the rational numbers

摘录 / EXCERPT: Let $\varphi:\mathbb{P}^1(\mathbb{Q})\to\mathbb{P}^1(\mathbb{Q})$ be a morphism of degree d, and let $\operatorname{Per}(\varphi,\mathbb{P}^1(\mathbb{Q}))$ denote its set of peri...

conjecture | open | MathDB 页面 | 原文线索

V2-071256 | MATHDB-369238

Shift-invariant optimal estimator conjecture for distributions on the real line

摘录 / EXCERPT: Let μ be a distribution on $\mathbb{R}$, and consider estimators for the associated location family. An estimator is shift-invariant if shifting every observation by c shifts...

conjecture | open | MathDB 页面 | 原文线索

V2-071257 | MATHDB-369247

Exceptional negative coefficients conjecture for $D'(2,3,n)$

摘录 / EXCERPT: Let $D'(k,m,n)$ be defined by ... where k and m are positive integers. A coefficient is negative when it is less than zero. The exceptional negative coefficients conjecture for...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-071258 | MATHDB-369248

Eventual positivity conjecture for $D'(k,m,n)$

摘录 / EXCERPT: Let $D'(k,m,n)$ be defined by ... where k and m are positive integers. A series is eventually positive when all coefficients are nonnegative from some index onward. The eventual...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-071259 | MATHDB-369262

Positivity conjecture for the random-cluster polynomial \(\mathcal{M}_{ef}(q)\)

摘录 / EXCERPT: Positivity conjecture for $\mathcal{M}_{ef}(q)$. For $0\leq q\leq1$ and positive edgeweights x_e, one has

conjecture | open | MathDB 页面 | 原文线索

V2-071260 | MATHDB-369263

The \(\alpha\beta\gamma\)-ansatz for random-cluster correlations

摘录 / EXCERPT: Let G be any graph and $0\leq q\leq1$. Let $\mathcal{M}_{ef}(q)$ be the random-cluster polynomial associated with distinct edges e,f, and let $A_{\beta,\gamma}$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-071261 | MATHDB-369266

Strong Larson's conjecture for functions tiling by translations and dilations

摘录 / EXCERPT: Let $f:\mathbb{R}\to\mathbb{R}$ be a nonnegative function that tiles $ obreak\mathbb{R}$ by translations and dilations, meaning ... A wavelet set is a measurable set...

conjecture | open | MathDB 页面 | 原文线索

V2-071262 | MATHDB-369275

Consistency conjecture for the generalized Hermite-process spectral density estimator

摘录 / EXCERPT: Let $Z(x,t)$ satisfy Condition B and the previous assumptions, and let $F_q({\Greekmath 0115}_m)$ and $\widehat{F}_q({\Greekmath 0115}_m)$ be the corresponding population and sampl...

conjecture | open | MathDB 页面 | 原文线索

V2-071263 | MATHDB-369280

The conjectured quadratic eigenvector for the position matrix

摘录 / EXCERPT: The quadratic eigenvector conjecture. M has an eigenvalue-eigenvector pair $(\frac{1}{4},\vec{v})$ with multiplicity at least 1. Numerical computation suggests this eigenvector...

conjecture | open | MathDB 页面 | 原文线索

V2-071264 | MATHDB-369281

The conjectured nonzero spectrum of the position matrix

摘录 / EXCERPT: Let M be the position matrix of the one-shelf shuffling machine. Its eigenvalues describe the associated Markov chain on card positions; the previously established eigenvalues in...

conjecture | open | MathDB 页面 | 原文线索

V2-071265 | MATHDB-369283

Generalized Cauchy-Schwarz inequality for real powers

摘录 / EXCERPT: Generalized Cauchy-Schwarz conjecture. For every such p, $\mathbf{v}$, and $\mathbf{w}$,

conjecture | open | MathDB 页面 | 原文线索

V2-071266 | MATHDB-369288

Uniform upper bound conjecture for diameters of random graphs with minimum degree

摘录 / EXCERPT: Minimum-degree diameter conjecture. For every $d\geq3$ and $ \epsilon>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-071267 | MATHDB-369291

Conjecture on the facet-normal characterization of near-maximal strong Carathéodory number

摘录 / EXCERPT: Facet-normal characterization conjecture. The normals of K are the same as either those of a simplex bounded by an extra facet, or those of a pyramid that is not a simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-071268 | MATHDB-369292

Conjecture on Carathéodory numbers for strong convexity and subsets of facet normals

摘录 / EXCERPT: Subset-normal bound conjecture. The inequalities above hold.

conjecture | open | MathDB 页面 | 原文线索

V2-071269 | MATHDB-369299

Kaplansky's idempotent conjecture for torsionfree groups

摘录 / EXCERPT: Let R be a field and let G be a group. Kaplansky's idempotent conjecture. If G is torsionfree, then the group ring RG has only the trivial idempotents, namely 0 and 1....

conjecture | open | MathDB 页面 | 原文线索

V2-071270 | MATHDB-369311

Negative-index root-amplitude bound for k-generalized Fibonacci polynomials

摘录 / EXCERPT: Negative-index amplitude-bound conjecture. Numerical evidence indicates that

conjecture | open | MathDB 页面 | 原文线索

V2-071271 | MATHDB-369322

Berestycki-Caffarelli-Nirenberg conjecture on symmetry in the half-space

摘录 / EXCERPT: Berestycki-Caffarelli-Nirenberg conjecture. (i) If there is a positive bounded solution u of this problem, then u depends only on x_n; consequently, $u_{x_n}>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071272 | MATHDB-369328

The dual Strong F-conjecture for the moduli space of pointed genus-zero curves

摘录 / EXCERPT: Let $\overline M_{0,n}$ be the moduli space of stable genus-zero curves with n marked points. Let $\operatorname{PBDcone}(n)$ be the cone associated with effective partition bloc...

conjecture | open | MathDB 页面 | 原文线索

V2-071273 | MATHDB-369329

Minimax exclusivity conjecture for power-type losses

摘录 / EXCERPT: Let $p,q>0$ be distinct exponents, and for each $r>0$ let $\mathcal{L}_r$ be the class of losses whose local behavior near $a=\theta$ is $c(\theta)|\theta-a|^r+o(|\theta-a|^r)$ wit...

conjecture | open | MathDB 页面 | 原文线索

V2-071274 | MATHDB-369330

Existence conjecture for global total exclusivity partitions

摘录 / EXCERPT: Let $\mathscr{L}$ be a rich loss space, for example the space of all continuous losses satisfying mild growth conditions, and let optimality be defined by a natural optimality noti...

conjecture | open | MathDB 页面 | 原文线索

V2-071275 | MATHDB-369336

The sharp lower bound conjecture for edge eigenvector Gaussian approximation

摘录 / EXCERPT: Let $\mathcal{G}_{d,N}$ denote the space of random d-regular graphs on N vertices, and let $T_N: \mathcal{G}_{d,N} \to \mathbb{R}$ be a sequence of statistics that distinguishe...

conjecture | open | MathDB 页面 | 原文线索

V2-071276 | MATHDB-369338

Fulton's weak nefness conjecture for the moduli space of stable pointed curves

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_{0,n}$ be the moduli space of stable n-pointed genus-zero curves. A divisor E on $\overline{\mathcal{M}}_{0,n}$ is F-nef if $C(a,b,c,d).E\geq 0$ f...

conjecture | open | MathDB 页面 | 原文线索

V2-071277 | MATHDB-369342

Polygonality conjecture for framing posets of strongly planar DAGs

摘录 / EXCERPT: Let G be a strongly planar directed acyclic graph, and let $\mathcal L$ be its framing poset. If x and y cover a common element z and there exists a least upper bound...

conjecture | open | MathDB 页面 | 原文线索

V2-071278 | MATHDB-369345

Eulerian odd-cover conjecture

摘录 / EXCERPT: Let G be an Eulerian graph, let $\Delta(G)$ be its maximum degree, and let a path odd-cover (respectively, cycle odd-cover) be a collection of paths (respectively, cycles) whose...

conjecture | open | MathDB 页面 | 原文线索

V2-071279 | MATHDB-369354

DeBiasio-Han-Lo-Molla-Piga-Treglown conjecture on Turánable oriented graphs

摘录 / EXCERPT: DeBiasio-Han-Lo-Molla-Piga-Treglown conjecture. The oriented graph H is Turánable if and only if there is an $s\in\mathbb{N}$ such that $H\subset D_s$.

conjecture | open | MathDB 页面 | 原文线索

V2-071280 | MATHDB-369355

Updated characterization conjecture for Turánable oriented graphs

摘录 / EXCERPT: Updated Turánability conjecture. The oriented graph H is Turánable if and only if there are integers r and k such that

conjecture | open | MathDB 页面 | 原文线索

V2-071281 | MATHDB-369363

Equidistribution conjecture for rank statistics on partitions into two part sizes

摘录 / EXCERPT: the observed rank-divisibility conjecture. The numbers of partitions of $An+B$ into two part sizes for which $rk_2(\lambda)$ is odd and for which $rk_2(\lambda)$ is even are both d...

conjecture | open | MathDB 页面 | 原文线索

V2-071282 | MATHDB-369372

Rationality conjecture for limiting invariants of Fermat hypersurface rings

摘录 / EXCERPT: For $d\geq2$, let ... be the degree-d Fermat hypersurface ring, and define c_n and c'_n by ... ... Here ξ_{2d} is a primitive $2d$-th root of unity. Rationality conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-071283 | MATHDB-369374

Simultaneous asymptotic equidistribution conjecture for free word tuples

摘录 / EXCERPT: Let $\mathbb F_r$ be the free group on r generators. A tuple $(w_1,\ldots,w_d)\in\mathbb F_r^d$ is simultaneously asymptotically equidistributed when its word-value distribution...

conjecture | open | MathDB 页面 | 原文线索

V2-071284 | MATHDB-369375

Weak twist conjecture for Artin generating sets

摘录 / EXCERPT: Weak twist conjecture. If $U\subseteq A$ is an Artin generating set with $R_S=R_U$, then Γ_S and Γ_U are twist equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-071285 | MATHDB-369376

Strong twist conjecture for Artin generating sets

摘录 / EXCERPT: Strong twist conjecture. If $U\subseteq A$ is an Artin generating set with $R_S=R_U$, then S and U are twist equivalent.

conjecture | open | MathDB 页面 | 原文线索

V2-071286 | MATHDB-369379

Stabilization conjecture for dominant affine Bruhat intervals

摘录 / EXCERPT: Stabilization conjecture. For any dominant coweight $\lambda\in\Lambda^\vee$ and dominant $x,y\in W$, there exists an integer $N_0=N_0(x,y,\lambda)$ such that for every $N\geq N_0$...

conjecture | open | MathDB 页面 | 原文线索

V2-071287 | MATHDB-369387

Classification conjecture for triply-transitive strongly-regular graphs

摘录 / EXCERPT: Let Γ be a graph as in Hypothesis; in particular, Γ is a strongly-regular graph in the setting of the paper. Complete classification conjecture. The graph Γ is...

conjecture | open | MathDB 页面 | 原文线索

V2-071288 | MATHDB-369391

Restricted-type orthonormal Strichartz conjecture at endpoint A

摘录 / EXCERPT: Let $n\geq1$, and let $A=(1/p,1/q)=((n-1)/(n+1),n/(n+1))$. For an orthonormal family $\{f_j\}_{j\in J}\subset L^2(\mathbb{R}^n)$ and a sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-071289 | MATHDB-369392

Restricted-type orthonormal Strichartz estimate on the segment (O,A]

摘录 / EXCERPT: Let $n\geq2$, let $(1/p,1/q)$ lie on the line segment $(O,A]$, and set s by ... For an orthonormal family $\{f_j\}_{j\in J}\subset\dot{H}^{s}(\mathbb{R}^n)$ and coefficients...

conjecture | open | MathDB 页面 | 原文线索

V2-071290 | MATHDB-369399

Erickson-Klein conjecture on weakly cordial trees

摘录 / EXCERPT: Let A be an Abelian group. A family of graphs is weakly A-cordial if all but finitely many graphs in the family are A-cordial. Erickson-Klein conjecture. For every Abelian gr...

conjecture | open | MathDB 页面 | 原文线索

V2-071291 | MATHDB-369405

Ichino-Ikeda type co-period formula for symmetric cube L-values

摘录 / EXCERPT: Let F be the number field and G the group in the preceding construction, let σ be a cuspidal automorphic $\operatorname{GL}_2(\mathbb{A})$-representation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071292 | MATHDB-369406

Yan's conjecture on isotropy groups of simple derivations

摘录 / EXCERPT: Yan's conjecture. If d is a simple derivation of R_n, then $\operatorname{Aut}(R_n)_d$ is conjugate in $\operatorname{Aut}(R_n)$ to a subgroup of the group of translations.

conjecture | open | MathDB 页面 | 原文线索

V2-071293 | MATHDB-369407

Zero-freeness in a tube for the Sherrington-Kirkpatrick partition function

摘录 / EXCERPT: Let $Z_{\bm{G}}(\beta)$ denote the partition function of the Sherrington-Kirkpatrick model, let $\varepsilon>0$, and let β_{2nd} be the second-moment threshold. A co...

conjecture | open | MathDB 页面 | 原文线索

V2-071294 | MATHDB-369422

Block Lanczos successive-iteration interlacing conjecture

摘录 / EXCERPT: Let $A\in\mathbb{R}^{n\times n}$ be a symmetric matrix and let $v\in\mathbb{R}^{n\times p}$ be a block vector. Let s be the largest index such that the block Krylov subspace...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-071295 | MATHDB-369429

Schröer's klt type and strong F-regularity conjecture

摘录 / EXCERPT: Let $R=\left(\mathbb{C}[X_1, \dots, X_n]/(f_1, \dots, f_r)\right)_{(X_1, \dots, X_n)}$, where each $f_i\in\mathbb{Z}[X_1,\dots,X_n]$, and let $R\bmod p$ denote its reduction modulo...

conjecture | open | MathDB 页面 | 原文线索

V2-071296 | MATHDB-369430

Gorenstein reduction conjecture for positive F-signature

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a d-dimensional Gorenstein local domain essentially of finite type over $\mathbb{C}$. Gorenstein positivity conjecture. If $\operatorname{Spec} R$ has kl...

conjecture | open | MathDB 页面 | 原文线索

V2-071297 | MATHDB-369436

Bosma-Dekking-Steiner metallic-mean iteration conjecture

摘录 / EXCERPT: For an integer $j\geq 1$, let α_j be the j-th metallic mean, so that $\alpha_j^2=j\alpha_j+1$. Let r_1 be the unique positive real number such that all terms defined by...

conjecture | open | MathDB 页面 | 原文线索

V2-071298 | MATHDB-369450

The Dehn-filling conjecture for mapping class groups

摘录 / EXCERPT: Let S be any surface of finite type, and let $g_1,\dots,g_l\in MCG(S)$. For a subset A of a group, write $\langle\langle A\rangle\rangle$ for its normal closure. Dehn-filling c...

conjecture | open | MathDB 页面 | 原文线索

V2-071299 | MATHDB-369481

Higher-order Eulerian triangle total positivity conjecture

摘录 / EXCERPT: For $r\geq 1$, let $E^{(r)} = \left(\genfrac{\langle}{\rangle}{0pt}{}{n}{k}^{\!(r)}\right)_{n,k\geq 0}$ be the rth-order Eulerian triangle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071300 | MATHDB-369482

Reversed higher-order quasi-Eulerian triangle total positivity conjecture

摘录 / EXCERPT: For $r\geq 2$, let $Q_{\rm C}^{(r)}$ and $Q_{\rm S}^{(r)}$ be the quasi-Eulerian cycle and subset triangles, respectively, and let $M^{\rm rev}$ denote reversal of the rows of a ma...

conjecture | open | MathDB 页面 | 原文线索

V2-071301 | MATHDB-369483

Left-half-plane conjecture for third-order Stirling subset polynomials

摘录 / EXCERPT: For $n\geq 0$, let $s_{3,n}(x)$ denote the third-order Stirling subset polynomial. Third-order Stirling subset zero conjecture. The zeros of $s_{3,n}(x)$ lie in the left half-plane...

conjecture | open | MathDB 页面 | 原文线索

V2-071302 | MATHDB-369484

Equivariant asymptotic Hodge conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let $\pi:\mathcal{Y}\to\Delta$ be a G-equivariant semistable degeneration of Calabi-Yau threefolds with ordinary double points, and let $\widetilde{Y}_0$ be a G-equivariant cre...

conjecture | open | MathDB 页面 | 原文线索

V2-071303 | MATHDB-369485

Node surgery principle for Calabi-Yau threefolds

摘录 / EXCERPT: Let Y_t be a smooth fibre of a semistable degeneration with nodal central fibre Y_0, let $\beta\in H^{2,2}(Y_t,\mathbb{Q})$ be rational, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071304 | MATHDB-369486

Non-local modulus of continuity on a bounded convex domain

摘录 / EXCERPT: Non-local modulus of continuity conjecture. Then $\varphi(\cdot,t)$ is a modulus of continuity for $u(\cdot,t)$ on $\Omega\times[0,\infty)$. This proposes that the higher-dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-071305 | MATHDB-369490

The boundedness conjecture for real finite-gap Boussinesq solutions

摘录 / EXCERPT: Let $u[K]$ denote the Abelian argument associated with a characteristic $[K]$, and let the finite-gap solution be the solution referred to in the source as the real finite-gap solu...

conjecture | open | MathDB 页面 | 原文线索

V2-071306 | MATHDB-369493

Furdui-Sîntămărian's consecutive harmonic-number product conjecture

摘录 / EXCERPT: Let $q\geq 1$ be an integer, and let $H_k=\sum_{j=1}^k\frac{1}{j}$ denote the kth harmonic number. For rational numbers $\alpha_{q,2},\alpha_{q,3},\ldots,\alpha_{q,q+2}$, conside...

conjecture | open | MathDB 页面 | 原文线索

V2-071307 | MATHDB-369494

Chowla-Milnor conjecture on linear independence of Hurwitz zeta values

摘录 / EXCERPT: Chowla-Milnor conjecture. The set $\mathfrak{L}$ is linearly independent over $\mathbb{Q}$. The conjecture is important because its consequences include conditional irrationality r...

conjecture | open | MathDB 页面 | 原文线索

V2-071308 | MATHDB-369505

Schneider's support conjecture for mixed Hessian measures

摘录 / EXCERPT: Schneider's conjecture. For $f_1,\ldots,f_n\in\operatorname{Conv}(\Omega)$, the mixed Hessian measure satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071309 | MATHDB-369508

Functorial Alperin weight conjecture

摘录 / EXCERPT: Functorial Alperin weight conjecture. In $K_0(\mathcal{F}\mathcal{B}\ell_{\mathbb{F},k})$,

conjecture | open | MathDB 页面 | 原文线索

V2-071310 | MATHDB-369513

The injectivity conjecture for relative braid homomorphisms

摘录 / EXCERPT: Let $f:\Sigma\to\Sigma'$ be a morphism as in Lemma, and let $f_*:Br_\Delta^f\to Br_{f(\Delta)}$ be the induced homomorphism. Relative braid homomorphism injectivity conjecture. In...

conjecture | open | MathDB 页面 | 原文线索

V2-071311 | MATHDB-369517

The non-injectivity conjecture for surface braid-group abelianization

摘录 / EXCERPT: Let Br_Δ be the braid group associated with a triangulation Δ, and let ... be its abelianization. Abelianization non-injectivity conjecture. The canonical homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-071312 | MATHDB-369520

Directed regular graph cycle-cover conjecture

摘录 / EXCERPT: Let G be a directed d-regular graph on n vertices, and regard a cycle as a directed cycle, with an individual edge allowed to count as a cycle of length two as in the source....

conjecture | open | MathDB 页面 | 原文线索

V2-071313 | MATHDB-369523

Selberg's orthogonality conjecture for cuspidal automorphic representations

摘录 / EXCERPT: Let $[?]$

conjecture | open | MathDB 页面 | 原文线索

V2-071314 | MATHDB-369525

Optimized full cyclic coordinate descent has a superlinear numerical rate

摘录 / EXCERPT: Let N denote the iteration count, and consider full fixed-step cyclic coordinate descent on a problem partitioned into 2 blocks, allowing the update to use all past gradient in...

conjecture | open | MathDB 页面 | 原文线索

V2-071315 | MATHDB-369526

Optimized full inexact gradient descent has a superlinear numerical rate for residual gradient minimization

摘录 / EXCERPT: Let N denote the iteration count and let the performance criterion be the minimum squared gradient norm, $\min_{k\in\{1,\ldots,N\}}\|\nabla f(x_k)\|^2$, for minimizing a smooth c...

conjecture | open | MathDB 页面 | 原文线索

V2-071316 | MATHDB-369537

The crepant-resolution conjecture for DT4 invariants of $[\mathbb C^4/\mathbb Z_r]$

摘录 / EXCERPT: Let r be a positive integer, let $[[200~\mathbb C^4/\mathbb Z_r]$ be the quotient Calabi-Yau 4-orbifold considered in the source, and let $q_0,\ldots,q_{r-1}$ be variables. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-071317 | MATHDB-369545

The barcode-process universality conjecture in the waterfall region

摘录 / EXCERPT: Let (t,x) be a macroscopic point in the waterfall region $\mathcal{W}$, and let q and κ be the model parameters. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071318 | MATHDB-369546

The two-periodic barcode-density conjecture

摘录 / EXCERPT: For $t\in\mathbb{Z}$ and $M\in\mathbb{Z}$, define the two truncation limits $\rho_0(t)$ and $\rho_1(t)$ by the displayed sums over $n\in\frac12\mathbb{Z}_{\ge0}$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071319 | MATHDB-369547

The block-Toeplitz limit conjecture for the barcode kernel

摘录 / EXCERPT: Let $K^{\mathrm{barcode}}_{(L)}(s,t)$ be the pre-limit barcode kernel for $s,t\in\mathbb{Z}$, and let q be the model parameter. Barcode-kernel limit conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-071320 | MATHDB-369548

The regularized-series conjecture for the barcode kernel

摘录 / EXCERPT: Let q and κ be the model parameters, let $\mathcal{F}_n^s(0)$ be the orthogonal functions appearing in the barcode kernel, and let $\mathcal{K}^{\mathrm{barcode}}(s,t)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-071321 | MATHDB-369555

Kalai's full flag conjecture for centrally symmetric polytopes

摘录 / EXCERPT: Let P be a centrally symmetric d-dimensional polytope. A flag of P is a chain of faces of P, one in each dimension. Kalai's full flag conjecture. P has at least...

conjecture | open | MathDB 页面 | 原文线索

V2-071322 | MATHDB-369561

Aubry-Castryck-Ghorpade-Lachaud-O'Sullivan-Ram conjecture for weighted projective spaces

摘录 / EXCERPT: Let $w=(w_0,w_1,\ots,w_m)\in\mathbb{N}_{\geq 1}^{m+1}$, let $\mathbb{P}(w)$ be the weighted projective space over $\mathbb{F}_q$, and let $e_q(d;w_0,w_1,\dots,w_m)$ denote the maxim...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071323 | MATHDB-369562

Painlevé I limit conjecture for the rescaled tau-function near the critical surface

摘录 / EXCERPT: Painlevé I limit conjecture. There exists a choice of constant $C = C(P_0,{\bf u})>0$ and a polynomial $Q(t_1,t_2,t_5)$ such that, defining $\hat{\tau}_0:= e^{Q}$, considered as a...

conjecture | open | MathDB 页面 | 原文线索

V2-071324 | MATHDB-369563

Pokrovskiy's semidegree conjecture for linked tournaments

摘录 / EXCERPT: Pokrovskiy's conjecture. For every $k\in\mathbb{N}$, there exists an integer $h(k)$ such that every $2k$-connected tournament with $\delta^0(D)\geq h(k)$ is k-linked.

conjecture | open | MathDB 页面 | 原文线索

V2-071325 | MATHDB-369564

Gire3o-Popielarz-Snyder linear semidegree conjecture for linked tournaments

摘录 / EXCERPT: Gir{a}o-Popielarz-Snyder conjecture. There exists a constant $C>0$ such that every $(2k+1)$-connected tournament with $\delta^+(T)\geq Ck$ is k-linked.

conjecture | open | MathDB 页面 | 原文线索

V2-071326 | MATHDB-369574

Linear degree-boundedness conjecture for pivot-minor exclusions

摘录 / EXCERPT: For a positive integer s, a graph is $K_{s,s}$-free if it contains no subgraph isomorphic to $K_{s,s}$. Linear pivot-minor degree-boundedness conjecture. For each bipartite graph...

conjecture | open | MathDB 页面 | 原文线索

V2-071327 | MATHDB-369579

Branch-point characterization of focusing mKdV finite-gap solutions

摘录 / EXCERPT: Branch-point characterization. All expressions given by the corresponding finite-gap formula comply with the first reality condition on...

conjecture | open | MathDB 页面 | 原文线索

V2-071328 | MATHDB-369580

The non-automorphy conjecture for truncated multiple Dirichlet series

摘录 / EXCERPT: Let A be the integer matrix defining the multiple Dirichlet series $D_{A,\boldsymbol{\Pi}}(\boldsymbol{s};\boldsymbol{\omega},\boldsymbol{\omega'})$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071329 | MATHDB-369581

The functional-equation existence conjecture for non-automorphic Dirichlet series

摘录 / EXCERPT: Let A, $\boldsymbol{\Pi}$, $\boldsymbol{\omega}$, and $\boldsymbol{\omega'}$ be parameters for the Dirichlet series...

conjecture | open | MathDB 页面 | 原文线索

V2-071330 | MATHDB-369582

The idempotent fim measure-stabilizer correspondence for automorphism groups

摘录 / EXCERPT: The automorphism-group idempotent fim measure conjecture. The following conditions are equivalent: (i) μ is idempotent with respect to the new convolution product; (ii) μ i...

conjecture | open | MathDB 页面 | 原文线索

V2-071331 | MATHDB-369588

Fayers' divisibility conjecture for $(s,s+t,s+2t,\dots)$-core polynomials

摘录 / EXCERPT: Let s and t be positive integers with $\gcd(s,t)=1$, and let $f_t(s)$ be the polynomial appearing in Fayers' formula for the number of $(s,s+t,s+2t,\dots)$-core partitions. Fay...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071332 | MATHDB-369590

Bapat-Sunder's permanent Hadamard-product inequality

摘录 / EXCERPT: Let $\mathcal{H}_n$ be the set of $n\times n$ positive semidefinite Hermitian matrices. For matrices $A=[a_{i,j}]$ and $B=[b_{i,j}]$ in $\mathcal{H}_n$, define the Hadamard product...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071333 | MATHDB-369591

Bapat-Sunder's largest-eigenvalue conjecture for the permanent matrix

摘录 / EXCERPT: Bapat-Sunder's conjecture. The largest eigenvalue of F_A should satisfy

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071334 | MATHDB-369592

Real-part largest-eigenvalue conjecture for the permanent matrix

摘录 / EXCERPT: Let $\mathcal{H}_n$ be the set of $n\times n$ positive semidefinite Hermitian matrices. For $A=[a_{i,j}]\in\mathcal{H}_n$, let $A(i,j)$ be obtained by deleting row i and column...

conjecture | open | MathDB 页面 | 原文线索

V2-071335 | MATHDB-369603

Ravenel's telescope conjecture for compactly generated triangulated categories

摘录 / EXCERPT: Ravenel's telescope conjecture. The decomposition $\mathbf{t}$ is compactly generated.

conjecture | open | MathDB 页面 | 原文线索

V2-071336 | MATHDB-369604

Krause's ideal-theoretic reformulation of the telescope conjecture

摘录 / EXCERPT: Krause's reformulated telescope conjecture. The ideal $I_{\mathbf{t}}$ is generated by identity morphisms of objects in $\mathcal{D}^c$.

conjecture | open | MathDB 页面 | 原文线索

V2-071337 | MATHDB-369605

Generalized telescope conjecture for t-structures with definable co-aisle

摘录 / EXCERPT: Generalized telescope conjecture. The assertion is exactly the stable telescope conjecture, or equivalently its ideal-theoretic reformulation, with $\mathbf{t}$ a t-structure in...

conjecture | open | MathDB 页面 | 原文线索

V2-071338 | MATHDB-369606

Critical-domain-length conjecture for controllability of competitive systems

摘录 / EXCERPT: Consider the diffusive Lotka-Volterra strongly competitive system on an interval $(0,L)$ with boundary-constrained controls, parameters $a>0$ and $b>1$ satisfying $b>a$, initial da...

conjecture | open | MathDB 页面 | 原文线索

V2-071339 | MATHDB-369618

Frantzikinakis-Lesigne-Weirdl conjecture on random differences in Szemerédi's theorem

摘录 / EXCERPT: Let k be a positive integer and let $S\subset\mathbb{N}$ be chosen independently at random with ... Here, $f=o(g)$ if and only if $g=\omega(f)$. Frantzikinakis-Lesigne-Weirdl conje...

conjecture | open | MathDB 页面 | 原文线索

V2-071340 | MATHDB-369620

Keith's congruence conjecture for coefficients of a reciprocal false theta function

摘录 / EXCERPT: Keith's conjecture. For every integer $n\geq 0$,

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071341 | MATHDB-369621

The polytopal realization conjecture for ornamentation lattices

摘录 / EXCERPT: Let D be a directed graph, and let $\operatorname{\mathcal{O}}(D)$ denote its ornamentation lattice. Polytopal realization conjecture. For any directed graph D, the ornamentati...

conjecture | open | MathDB 页面 | 原文线索

V2-071342 | MATHDB-369622

Repdigit Niven-number parametrization conjecture

摘录 / EXCERPT: Repdigit Niven-number parametrization conjecture. The solutions of the repdigit Niven-number equation are parameterized by

conjecture | open | MathDB 页面 | 原文线索

V2-071343 | MATHDB-369624

Optimal one-symbol redundancy conjecture for Lee-metric function-correcting codes

摘录 / EXCERPT: Let $t\geq 1$ and $q\geq 2t(t+1)+1$. For a function f, consider the encoding ... where $p\in[2t:\lfloor q/2\rfloor]$. The code is required to function-correct for the Lee weight fu...

conjecture | open | MathDB 页面 | 原文线索

V2-071344 | MATHDB-369629

Geometric Langlands connected-components conjecture

摘录 / EXCERPT: Let X be a smooth complete curve over a field k, let G be a reductive group, and let $\Shv_{\Nilp}(\Bun_G)$ be the category of ℓ-adic sheaves on $\Bun_G$ with nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-071345 | MATHDB-369632

The Functional Batch Conjecture for binary simplex codes

摘录 / EXCERPT: Functional Batch Conjecture. The binary $[k,2^k-1]$ simplex code is a 2^{k-1}-functional batch code.

conjecture | open | MathDB 页面 | 原文线索

V2-071346 | MATHDB-369633

Balister-Győri-Schelp partition conjecture for binary vector lists

摘录 / EXCERPT: Balister-Győri-Schelp conjecture. There exists a partition of $\mathbb F_2^k$ into 2-sets $\{w_i,z_i\}$, for $i\in\{1,\dots,2^{k-1}\}$, such that

conjecture | open | MathDB 页面 | 原文线索

V2-071347 | MATHDB-369634

Xu's conjecture on alternating double zeta values

摘录 / EXCERPT: For a positive integer k, let ... be the indicated combination of alternating double zeta values, where a bar denotes a negative sign in the corresponding alternating multiple zeta...

conjecture | open | MathDB 页面 | 原文线索

V2-071348 | MATHDB-369640

The intersection-number-zero criterion for plabic graphs

摘录 / EXCERPT: Fix a positive integer m, and let G be a plabic graph of type (k,n) with $\dim\Pi_G=km$. Let T_G denote the associated family of k-planes. Intersection-number-zero criter...

conjecture | open | MathDB 页面 | 原文线索

V2-071349 | MATHDB-369643

Backbone colouring conjecture for chordal graphs and spanning forests

摘录 / EXCERPT: Backbone colouring conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071350 | MATHDB-369644

Bounded-mad backbone colouring conjecture for chordal graphs

摘录 / EXCERPT: Bounded-mad backbone colouring conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071351 | MATHDB-369647

Localizability of the nonzero filter in a quantale

摘录 / EXCERPT: Nonzero-filter localizability conjecture. The filter $ mid 0$ is localizable over Q.

conjecture | open | MathDB 页面 | 原文线索

V2-071352 | MATHDB-369648

Beauville decomposition conjecture for compactified Jacobian fibrations

摘录 / EXCERPT: Let $\bar J_{\mathcal C} \xrightarrow{\pi} B$ be a compactified Jacobian fibration associated to a flat family of integral projective locally planar curves of arithmetic genus g...

conjecture | open | MathDB 页面 | 原文线索

V2-071353 | MATHDB-369669

Horizon-map determination of the set of minima

摘录 / EXCERPT: For a set of minima, let C_{max} be the largest set of curves containing C for which every nonfilling subset C' of C_{max} gives...

conjecture | open | MathDB 页面 | 原文线索

V2-071354 | MATHDB-369674

Binomial Ward numbers and central Stirling numbers conjecture

摘录 / EXCERPT: Central Stirling numbers conjecture. The binomial Ward numbers satisfy

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071355 | MATHDB-369679

Hirose-Watanabe-Yoshida conjecture on the F-pure threshold and the a-invariant

摘录 / EXCERPT: Hirose-Watanabe-Yoshida conjecture. One always has

conjecture | open | MathDB 页面 | 原文线索

V2-071356 | MATHDB-369686

Bressan's Fire Conjecture for isotropic fire blocking

摘录 / EXCERPT: Let the fire propagate with unit speed in all directions from $R_0=B_1(0)$, and let the barrier be constructed at constant speed $\sigma>0$, so that $F=\overline{B_1(0)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071357 | MATHDB-369695

Jack polynomial dominance conjecture

摘录 / EXCERPT: Jack polynomial dominance conjecture. The following statements are equivalent: λ dominates μ; for every $x\in[0,\infty)^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-071358 | MATHDB-369696

Jack polynomial weak-dominance conjecture

摘录 / EXCERPT: Jack polynomial weak-dominance conjecture. The following statements are equivalent: λ weakly dominates μ; for $x\in[1,\infty)^n$,

conjecture | open | MathDB 页面 | 原文线索

V2-071359 | MATHDB-369698

Nonexistence conjecture for rational periodic vectors of type (L,R,L,R)

摘录 / EXCERPT: Let C,D be distinct, nonzero rational numbers, and let $f(x,y)=Cx^2+Dy^2$ be a binary quadratic form. A rational periodic vector of type (L,R,L,R) is a rational vector undergoi...

conjecture | open | MathDB 页面 | 原文线索

V2-071360 | MATHDB-369704

Induced-minor-free product structure conjecture

摘录 / EXCERPT: Induced-minor-free product structure conjecture. There is a function $f\colon \mathbb{N}\to\mathbb{N}$ such that every H-induced-minor-free graph G with maximum degree at most...

conjecture | open | MathDB 页面 | 原文线索

V2-071361 | MATHDB-369705

Lokshtanov-McCarty bounded-degree region intersection conjecture

摘录 / EXCERPT: Lokshtanov-McCarty's conjecture. There exists a graph H' such that every H-induced-minor-free graph with maximum degree at most Δ is a region intersection graph over an...

conjecture | open | MathDB 页面 | 原文线索

V2-071362 | MATHDB-369706

Dimensional absolute stability implies diagonalizability for T-meshes

摘录 / EXCERPT: Dimensional absolute stability conjecture. $\mathscr{T}$ is dimensionally absolutely stable if and only if it is diagonalizable.

conjecture | open | MathDB 页面 | 原文线索

V2-071363 | MATHDB-369708

Large irreducible factors in iterated polynomial-value differences

摘录 / EXCERPT: Let $\alpha,\beta\in\mathbb{F}_q$, let $\lambda\in\overline{\mathbb F}_p$, and let $f_\lambda(z)=z^d+\lambda$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-071364 | MATHDB-369712

Maximal-dimension conjecture for constant-rank affine spaces of nilpotent matrices

摘录 / EXCERPT: Let $n\in\mathbb{N}-\{0\}$ and let K be a field. Let S be an affine subspace of $M(n\times n,K)$ such that every element of S is nilpotent, and assume that each element of...

conjecture | open | MathDB 页面 | 原文线索

V2-071365 | MATHDB-369713

The soficity conjecture for minor-excluded unimodular random rooted graphs

摘录 / EXCERPT: Minor-excluded soficity conjecture. Every such unimodular random rooted graph (G,ρ) is sofic.

conjecture | open | MathDB 页面 | 原文线索

V2-071366 | MATHDB-369716

The five-holed graph chromatic bound conjecture

摘录 / EXCERPT: Five-holed graph chromatic bound conjecture. If G is 5-holed, then

conjecture | open | MathDB 页面 | 原文线索

V2-071367 | MATHDB-369718

Level-rank conjecture for finite reductive groups

摘录 / EXCERPT: Level-rank conjecture. The two image sets are unions of $H_{L,\lambda}^G(\zeta_m)$- and $H_{M,\mu}^G(\zeta_l)$-blocks, respectively; the bijection between them induced by...

conjecture | open | MathDB 页面 | 原文线索

V2-071368 | MATHDB-369737

Nadirashvili's universal nodal measure lower-bound conjecture

摘录 / EXCERPT: Let u be a harmonic function in $B(0,1)\subset\mathbb{R}^n$ with $u(0)=0$, and let $Z(u)$ denote its nodal set. Nadirashvili's lower-bound conjecture. There exists a universal co...

conjecture | open | MathDB 页面 | 原文线索

V2-071369 | MATHDB-369747

Gerards-Seymour odd-minor coloring conjecture

摘录 / EXCERPT: Let G be a graph and let t be a positive integer. An odd-minor of a graph is a graph formed from a subgraph by contracting edge cuts; odd-minors preserve the parity of cycles....

conjecture | open | MathDB 页面 | 原文线索

V2-071370 | MATHDB-369751

Liechti-Strenner's minimal dilatation conjecture for closed nonorientable surfaces

摘录 / EXCERPT: Liechti-Strenner's conjecture. For all $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-071371 | MATHDB-369760

Strong modeling limit conjecture for monadically stable graph classes

摘录 / EXCERPT: Strong modeling limit conjecture. Every FO-convergent sequence of graphs in $\mathscr C$ has a strong modeling FO-limit. The paper proves existence of modeling FO-limits for monadi...

conjecture | open | MathDB 页面 | 原文线索

V2-071372 | MATHDB-369764

Batyrev-Manin conjecture for singular degree-one del Pezzo surfaces

摘录 / EXCERPT: Let $A,B\in\mathbb{Z}$ satisfy $4A^3-27B^2\neq 0$, let $Q(u,v)$ be a positive-definite binary quadratic form, and define ... For rational points represented by integral (x,y,u,v) w...

conjecture | open | MathDB 页面 | 原文线索

V2-071373 | MATHDB-369765

Dolbeault S-duality matching conjecture for tempered magical triples

摘录 / EXCERPT: Dolbeault S-duality matching conjecture. Over the $\diamondsuit$-locus, the following matching data hold under $\mathbf{S}_{\mathrm{Dol}}$: the dual of $A_{\mathrm{Dol}}(M_\rho')$...

conjecture | open | MathDB 页面 | 原文线索

V2-071374 | MATHDB-369782

Greenberg-Benois exceptional zero conjecture for $\mathrm{GL}(3)$

摘录 / EXCERPT: Let $L_p^{(c)}(\pi,s)=\mathscr{L}_p^{(c)}(\pi,\langle-\rangle^s)$ for $s\in\mathbb{Z}_p$, where $\langle-\rangle:\mathbb{Z}_p^{\times}\to1+2p\mathbb{Z}_p$ is the natural map. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071375 | MATHDB-369788

The conjecture on Steklov functional maximizers for critical ellipses

摘录 / EXCERPT: Let $\mathbb{D}$ be the disk, let h_t^+ denote the corresponding Steklov data for the critical ellipse, let $E_t=\{x^2+t y^2=1\}$, and let $co(E_t)=\{x^2+t y^2\leq 1\}$. Equip...

conjecture | open | MathDB 页面 | 原文线索

V2-071376 | MATHDB-369789

The conjecture on non-planar free boundary minimal disks in ellipsoids

摘录 / EXCERPT: For $q\geq 1$, consider the ellipsoid ... Ellipsoid existence conjecture. There is a non-planar free boundary minimal immersed disk into this ellipsoid if and only if $q>3$. This is...

conjecture | open | MathDB 页面 | 原文线索

V2-071377 | MATHDB-369791

Andrews' nonnegativity conjecture for the Alladi-Schur polynomial coefficients

摘录 / EXCERPT: Andrews' conjecture. For all n and j, $c(n,j)$ has nonnegative coefficients. The conjecture concerns coefficientwise nonnegativity in the quotient arising from the Alladi-Schur...

conjecture | open | MathDB 页面 | 原文线索

V2-071378 | MATHDB-369794

Flamingo-web injection conjecture for three-row increasing tableaux

摘录 / EXCERPT: A normal plabic graph is a plabic graph whose boundary vertices are black of degree 1, whose internal black vertices have degree 3, and in which no two vertices of the same col...

conjecture | open | MathDB 页面 | 原文线索

V2-071379 | MATHDB-369797

Lyapunov functional existence conjecture for the inertial and first-order Kuramoto models

摘录 / EXCERPT: Let $\mathcal{V}=(\nu_1,\ldots,\nu_N)$ be the natural-frequency vector, let $\mathcal{D}(\mathcal{V})$ denote its frequency-diameter quantity, and let $\kappa_c(\mathcal{V})$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-071380 | MATHDB-369799

Malleability characterization of exponentiable virtual double categories

摘录 / EXCERPT: Malleability conjecture. The following conditions are equivalent:

conjecture | solved | MathDB 页面 | 原文线索

V2-071381 | MATHDB-369819

Equality characterization for independent distance domination in connected bipartite graphs

摘录 / EXCERPT: Let $d\geq 2$, and let G be a connected bipartite graph of order n. Write C_{2d+2} for the cycle of length $2d+2$, and let $\mathcal{B}_d$ denote the graph family defined in...

conjecture | open | MathDB 页面 | 原文线索

V2-071382 | MATHDB-369830

Fixed-point conjecture for the Berezin transform on Fock-type spaces

摘录 / EXCERPT: Fixed-point conjecture. If for any $m>0$ we have $B_mf=f$, then f is harmonic.

conjecture | open | MathDB 页面 | 原文线索

V2-071383 | MATHDB-369831

Gasca-Maeztu conjecture for GC_n sets

摘录 / EXCERPT: Let a GC_n set be an n-correct set $\mathcal{X}$ in which the fundamental polynomial of every node is a product of n linear factors. Gasca-Maeztu conjecture. Any GC_n set c...

conjecture | solved | MathDB 页面 | 原文线索

V2-071384 | MATHDB-369832

Nonvanishing conjecture for graph amplitude functions

摘录 / EXCERPT: Let G be a non-oriented finite simple graph, let $C_G(\mathbb{R}^3)$ be its configuration space, and let $\mathcal{A}_G$ denote the associated G-amplitude function. Graph-ampli...

conjecture | open | MathDB 页面 | 原文线索

V2-071385 | MATHDB-369833

Real-part conjecture for tree amplitude functions

摘录 / EXCERPT: Let T be a tree, meaning a connected circuit-free non-oriented finite simple graph, and let $C_T(\mathbb{R}^3)$ be its configuration space. Let $\mathcal{A}_T$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-071386 | MATHDB-369836

Schur-polynomial formulation of the self-conjugate-grid pointwise conjecture

摘录 / EXCERPT: Schur-polynomial pointwise conjecture. For all $0\leq k<n\leq t$,

conjecture | open | MathDB 页面 | 原文线索

V2-071387 | MATHDB-369838

Adams et al.'s cyclic sieving conjecture for Statistic 373

摘录 / EXCERPT: Adams et al.'s conjecture. The number of weak excedances that are also mid-points of a decreasing subsequence of length 3 (Statistic 373 in FindStat) is equidistributed with th...

conjecture | open | MathDB 页面 | 原文线索

V2-071388 | MATHDB-369839

Stieltjes moment conjecture for vincular pattern Classes 1 and 2

摘录 / EXCERPT: Stieltjes moment conjecture. The sequences of polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-071389 | MATHDB-369840

Hankel total positivity conjecture for vincular pattern Classes 1 and 2

摘录 / EXCERPT: Hankel total positivity conjecture. The sequences of polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-071390 | MATHDB-369854

Conjectural frequency of regular twists with a rational point

摘录 / EXCERPT: Let $C/\mathbb Q$ be a genus-one hyperelliptic curve, let E be its Jacobian, and let $\mathcal F_b$ be the twist family specified by the local data b. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071391 | MATHDB-369856

Hasse-principle frequency conjecture for genus-one hyperelliptic twists

摘录 / EXCERPT: Let $C/\mathbb Q$ be a genus-one hyperelliptic curve, let E be its Jacobian, let Σ be the specified finite set of places, and let $\mathcal F_b$ be the corresponding twist...

conjecture | open | MathDB 页面 | 原文线索

V2-071392 | MATHDB-369857

Padé-biroot correspondence for square-root approximations

摘录 / EXCERPT: Let $P_{[p/q]}^{(c)}(x)$ denote the $[p/q]$ Padé approximant associated with the square-root expansion at c, and let $\beta_m^2(x,\sqrt{c})$ be the square biroot approximation. S...

conjecture | open | MathDB 页面 | 原文线索

V2-071393 | MATHDB-369858

Generalized Binomial Biroot Conjecture

摘录 / EXCERPT: Generalized Binomial Biroot Conjecture. The approximations converge according to

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071394 | MATHDB-369859

Factored square-biroot identity at one

摘录 / EXCERPT: For a positive expansion parameter c, let $\beta_m^2(1,c)$ denote the factored square biroot evaluated at 1. Factored square-biroot identity. The value is ... The source presents...

conjecture | open | MathDB 页面 | 原文线索

V2-071395 | MATHDB-369863

The sparse-graph reformulation of the fractional expectation threshold conjecture

摘录 / EXCERPT: For $q\in[0,1]$, call a graph J q-sparse if ... where $\mathbb{E}_qX_I$ is the expected number of copies of I in $G_{n,q}$. Sparse-graph reformulation. There is a fixed K suc...

conjecture | open | MathDB 页面 | 原文线索

V2-071396 | MATHDB-369870

Quadratic isogenies conjecture for quadratic points on modular curves

摘录 / EXCERPT: Quadratic isogenies conjecture. There exists an integer C, independent of K, such that if $N>C$, then every point

conjecture | open | MathDB 页面 | 原文线索

V2-071397 | MATHDB-369882

Generalized KMT conjecture for Delaunay circle patterns

摘录 / EXCERPT: Let S be a closed, oriented surface of genus at least 2. An admissible weighted graph (Γ,θ) in S is a graph embedded in S that is the 1-skeleton of a cell decom...

conjecture | open | MathDB 页面 | 原文线索

V2-071398 | MATHDB-369887

The two-copy coarse Erdős-Pósa conjecture for planar graphs

摘录 / EXCERPT: Let H be a planar graph. A K-fat minor-model of H is a K-fat model of H, and $B_G(Z,cK)$ denotes the ball of radius cK around Z in G. Then there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-071399 | MATHDB-369888

The far-apart planar Erdős-Pósa conjecture

摘录 / EXCERPT: Let H be a planar graph. An H minor-model is a model of H in G, and minor-models are pairwise distance at least d when every two distinct models are at that distance or f...

conjecture | open | MathDB 页面 | 原文线索

V2-071400 | MATHDB-369889

Cecotti-Vafa spectrum conjecture for distinguished matrices

摘录 / EXCERPT: Cecotti-Vafa spectrum conjecture. The Cecotti-Vafa construction can be made precise so that

conjecture | open | MathDB 页面 | 原文线索

V2-071401 | MATHDB-369890

Bernoulli-moment characterization conjecture for distinguished matrices

摘录 / EXCERPT: Bernoulli-moment characterization conjecture. There is a family of functions $r(k,n,\nu):\mathbb{R}_{>0}\to\mathbb{R}_{>0}$ such that a matrix S is a distinguished matrix of a si...

conjecture | open | MathDB 页面 | 原文线索

V2-071402 | MATHDB-369891

The weak happy edge conjecture for plane spanning trees

摘录 / EXCERPT: Let S be a convex point set, and let T_{in} and T_{tar} be plane spanning trees on S. An edge is happy if it belongs to both T_{in} and...

conjecture | open | MathDB 页面 | 原文线索

V2-071403 | MATHDB-369892

The parking edge conjecture for plane spanning trees

摘录 / EXCERPT: Let S be a convex point set, and let T_{in} and T_{tar} be plane spanning trees on S. A parking edge is an edge appearing in a flip sequence that is not conta...

conjecture | open | MathDB 页面 | 原文线索

V2-071404 | MATHDB-369893

Interlacing conjecture for Chow polynomials of simplicial posets

摘录 / EXCERPT: Let P be a simplicial poset, with associated polynomials $\operatorname{H}_{\widehat{P}}(x)$, $\operatorname{H}_{{\widehat{P}}^\ast}(x)$, and augmented polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-071405 | MATHDB-369911

The ordered-forest core conjecture for polynomial testability

摘录 / EXCERPT: Let F be an ordered graph, meaning a graph equipped with a linear order on its vertices. An ordered forest is an ordered graph whose underlying graph is a forest, and the core of...

conjecture | open | MathDB 页面 | 原文线索

V2-071406 | MATHDB-369912

The polynomial removal lemma conjecture for ordered binary matrices

摘录 / EXCERPT: Ordered matrix removal conjecture. For every $k\times k$ binary matrix A and every $\varepsilon>0$, there is a $\delta=\delta(k,\varepsilon)=\operatorname{poly}(\varepsilon)>0$ s...

conjecture | open | MathDB 页面 | 原文线索

V2-071407 | MATHDB-369919

Randomly shifted Gumbel limit for the maximum of the two-dimensional inhomogeneous DGFF

摘录 / EXCERPT: Let $d=2$ and suppose that Assumption-(i) and the moment condition hold, in particular $\operatorname{\mathbb{P}}[\omega(e)>0]=1$ for $e\in E_d$. For $n\in\mathbb{N}$, set ... and .....

conjecture | open | MathDB 页面 | 原文线索

V2-071408 | MATHDB-369945

Eigenvalue-weighted moment conjecture for holomorphic cusp forms

摘录 / EXCERPT: Let $\ell\geq1$ and $r\geq1$ be integers, and set ... For the forms $f\in H_k(1)$, let $\lambda_f(\ell)$ denote the Hecke eigenvalue, let $L(s,f)$ be the associated L-function, and...

conjecture | open | MathDB 页面 | 原文线索

V2-071409 | MATHDB-369946

Lipschitz continuity conjecture for the convex QCQP reformulation

摘录 / EXCERPT: Lipschitz continuity conjecture for the convex QCQP reformulation. If the paper's assumptions $Cref{assum:Ab}$ and $ref{assum:main}$ hold and $k>0$, then for every...

conjecture | open | MathDB 页面 | 原文线索

V2-071410 | MATHDB-369951

XXY equivalence conjecture for maximal equiangular tight frames

摘录 / EXCERPT: Let $d>3$. An equiangular tight frame (ETF) with parameters (r,n) is a set of n unit vectors in $\mathbb{S}^{r-1}$ whose coherence attains the Welch bound. XXY's ETF existence...

conjecture | open | MathDB 页面 | 原文线索

V2-071411 | MATHDB-369977

Sharp maximal estimate conjecture for orthonormal systems of wave equations

摘录 / EXCERPT: Let $n\geq2$ and $\frac12<s<\frac n2$. Let $(f_j)_j$ be an orthonormal system in $H^s(\mathbb{R}^n)$, and let β be the exponent in the maximal estimate denoted by ... holds for...

conjecture | open | MathDB 页面 | 原文线索

V2-071412 | MATHDB-369978

Conjecture on fractional domatic number of planar graphs by girth

摘录 / EXCERPT: Planar girth fractional-domatic conjecture. For every integer $k\ge 1$, if $3k-1\le g\le 3k+1$, then

conjecture | open | MathDB 页面 | 原文线索

V2-071413 | MATHDB-369979

Compatibility of the Dolbeault Langlands equivalence with quasi-BPS decompositions

摘录 / EXCERPT: Let G be a reductive group, let ^{L}G be its Langlands dual, and let $(\chi,w)\in\pi_1(G)\times Z(G)^{\vee}$. Consider the quasi-BPS semiorthogonal decompositions of the cohere...

conjecture | open | MathDB 页面 | 原文线索

V2-071414 | MATHDB-369984

Wilson-Hecke compatibility in Dolbeault geometric Langlands

摘录 / EXCERPT: Let C be a smooth projective curve, let G be a reductive group, and let ^{L}G be its Langlands dual. For a minuscule coweight μ and a point $x\in C$, let W_x^μ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-071415 | MATHDB-369993

Semistability of semi-log canonical hypersurface families

摘录 / EXCERPT: Consider the two-dimensional semi-log canonical hypersurface singularities in the families labelled (1)--(3) in the paper. Semistability conjecture. All singularities in these thre...

conjecture | open | MathDB 页面 | 原文线索

V2-071416 | MATHDB-369994

The nonpolygonality conjecture for the L-tetromino growth form

摘录 / EXCERPT: An L-tetromino tiling is a tiling by L-shaped tetrominoes, and its growth form is the asymptotic shape obtained from its coordination shells. L-tetromino nonpolygonality conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-071417 | MATHDB-369995

The area conjecture for growth forms of deformed hat tilings

摘录 / EXCERPT: For positive $b\neq 1$, let $Tile(1,b)$ be the deformed hat tile, and let a tiling by these tiles have growth form with area $area_{growth\ form}$. The tile has area...

conjecture | open | MathDB 页面 | 原文线索

V2-071418 | MATHDB-369996

The tilt conjecture for the hat tiling growth form

摘录 / EXCERPT: Let α satisfy ... where τ is the golden ratio, and let γ denote the tilt angle of the growth form of the hat tiling. Hat growth-form tilt conjecture. The growth fo...

conjecture | open | MathDB 页面 | 原文线索

V2-071419 | MATHDB-370003

Stembridge's monomial immanant nonnegativity conjecture

摘录 / EXCERPT: Let A) be a totally nonnegative $n\times n$ matrix, meaning that all its minors are nonnegative, and let $\lambda\vdash n$ be a partition. For a class function ϕ^{λ} o...

conjecture | open | MathDB 页面 | 原文线索

V2-071420 | MATHDB-370007

Existence of a bounded additive list-coloring constant for proper conflict-free coloring

摘录 / EXCERPT: Bounded additive list-coloring conjecture. There exists an absolute constant k such that every graph is proper conflict-free $(\mathrm{degree}+k)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-071421 | MATHDB-370008

Proper conflict-free degree-plus-two choosability conjecture

摘录 / EXCERPT: Proper conflict-free degree-plus-two choosability conjecture. Every connected graph other than the 5-cycle is proper conflict-free $(\mathrm{degree}+2)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-071422 | MATHDB-370025

Schmeichel-Hakimi conjecture on exceptional planar graphical sequences

摘录 / EXCERPT: A graphical sequence is planar graphical if it is the degree sequence of a planar graph. Using the shorthand a^bc^d for b repetitions of a followed by d repetitions of c,...

conjecture | open | MathDB 页面 | 原文线索

V2-071423 | MATHDB-370027

The rise-decorated rectangular shuffle conjecture

摘录 / EXCERPT: For $m,n\in\mathbb{N}$, let $e_{m,n}$ be the rectangular symmetric function, let Θ_{e_k} be the theta operator indexed by e_k, and let $\mathsf{LRD}(m+k,n+k)^{\ast k}$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-071424 | MATHDB-370028

The factor-free valley square conjecture

摘录 / EXCERPT: For $n,k\in\mathbb{N}$, let Θ_{e_k}, ∇, and ω be the indicated symmetric-function operators, let p_n be the power-sum symmetric function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071425 | MATHDB-370029

The Delta square conjecture in the fall version

摘录 / EXCERPT: For $n,k\in\mathbb{N}$ with $n>0$, let Θ_{e_k}, ∇, and ω be the indicated symmetric-function operators, let p_n be the power-sum symmetric function, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071426 | MATHDB-370030

The symmetric-function identity for the fall version

摘录 / EXCERPT: For $n,k\in\mathbb{N}$, let Θ_{e_k}, ∇, and ω be the indicated symmetric-function operators, let p_n and $p_{n,n(k+1)}$ be the relevant symmetric functions,...

conjecture | open | MathDB 页面 | 原文线索

V2-071427 | MATHDB-370033

Obstructions to the Hasse principle for variational problems

摘录 / EXCERPT: A variational problem may be formulated on chain spaces with coefficients in $\mathbb{R}$, $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ for $n\geq 2$. Obstructions to the Hasse princ...

open_problem | open | MathDB 页面 | 原文线索

V2-071428 | MATHDB-370040

The characterization of groups with singleton maximal-elasticity difference set

摘录 / EXCERPT: Let G be a finite abelian group with $|G|>4$. Write $\rho(G)$ for its elasticity and let $\Delta_{\rho}(G)$ be the set of all $d\in\mathbb N$ such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-071429 | MATHDB-370055

The factor conjecture for constant-length substitution speedups

摘录 / EXCERPT: Let θ be a constant-length left proper substitution and let $c>1$ satisfy $\gcd(c,|\theta|)=1$. If l is a factor of c with $1\leq l<c$, then (X_θ,T^l) and...

conjecture | open | MathDB 页面 | 原文线索

V2-071430 | MATHDB-370059

Completion-determination conjecture for the isolated closed balls of ultrametric spaces

摘录 / EXCERPT: Let (X,d) and (Y,ρ) be ultrametric spaces, with completions $(\tilde{X},\tilde{d})$ and $(\tilde{Y},\tilde{\rho})$. Let $\bar{\mathbf{B}}_X^0$ and $\bar{\mathbf{B}}_Y^0$ den...

conjecture | open | MathDB 页面 | 原文线索

V2-071431 | MATHDB-370069

Duchet-Meyniel minor conjecture

摘录 / EXCERPT: Duchet-Meyniel conjecture. Every graph G of order n has a $K_{\lceil n/\alpha(G)\rceil}$ minor.

conjecture | open | MathDB 页面 | 原文线索

V2-071432 | MATHDB-370070

Odd-minor Duchet-Meyniel conjecture of Ji, Song, Weiss, and Zhang

摘录 / EXCERPT: Odd-minor Duchet-Meyniel conjecture. For any graph G of order n,

conjecture | open | MathDB 页面 | 原文线索

V2-071433 | MATHDB-370071

Reid's Fantasy on the connectedness of Calabi-Yau moduli

摘录 / EXCERPT: Consider the directed graph whose nodes are deformation families of smooth, compact Calabi-Yau threefolds. For two nodes $\mathcal{M}_1$ and $\mathcal{M}_2$, draw an arrow ... when,...

conjecture | open | MathDB 页面 | 原文线索

V2-071434 | MATHDB-370077

Ballantine-Merca inequality for truncated overpartition sums

摘录 / EXCERPT: Let $\overline{M}_k(n)$ denote the number of overpartitions of n in which the first part larger than k appears at least $k+1$ times. For $n,k\geq 1$, Ballantine-Merca's conject...

conjecture | open | MathDB 页面 | 原文线索

V2-071435 | MATHDB-370078

Guo-Zeng's strengthened truncated Gauss inequality for overpartitions

摘录 / EXCERPT: Let $\overline{p}(n)$ denote the number of overpartitions of n. For $n,k\geq 1$, Guo-Zeng's conjecture. ... with strict inequality if $n\geq k^2$. The conjecture was proved indepen...

conjecture | open | MathDB 页面 | 原文线索

V2-071436 | MATHDB-370079

Bounded sliding circuit sets for powers of rigid elements

摘录 / EXCERPT: Let G be a Garside group and let $x\in G$ be rigid. For each positive integer n, let $SC(x^n)$ denote the sliding circuits set of x^n. Boundedness conjecture. For any rigid e...

conjecture | open | MathDB 页面 | 原文线索

V2-071437 | MATHDB-370087

Weak polynomiality conjecture for Chern classes of polynomial spaces

摘录 / EXCERPT: Let $\Pol^d(\mathbb{C}^n)$ denote the relevant polynomial-space bundle, and let $c_k\big(\Pol^d(\mathbb{C}^n)\big)$ be its kth Chern class. Weak conjecture. For any fixed k and...

conjecture | open | MathDB 页面 | 原文线索

V2-071438 | MATHDB-370088

Strong binomial-polynomiality conjecture for Chern classes of polynomial spaces

摘录 / EXCERPT: Strong conjecture. The class $c_k\big(\Pol^d(\mathbb{C}^n)\big)$ is of the form $P(B_1,\dots,B_s)$, where P is a polynomial. This strengthens the weak conjecture by proposing an...

conjecture | open | MathDB 页面 | 原文线索

V2-071439 | MATHDB-370090

Brownian loop-soup cluster expansion conjecture in fractional occupation fields

摘录 / EXCERPT: Fix $c\in(0,1)$. Let $\mathcal L_D^c$ be a Brownian loop soup of central charge c, let $\mathcal C$ be one of its clusters, and let $:\mathtt h_c\Theta_{\mathcal L_D^c}^{n-1}:$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-071440 | MATHDB-370093

The 6-regular three-2-factor conjecture

摘录 / EXCERPT: Let G be a 6-regular graph. A 2-factor is a spanning 2-regular subgraph, and a component of a 2-factor is one of its connected components. The 6-regular three-2-factor conj...

conjecture | open | MathDB 页面 | 原文线索

V2-071441 | MATHDB-370096

Higher transversality conjecture for the involution ω

摘录 / EXCERPT: Higher transversality conjecture. The involution ω is also n-transverse for every integer $n\ge 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-071442 | MATHDB-370098

Stability and ground-state transition conjecture for solitary waves

摘录 / EXCERPT: Stability conjecture. Let $1\leq d\leq 6$. For $\alpha\leq 2/d$, the solution φ_ω minimizes $\mathcal{E}$ at mass $M=M(\omega)$ for all $\omega\in(0,\omega^*)$ and is...

conjecture | open | MathDB 页面 | 原文线索

V2-071443 | MATHDB-370099

The APN distance-one conjecture

摘录 / EXCERPT: APN distance-one conjecture. If $F\neq G$, then $d(F,G)>1$. It is unknown whether two APN functions can have Hamming distance one; the conjecture is known for several classes, incl...

conjecture | open | MathDB 页面 | 原文线索

V2-071444 | MATHDB-370108

Character-sum formula for primary characters of unitary groups

摘录 / EXCERPT: Let $\chi=\chi^{\phi,\lambda}$ be a primary irreducible character of $\mathrm{GU}_n(q)$, where ϕ is an orbit other than $\{\mathbf{1}\}$ and $\mathbf{1}$ is the trivial charac...

conjecture | open | MathDB 页面 | 原文线索

V2-071445 | MATHDB-370113

Bounded barcode-chain conjecture for merge tree interleaving distance

摘录 / EXCERPT: Bounded barcode-chain conjecture. There exists a number $M(n)$ such that for any $y_0,y_1\in\mathrm{MT}_n$, there exist $x_0,\ldots,x_k$ with $k<M(n)$, $y_0=x_0$, and $y_1=x_1$ sat...

conjecture | open | MathDB 页面 | 原文线索

V2-071446 | MATHDB-370115

The N=1 conjecture for character sums over Riordan partitions

摘录 / EXCERPT: Let $\lambda \vdash n$ be a partition. Write $\operatorname{Ev}(\lambda)$ for the set of partitions obtained from λ in the construction used in the paper, let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071447 | MATHDB-370116

Amdeberhan's conjectured q-series identity involving Riordan numbers

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\lambda_2,\ldots,\lambda_r)$ be a partition, and define ... For a partition λ, let z_λ be the standard symmetric-function multiplicity factor;...

conjecture | open | MathDB 页面 | 原文线索

V2-071448 | MATHDB-370118

Strong and powerful tableau bounds for chromatic symmetric function coefficients

摘录 / EXCERPT: Let P) be a $(\mathbf{3}+\mathbf{1})$-free poset, let λ be a partition, and let c_λ^P denote the coefficient of $e_\lambda(\mathbf{x})$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-071449 | MATHDB-370119

Strong tableau lower bound for q-chromatic coefficients

摘录 / EXCERPT: Let P be a natural unit interval order with n elements, let $\lambda\vdash n$, and let $\operatorname{strongST}_P(\lambda)$ be the set of strong standard P-tableaux of shape...

conjecture | open | MathDB 页面 | 原文线索

V2-071450 | MATHDB-370120

Strong-tableau nonvanishing conjecture for elementary coefficients

摘录 / EXCERPT: Let P be a $(\mathbf{3}+\mathbf{1})$-free poset, let λ be a partition, and let c_λ^P be the corresponding elementary-basis coefficient of...

conjecture | open | MathDB 页面 | 原文线索

V2-071451 | MATHDB-370121

Powerful tableau noncommutative upper-bound conjecture

摘录 / EXCERPT: Let P be a $(\mathbf{3}+\mathbf{1})$-free poset, let $\lambda\vdash n$, and let $m_\lambda^P(\mathbf{u})$ be the relevant noncommutative elementary coefficient in...

conjecture | open | MathDB 页面 | 原文线索

V2-071452 | MATHDB-370122

Strong tableaux and Hikita tableaux equivalence conjecture

摘录 / EXCERPT: Let $\mathbf{m}\in\mathbb{E}_n$, let $P=P_{\mathbf{m}}$ be the associated natural unit interval order, and let $\lambda\vdash n$. Let $\operatorname{strongST}_P(\lambda)$ be the st...

conjecture | open | MathDB 页面 | 原文线索

V2-071453 | MATHDB-370124

Powerful-tableau formula for chromatic quasisymmetric functions of K-chains

摘录 / EXCERPT: Let $a,b\geq2$ be positive integers, let $\gamma=(a,2,2,\ldots,2,b)$, and let K_γ be the corresponding K-chain with associated poset P. K-chain powerful-tableau formula co...

conjecture | open | MathDB 页面 | 原文线索

V2-071454 | MATHDB-370131

Weak Goldfeld positive-proportion conjecture

摘录 / EXCERPT: Weak Goldfeld conjecture. For each $r\in\{0,1\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-071455 | MATHDB-370132

The stronger edge bound conjecture for stress-independent graphs

摘录 / EXCERPT: Stronger edge bound conjecture. The number of edges satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071456 | MATHDB-370134

The stress-linked pair component conjecture

摘录 / EXCERPT: Stress-linked pair component conjecture. If $\{u,v\}$ is d-stress-linked in G, then there is some $\mathcal{R}_d$-component G_0 of G such that $\{u,v\}$ is d-stress-linke...

conjecture | open | MathDB 页面 | 原文线索

V2-071457 | MATHDB-370135

The clique exclusion conjecture for minimally globally rigid graphs

摘录 / EXCERPT: Clique exclusion conjecture. The graph G does not contain a copy of K_{d+2}.

conjecture | open | MathDB 页面 | 原文线索

V2-071458 | MATHDB-370143

Conjugation quandle detection conjecture for dihedral groups

摘录 / EXCERPT: Conjugation quandle detection conjecture. The collection of all conjugation quandles over all dihedral groups is able to detect causality across $(2+1)$ globally hyperbolic spaceti...

conjecture | open | MathDB 页面 | 原文线索

V2-071459 | MATHDB-370145

Spin-parity refinement of the polynomiality conjecture for Witten's classes

摘录 / EXCERPT: Let a be fixed, and suppose that r is even and a is odd. Let $W_g^\pm(a,r)$ denote the refinement of Witten's class according to spin parity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071460 | MATHDB-370147

Stable integral abelianization conjecture for handlebody Torelli groups

摘录 / EXCERPT: Let $\mathcal{HI}_{g,p}^b$ be the handlebody Torelli group, and let $\tau(\mathcal{HI}_{g,p}^b)$ and $\sigma(\mathcal{HK}_{g,p}^b)$ denote the images of the Johnson and Birman--Cra...

conjecture | open | MathDB 页面 | 原文线索

V2-071461 | MATHDB-370149

Stapleton's asymptotic conjecture for the degree of irrationality of K3 surfaces

摘录 / EXCERPT: Stapleton's asymptotic conjecture. The degree of irrationality of a very general polarized $K3$ surface grows on the order of $\sqrt d$. This refines the known upper bound and is c...

conjecture | open | MathDB 页面 | 原文线索

V2-071462 | MATHDB-370150

Multiplicativity conjecture for the degree of irrationality of products of general curves

摘录 / EXCERPT: Product-of-curves conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071463 | MATHDB-370151

Density-degree conjecture for products of hyperelliptic curves

摘录 / EXCERPT: Density-degree conjecture. Most pairs of hyperelliptic curves (C,D) of sufficiently large genera satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-071464 | MATHDB-370156

The induced Grid Theorem for $K_{1,d}$-free graphs

摘录 / EXCERPT: The induced Grid Theorem. There exists a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that every $K_{1,d}$-free graph G with α-$\mathsf{tw}(G)\geq f(k,d)$ contains...

conjecture | open | MathDB 页面 | 原文线索

V2-071465 | MATHDB-370157

The induced Menger conjecture for $K_{1,d}$-free graphs

摘录 / EXCERPT: Induced Menger conjecture. There exists a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that there exists either k pairwise non-adjacent A-B paths, or a separator S b...

conjecture | open | MathDB 页面 | 原文线索

V2-071466 | MATHDB-370158

The induced double-wheel conjecture for $K_{1,d}$-free graphs

摘录 / EXCERPT: Double-wheel conjecture. There exists a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that every $K_{1,d}$-free graph G with...

conjecture | open | MathDB 页面 | 原文线索

V2-071467 | MATHDB-370159

The two induced cycles conjecture for $K_{1,d}$-free graphs

摘录 / EXCERPT: Two induced cycles conjecture. There exists a function $f:\mathbb{N}^2\rightarrow\mathbb{N}$ such that every $K_{1,d}$-free graph G with...

conjecture | open | MathDB 页面 | 原文线索

V2-071468 | MATHDB-370160

Kurokawa's Deep Riemann Hypothesis for automorphic Euler products

摘录 / EXCERPT: Let π be an irreducible cuspidal automorphic representation of GL_n over $\mathbb Q$, with associated L-function ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071469 | MATHDB-370167

A zero-part formula for logarithmic double ramification intersections

摘录 / EXCERPT: Let d be the degree, let $\boldsymbol{\nu}$ be the ramification data, and let $\operatorname{logDR}_0(d,-\boldsymbol{\nu})$ denote the logarithmic double ramification cycle. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071470 | MATHDB-370176

Equality of Maker-Breaker domination parameters for random graphs

摘录 / EXCERPT: Let G be a random graph, and let $\gamma_{MB}(G,b)$ and $s_{MB}(G,b)$ denote the two Maker-Breaker domination game parameters. For a Binomial random graph $G_{n,p}$ with constant...

conjecture | open | MathDB 页面 | 原文线索

V2-071471 | MATHDB-370181

Conjecture on the number of terms in the polynomials \(\psi_n\)

摘录 / EXCERPT: Term-count conjecture. For every $n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-071472 | MATHDB-370186

Simultaneously bounded and dense orbits for countably many diagonal flows

摘录 / EXCERPT: Let $d\geq 3$, let $X_d=\mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z})$, and let I_m,J_n be index sets. Choose nonzero vectors...

conjecture | open | MathDB 页面 | 原文线索

V2-071473 | MATHDB-370187

Margulis' bounded diagonal orbit conjecture

摘录 / EXCERPT: Let $X_d=\mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z})$ and let A be the connected diagonal subgroup ... An A-orbit is bounded if its image in X_d is bounded. Margulis' c...

conjecture | open | MathDB 页面 | 原文线索

V2-071474 | MATHDB-370188

Stable-perturbation conjecture for bounded cone orbits

摘录 / EXCERPT: Stable-perturbation conjecture. For every open cone $C\subseteq\mathfrak{a}$, every bounded A_C-orbit in X_d contains an A_C-stable perturbation of a compact A-orbit. This...

conjecture | open | MathDB 页面 | 原文线索

V2-071475 | MATHDB-370189

The harmonious labelling conjecture

摘录 / EXCERPT: The harmonious labelling conjecture. For every n-edge tree T, there exists a labelling $\phi\colon V(T)\to \mathbb{Z}_n$ such that the edge sums $\phi(x)+\phi(y)$ are all disti...

conjecture | open | MathDB 页面 | 原文线索

V2-071476 | MATHDB-370193

The Graham-Häggkvist tree decomposition conjecture

摘录 / EXCERPT: Graham-Häggkvist conjecture. The edge set of $K_{n,n}$ can be decomposed into copies of every n-edge tree T.

conjecture | open | MathDB 页面 | 原文线索

V2-071477 | MATHDB-370199

Kaiser's lower-bound conjecture for three perfect matchings

摘录 / EXCERPT: Kaiser's conjecture. For every bridgeless cubic graph G,

conjecture | open | MathDB 页面 | 原文线索

V2-071478 | MATHDB-370201

Non-degeneracy conjecture for the intersection pairing on Shtuka isotypic parts

摘录 / EXCERPT: Let C be a smooth projective curve, let I be a finite index set with $|I|=r$, and let $\underline{\e}\in\{\pm1\}^r_0$. For $d\in\ZZ$, the intersection pairing on the degree-d...

conjecture | open | MathDB 页面 | 原文线索

V2-071479 | MATHDB-370202

Categorical trace conjecture for the diagonal correspondence

摘录 / EXCERPT: Let H be a reductive group, let $G=H\times H$, and let $X=H\backslash(H\times H)$. Let $c:H\to H$ be the Cartan involution and let fs_H be a local system on the dual group. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-071480 | MATHDB-370219

Higher-rank Mistretta-Stoppino conjecture for Clifford-index bundles

摘录 / EXCERPT: Higher-rank Mistretta-Stoppino conjecture. The following assertions hold: (i) E is linearly (semi)stable; and (ii) E is linearly (semi)stable if and only if M_E is (semi)stab...

conjecture | open | MathDB 页面 | 原文线索

V2-071481 | MATHDB-370220

Brzostowski-Krasiński-Oleksik conjecture on the Łojasiewicz exponent

摘录 / EXCERPT: Brzostowski-Krasiński-Oleksik conjecture. If $\Gamma(f)$ contains a nonexceptional facet, then

conjecture | open | MathDB 页面 | 原文线索

V2-071482 | MATHDB-370221

Triangulation-dependent conjecture for the Łojasiewicz exponent

摘录 / EXCERPT: Triangulation-dependent Łojasiewicz conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-071483 | MATHDB-370223

David-Meisner conjecture for special values of higher-order Hecke L-functions

摘录 / EXCERPT: Let K be a number field containing the ℓ-th roots of unity, let $\mathcal{O}_K$ be its ring of integers, and let $\mathcal{F}_\ell$ be a family of Hecke characters of order...

conjecture | open | MathDB 页面 | 原文线索

V2-071484 | MATHDB-370229

Brooks-type conjecture for proper conflict-free coloring

摘录 / EXCERPT: Let G be a simple finite undirected graph with maximum degree $\Delta\geq 3$, and let $\chi_{\mathrm{pcf}}(G)$ denote its proper conflict-free chromatic number. Brooks-type conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071485 | MATHDB-370242

Cocoercive-subdifferential convergence-rate conjecture for Douglas-Rachford splitting

摘录 / EXCERPT: Let A and B be operators satisfying the paper's standing Assumption, let B be β-cocoercive, and let $A=\partial g$ for a closed proper convex function g. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071486 | MATHDB-370243

Exact convergence-rate conjecture for Douglas-Rachford splitting on smooth convex composites

摘录 / EXCERPT: Consider the convex composite problem ... where f and g are closed proper convex functions. Assume that the Douglas-Rachford operator has a fixed point $w^\star$, that f is L...

conjecture | open | MathDB 页面 | 原文线索

V2-071487 | MATHDB-370244

Silver-schedule convergence-rate conjecture for Douglas-Rachford splitting

摘录 / EXCERPT: In the convex composite problem with closed proper convex functions f,g, suppose the Douglas-Rachford operator has a fixed point $w^\star$ and f is L-smooth. Let ρ denot...

conjecture | open | MathDB 页面 | 原文线索

V2-071488 | MATHDB-370250

Koksma's analytic composition conjecture

摘录 / EXCERPT: Let $(u_n)_{n\in\mathbb{N}}$ satisfy the conditions of the cited Koksma theorem, and let $f_1,\dots,f_k:\mathbb{R}\to\mathbb{R}$ be analytic functions such that $1,f_1,\dots,f_k$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-071489 | MATHDB-370251

Jointly scattered derivative conjecture for analytic compositions

摘录 / EXCERPT: Let $I\subset\mathbb{R}$ be a compact interval and let $k\in\mathbb{N}$. For each $i=1,\dots,k$, let $(u_{i,n})_{n\in\mathbb{N}}$ be a sequence of functions in $C^1(I,\mathbb{R})$...

conjecture | open | MathDB 页面 | 原文线索

V2-071490 | MATHDB-370253

The noncomplete contraction-critical graph connectivity conjecture

摘录 / EXCERPT: Contraction-critical connectivity conjecture. Any noncomplete k-contraction-critical graph is k-connected. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-071491 | MATHDB-370254

Conjectured variation of the three-colored partition identity for $Q^_1(y)$

摘录 / EXCERPT: Let Γ be the set of three-colored partitions, and let $A^\ast_1(m,n)$ count the partitions of n in Γ having no occurrences of $1_{\textcolor{Red}{\bf R}}$, where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071492 | MATHDB-370268

The parabolic intersection conjecture for Artin groups

摘录 / EXCERPT: Let A be an Artin group, and let parabolic subgroups mean the parabolic subgroups of A. The parabolic intersection conjecture. The intersection of any two parabolic subgroups i...

conjecture | open | MathDB 页面 | 原文线索

V2-071493 | MATHDB-370269

Baur's magnetic Faber-Krahn conjecture for higher eigenvalues

摘录 / EXCERPT: Let $\Omega \subset \mathbb R^2$ be a bounded simply-connected planar domain with area $|\Omega|=1$, and let $n\in\mathbb N$. For a homogeneous magnetic field of strength $B\geq 0$...

conjecture | open | MathDB 页面 | 原文线索

V2-071494 | MATHDB-370272

The beta-representation formula for the stabilisability radius

摘录 / EXCERPT: Let $\beta\in\mathbb{R}_{>1}$, let c be the contraction factor associated with A_2, and let ${\cal M}(\theta,c,\beta)$ denote the corresponding switched-system matrix family. I...

conjecture | open | MathDB 页面 | 原文线索

V2-071495 | MATHDB-370282

Grothendieck-André periods conjecture for the 1-motive M

摘录 / EXCERPT: Let $M=[u:\mathbb{Z}\to G^n]$ be the 1-motive with $u(1)=(R_1,\dots,R_n)$ defined by the points in the source. Let k be the endomorphism field and let $\mathrm{LieBracket}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071496 | MATHDB-370310

The mixed-norm Strichartz conjecture on the real line times the circle

摘录 / EXCERPT: Let $N\geq 1$, let φ be the time cutoff appearing in the estimate, and let $\phi\in L^2(\mathbb{R}\times\mathbb{T})$. For $\xi_0\in\mathbb{R}\times\mathbb{Z}$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-071497 | MATHDB-370315

Goldberg-Milnor conjecture on perturbing parabolic polynomial cycles

摘录 / EXCERPT: Let P be a polynomial having a parabolic cycle, and let $J(P)$ denote its Julia set. A small perturbation of P should convert the immediate basin of the parabolic cycle into an...

conjecture | open | MathDB 页面 | 原文线索

V2-071498 | MATHDB-370319

Schlosser's sign-pattern conjecture for powers of the infinite Borwein product

摘录 / EXCERPT: Schlosser's conjecture. If δ is real and

conjecture | open | MathDB 页面 | 原文线索

V2-071499 | MATHDB-370320

Schlosser and Zhou's sign-pattern conjecture for powers of $Q_{10}(q)$

摘录 / EXCERPT: Let ... The sign patterns are coefficient patterns in the displayed order. Schlosser and Zhou's conjecture. For $\delta=1$, the coefficients of $Q_{10}(q)^{\delta}$ exhibit...

conjecture | open | MathDB 页面 | 原文线索

V2-071500 | MATHDB-370321

The Solomon-Terao polynomial order-degree conjecture for multiarrangements

摘录 / EXCERPT: Solomon-Terao order-degree conjecture. If $\eta\in U_{d+1}({\mathcal{A}})$, then $ST_{d+1}({\mathcal{A}};x)$ is monic of degree

conjecture | open | MathDB 页面 | 原文线索

V2-071501 | MATHDB-370329

Alon-Powierski-Savery-Scott-Wilmer's n-dijoin conjecture

摘录 / EXCERPT: Alon-Powierski-Savery-Scott-Wilmer's n-dijoin conjecture. If

conjecture | refuted | MathDB 页面 | 原文线索

V2-071502 | MATHDB-370330

Additivity conjecture for tournament minimum rank of dijoins

摘录 / EXCERPT: Additivity conjecture for tournament minimum rank. For all tournaments D_1 and D_2,

conjecture | open | MathDB 页面 | 原文线索

V2-071503 | MATHDB-370331

Lower-bound conjecture for inversion number of tournament dijoins

摘录 / EXCERPT: Lower-bound conjecture for tournament dijoins. One should have

conjecture | open | MathDB 页面 | 原文线索

V2-071504 | MATHDB-370336

Linear cop-number conjecture for claw-free path-free graphs

摘录 / EXCERPT: Let P_k denote the path on k vertices, let $\operatorname{claw}=K_{1,3}$, and let $c(G)$ denote the cop number of a graph G. A graph is $(P_k,\operatorname{claw})$-free if it...

conjecture | open | MathDB 页面 | 原文线索

V2-071505 | MATHDB-370346

Williams's presentation conjecture for pure braid groups of finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group. Write $d54b_c:=\{\mathbbm{t}_c:t\in T\}$ for the squares of the dual braid generators, and let $\mathbb{c}=\mathbf{w}_\circ^2=\mathbf{c}^h$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-071506 | MATHDB-370347

Semidistributivity conjecture for concordancy posets

摘录 / EXCERPT: Let W be a finite Coxeter group, let c be a Coxeter element, and for $u\in W_c^+$ let $\mathsf{CC}_c(u)$ be the poset of commutation classes of maximal c-noncrossing chains c...

conjecture | open | MathDB 页面 | 原文线索

V2-071507 | MATHDB-370348

Facet adjacency conjecture for concordancy classes

摘录 / EXCERPT: Let $u\in W_c^+$, let $\mathsf{CC}_c(u)$ be the concordancy poset, and let $\Delta(\mathcal{C})$ denote the collection of SBDW simplices associated with a class $\mathcal{C}$. Face...

conjecture | open | MathDB 页面 | 原文线索

V2-071508 | MATHDB-370349

Lattice conjecture for the c-noncrossing weak order

摘录 / EXCERPT: Let W be a finite Coxeter group and let c be a Coxeter element. The c-noncrossing weak order on W is the poset obtained by restricting the Cayley graph of the dual braid mo...

conjecture | open | MathDB 页面 | 原文线索

V2-071509 | MATHDB-370350

Isomorphism conjecture for the c-noncrossing weak and Bruhat orders

摘录 / EXCERPT: Let W be a finite Coxeter group and let c be a Coxeter element. Consider the c-noncrossing weak order and the c-noncrossing Bruhat order on W. Order-isomorphism conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-071510 | MATHDB-370351

Lattice conjecture for the c-noncrossing Bruhat order

摘录 / EXCERPT: Let W be a finite Coxeter group and let c be a Coxeter element. The c-noncrossing Bruhat order is the order on W defined from the noncrossing Bruhat construction. Bruhat-or...

conjecture | open | MathDB 页面 | 原文线索

V2-071511 | MATHDB-370355

The gap-to-entropy implication for posets

摘录 / EXCERPT: Let P be a finite poset, let $h(x)=\mathbb E f(x)$ for a uniformly random linear extension f of P, and let $\operatorname{gap}(P)$ be the maximum consecutive gap among the or...

conjecture | open | MathDB 页面 | 原文线索

V2-071512 | MATHDB-370356

The ideal average-height conjecture for posets

摘录 / EXCERPT: Let P be a finite poset, let A be an ideal of P, and let $h(x)$ denote the expected position of x in a uniformly random linear extension. Write $\max(A)$ for the maximal el...

conjecture | open | MathDB 页面 | 原文线索

V2-071513 | MATHDB-370360

The ODP conjecture on normalized volumes of klt singularities

摘录 / EXCERPT: ODP conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071514 | MATHDB-370376

The strong uniform time conjecture for Rubik's cube corners

摘录 / EXCERPT: Fix an arbitrary order on the positions of the corners of a Rubik's cube. Let X_t denote the configuration of the cube at time t, and call two corners unlinked when they are un...

conjecture | open | MathDB 页面 | 原文线索

V2-071515 | MATHDB-370387

The degeneracy conjecture for proper conflict-free list coloring

摘录 / EXCERPT: Degeneracy conjecture for proper conflict-free list coloring. If G is a d-degenerate graph for some positive integer d, then G is proper conflict-free ${\rm degree}+d)$-ch...

conjecture | open | MathDB 页面 | 原文线索

V2-071516 | MATHDB-370388

Garvan-Jennings-Shaffer nonnegativity conjecture for M_C1 and M_C5

摘录 / EXCERPT: Let $M_{C1}(m,n)$ and $M_{C5}(m,n)$ be the coefficients defined by the corresponding two-variable generating functions in the source, with $m\in\mathbb{Z}$ and $n\in\mathbb{N}$, wh...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071517 | MATHDB-370393

Tate's equivalence conjecture for BSD and Artin-Tate

摘录 / EXCERPT: Let $f\colon X\rightarrow C$ be a smooth projective surface fibration over a smooth projective curve over $k=\mathbb{F}_q$, with smooth generic fibre X_η over $K=k(C)$, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-071518 | MATHDB-370401

Ayoub's conservativity conjecture for Betti realization of geometric motives

摘录 / EXCERPT: Let $\mathbf{DM}_{\mathrm{gm}}(\mathbb{C};\mathbb{Q})$ be the category of geometric motives over $\mathbb{C}$ with rational coefficients, and let $\mathbf{D}(\mathbb{Q})$ be the de...

conjecture | open | MathDB 页面 | 原文线索

V2-071519 | MATHDB-370405

WDVV linear-reduction conjecture

摘录 / EXCERPT: Let N be the dimension of the WDVV system, let S_L denote its linear subsystem, and let $S_{lambda amu b}$ denote the remaining equations indexed by $2<\lambda<\mu\leq N$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071520 | MATHDB-370408

Equivariant Iwasawa main conjecture for Tate motives

摘录 / EXCERPT: Let k_∞/k be a $\mathbb Z_p$-power extension with no finite place splitting completely, let $K_\infty=Kk_\infty$, and use the source's decomposition and character χ to...

conjecture | open | MathDB 页面 | 原文线索

V2-071521 | MATHDB-370409

Frozen-corner determinant conjecture for alternating sign matrices

摘录 / EXCERPT: Frozen-corner determinant conjecture. The number $B_{n,s}$ may be evaluated as

conjecture | open | MathDB 页面 | 原文线索

V2-071522 | MATHDB-370412

Mkrtchyan-Hakobyan S_{12}-Conjecture

摘录 / EXCERPT: S_{12}-Conjecture. If G is a cubic graph with a perfect matching, then

conjecture | open | MathDB 页面 | 原文线索

V2-071523 | MATHDB-370431

The colliding orbits conjecture for non-additive polynomials

摘录 / EXCERPT: The colliding orbits conjecture. The set $C_f(\alpha_1,\alpha_2;\beta)$ is infinite if and only if $f(\alpha_1)=f(\alpha_2)$. This problem concerns when the strict forward orbits o...

conjecture | solved | MathDB 页面 | 原文线索

V2-071524 | MATHDB-370438

Log boundedness conjecture for K-moduli compactifications

摘录 / EXCERPT: Let $\mathcal{G}$ be a log bounded set of couples that is complete under dominant small modifications, and let $\widehat{\mathcal{G}}$ be its K-moduli completion: a couple belongs...

conjecture | open | MathDB 页面 | 原文线索

V2-071525 | MATHDB-370443

The improper 4-coloring conjecture for toroidal graphs with defect sequence $(0,0,0,3)$

摘录 / EXCERPT: Improper coloring conjecture for toroidal graphs. Every toroidal graph is $(0,0,0,3)$-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-071526 | MATHDB-370451

The global moduli fibration conjecture for special-holonomy QALE resolutions

摘录 / EXCERPT: Let V be a vector space, let $\Gamma\subset\operatorname{SU}(V)$ be finite, and assume that V/Γ admits a crepant resolution. Let $\Theta_{\operatorname{Im}(\mathbb{K})}$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-071527 | MATHDB-370454

Equitable coloring conjecture for outerplanar graphs with four or five colors

摘录 / EXCERPT: Equitable coloring conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071528 | MATHDB-370455

Five-color version of the planar equitable coloring theorem

摘录 / EXCERPT: Five-color planar theorem conjecture. The same statement remains true when the lower bound $s\ge 40$ is replaced by $s\ge 5$.

conjecture | open | MathDB 页面 | 原文线索

V2-071529 | MATHDB-370459

Asymptotic exponent-difference conjecture for Coxeter B_2 multiarrangements

摘录 / EXCERPT: Asymptotic exponent-difference conjecture. There exists a natural number $N\in\mathbb{N}$ such that, whenever $\min\{n_1,n_2\}\geq N$, one has $\Delta(\mathcal{A},m)\leq 1$. This c...

conjecture | open | MathDB 页面 | 原文线索

V2-071530 | MATHDB-370465

2d log-concavity conjecture for Links-Gould coefficients of alternating links

摘录 / EXCERPT: 2d log-concavity conjecture. The absolute values $(|a_{ij}|)$ form a 2d log-concave sequence with no interior zeros. This conjecture is proposed as a bidimensional analogue of log-...

conjecture | open | MathDB 页面 | 原文线索

V2-071531 | MATHDB-370472

Stanton's conjecture on dissections and cyclic sieving polynomials

摘录 / EXCERPT: Stanton's conjecture. The polynomials $f(n,k;q)$ and $a_\mu(q)$ satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-071532 | MATHDB-370479

Hibi-Li conjecture on the face numbers of order and chain polytopes

摘录 / EXCERPT: Hibi-Li conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071533 | MATHDB-370481

Das-Kaliraj coefficient-growth conjecture for univalent harmonic mappings

摘录 / EXCERPT: Let $\mathcal{S}_{H}$ be the family of sense-preserving planar harmonic univalent mappings $f=h+\overline{g}$ in $\mathbb{D}$ satisfying $h(0)=g(0)=h'(0)-1=0$, and let ... Assume the...

conjecture | open | MathDB 页面 | 原文线索

V2-071534 | MATHDB-370489

Gongyo's log-canonicity conjecture for ramification divisors

摘录 / EXCERPT: Gongyo's conjecture. After replacing f by an iterate, the pair

conjecture | open | MathDB 页面 | 原文线索

V2-071535 | MATHDB-370492

Hereditariness conjecture for the Černý bound

摘录 / EXCERPT: Let $\mathcal{A}$ be a synchronizing automaton, and let $\sigma\in\operatorname{Cong}(\mathcal{A})$ be a non-trivial congruence, so that the quotient automaton $\mathcal{A}/\sigma$...

open_problem | open | MathDB 页面 | 原文线索

V2-071536 | MATHDB-370502

The genus bound for the theta knot invariant

摘录 / EXCERPT: Genus bound conjecture. For every knot K, the displayed inequality holds. The conjecture has been verified for all knots with up to 13 crossings, but remains unproved in general;...

conjecture | open | MathDB 页面 | 原文线索

V2-071537 | MATHDB-370503

The fibered-knot coefficient conjecture for theta

摘录 / EXCERPT: Fibered-knot coefficient conjecture. This coefficient is an integer multiple of

conjecture | open | MathDB 页面 | 原文线索

V2-071538 | MATHDB-370504

The hexagonal symmetry conjecture for theta

摘录 / EXCERPT: Hexagonal symmetry conjecture. For every knot K, $\theta(K)$ is invariant under both substitutions

conjecture | open | MathDB 页面 | 原文线索

V2-071539 | MATHDB-370505

The mirror-symmetry conjecture for theta

摘录 / EXCERPT: Mirror-symmetry conjecture. The theta invariant changes sign under mirroring:

conjecture | open | MathDB 页面 | 原文线索

V2-071540 | MATHDB-370506

The orientation-reversal conjecture for theta

摘录 / EXCERPT: Orientation-reversal conjecture. Reversing the orientation does not change the invariant:

conjecture | open | MathDB 页面 | 原文线索

V2-071541 | MATHDB-370511

Beilinson-Bloch-Kato conjecture for polarized motives

摘录 / EXCERPT: Let F be a number field, E a number field, M a polarized motive in the pseudo-Abelian category of Chow motives over F with coefficients in E, and let $\lbd$ be a finite p...

conjecture | open | MathDB 页面 | 原文线索

V2-071542 | MATHDB-370513

Goulden-Jackson's Matching-Jack conjecture

摘录 / EXCERPT: Let $c^\rho_{\tau, u}(\alpha)$ be the coefficients defined by ... Here $b:=\alpha-1$ is the shifted Jack parameter. Goulden-Jackson's conjecture. The coefficients satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-071543 | MATHDB-370514

Monomial positivity conjecture for the super nabla operator

摘录 / EXCERPT: Fix $n\geq 1$. For partitions $\pi,\mu,\nu\vdash n$, define $d^\pi_{\mu,\nu}(\alpha)$ by ... where m_ν denotes the monomial symmetric function in the alphabet $\boldsymbol{q}$. M...

conjecture | open | MathDB 页面 | 原文线索

V2-071544 | MATHDB-370520

Local limit conjecture for conditioned sub-critical non-generic marked Galton-Watson trees

摘录 / EXCERPT: Let τ^{*} be a sub-critical and non-generic marked Galton-Watson tree with offspring distribution $\mathbf{p}$ satisfying the condition in ... . Let $M(\tau^*)$ denote the total...

conjecture | open | MathDB 页面 | 原文线索

V2-071545 | MATHDB-370521

Total-coloring extension conjecture for matchings

摘录 / EXCERPT: Total-coloring extension conjecture for matchings. If H is a matching, then every total-$(\Delta+3)$-coloring of H in G extends to a total-$(\Delta+3)$-coloring of G.

conjecture | open | MathDB 页面 | 原文线索

V2-071546 | MATHDB-370524

Kotani-Last conjecture for almost periodic Dirac operators

摘录 / EXCERPT: Let Λ_φ be the one-dimensional Dirac operator on $L^2({\mathbb{R}},{\mathbb{C}}^2)$, with potential $\varphi:{\mathbb{R}}\to{\mathbb{C}}$ satisfying ... Call...

conjecture | open | MathDB 页面 | 原文线索

V2-071547 | MATHDB-370528

Conifold Gap Conjecture for local \(\mathbb{P}^2\)

摘录 / EXCERPT: Conifold Gap property for $K_{\mathbb{P}^2}$. For $g \in \mathbb{Z}_{\geq 2}$, near $t_{\rm CF}=0$,

conjecture | open | MathDB 页面 | 原文线索

V2-071548 | MATHDB-370534

CGS conjecture for Jack polynomials

摘录 / EXCERPT: Let λ and μ be partitions with $|\lambda|=|\mu|$. For Jack polynomials $P_\lambda(x;\tau)$, let $\mathbb F_{\geqslant0}^{\mathbb R}$ denote the real cone of rational fu...

conjecture | open | MathDB 页面 | 原文线索

V2-071549 | MATHDB-370535

KT conjecture for Jack polynomials

摘录 / EXCERPT: Let λ and μ be partitions, and let $P_\lambda(x;\tau)$ denote the Jack polynomial. Write $\bm1=(1,\ldots,1)$. KT conjecture. The following are equivalent: 1. ... 2. For s...

conjecture | open | MathDB 页面 | 原文线索

V2-071550 | MATHDB-370536

Jack differences in the Muirhead semiring

摘录 / EXCERPT: Fix $\tau_0\in[0,\infty]$, and let λ and μ be partitions. Let $\mathcal M_S(\mathbb R_{\geqslant0})$ denote the Muirhead semiring over the nonnegative reals. Muirhead-s...

conjecture | refuted | MathDB 页面 | 原文线索

V2-071551 | MATHDB-370537

Jack-parameter monotonicity in the Jack semiring

摘录 / EXCERPT: Let λ and μ be partitions, and let $\mathcal J_C^\sigma(\mathbb R_{\geqslant0})$ be the Jack cone generated by normalized Jack differences with parameter σ. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-071552 | MATHDB-370538

Alexandersson-Haglund-Wang conjecture on Schur expansions of integral Jack polynomials

摘录 / EXCERPT: Alexandersson-Haglund-Wang conjecture. The coefficients $a_k(\lambda,\mu)$ and $b_{d-k}(\lambda,\mu)$ are nonnegative integers. Moreover, the polynomials

conjecture | open | MathDB 页面 | 原文线索

V2-071553 | MATHDB-370539

CGS and Muirhead conjecture for Macdonald polynomials

摘录 / EXCERPT: Let λ and μ be partitions with $|\lambda|=|\mu|$. Let $P_\lambda(x;q,t)$ be the Macdonald polynomial, $\mathcal M_S(\mathbb F_{\geqslant0})$ the Muirhead semiring over...

conjecture | open | MathDB 页面 | 原文线索

V2-071554 | MATHDB-370540

Containment via Macdonald polynomial differences

摘录 / EXCERPT: Let λ and μ be partitions, and let $P_\lambda(x;q,t)$ denote the Macdonald polynomial. Macdonald containment conjecture. The Macdonald counterpart over...

conjecture | open | MathDB 页面 | 原文线索

V2-071555 | MATHDB-370544

The Bergman-space characterization of absolutely summing Toeplitz operators

摘录 / EXCERPT: Let ... of $ ^n$, and let $1<p\backslashle 2$. Let A^p denote the Bergman space, let $T_$ be the Toeplitz operator induced by ... be its Berezin transform, and let $K(,)$ be the Berg...

conjecture | open | MathDB 页面 | 原文线索

V2-071556 | MATHDB-370550

Asymptotic form conjecture for partitions of a 3-row rectangle

摘录 / EXCERPT: Let $p(3,n)$ denote the number of partitions of a $3\times n$ rectangle into permitted rectangular blocks, as defined in the paper. Asymptotic form conjecture. There exist constant...

conjecture | open | MathDB 页面 | 原文线索

V2-071557 | MATHDB-370551

Quasi-polynomial conjecture for restricted rectangle partitions p_{k,1}(2,n)

摘录 / EXCERPT: Fix an integer $k\geq 1$. Let $p_{k,1}(2,n)$ denote the number of partitions of the rectangle $2\times n$ using the block-size restrictions indexed by k and 1 in the paper. A f...

conjecture | open | MathDB 页面 | 原文线索

V2-071558 | MATHDB-370555

Polytopal realizations of infinitely many spheres $S(s,t)$

摘录 / EXCERPT: For integers s,t with $t>s$, let $S(s,t)$ denote the shellable, dual shellable 3-spheres constructed in the paper. The family has unbounded fatness as s grows. Polytopal realiz...

conjecture | open | MathDB 页面 | 原文线索

V2-071559 | MATHDB-370561

The Lemma-Flach factorisation conjecture for the d-region

摘录 / EXCERPT: Let $\underline{\Pi}$ be a family on GSp_4 whose specialisations have root number -1, let $\underline{\sigma}$ be the accompanying GL_2 family, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-071560 | MATHDB-370574

Stone-parity conjecture for signed tilings of the hexagonal grid

摘录 / EXCERPT: Stone-parity conjecture. The sign of ∂R corresponds to the parity of the minimum number of stones needed for R to be signed tilable when adding tiles of either weight a...

conjecture | open | MathDB 页面 | 原文线索

V2-071561 | MATHDB-370576

Nonexistence of product-type half-flag-transitive automorphism groups

摘录 / EXCERPT: Let $\mathcal D$ be a 2-(v,k,λ) design with replication number r and $\lambda\geq (r,2\lambda)^2$. An automorphism group of $\mathcal D$ is half-flag-transitive when it...

conjecture | open | MathDB 页面 | 原文线索

V2-071562 | MATHDB-370579

The general vector-stability chromatic formula conjecture

摘录 / EXCERPT: Let n,k be positive integers with $k\geq2$, and let $\vec{s}=(s_1,\ldots,s_k)$ be a positive integer vector with $s_i\geq2$ for $1\leq i\leq k-1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071563 | MATHDB-370587

Huybrechts' Hodge-theoretic characterization of Fourier-Mukai partners

摘录 / EXCERPT: Let X and Y be smooth cubic fourfolds, and write $\widetilde{\mathrm{H}}(X)$ and $\widetilde{\mathrm{H}}(Y)$ for the Addington-Thomas Hodge structures associated with their Kuz...

conjecture | open | MathDB 页面 | 原文线索

V2-071564 | MATHDB-370590

Object-space equivalence conjecture for iterated enriched categories

摘录 / EXCERPT: Let $\mathcal{C}$ be an (∞,n)-category, and let O_{n-1} and Θ_n be the indexing shapes used to define the iterated constructions...

conjecture | open | MathDB 页面 | 原文线索

V2-071565 | MATHDB-370592

Strong Schur log-concavity conjecture for growing partition shapes

摘录 / EXCERPT: Let $lambda, beta$ be partitions and let $alpha$ be an integral vector. Suppose $ell(lambda) gg ell(alpha)$ and $lambda_{ell(lambda)} gg beta_1$. Growing-shape strong Schur log-con...

conjecture | open | MathDB 页面 | 原文线索

V2-071566 | MATHDB-370595

Strong Schur log-concavity conjecture for arithmetic progressions of Schur functions

摘录 / EXCERPT: Let λ be a partition and let α be an integral vector. Arithmetic-progression Schur log-concavity conjecture. The sequence ... is strongly Schur log-concave for every s...

conjecture | open | MathDB 页面 | 原文线索

V2-071567 | MATHDB-370596

Bajo et al.'s leading-coefficient conjecture for trees

摘录 / EXCERPT: Let G be a graph, and let $\widetilde{\chi}_G^{\mathbb{1}}(q,x)$ denote the polynomial introduced in the source for the linear form $\mathbb{1}=(1,1,\ldots,1)$. Bajo et al.'s con...

conjecture | open | MathDB 页面 | 原文线索

V2-071568 | MATHDB-370598

Numerical threshold conjecture for cubic overpartition Jensen polynomials

摘录 / EXCERPT: Let $\overline{a}(n)$ denote the cubic overpartition function, and define its Jensen polynomials by ... For each degree d, let $N_{\overline{a}}(d)$ be the minimum integer such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-071569 | MATHDB-370599

Morgan-Mullen conjecture on primitive completely normal elements

摘录 / EXCERPT: Let $\mathbb F_{q^n}/\mathbb F_q$ be a finite field extension. The extension possesses the MM property if it contains an element $a\in\mathbb F_{q^n}$ that is primitive and complet...

conjecture | open | MathDB 页面 | 原文线索

V2-071570 | MATHDB-370608

Positive-measure observability conjecture for Schrödinger eigenfunctions on tori

摘录 / EXCERPT: Let $d\geq 1$, let $\mathbb{T}^d$ be the d-dimensional torus, and let $V\in L^\infty_x(\mathbb{T}^d)$. Let $\mathcal{H}_V=-\Delta+V$, and let f be any eigenfunction of...

conjecture | open | MathDB 页面 | 原文线索

V2-071571 | MATHDB-370613

Stable atom positivity conjectures for flagged LLT and modified nonsymmetric polynomials

摘录 / EXCERPT: Let $\mathcal P(r)$ be the polynomial space in which the stable atoms form a basis, and let stable atom positivity mean that the coefficients in this basis belong to...

conjecture | open | MathDB 页面 | 原文线索

V2-071572 | MATHDB-370620

Positive-density annihilating-differential conjecture

摘录 / EXCERPT: Positive-density conjecture. The set of primes p of good reduction for C such that $\bar{\omega}_p$ does not vanish on $C(\mathbb F_p)$ has positive density, unless J splits...

conjecture | open | MathDB 页面 | 原文线索

V2-071573 | MATHDB-370622

Finite-time convergence conjecture for sandpiles in a silo

摘录 / EXCERPT: Finite-time convergence conjecture. Then $u(t,\cdot)$ converges to u_ϕ in finite time.

conjecture | open | MathDB 页面 | 原文线索

V2-071574 | MATHDB-370623

Restricted global geometric Langlands conjecture

摘录 / EXCERPT: Let X be a smooth complete curve, let G be a reductive group with Langlands dual cG, and let $dShv_\Nilp(\operatorname{Bun}_G)$ denote the nilpotent sheaf category. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071575 | MATHDB-370624

Restricted local geometric Langlands conjecture

摘录 / EXCERPT: Let G be a reductive group with Langlands dual cG, let $\fL(G)$ be its loop group, and let $\fL(G)\operatorname{Cat}_{\operatorname{restr}}$ be the restricted 2-category of cat...

conjecture | open | MathDB 页面 | 原文线索

V2-071576 | MATHDB-370625

Local trace conjecture for restricted loop-group categories

摘录 / EXCERPT: Let G be a reductive group and let $\fL(G)\operatorname{Cat}_{\operatorname{restr}}$ be the restricted 2-category of categories acted on by its loop group. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071577 | MATHDB-370627

Dualizability conjecture for Hecke-lisse sheaves with level structure

摘录 / EXCERPT: With X, G, $\ul{x}$, and $\ul{\operatorname{Shv}}_{\operatorname{HL}}(\operatorname{Bun}_G^{\operatorname{level}_{\ul{x}}})$ as above, regard this object as an object of...

conjecture | open | MathDB 页面 | 原文线索

V2-071578 | MATHDB-370629

Ramified global geometric Langlands correspondence

摘录 / EXCERPT: Let $\oX=X-\ul{x}$, let $\operatorname{LS}^{\operatorname{restr}}_{cG}(\oX)$ and $\operatorname{LS}^{\operatorname{restr}}_{cG}(\oD_{\ul{x}})$ be the restricted moduli stacks of l...

conjecture | open | MathDB 页面 | 原文线索

V2-071579 | MATHDB-370630

Trace-classification conjecture for relative ind-coherent categories

摘录 / EXCERPT: Let $g:\mathcal Z\to\mathcal Y$ be a quasi-smooth map, let ϕ be an endomorphism of $\mathcal Y$ lifting to $\mathcal Z$, and let $\mathcal N$ be a conical singular-support con...

conjecture | open | MathDB 页面 | 原文线索

V2-071580 | MATHDB-370632

\(\Gamma\)-generalized Bayer-Macrì-Toda inequality

摘录 / EXCERPT: Let (X,H) be a polarised smooth projective threefold. Let $\operatorname{CH}_1(X)_{\mathbb R}$ denote the real Chow group of 1-cycles, and let $\nu_{b,w}$, Δ_H,...

conjecture | open | MathDB 页面 | 原文线索

V2-071581 | MATHDB-370637

The periodic coloring conjecture

摘录 / EXCERPT: Periodic coloring conjecture. For any V, N, Σ, and Ω as above, the solution set $\operatorname{Coloring}(V,N,\Sigma,\Omega)$ is not aperiodic.

conjecture | open | MathDB 页面 | 原文线索

V2-071582 | MATHDB-370643

Tosatti's diameter-collapse conjecture for irrational nef isotropic K3 classes

摘录 / EXCERPT: Tosatti's diameter-collapse conjecture. For every sequence $[\alpha_i]\in\mathcal{C}$ with $[\alpha_i]\to[\alpha]$,

conjecture | open | MathDB 页面 | 原文线索

V2-071583 | MATHDB-370644

Density of the rational obstructed boundary in the obstructed boundary

摘录 / EXCERPT: Let $\mathcal{H}$ be a hyperbolic component of disjoint type. Define its obstructed boundary by ... Let $\partial^\infty_{\mathbb{Q}}\mathcal{H}$ be the subset of...

conjecture | open | MathDB 页面 | 原文线索

V2-071584 | MATHDB-370655

Empirical-process convergence conjecture for pseudo-observations

摘录 / EXCERPT: Let F be the distribution function of the transformed observations, let $\hat F_n$ be the empirical distribution function based on the rank-based pseudo-observations, and define...

conjecture | open | MathDB 页面 | 原文线索

V2-071585 | MATHDB-370656

Zeta-type functional equation conjecture for the exponential congruence symbol

摘录 / EXCERPT: Zeta-type relation. The completed function may satisfy a functional equation of the form

conjecture | open | MathDB 页面 | 原文线索

V2-071586 | MATHDB-370659

Eventual upper-bound conjecture for odd generalized Davenport constants

摘录 / EXCERPT: Eventual upper-bound conjecture. If ${\sf D}(G)\ge 4$, then there exists $k_0\in\mathbb N$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071587 | MATHDB-370661

The $(2,3,7)$ hypergeometric-family Hodge-structure conjecture

摘录 / EXCERPT: Let E_7 be an elliptic curve with complex multiplication by $\mathbb{Q}(\sqrt{-21})$, and let ... be a fiber of the hypergeometric family. The $(2,3,7)$ Hodge-structure conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071588 | MATHDB-370682

The generalized colored overpartition generating-function identity

摘录 / EXCERPT: Let $x_1,\dots,x_k,m,n\in\mathbb{N}$ and $j\in\{1,\dots,k\}$. Define $\overline{S_j}(x_1,\dots,x_k;m,n)$ as the colored overpartition count from Definition, with the fourth conditi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071589 | MATHDB-370684

Hamilton-Jacobi conjecture for the bipartite enriched free energy

摘录 / EXCERPT: Let $\mathcal Q$ be the space of paths used to parameterize overlaps, let $q=(q_1,q_2)\in\mathcal Q^2$, and let $\psi_2:\mathcal Q^2\to\mathbb R$ be the initial free-energy functio...

conjecture | open | MathDB 页面 | 原文线索

V2-071590 | MATHDB-370688

Switch-generation conjecture for the matching preorder

摘录 / EXCERPT: Let $\mathcal{M}_n$ be the set of the labelled perfect matchings under consideration, and let $\preceq$ be the preorder on this set. A switch is the local transformation defined ea...

conjecture | open | MathDB 页面 | 原文线索

V2-071591 | MATHDB-370689

Antisymmetry conjecture for the matching preorder

摘录 / EXCERPT: Let $\mathcal{M}_n$ be the set of labelled perfect matchings, equipped with the preorder $\preceq$. Antisymmetry conjecture. The preorder $(\mathcal{M}_n,\preceq)$ is a poset; equi...

conjecture | open | MathDB 页面 | 原文线索

V2-071592 | MATHDB-370690

Special labelled h-factor realization conjecture

摘录 / EXCERPT: Let $n\in\mathbb{N}$, let $d_1\ge d_2\ge\dots\ge d_n\ge1$ be integers, and let h be a positive integer. The labelled h-factor is ... where each tuple denotes the complete graph on...

conjecture | open | MathDB 页面 | 原文线索

V2-071593 | MATHDB-370699

Nonuniqueness conjecture for quotient-neck singularities in Ricci flow

摘录 / EXCERPT: Let $k\geq3$. Consider Ricci flow on $S^1\times S^3/\mathbb{Z}_k$, and call a singularity a quotient-neck singularity when its tangent flow is $\mathbb{R}\times S^3/\mathbb{Z}_k$....

conjecture | open | MathDB 页面 | 原文线索

V2-071594 | MATHDB-370707

Deligne's rationality conjecture for critical motives

摘录 / EXCERPT: Let M be a critical motive over K with coefficient field $\mathcal{F}$. Let γ and δ be $\mathcal{F}$-bases of ${\det}_\mathcal{F}(H_B(M)^+)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071595 | MATHDB-370708

Generalized Deligne conjecture for critical motives

摘录 / EXCERPT: Let M be a critical motive over K with coefficient field $\mathcal{F}$, let $\Omega_{\gamma,\delta}$ be its period, and assume the height pairing h_M is available. If x and...

conjecture | open | MathDB 页面 | 原文线索

V2-071596 | MATHDB-370715

The optimal exponential-sum conjecture for one-dimensional Strichartz estimates

摘录 / EXCERPT: Optimal-loss conjecture. Under these sharp exponential-sum bounds, the Strichartz estimates should hold with loss $1/6+\epsilon$ for every $\epsilon>0$ and arbitrary...

conjecture | open | MathDB 页面 | 原文线索

V2-071597 | MATHDB-370734

Raspaud-Wang conjecture on partitioning triangle-free planar graphs

摘录 / EXCERPT: Let G be a finite simple triangle-free planar graph. A partition of G into an independent set and a forest means that there is a partition (V_1,V_2) of $V(G)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-071598 | MATHDB-370735

Le Potier's strange duality conjecture for Hilbert squares of K3 surfaces

摘录 / EXCERPT: Strange duality conjecture. For any $g\geq6$, this map is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-071599 | MATHDB-370740

Refined Kessar-Schaps conjecture for spin blocks

摘录 / EXCERPT: Let $B^{\eta,\rho,d}_{R_p}$ and $B^{\eta,\rho',d'}_{R_p}$ be arbitrary spin blocks of symmetric groups, or let $(B^{\eta,\rho,d}_{R_p})_{0}$ and $(B^{\eta,\rho',d'}_{R_p})_{0}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-071600 | MATHDB-370752

The strong-product infinity conjecture for distance packing domination

摘录 / EXCERPT: Strong-product infinity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071601 | MATHDB-370757

The niceness conjecture for Boolean lattices

摘录 / EXCERPT: Let B_n denote the Boolean lattice of subsets of an n-element set, and let a poset be nice when its incomparability graph has the nice property. Boolean-lattice niceness conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071602 | MATHDB-370758

LQYZ's non-Schur-positivity conjecture for products of chains

摘录 / EXCERPT: Let $\textbf m$ and $\textbf n$ denote chains indexed by the positive integers m and n, and let $\textbf m\times\textbf n$ be their product lattice. A lattice is Schur positive...

conjecture | open | MathDB 页面 | 原文线索

V2-071603 | MATHDB-370764

The Baby Wiegold conjecture for characteristic finite simple quotients of free groups

摘录 / EXCERPT: Let F_n denote the free group on n generators. A subgroup $C\leq F_n$ is characteristic if it is preserved by every automorphism of F_n. The Baby Wiegold conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-071604 | MATHDB-370765

Eventual characteristic alternating quotients of the rank-two free group

摘录 / EXCERPT: Let F_2 be the free group on two generators, and let A_n denote the alternating group on n elements. A finite quotient of F_2 is characteristic when it is of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-071605 | MATHDB-370767

Residual finite simplicity of $\mathrm{Aut}(F_2)$

摘录 / EXCERPT: Let F_2 be the free group on two generators. A group is fully residually finite simple if, for every finite subset of its nontrivial elements, there is a homomorphism to a finite...

conjecture | open | MathDB 页面 | 原文线索

V2-071606 | MATHDB-370770

Zero-free conjecture for the Mittag-Leffler function on the imaginary axis

摘录 / EXCERPT: Let $calpha a\in\left(\frac{3}{5}, 1\right)$. The Mittag-Leffler function $E_{\alpha,1}(z)$ is defined by its standard power series, and the imaginary axis is the set...

conjecture | open | MathDB 页面 | 原文线索

V2-071607 | MATHDB-370771

Zero-free conjecture for the auxiliary function in fractional Schrödinger equations

摘录 / EXCERPT: Let $calpha\in(1,2)$, and let $calpsi(t)$ be the function defined in the paper by equation (psidf). Zero-free conjecture. The function $calpsi(t)$ has no zeros on $(0,\infty)$, nam...

conjecture | open | MathDB 页面 | 原文线索

V2-071608 | MATHDB-370772

Complete monotonicity or log-convexity of a Mittag-Leffler expression

摘录 / EXCERPT: Let $0<\alpha\leq\frac{1}{2}$ and define ... Here $E_{\rho,\beta}$ denotes the Mittag-Leffler function with parameters ρ and β. Complete-monotonicity conjecture. The funct...

conjecture | open | MathDB 页面 | 原文线索

V2-071609 | MATHDB-370773

Optimal Brauer p-dimension conjecture for complete discretely valued fields

摘录 / EXCERPT: Let $\mathcal{K}$ be a complete discrete valued field of characteristic zero with residual characteristic p, and let κ be its residue field. For a field E of characteris...

conjecture | open | MathDB 页面 | 原文线索

V2-071610 | MATHDB-370775

Bremner's conjecture on additive rigidity of elliptic-curve points

摘录 / EXCERPT: Let $E/\a0\mathbb{Q}$ be an elliptic curve of Mordella0a0Weil rank r, and let $\{P_1,\ldots,P_N\}$ be a sequence of rational points whose x-coordinates form an arithmetic progr...

conjecture | open | MathDB 页面 | 原文线索

V2-071611 | MATHDB-370781

Optimality conjecture for harmonic frames in real two-dimensional phase retrieval

摘录 / EXCERPT: Harmonic-frame optimality conjecture. If $m\geq 4$ is even, then

conjecture | open | MathDB 页面 | 原文线索

V2-071612 | MATHDB-370798

Conjectured removal of the logarithmic factor in tensor robust PCA recovery

摘录 / EXCERPT: Tensor robust PCA recovery conjecture. The factor

conjecture | open | MathDB 页面 | 原文线索

V2-071613 | MATHDB-370802

Fixed-point conjecture for the EFX allocation map

摘录 / EXCERPT: Let m agents and n goods be given, with valuation parameters $v_{\ell i}$ and $V=\sum_{\ell,i}v_{\ell i}$. Let $\epsilon>0$ be arbitrarily small and assume $V\leq\epsilon$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071614 | MATHDB-370805

Strong hub cover pebbling conjecture for cycles

摘录 / EXCERPT: Strong hub cover pebbling conjecture for cycles.

conjecture | open | MathDB 页面 | 原文线索

V2-071615 | MATHDB-370820

The scaling-limit conjecture for fuzzy Potts clusters and cluster measures

摘录 / EXCERPT: Fix $q\in(1,4)$, set $r=1/q$, and let ω^δ be the fuzzy Potts configuration on $\delta\mathbb{Z}^2$. For lattice approximations z_i^δ of...

conjecture | open | MathDB 页面 | 原文线索

V2-071616 | MATHDB-370824

Secondary braid group conjecture for trigonal abelian-differential strata

摘录 / EXCERPT: Let k be an integer for which the indicated trigonal stratum ${\mathbf P}{\mathcal H}^{tri}_{3k+1}(4k,2k)$ is defined, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071617 | MATHDB-370853

Generic two-component terminal boundary for charged one-ended black holes

摘录 / EXCERPT: Consider generic regular spherically symmetric one-ended asymptotically flat black hole solutions of the Einstein-Maxwell-charged-scalar-field equations $1.1\$-$5.1\$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-071618 | MATHDB-370854

C^2 Strong Cosmic Censorship conjecture for spherically symmetric collapse

摘录 / EXCERPT: Let the Einstein-Maxwell-charged-scalar-field system be given by equations ... , and consider generic, one-ended, asymptotically flat, complete spherically symmetric initial data. It...

conjecture | open | MathDB 页面 | 原文线索

V2-071619 | MATHDB-370855

Potent categorical Langlands duality for reductive groups

摘录 / EXCERPT: Let G be a reductive group and let $\check G$ be its Langlands dual group. Denote by $\mathcal{H}_G^{dR}$ and $\mathcal{H}_{\check G}^{dR}$ the de Rham potent Hecke monads, and b...

conjecture | open | MathDB 页面 | 原文线索

V2-071620 | MATHDB-370858

The binary polynomial Collatz conjecture

摘录 / EXCERPT: Let $A\in\mathbb{F}_2[x]\setminus\{0\}$. Define the associated sequences $(a_{2k})_k$, $(A_{2k})_k$, and $(A_{2k+1})_k$ by $A_0=A$, $A_{2k}=1+(x^2+x+1)A_{2k-1}$, and by removing al...

conjecture | open | MathDB 页面 | 原文线索

V2-071621 | MATHDB-370859

The Mersenne-type binary polynomial odd-sequence length conjecture

摘录 / EXCERPT: Let $M_1=x^2+x+1$, and let the odd sequence of a polynomial be the sequence of odd polynomials obtained by the binary polynomial Collatz transformations. Let $r\geq 1$ and...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-071622 | MATHDB-370860

The degree multiplicity conjecture for x^n+x+1

摘录 / EXCERPT: Let $A=x^n+x+1$ with $n\geq 4$, and let s be the greatest positive integer such that ... For each positive integer $t\leq s-1$, denote by d_t the common degree asserted for the r...

conjecture | open | MathDB 页面 | 原文线索

V2-071623 | MATHDB-370861

The even-and-odd sequence length conjecture for x^n+x+1

摘录 / EXCERPT: Let $A=x^n+x+1$ with $n\geq 4$, and let s be the greatest positive integer such that ... Consider the even and odd sequences associated with A under the binary polynomial Collatz...

conjecture | open | MathDB 页面 | 原文线索

V2-071624 | MATHDB-370872

Xin's partition-type q,t-symmetry conjecture for $\vec{k}$-Dyck paths

摘录 / EXCERPT: Let $\vec{k}$ be a vector of positive integers, let $\mathcal{K}$ be the associated parameter, and define the area-bounce polynomial ... Write $\lambda(\mathcal{K})$ for the partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-071625 | MATHDB-370874

The elliptic braid-group presentation conjecture

摘录 / EXCERPT: Let R be an elliptic root system, with underlying affine part R_a. Let $\tilde{W}(R)$ be its extended Weyl group, let $\tilde{c}$ be a hyperbolic Coxeter element satisfying ... a...

conjecture | open | MathDB 页面 | 原文线索

V2-071626 | MATHDB-370879

Cartesian multicategories as retromodels

摘录 / EXCERPT: Cartesian multicategories as retromodels. The category of cartesian models is the category of retromodels of $\ifstrequal{S}{1}{\mathds{1}}{\mathbb{S}}^\mathrm{op}$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-071627 | MATHDB-370881

The contractibility conjecture for stability spaces of acyclic quivers

摘录 / EXCERPT: Let Q be a finite connected acyclic quiver. Write $\operatorname{Stab}(\mathcal{D}^b(Q))$ for the space of stability conditions on $\mathcal{D}^b(Q)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071628 | MATHDB-370882

Genčev-Rucki conjecture for multiple Apéry-like series

摘录 / EXCERPT: Genčev-Rucki conjecture. For every $r\in\mathbb{N}$ with $r>1$,

conjecture | open | MathDB 页面 | 原文线索

V2-071629 | MATHDB-370886

Folklore conjecture on the reverse entropy power inequality for i.i.d. log-concave random vectors

摘录 / EXCERPT: Let X and Y be independent and identically distributed log-concave random vectors taking values in $\mathbb{R}^d$, and let h denote differential entropy. Folklore conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071630 | MATHDB-370905

The Cartier-Witt divisor description conjecture for $cwidehat{\bbF}_1^\mathrm{Syn}$

摘录 / EXCERPT: Cartier-Witt divisor conjecture. An R-point of $\widehat{\bbF}_1^\mathrm{Syn}$ is given by a formal group over R, a Cartier-Witt divisor on this pullback, and an isomorphism ov...

conjecture | open | MathDB 页面 | 原文线索

V2-071631 | MATHDB-370919

The iCoulomb branch and shifted iYangian conjecture

摘录 / EXCERPT: Let $Q=({\mathbb I},\Omega)$ be an ADE quiver, let $(V_i;W_i)_{i\in{\mathbb I}}$ be a representation of its framed double quiver, and impose the stated Satake-diagram symmetry and...

conjecture | open | MathDB 页面 | 原文线索

V2-071632 | MATHDB-370921

Conjecture on polynomial exponential sums becoming kloostermanian functions

摘录 / EXCERPT: Polynomial exponential-sum conjecture. If g and h satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-071633 | MATHDB-370931

Dyer's conjecture on joins of biclosed sets in extended weak order

摘录 / EXCERPT: Let (W,S) be a Coxeter system with associated root system Φ, positive roots Φ^+, and reflection s_α corresponding to each $\alpha\in\Phi^+$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071634 | MATHDB-370932

Dyer's finite Coxeter group conjecture on joins in weak order

摘录 / EXCERPT: Let (W,S) be a finite Coxeter system, let T be its set of reflections, and let ℓ be the Coxeter length function. For $w\in W$, define its left-reflection set by ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-071635 | MATHDB-370943

Integral unitary representations conjecture

摘录 / EXCERPT: Let G be a reductive group with Langlands parameter φ satisfying $I_F\subseteq\ker\varphi$ and ... where λ is dominant integral in the fundamental positive chamber....

conjecture | open | MathDB 页面 | 原文线索

V2-071636 | MATHDB-370948

Gamow's conjecture for the Coulomb liquid drop model

摘录 / EXCERPT: Let m_{ball} be the critical mass below which balls minimize the liquid drop energy, let m_ex be the critical mass up to which minimizers exist, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071637 | MATHDB-370949

Higher-dimensional Gamow conjecture for Riesz interactions

摘录 / EXCERPT: Let $n\geq 2$ and $0<\alpha<n$, and let m_{ball}, m_ex, and m_crit denote respectively the critical mass below which balls minimize the liqui...

conjecture | open | MathDB 页面 | 原文线索

V2-071638 | MATHDB-370962

Dominating Hadwiger's Conjecture

摘录 / EXCERPT: Dominating Hadwiger's Conjecture. For every integer $t\geq 1$, every graph with no dominating K_t minor is $(t-1)$-colorable.

conjecture | refuted | MathDB 页面 | 原文线索

V2-071639 | MATHDB-370967

Partially compactified Radon transform conjecture

摘录 / EXCERPT: Let P and $P^{\vee}$ be the projective spaces replacing the dual projective spaces in the Radon transform, and let $D(P)_{\boldsymbol\nu}$ and $D_{\boldsymbol\nu}$ be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-071640 | MATHDB-370969

Arnold-Khesin coadjoint-orbit density conjecture

摘录 / EXCERPT: Let X be a vector field in the setting of three-dimensional volume-preserving flows, and let its corresponding helicity level set be the set of vector fields with the same helici...

conjecture | open | MathDB 页面 | 原文线索

V2-071641 | MATHDB-370984

Asymptotic complete-graph-power conjecture for t-tone coloring

摘录 / EXCERPT: Complete-graph-power conjecture. For all positive integers t and b, there exists an $N=N(t,b)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071642 | MATHDB-370985

Stefanich's conjecture on limits of presentable n-categories

摘录 / EXCERPT: Let $n\mathsf{Pr}$ be the 1-category of presentable n-categories, defined as the filtered colimit ... where $n\mathsf{Pr}^{\kappa}$ is the category of κ-compactly generate...

conjecture | open | MathDB 页面 | 原文线索

V2-071643 | MATHDB-370987

Bouleau-Hirsch energy image density conjecture for Dirichlet structures

摘录 / EXCERPT: Bouleau-Hirsch energy image density conjecture. Every Dirichlet structure satisfies the energy image density property.

conjecture | open | MathDB 页面 | 原文线索

V2-071644 | MATHDB-370988

Chen-Zhang's optimal list-recovery conjecture for folded Reed-Solomon codes

摘录 / EXCERPT: Chen-Zhang's conjecture. For the stated parameters, folded Reed-Solomon codes achieve list-recovery radius arbitrarily close to ρ^* with input list size ℓ and output list s...

conjecture | open | MathDB 页面 | 原文线索

V2-071645 | MATHDB-370993

Baily-Borel-Hodge compactification conjecture for Calabi-Yau type hypersurfaces

摘录 / EXCERPT: Let (n,d) be a pair with ... Let M be the GIT moduli space of degree d hypersurfaces in $\mathbb{P}^n$ with m-rational singularities, and let $M^{\rm BBH}$ denote its Hodge-t...

conjecture | open | MathDB 页面 | 原文线索

V2-071646 | MATHDB-370996

The SSM-Thom polynomial conjecture for multisingularities

摘录 / EXCERPT: Assume $l\geq 1$. Let $\mathcal{C}_l$ be the class of maps under consideration, let $\underline{\eta}$ be a T- or S-multisingularity, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071647 | MATHDB-370997

The degree-bounded SSM-Thom polynomial conjecture for Mather multisingularities

摘录 / EXCERPT: Assume $l\geq 1$. Let $\underline{\eta}$ be a Mather T- or S-multisingularity, let $M(l)$ be the Mather bound, and let $\mathcal{C}_l$ be the class of maps under consideration. Wri...

conjecture | open | MathDB 页面 | 原文线索

V2-071648 | MATHDB-371010

Vu's codegree conjecture for chromatic number

摘录 / EXCERPT: Let G be a graph. Write $\operatorname{\Delta_2}(G)$ for its maximum codegree and $\operatorname{\Delta}(G)$ for its maximum degree. Vu's conjecture. Fix...

conjecture | open | MathDB 页面 | 原文线索

V2-071649 | MATHDB-371013

Square-coloring codegree conjecture

摘录 / EXCERPT: Let G be a graph. Its square G^2 is obtained by joining distinct vertices that are connected by a two-edge path in G. Let $\operatorname{\Delta}(G)$ be the maximum degree,...

conjecture | open | MathDB 页面 | 原文线索

V2-071650 | MATHDB-371026

Equality conjecture for the PRs cubed and PRS relaxations

摘录 / EXCERPT: Let $\mathcal{S}^\kappa_{P,R,s^3}$ and $\mathcal{S}^\kappa_{P,R,S}$ denote the relaxations associated respectively with the PRs^3 and PRS formulations. Equality conjecture. ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-071651 | MATHDB-371041

Low-temperature crystallization conjecture for one-dimensional Riesz gases

摘录 / EXCERPT: Let $\mathsf{P}_{s,\beta}$ denote a one-dimensional non-singular Riesz gas with homogeneity parameter s, and let $\mathsf{Leb}_{\mathbb{R}}$ be Lebesgue measure on $\mathbb{R}$....

conjecture | open | MathDB 页面 | 原文线索

V2-071652 | MATHDB-371069

The b-chromatic number of the tensor product of distinct star graphs

摘录 / EXCERPT: Let G be the tensor product of two distinct star graphs S_n and S_m. Star-graph tensor-product conjecture. The b-chromatic number of G is $m+n+1$. This conjecture propose...

conjecture | open | MathDB 页面 | 原文线索

V2-071653 | MATHDB-371070

The b-chromatic number inequality for powers of Cartesian graph products

摘录 / EXCERPT: Let G and H be graphs, let $k\text{?}$ be a positive integer, and write φ for the b-chromatic number and G^k for the k-th graph power. Cartesian-product power con...

conjecture | open | MathDB 页面 | 原文线索

V2-071654 | MATHDB-371082

Cocks's hereditary treewidth-clique boundedness conjecture

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary graph class. A graph H is H-free in $\mathcal{G}$ when graphs in the relevant subclass have no induced subgraph isomorphic to H. The class...

conjecture | open | MathDB 页面 | 原文线索

V2-071655 | MATHDB-371087

Folklore conjecture on better-quasi-ordering finite graphs by minors

摘录 / EXCERPT: Let a graph be better-quasi-ordered (BQO) under the minor relation when it satisfies the better-quasi-order property for that relation. Folklore conjecture. The finite graphs are b...

conjecture | open | MathDB 页面 | 原文线索

V2-071656 | MATHDB-371091

KPS Hecke-tree conjecture for rational branches

摘录 / EXCERPT: Let $P(t)\in\mathbb{F}_q[t]$ be irreducible. For a lattice $\Lambda_0\in\mathcal{L}_2$, let $\mathbb{T}_P(\Lambda_0)$ be its $P(t)$-Hecke tree, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071657 | MATHDB-371099

The induced-minor finite-asymptotic-dimension conjecture

摘录 / EXCERPT: For a graph F, a graph class excludes F as an induced minor when none of its graphs contains F as an induced minor. A graph class has finite asymptotic dimension when its asy...

conjecture | open | MathDB 页面 | 原文线索

V2-071658 | MATHDB-371104

Analyticity criterion for suitable C-Summation sequences

摘录 / EXCERPT: Analyticity criterion for C-Summation. If this series is analytic at $-\frac{k_2}{k_1}$, then I_n is suitable for C-Summation.

conjecture | open | MathDB 页面 | 原文线索

V2-071659 | MATHDB-371111

Exponential relaxation of FA-1f from exponentially nonempty initial laws

摘录 / EXCERPT: Exponential-relaxation conjecture. If there exists $\kappa>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-071660 | MATHDB-371113

The scaling-limit and dimension conjecture for three-dimensional loop clusters

摘录 / EXCERPT: Let $\mathcal{L}_{1/2}^{\widetilde{\mathbb{Z}}^3}$ be the loop soup on the metric graph of $\mathbb{Z}^3$, and let $\mathcal{L}_{1/2}^{\mathbb{R}^3}$ be the corresponding loop soup...

conjecture | open | MathDB 页面 | 原文线索

V2-071661 | MATHDB-371114

Negativity conjecture for the Jensen-Shannon and Kullback-Leibler divergence Jacobian

摘录 / EXCERPT: Let $\Omega^{\mathrm{o}}=\{(\mu,\nu)\in(0,1)\times(0,1)\mid \nu<\mu\}$ be the interior of Ω, and let J be the Jacobian defined in Eq.. Negativity conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-071662 | MATHDB-371115

Conjectured constrained moment-matching lower bound in the interpolation regime

摘录 / EXCERPT: Constrained moment-matching lower-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071663 | MATHDB-371144

Hoffmann-Laghribi's Pfister-neighbour conjecture for forms of small defect

摘录 / EXCERPT: Hoffmann-Laghribi's conjecture. If $s<2r$ and φ_1 is defined over F, then φ is a Pfister neighbour.

conjecture | open | MathDB 页面 | 原文线索

V2-071664 | MATHDB-371145

The Pfister-neighbour conjecture for forms with small defect parameter

摘录 / EXCERPT: Pfister-neighbour conjecture. If $s<\frac{2^{n+1}}{3}$ and φ_1 is defined over F, then φ is a Pfister neighbour.

conjecture | open | MathDB 页面 | 原文线索

V2-071665 | MATHDB-371146

Vishik's and Hoffmann's Pfister-neighbour conjectures for nondegenerate forms

摘录 / EXCERPT: Vishik-Hoffmann conjectures. The following assertions should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-071666 | MATHDB-371147

Excellent connections for degenerate forms

摘录 / EXCERPT: Excellent-connections conjecture. For integers a,b with $0\leq a,d_X-b<r$, if (a,b) is excellent, then a_{lo} and b^{up} are connected in $\Lambda(X)$.

conjecture | open | MathDB 页面 | 原文线索

V2-071667 | MATHDB-371154

Level-two congruence subgroup conjecture for non-spherical small Coxeter groups

摘录 / EXCERPT: Let Γ be a non-spherical small Coxeter graph, with associated Artin group $A[\Gamma]$, Coxeter group $W[\Gamma]$, and level-two congruence subgroup $A[\Gamma][2]$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-071668 | MATHDB-371169

Gao's conjecture on the Erdős-Ginzburg-Ziv constant

摘录 / EXCERPT: Gao's conjecture. For every finite abelian group G, one has

conjecture | open | MathDB 页面 | 原文线索

V2-071669 | MATHDB-371178

Gontier-Lewin-Nazar's finite-rank optimizer conjecture

摘录 / EXCERPT: Let $d\geq 2$, let $q>1$, and let γ be a density matrix satisfying $0\leq\gamma\leq 1$. Write ... for a rank-N projection, and let ρ_γ denote its density. The asso...

conjecture | open | MathDB 页面 | 原文线索

V2-071670 | MATHDB-371185

Bounded discrepancies between tie-breaking conventions

摘录 / EXCERPT: Let S be a subtraction set, and consider any two tie-breaking conventions in the associated self-interest cumulative subtraction game. For a heap size h, let the discrepancy be...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-071671 | MATHDB-371188

Numerical slope conjecture for ground states of the nonlinear Schrödinger equation on the unit ball

摘录 / EXCERPT: Let B_1 be the unit ball, let β denote the first Dirichlet eigenvalue parameter, and let Q_b be the least-energy (ground state) solution of the stationary nonlinear Schrö...

conjecture | open | MathDB 页面 | 原文线索

V2-071672 | MATHDB-371205

Equivariant log-concavity conjecture for rook placement loci

摘录 / EXCERPT: For $0\le r\le\min\{m,n\}$, let ${\mathcal{Z}}_{n,m,r}$ be the locus of $n\times m$ rook placements with exactly r rooks, and let $R({\mathcal{Z}}_{n,m,r})$ denote its orbit harm...

conjecture | open | MathDB 页面 | 原文线索

V2-071673 | MATHDB-371209

Ill-posedness conjecture for the two-dimensional Schrödinger equation with an L^1+L^∞ potential

摘录 / EXCERPT: Ill-posedness conjecture. For any $\gamma\in \mathbb{R}$, there exists $\eta \in L_x^1+L_x^{\infty}(\mathbb{R}^2)$ such that (eq:NLS) is ill-posed in $H_x^\gamma(\mathbb{R}^2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-071674 | MATHDB-371210

The Halanay-Smith criterion for strong stability with distributed delays

摘录 / EXCERPT: Halanay-Smith criterion. Let $M\in\mathcal G$. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-071675 | MATHDB-371215

Generalized Bell-polynomial relation for symmetric functions

摘录 / EXCERPT: Generalized Bell-polynomial relation. The relation

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071676 | MATHDB-371216

Homogeneous-polynomial extension of the generalized relation

摘录 / EXCERPT: Let ${\bf x}^m=\{x_1,\ldots,x_m\}$ and let ${\bf 1}^m$ denote the m-tuple all of whose entries are 1. Let $U_n({\bf x}^m,{\bf y}^m)$ be the function defined from the symmetric...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071677 | MATHDB-371217

Power-sum relation for arbitrary homogeneous symmetric polynomials

摘录 / EXCERPT: Let ${\bf x}^m=\{x_1,\ldots,x_m\}$ and ${\bf y}^m=\{y_1,\ldots,y_m\}$. For an arbitrary homogeneous symmetric polynomial $S_n({\bf x}^m)$ of degree n, write ... where...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071678 | MATHDB-371221

The Baxter permutation longest increasing subsequence exponent conjecture

摘录 / EXCERPT: A Baxter permutation of size n is a permutation avoiding the two patterns specified in the source, and bipolar-oriented planar maps with general face degrees correspond to Baxter...

conjecture | open | MathDB 页面 | 原文线索

V2-071679 | MATHDB-371231

Hausdorff convergence to the optimizing convex set

摘录 / EXCERPT: Let $\mathbf{K}$ be a compact convex set of area 1, let $\lambda>0$, and suppose that $\operatorname{argmax}\Phi_{\lambda,\mathbf{K}}$ consists of the single element $C^\star$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-071680 | MATHDB-371233

Conjectured distance threshold for $(4,s)$-Salem sets

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field, let $d\geq 2$, and let $E\subseteq\mathbb{F}_q^d$ be a $(4,s)$-Salem set. Let $\alpha(d,4,s)$ be the smallest exponent such that every $(4,s)$...

conjecture | open | MathDB 页面 | 原文线索

V2-071681 | MATHDB-371237

Tan-Lei similarity conjecture for cubic superattracting parameter curves

摘录 / EXCERPT: Let $F=F_{a,v}$ be a cubic polynomial in the superattracting period-p parameter curve $4\mathcal{S}_p4$, written in the normal form ... Assume that F is a Misiurewicz map, meanin...

conjecture | open | MathDB 页面 | 原文线索

V2-071682 | MATHDB-371244

Critical-temperature variance conjecture for the SK model

摘录 / EXCERPT: Let $F_N^{SK}(\beta)$ denote the free energy of the Sherrington-Kirkpatrick model with system size N at inverse temperature β, and let the critical value be $\beta_c=1$. Cr...

conjecture | open | MathDB 页面 | 原文线索

V2-071683 | MATHDB-371249

Fomichev-Karev conjecture on graph invariants

摘录 / EXCERPT: Let $\mathbf{G}$ be the set of isomorphism classes of finite simple graphs. For $G\in\mathbf{G}$, let $\varphi(G)$ be the graph invariant defined by ... where $\chi_3(G)$ is the numb...

conjecture | open | MathDB 页面 | 原文线索

V2-071684 | MATHDB-371254

The generalized-quasiorder characterization conjecture for partial orders

摘录 / EXCERPT: Let A be a set and let ρ be a partial order on A with least element 0 and greatest element 1. Let $\operatorname{Pol}\rho$ be the clone of operations preserving ρ...

conjecture | open | MathDB 页面 | 原文线索

V2-071685 | MATHDB-371256

Doyle's quadratic-polynomial preperiodic-point bound over quadratic fields

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a quadratic field, let $f\in K[z]$ be a quadratic polynomial, and let $\operatorname{PrePer}(f,K)$ denote its set of K-rational preperiodic points. Doyle's...

conjecture | open | MathDB 页面 | 原文线索

V2-071686 | MATHDB-371257

Periodic-point uniform boundedness conjecture over number fields

摘录 / EXCERPT: Let $\operatorname{Per}(\varphi,\mathbb{P}^1(K))$ denote the set of periodic points of a morphism $\varphi:\mathbb{P}^1\to\mathbb{P}^1$ defined over a number field K. Periodic-po...

conjecture | open | MathDB 页面 | 原文线索

V2-071687 | MATHDB-371272

Cutoff conjecture for the Burnside process

摘录 / EXCERPT: Fix $k\ge 2$, and for each n let K_n be the Burnside process on (C_k^n,S_n). Consider starting states whose limiting empirical distribution on C_k assigns a positive propor...

conjecture | open | MathDB 页面 | 原文线索

V2-071688 | MATHDB-371287

Willmore conjecture in codimension 1

摘录 / EXCERPT: Let $g>1$ be an integer, and consider smooth surfaces of $\mathbb{R}^3$ of genus g and their Willmore energy. Willmore conjecture in codimension 1. The only minimizers are the co...

conjecture | open | MathDB 页面 | 原文线索

V2-071689 | MATHDB-371288

The 16π conjecture for sphere eversion

摘录 / EXCERPT: The 16π conjecture. The number β_0 is attained by a path $\omega\in C^0([0,1],\operatorname{Imm}(\mathbb{S}^2,\mathbb{R}^3))$ such that both restrictions...

conjecture | open | MathDB 页面 | 原文线索

V2-071690 | MATHDB-371292

Nonvanishing Jacobian-minor conjecture for generalized constants

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{\geq2}$ be fixed. For each $m\in\{2,3,\ldots,n\}$ and each $\theta\in\left(\mathcal{G}_{m}^{\left(\operatorname{U}\right)}\cup\{\gamma\}\right)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071691 | MATHDB-371300

Christol's conjecture on globally bounded D-finite functions

摘录 / EXCERPT: Christol's conjecture. Every D-finite globally bounded function is the diagonal of a rational function.

conjecture | open | MathDB 页面 | 原文线索

V2-071692 | MATHDB-371306

Piecewise-affine factor-map conjecture for Lorenz-attractor perturbations

摘录 / EXCERPT: Let $\mathcal{R}$ be the collection of all maps $\mathcal{F}:[0,1]\to[0,1]$ that serve as factor maps for first-return maps $\psi_v:R\setminus W\to R$ of the Lorenz attractor, not...

conjecture | open | MathDB 页面 | 原文线索

V2-071693 | MATHDB-371308

The disjoint-cone conjecture for SL_2-plethysm generating functions

摘录 / EXCERPT: The bivariate generating function is ... A weighted lattice-point cone conjecture. The generating function $A_\mu(z,q)$ is a weighted generating function of lattice points of a disjo...

conjecture | open | MathDB 页面 | 原文线索

V2-071694 | MATHDB-371309

The denominator conjecture for SL_2-plethysm generating functions

摘录 / EXCERPT: Denominator conjecture. The expression

conjecture | open | MathDB 页面 | 原文线索

V2-071695 | MATHDB-371320

The GOE nodal-count central limit conjecture

摘录 / EXCERPT: GOE nodal-count central limit conjecture. For every $k\in[n]$,

conjecture | open | MathDB 页面 | 原文线索

V2-071696 | MATHDB-371324

Conjecture on common roots of sorted binomial polynomials

摘录 / EXCERPT: Let $P_n(z)$ denote the sorted binomial polynomial, and call a root nontrivial when it is not the trivial root shared by the relevant polynomials. Common-root conjecture. $P_n(z)$...

conjecture | open | MathDB 页面 | 原文线索

V2-071697 | MATHDB-371325

Irreducibility conjecture for even sorted binomial polynomials

摘录 / EXCERPT: Let $P_n(z)$ be the sorted binomial polynomial with parameter n. Irreducibility conjecture. $P_n(z)$ is irreducible over $\mathbb{Q}$ for all even n. The analogous factorizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-071698 | MATHDB-371331

Signed Gyárfás-Sumner conjecture

摘录 / EXCERPT: Let $\widehat{K}_2^- = \widehat{(K_2,-)}$ and let $\widetilde{G}$ denote the signed graph obtained from a graph G by replacing each edge with a digon. A finite set of signed grap...

conjecture | open | MathDB 页面 | 原文线索

V2-071699 | MATHDB-371332

K4-free signed Gyárfás-Sumner conjecture for linear forests

摘录 / EXCERPT: Let F be a linear forest, meaning a forest whose connected components are paths. Let $\widehat{(K_4,-)}$ be the specified signed complete graph, and call a finite set of signed g...

conjecture | open | MathDB 页面 | 原文线索

V2-071700 | MATHDB-371334

Equivalent local boundedness conjecture for signed graph classes

摘录 / EXCERPT: Let $(K_3,-)$ denote the signed triangle with the specified signature, let (K_4,M) denote the signed complete graph whose negative edges form a matching of size two, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071701 | MATHDB-371335

Minimal signed Gyárfás-Sumner sets containing a negative edge

摘录 / EXCERPT: Let a finite set of signed graphs be a GS set when its forbidden induced-subgraph class has bounded balanced chromatic number, and call it minimal if no proper subset is a GS set....

conjecture | open | MathDB 页面 | 原文线索

V2-071702 | MATHDB-371339

Uniform better-than-square-root cancellation conjecture for smooth numbers

摘录 / EXCERPT: Uniform smooth-number cancellation conjecture. The paper conjectures that its main estimate should hold uniformly for every $2\leqslant y\leqslant x$:

conjecture | open | MathDB 页面 | 原文线索

V2-071703 | MATHDB-371345

Vendi's conjecture that Vendi Score bounds LCR

摘录 / EXCERPT: Let Z be an $n\times n$ positive semidefinite similarity matrix with $n>0$, symmetric entries in $[0,1]$, and diagonal entries equal to 1. On the uniform distribution, write LC...

conjecture | open | MathDB 页面 | 原文线索

V2-071704 | MATHDB-371346

Equivalent trace and row-sum bounds for Vendi Score and LCR

摘录 / EXCERPT: Let Z be an $n\times n$ positive semidefinite similarity matrix as above, and let $\mathbf{1}$ denote the n-dimensional all-ones vector. Equivalent bounds conjecture. For $q>1$...

conjecture | open | MathDB 页面 | 原文线索

V2-071705 | MATHDB-371347

Symmetric matrix power inequality for row sums

摘录 / EXCERPT: Let M be any $n\times n$ symmetric matrix with entries satisfying $0\leq M_{ij}\leq1$, and let $\mathbf{1}$ be the n-dimensional all-ones vector. Matrix power inequality conjec...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071706 | MATHDB-371356

Non-subnormality conjecture for Cauchy duals on three-point atomic spaces

摘录 / EXCERPT: Let ζ_1,ζ_2,ζ_3 be any three distinct points on the unit circle, let c_1,c_2,c_3 be positive real numbers, and set ... Let M_z denote multiplication by z on...

conjecture | open | MathDB 页面 | 原文线索

V2-071707 | MATHDB-371357

Uniform critical-orbit comparison for bi-critical irrationally indifferent attractors

摘录 / EXCERPT: Uniform critical-orbit comparison conjecture. There exists a uniform constant $C_{}>0$ such that, for all $k\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-071708 | MATHDB-371359

Toric g-vector Kruskal-Katona conjecture for simple polytopes

摘录 / EXCERPT: Let P be a simple polytope, and let $(\gamma_i(P))$ and $(g_i(P))$ denote its γ-vector entries and toric g-vector entries, respectively. Toric g-vector Kruskal-Katona co...

conjecture | open | MathDB 页面 | 原文线索

V2-071709 | MATHDB-371370

The pathwidth-clique-number conjecture for bounded alpha-treewidth

摘录 / EXCERPT: Let $\mathcal{G}$ be a graph class. For graph parameters ρ and ω, say that $\mathcal{G}$ is (ρ,ω)-bounded when bounded clique number in $\mathcal{G}$ implies...

conjecture | open | MathDB 页面 | 原文线索

V2-071710 | MATHDB-371377

The type C K-Stanley expansion conjecture in GQ functions

摘录 / EXCERPT: Type C K-Stanley expansion conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071711 | MATHDB-371381

Free-boundary pseudocritical scaling conjecture for the weakly coupled $|\varphi|^4$ model

摘录 / EXCERPT: Consider the $|\varphi|^4$ model with free boundary conditions (FBC), and let $\chi_{g,\nu,N}^{\rm F}$ denote its susceptibility, while $\chi_{g,\nu,N}$ denotes the susceptibility...

conjecture | open | MathDB 页面 | 原文线索

V2-071712 | MATHDB-371392

Mazur-Tate's order-of-vanishing and weak main conjectures

摘录 / EXCERPT: Mazur-Tate conjecture. The following claims are valid:

conjecture | open | MathDB 页面 | 原文线索

V2-071713 | MATHDB-371393

Mazur-Tate's leading-term conjecture for elliptic curves

摘录 / EXCERPT: Mazur-Tate leading-term conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-071714 | MATHDB-371406

Weak-limit conjecture for maximal Schnyder woods of the UIHPT

摘录 / EXCERPT: Weak-limit conjecture. The unique maximal Schnyder wood of the UIHPT is the weak limit of the maximal Schnyder wood of a uniformly random triangulation from $\cT_n^m$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-071715 | MATHDB-371411

Central limit theorem for random polynomials with general coefficients

摘录 / EXCERPT: Let $N_{\mathbb{R},\boldsymbol{\xi}}$ be the number of real zeros of a random polynomial from any model considered in Section. Assume that the coefficient distribution is subgaussi...

conjecture | open | MathDB 页面 | 原文线索

V2-071716 | MATHDB-371418

Favaron-Shi minimum-degree conjecture for minimal factor-critical graphs

摘录 / EXCERPT: Favaron-Shi's conjecture. If G is a minimal k-factor-critical graph, then

conjecture | open | MathDB 页面 | 原文线索

V2-071717 | MATHDB-371426

Spectral-memory correspondence for coupled gravitational and electromagnetic fields

摘录 / EXCERPT: Let $(\mathcal M,g)$ be an asymptotically flat spacetime with Bondi coordinates (u,r,θ,ϕ) near future null infinity $\mathscr I^{+}$. Write the radiative data as ... and de...

conjecture | open | MathDB 页面 | 原文线索

V2-071718 | MATHDB-371433

Sokal's total monomial positivity conjecture for Hadamard products of dual Jacobi-Trudi matrices

摘录 / EXCERPT: Sokal's conjecture. The Hadamard product $M(\mathbf{x})*M(\mathbf{y})$ is totally monomial positive; equivalently, every minor

conjecture | open | MathDB 页面 | 原文线索

V2-071719 | MATHDB-371434

Sokal's factorial monomial-positivity conjecture for products of elementary symmetric functions

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\dots,\lambda_n)$ and $\mu=(\mu_1,\dots,\mu_n)$ be partitions with at most n parts satisfying $\mu\subseteq\lambda$. Let k and r be integers with...

conjecture | open | MathDB 页面 | 原文线索

V2-071720 | MATHDB-371445

Guo-Wang's conjecture for positive harmonic functions on the unit ball

摘录 / EXCERPT: Let $n\geq 3$, let $\mathbb{B}^n\subset\mathbb{R}^n$ be the unit ball with boundary sphere $\mathbb{S}^{n-1}$, and let $u\in C^{\infty}(\mathbb{B}^n)$ be positive and satisfy ... whe...

conjecture | open | MathDB 页面 | 原文线索

V2-071721 | MATHDB-371447

The exceptional-order conjecture for decomposing complete graphs into squares of Hamilton cycles

摘录 / EXCERPT: Let K_n be the complete graph on n vertices, let C_n be a Hamilton cycle on these vertices, and let C_n^2 denote its square, obtained by joining pairs at distance at most t...

conjecture | open | MathDB 页面 | 原文线索

V2-071722 | MATHDB-371456

Chow et al.'s Latin Young diagram conjecture

摘录 / EXCERPT: Let Y be a Young diagram with row lengths $l_1,\ldots,l_m$. It is Latin if its cells can be assigned integers so that row i contains $1,\ldots,l_i$, and the entries in each col...

conjecture | open | MathDB 页面 | 原文线索

V2-071723 | MATHDB-371457

Allocation conjecture for wide Young diagrams

摘录 / EXCERPT: Let Y be a Young diagram, and let an allocation mean a coarse filling obtained by subdividing Y into subrectangles and partitioning the symbol set into smaller subsets, with pr...

conjecture | open | MathDB 页面 | 原文线索

V2-071724 | MATHDB-371460

The k-ball cutoff conjecture for balanced generalised Bernoulli--Laplace chains

摘录 / EXCERPT: k-ball cutoff conjecture. For $k=o(n)$, the balanced chain exhibits cutoff around the time

conjecture | open | MathDB 页面 | 原文线索

V2-071725 | MATHDB-371461

The product-chain cutoff conjecture for ordered labeled urn chains

摘录 / EXCERPT: Ordered labeled product-chain conjecture. The chains $\mathbf{X}^{\operatorname{ord,lab}}$ and $\mathbf{X}^{\operatorname{ord,lab}}_{\operatorname{prod}}$ exhibit cutoff around the...

conjecture | open | MathDB 页面 | 原文线索

V2-071726 | MATHDB-371469

Guo-Wang-Zhang's Hörmander dichotomy conjecture

摘录 / EXCERPT: Hörmander dichotomy conjecture. The operators T_ϕ^λ satisfy the operator boundedness estimate for $p>\frac{2n}{n-1}$ if and only if ϕ satisfies Bourgain's condition...

conjecture | open | MathDB 页面 | 原文线索

V2-071727 | MATHDB-371470

Guo-Wang-Zhang's curved Kakeya conjecture for Bourgain's condition

摘录 / EXCERPT: Curved Kakeya conjecture. If ϕ satisfies Bourgain's condition, then every ϕ-Kakeya set has Hausdorff dimension n.

conjecture | open | MathDB 页面 | 原文线索

V2-071728 | MATHDB-371471

Discretized classical sticky Kakeya conjecture in $\mathbb{R}^n$

摘录 / EXCERPT: Discretized sticky Kakeya conjecture. For every $\varepsilon>0$ there exist $\eta=\eta_{n,\mathrm{rest}}(\varepsilon)>0$ and $\delta_0=\delta_{0,n,\mathrm{rest}}(\varepsilon)>0$ su...

conjecture | open | MathDB 页面 | 原文线索

V2-071729 | MATHDB-371474

Uniform fixed-vector growth conjecture for reductive p-adic groups

摘录 / EXCERPT: Let G be a reductive group over a p-adic field F, let x be a point in its Bruhat-Tits building, and let $K_\ell:=K_{G,x,\ell}$ be the ℓth Moy-Prasad subgroup at x. W...

conjecture | open | MathDB 页面 | 原文线索

V2-071730 | MATHDB-371482

Signed Iwasawa main conjecture for diagonal cycles

摘录 / EXCERPT: Let M be the residual-representation lattice and let $A=(M\otimes_{\mathbb{Z}_p}\mathbb{Q}_p)/M$. For each choice of signed local condition indexed by...

conjecture | open | MathDB 页面 | 原文线索

V2-071731 | MATHDB-371483

Spectral growth conjecture for Salem numbers in $W(E_{10})$

摘录 / EXCERPT: Let $W(E_{10})$ be the Coxeter group under consideration, and let $\rho(d4e)$ denote the spectral radius of an element $d4e\in W(E_{10})$. For each level n, let...

conjecture | open | MathDB 页面 | 原文线索

V2-071732 | MATHDB-371485

Equality of average-order constants for lucky omega functions

摘录 / EXCERPT: The lucky divisibility functions are defined by ... where $\overset{*}{\omega}(n)$ counts distinct lucky divisors and $\overset{*}{\Omega}(n)$ counts them with their lucky order. Luc...

open_problem | open | MathDB 页面 | 原文线索

V2-071733 | MATHDB-371489

Ji's length-spectrum rigidity conjecture for rational maps

摘录 / EXCERPT: Length-spectrum rigidity conjecture. If f and g have the same length spectrum, then

conjecture | open | MathDB 页面 | 原文线索

V2-071734 | MATHDB-371490

Ji's multiplier-spectrum injectivity conjecture

摘录 / EXCERPT: Multiplier-spectrum injectivity conjecture. If $\tau_d(f)=\tau_d(g)$, then either

conjecture | open | MathDB 页面 | 原文线索

V2-071735 | MATHDB-371491

Ji's small-period multiplier-spectrum injectivity conjecture

摘录 / EXCERPT: Small-period multiplier-spectrum injectivity conjecture. For every $d\geq 2$, the morphism

conjecture | open | MathDB 页面 | 原文线索

V2-071736 | MATHDB-371492

Ji's generic length-spectrum injectivity conjecture

摘录 / EXCERPT: Generic length-spectrum injectivity conjecture. There is a Zariski closed proper subset $E_d\subset {\mathcal M}_d$ defined over $\mathbb R$ such that, for every $[f]\notin E_d$, i...

conjecture | open | MathDB 页面 | 原文线索

V2-071737 | MATHDB-371494

Maturana-Rashmi bandwidth-optimality conjecture for split-regime MDS convertible codes

摘录 / EXCERPT: Maturana-Rashmi's conjecture. Under the Uniform Cost Assumption, every stable linear MDS $(n^{I},k^{I}={\lambda}^{F}k^{F};n^{F},k^{F})$ convertible code with $r^{F}<r^{I}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071738 | MATHDB-371504

Mixed-syzygy length conjecture for rational surfaces

摘录 / EXCERPT: Let X be a rational surface and let L be an ample and base point free line bundle on X. Suppose that $|L|$ is a linear system containing curves C of genus $g\geq 1$, and wr...

conjecture | open | MathDB 页面 | 原文线索

V2-071739 | MATHDB-371505

Mixed-syzygy length conjecture for line bundles on the quadric surface

摘录 / EXCERPT: Let $X=\mathbb{P}^1\times\mathbb{P}^1$ and let $L=aC_0+bf$. Let $\delta(a,b)$ denote the length of the region of mixed syzygies. Mixed-syzygy length conjecture. The length is ... Equ...

conjecture | open | MathDB 页面 | 原文线索

V2-071740 | MATHDB-371516

Ji-Xie conjecture on non-conjugate rational maps with equal multiplier spectra

摘录 / EXCERPT: Ji-Xie conjecture. If f and g have the same multiplier spectrum, then one of the following holds: f and g are Lattès maps, or $f\sim g$.

conjecture | open | MathDB 页面 | 原文线索

V2-071741 | MATHDB-371520

Deformation conjecture for semi-log canonical foliation singularities

摘录 / EXCERPT: Semi-log canonical singularities deform. Under these hypotheses, semi-log canonical singularities of the fibre at t persist for all fibres over some Zariski-open neighbourhood...

conjecture | open | MathDB 页面 | 原文线索

V2-071742 | MATHDB-371521

Inversion of adjunction conjecture for foliations

摘录 / EXCERPT: Inversion of adjunction. Log canonicity of the induced foliated pair on the normalisation of S should imply log canonicity of the original foliated pair near S.

conjecture | open | MathDB 页面 | 原文线索

V2-071743 | MATHDB-371525

Exact induced-degenerate-subgraph ratio conjecture for k-degenerate graphs

摘录 / EXCERPT: Let $\alpha_d(k)$ denote the infimum, over k-degenerate graphs, of the ratio of the order of a largest induced d-degenerate subgraph to the number of vertices. Induced-degenera...

conjecture | open | MathDB 页面 | 原文线索

V2-071744 | MATHDB-371526

Half-induced-forest conjecture for graphs on surfaces

摘录 / EXCERPT: For a graph G with n vertices and genus g, let $\alpha_1(G)$ denote the order of a largest induced 1-degenerate subgraph of G, equivalently a largest induced forest. Half...

conjecture | open | MathDB 页面 | 原文线索

V2-071745 | MATHDB-371527

Induced forest bound for graphs of prescribed girth

摘录 / EXCERPT: Let G be a graph on n vertices with m edges, and let $\alpha_1(G)$ denote the order of a largest induced 1-degenerate subgraph of G, equivalently a largest induced forest...

conjecture | open | MathDB 页面 | 原文线索

V2-071746 | MATHDB-371529

Brück's conjecture for entire functions sharing a finite value with their derivative

摘录 / EXCERPT: Brück's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071747 | MATHDB-371530

Liu-Laine difference-sharing conjecture for entire functions

摘录 / EXCERPT: Liu-Laine's conjecture. If f and $\Delta_c^n f$ share $a\in\mathbb{C}\backslash\{0\}$ CM, then

conjecture | open | MathDB 页面 | 原文线索

V2-071748 | MATHDB-371531

Zhang et al.'s exponential-form conjecture for difference sharing

摘录 / EXCERPT: Zhang et al.'s conjecture. If f and $\Delta^n f$ share 0 CM, then f has the form

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V2-071749 | MATHDB-371537

Gorenstein characterization conjecture for collections of cells

摘录 / EXCERPT: Let $\mathcal{P}$ be a collection of cells. The Gorenstein characterization conjecture asserts that the following are equivalent: 1. $K[\mathcal{P}]$ is Gorenstein; 2. the h-poly...

conjecture | open | MathDB 页面 | 原文线索

V2-071750 | MATHDB-371545

Discrete nilpotent loci conjecture for isomonodromic sections

摘录 / EXCERPT: Let $\mathcal T_g$ be the Teichmüller space of genus $g\geq 2$ compact Riemann surfaces, with universal family $\mathcal X/\mathcal T_g$. For an isomonodromic section...

conjecture | open | MathDB 页面 | 原文线索

V2-071751 | MATHDB-371546

The signed-graph K_5-minor conjecture for maps to signed projective cubes

摘录 / EXCERPT: Signed-graph K_5-minor conjecture. Any signed graph (G,σ) which has neither a $(K_5,-)$-minor nor induces an element of $\mathcal{O}_{k+1}$ maps to $SPC(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-071752 | MATHDB-371562

Polynomial row-treewidth conjecture for string graphs on fixed surfaces

摘录 / EXCERPT: Let G be a string graph drawn in a surface of fixed Euler genus g, and let Δ be the maximum degree of G. Polynomial row-treewidth conjecture. The row treewidth of G...

conjecture | open | MathDB 页面 | 原文线索

V2-071753 | MATHDB-371563

Bounded convex decomposition conjecture for rotation sets of surface homeomorphisms

摘录 / EXCERPT: Bounded convex decomposition conjecture. There exists a map

conjecture | open | MathDB 页面 | 原文线索

V2-071754 | MATHDB-371611

O'Grady's generalized Franchetta conjecture for polarized K3 surfaces

摘录 / EXCERPT: For each $g\ge 3$, let $\mathcal F_g'$ be the smooth dense open moduli space of primitively polarized K3 surfaces of degree $2g-2$ with trivial automorphism groups, and let ... be th...

conjecture | open | MathDB 页面 | 原文线索

V2-071755 | MATHDB-371616

The mod-2 persistence conjecture for prefix-reversal generating triples

摘录 / EXCERPT: Let n,m,k be integers with $2\leqslant k<m<n$. Mod-2 persistence conjecture. For any $n\equiv0,1\pmod{4}$, if ... then ... This was proposed from computational experiments for...

conjecture | open | MathDB 页面 | 原文线索

V2-071756 | MATHDB-371628

Conjecture on the limiting distribution of the maximum degree as p tends to zero

摘录 / EXCERPT: Let X denote the limiting random variable associated with the normalized maximum degree in Theorem 1.2, and let D_1 and D_2 be the constants appearing there. Maximum-degree l...

conjecture | open | MathDB 页面 | 原文线索

V2-071757 | MATHDB-371633

The product dimension conjecture for posets

摘录 / EXCERPT: Let P and Q be posets. Write $P\times Q$ for their Cartesian product, ordered componentwise, and let $\dim(P)$ denote the least integer n such that P is isomorphic to a sub...

conjecture | open | MathDB 页面 | 原文线索

V2-071758 | MATHDB-371640

The Nine Dragon Tree Conjecture for bounded forest decompositions

摘录 / EXCERPT: Let G be a graph, and let k,d be nonnegative integers. The Nine Dragon Tree Conjecture. If ... then G decomposes into $k+1$ forests, one of which is d-bounded. The conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-071759 | MATHDB-371642

The mixed Nine Dragon Tree conjecture for spanning mixed arborescences

摘录 / EXCERPT: Let $H=(V,E,A)$ be a mixed graph, let $\mathcal{D}(V)$ denote the relevant family of subpartitions, and let $d_A^-(X)$ count arcs entering X. Let $k\geq 0$ and $d\geq 1$ be integer...

conjecture | open | MathDB 页面 | 原文线索

V2-071760 | MATHDB-371649

The QUP polar-code functional inequality assumption

摘录 / EXCERPT: QUP polar-code functional inequality. Then

conjecture | open | MathDB 页面 | 原文线索

V2-071761 | MATHDB-371672

Shalom's spectral gap conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be a center-free irreducible higher-rank lattice in G, and let V be a unitary representation of Γ. Assume that the Γ-action on V does not extend to...

conjecture | open | MathDB 页面 | 原文线索

V2-071762 | MATHDB-371673

The strong spectral gap conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be a center-free irreducible higher-rank lattice in G, and let V be a unitary representation of Γ whose action does not extend to a G-action on any non-tr...

conjecture | open | MathDB 页面 | 原文线索

V2-071763 | MATHDB-371674

Lubotzky's property tau conjecture for higher-rank lattices

摘录 / EXCERPT: Let Γ be an irreducible higher-rank lattice. Lubotzky's property τ conjecture. The group Γ has property τ. This is listed among consequences of the spectral...

conjecture | open | MathDB 页面 | 原文线索

V2-071764 | MATHDB-371680

The acyclic hero characterization for tournaments

摘录 / EXCERPT: The acyclic hero conjecture. A tournament H is an acyclic hero if and only if $H\in\mathcal{AH}$.

conjecture | open | MathDB 页面 | 原文线索

V2-071765 | MATHDB-371681

The local-to-global conjecture for acyclic dichromatic number

摘录 / EXCERPT: Local-to-global conjecture. There exists a function $f\colon \mathbb{N}\to\mathbb{N}$ such that every tournament T satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071766 | MATHDB-371682

The acyclic 3-dicolouring conjecture for planar oriented graphs

摘录 / EXCERPT: Planar acyclic dichromatic conjecture. Every oriented planar graph D satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071767 | MATHDB-371693

NP optimality of geometric neutral zones

摘录 / EXCERPT: NP optimality of geometric neutral zones. The reliability score induces a monotone likelihood-ratio ordering: whenever $R_1>R_2$,

conjecture | open | MathDB 页面 | 原文线索

V2-071768 | MATHDB-371694

Pure transport limit for ascent and descent stability

摘录 / EXCERPT: Let Ω be a domain, let $\boldsymbol{\beta}\in [L^\infty(\Omega)]^d$ be divergence-free, and let L be the transport operator with inflow boundary conditions, ... For a discre...

conjecture | open | MathDB 页面 | 原文线索

V2-071769 | MATHDB-371706

Montesinos-Morton branched-cover conjecture for fibered links

摘录 / EXCERPT: Let L be a fibered link in S^3. A simple branched cover is a map $\pi\colon S^3\to S^3$ branched over a link L', and α and β are unknots that are braid axes of...

conjecture | open | MathDB 页面 | 原文线索

V2-071770 | MATHDB-371707

Bode's braided open-book conjecture for fibered links

摘录 / EXCERPT: Let L be a fibered link in S^3. A braided open book is an open book in S^3 whose binding is braided relative to the standard open book with unknotted binding; equivalently, i...

conjecture | open | MathDB 页面 | 原文线索

V2-071771 | MATHDB-371708

Mutual arc presentation conjecture for fibered links

摘录 / EXCERPT: Let L be a fibered link in S^3. A mutual arc presentation is the presentation defined in the paper that generalizes Cromwell's arc presentations. Mutual arc presentation conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071772 | MATHDB-371712

Toruńczyk's conjecture on blind flip-width and linear clique-width

摘录 / EXCERPT: Let $\mathscr{C}$ be a class of graphs. Toruńczyk's conjecture. The class $\mathscr{C}$ has bounded blind flip-width if and only if it has bounded linear clique-width. Toruńczyk in...

conjecture | open | MathDB 页面 | 原文线索

V2-071773 | MATHDB-371723

Navarra-Qureshi-Rinaldo domino-stability conjecture for Gorenstein cell ideals

摘录 / EXCERPT: Domino-stability conjecture. The ring $K[\mathcal{P}]$ is Gorenstein if and only if $\mathcal{P}$ is domino-stable.

conjecture | open | MathDB 页面 | 原文线索

V2-071774 | MATHDB-371734

The move-connectivity conjecture for rotationally symmetric plabic graphs

摘录 / EXCERPT: Move-connectivity conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071775 | MATHDB-371747

Universal Beilinson-Flach Euler system conjecture

摘录 / EXCERPT: Let U be a sufficiently small affinoid in the half-ordinary universal deformation family, let V_U be its associated representation, and let $\mathscr{D}^+_U$ be the chosen Panc...

conjecture | open | MathDB 页面 | 原文线索

V2-071776 | MATHDB-371754

Seymour's improvement conjecture for Hadwiger numbers

摘录 / EXCERPT: Let G be a finite simple graph, let $\alpha(G)$ denote its independence number, and let $\operatorname{had}(G)$ denote its Hadwiger number. Seymour's improvement conjecture. Ther...

conjecture | open | MathDB 页面 | 原文线索

V2-071777 | MATHDB-371764

L-equivalence implies Fourier-Mukai partnership for cubic fourfolds

摘录 / EXCERPT: Let Y and Y' be cubic fourfolds. Write $\mathcal{A}_Y$ for the Kuznetsov component of Y, defined by ... The cubic fourfolds are FM partners if there is a Fourier-Mukai equivale...

conjecture | open | MathDB 页面 | 原文线索

V2-071778 | MATHDB-371769

Keith's parity conjecture for reciprocals of false theta functions

摘录 / EXCERPT: Keith's conjecture. For $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-071779 | MATHDB-371775

Near-linear-time weak recovery by Glauber dynamics

摘录 / EXCERPT: In the spiked Wigner inference problem, let annealed Glauber dynamics refer to the Markov chain studied in the paper, and let weak recovery mean achieving nonzero correlation with...

conjecture | open | MathDB 页面 | 原文线索

V2-071780 | MATHDB-371776

Polynomial- or near-linear-time posterior sampling by annealed Glauber dynamics

摘录 / EXCERPT: Consider the spiked Wigner inference problem with signal-to-noise parameter $\beta_{\bM}$ and threshold $\beta_{\bM}$ as used in the paper. Let $\beta_{\bM}$ denote the scaled post...

conjecture | open | MathDB 页面 | 原文线索

V2-071781 | MATHDB-371784

The SOS conjecture for symmetric biquadratic M-tensors

摘录 / EXCERPT: Let $\mathcal{A}$ be a symmetric biquadratic M-tensor: a symmetric biquadratic tensor that can be written as $\mathcal{A}=\alpha\mathcal{I}-\mathcal{B}$, where $\mathcal{I}$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-071782 | MATHDB-371785

The SOS conjecture for symmetric B_0-biquadratic tensors

摘录 / EXCERPT: Let $\mathcal{A}$ be a symmetric B_{0}-biquadratic tensor, meaning a symmetric biquadratic tensor satisfying the B_{0} conditions specified in the source. An...

conjecture | open | MathDB 页面 | 原文线索

V2-071783 | MATHDB-371786

The five-square conjecture for 3 × 3 symmetric PSD biquadratic forms

摘录 / EXCERPT: Let $P(\mathbf{x},\mathbf{y})$ be a 3×3 symmetric PSD biquadratic form, so $P(\mathbf{x},\mathbf{y})\geq 0$ for all $\mathbf{x},\mathbf{y}\in\mathbb{R}^3$. A bilinear form...

conjecture | open | MathDB 页面 | 原文线索

V2-071784 | MATHDB-371788

Sharpness conjecture for the Hankel-Korn inequality

摘录 / EXCERPT: Let d be the dimension, let $cmathcal H(d)$ denote the Hankel subspace, and let $p^*=\max\{p,p/(p-1)\}$ be the conjugate exponent. Write $C(\mathcal H(d),p)$ for the optimal...

conjecture | open | MathDB 页面 | 原文线索

V2-071785 | MATHDB-371795

Chowla's sign-pattern conjecture for consecutive Liouville values

摘录 / EXCERPT: Let $s\geq 1$ and let $(\epsilon_1,\ldots,\epsilon_s)\in\{-1,1\}^s$ be a sign pattern. Chowla's sign-pattern conjecture. The set ... has natural density 2^{-s}. This is the sign-pa...

conjecture | open | MathDB 页面 | 原文线索

V2-071786 | MATHDB-371797

Ben-Zvi-Sakellaridis-Venkatesh reduced-dual period formula

摘录 / EXCERPT: Let $\hat{\Delta}_{\mathrm{red}}=(\hat{L},\hat{H}',1,\hat{\rho}_{\hat{H}',\hat{\iota}'})$, where ... Let π_L be a tempered automorphic representation of $L(\mathbb A)$ and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071787 | MATHDB-371798

Relative trace formula comparison for BZSV periods

摘录 / EXCERPT: Let $Q=LU$ be a parabolic subgroup of G, let N_L be a maximal unipotent subgroup of L, set $N=N_LU$, and let ξ_L be a generic character of $[N_L]$, extended trivially acr...

conjecture | open | MathDB 页面 | 原文线索

V2-071788 | MATHDB-371799

Sun-Zhang's K-polystability conjecture for polarized Fano fibrations

摘录 / EXCERPT: Sun-Zhang's conjecture. A polarized Fano fibration $(\pi:X\to Y,\xi)$ admits a Kähler-Ricci shrinker (ω,ξ), unique up to automorphisms of X preserving ξ, if and only...

conjecture | open | MathDB 页面 | 原文线索

V2-071789 | MATHDB-371800

The period-index conjecture for Brauer classes over C_d fields

摘录 / EXCERPT: Let K be a C_d field, and let $\alpha\in \operatorname{Br}(K)$ be a Brauer class. The period $\operatorname{per}(\alpha)$ is its order in the Brauer group, and the index...

conjecture | solved | MathDB 页面 | 原文线索

V2-071790 | MATHDB-371809

Xiao's hyperellipticity conjecture for Lefschetz fibrations

摘录 / EXCERPT: Let f be a genus-g Lefschetz fibration with slope λ_f, and call f hyperelliptic when its monodromy lies in the hyperelliptic mapping class group. Xiao's hyperelliptic...

conjecture | open | MathDB 页面 | 原文线索

V2-071791 | MATHDB-371818

Dominance conjecture for descent polynomials of spinal graphs

摘录 / EXCERPT: Dominance conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071792 | MATHDB-371821

Submaximal-rank classification conjecture for multiplicity-one operator identities

摘录 / EXCERPT: Let R be an operator identity of degree 3 and multiplicity 1 on an associative triple system. Let $v\in\mathbb{F}^4$ be the coefficient vector of R, and suppose that v pr...

conjecture | open | MathDB 页面 | 原文线索

V2-071793 | MATHDB-371827

Stein's minimum semi-degree conjecture for oriented paths

摘录 / EXCERPT: Stein's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-071794 | MATHDB-371844

The nonlinear harmonic bundle conjecture for relative nonabelian Hodge moduli

摘录 / EXCERPT: Let α_S be the canonical homeomorphism between the relative de Rham and Higgs moduli spaces, and assume that it is an isomorphism of differentiable fiber bundles. Denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-071795 | MATHDB-371853

Concentration conjecture for separating systoles of random triangulations

摘录 / EXCERPT: Fix $\theta\in(0,1/2)$ and a sequence (g_n) such that $g_n/n\rightarrow\theta$. Let $\mathbf{T}_{2n,g_n}$ be a uniform triangulation of genus g_n with $2n$ faces. The separatin...

conjecture | open | MathDB 页面 | 原文线索

V2-071796 | MATHDB-371856

Conjectural generalisation of the sewing lemma for approximate actions

摘录 / EXCERPT: Let P be a metric space, and let $\big((M_x),\,(\mu_{xy})\big)$ be an approximate action of the pair groupoid $P\times P$, with each M_x a complete metric space and μ_{xy}...

conjecture | open | MathDB 页面 | 原文线索

V2-071797 | MATHDB-371863

Converse parametrization conjecture for Brunovsky transformations

摘录 / EXCERPT: Converse parametrization conjecture. If a Brunovsky transformation (T,F,G) transforms the staircase linear system into Brunovsky form, then there exists an observation matrix C...

conjecture | open | MathDB 页面 | 原文线索

V2-071798 | MATHDB-371871

Intersection-product conjecture for odd-dimensional or Calabi-Yau hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb{P}^{n+1}(\mathbb{C})$ be a smooth hypersurface. Assume that either the dimension n is odd, or the degree of X is $n+2$, so that X is Calabi-Yau. Intersec...

conjecture | open | MathDB 页面 | 原文线索

V2-071799 | MATHDB-371875

The characterization of totally positive sequence preservers by their remainders

摘录 / EXCERPT: The remainder characterization conjecture. The sequence ${\mathbf A}$ is a TP_∞-preserver if and only if, for every $l\in\mathbb N\cup\{0\}$, $R_l[\mathbf A]$ is an...

conjecture | open | MathDB 页面 | 原文线索

V2-071800 | MATHDB-371881

Continuity conjecture for the frog-model phase transition

摘录 / EXCERPT: Let G be a transitive graph, and let the frog model on G have critical curves $\lambda_{\mathrm{c}}(t)$ and $t_{\mathrm{c}}(\lambda)$. Continuity conjecture. The phase transiti...

conjecture | open | MathDB 页面 | 原文线索

V2-071801 | MATHDB-371882

Strict monotonicity conjecture for the frog-model critical curves

摘录 / EXCERPT: Let G be a transitive graph of superlinear growth, and let $\lambda_{\mathrm{c}}(t)$ and $t_{\mathrm{c}}(\lambda)$ denote the critical curves of the frog model. Strict monotonici...

conjecture | open | MathDB 页面 | 原文线索

V2-071802 | MATHDB-371886

The multi-horizon angular-momentum Penrose inequality

摘录 / EXCERPT: Let the initial data contain n disjoint outermost marginally outer trapped surfaces (MOTS) $\{\Sigma_i\}_{i=1}^n$, with areas A_i and angular momenta J_i. Multi-horizon exten...

conjecture | open | MathDB 页面 | 原文线索

V2-071803 | MATHDB-371890

The entropy-mass bound for black holes

摘录 / EXCERPT: Let $S=A/(4G\hbar)$ be the Bekenstein-Hawking entropy of a black hole with mass M and horizon area A. Entropy-mass bound conjecture. The entropy should satisfy ... with equality...

conjecture | open | MathDB 页面 | 原文线索

V2-071804 | MATHDB-371894

The universal black-hole inequality with angular momentum and charges

摘录 / EXCERPT: Universal black-hole inequality conjecture. The mass should satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-071805 | MATHDB-371900

Baby Julia sets in parameter slices of generalized McMullen maps

摘录 / EXCERPT: Let a one-dimensional subfamily of the bifurcation locus of the generalized McMullen family be given, with two critical orbits having independent behavior. Suppose one critical val...

conjecture | open | MathDB 页面 | 原文线索

V2-071806 | MATHDB-371902

Spine approximation conjecture for the boundedness locus of generalized McMullen maps

摘录 / EXCERPT: For the subfamily $r_{n,a}$, let $M_n(r_{n,a})$ denote its boundedness locus, let $\mathcal{S}_n$ be its finite-n spine, let $\mathcal{S}_\infty$ be the limiting spine, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071807 | MATHDB-371903

Baby Mandelbrot centers in the generalized McMullen subfamily

摘录 / EXCERPT: For each $n\geq 3$, consider the family $r_{n,a}$ in the a-plane and the parameters a_k associated with the potential centers of hyperbolic components, where k is odd and...

conjecture | open | MathDB 页面 | 原文线索

V2-071808 | MATHDB-371904

Aubert-Zelevinsky duality conjecture for unipotent representations

摘录 / EXCERPT: Aubert-Zelevinsky duality conjecture. If $X=X(q^{\frac{1}{2}{\it h}},\mathcal{S},\mathcal{L})$ is an irreducible unipotent representation in...

conjecture | open | MathDB 页面 | 原文线索

V2-071809 | MATHDB-371905

Weak Arthur packets as unions of simplified Arthur packets

摘录 / EXCERPT: Weak Arthur packet decomposition conjecture. There is a set $\mathcal T$ and an appropriate notion of Arthur packet such that

conjecture | open | MathDB 页面 | 原文线索

V2-071810 | MATHDB-371909

Ebenfelt's SOS conjecture for ranks of prolonged sums of squares

摘录 / EXCERPT: SOS Conjecture. If $A(z,\bar{z})\|z\|^2\in {\rm{SOS}}_n$, then either

conjecture | open | MathDB 页面 | 原文线索

V2-071811 | MATHDB-371913

Pineda-Villavicencio's lower-bound conjecture for polytopes with $2d+\ell$ vertices

摘录 / EXCERPT: Let $d\geqslant 3$ and $1\leqslant\ell\leqslant d-1$. Let P be a d-polytope with $2d+\ell$ vertices, and let $T(r)$ denote the r-simplex. Pineda-Villavicencio's conjecture. 1...

conjecture | open | MathDB 页面 | 原文线索

V2-071812 | MATHDB-371916

Rational connectedness conjecture for quasi-Campana pairs

摘录 / EXCERPT: Let $(X,D_{\mathbfsymbol{m}})$ be a Campana pair, where $D_{\mathbfsymbol{m}}$ is its boundary divisor, and assume that the log-anticanonical divisor $-K_X-D_{\mathbfsymbol{m}}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-071813 | MATHDB-371917

The index and finiteness conjecture for embedded minimal surfaces in the three-sphere

摘录 / EXCERPT: Let $\gamma\in\mathbb{N}$, and let $\Sigma\subset\mathbb{S}^3$ be a closed embedded minimal surface of genus γ. Denote its Morse index by $\operatorname{Ind}(\Sigma)$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-071814 | MATHDB-371919

Selection at the pushed-to-pulled transition

摘录 / EXCERPT: Let u_{tr} be a pushed-to-pulled transition front. A pushed-to-pulled transition front is a front at the boundary between pushed and pulled invasion, characterized here by...

conjecture | open | MathDB 页面 | 原文线索

V2-071815 | MATHDB-371920

Selection of rigid pulled pattern-forming fronts

摘录 / EXCERPT: Let u_pl be a rigid pulled pattern-forming front, meaning a rigid front with marginal spectral stability that leaves behind a diffusively stable periodic pattern and has...

conjecture | open | MathDB 页面 | 原文线索

V2-071816 | MATHDB-371921

Selection of modulated pushed pattern-forming fronts

摘录 / EXCERPT: Let $u_\mathrm{ps}(\xi,\tau)$ be a modulated pushed pattern-forming front, namely a temporally modulated pushed front leaving a pattern-forming wake. Selection of modulated pushed...

conjecture | open | MathDB 页面 | 原文线索

V2-071817 | MATHDB-371922

Selection of modulated pulled pattern-forming fronts

摘录 / EXCERPT: Let $u_\mathrm{pl}(\xi,\tau)$ be a modulated pulled pattern-forming front satisfying the stated pinched-double-root, periodic-wake, exponential-asymptotic, marginal-Floquet-stabili...

conjecture | open | MathDB 页面 | 原文线索

V2-071818 | MATHDB-371945

L-equivalence conjecture for cubic fourfold Fourier-Mukai partners

摘录 / EXCERPT: Let X and Y be cubic fourfolds over the complex numbers. Their Kuznetsov components are the admissible subcategories $\mathcal{A}_X$ and $\mathcal{A}_Y$ in the semiorthogonal d...

conjecture | open | MathDB 页面 | 原文线索

V2-071819 | MATHDB-371949

Simply laced Bruhat preclosure closure conjecture

摘录 / EXCERPT: Simply laced Bruhat preclosure conjecture. The Bruhat preclosure is idempotent:

conjecture | open | MathDB 页面 | 原文线索

V2-071820 | MATHDB-371951

Esperet's conjecture on polynomially chi-bounded hereditary classes

摘录 / EXCERPT: Esperet's conjecture. Every chi-bounded hereditary class is poly-chi-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-071821 | MATHDB-371953

Esperet's conjecture on polynomial χ-bounds

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary graph class, and suppose it is χ-bounded: there is a function g such that $\chi(G)\leq g(\omega(G))$ for every $G\in\mathcal{G}$. Esperet's...

conjecture | open | MathDB 页面 | 原文线索

V2-071822 | MATHDB-371954

Polynomial χ-bounds for forest-free graphs

摘录 / EXCERPT: Let H be a forest, and let G be an H-free graph, meaning that G has no induced subgraph isomorphic to H. Write $\chi(G)$ for its chromatic number and $\omega(G)$ for its...

conjecture | open | MathDB 页面 | 原文线索

V2-071823 | MATHDB-371965

Critical-point Schwarzian boundary conjecture for univalent rational functions

摘录 / EXCERPT: Let $\mathcal{U}_R$ be the indicated class of univalent rational functions, let $\mathbb{T}$ be the unit circle, and let S_R denote the Schwarzian derivative of R. Critical-poi...

conjecture | open | MathDB 页面 | 原文线索

V2-071824 | MATHDB-371966

Schwarzian norm conjecture for maximizers of the functional Lambda

摘录 / EXCERPT: Let $\mathcal{S}$ be the class of normalized univalent functions, let $\overline{\mathbf{S}}$ be the indicated closed Schwarzian space, let S_f be the Schwarzian derivative of...

conjecture | open | MathDB 页面 | 原文线索

V2-071825 | MATHDB-371978

Minimal-dimension locus conjecture for type A quiver Grassmannians

摘录 / EXCERPT: Type A minimal-dimension conjecture. There exists a representation M^2 of dimension $\dim P$ such that $\Gr_{\dim P}(N)$, for $N\in{\rm Rep}_{\bf d}$, has dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-071826 | MATHDB-371990

Universal optimality of the logarithmic coordinate for heavy tails

摘录 / EXCERPT: Let $\mathcal{P}$ be a distribution family on $(0,\infty)$ whose members have power-law tails satisfying $P(X>x)\sim x^{-\alpha}$ for some $\alpha>0$. Let the logarithmic coordinat...

conjecture | open | MathDB 页面 | 原文线索

V2-071827 | MATHDB-371991

Adaptive optimality of data-driven concentration coordinates

摘录 / EXCERPT: Let μ be a distribution and let $\mathcal{F}[\mu,\psi]$ denote the relevant concentration objective for a coordinate ψ. Given n samples, let $\hat{\psi}_n$ be a data-dri...

conjecture | open | MathDB 页面 | 原文线索

V2-071828 | MATHDB-371995

6-color conjecture for cubic bipartite planar graph squares

摘录 / EXCERPT: Cubic bipartite planar square-coloring conjecture. The square satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071829 | MATHDB-371998

The bounded-size optimal local realizer conjecture

摘录 / EXCERPT: Let P be a finite poset. A local realizer of P is a family of partial linear extensions whose frequency is the maximum number of partial linear extensions in which any element...

conjecture | open | MathDB 页面 | 原文线索

V2-071830 | MATHDB-372001

Trailing Betti numbers for Sally type semigroups S^e(m)

摘录 / EXCERPT: Fix the multiplicity e and write $\beta_j(m):=\beta_j(S^e(m))$. Trailing Betti-number conjecture. For every $m\geq 2$, the Betti numbers satisfy ... and ... This proposes that the Be...

conjecture | open | MathDB 页面 | 原文线索

V2-071831 | MATHDB-372002

Trailing Betti numbers for S^e(2,n)

摘录 / EXCERPT: Fix the multiplicity e and write $\beta_j(a,b):=\beta_j(S^e(a,b))$. Trailing Betti-number conjecture for S^e(2,n). For $n\geq 6$, ... and ... Moreover, $S^e(2,e-1)$ is conjectured to...

conjecture | open | MathDB 页面 | 原文线索

V2-071832 | MATHDB-372003

Stability of initial Betti numbers for S^e(m,m+1)

摘录 / EXCERPT: Fix the multiplicity e. Betti-number stability conjecture. For every $m\geq 4$, the Betti numbers of $S^e(m,m+1)$ and $S^e(m+1,m+2)$ agree through the indicated range; equivalent...

conjecture | open | MathDB 页面 | 原文线索

V2-071833 | MATHDB-372007

Uniqueness conjecture for isotropic elastic strain fields from three components

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be a compact domain, and let $\epsilon\in C^2(\mathcal{S}^2,\Omega)$ be an elastic strain field in an isotropic medium that satisfies equilibrium an...

conjecture | open | MathDB 页面 | 原文线索

V2-071834 | MATHDB-372010

Fox-Himwich-Zhou TAS-tree conjecture

摘录 / EXCERPT: TAS-tree conjecture. For every undirected tree, there exists some orientation that has the tournament anti-Sidorenko property.

conjecture | open | MathDB 页面 | 原文线索

V2-071835 | MATHDB-372011

Reduced TAS-tree conjecture for trees without isomorphic pairs

摘录 / EXCERPT: Reduced TAS-tree conjecture. For every undirected tree that is not a caterpillar, has at least 2 vertices of even degree, and has no isomorphic pair, there exists some orientatio...

conjecture | open | MathDB 页面 | 原文线索

V2-071836 | MATHDB-372013

Direction-flip conjecture for oriented paths

摘录 / EXCERPT: Direction-flip conjecture. For all $k\geq 3$, an oriented path with k edges and one direction flip ($\rightarrow\dots\rightarrow\leftarrow\dots\leftarrow$) is TAS.

conjecture | open | MathDB 页面 | 原文线索

V2-071837 | MATHDB-372020

Nontriviality conjecture for the ideals associated with frame parameters

摘录 / EXCERPT: Nontriviality conjecture. For every such p, i, and γ,

conjecture | open | MathDB 页面 | 原文线索

V2-071838 | MATHDB-372021

Terminal-leaf conjecture for the frames of maximal-class p-groups

摘录 / EXCERPT: Terminal-leaf conjecture. The leaves of the frame $\mathcal F_i$ are terminal groups in $\mathcal B_i$.

conjecture | open | MathDB 页面 | 原文线索

V2-071839 | MATHDB-372022

Periodicity conjecture for the twigs of frame groups

摘录 / EXCERPT: Twig periodicity conjecture. There exist integers $e=e(p)$ and $f=f(p)$ with $(p-1)\mid f$ such that, for every $i\geq e$ and every $\gamma\in\hat{H}_i$,

conjecture | open | MathDB 页面 | 原文线索

V2-071840 | MATHDB-372023

Log-rigid class values as Gross-Stark logarithms

摘录 / EXCERPT: Let F be a totally real field in which p is inert. Let $\tau in X_p$ be a point attached to an oriented integral ideal as above, let H be the narrow Hilbert class field of F...

conjecture | open | MathDB 页面 | 原文线索

V2-071841 | MATHDB-372033

Categorical non-Abelian Hodge correspondence for moduli stacks of quiver connections

摘录 / EXCERPT: Categorical non-Abelian Hodge conjecture. There are mappings

conjecture | open | MathDB 页面 | 原文线索

V2-071842 | MATHDB-372035

Logunov-Lakshmi Priya-Sartori linear nodal-volume conjecture

摘录 / EXCERPT: Let $B\subset\mathbb{R}^{n}$ be a unit ball with $n\geq 3$. Let $u:4B\to\mathbb{R}$ be harmonic and satisfy $u(0)=0$, and let $N_u(0,\frac{1}{2})$ denote its doubling index at the...

conjecture | open | MathDB 页面 | 原文线索

V2-071843 | MATHDB-372040

Conjecture on the sixth Hankel determinant for convolution powers of Narayana polynomials

摘录 / EXCERPT: Conjecture 7.6. The sixth Hankel determinant satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-071844 | MATHDB-372048

Equality of the Schwartz index and the Euler obstruction

摘录 / EXCERPT: Let $(V,0)\subset(\mathbb{C}^n,0)$ be a complex analytic germ, let W be a local ambient representation, and let $\tilde{\nu}$ be a continuous vector field on W with an isolated...

conjecture | open | MathDB 页面 | 原文线索

V2-071845 | MATHDB-372050

Hollander's conjecture on regular numeration languages and dominant-root limits

摘录 / EXCERPT: Hollander's conjecture. If L is regular, there exists p such that the limit

conjecture | open | MathDB 页面 | 原文线索

V2-071846 | MATHDB-372060

Connamacher and Molloy's satisfiability-threshold conjecture for random uniquely extendable CSPs

摘录 / EXCERPT: Let $k\frac{}{}$ and r be fixed integers with $k\geq 3$ and $r\geq 2$. Let d_k be the constant defined in the random k-XORSAT satisfiability-threshold theorem. Let Φ be...

conjecture | open | MathDB 页面 | 原文线索

V2-071847 | MATHDB-372061

The threshold conjecture for distributions of reducible uniquely extendable constraints

摘录 / EXCERPT: Let $k\geq 3$, and let μ be a distribution over the set $\Lambda_{k,r}$ of k-ary uniquely extendable constraint functions. Suppose that, if $k\geq 4$, every function in...

conjecture | open | MathDB 页面 | 原文线索

V2-071848 | MATHDB-372063

The weakened-support conjecture for random uniquely extendable constraints

摘录 / EXCERPT: Let π be a distribution over $\Lambda_{k,r}$, and let $H_n(\pi,k,m)$ be the associated random uniquely extendable constraint system. Weakened-support conjecture. Theorem should...

conjecture | open | MathDB 页面 | 原文线索

V2-071849 | MATHDB-372064

The replica-symmetry-breaking transition conjecture for random uniquely extendable CSPs

摘录 / EXCERPT: Let π be a distribution over $\Lambda_{k,r}$ and let $H_n(\pi,k,m)$ be the associated random uniquely extendable constraint system. Suppose that either $k\geq 4$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071850 | MATHDB-372066

Lyapunov-exponent upper bound for permanent saturation of metric graphs

摘录 / EXCERPT: Let $G=(V,E,l)$ be a connected metric graph endowed with an interval exchange transformation T partitioning G into n intervals with lengths $\ell_1, \ell_2, \ldots, \ell_n$ l...

conjecture | open | MathDB 页面 | 原文线索

V2-071851 | MATHDB-372081

Soundness and completeness of UCF simplicial models for K45+NU

摘录 / EXCERPT: Let a UCF simplicial model be a simplicial model in the class called UCF in the paper, and let $d53f$ be the logic consisting of propositional logic together with the...

conjecture | open | MathDB 页面 | 原文线索

V2-071852 | MATHDB-372101

The generalized Markoff solution-count conjecture over finite fields

摘录 / EXCERPT: Let $a_{11},a_{22},a_{33},a_{12},a_{13},a_{23},d$ be integers, and consider the generalized Markoff equation ... Let D and A_{11},A_{22},A_{33} denote the quantities defined earl...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071853 | MATHDB-372103

Conjecture on non-existence of holomorphic isomonodromic deformations of non-nilpotent Higgs bundles

摘录 / EXCERPT: Let $\mathcal T_g$ denote the Teichmüller space of compact Riemann surfaces of genus $g\geq2$, let $\mathcal X/\mathcal T_g$ be its universal family, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071854 | MATHDB-372107

The Makar-Limanov invariant stability conjecture for polynomial extensions

摘录 / EXCERPT: Makar-Limanov invariant stability conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071855 | MATHDB-372115

Lehmer-type conjecture for adelic heights

摘录 / EXCERPT: Let K be a number field, let μ be an adelic measure defined over K, and let h_μ be its associated adelic height on $\mathbb{P}^1(\overline{K})$. Lehmer-type conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-071856 | MATHDB-372122

Harpaz's conjecture on the operadic nerve functor

摘录 / EXCERPT: Harpaz's conjecture. The functor N_{Lurie} induces an equivalence of ∞-categories after localizing at weak equivalences.

conjecture | open | MathDB 页面 | 原文线索

V2-071857 | MATHDB-372129

Palindromic decomposition conjecture for symmetric edge-polytope edge deletion

摘录 / EXCERPT: Let G be a 2-connected graph and let ij be any edge of G. Let Γ be a unimodular triangulation of the subcomplex of $P_{G\setminus ij}$ consisting of all facets visible...

conjecture | open | MathDB 页面 | 原文线索

V2-071858 | MATHDB-372130

Exact residual finiteness growth conjecture for torsion-free nilpotent groups

摘录 / EXCERPT: Let G be a torsion-free nilpotent group, and let δ be the constant in the previously established upper bound for the residual finiteness growth RF_G. Exac...

conjecture | open | MathDB 页面 | 原文线索

V2-071859 | MATHDB-372131

The adjacency-preservation conjecture for substitution systems

摘录 / EXCERPT: A substitution system is a tiling system equipped with tile substitutions; it preserves adjacency when, for any two tiles t,t' belonging to distinct supertiles s,s', the tiles...

conjecture | open | MathDB 页面 | 原文线索

V2-071860 | MATHDB-372143

Füredi's linear connected-matching conjecture

摘录 / EXCERPT: Let G be a finite simple graph. A connected matching is a matching $M\subseteq E(G)$ such that for every two edges $e,f\in M$, an endpoint of e is adjacent to an endpoint of...

conjecture | open | MathDB 页面 | 原文线索

V2-071861 | MATHDB-372149

Baruah and Sarma's sign conjecture for coefficients of Rogers-Ramanujan products

摘录 / EXCERPT: Baruah-Sarma sign conjecture. For all integers $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-071862 | MATHDB-372151

The 5-flow reconfiguration conjecture

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{F}(G,k)$ denote its reconfiguration graph whose vertices are nowhere-zero k-flows, with adjacency given by changing flow values only on a cy...

conjecture | open | MathDB 页面 | 原文线索

V2-071863 | MATHDB-372152

Existence of an abelian group guaranteeing flow reconfiguration connectivity

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{F}(G,A)$ denote its reconfiguration graph whose vertices are nowhere-zero A-flows, with adjacency given by changing flow values only on a cy...

conjecture | open | MathDB 页面 | 原文线索

V2-071864 | MATHDB-372153

Composite-flow coverage conjecture for elementary abelian 2-groups

摘录 / EXCERPT: Let G be a graph, and let $\mathcal{F}(G,\mathbb{Z}_2^k)$ denote the reconfiguration graph of nowhere-zero $\mathbb{Z}_2^k$-flows. A nowhere-zero $\mathbb{Z}_2^k$-flow...

conjecture | open | MathDB 页面 | 原文线索

V2-071865 | MATHDB-372158

Painlevé I tau-function connection formula across a B-cycle Stokes jump

摘录 / EXCERPT: Let $(\nu^\pm,\rho^\pm)$ be two parameter pairs, and let $\mathscr T^\pm(t,\nu^\pm,\rho^\pm,\hbar)$ be analytic τ-functions of $(P_{\rm I})$ defined on domains containing...

conjecture | open | MathDB 页面 | 原文线索

V2-071866 | MATHDB-372159

Stokes automorphism conjecture for the Painlevé I perturbative partition function

摘录 / EXCERPT: Let $Z(t,\nu,\hbar)$ be the perturbative partition function for the Painlevé I system, and let $\mathfrak S$ denote the Stokes automorphism associated with the B-periods. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-071867 | MATHDB-372199

The DI/DII longest-element conjecture for type D Weyl groups

摘录 / EXCERPT: Let $W=W_n^+$ be the type D Weyl group, let w_0 be its longest element, and let ... For a strict partition $\lambda=(\lambda_1>\dots>\lambda_m>0)$ and a part a, write...

conjecture | open | MathDB 页面 | 原文线索

V2-071868 | MATHDB-372200

The maximal fixed-point-free involution conjecture for type D Weyl groups

摘录 / EXCERPT: The maximal fixed-point-free involution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071869 | MATHDB-372201

The CII maximal-involution conjecture for type B Weyl groups

摘录 / EXCERPT: The CII maximal-involution conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071870 | MATHDB-372212

Universal bound conjecture for Gaussian entropic optimal transport

摘录 / EXCERPT: Let $\boldsymbol{A},\boldsymbol{B}\rhd 0$ be positive-definite covariance operators on $\mathcal{H}=\mathbb{R}^{d}$, and let π_{ε} and π_{0} denote the entropic a...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-071871 | MATHDB-372213

Superexponential chromatic evaluation spectrum at -1

摘录 / EXCERPT: For each integer $n\geq 1$, let $\mathsf{S}_n(-1)=\{P_G(-1):G\text{ is a simple graph on }n\text{ vertices}\}$. Since $|P_G(-1)|$ counts acyclic orientations of G, the spectrum c...

conjecture | open | MathDB 页面 | 原文线索

V2-071872 | MATHDB-372231

Existence conjecture for link-irregular tournaments

摘录 / EXCERPT: Let a link-irregular tournament be a tournament in which the directed links of every pair of distinct vertices are non-isomorphic. Link-irregular tournament conjecture. A link-irre...

conjecture | open | MathDB 页面 | 原文线索

V2-071873 | MATHDB-372245

Alon-Krivelevich choosability conjecture for bipartite graphs

摘录 / EXCERPT: Alon-Krivelevich conjecture. There is an absolute constant C such that

conjecture | open | MathDB 页面 | 原文线索

V2-071874 | MATHDB-372261

The K-theoretic no-poles conjecture for generalized invariants

摘录 / EXCERPT: Let $\mathcal A$ be the category under consideration, let α be a class and let μ be a stability condition, with ε_α^μ the associated generalized K-t...

conjecture | open | MathDB 页面 | 原文线索

V2-071875 | MATHDB-372263

The Burness-Giudici conjecture on common neighbours in Saxl graphs

摘录 / EXCERPT: Burness-Giudici conjecture. Every two vertices of $\Sigma(G)$ have a common neighbour.

conjecture | open | MathDB 页面 | 原文线索

V2-071876 | MATHDB-372264

Bollobás-Pebody-Riordan conjecture on almost-complete chromatic polynomials

摘录 / EXCERPT: Bollobás-Pebody-Riordan conjecture. For the model with $p(n)=1/2$, the chromatic polynomial is almost complete.

conjecture | open | MathDB 页面 | 原文线索

V2-071877 | MATHDB-372265

Nonexistence of almost-complete Harary polynomials

摘录 / EXCERPT: Harary almost-completeness conjecture. There is no Harary polynomial which is almost complete for $\mathbb{G}(n,1/2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-071878 | MATHDB-372266

Comparability conjecture for Harary polynomials

摘录 / EXCERPT: Harary-polynomial comparability conjecture. Either $\chi_{\mathcal P}$ and $\chi_{\mathcal Q}$ are d.p.-equivalent or they are d.p.-incomparable.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071879 | MATHDB-372271

The wrapped Fukaya category as localization along continuation maps

摘录 / EXCERPT: Let $\mathcal{S}$ be an abstract wrapped Floer setup, let $\mathcal{F}^{\text{pre}}_{\mathcal{S}}$ be its pre-wrapped Fukaya A_∞-category, let $\widetilde{C}$ be the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-071880 | MATHDB-372277

Kashima-Škrekovski-Xu conjecture on bounded degree-plus-k choosability

摘录 / EXCERPT: Kashima-Škrekovski-Xu conjecture. There exists an absolute constant k such that every graph is proper conflict-free $(\mathrm{degree}+k)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-071881 | MATHDB-372278

Kashima-Škrekovski-Xu conjecture on degree-plus-two choosability

摘录 / EXCERPT: Kashima-Škrekovski-Xu conjecture. Every connected graph other than C_5 is proper conflict-free $(\mathrm{degree}+2)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-071882 | MATHDB-372283

Conjecture on the average number of punctures for GRS covering

摘录 / EXCERPT: Puncture-count conjecture. For an $[n,k,d]_q$ GRS code, the average number of punctures needed for $\operatorname{GRS\text{-}cover}$ to succeed in returning a codeword within the c...

conjecture | open | MathDB 页面 | 原文线索

V2-071883 | MATHDB-372285

Cooper-Embree-Hà-Hoefel conjecture on Waldschmidt constants of monomial ideals

摘录 / EXCERPT: Let $I\subset K[x_1,\ldots,x_n]$ be a monomial ideal, and let h be its big height. Let $\alpha(I)$ be the least degree of a nonzero element of I, and let ... be its Waldschmidt c...

conjecture | open | MathDB 页面 | 原文线索

V2-071884 | MATHDB-372298

Bruzzo and Graña Otero conjecture for curve semistable Higgs bundles

摘录 / EXCERPT: Bruzzo and Graña Otero conjecture. The Higgs bundle $\mathfrak{E}$ is semistable with respect to some polarization H and $\Delta(E)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-071885 | MATHDB-372299

Chern-class vanishing conjecture for stable H-nflat Higgs bundles

摘录 / EXCERPT: Chern-class vanishing conjecture. All Chern classes of $\mathfrak{E}$ vanish.

conjecture | open | MathDB 页面 | 原文线索

V2-071886 | MATHDB-372305

The nice-knot characterization conjecture

摘录 / EXCERPT: Let K be a knot. A knot is nice when it admits a braid representative equipped with an inversion datum, as defined in the paper. It is fibered when its complement fibers over the...

conjecture | open | MathDB 页面 | 原文线索

V2-071887 | MATHDB-372306

Stability-series existence for strongly quasinegative knots

摘录 / EXCERPT: Let K be a strongly quasinegative knot, and let the colored Jones function denote the sequence of colored Jones polynomials of K. A stability series is a Laurent power series i...

conjecture | open | MathDB 页面 | 原文线索

V2-071888 | MATHDB-372311

Habiro's MMR expansion conjecture at roots of unity

摘录 / EXCERPT: Let K be a knot, let J_K be its colored Jones function, let ω be a primitive p-th root of unity with $p\geq 1$, and let Λ be the source's ring containing J_K...

conjecture | open | MathDB 页面 | 原文线索

V2-071889 | MATHDB-372317

The (Q_n) conjecture on comonotone jet interpolation

摘录 / EXCERPT: (Q_n) conjecture. (Q_n) is true for all nonnegative integers n.

conjecture | open | MathDB 页面 | 原文线索

V2-071890 | MATHDB-372323

Uniform a priori bounds conjecture for quadratic renormalizations

摘录 / EXCERPT: For $c\in\partial\mathcal{M}$ and $\eta>0$, let $p_c(z)=z^2+c$, and let $\mathcal{R}_\eta p_c$ be the rescaled first-return renormalization defined by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-071891 | MATHDB-372324

Hyperbolicity conjecture for cylinder renormalization

摘录 / EXCERPT: Let $\overline{\Theta}$ be the parameter space of the cylinder renormalization considered in the source, and let $\mathcal{R}_{\operatorname{cyl}}$ denote a cylinder renormalizatio...

conjecture | open | MathDB 页面 | 原文线索

V2-071892 | MATHDB-372325

Hyperbolicity conjecture for Molecule renormalization

摘录 / EXCERPT: Let the sped-up Molecule map be the map associated with the Molecule Renormalization, as in the source's Figure, and let a renormalization be parametrized by this map. Molecule ren...

conjecture | open | MathDB 页面 | 原文线索

V2-071893 | MATHDB-372334

Conjecture on symmetry of complements in the finite k-chain

摘录 / EXCERPT: Symmetry of complements. For all $\lambda\in\mathcal R_k$, $\pi(\lambda)=\pi(\lambda^{\mathrm{comp}})$. The conjecture is supported in the paper by computations for $k\leq 6$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071894 | MATHDB-372335

Conjecture that the finite-chain limit shape is D_{k+1}

摘录 / EXCERPT: Let $\mathcal C_k$ denote the limit shape of the random-core growth process for fixed k. Let D_k be the piecewise-linear curve with vertices ... where γ is the scaling con...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071895 | MATHDB-372337

Altshuller multiplier conjecture for periodic solutions of Lurye systems

摘录 / EXCERPT: Altshuller multiplier conjecture. If such a multiplier exists, then there exists a nonzero periodic solution with period T, which need not be unique or attracting.

conjecture | open | MathDB 页面 | 原文线索

V2-071896 | MATHDB-372338

OZF multiplier conjecture for bounded Lurye-system outputs

摘录 / EXCERPT: OZF multiplier conjecture. If such a multiplier exists, then $y_2\in\mathcal{L}_2\cap\mathcal{L}_{\infty}$.

conjecture | open | MathDB 页面 | 原文线索

V2-071897 | MATHDB-372339

Periodic-solution conjecture under uniform boundedness for Lurye systems

摘录 / EXCERPT: Periodic-solution conjecture. If all solutions are uniform-bounded and uniform ultimate-bounded, then there exists a nonzero periodic solution with period T, which need not be un...

conjecture | open | MathDB 页面 | 原文线索

V2-071898 | MATHDB-372370

Parity conjecture for multiple Hurwitz polylogarithms

摘录 / EXCERPT: Let $r>1$, let $x_1,\ldots,x_r$ be roots of unity, let $a\in\mathbb{C}\setminus\mathbb{Z}$, and let $k_1,\ldots,k_r\geq1$ with $(k_r,x_r)\neq(1,1)$. Parity conjecture. The combinat...

conjecture | open | MathDB 页面 | 原文线索

V2-071899 | MATHDB-372371

Symmetry conjecture for multiple Hurwitz polylogarithms

摘录 / EXCERPT: Let $x_1,\ldots,x_r$ be roots of unity and let $a\in\mathbb{C}\setminus\mathbb{Z}$. For $\boldsymbol{k}=(k_1,\ldots,k_r)\in\mathbb{N}^r$ with $(k_1,x_1)\neq(1,1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-071900 | MATHDB-372372

Stable compressed-rank conjecture for Stallings graphs

摘录 / EXCERPT: Stable compressed-rank conjecture. For every Stallings graph Γ and $d\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-071901 | MATHDB-372381

The Gaussian moments conjecture for Hecke-Maass cusp forms

摘录 / EXCERPT: Let $\mathbb{X}=\operatorname{SL}(2,\mathbb{Z})\backslash\mathbb{H}$, let $\{f_i\}$ be Hecke-Maass cusp forms normalized by ... and define the Gaussian moments by...

conjecture | open | MathDB 页面 | 原文线索

V2-071902 | MATHDB-372382

Hua-Huang-Li's statistical independence conjecture for powers of Hecke-Maass forms

摘录 / EXCERPT: Let $\{f_i\}_{i=1}^I$ be an orthogonal family of Hecke-Maass cusp forms, with $\langle f_i,f_{i'}\rangle=0$ for $i\ne i'$, and let C_n denote the nth moment of a standard real...

conjecture | open | MathDB 页面 | 原文线索

V2-071903 | MATHDB-372383

The joint statistical independence conjecture for Hecke-Maass forms and Eisenstein series

摘录 / EXCERPT: Let $\{f_i\}_{i=1}^I$ be an orthogonal family of Hecke-Maass cusp forms and let $\{\widetilde{E_{T_j}}\}_{j=1}^J$ be an orthogonal family of normalized real Eisenstein series with...

conjecture | open | MathDB 页面 | 原文线索

V2-071904 | MATHDB-372397

Virtual specialness for graphs of virtually special cube complexes with cyclonormal edges

摘录 / EXCERPT: Let X be a compact cube complex that splits as a graph of virtually special cube complexes with cyclonormal edges. Virtual-specialness conjecture. Then X is virtually special....

conjecture | open | MathDB 页面 | 原文线索

V2-071905 | MATHDB-372403

The polynomial-independence conjecture for normalized symmetric-group character values

摘录 / EXCERPT: For each partition $lambda$, let t_lambda denote a formal variable corresponding to the normalized character value $chi(lambda)/chi(1)$, and let T_lambda be the specified ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-071906 | MATHDB-372415

Han's decomposition and temperedness conjectures for Langlands-Shahidi parameters

摘录 / EXCERPT: Let $\varphi\colon W_F\to{{}^LG}({\overline{\mathbb{Q}}_\ell})$ be a semisimple L-parameter of Langlands-Shahidi type, and let ${\mathcal{F}}_\varphi$ be the corresponding Hecke...

conjecture | open | MathDB 页面 | 原文线索

V2-071907 | MATHDB-372416

Failure of duality for power weights over finite residue rings

摘录 / EXCERPT: Power-weight duality conjecture. The power weights $w^{(\ell)}$, $\ell \geq 2$, do not respect duality over any $\mathbb{Z}/m\mathbb{Z}$ with $m \geq 4$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071908 | MATHDB-372417

Wang-Feng-Kovács conjecture on maximum intersections of permutation Hamming balls

摘录 / EXCERPT: Wang-Feng-Kovács conjecture. For any $r\ge 5$ and $n\ge r$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-071909 | MATHDB-372420

Higher-spatial-regularity conjecture for the Augmented formula

摘录 / EXCERPT: Higher-spatial-regularity conjecture. The Augmented formula is well-posed in $\mathscr{C}^k\left([0,T),\mathbb{H}^n\times\mathbb{R}^4\right)$, with the corresponding weighted norm...

conjecture | open | MathDB 页面 | 原文线索

V2-071910 | MATHDB-372432

Nonnegativity conjecture for the q-binomial quotient $G_{k,n}(q)$

摘录 / EXCERPT: For $k,n\in\mathbb{N}_0$, define $G_{k,n}(q)$ by ... Here $\begin{bmatrix}a\\b\end{bmatrix}_q$ denotes the q-binomial coefficient, and $A(q)\succeq0$ means that all coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-071911 | MATHDB-372433

Monotonicity conjecture for consecutive q-binomial quotients

摘录 / EXCERPT: For $k,n\in\mathbb{N}$ with $k\ge n+1$, let $G_{k,n}(q)$ be defined by ... Write $A(q)\succeq0$ when all coefficients of $A(q)$ are nonnegative. Monotonicity conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-071912 | MATHDB-372434

Partial-fraction positivity conjecture for $G_{k,n}(q)$

摘录 / EXCERPT: For $k,n\in\mathbb{N}_0$, let $G_{k,n}(q)$ be defined by ... The polynomials $z_{k,j}(q)$ are to be chosen for $0\le j\le n$. Partial-fraction positivity conjecture for $G_{k,n}(q)$....

conjecture | open | MathDB 页面 | 原文线索

V2-071913 | MATHDB-372436

Subadditivity in t for all-symbol PIR and batch codes

摘录 / EXCERPT: Let $ASP(k,t,q)$ and $ASB(k,t,q)$ denote the minimum lengths of, respectively, all-symbol PIR and all-symbol batch codes with parameters k,t,q. For all k,q and $t_1,t_2\geq 1$,...

conjecture | open | MathDB 页面 | 原文线索

V2-071914 | MATHDB-372438

The rGD wave function conjecture of BCT3

摘录 / EXCERPT: Let $r\geq 2$, let L be the Gelfand-Dikii operator, and let $\Phi(T_*,\lambda)$ solve ... Write $\phi=\log\Phi=\sum_{g\in\mathbb Z}\lambda^{g-1}\phi_g$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071915 | MATHDB-372444

Nonvanishing conjecture for Abel-Jacobi product-cycle defects

摘录 / EXCERPT: Nonvanishing conjecture for Abel-Jacobi product-cycle defects. The class $d6a5_{g,s}$ lies in the Gorenstein kernel and is nonzero for all but finitely many pairs of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-071916 | MATHDB-372445

Generation conjecture for Gorenstein kernels of tautological rings

摘录 / EXCERPT: Generation conjecture for Gorenstein kernels. The following hold: (P1) $d6a5_{g,\mathsf A}\in\mathsf{K}_{g,n}$ for all g and all $s\times n$ matrices $d538$; and (P2) the system...

conjecture | open | MathDB 页面 | 原文线索

V2-071917 | MATHDB-372455

Lam-Postnikov-Pylyavskyy conjecture on Schur positivity

摘录 / EXCERPT: Let λ, μ, λ', and μ' be partitions in $\mathbb{Z}^n$. For vectors $x,y\in\mathbb{Z}^n$, define ... A symmetric polynomial is Schur nonnegative if it is a nonn...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071918 | MATHDB-372460

Flag nested-set complex real-rootedness conjecture

摘录 / EXCERPT: Flag nested-set real-rootedness conjecture. If $\mathbf{N}(\mathcal{L},\mathcal{G})$ is flag, then its h-polynomial is real-rooted.

conjecture | open | MathDB 页面 | 原文线索

V2-071919 | MATHDB-372470

Asymptotic optimality of coset shaping

摘录 / EXCERPT: Let Λ be the shaping lattice, let $G(\Lambda)$ denote its normalized second moment, let R denote the transmission rate, and let C denote the capacity of the additive wh...

conjecture | open | MathDB 页面 | 原文线索

V2-071920 | MATHDB-372473

Kobayashi's standard quotient conjecture for reductive homogeneous spaces

摘录 / EXCERPT: Kobayashi's conjecture. The reductive homogeneous space G/H admits compact quotients if and only if it admits standard ones.

conjecture | open | MathDB 页面 | 原文线索

V2-071921 | MATHDB-372475

The compact quotient-local fibration equivalence conjecture

摘录 / EXCERPT: Compact quotient-local fibration conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-071922 | MATHDB-372477

The compact quotient implies local geometric fibration conjecture

摘录 / EXCERPT: Compact quotient implies local fibration conjecture. If G/H admits compact quotients, then it admits local geometric fibrations.

conjecture | open | MathDB 页面 | 原文线索

V2-071923 | MATHDB-372482

The modulo-10 Collatz conjecture

摘录 / EXCERPT: For a positive integer $n\geq1$, if n is a multiple of 10, replace it by $n/10$; otherwise, let r be its last digit, replace it by $(12n+8r)/10$, and iterate this process. Mo...

conjecture | open | MathDB 页面 | 原文线索

V2-071924 | MATHDB-372483

The diagonal generalized Collatz conjecture

摘录 / EXCERPT: For $p\geq0$, let $(d_p,\alpha_p,\beta_p)_+=(2^{p+1},3\cdot2^p,2^p)_+$ and let T_p be its associated mapping. Diagonal generalized Collatz conjecture. If $p\geq0$ and $p\neq2$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-071925 | MATHDB-372485

The proposed formula for the three-deletion sequence reconstruction number

摘录 / EXCERPT: Let $N(n,3,t)$ denote the sequence reconstruction number for length-n sequences over a binary deletion channel with three deletions and trace length parameter t, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071926 | MATHDB-372486

Nian-Tsujie-Uchiumi-Yoshinaga limit conjecture for q-deformed graphic arrangements

摘录 / EXCERPT: Let $G=([[200~\ell], E)$ be a simple graph on ℓ vertices, and let $\mathcal{A}_G^q$ be its q-deformation of graphic arrangement. Write $\chi(\mathcal{A}_G^q,t)$ for its char...

conjecture | open | MathDB 页面 | 原文线索

V2-071927 | MATHDB-372487

Limit conjecture for q-deformations of simplicial-complex arrangements

摘录 / EXCERPT: Let $\Delta=([\ell],\mathcal{F})$ be a simplicial complex on ℓ vertices, and let $\mathcal{S}_\Delta^q$ be its generalized q-deformation. Let G be the underlying graph of...

conjecture | open | MathDB 页面 | 原文线索

V2-071928 | MATHDB-372490

The absorption-time and Brownian-bridge maximum distribution conjecture

摘录 / EXCERPT: For each $N\geq 1$, let T_{2N} and T_{2N+1} be the absorption times in the discrete Whittaker process. Let M_N be the maximum height of N non-intersecting reflected Brownia...

conjecture | open | MathDB 页面 | 原文线索

V2-071929 | MATHDB-372492

Bardakov-Etingof-Sosnovsky conjecture on idempotents in integral latin quandle rings

摘录 / EXCERPT: Let X be a finite latin quandle, and let $u\in\mathbb{Z}[X]$ be a nonzero idempotent, so $u^2=u$. An element of X is called a trivial idempotent, and ε denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-071930 | MATHDB-372493

Augmentation conjecture for odd powers of the fundamental ideal in dihedral quandle rings

摘录 / EXCERPT: Let R_{2n+1} denote the dihedral quandle of odd order $2n+1$, let $\Delta(R_{2n+1})$ be its fundamental ideal, and let $\Delta^{2n+1}(R_{2n+1})$ be its $(2n+1)$-st power. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-071931 | MATHDB-372501

Optimal augmentation size for BONuS under a point-mass prior

摘录 / EXCERPT: Let $\tilde m_{\textnormal{opt}}$ be the choice of augmentation size that maximizes the power of BONuS for a point-mass prior, with m fixed. Optimal augmentation-size conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-071932 | MATHDB-372502

Optimal augmentation size for BONuS under a subspace prior

摘录 / EXCERPT: Optimal augmentation-size conjecture. Under this regime,

conjecture | open | MathDB 页面 | 原文线索

V2-071933 | MATHDB-372503

Completeness of the rational-point list on the obstruction curve

摘录 / EXCERPT: Let $F(s,a)$ be the obstruction polynomial, let $\widehat{F}(S,A,Z)$ be its homogenization, and let ... be the projective closure. The computation lists the 8 rational points ... Com...

conjecture | open | MathDB 页面 | 原文线索

V2-071934 | MATHDB-372508

The scattering-transform conjecture for noncommutative instantons

摘录 / EXCERPT: Let $\widetilde{M}(r)$ be the rank-r moduli space, with matrices (B_1,B_2,I,J), and let $\widetilde{M}(r)^{\mathbb{C}^\times}$ be the fixed locus of the $\mathbb{C}^\times$-act...

conjecture | open | MathDB 页面 | 原文线索

V2-071935 | MATHDB-372522

The Ramanujan Machine continued fraction conjecture for $-\pi/4$

摘录 / EXCERPT: Ramanujan Machine conjecture. The continued fraction equals

conjecture | open | MathDB 页面 | 原文线索

V2-071936 | MATHDB-372524

Existence conjecture for plane curves with prescribed graded Betti partition

摘录 / EXCERPT: Let $C:f=0$ be a reduced plane curve that is not free, with $t(C)=\sum_{j=1}^{m-2}\epsilon_j\geq 1$, and let π_C be the ordered partition of $t(C)$ whose parts are the positive...

conjecture | open | MathDB 页面 | 原文线索

V2-071937 | MATHDB-372538

Geometric Manin conjecture on the nef section cone

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a good Fano fibration. An intersection profile λ specifies one generically smooth, geometrically integral irreducible component in each fiber,...

conjecture | open | MathDB 页面 | 原文线索

V2-071938 | MATHDB-372539

Thinness conjecture for the exceptional set

摘录 / EXCERPT: Let $\pi:\mathcal{X}\to B$ be a good Fano fibration, let $\mathcal{X}_{\eta}$ be its generic fiber, and let $\mathsf Z$ be the union of the images of all breaking thin maps into...

conjecture | open | MathDB 页面 | 原文线索

V2-071939 | MATHDB-372541

Proper constructibility conjecture for the exceptional set

摘录 / EXCERPT: Let $\pi:\mathcal X\to B$ be a good Fano fibration and let $\mathsf Z\subset\mathcal X_{\eta}(K(B))$ be the exceptional set. A properly constructible thin set is a finite union of...

conjecture | open | MathDB 页面 | 原文线索

V2-071940 | MATHDB-372542

All-height version of Manin's conjecture over global function fields

摘录 / EXCERPT: Let $\pi:\mathcal X\to B$ be a good Fano fibration, let λ be an intersection profile, let M_{α} be the unique Manin component for sufficiently large...

conjecture | open | MathDB 页面 | 原文线索

V2-071941 | MATHDB-372543

Conjectural leading constant in the all-height Manin conjecture

摘录 / EXCERPT: Let $\pi:\mathcal X\to B$ be a good Fano fibration, let λ be an intersection profile, and let $c(\mathcal X,\lambda)$ be the leading constant in the all-height Manin conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-071942 | MATHDB-372544

Uniform boundedness conjecture for Manin components

摘录 / EXCERPT: Let $\pi:\mathcal X\to B$ be a good Fano fibration, let λ be an intersection profile, and let $\mathsf{S}_{\lambda,\mathbb Z}$ be the corresponding integral section classes...

conjecture | open | MathDB 页面 | 原文线索

V2-071943 | MATHDB-372548

Elder's conjecture on one-element commutation classes

摘录 / EXCERPT: Let $n\geq 1$ and let $\sigma\in\mathfrak{S}_{n+1}$. Write $R(\sigma)$ for the set of reduced words of σ and $C(\sigma)$ for the set of commutation classes of reduced words....

conjecture | open | MathDB 页面 | 原文线索

V2-071944 | MATHDB-372550

Necessary and sufficient conditions for scatteredness of the generalized quadrinomial

摘录 / EXCERPT: Scatteredness conjecture. The polynomial $\psi_{m,h,s}$ is scattered if and only if m and h satisfy the assumptions in Theorem.

open_problem | open | MathDB 页面 | 原文线索

V2-071945 | MATHDB-372559

Kährström's equivalence conjecture for Kostant-negative involutions

摘录 / EXCERPT: For $n\geq 0$, let S_n be the symmetric group, let Inv_n be its set of involutions, and let $\mathbf{K}_n(w)=-$ mean that Kostant's problem for the simple highest weig...

conjecture | open | MathDB 页面 | 原文线索

V2-071946 | MATHDB-372571

Cell-complex realization conjecture for spheres with $g_k=1$

摘录 / EXCERPT: Let $k\geq 2$, $d\geq 2k$, and let Δ be a $(d-1)$-sphere with $g_k(\Delta)=1$. Cell-complex realization conjecture. There exists a d-dimensional cell complex $\mathcal{C}$...

conjecture | open | MathDB 页面 | 原文线索

V2-071947 | MATHDB-372575

Green-Griffiths implication for the second Chow-group filtration over number fields

摘录 / EXCERPT: Green-Griffiths implication. The induced second filtration step vanishes:

conjecture | open | MathDB 页面 | 原文线索

V2-071948 | MATHDB-372585

Le's conjecture for symmetric hyperbolic polynomials

摘录 / EXCERPT: Le's conjecture. Then $\operatorname{diag}(A)\in\Gamma(P)$ and

conjecture | open | MathDB 页面 | 原文线索

V2-071949 | MATHDB-372588

The Erdős-Rényi random graph maximum likelihood threshold conjecture

摘录 / EXCERPT: Let G be a random graph generated according to the Erdős-Rényi model $G(n,c/n)$, where c is a fixed positive real number, and let d be the minimum integer such that G is...

conjecture | open | MathDB 页面 | 原文线索

V2-071950 | MATHDB-372593

Positivity conjecture for the plane-partition D'Arcais differences

摘录 / EXCERPT: Let $\sigma_2(n):=\sum_{d\mid n}d^2$, and let $\Delta_n^{\sigma_2}(x)$ denote the difference associated with the D'Arcais polynomials for $g(n)=\sigma_2(n)$. Plane-partition positi...

conjecture | open | MathDB 页面 | 原文线索

V2-071951 | MATHDB-372594

D'Arcais polynomial positivity conjecture

摘录 / EXCERPT: D'Arcais positivity conjecture. For every integer $n\geq 6$ and every real $x\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-071952 | MATHDB-372597

The conjecture that the degree of an APN mapping is less than the dimension

摘录 / EXCERPT: Degree bound conjecture. The degree of an APN mapping is less than m.

conjecture | open | MathDB 页面 | 原文线索

V2-071953 | MATHDB-372606

Clemens's conjecture for rational curves on the generic quintic threefold

摘录 / EXCERPT: Let $X\subset\mathbb{P}^4$ be a generic quintic threefold. A rational curve means a curve on X birational to $\mathbb{P}^1$, and its degree is measured with respect to the embedd...

conjecture | open | MathDB 页面 | 原文线索

V2-071954 | MATHDB-372607

Relative Clemens conjecture for rational curves on cubic threefolds

摘录 / EXCERPT: Let $X\subset\mathbb{P}^4$ be a cubic threefold, let $Y\subset X$ be a hyperplane, and let $C\subset X$ be a smooth rational curve of degree d, with X, Y, and C all general...

conjecture | open | MathDB 页面 | 原文线索

V2-071955 | MATHDB-372611

Butler's conjecture for generated coherent systems

摘录 / EXCERPT: Let (E,V) be a generated coherent system of type (n,d,v) on a general curve X of genus $g\ge3$, where n,d,v are positive integers and $v>n$. Write $D_{E,V}$ for its dual sp...

conjecture | open | MathDB 页面 | 原文线索

V2-071956 | MATHDB-372612

Continued fraction conjecture for $-\pi/4$

摘录 / EXCERPT: A continued fraction has partial numerators a_n and partial denominators b_n, with the displayed initial terms determining the pattern. Continued fraction conjecture for...

conjecture | open | MathDB 页面 | 原文线索

V2-071957 | MATHDB-372621

Conjecture on the initial ideal associated with an A_{r-1} surface singularity

摘录 / EXCERPT: Let $r\in\mathbf{Z}_{\geq 2}$, let $\mathcal{S}$ be the ring in which the ideal $\mathfrak{a}$ is defined, and let J_r be the ideal generated by the monomials described above: th...

conjecture | open | MathDB 页面 | 原文线索

V2-071958 | MATHDB-372627

The exact L(3,2,1)-labeling number of a 4-valent circulant with steps 1 and 3

摘录 / EXCERPT: Exact labeling-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071959 | MATHDB-372628

The exact L(3,2,1)-labeling number of odd 4-valent circulants with steps 1 and 5

摘录 / EXCERPT: Exact labeling-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-071960 | MATHDB-372639

Normal-form classification conjecture for integrable RH conservation laws

摘录 / EXCERPT: Consider an integrable hierarchy of scalar conservation laws ... which is a perturbation of the Riemann--Hopf hierarchy, and transform it to the Arsie--Lorenzoni--Moro normal form ……

conjecture | open | MathDB 页面 | 原文线索

V2-071961 | MATHDB-372641

Camassa--Holm realization conjecture for the local DR hierarchy

摘录 / EXCERPT: Let $a_2\in\mathbb{C}$ and set ... Consider the corresponding local DR hierarchy and its particular flow. Camassa--Holm realization conjecture. There is a unique choice of...

conjecture | open | MathDB 页面 | 原文线索

V2-071962 | MATHDB-372660

Modal logics of Minkowski spacetime

摘录 / EXCERPT: For each n, let $\mathcal{F}^n = \langle \mathbb{R}^{1+n}, \mathrel{\scalebox{0.85}{$oldalpha$}} \rangle$ be the frame of $(1+n)$-dimensional Minkowski spacetime with causal rela...

conjecture | open | MathDB 页面 | 原文线索

V2-071963 | MATHDB-372665

The conformal welding injectivity conjecture

摘录 / EXCERPT: Let a Jordan curve have a conformal welding homeomorphism, and let a curve be conformally removable when every homeomorphism of the Riemann sphere conformal off it is a Möbius tran...

conjecture | open | MathDB 页面 | 原文线索

V2-071964 | MATHDB-372671

Refined Kurihara conjecture for modular-symbol invariants

摘录 / EXCERPT: Let $\boldsymbol{\delta}=\{\delta_n\}_{n\in\mathcal{N}}$, and let $\mathscr{M}_\infty(\boldsymbol{\delta})$ be the limiting divisibility invariant defined from these quantities. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-071965 | MATHDB-372672

Zhang's refined Kolyvagin conjecture for Heegner-point classes

摘录 / EXCERPT: Let K be the imaginary quadratic field in the setup, let $\boldsymbol{\kappa}^{\rm Heeg}$ be the Heegner-point Kolyvagin system, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-071966 | MATHDB-372690

Kitaoka's finiteness conjecture for universal ternary quadratic forms

摘录 / EXCERPT: A universal ternary classical quadratic form over a totally real number field is a positive definite classical quadratic form in three variables that represents every totally posit...

conjecture | open | MathDB 页面 | 原文线索

V2-071967 | MATHDB-372700

The maximal-rank conjecture for elementary symmetric polynomials

摘录 / EXCERPT: Maximal-rank conjecture for e_d. The element e_d is a maximal rank element on A if and only if either $d=2$, or $d=3$ and $n\not\equiv 2\pmod{3}$.

conjecture | open | MathDB 页面 | 原文线索

V2-071968 | MATHDB-372701

The maximal-rank conjecture for complete homogeneous symmetric polynomials

摘录 / EXCERPT: Maximal-rank conjecture for h_d. The element h_d is a maximal rank element on A if and only if either $d=2$, $d=3$ and $n\not\equiv 1\pmod{3}$, or $d=4$ and $n=3$.

conjecture | open | MathDB 页面 | 原文线索

V2-071969 | MATHDB-372702

The three-variable asymptotic maximal-rank conjecture for Schur polynomials

摘录 / EXCERPT: Three-variable Schur maximal-rank conjecture. For all sufficiently large d, the following holds: if $\lambda\in F$, then s_λ fails to be a maximal rank element on A; if...

conjecture | open | MathDB 页面 | 原文线索

V2-071970 | MATHDB-372703

The Betti-number conjecture for elementary and complete homogeneous symmetric cubics

摘录 / EXCERPT: Betti-number conjecture. The ideals I_h and I_e have the same Betti numbers if and only if $n=3a$ for some integer a.

conjecture | open | MathDB 页面 | 原文线索

V2-071971 | MATHDB-372704

The non-Borel-Cantelli sequence conjecture for drawable graphs

摘录 / EXCERPT: Non-Borel-Cantelli sequence conjecture. If $\textbf{p}\in(0,1)^\omega$ is not Borel-Cantelli, then

conjecture | open | MathDB 页面 | 原文线索

V2-071972 | MATHDB-372705

The invariant-set conjecture for random graph edge perturbations

摘录 / EXCERPT: Invariant-set conjecture. For any Borel-Cantelli sequence $\textbf{p}$ and $\textbf{p}$-random graph $G=(\omega,E_G)$, there exists an infinite set $A\subseteq[\omega]^2$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-071973 | MATHDB-372707

Dependent internal tree-ordering expansions and bounded twin-width

摘录 / EXCERPT: Let $\mathscr C$ be a hereditary weakly sparse class of graphs. An internal tree-ordering expansion of a graph is an expansion by the forest-ordering defined by a spanning forest;...

conjecture | open | MathDB 页面 | 原文线索

V2-071974 | MATHDB-372725

Kenyon-Okounkov conjecture for height fluctuations in random tilings

摘录 / EXCERPT: Let $\mathsf{Ht}_\mathcal N$ be the random height function of the model at scale $\mathcal N$, let $\mathbb E[\mathsf{Ht}_\mathcal N]$ be its expectation, let $\pounds$ denote the...

conjecture | open | MathDB 页面 | 原文线索

V2-071975 | MATHDB-372728

Spectral interpretation of the deformation parameter

摘录 / EXCERPT: Let $(A,\phi,\psi)\in\mathcal{M}_\alpha^G$ be a solution and let $b=\mathrm{Hit}_\alpha^G(A,\phi,\psi)$. Denote by Σ_b the associated spectral curve and by L the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-071976 | MATHDB-372736

Virtual retract conjecture for iterated monodromy groups of post-critically finite polynomials

摘录 / EXCERPT: Let $f\in \mathbb{C}[x]$ be a post-critically finite rational function. Write $\mathrm{IMG}(f)$ for its iterated monodromy group, and say that it has (LR) when it has th...

conjecture | open | MathDB 页面 | 原文线索

V2-071977 | MATHDB-372740

Sharp maximum conjecture for one-phase cones

摘录 / EXCERPT: Let $\Lambda_n:= \sup_{U \in \mathcal{S}\mathcal{C}(\mathbb{R}^n)} \lambda_1$ be the supremum of the first stability eigenvalue over singular minimizing cones in $\mathbb{R}^n$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-071978 | MATHDB-372743

Height bound conjecture for modular polynomials with level structures

摘录 / EXCERPT: Let N be the level parameter, let α be the invariant defining the modular polynomials with level structures, and let J_1 be the corresponding polynomial. For a polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-071979 | MATHDB-372744

Permutation conjecture for modular-polynomial evaluations outside the invariant image

摘录 / EXCERPT: Let N be the level parameter, let α be the invariant, let $\operatorname{image}\alpha$ denote its image, let Φ_m^{α} be the modular polynomial with level structu...

conjecture | open | MathDB 页面 | 原文线索

V2-071980 | MATHDB-372745

The convergence conjecture for the triplet $(10,12,8)_+$

摘录 / EXCERPT: Convergence conjecture for $(10,12,8)_+$. Repeating this process indefinitely, every starting number eventually reaches 4 after finitely many iterations.

conjecture | open | MathDB 页面 | 原文线索

V2-071981 | MATHDB-372757

The relative Langlands dual and derived Satake equivalence for symmetric spaces

摘录 / EXCERPT: Let G be a reductive group with involution θ, let $X_\theta=G^{\operatorname{inv}\circ\theta}=\{g\in G\mid\theta(g)=g^{-1}\}$ with its θ-twisted conjugation action,...

conjecture | open | MathDB 页面 | 原文线索

V2-071982 | MATHDB-372764

Perfect-power conjecture for the plane partition function

摘录 / EXCERPT: Let $pl(n)$ denote the plane partition function, counting plane partitions of n. A perfect power is an integer of the form a^b with integers $a,b>1$. The plane-partition perfec...

conjecture | open | MathDB 页面 | 原文线索

V2-071983 | MATHDB-372766

Stabilization conjecture for near-perfect powers of colored partition numbers

摘录 / EXCERPT: Let $p_r(n)$ denote the r-colored partition function, and define $L_{p_r}(d)$ and $M_{p_r,k}(d)$ analogously to the corresponding quantities for the partition and overpartition f...

conjecture | open | MathDB 页面 | 原文线索

V2-071984 | MATHDB-372776

The bounded clique-width conjecture for 2-well-quasi-ordered hereditary graph classes

摘录 / EXCERPT: Let $\mathcal{C}$ be a hereditary class of finite graphs. The bounded clique-width conjecture. If $\mathcal{C}$ is 2-well-quasi-ordered, then $\mathcal{C}$ has bounded clique-wid...

conjecture | open | MathDB 页面 | 原文线索

V2-071985 | MATHDB-372778

Roberson's weak conjecture for bounded Hadwiger number

摘录 / EXCERPT: For a graph G, its Hadwiger number is the largest integer t such that G contains the complete graph K_t as a minor. Two graphs are homomorphism indistinguishable over a gra...

conjecture | open | MathDB 页面 | 原文线索

V2-071986 | MATHDB-372783

The graph-product conjecture for linear sofic groups

摘录 / EXCERPT: The graph-product conjecture. The free product $\ast_{v\in V}\Gamma_v$ is $\mathbb{K}$-linear sofic, and, more generally, G_X is $\mathbb{K}$-linear sofic for every graph...

conjecture | open | MathDB 页面 | 原文线索

V2-071987 | MATHDB-372784

The graph-product conjecture for weakly sofic groups

摘录 / EXCERPT: The graph-product conjecture. The free product $\ast_{v\in V}\Gamma_v$ is weakly sofic, and, more generally, G_X is weakly sofic for every graph $X=(V,E)$ on V.

conjecture | open | MathDB 页面 | 原文线索

V2-071988 | MATHDB-372785

The LEF conjecture for nilpotent products

摘录 / EXCERPT: The LEF nilpotent-product conjecture. For each $k\geq 1$, the k-nilpotent product of a finite family of LEF groups is LEF.

conjecture | open | MathDB 页面 | 原文线索

V2-071989 | MATHDB-372788

Barnett-Betcke MFS convergence-rate conjecture for Helmholtz problems

摘录 / EXCERPT: Let Ω be an exterior sound-soft Helmholtz domain with boundary $\Gamma=\partial\Omega$, and let u be the scattered-field solution represented by the method of fundamental...

conjecture | open | MathDB 页面 | 原文线索

V2-071990 | MATHDB-372792

The biquadratic SOS rank conjecture for 3 by 3 forms

摘录 / EXCERPT: Let a biquadratic form be a homogeneous polynomial of degree two in each of two variable vectors, and let its SOS rank be the smallest number of squares of bilinear forms in a sum-...

conjecture | open | MathDB 页面 | 原文线索

V2-071991 | MATHDB-372794

The nonnegative-coefficient conjecture for the polynomial h(t,a;r)

摘录 / EXCERPT: Nonnegative-coefficient conjecture. If $r\in\mathbb N$ is arbitrary but fixed, then $h(t,a;r)$, viewed as a polynomial in (t,a), has nonnegative coefficients. This would consider...

conjecture | open | MathDB 页面 | 原文线索

V2-071992 | MATHDB-372799

The Asymptotic Shape Conjecture for Kostant-positive permutations

摘录 / EXCERPT: Let $\lambda=(\lambda_1,\ldots,\lambda_r)\in\Lambda_m$ be a Young diagram with λ_1 boxes in its first row, and let $\lambda^{(n)}$ be obtained by adding a first row of n...

conjecture | open | MathDB 页面 | 原文线索

V2-071993 | MATHDB-372802

Finite-time collapse criterion for generalized sticky particle solutions

摘录 / EXCERPT: Let (ρ,v) be a GSPS to the pressureless Euler-Poisson equation, with ρ_0 compactly supported, center of mass 0, and total momentum 0. Let y_L, y_R, and f be the...

conjecture | open | MathDB 页面 | 原文线索

V2-071994 | MATHDB-372812

Andrews-El Bachraoui positivity conjecture for the series $F_{k,1}(q)$

摘录 / EXCERPT: Andrews-El Bachraoui positivity conjecture. For every positive integer k, the series $F_{k,1}(q)$ is positive.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-071995 | MATHDB-372822

Optimal condition-number threshold conjecture for low-rank synchronization factorization

摘录 / EXCERPT: Optimal condition-number threshold conjecture. The optimization landscape of the low-rank factorization problem is benign if

conjecture | open | MathDB 页面 | 原文线索

V2-071996 | MATHDB-372824

Boston's virtually Golod-Shafarevich conjecture

摘录 / EXCERPT: Boston's virtually Golod-Shafarevich conjecture. The group $\operatorname{Gal}(L/K)$ contains an open pro-p subgroup G such that $d(G)$ and $r(G)$ are finite and

conjecture | open | MathDB 页面 | 原文线索

V2-071997 | MATHDB-372825

The open Golod-Shafarevich subgroup conjecture for pro-p extensions

摘录 / EXCERPT: Open Golod-Shafarevich subgroup conjecture. The group $\operatorname{Gal}(L/K)$ contains an open Golod-Shafarevich pro-p subgroup.

conjecture | open | MathDB 页面 | 原文线索

V2-071998 | MATHDB-372831

Universal local-limit conjecture for mandatory and blocking edges

摘录 / EXCERPT: Universal mandatory and blocking edge conjecture. The marked random graphs

conjecture | open | MathDB 页面 | 原文线索

V2-071999 | MATHDB-372857

The conjectured optimal bound for the logarithmic interpolation error

摘录 / EXCERPT: The conjectured optimal logarithmic interpolation bound. For all $(x,\alpha)\in\widehat{D}$,

conjecture | open | MathDB 页面 | 原文线索

V2-072000 | MATHDB-372858

The conjectured extension of the Burnside-Stirling-Gautschi bound

摘录 / EXCERPT: Let $\widehat{m}_{1/2}(x,\alpha)$ denote the logarithmic error term for the shifted interpolation parameter α, let $\widehat{D}_{1/2}$ be its stated domain, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072001 | MATHDB-372860

Deligne's period conjecture for motives of odd weight

摘录 / EXCERPT: Let M be a $\mathbb{Q}$-motive with coefficients in a number field F and odd weight. Let $H_B^+(M)$ be the $+1$-eigenspace of complex conjugation, let α_M^+ be the compa...

conjecture | open | MathDB 页面 | 原文线索

V2-072002 | MATHDB-372862

The SHGH-type conjecture for effective divisors on blow-ups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb F_{e,r}$ be the blow-up of the Hirzebruch surface $\mathbb F_e$ at r very general points, and let C_e be the negative section of $\mathbb F_e$ with strict transfor...

conjecture | open | MathDB 页面 | 原文线索

V2-072003 | MATHDB-372863

Laface's special-divisor conjecture for blow-ups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb F_{e,r}$ be the blow-up of the Hirzebruch surface $\mathbb F_e$ at r very general points. Let D be a divisor satisfying the paper's condition; assume D is effect...

conjecture | open | MathDB 页面 | 原文线索

V2-072004 | MATHDB-372864

The vanishing conjecture for integral curves on blow-ups of Hirzebruch surfaces

摘录 / EXCERPT: Let $\mathbb F_{e,r}$ be the blow-up of the Hirzebruch surface $\mathbb F_e$ at r very general points, and let $\widetilde{C}_e$ be the strict transform of the negative section...

conjecture | open | MathDB 页面 | 原文线索

V2-072005 | MATHDB-372883

Non-well-quasi-ordering of k-connected bipartite graphs by bipartite minors

摘录 / EXCERPT: Let k be a positive integer, and consider the class of k-connected bipartite graphs with the bipartite minor relation. Bipartite-minor non-well-quasi-order conjecture. There ex...

conjecture | open | MathDB 页面 | 原文线索

V2-072006 | MATHDB-372884

Shareshian-Wachs conjecture for chromatic quasisymmetric functions

摘录 / EXCERPT: Let $G=(V,E)$ be a unit interval graph, with its vertices in the natural order. Define its chromatic quasisymmetric function by ... where the sum is over proper colorings and...

conjecture | open | MathDB 页面 | 原文线索

V2-072007 | MATHDB-372885

Tree isomorphism conjecture for chromatic symmetric functions

摘录 / EXCERPT: Let G and H be trees, and let $\chi_G(x)$ denote the chromatic symmetric function of G. Tree isomorphism conjecture. If G and H are non-isomorphic trees, then ... The conje...

conjecture | open | MathDB 页面 | 原文线索

V2-072008 | MATHDB-372886

Aval's oriented tree isomorphism conjecture

摘录 / EXCERPT: Let G and H be trees equipped with orientations γ. For an orientation γ, let $\chi_G^\gamma(x;q)$ denote the oriented chromatic quasisymmetric function obtained b...

conjecture | open | MathDB 页面 | 原文线索

V2-072009 | MATHDB-372888

Equality of the sufficient and admissible parameter sets for conditional entropies

摘录 / EXCERPT: Let $\mathcal{D}_{\rm bulk}$ and $\mathcal{D}_{-\infty}$ denote the parameter sets for which the corresponding conditional entropies are monotone under conditionally mixing channel...

conjecture | open | MathDB 页面 | 原文线索

V2-072010 | MATHDB-372892

The even-multiplicity descent conjecture for decomposition lengths

摘录 / EXCERPT: Let S be a numerical semigroup of multiplicity $m=2k$, where $k\in\mathbb{Z}$. A decomposition of S is an irredundant intersection of irreducible numerical semigroups, and its...

conjecture | open | MathDB 页面 | 原文线索

V2-072011 | MATHDB-372893

Equivalence of local balance conditions for Speiser graphs and planar bipartite maps

摘录 / EXCERPT: The discussion concerns three local balance conditions: the condition for Speiser graphs in Theorem, the condition in section S, and the condition in section S. Equivalence conject...

conjecture | open | MathDB 页面 | 原文线索

V2-072012 | MATHDB-372898

The 4D Barth conjecture for framed sheaf and ADHM moduli spaces

摘录 / EXCERPT: Let $(\vec{r},n)$ be the discrete parameters for the 4D ADHM quiver, and let $M_{Q_4}(\vec{r},n)$ be its stable representation moduli space. Let $M_{(\mathbb P^1)^4}(\vec{r},n)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-072013 | MATHDB-372899

Stable/co-stable wall-crossing conjecture for 4D ADHM invariants

摘录 / EXCERPT: Let $M_{Q_4}(\vec{r},n)$ be the moduli space defined using the stable condition, and let $M_{Q_4}^c(\vec{r},n)$ be the moduli space defined using co-stability, namely the condition...

conjecture | open | MathDB 页面 | 原文线索

V2-072014 | MATHDB-372905

Sharp three-operator Clarkson-McCarthy inequality

摘录 / EXCERPT: Let S_p denote the Schatten p-class, and let $A,B,C\in S_p$. For $p>0$, write $\|\cdot\|_p$ for the Schatten p-quasi-norm. Sharp three-operator Clarkson-McCarthy conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072015 | MATHDB-372911

Milnor number-modality conjecture for isolated singularities in positive characteristic

摘录 / EXCERPT: Let K be a field of characteristic $p>0$, and let $f_k\in K\{x_1,\ldots,x_n\}$ be a sequence of isolated singularities whose Milnor numbers tend to infinity as $k\to\infty$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-072016 | MATHDB-372914

Cartesianity of exponentiation in infinity-categories

摘录 / EXCERPT: Cartesianity conjecture. If C is an infinity-category, then

conjecture | open | MathDB 页面 | 原文线索

V2-072017 | MATHDB-372915

Sup-norm conjecture for simultaneous eigenvectors of supersingular isogeny graphs

摘录 / EXCERPT: Let A_ℓ denote the weighted adjacency matrix of the supersingular ℓ-isogeny graph $\mathcal{G}(p,\ell)$ for all $\ell\neq p$. Let v be an L^2-normalized simultaneous...

conjecture | open | MathDB 页面 | 原文线索

V2-072018 | MATHDB-372923

Threshold transition conjecture for cubic focusing nonlinear waves

摘录 / EXCERPT: Consider the focusing nonlinear wave equation $P[\phi]=0$ with $p=3$ in spatial dimension $n\geq 5$. Let $(\phi_{0,\omega},\phi_{1,\omega})$, $\omega\in[0,1]$, be a one-parameter f...

conjecture | open | MathDB 页面 | 原文线索

V2-072019 | MATHDB-372924

Hickingbotham's conjecture on low-connectivity ghost-edges

摘录 / EXCERPT: Let $k\in\mathbb{N}$, let G be a connected graph with $tw(G)\leq k$, and let $xy\in E(G^c)$. A k-ghost-edge is a nonedge xy such that every tree decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-072020 | MATHDB-372926

Joyce's stability and special Lagrangian conjecture

摘录 / EXCERPT: Joyce's conjecture. There exists a Bridgeland stability condition $\sigma=(Z,\mathcal{P})$ on $\operatorname{DFuk}(M)$ such that the central charge is induced by

conjecture | open | MathDB 页面 | 原文线索

V2-072021 | MATHDB-372927

Finite-index finite-ends conjecture for Allen-Cahn solutions

摘录 / EXCERPT: Finite index implies finite ends. For $n\geq 3$, every finite Morse index solution has finitely many ends.

conjecture | open | MathDB 页面 | 原文线索

V2-072022 | MATHDB-372928

Del Pino-Kowalczyk-Wei conjecture on finite-index Allen-Cahn ends

摘录 / EXCERPT: Del Pino-Kowalczyk-Wei conjecture. Outside a large ball, each level set of u has finitely many components, each asymptotic either to a plane or to a catenoid; after a rotation of...

conjecture | open | MathDB 页面 | 原文线索

V2-072023 | MATHDB-372929

Finite-index De Giorgi conjecture in dimensions four through seven

摘录 / EXCERPT: Finite index De Giorgi conjecture. If $4\leq n\leq 7$, then u is one-dimensional.

conjecture | open | MathDB 页面 | 原文线索

V2-072024 | MATHDB-372937

The geometric string structure conjecture for Chern-Simons enhancements

摘录 / EXCERPT: Let K be a Lie group and let $P_K:M\to\mathrm{B}K$ classify a principal K-bundle. Let $\mathrm{B}^2\mathrm{U}(1)\to\mathrm{String}(K)\to K$ be the string extension, enhanced to...

conjecture | open | MathDB 页面 | 原文线索

V2-072025 | MATHDB-372939

Fel's conjecture on universal symmetric polynomials for numerical semigroups

摘录 / EXCERPT: Let S be the numerical semigroup under consideration, with multiplicity m, generators $d_1,\dots,d_m$, gap power sums $G_r(S)$, and invariants $K_p(S)$. Write...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072026 | MATHDB-372942

Neumann's fixed-point conjecture for primitive permutation groups

摘录 / EXCERPT: Let G be a finite primitive non-regular permutation group of degree n. Write $\operatorname{fix}(g)$ for the number of points fixed by $g\in G$. Neumann's conjecture. There exi...

conjecture | open | MathDB 页面 | 原文线索

V2-072027 | MATHDB-372944

Clarke-Larson-Vogt's conjecture on stability of normal bundles of BN-curves

摘录 / EXCERPT: A BN-triple is a triple (g,r,d) specifying a Brill-Noether curve of genus g, degree d, and dimension r. For a general BN-curve $X\subset\mathbb{P}^r$ of type (g,r,d), con...

conjecture | open | MathDB 页面 | 原文线索

V2-072028 | MATHDB-372945

Conjectured conormal regularity for Type I singularity formation

摘录 / EXCERPT: Let $\mathcal{X}=(\{v\leq 0\}\cup\{t\leq -0.9\})\cap\{-1\leq u<0\}$, and let ϕ be a solution arising from smooth initial data in the stated Type I self-similar stability resu...

conjecture | open | MathDB 页面 | 原文线索

V2-072029 | MATHDB-372946

Conjectured energy-critical Type II singularity formation

摘录 / EXCERPT: Let $\mathcal{X}=(\{v\leq 0\}\cup\{t\leq -0.9\})\cap\{-1\leq u<0\}$ and let $\mathcal{C}$ be the backward light cone. The symbols ϕ, r, ν, d, p, and k have the me...

conjecture | open | MathDB 页面 | 原文线索

V2-072030 | MATHDB-372947

Conjectured energy-supercritical Type II singularity formation

摘录 / EXCERPT: Let $\mathcal{X}=(\{v\leq 0\}\cup\{t\leq -0.9\})\cap\{-1\leq u<0\}$ and let $\mathcal{C}$ be the backward light cone. Let ϕ solve the relevant wave-map or semilinear equation...

conjecture | open | MathDB 页面 | 原文线索

V2-072031 | MATHDB-372955

The negative-invariant conjecture for nearly Gorenstein normal graded rings

摘录 / EXCERPT: Negative-invariant conjecture. If $\operatorname{char}(k)=0$ and $b(R)<0$, then R is of strongly F-regular type. If, moreover, R is $\mathbb Q$-Gorenstein, meaning that...

conjecture | open | MathDB 页面 | 原文线索

V2-072032 | MATHDB-372956

Polynomiality conjecture for low-degree coefficients of colored-triangle polynomials

摘录 / EXCERPT: For each n, write ... where $P_n(q)$ is the normalized polynomial associated with colored interlacing triangles. Polynomiality conjecture. For each fixed $k\geq 0$, there is an...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072033 | MATHDB-372957

Integrality and leading-coefficient conjecture for colored-triangle coefficient polynomials

摘录 / EXCERPT: For each fixed $k\geq 0$, let $a_k(n)$ be the coefficient of q^k in $P_n(q)$, regarded as a polynomial in n as in the polynomiality conjecture. Integrality and leading-coeffici...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072034 | MATHDB-372958

The 5/4 chromatic bound conjecture for even-hole-free graphs

摘录 / EXCERPT: Let G be an even-hole-free graph, meaning that G has no induced cycle of even length at least four. Write $\chi(G)$ for its chromatic number and $\omega(G)$ for its clique numb...

conjecture | open | MathDB 页面 | 原文线索

V2-072035 | MATHDB-372960

Decidability conjecture for non-hypothetical logic over Peano Arithmetic

摘录 / EXCERPT: The language and proof system of non-hypothetical logic are considered over the axioms of Peano Arithmetic, denoted by PA, in a purely relational language. One may also...

conjecture | open | MathDB 页面 | 原文线索

V2-072036 | MATHDB-372965

The conjecture that locally indicable groups are Lewin groups

摘录 / EXCERPT: A group G is a Lewin group if, for every field F with a G-action, the crossed product $F\ast G$ admits a Hughes-free $F\ast G$-division ring that is its universal division ri...

conjecture | open | MathDB 页面 | 原文线索

V2-072037 | MATHDB-372967

Wise's conjecture on locally indicable towers and vanishing L^2-Betti number

摘录 / EXCERPT: Let Y be a finite connected 2-dimensional CW-complex. A connected tower with target Y is a finite sequence of covering maps and embeddings ... Call such a tower non-positive...

conjecture | open | MathDB 页面 | 原文线索

V2-072038 | MATHDB-372968

Conjecture on the Costello-Li twist and the Courant contact model

摘录 / EXCERPT: Let $\mathcal E_{\mathrm{twist}}$ be the Courant algebroid over the Dolbeault complex associated to a twist datum. The Costello-Li twist of type I supergravity is the perturbative...

conjecture | open | MathDB 页面 | 原文线索

V2-072039 | MATHDB-372970

Conjecture on the Courant contact model and the extended BPS complex

摘录 / EXCERPT: Let $\mathcal E_{\mathrm{twist}}$ be the Courant algebroid associated to the twist, and let the extended BPS complex be the complex for $G=U(5)\times SO(10+k)$ associated to the co...

conjecture | open | MathDB 页面 | 原文线索

V2-072040 | MATHDB-372985

Ranked-poset conjecture for tilted Bruhat order

摘录 / EXCERPT: Ranked-poset conjecture for tilted Bruhat order. The poset $(S_n,\leq_{\mathbf{a}})$ is ranked.

conjecture | open | MathDB 页面 | 原文线索

V2-072041 | MATHDB-372986

Connectivity conjecture for tilted Bruhat posets

摘录 / EXCERPT: Connectivity conjecture for tilted Bruhat posets. The poset defined in the source by $\ell_{\mathbf{a}}(w)$ is connected, and $\lesssim_{\mathbf{a}}$ is connected when restricted t...

conjecture | open | MathDB 页面 | 原文线索

V2-072042 | MATHDB-372991

Ashkin-Teller magnetisation-field decomposition conjecture

摘录 / EXCERPT: Ashkin-Teller magnetisation-field decomposition conjecture. The statement of the main magnetisation-field decomposition theorem holds for the Ashkin-Teller magnetisation field at e...

conjecture | open | MathDB 页面 | 原文线索

V2-072043 | MATHDB-373006

Asymptotic gap conjecture for the KPZ line ensemble

摘录 / EXCERPT: Asymptotic gap conjecture. Almost surely as $n\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-072044 | MATHDB-373007

Parabolic rigidity conjecture for the KPZ line ensemble

摘录 / EXCERPT: Parabolic rigidity conjecture. As $n\to\infty$, the sequence of functions

conjecture | open | MathDB 页面 | 原文线索

V2-072045 | MATHDB-373008

Airy-type characterization conjecture for the KPZ line ensemble

摘录 / EXCERPT: Airy-type characterization conjecture. Then $\mathcal{L}$ is equal in distribution to the KPZ_t line ensemble, up to an independent affine shift.

conjecture | open | MathDB 页面 | 原文线索

V2-072046 | MATHDB-373019

Orthogonal Grassmannian decomposition conjecture for a pencil of quadrics

摘录 / EXCERPT: Let $\langle Q_1,Q_2\rangle$ be a pencil of quadrics in $\mathbb{P}^{n+2}$ whose base locus is smooth in an odd-dimensional projective space. Let C be the hyperelliptic curve ass...

conjecture | open | MathDB 页面 | 原文线索

V2-072047 | MATHDB-373020

Relative orthogonal Grassmannian decomposition conjecture for quadric fibrations

摘录 / EXCERPT: Let $p:\mathfrak{Q}\to S$ be a quadric fibration of relative dimension $n+1$ over a smooth base, with fibers $\mathfrak{Q}_s$ of rank at least 4, where n is odd. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072048 | MATHDB-373040

Floquet-isospectral antisymmetric periodic potentials

摘录 / EXCERPT: Let $q>3$, let V be a nonzero $q\mathbb{Z}$-periodic potential of the form ... and let Δ denote the discrete Laplacian. Floquet-isospectral potential conjecture. There exist...

conjecture | open | MathDB 页面 | 原文线索

V2-072049 | MATHDB-373049

Rationality conjecture for the Hilbert-Poincaré series of coefficients of a very-well-poised hypergeometric series

摘录 / EXCERPT: Hilbert-Poincaré rationality conjecture. The Hilbert-Poincaré series of these coefficient spaces is rational:

conjecture | open | MathDB 页面 | 原文线索

V2-072050 | MATHDB-373050

Rationality conjecture for the image series of the coefficient-summation map

摘录 / EXCERPT: Image Hilbert-Poincaré conjecture. The generating function of the dimensions of the images is rational, with initial expansion

conjecture | open | MathDB 页面 | 原文线索

V2-072051 | MATHDB-373051

Rationality conjectures for the coefficient and image dimensions of S_1

摘录 / EXCERPT: S1 dimension conjecture. The following two assertions are conjectured:

conjecture | open | MathDB 页面 | 原文线索

V2-072052 | MATHDB-373052

Rationality conjectures for three WZ-seed coefficient series

摘录 / EXCERPT: Let S_1,S_2,S_3 be the three summands defined from the WZ identities in the preceding construction, and let $V_N(S_i)$ denote the span of the coefficients of degree-N parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-072053 | MATHDB-373053

Rationality conjectures for two harmonic WZ-seed dimension series

摘录 / EXCERPT: Let $S_1=S_1(a,b,c,d;n)$ and $S_2=S_2(a,b,c,d;n)$ be the two summands obtained from the WZ seeds in the harmonic extension of the displayed series, and let $V_N(S_i)$ be their degr...

conjecture | open | MathDB 页面 | 原文线索

V2-072054 | MATHDB-373054

Rationality conjectures for three harmonic extensions of a Ramanujan series

摘录 / EXCERPT: Let S_1,S_2,S_3 be the three summands obtained from the WZ seeds used to construct harmonic extensions of the displayed Ramanujan $1/\pi$-formula, and let $V_N(S_i)$ denote the s...

conjecture | open | MathDB 页面 | 原文线索

V2-072055 | MATHDB-373055

Rationality conjecture for the coefficient dimension series of a WZ-seed extension

摘录 / EXCERPT: Let $S=S(a,b,c;n)$ be the summand obtained from the WZ pair in the harmonic extension of the displayed Ramanujan $1/\pi$-series, and let $V_N(S)$ be the span of the coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-072056 | MATHDB-373056

Rationality conjecture for the coefficient dimension series of a zeta-value WZ-seed extension

摘录 / EXCERPT: Let $S=S(a,b,c,d,e;n)$ be the summand obtained from the WZ-seed extension of the displayed $\zeta(5)$ series, and let $V_N(S)$ be the span of the coefficients of degree-N paramet...

conjecture | open | MathDB 页面 | 原文线索

V2-072057 | MATHDB-373057

Roberson's homomorphism-counting conjecture for proper minor-closed classes

摘录 / EXCERPT: Let $\mathcal{M}$ be a graph class that is proper, minor-closed, and union-closed. For graphs G and H, write $G =_{\mathcal{M}} H$ when $\hom(K,G)=\hom(K,H)$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-072058 | MATHDB-373058

Distinguishing conjecture for distinct immersion-closed classes

摘录 / EXCERPT: Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be two distinct immersion-closed and union-closed graph classes. For graphs G and H, write $G =_{\mathcal{M}} H$ when...

open_problem | open | MathDB 页面 | 原文线索

V2-072059 | MATHDB-373059

Linear odd-colouring bound for complete-graph immersions

摘录 / EXCERPT: Let G be a graph and let $G\xrightarrow{\operatorname{odd}} K_{ct}$ mean that G admits a t'-oddomorphism to K_{ct}, as defined in the source. Linear immersion conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-072060 | MATHDB-373066

Positive-definite function classification conjecture for finite spheres

摘录 / EXCERPT: Let $n>2$, let $f:[-1,1]\to\mathbb{R}$ satisfy $f(1)=1$, and define $\widehat f:SO(n)\to\mathbb{R}$ by $\widehat f(\gamma)=f(\gamma_{11})$. Assume that $\widehat f$ defines a posit...

conjecture | open | MathDB 页面 | 原文线索

V2-072061 | MATHDB-373071

Maillard et al.'s replica prediction for Bayes risk

摘录 / EXCERPT: Consider the generalized linear model with orthogonally invariant design, and let $\widehat{\beta}_{\mathrm{B}}=\mathbb{E}[\beta_*\mid X,y]$ be the posterior mean estimator. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072062 | MATHDB-373072

The depth-one decomposition hypothesis for solvable meromorphic equations

摘录 / EXCERPT: Let $X\subseteq\mathbb{C}$ be a connected open set and let $f:X\to\mathbb{C}$ be a meromorphic elementary function. A function is solvable when its equation with a constant right-h...

conjecture | open | MathDB 页面 | 原文线索

V2-072063 | MATHDB-373074

The flagged skew tableau realization conjecture for Demazure crystals

摘录 / EXCERPT: A Demazure crystal is denoted by $\mathcal{B}_{w,\lambda}$, where the crystal is of type A, normal, and finite. A flagged skew tableau crystal is the crystal of flagged skew tabl...

open_problem | open | MathDB 页面 | 原文线索

V2-072064 | MATHDB-373077

The Jordan-curve conjecture for banded Toeplitz symbols

摘录 / EXCERPT: Jordan-curve conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-072065 | MATHDB-373100

Fibonacci degree conjecture for Golombic and Levine polynomials

摘录 / EXCERPT: Fibonacci degree conjecture. Both polynomial expressions $\gamma_n(b_1^{m_1}\dots b_k^{m_k})$ and $\ell_n(b_1^{m_1}\dots b_k^{m_k})$ have degree F_n in m_j and degree F_{n-1}...

conjecture | open | MathDB 页面 | 原文线索

V2-072066 | MATHDB-373121

Orbit-harmonics containment conjecture for rectangular set partitions

摘录 / EXCERPT: Let $ Pi_{(b^a)}$ and $ Pi_{(a^b)}$ denote the loci of unordered set partitions whose block-size types are (b^a) and (a^b), respectively, and let ${\rm gr}{\bf I}(Pi)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072067 | MATHDB-373123

Symmetric-group log-concavity conjecture for orbit harmonics of bounded-block partitions

摘录 / EXCERPT: For integers n and m, let $ Pi_{n,m}$ be the locus of unordered set partitions of $[n]$ with block sizes bounded by m, and let $R(Pi_{n,m})$ be its orbit-harmonics graded...

conjecture | open | MathDB 页面 | 原文线索

V2-072068 | MATHDB-373144

Improved coefficient conjecture for univalent harmonic mappings

摘录 / EXCERPT: Let $f=h+\overline{g}\in\mathcal{S}_{\mathcal{H}}$, where $h(z)=z+\sum_{n=2}^{\infty}a_nz^n$ and $g(z)=\sum_{n=1}^{\infty}b_nz^n$ are the analytic and co-analytic parts of a normal...

conjecture | open | MathDB 页面 | 原文线索

V2-072069 | MATHDB-373145

El-Zahar's cycle-cover conjecture

摘录 / EXCERPT: El-Zahar's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072070 | MATHDB-373147

Extendability conjecture for sea grafting (g) in Penrose P2 tilings

摘录 / EXCERPT: Consider the sea grafting (g) shown in Figure of the paper. A bi-infinite fully leafed caterpillar is a fully leafed caterpillar that extends infinitely in both directions. Extenda...

conjecture | open | MathDB 页面 | 原文线索

V2-072071 | MATHDB-373150

The planar graph $(5m-1,m)$-choosability conjecture

摘录 / EXCERPT: Planar graph choosability conjecture. There exists a constant integer m such that every planar graph is $(5m-1,m)$-choosable.

conjecture | open | MathDB 页面 | 原文线索

V2-072072 | MATHDB-373152

Uniform observability conjecture for Schrödinger equations on hyperbolic coverings

摘录 / EXCERPT: Uniform observability conjecture. For every $T>0$, there exists a constant $C=C(X,S,T)>0$ such that the observability inequality holds. Moreover, the constant can be chosen to depe...

conjecture | open | MathDB 页面 | 原文线索

V2-072073 | MATHDB-373158

Mumford-goodness conjecture for higher direct-image metrics

摘录 / EXCERPT: Let $\mu:Y\to X$ be the birational modification in a semistable reduction, let $f:Y\to T$ be the induced family, let $F^{*}\xi$ be the pulled-back coefficient bundle, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072074 | MATHDB-373159

Musiker's positivity conjecture for super lambda-lengths

摘录 / EXCERPT: Consider the once-punctured torus with the spin structure induced by the cyclic orientation of the two triangles in the fundamental domain, clockwise on one triangle and counter-cl...

conjecture | open | MathDB 页面 | 原文线索

V2-072075 | MATHDB-373160

q-Markov function conjecture for mirror Markov-tree branches

摘录 / EXCERPT: Fix a Markov number $\ell\geq 5$. Let $m_\ell(q)$ be a mirror deformation of ℓ, and let $(p_n)_{n\geq 0}$ be the sequence of polynomials on a mirror Markov-tree branch fixing...

conjecture | open | MathDB 页面 | 原文线索

V2-072076 | MATHDB-373161

Coefficient-positivity conjecture for mirror deformed Markov numbers

摘录 / EXCERPT: Let m_n denote the mirror deformed Markov numbers. Positivity conjecture. Every coefficient of every m_n is positive. The claim is based on experimental computations and is sta...

conjecture | open | MathDB 页面 | 原文线索

V2-072077 | MATHDB-373162

q-deformed uniqueness conjecture for mirror Markov triples

摘录 / EXCERPT: Consider triples in the mirror deformation of the Markov tree, with each triple consisting of mirror deformed Markov numbers and with degree taken in the deformation parameter q....

conjecture | open | MathDB 页面 | 原文线索

V2-072078 | MATHDB-373165

Reckless conjecture on non-reduced cores and mirror sheaves

摘录 / EXCERPT: Let $\theta:\mathcal{M}\to\mathbb{A}$ be a smooth integrable system of weight 1 with Kostant section κ, and suppose it has a mirror-dual system...

conjecture | open | MathDB 页面 | 原文线索

V2-072079 | MATHDB-373166

Symmetry and Hankel structure of minimal matrices of fixed RSK shape

摘录 / EXCERPT: Let λ be a partition with n parts, and let $M=(m_{i,j})_{1\le i,j \le n}\in\mathcal{M}_\lambda$ be a minimal matrix of shape λ. A matrix is Hankel when its entrie...

conjecture | open | MathDB 页面 | 原文线索

V2-072080 | MATHDB-373172

Local deep congruence conjecture for semistable representations

摘录 / EXCERPT: Let $\mathcal{L},\mathcal{L}'\in\mathbb{Q}_p$ be admissible, let k be the weight, and let $C_{p,k}$ be the constant appearing in the conjecture. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072081 | MATHDB-373187

Length-set realization conjecture for non-torsion power monoids

摘录 / EXCERPT: Let H be a monoid that is not torsion, and let $\mathcal P_{\mathrm{fin},1}(H)$ be its finitary power monoid containing the identity. A length set is the set of factorization len...

conjecture | open | MathDB 页面 | 原文线索

V2-072082 | MATHDB-373190

Conjecture on the effect of quadratic phase on global existence and blow-up

摘录 / EXCERPT: Quadratic-phase conjecture. For the combined nonlinear Schrödinger equation, if $b>0$, the solution with initial condition v_0 exists globally and scatters, whereas if $b<0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072083 | MATHDB-373197

Rank-unimodality conjecture for principal permutation-pattern downsets

摘录 / EXCERPT: Let ρ be a permutation, and let $D(\rho)$ be the principal downset of all permutations contained in ρ, ordered by permutation-pattern containment and ranked by permutatio...

conjecture | open | MathDB 页面 | 原文线索

V2-072084 | MATHDB-373198

McNamara-Steingrímsson interval rank-unimodality conjecture

摘录 / EXCERPT: For permutations ρ and τ with $\rho oeq \tau$ in the permutation pattern poset, let $[\rho,\tau]$ denote the interval of permutations $ u$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-072085 | MATHDB-373199

Rank-unimodality conjecture for composition and layered-permutation downsets

摘录 / EXCERPT: Let w be a composition, ordered by the subword order, and let its principal downset consist of all compositions contained in w. A composition is associated with a layered permu...

conjecture | open | MathDB 页面 | 原文线索

V2-072086 | MATHDB-373200

Pantone's 123-avoiding derangement proportion conjecture

摘录 / EXCERPT: Let C_n be the nth Catalan number, and let d_n be the number of derangements among the 123-avoiding permutations of length n. The source considers an asymptotic form...

conjecture | open | MathDB 页面 | 原文线索

V2-072087 | MATHDB-373214

The correspondence coloring conjecture for K_5-minor-free graphs

摘录 / EXCERPT: Correspondence coloring conjecture. The graph G has 6 disjoint $\mathbf{M}$-colorings. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-072088 | MATHDB-373222

Hertling's variance conjecture for spectrum numbers

摘录 / EXCERPT: Hertling's conjecture. The spectrum numbers satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-072089 | MATHDB-373223

The maximal spectrum number lies outside the Tjurina spectrum

摘录 / EXCERPT: Maximal-exponent conjecture. The maximal spectrum number of f is not contained in its Tjurina spectrum.

conjecture | open | MathDB 页面 | 原文线索

V2-072090 | MATHDB-373224

Generalized Hertling conjecture for the Tjurina spectrum

摘录 / EXCERPT: Generalized Hertling conjecture for the Tjurina spectrum. The exponents satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-072091 | MATHDB-373225

The description of R(f) for the four exceptional modality-three cases

摘录 / EXCERPT: Exceptional-case conjecture. For these four cases, $\operatorname{R}(f)$ is exactly the set consisting of the largest and second-largest exponents.

conjecture | open | MathDB 页面 | 原文线索

V2-072092 | MATHDB-373250

The dimension formula for polytopes of quarter-turn symmetric alternating sign matrices

摘录 / EXCERPT: Let $n \geq 5$ be such that $n \not\equiv 2 \pmod 4$, and set $k = \floor*{n/2}$. The polytope P_QTSASM is the polytope of quarter-turn symmetric alternating sign matric...

conjecture | open | MathDB 页面 | 原文线索

V2-072093 | MATHDB-373265

Temperedness conjecture for global packets on basic inner forms

摘录 / EXCERPT: Temperedness conjecture. The set Π_ϕ always consists of irreducible, admissible, and tempered representations of $G(\mathbb A_F)$.

conjecture | open | MathDB 页面 | 原文线索

V2-072094 | MATHDB-373277

Congruence conjectures for DSOME modulo 8 and 16

摘录 / EXCERPT: Let $DSOME(n)$ denote the sum of all odd parts in the partitions of n into distinct parts minus the sum of all even parts. DSOME congruence conjecture. For all integers $n\geq0$,...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072095 | MATHDB-373286

Unimodality conjecture for coefficients of constructed polynomial Pythagorean triples

摘录 / EXCERPT: A polynomial solution $(\mathcal{A},\mathcal{B},\mathcal{C})$ is constructed for each standard Pythagorean triple, where $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-072096 | MATHDB-373311

McMullen's expander conjecture for origami orbit graphs in \mathcal{H}(2)

摘录 / EXCERPT: Let $(\mathcal{G}_{n})_{n}$ be the family of $\operatorname{SL}(2,\mathbb{Z})$-orbit graphs of primitive n-squared origamis in the stratum $\mathcal{H}(2)$. McMullen's expander c...

conjecture | open | MathDB 页面 | 原文线索

V2-072097 | MATHDB-373312

Parity conjecture for the loci W_{d^2}[n]

摘录 / EXCERPT: Let $W_{d^2}[n]$ be the locus under consideration, and for odd n let $W_{d^2}^{\epsilon}[n]$, with $\epsilon\in\{0,1\}$, denote its two parity-labelled subloci. Parity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072098 | MATHDB-373313

Delecroix-Lelièvre's orbit-count conjecture in \mathcal{H}(4)

摘录 / EXCERPT: Let n be an integer greater than 8, and consider primitive n-squared origamis in $\mathcal{H}(4)$, with $\mathcal{H}^{\mathrm{odd}}(4)$ denoting the odd component and...

conjecture | open | MathDB 页面 | 原文线索

V2-072099 | MATHDB-373317

Volčič's invariant-subspace Nullstellensatz conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $f_1,\ldots,f_l,g\in\mathbb{K}\langle\underline{x}\rangle$ be noncommutative polynomials, and for each $n\in\mathbb{N}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072100 | MATHDB-373322

The upper bound conjecture for wavefront sets of representations

摘录 / EXCERPT: Let H be a split classical group over F, let $ u$ denote the nilpotent orbit of the image of ... under the derivative of the relevant $\mathrm{SL}_2(\mathbb{C})$-factor, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072101 | MATHDB-373334

The large-values limsup conjecture for summatory k-free character functions

摘录 / EXCERPT: Let $k\geq 2$ be an integer, let χ be a real non-principal Dirichlet character modulo q, and let $f=\mu^{(k)}\chi$ or $f=\mu^{(k)}g_\chi$. Then the large-values limsup conje...

conjecture | open | MathDB 页面 | 原文线索

V2-072102 | MATHDB-373349

Artin's conjecture on forms over the p-adic numbers

摘录 / EXCERPT: For a field k, say that it is C_2 if, for every form $F\in k[X_1,\ldots,X_n]$ of degree d with $n>d^2$, the equation ... has a nontrivial solution in k^n. Artin's conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-072103 | MATHDB-373366

The Burning Number Conjecture for trees

摘录 / EXCERPT: Tree form of the Burning Number Conjecture. Every tree on n vertices satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072104 | MATHDB-373367

Greedy optimality conjecture for tooth-dominant comb graphs

摘录 / EXCERPT: Greedy optimality conjecture for comb graphs. If $n\leq m$, then

conjecture | open | MathDB 页面 | 原文线索

V2-072105 | MATHDB-373374

The leading-block and eigenvalue conjecture for E0_2-matrices

摘录 / EXCERPT: Leading-block and eigenvalue conjecture. This leading principal diagonal block of A^{-1} is a $\mathbf{Z}$-matrix. Moreover, A has exactly one negative eigenvalue.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072106 | MATHDB-373375

The determinant and inverse conjecture for E0_2-matrices

摘录 / EXCERPT: Determinant and inverse conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-072107 | MATHDB-373384

Goh's Möbius uncertainty conjecture for locally finite posets

摘录 / EXCERPT: Let P be a locally finite poset, and let $\mu_P(x,y)$ denote its Möbius function. Say that P has the Möbius uncertainty property when this property holds as defined in the pape...

conjecture | open | MathDB 页面 | 原文线索

V2-072108 | MATHDB-373387

Ballantine-Burson-Craig-Folsom-Wen hook-count conjecture

摘录 / EXCERPT: Let $a_t(n)$ be the total number of t-hooks in all partitions of n into odd parts, and let $b_t(n)$ be the total number of t-hooks in all partitions of n into distinct part...

conjecture | solved | MathDB 页面 | 原文线索

V2-072109 | MATHDB-373389

Okazaki-Smith conjecture on line defect half-indices for ${\rm SU}(3)_{-4}$

摘录 / EXCERPT: Let $\langle W_m\rangle_{\mathcal{N},1,0}^{{\rm SU}(3)_{-4}}$ denote the line defect half-index for the m-th Wilson line in the ${\rm SU}(3)_{-4}$ Chern-Simons theory with the in...

conjecture | open | MathDB 页面 | 原文线索

V2-072110 | MATHDB-373394

Local constancy of automorphic categories in contractible analytic families

摘录 / EXCERPT: Local constancy conjecture. If U is contractible, then the restriction functor i_u^* is an equivalence. This is the fiberwise formulation of local constancy for the automorphic...

conjecture | open | MathDB 页面 | 原文线索

V2-072111 | MATHDB-373396

The Links-Quivers Correspondence for rational links

摘录 / EXCERPT: Let L be a link with $|L|$ components, and let $P^{\bigwedge^j}_L(q,a)$ denote its $\bigwedge^j$-colored HOMFLY-PT polynomial. Define ... Let $P_Q(\overline{x})$ be the generating...

conjecture | open | MathDB 页面 | 原文线索

V2-072112 | MATHDB-373397

The Links-Quivers Correspondence for colored HOMFLY-PT polynomials

摘录 / EXCERPT: Links-Quivers Correspondence. For every link L, there is a symmetric quiver Q_L such that

conjecture | open | MathDB 页面 | 原文线索

V2-072113 | MATHDB-373399

Non-degeneracy of conditional limits for multi-spin systems with uniform mark distribution

摘录 / EXCERPT: Non-degeneracy conjecture. For every $c\in (\kappa h_{\min},\kappa h_{\max})$, U_n given $\operatorname{Conset}_n(h,c)$ lies asymptotically in $\operatorname{Gibbsnondeg}(c)$.

conjecture | open | MathDB 页面 | 原文线索

V2-072114 | MATHDB-373402

Conjecture on dyadic congruences for overcolored partition functions

摘录 / EXCERPT: Dyadic congruence conjecture. For every integer $n\geq 0$, the congruences

conjecture | open | MathDB 页面 | 原文线索

V2-072115 | MATHDB-373420

Aharoni-Berger-Kfir conjecture on acyclic sets in oriented graphs

摘录 / EXCERPT: Let D be an oriented graph with n vertices and m edges or arcs, and let $\vec{\alpha}(D)$ denote its largest acyclic set. Aharoni-Berger-Kfir conjecture. ... For tournaments th...

conjecture | open | MathDB 页面 | 原文线索

V2-072116 | MATHDB-373435

Conjecture on enumerating permutations avoiding classical and arrow patterns

摘录 / EXCERPT: Enumeration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072117 | MATHDB-373437

Nelson-Nguyen conjecture on sparse oblivious subspace embeddings

摘录 / EXCERPT: Fix a subspace dimension d and distortion parameters $\alpha = \beta \in (0,1)$. A random sparse matrix $\bm{\Phi}: \mathbb{R}^n \to \mathbb{R}^k$ has at most ζ nonzero entr...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072118 | MATHDB-373440

The strip energy ball comparison conjecture

摘录 / EXCERPT: Strip energy ball comparison conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072119 | MATHDB-373447

The co-diameter-two co-chordal characterization conjecture

摘录 / EXCERPT: Co-diameter-two characterization conjecture. The graph G is non-trivially minimally tough if and only if G is isomorphic to $S_{\ell,\ell,\ell}$ for some $\ell\geq 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-072120 | MATHDB-373450

Logarithmic threshold conjecture for the chromatic number of random Borsuk graphs

摘录 / EXCERPT: Let $G(n,\alpha)$ be the random Borsuk graph in dimension d, and let $\chi(G(n,\alpha))$ denote its chromatic number. Logarithmic threshold conjecture. For every $d\geq 2$ there...

conjecture | open | MathDB 页面 | 原文线索

V2-072121 | MATHDB-373451

Polynomial threshold conjecture for intermediate chromatic numbers of random Borsuk graphs

摘录 / EXCERPT: Let $G(n,\alpha)$ be the random Borsuk graph in dimension d, and let $\chi(G(n,\alpha))$ denote its chromatic number. Polynomial threshold conjecture. For each $d\geq 2$, there e...

conjecture | open | MathDB 页面 | 原文线索

V2-072122 | MATHDB-373453

Two-consecutive-values conjecture for the chromatic number of sparse random Borsuk graphs

摘录 / EXCERPT: Let $G(n,\alpha)$ be the random Borsuk graph in dimension d, let $\chi(G(n,\alpha))$ denote its chromatic number, and let α_0 be the constant supplied by the Erdős-Hajnal...

conjecture | open | MathDB 页面 | 原文线索

V2-072123 | MATHDB-373454

Full holonomy conjecture for higher-dimensional Laplacian dynamics

摘录 / EXCERPT: Full holonomy conjecture. For $d\geq 3$, $n>d$, and initial conditions with $D(P(0))$ not scalar (non-regular graph), the restricted holonomy group of Laplacian dynamics is

conjecture | open | MathDB 页面 | 原文线索

V2-072124 | MATHDB-373456

Cao-Maulik-Toda DT-GV correspondence for Calabi-Yau fourfolds

摘录 / EXCERPT: Let Y be a Calabi-Yau 4-fold, let $\beta\in H_2(Y,\mathbb{Z})$ be a curve class, and let $\mathbf{M}_{\beta}(Y)$ be the moduli space of one-dimensional stable sheaves F on...

conjecture | open | MathDB 页面 | 原文线索

V2-072125 | MATHDB-373464

Multiple minimizers via symmetry for a three-fold symmetric elliptic system

摘录 / EXCERPT: Multiple minimizers via symmetry. The constrained problem admits at least three distinct minimizers related by rotations through $2\pi/3$.

conjecture | open | MathDB 页面 | 原文线索

V2-072126 | MATHDB-373466

Weak Green-Griffiths-Lang conjecture for C-pairs

摘录 / EXCERPT: Let (X, Δ) be a smooth C-pair over $\mathbb{C}$, where a dense entire curve means a holomorphic C-pair map $\mathbb{C} \to (X, \Delta)^{\an}$ whose image is Zariski-dense in...

conjecture | open | MathDB 页面 | 原文线索

V2-072127 | MATHDB-373467

Weak Bombieri-Lang conjecture for C-pairs

摘录 / EXCERPT: Let (X, Δ) be a smooth C-pair over $\overline{\mathbb{Q}}$. Say that (X, Δ) satisfies potential density of integral points if, for some number field K, finite set o...

conjecture | open | MathDB 页面 | 原文线索

V2-072128 | MATHDB-373481

Stable odd-cycle trade conjecture with bounded isotopy depth

摘录 / EXCERPT: Let $\mathcal{L}_n$ be the set of Latin squares of even order n. A stable odd-cycle two-line trade is an odd-cycle trade whose canonical selection remains compatible with reappli...

conjecture | open | MathDB 页面 | 原文线索

V2-072129 | MATHDB-373489

Maulik-Yun's perverse-Lefschetz opposition conjecture for compactified Jacobians

摘录 / EXCERPT: Maulik-Yun's conjecture. The perverse filtration and the Lefschetz filtration are opposite:

conjecture | open | MathDB 页面 | 原文线索

V2-072130 | MATHDB-373505

The factorization conjecture for numerical ranges of compressed shifts

摘录 / EXCERPT: Factorization conjecture. If θ does not factor in this way, then $W(S_\theta^1)$ and $W(S_\theta^2)$ are open. If θ does factor in this way, then $W(S_\theta^1)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072131 | MATHDB-373517

Height boundedness conjecture for isolated periodic points of polynomial automorphisms

摘录 / EXCERPT: Height boundedness conjecture. The set of isolated periodic points of f has bounded height.

conjecture | open | MathDB 页面 | 原文线索

V2-072132 | MATHDB-373528

Balogh-Morris-Samotij diameter-game threshold conjecture

摘录 / EXCERPT: Let $s\geq 3$ and let $\mathcal{D}_{s,n}$ be the $(1:b)$ Maker-Breaker game on the edges of K_n in which Maker wins precisely when her spanning subgraph has diameter at most s....

conjecture | open | MathDB 页面 | 原文线索

V2-072133 | MATHDB-373529

Rainbow-connectivity game threshold conjecture

摘录 / EXCERPT: For the rainbow-connectivity game $\mathcal{C}_{s,n}$, let $b_{\mathcal{C}_{s,n}}$ denote its threshold bias. Assume that $s=\omega(\log(n))$. Rainbow-connectivity threshold conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-072134 | MATHDB-373530

Rainbow-spanning-tree game threshold conjecture

摘录 / EXCERPT: Let $\mathcal{RS}_n$ be the rainbow-spanning-tree game played on $n-1$ copies of K_n, where Maker wins by claiming a rainbow spanning tree, and let $b_{\mathcal{RS}_n}$ denote it...

conjecture | open | MathDB 页面 | 原文线索

V2-072135 | MATHDB-373531

Rainbow perfect-matching game threshold conjecture

摘录 / EXCERPT: Let n be even. In the rainbow perfect matching game $\mathcal{RP}_n$, played on $n/2$ copies of K_n, Maker wins by claiming a rainbow perfect matching; let $b_{\mathcal{RP}_n}$...

conjecture | open | MathDB 页面 | 原文线索

V2-072136 | MATHDB-373536

Hook-length bias conjecture for 5-core partitions

摘录 / EXCERPT: Let $a_{t,k}(n)$ count the hooks of length k in all the t-core partitions of n. Hook-length bias conjecture for 5-core partitions. For all $n\geq 0$, one has ... The inequaliti...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072137 | MATHDB-373548

Perälä-Virtanen conjecture on essential positivity of radial Bergman Toeplitz operators

摘录 / EXCERPT: Perälä-Virtanen conjecture. The operator T_f is essentially positive if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-072138 | MATHDB-373553

Loubaton's characterization of the image of the right adjoint embedding

摘录 / EXCERPT: Let $R\colon \Cat^L_\infty\hookrightarrow \Cat_\infty$ be the embedding of the category of presentable ∞-categories and colimit-preserving functors into the category of...

conjecture | open | MathDB 页面 | 原文线索

V2-072139 | MATHDB-373570

Weak Sard characterization of dissipation anomalies for passive scalar flows

摘录 / EXCERPT: Weak Sard characterization. The vector field v allows dissipation anomalies if and only if its stream function violates the weak Sard property.

conjecture | open | MathDB 页面 | 原文线索

V2-072140 | MATHDB-373576

Chapoton's simplicial-polytope conjecture for arbor polytopes

摘录 / EXCERPT: Let τ be an arbor on the ground set $[n]:=\{1,2,\dots,n\}$, and let $\mathcal Q_\tau$ be its associated n-dimensional arbor polytope. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-072141 | MATHDB-373582

Conjecture on reverse-colex concatenation of fixed-weight de Bruijn sequences

摘录 / EXCERPT: Let $n,t\geq 2$ and let $w<t$. Let $\mathcal{V}_t(n,w)$ be the universal cycle constructed from the relevant fixed-weight strings, and let $\mathcal{V'}_t(n,w)$ be the sequence obt...

conjecture | open | MathDB 页面 | 原文线索

V2-072142 | MATHDB-373588

Thejitha-Fathima overcolored partition congruence conjecture

摘录 / EXCERPT: Let an overcolored partition of n be a partition in which even parts may appear in one of r colors and odd parts may appear in one of s colors, with the first occurrence of e...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072143 | MATHDB-373598

Generalized Hamilton-connectedness conjecture for forbidden-star split graphs

摘录 / EXCERPT: Generalized Hamilton-connectedness conjecture. Every r-connected ${K_{1,r+1},K_{1,r+1}+e}$-free split graph of order at least $2r+7$ is Hamilton-connected.

conjecture | open | MathDB 页面 | 原文线索

V2-072144 | MATHDB-373601

Fraser-Li conjecture on the critical catenoid

摘录 / EXCERPT: Let $B^3\subset\mathbb{R}^3$ be the unit ball, and let Σ_{cc} denote the critical catenoid, the free boundary minimal annulus obtained by scaling the Euclidean cate...

conjecture | open | MathDB 页面 | 原文线索

V2-072145 | MATHDB-373602

Barbosa-Viana pinching conjecture in the four-dimensional ball

摘录 / EXCERPT: Let $\Sigma^3\subset B^4\subset\mathbb{R}^4$ be a capillary minimal hypersurface in the unit ball; the free boundary case corresponds to contact angle $\pi/2$. Assume that ... for al...

conjecture | open | MathDB 页面 | 原文线索

V2-072146 | MATHDB-373603

Yudovich's conjecture on generic vorticity-gradient growth

摘录 / EXCERPT: Let u be an inviscid incompressible flow on $\mathbb{R}^2$ with scalar vorticity $\omega=\partial_1u^2-\partial_2u^1$, recovered from ω by the Biot-Savart law...

conjecture | open | MathDB 页面 | 原文线索

V2-072147 | MATHDB-373606

Sedlar-Škrekovski local irregularity conjecture for colorable graphs

摘录 / EXCERPT: Let B be the bow-tie cactus with $\mathrm{lir}(B)=4$. For a connected graph G that is locally irregular colorable, let $\mathrm{lir}(G)$ denote the minimum number of colors in...

conjecture | open | MathDB 页面 | 原文线索

V2-072148 | MATHDB-373607

Shahidi's genericity conjecture for tempered L-packets

摘录 / EXCERPT: Shahidi Conjecture. For any quasi-split reductive group G, every tempered L-packet has a generic member.

conjecture | open | MathDB 页面 | 原文线索

V2-072149 | MATHDB-373631

The radiation compatibility conjecture for multi-soliton singularity formation

摘录 / EXCERPT: Fix soliton parameters A, $\boldsymbol{\lambda}$, ν, and σ. Let ϕ_{rad} be a smooth solution of the wave equation $\Box\phi_{\mathrm{rad}}=0$ satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-072150 | MATHDB-373635

Nesting conjecture for primitive normalized Witten intersection numbers

摘录 / EXCERPT: For $g\geq2$, $n\geq1$, and $d_1,\ldots,d_n\geq1$ satisfying $|\mathbf d|=3g-3+n$, let $C(\mathbf d)$ denote the primitive normalized Witten intersection number, and write...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072151 | MATHDB-373636

The conjectured dispersion-relation structure for sharp stratification

摘录 / EXCERPT: Let $delta>0$, and let $rhob_{\delta}$ and Vb_{δ} be the density and velocity profiles defined by the sharp stratification and shear-flow setup. Let $mathcal C_{\delta}(k)$...

conjecture | open | MathDB 页面 | 原文线索

V2-072152 | MATHDB-373637

Crouzeix's conjecture for scaled q-numerical ranges

摘录 / EXCERPT: Let A be an $n\times n$ matrix, let $0<|q|\leq 1$, and let Ω_q denote the scaled q-numerical range introduced in the source. For a polynomial p with complex coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-072153 | MATHDB-373642

Asymptotic uniformity conjecture for permanents over finite fields

摘录 / EXCERPT: Asymptotic uniformity conjecture. For every $x\inFbb{F}_{q}$,

conjecture | open | MathDB 页面 | 原文线索

V2-072154 | MATHDB-373664

Benedetto-Ih finiteness conjecture for S-integral post-critically finite parameters

摘录 / EXCERPT: Benedetto-Ih conjecture. If $f_{d,\alpha}$ is not post-critically finite, then there are finitely many parameters $c\in\overline{K}$ that are S-integral relative to (α) an...

conjecture | open | MathDB 页面 | 原文线索

V2-072155 | MATHDB-373671

Natural-definition problem for non-full-flag Higgs bundle moduli spaces

摘录 / EXCERPT: Let p_i be punctures with parabolic weights $\boldsymbol{\alpha}=(\alpha_i)$, and call the flags not all full when, for some p_i, the corresponding weight satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-072156 | MATHDB-373673

The universal hyperbolic bound for block-structured coded retrieval

摘录 / EXCERPT: Universal hyperbolic bound. For any partition $s_1+s_2=k$ with $\max\{s_1,s_2\}\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-072157 | MATHDB-373674

The monotonicity conjecture for achievability regions

摘录 / EXCERPT: Achievability-region monotonicity conjecture. For every $n\geq k$,

conjecture | open | MathDB 页面 | 原文线索

V2-072158 | MATHDB-373679

The power-of-two values conjecture for the Conway-type sequence d(n)

摘录 / EXCERPT: Let $d(n)$ be the Conway-type integer sequence, and let I_K denote its generation domains, with $I_{-1}=\{1,2\}$. Power-of-two values conjecture. The sequence satisfies … and … T…

conjecture | open | MathDB 页面 | 原文线索

V2-072159 | MATHDB-373680

The parent-spot generation conjecture for the Conway-type sequence d(n)

摘录 / EXCERPT: Let $d(n)$ be the Conway-type integer sequence. For $K\geq -1$, let … with initial domain $I_{-1}=\{1,2\}$. For each n, let n_1 and n_2 be the two parent spots occurring in t…

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072160 | MATHDB-373683

Conjecture on the independence of the maps Pi_m and pi_m

摘录 / EXCERPT: For each integer $m\geq 0$, let Π_m and π_m be the maps defined in the paper from the Ekedahl-Oort stratification data. Independence conjecture. The maps Π_m and…

conjecture | open | MathDB 页面 | 原文线索

V2-072161 | MATHDB-373687

Lévêque et al.'s 4-color conjecture for ISK4-free graphs

摘录 / EXCERPT: All graphs under consideration are finite and simple. A graph is ISK4-free if it contains no induced subdivision of K_4. Lévêque et al.'s 4-color conjecture. Every ISK4-free grap…

conjecture | open | MathDB 页面 | 原文线索

V2-072162 | MATHDB-373699

The backreacted Calabi-Yau category conjecture

摘录 / EXCERPT: Backreacted Calabi-Yau category conjecture. Given a holography package $\mathcal{P}_{\mathcal{C},\lambda,h,s}$ as above, there is a family of Calabi-Yau categories…

conjecture | open | MathDB 页面 | 原文线索

V2-072163 | MATHDB-373701

Asymptotic persistence density conjecture for Benford sequences

摘录 / EXCERPT: Let b range over integer bases greater than 1. A base is B-EQUI-PERS at sample size N when there exists a continued-fraction convergent denominator $q_k\leq N$ such that…

conjecture | open | MathDB 页面 | 原文线索

V2-072164 | MATHDB-373702

Beta-universality conjecture for bounded-type Benford sequences

摘录 / EXCERPT: Let (a_n) be a sequence satisfying Benford's Law through equidistribution dynamics, and suppose the continued fraction of its rotation number is of bounded type, meaning that…

conjecture | open | MathDB 页面 | 原文线索

V2-072165 | MATHDB-373716

Bickle-Phillips conjecture on the 2-tone chromatic number of cubic graphs

摘录 / EXCERPT: Let G) be a cubic graph. The Bickle-Phillips conjecture. … If G does not contain K_4, then … If G does not contain K_4-e, then … The first assertion is known, while the t…

conjecture | open | MathDB 页面 | 原文线索

V2-072166 | MATHDB-373717

The recursive bound for tone chromatic numbers of subcubic outerplanar graphs

摘录 / EXCERPT: Let G be a subcubic outerplanar graph, meaning that G is outerplanar and has maximum degree at most 3. For each positive integer t, let $\tau_t(G)$ denote the smallest numb…

conjecture | open | MathDB 页面 | 原文线索

V2-072167 | MATHDB-373718

Conjecture on the 2-tone chromatic number of cubic Halin graphs

摘录 / EXCERPT: Let G be a cubic Halin graph of order n, where a Halin graph is formed from a tree with no vertices of degree two and a cycle joining its leaves in their planar cyclic order. T…

conjecture | open | MathDB 页面 | 原文线索

V2-072168 | MATHDB-373722

Conforti-Cornuéjols conjecture for square-free monomial ideals

摘录 / EXCERPT: Conforti-Cornuéjols conjecture. All symbolic powers and ordinary powers of I coincide if and only if I has the packing property:

conjecture | open | MathDB 页面 | 原文线索

V2-072169 | MATHDB-373724

Liu-Pego's local convexity conjecture for piecewise affine Lipschitz maps

摘录 / EXCERPT: Liu-Pego's local convexity conjecture. Every such map ϕ must necessarily be locally convex in Ω.

conjecture | open | MathDB 页面 | 原文线索

V2-072170 | MATHDB-373725

Ai-Guo-Freschi-Lo Ore-type conjecture for oriented Hamilton cycles

摘录 / EXCERPT: Ai-Guo-Freschi-Lo conjecture. Let G be an oriented graph on $n\geq 3$ vertices. If $\sigma_2(G)\geq n$, then G contains a Hamilton cycle C such that

conjecture | open | MathDB 页面 | 原文线索

V2-072171 | MATHDB-373726

Cycle-factor extension of the oriented discrepancy conjecture

摘录 / EXCERPT: Cycle-factor extension conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072172 | MATHDB-373730

Gaussian torsional rigidity exponent conjecture for centrally symmetric convex sets

摘录 / EXCERPT: Let Ω_0 and Ω_1 be open, bounded, centrally symmetric subsets of $\mathbb{R}^n$, and let $t\in[0,1]$. Write ... The Gaussian torsional rigidity of a set Ω is den...

conjecture | open | MathDB 页面 | 原文线索

V2-072173 | MATHDB-373735

Non-Parry powers conjecture for roots of x^d-x-1

摘录 / EXCERPT: Non-Parry powers conjecture. For every $d\geq4$ and every integer $k\geq1$, the power $4\beta_d^k4$ is non-Parry.

conjecture | open | MathDB 页面 | 原文线索

V2-072174 | MATHDB-373736

Parry and non-Parry powers conjecture for roots of $x^d-x^{d-1}-1$

摘录 / EXCERPT: Parry-power conjecture. For every $d\geq6$, $4\theta_d4$ is Parry, and for every integer $k\geq2$, $4\theta_d^k4$ is non-Parry.

conjecture | open | MathDB 页面 | 原文线索

V2-072175 | MATHDB-373741

Connectivity of the reduced bumpless pipe dream graph

摘录 / EXCERPT: Connectivity conjecture. For every n, the set of reduced bumpless pipe dreams of size n is connected under the 2×2 flips that preserve reducedness.

conjecture | open | MathDB 页面 | 原文线索

V2-072176 | MATHDB-373742

Convergence to the Schubert permuton

摘录 / EXCERPT: Convergence-to-the-Schubert-permuton conjecture. The random permutations distributed according to this measure converge, as $n\to\infty$, to a deterministic permuton μ on...

conjecture | open | MathDB 页面 | 原文线索

V2-072177 | MATHDB-373743

The RBPD limit-shape and arctic-curve conjecture

摘录 / EXCERPT: RBPD limit-shape and arctic-curve conjecture. As $n\to\infty$, the height function converges to a deterministic limit shape consisting of four frozen regions adjacent to the domain...

conjecture | open | MathDB 页面 | 原文线索

V2-072178 | MATHDB-373752

Neutral-fermion conjecture for factorial gp-functions

摘录 / EXCERPT: Factorial gp-function neutral-fermion conjecture. The factorial gp-function satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072179 | MATHDB-373757

Eltzner's Type I finite sample smeariness conjecture on spheres

摘录 / EXCERPT: Let X be a random variable supported on a set $A\subset \mathbb{S}^m$ whose convex closure has nonzero volume and which has a unique Fréchet mean μ. A point μ is Type I f...

conjecture | open | MathDB 页面 | 原文线索

V2-072180 | MATHDB-373762

Higher-dimensional parallelogram decomposition conjecture for centrally symmetric polyhedral surfaces

摘录 / EXCERPT: Higher-dimensional parallelogram decomposition conjecture. Every surface in

conjecture | open | MathDB 页面 | 原文线索

V2-072181 | MATHDB-373769

Conway-Guttman asymptotics conjecture for occurrences of length-four patterns

摘录 / EXCERPT: Conway-Guttman conjecture. For $q \in \{4321,4312\}$, there is a constant K such that

conjecture | open | MathDB 页面 | 原文线索

V2-072182 | MATHDB-373772

Kelly's minimum semi-degree conjecture for directed cycles

摘录 / EXCERPT: Let $l\geq 4$ be an integer, and let k be the smallest integer greater than 2 that does not divide l. Let $\delta^0(G)$ denote the minimum semi-degree of an oriented graph...

conjecture | open | MathDB 页面 | 原文线索

V2-072183 | MATHDB-373775

Unitary Conjugation Groupoid Conjecture

摘录 / EXCERPT: Unitary Conjugation Groupoid Conjecture. There exists a natural assembly map

conjecture | open | MathDB 页面 | 原文线索

V2-072184 | MATHDB-373777

Stanton's conjecture on symmetric fake Gaussian sequences

摘录 / EXCERPT: Let $\mathbf{a}=(a_1,\ldots,a_n)$ be a sequence of non-negative integers, and let m be a non-negative integer. Consider ... A sequence $\mathbf{a}$ is symmetric when its entries ar...

conjecture | open | MathDB 页面 | 原文线索

V2-072185 | MATHDB-373778

Gatzweiler-Krattenthaler positivity conjecture for quotients of Gaussian binomials

摘录 / EXCERPT: Gatzweiler-Krattenthaler conjecture. If this quotient is a polynomial, then it has non-negative integer coefficients. The conjecture is a positivity assertion for polynomial quotie...

conjecture | open | MathDB 页面 | 原文线索

V2-072186 | MATHDB-373779

Gatzweiler-Krattenthaler conjecture on positivity of Stanton fractions

摘录 / EXCERPT: Gatzweiler-Krattenthaler conjecture. If this fraction is a polynomial in q, then it has non-negative coefficients. This is presented as a strengthening of Stanton's conjecture, r...

conjecture | open | MathDB 页面 | 原文线索

V2-072187 | MATHDB-373780

The unitary dual conjecture for general connected reductive groups

摘录 / EXCERPT: Let F be a non-Archimedean local field of characteristic zero, and let G be a general connected reductive group defined over F. Let $\Pi_A(G)$ denote the set of Arthur repres...

conjecture | open | MathDB 页面 | 原文线索

V2-072188 | MATHDB-373781

The good-parity unitary representation conjecture

摘录 / EXCERPT: Let G be a classical group, and let π be an irreducible unitary representation of G of good parity. A representation is of Arthur type when it belongs to an Arthur packet....

conjecture | open | MathDB 页面 | 原文线索

V2-072189 | MATHDB-373782

The unitary dual conjecture for classical groups

摘录 / EXCERPT: Let G_n be the classical group under consideration. Let $\Pi_{\overline{A}}^{\lim}(G_n)$ and $\Pi_{\overline{A}}^{\lim'}(G_n)$ denote the sets defined in the preceding discussion...

conjecture | open | MathDB 页面 | 原文线索

V2-072190 | MATHDB-373783

Universality of the Euclidean form on affine type C admissible tuples

摘录 / EXCERPT: For an integer $n\geq1$, define ... where the latter two conditions hold for $1\leq i<j\leq n$. The Euclidean form is $\sum_{i=1}^n x_i^2$. Affine type C universality conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-072191 | MATHDB-373785

Quasiconvexity conjecture for the Korn variational integrand

摘录 / EXCERPT: Quasiconvexity conjecture. The displayed function is quasiconvex at some matrix A_0. By homogeneity, quasiconvexity at A_0 implies quasiconvexity at 0. This proposed strength...

conjecture | open | MathDB 页面 | 原文线索

V2-072192 | MATHDB-373795

The trivial units conjecture for torsion-free groups

摘录 / EXCERPT: Let G be a torsion-free group and let R be an integral domain. The trivial units conjecture. G satisfies the trivial units property for R, meaning that the only units in th...

conjecture | open | MathDB 页面 | 原文线索

V2-072193 | MATHDB-373796

The L^2 anti-concentration conjecture for symmetric log-concave measures

摘录 / EXCERPT: Let $d\in\mathbb{N}$. Let H_n be the hyperoctahedral group acting on $\mathbb{R}^n$ by coordinate permutations and sign changes. A measure μ on $\mathbb{R}^n$ is isotropic if...

conjecture | open | MathDB 页面 | 原文线索

V2-072194 | MATHDB-373811

Staircase conjecture for first-occurrence layers in the partition graph

摘录 / EXCERPT: Let $L_r(n)$ be the vertices of local simplex dimension r in the partition graph, let $\tau(r)$ be the least n for which $L_r(n)$ is nonempty, and let...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072195 | MATHDB-373824

Brešar-Klavžar-Rall-Wash conjecture on subdivisions of subcubic graphs

摘录 / EXCERPT: For a graph G, its 1-subdivision is obtained by replacing every edge with a path of two edges. The packing chromatic number $\chi_p(G)$, or PCN, is the smallest positive integer...

conjecture | open | MathDB 页面 | 原文线索

V2-072196 | MATHDB-373829

Boundary regularity conjecture for Hawkes-process resolvent functions

摘录 / EXCERPT: Boundary regularity conjecture. For every $x\in[0,A]$, there exists a version of $s\mapsto p^{![2]}_{r,i,c,k}(x,s,l,j)$ that is continuous at 0 and at A. Moreover, there exist...

conjecture | open | MathDB 页面 | 原文线索

V2-072197 | MATHDB-373830

Kernel non-triviality conjecture for tautological classes of strata of differentials

摘录 / EXCERPT: Let $g\geq 2$, let ℓ be a positive integer, and let $\mu=(m_1,\ldots,m_k)$ be a partition of $\ell(2g-2)$ with $m_i\neq-\ell$ for every i. Let $\mathcal{P}^\ell(\mu)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072198 | MATHDB-373836

Kalai-Keller-Mossel correlation conjecture for increasing balanced families

摘录 / EXCERPT: Kalai-Keller-Mossel correlation conjecture. For every increasing and balanced family $\mathcal{A}\subseteq\Omega_n$, there exists an increasing linear-threshold family...

conjecture | open | MathDB 页面 | 原文线索

V2-072199 | MATHDB-373840

Planar Circular Flow Conjecture

摘录 / EXCERPT: Planar Circular Flow Conjecture. Every $4k$-edge-connected planar graph admits a circular $(2+\frac{1}{k})$-flow.

conjecture | open | MathDB 页面 | 原文线索

V2-072200 | MATHDB-373853

Odabaş's arithmetic exceptionality conjecture for Lattès maps

摘录 / EXCERPT: Let k be a positive integer, let $E/\mathbb{Q}$ be an elliptic curve, and let L_k be the k-th Lattès map attached to E. A rational function is arithmetically exceptional if...

conjecture | open | MathDB 页面 | 原文线索

V2-072201 | MATHDB-373854

Packing chromatic number conjecture for 0-saturated subcubic graphs

摘录 / EXCERPT: Let G be a 0-saturated subcubic graph, and let $g_3(G)$ denote its local girth parameter. Packing chromatic number conjecture. If ... then ... The paper proves the corresponding up...

conjecture | open | MathDB 页面 | 原文线索

V2-072202 | MATHDB-373855

Five-color packing coloring conjecture for 2-saturated subcubic graphs

摘录 / EXCERPT: Let G be a 2-saturated subcubic graph. Five-color packing coloring conjecture. The graph G is $(1,2,2,2,2)$-packing colorable. The paper proves a four-color variant when...

conjecture | open | MathDB 页面 | 原文线索

V2-072203 | MATHDB-373856

Packing coloring conjecture for (3,0)-saturated subcubic graphs

摘录 / EXCERPT: Let G be a $(3,0)$-saturated subcubic graph, and let $g_3(G)$ denote its local girth parameter. Packing coloring conjecture. If ... then G is $(1,1,2)$-packing colorable. The pap...

conjecture | open | MathDB 页面 | 原文线索

V2-072204 | MATHDB-373857

Exceptional packing coloring conjecture for claw-free subcubic graphs

摘录 / EXCERPT: Let G be a claw-free subcubic graph, and let $\mathcal{G}_{11}$ be the graph specified in the paper. Exceptional packing coloring conjecture. If $G\neq\mathcal{G}_{11}$, then G...

conjecture | open | MathDB 页面 | 原文线索

V2-072205 | MATHDB-373858

Seven-color packing coloring conjecture for claw-free subcubic graphs

摘录 / EXCERPT: Let G be a claw-free subcubic graph. Seven-color packing coloring conjecture. The graph G is $(2,2,2,2,2,2,2)$-packing colorable. The paper gives related packing-coloring resul...

conjecture | open | MathDB 页面 | 原文线索

V2-072206 | MATHDB-373863

Superbrick conjecture for globally linked pairs in body-hinge graphs

摘录 / EXCERPT: Let H be a multigraph, let G_H be the body-hinge graph induced by H, and let $d\geq 3$. A k-superbrick is the graph-theoretic object defined in the source for the multigrap...

conjecture | open | MathDB 页面 | 原文线索

V2-072207 | MATHDB-373864

Path-category realization conjecture for spectroids

摘录 / EXCERPT: Path-category realization conjecture. For every spectroid $\mathcal{C}$, there exists a choice of $(\mathbb{X},\Gamma{/}{\sim})$ such that $\mathcal{C}$ is the $\Bbbk$-linear path...

conjecture | open | MathDB 页面 | 原文线索

V2-072208 | MATHDB-373871

Gau-Wang-Wu conjecture about nilpotent partial isometries

摘录 / EXCERPT: Gau-Wang-Wu conjecture. If the numerical range $W(A)$ is a circular disc centered at the origin, then A is rotationally invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-072209 | MATHDB-373873

Asymptotic equidistribution conjecture for modular Ackermann maps

摘录 / EXCERPT: For fixed $m\ge3$, consider the map ... from uniformly chosen $n\in\mathbb Z_{2^k}$ to $\mathbb Z_{2^k}$. Asymptotic equidistribution conjecture. As $k\to\infty$, this map approaches...

conjecture | open | MathDB 页面 | 原文线索

V2-072210 | MATHDB-373874

Hierarchy mixing threshold conjecture for modular Ackermann maps

摘录 / EXCERPT: For powers-of-two moduli 2^k, consider the modular Ackermann map $A_{2^k}(m,n)$ and its output distribution on $\mathbb Z_{2^k}$ as the recursion level m varies. Hierarchy mixi...

conjecture | open | MathDB 页面 | 原文线索

V2-072211 | MATHDB-373881

Computational hardness of consistent moment and cumulant tensor estimation

摘录 / EXCERPT: Let p be the dimension, n the sample size, and d the tensor order. Consider estimating the order-d moment or cumulant tensors of a sub-Gaussian random vector, with consiste...

conjecture | open | MathDB 页面 | 原文线索

V2-072212 | MATHDB-373882

Optimal computational error rate for moment and cumulant tensor estimation

摘录 / EXCERPT: Optimal estimation rate conjecture. The computationally optimal error rate is

conjecture | open | MathDB 页面 | 原文线索

V2-072213 | MATHDB-373891

Functoriality conjecture for wrong-way maps in KK-bordism theory

摘录 / EXCERPT: Functoriality conjecture. The class $\Omega_!(g\circ f)$ should be identified, up to a canonical bordism, with

conjecture | open | MathDB 页面 | 原文线索

V2-072214 | MATHDB-373900

The non-dualizable quasi-monodromic categorical representation conjecture

摘录 / EXCERPT: Let G be a group, let $\Shv(G)^{\on{q-mon}}$ denote the quasi-monodromic sheaf category, and write $G\mmod$ for the category of categorical representations of G. Let ... be the i...

conjecture | open | MathDB 页面 | 原文线索

V2-072215 | MATHDB-373901

The core decomposition equality for polygonal presentations of compacta on surfaces

摘录 / EXCERPT: Core decomposition equality. The core decomposition of K satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072216 | MATHDB-373902

The critical-fugacity asymptotic conjecture for the hard-core model on the lattice

摘录 / EXCERPT: Critical-fugacity asymptotic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072217 | MATHDB-373911

Totient-threshold non-representability conjecture for anharmonic-oscillator Stokes multipliers

摘录 / EXCERPT: Totient threshold conjecture. If $\varphi(M-1)/2\geq 2$, then C_M does not admit a representation

conjecture | open | MathDB 页面 | 原文线索

V2-072218 | MATHDB-373912

Schützenberger's maximal-code commutative equivalence conjecture

摘录 / EXCERPT: Schützenberger's conjecture. Every maximal code is commutatively equivalent to a prefix-free code.

conjecture | open | MathDB 页面 | 原文线索

V2-072219 | MATHDB-373916

Main conjecture on log-concavity of local BPS invariants

摘录 / EXCERPT: Let $(C,0)$ be a reduced planar curve singularity over $\mathbf{k} = \mathbb{C}$, let δ be its δ-invariant, and let $n_h(C,0)$, $0\leq h\leq\delta$, be the BPS invari...

conjecture | open | MathDB 页面 | 原文线索

V2-072220 | MATHDB-373917

Global log-concavity conjecture for BPS invariants of locally planar curves

摘录 / EXCERPT: Let C be an integral projective locally planar curve over $\mathbf{k} = \mathbb{C}$, let $g=g(C)$ and $\widetilde{g}=\widetilde{g}(C)$ be its arithmetic and geometric genera, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072221 | MATHDB-373926

Universal odd critical exponent conjecture for exponential-kernel minimizers

摘录 / EXCERPT: Consider the minimum-energy problem for k points on the unit interval with pairwise kernel $e^{-|x-y|^q}$, and write an odd number of points as ... Let the global critical exponent...

conjecture | open | MathDB 页面 | 原文线索

V2-072222 | MATHDB-373933

Li-Li-Helleseth-Qu's permutation-inverse bent-function conjecture

摘录 / EXCERPT: Let $q=2^e$ with e an even positive integer, let ${\bb F}_{q^2}$ be the finite field of order q^2, and for each $\alpha\in{\bb F}_{q}^*$ define ... where $\sigma(X)=X+X^d+X^{dq}$...

conjecture | open | MathDB 页面 | 原文线索

V2-072223 | MATHDB-373934

Xie-Li-Wang-Zeng's cyclotomic bent-function conjecture

摘录 / EXCERPT: Let $q=2^e$ with e an even positive integer, let u be a generator of μ_{q+1}, and for $\alpha\in{\mathbb F}_{q}^*$ define g_α on ${\mathbb F}_{q^2}$ by...

conjecture | open | MathDB 页面 | 原文线索

V2-072224 | MATHDB-373945

The crystal Temperley-Lieb center conjecture

摘录 / EXCERPT: Crystal Temperley-Lieb center conjecture. The canonical functor G is a monoidal equivalence. This conjecture asserts that the only objects in the Drinfeld center are those induce...

conjecture | open | MathDB 页面 | 原文线索

V2-072225 | MATHDB-373965

Classification conjecture for Seymour Cayley orientations

摘录 / EXCERPT: Cayley classification conjecture. Every Seymour Cayley orientation can be constructed by taking, possibly repeatedly, lexicographic products of empty graphs, powers of directed cyc...

conjecture | open | MathDB 页面 | 原文线索

V2-072226 | MATHDB-373967

Converse-invariance conjecture for regular Seymour-tight orientations

摘录 / EXCERPT: Converse-invariance conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072227 | MATHDB-373968

Picard-rank-one smoothing conjecture for the reconstructed Fano threefolds

摘录 / EXCERPT: For each of the seven specified Noether--Lefschetz divisors in the degree-22 moduli space, let X be the Gorenstein canonical Fano threefold constructed from a general polarized...

conjecture | open | MathDB 页面 | 原文线索

V2-072228 | MATHDB-373986

Conformal-embedding conjecture for folded Dynkin-diagram CFTs

摘录 / EXCERPT: Let $G=ADE$ and let G' be a folding of G. Write $\mathrm{CFT}(A_1,G)$ and $\mathrm{CFT}(A_1,G')$ for the two-dimensional conformal field theories associated with these Dynkin d...

conjecture | open | MathDB 页面 | 原文线索

V2-072229 | MATHDB-373989

Nonnegative BH representation conjecture for Eigen-CG inequalities

摘录 / EXCERPT: For any $(v_0,v)\in\mathbb{R}^{n+1}$, let $\text{E-CG}(v_0,v)$ denote the corresponding Eigen-CG inequality, and let BH inequalities denote the Boros-Hammer inequalities. Nonnegati...

conjecture | open | MathDB 页面 | 原文线索

V2-072230 | MATHDB-373992

Infinite equalizer conjecture for free-group endomorphisms

摘录 / EXCERPT: Infinite equalizer conjecture. There is an upper bound, depending only on n, for the number of $\operatorname{Eq}(\phi,\psi)$-orbits of the regular points of...

conjecture | open | MathDB 页面 | 原文线索

V2-072231 | MATHDB-373996

Permanent anti-concentration conjecture for complex Gaussian matrices

摘录 / EXCERPT: Let W be an $n\times n$ matrix with independent entries distributed as $N(0,1)_{\mathbb C}$, and let p be a polynomial in its arguments. Permanent anti-concentration conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072232 | MATHDB-374006

Dallard et al.'s forbidden-path and biclique conjecture for tree-independence number

摘录 / EXCERPT: For a family $\mathcal{F}$ of graphs, a graph G is $\mathcal{F}$-free if no induced subgraph of G is isomorphic to a graph in $\mathcal{F}$. Let P_r be the path with r vert...

conjecture | open | MathDB 页面 | 原文线索

V2-072233 | MATHDB-374009

Congruence conjecture for c(32n+23) modulo 8

摘录 / EXCERPT: Let $c(n)$ denote the coefficient sequence under consideration, and let $n\in\mathbb{N}_0$. The congruence conjecture for $c(32n+23)$. ... This conjecture is proposed as an open ques...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072234 | MATHDB-374010

Infinite family of congruence conjectures for c(n)

摘录 / EXCERPT: Let $c(n)$ denote the coefficient sequence under consideration, and let $k,n\in\mathbb{N}_0$. The infinite-family congruence conjecture for $c(n)$. ... ... The authors propose these...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072235 | MATHDB-374019

Recursive criterion for record-equivalence of permutation patterns

摘录 / EXCERPT: Recursive criterion for record-equivalence. The permutations π and σ are record-equivalent if and only if all of the following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-072236 | MATHDB-374021

The toroidal efficient total coloring conjecture

摘录 / EXCERPT: Let $M(G)$ be a toroidal cubic map whose 1-skeleton is a simple cubic graph of girth 4. Call G toroidally 3-edge connected when the bicutout condition defined in the source hol...

conjecture | open | MathDB 页面 | 原文线索

V2-072237 | MATHDB-374024

Alexander polynomial conjecture for the monodromy of associated surface singularities

摘录 / EXCERPT: Let C_0 and C_1 be the curves and let g_0 and g_1 be the polynomial function germs associated with them. Write $V(g_j)$ for the zero locus of g_j and let the monodromy of...

conjecture | open | MathDB 页面 | 原文线索

V2-072238 | MATHDB-374026

The stable-Arthur-packet conjecture for the first failure of Adams functoriality

摘录 / EXCERPT: Let (G,H) be a dual pair, let π be a representation of G of Arthur type, and let $\psi^{\max}(\pi)$ be the maximal Arthur parameter attached to π. Let α_0 be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072239 | MATHDB-374027

The refined Adams conjecture for ABV-packets

摘录 / EXCERPT: The refined Adams conjecture for ABV-packets. If $\pi\in\Pi_\phi^{\mathrm{ABV}}$, then: (1) for $\alpha\gg0$, $\theta_{-\alpha}^\pm(\pi)\in\Pi_{\phi_\alpha}^{\mathrm{ABV}}$; (2) if...

conjecture | open | MathDB 页面 | 原文线索

V2-072240 | MATHDB-374028

The closure-monotonicity conjecture for Adams lifts

摘录 / EXCERPT: Let $G=G(W_n)$ and let $\psi,\psi'\in\Psi^+(G_n)$ be local Arthur parameters with $\psi\geq_C\psi'$. Suppose that $\pi\in\Pi_\psi\cap\Pi_{\psi'}$. The closure-monotonicity conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-072241 | MATHDB-374039

The strict descent conjecture for convex polygons

摘录 / EXCERPT: Strict descent conjecture. Every convex polygon satisfies the strict descent condition. The condition is introduced to construct electrostatic skeletons where different Schwarz ref...

conjecture | open | MathDB 页面 | 原文线索

V2-072242 | MATHDB-374042

Codimension bound conjecture for transmission loci

摘录 / EXCERPT: Codimension bound conjecture. Every component of $W^\tau(C,p,q)$ has codimension at most $\operatorname{inv}_k(\tau)$.

conjecture | open | MathDB 页面 | 原文线索

V2-072243 | MATHDB-374043

Larson-Larson-Vogt intersection-class conjecture for transmission loci

摘录 / EXCERPT: Larson-Larson-Vogt intersection-class conjecture. All statements of Larson-Larson-Vogt's Theorem 1.2 should generalize from splitting loci to transmission loci of a general point i...

conjecture | open | MathDB 页面 | 原文线索

V2-072244 | MATHDB-374044

Relative codimension bound conjecture for transmission loci

摘录 / EXCERPT: Relative codimension bound conjecture. Every component of $W^\tau(\pi,p,q)$ has codimension at most $\operatorname{inv}_k(\tau)$.

conjecture | open | MathDB 页面 | 原文线索

V2-072245 | MATHDB-374045

Hecke-word Euler characteristic conjecture for transmission loci

摘录 / EXCERPT: Hecke-word Euler characteristic conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072246 | MATHDB-374046

Fill's characterization conjecture for minimizers of the adjacent-transposition spectral gap

摘录 / EXCERPT: Fill's characterization conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-072247 | MATHDB-374047

Fill's multiplicity conjecture for minimizing adjacent-transposition walks

摘录 / EXCERPT: Let $n\geq 3$ and let $\vec{p}$ be a regular parameter vector. A label is neutral if $p_{c,i}=p_{i,c}=1/2$ for every $i\neq c$. Let ... be the number of neutral labels, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072248 | MATHDB-374052

Nevo-Petersen conjecture on gamma-vectors of flag simplicial spheres

摘录 / EXCERPT: Let Δ be a flag simplicial sphere of dimension $d-1$, and let ... be the coefficients in its gamma expansion. Nevo-Petersen conjecture. There exists a simplicial complex...

conjecture | open | MathDB 页面 | 原文线索

V2-072249 | MATHDB-374057

Universal error-function edge kernel conjecture for pluricomplex polynomial Bergman kernels

摘录 / EXCERPT: Let $d\geq 1$, let $\mathscr Q:\mathbb C^d\to\mathbb R$ be C^2 and strictly plurisubharmonic, and let $S_{\mathscr Q}$ be its droplet with smooth boundary. For...

conjecture | open | MathDB 页面 | 原文线索

V2-072250 | MATHDB-374058

Universal multivariate error-function edge kernel conjecture

摘录 / EXCERPT: Let $d\geq 1$, let $\mathscr Q:\mathbb C^d\to\mathbb R$ be C^2 and strictly plurisubharmonic, and suppose that its droplet $S_{\mathscr Q}$ has smooth boundary. For each...

conjecture | open | MathDB 页面 | 原文线索

V2-072251 | MATHDB-374073

Backward propagation conjecture for cubically hyponormal matrix-valued weighted shifts

摘录 / EXCERPT: Let $W_{\mathcal A}$ be a cubically hyponormal matrix-valued weighted shift, and let $x\in\mathcal H$. Suppose that ... for some $k\geq 1$. Backward propagation conjecture. Then ... fo...

conjecture | open | MathDB 页面 | 原文线索

V2-072252 | MATHDB-374074

Banerjee-Bringmann-Bachraoui's family of congruence conjectures for two-color partition numbers

摘录 / EXCERPT: Banerjee-Bringmann-Bachraoui's conjecture. For all integers n and k,

conjecture | open | MathDB 页面 | 原文线索

V2-072253 | MATHDB-374078

Asymptotic semistability conjecture for syzygy bundles of Ulrich bundles

摘录 / EXCERPT: Asymptotic semistability conjecture. There exists an integer $m_0\gg0$ such that, for every $m\geq m_0$, the syzygy bundle $S_{\mathcal{E}(m)}$ is semistable. This is proposed as a...

conjecture | open | MathDB 页面 | 原文线索

V2-072254 | MATHDB-374090

The identity relating the double Lambert series to $D_2(q)-D_1(q)$

摘录 / EXCERPT: Identity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072255 | MATHDB-374094

The exact-order conjecture for hypercube saturation

摘录 / EXCERPT: Hypercube saturation conjecture. For every $k\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-072256 | MATHDB-374107

Planted Jensen-gap conjecture for the Random Subsequence Model

摘录 / EXCERPT: Let $f_{\mathrm{pl}}(\alpha)$ and $f_{\mathrm{pl}}^{\mathrm{ann}}(\alpha)$ denote the planted quenched and annealed free energies, respectively, for aspect ratio α. Planted...

conjecture | open | MathDB 页面 | 原文线索

V2-072257 | MATHDB-374108

Mean-field and rank-one free-energy comparison conjecture

摘录 / EXCERPT: For $\alpha\in[0,1]$, let $f_{\mathrm{null}}(\alpha)$ be the quenched free energy of the null Random Subsequence Model, let $f^{\mathrm{BMM}}(\alpha)$ be the quenched free energy o...

conjecture | open | MathDB 页面 | 原文线索

V2-072258 | MATHDB-374109

Roé-Urbinati's b-divisorial valuation conjecture

摘录 / EXCERPT: Roé-Urbinati's conjecture. The valuation v is b-divisorial if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-072259 | MATHDB-374113

Andrews' growth conjecture for the coefficients of $v_1(q)$

摘录 / EXCERPT: Andrews' Conjecture 3. The coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-072260 | MATHDB-374118

Logarithmically corrected Edwards-Wilkinson scaling for the directed exclusion process

摘录 / EXCERPT: Let $({\mathcal Y}_t^\varepsilon)_{t\ge 0}$ be the fluctuation field of the directed exclusion process, with $s_\varepsilon(t)$ determined by ... The field ... for test functions...

conjecture | open | MathDB 页面 | 原文线索

V2-072261 | MATHDB-374120

Andrews-Dixit-Schultz-Yee oddness conjecture for a double Lambert series

摘录 / EXCERPT: Let ... Here q is a complex number with $|q|<1$, so the double Lambert series defines an analytic function in its domain of convergence.
Andrews-Dixit-Schultz-Yee conjecture. The f...

conjecture | open | MathDB 页面 | 原文线索

V2-072262 | MATHDB-374122

Parity-dependent real-root conjecture for domination polynomials of book graphs

摘录 / EXCERPT: Let B_n be the book graph with parameter n, and let $D(B_n,x)$ denote its domination polynomial. The numerical data suggest the following parity-dependent statement. Real-root...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072263 | MATHDB-374124

Sharp-threshold conjecture for reconstructible subsets near the random-graph critical point

摘录 / EXCERPT: Let $V\subseteq\mathbb{R}$ with $|V|=n$, and let the known distances be distributed as in $G(V,\lambda/n)$ for a parameter $\lambda>0$. Call a subset reconstructible if every injec...

conjecture | open | MathDB 页面 | 原文线索

V2-072264 | MATHDB-374136

Stable atom positivity conjectures for nonsymmetric LLT expressions

摘录 / EXCERPT: Let stable atoms be the basis elements ${\mathcal A}_{\eta|\lambda}$ defined from Demazure atoms, and say that an element is stable atom positive when it is a nonnegative linear co...

conjecture | open | MathDB 页面 | 原文线索

V2-072265 | MATHDB-374140

Jones-Reutter's conjectural fiber sequence for monoidal 2-categories

摘录 / EXCERPT: Let $\mathfrak C$ be a monoidal $\mathbb K$-linear 2-category, subject to suitable assumptions. Write $\mathfrak C^\times$ for its subgroupoid of invertible objects, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072266 | MATHDB-374143

The conjecture on Aubert-Zelevinsky involutions of ABV-packets

摘录 / EXCERPT: Let G be a reductive group over a non-Archimedean field, let ϕ be an L-parameter, and let $\Pi_{\phi}^{\mathrm{ABV}}$ denote its ABV-packet. Let $\widehat{\phi}$ be the Py...

conjecture | open | MathDB 页面 | 原文线索

V2-072267 | MATHDB-374144

Strict separation of Robin and Weyl critical parameters

摘录 / EXCERPT: Let $\beta(\gamma,d)$ be the critical Robin parameter from Theorem, and let $\beta_W(\gamma,d-1)$ be the point where the second term in the two-term Weyl formula changes sign. Stri...

conjecture | open | MathDB 页面 | 原文线索

V2-072268 | MATHDB-374145

Beatson-Castell-Xu conjecture on truncated powers on spheres

摘录 / EXCERPT: Let $d\geq 1$, set $\lambda=\frac{d-1}{2}$, and let ${\theta}-t)_+=\max\{{\theta}-t,0\}$. For $\delta\geq\lambda+1$ and any ${\theta}\in(0,\pi)$, define ... Beatson-Castell-Xu conje...

conjecture | open | MathDB 页面 | 原文线索

V2-072269 | MATHDB-374147

The eigenvalue-branch bound for the Dirichlet-to-Neumann map

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a bounded Lipschitz domain, and let $\sigma^{(\Lambda)}$, $\Lambda\le 0$, be a fixed analytic eigenvalue branch of the Dirichlet-to-Neumann opera...

conjecture | open | MathDB 页面 | 原文线索

V2-072270 | MATHDB-374148

Chaigneau's asymptotic conjecture for curvilinear polygons

摘录 / EXCERPT: Let a curvilinear polygon have n vertices with interior angles $\alpha_1,\dots,\alpha_n$. For $\alpha\in(0,2\pi)$, define ... and ... Form the multiset...

conjecture | open | MathDB 页面 | 原文线索

V2-072271 | MATHDB-374160

Emerton-Gee-Hellmann's categorical p-adic local Langlands conjecture

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}_p$. Write $\mathcal{D}(\ast/GL_n(K)^{\la})$ for the stable ∞-category of locally analytic representations of $GL_n(K)$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072272 | MATHDB-374162

Parity bias conjecture for missing integers in partitions

摘录 / EXCERPT: Let $P(n,m)$ denote the number of partitions of n with m missing integers. Define ... Thus, $\mathcal{M}_e(n)$ and $\mathcal{M}_o(n)$ count partitions of n with an even and odd...

conjecture | open | MathDB 页面 | 原文线索

V2-072273 | MATHDB-374163

Distinct-part bound conjecture for the missing-integer parity difference

摘录 / EXCERPT: Let $P(n,m)$ denote the number of partitions of n with m missing integers, and define ... Let $q(n)$ denote the number of partitions of n into distinct parts. Distinct-part bou...

conjecture | open | MathDB 页面 | 原文线索

V2-072274 | MATHDB-374164

Parity bias conjecture for missing integers in overpartitions

摘录 / EXCERPT: Let $\overline{P}(n,m)$ denote the number of overpartitions of n with m missing integers. Define $\overline{\mathcal{M}}_e(n)$ and $\overline{\mathcal{M}}_o(n)$ to be the numbe...

conjecture | open | MathDB 页面 | 原文线索

V2-072275 | MATHDB-374181

Meyer's conjecture on the equitable chromatic number

摘录 / EXCERPT: Meyer's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072276 | MATHDB-374182

Chen-Lih-Wu conjecture on the equitable chromatic threshold

摘录 / EXCERPT: Chen-Lih-Wu conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072277 | MATHDB-374183

Zhang-Zhang conjecture on equitable degenerate colouring

摘录 / EXCERPT: Zhang-Zhang conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072278 | MATHDB-374193

Chromatic number conjecture for powers of random graphs

摘录 / EXCERPT: Let $G_{n,p}$ be the random graph on n vertices, let $G_{n,p}^r$ denote its rth power, and write $\chi(G)$ and $\alpha(G)$ for the chromatic and independence numbers of a graph...

conjecture | open | MathDB 页面 | 原文线索

V2-072279 | MATHDB-374194

Cubic universality conjecture for quasi-complementary sequence sets

摘录 / EXCERPT: Let a quasi-complementary sequence set (QCSS) have set size M, flock size K, sequence length N, and tightness factor ρ. It is asymptotically near-optimal when its correl...

conjecture | open | MathDB 页面 | 原文线索

V2-072280 | MATHDB-374202

Simpson's completeness conjecture for the Lagrangians in the de Rham moduli space

摘录 / EXCERPT: Let $W^1(\bar\partial_0,\Phi_0)$ be one of the Lagrangians arising from the conformal-limit construction, and let M_{dR} be the de Rham moduli space. Simpson's conjecture. The La...

conjecture | open | MathDB 页面 | 原文线索

V2-072281 | MATHDB-374217

The 1-2-3 conjecture for locally finite polygonal tilings

摘录 / EXCERPT: Let $\mathcal T$ be a locally finite tiling with a finite number of polygonal prototiles. Suppose ... is a partition into translations of finitely many, not necessarily distinct, fin...

conjecture | open | MathDB 页面 | 原文线索

V2-072282 | MATHDB-374218

FK representation conjecture for Ashkin-Teller spins

摘录 / EXCERPT: The Ashkin-Teller model on a box Λ_n has configurations $(\sigma_1,\sigma_2):\Lambda_n\to\{-1,+1\}^2$ with density proportional to ... On the self-dual line, assume $J\geq0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-072283 | MATHDB-374223

Compactness of the minimal elements of quasicomplemented posets

摘录 / EXCERPT: Compactness conjecture. Is $Min(Q)$ compact?

open_problem | open | MathDB 页面 | 原文线索

V2-072284 | MATHDB-374233

The interval conjecture for general position spectra

摘录 / EXCERPT: Let G be a graph, and let ... be its general position spectrum. The interval-spectrum conjecture. The set $S_{{\rm gp}}(G)$ is an interval of integers. The spectrum is an interval...

conjecture | open | MathDB 页面 | 原文线索

V2-072285 | MATHDB-374235

Coefficient majorization conjecture for the class $cmathcal{B}_1(\alpha)$

摘录 / EXCERPT: Let S denote the class of normalized univalent functions. For $0\leq\alpha\leq1$, suppose that $g\in S$ has the expansion $g(z)=z+b_2z^2+\cdots$ and satisfies ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072286 | MATHDB-374255

Charney-Davis conjecture for flag odd-dimensional spheres

摘录 / EXCERPT: Let $k\geq 2$. A flag sphere is a simplicial sphere with no missing faces of dimension larger than one, and $g_0,g_1,\ldots,g_k$ are its g-numbers. Charney-Davis conjecture. Ever...

conjecture | open | MathDB 页面 | 原文线索

V2-072287 | MATHDB-374256

Affine stress reconstruction conjecture for spheres without large missing faces

摘录 / EXCERPT: Assume $u\leq d/2$. Let (Δ,p) be a $(d-1)$-sphere in $S(d-u,d-1)$, with p a generic or natural embedding in $\mathbb{R}^d$. Let $\mathcal{S}^a_i(\Delta,p)$ denote the spac...

conjecture | open | MathDB 页面 | 原文线索

V2-072288 | MATHDB-374257

Affine stress support conjecture for spheres without large missing faces

摘录 / EXCERPT: Let $2\leq i\leq d/2$, let $\Delta\in S(d-i,d-1)$, and let p be a generic or natural embedding of Δ in $\mathbb{R}^d$. An affine i-stress is an element of the affine str...

conjecture | open | MathDB 页面 | 原文线索

V2-072289 | MATHDB-374273

Modified Ballantine-Beck-Merca injectivity conjecture for pre_k

摘录 / EXCERPT: Modified Ballantine-Beck-Merca conjecture. For $k\geq 3$, pre_k is injective on partitions of n with j parts for each $j\geq k+1$.

conjecture | refuted | MathDB 页面 | 原文线索

V2-072290 | MATHDB-374288

Dimca-Greuel's $4/3$ conjecture for plane curve singularities

摘录 / EXCERPT: Let $f\in\mathbb{C}[[x,y]]$ be an isolated hypersurface singularity, with Milnor number μ and Tjurina number τ. Dimca-Greuel's conjecture. ... The conjecture was proved comp...

conjecture | open | MathDB 页面 | 原文线索

V2-072291 | MATHDB-374289

Almirón's $3/2$ conjecture for surface hypersurface singularities

摘录 / EXCERPT: Let $f\in\mathbb{C}[[x,y,z]]$ be an isolated hypersurface singularity, with Milnor number μ and Tjurina number τ. Almirón's conjecture. ... This is described as a weak versi...

conjecture | open | MathDB 页面 | 原文线索

V2-072292 | MATHDB-374290

Durfee's conjecture for isolated surface hypersurface singularities

摘录 / EXCERPT: Let $f\in\mathbb{C}[[x,y,z]]$ be an isolated hypersurface singularity. Let μ be its Milnor number and p_g its geometric genus. Durfee's conjecture. ... with equality only when...

conjecture | open | MathDB 页面 | 原文线索

V2-072293 | MATHDB-374294

Generic automorphism conjecture for hyperelliptic curves of 2-rank zero

摘录 / EXCERPT: Let C be a generic hyperelliptic curve of genus g and 2-rank 0 over a field of characteristic 2. Generic automorphism conjecture for 2-rank-zero hyperelliptic curves. If...

conjecture | open | MathDB 页面 | 原文线索

V2-072294 | MATHDB-374295

Refined Oort analogue for generic supersingular hyperelliptic curves

摘录 / EXCERPT: Let C be a generic supersingular hyperelliptic curve of genus g over a field of characteristic 2. Refined hyperelliptic Oort conjecture. If g is not a power of 2, then .....

conjecture | open | MathDB 页面 | 原文线索

V2-072295 | MATHDB-374296

Regular bipartite switch-connectivity threshold conjecture

摘录 / EXCERPT: Let G be a d-regular balanced bipartite graph on $2n$ vertices, and let $f4d8^{\mathbf{connect}}_{k,\mathrm{reg}}(n)$ denote the minimum-degree threshold for the k-switch gra...

conjecture | open | MathDB 页面 | 原文线索

V2-072296 | MATHDB-374319

Symmetric binary perceptron ultrametric OGP-parametric RDT equivalence conjecture

摘录 / EXCERPT: Let $\mathbf{\mathcal S}(G,\kappa,\alpha)$ be a statistical symmetric binary perceptron, let $\alpha_{ult_s}(\kappa)$ be its s-level ultrametric OGP critical density, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072297 | MATHDB-374320

Finite-level ultrametric OGP-parametric RDT comparison conjecture for the symmetric binary perceptron

摘录 / EXCERPT: Let $\mathbf{\mathcal S}(G,\kappa,\alpha)$ be a statistical symmetric binary perceptron, let $\alpha_{ult_s}(\kappa)$ be its s-level ultrametric OGP critical density, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072298 | MATHDB-374322

The continuous-circle logarithmic greedy-routing conjecture

摘录 / EXCERPT: Continuous-circle greedy-routing conjecture. For every $\varepsilon>0$, there exists a constant $K_0=K_0(\varepsilon)$ such that, for all $K\geq K_0$ and sufficiently large n, wi...

conjecture | open | MathDB 页面 | 原文线索

V2-072299 | MATHDB-374324

The conjecture that the entropy sum lower-bound constant can be improved to one-half

摘录 / EXCERPT: Entropy sum lower-bound conjecture. The constant $1/8$ can be improved to $1/2$, meaning that

conjecture | open | MathDB 页面 | 原文线索

V2-072300 | MATHDB-374331

Banded perturbation conjecture for twisted Toeplitz matrices

摘录 / EXCERPT: Banded perturbation conjecture. The measures μ_n converge weakly to

conjecture | open | MathDB 页面 | 原文线索

V2-072301 | MATHDB-374332

Randomised twisted Toeplitz spectral-measure conjecture

摘录 / EXCERPT: Randomised twisted Toeplitz spectral-measure conjecture. The measures μ_n converge weakly to

conjecture | open | MathDB 页面 | 原文线索

V2-072302 | MATHDB-374340

The mixed edge-deletion conjecture for majority C-chromatic edge-critical graphs

摘录 / EXCERPT: Let G be an $\overline{\chi}_{\geqslant}$-edge-critical graph with at least two edges, and let $e,f\in E(G)$. The mixed edge-deletion conjecture. It is impossible that ... The clai...

conjecture | open | MathDB 页面 | 原文线索

V2-072303 | MATHDB-374362

Treibich's uniform count conjecture for two moving poles

摘录 / EXCERPT: For nonnegative integer weights (n_0,n_1,n_2,n_3), let $T(n_0,n_1,n_2,n_3;2)$ denote the number of unordered configurations $\{\pm p_1,\pm p_2\}$ satisfying the finite-gap equili...

conjecture | open | MathDB 页面 | 原文线索

V2-072304 | MATHDB-374368

Monotonic growth-constant conjecture for 132-avoiding permutations with bounded adjacency

摘录 / EXCERPT: Monotonic growth-constant conjecture. For fixed m, the limit α_m exists, and

conjecture | open | MathDB 页面 | 原文线索

V2-072305 | MATHDB-374369

A conjectural divided-difference formula for products of operators

摘录 / EXCERPT: Let i and n be positive integers, let z_j be variables, let $\mathcal{D}_j$ be the operators appearing in the modified Schubert-polynomial formalism, and let ∂_j den...

conjecture | open | MathDB 页面 | 原文线索

V2-072306 | MATHDB-374373

Bisection threshold conjecture for p-orientations

摘录 / EXCERPT: For each $d\geq3$, let $p^*(d)$ be the minimum value of $p\in\{1,2,\ldots,\lfloor d/2\rfloor\}$ such that a.a.s. the random graph $\mathcal{G}_{n,d}$ has a p-orientation. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-072307 | MATHDB-374376

Stacking-estimating equality for finite connected trees

摘录 / EXCERPT: Let T be a finite connected tree. Stacking-estimating conjecture. ... The statement is presented as an immediate consequence of the preceding theorem under ASH, which gives the cor...

conjecture | open | MathDB 页面 | 原文线索

V2-072308 | MATHDB-374384

Visibility Density Conjecture for polynomial lines of sight

摘录 / EXCERPT: Visibility Density Conjecture. If $F(x)\in\mathbb{Z}[x]$ has at least two distinct roots, then the set of lattice points in $\mathbb{Z}_{>0}\times\mathbb{Z}_{>0}$ visible along...

conjecture | open | MathDB 页面 | 原文线索

V2-072309 | MATHDB-374390

Ding et al.'s Bose-distance conjecture for primitive BCH codes

摘录 / EXCERPT: Ding et al.'s conjecture. The minimum distance is always equal to the Bose distance:

conjecture | open | MathDB 页面 | 原文线索

V2-072310 | MATHDB-374394

Boundedness of complements for normalized foliated structures

摘录 / EXCERPT: Let d be a positive integer, and let $\Gamma\subset [0,1]$ be a DCC set of rational numbers whose closure is contained in $\mathbb Q$. A projective $\mathbb R$-complementary norm...

conjecture | open | MathDB 页面 | 原文线索

V2-072311 | MATHDB-374395

Chollet's permanent conjecture for positive semidefinite matrices

摘录 / EXCERPT: Let $A,B e 0$ be Hermitian positive semidefinite matrices of the same size, and let $A\circ B$ denote their Hadamard product. Chollet's conjecture. ... The conjecture is a permanent...

conjecture | open | MathDB 页面 | 原文线索

V2-072312 | MATHDB-374399

Sharp rank-$2n-2$ equalizer bound for monomorphisms

摘录 / EXCERPT: Let $n\ge 3$, let F_n be a free group of rank n, and let F be a free group. Let $g,h:F_n\to F$ be monomorphisms. Sharp equalizer rank conjecture. Then ... The paper constructs...

conjecture | open | MathDB 页面 | 原文线索

V2-072313 | MATHDB-374410

Elias's acyclicity and unique extrema conjecture for affine reduced-word graphs

摘录 / EXCERPT: For an affine permutation $w\in \widetilde{\mathfrak{S}}_n$, let $\mathcal{G}(w)$ be the directed graph whose vertices are commutation classes of reduced words for w, with edges...

conjecture | open | MathDB 页面 | 原文线索

V2-072314 | MATHDB-374411

Acyclicity conjecture for graphs of affine reversal sets

摘录 / EXCERPT: Let $w\in \widetilde{\mathfrak{S}}_n$, let $\mathcal{C}_w(n,k)$ denote the relevant collection of consistent subsets, and let $R\in \mathcal{C}_w(n,k)$. Define G_R by including e...

conjecture | open | MathDB 页面 | 原文线索

V2-072315 | MATHDB-374416

Uniform quasiconformality conjecture for CaTherine wheels from expanding Thurston maps

摘录 / EXCERPT: Let $f:S^1\to\mathbb{CP}^1$ be a CaTherine wheel arising from an expanding Thurston map. Uniform quasiconformality conjecture. There exists a finite constant K such that f is a...

conjecture | open | MathDB 页面 | 原文线索

V2-072316 | MATHDB-374434

Atom-positivity conjecture for row division of key polynomials

摘录 / EXCERPT: Atom-positivity conjecture. The row operator applied to a key polynomial is atom-positive:

conjecture | open | MathDB 页面 | 原文线索

V2-072317 | MATHDB-374436

The GOE level-crossing law for real orthogonal matrices

摘录 / EXCERPT: Let A and B be independently chosen from the real Gaussian orthogonal ensemble $GOE_n^{\mathbb R}$, consisting of real symmetric $n\times n$ matrices with the standard GOE dist...

conjecture | open | MathDB 页面 | 原文线索

V2-072318 | MATHDB-374437

Pseudocovariance-independence conjecture for GOE level crossings

摘录 / EXCERPT: Let A_n and B_n be independent GOE matrices, and write the normalized complex Gaussian pencil as $\widehat C_n(\lambda)=(A_n+\lambda B_n)/\sqrt{n(1+|\lambda|^2)}$. Its entry ps...

conjecture | open | MathDB 页面 | 原文线索

V2-072319 | MATHDB-374438

General real i.i.d. level-crossing law

摘录 / EXCERPT: Let $A_n=(a_{ij})$ and $B_n=(b_{ij})$ be independent real $n\times n$ matrices with i.i.d. entries satisfying ... and assume a standard moment condition strong enough to imply the ci...

conjecture | open | MathDB 页面 | 原文线索

V2-072320 | MATHDB-374439

Possible non-universal corrections for constrained real ensembles

摘录 / EXCERPT: Consider real matrix ensembles with additional linear constraints, including real symmetric or other structured real matrices. Let $\rho(\lambda)\,dx\,dy$ denote an absolutely cont...

conjecture | open | MathDB 页面 | 原文线索

V2-072321 | MATHDB-374440

Existence of the limiting GUE level-crossing density

摘录 / EXCERPT: For each n, let $\mathcal P_n^{GUE}(\theta,\phi)$ denote the level-crossing density for a pencil of independent GUE matrices, expressed in spherical coordinates (θ,ϕ) o...

conjecture | open | MathDB 页面 | 原文线索

V2-072322 | MATHDB-374442

Potential-theoretic form of the Hermitian level-crossing limit

摘录 / EXCERPT: Let $\lambda\in\mathbb C$ be an affine coordinate on $\mathbb{CP}^1$, let ... and let $G(q)$ denote the logarithmic energy of the elliptic law with elliptic parameter $\sqrt q$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072323 | MATHDB-374489

Puncturing-inclusion conjecture for MRD codes

摘录 / EXCERPT: Let $n,d\in\mathbb N$ satisfy $1<d<n$, and let $\mathbb F$ be a finite field. An MRD code is a maximum rank-distance code. Puncturing-inclusion conjecture. There exists an...

open_problem | open | MathDB 页面 | 原文线索

V2-072324 | MATHDB-374491

The maximal Bochner-Riesz conjecture on Métivier groups

摘录 / EXCERPT: Let G be a Métivier group with sub-Laplacian $\mathcal{L}$, and let $S_*^{\alpha}(\mathcal{L})$ denote the maximal Bochner-Riesz operator associated with $\mathcal{L}$. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-072325 | MATHDB-374501

Conjecture on simultaneous supersingular reduction for elliptic-curve pairs

摘录 / EXCERPT: Let E and E' be two elliptic curves defined over a number field K. Assume that E and E' are non-CM and non-isogenous over $\overline{\mathbb{Q}}$. Let ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-072326 | MATHDB-374504

Finite q-duality conjecture for words in three letters

摘录 / EXCERPT: Finite q-duality conjecture. For every word $w=u_{1}\cdots u_{k}\in\mathfrak{h}^{0}$,

conjecture | open | MathDB 页面 | 原文线索

V2-072327 | MATHDB-374510

Dallard-Milanič-Štorgel tree-independence conjecture for hereditary classes

摘录 / EXCERPT: Let $\mathcal{G}$ be a hereditary class of graphs. A class has bounded tree-independence number if there is a constant $c\in\mathbb{N}$ such that every graph in the class admits a...

conjecture | open | MathDB 页面 | 原文线索

V2-072328 | MATHDB-374511

Dallard-Krnc-Kwon-Milanič-Munaro-Štorgel-Wiederrecht conjecture for finitely forbidden induced subgraphs

摘录 / EXCERPT: Let $\mathcal{F}$ be a finite class of graphs. A graph is $\mathcal{F}$-free if it has no induced subgraph isomorphic to any member of $\mathcal{F}$. The class of all $\mathcal{F}$...

conjecture | open | MathDB 页面 | 原文线索

V2-072329 | MATHDB-374519

Prasad and Ram's subspace enumeration formula for three equal parts

摘录 / EXCERPT: Prasad and Ram's conjecture. For any $T\in M_{3m}(\mathbb{F}_q)$,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072330 | MATHDB-374527

The F1 classification conjecture for Niho type locally-APN power functions

摘录 / EXCERPT: Let $n=2m$. For integers $r<m$ with $\gcd(r,m)=1$, define s by ... Set $F_1(x)=x^{s(2^m-1)+1}$ over $\mathbb{F}_{2^{2m}}$. A power function is Niho type locally-APN if it is both o...

conjecture | open | MathDB 页面 | 原文线索

V2-072331 | MATHDB-374529

Associated-connection conjecture for Lie group fiber bundles

摘录 / EXCERPT: Associated-connection conjecture. The connection η can be seen as an associated connection to a principal connection on $(\hat{P}, M, \pi_{\hat{P}, M}, \operatorname{Aut}(G))$...

conjecture | open | MathDB 页面 | 原文线索

V2-072332 | MATHDB-374533

Curto-Hwang-Lee conjecture on hyponormal block Toeplitz operators

摘录 / EXCERPT: Curto-Hwang-Lee conjecture. There exists a finite Blaschke-Potapov product $B\in\mathcal{E}(\Phi)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-072333 | MATHDB-374534

Andrews et al.'s fermionic formula conjecture for the Macdonald index

摘录 / EXCERPT: Andrews et al.'s conjecture. The specialization of the Macdonald index at $z=1$ satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072334 | MATHDB-374537

Palindromic symmetry conjecture for recurrence coefficients in superelliptic central extensions

摘录 / EXCERPT: Let $A=\mathbb{C}[x^{\pm1},u\mid u^2=P(x)]$, where $P(x)\in\mathbb{C}[x]$ is palindromic of even degree $2r$ with simple roots. Let $A/\partial A$ denote the quotient by exact deri...

conjecture | open | MathDB 页面 | 原文线索

V2-072335 | MATHDB-374562

Phase transition in LAWS convergence

摘录 / EXCERPT: Let $P_{\mathcal{M}}$ be a stationary distribution with entropy H, let H_N denote the LAWS hit rate after N total queries, and let $N_{\min}$ be the per-node visit threshold....

conjecture | open | MathDB 页面 | 原文线索

V2-072336 | MATHDB-374564

Optimal chunking at surprisal peaks

摘录 / EXCERPT: Let t_i be the token at position i, let $H_i=H(t_i\mid t_{<i})$ be its conditional entropy, and let $\mathcal{T}$ be the LAWS trie with description length $L(\mathcal{T})$. Opt...

conjecture | open | MathDB 页面 | 原文线索

V2-072337 | MATHDB-374565

Cross-domain transfer via a shared LAWS trie

摘录 / EXCERPT: Let $\mathcal{M}_1$ be a language model and $\mathcal{M}_2$ a robotics model with a shared natural-language task representation. Let n^* be a shared PLT trie node and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072338 | MATHDB-374566

LAWS acquisition lower bound for stationary online caching

摘录 / EXCERPT: Let $P_{\mathcal{M}}$ range over stationary distributions with entropy H, and let N be the number of queries. LAWS acquisition lower bound. No online inference caching algorith...

conjecture | open | MathDB 页面 | 原文线索

V2-072339 | MATHDB-374573

Gan-Gross-Prasad conjecture for relevant pure inner forms

摘录 / EXCERPT: Gan-Gross-Prasad conjecture. If $n-m$ is odd, a nonzero Bessel period $\mathcal{B}_{k,\psi^{\varepsilon}}$ on $\pi_1 \boxtimes \pi_2$ implies that (M,N) is a relevant pair. For a...

conjecture | open | MathDB 页面 | 原文线索

V2-072340 | MATHDB-374578

Finite area determination conjecture for Kähler-Einstein metrics on a product of surfaces

摘录 / EXCERPT: Let Σ be the closed surface under consideration, let $\mathcal L^*(g)$ denote the relevant set of surface subgroups of genus g, and let $A(h)(\Pi)$ be the associated area...

conjecture | open | MathDB 页面 | 原文线索

V2-072341 | MATHDB-374585

Lin-Zhao stability conjecture for collective strength

摘录 / EXCERPT: Let $\mathbb{K}$ be a field with ... and let $P\in\mathbb{K}[x_1,\dots,x_n]_d$ be a homogeneous polynomial of degree d. Here $\operatorname{str}_{\mathbb{K}}(P)$ denotes the streng...

conjecture | open | MathDB 页面 | 原文线索

V2-072342 | MATHDB-374586

Adiprasito-Kazhdan-Ziegler partition-rank versus analytic-rank conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be a finite field, let $d\geq 2$ be an integer, and let ... Here $\operatorname{PR}_{\mathbb{K}}(f)$ is the partition rank and $\operatorname{AR}_{\mathbb{K}}(f)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-072343 | MATHDB-374587

Baily's strength versus Birch-rank conjecture

摘录 / EXCERPT: Let $\mathbb{K}$ be a field, let $d\geq 2$ be an integer, and let $P\in\mathbb{K}[x_1,\dots,x_n]_d$ be a homogeneous polynomial of degree d. Suppose that ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-072344 | MATHDB-374588

Asymptotic-velocity conjecture for BBM with uniformly positive rank-based killing

摘录 / EXCERPT: Let ψ be a selection function such that $\psi(x)\geq\delta$ for all $x\in[0,1]$ and $\psi(x)=\delta$ for all $x\in[1-p,1]$ for some $p\in(0,1)$. Asymptotic-velocity conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072345 | MATHDB-374593

Small-tail extrapolation conjecture for top-1 retrieval

摘录 / EXCERPT: Let n be the vocabulary size and d the embedding dimension, with $n,d\to\infty$. Top-1 retrieval requires the memory load n/d^2 to lie below the critical threshold obtained b...

conjecture | solved | MathDB 页面 | 原文线索

V2-072346 | MATHDB-374594

Maximum-generator conjecture for S-LCG

摘录 / EXCERPT: Let $n\in\mathbb{N}$ be odd, and let $\alpha_{\max}^{(n)}$ denote the maximum valid generator for S-LCG. Maximum-generator conjecture. The quantity $\alpha_{\max}^{(n)}$ has the cl...

conjecture | open | MathDB 页面 | 原文线索

V2-072347 | MATHDB-374605

The AIM low-degree conjecture for independent sets in dense random graphs

摘录 / EXCERPT: AIM low-degree conjecture. In $G(n,\frac12)$, no degree-$o(\log^2 n)$ polynomial can find an independent set of size $0.9\log_2 n$.

conjecture | open | MathDB 页面 | 原文线索

V2-072348 | MATHDB-374609

Asymptotic for the Pell-family consecutive powerful-number progressions

摘录 / EXCERPT: Let $\mathcal{A}$ be the set of positive integers N for which some $d\in\mathbb{N}$ makes $N,N+d,N+2d$ a three-term arithmetic progression of consecutive powerful numbers. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072349 | MATHDB-374610

Logarithmic growth conjecture for all classes of consecutive powerful-number progressions

摘录 / EXCERPT: Let $\mathcal{A}$ be the set of positive integers N for which the corresponding triple $N,N+d,N+2d$ is a three-term arithmetic progression of consecutive powerful numbers. Partit...

conjecture | open | MathDB 页面 | 原文线索

V2-072350 | MATHDB-374624

Weak knitwork immersion conjecture for bounded-treewidth classes

摘录 / EXCERPT: Weak knitwork immersion conjecture. The class $\mathbf{T}(k,d,\Delta,\ell;\Omega)$ is well-quasi-ordered by Ω-knitwork immersion.

conjecture | open | MathDB 页面 | 原文线索

V2-072351 | MATHDB-374625

The idealizer conjecture for stable polynomials

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let $X_n\subset\mathbb{R}_n^+$ be the largest subsemigroup containing $\mathcal{H}_n$ and $\mathbb{I}_n$ such that, for every $m\leq n$, Hadamard multipl...

conjecture | open | MathDB 页面 | 原文线索

V2-072352 | MATHDB-374626

The quasi-stable idealizer conjecture

摘录 / EXCERPT: For each $n\in\mathbb{N}$, let $X_n^\star$ be the idealizer associated with quasi-stable polynomials, and let $Y_n^\star$ be the corresponding family defined by $Y_n^\star=Y_n$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-072353 | MATHDB-374630

First-order nonvanishing obstruction for rank-three incidence lifts

摘录 / EXCERPT: Let Π be a projective plane of order q, let U be a rank-3 residue zero-pattern model satisfying the identity-pattern minor constraints, and let T_U denote its tangent s...

conjecture | open | MathDB 页面 | 原文线索

V2-072354 | MATHDB-374636

The C4-free subclass conjecture for chi-bounded graph classes

摘录 / EXCERPT: C4-free subclass conjecture. The C_4-free subclass of every χ-bounded class is linearly χ-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-072355 | MATHDB-374637

The cocktail-party-free subclass conjecture for chi-bounded graph classes

摘录 / EXCERPT: Cocktail-party-free subclass conjecture. For every integer $m\ge1$, the $\overline{mK_2}$-free subclass of every χ-bounded class is linearly χ-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-072356 | MATHDB-374639

Equivalence conjecture for weighted homogeneous hypersurface germs and vector fields

摘录 / EXCERPT: Equivalence conjecture. The following statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-072357 | MATHDB-374644

The strong-snark conjecture for nontrivial cubic graphs

摘录 / EXCERPT: Strong-snark conjecture. If the perfect matching index of G is greater than 4 and G is not the Petersen graph, then G is a strong snark.

conjecture | open | MathDB 页面 | 原文线索

V2-072358 | MATHDB-374645

The four-perfect-matchings conjecture for cubic graphs with a two-circuit 2-factor

摘录 / EXCERPT: Four-perfect-matchings conjecture. The perfect matching index of G is at most 4, unless G is the Petersen graph.

conjecture | open | MathDB 页面 | 原文线索

V2-072359 | MATHDB-374647

The Cambrian lattice mutation conjecture for finite Coxeter groups

摘录 / EXCERPT: Finite Coxeter Cambrian mutation conjecture. The Cambrian lattice $\operatorname{Camb}(Q)$ is mutable.

conjecture | open | MathDB 页面 | 原文线索

V2-072360 | MATHDB-374648

The Ordovician lattice structure conjecture

摘录 / EXCERPT: Ordovician lattice structure conjecture. The following hold:

conjecture | open | MathDB 页面 | 原文线索

V2-072361 | MATHDB-374649

Exhaustion conjecture for indecomposable parabolic homomorphisms of irreducible crystallographic Hecke monoids

摘录 / EXCERPT: Let $(W(M),\star)$ and $(W(M'),\star)$ be Hecke monoids of irreducible crystallographic types. A homomorphism is injective parabolic if it is both injective and parabolic, and it i...

conjecture | open | MathDB 页面 | 原文线索

V2-072362 | MATHDB-374650

Woldar's purely singular splitting conjecture

摘录 / EXCERPT: Let k be a positive integer and write $S(k)=\{1,2,\dots,k\}$. Let G be a finite abelian group. Say that $S(k)$ splits G if there is a subset $T\subseteq G$ such that the tran...

conjecture | open | MathDB 页面 | 原文线索

V2-072363 | MATHDB-374657

Gao's nonnegative pattern-expansion conjecture for principal specializations

摘录 / EXCERPT: Gao's conjecture. For any permutation w, $c_w\geq 0$. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-072364 | MATHDB-374658

Dennin's coefficient-positivity conjecture for Grothendieck pattern coefficients

摘录 / EXCERPT: Dennin's conjecture. For every permutation w,

conjecture | open | MathDB 页面 | 原文线索

V2-072365 | MATHDB-374659

Grothendieck inclusion-exclusion positivity conjecture

摘录 / EXCERPT: Grothendieck inclusion-exclusion positivity conjecture. For every permutation $w\in S_n$ and every subword u of w,

conjecture | open | MathDB 页面 | 原文线索

V2-072366 | MATHDB-374660

Irreducibility conjecture for reciprocal subsum numerators

摘录 / EXCERPT: Let $n\geq 1$, and let $\mathrm{num}(n,x)$ denote the numerator polynomial associated with sums of reciprocals of subsum polynomials over partitions of n. Irreducibility conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-072367 | MATHDB-374661

Coprimality conjecture for reciprocal subsum numerator and denominator

摘录 / EXCERPT: Let $n\geq 1$, and let $\mathrm{num}(n,x)$ and $\mathrm{den}(n,x)$ be the reduced numerator and denominator associated with the sum of reciprocals of subsum polynomials over partit...

conjecture | open | MathDB 页面 | 原文线索

V2-072368 | MATHDB-374662

Unimodality conjecture for the even part of reciprocal subsum numerators

摘录 / EXCERPT: Let $\mathrm{num}(n,x)$ be the numerator polynomial associated with sums of reciprocals of subsum polynomials over partitions of n, and write ... where $\mathrm{num}_0(n,x)$ is eve...

conjecture | open | MathDB 页面 | 原文线索

V2-072369 | MATHDB-374663

Log-concavity conjecture for reciprocal subsum denominators

摘录 / EXCERPT: Let $\mathrm{den}(n,x)$ be the denominator polynomial associated with sums of reciprocals of subsum polynomials over partitions of n. Log-concavity conjecture. The sequence of co...

conjecture | open | MathDB 页面 | 原文线索

V2-072370 | MATHDB-374664

Coprimality conjecture for binary-partition reciprocal polynomials

摘录 / EXCERPT: Let $\mathcal B(n)$ be the set of binary partitions of n, and let $\mathrm{num}_{\mathcal B}(n,x)$ and $\mathrm{den}_{\mathcal B}(n,x)$ denote the associated numerator and denomi...

conjecture | open | MathDB 页面 | 原文线索

V2-072371 | MATHDB-374665

Unimodality and log-concavity conjecture for binary-partition numerators

摘录 / EXCERPT: Let $\mathcal B(n)$ be the set of binary partitions of n, and let $\mathrm{num}_{\mathcal B}(n,x)$ be the associated numerator polynomial. Binary-partition coefficient conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072372 | MATHDB-374666

Cyclotomic nondivisibility conjecture for binary-partition numerators

摘录 / EXCERPT: Let $\mathcal B(n)$ be the set of binary partitions of n, and let $\mathrm{num}_{\mathcal B}(n,x)$ be the associated numerator polynomial. Binary-partition nondivisibility conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-072373 | MATHDB-374669

Adjacent-triple congruence conjecture for Jack Littlewood-Richardson coefficients

摘录 / EXCERPT: Let (μ_1,ν_1,λ_1) and (μ_2,ν_2,λ_2) be triples of partitions that differ by a single box move in one pair, with pivot boxes $b_1\in\sigma_1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072374 | MATHDB-374670

Strong Stanley conjecture for Jack Littlewood-Richardson coefficients

摘录 / EXCERPT: Let $c_{\mu\nu}^{\lambda}$ be the ordinary Littlewood-Richardson coefficient. For a triple of partitions (μ,ν,λ), let $\mathsf{SD}_{\mu\nu}^{\lambda}$ denote the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-072375 | MATHDB-374671

Stanley-sum rule conjecture for Jack Littlewood-Richardson coefficients

摘录 / EXCERPT: For partitions μ,ν,λ, let $\mathsf{SD}_{\mu\nu}^{\lambda}$ be the set of Stanley diagrams and let a Stanley sum be an integer linear combination of these diagrams. Stan...

conjecture | open | MathDB 页面 | 原文线索

V2-072376 | MATHDB-374672

Jack windowing conjecture

摘录 / EXCERPT: Let T be a triple of partitions and let W be a window for T. A rule is a Stanley sum whose evaluation gives the corresponding Jack Littlewood-Richardson coefficient. Jack win...

conjecture | open | MathDB 页面 | 原文线索

V2-072377 | MATHDB-374673

Stanley positivity conjecture for an eight-parameter family

摘录 / EXCERPT: Let ϕ be the displayed map from the five auxiliary variables into the specified linear space, and write $\{m,n,r\}=(m_1,m_2,m_3,n_1,n_2,n_4,r_1,r_2)$. Stanley positivity conje...

conjecture | open | MathDB 页面 | 原文线索

V2-072378 | MATHDB-374674

Homogeneous positivity conjecture for the Stanley polynomial

摘录 / EXCERPT: Homogeneous positivity conjecture. The polynomial $\rho(\{y\},\beta)$ has non-negative coefficients as a homogeneous polynomial in y_i and β, and it has degree 7 in...

conjecture | open | MathDB 页面 | 原文线索

V2-072379 | MATHDB-374675

Hook-correspondence conjecture for adjacent Jack coefficients

摘录 / EXCERPT: Let (μ_1,ν_1,λ_1) and (μ_2,ν_2,λ_2) be adjacent triples of partitions corresponding to a pivot at b. Let $\mathcal W'$ and $\mathcal W$ be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-072380 | MATHDB-374676

Shifted Jack congruence conjecture

摘录 / EXCERPT: Let shifted Jack Littlewood-Richardson coefficients be defined for partitions with $|\lambda|\leq|\mu|+|\nu|$. Shifted Jack congruence conjecture. The adjacent-triple congruence co...

conjecture | open | MathDB 页面 | 原文线索

V2-072381 | MATHDB-374677

Eventual p-adic periodicity conjecture for generalized j-Euler numbers

摘录 / EXCERPT: Eventual p-adic periodicity conjecture. The numbers $\mathcal{E}_{mpn}^{(mp,j)}$ are p-adic integers, and for every positive integer r there is a positive integer n_0 such th...

conjecture | open | MathDB 页面 | 原文线索

V2-072382 | MATHDB-374678

Singular-locus conjecture for dual periods of rational polypols

摘录 / EXCERPT: Let $\mathcal P$ be a regular rational polypol, and let $F_{\mathcal P}(u,v)$ denote its associated dual period. Consider the holonomic system annihilating $F_{\mathcal P}$, and th...

conjecture | open | MathDB 页面 | 原文线索

V2-072383 | MATHDB-374684

Mode-by-mode Morse index conjecture for the critical spherical catenoid

摘录 / EXCERPT: Let Σ_a be the critical spherical catenoid with parameter $a>1/2$. Decompose its Robin Morse index into Fourier modes $|k|=0$, $|k|=1$, and $|k|\geq 2$. Mode-by-mode index c...

conjecture | open | MathDB 页面 | 原文线索

V2-072384 | MATHDB-374713

Double-shortcut equivalence-class conjecture for type A hypercube decompositions

摘录 / EXCERPT: Double-shortcut triviality conjecture. The above equivalence relation is trivial, meaning that it has only one equivalence class. This is a conjecture of Ellenberg and McNamara and...

conjecture | open | MathDB 页面 | 原文线索

V2-072385 | MATHDB-374714

Amazing R-element conjecture for double-shortcut equivalence classes

摘录 / EXCERPT: Let W be a Coxeter group of type A, and fix a Bruhat interval $[u,v]$. Consider the equivalence relation on the amazing hypercube decompositions of $[u,v]$ generated by ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-072386 | MATHDB-374719

The generalized Bayer-Macrì-Toda inequality for stable objects

摘录 / EXCERPT: Let X be the threefold considered in the paper, let H be the chosen ample divisor, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072387 | MATHDB-374720

The quadratic Bayer-Macrì-Toda inequality for semistable objects

摘录 / EXCERPT: Let X, H, Γ, $\mathcal{U}$, $\mathcal{A}^b(X)$, $\nu_{b,w}$, $\mathbf{v}_i^b$, $\overline{\Delta}$, and $\mathbf{ch}^{\gamma}_1$ be as in the paper, with...

conjecture | open | MathDB 页面 | 原文线索

V2-072388 | MATHDB-374731

The product criterion for Type-2 isomorphic circulant graphs

摘录 / EXCERPT: Let $C_{n_1}(R_1)$ and $C_{n_2}(R_2)$ be circulant graphs such that ... for some R. Product criterion conjecture. The graph $C_{n_1n_2}(R)$ has Type-2 isomorphic circulant graphs i...

conjecture | open | MathDB 页面 | 原文线索

V2-072389 | MATHDB-374732

The lifting conjecture for Type-2 isomorphic circulant graphs

摘录 / EXCERPT: Lifting conjecture. Then, for some S,

conjecture | open | MathDB 页面 | 原文线索

V2-072390 | MATHDB-374738

The amended reducibility conjecture for finite Jacobi pencils

摘录 / EXCERPT: Amended reducibility conjecture. Every reducible factorization of χ_n is expected to be obtained by iterating the following operations on connected blocks: cutting the chain a...

conjecture | open | MathDB 页面 | 原文线索

V2-072391 | MATHDB-374741

The degree-eight distinct-diagonal reducibility conjecture for Jacobi spectral curves

摘录 / EXCERPT: Degree-eight distinct-diagonal test case. Every reducible connected degree-eight spectral curve is obtained from a constant branch or from reversal symmetry. In particular, there i...

conjecture | open | MathDB 页面 | 原文线索

V2-072392 | MATHDB-374742

Bogomolov-Fu-Tschinkel conjecture on torsion-coordinate intersections

摘录 / EXCERPT: Let E_1 and E_2 be elliptic curves defined over $\mathbb{C}$, given by Weierstrass equations ... where f_1 and f_2 do not have the same roots. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-072393 | MATHDB-374745

Connectivity conjecture for intermediate root subsystems

摘录 / EXCERPT: Let Δ be the root system under consideration, let I index the complete flag $\mathcal{S}_\bullet^1$, and for each i with $1\leq i\leq |I|$ define ... The roots...

conjecture | open | MathDB 页面 | 原文线索

V2-072394 | MATHDB-374748

The Links-Gould polynomial conjecture for detecting causality

摘录 / EXCERPT: Links-Gould polynomial conjecture. The invariant LG completely captures causality in such spacetimes.

conjecture | open | MathDB 页面 | 原文线索

V2-072395 | MATHDB-374750

Gross's newform conjecture for generic representations of split odd special orthogonal groups

摘录 / EXCERPT: Gross's conjecture. The following assertions hold: $\pi^{K_{n,m}}=0$ for $0\leq m<c_\pi$ and $\dim_\mathbb{C}\pi^{K_{n,c_\pi}}=1$; the action of $J_{n,c_\pi}/K_{n,c_\pi}$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-072396 | MATHDB-374756

Mixing-time bound for giant components of random graphs with given degrees

摘录 / EXCERPT: Let m be the number of edges, let $n_0(\mathcal D)$ denote the number of vertices of degree zero, let $m^{\neq 2}$ denote the number of edges incident to vertices whose degrees a...

conjecture | open | MathDB 页面 | 原文线索

V2-072397 | MATHDB-374758

Uniform positivity conjecture for the Frobenius-alpha invariant of log Fano pairs

摘录 / EXCERPT: Let (X,Δ) be a projective log Fano pair over $\mathbb{C}$, and let (X_p,Δ_p) denote reductions to characteristic p for sufficiently large primes. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072398 | MATHDB-374759

Bousquet-Mélou-Mishna D-finiteness conjecture for quarter-plane walks

摘录 / EXCERPT: A small-step quarter-plane lattice model is determined by a stepset $S\subseteq\{-1,0,1\}^2\setminus\{(0,0)\}$, and its counting generating function is ... where the sum ranges over...

conjecture | open | MathDB 页面 | 原文线索

V2-072399 | MATHDB-374760

Boundary generating-function non-D-finiteness conjecture for selected quarter-plane models

摘录 / EXCERPT: For a small-step quarter-plane model with stepset S, let $Q(x,y;t)$ be its generating function, and let $Q(0,1;t)$ and $Q(1,0;t)$ denote its two boundary specializations. The mod...

conjecture | open | MathDB 页面 | 原文线索

V2-072400 | MATHDB-374768

Equality of relative Davenport constants after removing a smaller coset

摘录 / EXCERPT: Relative Davenport equality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-072401 | MATHDB-374770

Kostochka-Pelsmajer-West conjecture on equitable list colorability

摘录 / EXCERPT: Let G be a graph, let k be a positive integer, and let L be a k-list assignment for G. Write $m:=\left\lceil\frac{|G|}{k}\right\rceil$. An L-coloring is m-bounded if...

conjecture | open | MathDB 页面 | 原文线索

V2-072402 | MATHDB-374780

Martin--Vial's Bloch--Beilinson criterion for sums of points

摘录 / EXCERPT: Martin--Vial's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072403 | MATHDB-374781

Ikeda's Petersson norm conjecture for the Ikeda-Yamana lift

摘录 / EXCERPT: Ikeda's Petersson norm conjecture. The Petersson norm of the Ikeda-Yamana lift is given, after division by the product of the completed adjoint L-values, by powers of 2 and the...

conjecture | open | MathDB 页面 | 原文线索

V2-072404 | MATHDB-374785

Recursive monodromy conjecture for irreducible discriminant slices

摘录 / EXCERPT: Let $X_{\zeta_{+}}$ be any Calabi-Yau threefold constructed by the GLSM flop procedure, with coordinates $z_{1},\ldots,z_{r}$ and divisors $\widehat D_{\alpha-1,\alpha}$. Assume th...

conjecture | open | MathDB 页面 | 原文线索

V2-072405 | MATHDB-374788

Conjecture on the number of distribution-equivalence classes of length-2 mesh patterns

摘录 / EXCERPT: Let p_1 and p_2 be mesh patterns of length 2. They are equivalent when $s_{n,k}(p_1)=s_{n,k}(p_2)$ for all $n,k\geq0$, where $s_{n,k}(p)$ counts n-permutations containing e...

conjecture | open | MathDB 页面 | 原文线索

V2-072406 | MATHDB-374789

Conjectured equivalence of eight specified length-2 mesh patterns

摘录 / EXCERPT: A mesh pattern of length 2 is a pair (τ,R), with τ a permutation of length 2 and $R\subseteq[0,2]\times[0,2]$ the set of shaded boxes. Two patterns are equivalent whe...

conjecture | open | MathDB 页面 | 原文线索

V2-072407 | MATHDB-374805

Positivity and unimodality for inflated characters of Weyl groups

摘录 / EXCERPT: Let G be a connected complex reductive group with Weyl group W, let $\psi\in\operatorname{Irr}(W)$ be inflated from a quotient ... and let $\alpha_{\psi,G}^{z}$ be the Laurent po...

conjecture | open | MathDB 页面 | 原文线索

V2-072408 | MATHDB-374815

The Chow tautological-generation conjecture for moduli spaces of stable curves

摘录 / EXCERPT: Chow tautological-generation conjecture. The Chow group

conjecture | open | MathDB 页面 | 原文线索

V2-072409 | MATHDB-374825

The misspecified Cramér-Rao bound conjecture

摘录 / EXCERPT: Misspecified Cramér-Rao bound conjecture. The mean-square error satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072410 | MATHDB-374831

Bergeron-Garsia-Haiman-Tesler Conjecture II on nabla and modified Hall-Littlewood polynomials

摘录 / EXCERPT: Let λ and μ be partitions, let $\widetilde{H}_{\mu}[X;0,t]$ be the modified Hall-Littlewood polynomial obtained by specializing the modified Macdonald polynomial at...

conjecture | open | MathDB 页面 | 原文线索

V2-072411 | MATHDB-374832

Bergeron-Garsia-Haiman-Tesler Conjecture III on nabla, omega, and Hall-Littlewood polynomials

摘录 / EXCERPT: Let λ and μ be partitions, let $\widetilde{H}_{\mu}[X;0,t^{-1}]$ be the modified Hall-Littlewood polynomial specialized at $q=0$, let ∇ be the nabla operator, an...

conjecture | open | MathDB 页面 | 原文线索

V2-072412 | MATHDB-374833

Conditioning conjecture for linear time-invariant system identification

摘录 / EXCERPT: Conditioning conjecture. The global optima of the underlying optimization problem are among the best-conditioned eigenvalues of the rectangular multiparameter eigenvalue problem.

conjecture | open | MathDB 页面 | 原文线索

V2-072413 | MATHDB-374835

P=W conjecture for affine Toda moduli spaces

摘录 / EXCERPT: Let G be a semisimple group over $\mathbb{C}$, let $\mathcal{M}^{\circ}_G$ be the neutral component of the affine Toda moduli space, and let $J^G_{G^{\mathrm{sc}}}$ be the associ...

conjecture | open | MathDB 页面 | 原文线索

V2-072414 | MATHDB-374836

Weight-filtration and hard-Lefschetz conjecture for affine Toda centralizers

摘录 / EXCERPT: Let G be a semisimple group over $\mathbb{C}$, let G^{sc} be its simply-connected cover, and let $J^G_{G^{\mathrm{sc}}}$ be the associated regular centralizer group sc...

conjecture | open | MathDB 页面 | 原文线索

V2-072415 | MATHDB-374837

Amdeberhan-Andrews-Ballantine coefficient identity for a double Lambert series

摘录 / EXCERPT: Let r be a positive integer. The notation $[q^n]A(q)$ denotes the coefficient of q^n in the series $A(q)$. Amdeberhan-Andrews-Ballantine coefficient identity. ... The identity wa...

conjecture | open | MathDB 页面 | 原文线索

V2-072416 | MATHDB-374847

Periodic-point density conjecture for finite-field quadratic polynomials

摘录 / EXCERPT: Let $f\in\mathbb{F}_q[x]$ be a monic quadratic polynomial, and say that two polynomials are conjugate when they are related by the relevant change of coordinates. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072417 | MATHDB-374851

Ballantine-Beck-Feigon-Maurischat's irreducibility conjecture for partition numerators

摘录 / EXCERPT: Let $\operatorname{num}(n,x)$ denote the ordinary partition numerator for $n\geq 1$. Ballantine-Beck-Feigon-Maurischat's irreducibility conjecture. For every $n\geq 1$, the polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-072418 | MATHDB-374852

Ballantine-Beck-Feigon-Maurischat's coprimality conjecture for partition polynomials

摘录 / EXCERPT: Let $\operatorname{num}(n,x)$ and $\operatorname{den}(n,x)$ be the ordinary numerator and denominator of the reciprocal of the partition polynomial for $n\geq 1$. Ballantine-Beck-F...

conjecture | open | MathDB 页面 | 原文线索

V2-072419 | MATHDB-374853

Ballantine-Beck-Feigon-Maurischat's unimodality conjecture for the even partition numerator

摘录 / EXCERPT: Write the ordinary partition numerator as $\operatorname{num}(n,x)=\operatorname{num}_0(n,x)+\operatorname{num}_1(n,x)$, where $\operatorname{num}_0(n,x)$ contains the even powers...

conjecture | open | MathDB 页面 | 原文线索

V2-072420 | MATHDB-374854

Ballantine-Beck-Feigon-Maurischat's log-concavity conjecture for partition denominators

摘录 / EXCERPT: Let $\operatorname{den}(n,x)$ be the ordinary denominator of the reciprocal of the partition polynomial for n. Ballantine-Beck-Feigon-Maurischat's log-concavity conjecture. The s...

conjecture | open | MathDB 页面 | 原文线索

V2-072421 | MATHDB-374855

Ballantine-Beck-Feigon-Maurischat's binary numerator-denominator coprimality conjecture

摘录 / EXCERPT: For binary partitions, let $\operatorname{num}_{\mathcal B}(n,x)$ and $\operatorname{den}_{\mathcal B}(n,x)$ denote the corresponding numerator and denominator, where binary parts...

conjecture | open | MathDB 页面 | 原文线索

V2-072422 | MATHDB-374856

Ballantine-Beck-Feigon-Maurischat's unimodality conjecture for binary partition numerators

摘录 / EXCERPT: For binary partitions, let $\operatorname{num}_{\mathcal B}(n,x)$ denote the binary numerator. Ballantine-Beck-Feigon-Maurischat's binary unimodality conjecture. For every $n>5$, t...

conjecture | open | MathDB 页面 | 原文线索

V2-072423 | MATHDB-374857

Ballantine-Beck-Feigon-Maurischat's ternary-partition evaluation conjecture

摘录 / EXCERPT: Let $\operatorname{num}_{\mathcal T}(n,x)$ denote the numerator associated with partitions into powers of 3, and let $v_3(M)$ be the exponent of 3 in the positive integer M....

conjecture | open | MathDB 页面 | 原文线索

V2-072424 | MATHDB-374858

Ballantine-Beck-Feigon-Maurischat's ternary-partition recurrence conjecture

摘录 / EXCERPT: For ternary partitions, let $\operatorname{num}_{\mathcal T}(n,x)$ denote the associated numerator. Define $t(n)\coloneq\operatorname{num}_{\mathcal T}(n,1)$ for $n\geq 0$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-072425 | MATHDB-374860

The MDS Conjecture for linear codes

摘录 / EXCERPT: MDS Conjecture. The stated upper bound holds for the length R of C. Despite significant progress, the MDS Conjecture remains open in general for larger values of k and q.

conjecture | open | MathDB 页面 | 原文线索

V2-072426 | MATHDB-374873

The logarithmic correlation-decay threshold for Poisson approximation of largest gaps

摘录 / EXCERPT: Let f be a smooth stationary Gaussian process, let $K(r)$ denote its correlation function, and let the Poisson approximation refer to the approximation for the locations and size...

conjecture | open | MathDB 页面 | 原文线索

V2-072427 | MATHDB-374874

CN-hyperenergeticity conjecture for finite CC-rings

摘录 / EXCERPT: Let R be a finite CC-ring, meaning that the centralizer of every non-central element of R is commutative, and let Γ_R be its commuting graph. A graph is CN-hyperenergeti...

conjecture | open | MathDB 页面 | 原文线索

V2-072428 | MATHDB-374877

Ishii's maximal-resolution conjecture for quotient surface resolutions

摘录 / EXCERPT: Let $G\subset GL_2({\mathbb{C}})$ be any finite subgroup, and let X be a resolution of $\mathbb{C}^2/G$. Let Y be the maximal resolution of the pair $(\mathbb{C}^2/G,B)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-072429 | MATHDB-374884

Transport-distance finiteness for uniformly hyperuniform random measures

摘录 / EXCERPT: Transport-distance finiteness conjecture. If Φ is $(p+\vartheta)$-uniform for some $\vartheta>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-072430 | MATHDB-374885

Signed 4-choosability of planar graphs of girth at least 4

摘录 / EXCERPT: Let (G,σ) be a signed planar graph, meaning that G is a planar graph equipped with a signature σ. Its girth is the length of its shortest cycle. Girth-four signed c...

conjecture | open | MathDB 页面 | 原文线索

V2-072431 | MATHDB-374895

Conjecture on the limiting eigenvalue measure for randomly twisted transfer operators

摘录 / EXCERPT: Let $d\geq 2$, and let ${\mathscr L}_N$ be the randomly twisted transfer operator defined using independently uniform permutation matrices. Write $\mathrm{Sp}({\mathscr L}_N)$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-072432 | MATHDB-374896

Prefix sum recurrence conjecture for Legendre-symbol walks

摘录 / EXCERPT: Prefix sum recurrence conjecture. For an absolute constant $A>0$, independently of $a\in\mathbb{F}_p$,

conjecture | open | MathDB 页面 | 原文线索

V2-072433 | MATHDB-374899

The transversely projective or surface pull-back conjecture for codimension-one foliations

摘录 / EXCERPT: Conjecture. Every codimension-one foliation on X either admits a singular transversely projective structure, or is the pull-back, by a rational map, of a foliation on a complex s...

conjecture | open | MathDB 页面 | 原文线索

V2-072434 | MATHDB-374902

Mader's tree-removability conjecture

摘录 / EXCERPT: For $k\ge 1$ and any tree T of order m, let G be a k-connected graph with minimum degree $\delta(G)\ge \left\lfloor 3k/2\right\rfloor+m-1$. A subtree T' is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-072435 | MATHDB-374903

Hasunuma's edge-removable tree conjecture

摘录 / EXCERPT: For $k\ge 1$, let T be a tree of order m, and let G be a k-connected or k-edge-connected graph with minimum degree $\delta(G)\ge k+m-1$. A subtree T' is isomorphic to...

conjecture | open | MathDB 页面 | 原文线索

V2-072436 | MATHDB-374905

The maximum removable matching conjecture in k-connected graphs

摘录 / EXCERPT: Let G be a k-connected graph on n vertices with minimum degree $\delta(G)\ge k+2$. A matching is k-removable when deleting its edges leaves a k-connected graph. The maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-072437 | MATHDB-374907

Laguerre-Wronskian completeness conjecture for symmetric harmonic-oscillator extensions

摘录 / EXCERPT: Let β be a momentum parameter, and consider rational extensions of the harmonic oscillator with momentum β that are symmetric under $x\mapsto -x$. A rational-extension...

conjecture | open | MathDB 页面 | 原文线索

V2-072438 | MATHDB-374928

Full-parameter properties conjecture for the discrete p-Hardy-Rellich-Birman weight

摘录 / EXCERPT: Full-parameter properties conjecture. The following assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-072439 | MATHDB-374934

Rank-two binomial positivity conjecture for Schur coefficients

摘录 / EXCERPT: For $n=2$, write the Chern class in the Schur basis as ... and expand each coefficient in the binomial basis: ... A polynomial is binomially positive when all its binomial-basis coeffi...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072440 | MATHDB-374937

Shifted-binomial log-concavity conjecture for stable CSM-Schur coefficients

摘录 / EXCERPT: Shifted-binomial log-concavity conjecture. The shifted polynomial

conjecture | open | MathDB 页面 | 原文线索

V2-072441 | MATHDB-374939

Positive-characteristic sparse, sporadic, lacunary, and generic architecture for quotient maps

摘录 / EXCERPT: Let $p>0$, let $p\nmid d$, and let h_d be the reduced separable quotient associated with $H_d^{\rm raw}$ in characteristic p. Assume the characteristic-zero case is settled and...

conjecture | refuted | MathDB 页面 | 原文线索

V2-072442 | MATHDB-374943

Frobenius formula for shifted square sequences beyond 30

摘录 / EXCERPT: Let a be an integer with $a>30$, and let ... Let $S(A(a))$ be the numerical semigroup generated by $A(a)$. For $n\in\mathbb{Z}^{+}$, let $\iota(n)$ be the least number of positive...

conjecture | open | MathDB 页面 | 原文线索

V2-072443 | MATHDB-374944

Alexandersson-Shapiro square-graph cone conjecture for derivative discriminants

摘录 / EXCERPT: Square-graph cone conjecture. For every $n\ge2$ and every $0\le k\le n-2$, the polynomial $\mathcal D_k^{(n)}$ belongs to the square-graph cone. Equivalently, $\mathcal D_k^{(n)}$...

conjecture | open | MathDB 页面 | 原文线索

V2-072444 | MATHDB-374945

Terminal stability conjecture for derivative discriminants

摘录 / EXCERPT: Terminal stability conjecture. For each fixed $r\ge2$ there exists a finite list of square multigraphs $G_1,\ldots,G_N$, depending only on r and not on n, such that for all...

conjecture | open | MathDB 页面 | 原文线索

V2-072445 | MATHDB-374946

Bruzzo's conjecture on semistability and vanishing discriminant for Higgs bundles

摘录 / EXCERPT: Bruzzo's conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-072446 | MATHDB-374947

Local smoothing conjecture for fractional Schrödinger operators in Bessel potential spaces

摘录 / EXCERPT: Fix $b1 eq 0$?

conjecture | open | MathDB 页面 | 原文线索

V2-072447 | MATHDB-374948

The microlocal-Whittaker equivalence for semistable Higgs bundles

摘录 / EXCERPT: Let C be a smooth projective curve, let G be a reductive group with Langlands dual group $G^{\vee}$, and let w denote the relevant stratum or weight. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-072448 | MATHDB-374955

Cha's simplicial-polytope conjecture for arbor h-polynomials

摘录 / EXCERPT: Let E be an n-element set and let τ be an arbor on E. Its h-polynomial is $h(\tau,t)=\sum_{i=0}^n h_i(\tau)t^i$, where $h_i(\tau)$ counts the lattice points of the arb...

conjecture | open | MathDB 页面 | 原文线索

V2-072449 | MATHDB-374956

Cha's real-rootedness conjecture for arbor Ehrhart polynomials

摘录 / EXCERPT: Let σ be an arbor, and let $\mathcal Q_\sigma$ be its arbor polytope. Cha's real-rootedness conjecture. All roots of the Ehrhart polynomial of $\mathcal Q_\sigma$ are real a...

conjecture | open | MathDB 页面 | 原文线索

V2-072450 | MATHDB-374957

Transpose duality conjecture for preorder multichain matrices

摘录 / EXCERPT: Let τ be a preorder, and for $k,\ell\in\mathbb N$ let $\nabla_\tau(k,\ell)$ denote the number of k-multichains in the poset of lattice points of $\ell\mathcal Q_\tau$. Trans...

conjecture | open | MathDB 页面 | 原文线索

V2-072451 | MATHDB-374958

Descent formula conjecture for preorder-polytope h-star polynomials

摘录 / EXCERPT: Let E be a totally ordered n-element set, let τ be a preorder on E, and let $\operatorname{des}(w)$ be the number of descents of a word $w\in E^n$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-072452 | MATHDB-374959

Magic-positivity conjecture for preorder-polytope Ehrhart polynomials

摘录 / EXCERPT: Let τ be a preorder and let $\mathcal Q_\tau$ be its preorder polytope. Magic-positivity conjecture. The Ehrhart polynomial $\operatorname{Ehr}(\mathcal Q_\tau,t)$ is magic po...

conjecture | open | MathDB 页面 | 原文线索

V2-072453 | MATHDB-374960

q-zeta reciprocity conjecture for preorder lattice-point posets

摘录 / EXCERPT: Let τ be a preorder of size n, let P_τ be the poset of lattice points associated with its preorder polytope, and let $\mathcal Z_q(P_\tau,t)$ be its q-zeta polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-072454 | MATHDB-374961

Simplicial and flag-simplicial h-vector conjectures for preorder polytopes

摘录 / EXCERPT: Let τ be a preorder of size n. For $0\leq i\leq n$, let $h_i(\tau)$ count the lattice points of $\mathcal Q_\tau$ with exactly i nonzero coordinates, and write...

conjecture | open | MathDB 页面 | 原文线索

V2-072455 | MATHDB-374962

Real-rootedness conjecture for preorder-polytope h-polynomials

摘录 / EXCERPT: Let τ be a preorder of size n and let $h(\tau,t)$ be its h-polynomial. Real-rootedness conjecture. The polynomial $h(\tau,t)$ has only real roots for every preorder τ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072456 | MATHDB-374963

Duality conjecture for preorder-polytope h-vectors

摘录 / EXCERPT: Let τ be a preorder and let $\tau^\ast$ denote its dual preorder. Preorder h-vector duality conjecture. One has ... for every preorder τ. The statement substantially genera...

conjecture | open | MathDB 页面 | 原文线索

V2-072457 | MATHDB-374964

Ordinal-sum factorization conjecture for polar preorder-polytope h-star polynomials

摘录 / EXCERPT: Let τ be the ordinal sum of preorders τ_1 and τ_2, and let $\mathcal R_\tau^\vee$ denote the polar dual of the translated reflexive preorder polytope. Ordinal-sum fa...

conjecture | open | MathDB 页面 | 原文线索

V2-072458 | MATHDB-374965

Bivariate Ehrhart reciprocity conjecture for preorder polytopes

摘录 / EXCERPT: Let τ be a preorder of size n, and let $\mathbb E_\tau(u,v)$ be the double Ehrhart polynomial counting lattice points of the two-parameter polytope $\mathcal Q_\tau(r,s)$. B...

conjecture | open | MathDB 页面 | 原文线索

V2-072459 | MATHDB-374966

Lattice-realization conjecture for preorder lattice-point posets

摘录 / EXCERPT: Lattice-realization conjecture. There exists a finite graded lattice L_τ of rank n such that

conjecture | open | MathDB 页面 | 原文线索

V2-072460 | MATHDB-374967

Duality conjecture for transmuted preorder M-triangles

摘录 / EXCERPT: Let τ be a preorder of size n, let P_τ be its lattice-point poset, and let $\overline M_{P_\tau}$ be the transmuted M-triangle. Transmuted M-triangle duality conjectu...

conjecture | open | MathDB 页面 | 原文线索

V2-072461 | MATHDB-374975

Demeter's continuum small-cap decoupling conjecture for the moment curve in \mathbb{R}^4

摘录 / EXCERPT: Let $11\le p\le 12$. Assume $a\ge b\ge 0$ and $a+b=\frac{p}{2}-3$. Let ... and let $\Phi(t)=(t,t^2,t^3,t^4)$ be the moment curve in $\mathbb{R}^4$. Here $n\sim N$ denotes summation o...

conjecture | open | MathDB 页面 | 原文线索

V2-072462 | MATHDB-374982

The double-cover equivalence for quadratic matrix factorizations

摘录 / EXCERPT: Let G be the group acting in the above setup, and let a_i and x_i be the coefficients and coordinates defining the quadratic potential ... Here MF denotes the...

conjecture | open | MathDB 页面 | 原文线索

V2-072463 | MATHDB-374983

Czerwiński et al.'s additive coloring conjecture

摘录 / EXCERPT: Let G be a finite simple graph. Write $\eta(G)$ for its additive chromatic number and $chi(G)$ for its chromatic number. Czerwiński et al.'s additive coloring conjecture. For all...

conjecture | open | MathDB 页面 | 原文线索

V2-072464 | MATHDB-374984

Wu-Xu-Xu conjecture on 3-colorability of even-hole-restricted graphs

摘录 / EXCERPT: Wu-Xu-Xu conjecture. Every graph in $\bigcup_{\ell\geq 2}{\cal H}_{\ell}$ is 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-072465 | MATHDB-374985

Lescure-Meyniel conjecture for strong immersions

摘录 / EXCERPT: Let G be a graph, let $chi(G)$ denote its chromatic number, and let $K_{chi(G)}$ be the complete graph on $chi(G)$ vertices. A strong immersion of a graph H in G consists of...

conjecture | open | MathDB 页面 | 原文线索

V2-072466 | MATHDB-374986

Vergara's conjecture for graphs with independence number two

摘录 / EXCERPT: Let G be a graph with independence number $\alpha(G)=2$, let $\chi(G)$ denote its chromatic number, and let $K_{\chi(G)}$ be the complete graph on $\chi(G)$ vertices. A weak imme...

conjecture | open | MathDB 页面 | 原文线索

V2-072467 | MATHDB-374987

Stanley's distinct-roots conjecture for dimer-covering denominators

摘录 / EXCERPT: Stanley's conjecture. The polynomial $Q_k(x)$ has distinct roots. This conjecture concerns the simplicity of the poles of the dimer-covering generating functions. It is refuted: th...

conjecture | open | MathDB 页面 | 原文线索

V2-072468 | MATHDB-374989

Gaussian Wasserstein bound for instantaneous preferential-attachment triangle counts

摘录 / EXCERPT: Fix $m\geqslant2$ and $\delta>0$. Let T_n be the number of triangles in the instantaneous variant of the linear preferential attachment model, with connections specified by the i...

conjecture | open | MathDB 页面 | 原文线索

V2-072469 | MATHDB-374992

Power conjecture for strange nilpotent orbits in ${\mathfrak{sl}}_n$

摘录 / EXCERPT: Let ${\mathcal N}\subset{\mathfrak{sl}}_n$ be the nilpotent cone, let $e\in{\mathcal N}$, and define ${\mathcal O}^{\langle k\rangle}:=SL_n\cdot e^k$ for $k\geqslant1$. Power conje...

conjecture | open | MathDB 页面 | 原文线索

V2-072470 | MATHDB-375003

Orbit-structure conjecture for alt ν-Tamari lattices

摘录 / EXCERPT: Orbit-structure conjecture. The orbit structure of the alt ν-Tamari lattice $\mathsf{Tam}_{\delta}(\nu)$ is independent of the increment vector δ.

conjecture | open | MathDB 页面 | 原文线索

V2-072471 | MATHDB-375004

Homometry conjecture for down-degree on alt ν-Tamari lattices

摘录 / EXCERPT: Alt ν-Tamari homometry conjecture. The down-degree statistic is homometric for rowmotion on alt ν-Tamari lattices $\mathsf{Tam}_{\delta}(\nu)$ and is independent of the inc...

conjecture | open | MathDB 页面 | 原文线索

V2-072472 | MATHDB-375015

Sharp threshold for PES free energy in dimension four

摘录 / EXCERPT: Let $N=4$ and let Ω be $\mathbb{R}^{4}$, $\mathbb{T}^{4}$, or a bounded C^{2} Neumann domain. Let $\mathcal{A}_M(\Omega)$ denote the admissible class of densities of mass...

conjecture | open | MathDB 页面 | 原文线索

V2-072473 | MATHDB-375017

Global unconditional well-posedness conjecture for the defocusing CCM equation

摘录 / EXCERPT: Let $L^2_{+}(\mathbb{T})$ and $L^2_{+}(\mathbb{R})$ denote the positive-frequency Hardy subspaces of $L^2(\mathbb{T})$ and $L^2(\mathbb{R})$, respectively. Consider the defocusing...

conjecture | open | MathDB 页面 | 原文线索

V2-072474 | MATHDB-375023

Steurer's conjecture on separated sets of nearly orthogonal vectors

摘录 / EXCERPT: Let $ obreak\forall obreak\text{?}$ For every $ obreak\text{?}$

conjecture | open | MathDB 页面 | 原文线索

V2-072475 | MATHDB-375034

Bsila-Cox-Hugo-Styron-Zhuang fixed-pixed points equidistribution conjecture

摘录 / EXCERPT: Let $\mathcal P$ be the family ... For $\Pi\in\mathcal P$, let $\mathfrak S_n(\Pi)$ denote the permutations of length n avoiding every pattern in Π, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072476 | MATHDB-375040

The permutation criterion for multivariate polynomial $g_{n,q}^{(\ell)}$ over finite fields

摘录 / EXCERPT: Let $q=p^m$ and $Q=p^e$. For integers n and $\ell\geq 1$, let $g_{n,q}^{(\ell)}$ be the multivariate polynomial obtained from $g_{n,q}$ by composing with...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072477 | MATHDB-375057

The twisted exceptional zero conjecture for elliptic curves

摘录 / EXCERPT: The twisted exceptional zero conjecture. For every such χ,

conjecture | open | MathDB 页面 | 原文线索

V2-072478 | MATHDB-375058

Mazur--Tate's refined exceptional zero conjecture

摘录 / EXCERPT: Mazur--Tate's refined exceptional zero conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-072479 | MATHDB-375071

The conjecture that chromatic noncommutative symmetric functions distinguish oriented paths

摘录 / EXCERPT: Let D_1 and D_2 be non-isomorphic oriented paths. Their chromatic noncommutative symmetric functions are denoted by $\mathbf{X}_{D_1}$ and $\mathbf{X}_{D_2}$, respectively. Ori...

conjecture | open | MathDB 页面 | 原文线索

V2-072480 | MATHDB-375087

Conjecture on point spectrum of self-adjoint extensions via Weyl relations

摘录 / EXCERPT: Point-spectrum conjecture. One has $\alpha\in\sigma_p(\widetilde A)$ if and only if either there exists a nontrivial element y_0 such that

conjecture | open | MathDB 页面 | 原文线索

V2-072481 | MATHDB-375090

Almost-everywhere zero-measure conjecture for random Julia sets

摘录 / EXCERPT: Almost-everywhere zero-measure conjecture. For Lebesgue almost all u, $\mathcal J_u$ has zero Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-072482 | MATHDB-375091

Positive-measure disconnectedness conjecture for random Julia sets

摘录 / EXCERPT: Positive-measure disconnectedness conjecture. There exists a set of positive Lebesgue measure $\Lambda\subset\mathbb T$ such that $\mathcal J_u$ is disconnected for all...

conjecture | open | MathDB 页面 | 原文线索

V2-072483 | MATHDB-375092

Unimodality conjecture for independence polynomials of zero-divisor graphs of integer rings

摘录 / EXCERPT: Zero-divisor graph unimodality conjecture. The independence polynomial $I\bigl(\varGamma(\mathbb{Z}_n),x\bigr)$ is unimodal. The paper presents this as a conjecture following compu...

conjecture | open | MathDB 页面 | 原文线索

V2-072484 | MATHDB-375100

Wavefront set conjecture for generic local Langlands parameters

摘录 / EXCERPT: Let F be a local field of characteristic zero, let G be a classical group over F, and let $\pi\in\Pi_F(G)$ have a generic local L-parameter. For each symbol $\square$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072485 | MATHDB-375101

Descent and generalized Whittaker wavefront set conjecture

摘录 / EXCERPT: Let F be a local field of characteristic zero, let G be a classical group over F, and let $\pi\in\Pi_F(G)$ have a generic local L-parameter. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072486 | MATHDB-375103

Cohen-Lenstra-Martinet conjecture for class groups of Γ-extensions

摘录 / EXCERPT: Let S) be a finite set of primes not containing any divisor of $2|\Gamma|$, let $\Gamma_\infty\subset\Gamma$ have order 1 or 2, and let $\mathcal{F}(\Gamma_\infty,X)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072487 | MATHDB-375111

Lyu et al.'s low-degree hardness conjecture for shared-subspace detection

摘录 / EXCERPT: Let $\mathbb{Q}_n$ denote the null distribution in which $\bm{A}^{(l)}\sim\bm{N}^{(l)}$ for all l, and let $\mathbb{P}_n$ denote the alternative in which...

conjecture | open | MathDB 页面 | 原文线索

V2-072488 | MATHDB-375114

Thomas-Yong-Mihalcea Grothendieck positivity inequality

摘录 / EXCERPT: Let $\mathcal P$ denote the set of partitions, let $\widetilde{G}_{\lambda}$ be the modified stable Grothendieck symmetric function, and let $\leqslant_{\operatorname{g}}$ denote G...

conjecture | open | MathDB 页面 | 原文线索

V2-072489 | MATHDB-375115

Gross-Robbins-Tucker log-concavity conjecture for genus distributions

摘录 / EXCERPT: Let G be a graph, and let $g_k(G)$ denote the number of equivalence classes of 2-cell embeddings of G on the orientable surface of genus k. The sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-072490 | MATHDB-375122

Cohen-Lenstra convergence criterion for general support patterns

摘录 / EXCERPT: General support-pattern conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072491 | MATHDB-375139

Aubian-Charbit-Lopes substitution conjecture for polynomially dichromatic-bounded tournaments

摘录 / EXCERPT: Let $ \mathcal{C}$ be a class of tournaments, where a tournament is an orientation of a complete graph. Let $ \mathcal{C}^{subst}$ denote the closure of $ \mathcal{C}$ under substi...

conjecture | open | MathDB 页面 | 原文线索

V2-072492 | MATHDB-375141

Chi-boundedness conjecture for poset tournaments

摘录 / EXCERPT: A poset tournament is a tournament T for which there exists a total ordering $<$ of $V(T)$ such that, for all $u<v<w$, if $u\to_T v$ and $v\to_T w$, then $u\to_T w$. Equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-072493 | MATHDB-375178

Triple-junction singularity conjecture for multi-arc integral equations

摘录 / EXCERPT: Triple-junction singularity conjecture. Along $4\Gamma_i$, the solution has a leading-order singularity of the form

conjecture | open | MathDB 页面 | 原文线索

V2-072494 | MATHDB-375187

Hedden's conjecture on homomorphisms induced by satellite patterns

摘录 / EXCERPT: A pattern $P\subset S^1\times D^2$ induces a map $[P]\colon\mathcal{C}\to\mathcal{C}$ on the knot concordance group, given by $[K]\mapsto[P(K)]$. This map is a homomorphism when...

conjecture | open | MathDB 页面 | 原文线索

V2-072495 | MATHDB-375201

HLS conjecture for Heegner divisors of Coble type quartics

摘录 / EXCERPT: Let $d4a4_2$, $d4a4_6$, $d4a4_8$, and $d4a4_d$ for $de5 16$ denote the indicated Heegner divisors, and call a Heegner divisor HLS if it has the HLS property used in the paper. HLS...

conjecture | open | MathDB 页面 | 原文线索

V2-072496 | MATHDB-375202

Smoothness-boundary conjecture for Coble type quartics

摘录 / EXCERPT: Let $\Sigma^\circ\subset |\mathcal O_{\mathbb P^9}(4)|$ be the locus of Coble type quartics singular exactly along a smooth hyperkähler fourfold in $\mathcal M_4^{(1)}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072497 | MATHDB-375204

Complement conjecture for the Coble fourfold moduli locus

摘录 / EXCERPT: Let $\mathcal M_4^{(1)}$ be the moduli space under consideration, let $\mathcal M_{\mathrm{Coble4}}^\circ$ be the open locus obtained from smooth stable Coble type quartics, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-072498 | MATHDB-375206

Andrews-Bachraoui conjecture on the signed generating function for $a_2(n)$

摘录 / EXCERPT: Andrews-Bachraoui conjecture. There holds

conjecture | open | MathDB 页面 | 原文线索

V2-072499 | MATHDB-375207

Andrews-Bachraoui conjecture on the values of $b_2''(n)$

摘录 / EXCERPT: Let $b_2”(n)$ be the signed count associated with partitions whose parts greater than 1 lie in $[2k,2l-1]$, whose smallest even part is $2k$, whose number of ones is $l-1$, and w...

conjecture | open | MathDB 页面 | 原文线索

V2-072500 | MATHDB-375214

Zaslavsky's multipartite graph conjecture for maximum frustration index

摘录 / EXCERPT: Let $K_{n_1,n_2,\ldots,n_k}$ be a complete multipartite graph with $k\geq 3$. Let $-K_{n_1,n_2,\ldots,n_k}$ denote the signature in which every edge is negative, and write $l(-G)$...

conjecture | open | MathDB 页面 | 原文线索

V2-072501 | MATHDB-375215

Zaslavsky's edge-disjoint negative-cycle conjecture

摘录 / EXCERPT: Let Σ be a signed graph. Denote by $p^-(\Sigma)$ the maximum number of edge-disjoint negative cycles, by $l(\Sigma)$ its frustration index, and by $p(\Sigma)$ the maximum nu...

conjecture | open | MathDB 页面 | 原文线索

V2-072502 | MATHDB-375226

Average-criterion PCL-indexability for restless bandits with imperfect feedback

摘录 / EXCERPT: Let $f(x,z)$ and $g(x,z)$ denote the average-criterion limiting marginal metrics, and define the diagonal MP index by ... The average-criterion analogues of $\textup{(PCLI1)}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072503 | MATHDB-375239

A super-Brownian representation conjecture for non-removable sets

摘录 / EXCERPT: Let $\Sigma\subset \mathbb{R}^d$ be a compact set that is not a removable set for Δ in $L^2(\mathbb{R}^d)$. A self-adjoint extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-072504 | MATHDB-375241

Brenti's conjecture on special matchings and Kazhdan-Lusztig R-polynomials

摘录 / EXCERPT: Let W be a Coxeter group of type A, let $u,v\in W$, and let M be a special matching of the Bruhat interval $[u,v]$. For $x,y\in [u,v]$ with $x\le y$, write $R_{x,y}(q)$ for t...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072505 | MATHDB-375245

Type B product conjecture for middle orders

摘录 / EXCERPT: Let B_n be the type B finite Coxeter group, with $n\geq3$, and let J be a maximal parabolic subgroup. A middle order is a distributive lattice that refines the weak order and...

conjecture | open | MathDB 页面 | 原文线索

V2-072506 | MATHDB-375246

Minuscule classification conjecture for middle orders on type D

摘录 / EXCERPT: Let D_n be the finite Coxeter group of type D_n. A minuscule middle order is a middle order arising from the minuscule construction described in the paper. Type D minuscule cla...

conjecture | open | MathDB 页面 | 原文线索

V2-072507 | MATHDB-375266

Dyer's Bruhat-preclosure conjecture for joins in finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group, and let $u,v\in W$. For $w\in W$, write $T_L(w)$ for its left-reflection set. Given $A\subseteq T$, define its Bruhat preclosure by ... Here an A...

conjecture | open | MathDB 页面 | 原文线索

V2-072508 | MATHDB-375278

Eskenazis-Ivanisvili conjecture on Gaussian Markov-Bernstein inequalities

摘录 / EXCERPT: Let γ denote the standard Gaussian measure on $\mathbb R$. For $n,d\in\mathbb N$, let $\mathcal P_d(\mathbb R^n)$ be the space of polynomials on $\mathbb R^n$ of degree at m...

conjecture | open | MathDB 页面 | 原文线索

V2-072509 | MATHDB-375279

The group-ring resolution conjecture for symmetric spaces

摘录 / EXCERPT: Group-ring resolution conjecture. There is a function $f(n)$ such that every n-generated ideal in $\mathbb K[G]$ has a free resolution of length $r-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-072510 | MATHDB-375289

Generalized Rains' positivity conjecture for specialized Koornwinder moments

摘录 / EXCERPT: Let Par_n denote the set of partitions with at most n parts. A $(0,1)$-substitution map is an operator φ on...

conjecture | open | MathDB 页面 | 原文线索

V2-072511 | MATHDB-375296

The four-pattern involution equidistribution conjecture for length-2 mesh patterns

摘录 / EXCERPT: Four-pattern involution equidistribution conjecture. The patterns

conjecture | open | MathDB 页面 | 原文线索

V2-072512 | MATHDB-375301

Flynn-Poonen-Schaefer conjecture on rational periods of quadratic polynomials

摘录 / EXCERPT: Flynn-Poonen-Schaefer conjecture. There is no quadratic polynomial $\varphi_{2,c}(z)=z^2+c\in\mathbb Q[z]$ with a rational point of exact period n.

conjecture | open | MathDB 页面 | 原文线索

V2-072513 | MATHDB-375302

Hutz's conjecture on rational periods of even-degree unicritical polynomials

摘录 / EXCERPT: Hutz's conjecture. There is no even degree $d>2$ and no $c\in\mathbb Q$ such that $\varphi_{d,c}$ has a rational point of exact period n. Moreover,

conjecture | open | MathDB 页面 | 原文线索

V2-072514 | MATHDB-375304

Nonexistence conjecture for permutation polynomials of the form x^{q+1}+bx^q+cx+d

摘录 / EXCERPT: Nonexistence conjecture. The polynomial $f(x)$ is not a permutation polynomial over $\mathbb{F}_{q^2}$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072515 | MATHDB-375332

Conjecture on degenerate lump waves in higher-order lump solutions

摘录 / EXCERPT: Degenerate-lump conjecture. An Mth-order lump solution produced by the generalized long-wave limit method contains exactly $\frac{M(M+1)}{2}$ degenerate lump waves.

conjecture | open | MathDB 页面 | 原文线索

V2-072516 | MATHDB-375361

Fortuna's entrywise nonnegativity conjecture for Lyapunov solutions with companion state matrices

摘录 / EXCERPT: Fortuna's conjecture. The solution X is entrywise nonnegative.

conjecture | open | MathDB 页面 | 原文线索

V2-072517 | MATHDB-375366

The DAHA prescription for boundary monopole bubbling of regular Nahm-pole 't Hooft lines

摘录 / EXCERPT: Let $U(N)$ have a dominant magnetic charge ... and let s_λ be the corresponding Schur character. Write $\mathbf e s_\lambda(Y)\mathbf e$ for the spherical DAHA element, whose...

conjecture | solved | MathDB 页面 | 原文线索

V2-072518 | MATHDB-375385

Multiplicability conjecture for finitary upho lattices

摘录 / EXCERPT: Multiplicability conjecture for finitary upho lattices. Every finitary upho lattice, and in particular every finite-type $\mathbb{N}$-graded upho lattice, is multiplicable.

conjecture | open | MathDB 页面 | 原文线索

V2-072519 | MATHDB-375386

Rank-generating realization conjecture for finite-type graded upho posets

摘录 / EXCERPT: Rank-generating realization conjecture. There exists a finitely generated LCH monoid M such that

conjecture | open | MathDB 页面 | 原文线索

V2-072520 | MATHDB-375390

Power-sum Laurent integrality for the Macdonald shift map

摘录 / EXCERPT: Let p_μ be a power-sum symmetric function, let P°_ν be the shifted power-sum basis, and define ... where $\operatorname{sh}_{q,t}$ is Lassalle's Macdonald shift map. Pow...

conjecture | open | MathDB 页面 | 原文线索

V2-072521 | MATHDB-375402

McKay-Wanless conjecture on subsquares in random Latin squares

摘录 / EXCERPT: Let L be a random Latin square of order n, and let $\mathbb{E}_a(n)$ denote the expected number of subsquares of order a in L. For $2\leqslant a\leqslant n/2$, McKay and Wa...

conjecture | open | MathDB 页面 | 原文线索

V2-072522 | MATHDB-375403

Density threshold conjecture for partial Latin squares in random Latin squares

摘录 / EXCERPT: For a non-empty partial Latin square P, define its density by ... Let $\mathbf{L}$ be a random Latin square of order n. The density threshold conjecture. As $n\to\infty$, ... When...

conjecture | open | MathDB 页面 | 原文线索

V2-072523 | MATHDB-375404

Sun's conjecture on the new constant $w_r(n)$

摘录 / EXCERPT: Let $n>1$ and $r>0$ be integers. Define $w_r(n)$ as the least positive integer k such that, for any k integer vectors ${\bf a}_i=(a_{i1},\ldots,a_{ir})$ not congruent to...

conjecture | open | MathDB 页面 | 原文线索

V2-072524 | MATHDB-375405

Sun's conjecture on the constants $s_1(n)$ and $s_2(n)$

摘录 / EXCERPT: Let $n>1$ and $r>0$ be integers. Define $s_r(n)$ as the least positive integer k such that every sequence of k integer vectors in the relevant rank-r setting contains a selec...

conjecture | open | MathDB 页面 | 原文线索

V2-072525 | MATHDB-375406

The logarithmic list-replicability conjecture for sign matrices

摘录 / EXCERPT: Let A be an $N \times M$ sign matrix, and let $\operatorname{\textnormal{\textsc{lr}}}(A)$ denote its list replicability number. List-replicability conjecture. Every $N \times M$...

conjecture | solved | MathDB 页面 | 原文线索

V2-072526 | MATHDB-375410

Kerz's commutativity conjecture for isomonodromic Higgs-bundle deformations

摘录 / EXCERPT: Kerz's commutativity conjecture. (i) For any $\lambda\in\mathbb R^*$, $\lambda\cdot\sigma_{\operatorname{Dol}}$ is also an isomonodromic deformation. (ii) For any...

conjecture | open | MathDB 页面 | 原文线索

V2-072527 | MATHDB-375420

Macroscopic boundary conjecture for the left side of the t-split two-periodic Aztec diamond

摘录 / EXCERPT: Macroscopic boundary conjecture. The macroscopic boundary of the t-split two-periodic Aztec diamond to the left of the interface is governed by this saddle function: the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-072528 | MATHDB-375421

Cusp-existence conjecture for the t-split two-periodic Aztec diamond

摘录 / EXCERPT: Let α and β be parameters satisfying ... Let t be the interface parameter, and let a smooth region mean a smooth region of the limit shape to the left of the interface...

conjecture | open | MathDB 页面 | 原文线索

V2-072529 | MATHDB-375429

Injectivity conjecture for skew Schur partition functions of two-row partitions

摘录 / EXCERPT: Let λ' be a two-row partition, and let $\mathrm{prs}_{\lambda'}$ be the associated skew Schur partition function, defined on those partitions for which the function is defi...

conjecture | open | MathDB 页面 | 原文线索

V2-072530 | MATHDB-375435

Dallard et al.'s induced-grid characterization conjecture for star-free graph classes

摘录 / EXCERPT: Let d be a positive integer and let $\mathcal{G}$ be a hereditary $K_{1,d}$-free graph class. Dallard et al.'s conjecture. The class $\mathcal{G}$ has bounded tree-independence n...

conjecture | open | MathDB 页面 | 原文线索

V2-072531 | MATHDB-375436

Polynomial bound for tree-independence number in star-free induced-grid-minor-free graphs

摘录 / EXCERPT: Let $d\geq 2$ be an integer, let G be a $K_{1,d}$-free graph, and let f be a polynomial. Polynomial-bound conjecture. There is a polynomial f such that, whenever G does not...

conjecture | open | MathDB 页面 | 原文线索

V2-072532 | MATHDB-375437

Bounded sim-width versus tree-independence number for star-free graph classes

摘录 / EXCERPT: Let $d\geq 2$, and let $\mathcal{G}$ be a $K_{1,d}$-free graph class. Sim-width conjecture. If $\mathcal{G}$ has bounded sim-width, then $\mathcal{G}$ has bounded tree-independence...

conjecture | open | MathDB 页面 | 原文线索

V2-072533 | MATHDB-375438

Bounded twin-width three versus tree-independence number for star-free graph classes

摘录 / EXCERPT: Let $d\geq 2$, and let $\mathcal{G}$ be a $K_{1,d}$-free graph class of twin-width at most 3. Twin-width conjecture. Does $\mathcal{G}$ have bounded tree-independence number? The...

conjecture | open | MathDB 页面 | 原文线索

V2-072534 | MATHDB-375440

Conditional positivity conjecture for Q_{01}

摘录 / EXCERPT: Conditional positivity conjecture. Q_{01} has nonnegative coefficients if and only if $p=2$ and $q\equiv 1 \mod 4$.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072535 | MATHDB-375444

Bøgvad's conjecture for smooth polytopes

摘录 / EXCERPT: Bøgvad's conjecture. The toric ideal I_P is quadratic.

conjecture | open | MathDB 页面 | 原文线索

V2-072536 | MATHDB-375446

Perfectly contractile graphs and quadratic stable-set ideals

摘录 / EXCERPT: Perfect-contractility conjecture. The following conditions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-072537 | MATHDB-375451

Meridian-link characterization of ribbon minimal knots

摘录 / EXCERPT: Let K be a knot in S^3, and let μ_K be a meridian of K. Define the two-component link ... A link is ribbon concordance minimal if it has no strictly smaller link under ribb...

conjecture | open | MathDB 页面 | 原文线索

V2-072538 | MATHDB-375452

Nonminimal knot as a component of a ribbon minimal link

摘录 / EXCERPT: A knot is ribbon minimal if it is minimal under ribbon concordance, and a link is ribbon concordance minimal if it is minimal under ribbon concordance. Component-separation conject...

conjecture | open | MathDB 页面 | 原文线索

V2-072539 | MATHDB-375459

Efficient sampling conjecture for Potts models on random regular bipartite graphs

摘录 / EXCERPT: Let G be a random Δ-regular bipartite graph on $2n$ vertices, let $\mu_{G,q;\beta}$ be the anti-ferromagnetic q-state Potts Gibbs distribution, and let an FPAUS mean a f...

conjecture | open | MathDB 页面 | 原文线索

V2-072540 | MATHDB-375460

FPTAS conjecture for Potts models on random regular bipartite graphs

摘录 / EXCERPT: Let G be a random Δ-regular bipartite graph, and let $Z_{G,q}(\beta)$ be the partition function of the anti-ferromagnetic q-state Potts model on G. Random-regular-grap...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072541 | MATHDB-375463

Hurwitz stability conjecture for Laguerre-polynomial resultants

摘录 / EXCERPT: Resultant stability conjecture. For any positive integers n,m such that $n\leq m$, and any $a>0$, the resultant $\operatorname{res}_z(P_n(s,z),P_m(s+a,z))$ is stable.

conjecture | open | MathDB 页面 | 原文线索

V2-072542 | MATHDB-375464

Carlet-Lou weight-spectrum conjecture for Reed-Muller codes

摘录 / EXCERPT: Let $RM(r,m)$ denote the Reed-Muller code of order r in m variables, and let its weight spectrum be the set of Hamming weights of its codewords. For a set S of weights, write...

conjecture | open | MathDB 页面 | 原文线索

V2-072543 | MATHDB-375469

Optimality conjecture for the Hausdorff-dimension bound on the singular set

摘录 / EXCERPT: Let f be a nonlinearity in the smooth asymptotic class of the dimensional-bound theorem, let T_f denote its blow-up level, and define ... For a corresponding stable solution...

conjecture | open | MathDB 页面 | 原文线索

V2-072544 | MATHDB-375470

Odd-cycle decomposition threshold conjecture

摘录 / EXCERPT: Odd-cycle decomposition threshold conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072545 | MATHDB-375472

Picard-Lefschetz/Alien correspondence for the heat kernel

摘录 / EXCERPT: Picard-Lefschetz/Alien correspondence. The pointed Alien derivative of $\widetilde K_0$ at ω is

conjecture | open | MathDB 页面 | 原文线索

V2-072546 | MATHDB-375484

Weak form of Greenberg's conjecture

摘录 / EXCERPT: Weak form of Greenberg's conjecture. If $X(k^\mathrm{c}_\infty)$ is nontrivial, then it has a nontrivial finite $bZ_p\llbracket\operatorname{Gal}(k^\mathrm{c}_\infty/k)\rrbracket$-...

conjecture | open | MathDB 页面 | 原文线索

V2-072547 | MATHDB-375487

Generalized Gross conjecture for cyclotomic coinvariants

摘录 / EXCERPT: Generalized Gross conjecture. The coinvariant group $X'(k^\mathrm{c}_\infty)_{\operatorname{Gal}(k^\mathrm{c}_\infty/k)}$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-072548 | MATHDB-375493

Categorical Langlands conjecture for the full derived category

摘录 / EXCERPT: Let G be a quasi-split reductive group over E, let $\check{G}$ be its Langlands dual group, and assume $\ell\nmid\lvert\pi_0(Z(G))\rvert$. Let W_E be the Weil group and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072549 | MATHDB-375518

Bumpless pipe dream transition compatibility

摘录 / EXCERPT: Let B be a k-row reduced bumpless pipe dream such that row k has no blanks (“weighty tiles”). Let $\phi(B)$ be the RC graph corresponding to B under the natural bijection b...

conjecture | open | MathDB 页面 | 原文线索

V2-072550 | MATHDB-375522

Perfect-power maximality conjecture for periodic grid quotients

摘录 / EXCERPT: Periodic perfect-power maximality conjecture. Its spanning-tree count is at most

conjecture | open | MathDB 页面 | 原文线索

V2-072551 | MATHDB-375525

Andrews-Merca-Guo-Zeng truncated Jacobi triple product positivity conjecture

摘录 / EXCERPT: Andrews-Merca-Guo-Zeng conjecture. This coefficient is nonnegative.

conjecture | open | MathDB 页面 | 原文线索

V2-072552 | MATHDB-375527

Finite obstruction conjecture for classes of a fixed limiting density

摘录 / EXCERPT: For a graph class $\mathcal{G}$, let $\operatorname{obs}(\mathcal{G})$ denote its minor obstruction set. For a class property $\mathbb{C}$, let $\operatorname{cobs}(\mathbb{C})$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-072553 | MATHDB-375540

Ai-Gutin-Lei-Yeo-Zhou's converse-invariant tree conjecture

摘录 / EXCERPT: Let T be an orientation of a tree with maximum degree at least 3. An oriented graph D is converse invariant if $f_T(D)=f_T(\overline{D})$ for every tournament T, where...

conjecture | open | MathDB 页面 | 原文线索

V2-072554 | MATHDB-375544

Reduced conjecture on conformal radii for non-Bruno rotation numbers

摘录 / EXCERPT: Reduced conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072555 | MATHDB-375545

Conjectural support and laminarity description for the bifurcation current

摘录 / EXCERPT: Conjectural support and laminarity description. Point (Q:support) holds: the support of T is homeomorphic to the support of $dd^c f$. Moreover, the difference...

conjecture | open | MathDB 页面 | 原文线索

V2-072556 | MATHDB-375557

Stable-factorization conjecture for quadratic polynomials over the Gaussian rationals

摘录 / EXCERPT: Stable-factorization conjecture. The polynomial f_r is eventually stable over $\mathbb{Q}(i)$ with constant $C(f_r,0)=4$, and exactly one of the following cases holds:

conjecture | open | MathDB 页面 | 原文线索

V2-072557 | MATHDB-375558

Second-iterate irreducibility conjecture for quadratic polynomials over the Gaussian rationals

摘录 / EXCERPT: Second-iterate irreducibility conjecture. If $f_r^2(x)$ is irreducible over K, then f_r is stable over K.

conjecture | open | MathDB 页面 | 原文线索

V2-072558 | MATHDB-375559

Non-square conjecture for the Gaussian-integer recurrence factors

摘录 / EXCERPT: Non-square conjecture. For every $n\ge3$, both $b_n^+(c)$ and $b_n^-(c)$ are non-square in $\mathbb{Z}[i]$.

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072559 | MATHDB-375562

Log-normal approximation for the Wasserstein distance between Gaussian samples

摘录 / EXCERPT: Let $\mathbf{W}^{(n)}=\mathrm{W}_2^2(\hat\mu^{1,(n)},\hat\nu^{1,(n)})$ be the squared Wasserstein distance between two empirical samples of size n from the standard normal distri...

conjecture | open | MathDB 页面 | 原文线索

V2-072560 | MATHDB-375563

Log-normal approximation for the Wasserstein normality-test statistic

摘录 / EXCERPT: Let $Y_1,\ldots,Y_n$ be independent identically distributed samples from a Gaussian distribution, let $\mathbf{y}=(Y_1,\ldots,Y_n)$, let $\bar{\mathbf{y}}$ be their sample mean, an...

conjecture | open | MathDB 页面 | 原文线索

V2-072561 | MATHDB-375564

Finitely generatedness of continuously homogeneous indecomposable planar continua

摘录 / EXCERPT: Finitely generatedness conjecture. Any continuously homogeneous indecomposable planar continuum is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-072562 | MATHDB-375569

Mazur-Swinnerton-Dyer's Iwasawa main conjecture for elliptic curves

摘录 / EXCERPT: Mazur-Swinnerton-Dyer's conjecture. There is an equality of ideals in Λ

conjecture | open | MathDB 页面 | 原文线索

V2-072563 | MATHDB-375583

Monotonicity conjecture for the optimal grouped MSE constant

摘录 / EXCERPT: Let Δ_k denote the optimal asymptotic MSE constant within the generalized grouping (GG) family for groups of size k. Monotonicity conjecture. ... i.e., the optimal asymptoti...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072564 | MATHDB-375584

Optimal injectivity conjecture for Artin monoid homomorphisms

摘录 / EXCERPT: Optimal injectivity conjecture. Any optimal and injective on generators homomorphism in $\mathscr{Ast}$ is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-072565 | MATHDB-375585

Generalized Tits injectivity conjecture for Artin monoids

摘录 / EXCERPT: Generalized Tits injectivity conjecture. All Tits homomorphisms $\mathbf T_{\mathbf d}$ are injective.

conjecture | open | MathDB 页面 | 原文线索

V2-072566 | MATHDB-375586

Injectivity criterion for dihedral standard homomorphisms into type A

摘录 / EXCERPT: Dihedral injectivity criterion. Both induced maps $\overline\Phi$ and $\overline\Phi_\star$ are injective if and only if either both Φ and Φ are self-orthogonal, or K s...

conjecture | open | MathDB 页面 | 原文线索

V2-072567 | MATHDB-375587

Classification conjecture for fully supported standard A2 homomorphisms

摘录 / EXCERPT: A2 classification conjecture. Up to diagram automorphisms, either $M=A_{3m-1}$ for some $m\ge1$ and $\Phi=\Phi_{m,J}$ with $J\subset[1,m-1]$ as in Theorem, or $M=E_6$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072568 | MATHDB-375588

Conjectural interval-family homomorphisms from dihedral type to type A

摘录 / EXCERPT: Interval-family homomorphism conjecture. The assignments

conjecture | open | MathDB 页面 | 原文线索

V2-072569 | MATHDB-375589

Conjectural type-A standard homomorphisms with a generator times the longest element

摘录 / EXCERPT: Type-A twelve-period homomorphism conjecture. Define

conjecture | open | MathDB 页面 | 原文线索

V2-072570 | MATHDB-375590

Conjectural standard homomorphism families from dihedral type to type D

摘录 / EXCERPT: Type-D family conjecture. The following assignments define standard homomorphisms:

conjecture | open | MathDB 页面 | 原文线索

V2-072571 | MATHDB-375600

Uniform equivariant quasi-isometry conjecture for geometrically rigid hyperbolic groups

摘录 / EXCERPT: Let G be a geometrically rigid hyperbolic group. An abstract commensurator of G is an isomorphism $\phi:H_1\to H_2$ between finite-index subgroups $H_1,H_2\leq G$. A map...

conjecture | open | MathDB 页面 | 原文线索

V2-072572 | MATHDB-375608

The conditional W-way MPST representation conjecture

摘录 / EXCERPT: Conditional W-way MPST conjecture. There is a unique inner- and outer-regular finite Borel measure $\widetilde\mu$ on $\widehat{\mathcal{A}}_{\mathrm{cond}}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-072573 | MATHDB-375610

Schur positivity conjecture for skew modified Hall-Littlewood functions

摘录 / EXCERPT: Schur positivity conjecture. The coefficients $a^\nu_{\lambda\mu}(t)$ are nonnegative integer polynomials in t.

conjecture | open | MathDB 页面 | 原文线索

V2-072574 | MATHDB-375626

Signature-native graph recovery for sparse multivariate Hawkes kernels

摘录 / EXCERPT: For a sparse multivariate Hawkes process, let A_T^{ij} denote the cross-area between coordinates i and j over a time horizon T, and let the associated excitation graph enco...

conjecture | open | MathDB 页面 | 原文线索

V2-072575 | MATHDB-375627

The Dolbeault geometric Langlands conjecture with nilpotent singular support

摘录 / EXCERPT: Let G be a reductive group, let $\chi\in\pi_1(G)$ and let $w\in Z_G^{\vee}$, where Z_G is the center of G. Let $\mathrm{Higgs}_G(\chi)$ denote the connected component of the...

conjecture | open | MathDB 页面 | 原文线索

V2-072576 | MATHDB-375628

The Whittaker normalization conjecture for the Fourier-Mukai transform

摘录 / EXCERPT: Let $G=\operatorname{GL}_r$. Let $\mathcal{B}\subset(\mathrm{B}_{\operatorname{GL}_r}^{\mathrm{red}})^{\mathrm{cl}}$ be an open subset, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072577 | MATHDB-375632

The $C^{1,b1}$-rectifiability conjecture for the singular set

摘录 / EXCERPT: Let Σ be the singular set of the anisotropic obstacle problem, and let A be the coefficient matrix with ellipticity constants. $C^{1,\alpha}$-rectifiability conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-072578 | MATHDB-375641

The Street-Roberts conjecture for complicial spaces

摘录 / EXCERPT: Let $(\bDelta^n)$ denote the oriented simplex, let $\tau_{n-1}(\bDelta^n)$ denote its $(n-1)$-truncation, and let $(\Delta^n)^t$ be the stratified simplicial space obtained from th...

conjecture | open | MathDB 页面 | 原文线索

V2-072579 | MATHDB-375643

Cyclic interval conjecture for odd cycles

摘录 / EXCERPT: Cyclic interval conjecture. The displayed inequality holds for every odd m and every nonempty proper subset $x\subseteq\mathbb{Z}_m$; consequently,

conjecture | open | MathDB 页面 | 原文线索

V2-072580 | MATHDB-375651

Polynomial point-count conjecture for moduli spaces of smooth curves

摘录 / EXCERPT: Let $\mathcal{M}_{g,n}$ be the moduli space of smooth curves of genus g with n marked points. Polynomial point-count conjecture. The space $\mathcal{M}_{g,n}$ has polynomial po...

conjecture | open | MathDB 页面 | 原文线索

V2-072581 | MATHDB-375652

Gal's nonnegativity conjecture for the second gamma-number of flag spheres

摘录 / EXCERPT: Let Δ be a flag simplicial d-sphere. Its second gamma-number $\gamma_2(\Delta)$ is defined by ... where $f_0(\Delta)$ and $f_1(\Delta)$ are the numbers of vertices and edges...

conjecture | open | MathDB 页面 | 原文线索

V2-072582 | MATHDB-375654

Tóth's phase-transition conjecture for long cycles in the three-dimensional interchange process

摘录 / EXCERPT: Tóth's conjecture. The quantity

conjecture | open | MathDB 页面 | 原文线索

V2-072583 | MATHDB-375657

Pure-point diffraction conjecture for visible points of irreducible cut-and-project sets

摘录 / EXCERPT: Let $d\geq 2$ and let $\Lambda=\Lambda(\mathcal{W},\mathcal{L})\subset\mathbb{R}^d$ be an irreducible cut-and-project set. Let $\mathcal{A}\subset\mathbb{R}^d$ be a van Hove sequen...

conjecture | open | MathDB 页面 | 原文线索

V2-072584 | MATHDB-375660

Conjecture on the unbounded coefficients of the companion series $C(q)$

摘录 / EXCERPT: Let $C(q)=\sum_{n\geq 0}c(n)q^n$ be the companion series whose coefficients are denoted by $c(n)$. Unboundedness conjecture. ... The conjecture asserts that the coefficients of...

conjecture | open | MathDB 页面 | 原文线索

V2-072585 | MATHDB-375671

Sagan-Swanson's monomial basis conjecture for superspace coinvariants

摘录 / EXCERPT: Sagan-Swanson's conjecture. The set of monomials ${\mathcal{A}}_{n,r}$ descends to a vector space basis of $SR_{n,r}$.

conjecture | open | MathDB 页面 | 原文线索

V2-072586 | MATHDB-375682

Erdős-Rubin-Taylor conjecture on planar graph choosability

摘录 / EXCERPT: A planar graph is a graph that can be drawn in the plane without crossings. Erdős-Rubin-Taylor's conjecture. Every planar graph is 5-choosable. Since planar graphs are 5-degenerate...

conjecture | open | MathDB 页面 | 原文线索

V2-072587 | MATHDB-375683

Erdős-Rubin-Taylor conjecture on non-4-choosable planar graphs

摘录 / EXCERPT: A planar graph is a graph that can be drawn in the plane without crossings. A graph is 4-choosable if every assignment of lists of four colors admits a proper coloring from those l...

conjecture | open | MathDB 页面 | 原文线索

V2-072588 | MATHDB-375684

Alon's logarithmic bound conjecture for bipartite graph list coloring

摘录 / EXCERPT: Let G be a bipartite graph, and let $\Delta(G)$ denote its maximum degree. Alon's conjecture. There exists a constant c such that ... This conjecture concerns whether bipartite g...

conjecture | open | MathDB 页面 | 原文线索

V2-072589 | MATHDB-375685

Ohba's chromatic-choosability conjecture

摘录 / EXCERPT: For a graph G, let $\chi(G)$ be its chromatic number, and let $\chi_\ell(G)$ be its list chromatic number. A graph is chromatic-choosable when $\chi_\ell(G)=\chi(G)$. Ohba's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-072590 | MATHDB-375686

He-et al. complete multipartite chromatic-choosability conjecture

摘录 / EXCERPT: A complete k-partite graph is a graph whose vertices are partitioned into k independent parts, with every pair of vertices in distinct parts adjacent. He et al.'s conjecture. A...

conjecture | open | MathDB 页面 | 原文线索

V2-072591 | MATHDB-375688

Gravier-Maffray list-coloring conjecture for claw-free graphs

摘录 / EXCERPT: A graph is claw-free if it has no induced subgraph isomorphic to $K_{1,3}$. Let $\chi(G)$ and $\chi_\ell(G)$ denote the chromatic and list chromatic numbers. Gravier-Maffray's conj...

conjecture | open | MathDB 页面 | 原文线索

V2-072592 | MATHDB-375690

Gravier et al.'s chromatic-choosability conjecture for claw-free perfect graphs

摘录 / EXCERPT: A graph is claw-free perfect if it is both claw-free and perfect. Let $\chi(G)$ and $\chi_\ell(G)$ denote its chromatic and list chromatic numbers. Gravier et al.'s conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-072593 | MATHDB-375694

Kostochka-Woodall list square coloring conjecture

摘录 / EXCERPT: For a graph G, its square G^2 is the graph on $V(G)$ in which two vertices are adjacent when their distance in G is at most two. Let χ and χ_ℓ denote the chromat...

conjecture | open | MathDB 页面 | 原文线索

V2-072594 | MATHDB-375695

Havet et al.'s list square coloring conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph, and let G^2 denote its square, in which two vertices are adjacent when their distance in G is at most two. Havet et al.'s conjecture. For any planar...

conjecture | open | MathDB 页面 | 原文线索

V2-072595 | MATHDB-375697

Borowiecki-Jozef asymptotic Cartesian product conjecture

摘录 / EXCERPT: Let G_1 and G_2 be graphs, and let Δ be a common upper bound for their maximum degrees, so that $\Delta(G_1)\leq\Delta$ and $\Delta(G_2)\leq\Delta$. Borowiecki-Jozef's a...

conjecture | open | MathDB 页面 | 原文线索

V2-072596 | MATHDB-375698

Lu et al.'s odd-cycle Cartesian product conjecture

摘录 / EXCERPT: Let C_r denote the cycle on r vertices. Lu et al.'s conjecture. For positive integers m,n, ... The supplied text establishes the corresponding values for products involving eve...

conjecture | open | MathDB 页面 | 原文线索

V2-072597 | MATHDB-375701

Pretransitivity conjecture for modal interpretations

摘录 / EXCERPT: Pretransitivity conjecture. If a normal modal logic Λ is interpretable by a modal-to-modal translation with parameters in a pretransitive normal logic, then Λ is pr...

conjecture | open | MathDB 页面 | 原文线索

V2-072598 | MATHDB-375712

Bergeron-Garsia-Haiman-Tesler signed Schur-positivity conjecture for nabla of monomial symmetric functions

摘录 / EXCERPT: Let n be a positive integer, and let mu and $lambda$ be partitions of n. Let m_μ be the monomial symmetric function, s_λ the Schur function, and ∇ the Berg...

conjecture | open | MathDB 页面 | 原文线索

V2-072599 | MATHDB-375721

Toda's chi-independence conjecture for BPS perverse sheaves

摘录 / EXCERPT: Toda's chi-independence conjecture. For any $\gamma\in B_S(\alpha)$, the invariant $\Phi(\alpha+n\delta)$ is independent of $n\in\mathbb{Z}$.

conjecture | open | MathDB 页面 | 原文线索

V2-072600 | MATHDB-375724

Real-rootedness conjecture for the squares of the Eulerian and Delannoy triangles

摘录 / EXCERPT: Let A and D denote the Eulerian and Delannoy triangles, respectively, and let A^2 and D^2 be their matrix squares. For a lower triangular matrix M, write its n-th row g...

conjecture | open | MathDB 页面 | 原文线索

V2-072601 | MATHDB-375725

Mu-Welker's total positivity conjecture for the barycentric subdivision transformation matrix

摘录 / EXCERPT: Let $\mathcal{F}$ be the barycentric subdivision, and let $H_{\mathcal{F}}$ be the transformation matrix encoding the transformation of the h-vector under an $\mathcal{F}$-unifor...

conjecture | open | MathDB 页面 | 原文线索

V2-072602 | MATHDB-375743

Low-codimension perfect-pairing conjecture for compact type moduli

摘录 / EXCERPT: Low-codimension pairing conjecture. For $i\leq 3$, these pairings are perfect. If $i=4$, the pairing has rank $\dim \mathsf{R}^4(\mathcal{M}_{g,n}^{\operatorname{ct}})$. The conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-072603 | MATHDB-375746

The double EPW sextic realization conjecture for Enriques moduli K3 surfaces

摘录 / EXCERPT: Let S be a polarized K3 surface in $\mathfrak{M}_{\operatorname{BE}}(2t)$ with $t\geq 2$. For some a satisfying $a^2\equiv -1\pmod{t}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072604 | MATHDB-375747

The Hodge-special Gushel-Mukai realization conjecture for Enriques moduli K3 surfaces

摘录 / EXCERPT: Let S be a K3 surface in $\mathfrak{M}_{\operatorname{BE}}(2t)$ with $t\geq 10$. An $\mathfrak{S}_2$-equivariant equivalence is an equivalence of derived categories compatible wi...

conjecture | open | MathDB 页面 | 原文线索

V2-072605 | MATHDB-375748

The Bridgeland conjecture for Enriques realizable K3 surfaces

摘录 / EXCERPT: Let S be an Enriques realizable K3 surface, meaning a K3 surface arising in the Enriques-related moduli setting considered in the source. Bridgeland conjecture. S is Bridgeland...

conjecture | open | MathDB 页面 | 原文线索

V2-072606 | MATHDB-375753

Faudree-Gyárfás-Schelp-Tuza strong clique index conjecture

摘录 / EXCERPT: For a finite simple graph G, let $L(G)$ denote its line graph, let G^2 denote the graph in which two vertices are adjacent exactly when they are at distance at most two in G,...

conjecture | open | MathDB 页面 | 原文线索

V2-072607 | MATHDB-375754

Faron-Postle Ore-degree conjecture for line-graph cliques

摘录 / EXCERPT: Let G be a finite simple graph. For a non-empty subgraph H of G, define its Ore-degree in G by ... with $\sigma_G(H)=0$ when H is empty. Suppose that H is a bipartite sub...

conjecture | open | MathDB 页面 | 原文线索

V2-072608 | MATHDB-375765

The tree product conjecture for graphs of polynomial growth

摘录 / EXCERPT: Let H be a finite graph, and write $f_H(r)$ for its growth function, namely the maximum number of vertices in a ball of radius r. For graphs $G_1,\dots,G_k$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072609 | MATHDB-375769

The vexillary Grothendieck Lam-Postnikov-Pylyavskyy inequality

摘录 / EXCERPT: Let $u,v\in S_n$ be vexillary permutations, let $\phi(w)$ be the Wachs flag of w, and write $a\preceq b$ when a is a prefix of b. Let $\mathfrak G_w^{(\beta)}(\mathbf x)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-072610 | MATHDB-375770

The stable Grothendieck Lam-Postnikov-Pylyavskyy inequality

摘录 / EXCERPT: Let λ and μ be partitions, and let $\widetilde G_\lambda(\mathbf x)$ denote the stable Grothendieck polynomial. Write $\lambda\vee\mu$ and $\lambda\wedge\mu$ for the jo...

conjecture | open | MathDB 页面 | 原文线索

V2-072611 | MATHDB-375799

Unconditional logarithmic rent conjecture

摘录 / EXCERPT: Unconditional logarithmic rent conjecture. The bound continues to hold when (L2)--(L4) are dropped: when the excitation is endogenous to the designer's policy, agents learn from th...

conjecture | open | MathDB 页面 | 原文线索

V2-072612 | MATHDB-375801

Conjecture that the involution of the Grothendieck ring is a group automorphism

摘录 / EXCERPT: Let $\mathfrak{r}$ be the group equipped with the involution d. The group-automorphism conjecture. The involution d of $\mathfrak{r}$ is a group automorphism. This is proposed...

conjecture | open | MathDB 页面 | 原文线索

V2-072613 | MATHDB-375803

Associativity conjecture for tensor products in shifted category O

摘录 / EXCERPT: Let V_1,V_2,V_3 be objects of the shifted category $\mathcal{O}_{sh}$. Associativity conjecture. There exists an isomorphism ... Thus, k-fold tensor products in...

conjecture | open | MathDB 页面 | 原文线索

V2-072614 | MATHDB-375806

Frenkel-Hernandez's Q-variable realization conjecture

摘录 / EXCERPT: Let $(i,a)\in I\times_2\mathbb Z$ and $w\in W$, and let $Q_{w\varpi_i,a}$ be the corresponding formal solution of the extended QQ-system. Let χ_{ℓ} denote the ℓ-char...

conjecture | open | MathDB 页面 | 原文线索

V2-072615 | MATHDB-375809

Congruence subgroup conjecture for groups generated by root unipotents

摘录 / EXCERPT: Fix a nonzero rational number $q=\tfrac{s}{t}\in\mathbb{Q}\cap(-4,4)$ in lowest terms, with $\gcd(s,t)=1$. Let Γ_q be the group generated by the root unipotents associated t...

conjecture | open | MathDB 页面 | 原文线索

V2-072616 | MATHDB-375810

A q-supercongruence for a truncated Appell series when n is 3 modulo 4

摘录 / EXCERPT: Let n be a positive odd integer with $n\equiv 3\pmod{4}$, let q be an indeterminate, and let $(a;q)_k$ denote the q-shifted factorial. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-072617 | MATHDB-375812

Erdős-Graham conjecture on monochromatic Pythagorean triples

摘录 / EXCERPT: Erdős-Graham conjecture. Any finite colouring of $\mathbb{N}$ has a monochromatic Pythagorean triple.

conjecture | open | MathDB 页面 | 原文线索

V2-072618 | MATHDB-375816

The $b[200~\Lambda\Sigma=\Delta$ character conjecture for cluster parking functions and diagonal coinvariantsb[201~

摘录 / EXCERPT: The $\Lambda\Sigma=\Delta$ conjecture. We have $\Lambda\Sigma=\Delta$.

conjecture | open | MathDB 页面 | 原文线索

V2-072619 | MATHDB-375818

The rotational case of the $\Lambda\Sigma=\Delta$ conjecture

摘录 / EXCERPT: Rotational $\Lambda\Sigma=\Delta$ conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072620 | MATHDB-375827

Deterministic constant-column-sparsity small-singular-value conjecture

摘录 / EXCERPT: Fix an integer $d\geq 2$ and a constant $C>0$. Let $r(k),n_k\in\mathbb{N}$ satisfy ... For each k, let $M^{(k)}\in\{-1,0,1\}^{k\times n_k}$ be a deterministic matrix whose every co...

open_problem | open | MathDB 页面 | 原文线索

V2-072621 | MATHDB-375841

The Kaneko-Zagier conjecture relating finite and symmetric multiple zeta values

摘录 / EXCERPT: Kaneko-Zagier conjecture. There exists an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-072622 | MATHDB-375852

The refined Malle conjecture for the Heisenberg group over \(\mathbb{F}_4\)

摘录 / EXCERPT: Refined Malle conjecture for ${\rm Heis}_4$. The group ${\rm Heis}_4$ fails to satisfy independence of local events, and

conjecture | open | MathDB 页面 | 原文线索

V2-072623 | MATHDB-375854

Eventual real-rootedness conjecture for GKP polynomial sequences

摘录 / EXCERPT: Eventual real-rootedness conjecture. For all sufficiently large n, every zero of $q_n^{u,v}$ is real.

conjecture | open | MathDB 页面 | 原文线索

V2-072624 | MATHDB-375866

Reed's chromatic bound conjecture

摘录 / EXCERPT: Let G be a graph, with maximum degree $\Delta(G)$ and clique number $\omega(G)$. Reed's conjecture. Every graph G satisfies ... This refines the Borodin-Kostochka direction and i...

conjecture | open | MathDB 页面 | 原文线索

V2-072625 | MATHDB-375868

Erdős-Neumann-Lara conjecture for oriented graphs

摘录 / EXCERPT: Let D be an oriented graph, and let Δ be the maximum degree of the underlying graph of D. Let $\vec{\chi}(D)$ denote its dichromatic number. Erdős-Neumann-Lara conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-072626 | MATHDB-375884

Analytic spread formula for pairs of cycles

摘录 / EXCERPT: Let C_n and C_m be cycles with $n,m\geq 3$, let J_{C_n,C_m} be their associated binomial edge ideal, and let $\ell(J_{C_n,C_m})$ denote its analytic spread. Assume that the c...

conjecture | open | MathDB 页面 | 原文线索

V2-072627 | MATHDB-375895

Equipartition duality for multi-degree-of-freedom kime states

摘录 / EXCERPT: Equipartition duality. Among all such states, the phase-equipartitioned product state, unique when it exists, simultaneously (i) maximizes the entropy, (ii) minimizes every within-...

conjecture | open | MathDB 页面 | 原文线索

V2-072628 | MATHDB-375898

The odd-cycle naive-dimension conjecture

摘录 / EXCERPT: Odd-cycle naive-dimension conjecture. The cyclic interval lemma holds for every odd m; consequently,

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072629 | MATHDB-375913

Conjectural rank distribution for Sylow subgroups of random bipartite graph sandpile groups

摘录 / EXCERPT: Rank distribution conjecture. If p is odd, then

conjecture | open | MathDB 页面 | 原文线索

V2-072630 | MATHDB-375914

Gateva-Ivanova's strong retraction conjecture for squarefree cycle sets

摘录 / EXCERPT: Gateva-Ivanova's strong conjecture. If X is a squarefree cycle set with $|X|>1$, then X is retractable.

conjecture | refuted | MathDB 页面 | 原文线索

V2-072631 | MATHDB-375915

Rump's retraction conjecture for quasilinear cycle sets

摘录 / EXCERPT: Rump's retraction conjecture. If A is a quasilinear cycle set with $|A|>1$, then A is retractable.

conjecture | open | MathDB 页面 | 原文线索

V2-072632 | MATHDB-375935

Asymptotic product formula conjecture for weighted binary-sum Hankel determinants

摘录 / EXCERPT: Let $\mathcal H(n,1)$ denote the Hankel determinant studied in the paper. For each fixed integer $k\geq 0$, consider the values with index $n=2^m+k$. Asymptotic product formula con...

conjecture | open | MathDB 页面 | 原文线索

V2-072633 | MATHDB-375936

Monotonicity conjecture for absolute values of weighted binary-sum Hankel determinants

摘录 / EXCERPT: Let (n_k) be the sequence defined earlier in the paper, and let $\mathcal H(n,1)$ be the associated Hankel determinant. Monotonicity conjecture. The sequence of absolute values...

conjecture | open | MathDB 页面 | 原文线索

V2-072634 | MATHDB-375941

Nivat's conjecture

摘录 / EXCERPT: Nivat's conjecture. Then c is 1-periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-072635 | MATHDB-375950

Joint recovery conjecture for coupled tensor-matrix estimation

摘录 / EXCERPT: Let $\mathcal{T}$ be a tensor whose mode-1 unfolding $\mathcal{T}_{(1)}$ is observed on $\Omega_{\mathcal{T}}$, let M be a matrix observed on Ω_M, and let G be the unkno...

conjecture | open | MathDB 页面 | 原文线索

V2-072636 | MATHDB-375958

Compatibility conjecture for structured decompositions of persistent narratives

摘录 / EXCERPT: Let $\textup{\textsf{T}}$ be a finite discrete time category, let $(\mathcal{C},\mathcal{G},\Omega)$ be a spined sd-category with the additional structure required for persistent n...

conjecture | open | MathDB 页面 | 原文线索

V2-072637 | MATHDB-375969

The nonvanishing and square-root bound conjecture for modular-symbol Ramanujan sums

摘录 / EXCERPT: Let $R_j^+(n)$ denote the scalar quantity associated with the embedded spectral contribution, and let $r(X)$ be the corresponding error term in the twisted Ramanujan-sum asymptotic...

conjecture | open | MathDB 页面 | 原文线索

V2-072638 | MATHDB-375973

Merca's positivity conjecture for truncated Jacobi theta series

摘录 / EXCERPT: Merca's conjecture. This theta series has non-negative coefficients. The conjecture refines the established positivity theorem for averaged truncations of Jacobi's triple product i...

conjecture | open | MathDB 页面 | 原文线索

V2-072639 | MATHDB-375974

Positivity conjecture for the Laurent polynomials \mathscr{L}_{k,n}^{d}(z)

摘录 / EXCERPT: Positivity conjecture. For all integers $k\ge1$, $d\ge2$, and $n\ge0$, the Laurent polynomial $\mathscr{L}_{k,n}^{d}(z)$ has positive coefficients. This conjecture is presented as...

conjecture | open | MathDB 页面 | 原文线索

V2-072640 | MATHDB-375981

The skew shape poset block-statistic conjecture

摘录 / EXCERPT: Let P be a skew shape poset of size n, let $S_n(P)$ denote the set of P-partitions of the labeling set represented by permutations, and let $\widetilde{\bl}_P(\sigma)$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072641 | MATHDB-375982

The circular fence poset block-statistic conjecture

摘录 / EXCERPT: Let P be a circular fence poset of size n, let $S_n(P)$ denote the relevant set of permutations, and let $\widehat{\bl}_P(\sigma)$ be the block statistic introduced for circular...

conjecture | open | MathDB 页面 | 原文线索

V2-072642 | MATHDB-375987

Champagnat-Bovier conjecture on the dimorphic state after fast branching

摘录 / EXCERPT: Champagnat-Bovier's dimorphic-state conjecture. As $(K,\mu_K,\sigma_K)\to(+\infty,0,0)$,

conjecture | open | MathDB 页面 | 原文线索

V2-072643 | MATHDB-375990

Hilbert-depth conjecture for intersections of disjoint variable ideals

摘录 / EXCERPT: The recalled conjecture. The Hilbert depth satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072644 | MATHDB-375999

Lin's multivariable matrix Young conjecture

摘录 / EXCERPT: Lin's conjecture. There exists a unitary matrix $U\in M_n$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-072645 | MATHDB-376008

Full conservativity of dRzk over RSTT

摘录 / EXCERPT: Full conservativity. The conservativity theorem should hold for every $\textsf{dRzk}$ derivation, without the ext-style-hypotheses condition.

conjecture | open | MathDB 页面 | 原文线索

V2-072646 | MATHDB-376011

Morton-Silverman uniform boundedness conjecture for preperiodic points

摘录 / EXCERPT: Morton-Silverman conjecture. There exists a constant $C=C(d,N,[k:{\mathbb Q}])>0$ such that every degree d endomorphism f of ${\mathbb P}^N$ over k satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072647 | MATHDB-376012

Pinned-spine conjecture for full binary trees

摘录 / EXCERPT: Pinned-spine conjecture. Every FBT T on n vertices admits a graceful labeling f such that, for some longest root-to-leaf path $v_0v_1\cdots v_\ell$, the restriction of f to...

conjecture | open | MathDB 页面 | 原文线索

V2-072648 | MATHDB-376037

Conjecture on conjugation-invariant matrix representations of truncated Toeplitz operators

摘录 / EXCERPT: Let M be a complex symmetric matrix, and suppose that M is unitarily equivalent to a truncated Toeplitz operator (TTO). Let B be an inner function, let K_B be the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-072649 | MATHDB-376038

Hopkins's rowmotion cyclic sieving meta-conjecture

摘录 / EXCERPT: Let P be a finite graded poset of rank r for which the roots of $\aΩ_P(m)$ are all integers or half-integers, and in the latter case possessing an order two automorphism. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072650 | MATHDB-376046

Huang-Xiao's uniformization conjecture for Stein spaces

摘录 / EXCERPT: Let $ \Omega$ be a normal Stein space with a compact, smooth, strongly pseudoconvex boundary. Let $ \Omega_{\operatorname{Reg}}$ denote its regular part, and let the Bergman metric...

conjecture | open | MathDB 页面 | 原文线索

V2-072651 | MATHDB-376047

Surjectivity conjecture for the minimal isogeny degree of supersingular elliptic curves

摘录 / EXCERPT: Let $\mathcal{E}$ be the set of isomorphism classes of supersingular elliptic curves defined over $\overline{\mathbb{F}}_p$ for all primes p, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072652 | MATHDB-376052

Gyárfás-Sumner conjecture on chi-bounded forest-free graphs

摘录 / EXCERPT: Let T be a forest, and let a graph be T-free if it has no induced subgraph isomorphic to T. A hereditary graph class is chi-bounded if there is a function...

conjecture | open | MathDB 页面 | 原文线索

V2-072653 | MATHDB-376053

Bang-Jensen, Picasarri-Arrieta, and Yeo's characterization of champions

摘录 / EXCERPT: A tournament H is a champion if there exists an integer c_H such that every H-free tournament has acyclic dichromatic number at most c_H. The notation TT_k denotes the tr...

conjecture | open | MathDB 页面 | 原文线索

V2-072654 | MATHDB-376071

Shortest 2-design curve conjecture for the 2-sphere

摘录 / EXCERPT: Shortest 2-design curve conjecture. The curve γ is the shortest 2-design curve in S^2, up to isometry and reparameterization. The conjecture is motivated by the precedin...

conjecture | open | MathDB 页面 | 原文线索

V2-072655 | MATHDB-376078

Magic positivity conjecture for generalized parking-function polytopes

摘录 / EXCERPT: For $\mathbf{b}=(b_1,\dots,b_n)\in\mathbb{Z}_{>0}^n$, let $\mathfrak{X}_n(\mathbf{b})$ denote the $\mathbf{b}$-parking-function polytope, the convex hull of all $\mathbf{b}$-parkin...

conjecture | open | MathDB 页面 | 原文线索

V2-072656 | MATHDB-376093

The strong embedding conjecture for 2-connected graphs

摘录 / EXCERPT: Strong embedding conjecture. Every 2-connected graph has a strong embedding on a surface.

conjecture | open | MathDB 页面 | 原文线索

V2-072657 | MATHDB-376098

The second Chern class conjecture for simple vector bundles on abelian surfaces

摘录 / EXCERPT: Let A be an abelian surface, and let F be a simple vector bundle on A with Mukai vector $v:=v(F)$. Suppose that $c_1(F)$ is symmetric. For a positive integer d, define ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-072658 | MATHDB-376105

Lecture hall position conjecture for the Euler partition bijection

摘录 / EXCERPT: Lecture hall position conjecture. If $\pi=\varphi(\lambda)$, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072659 | MATHDB-376112

Typical distance lower-bound conjecture for multipartite random graphs

摘录 / EXCERPT: Let G_n be the random graph and let o_1,o_2 be the two vertices whose distance is considered. Assume the hypotheses of Theorem. Then typical distance lower-bound conjecture. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-072660 | MATHDB-376124

Random totally simple graph conjecture

摘录 / EXCERPT: Random totally simple graph conjecture. One has ... Moreover, for fixed n, the function ... is nondecreasing. More quantitatively, there exist constants $c_k>0$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072661 | MATHDB-376127

Half-line hyperuniformity conjecture for Schrödinger eigenvalue counts

摘录 / EXCERPT: Let H be the Schrödinger operator considered in the paper, let $N(\lambda)$ be its eigenvalue counting function, and let $R_{\mathrm h}(\lambda)$ denote the half-line hyperunifor...

conjecture | open | MathDB 页面 | 原文线索

V2-072662 | MATHDB-376128

Eigenvalue-counting correlation asymptotics for white-noise Schrödinger operators

摘录 / EXCERPT: Let $N(\lambda)$ be the eigenvalue counting function of the Schrödinger operator considered in the paper, and define ... Here κ is the parameter appearing in the model. The ei...

conjecture | open | MathDB 页面 | 原文线索

V2-072663 | MATHDB-376130

Hasunuma's conjecture on removable trees in k-connected graphs

摘录 / EXCERPT: Hasunuma's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072664 | MATHDB-376141

Kuckuck's n-$(n+1)$-$(n+2)$ conjecture for group finiteness properties

摘录 / EXCERPT: The n-$(n+1)$-$(n+2)$ conjecture. If N_{1} is of type $\operatorname F_n$, G_1 and G_2 are of type $\operatorname F_{n+1}$, and Q is of type $\operatorname F_{n+2}$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-072665 | MATHDB-376160

Asymptotic linear-form conjecture for empirical transport-based quantiles

摘录 / EXCERPT: Asymptotic linear-form conjecture. Under the setting of Theorem InfluenceQuantiles-main,

conjecture | open | MathDB 页面 | 原文线索

V2-072666 | MATHDB-376177

Sharp operator-norm minimax rate for nonparanormal covariance estimation

摘录 / EXCERPT: Let $\mathcal C(\sigma_{\min},\sigma_{\max})$ be the bounded-spectrum correlation class, and let $\widehat{\bm{\Sigma}}$ estimate the latent correlation matrix $\bm{\Sigma}$. Sharp...

conjecture | open | MathDB 页面 | 原文线索

V2-072667 | MATHDB-376179

Conjecture on dense colorings for typical norms

摘录 / EXCERPT: Typical-norm dense-coloring conjecture. For a typical norm,

conjecture | open | MathDB 页面 | 原文线索

V2-072668 | MATHDB-376187

Erdős-Lovász-Tihany Conjecture

摘录 / EXCERPT: Erdős-Lovász-Tihany Conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072669 | MATHDB-376219

The switching-rook-polynomial conjecture for polyominoes

摘录 / EXCERPT: Switching-rook-polynomial conjecture. The h-polynomial of $\mathbb{K}[\mathcal{P}]$ is equal to $\widetilde{r}_{\mathcal{P}}(t)$, and its regularity is equal to $r(\mathcal{P})$.

conjecture | open | MathDB 页面 | 原文线索

V2-072670 | MATHDB-376221

Meng's Hessian conjecture

摘录 / EXCERPT: Let $\Bbbk$ be a field of characteristic zero and let $\varphi\in\Bbbk[x_1,\dots,x_n]$. Write ... If $\det\operatorname{Hess}\varphi\in\Bbbk^{\times}$, the gradient map has a formal...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-072671 | MATHDB-376228

The Bi-UF Positive Conjecture for positive semirings

摘录 / EXCERPT: Bi-UF Positive Conjecture. The prototypical semiring $\mathbb{N}_0$ is the only positive semiring that is a bi-UFS.

conjecture | open | MathDB 页面 | 原文线索

V2-072672 | MATHDB-376233

Bi-UFS Positive Conjecture for positive semidomains

摘录 / EXCERPT: Bi-UFS Positive Conjecture. A positive semidomain S is a bi-UFS if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-072673 | MATHDB-376237

Wegner's conjecture on the 2-distance chromatic number of planar graphs

摘录 / EXCERPT: Let G be a planar graph with maximum degree Δ, and let $\chi_2(G)$ denote the minimum number of colors in a coloring in which vertices at distance at most 2 receive dist...

conjecture | open | MathDB 页面 | 原文线索

V2-072674 | MATHDB-376240

Conjectural fourth relation among rank-four tadpole Nahm sums

摘录 / EXCERPT: For $h=1,2,3,4$, let $A^{(h)}(q)$, $B^{(h)}(q)$, $C^{(h)}(q)$, and $D^{(h)}(q)$ denote the four Nahm sums introduced in the source, and let a, b, and $R_4^{(h)}$ be the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-072675 | MATHDB-376244

Simon's quantization conjecture for closed minimal surfaces in spheres

摘录 / EXCERPT: Let M) be a closed surface minimally immersed in $\mathbb{S}^N$ such that its image is not contained in any hyperplane of $\mathbb{R}^{N+1}$. Set $S=|h|^2$, where h is the seco...

conjecture | open | MathDB 页面 | 原文线索

V2-072676 | MATHDB-376255

The corner-factor conjecture for Neumann jump determinants

摘录 / EXCERPT: Corner-factor conjecture. There is a canonical determinant $\operatorname{det}_{\angle}'\mathcal N_{\partial P}$ of the Neumann jump operator on a piecewise real-analytic curve wit...

conjecture | open | MathDB 页面 | 原文线索

V2-072677 | MATHDB-376273

The $K(\pi, 1)$ conjecture for Artin groups of spherical type

摘录 / EXCERPT: Let W be a Coxeter group, let G_W be its associated Artin group, and let ... be the orbit configuration space, where I is the Tits cone in the real reflection representation...

conjecture | open | MathDB 页面 | 原文线索

V2-072678 | MATHDB-376279

Stein's path-orientation conjecture for oriented graphs

摘录 / EXCERPT: Let G be an oriented graph, and let $4\delta^0(G)4$ denote its minimum semidegree, the minimum of the in-degree and out-degree over all vertices. An oriented path of length...

conjecture | open | MathDB 页面 | 原文线索

V2-072679 | MATHDB-376280

Stein's pseudo-semidegree conjecture for long antipaths

摘录 / EXCERPT: Let G be an oriented graph. Its minimum pseudo-semidegree $\bar{\delta}^0(G)$ is the minimum among all non-zero in-degrees and out-degrees of its vertices; equivalently, it is th...

conjecture | open | MathDB 页面 | 原文线索

V2-072680 | MATHDB-376298

Huang-Zhou-Chang's 5/4 coloring conjecture for even-hole-free graphs

摘录 / EXCERPT: A hole is an induced cycle of length at least four, and a graph is even-hole-free if it has no hole of even length. For a graph G, let $\chi(G)$ denote its chromatic number and...

conjecture | open | MathDB 页面 | 原文线索

V2-072681 | MATHDB-376318

Pairwise orthogonality conjecture for multivariate Hermitian Jacobi polynomials

摘录 / EXCERPT: Let Θ_k index the multivariate Hermitian Jacobi polynomials $\tilde{P}^{(\alpha,\beta)}_{\tau_1,\dots,\tau_k;\phi}$, defined by triangular expansion in the multivariate Schu...

conjecture | open | MathDB 页面 | 原文线索

V2-072682 | MATHDB-376346

Martinsson-Steiner conjecture on the fractional chromatic number of degenerate triangle-free graphs

摘录 / EXCERPT: Let d be sufficiently large, and let G be a d-degenerate triangle-free graph with fractional chromatic number $\chi_f(G)$. Martinsson-Steiner conjecture. The following two as...

conjecture | open | MathDB 页面 | 原文线索

V2-072683 | MATHDB-376354

Modified Lyons-Sidorova conjecture on entire log-signatures of tree-reduced paths

摘录 / EXCERPT: Modified Lyons-Sidorova conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-072684 | MATHDB-376358

Linear Erdős-Pósa bound for models of forests

摘录 / EXCERPT: Linear Erdős-Pósa conjecture. At least one of the following holds:

conjecture | open | MathDB 页面 | 原文线索

V2-072685 | MATHDB-376379

Pilgrim's finite curve attractor conjecture

摘录 / EXCERPT: Let $f\colon \widehat{\mathbb{C}}\to\widehat{\mathbb{C}}$ be a post-critically finite rational map that is not a flexible Lattès map. A simple closed curve on...

conjecture | open | MathDB 页面 | 原文线索

V2-072686 | MATHDB-376384

The compact induction conjecture for supercuspidal representations

摘录 / EXCERPT: Let F be a nonarchimedean local field, let $\boldsymbol{\mathrm{G}}$ be a connected reductive group defined over F, and let $G=\boldsymbol{\mathrm{G}}(F)$ with center Z. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072687 | MATHDB-376385

The conjecture on geometrically minimal K-types

摘录 / EXCERPT: Let (π,V) be an irreducible supercuspidal representation of G, and let ${\mathscr{S}}$ be the G-equivariant Schneider-Stuhler sheaf on the building X associated with...

conjecture | open | MathDB 页面 | 原文线索

V2-072688 | MATHDB-376386

The collapse conjecture for Schneider-Stuhler sheaves

摘录 / EXCERPT: Let X be the building of G, and let ${\mathscr{S}}$ be the Schneider-Stuhler G-equivariant sheaf associated with an irreducible supercuspidal representation. A basic elementa...

conjecture | open | MathDB 页面 | 原文线索

V2-072689 | MATHDB-376399

The Generalized Pushout Conjecture for infinity-groupoids and controlled theories

摘录 / EXCERPT: The Generalized Pushout Conjecture. The following statements hold: first, the map $p:X\to X_+$ is a weak equivalence; second, given a connected diagram of controlled theories J,...

conjecture | open | MathDB 页面 | 原文线索

V2-072690 | MATHDB-376406

Bridgeland stability conjecture for the Kuznetsov component of a singular cubic sevenfold

摘录 / EXCERPT: Bridgeland stability conjecture. The category $\operatorname{\mathcal{K}\mkern-1mu\mathit{u}}(Y)$ admits a Bridgeland stability condition.

conjecture | open | MathDB 页面 | 原文线索

V2-072691 | MATHDB-376413

Simplicity conjecture for nonzero roots of generalized Wronskian-Hermite polynomials

摘录 / EXCERPT: Simplicity conjecture. All nonzero roots of $W_N^{[m,k,l]}(z)$ are simple.

conjecture | open | MathDB 页面 | 原文线索

V2-072692 | MATHDB-376414

The entropic Solymosi inequality

摘录 / EXCERPT: Let X and X' be independent and identically distributed, discrete, real-valued random variables with finite entropy $H(X)<\infty$. Entropic Solymosi inequality. One has ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-072693 | MATHDB-376429

Conjecture on Beilinson and Cheeger-Chern classes of universal flat bundles

摘录 / EXCERPT: Beilinson-Cheeger-Chern class conjecture. For all $p \geq 1$ and $n \in {\mathbb N} \cup\{\infty\}$,

conjecture | open | MathDB 页面 | 原文线索

V2-072694 | MATHDB-376430

The blow-up conjecture for ruled surfaces

摘录 / EXCERPT: Let M be a ruled compact complex surface, and let $\widetilde{M}$ be its blow-up at sufficiently many points. Blow-up conjecture. The surface $\widetilde{M}$ admits a scalar-flat...

conjecture | open | MathDB 页面 | 原文线索

V2-072695 | MATHDB-376431

The quasi-stability conjecture for parabolic bundles on ruled surfaces

摘录 / EXCERPT: Let M be a blown-up ruled surface of genus at least 2, and let $[\omega]$ be an admissible Kähler class, meaning that $c_1\cup[\omega]=0$. Associate to $[\omega]$ its correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-072696 | MATHDB-376446

Integrality conjecture for rigid Zariski-dense representations of property T groups

摘录 / EXCERPT: Property T integrality conjecture. The representation ϱ is integral.

conjecture | open | MathDB 页面 | 原文线索

V2-072697 | MATHDB-376448

Shokurov's conjecture on accumulation of minimal discrepancies

摘录 / EXCERPT: Shokurov's conjecture. For every natural n minimal discrepancies of n-dimensional log-terminal singularities can accumulate only from above.

conjecture | open | MathDB 页面 | 原文线索

V2-072698 | MATHDB-376462

Odd-positive coefficient conjecture for the rational-surface polynomials B_{4n}

摘录 / EXCERPT: Let h and p be the generators used in the paper for the polynomial expressions B_{4n}. For every integer n, there is a nonnegative integer $l(n)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-072699 | MATHDB-376463

Universal polynomial and positivity conjecture for wall-crossing coefficients

摘录 / EXCERPT: Let S be a rational surface, let N,d,r be the parameters in the wall-crossing formula, let K^2 denote K_S^2, and let $Q_i(N,d,r,K^2)$ and $P_i(N,d,r,K^2)$ be the polynomial...

conjecture | open | MathDB 页面 | 原文线索

V2-072700 | MATHDB-376472

Reid-Watanabe weighted blow-up conjecture for canonical modifications

摘录 / EXCERPT: Let (X,x) be a 3-dimensional, isolated, non-canonical, log-canonical singularity of embedded dimension 4. Choose an embedding ... and let $\boldsymbol{w}=(w_0,w_1,\ldots,w_n)$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-072701 | MATHDB-376473

Twisted Spin Verlinde dimension formula

摘录 / EXCERPT: Let $m\geq 3$, and let $SC_m\rightarrow SL(V)$ be a representation satisfying the condition in. Write d_V for the height (Dynkin index) of V, and let $N_{d_V}^{+}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072702 | MATHDB-376474

Vistoli's conjecture on the kernel of equivariant Riemann-Roch

摘录 / EXCERPT: Vistoli's conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072703 | MATHDB-376476

The A_n-base conjecture for Mori conic bundles

摘录 / EXCERPT: The A_n-base conjecture. If $f\colon (X,C)\to (S,0)$ is a Mori conic bundle, then $(S,0)$ is a Du Val singularity of type A_n.

conjecture | open | MathDB 页面 | 原文线索

V2-072704 | MATHDB-376490

Tian-Todorov-type smoothness problem for X_ζ when $\Gamma\subset\operatorname{SU}(3)$

摘录 / EXCERPT: Let $\Gamma\subset\operatorname{SU}(3)$, let X_ζ be the moduli space associated with a generic parameter ζ, and let $(M^{0,*,\Gamma},\bar{\partial}_\alpha)$ be the diff...

open_problem | open | MathDB 页面 | 原文线索

V2-072705 | MATHDB-376491

The crepant smooth-resolution conjecture for abelian cspan class="math-inline"mathGammac/spancspan class="math-inline"mathsubseteq\operatorname{SU}(3)c/spancspan class="math-inline"math

摘录 / EXCERPT: Let $\Gamma\subset\operatorname{SU}(3)$ be an abelian group acting on $\mathbb{C}^3$ freely outside the origin. Let $X_0=\mathbb{C}^3/\Gamma$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072706 | MATHDB-376493

Sturmfels-Thomas bounded-facet conjecture for Gröbner fans

摘录 / EXCERPT: Let $A\in{\bf Z}^{d\times n}$ be a matrix of rank d. The Gröbner fan of A is the fan whose cones encode the reduced Gröbner bases of the toric ideal associated to A. Sturmfel...

conjecture | open | MathDB 页面 | 原文线索

V2-072707 | MATHDB-376495

The log-terminality conjecture for Mori conic bundles

摘录 / EXCERPT: Log-terminality conjecture. The divisor pair

conjecture | open | MathDB 页面 | 原文线索

V2-072708 | MATHDB-376496

The physicists' Euler number conjecture for stringy Euler characteristics

摘录 / EXCERPT: Physicists' Euler number conjecture. In appropriate circumstances,

conjecture | open | MathDB 页面 | 原文线索

V2-072709 | MATHDB-376497

The tautological-sheaf basis conjecture for quotient resolutions

摘录 / EXCERPT: Tautological-sheaf basis conjecture. Under appropriate circumstances, the tautological sheaves $\mathcal F_\rho$ form a $\mathbb Z$-basis of $K_0(\operatorname{Coh}Y)$; their Chern...

conjecture | open | MathDB 页面 | 原文线索

V2-072710 | MATHDB-376499

Infinitude conjecture for approximating fiber groups

摘录 / EXCERPT: Let X_0 be a nontrivial chain or ring of curves of genus greater than 0, let g be its arithmetic genus, and let X_I be the family of curves over an interval described in th...

conjecture | open | MathDB 页面 | 原文线索

V2-072711 | MATHDB-376502

The complete-intersection quotient conjecture for projective crepant resolutions

摘录 / EXCERPT: Let G be a finite subgroup of $\operatorname{SL}(d,\mathbb{C})$ such that $\mathbb{C}^d/G$ is minimally embeddable as a complete intersection in an affine space $\mathbb{C}^r$, w...

conjecture | open | MathDB 页面 | 原文线索

V2-072712 | MATHDB-376506

The limiting mixed twistor structure conjecture

摘录 / EXCERPT: Limiting mixed twistor structure conjecture. The filtered bundle (M,W) is a mixed twistor structure, and N is a morphism of mixed twistor structures such that the weight filtra...

conjecture | open | MathDB 页面 | 原文线索

V2-072713 | MATHDB-376515

The fully faithful Yoneda conjecture for higher categories

摘录 / EXCERPT: Yoneda conjecture. The functors α and β are fully faithful. This is the higher-categorical analogue of the corresponding fact for $n=0$; the paper gives no proof or re...

conjecture | open | MathDB 页面 | 原文线索

V2-072714 | MATHDB-376516

The representability conjecture for higher-categorical functors

摘录 / EXCERPT: Suppose A is an $n+1$-category admitting arbitrary direct and inverse limits. A functor $h:A^o\rightarrow nCAT'$ is called representable when it is induced by an object of A, a...

conjecture | open | MathDB 页面 | 原文线索

V2-072715 | MATHDB-376518

The universal morphism conjecture for properties of higher-categorical morphisms

摘录 / EXCERPT: Let $\psi:A\rightarrow nCAT'$ be a functor and let $\mathcal P$ be a property of morphisms from ψ to objects of $nCAT'$. A universal morphism with property $\mathcal P$ is a m...

conjecture | open | MathDB 页面 | 原文线索

V2-072716 | MATHDB-376519

The existence conjecture for direct images of higher-categorical families

摘录 / EXCERPT: Let $F:A\rightarrow B$ be a morphism of $n+1$-categories, and let $\varphi:A\rightarrow nCAT'$ be a family of n-categories over A. A direct image is a universal family...

conjecture | open | MathDB 页面 | 原文线索

V2-072717 | MATHDB-376525

Uniqueness conjecture for fundamental families of an admissible Kleinian group

摘录 / EXCERPT: Let a Kleinian group be called admissible if it has generators φ_i, $i\in I_+$, satisfying the conditions of the amplification. For an admissible Kleinian group, a fundamen...

conjecture | open | MathDB 页面 | 原文线索

V2-072718 | MATHDB-376530

The slope-bound interpolation conjecture for d-gonal loci

摘录 / EXCERPT: Let $\mathcal{F}_d(g)$ be the exact upper bound for slopes of families of stable curves with smooth d-gonal general member, and let $g_d=2d-3$ for $d\geq 3$ and $g_2=2$. Slope-bo...

conjecture | open | MathDB 页面 | 原文线索

V2-072719 | MATHDB-376531

The semistable slope bound for families of d-gonal curves

摘录 / EXCERPT: Let X be a family of d-gonal curves of genus g, with base B not contained in a certain codimension-one closed subset of $\overline{\mathcal D}_d$. Semistable slope-bound co...

conjecture | open | MathDB 页面 | 原文线索

V2-072720 | MATHDB-376532

The Clifford-index slope conjecture for special linear-series loci

摘录 / EXCERPT: Let $X\to B$ be a general family of genus g stable curves whose general member has Clifford index $\mathfrak{c}$, and let B be a general curve in $\overline{\mathcal D}^r_d$. C...

conjecture | open | MathDB 页面 | 原文线索

V2-072721 | MATHDB-376535

The conifold mirror conjecture for Calabi-Yau complete intersections in Grassmannians

摘录 / EXCERPT: Let X_0 be a conifold degeneration of a generic Calabi-Yau complete intersection $X\subset G(k,n)$, let p be the number of nodes of X_0, and set $\alpha=(k-1)(n-k-1)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072722 | MATHDB-376548

The hard-core uniqueness conjecture at the tree threshold

摘录 / EXCERPT: Let G be a countably infinite graph of maximum degree ... , the hard-core lattice gas on G has a unique Gibbs measure whenever ... for all vertices x; perhaps this remains true w...

conjecture | open | MathDB 页面 | 原文线索

V2-072723 | MATHDB-376549

The claw-free hard-core zero conjecture

摘录 / EXCERPT: Let G be a finite claw-free graph, meaning that it has no induced subgraph isomorphic to $K_{1,3}$. Let $Z(w)$ be the hard-core lattice-gas partition function of G. Hamidoune-S...

conjecture | open | MathDB 页面 | 原文线索

V2-072724 | MATHDB-376550

The complex-fugacity Glauber spectral-radius conjecture

摘录 / EXCERPT: Let G be a finite graph of maximum degree Δ, let P be the Glauber transition matrix at complex fugacity w, and let $\operatorname{spr}(P)$ denote its spectral radius o...

conjecture | open | MathDB 页面 | 原文线索

V2-072725 | MATHDB-376553

Sum-of-components formula conjecture for odd XXZ ground states

摘录 / EXCERPT: Let $N=2n+1$ be odd, let ${\mathcal N}_n$ be the norm of the ground state, and let the state vector have components in the spin-configuration basis. Sum-of-components conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-072726 | MATHDB-376554

Next-nearest-neighbor correlation formula conjecture

摘录 / EXCERPT: Let $N=2n+1$ be odd, and let a_j denote the local observable used for the spin-chain correlations. Correlation formula conjecture. The next-nearest-neighbor correlation is ... The...

conjecture | open | MathDB 页面 | 原文线索

V2-072727 | MATHDB-376558

Baik-Rains conjecture for the stationary TASEP two-point distribution

摘录 / EXCERPT: Let $\rho \in (0,1)$ be the stationary Bernoulli density, let $\mathbb{P}_\rho$ be the corresponding TASEP path measure, and let $h_t(j)$ be the height function. For...

conjecture | open | MathDB 页面 | 原文线索

V2-072728 | MATHDB-376559

The twisted XXZ-chain 1-sum conjecture

摘录 / EXCERPT: Let $L=2n$ and consider the twisted XXZ chain at $\Delta=-\frac{1}{2}$, with the groundstate wavefunction normalised as in the source. Let $S_{2n}^{(1)}$ denote the sum of its wave...

conjecture | open | MathDB 页面 | 原文线索

V2-072729 | MATHDB-376560

The twisted XXZ-chain 2-sum conjecture

摘录 / EXCERPT: Let $L=2n$ and consider the twisted XXZ chain at $\Delta=-\frac{1}{2}$, with the groundstate wavefunction normalised as in the source. Let $S_{2n}^{(2)}$ be the sum of the squares...

conjecture | open | MathDB 页面 | 原文线索

V2-072730 | MATHDB-376562

The open O(1) loop-model largest-component conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with open boundary conditions and system size L, and normalise its groundstate wavefunction so that its smallest element is 1. Let $N_8(2n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-072731 | MATHDB-376563

The open O(1) loop-model 1-sum conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with open boundary conditions and system size L, and let $S_L^{(1)}$ be the sum of the components of its groundstate wavefunction, normalised s...

conjecture | open | MathDB 页面 | 原文线索

V2-072732 | MATHDB-376564

The periodic even-size O(1) loop-model largest-component conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with periodic boundary conditions and even system size $L=2n$, and normalise its groundstate wavefunction so that its smallest element is 1. Le...

conjecture | open | MathDB 页面 | 原文线索

V2-072733 | MATHDB-376565

The periodic even-size O(1) loop-model 1-sum conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with periodic boundary conditions and even system size $L=2n$, and let $S_{2n}^{(1)}$ be the sum of the components of its groundstate wavefunctio...

conjecture | open | MathDB 页面 | 原文线索

V2-072734 | MATHDB-376566

The periodic odd-size O(1) loop-model largest-component conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with periodic boundary conditions and odd system size $L=2n-1$, and normalise its groundstate wavefunction so that its smallest element is 1. L...

conjecture | open | MathDB 页面 | 原文线索

V2-072735 | MATHDB-376567

The periodic odd-size O(1) loop-model 1-sum conjecture

摘录 / EXCERPT: Consider the dense O(1) loop model with periodic boundary conditions and odd system size $L=2n-1$, and let $S_{2n-1}^{(1)}$ be the sum of the components of its groundstate wavefunc...

conjecture | open | MathDB 页面 | 原文线索

V2-072736 | MATHDB-376569

Twisted XXZ component-ratio conjecture

摘录 / EXCERPT: Let $|\Psi^{(2n)}\rangle$ be the normalized ground-state vector of the twisted XXZ chain with $2n$ sites, and let A_n denote the number of alternating-sign $n\times n$ matrices....

conjecture | open | MathDB 页面 | 原文线索

V2-072737 | MATHDB-376570

Twisted XXZ even-chain norm conjecture

摘录 / EXCERPT: Let $|\Psi^{(2n)}\rangle$ be the normalized ground-state vector of the twisted XXZ chain with $2n$ sites, let ... and let A_n be the number of alternating-sign $n\times n$ matrices...

conjecture | open | MathDB 页面 | 原文线索

V2-072738 | MATHDB-376571

Batchelor-de Gier-Nienhuis squared-component-sum conjecture

摘录 / EXCERPT: Let $|\Psi^{(2n)}\rangle$ be the ground-state vector for the twisted XXZ chain with $2n$ sites, let $W^{(2n)}$ index its components, and let A_n be the number of alternating-sign...

conjecture | open | MathDB 页面 | 原文线索

V2-072739 | MATHDB-376572

Twisted XXZ next-nearest-neighbour correlation conjecture

摘录 / EXCERPT: For the even-site ground state $|\Psi^{(2n)}\rangle$, define the normalized expectation of an operator O by ... and define $\alpha_i=(1+\sigma_i^z)/2$. The twisted XXZ next-nearest...

conjecture | open | MathDB 页面 | 原文线索

V2-072740 | MATHDB-376574

Half-turn-symmetric ground-state conjecture for odd periodic dense O(1) loops

摘录 / EXCERPT: Half-turn-symmetric ground-state conjecture. The vector

conjecture | open | MathDB 页面 | 原文线索

V2-072741 | MATHDB-376575

Vertically symmetric ground-state conjecture for even open dense O(1) loops

摘录 / EXCERPT: Vertically symmetric ground-state conjecture. The vector

conjecture | open | MathDB 页面 | 原文线索

V2-072742 | MATHDB-376576

Log-squared weight conjecture for q-scaled recurrence coefficients

摘录 / EXCERPT: Log-squared weight conjecture. The weight satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-072743 | MATHDB-376577

Beraha-number vanishing-amplitude conjecture for triangular-lattice strips

摘录 / EXCERPT: Let L be the width of a triangular-lattice strip with free, cylindrical or “zig-zag” boundary conditions. Let $B_2,\ldots,B_{L+1}$ be the corresponding Beraha numbers, let...

conjecture | open | MathDB 页面 | 原文线索

V2-072744 | MATHDB-376578

Cylindrical-strip Beraha-zero pattern conjecture

摘录 / EXCERPT: Let L be the width of a triangular-lattice strip with cylindrical boundary conditions, let B_k denote the Beraha numbers, and let $D(q)$ be the amplitude matrix. Cylindrical-st...

conjecture | open | MathDB 页面 | 原文线索

V2-072745 | MATHDB-376580

Exact spectral-gap relation for the multi-urn backgammon model

摘录 / EXCERPT: Exact spectral-gap conjecture. The inequality relating the m-urn and two-urn gaps is an equality:

conjecture | open | MathDB 页面 | 原文线索

V2-072746 | MATHDB-376581

Norm bound for the contingency-table multi-decomposition

摘录 / EXCERPT: Contingency-table norm conjecture. For arbitrary table dimensions m and n, table sum Σ, and row and column sums r_k and c_p, this multi-decomposition is non-degenera...

conjecture | open | MathDB 页面 | 原文线索

V2-072747 | MATHDB-376584

Large-weight spanning-forest free-energy conjecture

摘录 / EXCERPT: Let $f(w)$ be the spanning-forest free energy per site on a regular lattice of dimension $d\geq2$ and coordination number r, and let f_0 denote the entropy per site for spannin...

conjecture | open | MathDB 页面 | 原文线索

V2-072748 | MATHDB-376586

The replicated duality conjecture for the multicritical point of the square-lattice 7 Ising model

摘录 / EXCERPT: Multicritical-point conjecture. The exact location of the multicritical point is determined by $x_0=x_0^{*}$ with $K=K_p$, equivalently

conjecture | open | MathDB 页面 | 原文线索

V2-072749 | MATHDB-376587

The entropy conjecture for the multicritical point of the square-lattice 7 Ising model

摘录 / EXCERPT: Entropy conjecture. The exact location of the multicritical point is given by

conjecture | open | MathDB 页面 | 原文线索

V2-072750 | MATHDB-376592

Square-lattice toroidal phase-diagram conjecture

摘录 / EXCERPT: For a zero-temperature square-lattice Potts antiferromagnet on toroidal strips, let q be the Potts parameter, let $q_0(L)$ be the relevant real crossing, and let ℓ denote th...

conjecture | open | MathDB 页面 | 原文线索

V2-072751 | MATHDB-376594

Triangular-lattice q0 convergence conjecture

摘录 / EXCERPT: Let $q_0(L)$ denote the crossing for a triangular-lattice strip of width L with fully periodic boundary conditions, and let $q_{\rm B}\simeq3.8196717312$ be the value where Baxte...

conjecture | open | MathDB 页面 | 原文线索

V2-072752 | MATHDB-376595

Triangular-lattice q2 convergence conjecture

摘录 / EXCERPT: Let $q_2(L)$ denote the crossing of the two largest eigenvalues in the $\ell=2$ sector for a triangular-lattice strip of width L with fully periodic boundary conditions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072753 | MATHDB-376596

Triangular-lattice toroidal phase-diagram conjecture

摘录 / EXCERPT: For a zero-temperature triangular-lattice Potts antiferromagnet on toroidal strips, let $q_0(L)$ and $q_2(L)$ be the relevant real crossings, and let ℓ denote the number of br...

conjecture | open | MathDB 页面 | 原文线索

V2-072754 | MATHDB-376597

Highest-sector eigenvalue conjecture for triangular-lattice toroidal strips

摘录 / EXCERPT: Let $T_{L,L}$ be the diagonal transfer-matrix sub-block for a triangular-lattice strip of width L with fully toroidal boundary conditions, and let $b^{(L)}_1$ denote the stated a...

conjecture | open | MathDB 页面 | 原文线索

V2-072755 | MATHDB-376598

Existence conjecture for Gross-Pitaevskii states with a linear counterpart

摘录 / EXCERPT: Let U_0 be the nonlinear interaction coefficient, let μ be the chemical potential, and let $(\mathcal{E}_n,\phi_n)$ be an eigenvalue and eigenfunction of the associated linea...

conjecture | open | MathDB 页面 | 原文线索

V2-072756 | MATHDB-376599

Zero distribution conjecture for the second-largest spinless-fermion eigenvalue

摘录 / EXCERPT: Zero distribution conjecture. $T^{(2)}_{n-1}(v)$ has one real zero ζ_{n-1} for n even with $2\leq n\leq p_0-1$, and two real zeros ζ_{n-1} and $-\zeta'_{n-1}$ for n...

conjecture | open | MathDB 页面 | 原文线索

V2-072757 | MATHDB-376606

Tightness conjecture for the MAP conditional-entropy bound

摘录 / EXCERPT: Let h_n denote the normalized conditional entropy in the code ensemble, and let τ and ρ be the degree sequences appearing in Theorem 1. For each admissible random varia...

conjecture | open | MathDB 页面 | 原文线索

V2-072758 | MATHDB-376607

Refined scaling conjecture for unconditionally stable LDPC ensembles

摘录 / EXCERPT: Refined scaling conjecture. As n tends to infinity,

conjecture | open | MathDB 页面 | 原文线索

V2-072759 | MATHDB-376610

The alpha-constraint label-cover hardness conjecture

摘录 / EXCERPT: Let $G={\rangle (V,E),2R,\Psi\langle}$ be a label-cover instance whose constraints are alpha-constraints, where an alpha-constraint is defined by permutations...

conjecture | open | MathDB 页面 | 原文线索

V2-072760 | MATHDB-376611

Khot's bipartite d-to-1 conjecture

摘录 / EXCERPT: Fix an integer $d>1$. Let $\Phi=(X,Y,\Psi,W)$ be a weighted bipartite d-to-1 label-cover instance with label sets $\{1,\ldots,R\}$ and $\{1,\ldots,R/d\}$ and total edge weight...

conjecture | open | MathDB 页面 | 原文线索

V2-072761 | MATHDB-376612

Symmetrical bound for the MDL code in terms of shortest-grammar length

摘录 / EXCERPT: Let v be a string, let $L^m(v)$ denote the length of its shortest grammar, and let $\boldsymbol{\operatorname{MDL}}(v)$ denote the MDL code for v. Let $\gamma(n):=n f_1(n)$. Sy...

conjecture | open | MathDB 页面 | 原文线索

V2-072762 | MATHDB-376615

Explicit inner and outer surfaces for Gaussian pattern recognition

摘录 / EXCERPT: Gaussian-case conjecture. In the Gaussian case, the surfaces of $\mathcal{R}_{in}$ and $\mathcal{R}_{out}$ are

conjecture | open | MathDB 页面 | 原文线索

V2-072763 | MATHDB-376621

MacWilliams identity conjecture for convolutional-code adjacency matrices

摘录 / EXCERPT: MacWilliams identity conjecture. The matrix

conjecture | open | MathDB 页面 | 原文线索

V2-072764 | MATHDB-376623

Consensus propagation mixing-time conjecture for the m-dimensional torus

摘录 / EXCERPT: Let the graph be the m-dimensional torus with n nodes, and let $\tau^\star$ denote the mixing time of the directed-edge process associated with consensus propagation. Consensus...

conjecture | open | MathDB 页面 | 原文线索

V2-072765 | MATHDB-376625

Existence of superregular matrices over fields of size $2^{\gamma-2}$

摘录 / EXCERPT: Superregular matrix existence conjecture. For every integer $b3\geq 5$, there exists a $b3\times b3$ superregular matrix over

conjecture | open | MathDB 页面 | 原文线索

V2-072766 | MATHDB-376627

Conjecture on the clustered multi-antenna relay DMT

摘录 / EXCERPT: Clustered-system DMT conjecture. The theorem stating that, for the clustered system, the DMT formula given for $m=1,2$ holds is true for arbitrary (m,n,k).

conjecture | open | MathDB 页面 | 原文线索

V2-072767 | MATHDB-376628

Monotonicity of proportional fair capacity for Gaussian broadcast channels

摘录 / EXCERPT: Let $C_{PF}(\mathbf{N},P_T)$ denote the proportional fair capacity of a Gaussian broadcast channel with noise vector $\mathbf{N}$ and total power P_T. Let $\preceq$ denote the co...

conjecture | open | MathDB 页面 | 原文线索

V2-072768 | MATHDB-376629

Finite excluded-minor characterization conjecture for codes over finite fields

摘录 / EXCERPT: Let ${\mathfrak C}$ be a minor-closed class of codes over a finite field ${\mathbb F}$. For a collection ${\mathcal F}$ of codes, let ${\mathfrak C}_{\mathcal F}$ denote the class...

conjecture | open | MathDB 页面 | 原文线索

V2-072769 | MATHDB-376630

Ahlswede-Han conjecture on capacity with rate-limited state information

摘录 / EXCERPT: Consider a state-dependent channel $p(y|x,s)$ with independent and identically distributed state S^n, where the receiver obtains rate-limited state information through a separate...

conjecture | open | MathDB 页面 | 原文线索

V2-072770 | MATHDB-376643

Polynomial growth conjecture for Vassiliev invariants

摘录 / EXCERPT: A Vassiliev invariant of order k is a knot invariant that vanishes on higher-order differences of knots. Let v_k be any Vassiliev invariant of order k, let K be a knot with...

conjecture | open | MathDB 页面 | 原文线索

V2-072771 | MATHDB-376669

Universal-formula conjecture for lower-level PU(2) monopole pairings

摘录 / EXCERPT: Let ℓ, F, L_1, $\mathfrak{s}_0$, $SW(\mathfrak{s}_0\otimes L_1)$, and Q_X denote the level, bundle and line-bundle data, Seiberg-Witten invariant, and intersection form...

conjecture | open | MathDB 页面 | 原文线索

V2-072772 | MATHDB-376670

Swiatak's second-derivative spanning conjecture for functional equations

摘录 / EXCERPT: Let $x\in R^n$ and $t\in R^r$, let $\Omega\subseteq R^r$ be open, and consider distribution solutions of the functional equation (1.1) satisfying assumptions A-1 through A-5. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072773 | MATHDB-376689

Horizon condition for spherically symmetric asymptotically flat Einstein-Vlasov data

摘录 / EXCERPT: Let $\mathcal{H}^+$ be the future event horizon and let ... is satisfied on $\mathcal{H}^+$. This is stated as a stronger conjecture implying the absence of future Cauchy horizons; t...

conjecture | open | MathDB 页面 | 原文线索

V2-072774 | MATHDB-376694

The chamber formula conjecture for the classical Gromov-Witten potentials

摘录 / EXCERPT: The chamber formula conjecture. In the fundamental chamber,

conjecture | open | MathDB 页面 | 原文线索

V2-072775 | MATHDB-376697

The generating-function conjecture for multiple Clausen value irreducibles

摘录 / EXCERPT: For weight n and depth k, let $P(n,k)$ denote the number of irreducible multiple Clausen values such that a rational basis at that weight and depth is formed from a minimum num...

conjecture | open | MathDB 页面 | 原文线索

V2-072776 | MATHDB-376699

Primary discriminant monodromy conjecture

摘录 / EXCERPT: Let X be the Calabi-Yau space under consideration, let Δ_0 be the primary component of its discriminant, and let ${\Scr K}^K_{{\Scr E}}$ denote the Fourier-Mukai transform...

conjecture | open | MathDB 页面 | 原文线索

V2-072777 | MATHDB-376701

Primary-discriminant criterion for 0-brane monodromy

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, let Δ_0 be the primary component of the discriminant, and let a 0-brane be an object $P\in\mathbf{D}(X)$ with Chern character equal to the...

conjecture | open | MathDB 页面 | 原文线索

V2-072778 | MATHDB-376702

Givental's conjecture for the projective line

摘录 / EXCERPT: Givental's conjecture. The generating function $G(\{T_k^+\},\{0\})$ is equal to one, or equivalently

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V2-072779 | MATHDB-376709

Kontsevich's spherical-object monodromy conjecture

摘录 / EXCERPT: Let $\mathbf{D}(X)$ be the bounded derived category of coherent sheaves on a Calabi-Yau threefold X, and let A be a spherical D-brane that becomes massless, together w...

conjecture | open | MathDB 页面 | 原文线索

V2-072780 | MATHDB-376710

The primary-component masslessness conjecture for the structure sheaf

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, and let $Delta$ be the discriminant locus in the moduli space of conformal field theories. Its primary component is the component corresponding t...

conjecture | open | MathDB 页面 | 原文线索

V2-072781 | MATHDB-376716

Asymptotic equivariant coercivity conjecture for harmonic map lumps

摘录 / EXCERPT: For the n-lump sector of the ${\mathbb{C}}{{P}}^1$ model on S^2, let $\tau^{eq}(\mu)$ be the equivariant coercivity constant for the map $\phi:z\mapsto \mu z^n$, with...

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V2-072782 | MATHDB-376717

The B-brane category conjecture for geometric N=2 SCFTs on tori

摘录 / EXCERPT: Let X be a torus with a geometric $N=2$ structure (I,J), and let $C^{\mathrm{B}}_{I,J}(X)$ denote the category of B-branes and $\mathcal{D}\operatorname{Coh}_{I,J}(X)$ the deri...

conjecture | open | MathDB 页面 | 原文线索

V2-072783 | MATHDB-376718

Functoriality of D-brane subcategories under generalised N=2 morphisms

摘录 / EXCERPT: Let Γ and Γ' be lattices with quadratic forms q and q', let $g:(\Gamma\oplus\Gamma^*,q)\to(\Gamma'\oplus{\Gamma'}^*,q')$ be a lattice isomorphism, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072784 | MATHDB-376728

The extended Picard-Fuchs Yukawa coupling conjecture for local mirror symmetry

摘录 / EXCERPT: Consider the A-model on a Calabi-Yau threefold X. Let u_i be the logarithm of the B-model complex deformation parameter z_i obtained from the toric construction of the mirror...

conjecture | open | MathDB 页面 | 原文线索

V2-072785 | MATHDB-376732

The torus-amplitude conjecture for logarithmic minimal models

摘录 / EXCERPT: Let p_+ and p_- be the parameters of the logarithmic (p_+,p_-) model. Let ${\bar{\mathscr{C}}}$ be the space of generalized characters, let $\mathscr{G}$ be the subspace desc...

conjecture | open | MathDB 页面 | 原文线索

V2-072786 | MATHDB-376743

The presentation conjecture for the pure braid group of points in the affine plane

摘录 / EXCERPT: Presentation conjecture. The homomorphism $\varphi_n:PL_n\to P_n^2$ is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-072787 | MATHDB-376746

Alexander-polynomial formula for the proper Abelian BF knot observable

摘录 / EXCERPT: Let K be a knot, let λ be the coupling parameter, and let $\bigcirc$ denote the unknot. Let $\Delta(K;z)$ be the Alexander polynomial evaluated at z, and set ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072788 | MATHDB-376750

The K(pi,1) conjecture for connected components of strata of holomorphic 1-forms

摘录 / EXCERPT: Let ${\cal M}_{\bf d}$ be a moduli space of holomorphic 1-forms with prescribed zero orders, and let ${\cal M}$ be one of its connected components. A group is commensurable with an...

conjecture | open | MathDB 页面 | 原文线索

V2-072789 | MATHDB-376757

Degeneracy conjecture for string, particle and membrane U-duality Eisenstein series

摘录 / EXCERPT: Let $d>2$, let $E_{d+1(d+1)}(\mathbb{Z})$ be the U-duality group, and let $\mathcal{E}_{\mathrm{string};3/2}$, $\mathcal{E}_{\mathrm{particle};d/2-1}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072790 | MATHDB-376760

M-theory R^4H^{4g-4} U-duality conjecture

摘录 / EXCERPT: Let M-theory be compactified on T^{d+1} with $d\leq4$, let H denote the resulting three-form field strengths, and let $M(\mathrm{string})$ be the mass matrix in the string repr...

conjecture | open | MathDB 页面 | 原文线索

V2-072791 | MATHDB-376762

Holomorphicity conjecture for Liouville fusion coefficients

摘录 / EXCERPT: Let $Q=b+b^{-1}$, and let $\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_{23}$ be complex Liouville momenta. The fusion coefficients are denoted by...

conjecture | open | MathDB 页面 | 原文线索

V2-072792 | MATHDB-376767

Bessis's positivity conjecture for the cubic PT-symmetric oscillator

摘录 / EXCERPT: Consider the eigenvalue problem ... where $H=-\frac{d^2}{dx^2}-(ix)^3$ acts on the real line. Bessis's conjecture. All eigenvalues of H are real and positive. This is a conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072793 | MATHDB-376768

Minimal-dimensional modular extension conjecture for finite braided tensor categories

摘录 / EXCERPT: Let $\2C$ be a braided tensor C^*-category of finite dimension, and let $\2Z(\2C)$ denote its center. A modular extension of $\2C$ is a modular category $\2M$ containing $\2C$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-072794 | MATHDB-376772

Miller's Aharonov-Casher lower-bound conjecture

摘录 / EXCERPT: Let $B(x)\ge 0$ be a magnetic field with flux ... which may be infinite, and let H denote the corresponding Pauli operator. Miller's conjecture. The dimension of its kernel should...

conjecture | refuted | MathDB 页面 | 原文线索

V2-072795 | MATHDB-376780

Closed-boundary rotor-model groundstate sum conjecture

摘录 / EXCERPT: For the rotor model with closed boundary conditions, let ψ_0 be its groundstate wavefunction and let $S(N)$ denote the sum of its elements. For odd size $N=2m-1$, normalize th...

conjecture | open | MathDB 页面 | 原文线索

V2-072796 | MATHDB-376795

Conjecture on degeneracies of Picard-group Laplace eigenvalues

摘录 / EXCERPT: Consider the Laplacian spectrum obtained by taking all four symmetry classes together. Steil's theorem explains certain degeneracies arising from Hecke eigenfunctions and the symme...

conjecture | open | MathDB 页面 | 原文线索

V2-072797 | MATHDB-376798

Gauge-matrix conjecture for monodromy matrices of the extended Lotka-Volterra lattice

摘录 / EXCERPT: Let $N\geq 4$, let $\mathbf{T}_{m;n_1,n_2}(z)$ be the monodromy matrix, and let L be the parity parameter appearing in its construction. Write $\vec{e}_1$ and $\vec{e}_N$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-072798 | MATHDB-376800

Lieb's Wehrl norm conjecture for spin coherent states

摘录 / EXCERPT: Let ${\mathcal F}_j$ be the spin-j function space on ${\mathbb S}^2={\mathbb C}\cup\{\infty\}$, and for $p\geq 1$ define the normalized norm by ... Let $K(z,w)$ denote the coherent...

conjecture | open | MathDB 页面 | 原文线索

V2-072799 | MATHDB-376802

The Hessian conjecture for formal Legendre transforms

摘录 / EXCERPT: Let K be a field of characteristic 0, let $\varphi\in K[x_1,\ldots,x_n]$ be a polynomial whose Hessian determinant is a nonzero constant, and let $\bar\varphi$ be its formal Le...

conjecture | open | MathDB 页面 | 原文线索

V2-072800 | MATHDB-376803

Cohn polynomial irreducibility and degree conjecture for Lamé spectra

摘录 / EXCERPT: Cohn polynomial conjecture. As a polynomial in J with integer coefficients, no Cohn polynomial has a nontrivial factor, except for the Type I Cohn polynomials with...

conjecture | open | MathDB 页面 | 原文线索

V2-072801 | MATHDB-376811

Distributional image characterization for the generalized Segal-Bargmann transform

摘录 / EXCERPT: Let K be the compact Lie group considered in the paper, let $K_{\mathbb C}$ be its complexification, let $t>0$, and let C_t denote the generalized Segal-Bargmann transform from...

conjecture | open | MathDB 页面 | 原文线索

V2-072802 | MATHDB-376819

Maxwell inequality for critical points and Voronoi cells

摘录 / EXCERPT: Let $V_{\alpha}(\bar x)$ be the potential of a generic configuration of point charges of the same sign, and let a_{α}^j denote the number of its critical points of index j...

conjecture | open | MathDB 页面 | 原文线索

V2-072803 | MATHDB-376821

Forrester-Frankel asymptotic conjecture for Fisher-Hartwig Hankel determinants

摘录 / EXCERPT: Forrester-Frankel conjecture. As $N\to\infty$ with m, q, and μ fixed,

conjecture | open | MathDB 页面 | 原文线索

V2-072804 | MATHDB-376824

Simon's absolutely continuous spectrum conjecture for multidimensional Schrödinger operators

摘录 / EXCERPT: Let $H=-\Delta+V(x)$ act on $\mathbb{R}^{d}$, where V is a potential satisfying ... Here $\sigma_{ac}(H)$ denotes the absolutely continuous spectrum of H, and $\mathbb{R}^{+}$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-072805 | MATHDB-376826

Generic gap-edge nondegeneracy conjecture for periodic elliptic operators

摘录 / EXCERPT: Generic gap-edge conjecture. Each endpoint of the gap is a unique, modulo the dual lattice, nondegenerate extremum of a single band function $\lambda_j(k)$.

conjecture | open | MathDB 页面 | 原文线索

V2-072806 | MATHDB-376828

Aldous-Diaconis spectral-gap conjecture for the symmetric simple exclusion process

摘录 / EXCERPT: Let Λ be a finite graph, and let L be the generator of the symmetric simple exclusion process with positive jump rates J_{xy} on its edges. Let $H_\Lambda^{(n)}$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-072807 | MATHDB-376831

Positivity conjecture for elliptic Bernoulli numbers

摘录 / EXCERPT: Let $\mathcal{B}_{2m}$ be the elliptic Bernoulli numbers, and let w, η, g_2, and g_3 denote the associated Weierstrass quantities. Positivity conjecture. The elliptic Be...

conjecture | open | MathDB 页面 | 原文线索

V2-072808 | MATHDB-376833

The minimal-coefficient formula for Jack superpolynomials

摘录 / EXCERPT: Let Λ be a superpartition, let $D[\Lambda]$ be its diagram, and let $a_{\Lambda}(s)$ and $\ell_{\Lambda}(s)$ denote respectively the arm and leg statistics of a cell...

conjecture | open | MathDB 页面 | 原文线索

V2-072809 | MATHDB-376837

Generic period conjecture for crystal cellular automata

摘录 / EXCERPT: Let $p\in P(m)$ be a state with soliton content m, and let $(T^{(r)}_l)^t(p)\neq0$ for every $t\geq0$. Denote by ${\mathcal P}^{(r)}_l$ the least common multiple defined earlier...

conjecture | open | MathDB 页面 | 原文线索

V2-072810 | MATHDB-376844

BdGN's largest-entry conjecture for the periodic model

摘录 / EXCERPT: Let ... , corresponding to the link pattern with arches connecting consecutive points, is $A(n-1)$. This refines the known result that the sum of entries is $A(n)$; the largest-entry...

conjecture | open | MathDB 页面 | 原文线索

V2-072811 | MATHDB-376845

BdGN's largest-entry conjecture for the open B model

摘录 / EXCERPT: At $q=e^{2i\pi/3}$ and in the homogeneous limit, let ... , with arches connecting consecutive points, is $A_V(r)$. The sum-of-entries formula is established in the supplied discussio...

conjecture | open | MathDB 页面 | 原文线索

V2-072812 | MATHDB-376846

The C-series sum-of-entries conjecture

摘录 / EXCERPT: At $q=e^{2i\pi/3}$, let ... is ... For odd $r=2n-1$, the preceding discussion presents the corresponding product as computed, while the even case follows by a specialization from the o...

conjecture | open | MathDB 页面 | 原文线索

V2-072813 | MATHDB-376847

The C-series largest-entry conjecture

摘录 / EXCERPT: At $q=e^{2i\pi/3}$ and in the homogeneous limit, let ... for C_r is the sum of entries for C_{r-1}. The assertion is presented after empirical sum and pairing formulas, and the s...

conjecture | open | MathDB 页面 | 原文线索

V2-072814 | MATHDB-376851

Aldous's spectral-gap conjecture for the symmetric simple exclusion process

摘录 / EXCERPT: Let $G=(V,E)$ be a finite graph with positive edge rates r_{xy}, and let L be the generator of the symmetric simple exclusion process on the n-particle configuration space...

conjecture | open | MathDB 页面 | 原文线索

V2-072815 | MATHDB-376861

Flat PNG universality conjecture for the Airy one process

摘录 / EXCERPT: Let $h(x,t)$ be the height function of the polynuclear growth (PNG) model on a one-dimensional substrate with flat initial conditions. Let ${\cal A}_{\rm 1}$ denote the Airy one pr...

conjecture | open | MathDB 页面 | 原文线索

V2-072816 | MATHDB-376866

Generalized Duistermaat-Heckman formula for compact groups

摘录 / EXCERPT: Generalized Duistermaat-Heckman conjecture. An analogous correlation-function formula should hold for every compact group, with the triangular integration domain identified with...

conjecture | open | MathDB 页面 | 原文线索

V2-072817 | MATHDB-376868

Convergent matrix-integral decomposition conjecture

摘录 / EXCERPT: Convergent matrix-integral decomposition conjecture. The convergent matrix integral $Z_{\rm conv}$ is a finite linear combination of convergent formal solutions of the loop equatio...

conjecture | open | MathDB 页面 | 原文线索

V2-072818 | MATHDB-376869

Mariño-Vafa conjecture for triple Hodge integrals

摘录 / EXCERPT: For the Deligne-Mumford moduli space of stable curves $\overline{{\mathcal M}}_{g,h}$, let $\Lambda_g^\vee(u)$ denote the total Chern polynomial of the Hodge bundle and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072819 | MATHDB-376880

Characterization of spectral curves of hyperbolic monopoles

摘录 / EXCERPT: Let S be a curve of bidegree (k,k) in ${\mathbb P}_1\times {\mathbb P}_1$. Let $L^r(a,b)$ denote the line bundle used in the hyperbolic-monopole spectral-curve construction, re...

conjecture | open | MathDB 页面 | 原文线索

V2-072820 | MATHDB-376885

The C_n elliptic hypergeometric transformation conjecture for the {}_{10}\Omega_9 series

摘录 / EXCERPT: Cn elliptic hypergeometric transformation conjecture. The following transformation should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-072821 | MATHDB-376895

The full strange duality conjecture for generalized Verlinde bundles

摘录 / EXCERPT: Full strange duality conjecture. The morphism SD is an isomorphism. This is the global formulation of the full strange-duality statement for the generalized Verlinde bundles; the...

conjecture | open | MathDB 页面 | 原文线索

V2-072822 | MATHDB-376896

Global generation conjecture for theta line bundles on moduli spaces of degree-zero bundles

摘录 / EXCERPT: Global generation conjecture. For any $r\geq 1$, $\mathcal{O}_{U}(k\Theta_N)$ is globally generated on $U_X(r,0)$ for $k\geq r+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-072823 | MATHDB-376903

The modular-flat freeness conjecture for hyperplane arrangements

摘录 / EXCERPT: Modular-flat freeness conjecture. If X is a modular flat and both ${\mathcal A}_X$ and $c{\mathcal A}_{{\overline{v}}}$ are free arrangements, then ${\mathcal A}$ is a free arran...

conjecture | open | MathDB 页面 | 原文线索

V2-072824 | MATHDB-376905

Arithmetic lattice discreteness conjecture for subgroups generated by unipotent elements

摘录 / EXCERPT: Arithmetic lattice discreteness conjecture. If $W\Gamma$ is discrete, then $W\cap\Gamma$ has finite index in W.

conjecture | open | MathDB 页面 | 原文线索

V2-072825 | MATHDB-376915

The generalized Eckardt point characterization for smooth hypersurfaces

摘录 / EXCERPT: Generalized Eckardt point conjecture. If the total log canonical threshold of X is $\frac{n-1}{n}$, then a wild tiger S for X is a cone in $\mathbb{P}^{n-1}$ over a smooth hy...

conjecture | open | MathDB 页面 | 原文线索

V2-072826 | MATHDB-376922

Uniform bound conjecture for Racah polynomials

摘录 / EXCERPT: Let T be a positive integer and define the Racah polynomial ... Here $R_n(s,T)$ is symmetric in n and s. Uniform Racah bound conjecture. For any integers n,s with...

conjecture | open | MathDB 页面 | 原文线索

V2-072827 | MATHDB-376928

Chmutov-Duzhin-Lando conjecture on intersection graphs of chord diagrams

摘录 / EXCERPT: Let D be a chord diagram, and let its intersection graph $\Gamma(D)$ be the simple graph whose vertices are the chords of D, with two vertices adjacent exactly when the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-072828 | MATHDB-376931

Relative Gromov-Witten invariant formula for plane curves satisfying (AL)

摘录 / EXCERPT: Let B be the plane cubic from the preceding setup, let m_d denote the number of degree-d plane curves satisfying (AL) and meeting B at a specified point of order $3d$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072829 | MATHDB-376935

Local square-integrability conjecture for first-order differential operators

摘录 / EXCERPT: Local regularity conjecture. Under these assumptions,

conjecture | open | MathDB 页面 | 原文线索

V2-072830 | MATHDB-376936

Frobenius descent conjecture for semi-stable bundles

摘录 / EXCERPT: Let X be the genus-two curve in characteristic 2, let X_1 be its Frobenius twist, let $F\colon X\to X_1$ be the relative Frobenius morphism, and let M_X and...

conjecture | open | MathDB 页面 | 原文线索

V2-072831 | MATHDB-376939

Conjecture on independence of essential tautological classes

摘录 / EXCERPT: Let g be a nonnegative integer, let P be a finite set of markings, and let k be a nonnegative integer. Consider the essential tautological classes in...

conjecture | open | MathDB 页面 | 原文线索

V2-072832 | MATHDB-376959

Spectral approximation conjecture for Harper operators on amenable graphs

摘录 / EXCERPT: Spectral approximation conjecture. The normalized eigenvalue-counting functions converge to the spectral density function:

conjecture | open | MathDB 页面 | 原文线索

V2-072833 | MATHDB-376962

MacPherson's conjecture on omitting the dimension axiom

摘录 / EXCERPT: Let X be a stratified space with stratification ${\cal S}$, let ${\cal P}(X,{\cal S})$ denote MacPherson's category of perverse sheaves, and let $D^b_{\cal S}(X)$ denote the boun...

conjecture | open | MathDB 页面 | 原文线索

V2-072834 | MATHDB-376964

Internal Hom conjecture for nonabelian mixed Hodge structures

摘录 / EXCERPT: Let ${\cal V}$ and ${\cal Y}$ be nonabelian mixed Hodge structures, and suppose that ${\cal Y}$ is 1-connected, meaning that its π_0 and π_1 are trivial. Internal Hom con...

conjecture | open | MathDB 页面 | 原文线索

V2-072835 | MATHDB-376972

The cancellation conjecture for effective Chow motives

摘录 / EXCERPT: Cancellation conjecture for effective Chow motives. If

conjecture | open | MathDB 页面 | 原文线索

V2-072836 | MATHDB-376975

Local indicability of left ordered groups without nonabelian free subgroups

摘录 / EXCERPT: Local indicability conjecture. Every left ordered NF-group is locally indicable.

conjecture | open | MathDB 页面 | 原文线索

V2-072837 | MATHDB-376978

Stolarsky bounds for the invariant logarithmic mean

摘录 / EXCERPT: Stolarsky's bounds conjecture. For all $(a,b,c)\in R_{+}^{3}$,

conjecture | open | MathDB 页面 | 原文线索

V2-072838 | MATHDB-376979

Monotonicity inequalities sufficient for the Stolarsky bounds

摘录 / EXCERPT: Sufficient inequalities for Stolarsky's bounds. The inequalities

open_problem | open | MathDB 页面 | 原文线索

V2-072839 | MATHDB-376980

Characterization of pairs invariant in both types

摘录 / EXCERPT: Both-types invariance conjecture. The only such pairs are the quasi-arithmetic transforms

conjecture | open | MathDB 页面 | 原文线索

V2-072840 | MATHDB-376987

The alternating sign matrix generating-function conjecture

摘录 / EXCERPT: Let $A_n(x,y)$ be the weight-generating function for n by n alternating sign matrices, where a matrix with r entries equal to -1 and its top-row 1 in position s has wei...

conjecture | open | MathDB 页面 | 原文线索

V2-072841 | MATHDB-376988

The flip-symmetric alternating sign matrix generating-function conjecture

摘录 / EXCERPT: Let $F_n(x)$ be the weight-generating function for flip-symmetric alternating sign matrices, with the exponent recording the number of -1 entries in the first half of the columns...

conjecture | open | MathDB 页面 | 原文线索

V2-072842 | MATHDB-376989

The half-turn-symmetric alternating sign matrix factorization conjecture

摘录 / EXCERPT: Let $H_n(x,y)$ be the weight-generating function for half-turn-invariant n by n alternating sign matrices, and let $Z_n(x,y,0)$ and $Z_{n-1}(x,y,1)$ be the corresponding plane-...

conjecture | open | MathDB 页面 | 原文线索

V2-072843 | MATHDB-376990

The odd half-turn-symmetric alternating sign matrix factorization conjecture

摘录 / EXCERPT: Let $H_n(x,y)$ be the generating function for half-turn-symmetric alternating sign matrices, and let $R_n(x,\mu)$, $T_n(x,\mu)$, and the polynomials $S_{4n+1}(x),S_{4n-1}(x)$ be as...

conjecture | open | MathDB 页面 | 原文线索

V2-072844 | MATHDB-376991

The quarter-turn-symmetric alternating sign matrix factorization conjecture

摘录 / EXCERPT: Let $Q_n(x,y)$ be the generating function for quarter-turn-symmetric n by n alternating sign matrices, and let H_n and A_n denote the half-turn-symmetric and unrestricted a...

conjecture | open | MathDB 页面 | 原文线索

V2-072845 | MATHDB-376992

The quarter-turn generating-polynomial factorization conjecture

摘录 / EXCERPT: Let $Q_n(x,y)$ be the generating function for quarter-turn-symmetric alternating sign matrices. Quarter-turn polynomial factorization conjecture. There exists a sequence of polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-072846 | MATHDB-376993

The plus-symmetric alternating sign matrix factorization conjecture

摘录 / EXCERPT: Let $P_n(x)$ be the generating function for alternating sign matrices invariant under flips in the vertical and horizontal axes, with the weight defined in the source. Such matrice...

conjecture | open | MathDB 页面 | 原文线索

V2-072847 | MATHDB-377004

The torus-to-plane convergence conjecture for domino configurations

摘录 / EXCERPT: Let (s,t) satisfy $|s|+|t| \leq 2$, and let a,b,c,d be weights satisfying $ab=cd$ such that the average height function for weighted torus tilings has tilt (s,t). Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072848 | MATHDB-377006

The variational-principle conjecture for finite domino configurations

摘录 / EXCERPT: Consider a sequence of finite regions converging to a fixed shape, and let (s,t) be the tilt given by the variational principle wherever that tilt is defined. Let $\mu_{s,t}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-072849 | MATHDB-377007

Semi-specializability of Chebyshev polynomial products for multiples of four

摘录 / EXCERPT: Let $T_m(x)$ denote the Chebyshev polynomial of index m, and let ℓ be a non-zero multiple of 4. Consider ... A continued fraction is semi-specializable if it has the partial...

conjecture | open | MathDB 页面 | 原文线索

V2-072850 | MATHDB-377054

Conjecture that the critical group is not determined by the Tutte polynomial

摘录 / EXCERPT: For a connected undirected graph G, let $T(G;x,y)$ denote its Tutte polynomial and let $\EuScript{K}(G)$ denote its critical group.
Non-determination conjecture. There exist conn...

conjecture | open | MathDB 页面 | 原文线索

V2-072851 | MATHDB-377059

Bessis-Digne-Michel's centralizer conjecture for regular braid-group elements

摘录 / EXCERPT: Bessis-Digne-Michel's centralizer conjecture. The natural morphism

conjecture | open | MathDB 页面 | 原文线索

V2-072852 | MATHDB-377060

A special-case identity for generalized Umemura polynomials

摘录 / EXCERPT: Special-case Umemura-polynomial identity. If $b_1=0$, then

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072853 | MATHDB-377065

Existence conjecture for distributional limits of uniform spherical triangulations

摘录 / EXCERPT: Let $M\in[6,\infty]$, and let T^M_j be chosen randomly uniformly among isomorphism classes of spherical triangulations with j vertices and maximum degree at most M. Given...

conjecture | open | MathDB 页面 | 原文线索

V2-072854 | MATHDB-377069

Strict-unit replacement conjecture for A_{∞}-pre-categories

摘录 / EXCERPT: Let an A_{∞}-pre-category be understood with the equivalence relation defined in the source, and let an A_{∞}-category with strict identity morphisms be a non-unital...

conjecture | open | MathDB 页面 | 原文线索

V2-072855 | MATHDB-377070

Directed A_{∞}-category correspondence conjecture

摘录 / EXCERPT: Let a directed A_{∞}-pre-category be an A_{∞}-pre-category whose objects are indexed by ${{\bf Z}}$ with transversal sequences ordered by the indices, and consider...

conjecture | open | MathDB 页面 | 原文线索

V2-072856 | MATHDB-377084

The First Borwein conjecture on the polynomials $A_n(q)$, $B_n(q)$ and $C_n(q)$

摘录 / EXCERPT: First Borwein conjecture. The polynomials $A_n(q)$, $B_n(q)$ and $C_n(q)$ have nonnegative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-072857 | MATHDB-377086

Goncharov's direct-sum conjecture for multiple zeta values

摘录 / EXCERPT: For each natural number w, let Z_w be the $\mathbb Q$-vector space generated by multiple zeta values of weight w, let $Z_0=\mathbb Q$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072858 | MATHDB-377095

Asymptotic corank growth conjecture for chromatic roots

摘录 / EXCERPT: Asymptotic corank-growth conjecture. As $k\to\infty$,

conjecture | open | MathDB 页面 | 原文线索

V2-072859 | MATHDB-377111

Klainerman's critical well-posedness conjecture for basic field theories

摘录 / EXCERPT: The systems under consideration are basic field theories with critical well-posedness exponent s_c, meaning that the $dot H^{s_c} \times dot H^{s_c-1}$ norm of the initial data i...

conjecture | open | MathDB 页面 | 原文线索

V2-072860 | MATHDB-377112

Horizontal-subspace conjecture for geometric Hodge structures

摘录 / EXCERPT: Let H be a geometric Hodge structure, and let H^{∇} denote its horizontal subspace. Horizontal-subspace conjecture. For any geometric Hodge structure H, the horizontal s...

conjecture | open | MathDB 页面 | 原文线索

V2-072861 | MATHDB-377133

Nonexistence conjecture for three odd-rank homogeneous spaces

摘录 / EXCERPT: Nonexistence conjecture. None of these homogeneous spaces has a tessellation. These are presented as three special cases of Kobayashi's general conjecture; the supplied source give...

conjecture | open | MathDB 页面 | 原文线索

V2-072862 | MATHDB-377134

One-dimensional subgroup tessellation conjecture

摘录 / EXCERPT: Let G be a Lie group, let H be a closed subgroup, and let $d(H)$ denote the invariant used in the paper to measure the dimension relevant to tessellations of G/H. One-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-072863 | MATHDB-377142

The degree and generic simple-root conjecture for the Heun spectral polynomial

摘录 / EXCERPT: Let $Q(E)$ be the polynomial whose zeros characterize those values of E for which the equation in the preceding theorem has a solution in the space ${\mathcal F}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072864 | MATHDB-377147

Strong ordering conjecture for pattern-avoidance classes

摘录 / EXCERPT: For a permutation τ, let $S_n(\tau)$ denote the set of permutations in S_n avoiding τ. Suppose $\tau,\sigma\in S_4$. Strong ordering conjecture. If ... for some k, then...

conjecture | open | MathDB 页面 | 原文线索

V2-072865 | MATHDB-377148

Asymptotic ordering conjecture for pattern-avoidance classes

摘录 / EXCERPT: Let $\tau,\sigma\in S_n$, and let $S_k(\tau)$ denote the set of permutations in S_k avoiding τ. Say that τ is asymptotically smaller than σ if...

conjecture | open | MathDB 页面 | 原文线索

V2-072866 | MATHDB-377157

Internal-zero conjecture for the pattern $1(l+1)l\dots 2$

摘录 / EXCERPT: Let $q=1(l+1)l\dots 2$ with $l\ge3$, and let the frequency sequence for q be the sequence of pattern frequencies indexed by n. A frequency sequence has internal zeros if it con...

conjecture | open | MathDB 页面 | 原文线索

V2-072867 | MATHDB-377166

Embeddedness conjecture for hexagonal circle patterns approximating power maps

摘录 / EXCERPT: Let $0<\alpha=\beta<\frac{1}{2}$ and set $\gamma=1-2\alpha$, so that $0<\gamma<1$. Consider the hexagonal circle patterns $z^{3\alpha}$ with an intersection point at the origin and...

conjecture | open | MathDB 页面 | 原文线索

V2-072868 | MATHDB-377170

Alignment conjecture for virtual crystals $V^{r,s}$

摘录 / EXCERPT: For $4\mathfrak{g}4$ of type $D_{n+1}^{(2)}$, $A_{2n}^{(2)}$, $A_{2n}^{(2)\dagger}$, or $C_n^{(1)}$, let $V^{r,s}$ be the virtual crystal embedded in an ambient type...

conjecture | open | MathDB 页面 | 原文线索

V2-072869 | MATHDB-377189

Naturalness conjecture for the D-brane support morphism

摘录 / EXCERPT: Let X be the Calabi-Yau threefold considered above, let $\beta\in H_2(X,\mathbb Z)$, and let ... be the surjective morphism from the normalization of the sheaf moduli space to the...

conjecture | open | MathDB 页面 | 原文线索

V2-072870 | MATHDB-377191

Higher-dimensional subharmonic gradient inequality

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be an open subset with smooth boundary. For $\lambda\in C^\infty(\overline{\Omega})$ with $\lambda>0$ on $\overline{\Omega}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072871 | MATHDB-377193

The subharmonic gradient-product inequality

摘录 / EXCERPT: Let $B(0,1)\subset\mathbb{R}^d$ be the unit ball. Let $\lambda\in C^\infty(\overline{B(0,1)})$ satisfy $\lambda>0$ on $\overline{B(0,1)}$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072872 | MATHDB-377231

Square-free Smith forms for q-weighted lozenge-tiling matrices

摘录 / EXCERPT: For the explicitly defined q-weighted Kasteleyn and Kasteleyn-Percus matrices $M(a,b,c;q)$, $M_\rho(a,a,a;q)$, $A_\tau(a,b,b;q)$, $\widetilde{A}_\tau(a,b,b;q)$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072873 | MATHDB-377232

q=-1 relations between Kasteleyn cokernels and Smith forms

摘录 / EXCERPT: Let G and G' be the symmetry groups specified in the claim, and write $M_G(a,b,c)_{-1}$ and $A_G(a,b,c)_{-1}$ for the corresponding matrices specialized at $q=-1$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072874 | MATHDB-377234

The elliptic Stark conjecture for CM curves

摘录 / EXCERPT: Let E be a CM elliptic curve, let χ be a character in the setting above, and assume the Dimension conjecture so that the Stark regulator ratio $A(E,\chi)$ is defined. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-072875 | MATHDB-377236

Coherence conjecture for higher-dimensional categories

摘录 / EXCERPT: Higher-dimensional coherence conjecture. Any small lax n-category is laxly n-equivalent to a small Gray n-category.

conjecture | open | MathDB 页面 | 原文线索

V2-072876 | MATHDB-377240

The MLS LCS formula for minimal linear strand arrangements

摘录 / EXCERPT: MLS LCS formula. If ${\mathcal A}$ is an MLS arrangement with group G, then, for all $k\ge 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-072877 | MATHDB-377258

Aldous-Diaconis continuum-limit conjecture for East-model excursions

摘录 / EXCERPT: Let ${\bf X}^0(t)$ be the East process on the positive integers, started from a single particle at site 0. Let $r({\bf x})$ be the rightmost occupied site in configuration...

conjecture | open | MathDB 页面 | 原文线索

V2-072878 | MATHDB-377264

Conjecture IV'_H on orbifold Kobayashi pseudometrics

摘录 / EXCERPT: Let $c_X:X\to C(X)$ be an admissible representative of the core of X, and let $\Delta(c_X)$ be its base orbifold. Conjecture IV'H. One should have ... In addition, if (X/Δ)...

conjecture | open | MathDB 页面 | 原文线索

V2-072879 | MATHDB-377265

Conjecture III_FF on potential density over function fields

摘录 / EXCERPT: Let $f:X\to C$ be a fibration over a function field, with K-core factorization $f=g_f\circ c_f$ and base orbifold $(C(f)/\Delta(c(f)))$. Conjecture III{FF}. The K'-rational poi...

conjecture | open | MathDB 页面 | 原文线索

V2-072880 | MATHDB-377266

Conjecture IV_FF on rational points for nonsplit fibrations

摘录 / EXCERPT: Let $f:X\to C$ be a fibration over a function field with K-core factorization $f=g_f\circ c_f$, and suppose the orbifold fibration ... is not split. Conjecture IV{FF}. There should...

conjecture | open | MathDB 页面 | 原文线索

V2-072881 | MATHDB-377267

Conjecture III_SD on dominated fibrations and rational quotients

摘录 / EXCERPT: Let $f:X\to C$ be a fibration over a function field, let $r_K:X\to R(X_K)$ be its rational quotient, and call X_K dominated when it admits the dominant product-type rational map...

conjecture | open | MathDB 页面 | 原文线索

V2-072882 | MATHDB-377294

Medvedev's solvable-subring conjecture for Lie p-rings

摘录 / EXCERPT: Medvedev's conjecture. L contains a solvable subring of m-bounded derived length whose index is (p,n,m)-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-072883 | MATHDB-377295

Steingrímsson's conjecture on partition statistics and q-Stirling numbers

摘录 / EXCERPT: Let ${\cal P}_n^k$ denote the set of partitions of $[n]$ into k blocks, and let $\mathop{\rm mak}$, $\mathop{\rm lmak}$, $\mathop{\rm mak}{}'$ and ${\mathop{\rm lmak}}'$ be the f...

conjecture | open | MathDB 页面 | 原文线索

V2-072884 | MATHDB-377296

Steingrímsson's Euler-Mahonian conjecture for ordered set partitions

摘录 / EXCERPT: Let ${\cal OP}_n^k$ be the set of ordered k-partitions of $[n]$, and let $\mathop{\rm bmaj}$ and $\mathop{\rm binv}$ be the block-major index and block-inversion statistics. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072885 | MATHDB-377313

The k-Schur involution conjecture

摘录 / EXCERPT: Let λ be a k-bounded partition, meaning that its largest part is at most k. Let λ^{ω_k} denote its k-conjugate, and let ω be the usual involution...

conjecture | open | MathDB 页面 | 原文线索

V2-072886 | MATHDB-377314

The k-Schur structure-coefficient bound conjecture

摘录 / EXCERPT: For k-bounded partitions λ, μ, and ν, write the product of k-Schur functions as ... where $c_{\lambda\mu}^{\nu}$ are the ordinary Littlewood-Richardson coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-072887 | MATHDB-377315

The t-involution conjecture for k-Schur functions

摘录 / EXCERPT: Let λ be a k-bounded partition, let λ^{ω_k} be its k-conjugate, and define $\omega_t f=(\omega f)|_{t\to1/t}$. The t-involution conjecture. For every k...

conjecture | open | MathDB 页面 | 原文线索

V2-072888 | MATHDB-377324

Macdonald polynomial Grassmannian vanishing conjecture for $U(m)\times U(n)$

摘录 / EXCERPT: Let m and n be integers with $0\leq m\leq n$, let $\mu\overline{\nu}$ be a dominant weight of $U(m+n)$, and let $P_{\mu\overline{\nu}}$ be the corresponding Macdonald polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-072889 | MATHDB-377325

Macdonald polynomial vanishing conjecture for the $U(2n)$ Grassmannian coset

摘录 / EXCERPT: Let $n\geq 0$, let $\mu\overline{\nu}$ be a dominant weight of $U(2n)$, and let Z normalize the integral so that the trivial polynomial integrates to 1. $U(2n)$ Grassmannian co...

conjecture | open | MathDB 页面 | 原文线索

V2-072890 | MATHDB-377326

Macdonald polynomial vanishing conjecture for a nonstandard $U(2n)$ weight

摘录 / EXCERPT: Let $n\geq 0$, let $\mu\overline{\nu}$ be a dominant weight of $U(2n)$, and let Z normalize the integral so that the trivial polynomial integrates to 1. Nonstandard $U(2n)$ van...

conjecture | open | MathDB 页面 | 原文线索

V2-072891 | MATHDB-377328

Conjectured generating function for permutations containing four occurrences of 321

摘录 / EXCERPT: The 321 four-occurrence generating-function conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-072892 | MATHDB-377352

The spectral-polynomial identity for the Heun equation

摘录 / EXCERPT: Spectral-polynomial identity. The two polynomials are equal:

conjecture | open | MathDB 页面 | 原文线索

V2-072893 | MATHDB-377388

Matrix-valued spherical-function product expansion conjecture

摘录 / EXCERPT: Matrix-valued product-expansion conjecture. The product admits a unique expansion of this form, with matrix coefficients A_k, for all nonnegative n.

conjecture | open | MathDB 页面 | 原文线索

V2-072894 | MATHDB-377389

Hook alternating property conjecture for matrix-valued spherical functions

摘录 / EXCERPT: Hook alternating property conjecture. The coefficients A_k with k in the range $j-i\leq k\leq j+i$ have the hook alternating property.

conjecture | open | MathDB 页面 | 原文线索

V2-072895 | MATHDB-377392

Nonnegative coefficients of the Q-Kostka polynomials

摘录 / EXCERPT: Let $L_{\lambda\mu}(q)$ denote the Q-Kostka polynomial, with λ and μ partitions in the indexing convention of the paper. Positivity conjecture. The Q-Kostka polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-072896 | MATHDB-377393

Growth monotonicity of Q-Kostka polynomials under adjoining a part

摘录 / EXCERPT: Let λ and μ be partitions, let r be an integer that is not a part of either partition, and let $L_{\lambda\mu}(q)$ denote the Q-Kostka polynomial. Growth conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-072897 | MATHDB-377394

Positivity of generalized Q-Kostka polynomials

摘录 / EXCERPT: Generalized Q-Kostka positivity conjecture. If $\bar{\mu^*}$ is a partition, then $L_{\lambda;\mu^*}(q)$ is a polynomial in q with non-negative integer coefficients. These coeffi...

conjecture | open | MathDB 页面 | 原文线索

V2-072898 | MATHDB-377401

The complexified Kashaev conjecture for hyperbolic links

摘录 / EXCERPT: Let L be a hyperbolic link, and let $J_N(L)$ be its colored Jones polynomial evaluated at $\exp\left(2\pi\sqrt{-1}/N\right)$. Write $\operatorname{vol}(L)$ for the hyperbolic vol...

conjecture | open | MathDB 页面 | 原文线索

V2-072899 | MATHDB-377422

The EKG linear-bound conjecture

摘录 / EXCERPT: Let $(a(n))_{n\geq 1}$ be the EKG sequence defined by $a(1)=1$, $a(2)=2$, and, for $n\geq 3$, choosing the smallest unused natural number sharing a nontrivial common divisor with...

conjecture | open | MathDB 页面 | 原文线索

V2-072900 | MATHDB-377425

Existence of a stability weight making Dynkin quiver indecomposables stable

摘录 / EXCERPT: Let Q be a quiver whose underlying unoriented graph is a disjoint union of Dynkin diagrams of type A, D, or E. A representation of Q is called stable with respect to a we...

conjecture | open | MathDB 页面 | 原文线索

V2-072901 | MATHDB-377428

The averaging-functor vanishing conjecture

摘录 / EXCERPT: The averaging-functor vanishing conjecture. Assume that E is irreducible and has rank greater than n. If

conjecture | open | MathDB 页面 | 原文线索

V2-072902 | MATHDB-377432

Higher-order discrete Laguerre inequality for real-rooted polynomials

摘录 / EXCERPT: Let p be a real-rooted polynomial of degree k, and let $M(p)$ denote its mesh, with $M(p)\geq 1$. Discrete Laguerre inequality. Then ... This conjecture proposes a discrete analo...

conjecture | open | MathDB 页面 | 原文线索

V2-072903 | MATHDB-377439

Winkelmann's upper-half-plane formulation of the isogeny criterion

摘录 / EXCERPT: Let H^+ be the upper half-plane, and define ... The groups $GL_2^+(\mathbb Q)$ and $B^+(\mathbb Q)$ act on H^+ by fractional linear transformations. Winkelmann's upper-half-plane...

conjecture | open | MathDB 页面 | 原文线索

V2-072904 | MATHDB-377454

Thomassen's conjecture for chromatic roots of 3-connected graphs

摘录 / EXCERPT: Let G be a loopless 3-connected graph with n vertices, and let $P(G,t)$ be its chromatic polynomial. Let $\alpha\approx 1.781$ be the chromatic root of $K_{3,4}$ in $(1,2)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-072905 | MATHDB-377456

Thomassen's conjecture on real chromatic roots of planar graphs

摘录 / EXCERPT: Let G be a loopless planar graph, and let $P(G,t)$ be its chromatic polynomial. Thomassen's conjecture. The real chromatic roots of planar graphs are dense everywhere in ... Real c...

conjecture | open | MathDB 页面 | 原文线索

V2-072906 | MATHDB-377457

Woodall's conjectures on chromatic roots of highly connected plane triangulations

摘录 / EXCERPT: A plane triangulation is a loopless plane graph in which every face has size three. Let $P(G,t)$ be its chromatic polynomial, let $\tau=(1+\sqrt{5})/2$ be the golden ratio, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072907 | MATHDB-377458

Beraha's conjecture on chromatic roots near the Beraha numbers

摘录 / EXCERPT: For each integer $r\geq2$, define the Beraha number ... A plane triangulation is a loopless plane graph whose faces all have size three. Beraha's conjecture. For every $r\geq2$ and e...

conjecture | open | MathDB 页面 | 原文线索

V2-072908 | MATHDB-377459

Salas-Sokal conjecture for planar bipartite graphs

摘录 / EXCERPT: Let G be a planar bipartite graph, and let $P(G,t)$ be its chromatic polynomial. Let $\tau=(1+\sqrt{5})/2$ be the golden ratio. Salas-Sokal conjecture. ... Equivalently, planar bip...

conjecture | open | MathDB 页面 | 原文线索

V2-072909 | MATHDB-377460

Sokal's half-plane conjecture for chromatic roots

摘录 / EXCERPT: Let G be a graph, let $\Delta(G)$ denote its maximum degree, and let $P(G,t)$ be its chromatic polynomial. Sokal's conjecture. ... for every complex number t satisfying ... This wo...

conjecture | open | MathDB 页面 | 原文线索

V2-072910 | MATHDB-377461

Sokal's bounded-edge-connectivity conjecture for chromatic roots

摘录 / EXCERPT: Let G be a graph and let $\Lambda(G)$ be the maximum number of edge-disjoint paths joining any pair of vertices of G. Sokal's conjecture. There exists a constant C such that...

conjecture | open | MathDB 页面 | 原文线索

V2-072911 | MATHDB-377462

Conjecture on flow roots of 3-connected graphs

摘录 / EXCERPT: Let G be a 3-connected graph with n vertices and m edges, and let $F(G,t)$ denote its flow polynomial. Let $\phi\approx1.749$ be the flow root of K_5 in $(1,2)$. 3-connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-072912 | MATHDB-377463

Finiteness conjecture for cyclically 4-connected cubic flow roots

摘录 / EXCERPT: Let G be a cyclically 4-connected cubic graph, and let $F(G,t)$ denote its flow polynomial. Let $\tau=(1+\sqrt{5})/2$. Finiteness conjecture. For every $\epsilon>0$, only finitel...

conjecture | open | MathDB 页面 | 原文线索

V2-072913 | MATHDB-377469

Torus-type conjecture for reduced sextics with large essential rank

摘录 / EXCERPT: Torus-type conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-072914 | MATHDB-377496

The Buchsbaum equality conjecture for symmetric and exterior iterated Betti numbers

摘录 / EXCERPT: Let Γ be a Buchsbaum simplicial complex, and let $b_{i,j}(\Gamma)$ and $b^e_{i,j}(\Gamma)$ denote its symmetric and exterior iterated Betti numbers. Let $\Delta(\Gamma)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072915 | MATHDB-377498

Conjecture on the dimension of pluriharmonic measure in complex domains

摘录 / EXCERPT: Pluriharmonic-measure dimension conjecture. The dimension of pluriharmonic measure of domains in $\mathbb{C}^n$ is at most

conjecture | open | MathDB 页面 | 原文线索

V2-072916 | MATHDB-377502

Arithmetic inclusion for normalized linear forms in odd zeta values

摘录 / EXCERPT: Arithmetic inclusion conjecture. There holds the inclusion

conjecture | open | MathDB 页面 | 原文线索

V2-072917 | MATHDB-377524

Generalized configuration-sum fermionic formula

摘录 / EXCERPT: Let $\eta\in\mathcal{P}$ satisfy $\eta_1\leq n-1$. For $s,u\in\{0,\dots,N\}$ and $r,v\in\{0,\dots,n-1\}$, set $\Lambda=(N-s)\Lambda_r+s\Lambda_{r+1}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072918 | MATHDB-377532

Optimality conjecture for the binary Varshamov-Tenengolts codes VT_0(n)

摘录 / EXCERPT: Let $A(n,1)$ denote the maximum size of a binary single-deletion-correcting code of length n, and let $VT_0(n)$ be the binary Varshamov-Tenengolts code with syndrome 0. A code...

conjecture | open | MathDB 页面 | 原文线索

V2-072919 | MATHDB-377534

Optimality of short alternating factorizations in the universal cover of SL(2,R)

摘录 / EXCERPT: Let $A_n(s)$ be the alternating factorization defined in the paper, and let $\widetilde{\operatorname{SL}}(2,\mathbb{R})$ be the universal covering group of...

conjecture | open | MathDB 页面 | 原文线索

V2-072920 | MATHDB-377538

The equivariant Bloch-Kato conjecture for motives

摘录 / EXCERPT: Equivariant Bloch-Kato conjecture. The Beilinson regulator induces an isomorphism $\Delta_f(G,M^\lor(1-k))\otimes {\Bbb{R}}\cong \1_{{\Bbb{R}}[G]}$, under which...

conjecture | open | MathDB 页面 | 原文线索

V2-072921 | MATHDB-377545

Deninger's exchanged-place Leopoldt conjecture

摘录 / EXCERPT: Assume that k is a CM field. Let $E_p(k)$ be the group of p-Weil numbers in k, let ... where ${\mathbb R}(1)_v$ is the purely imaginary subspace at the complex place v, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-072922 | MATHDB-377551

Density of uniformly exponential decay in generic one-parameter families

摘录 / EXCERPT: Consider a generic one-parameter family of maps. Density conjecture. Maps with exponential decay of correlations are Lebesgue density points of other maps with uniform exponential...

conjecture | open | MathDB 页面 | 原文线索

V2-072923 | MATHDB-377552

Ritoré-Ros isoperimetric conjecture for the cubic three-torus

摘录 / EXCERPT: Ritoré-Ros conjecture. The optima for the isoperimetric problem in a cubic T^3 are the sphere, cylinder, and slab.

conjecture | open | MathDB 页面 | 原文线索

V2-072924 | MATHDB-377557

Concavity conjecture for the double-bubble profile in the three-torus

摘录 / EXCERPT: Concavity conjecture. The least area to enclose and separate two given volumes in the three-torus is a concave function of the volumes.

conjecture | open | MathDB 页面 | 原文线索

V2-072925 | MATHDB-377558

Double Slab conjecture for very long three-tori

摘录 / EXCERPT: Very-long-torus Double Slab conjecture. In the special case of a very long T^3, the Double Slab is optimal for most volumes.

conjecture | open | MathDB 页面 | 原文线索

V2-072926 | MATHDB-377578

Gray-operad conjecture for semistrict n-categories

摘录 / EXCERPT: Gray-operad conjecture. There is a unique contractible n-operad G_n such that, for every k with $0\leq k\leq n-1$, $\text{$cal P$}_k(G_n)$ is the free k-fold monoid operad....

conjecture | open | MathDB 页面 | 原文线索

V2-072927 | MATHDB-377581

Analytic order conjecture for isolated critical-point germs

摘录 / EXCERPT: Let $n\ge 3$ and let $f:({\mathbb C}^n,0)\to({\mathbb C},0)$ be a germ of a holomorphic function with an isolated critical point, so that $\mu(f)<\infty$. Let $d(f)$ denote the min...

conjecture | open | MathDB 页面 | 原文线索

V2-072928 | MATHDB-377582

The interlace-polynomial gap conjecture

摘录 / EXCERPT: Let $q_G(x)$ denote the interlace polynomial of a graph G. There are constants $c_1,c_2,\ldots$ with ... such that, for every $k\geq 1$ and all sufficiently large n, there are gr...

conjecture | open | MathDB 页面 | 原文线索

V2-072929 | MATHDB-377584

The virtual R-matrix restriction conjecture

摘录 / EXCERPT: Let $(V,\widehat{V})$ and $(V',\widehat{V}')$ be simple aligned virtual crystals, where V and V' are X-crystals and the hatted crystals are Y-crystals. Let ... be the type...

conjecture | open | MathDB 页面 | 原文线索

V2-072930 | MATHDB-377586

Linear independence of rectangularly complementary Schur products

摘录 / EXCERPT: Linear-independence conjecture. Fix a rectangle R. The products

conjecture | open | MathDB 页面 | 原文线索

V2-072931 | MATHDB-377587

Witness-elimination conjecture for rectangularly complementary Schur products

摘录 / EXCERPT: Witness-elimination conjecture. With

conjecture | open | MathDB 页面 | 原文线索

V2-072932 | MATHDB-377589

The slope s_g is computed by the quotient of the lambda and nonseparating-boundary coefficients

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_g$ be the moduli space of stable curves of genus g, and let s_g be its slope. For an effective divisor D on $\overline{\mathcal{M}}_g$, write ... wh...

conjecture | open | MathDB 页面 | 原文线索

V2-072933 | MATHDB-377592

Perrin-Riou's integrality conjecture for the determinant of the big exponential map

摘录 / EXCERPT: Perrin-Riou's integrality conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-072934 | MATHDB-377593

Warnaar's $tilde A_n$ basic hypergeometric summation conjecture

摘录 / EXCERPT: Warnaar's conjecture. These two expressions are equal. This conjectural identity would extend the hierarchy of transformations between multiple basic hypergeometric series of diffe...

conjecture | open | MathDB 页面 | 原文线索

V2-072935 | MATHDB-377597

The divergence obstruction for the two-variable reduction

摘录 / EXCERPT: Let $r(x)$ be a monic polynomial of degree $N+1>1$ with distinct roots, and let $\lambda(x)$ and $\mu(x)$ be the unique polynomials satisfying ... with $\deg\mu(x)\leq N-1$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072936 | MATHDB-377598

Convergence conjecture for the associated formal differential operator

摘录 / EXCERPT: Let F be an analytic map defined in an open neighborhood of $0\in{\mathbb C}^n$, and let $a(z)$ denote the formal series associated with F through the formal differential opera...

conjecture | open | MathDB 页面 | 原文线索

V2-072937 | MATHDB-377606

The positive trace conjecture for words in two positive definite matrices

摘录 / EXCERPT: Let a word be a product formed from two real positive definite matrices. A word is symmetric if it is unchanged by reversal, and a product of two symmetric words means their juxtap...

conjecture | open | MathDB 页面 | 原文线索

V2-072938 | MATHDB-377622

Dydak's compactification conjecture in extension theory

摘录 / EXCERPT: Let K be a countable CW complex, and let $AE(X)$ mean that every map from a closed subset of X to K extends over X. A metrizable compactification $\widetilde{X}$ is a compa...

conjecture | open | MathDB 页面 | 原文线索

V2-072939 | MATHDB-377633

Exhaustivity of unipotent representations for rigid orbits

摘录 / EXCERPT: Let $\mathcal U$ be the relevant set of rigid orbits, and suppose that $G(1)$ is not $O(l)$ for any $l\geq2$. Unipotent-exhaustivity conjecture. The set...

conjecture | open | MathDB 页面 | 原文线索

V2-072940 | MATHDB-377639

The linear phase transition conjecture for random K-SAT

摘录 / EXCERPT: A random K-SAT instance has n Boolean variables and $m=cn$ clauses, where c is the clause-to-variable ratio and K is the number of literals per clause. The linear phase trans...

conjecture | open | MathDB 页面 | 原文线索

V2-072941 | MATHDB-377640

The maximum-satisfiable-clauses scaling conjecture for random K-SAT

摘录 / EXCERPT: Let $N(n,m)$ be the maximum number of clauses that can be satisfied by one assignment of n variables in a random K-SAT instance with m clauses, and let c^*_K be the threshold...

conjecture | open | MathDB 页面 | 原文线索

V2-072942 | MATHDB-377641

Erdős's colorability threshold conjecture for sparse random graphs

摘录 / EXCERPT: Fix an integer $k\geq 2$, and let $G(n,cn)$ be the sparse random graph with n vertices and cn edges. The graph is k-colorable when its vertices can be assigned colors...

conjecture | open | MathDB 页面 | 原文线索

V2-072943 | MATHDB-377642

The linear phase transition conjecture for random K-LSAT

摘录 / EXCERPT: Let c^*_K be the constant introduced in the source's theorem for the optimal value ${\cal LP}(n,c)$ of the random K-LSAT linear program, and consider the corresponding feasibilit...

conjecture | open | MathDB 页面 | 原文线索

V2-072944 | MATHDB-377645

Selmer cotangent-space isomorphism conjecture

摘录 / EXCERPT: Selmer cotangent-space conjecture. The inclusion is an isomorphism:

conjecture | open | MathDB 页面 | 原文线索

V2-072945 | MATHDB-377651

Orbifold Iitaka Cnm conjecture

摘录 / EXCERPT: Let Y^∂ be a smooth or e-admissible effective orbifold, let $g^\partial:Y^\partial\rightarrow Z$ be a fibration, and let Y_z^∂ be its orbifold general fiber. Orbi...

conjecture | open | MathDB 页面 | 原文线索

V2-072946 | MATHDB-377656

Geometric Langlands conjecture for generic normalized eigensheaves

摘录 / EXCERPT: Let X be the curve and G the reductive group of the paper. Let E be a $\check{G}$-local system on X, and put $W=V^{\gamma}_E$. Assume that W is irreducible and that whene...

conjecture | open | MathDB 页面 | 原文线索

V2-072947 | MATHDB-377657

Whittaker sheaf Hecke property conjecture

摘录 / EXCERPT: Let G be the group and let γ be the coweight used to define the stacks $\mathcal Y_d$. For a $\check{G}$-local system E on X, let $\mathcal W^d_{E,\psi}$ be the Whitta...

conjecture | open | MathDB 页面 | 原文线索

V2-072948 | MATHDB-377658

Frenkel-Gaitsgory-Vilonen vanishing conjecture for Levi averaging

摘录 / EXCERPT: Let G be the group in the paper, let P be a standard proper parabolic with Levi subgroup M, and let W be an irreducible local system on X of rank $r=\dim V^\gamma$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-072949 | MATHDB-377663

The face-subgraph characterization conjecture for compact visibility graphs

摘录 / EXCERPT: Let G be a planar graph. For each internal face F in a plane drawing of G, let G_F denote the subgraph induced by the vertices incident with F. Compact visibility conject...

conjecture | open | MathDB 页面 | 原文线索

V2-072950 | MATHDB-377664

The connected-graph visibility conjecture

摘录 / EXCERPT: A graph is representable as a visibility graph if it admits a visibility representation by regions whose unblocked sightlines correspond exactly to the edges of the graph. Connecte...

conjecture | open | MathDB 页面 | 原文线索

V2-072951 | MATHDB-377674

Ciucu-Krattenthaler enumeration conjecture for stationary-state coefficients

摘录 / EXCERPT: Write p repeated opening parentheses as $(^p$ and p repeated opening parentheses followed by p closing parentheses as $(\ldots)_p=(^p\ldots)^p$. For matchings of the form...

conjecture | open | MathDB 页面 | 原文线索

V2-072952 | MATHDB-377678

General reverse-unimodality conjecture for beta-values of lists

摘录 / EXCERPT: Let l be a positive integer, and let i and j be non-negative integers with $j>i$. For $1\leq k\leq l$, let $L_{i,j}^k$ be the list of length l whose entries are i except...

conjecture | open | MathDB 页面 | 原文线索

V2-072953 | MATHDB-377679

Skewed-pair inequality conjecture for the cd-index

摘录 / EXCERPT: Let $\beta(L)$ denote the beta-value associated with a list L, and let list concatenation be written by juxtaposition. Let m,n be non-negative integers with $n>m$, and let L...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072954 | MATHDB-377680

Better-balanced-pair inequality conjecture for the cd-index

摘录 / EXCERPT: Let (m_1,n_1) and (m_2,n_2) be pairs of non-negative integers, where (m_1,n_1) is strictly better balanced than (m_2,n_2), meaning that the first pair is strictly closer to...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-072955 | MATHDB-377681

Balanced-list maximization conjecture for the cd-index

摘录 / EXCERPT: Let $\beta(L)$ denote the beta-value associated with a list L. A list is balanced if its entries are either k or $k+1$ for some non-negative integer k. Let L, M, and...

conjecture | open | MathDB 页面 | 原文线索

V2-072956 | MATHDB-377687

Kuo's conjecture on blow-analytic sufficiency of jets

摘录 / EXCERPT: Kuo's conjecture. If $w\in J^r_{\mathbb{R}}(n,1)$ satisfies the Kuiper-Kuo condition as a complex r-jet, then w is blow-analytically sufficient in C^{ω}-functions.

conjecture | open | MathDB 页面 | 原文线索

V2-072957 | MATHDB-377704

Modular-3 congruence for the conjectural Pascal polynomial factors

摘录 / EXCERPT: Modular-3 conjecture. For every k,

conjecture | open | MathDB 页面 | 原文线索

V2-072958 | MATHDB-377707

The noncompactified blowup formula for framed sheaf moduli

摘录 / EXCERPT: Let X be a smooth projective surface, let $C\subset X$ be the framing divisor, and let $\pi:\tilde X\to X$ be the blowup of a point $x_0\notin C$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-072959 | MATHDB-377709

The positive-dimensional subgroup conjecture in automorphism groups of rooted trees

摘录 / EXCERPT: Let $\Gamma(p)$ be the profinite automorphism group of the rooted p-ary tree, and let positive-dimensional mean having positive Hausdorff dimension as a closed subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-072960 | MATHDB-377722

Giraud's theorem for Segal topoi

摘录 / EXCERPT: Giraud's theorem for Segal topoi. A is a $\mathbb{U}$-Segal topos if and only if:

conjecture | open | MathDB 页面 | 原文线索

V2-072961 | MATHDB-377729

Symmetry-group conjecture for graphs of type 3_n

摘录 / EXCERPT: Symmetry-group conjecture. (i) $3_n(D_{2h})$ exists if and only if $\frac{n}{4}$ is even and $n\geq 16$; there are no tight $3_n(D_{2h})$. (ii) $3_n(D_{2d})$ exists if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-072962 | MATHDB-377730

Existence and zigzag bound conjecture for tight polyhedra of type 4_n

摘录 / EXCERPT: 4_n existence and zigzag conjecture. (i) A z-knotted 4_n exists if and only if $n\geq 30$ and $n\equiv 2\pmod 4$. (ii) A tight 4_n exists if and only if $n\geq 8$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-072963 | MATHDB-377739

The double-dual isomorphism for finite tensor categories

摘录 / EXCERPT: Let ${\mathcal C}$ be a finite tensor category, let $V\in{\mathcal C}$, let V^{**} denote the double dual of V, and let L_ρ be the distinguished invertible object of...

conjecture | open | MathDB 页面 | 原文线索

V2-072964 | MATHDB-377741

Ramanujan's conjecture for half-integral weight automorphic forms

摘录 / EXCERPT: Ramanujan conjecture. As $|D|\mapsto\infty$, for every $\alpha>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-072965 | MATHDB-377745

The complete-intersection semigroup-condition conjecture for splice diagrams

摘录 / EXCERPT: Complete-intersection semigroup-condition conjecture. If X is a complete intersection, then Δ satisfies the semigroup condition.

conjecture | open | MathDB 页面 | 原文线索

V2-072966 | MATHDB-377751

Rapoport-Zink weight spectral sequence conjecture

摘录 / EXCERPT: Rapoport-Zink spectral sequence conjecture. For all r,w, the induced map on E_2-terms is an isomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-072967 | MATHDB-377752

The p-adic weight-monodromy conjecture

摘录 / EXCERPT: The p-adic weight-monodromy conjecture. The filtrations satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-072968 | MATHDB-377753

Mokrane's p-adic weight spectral sequence conjecture

摘录 / EXCERPT: Mokrane's spectral sequence conjecture. For all r,w, the map

conjecture | open | MathDB 页面 | 原文线索

V2-072969 | MATHDB-377767

Nash's conjecture on arcs and exceptional curves of surface singularities

摘录 / EXCERPT: Nash's conjecture. The number of exceptional curves in the minimal resolution of (S,s) equals the number of irreducible components of $\mathcal{H}$.

conjecture | open | MathDB 页面 | 原文线索

V2-072970 | MATHDB-377770

Conjecture on variations of mixed Hodge-Tate structures for multiple polylogarithms

摘录 / EXCERPT: Let $Li_{n_1,\ldots,n_k}(x_1,\ldots,x_k)$ be a multiple polylogarithm, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-072971 | MATHDB-377788

The double coset conjecture for finite Coxeter groups

摘录 / EXCERPT: Let W be a finite Coxeter group with simple generators S. For $I,J\subset S$, let X_{IJ} be the set of distinguished representatives used in the double-coset decomposition, a...

conjecture | open | MathDB 页面 | 原文线索

V2-072972 | MATHDB-377790

Coefficient-positivity conjecture for the refined Borwein polynomials

摘录 / EXCERPT: Refined Borwein coefficient-positivity conjecture. The polynomials $\mathcal{A}_{L,M}(q)$ and $\mathcal{C}_{L,M}(q)$ have nonnegative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-072973 | MATHDB-377791

Two-parameter Borwein positivity conjecture for $A_{L,m}(q)$ and $C_{L,m}(q)$

摘录 / EXCERPT: Two-parameter Borwein positivity conjecture. The polynomials $A_{L,m}(q)$ for $0\leq m\leq L$ and $C_{L,m}(q)$ for $0\leq m\leq L-1$ have nonnegative coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-072974 | MATHDB-377800

Virtual crossing realization for maximally virtualized isotopy sequences

摘录 / EXCERPT: Virtual crossing realization conjecture. Every maximally virtualized virtual isotopy sequence with switch-free classical initial and final diagrams admits a valid virtual crossing...

conjecture | open | MathDB 页面 | 原文线索

V2-072975 | MATHDB-377803

The projective-space generation conjecture for isolated fixed-point actions

摘录 / EXCERPT: Projective-space generation conjecture. Stably complex S^1 actions with isolated fixed points up to bordism are generated by linear actions on projective spaces

conjecture | open | MathDB 页面 | 原文线索

V2-072976 | MATHDB-377804

The geometric Gamma-operation generation conjecture for equivariant bordism

摘录 / EXCERPT: Geometric Gamma-operation generation conjecture. Ω^{S^1} is generated over geometric versions of the operations Γ_V by linear actions on projective spaces.

conjecture | open | MathDB 页面 | 原文线索

V2-072977 | MATHDB-377811

The conjectural elliptic beta integral for the A_n root system

摘录 / EXCERPT: Let n be a positive integer. Let z_i, $i=1,\dots,n$, t_k, $k=1,\dots,n+1$, and f_j, $j=1,\dots,n+2$, be independent complex variables, and set ... With...

conjecture | open | MathDB 页面 | 原文线索

V2-072978 | MATHDB-377812

The conjectural terminating elliptic hypergeometric sum for the A_n root system

摘录 / EXCERPT: Let N be a positive integer, let t and $t_1,\dots,t_{n+3}$ be parameters, and suppose that ... For an integer λ, write $\theta(a)_\lambda=\theta(a;p;q)_\lambda$. The term...

conjecture | open | MathDB 页面 | 原文线索

V2-072979 | MATHDB-377820

Existence of strongly MDS convolutional codes with maximum distance profile

摘录 / EXCERPT: Existence conjecture. For all $n>k>0$ and for all $\delta\geq 0$ there exists an (n,k,δ) code over a sufficiently large field which is both strongly MDS and has a maximum di...

conjecture | open | MathDB 页面 | 原文线索

V2-072980 | MATHDB-377825

Elementary abelian 2-group Lehmer constant conjecture

摘录 / EXCERPT: For an integer $n\geqslant 2$, let $\mathbb{Z}_{\!/2}^n$ denote the direct sum of n copies of the cyclic group $\mathbb{Z}_{\!/2}$. Let $\lambda(\mathbb{Z}_{\!/2}^n)$ denote its L...

conjecture | open | MathDB 页面 | 原文线索

V2-072981 | MATHDB-377838

Anisotropic variational principle for perturbed simply attractive potentials

摘录 / EXCERPT: Consider perturbed simply attractive potentials, including anisotropic potentials, meaning potentials that are not necessarily isotropic. Let the variational principle be the relat...

conjecture | open | MathDB 页面 | 原文线索

V2-072982 | MATHDB-377839

Exclusion of infinite-specific-free-energy gradient Gibbs measures

摘录 / EXCERPT: Let U_{ϕ} be the set appearing in the source, and let μ be a non-trivial gradient Gibbs measure, meaning that μ-almost surely $Z_{\Lambda}(\phi)<\infty$ for every fini...

conjecture | open | MathDB 页面 | 原文线索

V2-072983 | MATHDB-377850

Complex spectral-order conjecture for polynomial zeros

摘录 / EXCERPT: Complex spectral-order conjecture. If $P\in\mathcal{C}_{n}$, then

conjecture | open | MathDB 页面 | 原文线索

V2-072984 | MATHDB-377851

Common-orbit conjecture for real polynomials under differential operators

摘录 / EXCERPT: Common-orbit conjecture. If $P_{1},P_{2}\in\mathcal{R}_{n}$, then there exist differential operators $\Lambda_{1},\Lambda_{2}\in\mathcal{S}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-072985 | MATHDB-377856

Maximal nondegeneracy conjecture for the two-variable p-adic height

摘录 / EXCERPT: Maximal nondegeneracy conjecture. If j is even and $0\leq j\leq 2\min(r^+,r^-)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-072986 | MATHDB-377857

The two-variable main conjecture for the Selmer group of an elliptic curve

摘录 / EXCERPT: Two-variable main conjecture. The two-variable p-adic L-function $\mathbf{L}$ generates the ideal

conjecture | open | MathDB 页面 | 原文线索

V2-072987 | MATHDB-377858

The cyclotomic and anticyclotomic main conjectures

摘录 / EXCERPT: Cyclotomic and anticyclotomic main conjectures. Both assertions hold:

conjecture | open | MathDB 页面 | 原文线索

V2-072988 | MATHDB-377862

Invariant-form criterion for generic diameter-three tridiagonal pairs

摘录 / EXCERPT: Let Phi be a generic tridiagonal-pair parameter array with diameter $d=3$, acting through A,A^* on V, and let w_1,w_2,w_3 be its weights. Invariant-form conjecture. The fol...

conjecture | open | MathDB 页面 | 原文线索

V2-072989 | MATHDB-377882

The low-regularity local well-posedness conjecture for power nonlinear wave equations

摘录 / EXCERPT: Let $n\geq 4$, let $0\leq s\leq \frac12$, and consider the power nonlinear wave equation ... The initial data lie in $H^s\times H^{s-1}$. Local well-posedness conjecture. The equatio...

conjecture | open | MathDB 页面 | 原文线索

V2-072990 | MATHDB-377885

The null-condition well-posedness conjecture for quasilinear wave equations

摘录 / EXCERPT: Null-condition well-posedness conjecture. If the null condition holds, then (GNLW) is well-posed in $H^s\times H^{s-1}$ for some $s<\frac n2+\frac34$ when $n=2$, respectively for s...

conjecture | open | MathDB 页面 | 原文线索

V2-072991 | MATHDB-377892

The Berge graph decomposition conjecture

摘录 / EXCERPT: Let G be a finite simple graph. A Berge graph is a graph containing neither an odd hole nor the complement of an odd hole. A graph is basic if it belongs to one of the four class...

conjecture | open | MathDB 页面 | 原文线索

V2-072992 | MATHDB-377899

Monotonicity of relaxation time for ferromagnetic Ising Glauber dynamics

摘录 / EXCERPT: Let G be any graph equipped with a ferromagnetic Ising model, with couplings J_{xy}, and let τ_2 denote the relaxation time of its Glauber dynamics. Peres's monotonicity c...

conjecture | open | MathDB 页面 | 原文线索

V2-072993 | MATHDB-377902

Slope quotient principle for effective divisors on the moduli space of curves

摘录 / EXCERPT: Let s_g be the slope of $\overline{\mathcal{M}}_g$, and let $s(D)$ denote the slope of an effective divisor $D\equiv a\lambda-\sum_{i=0}^{[g/2]}b_i\delta_i$. Slope quotient princ...

conjecture | open | MathDB 页面 | 原文线索

V2-072994 | MATHDB-377905

Markov field property for local causal states

摘录 / EXCERPT: Markov field property. The local causal states form a Markov field in space-time.

conjecture | open | MathDB 页面 | 原文线索

V2-072995 | MATHDB-377907

The scaling limit conjecture for the conditioned Brownian bead process

摘录 / EXCERPT: Let the c3-finite bead process be conditioned to survive up to distance r, equivalently up to time r^2 or imaginary part r, and let $r\rightarrow\infty$. Scaling limit conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-072996 | MATHDB-377917

Vershik's uniqueness conjecture for the Poisson-Dirichlet distribution under CCF

摘录 / EXCERPT: Let Ω_1 be the space of decreasing nonnegative sequences with sum 1, and let $\widehat{\mu}_1$ denote the Poisson-Dirichlet distribution on Ω_1 with parameter...

conjecture | open | MathDB 页面 | 原文线索

V2-072997 | MATHDB-377920

Mestre bound sharpness conjecture over number fields

摘录 / EXCERPT: Fix a number field F. For each positive integer N, let $r_F(N)$ be the maximum of $\operatorname{Rank}_{\mathbb{Z}}E(F)$ over elliptic curves E/F whose conductor...

conjecture | open | MathDB 页面 | 原文线索

V2-072998 | MATHDB-377923

Polynomial growth conjecture for automorphic coefficients

摘录 / EXCERPT: Let (π,π') be fixed automorphic representations, let π_{λ_i} denote the automorphic representation associated with the spectral parameter λ_i, and let b_i...

conjecture | open | MathDB 页面 | 原文线索

V2-072999 | MATHDB-377926

Canonical singularities conjecture for Kontsevich moduli spaces

摘录 / EXCERPT: Let $X\subset \mathbb{P}^n$ be a general hypersurface of degree d, and let $\overline{\mathcal{M}}_{0,0}(X,e)$ be the Kontsevich moduli space of degree-e stable maps to X. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-073000 | MATHDB-377933

The MAX-SAT and SAT equivalence conjecture

摘录 / EXCERPT: MAX-SAT and SAT equivalence conjecture. For any $c<0$,

conjecture | open | MathDB 页面 | 原文线索

V2-073001 | MATHDB-377939

The eta-invariant torsion conjecture for boundary links

摘录 / EXCERPT: Let $q>1$ and let (L,φ) be an F_m-link concordance class in $C_{2q+1}(F_m)$. For each positive integer k, let $R_k(F_m)$ denote the space of k-dimensional unitary rep...

conjecture | open | MathDB 页面 | 原文线索

V2-073002 | MATHDB-377940

Căldăraru's twisted derived equivalence conjecture for K3 surfaces

摘录 / EXCERPT: Căldăraru's conjecture. There exists an exact equivalence between the derived categories of twisted coherent sheaves

conjecture | open | MathDB 页面 | 原文线索

V2-073003 | MATHDB-377943

Lassalle's explicit formula conjecture for Jack polynomial shifted symmetric functions

摘录 / EXCERPT: Let $lambda$ be a partition and let $rgeq 0$. For the Pieri coefficients $c_i^{\alpha}(lambda)$, define ... Let $d_\rho(lambda)$ and z_ρ denote the quantities associated with a...

conjecture | open | MathDB 页面 | 原文线索

V2-073004 | MATHDB-377944

Edelman-Reiner conjecture on freeness of extended Catalan and Shi arrangements

摘录 / EXCERPT: Edelman-Reiner conjecture. The following arrangements are free:

conjecture | open | MathDB 页面 | 原文线索

V2-073005 | MATHDB-377964

Conjecture that the principal arithmetic subgroup has no non-trivial symmetries

摘录 / EXCERPT: Let $N(r)$ denote the order of the outer automorphism group of the principal arithmetic subgroup Λ_0^n, in the setting of the arithmetic quotients considered in the paper....

conjecture | open | MathDB 页面 | 原文线索

V2-073006 | MATHDB-377967

Conjecture on Fatou components of the symmetric critically finite maps

摘录 / EXCERPT: Fatou-set conjecture. For $n\geq 5$, F_g consists of the basins of attraction of the p_m-points.

conjecture | open | MathDB 页面 | 原文线索

V2-073007 | MATHDB-377982

Rigidity criterion for standard conic bundles over the projective plane

摘录 / EXCERPT: Standard conic bundle rigidity conjecture. If $X\to\mathbb{P}^2$ satisfies condition $(\ast)$, then it is birationally rigid. This is proposed as a criterion for birational rigidit...

conjecture | open | MathDB 页面 | 原文线索

V2-073008 | MATHDB-377983

High-degree discriminant rigidity conjecture for standard conic bundles

摘录 / EXCERPT: Let $\pi\colon X\to\mathbb{P}^2$ be a standard conic bundle, and let its discriminant curve have degree d. High-degree discriminant rigidity conjecture. If $d\geq 9$, then...

conjecture | open | MathDB 页面 | 原文线索

V2-073009 | MATHDB-377984

Grinenko's rigidity conjecture for nonsingular dP3 fibrations

摘录 / EXCERPT: Grinenko's dP3 rigidity conjecture. If X satisfies condition $(\ast)$, then X is birationally rigid. This conjecture extends the known rigidity theorem under the stronger K^2...

conjecture | open | MathDB 页面 | 原文线索

V2-073010 | MATHDB-377985

Semistable form of Grinenko's dP3 rigidity conjecture

摘录 / EXCERPT: Semistable dP3 rigidity conjecture. Then $X/\mathbb{P}^1$ is birationally rigid. This is a semistable variant of Grinenko's conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073011 | MATHDB-377989

The generic rigidity characterization by pseudo-triangulation straightening

摘录 / EXCERPT: Let G be a plane graph. A plane graph is generically rigid if its generic realizations are infinitesimally rigid, and it can be straightened as a pseudo-triangulation if its embe...

conjecture | open | MathDB 页面 | 原文线索

V2-073012 | MATHDB-377992

Value-distribution conjecture for cyclotomic polynomial coefficients

摘录 / EXCERPT: Let $a_n(k)$ denote the coefficient of x^k in the cyclotomic polynomial of order n, let $\delta(a_n(k)=v)$ denote the density of integers n for which this coefficient equals...

conjecture | open | MathDB 页面 | 原文线索

V2-073013 | MATHDB-377993

Nienhuis's conformal invariance conjecture for self-avoiding walks

摘录 / EXCERPT: Let D be a domain with boundary points A and B, and let $P_{D,A,B}^{SAW}$ denote a scaling limit of measures on self-avoiding curves from A to B in D, assigning weight...

conjecture | open | MathDB 页面 | 原文线索

V2-073014 | MATHDB-377994

The SLE$_{8/3}$ scaling-limit conjecture for self-avoiding walks

摘录 / EXCERPT: Let γ denote a rescaled self-avoiding curve, and let $SLE_{8/3}$ denote chordal Schramm-Loewner Evolution with parameter $8/3$. Self-avoiding-walk scaling-limit conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-073015 | MATHDB-377995

SLE reversibility conjecture

摘录 / EXCERPT: Let $\mathbb H$ be the upper half-plane and let γ be a chordal SLE_κ path from the origin to infinity, with fixed $\kappa\leq 8$. SLE reversibility conjecture. The la...

conjecture | open | MathDB 页面 | 原文线索

V2-073016 | MATHDB-377996

Dubédat's SLE boundary duality conjecture

摘录 / EXCERPT: Dubédat's boundary duality conjecture. The right boundary of $SLE^\sigma(\kappa,\kappa-4)$ is an $SLE(\kappa',(\kappa'-4)/2)$. The conjecture is a precise proposed realization of S...

conjecture | open | MathDB 页面 | 原文线索

V2-073017 | MATHDB-377998

Parameter-space wake decomposition conjecture for exponential maps

摘录 / EXCERPT: Let W be a hyperbolic component in the parameter space of exponential maps, and let $\mathcal{W}(W)$ denote its wake. A subwake of W is a wake associated with a child component...

conjecture | open | MathDB 页面 | 原文线索

V2-073018 | MATHDB-378006

The ring compatibility conjecture for the modified Hochschild-Kostant-Rosenberg isomorphism

摘录 / EXCERPT: Let I be the Hochschild-Kostant-Rosenberg isomorphism and let I' be its modification obtained by composing it with wedging by $\operatorname{td}_X^{1/2}$. For a homogeneous ele...

conjecture | open | MathDB 页面 | 原文线索

V2-073019 | MATHDB-378008

Sturmfels' degree bound conjecture for graded normal configurations

摘录 / EXCERPT: Let ${\mathcal A}\subset\mathbb Z^d$ be a finite graded, normal configuration, and let $I_{\mathcal A}$ be its toric ideal. Write $\dim({\mathcal A})=d$ for the Krull dimension of...

conjecture | open | MathDB 页面 | 原文线索

V2-073020 | MATHDB-378010

Bounded positive Dehn-twist length conjecture

摘录 / EXCERPT: Let Γ_g^r be the mapping class group of a genus-g surface with $r>0$ boundary components. A right-handed Dehn twist is a Dehn twist with the positive orientation conventio...

conjecture | open | MathDB 页面 | 原文线索

V2-073021 | MATHDB-378012

Stable equality conjecture for double quiver polynomials and double quiver functions

摘录 / EXCERPT: Let $\mathbf r$ be a rank array, let $\mathbf x$ and $\mathbf y$ be the corresponding alphabets, and write $\mathcal Q_{m+\mathbf r}(\mathbf x-\mathring{\mathbf y})$ for the double...

conjecture | open | MathDB 页面 | 原文线索

V2-073022 | MATHDB-378021

The complete bipartite graph ideal degree conjecture

摘录 / EXCERPT: Complete bipartite graph ideal degree conjecture. The graph ideal for $G=K_{m,n}$ is generated in degree at most

conjecture | open | MathDB 页面 | 原文线索

V2-073023 | MATHDB-378030

Escaping set rigidity for exponential maps with singular value in the Julia set

摘录 / EXCERPT: Let $E_{\kappa}(z)=e^z+\kappa$ be an exponential map, let $J(E_{\kappa})$ denote its Julia set, and let $I(E_{\kappa})$ denote its escaping set. An order-preserving conjugacy betwe...

conjecture | open | MathDB 页面 | 原文线索

V2-073024 | MATHDB-378033

Conjecture on local maxima for Sendov's conjecture

摘录 / EXCERPT: Let S_n be the set of monic complex polynomials of degree $n\ge 2$ whose zeros lie in the closed unit disk, and let ... A polynomial is locally maximal for Sendov's conjecture if i...

conjecture | open | MathDB 页面 | 原文线索

V2-073025 | MATHDB-378049

Strong piecewise polynomiality conjecture for double Hurwitz numbers

摘录 / EXCERPT: Fix g, m, and n, and let $P^{g}_{m,n}(alpha_1,\dots,alpha_m,beta_1,\dots,beta_n)=H^{g}_{\alpha,\beta}$ on the tuples of positive integers satisfying...

conjecture | solved | MathDB 页面 | 原文线索

V2-073026 | MATHDB-378050

ELSV-type formula conjecture for one-part double Hurwitz numbers

摘录 / EXCERPT: For $g\geq 0$, $n\geq 1$, and $(g,n)\neq(0,1),(0,2)$, let $H^g_{(d),\beta}$ be the one-part double Hurwitz number, let $r^g_{(d),\beta}$ be its associated number of simple branch p...

conjecture | solved | MathDB 页面 | 原文线索

V2-073027 | MATHDB-378057

Cartier's simplicity conjecture for the Maass Laplace spectrum

摘录 / EXCERPT: Consider the Laplace spectrum of Maass forms on $SL(2,\mathbb Z)\backslash\mathbb H$. Cartier's simplicity conjecture. The spectrum should be simple: there should not be two linear...

open_problem | open | MathDB 页面 | 原文线索

V2-073028 | MATHDB-378066

The dual stacked upper-bound conjecture for cd-indices of Bruhat intervals

摘录 / EXCERPT: Let $[u,v]$ be an interval in the Bruhat order on a Coxeter group, with $l(u)=k$ and $l(v)=d+k+1$. Let $\Phi_{[u,v]}$ denote its cd-index, and let a dual stacked polytope be a poly...

conjecture | open | MathDB 页面 | 原文线索

V2-073029 | MATHDB-378067

Non-negativity conjecture for cd-indices of Bruhat intervals

摘录 / EXCERPT: Let W be a Coxeter group, let $u,v\in W$ satisfy $u\leq v$ in the Bruhat order, and let $\Phi_{[u,v]}$ denote the cd-index of the interval $[u,v]$. Bruhat-interval non-negativity...

conjecture | open | MathDB 页面 | 原文线索

V2-073030 | MATHDB-378072

Identification of generalized mirror-transformation polynomials with virtual invariants

摘录 / EXCERPT: Let $\sigma_m=d_1+\cdots+d_{l(\sigma_m)}$ be a partition, let $V_{d-m}^{N,k,d}(n;\sigma_m)$ be the normalized virtual Gromov-Witten invariant defined in the paper, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073031 | MATHDB-378074

Differentiation is better than midpoint

摘录 / EXCERPT: Differentiation is better than midpoint conjecture. The gaps between consecutive zeros of f' lie between the infimum and supremum of the corresponding two-step midpoint-process g...

conjecture | open | MathDB 页面 | 原文线索

V2-073032 | MATHDB-378091

The shuffle conjecture for the character of diagonal coinvariants

摘录 / EXCERPT: Fix n and let $\delta_n=(n-1,n-2,\ldots,1,0)$. For a partition $\lambda\subseteq\delta_n$, let T range over semistandard tableaux of skew shape $(\lambda+(1^n))/\lambda$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-073033 | MATHDB-378092

The refined shuffle conjecture for $\nabla E_{n,k}$

摘录 / EXCERPT: The refined shuffle conjecture. For $1\leq k\leq n$,

conjecture | open | MathDB 页面 | 原文线索

V2-073034 | MATHDB-378093

The m-shuffle conjecture

摘录 / EXCERPT: Fix positive integers m,n and set $\delta_n=(n-1,n-2,\ldots,1,0)$. For a partition $\lambda\subseteq m\delta_n$, let T be a semistandard tableau of shape...

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V2-073035 | MATHDB-378094

The refined m-shuffle conjecture for $\nabla^m E_{n,k}$

摘录 / EXCERPT: Fix positive integers m,n and set $\delta_n=(n-1,n-2,\ldots,1,0)$. Let $E_{n,k}$ be defined by ... and let $D_n^{(m),\lambda}(z;q)$ denote the generalized tableau sum associated wi...

conjecture | open | MathDB 页面 | 原文线索

V2-073036 | MATHDB-378101

Zudilin's strengthened integrality conjecture for Apéry-type sequences

摘录 / EXCERPT: For $n\ge 0$, let ${\bf u}_n$ and ${\bf v}_n$ be the sequences defined by the Apéry-type hypergeometric series ${\bf S}_{n,4,2,1,1}(1)={\bf u}_n\zeta(4)-{\bf v}_n$. Define ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-073037 | MATHDB-378114

The fundamental lemma for restricted orbital integrals

摘录 / EXCERPT: The fundamental lemma. The orbital integrals satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073038 | MATHDB-378123

Derived McKay correspondence for finite special-linear quotient resolutions

摘录 / EXCERPT: Let V be a vector space, let $G\subset SL(V)$ be a finite subgroup, and let $X\to V/G$ be a crepant resolution. Write $D^b_G(V)$ for the bounded derived category of G-equivaria...

conjecture | open | MathDB 页面 | 原文线索

V2-073039 | MATHDB-378137

The MF-conjecture for F-divisors on the moduli space of stable pointed rational curves

摘录 / EXCERPT: Let ${\overline{\mathnormal{M}}}_{0,N}$ be the moduli space of stable N-pointed genus-zero curves. An F-divisor is a divisor that nonnegatively intersects every F-curve; let...

conjecture | open | MathDB 页面 | 原文线索

V2-073040 | MATHDB-378142

The conjectural braid-group presentation for the complex reflection group G_{29}

摘录 / EXCERPT: The G_{29} braid-group conjecture. The braid group associated with G_{29} admits the presentation

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V2-073041 | MATHDB-378143

The conjectural braid-group presentation for the complex reflection group G_{31}

摘录 / EXCERPT: The G_{31} braid-group conjecture. The braid group associated with G_{31} admits the presentation

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V2-073042 | MATHDB-378144

The conjectural braid-group presentation for the complex reflection group G_{33}

摘录 / EXCERPT: The G_{33} braid-group conjecture. The braid group associated with G_{33} admits the presentation

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V2-073043 | MATHDB-378145

The conjectural braid-group presentation for the complex reflection group G_{34}

摘录 / EXCERPT: The G_{34} braid-group conjecture. The braid group associated with G_{34} admits the presentation

conjecture | open | MathDB 页面 | 原文线索

V2-073044 | MATHDB-378147

The fundamental lemma for matching orbital integrals

摘录 / EXCERPT: Let G be a reductive group over the local field F, let H be an endoscopic group, and let $\gamma\in H(F)$ be strongly G-regular semisimple. Write $\Lambda_{G,H}(\gamma)$ fo...

conjecture | open | MathDB 页面 | 原文线索

V2-073045 | MATHDB-378150

Continuous blooming conjecture for convex polyhedral boundaries

摘录 / EXCERPT: Continuous blooming conjecture. Every convex polyhedral boundary has a continuous blooming.

conjecture | open | MathDB 页面 | 原文线索

V2-073046 | MATHDB-378166

Bounded velocity conjecture for minimizers in random time-dependent potentials

摘录 / EXCERPT: Let ... where the $U_j(x)$ are fixed non-random potentials of class C^1 satisfying condition, and $(a_j^\omega(t),1\leq j\leq N)$ is a realization of a stationary vector-valued ran...

conjecture | open | MathDB 页面 | 原文线索

V2-073047 | MATHDB-378177

The nef abundance conjecture for Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a Calabi-Yau 3-fold and let L be a nef divisor on X. Nef abundance conjecture. The divisor L is semiample. The surrounding discussion derives this assertion from...

conjecture | open | MathDB 页面 | 原文线索

V2-073048 | MATHDB-378182

Auxiliary solution-count conjectures for the discrete logarithm equation

摘录 / EXCERPT: Auxiliary solution-count conjectures.

conjecture | open | MathDB 页面 | 原文线索

V2-073049 | MATHDB-378193

Soprunova's mean-value formula conjecture for exponential sums

摘录 / EXCERPT: Mean-value formula conjecture. The mean value is equal to

conjecture | open | MathDB 页面 | 原文线索

V2-073050 | MATHDB-378201

The Chow quotient and log canonical model conjecture for Grassmannians

摘录 / EXCERPT: Chow quotient and log canonical model conjecture. The inclusion $X(r,n)\subset \overline{X}(r,n)$ is the log canonical model precisely in the cases

conjecture | open | MathDB 页面 | 原文线索

V2-073051 | MATHDB-378202

Uniform-child toric ideals are quadratically generated

摘录 / EXCERPT: Let T be a rooted tree and let I_T be the toric ideal associated to the homogeneous binary Markov model. Suppose either that T is binary, or more generally that every non-lea...

conjecture | open | MathDB 页面 | 原文线索

V2-073052 | MATHDB-378203

Cubic generation conjecture for path phylogenetic ideals

摘录 / EXCERPT: Let T be a path of length n, and let I_T be its toric ideal of phylogenetic invariants. The ideal I_T should be generated in degree 3, with $2n-4$ generators of degree...

conjecture | open | MathDB 页面 | 原文线索

V2-073053 | MATHDB-378204

Unbounded vertices conjecture for arbitrary-tree phylogenetic polytopes

摘录 / EXCERPT: Let T be a rooted tree, let ${\mathcal A}_T$ be the configuration of exponent vectors from the toric parametrization, and let P_T be the convex hull of the columns of...

conjecture | open | MathDB 页面 | 原文线索

V2-073054 | MATHDB-378208

Symmetry conjecture for generating functions of even and odd 3412-avoiding involutions

摘录 / EXCERPT: Symmetry conjecture. For all $m \ge 1$ and all $l_1, \ldots, l_m \ge 1$, the generating functions for the even involutions in $I(3412, [l_1, \ldots, l_m])$ and for the odd involuti...

conjecture | open | MathDB 页面 | 原文线索

V2-073055 | MATHDB-378212

Hitchin-Goldman unique energy-minimum conjecture

摘录 / EXCERPT: Hitchin-Goldman conjecture. The function e has a unique minimum.

conjecture | open | MathDB 页面 | 原文线索

V2-073056 | MATHDB-378213

The F-M triangle transformation conjecture for crystallographic root systems

摘录 / EXCERPT: F-M triangle transformation conjecture. The two generating functions are related by the invertible transformation

conjecture | open | MathDB 页面 | 原文线索

V2-073057 | MATHDB-378214

Rigidity conjecture for fusion rings and tensor functors

摘录 / EXCERPT: Rigidity conjecture. (i) Any fusion ring has at most finitely many realizations over ${\bf k}$, up to equivalence (possibly none). (ii) The number of tensor functors between two fi...

conjecture | open | MathDB 页面 | 原文线索

V2-073058 | MATHDB-378215

Effective linearization conjecture for configurations of linear subspaces

摘录 / EXCERPT: Effective linearization conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073059 | MATHDB-378216

The Lefschetz formula for locally symmetric spaces

摘录 / EXCERPT: Let G be the reductive group, Γ the arithmetic subgroup, A the split torus, and A^- its negative chamber. Let $\hat G_{\rm adm}$ denote the admissible dual, let...

conjecture | open | MathDB 页面 | 原文线索

V2-073060 | MATHDB-378217

Refined Lefschetz formula with continuous-spectrum multiplicities

摘录 / EXCERPT: Let $\mathfrak a_{\mathbb R}^{*,+}$ be the positive dual cone, let $\overline{\mathfrak a_{\mathbb R}^{*,+}}$ be its closure, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073061 | MATHDB-378219

Conjecture on Hurwitz actions and reduced decompositions of braid Coxeter elements

摘录 / EXCERPT: In either of the two settings, let B be the associated braid group, let W be the corresponding Coxeter or complex reflection group, let $p:B\twoheadrightarrow W$ be the natural...

conjecture | open | MathDB 页面 | 原文线索

V2-073062 | MATHDB-378220

Lattice conjecture for prefixes of a braid Coxeter element

摘录 / EXCERPT: Let B be the associated braid group, let R be its set of braid reflections, and let B_+ be the submonoid of B generated by R. Define a relation $\preccurlyeq$ on B_+ by...

conjecture | open | MathDB 页面 | 原文线索

V2-073063 | MATHDB-378225

Limit periodicity conjecture for Jacobi matrices from expanding polynomials

摘录 / EXCERPT: Let $T(z)$) be an expanding polynomial of degree d with a real Julia set E, where $E\subset[-\xi,\xi]$ and $T^{-1}:[-\xi,\xi]\to[-\xi,\xi]$. For $x\in[-\xi,\xi]$, define the Ja...

conjecture | open | MathDB 页面 | 原文线索

V2-073064 | MATHDB-378228

Xu's conjecture on the bound for the recursively defined polynomial T_d

摘录 / EXCERPT: For $d\ge 3$, let T_d be the recursively defined polynomial in the elementary symmetric functions on the standard simplex T^d, and write $\|T_d\|_{C(T^d)}$ for its uniform norm...

conjecture | open | MathDB 页面 | 原文线索

V2-073065 | MATHDB-378234

Generalized common-divisor conjecture for elliptic curves over function fields

摘录 / EXCERPT: Let K be a characteristic-zero function field, let E_1/K and E_2/K be elliptic curves, and let $P_1\in E_1(K)$ and $P_2\in E_2(K)$ be K-independent, meaning that no nonzero...

conjecture | open | MathDB 页面 | 原文线索

V2-073066 | MATHDB-378235

Positive common-divisor growth conjecture in characteristic p

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field of characteristic p, let $E/\mathbb{F}_q(T)$ be an elliptic curve, and let $P,Q\in E(\mathbb{F}_q(T))$ be nontorsion points. Positive common-...

conjecture | open | MathDB 页面 | 原文线索

V2-073067 | MATHDB-378237

Existence of large half-canonical loci with injective Gaussian maps

摘录 / EXCERPT: Let $\mathcal{S}_g$ denote the moduli space of smooth genus-g spin curves, and let $\mathcal{S}_g^r$ denote the locus of spin curves whose theta-characteristic has at least $r+1$...

conjecture | open | MathDB 页面 | 原文线索

V2-073068 | MATHDB-378243

Cambrian fan polytopality conjecture

摘录 / EXCERPT: Let W be a finite Coxeter group and let $\stackrel{\to}{G}$ be an orientation of its Coxeter diagram. The Cambrian fan ... is the fan whose maximal cones are the unions of reflecti...

conjecture | open | MathDB 页面 | 原文线索

V2-073069 | MATHDB-378248

The real-part bound for roots of Ehrhart polynomials

摘录 / EXCERPT: Real-part bound. All roots α of Ehrhart polynomials of lattice d-polytopes satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073070 | MATHDB-378262

The 3-strand bracket polynomial conjecture for virtual knot diagrams

摘录 / EXCERPT: Let K be a virtual knot diagram. The 3-strand bracket polynomial conjecture. The 3-strand bracket polynomial detects all non-trivial virtual knot diagrams. This would establish t...

conjecture | open | MathDB 页面 | 原文线索

V2-073071 | MATHDB-378263

Family A's 3-strand bracket-polynomial detection conjecture

摘录 / EXCERPT: Let $K_D(t)$ denote the virtual knot diagrams in Family A depicted in the paper's schematic, and let $\langle K\rangle_N$ denote the N-strand bracket polynomial. The Family A det...

conjecture | open | MathDB 页面 | 原文线索

V2-073072 | MATHDB-378264

The modified Family B detection conjecture

摘录 / EXCERPT: Let $K_m'(t)$ be the modified family of virtual knot diagrams illustrated in the paper. Let $\langle K\rangle_N$ denote the N-strand bracket polynomial. The modified Family B con...

conjecture | open | MathDB 页面 | 原文线索

V2-073073 | MATHDB-378268

The converse Liouville property conjecture for amenable measured groupoids

摘录 / EXCERPT: Converse Liouville property conjecture. Any amenable measured groupoid is Liouville.

conjecture | open | MathDB 页面 | 原文线索

V2-073074 | MATHDB-378270

The Dumont permutation enumeration conjecture for pattern 4132

摘录 / EXCERPT: Enumeration conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073075 | MATHDB-378277

Polynomiality conjecture for Burgers' equations with nilpotent Hessian

摘录 / EXCERPT: Polynomiality conjecture. The solution $U_t(z)$ must be a polynomial in both z and t. This conjecture links the nilpotent-Hessian condition to polynomial solutions of the invis...

conjecture | open | MathDB 页面 | 原文线索

V2-073076 | MATHDB-378278

The real sample covariance analogue of the largest-eigenvalue phase transition

摘录 / EXCERPT: Real sample covariance conjecture. For real sample covariance, Theorem 1 should still hold, with different limiting distributions but the same scaling. In particular, the critical...

conjecture | open | MathDB 页面 | 原文线索

V2-073077 | MATHDB-378284

Unbounded rank conjecture for the family $E_k:x^3+y^3=k$

摘录 / EXCERPT: For positive cubefree integers k, consider the elliptic curves ... Their rank is the rank of the finitely generated group $E_k(\mathbb{Q})$. Unbounded rank conjecture for E_k. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-073078 | MATHDB-378285

Andrews's modified partition identity for $A_{4,3,3}(n)$

摘录 / EXCERPT: Andrews's conjecture. For every positive integer n,

conjecture | open | MathDB 页面 | 原文线索

V2-073079 | MATHDB-378293

The uniqueness conjecture for the densest lattice covering in dimension 6

摘录 / EXCERPT: Let Q^{c1}_6 be the positive quadratic form defining the lattice covering studied in the paper, and let lattice coverings in dimension 6 be compared by their covering density....

conjecture | open | MathDB 页面 | 原文线索

V2-073080 | MATHDB-378300

Generic geometric syzygy conjecture for canonical curves

摘录 / EXCERPT: Generic geometric syzygy conjecture. For every p, the space of p-th linear syzygies of C is spanned by scrollar syzygies.

conjecture | open | MathDB 页面 | 原文线索

V2-073081 | MATHDB-378302

Conjecture on coefficients attached to closed orbits in automorphic representations

摘录 / EXCERPT: Let $\mathcal O$ be a fixed closed orbit, and let $a_{n,\lambda}$ be the coefficients in the expansion associated with $\mathcal O$ and the representation parameter λ. Clos...

conjecture | open | MathDB 页面 | 原文线索

V2-073082 | MATHDB-378305

Equality characterization for the maximal Stanley-Wilf limit

摘录 / EXCERPT: A pattern q of length k is a permutation pattern, and let $S_n(q)$ denote the number of permutations of length n avoiding q. Define its Stanley-Wilf limit by ... A pattern is...

conjecture | open | MathDB 页面 | 原文线索

V2-073083 | MATHDB-378311

Fomin-Fulton-Li-Poon conjecture for skew partitions

摘录 / EXCERPT: Fomin-Fulton-Li-Poon conjecture for skew partitions. For any skew partitions μ/α and ν/β, if

conjecture | open | MathDB 页面 | 原文线索

V2-073084 | MATHDB-378312

Conjecture on maximizers for the wave Strichartz inequality

摘录 / EXCERPT: Wave maximizer conjecture. The set of maximizers for which equality holds in the sharp wave Strichartz inequality coincides with the set of initial data of solutions in the orbit o...

conjecture | open | MathDB 页面 | 原文线索

V2-073085 | MATHDB-378318

Equality of the contact-process critical values beyond survival

摘录 / EXCERPT: Let T be an infinite tree, and let ρ be its root. For the contact process on T, define ... ... and ... where $\overline\mu$ is the upper invariant measure and δ_0 is th...

conjecture | open | MathDB 页面 | 原文线索

V2-073086 | MATHDB-378323

Pemantle-Steif conjecture on symmetry breaking with plus boundary conditions

摘录 / EXCERPT: Pemantle-Steif conjecture. For any Heisenberg model on any graph, SB is equivalent to SB+.

conjecture | open | MathDB 页面 | 原文线索

V2-073087 | MATHDB-378324

Pemantle-Steif phase-transition conjecture for Heisenberg models

摘录 / EXCERPT: Pemantle-Steif conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-073088 | MATHDB-378341

The Craven-Csordas-Smith conjecture on real zeros of a polynomial Wronskian

摘录 / EXCERPT: Craven-Csordas-Smith conjecture. The polynomial $W(P,P')$ has at most $2s$ real zeros.

conjecture | open | MathDB 页面 | 原文线索

V2-073089 | MATHDB-378342

The Hawaiian conjecture on chords of the garden of a logarithmic derivative

摘录 / EXCERPT: Hawaiian conjecture. Each chord of $\mathcal{G}_{P}$ contains at least one nonreal zero of P.

conjecture | open | MathDB 页面 | 原文线索

V2-073090 | MATHDB-378343

The conjecture for sums of powers of quadratic polynomials

摘录 / EXCERPT: Conjecture for sums of powers of quadratic polynomials. If $f\in\mathcal{QP}_n$, then f has at most $2n-1$ real critical points. Moreover, if α is a negative integer, each...

conjecture | open | MathDB 页面 | 原文线索

V2-073091 | MATHDB-378346

Quantizability conjecture for Lie bialgebroids

摘录 / EXCERPT: Quantizability conjecture. Any Lie bialgebroid is quantizable.

conjecture | open | MathDB 页面 | 原文线索

V2-073092 | MATHDB-378351

The four-case integrality conjecture for characteristic elements

摘录 / EXCERPT: The four-case integrality conjecture. In each of the following four cases, the assertions (a), (b), (c), and (d) are equivalent for every $xiin K1(R3):

conjecture | open | MathDB 页面 | 原文线索

V2-073093 | MATHDB-378352

The noncommutative Selmer torsion conjecture for elliptic curves

摘录 / EXCERPT: The noncommutative Selmer torsion conjecture. The Pontryagin dual of the Selmer group,

conjecture | open | MathDB 页面 | 原文线索

V2-073094 | MATHDB-378353

The non-CM p-adic L-function conjecture for elliptic curves

摘录 / EXCERPT: The non-CM p-adic L-function conjecture. There exists ${\cal L}_E\in K_1(\Lambda(G)_{S^*})$ such that ${\cal L}_E(\rho)\ne\infty$ for every Artin representation ρ of G, and

conjecture | open | MathDB 页面 | 原文线索

V2-073095 | MATHDB-378354

The main conjecture for the noncommutative Selmer group of an elliptic curve

摘录 / EXCERPT: The main conjecture. The p-adic L-function ${\cal L}_E\in K_1(\Lambda(G)_{S^*})$ is a characteristic element of $X(E/F_{\infty})$.

conjecture | open | MathDB 页面 | 原文线索

V2-073096 | MATHDB-378358

Infinitude conjecture for elliptic curves with odd modular degree

摘录 / EXCERPT: An elliptic curve over $\mathbf{Q}$ has odd modular degree when the degree of its optimal modular parametrization is odd. The odd modular-degree infinitude conjecture. There are in...

conjecture | open | MathDB 页面 | 原文线索

V2-073097 | MATHDB-378362

Quasi-adjunction eigenspace formula for finite covers

摘录 / EXCERPT: Quasi-adjunction eigenspace conjecture. Then $h^{n,0}_{\chi}(\bar U_G)=0$ unless the lift of χ belongs to a contributing global face of quasi-adjunction $\mathcal F$; in the l...

conjecture | open | MathDB 页面 | 原文线索

V2-073098 | MATHDB-378368

The section and universal-bundle obstruction conjecture for dual abelian fibrations

摘录 / EXCERPT: Let $X\to\mathbb P^n$ be an abelian fibration, let $P\to\mathbb P^n$ be its relative Picard, and let X_0 be the relative Picard of P. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073099 | MATHDB-378375

Parabolic stability necessity conjecture for scalar-flat Kähler iterated blow-ups

摘录 / EXCERPT: Parabolic stability necessity conjecture. If $\widehat M$ admits a scalar-flat Kähler metric, then E must be parabolically stable. This is a necessary-condition conjecture for th...

conjecture | open | MathDB 页面 | 原文线索

V2-073100 | MATHDB-378382

Mod p main conjecture for branches of Hida families

摘录 / EXCERPT: Let $\bar{\rho}:G_{\mathbf{Q}}\to\operatorname{GL}_{2}(k)$ be the residual representation, let $\mathcal H(\bar{\rho})$ be its Hida family, and let Y be a branch of this family....

conjecture | open | MathDB 页面 | 原文线索

V2-073101 | MATHDB-378419

Denef-Sargos instability conjecture for oscillating integrals

摘录 / EXCERPT: Let f be a real analytic germ at $0\in\mathbb{R}^{n}$ with Newton polyhedron, and let τ_{0} be the face determining the relevant pole and coefficient $\mu(\varphi)$ of the a...

conjecture | open | MathDB 页面 | 原文线索

V2-073102 | MATHDB-378420

Denef-Sargos nonvanishing conjecture for stable Newton faces

摘录 / EXCERPT: Let f be non-degenerate with respect to its Newton polyhedron, and let τ_{0} be a codimension-one simplex whose only lattice points in its intersection with the support of...

conjecture | open | MathDB 页面 | 原文线索

V2-073103 | MATHDB-378423

The incipient infinite lattice tree scaling conjecture

摘录 / EXCERPT: Incipient infinite lattice tree scaling conjecture. There exist constants A,V,v and $\delta\in(0,1)$ such that (rhoscal) holds for $\rho^{(\mathrm{lt})}$.

conjecture | open | MathDB 页面 | 原文线索

V2-073104 | MATHDB-378425

The uniform spanning tree scaling conjecture to infinite canonical super-Brownian motion

摘录 / EXCERPT: Uniform spanning tree scaling conjecture. There exist constants A,V,v and $\delta\in(0,1)$ such that (rhoscal) holds for $\rho^{(\mathrm{st})}$.

conjecture | open | MathDB 页面 | 原文线索

V2-073105 | MATHDB-378427

Smoothing conjecture and degree bound for Fano threefolds with cDV singularities

摘录 / EXCERPT: cDV smoothing conjecture. V has a smoothing by a small deformation. In particular,

conjecture | open | MathDB 页面 | 原文线索

V2-073106 | MATHDB-378428

Indecomposability conjecture for traces of universal \(PGL_3\)-monodromy

摘录 / EXCERPT: Let S be the surface under consideration, let ${\cal X}_{PGL_3,S}$ be its moduli space of framed PGL_3-local systems, and let n be a positive integer. A good positive Laurent...

conjecture | open | MathDB 页面 | 原文线索

V2-073107 | MATHDB-378431

The type-preserving duality conjecture for full antichains

摘录 / EXCERPT: Let Φ be a finite root system of rank n, with positive roots $\Phi_{\geq 0}$ ordered by root order. An antichain is a set of pairwise incomparable positive roots. For an ant...

conjecture | open | MathDB 页面 | 原文线索

V2-073108 | MATHDB-378432

The full-root and maximal-antichain bijection conjecture

摘录 / EXCERPT: Let Φ be a finite root system of rank n, with positive roots ordered by root order. A full root is a positive root whose support is the full Dynkin diagram, and an antichain...

conjecture | open | MathDB 页面 | 原文线索

V2-073109 | MATHDB-378448

Bernoulli-moment sign conjectures for isolated hypersurface singularities

摘录 / EXCERPT: Bernoulli-moment sign conjectures. For every $k\in{\mathbb N}$, both the weak form

conjecture | open | MathDB 页面 | 原文线索

V2-073110 | MATHDB-378450

Equidistribution conjecture for dense lattice orbits on compact homogeneous spaces

摘录 / EXCERPT: Let G) be a connected semisimple Lie group with finite center, let Γ be a lattice in G, and let Y be a compact homogeneous space of G. Let K be a maximal compact s...

conjecture | open | MathDB 页面 | 原文线索

V2-073111 | MATHDB-378455

The Curto-Muhly-Xia conjecture on commuting subnormal operators

摘录 / EXCERPT: Curto-Muhly-Xia conjecture. The pair $\mathbf{T}$ is subnormal if and only if it is hyponormal.

conjecture | open | MathDB 页面 | 原文线索

V2-073112 | MATHDB-378458

Generalized Graham pebbling product conjecture

摘录 / EXCERPT: Let G_1 and G_2 be graphs, let p_1,p_2 be positive integers, and let $f_{(p_1,p_2)}$ denote the corresponding generalized pebbling number, with $f_{p_i}(G_i)$ the one-factor...

conjecture | open | MathDB 页面 | 原文线索

V2-073113 | MATHDB-378459

Threshold-range conjecture for graph sequences

摘录 / EXCERPT: Let t_1 and t_2 be functions such that $t_1\in\Omega(n^{1/2})$, $t_2\in O(n)$, and $t_1\ll t_2$. Threshold-range conjecture. There is a graph sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-073114 | MATHDB-378460

Threshold existence conjecture for graph sequences

摘录 / EXCERPT: Let ${\cal G}=\{G_1,\ldots,G_n,\ldots\}$ be a graph sequence, and let $th({\cal G})$ denote its threshold when one exists. Threshold existence conjecture. Every graph sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-073115 | MATHDB-378477

Grigorchuk-Lubotzky-Pyber conjecture on congruence subgroup growth

摘录 / EXCERPT: Grigorchuk-Lubotzky-Pyber conjecture. The two asymptotic constants coincide and equal

conjecture | open | MathDB 页面 | 原文线索

V2-073116 | MATHDB-378479

Uniqueness and stability conjecture for spherical graph representations

摘录 / EXCERPT: A stable representation is an ${\mathcal{M}}$-representation that is a local minimum with respect to the ordering relation defining stability. Consider a graph on the sphere and fi...

conjecture | open | MathDB 页面 | 原文线索

V2-073117 | MATHDB-378488

Füredi-Lagarias-Morgan exponential bound for equilateral sets in strictly convex spaces

摘录 / EXCERPT: Let X be a strictly convex n-dimensional Minkowski space. Füredi-Lagarias-Morgan conjecture. There exists some constant $\varepsilon>0$ such that ... The conjecture would improve...

conjecture | open | MathDB 页面 | 原文线索

V2-073118 | MATHDB-378511

Automorphism-action compatibility conjecture for analytic K3 surfaces

摘录 / EXCERPT: Let X^{an}/K be an analytic K3 surface with the lattice Λ_X, period homomorphism ρ_X, valuation vector $v_X:=val_K\circ\rho_X$, and affine piecewise-linear action on...

conjecture | open | MathDB 页面 | 原文线索

V2-073119 | MATHDB-378526

J-function conjecture for nonabelian quotients

摘录 / EXCERPT: J-function conjecture. $J^{X//G}(t_0,\mathbf t,\hbar)$ is obtained from I by an explicit change of variables (the “mirror transformation”). If $X//G$ is Fano of index at least...

conjecture | open | MathDB 页面 | 原文线索

V2-073120 | MATHDB-378527

Concavity conjecture for the ground state of the one-dimensional symmetric stable process

摘录 / EXCERPT: Let $I=(-1,1)$, and let φ_1^α be the ground state eigenfunction for the symmetric stable process of index $0<\alpha<2$ killed upon leaving I. Concavity conjecture. T...

conjecture | open | MathDB 页面 | 原文线索

V2-073121 | MATHDB-378540

Universal positive lower bound for the tau constant of metrized graphs

摘录 / EXCERPT: Let Γ be a metrized graph, with total length $\ell(\Gamma)$ and tau constant $\tau(\Gamma)$. The ratio $\tau(\Gamma)/\ell(\Gamma)$ is invariant under scaling all edge length...

conjecture | open | MathDB 页面 | 原文线索

V2-073122 | MATHDB-378554

Trench's asymptotic absolute equidistribution conjecture for random geometric graph spectra

摘录 / EXCERPT: Let $P(\mathcal{X}_n)$ and $P(\mathcal{D}_n)$ be the matrices associated with the random geometric graph and its deterministic comparison model, and let $\lambda_i(\mathcal{X}_n)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073123 | MATHDB-378568

Macdonald-polynomial odd-power logarithm conjecture

摘录 / EXCERPT: Let $P_{\lambda}(x;q,t)$ denote the Macdonald polynomial corresponding to a partition λ, and let ... where the sum runs over all partitions λ. Write p_k for the pow...

conjecture | open | MathDB 页面 | 原文线索

V2-073124 | MATHDB-378571

The Donaldson-Thomas/Gromov-Witten correspondence for Calabi-Yau threefolds

摘录 / EXCERPT: Let X be a Calabi-Yau threefold, meaning that K_X is trivial. For curve classes $\beta\ne0$, let $Z^{\mathrm{DT}}(X)'$ be the reduced Donaldson-Thomas partition function, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-073125 | MATHDB-378573

The nodal-line conjecture for the second Cauchy-process eigenfunction

摘录 / EXCERPT: Let $D\subset\mathbb{R}^2$ be a convex domain symmetric with respect to both coordinate axes. Let λ_n and φ_n be the eigenvalues and eigenfunctions of the Cauchy pr...

conjecture | open | MathDB 页面 | 原文线索

V2-073126 | MATHDB-378574

The rectangle lower-bound conjecture for the Cauchy-process spectral gap

摘录 / EXCERPT: Let $D\subset\mathbb{R}^2$ be a bounded convex domain symmetric with respect to both coordinate axes. Let R be the smallest rectangle containing D whose sides are parallel to t...

conjecture | open | MathDB 页面 | 原文线索

V2-073127 | MATHDB-378579

Guardia's Mumford isomorphism conjecture for hyperelliptic period matrices

摘录 / EXCERPT: Let X be a hyperelliptic Riemann surface of genus $g\geq 2$, let τ be a period matrix associated to X, and let ... be a fundamental system, where $\mathcal{F}$ is the set of...

conjecture | open | MathDB 页面 | 原文线索

V2-073128 | MATHDB-378582

The pairwise-Leonard criterion for Leonard triples

摘录 / EXCERPT: Let V be a finite-dimensional vector space over $\mathbb K$ of positive dimension, and let $A,A^*,A^\varepsilon:V\to V$ be linear transformations. A Leonard pair is a pair of tra...

conjecture | open | MathDB 页面 | 原文线索

V2-073129 | MATHDB-378588

The hyperpolarity conjecture for polar actions on compact symmetric spaces

摘录 / EXCERPT: Let M be a compact symmetric space of rank bigger than one, and let a compact Lie group act polarly on M. Hyperpolarity conjecture. The action is hyperpolar. This extends the p...

conjecture | open | MathDB 页面 | 原文线索

V2-073130 | MATHDB-378596

Uniqueness of the finite-energy boundary point for prescribed minimal surfaces

摘录 / EXCERPT: Uniqueness conjecture. It may be possible that there is only one boundary point of the compactified moduli space having a finite-energy limit of map energies for the given set of m...

conjecture | open | MathDB 页面 | 原文线索

V2-073131 | MATHDB-378597

Dependence of the limiting straight-segment complex on the direction of approach

摘录 / EXCERPT: Direction-of-approach conjecture. The surface parts of the limit image should depend only on the boundary point of moduli space approached, whereas the complex of straight line seg...

conjecture | open | MathDB 页面 | 原文线索

V2-073132 | MATHDB-378608

Geometric characterization of the Q-rank by closed flats

摘录 / EXCERPT: Let $\widetilde{X}$ be a symmetric space with no Euclidean local factors, let $X=\Gamma\backslash\widetilde{X}$ be a finite-volume locally symmetric space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073133 | MATHDB-378619

Shokurov's weak epsilon-lc complement conjecture

摘录 / EXCERPT: Weak $b5$-lc complement conjecture. For every $0<b4$ and d, there exist a finite set $a9_{b4,d,91b}$ of positive integers and $0<b5$ such that every such pair is $(b5,n)$-complem...

conjecture | open | MathDB 页面 | 原文线索

V2-073134 | MATHDB-378620

Shokurov's strong epsilon-lc complement conjecture

摘录 / EXCERPT: Strong $b5$-lc complement conjecture. For every $0<b5$ and d, there exists a finite set $a9_{b5,d}$ of positive integers such that every such pair has an $(b5,n)$-complement over...

conjecture | open | MathDB 页面 | 原文线索

V2-073135 | MATHDB-378626

Common-root conjecture for truncated binomials

摘录 / EXCERPT: For integers d and $0<k<m<d$, let $P_{k;d}(x)$ and $P_{m;d}(x)$ denote the truncated binomial polynomials used in the paper. Common-root conjecture. The polynomials $P_{k;d}(x)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073136 | MATHDB-378636

Conjectures on when the generalized rotor-router sequence is Sturmian

摘录 / EXCERPT: Sturmian-region conjectures. (i) If $-4\leq r-s\leq3$, then, except for $(r,s)=(4,1)$, $w_{r,s}$ is Sturmian. (ii) If $r-s=4$, then $w_{r,s}$ is Sturmian if and only if r is even...

conjecture | open | MathDB 页面 | 原文线索

V2-073137 | MATHDB-378646

The homogeneous vanishing conjecture for Hessian-nilpotent polynomials

摘录 / EXCERPT: Homogeneous vanishing conjecture. Both of the following vanishings should hold: 1. $\Delta^mP^{m+1}=0$ for every $m>\alpha_{[n,d]}$. 2. For every $k\geq1$, $\Delta^mP^{m+k}=0$ for...

conjecture | open | MathDB 页面 | 原文线索

V2-073138 | MATHDB-378652

The maximal-distance conjecture for tangent cones of harmonic maps

摘录 / EXCERPT: Let Y be the object under consideration, let p be a point, and let TC_pY denote its tangent cone at p. Write 0_p for the zero element in this tangent cone, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073139 | MATHDB-378655

The group-pair analogue of the PD3 splitting theorem

摘录 / EXCERPT: Group-pair splitting conjecture. There is an analogue for group pairs of the preceding theorem on $it{PD}^3$ groups.

conjecture | open | MathDB 页面 | 原文线索

V2-073140 | MATHDB-378656

The relative hyperbolicity conjecture for atoroidal acylindrical PD3 pairs

摘录 / EXCERPT: Relative hyperbolicity conjecture. The group G is hyperbolic relative to $\mathbf S$.

conjecture | open | MathDB 页面 | 原文线索

V2-073141 | MATHDB-378659

Morgan's conjecture on Steiner point degrees

摘录 / EXCERPT: Morgan's conjecture. For all $d\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-073142 | MATHDB-378660

Non-autonomous basin biholomorphism conjecture

摘录 / EXCERPT: Let $f_1,f_2,\ldots$ be a sequence of automorphisms of $\mathbb{C}^k$. Assume that there exist constants $0<a<b<1$ such that, for every n and every z in the unit ball, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-073143 | MATHDB-378661

Joyce's extended Donaldson-Thomas invariant conjecture

摘录 / EXCERPT: Fix $\mathbb K=\mathbb C$. Let P be a Calabi-Yau 3-fold, set $\mathcal A=\operatorname{coh}(P)$, and let $\mathfrak F_{\mathcal A}$ and $K(\mathcal A)$ be as in the source. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073144 | MATHDB-378664

McNamara's conjecture on cylindric Schur-positivity

摘录 / EXCERPT: Let $\lambda/d/\mu$ be a cylindric skew shape that is a subposet of $\mathfrak{C}_{k,n-k}$. For variables $x=(x_1,x_2,\ldots)$, call $s_{\lambda/d/\mu}(x)$ cylindric Schur-positive...

conjecture | open | MathDB 页面 | 原文线索

V2-073145 | MATHDB-378668

Rationality conjecture for Jacobian points and twisted Eisenstein series

摘录 / EXCERPT: Rationality conjecture. The sheaf ${\mathcal F}_0(h)$ gives a rational point on the Jacobian if and only if, for each k, there exists a nonzero formal power series $R_k(q)$ such...

conjecture | open | MathDB 页面 | 原文线索

V2-073146 | MATHDB-378684

Maltsev's vanishing conductivity conjecture for strongly chaotic trajectories

摘录 / EXCERPT: In a periodic energy surface, let the level-set trajectories induced by a magnetic field include strongly chaotic trajectories, meaning trajectories whose lifts to the plane have n...

conjecture | open | MathDB 页面 | 原文线索

V2-073147 | MATHDB-378688

Laface's modified Harbourne-Hirschowitz conjecture for Hirzebruch surfaces

摘录 / EXCERPT: Let ${\mathcal{L}}=|ah+bF-\sum_{i=1}^{h}m_iP_i|$ be a system on the Hirzebruch surface ${\mathbb{F}}_e$, and let ${\mathcal{M}}$ be its residual system after repeatedly subtracting...

conjecture | open | MathDB 页面 | 原文线索

V2-073148 | MATHDB-378689

De Volder-Laface conjecture for linear systems on K3 surfaces

摘录 / EXCERPT: Let X be a K3 surface with $H^2=n\in2{\mathbb{Z}}$, and let ${\mathcal{L}}={\mathcal{L}}^n(d,m_1,\dots,m_h)$ be the system of curves in $|dH|$ through general points...

conjecture | open | MathDB 页面 | 原文线索

V2-073149 | MATHDB-378698

Multiplicity of stable matchings under the strongest link rule

摘录 / EXCERPT: Strongest-link multiplicity conjecture. Under the strongest link rule, every 3G matching market has at least two stable matchings.

conjecture | open | MathDB 页面 | 原文线索

V2-073150 | MATHDB-378699

Stable matching existence under the circular rule

摘录 / EXCERPT: Circular-rule existence conjecture. Under the circular rule, every 3G matching market has a stable matching.

conjecture | open | MathDB 页面 | 原文线索

V2-073151 | MATHDB-378701

The (C,A) inversion formula

摘录 / EXCERPT: Let $p,q,t\in\mathbb C$ satisfy $M<|t|^{n+1}<1$. For fixed $w\in\mathbb T^n$, let C_w be a contour inside the annulus ... for infinitesimally small positive ϵ, with the po...

conjecture | open | MathDB 页面 | 原文线索

V2-073152 | MATHDB-378706

Face-distance conjecture for torus decompositions of cubic irrationality

摘录 / EXCERPT: A torus decomposition arising from a periodic two-dimensional continued fraction of cubic irrationality has faces, each equipped with an integer distance to the origin. Face-distan...

conjecture | open | MathDB 页面 | 原文线索

V2-073153 | MATHDB-378707

Unit-face-distance conjecture for torus decompositions of cubic irrationality

摘录 / EXCERPT: A torus decomposition arising from a periodic two-dimensional continued fraction of cubic irrationality has faces with integer distances to the origin. Unit-face-distance conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-073154 | MATHDB-378708

Triangular-face conjecture for torus decompositions of cubic irrationality

摘录 / EXCERPT: A torus decomposition arising from a periodic two-dimensional continued fraction of cubic irrationality consists of faces with integer affine types. Triangular-face conjecture. For...

conjecture | open | MathDB 页面 | 原文线索

V2-073155 | MATHDB-378709

Cochran-Orr-Teichner filtration rank and nontriviality conjecture

摘录 / EXCERPT: Let $\mathcal F_n$ and $\mathcal F_{n.5}$ denote the successive terms of the Cochran-Orr-Teichner filtration of the smooth knot concordance group, indexed by $n\in\mathbb N_0$. Coc...

conjecture | open | MathDB 页面 | 原文线索

V2-073156 | MATHDB-378715

The cooperative two-sided secretary asymptotic rank conjecture

摘录 / EXCERPT: In the cooperative two-sided secretary game, let N be the number of rounds and let $R_N(1)$ denote the expected N-rank of a player entering the game, under an optimal common st...

conjecture | open | MathDB 页面 | 原文线索

V2-073157 | MATHDB-378716

The universal-rank-symmetry conjecture for the two-sided secretary problem

摘录 / EXCERPT: In the two-sided secretary game, suppose there are U men and U women, each player meets N partners over N rounds, and preferences satisfy universal rank symmetry: if a man...

conjecture | open | MathDB 页面 | 原文线索

V2-073158 | MATHDB-378723

Segre's conjecture on special plane linear systems

摘录 / EXCERPT: Segre's conjecture. If ${\mathcal{L}}_{2,d}(-\sum_{i=1}^hm_iP_i)$ is special, then its general member is non-reduced; equivalently, by Bertini's theorem, the system has a multiple...

conjecture | open | MathDB 页面 | 原文线索

V2-073159 | MATHDB-378731

Finite self-similarity conjecture for Julia sets of rational functions

摘录 / EXCERPT: Let f be a complex rational function, and let $J(f)$ denote its Julia set. A compact metrizable space is finitely self-similar if it arises from a finite self-similarity system,...

conjecture | open | MathDB 页面 | 原文线索

V2-073160 | MATHDB-378733

Convex-polytope f-vector unimodality conjecture

摘录 / EXCERPT: Let P be a convex d-polytope with f-vector ... where f_k is the number of k-dimensional faces of P. The unimodality conjecture. For each d-polytope P there is an inte...

conjecture | open | MathDB 页面 | 原文线索

V2-073161 | MATHDB-378734

Bjrner's quarter-monotonicity conjecture for convex-polytope f-vectors

摘录 / EXCERPT: Let P be a convex d-polytope with f-vector $(f_0,f_1,\ldots,f_{d-1})$, where f_k denotes the number of k-dimensional faces. Bjrner's quarter-monotonicity conjecture. The...

conjecture | open | MathDB 页面 | 原文线索

V2-073162 | MATHDB-378735

Lower-bound conjecture for convex-polytope f-vectors

摘录 / EXCERPT: Let P be a convex d-polytope with f-vector $(f_0,f_1,\ldots,f_{d-1})$, where f_k is the number of k-dimensional faces. The face-number lower-bound conjecture. For every...

conjecture | open | MathDB 页面 | 原文线索

V2-073163 | MATHDB-378737

Conjecture on cyclic resultants of generic monic polynomials

摘录 / EXCERPT: Let K be the coefficient field, and let $f(x)\in K[x]$ be a generic monic polynomial of degree d. Its cyclic resultants are the associated sequence of resultants. Generic cycli...

conjecture | open | MathDB 页面 | 原文线索

V2-073164 | MATHDB-378748

The four-dimensional numerical non-vanishing conjecture

摘录 / EXCERPT: Let (X,B) be a log minimal model with $\dim(X)=4$, and let C be a very general curve in X. Four-dimensional numerical non-vanishing conjecture. If $(K+B)\cdot C>0$ for very g...

conjecture | open | MathDB 页面 | 原文线索

V2-073165 | MATHDB-378749

The sine formula conjecture for admissible-cover Hodge integrals

摘录 / EXCERPT: Sine formula conjecture. For all $d\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-073166 | MATHDB-378755

The diagonal quasi-symmetric Hilbert matrix recurrence conjecture

摘录 / EXCERPT: Let ${\rm DQ}_n={\mathbb Q}[{\bf x},{\bf y}]/\mathcal J_n$ be the quotient by the ideal generated by diagonally quasi-symmetric polynomials without constant term, and let M_n be...

conjecture | open | MathDB 页面 | 原文线索

V2-073167 | MATHDB-378759

Existence of a compact solenoid by holomorphic curves in the complex projective plane

摘录 / EXCERPT: A solenoid by holomorphic curves is a compact solenoid in the complex projective plane whose leaves are holomorphic curves. Holomorphic-solenoid conjecture. There exists a compact...

conjecture | open | MathDB 页面 | 原文线索

V2-073168 | MATHDB-378761

The Hurwitz transitivity conjecture for reduced reflection factorizations

摘录 / EXCERPT: Let B be the braid group associated with a well-generated complex reflection group, let $\mathcal{R}$ be its set of braid reflections, and let δ be the braid-group analogu...

conjecture | open | MathDB 页面 | 原文线索

V2-073169 | MATHDB-378762

The Ehrenpreis conjecture for closed Riemann surfaces

摘录 / EXCERPT: Let S_1 and S_2 be closed Riemann surfaces of the same genus, and let $\tilde S_i$ denote covers of S_i for $i=1,2$. A cover is conformal when the covering map is conformal,...

conjecture | open | MathDB 页面 | 原文线索

V2-073170 | MATHDB-378769

K-theoretic lace diagram formula for type A quiver orbit closures

摘录 / EXCERPT: Let T be a maximal torus in G, let μ be an orbit of representations of an arbitrary quiver of type A, and let $\overline{\mu}$ denote its closure. A K-theoretic lace diag...

conjecture | open | MathDB 页面 | 原文线索

V2-073171 | MATHDB-378772

Tchebyshev transform conjecture for spherical shellable posets

摘录 / EXCERPT: Let P be a poset that is spherical and shellable, and let $T(P)$ denote its Tchebyshev transform. Spherical shellability conjecture. If P is spherical and shellable, then...

conjecture | open | MathDB 页面 | 原文线索

V2-073172 | MATHDB-378773

Tchebyshev transform conjecture for Gorenstein* posets

摘录 / EXCERPT: Let P be a Gorenstein poset, meaning an Eulerian poset that is Cohen-Macaulay, and let $T(P)$ denote its Tchebyshev transform. Gorenstein transform conjecture. If P is Gorenste...

conjecture | open | MathDB 页面 | 原文线索

V2-073173 | MATHDB-378774

Lifting conjecture for instanton sheaves on projective spaces

摘录 / EXCERPT: Let E be a rank $r\geq 2n$ locally-free instanton sheaf of trivial splitting type on $\mathbb{P}^{2n}$ and charge c. Lifting conjecture. There is a rank r locally-free instan...

conjecture | open | MathDB 页面 | 原文线索

V2-073174 | MATHDB-378778

Preperiodicity conjecture for components of the p-adic Fatou set

摘录 / EXCERPT: Conjecture de pré-périodicité des composantes. Toutes les composantes de l'ensemble de Fatou sont pré-périodiques.

conjecture | open | MathDB 页面 | 原文线索

V2-073175 | MATHDB-378780

Conjecture on log-concavity and unimodality of involution statistics

摘录 / EXCERPT: The conjecture. For all $n\geq 4$:

conjecture | open | MathDB 页面 | 原文线索

V2-073176 | MATHDB-378784

Tilde-operation conjecture for products of skew Schur functions

摘录 / EXCERPT: Tilde-operation conjecture for skew Schur functions. For all μ/α and ν/β, if $(\mu/\alpha,\nu/\beta)^\sim=(\lambda/\sigma,\rho/\tau)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073177 | MATHDB-378785

The integrality conjecture for Jack superpolynomial coefficients

摘录 / EXCERPT: Let $J_\Lambda^{(1/\beta)}$ be a Jack superpolynomial, and write its supermonomial expansion in the form ... Here $\Omega\leq\Lambda$ denotes the ordering used for the triangular exp...

conjecture | open | MathDB 页面 | 原文线索

V2-073178 | MATHDB-378786

Coxeter's A_d-hypothesis on minimal-density perfect forms

摘录 / EXCERPT: A perfect form is a positive definite quadratic form whose associated Delone subdivision is determined by its minimal vectors; Voronoi's first perfect form is the first form in Vor...

conjecture | open | MathDB 页面 | 原文线索

V2-073179 | MATHDB-378787

Local optimality of the E_8 lattice for the packing-covering problem

摘录 / EXCERPT: Let E_8 denote the root lattice in dimension 8. A lattice is a locally optimal solution to the lattice packing-covering problem if no sufficiently small deformation im...

conjecture | open | MathDB 页面 | 原文线索

V2-073180 | MATHDB-378794

The lower bound for C-loops with associators of prescribed order

摘录 / EXCERPT: Lower-bound conjecture. The order of C satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073181 | MATHDB-378807

Local mirror symmetry for general Calabi-Yau threefolds

摘录 / EXCERPT: Let $Y=(Y,J,h)$ be a general Calabi-Yau threefold and let $\chi\in\mathcal{K}_{\mathbb{C}}(Y)$ satisfy $\operatorname{Im}\chi=-\frac12\operatorname{Im}h$. Call (Y,χ) a polariz...

conjecture | open | MathDB 页面 | 原文线索

V2-073182 | MATHDB-378808

Morrison's reverse geometric transition conjecture

摘录 / EXCERPT: Let $T(Y,\overline{Y},\widetilde{Y})$ be a geometric transition, and let Y° and $\widetilde{Y}^{\circ}$ be mirror partners of Y and $\widetilde{Y}$, respectively. Morris...

conjecture | open | MathDB 页面 | 原文线索

V2-073183 | MATHDB-378809

Deformation to a conifold transition

摘录 / EXCERPT: Let $T(Y,\overline{Y},\widetilde{Y})$ be a geometric transition satisfying a good condition, for example that its birational morphism ϕ contracts k exceptional divisors to a...

conjecture | open | MathDB 页面 | 原文线索

V2-073184 | MATHDB-378811

Type-C q-multiplicity and one-dimensional-sum identities

摘录 / EXCERPT: Type-C identities.

conjecture | open | MathDB 页面 | 原文线索

V2-073185 | MATHDB-378814

Generalization of the centralizer theorem for parabolic subgroups in Coxeter groups

摘录 / EXCERPT: Generalized centralizer theorem. For arbitrary $I\subseteq S$, if $\gamma\in\Pi^I$ and $s_\gamma\in W^{\perp I}{}_{\mathrm{fin}}$, then $w\cdot\gamma=\gamma$ for every $w\in Y_I$....

conjecture | open | MathDB 页面 | 原文线索

V2-073186 | MATHDB-378819

Scherk uniqueness conjecture for minimal surfaces

摘录 / EXCERPT: Let M be a connected properly immersed minimal surface in ${\mathbb R}^3$, and let $A(M)$ denote its area growth constant. Let S_θ, for $\theta \in (0,\frac{\pi}{2}]$, den...

conjecture | open | MathDB 页面 | 原文线索

V2-073187 | MATHDB-378822

Malle's polynomial bound conjecture for number fields

摘录 / EXCERPT: For a number field k, write $\mathcal{D}_k$ for its discriminant. Malle's polynomial bound conjecture. There exists an absolute constant B such that, for sufficiently large x...

conjecture | open | MathDB 页面 | 原文线索

V2-073188 | MATHDB-378823

Polynomial growth conjecture for maximal arithmetic subgroups

摘录 / EXCERPT: Let H be a connected semi-simple Lie group without almost simple factors of type A_1 and without compact factors. For sufficiently large x, consider maximal irreduci...

conjecture | open | MathDB 页面 | 原文线索

V2-073189 | MATHDB-378844

Finiteness conjecture for filtered lambda-ring structures with prescribed Adams linear coefficients

摘录 / EXCERPT: Let n be an integer with $n\geq 3$, and let ... be a family of nonzero integers. A filtered λ-ring structure on $\mathbf{Z}[x]/(x^n)$ has prescribed Adams linear coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-073190 | MATHDB-378845

Extension invariance conjecture for filtered lambda-ring structures

摘录 / EXCERPT: Let n be an integer with $n\geq 3$, and let R and S be isomorphic filtered λ-ring structures on ... Let $\sigma:R\xrightarrow{\cong}S$ be an isomorphism. A filtered...

conjecture | open | MathDB 页面 | 原文线索

V2-073191 | MATHDB-378857

The ratio inequality for expected exit times under symmetrization

摘录 / EXCERPT: Let U and R be domains in the setting of Theorem, let z be the specified starting point, and write τ_U, τ_R, τ_{U^+}, and τ_{R^+} for the corresponding fi...

conjecture | open | MathDB 页面 | 原文线索

V2-073192 | MATHDB-378869

The terminating Weyl-antisymmetry conjecture for the first eigenfunction

摘录 / EXCERPT: Weyl-antisymmetry conjecture. This polynomial satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073193 | MATHDB-378871

The quasi-eigenfunction transformation conjecture

摘录 / EXCERPT: Transformation conjecture. The function $F(\alpha)$ also satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073194 | MATHDB-378872

The product-formula conjecture for the quasi-eigenfunction at $\alpha=-t^{1/2}$ and $\alpha=t$

摘录 / EXCERPT: Product-formula conjecture. The following identities hold:

conjecture | open | MathDB 页面 | 原文线索

V2-073195 | MATHDB-378873

The Pfaffian product conjecture for the quasi-eigenfunction at $\alpha=q^{1/2}t^{1/2}$

摘录 / EXCERPT: Pfaffian product conjecture. At $\alpha=q^{1/2}t^{1/2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-073196 | MATHDB-378874

The rational-factor conjecture for the quasi-eigenfunction at $\alpha=-q^{\ell}t^{1/2}$

摘录 / EXCERPT: Rational-factor conjecture. At $\alpha=-q^\ell t^{1/2}$,

conjecture | open | MathDB 页面 | 原文线索

V2-073197 | MATHDB-378878

Thomas's trimness conjecture for Cambrian lattices

摘录 / EXCERPT: Let $C(\bar G)$ be a Cambrian lattice associated with an oriented Dynkin diagram $\bar G$ of a finite reflection group. A lattice is trim if it has a left modular maximal chain and...

conjecture | open | MathDB 页面 | 原文线索

V2-073198 | MATHDB-378879

Thomas's pre-Cambrian-Cambrian coincidence conjecture

摘录 / EXCERPT: Let W be a finite reflection group containing -1, let G be its Dynkin diagram, and let $\bar G$ be an orientation. Form the Coxeter element c from the oriented diagram, ord...

conjecture | open | MathDB 页面 | 原文线索

V2-073199 | MATHDB-378881

The conjectured formula for $d54d_w$ for special permutations

摘录 / EXCERPT: The conjectured formula for $\mathfrak{D}_w$. For special w,

conjecture | open | MathDB 页面 | 原文线索

V2-073200 | MATHDB-378883

Zuber's polynomiality conjecture for FPL configurations with nested arches

摘录 / EXCERPT: Zuber's conjecture. The number $A_X(m)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-073201 | MATHDB-378884

Zuber's two-matching polynomiality conjecture for FPL configurations

摘录 / EXCERPT: Zuber's conjecture. The number $A_{X,Y}(m)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-073202 | MATHDB-378885

Zhou's two-partition Hodge integral formula

摘录 / EXCERPT: Let $G^\bullet_{\mu^+,\mu^-}(\lambda;\tau)$ be the disconnected generating function of two-partition Hodge integrals for partitions μ^+ and μ^-. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073203 | MATHDB-378887

Fomin-Fulton-Li-Poon's Schur-positivity conjecture for sorted partitions

摘录 / EXCERPT: Let λ and μ be partitions, and let $\lambda\cup\mu=(\nu_1,\nu_2,\nu_3,\ldots)$ be the partition obtained by rearranging all parts of λ and μ in weakly decre...

conjecture | open | MathDB 页面 | 原文线索

V2-073204 | MATHDB-378893

Inductive generation of tautological equations

摘录 / EXCERPT: Inductive generation conjecture. Conjecture 2 will produce all tautological equations inductively.

conjecture | open | MathDB 页面 | 原文线索

V2-073205 | MATHDB-378899

Conjecture on the transcendence of Euler's constant

摘录 / EXCERPT: Transcendence conjecture for Euler's constant. The number γ is transcendental. The source notes that even irrationality of γ is not known, so this stronger conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-073206 | MATHDB-378900

Parabolic compatibility conjecture for the integrable tame harmonic bundle

摘录 / EXCERPT: Let X be a compact Riemann surface, let $P\subset X$ be a finite set, and let (V,∇_V) be a semisimple holomorphic flat bundle on $X\setminus P$. Let (H,D_V,h) be the tam...

conjecture | open | MathDB 页面 | 原文线索

V2-073207 | MATHDB-378911

The boundary-image conjecture for Poincaré-Einstein metrics

摘录 / EXCERPT: Boundary-image conjecture. For any component ${\mathcal E}_{0}$ of ${\mathcal E}$, $\bar{\Pi}(\partial {\mathcal E}_{0})$ has empty interior in ${\mathcal C}$.

conjecture | open | MathDB 页面 | 原文线索

V2-073208 | MATHDB-378934

Farkas's codimension conjecture for syzygy strata of curves

摘录 / EXCERPT: Let $\mathcal{M}_g^0$ be the open moduli space under consideration, and let $\mathcal{Z}_{g,i}$ be the locus of curves C for which there exists a pencil $A\in W^1_k(C)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-073209 | MATHDB-378936

Plurality Is Asymptotically Stablest conjecture

摘录 / EXCERPT: Let $0\leq\rho\leq1$ and let $q\geq2$ be an integer. For $f:[q]^n\to[0,1]$, let $\mathbb{S}_\rho(f)$ denote its noise stability and let $\mathrm{Inf}_i(f)$ denote the influence of...

conjecture | open | MathDB 页面 | 原文线索

V2-073210 | MATHDB-378947

The three-variable basic hypergeometric conjecture for Macdonald eigenfunctions

摘录 / EXCERPT: Three-variable basic hypergeometric conjecture. The series

conjecture | open | MathDB 页面 | 原文线索

V2-073211 | MATHDB-378964

Nonnegative symmetric decomposition conjecture for fixed-point-free involution Eulerian polynomials

摘录 / EXCERPT: Nonnegative decomposition conjecture. For $n\geq 9$ and $k\geq 1$, the coefficients $b_{2n,k}$ are nonnegative integers.

conjecture | open | MathDB 页面 | 原文线索

V2-073212 | MATHDB-378970

Rank-one optimal solution conjecture for dissipative bilinear control semidefinite programs

摘录 / EXCERPT: Rank-one optimal solution conjecture. If

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073213 | MATHDB-378972

Degree-singularity-root correspondence for nilpotent Toda lattices

摘录 / EXCERPT: Degree-singularity-root conjecture. The degree $\eta(w_*)$ of $p(q)$ is the multiplicity d at p_o, where d is given by the minimal degree of the product of the τ-functio...

conjecture | open | MathDB 页面 | 原文线索

V2-073214 | MATHDB-378976

The Lindelöf-type conjecture for triple periods of Maass forms

摘录 / EXCERPT: Let ϕ_i be a Hecke-Maass cusp form with spectral parameter $\lm_i$, and let d_i denote its associated triple period coefficient. For every $\eps>0$, the notation...

conjecture | open | MathDB 页面 | 原文线索

V2-073215 | MATHDB-378985

Geometric section conjecture for the boundary divisor D_2

摘录 / EXCERPT: Let $A^*({\overline{\mathcal{M}}}_{0,2}({\mathbb P}^1,2))$ denote the Chow ring of the moduli space of stable maps, let L_1 and L_2 be the cotangent line bundles associated wit...

conjecture | open | MathDB 页面 | 原文线索

V2-073216 | MATHDB-378986

Analyticity of the giant-component fraction away from the critical point

摘录 / EXCERPT: Let $\rho(c)$ denote the giant-component fraction as a function of the scaling parameter c, and let c_0 be the critical value. Theorem5(i) establishes analyticity under conditi...

conjecture | open | MathDB 页面 | 原文线索

V2-073217 | MATHDB-378987

Critical slope and leading-eigenfunction conjecture for inhomogeneous random graphs

摘录 / EXCERPT: Let ... be its associated integral operator with norm ... has an eigenfunction ... satisfying $<$; conversely, $=0$ if >0 but no such eigenfunction exists or an eigenfunction exists wi...

conjecture | open | MathDB 页面 | 原文线索

V2-073218 | MATHDB-378991

A recurrence-product conjecture for parametrized Euler sums

摘录 / EXCERPT: Recurrence-product conjecture. The sequence satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073219 | MATHDB-378997

The coherent component conjecture for McKay quiver moduli

摘录 / EXCERPT: Let $G\subset \operatorname{GL}(n,\Bbbk)$ be a finite subgroup. For all θ in the rational character space, let $\mathcal{M}_\theta=Z{\operatorname{/\tslash}}_\theta H$ be th...

conjecture | open | MathDB 页面 | 原文线索

V2-073220 | MATHDB-379000

Determinant growth conjecture for random symmetric matrices

摘录 / EXCERPT: Let Q_n be the random symmetric matrix whose entries take values 1 and -1, as in the stated model. Determinant growth conjecture. Almost surely, ... This is the symmetric-matri...

conjecture | open | MathDB 页面 | 原文线索

V2-073221 | MATHDB-379001

Lie's second theorem for Weinstein groupoids

摘录 / EXCERPT: Lie's second theorem. For any morphism of Lie algebroids $\phi:A\to B$, there is a unique morphism Φ from the Weinstein groupoid $\mathcal{H}(A)$ to any Weinstein groupoid...

conjecture | open | MathDB 页面 | 原文线索

V2-073222 | MATHDB-379017

Stanley's coefficient conjecture for Kerov character polynomials

摘录 / EXCERPT: For $n\geq 0$, let $\Sigma_{k,2n}$ denote the component of weight $k+1-2n$ in the Kerov character polynomial Σ_k, and write $[R_2^i]\Sigma_{k,2n}$ for the coefficient of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073223 | MATHDB-379018

C-positivity conjecture for the graded Kerov character polynomials

摘录 / EXCERPT: Let $R(x)=1+\sum_{i\geq 2}R_ix^i$, and define C_m by $C_0=1$, $C_1=0$, and ... For $n\geq 0$, let $\Sigma_{k,2n}$ be the sum of the terms of weight $k+1-2n$ in Σ_k. A polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-073224 | MATHDB-379021

Asymptotic minimaxity of the $2k\log(n/k)$ penalized model-selection procedure

摘录 / EXCERPT: Let model selection mean adaptive selection of nonzero means, and let the underlying estimand be bc, belonging to one of the parameter spaces and considered under one of the loss...

conjecture | open | MathDB 页面 | 原文线索

V2-073225 | MATHDB-379022

The Schwartz genus conjecture for polynomial equations

摘录 / EXCERPT: Let q_n denote the polynomial equation problem associated with the unordered configuration space of n points, let $g(q_n)$ be its Schwartz genus, and let L_n be the relevant...

conjecture | open | MathDB 页面 | 原文线索

V2-073226 | MATHDB-379027

The Stanley-Wilf limit and longest-pattern-avoiding-subsequence constant conjecture

摘录 / EXCERPT: The Stanley-Wilf and subsequence-constant conjecture. Both limit superiors are actual limits, and

conjecture | open | MathDB 页面 | 原文线索

V2-073227 | MATHDB-379031

The eight-point maximum conjecture for the three-dimensional angular sum

摘录 / EXCERPT: Let $\mathcal{Q}_N$ denote the configuration space of N points in d-dimensional space, and let F_N be the angular sum defined in the paper. The eight-point maximum conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-073228 | MATHDB-379037

Babson-Kozlov vanishing conjecture for odd-cycle Hom-complexes

摘录 / EXCERPT: Let C_{2r+1} be an odd cycle, let K_n be the complete graph on n vertices, and let $\varpi_1^{n-2}(\text{\tt Hom}(C_{2r+1},K_n))$ denote the indicated Stiefel-Whitney charact...

conjecture | open | MathDB 页面 | 原文线索

V2-073229 | MATHDB-379049

Sharpness conjecture for the lower semigroup exponent of complex elliptic operators

摘录 / EXCERPT: Let $n\ge 3$, let $\mathcal{E}$ be the class of all complex elliptic operators under study, and for $L\in\mathcal{E}$ let $p_-(L)$ be the lower endpoint of the maximal interval of...

conjecture | open | MathDB 页面 | 原文线索

V2-073230 | MATHDB-379050

Kontsevich's $K(\pi,1)$ conjecture for strata of quadratic differentials

摘录 / EXCERPT: Let $\mathcal H$ and $\mathcal Q_g$ denote the moduli spaces of Abelian and quadratic differentials, respectively, and consider a connected component of a stratum of ... A group...

conjecture | open | MathDB 页面 | 原文线索

V2-073231 | MATHDB-379073

Nielsen-type conjecture for the mapping class group of a hyperbolic surface

摘录 / EXCERPT: Nielsen-type conjecture. The natural monomorphism

conjecture | open | MathDB 页面 | 原文线索

V2-073232 | MATHDB-379074

Critical clustering for linearly interacting diffusions

摘录 / EXCERPT: Let $D\subset\mathbb R^d$ be open, bounded, and convex, let ${\cal W}$ be a renormalization class on $\overline D$, and suppose that the asymptotic fixed-point equation in the zero...

conjecture | open | MathDB 页面 | 原文线索

V2-073233 | MATHDB-379081

Kajiura-Saito-Takahashi conjecture for dual regular weight systems

摘录 / EXCERPT: Let W be a regular weight system and f be a quasi-homogeneous polynomial attached to W. Assume W has a dual regular weight system $W^*=(a^*,b^*,c^*;h)$ in the sense of K. S...

conjecture | open | MathDB 页面 | 原文线索

V2-073234 | MATHDB-379082

ADE singularity derived-category equivalence conjecture

摘录 / EXCERPT: Let W be a regular weight system corresponding to an ADE singularity, and let f be a quasi-homogeneous polynomial attached to W. Write $D^b_{\mathbb Z}({\mathcal A}_f)$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-073235 | MATHDB-379088

Duke's realization conjecture for triangulated tori

摘录 / EXCERPT: Duke's torus-realization conjecture. Every triangulated torus can be realized as a polyhedron in $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-073236 | MATHDB-379100

Conformal Ehrenpreis conjecture

摘录 / EXCERPT: Let Σ be a fixed compact surface of genus greater than 1, let $\mathcal{T}(\Sigma)$ be its Teichmüller space, and let $d_{\mathcal{T}(\Sigma)}$ be the Teichmüller metric....

conjecture | open | MathDB 页面 | 原文线索

V2-073237 | MATHDB-379118

Orlik's integral diagonalization conjecture for weighted homogeneous monodromy

摘录 / EXCERPT: Assume $Y={\mathbb C}^{n+1}$ and that f is weighted homogeneous. Let C be the matrix of the monodromy operator h_* on $H_n(X_\delta,{\mathbb C})$ with respect to a basis, let...

conjecture | open | MathDB 页面 | 原文线索

V2-073238 | MATHDB-379127

Non-subloop conjecture for Moufang elements in PACC-loops

摘录 / EXCERPT: Non-subloop conjecture. There exists a PACC-loop Q in which $M(Q)$ is not a subloop.

conjecture | open | MathDB 页面 | 原文线索

V2-073239 | MATHDB-379129

The maximal obstruction conjecture for the monic integer transfinite diameter

摘录 / EXCERPT: Let I be an interval of length less than 4. A maximal obstruction for I is an obstruction whose value is maximal among the obstructions considered for I; denote its value b...

conjecture | open | MathDB 页面 | 原文线索

V2-073240 | MATHDB-379130

The existence conjecture for maximal obstructions on short intervals

摘录 / EXCERPT: Let I be an interval. Maximal obstruction existence conjecture. Every interval of length less than 4 has a maximal obstruction. The paper proves existence when the interval has...

conjecture | open | MathDB 页面 | 原文线索

V2-073241 | MATHDB-379131

The critical polynomial conjecture for intervals of length less than four

摘录 / EXCERPT: Let I be an interval. An irreducible polynomial $Q(x)=a_dx^d+\cdots+a_0\in\mathbb{Z}[x]$ with $a_d>0$, all roots in I, and $a_d^{-1/d}>t_{\mathbb{Z}}(I)$ is a nonmonic critical...

conjecture | open | MathDB 页面 | 原文线索

V2-073242 | MATHDB-379132

The unique-solvability conjecture for symmetric word equations

摘录 / EXCERPT: Let $S(X,B_i)=P$ be a symmetric word equation, where X is an $n\times n$ positive semidefinite matrix, the B_i are fixed $n\times n$ positive definite matrices, and P is an...

conjecture | open | MathDB 页面 | 原文线索

V2-073243 | MATHDB-379133

The trace-positivity conjecture for words in positive definite matrices

摘录 / EXCERPT: Let W be a word in two letters A and B, and let A and B be real positive definite matrices. Trace-positivity conjecture. The word W has positive trace for every pair...

conjecture | open | MathDB 页面 | 原文线索

V2-073244 | MATHDB-379134

The two-by-two unique-solvability conjecture for symmetric word equations

摘录 / EXCERPT: Let $S(X,B)=P$ be a symmetric word equation in positive definite matrices, with X, B, and P of size 2×2. Two-by-two unique-solvability conjecture. Symmetric word equat...

conjecture | open | MathDB 页面 | 原文线索

V2-073245 | MATHDB-379142

The secant-flag monotone conjecture

摘录 / EXCERPT: Let γ be the rational normal curve, and for a set $S_i=\{s_{i1},\dotsc,s_{in}\}$ of n distinct points of $\mathbb{R}\mathbb{P}^1$, let $F_\bullet(S_i)$ be the flag whose...

conjecture | open | MathDB 页面 | 原文线索

V2-073246 | MATHDB-379151

Local rigidity conjecture for irreducible affine actions without rank-one factors

摘录 / EXCERPT: Let $G=G_1\times\cdots\times G_k$ be a semisimple Lie group whose factors are all non-compact, let $\Gamma<G$ be an irreducible lattice, and let ρ be an affine action of...

conjecture | open | MathDB 页面 | 原文线索

V2-073247 | MATHDB-379157

The Möbius-Chebyshev conjecture for subwords over the poset Λ

摘录 / EXCERPT: Möbius-Chebyshev conjecture. For all integers $i\leq j$, $\mu(a^i,c^j)$ is the coefficient of x^{j-i} in $T_{i+j}(x)$.

conjecture | open | MathDB 页面 | 原文线索

V2-073248 | MATHDB-379178

Non-closure conjecture for invertible elements in Bol magmas

摘录 / EXCERPT: Let L be a Bol magma with a neutral element, and let $\mathcal{J}(L)$ denote the set of elements of L possessing a two-sided inverse. Non-closure conjecture. There exists a Bol...

conjecture | open | MathDB 页面 | 原文线索

V2-073249 | MATHDB-379190

The Martin boundary conjecture for the graph of zigzag diagrams

摘录 / EXCERPT: Let ${\cal Z}$ be the graded graph of zigzag diagrams, with level-${n}$ vertices ${\cal Z}_n$. For a sequence $(\lambda_n\in{\cal Z}_n)$, let $\iota_n(\lambda_n)$ be its image in t...

conjecture | open | MathDB 页面 | 原文线索

V2-073250 | MATHDB-379193

Parker's factorization and palindromicity conjecture for cyclotomic coordinator polynomials

摘录 / EXCERPT: Let m be a positive integer. Write $\sqrt m$ for the squarefree part of m, let $\zeta_m=e^{2\pi i/m}$, and let $h_m(x)$ be the coordinator polynomial of the lattice...

conjecture | open | MathDB 页面 | 原文线索

V2-073251 | MATHDB-379195

Parker's explicit coordinator polynomial conjecture for the cyclotomic lattice of order 15

摘录 / EXCERPT: Let $\zeta_{15}=e^{2\pi i/15}$, and let $h_{15}(x)$ be the coordinator polynomial of the lattice $\mathbb Z[\zeta_{15}]$ with respect to the set of all 15th roots of unity. Parke...

conjecture | open | MathDB 页面 | 原文线索

V2-073252 | MATHDB-379196

The conjecture that the coordinator polynomial of order 105 is not palindromic

摘录 / EXCERPT: Let $h_{105}(x)$ be the coordinator polynomial of the lattice $\mathbb Z[\zeta_{105}]$ with respect to the set of all 105th roots of unity. A polynomial is palindromic if its coe...

conjecture | open | MathDB 页面 | 原文线索

V2-073253 | MATHDB-379212

Arouche's subgroup-index consequence for braid commutator subgroups

摘录 / EXCERPT: Let B_n be the braid group on n strands and let $K_n=[B_n,B_n]$ be its commutator subgroup. A subgroup has index r if its coset action gives a transitive representation...

conjecture | open | MathDB 页面 | 原文线索

V2-073254 | MATHDB-379214

The Lex Plus Powers conjecture for Hilbert functions

摘录 / EXCERPT: The Lex Plus Powers conjecture for Hilbert functions. Then

conjecture | open | MathDB 页面 | 原文线索

V2-073255 | MATHDB-379215

The Lex Plus Powers conjecture for the top socle degree

摘录 / EXCERPT: Top-socle Lex Plus Powers conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-073256 | MATHDB-379218

Gaussian-product unimodality conjecture

摘录 / EXCERPT: Gaussian-product unimodality conjecture. For every $0\leq k<n$, this polynomial is symmetric and unimodal. The source states that, using known results together with its preceding t...

conjecture | open | MathDB 页面 | 原文线索

V2-073257 | MATHDB-379233

The centralizer braid-group conjecture for regular elements

摘录 / EXCERPT: Let ${\bf w}$ be an F-root of ${\boldsymbol\pi}$, put $w=\beta({\bf w})$, and let $C_W(wF)$ be the centralizer of the corresponding regular element of WF. Let $B(w)$ be the bra...

conjecture | open | MathDB 页面 | 原文线索

V2-073258 | MATHDB-379251

Balabdaoui-Wellner conjecture on uniform Hermite spline interpolation error

摘录 / EXCERPT: Let k be the spline parameter, let f be a $(k-1)$-times differentiable function whose $(k-1)$st derivative has finite total variation, and let $\mathcal{E}_k(f)$ denote the int...

conjecture | open | MathDB 页面 | 原文线索

V2-073259 | MATHDB-379252

Stahl's real-negative-root conjecture for specified graph families

摘录 / EXCERPT: For each graph family listed in Example, let its genus polynomial be the polynomial whose coefficients count embeddings by genus. Stahl's conjecture. The zeros of the genus polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-073260 | MATHDB-379262

The coefficient inequality conjecture for rational cuspidal plane curves

摘录 / EXCERPT: Let $(C,p_i)_{i=1}^\nu$ be a collection of locally irreducible local plane curve singularities satisfying ... for some integer d. Let c_l be the coefficients associated with the...

conjecture | open | MathDB 页面 | 原文线索

V2-073261 | MATHDB-379263

The non-abelian Euler-characteristic criterion for twisted L-values

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p, let $F_\infty/\mathbb{Q}$ be the false Tate curve extension, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073262 | MATHDB-379268

The lattice theta-series root-index conjecture

摘录 / EXCERPT: Let $\Theta_{\Lambda}(x)$ be the theta series of a d-dimensional lattice, and let $\mathcal{P}_n$ denote the class of generating functions whose nth roots have integral coeffic...

conjecture | open | MathDB 页面 | 原文线索

V2-073263 | MATHDB-379269

The BCH weight-enumerator integrality conjecture

摘录 / EXCERPT: Let m and t be integers, and let C be obtained by adding an overall parity check to the primitive BCH code of length 2^m-1 and designed distance $2t-1$. Thus C has length...

conjecture | open | MathDB 页面 | 原文线索

V2-073264 | MATHDB-379270

Conjecture on differentiability of the ISE density

摘录 / EXCERPT: Let $f_{\text{\sc ise}}$ be the density of the integrated super-Brownian excursion (ISE), whose existence and Hölder continuity are established in the paper. ISE density differenti...

conjecture | open | MathDB 页面 | 原文线索

V2-073265 | MATHDB-379271

Almost-all graphs greedy and tree-solvable conjecture

摘录 / EXCERPT: A graph is greedy if every configuration of $\pi(G)$ pebbles can be solved at any specified root using only greedy pebbling steps. It is tree-solvable if every configuration of siz...

conjecture | open | MathDB 页面 | 原文线索

V2-073266 | MATHDB-379279

The maximal FAn subgroup conjecture for Coxeter groups

摘录 / EXCERPT: Maximal FAn subgroup conjecture. A subgroup $H\subset W$ is maximal FA_n if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-073267 | MATHDB-379285

Hindry's integral preperiodic-point conjecture for rational maps

摘录 / EXCERPT: Let k be a number field, let S be a finite set of places of k containing the archimedean places, and let ${\mathcal O}_{k,S}$ be the ring of S-integers. Let $R(x)\in k(x)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073268 | MATHDB-379292

Kobayashi's complex sphere conjecture

摘录 / EXCERPT: Let n be a positive integer, and let $SO(n+1,\mathbb{C})/SO(n,\mathbb{C})$ be the n-dimensional complex sphere. Kobayashi's complex sphere conjecture. The complex sphere admits...

conjecture | open | MathDB 页面 | 原文线索

V2-073269 | MATHDB-379293

Kobayashi's classification conjecture for irreducible complex symmetric spaces

摘录 / EXCERPT: Let $G_\mathbb{C}/K_\mathbb{C}$ be an irreducible complex symmetric space, where $G_\mathbb{C}$ is a complex semisimple Lie group and $K_\mathbb{C}$ is the fixed-point subgroup of...

conjecture | open | MathDB 页面 | 原文线索

V2-073270 | MATHDB-379295

The diameter-three $Omega_\omega$ upper-bound conjecture

摘录 / EXCERPT: Let G be a graph of diameter 3 with n vertices, and let $Omega_\omega(G)$ denote the $omega$-cover pebbling parameter used in the paper. For $omega\geq 1$, diameter-three...

conjecture | open | MathDB 页面 | 原文线索

V2-073271 | MATHDB-379299

Perrin-Riou's integral Iwasawa-theoretic conjecture for crystalline representations

摘录 / EXCERPT: $C_{\mathrm{Iw}}(K_\infty/K,V)$. The canonical trivialization satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073272 | MATHDB-379303

The nonexistence of a skein theory for the tangle group functor

摘录 / EXCERPT: Let $\mathcal{T}ang$ be the category of tangles, let $\Lambda\mathbf{Grp}$ be the category of groups, and let $\Gamma\colon\mathcal{T}ang\to\Lambda\mathbf{Grp}$ be the tangle group...

conjecture | open | MathDB 页面 | 原文线索

V2-073273 | MATHDB-379354

Categoricity and closure of amalgamation bases under unions

摘录 / EXCERPT: Let $\operatorname{\mathcal{K}}$ be an abstract elementary class, let $\operatorname{Hanf}(\operatorname{\mathcal{K}})$ be its Hanf number, and let $LS(\operatorname{\mathcal{K}})$...

conjecture | open | MathDB 页面 | 原文线索

V2-073274 | MATHDB-379355

Uniqueness of limit models of different lengths in abstract elementary classes

摘录 / EXCERPT: Let $\operatorname{\mathcal{K}}$ be an abstract elementary class, let $\operatorname{Hanf}(\operatorname{\mathcal{K}})$ and $LS(\operatorname{\mathcal{K}})$ denote its Hanf and Löw...

conjecture | open | MathDB 页面 | 原文线索

V2-073275 | MATHDB-379378

The period-index conjecture for C_d fields

摘录 / EXCERPT: Let K be a C_d field, meaning that every homogeneous form of degree e in more than e^d variables over K has a nontrivial zero, and let $\alpha\in\operatorname{Br}(K)$. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-073276 | MATHDB-379392

Self-avoiding loop measure scaling-limit conjecture

摘录 / EXCERPT: Self-avoiding loop scaling-limit conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073277 | MATHDB-379417

Albert's conjecture on pattern-avoiding subsequence expectations

摘录 / EXCERPT: Let X be a finite proper subset of $\bigcup_{j\geq 2}\mathfrak{S}_j$, meaning that X does not contain both an identity permutation $12\cdots j$ and a reverse identity permutati...

conjecture | open | MathDB 页面 | 原文线索

V2-073278 | MATHDB-379419

Nondegeneracy conjecture for the energy minimum of Hitchin representations

摘录 / EXCERPT: Let S be a closed oriented surface, let ρ be a Hitchin representation, and let e_ρ denote its energy functional on Teichmüller space. Nondegeneracy conjecture. The mini...

conjecture | open | MathDB 页面 | 原文线索

V2-073279 | MATHDB-379423

Promotion conjecture for tensor products of Kirillov-Reshetikhin crystals

摘录 / EXCERPT: Promotion conjecture. Theorem 4.11 remains true for any tensor product

conjecture | open | MathDB 页面 | 原文线索

V2-073280 | MATHDB-379428

The finite-characteristic reformulation of the Jacobian Conjecture

摘录 / EXCERPT: Let $n\ge 2$, let $d\ge 1$, and let ${\bf k}$ be a field. A polynomial map $\phi:{\mathbb A}^n_{\bf k}\to{\mathbb A}^n_{\bf k}$ has degree at most d if its coordinate polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-073281 | MATHDB-379442

Zero-free region conjecture for the hypergeometric function $F_k(z)$

摘录 / EXCERPT: Let n be the dimension parameter, let $\rho=(n-1)/2$, and define ... where $F(a,b;c;z)$ is the hypergeometric function and k is complex. The zero-free region conjecture. For ever...

conjecture | open | MathDB 页面 | 原文线索

V2-073282 | MATHDB-379443

Kalai-Kleinschmidt-Lee conjecture on empty simplices of simplicial polytopes

摘录 / EXCERPT: Kalai-Kleinschmidt-Lee conjecture. For every j, the number of j-dimensional empty simplices of P is at most the number of j-dimensional empty simplices of...

conjecture | open | MathDB 页面 | 原文线索

V2-073283 | MATHDB-379454

Uniform boundedness conjecture for semistable holomorphic chains

摘录 / EXCERPT: Uniform boundedness conjecture. For every type t, there is a bounded region

conjecture | open | MathDB 页面 | 原文线索

V2-073284 | MATHDB-379460

The higher-dimensional McKay correspondence for crepant desingularizations

摘录 / EXCERPT: Let G be a finite subgroup of $\operatorname{SL}(r,\mathbb{C})$, let $X=\mathbb{C}^{r}/G$, and let $\widehat{X}\longrightarrow X$ be a crepant desingularization. For each integer...

conjecture | open | MathDB 页面 | 原文线索

V2-073285 | MATHDB-379461

Existence conjecture for projective crepant resolutions of quotient complete intersections

摘录 / EXCERPT: Let $G\subset\operatorname{SL}(r,\mathbb{C})$ be finite, and consider quotient complete-intersection singularities $\mathbb{C}^{r}/G$ in arbitrary dimension. Existence conjecture....

conjecture | open | MathDB 页面 | 原文线索

V2-073286 | MATHDB-379470

Arnold's density-profile conjecture for numerical semigroups

摘录 / EXCERPT: Arnold's density-profile conjecture. Asymptotically for large ${\bf d}^m$, the density at s is

conjecture | open | MathDB 页面 | 原文线索

V2-073287 | MATHDB-379475

Julia-set containment of elliptic-curve attracting basins

摘录 / EXCERPT: Let f be a rational map of ${\mathbb P}^2({\mathbb C})$, let ${\mathcal C}\subset{\mathbb P}^2({\mathbb C})$ be an f-invariant elliptic curve, let ${\mathcal B}({\mathcal C})$...

conjecture | open | MathDB 页面 | 原文线索

V2-073288 | MATHDB-379476

Repelling periodic-point conjecture for invariant complex elliptic curves

摘录 / EXCERPT: Let f be a rational map with an invariant complex elliptic curve. A periodic point is repelling when the derivative of the corresponding return map is expanding. Repelling-point...

conjecture | open | MathDB 页面 | 原文线索

V2-073289 | MATHDB-379477

Positive-measure prevalence of attracting Herman-ring cycles in real Desboves maps

摘录 / EXCERPT: Consider the parameter space of complex Desboves maps with real coefficients. A cycle of attracting Herman rings is a periodic collection of Herman rings whose return dynamics is a...

conjecture | open | MathDB 页面 | 原文线索

V2-073290 | MATHDB-379482

Automorphic matrix-coefficient bound for low-rank absolutely almost simple groups

摘录 / EXCERPT: Let K be a global field, let $\mathbf G$ be a connected absolutely almost simple K-group, and let W_f be a compact open subgroup of $\mathbf G(\bA_f)$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-073291 | MATHDB-379486

Relative hyperbolicity of Artin groups from their Coxeter groups

摘录 / EXCERPT: Let A) be an Artin group, with corresponding Coxeter group W. The standard parabolic subgroups of A of finite type are the parabolic subgroups associated with finite standard...

conjecture | open | MathDB 页面 | 原文线索

V2-073292 | MATHDB-379500

Injectivity on connected components of positive unipotent bicrystals

摘录 / EXCERPT: Injectivity conjecture. For every connected component of $\mathcal{B}_{f,\Theta}$, the restriction of $\tilde {\bf f}_w$ is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-073293 | MATHDB-379502

Saturated-valuation construction of unipotent bicrystals

摘录 / EXCERPT: Saturated-valuation conjecture. Under these assumptions, such a strongly positive unipotent bicrystal exists and the valuation-induced map identifies the normal crystal of...

conjecture | open | MathDB 页面 | 原文线索

V2-073294 | MATHDB-379505

Tropical tensor-product factorization of Kashiwara crystals

摘录 / EXCERPT: Tropical tensor-product conjecture. The tropicalization of F is an isomorphism of torsion-free Kashiwara crystals

conjecture | open | MathDB 页面 | 原文线索

V2-073295 | MATHDB-379508

Bershadsky-Cecotti-Ooguri-Vafa quintic mirror genus-one and BCOV conjectures

摘录 / EXCERPT: Let $\pi\colon\mathcal W\to\mathbb P^{1}$ be a family of quintic mirror threefolds. Let $N_g(d)$ be the genus-g Gromov-Witten invariant of degree d of a general quintic threefo...

conjecture | open | MathDB 页面 | 原文线索

V2-073296 | MATHDB-379510

Affine Chevalley conjecture for affine Stanley symmetric functions

摘录 / EXCERPT: Let $\tilde{F}_w(X)$ be the affine Stanley symmetric function indexed by $w\text{,}$ and let $c^w_{u,v}$ be the affine Schubert structure constants defined by ... For...

conjecture | open | MathDB 页面 | 原文线索

V2-073297 | MATHDB-379535

Bounded digit-sum conjecture for non-classical zero sets of interpolated L-functions

摘录 / EXCERPT: Let k be a global function field, let w be any place of k, and let N_w denote the non-classical set for the interpolation of $L(\chi,s)$ at w; when $w=\infty$, this inter...

conjecture | open | MathDB 页面 | 原文线索

V2-073298 | MATHDB-379545

Mills-Robbins-Rumsey's monotone-triangle correspondence conjecture

摘录 / EXCERPT: Let $\TSPP{n}^{k}$ be the subset of triangular shifted plane partitions whose entries in the first $n-1-k$ columns attain their maximal values n. Let $\MT{n}^{k}$ be the set of m...

conjecture | open | MathDB 页面 | 原文线索

V2-073299 | MATHDB-379546

The refined monotone-triangle enumeration conjecture for cyclically symmetric plane partitions

摘录 / EXCERPT: Let $\CSPP{n}^{k}$ be the restricted class of cyclically symmetric plane partitions, let $\overline U_r(c)$ be its statistic for $1\leq r\leq n$, and define ... where $\MT{n}^{k}$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-073300 | MATHDB-379553

Finite-basis conjecture for grid classes

摘录 / EXCERPT: Let M be a finite matrix whose entries specify the monotone permutation classes placed in the cells of a grid, and let $\operatorname{Grid}(M)$ denote the resulting grid class. A...

conjecture | open | MathDB 页面 | 原文线索

V2-073301 | MATHDB-379554

Rational generating functions for grid classes with forest graphs

摘录 / EXCERPT: Let M be a finite matrix defining a grid class $\operatorname{Grid}(M)$, and let $G(M)$ be the bipartite graph whose vertices represent the rows and columns of M, with an edge...

conjecture | open | MathDB 页面 | 原文线索

V2-073302 | MATHDB-379592

The adjacency injection conjecture for discriminant-complement groups

摘录 / EXCERPT: Let f and g be singular functions such that f is adjacent to g, so that a versal unfolding of g is also a versal unfolding for f. Let $4mu$ and $tilde\mu$ be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-073303 | MATHDB-379593

The bifurcation-complement conjecture for braid monodromy groups

摘录 / EXCERPT: Let f be a singular function, let its bifurcation diagram be the relevant discriminant in the parameter space of a versal unfolding, and let the braid monodromy group be the imag...

conjecture | open | MathDB 页面 | 原文线索

V2-073304 | MATHDB-379598

Stone-regularity conjecture for boundary-value problems with polynomial resolvent growth

摘录 / EXCERPT: Let a boundary-value problem be subject to the estimate referenced in the source as ... . The conjecture proposes a general regularity consequence of polynomial resolvent growth, ext...

conjecture | open | MathDB 页面 | 原文线索

V2-073305 | MATHDB-379601

Ghrist's conjecture on planar graph braid groups

摘录 / EXCERPT: Let Γ be a planar graph, and let $B_n\Gamma$ and $P B_n\Gamma$ denote its n-strand braid group and pure braid group, respectively. A group is right-angled Artin if it admi...

conjecture | open | MathDB 页面 | 原文线索

V2-073306 | MATHDB-379604

Average jump number conjecture for grids

摘录 / EXCERPT: Let $[m]^n$ denote the n-dimensional grid poset, and let $\bar{s}([m]^n)$ be its average jump number. The parameters m and n range over values for which...

conjecture | open | MathDB 页面 | 原文线索

V2-073307 | MATHDB-379611

Seymour's projective-cube homomorphism conjecture for planar graphs

摘录 / EXCERPT: Seymour's conjecture. Every planar graph whose odd cycles all have length at least $2k+1$ has a homomorphism to PQ_{2k}.

conjecture | open | MathDB 页面 | 原文线索

V2-073308 | MATHDB-379612

The triangle-free cubic graph conjecture for the Clebsch graph

摘录 / EXCERPT: Triangle-free cubic graph conjecture. Every triangle-free cubic graph is homomorphic to PQ_4.

conjecture | open | MathDB 页面 | 原文线索

V2-073309 | MATHDB-379613

Nešetřil's Pentagon Conjecture for high-girth cubic graphs

摘录 / EXCERPT: Nešetřil's Pentagon Conjecture. If G is a cubic graph of sufficiently high girth, then G is homomorphic to C_5.

conjecture | open | MathDB 页面 | 原文线索

V2-073310 | MATHDB-379614

Quadratic h^*-polynomial inequality

摘录 / EXCERPT: Let Δ be an n-dimensional lattice polytope whose h^*-polynomial is quadratic, written as ... Quadratic h^*-polynomial conjecture. One has ... This is proposed as a more pr...

conjecture | open | MathDB 页面 | 原文线索

V2-073311 | MATHDB-379616

The characterization of self-conjugate skew-equivalent diagrams

摘录 / EXCERPT: Self-conjugate skew-equivalence conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-073312 | MATHDB-379631

Upper asymptotic matching conjecture

摘录 / EXCERPT: Let $\{G_k=(V_k,E_k)\}_{k\in\mathbb{N}}$ be a sequence of r-regular bipartite graphs with $\#V_k\to\infty$. Let $h_{\{G_k\}}(p)$ denote the associated monomer-dimer entropy, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073313 | MATHDB-379632

Entropy maximality conjecture for periodic regular graphs

摘录 / EXCERPT: Let G and G_0 satisfy the assumptions of the theorem preceding this statement, and let $h_G(p)$ and $h_{G_0}(p)$ denote their monomer-dimer entropies at density p. Entropy ma...

conjecture | open | MathDB 页面 | 原文线索

V2-073314 | MATHDB-379638

Kahn-Kalai optimal-scale conjecture

摘录 / EXCERPT: Optimal-scale conjecture. For each $C>0$ there is an $\varepsilon>0$ such that, for every monotone ${\cal F}$, there exists

conjecture | open | MathDB 页面 | 原文线索

V2-073315 | MATHDB-379645

Converse to the compact Clifford-Klein form criterion

摘录 / EXCERPT: Converse to Theorem 15. The converse of Theorem 15 also holds.

conjecture | open | MathDB 页面 | 原文线索

V2-073316 | MATHDB-379652

McKay's crepant-resolution conjecture

摘录 / EXCERPT: Let V be a finite-dimensional vector space over a field k, and let E be a finite subgroup of $\operatorname{GL}(V)$ whose order is invertible in k. A crepant resolution is...

conjecture | open | MathDB 页面 | 原文线索

V2-073317 | MATHDB-379653

The Asymptotic Lower Matching Conjecture

摘录 / EXCERPT: The Asymptotic Lower Matching Conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-073318 | MATHDB-379655

The Asymptotic Upper Matching Conjecture

摘录 / EXCERPT: The Asymptotic Upper Matching Conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-073319 | MATHDB-379656

The Asymptotic Lower r-Permanent Conjecture

摘录 / EXCERPT: The Asymptotic Lower r-Permanent Conjecture. Under these hypotheses,

conjecture | open | MathDB 页面 | 原文线索

V2-073320 | MATHDB-379665

The mixed TERP-structure correspondence conjecture

摘录 / EXCERPT: Let $(H,\nabla,H'_{\mathds R},P)$ be a TERP-structure. It requires no ramification when its formal decomposition can be made without ramification; it is a mixed TERP-structure when...

conjecture | open | MathDB 页面 | 原文线索

V2-073321 | MATHDB-379666

The positivity conjecture for semi-simple TERP-structures

摘录 / EXCERPT: Let $(w,u_1,\ldots,u_n,\xi,T)$ be data with $w\in{\mathds Z}$, $u_i\in{\mathds C}$ pairwise distinct, $\xi\in S^1$ satisfying $\Re((u_i-u_j)/\xi)<0$ for $i<j$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073322 | MATHDB-379667

The spectral interval conjecture for positive semi-simple TERP data

摘录 / EXCERPT: Fix $u_1,\ldots,u_n\in{\mathds C}$, $\xi\in S^1$ with $\Re((u_i-u_j)/\xi)<0$ for $i<j$, and $T\in M(n\times n,{\mathds R})$ upper triangular with $T_{ii}=1$ and $T+T^{tr}$ positive...

conjecture | open | MathDB 页面 | 原文线索

V2-073323 | MATHDB-379668

Orthogonal Igusa-function polynomial conjecture

摘录 / EXCERPT: Let ${\mathcal{V}}$ be an n-dimensional, non-degenerate quadratic space, let $[n-1]=\{1,\ldots,n-1\}$, let ${\mathcal{C}}_n$ be the subgroup of chessboard elements in the symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-073324 | MATHDB-379679

Hyperbolicity conjecture for general hypersurfaces of degree at least $2n-1$

摘录 / EXCERPT: Hyperbolicity conjecture for general hypersurfaces. A general hypersurface

conjecture | open | MathDB 页面 | 原文线索

V2-073325 | MATHDB-379687

The parity conjecture for elliptic curves over number fields

摘录 / EXCERPT: Let E be an elliptic curve over a number field K. Let $w(E/K)\in\{\pm 1\}$ be the root number, defined as the conjectural sign in the functional equation for $L(E/K,s)$ under...

conjecture | open | MathDB 页面 | 原文线索

V2-073326 | MATHDB-379692

Whitehead graph feature-region conjecture

摘录 / EXCERPT: Let F be a free group, and associate to each element of F the feature vector of weights of its Whitehead Graph in the corresponding feature space. A region is a bounded subset...

conjecture | open | MathDB 页面 | 原文线索

V2-073327 | MATHDB-379697

The R-invariance conjecture for tautological equations

摘录 / EXCERPT: Let $\sum_i c_i\Gamma_i=0$ be a tautological equation in codimension k on $\overline{M}_{g,n}$, and let $\mathfrak{r}_l$ be the operators sending decorated graphs on...

conjecture | open | MathDB 页面 | 原文线索

V2-073328 | MATHDB-379704

Pivot-orbit containment conjecture for the family ${\cal F}^{n,t}$

摘录 / EXCERPT: Let n and t satisfy $0\leq t\leq n-1$, and let ${\cal F}^{n,t}$ be the family of Boolean functions defined by ... where $\deg(a)\leq 1$, together with arbitrary Boolean functions...

conjecture | open | MathDB 页面 | 原文线索

V2-073329 | MATHDB-379707

Conformal-radius bound for polynomials with an indifferent fixed point

摘录 / EXCERPT: Let f be a polynomial of degree d with an indifferent fixed point at the origin. Let c_i be the critical points of f, let θ be the rotation number at the origin, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073330 | MATHDB-379708

Asymptotic count of reducible integer matrices

摘录 / EXCERPT: Asymptotic counting conjecture. The lower bound gives the true order of growth:

conjecture | open | MathDB 页面 | 原文线索

V2-073331 | MATHDB-379711

A bound for chromatic deficiency of graph joins

摘录 / EXCERPT: Join bound. If

conjecture | open | MathDB 页面 | 原文线索

V2-073332 | MATHDB-379713

Higher Castelnuovo conjecture for points in symmetric position

摘录 / EXCERPT: Higher Castelnuovo conjecture. Then Γ lies on a curve D of degree at most $n+m-2$. This is related to the Eisenbud-Harris conjecture for curves: the cases $m=1$ and $m=2$...

conjecture | open | MathDB 页面 | 原文线索

V2-073333 | MATHDB-379722

Conjecture on bounded generation of noncocompact lattices in SL(3)

摘录 / EXCERPT: Bounded-generation conjecture. If Γ is any noncocompact lattice in either

conjecture | open | MathDB 页面 | 原文线索

V2-073334 | MATHDB-379723

The geometric Langlands conjecture for semisimple groups

摘录 / EXCERPT: The geometric Langlands correspondence. There exists a canonical equivalence of categories

conjecture | open | MathDB 页面 | 原文线索

V2-073335 | MATHDB-379725

AMU conjecture on detecting pseudo-Anosov mapping classes by TQFT representations

摘录 / EXCERPT: AMU conjecture. For a mapping class $\phi\in\Gamma$, ϕ is Pseudo-Anosov or reducible with Pseudo-Anosov pieces if and only if $\rho_k(\phi)$ is infinite order for large enough...

conjecture | open | MathDB 页面 | 原文线索

V2-073336 | MATHDB-379733

Coincidence conjecture for the two Eisenstein-series isomorphisms

摘录 / EXCERPT: Coincidence conjecture. The isomorphisms supplied by Proposition 'IC calculation' and Theorem 'FFKM thm' (1) coincide. This is a compatibility assertion internal to the proof of th...

conjecture | open | MathDB 页面 | 原文线索

V2-073337 | MATHDB-379738

Hartl's finitely generated value-group conjecture for weakly admissible points

摘录 / EXCERPT: Hartl's value-group conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-073338 | MATHDB-379740

Hartl's maximality conjecture for the universal local system

摘录 / EXCERPT: Hartl's local-system maximality conjecture. The space $\breve{\cal{F}}^0_b$ is the largest open $\breve E$-analytic subspace of $\breve{\cal{F}}^{wa}_b$ carrying a local system...

conjecture | open | MathDB 页面 | 原文线索

V2-073339 | MATHDB-379742

Strong rational connectedness conjecture for smooth loci of log Del Pezzo surfaces

摘录 / EXCERPT: Strong rational connectedness conjecture. The smooth locus X^{sm} of a log Del Pezzo surface (X,Δ) is strongly rationally connected.

conjecture | open | MathDB 页面 | 原文线索

V2-073340 | MATHDB-379767

Character-sheaf central functor equivalence conjecture

摘录 / EXCERPT: Let ${\mathcal C}_\zeta^{{\underline c}_f}$ be the monoidal category associated with a finite two-sided cell ${\underline c}_f\subset W_f(\zeta)$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073341 | MATHDB-379772

Positive h-vector conjecture for generalized cluster complexes

摘录 / EXCERPT: Let Φ be an irreducible crystallographic root system of rank ℓ, let $m\geq 1$, and let ${\mathcal A}^m_\Phi$ be its generalized Catalan hyperplane arrangement. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073342 | MATHDB-379777

The asymptotic norm conjecture for Toeplitz matrices with multiple Fisher-Hartwig singularities

摘录 / EXCERPT: Let $R\geq 2$ and let ... where $t_1,\ldots,t_R$ are distinct points on ${\bf T}$, $0<\operatorname{Re}\alpha_r<1/2$, $-1/2<\operatorname{Re}\beta_r\leq 1/2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073343 | MATHDB-379786

Mills-Robbins-Rumsey's refined vertical-symmetry conjecture

摘录 / EXCERPT: Mills-Robbins-Rumsey's refined vertical-symmetry conjecture. The number of $b\in\TSPP{2n+1}$ satisfying $\gamma(b)=b$ and $U_2(b)=r-1$ should equal the number of vertically symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-073344 | MATHDB-379787

The twisted and column-even domino-plane-partition bijection conjecture

摘录 / EXCERPT: Twisted domino correspondence conjecture. For non-negative integers m and $n\geq1$, there should be a bijection

conjecture | open | MathDB 页面 | 原文线索

V2-073345 | MATHDB-379788

The determinant evaluations for symmetric alternating-sign-matrix enumerators

摘录 / EXCERPT: Determinant-evaluation conjecture. The following identities should hold:

conjecture | open | MathDB 页面 | 原文线索

V2-073346 | MATHDB-379799

Lattice-width conjecture for lattice k-point tetrahedra

摘录 / EXCERPT: Let T be a lattice tetrahedron with exactly k interior lattice points. In a minimal direction for its lattice width, consider the consecutive lattice planes meeting T. Lattic...

conjecture | open | MathDB 页面 | 原文线索

V2-073347 | MATHDB-379801

The intrinsic virtual knotting conjecture for intrinsically knotted graphs

摘录 / EXCERPT: Let G be a graph. Call G intrinsically knotted if every classical spatial embedding of G contains a non-trivial knot, and call it intrinsically virtually knotted of degree 1...

conjecture | open | MathDB 页面 | 原文线索

V2-073348 | MATHDB-379818

Folklore conjecture on density of pseudo-Anosovs in the mapping class group

摘录 / EXCERPT: Fix a finite generating set for ${\rm Mod}_g$, let $B(r)$ be the ball of radius r, and define the density of $S\subset {\rm Mod}_g$ by ... Let ${\cal P}$ be the set of pseudo-Anoso...

conjecture | open | MathDB 页面 | 原文线索

V2-073349 | MATHDB-379820

Maximality conjecture for the moduli orbifold

摘录 / EXCERPT: For $g\geq 3$, let $\M_g$ be the smooth moduli orbifold of genus-g Riemann surfaces. Maximality conjecture for $\M_g$. The orbifold $\M_g$ does not finitely orbifold-cover any ot...

conjecture | open | MathDB 页面 | 原文线索

V2-073350 | MATHDB-379823

Kähler-group conjecture for mapping class groups

摘录 / EXCERPT: For $g\geq 3$, let ${\rm Mod}_g$ be the mapping class group of a closed genus-g surface. Kähler-group conjecture. The group ${\rm Mod}_g$ is a Kähler group, meaning that it is is...

conjecture | open | MathDB 页面 | 原文线索

V2-073351 | MATHDB-379824

Finite-presentation conjecture for the Torelli group

摘录 / EXCERPT: Let $\T_g$ be the Torelli group of a closed genus-g surface. Torelli finite-presentation conjecture. The group $\T_g$ is finitely presented for $g\geq 4$. This is presented as a...

conjecture | open | MathDB 页面 | 原文线索

V2-073352 | MATHDB-379829

Stoll's very good curves conjecture

摘录 / EXCERPT: Let k be a number field, and let C be a smooth projective geometrically connected curve over k. In the paper, a curve is very good when its finite descent obstructions accoun...

conjecture | open | MathDB 页面 | 原文线索

V2-073353 | MATHDB-379830

Poonen's finite-place obstruction conjecture for curves

摘录 / EXCERPT: Let C be a smooth projective geometrically connected curve of genus at least 2 over a number field k, with $C(k)=\emptyset$. Assume that C has a rational divisor class of d...

conjecture | open | MathDB 页面 | 原文线索

V2-073354 | MATHDB-379837

Connectivity bound for graphs exceeding Reed's bound

摘录 / EXCERPT: Let G be a graph, with chromatic number χ, clique number ω, maximum degree Δ, order n, and let $\overline{\kappa}=\kappa(\overline{G})$ denote the connectivi...

conjecture | open | MathDB 页面 | 原文线索

V2-073355 | MATHDB-379841

Minor-closedness conjecture for maximal cut-ideal generating degree

摘录 / EXCERPT: Let G be a graph, let I_G be its cut ideal, and let $\mu(I_G)$ denote the maximal degree of a binomial in a minimal generating set of I_G. A graph H is a minor of G if it...

conjecture | open | MathDB 页面 | 原文线索

V2-073356 | MATHDB-379842

Quartic generation conjecture for cut ideals

摘录 / EXCERPT: Let G be a graph and let I_G be its cut ideal. A graph is K_5-minor-free when it has no minor isomorphic to K_5. Quartic generation conjecture. ... This extends the proposed...

conjecture | open | MathDB 页面 | 原文线索

V2-073357 | MATHDB-379845

Conjecture on diagonal-flip stabilization of surface triangulations

摘录 / EXCERPT: Conjecture on diagonal-flip stabilization. The only surfaces S for which

conjecture | open | MathDB 页面 | 原文线索

V2-073358 | MATHDB-379847

Hirschowitz-Harbourne conjecture for special plane linear systems

摘录 / EXCERPT: Hirschowitz-Harbourne conjecture. The system L is special if and only if there exists a -1-curve $C\subset\mathbb{P}^{2}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073359 | MATHDB-379854

Nilsoliton stability conjecture for Ricci flow near nilpotent homogeneous metrics

摘录 / EXCERPT: Let $(\mathcal{M}^{n},g(t))$ be a solution of Ricci flow such that $g(0)$ is sufficiently close, in a suitable norm, to a (locally) homogeneous metric on a quotient of a nilpotent...

conjecture | open | MathDB 页面 | 原文线索

V2-073360 | MATHDB-379855

The extended bi-graded Toda conjecture for orbifold descendent potentials

摘录 / EXCERPT: Let $\mathcal{C}_{k,m}$ be the orbifold obtained from $\mathbb{P}^1$ by replacing neighborhoods of 0 and ∞ with the orbifold discs $D_1/\mathbb{Z}_k$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-073361 | MATHDB-379868

Existence of a low-growth surjective cellular automaton

摘录 / EXCERPT: Low-growth existence conjecture. There exists $N>1$ and a surjective cellular automaton f on N symbols such that $\nu(f,S_N)<N$. The surrounding discussion poses related quanti...

conjecture | open | MathDB 页面 | 原文线索

V2-073362 | MATHDB-379887

Lattice comparison conjecture for representation growth

摘录 / EXCERPT: Let ... where each K_i is a local field, each G_i is an absolutely almost simple K_i-group, and ... Let $\rho(\Gamma)$ denote the abscissa of convergence of the representation ze...

conjecture | open | MathDB 页面 | 原文线索

V2-073363 | MATHDB-379900

Determinantal process conjecture for zeros of Gaussian matrix-valued power series

摘录 / EXCERPT: Let A_k, for $k\geq 0$, be independent $n\times n$ matrices whose entries are independent standard complex Gaussian random variables. The zeros in the unit disk of the determinan...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073364 | MATHDB-379901

Sodin's moderate and large deviation conjecture for planar GAF zeros

摘录 / EXCERPT: Let $n(r)$ be the number of zeros of the planar Gaussian analytic function $\mathbf g$ in the disk $D(0,r)$. Sodin's deviation conjecture. As $r\to\infty$, ... This predicts the loga...

conjecture | open | MathDB 页面 | 原文线索

V2-073365 | MATHDB-379904

Nonnegative q-series product conjectures

摘录 / EXCERPT: Nonnegative product conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073366 | MATHDB-379912

Johnson's real-rootedness conjecture for mixed determinant polynomials

摘录 / EXCERPT: Johnson's conjecture. If A is positive semidefinite, then the polynomial $\eta(zA,-B)$ has all real zeros.

conjecture | open | MathDB 页面 | 原文线索

V2-073367 | MATHDB-379913

Johnson-Bapat interlacing conjecture for mixed determinant polynomials

摘录 / EXCERPT: Johnson-Bapat interlacing conjecture. If A is positive semidefinite and $\eta(zA,-B)$ is not identically zero, then, for every $1\leq j\leq n$, the zeros of

conjecture | open | MathDB 页面 | 原文线索

V2-073368 | MATHDB-379914

Higher-rank homogeneous measure classification conjecture

摘录 / EXCERPT: Let $n \geq 3$, let $G \cong \operatorname{PGL}_n(\mathbb{R})$, and let $\Gamma < G$ be a lattice as in examples (L-1) or (L-2) of the source. Let H be the relevant diagonal subg...

conjecture | open | MathDB 页面 | 原文线索

V2-073369 | MATHDB-379939

The conjecture on boundary stabilization of non-primitive knots

摘录 / EXCERPT: Let $K\subset S^3$ be a knot that is not γ--primitive, and let a minimal genus Heegaard splitting of its exterior $E(K)$ be given. A boundary stabilization is the genus-incr...

conjecture | open | MathDB 页面 | 原文线索

V2-073370 | MATHDB-379946

Scaling conjecture for filling functions of Carnot groups

摘录 / EXCERPT: Scaling conjecture. There is a map $\alpha:S^k\to G$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073371 | MATHDB-379950

Multinomial monotonicity conjecture for intersection numbers of stable curves

摘录 / EXCERPT: Multinomial monotonicity conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073372 | MATHDB-379953

The ample-cone conjecture for symmetric products of very general curves

摘录 / EXCERPT: Ample-cone conjecture. If C is a very general curve of genus $g\ge 4$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073373 | MATHDB-379955

Asymptotic tightness of the lower bound for 1-perfect binary codes

摘录 / EXCERPT: Let $B(n-1)$ denote the number of 1-perfect binary codes of length $n-1=2^m-1$, and let $\widetilde K_{\scriptscriptstyle A}(n)$ be the lower-bound expression satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-073374 | MATHDB-379990

Antidiagonal mirror symmetry formulas for local curves

摘录 / EXCERPT: Let $X_k=\mathcal O(k)\oplus\mathcal O(-2-k)\rightarrow\mathbb{P}^1$ carry the antidiagonal torus action $(\lambda,-\lambda)$ on the bundle, and let t_k denote the Kähler class o...

conjecture | open | MathDB 页面 | 原文线索

V2-073375 | MATHDB-379991

Diagonal-action mirror symmetry equivalence for local curves

摘录 / EXCERPT: Let $X_k=\mathcal O(k)\oplus\mathcal O(-2-k)\rightarrow\mathbb{P}^1$ carry the diagonal torus action $\lambda_1=\lambda_2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073376 | MATHDB-379993

Bridgeland's conjecture on the derived autoequivalence group of twisted K3 surfaces

摘录 / EXCERPT: Let (X,α) be a projective twisted K3 surface, let ${\rm D}^{\rm b}(X,\alpha)$ be its bounded derived category of twisted coherent sheaves, let ${\mathcal P}_0(X,\alpha)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-073377 | MATHDB-379999

Permutation bootstrap consistency conjecture for the max self-similarity estimator

摘录 / EXCERPT: Let $X(1),\ldots,X(N)$ be an i.i.d. sample, and let $\widehat H_i$, $i=1,\ldots,M$, be a permutation bootstrap sample of the estimator $\widehat H(j_1,j_2)$. Let the scales j_1,...

conjecture | open | MathDB 页面 | 原文线索

V2-073378 | MATHDB-380001

Flossing-order conjecture for tree graph-associahedra

摘录 / EXCERPT: For each n, consider the set of unlabelled isomorphism classes of trees with n nodes. For a tree T, let $\mathcal{B}(T)$ be its graphical building set and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073379 | MATHDB-380003

Rational generating function conjecture for the class of permutations corresponding to pin words

摘录 / EXCERPT: A pin word is a word encoding a permutation through the pin sequence construction; the set of permutations corresponding to pin words forms a permutation class. Rational generating...

conjecture | open | MathDB 页面 | 原文线索

V2-073380 | MATHDB-380007

Characterization of maximal elements of the L-positivity poset

摘录 / EXCERPT: Let PC_n be the poset of unordered pairs $\{\alpha,\beta\}$ of compositions satisfying $|\alpha|+|\beta|=n$, ordered by ... For an unordered pair $\{\alpha,\beta\}$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-073381 | MATHDB-380008

Density of hyperbolicity for rational maps

摘录 / EXCERPT: Let $f:\hat{\mathbb{C}}\rightarrow \hat{\mathbb{C}}$ be a rational map of degree $d\geqslant 2$, and let Rat_{d} denote the space of Möbius equivalence classes of ration...

conjecture | open | MathDB 页面 | 原文线索

V2-073382 | MATHDB-380019

Conjecture A on Kazhdan-Lusztig cells and domino insertion

摘录 / EXCERPT: Conjecture A. Assume that $\Gamma=\mathbb{Z}$, $a=2$ and $b=2r+1$. Then, for $w,w'\in W_n$, they lie in the same Kazhdan-Lusztig left cell if and only if $Q^r(w)=Q^r(w')$, in the s...

conjecture | open | MathDB 页面 | 原文线索

V2-073383 | MATHDB-380020

Conjecture A+ on Kazhdan-Lusztig cells for unequal parameters

摘录 / EXCERPT: Conjecture A+. Let $r\geqslant 0$ and assume that $ra<b<(r+1)a$. Then the Kazhdan-Lusztig left, right, and two-sided cells for these parameters coincide with those obtained for the...

conjecture | open | MathDB 页面 | 原文线索

V2-073384 | MATHDB-380021

Conjecture on Kazhdan-Lusztig cells at integral parameter ratios

摘录 / EXCERPT: Integral-ratio cell conjecture. Assume that $b=ra$ for some $r\geqslant 1$. Then the Kazhdan-Lusztig left, right, and two-sided cells of W_n are the smallest subsets of W_n tha...

conjecture | open | MathDB 页面 | 原文线索

V2-073385 | MATHDB-380022

Kleppe-Ellia's stable degeneracy conjecture for curves on cubic surfaces

摘录 / EXCERPT: Kleppe-Ellia's conjecture. If C is linearly normal, then every small deformation C' of C in $\mathbb P^3$ is contained in a cubic surface $S'\subset\mathbb P^3$; equivalently...

conjecture | open | MathDB 页面 | 原文线索

V2-073386 | MATHDB-380050

Homomorphism-preserving bijections of finite graphs are trivial

摘录 / EXCERPT: Homomorphism cancellation conjecture. If, for all graphs $G,H\in\mathcal{G}$,

open_problem | open | MathDB 页面 | 原文线索

V2-073387 | MATHDB-380054

Uniqueness of terminal transversals under $(213)\rightarrow(123)$ moves

摘录 / EXCERPT: Let Y be the Young diagram under consideration, let S_Y denote its transversals, and let $S_Y(213)$ denote the transversals avoiding the pattern (213). For a transversal...

conjecture | open | MathDB 页面 | 原文线索

V2-073388 | MATHDB-380055

Equidistribution of generic analytic subsets toward powers of the Green current

摘录 / EXCERPT: Equidistribution conjecture. For every generic H, one has

conjecture | open | MathDB 页面 | 原文线索

V2-073389 | MATHDB-380056

The scattering conjecture for mass-critical NLS

摘录 / EXCERPT: Scattering conjecture. In the defocusing case ($\mu=+1$) we have $m_0=+\infty$, while in the focusing case ($\mu=-1$) we have $m_0=M(Q)$. This predicts global well-posedness and sc...

conjecture | open | MathDB 页面 | 原文线索

V2-073390 | MATHDB-380057

The spherically symmetric scattering conjecture for mass-critical NLS

摘录 / EXCERPT: Spherically symmetric scattering conjecture. In the defocusing case ($\mu=+1$) we have $m_{0,\operatorname{rad}}=+\infty$, while in the focusing case ($\mu=-1$) we have...

conjecture | open | MathDB 页面 | 原文线索

V2-073391 | MATHDB-380061

Modularity conjecture for character-sheaf packets

摘录 / EXCERPT: The modularity conjecture. The subcategory ${\mathcal M}_e$ is closed under convolution and hence is a monoidal category with unit e if the preceding perversity conjecture holds;...

conjecture | open | MathDB 页面 | 原文线索

V2-073392 | MATHDB-380064

Character-sheaf basis conjecture for class functions

摘录 / EXCERPT: Let G_0 be any connected unipotent group over ${\mathbb F}_q$, let $G=G_0\otimes_{{\mathbb F}_q}{\mathbb F}$ with Frobenius Fr, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073393 | MATHDB-380078

Sufficiency of weak Hilbert-transform estimates in the absolutely continuous case

摘录 / EXCERPT: Absolutely continuous sufficiency conjecture. Under these assumptions, the conditions and are sufficient for A to be similar to a selfadjoint operator.

conjecture | open | MathDB 页面 | 原文线索

V2-073394 | MATHDB-380079

Krasikov's stronger conjecture for Jacobi polynomial extrema

摘录 / EXCERPT: Krasikov's stronger conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-073395 | MATHDB-380081

Guo-Zeng's gamma-positivity conjecture for involution descent polynomials

摘录 / EXCERPT: Guo-Zeng conjecture. For every $n\in\mathbb{N}$, there exist coefficients $a_{n,i}\in\mathbb{N}$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073396 | MATHDB-380082

The nonvanishing conjecture for projective klt pairs

摘录 / EXCERPT: Let (X,Δ) be a projective kawamata log terminal pair. Nonvanishing conjecture. If $K_X+\Delta$ is pseudo-effective, then $\kappa(X,K_X+\Delta)\geq 0$. This is identified as...

conjecture | open | MathDB 页面 | 原文线索

V2-073397 | MATHDB-380083

The positive Kodaira dimension conjecture for projective klt pairs

摘录 / EXCERPT: Let (X,Δ) be a projective kawamata log terminal pair. If $K_X+\Delta$ is pseudo-effective and, for some ample divisor H, ... is not a bounded function of m, then...

conjecture | open | MathDB 页面 | 原文线索

V2-073398 | MATHDB-380084

The abundance conjecture for projective klt pairs

摘录 / EXCERPT: Let (X,Δ) be a projective kawamata log terminal pair. Abundance conjecture. If $K_X+\Delta$ is nef, then it is semiample. The source describes this as one of the main outsta...

conjecture | open | MathDB 页面 | 原文线索

V2-073399 | MATHDB-380085

The log canonical model conjecture for big log canonical divisors

摘录 / EXCERPT: Let (X,Δ) be a projective log canonical pair of dimension n. Log canonical model conjecture. If $K_X+\Delta$ is big, then (X,Δ) has a log canonical model. The sourc...

conjecture | open | MathDB 页面 | 原文线索

V2-073400 | MATHDB-380090

p-parity conjecture for elliptic curves

摘录 / EXCERPT: p-parity conjecture. The p^∞-Selmer rank $\operatorname{rk}_p E/K$ is even if and only if $w(E/K)=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-073401 | MATHDB-380091

BSD regulator-quotient square conjecture for elliptic curves

摘录 / EXCERPT: BSD regulator-quotient square conjecture. Then

conjecture | open | MathDB 页面 | 原文线索

V2-073402 | MATHDB-380092

Conference-construction maximal-parameter conjecture for complex Hadamard matrices

摘录 / EXCERPT: Let D be an arbitrary dephased, symmetric complex Hadamard matrix arising from the conference matrix construction, of order N. Apply Construction by repeatedly selecting suitab...

conjecture | open | MathDB 页面 | 原文线索

V2-073403 | MATHDB-380118

Stanley's coloured factorization conjecture for symmetric groups

摘录 / EXCERPT: Let $[m]=\{1,2,\dots,m\}$, let $\mu\vdash k$, and let ω_μ be a fixed element of the conjugacy class C_μ in $\mathfrak{S}_k$. Let $\mathfrak{S}^{(m)}_k$ be the set of p...

conjecture | open | MathDB 页面 | 原文线索

V2-073404 | MATHDB-380123

The $K(\pi,1)$-conjecture for infinite Coxeter groups

摘录 / EXCERPT: Let W be an infinite Coxeter group, let Y_W be the associated hyperplane complement in $\mathbb{C}\otimes V$ restricted to the Tits cone, and let A be the associated Artin gr...

conjecture | open | MathDB 页面 | 原文线索

V2-073405 | MATHDB-380136

Reiner-Chapoton cyclic sieving conjecture for chains in dual Garside categories

摘录 / EXCERPT: Let W be a well-generated finite complex reflection group with degrees $d_1,\dots,d_n$ and Coxeter number h, let d be a regular degree, let W' be the centralizer of a d-r...

conjecture | open | MathDB 页面 | 原文线索

V2-073406 | MATHDB-380143

The vanishing conjecture for complex valued Ray-Singer torsion

摘录 / EXCERPT: Vanishing conjecture. We have

conjecture | open | MathDB 页面 | 原文线索

V2-073407 | MATHDB-380145

The power hyponormality conjecture for commuting weighted shifts

摘录 / EXCERPT: Power hyponormality conjecture. If

conjecture | open | MathDB 页面 | 原文线索

V2-073408 | MATHDB-380158

The limsup volume conjecture for non-splittable links

摘录 / EXCERPT: Limsup volume conjecture. For every non-splittable link L,

conjecture | open | MathDB 页面 | 原文线索

V2-073409 | MATHDB-380166

Asymptotic maximal determinant conjecture for Bernoulli matrices

摘录 / EXCERPT: Maximal determinant conjecture. Almost surely,

conjecture | open | MathDB 页面 | 原文线索

V2-073410 | MATHDB-380167

Symmetric Bernoulli determinant conjecture

摘录 / EXCERPT: Let Q_n be the random symmetric matrix whose upper-diagonal entries are independent Bernoulli random variables. Symmetric determinant conjecture. Almost surely, ... This is the sym...

conjecture | open | MathDB 页面 | 原文线索

V2-073411 | MATHDB-380168

Resilience conjecture for nonsingularity of Bernoulli matrices

摘录 / EXCERPT: For a $\{-1,1\}$ matrix M, let $\operatorname{Res}(M)$ be the minimum number of entries that must be switched from 1 to -1 or vice versa in order to make M singular. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073412 | MATHDB-380169

Diagonal-switching conjecture for Bernoulli matrices

摘录 / EXCERPT: Let M be a $\{-1,1\}$ matrix. Call M good if every matrix obtained by independently switching any subset of its diagonal entries is nonsingular. Let M_n be a random...

conjecture | open | MathDB 页面 | 原文线索

V2-073413 | MATHDB-380173

The conjecture on weak homogeneity and moderate Schwartz pairings

摘录 / EXCERPT: The conjecture. If $u\in\mathcal G(\mathbb R^d)$ is weakly homogeneous of degree α, then there is a representative $(u_\varepsilon)_\varepsilon$ of u such that, for every...

conjecture | open | MathDB 页面 | 原文线索

V2-073414 | MATHDB-380190

The Mazur-Merel version of the Strong Main Conjecture

摘录 / EXCERPT: Mazur-Merel version of the Strong Main Conjecture. Assuming the hypotheses of the Strong Main Conjecture, for all $p\notin P_{\mathbf C}$, only finitely many $V\in\mathcal V_p(J)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073415 | MATHDB-380193

The uniform non-modularity conjecture for the Hurwitz-space construction

摘录 / EXCERPT: Uniform non-modularity conjecture. The same conclusion holds for every $p\neq 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-073416 | MATHDB-380202

The minimal book embedding conjecture for graphs

摘录 / EXCERPT: Let G be a graph, and let $mnl(G)$ denote the minimum number of links among spatial embeddings of G. A book embedding is a spatial embedding in which the graph is embedded with...

conjecture | open | MathDB 页面 | 原文线索

V2-073417 | MATHDB-380205

Faber's intersection number conjecture in ψ-form

摘录 / EXCERPT: Let ${\mathcal M}_{g,n}^{rt}$ be the moduli space of stable n-pointed genus-g curves with rational tails, and let ... be the forgetful morphism. For positive integers...

conjecture | open | MathDB 页面 | 原文线索

V2-073418 | MATHDB-380207

Exponential lower subdegree growth implies continuum many permutation-ends

摘录 / EXCERPT: Let G be an infinite primitive permutation group whose subdegrees are all finite, and let (m_r) be its lower subdegree sequence. Permutation-ends conjecture. If (m_r) grows e...

conjecture | open | MathDB 页面 | 原文线索

V2-073419 | MATHDB-380216

The Maslov-index formula for immersed Lagrangian cylinders

摘录 / EXCERPT: Let C be an immersed Lagrangian cylinder between smoothly concordant Legendrian knots K_1 and K_2, constructed as in the immersion construction discussed in the paper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073420 | MATHDB-380236

Meromorphic theta-function conjecture for generalized $H^{q,k}$

摘录 / EXCERPT: Let $k,q\in\mathbb{N}$, $v\in\mathbb{C}$, and $\omega,\lambda\in\mathbb{R}_{>0}^2$. Let $H^{q,k}(z,v,\omega,\lambda)$ be the integral defined by ... where J and $\mathfrak{g}$ are...

conjecture | open | MathDB 页面 | 原文线索

V2-073421 | MATHDB-380238

The low-genus Kodaira-dimension conjecture for $\overline{\mathcal{M}}_g$

摘录 / EXCERPT: Let $\overline{\mathcal{M}}_g$ be the moduli space of stable curves of genus g, and let $\kappa(\overline{\mathcal{M}}_g)$ denote its Kodaira dimension. Low-genus Kodaira-dimensi...

conjecture | open | MathDB 页面 | 原文线索

V2-073422 | MATHDB-380240

The exact Stechkin constant conjecture at the critical uniform-metric scale

摘录 / EXCERPT: The exact Stechkin constant conjecture. For every $r\in\mathbb{N}$,

conjecture | open | MathDB 页面 | 原文线索

V2-073423 | MATHDB-380246

Conjecture on uniqueness of root-of-unity conformal-block solutions

摘录 / EXCERPT: Let $q=\exp(\pi i/\ell)$ with $\ell\ge3$, and let $\operatorname{Conf}_{\vec\Lambda}^{(\ell)}$ be the space spanned by the lattice restrictions of the Weyl anti-symmetrized hyperge...

conjecture | open | MathDB 页面 | 原文线索

V2-073424 | MATHDB-380247

The strengthened 2-criticality conjecture for row-isotoped abelian 2-groups

摘录 / EXCERPT: Strengthened 2-criticality conjecture. The set $\operatorname{gcs}(\alpha_{k,k'}L_s)$ is 2-critical and strong, and completes top down to $\alpha_{k,k'}L_s$.

conjecture | open | MathDB 页面 | 原文线索

V2-073425 | MATHDB-380255

Goda--Teragaito's order bound conjecture for hyperbolic knots with lens space surgeries

摘录 / EXCERPT: Let $K\subset S^3$ be a hyperbolic knot of genus g, and suppose that a surgery on K yields a lens space of order r. Goda--Teragaito's conjecture. The order satisfies ... The co...

conjecture | open | MathDB 页面 | 原文线索

V2-073426 | MATHDB-380256

Bleiler--Litherland's lower bound conjecture for hyperbolic lens space surgeries

摘录 / EXCERPT: Let $K\subset S^3$ be a hyperbolic knot, and suppose that a surgery on K yields a lens space of order r. Bleiler--Litherland's conjecture. The order satisfies ... The paper prove...

conjecture | open | MathDB 页面 | 原文线索

V2-073427 | MATHDB-380275

Reiner-Stanton-White cyclic sieving conjecture for facets of generalized cluster complexes

摘录 / EXCERPT: Let s be a positive integer, let Φ be a root system of rank n with Coxeter number h and exponents $e_1,\ldots,e_n$, and let X be the set of facets of the generalized c...

conjecture | open | MathDB 页面 | 原文线索

V2-073428 | MATHDB-380277

Margulis's conjecture on Hilbert modular lattices

摘录 / EXCERPT: Let G be the product of $n\geq 2$ copies of $\operatorname{SL}(2,\mathbb{R})$. Let U_1 and U_2 be the products of, respectively, upper- and lower-unipotent one-parameter subg...

conjecture | open | MathDB 页面 | 原文线索

V2-073429 | MATHDB-380280

The permanent real-rootedness conjecture

摘录 / EXCERPT: Let $A=(a_{ij})$ be a real $n\times n$ matrix with non-negative entries, and let J denote the all-ones matrix. Assume that the entries of A are weakly increasing down columns....

conjecture | open | MathDB 页面 | 原文线索

V2-073430 | MATHDB-380282

The scalar strong BMV coefficient conjecture

摘录 / EXCERPT: The scalar strong BMV coefficient conjecture. Every f_i belongs to $\mathbf{P}^{\mathrm{pos}}$, and consecutive coefficients satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073431 | MATHDB-380283

The composition conjecture

摘录 / EXCERPT: The composition conjecture. Every coefficient f_i belongs to $\overline{\mathbf{P}}_d$, and

conjecture | open | MathDB 页面 | 原文线索

V2-073432 | MATHDB-380284

The weighted derivative-substitution conjecture

摘录 / EXCERPT: The weighted derivative-substitution conjecture. This mapping determines a map

conjecture | open | MathDB 页面 | 原文线索

V2-073433 | MATHDB-380285

The Faà di Bruno real-rootedness conjecture

摘录 / EXCERPT: The Faà di Bruno real-rootedness conjecture. For every $\alpha\in\mathbb{R}$ and $m=0,1,\dots$, $F_m(\alpha,y)$ has only real roots and

conjecture | open | MathDB 页面 | 原文线索

V2-073434 | MATHDB-380286

The totally stable matrix conjecture

摘录 / EXCERPT: Let $f(x,y)=\sum f_i(x)y^i\in\mathbf{P}^{\mathrm{pos}}_2$, and form the upper-triangular coefficient matrix ... A minor means the determinant of any finite square submatrix. The tota...

conjecture | open | MathDB 页面 | 原文线索

V2-073435 | MATHDB-380289

Shifted Schur Q-function lattice positivity conjecture

摘录 / EXCERPT: Let λ/μ and ν/ρ be skew shifted shapes. For partitions, define componentwise operations ... and extend them to skew shifted shapes by...

conjecture | open | MathDB 页面 | 原文线索

V2-073436 | MATHDB-380290

Shifted Fomin-Fulton-Li-Poon Schur Q-positivity conjecture

摘录 / EXCERPT: For two shifted shapes λ and μ, let $\lambda\cup\mu$ be the partition obtained by rearranging all parts of λ and μ in weakly decreasing order, and write ... D...

conjecture | open | MathDB 页面 | 原文线索

V2-073437 | MATHDB-380293

Gap conjecture for the circular chromatic index

摘录 / EXCERPT: Gap conjecture for the circular chromatic index. For every integer $k\geq 2$, there exists an $\epsilon>0$ such that no graph G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073438 | MATHDB-380295

Burns's strengthened Rubin-Stark congruence

摘录 / EXCERPT: Let I be the augmentation ideal of $\mathbb{Z}[\Gamma]$, let $epsilon$ be the element supplied by the Rubin-Stark conjecture, and let $Phi\in\iota(Lambda^n U^*)$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073439 | MATHDB-380303

Campana's conjecture on soft integral points of constellation curves

摘录 / EXCERPT: Let $(Y/\boldsymbol{\Delta})$ be a smooth projective Campana constellation supported by a firmament $\boldsymbol{\Gamma}$. Campana's conjecture. Points on Y integral with respect...

conjecture | open | MathDB 页面 | 原文线索

V2-073440 | MATHDB-380321

The classical Schottky conjecture for Kleinian groups with small Hausdorff dimension

摘录 / EXCERPT: Let Γ be a Kleinian group with limit set $\Lambda(\Gamma)$ of Hausdorff dimension less than 1. The classical Schottky conjecture. The group Γ is a Schottky-type gro...

conjecture | open | MathDB 页面 | 原文线索

V2-073441 | MATHDB-380322

The small-limit-set geometric finiteness conjecture

摘录 / EXCERPT: Let Γ be a finitely generated subgroup of $\operatorname{Mob}(\mathbb S^n)$ such that its limit set $\Lambda(\Gamma)$ has Hausdorff dimension less than 2. Small-limit-set...

conjecture | open | MathDB 页面 | 原文线索

V2-073442 | MATHDB-380345

The core conjecture for qualitative independence graphs

摘录 / EXCERPT: Core conjecture. For any positive integer k, the graph $QI(k^2,k)$ is a core.

conjecture | open | MathDB 页面 | 原文线索

V2-073443 | MATHDB-380354

The maximal-rigid-to-cluster-tilting conjecture

摘录 / EXCERPT: The maximal-rigid-to-cluster-tilting conjecture. Any maximal rigid object whose quiver has no loops or 2-cycles is a cluster tilting object.

conjecture | open | MathDB 页面 | 原文线索

V2-073444 | MATHDB-380362

The distance bound for loops with nonequivalent associator

摘录 / EXCERPT: Let $L=(G,\circ)$ and $M=(G,*)$ be Moufang loops on the same set G of order n, and define their distance by ... Suppose that L and M have nonequivalent associators. Distance-...

conjecture | open | MathDB 页面 | 原文线索

V2-073445 | MATHDB-380366

The hsop existence conjecture for well-generated reflection groups

摘录 / EXCERPT: The hsop existence conjecture. If $p\equiv 1\pmod h$, then S contains an hsop $\mathbf{\Theta}$ carrying $V^*(-p)$; if $p\equiv -1\pmod h$, then S contains an hsop...

conjecture | open | MathDB 页面 | 原文线索

V2-073446 | MATHDB-380367

Bessis's cyclic sieving conjecture for the modified action on m-divisible noncrossing partitions

摘录 / EXCERPT: Bessis's modified-action CSP conjecture. The triple

conjecture | open | MathDB 页面 | 原文线索

V2-073447 | MATHDB-380374

The conjecture on infinite essential dimension of elliptic curves

摘录 / EXCERPT: Let E be an elliptic curve over a number field. Its essential dimension is denoted by $\operatorname{ed}E$. Infinite essential-dimension conjecture for elliptic curves. Then ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-073448 | MATHDB-380384

Clustering distribution on the two-dimensional integer lattice

摘录 / EXCERPT: Let $D\subset\mathbb R^d$ be open, bounded, and convex, and let $\mathcal W$ be a renormalization class on $\overline D$. Assume that the asymptotic fixed-point equation has a uniq...

conjecture | open | MathDB 页面 | 原文线索

V2-073449 | MATHDB-380390

Conjecture on reconstruction from transposition-neighborhoods

摘录 / EXCERPT: Let $G=\operatorname{Sym}_n$ and let T be the set of transpositions, so that $\operatorname{Sym}_n(T)$ is the corresponding Cayley graph, $B_r(\sigma)$ denotes the radius-r bal...

conjecture | open | MathDB 页面 | 原文线索

V2-073450 | MATHDB-380394

The equivariant bundle conjecture for Hecke operators

摘录 / EXCERPT: Equivariant bundle conjecture. There exists an F-equivariant vector bundle $(\mathcal E,g)$ of rank N, defined over $\mathbb Q$, such that for $z\in X_n$ the eigenvalue of...

conjecture | open | MathDB 页面 | 原文线索

V2-073451 | MATHDB-380397

The two-dimensional Weil conjecture for lattice partition functions

摘录 / EXCERPT: Two-dimensional Weil conjecture. For every endomorphism $M\in\operatorname{End}_{\mathcal C_{\mathbb F_q}}(X)$ there exist super Boltzmann data (V_1,V_2,R) such that, for every f...

conjecture | open | MathDB 页面 | 原文线索

V2-073452 | MATHDB-380400

Minimal-period conjecture for coefficients of restricted-partition quasi-polynomials

摘录 / EXCERPT: Minimal-period conjecture. The minimum period of $c_j(k)$ is p_j for every $0\leq j\leq d$. This conjecture predicts that the divisibility data of the a_i determine the exact p...

conjecture | open | MathDB 页面 | 原文线索

V2-073453 | MATHDB-380401

Janicki and Woyczynski's Hausdorff-dimension conjecture for Lagrangian regular points

摘录 / EXCERPT: Janicki and Woyczynski's conjecture. For every $t>0$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-073454 | MATHDB-380402

Shi's first-passage tail conjecture for integrated stable processes

摘录 / EXCERPT: Shi's conjecture. Suppose that Z is symmetric and that $\alpha>1$. Then

conjecture | open | MathDB 页面 | 原文线索

V2-073455 | MATHDB-380403

The integrated stable-process first-passage tail conjecture

摘录 / EXCERPT: Integrated stable-process tail conjecture. One has

conjecture | open | MathDB 页面 | 原文线索

V2-073456 | MATHDB-380413

Existence of exponential growth rates for hereditary ordered graph properties

摘录 / EXCERPT: Exponential growth-rate conjecture. The limit

conjecture | open | MathDB 页面 | 原文线索

V2-073457 | MATHDB-380414

The smallest speed above 2 conjecture

摘录 / EXCERPT: Smallest-speed-above-2 conjecture. The smallest $c\in\mathcal{S}$ with $c>2$ is the largest real root of

conjecture | open | MathDB 页面 | 原文线索

V2-073458 | MATHDB-380415

The no-accumulation-from-above conjecture for ordered graph speeds

摘录 / EXCERPT: No-accumulation-from-above conjecture. For every $c\in\mathbb{R}$, there exists $\varepsilon=\varepsilon(c)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073459 | MATHDB-380421

The conjectured bases for hereditary properties of tournaments up to 1.755

摘录 / EXCERPT: Conjecture on the possible bases.

conjecture | open | MathDB 页面 | 原文线索

V2-073460 | MATHDB-380422

The polynomial-to-factorial speed jump for hereditary properties of oriented graphs

摘录 / EXCERPT: Conjecture on oriented-graph speeds. This predicts a jump from polynomial to at least factorial speed for unlabelled hereditary properties of oriented graphs. The paper poses it as...

conjecture | open | MathDB 页面 | 原文线索

V2-073461 | MATHDB-380429

Averaged moment inequality for conformal-centroid compact continua

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be compact and connected, with logarithmic capacity $\operatorname{cap}(K)=1$, $0\in K$, and conformal centroid at the origin. Let g be the Green's funct...

conjecture | open | MathDB 页面 | 原文线索

V2-073462 | MATHDB-380430

Convex-function moment inequality for conformal-centroid compact continua

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be compact and connected, with logarithmic capacity $\operatorname{cap}(K)=1$, $0\in K$, and conformal centroid at the origin. Let μ_K be the equilibri...

conjecture | open | MathDB 页面 | 原文线索

V2-073463 | MATHDB-380432

Farthest-point norm conjecture for logarithmic-capacity continua

摘录 / EXCERPT: Let $K\subset\mathbb{C}$ be compact and connected with more than one point. Let M_K be the smallest number such that, whenever p has degree $n\geq1$ and $p=p_1\cdots p_m$, ... He...

conjecture | open | MathDB 页面 | 原文线索

V2-073464 | MATHDB-380435

The binary-matrix finiteness conjecture

摘录 / EXCERPT: Binary-matrix finiteness conjecture. Every pair of binary matrices has the finiteness property.

conjecture | open | MathDB 页面 | 原文线索

V2-073465 | MATHDB-380448

The orientation-preserving Hodge-isometry conjecture for K3 autoequivalences

摘录 / EXCERPT: Let X be a K3 surface. Let ... be the homomorphism induced by the action of Fourier--Mukai autoequivalences on the Mukai lattice, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073466 | MATHDB-380467

Shibuya's cobordism conjecture for self Delta-equivalence

摘录 / EXCERPT: Let oriented links be related by self Δ-equivalence when one can be transformed into the other by self delta moves and ambient isotopies, where each self delta move has all...

conjecture | open | MathDB 页面 | 原文线索

V2-073467 | MATHDB-380468

Shibuya's boundary-link conjecture for self Delta-equivalence

摘录 / EXCERPT: Let an oriented link be a boundary link if its components bound pairwise disjoint Seifert surfaces in S^3. Let the trivial link be the link whose components are disjoint unknotte...

conjecture | open | MathDB 页面 | 原文线索

V2-073468 | MATHDB-380479

Révész-Sarantopoulos conjecture on the Bernstein factor for convex bodies

摘录 / EXCERPT: Let K be a convex body in a normed space, let $x\in\operatorname{int}K$, and let $B_n(K,x)$ denote the Bernstein factor at x for polynomials of degree at most n. Write $w(K)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073469 | MATHDB-380480

Global divisibility conjecture for Heegner points and Tamagawa numbers

摘录 / EXCERPT: Let $E_{/\mathbb{Q}}$ be an elliptic curve of conductor N, let K be the quadratic imaginary field associated with a Heegner discriminant, and let $y_K\in E(K)$ be the correspon...

conjecture | open | MathDB 页面 | 原文线索

V2-073470 | MATHDB-380481

Batyrev-Borisov stringy E-function conjecture for Gorenstein polytopes

摘录 / EXCERPT: Let Δ be a Gorenstein polytope of CY-dimension n, so that its stringy E-function $E_{\rm st}(\Delta;u,v)$ and CY-dimension are defined as in the preceding discussion. Stri...

conjecture | open | MathDB 页面 | 原文线索

V2-073471 | MATHDB-380490

Joyce's stability-independent counting invariant conjecture for K3 and abelian surfaces

摘录 / EXCERPT: Joyce's conjecture. For every $\sigma\in\operatorname{Stab}^{\ast}(X)$ and $\alpha\in N(X)$, there is an invariant $J^{\alpha}(\sigma)\in\Lambda$ such that $J^{\alpha}(\sigma)$ doe...

conjecture | open | MathDB 页面 | 原文线索

V2-073472 | MATHDB-380511

Universality conjecture for Hausdorff dimension of cubic critical circle maps

摘录 / EXCERPT: Let μ be the unique normalized invariant measure of a cubic critical homeomorphism of the circle, and let $HD(\mu)$ denote the Hausdorff dimension of its measure-theoretical su...

conjecture | open | MathDB 页面 | 原文线索

V2-073473 | MATHDB-380526

The density conjecture for the family lambda tan z

摘录 / EXCERPT: Let ${\cal F}$ be the holomorphic family defined over ${\bf C}-\{0\}$, and let J-stable maps be the maps whose Julia sets vary stably in this family. The Density Conjecture. The hy...

conjecture | open | MathDB 页面 | 原文线索

V2-073474 | MATHDB-380527

The invariant measurable line-field formulation of the density conjecture

摘录 / EXCERPT: Let ${\cal F}$ be the holomorphic family, let B be a disk in the set of quasiconformally stable parameters, and let J_λ denote the Julia set of the corresponding map. An...

conjecture | open | MathDB 页面 | 原文线索

V2-073475 | MATHDB-380531

Rational period conjecture for quadratic polynomials

摘录 / EXCERPT: Let N be an integer with $N\geq 4$, and let $f(z)\in\mathbb{Q}[z]$ be a quadratic polynomial. A point is of exact period N if its forward orbit under f has least period N....

conjecture | open | MathDB 页面 | 原文线索

V2-073476 | MATHDB-380546

Real-part conjecture for unbalanced truncated affine arrangements

摘录 / EXCERPT: Let R be an irreducible root system in $\mathbb{R}^n$, and let a and b be integers with $a+b\geq 2$. Define the truncated R-affine arrangement ${\cal A}^{ab}(R)$ by the hyp...

conjecture | open | MathDB 页面 | 原文线索

V2-073477 | MATHDB-380561

Arctic ellipse conjecture for random lozenge tilings

摘录 / EXCERPT: Consider a lozenge tiling of a hexagon with normalized side lengths α,β,γ, and define the arctic region to be the set of lozenges connected to the boundary by seque...

conjecture | open | MathDB 页面 | 原文线索

V2-073478 | MATHDB-380562

Mirror-family conjecture for the Pfaffian and Grassmannian Calabi-Yau sections

摘录 / EXCERPT: Mirror-family conjecture. The pair M_y+W_y is the mirror family of X_A+Y_A, with M_y and W_y forming different parts of the B-model moduli space and X_A and Y_A forming...

conjecture | open | MathDB 页面 | 原文线索

V2-073479 | MATHDB-380564

Malcev completion conjecture for the kernel of the evaluation map

摘录 / EXCERPT: Let $R={\mathbb Z}[t]$ or $R={\mathbb Z}[t,t^{-1}]$, let $K(R)$ be the kernel of the evaluation map $SL_2(R)\to SL_2({\mathbb Z})$, and let ${\mathcal W}$ be the kernel of the indu...

conjecture | open | MathDB 页面 | 原文线索

V2-073480 | MATHDB-380586

The generalized Kostka-Foulkes and ribbon-tableau polynomial conjecture

摘录 / EXCERPT: Let $R=(R_1,\ldots,R_p)$ be a dominant sequence of rectangular partitions, and let Λ be the partition with empty p-core and p-quotient $(R_1,\ldots,R_p)$. The polynomia...

conjecture | open | MathDB 页面 | 原文线索

V2-073481 | MATHDB-380587

Positivity conjecture for the cocharge-normalized generalized Kostka polynomial

摘录 / EXCERPT: For a sequence of rectangles R, let $r_{i,j}(R)$ be the number of rectangles containing (i,j), and define ... Set $\widetilde{K}_{\lambda;R}(q)=q^{n(R)}K_{\lambda;R}(q^{-1})$. Co...

conjecture | open | MathDB 页面 | 原文线索

V2-073482 | MATHDB-380588

Transpose symmetry conjecture for generalized Kostka polynomials

摘录 / EXCERPT: Let R be a dominant sequence of rectangles, and let R^t be obtained by transposing every rectangle in R. Let R' be a dominant rearrangement of R^t, and let m be the tot...

conjecture | open | MathDB 页面 | 原文线索

V2-073483 | MATHDB-380589

Statistic-preserving rigged-configuration bijection conjecture

摘录 / EXCERPT: Let R be a dominant sequence of rectangles. Let $LRT(\lambda;R)$ be the corresponding Littlewood-Richardson tableaux, let $RC(\lambda^t;R^t)$ be the rigged configurations, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-073484 | MATHDB-380590

Image conjecture for catabolizable tableaux

摘录 / EXCERPT: Let R be a sequence of rectangles. Let $b_R:RC(\lambda^t;R^t)\to CST(\lambda,\gamma(R))$ be the restriction of the inverse rigged-configuration bijection to the indicated subset,...

conjecture | open | MathDB 页面 | 原文线索

V2-073485 | MATHDB-380591

Rectangle-reordering symmetry conjecture for LR tableaux

摘录 / EXCERPT: Let R be a sequence of rectangles and let u be a permutation of the rectangle indices. The bijection $u_R:LRT(\lambda;R)\to LRT(\lambda;uR)$ is defined by composing the adjacen...

conjecture | open | MathDB 页面 | 原文线索

V2-073486 | MATHDB-380592

Evacuation-rigged-configuration compatibility conjecture

摘录 / EXCERPT: Let ev be evacuation on column-strict tableaux, restricting to a bijection $LRT(\lambda;R)\to LRT(\lambda;\operatorname{rev}(R))$. Let θ complement each rig...

conjecture | open | MathDB 页面 | 原文线索

V2-073487 | MATHDB-380593

Duality compatibility conjecture for LR and rigged-configuration bijections

摘录 / EXCERPT: Let R be a sequence of rectangles, let $\widetilde{R}$ and $\widetilde{\lambda}$ be the complementary dual data defined in the source, and let dual be the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-073488 | MATHDB-380594

Column-catabolizable image conjecture

摘录 / EXCERPT: Let R be a sequence of rectangles such that R^t is dominant. Let b_R be the map from rigged configurations to tableaux, and let $CCT(\lambda;R)$ denote the set of R-column-...

conjecture | open | MathDB 页面 | 原文线索

V2-073489 | MATHDB-380595

Transpose bijection conjecture for catabolizable tableaux

摘录 / EXCERPT: Let R be a sequence of rectangles, and let $CT(\lambda;R)$ and $CCT(\lambda^t;R^t)$ denote the row- and column-catabolizable tableaux defined in the source. Catabolizable transpo...

conjecture | open | MathDB 页面 | 原文线索

V2-073490 | MATHDB-380596

Path-independence conjecture for monotonicity embeddings

摘录 / EXCERPT: Let $R\triangleleftarrow R'$ be a covering relation in the dominance order on sequences of rectangles. Let $\theta_{R'}:LRT(\lambda;R)\to LRT(\lambda;R')$ be the embedding obtained...

conjecture | open | MathDB 页面 | 原文线索

V2-073491 | MATHDB-380606

Transitive monodromy conjecture for rational curves on primitive K3 surfaces

摘录 / EXCERPT: Let $S\subset {\mathbb P}^n$ be a primitive K3 surface, and let $d>0$. Consider the family of rational curves in the linear system $|{\mathcal O}_S(d)|$. Transitive monodromy conje...

conjecture | open | MathDB 页面 | 原文线索

V2-073492 | MATHDB-380607

Nodal and transverse intersection conjecture for trigonal K3 surfaces

摘录 / EXCERPT: Let S be a surface given in ${\bf{TK1}}$, ${\bf{TK2}}$, or ${\bf{TK3}}$. Let F and C denote the divisor classes used in these trigonal K3 surface constructions, and let l b...

conjecture | open | MathDB 页面 | 原文线索

V2-073493 | MATHDB-380615

Obsolescence of condition 1 in the inverse spectral theorem

摘录 / EXCERPT: Obsolescence conjecture. Condition 1 in Theorem 4.1 is obsolete in general, meaning that the theorem should remain valid without this condition.

conjecture | open | MathDB 页面 | 原文线索

V2-073494 | MATHDB-380617

Vanishing conjecture for L^2 harmonic forms on Hitchin's moduli space

摘录 / EXCERPT: Let ${\cal M}$ be Hitchin's moduli space of rank-2 Higgs bundles with fixed determinant of degree 1 over a Riemann surface Σ of genus $g>1$. It is a simply connected, no...

conjecture | open | MathDB 页面 | 原文线索

V2-073495 | MATHDB-380633

The matrix-kernel approximation conjecture for residual group towers

摘录 / EXCERPT: Matrix-kernel approximation conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073496 | MATHDB-380637

Drisko's uniqueness conjecture for transversal-free row-Latin rectangles

摘录 / EXCERPT: Drisko's conjecture. Either A has a transversal, or A can be transformed into $R_{2n-2,n}$ by permuting rows, columns, and symbols.

conjecture | open | MathDB 页面 | 原文线索

V2-073497 | MATHDB-380639

Thomason-type colimit conjecture for Segal n-categories

摘录 / EXCERPT: Let Y be a category and let A be a presheaf of Segal n-categories over Y^o. Let $horiz$ be the set of horizontal arrows in $\int_Y A$, namely the arrows from y to...

conjecture | open | MathDB 页面 | 原文线索

V2-073498 | MATHDB-380640

Strictification conjecture for weak sections of a left Quillen presheaf

摘录 / EXCERPT: Let Y be a small category and let $\mathbf{M}$ be a left Quillen presheaf on Y. Suppose that $\mathbf{M}$ satisfies hypothesis (o), introduced in Section 19, which in particula...

conjecture | open | MathDB 页面 | 原文线索

V2-073499 | MATHDB-380668

Comparison conjecture for strict non-unital and weak n-categories

摘录 / EXCERPT: Comparison conjecture for snucategories. The localization of the category of n-snucategories by equivalences is equivalent to the localizations of the categories of weak n-cate...

conjecture | open | MathDB 页面 | 原文线索

V2-073500 | MATHDB-380674

Chan-Robbins-Yuen volume conjecture for the polytope of restricted permutation matrices

摘录 / EXCERPT: Chan-Robbins-Yuen volume conjecture. The relative volume of P_n is

conjecture | open | MathDB 页面 | 原文线索

V2-073501 | MATHDB-380718

Unimodality conjecture for tableau types and the statistic $a_{(m2^a1^b)}$

摘录 / EXCERPT: Unimodality conjecture. For each fixed $(m2^a1^b)$-type (S,s), the sequence

conjecture | open | MathDB 页面 | 原文线索

V2-073502 | MATHDB-380730

The torus ramification polynomial conjecture

摘录 / EXCERPT: Let $m\ge 1$, let $\alpha=(\alpha_1,\ldots,\alpha_m)$ be a ramification type, and let $n=\alpha_1+\cdots+\alpha_m$. For $i=1,\ldots,m$, let e_i be the i-th elementary symmetric...

conjecture | open | MathDB 页面 | 原文线索

V2-073503 | MATHDB-380731

The degree conjecture for torus and higher-genus ramification polynomials

摘录 / EXCERPT: For $g=0$ and $m\ge 3$, or $g\ge 1$ and $m\ge 1$, let $f_m^{(g)}(\alpha_1,\ldots,\alpha_m)$ denote the rescaled ramification number associated with coverings of the sphere by a gen...

conjecture | open | MathDB 页面 | 原文线索

V2-073504 | MATHDB-380732

The closed formula conjecture for genus-one ramification polynomials

摘录 / EXCERPT: For $m\ge 1$, let e_i denote the i-th elementary symmetric function in the variables used to define $f_m^{(1)}$. The closed formula conjecture for genus-one ramification polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-073505 | MATHDB-380742

The central formal-series conjecture for shifted symmetric polynomials

摘录 / EXCERPT: Conjecture centrale. Pour tout entier $n\geq1$,

conjecture | open | MathDB 页面 | 原文线索

V2-073506 | MATHDB-380743

The full-measure basin conjecture for the 20-point orbit of h_{11}

摘录 / EXCERPT: Let h_{11} be the map under discussion on ${\mathbf{CP}^{3}}$, and let the 20-point orbit denote the orbit of the octahedral 20-points. A basin is the set of points whose ite...

conjecture | open | MathDB 页面 | 原文线索

V2-073507 | MATHDB-380744

The Julia-set conjecture for the real ${\mathcal S}_{5}$-invariant ${\mathbf{RP}^{3}}$ of h_{11}

摘录 / EXCERPT: Let h_{11} be the map under discussion, and let ${\mathcal R}$ be the ${\mathcal S}_{5}$-symmetric real projective subspace ${\mathbf{RP}^{3}}$ preserved by h_{11}. The Julia s...

conjecture | open | MathDB 页面 | 原文线索

V2-073508 | MATHDB-380745

The full-measure basin conjecture for the 5-point orbit of f_{6}

摘录 / EXCERPT: Let f_{6} be the map under discussion on ${\mathbf{CP}^{3}}$, and let the 5-point orbit denote the orbit of the 5-points. The basins of an attracting orbit are the sets of po...

conjecture | open | MathDB 页面 | 原文线索

V2-073509 | MATHDB-380756

The canonicality conjecture for the weight systems W_n

摘录 / EXCERPT: Let W_n be the weight system defined in the paper, let Z denote the Kontsevich integral, and let ∇_n denote the degree-n coefficient of the multivariable Conway potent...

conjecture | open | MathDB 页面 | 原文线索

V2-073510 | MATHDB-380757

The repelling 45-lines conjecture for the sextic map h_{19}

摘录 / EXCERPT: Repelling 45-lines conjecture. On the 45-lines h_{19} is repelling and, hence, $\{X=0\}$ resides in the Julia set $J_{h_{19}}$.

conjecture | open | MathDB 页面 | 原文线索

V2-073511 | MATHDB-380774

Shapiro-Shapiro conjecture for totally positive flags

摘录 / EXCERPT: Let $m,p>1$, let $\operatorname{Grass}(p,m+p)$ be the Grassmannian, and let $\alpha^1,\ldots,\alpha^n$ be Schubert data. A real upper-triangular matrix with diagonal entries 1 is...

conjecture | open | MathDB 页面 | 原文线索

V2-073512 | MATHDB-380780

The QALE Laplacian index-set and Fredholm conjecture

摘录 / EXCERPT: The QALE Laplacian index-set and Fredholm conjecture. The displayed description of $\mathcal{I}_X$ holds, and the displayed map is Fredholm, with finite-dimensional kernel and coke...

conjecture | open | MathDB 页面 | 原文线索

V2-073513 | MATHDB-380782

Integrality criterion for inverses of reciprocal binomial Hankel matrices

摘录 / EXCERPT: For a positive integer r, let $b_k(r)=\binom{k+r-1}{r}$, and let $R_n(b_k(r))$ denote the corresponding reciprocal Hankel matrix. Integrality criterion conjecture. The inverse of...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073514 | MATHDB-380783

Fibonomial reciprocal Hankel inverse formula conjecture

摘录 / EXCERPT: Let F_n be the nth Fibonacci number, let $e(n,i,k)$ be the exponent used to determine the sign in the source, and let $d_k(r)$ be the Fibonomial coefficient corresponding to...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073515 | MATHDB-380786

The inverse-limit homeomorphism conjecture for substitution complexes

摘录 / EXCERPT: Let $\mathcal{M}$ be the class of maps under consideration, let $\operatorname{invlim} f$ denote the inverse limit of f, let χ_f be the substitution associated with f, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073516 | MATHDB-380787

The substitution equivalence conjecture for substitution tiling spaces

摘录 / EXCERPT: Let $\mathcal{T}_{\chi}$ and $\mathcal{T}_{\psi}$ be the tiling spaces associated with substitutions χ and ψ, and let $\tilde{\chi}$ and $\tilde{\psi}$ denote the associa...

conjecture | open | MathDB 页面 | 原文线索

V2-073517 | MATHDB-380789

The K4 subgraph conjecture for uniquely 3-list-colorable planar graphs

摘录 / EXCERPT: K4 subgraph conjecture. Every U3LC planar graph has K_4 as a subgraph.

conjecture | open | MathDB 页面 | 原文线索

V2-073518 | MATHDB-380800

The uniquely list chromatic number bound and equality characterization

摘录 / EXCERPT: For a graph G and a positive integer k, let $\chi_u(G,k)$ be the minimum number t such that G is uniquely (k,t)-list colorable, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073519 | MATHDB-380803

The separable-extension conjecture for characteristic-p L-series

摘录 / EXCERPT: Let $L(s)$, with $s=(x,y)\in S_\infty$, be a characteristic-p L-series arising from arithmetic, and let $\mathbf K_L$ be the finite extension of $\mathbf K_{\mathbf V}$ generat...

conjecture | open | MathDB 页面 | 原文线索

V2-073520 | MATHDB-380804

The characteristic-p absolute-value conjecture for reciprocal zeroes

摘录 / EXCERPT: Let $L(x,y)=\prod_i(1-\beta_i^{(y)}/x)$, where the nonzero $\beta_i^{(y)}$ are the zeroes of $L(x,y)$, and let $\lambda_i^{(y)}:=1/\beta_i^{(y)}$ be the reciprocal zeroes. The char...

conjecture | open | MathDB 页面 | 原文线索

V2-073521 | MATHDB-380814

The weight-one rational-function conjecture for quotient singularities with crepant resolutions

摘录 / EXCERPT: Let V be a vector space and let $G\subset SL(V)$ be finite. Set $X=V/G$, and assume that X admits a smooth crepant resolution $Y\to X$. For a cyclic subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-073522 | MATHDB-380817

Equality of the Eulerian and doubled half-Eulerian flag-vector cones

摘录 / EXCERPT: Cone equality conjecture. The two cones are equal:

conjecture | open | MathDB 页面 | 原文线索

V2-073523 | MATHDB-380818

The half-Eulerian inequality-generation conjecture

摘录 / EXCERPT: Half-Eulerian inequality-generation conjecture. Every linear form that is nonnegative on the flag vectors of all half-Eulerian posets is the sum of a linear form that is nonnegativ...

conjecture | open | MathDB 页面 | 原文线索

V2-073524 | MATHDB-380819

Finite generation of the Eulerian flag-vector cone

摘录 / EXCERPT: Finite-generation conjecture. For every positive integer n, this closed cone is finitely generated.

conjecture | open | MathDB 页面 | 原文线索

V2-073525 | MATHDB-380824

Bijective decomposition conjecture for Schur-function Plücker relations

摘录 / EXCERPT: Let λ be a partition with n outside corners. Choose a corner k and corresponding weight ω_ℓ, and retain the notions of Theorem ... 1leq i1<cdots<irleq kleq jr<c...

conjecture | open | MathDB 页面 | 原文线索

V2-073526 | MATHDB-380838

The linear repetitivity conjecture for dense repetitivity

摘录 / EXCERPT: The linear repetitivity conjecture. Every aperiodic linearly repetitive Delone set is densely repetitive.

conjecture | open | MathDB 页面 | 原文线索

V2-073527 | MATHDB-380839

The patch-counting lower-bound conjecture for aperiodic Delone sets

摘录 / EXCERPT: The patch-counting lower-bound conjecture. For every dimension $n\geq 1$ and Delone constants (r,R), there is a positive constant $c=c(n,r,R)$ such that every aperiodic Delone se...

conjecture | open | MathDB 页面 | 原文线索

V2-073528 | MATHDB-380844

Abstract Birkhoff-regularity conjecture for completely continuous operators

摘录 / EXCERPT: Let G be an abstract completely continuous operator, let $Lambda:=\left\{\lambda_j\right\}$ be the set of its eigenvalues, and fix $delta>0$. For each $lambda$, define the circle...

conjecture | open | MathDB 页面 | 原文线索

V2-073529 | MATHDB-380855

Goulden-Jackson's structural conjecture for higher-genus Hurwitz generating series

摘录 / EXCERPT: Let ... where i is an integer, and let s be the unique formal power series in x and p satisfying ... For the Hurwitz generating series $H_g(x,p)$, and for a partition θ...

conjecture | open | MathDB 页面 | 原文线索

V2-073530 | MATHDB-380861

The q-tangent continued-fraction conjecture for parameter tuple $(1,0,1,0)$

摘录 / EXCERPT: Continued-fraction conjecture. The positive powers of q in this continued fraction follow the sequence $0,1,6,15,\dots$, given by $k(2k-1)$, while the negative powers follow...

conjecture | open | MathDB 页面 | 原文线索

V2-073531 | MATHDB-380862

The q-tangent continued-fraction conjecture for parameter tuple $(1,1,1,-1)$

摘录 / EXCERPT: Continued-fraction conjecture. The positive powers of q follow the sequence $0,3,10,21,\dots$, given by $(k-1)(2k-1)$, while the negative powers follow $0,5,14,27,\dots$, given b...

conjecture | open | MathDB 页面 | 原文线索

V2-073532 | MATHDB-380877

The generalized Bethe-Sommerfeld conjecture for orbifold Schrödinger operators

摘录 / EXCERPT: Let Γ be the orbifold fundamental group of a compact good orbifold, acting on its universal orbifold cover $\widetilde{M}$, and let $H_V=\Delta+V$ be a Γ-invariant Sc...

conjecture | open | MathDB 页面 | 原文线索

V2-073533 | MATHDB-380900

Irreducibility conjecture for geometric Eisenstein series

摘录 / EXCERPT: Irreducibility conjecture. Moreover, if ${\mathcal S}$ is irreducible, then so is $\Eis({\mathcal S})$.

conjecture | open | MathDB 页面 | 原文线索

V2-073534 | MATHDB-380903

Uniqueness and convexity conjecture for optimal configurations

摘录 / EXCERPT: Let Ω be a convex domain, and let $D=D_{\alpha,A}$ be an optimal configuration in Ω. Write D^c for its complement and let $\overline{\alpha}_\Omega(A)$ denote the t...

conjecture | open | MathDB 页面 | 原文线索

V2-073535 | MATHDB-380904

Regularity conjecture for the free boundary in dimension two

摘录 / EXCERPT: Let D be an optimal configuration in a planar domain, and let ∂D denote its free boundary. Two-dimensional free-boundary regularity conjecture. In dimension two, the fr...

conjecture | open | MathDB 页面 | 原文线索

V2-073536 | MATHDB-380905

One-component conjecture for the complement in dumbbell domains

摘录 / EXCERPT: Let Ω_h be a dumbbell domain, let $D=D_{\alpha,A}$ be an optimal configuration, and let ψ_{Ω} be the function whose maxima determine the relevant localization regi...

conjecture | open | MathDB 页面 | 原文线索

V2-073537 | MATHDB-380906

Symmetry-breaking threshold conjecture for dumbbell domains

摘录 / EXCERPT: Let Ω_h be a dumbbell domain, and let $D=D_{\alpha,A}$ be an optimal configuration. Dumbbell symmetry conjecture. For every $\alpha>0$, there is $\rho_1(\alpha,h)>0$ such th...

conjecture | open | MathDB 页面 | 原文线索

V2-073538 | MATHDB-380907

Symmetry-breaking threshold conjecture for annuli

摘录 / EXCERPT: Consider an annulus of thickness τ, and write $\delta=A/|\Omega|$ for the prescribed area fraction. Let $D=D_{\alpha,A}$ be an optimal configuration, and let symmetry breaking...

conjecture | open | MathDB 页面 | 原文线索

V2-073539 | MATHDB-380908

Symmetry-preservation conjecture for small penalization

摘录 / EXCERPT: Let Ω be a domain, let A be the prescribed parameter, and let $D=D_{\alpha,A}$ be an optimal configuration. Say that D has the same symmetries as Ω when every sym...

conjecture | open | MathDB 页面 | 原文线索

V2-073540 | MATHDB-380910

Nonnegative Laurent expansion conjecture for minors and chamber minors

摘录 / EXCERPT: Let M be a generic square matrix, and let a double wiring diagram be arbitrary. The chamber-minor Laurent positivity conjecture. Every minor of M can be written as a Laurent po...

conjecture | open | MathDB 页面 | 原文线索

V2-073541 | MATHDB-380911

Nonnegative Laurent phenomenon conjecture for Somos-5 sequences

摘录 / EXCERPT: Let $a_1,a_2,\dots$ be a Somos-5 sequence, meaning that every six consecutive terms satisfy ... The Somos-5 Laurent positivity conjecture. Every term a_n is a Laurent polynomial wi...

conjecture | open | MathDB 页面 | 原文线索

V2-073542 | MATHDB-380923

The tricolour graph construction of four-critical-value rational functions

摘录 / EXCERPT: Tricolour graph construction. For any tricolour graph there exists a function

conjecture | open | MathDB 页面 | 原文线索

V2-073543 | MATHDB-380929

The polynomial identity for the general rational Gaudin Lax matrix solution

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and let $c_1,\dots,c_k$ be arbitrary constants. For each i, let $\{\vec{J}_i\}$ denote the set of vectors satisfying ... For $\vec{J}_i$, write...

conjecture | open | MathDB 页面 | 原文线索

V2-073544 | MATHDB-380933

Universality conjecture for the unique non-trivial renormalization fixed point

摘录 / EXCERPT: Consider the renormalization operator acting on the class of twist maps under discussion, and call a property universal when it holds on an open set of functions. A fixed point is...

conjecture | open | MathDB 页面 | 原文线索

V2-073545 | MATHDB-380934

Universality conjecture for the regularity of critical invariant circles

摘录 / EXCERPT: Let R denote a critical invariant circle, and let $\kappa(R)$ denote its regularity. A numerical characteristic is universal when it takes the same value on an open set of functi...

conjecture | open | MathDB 页面 | 原文线索

V2-073546 | MATHDB-380935

Universality conjecture for stratified critical-circle conjugacies

摘录 / EXCERPT: Let g, h, and h^{-1} be the conjugacies associated with a critical invariant circle, and let $\kappa(\cdot)$ denote their regularity. A numerical characteristic is universal...

conjecture | open | MathDB 页面 | 原文线索

V2-073547 | MATHDB-380936

Universality conjecture for regularity between pairs of critical circles

摘录 / EXCERPT: Let γ_1 and γ_2 be critical invariant circles, and let G_{γ_1,γ_2} and H_{γ_1,γ_2} denote the associated maps between them. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073548 | MATHDB-380937

Strict pairwise regularity conjecture for critical-circle conjugacies

摘录 / EXCERPT: Let γ_1 and γ_2 be critical invariant circles in the same universality class, and let h_{γ_i} be their conjugacies to the reference dynamics. Write...

conjecture | open | MathDB 页面 | 原文线索

V2-073549 | MATHDB-380944

Ribbon-tableau positivity conjecture for H-functions

摘录 / EXCERPT: Positivity conjecture. The coefficients in these Schur-basis expansions are polynomials with nonnegative integer coefficients.

conjecture | open | MathDB 页面 | 原文线索

V2-073550 | MATHDB-380945

Andrews-Gordon q-series conjecture

摘录 / EXCERPT: Let $k\geq1$, $1\leq\ell\leq k$, $1\leq i\leq k+1$, $1\leq i'\leq\ell+1$, and $|q|<1$. Let C_k be the Cartan matrix of type A_k, let $B_{\ell-1}$ be the inverse Cartan matrix o...

conjecture | open | MathDB 页面 | 原文线索

V2-073551 | MATHDB-380947

Generalized basic hypergeometric formula for separated Laurent polynomials

摘录 / EXCERPT: Let n be a positive integer, let g and ℓ be parameters, and let $\lambda=(\lambda_1,\ldots,\lambda_n)$. Write $(y;q)_m$ for the q-Pochhammer symbol and...

conjecture | open | MathDB 页面 | 原文线索

V2-073552 | MATHDB-380951

Lapointe-Vinet's creation-operator formula for Macdonald polynomials

摘录 / EXCERPT: Lapointe-Vinet's creation-operator conjecture. The Macdonald polynomials are given by

conjecture | open | MathDB 页面 | 原文线索

V2-073553 | MATHDB-380952

Lapointe-Vinet's identification of Macdonald creation operators

摘录 / EXCERPT: Lapointe-Vinet's operator-identification conjecture. The creation operators satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073554 | MATHDB-380953

Lapointe-Vinet's Jack creation-operator properties

摘录 / EXCERPT: Lapointe-Vinet's Jack-operator conjecture. The creation operators share the following properties:

conjecture | open | MathDB 页面 | 原文线索

V2-073555 | MATHDB-380980

The conjectural model structure on A_{∞}-precat categories

摘录 / EXCERPT: Model-structure conjecture. For a wide range of A_{∞}-operads ${\cal C}$, the category of $\widehat{{\cal C}}$-precats admits a closed model structure with these cofibration...

conjecture | open | MathDB 页面 | 原文线索

V2-073556 | MATHDB-381012

Extension of the spectral radius formulae to all linear fractional self-maps

摘录 / EXCERPT: Spectral radius extension conjecture. The spectral radius formulae of Theorem $$ are valid for all $\varphi\in\mathcal{S}_m$.

conjecture | open | MathDB 页面 | 原文线索

V2-073557 | MATHDB-381013

Circle law for the Dirichlet Markov Ensemble

摘录 / EXCERPT: Circle law for the Dirichlet Markov Ensemble. Almost surely, the empirical spectral distribution converges weakly to $\mathcal{C}_1$:

conjecture | open | MathDB 页面 | 原文线索

V2-073558 | MATHDB-381017

Matrix Stahl-Totik regularity conjecture for matrix-valued measures

摘录 / EXCERPT: Matrix Stahl-Totik conjecture. Then μ is regular.

conjecture | open | MathDB 页面 | 原文线索

V2-073559 | MATHDB-381018

Linearity and group-theoretic properties of braid and pure braid groups of complex reflection groups

摘录 / EXCERPT: Conjecture. The groups B and P are linear and satisfy all the properties enumerated in that theorem whenever W is an irreducible complex reflection group.

conjecture | open | MathDB 页面 | 原文线索

V2-073560 | MATHDB-381030

Nonempty positive-rank fibers conjecture for the elliptic surface defined by H

摘录 / EXCERPT: Let $a,b,c,d,e,f\binom{}{} \in\mathbb{Z}$ and consider the elliptic surface ... where the coefficients $a_{i,j}$ are given by (ais). For $t\in\mathbb{Q}$, let $\mathcal{E}_t$ denote...

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V2-073561 | MATHDB-381033

The Seiberg-Witten invariant conjecture for constants of nice surface singularities

摘录 / EXCERPT: Seiberg-Witten invariant conjecture. The constants are the Seiberg-Witten invariants of the link, namely

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V2-073562 | MATHDB-381035

Shapiro-Tater conjecture on root-counting measures of Heun spectral polynomials

摘录 / EXCERPT: Associate to $Sp_n(\lambda)$ the finite measure ... where $\delta(z-a)$ is the Dirac measure supported at a. The measure μ_n is the root-counting measure of the spectral polyno...

conjecture | open | MathDB 页面 | 原文线索

V2-073563 | MATHDB-381036

The necessary condition for subcriticality in inhomogeneous random graphs

摘录 / EXCERPT: Let κ satisfy condition ... be the fixed-point equation for the corresponding branching-process generating function. Necessary-and-sufficient condition for subcriticality. One...

conjecture | open | MathDB 页面 | 原文线索

V2-073564 | MATHDB-381038

The finite test set conjecture for sums of triangular numbers representing odd integers

摘录 / EXCERPT: Finite test set conjecture. A sum of triangular numbers f represents all odd integers if and only if it represents the integers

conjecture | open | MathDB 页面 | 原文线索

V2-073565 | MATHDB-381041

Scattering conjecture for barely supercritical wave equations

摘录 / EXCERPT: Scattering conjecture. If u is the solution with data (u_0,u_1), then u exists for all time t and there is a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-073566 | MATHDB-381052

The branch-length conjecture for pods of conjugates in the Bass-Serre tree

摘录 / EXCERPT: Let f be the polynomial automorphism under consideration, and let an n-pod be a configuration formed by n conjugates of f in the Bass-Serre tree, with consecutive branches...

conjecture | open | MathDB 页面 | 原文线索

V2-073567 | MATHDB-381056

The $p=1$ case of the Feigin-Stoyanovsky identity

摘录 / EXCERPT: Let k,n be positive integers, set $N=2n+1$ and $p=1$, and let C_{ab} be the Cartan matrix of type $\textup{A}_{N-1}$. For nonnegative integers $m_i^{(a)}$, define the associate...

conjecture | open | MathDB 页面 | 原文线索

V2-073568 | MATHDB-381060

Polynomial-coefficient conjecture for the difference of special-boundary transfer-matrix eigenvalues

摘录 / EXCERPT: Let m be the lattice-width parameter, let q be the Potts-model state parameter, and let $\lambda_{\star,=}(m)$ and $\lambda_{\star,\neq}(m)$ denote the dominant transfer-matrix...

conjecture | open | MathDB 页面 | 原文线索

V2-073569 | MATHDB-381074

Khabibullin's integral inequality conjecture

摘录 / EXCERPT: Let λ be a positive real number and let n be an integer with $n\geqslant 2$. Let $S(x)$ be a nonnegative nondecreasing function on $[0,+\infty)$. If ... for every...

conjecture | refuted | MathDB 页面 | 原文线索

V2-073570 | MATHDB-381076

Lusztig's conjecture on the function a

摘录 / EXCERPT: Lusztig's conjecture on the function a. For every $w\in W$, one has

conjecture | open | MathDB 页面 | 原文线索

V2-073571 | MATHDB-381080

Vanishing Abelian integrals for balanced and unbalanced cycles

摘录 / EXCERPT: Vanishing-integral conjecture. (i) If $\gamma(t)$ is totally unbalanced, then

conjecture | open | MathDB 页面 | 原文线索

V2-073572 | MATHDB-381088

The graph characterization of generic minimal rigidity on concentric cones and tori

摘录 / EXCERPT: Graph characterization conjecture. Then (G,p) is generically minimally rigid on ${\mathcal{M}}$ if and only if $G=K_{2}$, K_3, K_4, or G is $(2,1)$-tight.

conjecture | open | MathDB 页面 | 原文线索

V2-073573 | MATHDB-381089

Doney's smoothness conjecture for scale functions of spectrally negative Lévy processes

摘录 / EXCERPT: Doney's conjecture. 1. If $\sigma^2>0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073574 | MATHDB-381095

The decreasing normalized count conjecture for binary self-dual codes

摘录 / EXCERPT: Decreasing normalized count conjecture. The sequence $\{a_k,\ k=10,11,12,\dots\}$ is decreasing. This conjecture concerns the growth of the number of inequivalent binary self-dual...

conjecture | open | MathDB 页面 | 原文线索

V2-073575 | MATHDB-381104

Monotonicity and uniqueness conjecture for the smallest eigenvalue

摘录 / EXCERPT: Let Λ_κ denote the smallest eigenvalue of the relevant weighted spherical operator, with parameter $\kappa>0$ in dimension n. Proposition gives three possible altern...

conjecture | open | MathDB 页面 | 原文线索

V2-073576 | MATHDB-381105

The conjecture that the reduced invariant defines a knot concordance homomorphism

摘录 / EXCERPT: Let $\mathcal{C}$ denote the smooth knot concordance group. The knot invariant $r(K)$ is defined for knots K and takes values in $\mathbb{Q}$. Concordance homomorphism conjecture...

conjecture | open | MathDB 页面 | 原文线索

V2-073577 | MATHDB-381112

The entropy-based measurement conjecture for binary sparse recovery

摘录 / EXCERPT: Entropy-based measurement conjecture. For sufficiently large n, the linear program referred to as LP exactly recovers x from $nH(k/n)/2$ measurements.

conjecture | open | MathDB 页面 | 原文线索

V2-073578 | MATHDB-381125

Polynomial-size grid drawing problem for planar graphs

摘录 / EXCERPT: Polynomial-size grid drawing problem. For every planar graph G, there is a proper grid drawing of G in a grid of polynomial size.

open_problem | open | MathDB 页面 | 原文线索

V2-073579 | MATHDB-381133

The converse compactness assertion for composition operators on analytic Campanato spaces

摘录 / EXCERPT: The converse compactness assertion. If the necessary condition (e1) and condition (e2) from Theorem ... remains valid throughout this remaining parameter range. The paper presents th...

conjecture | open | MathDB 页面 | 原文线索

V2-073580 | MATHDB-381135

Pointwise asymptotic conjecture for the optimal Hardy weight

摘录 / EXCERPT: Let $P u=-\Delta u+V(x)u+u$ be subcritical in $\mathbb{R}^n$, and assume that ... If u is a positive solution of $Pu=0$ in $\mathbb{R}^n$ satisfying the stated condition, then let...

conjecture | open | MathDB 页面 | 原文线索

V2-073581 | MATHDB-381138

Locally elliptic actions on Davis realizations have a global fixed point

摘录 / EXCERPT: Let G be a group acting by type-preserving simplicial isometries on a building $Delta$. Let X be the Davis realization of $Delta$, and call the action locally elliptic if every...

conjecture | open | MathDB 页面 | 原文线索

V2-073582 | MATHDB-381140

Optimality of the rates in Theorem 4

摘录 / EXCERPT: Optimality conjecture. The rates found in Theorem 4 are optimal.

conjecture | open | MathDB 页面 | 原文线索

V2-073583 | MATHDB-381141

Minimax error bound for the column sparse principal subspace case

摘录 / EXCERPT: Let $q\in[0,1]$, and let $\hat{\mathcal{S}}$ estimate the principal subspace $\mathcal{S}$ under the same conditions as in the column sparse upper-bound corollary, with R_q repla...

conjecture | open | MathDB 页面 | 原文线索

V2-073584 | MATHDB-381146

Aldous's scaling-limit conjecture for the stationary East process

摘录 / EXCERPT: Aldous's conjecture. As $q\downarrow 0$, the rescaled stationary East process converges to a limit point process X_t on $[0,+\infty)$ such that:

conjecture | open | MathDB 页面 | 原文线索

V2-073585 | MATHDB-381148

Uniqueness conjecture for half-plane differentials with fourth-order poles

摘录 / EXCERPT: Let X be a Riemann surface equipped with poles whose orders are all 4, and prescribe the residues and leading order terms at those poles. Theorem provides a half-plane differen...

conjecture | open | MathDB 页面 | 原文线索

V2-073586 | MATHDB-381151

The first signed path conjecture for finite binary trees

摘录 / EXCERPT: First signed path conjecture. For every pair of finite, binary trees (D,R) with the same number of leaves, there is a sign assignment of D and a word w of rotation symbols va...

conjecture | open | MathDB 页面 | 原文线索

V2-073587 | MATHDB-381156

Kraaijevanger-MacDonald-Gottlieb conjecture on the radius of absolute monotonicity

摘录 / EXCERPT: Kraaijevanger-MacDonald-Gottlieb conjecture. For every such method,

conjecture | solved | MathDB 页面 | 原文线索

V2-073588 | MATHDB-381157

Hatano-Tsunekawa conjecture on the optimal radius for diagonal implicit Runge-Kutta methods

摘录 / EXCERPT: Hatano-Tsunekawa conjecture. For every $s\ge 1$,

conjecture | solved | MathDB 页面 | 原文线索

V2-073589 | MATHDB-381159

Spectral eigenvalue conjecture for the Dirac soliton linearization

摘录 / EXCERPT: Let the spectral problem be the linearized spectral problem denoted by (third), with parameter $\omega\in(-1,1)$. Its isolated eigenvalues and end-point resonances are considered a...

conjecture | open | MathDB 页面 | 原文线索

V2-073590 | MATHDB-381162

Gepner-type stability conjecture for smooth quintic threefolds

摘录 / EXCERPT: Let $X\subset\mathbb{P}^4$ be a smooth quintic 3-fold, and assume the strong Bogomolov-Gieseker conjecture used to construct the heart $\mathcal{A}_G$ and central charge...

conjecture | open | MathDB 页面 | 原文线索

V2-073591 | MATHDB-381169

Conjecture on the global-null correlation maximum bound

摘录 / EXCERPT: Let $f_n(x)$ be the density of $\sqrt{n}|S_{ij}|$, let $\bar{F}_n(x)=\int_x^{\sqrt{n}}f_n(x)$, and let $G_{n,p}(x)$ be the cumulative distribution function of the maximum of an ind...

conjecture | open | MathDB 页面 | 原文线索

V2-073592 | MATHDB-381179

Finite descriptiveness conjecture for stepped surfaces of real cubic fields

摘录 / EXCERPT: Let $\bar{\bar \alpha}=\left(\alpha_{0},\alpha_{1},\alpha_{2}\right)$ be any $\mathbb{Q}$-basis of a real cubic number field. Let $\mathscr S\left(\bar{\bar \alpha}\right)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-073593 | MATHDB-381188

Conjectured distribution formula for the ascending Lévy process supremum

摘录 / EXCERPT: Conjectured supremum-distribution formula. For $t>t_0$ and $x>0$,

conjecture | open | MathDB 页面 | 原文线索

V2-073594 | MATHDB-381189

Product formula conjecture for Kazhdan-Lusztig R-polynomials

摘录 / EXCERPT: Let $n\geq 2$, and let $e\leq \sigma_1\leq \sigma_2\leq \Omega_n$. For the Kazhdan-Lusztig R-polynomial $\widetilde{R}_{\sigma_1,\sigma_2}(q)$, write $F_j(q)$ for the q-Fibonac...

conjecture | open | MathDB 页面 | 原文线索

V2-073595 | MATHDB-381193

Saks-West conjecture on semiantichains and unichain coverings

摘录 / EXCERPT: Saks-West conjecture. In every product $P\times Q$ of two posets, the size of a largest semiantichain equals the size of a smallest unichain covering.

conjecture | open | MathDB 页面 | 原文线索

V2-073596 | MATHDB-381196

The converse Hasse-Witt rank conjecture for curves in characteristic 2

摘录 / EXCERPT: Let k be a perfect field of characteristic 2, let Δ be a two-dimensional lattice polygon satisfying the conditions from Lemma, and let f be a Δ-non-degenerate (...

conjecture | open | MathDB 页面 | 原文线索

V2-073597 | MATHDB-381197

Polynomial recoloring diameter conjecture for planar graphs

摘录 / EXCERPT: Let G be a planar graph and let k be an integer. The k-recoloring graph $R_k(G)$ has vertices corresponding to proper k-colorings of G, with edges joining colorings that...

conjecture | open | MathDB 页面 | 原文线索

V2-073598 | MATHDB-381206

The conjecture that the equal-weight subgraph-count property is quasi-random

摘录 / EXCERPT: Let F be a graph with $e(F)>1$, and let $m=|F|$. For a graph property $\mathcal P(F;1/m,\dots,1/m)$ and its weaker version $\widetilde\mathcal P(F;1/m,\dots,1/m)$, the source con...

conjecture | open | MathDB 页面 | 原文线索

V2-073599 | MATHDB-381208

Gopalan-Klivans-Zuckerman list-decoding radius conjecture for Reed-Muller codes

摘录 / EXCERPT: Let $\mathcal{P}_{d}({\mathbb{F}}^n)$ be the class of polynomials of degree at most d from ${\mathbb{F}}^n$ to ${\mathbb{F}}$. For a function $g:{\mathbb{F}}^n\to{\mathbb{F}}$, l...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073600 | MATHDB-381211

The alternating-sum conjecture for zeros of Wronskians of orthogonal polynomials

摘录 / EXCERPT: Let $\{P_n\}_{n=0}^\infty$ be an orthogonal polynomial system with respect to an arbitrary positive measure $d\mu=W\,dx$ supported on an interval I of the real line. Assume that...

conjecture | open | MathDB 页面 | 原文线索

V2-073601 | MATHDB-381213

Bijectivity of the silting-to-cluster-tilting map

摘录 / EXCERPT: Bijectivity conjecture. The map $\pi\colon\operatorname{silt}\nolimits^{\mathcal F}{\mathcal T}\to{\it d}\strut\kern-.2em\operatorname{-ctilt}\nolimits{\mathcal C}$ is bijective fo...

conjecture | open | MathDB 页面 | 原文线索

V2-073602 | MATHDB-381221

Strong-disorder localisation under absence of supported formal solutions

摘录 / EXCERPT: Let $H(g)$ be a Γ-trimmed random Schrödinger operator on $\mathbb{Z}^d$ satisfying the Assumptions. Suppose that the complement of Γ is a union of finite connected co...

conjecture | open | MathDB 页面 | 原文线索

V2-073603 | MATHDB-381224

All-pass parametrization conjecture for minimal right-invertible spectral factors

摘录 / EXCERPT: All-pass parametrization conjecture. The set of all minimal right-invertible spectral factors of $\Phi(z)$ is

conjecture | open | MathDB 页面 | 原文线索

V2-073604 | MATHDB-381227

Polynomial spectral-radius lower-bound conjecture for p-CLI methods

摘录 / EXCERPT: Polynomial lower-bound conjecture. There exists $\eta\in[\mu,L]$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073605 | MATHDB-381235

Eventual odd circular mixing conjecture for bipartite graphs

摘录 / EXCERPT: Eventual odd circular mixing conjecture. There exists an integer q_0 such that, for every integer $q\geq q_0$, the graph G is $(2q+1,q)$-mixing.

conjecture | open | MathDB 页面 | 原文线索

V2-073606 | MATHDB-381240

Zero-Lelong-number conjecture for degenerate complex Monge-Ampère flows

摘录 / EXCERPT: Let $\Omega\subset\mathbb{C}^n$ be the strictly pseudoconvex domain and let u_0 be the initial datum considered in Theorem. Replace the assumption ... by the condition that u_0 h...

conjecture | open | MathDB 页面 | 原文线索

V2-073607 | MATHDB-381242

The prey-refuge conjecture for preventing blow-up in a three-species food chain

摘录 / EXCERPT: Consider the three-species food-chain model given by equations (1.3)-(1.1) on a bounded domain with Neumann boundary conditions. Let the parameters satisfy $c<\frac{w_3}{D_3}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-073608 | MATHDB-381248

Equivalence of loop-group representation categories and conformal-net categories

摘录 / EXCERPT: Equivalence conjecture. For every simple simply connected Lie group G and every level $k\geq 0$, the categories $\mathrm{Rep}^k(LG)$ defined via textit{a.} or textit{b.}, and via...

conjecture | open | MathDB 页面 | 原文线索

V2-073609 | MATHDB-381252

The generalized theta-to-q difference-equation conjecture

摘录 / EXCERPT: Let m-periodic variables θ_i^t satisfy the generalized discrete Toda equations referred to as, and define $q_{j,i}^t$ from them via. Theta-to-q conjecture. The quantities...

conjecture | open | MathDB 页面 | 原文线索

V2-073610 | MATHDB-381253

The essential-set containment conjecture for simply-laced root systems

摘录 / EXCERPT: Let Δ be a simply-laced root system with highest root θ, and let θ be fundamental. For a simple root α, let $\EuScript F_{\alpha}$ be the associated min...

conjecture | open | MathDB 页面 | 原文线索

V2-073611 | MATHDB-381255

The order-scattered ideal-lattice conjecture for join-semilattices

摘录 / EXCERPT: Let P be a join-semilattice, and let $\operatorname{Id} P$ denote its lattice of ideals. An ordered set is order-scattered if it does not contain a copy of the order of the ratio...

conjecture | open | MathDB 页面 | 原文线索

V2-073612 | MATHDB-381266

The conjecture identifying the boundary sheaf of the Drinfeld half-plane covering

摘录 / EXCERPT: Let $c1$ be the representation occurring in the definition of the sheaf $c4_n^{-1}(V-B_i)^{c1}$, and let $a3_n$ be the corresponding covering of the Drinfeld half-plane. For an ope...

conjecture | open | MathDB 页面 | 原文线索

V2-073613 | MATHDB-381271

Characterization conjecture for finite abelian groups by sets of lengths

摘录 / EXCERPT: Characterization conjecture. If $\mathsf D(G)\geq 4$, then G and G' are isomorphic.

conjecture | open | MathDB 页面 | 原文线索

V2-073614 | MATHDB-381275

The incorrect-path splitting conjecture for polar-code list decoding

摘录 / EXCERPT: Incorrect-path splitting conjecture. Under the Gaussian approximation, for any incorrect path that survives at some unfrozen bit u_i, it will split at some stage within...

conjecture | open | MathDB 页面 | 原文线索

V2-073615 | MATHDB-381283

The completion conjecture for the graph invariant c_2

摘录 / EXCERPT: A graph's completion is the graph obtained by the completion operation referred to in the source, and c_2 denotes the c_2-invariant of a connected graph with at least three ver...

conjecture | open | MathDB 页面 | 原文线索

V2-073616 | MATHDB-381285

Polynomial growth conjecture for reachable sets

摘录 / EXCERPT: Polynomial growth conjecture. There exist $N\in\mathbb{N}$ and $T,C>0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073617 | MATHDB-381289

Cao-Jiang conjecture on exceptional objects of weak del Pezzo surfaces

摘录 / EXCERPT: Cao-Jiang conjecture. For any exceptional object $\mathcal{E}\in D(X)$, there exists an autoequivalence $\Phi\in\operatorname{Auteq}(D(X))$ such that $\Phi(\mathcal{E})$ is an exce...

conjecture | open | MathDB 页面 | 原文线索

V2-073618 | MATHDB-381295

Uniqueness conjecture for minimizers in the defocusing antiperiodic problem

摘录 / EXCERPT: Let $b<0$, and consider the minimization problem in $A_{T/2}$ with fixed mass m, denoted by ... operatorname{sn}{alpha,beta,k}(x)=frac{1}{alpha}operatorname{sn}left(frac{x}{beta},k...

conjecture | open | MathDB 页面 | 原文线索

V2-073619 | MATHDB-381297

Uniqueness conjecture for gravitating vortices on the projective line

摘录 / EXCERPT: Let L be the line bundle and ϕ the vortex data on ${\mathbb P}^1$; let τ satisfy the inequality in equation, and let $\alpha>0$. The datum ϕ is polystable. Uniquen...

conjecture | open | MathDB 页面 | 原文线索

V2-073620 | MATHDB-381298

Existence of a function eliminating the asymptotic set when the bifurcation set is empty

摘录 / EXCERPT: Existence conjecture. Does there exist a function ρ such that

open_problem | open | MathDB 页面 | 原文线索

V2-073621 | MATHDB-381301

Scaling-limit conjecture for the metric-graph GFF local-time distance and CLE$_4$

摘录 / EXCERPT: Let D be an open bounded simply connected domain of $\mathbb{C}$. Let D^ε be a fine-mesh approximation of D, A^{ε} the approximation of the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-073622 | MATHDB-381313

The shortest-identity conjecture for the transformation semigroup T_5

摘录 / EXCERPT: Shortest-identity conjecture for T_5. This identity is the shortest identity of T_5. The preceding theorem establishes that this identity holds, while the conjecture concerns i...

conjecture | open | MathDB 页面 | 原文线索

V2-073623 | MATHDB-381316

Non-density conjecture for product polynomial maps

摘录 / EXCERPT: Let $f(z,w)=(p(z),q(w))$ be the product of two polynomial maps of degree d in $\mathbb{C}$. The non-density conjecture. If p or q belongs to the bifurcation locus in...

conjecture | open | MathDB 页面 | 原文线索

V2-073624 | MATHDB-381319

Faber's Gorenstein conjectures for the rational tails locus

摘录 / EXCERPT: Faber's Gorenstein conjectures. (D) The intersection pairings

conjecture | open | MathDB 页面 | 原文线索

V2-073625 | MATHDB-381321

Extension of the segregative pulsating fronts exclusivity theorem to non-constant extinction states

摘录 / EXCERPT: Segregative pulsating fronts exclusivity conjecture. Theorem still holds true in the non-constant case.

conjecture | open | MathDB 页面 | 原文线索

V2-073626 | MATHDB-381322

Identifiability conjecture for causal models with H_1 entropy

摘录 / EXCERPT: Consider the causal model $\mathcal{M}=(\{X,Y\},E,f,X\rightarrow Y,p_{X,E})$, where the discrete random variables X and Y have n states, E is independent of X and has...

conjecture | open | MathDB 页面 | 原文线索

V2-073627 | MATHDB-381325

Global minimality conjecture for FCC and BCC lattices under Lennard-Jones-type energies

摘录 / EXCERPT: Let f be the Lennard-Jones-type interaction defined by ... where $(a_1,a_2)\in(0,+\infty)^2$ and $0<x_1<x_2$. Let E_f denote the interaction energy among Bravais lattices, and le...

conjecture | open | MathDB 页面 | 原文线索

V2-073628 | MATHDB-381326

The Painlevé IV conformal-block expansion conjecture

摘录 / EXCERPT: Let θ, θ_t, β, and s be parameters, let G denote the Barnes G-function, and let Φ and the matrix elements appearing below be the conformal-block data...

conjecture | open | MathDB 页面 | 原文线索

V2-073629 | MATHDB-381344

Tightness conjecture for the rooted $K_{3,3}$ minor bound

摘录 / EXCERPT: The preceding theorem gives a bound on rooted $K_{3,3}$ minors; denote this bound by the quantity appearing in Theorem. Tightness conjecture. The bound in Theorem is tight. This co...

conjecture | open | MathDB 页面 | 原文线索

V2-073630 | MATHDB-381346

Co-primeness conjecture for odd higher-dimensional q-discrete Painlevé II equations

摘录 / EXCERPT: Let $k=2m+1\geq 7$ be odd, and define rational functions $y_n\in\mathbb{Q}(y_0,\dots,y_{2m-3},\alpha,\beta,\gamma)$ by the difference equation specified in the source. Odd-dimensio...

conjecture | open | MathDB 页面 | 原文线索

V2-073631 | MATHDB-381357

Limiting-profile conjecture for non-cooperative Fisher-KPP traveling waves

摘录 / EXCERPT: Let $\boldsymbol{p}_{eta}_{eta>0}$ and $c_{eta}_{eta\geq0}$ satisfy ... Assume (H_{7}). Stability is understood as local asymptotic stability with respect to (E_{KPP}^{0}). Limit...

conjecture | open | MathDB 页面 | 原文线索

V2-073632 | MATHDB-381360

The non-vanishing conjecture for projective lc pairs

摘录 / EXCERPT: Non-vanishing conjecture. There exists an effective $\mathbb{R}$-divisor $E\geq 0$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073633 | MATHDB-381362

Circuit-free permutations attaining the lower bound for reduced words

摘录 / EXCERPT: Circuit-free permutation conjecture. The permutation w is circuit-free, and thus achieves the lower bound in Theorem, if and only if at least one of the following conditions is s...

conjecture | open | MathDB 页面 | 原文线索

V2-073634 | MATHDB-381367

Nonconfinement conjecture for the coefficient-free R-system of the five-element poset

摘录 / EXCERPT: Let P be the poset shown in Figure, and let $G(P)$ be its associated graph. Consider the coefficient-free R-system associated with $G(P)$. A strong τ-sequence and the sing...

conjecture | open | MathDB 页面 | 原文线索

V2-073635 | MATHDB-381370

The inner-loop error-bound conjecture for unique information computation

摘录 / EXCERPT: Inner-loop error-bound conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073636 | MATHDB-381373

The presentation conjecture for the prop $\mathrm{LagRel}_k^{\circ}$

摘录 / EXCERPT: Let $\mathrm{LagRel}_k^{\circ}$ be the prop generated by the eight morphisms ... The theorem preceding the conjecture gives two extraspecial commutative Frobenius monoids and two bim...

conjecture | open | MathDB 页面 | 原文线索

V2-073637 | MATHDB-381378

Zero-infinity conjecture for packing measures of non-almost-deterministic random code-tree attractors

摘录 / EXCERPT: Let bbL be a family of iterated function systems satisfying the UOSC and Condition. Let bbP be a random code-tree measure, and assume that F_τ is not almost deterministic....

conjecture | open | MathDB 页面 | 原文线索

V2-073638 | MATHDB-381385

Existence of colliding action minimizers in the spatial N-body problem

摘录 / EXCERPT: Let $N>3$, let $\omega\in(0,N)$, and let $\xi\in\Xi_N$. Let q^{ω} be an action minimizer of $\mathcal{A}_{\omega}$ among all loops in $\Lambda^{D_N}_{\xi}$ satisfying the co...

conjecture | open | MathDB 页面 | 原文线索

V2-073639 | MATHDB-381387

The compactification conjecture for subsets of Euclidean space

摘录 / EXCERPT: Compactification conjecture. Every m-dimensional subset of $\mathbb{R}^n$ has an m-dimensional compactification that can be embedded into $\mathbb{R}^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-073640 | MATHDB-381391

Linear rank growth for points on Legendre elliptic curves over solvable extensions

摘录 / EXCERPT: Let K be a number field, and let E be a Legendre elliptic curve over K with parameter $\lambda\in K$. For each even integer $n\geq 4$, consider the n points constructed abo...

conjecture | open | MathDB 页面 | 原文线索

V2-073641 | MATHDB-381393

De Pierro's conjecture on underrelaxed projection limits

摘录 / EXCERPT: De Pierro's conjecture. The weak limits satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073642 | MATHDB-381394

The critical-exponent conjecture for semilinear wave equations with noneffective damping

摘录 / EXCERPT: Critical-exponent conjecture. The critical exponent of the equation is

conjecture | open | MathDB 页面 | 原文线索

V2-073643 | MATHDB-381402

Nonexistence conjecture for extremizers of the odd fractional Schrödinger inequality

摘录 / EXCERPT: Conjecture for the odd fractional Schrödinger inequality. For every $p\geq 2$,

conjecture | open | MathDB 页面 | 原文线索

V2-073644 | MATHDB-381408

The Catalan triangle Laurent polynomial and positivity conjecture

摘录 / EXCERPT: Catalan triangle Laurent polynomial and positivity conjecture. This expression is a Laurent polynomial in q, and it has non-negative integer coefficients when...

conjecture | open | MathDB 页面 | 原文线索

V2-073645 | MATHDB-381413

K-jdt and Buch insertion equivalence for increasing tableaux

摘录 / EXCERPT: K-jdt-Buch insertion conjecture. The K-jdt rectification of $T \ast S$ is the same tableau as the Buch insertion of T into S.

conjecture | open | MathDB 页面 | 原文线索

V2-073646 | MATHDB-381420

Polynomial-growth conjecture for Möbius values of the permutations $\pi_{n,k}$

摘录 / EXCERPT: For $k\le n$, let $\pi_{n,k}$ be the subpermutation of κ_n induced by the values $n+1, n+3,\dots,n+2k-1,1,3n+1,2,3n+2,\dots,n,4n,n+2,n+4,\dots, n+2k$. In particular,...

conjecture | open | MathDB 页面 | 原文线索

V2-073647 | MATHDB-381421

Existence of a singular open book without radiality

摘录 / EXCERPT: Let G satisfy the conditions of Theorem sf2, except that its discriminant need not be radial. For sufficiently small $\varepsilon>0$, consider the sphere S^{m-1}_{ε}...

conjecture | open | MathDB 页面 | 原文线索

V2-073648 | MATHDB-381424

Lattice-isomorphism conjecture for symmetric-power quotients of Q^*

摘录 / EXCERPT: Lattice-isomorphism conjecture. These lattices are isomorphic, compatibly with the identification predicted by the periodicity conjecture for the corresponding Grothendieck-group c...

conjecture | open | MathDB 页面 | 原文线索

V2-073649 | MATHDB-381425

Green-Griffiths-Lang conjecture for compactified ball quotients

摘录 / EXCERPT: Let $\overline{X} = \overline{{\left.\raisebox{-.2em}{$Gamma$}\middle\backslash\raisebox{.2em}{$mathbb B^n$}\right.}}$ be a compactification as in Theorem, with $n \geq 6$. Here...

conjecture | open | MathDB 页面 | 原文线索

V2-073650 | MATHDB-381434

Integral ISS conjecture for bilinear systems

摘录 / EXCERPT: Let $\Sigma(A,[0,B_1])$ and $\Sigma(A,[0,B_2])$ denote the two linear systems associated with a bilinear system, and suppose that the bilinear system satisfies the hypothesis of Se...

conjecture | open | MathDB 页面 | 原文线索

V2-073651 | MATHDB-381437

The circular homomorphism conjecture for planar graphs of girth at least $4k$

摘录 / EXCERPT: Circular homomorphism conjecture. Every planar graph G of girth at least $4k$ admits a homomorphism to C_{2k+1}. Equivalently,

conjecture | open | MathDB 页面 | 原文线索

V2-073652 | MATHDB-381438

Ehrhart positivity conjecture for generalized Gelfand-Tsetlin polytopes

摘录 / EXCERPT: Let $\mathcal{GT}(lambda,sigma)$ be the generalized Gelfand-Tsetlin polytope associated with a partition $lambda$ and a permutation $sigma$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073653 | MATHDB-381440

Conjecture on large four-genera of connected sums of non-torsion solvable knots

摘录 / EXCERPT: Large four-genus conjecture. The collection $\{\#^n K\}$ consists of h-solvable knots and contains knots with arbitrarily large smooth four-genera g_4.

conjecture | open | MathDB 页面 | 原文线索

V2-073654 | MATHDB-381446

The P-property conjecture under the conditions of Theorem main2

摘录 / EXCERPT: P-property conjecture. Under the conditions of Theorem main2, T has the P property. The source further says that it is enough to check whether T satisfies the condition

conjecture | open | MathDB 页面 | 原文线索

V2-073655 | MATHDB-381458

Maximum-degree conjecture for trees with a given neighbor-locating-chromatic number

摘录 / EXCERPT: Let $k\geq 2$ and let T be a tree. Write $\chi _{_{NL}}(T)$ for the neighbor-locating-chromatic number of T, and let $\Delta(T)$ denote its maximum degree. Maximum-degree conje...

conjecture | open | MathDB 页面 | 原文线索

V2-073656 | MATHDB-381461

Stark's rank one abelian conjecture for quadratic fields

摘录 / EXCERPT: Let F be a quadratic extension of $\mathbb{Q}$ and let K be a nontrivial finite abelian extension of F. If F is real quadratic, assume that one infinite place of F stays...

conjecture | open | MathDB 页面 | 原文线索

V2-073657 | MATHDB-381464

Higher-dimensional Logarithmic Extension Theorem conjecture for log canonical pairs

摘录 / EXCERPT: Let k be a perfect field of characteristic $p\geq 7$, and let $(X,\emptyset)$ be a log canonical pair over k of dimension at most $p-2$. Logarithmic Extension Theorem conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-073658 | MATHDB-381472

Mácajová-Raspaud-Škoviéra signed Four Color conjecture

摘录 / EXCERPT: All graphs are finite, simple, undirected graphs. Let G be a graph and let $\sigma:E(G)\to\{-1,+1\}$ be a signature; the pair (G,σ) is a signed graph, with underlying grap...

conjecture | open | MathDB 页面 | 原文线索

V2-073659 | MATHDB-381477

Effective determination conjecture for quintic FJRW invariants

摘录 / EXCERPT: Let $\Theta_{g,k}$ denote the genus-g FJRW invariants of the quintic with k insertions, and consider the relations obtained from the virtual-localization equations for mixed sp...

conjecture | open | MathDB 页面 | 原文线索

V2-073660 | MATHDB-381485

The denominator conjecture for pop-stacked permutation generating functions

摘录 / EXCERPT: Denominator conjecture. For all k, the rational generating function $F_k(x)$ can be written as

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073661 | MATHDB-381499

Finiteness conjecture for automorphism groups in case c) of Theorem on kernels

摘录 / EXCERPT: Let D be a derivation as in case $c)$ of Theorem, and let ${\rm Aut}(D)$ denote its automorphism group. Finiteness conjecture. In case $c)$ of Theorem, the group ${\rm Aut}(D)$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-073662 | MATHDB-381504

Three-family conjecture for admissible transformations of two-dimensional shallow water equations

摘录 / EXCERPT: Let $\mathcal{C}$ denote the class of two-dimensional shallow water equations, let $G^\sim$ be its equivalence group, and call an admissible transformation independent when it is i...

conjecture | open | MathDB 页面 | 原文线索

V2-073663 | MATHDB-381506

Hedetniemi's conjecture on the chromatic number of tensor products

摘录 / EXCERPT: Hedetniemi's conjecture.

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073664 | MATHDB-381511

Polynomial growth conjecture for the diagonal-edge invariant

摘录 / EXCERPT: Polynomial growth conjecture. The constant $D(t)$ above can be a polynomial of t.

conjecture | open | MathDB 页面 | 原文线索

V2-073665 | MATHDB-381512

Polynomial growth conjecture for the length-ratio invariant

摘录 / EXCERPT: Polynomial growth conjecture. The constant $L(t)$ above can be a polynomial of t.

conjecture | open | MathDB 页面 | 原文线索

V2-073666 | MATHDB-381520

Develin-Hartke lower-bound conjecture for firefighting on integer lattices

摘录 / EXCERPT: Develin-Hartke conjecture. There exists an outbreak on $\mathbb{Z}^d$ which cannot be contained by deploying $f(n)$ firefighters at time n.

conjecture | open | MathDB 页面 | 原文线索

V2-073667 | MATHDB-381530

The limit-terminality conjecture for pseudo- and lax-cones

摘录 / EXCERPT: Limit-terminality conjecture. The following two statements are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-073668 | MATHDB-381533

The Texan conjecture on dimensions of bounded-type continued fraction sets

摘录 / EXCERPT: For each non-empty finite set $\mathcal{I}\subset\mathbb{N}$, let ... where $a_n(x)$ are the continued-fraction digits of the irrational number x. Texan conjecture. The set ... is de...

conjecture | open | MathDB 页面 | 原文线索

V2-073669 | MATHDB-381535

Random puncturing conjecture for list recovery of Reed-Solomon codes

摘录 / EXCERPT: Let d be fixed, and consider the degree-d Reed-Solomon code with evaluation set $FbbF_q$. A random puncturing is obtained by retaining a random subset of evaluation coordinates...

conjecture | open | MathDB 页面 | 原文线索

V2-073670 | MATHDB-381541

Transcendence conjecture for strong Engel series with signs

摘录 / EXCERPT: Let ξ be a real number defined by a series of the form ... for arbitrary $p/q\in\mathbb{Q}$ and positive integer parameters $z_2,z_3,\ldots$. Transcendence conjecture. Every such...

conjecture | open | MathDB 页面 | 原文线索

V2-073671 | MATHDB-381558

Strengthened equivalence-class bound for increasing and decreasing pattern replacements

摘录 / EXCERPT: Let S_n be the set of permutations of $\\{1,2,\ldots,n\\\\}$, and let two permutations be equivalent when one can be obtained from the other by replacing an occurrence of the pat...

conjecture | refuted | MathDB 页面 | 原文线索

V2-073672 | MATHDB-381566

The shrubdepth characterization by paths

摘录 / EXCERPT: Let $\mathcal{C}$ be a class of graphs, let $\mathcal{P}$ denote the class of paths, and let $\sqsupseteq_\mathrm{FO}$ denote the first-order transduction quasiorder. A graph class...

conjecture | open | MathDB 页面 | 原文线索

V2-073673 | MATHDB-381567

The maximal-antichain conjecture for paths and tree classes

摘录 / EXCERPT: Let $\mathcal{P}$ denote the class of paths, let $\mathcal{T}_n$ denote the class of rooted trees of height at most n, and let $\sqsupseteq_\mathrm{FO}$ denote the first-order tr...

conjecture | open | MathDB 页面 | 原文线索

V2-073674 | MATHDB-381571

Kang and Park's modified Alder conjecture

摘录 / EXCERPT: Kang and Park's conjecture. For all $d,n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-073675 | MATHDB-381572

Higher-order modified Alder conjecture for a=3

摘录 / EXCERPT: Higher-order modified Alder conjecture. For all $d,n\geq 1$,

conjecture | open | MathDB 页面 | 原文线索

V2-073676 | MATHDB-381576

Berglund-Hübsch-Krawitz graded mirror symmetry conjecture for Milnor fibers

摘录 / EXCERPT: Let V be the Milnor fiber under consideration, let $W^\vee$ be its Berglund-Hübsch mirror polynomial, and let $\Gamma^\vee$ be the maximal symmetry group of $W^\vee$, ... Write...

conjecture | open | MathDB 页面 | 原文线索

V2-073677 | MATHDB-381577

Polynomial mixing-time conjecture for Glauber dynamics on graph colourings

摘录 / EXCERPT: Polynomial mixing-time conjecture. For any graph G, the Glauber dynamics on the k-colourings of G for $k\geq\Delta+2$ has polynomial mixing time.

conjecture | open | MathDB 页面 | 原文线索

V2-073678 | MATHDB-381583

Ponomarenko's index conjecture for minimal zero-sum sequences

摘录 / EXCERPT: Let G be the cyclic group of order n, and let S be a minimal zero-sum sequence over G of length 4. The index $\operatorname{ind}(S)$ is the smallest number in ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-073679 | MATHDB-381598

Lin and Kim's refinement conjecture for Schröder numbers

摘录 / EXCERPT: Let $S_{n,k}$ be the triangle defined by ... for $1\leq k\leq n-2$, with $S_{n,n}=S_{n,n-1}=S_{n,n-2}$ for $n\geq 3$ and $S_{1,1}=S_{2,1}=S_{2,2}=1$. For a pair of patterns...

conjecture | open | MathDB 页面 | 原文线索

V2-073680 | MATHDB-381614

The four-minimizer bound for homogeneous QCQPs with two constraints

摘录 / EXCERPT: Consider the homogeneous quadratically constrained quadratic optimization problem denoted by ... has at most four strict local non-global minimizers. The statement is presented after...

conjecture | open | MathDB 页面 | 原文线索

V2-073681 | MATHDB-381616

Constant-distance conjecture for spatial inhomogeneous random graphs

摘录 / EXCERPT: Let $\mathbb{G}_n=G(\mathbf{Y}^{(n)},\mathbf{W}^{(n)},\kappa_n)$ satisfy the assumptions of the local weak convergence theorem in the source, except for the assumption imposing reg...

conjecture | open | MathDB 页面 | 原文线索

V2-073682 | MATHDB-381617

Morris-Spiga-Roney-Dougal conjecture on intersection densities of transitive groups

摘录 / EXCERPT: Let G be a transitive permutation group of degree n. For a transitive permutation group K of degree n, let $I(n)$ be the maximum intersection density among such groups. If...

conjecture | open | MathDB 页面 | 原文线索

V2-073683 | MATHDB-381618

The (σ,ρ)-Poisson summation formula for automorphic representations

摘录 / EXCERPT: Let k be the number field and G a k-split reductive group, with complex dual group $G^\vee(\mathbb{C})$. Let $\rho\colon G^\vee(\mathbb{C})\to\operatorname{GL}_n(\mathbb{C})$...

conjecture | open | MathDB 页面 | 原文线索

V2-073684 | MATHDB-381627

A constant-factor bound for the list chromatic packing number

摘录 / EXCERPT: Let G be a graph. For a list-assignment L of G, an L-packing is a collection of mutually disjoint L-colourings, and it is proper if each colouring is proper. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073685 | MATHDB-381642

Integral congruence conjecture for skew-symmetric matrices

摘录 / EXCERPT: Integral congruence conjecture. There is an integer matrix Y such that

conjecture | open | MathDB 页面 | 原文线索

V2-073686 | MATHDB-381646

The non-termination conjecture for the lower central series of two-strand braid groups on the projective plane

摘录 / EXCERPT: Non-termination conjecture. The LCS of ${\mathbf{B}}_{2,m}({\mathbb{P}^2})$ does not stop.

conjecture | open | MathDB 页面 | 原文线索

V2-073687 | MATHDB-381668

Rank inequality for cocompact discontinuous groups

摘录 / EXCERPT: Let G/H be a homogeneous space in the reductive setting, and let K be a maximal compact subgroup of G. Rank inequality conjecture. If G/H admits a cocompact discontinuous g...

conjecture | open | MathDB 页面 | 原文线索

V2-073688 | MATHDB-381672

Galbraith's exceptional-point conjecture for Atkin-Lehner quotients

摘录 / EXCERPT: Galbraith's conjecture. If $2\leq g^+_N\leq 5$, then $X^+_0(N)$ contains exceptional rational points if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-073689 | MATHDB-381674

Eventual stability conjecture for reducible cubic polynomials

摘录 / EXCERPT: Let $f(z)=z^3+\frac{1}{c}$ with $c\in\mathbb{Z}\setminus\{0\}$. Here f^n denotes the n-fold iterate of f. Eventual stability conjecture. If $f(z)$ is reducible over...

conjecture | open | MathDB 页面 | 原文线索

V2-073690 | MATHDB-381675

Output-Lagrange stability and output asymptotic gain imply OL-IOS

摘录 / EXCERPT: Output-Lagrange stability and output asymptotic gain imply OL-IOS. Under these assumptions, the system is OL-IOS.

conjecture | open | MathDB 页面 | 原文线索

V2-073691 | MATHDB-381682

Conjecture on equal densities of densest plane group packings

摘录 / EXCERPT: Equal-density packing conjecture. Densities of the densest $p2$, pg, and $p2gg$ packings are equal for all, but centrally nonsymmetric n-gons with three-fold rotational symmetr...

conjecture | open | MathDB 页面 | 原文线索

V2-073692 | MATHDB-381684

Least-denominator conjecture for the Poincaré series of generic matrix invariants

摘录 / EXCERPT: Least-denominator conjecture. $\bar{C}(n,k)$ is a rational function with least denominator equal to the denominator of the $(n-1)(k-1)$-st derivative of $F(t)$.

conjecture | open | MathDB 页面 | 原文线索

V2-073693 | MATHDB-381690

The $P=W$ conjecture for GL_n

摘录 / EXCERPT: The Betti moduli space M_B parametrizes irreducible local systems on a punctured curve with prescribed central monodromy, while M_{Dol} is the corresponding Dolbeault...

conjecture | open | MathDB 页面 | 原文线索

V2-073694 | MATHDB-381694

Brevig-Ortega-Cerdà-Seip-Zhao conjecture on contractive Hardy-Dirichlet inclusions

摘录 / EXCERPT: Let $p>2$, let $\mathbb{D}$ be the unit disk, and let f be analytic in $\mathbb{D}$. Denote by $\|f\|_{H^p}$ the Hardy-space norm and by $\|f\|_{D_{p/2}}$ the norm of the corresp...

conjecture | open | MathDB 页面 | 原文线索

V2-073695 | MATHDB-381706

Asymptotic product formula conjecture for refined BPS invariants of local $\mathbb{P}^2$

摘录 / EXCERPT: Consider the generating series ... Set ... where $\mathsf{S}^\bullet$ denotes the plethystic exponential, and let $[H(q,t)]^{i,j}$ denote the coefficient of q^it^j in the expansion a...

conjecture | open | MathDB 页面 | 原文线索

V2-073696 | MATHDB-381710

Transitivity conjecture for exceptional bases of the Grothendieck group of projective space

摘录 / EXCERPT: Transitivity conjecture. The group spanned by mutations of exceptional bases and the isometries of $K_0(\mathbb{P}_n)$ acts transitively on the set of exceptional bases of...

conjecture | open | MathDB 页面 | 原文线索

V2-073697 | MATHDB-381712

Barnabei-Bonetti-Castronuovo-Silimbani's Motzkin enumeration conjecture for alternating involutions

摘录 / EXCERPT: Let $\mathcal{AI}_{m}(\rho)$ and $\mathcal{RAI}_{m}(\rho)$ denote, respectively, alternating and reverse alternating involutions of size m avoiding the pattern ρ, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073698 | MATHDB-381750

Essential positivity conjecture for radial Toeplitz operators

摘录 / EXCERPT: Essential positivity conjecture. The operator T_f is essentially positive if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-073699 | MATHDB-381769

Symmetric power functoriality conjecture for RAESDC Hilbert modular representations

摘录 / EXCERPT: Let $n\geq 1$. Let F^+ be a totally real field, and let (π,χ) be a RAESDC automorphic representation of $\mathrm{GL}_2({\mathbf A}_{F^+})$ without CM. Symmetric power func...

conjecture | open | MathDB 页面 | 原文线索

V2-073700 | MATHDB-381770

Local-global symmetric power functoriality conjecture for essentially discrete series

摘录 / EXCERPT: Let $n\geq 1$. Let F^+ be a totally real field, and let π be a cuspidal automorphic representation of $\mathrm{GL}_2({\mathbf A}_{F^+})$ without CM, with π_∞ essenti...

conjecture | open | MathDB 页面 | 原文线索

V2-073701 | MATHDB-381784

Five-point minimizer phase-transition conjecture for the frame potential

摘录 / EXCERPT: Let $\operatorname{FP}_{p,5,2}(X)$ be the pth frame potential of a five-point configuration X in $\mathbb{R}^2$, and let $X_5^\perp$, $X_5^{(h)}$, $Y(\alpha)$, and $Z(\alpha)$...

conjecture | open | MathDB 页面 | 原文线索

V2-073702 | MATHDB-381785

The equality conjecture for Colin de Verdière-type parameters and realizable dimension

摘录 / EXCERPT: Equality conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073703 | MATHDB-381786

The one-unit gap conjecture for realizable dimensions

摘录 / EXCERPT: One-unit gap conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073704 | MATHDB-381787

The forbidden-minor characterization conjecture for realizable dimension at most two

摘录 / EXCERPT: Forbidden-minor characterization conjecture. The following are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-073705 | MATHDB-381791

Hodge double ramification cycle conjecture

摘录 / EXCERPT: Let $g,k,u\geq 0$ and let $A=(a_1,\ldots,a_n)\in\mathbb{Z}^n$ satisfy ... Let $\overline{{\mathcal M}}_{g,A}^{k,r}$ be the moduli space of twisted r-spin structures, let...

conjecture | open | MathDB 页面 | 原文线索

V2-073706 | MATHDB-381792

Destabilization conjecture for diffusive boundary feedback

摘录 / EXCERPT: For $\eta>0$, consider the coupled transport-diffusion system with state variables y and $\hat y$ on $[0,1]$: ... with boundary conditions ... Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073707 | MATHDB-381805

Conjecture on the super edge-magic total strength of odd-cycle unicyclic graphs

摘录 / EXCERPT: Let $G(n;k_{1},\dots,k_{n})$ be the super edge-magic total unicyclic (p,q)-graph consisting of an odd cycle ... and k_i pendant vertices adjacent to each a_i, for...

conjecture | open | MathDB 页面 | 原文线索

V2-073708 | MATHDB-381812

Universal-element conjecture for componentwise finite-minor graphs

摘录 / EXCERPT: Let $\mathcal F$ be the class of graphs G such that no component of G contains every finite graph as a minor. A graph U is $<$-universal for a class if every graph in the cla...

conjecture | open | MathDB 页面 | 原文线索

V2-073709 | MATHDB-381824

The ribbon cobordism converse to cubiquitous embeddings

摘录 / EXCERPT: Ribbon cobordism converse. If $\Lambda(L_2)$ admits a cubiquitous embedding into a stabilization of $\Lambda(L_1)$, then there exists a ribbon cobordism from $\Sigma(L_1)$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-073710 | MATHDB-381827

Michel-Venkatesh mixing conjecture for shifted Heegner packets

摘录 / EXCERPT: Let $d\to\fty$ along ${\mathbb D}^\flat$, let ${\sf R}_d^\Delta([\mathfrak s])$ be the shifted diagonal Heegner packet, let μ be the quotient measure on ${\sf S}^2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073711 | MATHDB-381828

Michel-Venkatesh adelic mixing conjecture at level U

摘录 / EXCERPT: Michel-Venkatesh adelic mixing conjecture. If $|D_i|\to\infty$ and $q_i\to\infty$ as $i\to\infty$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073712 | MATHDB-381830

Cranston's sparse-graph odd coloring conjecture

摘录 / EXCERPT: Let G be a finite simple graph, let $c\geq 4$, and let $\operatorname{mad}(G)$ be its maximum average degree, defined by ... An odd c-coloring is a proper c-coloring in which e...

conjecture | open | MathDB 页面 | 原文线索

V2-073713 | MATHDB-381831

Strong-KPP uniqueness conjecture for positive bounded solutions in general domains

摘录 / EXCERPT: Let (Ω,ϱ) be a domain with the boundary conditions and elliptic problem specified by ... admits at most one positive bounded solution. The paper's methods do not treat c...

conjecture | open | MathDB 页面 | 原文线索

V2-073714 | MATHDB-381832

Critical-eigenvalue nonexistence conjecture for strong-KPP reactions

摘录 / EXCERPT: Let (Ω,ϱ) be a domain and boundary parameter for the problem ... lambda(Omega,varrho)=f'(0), ... has no positive bounded solution. This extends the established conclusion...

conjecture | open | MathDB 页面 | 原文线索

V2-073715 | MATHDB-381835

Hankel determinant bound for functions in the class U with vanishing second coefficient

摘录 / EXCERPT: Hankel determinant conjecture. For every integer $n\ge3$,

conjecture | open | MathDB 页面 | 原文线索

V2-073716 | MATHDB-381837

Regular unimodular triangulation conjecture for the parking function polytope

摘录 / EXCERPT: Let PF_n denote the parking function polytope. A triangulation of a lattice polytope is regular if it is induced by a lifting function, and unimodular if every simplex i...

conjecture | open | MathDB 页面 | 原文线索

V2-073717 | MATHDB-381838

Wildon's stability conjecture for plethysm coefficient sequences

摘录 / EXCERPT: Let $ u$ and τ be arbitrary partitions, and let $min\mathbb{N}$. The notation $\mathsf{A}_{m-1,m}(u,\tau)$ denotes the associated sequence of plethysm coefficients. Wildon's...

conjecture | open | MathDB 页面 | 原文线索

V2-073718 | MATHDB-381839

General monotonicity conjecture for plethysm coefficient sequences

摘录 / EXCERPT: Let $min\mathbb{N}$ and $lin\{0,1,\dotsc,m\}$. For partitions $ u$ and τ, let $\mathsf{A}_{l,m}(u,\tau)$ be the sequence of plethysm coefficients defined from the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-073719 | MATHDB-381858

The b5-conjecture for class group torsion

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a number field of degree s with discriminant D_K. For an integer $\ell\geq 1$, write $h_K[\ell]$ for the order of the ℓ-torsion subgroup of the class...

conjecture | open | MathDB 页面 | 原文线索

V2-073720 | MATHDB-381864

Torsion refined Gopakumar-Vafa stability conjecture for non-Kähler small resolutions

摘录 / EXCERPT: Let X be a Calabi-Yau threefold with only terminal conifold singularities and no Kähler small resolution. Let $\mathcal{S}$ be the set of all small resolutions $\widehat{X}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-073721 | MATHDB-381867

Caro-Petru1evski-skrekovski's conflict-free chromatic conjecture

摘录 / EXCERPT: Caro-Petru1evski-skrekovski's conjecture. For every connected graph G of maximum degree $\Delta\geq 3$,

conjecture | open | MathDB 页面 | 原文线索

V2-073722 | MATHDB-381871

Belkhechine et al.'s lower-bound conjecture for the inversion number

摘录 / EXCERPT: Belkhechine et al.'s inversion-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073723 | MATHDB-381872

The inversion number of the tournament Q_n

摘录 / EXCERPT: The Q_n inversion-number conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-073724 | MATHDB-381875

Alon's partition conjecture for inversion number

摘录 / EXCERPT: Alon's inversion-number partition conjecture. For every two positive integers k_1,k_2, there exists an integer $f(k_1,k_2)$ such that every oriented graph D with

conjecture | open | MathDB 页面 | 原文线索

V2-073725 | MATHDB-381876

The all-k counterexample conjecture for dijoin additivity

摘录 / EXCERPT: The all-k dijoin counterexample conjecture. For any $k\geq3$, there is a tournament T_k with $\operatorname{inv}(T_k)=k$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073726 | MATHDB-381881

Monotonicity conjecture for normalized mixed-code sphere sizes

摘录 / EXCERPT: Let n be the alphabet length, let s_r denote the size of the sphere of radius r, and let $\binom{n}{r}$ be the binomial coefficient. The quantities $\frac{s_r}{\binom{n}{r}}$...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073727 | MATHDB-381883

Butler's Schur-positivity conjecture for Macdonald intersection polynomials

摘录 / EXCERPT: Butler's conjecture. The symmetric function $\operatorname{I}_{\lambda,\mu}[X;q,t]$ is Schur positive.

conjecture | open | MathDB 页面 | 原文线索

V2-073728 | MATHDB-381889

The weaker Vizing conjecture for layer projections contained in minimum dominating sets

摘录 / EXCERPT: Let G and H be graphs, let $G\Box H$ be their Cartesian product, and let D be a dominating set of $G\Box H$. For each $h\in V(H)$, let G_h be the G-layer corresponding to...

conjecture | open | MathDB 页面 | 原文线索

V2-073729 | MATHDB-381890

Three-set extension of Vizing's conjecture for Cartesian products

摘录 / EXCERPT: Let G and H be graphs, let $G\Box H$ be their Cartesian product, and let D be a dominating set of $G\Box H$. Let $M_G(A)$ denote the family of dominating sets of G associat...

conjecture | open | MathDB 页面 | 原文线索

V2-073730 | MATHDB-381894

A linear upper bound for the Gaussian Benjamini-Hochberg false discovery rate

摘录 / EXCERPT: Gaussian BH-FDR bound. There exists $c\geqslant 1$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073731 | MATHDB-381895

Equality of the Gaussian worst-case FDR and Simes error bounds

摘录 / EXCERPT: Let $\mathcal G^K$ be the class of Gaussian-dependent vectors of K p-values, and let $S_K(\mathbf P)$ be the Simes p-value. Define ... For the Benjamini-Hochberg procedure at level...

conjecture | open | MathDB 页面 | 原文线索

V2-073732 | MATHDB-381912

The symmetry-principle link-existence conjecture for height four complete intersections

摘录 / EXCERPT: Link-existence conjecture. For any i, $3\leq i\leq d-2$, if the i-th link does not exist, so that there is a common factor in the initial degree of the ideal of Y_{i-1}, then...

conjecture | open | MathDB 页面 | 原文线索

V2-073733 | MATHDB-381919

The global period-index conjecture for orbifolds

摘录 / EXCERPT: Global period-index conjecture. For any such Brauer class α,

conjecture | open | MathDB 页面 | 原文线索

V2-073734 | MATHDB-381948

The dijoin conjecture for oriented graphs

摘录 / EXCERPT: Dijoin conjecture.

conjecture | refuted | MathDB 页面 | 原文线索

V2-073735 | MATHDB-381949

The full range of counterexamples to the dijoin conjecture

摘录 / EXCERPT: Dijoin counterexample-range conjecture. For all $\ell,r\in\mathbb{N}$ with $\ell\geq 3$ or $r\geq 3$, there exist oriented graphs L and R such that

conjecture | refuted | MathDB 页面 | 原文线索

V2-073736 | MATHDB-381952

Full transfer conjecture for local Arthur parameters

摘录 / EXCERPT: Let $\Delta=((T_i,d_i,\eta_i,\lambda_i))_i$ be a shape and let φ_v be a trace-positive test function on G_v. For every $\psi=\bigoplus_i\psi_i[d_i]$ in Δ, with...

conjecture | open | MathDB 页面 | 原文线索

V2-073737 | MATHDB-381956

Matherne-Morales-Selover conjecture on unit interval orders and the zeta bijection

摘录 / EXCERPT: Let $\mathcal U_n$ be the set of unit interval orders on n elements, and let $\mathcal D_n$ be the set of Dyck paths from $(0,0)$ to (n,n). For $U\in\mathcal U_n$, let $a(U)$ b...

conjecture | open | MathDB 页面 | 原文线索

V2-073738 | MATHDB-381960

Horak-AlBdaiwi conjecture on diameter perfect Lee codes

摘录 / EXCERPT: Let $DPL(n,d)$ denote a diameter d perfect Lee code in $\mathbb{Z}^{n}$. For odd d, this agrees with a perfect Lee code of radius $(d-1)/2$; for even d, it is defined using a...

conjecture | open | MathDB 页面 | 原文线索

V2-073739 | MATHDB-381968

Alternation conjecture for Euler characteristics of triple Schubert-cell intersections

摘录 / EXCERPT: Alternation conjecture. The Euler characteristic is alternating:

conjecture | open | MathDB 页面 | 原文线索

V2-073740 | MATHDB-381979

Kusner-McGrath's eigenspace conjecture for embedded free boundary minimal surfaces

摘录 / EXCERPT: Kusner-McGrath's conjecture. The first Steklov eigenspace of S coincides with the span of the coordinate functions of the embedding.

conjecture | open | MathDB 页面 | 原文线索

V2-073741 | MATHDB-381994

The conjecture that the common preperiodic-point bound equals $4d-1$

摘录 / EXCERPT: The $4d-1$ bound conjecture. One can take

conjecture | open | MathDB 页面 | 原文线索

V2-073742 | MATHDB-381996

The lonely pseudo-Anosov conjecture

摘录 / EXCERPT: For a finite type surface S, let $f\colon S\to S$ be a pseudo-Anosov homeomorphism, and let (X,q) be the associated complex structure and holomorphic quadratic differential. Ca...

conjecture | open | MathDB 页面 | 原文线索

V2-073743 | MATHDB-381999

The Lemke-Kleitman conjecture on index-one subsequences

摘录 / EXCERPT: Let n be a positive integer, let T be a sequence of length n over the cyclic group $\mathbb{Z}_n$, and call a subsequence of T index one when its index is one. Lemke-Kleitm...

conjecture | open | MathDB 页面 | 原文线索

V2-073744 | MATHDB-382001

The minimality conjecture for zero-sum sequences over elementary abelian 3-groups

摘录 / EXCERPT: Let G be a finite abelian group, let $\mathcal A(G)$ denote the set of minimal zero-sum sequences over G, and let ${\rm D}(G)$ denote the Davenport constant. A set...

conjecture | open | MathDB 页面 | 原文线索

V2-073745 | MATHDB-382007

Cyclic sieving conjecture for promotion on fans of Dyck paths

摘录 / EXCERPT: Let $D_n^{(r)}$ be the set of all r-fans of Dyck paths of length $2n$, and define ... where $[m]_q=1+q+q^2+\cdots+q^{m-1}$. Let C_{2n} be the cyclic group of order $2n$ acting on...

conjecture | open | MathDB 页面 | 原文线索

V2-073746 | MATHDB-382020

The nilpotent generalized near-group fusion-ring conjecture

摘录 / EXCERPT: Let $\mathcal{C}$ be a braided generalized near-group fusion category. Write $R(m,n)$ for the corresponding generalized near-group fusion ring, with $m,n\in\mathbb{Z}_{\geq 1}$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-073747 | MATHDB-382026

The rigidity conjecture for multiassociahedra along the moment curve

摘录 / EXCERPT: Let P be a convex n-gon, and let k-triangulations be its maximal sets of diagonals containing no $k+1$ mutually crossing diagonals. Place n generic points on the moment cur...

conjecture | open | MathDB 页面 | 原文线索

V2-073748 | MATHDB-382030

Pagani-Ricolfi-van Zelm's tautologicality conjecture for Brill-Noether pullbacks

摘录 / EXCERPT: Pagani-Ricolfi-van Zelm's conjecture. The classes $\mathsf{w}_{g,A,d}^r(\theta)$ lie in the tautological ring $R^*(\overline{\mathcal{M}}_{g,n})$ for any choice of g,d,A and gene...

conjecture | open | MathDB 页面 | 原文线索

V2-073749 | MATHDB-382031

The logarithmic double ramification cycle tautologicality conjecture

摘录 / EXCERPT: Logarithmic double ramification tautologicality conjecture. The logarithmic double ramification cycle belongs to

conjecture | open | MathDB 页面 | 原文线索

V2-073750 | MATHDB-382033

Ishii-Stade shift-equation conjecture for Mellin transforms

摘录 / EXCERPT: Let $m,n\in \mathbb Z$ with $1\le m\le n-1$ and let $\delta\in \mathbb Z_{\ge0}$. For $x\in\mathbb C$ and $q\in\mathbb Z_{\ge0}$, write ... Let $\what{n}{\alpha}{s}$ be the Mellin tr...

conjecture | open | MathDB 页面 | 原文线索

V2-073751 | MATHDB-382037

Gray-Wyner conjecture for the doubly symmetric binary source

摘录 / EXCERPT: Let (X,Y) be a doubly symmetric binary source ${\mathrm{DSBS}}(p)$ with $p\in(0,1/2)$. Let $R(D_{1},D_{2})$ denote its rate-distortion function under Hamming distortion, let h...

conjecture | open | MathDB 页面 | 原文线索

V2-073752 | MATHDB-382041

Ivančo's supermagic labeling conjecture for Cartesian products of cycles

摘录 / EXCERPT: Let C_n and C_m be cycles with $n,m\geq 3$. A supermagic labeling of a graph with q edges is a bijection from its edge set to $\{1,2,\dots,q\}$ such that the sums of the labe...

conjecture | open | MathDB 页面 | 原文线索

V2-073753 | MATHDB-382049

The chordal graph representation conjecture for clique and reduced clique graphs

摘录 / EXCERPT: Chordal graph representation conjecture. Let H be a chordal graph. There are chordal graphs G and G' such that H is isomorphic to both $C(G)$ and $C_R(G')$.

conjecture | open | MathDB 页面 | 原文线索

V2-073754 | MATHDB-382052

Binomial generating-series conjecture for orthogonal vector spaces

摘录 / EXCERPT: In the orthogonal vector-space situation, let $\binom{x}{n}$ denote the generalized binomial coefficient, let $\boldsymbol{D}_n$ and $\boldsymbol{B}_n$ denote the corresponding Dyn...

conjecture | open | MathDB 页面 | 原文线索

V2-073755 | MATHDB-382053

Generating-series conjecture for the self-dual \tilde{A}_1 quiver

摘录 / EXCERPT: Let $Q=(\bullet\rightrightarrows\bullet)$ be the self-dual $\tilde{A}_1$ quiver with its two vertices exchanged by the contravariant involution, and write $\tilde{A}_1^{u,v}$ for t...

conjecture | open | MathDB 页面 | 原文线索

V2-073756 | MATHDB-382054

Wall-crossing conjecture for self-dual Donaldson--Thomas invariants

摘录 / EXCERPT: Let (Q,W) be a self-dual quiver with potential, and let τ and $\widetilde{\tau}$ be self-dual weak stability conditions on the associated self-dual abelian category. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-073757 | MATHDB-382064

The flag-accuracy conjecture for extended Catalan arrangements

摘录 / EXCERPT: For a root system of Dynkin type, let Cat^m denote the corresponding extended Catalan arrangement. An arrangement is flag-accurate when it admits a flag satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-073758 | MATHDB-382065

The freeness conjecture for the interpolating arrangements $\mathscr{D}^r_\ell(a)$

摘录 / EXCERPT: Freeness conjecture for $\mathscr{D}^r_\ell(a)$. The cone $\mathbf{c}\mathscr{D}^r_\ell(a)$ is free with exponents

conjecture | open | MathDB 页面 | 原文线索

V2-073759 | MATHDB-382079

The diagonal determination conjecture for Bar-Natan-van der Veen polynomials

摘录 / EXCERPT: Diagonal determination conjecture. The family of polynomials $\rho_K^{i,i}$, $i>0$, is completely determined by the Alexander polynomial of K.

conjecture | open | MathDB 页面 | 原文线索

V2-073760 | MATHDB-382083

Spectral union conjecture for monodromy operators of neutral renewal equations

摘录 / EXCERPT: Let $A\colon \mathbb{R} \to \mathbb{R}^{d \times d}$ be a τ-periodic function whose components have bounded variation and are continuous from the right. Let U be the monodro...

conjecture | open | MathDB 页面 | 原文线索

V2-073761 | MATHDB-382084

Euler partition inequalities involving $Q_2(n)$

摘录 / EXCERPT: For $n,k\geqslant0$, let $p(n)$ denote Euler's partition function, let $Q_2(n)$ denote the partition function used in the source, and let $(\rho_j)_{j\geqslant0}$ be the coefficien...

conjecture | open | MathDB 页面 | 原文线索

V2-073762 | MATHDB-382085

6-regular partition truncated-sum inequality

摘录 / EXCERPT: For any integer n, define ... Let $b_6(n)$ denote the 6-regular partition function. 6-regular partition inequality. For $n\geqslant0$ and $k>0$, ... with strict inequality if...

conjecture | open | MathDB 页面 | 原文线索

V2-073763 | MATHDB-382086

6-regular partition inequalities involving β_n

摘录 / EXCERPT: For any integer n, define ... Let $b_6(n)$ denote the 6-regular partition function, and let $(\rho_j)_{j\geqslant0}$ be the coefficient sequence in the source. 6-regular partitio...

conjecture | open | MathDB 页面 | 原文线索

V2-073764 | MATHDB-382087

Weaker Q_2 truncated-sum inequality

摘录 / EXCERPT: For any nonnegative integer n, define ... Let $Q_2(n)$ denote the partition function in the source. Weaker Q_2 truncated-sum inequality. For $n\geqslant0$ and $k>0$, ... The source...

conjecture | open | MathDB 页面 | 原文线索

V2-073765 | MATHDB-382092

The cyclotomic-number conjecture for conflict-avoiding codes

摘录 / EXCERPT: Cyclotomic-number conjecture. If $\ell\geq 3$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073766 | MATHDB-382101

Sarnak's Weyl-law conjecture for congruence subgroups and arbitrary types

摘录 / EXCERPT: Let G be the reductive group considered in the paper, let $\Gamma\subset G(\mathbb{Q})$ be a congruence subgroup, and let $\Pi(K_\infty)$ denote the set of K_∞-types. For...

conjecture | open | MathDB 页面 | 原文线索

V2-073767 | MATHDB-382108

Asymptotic almost periodicity conjecture for the resolvent norm function

摘录 / EXCERPT: Let $\alpha(\omega)$ denote the norm of $W(-\nu_{0}+i\omega)$, where W is the operator-valued function defined in the preceding construction. Asymptotic almost periodicity conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-073768 | MATHDB-382113

Bogomolny-Schmit nodal domain conjecture for even Maass cusp forms

摘录 / EXCERPT: Let $\Gamma=SL_2(\mathbb{Z})$ be the full modular group, let $\mathbb{X}=\Gamma\backslash\mathbb{H}$ be the modular surface, and let ϕ be an even Hecke-Maass cusp form with ei...

conjecture | open | MathDB 页面 | 原文线索

V2-073769 | MATHDB-382114

Iwaniec-Sarnak local L^p-norm conjecture for Hecke-Maass forms

摘录 / EXCERPT: Let ϕ be a Hecke-Maass form for the full modular group, let K be a compact subset of the modular surface $\mathbb{X}$, let $2<p\le\infty$, and let $\epsilon>0$. Then Iwaniec...

conjecture | open | MathDB 页面 | 原文线索

V2-073770 | MATHDB-382115

Optimal tapered lightning approximation conjecture for x^α

摘录 / EXCERPT: Let α be the exponent in the approximation of x^α on $[0,1]$, and let $r(x)$ be a lightning + polynomial approximation with N_1 lightning terms,...

conjecture | open | MathDB 页面 | 原文线索

V2-073771 | MATHDB-382116

Optimal tapered lightning approximation conjecture for z^α on a V-shaped domain

摘录 / EXCERPT: Let α be the exponent, let $\beta\in[0,2)$, and consider the V-shaped domain $[0,1]e^{\pm i\beta\pi/2}$. Let $r(z)$ be a lightning + polynomial approximation with N_1 ligh...

conjecture | open | MathDB 页面 | 原文线索

V2-073772 | MATHDB-382122

Conjectured asymptotic MSE of the mismatched Bayes estimator

摘录 / EXCERPT: Consider the rank-one rectangular observation model $\bY$ and the mismatched Bayes estimator $M_{\mathrm{mis}}(\bY)$. Let λ and λ_* denote the estimation and true s...

conjecture | open | MathDB 页面 | 原文线索

V2-073773 | MATHDB-382123

Baer's conjecture on Noetherian integral group rings

摘录 / EXCERPT: Let G be a group. Its integral group ring is denoted by $ \mathbb{Z}G$. Baer's conjecture. $\mathbb{Z}G$ is Noetherian if and only if G is virtually polycyclic. The forward imp...

conjecture | open | MathDB 页面 | 原文线索

V2-073774 | MATHDB-382124

Conjecture on properly coherent group rings of amenable groups

摘录 / EXCERPT: Let G be a finitely generated amenable group. Its integral group ring is denoted by $\mathbb{Z}G$. A group ring is properly coherent when it is coherent but not Noetherian. A pro...

conjecture | open | MathDB 页面 | 原文线索

V2-073775 | MATHDB-382125

Nonlinear smoothing conjecture for dispersive equations

摘录 / EXCERPT: Let a nonlinear dispersive equation have dispersion of order l and a nonlinearity of order $k\ge 3$ with a loss of m derivatives. Let s_0 be such that, for $s>s_0$, the H^s...

conjecture | refuted | MathDB 页面 | 原文线索

V2-073776 | MATHDB-382132

The first main conjecture on fluctuations of Schensted row insertion

摘录 / EXCERPT: Fix $z\in[0,1]$. Let $(u_0,v_0)=(\operatorname{RSK\ cos}z,\operatorname{RSK\ sin}z)$ be the corresponding point on the Logan--Shepp--Vershik--Kerov limit curve Ω_*. For i.i....

conjecture | open | MathDB 页面 | 原文线索

V2-073777 | MATHDB-382133

The second main conjecture on jeu de taquin trajectories

摘录 / EXCERPT: Fix $z\in[0,1]$ and let (u_0,v_0) be the corresponding point on the Logan--Shepp--Vershik--Kerov limit curve Ω_*. Let $w_1,w_2,\ldots$ be i.i.d. uniform random variables o...

conjecture | open | MathDB 页面 | 原文线索

V2-073778 | MATHDB-382134

The Plancherel-TASEP second-class-particle Brownian-motion conjecture

摘录 / EXCERPT: Consider the Plancherel-TASEP interacting particle system, and let u_n be the position of its second-class particle at time n. Let μ_{SC} be the semicircle d...

conjecture | open | MathDB 页面 | 原文线索

V2-073779 | MATHDB-382135

The Plancherel transition-measure convergence-rate conjecture

摘录 / EXCERPT: Let $\lambda^{(n)}$ be a random Young diagram with n boxes distributed according to the Plancherel measure. Let $K_{\lambda^{(n)}}$ be its transition-measure distribution functio...

conjecture | open | MathDB 页面 | 原文线索

V2-073780 | MATHDB-382136

Uniform boundedness conjecture for minimal log discrepancies of Kollár components

摘录 / EXCERPT: Fix $n\in\mathbb{N}$, $\varepsilon>0$, and let $I\subseteq [0,1]$ be a finite set. For a klt singularity $x\in (X,\Delta)$, let $\operatorname{mld}^{\mathrm{K}}(x,X,\Delta)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-073781 | MATHDB-382149

Injectivity of symmetric compositions for Q-key polynomials

摘录 / EXCERPT: Let α and γ be distinct symmetric weak compositions. Injectivity conjecture. Then ... This conjecture is supported by computations, although the authors report no heuri...

conjecture | open | MathDB 页面 | 原文线索

V2-073782 | MATHDB-382150

Leading-term conjecture for P-key polynomials

摘录 / EXCERPT: Let α be a skew-symmetric weak composition, let $n=\ell(\alpha)$, and let $\widetilde\rho(\alpha)$ and $\widetilde\gamma(\alpha)$ denote its strict sub-diagonal row and colu...

conjecture | open | MathDB 页面 | 原文线索

V2-073783 | MATHDB-382151

Q-key expansion conjecture for orthogonal involution Schubert polynomials

摘录 / EXCERPT: Let I_∞ denote the involutions of the infinite symmetric group, let $\mathfrak S^{\mathsf{O}}_z$ be the corresponding orthogonal involution Schubert polynomial, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073784 | MATHDB-382152

Converse to the alpha-one theorem for Q-key polynomials

摘录 / EXCERPT: Let α be a symmetric weak composition, and let a P-key polynomial mean a polynomial of the form κ^P_η for a skew-symmetric weak composition η. Conver...

conjecture | open | MathDB 页面 | 原文线索

V2-073785 | MATHDB-382153

Key-positivity constraint for P-key expansions

摘录 / EXCERPT: Let α and γ be weak compositions, with α skew-symmetric. Say that $\kappa^\mathsf{P}_\alpha-\kappa_\gamma$ is key positive when it is a nonnegative linear comb...

conjecture | open | MathDB 页面 | 原文线索

V2-073786 | MATHDB-382155

Stable orthogonal Grothendieck specialization for involutions fixing 1

摘录 / EXCERPT: Let I_∞ be the involutions of the infinite symmetric group, let $\mathrm{Des}_V(z)$ be the visible descent set, and let $\mathfrak G^{\mathsf{O}}_z$ be the orthogonal involu...

conjecture | open | MathDB 页面 | 原文线索

V2-073787 | MATHDB-382161

The affine type B distance conjecture

摘录 / EXCERPT: Let w be an affine signed permutation in $\widetilde{S}^B_n$. An integer $j\in[n-1]$ is good if w restricts to a bijection from $\pm[j]$ to itself, and it is very good if this...

conjecture | open | MathDB 页面 | 原文线索

V2-073788 | MATHDB-382162

Bounds and equality characterization for the affine type D distance

摘录 / EXCERPT: Let $w\in\widetilde{S}^D_n$ be an affine signed permutation, and let $\$(w)$ and $operatorname{dis}(w)...

conjecture | open | MathDB 页面 | 原文线索

V2-073789 | MATHDB-382169

Strong-subsolution conjecture for solutions of the Z-critical equation

摘录 / EXCERPT: Let E be an object carrying a Z-critical equation, and let h be a metric on E. A strong subsolution is a metric satisfying the positivity conditions on all formal matrix de...

conjecture | open | MathDB 页面 | 原文线索

V2-073790 | MATHDB-382170

Gyárfás-Sumner analogue for eta-boundedness

摘录 / EXCERPT: Gyárfás-Sumner analogue for η-boundedness. For every forest H, there exists a function $f:\mathbb{N}\to\mathbb{N}$ such that every H-free graph G satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-073791 | MATHDB-382171

Polynomial eta-boundedness of P_5-free graphs

摘录 / EXCERPT: Polynomial P_5-free eta-boundedness conjecture. There exists $d\in\mathbb{N}$ such that every P_5-free graph G satisfies $\eta(G)\leq\omega(G)^d$.

conjecture | open | MathDB 页面 | 原文线索

V2-073792 | MATHDB-382172

Polynomial eta-boundedness of matching-free graphs

摘录 / EXCERPT: Matching-free polynomial eta-boundedness conjecture. For every integer $t\geq 3$, M_t-free graphs are polynomially η-bounded.

conjecture | open | MathDB 页面 | 原文线索

V2-073793 | MATHDB-382176

Lapid-Mao conjecture on Whittaker and L^2 normalizations for GSp_4

摘录 / EXCERPT: Let $G=\mathrm{GSp}_4$, let $\pi=\otimes_v\pi_v$ be an irreducible, unitary, cuspidal, generic automorphic representation of $G(\mathbb A)$ with trivial central character, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-073794 | MATHDB-382177

Liu's refined Gan-Gross-Prasad conjecture for $({\rm SO}(5),{\rm SO}(2))$

摘录 / EXCERPT: Let $L/\mathbb Q$ be an imaginary quadratic extension with discriminant $-D$, let Λ be a character of $L^\times\backslash\mathbb A_L^\times$ trivial on $\mathbb A^\times$,...

conjecture | open | MathDB 页面 | 原文线索

V2-073795 | MATHDB-382185

Ghosh-Sarnak conjecture on integral Hasse failures for Markoff surfaces

摘录 / EXCERPT: Let U_m be the affine Markoff surface ... where m is an integer, and let $U_m(\mathbb{A}_{\mathbb{Z}})$ denote its integral adelic points. Ghosh-Sarnak conjecture. The number of...

conjecture | refuted | MathDB 页面 | 原文线索

V2-073796 | MATHDB-382186

Nearly optimal spherical cap discrepancy for greedy sequences on the sphere

摘录 / EXCERPT: Let $\omega_{N,d} \subseteq \mathbb{S}^d$ be the first N elements of the greedy sequence with respect to K_{-1} as defined in the greedy construction. For d-dimensional spher...

conjecture | open | MathDB 页面 | 原文线索

V2-073797 | MATHDB-382191

Generalized exponential polynomial conjecture for mixed additive-function equations

摘录 / EXCERPT: Let n be a positive integer, let $\mathbb{F}\subset\mathbb{C}$ be a field, and let $p_{1},\ldots,p_{n},q_{1},\ldots,q_{n}$ be fixed positive integers satisfying conditions C(i)--...

conjecture | open | MathDB 页面 | 原文线索

V2-073798 | MATHDB-382192

Bistability conjecture for the Allee-effect fear competition system

摘录 / EXCERPT: Bistability conjecture. There exist initial data $[u_0(x),v_0(x)]$ such that the corresponding solution (u,v) converges uniformly to (u^*,v^*) as $t\to\infty$, while for some o...

conjecture | open | MathDB 页面 | 原文线索

V2-073799 | MATHDB-382197

Higashitani-Matsumoto conjecture on Kempe classes of almost bipartite graphs

摘录 / EXCERPT: Higashitani-Matsumoto conjecture.

conjecture | refuted | MathDB 页面 | 原文线索

V2-073800 | MATHDB-382198

Alternating subword complex equivalence conjecture

摘录 / EXCERPT: Let Q be a word in the simple generators of a finite Coxeter group, let ω be a Coxeter-group element, and let $[e,\omega]$ be the corresponding interval in the weak order....

conjecture | open | MathDB 页面 | 原文线索

V2-073801 | MATHDB-382205

Boundary unique continuation conjecture for harmonic functions on Lipschitz domains

摘录 / EXCERPT: Let Ω be a Lipschitz domain, and let $u\in H^1$ be a weak solution to the Dirichlet problem ... Suppose that the set...

conjecture | open | MathDB 页面 | 原文线索

V2-073802 | MATHDB-382234

Global bubbling conjecture for Kähler-Einstein orbifold smoothings

摘录 / EXCERPT: Let $\pi:\mathcal{X}\rightarrow\Delta$ be a smoothing of a Kähler-Einstein orbifold (X_0,ω_0) with A_k-singularities, and suppose that $\operatorname{Aut}(X_0)$ is discret...

conjecture | open | MathDB 页面 | 原文线索

V2-073803 | MATHDB-382244

Polynomial chi-boundedness conjecture for path-free graphs

摘录 / EXCERPT: For a graph G, let $\chi(G)$ denote its chromatic number and let $\omega(G)$ denote its clique number. A graph is P-free if it contains no induced copy of the path P. Path-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-073804 | MATHDB-382249

The completely monotone conjecture for Fisher information along the heat flow

摘录 / EXCERPT: Completely monotone conjecture. The function $t\longmapsto I(X,t)$ is completely monotonous, that is

conjecture | open | MathDB 页面 | 原文线索

V2-073805 | MATHDB-382257

Bipartition-respecting packing coloring conjecture for hypercubes

摘录 / EXCERPT: Bipartition-respecting packing coloring conjecture. For every $n\geq 0$,

conjecture | open | MathDB 页面 | 原文线索

V2-073806 | MATHDB-382265

Polynomial minimal separators or bounded hole length for induced-minor-free graphs

摘录 / EXCERPT: Polynomial-separator or bounded-hole conjecture. There exists a polynomial p and an integer k such that if G has no clique cutset and does not contain $K_{2,3}$ as an induced...

conjecture | open | MathDB 页面 | 原文线索

V2-073807 | MATHDB-382267

Ih's finiteness conjecture for S-integral preperiodic points

摘录 / EXCERPT: Let K be a number field, let S be a finite set of places of K containing all archimedean places, and let $f:\mathbb{P}^1\to\mathbb{P}^1$ be a rational map of degree at least...

conjecture | open | MathDB 页面 | 原文线索

V2-073808 | MATHDB-382274

Type 1 conjecture for fullerene nanodiscs

摘录 / EXCERPT: A fullerene nanodisc D_r is a fullerene nanodisc indexed by an integer r. Type 1 conjecture. The fullerene nanodiscs D_r, for every integer $r\geq 2$, are Type 1. The conject...

conjecture | open | MathDB 页面 | 原文线索

V2-073809 | MATHDB-382284

Jin-Szenes conjecture for perturbed genus-zero Gromov-Witten invariants

摘录 / EXCERPT: Let M_N^k be the degree-k projective Fano hypersurface considered in the paper, and let $\langle {\cal O}_{h^a}{\cal O}_{h^b}\rangle_0(t^0,t^1,\ldots,t^{N-2})$ denote the pertu...

conjecture | open | MathDB 页面 | 原文线索

V2-073810 | MATHDB-382286

The smoothed equivariant p-adic Artin conjecture with uniqueness

摘录 / EXCERPT: Let L/K be as above, and let S and T be finite non-empty disjoint sets of places of K, with S containing all places ramifying in L^+_∞/K and all infinite places. W...

conjecture | open | MathDB 页面 | 原文线索

V2-073811 | MATHDB-382287

Local antimagic chromatic number of even-by-odd flower-bouquet graphs

摘录 / EXCERPT: Let r and s be integers, with r even and $r\ge 2$, and s odd and $s\ge 3$. Let $rFB(s)$ denote the corresponding flower-bouquet graph, and let $\chi_{la}(G)$ be the local a...

conjecture | open | MathDB 页面 | 原文线索

V2-073812 | MATHDB-382293

The multi-wedge variational formula for the directed landscape

摘录 / EXCERPT: Let $Z\subset\mathbb{R}$ be a nonempty finite set, let $g:Z\to\mathbb{R}$, let $Y\subset\mathbb{R}$ be a nonempty finite set, and let $f:Y\to\mathbb{R}$ satisfy ... For $z\in Z$, def...

conjecture | open | MathDB 页面 | 原文线索

V2-073813 | MATHDB-382298

The signed projective cube homomorphism conjecture for signed planar graphs

摘录 / EXCERPT: Let (G,σ) be a signed planar graph, and let $g_{ij}(G,\sigma)$ denote the corresponding signed girth parameter; let $SPC(k)$ be the signed projective cube of dimension k,...

conjecture | open | MathDB 页面 | 原文线索

V2-073814 | MATHDB-382308

Optimality conjecture for the continuous Kaiser-Bessel shape parameter

摘录 / EXCERPT: Optimality conjecture. The parameter $\beta=m(\pi-\delta)$ is optimal for the Shannon sampling formula with the continuous Kaiser-Bessel window function, as it guarantees the maxim...

conjecture | open | MathDB 页面 | 原文线索

V2-073815 | MATHDB-382311

Castelnuovo bound conjecture for Gopakumar-Vafa invariants

摘录 / EXCERPT: Castelnuovo bound conjecture. The invariant vanishes when

conjecture | open | MathDB 页面 | 原文线索

V2-073816 | MATHDB-382332

The unique gridding conjecture for coils in cyclic partial multiplication matrices

摘录 / EXCERPT: Let M be a cyclic partial multiplication matrix containing ℓ non-empty cells. A coil is a permutation represented by a cyclic sequence of points following the cycle of non-e...

conjecture | open | MathDB 页面 | 原文线索

V2-073817 | MATHDB-382333

The refined arithmetic Gan-Gross-Prasad conjecture for p-adic heights

摘录 / EXCERPT: Let $V\in {\mathscr{V}}^{\circ}$ be an incoherent pair, and let π be a distinguished, stable, ordinary, cuspidal automorphic representation of $\mathrm{G}^{V}(\mathbf{A})$, tri...

conjecture | open | MathDB 页面 | 原文线索

V2-073818 | MATHDB-382352

The genericity conjecture for pure ABV-packets

摘录 / EXCERPT: Genericity conjecture. A pure L-packet $\Pi^{\text{pure}}_\phi(G)$ is generic if and only if ϕ is an open parameter for G.

conjecture | open | MathDB 页面 | 原文线索

V2-073819 | MATHDB-382360

Long-term behaviour conjecture for interacting reflected OU processes on graphs

摘录 / EXCERPT: Long-term behaviour conjecture. 1) If $\alpha<0$ and $\alpha+\beta\nu(G)<0$, then the process X is positive recurrent with the stationary distribution μ defined in Theorem. 2...

conjecture | open | MathDB 页面 | 原文线索

V2-073820 | MATHDB-382365

An analogue of Lehmer's conjecture for bounded-coefficient polynomial sequences

摘录 / EXCERPT: Let $N\geq 1$ be a natural number, and let P_{2n+2l} be a sequence of integer polynomials defined in the source's equation (with coefficients bounded in modulus by N). Its limi...

conjecture | open | MathDB 页面 | 原文线索

V2-073821 | MATHDB-382367

The conjecture that Theorem MainTechnical covers all rank-deficient permutations

摘录 / EXCERPT: Rank-deficient permutation coverage conjecture. For V generic Toeplitz, every permutation P for which

conjecture | open | MathDB 页面 | 原文线索

V2-073822 | MATHDB-382373

Classification conjecture for mutually unbiased basis triplets in dimension six

摘录 / EXCERPT: Classification conjecture. Any MUB-triplet (X_1,X_2,X_3) in dimension $d=6$ is permutationally unitary equivalent to an MUB-triplet obtained by the construction (hff)....

conjecture | open | MathDB 页面 | 原文线索

V2-073823 | MATHDB-382377

Generic dimension and non-smoothability conjecture for homogeneous curves

摘录 / EXCERPT: Generic dimension and non-smoothability conjecture. For generic L_r^n, equality holds in the stated bounds for $\dim T^1_l$ when $l>0$ and for $\dim T^1_{-1}$, with some exceptio...

conjecture | open | MathDB 页面 | 原文线索

V2-073824 | MATHDB-382387

Derickx-van Hoeij conjecture on modular units computing the gonality of X_1(p)

摘录 / EXCERPT: Derickx-van Hoeij conjecture. There is a modular unit f defined over $\mathbb{Q}$ on $X_1(p)$ such that

conjecture | open | MathDB 页面 | 原文线索

V2-073825 | MATHDB-382388

Uniform shifted-sum estimate for the discrete null cone

摘录 / EXCERPT: Uniform shifted-sum conjecture. The shifted sum satisfies estimate uniformly in z_0. Accordingly, for $d\geq 3$ and $a,b\in\mathbb{R}^d$, the sum

conjecture | open | MathDB 页面 | 原文线索

V2-073826 | MATHDB-382403

Echelonmotion realization of Dilworth's theorem for modular lattices

摘录 / EXCERPT: Let L be a modular lattice, and let σ be a linear extension of L. For $x\in L$, write $\operatorname{Cov}_L^{\uparrow}(x)$ for the elements covering x and...

conjecture | open | MathDB 页面 | 原文线索

V2-073827 | MATHDB-382405

Conjecture for Stirling cycle numbers of all orders

摘录 / EXCERPT: Let $C^{(r)}$ be the Stirling cycle triangle of order r, let ... be the triangle formed by reversing the rows of $C^{(r)}$, and let $c_{r,n}(x)$ be its row-generating polynomials....

conjecture | open | MathDB 页面 | 原文线索

V2-073828 | MATHDB-382407

Poonen's conjecture on rational periodic points of quadratic polynomials

摘录 / EXCERPT: Let $N\geq 4$. A quadratic polynomial is a polynomial $f(x)\in\mathbb{Q}[x]$ of degree 2, and a rational point of period N is an $x\in\mathbb{Q}$ such that $f^{(N)}(x)=x$ but...

conjecture | open | MathDB 页面 | 原文线索

V2-073829 | MATHDB-382410

General convergence conjecture for diffusion processes in narrow tubes

摘录 / EXCERPT: General convergence conjecture. The process $\Pi^\epsilon \circ Z^\epsilon$ converges weakly to $\bar{Z}$ in $C([0,\infty);\Gamma)$, uniformly with respect to the initial condition...

conjecture | open | MathDB 页面 | 原文线索

V2-073830 | MATHDB-382412

McCallum-Sharifi conjecture on cyclotomic p-unit cup products

摘录 / EXCERPT: McCallum-Sharifi conjecture. For all primes p, the pairing $\langle\cdot,\cdot\rangle_p$ is nontrivial.

conjecture | open | MathDB 页面 | 原文线索

V2-073831 | MATHDB-382415

Elliott-Pestun's multiplicative geometric Langlands conjecture

摘录 / EXCERPT: Let G be a semisimple simply-laced complex group, let θ be a diagram automorphism of G, and let C be a Calabi-Yau Riemann surface. For any...

conjecture | open | MathDB 页面 | 原文线索

V2-073832 | MATHDB-382418

Monomial positivity conjecture for the Jack super nabla operator on elementary functions

摘录 / EXCERPT: Let $n\geq 1$. For partitions $\pi,\mu,\nu\vdash n$, define the coefficient $f^\pi_{\mu,\nu}(\alpha)$ by ... where e_π denotes the elementary symmetric function. Monomial positiv...

conjecture | open | MathDB 页面 | 原文线索

V2-073833 | MATHDB-382421

Stembridge's conjecture on total nonnegativity of monomial trace immanants

摘录 / EXCERPT: Let n be a positive integer, let $\lambda\vdash n$ be a partition, and let ϕ^λ denote the corresponding monomial trace. For an $n\times n$ matrix x, write...

conjecture | open | MathDB 页面 | 原文线索

V2-073834 | MATHDB-382423

Conjecture on maximal third Chern character of rank two semistable sheaves on the degree-five Fano threefold

摘录 / EXCERPT: The maximal third Chern character conjecture. If $c=-1$ and $d=\frac12+\frac C5\ll0$, then

conjecture | open | MathDB 页面 | 原文线索

V2-073835 | MATHDB-382427

Classification conjecture for forms of nondefective height 1

摘录 / EXCERPT: Classification conjecture. Then φ is a Pfister neighbour.

conjecture | open | MathDB 页面 | 原文线索

V2-073836 | MATHDB-382438

The fusion-realisation condition for arbitrary fusion rings

摘录 / EXCERPT: Let R be a fusion ring and let $(r_{s,t})_{s,t\in S}$ be a matrix satisfying the geometric conditions for a Coxeter system $(\mathbb W,S)$, as in the fusion-realisation condition...

conjecture | open | MathDB 页面 | 原文线索

V2-073837 | MATHDB-382439

The unfolding theorem for arbitrary fusion rings

摘录 / EXCERPT: Let $(\mathbb W,S)$ be a Coxeter system with a geometric R-realisation over a fusion ring R, and let $(\check{\mathbb W},\check{S})$ be the corresponding unfolded Coxeter syste...

conjecture | open | MathDB 页面 | 原文线索

V2-073838 | MATHDB-382440

Rank conjecture for the Fisher information matrix of the Poisson canonical polyadic model

摘录 / EXCERPT: Fisher information rank conjecture. The rank is

conjecture | open | MathDB 页面 | 原文线索

V2-073839 | MATHDB-382446

Haettel's bowtie-free and upward-flag conjecture for type D_n subdivisions

摘录 / EXCERPT: Let $d6f2$ be a Coxeter diagram of type D_n with $n\geq 3$. Construct the (s_n,s_{n-1})-subdivision $d6f2'_d6f2$ of the simplicial complex $d6f2_d6f2$ by subdividing each edge...

conjecture | open | MathDB 页面 | 原文线索

V2-073840 | MATHDB-382455

The uniform rectangle-creation rate conjecture for the finite k-chain

摘录 / EXCERPT: Fix k and let $\mathcal{R}_k$ be the state space of the finite k-chain. For a partition $\lambda\in\mathcal{R}_k$, let λ^{+i} denote the state obtained by adding a box...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073841 | MATHDB-382466

The generating-function conjecture for identities from a five-parameter hypergeometric extension

摘录 / EXCERPT: Generating-function conjecture. The dimensions of these spaces satisfy

conjecture | open | MathDB 页面 | 原文线索

V2-073842 | MATHDB-382480

Energetic stability threshold conjecture for even and odd periodic waves

摘录 / EXCERPT: Energetic stability threshold conjecture. There is $\omega_* \in (\omega_L,1)$ and $\Omega_* \in (0,1)$ such that the even wave satisfying (even-wave) is energetically stable for...

conjecture | open | MathDB 页面 | 原文线索

V2-073843 | MATHDB-382481

The Prefix Conjecture for almost Golomb sequences

摘录 / EXCERPT: Let a_r be the order-r almost Golomb sequence, defined by the sliding-window greedy recurrence in the source, and let G denote Golomb's sequence. Prefix Conjecture. For all...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-073844 | MATHDB-382482

The domination lemma for almost Golomb sequences

摘录 / EXCERPT: Let a_r be the order-r almost Golomb sequence, let $N_r(n)$ denote the multiplicity of the value n, and define $M(r)=\sup_n N_r(n)$. Domination Lemma. For all $r\ge 5$, ... Tog...

conjecture | open | MathDB 页面 | 原文线索

V2-073845 | MATHDB-382485

Polarity-local times equivalence for Gaussian random fields

摘录 / EXCERPT: Polarity-local times equivalence. Points are polar for X if and only if

conjecture | open | MathDB 页面 | 原文线索

V2-073846 | MATHDB-382497

Global optimality of OFDM over the almost-sure set

摘录 / EXCERPT: Global optimality of OFDM over the almost-sure set. Under the same assumptions as the preceding asymptotic and local optimality theorems, OFDM is the global minimizer of the rangin...

conjecture | open | MathDB 页面 | 原文线索

V2-073847 | MATHDB-382501

Uniform expansion conjecture for quasirandom groups

摘录 / EXCERPT: Uniform expansion conjecture. If G is ε-quasirandom, then all connected Cayley graphs of G are ε-uniform expanders.

conjecture | open | MathDB 页面 | 原文线索

V2-073848 | MATHDB-382502

Generalized Gaussian decay conjecture for the harmonic oscillator

摘录 / EXCERPT: Let $a,b>0$ with $ab<1$. Let $\a0Omega_{a,b}$ be the function defined in equation ... with initial data u_0. Write ... If $u_0\in E^1(a,b)$, then the generalized Gaussian decay conje...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-073849 | MATHDB-382518

Wang's D-coloring conjecture for planar graphs

摘录 / EXCERPT: Wang's conjecture. The D-coloring number of G satisfies

conjecture | open | MathDB 页面 | 原文线索

V2-073850 | MATHDB-382519

The kinetic Schauder estimate for velocity Hessians

摘录 / EXCERPT: The kinetic Schauder estimate. The displayed estimate asserts a uniform interior Hölder bound for the velocity Hessian of smooth solutions in terms of the solution, the coefficient...

conjecture | refuted | MathDB 页面 | 原文线索

V2-073851 | MATHDB-382529

Harmonic chainlet characterization

摘录 / EXCERPT: Let A be a chainlet of class N^r, and let Ch map differential forms of class B^r to chainlets. A form is harmonic when it satisfies the relevant harmonic equation, and a ch...

conjecture | open | MathDB 页面 | 原文线索

V2-073852 | MATHDB-382530

The left-eigenvector weighted-sum conjecture for the level-one D_r qKZ solution

摘录 / EXCERPT: At the Razumov-Stroganov point $q=e^{2i\pi/3}$ and in the homogeneous limit $w_i=1$, let Ψ be normalized so that its smallest entry is $\Psi_{\pi_0}=1$. Let v be the left ei...

conjecture | open | MathDB 页面 | 原文线索

V2-073853 | MATHDB-382531

The half-turn symmetric alternating-sign-matrix enumeration conjecture for the D_r qKZ ground state

摘录 / EXCERPT: At the Razumov-Stroganov point $q=e^{2i\pi/3}$, take the homogeneous limit $w_i=1$ for all i in the level-one D_r qKZ solution, viewed as the Perron-Frobenius eigenvector...

conjecture | open | MathDB 页面 | 原文线索

V2-073854 | MATHDB-382533

Local density conjecture for domino tilings

摘录 / EXCERPT: Let p_a, p_b, p_c, and p_d be the edge-density quantities associated with the dimer model on an $n\times n$ torus in the thermodynamic limit. Consider an infinite sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-073855 | MATHDB-382536

The spanning conjecture for graded pieces of the arrangement quotient

摘录 / EXCERPT: Spanning conjecture. If $k \leq r \leq 2k-n-2$, then $(R_n/(E+(Q-1)))_r$ is spanned by the expressions in the generating family for which $i_1<(n-1)+(r-k)$; if $k-n<r<k$, the expre...

conjecture | open | MathDB 页面 | 原文线索

V2-073856 | MATHDB-382548

Eyink's enstrophy-defect conjecture for two-dimensional Euler flows

摘录 / EXCERPT: Eyink's conjecture. Both limits

conjecture | open | MathDB 页面 | 原文线索

V2-073857 | MATHDB-382551

Unrestricted-support conjecture for the low-lying zeros theorem

摘录 / EXCERPT: Let ϕ be a suitable test function, and let Theorem denote the density theorem established in the paper for the family of elliptic curves. Unrestricted-support conjecture. Theor...

conjecture | open | MathDB 页面 | 原文线索

V2-073858 | MATHDB-382554

Shokurov-Kawamata adjunction and effective adjunction conjectures

摘录 / EXCERPT: Adjunction and effective adjunction conjectures. One can choose an effective representative $M_Z\geq 0$ in its $\mathbb{R}$-linear equivalence class such that (Z,D_Z+M_Z) is log...

conjecture | open | MathDB 页面 | 原文线索

V2-073859 | MATHDB-382555

The eventual vanishing criterion for Hessian-nilpotent formal power series

摘录 / EXCERPT: Let $P(z)\in {\mathbb C}[[z]]$ satisfy $o(P(z))\geq 2$, and write $\operatorname{Hes}P(z)$ for its Hessian matrix. The series $P(z)$ is Hessian nilpotent when...

conjecture | open | MathDB 页面 | 原文线索

V2-073860 | MATHDB-382566

The classification conjecture for complete \Phi_{\kappa}-sequences

摘录 / EXCERPT: Classification conjecture. A Φ_{κ}-sequence $(a_n)_n$ is complete if and only if

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-073861 | MATHDB-382569

Extension of Rudolph's mod-2 relation to decorated Kauffman satellite invariants

摘录 / EXCERPT: Let L be a framed unoriented link with components L_i, and let y_{λ_i} and Q_{λ_i,λ_i} denote the corresponding Kauffman and Homfly annular decorations fo...

conjecture | open | MathDB 页面 | 原文线索

V2-073862 | MATHDB-382576

Uniform diameter bound for chess squares of the Feigenbaum map

摘录 / EXCERPT: Let ${\cal P}_0$ be the collection of chess squares in the level-zero puzzle for the Feigenbaum map, and let $\operatorname{diam}(P)$ denote the diameter of a chess square P. Uni...

conjecture | open | MathDB 页面 | 原文线索

V2-073863 | MATHDB-382577

Bennewitz's conjecture on higher-order HELP inequalities

摘录 / EXCERPT: Let M_N and M_D be the Neumann and Dirichlet Titchmarsh-Weyl matrices, respectively, and let $\operatorname{Res}(M_N,0)$ and $\operatorname{Res}(M_D,0)$ denote their residue ma...

conjecture | open | MathDB 页面 | 原文线索

V2-073864 | MATHDB-382579

Existence of deformation dessins for rational functions with three critical values

摘录 / EXCERPT: Let R be a rational function in z_1 with deformation dimension $m=1$, satisfying the conditions of Proposition[necessary condition], and suppose that its first three critical v...

conjecture | open | MathDB 页面 | 原文线索

V2-073865 | MATHDB-383058

Convergence problem for three-block ADMM with an identity third constraint block

摘录 / EXCERPT: Consider the separable constrained optimization problem $\min_{x_1,x_2,x_3}\ f_1(x_1)+f_2(x_2)+f_3(x_3)$ subject to $A_1x_1+A_2x_2+Ix_3=b$, where the first two objective functions...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-073866 | MATHDB-383213

Sidon set problem

摘录 / EXCERPT: For a set $A \subseteq \mathbb{N}$, call A a Sidon set if all pairwise sums of its elements are distinct, i.e. if for all $a_i, a_j, a_k, a_l \in A$, ... For a real number $x > 0$...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-073867 | MATHDB-383408

Balanced Boolean-function nonlinearity problem

摘录 / EXCERPT: For each integer $n\geq 1$, determine ... where ... In particular, determine the exact value of $M_{\mathrm{bal}}(8)$; the supplied report claims $M_{\mathrm{bal}}(8)=116$, excluding...

open_problem | claimed_solved | MathDB 页面 | 原文线索

V2-073868 | MATHDB-384690

How does vasculature in eye change with diabetes?

摘录 / EXCERPT: How does vasculature in eye change with diabetes?

open_problem | open | MathDB 页面 | 原文线索

V2-073869 | MATHDB-384691

How does change in vasculature can change oxygenation in the eye?

摘录 / EXCERPT: How does change in vasculature can change oxygenation in the eye?

open_problem | open | MathDB 页面 | 原文线索

V2-073870 | MATHDB-384692

Can we identify any early warning signs by monitoring microvasculature?

摘录 / EXCERPT: Can we identify any early warning signs by monitoring microvasculature?

open_problem | open | MathDB 页面 | 原文线索

V2-073871 | MATHDB-384693

Can we characterize and quantify the leakage factors from a clinical and modeling perspective in diabetes?

摘录 / EXCERPT: Can we characterize and quantify the leakage factors from a clinical and modeling perspective in diabetes?

open_problem | open | MathDB 页面 | 原文线索

V2-073872 | MATHDB-384694

How can we develop a predictive model for early imaging related warnings and diabetes progression?

摘录 / EXCERPT: How can we develop a predictive model for early imaging related warnings and diabetes progression?

open_problem | open | MathDB 页面 | 原文线索

V2-073873 | MATHDB-384695

How does aging influence intraocular pressure(IOP)of an eye?

摘录 / EXCERPT: How does aging influence intraocular pressure(IOP)of an eye? How does aging influence the structure of the eye?

open_problem | open | MathDB 页面 | 原文线索

V2-073874 | MATHDB-384696

How does velocity profile change in young vs old healthy population?

摘录 / EXCERPT: How does velocity profile change in young vs old healthy population?

open_problem | open | MathDB 页面 | 原文线索

V2-073875 | MATHDB-384697

Can we quantify and characterize the factors that determine the oxygenation in the eye?

摘录 / EXCERPT: Can we quantify and characterize the factors that determine the oxygenation in the eye? (e.g. Light + dark).

open_problem | open | MathDB 页面 | 原文线索

V2-073876 | MATHDB-384698

Can we develop kidney inspired models for aqueous humor production?

摘录 / EXCERPT: Can we develop kidney inspired models for aqueous humor production?

open_problem | open | MathDB 页面 | 原文线索

V2-073877 | MATHDB-385002

Ultrapowers (problem 2)

摘录 / EXCERPT: Assume A is separable and A embeds into $\prod_{\mathcal U}\mathcal O_2$. Is there a $\leq$-maximal embedding ϕ of A into $\prod_{\mathcal U}\mathcal O_2$?

open_problem | open | MathDB 页面 | 原文线索

V2-073878 | MATHDB-385003

Ultrapowers (problem 3)

摘录 / EXCERPT: Can one prove in ZFC that for some free ultrafilter $\mathcal U$ on $\mathbb N$ we have...

open_problem | open | MathDB 页面 | 原文线索

V2-073879 | MATHDB-385004

Ultrapowers (problem 4)

摘录 / EXCERPT: Is a nonprincipal ultrafilter such that $\mathcal{B}\left(H\right) '\cap \mathcal{B}\left(H\right) ^{\mathcal U}\neq \mathbb C I$ flat?

open_problem | open | MathDB 页面 | 原文线索

V2-073880 | MATHDB-385009

Tensor products (problem 1)

摘录 / EXCERPT: Does the number of C-norms on $A\otimes _{alg}B$ for begin{itemize} item $A=B=\mathcal{B}\left(H\right) $ item $A=B=C^{\ast }\left(\mathbb{F}_{\infty }\right) $, where...

open_problem | open | MathDB 页面 | 原文线索

V2-073881 | MATHDB-385025

What is the right set-theoretic framework to deal with functorial classification?

摘录 / EXCERPT: What is the right set-theoretic framework to deal with functorial classification?

open_problem | open | MathDB 页面 | 原文线索

V2-073882 | MATHDB-385026

Is there a constructive Borel proof of the ... embedding theorem?

摘录 / EXCERPT: Is there a constructive Borel proof of the $\mathcal{O}_{2}$ embedding theorem?

open_problem | open | MathDB 页面 | 原文线索

V2-073883 | MATHDB-385029

Is there a Borel inverse of the classification functor?

摘录 / EXCERPT: Is there a Borel inverse of the classification functor?

open_problem | open | MathDB 页面 | 原文线索

V2-073884 | MATHDB-385035

Between Santalo and Petty Projection Inequality (problem 1)

摘录 / EXCERPT: If K has barycenter at 0, is it true that $\Phi_{[n]}(K) \simeq \Phi_{[n]}(D_N)$? The lower bound is true but is it true with constant 1?

open_problem | open | MathDB 页面 | 原文线索

V2-073885 | MATHDB-385036

Between Santalo and Petty Projection Inequality (problem 2)

摘录 / EXCERPT: Is it true that $\Phi_{[1]}(K) \geq \Phi_{[1]}(D_N)$ for every compact K? Also for $N-1$ or any n?

open_problem | open | MathDB 页面 | 原文线索

V2-073886 | MATHDB-385037

Is there an argument or proof that gives Santalo and Petty projection inequality at the same time?

摘录 / EXCERPT: Is there an argument or proof that gives Santalo and Petty projection inequality at the same time?

open_problem | open | MathDB 页面 | 原文线索

V2-073887 | MATHDB-385038

The Case $\ell_2\otimes_{\pi}\ell_2\otimes_{\pi}\ell_2$

摘录 / EXCERPT: The result of Briet, Naor and Regev does not cover the case where $p=q=r=2$ and therefore, it remains open whether $\ell_2\otimes_{\pi}\ell_2\otimes_{\pi}\ell_2$ has non-trivial co...

open_problem | open | MathDB 页面 | 原文线索

V2-073888 | MATHDB-385039

This was a blank problem, and David added this sentence.

摘录 / EXCERPT: This was a blank problem, and David added this sentence.

open_problem | open | MathDB 页面 | 原文线索

V2-073889 | MATHDB-385040

Log-Brunn-Minkowski Inequality

摘录 / EXCERPT: Show that if K and L symmetric convex bodies in ${\\mathbb{R}}^n$, then for all $0 \leq t \leq 1$ the following hods ... with equality if and only if $K = L$ or K and L are p...

open_problem | open | MathDB 页面 | 原文线索

V2-073890 | MATHDB-385041

L_p-Brunn Minkowski inequality

摘录 / EXCERPT: Show that if K and L symmetric convex bodies in ${\\mathbb{R}}^n$, then for all $0 \leq t \leq 1$ and $p\geq 0$ ...

open_problem | open | MathDB 页面 | 原文线索

V2-073891 | MATHDB-385043

Cone-volume measure

摘录 / EXCERPT: If K and L symmetric convex bodies in ${\\mathbb{R}}^n$ such that $\bar{V}_K=\bar{V}_L$, then $K=L$ or K and L are parallelograms with parallel sides.

open_problem | open | MathDB 页面 | 原文线索

V2-073892 | MATHDB-385044

Brunn-Minkowski inequality for dual quermassintegrals

摘录 / EXCERPT: Show that for convex bodies K and L ... where $\tilde{W}_{n-i}(K) = \int\limits_{S^{n-1}} \rho_K(u)^i\,du$.

open_problem | open | MathDB 页面 | 原文线索

V2-073893 | MATHDB-385045

Brunn-Minkowski inequality for L_p surface area

摘录 / EXCERPT: Show that for convex bodies K and L in ${\\mathbb{R}}^n$ and for $0 \leq p \leq 1$ ... where $S_p(K) = \int\limits_{S^{n-1}} h_K(u)^{1-p} \, dS_K(u)$ is the L_p surface area.

open_problem | open | MathDB 页面 | 原文线索

V2-073894 | MATHDB-385046

Conjecture due to D. Cordero-Erausquin

摘录 / EXCERPT: If K and L symmetric convex bodies in ${\\mathbb{R}}^n$, then ... where $K^\circ =\{y \in {\\mathbb{R}}^n:\ \forall x \in K \, \langle x,y \rangle \leq 1 \}$ is a polar body of...

open_problem | open | MathDB 页面 | 原文线索

V2-073895 | MATHDB-385047

Dar's conjecture

摘录 / EXCERPT: Let $\mu(K, L)$ is defined by $\mu(K, L) = \max\limits_{x \in {\\mathbb{R}}^n} |K \cap (x + L)|.$ Then for convex bodies K and L in ${\\mathbb{R}}^n$ ...

open_problem | open | MathDB 页面 | 原文线索

V2-073896 | MATHDB-385048

Convex bodies in $\mathbb{C}^n$

摘录 / EXCERPT: Consider a convex body in $\mathbb{C}^n$ and fix a real subspace E of $\mathbb{C}^n$. Define $V_E(K) = \int\limits_{U(n)} |K|\phi(E)| \, d\phi$, where $U(n)$ is a unitary group....

open_problem | open | MathDB 页面 | 原文线索

V2-073897 | MATHDB-385049

Isoperimetric type problems for quermassintegrals in hyperbolic or spherical space.

摘录 / EXCERPT: Isoperimetric type problems for quermassintegrals in hyperbolic or spherical space.

open_problem | open | MathDB 页面 | 原文线索

V2-073898 | MATHDB-385051

Question by Makai and Martini

摘录 / EXCERPT: What if we pose the same question but replacing volumes by surface areas, that is $\widetilde{A}_{K,u}(t) = |\partial(K \cap \{tu + u^\perp\})|_{n-1}$, is body K still symmetric?

open_problem | open | MathDB 页面 | 原文线索

V2-073899 | MATHDB-385052

Symmetry and convexity (problem 3)

摘录 / EXCERPT: Let $p > 1$ and $H \subset {\\mathbb{R}}^n$ of dimension k. Consider a map ... where $B_p^n = \{ x\in {\\mathbb{R}}^n: \ \sqrt[p]{|x_1|^p + \cdots + |x_n|^p} \leq 1\}$. Is f log-...

open_problem | open | MathDB 页面 | 原文线索

V2-073900 | MATHDB-385053

Symmetry and convexity (problem 4)

摘录 / EXCERPT: Take $a_1, \dots, a_n \in {\\mathbb{R}}$ and $v_1, \dots, v_n \in {\\mathbb{R}}^n$ and define ... Then $f(t)$ is log-concave is equivalent to log-Brunn-Minkowski. What is possible...

open_problem | open | MathDB 页面 | 原文线索

V2-073901 | MATHDB-385054

If we replace condition in the theorem with ... , can we obtain better results?

摘录 / EXCERPT: If we replace condition in the theorem with $\nabla^2 W(x) \geq F(x) \, Id$, can we obtain better results?

open_problem | open | MathDB 页面 | 原文线索

V2-073902 | MATHDB-385055

Symmetry and convexity (problem 6)

摘录 / EXCERPT: For a smooth convex domain K in ${\\mathbb{R}}^n$, Laplace equation with Dirichlet boundary conditions is ... Choose $u_1 > 0$ in $\text{int}\, K$, then ... True for $K = B_2$. Quest...

open_problem | open | MathDB 页面 | 原文线索

V2-073903 | MATHDB-385056

Is there a JSJ decomposition for finitely presented groups over small groups?

摘录 / EXCERPT: Is there a JSJ decomposition for finitely presented groups over small groups?

open_problem | open | MathDB 页面 | 原文线索

V2-073904 | MATHDB-385057

JSJ Decompositions (problem 2)

摘录 / EXCERPT: Let G be a finitely presented group that does not split over a virtually abelian group. Can G have infinitely many unfolded splittings over distinct subgroups isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-073905 | MATHDB-385058

Is the isomorphism problem solvable for the graphs of groups for which all edges and vertices are labelled by ... ?

摘录 / EXCERPT: Is the isomorphism problem solvable for the graphs of groups for which all edges and vertices are labelled by $\mathbb Z$?

open_problem | open | MathDB 页面 | 原文线索

V2-073906 | MATHDB-385059

JSJ Decompositions (problem 4)

摘录 / EXCERPT: Let G be a $CAT(0)$ group with $Out(G)$ infinite. Is it true that G admits a Dehn twist of infinite order?

open_problem | open | MathDB 页面 | 原文线索

V2-073907 | MATHDB-385060

Accessibility Questions (problem 1)

摘录 / EXCERPT: Let ${\mathcal C}$ be the class of finite and virtually-$\mathbb Z$ groups. Are hyperbolic groups strongly accessible over ${\mathcal C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-073908 | MATHDB-385061

Accessibility Questions (problem 2)

摘录 / EXCERPT: Let ${\mathcal C}$ be the class of finite and virtually-$\mathbb Z$ groups. Are finitely presented groups strongly accessible over ${\mathcal C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-073909 | MATHDB-385062

Accessibility Questions (problem 3)

摘录 / EXCERPT: Is there a 1-ended finitely presented group G such that $G\simeq G*_{\mathbb Z}$?

open_problem | open | MathDB 页面 | 原文线索

V2-073910 | MATHDB-385063

Accessibility Questions (problem 4)

摘录 / EXCERPT: Let ${\mathcal C}$ be the class of finitely generated groups. Are finitely presented groups strongly accessible over ${\mathcal C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-073911 | MATHDB-385064

Accessibility Questions (problem 5)

摘录 / EXCERPT: Let $G=A*_{C_n}B_n,\, n\in \mathbb N$ where G is a finitely generated, C_n are slender and $C_n < C_{n+1}$ for all n. Is it true that $G=A*_C B$ where $C < C_n$ for all n?...

open_problem | open | MathDB 页面 | 原文线索

V2-073912 | MATHDB-385067

Are one-ended CAT(0) groups semi-stable at infinity?

摘录 / EXCERPT: Are one-ended CAT(0) groups semi-stable at infinity?

open_problem | open | MathDB 页面 | 原文线索

V2-073913 | MATHDB-385068

Are splittings of one-ended finitely generated groups over 2-ended groups invariant under quasi-isometries?

摘录 / EXCERPT: Are splittings of one-ended finitely generated groups over 2-ended groups invariant under quasi-isometries?

open_problem | open | MathDB 页面 | 原文线索

V2-073914 | MATHDB-385070

Are splittings over virtually- ... groups invariant under quasi-isometries?

摘录 / EXCERPT: Are splittings over virtually-$\mathbb Z ^2$ groups invariant under quasi-isometries? More precisely let G be a one-ended finitely presented group that does not split over a 2-en...

open_problem | open | MathDB 页面 | 原文线索

V2-073915 | MATHDB-385073

Approximation of L_2-torsion. (problem 1)

摘录 / EXCERPT: Do we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-073916 | MATHDB-385074

Approximation of L_2-torsion. (problem 2)

摘录 / EXCERPT: Is ...

open_problem | open | MathDB 页面 | 原文线索

V2-073917 | MATHDB-385075

Approximation of L_2-torsion. (problem 3)

摘录 / EXCERPT: Let $A\in M_l(\mathbb{Z} \Gamma)$, denote $A_k=A/\Gamma_k$. Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-073918 | MATHDB-385076

Is ... for any (not necessarily finitely generated) group?

摘录 / EXCERPT: Is ... for any (not necessarily finitely generated) group?

open_problem | open | MathDB 页面 | 原文线索

V2-073919 | MATHDB-385080

Orbit Equivalence of Measure Preserving Actions (problem 4)

摘录 / EXCERPT: Is it true that for any N there exist measurable subsets $A_g\subseteq X$ $(g\in\Gamma)$ such that $\prod\limits_{x\in A_g}(x,xg)$ is a forest with...

open_problem | open | MathDB 页面 | 原文线索

V2-073920 | MATHDB-385082

More problems (problem 2)

摘录 / EXCERPT: Let $\Gamma/\mathbb{Z}_2$ be a sofic group. Is Γ sofic?

open_problem | open | MathDB 页面 | 原文线索

V2-073921 | MATHDB-385083

More problems (problem 3)

摘录 / EXCERPT: Do we have $\beta_1^{(2)}(\Gamma)\leq nrk(\Gamma)-1$ for a torsion free group Γ?

open_problem | open | MathDB 页面 | 原文线索

V2-073922 | MATHDB-385084

The deep-fall property and the Atiyah Conjecture

摘录 / EXCERPT: Find many examples of left-orderable groups and their free actions on complexes that have the deep-fall property in some dimension i. Find free actions by left orderable groups t...

open_problem | open | MathDB 页面 | 原文线索

V2-073923 | MATHDB-385086

Thompson group F and groups of homeomorphisms of the interval and the circle (problem 2)

摘录 / EXCERPT: Is it true that a subgroup of F is elementary amenable if and only if it does not contain a copy of F?

open_problem | open | MathDB 页面 | 原文线索

V2-073924 | MATHDB-385087

What is the Cantor-Bendixon rank of the space of Subgroups of ... ?

摘录 / EXCERPT: What is the Cantor-Bendixon rank of the space of Subgroups of F?

open_problem | open | MathDB 页面 | 原文线索

V2-073925 | MATHDB-385090

Thompson group F and groups of homeomorphisms of the interval and the circle (problem 6)

摘录 / EXCERPT: What is the subgroup structure of H? Is there an infinite anti-chain of subgroups of H? Are there only finitely many obstructions to amenability? (i.e., is there a finite colle...

open_problem | open | MathDB 页面 | 原文线索

V2-073926 | MATHDB-385091

Thompson group F and groups of homeomorphisms of the interval and the circle (problem 7)

摘录 / EXCERPT: Is F is invariably generated by a finite set?

open_problem | open | MathDB 页面 | 原文线索

V2-073927 | MATHDB-385101

Grigorchuk's group, branch groups and groups of intermediate growth (problem 3)

摘录 / EXCERPT: Is the universal Grigorchuk group amenable?

open_problem | open | MathDB 页面 | 原文线索

V2-073928 | MATHDB-385104

Does the limit ... exist for the Grigorchuk group?

摘录 / EXCERPT: Does the limit ... exist for the Grigorchuk group?

open_problem | open | MathDB 页面 | 原文线索

V2-073929 | MATHDB-385105

Grigorchuk's group, branch groups and groups of intermediate growth (problem 7)

摘录 / EXCERPT: Is $\rho_c<1$ for all groups of intermediate growth?

open_problem | open | MathDB 页面 | 原文线索

V2-073930 | MATHDB-385106

Grigorchuk's group, branch groups and groups of intermediate growth (problem 8)

摘录 / EXCERPT: Is there a finitely presented textit{amenable} group that surjects onto a group of intermediate growth?

open_problem | open | MathDB 页面 | 原文线索

V2-073931 | MATHDB-385108

Amenability+ (problem 1)

摘录 / EXCERPT: begin{enumerate} item textbf{Q:} (Rosenblatt 74') Is there a supramenable group of exponential growth? item Is the direct product of 2 supramenable groups also supramenable? end{...

open_problem | open | MathDB 页面 | 原文线索

V2-073932 | MATHDB-385110

Amenability+ (problem 3)

摘录 / EXCERPT: Is there a f.p. amenable infinite simple group? A construction of Juschenko and Monod (2012) gives a finitely generated infinite simple amenable group. textbf{Conjecture:} The deri...

open_problem | open | MathDB 页面 | 原文线索

V2-073933 | MATHDB-385112

Amenability+ (problem 5)

摘录 / EXCERPT: begin{enumerate} item What are "good" SQ-closed families of amenable groups? item What is the SQ-closure of the set of bounded automata groups? end{enumerate}

open_problem | open | MathDB 页面 | 原文线索

V2-073934 | MATHDB-385114

Amenability+ (problem 7)

摘录 / EXCERPT: begin{enumerate} item is every ever locally compact amenable group in AE? item Find f.g. amenable groups with infinite commensurated subgroups with trivial normal core. item Is eve...

open_problem | open | MathDB 页面 | 原文线索

V2-073935 | MATHDB-385115

Is there a non-amenable group that has a Liouville action with amenable stabilizers?

摘录 / EXCERPT: Is there a non-amenable group that has a Liouville action with amenable stabilizers?

open_problem | open | MathDB 页面 | 原文线索

V2-073936 | MATHDB-385116

Amenability+ (problem 9)

摘录 / EXCERPT: Is there a transitive action $G \def\actson{\curvearrowright} X$ that is μ-Liouville, but the action on some orbit of $G\def\actson{\curvearrowright} \subset X \times X$ is not...

open_problem | open | MathDB 页面 | 原文线索

V2-073937 | MATHDB-385118

Is the Liouville property stable under the choice of of symmetric, finitely supported, generating measure?

摘录 / EXCERPT: Is the Liouville property stable under the choice of of symmetric, finitely supported, generating measure?

open_problem | open | MathDB 页面 | 原文线索

V2-073938 | MATHDB-385121

Is there an amenable group whose spectrum is a Cantor set?

摘录 / EXCERPT: Is there an amenable group whose spectrum is a Cantor set? textbf{Remark} (Dudko): For Grigorchuk's group G, the spectrum is a union of two intervals: $[-1,0]\cup[1/2,1]$; the qu...

open_problem | open | MathDB 页面 | 原文线索

V2-073939 | MATHDB-385123

Is there a finitely presented simple group that is not 2-generated?

摘录 / EXCERPT: Is there a finitely presented simple group that is not 2-generated?

open_problem | open | MathDB 页面 | 原文线索

V2-073940 | MATHDB-385129

Other problems (problem 8)

摘录 / EXCERPT: textbf{Conjecture:}If the Folner function of a group is sub-exponential, then the group is virtually nilpotent.

open_problem | open | MathDB 页面 | 原文线索

V2-073941 | MATHDB-385130

Other problems (problem 9)

摘录 / EXCERPT: Are the Hanoi tower groups H_n, $n \geq 4$ amenable?

open_problem | open | MathDB 页面 | 原文线索

V2-073942 | MATHDB-385131

Semi-stability conjecture

摘录 / EXCERPT: Let G be a finitely-presented one-ended group, and let K_G be a presentation 2-complex. Does $\text{pro}-\pi_1(E(\widetilde K_g), r)$ depend on the ray r?

open_problem | open | MathDB 页面 | 原文线索

V2-073943 | MATHDB-385133

Existence and uniqueness of Z-set compactifications

摘录 / EXCERPT: Does every group of type F admit a Z-set compactification? EZ-set compactification?

open_problem | open | MathDB 页面 | 原文线索

V2-073944 | MATHDB-385136

PI spaces

摘录 / EXCERPT: The standard Sierpinski carpet is obtained by removing middle thirds from the unit square. This space is not PI with the standard metric. Is there a metric in the quasi-symmetric g...

open_problem | open | MathDB 页面 | 原文线索

V2-073945 | MATHDB-385137

What is the conformal dimension ... of the standard Sierpinski carpet?

摘录 / EXCERPT: What is the conformal dimension q of the standard Sierpinski carpet? It is known ... The upper bound above is the Hausdorff dimension (Kigami). For an easier hard problem, prove th...

open_problem | open | MathDB 页面 | 原文线索

V2-073946 | MATHDB-385139

Analytic questions (problem 4)

摘录 / EXCERPT: Let G be a random hyperbolic group. Is there a metric in the quasi-symmetry gauge of ∂G so that ∂G is a PI space?

open_problem | open | MathDB 页面 | 原文线索

V2-073947 | MATHDB-385140

Conformal dimension

摘录 / EXCERPT: What can one say about a space B that attains its Ahlfors regular conformal dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-073948 | MATHDB-385141

Analytic questions (problem 6)

摘录 / EXCERPT: Let G be hyperbolic with $\partial G\subsetneq S^2$. Is its (Ahlfors regular) conformal dimension less than 2?

open_problem | open | MathDB 页面 | 原文线索

V2-073949 | MATHDB-385142

Analytic questions (problem 7)

摘录 / EXCERPT: If $G=A*_{\mathbb{Z}} B$ what is the relationship between the conformal dimension of A, B, and G?

open_problem | open | MathDB 页面 | 原文线索

V2-073950 | MATHDB-385145

Cannon and related conjectures

摘录 / EXCERPT: If G is hyperbolic and $\partial G\simeq S^2$, then G is virtually a Kleinian group.

open_problem | open | MathDB 页面 | 原文线索

V2-073951 | MATHDB-385146

Rigidity questions (problem 2)

摘录 / EXCERPT: If G is hyperbolic and $\partial G\simeq\mathscr{S}$ is a Sierpinski curve, then G is virtually a Kleinian group.

open_problem | open | MathDB 页面 | 原文线索

V2-073952 | MATHDB-385147

Rigidity questions (problem 3)

摘录 / EXCERPT: Let (G,P) relative hyperbolic with Bowditch boundary $\mathscr S$ or S^2. Is G virtually a Kleinian group?

open_problem | open | MathDB 页面 | 原文线索

V2-073953 | MATHDB-385149

Rigidity questions (problem 5)

摘录 / EXCERPT: Is a random hyperbolic group QI rigid? (i.e. any quasi-isometry is bounded distance from an isometry, i.e. emph{QI}$(G)=G$).

open_problem | open | MathDB 页面 | 原文线索

V2-073954 | MATHDB-385150

Morse boundaries

摘录 / EXCERPT: Take notions from the boundary of hyperbolic groups and extend them to Morse boundaries.

open_problem | open | MathDB 页面 | 原文线索

V2-073955 | MATHDB-385151

Does a quasi-symmetry' of the Morse boundary come from a quasi-isometry?

摘录 / EXCERPT: Does a quasi-symmetry' of the Morse boundary come from a quasi-isometry? (so long as Morse boundary is big enough)

open_problem | open | MathDB 页面 | 原文线索

V2-073956 | MATHDB-385152

What do connectivity properties of the Morse boundary tell about ... .

摘录 / EXCERPT: What do connectivity properties of the Morse boundary tell about G. Specifically mapping class group, right-angled Artin groups, right-angled Coxeter groups.

open_problem | open | MathDB 页面 | 原文线索

V2-073957 | MATHDB-385153

Morse and other boundaries (problem 4)

摘录 / EXCERPT: If G acts geometrically on two CAT(0) cube complexes X_0, X_1, are the Morse boundaries of X_0, X_1 homeomorphic? (What about the Croke--Kleiner group?)

open_problem | open | MathDB 页面 | 原文线索

V2-073958 | MATHDB-385154

Morse and other boundaries (problem 5)

摘录 / EXCERPT: Let G be a finitely presented group with Morse boundary with at least 3 points. Must G be acylindrically hyperbolic?

open_problem | open | MathDB 页面 | 原文线索

V2-073959 | MATHDB-385155

Does the Morse boundary have a quasi-isometry invariant compactification?

摘录 / EXCERPT: Does the Morse boundary have a quasi-isometry invariant compactification?

open_problem | open | MathDB 页面 | 原文线索

V2-073960 | MATHDB-385156

Morse and other boundaries (problem 7)

摘录 / EXCERPT: Let $H\subset G$ be almost malnormal and such that the limit set of H in the Morse boundary of G is the Morse boundary of H. Is H hyperbolically embedded?

open_problem | open | MathDB 页面 | 原文线索

V2-073961 | MATHDB-385157

Hierarchically hyperbolic groups

摘录 / EXCERPT: Is there a convergence group characterization of hierarchically hyperbolic groups? What about acylindrical hyperbolic groups?

open_problem | open | MathDB 页面 | 原文线索

V2-073962 | MATHDB-385158

Does the hierarchical hyperbolic boundary depend on the hierarchically hyperbolic structure of a (non-hyperbolic) hierarchically hyperbolic group?

摘录 / EXCERPT: Does the hierarchical hyperbolic boundary depend on the hierarchically hyperbolic structure of a (non-hyperbolic) hierarchically hyperbolic group?

open_problem | open | MathDB 页面 | 原文线索

V2-073963 | MATHDB-385159

Morse and other boundaries (problem 10)

摘录 / EXCERPT: Is the hierarchically hyperbolic boundary of Teichm"uller space (with the Teichm"uller metric) homeomorphic to PML?

open_problem | open | MathDB 页面 | 原文线索

V2-073964 | MATHDB-385160

The boundary of a hierarchically hyperbolic group is metrizable.

摘录 / EXCERPT: The boundary of a hierarchically hyperbolic group is metrizable. Is there a natural metric?

open_problem | open | MathDB 页面 | 原文线索

V2-073965 | MATHDB-385161

More questions

摘录 / EXCERPT: Characterize acylindrical hyperbolic groups in terms of the Poisson boundary or the Martin boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-073966 | MATHDB-385162

Morse and other boundaries (problem 13)

摘录 / EXCERPT: Does every group that admits some convergence action on a compact space admit a emph{universal} convergence action?

open_problem | open | MathDB 页面 | 原文线索

V2-073967 | MATHDB-385163

What can be said about existence of Cannon--Thurston maps for Morse boundaries?

摘录 / EXCERPT: What can be said about existence of Cannon--Thurston maps for Morse boundaries? hierarchically hyperbolic boundaries? other boundaries?

open_problem | open | MathDB 页面 | 原文线索

V2-073968 | MATHDB-385164

For various boundaries, what is the weakest condition on a map between spaces that ensures that there is a map between the boundaries?

摘录 / EXCERPT: For various boundaries, what is the weakest condition on a map between spaces that ensures that there is a map between the boundaries?

open_problem | open | MathDB 页面 | 原文线索

V2-073969 | MATHDB-385165

What are useful examples of boundaries of group pairs beyond the Bowditch boundary?

摘录 / EXCERPT: What are useful examples of boundaries of group pairs beyond the Bowditch boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-073970 | MATHDB-385168

What is the group theoretic analogue of taking a ramified cover along a codimension-2 subgroup?

摘录 / EXCERPT: What is the group theoretic analogue of taking a ramified cover along a codimension-2 subgroup?

open_problem | open | MathDB 页面 | 原文线索

V2-073971 | MATHDB-385169

CAT(0) groups

摘录 / EXCERPT: When does a Coxeter group have a unique CAT(0) boundary up to homeomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-073972 | MATHDB-385171

Characterization questions

摘录 / EXCERPT: What spaces can be the Gromov boundary of a hyperbolic group? How about the Bowditch or Morse boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-073973 | MATHDB-385172

Which metric spaces are quasi-symmetric to a Gromov boundary and a Bowditch boundary?

摘录 / EXCERPT: Which metric spaces are quasi-symmetric to a Gromov boundary and a Bowditch boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-073974 | MATHDB-385174

Miscellaneous (problem 9)

摘录 / EXCERPT: Let K be a Kleinian group with Sierpinski boundary. Is there a quasi-symmetry invariant that detects whether or not K has a rank-2 cusp?

open_problem | open | MathDB 页面 | 原文线索

V2-073975 | MATHDB-385178

Embedability

摘录 / EXCERPT: Let M be a II$_1$ factor with Property (T) and N a II$_1$ factor with Haagerup property. If $M\subseteq N^\omega $, prove there exists $N_n\subseteq N$, finite dimensional, so...

open_problem | open | MathDB 页面 | 原文线索

V2-073976 | MATHDB-385179

Fundamental Groups of Ultraproducts of II$_1$ Factors

摘录 / EXCERPT: Find a II$_1$ factor P so that $\mathcal{F}(P^\omega)\neq \mathbb{R} $.

open_problem | open | MathDB 页面 | 原文线索

V2-073977 | MATHDB-385180

Non-Isomorphic Ultrapowers

摘录 / EXCERPT: Find full type II$_1$ factors P_1 and P_2 so that $P_1^\omega\not\cong P_2^\omega $.

open_problem | open | MathDB 页面 | 原文线索

V2-073978 | MATHDB-385181

Non-Isomorphic Ultrapowers III

摘录 / EXCERPT: Construct type III factors with non-isomorphic ultrapowers.

open_problem | open | MathDB 页面 | 原文线索

V2-073979 | MATHDB-385182

Ultraproducts (problem 5)

摘录 / EXCERPT: Is $\mathcal{R}^\omega\cong L(G) $ for some group G.

open_problem | open | MathDB 页面 | 原文线索

V2-073980 | MATHDB-385183

Isomorphic Ultrapowers

摘录 / EXCERPT: Find a class of II_1 factors with isomorphic ultrapowers.

open_problem | open | MathDB 页面 | 原文线索

V2-073981 | MATHDB-385184

Elementary Equivalence of the Free Group Factors

摘录 / EXCERPT: Is $L(\mathbb{F}_n) $ elementarily equivalent to $L(\mathbb{F}_\infty) $?

open_problem | open | MathDB 页面 | 原文线索

V2-073982 | MATHDB-385185

Property (T)

摘录 / EXCERPT: Provide an example of an infinite icc property (T) group that is W* super-rigid.

open_problem | open | MathDB 页面 | 原文线索

V2-073983 | MATHDB-385189

The Class of $\mathcal{V}\mathcal{C} $

摘录 / EXCERPT: Characterize the class of groups $\Gamma $ such that for all II$_1$ factors N and all co-cycle actions $\Gamma \curvearrowright N$ can be perturbed to a genuine action.

open_problem | open | MathDB 页面 | 原文线索

V2-073984 | MATHDB-385191

McDuff

摘录 / EXCERPT: Assume that $M\cong P_1\bar\otimes P_2$ is McDuff. Does it follow that P_1 or P_2 is McDuff?

open_problem | open | MathDB 页面 | 原文线索

V2-073985 | MATHDB-385196

C* Super-Rigidity for Non-Discrete Groups Acting on Trees

摘录 / EXCERPT: Does $C^*_r(G_1)\cong C^*_r(G_2) $ imply that $G_1\cong G_2 $?

open_problem | open | MathDB 页面 | 原文线索

V2-073986 | MATHDB-385200

Prove cocycle superrigidity for Bernoulli actions of property (T) groups with totally disconnected target groups.

摘录 / EXCERPT: Prove cocycle superrigidity for Bernoulli actions of property (T) groups with totally disconnected target groups.

open_problem | open | MathDB 页面 | 原文线索

V2-073987 | MATHDB-385201

Characterize all groups which are measure equivalent to free groups, or find new examples of such groups.

摘录 / EXCERPT: Characterize all groups which are measure equivalent to free groups, or find new examples of such groups.

open_problem | open | MathDB 页面 | 原文线索

V2-073988 | MATHDB-385202

Groups (problem 2)

摘录 / EXCERPT: Find a weird example of a groups Γ that is measure equivalent to $\mathbb{F}_2\times \mathbb{F}_2 $. Is it true that every such group Γ is either a product of groups...

open_problem | open | MathDB 页面 | 原文线索

V2-073989 | MATHDB-385203

Groups (problem 3)

摘录 / EXCERPT: Let G be a groups with $\beta_1^{(2)}(G)>0 $. Then there is an infinite amenable subgroup, e.g. an element of infinite order, or a finite index subgroup $G_1\leqslant G $ so that...

open_problem | open | MathDB 页面 | 原文线索

V2-073990 | MATHDB-385204

Groups (problem 4)

摘录 / EXCERPT: Which groups Γ admit a Bernoulli action on $\prod_{g \in \Gamma} (\{0,1\},\mu_g)$ that is nonsingular and of type III?

open_problem | open | MathDB 页面 | 原文线索

V2-073991 | MATHDB-385205

Groups (problem 5)

摘录 / EXCERPT: Is treeability stable under finite index, i.e. if $S\subset R $ is a finite index subequivalence relation of a treaable relation R, is S treeable?

open_problem | open | MathDB 页面 | 原文线索

V2-073992 | MATHDB-385206

Does every equivalence relation with spectral gap or property (T) measurably contain ... ?

摘录 / EXCERPT: Does every equivalence relation with spectral gap or property (T) measurably contain $\mathbb{F}_2 $?

open_problem | open | MathDB 页面 | 原文线索

V2-073993 | MATHDB-385207

Stability of the Liouville property

摘录 / EXCERPT: For any group G and two measures μ_1, μ_2 which are symmetric and finitely supported, the Poisson boundary for (G, μ_1) is trivial if and only if the Poisson boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-073994 | MATHDB-385210

Singularity of the Harmonic Measure

摘录 / EXCERPT: Let Γ be a Fuchsian group, and let μ be a finitely supported measure on Γ. Is the harmonic measure singular with respect to the Lebesgue measure on the circle?

open_problem | open | MathDB 页面 | 原文线索

V2-073995 | MATHDB-385213

Exponential Genericity of Pseudo-Anosovs

摘录 / EXCERPT: Let S be a generating set for the mapping class group. Does the number of non-pseudo-Anosov elements in the R-ball with respect to S converge exponentially to 0 as R goes to in...

open_problem | open | MathDB 页面 | 原文线索

V2-073996 | MATHDB-385216

Is each Artin group isomorphic to its dual?

摘录 / EXCERPT: Is each Artin group isomorphic to its dual?

open_problem | open | MathDB 页面 | 原文线索

V2-073997 | MATHDB-385221

The word problem (problem 5)

摘录 / EXCERPT: If $g = a\overline{b}$ for a,b words in the positive monoid with no cancellation in the middle, are a and b unique?

open_problem | open | MathDB 页面 | 原文线索

V2-073998 | MATHDB-385223

Solve the isomorphism problem for 2-dimensional Artin groups.

摘录 / EXCERPT: Solve the isomorphism problem for 2-dimensional Artin groups.

open_problem | open | MathDB 页面 | 原文线索

V2-073999 | MATHDB-385226

The isomorphism problem (problem 5)

摘录 / EXCERPT: For an Artin group A, when is $\mathrm{Out}(A)$ finite/finitely generated? Find explicit generators.

open_problem | open | MathDB 页面 | 原文线索

V2-074000 | MATHDB-385228

The isomorphism problem (problem 7)

摘录 / EXCERPT: For fixed A, which spherical and/or dihedral Artin groups embed in A?

open_problem | open | MathDB 页面 | 原文线索

V2-074001 | MATHDB-385231

Does the ... Artin group act properly and cocompactly on a Helly graph?

摘录 / EXCERPT: Does the $\tilde{A}_2$ Artin group act properly and cocompactly on a Helly graph?

open_problem | open | MathDB 页面 | 原文线索

V2-074002 | MATHDB-385232

Further classify the systolic Artin groups.

摘录 / EXCERPT: Further classify the systolic Artin groups.

open_problem | open | MathDB 页面 | 原文线索

V2-074003 | MATHDB-385235

Describe the hyperbolic space on which the acylindrically hyperbolic Artin groups act.

摘录 / EXCERPT: Describe the hyperbolic space on which the acylindrically hyperbolic Artin groups act.

open_problem | open | MathDB 页面 | 原文线索

V2-074004 | MATHDB-385243

Complexes for Artin groups (problem 7)

摘录 / EXCERPT: Does the Br_4 triangle complex embed in a (locally finite?) $\tilde{A}_2$ building?

open_problem | open | MathDB 页面 | 原文线索

V2-074005 | MATHDB-385245

Which Artin groups embed in the mapping class group of a surface?

摘录 / EXCERPT: Which Artin groups embed in the mapping class group of a surface?

open_problem | open | MathDB 页面 | 原文线索

V2-074006 | MATHDB-385246

Relationships to mapping class groups (problem 2)

摘录 / EXCERPT: For an Artin group A and orbifold $\mathcal O$ with $A \cong MCG(\mathcal O)$, is there a (reasonably defined) curve complex for $\mathcal O$ which agrees with the complex of irr...

open_problem | open | MathDB 页面 | 原文线索

V2-074007 | MATHDB-385252

Which Artin groups are profinitely rigid?

摘录 / EXCERPT: Which Artin groups are profinitely rigid?

open_problem | open | MathDB 页面 | 原文线索

V2-074008 | MATHDB-385260

Quasi-convex subgroups

摘录 / EXCERPT: Let G be a hyperbolic free-by-cyclic group and H a non-quasiconvex subgroup of G. Then H contains a subgroup F which is a virtual fibre or a virtual semi-fibre of a free-...

open_problem | open | MathDB 页面 | 原文线索

V2-074009 | MATHDB-385264

Subgroups of free-by-cyclic groups (problem 5)

摘录 / EXCERPT: Let G be a hyperbolic free-by-cyclic groups. Which subgrups of G admit Cannon--Thurston maps?

open_problem | open | MathDB 页面 | 原文线索

V2-074010 | MATHDB-385267

Hyperbolic case

摘录 / EXCERPT: If G is a hyperbolic free-by-cyclic group with irreducible monodromy then G is LERF if and only if the BNS invariant of every finite index subgroup of G is symmetric.

open_problem | open | MathDB 页面 | 原文线索

V2-074011 | MATHDB-385269

Congruence subgroup property

摘录 / EXCERPT: Let Q be a finite quotient of $\mathrm{Out}(F_n)$ for $n > 2$. Does there exist a finite index characteristic subgroup K of F_n such that the quotient...

open_problem | open | MathDB 页面 | 原文线索

V2-074012 | MATHDB-385274

Quasi-isometric and measure equivalence rigidity (problem 4)

摘录 / EXCERPT: Let $F = F(a,b,c,d,e)$ be a free group of rank five. Define an automorphism $\phi \in \mathrm{Aut}(F)$ such that ... Let F' be the free factor of F generated by $\{a,b,c\}$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-074013 | MATHDB-385276

Quasi-isometric and measure equivalence rigidity (problem 6)

摘录 / EXCERPT: Let G and H be relatively hyperbolic non-hyperbolic free-by-cyclic groups with a trivial JSJ decomposition. If G and H are quasi-isometric then they are commensurable.

open_problem | open | MathDB 页面 | 原文线索

V2-074014 | MATHDB-385277

ℓ^2-torsion

摘录 / EXCERPT: If G is a free-by-cyclic group with exponentially growing monodromy then it has non-vanishing ℓ^2-torsion.

open_problem | open | MathDB 页面 | 原文线索

V2-074015 | MATHDB-385280

RFRS

摘录 / EXCERPT: Characterise the non-hyperbolic free-by-cyclic groups which are virtually RFRS.

open_problem | open | MathDB 页面 | 原文线索

V2-074016 | MATHDB-385697

Specific questions for graphs of small degree (problem 1)

摘录 / EXCERPT: What is the μ-measurable chromatic number of G?

open_problem | open | MathDB 页面 | 原文线索

V2-074017 | MATHDB-385698

Specific questions for graphs of small degree (problem 2)

摘录 / EXCERPT: What is the approximate μ-measurable chromatic number of G?

open_problem | open | MathDB 页面 | 原文线索

V2-074018 | MATHDB-385699

What is the Borel edge chromatic number of ... ?

摘录 / EXCERPT: What is the Borel edge chromatic number of G?

open_problem | open | MathDB 页面 | 原文线索

V2-074019 | MATHDB-385700

Specific questions for graphs of small degree (problem 4)

摘录 / EXCERPT: What is the μ-measurable edge chromatic number of G?

open_problem | open | MathDB 页面 | 原文线索

V2-074020 | MATHDB-385701

Is there a Borel partition of the edges of G, say into A, B, and C, such that ...

摘录 / EXCERPT: Is there a Borel partition of the edges of G, say into A, B, and C, such that begin{enumerate}[i.] item For all $x \in X$, $H \cdot x$ contains an edge from A (meaning it contain...

open_problem | open | MathDB 页面 | 原文线索

V2-074021 | MATHDB-385710

Relationships between different definable settings (problem 1)

摘录 / EXCERPT: Let $d \in \omega$. Given an SFT (sub-shift of finite type) of ${\\mathbb{Z}}^d$, say S (equivalently, given a LCL (locally checkable labelling problem), say Π), are any of t...

open_problem | open | MathDB 页面 | 原文线索

V2-074022 | MATHDB-385711

Relationships between different definable settings (problem 2)

摘录 / EXCERPT: Let Π be a LCL such that every highly computable graph admits a computable Π-coloring. Does every locally finite Borel graph admit a Baire measurable Π-coloring?

open_problem | open | MathDB 页面 | 原文线索

V2-074023 | MATHDB-385712

Graphings, Treeings, and CBERs (countable Borel equivalence relations) (problem 1)

摘录 / EXCERPT: Let E be treeable. begin{enumerate}[a.] item Does every Borel graphing, say, G of E contain a Borel subgraph, say T, which is a treeing of E? item What if we just want...

open_problem | open | MathDB 页面 | 原文线索

V2-074024 | MATHDB-385713

Graphings, Treeings, and CBERs (countable Borel equivalence relations) (problem 2)

摘录 / EXCERPT: Let E be pmp and G a Borel graphing of E. Does G contain Borel subgraphings (of E) whose cost is arbitrarily close to the cost of E?

open_problem | open | MathDB 页面 | 原文线索

V2-074025 | MATHDB-385714

Is every finite index extension of a hypersmooth equivalence relation hypersmooth?

摘录 / EXCERPT: Is every finite index extension of a hypersmooth equivalence relation hypersmooth?

open_problem | open | MathDB 页面 | 原文线索

V2-074026 | MATHDB-385715

Graphings, Treeings, and CBERs (countable Borel equivalence relations) (problem 4)

摘录 / EXCERPT: Let E be treeable. begin{enumerate}[a.] item Is every finite index extension of E treeable? item What if we only require a μ-measurable treeing. end{enumerate}

open_problem | open | MathDB 页面 | 原文线索

V2-074027 | MATHDB-385716

Graphings, Treeings, and CBERs (countable Borel equivalence relations) (problem 5)

摘录 / EXCERPT: Suppose E and F are treeable of type IIIA. do they have class-bijective lifts which are stably orbit equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-074028 | MATHDB-385717

Is there an “informative” notion of cost in the quasi-pmp setting?

摘录 / EXCERPT: Is there an “informative” notion of cost in the quasi-pmp setting?

open_problem | open | MathDB 页面 | 原文线索

V2-074029 | MATHDB-385718

Hyperfiniteness and related notions (problem 1)

摘录 / EXCERPT: Let $f_n: X \rightarrow [0,1]$ Borel for $n \in \omega$. Let $I \subset {Borel}(X)$ a σ-ideal. Let $F_n = \frac{1}{n} \sum_{m < n} f_m$. Does $(F_n)_{n \in \omega}$ admit a...

open_problem | open | MathDB 页面 | 原文线索

V2-074030 | MATHDB-385719

Is every orbit equivalence relation induced by a Borel action of the Grigorchuk group hyperfinite?

摘录 / EXCERPT: Is every orbit equivalence relation induced by a Borel action of the Grigorchuk group hyperfinite?

open_problem | open | MathDB 页面 | 原文线索

V2-074031 | MATHDB-385720

Hyperfiniteness and related notions (problem 3)

摘录 / EXCERPT: Let E be a CBER on X. Suppose there is a Borel binary relation $<$ on X such that the restriction of $<$ to each E-class is a linear order isomorphic to ${\\mathbb{Z}}^2$ w...

open_problem | open | MathDB 页面 | 原文线索

V2-074032 | MATHDB-385721

Is an increasing countable union of hyperfinite CBERs hyperfinite?

摘录 / EXCERPT: Is an increasing countable union of hyperfinite CBERs hyperfinite? What if we add some restrictions, e.g. involving Borel asymptotic dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-074033 | MATHDB-385722

Is a Borel graph with finite asi (asymptotic separation index) hyperfinite?

摘录 / EXCERPT: Is a Borel graph with finite asi (asymptotic separation index) hyperfinite? Does such a graph always have asi ≤ 1?

open_problem | open | MathDB 页面 | 原文线索

V2-074034 | MATHDB-385727

Is the measurable arboricity of a Borel graph always equal to its actual arboricity?

摘录 / EXCERPT: Is the measurable arboricity of a Borel graph always equal to its actual arboricity?

open_problem | open | MathDB 页面 | 原文线索

V2-074035 | MATHDB-385729

Games (problem 1)

摘录 / EXCERPT: begin{enumerate} item For what values of κ and γ are these games determined? item How can these games be generalized to, for example, extender sequences? What about s...

open_problem | open | MathDB 页面 | 原文线索

V2-074036 | MATHDB-385732

Compatcness properties of graphs (problem 1)

摘录 / EXCERPT: Let $\phi(n)$ be the statement “for every graph of size $\aleph_{\omega+1}$, if every subgraph of size $<\aleph_{\omega+1}$ has chromatic number $\le\aleph_n$, then the entire grap...

open_problem | open | MathDB 页面 | 原文线索

V2-074037 | MATHDB-386150

Prove existence and stability for pushed or pulled fronts for a toy model using a computer assisted proof.

摘录 / EXCERPT: Prove existence and stability for pushed or pulled fronts for a toy model using a computer assisted proof.

open_problem | open | MathDB 页面 | 原文线索

V2-074038 | MATHDB-386152

Develop rigorous computer assisted methods for computing the Maslov index in two dimensions.

摘录 / EXCERPT: Develop rigorous computer assisted methods for computing the Maslov index in two dimensions.

open_problem | open | MathDB 页面 | 原文线索

V2-074039 | MATHDB-386157

Prove existence of quasiperiodic patters for the 1d Swift-Hogenberg.

摘录 / EXCERPT: Prove existence of quasiperiodic patters for the 1d Swift-Hogenberg.

open_problem | open | MathDB 页面 | 原文线索

V2-074040 | MATHDB-386161

Continuation after blowup

摘录 / EXCERPT: If solutions of the wave kinetic equation blowup in L^∞, is there continuation of these solutions after the blowup?

open_problem | open | MathDB 页面 | 原文线索

V2-074041 | MATHDB-386162

Monotone quantities

摘录 / EXCERPT: Can we find any monotone hydrodynamic quantities for a wave kinetic equation?

open_problem | open | MathDB 页面 | 原文线索

V2-074042 | MATHDB-386163

Conditions for global well-posedness

摘录 / EXCERPT: Can we find conditions on the initial data so that the homogeneous wave kinetic equation is globally well-posed?

open_problem | open | MathDB 页面 | 原文线索

V2-074043 | MATHDB-386164

Regularity of free boundary/obstacle problem

摘录 / EXCERPT: Consider the following Cauchy problem, ... Determine the regularity of solutions for $\mathscr{L} = \nabla_v^s$ (This is known in $d = 1$) Determine the regularity of solutions of th...

open_problem | open | MathDB 页面 | 原文线索

V2-074044 | MATHDB-386165

Alt-Caffarelli type problem

摘录 / EXCERPT: Find a function that satisfies the Kolmogorov equation ... Derive and study the reasonable Alt-Caffarelli type problem.

open_problem | open | MathDB 页面 | 原文线索

V2-074045 | MATHDB-386166

Hele-Shaw: convergence from a particle system

摘录 / EXCERPT: Find a particle system that converges to ρ. That is, ...

open_problem | open | MathDB 页面 | 原文线索

V2-074046 | MATHDB-386167

Asymptotic behavior of nonlinear kinetic Fokker-Planck equations

摘录 / EXCERPT: Prove convergence... Determine convergence rates...

open_problem | open | MathDB 页面 | 原文线索

V2-074047 | MATHDB-386169

Consensus based optimization with time dependent temperature

摘录 / EXCERPT: Can we design a time-dependent global optimization such that you can still prove convergence to the global minimizer? Can we find an optimal cooling schedule?

open_problem | open | MathDB 页面 | 原文线索

V2-074048 | MATHDB-386170

Propagation of regularity

摘录 / EXCERPT: In the case of Maxwell molecules, can we propagate regularity of initial data for the cutoff Boltzmann equation?

open_problem | open | MathDB 页面 | 原文线索

V2-074049 | MATHDB-386171

Large data global well-posedness for nearly homogeneous non-cutoff Boltzmann

摘录 / EXCERPT: Prove large data global well-posedness for the inhomogeneous (3 +1) non-cutoff Boltzmann equation.

open_problem | open | MathDB 页面 | 原文线索

V2-074050 | MATHDB-386172

Construction of large shocks

摘录 / EXCERPT: Construct traveling wave solutions to the Boltzmann equation with large amplitude (at least amplitude 1) in dimension 1. Is it possible to observed textit{temperature overshoot}, a...

open_problem | open | MathDB 页面 | 原文线索

V2-074051 | MATHDB-386173

Caflisch construction for implosion singularity formation

摘录 / EXCERPT: Study the formation of implosion singularities for the Boltzmann equation via the Caflisch construction of solutions

open_problem | open | MathDB 页面 | 原文线索

V2-074052 | MATHDB-386175

Discrete operators to nonlocal operators (problem 1)

摘录 / EXCERPT: With the above notion of nonlocal derivative, characterize the behavior of the nonlocal transport equation ... as $\delta \to 0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074053 | MATHDB-386176

Convergence of graph operators to nonlocal operators in $\R^n$

摘录 / EXCERPT: Show convergence of graph operators to their nonlocal counterparts in the limit as the $\#$ of vertices “fills up" ${\\mathbb{R}}^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-074054 | MATHDB-386177

General $L^p \to L^\infty$ estimate

摘录 / EXCERPT: Prove $L^p \to L^\infty$ estimate for ... for $f \in L^p$ and $\mathscr{L}$ a non-divergence, nonlocal form.

open_problem | open | MathDB 页面 | 原文线索

V2-074055 | MATHDB-386178

Continuity of solutions for multi-dimensional p-Laplacian

摘录 / EXCERPT: Prove continuity of solutions for the system case of the fraction p- Laplacian.

open_problem | open | MathDB 页面 | 原文线索

V2-074056 | MATHDB-386179

Schauder estimate for general transport term

摘录 / EXCERPT: Obtain Schauder estimates for a general transport term, ... Additionally, kinetic formulation of systems of conservation laws yield transport operators that look like ... Can one obtai...

open_problem | open | MathDB 页面 | 原文线索

V2-074057 | MATHDB-386180

Uniqueness for porous medium in $d > 1$

摘录 / EXCERPT: Prove uniqueness of solutions for the porous medium equation in $d > 1$ with not too smooth initial data. Is the Wasserstein distance contractive?

open_problem | open | MathDB 页面 | 原文线索

V2-074058 | MATHDB-386181

Homogeneous relativistic Landau equation

摘录 / EXCERPT: Prove global existence of smooth solutions for the relativistic Landau equation.

open_problem | open | MathDB 页面 | 原文线索

V2-074059 | MATHDB-386182

Example of blowup for inhomogeneous Landau (or Boltzmann)

摘录 / EXCERPT: Can we find a set of conditions such that the homogeneous equation is globally well-posed, but the inhomogeneous equation blows up? Can this be down for the isotropic Landau equati...

open_problem | open | MathDB 页面 | 原文线索

V2-074060 | MATHDB-386183

Regularity up to boundary for kinetic equations

摘录 / EXCERPT: What is the relationship between the regularity of the coefficients and the boundary regularity? Is $C_x^{1/6}C_v^{1/2}$ regularity sharp in the case of constant coefficients?

open_problem | open | MathDB 页面 | 原文线索

V2-074061 | MATHDB-386184

Regularity of solutions of non-divergence form of a linear kinetic equation

摘录 / EXCERPT: Study the regularity of solutions a kinetic equation ... where $\mathscr{L}$ is linear and of non-divergence form. In the local case, $\mathscr{L} = Tr(AD^2_v\cdot)$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-074062 | MATHDB-386185

Inverse problem for interacting particle system

摘录 / EXCERPT: Assume that one can tune any macroscopic quantity (i.e. $\rho(t,x)$, E, B etc.) What is necessary to recover the initial distribution f_0 for $d \geq 2$? Is this possible? Th...

open_problem | open | MathDB 页面 | 原文线索

V2-074063 | MATHDB-386502

mapping problem 1

摘录 / EXCERPT: Let $n \ge 2$. begin{itemize} item Given a proper rational map $f: \mathbb{B}^{n} \rightarrow \mathbb{B}^{N}$, find the smallest number $k(n, N)$ such that f is determined by its...

open_problem | open | MathDB 页面 | 原文线索

V2-074064 | MATHDB-386505

Mappings between balls (problem 4)

摘录 / EXCERPT: Assume $\Gamma \subseteq \textrm{Aut}(\mathbb{B}^{N})$ is a discrete subgroup and $f: \mathbb{B}^{n} \rightarrow \mathbb{B}^{N}$ is a proper holomorphic embedding such that...

open_problem | open | MathDB 页面 | 原文线索

V2-074065 | MATHDB-386853

Interpretation of higher powers of descendant classes in terms of tangency conditions to contact hyperplanes, generalizing work of Okounkov-Pandharipande for ... .

摘录 / EXCERPT: Interpretation of higher powers of descendant classes in terms of tangency conditions to contact hyperplanes, generalizing work of Okounkov-Pandharipande for $V=S^1$.

open_problem | open | MathDB 页面 | 原文线索

V2-074066 | MATHDB-386855

Moduli (problem 1)

摘录 / EXCERPT: a) Is there an explicit cycle ... representing the class ... in $\overline{\mathcal{M}}_{g,1}(\mathbb{P}^1,d)$? Partial answer (Okounkov, Pandharipande): if...

open_problem | open | MathDB 页面 | 原文线索

V2-074067 | MATHDB-386856

Moduli (problem 2)

摘录 / EXCERPT: Consider ... It is known that (1)=(3) on $\mathcal{M}_{g,n}^{\rm ct}$. a) Find an expression for (1) and (2) in the tautological ring of $\overline{\mathcal{M}}_{g,n}$. b) Does (3) e...

open_problem | open | MathDB 页面 | 原文线索

V2-074068 | MATHDB-386858

Can SFT be used as an invariant to distinguish lens spaces?

摘录 / EXCERPT: Can SFT be used as an invariant to distinguish lens spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-074069 | MATHDB-386860

Top four problem areas (problem 1)

摘录 / EXCERPT: Characterizing and modeling tumor heterogeneity

open_problem | open | MathDB 页面 | 原文线索

V2-074070 | MATHDB-386861

Linking signaling models to phenotype (e.g., tumor growth)

摘录 / EXCERPT: Linking signaling models to phenotype (e.g., tumor growth)

open_problem | open | MathDB 页面 | 原文线索

V2-074071 | MATHDB-386862

Top four problem areas (problem 3)

摘录 / EXCERPT: Translating pre-clinical models to human

open_problem | open | MathDB 页面 | 原文线索

V2-074072 | MATHDB-386863

Other biology-centric problem areas (problem 1)

摘录 / EXCERPT: The role of cellular metabolism in cancer

open_problem | open | MathDB 页面 | 原文线索

V2-074073 | MATHDB-386864

Other biology-centric problem areas (problem 2)

摘录 / EXCERPT: Cell cycle-dependent variation in tumor cells.

open_problem | open | MathDB 页面 | 原文线索

V2-074074 | MATHDB-386865

Challenges: Time lines, focus, intellectual property

摘录 / EXCERPT: Challenges: Time lines, focus, intellectual property

open_problem | open | MathDB 页面 | 原文线索

V2-074075 | MATHDB-386866

Opportunities: Student clinics, Research Experience for Undergraduates (REU) programs, internships

摘录 / EXCERPT: Opportunities: Student clinics, Research Experience for Undergraduates (REU) programs, internships

open_problem | open | MathDB 页面 | 原文线索

V2-074076 | MATHDB-386867

Funding for industry/academic interface: US National Institutes of Health (1) R01 "Interface of the Life and Physical Sciences" grants (PAR-10-141, PAR-10-142), (2) Small Business Innovation research

摘录 / EXCERPT: Funding for industry/academic interface: US National Institutes of Health (1) R01 "Interface of the Life and Physical Sciences" grants (PAR-10-141, PAR-10-142), (2) Small Business...

open_problem | open | MathDB 页面 | 原文线索

V2-074077 | MATHDB-386868

General fusion category questions (problem 1)

摘录 / EXCERPT: Is any integral fusion category unitarizable?

open_problem | open | MathDB 页面 | 原文线索

V2-074078 | MATHDB-386870

General fusion category questions (problem 3)

摘录 / EXCERPT: Are all fusion categories pivotal? Are all fusion categories spherical? Does it depend on the ground field (k v.s. ${\\mathbb{C}}$)?

open_problem | open | MathDB 页面 | 原文线索

V2-074079 | MATHDB-386875

How many fusion categories... (problem 4)

摘录 / EXCERPT: What can you say about all fusion categories $\mathcal{C}$ for which $\#\{\dim(X)|X\in\mathcal{C}\}=2?$

open_problem | open | MathDB 页面 | 原文线索

V2-074080 | MATHDB-386879

Is there a sense in which a randomly chosen fusion graph doesn't have cylotomic dimensions?

摘录 / EXCERPT: Is there a sense in which a randomly chosen fusion graph doesn't have cylotomic dimensions?

open_problem | open | MathDB 页面 | 原文线索

V2-074081 | MATHDB-386884

A fusion category version of supertransitivity

摘录 / EXCERPT: In a fusion category, is there an upper bound on the N such that $X^{\otimes N}$ is a simple object (where $\dim(X)>1$)?

open_problem | open | MathDB 页面 | 原文线索

V2-074082 | MATHDB-386885

Objects in fusion categories (problem 4)

摘录 / EXCERPT: There are accumulation points from below for $\text{FPdim}(X)$ for an object in a fusion category or $[M\colon N]$ for finite depth subfactors. Are there any accumulation points fr...

open_problem | open | MathDB 页面 | 原文线索

V2-074083 | MATHDB-386891

Is there an extension theory for fusion categories extended by fusion rings?

摘录 / EXCERPT: Is there an extension theory for fusion categories extended by fusion rings? (For example, near group categories)

open_problem | open | MathDB 页面 | 原文线索

V2-074084 | MATHDB-386895

Is there a conceptual construction of the even half of 4442?

摘录 / EXCERPT: Is there a conceptual construction of the even half of 4442?

open_problem | open | MathDB 页面 | 原文线索

V2-074085 | MATHDB-386911

Assuming an appropriate generalization of PFA, determine the gaps spectrum of ... .

摘录 / EXCERPT: Assuming an appropriate generalization of PFA, determine the gaps spectrum of $P(\omega)/fin$.

open_problem | open | MathDB 页面 | 原文线索

V2-074086 | MATHDB-386918

Can we obtain the super tree property at ... ?

摘录 / EXCERPT: Can we obtain the super tree property at $\aleph_{\omega+1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-074087 | MATHDB-386929

Are Computer-assisted proofs real proofs?

摘录 / EXCERPT: Are computer-assisted proofs always real proofs when proprietary software and/or hardware is involved that is not accessible to all readers/reviewers?

open_problem | open | MathDB 页面 | 原文线索

V2-074088 | MATHDB-386931

How should limitations in technology for conjectures/discoveries be adressed?

摘录 / EXCERPT: Can we uncover quantitative properties such as periodicity from numerical methods for PDEs? - In particular for integrable systems. Does (or should) a correct notion of computabili...

open_problem | open | MathDB 页面 | 原文线索

V2-074089 | MATHDB-386932

Computer-assisted proof of Mochizuk's work on the ABC conjecture!

摘录 / EXCERPT: Can one set up a machine-verifiable proof of the work by Mochizuki trying to proof the ABC conjecture?

open_problem | open | MathDB 页面 | 原文线索

V2-074090 | MATHDB-386933

Make computer-assisted proofs and software more accessible to math community!

摘录 / EXCERPT: What do mathematicians find difficult about working with computer-assistance in Mathematics? How can software be improved from the view of users to ?

open_problem | open | MathDB 页面 | 原文线索

V2-074091 | MATHDB-386936

How to use AI for computer-assisted proofs?

摘录 / EXCERPT: How to use AI for computer-assisted proofs?

open_problem | open | MathDB 页面 | 原文线索

V2-074092 | MATHDB-386938

Exhibit a model for phase transitions!

摘录 / EXCERPT: Exhibit a model that has 3 phase transitions, as basis pursuit is known to have (at least) 2. For example one with a transition from polynomial to exponential time or unbounded tim...

open_problem | open | MathDB 页面 | 原文线索

V2-074093 | MATHDB-386940

Create computer-assisted eigenvalue solver for matrices!

摘录 / EXCERPT: Given a n by n matrix, can you output the spectrum/eigenvalues to given accuracy? - In polynomial time.

open_problem | open | MathDB 页面 | 原文线索

V2-074094 | MATHDB-386942

How many regular tetraheadra in $\mathbb R^3$ can touch at a single point?

摘录 / EXCERPT: How many regular tetraheadra in $\mathbb R^3$ can touch at a single point? Follow-up comments/question: How many regular tetrahedra with side length $=1$ fit into a unit ball?

open_problem | open | MathDB 页面 | 原文线索

V2-074095 | MATHDB-386944

Develop a pivotal/spherical version of extension theory as in ENO.

摘录 / EXCERPT: Develop a pivotal/spherical version of extension theory as in ENO.

open_problem | open | MathDB 页面 | 原文线索

V2-074096 | MATHDB-386945

List of all question (problem 3)

摘录 / EXCERPT: Determine Representations of motion groups of links in S^3. In particular, study the ones coming from Dijkgraff-Witten $(3+1)$-D TQFTs.

open_problem | open | MathDB 页面 | 原文线索

V2-074097 | MATHDB-386946

List of all question (problem 4)

摘录 / EXCERPT: If $\mathcal{Z}(\mathcal{C})\cong \mathcal{Z}(\mathcal{D})$ and $\mathcal{C},\mathcal{D}$ and they both admit a fiber functor, are they Morita equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-074098 | MATHDB-386947

List of all question (problem 5)

摘录 / EXCERPT: Do generalized spins in 2-categories need to be rational? This a 2-category version of Vafa's theorem.

open_problem | open | MathDB 页面 | 原文线索

V2-074099 | MATHDB-386948

List of all question (problem 6)

摘录 / EXCERPT: Suppose we have two BFCs $\mathcal{B},\mathcal{C} \subseteq \mathcal{M}$ with $\mathcal{M}$ modular such that $\mathcal{B},\mathcal{C}$ centralize each other. Then what is the Witt...

open_problem | open | MathDB 页面 | 原文线索

V2-074100 | MATHDB-386949

List of all question (problem 7)

摘录 / EXCERPT: What is the correct definition of the 2-center of categories so that the following statement holds: $\mathcal{Z}(\mathcal{C}) \cong \mathcal{Z}(\mathcal{B})$ if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-074101 | MATHDB-386950

What are the braided --- categories so that invertibles correspond to cobordisms?

摘录 / EXCERPT: What are the braided --- categories so that invertibles correspond to cobordisms?

open_problem | open | MathDB 页面 | 原文线索

V2-074102 | MATHDB-386952

List of all question (problem 10)

摘录 / EXCERPT: Develop an axiomatic theory of Fractons.

open_problem | open | MathDB 页面 | 原文线索

V2-074103 | MATHDB-386955

Compute examples of fusion ... -categories with fiber functors.

摘录 / EXCERPT: Compute examples of fusion 2-categories with fiber functors.

open_problem | open | MathDB 页面 | 原文线索

V2-074104 | MATHDB-386956

List of all question (problem 14)

摘录 / EXCERPT: Understand fiber 2-functors F of a fusion 2-category and understand boundary anomalies.

open_problem | open | MathDB 页面 | 原文线索

V2-074105 | MATHDB-386957

List of all question (problem 15)

摘录 / EXCERPT: Study representations of motion groups of links in S^3. For example, study the ones coming from Dijkgraff-Witten $(3+1)$-D TQFTs.

open_problem | open | MathDB 页面 | 原文线索

V2-074106 | MATHDB-386958

List of all question (problem 16)

摘录 / EXCERPT: Define and study G-crossed Braided Fusion categories (BFC) with anti-unitary G-action.

open_problem | open | MathDB 页面 | 原文线索

V2-074107 | MATHDB-386959

List of all question (problem 17)

摘录 / EXCERPT: In $(3+1)$D there is an $O(5)$ obstruction to the full G-crossed theory. Calculate it systematically. This can be rephrased as: calculate the G-extension theory of a braided fu...

open_problem | open | MathDB 页面 | 原文线索

V2-074108 | MATHDB-386962

Do all fusion categories admit a Hermitian structure (a ... -structure which is positive) ?

摘录 / EXCERPT: Do all fusion categories admit a Hermitian structure (a $\dagger$-structure which is positive) ?

open_problem | open | MathDB 页面 | 原文线索

V2-074109 | MATHDB-386963

If a fusion category admits a Hermitian structure, is it unique?

摘录 / EXCERPT: If a fusion category admits a Hermitian structure, is it unique?

open_problem | open | MathDB 页面 | 原文线索

V2-074110 | MATHDB-386965

List of all question (problem 23)

摘录 / EXCERPT: Do QCAs form Ω spectrum? Are the first four levels the same as the spectrum of modular tensor categories?

open_problem | open | MathDB 页面 | 原文线索

V2-074111 | MATHDB-386966

Starting with a trivial TQFT, what is the minimum number of abelian extensions that one can do to get reach some given TQFT?

摘录 / EXCERPT: Starting with a trivial TQFT, what is the minimum number of abelian extensions that one can do to get reach some given TQFT?

open_problem | open | MathDB 页面 | 原文线索

V2-074112 | MATHDB-386970

List of all question (problem 28)

摘录 / EXCERPT: Reconstruction theory for fusion 2-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-074113 | MATHDB-386971

If there is a BFC that has order ... in the Witt group, does it admit a time-reversal symmetry ?

摘录 / EXCERPT: If there is a BFC that has order 2 in the Witt group, does it admit a time-reversal symmetry ?

open_problem | open | MathDB 页面 | 原文线索

V2-074114 | MATHDB-386972

Do fusion ... -categories exhibit some version of Ocneanu rigidity?

摘录 / EXCERPT: Do fusion 2-categories exhibit some version of Ocneanu rigidity?

open_problem | open | MathDB 页面 | 原文线索

V2-074115 | MATHDB-386974

Is factorization of MTCs unique if there are no invertible objects?

摘录 / EXCERPT: Is factorization of MTCs unique if there are no invertible objects?

open_problem | open | MathDB 页面 | 原文线索

V2-074116 | MATHDB-386976

List of all question (problem 34)

摘录 / EXCERPT: Construct fermionic 2-categories. In other words, construct enriched fusion 2-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-074117 | MATHDB-386977

What is the proper notion of a higher tube category?

摘录 / EXCERPT: What is the proper notion of a higher tube category?

open_problem | open | MathDB 页面 | 原文线索

V2-074118 | MATHDB-386978

Is it possible to detect the trivial Witt Class from the S and T matrices?

摘录 / EXCERPT: Is it possible to detect the trivial Witt Class from the S and T matrices? (In other words, can we tell if a nondegenrate braided fusion category is a Drinfeld center just by looki...

open_problem | open | MathDB 页面 | 原文线索

V2-074119 | MATHDB-386981

What is the diagrammatic interpretation of Gauss sums for ... ?

摘录 / EXCERPT: What is the diagrammatic interpretation of Gauss sums for $\mathcal{C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-074120 | MATHDB-386982

Develop pivotal/spherical version of ENO extension theory.

摘录 / EXCERPT: Develop pivotal/spherical version of ENO extension theory.

open_problem | open | MathDB 页面 | 原文线索

V2-074121 | MATHDB-386984

Develop extension theory for rings (in place of groups).

摘录 / EXCERPT: Develop extension theory for rings (in place of groups). There is motivation to investigate this problem from the study of Haagerup fusion categories.

open_problem | open | MathDB 页面 | 原文线索

V2-074122 | MATHDB-386987

List of all question (problem 45)

摘录 / EXCERPT: Construct a $(p,\mathbf{Z}_p)$ near group fusion category. These are categories generated by two objects $\{\mathbf{1},X\}$ with X invertible and the fusion rule is...

open_problem | open | MathDB 页面 | 原文线索

V2-074123 | MATHDB-386988

List of all question (problem 46)

摘录 / EXCERPT: Interpret $(3+1)$D SETs using G-crossed 2-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-074124 | MATHDB-386991

Condense out everything below middle dimension excitations in TQFTs.

摘录 / EXCERPT: Condense out everything below middle dimension excitations in TQFTs. Then, what is left?

open_problem | open | MathDB 页面 | 原文线索

V2-074125 | MATHDB-386993

Give anomaly/TQFT/bordism interpretation of Gauss sums.

摘录 / EXCERPT: Give anomaly/TQFT/bordism interpretation of Gauss sums.

open_problem | open | MathDB 页面 | 原文线索

V2-074126 | MATHDB-386994

Look at Gauss' proof on positivity of Gauss sums (...).

摘录 / EXCERPT: Look at Gauss' proof on positivity of Gauss sums ($\sum_{k=0}^{p-1} e^{2\pi i k^2/p} $). Is there a TQFT interpretation for this positivity?

open_problem | open | MathDB 页面 | 原文线索

V2-074127 | MATHDB-386995

Effective Jumps (problem 1)

摘录 / EXCERPT: For which equivalence relation E is $E < E^+$? (In both the computable and Borel context.) begin{itemize} item For which orbit equivalence relation? item For which Σ^1_1 e...

open_problem | open | MathDB 页面 | 原文线索

V2-074128 | MATHDB-386996

Effective Jumps (problem 2)

摘录 / EXCERPT: Given an equivalence relation E on X and a countable group G, define an equivalence relation on X^G by $\alpha E^{[G]} \beta$ iff there is some $g\in G$ such that for every...

open_problem | open | MathDB 页面 | 原文线索

V2-074129 | MATHDB-386997

Effective Jumps (problem 3)

摘录 / EXCERPT: For countable groups G, H, and an equivalence relation E, is there a countable group K such that $(E^{[G]})^{[H]}$ is bi-reducible with $E^{[K]}$

open_problem | open | MathDB 页面 | 原文线索

V2-074130 | MATHDB-386998

Effective Jumps (problem 4)

摘录 / EXCERPT: For a “reasonable" computable group G and some arithmetical E is $E^{[G]}$ always universal at some Σ^0_n level?

open_problem | open | MathDB 页面 | 原文线索

V2-074131 | MATHDB-386999

Modify the Friedman--Stanley jump by using a fixed subgroup ... .

摘录 / EXCERPT: Modify the Friedman--Stanley jump by using a fixed subgroup $G \le S_\infty$. Ask the above questions.

open_problem | open | MathDB 页面 | 原文线索

V2-074132 | MATHDB-387000

Effective Jumps (problem 6)

摘录 / EXCERPT: Consider the Louveau jump: let F be a filter on ω and E an equivalence relation, define E^F to be the equivalence relation on X^ω by $\alpha E^F \beta$ iff...

open_problem | open | MathDB 页面 | 原文线索

V2-074133 | MATHDB-387003

Continuous degrees of smooth equivalence relations (problem 2)

摘录 / EXCERPT: Does every Π^0_1 smooth equivalence relation on ω^ω continuously reduce to $=^+$? Does every strictly ascending sequence has a supremum?

open_problem | open | MathDB 页面 | 原文线索

V2-074134 | MATHDB-387004

Reducibility on a cone (problem 1)

摘录 / EXCERPT: Given A and B classes of structures, define $A\le_{\text{eff}}^\triangledown B$ if there is a c such that for every $y\ge_T c$, there is a computable Φ such that for all...

open_problem | open | MathDB 页面 | 原文线索

V2-074135 | MATHDB-387006

Countable and finitary reducibility (problem 1)

摘录 / EXCERPT: Is $E_\infty \le^{<\omega} E_0$? Is E_{set} universal for $\le ^{<\omega}$ among Π^0_3 equivalence relations?

open_problem | open | MathDB 页面 | 原文线索

V2-074136 | MATHDB-387008

Countable and finitary reducibility (problem 3)

摘录 / EXCERPT: Consider versions of $\le^n$ or $\le^\omega$ obtained by playing a game for the reals. (I.e., you give the first input, the functional returns the first output, then you give the s...

open_problem | open | MathDB 页面 | 原文线索

V2-074137 | MATHDB-387009

Model-theoretic equivalence relations (problem 1)

摘录 / EXCERPT: Is there an $L_{\omega_1,\omega}$-elementary class K of structures which is Borel complete and there is an $X\subseteq \omega$ such that there is no X-computable model in K o...

open_problem | open | MathDB 页面 | 原文线索

V2-074138 | MATHDB-387010

Model-theoretic equivalence relations (problem 2)

摘录 / EXCERPT: Consider the equivalence relations on countable structures, by model theoretic bi-interpretability (in first-order logic). Is this equivalence relation Σ^1_1-complete? We ma...

open_problem | open | MathDB 页面 | 原文线索

V2-074139 | MATHDB-387011

Is some version of the hyperaperiodicity argument useful for the ... -content?

摘录 / EXCERPT: Is some version of the hyperaperiodicity argument useful for the ω-content?

open_problem | open | MathDB 页面 | 原文线索

V2-074140 | MATHDB-387012

Further questions (problem 1)

摘录 / EXCERPT: Consider the collection C of countable structures. Consider two equivalence relations on C: begin{enumerate} item Medvedev equivalence: $A\equiv_s B$ if there are two Turing fu...

open_problem | open | MathDB 页面 | 原文线索

V2-074141 | MATHDB-387013

Further questions (problem 2)

摘录 / EXCERPT: Is there a T_2 (or T_1) and second countable space such that the relation $x E y$ iff $\bigcup_n U_{X(n)} = \bigcup_n U_{Y(n)}$ has arbitrarily high Borel rank (but is Borel)?...

open_problem | open | MathDB 页面 | 原文线索

V2-074142 | MATHDB-387014

Further questions (problem 3)

摘录 / EXCERPT: Which equivalence relations on c.e. sets are induced by an action of a semi-group computable on the indices?

open_problem | open | MathDB 页面 | 原文线索

V2-074143 | MATHDB-387015

Are there Borel (or continuous) non-reducibility result proved using the recursion theorem?

摘录 / EXCERPT: Are there Borel (or continuous) non-reducibility result proved using the recursion theorem?

open_problem | open | MathDB 页面 | 原文线索

V2-074144 | MATHDB-387019

Classify real linear combinations of products of minors that are nonnegative for all ... matrices.

摘录 / EXCERPT: Classify real linear combinations of products of minors that are nonnegative for all TN matrices. More concretely, given a polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-074145 | MATHDB-387048

Macdonald polynomials (problem 2)

摘录 / EXCERPT: Categorify the modified Macdonald polynomials $\tilde{H_\mu}$.

open_problem | open | MathDB 页面 | 原文线索

V2-074146 | MATHDB-387060

Understand Maulik's proof of the Oblomkov-Shende conjecture, then categorify it!

摘录 / EXCERPT: Understand Maulik's proof of the Oblomkov-Shende conjecture, then categorify it!

open_problem | open | MathDB 页面 | 原文线索

V2-074147 | MATHDB-387070

This problem asks generally how to build a database of databases to help users navigate to the resource where the information they wish to obtain is.

摘录 / EXCERPT: This problem asks generally how to build a database of databases to help users navigate to the resource where the information they wish to obtain is. The concrete goal of this obje...

open_problem | open | MathDB 页面 | 原文线索

V2-074148 | MATHDB-387071

This group aims to find examples of open-source project groups that did and did not succeed at long-term sustainability.

摘录 / EXCERPT: This group aims to find examples of open-source project groups that did and did not succeed at long-term sustainability. The concrete goal is to discover principles that underlie t...

open_problem | open | MathDB 页面 | 原文线索

V2-074149 | MATHDB-387072

The goal of this group is to produce an example of an "on-ramp" which could be given to someone with little to no experience in software, from which they could learn to do some procedure.

摘录 / EXCERPT: The goal of this group is to produce an example of an "on-ramp" which could be given to someone with little to no experience in software, from which they could learn to do some pro...

open_problem | open | MathDB 页面 | 原文线索

V2-074150 | MATHDB-387073

The goal of this group is to discuss ways to create an online database of mathematical visualizations.

摘录 / EXCERPT: The goal of this group is to discuss ways to create an online database of mathematical visualizations. While creating such a database in a week is not feasible, the group aims to f...

open_problem | open | MathDB 页面 | 原文线索

V2-074151 | MATHDB-387074

The goals of this group are to reevaluate the aims of math communication, using specific examples, and introduce new concepts that they believe will impact the future of how mathematics is communicate

摘录 / EXCERPT: The goals of this group are to reevaluate the aims of math communication, using specific examples, and introduce new concepts that they believe will impact the future of how mathem...

open_problem | open | MathDB 页面 | 原文线索

V2-074152 | MATHDB-387076

The sole object of this group is to decide whether or not to finish the software SpaceMath.

摘录 / EXCERPT: The sole object of this group is to decide whether or not to finish the software SpaceMath.

open_problem | open | MathDB 页面 | 原文线索

V2-074153 | MATHDB-387077

The goal of this group is to create a survey of what work has been done on educational proof software and to describe the goals which spurred its creation.

摘录 / EXCERPT: The goal of this group is to create a survey of what work has been done on educational proof software and to describe the goals which spurred its creation. The group should aim to...

open_problem | open | MathDB 页面 | 原文线索

V2-074154 | MATHDB-387078

The goal of this group should be to investigate what tools exist for knowledge tracing and should outline what tools could be developed to continue this pursuit, now that interest in it has been renew

摘录 / EXCERPT: The goal of this group should be to investigate what tools exist for knowledge tracing and should outline what tools could be developed to continue this pursuit, now that interest...

open_problem | open | MathDB 页面 | 原文线索

V2-074155 | MATHDB-387079

This group should aim to understand to what extent LaTeX does and does not meet the needs of current mathematicians.

摘录 / EXCERPT: This group should aim to understand to what extent LaTeX does and does not meet the needs of current mathematicians. After such an understanding is had, it should aim to create a l...

open_problem | open | MathDB 页面 | 原文线索

V2-074156 | MATHDB-390544

Marton’s conjecture on broadcast-channel capacity

摘录 / EXCERPT: For every finite two-receiver discrete memoryless broadcast channel $p_{Y,Z\mid X}$, the capacity region equals the full Marton inner bound:...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-074157 | MATHDB-390615

Keller’s conjecture

摘录 / EXCERPT: For every integer $n\ge 1$ and every tiling $\mathcal{T}$ of $\mathbb{R}^n$ by unit n-dimensional cubes, there exist distinct cubes $C,C'\in\mathcal{T}$ such that $C\cap C'$ is a...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-074158 | MATHDB-392062

Exact value of the Bohr radius of the bidisc

摘录 / EXCERPT: Let ... and let S_2 be the Schur class of analytic functions $f:\mathbb D^2\to\mathbb C$ satisfying ... If ... define ... Determine the exact value of K_2.

open_problem | claimed_progress | Shivam Patel | MathDB 页面 | 原文线索

V2-074159 | MATHDB-392108

Han’s conjecture on distributed testing against dependence

摘录 / EXCERPT: Let Q_{XY} be a joint distribution on finite alphabets, and consider distributed hypothesis testing with null hypothesis $\mathcal{H}_0: P_{XY}=Q_XQ_Y$ and alternative hypothesis...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-074160 | MATHDB-392119

Strang’s conjecture

摘录 / EXCERPT: Let Δ be a triangulation of a simply connected polygonal domain $\Omega\subset\mathbb{R}^2$, let f_1° and f_0° denote the numbers of interior edges and interio...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-074161 | MATHDB-392308

Agnostic multiclass transductive learning versus PAC learning

摘录 / EXCERPT: For every multiclass hypothesis class $\mathcal H$ with arbitrary label space, determine the minimax agnostic transductive excess error from a population of size n in terms of th...

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-074162 | MATHDB-392617

Askey-Szegő problem

摘录 / EXCERPT: Determine all triples $(a,b,c)\in\mathbb{R}_{>0}^{3}$ such that ${}_1F_2\!\left(\begin{matrix}a\\ b,c\end{matrix};-x\right)>0$ for every $x>0$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-074163 | MATHDB-392619

Constant-factor approximation of the cut norm

摘录 / EXCERPT: For a real matrix $A=(a_{ij})_{i\in R,\,j\in S}$, define its cut norm by $\lVert A\rVert_C=\max_{I\subseteq R,\,J\subseteq S}\left|\sum_{i\in I,\,j\in J}a_{ij}\right|$. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-074164 | MATHDB-392620

Exact value of Grothendieck’s constant

摘录 / EXCERPT: Determine the exact value of Grothendieck’s constant K_G, defined as the least constant such that, for every real $m\times n$ matrix $A=(a_{ij})$,nn ...

open_problem | open | MathDB 页面 | 原文线索

V2-074165 | MATHDB-392621

Connectivity threshold for random k-nearest-neighbour graphs

摘录 / EXCERPT: Let P be a Poisson process of intensity 1 in a square S_n of area n, and let $G_{n,k}$ be the random geometric graph obtained by joining each point of P to its k neares...

open_problem | open | MathDB 页面 | 原文线索

V2-074166 | MATHDB-392622

Number of linear extensions of the Boolean lattice

摘录 / EXCERPT: Let $L(P)$ denote the number of linear extensions of a poset P, and let Q_t be the Boolean lattice of subsets of $[U+007F]\{1,2,\ldots,t\u007f\}$ ordered by inclusion. With all logari...

open_problem | open | MathDB 页面 | 原文线索

V2-074167 | MATHDB-392624

Coloring connected claw-free graphs containing a stable set of size three

摘录 / EXCERPT: Let G be a connected, claw-free graph that contains a stable set of size three. Then $\chi(G) \le 2\omega(G)$. This bound is best possible.

open_problem | open | MathDB 页面 | 原文线索

V2-074168 | MATHDB-392626

Traffic-light switches and dependence on one coordinate

摘录 / EXCERPT: Assume that at a given road junction there are n three-position switches controlling the red-yellow-green position of the traffic light. You are told that whenever you change the...

open_problem | open | MathDB 页面 | 原文线索

V2-074169 | MATHDB-392630

Correct second-order constant for fork-free families

摘录 / EXCERPT: For a fixed positive integer r, determine the correct second-order asymptotic term for the maximum size of a family $\mathcal{F}\subseteq 2^{[n]}$ containing no $(r+1)$-fork, whe...

open_problem | open | MathDB 页面 | 原文线索

V2-074170 | MATHDB-392632

Universal spanning triangulations at minimum degree $2n/3$

摘录 / EXCERPT: Given n, is there a triangulation P_n of order n that is contained in every graph G of order n with minimum degree at least $2n/3$?

open_problem | open | MathDB 页面 | 原文线索

V2-074171 | MATHDB-392633

Bollobás-Komlós bandwidth conjecture

摘录 / EXCERPT: For every $\gamma>0$ and all $r,\Delta\in\mathbb{N}$, there exist $\alpha>0$ and $n_0\in\mathbb{N}$ such that every graph G of order $n\ge n_0$ with minimum degree at least...

conjecture | open | MathDB 页面 | 原文线索

V2-074172 | MATHDB-392634

Finite-rank positive semidefinite moment matrices have unique atomic representing measures

摘录 / EXCERPT: If $y\in\mathbb{R}^{\mathbb{Z}_+^n}$, $M(y)\succeq 0$, and $M(y)$ has finite rank r, then y has a unique representing measure, which is r-atomic. Conversely, if y has an...

open_problem | open | MathDB 页面 | 原文线索

V2-074173 | MATHDB-392635

The F-moment problem: characterization of an atomic representing measure

摘录 / EXCERPT: Let $F:=\{x\in\mathbb{R}^n\mid h_1(x)\ge 0,\ldots,h_m(x)\ge 0\}$, where $h_1,\ldots,h_m\in\mathbb{R}[x_1,\ldots,x_n]$. For $d_j:=\lceil\deg(h_j)/2\rceil$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-074174 | MATHDB-392637

Bilu-Linial signing conjecture

摘录 / EXCERPT: Every d-regular graph has a signing with spectral radius at most $2\sqrt{d-1}$.

conjecture | open | MathDB 页面 | 原文线索

V2-074175 | MATHDB-392638

Sankoff-Mainville conjecture for the longest common subsequence

摘录 / EXCERPT: Let L be the length of the longest common subsequence of two uniformly and independently chosen words of length n over an alphabet of size k, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-074176 | MATHDB-392639

Expected length of the longest common subsequence for large alphabets

摘录 / EXCERPT: Let L be the length of the longest common subsequence of two uniformly and independently chosen random words of length n over an alphabet of size k. For every $\epsilon>0$, t...

open_problem | open | MathDB 页面 | 原文线索

V2-074177 | MATHDB-392640

Non-principal families of size two for even uniformities and for 3-graphs

摘录 / EXCERPT: For every even integer $k\ge 4$, and also for $k=3$, there exist k-graphs F and H such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-074178 | MATHDB-392644

Kohayakawa-Łuczak-Rödl conjecture

摘录 / EXCERPT: Let H be an arbitrary graph and let $\beta>0$. Then there exist positive constants $\\varepsilon_0$, C, and n_0 such that ... for all $m\\ge Cn^{2-1/d_2(H)}$, $n\\ge n_0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-074179 | MATHDB-392647

Brandt’s conjecture for triangle-free graphs at minimum degree n/3

摘录 / EXCERPT: S. Brandt conjectured that $f(1/3)=4$, where $f(c)$ is the function in the preceding assertion: every triangle-free graph G with n vertices and minimum degree at least cn has...

conjecture | open | MathDB 页面 | 原文线索

V2-074180 | MATHDB-392650

Sharp bounds for long arithmetic progressions in unrestricted sumsets

摘录 / EXCERPT: Let $A\subseteq[n]=\{1,\ldots,n\}$ be a finite set, and let $lA=\{a_1+\cdots+a_l:a_i\in A\}$, where repetitions are allowed. For fixed n, l, and $|A|$, determine or estimate th...

open_problem | open | MathDB 页面 | 原文线索

V2-074181 | MATHDB-392651

Infinite arithmetic progressions in subset sums of dense sequences

摘录 / EXCERPT: There exists a sufficiently large constant C such that every strictly increasing sequence of positive integers $A=\\{a_1<a_2<\\cdots\\}$ for which the number of elements of A b...

open_problem | open | MathDB 页面 | 原文线索

V2-074182 | MATHDB-392652

Kissing number in dimension four

摘录 / EXCERPT: The kissing number in dimension four is $\kappa(4)=24$.

open_problem | open | MathDB 页面 | 原文线索

V2-074183 | MATHDB-392653

Asymptotic distribution of the deviance statistic when testing for GARCH effects alone

摘录 / EXCERPT: Theorem 1. Assume that X_i satisfies model (1), with $\\phi=1$, and that the $\\varepsilon_i$ are i.i.d. with expectation 0, variance 1, and finite third and fourth moments...

open_problem | open | MathDB 页面 | 原文线索

V2-074184 | MATHDB-392655

Estimation of the power index in asymmetric power GARCH approximation

摘录 / EXCERPT: For the volatility model $X_t=\sigma_t\varepsilon_t$, estimate the power index by choosing $\hat\gamma\in[u_0,2]$ to minimize ... where $k\ge1$, $u_0>0$, and $\hat\theta(\gamma)$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-074185 | MATHDB-392656

Inverse earthquake modeling at engineering frequencies

摘录 / EXCERPT: Determine the distribution of elastic parameters in a large sedimentary basin, such as the Los Angeles Basin, from surface observations of past earthquakes. The inverse problem is...

open_problem | open | MathDB 页面 | 原文线索

V2-074186 | MATHDB-392657

Sufficiency of the second-order condition for parabolic state-constrained control in dimensions $N>2$

摘录 / EXCERPT: For the distributed-control semilinear parabolic problem on $Q=\Omega\times(0,T)$, with bounded $\Omega\subset\mathbb{R}^N$ having $C^{1,1}$ boundary, homogeneous Neumann boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-074187 | MATHDB-392660

Well-posedness of vortex sheets with surface tension

摘录 / EXCERPT: Does accounting for surface tension make the full initial-value problem for a vortex sheet well-posed?

open_problem | open | MathDB 页面 | 原文线索

V2-074188 | MATHDB-392661

Stability of periodic peakons

摘录 / EXCERPT: Theorem. The periodic peakons are stable.

open_problem | open | MathDB 页面 | 原文线索

V2-074189 | MATHDB-392663

Convergence of the successive approximations for the extreme Nekrasov wave

摘录 / EXCERPT: For $0\le s\le\pi$, define ϕ_0 by $\sin\phi_0(s)=\tfrac12\cos\tfrac{s}{2}$ and define $\phi_{n+1}=T_0\phi_n$, where ... Determine whether the sequence $(\phi_n)_{n\ge0}$ converg...

open_problem | open | MathDB 页面 | 原文线索

V2-074190 | MATHDB-392664

Global uniqueness of positive solutions of Nekrasov’s equation

摘录 / EXCERPT: For the Nekrasov equation ... does there exist at most one non-trivial solution satisfying $0\leq\theta(\sigma)\leq\pi$ for $0\leq\sigma\leq\pi$, for every $\mu\in(3,\infty)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074191 | MATHDB-392665

Inverse spectral uniqueness for the \ba=0 Camassa\Holm spectral problem

摘录 / EXCERPT: Consider the spectral problem ... with w locally integrable and $(1+e^x)w(x)\in L^1(\mathbb R)$. Then the spectrum is discrete. If λ_n are the eigenvalues and...

open_problem | open | MathDB 页面 | 原文线索

V2-074192 | MATHDB-392666

Well-posedness of the generalized Benjamin-Ono equation in the energy space

摘录 / EXCERPT: For the generalized Benjamin-Ono equation ... where H is the Hilbert transform and $k\ge 2$ is an integer, establish a well-posedness result in the energy space...

open_problem | open | MathDB 页面 | 原文线索

V2-074193 | MATHDB-392667

L^p-L^q decay estimates for wave equations with time-dependent coefficients

摘录 / EXCERPT: Can the L^p-L^q decay estimates described above be generalized to solutions of wave equations with time-dependent coefficients such as ...

open_problem | open | MathDB 页面 | 原文线索

V2-074194 | MATHDB-392668

Hardy spaces for general NCC symmetric spaces

摘录 / EXCERPT: How can one define Hardy spaces for general non-compactly causal (NCC) symmetric spaces Y?

open_problem | open | MathDB 页面 | 原文线索

V2-074195 | MATHDB-392669

Hardy spaces and Plancherel measure for general NCC symmetric spaces

摘录 / EXCERPT: How can one define Hardy spaces for general non-compactly causal (NCC) symmetric spaces and determine the Plancherel measure explicitly?

open_problem | open | MathDB 页面 | 原文线索

V2-074196 | MATHDB-392671

Realization of countable discrete groups as automorphism groups of Riemann surfaces

摘录 / EXCERPT: Let G be a countable discrete group. Then there exists a non-compact Riemann surface X, hyperbolic in the sense of Kobayashi, such that G is isomorphic to the automorphism gr...

open_problem | open | MathDB 页面 | 原文线索

V2-074197 | MATHDB-392672

Are continuous homomorphisms between infinite-dimensional Lie groups necessarily smooth?

摘录 / EXCERPT: Is every continuous homomorphism between infinite-dimensional Lie groups smooth?

open_problem | open | MathDB 页面 | 原文线索

V2-074198 | MATHDB-392674

Free boundary value problem for analytic functions (Problem 1)

摘录 / EXCERPT: Let $z_1,\ldots,z_n$ be finitely many points in a bounded simply connected domain $G\subset\mathbb{C}$, and let $\phi:\partial G\to(0,\infty)$ be continuous. Show that there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-074199 | MATHDB-392675

Infinitely many zeros of a meromorphic sum with positive coefficients

摘录 / EXCERPT: Let $z_k\in\mathbb{C}$ and $a_k>0$ satisfy ... and define ... Must f have infinitely many zeros in $\mathbb{C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-074200 | MATHDB-392677

Minimal-winding boundary interpolation for finite Blaschke products

摘录 / EXCERPT: Given interpolation data $t_j\in\mathbb T$ and $w_j\in M_{t_j}$ for $j=0,\ldots,k$, find a solution w of the boundary interpolation problem $w(t_j)=w_j$ with minimal winding numb...

open_problem | open | MathDB 页面 | 原文线索

V2-074201 | MATHDB-392678

Periodic rays of exponential maps land

摘录 / EXCERPT: Every periodic ray of every exponential map lands.

open_problem | open | MathDB 页面 | 原文线索

V2-074202 | MATHDB-392679

Landing of periodic dynamic rays for cosine maps

摘录 / EXCERPT: For a cosine map $f(z)=a\exp(z)+b\exp(-z)$, where $a,b\in\mathbb{C}$, does every periodic dynamic ray of f land?

open_problem | open | MathDB 页面 | 原文线索

V2-074203 | MATHDB-392680

Zeros of the solutions of a functional equation

摘录 / EXCERPT: The report states that an entire function $f(z)=\sum_{n=0}^{\infty}a_nz^n$ satisfying ... has zeros z_n, indexed in order of nondecreasing modulus, with asymptotic expansion ... wher...

open_problem | open | MathDB 页面 | 原文线索

V2-074204 | MATHDB-392681

Rhie’s $5n-5$ image bound for point-mass gravitational lenses

摘录 / EXCERPT: For $n>1$, an n-point gravitational lens in the report’s model gives at most $5n-5$ lensed images when the shear is $\gamma=0$ and the optical depth is $\sigma>0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074205 | MATHDB-392682

Bulk sine-kernel universality for the Gaussian external-source ensemble

摘录 / EXCERPT: For the Gaussian external-source ensemble with $V(x)=\tfrac12x^2$, external-source eigenvalues $a_1=a$ and $a_2=-a$ of equal multiplicity ($n_1=n_2$), and $a>1$, the local eigenval...

open_problem | open | MathDB 页面 | 原文线索

V2-074206 | MATHDB-392683

Strong asymptotics of Bergman polynomials for piecewise analytic Jordan curves

摘录 / EXCERPT: If L is a piecewise analytic Jordan curve without cusps, then ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-074207 | MATHDB-392684

Hinkkanen’s minimum-modulus regularity question

摘录 / EXCERPT: Is it true that every transcendental entire function f of order less than $1/2$ satisfies the following condition? There exist positive numbers R_0,L,δ,C with $R_0>e$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074208 | MATHDB-392686

Quasicircle characterization for post-critically finite rational maps

摘录 / EXCERPT: Let $R:\widehat{\mathbb C}\to\widehat{\mathbb C}$ be a rational function satisfying: 1. R is post-critically finite: if Ω_R is the set of critical points of R and...

open_problem | open | MathDB 页面 | 原文线索

V2-074209 | MATHDB-392687

Quasicircle containment problem for post-critically finite rational maps

摘录 / EXCERPT: Let R be a rational function satisfying (i)-(iii). Does there exist a quasicircle $C\subseteq\mathbb{C}$ such that $P_R\subseteq C$ and $C\subseteq R^{-1}(C)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074210 | MATHDB-392688

Ahlfors regular conformal dimension for subdivision-rule spheres

摘录 / EXCERPT: If X comes from a subdivision rule with associated expanding map f, then ... Moreover, $\dim_{AR}X$ is never attained unless $X\mathrel{\simeq_{qs}}S^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-074211 | MATHDB-392690

Classification of surfaces of general type with geometric genus zero

摘录 / EXCERPT: Give a complete classification of surfaces of general type with geometric genus $p_g=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074212 | MATHDB-392691

Classification of numerical Godeaux surfaces with an involution

摘录 / EXCERPT: Every numerical Godeaux surface S with an involution $\d70: S\to S$ is birationally equivalent to one of the following: (1) a double plane of Campedelli type; (2) a double plane...

open_problem | open | MathDB 页面 | 原文线索

V2-074213 | MATHDB-392692

Irreducibility of the genus-2 Albanese-fibre family

摘录 / EXCERPT: For minimal surfaces of general type with $p_g=q=1$ and $K_S^2=3$, the family whose Albanese fibre has genus 2-described as double covers of the second symmetric product of an el...

open_problem | open | MathDB 页面 | 原文线索

V2-074214 | MATHDB-392693

Existence of surfaces with Enriques bicanonical image

摘录 / EXCERPT: Can there exist a surface S with $p_g(S)=0$ and $K_S^2=3$ whose bicanonical image is birational to an Enriques surface? The report answers affirmatively: such surfaces exist.

open_problem | open | MathDB 页面 | 原文线索

V2-074215 | MATHDB-392694

Classification of the non-standard bicanonical case for surfaces with $p_g=q=1$

摘录 / EXCERPT: Classify all families of surfaces of general type with $p_g=q=1$ presenting the non-standard case: the bicanonical map is non-birational, but the surface contains no genus-2 penc...

open_problem | open | MathDB 页面 | 原文线索

V2-074216 | MATHDB-392695

Castelnuovo’s conjecture for irregular surfaces

摘录 / EXCERPT: For an irregular surface with $q \ge 4$ and Albanese map of degree 1, the number m of moduli satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-074217 | MATHDB-392696

Construction of wavelet sets for prescribed dilation and translation systems

摘录 / EXCERPT: Given a dilation set Δ and a translation set Γ, find measurable sets $\Omega\subset\mathbb{R}^d$ of positive and finite measure such that, with...

open_problem | open | MathDB 页面 | 原文线索

V2-074218 | MATHDB-392697

Linear independence of finite subsets of Gabor frames

摘录 / EXCERPT: Every finite subset of a Gabor frame is linearly independent.

open_problem | open | MathDB 页面 | 原文线索

V2-074219 | MATHDB-392699

Dependence of frame bounds on the interpolation parameter

摘录 / EXCERPT: For the family of intermediate -Gabor-wavelet frames $\{g^{\alpha}_{j,k}\}_{j,k\in\mathbb Z}$ described in the report, with $\alpha\in[0,1)$, determine the behavior of its frame bo...

open_problem | open | MathDB 页面 | 原文线索

V2-074220 | MATHDB-392700

HRT Conjecture for finite time-frequency shifts

摘录 / EXCERPT: If $g\in L^2(\mathbb{R})$ is nonzero and $\{(\alpha_k,\beta_k)\}_{k=1}^N$ is any set of N distinct points in $\mathbb{R}^2$, then $\{e^{2\pi i\beta_k x}g(x-\alpha_k)\}_{k=1}^N$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-074221 | MATHDB-392701

HRT Conjecture 2: a four-function linear-independence problem

摘录 / EXCERPT: If $g\in L^2(\mathbb R)$ is nonzero, then $\{g(x),\,g(x-1),\,g(x-\sqrt{2}),\,e^{2\pi i x}g(x)\}$ is a linearly independent set of functions in $L^2(\mathbb R)$.

conjecture | open | MathDB 页面 | 原文线索

V2-074222 | MATHDB-392702

Generic minimal generator sets for finitely generated shift-invariant subspaces of $L^p(\mathbb{R}^n)$

摘录 / EXCERPT: Does the theorem stated on the preceding page for finitely generated shift-invariant subspaces of $L^2(\mathbb{R}^n)$ also hold for finitely generated shift-invariant subspaces of...

open_problem | open | MathDB 页面 | 原文线索

V2-074223 | MATHDB-392704

Attainment of upper dimensions for wave-packet frames with integer modulations

摘录 / EXCERPT: For the case $M=B\times\mathbb Z$, is every value in $[1,3]$ attained as the upper dimension $\dim^+(B\times\mathbb Z)$ of a wave-packet frame on $L^2(\mathbb R)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074224 | MATHDB-392706

Finite projection decompositions of positive operators

摘录 / EXCERPT: Which positive operators admit finite projection decompositions?

open_problem | open | MathDB 页面 | 原文线索

V2-074225 | MATHDB-392707

Rapidly decreasing tight-frame generators for wavelet systems on symmetric cones

摘录 / EXCERPT: In the setup where $H=\mathrm{GL}(C)$ is the automorphism group of a symmetric cone $C\subset\mathbb{R}^d$, O_j is an open H-orbit in the frequency domain,...

open_problem | open | MathDB 页面 | 原文线索

V2-074226 | MATHDB-392708

Besov spaces on symmetric cones via coorbit theory and discrete wavelet transforms

摘录 / EXCERPT: Study the Besov spaces introduced in [2] using the theory of co-orbit spaces and the discrete wavelet transform, using the results from [22, 23] applied to the group $GL(C)$.

open_problem | open | MathDB 页面 | 原文线索

V2-074227 | MATHDB-392709

How close can an L^1-function be to a convolution idempotent?

摘录 / EXCERPT: For symmetric functions $g\in L^1(\mathbb{R})$ satisfying $\|g\|_1=1$, determine ... Does there exist a function g attaining this infimum? Since the problem is invariant under...

open_problem | open | MathDB 页面 | 原文线索

V2-074228 | MATHDB-392712

Exponential frames for $L^2(A)$

摘录 / EXCERPT: Let $A\subset\mathbb{R}$ be an open, dense set of finite Lebesgue measure, let $H=H_A$ be the closed translation-invariant subspace of $L^2(\mathbb{R})$ consisting of functions who...

open_problem | open | MathDB 页面 | 原文线索

V2-074229 | MATHDB-392713

Exponential frames for a Fourier-supported reproducing-kernel space

摘录 / EXCERPT: Choose an open, dense subset $A\subset\mathbb{R}$ of finite measure, and let ... Does there exist a subset $\Gamma\subset\mathbb{R}$ and a function $\eta\in H_A$ such that the family...

open_problem | open | MathDB 页面 | 原文线索

V2-074230 | MATHDB-392714

Olson-Zalik conjecture on Schauder bases of translates

摘录 / EXCERPT: Does there exist a function $g\in L^2(\mathbb{R})$ and a countable set $\Gamma\subset\mathbb{R}$ such that the system of pure translates $\{T_a g\}_{a\in\Gamma}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-074231 | MATHDB-392715

Nyquist-density conjecture for Gabor Schauder bases

摘录 / EXCERPT: Let $g\in L^2(\mathbb R)$ and let $\Lambda\subset\mathbb R^2$. Define ... If $G(g,\Lambda)$ is a Schauder basis for $L^2(\mathbb R)$, must ... Here ... with $Q_r(z)$ the square centered...

conjecture | open | MathDB 页面 | 原文线索

V2-074232 | MATHDB-392717

Compactness of the divergence of discrete divergence-free edge-element fields

摘录 / EXCERPT: For an L^2-bounded sequence of so-called discrete divergence-free vector fields $u_h\in X_h$, does the divergence of u_h have compactness in H^{-1}?

open_problem | open | MathDB 页面 | 原文线索

V2-074233 | MATHDB-392719

Polynomial-preserving commuting extension operators on a tetrahedron

摘录 / EXCERPT: Does there exist, on a tetrahedral element T, a commuting family of polynomial-preserving extension operators $E_{ abla}$, $E_{ abla\times}$, and $E_{\nabla\cdot}$ from the bound...

open_problem | open | MathDB 页面 | 原文线索

V2-074234 | MATHDB-392720

Uniqueness from one incident plane wave in inverse electromagnetic obstacle scattering

摘录 / EXCERPT: For scattering of a time-harmonic electromagnetic plane wave from an impenetrable bounded scatterer $D\subset\mathbb{R}^3$ with either a perfect-conductor or an impedance boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-074235 | MATHDB-392721

Tikhonov-Morozov justification of the electromagnetic linear sampling method

摘录 / EXCERPT: For electromagnetic inverse obstacle scattering, provide a mathematical justification that solving the far-field equation $F g(\cdot,z)=E^{\infty,\mathrm{dip}}(\cdot,z)p$ by Tikhon...

open_problem | open | MathDB 页面 | 原文线索

V2-074236 | MATHDB-392723

Classification of low triples

摘录 / EXCERPT: Let Φ be a root system with a basis of simple roots Δ, let Λ^+ be the corresponding monoid of dominant weights, and let $\le$ denote the dominant order. For...

open_problem | open | MathDB 页面 | 原文线索

V2-074237 | MATHDB-392728

Normality of the closure of the nilpotent orbit with Jordan block sizes $(4,4,2,2)$

摘录 / EXCERPT: The closure of the nilpotent orbit with Jordan block sizes $(4,4,2,2)$ in a special orthogonal group is normal.

open_problem | open | MathDB 页面 | 原文线索

V2-074238 | MATHDB-392731

Euclidean two-way balanced partition bound

摘录 / EXCERPT: For the $r=2$ and Euclidean norm case, it is conjectured that the best bound is of order $\sqrt d$.

open_problem | open | MathDB 页面 | 原文线索

V2-074239 | MATHDB-392732

Conjectured power-law one-sided discrepancy for fixed-radius circles

摘录 / EXCERPT: Under the preceding hypotheses-an even integer n, a globally balanced 2-coloring of the $n\times n$ square lattice viewed as a torus, and $2<R<n/2$-the report proves that some ci...

conjecture | open | MathDB 页面 | 原文线索

V2-074240 | MATHDB-392735

Parity of the binary sum of digits of squares

摘录 / EXCERPT: Let $s(n)$ denote the sum of the binary digits of n. Is the asymptotic frequency of even values of $s(n^2)$ equal to $1/2$? Equivalently, is ...

open_problem | open | MathDB 页面 | 原文线索

V2-074241 | MATHDB-392737

Weak \epsilon-nets for convex sets in the plane

摘录 / EXCERPT: Improve the quadratic upper bound on $f(2,\varepsilon)$, say, or provide a superlinear lower bound.

open_problem | open | MathDB 页面 | 原文线索

V2-074242 | MATHDB-392738

Improving bounds for weak epsilon-nets on the sphere

摘录 / EXCERPT: Improve bounds on the minimum size of a weak ε-net for convex sets in $\mathbb{R}^d$ with respect to μ, the uniform measure on S^{d-1}, in high-dimensional convex...

open_problem | open | MathDB 页面 | 原文线索

V2-074243 | MATHDB-392741

Low-discrepancy lattice point sets

摘录 / EXCERPT: Prove an upper bound, such as (1), for lattice point sets M_n, namely an estimate of the form ... where (1) is the bound stated earlier in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-074244 | MATHDB-392742

Optimal-order L_2 discrepancy for a digital $(0,1)$-sequence generated by an NUT matrix

摘录 / EXCERPT: Does there exist a digital $(0,1)$-sequence over $ \mathbb{Z}_2$ generated by an NUT matrix C such that, for some constant $c_1>0$, its L_2-discrepancy satisfies ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-074245 | MATHDB-392743

Convex-sequence additive-energy conjecture

摘录 / EXCERPT: Let $B=\{1,2,\ldots,N\}$ be a base set and let $S=\{s_j\}_{j=1}^N$ be a strictly convex sequence, meaning that the differences $s_{j+1}-s_j$ are strictly monotone in j. Let C_d...

conjecture | open | MathDB 页面 | 原文线索

V2-074246 | MATHDB-392744

Erdős's question on additive-intersective sets

摘录 / EXCERPT: Let $S=\\{s_1,s_2,\\ldots\\}\\subseteq\\mathbb N$ and define $S(n)=|S\\cap[n]|$, where $[n]=\\{1,2,\\ldots,n\\}$. For $d,r,n\\in\\mathbb N$, put ... Suppose that S satisfies ... for...

open_problem | open | MathDB 页面 | 原文线索

V2-074247 | MATHDB-392746

Optimal number of at-large representatives in the described voting-game model

摘录 / EXCERPT: In the described composite voting game, the sum of the Banzhaf powers of all voters-the sensitivity of the composite game-is maximized when the number of at-large representatives i...

open_problem | open | MathDB 页面 | 原文线索

V2-074248 | MATHDB-392747

Saari's conjecture on explaining differences among voting rules

摘录 / EXCERPT: The conjecture, which turned out to be true, is that all possible differences among voting rules can be explained (and examples constructed) simply by knowing these configurations...

conjecture | open | MathDB 页面 | 原文线索

V2-074249 | MATHDB-392750

Kimura-O'Sullivan finite-dimensionality conjecture for rational motives

摘录 / EXCERPT: Kimura conjectured that the category M_r of $\\mathbb{Q}$-linear motives modulo rational equivalence is Kimura-finite; that is, every object admits a decomposition into an even o...

conjecture | open | MathDB 页面 | 原文线索

V2-074250 | MATHDB-392753

The missing degree-two Chow-motive realization statement for products of surfaces

摘录 / EXCERPT: For smooth projective surfaces S and S' over a field k, prove the degree-two analogue ...

open_problem | open | MathDB 页面 | 原文线索

V2-074251 | MATHDB-392756

A posteriori upper error bound for the nonconforming Crouzeix-Raviart approximation of Stokes flow

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$, $d\in\{2,3\}$, be an open polygonal or polyhedral domain. For the stationary incompressible Stokes problem with homogeneous Dirichlet boundary cond...

open_problem | open | MathDB 页面 | 原文线索

V2-074252 | MATHDB-392759

Multidimensional transonic shocks in an infinite nozzle with arbitrary cross-sections

摘录 / EXCERPT: For an infinite nozzle $\Omega=\Phi(C)$ that is sufficiently close in $C^{3,\alpha}$ to the infinite cylinder $C=\Lambda\times(-a,\infty)$, where $a>0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-074253 | MATHDB-392760

Existence of weak solutions for the generalized Camassa-Holm equation

摘录 / EXCERPT: Let $g\in \operatorname{Lip}_{\mathrm{loc}}(\mathbb R)$ satisfy $g(0)=0$, and let $u_0\in H^1(\mathbb R)$. Then the Cauchy problem ... with $u(0,x)=u_0(x)$ has a global weak solution...

open_problem | open | MathDB 页面 | 原文线索

V2-074254 | MATHDB-392761

Global weak solutions and estimates for the generalized Camassa-Holm equation

摘录 / EXCERPT: Let ... with initial condition $u(0,x)=u_0(x)$, where ... Define $h(u)=g(u)-u^2/2$. A weak solution is a function $u:\mathbb{R}_+\times\mathbb{R}\to\mathbb{R}$ such that ... which satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-074255 | MATHDB-392762

Asymptotic Stability of Riemann Solutions for a Class of Multi-Dimensional Viscous Systems of Conservation Laws

摘录 / EXCERPT: For systems whose flux functions in different spatial directions share a common complete system of Riemann invariants whose level surfaces are hyperplanes (Temple fluxes), two-stat...

open_problem | open | MathDB 页面 | 原文线索

V2-074256 | MATHDB-392763

Uniqueness for weak solutions with spatially dependent diffusion

摘录 / EXCERPT: For the two-phase-flow problem (1)-(4), can the Kružkov techniques used for problems with spatially dependent diffusion coefficients be adapted to the case in which the diffusion f...

open_problem | open | MathDB 页面 | 原文线索

V2-074257 | MATHDB-392765

Complete steady-state solution for the back-order queueing-inventory model

摘录 / EXCERPT: For the single-server systems with an attached inventory under continuous review and back-ordering, obtain a complete steady-state solution of the system.

open_problem | open | MathDB 页面 | 原文线索

V2-074258 | MATHDB-392766

Repeated-auction extension of iterative multi-unit auction results

摘录 / EXCERPT: Extend the results for iterative multi-unit auctions to repeated auction settings.

open_problem | open | MathDB 页面 | 原文线索

V2-074259 | MATHDB-392767

Optimal allocation policy in a fully flexible production system

摘录 / EXCERPT: Theorem 3. Consider a fully flexible production system with n plants, each with capacity C units, and m products with current demands d_i, $i=1,\ldots,m$. Let I_i be the...

open_problem | open | MathDB 页面 | 原文线索

V2-074260 | MATHDB-392769

Polynomial-time solvability of the multilevel general-concave-cost lot-sizing problem

摘录 / EXCERPT: It would be interesting to determine if the multi-level case with general concave cost functions can be solved in polynomial time in both the time horizon and the number of levels.

open_problem | open | MathDB 页面 | 原文线索

V2-074261 | MATHDB-392770

Tighter formulations for Resource Capacity constraints

摘录 / EXCERPT: Finding tighter formulations for Resource Capacity RC constraints?

open_problem | open | MathDB 页面 | 原文线索

V2-074262 | MATHDB-392771

Facet status of cuts produced by the decomposition

摘录 / EXCERPT: We conjecture that in fact each cut from the decomposition must already be a facet.

open_problem | open | MathDB 页面 | 原文线索

V2-074263 | MATHDB-392772

NP-hardness of Roundy's assembly-problem variant

摘录 / EXCERPT: Is the variant of the assembly problem considered by Roundy-specifically, the version in which all cost parameters are fixed over time-NP-hard?

open_problem | open | MathDB 页面 | 原文线索

V2-074264 | MATHDB-392773

Integrality of type-\a string polytopes

摘录 / EXCERPT: For a group G of type A, any reduced decomposition $\underline{w}_0$ of the longest element w_0 of the Weyl group, and any dominant weight λ, the string polytope...

open_problem | open | MathDB 页面 | 原文线索

V2-074265 | MATHDB-392774

Generalized h-vector of a non-rational polytope

摘录 / EXCERPT: In the non-rational case, does the new generalized h-vector still have the same three properties: positivity, symmetry, and unimodality?

open_problem | open | MathDB 页面 | 原文线索

V2-074266 | MATHDB-392780

Green's finite-order conjecture for holomorphic curves omitting two lines and a conic

摘录 / EXCERPT: Every holomorphic curve $f:\mathbb{C}\to\mathbb{P}^2(\mathbb{C})$ that omits the two lines $\{x_1=0\}$ and $\{x_2=0\}$ and the conic $\{x_0^2+x_1^2+x_2^2=0\}$ has image contained i...

conjecture | open | MathDB 页面 | 原文线索

V2-074267 | MATHDB-392783

Heier's higher-dimensional formula for the order of contact of a holomorphic ideal

摘录 / EXCERPT: Conjecture. Let $I=(f_1,\ldots,f_N)\subset\mathcal O_{\mathbb C^n,0}$, where $V(I)=\{0\}$. Let $\mathcal P$ be the set of all subsets $S\subset\{1,\ldots,N\}$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-074268 | MATHDB-392787

Rogawski's arithmeticity conjecture for certain cocompact lattices in \(PU(2,1)\)

摘录 / EXCERPT: Let Γ be a torsion-free cocompact lattice in $PU(2,1)$, and let ... If $b_1(M)=0$ and ... then Γ is arithmetic.

conjecture | open | MathDB 页面 | 原文线索

V2-074269 | MATHDB-392788

Virtual positivity of the first Betti number for compact complex two-ball quotients

摘录 / EXCERPT: For every compact complex two-ball quotient $M=\Gamma\backslash \mathrm{PU}(2,1)/\mathrm{P}(\mathrm{U}(2)\times \mathrm{U}(1))$, with Γ a cocompact torsion-free lattice, the...

open_problem | open | MathDB 页面 | 原文线索

V2-074270 | MATHDB-392791

Demailly-Kollár Conjecture A on strong semicontinuity of complex singularity exponents

摘录 / EXCERPT: Let f be analytic on a ball $B\subset\mathbb{C}^n$. Then there exists $\varepsilon>0$ such that, whenever g is analytic on B and $\lVert g-f\rVert_{C^0}<\varepsilon$, one has...

conjecture | open | MathDB 页面 | 原文线索

V2-074271 | MATHDB-392801

Characterization of invariant measures by the Floquet exponent

摘录 / EXCERPT: To what extent does the Floquet exponent w characterize the measure $\$?

open_problem | open | MathDB 页面 | 原文线索

V2-074272 | MATHDB-392803

Conformal-process interpretations of Brownian-loop area variables

摘录 / EXCERPT: Are there, if any, interpretations of the random variable $S_0=\sum_{n\,\mathrm{odd}} S_n$, and more generally of the random variables S and A, in terms of conformally invarian...

open_problem | open | MathDB 页面 | 原文线索

V2-074273 | MATHDB-392804

Fractional-moment localization without the covering condition

摘录 / EXCERPT: Can the localization proofs based on the fractional moment method (FMM) be extended to single-site potentials with smaller support, thereby removing the covering assumption that th...

open_problem | open | MathDB 页面 | 原文线索

V2-074274 | MATHDB-392805

Finiteness dimension of one-valuation arithmetic subgroups

摘录 / EXCERPT: Let G be an almost simple Chevalley group over a function field F, and let $\[1m\Gamma\u001b[0m$ be an arithmetic subgroup defined using one valuation of F. Writing...

open_problem | open | MathDB 页面 | 原文线索

V2-074275 | MATHDB-392806

Dependence of asymptotic cones on the ultrafilter

摘录 / EXCERPT: Gromov asked whether there are finitely presented groups whose asymptotic cone depends on the ultrafilter used to define it.

open_problem | open | MathDB 页面 | 原文线索

V2-074276 | MATHDB-392807

Bilipschitz embeddings of hyperbolic spaces into products of locally compact trees

摘录 / EXCERPT: For $n\ge 3$, is there a bilipschitz embedding of $\mathbb{H}^n$ into an n-fold product of locally compact trees?

open_problem | open | MathDB 页面 | 原文线索

V2-074277 | MATHDB-392808

Closures of projections of cocompact lattices in products of trees

摘录 / EXCERPT: Let T_1 and T_2 be locally finite regular trees. For a cocompact lattice $\Gamma<\operatorname{Aut}(T_1)\times\operatorname{Aut}(T_2)$, let H_i be the closure of the projecti...

open_problem | open | MathDB 页面 | 原文线索

V2-074278 | MATHDB-392810

Isomorphism problem for Kac-Moody groups and twin root data

摘录 / EXCERPT: Determine all pairs of twin root data $D_i=(G_i;(U_{i,\alpha})_{\alpha\in\Phi(W_i,S_i)};H_i)$ $(i\in\{1,2\})$, where D_i comes from a Kac-Moody group G_i over a field K_i, su...

open_problem | open | MathDB 页面 | 原文线索

V2-074279 | MATHDB-392811

Isomorphism problem for Kac-Moody groups

摘录 / EXCERPT: Let $D_i=(G_i,(U_{i,\alpha})_{\alpha\in\Phi(W_i,S_i)},H_i)$ be twin root data arising from Kac-Moody groups G_i over fields K_i, where each K_i is either a finite field of ca...

open_problem | open | MathDB 页面 | 原文线索

V2-074280 | MATHDB-392813

Multiplicative McKay structures across crepant-resolution flops

摘录 / EXCERPT: For a finite subgroup $G\subset SL(3,\mathbb C)$ acting with an isolated singularity on $\mathbb C^3/G$, and for crepant resolutions whose tautological bundles correspond to irredu...

open_problem | open | MathDB 页面 | 原文线索

V2-074281 | MATHDB-392814

Triply extended Kac-Moody symmetry of maximally oxidised theories

摘录 / EXCERPT: It has been conjectured that the maximally oxidised actions associated with all simple groups G possess the larger triply extended Kac-Moody symmetry G^{+++}, where G^{+++} i...

open_problem | open | MathDB 页面 | 原文线索

V2-074282 | MATHDB-392819

Geometrizability of chamber systems arising from flips of twin buildings

摘录 / EXCERPT: Let $B=(B^+,B^-)$ be a twin building. A flip is an automorphism δ of B such that: (i) $\delta^2=\mathrm{Id}$; (ii) $B^\delta_+=B^-$; (iii)...

open_problem | open | MathDB 页面 | 原文线索

V2-074283 | MATHDB-392822

Most general integrands for local time-space integration

摘录 / EXCERPT: Specify, for continuous semimartingales, the most general class of functions G for which the double integral ... makes sense.

open_problem | open | MathDB 页面 | 原文线索

V2-074284 | MATHDB-392823

Extension of the local time-space Itô formula beyond two technical conditions

摘录 / EXCERPT: Let $X=(X_t)_{t\ge 0}$ be a semimartingale and let $F:\mathbb{R}\times[0,1)\to\mathbb{R}$ be a deterministic function admitting the stated Radon-Nikodym derivatives F_x and F_t...

open_problem | open | MathDB 页面 | 原文线索

V2-074285 | MATHDB-392824

Smooth pasting for the perpetual American put under Lévy processes

摘录 / EXCERPT: For a Lévy process $X=(X_t)_{t\ge 0}$, consider ... The report states that this problem is solved by a lower-threshold strategy $\tau^*=\inf\{t\ge 0:X_t<x^*\}$ for an optimal thresho...

open_problem | open | MathDB 页面 | 原文线索

V2-074286 | MATHDB-392826

Lebesgue-constant growth for polynomial interpolation on the sphere

摘录 / EXCERPT: Can one choose, for every degree n, a fundamental system of $d_n=(n+1)^2$ points on S^2 for $ \mathcal P_n(S^2)$ such that the Lebesgue constants satisfy $ \|\Lambda_n\|=O(n)$...

open_problem | open | MathDB 页面 | 原文线索

V2-074287 | MATHDB-392828

Worst-case cubature error rate on S^2

摘录 / EXCERPT: For $s>1$, there exists a constant $\widetilde c_s>0$ such that every sequence $(Q_n)_{n\in\mathbb N}$ of positive-weight cubature rules on S^2 satisfying (i) $Q_n p=I p$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-074288 | MATHDB-392829

Spectral radius of the double-layer operator on general polyhedrons

摘录 / EXCERPT: In three dimensions, for a general polyhedron, is the spectral radius of the double-layer operator L with respect to the $L^2(\Gamma)$ norm strictly less than 1?

open_problem | open | MathDB 页面 | 原文线索

V2-074289 | MATHDB-392830

Stability and convergence of fully discrete spectral methods for first-kind boundary integral equations

摘录 / EXCERPT: The report identifies as an open problem the task of proposing and proving stability and convergence of a fully discrete spectral method for first-kind boundary integral equations...

open_problem | open | MathDB 页面 | 原文线索

V2-074290 | MATHDB-392831

Polynomial interpolation on the sphere: characterization of fundamental systems

摘录 / EXCERPT: Let $S^2:=\{x\in\mathbb{R}^3:\lvert x\rvert=1\}$, let V_n be the space of spherical polynomials of degree at most n, and set $N:=(n+1)^2=\dim V_n$. Given N pairwise distinct...

open_problem | open | MathDB 页面 | 原文线索

V2-074291 | MATHDB-392832

Fundamental system for odd-degree spherical polynomial interpolation

摘录 / EXCERPT: Let $n\in\mathbb N$ be odd, and let ... with $\rho_{n+2-j}:=\pi-\rho_j$ for $j=1,\ldots,(n+1)/2$. Define ... where Φ is the spherical-coordinate parametrization of S^2 and ... wit...

open_problem | open | MathDB 页面 | 原文线索

V2-074292 | MATHDB-392833

Fundamental interpolation system on the sphere for odd degree

摘录 / EXCERPT: Let $n\in\mathbb N$ be odd, and choose $0<\rho_1<\rho_2<\cdots<\rho_{(n+1)/2}<\pi/2$, with $\rho_{n+2-j}:=\pi-\rho_j$ for $j=1,\ldots,(n+1)/2$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-074293 | MATHDB-392835

Best Chebyshev harmonic approximation on an unbounded strip

摘录 / EXCERPT: Conjecture: U_0 is a best Chebyshev harmonic approximant to h in the unbounded domain $S(b)$.

open_problem | open | MathDB 页面 | 原文线索

V2-074294 | MATHDB-392836

Nonexistence of first-order split operators for the Laplacian

摘录 / EXCERPT: For the Laplacian in three variables, there does not exist a nontrivial split operator of first-order form $\Delta_D=\sum_{i=1}^{3}B_i\partial_i+B$, where the coefficients B_i an...

open_problem | open | MathDB 页面 | 原文线索

V2-074295 | MATHDB-392843

Counting knapsack solutions

摘录 / EXCERPT: For the zero-one knapsack solution set $S=\{x\in\{0,1\}^{n}:\sum_{j=1}^{n}a_jx_j\le b\}$, where $0\le a_1\le a_2\le\cdots\le a_n\le b$ are integers, the report gives an $O(n^{3})$-...

open_problem | open | MathDB 页面 | 原文线索

V2-074296 | MATHDB-392846

Hardness of approximating maximum connected subgraphs with hereditary properties

摘录 / EXCERPT: For every graph property π that is nontrivial and hereditary on connected graphs, and for every $\epsilon>0$, the maximum connected subgraph with property π problem cannot...

open_problem | open | MathDB 页面 | 原文线索

V2-074297 | MATHDB-392847

Hardness of approximating maximum connected hereditary subgraphs

摘录 / EXCERPT: Let π be a nontrivial hereditary property on connected graphs that is satisfied by all paths and does not hold for some star. Then, for every $\epsilon>0$, the maximum connecte...

open_problem | open | MathDB 页面 | 原文线索

V2-074298 | MATHDB-392848

Hardness of approximating maximum subgraphs for hereditary properties on directed graphs

摘录 / EXCERPT: For every nontrivial hereditary property π on directed graphs and every $\varepsilon>0$, the maximum subgraph with property π problem cannot be approximated within a factor...

open_problem | open | MathDB 页面 | 原文线索

V2-074299 | MATHDB-392851

PTAS for the \epsilon-Dense Steiner Tree Problem

摘录 / EXCERPT: An epsilon-Dense Steiner Tree instance consists of a graph $G=(V,E)$ with unit edge weights and a terminal set $S\subseteq V$ such that, for every terminal $s\in S$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074300 | MATHDB-392852

Rapid mixing of Glauber dynamics for proper graph colorings

摘录 / EXCERPT: Let $G=(V,E)$ be a graph on n vertices with maximum degree Δ, and let Ω be the set of proper k-colorings of G. The Glauber dynamics updates a uniformly chosen v...

open_problem | open | MathDB 页面 | 原文线索

V2-074301 | MATHDB-392853

Rapid mixing of Glauber dynamics for graph colorings at the \k\ = \\ + 2\ threshold

摘录 / EXCERPT: For a graph G on n vertices with maximum degree $\$, and an integer $k \ \+2$, is it true that the Glauber dynamics on the proper k-colorings of G satisfies the rapid-conve...

open_problem | open | MathDB 页面 | 原文线索

V2-074302 | MATHDB-392857

Random Sampling and Approximation of MAX-CSP Problems - Theorem 16

摘录 / EXCERPT: Let r,n be positive integers, with r fixed, and let $\\epsilon$ be a positive real. A MAX-rCSP instance consists of a set F of m distinct Boolean functions...

open_problem | open | MathDB 页面 | 原文线索

V2-074303 | MATHDB-392858

Approximation hardness of balanced MAX-(3,B2)-SAT

摘录 / EXCERPT: Does the balanced class $\mathrm{MAX\text{-}(3,B2)\text{-}SAT}$ remain hard to approximate? Here a balanced MAX-$(3,Bk)$-SAT instance is a set of clauses of size exactly 3 in which...

open_problem | open | MathDB 页面 | 原文线索

V2-074304 | MATHDB-392859

Existence of a balanced unsatisfiable $(3,B2)$-SAT formula

摘录 / EXCERPT: Does there exist an unsatisfiable balanced $(3,B2)$-SAT formula, meaning a Boolean CNF formula in which every clause has exactly three literals and every literal occurs exactly twi...

open_problem | open | MathDB 页面 | 原文线索

V2-074305 | MATHDB-392861

Light multi-multiway cuts

摘录 / EXCERPT: Theorem 18 (Light multi-multiway cuts). Let I be an instance of the minimum multi-multiway cut problem. Let Opt_I be the weight of an optimal multi-multiway cut...

open_problem | open | MathDB 页面 | 原文线索

V2-074306 | MATHDB-392862

A PTAS for the minimization of polynomials of fixed degree over the simplex

摘录 / EXCERPT: Let $d\ge 2$ be fixed. For a polynomial $p(x)$ of degree d, consider the problem ... For each integer $k\ge 1$, define ... Then the grid bounds $p_{\Delta(k)}$ yield a polynomial-tim...

open_problem | open | MathDB 页面 | 原文线索

V2-074307 | MATHDB-392867

Constant-factor performance of the same-set shortest-path heuristic for Steiner Forest

摘录 / EXCERPT: For the Steiner Forest Problem, consider the shortest-path heuristic that, at each step, connects two terminals belonging to the same terminal set and at minimum distance by a shor...

open_problem | open | MathDB 页面 | 原文线索

V2-074308 | MATHDB-392869

PTAS for planar minimum bisection

摘录 / EXCERPT: Does the PLANAR MIN-BISECTION problem admit a PTAS?

open_problem | open | MathDB 页面 | 原文线索

V2-074309 | MATHDB-392870

Rigidity estimate for two strongly incompatible wells

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded Lipschitz domain, with $n\ge 2$, and let ... where $\lambda_i>0$ and ... Then there exists a positive constant $C(\Omega,H)$ such that, for...

open_problem | open | MathDB 页面 | 原文线索

V2-074310 | MATHDB-392871

Optimality of the supercritical upper bound for a discrete membrane with randomly distributed defects

摘录 / EXCERPT: In the supercritical regime, the upper bound in (0.6) is conjectured to be optimal: the $ ... with gp suitably defined by a least-path problem on the weak cluster.

open_problem | open | MathDB 页面 | 原文线索

V2-074311 | MATHDB-392872

Asymptotic shape of minimizers in the high-energy limit

摘录 / EXCERPT: Assume that the interaction potential satisfies (H1) and (H2). Let $W_0:=\{q\in W^{1,2}_{\mathrm{loc}}(\mathbb R)\mid q(0)=0,\ q'\in L^2(\mathbb R)\}$, let $r(x):=q(x+1)-q(x)$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-074312 | MATHDB-392874

Necessary and sufficient condition for the Nirenberg problem on S^2

摘录 / EXCERPT: Determine a condition on a function $f:S^2\to\mathbb{R}$ that is necessary and sufficient for the existence of a metric g on S^2, conformal to the standard spherical metric, wh...

open_problem | open | MathDB 页面 | 原文线索

V2-074313 | MATHDB-392875

Gamma-convergence result for the two-well problem with surface energy

摘录 / EXCERPT: The report states that, when $n=2$, $K=SO(2)A\cup SO(2)B$, and Ω is star-shaped, the functionals ... Γ-converge to ... where $J_{\nabla u}$ is the jump set of ∇u.

open_problem | open | MathDB 页面 | 原文线索

V2-074314 | MATHDB-392876

Asymptotic ground-state energy for two-dimensional crystallization

摘录 / EXCERPT: Let $d=2$. There exists a constant $a>0$ such that, for every $V\in C^2(0,\infty)$ satisfying $V(1)=-1$, $\lim_{r\to\infty}V(r)=0$, $V(r)\geq 1/a$ for all $r\in[0,1-a]$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074315 | MATHDB-392877

Ros’s conjecture on relative isoperimetric minimizers

摘录 / EXCERPT: For convex $\Omega\subset\mathbb{R}^3$, the boundary of the minimizer within the class of Caccioppoli sets should actually be a disk.

conjecture | open | MathDB 页面 | 原文线索

V2-074316 | MATHDB-392878

Existence of diffuse laminates in the differential inclusion associated with a polyconvex energy

摘录 / EXCERPT: For the energy $F(X)=\varepsilon |X|^2/2+\det(X)^2$ and the differential-inclusion set...

open_problem | open | MathDB 页面 | 原文线索

V2-074317 | MATHDB-392879

Minimal tunnelling cost for the nonlocal evolution equation

摘录 / EXCERPT: For the one-dimensional nonlocal evolution equation $u_t=f_L(u)=-\delta F_L(u)/\delta u$ on $[-L,L]$, with $L>1$, $\beta>1$, Neumann boundary conditions, and pure phases...

open_problem | open | MathDB 页面 | 原文线索

V2-074318 | MATHDB-392880

Non-standard phase diagrams for semi-insulating GaAs with a liquid droplet

摘录 / EXCERPT: Calculate the phase diagram for a semi-insulating GaAs solid containing a single spherical liquid droplet of radius r_l, for isothermal processes at constant outer pressure, by d...

open_problem | open | MathDB 页面 | 原文线索

V2-074319 | MATHDB-392881

Refuted Wulff-shape conjecture for droplet growth under local Kawasaki dynamics

摘录 / EXCERPT: In the two-dimensional local Kawasaki dynamics for the anisotropic Ising lattice gas described in the report, with particles conserved in the interior, exchanged with a boundary re...

conjecture | open | MathDB 页面 | 原文线索

V2-074320 | MATHDB-392882

Scope and mechanism of the van den Berg-Kahn conditional inequality

摘录 / EXCERPT: Why does the inequality $P(A\mid B\cap C)\ge P(A\mid C)$ hold when the primary conditioning event C is a disconnection event, but not when it is a connection event?

open_problem | open | MathDB 页面 | 原文线索

V2-074321 | MATHDB-392883

Classification of pairwise-intersecting families of planes in $\mathbb{P}^5$

摘录 / EXCERPT: What are the continuous families of planes in $\mathbb{P}^5$ such that any two planes of the family intersect?

open_problem | open | MathDB 页面 | 原文线索

V2-074322 | MATHDB-392885

Brauer obstruction for rational points over function fields of surfaces

摘录 / EXCERPT: What hypotheses on the geometric generic fiber $X\otimes_K \overline K$ guarantee that the only obstruction to existence of a K-point is the Brauer obstruction?

open_problem | open | MathDB 页面 | 原文线索

V2-074323 | MATHDB-392886

Determine the effective cone of the moduli space of stable curves

摘录 / EXCERPT: Determine the cone $\operatorname{Eff}(\mathcal M_g)$ of effective divisors on the Deligne-Mumford moduli space $\mathcal M_g$ of stable curves of genus g.

open_problem | open | MathDB 页面 | 原文线索

V2-074324 | MATHDB-392887

Harris-Morrison Slope Conjecture

摘录 / EXCERPT: Let $s_g:=\inf_{D\in\operatorname{Eff}(M_g)}s(D)$, where $s(D)$ is the slope of an effective divisor on the moduli space of stable curves of genus g. The Harris-Morrison Slope Co...

conjecture | open | MathDB 页面 | 原文线索

V2-074325 | MATHDB-392888

Uniqueness of Brill-Noether divisors at the conjectural minimal slope

摘录 / EXCERPT: Are combinations of Brill-Noether divisors the only effective divisors on $\mathcal M_g$ of slope $6+12/(g+1)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074326 | MATHDB-392891

Conjecture on bounded complements

摘录 / EXCERPT: Let $d>0$ be a positive integer and let $\b5\ge 0$ be a real number. Then there exists a finite set $N_d(\u0003b5)$ of positive integers such that, for any contraction of normal al...

conjecture | open | MathDB 页面 | 原文线索

V2-074327 | MATHDB-392894

Harbourne-Hirschowitz conjecture for special linear systems in \(\mathbb P^2\)

摘录 / EXCERPT: Let $m_1,\ldots,m_r$ be positive integers and let $p_1,\ldots,p_r\in\mathbb P^2$ be general points. For the linear system $L=L_d(m_1,\ldots,m_r)$ of degree-d plane curves having...

conjecture | open | MathDB 页面 | 原文线索

V2-074328 | MATHDB-392895

Harbourne-Hirschowitz conjecture for multiplicities at most 7

摘录 / EXCERPT: Let $L=L_d(m_1,\dots,m_k)$ be the linear system of degree-d plane curves through general points $p_1,\dots,p_k\in\mathbb{P}^2$, with multiplicity at least m_i at p_i. Its exp...

conjecture | open | MathDB 页面 | 原文线索

V2-074329 | MATHDB-392896

Connectedness of the moduli space for $p_g=4$, $K^2=6$

摘录 / EXCERPT: Is the moduli space of surfaces with $p_g=4$ and $K^2=6$ disconnected?

open_problem | open | MathDB 页面 | 原文线索

V2-074330 | MATHDB-392899

Theorem 5: Regularity bound for powers of an ideal

摘录 / EXCERPT: Let $I \subset R=k[X_0,\ldots,X_n]$ be an ideal generated in degrees $d_1\geq\cdots\geq d_s$. Set $X:=\operatorname{Proj}(R/I)\subset\mathbb{P}^n_k$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074331 | MATHDB-392900

Betti numbers of a classical ideal and an initial ideal

摘录 / EXCERPT: Given a classical ideal I, does there exist an initial ideal (in the given coordinates) H such that $\beta_{ij}(I)=\beta_{ij}(H)$ for all i,j?

open_problem | open | MathDB 页面 | 原文线索

V2-074332 | MATHDB-392903

Conjecture 2: regularity of the torsion in symmetric powers

摘录 / EXCERPT: Conjecture 2. Let $I\subset S=K[x_0,\ldots,x_r]$ have linear type on the punctured spectrum. If I is generated by forms of degree d and has a linear free resolution, then, with…

conjecture | open | MathDB 页面 | 原文线索

V2-074333 | MATHDB-392905

A third-term asymptotic for Hilbert-Kunz functions under codimension-two regularity

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be an excellent local normal ring of characteristic p with perfect residue field and dimension d, let $I\subset R$ be $\mathfrak m$-primary, let M be…

open_problem | open | MathDB 页面 | 原文线索

V2-074334 | MATHDB-392906

Multigraded Poincaré series of monomial rings via standard matchings

摘录 / EXCERPT: Let k be a field of arbitrary characteristic, let $S:=k[x_1,\ots,x_n]$ be the polynomial ring in commuting indeterminates with $\{\deg}(x_i)=e_i\in\mathbb N^n$, and let…

open_problem | open | MathDB 页面 | 原文线索

V2-074335 | MATHDB-392908

Lefschetz conjecture for Gorenstein* complexes

摘录 / EXCERPT: Let K be a $(d-1)$-dimensional Gorenstein simplicial complex, let $R(K)$ be its face ring, and let $\vartheta_1,\ldots,\vartheta_d$ be a generic linear system of parameters. Set…

conjecture | open | MathDB 页面 | 原文线索

V2-074336 | MATHDB-392909

A weaker graded-ring structure implying Macaulay and Kruskal-Katona inequalities

摘录 / EXCERPT: Define an abstract notion of a graded “ring” (with ambiguous product) that is strong enough to imply Macaulay’s inequalities (and Kruskal-Katona inequalities in the exterior or squ…

open_problem | open | MathDB 页面 | 原文线索

V2-074337 | MATHDB-392910

Extension of Khovanskii’s upper bound theorem

摘录 / EXCERPT: Problem 10. Extend Khovanskii’s theorem:

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V2-074338 | MATHDB-392911

Upper multiplicity bound in terms of maximal graded shifts

摘录 / EXCERPT: Let $S=K[x_1,\ldots,x_n]$ be a standard-graded polynomial ring over a field K, let $I\subset S$ be a graded ideal, and let $R=S/I$. If $c=\operatorname{codim}(R)$ and … then …

open_problem | open | MathDB 页面 | 原文线索

V2-074339 | MATHDB-392912

Minimal free resolution of the ideal J_ϕ

摘录 / EXCERPT: Let $R=K[x_1,x_2,x_3]$, where K is a field with $\operatorname{char}(K)=0$ or $\operatorname{char}(K)>\sum_{i=1}^3\phi(i)-2$. Let $\phi(r)=l+k(3-r)>0$ for $k,l\in\mathbb N$, and…

open_problem | open | MathDB 页面 | 原文线索

V2-074340 | MATHDB-392914

Scarf-type resolutions for the ideals J_φ

摘录 / EXCERPT: Let $R=K[x_1,x_2,x_3]$, let $\varphi(r)=l+k(3-r)>0$ for $k,l\in\mathbb N$, and assume $\operatorname{char}(K)=0$ or $\operatorname{char}(K)>\sum_{i=1}^3\varphi(i)-2$. For … determi…

open_problem | open | MathDB 页面 | 原文线索

V2-074341 | MATHDB-392916

Uniqueness theorem for the stabilized coupled variational problem

摘录 / EXCERPT: Theorem 5: Solutions $(u,\lambda)\in H^1(\Omega^-)\times H^{-1/2}(\Gamma)$ of the variational problem (8) are unique.

open_problem | open | MathDB 页面 | 原文线索

V2-074342 | MATHDB-392917

Uniqueness from one incident plane wave

摘录 / EXCERPT: Uniqueness for one incident plane wave remains a challenging open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-074343 | MATHDB-392918

Solvability criterion for the factorization-method far-field equation

摘录 / EXCERPT: In the stated acoustic inverse-obstacle-scattering setup, with far-field operator $F:L^2(\Omega)\to L^2(\Omega)$ and point-source far-field pattern $w_i^\infty(\cdot,z)$, the equat…

open_problem | open | MathDB 页面 | 原文线索

V2-074344 | MATHDB-392919

Wavenumber dependence of the Galerkin stability threshold for stabilized acoustic single-layer operators

摘录 / EXCERPT: How does the threshold for the stability of the Galerkin discretization quantitatively depend on the wave number k for a general stabilization approach for the acoustic single-la…

open_problem | open | MathDB 页面 | 原文线索

V2-074345 | MATHDB-392920

Minimax expected Kullback-Leibler redundancy for Bernoulli estimation

摘录 / EXCERPT: For a sample of size n from a Bernoulli distribution with unknown parameter $p\in[0,1]$, choose numbers $q_k\in(0,1)$, $k=0,1,\ldots,n$, where q_k is used when the symbol A_1…

open_problem | open | MathDB 页面 | 原文线索

V2-074346 | MATHDB-392923

Convergence and optimality of AdaBoost in the separable case

摘录 / EXCERPT: Given a separable classification problem with data points $x_1,\ldots,x_D$, hypotheses $h_j:X\to\{-1,1\}$ for $j=1,\ldots,H$, labels $y:X\to\{-1,1\}$, and the AdaBoost iterates des…

open_problem | open | MathDB 页面 | 原文线索

V2-074347 | MATHDB-392924

Size of the adjacent zone in simultaneous space-time wavelet solutions

摘录 / EXCERPT: Determine the correct size of the adjacent zone, that is, the appropriate number of neighbouring points to retain for a simultaneous space-time solution.

open_problem | open | MathDB 页面 | 原文线索

V2-074348 | MATHDB-392925

Shrinkage and democracy properties for warped wavelets

摘录 / EXCERPT: The report asks what properties a warped wavelet family can retain under suitable assumptions on the warping. The following result is given. Let $1<p<\infty$, let $\omega\in A_p$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074349 | MATHDB-392926

Countability dichotomy for sets definable using complex exponentiation

摘录 / EXCERPT: Every subset of $\mathbb{C}$ definable using exponentiation is either countable or co-countable.

open_problem | open | MathDB 页面 | 原文线索

V2-074350 | MATHDB-392927

Zil’ber’s conjecture on pseudo-exponentiation

摘录 / EXCERPT: Zil’ber’s conjecture is that the structure he has built is actually the complex field with the usual exponential function.

conjecture | open | MathDB 页面 | 原文线索

V2-074351 | MATHDB-392928

Tamm’s theorem on the boundary

摘录 / EXCERPT: Let k be a subfield of $\real$ and let $f:A\to\real$, where $A\subseteq\real^{m+n}$, be a function definable in the o-minimal expansion $\real_{k\mathrm{an}}$ of the real field b...

open_problem | open | MathDB 页面 | 原文线索

V2-074352 | MATHDB-392933

Model completeness of the expansion by Pfaffian chains

摘录 / EXCERPT: Let $\mathcal R$ be an o-minimal expansion of the real field. A sequence $f=(f_1,\ldots,f_k):U\to\mathbb R^k$ of differentiable functions on an open set $U\subseteq\mathbb R^n$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-074353 | MATHDB-392934

Ultrafilter dependence of asymptotic cones of finitely presented groups

摘录 / EXCERPT: Does there exist a finitely presented group whose asymptotic cone depends on the ultrafilter μ?

open_problem | open | MathDB 页面 | 原文线索

V2-074354 | MATHDB-392935

Functional equation and generalized Schanuel conditions for the Weierstrass function

摘录 / EXCERPT: Find the functional equation, GSCH and EC for the Weierstrass function $p(\omega,x)$ as a function of two variables.

open_problem | open | MathDB 页面 | 原文线索

V2-074355 | MATHDB-392936

Maximal regularity for analytic semigroup generators on $L^q(\Omega)$

摘录 / EXCERPT: Does every generator A of an analytic semigroup on an $L^q(\Omega)$-space, with $1<q<\infty$, have maximal L^p-regularity; that is, does the solution y of ... satisfy ... The rep...

open_problem | open | MathDB 页面 | 原文线索

V2-074356 | MATHDB-392938

Spectral mapping theorem for the logarithm of a sectorial operator

摘录 / EXCERPT: For an injective sectorial operator A with sectoriality angle $\omega(A)<\pi$, the extended spectrum satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-074357 | MATHDB-392939

Endpoint maximal estimate for the nonlinear Fourier transform

摘录 / EXCERPT: Conjecture 1. For the diagonal entry $a(k,x)$ of the solution $G(x)$ of equation (1), ...

open_problem | open | MathDB 页面 | 原文线索

V2-074358 | MATHDB-392940

Uniformity of the nonlinear Hausdorff-Young constants as $p\to2^-$

摘录 / EXCERPT: Conjecture 2. In the nonlinear Hausdorff-Young estimate (2), $\|\sqrt{\log|a|}\|_{p'}\le C_p\|F\|_p$ for $1<p<2$, can the constant C_p be chosen uniformly as p tends to 2?

open_problem | open | MathDB 页面 | 原文线索

V2-074359 | MATHDB-392943

The Hypercyclicity Criterion Problem

摘录 / EXCERPT: If T is hypercyclic on X, is the direct sum $T\oplus T$ of two copies of T hypercyclic on $X\oplus X$?

open_problem | open | MathDB 页面 | 原文线索

V2-074360 | MATHDB-392945

Kato square root problem for divergence-form operators on $\mathbb{R}^n$

摘录 / EXCERPT: Let $H_1=L^2(\mathbb{R}^n)$, $H_2=L^2(\mathbb{R}^n;\mathbb{C}^n)$, $V=W^1_2(\mathbb{R}^n)$, $S=\nabla$, and let $A=(a_{jk})\in L^\infty(\mathbb{R}^n;\mathcal{L}(\mathbb{C}^n))$ sat...

open_problem | open | MathDB 页面 | 原文线索

V2-074361 | MATHDB-392946

Analytic dependence of Kato square roots on a parameter

摘录 / EXCERPT: Suppose that $A_t\in L^\infty(\mathbb R^n;L(\mathbb C^n))$ is self-adjoint and depends analytically on $t\in(-1,1)$. Let $L_t=-\operatorname{div}A_t\nabla$. Then the operators...

open_problem | open | MathDB 页面 | 原文线索

V2-074362 | MATHDB-392947

Kato square-root problem on Lipschitz domains with mixed boundary conditions

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$, $n\in\mathbb{N}$, be a bi-Lipschitz image of a bounded smooth domain Ω', or of an unbounded smooth domain that coincides with the upper half...

open_problem | open | MathDB 页面 | 原文线索

V2-074363 | MATHDB-392950

Boundedness for the class \(\widetilde{BS}_{1,0}^{m}\)

摘录 / EXCERPT: Does the boundedness estimate established in (6) for symbols in $BS_{1,0}^{m}$ also hold for symbols in $\widetilde{BS}_{1,0}^{m}$, particularly at the endpoint $m=0$? More precise...

open_problem | open | MathDB 页面 | 原文线索

V2-074364 | MATHDB-392951

Composition of a linear pseudodifferential operator with a bilinear operator in $BS^m_{1,0}$

摘录 / EXCERPT: For $a\in S^k_{1,0}$ and $\sigma\in BS^m_{1,0}$, is it true that $L_aT_\sigma=T_\lambda$ with $\lambda\in BS^{m+k}_{1,0}$? The report answers negatively; already for x-independen...

open_problem | open | MathDB 页面 | 原文线索

V2-074365 | MATHDB-392952

Composition of a linear pseudodifferential operator with a bilinear operator in the tilde class

摘录 / EXCERPT: For $a\in S^k_{1,0}$ and $\sigma\in\widetilde{BS}^m_{1,0}$, is it true that $L_aT_\sigma=T_\lambda$ with $\lambda\in\widetilde{BS}^{m+k}_{1,0}$? The report states that the answer i...

open_problem | open | MathDB 页面 | 原文线索

V2-074366 | MATHDB-392953

Theorem 1 on asymptotic solutions under exponentially weighted perturbations

摘录 / EXCERPT: Assume the setup on pages 68-69: $y'=M(\cdot)y$ has an exponential dichotomy on $\mathbb R$ with dichotomy projector Q, and $\{Q_j\}_{j=1}^n$ is its ordered finest exponential sp...

open_problem | open | MathDB 页面 | 原文线索

V2-074367 | MATHDB-392955

Commutator preservation under scalar functional calculus in symmetrically normed ideals

摘录 / EXCERPT: In the special case bM=L(H) with the usual trace, let bSE be a symmetrically normed ideal, let ba:operatorname{Dom}(a)to H be self-adjoint, and let bx=x^in SEsubseteq K(H). For whi...

open_problem | open | MathDB 页面 | 原文线索

V2-074368 | MATHDB-392956

Characterization of r.i. spaces for the functional-calculus domain property

摘录 / EXCERPT: For which separable rearrangement-invariant function spaces $E(0,\)$ is the condition (NC), namely $f\in C_b^1(\mathbb{R})$ and $f(0)=0$, sufficient to ensure that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-074369 | MATHDB-392957

Weighted maximal regularity for the Dirichlet heat equation on wedges and cones

摘录 / EXCERPT: Let $m\in\mathbb N_0$, $n\ge 2$, let $\Omega\subset S^{n-1}$ be open with smooth nonempty boundary, let $C_\Omega=\{x\in\mathbb R^n\setminus\{0\}:x/|x|\in\Omega\}$, set...

open_problem | open | MathDB 页面 | 原文线索

V2-074370 | MATHDB-392959

Gabber’s purity conjecture for the Brauer group of punctured complete-intersection local rings

摘录 / EXCERPT: Let $(R,\mathfrak m)$ be a strictly henselian complete-intersection Noetherian local ring of dimension at least 4. Let $U_R=\operatorname{Spec}(R)\setminus\{\mathfrak m\}$, and l...

conjecture | open | MathDB 页面 | 原文线索

V2-074371 | MATHDB-392960

Brauer-group purity in a four-dimensional mixed-characteristic case

摘录 / EXCERPT: Conjecture 1 holds when R is of dimension 4 and of mixed characteristic $(0,p)$, and there is a sequence of birational blowing-ups with regular centers ... such that the componen...

open_problem | open | MathDB 页面 | 原文线索

V2-074372 | MATHDB-392963

Langlands Fundamental Lemma for Unitary Groups in Positive Characteristic

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{F}_p((t))$, let $G=U(n)$ be a unitary group, and assume $p>n$. Then the Langlands Fundamental Lemma holds for G over F.

open_problem | open | MathDB 页面 | 原文线索

V2-074373 | MATHDB-392975

Large-rescaling correspondence for TERP-structures and mixed TERP-structures

摘录 / EXCERPT: Conjecturally, for a TERP-structure [4m]T[0m, for all sufficiently large positive real numbers [4mr[0m, the pullback [4mpir^T[0m is a positive-definite, trivial TERP-structure if a...

open_problem | open | MathDB 页面 | 原文线索

V2-074374 | MATHDB-392978

Formal Kapustin-Orlov Fourier-Mukai equivalence

摘录 / EXCERPT: In the setup above, with a complex analytic torus X, its dual torus $X^\vee$, a holomorphic Poisson structure Π on X, the corresponding formal non-commutative space X_Π...

open_problem | open | MathDB 页面 | 原文线索

V2-074375 | MATHDB-392979

The general Lifting Problem for singular integral affine structures on S^2

摘录 / EXCERPT: Find a sheaf $\mathcal O_{S^2}$ whose restriction to $S^2\setminus\{x_1,\ldots,x_{24}\}$ is locally isomorphic to $\mathcal O^{\mathrm{can}}_{S^2\setminus\{x_1,\ldots,x_{24}\}}$, a...

open_problem | open | MathDB 页面 | 原文线索

V2-074376 | MATHDB-392980

Unimodular triangulation conjecture for the hive-polytope matrix configuration

摘录 / EXCERPT: Fix an integer $r>0$ and let B and R be the matrices in (1). Then the convex hull of the points whose coordinates are the columns of the block matrix $[B\;0\;R\;I]$ has a unimo...

conjecture | open | MathDB 页面 | 原文线索

V2-074377 | MATHDB-392982

Unimodality of the Ehrhart h^*-vector for standard Gorenstein semigroup rings

摘录 / EXCERPT: Let P be a d-dimensional integral polytope in $\mathbb{R}^N$, with Ehrhart polynomial $i(P,r)=\#(rP\cap\mathbb{Z}^N)$ and h^*-vector $h^*(P)=(h_0^*,h_1^*,\ldots,h_d^*)$ defin...

open_problem | open | MathDB 页面 | 原文线索

V2-074378 | MATHDB-392986

A two-leading-coefficient bound for the lattice-point enumerator

摘录 / EXCERPT: For a convex body $K\subset\mathbb{R}^n$, with $G(K)=\#(K\cap\mathbb{Z}^n)$, does there exist a constant $c(n)$ depending only on the dimension such that ... Here $V_i(K)$ denotes th...

open_problem | open | MathDB 页面 | 原文线索

V2-074379 | MATHDB-392987

Wills' conjecture for Ehrhart coefficients

摘录 / EXCERPT: Let $C_n=[-1,1]^n$, and let $G_i(P)$, $0\le i\le n$, denote the coefficients of the Ehrhart polynomial of a 0-symmetric lattice polytope P. Is ...

conjecture | open | MathDB 页面 | 原文线索

V2-074380 | MATHDB-392988

Ehrhart polynomial formula for integral cyclic polytopes

摘录 / EXCERPT: Conjecture. Let $P=C_d(n)$ be any cyclic polytope realized with integer vertices on the standard moment curve $t\mapsto(t,t^2,\ldots,t^d)$ in $\mathbb{R}^d$. Then the Ehrhart polyn...

open_problem | open | MathDB 页面 | 原文线索

V2-074381 | MATHDB-392990

Question 3: Boundary-to-center distance for analytic vector-valued functions

摘录 / EXCERPT: Let D be the closed unit disk, let N be a (usually very big) positive integer, and let $\|\cdot\|_p$ denote the ℓ^p norm on $\mathbb{C}^N$. Let $f_1,f_2,\ldots,f_k$ be k...

open_problem | open | MathDB 页面 | 原文线索

V2-074382 | MATHDB-392992

Question 4 on analytic vector-valued functions and finite-dimensional approximation

摘录 / EXCERPT: Let D be the closed unit disk, let N be a positive integer, and let $H^\infty(D,\mathbb C^N)$ denote the space of all $\mathbb C^N$-valued functions...

open_problem | open | MathDB 页面 | 原文线索

V2-074383 | MATHDB-392993

Logarithmically convex inequality for measures of non-compactness under real interpolation

摘录 / EXCERPT: For general couples, without assuming any approximation hypothesis, the report states and proves the logarithmically convex inequality ... When one restriction of T is compact, the...

open_problem | open | MathDB 页面 | 原文线索

V2-074384 | MATHDB-392995

Number of records or cuts in conditioned Galton-Watson trees

摘录 / EXCERPT: Let T_n be a conditioned Galton-Watson tree of size n, defined by an offspring distribution $\[1m\xi\u001b[0m$ with $\[1m\mathbb{E}\xi=1\u001b[0m$ and finite variance...

open_problem | open | MathDB 页面 | 原文线索

V2-074385 | MATHDB-392997

Existence of a survival-function solution for a max-type fixed-point equation

摘录 / EXCERPT: Let b[T=(Ti){ithosei I}b be the positive factors in the max-type fixed-point equation. Define $m(\alpha):=\mathbb{E}\sum_{i\in I}T_i^\alpha$. If, for some $\alpha>0$, $m(\alpha)=1$...

open_problem | open | MathDB 页面 | 原文线索

V2-074386 | MATHDB-392999

Non-extension of a homogeneous vector bundle on a small compactification

摘录 / EXCERPT: Let $X_0:=\mathbb{C}^n\setminus\{0\}$ with the natural action of $G:=\mathrm{GL}_n$, let $X:=\mathbb{P}^n$ be the equivariant compactification, and let V_0 be the pull-back of th...

open_problem | open | MathDB 页面 | 原文线索

V2-074387 | MATHDB-393000

Exceptional minimal set conjecture for holomorphic foliations on the complex projective plane

摘录 / EXCERPT: Every leaf of a holomorphic foliation of the complex projective plane accumulates on a singularity.

conjecture | open | MathDB 页面 | 原文线索

V2-074388 | MATHDB-393001

Openness of the real-group semistable set

摘录 / EXCERPT: Let Z be a complex space with a holomorphic action of $U^{\d7\mathbb C}$, where U is compact, and let $G\subset U^{\mathbb C}$ have Cartan decomposition $G=K\exp(\mathfrak p)$...

open_problem | open | MathDB 页面 | 原文线索

V2-074389 | MATHDB-393003

Deformation of almost Fano threefolds in the divisorial case

摘录 / EXCERPT: Let X be a smooth almost Fano threefold, meaning that $-K_X$ is nef and big, with $-K_X\cdot C=0$ for at least one curve $C\subset X$. Assume the divisorial case of the report:...

open_problem | open | MathDB 页面 | 原文线索

V2-074390 | MATHDB-393004

Automorphic form vanishing on the discriminant locus

摘录 / EXCERPT: Construct an automorphic form on B vanishing exactly on the discriminant locus H, if it exists.

open_problem | open | MathDB 页面 | 原文线索

V2-074391 | MATHDB-393005

Invariants and fundamental-group presentations for discriminant complements of Brieskorn-Pham polynomials

摘录 / EXCERPT: For a Brieskorn-Pham polynomial $f\in\mathbb{C}[x_1,\ldots,x_n]$ of the form ... determine the braid-monodromy invariants and group presentations for the fundamental group...

open_problem | open | MathDB 页面 | 原文线索

V2-074392 | MATHDB-393012

Critical static structure-function asymptotic at the critical temperature

摘录 / EXCERPT: The report records the conjecture that, at inverse critical temperature $\beta=\beta_c$, the static structure function satisfies ... with critical exponent $\eta=0.03$ for $d=3$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-074393 | MATHDB-393016

Diffusive/non-diffusive transition in KA models

摘录 / EXCERPT: We conjecture that the same result holds in any dimension, namely a diffusive/non-diffusive transition does not occur at any density.

open_problem | open | MathDB 页面 | 原文线索

V2-074394 | MATHDB-393020

Equilibrium fluctuations for a system of harmonic oscillators with conservative noise

摘录 / EXCERPT: Under the diffusive scaling described on page 45, the pair of interdistance and energy fluctuation fields converges in law to the solution of the generalized Ornstein-Uhlenbeck sys...

open_problem | open | MathDB 页面 | 原文线索

V2-074395 | MATHDB-393022

Hausdorff dimension of the boundary of the deformation space

摘录 / EXCERPT: What is the Hausdorff dimension of the boundary of the deformation space $AH(M)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074396 | MATHDB-393023

Thurston’s Ending Lamination Conjecture for Kleinian surface groups

摘录 / EXCERPT: For a Kleinian surface group, the invariants called “ending laminations” suffice to determine the surface group uniquely.

conjecture | open | MathDB 页面 | 原文线索

V2-074397 | MATHDB-393024

Malcev completions of quasi-isometric nilpotent groups

摘录 / EXCERPT: Are the Malcev completions of quasi-isometric nilpotent groups isomorphic?

open_problem | open | MathDB 页面 | 原文线索

V2-074398 | MATHDB-393027

Carlsson’s total mod-[U+007F]p Betti-number question

摘录 / EXCERPT: Suppose that $G=(\mathbb{Z}/p)^k$ acts freely on a finite complex X. Is ...

open_problem | open | MathDB 页面 | 原文线索

V2-074399 | MATHDB-393028

Free smooth actions on products of two spheres

摘录 / EXCERPT: If there is a finite free G-CW complex $X \\simeq S^n \\times S^n$, does G act freely and smoothly on $S^n \\times S^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-074400 | MATHDB-393036

Nonexistence of equivariant Dirac operators on $SU_q(n)$ for $n>2$

摘录 / EXCERPT: For $SU_q(n)$ with $n>2$, the L^2-space does not admit any equivariant Dirac operator with nontrivial sign acting on it.

open_problem | open | MathDB 页面 | 原文线索

V2-074401 | MATHDB-393037

Riemann-Hilbert classification of equisingular flat connections

摘录 / EXCERPT: The report presents a Riemann-Hilbert correspondence classifying equisingular flat bundles associated with equisingular flat G-connections on the total space B of a...

open_problem | open | MathDB 页面 | 原文线索

V2-074402 | MATHDB-393039

Universal existence of a γ-element

摘录 / EXCERPT: Is it true that every group G has a $\b3$-element?

open_problem | open | MathDB 页面 | 原文线索

V2-074403 | MATHDB-393040

Theorem 1.3: computation of the dual-Dirac group

摘录 / EXCERPT: Let G be a discrete group with a G-finite model for EG. There is a natural isomorphism ... Under this isomorphism, a dual Dirac class corresponds to any element in the domain o...

open_problem | open | MathDB 页面 | 原文线索

V2-074404 | MATHDB-393041

Existence of a \b3-element for every discrete group

摘录 / EXCERPT: Does every discrete group G have a $\b3$-element?

open_problem | open | MathDB 页面 | 原文线索

V2-074405 | MATHDB-393044

Connes’ problem on the character of the canonical K-cycle

摘录 / EXCERPT: For a lattice Γ in a semisimple Lie group G, let the K-cycle associated with the elliptic differential operator on the symmetric space be considered over $C_r^*(\Gamma)$....

open_problem | open | MathDB 页面 | 原文线索

V2-074406 | MATHDB-393046

Chern character for totally disconnected groups

摘录 / EXCERPT: Let G be a second-countable, locally compact, totally disconnected group. Let X and Y be finite-dimensional, locally finite G-simplicial complexes. If the action of G on...

open_problem | open | MathDB 页面 | 原文线索

V2-074407 | MATHDB-393047

Realization of Borel 3-cocycles by automorphic representations

摘录 / EXCERPT: Let G be a locally compact group and let $[w]\in H^3_{\mathrm{Borel}}(G;\mathbb T)$ be represented by a Borel 3-cocycle $w:G^3\to\mathbb T$. Is every such class realizable as t...

open_problem | open | MathDB 页面 | 原文线索

V2-074408 | MATHDB-393051

Strict decrease of the Nyman approximation distances

摘录 / EXCERPT: Let $g_\alpha(u)=\{\alpha/u\}$ for $u>0$ and $0<\alpha\le 1$, and let δ_n be the distance in $L^2(0,+\infty)$ from the characteristic function of $(0,1)$ to the set of all...

open_problem | open | MathDB 页面 | 原文线索

V2-074409 | MATHDB-393052

Characterization of admissible functions for translation-generated subspaces

摘录 / EXCERPT: Let G be a locally compact abelian group with Haar measure, let $H=L^2(G)$, and for $g\in H$ and $\gamma\in G$ define $g_\gamma(x)=g(x\gamma^{-1})$. Call g admissible if, for e...

open_problem | open | MathDB 页面 | 原文线索

V2-074410 | MATHDB-393053

Non-vanishing of class group L-functions at the central point

摘录 / EXCERPT: Let $K=\mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field of discriminant $-D$, let C be its class group, and write $h=\#C$. For each character $\chi\in\widehat C$, define .....

open_problem | open | MathDB 页面 | 原文线索

V2-074411 | MATHDB-393056

Obtaining the derivative moment asymptotic by Gonek's method

摘录 / EXCERPT: Can a result similar to Theorem 2 be obtained by Gonek's method, assuming only the Riemann Hypothesis?

open_problem | open | MathDB 页面 | 原文线索

V2-074412 | MATHDB-393057

An analogue of Turán's reformulation

摘录 / EXCERPT: What would an analogue of Turán's reformulation be?

open_problem | open | MathDB 页面 | 原文线索

V2-074413 | MATHDB-393058

Uniform convergence for the indicator functions of intervals in the distribution of zeta-zero ordinates

摘录 / EXCERPT: Fix a nonzero real number α. Let γ range over the imaginary parts of the nontrivial zeros $\rho=\beta+i\gamma$ of $\zeta(s)$, let $N(T)=\#\{\gamma:0<\gamma\le T\}$, a...

open_problem | open | MathDB 页面 | 原文线索

V2-074414 | MATHDB-393061

Nonvanishing of twists of automorphic L-functions on $GL(n)$

摘录 / EXCERPT: Let F be a number field, let π be an irreducible cuspidal automorphic representation of $GL(n,\mathbb{A}_F)$, let S be a finite set of places of F, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-074415 | MATHDB-393062

An improved conditional bound for the summatory Möbius function

摘录 / EXCERPT: Assuming the Riemann Hypothesis, with $M(x)=\sum_{n\le x}\mu(n)$, Helmut Maier and Hugh L. Montgomery report that the estimate $M(x)\ll x^{1/2+\varepsilon(x)}$ is valid with...

open_problem | open | MathDB 页面 | 原文线索

V2-074416 | MATHDB-393065

General converse theorem for the Selberg class

摘录 / EXCERPT: For every $d\in\mathbb{N}$, S_d is equal to the class of automorphic L-functions of degree d, where $S_d=\{F\in S:d_F=d\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-074417 | MATHDB-393068

Joint universality conjecture for orthogonal families of Selberg-class L-functions

摘录 / EXCERPT: Suppose that $L_1, obreak\ldots,L_m$ are elements of $\widetilde{\mathcal S}$ satisfying ... where $\kappa_j>0$ and $\delta_{jk}=1$ if $j=k$ and $\delta_{jk}=0$ otherwise. For...

conjecture | open | MathDB 页面 | 原文线索

V2-074418 | MATHDB-393070

Mean values of automorphic and quadratic-twist L-functions

摘录 / EXCERPT: Let f be an automorphic form over GL_n. Estimate the mean values ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-074419 | MATHDB-393075

Hyperbolic dimension of the product of two hyperbolic planes

摘录 / EXCERPT: Determine the hyperbolic dimension $\operatorname{hypdim}(\mathbb H^2\times\mathbb H^2)$. The report states that it is either 3 or 4, and conjectures that it is 4.

open_problem | open | MathDB 页面 | 原文线索

V2-074420 | MATHDB-393076

Classification of complete cmc surfaces with vanishing generalized Hopf differential

摘录 / EXCERPT: Let $(\kappa,H)\ne(0,0)$, let L be the symmetric bilinear form corresponding to the field of orthogonal projectors onto the vertical lines in $M^2_\kappa\times\mathbb R$, and def...

open_problem | open | MathDB 页面 | 原文线索

V2-074421 | MATHDB-393081

Zagier’s conjectural formula for \(\zeta_F(2)\)

摘录 / EXCERPT: Let F be a number field with ring of integers $\mathcal O_F$, discriminant d_F, r_1 real places, and r_2 complex places. Choose non-real embeddings...

open_problem | open | MathDB 页面 | 原文线索

V2-074422 | MATHDB-393084

Rationality of the quenched beta-function

摘录 / EXCERPT: Conjecture: the quenched β-function is a series in the fine-structure constant with rational coefficients.

open_problem | open | MathDB 页面 | 原文线索

V2-074423 | MATHDB-393086

Analytic continuation of the p-adic polylogarithm series

摘录 / EXCERPT: There is no analytic function on $\mathbb{C}_p\setminus\{1\}$ that agrees on the open unit disk with the polylogarithm series $\sum_{n=1}^{\infty} z^n/n^k$.

open_problem | open | MathDB 页面 | 原文线索

V2-074424 | MATHDB-393087

Irrationality of \(\zeta(5)\)

摘录 / EXCERPT: Is the Riemann zeta value $\zeta(5)$ irrational?

open_problem | open | MathDB 页面 | 原文线索

V2-074425 | MATHDB-393088

Double-shuffle generation conjecture for multiple zeta values

摘录 / EXCERPT: Let $\zeta(s_1,\ldots,s_r)$ denote the convergent multiple zeta values with $s_1\ge 2$, and let $H_0=\mathbb{Q}\,1\oplus x_0\mathbb{Q}\langle x_0,x_1\rangle x_1$ be the correspondi...

conjecture | open | MathDB 页面 | 原文线索

V2-074426 | MATHDB-393091

The p-part of the Bloch-Kato conjecture for Dirichlet motives

摘录 / EXCERPT: For a Dirichlet character χ and an integer r satisfying $\chi(-1)=(-1)^r$, the p-part of the Bloch-Kato conjecture for the Dirichlet motive $h(\chi)(r)$ is equivalent to t...

conjecture | open | MathDB 页面 | 原文线索

V2-074427 | MATHDB-393092

Iwasawa main-conjecture statement for Dirichlet characters

摘录 / EXCERPT: With Q_n the unique subfield of $\mathbb{Q}(\zeta_{p^{n+1}})$ satisfying $\operatorname{Gal}(Q_n/\mathbb{Q})\cong\mathbb{Z}/p^n\mathbb{Z}$, let Z_n be its ring of integers, let...

conjecture | open | MathDB 页面 | 原文线索

V2-074428 | MATHDB-393093

Distinguishing recent transmission from common evolutionary descent

摘录 / EXCERPT: Without an underlying evolutionary model, it is difficult to identify whether two bacterial strains are identical because of recent transmission or because of common evolutionary d...

open_problem | open | MathDB 页面 | 原文线索

V2-074429 | MATHDB-393094

Statistics of unconditioned connected components in the contact-tracing branching process

摘录 / EXCERPT: Determine the statistics of the size and structure of connected components when one does not condition on the age of the root. The report notes that this requires determining the M...

open_problem | open | MathDB 页面 | 原文线索

V2-074430 | MATHDB-393095

Identifiability of infection duration under immunity-dependent detection

摘录 / EXCERPT: When infection and clearance may depend on acquired immunity, and the sensitivity of parasitological detection is unknown, below 100%, and may also depend on acquired immunity, doe...

open_problem | open | MathDB 页面 | 原文线索

V2-074431 | MATHDB-393096

Local smoothing estimate for Schrödinger equations with repulsive potentials in dimensions $n>3$

摘录 / EXCERPT: Let V be a potential on $\mathbb{R}^n$, $n>3$, satisfying Hypotheses 1 and 2, and let $W(r)$ satisfy (2), (5), (6), and ... If $\varepsilon>0$ satisfies ... then the unique solution...

open_problem | open | MathDB 页面 | 原文线索

V2-074432 | MATHDB-393097

Theorem 3: local smoothing estimate for Schrödinger equations with nonradial potentials

摘录 / EXCERPT: The report states and proves the following theorem. Let V be a potential on $\mathbb{R}^n$, $n\ge 3$, satisfying Hypotheses 1 and 2. Suppose that a nonnegative function W sati...

open_problem | open | MathDB 页面 | 原文线索

V2-074433 | MATHDB-393098

Local well-posedness theorem for periodic modified KdV

摘录 / EXCERPT: The report states the following theorem for the modified KdV equation on the one-dimensional torus: let $-\frac12<s<-\frac13$. For every $u_0\in H^s$, there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-074434 | MATHDB-393099

Global fold structure of the Miura map on the circle

摘录 / EXCERPT: Theorem 4 ([10]). Let $\mathbb T=\mathbb R/\mathbb Z$ and let $B(r)=r_x+r^2$ be the Miura map. (i) For any $\alpha\ge 0$, the map $B:H^\alpha(\mathbb T)\to H^{\alpha-1}(\mathbb T)$...

open_problem | open | MathDB 页面 | 原文线索

V2-074435 | MATHDB-393101

Smoothing, Strichartz, and maximal-function estimates for one-dimensional Schrödinger equations with BV metrics

摘录 / EXCERPT: For the equation $i\partial_t u+\partial_x(a(x)\partial_xu)=0$, $u(x,0)=u_0(x)$, assume that a is a real-valued BV function-namely, a bounded function whose derivative is a Radon...

open_problem | open | MathDB 页面 | 原文线索

V2-074436 | MATHDB-393102

Ground-state mass concentration for finite-time blowup of L^2-critical cubic NLS

摘录 / EXCERPT: Determine whether every solution of the two-dimensional focusing cubic NLS equation (1) with initial data $u_0\in L^2(\mathbb R^2)$ and a finite maximal forward existence interval...

open_problem | open | MathDB 页面 | 原文线索

V2-074437 | MATHDB-393103

Sub-ground-state global existence and spacetime boundedness for the 2D focusing cubic NLS

摘录 / EXCERPT: For the two-dimensional focusing cubic nonlinear Schrödinger equation ... all solutions with $u_0\in L^2(\mathbb R^2)$ satisfying ... should be global-in-time and satisfy ... where Q i...

open_problem | open | MathDB 页面 | 原文线索

V2-074438 | MATHDB-393104

Endpoint local well-posedness for the one-dimensional derivative nonlinear Schrödinger equation

摘录 / EXCERPT: For the Cauchy problem ... in one space dimension, does local well-posedness hold for data $u_0\in \widehat H^{1/2}_{1}(\mathbb R)$, i.e. at the endpoint $r=1$, $s=\tfrac12$ of the t...

open_problem | open | MathDB 页面 | 原文线索

V2-074439 | MATHDB-393105

Uniform estimates and singular-limit convergence for generalized Zakharov systems

摘录 / EXCERPT: Obtain rigorous estimates uniform in the parameter (as tends to zero) and prove convergence of the Schrödinger component to the corresponding nonlinear Schrödinger limit for genera...

open_problem | open | MathDB 页面 | 原文线索

V2-074440 | MATHDB-393106

Uniqueness at the energy-critical exponent for the defocusing semilinear wave equation

摘录 / EXCERPT: For the defocusing semilinear wave equation $u_{tt}-\Delta u+|u|^{p-1}u=0$ on $\mathbb{R}\times\mathbb{R}^{d}$ with $d\ge 3$, where $p_{*}=\frac{2d}{d-2}$, does uniqueness hold in...

open_problem | open | MathDB 页面 | 原文线索

V2-074441 | MATHDB-393107

Unique continuation for partially elliptic operators with critical lower-dimensional potentials

摘录 / EXCERPT: Let $P(x',D_{x'},D_{x''})$ be a second-order differential operator with real principal symbol, defined on a neighbourhood $\Omega\subset\mathbb{R}^n$ of x_0, with C^1 coeffici...

open_problem | open | MathDB 页面 | 原文线索

V2-074442 | MATHDB-393108

Zygmund’s two-dimensional Fourier restriction theorem

摘录 / EXCERPT: For $p<4/3$, every $f\in L^p(\mathbb{R}^2)$ has a Fourier transform whose restriction to S^1 is a function in $L^q(S^1)$, where $q=p'/3$ and $1/p+1/p'=1$.

open_problem | open | MathDB 页面 | 原文线索

V2-074443 | MATHDB-393109

Long-range asymptotics for the Maxwell-Schrödinger system

摘录 / EXCERPT: The report states the following existence theorem for the Coulomb-gauge Maxwell-Schrödinger system in three space dimensions. Let $h(t)=t^{-1}(2+\ln t)^2$, and let (u_a,A_a) be t...

open_problem | open | MathDB 页面 | 原文线索

V2-074444 | MATHDB-393110

Finite-time blow-up for the radial semilinear wave equation in Schwarzschild spacetime

摘录 / EXCERPT: For every p with $1<p<1+\sqrt{2}$, there exists $\varepsilon_0>0$ such that, for every $\varepsilon\in(0,\varepsilon_0)$, there are $r_0=r_0(p,\varepsilon)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-074445 | MATHDB-393111

Local well-posedness for the modified Ishimori system

摘录 / EXCERPT: For the Cauchy problem associated with the schematic equation $u_t-i\Delta u+(a\cdot\nabla u)=F(u,\nabla u)$, with $u_0\in H^s(\mathbb{R}^2)$ and $s>\tfrac12$, there exists a time...

open_problem | open | MathDB 页面 | 原文线索

V2-074446 | MATHDB-393112

Global well-posedness and scattering for the defocusing quintic NLS in \(\mathbb{R}^{1+3}\)

摘录 / EXCERPT: For the defocusing quintic Schrödinger equation ... with energy ... if $E(u_0)<\infty$ (equivalently $u_0\in\dot H^1(\mathbb{R}^3)$), then there exists a unique global solution ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-074447 | MATHDB-393113

Lagrange’s claim that his series gives the smallest root

摘录 / EXCERPT: For the equation $\alpha-x+n\phi(x)=0$, Lagrange claimed that his series always represents the numerically smallest root.

open_problem | open | MathDB 页面 | 原文线索

V2-074448 | MATHDB-393114

Poincaré’s multidimensional intermediate value theorem

摘录 / EXCERPT: Let $X_1,X_2,\ldots,X_n$ be n continuous functions of n variables $x_1,x_2,\ldots,x_n$, where x_i varies between $-a_i$ and a_i. Suppose that X_i is constantly positive w...

open_problem | open | MathDB 页面 | 原文线索

V2-074449 | MATHDB-393115

Recovering edges of an image from noisy tomographic data

摘录 / EXCERPT: What is the best attainable accuracy in recovering edges from noisy observations of projections, and which methods can achieve this optimal performance?

open_problem | open | MathDB 页面 | 原文线索

V2-074450 | MATHDB-393117

Endpoint L^2 conjecture for one-dimensional Schrödinger scattering

摘录 / EXCERPT: For the half-line Schrödinger operator $H_V=-\frac{d^2}{dx^2}+V(x)$ on $\mathbb R_+=(0,\infty)$ with Dirichlet boundary condition at the origin, does the conclusion of Theorem 1 re...

conjecture | open | MathDB 页面 | 原文线索

V2-074451 | MATHDB-393118

Embedded singular continuous spectrum for slowly decaying one-dimensional Schrödinger potentials

摘录 / EXCERPT: Does there exist a potential V satisfying $|V(x)|\le C(1+|x|)^{-\alpha}$ for some $\alpha>\tfrac12$ such that the half-line Schrödinger operator $H_V=-\frac{d^2}{dx^2}+V(x)$, wit...

open_problem | open | MathDB 页面 | 原文线索

V2-074452 | MATHDB-393119

Embedded singular continuous spectrum at arbitrarily slow decay

摘录 / EXCERPT: Let $H_V=-\frac{d^2}{dx^2}+V(x)$ be the one-dimensional Schrödinger operator on $\mathbb R_+=(0,\infty)$ with Dirichlet boundary condition at the origin. For any function h satis...

open_problem | open | MathDB 页面 | 原文线索

V2-074453 | MATHDB-393126

Naive arbitrary-field magnetic Lieb-Thirring inequality

摘录 / EXCERPT: The conjecture was that, for an arbitrary magnetic field (B) in $\mathbb R^3$, the negative eigenvalues $e_j(H)$ of the Pauli operator ... would satisfy ... where $[a]_-:=\min\{0,a\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074454 | MATHDB-393128

Homogenization of the stationary periodic Maxwell system

摘录 / EXCERPT: For a lattice $\Gamma\subset\mathbb{R}^3$, let $\eta(x)$ and $\mu(x)$ be measurable, real, symmetric, Γ-periodic 3×3 matrix-valued functions satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-074455 | MATHDB-393129

Monotonicity of the spherically averaged atomic electron density

摘录 / EXCERPT: Is the spherically averaged electron density $\widetilde{\rho}(r)$ monotonically decreasing?

open_problem | open | MathDB 页面 | 原文线索

V2-074456 | MATHDB-393131

Gaussian maximizers for the Strichartz inequality in dimensions one and two

摘录 / EXCERPT: For the free homogeneous Schrödinger Strichartz inequality in dimension $d=1$ or $d=2$, with exponent $p=2+4/d$, an initial datum $f\in L^2(\mathbb R^d)$ is a maximizer-i.e. equali...

open_problem | open | MathDB 页面 | 原文线索

V2-074457 | MATHDB-393132

Bethe-Sommerfeld theorem for the three-dimensional periodic Landau operator

摘录 / EXCERPT: In dimension $d=3$, let $H_{B;V}=(D-A)^2+V$ be the periodic Landau operator, where $D=i\nabla$, $B=\nabla\times A$ is a constant magnetic field, V is periodic, and the flux-ratio...

open_problem | open | MathDB 页面 | 原文线索

V2-074458 | MATHDB-393133

Asymptotics of the magnetic spectral shift function above Landau levels

摘录 / EXCERPT: Assume condition (1) on the potential V. Fix $b>0$ and $q\in\mathbb Z_+$. For every $\varepsilon\in(0,1)$, as $\lambda\downarrow0$,nn ... n ... nnHere p_q is the Landau-level proje...

open_problem | open | MathDB 页面 | 原文线索

V2-074459 | MATHDB-393134

Corollary 3(iii): spectral-shift asymptotics at Landau levels for compactly supported effective potentials

摘录 / EXCERPT: Let $H_0=(i\nabla+A)^2-b$ be the three-dimensional magnetic Schrödinger operator with constant magnetic field $b>0$, let $H=H_0-V$, and let $\xi(\cdot;H,H_0)$ denote the spectral s...

open_problem | open | MathDB 页面 | 原文线索

V2-074460 | MATHDB-393137

Heath-Brown’s rational-point counting conjecture for nonsingular projective hypersurfaces

摘录 / EXCERPT: Suppose that $n\ge 2$, that $X\subset\mathbb{P}^n$ is a nonsingular hypersurface of degree $d\ge 2$ defined over $\mathbb{Q}$, and let $\varepsilon>0$. With...

conjecture | open | MathDB 页面 | 原文线索

V2-074461 | MATHDB-393138

Batyrev-Manin upper-bound conjecture for rational points on smooth hypersurfaces

摘录 / EXCERPT: Let $d\ge 3$ and $n\ge 4$, and suppose that $X\subset\mathbb{P}^n$ is a non-singular hypersurface of degree d. Then there exists $\delta>0$ such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-074462 | MATHDB-393139

Uniform rational-point bound for non-singular hypersurfaces

摘录 / EXCERPT: Conjecture 1. Let $X\subset\mathbb{P}^n$ be a non-singular hypersurface of degree $d\ge 2$, defined over $\mathbb{Q}$, and let $\varepsilon>0$. For ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-074463 | MATHDB-393141

NP-hardness of irreducibility testing for supersparse bivariate polynomials

摘录 / EXCERPT: Determining the irreducibility of a supersparse (lacunary) bivariate polynomial over a large finite field of any characteristic is NP-hard via randomized reductions.

open_problem | open | MathDB 页面 | 原文线索

V2-074464 | MATHDB-393143

Large maximal sum-free sets in ternary vector spaces

摘录 / EXCERPT: Let $r\ge 3$ and suppose that $A\subseteq \mathrm{GF}(3^r)$ is a maximal (by inclusion) sum-free set. If $|A|>(3^{r-1}+1)/2$, then A is a union of cosets of a non-zero subgroup o...

open_problem | open | MathDB 页面 | 原文线索

V2-074465 | MATHDB-393144

Threshold for structural classification of large sum-free subsets of \(\mathrm{GF}(p)\)

摘录 / EXCERPT: Determine how far the factor 0.33 in the structural theorem for sum-free subsets $A\subseteq\mathrm{GF}(p)$ can be reduced; specifically, bridge the gap between the example with...

open_problem | open | MathDB 页面 | 原文线索

V2-074466 | MATHDB-393145

Limiting orbit-length distributions for birational maps over finite fields

摘录 / EXCERPT: Let L be a birational map, and for a reduction of L over $\a0\mathbb F_p$ let $T(z)$ denote the length of the orbit of L through $z\in\mathbb F_p^2$. Define ... Conjecture 1. T...

open_problem | open | MathDB 页面 | 原文线索

V2-074467 | MATHDB-393148

Conjecture 2 on normalized orbit distributions of non-integrable birational maps

摘录 / EXCERPT: For every non-integrable bi-rational map which is not R-reversible and has no other symmetry, ...

conjecture | open | MathDB 页面 | 原文线索

V2-074468 | MATHDB-393151

List decodability of Reed-Solomon codes beyond the square-root radius

摘录 / EXCERPT: For the distance bound between $n-\sqrt{nk}$ and $n-\hat g(n,k,q)$, is the Reed-Solomon code $[n;k]_q$ list decodable?

open_problem | open | MathDB 页面 | 原文线索

V2-074469 | MATHDB-393155

Existence of good Artin-Schreier towers of Type III

摘录 / EXCERPT: Does there exist a good tower of Type III?

open_problem | open | MathDB 页面 | 原文线索

V2-074470 | MATHDB-393156

Quasiconvex relaxation of the rigid-elastic single-slip energy

摘录 / EXCERPT: For one slip system, define ... The report states that in two dimensions its quasiconvex envelope is ... where $\lambda_i(F)$ are the singular values of F.

open_problem | open | MathDB 页面 | 原文线索

V2-074471 | MATHDB-393157

Uniqueness of evolutions in the Gurson model

摘录 / EXCERPT: Given a time-dependent strain evolution $\varepsilon=\varepsilon(t)$, does the Gurson model (1)-(5), with suitable initial conditions, define a unique evolution of the stress...

open_problem | open | MathDB 页面 | 原文线索

V2-074472 | MATHDB-393158

Three-dimensional global existence and sharp-interface convergence for a configurational-force phase-field model

摘录 / EXCERPT: For the initial-boundary value problem for the system consisting of equations (1), (2), and (5), with positive regularization parameter $ u$, do solutions exist globally in time in...

open_problem | open | MathDB 页面 | 原文线索

V2-074473 | MATHDB-393159

Conservative configurational-forces phase-field equation

摘录 / EXCERPT: Determine whether the equation $S_t=c\,\operatorname{div}_x\!\left(\frac{\nabla_x(\psi_S-\mu\Delta_x S)}{|\nabla_xS|}\right)$, identified in the report as the conservative configura...

open_problem | open | MathDB 页面 | 原文线索

V2-074474 | MATHDB-393161

Uniqueness of an affine solution in nonlinear elastodynamics

摘录 / EXCERPT: Let a homogeneous compressible elastic material occupy a bounded smooth region $\Omega\subset\mathbb{R}^3$, with Ω star-shaped with respect to an interior point. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-074475 | MATHDB-393162

Incremental minimization problem for fatigue debonding

摘录 / EXCERPT: For each $i\in\mathbb{N}$, find (v_i,δ_i) such that ... with $v_0=\delta_0=0$, where ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-074476 | MATHDB-393163

Unique, discretization-independent solution of the incremental fatigue-debonding problem

摘录 / EXCERPT: For the thin-film fatigue-debonding incremental problem defined by ... ... with $v_0=\delta_0=0$, the minimization problem $E_i(v_i)=\min_{v\in V_i}E_i(v)$ has a unique solution at eve...

open_problem | open | MathDB 页面 | 原文线索

V2-074477 | MATHDB-393164

Completeness of the elastic-invariant defect densities

摘录 / EXCERPT: For two crystal states defined on the same body B, if the defect-density fields in list (2), namely b_a, α_{ab}, n, $\nabla(\alpha_{ab}/n)\cdot d_c$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-074478 | MATHDB-393165

Neutral deformations preserving elastic invariants

摘录 / EXCERPT: Are there non-elastic deformations that leave the elastic invariant integrals unchanged, and, if so, what form do such deformations have?

open_problem | open | MathDB 页面 | 原文线索

V2-074479 | MATHDB-393166

Generalizing COET to realistic molecular interaction models

摘录 / EXCERPT: How should the consistently ordered extended thermodynamics (COET) method be generalized from Maxwell-molecule and BGK models to more realistic molecular interaction models?

open_problem | open | MathDB 页面 | 原文线索

V2-074480 | MATHDB-393168

Decidability and reconstruction of Bacher-matrix classes

摘录 / EXCERPT: Can one decide whether a given integral matrix is the Bacher matrix for some class? How to construct this class if it exists?

open_problem | open | MathDB 页面 | 原文线索

V2-074481 | MATHDB-393170

Exact linear-programming auxiliary functions for the Leech lattice and E_8

摘录 / EXCERPT: Determine whether there exists an admissible function $f:\mathbb{R}^n\to\mathbb{R}$, not identically zero, satisfying … with $(n,r)=(24,2)$, respectively $(n,r)=(8,\sqrt{2})$, exac...

open_problem | open | MathDB 页面 | 原文线索

V2-074482 | MATHDB-393171

Exact Cohn-Elkies auxiliary function in dimension 24

摘录 / EXCERPT: Does there exist an admissible function $f:\mathbb{R}^{24}\to\mathbb{R}$, not identically zero, satisfying the criteria of Theorem 4 with $r=2$: (1) $f(0)=\widehat f(0)>0$; (2)...

open_problem | open | MathDB 页面 | 原文线索

V2-074483 | MATHDB-393172

Incomplete computation of $K_7(\mathbb{Z})$

摘录 / EXCERPT: The report states that the computations concerning $K_7(\mathbb{Z})$ are still incomplete. It expects that the perfect forms of rank 7 do not contribute to the odd torsion of...

open_problem | open | MathDB 页面 | 原文线索

V2-074484 | MATHDB-393174

Projective-drop bound (Katz’s Conjecture (2.7.4))

摘录 / EXCERPT: Let $V= … Moreover, equality occurs precisely when G is the Weyl group $W(E8)$ acting on its 8-dimensional reflection representation.

conjecture | open | MathDB 页面 | 原文线索

V2-074485 | MATHDB-393175

The lattice covering problem in every dimension

摘录 / EXCERPT: For each integer $d\ge 1$, let B_d be the Euclidean unit ball in $\mathbb{R}^d$. For a full-rank discrete subgroup (lattice) $L\subset\mathbb{R}^d$, define … where $\det(L)$ is t...

open_problem | open | MathDB 页面 | 原文线索

V2-074486 | MATHDB-393176

Optimality of the Leech lattice sphere covering in dimension 24

摘录 / EXCERPT: Conjecture: The Leech lattice gives the optimal sphere covering in dimension 24.

open_problem | open | MathDB 页面 | 原文线索

V2-074487 | MATHDB-393179

Invariance of the size of admissible arithmetic extensions under base change

摘录 / EXCERPT: Let K be a number field, let F and G be hermitian vector bundles on $S=\operatorname{Spec}\mathcal O_K$, and let E be an admissible extension of F by G. For a finite ex...

open_problem | open | MathDB 页面 | 原文线索

V2-074488 | MATHDB-393180

Basis problem for Siegel cusp forms of squarefree level

摘录 / EXCERPT: Assume that N is squarefree, that χ is a quadratic character satisfying $\chi(-1)=(-1)^k$, and that $k\geq 2n+1$. Then all cusp forms of degree n, weight k, and characte...

open_problem | open | MathDB 页面 | 原文线索

V2-074489 | MATHDB-393183

Global solution of the regularized total least-squares problem

摘录 / EXCERPT: The report formulates the regularized total least-squares problem as … where Q is positive definite. It then states that a global optimal solution can be found efficiently, using...

open_problem | open | MathDB 页面 | 原文线索

V2-074490 | MATHDB-393184

Convex reduction of the regularized total least-squares subproblem

摘录 / EXCERPT: The report considers the convex problem … It states that, by convex duality, a solution can be obtained by solving at most two single-variable convex optimization problems.

open_problem | open | MathDB 页面 | 原文线索

V2-074491 | MATHDB-393185

L^1-optimal Dirichlet boundary control of a string to rest in finite time

摘录 / EXCERPT: Given $T>0$, $L>0$, $c>0$, and initial data y_0,y_1, consider … The report states that, under $T\ge t_0=L/c$, $y_0\in L^1(0,L)$, and $Y_1(x)=\int_0^x y_1(s)\,ds\in L^1(0,L)$, The...

open_problem | open | MathDB 页面 | 原文线索

V2-074492 | MATHDB-393186

Canonical perturbations do not imply Kojima strong stability

摘录 / EXCERPT: The implication that local single-valuedness and continuity of the stationary solution map under canonical perturbations imply strong stability in Kojima’s sense is false. The repo...

open_problem | open | MathDB 页面 | 原文线索

V2-074493 | MATHDB-393187

Nearest stable polynomial near s^k

摘录 / EXCERPT: Does every degree-k polynomial $p(s)$ near the polynomial s^k have a unique nearest stable polynomial?

open_problem | open | MathDB 页面 | 原文线索

V2-074494 | MATHDB-393190

Chromatic number at the exact minimum-degree threshold for triangle-free graphs

摘录 / EXCERPT: What is the maximum possible chromatic number of a triangle-free graph on n vertices whose minimum degree is exactly $n/3$?

open_problem | open | MathDB 页面 | 原文线索

V2-074495 | MATHDB-393192

Classification of graphs with bounded minimum rank over arbitrary fields

摘录 / EXCERPT: Let F be a field, let G be a finite undirected simple graph, and let $S(F,G)$ be the set of symmetric matrices over F whose off-diagonal nonzero entries occur exactly in the...

open_problem | open | MathDB 页面 | 原文线索

V2-074496 | MATHDB-393193

Everett-Reed characterization of perfectly contractile graphs

摘录 / EXCERPT: A graph is perfectly contractile if and only if it contains no odd hole, no antihole, and no odd prism.

open_problem | open | MathDB 页面 | 原文线索

V2-074497 | MATHDB-393194

Real roots of the stable set polynomial of a clawfree graph

摘录 / EXCERPT: For a graph G, let $S(G)(x)=\sum_{i\ge 0} a_i x^i$, where a_i is the number of stable sets of G of size i. If G is clawfree, meaning that it has no induced claw, then all...

open_problem | open | MathDB 页面 | 原文线索

V2-074498 | MATHDB-393200

Wide chromatic number of orientable surfaces

摘录 / EXCERPT: For every integer $g\ge 0$, the wide chromatic number of the orientable surface S_g satisfies $\chi_w(S_g)=4$.

open_problem | open | MathDB 页面 | 原文线索

V2-074499 | MATHDB-393201

Grünbaum’s face-width conjecture for cubic graphs on orientable surfaces

摘录 / EXCERPT: Conjecture 2. For any embedding of a cubic graph in an orientable surface S_g such that $\operatorname{ew}(G^*)\ge 3$, we have $\phi_c(G)=4$. Equivalently, every cubic graph embe...

conjecture | open | MathDB 页面 | 原文线索

V2-074500 | MATHDB-393204

Hadwiger numbers of lifts of complete graphs

摘录 / EXCERPT: Do there exist lifts of the complete graphs K_r in which the Hadwiger number is $o(r)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074501 | MATHDB-393206

Higher-genus drawing characterization of Pfaffian graphs

摘录 / EXCERPT: For a graph G and a non-negative integer g, the following are equivalent: (1) There exists a drawing of G on an orientable surface of genus g such that...

open_problem | open | MathDB 页面 | 原文线索

V2-074502 | MATHDB-393207

Mohar’s finiteness conjecture for 3-connected critical graphs

摘录 / EXCERPT: There are only finitely many 3-connected k-color-critical graphs without a K_k minor.

conjecture | open | MathDB 页面 | 原文线索

V2-074503 | MATHDB-393208

Recursive construction of 3-connected redundantly rigid graphs

摘录 / EXCERPT: Every 3-connected and redundantly rigid graph $G=(V,E)$ can be obtained from K_4 by a sequence of edge additions and 1-extensions.

open_problem | open | MathDB 页面 | 原文线索

V2-074504 | MATHDB-393209

Odd-\K_5-free graphs are 4-colourable

摘录 / EXCERPT: A graph G contains a graph H as an odd minor if H can be obtained from G by first deleting edges and then contracting every edge on some cut. A graph is odd-K_5-free if i...

open_problem | open | MathDB 页面 | 原文线索

V2-074505 | MATHDB-393210

Hendrickson’s conjecture for generic planar frameworks

摘录 / EXCERPT: Let (G,p) be a generic framework in $\mathbb{R}^2$. Then (G,p) is globally rigid if and only if either $G=K_2$ or $G=K_3$, or G is 3-connected and redundantly rigid.

conjecture | open | MathDB 页面 | 原文线索

V2-074506 | MATHDB-393212

The “Bellizing” unitary

摘录 / EXCERPT: Classify all unitary operators U that connect a fixed separable orthonormal basis to a Bell orthonormal basis.

open_problem | open | MathDB 页面 | 原文线索

V2-074507 | MATHDB-393213

Clean POVMs

摘录 / EXCERPT: Which POVMs are “undisturbed,” namely they are not irreversibly connected to another POVM via a channel? For POVMs P and Q, define $P\succ Q$ iff there exists a channel E suc...

open_problem | open | MathDB 页面 | 原文线索

V2-074508 | MATHDB-393214

Separability indicator criterion

摘录 / EXCERPT: For a Hermitian matrix A admitting tensor-product decompositions $A=\sum_{i=1}^{r}B_i\otimes C_i$ into Hermitian matrices, let $m(X)$ denote the smallest eigenvalue of X, defin...

open_problem | open | MathDB 页面 | 原文线索

V2-074509 | MATHDB-393216

Uniqueness of mixed-state entanglement under PPT-preserving operations

摘录 / EXCERPT: In the mixed-state setting of asymptotic entanglement manipulation under positive-partial-transpose-preserving (PPT-preserving) operations, it was conjectured that all entanglement...

open_problem | open | MathDB 页面 | 原文线索

V2-074510 | MATHDB-393218

Error estimate for the fully discrete mixed finite-element approximation of Richards' equation

摘录 / EXCERPT: Assuming that the flux belongs to $H^1(\Omega)^d$ rather than merely $H(\operatorname{div},\Omega)$, the report states the estimate ...

open_problem | open | MathDB 页面 | 原文线索

V2-074511 | MATHDB-393219

L^∞-stability of the finite element method on irregular meshes

摘录 / EXCERPT: For the model problem $-\Delta u=f$ in a convex polygonal or polyhedral domain $\Omega\subset\mathbb{R}^d$, with $u=0$ on $\partial\Omega$, and continuous piecewise-linear (P_1)...

open_problem | open | MathDB 页面 | 原文线索

V2-074512 | MATHDB-393220

L^∞ stability of the P_1 finite element method on irregular meshes

摘录 / EXCERPT: For the Dirichlet problem $-\Delta u=f$ in a convex polygonal domain $\Omega\subset\mathbb R^2$, with $u=0$ on $\partial\Omega$, let R_h be the H_0^1-Ritz projection onto the c...

open_problem | open | MathDB 页面 | 原文线索

V2-074513 | MATHDB-393223

Maltsiniotis's additivity conjecture for derivator K-theory

摘录 / EXCERPT: For a derivator D, define D' by $D'(I)=D(1\times I)$, where 1 is the category $\bullet\to\bullet$. The two inclusions of the terminal one-point category into 1 induce natur...

conjecture | open | MathDB 页面 | 原文线索

V2-074514 | MATHDB-393230

Controlled wildness of the subcategory $S_4(\Lambda)$

摘录 / EXCERPT: Let Λ be a uniserial ring of length $n\ge 7$, and let k be its radical factor. Then the category $S_4(\Lambda)$ is controlled k-wild.

open_problem | open | MathDB 页面 | 原文线索

V2-074515 | MATHDB-393231

Transitivity of autoequivalences on spherical objects

摘录 / EXCERPT: Let E be a Calabi-Yau curve, i.e. $\omega_E\cong\mathcal O_E$, and let $\mathcal E\in D^b(\operatorname{Coh}E)$ be a spherical object, meaning a perfect complex satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-074516 | MATHDB-393239

Rational connectedness of moduli spaces of curves

摘录 / EXCERPT: For which values of $g\le 23$ is $\mathcal M_g$ rationally connected?

open_problem | open | MathDB 页面 | 原文线索

V2-074517 | MATHDB-393241

Green's conjecture for canonical syzygies

摘录 / EXCERPT: For a smooth curve X of genus g and Clifford index c, Green conjectured the vanishing $K_{g-c-1,1}(X,\omega_X)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-074518 | MATHDB-393242

Non-planar curves with Clifford index three less than gonality

摘录 / EXCERPT: It would be interesting to have an example of a non-planar curve of large genus with $\operatorname{Cliff}=\operatorname{gon}-3$ for which the Green conjecture be verified.

open_problem | open | MathDB 页面 | 原文线索

V2-074519 | MATHDB-393247

Asymptotic K-stability versus properness of the K-energy

摘录 / EXCERPT: Deduce the properness of the K-Energy from the assumption of asymptotic K-Stability.

open_problem | open | MathDB 页面 | 原文线索

V2-074520 | MATHDB-393249

Modularity criterion for Calabi-Yau threefolds containing elliptic ruled surfaces

摘录 / EXCERPT: Let X be a smooth projective threefold defined over $\bQ$, with $h^{3,0}(X)=h^{0,3}(X)=1$. Set $b:=h^{1,2}(X)$, and suppose that X contains birational elliptic ruled surfaces...

open_problem | open | MathDB 页面 | 原文线索

V2-074521 | MATHDB-393250

Unirational-component conjecture for the genus-2 Albanese family

摘录 / EXCERPT: Let M be the family in the moduli space of minimal surfaces of general type with $p_g=q=1$ and $K^2=3$. For the case in which the Albanese map is a genus-2 fibration, it was co...

conjecture | open | MathDB 页面 | 原文线索

V2-074522 | MATHDB-393253

Finite generation of a bigraded ring on the first Voronoi compactification

摘录 / EXCERPT: Is the bigraded subring ... finitely generated?

open_problem | open | MathDB 页面 | 原文线索

V2-074523 | MATHDB-393254

Plurigenera and intersection numbers of the first Voronoi compactification

摘录 / EXCERPT: (Characteristic zero.) Compute the plurigenera $P_n(A_F^g)$ and the top self-intersection number $(K_{A_F^g})^N$, where $N=g(g+1)/2$. More generally, find the intersection numbers...

open_problem | open | MathDB 页面 | 原文线索

V2-074524 | MATHDB-393256

Quasioptimal finite-element error estimates for Willmore flow of graphs

摘录 / EXCERPT: Suppose that the problem (2)-(4) has a smooth solution on the interval $[0,T]$. Then ... and ... Here (u_h,w_h) is the semidiscrete finite-element solution defined on page 17, and...

open_problem | open | MathDB 页面 | 原文线索

V2-074525 | MATHDB-393258

How to prove the stability of a nonlinear cascaded loop

摘录 / EXCERPT: Prove stability for nonlinear multicascaded nonlinear systems of the kind arising in reference-shaping filters for systems with input saturation. In particular, the report asks for...

open_problem | open | MathDB 页面 | 原文线索

V2-074526 | MATHDB-393259

Stability of nonlinear multicascaded systems with input saturation

摘录 / EXCERPT: Prove stability for the nonlinear multicascaded nonlinear systems arising in reference-shaping filters with input saturation. The report explains that, although an appropriate refe...

open_problem | open | MathDB 页面 | 原文线索

V2-074527 | MATHDB-393261

Nonlinear output regulation via nonlinear internal models

摘录 / EXCERPT: For the nonlinear system ... with exosystem $\dot w=s(w)$, compact admissible initial-condition sets W,Z,E, and Assumptions 1 and 2 of the report, construct an internal-model contr...

open_problem | open | MathDB 页面 | 原文线索

V2-074528 | MATHDB-393264

Choosing interpolation points for rational-Krylov model reduction

摘录 / EXCERPT: How to choose the interpolation points, so that the reduced system satisfies desired properties?

open_problem | open | MathDB 页面 | 原文线索

V2-074529 | MATHDB-393266

Non-adaptive polynomial stabilization of linear minimum-phase systems with a known upper bound on the relative degree

摘录 / EXCERPT: Consider the single-input, single-output systems ... where the entries of $A\in\mathbb{R}^{n\times n}$, $b,c^\top,x_0\in\mathbb{R}^n$ are unknown and satisfy: - for a known...

open_problem | open | MathDB 页面 | 原文线索

V2-074530 | MATHDB-393267

When is a Linear System Optimal?

摘录 / EXCERPT: Characterize the behaviors $B\in\mathcal{L}^w$ that are stationary or optimal. Equivalently, determine the conditions on $B\in\mathcal{L}^w$ under which there exists a polynomial m...

open_problem | open | MathDB 页面 | 原文线索

V2-074531 | MATHDB-393269

2-Moufang generalized polygons are Moufang

摘录 / EXCERPT: For every generalized n-gon Γ and every group $G\leq\operatorname{Aut}(\Gamma)$, if Γ is 2-Moufang with respect to G-that is, for every simple path...

open_problem | open | MathDB 页面 | 原文线索

V2-074532 | MATHDB-393272

Divisibility of automorphism-group order for finite non-cyclic p-groups

摘录 / EXCERPT: The conjecture that, for a finite non-cyclic p-group G with $|G|\ge p^3$, one has $|G|\mid|\operatorname{Aut}(G)|$.

open_problem | open | MathDB 页面 | 原文线索

V2-074533 | MATHDB-393273

Automorphisms of Suzuki-Tits Moufang buildings of rank one

摘录 / EXCERPT: Let $(X,(U_x)_{x\roughlyin X})$ be a Moufang line, with $G^\dagger=\langle U_x\mid x\in X\rangle$, and let $V_x\neq U_x$ be a nontrivial subgroup of U_x that is normal in...

open_problem | open | MathDB 页面 | 原文线索

V2-074534 | MATHDB-393274

Automorphism group of a rank-one Moufang building

摘录 / EXCERPT: For a proper Moufang building of rank one (X,Λ) arising from a Moufang line with little projective group $G^{\dagger}$, is...

open_problem | open | MathDB 页面 | 原文线索

V2-074535 | MATHDB-393280

Classification of finite simple groups of sporadic even type

摘录 / EXCERPT: Theorem C^+_5 (open in the report). Let G be a finite simple group all of whose proper simple sections are of known type. Suppose that G is of sporadic even type. Then either G...

open_problem | open | MathDB 页面 | 原文线索

V2-074536 | MATHDB-393281

Completion of the proof of Theorem C^+_6 for meager finite simple groups

摘录 / EXCERPT: Establish Theorem C^+_6: if G is a finite simple group all of whose proper simple sections are known and G is of meager even type, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-074537 | MATHDB-393282

Counting conjugacy classes of generated primitive subgroups

摘录 / EXCERPT: There exists a constant c such that the number of conjugacy classes of d-generated primitive subgroups of $\operatorname{Sym}(n)$ is at most n^{cd}.

open_problem | open | MathDB 页面 | 原文线索

V2-074538 | MATHDB-393284

Corollary on automorphisms of simple groups

摘录 / EXCERPT: Suppose that G is simple, R is abelian and acts faithfully on G, $p>3$, and $F^*(C_G(\alpha))$ is a p-group for every $\alpha\in R^\#$. Then R is cyclic.

open_problem | open | MathDB 页面 | 原文线索

V2-074539 | MATHDB-393287

Strict upper bound for the minimum subdegree of twisted-wreath-type primitive groups

摘录 / EXCERPT: Question 4. Is it true that, for all finite primitive permutation groups $G=\operatorname{Soc}(G)\rtimes P\leq H\wr S_k$ of O'Nan-Scott type TW, with P,H as in Theorem 3, the str…

open_problem | open | MathDB 页面 | 原文线索

V2-074540 | MATHDB-393289

Bounded elementary generation for universal lattices

摘录 / EXCERPT: For each integer $d\ge 3$, does the group $G_{d,1}:=\operatorname{SL}_d(\mathbb{Z}[x])$ have the bounded elementary generation property; that is, does there exist an integer N su…

open_problem | open | MathDB 页面 | 原文线索

V2-074541 | MATHDB-393290

Kazhdan property T for the universal lattices

摘录 / EXCERPT: For the groups $G_{d,k}:=\mathrm{SL}_d\bigl(\mathbb{Z}[x_1,\ldots,x_k]\bigr)$ described in the report, does $G_{d,k}$ have Kazhdan's property T?

open_problem | open | MathDB 页面 | 原文线索

V2-074542 | MATHDB-393292

Generation by root groups for Moufang quadrangles of type E8

摘录 / EXCERPT: Let K be an arbitrary commutative field, let E/K be a separable quadratic extension with norm N, and let q be an anisotropic quadratic form of type $E8$, meaning that, up t…

open_problem | open | MathDB 页面 | 原文线索

V2-074543 | MATHDB-393293

Equality in the leading-exponent bound for spherical-function singularities

摘录 / EXCERPT: We conjecture that the inequality $n_X^\omega \le n_\tau(m_X)$ is an equality for generic $\lambda\in a_\mathbb{C}^*$, where the notation is as defined in the surrounding discussio…

open_problem | open | MathDB 页面 | 原文线索

V2-074544 | MATHDB-393294

Saturation factor for simply laced and non-simply laced groups

摘录 / EXCERPT: Conjecture: In Theorem 2, one can take $k=1$ for simply laced groups and $k=2$ in the non-simply laced case.

open_problem | open | MathDB 页面 | 原文线索

V2-074545 | MATHDB-393295

Conjecture on a splitting bundle and the induced equivalence

摘录 / EXCERPT: For fixed $c\in Q_{\mathrm{good}}$, where $Q_{\mathrm{good}}=\{c\in\mathbb{Q}:c\ge 0\text{ and }c\notin \tfrac12+\mathbb{Z}\}$, Conjecture 1 asserts: (a) One can choose the splitti…

conjecture | open | MathDB 页面 | 原文线索

V2-074546 | MATHDB-393297

Richardson elements in type E_n parabolics and stabilizer equality

摘录 / EXCERPT: Conjecturally, if $G=E_n$, then every parabolic subgroup P having a Richardson element $x\in\mathfrak g_1$ satisfies $G_x=P_x$.

open_problem | open | MathDB 页面 | 原文线索

V2-074547 | MATHDB-393299

Subpolynomial growth of normalized triple-period coefficients

摘录 / EXCERPT: For any $\epsilon>0$, one has $d_i\ll|\lambda_i|^{\epsilon}$, where $d_i=\gamma(\lambda_i)|c_i|^2$ is the normalized triple-period coefficient defined on page 50.

open_problem | open | MathDB 页面 | 原文线索

V2-074548 | MATHDB-393302

Approximate fixed point property in product spaces

摘录 / EXCERPT: If (X,ρ) and (Y,d) are metric spaces having the approximate fixed point property (AFPP) for nonexpansive mappings, then when does the product metric space…

open_problem | open | MathDB 页面 | 原文线索

V2-074549 | MATHDB-393303

Properness of the constructive arithmetical hierarchy

摘录 / EXCERPT: The report states that T proves, for every $n\in\omega$, $\Pi^0_n\ne\Delta^0_{n+1}\ne\Sigma^0_{n+1}$ and $\Sigma^0_n\ne\Delta^0_{n+1}\ne\Pi^0_{n+1}$, for the constructive arithme…

open_problem | open | MathDB 页面 | 原文线索

V2-074550 | MATHDB-393304

Consistency of New Foundations (NF)

摘录 / EXCERPT: Is New Foundations (NF) consistent relative to any theory in which we have reasonable confidence?

open_problem | open | MathDB 页面 | 原文线索

V2-074551 | MATHDB-393306

Limiting covariance of traces of Wishart matrices

摘录 / EXCERPT: For the complex Wishart matrices described in the report, with asymptotic ratio $c=M/N$, the limiting covariance is … where $S_{NC}(m,n)$ is the set of non-crossing (m,n)-annular…

open_problem | open | MathDB 页面 | 原文线索

V2-074552 | MATHDB-393307

Finite free entropy does not force generation of a free group factor

摘录 / EXCERPT: Is it true that a set of noncommutative random variables with finite free entropy necessarily generate a free group factor?

open_problem | open | MathDB 页面 | 原文线索

V2-074553 | MATHDB-393308

Finite free entropy and the Haagerup approximation property

摘录 / EXCERPT: Does finite free entropy together with the Haagerup approximation property suffice to ensure that one is looking at a free group factor?

open_problem | open | MathDB 页面 | 原文线索

V2-074554 | MATHDB-393310

Atiyah conjecture for torsion-free groups

摘录 / EXCERPT: If Γ is a torsion-free group, then all L^2-Betti numbers of finite CW-complexes with fundamental group Γ are integers.

conjecture | open | MathDB 页面 | 原文线索

V2-074555 | MATHDB-393311

Limit shape of a subtableau of a random square Young tableau

摘录 / EXCERPT: Fix $0<\alpha<1$. Consider a randomly chosen Young tableau of the $n\times n$ square, filled with the numbers $1,\ldots,n^2$ so that the numbers increase along the prescribed diago...

open_problem | open | MathDB 页面 | 原文线索

V2-074556 | MATHDB-393316

Minimum spanning tree bound in the unit disk

摘录 / EXCERPT: Let S be a set of points in the unit disk centered at the origin, with the origin in S. Let $e_1,e_2,\\ldots,e_{|S|-1}$ be the edges of the Euclidean minimum spanning tree of...

open_problem | open | MathDB 页面 | 原文线索

V2-074557 | MATHDB-393317

Asymptotic maximum number of geometric permutations for disjoint translates in \(\mathbb R^3\)

摘录 / EXCERPT: Determine the asymptotic growth, as $n\to\infty$, of the maximum number of geometric permutations induced by line transversals of a family of n pairwise disjoint translates of a...

open_problem | open | MathDB 页面 | 原文线索

V2-074558 | MATHDB-393318

Maximum number of vertices in a convex lattice polygon contained in a planar convex body

摘录 / EXCERPT: Let $K\subset\mathbb{R}^2$ be a convex body, let $\mathbb{Z}_t=t^{-1}\mathbb{Z}^2$ for t large, and let $\mathcal{P}(K;t)$ be the family of all convex $\mathbb{Z}_t$-lattice poly...

open_problem | open | MathDB 页面 | 原文线索

V2-074559 | MATHDB-393319

Maximal convex lattice polygons inscribed in a plane convex set

摘录 / EXCERPT: Given a convex body $K\subset\mathbb{R}^2$, let $Z_t=t^{-1}\mathbb{Z}^2$ and let $P(K;t)$ be the set of all convex Z_t-lattice polygons contained in K. Define $A(K)$ as the sup...

open_problem | open | MathDB 页面 | 原文线索

V2-074560 | MATHDB-393320

Axially symmetric planar sections of convex bodies in \(\mathbb{R}^3\)

摘录 / EXCERPT: Conjecture: Let $K\subset\mathbb{R}^3$ be a convex body. Then K is a solid of revolution or an ellipsoid if and only if every planar section of K is axially symmetric.

open_problem | open | MathDB 页面 | 原文线索

V2-074561 | MATHDB-393321

Lebesgue’s universal cover problem

摘录 / EXCERPT: Determine the minimum area of a convex set U in the plane such that, for every set C of diameter 1, U contains a congruent copy C' of C.

open_problem | open | MathDB 页面 | 原文线索

V2-074562 | MATHDB-393322

Universal optimality conjecture for three Euclidean lattices

摘录 / EXCERPT: The analogous linear programming bounds for potential-energy minimization in Euclidean space prove universal optimality for the hexagonal lattice in $\mathbb{R}^2$, the E_8 root...

conjecture | open | MathDB 页面 | 原文线索

V2-074563 | MATHDB-393323

Characterization of 3-realizable graphs

摘录 / EXCERPT: A graph G is 3-realizable if and only if it has neither K_5 nor the edge graph (1-skeleton) of the octahedron as a minor.

open_problem | open | MathDB 页面 | 原文线索

V2-074564 | MATHDB-393325

Generalizing Scarf’s theorem to non-generic orthogonal surfaces in higher dimensions

摘录 / EXCERPT: For dimensions higher than 3, find a set of conditions on a non-generic orthogonal surface that is less restrictive than genericity and nevertheless allows a generalization of Sc...

open_problem | open | MathDB 页面 | 原文线索

V2-074565 | MATHDB-393326

Few-distance sets in normed spaces

摘录 / EXCERPT: For a normed space $N^d=(\mathbb{R}^d,\|\cdot\|)$, let $f(N^d,k)$ be the maximum size of a set having at most k distinct nonzero pairwise distances. It is conjectured that ...

open_problem | open | MathDB 页面 | 原文线索

V2-074566 | MATHDB-393327

Erdős’s conjectured bound for few-distance sets in the Euclidean plane

摘录 / EXCERPT: Erdős (1946) conjectured that the $\sqrt{n}\times\sqrt{n}$ lattice gives the minimum number of distinct distances for n-point sets in the plane; in the notation of the report, th...

conjecture | open | MathDB 页面 | 原文线索

V2-074567 | MATHDB-393328

Exact determination of Tverberg’s separation number

摘录 / EXCERPT: For each positive integer k, determine the minimal positive integer $s(k)$ such that every collection of at least $s(k)$ nonempty convex sets in the plane with pairwise disjoint...

open_problem | open | MathDB 页面 | 原文线索

V2-074568 | MATHDB-393329

The g-vector conjecture for doubly Cohen-Macaulay complexes

摘录 / EXCERPT: The g-vector of any 2-CM complex is an M-sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-074569 | MATHDB-393331

Edge bound for geometric graphs with no pair of parallel edges

摘录 / EXCERPT: A geometric graph is a graph drawn in the plane with its vertices as points and its edges as straight-line segments connecting corresponding points. Two edges are parallel if they...

open_problem | open | MathDB 页面 | 原文线索

V2-074570 | MATHDB-393332

Local or global optimality of the root lattice E_8 for the packing-covering problem

摘录 / EXCERPT: Is the root lattice E_8 a local minimum, or even a global minimum, of the packing-covering constant $\gamma(L)=\rho(L)/\lambda(L)$ among lattices L of rank 8, as conjectured...

open_problem | open | MathDB 页面 | 原文线索

V2-074571 | MATHDB-393333

Optimal universal constant for the biplanar crossing number

摘录 / EXCERPT: What is the smallest constant c such that, for every graph G, $\operatorname{cr}_2(G) \le c\,\operatorname{cr}(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074572 | MATHDB-393334

Unit-distance problem for translates of a convex body in the plane

摘录 / EXCERPT: Let $t_2(n)$ be the maximum number of pairs among n pairwise disjoint translates of one convex compact set in $\boldsymbol{R}^2$ that have mutual distance 1. The Erdős-Pach con...

open_problem | open | MathDB 页面 | 原文线索

V2-074573 | MATHDB-393335

Optimal static-output-feedback H^∞ control design

摘录 / EXCERPT: For the finite-dimensional control system described by equations (1)-(2) on the source page, with given system matrices and static output-feedback law $u(t)=Fy(t)$, find...

open_problem | open | MathDB 页面 | 原文线索

V2-074574 | MATHDB-393337

Exponential stability of a plate equation under Dirichlet boundary stabilization

摘录 / EXCERPT: The report states the following theorem. Let $\Omega=(0,\pi)\times(0,\pi)$ and let Γ be an open subset of $\partial\Omega$. Consider ... where ν is the outer unit normal a...

open_problem | open | MathDB 页面 | 原文线索

V2-074575 | MATHDB-393339

First-order necessary condition for the abstract convex-control problem

摘录 / EXCERPT: Consider the abstract problem (1), namely minimizing $f(u)$ subject to $u\in U_{\mathrm{ad}}\subset L^2(Q)^n$, where $f:L^2(Q)^n\to\mathbb{R}$ is twice Fréchet differentiable and...

open_problem | open | MathDB 页面 | 原文线索

V2-074576 | MATHDB-393344

Square-free initial ideals of minors of a linear matrix

摘录 / EXCERPT: Let t_{ij} be distinct variables over a field K, with $1\le i\le m$ and $1\le j\le n$. Let ... be an $m\times n$ matrix whose entries are linear forms ... Denote by $I_2(L)$ the id...

open_problem | open | MathDB 页面 | 原文线索

V2-074577 | MATHDB-393345

Square-free multigraded initial ideals of 2-minor ideals of row-graded matrices

摘录 / EXCERPT: Let t_{ij} be distinct variables over a field K, with $1\le i\le m$ and $1\le j\le n$. Let $L=(L_{ij})$ be an $m\times n$ matrix with ... Denote by $I_2(L)$ the ideal of the 2-...

open_problem | open | MathDB 页面 | 原文线索

V2-074578 | MATHDB-393346

One-root factorization conjecture for elementary matrices over monoid rings

摘录 / EXCERPT: Let R be a Dedekind domain, let M be a submonoid of a rational vector space, let $n\ge 3$ and $c\ge 2$, and let $A\in \operatorname{SL}_n(R[M])$. The conjecture is that...

conjecture | open | MathDB 页面 | 原文线索

V2-074579 | MATHDB-393348

Glicci property of 20 points in general position in $\mathbb{P}^3$

摘录 / EXCERPT: Is a set of 20 points in general position in $\mathbb{P}^3$ glicci?

open_problem | open | MathDB 页面 | 原文线索

V2-074580 | MATHDB-393350

Factorization of the adjoint of a generic matrix

摘录 / EXCERPT: Let k be a field, let $X=(x_{ij})$ be the generic $n\times n$ matrix over k, and put $S=k[x_{ij}]$. Determine whether the classical adjoint $\operatorname{adj}(X)$ can be writt...

open_problem | open | MathDB 页面 | 原文线索

V2-074581 | MATHDB-393352

Classification of extensions arising from adjoint factorizations

摘录 / EXCERPT: For $n\ge 4$, let k be a field, let $X=(x_{ij})$ be the generic $n\times n$ matrix, set $S=k[x_{ij}]$, $R=S/(\det X)$, $L=\operatorname{coker}X$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-074582 | MATHDB-393353

Jacobian conjecture for the complex affine plane

摘录 / EXCERPT: Let $\c6:\bb A^2_{\bb C}\to\bb A^2_{\bb C}$ be an étale endomorphism. Is $\c6$ an isomorphism?

conjecture | open | MathDB 页面 | 原文线索

V2-074583 | MATHDB-393354

Set-theoretic Cohen-Macaulayness of the fourth Veronese-type curve in characteristic zero

摘录 / EXCERPT: Let K be a field of characteristic 0, let $R=K[w,x,y,z]$, and consider the K-linear ring homomorphism … that sends w,x,y,z to s^4,s^3t,st^3,t^4, respectively. Let…

open_problem | open | MathDB 页面 | 原文线索

V2-074584 | MATHDB-393357

Uniform control across Ricci-flow surgeries

摘录 / EXCERPT: Establish a self-contained exposition of the uniform control of a fixed scale $\rho>0$ past all surgeries on a finite time interval, below which Perelman’s approximation theorem ap…

open_problem | open | MathDB 页面 | 原文线索

V2-074585 | MATHDB-393358

General steady-state selection for driven systems with multiple conservation laws

摘录 / EXCERPT: Find a general rule for steady-state selection in finite one-dimensional driven diffusive systems with open boundaries and more than one conserved density.

open_problem | open | MathDB 页面 | 原文线索

V2-074586 | MATHDB-393360

Percolating paths through random Poisson points

摘录 / EXCERPT: Fix $d\ge 2$ and let (ξ_i) be a rate-1 Poisson point process in $\mathbb{R}^d$. For $s>0$, define W_s as the minimum, over all $m\ge 0$ and all choices of m distinct Pois…

open_problem | open | MathDB 页面 | 原文线索

V2-074587 | MATHDB-393361

Conjectured distinction between two flows of kernels

摘录 / EXCERPT: The report conjectures that the n-point motions, for $n \ge 4$, of the two flows of kernels under discussion do not agree, and that the two flows of kernels are different.

conjecture | open | MathDB 页面 | 原文线索

V2-074588 | MATHDB-393362

Continuous scaling limit of random planar quadrangulations

摘录 / EXCERPT: For uniformly chosen pointed quadrangulations with n faces, and for rooted quadrangulations with the distribution P_n^D specified in the report, does rescaling graph distances…

open_problem | open | MathDB 页面 | 原文线索

V2-074589 | MATHDB-393364

Error estimate for the compressed up-scaled operator

摘录 / EXCERPT: The report states that there exist constants $\alpha,C_m>0$ such that the coarse-mesh solution $u^m\in V_h$, defined by $\mathcal U_h a[\varphi_i,u^m]=(\varphi_i,g)_{L^2(\Omega)}$…

open_problem | open | MathDB 页面 | 原文线索

V2-074590 | MATHDB-393365

Classification of scaling limits for finitely many interfaces

摘录 / EXCERPT: Classify the possible scaling limits for a finite number of interfaces in critical planar statistical-mechanical models. The report considers configurations in which each interface…

open_problem | open | MathDB 页面 | 原文线索

V2-074591 | MATHDB-393366

Alexander-Orbach conjecture for the incipient infinite cluster

摘录 / EXCERPT: For the incipient infinite cluster $\widetilde{\mathcal C}$ in $\mathbb{Z}^d$, for every $d\ge 2$, the transition density satisfies $q_t(x,x)\asymp t^{-2/3}$ as $t\to\infty$. Equiv…

conjecture | open | MathDB 页面 | 原文线索

V2-074592 | MATHDB-393367

Alexander-Orbach walk-dimension conjecture for the incipient infinite cluster

摘录 / EXCERPT: For the incipient infinite cluster $\widetilde{\mathcal C}$ in $\mathbb Z^d$, for every $d\ge 2$, the Alexander-Orbach conjecture is …

conjecture | open | MathDB 页面 | 原文线索

V2-074593 | MATHDB-393368

Transversal (volume) exponent for a 1+1-dimensional directed polymer

摘录 / EXCERPT: Conjecture: For the directed polymer model described in the report, the volume exponent satisfies $\xi=\frac{2}{3}$ when $d=1$ and $\beta>0$ is small.

open_problem | open | MathDB 页面 | 原文线索

V2-074594 | MATHDB-393371

Planar Brownian intersection exponent formula

摘录 / EXCERPT: For the planar Brownian intersection exponents defined in the report, Lawler, Schramm, and Werner proved that … for the integer package sizes M,N occurring in the preceding defin…

open_problem | open | MathDB 页面 | 原文线索

V2-074595 | MATHDB-393372

Existence and uniqueness of the truncated spherical PML problem

摘录 / EXCERPT: Let g admit an $H(\operatorname{curl};\Omega_1)$ extension $\widetilde g$ supported in Ω_1, where Ω_1 is the ball of radius r_1 with the scatterer Ω remove…

open_problem | open | MathDB 页面 | 原文线索

V2-074596 | MATHDB-393373

Uniqueness for the truncated PML Maxwell problem

摘录 / EXCERPT: For R_t sufficiently large, the only $\boldsymbol{\theta}\\boldsymbol{\theta}\\boldsymbol{\theta}\\Theta\in H_0(\operatorname{curl};\Omega_\infty)$ satisfying … is $\Theta=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074597 | MATHDB-393374

Approximation of the right-hand side for solutions in the approximation class A^s

摘录 / EXCERPT: For $s\in(0,\tfrac12]$, conjecture that if the solution u belongs to the approximation class A^s, then the corresponding right-hand side f admits, for every n, a piecewise-…

open_problem | open | MathDB 页面 | 原文线索

V2-074598 | MATHDB-393376

Energy-norm convergence of multigrid methods for the Stokes problem on general polygonal domains

摘录 / EXCERPT: For the two-dimensional Stokes problem on a bounded polygonal domain $\Omega\subset\mathbb{R}^2$, with stable nested finite-element pairs...

open_problem | open | MathDB 页面 | 原文线索

V2-074599 | MATHDB-393378

Oscillatory behavior of the vacuum G_2 silent-boundary system

摘录 / EXCERPT: Prove that the vacuum G_2 silent-boundary (SB) system has oscillatory behavior analogous to the oscillatory behavior proved for the Bianchi IX system.

open_problem | open | MathDB 页面 | 原文线索

V2-074600 | MATHDB-393379

Strong cosmic censorship for T^3-Gowdy spacetimes

摘录 / EXCERPT: For vacuum T^3-Gowdy spacetimes, let $G_{l,m}$ denote the set of solutions with l non-degenerate true spikes and m non-degenerate false spikes satisfying the asymptotic condi...

open_problem | open | MathDB 页面 | 原文线索

V2-074601 | MATHDB-393381

Behavior of semilinear waves on Schwarzschild or Reissner-Nordström exteriors for $1+\\sqrt{2}\le p\le 4$

摘录 / EXCERPT: For the semilinear wave equation $\Box_g\phi=|\phi|^p$ on a Schwarzschild or Reissner-Nordström exterior background, determine the behavior of solutions in the range...

open_problem | open | MathDB 页面 | 原文线索

V2-074602 | MATHDB-393383

Undirected s-t connectivity in deterministic logarithmic space

摘录 / EXCERPT: Given an undirected graph G and two vertices s,t, decide whether s and t are connected by a path in G (the USTCON problem). The report presents a deterministic logarithmi...

open_problem | open | MathDB 页面 | 原文线索

V2-074603 | MATHDB-393384

Explicit two-source extractors at arbitrary constant min-entropy rate

摘录 / EXCERPT: Construct a polynomial-time computable, seedless extractor using two independent sources such that, for every constant $\b4>0$, it works for all pairs of independent distributions...

open_problem | open | MathDB 页面 | 原文线索

V2-074604 | MATHDB-393389

One-way functions or pseudorandom generators in NC^0

摘录 / EXCERPT: Are there one-way functions, or even pseudorandom generators, in NC^0?

open_problem | open | MathDB 页面 | 原文线索

V2-074605 | MATHDB-393391

Conjecture that \(\mathrm{NP}\cap\mathrm{coNP}\not\subseteq\mathrm{BPP}\)

摘录 / EXCERPT: The conjecture that $\mathrm{NP}\cap\mathrm{coNP}\not\subseteq\mathrm{BPP}$.

conjecture | open | MathDB 页面 | 原文线索

V2-074606 | MATHDB-393392

Basing one-way functions on worst-case NP-hardness via arbitrary reductions

摘录 / EXCERPT: Assuming that $ obreak\text{coNP}\subseteq\text{AM}$, is it impossible to base the existence of any one-way function on worst-case NP-hardness via any reduction?

open_problem | open | MathDB 页面 | 原文线索

V2-074607 | MATHDB-393395

Concurrent Composition Using Super-Polynomial Simulation

摘录 / EXCERPT: For any multi-party functionality, construct a secure protocol that remains secure when executed concurrently with multiple copies of itself and with other protocols, without assum...

open_problem | open | MathDB 页面 | 原文线索

V2-074608 | MATHDB-393396

Sublinear authentication and online memory checking

摘录 / EXCERPT: For a file of size n, if a user wants to authenticate it while reading only t bits, and stores a private randomized fingerprint of size s, then against an unbounded adversary...

open_problem | open | MathDB 页面 | 原文线索

V2-074609 | MATHDB-393405

Volume conjecture for small complex angles

摘录 / EXCERPT: Theorem 1. For every knot K there exists a neighborhood U_K of $0\in\mathbb{C}$ such that, for every $a\in U_K$, the limit ... exists. Moreover, convergence is uniform on compact...

conjecture | open | MathDB 页面 | 原文线索

V2-074610 | MATHDB-393406

Compatibility of mod-[U+007F]p[U+007F] reductions with the mod-[U+007F]p[U+007F] Langlands correspondence

摘录 / EXCERPT: With the notation of the report, Breuil’s conjecture asserted the compatibility ... where $E/\mathbf Q_p$ is finite, $k\ge 2$, $a_p\in E$ satisfies $\operatorname{val}(a_p)>0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074611 | MATHDB-393407

Compatibility of the mod-[U+007F]p correspondences for crystalline representations

摘录 / EXCERPT: For the parameters and reductions defined in the report, the conjectured compatibility is ... Here $E/\mathbb{Q}_p$ is finite, $k\ge 2$, $a_p\in E$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-074612 | MATHDB-393409

Automic representations with prescribed finite-level type

摘录 / EXCERPT: Let G be a connected reductive group over $ \mathbb{Q}$, let J be a compact open subgroup of the finite adeles of G, and let τ be an irreducible complex smooth represent...

open_problem | open | MathDB 页面 | 原文线索

V2-074613 | MATHDB-393411

Automorphic representations with prescribed finite-adelic type for \(\mathrm{GL}_n\)

摘录 / EXCERPT: For $G=\mathrm{GL}_n$ over $\mathbf Q$, with $n>2$, let $J\subset G(\mathbf A_f)$ be a compact open subgroup and let ρ be an irreducible complex smooth representation of J....

open_problem | open | MathDB 页面 | 原文线索

V2-074614 | MATHDB-393412

Image inclusion in an exceptional Fontaine-Mazur-type L-invariant case

摘录 / EXCERPT: In the exceptional case $m=d$ is even and $i+1=d/2$, prove that $\operatorname{Im}(\lambda_{\mathrm{FM},\pi})\subseteq\operatorname{Im}(\nu)$; the source further expects equality f...

open_problem | open | MathDB 页面 | 原文线索

V2-074615 | MATHDB-393417

Schneider’s conjecture on the Hodge and covering filtrations

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}_p$, let X be Drinfel’d’s symmetric space of dimension d, let $\Gamma\subset\mathrm{GL}_{d+1}(K)$ be a cocompact discrete torsion-fr...

conjecture | open | MathDB 页面 | 原文线索

V2-074616 | MATHDB-393418

Local-system conjecture for Frobenius invariants on the admissible locus

摘录 / EXCERPT: We conjecture that the c(cvarphic)-invariants c($M^{\cvarphi=1}$c) form a local system of c($\cmathbb{Q}_p$c)-vector spaces on c(F_ac).

conjecture | open | MathDB 页面 | 原文线索

V2-074617 | MATHDB-393422

Existence of ground states for nonlinear Schrödinger equations with potentials vanishing at infinity

摘录 / EXCERPT: Assume $n\ge 3$, $0<\alpha<2$, and $\beta>0$. Let V and K satisfy ... for some $a_1,a_2,a_3>0$. Define ... If $\sigma_{n,\alpha,\beta}<p<\dfrac{n+2}{n-2}$, then for every...

open_problem | open | MathDB 页面 | 原文线索

V2-074618 | MATHDB-393424

Critical-case extension of Theorem 5

摘录 / EXCERPT: Can one extend the result of Theorem 5 to the critical case $p=\frac{n+2m}{n-2m}$ or not? For $m=1$, results were obtained by Schaaf [4].

open_problem | open | MathDB 页面 | 原文线索

V2-074619 | MATHDB-393426

Antisymmetry of sign-changing solutions with Morse index two

摘录 / EXCERPT: If u has Morse index two and changes sign, is it also antisymmetric?

open_problem | open | MathDB 页面 | 原文线索

V2-074620 | MATHDB-393427

Peak solutions off solutions with interior layers

摘录 / EXCERPT: Construct peak solutions off solutions with interior layers.

open_problem | open | MathDB 页面 | 原文线索

V2-074621 | MATHDB-393428

Energy bounds and the dimension of the concentration set

摘录 / EXCERPT: Given a solution u_ε of (3) satisfying an energy bound $E_\varepsilon[u_\varepsilon]\le C\varepsilon^{N-m}$, can one conclude that its concentration set is m-dimensio...

open_problem | open | MathDB 页面 | 原文线索

V2-074622 | MATHDB-393431

Number of critical points of the Green function

摘录 / EXCERPT: The Green function G defined by equation (5) has at most five critical points.

open_problem | open | MathDB 页面 | 原文线索

V2-074623 | MATHDB-393432

Multiplicity of solutions for a critical Hardy-potential problem

摘录 / EXCERPT: For the problem ... where $\Omega\subset\mathbb{R}^N$ is not necessarily bounded, $0\in\Omega$, $1<p<N$, $\lambda<\bigl(\frac{N-p}{p}\bigr)^p$, and $f\in(D_0^{1,p}(\Omega))'$, descri...

open_problem | open | MathDB 页面 | 原文线索

V2-074624 | MATHDB-393433

Liouville conjecture for the superlinear heat equation

摘录 / EXCERPT: Let $p>1$. For the equation ... positive solutions exist if and only if $p(N-2)\ge N+2$.

conjecture | open | MathDB 页面 | 原文线索

V2-074625 | MATHDB-393435

Energy-space regularity of bounded sign-changing entire critical solutions

摘录 / EXCERPT: For $N\ge 3$, is every bounded sign-changing solution of ... contained in the energy space $D^{1,2}(\mathbb{R}^N)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074626 | MATHDB-393436

Symmetry of minimax critical points under cap symmetrization

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a ball or an annulus, and consider ... Assume that $f(|x|,\cdot)$ is continuous and satisfies $|f(|x|,s)|\le C(1+|s|^p)$, with $1\le p<(N+2)/(N-2)$...

open_problem | open | MathDB 页面 | 原文线索

V2-074627 | MATHDB-393437

The two-dimensional polynomial case of the Lazer-McKenna conjecture

摘录 / EXCERPT: For the Dirichlet problem $-\Delta u=g(u)-s\phi_1(x)$ in a smooth bounded domain $\Omega\subset\mathbb{R}^N$, with $u=0$ on $\partial\Omega$, $s>0$, and $g(t)=t_+^p+\lambda t$ wher...

conjecture | open | MathDB 页面 | 原文线索

V2-074628 | MATHDB-393439

Bahri’s discrete inequality for arbitrary numbers of points

摘录 / EXCERPT: Establish whether there exists a constant $c(p)>0$ such that, for all $u\in\mathbb{R}^p$ and all $a_1,\ldots,a_p\in\mathbb{R}^3$, with $A=(\alpha_{ij})_{i,j=1}^p$ defined by ... one...

open_problem | open | MathDB 页面 | 原文线索

V2-074629 | MATHDB-393440

Radial symmetry at the critical exponent for a uniformly elliptic maximal operator

摘录 / EXCERPT: Let p_A^* be the critical exponent found in Theorem 1. Is every nonnegative C^2 solution of ... radially symmetric?

open_problem | open | MathDB 页面 | 原文线索

V2-074630 | MATHDB-393441

Multiplicity conjecture for positive solutions of (P_λ)

摘录 / EXCERPT: We conjecture that (P_λ) has at least $\operatorname{cat}(\Omega_1)+\operatorname{cat}(\Omega_2)+\operatorname{cat}(\Omega_1)\times\operatorname{cat}(\Omega_2)$ positive so...

conjecture | open | MathDB 页面 | 原文线索

V2-074631 | MATHDB-393442

Conjecture that Theorem 6 holds with $b^*=1$

摘录 / EXCERPT: Assume (A) holds and $m(x)$ is non-negative. For every $c\in(c^*,1)$, the conclusion of Theorem 6 should remain valid with $b^*=1$: whenever $b\in(0,1)$ and $(\mu,\nu)\in\Sigma_c$,...

conjecture | open | MathDB 页面 | 原文线索

V2-074632 | MATHDB-393443

Boundedness of maximal operators associated with analytic hypersurfaces

摘录 / EXCERPT: Let $S\subset\mathbb{R}^{n+1}$ be an analytic hypersurface, let ψ be a fixed nonnegative smooth function with compact support on S, and define ... Let $A(X)$ be the matrix of...

open_problem | open | MathDB 页面 | 原文线索

V2-074633 | MATHDB-393444

Global well-posedness of KP-I in the natural energy space

摘录 / EXCERPT: Is the KP-I initial value problem ... on $\mathbb{R}\times\mathbb{R}$ globally well-posed for initial data ϕ in the natural energy space $Z^1(\mathbb{R}\times\mathbb{R})$?

open_problem | open | MathDB 页面 | 原文线索

V2-074634 | MATHDB-393445

Global well-posedness of the KP-I initial value problem in $Z^2(S\times\mathbb T)$

摘录 / EXCERPT: Let $\mathbb T=\mathbb R/(2\pi\mathbb Z)$ and let $S=\mathbb T$ or $S=\mathbb R$. For $g\in L^2(\mathbb T\times\mathbb T)$, define ... and, for $g\in L^2(\mathbb R\times\mathbb T)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074635 | MATHDB-393446

Matrix-weighted Triebel-Lizorkin wavelet characterization

摘录 / EXCERPT: Let W be a matrix weight, $0<p<\infty$, $0<q\leq\infty$, and $\alpha\in\mathbb{R}$. For the homogeneous matrix-weighted Triebel-Lizorkin space $\dot F^{\alpha,q}_p(W)$ and its di...

open_problem | open | MathDB 页面 | 原文线索

V2-074636 | MATHDB-393447

Conditions for equivalence of matrix-weighted sequence Triebel-Lizorkin norms

摘录 / EXCERPT: Let $0<p<\infty$. For the reducing operators A_Q associated with the matrix weight W, the equivalence ... holds if $W\in (RH)_p$ when $1<p<\infty$, and if $W\in A_p$ when...

open_problem | open | MathDB 页面 | 原文线索

V2-074637 | MATHDB-393448

Boundedness of maximal averages over monomial polyhedra

摘录 / EXCERPT: Let $O_n=\{x=(x_1,\ldots,x_n)\in\mathbb R^n:x_j>0,\ 1\le j\le n\}$. For $q=(q_1,\ldots,q_n)\in\mathbb R^n$, define $F_q(x)=x_1^{q_1}\cdots x_n^{q_n}$. Given...

open_problem | open | MathDB 页面 | 原文线索

V2-074638 | MATHDB-393449

Uniform Bergman-kernel estimates for the cross of iron domain

摘录 / EXCERPT: For the cross of iron domain $\Omega^\dagger=\{z\in\mathbb C^3:\operatorname{Re} z_3>Q(z)\}$, where $Q(z)=|z_1|^6+|z_2|^6+|z_1z_2|^2$, reconcile the two diagonal Bergman-kernel reg...

open_problem | open | MathDB 页面 | 原文线索

V2-074639 | MATHDB-393450

Endpoint strong-type estimate for a degenerate Radon transform

摘录 / EXCERPT: Let $n\ge 3$, let $S(x,y)=\sum_{k=1}^{n-1}a_kx^ky^{n-k}$ with $a_1\ne0$ and $a_{n-1}\ne0$, and let ψ be a smooth cutoff supported near the origin in $\mathbb{R}^3$. Define ... T...

open_problem | open | MathDB 页面 | 原文线索

V2-074640 | MATHDB-393451

Endpoint L^p-L^q estimates for analytic degenerate Radon transforms

摘录 / EXCERPT: Let $S(x,y)=\sum_{\alpha,\beta>0}a_{\alpha,\beta}x^\alpha y^\beta$ be a real-analytic phase, and let ... If Δ is the closed convex hull of...

open_problem | open | MathDB 页面 | 原文线索

V2-074641 | MATHDB-393452

Generalization of the Carleson-Hunt theorem for translated multilinear symbols

摘录 / EXCERPT: Let $m\in L^\infty(\mathbb R^n)$ be smooth away from the origin and satisfy ... for sufficiently many multi-indices α. For $N\in\mathbb R^n$, define...

open_problem | open | MathDB 页面 | 原文线索

V2-074642 | MATHDB-393454

Absence of positive eigenfunctions under Koch-Tataru potential assumptions

摘录 / EXCERPT: Let V and W be potentials on $ ... and ... must satisfy $uequiv0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074643 | MATHDB-393455

Mixed boundary problem without the crease condition

摘录 / EXCERPT: Determine whether the mixed boundary value problem for Laplace’s equation can be solved without Brown’s crease condition. In the report’s setting,...

open_problem | open | MathDB 页面 | 原文线索

V2-074644 | MATHDB-393456

Variable-coefficient oscillatory-integral estimates in the Bochner-Riesz range

摘录 / EXCERPT: Let $T_\lambda f(z)=\int e^{i\lambda\phi(z,y)}a(z,y)f(y)\,dy$, where $(z,y)\in\mathbb{R}^{n+1}\times\mathbb{R}^n$, $n\ge 1$, $a\in C_0^\infty(\mathbb{R}^{2n+1})$, and ϕ is smo...

open_problem | open | MathDB 页面 | 原文线索

V2-074645 | MATHDB-393460

Chaotic second-species orbits in the elliptic three-body problem

摘录 / EXCERPT: Theorem 1 (Bolotin). For every $h\in(-3/2,\sqrt{2})$, there is a dense set $\{G_k\}_{k\in L}\subset(2-\sqrt{4h+6},-h)$ such that, for every finite set $K\subset L$, there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-074646 | MATHDB-393461

Four-body central configurations with zero total mass

摘录 / EXCERPT: For any set of masses $(x,-x,y,-y)$, where $x,y\ne 0$, there are exactly two non-collinear central configurations with $\xi\ne 0$. They are trapezia.

open_problem | open | MathDB 页面 | 原文线索

V2-074647 | MATHDB-393462

Continuum of five-body central configurations with vanishing total mass

摘录 / EXCERPT: Does there exist a continuum of five-body central configurations with total mass $M=m_1+m_2+m_3+m_4+m_5=0$?

open_problem | open | MathDB 页面 | 原文线索

V2-074648 | MATHDB-393463

Finiteness of central configurations

摘录 / EXCERPT: Is the number of central configurations, up to isometries and homotheties, finite?

open_problem | open | MathDB 页面 | 原文线索

V2-074649 | MATHDB-393472

Ulam’s floating-body problem

摘录 / EXCERPT: Is the round ball the only solid of uniform density that floats in equilibrium in all positions?

open_problem | open | MathDB 页面 | 原文线索

V2-074650 | MATHDB-393473

Actual deformations of bicycle polygons

摘录 / EXCERPT: For infinitesimal deformations of regular n-gons as bicycle (n,k)-gons, a non-trivial deformation exists if and only if ... for some $2\le r\le n-2$. The report gives further sol...

open_problem | open | MathDB 页面 | 原文线索

V2-074651 | MATHDB-393474

Justification of the nonlinear Schrödinger approximation for slowly varying water-wave trains

摘录 / EXCERPT: Prove that the Zakharov approximation ... where $A(X,T)$ solves ... gives an accurate approximation to the water-wave problem for slowly varying periodic wave trains on the time scale...

open_problem | open | MathDB 页面 | 原文线索

V2-074652 | MATHDB-393476

Uniqueness of the concordance class on the tight 3-sphere

摘录 / EXCERPT: Conjecture: There is only one concordance class of finite-energy foliations on the tight 3-sphere.

open_problem | open | MathDB 页面 | 原文线索

V2-074653 | MATHDB-393477

Uniqueness of the concordance class of foliations on the tight 3-sphere

摘录 / EXCERPT: There is only one concordance class of foliations on the tight 3-sphere.

open_problem | open | MathDB 页面 | 原文线索

V2-074654 | MATHDB-393478

Pop’s characterization of Kronecker dimension by Pfister forms

摘录 / EXCERPT: For each $n\in\mathbb{N}$ and each finitely generated field K, one has $\operatorname{Krdim}K\le n$ if and only if either: - $\operatorname{char}(K)=2$ and $[K:K^2]\le 2^n$; or -...

open_problem | open | MathDB 页面 | 原文线索

V2-074655 | MATHDB-393484

Average value of $2^{\text{$4$-rank}}$ for real quadratic class groups

摘录 / EXCERPT: Let D range over discriminants of quadratic fields, and let Cl_D denote the ordinary ideal class group of $\mathbb{Q}(\sqrt D)$. With...

open_problem | open | MathDB 页面 | 原文线索

V2-074656 | MATHDB-393485

Average 4-rank moment for narrow class groups of real quadratic fields

摘录 / EXCERPT: Let D range over positive discriminants of quadratic fields, and let C_D be the narrow class group. With $\operatorname{rk}_4(A):=\operatorname{rk}_2(A^2)$, the conjectural for...

open_problem | open | MathDB 页面 | 原文线索

V2-074657 | MATHDB-393487

Strong finite-covering conjecture for rational points on curves

摘录 / EXCERPT: Let C be a smooth projective curve over a number field k. Define a residue class on C to be a subset $X\subset C(\mathbb A_k)=\prod_v C(k_v)$ of the form ... where S is a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-074658 | MATHDB-393488

Stoll’s Main Conjecture, weak version

摘录 / EXCERPT: Let C be a smooth projective curve over a number field k. Write $C(\A_k)=\prod_v C(k_v)$ for its adelic points. A residue class on C is a subset ... where S is a finite set o...

conjecture | open | MathDB 页面 | 原文线索

V2-074659 | MATHDB-393489

Eventually Small Rank Conjecture

摘录 / EXCERPT: Let C be a curve of genus at least 2. Then there is some $n\ge 1$ such that, for all twists D_j of the multiplication-by-n covering of C with...

conjecture | open | MathDB 页面 | 原文线索

V2-074660 | MATHDB-393490

Solvability of polynomial equations over finite fields with a fixed number of variables

摘录 / EXCERPT: Given a finite field $[U+007F]mathbb{F}_q$ and polynomials $f_1,[U+007F]ots,f_m [U+007F]in [U+007F]mathbb{F}_q[x_1,[U+007F]ots,x_n]$ of total degree at most d, decide whether there exists a point...

open_problem | open | MathDB 页面 | 原文线索

V2-074661 | MATHDB-393491

Practicality of the modular Heegner-point method

摘录 / EXCERPT: Determine whether the proposed method for identifying odd twists likely to have analytic rank at least 3 is practical.

open_problem | open | MathDB 页面 | 原文线索

V2-074662 | MATHDB-393492

Frequency of even quadratic twists with analytic rank at least two

摘录 / EXCERPT: For a fixed rational elliptic curve E, the report conjectures that the number $N_E(D)$ of parameters d with $|d|<D$ for which the twist E_d has even analytic rank at least...

open_problem | open | MathDB 页面 | 原文线索

V2-074663 | MATHDB-393497

Finiteness of Tate-Shafarevich groups for rank-2 optimal elliptic curves

摘录 / EXCERPT: There are 18 optimal elliptic curves of conductor at most 1000 and rank 2 (and none of rank greater than 2). For these curves E, the report states that nobody had proved th...

open_problem | open | MathDB 页面 | 原文线索

V2-074664 | MATHDB-393498

Nonuniqueness of the constant in the Hadamard product for \(\xi_K\)

摘录 / EXCERPT: Let $\xi_K(s)$ be written in Hadamard-product form as ... where the nontrivial zeros ρ of $\zeta_K(s)$ are grouped so that $\sum_{\rho}'1/\rho$ converges conditionally, and ... Th...

open_problem | open | MathDB 页面 | 原文线索

V2-074665 | MATHDB-393499

The Willmore conjecture for tori in \(\mathbb{R}^3\)

摘录 / EXCERPT: Let $\beta_{n,p}$ denote the infimum of the Willmore energy among immersions of closed, oriented surfaces of genus p into $\mathbb{R}^n$. The Willmore conjecture states that...

conjecture | open | MathDB 页面 | 原文线索

V2-074666 | MATHDB-393500

Hausdorff dimension of the singular set for vectorial elliptic systems

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^n$ be bounded and open, with $n\ge 2$, and let $N\ge 1$. Consider ... where $a:\Omega\times\mathbb R^N\times\mathbb R^{nN}\to\mathbb R^{nN}$ satisfies the...

open_problem | open | MathDB 页面 | 原文线索

V2-074667 | MATHDB-393501

Equivalence of v-convexity and H-convexity in Carnot groups

摘录 / EXCERPT: Let G be a Carnot group, let $\\Omega\subset G$ be open, and let $u:\Omega\to\mathbb{R}$ be upper semicontinuous. The function u is H-convex when its restriction to every hor...

open_problem | open | MathDB 页面 | 原文线索

V2-074668 | MATHDB-393502

Measure-valued commutators of H-convex functions in Carnot groups

摘录 / EXCERPT: Is it true that $[X_i,X_j]u$ are measures when u is an H-convex function?

open_problem | open | MathDB 页面 | 原文线索

V2-074669 | MATHDB-393503

Validity of the mollifier identity in higher-step Carnot groups

摘录 / EXCERPT: Is the identity $X_iX_j\theta(x)=X_iX_j\theta(x^{-1})$ valid in Carnot groups of higher step for mollifiers θ satisfying $\theta(x)=\theta(x^{-1})$?

open_problem | open | MathDB 页面 | 原文线索

V2-074670 | MATHDB-393504

Smoothness of the support of the limit measure for stable phase-field critical points

摘录 / EXCERPT: Show that the support of the limit measure is smooth for $n \le 7$.

open_problem | open | MathDB 页面 | 原文线索

V2-074671 | MATHDB-393505

Modified De Giorgi conjecture for the area-plus-Willmore functional

摘录 / EXCERPT: Let $n=2,3$, let $\[1m\Omega\subset\mathbb{R}^n$[0m be a set, and let $W(t):=(1-t^2)^2$. For $\[1m\varepsilon>0$[0m define $F_\varepsilon:L^1(\Omega)\to\mathbb{R}$ by ... if...

conjecture | open | MathDB 页面 | 原文线索

V2-074672 | MATHDB-393506

Dimension bound for the singular set in the two-phase free-boundary problem

摘录 / EXCERPT: Let $B\subset\mathbb{R}^n$ be an open set, let Q^+ and Q^- be smooth functions on B with $Q^+-Q^-\ge c>0$, and let u be a minimizer over the class K of the functional ... F...

open_problem | open | MathDB 页面 | 原文线索

V2-074673 | MATHDB-393507

Modified De Giorgi conjecture for a phase-field approximation of the Willmore functional

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$, let $W(t)=(1-t^2)^2$, and for $\varepsilon>0$ define $F_\varepsilon:L^1(\Omega)\to\mathbb{R}\cup\{+\infty\}$ by ... if...

conjecture | open | MathDB 页面 | 原文线索

V2-074674 | MATHDB-393508

Uniqueness for solutions of the higher-dimensional Beltrami system

摘录 / EXCERPT: Let G on Ω and H on Ω' be bounded measurable maps into $S(n)$, the positive-definite $n\times n$ matrices of determinant 1. Suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-074675 | MATHDB-393510

Regularity theorem for stable minimal hypersurfaces with mass ratio close to 2

摘录 / EXCERPT: There exist fixed constants $\epsilon\in(0,1)$, $\sigma\in(0,1)$, $C\in(0,\infty)$, and $\alpha\in(0,1)$ such that the following holds. If $2\le n\le 6$, M is an orientable immer...

open_problem | open | MathDB 页面 | 原文线索

V2-074676 | MATHDB-393512

Pure unrectifiability and universal singular sets

摘录 / EXCERPT: Theorem 2. If any compact purely unrectifiable set $S\subset\mathbb{R}^{2}$ and any superlinearity ω are given, then a Lagrangian L with ω-growth, satisfying (LC) a...

open_problem | open | MathDB 页面 | 原文线索

V2-074677 | MATHDB-393513

Size of universal singular sets in Tonelli’s variational problem

摘录 / EXCERPT: Under the superlinearity, smoothness, and convexity assumptions on the Lagrangian stated in the report, can a universal singular set have infinite length, or even Hausdorff dimensi...

open_problem | open | MathDB 页面 | 原文线索

V2-074678 | MATHDB-393514

Transfer of ocean-state uncertainties to Lagrangian coherent structures

摘录 / EXCERPT: The transfer of these uncertainties from the ocean state to the LCSs is an unexplored but intriguing scientific problem.

open_problem | open | MathDB 页面 | 原文线索

V2-074679 | MATHDB-393516

Integrability of vortex problems on a rotating sphere

摘录 / EXCERPT: The 3-vortex problem on the rotating sphere is integrable for all vortex strengths. The 4-vortex problem is integrable if $J=0$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-074680 | MATHDB-393519

Closure of translation vectors and directional limit points

摘录 / EXCERPT: Let Γ be a nonelementary group of isometries as in the report. Let P_{Γ} denote the set of directions of limit points, and let ℓ_{Γ} denote the set of trans...

open_problem | open | MathDB 页面 | 原文线索

V2-074681 | MATHDB-393526

Strominger-Yau-Zaslow fibration conjecture

摘录 / EXCERPT: Near or at a large complex structure limit, a Calabi-Yau space M should be the total space of a fibration over a base $\bar B$ of real half-dimension, with generic fiber a specia...

conjecture | open | MathDB 页面 | 原文线索

V2-074682 | MATHDB-393527

Solvability of the \c3_k-Yamabe problem beyond the known cases

摘录 / EXCERPT: The c3k-Yamabe problem remains open for $2<k\le n/2$ when M is not locally conformally flat.

open_problem | open | MathDB 页面 | 原文线索

V2-074683 | MATHDB-393530

No proper gaps for the shift on the Bergman space A^2

摘录 / EXCERPT: Let A^2 be the Bergman space of holomorphic functions on $\[D$ with finite norm $\|f\|=(\int_{\mathbb D}|f(z)|^2\,dA(z))^{1/2}$, and let $S:A^2\to A^2$ be the shift...

open_problem | open | MathDB 页面 | 原文线索

V2-074684 | MATHDB-393532

Interpolating measures for subnormal operators

摘录 / EXCERPT: If S is a subnormal operator, does S have an interpolating measure? If not, which subnormal operators have interpolating measures?

open_problem | open | MathDB 页面 | 原文线索

V2-074685 | MATHDB-393535

Invariant Gaussian measure for operators with perfectly spanning unimodular eigenvectors

摘录 / EXCERPT: If $T\in\mathcal B(X)$ has a perfectly spanning set of unimodular eigenvectors, is it always true that T admits a non-degenerate Gaussian measure with respect to which it is weak...

open_problem | open | MathDB 页面 | 原文线索

V2-074686 | MATHDB-393536

Direct sums of frequently hypercyclic operators

摘录 / EXCERPT: Let X be a separable F-space and let $T:X\to X$ be a continuous linear frequently hypercyclic operator, meaning that there exists $x\in X$ such that, for every nonempty open se...

open_problem | open | MathDB 页面 | 原文线索

V2-074687 | MATHDB-393537

Category size of frequently hypercyclic vectors

摘录 / EXCERPT: Let X be a separable F-space and let T be a frequently hypercyclic continuous linear operator on X. Can the set of frequently hypercyclic vectors of T be of second catego...

open_problem | open | MathDB 页面 | 原文线索

V2-074688 | MATHDB-393538

Brennan’s conjecture for conformal maps

摘录 / EXCERPT: For all such conformal maps τ of the unit disk D, $\displaystyle\int_D \left(\frac{1}{|\tau'|}\right)^p dA<\infty$ holds for $-\frac{2}{3}<p<2$.

conjecture | open | MathDB 页面 | 原文线索

V2-074689 | MATHDB-393542

Intermediate closed ideals in H^∞

摘录 / EXCERPT: Let I_1,I_2 be closed ideals in H^∞ of the unit disc, with $I_1\subset I_2$ and $\dim(I_2/I_1)>1$. Does there exist a closed ideal J of H^∞ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-074690 | MATHDB-393543

Intermediate invariant subspaces for the backward shift on Bergman spaces

摘录 / EXCERPT: Assume that $1 \le p < \infty$, and let A_p denote the Bergman space of all holomorphic functions on the unit disc $\mathbb{D}$ that belong to $L^p(\mathbb{D})$. If M_1,M_2 are...

open_problem | open | MathDB 页面 | 原文线索

V2-074691 | MATHDB-393544

Capacity theory and boundary limits in the Drury-Arveson space

摘录 / EXCERPT: Work out the details of a capacity theory for H^2_d, and prove that functions in H^2_d have nontangential limits everywhere on $\[?]partial B_d$ except perhaps a set of “capaci...

open_problem | open | MathDB 页面 | 原文线索

V2-074692 | MATHDB-393545

Corona property for complements of Cartesian products of compact real sets

摘录 / EXCERPT: For compact subsets $K,S\subseteq\mathbb{R}$, is the domain $\Omega=\mathbb{C}\setminus(K\times S)$ guaranteed to have the corona property? Equivalently, is Ω dense in the m...

open_problem | open | MathDB 页面 | 原文线索

V2-074693 | MATHDB-393546

Distribution of mass by a Bergman function

摘录 / EXCERPT: Let L^1_a be the Bergman space of functions analytic on the unit disk $\D$ with norm $\\|f\|_1=\int_D |f|\,dA$. For $\varepsilon>0$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-074694 | MATHDB-393547

Garnett-Jones approximation problem for Blaschke products

摘录 / EXCERPT: Can every Blaschke product be uniformly approximated by interpolating Blaschke products?

open_problem | open | MathDB 页面 | 原文线索

V2-074695 | MATHDB-393548

Compact differences of composition operators on H^2

摘录 / EXCERPT: Let $[U+007F]\varphi,[U+007F]\psi:\mathbb{D}\to\mathbb{D}$ be analytic, and let $C_[U+007F]\varphi,C_[U+007F]\psi$ be the corresponding composition operators on the Hardy space H^2. Suppose ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-074696 | MATHDB-393549

Error expansion on locally refined meshes for the DWR method

摘录 / EXCERPT: Our hypothesis is that a similar error expansion as (6) holds on locally refined meshes, where the function w depends on the mesh but this dependency can be localized. To be more...

open_problem | open | MathDB 页面 | 原文线索

V2-074697 | MATHDB-393550

Stability of second-order differences on locally refined meshes

摘录 / EXCERPT: For $u\in C^2(\Omega)$, we conjecture that ...

open_problem | open | MathDB 页面 | 原文线索

V2-074698 | MATHDB-393552

Counterexample to error reduction under a particular mesh refinement

摘录 / EXCERPT: For the Poisson problem with constant right-hand side $f\equiv 1$ and homogeneous Dirichlet boundary conditions on the unit square, the P_1 finite element solutions on the meshes...

open_problem | open | MathDB 页面 | 原文线索

V2-074699 | MATHDB-393553

Energy reduction property for adaptive finite element approximations

摘录 / EXCERPT: For nonlinear elasticity such as Hencky elastoplasticity with hardening, let $E(v)$ denote the elastic energy and let $\delta_\ell := E(u_\ell)-E(u)$ be the difference between the...

open_problem | open | MathDB 页面 | 原文线索

V2-074700 | MATHDB-393555

T-admissibility of the natural K-theory extension to orbispaces

摘录 / EXCERPT: Is the natural extension of K-theory to orbispaces also T-admissible?

open_problem | open | MathDB 页面 | 原文线索

V2-074701 | MATHDB-393559

LeBrun’s positive-frequency conjecture for self-dual Poincaré-Einstein fillings

摘录 / EXCERPT: Let $M_\pm$ be the spaces of metrics g on S^n near the round metric $g_{(0)}$ such that $[g]$ is the conformal infinity of a $\pm$ self-dual Poincaré-Einstein metric. Let S b...

conjecture | open | MathDB 页面 | 原文线索

V2-074702 | MATHDB-393560

Explicit optimal vector field for the square

摘录 / EXCERPT: In the example of a square, the Cheeger constant and a minimizing subset S can be found explicitly (round off the corners by quarter circles and optimize over their radius). Howe...

open_problem | open | MathDB 页面 | 原文线索

V2-074703 | MATHDB-393563

Existence of finite classifying spaces

摘录 / EXCERPT: Determine when there exists a finite $B\Gamma$-space.

open_problem | open | MathDB 页面 | 原文线索

V2-074704 | MATHDB-393564

Joint spectral resolution on finite-volume locally symmetric spaces

摘录 / EXCERPT: Let $S=G/K$ be a globally Riemannian symmetric space of noncompact type, where G is a real semisimple Lie group and $K\subset G$ is a maximal compact subgroup. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-074705 | MATHDB-393565

Remainder term in Weyl’s law for cuspidal forms on principal congruence quotients of $SL(n)$

摘录 / EXCERPT: Let $X=SL(n,\mathbb{R})/SO(n)$ with $n\ge 2$, let $d=\dim X$, and let Γ be a principal congruence subgroup of $SL(n,\mathbb{Z})$. If $N_{\mathrm{cusp}}^{\Gamma}(\lambda)$ de...

open_problem | open | MathDB 页面 | 原文线索

V2-074706 | MATHDB-393566

Sarnak’s essential cuspidality conjecture

摘录 / EXCERPT: In the setup where $S=G/K$ is a globally Riemannian symmetric space of noncompact type and $\Gamma\subset G$ is a lattice, let $X_\Gamma=\Gamma\backslash S$, $d=\dim S$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-074707 | MATHDB-393567

Selberg eigenvalue conjecture for principal congruence subgroups

摘录 / EXCERPT: Let $\Gamma(N)\subset SL(2,\mathbb Z)$ be the principal congruence subgroup of level N, and let $X(N)=\Gamma(N)\backslash\mathbb H$. The Selberg conjecture asks whether the first...

conjecture | open | MathDB 页面 | 原文线索

V2-074708 | MATHDB-393568

Ramanujan conjecture for the spherical spectrum of $\mathrm{SL}(n)$

摘录 / EXCERPT: Let $G=\mathrm{SL}(n)$ in the report’s notation, with G denoting its real points. Let $\widehat G_{\mathrm{unit}}$ be the unitary dual of G, let $\widehat G_{\mathrm{temp}}$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-074709 | MATHDB-393571

Folklore conjecture for the global Weyl remainder on generic negatively curved surfaces

摘录 / EXCERPT: On a generic negatively curved surface, the global remainder in Weyl’s law, defined by $R(\lambda)=\int_X R_x(\lambda)$, satisfies $R(\lambda)=O(\lambda^{\varepsilon})$ for every...

conjecture | open | MathDB 页面 | 原文线索

V2-074710 | MATHDB-393573

Removing the root-system restriction in Saper’s L^2-harmonic-forms theorem

摘录 / EXCERPT: Remove the hypothesis that G has no $\mathbb{Q}$-simple factors with $\mathbb{Q}$-root system of type D_n, E_n, or F_4 from the theorem stated in the report: for $D=G/K$, a...

open_problem | open | MathDB 页面 | 原文线索

V2-074711 | MATHDB-393575

Phase transition for the distance in the random-transposition walk

摘录 / EXCERPT: For the random-transposition walk $(\sigma_t,t\ge 0)$ on the symmetric group S_n, let D_t be the minimum number of transpositions needed to return from σ_t to the identi...

open_problem | open | MathDB 页面 | 原文线索

V2-074712 | MATHDB-393577

Comparison of horizontal and vertical estimates of genetic variation

摘录 / EXCERPT: For a sample of n DNA sequences of arbitrary length L, with no recombination between sites, let $\theta=4Nu$, where N is the population size and u is the mutation rate to n...

open_problem | open | MathDB 页面 | 原文线索

V2-074713 | MATHDB-393578

Large-deviation principle for Poisson-Dirichlet distributions with heterozygote advantage

摘录 / EXCERPT: For each integer $m\ge 1$ and selection intensity $\sigma(\theta)>0$, let $H_m(p)=\sum_{i=1}^{\infty}p_i^m$ on...

open_problem | open | MathDB 页面 | 原文线索

V2-074714 | MATHDB-393579

Likelihood-ratio threshold for detecting heterozygote advantage

摘录 / EXCERPT: For each fixed $m\ge 1$, let $H_m(p)=\sum_{i=1}^{\infty}p_i^m$ and define ... Under Π_θ, as $\theta\to\infty$, ... where Z_2 is normally distributed with mean 0 and varian...

open_problem | open | MathDB 页面 | 原文线索

V2-074715 | MATHDB-393581

Asymptotic E-value formula for gapped local alignments

摘录 / EXCERPT: The conjecture is that, also for gapped local alignments, the E-value for score at least s is asymptotically of the form $nm k' e^{-\lambda' s}$, where n and m are the sequen...

open_problem | open | MathDB 页面 | 原文线索

V2-074716 | MATHDB-393582

Non-regenerativity of composition structures from simultaneous-multiple-collision coalescents

摘录 / EXCERPT: We conjecture that all the corresponding composition structures are non-regenerative, except for the case when Λ_j is the zero measure for all $j\ge 2$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-074717 | MATHDB-393584

Necessity and sufficiency of condition (6) outside \(\Theta_d^*\)

摘录 / EXCERPT: For general parameters $\theta\in\mathbb{R}^d\setminus\Theta_d^*$, is condition (6), namely ... for some positive measure dH on $[0,1]$, necessary and sufficient for the non-negati...

open_problem | open | MathDB 页面 | 原文线索

V2-074718 | MATHDB-393585

Survival versus survival of the reversed braco-process

摘录 / EXCERPT: Let Λ be a finite or countable set, and let G be a transitive group of bijections of Λ. Consider the (a,b,c,d)-braco-process, where particles jump from i to...

open_problem | open | MathDB 页面 | 原文线索

V2-074719 | MATHDB-393590

Fibering Theorem for uniform embeddings

摘录 / EXCERPT: Let the discrete group Γ act on the metric spaces X and Y by isometries, and let $p:X\to Y$ be a Lipschitz, Γ-equivariant map. Assume that Y is an exact space a...

open_problem | open | MathDB 页面 | 原文线索

V2-074720 | MATHDB-393591

Uniform embeddability of relatively hyperbolic groups

摘录 / EXCERPT: Let Γ be a finitely generated discrete group which is relatively hyperbolic with respect to a subgroup H. Then Γ is uniformly embeddable if and only if H is unifo...

open_problem | open | MathDB 页面 | 原文线索

V2-074721 | MATHDB-393592

Chern character for twisted K-theory of orbifolds

摘录 / EXCERPT: The report states the following main result: if G is an orbifold groupoid and $\[0m\alpha\in H^{3}(G,\mathbb Z)$ is represented by an S^{1}-central extension...

open_problem | open | MathDB 页面 | 原文线索

V2-074722 | MATHDB-393593

Relation between O-exactness and C*-exactness

摘录 / EXCERPT: What is the relation between O-exactness and C^*-exactness?

open_problem | open | MathDB 页面 | 原文线索

V2-074723 | MATHDB-393596

Popa's free-complementation conjecture for maximal hyperfinite masas

摘录 / EXCERPT: If $A\subset L(F_2)$ is a masa that is maximal hyperfinite, then A is freely complemented in $L(F_2)$.

conjecture | open | MathDB 页面 | 原文线索

V2-074724 | MATHDB-393598

Maximal torus conjecture for finite loop spaces

摘录 / EXCERPT: The classifying space functor, which assigns to a compact Lie group G the finite loop space $(G,BG,e:G\simeq\Omega BG)$, gives a one-to-one correspondence between compact Lie gro...

conjecture | open | MathDB 页面 | 原文线索

V2-074725 | MATHDB-393610

Effective Mordell / Vojta height conjecture for curves

摘录 / EXCERPT: Let K be a number field and X/K a smooth, projective curve defined over K. For every $\epsilon>0$, there exists a constant $\kappa(\epsilon)$ such that, for all $P\in X(K)$,...

conjecture | open | MathDB 页面 | 原文线索

V2-074726 | MATHDB-393611

Conjecture II on arithmetic self-intersection numbers

摘录 / EXCERPT: Let X be a regular model of X over $\operatorname{Spec}\mathcal O_K$, and let ω be the dualizing sheaf equipped with the Arakelov metric. Then the arithmetic self-inters...

conjecture | open | MathDB 页面 | 原文线索

V2-074727 | MATHDB-393612

Birational invariance of the BCOV invariant for Calabi-Yau threefolds

摘录 / EXCERPT: If Calabi-Yau threefolds X and X' are birationally equivalent, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-074728 | MATHDB-393613

The representability problem for the two-body density matrix of a many-fermion system

摘录 / EXCERPT: For fermionic one-particle space $H_1=L^2(V)$ with $V=\mathbb{R}^3\times\mathbb{Z}_2$, antisymmetric N-particle space $H_N=H_1\wedge\cdots\wedge H_1$, and two-body reduced densit...

open_problem | open | MathDB 页面 | 原文线索

V2-074729 | MATHDB-393614

Pure-state representability of one-body density matrices

摘录 / EXCERPT: Characterize the set of one-body density matrices arising from antisymmetric pure states of an N-fermion system; equivalently, determine exactly which one-body density matrices a...

open_problem | open | MathDB 页面 | 原文线索

V2-074730 | MATHDB-393615

Tarski’s problem for finitely generated non-abelian free groups

摘录 / EXCERPT: The first-order theories of all finitely generated non-abelian free groups are identical.

open_problem | open | MathDB 页面 | 原文线索

V2-074731 | MATHDB-393626

Stabilization of a weakly reducible splitting of a twisted torus knot complement

摘录 / EXCERPT: For the knots $K_m=T(7,17,10m-4)$, with $m\in\mathbb Z$ and sufficiently large $|m|$, is the weakly reducible genus-three Heegaard splitting of $S^3\setminus N(K_m)$ obtained by...

open_problem | open | MathDB 页面 | 原文线索

V2-074732 | MATHDB-393627

Hyperbolicity of graphs with pairwise intersecting holes

摘录 / EXCERPT: Conjecture 1. Suppose that $G(S)$ admits an action by the mapping class group and any pair of holes X and Y intersect. Then $G(S)$ is Gromov hyperbolic.

open_problem | open | MathDB 页面 | 原文线索

V2-074733 | MATHDB-393631

Localization theorem for model categories

摘录 / EXCERPT: Let C be a model category and $W\subseteq C$ the class of weak equivalences. Then the functor $\gamma:C\to\operatorname{Ho}(C)$ is a localization of C with respect to W. More...

open_problem | open | MathDB 页面 | 原文线索

V2-074734 | MATHDB-393633

Using regularization results to choose Bayesian priors

摘录 / EXCERPT: Can regularization results help Bayesians in choosing priors?

open_problem | open | MathDB 页面 | 原文线索

V2-074735 | MATHDB-393634

Performance guarantees for Bayesian procedures from oracle or risk inequalities

摘录 / EXCERPT: Do oracle or risk inequalities tell us about the performance of Bayesian procedures: in practice, for growing finite sample sizes, or asymptotically?

open_problem | open | MathDB 页面 | 原文线索

V2-074736 | MATHDB-393635

Prediction regions for the optimal nested-model selector

摘录 / EXCERPT: Let $X_n\sim N_n(\theta_n,\sigma^2 I_n)$ with known $\sigma>0$, and define $\widehat\theta_n^{(k)}=(X_{n,1},\ldots,X_{n,k},0,\ldots,0)^\top$ for $k\in\{0,\ldots,n\}$. With loss and...

open_problem | open | MathDB 页面 | 原文线索

V2-074737 | MATHDB-393636

Strongly consistent model selection for density estimation

摘录 / EXCERPT: Let $(F_k)_{k\ge 1}$ be nested classes of densities on $\mathbb{R}^d$, let $f\in F=\bigcup_{k\ge1}F_k$, and define $k^*=\min\{k\ge1:f\in F_k\}$. For an even sample of size $2n$, le...

open_problem | open | MathDB 页面 | 原文线索

V2-074738 | MATHDB-393637

Tolerance of a simpler model relative to a richer model

摘录 / EXCERPT: For cases where a narrow model with, say, p parameters is extended to a suitable richer model with, say, $p+q$ parameters, how much can the simpler model tolerate in order for th...

open_problem | open | MathDB 页面 | 原文线索

V2-074739 | MATHDB-393639

Sparsity transfer from the true penalized solution to the empirical solution

摘录 / EXCERPT: In the setup of the report, with λ_ε and $\widehat\lambda_\varepsilon$ denoting respectively the minimizers of the true and empirical penalized ℓ-risks, supp...

open_problem | open | MathDB 页面 | 原文线索

V2-074740 | MATHDB-393640

Oracle inequality for recursive mirror-averaging aggregation

摘录 / EXCERPT: Let $(\mathcal Z,\mathcal F)$ be the measurable space, let $\Theta=\{\theta\in\mathbb R^M:\sum_{j=1}^M\theta^{(j)}=1,\ \theta^{(j)}\ge 0\}$ with $M\ge 2$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-074741 | MATHDB-393641

Data-dependent adaptation to unknown regularity in higher-order estimation

摘录 / EXCERPT: Can the optimal values of the higher-order index k and approximating-subspace dimension m be chosen data-dependently through cross-validation, thereby yielding estimators that...

open_problem | open | MathDB 页面 | 原文线索

V2-074742 | MATHDB-393643

Local observables for factorizing S-matrices in the subfamily $\mathcal S_0$

摘录 / EXCERPT: For every factorizing S-matrix $S\in\mathcal S_0$, do the corresponding wedge-local model theories contain observables localized in arbitrarily small double cones? The report answe...

open_problem | open | MathDB 页面 | 原文线索

V2-074743 | MATHDB-393646

Lorentz invariance and multiplicativity in massless renormalization

摘录 / EXCERPT: The massless models pose the puzzling problem of reconciling manifest Lorentz invariance of the renormalization map with the multiplicative property.

open_problem | open | MathDB 页面 | 原文线索

V2-074744 | MATHDB-393649

Essential self-adjointness criterion for globally hyperbolic spectral triples

摘录 / EXCERPT: Conjecture: A timelike foliated Lorentzian spectral triple (Lost) is a globally hyperbolic spectral triple (Ghyst) if and only if there exists an admissible pair...

open_problem | open | MathDB 页面 | 原文线索

V2-074745 | MATHDB-393650

Effect of polydispersity on effective conductivity in a two-phase random composite

摘录 / EXCERPT: In the two-dimensional mathematical model of a composite with conducting inclusions (fibers) randomly embedded in a matrix, when the inclusions have higher conductivity than the ma...

open_problem | open | MathDB 页面 | 原文线索

V2-074746 | MATHDB-393652

Global existence and asymptotic stability for nonisotropic reactive-flow systems

摘录 / EXCERPT: Global existence of solutions and asymptotic stability is an open problem for such nonisotropic systems.

open_problem | open | MathDB 页面 | 原文线索

V2-074747 | MATHDB-393653

Godunov flux at an interface between two rock types

摘录 / EXCERPT: For a one-dimensional two-phase-flow conservation law with a discontinuity in the rock type, with rock type I on the left and II on the right, the Godunov numerical flux is extende...

open_problem | open | MathDB 页面 | 原文线索

V2-074748 | MATHDB-393654

Rigorous derivation of the compressible Reynolds equations for gas lubrication

摘录 / EXCERPT: Under the report’s hypotheses, let … where $O\subset\mathbb R^{n-1}$ is bounded and smooth, h is smooth, and $0<h_m\le h(x')\le h_M$. Assume the isothermal law…

open_problem | open | MathDB 页面 | 原文线索

V2-074749 | MATHDB-393655

Collision-driven selection of a unique self-similar state in the LSW model

摘录 / EXCERPT: It is conjectured that the effect of collisions between particles is the dominating effect that drives the particle system toward a unique self-similar state.

open_problem | open | MathDB 页面 | 原文线索

V2-074750 | MATHDB-393657

Zarankiewicz crossing-number conjecture

摘录 / EXCERPT: “Zarankiewicz conjecture” asserts that …

conjecture | open | MathDB 页面 | 原文线索

V2-074751 | MATHDB-393659

Planar graph requiring an optimal embedding of dimension greater than three

摘录 / EXCERPT: Construct a connected planar graph whose optimal embedding for problem (1) has minimal dimension greater than three, where problem (1) is …

open_problem | open | MathDB 页面 | 原文线索

V2-074752 | MATHDB-393660

Linear-time solvability of the reverse 1-median problem on a cycle

摘录 / EXCERPT: The reverse 1-median problem on a cycle can be solved in $O(n)$ time.

open_problem | open | MathDB 页面 | 原文线索

V2-074753 | MATHDB-393662

Polynomial-time implementation of DoublePivot for Universal Facility Location

摘录 / EXCERPT: Is it possible to implement the DoublePivot operation in polynomial time for the Universal Facility Location Problem?

open_problem | open | MathDB 页面 | 原文线索

V2-074754 | MATHDB-393666

Thick Lehman matrices and projective planes

摘录 / EXCERPT: Except for a number of sporadic examples, do all thick Lehman matrices arise from projective planes?

open_problem | open | MathDB 页面 | 原文线索

V2-074755 | MATHDB-393668

Strength of the Lovász-Schrijver relaxation on MIPLIB instances

摘录 / EXCERPT: Determine how strong the Lovász-Schrijver relaxation is on the MIPLIB instances.

open_problem | open | MathDB 页面 | 原文线索

V2-074756 | MATHDB-393670

Polynomial-time separation for the crossing-minimization ILP

摘录 / EXCERPT: For a graph $G=(V,E)$, consider the valid inequalities in the report’s ILP formulation: for every simple set D of unordered pairs of edges and every Kuratowski subgraph H of…

open_problem | open | MathDB 页面 | 原文线索

V2-074757 | MATHDB-393672

Fourier’s 17-line intersection problem

摘录 / EXCERPT: Arrange 17 lines in the same plane so that they give exactly 101 points of intersection. The lines are assumed to extend to infinity, and no point of intersection may belong to mor…

open_problem | open | MathDB 页面 | 原文线索

V2-074758 | MATHDB-393673

Borda-rule tour-construction dominance theorem

摘录 / EXCERPT: Using the Borda Rule, for any complete graph with $|V|\ge 5$, there exist preference profiles such that: (i) the result of the local tour construction is pairwise dominated by the…

open_problem | open | MathDB 页面 | 原文线索

V2-074759 | MATHDB-393675

Even/odd parity assignment problem

摘录 / EXCERPT: Given a matrix $A\in\mathbb{R}^{n\times n}$, let P_n^+ and P_n^- denote the even and odd permutations of $\{1,\ldots,n\}$, respectively, and let…

open_problem | open | MathDB 页面 | 原文线索

V2-074760 | MATHDB-393676

Even/odd parity assignment equality problem

摘录 / EXCERPT: Given a matrix $A=(a_{ij})\in\mathbb R^{n\times n}$, with the report’s convention $\mathbb R:=\mathbb R\cup\{-\infty\}$, let $N=\{1,\ldots,n\}$, let P_n^+ and P_n^- denote the…

open_problem | open | MathDB 页面 | 原文线索

V2-074761 | MATHDB-393677

Even-odd parity assignment equality

摘录 / EXCERPT: Given a matrix $A=(a_{ij})\in\mathbb{R}^{n\times n}$, with P_n^+ and P_n^- denoting the sets of even and odd permutations of $\{1,\ldots,n\}$ and…

open_problem | open | MathDB 页面 | 原文线索

V2-074762 | MATHDB-393679

Cantor spectrum for one-frequency quasiperiodic Schrödinger operators with positive Lyapunov exponent

摘录 / EXCERPT: When the frequency dimension is one (d=1), does the spectrum always have Cantor structure whenever the Lyapunov exponent is positive?

open_problem | open | MathDB 页面 | 原文线索

V2-074763 | MATHDB-393680

Residuality of Cantor-spectrum potentials for every irrational frequency

摘录 / EXCERPT: Is it true (or false) that, for each $\\gamma \\in \\Gamma$, the set $D_\\gamma$ is residual in $C(\\mathbb{T}^k)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074764 | MATHDB-393681

Almost-everywhere dichotomy for Liouville-frequency cocycles

摘录 / EXCERPT: Let $\alpha\in\mathbb R/\mathbb Z\setminus\mathbb Q/\mathbb Z$ be Liouville and let $A:\mathbb R/\mathbb Z\to\mathrm{SL}(2,\mathbb R)$ be smooth. Is it true that, for Lebesgue almo...

open_problem | open | MathDB 页面 | 原文线索

V2-074765 | MATHDB-393682

Almost rotations-reducibility for arbitrary irrational frequencies

摘录 / EXCERPT: In the setting of Theorem 3, let α be irrational and let A be a smooth cocycle that is C^0-conjugated to a cocycle of rotations. Is it true that, for Lebesgue-almost eve...

open_problem | open | MathDB 页面 | 原文线索

V2-074766 | MATHDB-393685

Boshernitzan condition on higher-dimensional tori

摘录 / EXCERPT: Study the validity of the Boshernitzan condition or its consequences when the circle is replaced by the d-dimensional torus and $f:T^d\to\mathbb{R}$ takes finitely many values.

open_problem | open | MathDB 页面 | 原文线索

V2-074767 | MATHDB-393686

Relaxation of the single variational problem at the first time step

摘录 / EXCERPT: How does one determine, exactly or approximately, the relaxation of the single variational problem corresponding to the first time step, equivalently the quasiconvex envelope of an...

open_problem | open | MathDB 页面 | 原文线索

V2-074768 | MATHDB-393687

Solvability of the energetic formulation in finite-strain plasticity

摘录 / EXCERPT: For the finite-strain plasticity model described in the report, establish solvability of the weak energetic formulation consisting of the global stability condition (S) and the ene...

open_problem | open | MathDB 页面 | 原文线索

V2-074769 | MATHDB-393688

Quasiconvex envelope for the two-dimensional rigid single-slip incremental problem

摘录 / EXCERPT: In two dimensions, for the perfectly rigid multiplicative crystal-plasticity model without hardening and one active slip system characterized by orthogonal unit vectors...

open_problem | open | MathDB 页面 | 原文线索

V2-074770 | MATHDB-393689

Relaxation of the linearized single-slip condensed energy

摘录 / EXCERPT: For orthonormal slip-system pairs ((si,mi)inmathbb R^3timesmathbb R^3), $i=1,\ldots,N$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-074771 | MATHDB-393690

Quasiconvex envelope of the rigid single-slip energy

摘录 / EXCERPT: Let $s,m\in\mathbb{R}^2$ be a fixed pair of orthonormal vectors. Define $W_r:\mathbb{R}^{2\times 2}\to[0,+\infty]$ by ... Then its quasiconvex envelope is ...

open_problem | open | MathDB 页面 | 原文线索

V2-074772 | MATHDB-393691

Finite-order laminate representation of the rank-one convex envelope

摘录 / EXCERPT: Find sufficient conditions, in terms of properties of the energy density W that are easy to check, such that W^{rc} can be computed with a laminate of finite order.

open_problem | open | MathDB 页面 | 原文线索

V2-074773 | MATHDB-393692

First-step brutal partial-damage minimization and quasiconvex relaxation

摘录 / EXCERPT: Under the report’s brutal partial-damage assumptions, with $\chi\in\{0,1\}$, $W(e,\chi)=\tfrac12(\chi A_w+(1-\chi)A_s)e\cdot e$, $A_w\leq A_s$, $A_w>0$, and $f_0=f(0)$, the initial...

open_problem | open | MathDB 页面 | 原文线索

V2-074774 | MATHDB-393695

A general construction of heterotic $(0,2)$ mirrors

摘录 / EXCERPT: Develop a general procedure for constructing heterotic $(0,2)$ mirrors.

open_problem | open | MathDB 页面 | 原文线索

V2-074775 | MATHDB-393696

Relevance of derived categories to heterotic strings

摘录 / EXCERPT: Are derived categories relevant for heterotic strings?

open_problem | open | MathDB 页面 | 原文线索

V2-074776 | MATHDB-393697

Whether the orientation-reversing Mukai-lattice involution is induced by an autoequivalence

摘录 / EXCERPT: For a general K3 surface X, let ... be the integral Hodge isometry of $H^*(X)$ that acts as $-\operatorname{id}$ on $H^2(X)$ and as the identity on $H^0(X)\oplus H^4(X)$. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-074777 | MATHDB-393698

Doublet-triplet splitting in a heterotic Standard Model construction

摘录 / EXCERPT: For the Calabi-Yau threefold $\widetilde X$ and equivariant vector bundle $\widetilde V$ described on the preceding pages, with the indicated $\mathbb Z_3\times\mathbb Z_3$ charact...

open_problem | open | MathDB 页面 | 原文线索

V2-074778 | MATHDB-393700

Mathematical definitions of the curve-moduli spaces

摘录 / EXCERPT: For both M_C and $\widehat{M}_C$, a mathematical definition is in general an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-074779 | MATHDB-393701

Extended moduli space of triangulated A_∞-categories

摘录 / EXCERPT: The extended moduli space is conjectured to be a moduli space of (triangulated) A_∞-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-074780 | MATHDB-393705

Geometric characterization of active boundary in the Robin problem on fractal domains

摘录 / EXCERPT: For a Euclidean domain $D\subset\mathbb{R}^d$, a fixed closed ball $B_*\subset D$ of non-zero radius, and the solution u of the mixed Dirichlet-Robin problem ... where $c>0$, the w...

open_problem | open | MathDB 页面 | 原文线索

V2-074781 | MATHDB-393707

Characterization of the trace process on a specified set

摘录 / EXCERPT: Suppose that X is a Lévy process in $\\mathbb{R}^n$ which is the sum of a Brownian motion in $\\mathbb{R}^n$ and an independent spherically symmetric $\\alpha$-symmetric stable p...

open_problem | open | MathDB 页面 | 原文线索

V2-074782 | MATHDB-393708

Riesz-transform boundedness on connected sums of nilpotent Lie groups

摘录 / EXCERPT: Let N be a simply connected nilpotent Lie group with a left-invariant metric, let ν be its homogeneous dimension,...

open_problem | open | MathDB 页面 | 原文线索

V2-074783 | MATHDB-393709

Non-well-posedness of the martingale problem for a class of non-local operators

摘录 / EXCERPT: The report states that, for a family of Lévy measures of the form $\mu(x,dh)=n(x,h)\,dh$ satisfying $c_0|h|^{-d-\alpha}\le n(x,h)\le c_1|h|^{-d-\beta}$ and $n(x,h)=n(x+h,-h)$, wher...

open_problem | open | MathDB 页面 | 原文线索

V2-074784 | MATHDB-393710

Classification of colored-order embeddability for wqo but non-bqo preorders

摘录 / EXCERPT: Let LO be the Polish space of total orders on $\mathbb{N}$. Given a preorder P on $\mathbb{N}$, define $\preccurlyeq_P$ on $LO\times\mathbb{N}^{\mathbb{N}}$ by...

open_problem | open | MathDB 页面 | 原文线索

V2-074785 | MATHDB-393711

Classification of dense-order-preserving domination for countable ordinals

摘录 / EXCERPT: For each countable ordinal $\alpha\in\omega_1$, let $\leq_\alpha$ be the usual order on α. For $\varphi,\psi\in{}^{\mathbb{Q}}\alpha$, define...

open_problem | open | MathDB 页面 | 原文线索

V2-074786 | MATHDB-393712

Continuous Fraïssé conjecture for closed subsets of \b0,1\d

摘录 / EXCERPT: For closed sets $A,B\subseteq[0,1]$, define $A\leq B$ iff there is a continuous, monotone, one-to-one embedding $f:A\to B$. Then $\leq$ is a better quasi-order, and its associated...

conjecture | open | MathDB 页面 | 原文线索

V2-074787 | MATHDB-393713

Continuous Fraïssé theorem for closed subsets of $[0,1]$

摘录 / EXCERPT: For two closed sets $A,B\subseteq[0,1]$, write $A\leq B$ if and only if there is a continuous monotone one-to-one embedding $f:A\to B$. Let $\sim$ be the associated equivalence rel...

open_problem | open | MathDB 页面 | 原文线索

V2-074788 | MATHDB-393719

Rosenthal’s unconditional-subsequence theorem for \[1m0\[0m-\[1m1\[0m-valued weakly null sequences

摘录 / EXCERPT: If [1m(xn)[0m is a weakly null sequence in [1mellinfty(Gamma)[0m, for some set [1mGamma[0m, and every [1m(xn)[0m takes only the values [1m0[0m and [1m1[0m, then [1m(xn)[0m has an u...

open_problem | open | MathDB 页面 | 原文线索

V2-074789 | MATHDB-393722

Closure of \(\Pi^1_3\) under quantification by \(\nu_n\)

摘录 / EXCERPT: Assuming AD, is Π^1_3 closed under quantification by the supercompactness measure ν_n on $P_{\omega_1}(\omega_n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074790 | MATHDB-393724

Characterization of non-polishable Borel groups via E_1

摘录 / EXCERPT: Let H be a Borel group. Is it true that H is not polishable if and only if there exists a Polish H-space X such that E_1 is Borel reducible to the orbit equivalence relat...

open_problem | open | MathDB 页面 | 原文线索

V2-074791 | MATHDB-393725

Consistency of the rainbow relation at \(\omega_1\)

摘录 / EXCERPT: It is consistent that $\omega_1 \to^*(\omega_1)^2_{2\text{-bdd}}$; that is, every 2-bounded coloring of $[\omega_1]^2$ has a polychromatic subset of order type ω_1.

open_problem | open | MathDB 页面 | 原文线索

V2-074792 | MATHDB-393726

Analytic P-ideal realization of the Borel-set isomorphism relation

摘录 / EXCERPT: Does there exist an assignment, Borel in the codes, of an analytic P-ideal I_A to each Borel set A of reals such that, for all Borel sets A and B, ...

open_problem | open | MathDB 页面 | 原文线索

V2-074793 | MATHDB-393730

Newton’s Proposition Six: velocity of the intersection of two moving curves

摘录 / EXCERPT: Let two curves c_1 and c_2 move so that they continuously intersect at a, and let v_1 and v_2 be representatives of the velocities of the two curves, represented in the f...

open_problem | open | MathDB 页面 | 原文线索

V2-074794 | MATHDB-393731

Tightness of the quadratic edge bound for geometric surfaces

摘录 / EXCERPT: Is it an open problem whether the upper bound $f_1=O(n^2)$ is tight?

open_problem | open | MathDB 页面 | 原文线索

V2-074795 | MATHDB-393732

Additive approximation of monotone edge-deletion problems

摘录 / EXCERPT: Let $\mathcal P$ be a monotone graph property, and let $E'_{\mathcal P}(G)$ be the minimum number of edge deletions needed to transform a graph G into a graph satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-074796 | MATHDB-393737

Angel escape for sufficiently large power

摘录 / EXCERPT: In the two-dimensional Angel-and-Devil game on $\mathbb{Z}^2$, the Angel starts at a square and may, on each move, jump to any square whose ℓ_∞-distance from his current...

open_problem | open | MathDB 页面 | 原文线索

V2-074797 | MATHDB-393740

Exact Turán function for the generalized triangle T_5

摘录 / EXCERPT: For the generalized triangle T_k whose edges are $\{1,\ldots,k\}$, $\{1,2,\ldots,k-1,k+1\}$, and $\{k,k+1,\ldots,2k-1\}$, determine the exact Turán function...

open_problem | open | MathDB 页面 | 原文线索

V2-074798 | MATHDB-393742

Generalized Upper Bound Conjecture for Levels in Arrangements of Great Hemispheres

摘录 / EXCERPT: If A is an arrangement of n great hemispheres in S^d and $0\le \ell < (n-d)/2$, then the number of vertices of A at level at most ℓ is at most $v(\ell,n,d)$, the corre...

conjecture | open | MathDB 页面 | 原文线索

V2-074799 | MATHDB-393743

Removing the factor 2 in the quadrupel-sequence bound

摘录 / EXCERPT: Can the factor 2 in Theorem 2 be removed? More precisely, under the theorem’s setup, does every ordered sequence of quadrupels $(A_r,B_r,C_r,D_r)_{r=1}^t$ satisfying (1)...

open_problem | open | MathDB 页面 | 原文线索

V2-074800 | MATHDB-393748

Erdős-Hajnal conjecture for bull-free graphs

摘录 / EXCERPT: Every bull-free graph G contains a stable set or a clique of size at least $|V(G)|^{1/4}$.

conjecture | open | MathDB 页面 | 原文线索

V2-074801 | MATHDB-393753

Matching points to hyperplanes

摘录 / EXCERPT: For every integer dimension $d\ge 1$ and every subset $P\subseteq\mathbb{R}^d$ whose affine span is $\mathbb{R}^d$, let H be the set of hyperplanes spanned by P. Does there exi...

open_problem | open | MathDB 页面 | 原文线索

V2-074802 | MATHDB-393754

Equality of maximum triangle-free and bipartite subgraphs in random graphs

摘录 / EXCERPT: Let $G=G(n,m)$ be a uniformly random graph with n vertices and $m=m(n)$ edges. Let $T(G)$ be a largest subset of $E(G)$ containing no triangle, and let $B(G)$ be a largest subset...

open_problem | open | MathDB 页面 | 原文线索

V2-074803 | MATHDB-393757

Almost-sure nonsingularity of random symmetric Bernoulli matrices

摘录 / EXCERPT: Let Q_n be the random symmetric $n\times n$ matrix whose upper-diagonal entries are i.i.d. Bernoulli random variables. Is Q_n almost surely nonsingular?

open_problem | open | MathDB 页面 | 原文线索

V2-074804 | MATHDB-393758

Perfect packings of complete graphs minus one edge

摘录 / EXCERPT: Kawarabayashi conjectured that if $H=K_\\ell^{-}$, the complete graph on $\\ell$ vertices with one edge removed, and $\\ell\\ge 4$, then in Theorem 1 one can replace the chromatic...

open_problem | open | MathDB 页面 | 原文线索

V2-074805 | MATHDB-393762

Binding-energy maximum and onset of instability in the Einstein-Vlasov system

摘录 / EXCERPT: Conjecture: For the spherically symmetric, asymptotically flat Einstein-Vlasov system, the binding-energy maximum along a steady-state sequence signals the onset of instability. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-074806 | MATHDB-393763

Examples with finite or vanishing Weyl tensor at an initial singularity

摘录 / EXCERPT: The report answers affirmatively the question of whether cosmological models can have an initial singularity at which the Weyl tensor is finite or zero. Let (M,g) be a spacetime,...

open_problem | open | MathDB 页面 | 原文线索

V2-074807 | MATHDB-393764

Converse characterization of conformal gauge singularities

摘录 / EXCERPT: Whether, or when, cosmological singularities with finite Weyl tensor are conformal gauge singularities.

open_problem | open | MathDB 页面 | 原文线索

V2-074808 | MATHDB-393765

The red-shift effect and decay rates for the wave equation on a Schwarzschild exterior

摘录 / EXCERPT: Let (M,g) be the maximally extended Schwarzschild spacetime with parameter $M>0$, let S be a complete Cauchy surface, and let φ be a locally C^6 solution of ... on M...

open_problem | open | MathDB 页面 | 原文线索

V2-074809 | MATHDB-393767

Existence and uniqueness of eternal acceleration

摘录 / EXCERPT: Theorem 1 (Existence and uniqueness of eternal acceleration). Let $D\ge 10$ (with $(m,n)\ne(2,7)$). Then there exists a unique solution $\gamma=-d\bar t^2+\bar a^2g$ of the $(1+m)$...

open_problem | open | MathDB 页面 | 原文线索

V2-074810 | MATHDB-393769

Equivalence of contractions with generalized Inönü-Wigner contractions

摘录 / EXCERPT: Every contraction (respectively, sequential contraction) is equivalent to a generalized Inönü-Wigner contraction with integer exponents.

open_problem | open | MathDB 页面 | 原文线索

V2-074811 | MATHDB-393772

Conditions preserving the number of invariants under generalized Inönü-Wigner contractions

摘录 / EXCERPT: Find sufficient and applicable conditions to ensure that a generalized Inönü-Wigner contraction $g\to g'$ preserves the number $N(g)$ of invariants.

open_problem | open | MathDB 页面 | 原文线索

V2-074812 | MATHDB-393774

Balance of canonical momentum without specifying the energy density

摘录 / EXCERPT: Can we formulate the balance equation of canonical momentum without specifying the energy density?

open_problem | open | MathDB 页面 | 原文线索

V2-074813 | MATHDB-393776

Mesoscale required to attain a representative volume element

摘录 / EXCERPT: For a specific type of random microstructure, on what mesoscale is the representative volume element (RVE) attained within a prescribed accuracy?

open_problem | open | MathDB 页面 | 原文线索

V2-074814 | MATHDB-393778

Constructing toric degenerations from polyhedral data

摘录 / EXCERPT: Can one construct a toric degeneration for any given pair (B,P)?

open_problem | open | MathDB 页面 | 原文线索

V2-074815 | MATHDB-393781

Uniform KKMS bounds for unimodular triangulations

摘录 / EXCERPT: Is there a constant $n=n(d)$ that works for all d-dimensional polytopes?

open_problem | open | MathDB 页面 | 原文线索

V2-074816 | MATHDB-393783

Unimodality of Ehrhart h-vectors

摘录 / EXCERPT: Let $P\subset\mathbb{R}^{n-1}$ be an integral convex polytope, and write its Ehrhart series as ... with $h_d\ne 0$. For which such polytopes is the h-vector $h(P)=(h_0,\ldots,h_d)$ u...

open_problem | open | MathDB 页面 | 原文线索

V2-074817 | MATHDB-393785

Unimodality of h-vectors of Gorenstein polytopes

摘录 / EXCERPT: Let P be an integral polytope such that P has a regular unimodular triangulation and $K[P]$ is Gorenstein. Then the h-vector of P satisfies ... More precisely, the vector ... i...

open_problem | open | MathDB 页面 | 原文线索

V2-074818 | MATHDB-393788

Exceptional parameters and exceptional quasidegrees for GKZ systems

摘录 / EXCERPT: Let $A\in\mathbb{Z}^{d\times n}$ have rank d, with its columns contained in a single open halfspace, and let $\mathbb{C}[\mathbb{N}A]$ have maximal homogeneous ideal...

open_problem | open | MathDB 页面 | 原文线索

V2-074819 | MATHDB-393789

Ottaviani-Paoletti conjecture for Veronese embeddings

摘录 / EXCERPT: For integers $d\ge 3$ and $n\ge 2$, the line bundle $\mathcal O_{\mathbb P^n}(d)$ satisfies Green-Lazarsfeld’s property N_{3d-3}; equivalently, the degree-d Veronese embedding...

conjecture | open | MathDB 页面 | 原文线索

V2-074820 | MATHDB-393791

Triangle groups as affine groups of Teichmüller curves

摘录 / EXCERPT: Let $\Gamma=\Delta(n,m,\infty)$, where $n,m\in\mathbb{Z}_{>1}\cup\{\infty\}$ satisfy $\frac{1}{n}+\frac{1}{m}<1$. Then there exists a Teichmüller curve C with projective affine g...

open_problem | open | MathDB 页面 | 原文线索

V2-074821 | MATHDB-393792

Optimal degree bounds for lifting points from reduced Hurwitz spaces

摘录 / EXCERPT: Given $\operatorname{prd}\in H^{\mathrm{rd}}(\mathcal C)(k)$, find an optimal upper bound for ...

open_problem | open | MathDB 页面 | 原文线索

V2-074822 | MATHDB-393796

Growth of ramification indices in four-branch-point modular towers

摘录 / EXCERPT: For reduced modular towers with four branch points, control the growth of the ramification indices-especially at points above infinity, the cusps-along the tower, in order to deter...

open_problem | open | MathDB 页面 | 原文线索

V2-074823 | MATHDB-393801

Shafarevich conjecture for global fields

摘录 / EXCERPT: If K is a global field, then $G_{K^{\mathrm{cycl}}}$ is free of countable rank.

conjecture | open | MathDB 页面 | 原文线索

V2-074824 | MATHDB-393802

Proper solutions for finite split embedding problems over large fields

摘录 / EXCERPT: Let F be a large field and let X be a smooth projective connected F-curve, with function field K. Every finite split embedding problem $(\alpha:G_K\to G,\ f:\Gamma\to G)$ f...

open_problem | open | MathDB 页面 | 原文线索

V2-074825 | MATHDB-393811

Poonen’s theorem on large subrings of \(\mathbb{Q}\)

摘录 / EXCERPT: There exist recursive sets of rational primes T_1 and T_2, both of natural density zero and with empty intersection, such that for any set S of rational primes containing...

open_problem | open | MathDB 页面 | 原文线索

V2-074826 | MATHDB-393814

Goodness of the mapping class group in the sense of Serre

摘录 / EXCERPT: The mapping class group is conjectured to be good in the sense of Serre.

open_problem | open | MathDB 页面 | 原文线索

V2-074827 | MATHDB-393815

Oda-Tamagawa criterion for good reduction of curves

摘录 / EXCERPT: Let S be the spectrum of a discrete valuation ring, and let U be its generic point. Then a curve of genus $g\ge 2$ over U extends uniquely over S if and only if the associa...

open_problem | open | MathDB 页面 | 原文线索

V2-074828 | MATHDB-393816

Ax’s conjecture on PAC fields

摘录 / EXCERPT: Is every perfect PAC field a C_1-field?

conjecture | open | MathDB 页面 | 原文线索

V2-074829 | MATHDB-393817

The ex-Ax conjecture (after Kollár)

摘录 / EXCERPT: Every PAC field of characteristic 0 is a C'_1-field; equivalently, if k is a PAC field of characteristic 0 and $f_1,\ldots,f_s\in k[x_0,\ldots,x_n]$ are homogeneous polynom...

conjecture | open | MathDB 页面 | 原文线索

V2-074830 | MATHDB-393818

A local proof of non-abelian Lubin-Tate theory for higher dimension

摘录 / EXCERPT: Give a local proof of Harris and Taylor’s theorem for $h>2$.

open_problem | open | MathDB 页面 | 原文线索

V2-074831 | MATHDB-393819

Characterization of the graph class admitting efficient chromatic-polynomial computation

摘录 / EXCERPT: Characterize the graph class permitting efficient computation of the chromatic polynomial-obtained by switching between a graph and its complement during the computation, and gener...

open_problem | open | MathDB 页面 | 原文线索

V2-074832 | MATHDB-393820

Convergence of the random-graph k-colorability threshold

摘录 / EXCERPT: For each fixed integer $k\ge 3$, let $d_k^*(n)$ be the Achlioptas-Friedgut threshold sequence such that, for every $\varepsilon>0$, a random graph with average degree...

open_problem | open | MathDB 页面 | 原文线索

V2-074833 | MATHDB-393821

Best possible induced-forest bound for 4-regular graphs

摘录 / EXCERPT: Examples in [4] suggest that the best-possible estimate could be $a(G) \ge \frac{4}{7}n$ for 4-regular graphs G of order n.

open_problem | open | MathDB 页面 | 原文线索

V2-074834 | MATHDB-393824

Precoloring extension for planar bipartite graphs with separated independent precolored vertices

摘录 / EXCERPT: Let G be a planar bipartite graph with a list assignment satisfying $|L(v)|\ge 3$ for every $v\in V(G)$, and let $W\subseteq V(G)$ be independent. Here $d(W)$ is the minimum dist...

open_problem | open | MathDB 页面 | 原文线索

V2-074835 | MATHDB-393827

Conjectured list-labeling number of paths for equal distance constraints

摘录 / EXCERPT: Conjecture 2. For every nonnegative integer d and every path P_n with $n\ge 3$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-074836 | MATHDB-393828

Equality of list and ordinary $L(2,1)$-labelling numbers for trees

摘录 / EXCERPT: For every tree T, it holds that $\chi^{\ell}_{2,1}(T)=\chi_{2,1}(T)$.

open_problem | open | MathDB 页面 | 原文线索

V2-074837 | MATHDB-393829

Characterization of non-normal 1-trees

摘录 / EXCERPT: Let $G_1+_vG_2$ denote the graph obtained from G_1 and G_2 by identifying them at node v. A 1-tree G is not normal if and only if one of the following holds: 1. $G=C_5$. 2....

open_problem | open | MathDB 页面 | 原文线索

V2-074838 | MATHDB-393830

Normal Graph Conjecture for cacti and their complements

摘录 / EXCERPT: The Normal Graph Conjecture is true for cacti and their complements.

conjecture | open | MathDB 页面 | 原文线索

V2-074839 | MATHDB-393831

Distributed coloring in \(\widetilde O(\sqrt{\log n})\) bit rounds

摘录 / EXCERPT: In the synchronous distributed model where each edge permits at most one transmitted bit per direction in each bit round, processors initially have identical states and know only t...

open_problem | open | MathDB 页面 | 原文线索

V2-074840 | MATHDB-393833

Finding a maximum-weight perfect matching in general graphs

摘录 / EXCERPT: The problem of finding a maximum-weight perfect matching in general graphs remains unsolved.

open_problem | open | MathDB 页面 | 原文线索

V2-074841 | MATHDB-393834

Improving the approximation factor for Maximum Acyclic Subgraph

摘录 / EXCERPT: Given a directed weighted graph $G=(V,A)$ with nonnegative edge weights, the maximum acyclic subgraph problem asks for a maximum-weight subset of edges inducing an acyclic subgraph...

open_problem | open | MathDB 页面 | 原文线索

V2-074842 | MATHDB-393836

Mixing rate of the triangulation walk

摘录 / EXCERPT: Determine the mixing rate of the Markov process that starts at some triangulation of a planar point set and repeatedly flips a random flippable edge.

open_problem | open | MathDB 页面 | 原文线索

V2-074843 | MATHDB-393838

Polynomial-time recognition of level-1 D-graphs

摘录 / EXCERPT: Is it possible to recognize level 1 D-graphs in polynomial time?

open_problem | open | MathDB 页面 | 原文线索

V2-074844 | MATHDB-393839

Counting homomorphisms to a fixed directed acyclic graph

摘录 / EXCERPT: For a fixed directed acyclic graph H, given an input directed graph G, determine the number of homomorphisms from G to H, where a homomorphism is a function from $V(G)$ to...

open_problem | open | MathDB 页面 | 原文线索

V2-074845 | MATHDB-393844

Distinctness of the intermediate ring classes

摘录 / EXCERPT: Do the intermediate conditions above define distinct classes of rings?

open_problem | open | MathDB 页面 | 原文线索

V2-074846 | MATHDB-393848

Class-invariance theorem for Bousfield classes of finite spectra

摘录 / EXCERPT: Let X and Y be finite p-local spectra. Then $\langle X\rangle\leq\langle Y\rangle$ if and only if $\operatorname{type}(X)\geq\operatorname{type}(Y)$, where $\langle-\rangle$...

open_problem | open | MathDB 页面 | 原文线索

V2-074847 | MATHDB-393849

Telescope conjecture for the Bousfield classes of $T(n)$ and $K(n)$

摘录 / EXCERPT: The telescope conjecture asserts that the spectra $T(n)$ and $K(n)$ referred to in the report are Bousfield equivalent, i.e. $\langle T(n)\rangle=\langle K(n)\rangle$.

conjecture | open | MathDB 页面 | 原文线索

V2-074848 | MATHDB-393856

Localization of well-generated triangulated categories

摘录 / EXCERPT: Is any well generated triangulated category a localization of a compactly generated category?

open_problem | open | MathDB 页面 | 原文线索

V2-074849 | MATHDB-393859

Van Hove asymptotics for the Neumann Laplacian in the supercritical phase

摘录 / EXCERPT: The percolating cluster produces the van Hove asymptotics ... in the Neumann case for $p>p_c$.

open_problem | open | MathDB 页面 | 原文线索

V2-074850 | MATHDB-393860

Skew-field embedding for torsion-free subgroups of $\mathrm{GL}_d(\mathbb{Z})$

摘录 / EXCERPT: Let G be a torsion-free subgroup of $\mathrm{GL}_d(\mathbb{Z})$. Does there exist a skew field D such that $\mathbb{C}G \subseteq D \subseteq U(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074851 | MATHDB-393864

Finite-element convergence for a state-constrained elliptic control problem

摘录 / EXCERPT: Let r $\Omega\subset\mathbb{R}^d$, $d\in\{2,3\}$, be a bounded, convex domain with smooth boundary. For $u\in L^2(\Omega)$, let $y=G(u)$ solve ... Given $\alpha>0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-074852 | MATHDB-393865

Efficient implementation of a generalized derivative for Preisach hysteresis identification

摘录 / EXCERPT: The efficient implementation of this generalized derivative in a semismooth Newton method for hysteresis identification is still an open problem, due to the arising integral as wel...

open_problem | open | MathDB 页面 | 原文线索

V2-074853 | MATHDB-393866

Completion of an elementary hexahedron for a discrete conjugate net

摘录 / EXCERPT: For a discrete conjugate net with $m=3$, given seven points f, f_i, and f_{ij} such that the quadrilaterals (f,f_i,f_{ij},f_j) are planar, find the eighth point f_{123} s...

open_problem | open | MathDB 页面 | 原文线索

V2-074854 | MATHDB-393869

Super stability of tensegrities from convex 3-dimensional polytopes

摘录 / EXCERPT: Suppose P is a convex polytope in E^3. Form a tensegrity $G(p)$ by taking the vertices of P as the configuration vertices, assigning the edges of P as cables, and assigning...

open_problem | open | MathDB 页面 | 原文线索

V2-074855 | MATHDB-393870

Densest packing of regular tetrahedra in Euclidean 3-space

摘录 / EXCERPT: Determine the densest packing of regular tetrahedra in Euclidean space $\mathbb{E}^3$.

open_problem | open | MathDB 页面 | 原文线索

V2-074856 | MATHDB-393871

Extension of polarity-preserved rigidity to smooth surfaces

摘录 / EXCERPT: Rigidity is preserved under polarity: if a non-convex triangulated polyhedron is infinitesimally rigid, then its polar polytope is infinitesimally rigid as well; the polar polytope...

open_problem | open | MathDB 页面 | 原文线索

V2-074857 | MATHDB-393872

Rigidity of equal-area triangulations of a quadrilateral

摘录 / EXCERPT: Given a triangulation of a quadrilateral in which all triangle areas are equal, is the configuration rigid under the constraint that the triangle areas remain constant?

open_problem | open | MathDB 页面 | 原文线索

V2-074858 | MATHDB-393873

Bobenko’s Delaunay-triangulation unfolding problem

摘录 / EXCERPT: Can the boundary of a convex polytope be unfolded into the plane without self-overlap by cutting the surface along edges of the Delaunay triangulation T of the boundary? (T can...

open_problem | open | MathDB 页面 | 原文线索

V2-074859 | MATHDB-393876

Strong quadrisecant-free embeddings versus $1/2$-planarity

摘录 / EXCERPT: For a graph $G=(V,E)$, with curved edges permitted, is it true that G admits an embedding in 3-space without a strong quadrisecant line if and only if it is $1/2$-planar? Here,...

open_problem | open | MathDB 页面 | 原文线索

V2-074860 | MATHDB-393880

A direct isotopy proof for piecewise-linear Morse theory

摘录 / EXCERPT: Let M be a triangulation of a convex region in $\[?]R^3$, and let $f:M\to\mathbb{R}$ be obtained by assigning pairwise distinct values to the vertices and extending linearly over...

open_problem | open | MathDB 页面 | 原文线索

V2-074861 | MATHDB-393883

Geometric realizability of triangulations of low-genus orientable surfaces

摘录 / EXCERPT: Every triangulation of an orientable surface of genus $1\le g\le 4$ is geometrically realizable.

open_problem | open | MathDB 页面 | 原文线索

V2-074862 | MATHDB-393884

Sufficiency of the area-constraint counting conditions for local uniqueness

摘录 / EXCERPT: The report conjectures that the counting conditions ... are sufficient for local uniqueness of a triangulated triangle with fixed triangle areas when the vertices are in generic posi...

open_problem | open | MathDB 页面 | 原文线索

V2-074863 | MATHDB-393886

Lovász conjecture for odd-cycle Hom-complexes

摘录 / EXCERPT: Let Γ be a finite graph and let C_{2r+1} be a cycle of odd length $2r+1$. If $\operatorname{Hom}(C_{2r+1},\Gamma)$ is k-connected, then $\chi(\Gamma) \ge k+4$.

conjecture | open | MathDB 页面 | 原文线索

V2-074864 | MATHDB-393888

Witt-group decomposition and vanishing-cycle surjectivity for perverse sheaves

摘录 / EXCERPT: Conjecture. The Witt group $W(\operatorname{Perv}(X))$ of perverse sheaves on X decomposes as a direct sum indexed by the simple objects of $\operatorname{Perv}(X)$. Furthermore,...

open_problem | open | MathDB 页面 | 原文线索

V2-074865 | MATHDB-393889

Splitting formula for noncommutative spectral flow along non-loop paths

摘录 / EXCERPT: Does such a splitting formula also exist for paths with invertible end points which are not necessarily loops?

open_problem | open | MathDB 页面 | 原文线索

V2-074866 | MATHDB-393891

Transfer maps in K-theory for smooth fibre bundles

摘录 / EXCERPT: Let $F\to E\xrightarrow{p}B$ be a smooth fibre bundle with n-dimensional closed fibre. The bundle projection induces a transfer map...

open_problem | open | MathDB 页面 | 原文线索

V2-074867 | MATHDB-393895

DVR sets from positive-weight cubature formulas

摘录 / EXCERPT: Let the configuration space be Ω, and let P_q^Ω denote the space of multivariable polynomials of degree at most q on Ω. Suppose we are given: (a) a cubature...

open_problem | open | MathDB 页面 | 原文线索

V2-074868 | MATHDB-393896

Optimal-control interpolation between falling-cat flip models

摘录 / EXCERPT: Construct relatively simple falling-cat motions between the rigid Kane-Scher model and the alternative, more acrobatic flip model, using an optimal-control formulation with several...

open_problem | open | MathDB 页面 | 原文线索

V2-074869 | MATHDB-393897

Explicit integration of the Euler top with three distinct moments of inertia

摘录 / EXCERPT: Find explicit, workable expressions for the exact flow of an Euler top with three distinct moments of inertia. The flow is known to be expressible in terms of theta functions, but...

open_problem | open | MathDB 页面 | 原文线索

V2-074870 | MATHDB-393904

Profinite relation rank of quasisimple groups with trivial Schur multiplier

摘录 / EXCERPT: Is it true that $\hat r(H)=2$ for every quasisimple group H with trivial Schur multiplier?

open_problem | open | MathDB 页面 | 原文线索

V2-074871 | MATHDB-393910

Beyond the Jantzen Region: Conjecture 3.2

摘录 / EXCERPT: Assume that $p>h$. For each $\\lambda\in X^+_{\mathrm{reg}}$, one has ...

conjecture | open | MathDB 页面 | 原文线索

V2-074872 | MATHDB-393921

Kazhdan-Lusztig left cells in the fractional-parameter case for type B_n

摘录 / EXCERPT: Let W be of type B_n, with weight-function parameters $a>0$ on $s_1,\dots,s_{n-1}$ and $b>0$ on t. In the case ... it is conjectured that the partition of W into KazhdanLuszt...

open_problem | open | MathDB 页面 | 原文线索

V2-074873 | MATHDB-393922

Kazhdan-Lusztig left cells at integral unequal parameters in type B_n

摘录 / EXCERPT: Let W be a finite Coxeter group of type B_n with weight function $L:W\to\mathbb{Z}$ satisfying $b:=L(t)>0$ and $a:=L(s_1)=\cdots=L(s_{n-1})>0$. If $b/a=r$ for some...

open_problem | open | MathDB 页面 | 原文线索

V2-074874 | MATHDB-393927

Universal validity of the higher analytic torsion formula (3)

摘录 / EXCERPT: It is conjectured that, for all fibre bundles $E\to B$ and all coefficient bundles $F\to E$ satisfying the conditions above,nn ...

open_problem | open | MathDB 页面 | 原文线索

V2-074875 | MATHDB-393930

Construct a massless \(\phi^4\) field theory in \(\mathbb{R}^3\)

摘录 / EXCERPT: Construct a massless ϕ^4 field theory in $\mathbb{R}^3$.

open_problem | open | MathDB 页面 | 原文线索

V2-074876 | MATHDB-393932

Extension of the limiting absorption principle to the full non-point spectrum

摘录 / EXCERPT: We conjecture that one can extend the method of this paper to the more general energy interval $\sigma(H)\setminus\sigma_{\mathrm{pp}}(H)$ for the Nelson model and to...

open_problem | open | MathDB 页面 | 原文线索

V2-074877 | MATHDB-393933

Limiting absorption principle on the full non-point spectrum for the Nelson model

摘录 / EXCERPT: We conjecture that one can extend the method of this paper to the energy interval $\sigma(H)\setminus\sigma_{\mathrm{pp}}(H)$ for the Nelson model.

open_problem | open | MathDB 页面 | 原文线索

V2-074878 | MATHDB-393935

Constructing tight convergent convex relaxations for dynamic-system functionals

摘录 / EXCERPT: How can one construct, in a computationally tractable manner, tight convergent convex relaxations of functionals with ODEs, DAEs, or PDAEs embedded?

open_problem | open | MathDB 页面 | 原文线索

V2-074879 | MATHDB-393937

Theorem 2: Solvability of the unconstrained two-point problem

摘录 / EXCERPT: Under the quadratic regulator setup on pages 25-27, define ... and ... Then ... so, for every initial condition $x_0\in X$, the two-point problem without the compatibility condition, ... i...

open_problem | open | MathDB 页面 | 原文线索

V2-074880 | MATHDB-393938

Necessity of the Reynolds-number restriction for stability

摘录 / EXCERPT: For the linearized degenerate hyperbolic PDAE system modeling the heat exchanger, the report proves stability when the Reynolds number is not too large. Is this Reynolds-number res...

open_problem | open | MathDB 页面 | 原文线索

V2-074881 | MATHDB-393939

Relation between the regular strangeness and tractability indices of linear DAEs

摘录 / EXCERPT: For a linear DAE of the form $A(t)(D(t)x(t))' + B(t)x(t)=q(t)$ with sufficiently smooth coefficient matrices and with the regular strangeness index $_s$ and tractability index...

open_problem | open | MathDB 页面 | 原文线索

V2-074882 | MATHDB-393941

Subdirect products of limit groups

摘录 / EXCERPT: Conjecture 1. Let $\Gamma_1,\ldots,\Gamma_n$ be limit groups, and let S be a type FP_n subgroup of their direct product. Then S is itself, up to finite index, a direct produc...

open_problem | open | MathDB 页面 | 原文线索

V2-074883 | MATHDB-393942

Exact optimal solution for the multi-peg Hanoi Towers problem

摘录 / EXCERPT: For each integer $k\ge 4$, with pegs labeled $0,\ldots,k-1$ and n disks of different sizes labeled $1,\ldots,n$ in increasing order of size, determine the smallest number of lega...

open_problem | open | MathDB 页面 | 原文线索

V2-074884 | MATHDB-393943

Higher-order isoperimetric spectra of groups

摘录 / EXCERPT: For each integer $n\ge 1$, which functions of the form x^α, with $\alpha\in[1,\infty)$, arise as order-n Dehn functions of groups of type F_{n+1}?

open_problem | open | MathDB 页面 | 原文线索

V2-074885 | MATHDB-393945

Amenable non-virtually cyclic groups with cut-points in asymptotic cones

摘录 / EXCERPT: Is there an amenable non-virtually cyclic group with cut-points in its asymptotic cones?

open_problem | open | MathDB 页面 | 原文线索

V2-074886 | MATHDB-393949

Special Aronszajn trees at \(\aleph_2\) or \(\aleph_3\)

摘录 / EXCERPT: Can this technique be used to reprove Theorem 5, or at least construct a model with no special Aronszajn trees on $\aleph_2$ or $\aleph_3$?

open_problem | open | MathDB 页面 | 原文线索

V2-074887 | MATHDB-393953

SU(3) Casson invariant conjecture for unions of torus-knot complements

摘录 / EXCERPT: For the indicated unions of certain $(2,p)$- and $(2,p')$-torus-knot pieces, $M=T_{2,p}\cup_T T_{2,p'}$, the conjecture is ...

conjecture | open | MathDB 页面 | 原文线索

V2-074888 | MATHDB-393954

Theoretical frameworks for integrating original sources into mathematics education

摘录 / EXCERPT: What are the possible epistemological/theoretical basis and frameworks for research and development towards the integration of original sources into the teaching and learning of ma...

open_problem | open | MathDB 页面 | 原文线索

V2-074889 | MATHDB-393955

Designing classroom activities for conceptual understanding of variables, unknowns, and parameters

摘录 / EXCERPT: An open problem is to design appropriate classroom activities that can provide students with rich mathematical experiences, allowing them to move along different layers of generali...

open_problem | open | MathDB 页面 | 原文线索

V2-074890 | MATHDB-393971

Preemptive donor lymphocyte infusion before relapse

摘录 / EXCERPT: Would preemptive donor lymphocyte infusion (DLI), administered before any evidence of relapse, be beneficial?

open_problem | open | MathDB 页面 | 原文线索

V2-074891 | MATHDB-393972

Singular-limit characterization for the tissue-degradation system

摘录 / EXCERPT: For the parabolic problem (P_k) on $Q_T=(0,T)\times\mathbb{R}^n_+$, ... with ... and the initial data specified in (P_k), the report proves that there are functions...

open_problem | open | MathDB 页面 | 原文线索

V2-074892 | MATHDB-393978

Classification of infinite tame groups

摘录 / EXCERPT: What is a classification of infinite tame groups?

open_problem | open | MathDB 页面 | 原文线索

V2-074893 | MATHDB-393979

Maximal length of strongly free quadratic sequences

摘录 / EXCERPT: For $h=2$, one can find strongly free sequences $r_1,\\ldots,r_m$ with $1\le m\le t(d)$, where ... It is conjectured that $t(d)$ is the largest possible value of m.

open_problem | open | MathDB 页面 | 原文线索

V2-074894 | MATHDB-393981

Settledness of irreducible quadratic polynomials over finite fields

摘录 / EXCERPT: Any irreducible quadratic polynomial $f\in\mathbb{F}_p[x]$ is settled.

open_problem | open | MathDB 页面 | 原文线索

V2-074895 | MATHDB-393985

Denominator of a Betti Eisenstein class for a weight-22 cusp form

摘录 / EXCERPT: Let f be the unique cusp form of weight 22, $f(q)=q-288q^2-128844q^3-2014208q^4+21640950q^5+\[0m\cdots=\sum_{n=1}^{\infty}a_nq^n$. The report notes the divisibility...

open_problem | open | MathDB 页面 | 原文线索

V2-074896 | MATHDB-393988

Ramification of primes in the maximal pro-\p extension

摘录 / EXCERPT: Can it ever happen that some mathfrak pin S does not ramify in $k_S(p)$?

open_problem | open | MathDB 页面 | 原文线索

V2-074897 | MATHDB-393990

Principal representatives in the noncommutative Iwasawa structure theorem

摘录 / EXCERPT: Let G be a noncommutative compact p-adic analytic group, with G assumed p-valued for the cited structure theorem, and let M be a finitely generated torsion...

open_problem | open | MathDB 页面 | 原文线索

V2-074898 | MATHDB-393992

Theorem 4: Equivalence involving the Tame Fontaine-Mazur Conjecture

摘录 / EXCERPT: Theorem 4. The Tame Fontaine-Mazur Conjecture holds for all number fields if and only if every infinite p-adic analytic extension of every number field is asymptotically bad.

conjecture | open | MathDB 页面 | 原文线索

V2-074899 | MATHDB-393997

RIGP question for virtually pro-\p universal Frattini towers

摘录 / EXCERPT: In the setup where G is a finite centerless p-perfect group, $p\mid |G|$, and $\widetilde{G}_p=\varprojlim_k G_k$ is its universal p-Frattini cover, let Ram_{r_0}...

open_problem | open | MathDB 页面 | 原文线索

V2-074900 | MATHDB-394000

Marković's non-realization theorem for mapping class groups

摘录 / EXCERPT: For every closed surface S of genus $g\ge 6$, the natural projection from the homeomorphism group to the mapping class group, $\operatorname{Homeo}(S)\to\operatorname{MC}(S)$, ha...

open_problem | open | MathDB 页面 | 原文线索

V2-074901 | MATHDB-394004

Non-realization of the mapping class group by homeomorphisms

摘录 / EXCERPT: Let M be a closed surface of genus $g\ge 2$, let $\mathrm{MC}(M)$ be the mapping class group of orientation-preserving homeomorphisms, let $\mathrm{Homeo}(M)$ be the group of ori...

open_problem | open | MathDB 页面 | 原文线索

V2-074902 | MATHDB-394006

Function space for generalized twist parameters on infinite-type surfaces

摘录 / EXCERPT: Let R be a Riemann surface of infinite type with a fixed maximal multicurve S and a fixed structure τ_0. Consider the map ... from the Teichmüller space of R. By equiconti...

open_problem | open | MathDB 页面 | 原文线索

V2-074903 | MATHDB-394007

Teichmüller ζ-functions

摘录 / EXCERPT: Let d denote the Teichmüller distance and let MC denote the mapping class group of some surface of finite type. Consider the ζ-type function ... Find the critical exponent s, i...

open_problem | open | MathDB 页面 | 原文线索

V2-074904 | MATHDB-394017

Eisenbud-Harris conjecture on high-genus projective curves

摘录 / EXCERPT: Let $C\subset\mathbb{P}^n$ be a reduced, irreducible, nondegenerate curve of degree d and arithmetic genus g, and let $\alpha\in\{0,1,\ldots,n-1\}$. If $g>\pi_\alpha(d,n)$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-074905 | MATHDB-394018

Conjecture 2 on symmetric point sets in projective space

摘录 / EXCERPT: If $\Gamma \subset \mathbb{P}^{n-1}$ is a set of $d \ge 2n+2m-1$ points in symmetric position, with $h_{\Gamma}(2) \ge 2n+m-2$ (where $1 \le m \le n-3$), then Γ must lie on...

conjecture | open | MathDB 页面 | 原文线索

V2-074906 | MATHDB-394020

Simple connectedness of a Godeaux surface under the stated bicanonical-pencil hypotheses

摘录 / EXCERPT: Let S be a surface with $p_g(S)=0$ and $K_S^2=1$, so that S is a Godeaux surface. Assume that its bicanonical pencil $|2K_S|$ consists of curves of genus 4, that $|2K_S|$ has...

open_problem | open | MathDB 页面 | 原文线索

V2-074907 | MATHDB-394024

Hot-spot asymptotics for a point load on a thin shell

摘录 / EXCERPT: In the shallow shell model for a shell of small positive thickness d, subject to a normal point load $Q(u)=F\,\delta_P$ at a point P of the midsurface, the zero-thickness asymp...

open_problem | open | MathDB 页面 | 原文线索

V2-074908 | MATHDB-394025

Uniqueness of the interfacial speed in the phase-field profile problem

摘录 / EXCERPT: For prescribed end-state densities ρ_1^0 and ρ_2^0, determine whether there exists a unique value λ, where $\lambda=(v_\Gamma-v)\mathbin{\cdot}\nu$, such that the...

open_problem | open | MathDB 页面 | 原文线索

V2-074909 | MATHDB-394026

Separate Γ-convergence and convergence of rate-independent solutions

摘录 / EXCERPT: Suppose that the energy and dissipation functionals (E_k,D_k) Γ-converge independently to (E,D), and that solutions $q_k:[0,T]\to Q$ converge pointwise,...

open_problem | open | MathDB 页面 | 原文线索

V2-074910 | MATHDB-394028

Precise time scales for screening and encounter corrections in Ostwald ripening

摘录 / EXCERPT: “The precise time scales over which what effect is most relevant have still to be identified.” Here, the effects are the screening/fluctuation correction and the particle-encounter...

open_problem | open | MathDB 页面 | 原文线索

V2-074911 | MATHDB-394030

Automorphisms of real affine spheres

摘录 / EXCERPT: Is the automorphism group of the real affine quadric (the n-sphere) $\{(x_0,\ldots,x_n)\in\mathbb{R}^{n+1}:x_0^2+\cdots+x_n^2=1\}$ equal to the orthogonal group $O(n+1)$ for arbi...

open_problem | open | MathDB 页面 | 原文线索

V2-074912 | MATHDB-394035

Computing the relative Brauer group of a genus 1 curve

摘录 / EXCERPT: Let k be a field and let C/k be a genus 1 curve. How can one compute the relative Brauer group ...

open_problem | open | MathDB 页面 | 原文线索

V2-074913 | MATHDB-394039

Simplicity testing for matrix and black-box groups

摘录 / EXCERPT: Decide whether a matrix group or black-box group of known characteristic is simple.

open_problem | open | MathDB 页面 | 原文线索

V2-074914 | MATHDB-394040

The explicit isomorphism problem for the Big Ree groups

摘录 / EXCERPT: Let $q=2^{2m+1}$ for some $m>0$, and let $S\leq \mathrm{GL}(26,q)$ be the standard copy of ${}^{2}F_{4}(q)$. Given $X\subseteq \mathrm{GL}(26,q)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-074915 | MATHDB-394041

Constructive Sylow problem IV for exceptional groups

摘录 / EXCERPT: For exceptional groups, solve problem (IV): given a nonabelian simple group of Lie type T, an automorphism θ of T with $|\theta|\nmid |T|$, and with p the characterist...

open_problem | open | MathDB 页面 | 原文线索

V2-074916 | MATHDB-394046

Two-generator, two-relator presentations for universal central extensions

摘录 / EXCERPT: The universal central extension of every finite simple group has a presentation with 2 generators and 2 relations.

open_problem | open | MathDB 页面 | 原文线索

V2-074917 | MATHDB-394047

The $F^{a,b,c}$ conjecture

摘录 / EXCERPT: Let $F^{a,b,c}=\\langle r,s\mid r^2=1,\,rs^ars^brs^c=1\\rangle$, let $n=a+b+c$, and let $d=(a-b,b-c)$. Suppose $(a,b,c)=1$ and $n\ne0$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-074918 | MATHDB-394051

Sublinear-time recognition of the natural action of S_n on k-sets

摘录 / EXCERPT: Given generators for a permutation group G acting on a set X of size $N=\binom{n}{k}$, where $k\le n/2$ is bounded and n is very large, and given that $G\cong S_n$ and that i...

open_problem | open | MathDB 页面 | 原文线索

V2-074919 | MATHDB-394055

Morrey’s conjecture in the planar matrix case

摘录 / EXCERPT: Whether rank-one convexity implies quasiconvexity in the case $n=m=2$.

conjecture | open | MathDB 页面 | 原文线索

V2-074920 | MATHDB-394056

Calderón-Zygmund estimates for the full range of exponents

摘录 / EXCERPT: For the model nonlinear equation ... extend Iwaniec’s implication ... from the established range $q\ge p$ to the full range $q\ge p-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-074921 | MATHDB-394057

Full Γ-limit for a singularly perturbed Eikonal functional

摘录 / EXCERPT: Determine the full Γ-limit of the functionals ... by removing the assumption $\nabla u_0\in BV$ in both Theorem 1 and Theorem 2 of the report. The report explicitly states tha...

open_problem | open | MathDB 页面 | 原文线索

V2-074922 | MATHDB-394058

Willmore conjecture for tori in Euclidean three-space

摘录 / EXCERPT: For an immersed surface $f:\Sigma\to\mathbb{R}^n$, the Willmore functional is $W(f)=\frac14\int_\Sigma |\vec H|^2\,d\mu_g$. Willmore conjectured that $W(f)\ge 2\pi^2$ for any torus...

conjecture | open | MathDB 页面 | 原文线索

V2-074923 | MATHDB-394059

Connectedness of optimal windows in convex domains

摘录 / EXCERPT: The report conjectures that optimal windows in convex domains should be connected.

open_problem | open | MathDB 页面 | 原文线索

V2-074924 | MATHDB-394060

Large-time behavior of the Bernoulli free-boundary gradient flow

摘录 / EXCERPT: Determine the large-time behavior of solutions $t\mapsto\Omega(t)$ of the evolution equation ... where $t\mapsto\Omega(t)$ is an unknown family of bounded open subsets of...

open_problem | open | MathDB 页面 | 原文线索

V2-074925 | MATHDB-394061

Monotonicity of the energy along the generalized Bernoulli flow

摘录 / EXCERPT: Prove that the energy ... where ... is non-increasing along the generalized viscosity solutions of the geometric flow ... with $\bar h(x,\Omega)=|\nabla u(x)|^2$, where u is the capaci...

open_problem | open | MathDB 页面 | 原文线索

V2-074926 | MATHDB-394062

Hall’s quantitative isoperimetric conjecture

摘录 / EXCERPT: For every integer $n\ge 2$, there exists a constant $C(n)$ such that every Borel set $E\subset\mathbb{R}^n$ with $0<|E|<\infty$ satisfies ... where B is the unit ball,...

conjecture | open | MathDB 页面 | 原文线索

V2-074927 | MATHDB-394063

One-bound-state conjecture for the one-dimensional Lieb-Thirring inequality

摘录 / EXCERPT: For the scalar Lieb-Thirring inequality for the Schrödinger operator $-\Delta-V(x)$ on $\mathbb{R}^n$, in the case $n=1$ and $\frac12\leq\gamma\leq\frac32$, Lieb and Thirring conje...

conjecture | open | MathDB 页面 | 原文线索

V2-074928 | MATHDB-394064

Semiclassical sharpness of the Lieb-Thirring constant for \(\gamma=1\) in dimensions \(n\ge 3\)

摘录 / EXCERPT: For the scalar Schrödinger operator $-\Delta-V$ on $\mathbb{R}^n$, is it true that ... Here $L(n,\gamma)$ is the sharp constant in the Lieb-Thirring inequality and...

open_problem | open | MathDB 页面 | 原文线索

V2-074929 | MATHDB-394065

Conditional duality theorem over the spectrum of a number ring

摘录 / EXCERPT: Assuming the Beilinson-Lichtenbaum conjecture, let B be the spectrum of the ring of integers of a number field. For a proper morphism $f:X\to B$ and a torsion sheaf G on X, d...

open_problem | open | MathDB 页面 | 原文线索

V2-074930 | MATHDB-394069

Gangl’s conjectural implication for the tame kernel when $d\equiv 6\pmod 9$

摘录 / EXCERPT: Let d be a negative square-free integer with $d\equiv 6\pmod 9$, and let $F=\mathbb{Q}(\sqrt d)$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-074931 | MATHDB-394072

Injectivity in the Gersten conjecture for Milnor K-theory

摘录 / EXCERPT: Let A be a regular connected semi-local ring containing an infinite field, and let $F=Q(A)$ be its fraction field. Then the natural homomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-074932 | MATHDB-394075

Products of maximal tori in a simple Lie group

摘录 / EXCERPT: Let G_i, $i=1,\ldots,n$, be maximal tori in a simple Lie group G. Determine the image of the map ...

open_problem | open | MathDB 页面 | 原文线索

V2-074933 | MATHDB-394076

Does exponential growth imply uniform exponential growth?

摘录 / EXCERPT: Does exponential growth imply uniform exponential growth?

open_problem | open | MathDB 页面 | 原文线索

V2-074934 | MATHDB-394077

Actions of irreducible higher-rank lattices on the real line

摘录 / EXCERPT: Suppose: - G is a connected, semisimple Lie group with finite center; - $\operatorname{rank}_{\mathbb{R}} G \ge 2$; and - Γ is any irreducible lattice in G. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-074935 | MATHDB-394078

Actions of fundamental groups of higher-rank irreducible locally symmetric spaces on the real line

摘录 / EXCERPT: Suppose Γ is the fundamental group of an irreducible locally symmetric space M of finite volume, such that: - M is of noncompact type and has no Euclidean factors locall...

open_problem | open | MathDB 页面 | 原文线索

V2-074936 | MATHDB-394079

Bounded generation of noncocompact lattices in $\operatorname{SL}(3,\mathbb{R})$ and $\operatorname{SL}(3,\mathbb{C})$

摘录 / EXCERPT: If Γ is any noncocompact lattice in either $\operatorname{SL}(3,\mathbb{R})$ or $\operatorname{SL}(3,\mathbb{C})$, then some finite-index subgroup of Γ is a product o...

open_problem | open | MathDB 页面 | 原文线索

V2-074937 | MATHDB-394082

Arnold-Nielsen conjecture

摘录 / EXCERPT: Let 4M4 be the surface and 4omega4 the area form in the preceding discussion. If 4phiinoperatorname{Symp}m(M,omega)4 has only non-degenerate fixed points, then the conjectured cons...

conjecture | open | MathDB 页面 | 原文线索

V2-074938 | MATHDB-394087

Cannon’s conjecture for hyperbolic groups with spherical boundary

摘录 / EXCERPT: If the ideal boundary of a hyperbolic group Γ is homeomorphic to the 2-sphere, then Γ acts isometrically on the 3-dimensional hyperbolic space $\mathbb{H}^3$ prop...

conjecture | open | MathDB 页面 | 原文线索

V2-074939 | MATHDB-394088

Isomorphism problem for hyperbolic groups with torsion

摘录 / EXCERPT: The isomorphism problem for hyperbolic groups with torsion remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-074940 | MATHDB-394089

Uniform Kazhdan property for \d4a5\d4f1_n(\mathbb{Z})

摘录 / EXCERPT: Is the group $\mathrm{SL}_n(\mathbb{Z})$, for $n\ge 3$, uniform Kazhdan; that is, does its uniform Kazhdan constant...

open_problem | open | MathDB 页面 | 原文线索

V2-074941 | MATHDB-394090

Congruence of maximal hyperbolic arithmetic reflection groups

摘录 / EXCERPT: Is it true that every maximal hyperbolic arithmetic reflection group is congruence?

open_problem | open | MathDB 页面 | 原文线索

V2-074942 | MATHDB-394094

Asymptotically equivalent invariant metrics and bounded difference

摘录 / EXCERPT: Let M be a complete length (or coarse length) space with respect to two metrics d_1,d_2. Let Γ act cocompactly by isometries with respect to both metrics. Assume ... Does...

open_problem | open | MathDB 页面 | 原文线索

V2-074943 | MATHDB-394100

Two-dimensional region-of-interest tomography problem

摘录 / EXCERPT: Let f be a smooth function with known compact support $\cOmega$. Define its 2D Radon transform by $Rf(\phi,s)=\int f(r\alpha+s\alpha^\perp)\,dr$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-074944 | MATHDB-394101

Stable ROI reconstruction for internal components of region A

摘录 / EXCERPT: Under the report’s 2D ROI-tomography setup, with convex support $ $OmegasubsetOmega+$, measured-line set M, unavailable-line set N, region $A=Omega+setminus B$, and $B={xinOmeg...

open_problem | open | MathDB 页面 | 原文线索

V2-074945 | MATHDB-394102

Classification of non-uniqueness sets for the spherical mean Radon transform

摘录 / EXCERPT: Conjecture. For $S\subset\mathbb{R}^d$, S is a non-uniqueness set for the spherical mean Radon transform if and only if ... where Σ is the surface of zeros of a homogeneous...

open_problem | open | MathDB 页面 | 原文线索

V2-074946 | MATHDB-394103

Range characterization via moment and orthogonality conditions

摘录 / EXCERPT: The report’s Theorem 1 asserts that, for $g\in C_0^\infty(S\times[0,2])$, the function g is representable as $g=R_Sf$ for some $f\in C_0^\infty(B)$ if and only if: (a) for every...

open_problem | open | MathDB 页面 | 原文线索

V2-074947 | MATHDB-394104

Inverse problem for non-constant wave speed in thermoacoustic tomography

摘录 / EXCERPT: Solve the inverse problem for non-constant wave speed.

open_problem | open | MathDB 页面 | 原文线索

V2-074948 | MATHDB-394107

Infinitely many exotic smooth structures on rational surfaces

摘录 / EXCERPT: Define $X_k=[U+007F]mathbb{CP}^2\#k\,\overline{[U+007F]mathbb{CP}}^{\,2}$. The rational surface X_k admits infinitely many exotic smooth structures for every integer $k\ge 5$.

open_problem | open | MathDB 页面 | 原文线索

V2-074949 | MATHDB-394109

Bing-double sliceness converse

摘录 / EXCERPT: Let K be a knot in S^3, and let $BD(K)$ denote its Bing double. If $BD(K)$ is a slice link, must K be a slice knot?

open_problem | open | MathDB 页面 | 原文线索

V2-074950 | MATHDB-394115

Strong version of Freedman’s conjecture

摘录 / EXCERPT: The conjecture was that the Borromean rings $L\subset S^3$ are not weakly A-B slice. Here, weakly A-B slice means that there exist decompositions $D^4=A_i\cup B_i$ and pair…

conjecture | open | MathDB 页面 | 原文线索

V2-074951 | MATHDB-394116

Freedman’s conjecture for the Borromean rings

摘录 / EXCERPT: Problem (Freedman’s conjecture). The Borromean rings are not $A\!-\B$ slice. Here, for a link $L=(l_1,\ldots,l_n)\subset S^3$, $A\!-\B$ sliceness means that there are decomposition…

conjecture | open | MathDB 页面 | 原文线索

V2-074952 | MATHDB-394118

Weak A-B sliceness of links with trivial linking numbers

摘录 / EXCERPT: For the Borromean rings, and more generally for every link $L\subset S^3$ with trivial linking numbers, L is weakly A-B slice.

open_problem | open | MathDB 页面 | 原文线索

V2-074953 | MATHDB-394121

Higher-order Arf invariant for the Bing double of the Figure-eight knot

摘录 / EXCERPT: Let $L=l_1\cup l_2\subset S^3$ be the Bing double of the Figure-eight knot. The link L bounds in B^4 two embedded disks D_i bounded by l_i, an immersed order-1 Whitney di…

open_problem | open | MathDB 页面 | 原文线索

V2-074954 | MATHDB-394124

Herrero’s direct-sum problem for hypercyclic operators

摘录 / EXCERPT: Is every hypercyclic operator T such that $T\oplus T$ is hypercyclic?

open_problem | open | MathDB 页面 | 原文线索

V2-074955 | MATHDB-394125

Herrero’s direct-sum hypercyclicity question

摘录 / EXCERPT: Must $T\oplus T$ be hypercyclic whenever T is?

open_problem | open | MathDB 页面 | 原文线索

V2-074956 | MATHDB-394126

Scalar rotations and disjoint hypercyclicity

摘录 / EXCERPT: Let X be a Fréchet space, let $N\ge 2$, and let $T_1,\ldots,T_N\in B(X)$ be d-hypercyclic: that is, there exists $z\in X$ such that $\{(T_1^n z,\ldots,T_N^n z):n=0,1,\ldots\}$…

open_problem | open | MathDB 页面 | 原文线索

V2-074957 | MATHDB-394130

Hypercyclic subspaces for unbounded operators and failure of the B_0 condition

摘录 / EXCERPT: Let T be an unbounded operator defined on the whole space E such that T has a hypercyclic subspace. Can T be such that for any infinite-dimensional closed subspace E_0 an…

open_problem | open | MathDB 页面 | 原文线索

V2-074958 | MATHDB-394133

Replacing hypercyclic vector subspaces by hypercyclic subspaces

摘录 / EXCERPT: To what degree can we replace the hypercyclic vector subspaces U_i in Theorem 1 by hypercyclic subspaces, i.e., closed hypercyclic vector subspaces for T_i?

open_problem | open | MathDB 页面 | 原文线索

V2-074959 | MATHDB-394134

Chaotic polynomials on separable infinite-dimensional Fréchet spaces

摘录 / EXCERPT: Does every separable infinite-dimensional Fréchet space admit a chaotic polynomial of degree $d>1$?

open_problem | open | MathDB 页面 | 原文线索

V2-074960 | MATHDB-394136

Does relative approximate density of one orbit imply hypercyclicity?

摘录 / EXCERPT: Let $T \in B(X)$ and $x \in X$. Suppose that for each $y \in X$, $y \ne 0$, there exists n such that … where ε is a fixed positive number (possibly $\varepsilon<1/3$)…

open_problem | open | MathDB 页面 | 原文线索

V2-074961 | MATHDB-394137

The great open problem in hypercyclicity

摘录 / EXCERPT: Does every hypercyclic operator satisfy the Hypercyclicity Criterion? Equivalently, is $T\oplus T$ hypercyclic for every hypercyclic operator T?

open_problem | open | MathDB 页面 | 原文线索

V2-074962 | MATHDB-394139

Mechanisms maintaining vortex coherence in vertical shear flows

摘录 / EXCERPT: What mechanisms cause such asymmetries and in which manner are they responsible for maintaining the vortex coherence in vertical shear flows?

open_problem | open | MathDB 页面 | 原文线索

V2-074963 | MATHDB-394149

Effective upper bound for the slope of Kodaira fibrations

摘录 / EXCERPT: Is there an effective upper bound for the slope of Kodaira fibrations strictly smaller than 3?

open_problem | open | MathDB 页面 | 原文线索

V2-074964 | MATHDB-394150

Maximum possible packing number for automorphisms of curves

摘录 / EXCERPT: What is the maximal possible value for the packing number α? Here, for a curve B of genus $b\ge 2$, a good set is a subset $S\subset\operatorname{Aut}(B)$ whose graphs for...

open_problem | open | MathDB 页面 | 原文线索

V2-074965 | MATHDB-394153

Kobayashi’s hyperbolicity conjecture for generic hypersurfaces

摘录 / EXCERPT: For $n\ge 3$, a generic hypersurface $X\subset \mathbb{P}^n_{\mathbb{C}}$ is hyperbolic provided that $d=\deg(X)\ge 2n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-074966 | MATHDB-394155

Real-analytic Levi-flat boundary in the complex projective plane

摘录 / EXCERPT: Does there exist a domain in $\mathbb{P}^2$ with real-analytic Levi-flat boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-074967 | MATHDB-394156

True order of the transition for disordered directed-polymer pinning

摘录 / EXCERPT: For the disordered renewal pinning model described on the preceding page, determine the true order of the phase transition in the small-disorder regime 4betau0004 small, particular...

open_problem | open | MathDB 页面 | 原文线索

V2-074968 | MATHDB-394157

True order of the localization transition for disordered pinning models

摘录 / EXCERPT: A natural and interesting open question concerns the true order of the transition in the presence of disorder. There is rather general consensus that for small β and...

open_problem | open | MathDB 页面 | 原文线索

V2-074969 | MATHDB-394165

Decidability of hypersurface resolution-graph realizability

摘录 / EXCERPT: Determine whether there exists a general procedure that, given a resolution graph, decides whether it can be realized as the resolution graph of an isolated hypersurface singularit...

open_problem | open | MathDB 页面 | 原文线索

V2-074970 | MATHDB-394166

The Nash problem on arc families of singularities

摘录 / EXCERPT: When is the Nash map $N_{X,0}$ bijective?

open_problem | open | MathDB 页面 | 原文线索

V2-074971 | MATHDB-394172

Full strongly exceptional collection for graded matrix-factorization categories

摘录 / EXCERPT: Let $R=A/(f)$ be an isolated weighted homogeneous hypersurface singularity of dimension 2, with $A=\mathbb{C}[x,y,z]$, and let T be one of the equivalent triangulated categorie...

open_problem | open | MathDB 页面 | 原文线索

V2-074972 | MATHDB-394174

Analogues of Proposition 3.1 and Theorem 3.2 for compact Lie groups

摘录 / EXCERPT: Do results analogous to Proposition 3.1 and to Theorem 3.2 hold for any compact Lie group?

open_problem | open | MathDB 页面 | 原文线索

V2-074973 | MATHDB-394178

Farrell-Jones isomorphism conjecture in L-theory for \(\mathbb Z_2 * \mathbb Z_3\)

摘录 / EXCERPT: The Farrell-Jones isomorphism conjecture in L-theory for $\mathbb Z_2 * \mathbb Z_3$.

conjecture | open | MathDB 页面 | 原文线索

V2-074974 | MATHDB-394180

Quasi-isometric classification of RAAGs with presentation trees

摘录 / EXCERPT: What happens for right-angled Artin groups whose presentation graphs are trees?

open_problem | open | MathDB 页面 | 原文线索

V2-074975 | MATHDB-394182

Nontriviality of all Morita classes

摘录 / EXCERPT: All of the Morita classes are non-trivial.

open_problem | open | MathDB 页面 | 原文线索

V2-074976 | MATHDB-394184

Infinite stable K-area and the maximal Rosenberg index

摘录 / EXCERPT: Does Gromov’s notion of infinite (stable) K-area imply that $\alpha_{\max}(M)\ne 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-074977 | MATHDB-394186

Tartar’s conjecture on compactness for gradient inclusions

摘录 / EXCERPT: Let aΩa be a bounded open subset of amathbb{R}^2a and let aKsubsetmathbb{R}^{2times2}a be compact. Tartar conjectured that if aKa contains no rank-one connections-that is, there ar...

conjecture | open | MathDB 页面 | 原文线索

V2-074978 | MATHDB-394187

Morrey’s conjecture for 2×2 matrices

摘录 / EXCERPT: Is every rank-one convex function $f:\mathbb{R}^{2\times 2}\to\mathbb{R}$ quasiconvex?

conjecture | open | MathDB 页面 | 原文线索

V2-074979 | MATHDB-394188

Asymptotic limits for rescaled oscillatory quasilinear systems

摘录 / EXCERPT: Proposition 2. Suppose that $A^{\varepsilon_k}_{\eta_k}[v_k]=0$ in B while $\varepsilon_k\to 0$, $\eta_k\to 0$, and $v_k\rightharpoonup v$ weakly in $W^{1,2}(B;\mathbb{R}^N)$. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-074980 | MATHDB-394194

Minimal volumes of cusped hyperbolic orbifolds

摘录 / EXCERPT: Let v_n denote the minimum of the volume spectrum of finite-volume cusped hyperbolic n-orbifolds, where a cusped hyperbolic n-orbifold is a quotient of $\d539^n$ by a cofinit...

open_problem | open | MathDB 页面 | 原文线索

V2-074981 | MATHDB-394196

Dirac family map for twisted loop-group representations

摘录 / EXCERPT: For a principal G-bundle $P\to S^1$, let L_PG be the group of gauge transformations of P, let $\mathcal A$ be the space of connections on P, and fix a projective cocycle...

open_problem | open | MathDB 页面 | 原文线索

V2-074982 | MATHDB-394197

Asymptotic expansion of the critical magnetic field

摘录 / EXCERPT: Establish an asymptotic expansion of $B(\nu)$ as $\nu\to0$.

open_problem | open | MathDB 页面 | 原文线索

V2-074983 | MATHDB-394198

Small-coupling asymptotics of the critical magnetic field

摘录 / EXCERPT: The critical magnetic field satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-074984 | MATHDB-394200

Naive right convergence does not characterize left convergence

摘录 / EXCERPT: For a dense sequence of simple graphs (G_n) with $|V(G_n)|\to\infty$, left convergence implies naive right convergence, but the converse is false. Thus the notions of left conver...

open_problem | open | MathDB 页面 | 原文线索

V2-074985 | MATHDB-394202

Intersecting-family junta conjecture

摘录 / EXCERPT: The family $\mathcal{J}^*$ guaranteed by Theorem 1 can be taken to be an intersecting family.

conjecture | open | MathDB 页面 | 原文线索

V2-074986 | MATHDB-394204

Connectivity of random addable graphs

摘录 / EXCERPT: Let $\mathcal A$ be a non-empty class of labelled graphs such that, for every $G\in\mathcal A$ and any two vertices u and v in distinct components of G, the graph obtained by...

open_problem | open | MathDB 页面 | 原文线索

V2-074987 | MATHDB-394209

Exact value of the maximum product for uniformly cross-intersecting families

摘录 / EXCERPT: Let $\mathcal A,\mathcal B\subseteq 2^{[n]}$ be ℓ-cross-intersecting, meaning that $|A\cap B|=\ell$ for every $A\in\mathcal A$ and $B\in\mathcal B$, and let $P_\ell(n)$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-074988 | MATHDB-394210

Uniform cross-intersection bound for all values of ℓ

摘录 / EXCERPT: Is the inequality from Theorem 1 valid for all values of ℓ, rather than only for sufficiently large ℓ? Namely, for every ℓ-cross-intersecting pair...

open_problem | open | MathDB 页面 | 原文线索

V2-074989 | MATHDB-394211

Erdős-Ko-Rado for three sets: the Frankl-Füredi conjecture

摘录 / EXCERPT: Fix integers $k\ge 3$ and $n\ge 3k/2$. Let $\mathcal G$ be a family of k-element subsets of an n-element set such that whenever $A,B,C\in\mathcal G$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-074990 | MATHDB-394212

Restricted cross-intersections with a set of legal intersection sizes

摘录 / EXCERPT: For two families of subsets $\mathcal A,\mathcal B$ of an n-element set, replace the condition of having one possible cross-intersection size by allowing a set L of legal cross...

open_problem | open | MathDB 页面 | 原文线索

V2-074991 | MATHDB-394213

Mubayi's exact bound for K(d)-families

摘录 / EXCERPT: Fix integers $k\ge d\ge 3$. A family $\mathcal G$ of k-element subsets of an n-element set $[n]$ is a $K(d)$-family if, whenever distinct sets $A_1,\ldots,A_d\in\mathcal G$ sat...

open_problem | open | MathDB 页面 | 原文线索

V2-074992 | MATHDB-394217

Exact determination of the cubic relaxed-coloring threshold

摘录 / EXCERPT: Let $f(3)$ be the smallest integer C such that every graph G with maximum degree at most 3 admits a $(1,C)$-relaxed coloring-that is, a partition of $V(G)$ into an independen...

open_problem | open | MathDB 页面 | 原文线索

V2-074993 | MATHDB-394218

Sudden hardness jump for symmetric relaxed colorings of maximum-degree-4 graphs

摘录 / EXCERPT: The report conjectures that the decision problem $(4,C)$-SymRelCol exhibits a similar sudden hardness jump, where (Δ,C)-SymRelCol asks whether a given graph of maximum degre...

open_problem | open | MathDB 页面 | 原文线索

V2-074994 | MATHDB-394220

Three-coloring graphs of maximum degree 9 with bounded monochromatic components

摘录 / EXCERPT: Is there a constant C such that every graph with maximum degree 9 can be three-colored so that every monochromatic component is of order at most C?

open_problem | open | MathDB 页面 | 原文线索

V2-074995 | MATHDB-394223

Other rearrangement maps and analogous maps for convexity-constrained estimation

摘录 / EXCERPT: Find other maps having the properties listed for the monotone rearrangement map, and study analogous maps for other estimation problems, such as estimation under convexity assumpti...

open_problem | open | MathDB 页面 | 原文线索

V2-074996 | MATHDB-394227

Global well-posedness and uniqueness of minimisers for the water-wave problem

摘录 / EXCERPT: It is an open question whether the following two conditions are satisfied for the water-wave problem: (i) the equation is globally well posed in its energy space (the space in whic...

open_problem | open | MathDB 页面 | 原文线索

V2-074997 | MATHDB-394228

Spectral instability of traveling water waves under amplitude variation

摘录 / EXCERPT: Can the spectrum on the imaginary axis (e.g. at $\varepsilon=0$) move into the right-half of the complex plane, resulting in instability of the base traveling wave?

open_problem | open | MathDB 页面 | 原文线索

V2-074998 | MATHDB-394229

Spectral instability of three-dimensional traveling water waves

摘录 / EXCERPT: For the traveling water waves considered in the report-periodic Stokes waves in two dimensions and periodic short-crested waves in three dimensions-with quite general quasiperiodic...

open_problem | open | MathDB 页面 | 原文线索

V2-074999 | MATHDB-394230

Well-posedness of the KdV initial-boundary-value problem on a quarter-plane

摘录 / EXCERPT: Consider the Korteweg-de Vries equation ... for $x>0$ and $t>0$, with initial and boundary data ... For given $-\tfrac34<s\le 1$, there exists $b\in(0,\tfrac12)$ such that, if...

open_problem | open | MathDB 页面 | 原文线索

V2-075000 | MATHDB-394232

Minimal measurements for instance-optimal compressed sensing in ℓ_p

摘录 / EXCERPT: For a given norm $X=\ell_p$, what is the minimal value of n for which there exists an encoding-decoding pair (Φ,Δ) such that...

open_problem | open | MathDB 页面 | 原文线索

V2-075001 | MATHDB-394233

Relative-entropy regularization of Lévy-model calibration

摘录 / EXCERPT: Given market option prices $C^M(T_i,K_i)$ for $(T_i,K_i)_{i\in I}$, weights w_i, and a reference Lévy process $P(\sigma_P,\nu_P,\gamma_P)$, find a risk-neutral Lévy process...

open_problem | open | MathDB 页面 | 原文线索

V2-075002 | MATHDB-394234

Natural term-rewriting systems with termination unprovable in PA

摘录 / EXCERPT: The report asks whether there are natural, simple-to-formulate, and memorable term-rewriting systems whose termination is not provable in Peano arithmetic (PA). It answers affirmat...

open_problem | open | MathDB 页面 | 原文线索

V2-075003 | MATHDB-394235

Unprovability of Hypothesis H

摘录 / EXCERPT: Try to prove the unprovability of Hypothesis H: for any finite collection of irreducible polynomials $F_1(x),F_2(x),\ldots,F_n(x)$ with integer coefficients such that...

conjecture | open | MathDB 页面 | 原文线索

V2-075004 | MATHDB-394236

Exact provability threshold for the graph-minor statement

摘录 / EXCERPT: For a function f, let GM_f be the statement: “for every K there is N such that for any sequence of simple graphs $G_1,G_2,\ldots,G_N$ such that $|G_i|<K+f(i)$, there are...

open_problem | open | MathDB 页面 | 原文线索

V2-075005 | MATHDB-394237

Existence of full satisfaction classes for models of PA

摘录 / EXCERPT: Every countable recursively saturated model M of Peano arithmetic has a full satisfaction class. Conversely, if M is a nonstandard model of Peano arithmetic admitting a full sa...

open_problem | open | MathDB 页面 | 原文线索

V2-075006 | MATHDB-394238

Exact reverse-mathematical strength of Fraïssé’s conjecture

摘录 / EXCERPT: Determine the exact axiomatic strength, in reverse mathematics, of Fraïssé’s conjecture FC: the assertion that the class LO of countable linear orders is well-quasi-ordered by...

conjecture | open | MathDB 页面 | 原文线索

V2-075007 | MATHDB-394239

Maximal order type of finite-Hausdorff-rank linear orders

摘录 / EXCERPT: Theorem: $o(LO_\\omega,\preceq)=\varphi_2(0)$, where LO_ω is the class of countable linear orders of finite Hausdorff rank and $o(Q,\leq)$ is the maximal order type of the w...

open_problem | open | MathDB 页面 | 原文线索

V2-075008 | MATHDB-394241

Nonexistence of an ECST proof of addition on \(\omega\)

摘录 / EXCERPT: ECST does not prove the existence of the addition function on ω.

open_problem | open | MathDB 页面 | 原文线索

V2-075009 | MATHDB-394244

Provability of the weak pigeonhole principle in bounded arithmetic

摘录 / EXCERPT: Does $I\Delta_0(f)\vdash PHP^{2n}_n(f)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075010 | MATHDB-394245

Provability of the Δ_0 pigeonhole principle in $I\Delta_0$

摘录 / EXCERPT: Does $I\Delta_0 \vdash PHP^{n+1}_n(\Delta_0)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075011 | MATHDB-394247

Polynomial-size constant-depth LK proofs for weak pigeonhole principles

摘录 / EXCERPT: Do PHP^{2n}_n and $PHP^{n^2}_n$ have polynomial-size, constant-depth LK proofs?

open_problem | open | MathDB 页面 | 原文线索

V2-075012 | MATHDB-394248

Provability of restricted Herbrand consistency at the \a0log\b2_{i+2} cut

摘录 / EXCERPT: For at least one $i\ge 1$, does either of the following hold? (1) $I\Delta_0+\Omega_i\vdash \operatorname{HCons}(I\Delta_0+\Omega_i,\log_{i+2})$; or (2)...

open_problem | open | MathDB 页面 | 原文线索

V2-075013 | MATHDB-394255

Quasi-invariance of heat-kernel measures on abstract Wiener groups

摘录 / EXCERPT: For the heat-kernel measures $\nu_t^{H\subset G}$, $t>0$, is quasi-invariance under left translation by $h\in H$ equivalent to the condition that AA^*-1 be a Hilbert-Schmidt oper...

open_problem | open | MathDB 页面 | 原文线索

V2-075014 | MATHDB-394263

Conjecture on the real parts of Steiner-polynomial roots

摘录 / EXCERPT: Conjecture 1. Let $K\in\mathcal K^n$. If $a_1\le\cdots\le a_n$ are the real parts of the roots of the Steiner polynomial, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-075015 | MATHDB-394264

Existence of hyperbolic virtual polytopes with any number of horns

摘录 / EXCERPT: Theorem [8, 10]. For any number $N \ge 4$, there exists a hyperbolic virtual polytope with N horns and its smooth version-a hyperbolic hérisson with N horns.

open_problem | open | MathDB 页面 | 原文线索

V2-075016 | MATHDB-394266

Matheron’s conjecture for the planar covariogram problem

摘录 / EXCERPT: Let K be a convex body in $\a0R^2$, with covariogram $g_K(x)=\operatorname{vol}(K\cap(K+x))$ for $x\in\u0000a0R^2$. Is every planar convex body determined by its covariogram, amo...

conjecture | open | MathDB 页面 | 原文线索

V2-075017 | MATHDB-394268

Symmetric Mahler conjecture for the volume product

摘录 / EXCERPT: For every symmetric convex set $K\subset\mathbb{R}^n$, $|K|\,|K^*|\ge |B_1^n|\,|B_\infty^n|=\frac{4^n}{n!}$, where K^* denotes the polar set.

conjecture | open | MathDB 页面 | 原文线索

V2-075018 | MATHDB-394271

Seven-direction uniqueness for convex bodies in three dimensions

摘录 / EXCERPT: Are convex bodies in $\mathbb{R}^3$ determined uniquely by parallel X-rays in any set of seven directions in general position?

open_problem | open | MathDB 页面 | 原文线索

V2-075019 | MATHDB-394272

Optimal asymptotic behavior of the Dvoretzky dimension constant

摘录 / EXCERPT: Determine the optimal asymptotic behavior, as $\varepsilon\to 0$, of the largest function $c(\varepsilon)$ such that, for every N-dimensional normed space Y and every...

open_problem | open | MathDB 页面 | 原文线索

V2-075020 | MATHDB-394273

Best possible Hölder exponents in stability inequalities

摘录 / EXCERPT: It is an open problem to find the best possible Hölder exponents in stability inequalities. In particular, the Hölder exponent in (1), and also the improved exponent $2/(n(n+1))$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075021 | MATHDB-394275

Slicing problem for L_p-balls with negative p

摘录 / EXCERPT: Does the slicing inequality ... with the slicing constant independent of dimension, hold for L_p-balls with $p<0$?

open_problem | open | MathDB 页面 | 原文线索

V2-075022 | MATHDB-394276

Slicing inequality for convex p-intersection bodies

摘录 / EXCERPT: For $0<p<2$, if K is an origin-symmetric convex p-intersection body in $\d53f^n$, then ... where ...

open_problem | open | MathDB 页面 | 原文线索

V2-075023 | MATHDB-394277

Exact distribution of the number of vertices of a random convex chain in a triangle

摘录 / EXCERPT: Let $P_1,\ldots,P_n$ be distributed independently and uniformly in the triangle with vertices $(0,1)$, $(0,0)$, and $(1,0)$. Consider the convex hull of...

open_problem | open | MathDB 页面 | 原文线索

V2-075024 | MATHDB-394278

Nonsymmetric L_p Busemann-Petty problem

摘录 / EXCERPT: For $p<1$ with $p\ne 0$, let I_p^+ be defined for a star body $K\subset\mathbb{R}^n$ by ... where $u^+=\{x\in\mathbb{R}^n:x\cdot u\ge 0\}$. Is it true that, for convex bodies K,L...

open_problem | open | MathDB 页面 | 原文线索

V2-075025 | MATHDB-394279

Volume lower bound for 0-symmetric lattice polytopes

摘录 / EXCERPT: Conjecture 4. Let P be a 0-symmetric lattice polytope. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-075026 | MATHDB-394280

Fourier-Mukai description associated with a polyhedral divisor

摘录 / EXCERPT: Conjecture 5. The polyhedral divisor D induces an object K_D of the derived category $D^b(Y\times Z)$, and the sheaf $\mathcal{O}_Y(D)$ is obtained from $U\in D^b(Z)$ via Fouri...

open_problem | open | MathDB 页面 | 原文线索

V2-075027 | MATHDB-394281

Two-dimensional quotients of affine space by reductive groups

摘录 / EXCERPT: Let G be a reductive group acting on the affine space $\mathbb{C}^n$. If $\mathop{\mathrm{dim}}(\mathbb{C}^n//G)=2$, then $V=\mathbb{C}^n//G$ is isomorphic to a quotient...

open_problem | open | MathDB 页面 | 原文线索

V2-075028 | MATHDB-394282

Existence of an essentially non-tame automorphism with a point-centered first Sarkisov blow-up

摘录 / EXCERPT: Is there an essentially non-tame automorphism θ on the affine 3-space $\mathbb{C}^3$ such that the Sarkisov factorization of the Cremona transformation Φ_θ on...

open_problem | open | MathDB 页面 | 原文线索

V2-075029 | MATHDB-394283

Dolgachev-Weisfeiler question for affine-space fibrations

摘录 / EXCERPT: Let $f:X\to Y$ be an A^n-fibration. Assume that f is affine and faithfully flat and that all closed fibers are geometrically integral. Is f an A^n-bundle?

open_problem | open | MathDB 页面 | 原文线索

V2-075030 | MATHDB-394286

Variables over the subring $R[t^m,t^{m+1},\ldots,t^{2m-1}]$

摘录 / EXCERPT: Let R be a ring containing $\mathbb{Q}$ and let m be a positive integer. If $f\in R[t^m,t^{m+1},\ldots,t^{2m-1}][x_1,\ldots,x_n]$ is a variable of $R[t][x_1,\ldots,x_n]$, then...

open_problem | open | MathDB 页面 | 原文线索

V2-075031 | MATHDB-394289

Vanishing Conjecture (VC)

摘录 / EXCERPT: For every homogeneous polynomial $P\in\mathbb{C}[X_1,\ldots,X_n]$ of degree 4, ...

conjecture | open | MathDB 页面 | 原文线索

V2-075032 | MATHDB-394290

Stability of the AK invariant under adjoining one variable

摘录 / EXCERPT: Let A be a domain which is either finitely generated as a ring or finitely generated over a field k. If $AK(A)=A$, then $AK(A[x])=A$.

open_problem | open | MathDB 页面 | 原文线索

V2-075033 | MATHDB-394291

External-codimension inequality for rational plane curves

摘录 / EXCERPT: For a curve in the space $\mathrm{Curv}_{a,b,c,d}$ parametrized by ... let g be the dimension of the group of transformations of $\mathbb C^2$ preserving $\mathrm{Curv}_{a,b,c,d}$....

open_problem | open | MathDB 页面 | 原文线索

V2-075034 | MATHDB-394293

Coordinate status of the first Vénéreau polynomial

摘录 / EXCERPT: Is the polynomial $v=y+x^{2}z+xy(yu+z^{2})\in\mathbb{C}[x,y,z,u]$ a coordinate? Equivalently, is v equivalent to one of the variables under a polynomial automorphism of...

open_problem | open | MathDB 页面 | 原文线索

V2-075035 | MATHDB-394294

Gizatullin-Harbourne-McMullen question on infinite automorphism groups of rational surfaces

摘录 / EXCERPT: Let X be a smooth projective (complex) rational surface. If $\operatorname{Aut}^{*}(X)$ is infinite, is there then a birational morphism ϕ of X to a surface Y having an...

open_problem | open | MathDB 页面 | 原文线索

V2-075036 | MATHDB-394295

Locally nilpotent derivations of a four-variable polynomial ring

摘录 / EXCERPT: Let k be a field of characteristic 0. Does there exist a locally nilpotent k-derivation $D:k[X_1,X_2,X_3,X_4]\to k[X_1,X_2,X_3,X_4]$ such that $\ker D$ is not finitely genera...

open_problem | open | MathDB 页面 | 原文线索

V2-075037 | MATHDB-394296

A conceptual proof for the non-polynomial semigroup generated by $(6,21,4)$

摘录 / EXCERPT: Give a conceptual proof, rather than a proof based on extensive computer calculations, that the planar semigroup with δ-generator sequence $(6,21,4)$ is not polynomial; equi...

open_problem | open | MathDB 页面 | 原文线索

V2-075038 | MATHDB-394297

Minimum degree of a nonzero difference of powers

摘录 / EXCERPT: Let k be a ground field of characteristic zero, and let P,Q be monic polynomials over k. When the top degrees cancel, so that $m\deg P=n\deg Q$, the report asks for an estima...

open_problem | open | MathDB 页面 | 原文线索

V2-075039 | MATHDB-394299

Abhyankar-Sathaye conjecture on polynomial embeddings

摘录 / EXCERPT: Conjecture 2 (Abhyankar-Sathaye). Every polynomial embedding $\phi:\mathbb{C}^{n-1}\hookrightarrow\mathbb{C}^{n}$ of $\mathbb{C}^{n-1}$ into $\mathbb{C}^{n}$ is rectifiable.

conjecture | open | MathDB 页面 | 原文线索

V2-075040 | MATHDB-394300

Proper holomorphic embeddings of finitely connected subsets into $\mathbb{C}^2$

摘录 / EXCERPT: Any finitely connected subset of $\mathbb{C}$ and any finitely connected subset of any torus (no punctures) can be properly holomorphically embedded into $\mathbb{C}^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-075041 | MATHDB-394303

Classification of finite subgroups of the plane Cremona group

摘录 / EXCERPT: Classify the conjugacy classes of finite subgroups of the plane Cremona group $\operatorname{Cr}(2)$ over the field of complex numbers.

open_problem | open | MathDB 页面 | 原文线索

V2-075042 | MATHDB-394305

Holomorphic equivalence of the embeddings U_n and V_n

摘录 / EXCERPT: For $n\ge 3$, does there exist a holomorphic automorphism of $\mathbb{C}^3$ that maps U_n onto V_n?

open_problem | open | MathDB 页面 | 原文线索

V2-075043 | MATHDB-394306

Classification of curves associated with nontrivial \(\mathbb{C}_+\)-actions

摘录 / EXCERPT: Let X be a smooth affine contractible threefold over the complex numbers equipped with a nontrivial $\mathbb{C}_+$-action. Suppose that $\pi:X\to S=X//\mathbb{C}_+$ is the quotie...

open_problem | open | MathDB 页面 | 原文线索

V2-075044 | MATHDB-394308

Does the Seiberg-Witten conjecture hold for one-cuspidal superisolated singularities?

摘录 / EXCERPT: Let $f:(\mathbb{C}^{3},0)\to(\mathbb{C},0)$ be a superisolated hypersurface singularity, $f=f_d+f_{d+1}+\cdots$, where each f_j is homogeneous of degree j, the projective plane...

conjecture | open | MathDB 页面 | 原文线索

V2-075045 | MATHDB-394309

Rigidity conjecture for rational cuspidal plane-curve complements

摘录 / EXCERPT: Fix a minimal logarithmic compactification (V,D) of a complement $\mathbb{P}^{2}\setminus C$, where V is a smooth projective surface, D is a normal-crossing divisor, and...

conjecture | open | MathDB 页面 | 原文线索

V2-075046 | MATHDB-394310

Compatibility-property conjecture (Conjecture CP) for rational cuspidal plane curves

摘录 / EXCERPT: Let $(C,p_i)_{i=1}^{\nu}$ be a collection of local plane curve singularities, all of them locally irreducible, and let $\Delta_i(t)$ be the characteristic polynomial of the monodro...

conjecture | open | MathDB 页面 | 原文线索

V2-075047 | MATHDB-394314

Cayley degree of \(\mathrm{SL}_5\)

摘录 / EXCERPT: Find the Cayley degree $\operatorname{Cay}(\mathrm{SL}_5)$, where $\operatorname{Cay}(G)$ is the minimum degree of a dominant G-equivariant rational map...

open_problem | open | MathDB 页面 | 原文线索

V2-075048 | MATHDB-394316

Existence of groups with arbitrarily large Cayley degree

摘录 / EXCERPT: Given $d\in\mathbb{N}$, is there G such that $\operatorname{Cay}(G)>d$? Is there a simple such G?

open_problem | open | MathDB 页面 | 原文线索

V2-075049 | MATHDB-394318

Stable tameness of two-dimensional polynomial automorphisms over a UFD

摘录 / EXCERPT: Let R be a UFD, and let $GA_2(R)=\operatorname{Aut}_R(R^{[2]})$. Show that all elements of $GA_2(R)$ are stably tame.

open_problem | open | MathDB 页面 | 原文线索

V2-075050 | MATHDB-394320

Ore conjecture on commutator width of finite simple groups

摘录 / EXCERPT: Every non-cyclic finite simple group has commutator width 1.

conjecture | open | MathDB 页面 | 原文线索

V2-075051 | MATHDB-394321

Isomorphism of complete local noetherian integral domains from all infinitesimal quotients

摘录 / EXCERPT: Let A and B be complete local noetherian integral domains, with maximal ideals $\d524$ and $\d525$, respectively. If $A/\d524^r \cong B/\d525^r$ for every natural number r, m...

open_problem | open | MathDB 页面 | 原文线索

V2-075052 | MATHDB-394325

Does the Baldwin-Lachlan theorem hold for thin universes?

摘录 / EXCERPT: The Baldwin-Lachlan theorem says that if a structure in a finite or countable language is uncountably categorical but not ω-categorical, then its countable elementarily equi...

open_problem | open | MathDB 页面 | 原文线索

V2-075053 | MATHDB-394328

Elementary substructure of a free product for models of the universal theory of free groups

摘录 / EXCERPT: If M is a nonabelian model of the universal theory of free groups, is $M \preccurlyeq M * \mathbb{Z}$?

open_problem | open | MathDB 页面 | 原文线索

V2-075054 | MATHDB-394330

Definable subgroups of models of the universal theory of free groups

摘录 / EXCERPT: If M is a nonabelian model of the universal theory of free groups such that $M \preccurlyeq M * \mathbb{Z}$, is any proper definable subgroup of M abelian?

open_problem | open | MathDB 页面 | 原文线索

V2-075055 | MATHDB-394333

Finite-rank trace theorem (Fact 3), a special case of generalized Mordell-Lang

摘录 / EXCERPT: Let Δ be a finite-rank subgroup of $(\mathbb{C}^{*})^{n}$, meaning that Δ contains a finitely generated subgroup G such that Δ/G is torsion. For every algebr...

open_problem | open | MathDB 页面 | 原文线索

V2-075056 | MATHDB-394339

Nonnegativity of the coefficients of the complete cd-index of a Bruhat interval

摘录 / EXCERPT: For every Bruhat interval $[u,v]$ in a Coxeter system (W,S), all coefficients of its complete cd-index $\widetilde{\Psi}_{u,v}$ are nonnegative; equivalently, in...

open_problem | open | MathDB 页面 | 原文线索

V2-075057 | MATHDB-394341

Nonnegativity of the coefficients $a_i(u,v)$

摘录 / EXCERPT: For every Coxeter system (W,S), every Bruhat interval $[u,v]$ with $u\le v$, and every integer $0\le i\le\lfloor n/2\rfloor$, where $n=\ell(v)-\ell(u)$, the coefficients ... are no...

open_problem | open | MathDB 页面 | 原文线索

V2-075058 | MATHDB-394342

Total nonnegativity of the g-theorem matrices

摘录 / EXCERPT: For each integer dimension parameter d, let M_d be the $(\lfloor d/2\rfloor+1)\times d$ matrix with entries ... Björner’s conjecture states that all minors of M_d are nonnegati...

open_problem | open | MathDB 页面 | 原文线索

V2-075059 | MATHDB-394343

Gradedness of Bruhat order on twisted identities

摘录 / EXCERPT: For which Coxeter systems (W,S) and involutive automorphisms θ is $\operatorname{Br}(\iota(\theta))$ graded, where $\iota(\theta)=\{\theta(w^{-1})w\mid w\in W\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075060 | MATHDB-394344

Ball replacement conjecture for intervals in twisted-identity Bruhat orders

摘录 / EXCERPT: Let (W,S) be a Coxeter system equipped with an involutive automorphism θ, and let $\iota(\theta)=\{\theta(w^{-1})w\mid w\in W\}$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-075061 | MATHDB-394345

Classical Littlewood-Offord maximum for sign sums

摘录 / EXCERPT: For real numbers $a_1,\dots,a_n a_ n\ge 1$, define ... The question of maximizing $\lvert N_r(\{a_i\}_{i=1}^n)\rvert$ over all such choices of $a_1,\dots,a_n$ is solved: for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075062 | MATHDB-394346

A concentration bound under a generalized-inverse norm condition

摘录 / EXCERPT: Let W be a real $d\times n$ matrix with rank n. Assume that every row of $(W^{\mathsf T}W)^{-1}W^{\mathsf T}$ has Euclidean norm at most 1. Then, for every $r>0$, ... The bound...

open_problem | open | MathDB 页面 | 原文线索

V2-075063 | MATHDB-394348

Optimality of diagonal-layer colorings for triangulated grids

摘录 / EXCERPT: For the family of d-dimensional triangulated grids T_n^d, the conjectured optimal worst-case value of the two-color parameter is the size of the largest diagonal layer, namely...

open_problem | open | MathDB 页面 | 原文线索

V2-075064 | MATHDB-394349

Monochromatic components in multi-colored triangulated grids

摘录 / EXCERPT: For fixed dimension d, and for every $k=3,4,\ldots,d-1$, the report conjectures that $\xi_k(T_n^d)$ is of order n^{d-k+1}, where T_n^d is the graph of a triangulation of the...

open_problem | open | MathDB 页面 | 原文线索

V2-075065 | MATHDB-394351

Embeddability after adjoining a missing face

摘录 / EXCERPT: Let K be the d-skeleton of a triangulated $2d$-sphere, and let T be a missing d-face of K, i.e. $T\notin K$ and the boundary of T is contained in K. Then $K\cup\{T\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-075066 | MATHDB-394355

Bleecker’s volume-increasing deformation conjecture

摘录 / EXCERPT: Every polyhedral surface in $\mathbb{R}^3$ has a volume-increasing isometric deformation.

conjecture | open | MathDB 页面 | 原文线索

V2-075067 | MATHDB-394356

Postnikov’s conjecture on Bruhat intervals and inversion-arrangement regions

摘录 / EXCERPT: For each inversion $i<j$ of a permutation $\pi\in S_n$, meaning $\pi(i)>\pi(j)$, let the associated hyperplane in $\mathbb{R}^n$ be $x_i=x_j$. Let $r(\pi)$ be the number...

conjecture | open | MathDB 页面 | 原文线索

V2-075068 | MATHDB-394359

Frobenius-characteristic conjecture for the symmetric functions $Q_{\lambda,j}$

摘录 / EXCERPT: For a partition λ of n, it is conjectured that the symmetric function $Q_{\lambda,j}$ is the Frobenius characteristic of some representation.

conjecture | open | MathDB 页面 | 原文线索

V2-075069 | MATHDB-394360

Maximum number of k-edges in a planar point set

摘录 / EXCERPT: Let S be a set of $n\ge 3$ points in the affine plane $\mathbb{R}^2$, no three collinear. For an integer parameter k, a k-edge of S is a directed segment pq, spanned by t...

open_problem | open | MathDB 页面 | 原文线索

V2-075070 | MATHDB-394361

Maximum number of vertices in a 2-dimensional affine section of a zonotope

摘录 / EXCERPT: Let $Z\subseteq\mathbb{R}^d$ be a zonotope with n zones, where, in other words, Z is the Minkowski sum of n line segments $[-v_i,v_i]$, with $v_i\in\mathbb{R}^d$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-075071 | MATHDB-394365

A generalized Holland model for wave diffraction by thin wires

摘录 / EXCERPT: The report formulates the two-dimensional acoustic diffraction problem in $ \Omega_R=D(0,R)$ containing one small obstacle $ \Omega_\varepsilon=D(0,\varepsilon)$, with Dirichlet bo...

open_problem | open | MathDB 页面 | 原文线索

V2-075072 | MATHDB-394366

Optimal convergence of the augmented finite-element formulation

摘录 / EXCERPT: Prove optimal convergence for the augmented discrete formulation $(u_{\varepsilon,h}^{\mathrm{ap}},p_{\varepsilon,h}^{\mathrm{ap}})\in V_h^e\times P_0$ for the small-obstacle probl...

open_problem | open | MathDB 页面 | 原文线索

V2-075073 | MATHDB-394367

Convergence of the fixed-mesh PUFEM under plane-wave enrichment

摘录 / EXCERPT: For the Partition of Unity Finite Element Method (PUFEM) described in the report, with the mesh fixed and the number of plane waves per node increased, prove convergence of the met...

open_problem | open | MathDB 页面 | 原文线索

V2-075074 | MATHDB-394368

Existence of subsonic heteroclinic travelling waves in a piecewise-quadratic lattice model

摘录 / EXCERPT: For the nearest-neighbour lattice travelling-wave equation ... with ... on the real line, the report proves the existence of heteroclinic waves whose strain satisfies ... for sufficientl...

open_problem | open | MathDB 页面 | 原文线索

V2-075075 | MATHDB-394369

Free infinite divisibility of the Gaussian distribution

摘录 / EXCERPT: Is the standard Gaussian distribution infinitely divisible in the free sense?

open_problem | open | MathDB 页面 | 原文线索

V2-075076 | MATHDB-394373

Optimal control of an eikonal equation

摘录 / EXCERPT: Let $I=[-1,1]$ and let $-1=\hat x_0<\hat x_1<\cdots<\hat x_{L-1}<\hat x_L=1$ be a partition of I. Given constants $0<\alpha<\beta$ and $y_T\in L^2(I)$, consider ... subject to...

open_problem | open | MathDB 页面 | 原文线索

V2-075077 | MATHDB-394374

Eremenko's conjecture on components of the escaping set

摘录 / EXCERPT: Let f be an entire transcendental function, and define its escaping set by $I(f):=\{z\in\mathbb{C}: f^n(z)\to\infty\text{ as }n\to\infty\}$. Eremenko conjectured that every compo...

conjecture | open | MathDB 页面 | 原文线索

V2-075078 | MATHDB-394375

Order of an entire function associated with a Newton map

摘录 / EXCERPT: Let f be an entire transcendental function and let $N_f(z)=z-f(z)/f'(z)$ be its associated Newton function. Suppose that N_f has an invariant Baker domain, meaning that there i...

open_problem | open | MathDB 页面 | 原文线索

V2-075079 | MATHDB-394376

Growth along a curve for Newton maps with an invariant Baker domain

摘录 / EXCERPT: Let f be a transcendental entire function and let $N_f(z)=z-f(z)/f'(z)$ be the associated Newton function. Suppose that N_f has an invariant Baker domain; that is, there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-075080 | MATHDB-394377

Finite connectivity greater than one for wandering domains of transcendental entire functions

摘录 / EXCERPT: Can the connectivity of a wandering domain for a function $f\in E$ be finite but greater than one, where E is the class of transcendental entire functions?

open_problem | open | MathDB 页面 | 原文线索

V2-075081 | MATHDB-394379

Value-sharing uniqueness under a modified second-derivative condition

摘录 / EXCERPT: Let f be a non-constant entire function and let a be a nonzero constant. Suppose that $E(a,f)=E(a,f')$, where equality means that $f-a$ and $f'-a$ have the same zeros with the...

open_problem | open | MathDB 页面 | 原文线索

V2-075082 | MATHDB-394380

Converse to the uniform-separation criterion for type-stability

摘录 / EXCERPT: Let (X,f) be a surface spread over the Riemann sphere, with X open and simply connected. Is every parabolic type-stable surface (X,f) necessarily a surface with uniformly sep...

open_problem | open | MathDB 页面 | 原文线索

V2-075083 | MATHDB-394381

Uniqueness of entire functions from shared differential-polynomial values

摘录 / EXCERPT: Let f and g be non-constant entire functions, let $a,b\in\mathbb{C}\setminus\{0\}$, and let n,k be positive integers with $n\ge 11$ and $n\ge k+2$. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-075084 | MATHDB-394382

Majorization of the modulus of continuity of analytic functions

摘录 / EXCERPT: Let G be a bounded domain, let f be analytic in G and continuous on $\overline{G}$, and let $\mu:[0,\infty)\to[0,\infty)$ be a non-negative non-decreasing majorant, meaning t...

open_problem | open | MathDB 页面 | 原文线索

V2-075085 | MATHDB-394383

Boundary majorization of the modulus of continuity for analytic functions

摘录 / EXCERPT: Let G be a bounded domain, let f be analytic in G and continuous on $\\overline{G}$, and let $\mu:[0,\infty)\to[0,\infty)$ be non-negative and non-decreasing with...

open_problem | open | MathDB 页面 | 原文线索

V2-075086 | MATHDB-394384

Mues' deficiency-sum conjecture for derivatives of meromorphic functions

摘录 / EXCERPT: Every transcendental meromorphic function f satisfies, for every positive integer n, ...

conjecture | open | MathDB 页面 | 原文线索

V2-075087 | MATHDB-394385

Density of bounded holomorphic functions

摘录 / EXCERPT: Let $\Omega \subseteq \mathbb{C}$ be a domain. Under which conditions is $H^\infty(\Omega)$ locally uniformly dense in $H(\Omega)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075088 | MATHDB-394386

X-universal functions for sequences of holomorphic self-maps

摘录 / EXCERPT: Let Ω be a domain in $\mathbb C$, and let X denote either the Fréchet space $H(\Omega)$ of all holomorphic functions on Ω, or the unit ball...

open_problem | open | MathDB 页面 | 原文线索

V2-075089 | MATHDB-394387

Minimal boundary regularity for equivalence of local completeness conditions

摘录 / EXCERPT: What are the minimal regularity conditions on the boundary set Γ such that the two conditions (a) and (b) in Corollary 1 are still equivalent? Here Corollary 1 concerns a do...

open_problem | open | MathDB 页面 | 原文线索

V2-075090 | MATHDB-394388

Eremenko’s conjecture: connected components of the escaping set

摘录 / EXCERPT: For every entire function f, every connected component of the escaping set $I(f)$ is unbounded.

conjecture | open | MathDB 页面 | 原文线索

V2-075091 | MATHDB-394389

Unbounded Fatou components for entire functions of small growth

摘录 / EXCERPT: Is it true that, for every transcendental entire function f of order at most $1/2$ and minimal type, the Fatou set $F(f)$ has no unbounded component?

open_problem | open | MathDB 页面 | 原文线索

V2-075092 | MATHDB-394391

Julia sets of polynomial semigroups when one element has the semigroup Julia set

摘录 / EXCERPT: Let G be a polynomial semigroup such that $J_h=J(G)$ for some $h gin G$, where $J(G)$ is not a line segment. Then $J_f=J_g$ for all $f,g\in G$.

open_problem | open | MathDB 页面 | 原文线索

V2-075093 | MATHDB-394392

Nevanlinna four-value problem with exactly one CM shared value

摘录 / EXCERPT: Do there exist different meromorphic functions f and g sharing four values, exactly one of them by “CM”? And if so, for which value s does this happen?

open_problem | open | MathDB 页面 | 原文线索

V2-075094 | MATHDB-394394

Optimal dividend and reinvestment policies with fixed and proportional costs

摘录 / EXCERPT: Under assumptions A1-A4 on the diffusion income process $dY_t=\mu(Y_t)\,dt+\sigma(Y_t)\,dW_t$, with generator $Lg(y)=\frac12\sigma^2(y)g''(y)+\mu(y)g'(y)-rg(y)$, the report conside...

open_problem | open | MathDB 页面 | 原文线索

V2-075095 | MATHDB-394395

Finite-element error estimates for dG gradient plasticity

摘录 / EXCERPT: Whether the finite-element error estimate for a discontinuous Galerkin formulation of gradient plasticity can be improved to $O(h)$, under assumptions on solution regularity and pi...

open_problem | open | MathDB 页面 | 原文线索

V2-075096 | MATHDB-394396

Uniqueness of energetic solutions for a thermally driven shape-memory model

摘录 / EXCERPT: For the thermally driven phase-transformation model defined by the energy E and dissipation potential R on $\mathcal F\times\mathcal Z$, and for a given stable initial datum...

open_problem | open | MathDB 页面 | 原文线索

V2-075097 | MATHDB-394397

Existence of time-space periodic solutions for the general martensitic phase-boundary problem

摘录 / EXCERPT: Establish an existence result for time-space periodic solutions of the general free-boundary problem governing the evolution of martensitic phase boundaries in heterogeneous media.

open_problem | open | MathDB 页面 | 原文线索

V2-075098 | MATHDB-394398

Pure-point spectrum on the single-ended canopy graph

摘录 / EXCERPT: For the single-ended canopy graph, corresponding to a horoball in the regular tree, the random Schrödinger operator has only pure-point spectrum at any strength of the disorder.

open_problem | open | MathDB 页面 | 原文线索

V2-075099 | MATHDB-394399

Density of the density-of-states measure

摘录 / EXCERPT: Does the positive measure ν have a density?

open_problem | open | MathDB 页面 | 原文线索

V2-075100 | MATHDB-394400

Uniform Wegner-type bound for random band matrices

摘录 / EXCERPT: Find a good constant C in the bound ... for functions f supported in an interval (a,b). The report notes that exploiting the diagonal terms yields such a bound with...

open_problem | open | MathDB 页面 | 原文线索

V2-075101 | MATHDB-394402

Detecting corrosion by thermal measurements

摘录 / EXCERPT: For $d\in\{2,3\}$, let $\mathbb{R}^d_-:=\{x\in\mathbb{R}^d:x_d<0\}$ and let $\Omega_-\subset\mathbb{R}^d_-$ be a bounded open domain with C^2 boundary $\Gamma:=\partial\Omega_-$....

open_problem | open | MathDB 页面 | 原文线索

V2-075102 | MATHDB-394403

Asymptotic Picard criterion for approximate data operators

摘录 / EXCERPT: Let $F=GTG^*$ be a compact data operator with factorization $F=GTG^*$, where G characterizes an inclusion $D\subset\mathbb{R}^d$, $d=2$ or 3, by $y\in D$ if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-075103 | MATHDB-394404

Global uniqueness for inverse periodic transmission problems with one incident wave

摘录 / EXCERPT: For the two-dimensional inverse TE and TM transmission problems for diffraction gratings, does one incident plane wave uniquely determine a grating whose profile is the graph of a...

open_problem | open | MathDB 页面 | 原文线索

V2-075104 | MATHDB-394408

A priori choice of the regularization parameter in the noisy linear sampling method

摘录 / EXCERPT: For a perturbed far-field operator F^δ satisfying $\lVert F-F^\delta\rVert\leq\delta$, determine whether the regularization parameter can be chosen a priori as...

open_problem | open | MathDB 页面 | 原文线索

V2-075105 | MATHDB-394409

Existence of transmission eigenvalues for anisotropic media

摘录 / EXCERPT: Except for the spherically stratified case, do transmission eigenvalues exist for the anisotropic interior transmission problem?

open_problem | open | MathDB 页面 | 原文线索

V2-075106 | MATHDB-394411

Factorization-method detection of diamagnetic conducting objects

摘录 / EXCERPT: The Factorization Method also works for the detection of diamagnetic, conducting objects.

open_problem | open | MathDB 页面 | 原文线索

V2-075107 | MATHDB-394414

Connes' embedding conjecture for separable type II$_1$ factors

摘录 / EXCERPT: Let R denote the hyperfinite type II$_1$ factor, and let F be a separable type II$_1$ factor. If U is a free ultrafilter on $\mathbb{N}$, then F can be embedded, as a ring...

conjecture | open | MathDB 页面 | 原文线索

V2-075108 | MATHDB-394415

BMV conjecture for exponents $m\le 9$

摘录 / EXCERPT: Whenever $A,B\in\operatorname{Sym}(\mathbb{C}^{s\times s})$ are positive semidefinite, the polynomial $\operatorname{tr}((A+tB)^m)\in\mathbb{R}[t]$ has nonnegative coefficients for...

conjecture | open | MathDB 页面 | 原文线索

V2-075109 | MATHDB-394417

Adaptive fixed-degree Bernstein certificates of positivity

摘录 / EXCERPT: Given a polynomial P of degree p that is positive on $[-1,1]$, find a subdivision $L: -1=\r\ell_0<\ell_1<\cdots<\ell_n=1$ such that, for every subinterval $[\ell_{i-1},\ell_i]$...

open_problem | open | MathDB 页面 | 原文线索

V2-075110 | MATHDB-394419

Connectedness conjecture for the real spectrum

摘录 / EXCERPT: Let $A=R[x_1,\ots,x_n]$, where R is a real closed field. For $\lpha,\eta i\\operatorname{Sper} A$ and a finite collection $g_1,\ots,g_s$ of elements of A not belonging to the s...

conjecture | open | MathDB 页面 | 原文线索

V2-075111 | MATHDB-394420

Gradient-ideal replacement conjecture

摘录 / EXCERPT: Let always $f\in\mathbb{R}[\bar X]:=\mathbb{R}[X_1,\ldots,X_n]$ and let $f^*:=\inf f(\mathbb{R}^n)$. Conjecture 2. In Theorem 1, the assumption $f^*\in f(\mathbb{R}^n)$ can be remo...

conjecture | open | MathDB 页面 | 原文线索

V2-075112 | MATHDB-394421

Removing condition (*) from Theorem 4

摘录 / EXCERPT: Let $f\in\mathbb{R}[X_1,\ldots,X_n]$ be bounded from below on $\mathbb{R}^n$, and let $d=\deg f$ with f_k denoting the k-homogeneous part of f. Condition $(*)$ is that there...

open_problem | open | MathDB 页面 | 原文线索

V2-075113 | MATHDB-394422

Asymptotic efficiency of plug-in estimators for linear functionals in endogenous nonparametric regression

摘录 / EXCERPT: For the nonparametric regression model $Y=\mu^*(X)+\varepsilon$ with $E[\varepsilon\mid W]=0$ w.p.1, where X may contain endogenous regressors, W is the vector of instrumental...

open_problem | open | MathDB 页面 | 原文线索

V2-075114 | MATHDB-394424

Berge’s strong path partition conjecture

摘录 / EXCERPT: Let $G=(V,E)$ be a directed graph with no loops or multiple edges, and let k be a positive integer. A path is a sequence of distinct vertices $(v_1,[U+007F]v_2,[U+007F]\ldots,v_l)$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-075115 | MATHDB-394427

Improving the separator bound for graphs with no K_h minor

摘录 / EXCERPT: For every graph G with n vertices, no K_h minor, and a weight function w on G, can the separator bound $O(h^{3/2}n^{1/2})$ be improved to $O(hn^{1/2})$?

open_problem | open | MathDB 页面 | 原文线索

V2-075116 | MATHDB-394428

Extension of the linear edge bound to Jordan-region intersection graphs

摘录 / EXCERPT: For each bipartite graph H, does there exist a constant $c(H)$ such that every intersection graph of n Jordan regions in the plane that does not contain H as a subgraph has a...

open_problem | open | MathDB 页面 | 原文线索

V2-075117 | MATHDB-394430

Hamilton circle in the line graph of a 4-edge-connected locally finite graph

摘录 / EXCERPT: If G is a 4-edge-connected locally finite graph, then $|L(G)|$ contains a Hamilton circle.

open_problem | open | MathDB 页面 | 原文线索

V2-075118 | MATHDB-394431

Hamilton circles in locally finite 7-connected line graphs

摘录 / EXCERPT: Every locally finite 7-connected line graph has a Hamilton circle.

open_problem | open | MathDB 页面 | 原文线索

V2-075119 | MATHDB-394433

Upper bound for the rotational dimension by the Colin de Verdière number

摘录 / EXCERPT: We conjecture that ... where $\mu(G)$ denotes the Colin de Verdière number of G.

open_problem | open | MathDB 页面 | 原文线索

V2-075120 | MATHDB-394436

The Molecular Conjecture for body-and-hinge frameworks

摘录 / EXCERPT: Let G be a multigraph. Then G can be realized as an infinitesimally rigid body-and-hinge framework in $\\mathbb{R}^d$ if and only if G can be realized as an infinitesimally r...

conjecture | open | MathDB 页面 | 原文线索

V2-075121 | MATHDB-394438

Thomassen’s conjecture on triangle-free torus quadrangulations

摘录 / EXCERPT: Every triangle-free quadrangulation of the torus is 3-colorable.

conjecture | open | MathDB 页面 | 原文线索

V2-075122 | MATHDB-394439

3-colorability of even-faced drawings in the Klein bottle

摘录 / EXCERPT: Every even-faced drawing in the Klein bottle with edge-width at least five and no odd meridian walk is 3-colorable.

open_problem | open | MathDB 页面 | 原文线索

V2-075123 | MATHDB-394441

Equivalence of \bμ(G) \dle 5, \bσ(G) \dle 5, and embeddability of the panelling

摘录 / EXCERPT: Let G be an undirected graph. Then the following are equivalent: (1) $\b\mu(G) \le 5$; (2) $\b\sigma(G) \le 5$; and (3) $\b\widetilde{G}$ can be embedded in $\b\mathbb{R}^4$.

open_problem | open | MathDB 页面 | 原文线索

V2-075124 | MATHDB-394445

Existence of many internally disjoint non-separating paths

摘录 / EXCERPT: Does there exist a function $f_1(k)$ such that, for every $f_1(k)$-connected graph G and every pair s and t of vertices in G, there exist k internally disjoint non-separa...

open_problem | open | MathDB 页面 | 原文线索

V2-075125 | MATHDB-394446

Conjecture 2: a path disjoint from a highly connected subgraph

摘录 / EXCERPT: There exists a function $f_3(k)$ such that for every $f_3(k)$-connected graph G and every three vertices s, t, and x in G, there exists an s-t path P and a k-conn...

conjecture | open | MathDB 页面 | 原文线索

V2-075126 | MATHDB-394447

Kriesell’s removable-path conjecture

摘录 / EXCERPT: There exists a function $f_2(k)$ such that, for every $f_2(k)$-connected graph G and every pair of vertices u and v of G, there exists a path P such that $G-E(P)$ is k-...

conjecture | open | MathDB 页面 | 原文线索

V2-075127 | MATHDB-394449

Lipschitz continuity of higher optimal eigenfunctions

摘录 / EXCERPT: In the report’s setup, where a0$D\subset\mathbb{R}^d$ is a bounded open set and $\hat\Omega\subset D$ is an optimal set for minimizing the k-th Dirichlet-Laplacian eigenvalue und...

open_problem | open | MathDB 页面 | 原文线索

V2-075128 | MATHDB-394450

Troesch’s stadium minimization question

摘录 / EXCERPT: Among convex sets of fixed area in $\mathbb{R}^2$, does the stadium minimize the second Dirichlet eigenvalue $\lambda_2(\Omega)$ of the Laplacian? The report answers no: the stadiu...

open_problem | open | MathDB 页面 | 原文线索

V2-075129 | MATHDB-394451

Faber-Krahn inequality for the first Robin eigenvalue

摘录 / EXCERPT: Let $N \ge 2$, let $\beta>0$ be constant, and let $\Omega\subset\mathbb{R}^N$ be a bounded Lipschitz domain. Let B be a ball with $|B|=|\Omega|$. If $\lambda_1(\Omega)$ denotes t...

open_problem | open | MathDB 页面 | 原文线索

V2-075130 | MATHDB-394452

Polya-Szegő conjecture for the first buckling eigenvalue of a clamped plate

摘录 / EXCERPT: Among all planar domains of area 1, the first buckling eigenvalue ... is minimized by a disc of area 1; equivalently, there exists a domain $\Omega^*\subset\mathbb{R}^2$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-075131 | MATHDB-394453

Shape optimization of the positivity threshold for convex domains

摘录 / EXCERPT: Which convex domains of given volume maximize $b_c(\d4aa)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075132 | MATHDB-394454

Global minimization of the second Dirichlet eigenvalue under a diameter constraint

摘录 / EXCERPT: The conjecture is that the disk is the global minimizer of the second Dirichlet eigenvalue [?]$(\lambda_2(\Omega))$ among plane domains of a given diameter.

open_problem | open | MathDB 页面 | 原文线索

V2-075133 | MATHDB-394455

Pólya conjecture for Dirichlet Laplace eigenvalues

摘录 / EXCERPT: Determine whether the Dirichlet eigenvalues of the fixed membrane problem on a domain $\Omega\subset\mathbb{R}^n$ satisfy ...

conjecture | open | MathDB 页面 | 原文线索

V2-075134 | MATHDB-394456

The fundamental gap problem (van den Berg’s conjecture)

摘录 / EXCERPT: For a bounded convex domain Ω, let λ_1 and λ_2 be the first two eigenvalues of the Dirichlet Laplacian $-\Delta$, and let $d=\sup\{|x-y|:x,y\in\Omega\}$ be t...

conjecture | open | MathDB 页面 | 原文线索

V2-075135 | MATHDB-394457

Payne-Pólya-Weinberger consecutive-eigenvalue ratio conjecture

摘录 / EXCERPT: For a domain $\Omega\subset\mathbb{R}^n$ and a positive integer k, conjecture that ...

conjecture | open | MathDB 页面 | 原文线索

V2-075136 | MATHDB-394458

Conjecture concerning a Faber-Krahn inequality for Robin problems

摘录 / EXCERPT: For fixed $\beta<0$, among all domains Ω of equal measure, the ball maximizes the first eigenvalue λ_1 of ...

conjecture | open | MathDB 页面 | 原文线索

V2-075137 | MATHDB-394459

Support of equilibrium measures on surfaces of revolution

摘录 / EXCERPT: For a set of revolution in $\mathbb{R}^3$ and a Riesz potential with $0<s<1$, when is the support of the equilibrium measure a proper subset of the outer boundary of the surface of...

open_problem | open | MathDB 页面 | 原文线索

V2-075138 | MATHDB-394460

Geometric characterization of Denjoy domains with \(\dim P_\infty=2\)

摘录 / EXCERPT: Under the standing assumptions that Ω is a Denjoy domain, $E:=\widehat{\mathbb C}\setminus\Omega\subset\widehat{\mathbb R}$, and every point of E, including ∞, is r...

open_problem | open | MathDB 页面 | 原文线索

V2-075139 | MATHDB-394461

Proper support of Riesz equilibrium measures on sets of revolution

摘录 / EXCERPT: Conjecture. For every s with $0<s<1$, there exists a compact set $A\subset H^+$ such that, writing $\Gamma(A)$ for the subset of $\mathbb{R}^3$ obtained by rotating A about the...

open_problem | open | MathDB 页面 | 原文线索

V2-075140 | MATHDB-394462

Asymptotic distribution of minimal Riesz-energy configurations

摘录 / EXCERPT: Let $A\subset\mathbb{R}^n$ be compact, let $d=\dim_H(A)$, and fix $0<s<d$. For each N, let $\omega_N^s=\{x_1,\ldots,x_N\}\subset A$ minimize ... and define ... If M_A is the set of...

open_problem | open | MathDB 页面 | 原文线索

V2-075141 | MATHDB-394463

Existence of asymptotic limits for Poincaré functions and harmonic measure

摘录 / EXCERPT: Let $p(x)=x^d+a_{d-1}x^{d-1}+\\cdots+a_1x$ be a real polynomial with $d>1$ and $a_1=p'(0)=\lambda>1$, and let f be the entire solution of $f(\lambda z)=p(f(z))$, $f(0)=0$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-075142 | MATHDB-394465

Nonexistence of asymptotically minimal Riesz-energy sequences on the circle

摘录 / EXCERPT: For every $s>1$, there exists no sequence $[U+007F]b_n[U+007F]_{n=1}^{\infty}\subset S^1$ such that the sequence of configurations $[U+007F][U+007F][U+007F]b_n[U+007F]_{n=1}^N[U+007F]_{N=1}^{\infty}$ is AEM$(S^1;s)$.

open_problem | open | MathDB 页面 | 原文线索

V2-075143 | MATHDB-394466

Falconer distance set conjecture at the critical dimension threshold

摘录 / EXCERPT: It is believed that, for every integer $n\ge 2$ and every Borel set $A\subset\mathbb{R}^n$, ... where $D(A)=\{|x-y|:x,y\in A\}$ and L^1 denotes one-dimensional Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-075144 | MATHDB-394467

Solvability of the Gauss variational problem

摘录 / EXCERPT: Let X be a locally compact Hausdorff space, let M be the linear space of all real-valued Radon measures on X, and let $\kappa:X\times X\to[0,\infty]$ be a lower semicontinuou...

open_problem | open | MathDB 页面 | 原文线索

V2-075145 | MATHDB-394468

Solvability criterion for the Gauss variational problem when one plate has infinite capacity

摘录 / EXCERPT: In the Gauss variational setup of the report, let $A=(A_i)_{i\in I}$ be a finite ordered collection of closed signed plates, with oppositely signed plates disjoint, and let $g>0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075146 | MATHDB-394470

Equality of Avoider-Enforcer hitting times for minimum positive degree and connectivity

摘录 / EXCERPT: Conjecture 3.1. $\tau_E(D_n)=\tau_E(T_n)$

open_problem | open | MathDB 页面 | 原文线索

V2-075147 | MATHDB-394473

Game chromatic index upper bound for arbitrary graphs

摘录 / EXCERPT: There exists $[U+007F]\varepsilon>0$ such that, for every non-empty graph G, $[U+007F]\chi'_g(G)\le(2-[U+007F]\varepsilon)\Delta(G)$, where $[U+007F]\chi'_g(G)$ is the game chromatic index and $[U+007F]\Delta(G)$...

open_problem | open | MathDB 页面 | 原文线索

V2-075148 | MATHDB-394476

Chooser-Picker 7-in-a-row game

摘录 / EXCERPT: Picker wins the Chooser-Picker version of the 7-in-a-row game.

open_problem | open | MathDB 页面 | 原文线索

V2-075149 | MATHDB-394479

Avoider’s winning strategy in the \(1:n/10\) Hamiltonicity game

摘录 / EXCERPT: Prove that Avoider has a winning strategy in the $1:n/10$ Hamiltonicity game.

open_problem | open | MathDB 页面 | 原文线索

V2-075150 | MATHDB-394480

Asymptotic comparison of lower and upper threshold biases in Avoider-Enforcer games

摘录 / EXCERPT: Prove that for the perfect matching, Hamiltonicity, non-planarity, non-k-colorability, and K_k-minor games, the lower and upper threshold biases satisfy $f_F^-=\\Theta\\(f_F^+)$.

open_problem | open | MathDB 页面 | 原文线索

V2-075151 | MATHDB-394484

Schinzel’s reduced-length inequality

摘录 / EXCERPT: Is the inequality ... true for all $t\in[-2,2]$ and all $P\in\mathbb{R}[x]$?

open_problem | open | MathDB 页面 | 原文线索

V2-075152 | MATHDB-394486

A complete metric space arising from the logarithmic Weil height

摘录 / EXCERPT: Describe the completion $\widehat G$ of $G=\overline{\mathbb{Q}}^{\times}/\operatorname{Tor}(\overline{\mathbb{Q}}^{\times})$ with respect to the metric...

open_problem | open | MathDB 页面 | 原文线索

V2-075153 | MATHDB-394487

Theorem 2: conditional nonexistence for the rational Catalan equation

摘录 / EXCERPT: Let $p,q>3$ be distinct primes satisfying: 1. $-1\in\langle p\bmod q\rangle$ and $-1\in\langle q\bmod p\rangle$; 2. $(pq,h^-_{pq})=1$, where h^-_{pq} is the relative class number...

open_problem | open | MathDB 页面 | 原文线索

V2-075154 | MATHDB-394491

One-dimensional Duffin-Schaeffer conjecture

摘录 / EXCERPT: For every function $\psi:[0,\infty)\to[0,\infty)$, let ... If ... then $|S_1(\psi)|=1$, where $|\cdot|$ denotes Lebesgue measure.

conjecture | open | MathDB 页面 | 原文线索

V2-075155 | MATHDB-394492

Euler’s question on four positive integers with the D(-1) property

摘录 / EXCERPT: Does there exist a set $D\subseteq\mathbb{Z}_{>0}$ consisting of four positive integers such that, for all distinct $x,y\in D$, the number $xy+x+y$ is a perfect square?

open_problem | open | MathDB 页面 | 原文线索

V2-075156 | MATHDB-394494

Classification of four-integer sets with pairwise square condition

摘录 / EXCERPT: Determine all sets $D\subseteq\mathbb{Z}$ with $|D|=4$ such that, for every pair of distinct elements $x,y\in D$, the integer $xy+x+y$ is a perfect square. The conjecture stated in...

open_problem | open | MathDB 页面 | 原文线索

V2-075157 | MATHDB-394496

Mahler-Mendès France binary-sequence problem

摘录 / EXCERPT: For every non-eventually periodic binary sequence $(a_n)_{n\ge 0}$, prove that at least one of the two real numbers ... is transcendental.

open_problem | open | MathDB 页面 | 原文线索

V2-075158 | MATHDB-394502

Lusztig’s diagonal-transition-matrix conjecture

摘录 / EXCERPT: Lusztig conjectured that the transition matrix between the almost characters of G^F and the characteristic functions of F-stable character sheaves should be diagonal.

conjecture | open | MathDB 页面 | 原文线索

V2-075159 | MATHDB-394503

Lusztig’s conjecture for split even-dimensional special orthogonal groups in characteristic 2

摘录 / EXCERPT: Determine the scalars occurring in Lusztig’s conjecture for the finite split special orthogonal groups $G^F=\operatorname{SO}_{2n}(\mathbb{F}_q)^F$ when the characteristic is $p=2$...

conjecture | open | MathDB 页面 | 原文线索

V2-075160 | MATHDB-394504

Determination of Lusztig-conjecture scalars for G^{F^2}

摘录 / EXCERPT: Let G be a classical group with connected center, defined over a finite field $ F_q$ of characteristic p with Frobenius map F, and assume that $DG\neq\operatorname{Spin}_N$....

conjecture | open | MathDB 页面 | 原文线索

V2-075161 | MATHDB-394508

Relations among cluster mutations

摘录 / EXCERPT: For mutations of seeds, and likewise for the corresponding seed tori, all relations between mutations are consequences of the relations - $\mu_i\mu_i=\mathrm{id}$; - if...

open_problem | open | MathDB 页面 | 原文线索

V2-075162 | MATHDB-394509

Quantization of the BRST charge in the Koszul-Tate case

摘录 / EXCERPT: The question of quantization of the charge in the Koszul-Tate case is unsolved.

open_problem | open | MathDB 页面 | 原文线索

V2-075163 | MATHDB-394510

Relation between modular classes of a Poisson Lie algebroid and its dual

摘录 / EXCERPT: Let A be a Lie algebroid with a Poisson structure $\c0\in\Gamma(\wedge^2 A)$ satisfying $[\u0003c0,\u0003c0]_A=0$, and equip A^* with the Lie algebroid structure defined by...

open_problem | open | MathDB 页面 | 原文线索

V2-075164 | MATHDB-394514

Meeks's conjecture on proper minimal immersions of open surfaces

摘录 / EXCERPT: Any open surface can be properly and minimally immersed in $\mathbb{R}^3$.

conjecture | open | MathDB 页面 | 原文线索

V2-075165 | MATHDB-394515

Constrained-Willmore property of Darboux transforms

摘录 / EXCERPT: The report conjectures that the Darboux transforms $L_\mu=\psi\mathbb H$ arising in Proposition 1 from parallel sections ψ of the flat connection d_μ are constrained Willm...

open_problem | open | MathDB 页面 | 原文线索

V2-075166 | MATHDB-394517

Non-existence of complete isometric immersions of negatively curved hyperbolic space forms

摘录 / EXCERPT: If $c<-1$, there is no complete isometric immersion with flat normal bundle of $H^k(c)$ into $H^n(-1)$ for any $k>1$ and any n.

open_problem | open | MathDB 页面 | 原文线索

V2-075167 | MATHDB-394519

Conformal type of homogeneous spaces with Cotton-tensor eigenvalue multiplicity two

摘录 / EXCERPT: How much of the conformal type of the spaces $M^3_{\\kappa,\\tau}$ is determined when requiring the Cotton tensor to have an eigenvalue of multiplicity 2 everywhere?

open_problem | open | MathDB 页面 | 原文线索

V2-075168 | MATHDB-394520

Lower boundedness of the spinor energy in Nil

摘录 / EXCERPT: Is the spinor energy E bounded from below on closed surfaces in Nil?

open_problem | open | MathDB 页面 | 原文线索

V2-075169 | MATHDB-394522

Sharp Willmore-energy estimate for spectral-genus-zero conformal tori

摘录 / EXCERPT: Let $f:T^2\to S^4$ be a conformal immersion of spectral genus zero. Then there is an n-fold conformal cover $T^2\to\widetilde{T}^2$ to a rectangular torus of width $\widetilde w$...

open_problem | open | MathDB 页面 | 原文线索

V2-075170 | MATHDB-394523

Trigonal minimal surfaces in 3-dimensional flat tori

摘录 / EXCERPT: Let $f:M_g\to\mathbb{R}^3/\Lambda$ be a conformal minimal immersion from a trigonal Riemann surface M_g of genus g into a 3-dimensional flat torus. Then $g=3r+1$ for some integ...

open_problem | open | MathDB 页面 | 原文线索

V2-075171 | MATHDB-394524

KKS inequalities in conformal coordinates

摘录 / EXCERPT: For the embedded CMC-ends under consideration, can one obtain the Korevaar-Kusner-Solomon inequalities in conformal coordinates?

open_problem | open | MathDB 页面 | 原文线索

V2-075172 | MATHDB-394525

Stable in-place 0/1-sorting with constant extra space

摘录 / EXCERPT: Given a sequence S of elements from a universe E and a characteristic function $f:E\to\{0,1\}$, stably rearrange S so that every element with $f(x)=0$ precedes every element...

open_problem | open | MathDB 页面 | 原文线索

V2-075173 | MATHDB-394526

Shortest Path with Time-Windows in the Elapsed-Time Cost Case

摘录 / EXCERPT: In the described AGV-routing setting, with time-windows on graph arcs and cost equal to elapsed time (travel time plus waiting time), the shortest-path problem with time-windows is...

open_problem | open | MathDB 页面 | 原文线索

V2-075174 | MATHDB-394527

Resilient sorting in the presence of memory faults

摘录 / EXCERPT: Given a set of n keys to be sorted, where the values of some keys may be arbitrarily corrupted during the sorting process, order correctly the set of uncorrupted keys. The report...

open_problem | open | MathDB 页面 | 原文线索

V2-075175 | MATHDB-394528

Proper regular solutions of finite split embedding problems over a Henselian quotient field

摘录 / EXCERPT: Let R be a domain which is Henselian with respect to some non-trivial ideal $\a$, and let $K=\operatorname{Quot}(R)$ be its field of fractions. Then K is a large field. In part...

open_problem | open | MathDB 页面 | 原文线索

V2-075176 | MATHDB-394532

Full faithfulness of the de Rham-Betti functor on motives

摘录 / EXCERPT: The de Rham-Betti functor on motifs is fully faithful.

open_problem | open | MathDB 页面 | 原文线索

V2-075177 | MATHDB-394535

Weak torsion conjecture for Jacobians of cyclic covers

摘录 / EXCERPT: For any number field k, any integer $r\ge 2$, and primes p,q with $p\nmid 2r$, there exists a constant $n_k(p,q,r)\ge 1$ such that, for every $f:Y\to\mathbb{P}^1_k$ in...

conjecture | open | MathDB 页面 | 原文线索

V2-075178 | MATHDB-394540

Non-Zariski density of S-integral points on complements of plane divisors

摘录 / EXCERPT: Let $D\subset \mathbb{P}^2_{\mathbb{Q}}$ be a normal crossing divisor. Set $X:=\mathbb{P}^2_{\mathbb{Q}}\setminus D$ and choose an affine embedding...

open_problem | open | MathDB 页面 | 原文线索

V2-075179 | MATHDB-394542

Basis theorem for 01-cell forms on the moduli space of genus-zero curves

摘录 / EXCERPT: The 01-cell forms represented by polygons of the form $[...,0,1,...,\infty]$ form a basis for $H^{n-3}(\mathfrak M_{0,n})$ among differential $(n-3)$-forms that are convergent on t...

open_problem | open | MathDB 页面 | 原文线索

V2-075180 | MATHDB-394544

Strong zero-one law for multidimensional diffusions in random environment

摘录 / EXCERPT: For the stationary, bounded, Lipschitz-continuous, uniformly elliptic, finite-range-dependent diffusion in random environment described on the source pages, does the strong zero-on...

open_problem | open | MathDB 页面 | 原文线索

V2-075181 | MATHDB-394545

Transient but zero-velocity diffusions in higher-dimensional random environments

摘录 / EXCERPT: For diffusions in random environments satisfying the model assumptions stated in the report-stationary, bounded, Lipschitz-continuous local characteristics, uniform ellipticity, an...

open_problem | open | MathDB 页面 | 原文线索

V2-075182 | MATHDB-394546

Does ballistic behavior imply condition (T') in dimensions $d\ge 2$?

摘录 / EXCERPT: For $d\ge 2$, the report conjectures the converse of its stated implication $(T')\Rightarrow$ ballistic behavior; equivalently, ballistic behavior should imply condition (T').

open_problem | open | MathDB 页面 | 原文线索

V2-075183 | MATHDB-394547

Coincidence of quenched and annealed Lyapunov exponents

摘录 / EXCERPT: Suppose $d \ge 4$ and $h>0$. Then there exists $\beta_0>0$ such that $m^{\mathrm{qu}}(h,\beta)=m^{\mathrm{an}}(h,\beta)$ for all $\beta\le\beta_0$. Moreover, when V_x is bounded,...

open_problem | open | MathDB 页面 | 原文线索

V2-075184 | MATHDB-394550

Scaling limit of the rescaled K-process at vanishing times

摘录 / EXCERPT: For almost every realization γ, where Y is the $K(\gamma,0)$ process and $\gamma=(\gamma_i)_{i\ge1}$ is the decreasing sequence of increments of an α-stable subordi...

open_problem | open | MathDB 页面 | 原文线索

V2-075185 | MATHDB-394552

Exact lower-bound exponent for biased-walk disconnection of a discrete cylinder

摘录 / EXCERPT: Can one prove the same result with $\varphi\equiv 0$ in (1)?

open_problem | open | MathDB 页面 | 原文线索

V2-075186 | MATHDB-394557

Validity of the bulk- and interface-matching expansions

摘录 / EXCERPT: Is the validity of the bulk- and interface-matching expansions in the mean-free-path limit established for the sharp-interface hydrodynamic description of the binary-fluid phase-se...

open_problem | open | MathDB 页面 | 原文线索

V2-075187 | MATHDB-394558

Minimum and uniqueness of one-dimensional phase-boundary fronts

摘录 / EXCERPT: Let $F:\mathbb{R}\to\braket{0,+\infty}$ be continuous and strictly positive except at $m=a$ and $m=b$, where $a<b$. Let ... and define ... where J is a non-negative integrable functi...

open_problem | open | MathDB 页面 | 原文线索

V2-075188 | MATHDB-394559

Uniqueness of binary-fluid critical fronts up to translation

摘录 / EXCERPT: For the binary-fluid excess free-energy functional G, if (w_1,w_2) and (v_1,v_2) are any two critical points of G in $M_{\rho_-,\rho_+}\times M_{\rho_+,\rho_-}$, then there...

open_problem | open | MathDB 页面 | 原文线索

V2-075189 | MATHDB-394560

Nonquadratic generalization of the complete-damage limit theorem

摘录 / EXCERPT: It remains open to generalize the above result to more general densities W allowing for nonquadratic cases.

open_problem | open | MathDB 页面 | 原文线索

V2-075190 | MATHDB-394561

Translation invariance of the two-dimensional hard-sphere Gibbs model

摘录 / EXCERPT: Let $r>0$, let $|\cdot|$ be any norm on $\mathbb{R}^{2}$, and define the pure hard-core potential ... Then, for every activity $z>0$, all equilibrium states corresponding to...

open_problem | open | MathDB 页面 | 原文线索

V2-075191 | MATHDB-394562

Existence of subsonic heteroclinic travelling waves in a bistable lattice

摘录 / EXCERPT: For the piecewise-quadratic interaction potential $V(\varepsilon)=\frac12\min\{(\varepsilon+1)^2,(\varepsilon-1)^2\}$, the travelling-wave equation ... has a heteroclinic solution wi...

open_problem | open | MathDB 页面 | 原文线索

V2-075192 | MATHDB-394563

Instance-optimal adaptive tree approximation

摘录 / EXCERPT: Can we achieve instance optimality in the sense of recovering the best N-term rate for each individual function?

open_problem | open | MathDB 页面 | 原文线索

V2-075193 | MATHDB-394564

Sharpness of the near-best tree approximation constant

摘录 / EXCERPT: Under the setting of Theorem 1, let T be a proper subtree of a fixed infinite master tree whose nodes have at most K children, let the local errors satisfy the weak subadditivi...

open_problem | open | MathDB 页面 | 原文线索

V2-075194 | MATHDB-394565

Local pointwise gradient estimator on nonconvex polyhedral domains

摘录 / EXCERPT: The report conjectures that, even when (Omega) is a nonconvex polyhedral domain, the local pointwise gradient error satisfies ... Here $D\subset\Omega$, u is the exact solution,...

open_problem | open | MathDB 页面 | 原文线索

V2-075195 | MATHDB-394566

Functional majorants for distributed optimal control problems with control constraints

摘录 / EXCERPT: For any control $u\in K$ and state $y\in H^1_0(\Omega)$, with arbitrary $w\in H^1_0(\Omega)$, $\tau\in H_{\mathrm{div}}(\Omega,\mathbb{R}^n)$, and $\beta>0$, the report asserts ...

open_problem | open | MathDB 页面 | 原文线索

V2-075196 | MATHDB-394572

Grothendieck’s problem on the period space of a fixed p-divisible group

摘录 / EXCERPT: Fix a Barsotti-Tate group X_0 over $\overline{\mathbb F}_p$ of height h and dimension d. Let $W=W(\overline{\mathbb F}_p)$, $K_0=W[1/p]$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075197 | MATHDB-394573

Compatibility of the p-adic Borel regulator with the étale Chern class

摘录 / EXCERPT: For $n\ge 1$, the Bloch-Kato exponential map ... maps the p-adic Borel regulator b_p to the étale Chern class c_n.

open_problem | open | MathDB 页面 | 原文线索

V2-075198 | MATHDB-394579

Almost-linear-time modular polynomial composition

摘录 / EXCERPT: Given three univariate polynomials f, g, and h of maximum degree n, compute the coefficients of $f\circ g \bmod h$. Can this modular polynomial composition problem be solve...

open_problem | open | MathDB 页面 | 原文线索

V2-075199 | MATHDB-394583

Difficult Point Conjecture for monadic-second-order-definable graph polynomials

摘录 / EXCERPT: For every graph polynomial that is definable in monadic second-order logic and that is $#P$-hard to evaluate for at least one point, its evaluation is $#P$-hard for almost all poin...

conjecture | open | MathDB 页面 | 原文线索

V2-075200 | MATHDB-394584

Subexponentiality of the elliptic-curve discrete logarithm problem over extension fields

摘录 / EXCERPT: There exists some $c>0$ such that the following holds: the discrete logarithm problem in $E(\mathbb{F}_{q^n})$, where $n\le c\log(q)$ and E is any elliptic curve over...

open_problem | open | MathDB 页面 | 原文线索

V2-075201 | MATHDB-394585

Foam surface-area problem in Euclidean space

摘录 / EXCERPT: What is the least surface area of a cell that tiles $\mathbb{R}^d$ by the lattice $\mathbb{Z}^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-075202 | MATHDB-394591

Constructing an elusive polynomial mapping against a given low-degree mapping

摘录 / EXCERPT: Let F be a field. Let $s=s(n)$, $m=m(n)$, and r be integers such that $s<m=n^{r+1}$ and $2\le r\le O(1)$. Given as input a polynomial mapping $\Gamma:F^s\to F^m$ of total degre...

open_problem | open | MathDB 页面 | 原文线索

V2-075203 | MATHDB-394592

Generalization of the minimal clique-density problem to K_r

摘录 / EXCERPT: For every fixed integer $r>3$, determine the minimal possible asymptotic density of copies of K_r in graphs with a prescribed edge density, in the regime where the edge density e...

open_problem | open | MathDB 页面 | 原文线索

V2-075204 | MATHDB-394596

Gluing constructions of special Lagrangian cones

摘录 / EXCERPT: For any odd genus $g=2d+1$, there exist infinitely many special Lagrangian cones C in $\mathbb{C}^3$ whose link $C\cap S^5(1)$ is an orientable surface of genus g.

open_problem | open | MathDB 页面 | 原文线索

V2-075205 | MATHDB-394597

Optimal constant in the intrinsic filling radius inequality

摘录 / EXCERPT: Conjecture: The optimal value for the intrinsic filling radius inequality is $\frac{1}{8}$.

open_problem | open | MathDB 页面 | 原文线索

V2-075206 | MATHDB-394600

Conjectural sharp threshold in Theorem 1

摘录 / EXCERPT: The constant n_0 in Theorem 1 is at least 12, and the authors conjecture that equality holds.

open_problem | open | MathDB 页面 | 原文线索

V2-075207 | MATHDB-394606

Vector-bundle structure of the Hitchin-component quotient for \(\mathrm{PSL}(4,\mathbb R)\)

摘录 / EXCERPT: Conjecturally, the quotient of the Hitchin component for $\mathrm{PSL}(4,\mathbb R)$ by the mapping class group is a vector bundle over the Riemannian moduli space.

open_problem | open | MathDB 页面 | 原文线索

V2-075208 | MATHDB-394613

Multiplicity conjecture for closed characteristics on compact convex hypersurfaces

摘录 / EXCERPT: The report states the conjecture ... Here $\mathcal H(2n)$ is the collection of C^3 compact convex hypersurfaces in $\mathbb R^{2n}$ described in the report, and $T(\Sigma)$ denote...

conjecture | open | MathDB 页面 | 原文线索

V2-075209 | MATHDB-394615

Smooth invariant measures for commuting circle diffeomorphisms

摘录 / EXCERPT: Let $d\ge 2$ and let $(f_1,\ldots,f_d)$ be a finite commuting family of smooth orientation-preserving circle diffeomorphisms with rotation numbers...

open_problem | open | MathDB 页面 | 原文线索

V2-075210 | MATHDB-394617

Optimal arithmetic condition for analytic linearization of commuting circle diffeomorphisms

摘录 / EXCERPT: Determine the optimal arithmetic condition on the rotation numbers under which any commuting family of real-analytic circle diffeomorphisms is simultaneously analytically conjugate...

open_problem | open | MathDB 页面 | 原文线索

V2-075211 | MATHDB-394619

Distribution and bounds for quintic S_5-extensions of fixed squarefree discriminant

摘录 / EXCERPT: Let D be a squarefree integer, and let $n(D)$ be the number of S_5-extensions of discriminant D. What is the average value of $n(D)$ as D ranges between $-X$ and X? What...

open_problem | open | MathDB 页面 | 原文线索

V2-075212 | MATHDB-394620

Experimental test of a Poisson law for counts of S_5-extensions

摘录 / EXCERPT: For squarefree discriminant D, let $n(D)$ denote the number of S_5-extensions of discriminant D. The report asks whether the claim that $n(D)$ obeys a Poisson distribution wi...

open_problem | open | MathDB 页面 | 原文线索

V2-075213 | MATHDB-394622

Vanishing central values among even quadratic twists of an elliptic curve

摘录 / EXCERPT: For a fixed elliptic curve E, the predicted asymptotic for the number of even-parity quadratic twists E_d with $|d|\le D$ and $L(E_d,1)=0$ is ... where c_E is a constant and...

open_problem | open | MathDB 页面 | 原文线索

V2-075214 | MATHDB-394625

Whittaker-coefficient interpretation of Weyl-group multiple Dirichlet series

摘录 / EXCERPT: Conjecturally, the Dirichlet series constructed for an irreducible, reduced root system Φ of rank r, with a regular weight θ, are the Fourier-Whittaker coefficients o...

open_problem | open | MathDB 页面 | 原文线索

V2-075215 | MATHDB-394629

Unbounded Mordell-Weil ranks for elliptic curves with prescribed rational torsion

摘录 / EXCERPT: For the fifteen torsion groups T occurring in Mazur’s theorem-namely $\mathbb Z/N\mathbb Z$ for $1\le N\le 10$ or $N=12$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-075216 | MATHDB-394630

Mordell-Weil rank 18 for elliptic K3 surfaces over \(\mathbb{Q}(T)\)

摘录 / EXCERPT: Can an elliptic K3 surface have Mordell-Weil rank 18 over $\mathbb{Q}(T)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075217 | MATHDB-394631

Artin’s primitive root conjecture for rank one tori

摘录 / EXCERPT: Let K be a number field, let T be a rank-one torus over K, and let $A\in T(K)$ be a non-torsion point. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-075218 | MATHDB-394636

Distributional validity of ε-limits for general initial data

摘录 / EXCERPT: For general initial data, is the ε-limit of a sequence of ε-solutions a solution of the free-boundary problem in the sense of distributions?

open_problem | open | MathDB 页面 | 原文线索

V2-075219 | MATHDB-394637

Estimates for the \c3\b2 equation in dimensions n \\

摘录 / EXCERPT: Estimates for $\sigma_2(D^2u)=1$ when $n\ge 4$?

open_problem | open | MathDB 页面 | 原文线索

V2-075220 | MATHDB-394638

Willmore minimization below the threshold for every conformal class

摘录 / EXCERPT: Let Σ be a closed surface of genus $p\ge 1$, let $n\in\{3,4\}$, and let $\tau\in\mathcal T(\Sigma)$. Define ... where $W(f)=\frac14\int_\Sigma|\vec H|^2\,d\mu_g$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075221 | MATHDB-394640

Regularity and singular-set bounds for stationary two-valued minimal graphs

摘录 / EXCERPT: There exists $\varepsilon_0=\varepsilon_0(n)\in(0,1/2)$ such that the following holds. Let $\alpha\in(0,1)$, let u be a 2-valued $C^{1,\alpha}(B_\rho(x_0))$ function with...

open_problem | open | MathDB 页面 | 原文线索

V2-075222 | MATHDB-394641

Singular-set bounds for multiplicity-two limits of stable minimal hypersurfaces

摘录 / EXCERPT: The report asks how large the singular set can be near a point where a stationary integral varifold V has a multiplicity-2 tangent plane. Theorem 2 answers this: if V is an...

open_problem | open | MathDB 页面 | 原文线索

V2-075223 | MATHDB-394642

Strong quantization for equation (1)

摘录 / EXCERPT: Under the hypotheses of Theorem 1-namely, for a bounded domain $\Omega\subset\mathbb{R}^4$, positive solutions u_k of $\Delta^2u_k=\lambda_k u_k e^{2u_k^2}$ with $\lambda_k>0$, N...

open_problem | open | MathDB 页面 | 原文线索

V2-075224 | MATHDB-394643

Instance optimality in compressed sensing for a general norm

摘录 / EXCERPT: Given a norm $\|\cdot\|_X$ on $\mathbb{R}^N$ and $k<N$, determine the minimal value of n for which there exists an encoding-decoding pair $\Phi:\mathbb{R}^N\to\mathbb{R}^n$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075225 | MATHDB-394644

The fair-duel problem for a universal ordering

摘录 / EXCERPT: Given a sequence $q=(q_n)_{n\ge 0}$ with $q_n\in\{-1,+1\}$, where q specifies whose turn it is in the duel, and where each shot independently hits with an unknown but fixed proba...

open_problem | open | MathDB 页面 | 原文线索

V2-075226 | MATHDB-394645

Exact projected convolution of hp-functions on locally refined grids

摘录 / EXCERPT: Given bounded-support functions $f\in S^f$ and $g\in S^g$, compute the exact function $\omega=P(f*g)$, where $(f*g)(x)=\int_{\mathbb R}f(y)g(x-y)\,dy$ and P is the...

open_problem | open | MathDB 页面 | 原文线索

V2-075227 | MATHDB-394647

Rotation-invariance question for the spherical random field

摘录 / EXCERPT: For the random field $(T_\xi)_{\xi\in S^2}$ describing the CMB, is $\xi\mapsto T_\xi$ rotation-invariant in the sense that, for every $\rho\in O(3)$ and every $\xi,\eta\in S^2$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075228 | MATHDB-394648

Alternative directional constructions for reproducible signals

摘录 / EXCERPT: Are there other constructions, involving operations on the plane other than rotation or shearing, that guarantee the reproducibility of a signal and have more flexible properties t...

open_problem | open | MathDB 页面 | 原文线索

V2-075229 | MATHDB-394650

Cameron-Kantor conjecture on base sizes

摘录 / EXCERPT: For every finite almost simple group acting faithfully and primitively in a non-standard permutation action, the minimum base size is at most 7; equality occurs only for the Math...

conjecture | open | MathDB 页面 | 原文线索

V2-075230 | MATHDB-394651

Thompson’s conjecture on products of conjugacy classes

摘录 / EXCERPT: Every finite simple group G has a conjugacy class C such that $C^2=G$.

conjecture | open | MathDB 页面 | 原文线索

V2-075231 | MATHDB-394652

Two-factor Waring conjecture for finite simple groups

摘录 / EXCERPT: For any nontrivial group word w, there is a number N such that, if G is a finite simple group with $|G|\ge N$, then $w(G)^2=G$.

conjecture | open | MathDB 页面 | 原文线索

V2-075232 | MATHDB-394654

Babai-Kantor-Lubotzky conjecture on expanders for finite simple groups

摘录 / EXCERPT: There exist constants $L>0$ and $\\epsilon>0$ such that any non-abelian finite simple group G has a generating set F such that $|F|\le L$ and the Cayley graphs $C(G;F)$ form a...

conjecture | open | MathDB 页面 | 原文线索

V2-075233 | MATHDB-394656

The Ore conjecture

摘录 / EXCERPT: Every element of every finite non-abelian simple group is a commutator.

conjecture | open | MathDB 页面 | 原文线索

V2-075234 | MATHDB-394658

The Kollár-Larsen problem on irreducibility of symmetric powers

摘录 / EXCERPT: Which closed subgroups of $GL(W)$ are irreducible on some symmetric power $\\operatorname{Sym}^k(W)$ with $k\ge 4$?

open_problem | open | MathDB 页面 | 原文线索

V2-075235 | MATHDB-394665

Completeness of the affine extremely primitive group list

摘录 / EXCERPT: We conjecture that the list in Theorem 3 is complete.

open_problem | open | MathDB 页面 | 原文线索

V2-075236 | MATHDB-394666

Completeness of the list of affine extremely primitive groups

摘录 / EXCERPT: Is every insoluble, simply primitive, extremely primitive group of affine type of the form $G=\mathbb Z_2^d:H$ isomorphic to one of the groups arising from the pairs (d,H) listed...

open_problem | open | MathDB 页面 | 原文线索

V2-075237 | MATHDB-394667

Existence of exponential growth-rate limits for oligomorphic permutation groups

摘录 / EXCERPT: Let G be an oligomorphic permutation group on $\cOmega$, and let f_n be the number of G-orbits on n-element subsets of $\cOmega$, while F_n is the number of G-orbits on...

open_problem | open | MathDB 页面 | 原文线索

V2-075238 | MATHDB-394670

Oda’s conjecture on lattice points in Minkowski sums

摘录 / EXCERPT: Let P be a smooth lattice polytope and let Q be a lattice polytope whose normal fan is a coarsening of that of P. Is every lattice point z in the Minkowski sum...

conjecture | open | MathDB 页面 | 原文线索

V2-075239 | MATHDB-394672

Minkowski sums of lattice polygons without unimodularity

摘录 / EXCERPT: For convex lattice polygons P,Q such that the normal fan of P refines that of Q, does ... hold without assuming that the normal cones of P are unimodular? Here...

open_problem | open | MathDB 页面 | 原文线索

V2-075240 | MATHDB-394673

Hadamard conjecture for Hadamard simplices

摘录 / EXCERPT: For every integer dimension $d>1$ such that $d+1$ is divisible by 4, there exists a Hadamard simplex in $[-1,1]^d$, i.e. a regular simplex whose $d+1$ vertices are vertices of th...

conjecture | open | MathDB 页面 | 原文线索

V2-075241 | MATHDB-394676

Normality of simple lattice polytopes with sufficiently long edges

摘录 / EXCERPT: Let P be a simple lattice polytope. Let k be the maximum of the heights of Hilbert basis elements of the tangent cones at the vertices of P. If every edge of P has lattice...

open_problem | open | MathDB 页面 | 原文线索

V2-075242 | MATHDB-394678

Invariants detecting non-normality of lattice polytopes

摘录 / EXCERPT: Are there some nice invariants that detect the non-normality of a polytope? Do the Ehrhart functions help?

open_problem | open | MathDB 页面 | 原文线索

V2-075243 | MATHDB-394679

Very ampleness of the divisor in the Ohsugi-Hibi embedding

摘录 / EXCERPT: Is the divisor corresponding to the embedding very ample?

open_problem | open | MathDB 页面 | 原文线索

V2-075244 | MATHDB-394680

The h^*-vector criterion for smooth polytopes

摘录 / EXCERPT: A necessary numerical criterion for normality is that the h^* vector be an M-sequence. Is at least this true for smooth polytopes?

open_problem | open | MathDB 页面 | 原文线索

V2-075245 | MATHDB-394681

Unimodular covers for lattice zonotopes

摘录 / EXCERPT: Does every lattice zonotope admit a unimodular cover?

open_problem | open | MathDB 页面 | 原文线索

V2-075246 | MATHDB-394685

Polyhedrality of the invertibility set for matrices over group rings

摘录 / EXCERPT: Let G be a finitely generated group. Let $A\in M_{p,p}(\mathbb{Z}[G])$, and let $N\subset\operatorname{Hom}(G,\mathbb{R})\setminus\{0\}$ be the set of those ξ such that A i...

open_problem | open | MathDB 页面 | 原文线索

V2-075247 | MATHDB-394686

Analogue of Theorem 3 for non-simply-connected spaces

摘录 / EXCERPT: Is there an analogue of Theorem 3 in the non-simply-connected case?

open_problem | open | MathDB 页面 | 原文线索

V2-075248 | MATHDB-394687

Global plane curve singularity groups

摘录 / EXCERPT: How many plane curve singularity groups (besides those coming from quasi-homogeneous curves) are global?

open_problem | open | MathDB 页面 | 原文线索

V2-075249 | MATHDB-394690

Bounds for curves of degree greater than one in spaces of hypersurfaces

摘录 / EXCERPT: What are the similar upper bounds for curves of degree larger than 1 in the space of hypersurfaces (or homogeneous polynomials)?

open_problem | open | MathDB 页面 | 原文线索

V2-075250 | MATHDB-394691

Pencils in \(\mathbb{CP}^4\) with more than two completely reducible fibers

摘录 / EXCERPT: Do there exist pencils in $\mathbb{CP}^4$ with $k>2$ that are not linear pullbacks of pencils from $\mathbb{CP}^3$?

open_problem | open | MathDB 页面 | 原文线索

V2-075251 | MATHDB-394692

Conjecture on the codimension and number of components of critical loci

摘录 / EXCERPT: If ω_λ is resonant in degree p, then $\operatorname{crit}(\Phi_\lambda)$ has codimension equal to p, and the number of its components equals...

conjecture | open | MathDB 页面 | 原文线索

V2-075252 | MATHDB-394694

Minimality and linearization: Theorem 6

摘录 / EXCERPT: Suppose that X has a minimal cell structure, meaning that the number of q-cells of X equals its rational Betti number $b_q(X)$ for every $q\ge 0$; in particular,...

open_problem | open | MathDB 页面 | 原文线索

V2-075253 | MATHDB-394701

Conjecture 1 on Weyl remainders for generic surfaces of revolution

摘录 / EXCERPT: Let M be a sphere with a generic (cf. [CdV, section 6]) metric of revolution. Then $R(\lambda)=O(\lambda^{1/2+\varepsilon})$ for any $\varepsilon>0$. Moreover, if x is not a po...

conjecture | open | MathDB 页面 | 原文线索

V2-075254 | MATHDB-394702

Conjecture 2: Remainder estimates for Liouville tori

摘录 / EXCERPT: Let M be a torus with a Liouville metric. Then ... for any $\varepsilon>0$ and for all $x\ne y\in M$.

conjecture | open | MathDB 页面 | 原文线索

V2-075255 | MATHDB-394703

Conjecture 3 on spectral remainders for generic negatively curved surfaces

摘录 / EXCERPT: Let M be a generic negatively curved surface. Then, as $ ... for any $varepsilon0$ and for all $xne yin M$. Here $N{x,y}$ is the spectral function, $Rx...

conjecture | open | MathDB 页面 | 原文线索

V2-075256 | MATHDB-394707

Decay of linear waves on Schwarzschild spacetime

摘录 / EXCERPT: Let (M,g) be Schwarzschild, let S be an arbitrary Cauchy surface for (M,g), and let $\\phi$ be the unique solution of the wave equation $\Box_g\phi=0$ on (M,g) with suffici...

open_problem | open | MathDB 页面 | 原文线索

V2-075257 | MATHDB-394708

Persistence of the black-hole region under vacuum perturbations of Schwarzschild

摘录 / EXCERPT: Given a sufficiently small perturbation of Schwarzschild initial data on a Cauchy surface S for the vacuum Einstein equations with $ \Lambda=0$, does the resulting maximal develo...

open_problem | open | MathDB 页面 | 原文线索

V2-075258 | MATHDB-394711

Generalized gluing theorem for Einstein constraint equations

摘录 / EXCERPT: Let $(M_1,\tilde g_1,\tilde\Pi_1)$ and $(M_2,\tilde g_2,\tilde\Pi_2)$ be two compact m-dimensional CMC solutions to the Einstein constraint equations (1)-(2), having the same con...

open_problem | open | MathDB 页面 | 原文线索

V2-075259 | MATHDB-394714

Derivation of Fourier’s law from microscopic dynamics

摘录 / EXCERPT: Derive Fourier’s law of heat conduction starting from deterministic dynamics of a system with a large number of degrees of freedom coupled to noise at its boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-075260 | MATHDB-394716

Kinetic conjecture for the FPU-\\ chain energy-current correlation

摘录 / EXCERPT: Let $I=[0,2\pi)$, let $\omega(x)=\lvert\sin(x/2)\rvert$ for $x\in I$ and $x\ne0$, with $\omega'(x)=\tfrac12\cos(x/2)$, and define ... For the linearized collision operator ... with f...

conjecture | open | MathDB 页面 | 原文线索

V2-075261 | MATHDB-394717

Mass-dependence of the limiting variance in the one-dimensional Rayleigh gas

摘录 / EXCERPT: Determine the dependence on the tagged-particle mass M of the limiting variance of the tagged particle, represented in the report by $\operatorname{Var}(t^{-1/2}Q_t)$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-075262 | MATHDB-394718

Hydrodynamic limit for the creation-annihilation model

摘录 / EXCERPT: For the creation-annihilation model with generator $L_2=L_0+\beta G^*+\sigma S$ described in the report, let $u_\varepsilon(t,x)$ be the piecewise-constant spin/charge field define...

open_problem | open | MathDB 页面 | 原文线索

V2-075263 | MATHDB-394719

Existence of Audusse-Perthame entropy solutions for discontinuous-flux conservation laws

摘录 / EXCERPT: For the scalar conservation law $\partial_t\rho+\partial_xF(x,\rho(t,x))=0$ with initial data $\rho|_{t=0}=\rho_0(x)$, where $F(\cdot,\rho)$ is continuous except on a set of measur...

open_problem | open | MathDB 页面 | 原文线索

V2-075264 | MATHDB-394720

Tracy-Widom-type limits for characteristic ASEP fluctuations

摘录 / EXCERPT: For the stationary one-dimensional asymmetric simple exclusion process at density $\rho\in(0,1)$, with particle flux $f(\rho)=(p-q)\rho(1-\rho)$ and characteristic speed...

open_problem | open | MathDB 页面 | 原文线索

V2-075265 | MATHDB-394721

Current-variance scaling for stationary asymmetric simple exclusion

摘录 / EXCERPT: Consider the one-dimensional nearest-neighbor asymmetric simple exclusion process on $\mathbb{Z}$, with right-jump rate p and left-jump rate q, where $p>q$ and $p+q=1$. Start t...

open_problem | open | MathDB 页面 | 原文线索

V2-075266 | MATHDB-394722

Superdiffusive diffusivity for finite-range asymmetric exclusion processes

摘录 / EXCERPT: For the stationary finite-range asymmetric exclusion process on $\mathbb Z$, with jump law p satisfying $p(z)\ge 0$, $p(z)=0$ for $|z|>R$ with $R<\infty$, $\sum_z p(z)=1$, and no...

open_problem | open | MathDB 页面 | 原文线索

V2-075267 | MATHDB-394723

Existence of the $t^{1/3}$-scaled TASEP diffusivity limit

摘录 / EXCERPT: For the stationary totally asymmetric simple exclusion process on $\mathbb{Z}$ with jump law $p(1)=1$ and $p(z)=0$ for $z\ne 1$, at fixed density $\rho\in(0,1)$, prove that the lim...

open_problem | open | MathDB 页面 | 原文线索

V2-075268 | MATHDB-394724

Correct asymptotics in the delocalized no-wall regime

摘录 / EXCERPT: For the polymer model without the wall and with $\lambda<1$, determine the correct asymptotics, as $L\to\infty$, of the spectral gap and the total-variation mixing time of the asso...

open_problem | open | MathDB 页面 | 原文线索

V2-075269 | MATHDB-394725

Sharp relaxation orders in the localized hard-wall polymer model

摘录 / EXCERPT: For the continuous-time heat-bath dynamics of the hard-wall polymer model, with finite pinning parameter $ \lambda>2$ and $L\to\infty$, prove the complementary bounds needed to est...

open_problem | open | MathDB 页面 | 原文线索

V2-075270 | MATHDB-394729

Definition of a Godbillon-Vey eta invariant

摘录 / EXCERPT: Define secondary invariants in the general foliated case, in particular a Godbillon-Vey eta invariant associated with the Godbillon-Vey cyclic cocycle.

open_problem | open | MathDB 页面 | 原文线索

V2-075271 | MATHDB-394730

Vanishing of the beta function for noncommutative ϕ^4_4 theory

摘录 / EXCERPT: Theorem 1. The equation $\Gamma_4(0,0,0,0)=\lambda\bigl(1-\partial_L\Sigma(0,0)\bigr)^2$ holds up to irrelevant terms to all orders of perturbation, either as a bare equation with...

open_problem | open | MathDB 页面 | 原文线索

V2-075272 | MATHDB-394733

Bounded-or-proper dichotomy for cocycles of amenable groups

摘录 / EXCERPT: Let G be a countable discrete amenable group. Every 1-cocycle with values in $\\ell^2(G)$ is either bounded or proper.

open_problem | open | MathDB 页面 | 原文线索

V2-075273 | MATHDB-394735

Dispersion below the traveling-wave energy threshold for the 3D Gross-Pitaevskii equation

摘录 / EXCERPT: There is a lower bound on the energy of traveling waves on $\mathbb{R}^3$, and it is conjectured that solutions with energy below that bound should disperse for large time.

open_problem | open | MathDB 页面 | 原文线索

V2-075274 | MATHDB-394736

Uniqueness in the Kerr spaces

摘录 / EXCERPT: Let $K^4(m,a)$ be the Kerr space with $m>0$ and $a=J/m\in[0,m)$, let E^4 be its exterior region, let $H=\delta E^4$ be the event horizon, and let $\xi=\partial_t$ be the Killing...

open_problem | open | MathDB 页面 | 原文线索

V2-075275 | MATHDB-394738

Global well-posedness and scattering for radial cubic NLS in two dimensions

摘录 / EXCERPT: For the cubic nonlinear Schrödinger equation ... with spherically symmetric initial data $u_0\in L_x^2(\mathbb R^2)$, there exists a unique global strong solution u. In the focusin...

open_problem | open | MathDB 页面 | 原文线索

V2-075276 | MATHDB-394739

Attainment of the one-dimensional sharp Strichartz constant

摘录 / EXCERPT: For the one-dimensional Strichartz inequality ... the best constant $S>0$ is attained. The report further states that ... and that a corresponding minimizer/maximizer is...

open_problem | open | MathDB 页面 | 原文线索

V2-075277 | MATHDB-394740

Loss of regularity for energy-supercritical defocusing nonlinear Schrödinger equations

摘录 / EXCERPT: For the defocusing nonlinear Schrödinger equation ... assume $n\ge 3$ and $\sigma>\frac{2}{n-2}$. Then there exist initial data $(\varphi_\lambda)_{0<\lambda\le 1}$ in the Schwartz c...

open_problem | open | MathDB 页面 | 原文线索

V2-075278 | MATHDB-394742

Nonlinear wave operators for subcritical gKdV in the full range

摘录 / EXCERPT: Can one construct a nonlinear wave operator for the generalized Korteweg-de Vries equation throughout the range $p\in(3,5)$, rather than only in the case $p=4$ treated in the repor...

open_problem | open | MathDB 页面 | 原文线索

V2-075279 | MATHDB-394743

Strichartz estimates for the Schrödinger equation with time-dependent magnetic potentials

摘录 / EXCERPT: Under what conditions on $A(t,x)$ does the magnetic Schrödinger equation with time-dependent potential ... have Strichartz estimates for u?

open_problem | open | MathDB 页面 | 原文线索

V2-075280 | MATHDB-394744

Local well-posedness for periodic DNLS when $r\le 4/3$

摘录 / EXCERPT: For the periodic derivative nonlinear Schrödinger Cauchy problem ... with $u(0,x)=u_0(x)$ for $x\in\mathbb T$ and $u_0\in\widehat H^s_r(\mathbb T)$, is there local well-posedness whe...

open_problem | open | MathDB 页面 | 原文线索

V2-075281 | MATHDB-394745

Local well-posedness for periodic DNLS below the Fourier-Lebesgue threshold

摘录 / EXCERPT: For the periodic derivative nonlinear Schrödinger Cauchy problem ... with data in $\widehat H_r^s(\mathbb T)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-075282 | MATHDB-394746

Well-posedness of the Maxwell-Schrödinger system in the energy space

摘录 / EXCERPT: For the Coulomb-gauge Maxwell-Schrödinger system ... with $\phi=(-\Delta)^{-1}(|u|^2)$, $J=2\operatorname{Im}(\overline u\,\nabla_Au)$, and P the Helmholtz projection, is the syste...

open_problem | open | MathDB 页面 | 原文线索

V2-075283 | MATHDB-394747

Global existence and spacetime bound for radial focusing energy-critical NLS below the ground-state gradient threshold

摘录 / EXCERPT: For the focusing energy-critical nonlinear Schrödinger equation ... let u be a spherically symmetric solution with maximal lifespan I. If ... where ... then $I=(-\infty,\infty)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075284 | MATHDB-394748

Decomposition of representation spaces into Stiefel-Whitney sectors

摘录 / EXCERPT: For a finitely generated discrete group π, there is a decomposition into nonempty, disjoint closed subsets (called w-sectors): ... where the w-sector consists of those homomo...

open_problem | open | MathDB 页面 | 原文线索

V2-075285 | MATHDB-394754

Spherical-class/Hurewicz-image conjecture for the sphere spectrum

摘录 / EXCERPT: For $Z=S^0$, the image of the Hurewicz map ... should consist only of Hopf-invariant-one elements and Kervaire-invariant-one elements, if they exist.

conjecture | open | MathDB 页面 | 原文线索

V2-075286 | MATHDB-394757

Finite resolutions of the $K(n)$-local sphere by higher real K-theories

摘录 / EXCERPT: Construct finite resolutions of $L_{K(n)}S^0$ by fixed-point spectra of E_n with respect to finite subgroups of G_n ("higher real K-theories").

open_problem | open | MathDB 页面 | 原文线索

V2-075287 | MATHDB-394758

Weak limiting behavior of cuts needed to isolate the root of a random recursive tree

摘录 / EXCERPT: Let T_n be a uniformly distributed random recursive tree with n vertices, and let X_n be the number of random edge cuts needed to isolate its root. Then, as $n\to\infty$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075288 | MATHDB-394760

The speed of coalescence when a Λ-coalescent comes down from infinity

摘录 / EXCERPT: Let $(\Pi(t),t\ge 0)$ be the Λ-coalescent associated with a finite measure Λ on $[0,1]$, and let N_t be its number of clusters. Define ... Let δ be the lower...

open_problem | open | MathDB 页面 | 原文线索

V2-075289 | MATHDB-394761

Central limit theorem for the Smoluchowski coagulation model

摘录 / EXCERPT: The report claims to solve the problem of constructing a well-defined infinite-dimensional Gaussian Ornstein-Uhlenbeck process, in a suitable distribution space, with the fluctuati...

open_problem | open | MathDB 页面 | 原文线索

V2-075290 | MATHDB-394762

Fragmentation process for cycles of a conditioned uniform random permutation

摘录 / EXCERPT: For each fixed n, does there exist a fragmentation process $(\[1m\Pi\[0m^1,\Pi^2,\ldots,\Pi^n)$ such that: (i) for each k, $\[1m\Pi\[0m^k$ has the same distribution as the part...

open_problem | open | MathDB 页面 | 原文线索

V2-075291 | MATHDB-394763

Criterion for the field of definition of a Hodge-locus component

摘录 / EXCERPT: In the setup of the report, let B_α be the projection to the parameter space of the connected component Z_α of the locus of Hodge classes passing through a Hodge clas...

open_problem | open | MathDB 页面 | 原文线索

V2-075292 | MATHDB-394767

Enriques quotients of a K3 surface

摘录 / EXCERPT: Given a K3 surface X, how many Enriques surfaces are obtained by taking quotients of X? Equivalently, how many conjugacy classes of free involutions are there in the automorphi...

open_problem | open | MathDB 页面 | 原文线索

V2-075293 | MATHDB-394768

Enriques quotients of a Picard-general Jacobian Kummer surface

摘录 / EXCERPT: Let C be a smooth projective curve of genus 2 whose Jacobian $J(C)$ has Picard number 1, and let $\operatorname{Km}C$ be the minimal resolution of its Jacobian Kummer quartic...

open_problem | open | MathDB 页面 | 原文线索

V2-075294 | MATHDB-394770

Vanishing conjecture for intersection numbers on the perfect cone compactification of \(\mathcal A_g\)

摘录 / EXCERPT: Let $G=g(g+1)/2=\dim \mathcal A_g$, and let ... where L is the Hodge line bundle and D is the boundary divisor. Conjecture: For any N, $a_N^{(g)}$ vanishes unless ... for some...

conjecture | open | MathDB 页面 | 原文线索

V2-075295 | MATHDB-394771

Equality of the power-sum component with the constrained VSP

摘录 / EXCERPT: In the setting of Theorem 4 for $d\ge 6$-where B is the smooth quintic del Pezzo 3-fold, $f:A\to B$ is the blow-up along a general smooth rational curve C of degree d, H_2...

open_problem | open | MathDB 页面 | 原文线索

V2-075296 | MATHDB-394772

Abelianization: is the pull-back map on level-one Verlinde spaces an isomorphism?

摘录 / EXCERPT: In the setup of the report, let X be a smooth projective curve, let G be a simple, simply-connected complex Lie group, let $\[W]$ be its Weyl group, and let $\[Z\to X]$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-075297 | MATHDB-394782

Deformations in the large of iterated principal holomorphic torus bundles

摘录 / EXCERPT: Can we determine all deformations in the large of (iterated) principal holomorphic torus bundles?

open_problem | open | MathDB 页面 | 原文线索

V2-075298 | MATHDB-394791

Exponential growth and the contact-process growth rate

摘录 / EXCERPT: If the contact process survives (i.e. percolates) on an exponentially growing graph, is the exponential growth rate ... necessarily positive?

open_problem | open | MathDB 页面 | 原文线索

V2-075299 | MATHDB-394792

Cluster uniqueness at zero exponential growth rate

摘录 / EXCERPT: For the nearest-neighbor contact process on a more general transitive graph G, with $\[?]$ exponential growth rate ... does cluster uniqueness hold when $r=0$, either in the weak s...

open_problem | open | MathDB 页面 | 原文线索

V2-075300 | MATHDB-394797

Multiplicity-one theorem for orthogonal and unitary groups

摘录 / EXCERPT: Let F be a non-Archimedean local field of characteristic 0, and let D be either F or a quadratic extension of F, with the nontrivial conjugation on D in the latter case...

open_problem | open | MathDB 页面 | 原文线索

V2-075301 | MATHDB-394798

Periods and global Gross-Prasad conjecture for special orthogonal groups

摘录 / EXCERPT: Let k be a global field with $\operatorname{char}(k)\ne 2$. Let $(V_0,Q_0)\subset (V_1,Q_1)$ be quadratic forms over k of ranks n and $n+1$, respectively, with $n\ge 2$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-075302 | MATHDB-394799

Formal-degree conjecture for compact Lie groups

摘录 / EXCERPT: The report formulates the conjecture that, if G is a compact Lie group of rank l and $\c0$ is an irreducible finite-dimensional representation of G, then ... where ...

conjecture | open | MathDB 页面 | 原文线索

V2-075303 | MATHDB-394800

Cusp-form characterizations of the Plancherel components for reductive symmetric spaces

摘录 / EXCERPT: Let G/H be a semisimple symmetric space with Schwartz space $[U+007F]mathcal{C}(G/H)$ and Plancherel decomposition $[U+007F]mathcal{C}(G/H)=\bigoplus_i [U+007F]mathcal{C}_i(G/H)$, where each componen...

open_problem | open | MathDB 页面 | 原文线索

V2-075304 | MATHDB-394801

Cusp-form characterization of the discrete series by unipotent integrals

摘录 / EXCERPT: For a semisimple symmetric space G/H, can the discrete series be characterized by a reasonable cusp-form condition defined through integrals over subgroups N (or $N/(N\cap H)$)...

open_problem | open | MathDB 页面 | 原文线索

V2-075305 | MATHDB-394802

Lipsman’s Conjecture on Proper Actions for Nilpotent Lie Groups

摘录 / EXCERPT: Suppose G is a simply connected nilpotent Lie group and L,H are its closed subgroups. Then the L-action on the homogeneous space G/H is proper if and only if it is fixed po...

conjecture | open | MathDB 页面 | 原文线索

V2-075306 | MATHDB-394803

Lipsman’s conjecture on proper actions

摘录 / EXCERPT: Conjecture 1 (Lipsman).

conjecture | open | MathDB 页面 | 原文线索

V2-075307 | MATHDB-394804

Lipsman’s Conjecture (A) for affine unipotent nilpotent actions

摘录 / EXCERPT: Suppose a simply connected nilpotent Lie group L acts on $\mathbb{R}^n$ affinely and unipotently. Then the action is proper if and only if it is fixed-point-free.

conjecture | open | MathDB 页面 | 原文线索

V2-075308 | MATHDB-394805

Multiplicity one for local Klyachko models

摘录 / EXCERPT: Let F be a local non-archimedean field and let π be an irreducible admissible representation of $\mathrm{GL}_n(F)$. For the Klyachko models...

open_problem | open | MathDB 页面 | 原文线索

V2-075309 | MATHDB-394806

Cramér-von Mises testing for functional equilibrium job-search models

摘录 / EXCERPT: Given a random function Y whose distribution depends on an unknown finite-dimensional parameter θ, and a random sample $\{X_1,\ldots,X_n\}$ of realizations of a random fun...

open_problem | open | MathDB 页面 | 原文线索

V2-075310 | MATHDB-394807

Optimal separation parameter for distinguishable models

摘录 / EXCERPT: For a parametric model class $M_k=\{f(\cdot;\theta):\theta\in\Omega\subseteq\mathbb{R}^k\}$ satisfying the central limit theorem, with a compact parameter space partitioned into eq...

open_problem | open | MathDB 页面 | 原文线索

V2-075311 | MATHDB-394808

Adaptive estimation theorem for a linear inverse problem

摘录 / EXCERPT: In the linear inverse-problem setup described in the report, let the variances v_k satisfy condition (MD): for constants $1<u_0\leq u$, $v_{k-1}\leq u v_k$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075312 | MATHDB-394809

A more precise relationship between twisted L-function derivatives and harmonic Maass-form coefficients

摘录 / EXCERPT: It would be very interesting to obtain a more precise relationship between these derivatives and the coefficients $c_g^+(\Delta)$, where the derivatives refer to...

open_problem | open | MathDB 页面 | 原文线索

V2-075313 | MATHDB-394811

Direct modularity of the partition generating function from its hypergeometric expansion

摘录 / EXCERPT: It is an open problem to show directly the modularity of $P(q)$ using the expansion ...

open_problem | open | MathDB 页面 | 原文线索

V2-075314 | MATHDB-394812

Relation between an overpartition rank-difference function and Hurwitz class numbers

摘录 / EXCERPT: Theorem 3 ([5]) states that ... Here the report defines $f(q)=\sum_n\alpha(n)q^n=\sum_n(p_e(n)-p_o(n))q^n$ using the even- and odd-rank overpartition counts, and...

open_problem | open | MathDB 页面 | 原文线索

V2-075315 | MATHDB-394813

Explicit Fourier coefficients of the map $L_{t,\chi}$

摘录 / EXCERPT: Let N be a natural number and let t be a positive squarefree integer. For $c=\sum_s c_s(s)\in C(2N,\chi^2)$ and a natural number D whose squarefree part does not divide $4Nt$...

open_problem | open | MathDB 页面 | 原文线索

V2-075316 | MATHDB-394814

Existence of a degree-4 Hermitian cusp form over \(\mathbb{Q}(\sqrt{-3})\)

摘录 / EXCERPT: It is conjectured that there is a non-trivial cusp form of degree 4 which is a linear combination of theta series for the Hermitian lattices over $K=\mathbb{Q}(\sqrt{-3})$ consid...

open_problem | open | MathDB 页面 | 原文线索

V2-075317 | MATHDB-394818

Mumford-Tate-Serre conjecture for motives with coefficients

摘录 / EXCERPT: Let E and F be number fields, and let M be a pure, polarizable motive over F with coefficients in E. For each finite place λ of E, let ... be the associated...

conjecture | open | MathDB 页面 | 原文线索

V2-075318 | MATHDB-394821

Ikeda’s period conjecture for the Ikeda lift

摘录 / EXCERPT: Let k and n be positive even integers, let $f\in S_{2k-n}(\Gamma_1)$ be a primitive form, and let $\widetilde f\in S^+_{k-n/2+1/2}(\Gamma_0(4))$ be the Hecke eigenform correspo...

conjecture | open | MathDB 页面 | 原文线索

V2-075319 | MATHDB-394825

Local theta-correspondence obstruction for generalized Duke-Imamoğlu-Ikeda lifts

摘录 / EXCERPT: Theorem. Let V be a vector space over E of even dimension $2r$ with a nondegenerate quadratic form q. (1) If $2n>r-1$ and there is a finite place v of E for which the com...

open_problem | open | MathDB 页面 | 原文线索

V2-075320 | MATHDB-394826

Teachers’ development of the concept of negotiated classroom norms

摘录 / EXCERPT: To what extent can teachers at different stages come to understand the concept that classroom norms are negotiated rather than imposed?

open_problem | open | MathDB 页面 | 原文线索

V2-075321 | MATHDB-394827

Evaluation of teachers’ responses to test items

摘录 / EXCERPT: “Whatever aspect one takes into account, an open question is how to evaluate responses or reactions of teachers on given test items.”

open_problem | open | MathDB 页面 | 原文线索

V2-075322 | MATHDB-394828

Hopf bifurcation in oscillating sedimentation

摘录 / EXCERPT: Is there a Hopf bifurcation to steady oscillations of a sphere or other object sedimenting in a non-Newtonian fluid? This should be constitutive-equation dependent, and not occur f...

open_problem | open | MathDB 页面 | 原文线索

V2-075323 | MATHDB-394829

Theory of the tilt-angle phenomenon for non-Newtonian liquids

摘录 / EXCERPT: “There is until now no satisfactory theory to explain the tilt angle phenomenon for non-Newtonian liquids.” Here, the tilt angle is the angle formed by the long axis of symmetry of...

open_problem | open | MathDB 页面 | 原文线索

V2-075324 | MATHDB-394834

Moving-surface ideal for tensor-product surfaces of bidegree $(2,1)$

摘录 / EXCERPT: Determine what can be said about the moving-surface ideal of a tensor-product surface of bidegree $(2,1)$, parametrized by four bihomogeneous polynomials of degree 2 in x,y and...

open_problem | open | MathDB 页面 | 原文线索

V2-075325 | MATHDB-394835

Computing the Newton polygon of the implicit equation

摘录 / EXCERPT: For planar curves parametrized by polynomials with fixed supports, determine the vertices of the Newton polygon $N(\phi)$ of the implicit equation, without computing $\phi(x_i)$, a...

open_problem | open | MathDB 页面 | 原文线索

V2-075326 | MATHDB-394837

Minimal-degree families of rational curves on rational surfaces

摘录 / EXCERPT: Given a rational surface, not necessarily regular, embedded in projective space of arbitrary dimension, find a family of rational curves on it with degree as small as possible.

open_problem | open | MathDB 页面 | 原文线索

V2-075327 | MATHDB-394838

The modular isomorphism problem

摘录 / EXCERPT: Whether a finite p-group is determined by $\b_mathbb F_pG$ up to isomorphism.

open_problem | open | MathDB 页面 | 原文线索

V2-075328 | MATHDB-394842

The conjecture that the projective Schur group equals the Brauer group

摘录 / EXCERPT: It was conjectured that $PS(k)=Br(k)$ in general, where $PS(k)$ is the projective Schur group of the field k and $Br(k)$ is its Brauer group.

conjecture | open | MathDB 页面 | 原文线索

V2-075329 | MATHDB-394849

Partial augmentation at the image of a torsion unit in \(\mathrm{GL}_n(k)\)

摘录 / EXCERPT: Let $G=\mathrm{GL}_n(k)$, where k is a finite field, and let $M_n(k)$ be the $n\times n$ matrix ring over k. Let $V(\mathbb ZG)$ be the group of augmentation-one units of...

open_problem | open | MathDB 页面 | 原文线索

V2-075330 | MATHDB-394852

Normalizer property for groups with finite normal subgroups satisfying NP

摘录 / EXCERPT: The normalizer problem (NP) for a group G asks whether the normalizer of G in the unit group of $ \mathbb{Z}G$ consists only of the obvious units, namely the subgroup generated...

open_problem | open | MathDB 页面 | 原文线索

V2-075331 | MATHDB-394853

Finite-index generators for the unit group of \(\mathbb Z[Q_8\times C_7]\)

摘录 / EXCERPT: Let Q_8 be the quaternion group of order 8, and let C_7 be the cyclic group of order 7. Find finitely many units in the integral group ring $\mathbb Z[Q_8\times C_7]$ that...

open_problem | open | MathDB 页面 | 原文线索

V2-075332 | MATHDB-394854

Hypercentre and finite conjugacy centre of unit groups of group rings

摘录 / EXCERPT: Let G be a group, let $Z^{\infty}(G)$ denote the union of the terms of the upper central series of G, and set $U=U(\mathbb ZG)$. Is $Z^{\infty}(U)\leq \Delta(U)$? Is...

open_problem | open | MathDB 页面 | 原文线索

V2-075333 | MATHDB-394862

NP-completeness of the Rank Decoding problem

摘录 / EXCERPT: Can one find a reduction proving that the following Rank Decoding (RD) problem is NP-complete? Given $N,n,k\in\mathbb{N}^*$, a generator matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-075334 | MATHDB-394863

Computing the period of the continued fraction of \(\sqrt N\)

摘录 / EXCERPT: Is it possible to compute exactly or approximately, with an m-precision of the order $O((\log \tau)^{\alpha_1})$, the period τ from N in deterministic polynomial time...

open_problem | open | MathDB 页面 | 原文线索

V2-075335 | MATHDB-394867

Computing the Newton polytope of the projected hypersurface

摘录 / EXCERPT: Given the equations for $X\subset (\mathbb{C}^*)^n$, how can the Newton polytope of $Y=\alpha(X)\subset (\mathbb{C}^*)^{n-c+1}$ be computed?

open_problem | open | MathDB 页面 | 原文线索

V2-075336 | MATHDB-394872

Dimension of the tropical-multiplicative closure of metric matrices

摘录 / EXCERPT: Let $\mathcal M_n$ be the set of matrices $D=(d_{ij})$ satisfying $d_{ii}=0$, $d_{ij}=d_{ji}$, and $d_{ij}+d_{jk}\ge d_{ik}$. Is the closure of $\mathcal M_n$ under tropical matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-075337 | MATHDB-394873

Maximality of the real Zeuthen problem for two lines

摘录 / EXCERPT: For every integer $d\ge 2$, does there exist a configuration C of 2 real lines and $d(d+3)/2-2$ real points such that all degree-d curves passing through the points and tange...

open_problem | open | MathDB 页面 | 原文线索

V2-075338 | MATHDB-394874

Representativeness of objective molecular-dynamics solutions

摘录 / EXCERPT: How representative are these solutions of the equations of molecular dynamics? That is a key question that may be best approached by methods of nonlinear dynamics. One may conjectu...

open_problem | open | MathDB 页面 | 原文线索

V2-075339 | MATHDB-394875

Creating a material with a desired wave-focusing property

摘录 / EXCERPT: Can one create a new material in D with a desired wave-focusing property? This means that a plane wave at a fixed wavenumber $k>0$ with a fixed incident direction α will h...

open_problem | open | MathDB 页面 | 原文线索

V2-075340 | MATHDB-394876

Creating a material with a prescribed refraction coefficient

摘录 / EXCERPT: Let $D\subset\mathbb{R}^3$ be bounded, let $k>0$, and let the initial refraction coefficient $n_0^2\in C(\mathbb{R}^3)$ satisfy $n_0^2=1$ on $\mathbb{R}^3\setminus D$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075341 | MATHDB-394877

Rank-one convex envelope for a two-slip-system crystal-plasticity energy

摘录 / EXCERPT: The report states that the rank-one convex envelope of ... is ... where ...

open_problem | open | MathDB 页面 | 原文线索

V2-075342 | MATHDB-394878

Quasiconvex envelope of the condensed energy in linearized crystal plasticity

摘录 / EXCERPT: In the linearized three-dimensional model with a sufficiently large set of slip systems $S=\{s_i\otimes m_i\}_{i=1}^N$, no self-hardening, positive latent hardening, plastic strain...

open_problem | open | MathDB 页面 | 原文线索

V2-075343 | MATHDB-394879

Nonvanishing asymptotic velocity for periodic drifts

摘录 / EXCERPT: For the first diffusion model, with $D>0$ and a C^{1} drift $b(t,x)$ that is periodic in time and space and has zero average in both variables, the limit ... exists, is independent...

open_problem | open | MathDB 页面 | 原文线索

V2-075344 | MATHDB-394880

Independence numbers of powers of odd cycles

摘录 / EXCERPT: Determine the independence number $\alpha(G^d)$ for powers G^d of odd cycles G, in the general cases not covered by the known formulas and bounds.

open_problem | open | MathDB 页面 | 原文线索

V2-075345 | MATHDB-394881

Independence numbers of arbitrary powers of odd cycles

摘录 / EXCERPT: The problem of determining the independence numbers of arbitrary powers of odd cycles remains widely open.

open_problem | open | MathDB 页面 | 原文线索

V2-075346 | MATHDB-394884

Intersecting families of permutations

摘录 / EXCERPT: Let $X\subset S_n$ be an intersecting family, meaning that for every $\sigma,\tau\in X$ there exists i such that $\sigma(i)=\tau(i)$. The maximum possible size of X is $(n-1)!$...

open_problem | open | MathDB 页面 | 原文线索

V2-075347 | MATHDB-394887

Rank-selected chain enumeration in $NC^{(m)}(A_n)$

摘录 / EXCERPT: The number of (multi-)chains $x_1\le x_2\le\cdots\le x_{\ell-1}$ in $NC^{(m)}(A_n)$ satisfying $\operatorname{rk}(x_i)=s_1+s_2+\cdots+s_i$ for $i=1,2,\ldots,\ell-1$ is ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-075348 | MATHDB-394891

Friedman’s spectral-radius conjecture for random graph lifts

摘录 / EXCERPT: Let G be a finite connected graph, let T be the infinite universal-cover tree of G, let D be the spectral radius of the adjacency matrix of G, and let ρ be the spect...

conjecture | open | MathDB 页面 | 原文线索

V2-075349 | MATHDB-394896

Density theorem for bipartite graphs

摘录 / EXCERPT: Let H be a bipartite graph with n vertices and maximum degree $\Delta\ge 1$. If $\varepsilon>0$ and G is a graph with $N\ge 32\Delta\varepsilon^{-\Delta}n$ vertices and at le...

open_problem | open | MathDB 页面 | 原文线索

V2-075350 | MATHDB-394901

Asymptotic threshold bias for the biased Maker-Breaker connectivity game

摘录 / EXCERPT: Is the random graph intuition valid for the connectivity game asymptotically? That is, is it true that $b_T=(1-o(1))\frac{n}{\ln n}$?

open_problem | open | MathDB 页面 | 原文线索

V2-075351 | MATHDB-394902

Asymptotic threshold bias for the Maker-Breaker Hamiltonicity game

摘录 / EXCERPT: Can Maker build a Hamilton cycle while playing against a Breaker of bias $(1-o(1))n/\ln n$ in the biased Maker-Breaker game on the edges of K_n?

open_problem | open | MathDB 页面 | 原文线索

V2-075352 | MATHDB-394903

Enforcer’s bias in the Avoider-Enforcer Hamiltonicity game

摘录 / EXCERPT: Can Enforcer, playing with bias $(1-o(1))\,n/\ln n$, play so that Avoider, playing with bias 1, builds a Hamilton cycle?

open_problem | open | MathDB 页面 | 原文线索

V2-075353 | MATHDB-394905

Embedding spanning subgraphs with a zero-free colouring

摘录 / EXCERPT: For all $r,\Delta\in\mathbb{N}$ and $\gamma>0$, there exist constants $\beta>0$ and $n_0\in\mathbb{N}$ such that for every $n\ge n_0$ the following holds. Let H be a graph with...

open_problem | open | MathDB 页面 | 原文线索

V2-075354 | MATHDB-394906

Box-distance between symmetric and monotone distributions

摘录 / EXCERPT: Let D_1 and D_2 be distributions on the product set $S_1\times S_2\times\cdots\times S_k$. Define their box-distance as the maximum of $\lvert D_1(A)-D_2(A)\rvert$ over boxes...

open_problem | open | MathDB 页面 | 原文线索

V2-075355 | MATHDB-394907

Dense subsets in non-abelian groups with only trivial six-term solutions

摘录 / EXCERPT: Do there exist infinitely many groups G_n of order n containing a subset S_n such that the equation ... has only trivial solutions and such that $|S_n|\gg n^{1/3}$?

open_problem | open | MathDB 页面 | 原文线索

V2-075356 | MATHDB-394909

Unavoidability of powers-of-two cycle lengths

摘录 / EXCERPT: There exists a constant d such that every graph with average degree at least d contains a cycle whose length is a power of 2.

open_problem | open | MathDB 页面 | 原文线索

V2-075357 | MATHDB-394910

Erdős-Hajnal odd-cycle reciprocal-sum conjecture

摘录 / EXCERPT: Does a graph of infinite chromatic number have an infinite sum of reciprocals of its odd cycle lengths?

conjecture | open | MathDB 页面 | 原文线索

V2-075358 | MATHDB-394912

Multiplicity Conjecture

摘录 / EXCERPT: Let I be a homogeneous ideal in the polynomial ring $S=k[x_1, \ldots,x_n]$, and let $A=S/I$. Let $h=\operatorname{codim}(I)$, and let $\beta_{i,j}$ be the graded Betti numbers of...

conjecture | open | MathDB 页面 | 原文线索

V2-075359 | MATHDB-394915

Real-rooted coefficient vectors and the g-theorem

摘录 / EXCERPT: Let $h(t)=h_0+h_1t+\3+h_dt^d\in\mathbb{N}[t]$ be a polynomial of degree d with $h_0=1$ and $h_i\ge 1$ for $1\le i\le d$. If $h(t)$ has only real roots, then $(h_0,\ldots,h_d)$ sa...

open_problem | open | MathDB 页面 | 原文线索

V2-075360 | MATHDB-394916

Real-rootedness of barycentric subdivisions of polytope boundaries

摘录 / EXCERPT: Conjecture 6. If ∂P is the boundary complex of a (not necessarily simplicial) d-polytope P, then the h-polynomial of $\operatorname{sd}(\partial P)$ has only real r...

open_problem | open | MathDB 页面 | 原文线索

V2-075361 | MATHDB-394917

Kalai’s Conjecture A for centrally symmetric polytopes

摘录 / EXCERPT: For every centrally symmetric convex polytope P of dimension d, with $f_i(P)$ denoting its number of i-dimensional faces, one has ... The report states that this conjecture is...

conjecture | open | MathDB 页面 | 原文线索

V2-075362 | MATHDB-394918

Kalai’s 3^d-conjecture for centrally symmetric polytopes

摘录 / EXCERPT: For every centrally symmetric convex d-polytope P with $d\ge 5$, the total number of nonempty faces satisfies $\displaystyle\sum_{i=0}^{d} f_i(P)\ge 3^d$.

conjecture | open | MathDB 页面 | 原文线索

V2-075363 | MATHDB-394919

Consistency of a strict inequality between almost-disjointness numbers

摘录 / EXCERPT: Is $a_\kappa<a_{\operatorname{cf}(\kappa)}$ consistent?

open_problem | open | MathDB 页面 | 原文线索

V2-075364 | MATHDB-394920

Consistency of separating the almost-disjointness numbers at \(\aleph_\omega\) and \(\omega\)

摘录 / EXCERPT: Is $\mathfrak a_{\aleph_\omega}<\mathfrak a$ consistent with $\mathfrak a>\aleph_\omega$?

open_problem | open | MathDB 页面 | 原文线索

V2-075365 | MATHDB-394922

The strength of indestructible weak compactness

摘录 / EXCERPT: What exactly is the strength of indestructible weak compactness?

open_problem | open | MathDB 页面 | 原文线索

V2-075366 | MATHDB-394923

Does Martin’s Maximum++ imply Woodin’s axiom (*)?

摘录 / EXCERPT: Does Martin’s Maximum $^{++}$ imply $(*)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075367 | MATHDB-394925

Well-quasi-ordering of Aronszajn lines by embeddability

摘录 / EXCERPT: Conjecture 6 (PFA). The Aronszajn lines are well quasi-ordered by embeddability.

open_problem | open | MathDB 页面 | 原文线索

V2-075368 | MATHDB-394927

Higher-level separation of projective measurability and the Baire property

摘录 / EXCERPT: Can we prove a variant of Theorem 6, where the set without the Baire property is a $\[1m\Delta^1_{k+1}\u001b[0m$ set, and we have, in addition, the Baire property for all...

open_problem | open | MathDB 页面 | 原文线索

V2-075369 | MATHDB-394928

Universality of the quasi-isometry relation for finitely generated groups

摘录 / EXCERPT: The quasi-isometry relation $\approx_{QI}$ on the space $\mathcal G$ of finitely generated groups is a universal K_σ equivalence relation.

open_problem | open | MathDB 页面 | 原文线索

V2-075370 | MATHDB-394929

Stationary non-approachable sets at higher cofinalities

摘录 / EXCERPT: Is it consistent that there is a stationary set on $\aleph_{\omega+1}$ concentrating on cofinalities larger than $\aleph_1$ and not in $I[\aleph_{\omega+1}]$?

open_problem | open | MathDB 页面 | 原文线索

V2-075371 | MATHDB-394931

Construction of RGD-systems without Levi decomposition

摘录 / EXCERPT: Construct new RGD-systems for which condition (8) in Theorem 1 fails (and hence, in particular, which do not satisfy (DRJ1){lin}); equivalently, construct new Moufang twin building...

open_problem | open | MathDB 页面 | 原文线索

V2-075372 | MATHDB-394935

Fundamental group of a semistable period domain

摘录 / EXCERPT: Let G be absolutely simple adjoint over a finite field k, let $\mathcal N$ be a conjugacy class of $\mathbb Q$-one-parameter subgroups, let ℓ be the absolute rank of G,...

open_problem | open | MathDB 页面 | 原文线索

V2-075373 | MATHDB-394936

Linear divergence of lattices in higher-rank semisimple Lie groups

摘录 / EXCERPT: We conjecture that lattices in semi-simple Lie groups of higher rank always have linear divergence.

open_problem | open | MathDB 页面 | 原文线索

V2-075374 | MATHDB-394937

Centre Conjecture for convex chamber subcomplexes of buildings of types E_6, E_7, and E_8

摘录 / EXCERPT: For a spherical building Δ of type E_6, E_7, or E_8, and a convex chamber subcomplex $\Omega\subseteq\Delta$, either every residue R in Ω has a residue R' i...

conjecture | open | MathDB 页面 | 原文线索

V2-075375 | MATHDB-394938

Quasi-isometric embedding of the hyperbolic plane into an affine building of type $\widetilde{A}_2$

摘录 / EXCERPT: Let X be a given affine building of type $\widetilde{A}_2$. Is it possible to construct a quasi-isometric embedding $\mathbb{H}^2 \to X$?

open_problem | open | MathDB 页面 | 原文线索

V2-075376 | MATHDB-394939

Infinite-horizon risk-sensitive portfolio optimization with a floor

摘录 / EXCERPT: For the infinite-horizon problem with initial capital $x>0$, risk parameter $\gamma\in(-\infty,1)$, and floor constraint $X^{x,\pi}\ge K$, the report defines...

open_problem | open | MathDB 页面 | 原文线索

V2-075377 | MATHDB-394940

Finite-horizon optimal portfolio liquidation for CARA investors

摘录 / EXCERPT: Given an initial position $x>0$ and a finite or infinite horizon $T\in[0,\infty]$, let $\mathcal X(x,T)$ be the class of bounded, predictable, absolutely continuous liquidation str...

open_problem | open | MathDB 页面 | 原文线索

V2-075378 | MATHDB-394941

Risk-sharing optimization for divergence utilities

摘录 / EXCERPT: Suppose there are N agents, and agent n assesses risk by the risk measure $\rho_{G_n}(X)=-u_{G_n}(X)$. If X is the total endowment, the risk-sharing problem is ... The sum is u...

open_problem | open | MathDB 页面 | 原文线索

V2-075379 | MATHDB-394942

Uniqueness of the local descent to the quasi-split even orthogonal group

摘录 / EXCERPT: Let F be a p-adic field and let τ be an irreducible, supercuspidal representation of $\mathrm{GL}_{2n}(F)$ such that $L(\tau,\operatorname{sym}^2,s)$ has a pole at $s=0$....

open_problem | open | MathDB 页面 | 原文线索

V2-075380 | MATHDB-394951

Mœglin’s conjectural criterion for noncuspidality of square-integrable automorphic representations

摘录 / EXCERPT: Let G be a classical group over a number field k, and assume the Arthur description of its square-integrable automorphic forms using ordinary and twisted endoscopy. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-075381 | MATHDB-394953

Dimension of the p-rank-zero locus in the Siegel moduli space with Iwahori level structure

摘录 / EXCERPT: For the Siegel moduli space $\mathcal A_{g,I}$ with Iwahori level structure at p, the dimension of its p-rank-0 locus is conjectured to be … for every integer $gge 1.

open_problem | open | MathDB 页面 | 原文线索

V2-075382 | MATHDB-394954

Universality for every Faber-expansion set versus every Taylor-series center

摘录 / EXCERPT: Let $K\subset\mathbb C$ be compact and connected, with connected complement, and let $\Omega=\mathbb C\setminus K$. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-075383 | MATHDB-394956

Strict cyclicity of the Volterra operator on $L^p[0,1]$

摘录 / EXCERPT: Question: Is V strictly cyclic on $L^p[0,1]$?

open_problem | open | MathDB 页面 | 原文线索

V2-075384 | MATHDB-394957

Characterization of the commutant of the Volterra operator

摘录 / EXCERPT: The problem of characterizing the commutant of the Volterra operator V remains open for $p\ne 1,2,\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-075385 | MATHDB-394958

Preservation of universal Taylor series under multiplication by $(z-a)$

摘录 / EXCERPT: For $a\that belongs to \mathbb{C}$ with $|a|\ge 1$ and $f\in U(\mathbb{D},0)$, is it true that $(z-a)f(z)\in U(\mathbb{D},0)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075386 | MATHDB-394959

Derivative preservation for Nestoridis universal Taylor series

摘录 / EXCERPT: Let $f \in U(\mathbb{D},0)$. Is it true that $f' \in U(\mathbb{D},0)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075387 | MATHDB-394960

Polynomial multiplication of entire functions universal with respect to derivatives

摘录 / EXCERPT: Let $f\in H_C(D)$ and let p be a non-constant polynomial. Is it true that $pf\in H_C(D)$? Here $H_C(D)$ denotes the class of entire functions whose sequence of derivatives is den...

open_problem | open | MathDB 页面 | 原文线索

V2-075388 | MATHDB-394961

Characterizing bounded universality for self-maps of non-simply connected domains

摘录 / EXCERPT: Characterize, for a non-simply connected planar domain Ω, those sequences (ϕ_n) of self-maps of Ω for which there exists a B-universal function, where...

open_problem | open | MathDB 页面 | 原文线索

V2-075389 | MATHDB-394962

Uniform approximation of inner functions by interpolating Blaschke products

摘录 / EXCERPT: For any inner function B and $\\varepsilon>0$, does there exist an interpolating Blaschke product I such that $\|B-I\|_{\infty}<\varepsilon$?

open_problem | open | MathDB 页面 | 原文线索

V2-075390 | MATHDB-394963

Approximation of the argument of an inner function by an interpolating Blaschke product

摘录 / EXCERPT: Let B be an inner function. Given $\varepsilon>0$, is there an interpolating Blaschke product I such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-075391 | MATHDB-394964

Hypercyclicity of compositions of affine-derivative operators

摘录 / EXCERPT: Let $\lambda,b,\mu,c\in\mathbb{C}$ with $|\lambda|\ge 1$ and $|\mu|\ge 1$, and define $T_{\lambda,b}:H(\mathbb{C})\to H(\mathbb{C})$ by ... Is the composition of the two operators...

open_problem | open | MathDB 页面 | 原文线索

V2-075392 | MATHDB-394965

Analytic continuation of universal Taylor series on the complement of a disk

摘录 / EXCERPT: Let $D(\zeta_1,r)$, $r>0$, be a closed disk in the complex plane and let $\Omega=\mathbb{C}\setminus D(\zeta_1,r)$. Fix any point $\zeta_0\in\Omega$. It is known that there exist f...

open_problem | open | MathDB 页面 | 原文线索

V2-075393 | MATHDB-394966

Dependence modeling in valuation portfolios for non-life insurance run-offs

摘录 / EXCERPT: How should dependence be modeled between different risks and different accident years in the construction of the valuation portfolio for a non-life insurance run-off?

open_problem | open | MathDB 页面 | 原文线索

V2-075394 | MATHDB-394967

Global attraction of finite-energy solutions to the solitary-wave set

摘录 / EXCERPT: A natural hypothesis is that the set of all solitary waves forms a global attractor for all finite-energy solutions.

open_problem | open | MathDB 页面 | 原文线索

V2-075395 | MATHDB-394968

Stringent linear stability as a necessary condition for orbital stability of NLS standing waves

摘录 / EXCERPT: For the nonlinear Schrödinger equation $i u_t+\c1 u+\b2(|u|^2)u=0$ on $\mathbb{R}\times\mathbb{R}^3$, with $\beta(0)=0$ and β sufficiently regular with not too fast growth at...

open_problem | open | MathDB 页面 | 原文线索

V2-075396 | MATHDB-394969

Massless-photon and point-charge limit of the nonrelativistic QED g-factor formula

摘录 / EXCERPT: Does (14) have a finite limit when $m_{\mathrm{ph}}\to 0$ and $\widehat{\varphi}(k)\to (2\pi)^{-3/2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-075397 | MATHDB-394970

Kac’s conjecture on the constant term of the counting polynomial

摘录 / EXCERPT: Kac’s conjecture asserts that the constant term of the polynomial counting absolutely indecomposable representations over a finite field is the corresponding root multiplicity.

conjecture | open | MathDB 页面 | 原文线索

V2-075398 | MATHDB-394972

Integrality of the wall-crossing factorization exponents

摘录 / EXCERPT: On the formal power-series ring $\mathbb{Q}[[x,y]]$ with Poisson bracket $\{x,y\}=xy$, define, for $(a,b)\neq(0,0)$, the Poisson automorphisms ... For the positive integer k occurr...

open_problem | open | MathDB 页面 | 原文线索

V2-075399 | MATHDB-394973

Multiplication formula for evaluation forms in the dual semicanonical basis

摘录 / EXCERPT: Theorem 1. For $X,Y\in\Lambda\text{-mod}_0$:

open_problem | open | MathDB 页面 | 原文线索

V2-075400 | MATHDB-394975

Do the constructed polynomial invariants generate all \\-invariants?

摘录 / EXCERPT: For a given n, the report conjectures that the polynomials F_i, $i=1,\\\ldots,\\\lfloor n/2\rfloor$, constructed there form a generating set of all $\\$-invariants.

open_problem | open | MathDB 页面 | 原文线索

V2-075401 | MATHDB-394981

Global BV existence for a hyperbolic phase-mixing flow under small weighted phase variation

摘录 / EXCERPT: For the system ... with $t>0$, $x\in\mathbb{R}$, $v>0$, and ... where a is smooth, $a(\lambda)>0$, and $a'(\lambda)>0$ for $\lambda\in[0,1]$, consider initial data ... Writing...

open_problem | open | MathDB 页面 | 原文线索

V2-075402 | MATHDB-394982

Selection criterion for collapsing and explosion waves in the Riemann problem

摘录 / EXCERPT: The numerical tests indicate that, when a pure phase is in contact with a mixture or the other pure phase, collapsing and explosion waves of the slowest speed should be preferred i...

open_problem | open | MathDB 页面 | 原文线索

V2-075403 | MATHDB-394983

Stabilization of thin-film traveling waves with larger surfactant mass

摘录 / EXCERPT: Can the waves be stabilized if there is more surfactant present, i.e., with larger values of m?

open_problem | open | MathDB 页面 | 原文线索

V2-075404 | MATHDB-394984

Consistency of the estimated number of change points in AutoPARM

摘录 / EXCERPT: It remains to show that the estimate $\hat m$ of the true number of change points m, obtained by minimizing the MDL criterion also with respect to m, is consistent in the gener...

open_problem | open | MathDB 页面 | 原文线索

V2-075405 | MATHDB-394985

Lipschitz regularity of optimal controls for pointwise state-constrained elliptic control problems

摘录 / EXCERPT: For the problem (P) described in the report, does the optimal control necessarily have the Lipschitz regularity known in the case where the optimal state is active at only finite...

open_problem | open | MathDB 页面 | 原文线索

V2-075406 | MATHDB-394986

Optimal Switching Boundary Control

摘录 / EXCERPT: For given data $y_0\in H^1(0,1)$, $y_1\in L^2(0,1)$, and $T\ge 2$, problem S is to minimize ... subject to ... ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-075407 | MATHDB-394987

Minimax feedback control for a constrained uncertain parabolic system

摘录 / EXCERPT: Given the parabolic system ... where A is the self-adjoint, uniformly strongly elliptic operator ... with $\Omega\subset\mathbb R^n$ open and bounded, $T>0$, and admissible controls...

open_problem | open | MathDB 页面 | 原文线索

V2-075408 | MATHDB-394988

Optimal control of conductive-radiative temperature fields generated via electromagnetic heating

摘录 / EXCERPT: Establish theoretical results-such as first-order necessary optimality conditions-for the full coupled optimal-control system consisting of the conductive-radiative heat equations,...

open_problem | open | MathDB 页面 | 原文线索

V2-075409 | MATHDB-394989

Unique continuation for a biharmonic problem near a corner

摘录 / EXCERPT: Let $G=\{(r,\theta):0<r<r_0,\ 0<\theta<\theta_0\}$ be a circular sector, and let $\Omega\subset\mathbb{R}^2$ be a Lipschitz bounded open set with a corner at the origin such that...

open_problem | open | MathDB 页面 | 原文线索

V2-075410 | MATHDB-394990

Error estimate for virtual-control regularization and finite-element discretization

摘录 / EXCERPT: The report states that, for the virtual-control regularization and conforming finite-element semidiscretization described there, the estimate ... holds, where h is the finite-eleme...

open_problem | open | MathDB 页面 | 原文线索

V2-075411 | MATHDB-394991

Nonexistence of Selberg-class elements of degree between 1 and 2

摘录 / EXCERPT: There is no member of the Selberg Class with degree in the open interval $(1,2)$.

open_problem | open | MathDB 页面 | 原文线索

V2-075412 | MATHDB-394992

Gelfond’s conjecture for the binary sum-of-digits function at primes

摘录 / EXCERPT: Let $w(n)$ be the number of 1’s in the binary expansion of the nonnegative integer n. Then, as $x\to\infty$, ... where the sum is over primes p and $\pi(x)$ denotes the number...

conjecture | open | MathDB 页面 | 原文线索

V2-075413 | MATHDB-394994

Manin asymptotic for a specific nonsingular quartic del Pezzo surface

摘录 / EXCERPT: Let $X\subset\mathbb{P}^4$ be the nonsingular surface defined by ... let $U\subset X$ be the Zariski-open subset obtained by deleting the 16 lines from X, and let H be the heig...

open_problem | open | MathDB 页面 | 原文线索

V2-075414 | MATHDB-394995

Typical depth of Pratt trees

摘录 / EXCERPT: For almost all primes p, the depth $D(p)$ of the Pratt tree with root p has order $\\log\\log p$.

open_problem | open | MathDB 页面 | 原文线索

V2-075415 | MATHDB-394996

Local injectivity of the Carmichael \-function

摘录 / EXCERPT: For every positive integer m, there is an integer $n\ne m$ such that $\lambda(n)=\lambda(m)$.

open_problem | open | MathDB 页面 | 原文线索

V2-075416 | MATHDB-394998

Uniform $O(N^{-1})$ rate in the multivariate CLT for quadratic forms

摘录 / EXCERPT: Let $X,X_1,X_2,\dots$ be i.i.d. random vectors in $\mathbb R^d$ with $\mathbf E X=0$ and $\beta_4=\mathbf E\|X|^4<\infty$. Let $S_N=N^{-1/2}(X_1+\cdots+X_N)$ converge weakly to a n...

open_problem | open | MathDB 页面 | 原文线索

V2-075417 | MATHDB-394999

Selberg’s asymptotic conjecture for the sifting limit

摘录 / EXCERPT: Let R be a positive integer, and consider the ideal-sieve model in which P is a finite set of primes, f is a non-negative multiplicative function, every $p\in P$ satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-075418 | MATHDB-395000

Growth of product sets in arbitrary groups

摘录 / EXCERPT: In general, let G be a group and let $A\subset G$. How large is $A\cdot A\cdot A$ compared to A, and why?

open_problem | open | MathDB 页面 | 原文线索

V2-075419 | MATHDB-395001

Local solubility of quartic forms versus integral solubility

摘录 / EXCERPT: Is the following assertion true for quartic forms? If a quartic form $F\in\mathbb{Z}[x_1,\ldots,x_n]$ has non-singular zeros over $\mathbb{R}$ and over every $\mathbb{Q}_p$, then...

open_problem | open | MathDB 页面 | 原文线索

V2-075420 | MATHDB-395002

Soundararajan’s conjecture on the horizontal distribution of zeros of \(\zeta'(s)\)

摘录 / EXCERPT: Let $N_1(T)$ be the number of non-trivial zeros of $\zeta'(s)$ up to height T. Define ... and define $m_-(x)$ similarly, with $\liminf$ in place of $\limsup$. Soundararajan conject...

conjecture | open | MathDB 页面 | 原文线索

V2-075421 | MATHDB-395003

Counting lattice-point configurations by weight

摘录 / EXCERPT: How many different configurations $J(r,P)$ have $N(r,P)=n$ (there are $K(n)$ such configurations, say), or $N(r,P)\le N$ (there are $M(N)$ such configurations, say)? As in Problem...

open_problem | open | MathDB 页面 | 原文线索

V2-075422 | MATHDB-395005

Conjectured optimal order of the primitive of Hardy’s function

摘录 / EXCERPT: Let $Z(t)=\chi^{-1/2}(\tfrac12+it)\zeta(\tfrac12+it)$ be Hardy’s function and let $F(T)=\int_0^T Z(t)\,dt$. Ivić conjectured that the estimate $F(T)\ll T^{1/4+\varepsilon}$ for eve...

conjecture | open | MathDB 页面 | 原文线索

V2-075423 | MATHDB-395007

A converse theorem for Dirichlet L-functions

摘录 / EXCERPT: For a primitive character $\chi\pmod q$, let $W(\chi)$ be the set of Dirichlet series $F(s)$ satisfying: 1. The coefficients $a(n)$ of $F(s)$ satisfy $a(n)\ll n^{\varepsilon}$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-075424 | MATHDB-395008

Bounded Gap Conjecture for primes

摘录 / EXCERPT: There exists an absolute constant C such that $p_{n+1}-p_n\le C$ for infinitely many n.

conjecture | open | MathDB 页面 | 原文线索

V2-075425 | MATHDB-395009

Positive density of prescribed digit sums of squares

摘录 / EXCERPT: Let $q\ge 2$ be an integer, and write $\sigma(n)$ for the sum of the base-q digits of n. For every integer $m\ge 2$ with $(m,q-1)=1$, there exist constants $C(q,m)>0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075426 | MATHDB-395011

Gao-Geroldinger conjecture on zero-sum-free sequences in $\mathbb{Z}_n^2$

摘录 / EXCERPT: Every zero-sum-free subsequence A of $\mathbb{Z}_n^2$ of length $2n-2$ contains an element with multiplicity at least $n-2$.

conjecture | open | MathDB 页面 | 原文线索

V2-075427 | MATHDB-395014

Soundararajan’s equivalence conjecture for zeros of \(\zeta\) and \(\zeta'\)

摘录 / EXCERPT: Assume the Riemann Hypothesis. Soundararajan conjectured that the following two assertions are equivalent: ... where $\beta'+i\gamma'$ ranges over zeros of $\zeta'(s)$, and ... where...

conjecture | open | MathDB 页面 | 原文线索

V2-075428 | MATHDB-395015

Contracting for optimal investment under risk constraints

摘录 / EXCERPT: The talk claims that, in a simple complete-market setting, a principal constrained by a law-invariant coherent risk measure on terminal wealth can construct a contract with an agen...

open_problem | open | MathDB 页面 | 原文线索

V2-075429 | MATHDB-395016

Asymptotic properties of maximum-likelihood estimators for nonlinear time-varying-parameter GARCH models

摘录 / EXCERPT: Determine the asymptotic properties of the maximum-likelihood estimators of the nonlinear (time-varying-parameter) GARCH models introduced in the report, comprising the additive an...

open_problem | open | MathDB 页面 | 原文线索

V2-075430 | MATHDB-395017

Maximum antichain size in the divisor poset of an integer

摘录 / EXCERPT: Let $p_1,\ldots,p_N$ be distinct primes and let $M=p_1^{k_1}\cdots p_N^{k_N}$, where $k_i\ge 1$. Identify the set D of divisors of M with $\prod_{i=1}^N\{0,1,\ldots,k_i\}$, ord...

open_problem | open | MathDB 页面 | 原文线索

V2-075431 | MATHDB-395018

Critical almost Mathieu spectrum Hausdorff-dimension conjecture

摘录 / EXCERPT: For every $\[?1m\alpha\u001b[0m \in \mathbb{R}\setminus\mathbb{Q}$, does the critical almost Mathieu spectrum satisfy ...

conjecture | open | MathDB 页面 | 原文线索

V2-075432 | MATHDB-395019

Nonuniform Upper Estimates of the Density of States in Dimension One

摘录 / EXCERPT: For the one-dimensional Anderson model $H_\omega=-\Delta+V_\omega$ on $\ell^2(\mathbb Z)$, where $V_\omega(n)$ are i.i.d. with uniform distribution...

open_problem | open | MathDB 页面 | 原文线索

V2-075433 | MATHDB-395020

Individual unfolded eigenvalue statistics for the Anderson model

摘录 / EXCERPT: Under the assumptions of the preceding section, let N be the integrated density of states, with $N\in C^1$ and $n(E):=N'(E)>0$ on $(\inf\Sigma,\sup\Sigma)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075434 | MATHDB-395021

Almost-sure Poisson statistics for globally unfolded Anderson eigenvalues

摘录 / EXCERPT: Under the assumptions of the preceding section-namely, the Anderson tight-binding model $H_\omega=-\Delta+V_\omega$ on $\mathbb Z^d$, with i.i.d. random variables $V_x(\omega)$ hav...

open_problem | open | MathDB 页面 | 原文线索

V2-075435 | MATHDB-395024

Asymptotic eigenvalue statistics for real spiked populations

摘录 / EXCERPT: Analyze the statistics of the eigenvalues $l_1,l_2,\ldots$ of the sample matrix $S=\frac{1}{n}\widehat M\widehat M^T$ as $p,n\to\infty$ with $p/n\to\gamma\in(0,\infty)$ for spiked...

open_problem | open | MathDB 页面 | 原文线索

V2-075436 | MATHDB-395025

Lifshitz tails for non-monotonous alloy-type random Schrödinger operators

摘录 / EXCERPT: Unless the distribution is Bernoulli and $E_{-}(a)=E_{-}(b)$, we have Lifshitz tails at the bottom of the spectrum.

open_problem | open | MathDB 页面 | 原文线索

V2-075437 | MATHDB-395026

Continuity of filled-in Julia sets and Julia sets under polynomial perturbation

摘录 / EXCERPT: For a complex polynomial f of degree $d\ge 2$ and a sequence of complex polynomials f_n of the same degree with $f_n\to f$, determine whether, and in what sense,...

open_problem | open | MathDB 页面 | 原文线索

V2-075438 | MATHDB-395027

Necessity of the Bryuno-Rüssmann condition for rational maps

摘录 / EXCERPT: Let b1 b (0,1) be irrational and set bb = e^{2c0 ib1}. Is the Bryuno-Rüssmann condition ... where p_n/q_n are the continued-fraction convergents of α, necessary for a ration...

open_problem | open | MathDB 页面 | 原文线索

V2-075439 | MATHDB-395028

Measurable Riemann Mapping Theorem

摘录 / EXCERPT: Let $U\subset\mathbb{R}^2$ be open and let $\mu\in L^\infty(U)$ satisfy $\|\mu\|_\infty<1$. Then there exists a quasiconformal homeomorphism $f:U\to f(U)$ such that ... almost everyw...

open_problem | open | MathDB 页面 | 原文线索

V2-075440 | MATHDB-395030

Non-collapse of the bounded-arithmetic hierarchy

摘录 / EXCERPT: Show that the full theory $T_2:=\bigcup_i T_i^2$ does not collapse to some finite level T_j^2.

open_problem | open | MathDB 页面 | 原文线索

V2-075441 | MATHDB-395031

Recursive acceptability of the theory T_2

摘录 / EXCERPT: Let T_2 be the Brouwerian theory extending FIM constructed in the preceding theorem, satisfying properties (i)-(iii) stated on the source page. Is T_2 recursively acceptable, i...

open_problem | open | MathDB 页面 | 原文线索

V2-075442 | MATHDB-395032

Mapping properties and spectral theory of the strongly singular VIE operator

摘录 / EXCERPT: Determine more general mapping properties of the strongly singular integral operator in the volume integral equation (E2) on $L^2(\Omega_-)$ or $H(\operatorname{div},\Omega_-)$, in...

open_problem | open | MathDB 页面 | 原文线索

V2-075443 | MATHDB-395033

Discrete local reliability for adaptive Galerkin BEM

摘录 / EXCERPT: Can one prove a discrete local reliability estimate for the residual error estimator used to steer adaptive Galerkin boundary element methods, sufficient to establish optimal conve...

open_problem | open | MathDB 页面 | 原文线索

V2-075444 | MATHDB-395034

Uniqueness of two Majorana representations of A_5

摘录 / EXCERPT: The alternating group A_5 of degree five possesses exactly two Majorana representations, of dimensions 26 and 21, corresponding respectively to $(2A,3A,5A)$- and $(2A,3C,5A)$...

open_problem | open | MathDB 页面 | 原文线索

V2-075445 | MATHDB-395039

Closed verbal subgroups in finitely generated prosoluble groups

摘录 / EXCERPT: The words $w\in F$ such that $w(G)$ is closed in all finitely generated prosoluble groups are precisely those outside the union of $F''(F')^p$ over all primes p.

open_problem | open | MathDB 页面 | 原文线索

V2-075446 | MATHDB-395042

Codistance-to-twinning theorem for 3-spherical buildings

摘录 / EXCERPT: Let $B^-=(C^-,\delta^-)$ be a thick building of 3-spherical type (W,S). Assume the following two conditions hold: - (lco) If R is a rank-2 residue containing a chamber c, t...

open_problem | open | MathDB 页面 | 原文线索

V2-075447 | MATHDB-395043

Does a codistance on a 2-spherical building imply a twinning?

摘录 / EXCERPT: For every 2-spherical building $\mathcal B$, does the existence of a codistance on $\mathcal B$ imply that $\mathcal B$ is one half of a twin building (equivalently, that the codis...

open_problem | open | MathDB 页面 | 原文线索

V2-075448 | MATHDB-395044

Γ-limit for atomistic ultra-thin films

摘录 / EXCERPT: Under the stated assumptions on the frame-indifferent atomistic energies $E^{(k)}$, including the decay estimate with some $q>3$ and the uniform bound on the atomic derivatives, th...

open_problem | open | MathDB 页面 | 原文线索

V2-075449 | MATHDB-395046

Γ-limit of a vector-valued dislocation phase-field energy

摘录 / EXCERPT: The report answers affirmatively the question of whether the BV-relaxation of the one-dimensional line-tension energy is the Γ-limit. Suppose that ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-075450 | MATHDB-395048

Polyak-Viro formula for coefficients of the Conway polynomial

摘录 / EXCERPT: For every integer $n\ge 1$, if K is a knot with Gauss diagram G, then the coefficient c_{2n} of t^{2n} in the Conway polynomial $C(K)$ is ... where $\mathfrak{C}_{2n}$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-075451 | MATHDB-395049

Determining the virtual crossing number

摘录 / EXCERPT: Determine the virtual crossing number $v(L)$ of a virtual knot or link L, where $v(L)$ is the minimum number of virtual crossings among all diagrams equivalent to L.

open_problem | open | MathDB 页面 | 原文线索

V2-075452 | MATHDB-395050

Virtual knots undetected by the linear biquandle

摘录 / EXCERPT: Does there exist a non-trivial virtual knot that cannot be detected by the linear biquandle? A proposed candidate is the closure of the virtual braid commutator...

open_problem | open | MathDB 页面 | 原文线索

V2-075453 | MATHDB-395052

Span bound for the Kauffman bracket of a virtual diagram

摘录 / EXCERPT: For a non-split connected virtual diagram K, one has ... where n is the number of classical crossings of K and g is the genus of the corresponding atom.

open_problem | open | MathDB 页面 | 原文线索

V2-075454 | MATHDB-395053

TQFT representations and pseudo-Anosov pieces

摘录 / EXCERPT: Let Σ be a compact orientable surface with negative Euler characteristic, and let ρ_k be the TQFT representation of the mapping class group of Σ at level k (fo...

open_problem | open | MathDB 页面 | 原文线索

V2-075455 | MATHDB-395057

Finite determination of the \mathfrak{g}-Witten\\Reshetikhin\\Turaev function

摘录 / EXCERPT: For every $\mathfrak{g}$ and knot K, $J_{\mathfrak{g},K}$ is uniquely determined among W-equivariant functions (that is, functions satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-075456 | MATHDB-395058

Residual error estimator for geometrically nonlinear large-strain elasticity

摘录 / EXCERPT: The report presents the residual-type estimator ... where ... with element residual $r_T=\operatorname{Div}T_1+f$ and edge residuals $r_E=[n\mathbin{\cdot}T_1]$, using the first Piola-...

open_problem | open | MathDB 页面 | 原文线索

V2-075457 | MATHDB-395059

Moderate-deviation asymptotics for chaotic analytic zero points in an intermediate regime

摘录 / EXCERPT: Let $\psi(z)=\sum_{k=0}^{\infty}\zeta_k z^k/\sqrt{k!}$, where the ζ_k are independent standard complex Gaussian variables, and let $Z_0(h_r)$ be the centered smooth linear st...

open_problem | open | MathDB 页面 | 原文线索

V2-075458 | MATHDB-395060

Typical number of nodal components of a random spherical harmonic

摘录 / EXCERPT: Let f be a random spherical harmonic of degree n on the 2-dimensional sphere, written as $f=\sum_{k=-n}^{n}\xi_kY_k$, where the ξ_k are independent identically distribute...

open_problem | open | MathDB 页面 | 原文线索

V2-075459 | MATHDB-395062

Coincidence of the Barlow-Bass and Kusuoka-Zhou Dirichlet forms

摘录 / EXCERPT: For the two local regular Dirichlet forms $(\[E_{\mathrm{BB}},\mathcal F\u001b]$ and $(\mathcal E_{\mathrm{KZ}},\mathcal F)$ constructed on the Sierpiński carpet F, do they coinc...

open_problem | open | MathDB 页面 | 原文线索

V2-075460 | MATHDB-395064

Almost-sure existence of the PASEP speed

摘录 / EXCERPT: For the PASEP with parameter $p\in(1/2,1)$, started from the configuration $Y_{0,k}=k$, let $X_{t,0}$ denote the position at time t of the particle of class 0. Is it true that...

open_problem | open | MathDB 页面 | 原文线索

V2-075461 | MATHDB-395068

Classification of $(d,4)$-nets

摘录 / EXCERPT: Conjecture. The only $(d,4)$-net is the Hesse arrangement of 12 lines.

open_problem | open | MathDB 页面 | 原文线索

V2-075462 | MATHDB-395072

Fundamental groups of smooth loci of log del Pezzo surfaces

摘录 / EXCERPT: For every group G appearing in step (2), determine whether there exists a pair $(\widetilde S,G)$ satisfying $\pi_1(\widetilde S^{\mathrm{sm}})=\{e\}$, where $(\widetilde S,G)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-075463 | MATHDB-395073

Geometric theta duality for moduli spaces of sheaves

摘录 / EXCERPT: Let S be a smooth complex projective surface with a generic polarization H. For $v\in K^{\mathrm{top}}(S)$, let M_v be the moduli space of Gieseker H-semistable sheaves of...

open_problem | open | MathDB 页面 | 原文线索

V2-075464 | MATHDB-395075

Equality with the normalization of the reduced coarse moduli space

摘录 / EXCERPT: Is $M_{1,n}(m)=\widetilde{M}_{1,n}(m)$ for all m, where $\widetilde{M}_{1,n}(m)$ is the normalization of the reduced space $M_{1,n}(m)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075465 | MATHDB-395081

Degree bound for canonical covers of singular surfaces of minimal degree

摘录 / EXCERPT: Let X be a surface of general type with at worst canonical singularities, with K_X ample and base-point-free, and let $\varphi:X\to W$ be the morphism induced by the complete l...

open_problem | open | MathDB 页面 | 原文线索

V2-075466 | MATHDB-395084

Global regular reflection for potential flow at the sonic-angle threshold

摘录 / EXCERPT: For the potential-flow shock-reflection Problem 1 with $\gamma>1$ and $\rho_1>\rho_0>0$, and for wedge angles $\theta_w\in(\theta_{\mathrm{sonic}},\pi/2)$, establish the expected g...

open_problem | open | MathDB 页面 | 原文线索

V2-075467 | MATHDB-395086

Global well-posedness for the cubic defocusing NLS in $H^s(\\bb R^2)$

摘录 / EXCERPT: For the cubic defocusing nonlinear Schrdinger equation ... it is conjectured that the Cauchy problem is globally well-posed in $H^s(\mathbb R^2)$ for every $s\ge 0$ (for arbitrarily...

open_problem | open | MathDB 页面 | 原文线索

V2-075468 | MATHDB-395087

Global well-posedness and scattering for the focusing mass-critical cubic NLS below the ground-state mass

摘录 / EXCERPT: For the focusing cubic nonlinear Schrödinger equation on $\mathbb{R}^2$, is the problem globally well-posed and does it scatter for L^2 initial data whose mass is less than the g...

open_problem | open | MathDB 页面 | 原文线索

V2-075469 | MATHDB-395088

Nonlinear stability of the KP-II line soliton

摘录 / EXCERPT: Establish a satisfactory nonlinear stability result for the line soliton of the Kadomtsev-Petviashvili II (KP-II) equation.

open_problem | open | MathDB 页面 | 原文线索

V2-075470 | MATHDB-395093

Improved uniqueness for the harmonic map heat flow in two dimensions

摘录 / EXCERPT: Let M be a closed Riemannian surface and let $N\subset\mathbb{R}^k$ be compact. For weak solutions $u\in H^1([0,T]\times M)$ of the harmonic map heat flow ... let ... and define ... If...

open_problem | open | MathDB 页面 | 原文线索

V2-075471 | MATHDB-395094

Topping’s conjecture on uniqueness for the two-dimensional harmonic map heat flow

摘录 / EXCERPT: Let M be a closed Riemannian surface and let $N\subset\mathbb{R}^k$ be compact. For weak solutions $u\in H^1([0,T]\times M)$ of the harmonic map heat flow ... where A is the seco...

conjecture | open | MathDB 页面 | 原文线索

V2-075472 | MATHDB-395097

Well-posedness for the Euler-Nordström system with cosmological constant

摘录 / EXCERPT: Theorem (Well-Posedness for EN_{κ}). Let $N\ge 3$ be an integer. Assume that the initial data $\mathring{V}$ for the EN_{κ} system are an...

open_problem | open | MathDB 页面 | 原文线索

V2-075473 | MATHDB-395098

Existence of blow-up solutions for energy-critical semilinear wave equations

摘录 / EXCERPT: For either of the following two models: 1. The co-rotational wave-map equation … with $Q(r)=2\arctan r$ and $Q_\lambda(r)=Q(\lambda r)$; or 2. The focusing quintic semilinear wave…

open_problem | open | MathDB 页面 | 原文线索

V2-075474 | MATHDB-395099

Almost-global well-posedness of the two-dimensional full water-wave system

摘录 / EXCERPT: Let the initial interface and initial velocity be … where f and g are smooth and rapidly decaying at infinity, and $\overline g=H_0\overline g$, with H_0 the Hilbert transfor…

open_problem | open | MathDB 页面 | 原文线索

V2-075475 | MATHDB-395100

Global-in-time behavior of small two-dimensional water waves

摘录 / EXCERPT: Determine the global-in-time behavior of solutions of the two-dimensional, infinite-depth water-wave system (1) for small initial data, where the fluid is incompressible and irrota…

open_problem | open | MathDB 页面 | 原文线索

V2-075476 | MATHDB-395101

Global-in-time well-posedness for the 2-D full water-wave problem

摘录 / EXCERPT: For the two-dimensional, infinite-depth, incompressible, irrotational water-wave system with zero surface tension and gravity field $(0,-1)$, does sufficiently small initial data o…

open_problem | open | MathDB 页面 | 原文线索

V2-075477 | MATHDB-395104

Pairwise-generating sets in maximal 2-generated direct powers of finite simple groups

摘录 / EXCERPT: Let S be a non-abelian finite simple group, and let n be the largest positive integer such that the direct power $G=S^n$ is 2-generated. Does there exist such an S for whic…

open_problem | open | MathDB 页面 | 原文线索

V2-075478 | MATHDB-395105

Universal clique-covering estimate for finite simple groups

摘录 / EXCERPT: Does there exist a universal constant $c\ge 1$ such that, for every finite simple group G, the estimate … holds, where $\omega(G)$ is the clique number of the generating graph,…

open_problem | open | MathDB 页面 | 原文线索

V2-075479 | MATHDB-395106

Ore's conjecture

摘录 / EXCERPT: Every element of every finite nonabelian simple group is a commutator.

conjecture | open | MathDB 页面 | 原文线索

V2-075480 | MATHDB-395107

Is property τ a profinite property?

摘录 / EXCERPT: There exist finitely generated groups Γ_1 and Γ_2 with $\widehat{\Gamma}_1 \cong \widehat{\Gamma}_2 \cong \prod_{n>3}\operatorname{Alt}(n)$ such that Γ_1 does…

open_problem | open | MathDB 页面 | 原文线索

V2-075481 | MATHDB-395108

Lubotzky-Zuk question on property (τ) and finite intersections

摘录 / EXCERPT: Let Γ be a group and let $\{H_n\mid n\ge 1\}$ be a family of finite-index subgroups of Γ having property (τ). Must the family … where $H^{g}=gHg^{-1}$, also have…

open_problem | open | MathDB 页面 | 原文线索

V2-075482 | MATHDB-395112

Polynomiality of the number of conjugacy classes of unitriangular groups

摘录 / EXCERPT: Does there exist a polynomial $t\in\mathbb{Z}[T]$ such that $h_{U_n(q)}=t(q)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075483 | MATHDB-395114

Triple-product covering in \(\operatorname{PSL}(n,q)\)

摘录 / EXCERPT: Proposition 1. Let $L=\operatorname{PSL}(n,q)$. If $B\subseteq L$ has size at least $\dfrac{2|L|}{q^{(n-1)/3}}$, then $B^3=L$.

open_problem | open | MathDB 页面 | 原文线索

V2-075484 | MATHDB-395115

Maximal subgroups of finitely generated branch groups

摘录 / EXCERPT: Is it true that a finitely generated branch group has maximal subgroups only of finite index?

open_problem | open | MathDB 页面 | 原文线索

V2-075485 | MATHDB-395117

Periodic points of the arithmetic trace-map reductions

摘录 / EXCERPT: For a sequence defining the map $\varphi(x,y)=[y^{-1}xy,x^{-1}]$ and its trace-map factor $\widetilde\psi_{\varphi}:\mathbb A^3_{s,u,t}\to\mathbb A^3_{s,u,t}$, determine conditions…

open_problem | open | MathDB 页面 | 原文线索

V2-075486 | MATHDB-395118

Meromorphic continuation of Euler products

摘录 / EXCERPT: Let $W(x,y)=\sum_{n,m}a_{n,m}x^ny^m$ be an integral polynomial with $W(x,0)=1$, and define … Conjecture: $D(s)$ is meromorphically continuable to the whole complex plane if and onl…

open_problem | open | MathDB 页面 | 原文线索

V2-075487 | MATHDB-395119

Finitely presented orbit-undecidable subgroups of automorphism groups of free groups

摘录 / EXCERPT: Does there exist a finitely presented subgroup $A \leqslant \operatorname{Aut}(F_n)$ that is orbit undecidable?

open_problem | open | MathDB 页面 | 原文线索

V2-075488 | MATHDB-395120

Euler-characteristic target enumeration

摘录 / EXCERPT: Given a finite collection of tame target supports $\{U_\alpha\}\subset X$ with uniform Euler characteristic $\chi(U_\alpha)=N\ne 0$ for every α, and counting function...

open_problem | open | MathDB 页面 | 原文线索

V2-075489 | MATHDB-395122

Conjecture 2: Gradient paths for non-face persistence pairs

摘录 / EXCERPT: If $\{\sigma,\tau\}\in P$ are such that σ is not a face of τ, then there is a single gradient path from τ to σ in the vector field V_P.

conjecture | open | MathDB 页面 | 原文线索

V2-075490 | MATHDB-395123

Realizing a persistence pairing by a discrete Morse gradient

摘录 / EXCERPT: Suppose P is a persistence pairing on K. Then with the correct choice of f, we have $P_{V_P,f}=P$.

open_problem | open | MathDB 页面 | 原文线索

V2-075491 | MATHDB-395124

Vanishing of a Stiefel-Whitney power for odd-cycle Hom-complexes

摘录 / EXCERPT: The conjectural claim that $w_1^{n-2}(\operatorname{Hom}(C_{2r+1},K_n))=0$ holds.

open_problem | open | MathDB 页面 | 原文线索

V2-075492 | MATHDB-395125

Universality of the long-time limit of grain-growth cell complexes

摘录 / EXCERPT: For generic initial conditions, the cell complex of grains evolves toward a universal random cell complex, defined up to a scaling factor, independent of the material. If a cell co...

open_problem | open | MathDB 页面 | 原文线索

V2-075493 | MATHDB-395127

Applicability of variance estimates for high-dimensional learning

摘录 / EXCERPT: How can the usual variance estimates in learning processes, which essentially involve the capacity of function classes and properties of function spaces from approximation theory,...

open_problem | open | MathDB 页面 | 原文线索

V2-075494 | MATHDB-395128

Proposition 1: Gradient outer product and inverse-regression covariance

摘录 / EXCERPT: Suppose the linear regression model $y=\beta^T x+\varepsilon$ holds with $E(\varepsilon)=0$. Let $\Omega_{X\mid Y}=\operatorname{cov}_Y\!\bigl(E_X(X\mid Y)\bigr)$ be the covariance...

open_problem | open | MathDB 页面 | 原文线索

V2-075495 | MATHDB-395129

A Poor Man’s Wilks Phenomenon

摘录 / EXCERPT: The report asks whether, beyond regular parametric maximum-likelihood models, the increment of a centered empirical process between a fixed point and a random point can exhibit a n...

open_problem | open | MathDB 页面 | 原文线索

V2-075496 | MATHDB-395130

A Wilks phenomenon for bounded contrast minimization

摘录 / EXCERPT: In settings beyond the classical parametric likelihood framework, does the increment of a centered empirical process between a fixed point and a random point exhibit a non-trivial...

open_problem | open | MathDB 页面 | 原文线索

V2-075497 | MATHDB-395132

Counterexample to De Giorgi’s symmetry conjecture in large dimensions

摘录 / EXCERPT: For the Allen-Cahn equation $\Delta u+u(1-u^2)=0$ in $\mathbb{R}^N$, there exist bounded solutions satisfying $\partial_{x_N}u>0$ whose level sets $\{u=\lambda\}$ are not hyperplan...

conjecture | open | MathDB 页面 | 原文线索

V2-075498 | MATHDB-395133

Ball’s conjecture on sufficient conditions for strong local minima

摘录 / EXCERPT: Ball conjectured that, for the classical variational problem described in the report, uniform quasiconvexity together with sufficient conditions for weak local minimality should be...

conjecture | open | MathDB 页面 | 原文线索

V2-075499 | MATHDB-395134

Counterexamples to the De Giorgi conjecture in dimensions $N\ge 9$

摘录 / EXCERPT: For every $N\ge 9$ and every sufficiently small $\alpha>0$, there exists a bounded solution $u_\alpha:\mathbb{R}^N\to\mathbb{R}$ of ... such that $u_\alpha(0)=0$,...

conjecture | open | MathDB 页面 | 原文线索

V2-075500 | MATHDB-395137

Uniqueness of the asymptotic foliation by surfaces of Willmore type

摘录 / EXCERPT: Assume that (M,g) satisfies the hypotheses of the existence theorem: it is (m,η,σ)-asymptotically Schwarzschild, satisfies $r^{-5}|\operatorname{Scal}|\le C'$, and has...

open_problem | open | MathDB 页面 | 原文线索

V2-075501 | MATHDB-395139

Maximum electron number for existence of Müller-functional minimizers

摘录 / EXCERPT: For the Müller density-matrix functional for a Coulomb system with M nuclei of charge $+Z$, is there a finite maximum electron number N for which the minimization problem over...

open_problem | open | MathDB 页面 | 原文线索

V2-075502 | MATHDB-395140

Optimal passenger boarding policy for an airplane

摘录 / EXCERPT: Given an airplane with known leg room and number of passengers per row, how should the airline control the queueing of passengers? Should it allow passengers from the back of the a...

open_problem | open | MathDB 页面 | 原文线索

V2-075503 | MATHDB-395142

Existence of minimizing Willmore surfaces of prescribed genus

摘录 / EXCERPT: For every $q\in\mathbb{N}_0$, there exists a smooth Willmore embedding $f:\Sigma\to\mathbb{R}^n$ with $\operatorname{genus}(\Sigma)=q$ and $W(f)=\beta_q$, where ...

open_problem | open | MathDB 页面 | 原文线索

V2-075504 | MATHDB-395143

Next-order vortex-pattern theorem for the Ginzburg-Landau model

摘录 / EXCERPT: Assume that ... and let $(u_\varepsilon,A_\varepsilon)$ be a minimizer of the Ginzburg-Landau energy G_ε. Choose x_ε at random in ω_λ, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075505 | MATHDB-395144

Weakening decay assumptions in symmetry and nonexistence theorems

摘录 / EXCERPT: We conjecture that the decay assumptions on u in Theorems 3 and 4 can be replaced by the weaker assumption $|\nabla u| \in L^2(\Sigma)$, where Σ is the unbounded radial do...

open_problem | open | MathDB 页面 | 原文线索

V2-075506 | MATHDB-395145

Does the orthogonal MTW condition imply the full MTW condition for squared Riemannian distance?

摘录 / EXCERPT: For the Riemannian distance-squared cost, does $\mathrm{MTW}^{\perp}\ge 0$ imply $\mathrm{MTW}\ge 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-075507 | MATHDB-395149

Furstenberg’s invariant-measure conjecture for the ×a, ×b action

摘录 / EXCERPT: Let a and b be multiplicatively independent integers, and let $t_a,t_b:\mathbb{R}/\mathbb{Z}\to\mathbb{R}/\mathbb{Z}$ be given by $t_a(x)=ax$ and $t_b(x)=bx$ modulo 1. Furste...

conjecture | open | MathDB 页面 | 原文线索

V2-075508 | MATHDB-395153

Global existence for semilinear wave equations under Hypothesis B

摘录 / EXCERPT: Theorem. Let $n=3$ or 4, and fix the exterior domain $\Omega\subset\mathbb{R}^n$ and boundary operator B as above. Assume that Hypothesis B is valid. Let $p=p_c$ be the positive ro...

conjecture | open | MathDB 页面 | 原文线索

V2-075509 | MATHDB-395154

Full-range uniform Fourier restriction estimates for polynomial curves

摘录 / EXCERPT: For curves $\Gamma(t)=(P_1(t),\ldots,P_d(t))$ whose coordinate functions are real polynomials, the report asks whether the uniform Fourier restriction estimate ... continues to hold...

open_problem | open | MathDB 页面 | 原文线索

V2-075510 | MATHDB-395155

Stein’s conjecture on the polynomial Carleson operator

摘录 / EXCERPT: For $d\in\mathbb N$, define ... where $\mathcal Q_d$ is the class of integer-(real-)coefficient polynomials Q with $\deg(Q)\le d$, and $f\in C^1(\mathbb T)$. Stein’s conjecture ass...

conjecture | open | MathDB 页面 | 原文线索

V2-075511 | MATHDB-395156

Existence of the wave operator in the low-exponent Euclidean NLS range

摘录 / EXCERPT: For the Euclidean analogue of the defocusing semilinear Schrödinger equation (1), does the wave operator exist in the range $\sigma\in(1/d,2/d]$?

open_problem | open | MathDB 页面 | 原文线索

V2-075512 | MATHDB-395158

Wolff’s inequality for the cone multiplier

摘录 / EXCERPT: For every integer $d\ge 2$, every $p>\frac{2(d+1)}{d-1}$, and every $\varepsilon>0$, does there exist a constant C_ε such that, for all sufficiently small $\delta>0$, e...

open_problem | open | MathDB 页面 | 原文线索

V2-075513 | MATHDB-395159

Mixed-norm Wolff inequality for cone plate decompositions

摘录 / EXCERPT: Let $d\ge 2$ and, for sufficiently small $\delta>0$, let ... Let $\{\omega_k\}\subset S^{d-1}$ be a maximal $\sqrt\delta$-separated sequence, and let ... where $c\approx1$. Is it true...

open_problem | open | MathDB 页面 | 原文线索

V2-075514 | MATHDB-395160

Conjectured optimal weight range for Bergman projections over light-cone tubes

摘录 / EXCERPT: Let $T=\mathbb{R}^{d+1}+i\Omega$ be the tube domain over the open light-cone, with Lorentz form $Q(y)=y_0^2-|y'|^2$. The weighted Bergman projections P_γ are conjectured to...

conjecture | open | MathDB 页面 | 原文线索

V2-075515 | MATHDB-395165

Sharp strong weighted estimate for the fractional integral operator

摘录 / EXCERPT: For $0<\alpha<n$ and $1<p<q<\infty$ satisfying $\frac1p-\frac1q=\frac\alpha n$, if $w\in A_{p,q}$, the report states that the corrected strong estimate is ...

open_problem | open | MathDB 页面 | 原文线索

V2-075516 | MATHDB-395166

Global existence for the defocusing L^2-critical nonlinear Schrödinger equation

摘录 / EXCERPT: For the defocusing case $\ba=-1$ of ... the corresponding maximal L^2 solution is global, i.e. its maximal forward and backward existence times satisfy $T^*=T_*=\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-075517 | MATHDB-395167

Conjectured multiplier characterization in the optimal radial range

摘录 / EXCERPT: Let $d>1$, let ϕ be a nontrivial C^∞ function supported in $(1,2)$, and let $m\in L^\infty([0,\infty))$. Define $K_t[m]=\mathcal F^{-1}[\phi(|\cdot|)m(t|\cdot|)]$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-075518 | MATHDB-395168

Radial Fourier multiplier characterization in the optimal range

摘录 / EXCERPT: The report conjectures that, for the radial-function operator norm, the equivalence ... holds throughout the same optimal range ... where φ is a nontrivial C^∞ function...

open_problem | open | MathDB 页面 | 原文线索

V2-075519 | MATHDB-395169

Mizohata-Takeuchi weighted restriction inequality for the sphere

摘录 / EXCERPT: The report asks whether, for the Fourier extension operator associated with the unit sphere, one always has ... where the supremum is over all 1-tubes T, i.e. unit neighborhoods...

open_problem | open | MathDB 页面 | 原文线索

V2-075520 | MATHDB-395170

Large sets with limited tube occupancy: determine admissible dimension pairs

摘录 / EXCERPT: Determine the set of pairs (β,γ) for which there exists a set $E\subseteq B$ with $0<\mathcal H^\beta(E)<\infty$ such that ... Here B is the unit ball in $\mathbb R^d$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075521 | MATHDB-395172

Polynomial Freiman-Ruzsa conjecture in characteristic 2

摘录 / EXCERPT: Let $A\subseteq\mathbb{F}_2^N$ (with N presumably very large) satisfy $|A+A|\leq K|A|$, where K is a relatively small constant. Then there is a subspace...

conjecture | open | MathDB 页面 | 原文线索

V2-075522 | MATHDB-395178

Equality between metric currents and locally flat chains

摘录 / EXCERPT: For an open set $U\subset\mathbb{R}^n$ with $n\ge 1$ and $m\ge 0$, is $C_m(\mathcal D_m(U))=\mathcal F^{\mathrm{loc}}_m(U)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075523 | MATHDB-395180

Hamilton’s conjecture on type-I Ricci-flow blow-up limits

摘录 / EXCERPT: Hamilton’s conjecture: rescaling limits around type-I singularities of the Ricci flow are gradient shrinking solitons.

conjecture | open | MathDB 页面 | 原文线索

V2-075524 | MATHDB-395181

Nonexistence of hyperbolic surface bundles over closed surfaces

摘录 / EXCERPT: Conjecture: There is no surface bundle $M\to S'$ with fiber S, where S' is a closed surface of genus $g'\ge 2$, such that the bundle has word-hyperbolic fundamental group...

open_problem | open | MathDB 页面 | 原文线索

V2-075525 | MATHDB-395186

Relation between the p-adic Borel regulator and the Soulé regulator

摘录 / EXCERPT: Let $K/\mathbb{Q}_p$ be a finite extension with ring of integers O_K. For $N\ge n\ge 2$, the Soulé regulator is given by the p-adic Borel regulator: under the isomorphism induc...

open_problem | open | MathDB 页面 | 原文线索

V2-075526 | MATHDB-395187

Global semistable reduction for overconvergent F-isocrystals

摘录 / EXCERPT: Let k be a field of characteristic $p>0$, and let K be a complete discretely valued field of characteristic 0 with residue field k, equipped with a continuous endomorphism....

open_problem | open | MathDB 页面 | 原文线索

V2-075527 | MATHDB-395190

Dominance of the moduli map from higher-genus pure-cycle Hurwitz spaces

摘录 / EXCERPT: Fix $r,g\ge 0$, $d\ge 1$, and $(e_1,\ldots,e_r)$ satisfying $2d-2-g=\sum_i(e_i-1)$. In characteristic p with $e_i<p$ for all i, and in characteristic 0, let H be the Hurwit...

open_problem | open | MathDB 页面 | 原文线索

V2-075528 | MATHDB-395196

Convergence of the Regge discretization of the curl-transpose-curl operator

摘录 / EXCERPT: The report states that it proves convergence of the eigenpairs of the $ abla\times^{\mathsf T}\nabla\times$ operator when the operator is discretized using Regge elements.

open_problem | open | MathDB 页面 | 原文线索

V2-075529 | MATHDB-395197

A posteriori error control for the discrete operator problem

摘录 / EXCERPT: What is the appropriate a-posteriori error control for the discrete operator problem?

open_problem | open | MathDB 页面 | 原文线索

V2-075530 | MATHDB-395199

Decomposition-based estimate for the adjoint approximation property

摘录 / EXCERPT: The report states that, under the hypotheses of its decomposition lemma, for $d\in\{1,2,3\}$ and every finite-dimensional approximation space $S\subset H^1(\Omega)$, the adjoint ap...

open_problem | open | MathDB 页面 | 原文线索

V2-075531 | MATHDB-395200

Convergence of nonconforming Maxwell eigensolvers

摘录 / EXCERPT: Let $0\le \lambda_1\le \lambda_2\le\cdots$ be the Maxwell eigenvalues, and let $\lambda=\lambda_j=\lambda_{j+1}=\cdots=\lambda_{j+m-1}$ be a Maxwell eigenvalue with multiplicity...

open_problem | open | MathDB 页面 | 原文线索

V2-075532 | MATHDB-395201

Normality relations in unitary free-group liberation

摘录 / EXCERPT: For a compact group $G\subset U_n$, determine whether the normality of the generators u_{ij} should be included among the defining relations R in the liberation construction...

open_problem | open | MathDB 页面 | 原文线索

V2-075533 | MATHDB-395204

Occurrence of uncountable proper fundamental groups of II_1 factors and equivalence relations

摘录 / EXCERPT: Can an uncountable subgroup of $\d53d_{+}$ different from $\d53d_{+}$ occur as the fundamental group of a $\d53d_{1}$ factor or of a $\d53d_{1}$ equivalence relation?

open_problem | open | MathDB 页面 | 原文线索

V2-075534 | MATHDB-395205

Permanence of the Bost conjecture under open subgroups

摘录 / EXCERPT: Let G be a locally compact second countable Hausdorff group. If the Bost conjecture is true for G and arbitrary separable C^*-coefficients, then the same is true for all open...

conjecture | open | MathDB 页面 | 原文线索

V2-075535 | MATHDB-395207

Invariance of the Bost conjecture under equivalent groupoids

摘录 / EXCERPT: Let G and H be locally compact Hausdorff groupoids admitting Haar systems and having $\c3$-compact unit spaces. If G and H are equivalent, then the Bost conjecture with arb...

conjecture | open | MathDB 页面 | 原文线索

V2-075536 | MATHDB-395211

Automatic triviality of continuous fields with fibre $\mathcal O_2$

摘录 / EXCERPT: Every separable unital continuous field with all fibres isomorphic to $\mathcal O_2$ over a finite-dimensional compact metrizable space X is trivial; that is, it is isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-075537 | MATHDB-395216

Horn’s characterization of eigenvalues of sums of Hermitian matrices

摘录 / EXCERPT: Let $A,B,C\in M_n(\mathbb C)^{\mathrm{s.a.}}$, and let ... be their eigenvalue sequences. Then these three sequences arise from Hermitian matrices satisfying $A+B+C=0$ if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-075538 | MATHDB-395217

Horn-type spectral-data problem for self-adjoint elements of a II$_1$ factor

摘录 / EXCERPT: Given a II$_1$-factor M with trace τ, characterize the possible spectral data of self-adjoint elements $a,b,c\in M$ satisfying $a+b+c=0$, when the spectral data of a and...

open_problem | open | MathDB 页面 | 原文线索

V2-075539 | MATHDB-395218

Horn inequalities in arbitrary finite factors

摘录 / EXCERPT: For every integer $n\ge 1$, every type II_1 factor M equipped with a trace τ normalized by $\tau(1)=n$, every self-adjoint $a,b,c\in M$ satisfying $a+b+c=0$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-075540 | MATHDB-395227

Property P for stable base loci of adjoint divisors

摘录 / EXCERPT: Fix a general small ample $\Q$-divisor A on X. Let $B=\sum_i B_i$ be a simple normal-crossings divisor and let $\mathcal B=\{\Theta=\sum_i b_iB_i\mid 0\le b_i\le 1\}$. Property...

open_problem | open | MathDB 页面 | 原文线索

V2-075541 | MATHDB-395228

Birational type of moduli spaces of polarised K3 surfaces

摘录 / EXCERPT: Determine the birational type of the moduli spaces of polarised K3 surfaces.

open_problem | open | MathDB 页面 | 原文线索

V2-075542 | MATHDB-395238

Existence of rational curves on minimal class VII surfaces

摘录 / EXCERPT: Every minimal compact complex surface X of class VII with $b_2(X)>0$ contains $b_2(X)$ rational curves.

open_problem | open | MathDB 页面 | 原文线索

V2-075543 | MATHDB-395240

Rogawski’s conjecture on the remaining fake projective-plane classes

摘录 / EXCERPT: There should be no fake projective planes of the type listed in Theorem 1(b).

conjecture | open | MathDB 页面 | 原文线索

V2-075544 | MATHDB-395241

Compactness of automorphism groups of relatively compact invariant domains

摘录 / EXCERPT: Let K be a connected compact Lie group and L a closed subgroup of K. Let $K^{\mathbb C}$ and $L^{\mathbb C}$ be the universal complexifications of K and L, respectively....

open_problem | open | MathDB 页面 | 原文线索

V2-075545 | MATHDB-395242

Identification of solid-fluid interaction terms in mixture theories of soft-tissue growth

摘录 / EXCERPT: How should mixture theory be properly used to identify the interaction terms between the solid and fluid phases of soft tissue?

open_problem | open | MathDB 页面 | 原文线索

V2-075546 | MATHDB-395243

Stability of coupled growth models with reaction terms

摘录 / EXCERPT: If uncontrolled growth is observed in coupled growth models solved by operator-splitting techniques, is it a result of instabilities inherent in the equations, or of a spurious nat...

open_problem | open | MathDB 页面 | 原文线索

V2-075547 | MATHDB-395244

Stabilization of chemical patterning by biophysical feedback

摘录 / EXCERPT: Could a chemical-based patterning mechanism be stabilized upon feedback with a biophysical mechanism?

open_problem | open | MathDB 页面 | 原文线索

V2-075548 | MATHDB-395245

Role of soft modes in growing shells

摘录 / EXCERPT: Might soft modes play a role in some growing shells, where spatially nonuniform stress states are to be expected?

open_problem | open | MathDB 页面 | 原文线索

V2-075549 | MATHDB-395246

Gross-Pitaevskii description of the ground state of a rotating trapped Bose gas

摘录 / EXCERPT: Whether the Gross-Pitaevskii equation correctly describes the ground state of a rotating, trapped Bose gas.

open_problem | open | MathDB 页面 | 原文线索

V2-075550 | MATHDB-395247

Matching lower bounds in the giant-vortex regime for rapidly rotating Bose gases

摘录 / EXCERPT: For the two-dimensional Gross-Pitaevskii energy in the fast-rotation Thomas-Fermi regime described in the report, when the angular velocity reaches the scale...

open_problem | open | MathDB 页面 | 原文线索

V2-075551 | MATHDB-395249

Complete delocalization of eigenvectors of random matrices

摘录 / EXCERPT: Are the eigenvectors of random matrices extended? More precisely, if $v=(v_1,\ldots,v_N)$ is an ℓ^2-normalized eigenvector, with $\|v\|=1$, is it completely delocalized in the...

open_problem | open | MathDB 页面 | 原文线索

V2-075552 | MATHDB-395250

String hypothesis for the Bethe Ansatz in the general multi-string case

摘录 / EXCERPT: For the Bethe Ansatz equations for the periodic isotropic Heisenberg chain, ... with ... and variables $\Lambda_j=\cot(k_j/2)$, resolve the general case of several strings: determine w...

conjecture | open | MathDB 页面 | 原文线索

V2-075553 | MATHDB-395251

String hypothesis for Bethe Ansatz solutions with several strings

摘录 / EXCERPT: Whether the string hypothesis holds in the general case of solutions of the Bethe Ansatz equations with several strings remains under investigation.

conjecture | open | MathDB 页面 | 原文线索

V2-075554 | MATHDB-395252

Validity of the Lee-Huang-Yang asymptotics throughout the full interaction-range regime

摘录 / EXCERPT: For the dilute Bose gas considered in Example (a), does the asymptotic formula ... hold for all $(\rho a^{3})^{1/2}\ll a/R$, rather than only in the narrower parameter ranges stated...

open_problem | open | MathDB 页面 | 原文线索

V2-075555 | MATHDB-395253

Occurrence and size of infinite cycles in the noninteracting spatial-permutation model

摘录 / EXCERPT: Let $\Lambda\subset\mathbb R^d$ be a cube of volume V, let $\rho_{n,n'}(\pi)=V^{-1}\#\{i: n\leq \ell_i(\pi)\leq n'\}$, and consider the noninteracting case $V\equiv0$. Assume tha...

open_problem | open | MathDB 页面 | 原文线索

V2-075556 | MATHDB-395255

Characterization of local minima of the mesoscale energies

摘录 / EXCERPT: Characterize the local minima of the functionals F_ε.

open_problem | open | MathDB 页面 | 原文线索

V2-075557 | MATHDB-395257

Besicovitch almost-periodic expansion of the rescaled spectral function

摘录 / EXCERPT: Let M be a compact Riemannian surface. Does there exist $p\ge 1$ such that, for all $x\in M$ and almost all $y\in M$, ... in the Besicovitch space B_p? Here $\Gamma_{x,y}$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-075558 | MATHDB-395263

Strong Tits Alternative

摘录 / EXCERPT: For every $d\in\mathbb{N}$ there is $N=N(d)\in\mathbb{N}$ such that, if K is any field and F is a finite symmetric subset of $GL_d(K)$ containing 1, then either the product s...

open_problem | open | MathDB 页面 | 原文线索

V2-075559 | MATHDB-395264

Uniform exponential growth for non-virtually-nilpotent subgroups of $\mathrm{GL}_2(\mathbb C)$

摘录 / EXCERPT: The report asks whether ρ_F is bounded away from 0 under the weaker assumption that a finite set $F\subset \mathrm{GL}_2(\mathbb C)$ containing 1 generates a non-virtually...

open_problem | open | MathDB 页面 | 原文线索

V2-075560 | MATHDB-395268

Borel nature of the self-similarity group of a measurable flow

摘录 / EXCERPT: For a measurable flow $T=(T_t)_{t\in\mathbb{R}}$, define $I(T)=\{s\in\mathbb{R}\setminus\{0\}:T\text{ and }T_s=(T_{st})_{t\in\mathbb{R}}\text{ are isomorphic}\}$. Is $I(T)$ a Borel...

open_problem | open | MathDB 页面 | 原文线索

V2-075561 | MATHDB-395269

Generic absence of self-similarity for measure-preserving flows

摘录 / EXCERPT: Is the absence of self-similarity generic in the set of measure-preserving flows?

open_problem | open | MathDB 页面 | 原文线索

V2-075562 | MATHDB-395270

Simplicity of the area-preserving homeomorphism group of the disc

摘录 / EXCERPT: Let $G=\operatorname{Homeo}(D^2,\partial D^2,\mathrm{Area})$ be the group of area-preserving homeomorphisms of the 2-disc that are the identity in some neighbourhood of the bound...

open_problem | open | MathDB 页面 | 原文线索

V2-075563 | MATHDB-395271

Finite width conjecture for the lower central series of a group acting on a building

摘录 / EXCERPT: Conjecture: Let G) be the group described in the report and let $G=\gamma_0(G)\geq\gamma_1(G)\geq\gamma_2(G)\geq\cdots$ denote its lower central series, with...

conjecture | open | MathDB 页面 | 原文线索

V2-075564 | MATHDB-395272

Isomorphism problem for hyperbolic groups

摘录 / EXCERPT: The isomorphy problem is solvable in the class of all hyperbolic groups.

open_problem | open | MathDB 页面 | 原文线索

V2-075565 | MATHDB-395274

Relative hyperbolicity of the group H

摘录 / EXCERPT: Is H not relatively hyperbolic with respect to any system of finitely generated subgroups?

open_problem | open | MathDB 页面 | 原文线索

V2-075566 | MATHDB-395276

Bi-Lipschitz bijections for finitely generated groups

摘录 / EXCERPT: Let G be a finitely generated group. Is there a bi-Lipschitz bijection $G \to G \times \mathbb{Z}^2$? The same question is asked for $G_1 \to G$, where G_1 is a subgroup of G...

open_problem | open | MathDB 页面 | 原文线索

V2-075567 | MATHDB-395278

Simon’s conjecture for knot groups

摘录 / EXCERPT: Simon’s conjecture asserts that a knot group (i.e. the fundamental group of the complement of a knotted S^1 in S^3) admits surjective homomorphisms onto only finitely many othe...

conjecture | open | MathDB 页面 | 原文线索

V2-075568 | MATHDB-395280

Classification of locally compact targets for hyperbolic lattices

摘录 / EXCERPT: Let Γ be a lattice in $G=\operatorname{Isom}(\mathbb{H}^n)$, with $n\ge 3$. Determine all locally compact groups H, apart from G, in which Γ embeds as a lattice;...

open_problem | open | MathDB 页面 | 原文线索

V2-075569 | MATHDB-395282

Nonvanishing of simplicial volume in Q-rank 1

摘录 / EXCERPT: Is the simplicial volume of locally symmetric spaces of non-compact type with finite volume non-zero provided that the $\mathbb{Q}$-rank is equal to 1?

open_problem | open | MathDB 页面 | 原文线索

V2-075570 | MATHDB-395286

Refined realization conjecture

摘录 / EXCERPT: Refined realization conjecture (N. Kuhn). The unbounded strong realization conjecture holds, and if $H^*X\notin U_0$, then $S(H^*X)\le N(H^*X)$, where $S(H^*X)$ and $N(H^*X)$ are t...

conjecture | open | MathDB 页面 | 原文线索

V2-075571 | MATHDB-395289

Vanishing of the first L^2-Betti number under an infinite normal subgroup hypothesis

摘录 / EXCERPT: Let G be a group with an infinite normal subgroup of infinite index, which is either finitely generated or amenable. Then $\beta^{(2)}_1(G)=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-075572 | MATHDB-395294

Forbidden-order-type Erdős-Szekeres problem

摘录 / EXCERPT: Assume that all point sets are in general position in the plane. Two point configurations have the same order type if there is a one-to-one correspondence preserving the orientatio...

open_problem | open | MathDB 页面 | 原文线索

V2-075573 | MATHDB-395295

Realization of allowable double-permutation sequences

摘录 / EXCERPT: Is every allowable double-permutation sequence the double-permutation sequence arising from a family of connected sets $\mathcal C$ together with a family $\mathcal A$ of pseudolin...

open_problem | open | MathDB 页面 | 原文线索

V2-075574 | MATHDB-395298

Improving the asymptotic bound for the number of Latin squares

摘录 / EXCERPT: Improve the bound ... which determines L_n only up to a factor of $e^{o(n^2)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-075575 | MATHDB-395299

Maximum rank of real n × n × n tensors

摘录 / EXCERPT: For a three-dimensional real tensor, rank one means that its entries have the form $a_{ijk}=x_i y_j z_k$, and the tensor rank is the least number of rank-one tensors whose sum is t...

open_problem | open | MathDB 页面 | 原文线索

V2-075576 | MATHDB-395302

Optimal constants in Theorem 2 and Corollary 3

摘录 / EXCERPT: It is an open problem to determine the optimal choices for the integers m_d and n_d in Theorem 2 and Corollary 3.

open_problem | open | MathDB 页面 | 原文线索

V2-075577 | MATHDB-395303

Dimension-uniform dilation for unimodular triangulations

摘录 / EXCERPT: For each positive integer d, there exists a positive integer n_d such that every d-dimensional lattice polytope P has the dilate n_dP admitting a unimodular triangulation...

open_problem | open | MathDB 页面 | 原文线索

V2-075578 | MATHDB-395304

Characterization of x-good ovals

摘录 / EXCERPT: For an oval $K\subset\mathbb R^2$ and $x\in\{1,\ldots,6\}$, determine whether K is x-good, where K is called x-good if arbitrarily large families F_K of Type x exist.

open_problem | open | MathDB 页面 | 原文线索

V2-075579 | MATHDB-395306

Ellipse characterization from the Type 4 geometric-permutation property

摘录 / EXCERPT: Let K be a centrally symmetric oval in $\mathbb{R}^2$ with smooth boundary, and suppose that K does not admit Type 4; equivalently, no finite family of pairwise disjoint tran...

open_problem | open | MathDB 页面 | 原文线索

V2-075580 | MATHDB-395308

Undecidability of PL embeddability in codimension one

摘录 / EXCERPT: For every fixed integer $d\ge 5$, the decision problem whose input is a finite simplicial complex K of dimension at most $d-1$ and whose output is YES exactly when K admits a p...

open_problem | open | MathDB 页面 | 原文线索

V2-075581 | MATHDB-395310

Average upper bound for Frobenius numbers

摘录 / EXCERPT: For a positive integral vector $\boldsymbol a=(a_1,\ldots,a_N)\in\mathbb Z_{>0}^N$ with $\gcd(a_1,\ldots,a_N)=1$, derive a good upper estimate for the Frobenius number in average....

open_problem | open | MathDB 页面 | 原文线索

V2-075582 | MATHDB-395313

Finite-field congruent-configuration threshold problem (Problem 5)

摘录 / EXCERPT: Let $E\subset\mathbb{F}_q^d$, $d\ge 2$, where $\mathbb{F}_q$ is a finite field and $\mathbb{F}_q^d$ is the d-dimensional vector space over $\mathbb{F}_q$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-075583 | MATHDB-395316

NP-hardness of integrating general polynomials over a simplex

摘录 / EXCERPT: Theorem 1 (Integrating general polynomials over a simplex is hard). The following problem is NP-hard.

open_problem | open | MathDB 页面 | 原文线索

V2-075584 | MATHDB-395317

Characterization of point sets admitting a cubic plane graph

摘录 / EXCERPT: Characterize the sets S of $n\ge 4$ points in general position in the plane that admit a cubic plane graph on S.

open_problem | open | MathDB 页面 | 原文线索

V2-075585 | MATHDB-395318

Equality of translative and lattice packing densities

摘录 / EXCERPT: Is it true that $\delta(K)=\delta^*(K)$ for every convex body K?

open_problem | open | MathDB 页面 | 原文线索

V2-075586 | MATHDB-395320

Balanced non-group-balanced spherical codes in dimensions 4, 5, or 6

摘录 / EXCERPT: Find balanced, non-group-balanced spherical codes in dimensions 4, 5, or 6, or prove that such codes do not exist.

open_problem | open | MathDB 页面 | 原文线索

V2-075587 | MATHDB-395321

Extension of the isolated-transversal theorem to disjoint convex sets

摘录 / EXCERPT: Can the local Helly-type result for an isolated line transversal be generalized to arbitrary collections of disjoint convex sets in $\mathbb{R}^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-075588 | MATHDB-395323

Least-squares matching of point sets under rotation: number of optimal-permutation intervals

摘录 / EXCERPT: Given points $A_1,\ldots,A_n,B_1,\ldots,B_n$ in the plane, define ... where π ranges over permutations of $\{1,\ldots,n\}$ and $R(\theta)$ is rotation about the origin through an...

open_problem | open | MathDB 页面 | 原文线索

V2-075589 | MATHDB-395324

Tight polytope around the unit ball

摘录 / EXCERPT: Let B_n be the n-dimensional unit ball centered at the origin, and let P_{2n} be an n-dimensional polytope with $2n$ facets containing B_n. Then ... where equality holds if...

open_problem | open | MathDB 页面 | 原文线索

V2-075590 | MATHDB-395325

Essential covers of the cube by hyperplanes

摘录 / EXCERPT: Cover the vertices of the discrete cube $\{0,1\}^n$ with as few affine hyperplanes $H_1,\ldots,H_k$ as possible, subject to: 1. Every H_i is essential: for every i, there is a...

open_problem | open | MathDB 页面 | 原文线索

V2-075591 | MATHDB-395327

Representing finite projective planes by convex sets

摘录 / EXCERPT: Does there exist a universal constant d_0 such that every finite projective plane, considered as a set system $(X,\mathcal F)$, is d_0-representable; that is, there are convex...

open_problem | open | MathDB 页面 | 原文线索

V2-075592 | MATHDB-395329

Measure-zero conjecture for the union of lines tangent to three convex sets in $\mathbb{R}^3$

摘录 / EXCERPT: Given three pairwise disjoint convex sets $A,B,C\subset\mathbb{R}^3$, let $S(A,B,C)$ be the set of points in $\mathbb{R}^3$ through which passes a line tangent to all three sets. H...

conjecture | open | MathDB 页面 | 原文线索

V2-075593 | MATHDB-395335

GKZ-rational configurations and essential Cayley configurations

摘录 / EXCERPT: Conjecture 1.3 in [5]: the only GKZ-rational configurations are those affinely equivalent to an essential Cayley configuration.

open_problem | open | MathDB 页面 | 原文线索

V2-075594 | MATHDB-395336

Toric-residue derivatives of stable rational A-hypergeometric functions

摘录 / EXCERPT: Conjecture 5.9 in [5] asserts that a suitable derivative of every stable rational A-hypergeometric function is a constant multiple of an explicit toric residue. The report states t...

open_problem | open | MathDB 页面 | 原文线索

V2-075595 | MATHDB-395337

Allcock’s monstrous proposal for a complex hyperbolic braid-group quotient

摘录 / EXCERPT: Let $G=\pi_1^{\mathrm{orb}}(X_0)$, where $X_0=(B^{13}\setminus H)/\Gamma$, H is the union of the mirrors of the triflections, and $\Delta=H/\Gamma$. The quotient obtained from...

open_problem | open | MathDB 页面 | 原文线索

V2-075596 | MATHDB-395338

Generation of the orbifold fundamental group by the 16 meridian loops

摘录 / EXCERPT: Let $X_0=(B^{13}\setminus H)/G$, where H is the union of the mirrors of the triflections and $\[1m\pi_1\u001b[0m$ denotes the orbifold fundamental group. For a suitable point...

open_problem | open | MathDB 页面 | 原文线索

V2-075597 | MATHDB-395341

Formal degrees and packets for unramified discrete Langlands data

摘录 / EXCERPT: Let ϕ_G be a $Z(G^\vee)$-equivariant spectral transfer morphism as in Theorem 3. Let Φ be an unramified discrete local Langlands datum for G^F, with corresponding centr...

open_problem | open | MathDB 页面 | 原文线索

V2-075598 | MATHDB-395342

Siu’s conjecture on maps between ball quotients

摘录 / EXCERPT: No surjective holomorphic map should exist between ball quotients of complex dimensions m and n when $m>n\ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-075599 | MATHDB-395345

Uniqueness of the standard Ricci flow

摘录 / EXCERPT: Fix a standard initial metric g_0 on $\cmathbb{R}^3$ as defined on PDF page 23, and let $(\cmathbb{R}^3,g(t))_{0\le t<T}$ be a standard Ricci flow. If...

open_problem | open | MathDB 页面 | 原文线索

V2-075600 | MATHDB-395346

Existence of surgery control parameters for Ricci flow with surgery

摘录 / EXCERPT: For any normalized initial metric $(M^3_0,g(0))$, there exist surgery control parameters r, $\bdelta\$ and $\bkappa\$ such that a Ricci flow with surgery can be defined and satis...

open_problem | open | MathDB 页面 | 原文线索

V2-075601 | MATHDB-395347

Closing the AWGN error-exponent gap using repulsive stationary point processes

摘录 / EXCERPT: Close the gap between the best currently known upper and lower bounds on the error exponent of the AWGN channel, for instance by considering sequences of stationary point processes...

open_problem | open | MathDB 页面 | 原文线索

V2-075602 | MATHDB-395348

Visibility with partially absorbing obstacles

摘录 / EXCERPT: What can be said about the visibility in one particular direction when the obstacles of the Boolean model absorb only a fraction of the “light”, proportional to the length of the p...

open_problem | open | MathDB 页面 | 原文线索

V2-075603 | MATHDB-395351

Asymptotic properties of Lorenz-curve statistics for testing complete spatial randomness

摘录 / EXCERPT: For the nonnegative random measures Z and Z_0 observed through cubic windows of fixed size $r>0$ centered at lattice points in $\lambda\mathbb{Z}^d\cap[0,n]^d$, with empirical...

open_problem | open | MathDB 页面 | 原文线索

V2-075604 | MATHDB-395352

Characterization of Palm measures via randomized invariant transports

摘录 / EXCERPT: For an invariant random measure ξ, if the definition of mass-stationarity is formulated using shifts induced by invariant ξ-preserving transport-kernels-equivalently, by al...

open_problem | open | MathDB 页面 | 原文线索

V2-075605 | MATHDB-395353

Branner-Hubbard conjecture on Cantor Julia sets

摘录 / EXCERPT: Assume that g is a polynomial. Denote by K_g its filled Julia set. Assume that every component of K_g containing a critical point is aperiodic under the iteration of g. The...

conjecture | open | MathDB 页面 | 原文线索

V2-075606 | MATHDB-395355

Douady’s Brjuno-condition conjecture for higher-degree polynomials

摘录 / EXCERPT: The Brjuno condition is also necessary for polynomials of degree $d\ge 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-075607 | MATHDB-395356

Conformal-radius upper bound for degree-\d polynomials

摘录 / EXCERPT: For $d\ge 3$, there exists $C=C(d)\in\mathbb{R}$ such that, for every polynomial f of degree d with an indifferent fixed point at the origin, ... where the c_i are the critical...

open_problem | open | MathDB 页面 | 原文线索

V2-075608 | MATHDB-395357

Quadratic polynomials obtainable from the Chebyshev polynomial by generalized regluing

摘录 / EXCERPT: Which quadratic polynomials can be obtained from the Chebyshev polynomial $z \mapsto z^2-2$ by generalized regluing?

open_problem | open | MathDB 页面 | 原文线索

V2-075609 | MATHDB-395361

Bounded limit functions on a wandering domain

摘录 / EXCERPT: Suppose f is a transcendental entire function with wandering domain U. Can the limit functions of $\{f^n|_U\}$ be bounded?

open_problem | open | MathDB 页面 | 原文线索

V2-075610 | MATHDB-395363

Dynamic-ray connectivity for escaping points of transcendental entire functions

摘录 / EXCERPT: For every transcendental entire function f, can every escaping point $z\in I(f)$, where $I(f)=\{z\in\mathbb{C}: f^n(z)\to\infty\text{ as }n\to\infty\}$, be connected to ∞...

open_problem | open | MathDB 页面 | 原文线索

V2-075611 | MATHDB-395364

Theorem 2 (structure) for near-parabolic renormalization

摘录 / EXCERPT: For $f(z)=e^{2\pi i\alpha h(z)}$, where $h\in F_1$ or $h(z)=z+z^2$ and α is of high type, there exist domains...

open_problem | open | MathDB 页面 | 原文线索

V2-075612 | MATHDB-395365

Decoration Conjecture on the Mandelbrot set

摘录 / EXCERPT: Let M be the Mandelbrot set, and let M_1 be a small copy of the Mandelbrot set. Then, for any $\\varepsilon>0$, there are at most finitely many connected components of...

conjecture | open | MathDB 页面 | 原文线索

V2-075613 | MATHDB-395366

Classification problem for Hadamard subfactors outside the stated classes

摘录 / EXCERPT: For Hadamard subfactors which are not depth 2 or constructed via twisted tensor product, the classification problem remains almost completely open.

open_problem | open | MathDB 页面 | 原文线索

V2-075614 | MATHDB-395367

Measurable version of von Neumann’s problem

摘录 / EXCERPT: Every non-amenable group Γ contains F_2 measurably: for the Bernoulli action $\Gamma\curvearrowright X=(S^1)^\Gamma$ with Lebesgue measure, there exists a free p.m.p. ergo...

open_problem | open | MathDB 页面 | 原文线索

V2-075615 | MATHDB-395368

Singularity versus 1-strong singularity for subfactors

摘录 / EXCERPT: For subfactors $B\subseteq M$ of a type II_1 factor, does singularity imply 1-strong singularity, i.e. does every singular subfactor satisfy ... The report answers this...

open_problem | open | MathDB 页面 | 原文线索

V2-075616 | MATHDB-395370

Redundancy of residual finiteness in the rigidity corollary

摘录 / EXCERPT: Conjecturally, in the preceding rigidity corollary the assumption that Λ is residually finite is unnecessary. More precisely, let...

open_problem | open | MathDB 页面 | 原文线索

V2-075617 | MATHDB-395371

Rapid decay for Wise’s non-Hopfian group

摘录 / EXCERPT: Let $G_W=\langle a,b,c,s,t\mid c=ab=ba,\ c^2=sas^{-1}=tbt^{-1}\rangle$. Then G_W has property RD (rapid decay).

open_problem | open | MathDB 页面 | 原文线索

V2-075618 | MATHDB-395372

An analog of Margulis-Zimmer superrigidity for hyperbolic lattices

摘录 / EXCERPT: Does there exist an analog of Margulis-Zimmer superrigidity for hyperbolic lattices, in the setting $G=\operatorname{Isom}(\mathbb{H}^n)$ with $n\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-075619 | MATHDB-395376

Stabilization with degenerate rotation

摘录 / EXCERPT: The report states that the question has a positive answer: for a deterministic linear system $\dot{x}=Ax$ with A having negative trace, there exists a skew-symmetric matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-075620 | MATHDB-395377

Small-viscosity behavior of the KdV action process

摘录 / EXCERPT: Consider the damped-driven KdV equation ... where $0<\nu\le 1$ and η is smooth in x, white in t, and satisfies mild non-degeneracy assumptions. The solution has either a pre...

open_problem | open | MathDB 页面 | 原文线索

V2-075621 | MATHDB-395379

Strong completeness under local Lipschitz and linear-growth assumptions

摘录 / EXCERPT: For the SDE ... where the coefficients are locally Lipschitz and have at most linear growth at infinity, does weak completeness imply strong completeness? Here strong completeness me...

open_problem | open | MathDB 页面 | 原文线索

V2-075622 | MATHDB-395380

Global rigidity of generic two-dimensional mixed frameworks

摘录 / EXCERPT: Let $G=(V;D,L)$ be a mixed graph, where D and L partition the edge set into direction edges and length edges, respectively, and let $p:V\to\mathbb{R}^2$ be a generic realizatio...

open_problem | open | MathDB 页面 | 原文线索

V2-075623 | MATHDB-395381

Global rigidity of generic mixed frameworks

摘录 / EXCERPT: The problem of characterizing when a generic mixed framework (G,p) is globally rigid is still an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-075624 | MATHDB-395382

Goemans’s conjecture for single-source unsplittable min-cost flow

摘录 / EXCERPT: Let $G=(V,E)$ be a directed graph with arc capacities $u=(u_e)_{e\\in E}$ and costs $c=(c_e)_{e\\in E}$, a source s, sink nodes $t_1,\\ldots,t_p$, and demands...

conjecture | open | MathDB 页面 | 原文线索

V2-075625 | MATHDB-395383

Goemans’ convex-combination conjecture for single-source unsplittable flows

摘录 / EXCERPT: Any flow x satisfying demands d can be written as a convex combination of unsplittable flows y^{ℓ}, $\ell\in L$, each satisfying demands d and such that...

conjecture | open | MathDB 页面 | 原文线索

V2-075626 | MATHDB-395384

Goemans’ conjecture for single-source unsplittable min-cost flow

摘录 / EXCERPT: Given a directed graph $G=(V,E)$ with arc capacities $u=(u_e)_{e\\in E}$ and costs $c=(c_e)_{e\\in E}$, a source $s\\in V$, sink nodes $t_1,\\ldots,t_p\\in V$, and demands...

conjecture | open | MathDB 页面 | 原文线索

V2-075627 | MATHDB-395387

Even Cycle Isomorphism Problem

摘录 / EXCERPT: Let (G,Σ) and (G',Σ') be signed graphs. When is the relation $\operatorname{ecycle}(G,\Sigma)=\operatorname{ecycle}(G',\Sigma')$ satisfied?

open_problem | open | MathDB 页面 | 原文线索

V2-075628 | MATHDB-395390

Membership in the convex hull of a mixed-integer bipartite graph set

摘录 / EXCERPT: Given a bipartite graph $G=(U\cup V,E)$, a set $I\subseteq U\cup V$, rational numbers b_{ij} for $ij\in E$, and ... the problem is: given $\bar{x}\in\mathbb{R}^{V(G)}$, decide whet...

open_problem | open | MathDB 页面 | 原文线索

V2-075629 | MATHDB-395393

Crossing number of the toroidal grid graph $T_{8,8}$

摘录 / EXCERPT: The crossing number of the smallest toroidal grid graph $T_{8,8}$ is still unknown and is conjectured to be 48.

open_problem | open | MathDB 页面 | 原文线索

V2-075630 | MATHDB-395396

Constant-factor approximation for Minimum Linear Arrangement

摘录 / EXCERPT: Let $G=(V,E)$) with $V=\{1,\ldots,n\}$, and let an arrangement be a bijection $\psi:V\to V$. The Minimum Linear Arrangement problem (MinLA) minimizes...

open_problem | open | MathDB 页面 | 原文线索

V2-075631 | MATHDB-395398

Woodall’s conjecture on directed cut covers

摘录 / EXCERPT: Let $D=(V,E)$ be a directed graph. Woodall’s conjecture asserts that the maximum number of edge-disjoint directed cut covers in D equals the minimum size of a directed cut.

conjecture | open | MathDB 页面 | 原文线索

V2-075632 | MATHDB-395400

Polyhedral reduction of pseudo-Boolean optimization to the cut polytope

摘录 / EXCERPT: Theorem 2. Assume that no operator in the objective function is an exclusive disjunction or an equivalence. Then the polytope P is isomorphic to a face of a cut polytope, where t...

open_problem | open | MathDB 页面 | 原文线索

V2-075633 | MATHDB-395401

Face containment of the quadratic linear ordering polytope

摘录 / EXCERPT: For all $i<j<k$, the constraint $x_{ik}-y_{ijik}-y_{ikjk}+y_{ijjk}=0$ induces a face of the polytope P_n. In particular, $QLO(n)$ is a face of P_n.

open_problem | open | MathDB 页面 | 原文线索

V2-075634 | MATHDB-395402

Classification of conjugacy classes of \c3-stable maximal k-split tori

摘录 / EXCERPT: For fields other than $\mathbb{R}$, classify the H_k-conjugacy classes of σ-stable maximal k-split tori, apart from the specific cases in which such a classification is...

open_problem | open | MathDB 页面 | 原文线索

V2-075635 | MATHDB-395405

Serre’s conjecture on complete reducibility over separable field extensions

摘录 / EXCERPT: Let k be any field, let G be a group defined over k, and let k_1/k be a separable field extension. A subgroup H of G is called G-completely reducible over k if, whe...

conjecture | open | MathDB 页面 | 原文线索

V2-075636 | MATHDB-395406

H(A)-distinguished representations of G(A)

摘录 / EXCERPT: Which representations π of $G(\mathbb A)$ admit a non-zero $H(\mathbb A)$-invariant functional?

open_problem | open | MathDB 页面 | 原文线索

V2-075637 | MATHDB-395407

Euler-product factorization of nonzero automorphic period functionals

摘录 / EXCERPT: In the setting of the report, if ((pi,nu)u0000be an automorphic representation and $P\circ\nu\neq 0$, can the functional $P\circ\nu$ be expressed as an Euler product, and how?

open_problem | open | MathDB 页面 | 原文线索

V2-075638 | MATHDB-395409

Center Conjecture

摘录 / EXCERPT: If A is convex and contractible, then $\a0operatorname{Aut}(A)$ or $\a0operatorname{Aut}(\bar X,A)$ has a fixed point in A.

conjecture | open | MathDB 页面 | 原文线索

V2-075639 | MATHDB-395420

Characterization of the image of the fixed-point divisor map

摘录 / EXCERPT: For a faithful action of $\mathbb{Z}/p\mathbb{Z}$ on $X=\operatorname{Spec} k[[x]]$, with first ramification jump m and versal equivariant deformation hull R_X, the map ... embed...

open_problem | open | MathDB 页面 | 原文线索

V2-075640 | MATHDB-395422

Artin-Schreier parametrization for big actions with elementary abelian second ramification group

摘录 / EXCERPT: Let (C,G) be a big action with $g\ge 2$ such that $G_2\simeq(\mathbb Z/p\mathbb Z)^n$, where $n\ge 1$. For every $t\ge 1$, let Σ_t be the k-vector subspace of $k[X]$ gen...

open_problem | open | MathDB 页面 | 原文线索

V2-075641 | MATHDB-395425

Existence of infinitely many good deformation data for the dihedral group

摘录 / EXCERPT: Suppose that G is dihedral of order $2p$ (that is, $m=2$ and $\\chi$ is injective). Then there exist infinitely many good deformation data with conductor $h=p+2$.

open_problem | open | MathDB 页面 | 原文线索

V2-075642 | MATHDB-395427

Validity of the semiclassical limit at eigenvalue crossings

摘录 / EXCERPT: Do the above results remain valid also for regions which contain points of jump singularities of ∇U corresponding to eigenvalue crossings?

open_problem | open | MathDB 页面 | 原文线索

V2-075643 | MATHDB-395430

Relevance of disorder at marginality in pinning models

摘录 / EXCERPT: For the generalized pinning model with $\alpha=1/2$, equivalently $K(n)\mathop{\sim}_{n\to\infty} Cn^{-3/2}$ for some $C>0$, disorder is relevant at marginality:...

open_problem | open | MathDB 页面 | 原文线索

V2-075644 | MATHDB-395431

Hard-core potentials for Gibbs measures with Delaunay triangle interactions

摘录 / EXCERPT: Can the preceding existence and variational characterization theorem for Gibbs measures with shift-invariant Delaunay tile potentials be extended to potentials with a hard core, al...

open_problem | open | MathDB 页面 | 原文线索

V2-075645 | MATHDB-395433

Scaling limits at the critical point with two large-deviation minimizers

摘录 / EXCERPT: Assume the weakly pinned Markov chains and rate functional described in the report, with $\xi^\varepsilon>0$, and suppose the two minimizers $\bar h$ and $\hat h$ satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-075646 | MATHDB-395436

A decreasing Glimm interaction functional for general non-genuinely nonlinear systems

摘录 / EXCERPT: For a general strictly hyperbolic, nonconservative system $u_t+A(u)u_x=0$ with $A\2$ regularity, with no genuine-nonlinearity or linear-degeneracy assumption, define ... where ... The...

open_problem | open | MathDB 页面 | 原文线索

V2-075647 | MATHDB-395438

Existence and uniqueness for the self-similar viscous boundary profile

摘录 / EXCERPT: Let $f:\mathbb{R}^n\to\mathbb{R}^n$ be smooth and let $A(u)=Df(u)$ be strictly hyperbolic, with $\det A(u)\ne 0$. For constant states $u_0,\bar u_b\in\mathbb{R}^n$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-075648 | MATHDB-395439

Existence and uniqueness of self-similar viscous boundary profiles

摘录 / EXCERPT: For the initial-boundary problem $\partial_tu+\partial_xf(u)=0$ with constant states $u_0,\bar u_b\in\mathbb R^n$, assuming strict hyperbolicity of $A(u)=Df(u)$ and the noncharacte...

open_problem | open | MathDB 页面 | 原文线索

V2-075649 | MATHDB-395440

Global self-similar shock reflection by a wedge for potential flow

摘录 / EXCERPT: For the self-similar potential-flow problem in the domain $\Lambda=\mathbb{R}^2_+\setminus W$, where $W=\{(x_1,x_2): |x_2|<x_1\tan\theta_w,\ x_1>0\}$, seek a solution φ of...

open_problem | open | MathDB 页面 | 原文线索

V2-075650 | MATHDB-395441

Global weak solutions for regular shock reflection by large-angle wedges

摘录 / EXCERPT: For every 4gamma14 and 4rho1rho004, there exist 4thetac=thetac(rho0,rho1,gamma)in(0,pi/2)4 and 4alpha=alpha(rho0,rho1,gamma)in(0,1)4 such that, for 4thetawin[thetac,pi/2)4, Problem...

open_problem | open | MathDB 页面 | 原文线索

V2-075651 | MATHDB-395442

Optimal regularity across the sonic arc in regular shock reflection

摘录 / EXCERPT: Let ϕ be a solution of Problem 1 satisfying properties (i)-(v) of Theorem 1. Then: (i) ϕ is $C^{2,\alpha}$ in Ω up to Γ_{sonic} away from the point...

open_problem | open | MathDB 页面 | 原文线索

V2-075652 | MATHDB-395443

Sound-proof two-scale model for strongly stratified atmospheric flow

摘录 / EXCERPT: Construct a sound-proof two-scale model from the full compressible flow equations that eliminates the fastest sound-propagation time scale while retaining both the internal-gravity...

open_problem | open | MathDB 页面 | 原文线索

V2-075653 | MATHDB-395445

Global smooth solvability of the 2D viscous Boussinesq system on a bounded domain

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded domain with smooth boundary $\partial\Omega$, let $\nu>0$, and let $e_2=(0,1)^T$. Consider ... with ... If $(\rho_0,U_0)\in H^3(\Omega)$ sa...

open_problem | open | MathDB 页面 | 原文线索

V2-075654 | MATHDB-395448

Patterns generated by local cell-cell interactions

摘录 / EXCERPT: Which pattern can result from local interactions, like e.g. from direct cell-cell contact, and not from cell growth and death?

open_problem | open | MathDB 页面 | 原文线索

V2-075655 | MATHDB-395449

Existence of monotone fronts in convex FPU-type chains

摘录 / EXCERPT: For all convex and twice continuously differentiable potentials $\Phi:\mathbb{R}\to\mathbb{R}$, the following assertions hold: (1) each front in the chain corresponds to a conserva...

open_problem | open | MathDB 页面 | 原文线索

V2-075656 | MATHDB-395450

Occurrence and scale of turbulent-laminar patterns in shear flows

摘录 / EXCERPT: Why do turbulent-laminar patterns occur near the critical Reynolds number Re_c in so many shear flows?

open_problem | open | MathDB 页面 | 原文线索

V2-075657 | MATHDB-395451

Orbital stability of KdV cnoidal waves under integer-multiple-period perturbations

摘录 / EXCERPT: The orbital stability question for KdV cnoidal waves with respect to periodic perturbations whose period is an integer multiple of the wave period is settled affirmatively.

open_problem | open | MathDB 页面 | 原文线索

V2-075658 | MATHDB-395452

Invariant torus for the toy problem with essential spectrum up to the imaginary axis

摘录 / EXCERPT: For the toy problem (1)-(2) described on the source pages, with spatial variable in $\mathbb{R}^d$, if $d\ge 5$, then there exists $\alpha _0>0$ such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075659 | MATHDB-395454

Nonlinear stability under larger classes of perturbations

摘录 / EXCERPT: Is the spectral stability assumption sufficient to ensure nonlinear stability with respect to perturbations in a larger class (for instance, bounded perturbations)?

open_problem | open | MathDB 页面 | 原文线索

V2-075660 | MATHDB-395456

Existence of near-planar waves for nonperiodic stratified KPP media

摘录 / EXCERPT: For the KPP-type equation $u_t=u_{xx}+u_{yy}+b(x)f(u)$ on $\mathbb{R}^2$, with f a KPP-type monostable nonlinearity and $b(x)>0$ no longer assumed periodic, is there a near-plana...

open_problem | open | MathDB 页面 | 原文线索

V2-075661 | MATHDB-395458

Nondisplaceability of points not displaced by probes

摘录 / EXCERPT: Some 2-dimensional polytopes have open subsets of points that cannot be displaced by a probe. Are these points really nondisplaceable?

open_problem | open | MathDB 页面 | 原文线索

V2-075662 | MATHDB-395459

Displaceability of points in Fano moment polytopes by probes

摘录 / EXCERPT: Can every point in a Fano moment polytope except for this special point be displaced by a probe?

open_problem | open | MathDB 页面 | 原文线索

V2-075663 | MATHDB-395460

Admissible-face representations of smooth Schubert cycles

摘录 / EXCERPT: I conjecture that all Schubert cycles defined by Schubert cells with smooth closures can be represented by an admissible face.

open_problem | open | MathDB 页面 | 原文线索

V2-075664 | MATHDB-395461

Cayley decompositions of lattice polytopes of fixed degree

摘录 / EXCERPT: Given a positive integer d, does there exist an integer $N(d)$, depending on d, such that every lattice polytope of degree d and dimension at least $N(d)$ is affinely equival...

open_problem | open | MathDB 页面 | 原文线索

V2-075665 | MATHDB-395462

Fulton’s F-conjecture for nef divisors on \(\overline{M}_{0,n}\)

摘录 / EXCERPT: A divisor on $\overline{M}_{0,n}$ is nef if and only if it has nonnegative intersection with every F-curve.

conjecture | open | MathDB 页面 | 原文线索

V2-075666 | MATHDB-395463

Finite generation of Cox rings of K3 surfaces

摘录 / EXCERPT: A K3 surface has a finitely generated Cox ring if and only if its cone of effective rational divisor classes is polyhedral.

open_problem | open | MathDB 页面 | 原文线索

V2-075667 | MATHDB-395467

Generators of the invariant ring of Cartan-matrix polynomials

摘录 / EXCERPT: Conjecture. The polynomials D_i, for $1\le i\le\lfloor n/2\rfloor$, generate the ring of all invariants when n is odd. When n is even, they generate the ring of all invariant...

open_problem | open | MathDB 页面 | 原文线索

V2-075668 | MATHDB-395468

Bound on the length of cyclic strongly exceptional sequences

摘录 / EXCERPT: If a rational surface X admits a full, cyclic strongly exceptional sequence of length n, then $n\le 9$.

open_problem | open | MathDB 页面 | 原文线索

V2-075669 | MATHDB-395474

Continuous extension of the volume function for angle structures

摘录 / EXCERPT: The volume function on the space of angle structures extends continuously to the compact closure of the moduli space of angle structures. This gives an affirmative answer to a ques...

open_problem | open | MathDB 页面 | 原文线索

V2-075670 | MATHDB-395475

Discrete square peg problem

摘录 / EXCERPT: Does every Jordan curve $C \subset \mathbb{R}^2$ have four points which form a square?

open_problem | open | MathDB 页面 | 原文线索

V2-075671 | MATHDB-395477

Complete integrability of the pentagram map on convex polygons

摘录 / EXCERPT: For the pentagram map $T:\mathcal C_n\to\mathcal C_n$ on the space $\mathcal C_n$ of convex n-gons in the projective plane modulo projective transformations, is T completely in...

open_problem | open | MathDB 页面 | 原文线索

V2-075672 | MATHDB-395478

Connectedness of the space of smooth embedded saddle tori in \(\mathbb{RP}^3\)

摘录 / EXCERPT: Is the set of all smooth embedded saddle tori in $\mathbb{RP}^3$ connected?

open_problem | open | MathDB 页面 | 原文线索

V2-075673 | MATHDB-395479

Alexandrov’s uniqueness conjecture for smooth convex surfaces

摘录 / EXCERPT: Let $K \subset \mathbb{R}^3$ be a smooth convex body. If, for a constant C, at every point of ∂K one has $R_1 \le C \le R_2$, then K is a ball, where R_1 and R_2...

conjecture | open | MathDB 页面 | 原文线索

V2-075674 | MATHDB-395483

Linked and non-linked embeddings of the graph of a convex 4-polytope

摘录 / EXCERPT: An embedded graph in S^3 is called linked if it contains two disjoint linked cycles. Is there a convex 4-polytope P such that its 1-skeleton $P^{(1)}$ is linked as a graph...

open_problem | open | MathDB 页面 | 原文线索

V2-075675 | MATHDB-395484

Characterization of functions positive on all planar embeddings of K_4

摘录 / EXCERPT: Let f assign a number to each line segment in the plane. For every embedding of K_4 in general position, with the self-stress $\\omega_{ij}$ and quantities f_{ij} from (1), r...

open_problem | open | MathDB 页面 | 原文线索

V2-075676 | MATHDB-395485

Uniqueness of inversive-distance packings on a doubly periodic hexagonal lattice

摘录 / EXCERPT: Let K be a doubly periodic hexagonal lattice in the plane, meaning an affine image of a regular hexagonal lattice. Define a circle packing P for K by centering a circle of ra...

open_problem | open | MathDB 页面 | 原文线索

V2-075677 | MATHDB-395488

Mixed infinitesimal rigidity of simplicial convex 3-polytopes

摘录 / EXCERPT: Given a simplicial convex 3-polytope, is it infinitesimally rigid when at each edge either length or dihedral angle is fixed and at least one edge length is given?

open_problem | open | MathDB 页面 | 原文线索

V2-075678 | MATHDB-395493

Rectangles on Curves Conjecture

摘录 / EXCERPT: Does every C^{∞}-embedded circle in $\mathbb{R}^2$ contain four points spanning a rectangle with prescribed edge-ratios?

conjecture | open | MathDB 页面 | 原文线索

V2-075679 | MATHDB-395495

Distribution of the martingale limit in the large-\m m-ary search tree

摘录 / EXCERPT: Determine the distribution of the complex random variable W^{DT} appearing as the martingale limit in the m-ary search-tree composition-vector asymptotics for $m\ge 27$.

open_problem | open | MathDB 页面 | 原文线索

V2-075680 | MATHDB-395497

Reduced-order shape optimization problem for the Stokes equations

摘录 / EXCERPT: For the shape optimization problem governed by the Stokes equations, one can construct a reduced-order shape optimization problem whose first-order optimality conditions are identi...

open_problem | open | MathDB 页面 | 原文线索

V2-075681 | MATHDB-395498

PDAS methods for mass-conserving vector-valued Allen-Cahn systems

摘录 / EXCERPT: Develop and analyze a primal-dual active set (equivalently, semismooth Newton) method for the vector-valued, nonlocal mass-conserving Allen-Cahn variational inequality, including t...

open_problem | open | MathDB 页面 | 原文线索

V2-075682 | MATHDB-395500

Unbounded retardation factor for one-shot optimization

摘录 / EXCERPT: For a fixed state equation in fixed-point form $G(y,u)=y$, let ρ denote the spectral radius governing the primal iteration and let $\widehat{\rho}$ denote the spectral radius...

open_problem | open | MathDB 页面 | 原文线索

V2-075683 | MATHDB-395501

Shape optimization of the acoustic transfer function for coupled elastic shells

摘录 / EXCERPT: For an admissible set of domain boundary shapes Γ_{ad}, with transfer function ... seek a solution of ... where T^* is the desired transfer function. The shapes conce...

open_problem | open | MathDB 页面 | 原文线索

V2-075684 | MATHDB-395502

A priori error estimate for a discretized parabolic optimal control problem with pointwise state constraints

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$, $d\in\{2,3\}$, be a bounded convex polygonal domain, let $T>0$, and set $\Omega_T=\Omega\times(0,T)$. Assume $f\in L^2(0,T;H^1(\Omega))$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075685 | MATHDB-395503

Existence of a solution for the regularized free-material-optimization problem

摘录 / EXCERPT: The regularized FMO problem ... has at least one solution.

open_problem | open | MathDB 页面 | 原文线索

V2-075686 | MATHDB-395504

Convergence of partition-based free-material optimization approximations

摘录 / EXCERPT: Assume the admissible piecewise-constant classes satisfy the nonemptiness condition $(*)$, and let J satisfy $(**)$: whenever $E_\kappa \mathrel{H\rightharpoonup} E$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-075687 | MATHDB-395506

Efficient iterative solvers converging in the scaled barrier norm

摘录 / EXCERPT: Construct efficient linear iterative solvers for the Newton-step systems arising in the function-space interior-point method for state-constrained optimal control, with convergence...

open_problem | open | MathDB 页面 | 原文线索

V2-075688 | MATHDB-395507

Bang-bang structure in the Robin parabolic boundary-control example

摘录 / EXCERPT: For the Robin boundary-control problem (1)-(3) described in the report, with $T=1.58$, $\b1=1$, and $y_d(x)=\frac{1}{2}(1-x^2)$, the report proves that the optimal control is bang-...

open_problem | open | MathDB 页面 | 原文线索

V2-075689 | MATHDB-395508

Mesh-independent superlinear convergence of semismooth Newton methods

摘录 / EXCERPT: For the complementarity reformulations $F_h(u_h)=u_h-G\!\left(-\frac{1}{\lambda}A_h(u_h)\right)$, where $G(z)=\max(0,z)$, suppose that for some $p\in(2,\infty]$, $\gamma>0$, $C>0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075690 | MATHDB-395509

A priori error estimate for a state-constrained parabolic optimal-control discretization

摘录 / EXCERPT: Theorem 4 (see [5]). Let $\a0\Omega \subset \mathbb{R}^d$ $(d=2,3)$ be a convex polygonal domain, $\a0\hat{u} \in L^2(0,T;L^2(\Omega))$, $u_0 \in H^2(\Omega)\cap H_0^1(\Omega)$, an...

open_problem | open | MathDB 页面 | 原文线索

V2-075691 | MATHDB-395512

Grothendieck’s section conjecture for curves over number fields

摘录 / EXCERPT: Let k be a number field and let X/k be a smooth, projective, geometrically connected curve of genus at least 2. For the exact sequence...

conjecture | open | MathDB 页面 | 原文线索

V2-075692 | MATHDB-395513

Period-index consequences of a section over a p-adic field

摘录 / EXCERPT: Let k be a finite extension of $\Q_p$ and let X/k be a smooth, projective curve of positive genus such that the fundamental-group extension $\pi_1(X/k)$ admits a section. Then:...

open_problem | open | MathDB 页面 | 原文线索

V2-075693 | MATHDB-395515

Real section conjecture for smooth projective curves over \(\mathbb{R}\)

摘录 / EXCERPT: Theorem 2 (Real section conjecture). Let $X/\mathbb{R}$ be a smooth, projective curve of genus ≥ 1. Then the map ... which maps a connected component of the real locus...

conjecture | open | MathDB 页面 | 原文线索

V2-075694 | MATHDB-395517

Theorem 2: Fully Hilbertian function fields

摘录 / EXCERPT: Let K be an arbitrary field and let F/K be a finitely generated transcendental extension. Then F is fully Hilbertian.

open_problem | open | MathDB 页面 | 原文线索

V2-075695 | MATHDB-395518

Universal definability of \(\mathbb{Z}\) in \(\mathbb{Q}\)

摘录 / EXCERPT: There is a polynomial $g\in\mathbb{Z}[t;x_1,\ldots,x_{58}]$ such that, for all $t\in\mathbb{Q}$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-075696 | MATHDB-395520

Patching equivalence for geometric field patching setups

摘录 / EXCERPT: Let T be a complete discrete valuation ring with uniformizer t, let $\widehat X$ be a smooth projective T-curve with closed fibre X, and let $U_1,U_2\subseteq X$ cover X,...

open_problem | open | MathDB 页面 | 原文线索

V2-075697 | MATHDB-395527

Elliptic curves of rank zero over number fields

摘录 / EXCERPT: For every number field K, does there exist an elliptic curve E/K such that $\operatorname{rank}(E(K))=0$?

open_problem | open | MathDB 页面 | 原文线索

V2-075698 | MATHDB-395531

Divisible elements of Tate-Shafarevich groups

摘录 / EXCERPT: Let X be a smooth projective geometrically integral curve over a number field k with points everywhere locally, and let J be its Jacobian. Can a nonzero element of the Tate-S...

open_problem | open | MathDB 页面 | 原文线索

V2-075699 | MATHDB-395535

Pólya’s conjecture in the presence of a constant magnetic field

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{2}$ be a domain of finite measure, let $B>0$, and let ... with Dirichlet boundary conditions. For $\gamma\ge 0$, $\lambda>0$, and ... one has ... where ... Th...

conjecture | open | MathDB 页面 | 原文线索

V2-075700 | MATHDB-395536

Magnetic eigenvalue-moment estimate (Theorem 2)

摘录 / EXCERPT: Let $\b3\ge 0$ and let $\Omega\subset\mathbb{R}^2$ be a domain of finite measure which is tiling. For the Dirichlet magnetic operator $H^\Omega(B)=(-i\nabla-B A)^2$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-075701 | MATHDB-395537

Pólya’s conjecture for Dirichlet eigenvalues on planar domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{2}$ be a domain of finite measure, and let λ_{n}^{Ω} be the n-th Dirichlet-Laplacian eigenvalue on Ω, counted with multiplicity....

conjecture | open | MathDB 页面 | 原文线索

V2-075702 | MATHDB-395538

Strictness of the Hersch-Payne-Schiffer inequality for higher Steklov eigenvalues

摘录 / EXCERPT: Determine whether, for every simply-connected bounded planar domain Ω with Lipschitz boundary, the Hersch-Payne-Schiffer inequality is strict for every integer $k\ge 2$, i.e...

open_problem | open | MathDB 页面 | 原文线索

V2-075703 | MATHDB-395539

Regular polygons and the first nonzero Neumann eigenvalue

摘录 / EXCERPT: It is plausible that, among polygons with four or more sides, the polygon maximizing the scale-invariant quantity $\mu_1 A$ is regular, where μ_1 is the first nonzero Neumann L...

open_problem | open | MathDB 页面 | 原文线索

V2-075704 | MATHDB-395540

Relaxed Robin Faber-Krahn inequality

摘录 / EXCERPT: For $\beta>0$, let $B\subset\mathbb{R}^N$ be a ball of measure m. Then ... for every measurable $u\geq 0$ such that $u^2\in SBV(\mathbb{R}^N)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075705 | MATHDB-395541

Faber-Krahn minimization for the first Dirichlet eigenvalue

摘录 / EXCERPT: For open sets $\Omega \subseteq \mathbb{R}^N$ with $|\Omega|=m$, the problem ... for the first eigenvalue of the Dirichlet Laplacian has the ball as its unique solution.

open_problem | open | MathDB 页面 | 原文线索

V2-075706 | MATHDB-395544

Strong minimum principle for a translated potential in a strictly convex domain

摘录 / EXCERPT: Let $D\subset\mathbb{R}^d$ be a bounded, strictly convex open domain with smooth boundary. Let $q\in C_c^\infty(D)$, and for $a\in\mathbb{R}^d$ consider the Schrödinger operator...

open_problem | open | MathDB 页面 | 原文线索

V2-075707 | MATHDB-395545

A conjectural sharp Weyl-transform upper bound for Dirichlet-Laplacian spectral sums

摘录 / EXCERPT: Let a0$\Omega\subset\mathbb{R}^d$ have Dirichlet-Laplacian eigenvalues $0<\lambda_1<\lambda_2\leq\lambda_3\leq\cdots$. For a nonnegative function f on $\mathbb{R}_+$ satisfying .....

open_problem | open | MathDB 页面 | 原文线索

V2-075708 | MATHDB-395546

Explicit remainder bound for the corrected Kac inequality on convex domains

摘录 / EXCERPT: At least for convex domains, can one improve Van den Berg’s result to ... where ϕ is an explicit function?

open_problem | open | MathDB 页面 | 原文线索

V2-075709 | MATHDB-395547

Eigenvalue bounds for complex Schrödinger operators

摘录 / EXCERPT: Let $d\ge 2$ and let $\gamma>0$ be given. Is there a positive constant C such that ... for every complex-valued potential $V\in C_0^\infty(\mathbb{R}^d)$ and every eigenvalue...

open_problem | open | MathDB 页面 | 原文线索

V2-075710 | MATHDB-395548

Blaschke-Lebesgue theorem for planar convex sets of constant width

摘录 / EXCERPT: Among all plane convex sets of constant width d, the Reuleaux triangle minimizes area. The Reuleaux triangle is the intersection of three discs of radius d whose centers are at...

open_problem | open | MathDB 页面 | 原文线索

V2-075711 | MATHDB-395550

Van den Berg’s spectral-gap conjecture for planar convex domains

摘录 / EXCERPT: Conjecture 1. For any planar convex domain K, we have ... where $\gamma(K):=\lambda_2(K)-\lambda_1(K)$ and d denotes the diameter of K.

conjecture | open | MathDB 页面 | 原文线索

V2-075712 | MATHDB-395551

Bounds for the first Dirichlet eigenvalue of a quadrilateral

摘录 / EXCERPT: Let Q be a quadrilateral with side lengths ℓ_i, $i=1,\ldots,4$, and diagonals of lengths d_1 and d_2 forming an angle $\theta\in(0,\pi/2]$, where d_2 is the length of...

open_problem | open | MathDB 页面 | 原文线索

V2-075713 | MATHDB-395552

Area-dependent spectral gap conjecture for planar convex domains

摘录 / EXCERPT: Let K be a planar convex domain, let d denote its diameter, let A denote its area, and let B_1 be the ball of unit area. With $\gamma(K):=\lambda_2(K)-\lambda_1(K)$ for the...

conjecture | open | MathDB 页面 | 原文线索

V2-075714 | MATHDB-395553

Isoperimetric inequality for the fundamental tone of a free plate under tension

摘录 / EXCERPT: Let $\a9\first be a smoothly bounded region in $a1^d$, with $dge 2$, and let $a9^$ be the ball having the same volume as $a9$. For $c40...

open_problem | open | MathDB 页面 | 原文线索

V2-075715 | MATHDB-395554

Payne-Pólya-Weinberger conjecture for ratios of consecutive Dirichlet eigenvalues

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be an arbitrary bounded domain, and let $\lambda_j(\Omega)$ denote the Dirichlet eigenvalues of $-\Delta$ on Ω. Prove that, for every m, ...

conjecture | open | MathDB 页面 | 原文线索

V2-075716 | MATHDB-395555

Payne-Pólya-Weinberger-type bound for doubled Dirichlet eigenvalue indices

摘录 / EXCERPT: For an arbitrary bounded domain $\Omega\subset\mathbb{R}^n$, prove that, for all m, ... Here λ_j denotes the j-th Dirichlet eigenvalue of the Laplacian on Ω.

open_problem | open | MathDB 页面 | 原文线索

V2-075717 | MATHDB-395556

Payne-Pólya-Weinberger conjecture for the sum of the first n eigenvalue ratios

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded domain, and let $0<\lambda_1(\Omega)\leq\lambda_2(\Omega)\leq\cdots$ be the Dirichlet Laplacian eigenvalues. Prove that, for $n\geq 2$,...

conjecture | open | MathDB 页面 | 原文线索

V2-075718 | MATHDB-395557

Sharp upper bound for the first Dirichlet eigenvalue in terms of area and perimeter

摘录 / EXCERPT: Conjecture. For any planar simply connected domain Ω with area A and perimeter L, ... where $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue and $j_{0,1}$ is the firs...

open_problem | open | MathDB 页面 | 原文线索

V2-075719 | MATHDB-395558

Minimization of the sum of the second and third Dirichlet eigenvalues

摘录 / EXCERPT: Prove that the disk minimizes the sum $\lambda_2+\lambda_3$ of the second and third eigenvalues of the Laplace-Dirichlet operator among domains of given area.

open_problem | open | MathDB 页面 | 原文线索

V2-075720 | MATHDB-395559

Polygonal Faber-Krahn inequality

摘录 / EXCERPT: Prove that the regular n-gon minimizes the first eigenvalue of the Laplace-Dirichlet operator among polygons with n sides and given area.

open_problem | open | MathDB 页面 | 原文线索

V2-075721 | MATHDB-395560

Existence of minimizers for higher Dirichlet eigenvalues

摘录 / EXCERPT: Let $ \Omega\subset\mathbb{R}^N$, $N\ge 2$, be a domain of finite volume V. For each $k\in\mathbb{N}$, prove that the isoperimetric problem ... has at least one solution Ω^*...

open_problem | open | MathDB 页面 | 原文线索

V2-075722 | MATHDB-395561

Neumann eigenvalues and loss of measure under Hausdorff convergence

摘录 / EXCERPT: Assume that $(\Omega_n)_{n\in\mathbb N}$ is a sequence of simply connected subsets of the unit ball $B_1\subset\mathbb R^N$, and that $\Omega_*\subset B_1$ satisfies ... and ... in the...

open_problem | open | MathDB 页面 | 原文线索

V2-075723 | MATHDB-395562

Maximization of the first Neumann eigenvalue among planar k-gons

摘录 / EXCERPT: Fix an integer $k\ge 4$. For planar k-gons Ω, maximize $\mu_1(\Omega)A(\Omega)$, where μ_1 is the first positive Neumann eigenvalue of the Laplacian and A is the are...

open_problem | open | MathDB 页面 | 原文线索

V2-075724 | MATHDB-395563

Geometric mean of the first two positive Neumann eigenvalues

摘录 / EXCERPT: For a bounded planar domain Ω with area A, let μ_1 and μ_2 be the first two positive Neumann eigenvalues, and let D be the unit disk. Prove or disprove ... with eq...

open_problem | open | MathDB 页面 | 原文线索

V2-075725 | MATHDB-395564

Dirichlet spectral-gap minimization among triangles

摘录 / EXCERPT: For a planar triangle Ω, let d be its diameter and let $\lambda_1(\Omega)\leq \lambda_2(\Omega)$ be the first two Dirichlet eigenvalues. Determine the minimum of ... among a...

open_problem | open | MathDB 页面 | 原文线索

V2-075726 | MATHDB-395565

Optimal window placement in a square

摘录 / EXCERPT: For a square room c(Omegasubsetmathbb{R}^{2}u0005c), among windows $D\subset\partial\Omega$ of prescribed boundary length that minimize the lowest eigenvalue $\lambda(D)$ of the La...

open_problem | open | MathDB 页面 | 原文线索

V2-075727 | MATHDB-395567

Smoothness of local minimizers for the first Dirichlet eigenvalue

摘录 / EXCERPT: On the space of plane domains of fixed area, with distance given by the area of the symmetric difference, is a local minimizer of the first Dirichlet eigenvalue necessarily a smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-075728 | MATHDB-395570

Pólya conjecture for Dirichlet Laplacian eigenvalues

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded domain with $n\ge 2$, and let E_j be the Dirichlet Laplacian eigenvalues defined by ... Is it true that ...

conjecture | open | MathDB 页面 | 原文线索

V2-075729 | MATHDB-395571

Existence of optimal potentials in Lieb-Thirring inequalities

摘录 / EXCERPT: Must an optimal potential V exist for those Lieb-Thirring inequalities in which the best constant is not known? In particular, this question is open for $n=1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075730 | MATHDB-395572

Bound of all bound-state moments by the ground-state energy

摘录 / EXCERPT: In the report’s setup, let $E_1<E_2\le E_3\le\cdots\le 0$ be the eigenvalues of $-\Delta-V$ on $L^2(\mathbb{R}^n)$. Does there exist a bound of the form ... Here C could depend on...

open_problem | open | MathDB 页面 | 原文线索

V2-075731 | MATHDB-395574

Pólya’s conjecture for tiling domains with magnetic fields

摘录 / EXCERPT: For $\b3=0$, does the Pólya conjecture hold true for tiling domains in the presence of magnetic fields? The report answers this negatively for constant magnetic fields.

conjecture | open | MathDB 页面 | 原文线索

V2-075732 | MATHDB-395575

Magnetic counterexample to the Lieb-Thirring conjecture in dimension 3

摘录 / EXCERPT: Can one construct a counterexample to the Lieb-Thirring conjecture that the optimal constant is the classical one for $n=3,\gamma=1$, by using a suitable magnetic field?

conjecture | open | MathDB 页面 | 原文线索

V2-075733 | MATHDB-395577

Morpurgo’s heat-kernel trace conjecture for the Yamabe operator

摘录 / EXCERPT: Let Y be the Yamabe operator on the round sphere (S^n,g), namely ... where Δ_{S^n} is the Laplace-Beltrami operator. For a positive smooth function W on S^n satisfying...

conjecture | open | MathDB 页面 | 原文线索

V2-075734 | MATHDB-395578

Monotonicity of a conformal heat-trace functional

摘录 / EXCERPT: Let (S^n,g) be the round sphere, let $Y=\Delta_{S^n}+\frac n2\left(\frac n2-1\right)$ be the Yamabe operator, and let W be a positive smooth function on S^n normalized by...

open_problem | open | MathDB 页面 | 原文线索

V2-075735 | MATHDB-395579

Van den Berg’s Gap Conjecture

摘录 / EXCERPT: For every bounded convex domain $\Omega\subset\mathbb{R}^n$ with $n\ge 1$ and every convex potential V on Ω, let $\lambda_1\le\lambda_2\le\cdots$ be the Dirichlet eigenval...

conjecture | open | MathDB 页面 | 原文线索

V2-075736 | MATHDB-395580

Nonattainment of the fundamental-gap infimum

摘录 / EXCERPT: Prove that the gap infimum $\inf_{\Omega\in\mathcal O}(\lambda_2-\lambda_1)$ is not attained, where $\mathcal O$ is the class of convex domains with diameter 1.

open_problem | open | MathDB 页面 | 原文线索

V2-075737 | MATHDB-395581

Monotonicity of potential means along Dirichlet eigenfunctions

摘录 / EXCERPT: For the Dirichlet eigenfunctions u_j of $-\Delta+V$ on a bounded convex domain $\Omega\subset\mathbb{R}^n$ with convex potential V, normalize the eigenfunctions in L^2 and de...

open_problem | open | MathDB 页面 | 原文线索

V2-075738 | MATHDB-395582

Gap inequality for magnetic Schrödinger operators

摘录 / EXCERPT: For magnetic Schrödinger operators, the Gap Conjecture cannot hold as stated because the eigenvalue gap can be reduced to zero by the introduction of a magnetic field. Can one stil...

open_problem | open | MathDB 页面 | 原文线索

V2-075739 | MATHDB-395583

Boundary-extremum conjecture for the Dirichlet eigenfunction ratio

摘录 / EXCERPT: For a convex domain Ω with smooth boundary, does the quotient u_2/u_1 of the first two Dirichlet eigenfunctions attain its maximum and minimum values on $\partial\Omega$?

conjecture | open | MathDB 页面 | 原文线索

V2-075740 | MATHDB-395584

Local well-posedness and local smoothing for a surface water-wave equation with surface tension

摘录 / EXCERPT: Let $S>0$ and $g\ge 0$. Consider ... where $\widehat{Hf}(\xi)=-i\operatorname{sgn}(\xi)\widehat f(\xi)$ and ... For $s>2+\frac12$, the initial-value problem with...

open_problem | open | MathDB 页面 | 原文线索

V2-075741 | MATHDB-395586

Stokes conjecture for irrotational extreme water waves

摘录 / EXCERPT: For irrotational travelling water waves $(\gamma\equiv 0)$, the profile of any extreme wave has corners with included angle 120° at stagnation points.

conjecture | open | MathDB 页面 | 原文线索

V2-075742 | MATHDB-395587

Stokes-conjecture slope alternatives at a stagnation point (Theorem 3)

摘录 / EXCERPT: Let $S,\psi)\u001b be an extreme wave with \(Q=0$, and suppose that the origin is a stagnation point. Assume ... where $\eta:\mathbb R\to\mathbb R$ is continuous and locally of bound...

conjecture | open | MathDB 页面 | 原文线索

V2-075743 | MATHDB-395588

Homogeneity of blow-up limits in the limiting water-wave problem

摘录 / EXCERPT: Any nontrivial solution $(\widetilde S,\widetilde\psi)$ of the limiting problem that arises as a blow-up limit of a solution of the original problem is homogeneous of degree $3/2$...

open_problem | open | MathDB 页面 | 原文线索

V2-075744 | MATHDB-395589

Uniform regularity and breakdown alternatives along global bifurcation curves

摘录 / EXCERPT: If a bifurcation curve is unbounded while remaining in a subset of the function space where the problem is uniformly elliptic and oblique, then there exists a sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-075745 | MATHDB-395590

Four local bifurcation curves for stratified capillary-gravity waves

摘录 / EXCERPT: The report states that, under the Local Bifurcation Condition and additional size assumptions, if the two eigenvalues satisfy $0<n_1<n_2$ and $n_2/n_1\ne 2$, then precisely four...

open_problem | open | MathDB 页面 | 原文线索

V2-075746 | MATHDB-395593

Generation by the intersection tensor-product subsystem

摘录 / EXCERPT: Let a0$E^{\ast}=(E_t)$ be a free product system with $E_s\odot E_t\subseteq E_{s+t}$. For each t, let ... the largest tensor product subsystem specified in the report. Does...

open_problem | open | MathDB 页面 | 原文线索

V2-075747 | MATHDB-395599

Units of n-parameter free product systems

摘录 / EXCERPT: Given n one-parameter free product systems E_i^{*}, form the n-parameter free product system $E_{(t_1,\c5,t_n)}:=E_1^{*}\ast\cdots\ast E_n^{*}$ and the associated E_0-semig...

open_problem | open | MathDB 页面 | 原文线索

V2-075748 | MATHDB-395605

Decomposition of an object with disjoint Thomason support

摘录 / EXCERPT: Let K be a tensor-triangulated category satisfying Hypothesis 7: K is rigid, meaning there is an equivalence $D:K^{\mathrm{op}}\xrightarrow{\sim}K$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-075749 | MATHDB-395607

Classification of thick subcategories of $D_f(A)$

摘录 / EXCERPT: Classify the thick subcategories of $D_f(A)$.

open_problem | open | MathDB 页面 | 原文线索

V2-075750 | MATHDB-395615

Fourth-order sampled-data redesign for single-input control-affine systems

摘录 / EXCERPT: Consider the single-input closed-loop system ... with smooth feedback controller $u:\mathbb R^n\to\mathbb R$, and its sampled-data implementation with a family of controllers...

open_problem | open | MathDB 页面 | 原文线索

V2-075751 | MATHDB-395616

Funnel-control feasibility for a positive Volterra-Stieltjes system

摘录 / EXCERPT: Theorem 6. Let a funnel $F(\sigma,\psi)$ be determined by its boundary $\psi(\cdot)$ with Lipschitz constant $\ell\ge 0$ and lower bound $\lambda>0$, with $\sigma\ge 0$ prescribed....

open_problem | open | MathDB 页面 | 原文线索

V2-075752 | MATHDB-395617

Auxiliary triplet design problem for nonlinear output regulation

摘录 / EXCERPT: Find, if possible, a triplet $\{L(\varphi),M,N(\varphi)\}$ such that the system ... possesses a compact invariant set $\mathcal A$ which is locally exponentially stable, with a domai...

open_problem | open | MathDB 页面 | 原文线索

V2-075753 | MATHDB-395618

Sufficiency of the Hautus test for exact observability of exponentially stable systems

摘录 / EXCERPT: In general, the Hautus test ... is not sufficient for exact observability of exponentially stable systems.

open_problem | open | MathDB 页面 | 原文线索

V2-075754 | MATHDB-395619

The moment problem for rational measures

摘录 / EXCERPT: Given a sequence of complex numbers $c_0,c_1,\dots,c_n$ and a finite-dimensional subspace $\mathcal P\subset C([a,b])$ with basis $(u_0,u_1,\dots,u_n)$, parameterize all positive r...

open_problem | open | MathDB 页面 | 原文线索

V2-075755 | MATHDB-395621

Choice of the expansion point in rational Krylov/Laguerre model reduction

摘录 / EXCERPT: Choose the expansion point in rational Krylov-subspace model reduction so as to obtain a good approximation of the system’s response to arbitrary input signals. The report reformul...

open_problem | open | MathDB 页面 | 原文线索

V2-075756 | MATHDB-395622

Characterization of covariance matrices of stationary reciprocal processes

摘录 / EXCERPT: Characterize the covariance matrix of a reciprocal process of index n; equivalently, determine when a full-rank symmetric block-circulant covariance matrix has a symmetric banded...

open_problem | open | MathDB 页面 | 原文线索

V2-075757 | MATHDB-395624

Achievable bisimulations by control interconnection

摘录 / EXCERPT: Let the plant P be the linear system ... the desired system S be ... and the controller C be ... The interconnection is defined by ... where Π is a permutation matrix. Define the...

open_problem | open | MathDB 页面 | 原文线索

V2-075758 | MATHDB-395625

Validity of circular assume-guarantee reasoning for linear systems

摘录 / EXCERPT: For the feedback interconnection of linear input-output systems, with the external signals specialized to $e_1=e_2=0$, the report states that the circular assume-guarantee-reasonin...

open_problem | open | MathDB 页面 | 原文线索

V2-075759 | MATHDB-395631

Optimal fixed-dimensional coarse space

摘录 / EXCERPT: For a positive-definite weak problem on a Lipschitz domain $\Omega\subset\mathbb{R}^2$ or $\mathbb{R}^3$, with $V=H_0^1(\Omega)$ or $V=H^1(\Omega)$ and energy norm...

open_problem | open | MathDB 页面 | 原文线索

V2-075760 | MATHDB-395632

Best-approximation property of the energy-minimizing sparse interpolation matrix

摘录 / EXCERPT: The report states the following theorem. Let P be the unique solution of the energy minimization problem (1), with the prescribed sparsity pattern, and let $P^{\star}$ denote the...

open_problem | open | MathDB 页面 | 原文线索

V2-075761 | MATHDB-395636

Self-dual tile substitutions satisfy the Pisot conjecture

摘录 / EXCERPT: Theorem 2. Let D be the digit set matrix of a tile substitution. If ... for some $c>0$, $t\in\mathbb{R}^{n}$, and permutation matrix P, then the tile substitution fulfils the Pis...

conjecture | open | MathDB 页面 | 原文线索

V2-075762 | MATHDB-395640

Relative Pisot conjecture for absence of eigenvalues

摘录 / EXCERPT: Let $\sigma:A\to A^*$ be a primitive substitution, let $u\in A^{\mathbb N}$ be a fixed point of some power of σ, and let $\Omega=\{\omega\in A^{\mathbb Z}:F_\omega=F_u\}$. G...

conjecture | open | MathDB 页面 | 原文线索

V2-075763 | MATHDB-395641

Pisot Conjecture for unimodular Pisot substitutions

摘录 / EXCERPT: Let σ be a unimodular Pisot substitution on $A=\{1,\ldots,d\}$, meaning that its incidence matrix M has an irreducible characteristic polynomial whose dominant root is a P...

conjecture | open | MathDB 页面 | 原文线索

V2-075764 | MATHDB-395642

Automorphism generating the boundary of a generalized Rauzy fractal

摘录 / EXCERPT: For a substitution σ on d letters satisfying the Pisot, unimodular, primitive, and irreducible conditions, if σ is invertible as an endomorphism of the free group,...

open_problem | open | MathDB 页面 | 原文线索

V2-075765 | MATHDB-395651

Mirković-Vilonen torsion-freeness conjecture

摘录 / EXCERPT: Conjecture 2.2 (Mirković-Vilonen). The stalks of $J_{!*}(\lambda,\mathbb{Z})$ are torsion-free.

conjecture | open | MathDB 页面 | 原文线索

V2-075766 | MATHDB-395657

Cluster monomials and simple objects in the categories \(\mathcal C_\ell\)

摘录 / EXCERPT: For arbitrary $\mathfrak g$ and ℓ, every cluster monomial of R_ℓ is the class of a simple object.

open_problem | open | MathDB 页面 | 原文线索

V2-075767 | MATHDB-395659

Batyrev’s embedding conjecture for del Pezzo surfaces

摘录 / EXCERPT: Let T be a universal torsor over a del Pezzo surface X, with torus T. There exists a T-equivariant embedding $\varphi:T\to (G/P)_a$ inducing an isomorphism...

conjecture | open | MathDB 页面 | 原文线索

V2-075768 | MATHDB-395662

Complete bandwidth selection for L^s-norm risks

摘录 / EXCERPT: The problem of bandwidth selection for risks described by L^s-norms is completely solved.

open_problem | open | MathDB 页面 | 原文线索

V2-075769 | MATHDB-395663

Uniform density-independent bounds and bandwidth selection in multivariate density estimation

摘录 / EXCERPT: For the empirical process $\xi_w(t)=\sum_{i=1}^n\bigl(w(t,X_i)-\mathbb{E}w(t,X)\bigr)$, find a uniform upper bound for $\|\xi_w\|_{s,\tau}$ that is independent of the unknown densi...

open_problem | open | MathDB 页面 | 原文线索

V2-075770 | MATHDB-395664

Characterization of exact $\[1$-recovery under sign restrictions

摘录 / EXCERPT: Given a matrix $A\in\ R^{m\\times n}$ with $m<n$, subsets $P_{+},P_{-}\subseteq\robrace{1,\ldots,n}$ satisfying $P_{+}\cap P_{-}=\varnothing$, and an integer sparsity level s, de...

open_problem | open | MathDB 页面 | 原文线索

V2-075771 | MATHDB-395665

Classification boundary for sparse high-dimensional vectors

摘录 / EXCERPT: Let $X=(X_1,\ldots,X_n)$ and $Y=(Y_1,\ldots,Y_m)$ be independent i.i.d. samples from distributions P_X and P_Y on $\mathbb{R}^d$, with $X_i^k=\xi_i^k$ and $Y_j^k=u_k+\eta_j^k$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075772 | MATHDB-395666

Tracy-Widom limit for the greatest-root statistic

摘录 / EXCERPT: Let $A\sim W_p(I,m)$ and $B\sim W_p(I,n)$ be independent, with $m\ge p$, and let $\theta_{1,p}$ be the largest eigenvalue of $(A+B)^{-1}B$. Assume that p is even,...

open_problem | open | MathDB 页面 | 原文线索

V2-075773 | MATHDB-395667

Group Lasso union-support recovery

摘录 / EXCERPT: In the multivariate regression setting considered in [10], the block-regularized $\[1m\ell_1/\ell_2$ Lasso recovers the exact row pattern of B^*-the set of k rows for which...

open_problem | open | MathDB 页面 | 原文线索

V2-075774 | MATHDB-395668

Phase transition for high-dimensional joint support recovery with \(\ell_1/\ell_\infty\)-regularization

摘录 / EXCERPT: For $r=2$ linear regression problems whose supports overlap in a fraction $\alpha\in[0,1]$ of their entries and whose design matrices are drawn from standard Gaussian ensembles, th...

open_problem | open | MathDB 页面 | 原文线索

V2-075775 | MATHDB-395669

Phase structure of the asymptotic rare/weak classification model

摘录 / EXCERPT: For the asymptotic rare/weak model $\operatorname{ARW}(\beta,r,n_p)$, with $\epsilon_p=p^{-\beta}$, $\tau_p=\sqrt{2r\log p}$, $(\beta,r)\in(0,1)^2$, and the number of observations...

open_problem | open | MathDB 页面 | 原文线索

V2-075776 | MATHDB-395670

Exact and robust matrix completion by nuclear-norm minimization

摘录 / EXCERPT: Let $M\in\mathbb{R}^{n_1\times n_2}$ be a fixed matrix of rank $r=O(1)$ with singular value decomposition $M=\sum_{k\in[r]}\sigma_k u_kv_k^*$, where $\sigma_k\ge 0$, and suppose it...

open_problem | open | MathDB 页面 | 原文线索

V2-075777 | MATHDB-395671

Asymptotic efficiency of simple decisions for the compound decision problem

摘录 / EXCERPT: For the compound decision problem with $Y_i\sim F_{\mu_i}$ independently and with $\mu_{1:n}$ known up to a permutation, let $\mathcal D^S_n$ be the class of simple symmetric rule...

open_problem | open | MathDB 页面 | 原文线索

V2-075778 | MATHDB-395672

Existence of solutions to the 2-cocycle gluing problem

摘录 / EXCERPT: Does the 2-cocycle gluing problem always have at least one solution?

open_problem | open | MathDB 页面 | 原文线索

V2-075779 | MATHDB-395673

Existence of centric linking systems for fusion systems

摘录 / EXCERPT: Whether every fusion system has a centric linking system.

open_problem | open | MathDB 页面 | 原文线索

V2-075780 | MATHDB-395690

Control of transfer in saturated fusion systems

摘录 / EXCERPT: For a saturated fusion system $\mathcal F$ on a finite p-group P, find $1\ne W(P)\unlhd P$ such that $[P,\mathcal F]=[P,N_{\mathcal F}(W(P))]$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-075781 | MATHDB-395695

Characterization of optimal plans and dual optimizers by a closed c-cyclically monotone set

摘录 / EXCERPT: Under the preceding hypotheses, if a, b, and c are continuous, then there exists a closed c-cyclically monotone set $\Gamma\subset X\times Y$ such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075782 | MATHDB-395697

Caffarelli regularity under convexity of the supports

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded convex body and let φ be a convex function on Ω satisfying $\varphi|_{\partial\Omega}\equiv 0$ and, in the viscosity sen...

open_problem | open | MathDB 页面 | 原文线索

V2-075783 | MATHDB-395698

Quantitative convergence of rescaled porous-medium solutions

摘录 / EXCERPT: For solutions $\rho(t)$ of the porous-medium equation ... with $m\ge 1-N^{-1}$ and $m>N(N+2)^{-1}$, let $\rho^*(t)$ be the corresponding self-similar solution. Under the suitable sel...

open_problem | open | MathDB 页面 | 原文线索

V2-075784 | MATHDB-395699

Recognition criterion for Miller spaces

摘录 / EXCERPT: For a space X, the following are equivalent: (1) X is a Miller space, meaning that $\operatorname{map}_*(X,K)\simeq *$ for every finite nilpotent CW complex K; (2)...

open_problem | open | MathDB 页面 | 原文线索

V2-075785 | MATHDB-395702

Minimal reducing subspaces of a finite Blaschke-product multiplication operator

摘录 / EXCERPT: Let B be a finite Blaschke product with n zeros, counting multiplicity, and let M_B be the induced multiplication operator on the Bergman space $L^2_a(\[0m\mathbb D)$. The re...

open_problem | open | MathDB 页面 | 原文线索

V2-075786 | MATHDB-395705

Normal extension criterion via the E-Cayley transform

摘录 / EXCERPT: Let $T\in SIC(H^2)$ be densely defined. Then T has an extension in $NIC(H^2)$ if and only if there exists a $W\in PC(H^2)$ such that $D(W)=R(T+E)^\perp$.

open_problem | open | MathDB 页面 | 原文线索

V2-075787 | MATHDB-395708

General Operator Corona Problem

摘录 / EXCERPT: Find a preferably local necessary and sufficient condition for a bounded operator-valued function $F\in H^\infty_{E^*\to E}$ to have a left inverse in $H^\infty_{E^*\to E}$; that i...

open_problem | open | MathDB 页面 | 原文线索

V2-075788 | MATHDB-395710

Finite Codimension Operator Corona Problem

摘录 / EXCERPT: Let $F\in H^{\infty}_{E_{*}\to E}$ satisfy the Corona condition ... where $\delta>0$, and suppose that $\operatorname{codim}\operatorname{Ran}F(z)<\infty$ for $z\in\mathbb D$. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-075789 | MATHDB-395714

Classification of complex exotic quadrics

摘录 / EXCERPT: The report states the folklore conjecture that Hirzebruch surfaces of even type are the only complex exotic quadrics. Here a complex exotic quadric means a complex surface that is...

open_problem | open | MathDB 页面 | 原文线索

V2-075790 | MATHDB-395718

Eventual point-independence of jets generated by multiples of a theta divisor

摘录 / EXCERPT: Let (X,Θ) be a principally polarized abelian surface. For a point $x\in X$, let $s(k\Theta,x)$ be the maximal integer s such that the evaluation map ... is surjective. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-075791 | MATHDB-395721

Rationality and discreteness of F-thresholds

摘录 / EXCERPT: Let R be a regular, F-finite ring of positive characteristic and let $\\mathfrak a$ be an ideal of R. If either (1) R is essentially of finite type over a field, or (2)...

open_problem | open | MathDB 页面 | 原文线索

V2-075792 | MATHDB-395722

Uniqueness condition for optimal solutions of the binomial-ideal linear program

摘录 / EXCERPT: Let k be a field of characteristic zero, let $S=k[x_1,\00,x_n]$, and let $\mathfrak a\subseteq (x_1,\00,x_n)$ be a binomial ideal containing no monomials. Suppose $f_1,\00,f_r$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-075793 | MATHDB-395725

Lex-plus-powers conjecture

摘录 / EXCERPT: Let $S=k[x_1,\dots,x_n]$, and let $F=(f_1,\dots,f_r)\ S$ be a regular sequence of homogeneous elements with $\deg f_i=e_i$ for an increasing sequence $2\le e_1\le\dots\le e_r$. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-075794 | MATHDB-395731

Finite-union conjecture for winning positions in lattice games

摘录 / EXCERPT: If W is the set of winning positions for a lattice game, then W is a finite union of translates of affine semigroups.

conjecture | open | MathDB 页面 | 原文线索

V2-075795 | MATHDB-395732

Finite affine-semigroup decomposition of winning positions in lattice games

摘录 / EXCERPT: Let $C\subseteq\mathbb{Z}^d$ be a fixed pointed affine semigroup, let $\Gamma\subseteq\mathbb{Z}^d\setminus\{0\}$ be a finite rule set with $\mathbb{N}\Gamma$ pointed and...

open_problem | open | MathDB 页面 | 原文线索

V2-075796 | MATHDB-395733

Irrational F-jumping exponents in the non-\[math]\[0mQ-Gorenstein case

摘录 / EXCERPT: If the ring is not [math]$\mathbb{Q}$-Gorenstein, does there exist an irrational [math]F-jumping exponent?

open_problem | open | MathDB 页面 | 原文线索

V2-075797 | MATHDB-395736

Stratification conjecture for the derived category of a complete intersection

摘录 / EXCERPT: Let k be a field and let $A=k[x_1,\\ldots,x_n]/I$, where I is generated by a regular sequence. Set $c=n-\dim A$, let $R=A[\chi_1,\ldots,\chi_c]$, with the χ_i indeterminat...

conjecture | open | MathDB 页面 | 原文线索

V2-075798 | MATHDB-395739

Nagata conjecture for general points in the complex projective plane

摘录 / EXCERPT: Let $s_1,\ldots,s_r$ be general closed points in $\mathbb{P}^2_{\mathbb{C}}$. If $r\ge 10$ and $d/m\le\sqrt{r}$, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-075799 | MATHDB-395742

Erdős’s enumeration conjecture for graphs avoiding a fixed subgraph

摘录 / EXCERPT: For a graph H containing a cycle, Erdős conjectured that the number $f_n(H)$ of labeled H-free graphs on n vertices satisfies ... as $n\to\infty$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-075800 | MATHDB-395745

Required strength of universal hash classes for cuckoo hashing

摘录 / EXCERPT: It is an open problem to determine how strong universal classes are needed to guarantee that cuckoo hashing works.

open_problem | open | MathDB 页面 | 原文线索

V2-075801 | MATHDB-395746

Largest induced trees in connected triangle-free graphs

摘录 / EXCERPT: Let $t(G)$ be the maximum number of vertices in an induced subgraph of G that is a tree, and let $t_3(n)$ be the minimum value of $t(G)$ over all connected triangle-free graphs...

open_problem | open | MathDB 页面 | 原文线索

V2-075802 | MATHDB-395747

Guaranteed induced trees in connected triangle-free graphs

摘录 / EXCERPT: For a graph G, let $t(G)$ denote the maximum number of vertices in an induced subgraph of G that is a tree. Every connected triangle-free graph G on n vertices satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-075803 | MATHDB-395748

Asymptotic order of the largest induced tree in connected triangle-free graphs

摘录 / EXCERPT: For a graph G, let $t(G)$ be the maximum number of vertices in an induced subgraph of G that is a tree, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075804 | MATHDB-395751

A moment inequality for sums of weakly negatively correlated random variables

摘录 / EXCERPT: Let $X_1,X_2,\ldots,X_n$ be real-valued random variables and let p be an even positive integer satisfying ... and ... Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-075805 | MATHDB-395756

Polynomial-space representation and computation of ε-optimal strategies

摘录 / EXCERPT: Can some non-standard representation be devised that enables the representation and computation of ϵ-optimal strategies in polynomial space?

open_problem | open | MathDB 页面 | 原文线索

V2-075806 | MATHDB-395759

Minimal percolating sets in the hypercube for threshold 3

摘录 / EXCERPT: It is easy to show that ... Determine $m(n)$. In particular, is it true that $m(n)=\Theta(n^2)$?

open_problem | open | MathDB 页面 | 原文线索

V2-075807 | MATHDB-395762

Removal property for arbitrary systems of linear equations

摘录 / EXCERPT: Let M be an $\times p$ matrix of integers and let $b\in\mathbb{N}^{\ell}$. For every $\delta>0$, there exists $\varepsilon=\varepsilon(\delta,M,b)>0$ such that, whenever...

open_problem | open | MathDB 页面 | 原文线索

V2-075808 | MATHDB-395764

Asymptotic Vickrey cost for the random non-bipartite matching problem

摘录 / EXCERPT: For a minimum-weight perfect matching in the complete graph K_n with i.i.d. edge weights distributed either uniformly on $[0,1]$ or exponentially with parameter 1, is the expec...

open_problem | open | MathDB 页面 | 原文线索

V2-075809 | MATHDB-395768

Stabilization time of random better-response dynamics

摘录 / EXCERPT: How long does the random better-response dynamics take to stabilize in uncoordinated two-sided matching markets?

open_problem | open | MathDB 页面 | 原文线索

V2-075810 | MATHDB-395770

Lower bound for greedy routing on uniform augmented rings

摘录 / EXCERPT: Giakkoupis and Hadzilacos conjectured a lower bound of $\Omega((\log n)^2/\ell)$ for greedy routing in both the one-sided and the two-sided case, where ℓ is the expected numbe...

open_problem | open | MathDB 页面 | 原文线索

V2-075811 | MATHDB-395771

Tight lower bound for two-sided greedy routing on uniform small-world rings

摘录 / EXCERPT: Consider the token process on cells $-n,-n+1,\ots,n$ with target 0. Initially, X_0 is chosen uniformly from $\{1,\ots,n}$. At each step, independently choose a set...

open_problem | open | MathDB 页面 | 原文线索

V2-075812 | MATHDB-395772

Steady states, boundedness, and global attractivity for the simplified CML model

摘录 / EXCERPT: Consider the system ... ... with boundary condition ... Assume that, for every x, $\alpha(x,\cdot)$ is continuous, positive, and decreasing, and that $\omega(\cdot)$ is bounded. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-075813 | MATHDB-395773

Stability of symmetric multi-peak steady states

摘录 / EXCERPT: Under suitable assumptions on the smooth, 1-periodic, odd orientational angle V, and for $\c3>0$ sufficiently small, all N-peak-like steady states with $N\ge 3$, symmetrically...

open_problem | open | MathDB 页面 | 原文线索

V2-075814 | MATHDB-395774

Optimal habitat configuration for species survival

摘录 / EXCERPT: In a binary periodic environment in which the growth rate $f'_u(x,0)$ takes only two values with a prescribed average, and diffusion is given by the Laplace operator, how should th...

open_problem | open | MathDB 页面 | 原文线索

V2-075815 | MATHDB-395775

Empirical stability: diameter of measures with fixed aggregated data

摘录 / EXCERPT: Given the sequence of aggregated observations $\{a_n\}_{n=1}^{\infty}$ defined by $a_n=\int_{nh}^{(n+1)h}d\mu$, let ... What is $\operatorname{diam}(A)$, and how does it depend on...

open_problem | open | MathDB 页面 | 原文线索

V2-075816 | MATHDB-395776

Local-global principle for homogeneous spaces over a function field of a complete discretely valued field

摘录 / EXCERPT: Let A be a complete discrete valuation ring of rank 1, with fraction field K and residue field k. Let F be a one-variable function field over K. For every rank-1 disc...

open_problem | open | MathDB 页面 | 原文线索

V2-075817 | MATHDB-395777

u-invariant of finitely generated function fields over p-adic fields

摘录 / EXCERPT: Let K be a p-adic field, and let L be a finitely generated function field of transcendence degree m over K. Then $u(L)=2^{m+2}$, where $u(L)$ is the maximal dimension of...

open_problem | open | MathDB 页面 | 原文线索

V2-075818 | MATHDB-395780

Grothendieck-Serre conjecture for adjoint strongly inner groups of types E6 and E7

摘录 / EXCERPT: Let R be a semi-local domain. Assume moreover that either (i) R is regular and contains an infinite perfect field k, or (ii) R is a semi-local ring of several points on a...

conjecture | open | MathDB 页面 | 原文线索

V2-075819 | MATHDB-395783

Excellent-connections conjecture for motives of quadrics

摘录 / EXCERPT: Let Q and P be anisotropic quadrics of dimension n, with P excellent. Identify $\Lambda(Q)=\Lambda(n)=\Lambda(P)$. Then, for $\lambda,\mu\in\Lambda(n)$, if λ and...

conjecture | open | MathDB 页面 | 原文线索

V2-075820 | MATHDB-395788

S-arithmeticity of the groups $G(4,2^n)$

摘录 / EXCERPT: Let $n\ge 3$, let ω_{2^n} be a primitive 2^n-th root of unity, let $K_n=\mathbf Q(\omega_{2^n}+\omega_{2^n}^{-1})$, and let $A_n=\mathbf Z[\omega_{2^n}+\omega_{2^n}^{-1}]$...

open_problem | open | MathDB 页面 | 原文线索

V2-075821 | MATHDB-395789

Gauge dependence of periodic solutions to a nonlinear magnetic Schrödinger equation

摘录 / EXCERPT: Let ... where $u\in L^2_{\mathrm{loc}}(\mathbb R^N,\mathbb C)$, $\nabla u+iAu\in L^2_{\mathrm{loc}}(\mathbb R^N,\mathbb C^N)$, $A\in C^{1,\alpha}(\mathbb R^N,\mathbb R^N)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075822 | MATHDB-395790

Multiplicity of Bloch solutions for a magnetic nonlinear Schrödinger equation

摘录 / EXCERPT: Theorem 2. Assume there exists $\varepsilon_0>0$ such that, for every $|z|<\varepsilon_0$, problem $(\wp_{A+z})$ has a nowhere-vanishing 2π-periodic solution u_z. Then proble...

open_problem | open | MathDB 页面 | 原文线索

V2-075823 | MATHDB-395791

Counterexample to De Giorgi’s conjecture in dimension 9

摘录 / EXCERPT: Let $N=9$, and let $\Gamma=\{x_9=F(x')\}$ be the nonflat Bombieri-De Giorgi-Giusti minimal graph, with dilation $\Gamma_\alpha=\alpha^{-1}\Gamma$. For every sufficiently small...

conjecture | open | MathDB 页面 | 原文线索

V2-075824 | MATHDB-395792

Symmetric quadruple-junction solution for a tetrahedral four-well potential

摘录 / EXCERPT: Let $Q:\mathbb{R}^3\to\mathbb{R}$ satisfy (Q1)-(Q3): Q is nonnegative and of class C^3, invariant under Γ (that is, $Q\circ\gamma=Q$ for every $\gamma\in\Gamma$); Q v...

open_problem | open | MathDB 页面 | 原文线索

V2-075825 | MATHDB-395793

Radial symmetry and monotonicity of least-energy solutions

摘录 / EXCERPT: Under the same conditions that guarantee existence of solutions (as in [1]), all least-energy solutions of (1), $-\Delta u=g(u)$ with $u\in H^1(\mathbb{R}^N)$, are radially symmetr...

open_problem | open | MathDB 页面 | 原文线索

V2-075826 | MATHDB-395795

Structure of the singular set of a C^1 p-minimal graph

摘录 / EXCERPT: Theorem C. Consider a C^1 smooth p-minimal graph over a plane domain Ω. Let p be a singular point in Ω. Then either p is an isolated singular point, i.e., there...

open_problem | open | MathDB 页面 | 原文线索

V2-075827 | MATHDB-395796

Maximal solutions of a singular elliptic Dirichlet problem

摘录 / EXCERPT: Consider ... There exists a maximal solution for every $\lambda>0$. There is a constant $\lambda^*>0$ such that, for $\lambda>\lambda^*$, the maximal solution is positive, whereas, f...

open_problem | open | MathDB 页面 | 原文线索

V2-075828 | MATHDB-395797

Extension of the Lane-Emden asymptotics to the sublinear regime

摘录 / EXCERPT: Do the above-mentioned results also hold when $1<q<p$?

open_problem | open | MathDB 页面 | 原文线索

V2-075829 | MATHDB-395798

Sign-changing bubble towers for the slightly subcritical problem

摘录 / EXCERPT: For the slightly subcritical problem ... where Ω is a smooth bounded domain in $\mathbb{R}^N$, $N\ge 3$, and $\varepsilon>0$ is small, assume $k\ge1$ is an integer. Then there...

open_problem | open | MathDB 页面 | 原文线索

V2-075830 | MATHDB-395799

Tower-of-bubbles solutions in arbitrary smooth bounded domains

摘录 / EXCERPT: Assume $N\ge 5$. Given an integer $k\ge 1$, there exists a number $\bar\lambda_k>0$ such that, if $\lambda>\bar\lambda_k$, there exists $\varepsilon_k>0$ for which, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075831 | MATHDB-395800

Global existence for complex-valued viscous Burgers solutions under an imaginary-part bound

摘录 / EXCERPT: For the viscous Burgers equation ... with initial condition $u(x,0)=u_0(x)$, let $u_0\in L^1(\mathbb{R})$ be complex-valued. If ... then the equation has a global smooth solution u w...

open_problem | open | MathDB 页面 | 原文线索

V2-075832 | MATHDB-395801

Asymptotic behavior of radial minimizers for the Schrödinger-Poisson-Slater problem

摘录 / EXCERPT: For $p\in(18/7,3)$, let u_λ be a minimizer of $I_\lambda\rvert_{H_r^1}$, where ... As $\lambda\to0$, ... where $d(v_\varepsilon,K)\to0$. Here ... $K\subset E$ is the set of minim...

open_problem | open | MathDB 页面 | 原文线索

V2-075833 | MATHDB-395802

Characterization of higher 1-Laplacian eigenfunctions via inner variations

摘录 / EXCERPT: For an open bounded set $\a9\subset\mathbb{R}^n$ with Lipschitz boundary, the report states that it remains open in general to what extent solutions of the single eigenvalue equati...

open_problem | open | MathDB 页面 | 原文线索

V2-075834 | MATHDB-395803

Lane-Emden conjecture for positive classical solutions

摘录 / EXCERPT: Conjecture. For $p,q>0$, the elliptic system ... has no positive classical solution if and only if ...

conjecture | open | MathDB 页面 | 原文线索

V2-075835 | MATHDB-395804

Lane-Emden conjecture in four space dimensions

摘录 / EXCERPT: For every $p,q>0$, the elliptic system ... has no positive classical solution if and only if ...

conjecture | open | MathDB 页面 | 原文线索

V2-075836 | MATHDB-395805

Large-time convergence for a one-dimensional viscous Hamilton-Jacobi equation with degenerate diffusion

摘录 / EXCERPT: For $R>0$, $p>2$, and $1<q<p-1$, consider weak solutions of ... with homogeneous Dirichlet boundary conditions. The report states that there is a one-parameter family of nonnegative...

open_problem | open | MathDB 页面 | 原文线索

V2-075837 | MATHDB-395806

Radial bound states for the symmetric nonlinear Schrödinger system in the full space

摘录 / EXCERPT: For the fully symmetric system ... with radial functions used to restore compactness, the report states that for $\beta\le -1$ the system admits infinitely many radial bound-state so...

open_problem | open | MathDB 页面 | 原文线索

V2-075838 | MATHDB-395807

Multiplicity of solutions for a symmetric nonlinear Schrödinger system

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a smooth domain with $N\le 3$. For the Dirichlet system $-\Delta u+u=u^3+\beta v^2u$, $-\Delta v+v=v^3+\beta u^2v$ in Ω, with $u,v>0$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-075839 | MATHDB-395808

Counterexamples to the Lin-Ni conjecture in convex domains

摘录 / EXCERPT: Suppose that $N\ge 4$. Then there are convex domains $\Omega\subset\mathbb{R}^N$ such that, for any fixed $\lambda>0$, the Neumann problem ... has infinitely many solutions. Thus the...

conjecture | open | MathDB 页面 | 原文线索

V2-075840 | MATHDB-395809

Lin-Ni conjecture for the critical Neumann problem

摘录 / EXCERPT: For a bounded domain $\Omega\subset\mathbb{R}^N$, consider ... The Lin-Ni conjecture stated that, for sufficiently small $\lambda>0$, this problem has only constant solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-075841 | MATHDB-395811

Smooth action of S_5 on a sphere with rank-one isotropy

摘录 / EXCERPT: Does S_5 act smoothly on a sphere with rank 1 isotropy?

open_problem | open | MathDB 页面 | 原文线索

V2-075842 | MATHDB-395812

Juan-Pineda-Leary conjecture for finite models of the virtually cyclic classifying space

摘录 / EXCERPT: Let G be a group for which there is a finite model for $E_{\mathcal{VC}}G$, the classifying space for the family of virtually cyclic subgroups. Then G is virtually cyclic.

conjecture | open | MathDB 页面 | 原文线索

V2-075843 | MATHDB-395814

Vanishing of negative K-theory, the reduced projective class group, and the Whitehead group

摘录 / EXCERPT: Suppose that G is torsionfree. Then $K_n(\mathbb{Z}G)$ for $n\leq -1$, $\widetilde{K}_0(\mathbb{Z}G)$, and $\operatorname{Wh}(G)$ vanish.

open_problem | open | MathDB 页面 | 原文线索

V2-075844 | MATHDB-395817

Class of the cocycle for an undistorted central \(\mathbb{Z}\)-extension

摘录 / EXCERPT: If a central $\mathbb{Z}$-extension of a group G is undistorted, what can we say about the class of the 2-cocycle defining it?

open_problem | open | MathDB 页面 | 原文线索

V2-075845 | MATHDB-395818

Undistorted central \(\mathbb Z\)-extensions and boundedness of the defining cocycle

摘录 / EXCERPT: Let G be a locally compact and compactly generated group. If a central $\mathbb Z$-extension $0\to\mathbb Z\to E\to G\to\{e\}$ is undistorted, what can be said about the class of...

open_problem | open | MathDB 页面 | 原文线索

V2-075846 | MATHDB-395819

Bounded Z-valued cocycles on connected Lie groups

摘录 / EXCERPT: Let G be a connected Lie group. The following conditions are equivalent: 1. The radical $\sqrt{G}$ of G is linear (equivalently, the closure of the commutator subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-075847 | MATHDB-395826

Classification of twisted rabbit branched coverings

摘录 / EXCERPT: For an integer m, write its base-4 expansion as $m=\sum_{i=0}^{\infty}m_i4^i$, where $m_i\in\{0,1,2,3\}$, with almost all $m_i=0$ if $m\ge 0$ and almost all $m_i=3$ if $m<0$. I...

open_problem | open | MathDB 页面 | 原文线索

V2-075848 | MATHDB-395830

Higher-dimensional Deligne conjecture for the fat graph operad

摘录 / EXCERPT: The singular chains of the n-dimensional fat graph operad act on blob cochains.

conjecture | open | MathDB 页面 | 原文线索

V2-075849 | MATHDB-395836

Sparse coarse-grid approximation of a partially evaluated inverse

摘录 / EXCERPT: Is there a sparse $\widetilde L_H$ such that $\Pi_{2,H}L_h^{-1}\Pi_{1,H}^*$ and $(\widetilde L_H)^{-1}$ coincide up to $O(H^\sigma)$, where σ is a suitable consistency order...

open_problem | open | MathDB 页面 | 原文线索

V2-075850 | MATHDB-395837

Validity of the dynamic quasicontinuum approximation

摘录 / EXCERPT: “Then an interesting problem is this: in what sense may the dynamic QC be good?”

open_problem | open | MathDB 页面 | 原文线索

V2-075851 | MATHDB-395838

Conditional parity formula for elliptic-curve ranks

摘录 / EXCERPT: Let E be an elliptic curve over a number field K, and set $F=K(E[2])$. If the Tate-Shafarevich group $X(E/F)$ is finite, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-075852 | MATHDB-395841

Gross-Zagier-type formula for the Néron-Tate height of $Z(f)$

摘录 / EXCERPT: Let V be the rational quadratic space of signature $(1,2)$ consisting of trace-zero 2×2 matrices, with quadratic form $Q(x)=N\det(x)$ for a fixed positive integer N. Cho...

open_problem | open | MathDB 页面 | 原文线索

V2-075853 | MATHDB-395842

Conjecture XT2 for elliptic curves over a number field

摘录 / EXCERPT: Let K be a number field. For every elliptic curve E/K, the dimension $\dim_{\mathbf F_2}X(E/K)[2]$ is even, where $X(E/K)$ denotes the Shafarevich-Tate group.

conjecture | open | MathDB 页面 | 原文线索

V2-075854 | MATHDB-395843

Gross-Kudla conjecture for the height of a modified diagonal cycle

摘录 / EXCERPT: Assume that $F=\mathbb{Q}$ and that π_i corresponds to an elliptic curve $E_i/\mathbb{Q}$, with the three curves having the same square-free conductor N. Assume that the prod...

conjecture | open | MathDB 页面 | 原文线索

V2-075855 | MATHDB-395845

Explicit non-potentially trianguline p-adic representation

摘录 / EXCERPT: Find an explicit example of a finite-dimensional E-linear p-adic representation V of $G_K=\operatorname{Gal}(\overline{\mathbf{Q}}_p/K)$, where $K/\mathbf{Q}_p$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075856 | MATHDB-395849

Newton polygons occurring in an Ekedahl-Oort stratum

摘录 / EXCERPT: Determine which Newton polygons occur in a given Ekedahl-Oort stratum.

open_problem | open | MathDB 页面 | 原文线索

V2-075857 | MATHDB-395850

Improved ramification bound for torsion semistable representations

摘录 / EXCERPT: Let $p>2$, let K be a finite totally ramified extension of $K_0=\operatorname{Frac}W(k)$, where k is a perfect field of characteristic p, and let V be a semistable represen...

open_problem | open | MathDB 页面 | 原文线索

V2-075858 | MATHDB-395855

Main conjecture for the specified noncommutative Iwasawa extensions

摘录 / EXCERPT: Theorem 2.3. With the assumptions on F_∞/F stated above, the tuple $\bigl(\zeta(K_i/F_i)\bigr)_{i\ge 0}$ belongs to Φ_S. By Proposition 2.2, this proves the main conjec...

conjecture | open | MathDB 页面 | 原文线索

V2-075859 | MATHDB-395857

Nonexistence of a section for the affine Reichardt-Lind curve

摘录 / EXCERPT: Let $U/\mathbb{Q}$ be the affine Reichardt-Lind curve defined by $2Y^2=Z^4-17$ with $Y\neq 0$, and let $X/\mathbb{Q}$ be its smooth completion. The fundamental-group extension...

open_problem | open | MathDB 页面 | 原文线索

V2-075860 | MATHDB-395867

Connections in motives of excellent quadrics

摘录 / EXCERPT: All connections between Tate motives which are present in the motive of an excellent quadric are also present in the motive of any anisotropic quadric of the same dimension.

open_problem | open | MathDB 页面 | 原文线索

V2-075861 | MATHDB-395873

Smale’s conjecture on density of Axiom A surface diffeomorphisms

摘录 / EXCERPT: Smale conjectured that Axiom A surface diffeomorphisms are open and dense in $\operatorname{Diff}^{1}(M)$.

conjecture | open | MathDB 页面 | 原文线索

V2-075862 | MATHDB-395874

Connectivity and polytope shape of the invariant fundamental set

摘录 / EXCERPT: Is the invariant fundamental set connected, and is its shape a (possibly non-convex) polytope?

open_problem | open | MathDB 页面 | 原文线索

V2-075863 | MATHDB-395877

Two-sided tail-free two-soliton solutions for general FPU potentials

摘录 / EXCERPT: For general Fermi-Pasta-Ulam potentials, do there exist solutions in which the dispersive tails are absent as time tends both to $+\infty$ and to $-\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-075864 | MATHDB-395879

Dependence of escape rate and limiting a.c.c.i.m. on the hole parameter

摘录 / EXCERPT: Theorem 25 (Dependence on a parameter [DWY1]). Let H_ε, $\varepsilon\ge 0$, be a contracting sequence of holes. Assume:

open_problem | open | MathDB 页面 | 原文线索

V2-075865 | MATHDB-395880

Perfect powers expressible as sums of two cubes

摘录 / EXCERPT: Conjecture. Let $n\ge 3$. The equation $a^3+b^3=c^n$ does not have any non-trivial primitive solutions, where $a,b,c\in\mathbb Z$, a solution is non-trivial if $abc\ne 0$, and it i...

open_problem | open | MathDB 页面 | 原文线索

V2-075866 | MATHDB-395883

Expected power-saving asymptotic for the second moment of twisted Hecke L-functions

摘录 / EXCERPT: Let f be a fixed holomorphic primitive cusp form of integer weight $k_f\ge 1$, level $D_f\ge 1$, and nebentypus ψ_f of modulus D_f, with $(D_f,q)=1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075867 | MATHDB-395889

Visibility of 3-torsion in genus-2 Jacobians

摘录 / EXCERPT: Any 3-torsion element of $X(E/k)$, where E is an elliptic curve over a number field k, can be made visible in the jacobian of a curve of genus 2.

open_problem | open | MathDB 页面 | 原文线索

V2-075868 | MATHDB-395891

Density of equality in the reflection principle for 4-ranks

摘录 / EXCERPT: Let C_d denote the narrow ideal class group of $\\mathbb{Q}(\\sqrt d)$, and let $\\operatorname{rk}_4(C_d):=\\operatorname{rk}_2(C_d^2)$. Let $A^-(X)$ be the set of negative fund...

open_problem | open | MathDB 页面 | 原文线索

V2-075869 | MATHDB-395903

Rank-three conjecture for seven elliptic curves with torsion subgroup $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/8\mathbb{Z}$

摘录 / EXCERPT: For each λ in … let … The conjecture is that $E_\lambda(\mathbb{Q})$ has Mordell-Weil rank 3 for all seven values of λ.

conjecture | open | MathDB 页面 | 原文线索

V2-075870 | MATHDB-395904

Determining the rank of a specific elliptic curve with torsion subgroup \(\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/8\mathbb{Z}\)

摘录 / EXCERPT: For $\lambda=15/272$, determine the Mordell-Weil rank of the elliptic curve … where … The report gives an upper bound of 3 and two independent points of infinite order, so the un...

open_problem | open | MathDB 页面 | 原文线索

V2-075871 | MATHDB-395906

Integral version of Shimura’s conjecture on Petersson inner products

摘录 / EXCERPT: Let F be a totally real field, let f be a Hilbert modular newform over F of weights $(k_1,\ldots,k_d)$ with $k_1\equiv\cdots\equiv k_d\pmod 2$, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-075872 | MATHDB-395908

Greenberg’s Tate-lattice commensurability conjecture

摘录 / EXCERPT: In the setup of the report, let F be a totally real number field of narrow class number one, let N be a square-free ideal of $\mathcal O_F$, let E/F be an elliptic curve of c...

conjecture | open | MathDB 页面 | 原文线索

V2-075873 | MATHDB-395909

Sums of three integer cubes

摘录 / EXCERPT: For a nonzero integer n with $n\not\equiv \pm 4 \pmod 9$, can one write $n=x^3+y^3+z^3$ with $x,y,z\in\mathbb Z$?

open_problem | open | MathDB 页面 | 原文线索

V2-075874 | MATHDB-395910

Modularity of elliptic curves over $\mathbb{Q}(\zeta_5)$

摘录 / EXCERPT: Let $F=\mathbb{Q}(\zeta_5)$. For a given elliptic curve E/F with conductor N, there should exist an automorphic form f on GL_2, also of conductor N, such that .....

open_problem | open | MathDB 页面 | 原文线索

V2-075875 | MATHDB-395912

Three-dimensional stability of Burgers vortices for arbitrary circulation

摘录 / EXCERPT: Determine the three-dimensional stability of Burgers vortices with respect to three-dimensional perturbations for arbitrary circulation numbers, in both the symmetric and non-symme...

open_problem | open | MathDB 页面 | 原文线索

V2-075876 | MATHDB-395915

Parametrization by minimal Lagrangian solutions in complex hyperbolic space

摘录 / EXCERPT: We conjecture that these solutions parametrize a portion of the representation space $\operatorname{Hom}(\pi_1\Sigma,\mathrm{SU}(2,1))/\mathrm{SU}(2,1)$ similar to the subset of th...

open_problem | open | MathDB 页面 | 原文线索

V2-075877 | MATHDB-395916

Gromov hyperbolicity versus CAT\((-1)\)-spaces

摘录 / EXCERPT: Given a Gromov hyperbolic space X, to what extent can X be roughly isometrically embedded into a CAT(-1)-space?

open_problem | open | MathDB 页面 | 原文线索

V2-075878 | MATHDB-395917

De Giorgi’s conjecture for bounded monotone entire Allen-Cahn solutions

摘录 / EXCERPT: Let u be a smooth bounded solution of ... in $\mathbb{R}^N$ satisfying $\partial_{x_1}u>0$ in $\mathbb{R}^N$. Is it true, for $N\le 8$, that every level set $u^{-1}(\lambda)$ is ei...

conjecture | open | MathDB 页面 | 原文线索

V2-075879 | MATHDB-395918

Removal of properness in the genus-zero classification of embedded minimal surfaces

摘录 / EXCERPT: Conjecture: Every complete embedded minimal surface of genus 0 in $\mathbb{R}^3$ is one of the catenoid, the helicoid, or a Riemann example; that is, the word “properly” can be r...

open_problem | open | MathDB 页面 | 原文线索

V2-075880 | MATHDB-395921

Quantitative stability of the anisotropic isoperimetric inequality

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be an open, bounded, convex set containing the origin. Define the dual norm by ... and, for a smooth open set $E\subset\mathbb{R}^n$, define its anisotropi...

open_problem | open | MathDB 页面 | 原文线索

V2-075881 | MATHDB-395922

Convexity of minimizers for small anisotropic crystals

摘录 / EXCERPT: For small masses, all minima of the problem $E\mapsto P_K(E)+\int_E g$, subject to $|E|=m$, are actually convex, with no regularity assumptions on g.

open_problem | open | MathDB 页面 | 原文线索

V2-075882 | MATHDB-395923

Relative minimizers of the Dirichlet integral and area for weaker norms

摘录 / EXCERPT: The report asks what remains true for the converse implication from relative minimality of the Dirichlet integral D to relative minimality of the area A when closeness is measu...

open_problem | open | MathDB 页面 | 原文线索

V2-075883 | MATHDB-395925

Quantization of the concentration energy for higher-order Adams-Moser-Trudinger equations

摘录 / EXCERPT: Let $m>2$ be an integer, let $\Omega\subset\mathbb{R}^{2m}$ be open, bounded, and have smooth boundary, and let $\lambda_k\to0^+$. Suppose (u_k) is a sequence of positive smooth...

open_problem | open | MathDB 页面 | 原文线索

V2-075884 | MATHDB-395926

Standardness of the blow-up limit in the higher-order Liouville equation

摘录 / EXCERPT: The function η_0 obtained as the limit in (5) is a standard solution of (6). Equivalently, there exist $\lambda>0$ and $x_0\in\mathbb{R}^{2m}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-075885 | MATHDB-395927

Gradient estimates via nonlinear potentials for quasilinear equations

摘录 / EXCERPT: Let $p\ge 2$, let $\Omega\subset\mathbb{R}^n$, and let $u\in C^1(\Omega)$ solve ... where $\mu\in L^1(\Omega)$ and a satisfies the growth, ellipticity, continuity, and Dini-type as...

open_problem | open | MathDB 页面 | 原文线索

V2-075886 | MATHDB-395928

Pointwise gradient estimate via nonlinear Wolff potentials

摘录 / EXCERPT: Let $p\ge 2$, and let $u\in C^1(\Omega)$ be a weak solution of ... where $\mu\in L^1(\Omega)$ and a satisfies (for fixed $0<\nu\le 1\le L$, $s\ge0$, and $L_1\ge1$) ... for...

open_problem | open | MathDB 页面 | 原文线索

V2-075887 | MATHDB-395929

Endpoint Lorentz gradient estimate for the model p-Laplacian

摘录 / EXCERPT: For the model equation $-\operatorname{div}(|Du|^{p-2}Du)=\mu$, when $p<n$, the borderline Lorentz-space implication is ...

open_problem | open | MathDB 页面 | 原文线索

V2-075888 | MATHDB-395930

C^1 regularity of Harvey-Lawson solutions of the special Lagrangian equation

摘录 / EXCERPT: Conjecture. Any Harvey-Lawson solution of SLE_θ on a ball B lies in $C^1(B)$ (if φ is sufficiently smooth).

open_problem | open | MathDB 页面 | 原文线索

V2-075889 | MATHDB-395931

Regularity theorem for minimizers in the class $M(\delta)$

摘录 / EXCERPT: The report’s Theorem 3 states that, for some $\delta>0$, if $E(S,u)=\inf_{\widetilde S\in M(\delta)}E$ and S satisfies condition (8), namely … then $S\in C^\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-075890 | MATHDB-395932

Critical points of the Moser-Trudinger energy in the supercritical regime

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded domain and define … There exists $\alpha_1\in(4\pi,8\pi]$ such that, for every $4\pi<\alpha<\alpha_1$, equation … has a pair of solutio…

open_problem | open | MathDB 页面 | 原文线索

V2-075891 | MATHDB-395933

Existence of a transported Brakke-flow limit from phase-field approximations

摘录 / EXCERPT: Let $d=2,3$ and $p>(d+2)/2$. There exists a subsequential limit of the measures … for all $t\in[0,T]$, such that: (a) $\frac1\sigma\mu_t$ is integral; (b) μ_t satisfies Brakke’…

open_problem | open | MathDB 页面 | 原文线索

V2-075892 | MATHDB-395934

Stokes-corner behavior at stagnation points of extreme waves

摘录 / EXCERPT: Let $(S,\\psi)$ be an extreme wave with $Q=0$. Suppose that $S=\{(X,\eta(X)):X\in\mathbb R\}$, where $\eta:\mathbb R\to\mathbb R$ is continuous and locally of bounded variation, an…

open_problem | open | MathDB 页面 | 原文线索

V2-075893 | MATHDB-395935

Existence of extreme waves for nonpositive vorticity (Theorem 1)

摘录 / EXCERPT: Suppose that $\gamma(r) \le 0$ for all $r \in [0,B]$. Let $\{(S_j,B_0,\psi_j,Q_j)\}_{j\u0001ge 1}$ be a sequence of regular waves of type (SMG) such that…

open_problem | open | MathDB 页面 | 原文线索

V2-075894 | MATHDB-395936

Parameter selection for locating resonances in truncated PML approximations

摘录 / EXCERPT: How to choose all of these parameters in such a way to guarantee that the resonances of interest in an actual application have been located remains a difficult question.

open_problem | open | MathDB 页面 | 原文线索

V2-075895 | MATHDB-395937

Convergence of an adaptive finite-element method for elliptic eigenvalue problems

摘录 / EXCERPT: For a scalar symmetric elliptic eigenproblem on a bounded polygonal domain in $\mathbb{R}^{2}$ or polyhedral domain in $\mathbb{R}^{3}$, with homogeneous Dirichlet boundary conditi…

open_problem | open | MathDB 页面 | 原文线索

V2-075896 | MATHDB-395938

Uniform banded approximation of spectral projectors for gapped Hermitian matrices

摘录 / EXCERPT: Let N_b be a fixed positive integer and let $N=N_bN_e$, where the integers N_e form a monotonically increasing sequence. Let $\{H_N\}$ be a sequence of Hermitian $N\times N$ ma…

open_problem | open | MathDB 页面 | 原文线索

V2-075897 | MATHDB-395939

Inverse spectral determination of damping in a viscously damped string

摘录 / EXCERPT: Can one determine the damping properties of a vibrating string from its spectrum, in the context of the viscously damped wave equation … with homogeneous Dirichlet conditions … and…

open_problem | open | MathDB 页面 | 原文线索

V2-075898 | MATHDB-395940

Computer-assisted proof of a photonic band gap

摘录 / EXCERPT: Let $\Omega=(0,1)^2$, let $\varepsilon(x)=1$ for $x\in[1/16,15/16]^2$ and $\varepsilon(x)=5$ in the rest of Ω, and let $I_n=\{\lambda_{k,n}:k\in[-\pi,\pi]^2\}$ denote the sp…

open_problem | open | MathDB 页面 | 原文线索

V2-075899 | MATHDB-395941

Existence of an H-matrix path of iterates

摘录 / EXCERPT: Is there a “path of iterates” within the class of H-matrices?

open_problem | open | MathDB 页面 | 原文线索

V2-075900 | MATHDB-395943

Energy density in the critical two-dimensional Ising model

摘录 / EXCERPT: Let Ω be a Jordan domain, let Ω^δ be its square-grid discretization of mesh δ, and consider the critical Ising model on Ω^δ with $+$ boundary…

open_problem | open | MathDB 页面 | 原文线索

V2-075901 | MATHDB-395944

Self-dual point versus the critical point for the random-cluster model

摘录 / EXCERPT: For the random-cluster process on $\mathbb{Z}^2$, prove the conjecture that its critical point equals the self-dual point …

open_problem | open | MathDB 页面 | 原文线索

V2-075902 | MATHDB-395945

Exponential law for metastable exit times in the random-field Curie-Weiss model

摘录 / EXCERPT: In the random-field Curie-Weiss model, under the hypothesis of Theorem 1, almost surely with respect to the environment, ... for all $\sigma\in A$ and for all $t\in\mathbb{R}_+$.

open_problem | open | MathDB 页面 | 原文线索

V2-075903 | MATHDB-395949

Discrete approximation of Hunt processes by Markov chains

摘录 / EXCERPT: For a Hunt process X on $\mathbb{R}^d$, can X be approximated by a family of Markov chains $X^{(n)}$ on $n^{-1}\mathbb{Z}^d$?

open_problem | open | MathDB 页面 | 原文线索

V2-075904 | MATHDB-395952

Orthogonality of Matérn Gaussian random fields in dimension 4

摘录 / EXCERPT: Determine whether Matérn Gaussian random fields with different parameters are orthogonal in dimension $d=4$; the report states that this case is still open.

open_problem | open | MathDB 页面 | 原文线索

V2-075905 | MATHDB-395953

Sharp minimax risks for estimating the two scales in the constant Gaussian microstructure-noise model

摘录 / EXCERPT: Under the constant-parameter model $Y{i,n}= ... ...

open_problem | open | MathDB 页面 | 原文线索

V2-075906 | MATHDB-395954

Reconstructing neuronal connectivity from noisy, incomplete data

摘录 / EXCERPT: “Reconstructing the connectivity from the available noisy, incomplete data remains a challenging open problem.”

open_problem | open | MathDB 页面 | 原文线索

V2-075907 | MATHDB-395955

Efficient filtering and smoothing of noisy voltage signals on dendritic trees

摘录 / EXCERPT: Construct an efficient filter for optimal filtering and smoothing of noisy voltage signals on dendritic trees. The state is the vector of voltages at all compartments, with realist...

open_problem | open | MathDB 页面 | 原文线索

V2-075908 | MATHDB-395956

Weak limit of normalized power-variation functionals of a Gaussian process

摘录 / EXCERPT: Let $(G_t)_{t\ge 0}$ be a Gaussian process with centered and stationary increments. What is the weak limit of the functionals ... where $\Delta_i^nG=G_{i/n}-G_{(i-1)/n}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-075909 | MATHDB-395957

Asymptotic theory for smoothed absolute-loadings principal components

摘录 / EXCERPT: Under suitable assumptions, what are the asymptotic properties of the smoothed absolute-loadings principal components approach when the sample size n, the number of SNPs p, and...

open_problem | open | MathDB 页面 | 原文线索

V2-075910 | MATHDB-395958

Asymptotic renormalization of the physical density

摘录 / EXCERPT: Theorem 0.3 (Asymptotic renormalization). Let $\\mu=0$ and let $\\nu\in L^2(\mathbb{R}^3)\cap\mathcal{C}$ satisfy, for some integer $m\ge 1$, ... For every $\varepsilon>0$, there exi...

open_problem | open | MathDB 页面 | 原文线索

V2-075911 | MATHDB-395962

Solution of the Infrared Catastrophe Problem in Non-Relativistic QED

摘录 / EXCERPT: The report presents a solution in the standard model of non-relativistic QED: infraparticle scattering states are constructed for Compton scattering after completely removing the i...

open_problem | open | MathDB 页面 | 原文线索

V2-075912 | MATHDB-395966

Minimal-energy theorem for unitaries in noncommutative-torus components

摘录 / EXCERPT: For integers m,n, if W lies in the connected component of $U(A^{\infty}_{\theta})$ containing $u^{m}v^{n}$, then ... with equality if and only if $W=\lambda u^{m}v^{n}$ for some...

open_problem | open | MathDB 页面 | 原文线索

V2-075913 | MATHDB-395967

RKK-equivalence of noncommutative principal torus bundles with the same nontrivial K-theory group bundle

摘录 / EXCERPT: Can one decide, in general, whether two given non-trivial noncommutative principal $\mathbb{T}^n$-bundles $A(X)$ and $B(X)$ are RKK-equivalent when both have non-trivial isomorph...

open_problem | open | MathDB 页面 | 原文线索

V2-075914 | MATHDB-395968

RKK-equivalence of an NCP torus bundle to a commutative principal bundle

摘录 / EXCERPT: Suppose that $A(X)$ is an NCP $\T^n$-bundle over the path connected space X. Then $A(X)$ is $\RKK$-equivalent to a commutative principal $\T^n$-bundle if and only if the associat...

open_problem | open | MathDB 页面 | 原文线索

V2-075915 | MATHDB-395977

Polynomial structure of the perturbative two-point function

摘录 / EXCERPT: The report conjectures that the renormalised planar connected two-point function $G_{\[?]alpha\beta}$ is, at any order, a polynomial with rational coefficients in α, β...

open_problem | open | MathDB 页面 | 原文线索

V2-075916 | MATHDB-395978

Endomotives for real-multiplication noncommutative tori

摘录 / EXCERPT: Is it possible to associate endomotives to real-multiplication noncommutative tori and their rings of Morita endomorphisms?

open_problem | open | MathDB 页面 | 原文线索

V2-075917 | MATHDB-395980

Map-free reformulation of order- n quantization axioms

摘录 / EXCERPT: Can the axioms for an order n quantization be reexpressed without reference to a quantization map Q?

open_problem | open | MathDB 页面 | 原文线索

V2-075918 | MATHDB-395981

Relation between temperature shift and initial martensitic microstructure in mechanically stabilized shape-memory alloys

摘录 / EXCERPT: Can one establish an understandable relation between the temperature shift and the initial martensitic microstructure?

open_problem | open | MathDB 页面 | 原文线索

V2-075919 | MATHDB-395982

Possibility of fully stabilizing martensite against austenite nucleation

摘录 / EXCERPT: Is it possible to obtain a fully stabilized martensite, i.e. to tune the microstructure, the crystallographic orientation, and the dimensions of the specimen such that the austenit...

open_problem | open | MathDB 页面 | 原文线索

V2-075920 | MATHDB-395983

Completeness of the classical jump conditions at a phase boundary

摘录 / EXCERPT: For a smooth, highly non-convex energy density $W:\mathbb{R}^{m\times d}\to\mathbb{R}$ and a smooth strain-discontinuity surface, the classical jump conditions ... where $P=W_F$,...

open_problem | open | MathDB 页面 | 原文线索

V2-075921 | MATHDB-395984

How many conditions are there on a phase boundary?

摘录 / EXCERPT: For a smooth, highly non-convex energy density $W:\mathbb{R}^{m\times d}\to\mathbb{R}$, let $P=W_F$, $[[F]]=F^+-F^-=a\otimes n$, and $[[P]]=P^+-P^-$. In addition to the classical j...

open_problem | open | MathDB 页面 | 原文线索

V2-075922 | MATHDB-395985

Uniform positive definiteness of the force-based QCF operator

摘录 / EXCERPT: For the periodic one-dimensional atomistic/continuum model described on PDF page 32, the atomistic and local QC Hessians are uniformly positive definite as $N\to\infty$ if and only...

open_problem | open | MathDB 页面 | 原文线索

V2-075923 | MATHDB-395986

Conditions on the orientational angle for one- or two-peak steady states

摘录 / EXCERPT: What are the conditions on V for either one or two peaks of equal size to occur?

open_problem | open | MathDB 页面 | 原文线索

V2-075924 | MATHDB-395988

Degeneracy of the quasiconvexified increasingly rigid single-slip model for subquadratic hardening

摘录 / EXCERPT: For all $p\in[1,2)$, one has $W^{qc}_{\varepsilon}(F)=0$ for all $F\in N$, where ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-075925 | MATHDB-395989

Γ-convergence of the rigid-elasticity approximation for single-slip plasticity

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded open Lipschitz domain, and let $s,m\in\mathbb{R}^2$ be orthonormal. Define ... ... and ... Then, for $p=q=2$, the functionals...

open_problem | open | MathDB 页面 | 原文线索

V2-075926 | MATHDB-395990

Periodicity of the entire-space ground state for a nonlocal pattern-forming energy

摘录 / EXCERPT: Is the entire-space ground state (u^*,b^*) associated with the energy ... periodic? Here u^* takes values in $\{\pm1\}$, b^* satisfies $\nabla\cdot b^*=u^*-m$ (with the corresp...

open_problem | open | MathDB 页面 | 原文线索

V2-075927 | MATHDB-395992

Nash mapping for normal surface singularities

摘录 / EXCERPT: For a normal surface singularity (X,O) defined over a field of characteristic 0, is the Nash mapping bijective?

open_problem | open | MathDB 页面 | 原文线索

V2-075928 | MATHDB-395993

Nash problem for simple surface singularities of type E_6

摘录 / EXCERPT: Is the Nash mapping bijective for a simple surface singularity of type E_6?

open_problem | open | MathDB 页面 | 原文线索

V2-075929 | MATHDB-395994

Nash bijectivity for the simple surface singularities E_6, E_7, and E_8

摘录 / EXCERPT: Is the Nash mapping bijective for the simple surface singularities of types E_6, E_7, and E_8?

open_problem | open | MathDB 页面 | 原文线索

V2-075930 | MATHDB-395997

Diagonal entries of A'_{∞} and the Bernstein polynomial

摘录 / EXCERPT: For a reductive linear free divisor $D=h^{-1}(0)\subset V=\mathbb{C}^{n}$, with $f\in V^{*}\setminus D^{*}$ and $A'_{\infty}=\operatorname{diag}(0,1,\ldots,n-1)-A_{\infty}$ as in T...

open_problem | open | MathDB 页面 | 原文线索

V2-075931 | MATHDB-395999

General computation of the polynomial P_α in Thom-series generating functions

摘录 / EXCERPT: For a contact singularity type α, compute the polynomial $P_\alpha(t_1,\ldots,t_\mu)$ in general, where it occurs in the rational generating-series formula ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-075932 | MATHDB-396010

Special parameters yielding a Hilbert square

摘录 / EXCERPT: Find specific σ for which Y_σ is actually isomorphic to an $S[2]$.

open_problem | open | MathDB 页面 | 原文线索

V2-075933 | MATHDB-396014

Abelianisations of stable orthogonal groups

摘录 / EXCERPT: Let L be an even lattice containing at least two hyperbolic planes such that $\rank_2(L)\ge 6$ and $\rank_3(L)\ge 5$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-075934 | MATHDB-396015

Full strong exceptional collections on smooth toric weak Fano stacks

摘录 / EXCERPT: Let Σ be a stacky fan in $N=\mathbb{Z}^d$, namely, a rational simplicial fan together with a lattice point for each ray, and let X_Σ be the associated d-dimensional...

open_problem | open | MathDB 页面 | 原文线索

V2-075935 | MATHDB-396016

Cohen-Macaulayness of fibers with log canonical fibers

摘录 / EXCERPT: Let $\phi:X\to B$ be a flat projective morphism such that all fibers are log canonical. Under this hypothesis, if one fiber of ϕ is Cohen-Macaulay, then all the fibers are Coh...

open_problem | open | MathDB 页面 | 原文线索

V2-075936 | MATHDB-396019

Quasi-monomiality of a valuation minimizing the discrepancy expression

摘录 / EXCERPT: Let ν be a valuation of the function field of X centered at x, such that for every other such valuation μ (that can be taken quasi-monomial) with the property that...

open_problem | open | MathDB 页面 | 原文线索

V2-075937 | MATHDB-396026

Classification of doubly periodic minimal tori with parallel ends

摘录 / EXCERPT: If M is a doubly periodic minimal surface of $\cstrong\eR^3\c/strong\e$ with parallel ends and genus one in the quotient, then $M\in\mathcal K$, where $\mathcal K$ is the family...

open_problem | open | MathDB 页面 | 原文线索

V2-075938 | MATHDB-396028

Convex Curve Conjecture

摘录 / EXCERPT: Two convex Jordan curves in parallel planes cannot bound a compact minimal surface of positive genus.

conjecture | open | MathDB 页面 | 原文线索

V2-075939 | MATHDB-396029

Connected Graph Conjecture

摘录 / EXCERPT: A minimal graph in $\mathbb{R}^3$ with zero boundary values over a proper, possibly disconnected domain in $\mathbb{R}^2$ can have at most two non-planar components. If the graph a...

conjecture | open | MathDB 页面 | 原文线索

V2-075940 | MATHDB-396032

One-Flux Conjecture

摘录 / EXCERPT: Conjecture 12 (One-Flux Conjecture, Meeks, Pérez, Ros). Let $M\in\mathcal C$, where $\mathcal C$ is the space of connected, complete, embedded minimal surfaces in $\mathbb R^3$, an...

conjecture | open | MathDB 页面 | 原文线索

V2-075941 | MATHDB-396033

Scherk Uniqueness Conjecture

摘录 / EXCERPT: If M is a connected, properly immersed minimal surface in $\mathbb{R}^3$ and $\operatorname{Area}(M \cap B(R)) \le 2\pi R^2$ holds in balls $B(R)$ of radius R, then M is a pl...

conjecture | open | MathDB 页面 | 原文线索

V2-075942 | MATHDB-396035

Unique Limit Tangent Cone Conjecture

摘录 / EXCERPT: Conjecture 14 (Unique Limit Tangent Cone Conjecture, Meeks). If $M\in P$ is not a plane and has quadratic area growth, then $\lim_{t\to\infty}\frac{1}{t}M$ exists and is a minimal,...

conjecture | open | MathDB 页面 | 原文线索

V2-075943 | MATHDB-396036

Embedded Calabi-Yau Conjecture, clause (1)

摘录 / EXCERPT: A connected, open surface M properly minimally embeds in every smooth bounded domain of $\mathbb{R}^3$ as a complete surface if and only if it is orientable and every end of M...

conjecture | open | MathDB 页面 | 原文线索

V2-075944 | MATHDB-396037

Four Point Conjecture

摘录 / EXCERPT: Let $\mathcal C$ be the space of connected, complete, embedded minimal surfaces in $\mathbb R^3$. If $M\in\mathcal C$ and the Gauss map of M omits 4 points on $S^2(1)$, then M...

conjecture | open | MathDB 页面 | 原文线索

V2-075945 | MATHDB-396038

Standard Middle End Conjecture

摘录 / EXCERPT: If $M\in\mathcal M$ and $E\subset M$ is a one-ended representative for a middle end of M, then E is C^0-asymptotic to the end of a plane or catenoid. In particular, if M ha...

conjecture | open | MathDB 页面 | 原文线索

V2-075946 | MATHDB-396039

Finite Genus Properness Conjecture

摘录 / EXCERPT: Let $\mathcal C$ be the space of connected, complete, embedded minimal surfaces in $\mathbb R^3$, and let $\mathcal P\subset\mathcal C$ be the subspace of properly embedded surface...

conjecture | open | MathDB 页面 | 原文线索

V2-075947 | MATHDB-396040

Calabi-Yau half-space conjecture for complete embedded minimal surfaces

摘录 / EXCERPT: Does there exist a complete non-flat embedded minimal surface $\Sigma\subset\mathbb{R}^3$ such that $\Sigma\subset\{x_3\ge 0\}$?

conjecture | open | MathDB 页面 | 原文线索

V2-075948 | MATHDB-396041

Strong chord-arc bound for embedded minimal disks

摘录 / EXCERPT: There exists a constant $C>0$ such that, if $0\in\Sigma$ is an embedded minimal disk and $B_R(0)\subset\Sigma\setminus\partial\Sigma$, then whenever $R/2\ge r_0>0$ and ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075949 | MATHDB-396042

Calabi’s halfspace conjecture for complete minimal immersions

摘录 / EXCERPT: Calabi conjectured that there is no complete, non-flat minimal immersion $F:\Sigma\to\mathbb{R}^3$ such that $F(\Sigma)\subset\{x_3\ge 0\}$, i.e. a complete non-flat minimal immers...

conjecture | open | MathDB 页面 | 原文线索

V2-075950 | MATHDB-396044

Weak chord-arc bound for embedded minimal disks

摘录 / EXCERPT: Let $\Sigma\subset\mathbb{R}^3$ be an embedded minimal disk. Then there exists a constant $\delta>0$, independent of Σ, such that whenever the intrinsic ball...

open_problem | open | MathDB 页面 | 原文线索

V2-075951 | MATHDB-396050

Characterization of properly embedded minimal annuli in \(\mathrm{Nil}_3\)

摘录 / EXCERPT: Can properly embedded annuli in Nil_3 be characterized in general?

open_problem | open | MathDB 页面 | 原文线索

V2-075952 | MATHDB-396051

Harmonic diffeomorphisms from the complex plane to the hyperbolic plane

摘录 / EXCERPT: There exist harmonic diffeomorphisms from the complex plane $\mathbb{C}$ onto the hyperbolic plane $\mathbb{H}^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-075953 | MATHDB-396052

Local energy decay for stationary nontrapping asymptotically flat wave operators

摘录 / EXCERPT: For Case A, where g is a smooth stationary Lorentzian metric on $\mathbb{R}\times\mathbb{R}^3$ with spacelike slices and $g=m+o_r(1)$, and $V=o(r^{-2})$ as $r\to\infty$, prove th...

open_problem | open | MathDB 页面 | 原文线索

V2-075954 | MATHDB-396053

Uniform boundedness for massive waves on slowly rotating Kerr-AdS spacetimes

摘录 / EXCERPT: Fix $\alpha<\frac{9}{4}$, a mass M, and a negative cosmological constant Λ. Then there exists $\tilde a>0$, depending on M and Λ, such that for any fixed Kerr-A...

open_problem | open | MathDB 页面 | 原文线索

V2-075955 | MATHDB-396056

Extension of global well-posedness to non-symmetric data

摘录 / EXCERPT: Can Theorem 1 be extended also to non-symmetric data?

open_problem | open | MathDB 页面 | 原文线索

V2-075956 | MATHDB-396057

Future global nonlinear stability of FLRW-type solutions for the irrotational Euler-Einstein system

摘录 / EXCERPT: For the irrotational Euler-Einstein system (1)-(2) on the background family of FLRW-type solutions described in Section 2, with positive cosmological constant and equation of state...

open_problem | open | MathDB 页面 | 原文线索

V2-075957 | MATHDB-396058

Global areal foliation for non-polarized vacuum T^2-symmetric spacetimes with positive cosmological constant

摘录 / EXCERPT: Let $(\mathcal{M},g)$ be the maximal Cauchy development of vacuum T^2-symmetric initial data with $\Lambda>0$, and suppose that the spacetime is not polarized. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-075958 | MATHDB-396059

“Matter does not matter” in Bianchi type IX cosmologies

摘录 / EXCERPT: The expected principle that “matter does not matter” is that the dynamics of solutions of the Einstein-matter equations is asymptotically the same as the dynamics of vacuum solutio...

open_problem | open | MathDB 页面 | 原文线索

V2-075959 | MATHDB-396060

Uniqueness for small-transverse-traceless conformal data

摘录 / EXCERPT: It is not known whether small-transverse-traceless (small-TT) conformal data determine a unique solution of the Einstein constraint equations.

open_problem | open | MathDB 页面 | 原文线索

V2-075960 | MATHDB-396061

Limiting behavior of Ricci flow on compact surfaces with isolated conic singularities

摘录 / EXCERPT: For an incomplete metric with isolated conic singularities on a compact surface Σ, define the Ricci flow so that the conic singularities and their individual cone angles are...

open_problem | open | MathDB 页面 | 原文线索

V2-075961 | MATHDB-396063

Hitting-time logarithm law for Lipschitz sublevel sets

摘录 / EXCERPT: Proposition 1. If a system has superpolynomial (faster than any power law) decay of correlations with respect to Lipschitz observables, then the time needed for a typical point x...

open_problem | open | MathDB 页面 | 原文线索

V2-075962 | MATHDB-396064

Common-boundary conjecture for the bistable self-consistent Perron-Frobenius operator

摘录 / EXCERPT: In the bistable regime of Theorem 2, the basin of attraction of the fixed point u_0 is conjectured to be the common boundary of the basins of attraction of the two other fixed po...

conjecture | open | MathDB 页面 | 原文线索

V2-075963 | MATHDB-396065

Vanishing Lebesgue measure of continued-fraction sum-level sets

摘录 / EXCERPT: For $n\in\mathbb{N}$, define ... The Fiala-Kleban conjecture asserts that ... where λ denotes Lebesgue measure.

open_problem | open | MathDB 页面 | 原文线索

V2-075964 | MATHDB-396068

Constructive coherence of polynomial rings over Prüfer domains

摘录 / EXCERPT: Provide a constructive proof that if A is a Prüfer domain, then $A[X_1,\ldots,X_n]$ is a coherent ring.

open_problem | open | MathDB 页面 | 原文线索

V2-075965 | MATHDB-396069

Parametrization of linear systems over rings

摘录 / EXCERPT: Given a homogeneous linear system over a ring (not necessarily a field), represented by a matrix operator, can one construct an operator over the same ring whose image is exactly t...

open_problem | open | MathDB 页面 | 原文线索

V2-075966 | MATHDB-396071

Consistency strength of a surjective failure of SCH without the axiom of choice

摘录 / EXCERPT: For a fixed $\\alpha \ge 2$, the following theories are equiconsistent: 1. $\\mathrm{ZFC} + \exists\kappa\,[\kappa\text{ is measurable}]$; 2....

open_problem | open | MathDB 页面 | 原文线索

V2-075967 | MATHDB-396072

Polylogarithmic independence fools AC^0 circuits

摘录 / EXCERPT: It was conjectured that poly-sized constant-depth circuits of unbounded fan-in cannot distinguish between any two polylogarithmically independent distributions; equivalently, every...

open_problem | open | MathDB 页面 | 原文线索

V2-075968 | MATHDB-396073

Independence required to fool bounded-depth AC^0 circuits

摘录 / EXCERPT: How large does $r=r(m,d,\varepsilon)$ have to be in order for every r-independent distribution μ on ${\{0,1\}}^n$ to ε-fool every function F that is computed by...

open_problem | open | MathDB 页面 | 原文线索

V2-075969 | MATHDB-396074

Permanent versus determinant orbit-closure separation

摘录 / EXCERPT: Conjecture 1: For $m<n$ with $n=m^{O(1)}$, $z^{n-m}\operatorname{perm}_m\notin \mathrm{GL}_{n^2}\cdot\det_n$, where $z=x_{1,m+1}$ and the group acts by linear substitution.

open_problem | open | MathDB 页面 | 原文线索

V2-075970 | MATHDB-396075

Polynomial-time decidability of positivity of Kronecker coefficients

摘录 / EXCERPT: Conjecture 2. Deciding whether the Kronecker coefficient $g_{\bb\mu\nu}\ne 0$ can be done in polynomial time.

open_problem | open | MathDB 页面 | 原文线索

V2-075971 | MATHDB-396076

Finite-field Kakeya conjecture

摘录 / EXCERPT: Let $\mathbb{F}$ be a finite field of size q. A set $K\subseteq\mathbb{F}^n$ is Kakeya if, for every direction $b\in\mathbb{F}^n$, there exists $a\in\mathbb{F}^n$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-075972 | MATHDB-396077

Unconditional 3-query locally decodable code construction

摘录 / EXCERPT: The report presents an unconditional construction of a 3-query locally decodable code with codeword length $\exp(\exp(O(\sqrt{\log n\,\log\log n})))$.

open_problem | open | MathDB 页面 | 原文线索

V2-075973 | MATHDB-396078

Historical exponential-length conjecture for locally decodable codes

摘录 / EXCERPT: For many years it was conjectured that LDCs should have an exponential dependence on n for any constant number of queries.

conjecture | open | MathDB 页面 | 原文线索

V2-075974 | MATHDB-396079

Bulatov et al.’s polymorphism characterization of tractable CSPs

摘录 / EXCERPT: Conjecture 3. Let Λ be a “core”. Exact-Λ is polynomial-time tractable if there exists $F\in\operatorname{Poly}(\Lambda)$ that is “not a junta”; otherwise Exact-...

open_problem | open | MathDB 页面 | 原文线索

V2-075975 | MATHDB-396080

Unique Games characterization of CSP approximation thresholds

摘录 / EXCERPT: For every finite-domain CSP Λ and every $c>0$, is $\alpha_{\Lambda,c}$ the correct approximation threshold for instances of value c on Λ, where...

open_problem | open | MathDB 页面 | 原文线索

V2-075976 | MATHDB-396081

Average maximal sensitivity under products of Boolean functions

摘录 / EXCERPT: For Boolean functions $f_1,\ldots,f_n:\{0,1\}^n\to\{0,1\}$, define ... where ... Suppose $f_1,\ldots,f_n$ satisfy $\operatorname{ams}(f_1,\ldots,f_n)=O((\log n)^d)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075977 | MATHDB-396082

Nonexistence of balanced AC^0 graph properties

摘录 / EXCERPT: Conjecture: There is no balanced AC^0 graph property; that is, there is no AC^0 property of graphs that is invariant under permutations of vertices and holds for between an...

open_problem | open | MathDB 页面 | 原文线索

V2-075978 | MATHDB-396083

Derandomization of PIT for multilinear depth-4 circuits

摘录 / EXCERPT: Does derandomizing polynomial identity testing for depth-4 multilinear arithmetic circuits imply a derandomization of PIT for general multilinear arithmetic circuits?

open_problem | open | MathDB 页面 | 原文线索

V2-075979 | MATHDB-396085

Improving the exponent of polynomial factorization

摘录 / EXCERPT: Improve the exponent of polynomial factorization, ideally to 1.

open_problem | open | MathDB 页面 | 原文线索

V2-075980 | MATHDB-396089

Monotonicity of the expected volume of a random simplex

摘录 / EXCERPT: For convex domains K and L with $K \subset L$, is it true that a random simplex in K-a simplex whose corners are chosen independently at random from K-has smaller expected...

open_problem | open | MathDB 页面 | 原文线索

V2-075981 | MATHDB-396091

Characterization of convex bodies in the class R_{n-2}

摘录 / EXCERPT: Let $E\in K_n^0$ be regular and strictly convex. The only sets in R_{n-2} are 1-tangential bodies of convex bodies lying in R_{n-1} which satisfy ... and their outer parallel bod...

open_problem | open | MathDB 页面 | 原文线索

V2-075982 | MATHDB-396092

Monotonicity of the expected volume of a random simplex in dimension 3

摘录 / EXCERPT: Determine whether, for every pair of 3-dimensional convex bodies $K,L\subseteq\mathbb{R}^3$ with $K\subseteq L$, one has ... Here the X_i are random points from the indicated conve...

open_problem | open | MathDB 页面 | 原文线索

V2-075983 | MATHDB-396093

Monotonicity of the expected volume of a random simplex in dimension three

摘录 / EXCERPT: Let B^3 be the unit ball, let L be the convex hull of the half-ball $\{x\in B^3:x_1\ge 0\}$ and the point $(-\varepsilon,0,0)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-075984 | MATHDB-396094

Unique determination of convex bodies by non-central sections

摘录 / EXCERPT: Let K and L be convex bodies in $\mathbb{R}^n$ containing a sphere of radius t in their interiors. Suppose that ... for every hyperplane H tangent to the sphere. Does this im...

open_problem | open | MathDB 页面 | 原文线索

V2-075985 | MATHDB-396095

Fixed or periodic points of the intersection-body operator

摘录 / EXCERPT: Do there exist any other fixed or periodic points of the intersection-body operator I in S_d, for $d\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-075986 | MATHDB-396096

Characterization of finite-dimensional versions of random variables

摘录 / EXCERPT: Let $X=(X_1,\\ldots,X_n)$ be an n-dimensional version of a random variable Y if there is a function $\\gamma:\mathbb{R}^n\\to\mathbb{R}$ such that $\\gamma(a)>0$ for every...

open_problem | open | MathDB 页面 | 原文线索

V2-075987 | MATHDB-396097

Positive-definite norm-dependent functions imply embedding in L^0

摘录 / EXCERPT: Let K be an origin-symmetric star body in $ obreak\boldsymbol{R}^n$, with $n\ge 2$. Suppose there exists an even, non-constant, continuous function...

open_problem | open | MathDB 页面 | 原文线索

V2-075988 | MATHDB-396098

Misiewicz’s conjecture on positive-definite norm-dependent functions

摘录 / EXCERPT: For an origin-symmetric star body $K\subset\mathbb{R}^n$, let $\Phi(K)$ be the class of continuous functions $f:[0,\infty)\to\mathbb{R}$ such that $f(\|\cdot\|_K)$ is positive defi...

conjecture | open | MathDB 页面 | 原文线索

V2-075989 | MATHDB-396099

Empirical covariance approximation for isotropic convex bodies

摘录 / EXCERPT: Let K be an isotropic convex body in $\mathbb{R}^n$. Given $\varepsilon>0$, how many independent points X_i uniformly distributed on K are needed for the empirical covariance...

open_problem | open | MathDB 页面 | 原文线索

V2-075990 | MATHDB-396100

Brunn-Minkowski inequality for even Minkowski valuations

摘录 / EXCERPT: Let $\Phi_i\in \mathrm{MVal}_i^{+}$, where $i\in\{2,\ldots,n-1\}$. If $K,L\in\mathcal K^n$ have non-empty interiors, then ... If Φ_i maps convex bodies with non-empty interiors...

open_problem | open | MathDB 页面 | 原文线索

V2-075991 | MATHDB-396101

Reconstruction from covariograms

摘录 / EXCERPT: For a convex body K in $\mathbb{R}^n$, construct an approximation to K from a finite number of noisy (that is, error-corrupted) measurements of its covariogram g_K.

open_problem | open | MathDB 页面 | 原文线索

V2-075992 | MATHDB-396102

Phase retrieval for characteristic functions of convex bodies: modulus

摘录 / EXCERPT: Let K be a convex body in $\\mathbb{R}^n$. Construct an approximation to K from a finite number of noisy (that is, taken with error) measurements of $\lvert\widehat{1_K}\rvert$...

open_problem | open | MathDB 页面 | 原文线索

V2-075993 | MATHDB-396103

Covariogram determination of three-dimensional convex bodies

摘录 / EXCERPT: Let K_n^o denote the class of convex bodies in $\a0\mathbb{R}^n$, and let U_n denote the class of convex bodies in $\a0\mathbb{R}^n$ that are determined, up to translation and...

open_problem | open | MathDB 页面 | 原文线索

V2-075994 | MATHDB-396104

Phase retrieval for characteristic functions of convex bodies: squared modulus

摘录 / EXCERPT: Let K be a convex body in $\\mathbb{R}^n$. Construct an approximation to K (or, equivalently, to 1_K) from a finite number of noisy measurements of $|\widehat{1_K}|^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-075995 | MATHDB-396105

Slicing problem

摘录 / EXCERPT: There exists an absolute constant C such that for any convex body K, $L_K \le C$.

open_problem | open | MathDB 页面 | 原文线索

V2-075996 | MATHDB-396106

Isotropy constant bound for polytopes with few vertices

摘录 / EXCERPT: There is an absolute constant $C>0$ such that, for every convex polytope $K\subseteq\mathbb{R}^n$ with N vertices, ...

open_problem | open | MathDB 页面 | 原文线索

V2-075997 | MATHDB-396108

Mahler’s conjecture for convex bodies with many symmetries

摘录 / EXCERPT: Mahler’s conjecture is confirmed for convex bodies whose hyperplanes of symmetry have intersection reduced to a point.

conjecture | open | MathDB 页面 | 原文线索

V2-075998 | MATHDB-396109

The hyperplane conjecture (slicing problem)

摘录 / EXCERPT: There exists an absolute constant $C>0$ such that $L_K\le C$ for every convex body K, where L_K is the isotropic constant defined for an isotropic convex body...

conjecture | open | MathDB 页面 | 原文线索

V2-075999 | MATHDB-396110

Strong polarization problem

摘录 / EXCERPT: For any set $u_1,\ldots,u_n$ of unit vectors in $\mathbb{R}^d$, there exists a unit vector $v\in\mathbb{R}^d$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-076000 | MATHDB-396111

Euclidean-ball maximal-function density question

摘录 / EXCERPT: Let $B(x,r)=x+rB_2^n$. For $A>0$, define ... Is it true that, for any $A>0$, $\operatorname{vol}(\widetilde E_A)=1-o(1)$ as $n\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-076001 | MATHDB-396112

Mahler’s conjecture for origin-symmetric convex bodies

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$, $n\ge 2$, be a convex origin-symmetric body, and let ... be its polar body. Define ... Is it always true that ...

conjecture | open | MathDB 页面 | 原文线索

V2-076002 | MATHDB-396113

Variational characterization of the limiting convex set for Jarník’s polygon problem

摘录 / EXCERPT: Let $D\subset\mathbb{R}^{2}$ be a convex compact set with $0\in\operatorname{int}D$ and $\operatorname{Area}D=1$, and let $r_{0}\in\mathcal{F}^{c}$ be its radial function. Here...

open_problem | open | MathDB 页面 | 原文线索

V2-076003 | MATHDB-396114

Kashin-type embedding from ℓ_p^n into ℓ_r^N

摘录 / EXCERPT: For $0<r<p<2$, with $r\le 1$, and $\eta>0$, there exists an embedding of ℓ_p^n into ℓ_r^N with distortion at most $C(\eta)$, where $N=\lfloor(1+\eta)n\rfloor$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076004 | MATHDB-396115

Characterization of the mixed volume for non-symmetric convex bodies

摘录 / EXCERPT: Conjecture (informally stated in the report). For a fixed dimension n, let $F:(obreak\text{$\mathcal Kn$})^n\to\mathbb R$ satisfy (P1) Minkowski additivity in each variable, (P2)...

open_problem | open | MathDB 页面 | 原文线索

V2-076005 | MATHDB-396116

Connectivity of the escaping set for complex exponential Misiurewicz parameters

摘录 / EXCERPT: Let $E_\lambda(z)=\lambda\exp(z)$, with $\lambda\in\mathbb{C}$, and let $I(E_\lambda)=\{z\in\mathbb{C}:E_\lambda^n(z)\to\infty\text{ as }n\to\infty\}$. If λ is a Misiurewic...

open_problem | open | MathDB 页面 | 原文线索

V2-076006 | MATHDB-396117

Expanding property of boundary points of quasiregular escaping sets

摘录 / EXCERPT: For a quasiregular mapping $f:\mathbb{R}^{n}\to\mathbb{R}^{n}$, do points of $\partial I(f)$ possess the expanding property? That is, for every $x\in\partial I(f)$ and every open s...

open_problem | open | MathDB 页面 | 原文线索

V2-076007 | MATHDB-396118

Connectivity of the wandering domains in Baker’s original example

摘录 / EXCERPT: It is still not known whether the wandering domains in Baker’s original example are finitely or infinitely connected.

open_problem | open | MathDB 页面 | 原文线索

V2-076008 | MATHDB-396119

Simply connected wandering domains contained in the fast escaping set

摘录 / EXCERPT: Does there exist an entire function f for which $A(f)$ contains a simply connected wandering domain, where ... $M(r,f):=\max_{|z|=r}|f(z)|$, and $R>\min_{z\in J(f)}|z|$?

open_problem | open | MathDB 页面 | 原文线索

V2-076009 | MATHDB-396120

Connectivity of Julia sets of transcendental meromorphic maps and weakly repelling fixed points

摘录 / EXCERPT: If the Julia set of a transcendental meromorphic map f is disconnected (equivalently, if the Fatou set $F(f)$ is multiply connected), then f has at least one weakly repelling f...

open_problem | open | MathDB 页面 | 原文线索

V2-076010 | MATHDB-396121

Connectedness of the escaping set for $f(z)=\pi\sinh z$

摘录 / EXCERPT: For $f(z)=\pi\sinh z$, is the escaping set $I(f)=\{z\in\mathbb{C}:f^n(z)\to\infty\text{ as }n\to\infty\}$ connected? The report answers no: $I(f)$ is disconnected.

open_problem | open | MathDB 页面 | 原文线索

V2-076011 | MATHDB-396123

Can the annulus exponent in Theorem 1 tend to infinity?

摘录 / EXCERPT: In the setting of Theorem 1-where f is a transcendental meromorphic function with a direct tract, U is a nested wandering domain, and $A\subset U$ is a domain such that every...

open_problem | open | MathDB 页面 | 原文线索

V2-076012 | MATHDB-396124

Invariant Jordan analytic curves of rational functions

摘录 / EXCERPT: Which Jordan analytic curves γ in the Riemann sphere can be invariant under rational functions?

open_problem | open | MathDB 页面 | 原文线索

V2-076013 | MATHDB-396126

Connectivity of the escaping set for exponential maps with only repelling periodic points

摘录 / EXCERPT: Let $f(z)=\exp(z)+\kappa$ be an exponential map such that all periodic points of f are repelling. Then the escaping set $I(f)$ is connected.

open_problem | open | MathDB 页面 | 原文线索

V2-076014 | MATHDB-396127

Complete invariance of an immediate attracting basin

摘录 / EXCERPT: Let f be an entire function and D an immediate basin of an attracting fixed point. Suppose that D contains all critical and asymptotic values of f. Does it follow that D...

open_problem | open | MathDB 页面 | 原文线索

V2-076015 | MATHDB-396128

Loop in the escaping set of the Fatou function

摘录 / EXCERPT: For the Fatou function $f(z)=z+1+e^{-z}$, does there exist a bounded domain $G\subset\mathbb{C}$ such that $\partial G\subset I(f)$ and $G\cap J(f)\neq\varnothing$?

open_problem | open | MathDB 页面 | 原文线索

V2-076016 | MATHDB-396130

Gauge-function realization of a conformal measure for a parabolic elliptic function

摘录 / EXCERPT: Suppose that $f:\mathbb C\to\mathbb C$ is an elliptic function whose all critical points are attracted either by attracting periodic points or rationally indifferent periodic (para...

open_problem | open | MathDB 页面 | 原文线索

V2-076017 | MATHDB-396132

Spectrum formula for the discrete Bernoulli displacement model when $|\lambda|>2$

摘录 / EXCERPT: For the one-dimensional discrete Bernoulli displacement model on $\ell^2(\mathbb Z)$, with $h_{\omega,\lambda}=h_0+V_\omega$ and $q=\lambda\delta_1$, let $\nu^*_k=\delta_1((-1)^k)$...

open_problem | open | MathDB 页面 | 原文线索

V2-076018 | MATHDB-396137

Characterization of the N-representable set for $m=2$

摘录 / EXCERPT: Characterize the convex set of density matrices called N-representable in the case $m=2$.

open_problem | open | MathDB 页面 | 原文线索

V2-076019 | MATHDB-396138

Block positivity implies a trace inequality

摘录 / EXCERPT: Let $M=(M_{ij})_{i,j=1}^N$ be a Hermitian block matrix with $K\times K$ blocks, so $M_{ij}=M_{ji}^{\dagger}$. Assume it is block positive, meaning ... Then ... equivalently, the second...

open_problem | open | MathDB 页面 | 原文线索

V2-076020 | MATHDB-396140

Quasi-static rectilinear growth of shear bands under mode II loading

摘录 / EXCERPT: The fact that shear bands grow quasi-statically and rectilinearly for remarkably long distances under mode II loading conditions, while the same feature is not observed in the akin...

open_problem | open | MathDB 页面 | 原文线索

V2-076021 | MATHDB-396141

Compatibility of the Gmax and PLS crack-kinking criteria

摘录 / EXCERPT: The conjecture is that the classical “Gmax” and Principle of Local Symmetry (“PLS”) criteria for determining the crack-kinking angle in the described planar linear-elastic brittle-...

open_problem | open | MathDB 页面 | 原文线索

V2-076022 | MATHDB-396142

Non-existence of a global nonnegative stationary solution in the random obstacle model

摘录 / EXCERPT: For the random-obstacle model defined in the report, in dimension $n=1$ and in the large-force regime $F>F_0$ of Theorem 1, there is almost surely no global nonnegative stationary...

open_problem | open | MathDB 页面 | 原文线索

V2-076023 | MATHDB-396143

Scaling laws for non-Euclidean plates and the $W^{2,2}$ isometric immersions of Riemannian metrics

摘录 / EXCERPT: For a smooth tangential metric $[g_{\b1\b\]$ on a bounded simply connected mid-plate $\a9\\\\\\\\\subset\mathbb R^2$, independent of the thickness variable, the report claims and p...

open_problem | open | MathDB 页面 | 原文线索

V2-076024 | MATHDB-396145

3d-Crystallization: FCC and HCP

摘录 / EXCERPT: Let $L\subset\mathbb{R}^3$ be either an fcc or an hcp lattice with lattice parameter 1. For an n-particle configuration $y=(y_1,\ldots,y_n)\in\mathbb{R}^{3\times n}$ and a pote...

open_problem | open | MathDB 页面 | 原文线索

V2-076025 | MATHDB-396146

Characterization of definable subsets of projective space

摘录 / EXCERPT: What is the characterization of all definable sets in P^n?

open_problem | open | MathDB 页面 | 原文线索

V2-076026 | MATHDB-396147

Borel definability of small instances in NIP theories

摘录 / EXCERPT: Assume that T has NIP. If $p(x)\in S(M)$ and $\varphi(x,y)\in L$, is $\{b : \varphi(x,b)\text{ is small for }p\}$ Borel definable over M?

open_problem | open | MathDB 页面 | 原文线索

V2-076027 | MATHDB-396149

Stable fields of finite weight

摘录 / EXCERPT: Conjecture 1. An infinite, stable field of finite weight is separably closed. Here, “of finite weight” means that its generic type has finite weight.

open_problem | open | MathDB 页面 | 原文线索

V2-076028 | MATHDB-396154

Extension of the stable-theory all-\n amalgamation result to simple theories

摘录 / EXCERPT: Does the stable-theory result that one can essentially assume n-amalgamation for all n extend to simple theories?

open_problem | open | MathDB 页面 | 原文线索

V2-076029 | MATHDB-396155

Looijenga’s conjecture on smoothability of cusp singularities

摘录 / EXCERPT: A cusp singularity is smoothable if and only if the exceptional set of the dual cusp lies on a rational surface as an anticanonical divisor.

conjecture | open | MathDB 页面 | 原文线索

V2-076030 | MATHDB-396156

Perfect pairing in Faber’s conjecture for the tautological ring

摘录 / EXCERPT: Is it true that the tautological ring of the moduli space M_g of smooth curves has a perfect pairing between complementary tautological degrees, as predicted by Faber’s conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-076031 | MATHDB-396157

Gorenstein conjecture for the tautological ring in genus 24

摘录 / EXCERPT: Is the tautological ring $R^\bullet(M_{24})$ Gorenstein with socle in degree 22, as asserted by the Gorenstein conjecture?

conjecture | open | MathDB 页面 | 原文线索

V2-076032 | MATHDB-396158

Relations in the even unit-weight case

摘录 / EXCERPT: For the even unit-weight case $\vec w=\frac{1}{n}$, if $n\ne 6$, the simple quadrics-that is, the S_n-orbit of the quadric displayed above-generate the ideal of relations.

open_problem | open | MathDB 页面 | 原文线索

V2-076033 | MATHDB-396161

Göttsche’s universal polynomial for nodal curves

摘录 / EXCERPT: For every integer $r\ge 0$, there exists a universal polynomial $T_r(x,y,z,t)$ of degree r with the following property: given a smooth projective surface X and a $(5r-1)$-very...

open_problem | open | MathDB 页面 | 原文线索

V2-076034 | MATHDB-396163

Rationality of the remaining moduli spaces of plane curves

摘录 / EXCERPT: For $G=SL_3(\mathbb C)$, let $V(d)=\operatorname{Sym}^d(\mathbb C^3)^\vee$ and $C(d)=\mathbb P(V(d))/G$, the moduli space of plane curves of degree d under projectivities. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-076035 | MATHDB-396164

Generic surjectivity of the projection fibres under the covariant S_d

摘录 / EXCERPT: For $d=3n+1$, with $V^{(d)}=\operatorname{Sym}^d(\mathbb{C}^3)^\vee$, the covariant S_d induces a map to $\mathbb{P}(V^{(4)})$. The required genericity statement is that a genera...

open_problem | open | MathDB 页面 | 原文线索

V2-076036 | MATHDB-396165

Conjectural genus-three trace formula for vector-valued Siegel cusp forms

摘录 / EXCERPT: For integers $a\ge b\ge c\ge 0$, the report conjectures that the trace of the Hecke operator $T(p)$ on $S_{a-b,b-c,c+4}$ is … where F is Frobenius and …

open_problem | open | MathDB 页面 | 原文线索

V2-076037 | MATHDB-396167

Determine the nilpotence-order parameter for nodal Burniat surfaces

摘录 / EXCERPT: In Theorem 7, for nodal Burniat surfaces with $K_S^2=4$, the Kuranishi base of the corresponding canonical model is locally analytically isomorphic to...

open_problem | open | MathDB 页面 | 原文线索

V2-076038 | MATHDB-396168

Classification of affinely contravariant valuations on $W^{1,1}(\a0\mathbb{R}^n)$

摘录 / EXCERPT: For $n\ge 3$, let $Z:W^{1,1}(\mathbb{R}^n)\to\mathcal K_c^n$ be a continuous, affinely contravariant valuation, where $\mathcal K_c^n$ is the space of origin-symmetric convex bodie...

open_problem | open | MathDB 页面 | 原文线索

V2-076039 | MATHDB-396169

Determine the minimizer of maximal affine surface area

摘录 / EXCERPT: Determine the convex set $K_{\\min}$ such that, for all n-dimensional convex $K \subset \mathbb{R}^n$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-076040 | MATHDB-396170

Classification of simple weak $SL(n)$-invariant valuations

摘录 / EXCERPT: Conjecture. If $\Phi:\mathcal K_0^n\to\mathbb R$ is an $SL(n)$-invariant, simple weak valuation of first Baire class, homogeneous of degree q, and vanishes on polytopes, then...

open_problem | open | MathDB 页面 | 原文线索

V2-076041 | MATHDB-396171

Density of Minkowski spaces without positive Crofton measures for Busemann area

摘录 / EXCERPT: In the space of n-dimensional Minkowski spaces, there is a dense subset of spaces in which there is no positive Crofton measure for the Busemann $(n-1)$-area.

open_problem | open | MathDB 页面 | 原文线索

V2-076042 | MATHDB-396173

Classification of local extensions of local functionals

摘录 / EXCERPT: Let $\mathcal P$ be the class of convex polytopes in $\mathbb R^d$, and let φ be a local functional on $\mathcal P$, meaning that it has a measurable, measure-valued local...

open_problem | open | MathDB 页面 | 原文线索

V2-076043 | MATHDB-396174

Measuring the additional predictive value of genetic information

摘录 / EXCERPT: How best to measure the additional value of genetic information over perhaps simpler to measure characteristics?

open_problem | open | MathDB 页面 | 原文线索

V2-076044 | MATHDB-396175

Pre-selection procedures for SCAD and adaptive lasso in high-dimensional Cox models

摘录 / EXCERPT: Develop better practical pre-selection procedures for high-dimensional Cox proportional-hazards models so that SCAD and adaptive lasso can be used without the high risk of missing...

open_problem | open | MathDB 页面 | 原文线索

V2-076045 | MATHDB-396176

Global weak solutions for the classical sharp-interface two-phase flow model

摘录 / EXCERPT: For two viscous incompressible Newtonian fluids of equal density in a suitable bounded domain in dimension $d\in\{2,3\}$, does the classical sharp-interface system obtained from eq...

open_problem | open | MathDB 页面 | 原文线索

V2-076046 | MATHDB-396177

Existence of weak solutions for the Stefan problem with anisotropic Gibbs-Thomson law

摘录 / EXCERPT: Let $ \text{(A1)} $ $ \text{$Omegasubsetmathbb{R}^n$ is a bounded domain with C^1 boundary and $T0$}. $ Assume $ \text{(A2)} $ $u_0\in L^\infty(\Omega)\cap H^{1,2}(\Omega)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076047 | MATHDB-396178

Theorem 3.2: convergence of stationary points of the smoothed penalized problem

摘录 / EXCERPT: Consider the smooth penalized problem with bounds $a,b\in L^2(\Omega)$ and a C^1-regularization of the max-operator. Let $\{\alpha\}$ and $\{\varepsilon(\alpha)\}$ satisfy Assump...

open_problem | open | MathDB 页面 | 原文线索

V2-076048 | MATHDB-396179

Convergence of the associated nonlocal functional for regularly varying initial data

摘录 / EXCERPT: Determine whether the additional assumption that an associated nonlocal functional converges sufficiently fast is satisfied for regularly varying data.

open_problem | open | MathDB 页面 | 原文线索

V2-076049 | MATHDB-396181

Common harmonic and 3-harmonic functions

摘录 / EXCERPT: Let u be both harmonic and 3-harmonic, i.e. u satisfies $\[1m\Delta u=0\u001b[0m$ and $\[1m\Delta_3 u=0\u001b[0m$ on the unit ball in $\[1m\mathbb{R}^n\u001b[0m$. Prove that...

open_problem | open | MathDB 页面 | 原文线索

V2-076050 | MATHDB-396182

Foliated Schwarz symmetry of minimizers for general p

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a radial bounded domain, and consider ... For general $p\ne 2$, under the existence assumptions stated in the report-namely, for some $\beta>0$, ei...

open_problem | open | MathDB 页面 | 原文线索

V2-076051 | MATHDB-396186

Boundedness of the centered Hardy-Littlewood maximal gradient at the endpoint $p=1$

摘录 / EXCERPT: Is the operator $u\mapsto |\nabla Mu|$ bounded from $W^{1,1}(\mathbb{R}^n)$ to $L^1(\mathbb{R}^n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076052 | MATHDB-396187

Approximate differentiability of the Hardy-Littlewood maximal function

摘录 / EXCERPT: There exists an integrable function $u\in L^1(\mathbb{R}^n)$ whose Hardy-Littlewood maximal function Mu is not approximately differentiable almost everywhere.

open_problem | open | MathDB 页面 | 原文线索

V2-076053 | MATHDB-396188

Higher-[U+007F]k bubbling solutions in the Trudinger[U+007F]\Moser supercritical problem

摘录 / EXCERPT: For the setup above, with [U+007F]$\Omega\subset\mathbb{R}^{2}[U+007F]$ a smooth bounded domain, the report conjectures that Theorem 2 should hold for every integer [U+007F]$k\ge 1[U+007F]$ when [U+007F]$\Omega[U+007F]$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-076054 | MATHDB-396190

Fractional Hardy inequality with the distance-to-boundary weight on convex domains

摘录 / EXCERPT: Conjecture: For every convex domain $\Omega\subset\mathbb{R}^n$, every $0<\alpha<2$, and every $f\in C_c^\infty(\Omega)$, ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-076055 | MATHDB-396191

Nodal line of the second p-Laplacian eigenfunction on a disk

摘录 / EXCERPT: For Ω a disk and any $p\in(1,\infty)$, the nodal line is a diameter.

open_problem | open | MathDB 页面 | 原文线索

V2-076056 | MATHDB-396192

Best constant and isoperimetric interpretation for the Strauss inequality

摘录 / EXCERPT: Problem 3 asks: What is the best constant in ... where $u\in C_0^\infty(\mathbb{R}^n,\mathbb{R}^n)$ and $\varepsilon(u)=\tfrac12(\nabla u+(\nabla u)^T)$? Is there any isoperimetric i...

open_problem | open | MathDB 页面 | 原文线索

V2-076057 | MATHDB-396194

Optimality of the Laplace-domain estimate for the inverse single-layer operator

摘录 / EXCERPT: Is the estimate ... sub-optimal for some or all domains? Alternatively, should the convergence theory use this more complex bound (9), rather than the simpler bound (3), namely ... for...

open_problem | open | MathDB 页面 | 原文线索

V2-076058 | MATHDB-396198

Multiple Traces Boundary Integral Formulation for Helmholtz Transmission Problems

摘录 / EXCERPT: The report formulates, and claims to establish uniqueness, continuity, and coercivity for, the Helmholtz transmission problem on mutually disjoint isotropic curvilinear Lipschitz p...

open_problem | open | MathDB 页面 | 原文线索

V2-076059 | MATHDB-396199

Uniqueness of the multi-trace formulation for the Helmholtz transmission problem

摘录 / EXCERPT: The solution of the multi-trace formulation is unique.

open_problem | open | MathDB 页面 | 原文线索

V2-076060 | MATHDB-396200

Explicit three-dimensional convergence rate for plane-wave approximation

摘录 / EXCERPT: For a three-dimensional star-shaped domain, determine or prove an explicit positive lower bound, depending only on the shape of the domain, for the parameter λ in the plane...

open_problem | open | MathDB 页面 | 原文线索

V2-076061 | MATHDB-396201

Well-posedness of the mixed formulation for an H(curl)-elliptic problem

摘录 / EXCERPT: For a bounded domain $\Omega\subset\mathbb R^3$ with boundary $\Gamma=\overline{\Gamma_D}\cup\overline{\Gamma_N}$, $\Gamma_D\cap\Gamma_N=\varnothing$, consider ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-076062 | MATHDB-396202

Coercivity of the combined boundary integral operator on a trapping domain

摘录 / EXCERPT: For the non-convex, non-starlike trapping domain in two dimensions considered in [9], is the combined boundary integral operator $A_{k,k}$ coercive?

open_problem | open | MathDB 页面 | 原文线索

V2-076063 | MATHDB-396203

Two-step gauging strategy for the discrete magnetostatic Maxwell problem

摘录 / EXCERPT: Under the report’s assumptions that a0bf a0Ω is contractible, homogeneous tangential Dirichlet conditions are imposed, $\nu\in L^\infty(\Omega)$ satisfies $0<\nu_0\leq \nu$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076064 | MATHDB-396204

Canonical nested separations for higher graph connectivity

摘录 / EXCERPT: For any fixed integer k, find a canonical set of nested k-separations that shapes the graph into a coarse tree structure made of $(k+1)$-connected components.

open_problem | open | MathDB 页面 | 原文线索

V2-076065 | MATHDB-396205

Unfriendly partition for countable graphs without a subdivision of an infinite clique

摘录 / EXCERPT: Let $G=(V,E)$ be a countable graph. If G contains no subdivision of the complete countable graph as a subgraph, then there exists a partition $p:V\to\{1,2\}$ such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-076066 | MATHDB-396206

Clique obstruction for holes in claw-free graphs

摘录 / EXCERPT: Let G be a claw-free graph, and let x and y be two non-adjacent vertices without common neighbours. Then there exists a hole through x and y if and only if no clique sepa...

open_problem | open | MathDB 页面 | 原文线索

V2-076067 | MATHDB-396207

Three-in-a-Tree problem

摘录 / EXCERPT: Given a graph G and three vertices x,y,z, decide whether there exists an induced subtree of G containing x,y,z. The report states that this problem can be solved in...

open_problem | open | MathDB 页面 | 原文线索

V2-076068 | MATHDB-396211

Determining the minimum semidefinite rank of a graph

摘录 / EXCERPT: The problem is to determine $\operatorname{mr}_+(G)$ for each graph G. Here $\operatorname{mr}_+(G)$ is the smallest dimension d in which G has a faithful orthogonal represen...

open_problem | open | MathDB 页面 | 原文线索

V2-076069 | MATHDB-396212

Goldberg-Seymour conjecture for equality in Vizing’s multigraph bound

摘录 / EXCERPT: Let G be a multigraph with maximum edge multiplicity $\\mu(G)\ge 2$. Then ... if and only if there exists an odd subset $S\subseteq V(G)$ with $|S|\ge 3$ such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-076070 | MATHDB-396213

Generic global rigidity of body-and-bar frameworks

摘录 / EXCERPT: For a body-and-bar framework in $\mathbb{R}^d$, the framework is generically globally rigid if and only if it is generically redundantly rigid. Here the bodies are disjoint collect...

open_problem | open | MathDB 页面 | 原文线索

V2-076071 | MATHDB-396215

Coloring $K_{3,k}$-minor-free graphs

摘录 / EXCERPT: There is a computable constant $f(k)$ such that every forbidden minor with respect to 5-colorability in $K_{3,k}$-minor-free graphs has at most $f(k)$ vertices.

open_problem | open | MathDB 页面 | 原文线索

V2-076072 | MATHDB-396216

Complete characterization of finitely forcible graphons

摘录 / EXCERPT: Give a complete characterization of graphons that are finitely forcible, in the sense that they are determined by a finite number of prescribed subgraph densities.

open_problem | open | MathDB 页面 | 原文线索

V2-076073 | MATHDB-396217

Fujita-Kawarabayashi connectivity-keeping subgraph conjecture

摘录 / EXCERPT: For all positive integers k,m, there is a least non-negative integer $f_k(m)$ such that every finite simple k-connected graph G with minimum degree ... contains a connected sub...

conjecture | open | MathDB 页面 | 原文线索

V2-076074 | MATHDB-396218

Existence of connectivity-keeping subgraphs isomorphic to a prescribed tree

摘录 / EXCERPT: For every positive integer k and every tree T, there exists a least non-negative integer $g_k(T)$ such that every k-connected graph G with minimum degree ... contains a subgr...

open_problem | open | MathDB 页面 | 原文线索

V2-076075 | MATHDB-396219

Mader’s connectivity-keeping trees conjecture

摘录 / EXCERPT: For every positive integer k and every tree T, let $g_k(T)$ be the least non-negative integer such that every k-connected graph G with minimum degree...

conjecture | open | MathDB 页面 | 原文线索

V2-076076 | MATHDB-396220

Richter’s bounded maximum-degree conjecture for crossing-critical graphs

摘录 / EXCERPT: For every positive integer k, there exists an integer $D(k)$ such that every k-crossing-critical graph has maximum degree less than $D(k)$, where a graph is k-crossing-critic...

conjecture | open | MathDB 页面 | 原文线索

V2-076077 | MATHDB-396221

Geometric reduction for the cubic-graph crossing-number problem

摘录 / EXCERPT: Can the NP-completeness of computing the crossing number for cubic graphs be proved by a reduction from an NP-complete geometric problem, rather than from the Linear Arrangement pr...

open_problem | open | MathDB 页面 | 原文线索

V2-076078 | MATHDB-396222

Weighted Disjoint S-Paths Problem for Tree-Distance Weights

摘录 / EXCERPT: Given an undirected graph $G=(V,E)$, a terminal set $S\subseteq V$, and a weight function $w:\binom{S}{2}\to\mathbb{N}$, define ... where the maximum is over all sets $\mathcal P$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-076079 | MATHDB-396223

Maximality of tree-distance-representable weights

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected graph, $S\subseteq V$ a set of terminals, and $w:\binom{S}{2}\to\mathbb{N}$ a weight function. A weight function w is representable by tree-distanc...

open_problem | open | MathDB 页面 | 原文线索

V2-076080 | MATHDB-396225

3-colorability of planar graphs with distant triangles

摘录 / EXCERPT: There exists an absolute constant d such that if G is a planar graph and every two triangles in G are at distance at least d, then G is 3-colorable.

open_problem | open | MathDB 页面 | 原文线索

V2-076081 | MATHDB-396226

Erdős-Simonovits minimum-degree question for triangle-free graphs

摘录 / EXCERPT: Every triangle-free graph G with n vertices and minimum degree $\b\delta(G)>n/3\u0007d$ has chromatic number $\b\chi(G)\le 4\u0007d$.

open_problem | open | MathDB 页面 | 原文线索

V2-076082 | MATHDB-396228

Unique Linkage Theorem

摘录 / EXCERPT: For all $k \ge 1$, there exists a value $w(k)$ such that the following holds. Let P be a linkage of order k in a graph G with $V(G)=V(P)$. If there does not exist a linkage...

open_problem | open | MathDB 页面 | 原文线索

V2-076083 | MATHDB-396229

Burnside’s odd-order conjecture

摘录 / EXCERPT: There are no finite simple groups of odd order.

conjecture | open | MathDB 页面 | 原文线索

V2-076084 | MATHDB-396231

W*-superrigidity of Bernoulli actions of groups with relative property (T)

摘录 / EXCERPT: Let $\amma$ be a countable ICC group admitting an infinite normal subgroup $\amma_0$ such that the inclusion $(\amma_0\subset \amma)$ has relative property (T). Let (X_0,μ_0) b...

open_problem | open | MathDB 页面 | 原文线索

V2-076085 | MATHDB-396232

Connes’ rigidity conjecture for property (T) groups

摘录 / EXCERPT: Is $L G_1 \cong L G_2$ equivalent to $G_1 \cong G_2$ in the world of property (T) groups?

conjecture | open | MathDB 页面 | 原文线索

V2-076086 | MATHDB-396234

ZFC construction of a counterexample to Naimark’s problem

摘录 / EXCERPT: Can a counterexample to Naimark’s problem be constructed using [31mZFC+CH[0m, or using [31mZFC[0m alone?

open_problem | open | MathDB 页面 | 原文线索

V2-076087 | MATHDB-396236

Whether every group is sofic

摘录 / EXCERPT: Is every group Γ sofic, where sofic means that for every finite subset $F\subseteq\Gamma$ and every $\varepsilon>0$, there exist an integer $n\ge 1$ and a map...

open_problem | open | MathDB 页面 | 原文线索

V2-076088 | MATHDB-396243

Anderson’s conjecture on pure states of $B(\cell^2)$

摘录 / EXCERPT: Every pure state on $B(\cell^2)$ restricts to a pure state on some atomic masa.

conjecture | open | MathDB 页面 | 原文线索

V2-076089 | MATHDB-396246

Existence of non-factorizable c.p.u.t. maps

摘录 / EXCERPT: There are examples of non-factorizable c.p.u.t. maps on the $n\times n$ matrices for $n\ge 3$.

open_problem | open | MathDB 页面 | 原文线索

V2-076090 | MATHDB-396247

Stability of unitary representations and amenability

摘录 / EXCERPT: Does the conjunction of the two stability properties described in the report characterize amenability for discrete groups: (i) unitary ε-representations are uniformly c...

open_problem | open | MathDB 页面 | 原文线索

V2-076091 | MATHDB-396251

Vanishing quasiconvex envelope in the absence of hardening

摘录 / EXCERPT: If $p=1$ and $ obreak\reepsilon>0$, then the quasiconvex envelope of $W_ obreak\reepsilon$ vanishes identically on N: $W_ obreak\reepsilon^{ obreak\rm qc}(F)=0$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-076092 | MATHDB-396254

Haglund’s Schur-positivity conjecture for specialized integral Macdonald polynomials

摘录 / EXCERPT: For every partition λ, every partition μ, and $k\in\mathbb N$, the Schur coefficient ... belongs to $\mathbb N[q]$. Equivalently, $J_\lambda(X;q,q^k)/(1-q)^{|\lambda|}$ i...

conjecture | open | MathDB 页面 | 原文线索

V2-076093 | MATHDB-396255

A tableau formula for products of symmetric Macdonald polynomials

摘录 / EXCERPT: Given the alcove-walk product formula $P_\mu P_\lambda=\sum_p c_{\mathrm{wt}(p)}(q,t)P_{-w_0\mathrm{wt}(p)}$, where the sum is over alcove walks of the prescribed type contained in...

open_problem | open | MathDB 页面 | 原文线索

V2-076094 | MATHDB-396256

A tableau version of the Macdonald-polynomial product formula

摘录 / EXCERPT: Is there a tableau version of equation (1) that generalizes the Pieri rule in Macdonald [M88, (6.24)]?

open_problem | open | MathDB 页面 | 原文线索

V2-076095 | MATHDB-396258

Deligne’s universal-property conjecture for \(\operatorname{Rep}^{ab}(S_t)\)

摘录 / EXCERPT: Given an abelian tensor category $\mathcal T$ and a tensor functor $F:\operatorname{Rep}(S_t)\to\mathcal T$, either F is an extension of the inclusion functor...

conjecture | open | MathDB 页面 | 原文线索

V2-076096 | MATHDB-396259

Expansion of the maximal M-basis element in the parabolic Kazhdan-Lusztig basis

摘录 / EXCERPT: For the maximal staircase shape $\mu=[n-1,\ldots,1,0]$, the maximal M-basis element expands as ... Here c_{λ} is the signed sum of the boxes between the maximal staircase and...

open_problem | open | MathDB 页面 | 原文线索

V2-076097 | MATHDB-396268

Factorization of the canonical bases for higher-level Fock spaces

摘录 / EXCERPT: For the canonical-basis transition matrices (the “v-decomposition matrices”) associated with the irreducible highest-weight $U_v(\widehat{\mathfrak{sl}}_e)$- and...

open_problem | open | MathDB 页面 | 原文线索

V2-076098 | MATHDB-396270

Charge and energy functions in classical-type crystals

摘录 / EXCERPT: In type C, the conjecture is that for every filling $F\in B_\mu$, the energy function and the newly defined charge agree: $D_{B_\mu}(F)=\operatorname{ch}(F)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-076099 | MATHDB-396272

Bandwidth selection for kernel density estimators

摘录 / EXCERPT: Given an i.i.d. sample $X_1,\ldots,X_n$ from a density f on $\mathbb{R}^d$, a kernel K with $\int K(x)\,dx=1$, and the family of kernel estimators...

open_problem | open | MathDB 页面 | 原文线索

V2-076100 | MATHDB-396273

Separation distance for testing homogeneity of a Poisson process

摘录 / EXCERPT: Under the alternative hypothesis, how far from S_0 should s be to obtain $P(\text{accept }H_0)\le \beta$?

open_problem | open | MathDB 页面 | 原文线索

V2-076101 | MATHDB-396274

Universal aggregation with exact oracle inequalities

摘录 / EXCERPT: In the fixed-design Gaussian regression model $Y_i=\eta(x_i)+\xi_i$, $i=1,\ldots,n$, with independent $\xi_i\sim N(0,\sigma^2)$, deterministic covariates $x_1,\ldots,x_n$, and dict...

open_problem | open | MathDB 页面 | 原文线索

V2-076102 | MATHDB-396275

Oracle properties of aggregation methods for normal-mean estimation

摘录 / EXCERPT: It is unclear whether aggregation methods possess oracle properties similar to or stronger than those of the maximum likelihood empirical Bayes and adaptive soft-threshold estimato...

open_problem | open | MathDB 页面 | 原文线索

V2-076103 | MATHDB-396276

Optimal operator-norm estimation rate for large covariance matrices

摘录 / EXCERPT: Suppose $X_1,\ldots,X_n$ are independent and identically distributed p-variate random variables with covariance matrix $\Sigma=(\sigma_{ij})_{1\le i,j\le p}$. Let ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-076104 | MATHDB-396285

Finite blocking property versus finite branched torus coverings

摘录 / EXCERPT: For general translation surfaces, is having the finite blocking property equivalent to being a finite branched covering of a torus?

open_problem | open | MathDB 页面 | 原文线索

V2-076105 | MATHDB-396288

Number of internal measurements needed for inverse transport

摘录 / EXCERPT: Assuming that σ_a and σ_s depend only on the spatial variable x, how many measurements $H(x)$-corresponding to equally many illuminations $g(x,v)$-are necessary for...

open_problem | open | MathDB 页面 | 原文线索

V2-076106 | MATHDB-396289

Reconstruction on $SO(3)$ from nonredundant circle-integral data

摘录 / EXCERPT: Let Γ be an arbitrary noncontractible plane curve in $SO(3)$. Any square-integrable function f on $SO(3)$ can be explicitly reconstructed from the data $Mf(C(h,gh))$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-076107 | MATHDB-396290

Maximal stably reconstructible region of interest from incomplete sinogram data

摘录 / EXCERPT: If the Radon data $p(\phi,s)$ are known only for all (ϕ,s) in a proper subset of $[0,\pi)\times[-\infty,\infty]$, what is the maximal region of interest that can be stably rec...

open_problem | open | MathDB 页面 | 原文线索

V2-076108 | MATHDB-396291

Exact shape reconstruction by one-step linearization in electrical impedance tomography

摘录 / EXCERPT: Let ... then ... Furthermore, at every point on the boundary of $\operatorname{supp}_{\partial\Omega}(\sigma-\sigma_0)$ having a neighborhood within that outer support on which...

open_problem | open | MathDB 页面 | 原文线索

V2-076109 | MATHDB-396292

Existence of an exact solution to the linearized EIT equation

摘录 / EXCERPT: For the piecewise-analytic conductivities σ and σ_0 considered in Theorem 1, whether there exists an exact solution κ of the linearized equation ...

open_problem | open | MathDB 页面 | 原文线索

V2-076110 | MATHDB-396293

Determination of the contrast function from Maxwell scattering data

摘录 / EXCERPT: At fixed frequency $\omega>0$, consider a nonmagnetic, isotropic medium in a bounded smooth domain $\Omega\subset\mathbb{R}^3$, with electric contrast function ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-076111 | MATHDB-396294

Uniqueness of a single inclusion from backscatter data

摘录 / EXCERPT: Let D be the unit disk and let $\[?m\Omega\subset D$ be a single simply connected inclusion with C^2 boundary and known constant conductivity $\[?m\sigma\ne 0$. Is $\[?m\Omega$...

open_problem | open | MathDB 页面 | 原文线索

V2-076112 | MATHDB-396295

Uniqueness of a non-insulating inclusion from backscatter data

摘录 / EXCERPT: Let D be the unit disk, and let $ Omega\trimmed$ be a single simply connected inclusion with C^2 boundary, $ Omegatrimmedboldsymbol{ e} D$ and constant known conductivity...

open_problem | open | MathDB 页面 | 原文线索

V2-076113 | MATHDB-396296

Continuous numerical estimation of the directional distribution of stationary Poisson cylinder processes

摘录 / EXCERPT: For a stationary Poisson cylinder process Φ^d in dimensions $d=2,3$, with intensity λ and symmetric directional distribution α on S^{d-1}, develop a numerical...

open_problem | open | MathDB 页面 | 原文线索

V2-076114 | MATHDB-396297

Sufficiency of reflection data in one-dimensional inverse scattering

摘录 / EXCERPT: For the one-dimensional Schrödinger equation $\left[-\frac{d^2}{dx^2}+V(x)\right]g(k,x)=k^2g(k,x)$ with real potential $V\in L^1_1(\mathbb R)$, knowledge of the reflection coeffici...

open_problem | open | MathDB 页面 | 原文线索

V2-076115 | MATHDB-396300

Facet connectivity of the fan governing dense parabolic-group orbits

摘录 / EXCERPT: Fix a natural number $t>1$ and a function $h:[U+007F]{1,[U+007F]ldots,t}[U+007F]\to[U+007F]{0,[U+007F]ldots,t-1}[U+007F]$ satisfying $h(i)<i$. For a flag $V_0=[U+007F]{0}[U+007F]\subseteq V_1\subseteq\cdots\subseteq V_t$ with dimension...

open_problem | open | MathDB 页面 | 原文线索

V2-076116 | MATHDB-396305

Horn's inductive characterization of spectra of sums of Hermitian matrices

摘录 / EXCERPT: Let $\operatorname{Horn}(n)=\{(\lambda(A),\lambda(B),\lambda(C)): A,B,C\text{ are Hermitian }n\times n\text{ matrices and }A+B+C=0\}$. For...

open_problem | open | MathDB 页面 | 原文线索

V2-076117 | MATHDB-396307

Invariance of G-complete reducibility under separable field extensions

摘录 / EXCERPT: Let k_1/k be a separable field extension, let G be defined over k, and let $H\leq G$ be k-defined. Then H is G-completely reducible over k if and only if H is G-c...

open_problem | open | MathDB 页面 | 原文线索

V2-076118 | MATHDB-396308

The Giambelli problem for classical homogeneous spaces G/P

摘录 / EXCERPT: For a classical complex Lie group G and parabolic subgroup P, find an explicit formula in $H^*(G/P,\mathbb{Z})$ expressing every Schubert class as a polynomial in the specially...

open_problem | open | MathDB 页面 | 原文线索

V2-076119 | MATHDB-396312

Finite asymptotic dimension of mapping class groups

摘录 / EXCERPT: Let Σ be a surface of finite type and let $\mathrm{MCG}(\Sigma)$ be its mapping class group. Then the asymptotic dimension of $\mathrm{MCG}(\Sigma)$ is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-076120 | MATHDB-396315

Gromov-boundary realization of the tree of copies of \(\mathbb{CP}^2\)

摘录 / EXCERPT: Is $X(\mathbb{CP}^2)$ homeomorphic to the Gromov boundary $\partial\Gamma$ of any hyperbolic group Γ?

open_problem | open | MathDB 页面 | 原文线索

V2-076121 | MATHDB-396316

Isomorphism problem for relatively hyperbolic groups with virtually polycyclic parabolic subgroups

摘录 / EXCERPT: The isomorphism problem is solvable for the class of relatively hyperbolic groups with virtually polycyclic parabolic groups, and which do not split over virtually polycyclic group...

open_problem | open | MathDB 页面 | 原文线索

V2-076122 | MATHDB-396318

Almost automorphisms of trees and lattices

摘录 / EXCERPT: Let $d\ge 2$ be a fixed integer, let T be a non-rooted $(d+1)$-regular tree, and let G be the group of almost automorphisms of T. Then G does not contain any lattice.

open_problem | open | MathDB 页面 | 原文线索

V2-076123 | MATHDB-396320

Cannon’s conjecture for hyperbolic Coxeter groups with spherical boundary

摘录 / EXCERPT: Let Γ be a word hyperbolic Coxeter group whose boundary is homeomorphic to the 2-sphere. Then there is a properly discontinuous, cocompact, and isometric action of...

conjecture | open | MathDB 页面 | 原文线索

V2-076124 | MATHDB-396326

Contractibility of the incompressible spanning-surface complex

摘录 / EXCERPT: Let $IS(L)$ be Kakimizu’s simplicial complex whose vertices are isotopy classes of spanning surfaces for L that are incompressible and $\\partial$-incompressible, but not necessa...

open_problem | open | MathDB 页面 | 原文线索

V2-076125 | MATHDB-396328

Flat Closing consequence for CAT(0) cube complexes

摘录 / EXCERPT: Let X be a locally compact CAT(0) cube complex and let $\a3\leq \operatorname{Aut}(X)$ be a discrete group acting properly and cocompactly. If X splits as a product of n non-...

open_problem | open | MathDB 页面 | 原文线索

V2-076126 | MATHDB-396333

Uniqueness of the equivariant embedded minimal torus in S^3

摘录 / EXCERPT: The Clifford torus is the only equivariant embedded minimal torus in S^3, as conjectured by Lawson in 1971.

open_problem | open | MathDB 页面 | 原文线索

V2-076127 | MATHDB-396335

Unitarizability of CMC loop-group connections for k > 3

摘录 / EXCERPT: For a concrete choice of all parameters in the construction, and when $k>3$, determine whether the connection $d+\[3m\xi$ is unitarizable.

open_problem | open | MathDB 页面 | 原文线索

V2-076128 | MATHDB-396337

Uniqueness of the helicoid

摘录 / EXCERPT: Let Σ be a complete, embedded minimal disk in $\mathbb{R}^3$. Then Σ is the plane or the helicoid.

open_problem | open | MathDB 页面 | 原文线索

V2-076129 | MATHDB-396339

Jenkins-Serrin criterion for ideal polygons in $\mathbb H^2\times\mathbb R$

摘录 / EXCERPT: Let Γ be an ideal polygon of $\mathbb H^2$ bounding a polygonal domain P, with an even number of sides $A_1,B_1,A_2,B_2,\ldots,A_k,B_k$ ordered clockwise. Choose pairwise...

open_problem | open | MathDB 页面 | 原文线索

V2-076130 | MATHDB-396340

Harmonic diffeomorphism from the complex plane to the hyperbolic plane

摘录 / EXCERPT: There exists a harmonic diffeomorphism from $\mathbb{C}$ to $\mathbb{H}^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-076131 | MATHDB-396341

The large genus limit of the infimum of the Willmore energy

摘录 / EXCERPT: For an oriented, closed (compact without boundary), smoothly immersed surface $\Sigma\hookrightarrow\mathbb{R}^n$ with genus p, define ... where $\mathcal C(n,p)$ is the class of s...

open_problem | open | MathDB 页面 | 原文线索

V2-076132 | MATHDB-396342

Monotonicity of the genus-dependent Willmore infimum

摘录 / EXCERPT: It was conjectured that the quantities β_p^n, defined as the infimum of the Willmore energy among oriented, closed, smoothly immersed surfaces of genus p in $\mathbb{R}^n$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076133 | MATHDB-396343

Embedded Calabi-Yau conjecture: existence in some smooth bounded domain

摘录 / EXCERPT: Let M be an open surface. There exists a complete proper minimal embedding of M in some smooth bounded domain $D\subset\mathbb{R}^3$ if and only if the number of nonorientable...

conjecture | open | MathDB 页面 | 原文线索

V2-076134 | MATHDB-396344

Embedded Calabi-Yau conjecture for open surfaces

摘录 / EXCERPT: Let M be an open surface. 1. There exists a complete proper minimal embedding of M in some smooth bounded domain $D\subset\mathbb{R}^3$ if and only if the number of nonorientab...

conjecture | open | MathDB 页面 | 原文线索

V2-076135 | MATHDB-396345

Embedded Calabi-Yau conjecture for the domain \(D_\infty\)

摘录 / EXCERPT: Let D_∞ be the bounded domain in $\mathbb{R}^3$ described in Figure 1, which is smooth except at one point. A necessary and sufficient condition for an open surface M to a...

conjecture | open | MathDB 页面 | 原文线索

V2-076136 | MATHDB-396346

Existence of compact flat surfaces with finitely many isolated singularities in hyperbolic 3-space

摘录 / EXCERPT: Does there exist a compact flat surface in H^3 with a finite number of isolated singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-076137 | MATHDB-396347

Geometric Cauchy problem for $H=1$ surfaces in hyperbolic 3-space

摘录 / EXCERPT: Given a regular curve β in $\mathbb H^3$ and a vector field V along β orthogonal to the curve, find an immersed surface $\Sigma\subset\mathbb H^3$ of constant mean cu...

open_problem | open | MathDB 页面 | 原文线索

V2-076138 | MATHDB-396349

Classification of exceptional domains in the plane

摘录 / EXCERPT: Conjecture: In dimension $m=2$, up to similarity, the only exceptional domains are the half-spaces, the complements of balls, and the domain ... Here an exceptional domain is a smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-076139 | MATHDB-396351

Higher-order variational symmetries associated with Hopf-type differentials

摘录 / EXCERPT: “The question is then: to which higher order variational symmetry do these 1-forms correspond to?” Here “these 1-forms” refers to the two real 1-forms obtained from the square root...

open_problem | open | MathDB 页面 | 原文线索

V2-076140 | MATHDB-396357

Macdonald positivity conjecture

摘录 / EXCERPT: For the transformed Macdonald polynomials $\widetilde H_\mu(z;q,t)$, expanded in the Schur basis as ... all coefficients $\widetilde K_{\lambda,\mu}(q,t)$ belong to $\mathbb N[q,t]$.

conjecture | open | MathDB 页面 | 原文线索

V2-076141 | MATHDB-396359

Families of minors defining nonempty totally nonnegative cells

摘录 / EXCERPT: Which families of minors define nonempty cells in the space of totally nonnegative matrices?

open_problem | open | MathDB 页面 | 原文线索

V2-076142 | MATHDB-396360

Rogalski-Stafford conjecture on birationally commutative surfaces of GK-dimension 4

摘录 / EXCERPT: Conjecture 1 (as stated in the report): Suppose that S is a connected graded domain of GK-dimension 4, generated in degree 1, that is birational to a commutative surface. The...

conjecture | open | MathDB 页面 | 原文线索

V2-076143 | MATHDB-396364

Conjecture 6: finiteness criterion for contraction of curves

摘录 / EXCERPT: Let R be a complete normal Gorenstein 3-fold and suppose there exists a projective birational morphism $Y \xrightarrow{f} X=\operatorname{Spec}R$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-076144 | MATHDB-396373

BEC and loss of mass in infinite cycles for an interacting Bose gas

摘录 / EXCERPT: For fixed $\beta>0$ and dimension $d\ge 3$, determine whether the minimizer(s) P in $\chi^{(\le)}(\beta,\rho)$ satisfy $\langle P,N^{(\ell)}_U\rangle<\rho$ for sufficiently small...

open_problem | open | MathDB 页面 | 原文线索

V2-076145 | MATHDB-396374

Evolution of Markov initial data under scalar conservation laws

摘录 / EXCERPT: For the scalar conservation law ... with a C^1 convex flux function f and random initial data $u(x,0)=u_0(x)$, determine the evolution of the law of u_0 induced by the entropy...

open_problem | open | MathDB 页面 | 原文线索

V2-076146 | MATHDB-396375

Optimal coarsening upper bounds for thin films with superlinear mobility

摘录 / EXCERPT: For the evolution of thin liquid films with superlinear mobility, obtain estimates on the steepness of the energy along the solution paths themselves that are sufficient to prove o...

open_problem | open | MathDB 页面 | 原文线索

V2-076147 | MATHDB-396376

Mixing time of the low-temperature Ising Glauber dynamics with plus boundary conditions

摘录 / EXCERPT: For the single-spin-flip Glauber dynamics of the zero-field Ising model in a d-dimensional cube of side length L, at inverse temperature $\beta>\beta_c$ and with “$+$” boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-076148 | MATHDB-396377

Poisson-Dirichlet cycle structure in the random stirring model on $\mathbb{Z}^d$

摘录 / EXCERPT: Conjecture: Let $d\ge 3$. For the random stirring model on a cubic box with n sites in $\mathbb{Z}^d$, there exists a critical time t_c such that the fraction $\nu=\nu_t$ of si...

open_problem | open | MathDB 页面 | 原文线索

V2-076149 | MATHDB-396378

Control of scale-to-scale changes in normalized harmonic projections at singular points

摘录 / EXCERPT: At each singular point x_0, obtain an estimate controlling, as r varies, the scale-to-scale change ...

open_problem | open | MathDB 页面 | 原文线索

V2-076150 | MATHDB-396380

Derived explanation of non-exponentiating marginal deformations

摘录 / EXCERPT: Can the phenomenon that certain deformations of $N=(2,2)$ CFTs along marginal fields do not exponentiate within perturbative CFT be explained in terms of “simplicial” or “derived”...

open_problem | open | MathDB 页面 | 原文线索

V2-076151 | MATHDB-396382

Möbius characterization of rank-one symmetric-space boundaries

摘录 / EXCERPT: Conjecture 1. Let X be a compact Ptolemy space satisfying the following four axioms:

open_problem | open | MathDB 页面 | 原文线索

V2-076152 | MATHDB-396383

Conjecture 2: Euclidean submetries and K-circles in compact Ptolemy spaces

摘录 / EXCERPT: Let X be a compact Ptolemy space satisfying Axioms (1)-(4) on page 6. For every $\c9\in X$, equip $X_\omega=X\setminus\{\omega\}$ with a metric d from the Möbius structure of...

conjecture | open | MathDB 页面 | 原文线索

V2-076153 | MATHDB-396392

Canonical surgical contraction of the exceptional divisor under the Kähler-Ricci flow

摘录 / EXCERPT: Let X be $\[1m\mathbb{P}^n\[0m$ blown up at one point y_0, let $\[1m\pi:X\to\mathbb{P}^n\[0m$ be the blow-down map, and let $E=\pi^{-1}(y_0)\cong\mathbb{P}^{n-1}$. For the Kähl...

open_problem | open | MathDB 页面 | 原文线索

V2-076154 | MATHDB-396394

Green’s Conjecture for curves on K3 surfaces

摘录 / EXCERPT: Theorem. Green’s Conjecture holds for every smooth curve C lying on an arbitrary K3 surface S. In the notation of the report, this means that $K_{p,2}(C,K_C)=0$ for all...

conjecture | open | MathDB 页面 | 原文线索

V2-076155 | MATHDB-396398

Determining whether a principally polarized abelian threefold is a Jacobian

摘录 / EXCERPT: Let (A,θ) be an indecomposable principally polarized abelian threefold over a perfect field k. The associated curve C over k has the property that either (A,θ)...

open_problem | open | MathDB 页面 | 原文线索

V2-076156 | MATHDB-396399

Fulton’s conjecture on automorphisms of \(\overline{M}_{0,n}\)

摘录 / EXCERPT: For $n\ge 5$, the only biregular automorphisms of $\overline{M}_{0,n}$ are those induced by permutations of the markings; equivalently,...

conjecture | open | MathDB 页面 | 原文线索

V2-076157 | MATHDB-396400

Counterexamples to Prym-Torelli for curves of high Clifford index

摘录 / EXCERPT: Is the Prym map (operatorname{pr}g) injective at every non-trivial étale double cover $\pi:\widetilde{X}\to X$ of a smooth projective curve X of genus $g\ge 2$, whenever X has...

open_problem | open | MathDB 页面 | 原文线索

V2-076158 | MATHDB-396404

Derived equivalence of the noncommutative Calabi-Yau categories associated with dual Cayley-plane sections

摘录 / EXCERPT: Conjecture: $\d4a0_X$ and $\d4a0_Y$ are equivalent.

open_problem | open | MathDB 页面 | 原文线索

V2-076159 | MATHDB-396407

Surjectivity of continuous sections over the scions of a descent problem

摘录 / EXCERPT: Let $D=(p:Y\to X,f:p^*E\to F)$ be a descent problem over $\mathbb C$, with $\mathcal C$ the sheaf of continuous functions. Then the natural map ... is surjective.

open_problem | open | MathDB 页面 | 原文线索

V2-076160 | MATHDB-396413

Asymptotic behavior of harmonic forms under cylindrical stretching

摘录 / EXCERPT: How does the space $H^*(r)$ of harmonic (differential) forms on $X(r)$ behave as $r\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-076161 | MATHDB-396415

Non-commutative Freiman problem

摘录 / EXCERPT: Let G be a group and A a finite subset of G. What can be said about A if $|AA|$ is small compared to $|A|^2$, that is, if $|AA| \le K|A|$, where K is a fixed parameter?

open_problem | open | MathDB 页面 | 原文线索

V2-076162 | MATHDB-396421

Characterization of functions with maximal Gowers U^3-norm

摘录 / EXCERPT: Suppose that $|f(x)|\le 1$ for all x. When is $\|f\|_{U^3}=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-076163 | MATHDB-396423

Oppenheim conjecture

摘录 / EXCERPT: Let Q be a real, nondegenerate, indefinite quadratic form in n variables, and call Q irrational if it is not proportional to a form with rational coefficients. For $n\ge 3$,...

conjecture | open | MathDB 页面 | 原文线索

V2-076164 | MATHDB-396424

Limit distribution of Frobenius numbers

摘录 / EXCERPT: The report states the following theorem. Let $d\ge 3$, let $\widehat{\mathbb Z}_{\ge 2}^{d}=\{a=(a_1,\ldots,a_d)\in\mathbb Z^d:\gcd(a_1,\ldots,a_d)=1,\ a_i\ge 2\}$, and let $F(a)$...

open_problem | open | MathDB 页面 | 原文线索

V2-076165 | MATHDB-396425

Equidistribution conjecture for irrationally sloped Farey points

摘录 / EXCERPT: The report conjectures that, for every $\alpha\notin\mathbb{Q}$ and every bounded continuous function f as in the preceding equation,nn ...

conjecture | open | MathDB 页面 | 原文线索

V2-076166 | MATHDB-396426

Asymptotic distribution of circles in an arbitrary Apollonian circle packing

摘录 / EXCERPT: For an arbitrary Apollonian circle packing $\mathcal P$, does there exist a Borel measure $\omega_{\mathcal P}$ on $\mathbb C$ such that, for every nice bounded Borel subset...

open_problem | open | MathDB 页面 | 原文线索

V2-076167 | MATHDB-396428

Expansion in arithmetic groups

摘录 / EXCERPT: Let $S\subset \mathrm{SL}_d(\mathbb Z)$ be finite and symmetric, and suppose that S generates a subgroup $G<\mathrm{SL}_d(\mathbb Z)$ that is Zariski-dense in SL_d. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-076168 | MATHDB-396429

Counting rational matrices with a fixed irreducible characteristic polynomial

摘录 / EXCERPT: Let $P(\lambda)=\lambda^n+a_{n-1}\lambda^{n-1}+\cdots+a_0$ be a monic polynomial with rational coefficients, irreducible over $\mathbb{Q}$, and, as assumed for simplicity in the re...

open_problem | open | MathDB 页面 | 原文线索

V2-076169 | MATHDB-396430

Second-highest eigenvalue identity for finite irreducible Coxeter groups

摘录 / EXCERPT: For every finite, irreducible Coxeter group $\Gamma=\langle s_1,\ldots,s_k\mid (s_i s_j)^{m_{ij}}=e\rangle$, with the maps $\Psi_\Delta:F_0\to\Delta_\Gamma$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076170 | MATHDB-396434

Amenability of subgroups of the interval exchange group

摘录 / EXCERPT: For every subgroup $G < \mathrm{IET}$, is G amenable?

open_problem | open | MathDB 页面 | 原文线索

V2-076171 | MATHDB-396437

Isomorphism problem for relatively hyperbolic groups with parabolics in \(\mathcal C\)

摘录 / EXCERPT: The isomorphism problem is solvable for the class of non-elementary relatively hyperbolic groups with parabolic groups in the class $\mathcal C$ of semidirect products...

open_problem | open | MathDB 页面 | 原文线索

V2-076172 | MATHDB-396438

Isomorphism problem for relative hyperbolic groups

摘录 / EXCERPT: Given two finite presentations of groups $G_1=\langle S_1\mid R_1\rangle$ and $G_2=\langle S_2\mid R_2\rangle$, and two finite generating sets of subgroups $P_1<G_1$ and $P_2<G_2$...

open_problem | open | MathDB 页面 | 原文线索

V2-076173 | MATHDB-396439

Coherence of \(\mathrm{SL}(3,\mathbb{Z})\)

摘录 / EXCERPT: Is $\mathrm{SL}(3,\mathbb{Z})$ coherent; that is, is every finitely generated subgroup of $\mathrm{SL}(3,\mathbb{Z})$ finitely presented?

open_problem | open | MathDB 页面 | 原文线索

V2-076174 | MATHDB-396440

Non-coherence of lattices in semisimple Lie groups

摘录 / EXCERPT: Let G be a semisimple Lie group which is not locally isomorphic to $\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SL}(2,\mathbb{C})$. Then every lattice in G is noncoherent.

open_problem | open | MathDB 页面 | 原文线索

V2-076175 | MATHDB-396442

Noncoherence of lattices in semisimple Lie groups

摘录 / EXCERPT: Let G be a semisimple Lie group without compact factors which is not locally isomorphic to $\operatorname{SL}(2,\mathbb{R})$ or $\operatorname{SL}(2,\mathbb{C})$. Then every latt...

open_problem | open | MathDB 页面 | 原文线索

V2-076176 | MATHDB-396443

Punctured Ehrenpreis conjecture for the Weil-Petersson metric

摘录 / EXCERPT: Let S and R be two finite-type Riemann surfaces that both have at least one puncture. Then, given $\5>0$, one can find covers $S_{\5}$ and $R_{\5}$ of S and R, respectively...

conjecture | open | MathDB 页面 | 原文线索

V2-076177 | MATHDB-396444

Existence of a group with an unbounded norm but no stably unbounded norm

摘录 / EXCERPT: Does there exist a group that does not admit a stably unbounded conjugation-invariant norm and yet admits a conjugation-invariant norm unbounded on some cyclic subgroup?

open_problem | open | MathDB 页面 | 原文线索

V2-076178 | MATHDB-396445

Legendrian linking of skies detects causality under a noncompact universal-cover hypothesis

摘录 / EXCERPT: Let X be a globally hyperbolic spacetime with Cauchy surface M. If the universal cover of M is non-compact, then two points $x,y\in X$ are causally related if and only if the...

conjecture | open | MathDB 页面 | 原文线索

V2-076179 | MATHDB-396446

Knot/link invariants yielding quasi-morphisms on pure braid groups

摘录 / EXCERPT: Which knot/link invariants may be used to define quasi-morphisms on P_n in a similar way?

open_problem | open | MathDB 页面 | 原文线索

V2-076180 | MATHDB-396447

Rasmussen’s conjecture that the invariants coincide

摘录 / EXCERPT: For every knot K, $s(K)=2\tau(K)$, where s is Rasmussen’s knot invariant and τ is the Ozsváth-Szabó knot invariant.

conjecture | open | MathDB 页面 | 原文线索

V2-076181 | MATHDB-396448

Global minimum of the height of flat tori and densest regular sphere packings

摘录 / EXCERPT: Conjecturally, the height of $A\mathbb Z^r\backslash\mathbb R^r$ has a global minimum when $A\mathbb Z^r$ is the densest regular sphere packing.

open_problem | open | MathDB 页面 | 原文线索

V2-076182 | MATHDB-396449

Unbounded Weyl-law remainder on surfaces

摘录 / EXCERPT: Is it true that, on any surface, $\displaystyle \limsup_{\lambda\to\infty}|R(\lambda)|=\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-076183 | MATHDB-396456

Asymptotic constant for stable commutator length of random elements

摘录 / EXCERPT: Let G be a hyperbolic group with a fixed generating set S. Let $\\operatorname{scl}_n$ denote the stable commutator length of a random element of word length n in the generat...

open_problem | open | MathDB 页面 | 原文线索

V2-076184 | MATHDB-396458

Functionals on Lie transformation groups that are homomorphisms on abelian subgroups

摘录 / EXCERPT: Do there exist continuous functionals $\mu:G\to\mathbb{R}$ on Lie transformation groups which are not quasi-morphisms but such that the restriction of μ to any abelian subgroup...

open_problem | open | MathDB 页面 | 原文线索

V2-076185 | MATHDB-396460

Continuity of entropy for convergent symmetric measures on a finitely generated group

摘录 / EXCERPT: Let G be a finitely generated group. Suppose there exist a finite subset V of G and a sequence of symmetric measures (μ_i) on G such that...

open_problem | open | MathDB 页面 | 原文线索

V2-076186 | MATHDB-396462

Embedding mapping-torus groups into $\mathrm{SL}_2(\mathbb{C})$

摘录 / EXCERPT: Let F_n be the free group of rank $n\ge 2$, and let $\varphi:F_n\to F_n$ be an irreducible proper injective endomorphism. Let G be the fundamental group of the mapping torus of...

open_problem | open | MathDB 页面 | 原文线索

V2-076187 | MATHDB-396463

Vector-valued optimal Lipschitz extension problem

摘录 / EXCERPT: Construct a vector-valued mapping whose Lipschitz constant is the minimum possible, relative to its boundary conditions, on every subdomain of a given domain.

open_problem | open | MathDB 页面 | 原文线索

V2-076188 | MATHDB-396465

Sharpening the weak-\$L^{4/3}$ interpolation estimate

摘录 / EXCERPT: In the superconducting interpolation estimate (2), stated for nonnegative u in the regime $ u^{6/7}\\ll\Phi\ll1$, can the weak space $w\!-\L^{4/3}$ be sharpened to the strong spa...

open_problem | open | MathDB 页面 | 原文线索

V2-076189 | MATHDB-396467

Absence of binding in Pekar’s polaron model

摘录 / EXCERPT: For the Pekar energy ... where $\psi\in L^2(\mathbb R^{3N})$, $\|\psi\|_2=1$, $U>0$, $\alpha>0$, and ... there exists a constant $\widetilde C>0$ such that, whenever...

open_problem | open | MathDB 页面 | 原文线索

V2-076190 | MATHDB-396468

Almost-everywhere boundary regularity for variational minima

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded $C^{1,\alpha}$ domain with $n\ge 3$, let $N>1$, and let $u_0\in C^{1,\alpha}(\overline{\Omega},\mathbb{R}^N)$ for some $\alpha\in(0,1]$...

open_problem | open | MathDB 页面 | 原文线索

V2-076191 | MATHDB-396469

Boundary regularity for vectorial variational problems

摘录 / EXCERPT: Under assumptions (2) and (4), with $\gamma<p$, let $\frac{2}{3}\le s\le\frac{p}{p-1}$ and assume $\alpha>\frac12$, $\beta>s$, and $\gamma\le ps+\frac{2ps}{n-2}$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-076192 | MATHDB-396471

Solomon’s conjecture on the least-density Simons cone

摘录 / EXCERPT: For $m\ge 1$, prove that the Simons cone ... is the $(2m+1)$-dimensional minimal (or area-minimizing) hypercone of least possible density. For $m=1$, cones with soap-film-like triple...

conjecture | open | MathDB 页面 | 原文线索

V2-076193 | MATHDB-396473

Continuity of the optimal-control cost under a 3-generating condition

摘录 / EXCERPT: Assume that the Lagrangian L satisfies ... for some constants $C_1,C_2,C_3>0$, and that the Hessian of L in the u variable is positive definite. If the family of vector fields...

open_problem | open | MathDB 页面 | 原文线索

V2-076194 | MATHDB-396474

Existence of bodies invisible in three or more directions

摘录 / EXCERPT: Do there exist bodies invisible in 3 or more directions?

open_problem | open | MathDB 页面 | 原文线索

V2-076195 | MATHDB-396475

Regularity of optimal transport maps on products of spheres

摘录 / EXCERPT: The report asks whether the optimal transport map satisfies $T\in C^\infty/C^0$ when $\log\rho,\log\bar\rho\in C^\infty/L^\infty$. It then states that the answer is affirmative whe...

open_problem | open | MathDB 页面 | 原文线索

V2-076196 | MATHDB-396477

Higher integrability of the excess density for area-minimizing currents

摘录 / EXCERPT: There exist constants $p>1$ and $C,\varepsilon_0>0$ with the following property. Let T be an area-minimizing, integer rectifiable m-dimensional current in ${\mathbf C}_4$ satis...

open_problem | open | MathDB 页面 | 原文线索

V2-076197 | MATHDB-396478

Nonuniqueness of weak solutions for the two-dimensional incompressible porous media equation

摘录 / EXCERPT: For every $T>0$, there exist infinitely many nontrivial weak solutions $(\rho,v)\in L^{\infty}(\mathbb{T}^{2}\times[0,T])$ to the two-dimensional incompressible porous media (IPM)...

open_problem | open | MathDB 页面 | 原文线索

V2-076198 | MATHDB-396479

Global existence for the two-dimensional incompressible porous media equation

摘录 / EXCERPT: Does every initial datum $\rho_0\in H^s(\mathbb T^2)$ with $s>2$ generate a global-in-time classical solution of the two-dimensional incompressible porous media (IPM) system?

open_problem | open | MathDB 页面 | 原文线索

V2-076199 | MATHDB-396493

General settings for Hopf (semi-)rings

摘录 / EXCERPT: Find the most general setting in which such Hopf (semi-)rings occur. Conjecturally, at least for “series groups with products and good intersections thereof,” including symmetric g...

open_problem | open | MathDB 页面 | 原文线索

V2-076200 | MATHDB-396496

Low hierarchical rank of tensorized polynomial samples

摘录 / EXCERPT: Let $d\ge 1$, $N:=\prod_{\mu=1}^{d}n_\mu$, and let $(x_i)_{i=1}^{N}$ be the discrete evaluation $x_i=f(\xi_i)$ of a polynomial f of degree at most p on the equispaced grid...

open_problem | open | MathDB 页面 | 原文线索

V2-076201 | MATHDB-396498

Uniformly bounded solutions of the divergence equation

摘录 / EXCERPT: For every $f\\in L^{d}_{\\#}(\\mathbb{T}^{d})$, there exists a vector field $U\\in L^{\\infty}(\\mathbb{T}^{d},\\mathbb{R}^{d})$ satisfying ...

open_problem | open | MathDB 页面 | 原文线索

V2-076202 | MATHDB-396499

Bounded divergence solutions with joint critical regularity

摘录 / EXCERPT: For every $f\in L^d_\#(\mathbb T^d):=\{g\in L^d(\mathbb T^d):\int_{\mathbb T^d}g=0\}$, the equation ... admits a solution ... Moreover, the solution is given by a hierarchical decompos...

open_problem | open | MathDB 页面 | 原文线索

V2-076203 | MATHDB-396500

Short-time predictability of thermostated dynamics

摘录 / EXCERPT: Can the thermostated dynamics be considered predictive on short time scales?

open_problem | open | MathDB 页面 | 原文线索

V2-076204 | MATHDB-396501

Nonlinear coupling of buoyancy reversal with external forcings

摘录 / EXCERPT: Can there still be a non-linear coupling between buoyancy reversal and other external forcings, like radiation?

open_problem | open | MathDB 页面 | 原文线索

V2-076205 | MATHDB-396502

Impact of surface-roughness uncertainty on Ekman/QG model outputs

摘录 / EXCERPT: What is the impact of uncertainty in the surface roughness on model outputs such as wind profiles, surface stress, and Ekman pumping?

open_problem | open | MathDB 页面 | 原文线索

V2-076206 | MATHDB-396503

Parameterization of eddy momentum fluxes in zonal-jet formation

摘录 / EXCERPT: The problem is to parameterize the eddy flux on the right-hand side of the zonal-mean momentum balance ... The suggested potential-vorticity-mixing parameterization is ... with positiv...

open_problem | open | MathDB 页面 | 原文线索

V2-076207 | MATHDB-396506

Odd higher Mahler measure along the geometric-sum family

摘录 / EXCERPT: Let $P_n(x)=\dfrac{x^n-1}{x-1}$. For fixed $h\ge 0$, $\displaystyle\lim_{n\to\infty}m_{2h+1}(P_n)=0$. For $h\ge 1$, the sequence $\bigl(m_{2h+1}(P_n)\bigr)_{n\ge 1}$ is nonconstant...

open_problem | open | MathDB 页面 | 原文线索

V2-076208 | MATHDB-396507

Lehmer’s question for the classical Mahler measure

摘录 / EXCERPT: Given $\epsilon>0$, can we find a polynomial $P(x)\in\mathbb{Z}[x]$ such that $0<m(P)<\epsilon$?

open_problem | open | MathDB 页面 | 原文线索

V2-076209 | MATHDB-396512

Precise cobordism distance between torus links

摘录 / EXCERPT: What is the precise value of $d_\chi\bigl(T(a,b),T(c,d)\bigr)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076210 | MATHDB-396513

Precise value of the rectangle transformation cost

摘录 / EXCERPT: For rectangles with integer side lengths, define $E(a\times b,c\times d)$ as the minimum cost of transforming a rectangle of size $a\times b$ into one of size $c\times d$ using: (1...

open_problem | open | MathDB 页面 | 原文线索

V2-076211 | MATHDB-396517

Real Places and Surface Bundles: Conjecture 0.3

摘录 / EXCERPT: For each pair $(-\chi,p)$ with $-\chi>1$, there exists a hyperbolic mapping torus of type $(-\chi,p)$ with trace field having a real place.

conjecture | open | MathDB 页面 | 原文线索

V2-076212 | MATHDB-396518

No M-factor in the A-polynomial of 2-bridge knots

摘录 / EXCERPT: For every 2-bridge knot K, the A-polynomial of K has no factor depending only on M; that is, Conjecture 0.44 holds for all 2-bridge knots.

open_problem | open | MathDB 页面 | 原文线索

V2-076213 | MATHDB-396519

Strong AJ conjecture for knots

摘录 / EXCERPT: For every knot K, the specialization $\alpha_K\big|_{q^{1/2}=1}$ of the recurrence polynomial is equal to the A-polynomial of K, up to multiplication by a factor depending on...

conjecture | open | MathDB 页面 | 原文线索

V2-076214 | MATHDB-396520

Absence of M-factors in the A-polynomial

摘录 / EXCERPT: For every knot K, the A-polynomial does not have any M-factor.

open_problem | open | MathDB 页面 | 原文线索

V2-076215 | MATHDB-396521

Complex-volume asymptotics for the colored Jones polynomial

摘录 / EXCERPT: Let K be a hyperbolic knot in S^3. If N is large, then ... where $J_K(N;q)$ is the N-colored Jones polynomial of K.

open_problem | open | MathDB 页面 | 原文线索

V2-076216 | MATHDB-396522

Reid-Walsh commensurability conjecture for hyperbolic knots

摘录 / EXCERPT: For every hyperbolic knot $K\subset S^3$, let $C(K)$ be the set of knots $K'\subset S^3$ whose complements are commensurable with the complement of K. Then $|C(K)|\le 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-076217 | MATHDB-396523

Characterization of periodic hyperbolic knots without hidden symmetries

摘录 / EXCERPT: Let K be a periodic hyperbolic knot without hidden symmetries. If $|C(K)|\ge 2$, then: 1. K has a unique axis of symmetry disjoint from K. 2. K is obtained by unwrapping a...

open_problem | open | MathDB 页面 | 原文线索

V2-076218 | MATHDB-396525

Halbeisen-Hungerbuehler-Pirillo-Varicchio problem

摘录 / EXCERPT: Find a word over some finite alphabet $\{0,1,\ldots,k\}$ such that two consecutive factors of the same length never have the same sum.

open_problem | open | MathDB 页面 | 原文线索

V2-076219 | MATHDB-396527

Structure of the Heinis spectrum

摘录 / EXCERPT: Let $u\in A^{\mathbb N}$, let $p(n)$ be the number of length-n factors occurring in u, and define ... Set ... What is the structure of Ω?

open_problem | open | MathDB 页面 | 原文线索

V2-076220 | MATHDB-396528

Perfectness of the language of infinite pattern-avoiding words

摘录 / EXCERPT: Let L be the set of infinite words over $\a3$ that avoid a pattern p. Is L perfect?

open_problem | open | MathDB 页面 | 原文线索

V2-076221 | MATHDB-396530

Additive refinement of the Ehrenfeucht-Silberger bound

摘录 / EXCERPT: Does $|w| > \frac{7}{3}\tau(w) - 4$ imply $\tau(w) = \pi(w)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076222 | MATHDB-396532

Independent three-equation systems with a nonperiodic solution

摘录 / EXCERPT: Does there exist an independent system of three equations with three unknowns such that it possesses a nonperiodic solution?

open_problem | open | MathDB 页面 | 原文线索

V2-076223 | MATHDB-396533

Length of descending chains of word equations on three unknowns

摘录 / EXCERPT: For equations over the free monoid Σ^* or the free semigroup Σ^+ with three unknowns, can there be a descending chain of length greater than 7?

open_problem | open | MathDB 页面 | 原文线索

V2-076224 | MATHDB-396535

Bounded non-repetitive chromatic number of planar graphs

摘录 / EXCERPT: Is the non-repetitive chromatic number of planar graphs bounded?

open_problem | open | MathDB 页面 | 原文线索

V2-076225 | MATHDB-396537

Repetition thresholds for trees

摘录 / EXCERPT: For an integer k and the class $ T$ of trees, the report states: 1. $RT(2,T)=\frac{7}{2}$; 2. $RT(3,T)=\frac{5}{2}$; 3. $RT(k,T)=\frac{3}{2}$ for $k\ge 4$.

open_problem | open | MathDB 页面 | 原文线索

V2-076226 | MATHDB-396539

Decidability of completeness for finite sets of words

摘录 / EXCERPT: Given a finite set S of finite words over an alphabet Σ, is it decidable whether S is complete, i.e. whether $\operatorname{Fact}(S^*)=\Sigma^*$?

open_problem | open | MathDB 页面 | 原文线索

V2-076227 | MATHDB-396540

Minimality of the parametrized uncompletable word for S_k

摘录 / EXCERPT: For each integer $k>6$, let $J_k=\bigcup_{i=1}^{k-3}(ba^i\Sigma\cup a^ib)$ and $S_k=\Sigma^k\setminus\{a^{k-2}bb\}\cup\Sigma ba^{k-4}\Sigma\cup\Sigma ba\cup\{b^4\}\cup J_k$. Let .....

open_problem | open | MathDB 页面 | 原文线索

V2-076228 | MATHDB-396542

Pattern characterization of occurrences of nonintegral powers

摘录 / EXCERPT: Let $\alpha>1$ be a real number that is not an integer. Is there any set of patterns P, finite or infinite, such that x has an occurrence of an α-power if and only if so...

open_problem | open | MathDB 页面 | 原文线索

V2-076229 | MATHDB-396543

Fragile infinite ternary squarefree words

摘录 / EXCERPT: Problem 39. Are there infinite words over $\{0,1,2\}$ that are fragile with respect to the property of squarefreeness?

open_problem | open | MathDB 页面 | 原文线索

V2-076230 | MATHDB-396544

Nonzero Hankel determinants at every order and position

摘录 / EXCERPT: Let $a=(a_i)_{i\ge 0}$ be an infinite sequence of integers. The Hankel determinant of order n beginning at position k is ... Is there an infinite sequence a over a finite subs...

open_problem | open | MathDB 页面 | 原文线索

V2-076231 | MATHDB-396546

The Pirillo-Varricchio-Halbeisen-Hungerbühler problem

摘录 / EXCERPT: Is there an infinite sequence over a finite subset of $\mathbb{Z}$ with the property that it has no factors of the form xx' with $|x|=|x'|$ and $\sum x=\sum x'$? Such a factor is...

open_problem | open | MathDB 页面 | 原文线索

V2-076232 | MATHDB-396547

Conjecture 48: avoidance of additive cubes

摘录 / EXCERPT: The fixed point generated by the morphism γ defined by $\gamma(0)=03$, $\gamma(1)=43$, $\gamma(3)=1$, and $\gamma(4)=01$ avoids $xx'x''$ with $|x|=|x'|=|x''|$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-076233 | MATHDB-396548

Stolarsky’s conjecture on the binary digit sum of powers

摘录 / EXCERPT: For every fixed integer $h\ge 2$, Stolarsky conjectured that ... where $s_2(n)$ is the sum of the binary digits of n. The report states that this conjecture was settled: more gener...

conjecture | open | MathDB 页面 | 原文线索

V2-076234 | MATHDB-396549

Sharp upper bound for the first positive Thue-Morse value along multiples

摘录 / EXCERPT: Let $t=(t_n)_{n\ge 0}$ be the Thue-Morse sequence, with $t_n=(-1)^{s_2(n)}$, and for $k\ge 1$ define $N_k=\{n:t_{kn}=1\}$ and $f(k)=\min\{n:n\in N_k\}$. Theorem 49 asserts that, fo...

open_problem | open | MathDB 页面 | 原文线索

V2-076235 | MATHDB-396550

Frequency of the Thue-Morse signs along cubic indices

摘录 / EXCERPT: Determine the frequency of $+1$ and -1 in the Thue-Morse word for the subsequence of indices $P(n)=n^3$.

open_problem | open | MathDB 页面 | 原文线索

V2-076236 | MATHDB-396551

Equal-length monochromatic block factors in a non-periodic word (Question 53)

摘录 / EXCERPT: Let A be a non-empty finite alphabet and let $W\in A^{\mathbb N}$ be a non-periodic infinite word. Let $c:F^+(W)\to C$ be any finite coloring of the set of all non-empty factors...

open_problem | open | MathDB 页面 | 原文线索

V2-076237 | MATHDB-396552

Finite colorings preventing monochromatic factorizations of Sturmian words

摘录 / EXCERPT: Given a Sturmian word W, does there exist a finite non-empty color set C and a coloring $c:F^{+}(W)\to C$ such that, for every factorization $W=U_0U_1U_2\cdots$, there exist in...

open_problem | open | MathDB 页面 | 原文线索

V2-076238 | MATHDB-396553

General computation of projective symmetry groups of polyhedral cones

摘录 / EXCERPT: Given a polyhedral cone $C\subset\mathbb{R}^n$ generated by rays $(v_i)_{1\le i\le N}$, find a general method for computing its projective symmetry group $\operatorname{Proj}(C)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076239 | MATHDB-396554

Choice of a minimal fundamental domain

摘录 / EXCERPT: There can be more than one minimal fundamental domain. Does it matter which fundamental domain is used?

open_problem | open | MathDB 页面 | 原文线索

V2-076240 | MATHDB-396555

Normality of semigroups associated with contingency tables

摘录 / EXCERPT: Let $r_1,\dots,r_N$ be natural numbers. An $r_1\times r_2\times\cdots\times r_N$ contingency table is a function $T:\{1,\dots,r_1\}\times\cdots\times\{1,\dots,r_N\}\to\mathbb Z_+$....

open_problem | open | MathDB 页面 | 原文线索

V2-076241 | MATHDB-396556

Classification of symmetry types in dimension at least three

摘录 / EXCERPT: Classify the symmetry types in dimension $n\ge 3$.

open_problem | open | MathDB 页面 | 原文线索

V2-076242 | MATHDB-396557

Orbitwise evaluation of symmetric rational generating functions

摘录 / EXCERPT: If a symmetric polyhedron P is represented by an orbitwise rational generating function obtained using symmetry, can the evaluation problem be solved in an orbitwise way so as to...

open_problem | open | MathDB 页面 | 原文线索

V2-076243 | MATHDB-396559

Circle Packing in a Square (CPS)

摘录 / EXCERPT: Given $N \in \mathbb{N}$ and $S \in \mathbb{Q}^{+}$, can N non-overlapping circles of unit radius be arranged in a square of side $2S$?

open_problem | open | MathDB 页面 | 原文线索

V2-076244 | MATHDB-396563

Almost \(\aleph_1\)-categoricity of the class \(\mathcal C\)

摘录 / EXCERPT: We conjecture that the first-order theory of the class $\mathcal C$ of compact complex Kähler-type spaces, viewed as a many-sorted structure, is “almost $\aleph_1$-categorical”: it...

open_problem | open | MathDB 页面 | 原文线索

V2-076245 | MATHDB-396576

Finiteness and decomposition of the movable cone for terminal Gorenstein threefolds

摘录 / EXCERPT: Let X be a terminal Gorenstein 3-fold with $-K_X$ semiample of positive Iitaka dimension. Then $M(X)^e$ decomposes as the union of the effective nef cones of the small modifica...

open_problem | open | MathDB 页面 | 原文线索

V2-076246 | MATHDB-396582

Fixed Price Question

摘录 / EXCERPT: Does $\operatorname{cost}(R_{\Gamma\curvearrowright (X,\mu)})$ depend on the particular free p.m.p. action?

open_problem | open | MathDB 页面 | 原文线索

V2-076247 | MATHDB-396584

Aldous-Lyons converse for graphing limits

摘录 / EXCERPT: Every graphing of maximum degree at most d is the local limit of a sequence of finite simple graphs of maximum degree at most d (with the numbers of vertices tending to infinit...

open_problem | open | MathDB 页面 | 原文线索

V2-076248 | MATHDB-396590

Expansion of congruence Cayley graphs for Zariski-dense subgroups

摘录 / EXCERPT: Let $S\subseteq \mathrm{SL}(n,\mathbb{Z})$ be a finite symmetric subset, and let $\Lambda=\langle S\rangle$ be Zariski dense in $\mathrm{SL}(n)$. It was conjectured that the Cayley...

open_problem | open | MathDB 页面 | 原文线索

V2-076249 | MATHDB-396591

Wiegold’s question on quotients of a Coxeter group

摘录 / EXCERPT: Do there exist uncountably many infinite 2-groups that are quotients of the group $\Delta=\langle x,y\mid x^2,y^4,(xy)^8\rangle$?

open_problem | open | MathDB 页面 | 原文线索

V2-076250 | MATHDB-396592

Torsion quotients of Coxeter groups

摘录 / EXCERPT: Let $\mathcal{C}$ be a non-virtually-abelian Coxeter group defined by a connected Coxeter graph $\mathcal{Z}$, with all edge labels m_{ij} being powers of 2 or ∞. If...

open_problem | open | MathDB 页面 | 原文线索

V2-076251 | MATHDB-396593

Existence of an infinite 2-generated 2-group with strengthened relations

摘录 / EXCERPT: Does there exist an infinite 2-generated 2-group, generated by x,y, satisfying ...

open_problem | open | MathDB 页面 | 原文线索

V2-076252 | MATHDB-396596

Characterization of irrational \[?25ll^2\[?25h-Betti numbers by lamplighter subgroups

摘录 / EXCERPT: For a countable discrete group G, is it the case that $C(G)\not\subseteq\mathbb{Q}$ is equivalent to $\mathbb{Z}/p\wr\mathbb{Z}\subset G$ for some p?

open_problem | open | MathDB 页面 | 原文线索

V2-076253 | MATHDB-396597

The Atiyah problem for the lamplighter group

摘录 / EXCERPT: What is $C\!\left(\mathbb{Z}/2\mathbb{Z}\wr\mathbb{Z}\right)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076254 | MATHDB-396598

Possible values of automaton-defined kernel-dimension series

摘录 / EXCERPT: Let n and C be integers, and let A be an automaton whose letters are $n\times n$ matrices with integer coefficients from the set $\{-C,-C+1,\ldots,C\}$. A matrix M of dimen...

open_problem | open | MathDB 页面 | 原文线索

V2-076255 | MATHDB-396599

Decidability of solvability of quadratic equations in positive integers

摘录 / EXCERPT: Can one decide, for quadratic equations in arbitrarily many unknowns, whether a solution exists in the positive integers?

open_problem | open | MathDB 页面 | 原文线索

V2-076256 | MATHDB-396601

Congruence kernel of an arithmetic Kleinian lattice

摘录 / EXCERPT: Let Γ be an arithmetic lattice in $\mathrm{SL}_2(\mathbb{C})$. Is the congruence kernel of Γ isomorphic to $\widehat{F}_{\omega}$?

open_problem | open | MathDB 页面 | 原文线索

V2-076257 | MATHDB-396603

Kesten’s theorem for invariant random subgroups

摘录 / EXCERPT: Let Γ be a group generated by a finite symmetric set S, and let H be an invariant random subgroup of Γ such that H has infinite index in Γ almost surely....

open_problem | open | MathDB 页面 | 原文线索

V2-076258 | MATHDB-396604

Flat closing problem for CAT(0) spaces

摘录 / EXCERPT: Let X be a CAT(0) space endowed with a proper action of a countable group G with compact quotient. If X contains an isometric copy of $\mathbb{R}^2$, must G contain a copy...

open_problem | open | MathDB 页面 | 原文线索

V2-076259 | MATHDB-396605

Existence and uniqueness of minimal blow-up solutions for inhomogeneous mass-critical NLS

摘录 / EXCERPT: For the two-dimensional equation ... with smooth positive bounded k, normalized by $\sup_{x\in\mathbb R^2}k(x)=1$, and with $k\in C^5\cap W^{1,\infty}$, let Q be the positive rad...

open_problem | open | MathDB 页面 | 原文线索

V2-076260 | MATHDB-396606

Behavior at the critical parameter \(\lambda=\tilde\lambda\)

摘录 / EXCERPT: The behavior of the solution in the case $\lambda=\tilde\lambda$ remains an interesting open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-076261 | MATHDB-396607

Finite-time singularity dichotomy for the full Landau-Lifshitz family

摘录 / EXCERPT: Does the dichotomy established for the harmonic-map heat flow extend to the full Landau-Lifshitz family...

open_problem | open | MathDB 页面 | 原文线索

V2-076262 | MATHDB-396608

Finite-time singularity formation near the degree-one Schrödinger-map soliton

摘录 / EXCERPT: Does finite-time singularity formation occur for 1-equivariant Schrödinger maps whose initial data is sufficiently close in $\\dot H^1$ to the two-parameter family...

open_problem | open | MathDB 页面 | 原文线索

V2-076263 | MATHDB-396609

Existence of characteristic blow-up points for the one-dimensional semilinear wave equation

摘录 / EXCERPT: Consider the one-dimensional semilinear wave equation ... with initial data $(u_0,u_1)\in H^1_{\mathrm{loc},u}(\mathbb R)\times L^2_{\mathrm{loc},u}(\mathbb R)$. If (u_0,u_1) is od...

open_problem | open | MathDB 页面 | 原文线索

V2-076264 | MATHDB-396610

Global well-posedness and scattering under bounded critical norm for the defocusing energy-supercritical wave equation

摘录 / EXCERPT: Let $u:I\times\mathbb{R}^3\to\mathbb{R}$ be a maximal-lifespan solution of ... with initial data satisfying ... Then either $E(u(t))$ is unbounded on I, or u is global and there ex...

open_problem | open | MathDB 页面 | 原文线索

V2-076265 | MATHDB-396611

Global critical-norm bound for the defocusing energy-supercritical wave equation

摘录 / EXCERPT: For the defocusing equation ... where ... there is a fixed, generally ineffective function $C:[0,\infty)\to[0,\infty)$ such that every local solution satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-076266 | MATHDB-396612

Local well-posedness for higher-order mKdV equations

摘录 / EXCERPT: The Cauchy problem for the higher-order mKdV equation of order j is locally well-posed in $\widehat H_s^r(\mathbb R)$ for $1<r\le 2$ and $s\ge \frac{2j-1}{2r'}$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-076267 | MATHDB-396613

Global regularity for the Maxwell-Dirac system in two space dimensions

摘录 / EXCERPT: For the full Maxwell-Dirac system in two space dimensions, the solution arising from the local theory extends globally in time; in particular, for smooth initial data, the solution...

open_problem | open | MathDB 页面 | 原文线索

V2-076268 | MATHDB-396615

Uniqueness of stationary vacuum black holes (the no-hair conjecture)

摘录 / EXCERPT: The no-hair conjecture asserts that the domain of outer communication of any regular, stationary, four-dimensional, vacuum black hole is isometrically diffeomorphic to the domain o...

conjecture | open | MathDB 页面 | 原文线索

V2-076269 | MATHDB-396616

Maltsiniotis' comparison and localization conjectures cannot both hold

摘录 / EXCERPT: Maltsiniotis' comparison and localization conjectures for derivator K-theory cannot be simultaneously true.

conjecture | open | MathDB 页面 | 原文线索

V2-076270 | MATHDB-396617

Farrell-Jones conjecture for cocompact lattices in connected Lie groups

摘录 / EXCERPT: The Farrell-Jones conjecture holds for cocompact lattices in connected Lie groups.

conjecture | open | MathDB 页面 | 原文线索

V2-076271 | MATHDB-396636

Existence of compactly supported tight shearlet frames

摘录 / EXCERPT: Does there exist a compactly supported tight shearlet frame?

open_problem | open | MathDB 页面 | 原文线索

V2-076272 | MATHDB-396637

Matrix extension of a Laurent-polynomial matrix to a unitary (paraunitary) matrix

摘录 / EXCERPT: Let P be an $n\times r$ matrix of Laurent polynomials satisfying $P^*P=I_r$. How can it be extended to an $n\times n$ unitary (paraunitary) matrix $A=[P\ Q]$?

open_problem | open | MathDB 页面 | 原文线索

V2-076273 | MATHDB-396638

Determining semi-effectivity of divisors on blow-ups of projective space

摘录 / EXCERPT: Let $\pi:X\to\mathbb{P}^N$ be the blow-up of a finite set of points $p_1,\ldots,p_s$, let $E_i=\pi^{-1}(p_i)$, and let H be the pullback of a hyperplane. Given ... is there a proce...

open_problem | open | MathDB 页面 | 原文线索

V2-076274 | MATHDB-396639

Classification of point configurations of type $(d-1,d)$

摘录 / EXCERPT: Let Z be a set of points in $\mathbb{P}^2$, let $I=I(Z)$ be its homogeneous ideal, and let $I^{(2)}=I(2Z)$. Say that Z has type $(d-t,d)$ if the minimal degrees of generators o...

open_problem | open | MathDB 页面 | 原文线索

V2-076275 | MATHDB-396642

Greuel-Lossen-Shustin conjecture for linear systems associated with plane-curve singularities

摘录 / EXCERPT: Let $S_1,\\ldots,S_r$ be equisingular singularity types, and let m_1,m_2,m_3 be the three largest multiplicities occurring in their associated multiplicity sequences. Define ... Fo...

conjecture | open | MathDB 页面 | 原文线索

V2-076276 | MATHDB-396644

Greuel-Lossen-Shustin maximal-rank conjecture for plane curves with prescribed singularities

摘录 / EXCERPT: Let $S_1,\ldots,S_r$ be equisingular singularity types, and let m_1,m_2,m_3 be the three largest associated multiplicities. For a composition of point blowups...

conjecture | open | MathDB 页面 | 原文线索

V2-076277 | MATHDB-396648

Ganea’s conjecture on Lusternik-Schnirelmann category

摘录 / EXCERPT: Ganea conjectured that $\operatorname{cat}(X\times S^k)=\operatorname{cat}(X)+1$ for any space X.

conjecture | open | MathDB 页面 | 原文线索

V2-076278 | MATHDB-396652

Immersion dimension of real projective spaces

摘录 / EXCERPT: Determine the immersion dimension $I(n)$ as a function of n for all natural numbers $n\ge 1$, where $I(n)$ is the smallest natural number k such that the real projective space...

open_problem | open | MathDB 页面 | 原文线索

V2-076279 | MATHDB-396654

Jones polynomial detection of the unknot

摘录 / EXCERPT: Does the Jones polynomial detect the unknot?

open_problem | open | MathDB 页面 | 原文线索

V2-076280 | MATHDB-396655

Convexification of planar polygons

摘录 / EXCERPT: Can any polygon in the plane be opened to a convex position via a continuous rigid motion that preserves the bar lengths and avoids self-intersections?

open_problem | open | MathDB 页面 | 原文线索

V2-076281 | MATHDB-396660

Fundamental groups of random 2-complexes in the range $p=n^\alpha$

摘录 / EXCERPT: For random 2-complexes in the Linial-Meshulam model with $p=n^\alpha$, where $\alpha\in[-1,-1/2]$, study the behaviour and show that the fundamental group $\pi_1(Y)$ is not free (h...

open_problem | open | MathDB 页面 | 原文线索

V2-076282 | MATHDB-396664

Wide-sense Regeneration for Harris Recurrent Markov Processes

摘录 / EXCERPT: Is every Harris recurrent Markov process wide-sense regenerative?

open_problem | open | MathDB 页面 | 原文线索

V2-076283 | MATHDB-396676

Cyclicity of shifts on weighted \(\ell^p(\mathbb Z)\) spaces

摘录 / EXCERPT: Given $p\ge 1$, characterize the positive weights $\omega:\mathbb Z\to(0,+\infty)$ satisfying $\sup_{n\in\mathbb Z}\frac{\omega(n+1)}{\omega(n)}<+\infty$ for which there exists a s...

open_problem | open | MathDB 页面 | 原文线索

V2-076284 | MATHDB-396678

Geometric characterizations of the Dirichlet-theoretic spaces χ and $\mathcal D\odot\mathcal D$

摘录 / EXCERPT: Is there a better, more geometric characterization of the functions belonging to χ and $\mathcal D\odot\mathcal D$?

open_problem | open | MathDB 页面 | 原文线索

V2-076285 | MATHDB-396679

Existence of bounded symbols for bounded truncated Toeplitz operators

摘录 / EXCERPT: Let Θ be an inner function, let $K_\Theta=H^2\ominus\Theta H^2$, and let A be a bounded truncated Toeplitz operator on K_Θ. Must there exist a symbol...

open_problem | open | MathDB 页面 | 原文线索

V2-076286 | MATHDB-396680

Sarason’s quasisymbol representation question for bounded truncated Toeplitz operators

摘录 / EXCERPT: Given an inner function Θ, let K_Θ be the corresponding model space and let $\mathcal C_2(\Theta)$ denote the complex Borel measures μ on the closed disk for whic...

open_problem | open | MathDB 页面 | 原文线索

V2-076287 | MATHDB-396681

Completeness of biorthogonal systems of reproducing kernels

摘录 / EXCERPT: Let H be a space of entire functions with reproducing kernels, and let $[U+007F]k_{\lambda_n}\u007f$ be an exact system of reproducing kernels in H. Is the biorthogonal system to...

open_problem | open | MathDB 页面 | 原文线索

V2-076288 | MATHDB-396682

Finite codimension of biorthogonal systems under a quasianalyticity condition

摘录 / EXCERPT: Suppose $t_n=n$ for $n\\in\\mathbb{Z}$, and suppose the weights b_n are so small that there is no nontrivial sequence (c_n) satisfying $|c_n|\\le b_n^{1/2}$ and ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-076289 | MATHDB-396685

Classification of syndetic Riesz-sequence decompositions for exponential families

摘录 / EXCERPT: Classify the measurable sets $E\subset[0,1]$ for which the family $\{e^{2\pi i n t}\chi_E\}_{n\in\mathbb Z}$ can be written as a finite union of Riesz sequences whose index sets ar...

open_problem | open | MathDB 页面 | 原文线索

V2-076290 | MATHDB-396686

Heisenberg uniqueness pairs for a hyperbola and a lattice-cross

摘录 / EXCERPT: Let $\Gamma=\{(x_1,x_2)\in\mathbb{R}^2:x_1x_2=1\}$ and, for positive reals α,β, let $\Lambda=(\alpha\mathbb{Z}\times\{0\})\cup(\{0\}\times\beta\mathbb{Z})$. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-076291 | MATHDB-396687

Riesz bases of exponentials for multiband subsets of the circle

摘录 / EXCERPT: Let b1 be an irrational real number, let $I=[a,b)\subset\mathbb T=\mathbb R/\mathbb Z$, and define $\Lambda(\alpha,I)=\{n\in\mathbb Z:a\le n\alpha<b\}$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-076292 | MATHDB-396688

Erdős-Szűsz-Kesten discrepancy criterion for irrational rotations

摘录 / EXCERPT: Let $\b1$ be irrational, let $S\subset\mathbb T$ be a single interval, and define ... where $\nu(n,S)$ is the number of integers k with $0\le k\le n-1$ such that $k\alpha\in S$. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-076293 | MATHDB-396689

BMO criterion for discrepancy of irrational rotations

摘录 / EXCERPT: Let α be an irrational number, and let $S\subset\mathbb{T}$ be a finite union of intervals. Let $\nu(n,S)$ denote the number of integers k with $0\le k\le n-1$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-076294 | MATHDB-396691

Frame bounds of Gaussian Gabor frames near critical density

摘录 / EXCERPT: Let $g_0(t)=e^{-t^2}$ and let $\\Lambda(a)=a\mathbb{Z}\times a\mathbb{Z}$. For the Gaussian Gabor frame $\\mathcal{G}(g_0,\\Lambda(a))$, let $A(a)$ and $B(a)$ denote its lower and...

open_problem | open | MathDB 页面 | 原文线索

V2-076295 | MATHDB-396692

The Type Problem for completeness of exponentials

摘录 / EXCERPT: Let bc be any finite positive Borel measure on r. For bb in an interval 0,a, let ... The Type Problem asks: when is $E_{[0,a]}$ complete in $L^2(\mu)$, that is, when are finite linea...

open_problem | open | MathDB 页面 | 原文线索

V2-076296 | MATHDB-396693

Optimal growth of frequently hypercyclic entire functions

摘录 / EXCERPT: For every $c>0$, there exists an entire function f on $ \mathbb{C}$ that is frequently hypercyclic for the differentiation operator D, where $Df=f'$, on the space of entire fun...

open_problem | open | MathDB 页面 | 原文线索

V2-076297 | MATHDB-396695

Leech-type interpolation with identity target on the polydisk

摘录 / EXCERPT: Can the example of the failure of Leech's theorem for $H^2(\mathbb{D}^n)$, $n>1$, be chosen so that $T_H=I$ in (1)?

open_problem | open | MathDB 页面 | 原文线索

V2-076298 | MATHDB-396697

Corona problem for $H^{\infty}(\mathbb B_n)$ in higher dimensions

摘录 / EXCERPT: For $n>1$, if $g_1,\ldots,g_N\in H^{\infty}(\mathbb B_n)$ satisfy ... must there exist $f_1,\ldots,f_N\in H^{\infty}(\mathbb B_n)$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-076299 | MATHDB-396700

Sharp dimension dependence in the strong Kreiss resolvent estimate

摘录 / EXCERPT: Is the factor $n^{3/2}$ in the estimate ... sharp for power-bounded operators $T:(\mathbb{C}^n,|\cdot|)\to(\mathbb{C}^n,|\cdot|)$, or can its dependence on n be improved? The repor...

open_problem | open | MathDB 页面 | 原文线索

V2-076300 | MATHDB-396701

Geometric characterization of multipliers and Carleson measures for infinite-dimensional Drury-Arveson space

摘录 / EXCERPT: Find a geometric characterization of the multipliers and of the Carleson measures for the infinite-dimensional Drury-Arveson space.

open_problem | open | MathDB 页面 | 原文线索

V2-076301 | MATHDB-396702

Polynomial completeness on crescent domains

摘录 / EXCERPT: Let coverline{Ω} be a compact set such that Ω is a bounded simply connected region of crescent type: the complement of $\overline{\Omega}$ has two components, a bound...

open_problem | open | MathDB 页面 | 原文线索

V2-076302 | MATHDB-396703

Brennan Problem 2: integrability exponent for conformal maps of crescent domains

摘录 / EXCERPT: Let Ω be a bounded simply connected region of crescent type in $\mathbb{C}$, meaning that its compact closure has two complementary components, a bounded component G and a...

open_problem | open | MathDB 页面 | 原文线索

V2-076303 | MATHDB-396704

Extreme points of the unit ball of a star-invariant subspace

摘录 / EXCERPT: Given an inner function θ on the unit disk $\mathbb D$, characterize the extreme points of the unit ball of K^∞_θ, where...

open_problem | open | MathDB 页面 | 原文线索

V2-076304 | MATHDB-396705

Weighted negative Fourier coefficients and Hardy-space membership

摘录 / EXCERPT: Let f belong to the Nevanlinna class of the unit disc $\D$, and assume that its radial limit f^* belongs to $L^2(\partial\u0001D)$. Write $f(z)=\sum_{n\ge 0}a_nz^n$ and suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-076305 | MATHDB-396706

A dyadic square-function oscillation question

摘录 / EXCERPT: For $k=1,2,\ldots$, let $I_k(x)$ be the dyadic interval of length 2^{-k} containing $x\in\mathbb{R}$. Let f be a locally integrable function on the real line, and let f_I den...

open_problem | open | MathDB 页面 | 原文线索

V2-076306 | MATHDB-396707

Inverting functions in the Drury-Arveson space

摘录 / EXCERPT: For an integer $d\ge 1$, let H^2_d be the Drury-Arveson space on the unit ball $B_d\subset\mathbb{C}^d$. If $f\in H^2_d$ and $|f(z)|\ge 1$ for every $z\in B_d$, must...

open_problem | open | MathDB 页面 | 原文线索

V2-076307 | MATHDB-396708

Inverting functions in the Drury-Arveson space for dimension 5

摘录 / EXCERPT: Let H^2_5 be the Drury-Arveson space of analytic functions on the unit ball $B_5\subset\mathbb{C}^5$ such that, for $f(z)=\sum_{n\in\mathbb{N}^5}a_nz^n$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076308 | MATHDB-396712

KPZ scaling limit for one-dimensional weakly asymmetric conservative particle systems

摘录 / EXCERPT: For the continuous-time process on $\Omega=\{0,1\}^{\mathbb Z}$ generated by the operator described on PDF pages 28-29, assume that the jump-rate function c satisfies the stated...

open_problem | open | MathDB 页面 | 原文线索

V2-076309 | MATHDB-396713

Full space-time scaling limit for corner-growth height fluctuations

摘录 / EXCERPT: A full non-trivial space-time limit for the height-function fluctuations of the totally asymmetric corner growth model ($q=1$, $p=0$) remains open, beyond the spatial limit process...

open_problem | open | MathDB 页面 | 原文线索

V2-076310 | MATHDB-396714

Rigorous universality beyond nearest-neighbor exclusion processes

摘录 / EXCERPT: It remains an important problem to extend the scope of rigorous universality beyond the nearest neighbor exclusion process.

open_problem | open | MathDB 页面 | 原文线索

V2-076311 | MATHDB-396716

Two-dimensional self-repellent Brownian polymer scaling

摘录 / EXCERPT: For the self-repellent Brownian polymer process in $d=2$, it is conjectured that $X(t)\sim t^{1/2}(\log t)^{1/4}$ and that the scaling limit is Gaussian (that is, Wiener).

open_problem | open | MathDB 页面 | 原文线索

V2-076312 | MATHDB-396717

Monotonicity of the diffusion coefficient in the particle density

摘录 / EXCERPT: For every fixed b30, the diffusion coefficient $D_\b3(\d70)$ is decreasing as a function of the density $\d70$ on $[0,1]$.

open_problem | open | MathDB 页面 | 原文线索

V2-076313 | MATHDB-396719

Efficient sampling of metastable equilibrium trajectories

摘录 / EXCERPT: How to generate efficiently equilibrium trajectories which leave one well of the potential V to reach another well.

open_problem | open | MathDB 页面 | 原文线索

V2-076314 | MATHDB-396720

Rigorous scaling law for the heat current in a disordered harmonic chain

摘录 / EXCERPT: For a one-dimensional disordered harmonic chain of n oscillators with i.i.d. random masses, identical nearest-neighbor springs, and Langevin heat baths at the two boundaries with...

open_problem | open | MathDB 页面 | 原文线索

V2-076315 | MATHDB-396721

Tightness of the interface in the one-dimensional multitype contact process

摘录 / EXCERPT: Consider the multitype contact process on $\mathbb{Z}$ with state space $\{0,1,2\}^{\mathbb{Z}}$, infection kernel p uniform on $\{-R,\ldots,-1,1,\ldots,R\}$, and equal infection...

open_problem | open | MathDB 页面 | 原文线索

V2-076316 | MATHDB-396722

Zero-temperature 3D Ising interface disappearance time

摘录 / EXCERPT: For the zero-temperature heat-bath dynamics of the three-dimensional Ising model on the cube C_L of side L, with spins outside C_L fixed at $+1$ and all spins in C_L initia...

open_problem | open | MathDB 页面 | 原文线索

V2-076317 | MATHDB-396724

Extension of microscopic-concavity fluctuation arguments to an extended exclusion process

摘录 / EXCERPT: Can the arguments for proving $t^{1/3}$ current-fluctuation scaling and $t^{2/3}$ second-class-particle fluctuation scaling be made to work for an extended exclusion process with p...

open_problem | open | MathDB 页面 | 原文线索

V2-076318 | MATHDB-396725

Optimal variance-decay exponent for the environment viewed by the particle

摘录 / EXCERPT: Let $(\omega_e)_{e\in B}$ be i.i.d. conductances on the edges of $\mathbb Z^d$, with $\omega_e\ge 1$, and let $\omega(t)=\theta_{Z_t}\omega$ be the environment viewed by the partic...

open_problem | open | MathDB 页面 | 原文线索

V2-076319 | MATHDB-396726

Optimal spectral-decay exponent for the gradient observable

摘录 / EXCERPT: For i.i.d. conductances on the edges of mathbb{Z}^d that are uniformly bounded above and below, let mathcal{L} be the positive self-adjoint generator of the environment viewed by t...

open_problem | open | MathDB 页面 | 原文线索

V2-076320 | MATHDB-396730

Choosing inducing representations and function spaces for infinite-dimensional principal series

摘录 / EXCERPT: Determine a good class of representations μ of M_R and a correspondingly appropriate space of functions $h:G_R\to H_\mu$ satisfying ... Here $P_R=M_R A_R N_R$ is the relevant m...

open_problem | open | MathDB 页面 | 原文线索

V2-076321 | MATHDB-396733

Classification of strongly visible linear actions

摘录 / EXCERPT: Let G^C be a connected complex reductive Lie group acting holomorphically and linearly on a complex vector space V, and let G_u be a compact real form of G^C. Then the foll...

open_problem | open | MathDB 页面 | 原文线索

V2-076322 | MATHDB-396736

Explicit spectral decomposition of the adelic regular representation

摘录 / EXCERPT: For the adelic quotient $G(\b[?]$Q)G(mathbb A)$, describe explicitly the spectral decomposition of the regular representation of $G(mathbb A)$ on $L^2(G(mathbb Q)backslash G(mathbb...

open_problem | open | MathDB 页面 | 原文线索

V2-076323 | MATHDB-396737

Phillips-Sarnak conjecture on embedded eigenvalues

摘录 / EXCERPT: For a generic non-compact hyperbolic surface of finite area, where generic refers to a point in the moduli space, the Laplacian has no embedded eigenvalues.

conjecture | open | MathDB 页面 | 原文线索

V2-076324 | MATHDB-396738

Limit multiplicities in towers of finite-index normal subgroups

摘录 / EXCERPT: Let G be the relevant semisimple group and let ... be a tower of normal subgroups of finite index such that $\bigcap_n \Gamma_n=\{e\}$. For every bounded Jordan-measurable set...

open_problem | open | MathDB 页面 | 原文线索

V2-076325 | MATHDB-396743

Basepoint independence of the reduced Bers boundary

摘录 / EXCERPT: Let $\partial_B^{m_0}T(S)$ be the Bers boundary of $T(S)$ based at $m_0\in T(S)$. Collapse each quasiconformal deformation space in this boundary to a point, obtaining the reduced...

open_problem | open | MathDB 页面 | 原文线索

V2-076326 | MATHDB-396744

The classical Plateau Problem

摘录 / EXCERPT: Let $\triangle$ be the unit disk in $\mathbb{R}^2$. For a map $\alpha:\triangle\to\mathbb{R}^n$, define its area by ... Given a Jordan curve Υ, let ... Find $\alpha^*\in F$ su...

open_problem | open | MathDB 页面 | 原文线索

V2-076327 | MATHDB-396746

Vanishing of the simplicial volume of the moduli space of closed Riemann surfaces

摘录 / EXCERPT: For every integer $g\ge 1$, the simplicial volume of the moduli space $\mathcal{M}_g$ is equal to 0.

open_problem | open | MathDB 页面 | 原文线索

V2-076328 | MATHDB-396747

Non-varying sum of Lyapunov exponents in hyperelliptic strata

摘录 / EXCERPT: For all hyperelliptic strata, the sum of Lyapunov exponents is non-varying.

open_problem | open | MathDB 页面 | 原文线索

V2-076329 | MATHDB-396748

Nielsen realization problem for finite quasiconformal mapping class subgroups

摘录 / EXCERPT: Let R be a Riemann surface admitting a hyperbolic structure, let $\operatorname{MCG}(R)$ be its quasiconformal mapping class group, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-076330 | MATHDB-396751

Sharp asymptotic of the Takhtajan-Zograf metric near puncture-adjacent nodes

摘录 / EXCERPT: Conjecture ([5]): ... if $i\le k$ and the node q_i is adjacent to punctures.

open_problem | open | MathDB 页面 | 原文线索

V2-076331 | MATHDB-396752

Self-adjoint extension of the Takhtajan-Zograf Laplace operator

摘录 / EXCERPT: Does the Laplace operator Δ_{TZ} of the Takhtajan-Zograf metric have a self-adjoint extension?

open_problem | open | MathDB 页面 | 原文线索

V2-076332 | MATHDB-396754

Characterization of shear functions induced by circle homeomorphisms

摘录 / EXCERPT: Let $s:\mathcal F\to\mathbb R$ be arbitrary, where $\mathcal F$ is the set of edges of the Farey tessellation. A function s is induced by a homeomorphism of S^1 if and only if,...

open_problem | open | MathDB 页面 | 原文线索

V2-076333 | MATHDB-396755

Characterization of shear functions arising from quasisymmetric maps

摘录 / EXCERPT: Let $\mathcal F$ be the Farey tessellation of $\mathbb H$, and let $\mathcal F^0$ be the set of endpoints of its edges. For each $p\in\mathcal F^0$, enumerate the fan...

open_problem | open | MathDB 页面 | 原文线索

V2-076334 | MATHDB-396756

Polyhedral realization of the Thurston compactification

摘录 / EXCERPT: Whether the Thurston compactification of $\operatorname{Teich}(\Sigma)$ can be realized as a polyhedron in a finite-dimensional projective space $\mathbb{P}(\mathbb{R}^N)$ by choos...

open_problem | open | MathDB 页面 | 原文线索

V2-076335 | MATHDB-396758

Andreadakis conjecture for the automorphism group of a free group

摘录 / EXCERPT: Let F_n be a free group of rank $n\ge 2$, let $IA_n=A'_n(1)$ be its IA-automorphism group, let $A'_n(1),A'_n(2),\ldots$ be the lower central series of IA_n, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-076336 | MATHDB-396762

Isometries of the length-spectrum metric on Teichmüller space

摘录 / EXCERPT: Let S be a closed surface of genus greater than 1, and let d_{ls} be the symmetrization of the non-symmetric Thurston metric d_{Thurs} on Teichmüller spac...

open_problem | open | MathDB 页面 | 原文线索

V2-076337 | MATHDB-396764

Positivity of the first eigenvalue for the length-spectrum or Thurston metric

摘录 / EXCERPT: For a closed surface S of genus >1, let d_{ls} be the symmetrization of the non-symmetric Thurston metric d_{Thurs} on Teichmüller space $\mathcal T(S)$....

open_problem | open | MathDB 页面 | 原文线索

V2-076338 | MATHDB-396767

Steinness of the thick part of Teichmüller space

摘录 / EXCERPT: For a surface S_g with $g>1$, let $\mathcal T^{>\epsilon}(S_g)$ denote the thick part of Teichmüller space in which every closed essential curve on S_g has length greater than...

open_problem | open | MathDB 页面 | 原文线索

V2-076339 | MATHDB-396769

Mass conservation for discrete coagulation-fragmentation equations with diffusion

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be an open, bounded, regular domain, and consider the discrete diffusive coagulation-fragmentation system ... with nonnegative initial data...

open_problem | open | MathDB 页面 | 原文线索

V2-076340 | MATHDB-396770

Coexistence and continuation of rotating stellar-dynamics equilibria

摘录 / EXCERPT: It would be of interest to determine whether the additional branch of localized minimizer solutions, analogous to the flat rotating case, exists in the full three-dimensional setti...

open_problem | open | MathDB 页面 | 原文线索

V2-076341 | MATHDB-396771

Exponential total-variation convergence for spatially homogeneous Maxwellian molecules

摘录 / EXCERPT: Let $\mu(\cdot,t)$ be a measure solution of the spatially homogeneous Boltzmann equation for Maxwellian molecules in $\mathbb{R}^3$, with angular collision kernel...

open_problem | open | MathDB 页面 | 原文线索

V2-076342 | MATHDB-396773

Finite-time Bose-Einstein condensation for the isotropic Uehling-Uhlenbeck equation

摘录 / EXCERPT: For the isotropic homogeneous Uehling-Uhlenbeck equation (1), with an initial datum having no singular part, $dF_0(x)=f_0(x)\sqrt{x}\,dx$, and with f_0 sufficiently large near...

open_problem | open | MathDB 页面 | 原文线索

V2-076343 | MATHDB-396774

Scale-mixture structure of stationary states for mean-conservative Maxwell-type models

摘录 / EXCERPT: The report states that, for the Maxwell-type collision model with random matrices and conservation of total kinetic energy only in ensemble average, the stationary state is general...

open_problem | open | MathDB 页面 | 原文线索

V2-076344 | MATHDB-396775

Exponential or almost-exponential convergence for degenerate transport models

摘录 / EXCERPT: It is not known whether exponential (or even “almost exponential”) convergence holds for these models.

open_problem | open | MathDB 页面 | 原文线索

V2-076345 | MATHDB-396776

Traveling waves of a kinetic transport model for the KPP-Fisher equation

摘录 / EXCERPT: For $s\ge s_0$ and ε sufficiently small, there exists a locally unique, up to translations, traveling-wave solution f_{TW} satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-076346 | MATHDB-396780

Biextensions of 1-motives and their tensor product

摘录 / EXCERPT: The report states and sketches a proof of the isomorphism ... obtained from ...

open_problem | open | MathDB 页面 | 原文线索

V2-076347 | MATHDB-396783

Failure of the natural Carleson embedding for $p>2$

摘录 / EXCERPT: For every $p\\in(2,\\infty)$, the estimate ... with ... is false. Here $\\mathcal D$ is the dyadic collection in $\\mathbb R^n$ and $E_Qf=1_Q\\langle f\\rangle_Q$. Already in dimension...

open_problem | open | MathDB 页面 | 原文线索

V2-076348 | MATHDB-396784

Kato square root problem for second-order elliptic operators on $\mathbb{R}^n$

摘录 / EXCERPT: Let $L=-\operatorname{div}A\nabla$ be a second-order divergence-form operator on $\mathbb{R}^n$, where $A\in L^\infty(\mathbb{R}^n;\mathcal{L}(\mathbb{C}^n))$ satisfies ... for all...

open_problem | open | MathDB 页面 | 原文线索

V2-076349 | MATHDB-396785

Reachability space for a boundary-controlled difference process

摘录 / EXCERPT: For the transport system ... with all speeds $c_j=1$, the reachability space at every time $t\ge m$ is ...

open_problem | open | MathDB 页面 | 原文线索

V2-076350 | MATHDB-396786

Polynomial stability of bounded difference semigroups

摘录 / EXCERPT: Let $(A,D(A))$ be a difference operator on $L^2([0,1],\aCm)$ generating the bounded difference semigroup $(T(t))_{t\ge 0}$. If there exist $C,\bb_0>0$ and $N\in\eN$ such that ... the...

open_problem | open | MathDB 页面 | 原文线索

V2-076351 | MATHDB-396787

Boundedness and compactness of Hankel-type bilinear forms on the Dirichlet space

摘录 / EXCERPT: Let $\mathcal D$ be the classical Dirichlet space of holomorphic functions on the unit disk, with ... For a holomorphic symbol b, define initially for polynomials...

open_problem | open | MathDB 页面 | 原文线索

V2-076352 | MATHDB-396788

Infinitely many Riccati-equation solutions from skew-conjugate spectral subspaces

摘录 / EXCERPT: Under the setting (a)-(c) on page 35, suppose that (A,B) is approximately controllable and that A has no non-observable eigenvalues on the imaginary axis. Then: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-076353 | MATHDB-396789

Whether vanishing reciprocal moments imply persistence of an Abel center

摘录 / EXCERPT: Let $n>m$ and let $A(\theta)$ and $B(\theta)$ be trigonometric polynomials. For ... write ... Assume this equation has a center, meaning that every solution with sufficiently small ini...

open_problem | open | MathDB 页面 | 原文线索

V2-076354 | MATHDB-396791

Necessity of the composition condition under vanishing moments

摘录 / EXCERPT: Let $A(\theta)$ and $B(\theta)$ be trigonometric polynomials and let $n>m$. Assume that the equation ... has a center, and that all moments of A with respect to B and of B with...

open_problem | open | MathDB 页面 | 原文线索

V2-076355 | MATHDB-396795

Uniform finiteness in Hilbert’s sixteenth problem

摘录 / EXCERPT: Does there exist, for each degree n, a finite uniform upper bound $H(n)$ for the number of limit cycles of every planar polynomial vector field of degree n with real coefficien...

open_problem | open | MathDB 页面 | 原文线索

V2-076356 | MATHDB-396796

Lins-de Melo-Pugh conjecture for classical Liénard equations

摘录 / EXCERPT: The Lins-de Melo-Pugh conjecture states that the maximal number of limit cycles is equal to l when $g(x)=x$ and the degree of f is either $2l$ or $2l+1$.

conjecture | open | MathDB 页面 | 原文线索

V2-076357 | MATHDB-396798

The center-focus problem for polynomial planar systems

摘录 / EXCERPT: Determine criteria deciding whether a critical point of a polynomial planar system whose linear part is a center is actually a center rather than a focus. In the local form discuss...

open_problem | open | MathDB 页面 | 原文线索

V2-076358 | MATHDB-396799

Finiteness of the quadratic Hilbert number

摘录 / EXCERPT: For real polynomials P_n,Q_n of degree at most $n\ge 2$, consider the system ... A limit cycle is an isolated closed orbit, and $H(n)$ denotes the maximum number of limit cycles am...

open_problem | open | MathDB 页面 | 原文线索

V2-076359 | MATHDB-396800

Maximum number of zeros of Abelian integrals

摘录 / EXCERPT: For fixed integers m,n, determine $Z(m,n)$, the maximum number of isolated zeros, counted with multiplicities, of the Abelian integrals ... where H,f,g are real polynomials of de...

open_problem | open | MathDB 页面 | 原文线索

V2-076360 | MATHDB-396807

Cauchy-Schwarz generation of valid inequalities between subgraph densities

摘录 / EXCERPT: Can all valid inequalities be proved using just the Cauchy-Schwarz inequality?

open_problem | open | MathDB 页面 | 原文线索

V2-076361 | MATHDB-396808

Moser’s union-free subfamily conjecture

摘录 / EXCERPT: For the function $f(m)$ defined as the minimum, over all families of m sets, of the size of the largest union-free subfamily, ...

conjecture | open | MathDB 页面 | 原文线索

V2-076362 | MATHDB-396809

Sharp constant for sum-free subsets of nonzero integers

摘录 / EXCERPT: Is the constant factor $1/3$ best possible in the assertion that every set of n nonzero integers contains a sum-free subset of size at least $(n+2)/3$, where sum-free means that...

open_problem | open | MathDB 页面 | 原文线索

V2-076363 | MATHDB-396810

Growth of maximum a-union-free subfamilies

摘录 / EXCERPT: Let $a\ge 2$ be an integer. A family of sets is called a-union-free if it contains no $a+1$ distinct sets $X_1,\ldots,X_a,X_{a+1}$ such that $X_1\cup\cdots\cup X_a=X_{a+1}$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-076364 | MATHDB-396811

Asymptotic determination of the bounded-occurrence SAT threshold

摘录 / EXCERPT: For each integer $k\ge 3$, let $f(k)$ be the largest integer s such that every (k,s)-CNF formula is satisfiable, where a k-CNF formula has clauses consisting of exactly k d...

open_problem | open | MathDB 页面 | 原文线索

V2-076365 | MATHDB-396812

Asymptotics of the clause-intersection satisfiability threshold

摘录 / EXCERPT: For an integer k, let $l(k)$ be the largest integer such that every k-CNF formula-whose clauses are disjunctions of exactly k distinct literals-in which each clause has a com...

open_problem | open | MathDB 页面 | 原文线索

V2-076366 | MATHDB-396814

Optimality of the two cycle-(semi)saturation constructions

摘录 / EXCERPT: Conjecture 4. Both of our constructions for $\operatorname{sat}(n,C_k)$ and $\operatorname{ssat}(n,C_k)$ are optimal whenever $n>n_0(k)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076367 | MATHDB-396815

The Lipschitz constant of the RSK correspondence

摘录 / EXCERPT: Determine, for permutations $\pi,\tau\in S_n$ at adjacent-transposition distance $d(\pi,\tau)=t$, the maximum value of ... where $\lambda(\pi)$ and $\lambda(\tau)$ are the Young-diag...

open_problem | open | MathDB 页面 | 原文线索

V2-076368 | MATHDB-396816

Erdős-Hajnal tournament-timidness conjecture

摘录 / EXCERPT: For every finite tournament H, there exist a real number ε with $0\leq\varepsilon<1$ and a constant c such that every finite tournament G containing no subtournam...

conjecture | open | MathDB 页面 | 原文线索

V2-076369 | MATHDB-396820

Conjectured functional inequality on the Hamming cube

摘录 / EXCERPT: For any subset $A\subseteq\{0,1\}^n$ and any function $f:\{0,1\}^n\to\mathbb{R}$, Samorodnitsky conjectures that ... where $\mathcal{E}(f,f)=\sum_{x\sim y}(f(x)-f(y))^2$ is the Diric...

conjecture | open | MathDB 页面 | 原文线索

V2-076370 | MATHDB-396821

Lovász-Plummer conjecture on perfect matchings in cubic bridgeless graphs

摘录 / EXCERPT: There exists a universal constant $\epsilon>0$ such that for any cubic bridgeless graph G, ... where $M(G)$ denotes the set of perfect matchings of G.

conjecture | open | MathDB 页面 | 原文线索

V2-076371 | MATHDB-396822

Simonovits-Sós conjecture on triangle-intersecting families of graphs

摘录 / EXCERPT: A family of graphs $\mathcal F$ on a fixed set of n vertices is triangle-intersecting if, for every $G,H\in\mathcal F$, the intersection $G\cap H$ contains a triangle. The Simono...

conjecture | open | MathDB 页面 | 原文线索

V2-076372 | MATHDB-396827

Directed \d+1\-sets in higher-order tournaments

摘录 / EXCERPT: For a fixed integer $d\ge 4$, let $M_d(n)$ be the maximum number of directed $(d+1)$-sets in a d-tournament on n vertices. An orientation of a 1-set is an assignment of...

open_problem | open | MathDB 页面 | 原文线索

V2-076373 | MATHDB-396828

Maximum number of directed simplices in a d-tournament

摘录 / EXCERPT: For each integer $d\ge 2$, determine the greatest number of directed $(d+1)$-sets in a d-tournament on n vertices. Here, an orientation of a 1-set assigns it a sign; an orien...

open_problem | open | MathDB 页面 | 原文线索

V2-076374 | MATHDB-396831

Havet-Thomassé generalization of Sumner’s conjecture

摘录 / EXCERPT: Let T be a directed tree on n vertices with k leaves. Then every tournament on $n+k-1$ vertices contains a copy of T.

conjecture | open | MathDB 页面 | 原文线索

V2-076375 | MATHDB-396832

Optimality and essential uniqueness of the cofilling-profile upper-bound example

摘录 / EXCERPT: The report conjectures that the upper bound supplied by the complete d-partite example is optimal, namely that $\varphi_d(\alpha)=\frac{d+1}{d}\alpha$ throughout the stated range...

open_problem | open | MathDB 页面 | 原文线索

V2-076376 | MATHDB-396834

Chromatic threshold of near-acyclic 3-chromatic graphs

摘录 / EXCERPT: Let H be a 3-chromatic graph. Suppose that its decomposition family contains a forest F such that, with $S=H-F$, every odd cycle of H meets S in at least two vertices (equi...

open_problem | open | MathDB 页面 | 原文线索

V2-076377 | MATHDB-396835

Characterization of chromatic thresholds of graphs

摘录 / EXCERPT: Let H be an r-chromatic graph with $r>2$, and let ... Let $\mathcal{M}(H)$ be the decomposition family of H, consisting of the bipartite induced subgraphs obtained by deleting...

open_problem | open | MathDB 页面 | 原文线索

V2-076378 | MATHDB-396837

Structure of the solutions to the fractional balanced-cut property

摘录 / EXCERPT: The only solutions to the fractional property $P^*_{1/2}$ are the affine combinations of $G(n,p)$ and $C_2(n,p)$, where $G(n,p)$ is the constant edge-weight assignment $x_{i,j}=p$...

open_problem | open | MathDB 页面 | 原文线索

V2-076379 | MATHDB-396840

Multidimensional Kruskal-Katona theorem

摘录 / EXCERPT: Let X be a set, and let $\binom{X}{r}$ denote the family of all r-element subsets of X. If $\r\mathcal{F}\subseteq\binom{X}{r}^{d}$ is a d-dimensional r-uniform family wi...

open_problem | open | MathDB 页面 | 原文线索

V2-076380 | MATHDB-396842

Chromatic number of the chamber graph $G_{q,d}$

摘录 / EXCERPT: Consider the graph $G_{q,d}$ defined as follows: - Vertices: the maximal chains $S_0 \subset S_1 \subset \cdots \subset S_{d+1}$ in the subspace lattice of all linear subspaces of...

open_problem | open | MathDB 页面 | 原文线索

V2-076381 | MATHDB-396843

Eckhoff’s conjecture for Radon numbers

摘录 / EXCERPT: A convexity space on a ground set X is a family $\\mathcal{F}\\subseteq 2^X$ containing $\\varnothing$ and X and closed under intersections. For $P\\subseteq X$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-076382 | MATHDB-396846

Nešetřil-Ossona de Mendez sparse-graph dichotomy question

摘录 / EXCERPT: Is it true that for any two positive integers $k,\\ell$ there exist $f(k,\\ell)$ and $s(k)$ such that every graph G with chromatic number at least $f(k,\\ell)$ contains a subgrap...

open_problem | open | MathDB 页面 | 原文线索

V2-076383 | MATHDB-396849

Raghavan’s inequality for the Borel almost-disjointness number

摘录 / EXCERPT: Is $\mathfrak h \leq \mathfrak a_{\mathrm{Borel}}$, where $\mathfrak h$ denotes the distributivity number?

open_problem | open | MathDB 页面 | 原文线索

V2-076384 | MATHDB-396864

Følner-set conjecture for the sets $A_{n,p}$ in Thompson’s group F

摘录 / EXCERPT: Let $\mathcal T$ be the set of variable-free terms in the binary operation $\hat{}$ and constant 1, and suppose there is a system of functions $f_T:\mathbb N\to\mathbb N$ satisfy...

conjecture | open | MathDB 页面 | 原文线索

V2-076385 | MATHDB-396866

Bergman and Steinhaus properties for products of finite special linear groups

摘录 / EXCERPT: (LC) If $(p_n\mid n\in\omega)$ is a sufficiently fast growing sequence of primes, then $\prod_{n\in\omega}\mathrm{SL}(2,p_n)$ has both the Bergman property and the Steinhaus proper...

open_problem | open | MathDB 页面 | 原文线索

V2-076386 | MATHDB-396872

Eliminating measurability from the hypothesis of Theorem 1

摘录 / EXCERPT: Question 1. Is it possible to eliminate the assumption on measurability of κ from Theorem 1?

conjecture | open | MathDB 页面 | 原文线索

V2-076387 | MATHDB-396873

Tree property at \cale\f{\aleph}_{\omega+1} with failure of SCH at \cale\f{\aleph}_\omega

摘录 / EXCERPT: Is it consistent to have the tree property at $\aleph_{\omega+1}$ and not SCH at $\aleph_\omega$?

open_problem | open | MathDB 页面 | 原文线索

V2-076388 | MATHDB-396877

Local well-posedness for a general one-dimensional nonlinear peridynamic Cauchy problem

摘录 / EXCERPT: Consider the Cauchy problem ... Assume that $f(0,\eta)=0$, that $f(\xi,\eta)$ is measurable in η for each ξ and continuously differentiable in ξ for almost all η. M...

open_problem | open | MathDB 页面 | 原文线索

V2-076389 | MATHDB-396878

Well-posedness of the nonlocal Dirichlet variational problem

摘录 / EXCERPT: Under the preceding assumptions on the bounded connected domain $\Omega\subset\mathbb{R}^d$, its nonlocal boundary $B\Omega\subset\mathbb{R}^d\setminus\Omega$, and the kernel C,...

open_problem | open | MathDB 页面 | 原文线索

V2-076390 | MATHDB-396879

Well-posedness of the mean-zero nonlocal variational problem

摘录 / EXCERPT: Let L be the nonlocal operator ... Let $C(r)=\gamma(r/\delta)$, where γ is nonnegative, radial, and compactly supported, with...

open_problem | open | MathDB 页面 | 原文线索

V2-076391 | MATHDB-396880

Injectivity of Integral Invariants for Shape Identification

摘录 / EXCERPT: Determine whether the two presented integral invariants-the circle-area invariant ... for $\Omega\subset\mathbb R^n$ and $r>0$, and the cone-area invariant ... where ... for a non-negati...

open_problem | open | MathDB 页面 | 原文线索

V2-076392 | MATHDB-396881

Sufficiency of L^2 tracking for the Dirichlet data

摘录 / EXCERPT: Tracking the Dirichlet data in the L^2-norm is not sufficient.

open_problem | open | MathDB 页面 | 原文线索

V2-076393 | MATHDB-396882

Ill-posedness of the L^2 Dirichlet-tracking formulation

摘录 / EXCERPT: In the generalized Bernoulli free-boundary setup, let $J_4(\\Omega)=\frac12\int_{\Gamma}w^2\,d\sigma\to\inf$, where w solves the mixed boundary-value problem $-\Delta w=f$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-076394 | MATHDB-396883

Shape reconstruction for a fully conductive obstacle from the far-field operator

摘录 / EXCERPT: Let the entire boundary of a bounded domain $D\subset\mathbb{R}^3$ be covered by the conductive layer, so that $\lambda>0$ on ∂D, and let $F:L^2(S^2)\to L^2(S^2)$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-076395 | MATHDB-396884

Injectivity of the Fréchet derivative for the complete electrode model

摘录 / EXCERPT: Let $B\subset\mathbb{R}^2$ be a Lipschitz domain, let $\mathcal{T}$ be a mesh, and let $V^+_{\mathcal{T}}$ be the class of restrictions to B of piecewise-polynomial functions on...

open_problem | open | MathDB 页面 | 原文线索

V2-076396 | MATHDB-396885

Shape reconstruction from electromagnetic far-field data

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^3$ be a bounded domain whose boundary Γ is a simply connected closed surface, with Ω diffeomorphic to a ball. For a fixed incident plane...

open_problem | open | MathDB 页面 | 原文线索

V2-076397 | MATHDB-396886

Identification of unknown non-separated inclusions in active thermography

摘录 / EXCERPT: It is an open problem to identify unknown non-separated inclusions.

open_problem | open | MathDB 页面 | 原文线索

V2-076398 | MATHDB-396887

Necessary and sufficient conditions for FS decompositions and structure conditions in defaultable markets

摘录 / EXCERPT: Determine necessary and sufficient conditions on the default time under which the structure conditions are preserved and/or the Föllmer-Schweizer decomposition exists for the progr...

open_problem | open | MathDB 页面 | 原文线索

V2-076399 | MATHDB-396888

Equilibrium considerations in a financial market with interacting investors

摘录 / EXCERPT: Consider a financial market with n agents who trade subject to individual investment restrictions, whose preferences are exponential utilities applied to a weighted sum of an ind...

open_problem | open | MathDB 页面 | 原文线索

V2-076400 | MATHDB-396889

Fixed point and law of large numbers for self-exciting correlated defaults

摘录 / EXCERPT: Under all assumptions of Theorem 2, find an $\mathcal F^0$-adapted process α such that $\alpha=\Gamma(\alpha)$, where ... The report’s Theorem 3 asserts that such an α...

open_problem | open | MathDB 页面 | 原文线索

V2-076401 | MATHDB-396890

Implicit characterization of the liquidity premium

摘录 / EXCERPT: In the stated portfolio-optimization model with proportional transaction costs, for small transaction costs ε, the liquidity premium λ is identified as the uniq...

open_problem | open | MathDB 页面 | 原文线索

V2-076402 | MATHDB-396893

Optimal investment with high-watermark performance fees

摘录 / EXCERPT: For initial wealth $x>0$, fee rate $\\lambda>0$, discount factor $\\beta>0$, fund excess return $\\alpha>0$, volatility $\\sigma$, and $p>0$, $p\neq1$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-076403 | MATHDB-396894

Distinguishing inequivalent surfaces from their apparent contours

摘录 / EXCERPT: Let ∂E_1 be a knotted torus, and let ∂E_2 be a connected surface obtained by removing two small holes from a sphere and connecting them with a knotted tube inte...

open_problem | open | MathDB 页面 | 原文线索

V2-076404 | MATHDB-396895

Preferred representative of a 3D shape class

摘录 / EXCERPT: Finding a “preferred” 3D shape in the class of all shapes differing from one another by a transformation of $\mathbb{R}^3$ mentioned in Theorem 2 is a problem that can be settled i...

open_problem | open | MathDB 页面 | 原文线索

V2-076405 | MATHDB-396897

Group dependence of shapes in computational anatomy

摘录 / EXCERPT: Assume that a population of N subjects is given, belonging to two (or more) groups. Each subject is represented by an image or shape, $S_1,\ldots,S_N$. The question is whether th...

open_problem | open | MathDB 页面 | 原文线索

V2-076406 | MATHDB-396898

Existence of a TGV-regularized deblurring solution

摘录 / EXCERPT: For a blurring kernel $k\in L^{1}(\Omega_{0})$ satisfying $\bar{k}=\int_{\Omega_{0}}k\,dx\ne 0$, with $\Omega\subset\mathbb{R}^{2}$ a Lipschitz domain,...

open_problem | open | MathDB 页面 | 原文线索

V2-076407 | MATHDB-396901

Hähnle’s diameter conjecture for polytopes

摘录 / EXCERPT: The diameter of a d-polytope with n facets cannot exceed $d(n-d)$.

conjecture | open | MathDB 页面 | 原文线索

V2-076408 | MATHDB-396902

Polynomial upper bound for polytope diameters

摘录 / EXCERPT: Is there a polynomial upper bound to the maximum diameter of a d-polytope with n facets? (Equivalently, of a d-polytope with $2d$ facets.)

open_problem | open | MathDB 页面 | 原文线索

V2-076409 | MATHDB-396903

Bárány-Larman Conjecture for the colored Tverberg problem

摘录 / EXCERPT: Let $d\ge 1$ and $r\ge 2$ be integers. Define $n(d,r)$ to be the smallest number n such that, for every affine map $f:\Delta^{n-1}\to\mathbb{R}^d$ and every partition...

conjecture | open | MathDB 页面 | 原文线索

V2-076410 | MATHDB-396904

Bárány-Larman conjecture for the alternative colored Tverberg function

摘录 / EXCERPT: Let $t(d,r)$ be the function defined in the report’s alternative colored Tverberg problem. The Bárány-Larman conjecture asserts that ...

conjecture | open | MathDB 页面 | 原文线索

V2-076411 | MATHDB-396906

A conjecture on polyominoes and inscribed squares

摘录 / EXCERPT: Let P be a polyomino with boundary a closed Jordan curve J. Assume that P is built from unit squares with vertices at lattice points. Let the greatest open square OS, with...

conjecture | open | MathDB 页面 | 原文线索

V2-076412 | MATHDB-396907

Toeplitz' square-on-a-Jordan-curve problem

摘录 / EXCERPT: Let J be any closed Jordan curve in $\mathbb{R}^2$. Are there always four points on J which form the vertices of a square?

open_problem | open | MathDB 页面 | 原文线索

V2-076413 | MATHDB-396908

Beck’s three-permutations conjecture

摘录 / EXCERPT: Given any 3 permutations on n symbols, one can color the symbols with red and blue such that, in any interval of any of those permutations, the numbers of red and blue symbols di...

conjecture | open | MathDB 页面 | 原文线索

V2-076414 | MATHDB-396909

Modified Integer Round-up Conjecture for bin packing

摘录 / EXCERPT: For the Gilmore-Gomory bin-packing linear program, with integer optimum OPT and fractional optimum OPT_f, Scheithauer and Terno conjectured that ... They further conjectured that...

conjecture | open | MathDB 页面 | 原文线索

V2-076415 | MATHDB-396911

False Grassmannian bundle conjecture for symmetric powers

摘录 / EXCERPT: Suppose d and k are even positive integers. Then there exists $n(d,k)$ such that, for every section of the bundle $\Sigma^d(\gamma_n^k)$ over G_n^k, with $n\ge n(d,k)$, there...

conjecture | open | MathDB 页面 | 原文线索

V2-076416 | MATHDB-396912

Gromov-Milman conjecture for homogeneous polynomials

摘录 / EXCERPT: For every even positive integer d and every positive integer k, there exists an integer $n(d,k)$ such that, for every homogeneous polynomial f of degree d on $\fR^n$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-076417 | MATHDB-396914

Fomin-Shapiro conjecture on the Bruhat stratification of the totally nonnegative unipotent radical

摘录 / EXCERPT: Conjecture 2. The neighborhood of the origin in the Bruhat stratification of the totally nonnegative part of the unipotent radical of a Borel in a semisimple, simply connected grou...

conjecture | open | MathDB 页面 | 原文线索

V2-076418 | MATHDB-396918

Polynomial pivoting rule for linear programming

摘录 / EXCERPT: Does there exist a simplex pivoting rule that requires a polynomial number of pivoting steps to solve every linear program?

open_problem | open | MathDB 页面 | 原文线索

V2-076419 | MATHDB-396919

Subexponential lower bound for Random-Facet on concrete linear programs

摘录 / EXCERPT: There exist concrete linear programs on which the Random-Facet pivoting rule performs an expected number of iterations of $2^{\Omega(\sqrt{n}/\log^c n)}$, where n is the number o...

open_problem | open | MathDB 页面 | 原文线索

V2-076420 | MATHDB-396921

Face numbers of flag simplicial spheres

摘录 / EXCERPT: What are the possible face numbers of flag simplicial spheres?

open_problem | open | MathDB 页面 | 原文线索

V2-076421 | MATHDB-396923

Face vectors of simplicial cell decompositions of complex projective spaces

摘录 / EXCERPT: Do conditions (1)-(3) characterize all possible face vectors of simplicial cell decompositions of $\mathbb{CP}^{3}$ (and more generally $\mathbb{CP}^{n}$)?

open_problem | open | MathDB 页面 | 原文线索

V2-076422 | MATHDB-396924

Lower bound for the second h-number of a simplicial cell decomposition of the 3-torus

摘录 / EXCERPT: Prove that if P is a simplicial cell decomposition of the 3-dimensional torus $T^3=S^1\times S^1\times S^1$, then $h_2(P)\ge 22$.

open_problem | open | MathDB 页面 | 原文线索

V2-076423 | MATHDB-396925

Equality between inversion-arrangement regions and Bruhat intervals

摘录 / EXCERPT: For a finite reflection group W and $w\in W$, characterize the elements w for which ... where $\operatorname{re}(w)$ is the set of regions of the inversion arrangement...

open_problem | open | MathDB 页面 | 原文线索

V2-076424 | MATHDB-396926

Equations defining the central curve of a linear program

摘录 / EXCERPT: For the linear program $\max\{c^{T}x:Ax=b,\ x\ge 0\}$, where A is a fixed rank-d matrix with n columns, determine polynomials defining the primal central curve without extran...

open_problem | open | MathDB 页面 | 原文线索

V2-076425 | MATHDB-396928

Generation of equivariant K-theory by elliptic-subsphere classes

摘录 / EXCERPT: In the setting of the report, let $X=S^{2N-1}\subset V=\mathbb C^N$ carry the indicated diagonal action of the torus $G=\mathbb R^m/\mathbb Z^m$, with no fixed point, and let Z b...

open_problem | open | MathDB 页面 | 原文线索

V2-076426 | MATHDB-396929

Equations of moment polyhedra of orbit projections

摘录 / EXCERPT: Let G be a connected Lie group, let K be a compact connected Lie subgroup, and let $\r\mathcal{O}\subset\mathfrak{g}^{*}$ be a coadjoint orbit. Assume that the orbit projection...

open_problem | open | MathDB 页面 | 原文线索

V2-076427 | MATHDB-396930

Extension of the McDuff-type symplectomorphism to all elliptic coadjoint orbits

摘录 / EXCERPT: Let G be a connected, noncompact, semisimple real Lie group with finite center, let K be a maximal compact subgroup such that G/K is Hermitian, and let θ give the Cart...

open_problem | open | MathDB 页面 | 原文线索

V2-076428 | MATHDB-396931

Extension of the holomorphic-orbit results to all elliptic coadjoint orbits

摘录 / EXCERPT: Can the results described above, in particular the generalization of McDuff’s theorem for holomorphic coadjoint orbits, be extended to all elliptic coadjoint orbits of G?

open_problem | open | MathDB 页面 | 原文线索

V2-076429 | MATHDB-396932

Recovering discrete-series multiplicities from a pushed-forward Liouville measure

摘录 / EXCERPT: Let G be a connected real reductive group with compact Cartan subgroup T, let G' be another connected reductive group with a compact Cartan subgroup, and suppose that G is...

open_problem | open | MathDB 页面 | 原文线索

V2-076430 | MATHDB-396937

Quantization commutes with reduction for proper cocompact actions

摘录 / EXCERPT: Under the hypotheses of the Hochs-Landsman conjecture, there exists a positive integer p_0 such that, for every integer $p\ge p_0$, ... If G has an Ad-invariant...

open_problem | open | MathDB 页面 | 原文线索

V2-076431 | MATHDB-396938

Fuchs-Schweigert formula for the unmarked moduli space

摘录 / EXCERPT: Let $G'=G/Z$, where $Z=Z(G)$, and let Σ be a closed surface of genus g. For a level k satisfying $P^\vee\!\cdot P^\vee\subset k^{-1}\mathbb Z$, and a choice of pre-quan...

open_problem | open | MathDB 页面 | 原文线索

V2-076432 | MATHDB-396940

Deformations of the Fourier transform: unitary representation theorem

摘录 / EXCERPT: Let C be the Coxeter group associated with a root system R in $\mathbb{R}^N$, let $k=(k_\alpha)$ be a C-invariant real multiplicity function, let Δ_k be the Dunkl Lapl...

open_problem | open | MathDB 页面 | 原文线索

V2-076433 | MATHDB-396945

Kac’s non-negativity conjecture

摘录 / EXCERPT: Kac’s conjecture: the coefficients of the Kac A-polynomial are non-negative.

conjecture | open | MathDB 页面 | 原文线索

V2-076434 | MATHDB-396952

Letellier’s conjecture on generic tensor-product multiplicities

摘录 / EXCERPT: Let $X_i=(\alpha_i\circ\det)\cdot R_{\lambda_i}$ for $i=1,\ldots,k$, where R_{λ_i} is the unipotent character of $\mathrm{GL}_n(\mathbb F_q)$ indexed by a partition...

conjecture | open | MathDB 页面 | 原文线索

V2-076435 | MATHDB-396953

Positivity of Donaldson-Thomas invariants for symmetric quivers

摘录 / EXCERPT: For a symmetric quiver Q and every $\boldsymbol{\alpha}\in\mathbb{N}^{Q_0}$, the function $\Omega_{\boldsymbol{\alpha}}(-q^{1/2})$ is a Laurent polynomial in $q^{1/2}$ with non-n...

open_problem | open | MathDB 页面 | 原文线索

V2-076436 | MATHDB-396954

Positivity conjecture for refined Donaldson-Thomas invariants

摘录 / EXCERPT: Assume that Q is a symmetric quiver. For every stability parameter $\theta\in\mathbb{R}^{Q_0}$ and every $\alpha\in\mathbb{N}^{Q_0}$, the function...

conjecture | open | MathDB 页面 | 原文线索

V2-076437 | MATHDB-396957

Positivity of quantized Donaldson-Thomas invariants for symmetric quivers

摘录 / EXCERPT: For a symmetric quiver Q and dimension vector $d\in\mathbb{N}^I$, the quantized Donaldson-Thomas invariant satisfies $\mathrm{DT}_d(q)\in\mathbb{N}[q]$.

open_problem | open | MathDB 页面 | 原文线索

V2-076438 | MATHDB-396960

Transitivity of autoequivalences on spherical objects of singular elliptic curves

摘录 / EXCERPT: Let E be a cycle of $n\ge 2$ projective lines, with the singular elliptic-curve conventions of §3, and let $\operatorname{Perf}(E)\subset D^b(E)$ be the full subcategory of perfe...

open_problem | open | MathDB 页面 | 原文线索

V2-076439 | MATHDB-396964

Bayes-risk consistency of depth-based procedures for functional data

摘录 / EXCERPT: It would be interesting to determine conditions under which depth-based procedures are also Bayes-risk consistent.

open_problem | open | MathDB 页面 | 原文线索

V2-076440 | MATHDB-396966

Local Gross-Prasad conjecture for generic orthogonal Langlands parameters

摘录 / EXCERPT: Let F be a finite extension of $\Q_p$. Fix an odd integer $d\ge 5$ and two non-equivalent non-degenerate quadratic spaces V_i and V_a of dimension d and fixed discriminant,...

conjecture | open | MathDB 页面 | 原文线索

V2-076441 | MATHDB-396967

Jacquet’s conjecture on distinction by quasi-split unitary groups

摘录 / EXCERPT: Let E/F be a quadratic extension of local fields. Then a representation $\c0$ of $\mathrm{GL}_n(E)$ is distinguished by the quasi-split unitary group if and only if...

conjecture | open | MathDB 页面 | 原文线索

V2-076442 | MATHDB-396968

Archimedean converse for unitary distinction

摘录 / EXCERPT: Conjecture: Let π be a representation of $\mathrm{GL}_n(\mathbb{C})$ that is the Langlands quotient of characters $\chi_1,\ldots,\chi_n$ of $\mathbb{C}^*$. If π is distingu...

open_problem | open | MathDB 页面 | 原文线索

V2-076443 | MATHDB-396969

Multiplicity for τ-invariant induced representations

摘录 / EXCERPT: Conjecture 3. If $\pi=\delta_1\times\cdots\times\delta_k$, where the δ_i are essentially square-integrable and τ-invariant, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-076444 | MATHDB-396970

Multiplicity of invariant functionals for imprimitive representations

摘录 / EXCERPT: Conjecture 4. If $\pi_1,\ldots,\pi_k$ are imprimitive, τ-invariant, and $\pi=\pi_1\times\cdots\times\pi_k$ is irreducible, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-076445 | MATHDB-396971

Kernel generation for the longest intertwining operator

摘录 / EXCERPT: Let $\pi=L_Q(\delta_1,\ldots,\delta_k)$ be an arbitrary irreducible representation, and let $K=\ker M(w_0)$ be the kernel of the longest intertwining operator, equivalently the ker...

open_problem | open | MathDB 页面 | 原文线索

V2-076446 | MATHDB-396976

Uniqueness of the normalized local zeta integral

摘录 / EXCERPT: Let V_v be the unique (up to isomorphism) hermitian space over E_v of rank $2n$ for which $I(V_v,[U+007F]pi_v):=\operatorname{Hom}_{G_v\times G_v}(R(V_v),[U+007F]pi_v^\vee\times[U+007F]pi_v)\neq 0$...

open_problem | open | MathDB 页面 | 原文线索

V2-076447 | MATHDB-396980

Uniqueness of the condensate evolution

摘录 / EXCERPT: In case a condensate is generated, does equation (1) have a unique solution which would describe the condensation?

open_problem | open | MathDB 页面 | 原文线索

V2-076448 | MATHDB-396981

Construction of a rough path over fractional Brownian motion for \(\alpha\in(1/8,1/4)\)

摘录 / EXCERPT: Construct second-order iterated integrals, and hence the corresponding rough-path lift, for two-dimensional fractional Brownian motion with Hurst index $\alpha\in(1/8,1/4)$. The re...

open_problem | open | MathDB 页面 | 原文线索

V2-076449 | MATHDB-396983

Weak trajectory approximation for thermostated model reduction

摘录 / EXCERPT: To what degree may the trajectories of the thermostated variables (for example, the strong point vortices) still approximate, in an appropriate weak sense, the trajectories of the...

open_problem | open | MathDB 页面 | 原文线索

V2-076450 | MATHDB-396987

Sufficiency of the discrete variational structure for inequality-constrained integration

摘录 / EXCERPT: For the constrained Discrete Euler-Lagrange Inclusion ... standard discrete variational structure alone is not sufficient to define a well-posed integrator in the inequality-constrai...

open_problem | open | MathDB 页面 | 原文线索

V2-076451 | MATHDB-396989

Classification of norm patterns under invertible operators

摘录 / EXCERPT: Given a finite frame [1m$\{f_i\}_{i\in I}$u001b[0m for H_N, classify the collection of norm sequences $\bigl\{(\|Tf_i\|)_{i=1}^M:T:H_N\to H_N\text{ is invertible}\bigr\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-076452 | MATHDB-396991

Wavelet-basis independence for leader scaling functions with finitely smooth wavelets

摘录 / EXCERPT: For a bounded function f, does the leader scaling function $\[1m\eta_f(p)\u001b[0m$ remain independent of the choice of wavelet basis for every $p\in\mathbb{R}$ when the wavelets...

open_problem | open | MathDB 页面 | 原文线索

V2-076453 | MATHDB-396996

Partitioning unit-norm frames with a nonmultiple number of vectors

摘录 / EXCERPT: Is there a result analogous to the following known result: a unit-norm tight frame $\{f_i\}_{i=1}^{KM}$ in an M-dimensional space can be partitioned into K linearly independent...

open_problem | open | MathDB 页面 | 原文线索

V2-076454 | MATHDB-396998

Fourier frame for the middle-third Cantor measure

摘录 / EXCERPT: Does there exist a frame of exponentials in $L^2(\bc_3)$, where $\bc_3$ is the Hausdorff measure of dimension $\bb=\ln 2/\u0003ln 3$ on the middle-third Cantor set?

open_problem | open | MathDB 页面 | 原文线索

V2-076455 | MATHDB-397000

Feichtinger’s frame conjecture for frame vectors of free-group representations

摘录 / EXCERPT: For the free group $G=F_n$ with $n>1$, does every frame vector satisfy Feichtinger’s frame conjecture?

conjecture | open | MathDB 页面 | 原文线索

V2-076456 | MATHDB-397012

Cerf’s theorem: \(\Gamma_4=0\)

摘录 / EXCERPT: Let Γ_n be the abelian group of orientation-preserving diffeomorphisms of S^{n-1}, modulo those that extend to a diffeomorphism of D^n. Cerf’s theorem states that...

open_problem | open | MathDB 页面 | 原文线索

V2-076457 | MATHDB-397013

Holomorphic fixed points of rational self-maps of \(\mathbb{P}^2\)

摘录 / EXCERPT: Does every rational self-map $f:\mathbb{P}^2\to\mathbb{P}^2$ have a holomorphic fixed point, meaning a point $p\in\mathbb{P}^2$ at which f is holomorphic and $f(p)=p$?

open_problem | open | MathDB 页面 | 原文线索

V2-076458 | MATHDB-397014

Gromov’s Vaserstein factorization problem

摘录 / EXCERPT: Does every holomorphic map $f:\mathbb{C}^{n}\to SL_{m}(\mathbb{C})$ decompose as a finite product of holomorphic maps sending $\mathbb{C}^{n}$ into unipotent subgroups of...

open_problem | open | MathDB 页面 | 原文线索

V2-076459 | MATHDB-397018

Holomorphic extendibility from complex lines through two points

摘录 / EXCERPT: Let B be the open unit ball in $\\mathbb{C}^2$, let bB be its boundary, and let $\mathcal{L}(a)$ denote the set of all complex lines passing through a. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-076460 | MATHDB-397019

Holomorphic mappings realizing a braid-group homomorphism

摘录 / EXCERPT: Let d4f5(X) be the Teichmüller space modeled on an open Riemann surface X with smooth boundary, and let $\varphi:\pi_1(X)\to\pi_1(\mathbb{C}^n\setminus V_{\Delta_n})=B_n$ be a ho...

open_problem | open | MathDB 页面 | 原文线索

V2-076461 | MATHDB-397020

Bogomolov finiteness theorem for rational and elliptic curves

摘录 / EXCERPT: There are only finitely many rational and elliptic curves on a surface of general type with $c_1^2>c_2$.

open_problem | open | MathDB 页面 | 原文线索

V2-076462 | MATHDB-397022

Characterization of Kähler infinitesimal deformations of a complex torus

摘录 / EXCERPT: Theorem 4. Let T be a complex torus of dimension at least 2. With respect to a standard identification of $H^1(T,\Theta)$ with the set of $n\times n$ matrices,...

open_problem | open | MathDB 页面 | 原文线索

V2-076463 | MATHDB-397023

Runge approximation for complex tori from elliptic curves

摘录 / EXCERPT: Could Runge approximation nevertheless hold for other Riemann surfaces, for example $\Sigma=\mathbb T^1$, when $M=\mathbb T^n$ has its usual complex structure?

open_problem | open | MathDB 页面 | 原文线索

V2-076464 | MATHDB-397024

Minimal energy in Runge approximation for pseudoholomorphic maps

摘录 / EXCERPT: Given the Runge-approximation data consisting of a compact Riemann surface Σ, an open set $U\subset\Sigma$, a compact set $K\subset U$, a J-holomorphic map $f:U\to(M,J)$ t...

open_problem | open | MathDB 页面 | 原文线索

V2-076465 | MATHDB-397029

Uniqueness of limit sets for non-recurrent Fatou components

摘录 / EXCERPT: For a non-recurrent Fatou component Ω of a generalized Hénon map, if a sequence of iterates $\{f^{n_j}\}$ converges on Ω to a limit map $h:\Omega\to\partial\Omega$, i...

open_problem | open | MathDB 页面 | 原文线索

V2-076466 | MATHDB-397031

Existence of a real rational plane quartic through 11 generic real points

摘录 / EXCERPT: There exists a real rational plane quartic through 11 generic real points.

open_problem | open | MathDB 页面 | 原文线索

V2-076467 | MATHDB-397033

Approximating fan tropical curves in the standard tropical plane

摘录 / EXCERPT: For the standard tropical plane $P\subset\mathbb{R}^3$, defined by $\max\{x,y,z,0\}$, and the complex plane $P\subset(\mathbb{C}^*)^3$ given by $x+y+z+1=0$, a fan tropical curve...

open_problem | open | MathDB 页面 | 原文线索

V2-076468 | MATHDB-397034

Enumerating real rational curves with tangency conditions

摘录 / EXCERPT: Given a set $P=\{p_1,\ldots,p_{3d-2}\}\subset\mathbb{R}^2$ and a generic immersed oriented curve $\Gamma\subset\mathbb{R}^2$, both in general position, let...

open_problem | open | MathDB 页面 | 原文线索

V2-076469 | MATHDB-397035

Chambers of polynomiality for double Hurwitz numbers

摘录 / EXCERPT: Let $H_g(x)$ be the double Hurwitz number regarded as a function on $H\cap\mathbb{Z}^n$, where $H=\{x\in\mathbb{R}^n:\sum_{i=1}^n x_i=0\}$. The chambers of polynomiality of...

open_problem | open | MathDB 页面 | 原文线索

V2-076470 | MATHDB-397037

Tropical investigation of the hypersurface governing non-invariance of Welschinger signs

摘录 / EXCERPT: For the moduli space $M^d_{g,3d-1+g}(\mathbb{R}P^2)$ and evaluation map $\operatorname{ev}:M^d_{g,3d-1+g}(\mathbb{R}P^2)\to(\mathbb{R}P^2)^{3d-1+g}$, investigate by tropical means...

open_problem | open | MathDB 页面 | 原文线索

V2-076471 | MATHDB-397038

Kleiman-Piene formula for nodal curves in generic families of surfaces

摘录 / EXCERPT: For an arbitrary generic family of curves on surfaces, given by a diagram $H \xrightarrow{j} W \xrightarrow{\pi} B$ in which π is a locally trivial fibration with compact smoot...

open_problem | open | MathDB 页面 | 原文线索

V2-076472 | MATHDB-397039

Beukers’ full-monodromy conjecture for A-hypergeometric systems

摘录 / EXCERPT: Let $H_A(\beta)$ be the A-hypergeometric system in the report, where A is an integer $n\times N$ matrix whose columns generate $\mathbb Z^n$, the rows of A have...

conjecture | open | MathDB 页面 | 原文线索

V2-076473 | MATHDB-397047

Kleshchev-Martin self-extension conjecture

摘录 / EXCERPT: For $p \ge 3$, is $\operatorname{Ext}^{1}_{k\Sigma_d}(D_{\lambda},D_{\lambda})=0$?

conjecture | open | MathDB 页面 | 原文线索

V2-076474 | MATHDB-397049

Parity of a diagonal Cartan invariant in a 2-block of a symmetric group

摘录 / EXCERPT: Given $n\ge 1$ and $p=2$, let $\lambda=(n+1,n-1)\vdash 2n$. Is the Cartan invariant $C_{\lambda\lambda}$ odd if and only if n is even?

open_problem | open | MathDB 页面 | 原文线索

V2-076475 | MATHDB-397054

Criterion for selecting the dynamic steady-state crack speed

摘录 / EXCERPT: There remains the key question of defining a criterion for predicting the dynamic, steady-state crack speed.

open_problem | open | MathDB 页面 | 原文线索

V2-076476 | MATHDB-397055

Counterexamples to the Eshelby-Kostrov property for smooth subsonic cracks

摘录 / EXCERPT: Can one obtain counterexamples to the Eshelby-Kostrov property for cracks moving at subsonic speeds along smooth curves, perhaps using perturbative arguments?

open_problem | open | MathDB 页面 | 原文线索

V2-076477 | MATHDB-397057

Asymptotic vanishing of k-dominant skylines

摘录 / EXCERPT: For a fixed dimension d, let D consist of n points sampled independently and uniformly either from the hypercube $[0,1]^d$ or from the simplex...

open_problem | open | MathDB 页面 | 原文线索

V2-076478 | MATHDB-397059

Classification of two-sided Fourier-transform fixed points for the homogeneous smoothing transform

摘录 / EXCERPT: Assume the homogeneous smoothing-transform setting with nonnegative random variables $T=(T_k)_{k\ge1}$, $N:=\sum_{k\ge1}\mathbf 1_{\{T_k>0\}}<\infty$ almost surely,...

open_problem | open | MathDB 页面 | 原文线索

V2-076479 | MATHDB-397060

Limit process for online Quicksort

摘录 / EXCERPT: For Quicksort run online on a list of n distinct real numbers, let $X_n(l)$ be the number of comparisons made until the l-th smallest element appears, where $1\le l\le n$. With...

open_problem | open | MathDB 页面 | 原文线索

V2-076480 | MATHDB-397062

Functional limit law for partial-match queries in random quadtrees

摘录 / EXCERPT: Consider a random two-dimensional quadtree built from independent uniformly distributed points in $[0,1]^2$. Let $C_n(s)$ be the number of nodes visited by the partial-match query...

open_problem | open | MathDB 页面 | 原文线索

V2-076481 | MATHDB-397063

Linear sample size for covariance estimation under finite $2+\\varepsilon$ moments

摘录 / EXCERPT: A conjecture is that the logarithmic oversampling is almost never needed; in particular, that $N=O(n)$ for distributions with $2+\varepsilon$ moments.

open_problem | open | MathDB 页面 | 原文线索

V2-076482 | MATHDB-397068

Lieb’s concavity theorem for the trace exponential

摘录 / EXCERPT: Let H be a self-adjoint matrix. Then the map $A \mapsto \operatorname{tr}\exp\{H+\log A\}$ is concave on the positive-definite cone.

open_problem | open | MathDB 页面 | 原文线索

V2-076483 | MATHDB-397070

Thin-shell (variance) conjecture for isotropic log-concave random vectors

摘录 / EXCERPT: Let X be an isotropic random vector in $\mathbb{R}^n$ with log-concave density f. Does there exist a universal constant C such that … or, equivalently as stated in the report...

conjecture | open | MathDB 页面 | 原文线索

V2-076484 | MATHDB-397071

Interpolating thin-shell and sharp large-deviation concentration for isotropic log-concave measures

摘录 / EXCERPT: Let X be an isotropic random vector in $\mathbb{R}^n$ with log-concave density, which is additionally ψ_α for some $\alpha\in[1,2]$ with constant b_α, meaning th...

open_problem | open | MathDB 页面 | 原文线索

V2-076485 | MATHDB-397072

Thin-shell concentration conjecture of Anttila, Ball, and Perissinaki

摘录 / EXCERPT: There exists a sequence (ε_n) with $\varepsilon_n\to 0$ as $n\to\infty$ such that, for every isotropic log-concave random vector X in $\mathbb{R}^n$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-076486 | MATHDB-397073

Four Moment Theorem for Wigner Hermitian matrices

摘录 / EXCERPT: For any small positive constant c, there is a small positive constant c_0 such that, for every integer $k\ge 1$, the following holds. Let $M_n=(\zeta_{ij})_{1\le i,j\le n}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076487 | MATHDB-397074

Hardness of approximation for the Quadratic Programming Problem

摘录 / EXCERPT: Theorem 3. For the Quadratic Programming Problem of maximizing $\sum_{i,j=1}^{n} a_{ij}x_i x_j$ over vectors x satisfying $\lVert x\rVert_{\infty}\le 1$, with diagonal coefficien...

open_problem | open | MathDB 页面 | 原文线索

V2-076488 | MATHDB-397075

Conjecture 1 on the largest singular value of random DFT submatrices

摘录 / EXCERPT: Let F be the $n\times n$ discrete Fourier transform matrix, with entries $F_{jk}=n^{-1/2}e^{-2\pi i(j-1)(k-1)/n}$. Let $\Omega_n,T_n\subset\{1,\ldots,n\}$ be formed by including...

conjecture | open | MathDB 页面 | 原文线索

V2-076489 | MATHDB-397076

General-potential soft-edge limit for beta ensembles

摘录 / EXCERPT: For a strictly convex polynomial potential V, replacing the Gaussian weight $e^{-\beta\\lambda^2/4}$ by $e^{-\beta V(\\lambda)}$ yields constants c_0,c_1 such that ... The report...

open_problem | open | MathDB 页面 | 原文线索

V2-076490 | MATHDB-397078

Learning a k-subspace junta

摘录 / EXCERPT: Problem 1 (Learning a k-subspace junta). Given examples drawn from a factorizable distribution $F=F_VF_W$ over $\mathbb{R}^n$, and labeled by ℓ (a k-subspace junta with re...

open_problem | open | MathDB 页面 | 原文线索

V2-076491 | MATHDB-397079

Explicit random operators for non-Euclidean embeddings

摘录 / EXCERPT: Can one give an explicit construction of a random operator that embeds $\cell\ell_p^n$ into $\cell\ell_r^N$, with the isomorphism constant depending only on p and r?

open_problem | open | MathDB 页面 | 原文线索

V2-076492 | MATHDB-397080

Compressed-sensing recovery from a kernel-diameter condition

摘录 / EXCERPT: Let $0<r\le p<\infty$, let $A\in\mathbb{R}^{n\times N}$, and let $1\le m\le N$. Writing q so that $\frac1p+\frac1q=\frac1r$, if ... then A satisfies the ℓ_r-minimization pro...

open_problem | open | MathDB 页面 | 原文线索

V2-076493 | MATHDB-397083

Linear growth of an entry in a random contingency table

摘录 / EXCERPT: Suppose that $m=n$ and $R=C=(3n,n,\ldots,n)$. Let D be chosen uniformly at random from $A_+(R,C)$, the set of $n\times n$ matrices with non-negative integer entries and row sums...

open_problem | open | MathDB 页面 | 原文线索

V2-076494 | MATHDB-397084

Multivariate normal approximation for traces of powers on compact classical groups

摘录 / EXCERPT: For a Haar-distributed matrix M_n in U_n, SO_n, or USp_{2n}, the report states that the Wasserstein distance between...

open_problem | open | MathDB 页面 | 原文线索

V2-076495 | MATHDB-397085

Universal diffusion exponent for the wind-tree model

摘录 / EXCERPT: For every $(a,b)\in(0,1)^2$, for almost every direction θ and every point $x\in T(a,b)$ with an infinite trajectory, ...

open_problem | open | MathDB 页面 | 原文线索

V2-076496 | MATHDB-397086

Simplicity of the Kontsevich-Zorich Lyapunov spectrum for Masur-Veech measure

摘录 / EXCERPT: For the Masur-Veech measure μ_{MV} on a connected component of a stratum $\mathcal H(k_1,\ldots,k_\sigma)$ of unit-area Abelian differentials, the Lyapunov spectrum of...

open_problem | open | MathDB 页面 | 原文线索

V2-076497 | MATHDB-397090

Classification of Teichmüller disks with completely degenerate Kontsevich-Zorich spectrum

摘录 / EXCERPT: It is conjectured that the genus-three Eierlegende Wollmilchsau example and the genus-four Forni-Matheus example represent the only Teichmüller disks contained in $D_g(1)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076498 | MATHDB-397091

Computing individual Lyapunov exponents of the Kontsevich-Zorich cocycle

摘录 / EXCERPT: How can one compute the individual Lyapunov exponents of the Kontsevich-Zorich cocycle over a Teichmüller curve, namely the numbers...

open_problem | open | MathDB 页面 | 原文线索

V2-076499 | MATHDB-397093

Pure periodicity versus torus branched coverings

摘录 / EXCERPT: Does pure periodicity imply that a translation surface is a torus branched covering? Here, a translation surface is purely periodic if, for every direction $\theta\in S^1$, the exi...

open_problem | open | MathDB 页面 | 原文线索

V2-076500 | MATHDB-397095

Density of the orbit of a particular translation surface

摘录 / EXCERPT: The problem of determining whether the $SL(2,\mathbb{R})$-orbit of a particular translation surface is dense in its connected component is wide open.

open_problem | open | MathDB 页面 | 原文线索

V2-076501 | MATHDB-397097

Quadratic Dehn function for the genus-2 handlebody group

摘录 / EXCERPT: The Dehn function for genus-2 handlebody groups is of quadratic growth type.

open_problem | open | MathDB 页面 | 原文线索

V2-076502 | MATHDB-397102

Atiyah problem in the torsion case

摘录 / EXCERPT: Every non-negative real number can be the kernel-dimension of an element in the integral group ring of a countable group.

open_problem | open | MathDB 页面 | 原文线索

V2-076503 | MATHDB-397103

Bernoulli-shift factor universality for non-amenable groups

摘录 / EXCERPT: I conjecture that if G is any non-amenable group, then every Bernoulli shift over G factors onto every Bernoulli shift over G.

open_problem | open | MathDB 页面 | 原文线索

V2-076504 | MATHDB-397104

Approximation Conjecture for L^2-torsion

摘录 / EXCERPT: Let G be a group together with an inverse system $\{G_i\mid i\in I\}$ of normal subgroups of G, directed by inclusion, such that $[G:G_i]$ is finite for all $i\in I$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-076505 | MATHDB-397105

The Atiyah problem for a countable discrete group

摘录 / EXCERPT: Given a countable discrete group G, what is the set $C(G)$ of all l^2-Betti numbers arising from G? Here a real number r arises from G if and only if there exist an integ...

open_problem | open | MathDB 页面 | 原文线索

V2-076506 | MATHDB-397108

Structural Atiyah conjecture for the division closure

摘录 / EXCERPT: Let Γ be a group whose finite subgroups have bounded orders, and suppose that Γ contains no finite normal subgroup. Let l be the least common multiple of the orders...

conjecture | open | MathDB 页面 | 原文线索

V2-076507 | MATHDB-397109

Structural Lück Approximation Conjecture

摘录 / EXCERPT: For any sofic approximation of a sofic group Γ, the regular closure in $\mathcal{M}^{\mathrm{alg}}_{\mu}$ is isomorphic, as a rank-regular $\star$-ring, to the Linnell-Schic...

conjecture | open | MathDB 页面 | 原文线索

V2-076508 | MATHDB-397110

Isomorphism of ultrapowers of a fixed II_1 factor

摘录 / EXCERPT: For a fixed II_1 factor M, are all ultrapowers $M^{\\mathcal U}$, with respect to nonprincipal ultrafilters $\\mathcal U$ on $\\mathbb N$, isomorphic? The report states that assum...

open_problem | open | MathDB 页面 | 原文线索

V2-076509 | MATHDB-397111

Existence of a non-weakly-sofic group

摘录 / EXCERPT: There exists a group that is not weakly sofic (w-sofic).

open_problem | open | MathDB 页面 | 原文线索

V2-076510 | MATHDB-397112

Closure of a product of conjugacy classes in a free group

摘录 / EXCERPT: For a finitely generated free group F, there exists a finite sequence $g_1,g_2,\ldots,g_k\in F$ such that ... where $[g_i]^F$ is the conjugacy class of g_i in F, the overline d...

open_problem | open | MathDB 页面 | 原文线索

V2-076511 | MATHDB-397113

Coarse embeddability versus coarse amenability for countable discrete groups

摘录 / EXCERPT: Does there exist a countable discrete group which is coarsely embeddable and coarsely non-amenable (equivalently, does not have Property A)?

open_problem | open | MathDB 页面 | 原文线索

V2-076512 | MATHDB-397117

Classification of sofic approximations for a direct product of two free groups

摘录 / EXCERPT: Is there an analogous result for the direct product of two free groups?

open_problem | open | MathDB 页面 | 原文线索

V2-076513 | MATHDB-397118

Soficity or hyperlinearity of several groups

摘录 / EXCERPT: Are the following groups sofic or hyperlinear: Thompson’s group F, one-relator groups, Burnside’s group, Burger-Mozes groups, and Gromov’s monster (a finitely generated group coa...

open_problem | open | MathDB 页面 | 原文线索

V2-076514 | MATHDB-397119

Extension of sofic approximations from a free subgroup of $\mathrm{SL}_3(\mathbb{Z})$

摘录 / EXCERPT: If F is a free subgroup of $\mathrm{SL}_3(\mathbb{Z})$, can every sofic approximation to F be obtained by restricting a sofic approximation to $\mathrm{SL}_3(\mathbb{Z})$?

open_problem | open | MathDB 页面 | 原文线索

V2-076515 | MATHDB-397124

Hypothesis A on the Euler and Pontryagin classes of $B\mathrm{STOP}(2m)$

摘录 / EXCERPT: Hypothesis A asserts that, for every positive integer m, $p_m=e^2\in H^{4m}(B\mathrm{STOP}(2m);\mathbb{Q})$, where e is the Euler class and p_m is the Pontryagin class.

conjecture | open | MathDB 页面 | 原文线索

V2-076516 | MATHDB-397126

Farrell-Jones conjecture for $\mathrm{SL}_n(\mathbf{Z})$

摘录 / EXCERPT: The K- and L-theoretic Farrell-Jones conjecture, in the more general version with coefficients in an additive category (with involutions), holds for $\mathrm{SL}_n(\mathbf{Z})$...

conjecture | open | MathDB 页面 | 原文线索

V2-076517 | MATHDB-397132

Theorem 1 (Lurie): preorientations of the derived multiplicative group

摘录 / EXCERPT: For any abelian monoid A in symmetric spectra Sp^{Σ} based on simplicial sets, and any simplicial abelian group N, there is a natural isomorphism of abelian gro...

open_problem | open | MathDB 页面 | 原文线索

V2-076518 | MATHDB-397134

Farrell-Wall question on cocompact actions on $S^n\times\mathbb R^k$

摘录 / EXCERPT: Which discrete groups can act freely, properly and cocompactly on some product $S^n\times\mathbb R^k$?

open_problem | open | MathDB 页面 | 原文线索

V2-076519 | MATHDB-397135

Adaptive eigensolver for large-scale nonlinear eigenvalue problems

摘录 / EXCERPT: Develop an adaptive eigensolver for the described large-scale nonlinear eigenvalue problem class, particularly for industrial applications; the report states that this remains main...

open_problem | open | MathDB 页面 | 原文线索

V2-076520 | MATHDB-397137

Extension to nonreal and multiple eigenvalues

摘录 / EXCERPT: The results need to be extended to problems with nonreal and multiple eigenvalues.

open_problem | open | MathDB 页面 | 原文线索

V2-076521 | MATHDB-397138

Convergence of adaptive methods for the hierarchical-type estimator

摘录 / EXCERPT: Can convergence of an adaptive method be proven for this type of estimator?

open_problem | open | MathDB 页面 | 原文线索

V2-076522 | MATHDB-397139

A theoretical bound for the time-dependent Henry-condition error

摘录 / EXCERPT: For the time-dependent Nitsche-XFEM discretization described in the report, derive a theoretical bound for the error quantity $\|[\beta u_h^*]\|_{L^2(\Gamma)}$, where u_h^* is th...

open_problem | open | MathDB 页面 | 原文线索

V2-076523 | MATHDB-397140

Delocalization criterion for random Schrödinger operators on regular trees

摘录 / EXCERPT: The report presents the following result: for the random operators $H_\lambda=T+\lambda V$ on regular tree graphs, under the assumptions stated before Theorem 1 (in particular, an...

open_problem | open | MathDB 页面 | 原文线索

V2-076524 | MATHDB-397144

Edge universality for generalized Wigner matrices under a four-plus-epsilon moment condition

摘录 / EXCERPT: Establish edge universality for generalized Wigner matrices assuming only a finite $(4+\varepsilon)$ moment of the matrix entries, rather than the twelfth-moment hypothesis used in...

open_problem | open | MathDB 页面 | 原文线索

V2-076525 | MATHDB-397145

Scaling limit of uniform random triangulations (Schramm’s problem)

摘录 / EXCERPT: Let M_n be uniformly distributed over the set of rooted planar triangulations with n faces, for even integers $n\ge 2$. With $V(M_n)$ equipped with graph distance...

open_problem | open | MathDB 页面 | 原文线索

V2-076526 | MATHDB-397146

Uniqueness of the Brownian map

摘录 / EXCERPT: Can different subsequential Gromov-Hausdorff limits of rescaled uniformly distributed rooted planar q-angulations, or different admissible values of q (namely $q=3$ or even...

open_problem | open | MathDB 页面 | 原文线索

V2-076527 | MATHDB-397147

Fluctuation exponents for a 1+1-dimensional directed random polymer

摘录 / EXCERPT: For the directed polymer with admissible up-right paths from $(0,0)$ to (m,n) and independent log-gamma weights satisfying $Y_{i,0}^{-1}\sim\operatorname{Gamma}(\theta)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076528 | MATHDB-397151

Finiteness of the deformation orbit space for unipotently connected groups

摘录 / EXCERPT: Let G be a simply connected, solvable real Lie group, and let Γ be a Zariski-dense lattice in G. If G is unipotently connected, then the orbit space...

open_problem | open | MathDB 页面 | 原文线索

V2-076529 | MATHDB-397154

Cofinite proper classifying spaces for lattices

摘录 / EXCERPT: Let Γ be a lattice in a semisimple Lie group G, and let X be the corresponding symmetric space. There exists a partial compactification X_Γ of X that is a cofin...

open_problem | open | MathDB 页面 | 原文线索

V2-076530 | MATHDB-397155

Geometric criterion for the generalized integral Novikov conjecture

摘录 / EXCERPT: Let Γ be a discrete subgroup of finite asymptotic dimension in a Lie group with finitely many connected components. If Γ admits a cofinite model X of the proper cla...

conjecture | open | MathDB 页面 | 原文线索

V2-076531 | MATHDB-397157

Veech groups among finite-index subgroups of $SL_2(\d53d)$

摘录 / EXCERPT: Is it true that every finite-index subgroup of $SL_2(\mathbb{Z})$ is a Veech group?

open_problem | open | MathDB 页面 | 原文线索

V2-076532 | MATHDB-397162

Convex cocompactness of finitely generated purely pseudo-Anosov subgroups

摘录 / EXCERPT: Is every finitely generated, purely pseudo-Anosov subgroup convex cocompact?

open_problem | open | MathDB 页面 | 原文线索

V2-076533 | MATHDB-397165

Kernel of the map between profinite completions

摘录 / EXCERPT: Determine the kernel of the map between the profinite completions $\widehat{\widetilde{\rho}_p(\widetilde{M}_g)}\to\widehat{G_p(\mathbb{Z})}$.

open_problem | open | MathDB 页面 | 原文线索

V2-076534 | MATHDB-397166

Dimension-three comparison of functorial seminorms

摘录 / EXCERPT: Does the functorial semi-norm on $H_3(\,\cdot\,;\mathbb{R})$ associated with the simplicial volume coincide with the ℓ^1-semi-norm?

open_problem | open | MathDB 页面 | 原文线索

V2-076535 | MATHDB-397169

Left-orderability of irreducible arithmetic subgroups in higher real rank

摘录 / EXCERPT: Suppose Γ is an irreducible arithmetic subgroup of a connected, semisimple Lie group G, such that $\operatorname{rank}_{\mathbb{R}} G \ge 2$. Then Γ does not have a...

open_problem | open | MathDB 页面 | 原文线索

V2-076536 | MATHDB-397177

Generic Serre-weight equality for a compact unitary group of rank 3

摘录 / EXCERPT: In the setup of the report, let F be a CM field with p splitting completely in F, assume $[F:\mathbb{Q}]=2$, and let G be a compact unitary group of rank 3 over...

open_problem | open | MathDB 页面 | 原文线索

V2-076537 | MATHDB-397178

Sharifi’s inverse-map conjecture

摘录 / EXCERPT: For the homomorphisms $\Psi:X^-\to H^-/IH^-$ and $\Phi:H^-/IH^-\to X^-$ constructed in the report, $\Phi\circ\Psi=1$ and $\Psi\circ\Phi=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-076538 | MATHDB-397183

Kudla's modularity conjecture for orthogonal Shimura-cycle generating series (K1)

摘录 / EXCERPT: Assuming that an Arakelov theory can be set up to deal with all the different singularities occurring, Kudla conjectures that $\Theta_n(L)$ and $\widehat{\Theta}_n(L)$ are, respect...

conjecture | open | MathDB 页面 | 原文线索

V2-076539 | MATHDB-397184

Positive-density Lieb-Thirring inequality for a Fermi sea

摘录 / EXCERPT: For space dimensions $d\ge 2$, let $\rho_0>0$ and let μ be related to ρ_0 by ... With $\Pi^-:=\chi_{(-\infty,\mu)}(-\Delta)$, there is a positive constant $K(d)$ such that,...

open_problem | open | MathDB 页面 | 原文线索

V2-076540 | MATHDB-397185

Closedness of tensor-format rank sets

摘录 / EXCERPT: The report states that the tensor-format sets T_r and H_r are closed, even weakly closed, whereas R_r is not closed. It further states that replacing the tree defining H_r...

open_problem | open | MathDB 页面 | 原文线索

V2-076541 | MATHDB-397187

Linear-scaling truncation of the Slater-determinant expansion for Hartree-Fock energy

摘录 / EXCERPT: Are all the $N!$ permutation terms necessary to compute the exchange term accurately, or can terms be neglected according to some heuristic criterion so that, in $O(N)$ steps, the...

open_problem | open | MathDB 页面 | 原文线索

V2-076542 | MATHDB-397190

Global isoperimetricity of the Huisken-Yau surfaces

摘录 / EXCERPT: Are the surfaces considered by Huisken and Yau also global minimizers, i.e. isoperimetric regions?

open_problem | open | MathDB 页面 | 原文线索

V2-076543 | MATHDB-397192

Possible \-valued side lengths of polygons in Euclidean buildings

摘录 / EXCERPT: Given a symmetric space of noncompact type or a thick Euclidean building modeled on a Euclidean Coxeter complex, determine the set $P_n(X)\subset\Delta_{\mathrm{euc}}^n$ of all po...

open_problem | open | MathDB 页面 | 原文线索

V2-076544 | MATHDB-397193

Stability of locally symmetric spaces of noncompact type under rescaled Ricci flow

摘录 / EXCERPT: Let (M_{sym},g) be a locally symmetric space of noncompact type which is Einstein with Einstein constant λ, so that g is a fixed point of the rescaled Ricci fl...

open_problem | open | MathDB 页面 | 原文线索

V2-076545 | MATHDB-397195

Classification of gravitational instantions on complements of anticanonical divisors

摘录 / EXCERPT: If $X \setminus D$ carries a gravitational instanton metric, then $X \setminus D$ is biholomorphic to one of the known examples up to hyperkähler rotation.

open_problem | open | MathDB 页面 | 原文线索

V2-076546 | MATHDB-397199

Image of the frequency mapping for central relative equilibria

摘录 / EXCERPT: Let S_0 be a real symmetric $2p\times 2p$ matrix with eigenvalues $\sigma_1\geq\cdots\geq\sigma_{2p}$, and let ... For $R\in SO(2p)$, set $S=RS_0R^{-1}$ and define the frequency ma...

open_problem | open | MathDB 页面 | 原文线索

V2-076547 | MATHDB-397202

A classical Hadamard-quasianalytic function space for KAM-curve parametrizations

摘录 / EXCERPT: An interesting open problem would be to find a natural space of functions which contains the parametrization of the KAM curves and which is quasianalytic in the classical Hadamard...

open_problem | open | MathDB 页面 | 原文线索

V2-076548 | MATHDB-397205

Area-preserving rational-rotation approximations of pseudo-rotations

摘录 / EXCERPT: Let φ be a smooth, area-preserving diffeomorphism of the closed disk D with precisely one periodic point, namely 0. Theorem 1 gives a sequence of smooth diffeomorphisms...

open_problem | open | MathDB 页面 | 原文线索

V2-076549 | MATHDB-397207

Existence of the fourth-order Birkhoff normal-form flow

摘录 / EXCERPT: For the nonlinear Schrödinger equation on $\mathbb{R}^d$, with $X_{G^{(4)}}$ defined by equation (5), does the flow $\psi_s(u)$ exist, and on which function space? The report answe...

open_problem | open | MathDB 页面 | 原文线索

V2-076550 | MATHDB-397208

Existence of scattering limits for the defocusing cubic nonlinear Schrödinger equation

摘录 / EXCERPT: For the defocusing cubic nonlinear Schrödinger equation on $\mathbb{R}^d$ defined on page 45, with $d>1$, the report states that ... exists in $H^1(\mathbb{R}^d)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076551 | MATHDB-397209

Arnold diffusion for typical perturbations of the pendulum-rotator product

摘录 / EXCERPT: For a typical $\[?1m\varepsilon\u001b[0m$-perturbation of the product of the pendulum and the rotator, a positive conditional measure of initial conditions concentrated in the main...

open_problem | open | MathDB 页面 | 原文线索

V2-076552 | MATHDB-397217

Mahler measure and the conductor-40 elliptic curve

摘录 / EXCERPT: Let $E/\mathbb Q$ be the elliptic curve defined by ... The report states the identity ... where, for $P\in\mathbb C[x,y]$ as written in the report, ...

open_problem | open | MathDB 页面 | 原文线索

V2-076553 | MATHDB-397218

A positive proportion of elliptic curves satisfy the BSD rank conjecture

摘录 / EXCERPT: A positive proportion of elliptic curves satisfy the Birch-Swinnerton-Dyer rank conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-076554 | MATHDB-397226

Goncharov’s conjecture on a four-variable functional equation for the 4-logarithm

摘录 / EXCERPT: There exists a combination $S_4(x,y;z)\in\mathbb{Z}[F]$ whose image under $\partial^{(4)}_4$ is $\kappa(\xi(x,y),z)$.

conjecture | open | MathDB 页面 | 原文线索

V2-076555 | MATHDB-397228

Function-field analogue of Vandiver’s conjecture

摘录 / EXCERPT: Does $H_p(\omega^{1-n})=\{0\}$ hold for n not divisible by $q-1$?

conjecture | open | MathDB 页面 | 原文线索

V2-076556 | MATHDB-397230

Nahm’s conjecture about modularity of q-series

摘录 / EXCERPT: For a positive integer r, a real positive-definite symmetric $r\times r$ matrix A, a vector B of length r, and a scalar C, define ... The problem is to describe all triples...

conjecture | open | MathDB 页面 | 原文线索

V2-076557 | MATHDB-397234

Erdős distinct-distances bound as reported at the workshop

摘录 / EXCERPT: Given a set of N points in the plane, the number of distinct distances between these points is at least $C\sqrt{N}/\log N$.

open_problem | open | MathDB 页面 | 原文线索

V2-076558 | MATHDB-397236

Maximal Riesz-transform boundedness versus variational boundedness

摘录 / EXCERPT: Given an arbitrary nonatomic measure μ on $\mathbb{R}^d$, does boundedness of the maximal Riesz transform R_*^μ on $L^2(\mu)$ imply boundedness of...

open_problem | open | MathDB 页面 | 原文线索

V2-076559 | MATHDB-397237

HRT conjecture for arbitrary time-frequency shifts

摘录 / EXCERPT: Let $(t_j,_j)_{j=1}^n$ be $n\ge 2$ distinct points in the plane. Then there is no nontrivial Schwartz function $f:\mathbb{R}\to\mathbb{C}$ satisfying a nontrivial linear dependenc...

conjecture | open | MathDB 页面 | 原文线索

V2-076560 | MATHDB-397238

HRT linear independence for $(1,3)$ and $(2,2)$ configurations

摘录 / EXCERPT: For every $(1,3)$ or $(2,2)$ configuration of distinct points $(t_j,\b6_j)$ in the plane, there is no nontrivial Schwartz function $f:\mathbb{R}\to\mathbb{C}$ satisfying ... for almo...

open_problem | open | MathDB 页面 | 原文线索

V2-076561 | MATHDB-397239

Schwartz solutions of a three-line time-frequency recurrence

摘录 / EXCERPT: Does ... have a Schwartz solution?

open_problem | open | MathDB 页面 | 原文线索

V2-076562 | MATHDB-397240

Continuous dependence for the one-dimensional cubic NLS below L^2

摘录 / EXCERPT: For the one-dimensional cubic nonlinear Schrödinger equation ... with either the focusing or defocusing sign, does the solution depend continuously on its initial data when the initi...

open_problem | open | MathDB 页面 | 原文线索

V2-076563 | MATHDB-397241

Sharp Fourier restriction range in linearly adapted coordinates

摘录 / EXCERPT: Let S be a smooth, finite-type hypersurface in $\e\d\mathbb{R}^3$ with Riemannian surface measure $d\sigma$, and let $d\mu=\rho\,d\sigma$, where $0\leq\rho\in C_0^\infty(S)$. Aft...

open_problem | open | MathDB 页面 | 原文线索

V2-076564 | MATHDB-397242

Equivalence of the k-Hessian and fractional Laplacian energies without the extra positivity assumption

摘录 / EXCERPT: Let $u\in\Phi_0^k(\mathbb R^n)$ with $1\le k<n/2$. The conjecture is that the additional assumption ... in Theorem 2 can be dropped; equivalently, the k-Hessian energy ... and the fr...

open_problem | open | MathDB 页面 | 原文线索

V2-076565 | MATHDB-397243

Extension of the one-variable vector-field theorem to the maximal operator

摘录 / EXCERPT: For a nonvanishing vector field $v:\mathbb{R}^2\to\mathbb{R}^2\setminus\{0\}$ depending on one variable, $v(x,y)=v(x)$, the report asks whether the preceding L^p boundedness theo...

open_problem | open | MathDB 页面 | 原文线索

V2-076566 | MATHDB-397244

Power-law decay for nondegenerate multilinear oscillatory integrals

摘录 / EXCERPT: Let ... where each $\ell_j:\mathbb R^d\to\mathbb R^{d_j}$ is a surjective linear map, $P:\mathbb R^d\to\mathbb R$ is a real-valued polynomial, and η is compactly supported and s...

open_problem | open | MathDB 页面 | 原文线索

V2-076567 | MATHDB-397245

Power-law bounds for multilinear sublevel sets

摘录 / EXCERPT: Let ... The report conjectures that if the polynomial P is nondegenerate relative to the linear maps ℓ_j, then there exist constants C and $\delta>0$ such that ... uniformly f...

open_problem | open | MathDB 页面 | 原文线索

V2-076568 | MATHDB-397247

Weighted dimension-free bounds for Riesz transforms of arbitrary-type Laguerre expansions

摘录 / EXCERPT: Does the weighted norm inequality stated in Theorem 5 remain valid for Laguerre expansions of arbitrary type studied by Nowak and Stempak [2]? The result proved there is: for every...

open_problem | open | MathDB 页面 | 原文线索

V2-076569 | MATHDB-397248

David-Semmes question for integer dimensions other than codimension one

摘录 / EXCERPT: Let $n\in\mathbb{Z}_{+}$ satisfy $0<n<d$ and $n\ne d-1$. Let $E\subset\mathbb{R}^{d}$ be compact with $0<\mathcal{H}^{n}(E)<\infty$, and assume that E is n-Ahlfors regular: the...

open_problem | open | MathDB 页面 | 原文线索

V2-076570 | MATHDB-397249

David-Semmes question for codimension-one Riesz transforms

摘录 / EXCERPT: Let E be a compact subset of $[U+007F]\mathbb{R}^d$ with $0<\mathcal{H}^s(E)<\infty$. Assume that E is Ahlfors s-regular: ... Let $R^s=(R_1^s,\ldots,R_d^s)$ be the vector Riesz transf...

open_problem | open | MathDB 页面 | 原文线索

V2-076571 | MATHDB-397250

Non-existence of lower-irregular sets with bounded non-integer Riesz transforms

摘录 / EXCERPT: For $s\in(d-1,d)$, there is no compact set $E\subset\mathbb{R}^d$ with $0<\mathcal{H}^s(E)<\infty$ such that the vector s-Riesz transform ... is bounded and ... (The report states th...

open_problem | open | MathDB 页面 | 原文线索

V2-076572 | MATHDB-397251

Non-integer-dimensional Riesz-transform boundedness

摘录 / EXCERPT: Conjecture: Let $0<s<d$ with $s\notin\mathbb Z$, and let $E\subset\mathbb R^d$ be compact with $0<\mathcal H^s(E)<\infty$. Then the vector s-Riesz transform R^s cannot define a...

open_problem | open | MathDB 页面 | 原文线索

V2-076573 | MATHDB-397252

Finiteness testing for finitely generated linear groups over infinite fields

摘录 / EXCERPT: Given a finitely generated linear group $G \leq \mathrm{GL}(n,F)$, where F is an infinite field, decide whether G is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-076574 | MATHDB-397256

Higman’s PORC conjecture

摘录 / EXCERPT: For each fixed n, let $f_n(p)$ denote the number of groups of order p^n, where p ranges over the primes. Higman’s conjecture is that $f_n(p)$ is polynomial on residue classes...

conjecture | open | MathDB 页面 | 原文线索

V2-076575 | MATHDB-397258

Local invariant-ring defects are controlled by pointwise stabilizers

摘录 / EXCERPT: Let $G\subseteq \mathrm{GL}(V)$ be a finite subgroup acting linearly on a finite-dimensional vector space V over a field K. For $x\in V$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-076576 | MATHDB-397264

Cancellation for natural central products of centrally indecomposable class-2 exponent-\p groups

摘录 / EXCERPT: For a group G, define its natural central product with itself by ... If G and H are p-groups of class 2 and exponent p that are not themselves central products of proper...

open_problem | open | MathDB 页面 | 原文线索

V2-076577 | MATHDB-397265

Undecidable word problem for a self-similar quotient of a free group

摘录 / EXCERPT: Let F be a free group of finite rank. Let $S=\{1,\ldots,k\}$ be a finite set, and let $\varphi:F\to F\wr_S\operatorname{Sym}(S)$ be a homomorphism. Iterating φ, obtain ho...

open_problem | open | MathDB 页面 | 原文线索

V2-076578 | MATHDB-397266

Construct low-degree irreducible representations of alternating groups

摘录 / EXCERPT: Construct all irreducible representations of A_n over $\c9bb F_q$ of degree d, where $n \le 25$ and $d \le n^3$.

open_problem | open | MathDB 页面 | 原文线索

V2-076579 | MATHDB-397267

Local uniqueness of the Clifford torus as a Willmore minimizer

摘录 / EXCERPT: The Clifford torus T_{Cliff} is the unique minimizer in $\mathbb{R}^3$, up to Möbius transformations, of the Willmore energy in an open neighbourhood in moduli space of...

open_problem | open | MathDB 页面 | 原文线索

V2-076580 | MATHDB-397270

Regularity of optimal transport when standard structural conditions fail

摘录 / EXCERPT: What if any of the standard conditions do not hold? Can we give conditions on the densities which are sufficient to conclude that the map is regular?

open_problem | open | MathDB 页面 | 原文线索

V2-076581 | MATHDB-397272

Regime of applicability of the Fokker-Planck treatment for random Hill equations

摘录 / EXCERPT: Determine the regime of applicability of the Fokker-Planck treatment for the random Hill equation class considered in the report, given that the derivation is valid for small stoch...

open_problem | open | MathDB 页面 | 原文线索

V2-076582 | MATHDB-397273

Minimum-energy configurations of Full Body problems

摘录 / EXCERPT: A reasonable hypothesis, still to be fully proven, is that all minimum-energy configurations of Full Body problems result in one of two general states: (i) all the particles rest o...

open_problem | open | MathDB 页面 | 原文线索

V2-076583 | MATHDB-397275

Extension of the FitzHugh-Nagumo estimates to other parameter regimes

摘录 / EXCERPT: Obtain results similar to Theorems 1 and 2 for parameter regimes other than the weak-noise regime described in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-076584 | MATHDB-397276

Similar interspike-statistics results beyond the weak-noise FitzHugh-Nagumo regime

摘录 / EXCERPT: The report identifies as an open problem obtaining similar results in other parameter regimes, as well as for other models such as the Morris-Lecar equations.

open_problem | open | MathDB 页面 | 原文线索

V2-076585 | MATHDB-397280

Gal’s conjecture for flag simple polytopes

摘录 / EXCERPT: Conjecture 1 (Gal, 2005). Any flag simple n-polytope P satisfies $\\gamma_i(P^n) \\ge 0$ for $i=0,\\ldots,\\left\\lfloor n/2\\right\\rfloor$.

conjecture | open | MathDB 页面 | 原文线索

V2-076586 | MATHDB-397281

Gal’s conjecture for graph cubeahedra

摘录 / EXCERPT: Gal’s conjecture holds for every graph cubeahedron.

conjecture | open | MathDB 页面 | 原文线索

V2-076587 | MATHDB-397285

Insuring against loss of evidence and capital

摘录 / EXCERPT: Under the protocol in which $X_0=1$, ${}^{I}K_0=1$, Investor announces $p_t\in\mathbb{R}$, Market announces $X_t\in[0,\infty)$, and ${}^{I}K_t:={}^{I}K_{t-1}+p_t(X_t-X_{t-1})$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-076588 | MATHDB-397289

Meromorphic continuation of a double Dirichlet series attached to quadratic Hecke characters

摘录 / EXCERPT: Let $E=\mathbb{Q}(i)$ and, for $n\in\mathbb{Z}$, let χ_n be the Hecke character of E defined by ... For a Hecke character χ of E, define ... where $I^*(\mathcal O_E)$ den...

open_problem | open | MathDB 页面 | 原文线索

V2-076589 | MATHDB-397290

Logarithm law for one-parameter unipotent flows

摘录 / EXCERPT: Let $G=\mathrm{SL}_2(\mathbb{R})^{r_1}\times\mathrm{SL}_2(\mathbb{C})^{r_2}$, and let $\Gamma\subseteq G$ be a non-uniform irreducible lattice. Let $K\subset G$ be a maximal compac...

open_problem | open | MathDB 页面 | 原文线索

V2-076590 | MATHDB-397291

Theta-function norm bounds for general groups and lattices

摘录 / EXCERPT: For what groups G and lattices Γ can one prove such bounds for all theta functions?

open_problem | open | MathDB 页面 | 原文线索

V2-076591 | MATHDB-397292

Lower bounds for sup norms of cusp forms on general non-compact locally symmetric spaces

摘录 / EXCERPT: For cusp forms on more general non-compact locally symmetric spaces, obtain lower bounds on the sup norm $[U+007F] f [U+007F]_\infty$ in terms of the Laplace eigenvalue λ, or more genera...

open_problem | open | MathDB 页面 | 原文线索

V2-076592 | MATHDB-397293

Sup-norm bounds for Hecke-Maass cuspidal newforms

摘录 / EXCERPT: Let f be a Hecke-Maass cuspidal newform of level N and Laplacian eigenvalue $\[1m\lambda\u001b[0m$. Assume that $\|f\|_2=1$ with respect to $dx\,dy/y^2$. Bound $\|f\|_\infty$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-076593 | MATHDB-397294

Sup-norm bound for Hecke-Maass newforms of square-free level

摘录 / EXCERPT: Let f be an L^2-normalized Hecke-Maass cuspidal newform of square-free level N, with trivial nebentypus and Laplacian eigenvalue $\$. Prove that, for every $\>0$, ... where the...

open_problem | open | MathDB 页面 | 原文线索

V2-076594 | MATHDB-397295

Uniform distribution of eigenfunctions for a point scatterer on the flat torus

摘录 / EXCERPT: Fix $\phi\in(-\pi,\pi)$. There is a subset $\Lambda_{\phi,\infty}\subset\Lambda_\phi$ of density one such that, for every observable $a\in C^\infty(\mathbb T^2)$, ... as...

open_problem | open | MathDB 页面 | 原文线索

V2-076595 | MATHDB-397297

Asymptotics for the coefficients g_n

摘录 / EXCERPT: The coefficients g_n defined by ... with $b_n=\zeta(n)B_n/n$, satisfy ...

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V2-076596 | MATHDB-397299

Distinct values of two Dirichlet L-functions at Riemann-zero heights

摘录 / EXCERPT: Assume the Riemann Hypothesis. Let $\c7_i$ ($i=1,2$) be primitive real nonprincipal Dirichlet characters of conductors q_i, with $q_1\ne q_2$. For $\c3\in(1/2,1]$, for a positive...

open_problem | open | MathDB 页面 | 原文线索

V2-076597 | MATHDB-397300

Grand Simplicity Hypothesis for primitive Dirichlet L-functions

摘录 / EXCERPT: The set $\{\gamma\ge 0 : L(1/2+i\gamma,\chi)=0,\ \chi\text{ primitive}\}$ is linearly independent over $\mathbb{Q}$.

conjecture | open | MathDB 页面 | 原文线索

V2-076598 | MATHDB-397301

Distinct Dirichlet L-function values at ordinates of zeta zeros

摘录 / EXCERPT: Let χ_1 and χ_2 be primitive real nonprincipal Dirichlet characters with distinct conductors $q_1\ne q_2$. For each fixed $\sigma\in(1/2,1]$, a positive proportion of ord...

open_problem | open | MathDB 页面 | 原文线索

V2-076599 | MATHDB-397302

Selberg’s small eigenvalue conjecture

摘录 / EXCERPT: Conjecture 1 (Selberg [6]): ...

conjecture | open | MathDB 页面 | 原文线索

V2-076600 | MATHDB-397304

Erdős distinct-distances conjecture in the plane

摘录 / EXCERPT: For every set S of s points in the plane, the number of distinct pairwise distances determined by points of S is at least $\Omega\!\left(\frac{s}{\sqrt{\log s}}\right)$; equi...

conjecture | open | MathDB 页面 | 原文线索

V2-076601 | MATHDB-397305

Three-dimensional line-incidence bound

摘录 / EXCERPT: Given a set of n lines in $\bb{R}^3$, the number of points incident to at least $k\ge 3$ of these lines is $O\!\left(n^{3/2}/k^2\right)$, provided that no plane contains more tha...

open_problem | open | MathDB 页面 | 原文线索

V2-076602 | MATHDB-397306

Large monochromatic components in a colored grid of small cubes

摘录 / EXCERPT: Suppose a d-dimensional cube Q^d is partitioned into n^d small cubes in the obvious way, where $0\le m<d$. If the set of small cubes of Q^d is colored with $m+1$ colors, th...

open_problem | open | MathDB 页面 | 原文线索

V2-076603 | MATHDB-397308

Purdy’s conjecture on planes versus lines determined by points in 3-space

摘录 / EXCERPT: For all sufficiently large n, let P be a set of n points in 3-space such that the points do not all lie on two skew lines and no plane contains $n-1$ of the points. If p...

conjecture | open | MathDB 页面 | 原文线索

V2-076604 | MATHDB-397309

Purdy’s Conjecture \(P_d^*\)

摘录 / EXCERPT: If W_k is the number of flats of dimension k determined by n points in d-space, then ... provided the n points cannot be covered by a collection of flats whose dimensions a...

conjecture | open | MathDB 页面 | 原文线索

V2-076605 | MATHDB-397310

Purdy’s Conjecture Q*(d) on incident hyperplanes

摘录 / EXCERPT: Conjecture Q_d^*. Let W_k be the number of k-dimensional flats determined by n points in d-space. If the points cannot be covered by a collection of flats whose dimension...

conjecture | open | MathDB 页面 | 原文线索

V2-076606 | MATHDB-397311

Strong Dirac conjecture for points in the plane

摘录 / EXCERPT: Every set of n points in $\bb R^2$ contains a point incident to at least $n/2-c$ lines spanned by the set, for some universal constant c.

conjecture | open | MathDB 页面 | 原文线索

V2-076607 | MATHDB-397312

Smallest-area keyhole for an equilateral triangle

摘录 / EXCERPT: Determine the smallest-area compact convex keyhole for an equilateral triangle, where a compact convex figure C is a keyhole for P if every orientation P' of P can be trans...

open_problem | open | MathDB 页面 | 原文线索

V2-076608 | MATHDB-397314

Danzer-Rogers piercing problem for convex sets

摘录 / EXCERPT: Given $\varepsilon>0$, let $f(\varepsilon)$ be the minimum size of a set of points in the unit square such that every compact convex set of area ε within the unit squar...

open_problem | open | MathDB 页面 | 原文线索

V2-076609 | MATHDB-397315

When is a disk trapped by four lines?

摘录 / EXCERPT: Let L_1,L_2,L_3,L_4 be four lines in $\\mathbb{R}^3$. A disk of diameter slightly smaller than h_0 is trapped by the four lines if and only if there is a direction v_0 such t...

open_problem | open | MathDB 页面 | 原文线索

V2-076610 | MATHDB-397316

Conway’s monostatic polyhedron problem

摘录 / EXCERPT: Does there exist a homogeneous, convex polyhedron in three dimensions that rests in a stable position when lying on only one of its facets? Here, stability on a facet means that th...

open_problem | open | MathDB 页面 | 原文线索

V2-076611 | MATHDB-397317

Minimum number of faces of a homogeneous monostatic convex polyhedron

摘录 / EXCERPT: Determine the minimum number of faces of a homogeneous convex polyhedron in three-dimensional space that is stable in exactly one face.

open_problem | open | MathDB 页面 | 原文线索

V2-076612 | MATHDB-397318

Grid-free Steiner systems

摘录 / EXCERPT: Given r, construct infinitely many grid-free Steiner systems $S(v,r,2)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076613 | MATHDB-397320

Bounds for generalized Frobenius numbers

摘录 / EXCERPT: Let $n\ge 2$, $s\ge 1$, and let $a=(a_1,\ldots,a_n)^\mathsf{T}\in\mathbb Z_{>0}^n$ be primitive, with $0<a_1<a_2<\cdots<a_n$. Define ... Then ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-076614 | MATHDB-397321

Circle-freeness of Minkowski sums with a ball

摘录 / EXCERPT: Is it true that for every circle-free convex body K, the set $K+B$ is also circle-free?

open_problem | open | MathDB 页面 | 原文线索

V2-076615 | MATHDB-397322

Crossing-number lower bound for arbitrary drawings of complete graphs

摘录 / EXCERPT: For every integer $n\ge 4$, does every drawing of the complete graph K_n have at least ... crossings?

open_problem | open | MathDB 页面 | 原文线索

V2-076616 | MATHDB-397324

Raz’s exponential-bound question for permutation families of bounded VC-dimension

摘录 / EXCERPT: Let 4Sn4 be the set of all permutations of 4[n]4. For a set 4Psubseteq Sn4, define its VC-dimension using restrictions to ordered tuples of positions, and let 4rk(n)4 be the larges...

open_problem | open | MathDB 页面 | 原文线索

V2-076617 | MATHDB-397326

Existence of $m_k(P)$ for cover-decomposable planar sets

摘录 / EXCERPT: For every cover-decomposable planar set P and every integer $k\ge 2$, the quantity $m_k(P)$ exists, where $m_k(P)$ is the smallest positive integer m such that every m-fold c...

open_problem | open | MathDB 页面 | 原文线索

V2-076618 | MATHDB-397327

Optimality of the tolerated Tverberg bound

摘录 / EXCERPT: For any nonnegative integers r,k,d with $d\ge 1$, there exists a set of $(r+1)(k-1)(d+1)$ points in $\mathbb{R}^d$ with no k-Tverberg partition with tolerance r.

open_problem | open | MathDB 页面 | 原文线索

V2-076619 | MATHDB-397328

Conjecture on line transversals to homothetic disks

摘录 / EXCERPT: Let a0$D\subset\mathbb{R}^2$ be a disk. For a finite family of homothets $\mathcal F=\{x_1+t_1D,\ldots,x_n+t_nD\}$, let $T(k)$ mean that every subfamily with at most k members ha...

conjecture | open | MathDB 页面 | 原文线索

V2-076620 | MATHDB-397329

Eckhoff’s conjecture for line transversals to three translates of a disk

摘录 / EXCERPT: Let D be a disk in $\\mathbb{R}^2$. For a finite family $\\mathcal{F}$ of translates of D, let $T(3)$ mean that every subfamily of at most three members has a common line trans...

conjecture | open | MathDB 页面 | 原文线索

V2-076621 | MATHDB-397331

Completeness of Shephard’s list of projectively unique 4-polytopes

摘录 / EXCERPT: The report states: “If the conjecture is correct, then the following list of eleven projectively unique 4-polytopes (all of them generated by McMullen’s techniques, duplicates remo...

open_problem | open | MathDB 页面 | 原文线索

V2-076622 | MATHDB-397332

Anchored rectangle packing conjecture

摘录 / EXCERPT: For every finite point set $S\subseteq U=[0,1]^2$ with $(0,0)\in S$, where Bob must choose for each $s\in S$ an axis-parallel rectangle $r(s)\subseteq U$ having s as its lower-le...

conjecture | open | MathDB 页面 | 原文线索

V2-076623 | MATHDB-397333

Maximum density of packings by congruent copies of a compact convex body

摘录 / EXCERPT: Given a compact convex body K in Euclidean space $\mathbb{R}^n$, determine the maximum density of a packing of congruent copies of K.

open_problem | open | MathDB 页面 | 原文线索

V2-076624 | MATHDB-397334

Kadets-type inequality for convex partitions of a planar convex body

摘录 / EXCERPT: Let $B\subset\mathbb{R}^{2}$ be a convex body, and let $C_{1}\cup\cdots\cup C_{k}=B$ be a convex partition, meaning a covering of B by closed convex sets with disjoint interiors....

open_problem | open | MathDB 页面 | 原文线索

V2-076625 | MATHDB-397335

Subset realization of approximate-to-minimum ellipsoids

摘录 / EXCERPT: Let S be a set of n points in $\d$ dimensions and let $\epsilon>0$. An ellipsoid E is called approximate to minimum if...

open_problem | open | MathDB 页面 | 原文线索

V2-076626 | MATHDB-397337

Integral lattices and embeddings into integer lattices

摘录 / EXCERPT: A lattice L in a Euclidean space is called an integral lattice if the inner product $\langle x,y\rangle$ is an integer for every $x,y\in L$. Is every integral lattice L congrue...

open_problem | open | MathDB 页面 | 原文线索

V2-076627 | MATHDB-397339

Bounded monotone paths in convex plane subdivisions with three unbounded rays

摘录 / EXCERPT: Is there a constant c such that, for every $n\ge 3$, there is a partition of the plane into convex regions whose boundaries form a connected graph with n vertices and 3 unbou...

open_problem | open | MathDB 页面 | 原文线索

V2-076628 | MATHDB-397340

Existence and value of the exponential growth limit in ApSimon’s mints problem

摘录 / EXCERPT: For each positive integer n, let $N(n)$ be the least integer N for which there exists a set V of n nonzero vectors in the plane with non-negative integer coordinates, conta...

open_problem | open | MathDB 页面 | 原文线索

V2-076629 | MATHDB-397341

Solymosi’s dense-arrangement triangle conjecture

摘录 / EXCERPT: There are constants $c>0$ and n_0 such that, if $n>n_0$ lines and n points in the plane determine at least $n^{4/3-c}$ incidences, then there is a triangle in the arrangement.

conjecture | open | MathDB 页面 | 原文线索

V2-076630 | MATHDB-397343

Unbounded maximum degree of a pair in a planar point set

摘录 / EXCERPT: Let X be a finite point set in the plane with no three points collinear. The degree of a pair $x,y\in X$ is the number of $z\in X$ such that the triangle with vertices x,y,z co...

open_problem | open | MathDB 页面 | 原文线索

V2-076631 | MATHDB-397344

Doubly covered paths for coverings by unit discs

摘录 / EXCERPT: Is there a constant c such that, for every covering of the plane by closed unit circular discs, any two points situated at distance d from one another and covered by at least t...

open_problem | open | MathDB 页面 | 原文线索

V2-076632 | MATHDB-397345

A constant-factor bound for the monotone crossing number

摘录 / EXCERPT: Is there a constant c such that, for every graph G, $\operatorname{mon\text{-}cr}(G) < c\,\operatorname{cr}(G)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076633 | MATHDB-397346

Boundary-arc path conjecture for unit-disc coverings

摘录 / EXCERPT: There exists a constant c_1 such that, whenever closed unit circular discs cover the plane and a,b are points at distance d apart, both lying on the boundaries of discs in th...

conjecture | open | MathDB 页面 | 原文线索

V2-076634 | MATHDB-397347

Order of magnitude of the longest guaranteed cycle in 3-connected cubic planar graphs

摘录 / EXCERPT: Let $f(n)$ be the maximum integer such that every 3-connected cubic planar graph with n vertices contains a simple cycle of length at least $f(n)$. Determine the order of magnitu...

open_problem | open | MathDB 页面 | 原文线索

V2-076635 | MATHDB-397359

Higher Atiyah-Patodi-Singer index formula for the Godbillon-Vey cocycle

摘录 / EXCERPT: For a foliated bundle with cylindrical ends and a -equivariant family of Dirac operators with invertible boundary family , the higher Atiyah-Patodi-Singer index problem is resolved...

open_problem | open | MathDB 页面 | 原文线索

V2-076636 | MATHDB-397362

Graphs with large girth and the coarse Baum-Connes conjecture

摘录 / EXCERPT: Let (X_n) be a sequence of finite graphs with bounded degrees such that $\operatorname{girth}(X_n)\to\infty$, and let $X=\bigsqcup X_n$ be equipped with a metric restricting to t...

conjecture | open | MathDB 页面 | 原文线索

V2-076637 | MATHDB-397363

Transcendental L^2-Betti numbers from group-ring operators

摘录 / EXCERPT: Find groups Γ and elements $A\in\mathbb{Z}[\Gamma]$ for which $\ker(A)$ and the associated von Neumann dimension $b^{(2)}(A)$ are explicitly calculable and have transcendent...

open_problem | open | MathDB 页面 | 原文线索

V2-076638 | MATHDB-397368

Collapse of the Goodwillie spectral sequence for S^1

摘录 / EXCERPT: The Goodwillie spectral sequence (GSS) for S^1 collapses at the E_2 page.

open_problem | open | MathDB 页面 | 原文线索

V2-076639 | MATHDB-397369

Morava E-theory attaching-map/Koszul differential conjecture

摘录 / EXCERPT: Let E be height-n Morava E-theory and let q be odd. Under the canonical isomorphisms $E_*^{\,q}(\Sigma^k D_{2^k}(\mathrm{Id})(S^q))\cong C[k]_{-q}^{\vee}$, the map induced...

conjecture | open | MathDB 页面 | 原文线索

V2-076640 | MATHDB-397372

Comparison of models for \b7\b7-categories

摘录 / EXCERPT: Construct a chain of Quillen equivalences between the model structure for b7b7-spaces and a model structure on small categories enriched in b7b7-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-076641 | MATHDB-397373

Comparison of \ceTheta_n-spaces with enriched higher categories

摘录 / EXCERPT: Show that the model structure on ceThetan-spaces is Quillen equivalent to one on the category of small categories enriched in c(infty,n-1c)-categories.

open_problem | open | MathDB 页面 | 原文线索

V2-076642 | MATHDB-397374

Localising subcategories for cochains on compact Lie groups

摘录 / EXCERPT: Let G be a compact Lie group and let k be a field of characteristic p. Is there a natural one-to-one correspondence between the localising subcategories of $D(C^*(BG;k))$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076643 | MATHDB-397376

Rationality of the moduli space of plane sextics

摘录 / EXCERPT: Is the quotient $\mathbb{P}(\operatorname{Sym}^{6}(\mathbb{C}^{3})^{\vee})/\operatorname{SL}_{3}(\mathbb{C})$, equivalently the moduli space of plane curves of degree 6 up to pro...

open_problem | open | MathDB 页面 | 原文线索

V2-076644 | MATHDB-397377

Factorization of birational maps of threefolds

摘录 / EXCERPT: An optimistic guess in the report is that, for any two birational threefolds X and X', there is a sequence of birational maps ... such that each $f_i:X_i\dashrightarrow X_{i+1}$...

open_problem | open | MathDB 页面 | 原文线索

V2-076645 | MATHDB-397382

Full exceptional collections for invertible-polynomial quotient stacks

摘录 / EXCERPT: Let $A=(a_{ij})$ be an $n\times n$ integer matrix with $a_{ij}\ge 0$ and $\det A\ne 0$, and set ... Assume that W has an isolated critical point at the origin. Define ... and the quo...

open_problem | open | MathDB 页面 | 原文线索

V2-076646 | MATHDB-397386

Movable cone conjecture for generic $(2, \ldots,2)$ Calabi\Yau hypersurfaces

摘录 / EXCERPT: Let $n\ge 3$, and let X be a generic hypersurface of multi-degree $(2,\ldots,2)$ in $(\mathbb{P}^{1})^{n+1}$. Then the abstract MorrisonKawamata movable cone conjecture is true f...

conjecture | open | MathDB 页面 | 原文线索

V2-076647 | MATHDB-397388

Hodge-number symmetry for the proposed toric Landau-Ginzburg mirror pair

摘录 / EXCERPT: If $\widetilde X_\sigma$ and $\widetilde X_{\check\sigma}$ are smooth, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-076648 | MATHDB-397389

Conjecture 1: Hodge-number mirror symmetry for dual Landau-Ginzburg models

摘录 / EXCERPT: Let $\sigma\subseteq M\otimes_{\mathbb Z}\mathbb R$ be a full-dimensional rational polyhedral cone for $M\cong\mathbb Z^n$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-076649 | MATHDB-397391

Equivalence of reduced residue Gromov-Witten and stable-pair invariants

摘录 / EXCERPT: The reduced residue Gromov-Witten and reduced residue stable-pair invariants of the canonical bundle of a nonsingular projective surface are conjecturally equivalent, via the appro...

open_problem | open | MathDB 页面 | 原文线索

V2-076650 | MATHDB-397392

Boundedness of log pairs: DCC for volumes and uniform birationality

摘录 / EXCERPT: Fix a positive integer n and a set $I\subset[0,1]$ satisfying the descending chain condition (DCC), and let ... Then ... satisfies the DCC. Furthermore, there exists a constant N,...

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V2-076651 | MATHDB-397393

DCC for volumes of log canonical pairs

摘录 / EXCERPT: Fix a positive integer n and a set $I\subset [0,1]$ satisfying the descending chain condition. Let ... Then ... satisfies the descending chain condition.

open_problem | open | MathDB 页面 | 原文线索

V2-076652 | MATHDB-397395

Weak dependence is insufficient for asymptotic FDR control in the null-problem

摘录 / EXCERPT: The report presents a counterexample showing that weak dependence alone does not guarantee asymptotic false-discovery-rate (FDR) control when the null-problem occurs, namely when t...

open_problem | open | MathDB 页面 | 原文线索

V2-076653 | MATHDB-397396

Conjectured minimax lower bound for expected conditional covariance

摘录 / EXCERPT: Under the report’s irregular-case simplification 4betap=betab=betau00024 (with 4beta/d<1/4u00024), when 4Xu00024 is random, its density 4gu00024 is absolutely continuous with respe...

conjecture | open | MathDB 页面 | 原文线索

V2-076654 | MATHDB-397397

Conjectured optimal rate in the random-design semiparametric model

摘录 / EXCERPT: Assume the irregular regime $\beta/d<1/4$, $\beta_p=\beta_b=\beta$, and observe i.i.d. data $O=(Y,A,X)$ with $X\in[0,1]^d$. In the random-design semiparametric model, suppose the d...

conjecture | open | MathDB 页面 | 原文线索

V2-076655 | MATHDB-397404

Long periods for quantised Anosov toral automorphisms for almost all levels

摘录 / EXCERPT: Let $\kappa\in\Gamma_\theta=\left\{\begin{pmatrix}a&b\\c&d\end{pmatrix}\in\mathrm{SL}_2(\mathbb Z):ab\equiv dc\equiv0\pmod 2\right\}$ be an Anosov map, and define ... Then, for a sub...

open_problem | open | MathDB 页面 | 原文线索

V2-076656 | MATHDB-397407

Lipschitz regularity of the IDS for nonnegative random block operators

摘录 / EXCERPT: For the random block operator $\mathds H=\begin{pmatrix}H&B\\ B&-H\end{pmatrix}$ on $\ell^2(\mathbb Z^d)\oplus\ell^2(\mathbb Z^d)$, with $H=H_0+V$, where V and B are multiplica...

open_problem | open | MathDB 页面 | 原文线索

V2-076657 | MATHDB-397409

Whether permutable-series absolute convergence implies BD-N over BISH

摘录 / EXCERPT: Over BISH, does the assertion that every permutably convergent series of real numbers is absolutely convergent imply BD-N, where a series is permutably convergent if every permutat...

open_problem | open | MathDB 页面 | 原文线索

V2-076658 | MATHDB-397410

Weak-permutable convergence implies BD-N over BISH

摘录 / EXCERPT: In BISH, if every weak-permutably convergent series in $\mathbb{R}$ is absolutely convergent, then BD-N holds:...

open_problem | open | MathDB 页面 | 原文线索

V2-076659 | MATHDB-397412

The Gomory-Chvátal Closure of a Non-Rational Polytope is a Rational Polytope

摘录 / EXCERPT: For every polytope $P\subseteq\mathbb{R}^n$, including a non-rational polytope, its Gomory-Chvátal closure P'-the intersection of all half-spaces defined by Gomory-Chvátal cuts...

open_problem | open | MathDB 页面 | 原文线索

V2-076660 | MATHDB-397413

Polynomial bound on the diameter of polyhedra

摘录 / EXCERPT: Let $P=\{x\in\mathbb{R}^n:Ax\le b\}$, where $A\in\mathbb{R}^{m\times n}$ and $b\in\mathbb{R}^m$, and let the diameter of P be the diameter of its polyhedral graph whose vertices...

open_problem | open | MathDB 页面 | 原文线索

V2-076661 | MATHDB-397414

Yannakakis’s open problem on nonsymmetric LP formulations for the TSP

摘录 / EXCERPT: Show that the traveling-salesman problem admits no polynomial-size linear program (LP), symmetric or not.

open_problem | open | MathDB 页面 | 原文线索

V2-076662 | MATHDB-397421

Classification of polynomial-time solvability for weighted integer multiflow

摘录 / EXCERPT: Let S be a terminal set and let $\mu:(S\choose 2)\to\mathbb{Q}_+$ be a weight. The integer multiflow problem $\mu\text{-IMP}$ is the problem of maximizing $\mu\cdot f$ over all i...

open_problem | open | MathDB 页面 | 原文线索

V2-076663 | MATHDB-397423

Fractionality classification for weighted fractional multiflow problems

摘录 / EXCERPT: Let $G=(V,E)$ be an undirected graph with possible parallel edges and loops, let $S\subseteq V$ be a terminal set, and let $\mu:\binom{S}{2}\to\mathbb{Q}_+$ be a terminal-weight fu...

open_problem | open | MathDB 页面 | 原文线索

V2-076664 | MATHDB-397425

Maximum Non-\k\-Linked Induced Subgraph

摘录 / EXCERPT: A graph is said to be k-linked if it has at least $2k$ vertices and, for any ordered k-tuples $(s_1,\ldots,s_k)$ and $(t_1,\ldots,t_k)$ of $2k$ distinct vertices, it contains p...

open_problem | open | MathDB 页面 | 原文线索

V2-076665 | MATHDB-397426

Exact value of the real Grothendieck constant K(1)

摘录 / EXCERPT: Determine the exact value of $K(1)$, where, for each integer $r\ge 1$, $K(r)$ is the smallest constant such that for every real matrix $A\in\mathbb{R}^{m\times n}$, ... with ......

open_problem | open | MathDB 页面 | 原文线索

V2-076666 | MATHDB-397429

Universality and polynomial-time solvability of fixed-parameter three-dimensional table integer programs

摘录 / EXCERPT: Consider the integer optimization problem ... Every integer program can be put precisely into this form; and, for any fixed parameter m, this problem can be solved in polynomial ti...

open_problem | open | MathDB 页面 | 原文线索

V2-076667 | MATHDB-397430

Cubic-time variable-entry n-fold integer programming theorem

摘录 / EXCERPT: The report states that the following n-fold integer programming problem is solvable in time cubic in n, linear in $\log(f,b,l,u)$, and polynomial in $\max_{i,j}|A_{i,j}|$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076668 | MATHDB-397433

Non-symmetric compact formulations for the matching polytope

摘录 / EXCERPT: Show that the matching polytope does not admit any non-symmetric compact formulation.

open_problem | open | MathDB 页面 | 原文线索

V2-076669 | MATHDB-397437

Fractional 2-matchings and the worst-case subtour-LP integrality gap

摘录 / EXCERPT: Conjecture: The worst-case integrality gap of the subtour LP is achieved by solutions to the subtour LP that are fractional 2-matchings; that is, by instances for which adding the...

open_problem | open | MathDB 页面 | 原文线索

V2-076670 | MATHDB-397438

Boyd-Carr conjecture on the 2-matching/subtour-LP ratio

摘录 / EXCERPT: For symmetric traveling-salesman instances whose costs satisfy the triangle inequality, the worst-case ratio of the cost of an optimal 2-matching to the cost of an optimal solution...

conjecture | open | MathDB 页面 | 原文线索

V2-076671 | MATHDB-397439

Error estimate for the semidiscrete C^1-finite-element approximation of Willmore flow of graphs

摘录 / EXCERPT: Suppose that $u:\overline{\Omega}\times[0,T]\to\mathbb{R}$ is a smooth solution of the Willmore-flow problem (2), with the Dirichlet and normal-derivative boundary conditions (3) a...

open_problem | open | MathDB 页面 | 原文线索

V2-076672 | MATHDB-397442

Existence of degenerate neckpinch Ricci flows with prescribed Type-II rates

摘录 / EXCERPT: The report conjectures that there is a family of rotationally symmetric normalized Ricci-flow solutions, parametrized by each integer $k\ge 3$, which develop degenerate neckpinch s...

open_problem | open | MathDB 页面 | 原文线索

V2-076673 | MATHDB-397444

Threshold dynamics for equal surface tensions

摘录 / EXCERPT: How can one simulate the L^2 gradient descent of ... where $\Sigma_i\cap\Sigma_j=(\partial\Sigma_i)\cap(\partial\Sigma_j)$ for all $i\ne j$, $\bigcup_{j=1}^N\Sigma_j=D$, and D is...

open_problem | open | MathDB 页面 | 原文线索

V2-076674 | MATHDB-397445

Other transformations and energy functionals in lifted Noether theory

摘录 / EXCERPT: Are there other transformations leading to new energy functionals?

open_problem | open | MathDB 页面 | 原文线索

V2-076675 | MATHDB-397446

Minimization of the renormalized energy

摘录 / EXCERPT: Determine the minimizer of the renormalized energy W on the admissible class A; in particular, determine whether the triangular lattice is a minimizer.

open_problem | open | MathDB 页面 | 原文线索

V2-076676 | MATHDB-397447

General heterogeneous admissible stress sets in elasto-plasticity

摘录 / EXCERPT: The general case of a set of admissible stresses $K(x)$ that depends pointwise on the point x of the domain is open, unless the multifunction $x \mapsto K(x)$ is piecewise contin...

open_problem | open | MathDB 页面 | 原文线索

V2-076677 | MATHDB-397449

Least barrier containing a set for minimal hypersurfaces

摘录 / EXCERPT: Given a closed set $\Omega\subset\mathbb{R}^{3}$ with $\partial\Omega\in C^{1,1}$, characterize the least set containing all minimal hypersurfaces whose boundary lies in Ω....

open_problem | open | MathDB 页面 | 原文线索

V2-076678 | MATHDB-397450

Uniqueness of distributional solutions for the semi-geostrophic systems

摘录 / EXCERPT: Are distributional solutions unique? The report states that this is unknown for both the semi-geostrophic system (SG) and its dual system (SGD).

open_problem | open | MathDB 页面 | 原文线索

V2-076679 | MATHDB-397452

Global solvability of the Landau-Lifshitz-Gilbert equation

摘录 / EXCERPT: For director fields $m:(0,\a0\infty)\times\mathbb{R}^n\to\mathbb{S}^2$, consider ... where $\lambda>0$. Suppose $n\ge 3$. Then there exist constants $\rho>0$ and $c>0$ with the follo...

open_problem | open | MathDB 页面 | 原文线索

V2-076680 | MATHDB-397453

Vanishing-\ε limit for the one-dimensional Cahn-Hilliard equation

摘录 / EXCERPT: Let ε∈(0,1], let T be the one-dimensional torus, and let W(z)=frac14(1-z^2)^2. For initial data u∈C^1(mathbb T), let u^ε(cdot,cdot,u) denote the solution of ... Does the limit of u^ε...

open_problem | open | MathDB 页面 | 原文线索

V2-076681 | MATHDB-397454

Improving the Babuška-Kondratiev weight range for eigenfunctions

摘录 / EXCERPT: Let H be the single-nucleus k-electron Schrödinger operator described in the report, and let $u\in L^2(\mathbb{R}^{3k})$ satisfy $Hu=\lambda u$ in the distributional sense. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-076682 | MATHDB-397460

Residual-chain approximation for normalized first Betti numbers

摘录 / EXCERPT: For which groups G does the limit ... exist for all systems $(G_i)_{i\in I}$ with $\bigcap_{i\in I}G_i=\{1\}$ and fields K, and is independent of the choice of $(G_i)_{i\in I}$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-076683 | MATHDB-397461

Normalized first Betti numbers in finite-index subgroup systems

摘录 / EXCERPT: For which groups G does the limit ... exist for every inverse system $(G_i)_{i\in I}$ of normal finite-index subgroups of G, directed by inclusion, satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-076684 | MATHDB-397462

Integral agreement of symmetric-signature constructions

摘录 / EXCERPT: Whether all known constructions of the symmetric signature for Witt spaces agree over the integers.

open_problem | open | MathDB 页面 | 原文线索

V2-076685 | MATHDB-397463

Well-posedness of mixed Poisson problems on higher-dimensional polyhedral domains with adjacent Neumann faces

摘录 / EXCERPT: For a stratified curvilinear polyhedral domain $\Omega\subset\mathbb{R}^d$ with mixed boundary decomposition $\partial\Omega=\partial_D\Omega\cup\partial_N\Omega$, the report prove...

open_problem | open | MathDB 页面 | 原文线索

V2-076686 | MATHDB-397464

Equivalence of qsf groups and easily representable groups

摘录 / EXCERPT: Is every finitely presented group Γ qsf if and only if it is easily representable, meaning that Γ admits an easy gsc-representation? Here, according to the report, a...

open_problem | open | MathDB 页面 | 原文线索

V2-076687 | MATHDB-397465

Equivalence of the QSF property and easy representability

摘录 / EXCERPT: Is the qsf property equivalent, for groups, to the condition to be easily-representable?

open_problem | open | MathDB 页面 | 原文线索

V2-076688 | MATHDB-397467

Existence of a Z-oriented compact \(\mathbb{Z}_2\)-Witt space with non-trivial middle-dimensional pairing

摘录 / EXCERPT: Is there a $\mathbb{Z}$-oriented compact $\mathbb{Z}_2$-Witt space of dimension $4j+2$ whose middle-dimensional pairing represents the non-trivial element of $W(\mathbb{Z}_2)$?

open_problem | open | MathDB 页面 | 原文线索

V2-076689 | MATHDB-397479

Generalized Hopf conjecture for symmetric spaces

摘录 / EXCERPT: The report presents the conjecture that no symmetric space of rank greater than one admits a positively curved metric.

conjecture | open | MathDB 页面 | 原文线索

V2-076690 | MATHDB-397482

Uniqueness of 7-dimensional regular Riemannian foliations of spheres

摘录 / EXCERPT: The report conjectured that there are no 7-dimensional regular Riemannian foliations of spheres other than the Hopf fibration $S^7 \to S^{15} \to S^8$.

open_problem | open | MathDB 页面 | 原文线索

V2-076691 | MATHDB-397484

Resolvent-difference estimates for Schrödinger operators with δ-interactions

摘录 / EXCERPT: For the setup of a compact smooth closed hypersurface $\Sigma\subset\mathbb{R}^n$ separating $\mathbb{R}^n$ into Ω_i and Ω_e, with $n\ge 2$ and real-valued bounded...

open_problem | open | MathDB 页面 | 原文线索

V2-076692 | MATHDB-397485

Gap opening and split band edges in coupled waveguides

摘录 / EXCERPT: Assume that $h>\frac{\pi}{\sqrt{2}}$ and that ... For the Laplacian on the domain formed by two strips of widths $d_+=\pi$ and $d_-=d$, coupled by a $2h$-periodic system of windows o...

open_problem | open | MathDB 页面 | 原文线索

V2-076693 | MATHDB-397487

Hyperbolic circle problem for a Schottky-group orbit in \(\mathbb H^3\)

摘录 / EXCERPT: Let $\Gamma=\langle g_1,\ldots,g_d\rangle$ be a Schottky group acting on $\mathbb H^3$, and let $p_0\in\mathbb H^3$. Define ... Then there exists $C>0$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-076694 | MATHDB-397490

Erdős's conjecture on continued-fraction partial quotients

摘录 / EXCERPT: Philipp proved that, for Lebesgue almost every $x\in(0,1)$, if $a_k(x)$ denotes the k-th entry in the regular continued-fraction expansion of x, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-076695 | MATHDB-397492

Asymptotic measure of preimages under the Farey map

摘录 / EXCERPT: For the Farey map $F:[0,1]\to[0,1]$, defined by ... Theorem 1 of the report states that, as $n\to\infty$, for arbitrary $x\in(0,1]$, ... where λ is Lebesgue measure and μ i...

open_problem | open | MathDB 页面 | 原文线索

V2-076696 | MATHDB-397493

Hausdorff dimensions of differentiability sets of Minkowski’s question-mark function

摘录 / EXCERPT: The unit interval is the disjoint union of ... Moreover, ... where $h_{\mathrm{top}}=\log 2$.

open_problem | open | MathDB 页面 | 原文线索

V2-076697 | MATHDB-397494

Explicit density of the invariant measure for the even continued-fraction set

摘录 / EXCERPT: Let $E_{\rm even}$ be the set of all irrational numbers in $[0,1]$ whose continued-fraction expansion entries are all even, and let h be its Hausdorff dimension. It was known tha...

open_problem | open | MathDB 页面 | 原文线索

V2-076698 | MATHDB-397498

Our Knowledge of Standard Models: A Case of Tacit Knowledge?

摘录 / EXCERPT: Can one single out the standard models of arithmetic, in particular by appealing to knowledge of what is meant by “natural number”?

open_problem | open | MathDB 页面 | 原文线索

V2-076699 | MATHDB-397499

Selection of adaptive artificial-viscosity coefficients

摘录 / EXCERPT: How should the coefficients of proportionality in the adaptive artificial-viscosity method be selected so that the computed solution is non-oscillatory while its discontinuous part...

open_problem | open | MathDB 页面 | 原文线索

V2-076700 | MATHDB-397500

Semiclassical limit of the Hohenberg-Kohn functional for two electrons

摘录 / EXCERPT: Theorem 1. For $N=2$ and all $\rho\\in R_N$, ... where $E_{OT}[\rho]$ is the optimal cost of the Coulomb optimal-transport problem ... subject to both the right and left marginals of...

open_problem | open | MathDB 页面 | 原文线索

V2-076701 | MATHDB-397502

Positive-temperature forbidden-density interval for continuum Lennard-Jones particles

摘录 / EXCERPT: For a system of point particles in $\mathbb{R}^2$ with Lennard-Jones interactions, does the zero-temperature result establishing a forbidden density interval $(0,\rho_c)$ extend to...

open_problem | open | MathDB 页面 | 原文线索

V2-076702 | MATHDB-397503

Liquid-vapor phase transition for the hard-core-free Kac model

摘录 / EXCERPT: For the hard-core-free Kac model described in the report, with energy density $e(s)=-s^2/2+s^4/4!$, for every $d\ge 2$ there exist constants $0<\beta_c<\beta_0$ such that, for each...

open_problem | open | MathDB 页面 | 原文线索

V2-076703 | MATHDB-397507

Gradient flows driven by a non-smooth repulsive interaction potential

摘录 / EXCERPT: For the one-dimensional problem with interaction potential $W(x)=|x|$, the report states that a gradient-flow solution exists and is unique, selecting one evolution from the infini...

open_problem | open | MathDB 页面 | 原文线索

V2-076704 | MATHDB-397509

Computable density-1 subsets of c.e. density-1 sets

摘录 / EXCERPT: Does every c.e. set of density 1 have a computable subset of density 1?

open_problem | open | MathDB 页面 | 原文线索

V2-076705 | MATHDB-397510

Absolutely undecidable sets in nonzero Turing degrees

摘录 / EXCERPT: Does every nonzero Turing degree contain an absolutely undecidable set?

open_problem | open | MathDB 页面 | 原文线索

V2-076706 | MATHDB-397511

K-triviality and weak ML-cuppability

摘录 / EXCERPT: For every set A, A is weakly ML-cuppable if and only if A is not K-trivial, where A is weakly ML-cuppable if there is an incomplete Martin-Löf random set X such that...

open_problem | open | MathDB 页面 | 原文线索

V2-076707 | MATHDB-397512

Cupping with a Martin-Löf random set while preserving only K-trivial common lower bounds

摘录 / EXCERPT: For any non-computable set $A <_T \emptyset'$ that is not K-trivial, there is a Martin-Löf random $R <_T \emptyset'$ such that $A \oplus R \equiv_T \emptyset'$ and, if...

open_problem | open | MathDB 页面 | 原文线索

V2-076708 | MATHDB-397514

Nondefinability of the array-recursive degrees at the \a3\0 level

摘录 / EXCERPT: Prove that the class d538 of array recursive Turing degrees is not definable by any a30 formula in the language $(\leq)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076709 | MATHDB-397515

Nondefinability of the array recursive degrees at the \[1m\[0mΣ² level

摘录 / EXCERPT: Show that the class of array recursive degrees [1m[0mAR is not definable by any Σ^0_2 formula in the language containing only the Turing reducibility relation...

open_problem | open | MathDB 页面 | 原文线索

V2-076710 | MATHDB-397516

The ML-covering problem

摘录 / EXCERPT: Is every K-trivial set computable from an incomplete Martin-Löf random set?

open_problem | open | MathDB 页面 | 原文线索

V2-076711 | MATHDB-397517

Incomplete Oberwolfach-random sets

摘录 / EXCERPT: Is there an incomplete Oberwolfach-random set?

open_problem | open | MathDB 页面 | 原文线索

V2-076712 | MATHDB-397518

Uniform computation of maximal chains in computable well partial orders

摘录 / EXCERPT: There is no hyperarithmetic procedure f such that, for every index e for a computable well partial order P, the function $n\mapsto f(e,n)$ is the characteristic function of a...

open_problem | open | MathDB 页面 | 原文线索

V2-076713 | MATHDB-397525

Mathias n-generics and Cohen n-generics

摘录 / EXCERPT: Does every Mathias n-generic set compute a Cohen n-generic set?

open_problem | open | MathDB 页面 | 原文线索

V2-076714 | MATHDB-397528

Finite-lattice embeddability in the c.e. cl- and ibT-degrees

摘录 / EXCERPT: Is there a finite lattice which cannot be embedded into $\mathcal{R}_{\mathrm{cl}}$ or $\mathcal{R}_{\mathrm{ibT}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-076715 | MATHDB-397529

Superconvergence of the AW30 and AW24 mixed finite elements

摘录 / EXCERPT: For the L-shaped-domain experiments with $f\equiv 0$, the AW30 and AW24 mixed finite elements exhibit a superconvergence phenomenon of order $O(N^{-2})$.

open_problem | open | MathDB 页面 | 原文线索

V2-076716 | MATHDB-397530

Converse comparison between Crouzeix-Raviart and Raviart-Thomas errors

摘录 / EXCERPT: For the Poisson problem on a bounded polygonal Lipschitz domain $\Omega\subset\mathbb R^2$, with $f\in L^2(\Omega)$, weak solution $u\in H^1_0(\Omega)$, and a shape-regular triangu...

open_problem | open | MathDB 页面 | 原文线索

V2-076717 | MATHDB-397531

Comparison result for first-order finite element methods in linear elasticity

摘录 / EXCERPT: For the Navier-Lamé problem (2), comparing the conforming Courant FEM, the nonconforming Kouhia-Stenberg FEM, and the nonconforming Crouzeix-Raviart FEM, with respective stress sol...

open_problem | open | MathDB 页面 | 原文线索

V2-076718 | MATHDB-397534

Stability of finite-element approximations for the finite-strain rate problem

摘录 / EXCERPT: Establish conditions for the stability of finite-element approximations of the finite-strain rate problem whose three-field functional is ... where F is the deformation gradient,...

open_problem | open | MathDB 页面 | 原文线索

V2-076719 | MATHDB-397535

Justification of the reference thickness in the local problem

摘录 / EXCERPT: Properly choose and justify the reference thickness t in the local problem. The choice affects the construction of the homogenized energy and the definition of the region in whic...

open_problem | open | MathDB 页面 | 原文线索

V2-076720 | MATHDB-397536

Convergence of an adaptive C^0-IPDG method for the biharmonic problem

摘录 / EXCERPT: Let $u\in H^2_0(\Omega)$ be the unique solution of the biharmonic weak problem (2). Let $\mathcal T_h(\Omega)$ be a simplicial triangulation obtained by refinement from...

open_problem | open | MathDB 页面 | 原文线索

V2-076721 | MATHDB-397537

BDDC condition-number estimate for the SIPG subproblem

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a bounded polygonal domain, let $\Omega_1,\ldots,\Omega_J$ be a shape-regular nonoverlapping decomposition, and let $\rho_j>0$ be constant on...

open_problem | open | MathDB 页面 | 原文线索

V2-076722 | MATHDB-397538

Bayesian estimation of the Hurst parameter from discrete fractional Brownian-motion increments

摘录 / EXCERPT: For a fractional Brownian motion with Hurst parameter $H\in(0,1)$, observe a single realization $X^{\widehat H}$ with true parameter $\widehat H$ at the equidistant times $t_j=j/n$...

open_problem | open | MathDB 页面 | 原文线索

V2-076723 | MATHDB-397539

Asymptotic Bayesian estimation of the Hurst parameter of fractional Brownian motion

摘录 / EXCERPT: Let Z^H be fractional Brownian motion with Hurst parameter $H\in(0,1)$, and suppose a single realization with true parameter $\widehat H$ is observed at the equidistant times...

open_problem | open | MathDB 页面 | 原文线索

V2-076724 | MATHDB-397540

Uniqueness of identifying a mixed-boundary obstacle and its impedance from full far-field data

摘录 / EXCERPT: For the two-dimensional exterior Helmholtz problem described in (1), with $\partial D=\partial D_D\cup\partial D_I$, the boundary components ∂D_D, ∂D_I, and the...

open_problem | open | MathDB 页面 | 原文线索

V2-076725 | MATHDB-397541

Defining the potential map for general nonnegative conductivity

摘录 / EXCERPT: Let (sigmain L^infty(mathbb R^3)) be nonnegative. In the unified variational formulation, define φ_A through ... so that $A+\nabla\varphi_A$ represents the electric field. It...

open_problem | open | MathDB 页面 | 原文线索

V2-076726 | MATHDB-397543

Transmission eigenvalues for anisotropic media containing an impenetrable Dirichlet obstacle

摘录 / EXCERPT: For the inverse-scattering problem for an inhomogeneous anisotropic medium containing an impenetrable obstacle with Dirichlet boundary conditions, prove that the associated transmi...

open_problem | open | MathDB 页面 | 原文线索

V2-076727 | MATHDB-397544

Nonlinear regularization for the two-dimensional Calderón inverse conductivity problem

摘录 / EXCERPT: Construct a family of continuous mappings $R_\alpha:Y\to X$, $0<\alpha<\infty$, for the forward map $F(\sigma)=\Lambda_\sigma$ such that, for each fixed $\sigma\in D(F)$, ... and cho...

open_problem | open | MathDB 页面 | 原文线索

V2-076728 | MATHDB-397545

Post-processing internal models for robust nonlinear output regulation

摘录 / EXCERPT: “In view of a (systematic) extension of the theory of nonlinear output regulation to general multivariable systems, it seems appropriate to investigate to what extent controllers i...

open_problem | open | MathDB 页面 | 原文线索

V2-076729 | MATHDB-397546

Certainty-equivalence control is optimal for the three event-triggered control settings

摘录 / EXCERPT: Let the system be given by (1) and (2). The class of policies $\mathcal U_{\mathrm{CE}}\subset\mathcal U$ defined by ... ... is a dominating class of policies for problem settings A-C.

open_problem | open | MathDB 页面 | 原文线索

V2-076730 | MATHDB-397547

Necessary and sufficient dissipativity conditions for stability of interconnected systems

摘录 / EXCERPT: Let $N\in\mathbb Z_{\ge 2}$ and consider the interconnected system ... where $x_i\in\mathbb R^{n_i}$, $g_i:\mathbb R^{n_1}\times\cdots\times\mathbb R^{n_N}\to\mathbb R^{n_i}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076731 | MATHDB-397548

Hybrid Linear Regulation

摘录 / EXCERPT: For the hybrid linear plant and exosystem with flow dynamics ... whenever $((\tau,w),x,u)\in W\times\mathbb R^n\times\mathbb R$, and jump dynamics ... whenever...

open_problem | open | MathDB 页面 | 原文线索

V2-076732 | MATHDB-397550

Almost-sure finite-time convergence of interval consensus

摘录 / EXCERPT: Consider the quantized gossip process in which activated nodes have states ordered by $\prec$. Let T be the first time at which there are two consecutive states x and y such...

open_problem | open | MathDB 页面 | 原文线索

V2-076733 | MATHDB-397553

q-Whittaker-function Conjecture 0.2

摘录 / EXCERPT: Let G be a semisimple, simply connected group over $\mathbb C$, with maximal torus T, Weyl group W, and weight lattice $\check\Lambda$. Does there exist a unique collection o...

conjecture | open | MathDB 页面 | 原文线索

V2-076734 | MATHDB-397554

Macdonald polynomials at $t=0$ and normalized q-Whittaker functions

摘录 / EXCERPT: Conjecture 0.9 asserts that, for every weight $\check\lambda$ under consideration, ...

open_problem | open | MathDB 页面 | 原文线索

V2-076735 | MATHDB-397556

Rouquier’s derived-equivalence conjecture for category \(\mathcal O\)

摘录 / EXCERPT: Let W be a finite complex reflection group acting on its reflection representation $\mathfrak h$, and let p,p' be parameters attached to W such that $q(p)=q(p')$ and all rank...

conjecture | open | MathDB 页面 | 原文线索

V2-076736 | MATHDB-397557

Determine the multiplicities in the p-canonical basis

摘录 / EXCERPT: Describe the coefficients p m{x,y} in general.

open_problem | open | MathDB 页面 | 原文线索

V2-076737 | MATHDB-397558

Rouquier’s derived-equivalence conjecture for category \(O\)

摘录 / EXCERPT: Let W be a finite complex reflection group, and let p,p' be parameters attached to W with $q(p)=q(p')$. Rouquier conjectures that, without requiring the orderings $\prec_p$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-076738 | MATHDB-397559

Determination of irreducible unitary representations of a real reductive group

摘录 / EXCERPT: Let G_R be a real reductive group, assumed semisimple in the exposition. Determine all irreducible unitary representations of G_R.

open_problem | open | MathDB 页面 | 原文线索

V2-076739 | MATHDB-397571

Uniform empirical-Bayes approximation for Poisson means

摘录 / EXCERPT: For independent observations $Y_i\sim\operatorname{Po}(\lambda_i)$, $i=1,\ldots,p$, estimate $\boldsymbol\lambda=(\lambda_1,\ldots,\lambda_p)$ under average quadratic risk...

open_problem | open | MathDB 页面 | 原文线索

V2-076740 | MATHDB-397572

Accuracy of empirical reduced-rank projections for noncentered Gaussian matrices

摘录 / EXCERPT: Let $\widehat\pi_r:=\operatorname*{arg\,max}_{\widetilde\pi_r\in\mathcal S_{M,r}}\|\widetilde\pi_rX\|_{S_2}^2$ and let...

open_problem | open | MathDB 页面 | 原文线索

V2-076741 | MATHDB-397573

Optimality of the approximation bound for rectangular singular-value spectra

摘录 / EXCERPT: The question remains whether the upper bound in Theorem 8 is optimal in the case of rank-r matrices with rectangular singular-value spectrum.

open_problem | open | MathDB 页面 | 原文线索

V2-076742 | MATHDB-397574

Existence of procedures adaptive for both global and local losses

摘录 / EXCERPT: Does there exist an optimal procedure that adapts, up to logarithmic factors in n, simultaneously for the global and local losses discussed in the preceding examples?

open_problem | open | MathDB 页面 | 原文线索

V2-076743 | MATHDB-397575

Extension of Γ-convergence to rate-dependent dissipative problems

摘录 / EXCERPT: Extension of Γ-convergence to rate-dependent problems and dissipation.

open_problem | open | MathDB 页面 | 原文线索

V2-076744 | MATHDB-397577

The missing interfacial flow rule in heterogeneous small-strain elastoplasticity

摘录 / EXCERPT: In a perfect small-strain elastoplastic evolution without inertia or body loads, driven by a boundary displacement, consider an interface between phases i and j, with associate...

open_problem | open | MathDB 页面 | 原文线索

V2-076745 | MATHDB-397578

Origin of the characteristic length of persistent shear bands

摘录 / EXCERPT: For granular materials undergoing shear, what determines the characteristic rolling-correlation length associated with persistent shear bands, whose typical width is about 10-20 pa...

open_problem | open | MathDB 页面 | 原文线索

V2-076746 | MATHDB-397579

Micromechanisms of dilatancy and compaction in dense granular materials

摘录 / EXCERPT: What are the micromechanisms that produce dilatancy and compaction in dense granular materials?

open_problem | open | MathDB 页面 | 原文线索

V2-076747 | MATHDB-397588

Equivariant Alperin-Robinson conjecture for blocks

摘录 / EXCERPT: For a finite k^{*}-group $\hat{G}$ of finite k^{*}-quotient, a block b of $\hat{G}$, and the associated Frobenius category $\F(b,\hat{G})$, does there exist a...

conjecture | open | MathDB 页面 | 原文线索

V2-076748 | MATHDB-397589

Conjugacy of maximal invariant 2-subgroups in {2,3}-groups

摘录 / EXCERPT: Let G be a ${2,3}$-group whose Sylow 3-subgroups have exponent 3. Then, for every element $x\in G$ of order 3, all maximal x-invariant 2-subgroups of G are conjugate...

open_problem | open | MathDB 页面 | 原文线索

V2-076749 | MATHDB-397590

Brauer’s $k(B)$-conjecture for defect groups with a central cyclic subgroup

摘录 / EXCERPT: Brauer’s $k(B)$-conjecture holds for defect groups which contain a central, cyclic subgroup of index at most 9.

conjecture | open | MathDB 页面 | 原文线索

V2-076750 | MATHDB-397592

Products of powers in finite quasisimple groups

摘录 / EXCERPT: Let $k>2$ be any integer and let $w(x)=x^k$. Then there are finite quasisimple groups G of arbitrarily large order such that $\{1\}\ne w(G)^2\ne G$.

open_problem | open | MathDB 页面 | 原文线索

V2-076751 | MATHDB-397593

Non-commutative Waring problem for finite non-abelian simple groups

摘录 / EXCERPT: Let $w\in F_d$ be a non-trivial word in the free group on d generators. Then there exists a constant $N=N(w)$ depending on w such that, for every finite non-abelian simple grou...

open_problem | open | MathDB 页面 | 原文线索

V2-076752 | MATHDB-397594

Products of two squares in finite quasisimple groups

摘录 / EXCERPT: Every element in any finite quasisimple group G is a product of two squares; equivalently, for every $g\in G$, there exist $x,y\in G$ such that $g=x^2y^2$.

open_problem | open | MathDB 页面 | 原文线索

V2-076753 | MATHDB-397595

Toy-model Lusztig conjecture around the Steinberg weight

摘录 / EXCERPT: Assume $p>h$. Let W be the Weyl group of the root system acting on the weight lattice X, and let $\mathrm{st}=(p-1)\rho$. Define ... For the quotient category...

conjecture | open | MathDB 页面 | 原文线索

V2-076754 | MATHDB-397596

Soergel’s toy-model multiplicity conjecture around the Steinberg weight

摘录 / EXCERPT: Assume that $p>h$. Let W be the Weyl group acting on X, let $\rho=\frac12\sum_{\alpha\in R^+}\alpha$, and let $\mathrm{st}=(p-1)\rho$. Define...

conjecture | open | MathDB 页面 | 原文线索

V2-076755 | MATHDB-397597

Almost-every-energy reducibility alternative for analytic quasiperiodic Schrödinger cocycles

摘录 / EXCERPT: Theorem 0.5. Let α be irrational and $v:\mathbb{T}\to\mathbb{R}$ be analytic. Then for almost every energy $E\in\mathbb{R}$ the corresponding Schrödinger cocycle has either...

open_problem | open | MathDB 页面 | 原文线索

V2-076756 | MATHDB-397598

Renormalization convergence for zero-Lyapunov-energy Schrödinger cocycles

摘录 / EXCERPT: Let $\alpha\in\mathbb{R}\setminus\mathbb{Q}$ and let $v:\mathbb{R}/\mathbb{Z}\to\mathbb{R}$ be real analytic. For Lebesgue almost every $E\in\{E\in\mathbb{R}:L(\alpha,S_{v,E})=0\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-076757 | MATHDB-397599

Bourgain’s negative result for nonperturbative purely absolutely continuous spectrum in two frequencies

摘录 / EXCERPT: Let $v=v(x_1,x_2)$ be a trigonometric polynomial on $\mathbb{T}^2$ with a nondegenerate local maximum. Fix $x=(x_1,x_2)$. For every $\mathord{\lambda}\ne 0$, there exists a set...

open_problem | open | MathDB 页面 | 原文线索

V2-076758 | MATHDB-397600

Absence of singular continuous spectrum for a typical operator

摘录 / EXCERPT: For a typical operator H, the spectrum measures have no singular continuous component.

open_problem | open | MathDB 页面 | 原文线索

V2-076759 | MATHDB-397601

Acriticality for measure-theoretically typical quasiperiodic Schrödinger potentials

摘录 / EXCERPT: Let α be an irrational frequency. Then, for a measure-theoretically typical potential v, the quasiperiodic Schrödinger operator $H_{\alpha,v}$ is acritical; that is, no en...

open_problem | open | MathDB 页面 | 原文线索

V2-076760 | MATHDB-397602

Dry Ten Martini problem for the non-critical almost Mathieu operator

摘录 / EXCERPT: Let $v(x)=2\cos 2\pi x$, $|\lambda|\ne 0,1$, and $\alpha\in\mathrm{DC}$. If E is in the spectrum Σ of H and $N(E)\in\alpha\mathbb{Z}+\mathbb{Z}$, then E belongs to the...

open_problem | open | MathDB 页面 | 原文线索

V2-076761 | MATHDB-397603

Almost Reducibility Conjecture

摘录 / EXCERPT: For a one-frequency analytic cocycle $T=(\alpha,A)$ with $\alpha\in\mathbb T\setminus\mathbb Q$ and $A\in C^\omega(\mathbb T,\mathrm{SL}_2(\mathbb R))$, subcriticality should imply...

conjecture | open | MathDB 页面 | 原文线索

V2-076762 | MATHDB-397607

Local version of a conjecture of Lusztig

摘录 / EXCERPT: If O is special and O' is the unique minimal orbit in its special piece, then the singularity associated to (O,O') is a quotient $\mathbb{C}^{2m}/\Gamma$, where Γ is L...

conjecture | open | MathDB 页面 | 原文线索

V2-076763 | MATHDB-397608

Determine emptiness of the orbit-intersection test for a \b8-group

摘录 / EXCERPT: In the setup of the report, prove that $U(w)\cap O'=\varnothing$, where $U(w)=V_2(h')\cap V_{\ge 2}(wh)$ for $w\in W_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-076764 | MATHDB-397609

Certifying emptiness in the closure-ordering test for nilpotent orbits of a theta-group

摘录 / EXCERPT: In the setting of a theta-group, let O' and O be nilpotent orbits with corresponding homogeneous $\mathfrak{sl}_2$-triples (h',e',f') and (h,e,f), let $V=\mathfrak g_1$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076765 | MATHDB-397612

Surjectivity of the comparison map in the general case

摘录 / EXCERPT: Without assuming that the real rank of G equals the rank of its maximal compact subgroup, is the natural comparison map...

open_problem | open | MathDB 页面 | 原文线索

V2-076766 | MATHDB-397614

Cartan decomposition and sphericality for Kac-Moody groups

摘录 / EXCERPT: Under the section’s standing assumptions on the Kac-Moody group G, the involution θ, and K and A, if $G=KAK$ holds, does it follow that G is spherical?

open_problem | open | MathDB 页面 | 原文线索

V2-076767 | MATHDB-397615

Cartan/Kostant-type decomposition for loop groups

摘录 / EXCERPT: When does $G=KUK$ hold for $G=\mathrm{SL}_n(F[t,t^{-1}])$?

open_problem | open | MathDB 页面 | 原文线索

V2-076768 | MATHDB-397616

Schwartz-space isomorphism for the Gelfand transform of a nilpotent Gelfand pair

摘录 / EXCERPT: For a nilpotent Gelfand pair (G,K) with nilpotent group N and Gelfand spectrum $\Sigma_d\subset\mathbb{R}^d$, does the Gelfand transform ... provide an isomorphism between...

open_problem | open | MathDB 页面 | 原文线索

V2-076769 | MATHDB-397623

Rainbow matching conjecture

摘录 / EXCERPT: Let $F_1,\\ldots,F_t$ be families of subsets in $\\binom{[n]}{k}$. If $|F_i|>\\max\\left\\{\\binom{n}{k}-\\binom{n-t+1}{k},\\binom{kt-1}{k}\\right\\}$ for every $1\\le i\\le t$, th...

conjecture | open | MathDB 页面 | 原文线索

V2-076770 | MATHDB-397628

Asymptotically minimal number of monochromatic 3-term arithmetic progressions in an interval

摘录 / EXCERPT: Determine the asymptotically minimal number of monochromatic 3-term arithmetic progressions in a 2-colouring of the interval $\{1,2,\ldots,n\}$, as n tends to infinity.

open_problem | open | MathDB 页面 | 原文线索

V2-076771 | MATHDB-397630

Graham’s quasirandom-permutation symmetry question

摘录 / EXCERPT: Let (π_j) be a sequence of permutations whose orders n_j tend to infinity. Call the sequence asymptotically k-symmetric if, for every k-point permutation τ,...

open_problem | open | MathDB 页面 | 原文线索

V2-076772 | MATHDB-397633

Nagle’s codegree-density conjecture for the complete 3-graph on four vertices

摘录 / EXCERPT: Nagle conjectured that $\gamma(K_4)=\frac{1}{2}$, where K_4 is the 3-graph on four vertices with four edges.

conjecture | open | MathDB 页面 | 原文线索

V2-076773 | MATHDB-397639

Kernel criterion in a fractional skew monoid ring

摘录 / EXCERPT: Let A be a unital ring, let T be a monoid acting on A by possibly nonunital endomorphisms α_t, and let $S\subseteq T$ be a submonoid satisfying the left denominator co...

open_problem | open | MathDB 页面 | 原文线索

V2-076774 | MATHDB-397640

Elashvili’s index conjecture for coadjoint stabilizers

摘录 / EXCERPT: For every $p\in\mathfrak g^*$, $\operatorname{ind}\mathfrak g_p=\operatorname{ind}\mathfrak g$; here $\mathfrak g_p$ denotes the stabilizer of p for the coadjoint action.

conjecture | open | MathDB 页面 | 原文线索

V2-076775 | MATHDB-397647

Non-negativity of the coefficients of $I_{P,T,\c3}(t)$

摘录 / EXCERPT: Let P be a lattice polytope of dimension d, let T be a regular lattice triangulation of P, and let $\c3\in T$. With ... the conjecture is that $I_{P,T,\u00\c3}(t)$ has non-ne...

open_problem | open | MathDB 页面 | 原文线索

V2-076776 | MATHDB-397648

Expected degree of the stringy E-function of a Gorenstein polytope

摘录 / EXCERPT: Let P) be a Gorenstein polytope of index r. Is it still open whether the degree of its stringy E-function $E_{st}(P;u,v)$ equals the expected degree $\dim(P)+1-2r$?

open_problem | open | MathDB 页面 | 原文线索

V2-076777 | MATHDB-397649

Exceptional solutions in the Bilu-Tichy criterion

摘录 / EXCERPT: Given a Bilu-Tichy decomposition ... with $f,g\in\mathbb Q[X]\setminus\mathbb Q$, $\lambda,\mu\in\mathbb Q[X]$ linear, and (f_1,g_1) a standard pair such that $f_1(x)=g_1(y)$ has i...

open_problem | open | MathDB 页面 | 原文线索

V2-076778 | MATHDB-397651

On the coefficients of linear forms in polylogarithms

摘录 / EXCERPT: Let $n\ge 1$. If an n-dimensional integral of a rational function over the product of n paths having distinct endpoints in $\mathbb{C}$ represents a linear form in zeta-values...

open_problem | open | MathDB 页面 | 原文线索

V2-076779 | MATHDB-397652

Fuchs-Zannier theorem on lacunary composite rational functions

摘录 / EXCERPT: Given $\l$ and $f(X)=g(h(X))=P(X)/Q(X)$ with $g,h display?

open_problem | open | MathDB 页面 | 原文线索

V2-076780 | MATHDB-397653

Integral representations for intermediate coefficients in zeta-value linear forms

摘录 / EXCERPT: For the coefficients $a_2,\ldots,a_{n-1}$ occurring in the linear form … under the hypotheses $n\ge 2$, $h_1,\ldots,h_n,j_1,\ldots,j_n,k\ge 0$, and $h_r+k-h_1\ge 0$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-076781 | MATHDB-397654

Determine the optimal simultaneous-approximation exponent for successive powers

摘录 / EXCERPT: For a point $\boldsymbol{\xi}=(\xi_1,\ldots,\xi_n)\in\mathbb{R}^n$, let $\lambda(\boldsymbol{\xi})$ be the supremum of the real numbers $\lambda\ge 0$ such that, for every sufficie...

open_problem | open | MathDB 页面 | 原文线索

V2-076782 | MATHDB-397655

Simultaneous approximation with a number-field basis

摘录 / EXCERPT: Determine $\sup\{\lambda(\omega_1:\cdots:\omega_n:\xi)\ ;\ \xi\in\mathbb{R}\setminus K\}$, where K is a number field of degree n over $\mathbb{Q}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076783 | MATHDB-397659

Schmidt’s conjecture on the intersection of weighted badly approximable sets

摘录 / EXCERPT: Let ... where $i,j\ge 0$, $i+j=1$, and $\|x\|$ denotes the distance from x to the nearest integer. Schmidt’s problem asks whether ...

conjecture | open | MathDB 页面 | 原文线索

V2-076784 | MATHDB-397660

Schmidt’s problem on badly approximable points

摘录 / EXCERPT: For $i,j\ge 0$ with $i+j=1$, define ... Schmidt’s problem was whether ...

open_problem | open | MathDB 页面 | 原文线索

V2-076785 | MATHDB-397663

Mahler’s conjecture on metric approximation by integer polynomials

摘录 / EXCERPT: For every fixed $n\in\mathbb{N}$ and every $\varepsilon>0$, for almost all real numbers x there are only finitely many polynomials $P\in\mathbb{Z}[x]$ with $\deg P\le n$ such tha...

conjecture | open | MathDB 页面 | 原文线索

V2-076786 | MATHDB-397666

Bogomolov property for the torsion field of an elliptic curve

摘录 / EXCERPT: Let E be an elliptic curve given by a Weierstrass equation $y^2=x^3+ax+b$ with $a,b\\in\\mathbb{Q}$. Let E_{tors} be the group of all points in...

open_problem | open | MathDB 页面 | 原文线索

V2-076787 | MATHDB-397667

Extension of the Kac-Wakimoto hierarchy for the full Gromov-Witten theory

摘录 / EXCERPT: The conjecture is that the integrable hierarchy governing the full Gromov-Witten theory of the orbifold projective line $\mathbb{P}^{1}_{r_1,r_2,1}$ is an extension of the Kac-Waki...

open_problem | open | MathDB 页面 | 原文线索

V2-076788 | MATHDB-397669

Calabi-Yau/Landau-Ginzburg correspondence for admissible orbifold pairs

摘录 / EXCERPT: Let $W\in\mathbb{C}[x_1,\ldots,x_n]$ be a quasi-homogeneous polynomial with an isolated singularity at the origin, and let G be an admissible group of automorphisms of W. Set...

open_problem | open | MathDB 页面 | 原文线索

V2-076789 | MATHDB-397670

Bijectivity of the Lyashko-Looijenga fiber-to-bases map

摘录 / EXCERPT: In the general hypersurface-singularity setting described by Hertling, is the map $LL^{-1}(u)\to B/(\mathrm{signs})$ a bijection?

open_problem | open | MathDB 页面 | 原文线索

V2-076790 | MATHDB-397672

Lattice sphere covering problem

摘录 / EXCERPT: Find a lattice $L=A\mathbb{Z}^n$, with $A\in GL_n(\mathbb{R})$, for which the density of the sphere covering-equal-sized spheres centered at the lattice points, with radius chosen...

open_problem | open | MathDB 页面 | 原文线索

V2-076791 | MATHDB-397675

Generalized lower bound conjecture: equality case for simplicial polytopes

摘录 / EXCERPT: Let P be a simplicial d-polytope. For an integer $1\le r\le d/2$, the following are equivalent: (i) $h_{r-1}(P)=h_r(P)$; (ii) P is $(r-1)$-stacked, namely, there is a triangu...

conjecture | open | MathDB 页面 | 原文线索

V2-076792 | MATHDB-397677

The equality case of the generalized lower-bound conjecture for simplicial polytopes

摘录 / EXCERPT: For an integer $1\le r\le d/2$, if P is a simplicial d-polytope with $h_{r-1}(P)=h_r(P)$, then P is $(r-1)$-stacked; that is, there is a triangulation K of P all of whose...

conjecture | open | MathDB 页面 | 原文线索

V2-076793 | MATHDB-397681

Nandakumar-Ramana Rao convex equipartition problem

摘录 / EXCERPT: Does every bounded convex set P in the plane admit a division into n convex pieces having equal area and equal perimeter?

open_problem | open | MathDB 页面 | 原文线索

V2-076794 | MATHDB-397684

Diameter conjecture for connected layer families of multisets

摘录 / EXCERPT: Let $\widetilde f(d,n)$ be the maximum diameter ℓ of a connected layer family of d-element multisets drawn from an n-element set. Thus, the multiset family is partitioned...

conjecture | open | MathDB 页面 | 原文线索

V2-076795 | MATHDB-397686

Virtual branched coverings over the figure-eight knot

摘录 / EXCERPT: Does M virtually cover S^3 branching over the figure-eight knot with arbitrarily high degree? More precisely, does, for any $L>0$, there exist a finite covering...

open_problem | open | MathDB 页面 | 原文线索

V2-076796 | MATHDB-397690

Vertices of hom-polytopes

摘录 / EXCERPT: “The challenge here is to understand the vertices of hom-polytopes. But, currently, even the case of general regular source and target polygons seems to be out of reach.”

open_problem | open | MathDB 页面 | 原文线索

V2-076797 | MATHDB-397692

Representability of the fiber-polytope hom-functor

摘录 / EXCERPT: For a polytope P and an affine map f between two polytopes, prove that there is a natural isomorphism ... where the fiber construction is formally extended to maps that are not n...

open_problem | open | MathDB 页面 | 原文线索

V2-076798 | MATHDB-397695

Subdivision invariance of collapsibility relative to a subcomplex

摘录 / EXCERPT: Assume that C is a simplicial complex that collapses to a subcomplex C'. If D is a subdivision of C such that D restricts to a subdivision D' of C', is it true that...

open_problem | open | MathDB 页面 | 原文线索

V2-076799 | MATHDB-397697

Maximum number of faces of a centrally symmetric polytope

摘录 / EXCERPT: What is the maximum number of k-dimensional faces that a centrally symmetric d-dimensional polytope with N vertices can have? This number is denoted by $f_k^{\max}(d,N)$.

open_problem | open | MathDB 页面 | 原文线索

V2-076800 | MATHDB-397698

Polynomial bound for polytope diameter

摘录 / EXCERPT: Does there exist a polynomial bound on the diameter of a polytope in terms of its dimension and number of vertices?

open_problem | open | MathDB 页面 | 原文线索

V2-076801 | MATHDB-397700

Minimal weak decomposability of simplicial polytopes

摘录 / EXCERPT: What can be said about the minimal value $k=k(d)$ such that all simplicial d-polytopes are (weakly) k-decomposable, but there exist simplicial d-polytopes that are not (weakl...

open_problem | open | MathDB 页面 | 原文线索

V2-076802 | MATHDB-397705

Common stellar refinements of linear triangulations of a simplex

摘录 / EXCERPT: Do any two linear triangulations of the d-simplex have a common stellar refinement?

open_problem | open | MathDB 页面 | 原文线索

V2-076803 | MATHDB-397709

Characterization of universal triangulations

摘录 / EXCERPT: Let T be a simplicial $(d-1)$-sphere linearly embeddable in $\\mathbb{R}^d$. A simplicial d-ball U with boundary $\\partial U=T$ is called universal for T if every linear e...

open_problem | open | MathDB 页面 | 原文线索

V2-076804 | MATHDB-397715

Existence of trace-free constraint tensors on Riemannian surfaces

摘录 / EXCERPT: Given a Riemannian surface (M,g), determine whether there exist trace-free tensors W satisfying, for $\bbda=0$, the constraints ... In particular, this existence problem is unres...

open_problem | open | MathDB 页面 | 原文线索

V2-076805 | MATHDB-397716

Yamabe invariant of cyclic quotients of the 5-sphere

摘录 / EXCERPT: It is conjectured that, for $M=S^5/(\cmathbb{Z}/m)$ with $m\ge 5$, the supremum in the definition of the smooth Yamabe invariant $\sigma(M)$ is attained by the round metric. This w...

open_problem | open | MathDB 页面 | 原文线索

V2-076806 | MATHDB-397718

Preservation of asymptotics under the normalized Kähler-Ricci flow

摘录 / EXCERPT: If g_ω is an asymptotically tame polyfibred cusp Kähler metric on X and ... is the solution to the normalized Ricci flow for $t\in[0,T)$ with ... then $g_{\widetilde\omega_t}$...

open_problem | open | MathDB 页面 | 原文线索

V2-076807 | MATHDB-397719

Short-time Yamabe flow on admissible incomplete edge spaces

摘录 / EXCERPT: There exists a solution to the Yamabe flow starting within a class of compact Riemannian spaces with admissible simple edge singularities that remains asymptotically admissible for...

open_problem | open | MathDB 页面 | 原文线索

V2-076808 | MATHDB-397721

Compactness and bubbling for minimal surfaces in hyperbolic three-space

摘录 / EXCERPT: Determine the compactness properties of the space $M_{k,g}$ of complete minimal surfaces in $\mathbb{H}^3$ with k well-separated asymptotic ends and genus g, under a uniform up...

open_problem | open | MathDB 页面 | 原文线索

V2-076809 | MATHDB-397728

Global existence and scattering for the quintic resonant system

摘录 / EXCERPT: Let $E\in(0,\infty)$. For any smooth initial data $\vec u_0=\{u_{0,j}\}_{j\in\mathbb Z^2}$ satisfying ... there exists a global solution $\vec u(t)=\{u_j(t)\}_{j\in\mathbb Z^2}$ of .....

open_problem | open | MathDB 页面 | 原文线索

V2-076810 | MATHDB-397729

Global well-posedness and scattering for the quintic resonant system

摘录 / EXCERPT: For every $E\in(0,\infty)$, and for every smooth initial datum $\vec u_0=\{u_{0,j}\}_{j\in\mathbb Z^2}$ satisfying ... the quintic resonant system ... where ... with $\vec u(0)=\vec u_0$...

open_problem | open | MathDB 页面 | 原文线索

V2-076811 | MATHDB-397734

Bubbling-off solutions and threshold behavior for the critical focusing nonlinear wave equation

摘录 / EXCERPT: For radial solutions of the energy-critical focusing nonlinear wave equation $\Box u=-u_{tt}+\Delta u=-u^5$ on $\mathbb{R}^{3+1}$ with energy close to, but slightly larger than, th...

open_problem | open | MathDB 页面 | 原文线索

V2-076812 | MATHDB-397739

Dynamics near the degree-one soliton for equivariant wave and Schrödinger maps

摘录 / EXCERPT: For finite-energy degree-one equivariant wave-map data of the form $(r,\theta)\mapsto(\phi(r),\theta)$ with $\phi(0)=0$ and $\phi(\infty)=\pi$, and for the corresponding enlarged e...

open_problem | open | MathDB 页面 | 原文线索

V2-076813 | MATHDB-397741

Global attractor for the weakly damped and forced periodic KdV equation

摘录 / EXCERPT: Consider the forced and weakly damped KdV equation ... with initial condition $u(x,0)=u_0(x)\in\dot L^2(\mathbb T)$, where $\gamma>0$, $f\in\dot L^2(\mathbb T)$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-076814 | MATHDB-397745

Uniqueness of instantaneously complete Ricci flows on surfaces

摘录 / EXCERPT: Conjecture. Let M^2 be a connected Riemannian surface without boundary and let g_0 be a smooth initial metric. Then an instantaneously complete Ricci flow $g(t)$ on M^2 with...

open_problem | open | MathDB 页面 | 原文线索

V2-076815 | MATHDB-397749

Weyl vanishing for boundary Yamabe blow-up

摘录 / EXCERPT: Let g be a smooth Riemannian metric defined in the unit half n-ball B_1^+, with $6\le n\le 24$. Suppose there is a sequence of positive solutions $\{u_i\}$ of ... where $K>0$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-076816 | MATHDB-397750

Compactness of the boundary Yamabe problem in dimensions n ≥ 25

摘录 / EXCERPT: In analogy with the case without boundary, one wonders whether Theorem 1 is false when $n \ge 25$; namely, whether the stated compactness conclusion for the boundary Yamabe problem...

open_problem | open | MathDB 页面 | 原文线索

V2-076817 | MATHDB-397751

Technical gradient condition for four-dimensional shrinking Ricci solitons

摘录 / EXCERPT: For every four-dimensional shrinking gradient Ricci soliton (M^4,g,f), prove that there exists a technical constant $C_{\mathrm{tech}}<\infty$ such that ... where p is a minimum...

open_problem | open | MathDB 页面 | 原文线索

V2-076818 | MATHDB-397752

Planar discs minimize area in finite-dimensional normed spaces

摘录 / EXCERPT: Let $X=(X,\|\cdot\|)$ be a finite-dimensional normed space, with area understood as two-dimensional Hausdorff measure. If $D\subset X$ is a two-dimensional disc contained in a two-...

open_problem | open | MathDB 页面 | 原文线索

V2-076819 | MATHDB-397753

Asymptotic Hausdorff area of translation-invariant metric balls

摘录 / EXCERPT: Let d be a metric on $\mathbb{R}^2$ invariant under translations by $\mathbb{R}^2$. Let B_R denote the metric ball of radius R centered at the origin in $(\mathbb{R}^2,d)$....

open_problem | open | MathDB 页面 | 原文线索

V2-076820 | MATHDB-397756

Degeneration of shrinking Ricci solitons

摘录 / EXCERPT: Let (M_i,g_i,f_i) be a sequence of closed shrinking Ricci solitons satisfying ... with ... for fixed positive constants v,D independent of i. After passing to a subsequence,...

open_problem | open | MathDB 页面 | 原文线索

V2-076821 | MATHDB-397758

An A_∞-structure for the proposed Fukaya-Seidel category

摘录 / EXCERPT: For the construction described, the differential, compositions, and higher operations (“Massey products”) conjecturally combine to an A_∞-structure.

open_problem | open | MathDB 页面 | 原文线索

V2-076822 | MATHDB-397760

Parity theory for higher-dimensional objects

摘录 / EXCERPT: How can one construct a parity theory for higher-dimensional objects, for example for 2-surfaces in 4-space?

open_problem | open | MathDB 页面 | 原文线索

V2-076823 | MATHDB-397761

Categorification of the parity bracket

摘录 / EXCERPT: How to construct a complex whose Euler characteristic is the parity bracket $[K]$, with all graphs treated, say, as gradings?

open_problem | open | MathDB 页面 | 原文线索

V2-076824 | MATHDB-397763

Bernhard-Jablan Conjecture on unlinking numbers

摘录 / EXCERPT: For a link diagram D, let $u(D)$ be the minimum number of crossing changes in D required to obtain a diagram representing an unlink, and let $u(L)=\min_D u(D)$ over all diagram...

conjecture | open | MathDB 页面 | 原文线索

V2-076825 | MATHDB-397764

Critical-polynomial degree conjecture for knot and link families

摘录 / EXCERPT: Let a knot-family Conway symbol have parameters $p_1,p_2,\ldots,p_n$ denoting twists, with $p_i>1$ for $i=1,\ldots,n$. The polynomial in the variables $p_1,p_2,\ldots,p_n$ with int...

conjecture | open | MathDB 页面 | 原文线索

V2-076826 | MATHDB-397765

Exotic smooth structures from twisting along embedded surfaces

摘录 / EXCERPT: Given an embedding $F_g\subset M^4$, determine when-and whether-twisting M along F_g produces an exotic copy of M; when $g=0$, this operation is called Gluck twisting.

open_problem | open | MathDB 页面 | 原文线索

V2-076827 | MATHDB-397768

Information determined by lower central series quotients of metabelian groups

摘录 / EXCERPT: Given a residually nilpotent, finitely presented, metabelian group G, what can I conclude about the group G from its lower central series quotients?

open_problem | open | MathDB 页面 | 原文线索

V2-076828 | MATHDB-397769

Cycle conjecture for minimal class VII surfaces

摘录 / EXCERPT: Conjecture 2. If $X\in \mathrm{VII}^{\min}_{b_2>0}$, then X has a cycle of rational curves. Here a cycle of rational curves is an effective divisor $C=\sum_{i\in\mathbb Z_k}C_i$...

conjecture | open | MathDB 页面 | 原文线索

V2-076829 | MATHDB-397770

Cycle of rational curves on minimal class VII surfaces with $b_2=2$

摘录 / EXCERPT: Every minimal class VII surface X with $b_2(X)=2$ has a cycle of rational curves.

open_problem | open | MathDB 页面 | 原文线索

V2-076830 | MATHDB-397772

Whitehead doubling and smooth sliceness

摘录 / EXCERPT: For the Whitehead doubling operator D on the smooth knot concordance group, $D^{-1}(\\\mathcal O)=\mathcal O$, where $\mathcal O$ is the class of the unknot. Equivalently, the Wh...

open_problem | open | MathDB 页面 | 原文线索

V2-076831 | MATHDB-397773

Trace-invariant representations of free braid groups under L-moves

摘录 / EXCERPT: Find representations of the free braid groups of the displayed form (2), with a linear map $R:V\otimes V\to V\otimes V$ satisfying the Yang-Baxter equation (1) and conditions (3) a...

open_problem | open | MathDB 页面 | 原文线索

V2-076832 | MATHDB-397774

Realisability of graphs under Reidemeister equivalence

摘录 / EXCERPT: Is every graph Reidemeister equivalent-using the Reidemeister moves of the relevant theory-to the intersection graph of a virtual knot diagram?

open_problem | open | MathDB 页面 | 原文线索

V2-076833 | MATHDB-397778

Stable approximation of oversampled thresholding processes

摘录 / EXCERPT: Characterize the systems that can be stably approximated with approximation processes that use oversampling.

open_problem | open | MathDB 页面 | 原文线索

V2-076834 | MATHDB-397779

RIP-preserving orthonormalization of subsets of a frame

摘录 / EXCERPT: Let $\{\varphi_i\}_{i=1}^M$ be a family of vectors in ℓ_2^N having the restricted isometry property with constant $0<\varepsilon<1$ for sets of size s. Partition...

open_problem | open | MathDB 页面 | 原文线索

V2-076835 | MATHDB-397780

Exact recovery of separated point sources by total-variation minimization

摘录 / EXCERPT: Let $x=\csum_j a_j ej\delta_{\tau_j}$ be a signal with amplitudes $a_j\in\mathbb{C}$ and locations $\tau_j\in T\subset[0,1]$. Suppose the observed data are the $n=2f_c+1$ low-frequ...

open_problem | open | MathDB 页面 | 原文线索

V2-076836 | MATHDB-397781

Stable super-resolution error bound under noisy data

摘录 / EXCERPT: Suppose φ_{λ_f} is a low-pass filter that suppresses frequencies above $f=1/\lambda_f$. Under the same hypotheses as above, the solution $\hat{x}$ to (5), namely ... ob...

open_problem | open | MathDB 页面 | 原文线索

V2-076837 | MATHDB-397782

Sparse control for consensus emergence in the Cucker-Smale model

摘录 / EXCERPT: For every initial condition $(x_0,v_0)\in(\mathbb{R}^d)^N\times(\mathbb{R}^d)^N$ and every $M>0$, there exist $T>0$ and a sparse control $u:[0,T]\to(\mathbb{R}^d)^N$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-076838 | MATHDB-397783

Frame-set conjecture for continuous totally positive functions

摘录 / EXCERPT: The modified conjecture is that the frame set of every continuous totally positive function is ...

conjecture | open | MathDB 页面 | 原文线索

V2-076839 | MATHDB-397784

Restricted isometry property for the random scattering matrix

摘录 / EXCERPT: Establish the so-called restricted isometry property for the structured random scattering matrix A defined in (5) on PDF page 43.

open_problem | open | MathDB 页面 | 原文线索

V2-076840 | MATHDB-397785

Critical gap-size bound for wavelet-based \[1m\[0m-minimization inpainting

摘录 / EXCERPT: To find the critical bound for the size of the gap that wavelets can asymptotically inpaint using ℓ_1 minimization (the known critical bound is for thresholding).

open_problem | open | MathDB 页面 | 原文线索

V2-076841 | MATHDB-397788

Block-iterative solution of an $\[1m\[0m\ell^1$-minimization problem under exact linear constraints

摘录 / EXCERPT: Develop a block-iterative method for finding ...

open_problem | open | MathDB 页面 | 原文线索

V2-076842 | MATHDB-397790

Random-code blind deconvolution / channel protection theorem

摘录 / EXCERPT: The report states the following result. Let $m\in\mathbb{R}^{N}$ be a message vector and $h\in\mathbb{R}^{K}$ a channel vector with channel coherence μ_h, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-076843 | MATHDB-397791

PhaseLift exact recovery by trace minimization

摘录 / EXCERPT: Theorem 1. Consider an arbitrary signal $x\in\mathbb{C}^n$ and suppose that the number of measurements obeys $m\ge c_0 n\log n$, where c_0 is a sufficiently large constant. Then,...

open_problem | open | MathDB 页面 | 原文线索

V2-076844 | MATHDB-397793

Fidelity guarantees for one-dimensional total variation minimization

摘录 / EXCERPT: “Guarantees on the fidelity of total variation minimization in the one-dimensional setting” remain an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-076845 | MATHDB-397794

Integral Hodge conjecture for smooth projective threefolds of nonpositive Kodaira dimension

摘录 / EXCERPT: The integral Hodge conjecture should hold for smooth complex projective threefolds of Kodaira dimension $\kappa\le 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-076846 | MATHDB-397796

Prym-Green Conjecture for even-genus paracanonical curves

摘录 / EXCERPT: Let bC,η\u0005din R{g,ell}$ be a general level-ell curve, where $gge 6$ is even, and set $L=KCotimeseta...

conjecture | open | MathDB 页面 | 原文线索

V2-076847 | MATHDB-397797

Kodaira dimensions of Prym moduli spaces in genera 9-12

摘录 / EXCERPT: Determine the Kodaira dimension of the Prym moduli space $\cR_g$ for each $g tu0005cin\tu0005b9,12\d$.

open_problem | open | MathDB 页面 | 原文线索

V2-076848 | MATHDB-397798

Hodge conjecture for 2-cycles on abelian 4-folds

摘录 / EXCERPT: The Hodge conjecture for 2-cycles on a 4-fold is not known, even for an abelian 4-fold.

conjecture | open | MathDB 页面 | 原文线索

V2-076849 | MATHDB-397799

The integral Hodge conjecture for smooth complex projective 3-folds with trivial canonical bundle

摘录 / EXCERPT: For every smooth complex projective 3-fold with trivial canonical bundle, the integral Hodge conjecture is true.

conjecture | open | MathDB 页面 | 原文线索

V2-076850 | MATHDB-397800

Potential density of a general K3 surface over a number field

摘录 / EXCERPT: It is not known whether a general K3 surface is potentially dense over a number field.

open_problem | open | MathDB 页面 | 原文线索

V2-076851 | MATHDB-397803

Natural parametrization of invariant curves

摘录 / EXCERPT: Can such invariant curves be naturally parameterized by the elements of $\overline{\mathbb{Q}}$ rather than $\mathbb{Q}$?

open_problem | open | MathDB 页面 | 原文线索

V2-076852 | MATHDB-397805

Moduli interpretations of the log canonical models of the moduli space of curves

摘录 / EXCERPT: Does $M_g(\alpha)$ inherit a moduli description? More precisely, is $M_g(\alpha)$ a moduli space of singular curves, and can the divisorial contractions and flips arising in the mi...

open_problem | open | MathDB 页面 | 原文线索

V2-076853 | MATHDB-397811

Extension of the Chentsov uniqueness theorem to general sample spaces

摘录 / EXCERPT: For any space Ω, the Fisher metric and the Amari-Chentsov tensors are uniquely characterized by their invariance under sufficient statistics.

open_problem | open | MathDB 页面 | 原文线索

V2-076854 | MATHDB-397812

Change point estimation in a regression model with a smooth change

摘录 / EXCERPT: For i.i.d. observations (X_i,Y_i) having the same distribution as (X,Y), consider the regression model $Y_i=f_{\theta_0}(X_i)+\varepsilon_i$, where $\theta_0\in[0,1]$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076855 | MATHDB-397813

Density estimation from an unweighted random k-nearest-neighbor graph

摘录 / EXCERPT: Assume we draw points $X_1,\dots,X_n$ independently and identically distributed from some nice density p on $\mathbb{R}^d$. Construct the unweighted, undirected k-nearest-neigh...

open_problem | open | MathDB 页面 | 原文线索

V2-076856 | MATHDB-397814

Explicit compactly supported infinitely differentiable radial positive-definite kernel

摘录 / EXCERPT: Find an explicit formula for a radial, positive definite, compactly supported, and infinitely differentiable kernel.

open_problem | open | MathDB 页面 | 原文线索

V2-076857 | MATHDB-397817

Entire-version conjecture for Harris-Taylor perverse sheaves

摘录 / EXCERPT: Let π_v be an irreducible cuspidal representation of $\mathrm{GL}_g(F_v)$ such that its modular reduction is supercuspidal. Then, for all...

conjecture | open | MathDB 页面 | 原文线索

V2-076858 | MATHDB-397818

Regularity of special difference divisors

摘录 / EXCERPT: In the unramified unitary Rapoport-Zink space of signature $(1,n-1)$, with $n\ge 1$ and $p\ge 3$, let j be a special homomorphism. For $D(j):=Z(j)-Z(j/p)$, the special difference...

open_problem | open | MathDB 页面 | 原文线索

V2-076859 | MATHDB-397819

Intersection of minuscule special cycles lies in the special fiber

摘录 / EXCERPT: Let $j_1,\ldots,j_n$ be special homomorphisms such that their fundamental matrix $T(j_1,\ldots,j_n)$ is equivalent to a matrix of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-076860 | MATHDB-397824

Asymptotic bounds for the \-spectrum of a sign-indefinite potential

摘录 / EXCERPT: For a real-valued, locally L^2 and integrable potential $V\in\mathcal V_1$ that is not single-signed, conjecture the leading-order asymptotic for ... as $R\to\infty$ lies between ……

open_problem | open | MathDB 页面 | 原文线索

V2-076861 | MATHDB-397825

Existence of formal asymptotic solutions for two almost-touching domains

摘录 / EXCERPT: There exist formal solutions, i.e. sets of asymptotic data $(\varepsilon_i,u_j^H)_{i,j,H}$ which solve the model problems on H_{int} and H_{ext} and satisfy t...

open_problem | open | MathDB 页面 | 原文线索

V2-076862 | MATHDB-397829

Dimension-only additive upper bound for hypersurface Laplace eigenvalues

摘录 / EXCERPT: For a bounded domain $\Omega\subset\mathbb{R}^{m+1}$ with smooth boundary $\Sigma=\partial\Omega$, and for all $k\ge 1$, is it possible to find constants A_m,B_m depending only o...

open_problem | open | MathDB 页面 | 原文线索

V2-076863 | MATHDB-397830

Ionization conjecture for pseudorelativistic atoms in Hartree-Fock theory

摘录 / EXCERPT: Let $Z\ge 1$ and $\alpha>0$, set $\kappa=Z\alpha$, and assume $0\le\kappa<2/\pi$. For the pseudorelativistic Hartree-Fock atom described in the report, there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-076864 | MATHDB-397832

Third Dirichlet eigenvalue minimization in the plane

摘录 / EXCERPT: The conjecture is that, among open sets $\Omega\subset\mathbb{R}^2$ with volume $|\Omega|=1$, the minimum of the third Dirichlet Laplacian eigenvalue $\lambda_3(\Omega)$ is attaine...

open_problem | open | MathDB 页面 | 原文线索

V2-076865 | MATHDB-397833

Sharp quantitative Faber-Krahn inequality with Fraenkel asymmetry exponent 2

摘录 / EXCERPT: For sets of unit volume in $\mathbb{R}^N$, the known estimate ... should hold with the exponent 4 replaced by 2; equivalently, one expects a dimensional constant $C(N)>0$ such th...

open_problem | open | MathDB 页面 | 原文线索

V2-076866 | MATHDB-397834

Sharp upper bounds for Dirichlet eigenvalue functionals on starlike planar domains

摘录 / EXCERPT: Let $R(\theta)$ be 2π-periodic, positive, and Lipschitz continuous, and let ... be the corresponding starlike plane domain. Let λ_j be the Dirichlet Laplacian eigenvalues...

open_problem | open | MathDB 页面 | 原文线索

V2-076867 | MATHDB-397835

Asymptotic shape of optimal fixed-area rectangles

摘录 / EXCERPT: Let R_a be a rectangle with side lengths a and $1/a$, where $a\geq 1$, and let $\lambda_k(a)$ be its kth Dirichlet Laplacian eigenvalue. Define ... Then, if a_k^* denotes an...

open_problem | open | MathDB 页面 | 原文线索

V2-076868 | MATHDB-397836

Optimal constant in the classical Lieb-Thirring inequality

摘录 / EXCERPT: Determine the optimal constant $K_d>0$, depending only on the dimension d, such that for every $N\in\mathbb{N}$, every collection of functions...

open_problem | open | MathDB 页面 | 原文线索

V2-076869 | MATHDB-397837

Pólya’s conjecture for Dirichlet Laplacian counting functions

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be any open set of finite volume, and let λ_j be the eigenvalues of the Dirichlet Laplacian on Ω, counted with multiplicities. Writin...

conjecture | open | MathDB 页面 | 原文线索

V2-076870 | MATHDB-397838

Some inequalities involving harmonic functions and Dirichlet eigenfunctions - Conjecture 1

摘录 / EXCERPT: Let $m\ge 2$ and let $\Omega\subset\mathbb{R}^m$ be a bounded open set with sufficiently smooth boundary. Let λ_j and u_j be the eigenvalues of the Dirichlet Laplacian on...

conjecture | open | MathDB 页面 | 原文线索

V2-076871 | MATHDB-397839

Bounds for Dirichlet eigenvalues

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^{m+1}$ be a bounded domain with smooth boundary $\Sigma=\partial\Omega$. Is it possible to obtain a bound of the form ... for all $k\ge 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-076872 | MATHDB-397840

Existence of a λ-harmonic function satisfying a spectral inequality

摘录 / EXCERPT: Let $m\ge 2$ and let $\Omega\subset\mathbb{R}^m$ be a bounded open set with a sufficiently smooth boundary. Let (λ_j,u_j) be the Dirichlet-Laplacian eigenvalues and corresp...

open_problem | open | MathDB 页面 | 原文线索

V2-076873 | MATHDB-397841

Minimizer of the first eigenvalue of the three-dimensional magnetic Laplacian

摘录 / EXCERPT: For a real number β and a bounded open set $\Omega\subset\mathbb{R}^3$, consider the first eigenvalue E_1 of the magnetic Laplacian defined by ... Can one find a domain...

open_problem | open | MathDB 页面 | 原文线索

V2-076874 | MATHDB-397842

Sharp constants in L^∞ bounds for Dirichlet eigenfunctions

摘录 / EXCERPT: Find the sharp constant $C_{j,m}$ in the estimate $\|\phi_j\|_\infty\le C_{j,m}\lambda_j(\Omega)^{m/4}$ for the jth Dirichlet eigenfunction of an open set...

open_problem | open | MathDB 页面 | 原文线索

V2-076875 | MATHDB-397844

Optimal power in upper bounds for the Steklov spectrum

摘录 / EXCERPT: For every integer $m\ge 2$, does there exist a constant $c'(m)$ such that, for every bounded Euclidean domain $\Omega\subset\mathbb{R}^{m+1}$ with smooth boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-076876 | MATHDB-397845

Surfaces with large Steklov eigenvalues

摘录 / EXCERPT: Does there exist a sequence Ω_l of compact Riemannian surfaces such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-076877 | MATHDB-397846

Bounded limsup of nodal-domain counts for Laplacians

摘录 / EXCERPT: For $-\Delta$ on some $\Omega a\subset\R^d$, is it possible that ...

open_problem | open | MathDB 页面 | 原文线索

V2-076878 | MATHDB-397847

Minimizer of the first Dirichlet eigenvalue on a simply connected flat torus domain

摘录 / EXCERPT: Determine the minimizer of ... where $\lambda_1(\Omega)$ is the first eigenvalue of the Dirichlet Laplacian acting in $L^2(\Omega)$. Here $\mathbb T(a,b)$ is the flat two-dimensional...

open_problem | open | MathDB 页面 | 原文线索

V2-076879 | MATHDB-397848

Positive-boundary Neumann eigenfunction on the square

摘录 / EXCERPT: Let $Q=(0,1)^2$ with Neumann boundary conditions for the Laplacian. Does there exist a non-constant Neumann eigenfunction u satisfying $- \Delta u=\lambda u$ on Q and...

open_problem | open | MathDB 页面 | 原文线索

V2-076880 | MATHDB-397849

Odd numbers of nodal domains on a flat rectangular torus

摘录 / EXCERPT: Let $T(a,b)$ be the flat torus obtained from the rectangle $(0,a)\times(0,b)$ with periodic boundary conditions. Its eigenvalues are...

open_problem | open | MathDB 页面 | 原文线索

V2-076881 | MATHDB-397850

Ground-state energy minima for a random displacement in an equilateral triangle

摘录 / EXCERPT: Let D be an equilateral triangle, and let q be a smooth, compactly supported potential whose support has no holes and which has the same reflection symmetries as D (for examp...

open_problem | open | MathDB 页面 | 原文线索

V2-076882 | MATHDB-397852

Legendre-Steinitz Problem on realization-space dimension

摘录 / EXCERPT: How does, for d-dimensional polytopes, the dimension of the realization space grow with the size of the polytopes? Here, the size of a polytope is the combined number of its vert...

open_problem | open | MathDB 页面 | 原文线索

V2-076883 | MATHDB-397854

Consistency of discrete Willmore energies

摘录 / EXCERPT: Theorem 2. Let M be a smooth surface in $\\mathbb{R}^3$. Then there exists an $h_0\in\mathbb{R}_+$ such that, for every pair consisting of a polyhedral surface M_h that is insc...

open_problem | open | MathDB 页面 | 原文线索

V2-076884 | MATHDB-397855

Discrete uniformization of triangulated surfaces

摘录 / EXCERPT: Given a discrete metric on a triangulated surface, find a discretely conformally equivalent Euclidean, spherical, or hyperbolic metric without cone-like singularities at the vertic...

open_problem | open | MathDB 页面 | 原文线索

V2-076885 | MATHDB-397856

Quantitative convergence of discrete elastica minimizers

摘录 / EXCERPT: For fixed length and clamped boundary data, let F be the continuous Euler-Bernoulli bending energy and F_n the discrete bending energy on the evenly segmented polygonal ansatz...

open_problem | open | MathDB 页面 | 原文线索

V2-076886 | MATHDB-397857

Shellability of the barycentric subdivision of a convex simplicial complex

摘录 / EXCERPT: Let C be a convex simplicial complex, meaning that it has a geometric realization in some $\d53f^k$ that is convex (equivalently, C is a linear triangulation of a convex polyto...

open_problem | open | MathDB 页面 | 原文线索

V2-076887 | MATHDB-397858

Minimal-area branched surface associated with a knot or link isotopy type

摘录 / EXCERPT: For a given non-trivial isotopy type of knots (or links), find, approximately and numerically, a 2-dimensional branched surface M (a standard spine, possibly with 1-dimensional p...

open_problem | open | MathDB 页面 | 原文线索

V2-076888 | MATHDB-397859

Diameter of two glued random polygon triangulations

摘录 / EXCERPT: Let T_1,T_2 be two independent and uniformly random triangulations of an n-gon (without internal vertices). Glue these triangulations together along the boundary n-gon to for...

open_problem | open | MathDB 页面 | 原文线索

V2-076889 | MATHDB-397861

Classification of bicycle (n,k)-gons

摘录 / EXCERPT: An equilateral plane n-gon $V_1V_2\ldots V_n$ is called a bicycle (n,k)-gon if, for all i, the quadrilateral $V_iV_{i+1}V_{i+k}V_{i+k+1}$ is an isosceles trapezoid:...

open_problem | open | MathDB 页面 | 原文线索

V2-076890 | MATHDB-397862

Classification of ambiguous closed bicycle tracks

摘录 / EXCERPT: The problem is to describe pairs of closed plane curves, the rear and front bicycle tracks, such that one cannot tell which way the bicycle went; a pair of concentric circles provi...

open_problem | open | MathDB 页面 | 原文线索

V2-076891 | MATHDB-397868

Local-in-time middle-dimensional non-squeezing for symplectomorphisms

摘录 / EXCERPT: For $1<k<n$, let $\{\varphi_t\}_{t\in[0,1]}$ be a smooth family of symplectomorphisms of $\mathbb{R}^{2n}$ such that φ_0 is linear. Is it true that ... for every sufficiently...

open_problem | open | MathDB 页面 | 原文线索

V2-076892 | MATHDB-397870

Smooth structures on orientable globally hyperbolic 4-dimensional spacetimes

摘录 / EXCERPT: If two orientable globally hyperbolic 4-dimensional spacetimes are homeomorphic, then they are diffeomorphic.

open_problem | open | MathDB 页面 | 原文线索

V2-076893 | MATHDB-397871

Slippery Conjecture for rotation-number ziggurats

摘录 / EXCERPT: Let F) be the free group of rank 2 with generators a,b, and let w be a positive word, i.e. a word in the semigroup generated by a and b. For $r,s\in\mathbb{R}$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-076894 | MATHDB-397872

Refined Slippery Conjecture

摘录 / EXCERPT: Refined Slippery Conjecture. Suppose $w=a^{\alpha_1}b^{\beta_1}\cdots b^{\beta_m}$ is positive. If $R(w,r,s)=p/q$, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-076895 | MATHDB-397873

Real-analytic attainment of the supremum rotation number

摘录 / EXCERPT: Let $w\in F$ be arbitrary. Is there a C^ω representation ρ so that $\operatorname{rot}_{\sim}(\rho(w))=R(w,r,s)$? Here F is the rank-two free group generated by a,b...

open_problem | open | MathDB 页面 | 原文线索

V2-076896 | MATHDB-397874

Universal vanishing of rotation number for a nontrivial free-group word

摘录 / EXCERPT: Let F be the free group of rank 2, and let $\widetilde{\mathrm{Homeo}}^+(S^1)$ denote the universal cover of the group of orientation-preserving homeomorphisms of the circle, e...

open_problem | open | MathDB 页面 | 原文线索

V2-076897 | MATHDB-397875

Classification of faithful homomorphisms from lattices in \(\mathrm{SO}(n,1)\) and \(\mathrm{SU}(n,1)\) to the Cremona group

摘录 / EXCERPT: The problem of describing all faithful homomorphisms from lattices in the groups $\mathrm{SO}(n,1)$ and $\mathrm{SU}(n,1)$ to the Cremona group $\operatorname{Bir}(\mathbb{P}^2)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-076898 | MATHDB-397883

Lafont-Schmidt blocking characterization of CROSSes

摘录 / EXCERPT: Conjecture: The compact rank-one symmetric spaces (CROSSes) are characterized by the following blocking property: whenever p and q are distinct points with...

open_problem | open | MathDB 页面 | 原文线索

V2-076899 | MATHDB-397884

Compatibility of the discriminant metric with contact orderability

摘录 / EXCERPT: Assuming that (M,ξ) is orderable, is the discriminant metric on $\widetilde{\mathrm{Cont}}_0(M,\xi)$ compatible with the partial order $\le$?

open_problem | open | MathDB 页面 | 原文线索

V2-076900 | MATHDB-397886

$SL_3(\mathbb{C})$-character-equivalent but non-conjugate elements of F_2

摘录 / EXCERPT: Do there exist elements $g,g'\in F_2$ that are $\mathrm{SL}_3(\mathbb C)$-character equivalent but not conjugate? Here, two elements $g,g'\in G$ are $\mathrm{SL}_n(\mathbb C)$-char...

open_problem | open | MathDB 页面 | 原文线索

V2-076901 | MATHDB-397889

Quasi-isometry of horospheres under a hyperbolic metric and another metric in dimension 3

摘录 / EXCERPT: In dimension 3, are the horospheres in $(\widetilde X,\widetilde g_0)$ and $(\widetilde X,\widetilde g)$ quasi-isometric, where g_0 is the hyperbolic metric? More generally, what...

open_problem | open | MathDB 页面 | 原文线索

V2-076902 | MATHDB-397893

Extension of prequantization quasimorphisms to contactomorphism groups

摘录 / EXCERPT: In the case of prequantization spaces there are quasi-morphisms on $\operatorname{Cont}_0(M,\alpha)$ coming from $\operatorname{Ham}(X,\omega)$. Can such quasi-morphisms be extende...

open_problem | open | MathDB 页面 | 原文线索

V2-076903 | MATHDB-397895

Lawson Conjecture for embedded minimal tori in the standard 3-sphere

摘录 / EXCERPT: The Lawson Conjecture characterizes the Clifford torus as the unique minimal embedded torus in the standard S^3.

conjecture | open | MathDB 页面 | 原文线索

V2-076904 | MATHDB-397896

Lawson conjecture for embedded minimal tori in S^3

摘录 / EXCERPT: Every embedded minimal surface in S^3 of genus 1 is congruent to the Clifford torus; equivalently, if $F:T^2\to S^3$ is a minimal embedding, then $F(T^2)$ is congruent to the C...

conjecture | open | MathDB 页面 | 原文线索

V2-076905 | MATHDB-397897

Sharp quantitative isoperimetric inequality in higher codimension

摘录 / EXCERPT: Let $n\ge 2$ and $k\ge 0$. There exists a constant $C>0$, depending only on n and k, such that for every $T\in\mathcal R_{n-1}(\mathbb R^{n+k})$ with $\partial T=0$, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-076906 | MATHDB-397898

Stability of optimal shapes for the Steklov Laplacian

摘录 / EXCERPT: Consider the volume-constrained optimization problem ... where $\sigma_2(\Omega)$ is the first nontrivial Steklov eigenvalue, characterized by ... The report states that balls solve th...

open_problem | open | MathDB 页面 | 原文线索

V2-076907 | MATHDB-397903

Openness of minimizers in spectral shape optimization

摘录 / EXCERPT: Let $k,N\in\mathbb{N}$ and let $F:\mathbb{R}^k\to\mathbb{R}$ be lower semicontinuous and non-decreasing in each variable. For quasi-open sets $\Omega\subset\mathbb{R}^N$ of unit vo...

open_problem | open | MathDB 页面 | 原文线索

V2-076908 | MATHDB-397905

Gamma-convergence of the Aviles-Giga energy

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be an open domain and, for divergence-free $u\in H^1(\Omega;\mathbb{R}^2)$, let ... For limiting configurations u_0 satisfying $|u_0|=1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-076909 | MATHDB-397906

Γ-convergence of the Bloch-wall energy

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$, and for $m=(u,m_3)\in H^1(\Omega,\mathbb{S}^2)$ with $u=(m_1,m_2)$ satisfying $\nabla\cdot u=0$ in Ω, define ... As $\varepsilon\downarrow0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076910 | MATHDB-397909

Scope of the conformal method for Einstein constraint equations

摘录 / EXCERPT: Determine the extent to which the conformal method can be used to parametrize and construct the range of solutions of the Einstein constraint equations, beyond the constant-mean-cu...

open_problem | open | MathDB 页面 | 原文线索

V2-076911 | MATHDB-397912

Bray’s conjecture on the isoperimetric character of the Huisken-Yau foliation

摘录 / EXCERPT: Let (M,g) be a complete three-dimensional asymptotically flat initial data set, possibly with compact minimal boundary, that is C^4-asymptotic to Schwarzschild of mass $m>0$. F...

conjecture | open | MathDB 页面 | 原文线索

V2-076912 | MATHDB-397914

Local propagation of impulsive gravitational waves

摘录 / EXCERPT: Theorem 1 (Luk-Rodnianski). Given the characteristic initial data described above, consisting of data on a truncated outgoing cone H_0 and a truncated incoming cone...

open_problem | open | MathDB 页面 | 原文线索

V2-076913 | MATHDB-397915

Black hole formation from a complete regular past for collisionless matter

摘录 / EXCERPT: There exists a class of initial data J for the spherically symmetric Einstein-Vlasov system with the property that black holes form in the future time direction and, in the past...

open_problem | open | MathDB 页面 | 原文线索

V2-076914 | MATHDB-397916

Bianchi VIII and IX singularity approximation by heteroclinic chains

摘录 / EXCERPT: The natural conjecture is that generic solutions of Bianchi types VIII and IX are approximated in the past by heteroclinic chains.

open_problem | open | MathDB 页面 | 原文线索

V2-076915 | MATHDB-397917

Relaxing the strict outer-trapped-boundary condition to a MOTS boundary

摘录 / EXCERPT: It would be interesting to relax the strict inequality $H + \operatorname{tr}_{\partial\Sigma}K < 0$ to $H + \operatorname{tr}_{\partial\Sigma}K = 0$, i.e. to the case of a MOTS bo...

open_problem | open | MathDB 页面 | 原文线索

V2-076916 | MATHDB-397918

Uniqueness for axially symmetric stationary KIDs in three dimensions

摘录 / EXCERPT: Whether uniqueness extends to axially symmetric, stationary KID when $m=3$ (and e.g. vacuum).

open_problem | open | MathDB 页面 | 原文线索

V2-076917 | MATHDB-397921

Critical solution for equal-mass ultrarelativistic black-hole collisions

摘录 / EXCERPT: For two equal-mass, nonspinning black holes, does there exist a unique critical solution that starts at infinite energy and ends in merger only after all the kinetic energy has bee...

open_problem | open | MathDB 页面 | 原文线索

V2-076918 | MATHDB-397922

Sharp logarithmic lower bound for energy decay on Schwarzschild-AdS

摘录 / EXCERPT: Let $(\mathcal M,g)$ be a Schwarzschild-AdS spacetime with mass $M>0$ and cosmological constant $\Lambda=-3/l^{2}$. Consider the equation $\Box_g\psi+(\alpha/l^{2})\psi=0$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-076919 | MATHDB-397926

Spacetime positive mass theorem in dimensions less than eight

摘录 / EXCERPT: The report states that, in dimensions less than eight, every asymptotically flat initial data set satisfying the dominant energy condition obeys the inequality $E\ge |P|$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-076920 | MATHDB-397927

Regular conformal field equations for matter with non-vanishing trace

摘录 / EXCERPT: In the case of an energy-momentum tensor with non-vanishing trace, is there a suitable choice of conformal variables that leads to a regular set of conformal field equations with m...

open_problem | open | MathDB 页面 | 原文线索

V2-076921 | MATHDB-397928

Nonlinear future stability of a pure-radiation FLRW cosmology

摘录 / EXCERPT: Suppose one is given Cauchy initial data for the Einstein-Euler system with a de Sitter-like cosmological constant and equation of state for pure radiation. If the initial data is...

open_problem | open | MathDB 页面 | 原文线索

V2-076922 | MATHDB-397930

Equipartition of energy for turbulent instability of AdS$_3$

摘录 / EXCERPT: In the critical case $d=2$, the turbulent instability of small perturbations of anti-de Sitter space is conjectured to lead to equipartition of energy as $t\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-076923 | MATHDB-397931

Local well-posedness of the Einstein vacuum Cauchy problem in CMCSH gauge for $H^s\times H^{s-1}$ data

摘录 / EXCERPT: The report states the following theorem: for every $s>2$, $t_0<0$, and $M_0>0$, there exist positive constants T_*, M_1, and M_2 such that, for every initial data set...

open_problem | open | MathDB 页面 | 原文线索

V2-076924 | MATHDB-397932

Electromagnetic Christodoulou memory formula

摘录 / EXCERPT: In the Einstein-Maxwell case, let $\Sigma^{+}(\cdot)=\lim_{u\to\infty}\Sigma(u,\cdot)$ and $\Sigma^{-}(\cdot)=\lim_{u\to-\infty}\Sigma(u,\cdot)$. Define ... Let Φ be the solutio...

open_problem | open | MathDB 页面 | 原文线索

V2-076925 | MATHDB-397934

Modularity criterion for sections over \(\mathbb Q\)

摘录 / EXCERPT: Let $X/\mathbb Q$ be a smooth, geometrically connected curve with non-abelian $\pi_1(X_{\overline{\mathbb Q}})$. A section $s:\operatorname{Gal}_{\mathbb Q}\to\pi_1(X)$ comes from...

open_problem | open | MathDB 页面 | 原文线索

V2-076926 | MATHDB-397939

Breuil-Mézard conjecture in the crystalline two-dimensional case

摘录 / EXCERPT: Let $p\geq 5$, let $L/\mathbb{Q}_p$ be finite with ring of integers $\mathcal O$, uniformizer ϖ, and residue field k, and let $\rho:G_{\mathbb{Q}_p}\to\mathrm{GL}_2(k)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-076927 | MATHDB-397943

Breuil-Mézard conjecture for potentially semistable representations

摘录 / EXCERPT: Let $p>2$, let $K/\mathbb{Q}_p$ be a finite extension, let $E/\mathbb{Q}_p$ be sufficiently large with ring of integers $\mathcal O$ and uniformizer π_E, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-076928 | MATHDB-397949

Stability and existence of compressible current-vortex sheets in three-dimensional MHD

摘录 / EXCERPT: For compressible current-vortex sheets in three-dimensional magnetohydrodynamics, the magnetic effect makes the linearized problem weakly stable; a suitable Nash-Moser-Hörmander it...

open_problem | open | MathDB 页面 | 原文线索

V2-076929 | MATHDB-397950

Necessity of far-field vacuum and square-integrable velocity for blow-up

摘录 / EXCERPT: Are the assumptions that the initial density approaches vacuum at infinity and that $u(\3,t)\in L^2(\mathbb{R}^N)$ necessary for finite-time blow-up of smooth solutions to the full...

open_problem | open | MathDB 页面 | 原文线索

V2-076930 | MATHDB-397951

Local solvability of a free-boundary magnetohydrodynamics problem in multi-connected domains

摘录 / EXCERPT: For the free-boundary magnetohydrodynamics problem described on the source pages-with viscous incompressible conducting fluid in $\Omega_1(t)$, vacuum in $\Omega_2(t)$, a fixed con...

open_problem | open | MathDB 页面 | 原文线索

V2-076931 | MATHDB-397952

Breakdown of smoothness for the Muskat problem

摘录 / EXCERPT: For the Muskat equation ... where $z=(z_1,z_2)$, there exists a solution $z(\alpha,t)$, defined for $t\in[t_0,t_2]$ and $\alpha\in\mathbb R$, together with a time $t_1\in(t_0,t_2)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076932 | MATHDB-397954

Existence parameters in the \bb=1, g=k family of strongly regular graphs

摘录 / EXCERPT: Suppose that there exists a strongly regular graph with parameters $((n^2+3n-1)^2,\,n^2(n+3),\,1,\,n(n+1))$, where $n\in\mathbb{N}$ is the positive eigenvalue in the family under i...

open_problem | open | MathDB 页面 | 原文线索

V2-076933 | MATHDB-397955

Bost’s tensor-product conjecture for the minimal slope invariant

摘录 / EXCERPT: For any Euclidean lattices L and M, let ... where $L^{(k)}$ denotes the set of k-dimensional sublattices of L. Bost’s conjecture asserts that ...

conjecture | open | MathDB 页面 | 原文线索

V2-076934 | MATHDB-397956

Optimal logarithmic configurations on spheres

摘录 / EXCERPT: Determine the logarithmic configurations of N points on the unit sphere $S^{d-1}\subset\mathbb{R}^d$, namely the configurations maximizing ... (or equivalently minimizing the discr...

open_problem | open | MathDB 页面 | 原文线索

V2-076935 | MATHDB-397957

Reconstruction from norms of subspace components by a closed formula

摘录 / EXCERPT: Given k-dimensional linear subspaces $[U+007F]\{V_j\}_{j=1}^n$ in $\mathbb{R}^d$, can the original signal $x\in S^{d-1}$ be reconstructed from the norms $\{\|P_{V_j}(x)\|\}_{j=1}^n$ by...

open_problem | open | MathDB 页面 | 原文线索

V2-076936 | MATHDB-397958

Poppy-Seed Bagel Theorem for minimal Riesz energy on rectifiable sets

摘录 / EXCERPT: Let $s>d$ and $p\ge d$, where d and p are integers. For every infinite compact d-rectifiable set $A\subset\mathbb{R}^p$, define ... Then ... where $C_{s,d}$ is a positive finite...

open_problem | open | MathDB 页面 | 原文线索

V2-076937 | MATHDB-397959

Polarization optimality of equally spaced points on the circle

摘录 / EXCERPT: For every integer $N\ge 1$ and every non-increasing, strictly convex function $f:[0,\pi]\to[0,\infty]$, let $\omega_N=(z_1,\ldots,z_N)$ be a configuration on S^1 and define ... whe...

open_problem | open | MathDB 页面 | 原文线索

V2-076938 | MATHDB-397960

Representations by a constrained quadratic form Q_5

摘录 / EXCERPT: For every integer $n\ge 1$ with $n\notin\{1,2,3,6\}$, there exist x_1,x_2,x_3,x_4 satisfying $x_j\in\mathbb Z+\frac{j}{5}$ for $1\le j\le 4$, $\max_{j\ne k}\frac{x_j}{x_k}\le 2$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076939 | MATHDB-397961

Asymptotic maximum polarization on compact rectifiable sets

摘录 / EXCERPT: For integers $s>d$ and $p\ge d$, and every infinite compact d-rectifiable set $A\subset\mathbb{R}^p$, define ... where $\omega_N=(x_1,\ldots,x_N)$ is an N-point configuration in...

open_problem | open | MathDB 页面 | 原文线索

V2-076940 | MATHDB-397962

Numerical semigroups and 5-core partitions of every size at least 6

摘录 / EXCERPT: For each integer $n\ge 6$, there exists a numerical semigroup S with multiplicity 5 and $w(S)+g(S)=n$. For each $n\ge 6$, there exists a 5-core partition λ of n whi...

open_problem | open | MathDB 页面 | 原文线索

V2-076941 | MATHDB-397963

Does rigidity imply infinitesimal rigidity for spanning spherical codes?

摘录 / EXCERPT: Let $C=\{x_1,\ldots,x_N\}\subset S^{n-1}$ be a spherical code that spans the ambient space. Call C rigid (or jammed) if the only continuous motions of its points that do not decr...

open_problem | open | MathDB 页面 | 原文线索

V2-076942 | MATHDB-397964

Uniqueness of the 24-point kissing configuration in four dimensions

摘录 / EXCERPT: The kissing configuration of 24 points in $[?]R^4$ is expected to be unique, but this question is open.

open_problem | open | MathDB 页面 | 原文线索

V2-076943 | MATHDB-397965

Meeting of semidefinite-programming and geometric bounds for spherical cap packings

摘录 / EXCERPT: For one cap size in dimension $n=3$ (that is, $N=1$), the semidefinite-programming upper bound for spherical cap packing density and Florian’s geometric upper bound appear to meet...

open_problem | open | MathDB 页面 | 原文线索

V2-076944 | MATHDB-397967

The Tammes problem for spherical point configurations

摘录 / EXCERPT: For a given positive integer N, determine ... that is, determine the largest possible minimum angular separation among N points on the unit sphere S^2.

open_problem | open | MathDB 页面 | 原文线索

V2-076945 | MATHDB-397968

Tightness of the Cohn-Elkies bound in dimensions 2, 8, and 24

摘录 / EXCERPT: The Cohn-Elkies bound is tight in dimensions 2, 8, and 24.

open_problem | open | MathDB 页面 | 原文线索

V2-076946 | MATHDB-397969

Korevaar-Meyers asymptotic bound for spherical designs

摘录 / EXCERPT: Korevaar and Meyers conjectured that $N(d,t) \le c_d t^d$, where c_d depends only on d.

open_problem | open | MathDB 页面 | 原文线索

V2-076947 | MATHDB-397970

Existence of spherical t-designs with optimal asymptotic order

摘录 / EXCERPT: For each $N\ge c_d t^d$, there exists a spherical t-design on S^d consisting of N points, where c_d is a constant depending only on d. Here $S^d\subset\mathbb{R}^{d+1}$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-076948 | MATHDB-397971

Asymptotically optimal spherical designs on the 2-sphere

摘录 / EXCERPT: Hardin and Sloane conjectured that the minimum number of points in a spherical t-design on S^2 satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-076949 | MATHDB-397972

Existence in the Gauss variational problem for infinite-dimensional vector measures

摘录 / EXCERPT: Let $0<\alpha<n$ and $n\ge 2$, and let $\mathcal A=(A_i)_{i\in I}$ be a countable collection of pairwise disjoint, nonempty, closed subsets of $\mathbb R^n$, with a sign...

open_problem | open | MathDB 页面 | 原文线索

V2-076950 | MATHDB-397973

Characterization of solvable mass vectors for the Gauss variational problem

摘录 / EXCERPT: Let $\kappa_\alpha(x,y)=|x-y|^{\alpha-n}$ with $0<\alpha<n$ in $\mathbb R^n$, $n\ge 2$. Let $\mathbf A=(A_i)_{i\in I}$ be a countable collection of pairwise disjoint, nonempty clos...

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V2-076951 | MATHDB-397974

Corollary 2: solvability criterion for the two-plate Gauss variational problem

摘录 / EXCERPT: Let $I=\{1,2\}$, $s_1=1$, $s_2=-1$, $C_\alpha(A_1)<\infty$, $g=1$, $\sigma\geq 0$, $a_2=a_1+\sigma(\mathbb{R}^n)$, and suppose that $\mathbb{R}^n\setminus A_2$ is connected. Then t...

open_problem | open | MathDB 页面 | 原文线索

V2-076952 | MATHDB-397977

Maximal measure of a measurable subset of S^2 avoiding orthogonal pairs

摘录 / EXCERPT: Let $A\subset S^2$ be measurable and satisfy $\langle x,y\rangle\ne 0$ for all $x,y\in A$. What is the largest possible value of $\mu(A)/\mu(S^2)$? The union of two antipodal spher...

open_problem | open | MathDB 页面 | 原文线索

V2-076953 | MATHDB-397978

A conjecture on the optimal approximation of the fractional Lévy area

摘录 / EXCERPT: Let $(B_t)_{t\ge 0}=(B_t^{(1)},B_t^{(2)})_{t\ge 0}$ be a two-dimensional fractional Brownian motion with Hurst parameter $H\in(1/4,1)$. For the fractional Lévy area X_T and the...

conjecture | open | MathDB 页面 | 原文线索

V2-076954 | MATHDB-397979

Equality of rough-path and lattice-path reachability sets

摘录 / EXCERPT: Let E) be a finite-dimensional vector space, let $\xi\in E$, let d be a positive integer and $T>0$, and let $A_1,\ldots,A_d$ be γ-Lipschitz vector fields on E with...

open_problem | open | MathDB 页面 | 原文线索

V2-076955 | MATHDB-397980

Quantifying the variation of a lattice-path realization

摘录 / EXCERPT: Quantify the p-variation of the lattice path which gives the same terminal value of the solution to (1) driven by a given geometric p-rough path.

open_problem | open | MathDB 页面 | 原文线索

V2-076956 | MATHDB-397982

Spatial rough-path construction for the nonlinear fractional path-sampling SPDE

摘录 / EXCERPT: Construct a suitable spatial rough path associated with the linear problem underlying the resulting fractional SPDE, as needed to treat the nonlinear case.

open_problem | open | MathDB 页面 | 原文线索

V2-076957 | MATHDB-397983

Optimal approximation rate for the fractional Lévy area

摘录 / EXCERPT: Let $(B_t)_{t\ge 0}$ be a two-dimensional fractional Brownian motion with Hurst parameter $H\in(1/4,1)$, and let X_T denote its fractional Lévy area and...

open_problem | open | MathDB 页面 | 原文线索

V2-076958 | MATHDB-397985

Knot-theoretical interpretation of the genus-zero quotient condition

摘录 / EXCERPT: What is the knot-theoretical interpretation of the condition that the quotient curve C/ι has genus $g_\iota=0$, where $\iota:(m,l)\mapsto(1/m,1/l)$ is the involution describe...

open_problem | open | MathDB 页面 | 原文线索

V2-076959 | MATHDB-397991

Connectivity of triangulations of 3-dimensional polytopes by 2-3 moves

摘录 / EXCERPT: Is it true that any two triangulations of a 3-dimensional polytope are related by a sequence of 2-3 moves?

open_problem | open | MathDB 页面 | 原文线索

V2-076960 | MATHDB-397994

Asymptotic Minimum Dilatation Problem

摘录 / EXCERPT: Let $\mathcal P=\bigcup_S\mathcal P(S)$ be the set of pairs (S,φ), where $S=S_{g,n}$ is an oriented surface of finite type and φ is a pseudo-Anosov mapping class on...

open_problem | open | MathDB 页面 | 原文线索

V2-076961 | MATHDB-397995

Minimum Dilatation Problem I

摘录 / EXCERPT: What is the behavior of $\delta(S_{g,n})$ as a function of g and n?

open_problem | open | MathDB 页面 | 原文线索

V2-076962 | MATHDB-397996

Minimum Dilatation Problem II

摘录 / EXCERPT: What do small dilatation mapping classes look like?

open_problem | open | MathDB 页面 | 原文线索

V2-076963 | MATHDB-397998

Lower bound for sums of roots of unity

摘录 / EXCERPT: For a fixed positive integer m, there exists a constant $E=E(m)$ such that, as $d\to\infty$, $\min_0\bigl((C_d)^{\# m}\bigr)>d^{-E}$. Here...

open_problem | open | MathDB 页面 | 原文线索

V2-076964 | MATHDB-397999

Approximation conjecture for first Betti numbers in arbitrary characteristic

摘录 / EXCERPT: Let G be finitely presented. For every inverse system $(G_i)_{i\in I}$ of normal subgroups of G, directed by inclusion, with $[G:G_i]<\infty$ for all $i\in I$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-076965 | MATHDB-398003

Nahm’s conjecture

摘录 / EXCERPT: Let $r\ge 1$ and let A be a real positive-definite symmetric $r\times r$ matrix. For the Nahm sum ... with rational data A,B,C, Nahm’s conjecture asserts that, for a given A, t...

conjecture | open | MathDB 页面 | 原文线索

V2-076966 | MATHDB-398004

Critical values in the genus-four Hassett-Keel program

摘录 / EXCERPT: In the remaining interval $\alpha\in\left(\frac{5}{9},\frac{2}{3}\right)$ of the genus-four Hassett-Keel program, there are exactly two further critical values, namely...

open_problem | open | MathDB 页面 | 原文线索

V2-076967 | MATHDB-398005

Ball quotient model and the GIT quotient for genus-four canonical curves

摘录 / EXCERPT: The period map $M^{\mathrm{GIT}}_4 \dashrightarrow (B^9/\Gamma)^*$ is resolved by blowing up a single point, which corresponds to a genus-four ribbon curve in $\mathbb{P}^3$. Moreo...

open_problem | open | MathDB 页面 | 原文线索

V2-076968 | MATHDB-398009

Bound on common zeroes of real-normalized differentials

摘录 / EXCERPT: There exists an explicit function $N(g,d)$ such that, for any g sufficiently large with respect to d, and for any smooth genus-g curve with d marked points...

open_problem | open | MathDB 页面 | 原文线索

V2-076969 | MATHDB-398013

Geometric realization of the exceptional E_7 Calabi-Yau-type variation

摘录 / EXCERPT: One of the main open questions related to Theorem 2 is the geometric realization of VHS of Calabi-Yau threefold type over the exceptional domain of type E_7 given by the theorem.

open_problem | open | MathDB 页面 | 原文线索

V2-076970 | MATHDB-398015

Connected components of strata of the Hodge bundle

摘录 / EXCERPT: For the strata ... apart from hyperelliptic strata, the strata are connected except when all m_i are even. In that case there are two connected components, distinguished by the par...

open_problem | open | MathDB 页面 | 原文线索

V2-076971 | MATHDB-398016

Classification of connected components of quadratic-differential strata in genera 3 and 4

摘录 / EXCERPT: Besides the hyperelliptic strata, the strata $Q_g(d_1,\ldots,d_k)$ are connected except for -$Q_3(9,-1)$, $Q_3(6,3,-1)$, and $Q_3(3,3,3,-1)$; and -$Q_4(12)$, $Q_4(9,3)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-076972 | MATHDB-398020

Gibbs states of the non-critical planar Potts model

摘录 / EXCERPT: For any integer $q\ge 2$ and any inverse temperature $\beta>\beta_c(q)$, the space of infinite-volume Gibbs measures for the nearest-neighbor ferromagnetic q-state Potts model on...

open_problem | open | MathDB 页面 | 原文线索

V2-076973 | MATHDB-398021

Conformal invariance of spin correlations in the planar Ising model

摘录 / EXCERPT: Let $\Omega\subset\mathbb C$ be a bounded multiply connected domain with two marked points a,b on the outer boundary component $\gamma_0\subset\partial\Omega$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-076974 | MATHDB-398026

Lipschitz embedding of independent Bernoulli sequences

摘录 / EXCERPT: Let $(X_i)_{i\in\mathbb Z}$ and $(Y_i)_{i\in\mathbb Z}$ be independent sequences of independent identically distributed $\operatorname{Ber}(1/2)$ random variables. Is it true that,...

open_problem | open | MathDB 页面 | 原文线索

V2-076975 | MATHDB-398028

Chemical distance on random interlacements

摘录 / EXCERPT: For every $u>0$ and $d\ge 3$, let $\mathbb P_u^0=\mathbb P[\,\cdot\mid 0\in I_u]$ and, for $x\in I_u$ and $n\ge 0$, let $\Lambda_u(x,n)=\{y\in I_u:\rho_u(x,y)\le n\}$. Then there e...

open_problem | open | MathDB 页面 | 原文线索

V2-076976 | MATHDB-398030

Stability of the greedy server on the circle

摘录 / EXCERPT: Consider the greedy single-server system on the unit-length circle $\mathbb{R}/\mathbb{Z}$: customers arrive according to a Poisson process with rate λ, each chooses a unif...

open_problem | open | MathDB 页面 | 原文线索

V2-076977 | MATHDB-398031

Scaling limits for directed polymers under slower-than-critical environment scaling

摘录 / EXCERPT: For environment scalings slower than $n^{-1/4}$-equivalently, for $\beta_n=\beta n^{-\\alpha}$ with $0\leq \\alpha<1/4$-does the rescaled environment-path pair $(\\omega_n,S_n)$ ha...

open_problem | open | MathDB 页面 | 原文线索

V2-076978 | MATHDB-398033

Minkowski content and natural parametrization for SLE

摘录 / EXCERPT: For an SLE curve with $\\kappa<8\$, write $\d=1+\kappa/8\$. For a curve piece $\\gamma_{s,t}=\gamma[s,t]\$, the conjectured natural length is, up to a multiplicative constant, the...

open_problem | open | MathDB 页面 | 原文线索

V2-076979 | MATHDB-398034

Unsolvability of the isomorphism problem for finitely presented subgroups of mapping class groups

摘录 / EXCERPT: Let S be a compact surface of genus $g>1$. If the genus of S is sufficiently large, then the isomorphism problem for the finitely presented subgroups of $\operatorname{Mod}(S)$...

open_problem | open | MathDB 页面 | 原文线索

V2-076980 | MATHDB-398035

Embedding direct products of Rips output groups in mapping class groups

摘录 / EXCERPT: Find a version of the Rips construction so that direct products of the output groups H can be embedded in mapping class groups. The constructions given above will then prove Theo...

open_problem | open | MathDB 页面 | 原文线索

V2-076981 | MATHDB-398036

Embedding fundamental groups of compact NPC cube complexes into RAAGs

摘录 / EXCERPT: Does every compact non-positively curved cube complex X have fundamental group $\\\pi_1X$ that embeds in some right-angled Artin group A_Γ?

open_problem | open | MathDB 页面 | 原文线索

V2-076982 | MATHDB-398040

Oblomkov-Shende conjecture for colored PT spaces

摘录 / EXCERPT: Conjecture 4 (Oblomkov-Shende). If $L_{C,(0,0)}$ is a link of the singularity of C at $(0,0)$, then

conjecture | open | MathDB 页面 | 原文线索

V2-076983 | MATHDB-398042

Symmetry of the compactified-Jacobian polynomial $c_{m,n}(q,t)$

摘录 / EXCERPT: For the plane curve singularity with one Puiseux pair (m,n), let J_C be its compactified Jacobian, and let the affine-cell decomposition of J_C have cells $\Sigma(D)$ indexed...

open_problem | open | MathDB 页面 | 原文线索

V2-076984 | MATHDB-398049

Faber’s conjecture on free divisors with reduced Jacobian ideal

摘录 / EXCERPT: Let $D=\{h=0\}\subset(\mathbb{C}^n,0)$ be a reduced complex analytic hypersurface germ. If D is free and the ideal $J_h\subset\mathcal{O}_S$ generated by the partial derivatives...

conjecture | open | MathDB 页面 | 原文线索

V2-076985 | MATHDB-398056

Classification of finite simple racks of type D

摘录 / EXCERPT: Classify all finite simple racks of type D.

open_problem | open | MathDB 页面 | 原文线索

V2-076986 | MATHDB-398064

Reduction of the real Milnor-fibre Euler characteristic to the one-dimensional target case

摘录 / EXCERPT: Let $f=(f_1,\dots,f_k):(X,0)\to(\mathbb{R}^k,0)$ be a real analytic mapping between real analytic space germs with an isolated singularity, with $\dim X=n$. For $k\ge 2$, set...

open_problem | open | MathDB 页面 | 原文线索

V2-076987 | MATHDB-398070

Structural theorem for irreducible euclidean Artin groups

摘录 / EXCERPT: Every irreducible euclidean Artin group is a torsion-free, centerless group with a solvable word problem and a finite-dimensional classifying space.

open_problem | open | MathDB 页面 | 原文线索

V2-076988 | MATHDB-398071

The classical $K(\c0,1)$ conjecture for irreducible euclidean Artin groups

摘录 / EXCERPT: Does the classical $K(\c0,1)$ conjecture hold for every irreducible euclidean Artin group?

conjecture | open | MathDB 页面 | 原文线索

V2-076989 | MATHDB-398073

Formality criterion for complements of subspace arrangements

摘录 / EXCERPT: Given a subspace arrangement, find a necessary and sufficient condition for the formality of its complement.

open_problem | open | MathDB 页面 | 原文线索

V2-076990 | MATHDB-398077

Problem 7: Questions about tropicalization and arrangements

摘录 / EXCERPT: Given an arrangement $A=\{H_0,\ldots,H_n\}$ of hyperplanes in complex projective space $\mathbb{P}^{\ell}$, let $\mathcal{S}=\mathcal{S}(A)$ be the arrangement of subspaces of...

open_problem | open | MathDB 页面 | 原文线索

V2-076991 | MATHDB-398078

Multidimensional polynomial Szemerédi conjecture for jointly intersective polynomials

摘录 / EXCERPT: Conjecture. Let $p_1,\ldots,p_r:\mathbb{Z}^m\to\mathbb{Z}^k$ be polynomials. Then the following assertions are equivalent:

conjecture | open | MathDB 页面 | 原文线索

V2-076992 | MATHDB-398080

Necessity of thickening for nontrivial triangular configurations

摘录 / EXCERPT: For nontrivial triangular configurations in the plane, it is still unknown whether the thickening of the set E is necessary.

open_problem | open | MathDB 页面 | 原文线索

V2-076993 | MATHDB-398083

Haar-measure closure and piecewise syndeticity of sumsets

摘录 / EXCERPT: Determine whether the following implication holds: If $A \subset \mathbb{Z}$, $\widetilde{A}$ denotes the closure of A in $b\mathbb{Z}$, $m_{b\mathbb{Z}}(\widetilde{A})>0$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-076994 | MATHDB-398085

ROM approximation error and the ROM Jacobian

摘录 / EXCERPT: For a special class of H^2 (Hardy-space)-optimal approximations, the reduced-order-model approximation error Y_n-Y lies in the null space of the ROM Jacobian.

open_problem | open | MathDB 页面 | 原文线索

V2-076995 | MATHDB-398086

Choice of a proper regularization space

摘录 / EXCERPT: The question about the proper choice of a regularization space is, in general, open until now.

open_problem | open | MathDB 页面 | 原文线索

V2-076996 | MATHDB-398087

Necessity of compactness for URE oracle inequalities in inverse problems

摘录 / EXCERPT: The report asks whether the compactness assumption on the linear operator in the noisy inverse problem is genuinely necessary, or merely convenient. It states that compactness is n...

open_problem | open | MathDB 页面 | 原文线索

V2-076997 | MATHDB-398091

Higher-rank strong orbit equivalence

摘录 / EXCERPT: Is there a higher-rank version of the strong orbit equivalence relation?

open_problem | open | MathDB 页面 | 原文线索

V2-076998 | MATHDB-398092

K-theory isomorphism for crossed products associated with a basis of projections

摘录 / EXCERPT: Let G be a countable group acting on a totally disconnected space $\a9$, and let $\{p_i:i\in I\}$ be a G-invariant linear basis of $C_c^\infty(\Omega)$ consisting of projection...

open_problem | open | MathDB 页面 | 原文线索

V2-076999 | MATHDB-398116

Decomposition rank of $C(X)\otimes\mathcal Z$

摘录 / EXCERPT: The decomposition rank of $C(X,\mathcal Z)$ is at most 2, independent of X.

open_problem | open | MathDB 页面 | 原文线索

V2-077000 | MATHDB-398123

Lifting tame morphisms of metrized complexes

摘录 / EXCERPT: Let (X,D,V) be a triangulated punctured K-curve with skeleton Σ. Let $f:\Sigma'\to\Sigma$ be a finite morphism of metrized complexes of curves such that, for every verte...

open_problem | open | MathDB 页面 | 原文线索

V2-077001 | MATHDB-398124

Surjectivity of component-group maps for finite morphisms of curves

摘录 / EXCERPT: Let K be a complete field equipped with a discrete valuation normalized so that $\operatorname{val}(K^\times)=\mathbb{Z}$. Let $f:X'\to X$ be a finite morphism of smooth, proper,...

open_problem | open | MathDB 页面 | 原文线索

V2-077002 | MATHDB-398125

Consistency of the types p_L in the comparison of stable completions and Berkovich analytifications

摘录 / EXCERPT: Show that, for any $h_1,\ldots,h_k\in H_d(L)$, there exists x such that ... so that the type p_L defined by these formulas is consistent.

open_problem | open | MathDB 页面 | 原文线索

V2-077003 | MATHDB-398127

Other natural toroidal compactifications with the Lemma 32 property

摘录 / EXCERPT: Find other “natural” toroidal compactifications satisfying the following property: the map ... is a closed embedding and the images of distinct cones span distinct linear spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-077004 | MATHDB-398128

The Real Tau Conjecture

摘录 / EXCERPT: The Real Tau Conjecture claims that the number of real zeros of a polynomial given by a depth-four circuit is bounded by a polynomial in the size of the circuit.

conjecture | open | MathDB 页面 | 原文线索

V2-077005 | MATHDB-398131

Lower bounds for linear-programming relaxations of constraint satisfaction problems

摘录 / EXCERPT: For some constraint satisfaction problem, rule out that there exists a linear program of size polynomial in n that achieves an approximation factor of 0.999 for all instances o...

open_problem | open | MathDB 页面 | 原文线索

V2-077006 | MATHDB-398135

Does the s-rank exponent imply a bound on the matrix-multiplication exponent?

摘录 / EXCERPT: With $\omega_s=\inf\{\tau:R_s(\langle n,n,n\rangle)\le O(n^\tau)\}$, does an upper bound on ω_s imply anything about ω? The report answers yes:...

open_problem | open | MathDB 页面 | 原文线索

V2-077007 | MATHDB-398136

Consistency-error estimate for the practical DPG method

摘录 / EXCERPT: Demonstrate that the consistency error in estimate (8) is of order $\omega(\omega h)^r$, where the exponent r is higher than the polynomial order used for traces and fluxes.

open_problem | open | MathDB 页面 | 原文线索

V2-077008 | MATHDB-398138

Choosing the absorption shift for shifted-Laplacian preconditioning

摘录 / EXCERPT: Determine, in general and with rigorous justification, how the absorption parameter $\varepsilon>0$ should be chosen so that the shifted-Laplacian discretization A_ε gi...

open_problem | open | MathDB 页面 | 原文线索

V2-077009 | MATHDB-398139

A coercive variational formulation for the interior impedance Helmholtz problem

摘录 / EXCERPT: Let Ω be a Lipschitz domain of diameter L that is star-shaped with respect to a ball: there exists $\gamma>0$ such that $x\cdot n(x)\geq \gamma L$ wherever the outward nor...

open_problem | open | MathDB 页面 | 原文线索

V2-077010 | MATHDB-398140

Extension of jackknife asymptotics to censored regression quantiles

摘录 / EXCERPT: Can the asymptotic justification outlined for the usual regression quantiles be extended to the censored regression quantiles of Portnoy (2003), using the inductive argument of Por...

open_problem | open | MathDB 页面 | 原文线索

V2-077011 | MATHDB-398142

Parametric shape optimization for multiphase linear elasticity

摘录 / EXCERPT: For a tuple of admissible deformations $\varphi=(\varphi_1,\ldots,\varphi_K)\in U_{\mathrm{ad}}$, the report associates the weak linear-elasticity state problem (P_φ): find...

open_problem | open | MathDB 页面 | 原文线索

V2-077012 | MATHDB-398143

Existence of an optimal two-phase conductor distribution

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a bounded design region, let $0<m<|\Omega|$, and let two conductivities satisfy $0<\alpha<\beta$. For measurable $B\subset\Omega$ with $|B|=m$, s...

open_problem | open | MathDB 页面 | 原文线索

V2-077013 | MATHDB-398144

Explicit optimizer for the first-order low-contrast functional in a ball

摘录 / EXCERPT: Let $\Omega=B(0,1)\subset\mathbb{R}^d$, with $d\in\{2,3\}$, and let $\mathcal{B}=\{B\subset\Omega:B\text{ measurable},\ |B|=m\}$. Let v_0 be the positive, L^2-normalized soluti...

open_problem | open | MathDB 页面 | 原文线索

V2-077014 | MATHDB-398145

Failure of centered-disk optimality for a two-phase conductor in a ball

摘录 / EXCERPT: When $\Omega=B(0,1)$, for $\beta=\alpha+\varepsilon$ sufficiently close to α, and for a prescribed amount $m>\omega_d(r_{0d})^d$, the distribution in which the material with...

open_problem | open | MathDB 页面 | 原文线索

V2-077015 | MATHDB-398146

Existence of an optimal boundary control for the chemotaxis system

摘录 / EXCERPT: Under the report's preceding hypotheses on the chemotaxis system- in particular, $f\in C^1(\mathbb R)$ is globally bounded and globally Lipschitz, and the state (u,v) solves ... wi...

open_problem | open | MathDB 页面 | 原文线索

V2-077016 | MATHDB-398147

Existence of a minimizer for matrix-parameter identification

摘录 / EXCERPT: Let ced0b8ced0 be an open bounded domain, let ced0 be fixed, let ced0 be either ced0 or ced0, and define ... where ... For each $A\in M$, let $T(A,g)$ be the unique weak solution of...

open_problem | open | MathDB 页面 | 原文线索

V2-077017 | MATHDB-398148

Isotropic constant conjecture (hyperplane conjecture)

摘录 / EXCERPT: Does there exist an absolute constant $C>0$ such that, for every dimension $n\\ge 1$ and every isotropic convex body $K\\subset\\mathbb{R}^n$ of volume 1, with ... for every...

conjecture | open | MathDB 页面 | 原文线索

V2-077018 | MATHDB-398149

Variance conjecture for hyperplane projections of the cross-polytope

摘录 / EXCERPT: There exists an absolute constant C such that for every hyperplane $H\subset\mathbb{R}^n$, if X is uniformly distributed on the orthogonal projection $P_H B_1^n$ of the unit ba...

conjecture | open | MathDB 页面 | 原文线索

V2-077019 | MATHDB-398152

Convergence in shape of iterated Steiner symmetrals without assumptions on directions

摘录 / EXCERPT: Do iterated Steiner symmetrals $S_{u_m}\cdots S_{u_1}K$ always converge in shape, without any assumptions on the sequence of directions?

open_problem | open | MathDB 页面 | 原文线索

V2-077020 | MATHDB-398153

Steiner symmetrals of compact sets under sequences that ball-symmetrize convex bodies

摘录 / EXCERPT: Assume that a sequence of directions (u_m) is such that $(\mathcal{S}_{u_m}\dots\mathcal{S}_{u_1}C)$ converges to C^* for each convex body C. Is it true that...

open_problem | open | MathDB 页面 | 原文线索

V2-077021 | MATHDB-398154

Logarithmic Minkowski inequality

摘录 / EXCERPT: Conjecture 1 (Logarithmic Minkowski inequality, BLYZ [2]). If K and C are o-symmetric convex bodies with $V(K)=V(C)$, then ... with equality if and only if K and C have dil...

open_problem | open | MathDB 页面 | 原文线索

V2-077022 | MATHDB-398155

The (B)-conjecture for Gaussian measures of dilates of symmetric convex sets

摘录 / EXCERPT: For a symmetric convex set $K\subset\mathbb{R}^d$ and standard Gaussian measure γ, the function $t\mapsto\gamma(e^tK)$ is log-concave on $\mathbb{R}$ (equivalently, the Gaus...

conjecture | open | MathDB 页面 | 原文线索

V2-077023 | MATHDB-398156

The Busemann-Petty problem in the complex hyperbolic space

摘录 / EXCERPT: Let $H_{\c9}^n$ be complex hyperbolic n-space. Identify $\d53d^n$ with $\d53d^{2n}$ as in the report, and let H_ξ be the complex hyperplane through the origin perpendicular t...

open_problem | open | MathDB 页面 | 原文线索

V2-077024 | MATHDB-398157

The Busemann-Petty problem for central sections of origin-symmetric convex bodies

摘录 / EXCERPT: Given two origin-symmetric convex bodies K and L in $ ... for every hyperplane H in $\mathbb{R}^n$ containing the origin, does it follow that ... The answer is affirmative for $nl...

open_problem | open | MathDB 页面 | 原文线索

V2-077025 | MATHDB-398158

Eventual concavity of the parallel volume

摘录 / EXCERPT: For a bounded measurable set $A\subset\mathbb{R}^n$, is the function $t\mapsto |A+tB_2^n|^{1/n}$ concave for all sufficiently large t?

open_problem | open | MathDB 页面 | 原文线索

V2-077026 | MATHDB-398159

The hyperplane conjecture

摘录 / EXCERPT: For each $n\ge 1$, let $L_n:=\max\{L_K:K\subset\mathbb R^n\text{ is an isotropic convex body}\}$, where an isotropic convex body K has volume 1, barycenter at the origin, and s...

conjecture | open | MathDB 页面 | 原文线索

V2-077027 | MATHDB-398160

Representation of the Steiner functional as a generalized Wills functional

摘录 / EXCERPT: For convex bodies $K,E\in\mathcal K^n$ with $0\in\operatorname{relint}E$, can the Steiner functional ... be obtained as a particular case of the generalized Wills functional? The rep...

open_problem | open | MathDB 页面 | 原文线索

V2-077028 | MATHDB-398161

Line segments in all directions on the boundary of a convex body

摘录 / EXCERPT: Can the boundary of a d-dimensional convex body contain line segments in all directions?

open_problem | open | MathDB 页面 | 原文线索

V2-077029 | MATHDB-398162

Kannan-Lovász-Simonovits empirical covariance question

摘录 / EXCERPT: Let K be a convex body in $\d53d^n$. Given $\d700>0$, how many independent points $X_1,\d... ,X_N$ uniformly distributed on K are needed so that the empirical covariance matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-077030 | MATHDB-398163

Classification of SL(n)-covariant continuous Minkowski valuations

摘录 / EXCERPT: Conjecture 3. Let $n \ge 3$. A map $\mu: K_n^o \to K_n$ is an $\mathrm{SL}(n)$-covariant continuous Minkowski valuation if and only if there exist constants $c_1,\ldots,c_4 \ge 0$...

open_problem | open | MathDB 页面 | 原文线索

V2-077031 | MATHDB-398164

Universal upper bound for the slicing constant

摘录 / EXCERPT: Does there exist a universal constant $C>0$ such that, for every dimension d and every d-dimensional convex body K, $L_K\le C$, where the report defines L_K by...

open_problem | open | MathDB 页面 | 原文线索

V2-077032 | MATHDB-398165

Many empty triangles have a common edge: Poisson lower bound

摘录 / EXCERPT: Let η_t be a homogeneous Poisson point process in $\mathbb{R}^2$ with intensity $t>0$, let $W\subset\mathbb{R}^2$ be a convex body, and set $X=\eta_t\cap W$. For a pair...

open_problem | open | MathDB 页面 | 原文线索

V2-077033 | MATHDB-398167

Existence of a uniformly subgaussian direction in every centered convex body

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a centered convex body. Does there exist a direction $\theta\in S^{n-1}$ such that ... with a universal constant $r=O(1)$, independent of K and n? H...

open_problem | open | MathDB 页面 | 原文线索

V2-077034 | MATHDB-398168

Breathers for the constant-coefficient nonlinear wave equation

摘录 / EXCERPT: For the constant-coefficient specialization of ... determine whether there exist nonzero solutions that are time-periodic and spatially localized (breathers), where s and q are c...

open_problem | open | MathDB 页面 | 原文线索

V2-077035 | MATHDB-398169

Existence of breathers for a nonlinear wave equation with periodic coefficients

摘录 / EXCERPT: For real-valued, 1-periodic, even coefficients s,q bounded below by a positive constant and $\b3\in\mathbb{R}$, find time-periodic and spatially localized solutions (breathers) o...

open_problem | open | MathDB 页面 | 原文线索

V2-077036 | MATHDB-398170

Hypothesis 1 for the radial Swift-Hohenberg equation

摘录 / EXCERPT: Does the equation ... have a bounded nontrivial solution $A(s)=q(s)$ on $[0,\infty)$, such that the linearization of this equation about $q(s)$ has no nontrivial solution that is bou...

conjecture | open | MathDB 页面 | 原文线索

V2-077037 | MATHDB-398172

Planar fronts and anisotropic lattice effects for general initial data

摘录 / EXCERPT: What is the role of planar fronts and anisotropic effects of the lattice in the evolution of general initial data far from a planar front?

open_problem | open | MathDB 页面 | 原文线索

V2-077038 | MATHDB-398174

Butterfly-type double-homoclinic organizing center for heterogeneity-induced pulse generators

摘录 / EXCERPT: It is conjectured that the hidden organizing center of the pulse-generator branches is a double-homoclinic point of butterfly type at the unstable stationary pattern $SP2$.

open_problem | open | MathDB 页面 | 原文线索

V2-077039 | MATHDB-398175

Existence of connected quotient groups in the simple setting

摘录 / EXCERPT: Do connected quotient groups actually occur in the simple setting?

open_problem | open | MathDB 页面 | 原文线索

V2-077040 | MATHDB-398176

NTP2 preservation for contractive σ-henselian valued difference fields

摘录 / EXCERPT: Let $K=(K,\Gamma,k,\sigma)$ be a contractive σ-henselian valued difference field of residue characteristic 0, with residue field k and value group Γ. Assume that...

open_problem | open | MathDB 页面 | 原文线索

V2-077041 | MATHDB-398177

Borel non-smoothness of Lascar strong types

摘录 / EXCERPT: Suppose T is a complete countable first-order theory, α a countable ordinal, and Y is a G_δ subset of C^α which is closed under $\equiv^\alpha_L$. If for s...

open_problem | open | MathDB 页面 | 原文线索

V2-077042 | MATHDB-398179

Uniform VC*-density bounds from the one-variable case

摘录 / EXCERPT: Assume T has NIP. Assume there is k such that for all formulas $\varphi(x;y)$ with $\ell(x)=1$ we have $\operatorname{vc}^{*}(\varphi)\leq k$. Is it true that for all n there...

open_problem | open | MathDB 页面 | 原文线索

V2-077043 | MATHDB-398180

Definable (k,k)-theorem for NIP theories

摘录 / EXCERPT: Assume T has NIP. Let $\theta(y)$ and $\formulax$ be formulas, and let $d\in\boldsymbol{N}$ satisfy $d>\vc^*(\formulax)$. Suppose that the family...

open_problem | open | MathDB 页面 | 原文线索

V2-077044 | MATHDB-398181

Pillay’s automorphism-group question for NIP formulas

摘录 / EXCERPT: Given a saturated model M of a NIP theory and a formula $\cvarphi(x,y)$, does there exist a homomorphism $h:\cmathrm{Aut}(M)\cto G$ into a nearly-Lie group G, whose kernel fixe...

open_problem | open | MathDB 页面 | 原文线索

V2-077045 | MATHDB-398182

The noncommutative Freiman problem / structure of approximate groups

摘录 / EXCERPT: For a fixed parameter $K\ge 1$, classify finite subsets A of an ambient group G satisfying $|AA|\le K|A|$, up to rough equivalence. The report states the solution in the equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-077046 | MATHDB-398183

Embedding a large subset of an approximate subgroup in a global group

摘录 / EXCERPT: Any K-approximate subgroup A of a local group has a subset of size at least $|A|/C(K)$ which can be embedded in a global group.

open_problem | open | MathDB 页面 | 原文线索

V2-077047 | MATHDB-398185

Existence of a model companion for the theory of globally valued fields

摘录 / EXCERPT: Does the theory $[U+007F]\mathrm{GVF}$ admit a model companion?

open_problem | open | MathDB 页面 | 原文线索

V2-077048 | MATHDB-398186

Elementary equivalence of free products

摘录 / EXCERPT: Let A_1,B_1,A_2,B_2 be groups. Suppose that A_1 is elementarily equivalent to A_2, and B_1 is elementarily equivalent to B_2. Then $A_1 * B_1$ is elementarily equivalent...

open_problem | open | MathDB 页面 | 原文线索

V2-077049 | MATHDB-398187

Groups definable in two orthogonal sorts: extension question

摘录 / EXCERPT: Is every group G interpretable in the two-sorted structure (Z,R)-equivalently, definable in $(Z,R)^{\rm eq}$-an extension of a group definable in one sort by a group definable...

open_problem | open | MathDB 页面 | 原文线索

V2-077050 | MATHDB-398189

Ampleness of theories of nonabelian torsion-free hyperbolic groups

摘录 / EXCERPT: The elementary theory of any nonabelian torsion-free hyperbolic group is ample.

open_problem | open | MathDB 页面 | 原文线索

V2-077051 | MATHDB-398190

Nonexistence of minimal ordered fields of characteristic zero

摘录 / EXCERPT: There is no minimal ordered field of characteristic zero, where “ordered” means that the field, viewed as a structure, has a definable order on singletons with an infinite chain, r...

open_problem | open | MathDB 页面 | 原文线索

V2-077052 | MATHDB-398194

Interior-or-nowhere-dense property for expansions of the real field

摘录 / EXCERPT: Let $\mathcal R$ be an expansion of the real field $\mathbb R=(\mathbb R,+,\cdot)$ by constructible sets, where each added set is a Boolean combination of open sets. If...

open_problem | open | MathDB 页面 | 原文线索

V2-077053 | MATHDB-398197

Clique Density Theorem

摘录 / EXCERPT: If $r\ge 3$ and $\gamma\in[0,1/2)$, then every graph on n vertices with at least γn^2 edges contains at least ... cliques of size r, where $s\ge 1$ is an integer satisfyi...

open_problem | open | MathDB 页面 | 原文线索

V2-077054 | MATHDB-398199

Linear bound for 6-list-critical graphs on surfaces

摘录 / EXCERPT: If G is a 6-list-critical graph embedded on a surface $\a3$ of genus g, then $|V(G)|=O(g)$.

open_problem | open | MathDB 页面 | 原文线索

V2-077055 | MATHDB-398200

Albertson’s precoloring-extension conjecture for planar graphs

摘录 / EXCERPT: There exists a constant D such that the following holds: if G is a plane graph with a 5-list-assignment L and $X\subseteq V(G)$ satisfies $d(u,v)\ge D$ for all distinct...

conjecture | open | MathDB 页面 | 原文线索

V2-077056 | MATHDB-398201

Exponentially many 5-list-colorings on a fixed surface

摘录 / EXCERPT: Conjecture 2. Let G be a graph embedded in a surface Σ, and let L be a 5-list-assignment for G. If G is L-colorable, then G has at least $2^{c|V(G)|}$ L-colo...

open_problem | open | MathDB 页面 | 原文线索

V2-077057 | MATHDB-398202

Fixed-parameter tractability of directed planar disjoint paths

摘录 / EXCERPT: The k-Disjoint Paths problem on directed planar graphs can be solved in time $f(k)\cdot n^{O(1)}$ for some computable function f.

open_problem | open | MathDB 页面 | 原文线索

V2-077058 | MATHDB-398205

Bounded size of critical multicommodity-flow instances

摘录 / EXCERPT: For fixed integers k,p with $p>1$, all critical instances of the k-commodity flow problem with edge-flow values restricted to rationals with denominator p have bounded size.

open_problem | open | MathDB 页面 | 原文线索

V2-077059 | MATHDB-398207

Heroic sets characterization conjecture

摘录 / EXCERPT: A finite set of graphs is heroic if and only if it contains a clique partition graph, a complete multipartite graph, a forest, and the complement of a forest.

conjecture | open | MathDB 页面 | 原文线索

V2-077060 | MATHDB-398209

Hajós’ conjecture on clique subdivisions

摘录 / EXCERPT: For every graph G, the order $[U+007F]\sigma(G)$ of its largest clique subdivision satisfies $\sigma(G)\ge \chi(G)$, where $\chi(G)$ is the chromatic number.

conjecture | open | MathDB 页面 | 原文线索

V2-077061 | MATHDB-398211

Superlocal strengthening of Reed’s conjecture

摘录 / EXCERPT: Conjecture 6 (Edwards-King). Every graph G satisfies ... where ... $\widetilde N(v)$ denotes the closed neighbourhood of v, and ω is the clique number.

conjecture | open | MathDB 页面 | 原文线索

V2-077062 | MATHDB-398212

Clique-averaged local Reed conjecture

摘录 / EXCERPT: Every graph G satisfies ... where $\mathcal C(G)$ is the set of maximal cliques of G and ...

conjecture | open | MathDB 页面 | 原文线索

V2-077063 | MATHDB-398214

Graham’s question on quasirandom permutations

摘录 / EXCERPT: For a sequence of permutations, if the limiting density of every permutation on at most some fixed number k_0 of elements equals the corresponding random value $1/k!$, must the s...

open_problem | open | MathDB 页面 | 原文线索

V2-077064 | MATHDB-398217

Undecidability of linear inequalities in graph homomorphism densities

摘录 / EXCERPT: The following problem is undecidable. - Instance: A positive integer k, finite graphs $H_1,\ldots,H_k$, and integers $a_1,\ldots,a_k$. - Question: Does the inequality...

open_problem | open | MathDB 页面 | 原文线索

V2-077065 | MATHDB-398218

Maximum number of 5-cycles in triangle-free graphs

摘录 / EXCERPT: For all sufficiently large positive integers $\l$ and every integer a with $0\le a\le 4$, the maximum number of copies of the cycle of length 5 in a triangle-free graph with...

open_problem | open | MathDB 页面 | 原文线索

V2-077066 | MATHDB-398220

Erdős-Sós conjecture (1963)

摘录 / EXCERPT: For every graph G of order n with average degree greater than $k-1$, and every tree T on $k+1$ vertices, $T\subseteq G$.

conjecture | open | MathDB 页面 | 原文线索

V2-077067 | MATHDB-398223

Polynomial stubborn 2-list cover conjecture

摘录 / EXCERPT: Can the set of solutions of every instance of the stubborn problem be covered by $O(n^c)$ instances consisting of lists of size 2?

conjecture | open | MathDB 页面 | 原文线索

V2-077068 | MATHDB-398226

Optimality of the proposed phase synchronization for space-time focusing

摘录 / EXCERPT: Suppose that, on a frequency band $[\omega_1,\omega_2]$, a right singular vector $\hat q_p(\omega)\in\mathbb C^N$ associated with the pth obstacle and a simple singular value...

open_problem | open | MathDB 页面 | 原文线索

V2-077069 | MATHDB-398227

Well-posed variational formulations for acoustic scattering by arbitrary planar screens

摘录 / EXCERPT: Let $\\Gamma\subset\Gamma_R$ be a bounded subset of the planar screen, where $\\Gamma$ is either closed or open. For closed $\\Gamma$, the variational formulations in (6), with...

open_problem | open | MathDB 页面 | 原文线索

V2-077070 | MATHDB-398228

Equality of the SPN solutions for two planar screen descriptions

摘录 / EXCERPT: For the Cantor-screen construction described in the report, the variational-formulation solutions satisfy $u_2=u_3$ for every incident field (1) if and only if $\alpha\ge 4$.

open_problem | open | MathDB 页面 | 原文线索

V2-077071 | MATHDB-398229

Nonexistence of the pseudo-periodic Green function at Wood anomalies

摘录 / EXCERPT: If there exist integers $j,\ell\in\mathbb{Z}$ such that ... then no (α,β)-pseudo-periodic Green function exists, even if the scattering problem for a particular periodic s...

open_problem | open | MathDB 页面 | 原文线索

V2-077072 | MATHDB-398232

Tractability of numerical integration for infinitely differentiable functions

摘录 / EXCERPT: Consider the class $F_d=\{f:[0,1]^d\to\mathbb{R}\mid \|D^\alpha f\|_\infty\le 1\text{ for all }\alpha\in\mathbb{N}_0^d\}$ and the numerical integration problem...

open_problem | open | MathDB 页面 | 原文线索

V2-077073 | MATHDB-398233

Kinematic formulas for sets definable by differences of convex functions

摘录 / EXCERPT: We conjecture that kinematic formulas are valid for sets definable by means of functions expressible as differences of convex functions.

open_problem | open | MathDB 页面 | 原文线索

V2-077074 | MATHDB-398234

Limit of anisotropic fractional perimeters as $s\to1^-$

摘录 / EXCERPT: For a bounded Borel set $E\subset\mathbb{R}^n$ of finite perimeter and an origin-symmetric convex body $K\subset\mathbb{R}^n$, the natural conjecture that ... holds in general is fal...

open_problem | open | MathDB 页面 | 原文线索

V2-077075 | MATHDB-398235

Positivity and the inverse Klain map

摘录 / EXCERPT: Assume that K is a centrally symmetric convex body. Then $K\in K(i)$ if and only if $\varphi(K)\ge 0$ for every even, i-homogeneous, continuous, translation-invariant valuation...

open_problem | open | MathDB 页面 | 原文线索

V2-077076 | MATHDB-398237

Finite affine basis of affine invariant points

摘录 / EXCERPT: For $n\ge 2$, let $\mathcal K_n$ be the set of compact convex subsets of $\mathbb R^n$ with nonempty interior, equipped with the Hausdorff metric...

open_problem | open | MathDB 页面 | 原文线索

V2-077077 | MATHDB-398238

Density of convex bodies with all affine invariant points

摘录 / EXCERPT: The set of all $K\in\mathcal K^n$ such that $P_n(K)=\mathbb R^n$ is open and dense in $(\mathcal K^n,d_H)$, where $\mathcal K^n$ is the space of convex bodies in $\mathbb R^n$ equi...

open_problem | open | MathDB 页面 | 原文线索

V2-077078 | MATHDB-398240

Multiplicity interpretation of the BPS integers

摘录 / EXCERPT: Let C be the curve considered in the report, with arithmetic genus $g_{\rm ar}$ and geometric genus $g_{\rm geom}$. For each h with $g_{\rm geom}\le h\le g_{\rm ar}$, let V_h...

open_problem | open | MathDB 页面 | 原文线索

V2-077079 | MATHDB-398241

Projective moduli space for slope-semistable sheaves on smooth projective threefolds

摘录 / EXCERPT: Theorem. Let X be a smooth projective threefold, let $H_1,H_2\in\operatorname{Pic}(X)$ be ample divisors, let $c_i\in H^{2i}(X,\mathbb Z)$ for $i=1,2,3$, let r be a positive in...

open_problem | open | MathDB 页面 | 原文线索

V2-077080 | MATHDB-398249

Conformal-blocks divisors and Veronese quotient morphisms

摘录 / EXCERPT: Every birational morphism from $M_{0,n}$ to one of the described GIT quotients of pointed rational normal curves of fixed degree, with the curves and marked points weighted by nonn...

open_problem | open | MathDB 页面 | 原文线索

V2-077081 | MATHDB-398252

Non-ordinary double-point loci in Andreotti-Mayer divisors

摘录 / EXCERPT: Describe the loci $N_0^{g-1}\\subset N_0$ such that the quadratic tangent cone Q_x has rank $\\leg-1$ for some quadratic singularity $x\\in\\operatorname{Sing}\\Theta$.

open_problem | open | MathDB 页面 | 原文线索

V2-077082 | MATHDB-398256

Existence of an optimal matching measure supported on the target boundary

摘录 / EXCERPT: Let $f^+,f^-\in L^\infty$ be nonnegative compactly supported functions with supports X^+ and X^-, respectively, satisfying ... Let D be a compact target set and let Ω be...

open_problem | open | MathDB 页面 | 原文线索

V2-077083 | MATHDB-398257

Regularity conjecture for the transformed gradient of p-harmonic functions

摘录 / EXCERPT: Conjecture: For a p-harmonic solution u, the quantity $V(\nabla u):=|\nabla u|^{(p-2)/2}\nabla u$ is locally of class C^1, i.e. $V(\nabla u)\in C^1_{\mathrm{loc}}$, in all dime...

conjecture | open | MathDB 页面 | 原文线索

V2-077084 | MATHDB-398259

Equality in the support and maximum-set inclusions for the limiting infinity-ground-state objects

摘录 / EXCERPT: Prove that the inclusions in (iii), namely $\operatorname{spt}(f_\infty)\subset \operatorname{argmax} u_\infty\subset \operatorname{argmax} d_\Omega$, are equalities.

open_problem | open | MathDB 页面 | 原文线索

V2-077085 | MATHDB-398260

Simplicity of the principal \-infinity eigenvalue on convex domains

摘录 / EXCERPT: Let Omega be a bounded open convex subset of mathbb{R}^n. Is the principal infty-eigenvalue lambdainfty(Omega)=1/max{xinOmega}d(x,partialOmega) simple; equivalently, are all positi...

open_problem | open | MathDB 页面 | 原文线索

V2-077086 | MATHDB-398261

Nonuniqueness of infinity ground states on a dumbbell domain

摘录 / EXCERPT: There exists $\delta_0>0$ such that, for every $\delta\u0003leq\delta_0$, the dumbbell domain ... possesses an infinity ground state u_∞ satisfying ... Consequently, u_∞...

open_problem | open | MathDB 页面 | 原文线索

V2-077087 | MATHDB-398262

Characterization of eigensolutions of the 1-Laplace operator

摘录 / EXCERPT: For the variational problem ... with $\Omega\subset\mathbb R^n$ open, bounded, and having Lipschitz boundary, the report asks whether every solution of the single 1-Laplace eigenvalu...

open_problem | open | MathDB 页面 | 原文线索

V2-077088 | MATHDB-398263

Theorem 1: principal eigenvalue for the p-Laplacian with convection

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a bounded domain with smooth boundary, let $1<p<\infty$, and consider ... where $B:\Omega\times\mathbb{R}^N\to\mathbb{R}$ is continuous, homogeneou...

open_problem | open | MathDB 页面 | 原文线索

V2-077089 | MATHDB-398265

Simplicity of the first eigenvalue for a variable exponent

摘录 / EXCERPT: For a smooth variable exponent $p(x)$ satisfying $1<p(x)<\\infty$, in the eigenvalue problem obtained by minimizing the Luxemburg-norm Rayleigh quotient...

open_problem | open | MathDB 页面 | 原文线索

V2-077090 | MATHDB-398268

A reasonable two-phase free-boundary condition for the infinity-Laplacian

摘录 / EXCERPT: Is the limiting two-phase free-boundary condition $u^+_\nu(x)=Q(x)\lambda_2$ a reasonable free-boundary condition for a two-phase problem?

open_problem | open | MathDB 页面 | 原文线索

V2-077091 | MATHDB-398269

Mixed boundary value problem for the infinity Laplacian

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^N$ be a bounded smooth domain, let $\Gamma_D\subset\partial\Omega$, set $\Gamma_N=\partial\Omega\setminus\Gamma_D$, and let $F:\Gamma_D\to\mathbb{R}$ b...

open_problem | open | MathDB 页面 | 原文线索

V2-077092 | MATHDB-398270

Higher-dimensional regularity of the nonlinear gradient field for p-harmonic functions

摘录 / EXCERPT: Conjecture: For every dimension $n\ge 3$, every $p\in(1,\infty)$, and every scalar p-harmonic function u in $\mathbb{R}^n$, the field ... belongs to C^1.

open_problem | open | MathDB 页面 | 原文线索

V2-077093 | MATHDB-398271

Strong comparison principle for p-harmonic functions

摘录 / EXCERPT: Suppose that u_1 and u_2 are two solutions of the p-Laplace equation ... in a domain Ω in n-dimensional space. If $u_2\ge u_1$ in Ω, is it then true that either...

open_problem | open | MathDB 页面 | 原文线索

V2-077094 | MATHDB-398272

Two-dimensional dispersive estimate for the Coriolis linear group

摘录 / EXCERPT: For all $t\ge 0$, $\Omega\in\mathbb{R}$, and $f\in L^1(\mathbb{R}^3)$ satisfying ... there exists a constant C such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-077095 | MATHDB-398276

Finite-time blow-up for the inviscid primitive equations

摘录 / EXCERPT: There exists smooth initial data for which the corresponding smooth solution of equations (6)-(10)-the three-dimensional inviscid primitive equations without Coriolis force, in the...

open_problem | open | MathDB 页面 | 原文线索

V2-077096 | MATHDB-398278

First-order necessary condition for a coupled data-assimilation problem

摘录 / EXCERPT: Theorem 2. Let (psi0^in H^s) be an optimal initial condition of the data-assimilation problem and let (barpsi=(bar ua,bar ha,bar uo,bar ho)) be the associated solution of the coupl...

open_problem | open | MathDB 页面 | 原文线索

V2-077097 | MATHDB-398280

Uniform small-\b1 decay estimates for the two-dimensional Oseen semigroup

摘录 / EXCERPT: Let a9 be a smooth exterior domain in $\^2$, let $\b1>0$, and let $L=L_\b1$ be the Oseen operator on $L^q_\c3(\a9)$. Can the estimates ... and ... with $n=2$, be improved in the two-di...

open_problem | open | MathDB 页面 | 原文线索

V2-077098 | MATHDB-398281

Existence and construction of multivariate compactly supported tight wavelet frames

摘录 / EXCERPT: In the multivariate case, determine the existence of compactly supported tight wavelet frames and develop general efficient methods for constructing them; the question is connected...

open_problem | open | MathDB 页面 | 原文线索

V2-077099 | MATHDB-398282

Estimates for the number of multivariate frame generators

摘录 / EXCERPT: Another open problem is to obtain the estimates for the number of frame generators in the multivariate case. This problem of finding an upper bound for the number of frame generato...

open_problem | open | MathDB 页面 | 原文线索

V2-077100 | MATHDB-398283

Existence results for tight wavelet frames via the Oblique Extension Principle

摘录 / EXCERPT: It is still not known whether analogous existence results hold for the tight wavelet frames constructed using the so-called Oblique Extension Principle in [5, 6].

open_problem | open | MathDB 页面 | 原文线索

V2-077101 | MATHDB-398284

Gaussian Gabor-frame conjecture for the lattice \913\

摘录 / EXCERPT: Let $g(t)=2^{d/4}e^{-\c0 t\b7t/2}$ be the normalized Gaussian, and let $\Lambda_3=A_3(\mathbb Z+i\mathbb Z)^2\subset\mathbb C^2\cong\mathbb R^4$, where ... Then the conjecture is tha...

conjecture | open | MathDB 页面 | 原文线索

V2-077102 | MATHDB-398285

Practical multivariate inversion for truncated moment measures

摘录 / EXCERPT: A practical generalization of the inversion procedure to measures supported in $\mathbb{R}^D$ is not known.

open_problem | open | MathDB 页面 | 原文线索

V2-077103 | MATHDB-398286

Boundary-value descriptions for growth spaces in the unit ball

摘录 / EXCERPT: Give a description of the boundary values of functions in growth spaces in the unit ball. In particular, determine whether there is an appropriate wavelet decomposition (or a more...

open_problem | open | MathDB 页面 | 原文线索

V2-077104 | MATHDB-398289

Detection of sparse principal components via a semidefinite relaxation

摘录 / EXCERPT: The report states the following proved result. Suppose $n=\b5m$ and $k^*=\bn$, where $\b5>0$ and $\b\\(0,1)$ are fixed and n is sufficiently large. Let $\b2_\b1$ be a detection t...

open_problem | open | MathDB 页面 | 原文线索

V2-077105 | MATHDB-398291

A noncommutative relative-entropy variational principle for rational matrix-valued moment solutions

摘录 / EXCERPT: For matrix-valued moment problems of the form $R=\int_S g_{\mathrm{left}}(x)\rho(x)g_{\mathrm{right}}(x)\,dx$, the rational representation of solutions is...

open_problem | open | MathDB 页面 | 原文线索

V2-077106 | MATHDB-398292

An extension problem for wavelet frames

摘录 / EXCERPT: Assume that $\psi_1,\widetilde\psi_1\in L^2(\mathbb{R})$ and that the wavelet systems $\{2^{j/2}\psi_1(2^j x-k)\}_{j,k\in\mathbb{Z}}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077107 | MATHDB-398293

Deguang Han’s wavelet-frame extension conjecture

摘录 / EXCERPT: Let $\{2^{j/2}\psi_1(2^j x-k)\}_{j,k\in\mathbb Z}$ be a wavelet frame for $L^2(\mathbb R)$ with upper frame bound B. Then there exists $D>B$ such that, for each $K\ge D$, there e...

conjecture | open | MathDB 页面 | 原文线索

V2-077108 | MATHDB-398294

Factorization of nonnegative matrix polynomials on the 2D torus

摘录 / EXCERPT: Let $M=M(\omega_1,\omega_2)$ be a symmetric trigonometric matrix polynomial in two variables, namely, a symmetric $m\times m$ matrix whose coefficients are trigonometric polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-077109 | MATHDB-398296

Schauder bases of translates and A_p-weights

摘录 / EXCERPT: Let $\psi \in L^2(\mathbb{R})$ and $B_\psi:=\{\psi_k(x):=\psi(x-k)\mid k\in\mathbb{Z}\}$. Define $\langle\psi\rangle:=\operatorname{span}B_\psi$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077110 | MATHDB-398297

Explicit closed formulas for the radially deformed Fourier-transform kernel

摘录 / EXCERPT: Find explicit closed formulas for the integral kernel $K_a(x,y)$ of the radially deformed Fourier transform F_a for arbitrary real parameter $a>0$, beyond the cases where closed...

open_problem | open | MathDB 页面 | 原文线索

V2-077111 | MATHDB-398299

Regularity of weak direct products in the non-bicontinuous cases

摘录 / EXCERPT: Let $(G_j)_{j\in J}$ be a family of non-discrete Lie groups and let ... Assuming the factors have the relevant C^k-regularity, is G regular in the cases where the inverse of the...

open_problem | open | MathDB 页面 | 原文线索

V2-077112 | MATHDB-398300

Conjecture that the diffeomorphism group is L^∞-regular

摘录 / EXCERPT: Conjecture: $\operatorname{Diff}(M)$ is L^∞-regular.

conjecture | open | MathDB 页面 | 原文线索

V2-077113 | MATHDB-398301

Periodic MsFEM error estimate

摘录 / EXCERPT: Under the hypothesis $a^{\varepsilon}(x)=a_1\text{-periodic}(x/\varepsilon)$, the MsFEM solution u_H satisfies ... where C is independent of H and $u^\star$ is the homogenized...

open_problem | open | MathDB 页面 | 原文线索

V2-077114 | MATHDB-398302

Mathematical justification of Boltzmann kinetic limits for weakly nonlinear wave equations

摘录 / EXCERPT: What is the precise mathematical meaning and domain of applicability of approximating weakly nonlinear wave equations by kinetic transport equations of Boltzmann type?

open_problem | open | MathDB 页面 | 原文线索

V2-077115 | MATHDB-398303

Validity and scope of kinetic Boltzmann approximations for weakly nonlinear wave equations

摘录 / EXCERPT: The precise mathematical meaning and domain of applicability of approximating weakly nonlinear wave equations by kinetic transport equations of Boltzmann type remain to be establis...

open_problem | open | MathDB 页面 | 原文线索

V2-077116 | MATHDB-398304

Optimality of the Plateau-type upper bound for the triple-junction map

摘录 / EXCERPT: For the triple-junction map u taking the three vertices of an equilateral triangle as values on three sectors of the source plane meeting at 120° angles, is the upper boun...

open_problem | open | MathDB 页面 | 原文线索

V2-077117 | MATHDB-398305

Nonlocality of relaxed area for qualitatively different discontinuous maps

摘录 / EXCERPT: Can the nonlocality of the relaxed area functional be obtained for maps v that are qualitatively different from the triple junction map u?

open_problem | open | MathDB 页面 | 原文线索

V2-077118 | MATHDB-398306

Existence of viscosity solutions for generalized perimeter flows

摘录 / EXCERPT: Let $P(E)$ be a perimeter on subsets $E\subset\mathbb{R}^d$ satisfying $P(E)\ge 0$, $P(\varnothing)=P(\mathbb{R}^d)=0$, lower semicontinuity under L^1 convergence of characterist...

open_problem | open | MathDB 页面 | 原文线索

V2-077119 | MATHDB-398307

Lower bound on asymptotic support propagation for the thin-film equation

摘录 / EXCERPT: For the thin-film equation ... given $n\in(1.5,3)$ and $x_s\in\operatorname{supp}u_0$, a solution u satisfies, for every $t\ge 0$, ... where ...

open_problem | open | MathDB 页面 | 原文线索

V2-077120 | MATHDB-398309

Existence for a bacteria-temperature-fluid system with environmental and velocity constraints

摘录 / EXCERPT: Theorem. Assume that $\theta_0\in H^2(\Omega)$, $w_0\in H^1(\Omega)^2$, and $v_0\in H^1_{0,\sigma}(\Omega)$ satisfy $\theta_0=\theta_\Gamma$ on Γ, $w_0\in E(\theta_0)$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-077121 | MATHDB-398310

A priori exclusion of overpacking versus formation of an interface in a tissue-growth model

摘录 / EXCERPT: For the tissue-growth equation $\partial_t n=\nabla\cdot(D(n)\nabla n)+kn$, with $n(x,t)\in[0,1]$, $k>0$, $D(n)>0$, and $D(n)\to+\infty$ as $n\to1^-$-in particular,...

open_problem | open | MathDB 页面 | 原文线索

V2-077122 | MATHDB-398311

Necessity of strong stationarity for boundary control of the obstacle problem

摘录 / EXCERPT: Consider the optimal control problem … subject to … where $\Omega\subset\mathbb R^d$, $d=2,3$, is a bounded Lipschitz domain with boundary Γ,…

open_problem | open | MathDB 页面 | 原文线索

V2-077123 | MATHDB-398312

No-growth bound for two oppositely oriented spirals

摘录 / EXCERPT: Assume the hypotheses of Theorem 2: $N=2$, $m_1=-m_2=1$, $a_1=(-\alpha,0)$, $a_2=(\alpha,0)$ with $\alpha\in(0,1/C)$, $W=\Omega\setminus(B_\rho(a_1)\cup B_\rho(a_2))$, and…

open_problem | open | MathDB 页面 | 原文线索

V2-077124 | MATHDB-398313

Well-posedness of the space-time variational formulation for diffusion on an evolving surface

摘录 / EXCERPT: For the evolving-surface problem, let H, $W:=\{v\in H\mid \dot v\in H'\}$, and $\mathring W:=\{v\in W\mid v(\cdot,0)=0\text{ on }\Gamma_0\}$ be as defined in the report. With…

open_problem | open | MathDB 页面 | 原文线索

V2-077125 | MATHDB-398314

Dependence of the confined Willmore minimum on the prescribed area

摘录 / EXCERPT: Let $a>0$, let B be the unit ball in $\\mathbb{R}^{3}$, and let $\mathcal{M}_{a}$ be the class of smoothly embedded surfaces $\Sigma\subset B$ of sphere type with…

open_problem | open | MathDB 页面 | 原文线索

V2-077126 | MATHDB-398315

Inverse problem of determining subboundaries from partial Cauchy data

摘录 / EXCERPT: Let (Omega0subsetmathbb{R}^n), $n=2,3$, be a bounded domain with smooth boundary $\partial\Omega_0$, and let … where $a_{ij}=a_{ji}\in C^1(\Omega_0)$, $c\in L^\infty(\Omega)$, and…

open_problem | open | MathDB 页面 | 原文线索

V2-077127 | MATHDB-398316

Finite permutation-pattern densities determine quasirandomness

摘录 / EXCERPT: Let S_k be the set of permutations of $[k]$. For a sequence of permutations (σ_j) with $|\sigma_j|\to\infty$, let $t(\pi,\sigma_j)$ denote the density of $\pi\in S_k$ in…

open_problem | open | MathDB 页面 | 原文线索

V2-077128 | MATHDB-398317

Are all unimodular measures sofic?

摘录 / EXCERPT: Is every unimodular measure $\mu\in\mathcal R_N$ sofic, where “sofic” means that μ is a Benjamini-Schramm limit of a sequence of finite graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-077129 | MATHDB-398322

Infinite families of nonbipartite Ramanujan graphs in every degree

摘录 / EXCERPT: It is still an open problem to construct an infinite family of d-regular nonbipartite Ramanujan graphs for all d.

open_problem | open | MathDB 页面 | 原文线索

V2-077130 | MATHDB-398323

Nathan Linial’s signing conjecture

摘录 / EXCERPT: For every d-regular graph G, it is always possible to put values ±1 on the edges such that all eigenvalues of the obtained signed matrix lie in the interval…

conjecture | open | MathDB 页面 | 原文线索

V2-077131 | MATHDB-398325

Margulis’ uniform-discreteness conjecture

摘录 / EXCERPT: Margulis conjectured that the family of all cocompact torsion-free arithmetic lattices in G is uniformly discrete.

conjecture | open | MathDB 页面 | 原文线索

V2-077132 | MATHDB-398335

Saturation constant one for simply laced groups

摘录 / EXCERPT: In the tensor-product case, with G diagonally embedded in $G\times G$ and $\Lambda(G):=\Lambda(G,G\times G)$, Conjecture 2 states: If G is simple and simply laced, then 1 is…

open_problem | open | MathDB 页面 | 原文线索

V2-077133 | MATHDB-398341

Partitioning the symmetric group into cosets

摘录 / EXCERPT: A subset X of S_n is called a t-coset if it is a coset of a stabilizer of t points. For example, the set of all permutations that send 1 to itself and 3 to 7 is a 2…

open_problem | open | MathDB 页面 | 原文线索

V2-077134 | MATHDB-398343

NP-completeness of weighted winning-agenda recognition for at least four voters

摘录 / EXCERPT: Let $A=(a_{ij})$ be an $n\times m$ matrix with entries $a_{ij}\in\{-1,1\}$, and let $w_1,\ldots,w_n$ and $z_1,\ldots,z_m$ be positive integer weights. For $Q\subseteq[m]$, call Q…

open_problem | open | MathDB 页面 | 原文线索

V2-077135 | MATHDB-398345

Polynomial upper and lower bounds for the mean-square displacement of self-avoiding walks

摘录 / EXCERPT: In two and three dimensions, prove that there exists $\varepsilon>0$ such that … for all sufficiently large n.

open_problem | open | MathDB 页面 | 原文线索

V2-077136 | MATHDB-398346

Asymptotic threshold for random graph k-colorability

摘录 / EXCERPT: Conjecture: the k-colorability threshold satisfies $d_{k\text{-col}}=2k\ln k-\ln k-1+o_k(1)$.

open_problem | open | MathDB 页面 | 原文线索

V2-077137 | MATHDB-398347

Existence of the limiting normalized number of k-colorings

摘录 / EXCERPT: For every integer $k\ge 3$ and every real $d>0$, the limit ... exists, where $G(n,m)$ is the random-graph model at average degree d as specified in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-077138 | MATHDB-398350

MSO zero-one law for connected graphs embeddable on a fixed surface

摘录 / EXCERPT: For each fixed surface S, let $\cmathcal{G}_S$ be the class of graphs embeddable in S. The conjecture is that the zero-one law for connected graphs in $\cmathcal{G}_S$, proved...

open_problem | open | MathDB 页面 | 原文线索

V2-077139 | MATHDB-398351

Theorem 4: counting labeled G-free graphs with m edges

摘录 / EXCERPT: Let G be a 2-balanced graph and let δ be a positive constant. There exists a constant C such that for every $n aq\in\mathbb{N}$, if $m\ge Cn^{2-1/m_2(G)}$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077140 | MATHDB-398357

Pentagon Problem

摘录 / EXCERPT: Does there exist ℓ with the following property: for any cubic graph G with girth ℓ, there exists a homomorphism $G \to C_5$?

open_problem | open | MathDB 页面 | 原文线索

V2-077141 | MATHDB-398358

Arithmetic progressions in the range of integer polynomials

摘录 / EXCERPT: Let $f(x)$ be a polynomial over the integers having degree at least two. Show that for every such f there is a bound B_f such that the range, $f(\mathbb{Z})$, contains no arith...

open_problem | open | MathDB 页面 | 原文线索

V2-077142 | MATHDB-398359

Erdős-Fajtlowicz conjecture on chromatic number versus clique subdivisions

摘录 / EXCERPT: For every integer $n\ge 2$, let $H(n)$ be the maximum of $\chi(G)/\sigma(G)$ over all graphs G on n vertices, where $\chi(G)$ is the chromatic number and $\sigma(G)$ is the lar...

conjecture | open | MathDB 页面 | 原文线索

V2-077143 | MATHDB-398360

Axial random assignment cost conjecture

摘录 / EXCERPT: The Axial problem has an asymptotic expected cost of c/n.

conjecture | open | MathDB 页面 | 原文线索

V2-077144 | MATHDB-398362

Growth rates of multidimensional random assignment costs

摘录 / EXCERPT: For $D \ge 3$, what are the true growth rates of the expected costs of optimal D-dimensional axial and planar assignments?

open_problem | open | MathDB 页面 | 原文线索

V2-077145 | MATHDB-398363

Tight lower bound for greedy routing in higher-dimensional small-world grids

摘录 / EXCERPT: For the described D-dimensional Kleinberg small-world grid model, greedy routing does not perform asymptotically better for any uniform and isotropic augmenting distribution than...

open_problem | open | MathDB 页面 | 原文线索

V2-077146 | MATHDB-398365

Replica symmetry for triangle upper tails in Erdős-Rényi random graphs

摘录 / EXCERPT: Fix $0<p<r<1$. Let G_n be an instance of the Erdős-Rényi random graph $G(n,p)$ conditioned on the event that it has at least $\binom{n}{3}r^3$ triangles. Is G_n, for large n,...

open_problem | open | MathDB 页面 | 原文线索

V2-077147 | MATHDB-398368

Square-root separator conjecture for string graphs

摘录 / EXCERPT: A string graph is the intersection graph of a collection of curves in the plane. A separator of a graph $G=(V,E)$ is a set $S\subseteq V$ such that every connected component of...

conjecture | open | MathDB 页面 | 原文线索

V2-077148 | MATHDB-398371

Erdős-Hajnal property for string graphs

摘录 / EXCERPT: Does the family of string graphs have the Erdős-Hajnal property; that is, does there exist a constant $c>0$ such that every string graph on n vertices contains a clique or an ind...

open_problem | open | MathDB 页面 | 原文线索

V2-077149 | MATHDB-398372

Homomorphism and chromatic thresholds of clique-free graphs

摘录 / EXCERPT: For every integer $r \ge 3$, the homomorphism and chromatic thresholds of K_r satisfy ...

open_problem | open | MathDB 页面 | 原文线索

V2-077150 | MATHDB-398373

Inference for functionals under partial identification

摘录 / EXCERPT: The properties of the likelihood and other methods for performing inference on functionals of the parameter under partial identification are not yet well understood.

open_problem | open | MathDB 页面 | 原文线索

V2-077151 | MATHDB-398374

Specification tests for partially identified models defined by moment inequalities

摘录 / EXCERPT: For an i.i.d. sample $(W_i)_{i=1}^n$ with distribution F, a finite-dimensional parameter $\theta\in\Theta$, and a known measurable function...

open_problem | open | MathDB 页面 | 原文线索

V2-077152 | MATHDB-398375

Untestability of completeness in a nonparametric IV model

摘录 / EXCERPT: Let ν be a positive σ-finite Borel measure on $\mathbb{R}^{d_x}\times\mathbb{R}^{d_z}$ such that $\nu=\nu_x\times\nu_z$, where ν_x is atomless. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-077153 | MATHDB-398376

Identification of the hedonic-model function via a variational problem

摘录 / EXCERPT: Theorem 1. Under Assumptions 1 and 3, V is identified up to a constant as the solution to the variational problem ... where ...

open_problem | open | MathDB 页面 | 原文线索

V2-077154 | MATHDB-398377

Asymptotic normality theorem for the nonparametric regression-quantile test statistic

摘录 / EXCERPT: Under the preceding setup and assumptions of the report, including $r_\alpha(x)=r_{\alpha,\theta_0(\alpha)}(x)+n^{-1/2}h^{-d/4}\Delta_\alpha(x)$ and $nh^{3d/2}\to\infty$, Theorem 1...

open_problem | open | MathDB 页面 | 原文线索

V2-077155 | MATHDB-398379

Genealogy of the fixed-size Brunet-Derrida particle system

摘录 / EXCERPT: For a branching population with a selection mechanism maintaining a fixed population size and allowing only the fittest (extreme) particles to survive, is the genealogical timescal...

open_problem | open | MathDB 页面 | 原文线索

V2-077156 | MATHDB-398383

Positivity at the critical barrier for consistent maximal displacement

摘录 / EXCERPT: For standard branching Brownian motion started from one particle at the origin, with Brownian motion between binary splittings at rate 1, define ... ... The report asks whether...

open_problem | open | MathDB 页面 | 原文线索

V2-077157 | MATHDB-398384

Most recent common ancestor in the Brunet-Derrida-Mueller-Munier selection model

摘录 / EXCERPT: In the fixed-size selection model of Brunet, Derrida, Mueller, and Munier, where the population has size N, each individual produces $k\ge 2$ offspring whose fitness equals the p...

open_problem | open | MathDB 页面 | 原文线索

V2-077158 | MATHDB-398386

Gromov-Hausdorff scaling limit of the dual tree of a recursive triangulation

摘录 / EXCERPT: Let $\beta=(\sqrt{17}-3)/2$, let T_n be the dual tree of the recursive triangulation at time n, equipped with its graph distance, and let T_M be the compact real tree encoded...

open_problem | open | MathDB 页面 | 原文线索

V2-077159 | MATHDB-398388

Near-critical conditioned limit for branching Brownian motion in a strip

摘录 / EXCERPT: For any finite time horizon, the law of the BBM in $(0,K)$ under $\lim_{K\downarrow K_0} P_x^K(\cdot\mid\text{survival})$ is equal to the law of a particle system consisting of: -...

open_problem | open | MathDB 页面 | 原文线索

V2-077160 | MATHDB-398391

An infinite-dimensional integrable system for the Plateau problem

摘录 / EXCERPT: Does there exist an analog of the dispersionless 2D Toda-hierarchy resolution for the Plateau problem-namely, an infinite-dimensional integrable system that generalizes the Schlesi...

open_problem | open | MathDB 页面 | 原文线索

V2-077161 | MATHDB-398392

Existence and regularity of Willmore minimizers in arbitrary torus conformal classes

摘录 / EXCERPT: For every conformal class of a torus, is the infimum of the Willmore energy attained by a smooth minimizer, without assuming that the infimum in that conformal class is below...

open_problem | open | MathDB 页面 | 原文线索

V2-077162 | MATHDB-398394

Existence of a smooth strictly convex sphere with an umbilic of index $3/2$

摘录 / EXCERPT: There exists a surface $S_E\in\mathcal S$ with an umbilic point $s_E\in S_E$ such that ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-077163 | MATHDB-398397

Isoperimetric domains in a quotient of the universal cover of \(\widetilde{\mathrm{SL}}(2,\mathbb{R})\)

摘录 / EXCERPT: In the three-dimensional-isometry-group case for a left-invariant metric on $\widetilde{\mathrm{SL}}(2,\mathbb{R})$, the quotient surfaces $\phi(\Sigma)/\Delta$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-077164 | MATHDB-398398

Non-holomorphic hyperelliptic minimal surfaces of even genus in flat tori

摘录 / EXCERPT: Are there any non-holomorphic hyperelliptic minimal surfaces of even genus in n-dimensional flat tori for $n>3$?

open_problem | open | MathDB 页面 | 原文线索

V2-077165 | MATHDB-398399

Minimum ropelength for a given knot or link

摘录 / EXCERPT: How much rope is needed to tie a given knot or link, where the rope has fixed circular cross-section and the quantity to minimize is ropelength, the scale-invariant quotient of len...

open_problem | open | MathDB 页面 | 原文线索

V2-077166 | MATHDB-398400

Single multi-crossing projection for every knot

摘录 / EXCERPT: Does every knot possess a projection with just a single multi-crossing? Equivalently, does every knot K have an integer n such that $c_n(K)=1$, where $c_n(K)$ is the minimal nu...

open_problem | open | MathDB 页面 | 原文线索

V2-077167 | MATHDB-398401

Knots realizing the lower bound for triple crossing number

摘录 / EXCERPT: Does there exist a knot K such that $c_3(K)=\frac{c_2(K)}{3}$? Here $c_2(K)$ is the traditional crossing number and $c_3(K)$ is the minimum number of triple crossings in a projec...

open_problem | open | MathDB 页面 | 原文线索

V2-077168 | MATHDB-398402

Behavior of the crossing-number spectrum of a knot

摘录 / EXCERPT: For a knot K, what is the behavior of its spectrum of crossing-number values $\{c_n(K)\}$, where $c_n(K)$ is the minimal number of n-crossings in a projection of K?

open_problem | open | MathDB 页面 | 原文线索

V2-077169 | MATHDB-398403

Global minimum ropelength of non-trivial knots versus 24

摘录 / EXCERPT: Let R_g denote the global minimum ropelength over all non-trivial knots. The question is whether $R_g \le 24$, equivalently, whether one can tie a non-trivial knot with a one-foo...

open_problem | open | MathDB 页面 | 原文线索

V2-077170 | MATHDB-398404

Superlinear ropelength growth versus crossing number

摘录 / EXCERPT: Prove or disprove the existence of infinitely many knots $\{K_n\}$ whose ropelength grows faster than $O(\operatorname{Cr}(K_n))$.

open_problem | open | MathDB 页面 | 原文线索

V2-077171 | MATHDB-398405

Existence of a Gordian Unknot

摘录 / EXCERPT: Does there exist a loop of fixed thickness and length whose core is unknotted, but which cannot be deformed to a round circle by an isotopy fixing its length and thickness?

open_problem | open | MathDB 页面 | 原文线索

V2-077172 | MATHDB-398406

Freedman-He-Wang conjecture on Möbius energy of links

摘录 / EXCERPT: It was conjectured that the Möbius energy is minimized, among all nontrivial links in $\mathbb{R}^3$, by the stereographic projection of the standard Hopf link. The standard Hopf l...

conjecture | open | MathDB 页面 | 原文线索

V2-077173 | MATHDB-398407

Freedman-He-Wang conjecture on Möbius cross energy

摘录 / EXCERPT: Among all nontrivial 2-component links in $\[3m]R^3$, the Möbius cross energy ... is minimized by the stereographic projection of the standard Hopf link.

conjecture | open | MathDB 页面 | 原文线索

V2-077174 | MATHDB-398408

Energy version of the Willmore conjecture

摘录 / EXCERPT: Is it true that if S is an immersed torus in $\mathbb{R}^3$ then $E(S) \ge E(T_{\sqrt{2}})$ and equality holds if and only if S is the Clifford torus up to Möbius transformatio...

conjecture | open | MathDB 页面 | 原文线索

V2-077175 | MATHDB-398410

Pulling a hair tight

摘录 / EXCERPT: Tight open knots are usually modelled mathematically with the two ends pulled straight in opposite directions. But physically, when a knot is tied tight in a hair or thread or cabl...

open_problem | open | MathDB 页面 | 原文线索

V2-077176 | MATHDB-398411

Ropelength versus crossing number

摘录 / EXCERPT: Given the crossing number n of a knot or link type, are there families for which the minimum ropelength grows more than linearly in n?

open_problem | open | MathDB 页面 | 原文线索

V2-077177 | MATHDB-398416

Local Langlands conjecture for split reductive p-adic groups

摘录 / EXCERPT: Let F be a p-adic field and let G be F-split. With $WD_F=W_F\times \mathrm{SL}_2(\mathbb{C})$ and $G^\vee$ the Langlands dual group, the local Langlands conjecture postulat...

conjecture | open | MathDB 页面 | 原文线索

V2-077178 | MATHDB-398418

Langlands-type conjectural Plancherel decomposition in the group case

摘录 / EXCERPT: Let H be a reductive group over a local field k, and let $X=H$ with $G=H\times H$ acting by left and right multiplication. The report presents as conjectural a description of t...

open_problem | open | MathDB 页面 | 原文线索

V2-077179 | MATHDB-398421

Matsuki conjecture for real spherical homogeneous spaces

摘录 / EXCERPT: Let G be a real reductive group, let $H<G$ be a closed subgroup with finitely many connected components, and set $Z=G/H$. If $P=MAN<G$ is a minimal parabolic subgroup, then Z i...

conjecture | open | MathDB 页面 | 原文线索

V2-077180 | MATHDB-398425

Boij-Söderberg cone theorem for finite-length Betti tables

摘录 / EXCERPT: Let $S=k[x_1,\\ldots,x_n]$ have its standard $\\mathbb{Z}$-grading, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-077181 | MATHDB-398426

Corollary 1(1): equivalence of two block identities

摘录 / EXCERPT: Under the notation and assumptions of Theorem 2, let D be a stably finite ring (for example, a noetherian domain), and assume $q+t=p+s$. For $R_1\in D^{q\times p}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-077182 | MATHDB-398427

Indecomposable rank-\a0$n-2$ vector bundles on projective space

摘录 / EXCERPT: Do there exist indecomposable vector bundles of rank $n-2$ on $\a0\mathbb{P}^n$ when $n\ge 6$?

open_problem | open | MathDB 页面 | 原文线索

V2-077183 | MATHDB-398434

Telescope conjecture for commutative semi-hereditary rings

摘录 / EXCERPT: For a commutative semi-hereditary ring R, the Telescope Conjecture holds for $\d53f(R)$ if and only if every localization of R at a maximal ideal has no nonzero idempotent idea...

conjecture | open | MathDB 页面 | 原文线索

V2-077184 | MATHDB-398439

Classification in the case $p_g=q=2$ and $K_S^2=4$

摘录 / EXCERPT: Every minimal irregular surface S of general type with $p_g(S)=q(S)=2$ and $K_S^2=4$ is a double cover of a principally polarised abelian surface (A,Θ) branched along a di...

open_problem | open | MathDB 页面 | 原文线索

V2-077185 | MATHDB-398445

Very ampleness of the fifth plurilog-canonical multiple

摘录 / EXCERPT: Is there a stable log surface (X,Δ) such that $5I(K_X+\Delta)$ is not very ample?

open_problem | open | MathDB 页面 | 原文线索

V2-077186 | MATHDB-398446

Geography conjecture for general stable log surfaces

摘录 / EXCERPT: For a general stable log surface (X,Δ), is ...

conjecture | open | MathDB 页面 | 原文线索

V2-077187 | MATHDB-398452

Kuznetsov’s conjecture on rational cubic fourfolds

摘录 / EXCERPT: For a smooth cubic fourfold W, let A_W be the component in the semiorthogonal decomposition $D^b(W)=\langle A_W,\mathcal O_W,\mathcal O_W(1),\mathcal O_W(2)\rangle$. Then W i...

conjecture | open | MathDB 页面 | 原文线索

V2-077188 | MATHDB-398453

Exceptional sequence and quasi-phantom component for the classical Godeaux surface

摘录 / EXCERPT: Let $Y=\{x_1^5+x_2^5+x_3^5+x_4^5=0\}\subset \mathbb{P}^3_{\mathbb{C}}$ and let $G=\mathbb{Z}/5=\langle\xi\rangle$ act by $x_i\mapsto \xi^i x_i$. For the classical Godeaux surface...

open_problem | open | MathDB 页面 | 原文线索

V2-077189 | MATHDB-398456

Strong semistability of semistable Higgs bundles with trivial Chern classes

摘录 / EXCERPT: Any semistable Higgs bundle with trivial Chern classes is always strongly semistable.

open_problem | open | MathDB 页面 | 原文线索

V2-077190 | MATHDB-398457

Classification of growth types of finitely generated groups

摘录 / EXCERPT: For a finitely generated group G with generating set S, define ... For functions v,w, write $v\preceq w$ if there is a constant C such that $v(n)\le w(Cn)$ for all n, and w...

open_problem | open | MathDB 页面 | 原文线索

V2-077191 | MATHDB-398458

Gap between growth type and infimal growth function

摘录 / EXCERPT: Let G be a finitely generated group. For a finite generating set S, define ... Let v_G denote the growth type of G, and define the infimal growth function by ... Question: How...

open_problem | open | MathDB 页面 | 原文线索

V2-077192 | MATHDB-398460

Orbit problem for fiber-and-orientation-preserving isomorphisms of hyperbolic suspensions

摘录 / EXCERPT: Given a finitely presented group F, automorphisms $\b1,\b2$ of F, and hyperbolic suspensions $F\rtimes_{\alpha}\langle t\rangle$ and $F\rtimes_{\beta}\langle t'\rangle$, suppos...

open_problem | open | MathDB 页面 | 原文线索

V2-077193 | MATHDB-398461

Conjugacy of atoroidal automorphisms in \[1mOut\[0m(\[1mF\[0m_n)

摘录 / EXCERPT: For a free group F_n of finite rank, the conjugacy problem in $\operatorname{Out}(F_n)$ is decidable for atoroidal automorphisms, i.e. automorphisms that do not preserve any conj...

open_problem | open | MathDB 页面 | 原文线索

V2-077194 | MATHDB-398462

The Malnormal Special Quotient Theorem

摘录 / EXCERPT: Let G be a hyperbolic group which acts properly and cocompactly on a CAT(0) cube complex with quotient admitting a finite special (orbi-)cover. Let $[U+007F]P_1,\ldots,P_m$ be an almo...

open_problem | open | MathDB 页面 | 原文线索

V2-077195 | MATHDB-398464

Unstable Gromov-Lawson-Rosenberg conjecture for crystallographic groups

摘录 / EXCERPT: For which crystallographic groups is the unstable Gromov-Lawson-Rosenberg Conjecture true?

conjecture | open | MathDB 页面 | 原文线索

V2-077196 | MATHDB-398465

Adem-Ge-Pan-Petrosyan conjecture on the Lyndon-Hochschild-Serre spectral sequence

摘录 / EXCERPT: Conjecture 2 (Adem-Ge-Pan-Petrosyan): The Lyndon-Hochschild-Serre spectral sequence associated to the semidirect product $\Gamma := \mathbb{Z}^n \rtimes_{\rho} \mathbb{Z}/m$ collap...

conjecture | open | MathDB 页面 | 原文线索

V2-077197 | MATHDB-398466

Abstract simplicity of irreducible complete Kac-Moody groups over arbitrary fields

摘录 / EXCERPT: Is every irreducible complete Kac-Moody group $G(k)$ over an arbitrary field k abstractly simple?

open_problem | open | MathDB 页面 | 原文线索

V2-077198 | MATHDB-398467

Finiteness of discrete overgroups of cocompact lattices in products of trees

摘录 / EXCERPT: Let T_1,T_2 be regular trees, let $\Gamma<\operatorname{Aut}(T_1)\times\operatorname{Aut}(T_2)$ be a cocompact lattice, and let G_i be the closure of the projection of Γ...

open_problem | open | MathDB 页面 | 原文线索

V2-077199 | MATHDB-398469

Silver-Williams separability conjecture for embedded surface subgroups

摘录 / EXCERPT: If [1m[0mπ_1S is not a fiber, then π_1S can be separated from some element of $[\pi_1M,\pi_1M]\setminus\pi_1S$.

conjecture | open | MathDB 页面 | 原文线索

V2-077200 | MATHDB-398471

CAT(0) conjecture for all braid groups

摘录 / EXCERPT: For all integers n, the n-strand braid group B_n is CAT(0).

conjecture | open | MathDB 页面 | 原文线索

V2-077201 | MATHDB-398472

Construction of additional radial solutions for dynamic cavitation

摘录 / EXCERPT: Can additional solutions of the initial-value problem (1) be constructed by solving the radial problem (2)-(3)?

open_problem | open | MathDB 页面 | 原文线索

V2-077202 | MATHDB-398473

Existence of self-similar weak solutions describing dynamic cavitation

摘录 / EXCERPT: For spatial dimension $d\ge 3$ and stored energy ... where $h:\mathbb{R}_+\to\mathbb{R}_+$ satisfies ... there exists a self-similar weak radial solution of the dynamic problem that de...

open_problem | open | MathDB 页面 | 原文线索

V2-077203 | MATHDB-398474

Global ill-posedness of the two-dimensional isentropic Euler system

摘录 / EXCERPT: Let $p(\rho)=\rho^{2}$. Then there exist Lipschitz initial data ρ_{0} and v_{0}, with $\rho_{0}\ge c_{0}>0$, for which the Cauchy problem for the two-dimensional isentropic...

open_problem | open | MathDB 页面 | 原文线索

V2-077204 | MATHDB-398475

Well-posedness of entropy solutions for systems of conservation laws

摘录 / EXCERPT: Are entropy solutions of systems of conservation laws well posed?

open_problem | open | MathDB 页面 | 原文线索

V2-077205 | MATHDB-398476

Cauchy problem for the two-component Hunter-Saxton system

摘录 / EXCERPT: For the generalized Hunter-Saxton system on the real line ... the report states that there exists a Lipschitz-continuous semigroup of conservative solutions, obtained by passing to L...

open_problem | open | MathDB 页面 | 原文线索

V2-077206 | MATHDB-398478

Gradient-constrained conservation law via a nonlocal formulation

摘录 / EXCERPT: Let $f,g:\mathbb{R}\to\mathbb{R}$ be given, and suppose that the function g in the nonlocal problem ... has the form $g(w)=w h_\varepsilon(w)$, where $h=h_\varepsilon$ is a regular...

open_problem | open | MathDB 页面 | 原文线索

V2-077207 | MATHDB-398479

Nonlocal conservation law arising from a gradient constraint

摘录 / EXCERPT: The report considers the Cauchy problem ... for $x\in\mathbb R$, $t\ge 0$, with $f,g:\mathbb R\to\mathbb R$ given. It states existence, uniqueness, and continuous dependence on the d...

open_problem | open | MathDB 页面 | 原文线索

V2-077208 | MATHDB-398480

Well-posedness and stability for balance laws coupled with ODEs

摘录 / EXCERPT: Assume that $f,g_1,g_2,b,B_1,B_2,F_1,F_2,\\Pi_1,\Pi_2$ satisfy, respectively, conditions (f), (g), (b), (B), (F), and (Π) together with (NC), as stated on the prece...

open_problem | open | MathDB 页面 | 原文线索

V2-077209 | MATHDB-398482

Minkowski’s conjecture on Euclidean minima of totally real number fields

摘录 / EXCERPT: For every totally real number field K of degree $n=[K:\mathbb{Q}]$, with ring of integers $\mathcal{O}_K$, absolute discriminant d_K, and Euclidean minimum ... where N is the a...

conjecture | open | MathDB 页面 | 原文线索

V2-077210 | MATHDB-398491

The Conjecture of Bogomolov-Pop for Function Fields of Transcendence Degree at Least 2

摘录 / EXCERPT: Let F_1 and F_2 be fields finitely generated and of transcendence degree at least 2 over k_1 and k_2, respectively, where k_1 is either $\\mathbb{Q}$ or $\\mathbb{F}_p$...

conjecture | open | MathDB 页面 | 原文线索

V2-077211 | MATHDB-398494

Unconditional lower bound for higher Pythagoras numbers

摘录 / EXCERPT: Show unconditionally that, for every real function field F/K in d variables and every $m\in\mathbb{N}$, the higher Pythagoras number satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-077212 | MATHDB-398499

Possible values of characteristic-2 quadratic-form invariants

摘录 / EXCERPT: Let F be a field of characteristic 2. With $Q(F)$ denoting the set of F-quadratic forms up to isometry, $T(F)$ the set of totally singular F-quadratic forms, and...

open_problem | open | MathDB 页面 | 原文线索

V2-077213 | MATHDB-398500

Universality of maximal-dimensional anisotropic nonsingular quadratic forms

摘录 / EXCERPT: Suppose that $u(F)<\infty$. Is any anisotropic nonsingular F-quadratic form of dimension $u(F)$ universal?

open_problem | open | MathDB 页面 | 原文线索

V2-077214 | MATHDB-398501

Possible values of the characteristic-two invariant \(\widetilde u\)

摘录 / EXCERPT: Does there exist a field F of characteristic 2 such that $\widetilde u(F)=9$?

open_problem | open | MathDB 页面 | 原文线索

V2-077215 | MATHDB-398502

The u-invariant of a rational function field

摘录 / EXCERPT: Let k be a field with $\\operatorname{char} k \\ne 2$, and let $k(t)$ be the rational function field over k. Compute $u(k(t))$.

open_problem | open | MathDB 页面 | 原文线索

V2-077216 | MATHDB-398506

Finiteness of the Brauer p-dimension for function fields of p-adic curves

摘录 / EXCERPT: For a function field F in one variable over a p-adic field, is the Brauer p-dimension $\Br_p\mathrm{dim}(F)$ finite?

open_problem | open | MathDB 页面 | 原文线索

V2-077217 | MATHDB-398507

Morita-independent signatures via an auxiliary hermitian form

摘录 / EXCERPT: There exists $H\in W(A,\sigma)$ such that $\operatorname{sign}^{M_P}_P H\ne 0$ for every $P\in\widetilde X_F$. Consequently, if $\delta\in\{-1,1\}$ is the sign of...

open_problem | open | MathDB 页面 | 原文线索

V2-077218 | MATHDB-398508

Integral Tate conjecture for cubic fourfolds over finite fields

摘录 / EXCERPT: The suitably defined integral Tate conjecture holds for cubic fourfolds over a finite field of characteristic at least 5.

conjecture | open | MathDB 页面 | 原文线索

V2-077219 | MATHDB-398513

Classification of additive unstable operations in theories of rational type

摘录 / EXCERPT: Let A^* be a theory of rational type and B^* any theory. There is a one-to-one correspondence between additive unstable operations $A^n\to B^m$ and transformations ... commuting...

open_problem | open | MathDB 页面 | 原文线索

V2-077220 | MATHDB-398515

Integral Tate conjecture for quadric fibrations with two-dimensional general fibre

摘录 / EXCERPT: Let F be a finite field, let S be a surface over F, and let $X\to S$ be a quadric fibration with two-dimensional general fibre. Is the cycle class map ... surjective?

conjecture | open | MathDB 页面 | 原文线索

V2-077221 | MATHDB-398520

Vanishing of K-groups containing two additive reciprocity functors

摘录 / EXCERPT: Let F be a perfect field with $\\operatorname{char}(F)\\ne 2$, and let $M_1,\\ldots,M_n$ be reciprocity functors. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077222 | MATHDB-398525

Finiteness of surgeries in Perelman’s Ricci flow

摘录 / EXCERPT: There is a continuous function $\delta:[0,\infty)\to(0,\infty)$ such that the following holds: Let M be a Ricci flow with surgery with normalized initial conditions which is perf...

open_problem | open | MathDB 页面 | 原文线索

V2-077223 | MATHDB-398534

Global Arnold diffusion across action-angle domains

摘录 / EXCERPT: Under generic conditions, do there exist diffusing orbits that travel arbitrarily far across the different action-angle regions of the phase space and also follow prescribed freque...

open_problem | open | MathDB 页面 | 原文线索

V2-077224 | MATHDB-398537

Image of the angular-momentum map and Horn’s problem

摘录 / EXCERPT: Let S_0 be a symmetric nonnegative $2p\times 2p$ matrix. For each complex structure J on the Euclidean space E compatible with its Euclidean structure-that is, an isometry sa...

open_problem | open | MathDB 页面 | 原文线索

V2-077225 | MATHDB-398539

Bourgain’s subpolynomial-growth conjecture for cubic defocusing NLS

摘录 / EXCERPT: For the periodic cubic defocusing nonlinear Schrödinger equation (0.1) with convolution potential $V=0$ in dimension 2 or higher, Bourgain conjectured that the growth of the...

conjecture | open | MathDB 页面 | 原文线索

V2-077226 | MATHDB-398543

Annulus-like global surface of section for a Hopf-linked Reeb flow on \(\mathbb{RP}^3\)

摘录 / EXCERPT: Assume that a tight Reeb flow on $\mathbb{RP}^3$ admits a pair of closed orbits P_0,P_1 forming a Hopf link. If all closed Reeb orbits have Conley-Zehnder index at least 1, and...

open_problem | open | MathDB 页面 | 原文线索

V2-077227 | MATHDB-398546

Hurwitz constant for generalized Lagrange spectra

摘录 / EXCERPT: Let $I\subset\mathcal M_g$ be a closed $\mathrm{SL}(2,\mathbb R)$-invariant locus of area-one translation surfaces, and define ... where $a(X)$ is the geometric quantity defined in (...

open_problem | open | MathDB 页面 | 原文线索

V2-077228 | MATHDB-398547

Large-coupling lower bounds for Lyapunov exponents of quasiperiodic Schrödinger cocycles

摘录 / EXCERPT: Find conditions on $f:\mathbb T\to\mathbb R$ under which ... for all $E\in\mathbb R$, provided that λ is sufficiently large.

open_problem | open | MathDB 页面 | 原文线索

V2-077229 | MATHDB-398548

Virtual abelianness of centralizers of analytic surface diffeomorphisms

摘录 / EXCERPT: Suppose M is a closed surface and $f\in\operatorname{Diff}^{\omega}(M)$ has infinite order. Is the centralizer $\operatorname{Cent}^{\omega}(f)$ in...

open_problem | open | MathDB 页面 | 原文线索

V2-077230 | MATHDB-398549

Tits alternative for real-analytic area-preserving surface diffeomorphisms

摘录 / EXCERPT: Conjecture 15 (Tits alternative). If M is a compact surface, then every finitely generated subgroup of $\operatorname{Symp}^{\omega}_{\mu}(M)$ is either virtually solvable or con...

open_problem | open | MathDB 页面 | 原文线索

V2-077231 | MATHDB-398552

Extending the proven running time for elliptic-curve discrete logarithms

摘录 / EXCERPT: The problem consists in showing that the heuristic expected time given in (1) can be achieved for larger input classes of elliptic curves, or perhaps even for curves of any fixed g...

open_problem | open | MathDB 页面 | 原文线索

V2-077232 | MATHDB-398553

Completeness of Zagier’s list of non-degenerate integral solutions

摘录 / EXCERPT: For rational parameters $(\alpha_1,\alpha_0,\beta)$, consider the normalized power-series solution $u(t)=\sum_{n\ge 0}u_nt^n$ with $u_0=1$ of ... Restrict to the non-degenerate case...

open_problem | open | MathDB 页面 | 原文线索

V2-077233 | MATHDB-398557

Boyd’s Mahler-measure identity for the conductor-15 elliptic curve

摘录 / EXCERPT: For the elliptic curve E defined by $x+x^{-1}+y+y^{-1}+1=0$ (of conductor 15), ...

open_problem | open | MathDB 页面 | 原文线索

V2-077234 | MATHDB-398560

Determination of the arithmetic invariant-theory obstruction class

摘录 / EXCERPT: Let k be a field of characteristic different from 2, let $G=\operatorname{SL}(W)$ act on pairs (A,B) of symmetric bilinear forms on W, and let $n=\dim W$. The invariant bin...

open_problem | open | MathDB 页面 | 原文线索

V2-077235 | MATHDB-398562

Computing genus-2 curves with complex multiplication

摘录 / EXCERPT: Let C be a genus 2 curve over a field k of characteristic 0, and let K be a totally imaginary quadratic extension of a real quadratic field F. We say that C has compl...

open_problem | open | MathDB 页面 | 原文线索

V2-077236 | MATHDB-398567

Effective determination of general homogeneous S-unit equations with more than three variables

摘录 / EXCERPT: For homogeneous S-unit equations $a_1x_1+\cdots+a_nx_n=0$ over a number field, with nonzero coefficients and more than three S-unit variables, the report states that nontrivial sol...

open_problem | open | MathDB 页面 | 原文线索

V2-077237 | MATHDB-398568

Square-summability for Szegő-class Jacobi matrices

摘录 / EXCERPT: Let E be the fixed Parreau-Widom set, let $J' = \{a'_n,b'_n\}_{n=-\infty}^{\infty}\in \mathcal{T}_E$, and let $J = \{a_n,b_n\}_{n=1}^{\infty}$ be an arbitrary Jacobi matrix. If...

open_problem | open | MathDB 页面 | 原文线索

V2-077238 | MATHDB-398569

Inverse spectral estimate for quasi-periodic Schrödinger operators

摘录 / EXCERPT: Consider the Schrödinger equation $-\psi''(x)+\varepsilon V(x)\psi(x)=E\psi(x)$ on $\mathbb{R}$, where $V(x)=\sum_{n\in\mathbb{Z}^{\nu}}c(n)e^{2\pi i x\,n\cdot\omega}$ is quasi-per...

open_problem | open | MathDB 页面 | 原文线索

V2-077239 | MATHDB-398570

Conjecture 1: ℓ^1 perturbations and the Szegő class

摘录 / EXCERPT: Suppose $J' = \{a'_n,b'_n\}_{n=-\infty}^{\infty}$ belongs to T_E, and let $J = \{a_n,b_n\}_{n=1}^{\infty}$ be an arbitrary Jacobi matrix. If ... then J belongs to the Szegő class...

conjecture | open | MathDB 页面 | 原文线索

V2-077240 | MATHDB-398571

Elliptic reducibility for a kicked one-frequency quasiperiodic Schrödinger cocycle

摘录 / EXCERPT: For the peaky potential $V(x)=\dfrac{K}{1+4\lambda\sin^2(\pi x)}$, with $\\lambda$ and K big positive constants satisfying $\lambda\gg K$, and with α equal to the golden m...

open_problem | open | MathDB 页面 | 原文线索

V2-077241 | MATHDB-398572

Conjecture 9.5: Equality for the Absolutely Continuous Spectrum of Jacobi Matrices

摘录 / EXCERPT: Let ... be a half-line Jacobi matrix with $a_n>0$, $b_n\in\mathbb R$, and ... Assume that, for some $q\in\mathbb N$, ... Let $\mathcal R$ be the set of two-sided right limits $J^{(r)}$ o...

conjecture | open | MathDB 页面 | 原文线索

V2-077242 | MATHDB-398573

Variance lower bound for Green-function logarithms on an Anderson strip

摘录 / EXCERPT: Let H be the random strip operator on $\mathbb{Z}\times\{1,\ldots,W\}$ given by $(H\psi)_n=-\psi_{n-1}-\psi_{n+1}+S_n\psi_n$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-077243 | MATHDB-398574

Kotani-Last conjecture on almost periodicity of reflectionless Jacobi matrices

摘录 / EXCERPT: Absolutely continuous spectrum forces reflectionless Jacobi matrices to be almost periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-077244 | MATHDB-398575

Lawson-surface conjecture for the genus-\{g} minimal and Willmore infima

摘录 / EXCERPT: For genus $g>1$, it has been conjectured that $\alpha_g=\beta_g=\operatorname{area}(\Sigma_g)$, where α_g is the infimal area of compact embedded minimal surfaces of genus...

conjecture | open | MathDB 页面 | 原文线索

V2-077245 | MATHDB-398576

Compactness below the 8π threshold for $\mathcal W^p$-bounded immersions

摘录 / EXCERPT: Let Σ be a closed surface, let $2<p<\infty$, and let $f_k\in W^{2,p}_{\mathrm{im}}(\Sigma,\mathbb R^n)$ be a sequence of immersions such that $0\in f_k(\Sigma)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-077246 | MATHDB-398577

Sharp regularity theorem for critical Cauchy-Riemann-type systems

摘录 / EXCERPT: Let [?2004h(E^m,Σ,∇) be a rank-[?2004lm vector bundle over a Riemann surface, equipped with an L^2 connection, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-077247 | MATHDB-398581

Theorem 2: Smoothness under vanishing residues at a branch point

摘录 / EXCERPT: For the constrained Willmore immersion considered in the report, suppose that both residues at the origin-the second residue β and the first-residue vector $\vec{\beta}_0$-va...

open_problem | open | MathDB 页面 | 原文线索

V2-077248 | MATHDB-398582

Li-Yau-type theorem for compact free-boundary surfaces

摘录 / EXCERPT: For an immersion $F:\Sigma\to\mathbb R^n$ of a compact free-boundary surface with respect to the unit ball in $\mathbb R^n$, let $\mu=\theta\,\mathcal H^2\llcorner F(\Sigma)$ be th...

open_problem | open | MathDB 页面 | 原文线索

V2-077249 | MATHDB-398584

Sharp lower bound for the frame energy of immersed tori

摘录 / EXCERPT: Let $\vec\Phi:T^2\hookrightarrow\mathbb{R}^m$ be a smooth immersion of the 2-torus into Euclidean space, with $m\ge 3$, and let $\vec e=(\vec e_1,\vec e_2)$ be any moving frame a...

open_problem | open | MathDB 页面 | 原文线索

V2-077250 | MATHDB-398586

Explicit construction of Hadamard states

摘录 / EXCERPT: The explicit construction of Hadamard states is still an open issue.

open_problem | open | MathDB 页面 | 原文线索

V2-077251 | MATHDB-398588

Propagation of connectivity in local self-alignment dynamics

摘录 / EXCERPT: For the general class of local self-alignment models in which some interaction coefficients may vanish, and in which $a_{ij}=a_{ij}(P(t))$ with $P(t):=\{p_k(t)\}_k$, trace the prop...

open_problem | open | MathDB 页面 | 原文线索

V2-077252 | MATHDB-398591

De Giorgi’s higher-integrability conjecture for Mumford-Shah minimizers

摘录 / EXCERPT: Let $ Omega\subset\mathbb{R}^n$ be open. For a relatively closed set $K\subset\Omega$ and $u\in W^{1,2}(\Omega\setminus K)$, define ... Assume that (u,K) is a local minimizer: for...

conjecture | open | MathDB 页面 | 原文线索

V2-077253 | MATHDB-398594

Differential Harnack inequalities on evolving domains

摘录 / EXCERPT: Investigate which new expressions arise in the differential Harnack inequalities of Li-Yau [LY] and Hamilton [Ha2] when their settings are generalized-for instance, for positive so...

open_problem | open | MathDB 页面 | 原文线索

V2-077254 | MATHDB-398599

Degree theory for second-order fully nonlinear elliptic operators with nonlinear oblique boundary conditions

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded smooth domain, let ... where $f\in C^{3,\alpha}(\Omega\times\mathbb{R}\times\mathbb{R}^n\times S^n)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077255 | MATHDB-398600

Rigidity of equality cases in Steiner’s perimeter inequality

摘录 / EXCERPT: For a Borel function $v:\mathbb{R}^{n-1}\to[0,\infty)$ satisfying $v\in BV(\mathbb{R}^{n-1})$ and $\mathcal{H}^{n-1}(\{v>0\})<\infty$, define ... A set $E\subset\mathbb{R}^n$ is v-...

open_problem | open | MathDB 页面 | 原文线索

V2-077256 | MATHDB-398601

Existence of solutions to a singular Liouville equation under a noncontractibility condition

摘录 / EXCERPT: Let Σ be a compact surface with metric g, and let $p_1,\ldots,p_m\in\Sigma$ with $\alpha_j>-1$. Consider ... Define ... For $q\in\Sigma$, set ... and extend χ to finite sub...

open_problem | open | MathDB 页面 | 原文线索

V2-077257 | MATHDB-398604

Stability and geometric description of finite-time singularity formation for perturbed self-similar binormal-flow solutions

摘录 / EXCERPT: If one considers a small perturbation of the self-similar solution $\chi_a(1)$ and lets it evolve through the binormal flow (BF), does it also generate a singularity in...

open_problem | open | MathDB 页面 | 原文线索

V2-077258 | MATHDB-398605

Global well-posedness and scattering for the cubic Dirac equation in the critical space

摘录 / EXCERPT: For $M>0$, the cubic Dirac equation for $ \bigl\langle\gamma^0\psi,\psi\bigr\rangle\psi, \qquad \psi:\mathbb{R}^{1+3}\to\mathbb{C}^4, $$ with the Dirac matrices and scalar product...

open_problem | open | MathDB 页面 | 原文线索

V2-077259 | MATHDB-398606

Existence of a multi-soliton solution with $J\ge 2$

摘录 / EXCERPT: For the finite-energy corotational wave maps considered in Theorem 1 under assumptions (A1), (A2), and (A3′)-in particular, in the sphere case covered by (A3)-does there exist a so...

open_problem | open | MathDB 页面 | 原文线索

V2-077260 | MATHDB-398607

Global solutions and asymptotic behavior for two-dimensional gravity water waves

摘录 / EXCERPT: For sufficiently small initial data that decay sufficiently at infinity, the Cauchy problem for the Craig-Sulem-Zakharov system ... is globally well-posed in time. Moreover, with ... t...

open_problem | open | MathDB 页面 | 原文线索

V2-077261 | MATHDB-398608

Stability of the nondispersive solution for the focusing cubic wave equation

摘录 / EXCERPT: The report records a stability theorem for the explicit solution $v_0(t,x)=\sqrt{2}/t$ of $(-\partial_t^2+\Delta_x)v+v^3=0$ in three spatial dimensions. Define ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-077262 | MATHDB-398609

Relaxation of 1-equivariant exterior wave maps to harmonic maps

摘录 / EXCERPT: Let $n\ge 0$ and let $\vec\psi_0=(\psi_0,\psi_1)\in\mathcal E_n$ be smooth finite-energy data, where ... with ... Then the initial-boundary value problem ... with $\psi(t,1)=0$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-077263 | MATHDB-398610

Uniqueness and non-degeneracy of fractional-Laplacian ground states

摘录 / EXCERPT: Let ... where $N\ge 1$, $s\in(0,1)$, $Q=Q(|x|)>0$ is a ground state, and $\alpha\in(0,\alpha^*)$, with ... Then Q is unique, and the linearized operator ... is non-degenerate in the se...

open_problem | open | MathDB 页面 | 原文线索

V2-077264 | MATHDB-398611

Low regularity local well-posedness for the Chern-Simons-Schrödinger system

摘录 / EXCERPT: Consider the Chern-Simons-Schrödinger system on $\mathbb{R}^{1+2}$, with $D_\alpha=\partial_\alpha+iA_\alpha$, $J_i=\operatorname{Im}(\overline{\phi}D_i\phi)$, and coupling constan...

open_problem | open | MathDB 页面 | 原文线索

V2-077265 | MATHDB-398612

H^1 stability of mKdV breathers

摘录 / EXCERPT: Let $\alpha,\beta>0$ be fixed scalings. There exist parameters η_0,A_0, depending on α,β only, such that the following holds. If $u_0\in H^1(\mathbb{R})$ and, for s...

open_problem | open | MathDB 页面 | 原文线索

V2-077266 | MATHDB-398613

Openness of the blow-up set in the critical wave equation

摘录 / EXCERPT: In the critical case, is set (2) open? Here set (2) consists of initial data satisfying $\vec u(0)\notin \pm\mathcal M$, with maximal existence time $T<\infty$, and ...

open_problem | open | MathDB 页面 | 原文线索

V2-077267 | MATHDB-398614

Periodic 3D Gross-Pitaevskii hierarchy spacetime estimate

摘录 / EXCERPT: Let $\Lambda=\mathbb{T}^3$. For every $\alpha>1$, there exists $C_1=C_1(\alpha)>0$ such that, for every $k\in\mathbb{N}$, $j\in\{1,\ldots,k\}$, and $(k+1)$-particle initial datum...

open_problem | open | MathDB 页面 | 原文线索

V2-077268 | MATHDB-398615

Margulis’s uniform-discreteness conjecture for arithmetic uniform lattices

摘录 / EXCERPT: For every fixed semisimple Lie group G, the family of all torsion-free arithmetic uniform lattices in G is uniformly discrete; equivalently, the minimal injectivity radii of th...

conjecture | open | MathDB 页面 | 原文线索

V2-077269 | MATHDB-398616

Uniform discreteness of cocompact torsion-free arithmetic lattices

摘录 / EXCERPT: Let G be a connected center-free semisimple Lie group without compact factors. Is the family of all cocompact torsion-free arithmetic lattices in G uniformly discrete; that is,...

open_problem | open | MathDB 页面 | 原文线索

V2-077270 | MATHDB-398617

Deninger’s Fuglede-Kadison determinant question

摘录 / EXCERPT: For a countable discrete group Γ, if $f \in \mathbb{Z}\Gamma$ is invertible in $\ell^1(\Gamma)$ but not invertible in $\mathbb{Z}\Gamma$, then do we have...

open_problem | open | MathDB 页面 | 原文线索

V2-077271 | MATHDB-398618

Is the free uniform spanning forest a factor of IID?

摘录 / EXCERPT: In the setup where G is a Cayley graph of a group Γ with respect to a finite symmetric generating set S, is the free uniform spanning forest (FUSF) a Γ-equivarian...

open_problem | open | MathDB 页面 | 原文线索

V2-077272 | MATHDB-398619

Existence of finitely generated simple amenable groups

摘录 / EXCERPT: There exist $2^{\aleph_0}$ pairwise non-isomorphic infinite amenable simple finitely generated groups.

open_problem | open | MathDB 页面 | 原文线索

V2-077273 | MATHDB-398621

Limit multiplicity for arbitrary subgroup collections of irreducible arithmetic lattices

摘录 / EXCERPT: Can one expect that, for irreducible arithmetic lattices, the limit multiplicity property holds for any collection of subgroups not containing non-trivial central elements?

open_problem | open | MathDB 页面 | 原文线索

V2-077274 | MATHDB-398622

Limit multiplicity for arbitrary finite-index subgroups without central elements

摘录 / EXCERPT: Let Γ be an irreducible arithmetic lattice. Does the limit multiplicity property hold for every collection $\mathcal D$ of finite-index subgroups $\Delta\leq\Gamma$ such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-077275 | MATHDB-398623

Limit multiplicity for subgroups of irreducible arithmetic lattices

摘录 / EXCERPT: Let Γ be an irreducible arithmetic cocompact lattice. Does the limit multiplicity property hold for every collection $\mathcal D$ of finite-index subgroups Δ of...

open_problem | open | MathDB 页面 | 原文线索

V2-077276 | MATHDB-398624

Generalization of the canonical Pimsner-Voiculescu embedding

摘录 / EXCERPT: Can the construction be generalized to higher-dimensional noncommutative tori, or to tori based on an even more general discrete abelian group?

open_problem | open | MathDB 页面 | 原文线索

V2-077277 | MATHDB-398627

Persistence Approximation Property for reduced crossed products

摘录 / EXCERPT: Let Γ be a finitely generated group that satisfies the Baum-Connes conjecture with coefficients and admits a cocompact universal example for proper actions. Then there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-077278 | MATHDB-398631

Voiculescu’s free Gibbs factor conjecture

摘录 / EXCERPT: If τ is a free Gibbs state with a polynomial potential, then $W^*(\tau)$ is a free group factor.

conjecture | open | MathDB 页面 | 原文线索

V2-077279 | MATHDB-398633

Deriving atomlessness from operator-valued free convolution

摘录 / EXCERPT: It is an open problem to derive the absence of atoms for the distribution of selfadjoint polynomials in free semicirculars (or, more generally, in free variables without atoms), an...

open_problem | open | MathDB 页面 | 原文线索

V2-077280 | MATHDB-398634

Universality of the simple quotient of an ultraproduct of alternating groups

摘录 / EXCERPT: Is it true that every countable group embeds into the unique simple quotient of $\prod_{\omega} A_n$?

open_problem | open | MathDB 页面 | 原文线索

V2-077281 | MATHDB-398638

Comparison of parabolic category-\a0O functors with stabilized Rouquier complexes

摘录 / EXCERPT: Determine whether, and in what sense, restricting the relevant shuffling functors to parabolic subcategories of BernsteinGelfandGelfand category $\mathcal O$ is equivalent to stabi...

open_problem | open | MathDB 页面 | 原文线索

V2-077282 | MATHDB-398639

Categorification of the Kauffman polynomial

摘录 / EXCERPT: Is there a categorification of the Kauffman polynomial?

open_problem | open | MathDB 页面 | 原文线索

V2-077283 | MATHDB-398641

Equivalence of D-brane categories across phases of the new non-abelian LSMs

摘录 / EXCERPT: For each of the five new one-parameter linear sigma models listed in the report, the categories associated with the respective phases are conjectured to be equivalent; equivalently...

open_problem | open | MathDB 页面 | 原文线索

V2-077284 | MATHDB-398642

Equivalence of D-brane categories for the phases of the five one-parameter LSMs

摘录 / EXCERPT: For the five one-parameter linear sigma models listed in the report, the categories associated to their respective phases are conjectured to be equivalent.

open_problem | open | MathDB 页面 | 原文线索

V2-077285 | MATHDB-398649

Existence of conjugation-invariant continuous measures on subgroup lattices

摘录 / EXCERPT: For what groups G do there exist conjugation-invariant continuous measures on $L(G)$, the lattice of subgroups of G?

open_problem | open | MathDB 页面 | 原文线索

V2-077286 | MATHDB-398655

Boltzmann-limit convergence for unscattered hard spheres

摘录 / EXCERPT: Let n hard spheres of diameter a evolve by force-free Newtonian dynamics under the Boltzmann-Grad scaling $n a^{d-1}=1$, with independent, identically distributed initial data...

open_problem | open | MathDB 页面 | 原文线索

V2-077287 | MATHDB-398656

Exact traveling breather solutions of the discrete p-Schrödinger equation

摘录 / EXCERPT: For the discrete p-Schrödinger equation ... with $\alpha>1$ and τ_0 depending on α, the existence of exact traveling breather solutions is an open problem. Such solutions...

open_problem | open | MathDB 页面 | 原文线索

V2-077288 | MATHDB-398658

Extensive adiabatic invariant with stretched-exponential time scale

摘录 / EXCERPT: Theorem 2. There exist positive constants a_*, β_*, C_*, C_1, C_2, and C_3 such that, for all $\beta\geq\beta_*$ and $0<a<a_*$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-077289 | MATHDB-398659

Sharpness of the lower bounds for the soliton norm threshold

摘录 / EXCERPT: The report asks for a better understanding of the L^2-norm threshold for the existence of solitons; specifically, determine which of the two estimates in Proposition 2 is sharper...

open_problem | open | MathDB 页面 | 原文线索

V2-077290 | MATHDB-398660

Existence and bifurcations of localized equilibria in a two-dimensional bistable lattice

摘录 / EXCERPT: For the square-lattice system ... where ... and ... give a detailed description of the existence and bifurcations of localized equilibria as the nearest-neighbor and next-nearest-neighbo...

open_problem | open | MathDB 页面 | 原文线索

V2-077291 | MATHDB-398661

Validity of the KdV approximation for a spatially periodic Boussinesq model with non-small contrast

摘录 / EXCERPT: Let ... where $u(t,x)\in\mathbb R$ and a and b are strictly positive periodic coefficient functions. Fix $s>\tfrac12$. Then there exist constants μ_1 and μ_2 such that, f...

open_problem | open | MathDB 页面 | 原文线索

V2-077292 | MATHDB-398662

Identifying the proper involutions in involution-constrained PDEs

摘录 / EXCERPT: “How are we sure that we have the proper involutions?” Examples of this issue are seen in both MHD and nonlinear elasticity.

open_problem | open | MathDB 页面 | 原文线索

V2-077293 | MATHDB-398663

Role of the involution-free Lagrangian structure in numerical ideal-MHD calculations

摘录 / EXCERPT: It is an open problem to understand the role of this structure in numerical calculations, and to determine whether it could have an influence on the stability of Riemann solvers fo...

open_problem | open | MathDB 页面 | 原文线索

V2-077294 | MATHDB-398664

Estimation in multi-zoom autoregressive time-series models

摘录 / EXCERPT: For the multi-zoom AR model $X_t=\alpha_1 \frac{1}{\tau_1}(X_{t-1}+\cdots+X_{t-\tau_1})+\cdots+\alpha_p\frac{1}{\tau_p}(X_{t-1}+\cdots+X_{t-\tau_p})+\varepsilon_t$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-077295 | MATHDB-398665

Efficient estimation of integrated volatility with jump activity greater than one

摘录 / EXCERPT: Given regularly spaced observations of a one-dimensional Itô semimartingale X on $[0,T]$, estimate the continuous part C_T of its quadratic variation when the jump activity ind...

open_problem | open | MathDB 页面 | 原文线索

V2-077296 | MATHDB-398666

Generalization of the GARCH(1,1) testing theory

摘录 / EXCERPT: Finally, we conjecture that our theory can be generalized to GARCH models of higher order and to augmented GARCH processes.

open_problem | open | MathDB 页面 | 原文线索

V2-077297 | MATHDB-398667

Discriminating between long-range dependence and nonstationarity

摘录 / EXCERPT: How can one discriminate whether an observed process comes from a stationary long-range-dependent model or from a nonstationary process, in particular by constructing a nonparametr...

open_problem | open | MathDB 页面 | 原文线索

V2-077298 | MATHDB-398668

Small ball conjecture for hyperbolic sums of multiparameter Haar functions

摘录 / EXCERPT: For dimensions $d\ge 3$, it is conjectured that, for all relevant integers n and arbitrary real coefficients α_R, ... where h_R is an L^∞-normalized d-dimensional...

conjecture | open | MathDB 页面 | 原文线索

V2-077299 | MATHDB-398669

Signed small ball conjecture

摘录 / EXCERPT: Does the signed small ball lower bound ... hold for every n, every choice of signs $\alpha_R\in\{-1,1\}$, and every dimension $d\ge 3$, where R ranges over dyadic rectangles in...

conjecture | open | MathDB 页面 | 原文线索

V2-077300 | MATHDB-398670

Exact asymptotic order of the minimal L^∞ discrepancy in dimensions $s\ge 3$

摘录 / EXCERPT: For $s\ge 3$, determine, as $N\to\infty$, the exact asymptotic order of the smallest possible L^∞ discrepancy among all N-element point sets $P\subset[0,1)^s$. Here, for...

open_problem | open | MathDB 页面 | 原文线索

V2-077301 | MATHDB-398671

Optimal-order conjecture for the cited constructions of the L^∞ discrepancy

摘录 / EXCERPT: We conjecture that the constructions in [4] and [6] achieve the optimal order of the L^∞ discrepancy.

conjecture | open | MathDB 页面 | 原文线索

V2-077302 | MATHDB-398672

Improve the one-dimensional uniform-distribution discrepancy constant

摘录 / EXCERPT: Let c^* be the supremum of the constants c such that, for every sequence $(x_n)_{n\ge 0}$ in $[0,1)$, the discrepancy satisfies $D_N \ge c\,\frac{\log N}{N}$ for infinitely man...

open_problem | open | MathDB 页面 | 原文线索

V2-077303 | MATHDB-398674

Adaptive quadrature for functions whose fourth derivative changes sign

摘录 / EXCERPT: Determine how to effectively deal with functions for which $f^{(4)}$ changes sign in adaptive Simpson quadrature.

open_problem | open | MathDB 页面 | 原文线索

V2-077304 | MATHDB-398678

Polynomial tractability of integration for permutation-invariant functions

摘录 / EXCERPT: For the integration problem on the permutation-invariant subspaces described in the report, determine under which conditions polynomial tractability holds; that is, determine when...

open_problem | open | MathDB 页面 | 原文线索

V2-077305 | MATHDB-398679

Factor maps between Bernoulli shifts over countable non-amenable groups

摘录 / EXCERPT: Is it true that, for every countable non-amenable group G, all Bernoulli shifts over G factor onto one another?

open_problem | open | MathDB 页面 | 原文线索

V2-077306 | MATHDB-398685

Linear growth for certain elliptic fibrations

摘录 / EXCERPT: Let $\varepsilon>0$ be fixed. Let $e_1,e_2,e_3\in\mathbb Z$ be three distinct integers, put $e=(e_1,e_2,e_3)$, and let $Q\in\mathbb Z[u,v]$ be a non-degenerate quadratic form. Defi...

open_problem | open | MathDB 页面 | 原文线索

V2-077307 | MATHDB-398686

Finiteness of the power-free saturation number on affine quadrics

摘录 / EXCERPT: Let $n\ge 3$, let $Q\in\mathbb Z[X_1,\ldots,X_n]$ be a non-singular indefinite quadratic form, and let $m\in\mathbb Z\setminus\{0\}$. If $n=3$, assume that $-m\det(Q)$ is not the s...

open_problem | open | MathDB 页面 | 原文线索

V2-077308 | MATHDB-398687

Lang-Trotter conjecture for Frobenius fields of non-CM elliptic curves

摘录 / EXCERPT: Let K be an imaginary quadratic field. Let $E/\mathbb{Q}$ be an elliptic curve such that $\operatorname{End}_{\mathbb{Q}}(E)\simeq\mathbb{Z}$. Then there exists a constant...

conjecture | open | MathDB 页面 | 原文线索

V2-077309 | MATHDB-398688

Proportion of critical-line zeros of Dirichlet L-functions

摘录 / EXCERPT: For a large positive parameter Q, let $B(Q)$ be the rectangle in the s-plane with vertices $0,1,1+iQ,iQ$. Let $\mathcal F(Q)$ be the family of L-functions $L(s,\chi)$ for pri...

open_problem | open | MathDB 页面 | 原文线索

V2-077310 | MATHDB-398689

Conjectural exact range for Hooley’s variance asymptotic

摘录 / EXCERPT: The report proposes that Hooley’s conjecture should hold in the extended range ... and says that the exponent threshold is best possible.

open_problem | open | MathDB 页面 | 原文线索

V2-077311 | MATHDB-398691

Strongly diagonal behaviour in Vinogradov’s mean value theorem

摘录 / EXCERPT: Let $J_{s,k}(X)$ denote the number of integral solutions of ... with $1\leq x_i,y_i\leq X$ $(1\leq i\leq s)$. Then, if $k\geq 4$ and $1\leq s\leq \tfrac14(k+1)^2$, for each...

open_problem | open | MathDB 页面 | 原文线索

V2-077312 | MATHDB-398692

Gaussian distribution under a weaker growth condition for the divisor function in arithmetic progressions

摘录 / EXCERPT: Conjecture: Theorem 1 should remain valid when the function $\Phi:\mathbb{R}_{+}\to\mathbb{R}_{+}$, with $\Phi(x)\to\infty$ as $x\to\infty$, satisfies the weaker condition...

open_problem | open | MathDB 页面 | 原文线索

V2-077313 | MATHDB-398694

Eight cubes of primes for large even integers

摘录 / EXCERPT: Every sufficiently large even integer is the sum of eight cubes of primes; equivalently, for every sufficiently large even integer n, there exist primes $p_1,\dots,p_8$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-077314 | MATHDB-398695

GRH bound for the maximum of character sums

摘录 / EXCERPT: Let χ be a non-principal Dirichlet character modulo q, and define ... Assuming the Generalized Riemann Hypothesis, Montgomery and Vaughan proved ...

open_problem | open | MathDB 页面 | 原文线索

V2-077315 | MATHDB-398697

Norm forms as products of linear polynomials

摘录 / EXCERPT: Let $K/\mathbb{Q}$ be a finite field extension of degree n, let $\{\omega_1,\ldots,\omega_n\}$ be a $\mathbb{Z}$-basis of $\mathcal{O}_K$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-077316 | MATHDB-398700

Function-field analogue of Chowla’s conjecture in the large finite-field limit

摘录 / EXCERPT: Fix integers $r>1$ and $n>1$. For any choice of distinct polynomials $\alpha_1,\ldots,\alpha_r\in\mathbb{F}_q[t]$ with $\max_j\deg\alpha_j<n$, where q is odd, and any...

conjecture | open | MathDB 页面 | 原文线索

V2-077317 | MATHDB-398701

Higher additive divisor problem

摘录 / EXCERPT: For every integer $r\ge 3$ and every fixed nonzero integer h, let ... where $d_r(n)$ is the number of representations of n as a product of r positive integers. It is conjecture...

open_problem | open | MathDB 页面 | 原文线索

V2-077318 | MATHDB-398702

Integral points of cubic hypersurfaces

摘录 / EXCERPT: Let $f(x_1,\ldots,x_n)\in\mathbb Z[x_1,\ldots,x_n]$, $n\ge 3$, be a polynomial such that its homogeneous part F of maximal degree d is absolutely irreducible over $\mathbb Q$....

open_problem | open | MathDB 页面 | 原文线索

V2-077319 | MATHDB-398704

Function-field analogue of the Bombieri-Pila bound

摘录 / EXCERPT: Let $K=\mathbb{F}_q(T)$ and let $F(X,Y)\in K[X,Y]$ be absolutely irreducible with $\deg F=d$. Obtain an upper bound for the number of solutions of $F(x,y)=0$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-077320 | MATHDB-398705

Stronger conjecture on the limiting distribution of Rudin-Shapiro polynomials

摘录 / EXCERPT: Let $c(0)=1$, $c(2n)=c(n)$, and $c(2n+1)=(-1)^n c(n)$, and define $P_k(z)=\sum_{0\le n<2^k}c(n)z^n$. The conjecture is that the curve $P_k(e(\theta))$ is asymptotically uniformly d...

conjecture | open | MathDB 页面 | 原文线索

V2-077321 | MATHDB-398707

Lattice points on short arcs of circles

摘录 / EXCERPT: Conjecture: If $\theta<1$, then there is a constant $C(\theta)$ such that an arc A of length R^θ on a circle of radius R centered at 0 contains at most $C(\theta)$ lat...

open_problem | open | MathDB 页面 | 原文线索

V2-077322 | MATHDB-398708

Determine the true order of an incomplete exponential sum

摘录 / EXCERPT: Let $S(q,a)=\sum_{r=1}^{q}e(ar^k/q)$ and, for odd k with $(a,q)=1$, let $T(q,a)=\sum_{r=1}^{q}(1/2-r/q)e(ar^k/q)$. The report states that...

open_problem | open | MathDB 页面 | 原文线索

V2-077323 | MATHDB-398712

Mixing time of the adjacent transposition shuffle

摘录 / EXCERPT: For the adjacent transposition shuffle of a deck of N cards, Wilson’s conjecture that the lower bound is sharp is solved: the deck is mixed after...

open_problem | open | MathDB 页面 | 原文线索

V2-077324 | MATHDB-398713

Quenched functional central limit theorem under first and inverse-first moment conditions

摘录 / EXCERPT: Conjecture. Let $d\ge 1$ and consider the continuous-time random conductance model on $(\mathbb Z^d,\mathbb E_d)$ with conductances $\omega(e)\in(0,\infty)$. Assume that the conduc...

open_problem | open | MathDB 页面 | 原文线索

V2-077325 | MATHDB-398714

Zero-temperature two-dimensional Ising droplets and anisotropic curve-shortening flow

摘录 / EXCERPT: For the continuous-time zero-temperature heat-bath nearest-neighbor Ising dynamics on $(\mathbb{Z}/L)^2$, with each spin updated at unit rate and adopting the majority of its four...

open_problem | open | MathDB 页面 | 原文线索

V2-077326 | MATHDB-398715

Energy fluctuation field for the harmonic Bernardin-Stoltz model

摘录 / EXCERPT: For the harmonic Bernardin-Stoltz model described on PDF page 40, with $\rho=0$, fixed $\beta>0$, and initial distribution $\mu_{0,\beta}$, define, for $f\in C_c^\infty(\mathbb R)$...

open_problem | open | MathDB 页面 | 原文线索

V2-077327 | MATHDB-398731

Energetic evolutions are threshold evolutions

摘录 / EXCERPT: For the broad class of materials considered, energetic evolutions are also threshold evolutions, for a threshold related to the energetic cost of damage (the energy dissipated per...

open_problem | open | MathDB 页面 | 原文线索

V2-077328 | MATHDB-398732

Derivation of rigid-elasticity crystal-plasticity models by variational convergence

摘录 / EXCERPT: Can the two-dimensional finite-plasticity models with rigid elasticity described in the report-using the stated one- or two-slip-system plastic energies-be derived from correspondi...

open_problem | open | MathDB 页面 | 原文线索

V2-077329 | MATHDB-398733

Gamma-convergence of single-slip finite-plasticity models as elasticity becomes rigid

摘录 / EXCERPT: For a bounded open domain $\Omega\subset\mathbb{R}^2$, let $p,q\ge 1$, and define the one-slip condensed energy ... Set ... where ... The report states that if $q\ge p/(p-1)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077330 | MATHDB-398734

Energy scaling for a clamped shear experiment with cross-hardening

摘录 / EXCERPT: For the rectangular shear experiment with prescribed displacement $u(\cdot,0,\cdot)=(0,0,0)$ and $u(\cdot,L,\cdot)=\gamma(1,0,0)$, where $\gamma\ge 0$, consider the energy...

open_problem | open | MathDB 页面 | 原文线索

V2-077331 | MATHDB-398736

Rigorous derivation of GENERIC from microscopically conservative systems

摘录 / EXCERPT: The report identifies as completely open the problem of developing a rigorous theory for deriving the GENERIC evolution equation ... from a microscopically conservative system.

open_problem | open | MathDB 页面 | 原文线索

V2-077332 | MATHDB-398738

Backward-error interpretation of inaccuracies in Arnoldi’s method

摘录 / EXCERPT: Can the inaccuracies in Arnoldi’s method be interpreted as a backward error-namely, can the resulting matrices V_{k+1} and H_k be exact for a perturbed matrix $A+E$ for some ma...

open_problem | open | MathDB 页面 | 原文线索

V2-077333 | MATHDB-398739

A posteriori error estimate for nonconforming approximation of multiple eigenvalues

摘录 / EXCERPT: The report states the following theorem. Let λ_i be an eigenvalue of multiplicity q, with ... Let $\lambda_{i,h}\le\cdots\le\lambda_{i+q-1,h}$ be approximations of...

open_problem | open | MathDB 页面 | 原文线索

V2-077334 | MATHDB-398741

A posteriori error estimator for the C^0 interior penalty method for biharmonic eigenvalue problems

摘录 / EXCERPT: For the biharmonic plate eigenvalue problem on a polygonal Lipschitz domain $\Omega\subset\mathbb{R}^2$, with a simple eigenvalue and eigenfunction normalized by $b(u,u)=1$, the qu...

open_problem | open | MathDB 页面 | 原文线索

V2-077335 | MATHDB-398743

Rigorous enclosure for the second eigenvalue of a rotating-beam operator

摘录 / EXCERPT: For the rotating-beam eigenvalue problem ... with $\alpha=10$, write $A=A_1+A_2$, where $A_1u=u^{(4)}$ on ... and $A_2u=-\frac{\alpha^2}{2}\frac{d}{dx}((1-x^2)u')$ on ... The report obta...

open_problem | open | MathDB 页面 | 原文线索

V2-077336 | MATHDB-398744

Global-in-time existence for the spatially homogeneous Coulomb Landau equation

摘录 / EXCERPT: For the true Landau equation corresponding to the Coulomb potential, does a global-in-time solution exist in the spatially homogeneous case?

open_problem | open | MathDB 页面 | 原文线索

V2-077337 | MATHDB-398745

Cercignani’s conjectured entropy-dissipation inequality for the Boltzmann equation

摘录 / EXCERPT: The Boltzmann analogue of an inequality of the form $H \le C D$, relating the entropy H to its production rate D for the Boltzmann equation, does not hold in general.

conjecture | open | MathDB 页面 | 原文线索

V2-077338 | MATHDB-398746

Global classical solutions for a four-species reaction-diffusion system in dimensions greater than two

摘录 / EXCERPT: For the four-species reaction-diffusion system ... with homogeneous Neumann boundary conditions and positive diffusion coefficients $d_1,d_2,d_3,d_4>0$, determine whether global clas...

open_problem | open | MathDB 页面 | 原文线索

V2-077339 | MATHDB-398747

Duality estimate for a parabolic problem with bounded diffusion coefficient

摘录 / EXCERPT: Let d4a6d4a6 be a bounded domain in d4a6^N with d4a6d4a6 d4a6 C^{2+alpha}, and let T0. Assume that $0<a\le M(t,x)\le b<\infty$ on $\Omega_T=\Omega\times(0,T)$. Consider ... In dimens...

open_problem | open | MathDB 页面 | 原文线索

V2-077340 | MATHDB-398748

Convergence to a Dirac mass for initial data in a higher-index stable domain of attraction

摘录 / EXCERPT: The report conjectures that if f_0 belongs to the normal domain of attraction (NDA) of some $\b1'$-stable law with $\b1' > \b1$, then the corresponding transient solution converg...

open_problem | open | MathDB 页面 | 原文线索

V2-077341 | MATHDB-398750

Rigorous justification of the small-droplet formal limit

摘录 / EXCERPT: Can the formal limit from system (1)-(4) to the limit system (11)-(8) be rigorously justified, at least locally in time for well-prepared data, possibly by a relative-energy method...

open_problem | open | MathDB 页面 | 原文线索

V2-077342 | MATHDB-398751

Rigorous derivation of the Landau equation in the weak-coupling limit

摘录 / EXCERPT: Prove rigorously that the kinetic description of a dense gas of weakly interacting particles converges, in the weak-coupling limit, to the Landau equation, starting from the underl...

open_problem | open | MathDB 页面 | 原文线索

V2-077343 | MATHDB-398752

Exponential convergence to the steady state for a biased-velocity-jump kinetic equation

摘录 / EXCERPT: Let $V=[-1/2,1/2]$, $0<\chi<1$, and let f solve ... where $x\in\mathbb R$ and $K(x,v)=1+\chi\,\operatorname{sign}(xv)$. Let $g(x,v)$ be the positive steady state from Theorem 1. Fo...

open_problem | open | MathDB 页面 | 原文线索

V2-077344 | MATHDB-398753

The positivity conjecture for cluster variables

摘录 / EXCERPT: All cluster variables are positive Laurent polynomials in any given cluster.

conjecture | open | MathDB 页面 | 原文线索

V2-077345 | MATHDB-398755

Uniqueness of non-degenerate potentials for two-punctured closed positive-genus surfaces

摘录 / EXCERPT: Conjecture. Let Q be a quiver arising from a triangulation of a closed surface of positive genus with exactly two punctures. Then there exists, up to weak right equivalence, a un...

open_problem | open | MathDB 页面 | 原文线索

V2-077346 | MATHDB-398761

Laurentness and positivity of a noncommutative Kontsevich recursion

摘录 / EXCERPT: Let noncommutative variables U_n, $n\in\mathbb{Z}$, satisfy ... where $k\ge 3$ is a fixed odd natural number. The conjecture is that each U_n is a noncommutative Laurent polynomi...

open_problem | open | MathDB 页面 | 原文线索

V2-077347 | MATHDB-398762

Tropical Donaldson-Thomas invariants for Hitchin integrable systems

摘录 / EXCERPT: Under the conditions on the discriminant locus stated in the report, for a Hitchin integrable system and for each $b\in B^0$ and $\gamma\in\Gamma_b$, the integers...

open_problem | open | MathDB 页面 | 原文线索

V2-077348 | MATHDB-398765

Local smooth solvability of Brenier’s rearrangement-constrained gradient flow

摘录 / EXCERPT: On the periodic cube $D=\mathbb T^d$, let $E[\phi]=\frac12\int_{\mathbb T^d}|\nabla\phi|^2\,dx$ and $K[v]=\frac12\int_{\mathbb T^d}|\nabla^m v|^2\,dx$, where $m=0,1,2,\ldots$. With...

open_problem | open | MathDB 页面 | 原文线索

V2-077349 | MATHDB-398766

Optimal scaling laws for ductile fracture by void-sheet formation

摘录 / EXCERPT: For an incompressible, deformation-theory plasticity model on an infinite slab of in-plane area L^2 and thickness H, subject to prescribed opening displacement δ on its...

open_problem | open | MathDB 页面 | 原文线索

V2-077350 | MATHDB-398767

Existence of special solutions for the thin-rod variational problem

摘录 / EXCERPT: Let D be a Jordan domain in $\\mathbb{R}^2$ and let $s\in\mathbb{R}$. Define ... and ... A minimizer u is called a special solution if $|\nabla u|\in\{0\}\cup(1,+\infty)$ almost ev...

open_problem | open | MathDB 页面 | 原文线索

V2-077351 | MATHDB-398768

Metastability of the discrete XY model

摘录 / EXCERPT: For the discrete XY energy $XY_\varepsilon(v):=\frac12\sum_{i,j\in\varepsilon\mathbb Z^2\cap\Omega,\ |i-j|=\varepsilon}|v(i)-v(j)|^2$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-077352 | MATHDB-398769

Read’s conjecture on log-concavity of chromatic-polynomial coefficients

摘录 / EXCERPT: For a finite graph G, let its chromatic polynomial be written as ... Then the sequence $a_0,a_1,\ldots,a_n$ is log-concave; that is, for every integer i with $0<i<n$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-077353 | MATHDB-398770

Erdős-Simonovits conjecture for bipartite graphs

摘录 / EXCERPT: For every bipartite graph H, there exists a positive constant c_H, depending only on H, such that for all graphs G, ...

conjecture | open | MathDB 页面 | 原文线索

V2-077354 | MATHDB-398774

Lovász-Szegedy compactness conjecture for finitely forcible graphons

摘录 / EXCERPT: If W is a finitely forcible graphon, then $T(W)$ is a compact space.

conjecture | open | MathDB 页面 | 原文线索

V2-077355 | MATHDB-398778

Renewals without replacement

摘录 / EXCERPT: Let $X=\{x_1,\ldots,x_m\}$ be a multiset of positive integers with sum n. Choose a uniform permutation $\sigma\in S_m$ and define $S_t=\sum_{i=1}^t x_{\sigma(i)}$. Determine or e...

open_problem | open | MathDB 页面 | 原文线索

V2-077356 | MATHDB-398779

Hamilton cycles in nearly regular oriented graphs

摘录 / EXCERPT: For every $c > 3/8$ and every $\[1m\eta\[0m > 0$ there exist a positive constant $\[1m\varepsilon\[0m := \[1m\varepsilon\[0m(c,\eta) > 0$ and an integer $n_0 := n_0(c,\eta)$ such t...

open_problem | open | MathDB 页面 | 原文线索

V2-077357 | MATHDB-398780

Kühn-Lapinskas-Osthus conjecture on packing Hamilton cycles

摘录 / EXCERPT: For every graph G with minimum degree $\b\delta(G)\ge n/2\d$, where $n=|V(G)|$, let $\b\operatorname{reg}_{\mathrm{even}}(G)\d$ be the maximum even degree of a spanning regular s...

conjecture | open | MathDB 页面 | 原文线索

V2-077358 | MATHDB-398785

The asymptotic random k-SAT threshold

摘录 / EXCERPT: For integers $k\ge 3$, the random k-SAT satisfiability threshold r_k satisfies ... where $o_k(1)\to 0$ as $k\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-077359 | MATHDB-398787

Hamilton-based flow-lattices under dense minimum-degree conditions

摘录 / EXCERPT: Conjecture (C.1). Theorem 1.(I.1) and Theorem 1.(I.3) remain true when “is Hamilton-generated” is replaced by “has a Hamilton-based flow-lattice.” Explicitly, for every $\gamma>0$...

open_problem | open | MathDB 页面 | 原文线索

V2-077360 | MATHDB-398789

Hamilton-based flow-lattice under a dense minimum-degree condition

摘录 / EXCERPT: For every real number $\gamma>0$, there exists $n0",

open_problem | open | MathDB 页面 | 原文线索

V2-077361 | MATHDB-398790

Morphing planar graphs

摘录 / EXCERPT: Given two straight-line planar drawings Γ and Γ' of a planar triangulation G with the outer face fixed, find a planar morph from Γ to Γ' consisting of...

open_problem | open | MathDB 页面 | 原文线索

V2-077362 | MATHDB-398792

Maximizing the number of independent sets of a fixed size

摘录 / EXCERPT: For every integer $t \ge 3$ and every integer $\delta \le n/2$, the complete bipartite graph $K_{\delta,n-\delta}$ maximizes the number $i_t(G)$ of independent sets of size t ove...

open_problem | open | MathDB 页面 | 原文线索

V2-077363 | MATHDB-398793

Unit distances with rational angle

摘录 / EXCERPT: Let $\varepsilon>0$. Given n points in the plane, the number of unit distances with rational angle between pairs of points is less than $n^{1+\varepsilon}$. Two points in the pla...

open_problem | open | MathDB 页面 | 原文线索

V2-077364 | MATHDB-398796

Revised random-graph bipartite decomposition conjecture

摘录 / EXCERPT: For a graph $G=(V,E)$, let $\tau(G)$ be the minimum number of pairwise edge-disjoint complete bipartite subgraphs of G such that every edge of G belongs to exactly one of them....

conjecture | open | MathDB 页面 | 原文线索

V2-077365 | MATHDB-398797

Hamilton circuits generating the cycle space at the Dirac threshold

摘录 / EXCERPT: For every odd $n\in\mathbb{N}$, every graph G on n vertices with minimum degree at least $\lceil n/2\rceil$ has the property that every cycle in G can be constructed as a sym...

open_problem | open | MathDB 页面 | 原文线索

V2-077366 | MATHDB-398798

Kalai-Meshulam chromatic boundedness conjecture

摘录 / EXCERPT: For every integer $l\ge 0$ there exists an integer k such that, if $I(H)\le l$ for every induced subgraph H of a graph G, then $\chi(G)\le k$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-077367 | MATHDB-398804

Hrušák’s conjecture on Rothberger numbers of fragmented ideals

摘录 / EXCERPT: Is it true that for every ideal I on ω which is fragmented but not gradually fragmented, its Rothberger number satisfies $\mathfrak b(I)=\aleph_1$?

conjecture | open | MathDB 页面 | 原文线索

V2-077368 | MATHDB-398805

Rothberger numbers of non-gradually fragmented ideals

摘录 / EXCERPT: Conjecture reported by Hrušák. Let I be a fragmented ideal on $\\omega$: there are a partition $(a_i)_{i\in\omega}$ of $\\omega$ into finite sets and submeasures φ_i on...

open_problem | open | MathDB 页面 | 原文线索

V2-077369 | MATHDB-398806

Possible initial Tukey structures of ultrafilters

摘录 / EXCERPT: What are the possible initial structures in the Tukey types of ultrafilters?

open_problem | open | MathDB 页面 | 原文线索

V2-077370 | MATHDB-398809

A universal tree at \(\aleph_1\) under reductions

摘录 / EXCERPT: Let $\mathcal T_\kappa$ be the structure of trees of size κ having no branch of size κ, ordered by reductions $f:T\to T'$ that preserve strict order:...

open_problem | open | MathDB 页面 | 原文线索

V2-077371 | MATHDB-398812

Whether the Ostaszewski club principle implies a Souslin tree

摘录 / EXCERPT: Does the Ostaszewski club principle imply the existence of a Souslin tree?

open_problem | open | MathDB 页面 | 原文线索

V2-077372 | MATHDB-398817

Failure of weak square and failure of SCH at \(\aleph_\omega\)

摘录 / EXCERPT: Is it consistent to have failure of weak square at $\aleph_\omega$ and not SCH at $\aleph_\omega$?

open_problem | open | MathDB 页面 | 原文线索

V2-077373 | MATHDB-398819

No maximal eventually different families in Solovay’s model

摘录 / EXCERPT: There are no med families in Solovay’s model.

open_problem | open | MathDB 页面 | 原文线索

V2-077374 | MATHDB-398823

Removal of the condition \(\tau\ne\tau^\theta\) in analytic-torsion asymptotics

摘录 / EXCERPT: Let $G=\mathrm{SO}_0(d,1)$, $K=\mathrm{SO}(d)$, $d>1$, and $\widetilde X=G/K$. Let $\Gamma_0\subset G$ be a lattice and let $\Gamma_i\triangleleft\Gamma_0$ be finite-index normals...

open_problem | open | MathDB 页面 | 原文线索

V2-077375 | MATHDB-398825

Finite multiplicities for spherical pairs in characteristic zero

摘录 / EXCERPT: It is conjectured that if $\operatorname{char}F=0$, then all spherical pairs have finite multiplicities.

open_problem | open | MathDB 页面 | 原文线索

V2-077376 | MATHDB-398829

Vanishing of higher Ext groups for generic \GL_{n+1}(F)\-to-\GL_n(F)\ branching

摘录 / EXCERPT: Let F be a non-Archimedean local field. Let π_1 be an irreducible generic representation of $\mathrm{GL}_{n+1}(F)$, and let π_2 be an irreducible generic representation o...

open_problem | open | MathDB 页面 | 原文线索

V2-077377 | MATHDB-398830

Euler-Poincaré pairing and tempered Ext vanishing for orthogonal branching

摘录 / EXCERPT: Let V be a quadratic space over the non-Archimedean local field F, with W a quadratic subspace of V of codimension 1. For any irreducible admissible representation...

open_problem | open | MathDB 页面 | 原文线索

V2-077378 | MATHDB-398832

Conservation relations for local theta correspondence

摘录 / EXCERPT: In the non-archimedean case, let $\mathbf t_1$ and $\mathbf t_2$ be two Witt towers in $\mathcal W_U$ whose difference is the anti-split Witt tower $\mathbf t_U^\circ$. For any...

open_problem | open | MathDB 页面 | 原文线索

V2-077379 | MATHDB-398833

Uniqueness of a locally smallest burnstile lobe

摘录 / EXCERPT: Is there just one locally smallest lobe?

open_problem | open | MathDB 页面 | 原文线索

V2-077380 | MATHDB-398834

Definition of a flow barrier in an unsteady flow

摘录 / EXCERPT: What is a flow barrier in an unsteady flow anyway?

open_problem | open | MathDB 页面 | 原文线索

V2-077381 | MATHDB-398835

Necessity of frame invariance for flow-barrier detection

摘录 / EXCERPT: Is frame invariance necessary for methods locating supposed flow barriers?

open_problem | open | MathDB 页面 | 原文线索

V2-077382 | MATHDB-398837

Sato-Tate equidistribution for the Legendre family

摘录 / EXCERPT: Let $\mathbb{F}_q$ be a finite field of odd characteristic, and for each $\lambda\in\mathbb{F}_q\setminus\{0,1\}$ let E_λ be the Legendre elliptic curve...

open_problem | open | MathDB 页面 | 原文线索

V2-077383 | MATHDB-398838

Geometric description and stability of stable characters for tame Deligne-Lusztig parameters

摘录 / EXCERPT: For a positive-characteristic local field F, an unramified connected reductive group G, and a tamely ramified Langlands parameter $ ... Here $H{ast,gamma}...

open_problem | open | MathDB 页面 | 原文线索

V2-077384 | MATHDB-398846

Genus growth for abstract modular covers

摘录 / EXCERPT: Let k be a finitely generated field of characteristic $p\ge 0$, let X/k be a smooth, separated, geometrically connected curve, and let L be an infinite set of primes with...

open_problem | open | MathDB 页面 | 原文线索

V2-077385 | MATHDB-398848

Coefficient descent for bounded families of mod-\ell representations

摘录 / EXCERPT: Conjecture. Let $\rho_\ell:\pi_1(X)\to\mathrm{GL}_r(F_\ell)$, $\ell\in L$, be a bounded family of continuous representations satisfying (T) and (P). Then there exists an integer...

open_problem | open | MathDB 页面 | 原文线索

V2-077386 | MATHDB-398852

Exact characterization of the monomial-convergence set for bounded holomorphic functions on B_{c_0}

摘录 / EXCERPT: Determine the set ... The report establishes the bounds ... where $(z_1^\star,z_2^\star,\ldots)$ is the decreasing rearrangement of z.

open_problem | open | MathDB 页面 | 原文线索

V2-077387 | MATHDB-398856

Bohr's absolute convergence problem for scalar Dirichlet series

摘录 / EXCERPT: For a Dirichlet series $D(s)=\sum_{n=1}^{\infty}a_n n^{-s}$ with $a_n\in\mathbb{C}$, let $\sigma_u(D)$ and $\sigma_a(D)$ be respectively the infima of the real numbers σ suc...

open_problem | open | MathDB 页面 | 原文线索

V2-077388 | MATHDB-398858

Optimality of Koksma’s criterion for Khinchin’s averages

摘录 / EXCERPT: Let $w(n)\ge 1$ be a function on the natural numbers that is submultiplicative, meaning $w(mn)\le w(m)w(n)$ for all m,n, and bounded in mean, meaning...

open_problem | open | MathDB 页面 | 原文线索

V2-077389 | MATHDB-398859

Sarason’s full Poisson-\A2 sufficiency conjecture for the two-weight Hilbert transform

摘录 / EXCERPT: Sarason conjectured that the full Poisson A_2 condition would be sufficient for the two-weight inequality for the Hilbert transform. The report states that F. Nazarov disproved t...

conjecture | open | MathDB 页面 | 原文线索

V2-077390 | MATHDB-398862

Optimality of the exponential factor in a polynomial inequality

摘录 / EXCERPT: For $1\le r\le\infty$, let $\Lambda(N)=\{\alpha\in\mathbb{N}_0^{(\mathbb{N})}:2^{\alpha_1}3^{\alpha_2}5^{\alpha_3}\cdots\le N\}$. Determine whether the exponent in the exponential...

open_problem | open | MathDB 页面 | 原文线索

V2-077391 | MATHDB-398863

Characterization of integrable functions with almost-everywhere convergent Fourier series

摘录 / EXCERPT: Let $\mathbb T=\mathbb R/\mathbb Z$, identified with $I=[0,1)$ equipped with Lebesgue measure m. For $f\in L^1(\mathbb T)$, define ... Characterize the set of all...

open_problem | open | MathDB 页面 | 原文线索

V2-077392 | MATHDB-398864

Extension of Guo’s Lipschitz-curve result beyond L^2

摘录 / EXCERPT: Let $B=B_{1/2}(1,0)\subset\mathbb R^2$. Suppose that $h:\mathbb R^2\to\mathbb R$ is Lipschitz and satisfies $\|\nabla h(x_1,x_2)-(1,0)\|\leq \tfrac12$ for almost every...

open_problem | open | MathDB 页面 | 原文线索

V2-077393 | MATHDB-398865

Symmetric polynomial zero sets in the bidisk

摘录 / EXCERPT: Produce an example of a polynomial $P=P(z_1,z_2)$ such that $P(z_1,z_2)=P(z_2,z_1)$ and the set $\{P=0\}$ is connected in D^2 but does not intersect the main diagonal of D^2.

open_problem | open | MathDB 页面 | 原文线索

V2-077394 | MATHDB-398866

Asymptotics of the zero count for harmonic polynomials

摘录 / EXCERPT: For harmonic polynomials of the form $h=p-q$ with $n=\deg p>\deg q=m$, the report defines $Z(m,n):=\#\{h=0\}$ and asks: find the asymptotics of $Z(m,n)$ for large m,n.

open_problem | open | MathDB 页面 | 原文线索

V2-077395 | MATHDB-398867

Asymptotic rate of the Bohr radius for degree-\N polynomials

摘录 / EXCERPT: Determine the asymptotic behavior of $R_N-\frac{1}{3}$ as $N\to\u0000infty$; in particular, Fournier suggested, based on numerical evidence, that...

open_problem | open | MathDB 页面 | 原文线索

V2-077396 | MATHDB-398868

Characterize cyclic polynomials in the Dirichlet spaces of the bidisk

摘录 / EXCERPT: Characterize cyclic polynomials in the Dirichlet spaces $\mathcal{D}_\alpha$ of the bidisk, for $-\infty<\alpha<\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-077397 | MATHDB-398869

The Lindelöf Hypothesis and BMO

摘录 / EXCERPT: For a finite Dirichlet series, define, for $1\le p<\infty$, the Hardy-space mean norm by ... and let H^p be the completion of the finite Dirichlet series in this norm. Let ... The Li...

conjecture | open | MathDB 页面 | 原文线索

V2-077398 | MATHDB-398870

Helson’s conjecture for the Dirichlet polynomial with coefficients 1

摘录 / EXCERPT: Show that ... where, for a Dirichlet polynomial f, ...

conjecture | open | MathDB 页面 | 原文线索

V2-077399 | MATHDB-398871

Stationarity of the set of perfect squares

摘录 / EXCERPT: Let S be the set of perfect squares. Is S stationary; that is, does there exist a constant C such that $[f]\le C\|f\|_\infty$ for every trigonometric polynomial f with spec...

open_problem | open | MathDB 页面 | 原文线索

V2-077400 | MATHDB-398872

Growth of the averaging length for Dirichlet-polynomial L^p norms

摘录 / EXCERPT: For a Dirichlet polynomial $f(s)=\sum_{n=1}^{N}a_n n^{-s}$ and $1\le p<\infty$, define ... For a fixed large N and a constant $C>1$, let $T_p(N,C)$ be the smallest T such that ……

open_problem | open | MathDB 页面 | 原文线索

V2-077401 | MATHDB-398873

The triangle Hilbert integral: an L^3 bound

摘录 / EXCERPT: Let $f_1,f_2,f_3\in C_c^\infty(\mathbb{R}^2)$. Define ... where the principal value is the limit as $\varepsilon\to0$ of the integral restricted to $|x_1+x_2+x_3|>\varepsilon$. Conje...

open_problem | open | MathDB 页面 | 原文线索

V2-077402 | MATHDB-398876

Optimal-dimensional spine of Teichmüller space

摘录 / EXCERPT: Construct a spine of T_g of dimension $4g-5$.

open_problem | open | MathDB 页面 | 原文线索

V2-077403 | MATHDB-398877

Density of holonomy fibers in moduli space

摘录 / EXCERPT: Let S be a closed oriented surface of genus $g\ge 2$, let $\mathcal P$ be the space of marked complex projective structures on S, let $\operatorname{hol}:\mathcal P\to\chi$ be...

open_problem | open | MathDB 页面 | 原文线索

V2-077404 | MATHDB-398878

Arbitrarily long positive factorizations of boundary multitwists

摘录 / EXCERPT: Let Σ_g^2 be a compact connected oriented surface of genus g with two boundary components δ_1 and δ_2, and let a be a nonseparating simple closed curve on...

open_problem | open | MathDB 页面 | 原文线索

V2-077405 | MATHDB-398882

Asymptotic behavior of the representations and parallel transport

摘录 / EXCERPT: Given a semistable Higgs bundle (E,ϕ), consider the family $(E,t\phi)$ with $t\in\mathbb C$, its associated harmonic metrics and flat connections, representations ρ_t, an...

open_problem | open | MathDB 页面 | 原文线索

V2-077406 | MATHDB-398883

Uniform domination for proper left-right actions on a real rank-one Lie group

摘录 / EXCERPT: Let Gcolon be a real rank-one semisimple Lie group, and let $\93< G\times G$ be finitely generated and act properly discontinuously on G by left-right multiplication,...

open_problem | open | MathDB 页面 | 原文线索

V2-077407 | MATHDB-398884

Structure of the image of the Fricke-character Johnson homomorphism

摘录 / EXCERPT: For a free group F_n of rank $n\ge 3$, let J be the specified $ \operatorname{Aut}(F_n)$-invariant ideal in the ring of Fricke characters, let...

open_problem | open | MathDB 页面 | 原文线索

V2-077408 | MATHDB-398885

Invariant complete CAT(0) metric on Teichmüller space

摘录 / EXCERPT: Does the Teichmüller space $T_{g,n}$ admit a complete CAT(0) metric invariant under the mapping class group $\\mathrm{MCG}$?

open_problem | open | MathDB 页面 | 原文线索

V2-077409 | MATHDB-398886

Positive Dehn-twist factorizations of boundary twists

摘录 / EXCERPT: Let S be a surface of genus $g \ge 3$ with boundary components $\delta_1,\ldots,\delta_n$, and let τ_γ denote the Dehn twist about a simple closed curve γ. Is it...

open_problem | open | MathDB 页面 | 原文线索

V2-077410 | MATHDB-398887

Actions of mapping-class-group quotients by powers of Dehn twists

摘录 / EXCERPT: Let a0$\operatorname{Mod}_{g,n}$ act on Teichm[U+007F]ller space and on the curve complex. Let $t(k)$ be the normal subgroup generated by all k-th powers of Dehn twists. Is it possible...

open_problem | open | MathDB 页面 | 原文线索

V2-077411 | MATHDB-398893

Positivity of Kac polynomials

摘录 / EXCERPT: For every finite quiver Γ and dimension vector v, the Kac polynomial counting absolutely indecomposable representations of (Γ,v) over finite fields has non-negative...

open_problem | open | MathDB 页面 | 原文线索

V2-077412 | MATHDB-398895

Polynomiality and mixed Hodge interpretation of $H_\mu(z,w)$

摘录 / EXCERPT: We conjecture that the functions $H_\mu(z,w)$ defined by ... are polynomials in z,w that encode the mixed Hodge numbers of M_μ.

open_problem | open | MathDB 页面 | 原文线索

V2-077413 | MATHDB-398901

Fourier-Mukai representability without full faithfulness

摘录 / EXCERPT: Does Orlov’s Fourier-Mukai representability theorem remain true if the full-faithfulness hypothesis is removed; that is, is every exact functor between the bounded derived categori...

open_problem | open | MathDB 页面 | 原文线索

V2-077414 | MATHDB-398903

Minimal free resolutions of equivariant ideals generated by irreducible representations

摘录 / EXCERPT: Let K be a field of characteristic zero, let F and G be K-vector spaces of dimensions m and n with $m\ge n$, and set $S=\operatorname{Sym}(F\otimes G)$. For a partition...

open_problem | open | MathDB 页面 | 原文线索

V2-077415 | MATHDB-398905

Infinite Kähler groups without infinite-image linear representations

摘录 / EXCERPT: Are there infinite Kähler groups with no linear representation with infinite image?

open_problem | open | MathDB 页面 | 原文线索

V2-077416 | MATHDB-398908

Generic simplicity of Dirichlet tilings in hyperbolic 3-space

摘录 / EXCERPT: Conjecture 0.9. Suppose that $\Gamma<\mathrm{PO}(3,1)$ is a discrete torsion-free subgroup. Then, for a generic choice of a base-point $x\in\mathbb{H}^3$, the corresponding tiling...

open_problem | open | MathDB 页面 | 原文线索

V2-077417 | MATHDB-398911

Every finitely-presented group embeds in a Kähler group

摘录 / EXCERPT: Every finitely-presented group appears as a subgroup of some Kähler group.

open_problem | open | MathDB 页面 | 原文线索

V2-077418 | MATHDB-398913

Kähler groups versus projective groups

摘录 / EXCERPT: Is every torsion-free Kähler group a projective group?

open_problem | open | MathDB 页面 | 原文线索

V2-077419 | MATHDB-398914

Minimum total multiplicity of stable line arrangements

摘录 / EXCERPT: For an arrangement of n lines in $\mathbb{CP}^2$, let $\mu(x)$ be the number of lines passing through a multiple point x, and let the total multiplicity be $\sum_x \mu(x)$, sum...

open_problem | open | MathDB 页面 | 原文线索

V2-077420 | MATHDB-398915

Quasiprojective free products

摘录 / EXCERPT: Suppose A and B are groups such that the free product $G=A*B$ is a quasiprojective group. Suppose, in addition, that both A and B admit nontrivial finite-index subgroups, a...

open_problem | open | MathDB 页面 | 原文线索

V2-077421 | MATHDB-398916

SHGH-related negative-curve conjecture for blow-ups of the projective plane

摘录 / EXCERPT: The only negative curves on the blow-up of the projective plane $\mathbb{P}^2$ in $s \ge 10$ points are the (-1)-curves.

conjecture | open | MathDB 页面 | 原文线索

V2-077422 | MATHDB-398919

Containment of the third symbolic power in the square of a point ideal

摘录 / EXCERPT: Let $Z=\{P_1,\ldots,P_s\}\subset\mathbb{P}^2$ be a set of points over an arbitrary field, and let $I=I(P_1)\cap\cdots\cap I(P_s)$, where $I(P_i)$ is the ideal of P_i. For points,...

open_problem | open | MathDB 页面 | 原文线索

V2-077423 | MATHDB-398920

When and why symbolic-power containments fail

摘录 / EXCERPT: Understand when and why failures of the containment $I^{(Nr-N+1)}\subseteq I^r$ occur, for ideals I of points in $\mathbb{P}^N$.

open_problem | open | MathDB 页面 | 原文线索

V2-077424 | MATHDB-398922

Finiteness of minimal models reached by the K_Y-MMP

摘录 / EXCERPT: Suppose that Y is a terminal projective threefold. Then the number of minimal models X_i that can be reached via the K_Y-MMP is finite.

open_problem | open | MathDB 页面 | 原文线索

V2-077425 | MATHDB-398923

Finiteness of outcomes of the threefold MMP with scaling

摘录 / EXCERPT: Suppose that Y is a terminal projective threefold. Then the number of minimal models X_i that can be the outcome of the K_Y-MMP with scaling by an ample divisor H on Y is...

open_problem | open | MathDB 页面 | 原文线索

V2-077426 | MATHDB-398927

Mori dream spaces among very general hypersurfaces in $\mathbb{P}^1\times\mathbb{P}^n$

摘录 / EXCERPT: Let X be a very general hypersurface of bidegree (d,e) in $\mathbb{P}^1\times\mathbb{P}^n$, and let $H_i=\operatorname{pr}_i^*\mathcal{O}(1)$. Then X is a Mori dream space if...

open_problem | open | MathDB 页面 | 原文线索

V2-077427 | MATHDB-398931

The effect of fattening in higher-dimensional projective space

摘录 / EXCERPT: Let Z be a finite set of points in projective space $\\text{P}^n$ and let $I=I_Z$ be the radical ideal defining Z. If ... then either $\alpha(I)=1$, i.e. Z is contained in a si...

open_problem | open | MathDB 页面 | 原文线索

V2-077428 | MATHDB-398933

Chain-model identification for the universal operations in the cyclic Deligne setting

摘录 / EXCERPT: Prove that the universal-operad construction for cyclic operads with a cyclic multiplication-an element of $O(2)$ inducing an associative, cyclically invariant multiplication-gives...

open_problem | open | MathDB 页面 | 原文线索

V2-077429 | MATHDB-398935

Framed-[U+007F]E\5f2 action compatible with the double cobar construction

摘录 / EXCERPT: Can one find an fE5f2 operad, equivalent to chains on the framed little-disks operad, that acts on fOmega S(Omega X) and S(Omega^2X) so that these are all equivalent as f5fE5f2-alg...

open_problem | open | MathDB 页面 | 原文线索

V2-077430 | MATHDB-398936

Bijective interpretation of the map-counting recurrence

摘录 / EXCERPT: Find a bijective interpretation of the displayed recurrence for the numbers $M^g_{i,j}$ of rooted maps of genus g with i vertices and j faces: ... for $i,j\ge1$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-077431 | MATHDB-398937

Fischer-Riegler multivariate Laurent polynomial identity

摘录 / EXCERPT: For integers $s,t\ge 0$, define the rational function ... Let ... If $s\le t$, then for every $i\in\{1,2,\ldots,s+t-1\}$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-077432 | MATHDB-398938

Refined enumeration of vertically symmetric alternating sign matrices

摘录 / EXCERPT: Conjecture 2 (Fischer). The number of $(2n+1)\times(2n+1)$ vertically symmetric alternating sign matrices whose first 1 in the second row is in column i is ...

open_problem | open | MathDB 页面 | 原文线索

V2-077433 | MATHDB-398939

Four-variable symmetry of the generalized q,t-Schröder polynomial

摘录 / EXCERPT: Let ... where the sum is over Dyck paths π from $(0,0)$ to (n,n), $a_i=a_i(\pi)$ is the number of squares in row i to the right of π and left of $x=y$,...

open_problem | open | MathDB 页面 | 原文线索

V2-077434 | MATHDB-398940

Parking-function formula for the Macdonald-polynomial Delta operator

摘录 / EXCERPT: Conjecture. For any integer k with $0\le k\le n$, ... Here $\operatorname{PF}(\pi)$ is the set of parking functions associated with the Dyck path π, and F denotes the quasisy...

open_problem | open | MathDB 页面 | 原文线索

V2-077435 | MATHDB-398941

Classification of D-finiteness for three-dimensional octant-walk generating functions

摘录 / EXCERPT: For every step set $S\subseteq\{-1,0,1\}^3\setminus\{(0,0,0)\}$, let ... where $a_{n,i,j,k}$ is the number of length-n lattice walks that start at $(0,0,0)$, remain in the nonnegat...

open_problem | open | MathDB 页面 | 原文线索

V2-077436 | MATHDB-398942

Scaling limit of uniform random plane quadrangulations

摘录 / EXCERPT: There exists a random compact metric space $\mathbf S$, called the Brownian map, such that, if Q_n denotes a uniform random rooted quadrangulation with n faces, then ... where th...

open_problem | open | MathDB 页面 | 原文线索

V2-077437 | MATHDB-398943

Volume of the signed Chan-Robbins-Yuen polytope

摘录 / EXCERPT: Conjecture (Mészáros-Morales): $\operatorname{vol}(\mathrm{CRY}^{\pm}_n)=2^{(n-1)^2}\,\operatorname{Cat}_1\operatorname{Cat}_2\cdots\operatorname{Cat}_{n-1}$.

open_problem | open | MathDB 页面 | 原文线索

V2-077438 | MATHDB-398945

Deciding whether a permutation is sortable by two stacks in series

摘录 / EXCERPT: Given a permutation $\sigma=\sigma_1\cdots\u0001sigma_n$ of $[1..n]$, read its entries from σ_1 to σ_n as input to two stacks H and V in series. The allowed opera...

open_problem | open | MathDB 页面 | 原文线索

V2-077439 | MATHDB-398946

Enumeration of labelled 4-regular planar graphs

摘录 / EXCERPT: Enumerate labelled 4-regular planar graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-077440 | MATHDB-398947

Growth constant for total walks in a proper step set

摘录 / EXCERPT: Let $S\subset\mathbb{Z}^d$ be a proper step set, meaning that: (1) S is not contained in a linear hyperplane; (2) S is not contained in a half-space u^{-} for any...

open_problem | open | MathDB 页面 | 原文线索

V2-077441 | MATHDB-398948

Stanley’s Conjecture 2 on maximum compatible families

摘录 / EXCERPT: For integers $1\le i<j<k\le n$ and $1\le a<b<c\le n$, call $\{i,j,k\}$ and $\{a,b,c\}$ compatible if there exist integers $x_1<x_2<\cdots<x_n$ such that x_i,x_j,x_k and...

conjecture | open | MathDB 页面 | 原文线索

V2-077442 | MATHDB-398949

Enumeration of pairwise compatible triples

摘录 / EXCERPT: Let $\mathcal M_n$ be the collection of all subsets S of $\binom{[n]}{3}$ such that any two elements of S are compatible, where $\{i,j,k\}$ and $\{a,b,c\}$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-077443 | MATHDB-398950

Equality of floor and ceiling distributions in the k-Catalan arrangement

摘录 / EXCERPT: For a crystallographic root system Φ with ambient space V, a chosen simple system and positive system Φ^+, and a positive integer k, consider the k-Catalan arrangem...

open_problem | open | MathDB 页面 | 原文线索

V2-077444 | MATHDB-398951

Bijection between floor- and ceiling-constrained dominant regions

摘录 / EXCERPT: Let Φ be a crystallographic root system with ambient space V, simple system S, and associated positive system Φ^+. For a positive integer k, let the k-Catalan arr...

open_problem | open | MathDB 页面 | 原文线索

V2-077445 | MATHDB-398953

Expected length of the longest k-alternating subsequence

摘录 / EXCERPT: For each permutation $w\in S_n$, let $\operatorname{as}_k(w)$ be the length of the longest sequence $w_{i_1}>w_{i_2}<w_{i_3}>w_{i_4}<\cdots$ such that $|w_{i_j}-w_{i_{j+1}}|\ge k$...

open_problem | open | MathDB 页面 | 原文线索

V2-077446 | MATHDB-398954

Expected fragment size for uniformly random unlabelled graphs in a decomposable bridge-addable class

摘录 / EXCERPT: There is a constant c such that, if $\widetilde{\mathcal A}$ is decomposable and bridge-addable and $\widetilde R_n\in_u\widetilde{\mathcal A}$, then...

open_problem | open | MathDB 页面 | 原文线索

V2-077447 | MATHDB-398955

Connectivity of random unlabelled bridge-addable graph classes

摘录 / EXCERPT: There is a constant $\delta>0$ such that, if $\tilde A$ is bridge-addable and $\tilde R_n\in_u\u0001tilde A$ is sampled uniformly from the unlabelled graphs in $\tilde A$ on n ve...

open_problem | open | MathDB 页面 | 原文线索

V2-077448 | MATHDB-398956

Connectivity for an unlabelled bridge-addable graph class

摘录 / EXCERPT: Let $\mathcal A$ be a class of graphs that is bridge-addable, meaning that whenever $G\in\mathcal A$ has vertices in distinct components, adding an edge between them produces a gra...

open_problem | open | MathDB 页面 | 原文线索

V2-077449 | MATHDB-398957

\(a+1)-positivity of the generating polynomials for quarter-plane loops

摘录 / EXCERPT: Consider square-lattice loops of length $2n$ that start and end at the origin and remain in the nonnegative quadrant. Let $Q_n(a)$ be the polynomial counting these loops by the num...

open_problem | open | MathDB 页面 | 原文线索

V2-077450 | MATHDB-398958

Reduction of degree in the coefficients of a generating function

摘录 / EXCERPT: Let $[y]_k=y(y-1)\cdots(y-k+1)$, and let $A_t(y,z)$ be the coefficient of x^t in ... Clearly, A_t has total degree at most $2t$. Show that A_t has total degree at most $t+1$ fo...

open_problem | open | MathDB 页面 | 原文线索

V2-077451 | MATHDB-398960

Narayana numbers in an identity involving semistandard Young tableaux

摘录 / EXCERPT: For integers $m\ge r\ge l\ge 1$ and $1\le \beta<m$, let $\operatorname{SSYT}_{r,l}(m)$ denote the number of semistandard Young tableaux of shape consisting of two columns of length...

open_problem | open | MathDB 页面 | 原文线索

V2-077452 | MATHDB-398961

An open problem on strongly consistent learning of the best predictor for Gaussian processes

摘录 / EXCERPT: For stationary Gaussian time series: “Is it possible to learn the best predictor almost surely in a strongly consistent way?”

open_problem | open | MathDB 页面 | 原文线索

V2-077453 | MATHDB-398962

Fast prediction rates for the Lasso beyond extreme designs

摘录 / EXCERPT: How far from these two extreme cases can a design be such that it still permits fast rates of prediction with the Lasso?

open_problem | open | MathDB 页面 | 原文线索

V2-077454 | MATHDB-398963

The restricted eigenvalue assumption under weak moment assumption

摘录 / EXCERPT: The report states the following result. There are absolute constants c_0,c_1,c_2 such that, for every $\b1\ge \tfrac12$, there is a constant $c_3(\alpha)$ depending only on...

open_problem | open | MathDB 页面 | 原文线索

V2-077455 | MATHDB-398964

Sharp full adaptation to unknown smoothness and ellipsoid size

摘录 / EXCERPT: In the Gaussian white-noise sequence model defined on page 35, let $0<\beta_1<\beta_2$ and $0<M_1<M_2<\infty$ be fixed, and let D be arbitrary. For $\beta\in[\beta_1,\beta_2]$ an...

open_problem | open | MathDB 页面 | 原文线索

V2-077456 | MATHDB-398966

Whether minimax regression, statistical learning, and online regression differ in rate

摘录 / EXCERPT: Let $\mathcal{X}$ be a set and let $\mathcal{F}$ be a class of functions from $\mathcal{X}$ to $[-1,1]$. The report considers the minimax risk $W_n(\mathcal{F})$ for correctly spec...

open_problem | open | MathDB 页面 | 原文线索

V2-077457 | MATHDB-398967

Simultaneous adaptation over all L^p-norms

摘录 / EXCERPT: Is it possible to adapt over all L^p-norms simultaneously?

open_problem | open | MathDB 页面 | 原文线索

V2-077458 | MATHDB-398968

Parametric-rate estimation of an individual precision-matrix entry

摘录 / EXCERPT: Let ... where $M>1$ is constant. If $X^{(1)},\ldots,X^{(n)}$ are i.i.d. $N_p(\mu,\Sigma)$ observations and $\Omega=\Sigma^{-1}$, then the report states that the necessary and suffici...

open_problem | open | MathDB 页面 | 原文线索

V2-077459 | MATHDB-398969

Lions’ maximal regularity problem for symmetric non-autonomous forms

摘录 / EXCERPT: Assume that the bounded coercive non-autonomous form satisfies ... Does the Cauchy problem ... have maximal regularity in H? In the formulation introduced immediately beforehand, thi...

open_problem | open | MathDB 页面 | 原文线索

V2-077460 | MATHDB-398974

Long-time stability of plane waves for the split-step Fourier method

摘录 / EXCERPT: For the cubic nonlinear Schrödinger equation on the torus, the split-step Fourier method-Fourier collocation in space combined with Strang splitting in time-can reproduce the long-...

open_problem | open | MathDB 页面 | 原文线索

V2-077461 | MATHDB-398975

Critical small-data well-posedness and scattering for the cubic Dirac equation

摘录 / EXCERPT: For $M>0$, consider the cubic Dirac equation ... for a spinor field $\psi:\mathbb{R}^{1+3}\to\mathbb{C}^4$, with summation over $\mu=0,\dots,3$, where the γ^μ are Dirac matr...

open_problem | open | MathDB 页面 | 原文线索

V2-077462 | MATHDB-398976

Exponential decay of dispersion-management solitons

摘录 / EXCERPT: Let $f\in L^2(\mathbb{R})$ be a weak solution of ... with $d_{\mathrm{av}}=0$ and the model dispersion profile $d_0=\mathbf{1}_{[0,1)}-\mathbf{1}_{[1,2)}$, where ... Then there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-077463 | MATHDB-398980

Existence of p-adic origamis associated with a given origami

摘录 / EXCERPT: Given an origami O, does $C_{O,p}=C_{O,0}\otimes\overline{\mathbb{Q}}_p$ intersect the locus of Mumford curves in $M_g(\overline{\mathbb{Q}}_p)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077464 | MATHDB-398983

Mirzakhani’s arithmeticity conjecture

摘录 / EXCERPT: If an orbit closure M has higher rank, i.e. $\operatorname{rank}(M)>1$, then M is arithmetic, i.e. $k(M)=\mathbb{Q}$, where $k(M)$ is the affine field of definition.

conjecture | open | MathDB 页面 | 原文线索

V2-077465 | MATHDB-398986

Classification conjecture for arithmetic Teichmüller curves in $\Omega\mathcal{M}_2(1,1)$

摘录 / EXCERPT: Conjecturally, the degree d of a minimal map to a torus, the torsion order M of the difference of the images of the two ramification points in the elliptic curve, and the spin...

conjecture | open | MathDB 页面 | 原文线索

V2-077466 | MATHDB-398987

Classification of Shimura curves in the genus-3 hyperelliptic Jacobian locus

摘录 / EXCERPT: Conjecture: The Shimura curves contained in the locus of hyperelliptic Jacobians of genus 3 are precisely the curves described in Theorem 1: for each fixed $u\in\mathbb{Q}[i]$, t...

open_problem | open | MathDB 页面 | 原文线索

V2-077467 | MATHDB-398988

Irreducibility of the arithmetic Teichmüller-curve locus with M=1

摘录 / EXCERPT: For odd d, is the union of Teichmüller curves in $\Omega\mathcal M_2(1,1)$ with invariants d, ε, and $M=1$ irreducible?

open_problem | open | MathDB 页面 | 原文线索

V2-077468 | MATHDB-398991

Asymptotic normality of the Hultman numbers

摘录 / EXCERPT: As $n\to\infty$, the numbers $h_{n,k}$ become normally distributed in k, with both mean and variance equal to $\log n$.

open_problem | open | MathDB 页面 | 原文线索

V2-077469 | MATHDB-398992

Lyapunov polygon versus Harder-Narasimhan polygon on Teichmüller curves

摘录 / EXCERPT: Conjecture. For any Teichmüller curve in the stratum under consideration, the polygon of Lyapunov exponents lies below the Harder-Narasimhan polygon. Equivalently, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-077470 | MATHDB-398993

Lyapunov polygon versus the Harder-Narasimhan polygon on Teichmüller curves

摘录 / EXCERPT: Let C be a Teichmüller curve in a stratum of Abelian differentials of genus g, with Lyapunov exponents $1=\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_g\geq0$ and normalized Har...

open_problem | open | MathDB 页面 | 原文线索

V2-077471 | MATHDB-398995

Character rigidity for commensurators

摘录 / EXCERPT: Let Λ be an irreducible lattice in a product $G\times H$, where G is a non-compact simple Lie group with property (T) and H is a totally disconnected, non-discrete si...

open_problem | open | MathDB 页面 | 原文线索

V2-077472 | MATHDB-398998

Existence of switched systems requiring arbitrarily complex Lyapunov functions

摘录 / EXCERPT: For every positive integer d, there are families of switched linear systems-in fixed dimension and defined by two matrices only-that are stable under arbitrary switching but do n...

open_problem | open | MathDB 页面 | 原文线索

V2-077473 | MATHDB-399000

The rational finiteness conjecture for the joint spectral radius

摘录 / EXCERPT: Let $A_1,\ldots,A_m$ be real $n\times n$ matrices with rational entries, and let $\rho(A_1,\ldots,A_m)$ denote their joint spectral radius. Is it always true that there exist a fin...

conjecture | open | MathDB 页面 | 原文线索

V2-077474 | MATHDB-399002

Conjecture on the maximizer of the largest zero of a mixed characteristic polynomial

摘录 / EXCERPT: In Problem 0.4, the maximal zero is achieved for vectors that are all in the interior of Λ^+, or identically zero. Here Problem 0.4 asks to maximize the largest zero of...

conjecture | open | MathDB 页面 | 原文线索

V2-077475 | MATHDB-399003

Definite symmetric determinantal representations of ternary hyperbolic polynomials

摘录 / EXCERPT: Let $h\in\mathbb{R}[X,Y,Z]$ be homogeneous of degree d and hyperbolic with respect to a direction $e\in\mathbb{R}^3$, meaning that $h(e)>0$ and, for every $a\in\mathbb{R}^3$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-077476 | MATHDB-399004

General moment vanishing problem for polynomial moments

摘录 / EXCERPT: Given a bounded domain $\Omega\subset\mathbb{R}^n$ and polynomials P,q in $x\in\mathbb{R}^n$, determine necessary and sufficient conditions on Ω,P,q for ...

open_problem | open | MathDB 页面 | 原文线索

V2-077477 | MATHDB-399005

Efficient implementations of critical-point methods for real-world problems

摘录 / EXCERPT: “It has been a long-standing problem to obtain efficient implementations for real-world problems based on critical point methods. Indeed, it requires improving the exponents in the...

open_problem | open | MathDB 页面 | 原文线索

V2-077478 | MATHDB-399006

Lower-bound conjecture for \(\alpha(k)\)

摘录 / EXCERPT: We conjecture that $\alpha(k)\ge k^2+1$ for all $k\ge 3$.

conjecture | open | MathDB 页面 | 原文线索

V2-077479 | MATHDB-399008

The autoregressive filter problem for two variables

摘录 / EXCERPT: Given $c_k=c_{-k}\in\mathbb C$ for $k=(k_1,k_2)\in\Lambda:=\{0,\ldots,n\}\times\{0,\ldots,m\}$, find a polynomial ... such that p has no zeros on the closed bidisk...

open_problem | open | MathDB 页面 | 原文线索

V2-077480 | MATHDB-399009

Horn's additive eigenvalue characterization

摘录 / EXCERPT: For a positive integer n, a triple of nonincreasing real sequences $\alpha=(\alpha_1,\ldots,\alpha_n)$, $\beta=(\beta_1,\ldots,\beta_n)$, and $\gamma=(\gamma_1,\ldots,\gamma_n)$...

open_problem | open | MathDB 页面 | 原文线索

V2-077481 | MATHDB-399010

Multiplicative Horn problem for singular numbers of three invertible matrices

摘录 / EXCERPT: For decreasing, strictly positive sequences of length n, α, β, and γ, there exist matrices $A,B,C\in M_n(\mathbb C)$ whose singular numbers are α,...

open_problem | open | MathDB 页面 | 原文线索

V2-077482 | MATHDB-399012

Degree bounds for rational sum-of-squares multipliers on curves

摘录 / EXCERPT: Given an even degree $2s$, find an integer $k\in\mathbb{N}$ depending only on s and invariants of an irreducible, non-degenerate real projective curve $X\subset\mathbb{P}^n$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-077483 | MATHDB-399016

Finite-dimensional optimizers for noncommutative trace optimization

摘录 / EXCERPT: Given symmetric noncommutative polynomials $p,g_1,\dots,g_r\in\mathbb{R}\langle X\rangle$, consider the trace-optimization problem $p_{\min}=\inf_{(\mathcal{N},\tau),A}\tau(p(A))$...

open_problem | open | MathDB 页面 | 原文线索

V2-077484 | MATHDB-399017

Determining the Theta rank of finite point configurations

摘录 / EXCERPT: Determine the Theta rank of finite point configurations in general, and understand point configurations with given or bounded Theta rank.

open_problem | open | MathDB 页面 | 原文线索

V2-077485 | MATHDB-399018

Stability of PhaseLift under noisy random coded diffraction measurements

摘录 / EXCERPT: Prove stability guarantees for PhaseLift in the presence of noise for certain measurement setups, most notably random coded diffraction patterns.

open_problem | open | MathDB 页面 | 原文线索

V2-077486 | MATHDB-399019

Explicit restricted-isometry constructions comparable to random matrices

摘录 / EXCERPT: Find explicit constructions of sensing matrices satisfying the restricted isometry property whose performance is comparable to that of known random constructions.

open_problem | open | MathDB 页面 | 原文线索

V2-077487 | MATHDB-399020

Generalization of the first theta-body relaxation to arbitrary-sized matrices

摘录 / EXCERPT: Does the first theta body coincide with the nuclear-norm unit ball for matrices of arbitrary size, as the report states it does for 2×2 matrices?

open_problem | open | MathDB 页面 | 原文线索

V2-077488 | MATHDB-399021

Existence of a cyclic-group window attaining the support-pair correspondence

摘录 / EXCERPT: For cyclic G, does there exist $\phi \in \mathbb{C}^{G}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-077489 | MATHDB-399023

DTI optimization and the value of Q-ball modelling

摘录 / EXCERPT: What advantage would Q-ball modelling bring, and is it worth the additional effort?

open_problem | open | MathDB 页面 | 原文线索

V2-077490 | MATHDB-399024

Comparison of cancer-cell movement strategies

摘录 / EXCERPT: Comparison of movement strategies, individual amoeboid, individual mesenchymal, collective. Which strategy is advantageous in what situations?

open_problem | open | MathDB 页面 | 原文线索

V2-077491 | MATHDB-399025

Bounds for a haptotaxis-chemotaxis model

摘录 / EXCERPT: For the haptotaxis-chemotaxis system with tissue remodelling ... is the cancer-cell density bounded in a bounded two-dimensional domain, i.e. does...

open_problem | open | MathDB 页面 | 原文线索

V2-077492 | MATHDB-399028

Frobenius-Schur indicators of Drinfel’d doubles of finite simple groups

摘录 / EXCERPT: Is every finite simple group G such that all Frobenius-Schur indicator values $ u_n(V)$ for representations V of the Drinfel’d double $D(G)$ are integers?

open_problem | open | MathDB 页面 | 原文线索

V2-077493 | MATHDB-399046

Converse to the weighted matricial-function theorem

摘录 / EXCERPT: Fix an admissible sequence X and its associated sequence of weights Z. For each normal representation $\sigma\in\mathrm{NRep}(M)$, let $D_{X,\sigma}$ be the corresponding domai...

open_problem | open | MathDB 页面 | 原文线索

V2-077494 | MATHDB-399050

Existence of the Shilov boundary ideal for an operator system

摘录 / EXCERPT: Let S be an operator system and let $A=C^*(S)$. A closed two-sided ideal I in A is called a boundary ideal if the quotient map $A\to A/I$ is completely isometric on S. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-077495 | MATHDB-399051

Arveson’s noncommutative Choquet-boundary conjecture

摘录 / EXCERPT: Let S be an operator system, let $A=C^*(S)$, and let a boundary representation of S mean an irreducible representation $\pi:A\to B(H)$ such that $\pi|_S$ has a unique completel...

conjecture | open | MathDB 页面 | 原文线索

V2-077496 | MATHDB-399054

Characterization of Carleson measures for model spaces K_{ϑ}

摘录 / EXCERPT: The report presents the following established theorem of Lacey, Sawyer, Shen, Uriarte-Tuero, and Wick. Let μ be a nonnegative Borel measure supported on $\mathbb D$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-077497 | MATHDB-399057

Characterization of multipliers for the Drury-Arveson space

摘录 / EXCERPT: Let [?1m[?1m$\mathbb B$ be the open unit ball in $\mathbb C^n$, with $n\ge 2$, and let H^2_n be the Drury-Arveson space with reproducing kernel...

open_problem | open | MathDB 页面 | 原文线索

V2-077498 | MATHDB-399058

Dilation of all tetrablock contractions

摘录 / EXCERPT: Is it the case that all tetrablock contractions dilate?

open_problem | open | MathDB 页面 | 原文线索

V2-077499 | MATHDB-399059

Exhaustion of rational matrix-inner functions by unitary colligations

摘录 / EXCERPT: Conjecture. To obtain all rational inner functions of matrix arguments $\operatorname{Mat}(m)\to\operatorname{Mat}(m\alpha)$, it is sufficient to eliminate x from the equation ……

open_problem | open | MathDB 页面 | 原文线索

V2-077500 | MATHDB-399060

Arthur conjecture for the discrete spectrum of \(\mathrm{Mp}_{2n}\)

摘录 / EXCERPT: Fix a nontrivial additive automorphic character ψ. Let … be a global discrete A-parameter for Mp_{2n} (equivalently, for SO_{2n+1}), where the Π_i...

conjecture | open | MathDB 页面 | 原文线索

V2-077501 | MATHDB-399063

Half-integral-weight lift and its Hecke and L-function properties

摘录 / EXCERPT: Assume that k is even, let $f\in S_{2k-2n}(\Gamma_1)$ be a Hecke eigenform, and let $I(f)\in S_k(\Gamma_{2n})$ be its Ikeda lift. Let Φ_1 be the first Fourier-Jacobi coeffic...

open_problem | open | MathDB 页面 | 原文线索

V2-077502 | MATHDB-399065

Regularized pairing with a meromorphic quadratic-form Poincaré series

摘录 / EXCERPT: The report answers the question of integrating a weight-k meromorphic cusp form against the meromorphic form $f_{k,-D,[Q]}$. For $D>0$, if $[Q]$ is a class of binary quadratic form...

open_problem | open | MathDB 页面 | 原文线索

V2-077503 | MATHDB-399066

Bloch-Kato conjecture for polarized pure motives of weight \

摘录 / EXCERPT: Let K be a number field, and let M be a Chow motive over K with rational coefficients, equipped with a polarization $M\times M^{\vee}\to\mathbb{Q}(1)$ and of pure weight -1...

conjecture | open | MathDB 页面 | 原文线索

V2-077504 | MATHDB-399068

Characterization of poles of symmetric-square L-functions

摘录 / EXCERPT: Let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb A)$ with central character ω_π. The function $L(s,\pi,\operatorname{sym}^2)$ has...

open_problem | open | MathDB 页面 | 原文线索

V2-077505 | MATHDB-399069

Whittaker-Fourier coefficients of cuspidal representations of the metaplectic group: Conjecture 1

摘录 / EXCERPT: For any elliptic Arthur parameter ϕ of Ramanujan type, one has …

conjecture | open | MathDB 页面 | 原文线索

V2-077506 | MATHDB-399070

Whittaker-Fourier coefficient conjecture for metaplectic descents

摘录 / EXCERPT: Conjecture 3. Assume that σ is the ψ_N-descent of $\{\pi_1,\ldots,\pi_k\}$ to $\widetilde{\mathrm{Sp}}_n$. Let π be the isobaric sum...

conjecture | open | MathDB 页面 | 原文线索

V2-077507 | MATHDB-399072

Finiteness of simple even lattices of signature $(2,n)$

摘录 / EXCERPT: For every integer $n\ge 1$, there exist only finitely many isomorphism classes of even simple lattices L of signature $(2,n)$. Here L is simple when the space $S_{1+n/2,L'/L}$...

open_problem | open | MathDB 页面 | 原文线索

V2-077508 | MATHDB-399074

Diffusion bridge problem for the Kyle-Glosten-Milgrom models

摘录 / EXCERPT: Suppose that $Z_t=Z_0+\a0\int_0^t\sigma(s)a(Z_s)\,dB_s$ is a diffusion, where a is a sufficiently regular function and σ is deterministic. Let W be another Brownian moti...

open_problem | open | MathDB 页面 | 原文线索

V2-077509 | MATHDB-399075

Optimal investment in a Black-Scholes model with a bubble

摘录 / EXCERPT: Let the bond be normalized to $B_t\\equiv 1$, and let the stock satisfy ... where $\\mu,\\sigma>0$, W is a Brownian motion, and $M^G\\cdot\\phi$ is a finite-variation martingale wi...

open_problem | open | MathDB 页面 | 原文线索

V2-077510 | MATHDB-399076

Existence and uniqueness for the infinite-jump BSDE

摘录 / EXCERPT: Let N be a non-explosive point process with jump times $S_1,S_2,\ldots$, continuous predictable compensator A, and filtration generated by N. Assume...

open_problem | open | MathDB 页面 | 原文线索

V2-077511 | MATHDB-399078

Hierarchical a posteriori error estimation for an inverse-square Schrödinger operator

摘录 / EXCERPT: For the operator $-\nabla^2+c^2/r^2$ on $\Omega\subset\R^2$, with $c\ge0$, $r=|x|$, and the origin contained in Ω, the report considers a source problem with solution u. L...

open_problem | open | MathDB 页面 | 原文线索

V2-077512 | MATHDB-399085

Serre functors and derived equivalences for hearts of t-structures

摘录 / EXCERPT: Let $\mathcal{A}$ be an abelian category such that $D^b\mathcal{A}$ admits a Serre functor $S:D^b\mathcal{A}\to D^b\mathcal{A}$, and let $(\mathcal{U},\mathcal{V})$ be a t-struct...

open_problem | open | MathDB 页面 | 原文线索

V2-077513 | MATHDB-399086

Perfect groups with vanishing stable commutator length and stably unbounded norms

摘录 / EXCERPT: Does there exist a perfect group whose stable commutator length is zero, but which admits a stably unbounded conjugation-invariant norm?

open_problem | open | MathDB 页面 | 原文线索

V2-077514 | MATHDB-399090

Alternative volume conjecture for knots

摘录 / EXCERPT: Conjecture 1. Let K be a knot. Then ... where $\operatorname{Vol}(S^3\setminus K)$ is the simplicial volume.

conjecture | open | MathDB 页面 | 原文线索

V2-077515 | MATHDB-399091

Entropy conjecture for pseudo-Anosov braids

摘录 / EXCERPT: Let β be a pseudo-Anosov n-braid whose closure is a knot. Then ... Moreover, if the associated invariant measured foliation is transversely orientable, then already ...

conjecture | open | MathDB 页面 | 原文线索

V2-077516 | MATHDB-399092

Li-Roberts conjecture on taut-foliation slopes

摘录 / EXCERPT: For every non-trivial knot $K\subset S^3$, the interval of slopes $(-a,b)$, with $a,b>0$, for which the exterior of K admits a co-orientable taut foliation meeting the boundary...

conjecture | open | MathDB 页面 | 原文线索

V2-077517 | MATHDB-399093

Classification of pass classes of virtual knots

摘录 / EXCERPT: Classify virtual knots under band-pass (pass) equivalence; equivalently, determine the pass classes of virtual knots.

open_problem | open | MathDB 页面 | 原文线索

V2-077518 | MATHDB-399094

Pattern parities for 2-knots and cobordism invariants

摘录 / EXCERPT: Are there any parities coming from patterns which can be extended to 2-knots and lead to cobordism invariants?

open_problem | open | MathDB 页面 | 原文线索

V2-077519 | MATHDB-399101

Conjecture 17 on common zeros of exponential sums

摘录 / EXCERPT: Let f,g be ESs of one variable without nontrivial common divisor in the ring of ES. Then the set of zeros of the system $f=g=0$ is finite.

conjecture | open | MathDB 页面 | 原文线索

V2-077520 | MATHDB-399104

Fixed-energy universality for Wigner matrices

摘录 / EXCERPT: For Wigner matrices with subgaussian entries, for either the real symmetric or complex Hermitian symmetry class, any $k\ge 1$, any continuous compactly supported function...

open_problem | open | MathDB 页面 | 原文线索

V2-077521 | MATHDB-399106

Distribution of the particle location in q-TASEP

摘录 / EXCERPT: A natural question for q-TASEP is to compute the distribution of $x_n(t)$, the location of particle n at time t.

open_problem | open | MathDB 页面 | 原文线索

V2-077522 | MATHDB-399107

Linearity of the shape function near the origin in a product-type random environment

摘录 / EXCERPT: For the one-dimensional product-type environment $V(x,i)=F(x)b(i)$, where $\{F(x):x\in\mathbb Z\}$ and $\{b(i):i\in\mathbb Z\}$ are independent i.i.d. fields, the law of F has a...

open_problem | open | MathDB 页面 | 原文线索

V2-077523 | MATHDB-399108

Nonidentity of annihilating Brownian motion and large-\tN Ginibre eigenvalue evolutions

摘录 / EXCERPT: The source asks whether annihilating Brownian motions (ABM) and the evolution of real eigenvalues of the real matrix-valued Ginibre process become identical as processes in the lar...

open_problem | open | MathDB 页面 | 原文线索

V2-077524 | MATHDB-399112

Gromov’s surface subgroup conjecture

摘录 / EXCERPT: Every one-ended hyperbolic group contains a surface group, i.e. a subgroup isomorphic to the fundamental group of a closed oriented surface of genus $g\ge 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-077525 | MATHDB-399114

Polynomial relation between polyhedron volume and areas of two-dimensional faces

摘录 / EXCERPT: Given a polyhedron, does there exist a relation ... where each $b_j(A)$ is a polynomial in the squares of the areas of the two-dimensional faces of the polyhedron?

open_problem | open | MathDB 页面 | 原文线索

V2-077526 | MATHDB-399124

Upscaled Cahn-Hilliard equation in a periodically perforated medium

摘录 / EXCERPT: The report states the following main result. Let the initial datum satisfy $\psi(x)\in H^2_E(\Omega)$, and assume that the chemical potential $\mu_0=\nabla\phi_E(\phi_0)$ is scale...

open_problem | open | MathDB 页面 | 原文线索

V2-077527 | MATHDB-399125

Computational verification of molecular-dynamics accuracy without solving the Schrödinger equation

摘录 / EXCERPT: How can the accuracy of molecular-dynamics simulations be verified computationally without solving a Schrödinger equation?

open_problem | open | MathDB 页面 | 原文线索

V2-077528 | MATHDB-399126

Do the Faber-Zagier relations span all relations in the tautological ring?

摘录 / EXCERPT: The Faber-Zagier relations span all the relations in the tautological ring of $\mathcal M_g$.

open_problem | open | MathDB 页面 | 原文线索

V2-077529 | MATHDB-399129

Rank-two Ulrich bundles on linearly normal K3 surfaces

摘录 / EXCERPT: Let $S\subset\mathbb{P}^N$ be a linearly normal K3 surface such that the Clifford index of cubic sections is computed by $\mathcal{O}_S(1)$. Then S carries a $(2N+8)$-dimensional...

open_problem | open | MathDB 页面 | 原文线索

V2-077530 | MATHDB-399130

Absolute continuity of the maximal-entropy measure implies the Kummer property

摘录 / EXCERPT: Let X be a smooth complex projective surface and let f be an automorphism of X with positive entropy. Let $\bcm_f$ be the measure of maximal entropy of f. If $\bcm_f$ is ab...

open_problem | open | MathDB 页面 | 原文线索

V2-077531 | MATHDB-399133

Higher-rank interpolation for sheaves on \(\mathbb{P}^2\)

摘录 / EXCERPT: Let ξ be the Chern character of positive rank of a stable sheaf on $\mathbb{P}^2$, and let $F\in M(\xi)$. A sheaf E satisfies interpolation with respect to F if...

open_problem | open | MathDB 页面 | 原文线索

V2-077532 | MATHDB-399139

Dimension of the base of the maximal rationally connected fibration

摘录 / EXCERPT: Conjecture 3. The maximal rationally connected fibration of $\widetilde{\mathcal M}^{4}_{15,16}$ has a three-dimensional base.

open_problem | open | MathDB 页面 | 原文线索

V2-077533 | MATHDB-399141

Rational points of derived-equivalent twisted K3 surfaces

摘录 / EXCERPT: Suppose that (X,α) and (Y,β) are derived-equivalent twisted K3 surfaces over a field F. Is ... Here $(X,\alpha)(F)=\{x\in X(F):\alpha|_x=0\in\operatorname{Br}(F)\}$, a...

open_problem | open | MathDB 页面 | 原文线索

V2-077534 | MATHDB-399142

Equivariant Birch-Swinnerton-Dyer conjecture for Hasse-Weil-Artin L-functions

摘录 / EXCERPT: For an elliptic curve $E/\mathbb{Q}$ and an Artin representation $\rho:G_{\mathbb{Q}}\to\operatorname{Aut}(V_\rho)\simeq\mathrm{GL}_n(\mathbb{C})$, let H be a finite extension of...

conjecture | open | MathDB 页面 | 原文线索

V2-077535 | MATHDB-399144

Explicit reciprocity laws for Euler systems

摘录 / EXCERPT: One expects that the classes in the Euler system should be related to values of the L-function $L(V,s)$, via an explicit reciprocity law describing the image of the Euler system cl...

open_problem | open | MathDB 页面 | 原文线索

V2-077536 | MATHDB-399145

Vanishing of the generalized Kato class beyond analytic rank two

摘录 / EXCERPT: Let $E/\mathbb{Q}$ be an elliptic curve and let $\rho=\rho_1\otimes\rho_2$, where ρ_1 and ρ_2 are odd, irreducible, two-dimensional Artin representations of...

open_problem | open | MathDB 页面 | 原文线索

V2-077537 | MATHDB-399150

Whether the patched construction gives a purely local p-adic Langlands correspondence

摘录 / EXCERPT: For a finite extension $F/\mathbf{Q}_p$, is the construction $r\mapsto V(r)$ a purely local correspondence for $\mathrm{GL}_n(F)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077538 | MATHDB-399152

Kudla’s modularity conjecture for special cycles

摘录 / EXCERPT: Let (V,Q) be a quadratic space over $\mathbb{Q}$ of signature $(n,2)$, let $L\subset V$ be an even unimodular lattice, let $\Gamma\subset\mathrm{SO}(L)$ be a finite-index subgro...

conjecture | open | MathDB 页面 | 原文线索

V2-077539 | MATHDB-399154

Exactness of the p-adic invariant-cycles sequence for coefficients

摘录 / EXCERPT: Let k be a perfect field of characteristic $p>0$, let $W:=W(k)$ be its ring of Witt vectors, and let K be the fraction field of W. Let X be a semistable curve over W such...

open_problem | open | MathDB 页面 | 原文线索

V2-077540 | MATHDB-399157

Absolute continuity of the measure in a Heisenberg normal current

摘录 / EXCERPT: Let $\xi=X\wedge Y$ be the horizontal non-involutive simple 2-vector field in $H^1\cong\mathbb R^3$, where ... Given a normal current $T=\xi\wedge\mu\in N_2(\mathbb R^3)$, is it po...

open_problem | open | MathDB 页面 | 原文线索

V2-077541 | MATHDB-399159

Bernstein problem for entire minimal graphs

摘录 / EXCERPT: Are affine functions the only entire minimal graphs over $\mathbb{R}^{n-1}$ in $\mathbb{R}^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-077542 | MATHDB-399163

Quantitative Faber-Krahn inequality with Fraenkel asymmetry

摘录 / EXCERPT: For a fixed dimension N, there exists a constant $\c3=\c3(N)>0$ such that every open set $\a9\subset\mathbb{R}^N$ with finite measure satisfies ... where $\bbda(\a9)$ is the first...

open_problem | open | MathDB 页面 | 原文线索

V2-077543 | MATHDB-399164

Existence and uniqueness of fractional Lawson cones

摘录 / EXCERPT: For every $m,n\ge 1$ and $0<s<1$, there exists a unique $\alpha=\alpha(s,m,n)>0$ such that the Lawson cone ... is an s-fractional minimal surface in...

open_problem | open | MathDB 页面 | 原文线索

V2-077544 | MATHDB-399165

Boundary regularity and anisotropic Young’s law for capillarity minimizers

摘录 / EXCERPT: Let $A\subset\mathbb{R}^n$ with $n\ge 2$ and $\partial A\in C^{1,1}$. Let $\Phi:A\times\mathbb{R}^n\to[0,\infty)$ be positively one-homogeneous and convex in its second variable,...

open_problem | open | MathDB 页面 | 原文线索

V2-077545 | MATHDB-399166

Tree-like pseudometric preserving the minimum matching value

摘录 / EXCERPT: Let $n\in\mathbb{N}$, let $X=\{x_1,\ldots,x_{2n}\}$ be equipped with a pseudometric d, and let $\mathcal{M}(X)$ denote the set of partitions of X into n unordered pairs. Defi...

open_problem | open | MathDB 页面 | 原文线索

V2-077546 | MATHDB-399168

Existence of solutions of the type in case ii.b)

摘录 / EXCERPT: Do solutions of the type arising in case ii.b) of Theorem 2 exist?

open_problem | open | MathDB 页面 | 原文线索

V2-077547 | MATHDB-399169

Dimension-free, ball-independent bounds for maximal operators associated with convex bodies

摘录 / EXCERPT: Prove the general case: for every $p>1$, the centered Hardy-Littlewood maximal operators associated with balls in $\mathbb{R}^d$ have strong-type (p,p) bounds whose constants are...

open_problem | open | MathDB 页面 | 原文线索

V2-077548 | MATHDB-399170

Dimension-free weak type $(1,1)$ bounds for the centered maximal operator over Euclidean balls

摘录 / EXCERPT: Let M_d be the centered Hardy-Littlewood maximal operator over Euclidean balls in $\mathbb{R}^d$ with Lebesgue measure, and let $c_{1,d}$ be the optimal constant in the weak typ...

open_problem | open | MathDB 页面 | 原文线索

V2-077549 | MATHDB-399171

Leinster-Willerton convex magnitude conjecture

摘录 / EXCERPT: If $X\subseteq\mathbb{R}^n$ is compact and convex, then $t\mapsto |tX|$ is a polynomial of degree n, and moreover ... where ω_i is the volume of the unit ball in...

conjecture | open | MathDB 页面 | 原文线索

V2-077550 | MATHDB-399175

Two-weight inequality for the Hilbert transform

摘录 / EXCERPT: For two Radon measures σ,w on $\mathbb R$ with no common atoms, meaning $\sigma(\{a\})w(\{a\})=0$ for every $a\in\mathbb R$, the inequality ... holds if and only if, uniformly...

open_problem | open | MathDB 页面 | 原文线索

V2-077551 | MATHDB-399176

Two-weight characterization of the Hilbert transform for general Radon measures

摘录 / EXCERPT: Let c3 and w be Radon measures on $\cmathbb{R}$. The estimate ... holds if and only if, for every interval $I\subset\mathbb{R}$, the local testing conditions ... and the one-sided co...

open_problem | open | MathDB 页面 | 原文线索

V2-077552 | MATHDB-399178

Uniform endpoint constant in the Heisenberg-group inequality (1)

摘录 / EXCERPT: For every smooth, compactly supported function $f\colon\mathbb{R}^{3}\to\mathbb{R}$ and every $p\in(1,2]$, inequality (1) holds: ... The conjecture is that the implicit constant rema...

open_problem | open | MathDB 页面 | 原文线索

V2-077553 | MATHDB-399179

Uniform lower bounds for Bergman kernels of Kähler-Ricci solitons

摘录 / EXCERPT: Let n and F be positive numbers, and let $\mathrm{KR}(n,F)$ be the space of compact n-dimensional Kähler-Ricci solitons (X,ω) with $\mathrm{Fut}(X,\omega)\le F$. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-077554 | MATHDB-399180

Main conjecture for Vinogradov's Mean Value Theorem

摘录 / EXCERPT: For all integers $r,d\ge 1$ and every $\varepsilon>0$, let $J_{r,d}(X)$ be the number of solutions in integers $1\le x_1,\ldots,x_{2r}\le X$ to ... Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-077555 | MATHDB-399181

Fixed-energy inverse scattering in the plane

摘录 / EXCERPT: Given a compactly supported real-valued potential $V:\mathbb{R}^{2}\to\mathbb{R}$ and a fixed frequency $k>0$, let $u(\cdot,\theta)$ be the outgoing solution, for each...

open_problem | open | MathDB 页面 | 原文线索

V2-077556 | MATHDB-399183

Higher-dimensional analogue of Tolsa's square-function criterion

摘录 / EXCERPT: For integers $n>1$, does the following analogue of Theorem 9 hold? If μ is a Radon measure in $\mathbb{R}^d$ such that $\Theta^{n,*}(x,\mu)>0$ for μ-almost every...

open_problem | open | MathDB 页面 | 原文线索

V2-077557 | MATHDB-399184

Higher-dimensional square-function characterization of rectifiability

摘录 / EXCERPT: For an integer $n>1$ and a Borel set $E\subset\mathbb{R}^d$ with $\mathcal{H}^n(E)<\infty$, is E n-rectifiable if and only if ... for $\mathcal{H}^n$-almost every $x\in E$?

open_problem | open | MathDB 页面 | 原文线索

V2-077558 | MATHDB-399185

Removing circuit-depth dependence from evaluated homomorphic signatures

摘录 / EXCERPT: Can one remove the dependency on the circuit depth in the evaluated signature?

open_problem | open | MathDB 页面 | 原文线索

V2-077559 | MATHDB-399186

General approximate common divisors (GACD) problem

摘录 / EXCERPT: For parameters γ, η, and ρ, given polynomially many (in γ, η, and ρ) different integers of the form $x_i=pq_i+r_i$ for $i=1,\dots,n$, where the...

open_problem | open | MathDB 页面 | 原文线索

V2-077560 | MATHDB-399188

Hardness of binary-LWE for smaller dimensions

摘录 / EXCERPT: Is it possible to prove binary-LWE is hard for smaller values of n (in other words, give a tighter reduction of LWE to binary-LWE)?

open_problem | open | MathDB 页面 | 原文线索

V2-077561 | MATHDB-399189

No-signaling multi-prover interactive proofs for time-bounded languages

摘录 / EXCERPT: For any language computable in time $t=t(n)$, one can construct a multi-prover interactive proof that is sound against no-signaling strategies, with prover running time...

open_problem | open | MathDB 页面 | 原文线索

V2-077562 | MATHDB-399190

Garbled RAM from one-way functions without circularity

摘录 / EXCERPT: Construct a garbled RAM based only on one-way functions, without a circularity assumption, with overhead $\operatorname{poly}(\log(n),\kappa)$, where n is the data size and...

open_problem | open | MathDB 页面 | 原文线索

V2-077563 | MATHDB-399191

Reuse of holdout sets under adaptive computations

摘录 / EXCERPT: When can a holdout set be reused under adaptive computations?

open_problem | open | MathDB 页面 | 原文线索

V2-077564 | MATHDB-399192

Closing the locality-storage trade-off gap for searchable symmetric encryption

摘录 / EXCERPT: The lower bound for searchable symmetric encryption suggests the question of a matching upper bound: can the gap between that lower bound and the constructions described in the rep...

open_problem | open | MathDB 页面 | 原文线索

V2-077565 | MATHDB-399193

Natural-Assumption Security Reduction for General-Purpose Indistinguishability Obfuscation

摘录 / EXCERPT: Can the security of general-purpose indistinguishability obfuscators be reduced to some "natural" intractability assumption?

open_problem | open | MathDB 页面 | 原文线索

V2-077566 | MATHDB-399194

Extending Tiny-OT to the n-party setting for binary circuits

摘录 / EXCERPT: Can the two-party Tiny-OT protocol be extended to the n-party setting for binary circuits?

open_problem | open | MathDB 页面 | 原文线索

V2-077567 | MATHDB-399195

Non-Black-Box Simulation in the Fully Concurrent Setting

摘录 / EXCERPT: There exists a public-coin concurrent zero-knowledge argument protocol based on the existence of collision-resistant hash functions that is secure even with any unbounded (polynomi...

open_problem | open | MathDB 页面 | 原文线索

V2-077568 | MATHDB-399204

Index bound for complete two-sided minimal surfaces in \(\mathbb{R}^3\)

摘录 / EXCERPT: Suppose that $\Sigma\to\mathbb{R}^{3}$ is an immersed complete two-sided minimal surface of genus g and with r ends. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077569 | MATHDB-399207

Solutions of the Yamabe equation on products of round spheres

摘录 / EXCERPT: For the conformal classes $[g_0^n+Tg_0^m]$ on $S^n\times S^m$, with $n,m\ge 2$ and $T>0$, are there solutions of the Yamabe equation $-a_n\Delta f+s_gf=\lambda f^{p-1}$ that depend...

open_problem | open | MathDB 页面 | 原文线索

V2-077570 | MATHDB-399209

Dranishnikov’s Rationality Conjecture

摘录 / EXCERPT: If M is rationally inessential, then it is macroscopically small.

conjecture | open | MathDB 页面 | 原文线索

V2-077571 | MATHDB-399213

Constructing and classifying solutions of the Einstein constraint system

摘录 / EXCERPT: Constructing and classifying solutions to the system (1).

open_problem | open | MathDB 页面 | 原文线索

V2-077572 | MATHDB-399216

Lens rigidity with partial data

摘录 / EXCERPT: The lens rigidity problem with partial data for a sound speed is whether one can determine the speed near some p from L known near $S_p\partial M$. For general metrics, the goa...

open_problem | open | MathDB 页面 | 原文线索

V2-077573 | MATHDB-399219

Global stability estimate for lens rigidity under a foliation condition

摘录 / EXCERPT: Assume that $M_0\subset M$ can be foliated by strictly convex hypersurfaces. Let $D\subset\partial_-SM$ be a neighborhood of the compact set of all...

open_problem | open | MathDB 页面 | 原文线索

V2-077574 | MATHDB-399220

Uniqueness up to scale in simultaneous TOF-PET estimation

摘录 / EXCERPT: Under weak smoothness assumptions, requiring the corrected data $p=m/a$ to satisfy the TOF-PET range condition determines the gradient $\nabla R\mu$ from the measured data m on e...

open_problem | open | MathDB 页面 | 原文线索

V2-077575 | MATHDB-399221

Convergence of TV-regularized tomographic reconstructions under mesh refinement

摘录 / EXCERPT: For the pencil-beam X-ray tomography model with a fixed data vector m, if the minimization problem ... is solved at several discrete resolutions n, then the resulting reconstruct...

open_problem | open | MathDB 页面 | 原文线索

V2-077576 | MATHDB-399222

Resolution-controlled conductivity discretization in electrical impedance tomography

摘录 / EXCERPT: An explicit resolution-based quantification of the discretization size for the complete electrode model (CEM) across Ω is still an open task.

open_problem | open | MathDB 页面 | 原文线索

V2-077577 | MATHDB-399224

Uniqueness and inversion of the V-line Radon transform in the circular setup

摘录 / EXCERPT: For the circular V-line Radon transform setup described in the report, with fixed scattering angle $\alpha\in(0,\pi/2)$, if $f(x,y)$ is a smooth function supported in the disc...

open_problem | open | MathDB 页面 | 原文线索

V2-077578 | MATHDB-399225

Exact inversion formula for the conical Radon transform

摘录 / EXCERPT: For a smooth function $f(x,y,z)$, let $g(x_v,y_v,z_v)$ denote its conical Radon transform, integrating f over the 3-dimensional family of circular cones with fixed opening angle...

open_problem | open | MathDB 页面 | 原文线索

V2-077579 | MATHDB-399226

Cone-beam transverse ray transform with source points on a curve

摘录 / EXCERPT: Given the transverse ray transform ... of a compactly supported vector field f in three-dimensional Euclidean space, recover f from $T_f(a,\theta)$ for all $a\in\Gamma$ and all d...

open_problem | open | MathDB 页面 | 原文线索

V2-077580 | MATHDB-399229

Lemmermeyer’s GI-extension conjecture for finite 2-groups

摘录 / EXCERPT: Every finite 2-group H embeds as a subgroup of index 2 in a group G generated by involutions outside H (equivalently, H has a GI-extension).

conjecture | open | MathDB 页面 | 原文线索

V2-077581 | MATHDB-399231

Uniform bound for negative curves of small arithmetic genus

摘录 / EXCERPT: There exists a universal constant $t>1$ such that, for every smooth projective surface X over a field k, with first Betti number $b_1(X)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077582 | MATHDB-399232

Determinant-density upper-bound conjecture

摘录 / EXCERPT: If K is any knot or link, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-077583 | MATHDB-399233

Geometric versus diagrammatic maximality of knot sequences

摘录 / EXCERPT: Is any diagrammatically maximal sequence of knots geometrically maximal, and vice versa? Here, according to the report's preceding definitions, a sequence (K_n) is diagrammatical...

open_problem | open | MathDB 页面 | 原文线索

V2-077584 | MATHDB-399244

Volume non-collapsing implies finite Weil-Petersson distance

摘录 / EXCERPT: In Setup 1, let X be a normal complex space with a projective surjective holomorphic map $f\colon X\to C$ with connected fibers, where C is a Riemann surface with distinguished...

open_problem | open | MathDB 页面 | 原文线索

V2-077585 | MATHDB-399245

Canonical central fiber at finite Weil-Petersson distance

摘录 / EXCERPT: Let X be a normal complex space with a projective surjective holomorphic map $f:X\to C$ with connected fibers, where C is a Riemann surface with distinguished point 0. Assume...

open_problem | open | MathDB 页面 | 原文线索

V2-077586 | MATHDB-399249

Bounded negativity of Shimura curves

摘录 / EXCERPT: For any compact smooth Shimura surface X not isogenous to a product and for any real number M, there are only finitely many Shimura curves C on X with $C^2<M$.

open_problem | open | MathDB 页面 | 原文线索

V2-077587 | MATHDB-399250

Semi-continuity of the minimal log discrepancy

摘录 / EXCERPT: Let (X,Δ) be a log pair. The minimal log discrepancy of (X,Δ) is lower semi-continuous.

open_problem | open | MathDB 页面 | 原文线索

V2-077588 | MATHDB-399251

Economical toroidal resolutions / qdlt modifications

摘录 / EXCERPT: Fix a positive integer n. Then there is a positive integer s with the following property: If (X,Δ) is any log canonical pair, where X has dimension n, then there is...

open_problem | open | MathDB 页面 | 原文线索

V2-077589 | MATHDB-399256

Ein-Lazarsfeld asymptotic syzygy criterion

摘录 / EXCERPT: Let C be a smooth projective curve of genus $g\ge 2$, let L be a very ample line bundle on C with $\deg L\gg 0$, and let B be a line bundle on C. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077590 | MATHDB-399263

Grünbaum mass partition problem

摘录 / EXCERPT: Determine the function $\Delta:\mathbb{N}^2\to\mathbb{N}$ defined by $\Delta(j,k)$ equal to the smallest dimension d such that every collection $M=\{\mu_1,\ldots,\mu_j\}$ of j...

open_problem | open | MathDB 页面 | 原文线索

V2-077591 | MATHDB-399264

Geometric permutations of non-overlapping congruent balls

摘录 / EXCERPT: Conjecture: For every $n\ge 4$, n congruent balls in $\mathbb{R}^d$ with pairwise disjoint interiors admit at most two geometric permutations.

open_problem | open | MathDB 页面 | 原文线索

V2-077592 | MATHDB-399268

A Hofer-Zehnder capacity inequality for plank coverings

摘录 / EXCERPT: Conjecture 3. If K can be covered with a finite set of planks with the sum of relative widths equal to w, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077593 | MATHDB-399270

Bang’s conjecture for planks in two directions

摘录 / EXCERPT: Let a convex body $K\subset\mathbb{R}^n$ be covered by a family of planks $P_1,\ldots,P_m$ whose normals have only two distinct directions. Then the sum of the widths of the planks...

conjecture | open | MathDB 页面 | 原文线索

V2-077594 | MATHDB-399271

Steinhaus’ circle lattice-point problem for planar regions

摘录 / EXCERPT: Is there a planar region of area n such that no congruent copy of it can contain exactly n lattice points?

open_problem | open | MathDB 页面 | 原文线索

V2-077595 | MATHDB-399273

Uniform cover-decomposability bounds for convex n-gons

摘录 / EXCERPT: Does there exist, for any integer $n>3$, an integer $m(n)$ such that every convex n-gon Q satisfies $m(Q)\le m(n)$, where $m(Q)$ is the smallest positive integer such that ever...

open_problem | open | MathDB 页面 | 原文线索

V2-077596 | MATHDB-399274

Points with distinct circumradii: polynomial bound

摘录 / EXCERPT: There is a constant C such that, for every positive integer k, any set of Ck^9 points in the plane in general position (no four on a line or circle) contains k points such...

open_problem | open | MathDB 页面 | 原文线索

V2-077597 | MATHDB-399275

Distinct circumradii among triples of planar points

摘录 / EXCERPT: For every positive integer k, does there exist an integer n_k such that, given n_k points in the plane in general position (i.e. no three on a line and no four on a circle),...

open_problem | open | MathDB 页面 | 原文线索

V2-077598 | MATHDB-399277

Erickson’s 4-polytope non-simplicial-facet question

摘录 / EXCERPT: Can a 4-dimensional polytope with n vertices have $\Omega(n^2)$ non-simplicial facets?

open_problem | open | MathDB 页面 | 原文线索

V2-077599 | MATHDB-399278

Membership of the rectilinear crossing number problem in NP

摘录 / EXCERPT: Does the rectilinear crossing number problem-deciding whether a given graph G has a straight-line drawing with at most k crossings-belong to $\textsf{NP}$?

open_problem | open | MathDB 页面 | 原文线索

V2-077600 | MATHDB-399279

Stretchability of pseudoline arrangements is ∃R-complete

摘录 / EXCERPT: A pseudoline arrangement is called stretchable if it has an equivalent straight-line arrangement. Theorem (Mnëv): deciding whether a pseudoline arrangement is stretchable is $\R$-c...

open_problem | open | MathDB 页面 | 原文线索

V2-077601 | MATHDB-399280

4-Feasibility is ∃R-complete

摘录 / EXCERPT: 4-Feasibility: Given an explicitly represented multivariate polynomial $f\colon\mathbb{R}^n\to\mathbb{R}$ of total degree at most 4, does it have a root? The report states that t...

open_problem | open | MathDB 页面 | 原文线索

V2-077602 | MATHDB-399283

Decidability of embeddability of 2-complexes in $\mathbb{R}^3$

摘录 / EXCERPT: Given a 2-dimensional complex K, a collection of triangular faces along with identifications of their edges and/or vertices, does the complex K embed in $\mathbb{R}^3$?

open_problem | open | MathDB 页面 | 原文线索

V2-077603 | MATHDB-399286

Monotonicity of an angle in a spherical triangle with fixed side lengths

摘录 / EXCERPT: Let ABC be a spherical triangle with fixed arc lengths a,b,c on a sphere of variable radius $r>\frac{a+b+c}{2\pi}$. By the spherical cosine law, ... Find an elementary proof that...

open_problem | open | MathDB 页面 | 原文线索

V2-077604 | MATHDB-399289

Contractibility of the space of convex-position polygon arrangements

摘录 / EXCERPT: We say an arrangement of convex sets is in convex position when the sets are indexed by $1,\ldots,n$ and, for every triple $\{K_i,K_j,K_k\}$, $i<j<k$, each set intersects the bound...

open_problem | open | MathDB 页面 | 原文线索

V2-077605 | MATHDB-399290

Covering an acute triangle by acute triangles and orthic perimeters

摘录 / EXCERPT: For an acute triangle T, denote by $\xi(T)$ the perimeter of its orthic triangle, i.e. the triangle formed by the bases of T’s altitudes. Is it true that whenever T is covere...

open_problem | open | MathDB 页面 | 原文线索

V2-077606 | MATHDB-399291

Crossing-Maximal Order Types

摘录 / EXCERPT: Given two finite equal-size point sets P) and Q in general position in the plane, a bijection $P\to Q$, $p\mapsto p'$, is called crossing-preserving if whenever the segment...

open_problem | open | MathDB 页面 | 原文线索

V2-077607 | MATHDB-399299

Refutation of the conjecture that Novikov-Shubin invariants are always positive

摘录 / EXCERPT: The conjecture of Lott and Lück that the Novikov-Shubin invariants are always positive is false.

conjecture | open | MathDB 页面 | 原文线索

V2-077608 | MATHDB-399300

Conjecture A on solvability of non-singular group equations

摘录 / EXCERPT: Let Γ be a countable group and let $n\in\mathbb{N}$ be positive. For $w\in\Gamma*F_n$, let $\varepsilon:\Gamma*F_n\to F_n$ be the natural augmentation homomorphism, and call...

conjecture | open | MathDB 页面 | 原文线索

V2-077609 | MATHDB-399301

Residual finiteness of one-relator groups with torsion

摘录 / EXCERPT: Every one-relator group with torsion presented as $\langle a,b\mid W^n\rangle$ is residually finite when $n\ge 2$.

open_problem | open | MathDB 页面 | 原文线索

V2-077610 | MATHDB-399303

Existence of solutions to the Kashiwara-Vergne equations

摘录 / EXCERPT: Does there exist an element $F\in \mathrm{TAut}_2$ satisfying the Kashiwara-Vergne equations, in particular the equation $F(x+y)=\log(e^x e^y)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077611 | MATHDB-399304

Undecidability of the finite development problem for permutoids

摘录 / EXCERPT: Given partial permutations $p_1,\ldots,p_m$ of a finite set X (bijections between subsets of X) satisfying: (1) $p_1=\operatorname{id}_X$; and (2) for all i,j with...

open_problem | open | MathDB 页面 | 原文线索

V2-077612 | MATHDB-399305

Existence of a non-trivial hyperbolic group with no finite quotients

摘录 / EXCERPT: Does there exist a non-trivial hyperbolic group with no finite quotients?

open_problem | open | MathDB 页面 | 原文线索

V2-077613 | MATHDB-399308

Recognition of knot complements lies in NP

摘录 / EXCERPT: The problem of recognizing knot complements lies in NP.

open_problem | open | MathDB 页面 | 原文线索

V2-077614 | MATHDB-399311

Variational inequality for the Laplacian with nonlinear flux restriction on tiny holes

摘录 / EXCERPT: Let $\Omega\subset\mathbb R^n$ be a bounded domain with smooth boundary, $n\ge 3$. Let G_0 be the unit ball centered at the origin, let...

open_problem | open | MathDB 页面 | 原文线索

V2-077615 | MATHDB-399312

Homogenized variational inequality for critically scaled tiny holes

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be bounded with smooth boundary, $n\ge 3$, and let Ω_ε be the domain perforated by periodically distributed balls...

open_problem | open | MathDB 页面 | 原文线索

V2-077616 | MATHDB-399314

Fingering in the Prandtl-Ishlinskii-Richards model

摘录 / EXCERPT: Does the Prandtl-Ishlinskii-Richards model given by ... with ... also show the fingering effect? Here p_c is monotonically increasing, $y\in[0,1]$, and the system is supplemented by...

open_problem | open | MathDB 页面 | 原文线索

V2-077617 | MATHDB-399318

ROOF optimal projection for reducing landmark shape dimension

摘录 / EXCERPT: Given a dataset $X=\{x_1,\ldots,x_n\}$ in the planar Kendall shape space Σ_d^2, and feature spaces $T_k=\Sigma_k^2$ for $k=3,\ldots,d$, the ROOF formulation seeks projection...

open_problem | open | MathDB 页面 | 原文线索

V2-077618 | MATHDB-399319

Defining second principal components in BHV tree-space

摘录 / EXCERPT: How should second principal components be defined in BHV tree-space, given the difficulty of defining parallel transport along first principal components lying in stratified tree-s...

open_problem | open | MathDB 页面 | 原文线索

V2-077619 | MATHDB-399320

Characterization of rank-2 circuits and circuit polynomials

摘录 / EXCERPT: Characterize circuits and circuit polynomials in rank $r=2$.

open_problem | open | MathDB 页面 | 原文线索

V2-077620 | MATHDB-399321

Parametrization of the diffusion-PCA density family

摘录 / EXCERPT: We conjecture that Γ in the Riemannian case can be parametrized by the bundle of symmetric positive-definite covariant tensors of order 2.

open_problem | open | MathDB 页面 | 原文线索

V2-077621 | MATHDB-399322

Alekseevskii conjecture

摘录 / EXCERPT: Every noncompact homogeneous Einstein space is diffeomorphic to a Euclidean space.

conjecture | open | MathDB 页面 | 原文线索

V2-077622 | MATHDB-399325

Classification of non-compact homogeneous Einstein spaces

摘录 / EXCERPT: What are the non-compact, non-flat, homogeneous spaces G/K which admit G-invariant Einstein metrics?

open_problem | open | MathDB 页面 | 原文线索

V2-077623 | MATHDB-399327

Strong Alekseevskii Conjecture

摘录 / EXCERPT: A non-compact, non-flat, homogeneous Einstein space G/K must be isometric to a solvable Lie group with a left-invariant metric. Equivalently, G is linear and K is a maximal c...

conjecture | open | MathDB 页面 | 原文线索

V2-077624 | MATHDB-399328

Self-adjointness of the U-action on the nilradical

摘录 / EXCERPT: What is the nature of the U-action on $\operatorname{Lie} N$? Is this action “self-adjoint”; i.e., is $\operatorname{Ad}(U)|_{\operatorname{Lie} N}$ closed under transpose?

open_problem | open | MathDB 页面 | 原文线索

V2-077625 | MATHDB-399329

Classification of left-invariant Einstein metrics on compact semisimple Lie groups

摘录 / EXCERPT: Find all left-invariant Einstein metrics on compact semisimple Lie groups.

open_problem | open | MathDB 页面 | 原文线索

V2-077626 | MATHDB-399340

Finite stability of the Fourier dimension under unions

摘录 / EXCERPT: For a finite family of Borel sets $[U+007F]\{A_k\}$ in $[U+007F]\mathbb{R}^d$, is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-077627 | MATHDB-399344

Characterization of minimizing compact open subgroups

摘录 / EXCERPT: Let G be a totally disconnected locally compact group, let $\alpha\in\operatorname{Aut}(G)$, and let U be a compact open subgroup. Define ... so that U is minimizing when...

open_problem | open | MathDB 页面 | 原文线索

V2-077628 | MATHDB-399345

Margulis-Zimmer question for commensurated subgroups of $\mathrm{SL}_n(\mathbb{Z}[1/S])$

摘录 / EXCERPT: Let $S=\{p_1,\ldots,p_s\}$ be a finite set of primes, let $n\ge 3$, and set $\Gamma=\mathrm{SL}_n\!\left(\mathbb{Z}\left[\frac{1}{p_1},\ldots,\frac{1}{p_s}\right]\right)$. Are all...

open_problem | open | MathDB 页面 | 原文线索

V2-077629 | MATHDB-399346

Commensurated subgroups of finite-index subgroups of $SL_3(\mathbb{Z})$

摘录 / EXCERPT: For every finite-index subgroup $\Gamma\le SL_3(\mathbb{Z})$, is every subgroup $\Lambda\le\Gamma$ that is commensurated by Γ finite or finite-index in Γ?

open_problem | open | MathDB 页面 | 原文线索

V2-077630 | MATHDB-399352

Explicit construction of the diffusion associated with the alternative Laplace operator on the cone of discrete Radon measures

摘录 / EXCERPT: A different natural choice of tangent space yields a different, well-defined Laplace operator in $L^{2}(\mathbb K(X),\mu)$. Using Dirichlet-form theory, one can prove that the corr...

open_problem | open | MathDB 页面 | 原文线索

V2-077631 | MATHDB-399353

Global Dirichlet heat-kernel estimates for rotationally symmetric Lévy processes

摘录 / EXCERPT: Conjecture: For a large class of rotationally symmetric Lévy processes X with Lévy exponent $ ... ... where $p(t,x)$ is the transition density of X.

open_problem | open | MathDB 页面 | 原文线索

V2-077632 | MATHDB-399354

Global Dirichlet heat-kernel estimate in a half-space

摘录 / EXCERPT: Let X be a symmetric Lévy process whose jumping kernel satisfies the assumptions of [5, 6, 14], and let $D=H$ be the upper half-space. The report states that there exist constant...

open_problem | open | MathDB 页面 | 原文线索

V2-077633 | MATHDB-399355

Strong uniqueness for the finite-volume \(\exp(\Phi)_2\) Dirichlet operator

摘录 / EXCERPT: Under $\alpha,\delta>\tfrac12$, $2\alpha+\delta>1$, and $-\sqrt{\pi/e}<a<\sqrt{\pi/e}$, the pre-Dirichlet operator $(L_0,\mathcal{FC}^{\infty}_b)$ associated with the finite-volume...

open_problem | open | MathDB 页面 | 原文线索

V2-077634 | MATHDB-399356

Uniqueness of the Friedrichs extension for the \exp(\(\phi\))\(_2\)-measure

摘录 / EXCERPT: Is the Friedrichs extension the only closed extension of the pre-Dirichlet operator $(\mathcal L_0,\mathcal{FC}_b^\infty)$ generating a C_0-semigroup on $L^p(\mu)$ for $p\ge 1$?...

open_problem | open | MathDB 页面 | 原文线索

V2-077635 | MATHDB-399360

Uniqueness of maximal immediate extensions under (K1) and (K2)

摘录 / EXCERPT: Let (K,v) be a henselian valued field such that: - if $p=\operatorname{char}Kv>0$, then the value group vK is p-divisible (K1); and - the residue field Kv is perfect (K2)....

open_problem | open | MathDB 页面 | 原文线索

V2-077636 | MATHDB-399366

Classification of connected groups satisfying a local-global principle

摘录 / EXCERPT: Which class of connected groups satisfy a local-global principle?

open_problem | open | MathDB 页面 | 原文线索

V2-077637 | MATHDB-399368

A problem of Zilber about residue rings of models of arithmetic

摘录 / EXCERPT: Suppose M is a nonstandard model of true arithmetic, and $\bc$ is an infinite element of M with $\bc \equiv 1 \pmod n$ for each standard integer n. Can one interpret arithmet...

open_problem | open | MathDB 页面 | 原文线索

V2-077638 | MATHDB-399369

Zilber’s question about residue rings of models of arithmetic

摘录 / EXCERPT: Suppose M is a nonstandard model of true arithmetic, and $\bc$ is an infinite element of M such that $\bc \equiv 1 \pmod n$ for each standard integer n. Can one interpret ari...

open_problem | open | MathDB 页面 | 原文线索

V2-077639 | MATHDB-399371

Bertin obstruction for metacyclic local lifting

摘录 / EXCERPT: Conjecture 5. For groups of the form $G:=\mathbb{Z}/p^n\rtimes\mathbb{Z}/m$, where $p\nmid m$, the Bertin obstruction is the only obstruction to the local lifting problem.

open_problem | open | MathDB 页面 | 原文线索

V2-077640 | MATHDB-399375

Geometric reconstruction in the remaining transcendence-degree-two case

摘录 / EXCERPT: Complete the reconstruction of the field F from its maximal 2-step nilpotent pro-ℓ quotient $G_F^{c,\ell}$ in the remaining case $\widetilde{k}=\widetilde{\mathbb{Q}}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077641 | MATHDB-399376

Recovering valuations on Demushkin fields: Koenigsmann-Strømmen Conjecture 1

摘录 / EXCERPT: Let $F/\mathbb{Q}_p$ be a finite extension with $\zeta_p\in F$. Let K be a field with $\zeta_p\in K$, and suppose $G_F^{(p)}\simeq G_K^{(p)}$. Then K has characteristic 0. Fu...

conjecture | open | MathDB 页面 | 原文线索

V2-077642 | MATHDB-399377

Existence of a field in the exceptional case for recovering 2-adic valuations

摘录 / EXCERPT: Does there exist a field K of characteristic 0 and transcendence degree 1 over $k:=K\cap\mathbb{Q}$ such that $G_K(2)\simeq G_{\mathbb{Q}_2}(2)$ and k admits both a nontriv...

open_problem | open | MathDB 页面 | 原文线索

V2-077643 | MATHDB-399378

Structure and radialization of multiplicity loci for morphisms of Berkovich curves

摘录 / EXCERPT: For a morphism $f:Y\to X$ of nice Berkovich curves, describe the sets $N_{f,\ge d}=\{y\in Y:[\mathcal H(y):\mathcal H(f(y))]\ge d\}$ and the associated metric structure. The report...

open_problem | open | MathDB 页面 | 原文线索

V2-077644 | MATHDB-399381

Positivity of the coefficients of the leading root for the deformed exponential

摘录 / EXCERPT: For the deformed exponential formal power series ... let $x_0(y)$ denote its leading (unique power-series) root, and set $U(y):=-x_0(y)$. The conjecture is that $U(y)$ has all strict...

open_problem | open | MathDB 页面 | 原文线索

V2-077645 | MATHDB-399382

Dependence between space requirement and total node path length in random m-ary search trees

摘录 / EXCERPT: For the random m-ary search tree defined on page 33, let S_n be its space requirement and let T_n be its total node path length. The report asks to what extent T_n depends...

open_problem | open | MathDB 页面 | 原文线索

V2-077646 | MATHDB-399383

Behavior of Boolean functions on Ford trees

摘录 / EXCERPT: For the Ford (-model) random tree of size n, label each internal node independently and uniformly from $\{\wedge,\vee\}$ and each leaf independently and uniformly from...

open_problem | open | MathDB 页面 | 原文线索

V2-077647 | MATHDB-399385

Integrality gap of the Lovász \[?25l\vartheta\[?25l-function SDP for sparse independent sets

摘录 / EXCERPT: On graphs with maximum degree d, the standard $\[?25l\vartheta\[?25l$-function-based SDP formulation for the independent set problem has an integrality gap of...

open_problem | open | MathDB 页面 | 原文线索

V2-077648 | MATHDB-399386

Integrality gap for the SA+ relaxation on sparse graphs

摘录 / EXCERPT: For graphs of maximum degree d, the $O(\log^4 d)$-level SA+ relaxation has integrality gap $\widetilde{O}\!\left(d/\log^2 d\right)$, where $\widetilde{O}(\cdot)$ suppresses facto...

open_problem | open | MathDB 页面 | 原文线索

V2-077649 | MATHDB-399387

Minimality of cut-generating pairs for translated lattices intersected with rational polyhedra

摘录 / EXCERPT: Let $S=(b+\bb Z^n)\cap Q$, where $b\in\bb R^n\setminus\bb Z^n$ and Q is a rational polyhedron. Given a maximal S-free convex set B containing the origin in its interior and r...

open_problem | open | MathDB 页面 | 原文线索

V2-077650 | MATHDB-399388

Sparse cutting-planes and extended formulations

摘录 / EXCERPT: Theorem. Let $n\in\mathbb{Z}_{>0}$. Let $P\subseteq\mathbb{R}^n$ be a polyhedron and let $Q\subseteq\mathbb{R}^n\times\mathbb{R}^m$ be an extension of P. Then ... Moreover, if n...

open_problem | open | MathDB 页面 | 原文线索

V2-077651 | MATHDB-399396

Every h-perfect graph is $(\omega+1)$-colorable

摘录 / EXCERPT: Conjecture 3. Every h-perfect graph is $(\omega+1)$-colorable, where ω is the clique-number.

open_problem | open | MathDB 页面 | 原文线索

V2-077652 | MATHDB-399397

Improving the convex-hull bound for tours

摘录 / EXCERPT: For a 3-edge-connected graph, can the bound in the fact that the all- edge function $\mathbf{1}$ lies in the convex hull of tours be improved? Here a tour is an Eulerian...

open_problem | open | MathDB 页面 | 原文线索

V2-077653 | MATHDB-399404

Optimality certificate for mixed-integer convex minimization

摘录 / EXCERPT: Let $f:\mathbb{R}^{n+d}\to\mathbb{R}$ be a continuous convex function. A point $x^{\star}\in\mathbb{Z}^{n}\times\mathbb{R}^{d}$ satisfies ... if and only if there exist $k\le 2^{n}$...

open_problem | open | MathDB 页面 | 原文线索

V2-077654 | MATHDB-399407

Maximal order type of generalized trees generated by a recursive construction

摘录 / EXCERPT: Let W be an effective construction that maps effectively given well-quasi-orders to effectively presented well-quasi-orders. Define $T(W)$ to be the set of generalized trees whos...

open_problem | open | MathDB 页面 | 原文线索

V2-077655 | MATHDB-399408

Polynomial-size Lovász-Schrijver proofs for the all-variable weak-duality principle

摘录 / EXCERPT: Let A, b, and c be variable matrix/vector entries in the weak-duality principle for linear programming,...

open_problem | open | MathDB 页面 | 原文线索

V2-077656 | MATHDB-399409

Length-width strengthening for resolution refutations

摘录 / EXCERPT: Can Theorem 1 be strengthened to guarantee that the narrow refutation Π' is not substantially longer than the initial short refutation Π?

open_problem | open | MathDB 页面 | 原文线索

V2-077657 | MATHDB-399410

Dependent choice for computable elements in the coherence-space domain

摘录 / EXCERPT: Does DC still hold when restricting D to computable elements?

open_problem | open | MathDB 页面 | 原文线索

V2-077658 | MATHDB-399416

Entropy estimate for weak model sets

摘录 / EXCERPT: The report states the following result: If H is second countable and all A_n are simply connected, then the pattern entropy of the weak model set satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-077659 | MATHDB-399417

Uniform boundary-layer control of the Ginzburg-Landau order parameter

摘录 / EXCERPT: For a smooth, bounded, simply connected sample r$\Omega\subset\mathbb R^2$ in the surface-superconductivity regime $1<b<\Theta_0^{-1}$, can the previously established L^2 closene...

open_problem | open | MathDB 页面 | 原文线索

V2-077660 | MATHDB-399419

Reverse Faber-Krahn conjecture for the first Robin eigenvalue

摘录 / EXCERPT: For $\[4m\alpha>0\u001b[0m$, among domains $\[4m\Omega\u001b[0m$ of fixed volume, balls are the maximizers of the first Robin eigenvalue $E_1(\alpha,\Omega)$.

conjecture | open | MathDB 页面 | 原文线索

V2-077661 | MATHDB-399420

Lower bound for the first magnetic Dirichlet eigenvalue in a constant field

摘录 / EXCERPT: Let a9subsetmathbb{R}^{2} be a bounded open set, and let the magnetic field be constant, $B(x)=B_{0}>0$. With $\lambda_{1}(\Omega,B_{0})$ denoting the first eigenvalue of the magne...

open_problem | open | MathDB 页面 | 原文线索

V2-077662 | MATHDB-399421

Transience between local and global magnetic fields

摘录 / EXCERPT: Is there a super-polynomial decay rate for the heat semigroup if B decays to zero very slowly at infinity?

open_problem | open | MathDB 页面 | 原文线索

V2-077663 | MATHDB-399422

Strong-confinement limit for the cubic nonlinear Schrödinger equation in a curved waveguide

摘录 / EXCERPT: Let $\Gamma\subset\mathbb R^2$ be a smooth simple curve with normal parametrization γ, let $S=M\times(-1,1)$, where $M=\mathbb R$ for an unbounded curve and...

open_problem | open | MathDB 页面 | 原文线索

V2-077664 | MATHDB-399424

Classification of positive-energy representations of gauge groups

摘录 / EXCERPT: Let M be compact, let $P\to M$ be a principal compact simple Lie-group bundle, and let $T:\mathbb R\to\operatorname{Aut}(P)$ be a time-translation action whose induced action...

open_problem | open | MathDB 页面 | 原文线索

V2-077665 | MATHDB-399427

Super versions of massive free fields

摘录 / EXCERPT: Describe super versions of massive free fields.

open_problem | open | MathDB 页面 | 原文线索

V2-077666 | MATHDB-399428

A Lax-Phillips normal form for one-parameter groups and standard real subspaces

摘录 / EXCERPT: Is there a variant of the Lax-Phillips Representation Theorem for one-parameter groups $(U_t)_{t\in\mathbb R}$ whose positive part leaves a standard real subspace...

open_problem | open | MathDB 页面 | 原文线索

V2-077667 | MATHDB-399429

Euclidean realizations of Borchers triples

摘录 / EXCERPT: Find natural Euclidean realizations of Borchers triples, in the sense of [Ne15].

open_problem | open | MathDB 页面 | 原文线索

V2-077668 | MATHDB-399430

BMV conjecture for traces of matrix exponentials

摘录 / EXCERPT: For two Hermitian $n\times n$ matrices A and B, the function $t\mapsto \operatorname{tr}(e^{A-tB})$ is a Laplace transform of a positive measure on the interval between the min...

conjecture | open | MathDB 页面 | 原文线索

V2-077669 | MATHDB-399431

Fixed points of increasing maps on finite decimals

摘录 / EXCERPT: The assertion that every increasing map $f:D\cap[0,1]\to D\cap[0,1]$, where D is the set of finite decimal numbers, has a fixed point is false: the report states that a counterex...

open_problem | open | MathDB 页面 | 原文线索

V2-077670 | MATHDB-399432

Increasing maps of finite decimal numbers and diagonal crossings

摘录 / EXCERPT: Let D be the set of finite decimal numbers. Is it possible for the graph of an increasing map $f:D\to D$ to cross the first bisector outside D?

open_problem | open | MathDB 页面 | 原文线索

V2-077671 | MATHDB-399433

Fixed-point property for increasing self-maps of the unit interval

摘录 / EXCERPT: Every increasing map $f:[0,1]\to[0,1]$ has a fixed point; that is, there exists $x\in[0,1]$ such that $f(x)=x$.

open_problem | open | MathDB 页面 | 原文线索

V2-077672 | MATHDB-399434

Parity of the sum of six consecutive natural numbers

摘录 / EXCERPT: “The sum of 6 consecutive natural numbers is always odd”.

open_problem | open | MathDB 页面 | 原文线索

V2-077673 | MATHDB-399435

Proof scripts as a vehicle for investigating and enhancing proof comprehension

摘录 / EXCERPT: The report invites colleagues to explore the conjecture that proof scripts provide an excellent vehicle for investigating and enhancing proof comprehension.

open_problem | open | MathDB 页面 | 原文线索

V2-077674 | MATHDB-399436

De Giorgi conjecture on elliptic-in-time variational regularization of semilinear waves

摘录 / EXCERPT: De Giorgi conjectured that the WIDE variational approach could be implemented in the hyperbolic setting of semilinear waves, namely for $\rho>0$, $D=0$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-077675 | MATHDB-399437

Unique irreversible quasi-static evolution for penny-shaped hydraulic cracks

摘录 / EXCERPT: Assume that the initial crack is the disk $\Gamma(0)=B_{R_0}:=\{(x_1,x_2,0):x_1^2+x_2^2\le R_0^2\}$, that the rock is homogeneous and isotropic, and that all admissible cracks have...

open_problem | open | MathDB 页面 | 原文线索

V2-077676 | MATHDB-399438

Existence of an irreversible quasi-static hydraulic-crack evolution in $\operatorname{Adm}_r(\Sigma)$

摘录 / EXCERPT: Let $r>0$, let $V\in AC([0,T];[0,+\infty))$, and let $\Gamma_0\in\operatorname{Adm}_r(\Sigma)$. Assume that ... for every $\Gamma\in\operatorname{Adm}_r(\Sigma)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-077677 | MATHDB-399439

Rigorous \-convergence derivation of the nematic-shell surface energy

摘录 / EXCERPT: Give a rigorous derivation, using -convergence, of the extrinsic nematic-shell surface energy proposed by Napoli and Vergori as a formal dimension reduction: ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-077678 | MATHDB-399441

Uniqueness in the exceptional biaxial-test case for 3D elasto-plasticity

摘录 / EXCERPT: For the biaxial test on $\Omega = (0,\ell) \times (0,d) \times (0,\ell)$, with pressure $\bar{p} \geq 0$ on the faces $x_2 = 0,d$, displacement $u_3 = t\, x_3$ on the frictionless faces $x_3 = 0,\ell$, traction-free faces $x_1 = 0$...

open_problem | open | MathDB 页面 | 原文线索

V2-077679 | MATHDB-399442

Spectral-measure parametrization for a canonical system

摘录 / EXCERPT: Consider the canonical system $-Jy'(x)=zH(x)y(x)$ on $[0,\ell)$ with $y_2(0)=0$, where H is a real symmetric non-negative measurable 2×2 matrix, $\operatorname{tr}H(x)=1$...

open_problem | open | MathDB 页面 | 原文线索

V2-077680 | MATHDB-399443

Sharp eigenvalue estimates under rank-one perturbations in Krein spaces

摘录 / EXCERPT: Let A and B be self-adjoint operators in the same Krein space, assume that A has κ negative squares, and suppose that, for some $\lambda_0\in\rho(A)\cap\rho(B)$, ... L...

open_problem | open | MathDB 页面 | 原文线索

V2-077681 | MATHDB-399444

Trace uniqueness under universal trace existence

摘录 / EXCERPT: Let ce(Omega) be a bounded, open, connected subset of $\mathbb{R}^d$, where $d\ge 2$, let $\Gamma=\partial\Omega$, and suppose that the $(d-1)$-dimensional Hausdorff measure satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-077682 | MATHDB-399445

Are isospectral Lipschitz domains necessarily congruent?

摘录 / EXCERPT: Let $\Omega_1,\Omega_2\subset\mathbb{R}^d$ be Lipschitz domains, where a domain is a bounded, open, connected set. They are called isospectral if their Dirichlet eigenvalues satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-077683 | MATHDB-399446

Spectral determination from all Robin parameters

摘录 / EXCERPT: Let Ω_1,Ω_2 be two Lipschitz domains. Assume that $\Delta^{\beta}_{\Omega_1}$ and $\Delta^{\beta}_{\Omega_2}$ have the same sequence of eigenvalues for each...

open_problem | open | MathDB 页面 | 原文线索

V2-077684 | MATHDB-399447

Strong-coupling asymptotics for a singular δ' interaction on a planar loop

摘录 / EXCERPT: The report states the following theorem. Let Γ be a C^4-smooth closed curve without self-intersections, and let $H_{\beta,\Gamma}$ be the operator associated with ... Then...

open_problem | open | MathDB 页面 | 原文线索

V2-077685 | MATHDB-399449

Detectable subspace for a matrix differential operator in the case $w\widetilde w=0$

摘录 / EXCERPT: The report states the following theorem. Let $q,u,w,\widetilde w\in L^\infty(0,1)$, assume $w\widetilde w=0$, and let $\theta(\cdot,\lambda)$ and $\phi(\cdot,\lambda)$ be a fundame...

open_problem | open | MathDB 页面 | 原文线索

V2-077686 | MATHDB-399450

Normalizations at non-regular-singular LP/N endpoints

摘录 / EXCERPT: Formulate satisfactory normalizations of the fundamental system $\{\varphi,\psi\}$ near endpoints of LP/N type that are not regular singular points; for example, Lennard-Jones pote...

open_problem | open | MathDB 页面 | 原文线索

V2-077687 | MATHDB-399451

High-energy spectral concentration for a radial Dirac operator

摘录 / EXCERPT: For the radial Dirac operator ... with $m\in\mathbb R$, $k\ge \tfrac12$, and real-valued $q\in L^1_{\mathrm{loc}}[0,\infty)$ satisfying the regularity and growth assumptions stated i...

open_problem | open | MathDB 页面 | 原文线索

V2-077688 | MATHDB-399452

Constancy of the resolvent norm on an open subset of the resolvent set

摘录 / EXCERPT: Can the resolvent norm $\lambda\mapsto\|(T-\lambda)^{-1}\|$ of an operator T be constant on an open subset of the resolvent set $\rho(T)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077689 | MATHDB-399453

Principal symbol of the Dirichlet-to-Neumann operator

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be bounded and smooth, with boundary $\partial\Omega=\Sigma$, and let A be a symmetric strongly elliptic second-order differential operator with r...

open_problem | open | MathDB 页面 | 原文线索

V2-077690 | MATHDB-399454

L^1 bounds for Dirichlet eigenfunctions

摘录 / EXCERPT: For every $d\in\mathbb N$ there is a constant C, depending only on d, such that, for every open set $\Omega\subset\mathbb R^d$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-077691 | MATHDB-399455

Dirichlet problem for a fractional elliptic operator

摘录 / EXCERPT: For a bounded smooth open set $U\subset\mathbb{R}^{n'}$ and the fractional power M^a of a second-order strongly elliptic differential operator M on $\mathbb{R}^{n'}$, the repor...

open_problem | open | MathDB 页面 | 原文线索

V2-077692 | MATHDB-399456

Parametrix for the fractional-order Dirichlet problem

摘录 / EXCERPT: Let $\widetilde Q_+ + G_0$ be a parametrix for the well-posed problem $Q_+v=g$, with G_0 a singular Green operator of order and class 0 in the Boutet de Monvel calculus. Then p...

open_problem | open | MathDB 页面 | 原文线索

V2-077693 | MATHDB-399457

Cusp formation in the nonlocal transport model

摘录 / EXCERPT: For the transport equation $\theta_t(x,t)+u(x,t)\theta_x(x,t)=0$ on $\mathbb{R}\times(0,\infty)$, with $u(x,t)=\int_{\mathbb{R}}|y-x|^{-\gamma}\theta(y,t)\,dy$ for $\gamma\in(0,1)$...

open_problem | open | MathDB 页面 | 原文线索

V2-077694 | MATHDB-399458

Isospectral characterization of global conservative multi-peakon solutions

摘录 / EXCERPT: For the Camassa-Holm equation ... consider the generalized spectral problem ... where $\upsilon(\cdot,t)$ is the singular part of the energy measure $\mu(\cdot,t)$. Let $\sigma(t)$ den...

open_problem | open | MathDB 页面 | 原文线索

V2-077695 | MATHDB-399464

General methods for solving rational and twisted reflection equations

摘录 / EXCERPT: Find a universal method for obtaining general matrix-valued solutions $K(u)$ of either of the reflection equations ... or ... for the rational R-matrices under consideration.

open_problem | open | MathDB 页面 | 原文线索

V2-077696 | MATHDB-399465

Quantization of solutions of the classical reflection equation

摘录 / EXCERPT: Given a solution $(\mathfrak g,\sigma,r)$ of the classical reflection equation (CRE), is it possible to lift it to construct a corresponding coideal in $U_q(\mathfrak g)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077697 | MATHDB-399469

Relations between different notions of convergence in deformation quantization

摘录 / EXCERPT: The precise relations between the different versions of convergence remain unclear.

open_problem | open | MathDB 页面 | 原文线索

V2-077698 | MATHDB-399472

The Isomorphism Problem for Kinematic Chains

摘录 / EXCERPT: Let the variables x_i and y_i range over the real line. Given an ordered set of $n\times n$ matrices $A_1,A_2,\ldots,A_m$, characterize the collection of all ordered sets...

open_problem | open | MathDB 页面 | 原文线索

V2-077699 | MATHDB-399473

Feasibility of deadline-constrained communication over unreliable links

摘录 / EXCERPT: Consider one central node and N peripheral nodes. At each discrete time instant, at most one packet may be sent from the central node to one peripheral node. A packet sent to nod...

open_problem | open | MathDB 页面 | 原文线索

V2-077700 | MATHDB-399474

Provable safety control for automated transportation systems

摘录 / EXCERPT: Consider a system of roadways, each consisting of multiple lanes, with traffic intersections, in which cars are modeled as unicycles and can communicate with each other and with th...

open_problem | open | MathDB 页面 | 原文线索

V2-077701 | MATHDB-399475

Commutation of discretization and optimization for general variational integrators

摘录 / EXCERPT: For general classes of variational integrators, does discretization and optimization commute?

open_problem | open | MathDB 页面 | 原文线索

V2-077702 | MATHDB-399477

Global convergence to a four-vehicle formation shape for nonrectangular formations

摘录 / EXCERPT: For four point agents with positions $p=[p_1,p_2,p_3,p_4]^{\top}\in\mathbb{R}^8$, desired inter-agent distances $\bar d_{ij}$, instantaneous distances...

open_problem | open | MathDB 页面 | 原文线索

V2-077703 | MATHDB-399478

Converse implication from the turnpike property to strict dissipativity

摘录 / EXCERPT: Under which assumptions does the converse implication to (2) hold?

open_problem | open | MathDB 页面 | 原文线索

V2-077704 | MATHDB-399479

Strict Dissipativity and the Turnpike Property

摘录 / EXCERPT: Consider discrete-time optimal control problems ... subject to ... where $f:X\times U\to\mathbb R^n$, $\ell:X\times U\to\mathbb R$, and $X\subset\mathbb R^n$, $U\subset\mathbb R^m$ are...

open_problem | open | MathDB 页面 | 原文线索

V2-077705 | MATHDB-399481

Whether circumcircle-preserving interpolation has lower quasiconformal distortion

摘录 / EXCERPT: Given circumcircle-preserving piecewise projective interpolation between triangulations, which is continuous across edges if and only if the triangulations are discretely conformal...

open_problem | open | MathDB 页面 | 原文线索

V2-077706 | MATHDB-399483

Discrete Riemann mapping conjecture

摘录 / EXCERPT: Suppose Ω is a Jordan domain in the complex plane. Take three distinct points p,q,r in the boundary of Ω. For each $\epsilon>0$, let Ω_ϵ be a maximum...

conjecture | open | MathDB 页面 | 原文线索

V2-077707 | MATHDB-399484

Hyperideal anti-de Sitter polyhedra: dihedral angles and induced metrics

摘录 / EXCERPT: Do the descriptions of the dihedral angles and induced metrics of ideal anti-de Sitter polyhedra extend to hyperideal anti-de Sitter polyhedra, that is, polyhedra with all vertices...

open_problem | open | MathDB 页面 | 原文线索

V2-077708 | MATHDB-399486

Existence of generalized Miura tessellations for positively curved surfaces

摘录 / EXCERPT: Does every positively curved surface admit a generalized Miura tessellation that approximates it?

open_problem | open | MathDB 页面 | 原文线索

V2-077709 | MATHDB-399487

A notion of “nested enough” for multiresolution hierarchies

摘录 / EXCERPT: Determine whether there exists a concept of “nested enough” for the multiresolution solver.

open_problem | open | MathDB 页面 | 原文线索

V2-077710 | MATHDB-399488

General feasibility test for nested cages

摘录 / EXCERPT: Determine a general feasibility test for the nested-cage pairwise problem: given a fine mesh F and a coarse mesh C, decide whether there exists an embedding of C that strictl...

open_problem | open | MathDB 页面 | 原文线索

V2-077711 | MATHDB-399490

Wilson’s conjecture on double-Ising interfaces

摘录 / EXCERPT: At the critical temperature $x=x_{\mathrm{crit}}$, the $13\mid24$ domain walls converge to Schramm’s $\mathrm{SLE}(4)$ curves with drifts depending on the boundary conditions.

conjecture | open | MathDB 页面 | 原文线索

V2-077712 | MATHDB-399492

Optimal aspect ratio of a smooth paper Möbius band

摘录 / EXCERPT: What is the smallest length λ such that a smooth Möbius band can be made from a $1\times a$ paper rectangle if $a>\lambda$? The known bounds are...

open_problem | open | MathDB 页面 | 原文线索

V2-077713 | MATHDB-399494

Existence of offset polytopes

摘录 / EXCERPT: We are given a non-convex three-dimensional polytope P whose boundary is homeomorphic to a sphere. We want to construct an offset polytope P_ε in which every face is...

open_problem | open | MathDB 页面 | 原文线索

V2-077714 | MATHDB-399495

Subdivision of discrete conformal structures

摘录 / EXCERPT: Let T be a triangulated surface, and let $\bblambda,\bbmu:E_T\to\mathbb{R}_{>0}$ be two discrete conformally equivalent metrics on T, represented as edge lengths. Here, discret...

open_problem | open | MathDB 页面 | 原文线索

V2-077715 | MATHDB-399504

Universal property of global spectra

摘录 / EXCERPT: Theorem (Gepner, N.). The functor $\Sigma^\infty_+ : \mathrm{SOrb} \to \mathrm{SpOrb}$ exhibits the ∞-category of Orbispectra as the universal SOrb-stable...

open_problem | open | MathDB 页面 | 原文线索

V2-077716 | MATHDB-399509

Donaldson-Thomas invariants for Bridgeland-semistable objects on Calabi-Yau 3-folds

摘录 / EXCERPT: Let X be a smooth projective Calabi-Yau 3-fold satisfying Conjecture 1 (the BG-type inequality stated on PDF page 11). Then, for every $v\in H^*(X,\mathbb{Q})$, there is a map...

open_problem | open | MathDB 页面 | 原文线索

V2-077717 | MATHDB-399510

Stable irrationality of very general complex hypersurfaces

摘录 / EXCERPT: For all $n\ge 3$ and all $d\ge 2\left\lceil\frac{n+2}{3}\right\rceil$, a very general complex hypersurface of degree d in $\mathbf{P}^{n+1}$ is not stably rational.

open_problem | open | MathDB 页面 | 原文线索

V2-077718 | MATHDB-399514

Logarithmic subadditivity under a big generic log-canonical divisor

摘录 / EXCERPT: Let $f:(X,D)\to(Y,E)$ be a surjective map of log-smooth projective pairs with coefficients at most 1, such that $D\ge f^*E$ and $K_{X_\eta}+D_\eta$ is big, where η is the ge...

open_problem | open | MathDB 页面 | 原文线索

V2-077719 | MATHDB-399515

Morrison’s conjecture on mirror symmetry and conifold transitions

摘录 / EXCERPT: For a conifold transition $X\rightsquigarrow Z$ and mirror-symmetry duals $\check{X}$ and $\check{Z}$ of X and Z, respectively, $\check{X}$ and $\check{Z}$ are related by a con...

conjecture | open | MathDB 页面 | 原文线索

V2-077720 | MATHDB-399516

Claimed proof of Morrison’s conjecture for projective conifold transitions

摘录 / EXCERPT: Given a conifold transition $X\rightsquigarrow Z$ with both ends projective, we thus obtain the mirror dual one $\check{Z}\rightsquigarrow\check{X}$, proving Morrison’s conjecture.

conjecture | open | MathDB 页面 | 原文线索

V2-077721 | MATHDB-399517

Wall crossing for derived categories of moduli spaces of sheaves on rational surfaces

摘录 / EXCERPT: Theorem 1 asserts the following. Let S be a smooth projective rational surface over $\[?]C$ with effective anti-canonical bundle, and let L_- and L_+ be ample line bundles on...

open_problem | open | MathDB 页面 | 原文线索

V2-077722 | MATHDB-399520

Formality conjecture for polystable sheaves on K3 surfaces

摘录 / EXCERPT: It is conjectured that every polystable sheaf $F=\bigoplus_{i=1}^{s}F_i^{n_i}$ on a K3 surface has formal dgla $R\operatorname{Hom}(F,F)$; equivalently, $R\operatorname{Hom}(F,F)$...

conjecture | open | MathDB 页面 | 原文线索

V2-077723 | MATHDB-399521

Bogomolov-Gieseker-type inequality for tilt-semistable objects on polarized threefolds

摘录 / EXCERPT: Conjecture. Let (X,H) be a smooth polarized threefold, and let $\alpha,\beta\in\mathbb{R}$ with $\alpha>0$. If $E\in\operatorname{Coh}^{\beta}(X)$ is $\nu_{\alpha,\beta}$-semista...

open_problem | open | MathDB 页面 | 原文线索

V2-077724 | MATHDB-399522

Auxiliary inequality (*) in the proof of the Bogomolov-Gieseker conjecture for abelian threefolds

摘录 / EXCERPT: () If E is $u_{[U+007F]\alpha,\beta}$-stable for (α,β) near $(0,0)$, and if E satisfies $H\operatorname{ch}_2(E)=0$, then it also satisfies $\operatorname{ch}_3(E)\le 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-077725 | MATHDB-399523

Mirror map and period identity for Fano polygons

摘录 / EXCERPT: Conjecture 2. Let P be a Fano polygon and let X be a generic qG-deformation of the toric surface X_P. Let L_P^T denote the affine space of maximally-mutable Laurent polynom...

open_problem | open | MathDB 页面 | 原文线索

V2-077726 | MATHDB-399524

Mutation equivalence determined by classical periods

摘录 / EXCERPT: Conjecture 3. Let P_1 and P_2 be Fano polygons with the same singularity content. Suppose that there is an affine-linear isomorphism $\varphi: L_TP_1\to L_TP_2$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-077727 | MATHDB-399530

Bulk-field object for non-semisimple full conformal field theories

摘录 / EXCERPT: For a non-semisimple factorizable finite ribbon category C, find an object F in the enveloping category $D := C \boxtimes C^{\mathrm{rev}}$ that gives the space of bulk fields...

open_problem | open | MathDB 页面 | 原文线索

V2-077728 | MATHDB-399531

Mapping-class-group and sewing invariants for arbitrary factorizable finite ribbon categories

摘录 / EXCERPT: Let $\mathcal C$ be an arbitrary factorizable finite ribbon category, let $\mathcal D:=\mathcal C\boxtimes\mathcal C^{\mathrm{rev}}$, and let φ be a ribbon automorphism of t...

open_problem | open | MathDB 页面 | 原文线索

V2-077729 | MATHDB-399534

Tannakian duality for unitary modular tensor categories

摘录 / EXCERPT: Conjecture (Tannakian duality for UMTCs). Let N be a type III factor and let $\mathcal{C}\subset\operatorname{End}(N)$ be a full and replete unitary modular tensor category. Ever...

open_problem | open | MathDB 页面 | 原文线索

V2-077730 | MATHDB-399535

Surjectivity of the weakly spherical-to-fusion-category classification map

摘录 / EXCERPT: Is the indicated surjectivity assertion true in dimension three: does every fusion category lie in the image of the forgetful map from weakly spherical fusion categories to fusion...

open_problem | open | MathDB 页面 | 原文线索

V2-077731 | MATHDB-399536

Anderson’s infinite-dimensional paving conjecture

摘录 / EXCERPT: For every $\varepsilon>0$ there exists $r\in\mathbb{N}$ such that, for every self-adjoint $H\in B(\ell^2)$ with zero diagonal, there exists a partition $\{A_1,\ldots,A_r\}$ of...

conjecture | open | MathDB 页面 | 原文线索

V2-077732 | MATHDB-399537

Unimodality of cycle-counting coefficients for functions

摘录 / EXCERPT: For $k,n\in\mathbb{N}_0$ with $0\le k\le n$, define $\left\{\begin{smallmatrix}n\\k\end{smallmatrix}\right\}$ to be the number of functions $f:\{1,\ldots,n\}\to\{1,\ldots,n\}$ havi...

open_problem | open | MathDB 页面 | 原文线索

V2-077733 | MATHDB-399538

Largest-root bound for mixed characteristic polynomials

摘录 / EXCERPT: Theorem. Let $A_1,\ldots,A_m$ be Hermitian positive semidefinite matrices satisfying $\sum_{i=1}^m A_i=I$ and $\operatorname{tr}(A_i)\le\varepsilon$ for all i. Define ... Then the...

open_problem | open | MathDB 页面 | 原文线索

V2-077734 | MATHDB-399539

Random-vector norm bound (Theorem 5)

摘录 / EXCERPT: Let $\varepsilon>0$ and let $v_1,\ldots,v_m\in\mathbb{C}^d$ be independent random vectors with finite support such that ... and ... Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077735 | MATHDB-399540

Weaver’s conjecture (Kadison-Singer partition conjecture)

摘录 / EXCERPT: There exist universal constants $\b7 \ge 2$ and $\b8>0$ such that, for vectors $w_1,\ldots,w_m\cin \bb C^d$ satisfying $\cnorm{w_i}\cle 1$ for all i and ... for every unit vector...

conjecture | open | MathDB 页面 | 原文线索

V2-077736 | MATHDB-399541

Marcus-Spielman-Srivastava partition theorem

摘录 / EXCERPT: Let $k\ge 2$ be an integer. Suppose $v_1,\ldots,v_m\in\mathbb{R}^d$ satisfy $\sum_{i=1}^m v_i v_i^*=I$, where I is the identity matrix, and $\|v_i\|^2\le\varepsilon$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-077737 | MATHDB-399542

Brändén’s hyperbolic-polynomial partition theorem

摘录 / EXCERPT: Let $k\ge 2$ be an integer and let $\varepsilon>0$. Suppose h is hyperbolic with respect to $e\in\mathbb{R}^n$, and let $u_1,\ldots,u_m\in\Lambda^+(e)$ satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-077738 | MATHDB-399544

Strongly Rayleigh measures are negatively associated

摘录 / EXCERPT: Strongly Rayleigh measures are negatively associated.

open_problem | open | MathDB 页面 | 原文线索

V2-077739 | MATHDB-399545

Negative association for symmetric exclusion processes with deterministic initial data

摘录 / EXCERPT: Let S be a finite set, and let $(\eta_t)_{t\ge 0}$ be a symmetric exclusion process on S, in which sites $i,j\in S$ exchange their particles at rates...

open_problem | open | MathDB 页面 | 原文线索

V2-077740 | MATHDB-399547

Constant integrality gap for the Held-Karp relaxation of ATSP

摘录 / EXCERPT: Is the integrality gap of the natural Held-Karp linear-programming relaxation for the Asymmetric Traveling Salesman Problem bounded above by a constant?

open_problem | open | MathDB 页面 | 原文线索

V2-077741 | MATHDB-399549

Matrix version of Spencer’s “six standard deviations suffice” theorem

摘录 / EXCERPT: Given $n\times n$ matrices $A_1,\ldots,A_n$ with $\|A_i\|\le 1$, does there exist a signing $s_1,\ldots,s_n\in\{\pm1\}$ such that $\left\|\sum_{i=1}^n s_iA_i\right\|\le O(\sqrt n)$...

open_problem | open | MathDB 页面 | 原文线索

V2-077742 | MATHDB-399550

Hyperbolicity of the volume polynomial of a Minkowski sum of line segments

摘录 / EXCERPT: If $u_1,\ldots,u_n$ are line segments in $\mathbb{R}^n$, is the polynomial ... hyperbolic? Here the sum is a Minkowski sum.

open_problem | open | MathDB 页面 | 原文线索

V2-077743 | MATHDB-399558

Morita equivalence for the blocks in Corollary 8

摘录 / EXCERPT: Conjecture 9. The blocks in the above corollary are Morita equivalent.

open_problem | open | MathDB 页面 | 原文线索

V2-077744 | MATHDB-399564

Bounds from the largest p-part of ordinary character degrees

摘录 / EXCERPT: Conjecture 3 of the report states: If p^a is the largest power of p dividing the irreducible characters of G, then $|G:O_p(G)|_p$ is bounded by $p^{f(a)}$, where $f(a)$ is a...

open_problem | open | MathDB 页面 | 原文线索

V2-077745 | MATHDB-399565

Functoriality of perfect localities for injective homomorphisms

摘录 / EXCERPT: Let P' and P be finite p-groups with Frobenius categories F' and F, respectively. If an injective group homomorphism $f:P'\to P$ induces a functor $f:F'\to F$, does there...

open_problem | open | MathDB 页面 | 原文线索

V2-077746 | MATHDB-399573

Origin of the extra normalized stable p-permutation equivalence in the cyclic 2-power defect case

摘录 / EXCERPT: In the situation of Broué’s Abelian Defect Group Conjecture, when the defect group D is cyclic and $|D|=2^n$ with $n\ge 2$, there are two normalized stable p-permutation equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-077747 | MATHDB-399575

Existence of two-hump surface waves with surface tension

摘录 / EXCERPT: Do there exist two-dimensional two-hump traveling surface waves on water of finite depth with surface tension?

open_problem | open | MathDB 页面 | 原文线索

V2-077748 | MATHDB-399577

Validity of the NLS approximation for the two-dimensional water-wave problem

摘录 / EXCERPT: The report states and proves the following approximation theorem. Let $b=0$ and $s\ge 7$. For every $k_0>0$ and $C_1,\tau_0>0$, there exist $\tau_1>0$ and $\varepsilon_0>0$ such th...

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V2-077749 | MATHDB-399578

Existence of a highest cusped periodic travelling wave for the Whitham equation

摘录 / EXCERPT: The report presents the existence of a highest, cusped, periodic travelling-wave solution of the Whitham equation ... where $L f=K*f$ and ... In steady variables...

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V2-077750 | MATHDB-399579

Wave breaking for the Whitham equation

摘录 / EXCERPT: Whitham’s conjecture was that the equation ... combining the surface-water-wave dispersion relation with shallow-water nonlinearity, exhibits wave breaking for sufficiently steep ini...

open_problem | open | MathDB 页面 | 原文线索

V2-077751 | MATHDB-399580

Local solvability of the initial-value problem for a model system for water waves

摘录 / EXCERPT: For the displayed nonlinear dispersive model system for water waves, with initial conditions $(\eta,\phi_0,\phi_1)=(\eta_0,\phi_0^0,\phi_1^0)$ at $t=0$, if the initial data and bot...

open_problem | open | MathDB 页面 | 原文线索

V2-077752 | MATHDB-399581

Energy density and flux for the general Boussinesq-type system (3)

摘录 / EXCERPT: For the approximate system ... with $0\leq\theta\leq1$, $b=\tfrac12(\theta^2-\tfrac13)h_0^2$, and $d=\tfrac12(1-\theta^2)h_0^2$, express the wave-motion energy density and energy flu...

open_problem | open | MathDB 页面 | 原文线索

V2-077753 | MATHDB-399582

Non-existence of solitary waves of depression in the presence of vorticity

摘录 / EXCERPT: Let (Ψ,η) be a solution of the free-boundary problem (1)-(4), namely ... ... where $D=\{(x,y)\in\mathbb R^2:0<y<\eta(x)\}$. Suppose that $\eta(x)\le d$ for all $x\in\mathbb R$...

open_problem | open | MathDB 页面 | 原文线索

V2-077754 | MATHDB-399583

Asymptotic stability of solitary waves in the Benney-Luke model

摘录 / EXCERPT: The report presents the following theorem. Let $c_0>\sigma>1$ and $0<\alpha<\frac{1}{2}\alpha_{c_0}$, where ... Assume that, in the exponentially weighted space ... the linearized oper...

open_problem | open | MathDB 页面 | 原文线索

V2-077755 | MATHDB-399584

Two-hump surface waves for the exact finite-depth water-wave problem

摘录 / EXCERPT: Do the exact two-dimensional finite-depth water-wave equations for an inviscid, incompressible fluid of constant density, bounded below by a horizontal rigid bottom and above by a...

open_problem | open | MathDB 页面 | 原文线索

V2-077756 | MATHDB-399586

Matrix probing recovery theorem

摘录 / EXCERPT: Let $A\in\mathbb{R}^{n\times n}$ be recoverable from $y_k=Ax_k$, where $x_1,\ldots,x_q\sim N(0,I_n)$, and suppose that A belongs to the known p-dimensional linear subspace span...

open_problem | open | MathDB 页面 | 原文线索

V2-077757 | MATHDB-399587

Efficient numerical tools for discovering useful compositions

摘录 / EXCERPT: Can we provide efficient numerical tools for finding such compositions?

open_problem | open | MathDB 页面 | 原文线索

V2-077758 | MATHDB-399588

Regularity of the inverse Born approximation

摘录 / EXCERPT: Let $\varepsilon>0$, let $s\in\mathbb{N}$ with $s\ge 2$, and let, for some $x_0\in\mathbb{R}^2$, $f\in L^2(\mathbb{R}^2)\cap H^{s+\varepsilon}(x_0)$ be compactly supported and real...

open_problem | open | MathDB 页面 | 原文线索

V2-077759 | MATHDB-399589

BV bounds for multidimensional graph-viscosity approximations

摘录 / EXCERPT: In the multidimensional case, it is not known whether the numerical solution $u_h(\boldsymbol{\3},t)$ generated by the graph-viscosity stabilization satisfies a bounded-variation b...

open_problem | open | MathDB 页面 | 原文线索

V2-077760 | MATHDB-399592

Multivariate Archimax copula conjecture (Open Problem 4.1)

摘录 / EXCERPT: For the d-variate setting ($d\ge 3$), let $\psi:[0,\infty)\to[0,1]$ be the generator of a d-variate Archimedean copula and let $\ell:[0,\infty)^d\to[0,\infty)$ be a d-variate...

conjecture | open | MathDB 页面 | 原文线索

V2-077761 | MATHDB-399593

Uniqueness of an Archimedean copula from its Kendall function

摘录 / EXCERPT: For any dimension $d\ge 2$, the Kendall function K uniquely determines an Archimedean copula family.

open_problem | open | MathDB 页面 | 原文线索

V2-077762 | MATHDB-399596

Dimension of multivariate spline spaces

摘录 / EXCERPT: Determine the dimension of $S_k^r(\Delta)$, the vector space of C^r splines on a fixed d-dimensional simplicial complex $\Delta\subseteq\mathbb{R}^d$, where each polynomial pie...

open_problem | open | MathDB 页面 | 原文线索

V2-077763 | MATHDB-399598

Interpolation at triangulation vertices by cubic C^1 splines

摘录 / EXCERPT: It is unknown whether it is always possible to interpolate to the function values at the vertices of the triangulation with a spline in S^1_3. The author conjectures that it is p...

open_problem | open | MathDB 页面 | 原文线索

V2-077764 | MATHDB-399599

Postulation-number bound for mixed splines on central polytopal complexes

摘录 / EXCERPT: Let $\mathcal{P}\subset\mathbb{R}^{3}$ be a central, pure, hereditary three-dimensional polytopal complex. Given smoothness parameters $\alpha(\tau)$ associated to each codimension...

open_problem | open | MathDB 页面 | 原文线索

V2-077765 | MATHDB-399600

Strong form of Schumaker’s partial interpolation conjecture

摘录 / EXCERPT: Let T be a triangle with vertices $v_1,v_2,v_3\in\mathbb{R}^2$, and let $d\ge 1$. Define ... For $v\in T$, let ... where (x,y,z) are the barycentric coordinates of v. Schumaker’s...

conjecture | open | MathDB 页面 | 原文线索

V2-077766 | MATHDB-399602

Dimension of the cubic C^1 bivariate spline space

摘录 / EXCERPT: Let Δ be a triangulation of a polygonal region $\Omega\subset\mathbb{R}^2$. Define ... The report presents as a conjecture that ... where $V_B(\Delta)$ and $V_I(\Delta)$ are res...

open_problem | open | MathDB 页面 | 原文线索

V2-077767 | MATHDB-399603

Characterization of non-interpolating hexagonal cells for C^1 quadratic splines

摘录 / EXCERPT: Characterize all hexagonal cells Δ such that $S^1_2(\Delta)$ does not interpolate at the exterior vertices $v_1,\ldots,v_6$, where $S^1_2(\Delta)$ is the vector space of all...

open_problem | open | MathDB 页面 | 原文线索

V2-077768 | MATHDB-399604

Higher-dimensional approximation estimates for tensor-product splines

摘录 / EXCERPT: A major challenge for future research concerns estimates in higher-dimensional cases. In particular, the dependence of the constant C on the shape of the domain and the aspect ra...

open_problem | open | MathDB 页面 | 原文线索

V2-077769 | MATHDB-399608

Dimension of trivariate C^1 splines on generic bipyramid cells

摘录 / EXCERPT: Let Δ be a bipyramid cell with n coplanar boundary vertices and let m be the number of distinct slopes determined by the coplanar interior edges $[v_0,v_i]$. In the gene...

open_problem | open | MathDB 页面 | 原文线索

V2-077770 | MATHDB-399611

Weight filtration of the GKZ system

摘录 / EXCERPT: Conjecture. Let NA be a saturated semigroup. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077771 | MATHDB-399613

Orbifold equivalence of Q_{10} and K_{14}

摘录 / EXCERPT: Conjecture: $Q_{10}\sim_{\mathrm{orb}} K_{14}$, where Q_{10} and K_{14} are exceptional unimodular singularities from Arnold’s classification.

open_problem | open | MathDB 页面 | 原文线索

V2-077772 | MATHDB-399614

Realizability of trivalent immersed tropical curves in power-type abelian surfaces

摘录 / EXCERPT: Let $k:=\mathbb{C}((t))$. Let $A=\begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}$ be a 2×2 integer matrix with non-zero determinant, and let ϖ_1,ϖ_2 be...

open_problem | open | MathDB 页面 | 原文线索

V2-077773 | MATHDB-399615

Batyrev’s conjecture relating Laurent-polynomial periods to Gromov-Witten invariants

摘录 / EXCERPT: For a toric Y_{sing} (or Y_{res}), associate the Laurent polynomial $W=\sum_v z^v$, where the sum runs over all vertices v of the fan polytope of...

conjecture | open | MathDB 页面 | 原文线索

V2-077774 | MATHDB-399618

Viro's conjecture on the first Betti number of the real part

摘录 / EXCERPT: Let X be a compact, connected, simply-connected projective real surface. Then $b_1(RX)\leq h^{1,1}(X)$, where RX is the real part of X.

conjecture | open | MathDB 页面 | 原文线索

V2-077775 | MATHDB-399620

Correspondence theorem for real tropical double Hurwitz numbers

摘录 / EXCERPT: Fix integers d,g, partitions σ,γ of d, and n_+,n_- marked simple branch points on the two arcs $\mathbb{R}\mathbb{P}^1\setminus\{0,\infty\}$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-077776 | MATHDB-399621

Nonnegativity versus solidity for circuit-supported polynomials

摘录 / EXCERPT: Let ... where $\Delta=\operatorname{conv}\{\alpha(0),\ldots,\alpha(n)\}\subset\mathbb R^n$ is a lattice simplex, $\alpha(j)\in(2\mathbb N)^n$, $y\in\operatorname{int}(\Delta)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-077777 | MATHDB-399631

Reduction from an arbitrary point to a closed point after field extension

摘录 / EXCERPT: Theorem 8. Let p be a point on $\operatorname{Proj} H^*(G,k)$. Let K be the residue field at p and let $\mathfrak m$ be a closed point in $\operatorname{Proj} H^*(G_K,K)$ “ly...

open_problem | open | MathDB 页面 | 原文线索

V2-077778 | MATHDB-399635

A conjectural blue-shift statement for Tate constructions

摘录 / EXCERPT: Let G be a p-group and let $X\in SH$ be a nonequivariant spectrum. If $X\otimes C_{p,n}=0$, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077779 | MATHDB-399642

Noetherianity of the projective generic representations

摘录 / EXCERPT: Lionel Schwartz conjectured that, for every integer $n\geq 0$, the functor $P_n\in\operatorname{Rep}(\mathbb F;\mathbb F)$ defined by...

open_problem | open | MathDB 页面 | 原文线索

V2-077780 | MATHDB-399643

Intersection criterion for irreducible components of two-row type-D Springer fibers

摘录 / EXCERPT: Fix an even positive integer $n=2m$ and $k\in\{1,\ldots,m\}$ such that either $k=m$ or k is odd. Let $F\ell_{n-k,k}$ be the type-D Springer fiber associated with a nilpotent endo...

open_problem | open | MathDB 页面 | 原文线索

V2-077781 | MATHDB-399644

Categorical diagonalizability of Rouquier full twists

摘录 / EXCERPT: Conjecture 3. Let F_T denote the Rouquier complex attached to the full twist of any finite Coxeter group W. Then F_T is categorically diagonalizable, and its spectrum is in b...

open_problem | open | MathDB 页面 | 原文线索

V2-077782 | MATHDB-399650

Finite parabolic conjugation on the nilpotent cone

摘录 / EXCERPT: For which standard parabolic subgroups $P\subset GL_n(\mathbb{C})$, with block sizes $(b_1,\ldots,b_p)$, is the number of P-orbits in...

open_problem | open | MathDB 页面 | 原文线索

V2-077783 | MATHDB-399655

Koszul duals of the categorical-action functors

摘录 / EXCERPT: Determine the Koszul dual functors to the functors E and F acting on the parabolic category $\mathcal{O}(\widehat{\mathfrak{gl}}_n)$ at level $-e-n$; the finite-type case sugg...

open_problem | open | MathDB 页面 | 原文线索

V2-077784 | MATHDB-399657

Irredundant inequalities for the multiplicative eigenpolytope

摘录 / EXCERPT: Let $n\ge 2$. As P ranges over the standard maximal parabolic subgroups, $(u_1,\ldots,u_n)\in (W^P)^n$, and non-negative integers d satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-077785 | MATHDB-399658

Deligne’s conjecture on universal abelian tensor categories

摘录 / EXCERPT: Let F be a field of characteristic zero. For a rigid symmetric F-linear tensor category A and an object $X\in A$ of dimension t, the tensor functor $F_X:D_t\to A$ uniquely...

conjecture | open | MathDB 页面 | 原文线索

V2-077786 | MATHDB-399659

Achievable estimation accuracy in the $M/G/\infty$ problem

摘录 / EXCERPT: What is the achievable estimation accuracy for estimating the service-time distribution G in the $M/G/\infty$ queue, and can one construct optimal estimators?

open_problem | open | MathDB 页面 | 原文线索

V2-077787 | MATHDB-399661

Optimal rate for estimating a size-dependent division rate from temporal data

摘录 / EXCERPT: In the size-dependent model, establish in full generality that the optimal rate for recovering the division rate $B(x)$ from data $\U_T$ is $e^{-s\lambda_B T/(2s+3)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-077788 | MATHDB-399662

Minimax detection bound for changes in volatility smoothness

摘录 / EXCERPT: Suppose the volatility at time θ either has a jump of absolute size larger than b_n, or its smoothness drops to a' on an interval of length at least $b_n^{1/a}$. For tes...

open_problem | open | MathDB 页面 | 原文线索

V2-077789 | MATHDB-399663

Adaptive estimation of the linear functional under sparsity constraints

摘录 / EXCERPT: In the Gaussian sequence model $y_j=\theta_j+\sigma\xi_j$, with $\theta\in\mathbb{R}^d$, the report asks whether the minimax rates for estimating $L(\theta)=\sum_{j=1}^d\theta_j$ c...

open_problem | open | MathDB 页面 | 原文线索

V2-077790 | MATHDB-399668

Conceptual explanation of the covariance for tensor-product fluctuations

摘录 / EXCERPT: Find a conceptual explanation for the covariance of the Gaussian global fluctuations arising from tensor products of irreducible representations; the covariance is structurally sim...

open_problem | open | MathDB 页面 | 原文线索

V2-077791 | MATHDB-399670

Realization of nc rational functions around a matrix point

摘录 / EXCERPT: Conjecture: If a noncommutative rational function r is regular at $Y\in (K^{s\times s})^d$, then r admits a unique, up to unique similarity, minimal (controllable and observabl...

open_problem | open | MathDB 页面 | 原文线索

V2-077792 | MATHDB-399672

The Douglas-Kazakov third-order phase transition

摘录 / EXCERPT: For every $T>0$, the limit ... exists. The function F is smooth on $\mathbb{R}_+^*\setminus\{\pi^2\}$, belongs to $C^2(\mathbb{R}_+^*)$, and its third derivative F''' has a jump...

open_problem | open | MathDB 页面 | 原文线索

V2-077793 | MATHDB-399675

Asymptotic freeness of partial transposes of Gaussian Wishart matrices

摘录 / EXCERPT: Let $G_1,\\ldots,G_{d_1}$ be independent complex Gaussian $d_2\\times p$ random matrices with entry mean 0 and complex variance 1, and set ... Write...

open_problem | open | MathDB 页面 | 原文线索

V2-077794 | MATHDB-399676

Spectral gap for dense subgroup translations on compact groups

摘录 / EXCERPT: Let G be a second-countable locally compact group and let $\Gamma<G$ be a countable dense subgroup. Consider the left translation action $\Gamma\curvearrowright G$, given by...

open_problem | open | MathDB 页面 | 原文线索

V2-077795 | MATHDB-399677

Ruziewicz’s problem for the sphere

摘录 / EXCERPT: Is Lebesgue measure the unique finitely additive measure on the n-dimensional sphere S^n that is invariant under rotations and defined on all Lebesgue measurable sets?

open_problem | open | MathDB 页面 | 原文线索

V2-077796 | MATHDB-399678

Eigenvalue at the top of the spectrum of an averaging operator

摘录 / EXCERPT: In the setting described in the report, with $G=\mathrm{SU}(2)$ and an averaging operator $P:L^2(G)\to L^2(G)$ of the form $P(\xi)=\frac{1}{|S|}\sum_{g\in S}g\cdot\xi$ for a finite...

open_problem | open | MathDB 页面 | 原文线索

V2-077797 | MATHDB-399681

Kaplansky’s direct finiteness conjecture

摘录 / EXCERPT: For any field R and any group Γ, the group ring $R\Gamma$ is directly finite: for any $a,b\in R\Gamma$, if $ab=1$, then $ba=1$.

conjecture | open | MathDB 页面 | 原文线索

V2-077798 | MATHDB-399682

Trivial amenable radical versus C*-simplicity

摘录 / EXCERPT: For a discrete group G, does $R(G)=1$ imply that G is C^*-simple, where $R(G)$ is the amenable radical of G?

open_problem | open | MathDB 页面 | 原文线索

V2-077799 | MATHDB-399683

Classification problem of Bost-Connes systems

摘录 / EXCERPT: For number fields K and L, if the associated Bost-Connes systems $(A_K,\sigma_{t,K})$ and $(A_L,\sigma_{t,L})$ are isomorphic, must K and L be isomorphic?

open_problem | open | MathDB 页面 | 原文线索

V2-077800 | MATHDB-399684

K-theoretic Farrell-Jones assembly conjecture

摘录 / EXCERPT: For a discrete group G and a ring R equipped with an action of G by automorphisms, the assembly map ... should be an isomorphism, where Vcyc is the family of virtual...

conjecture | open | MathDB 页面 | 原文线索

V2-077801 | MATHDB-399685

K-theory Novikov conjecture

摘录 / EXCERPT: For every discrete group G, the assembly map ... is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-077802 | MATHDB-399689

Finite-part operator K-theory conjecture

摘录 / EXCERPT: Let G be a countable group. For each nonidentity finite-order element $g\in G$ of order d, define ... and let $K^{\mathrm{fin}}_0(C^*(G))$ be the subgroup of $K_0(C^*(G))$ genera...

conjecture | open | MathDB 页面 | 原文线索

V2-077803 | MATHDB-399691

Polynomial diameter bound for symmetric groups (Babai’s conjecture)

摘录 / EXCERPT: Babai’s conjecture: the diameter of the symmetric group $\operatorname{Sym}(n)$ is polynomially bounded in n.

conjecture | open | MathDB 页面 | 原文线索

V2-077804 | MATHDB-399692

Coarse Baum-Connes conjecture for the warped cones in Theorem 3

摘录 / EXCERPT: We conjecture that the coarse Baum-Connes conjecture fails for warped cones provided by the above theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-077805 | MATHDB-399693

Warped cones and coarsely embedded sequences of expanders

摘录 / EXCERPT: Is there an action with a spectral gap such that the corresponding warped cone does not contain a coarsely embedded sequence of expanders?

open_problem | open | MathDB 页面 | 原文线索

V2-077806 | MATHDB-399694

Geometric CAT(-1) actions for hyperbolic groups

摘录 / EXCERPT: Does every hyperbolic group admit a geometric action on a CAT(-1) space?

open_problem | open | MathDB 页面 | 原文线索

V2-077807 | MATHDB-399696

Finitely presented groups violating a cyclic-subgroup dichotomy

摘录 / EXCERPT: Find a finitely presented group which violates either dichotomy.

open_problem | open | MathDB 页面 | 原文线索

V2-077808 | MATHDB-399697

Action Dimension Conjecture

摘录 / EXCERPT: For a group G, the action dimension satisfies ... where $\ell^2\!\dim G:=\sup\{i\mid b_i^{(2)}(G)\ne 0\}$ is the largest degree in which an ℓ^2-Betti number of G is nonzero.

conjecture | open | MathDB 页面 | 原文线索

V2-077809 | MATHDB-399698

Lott-Lück spectral-density optimality conjecture

摘录 / EXCERPT: For every continuous computable function $g:\mathbb{R}_{+}\to\mathbb{R}_{+}$ such that $g(\varepsilon)\to 0$ as $\varepsilon\to 0$, there exist a group G and a self-adjoint eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-077810 | MATHDB-399699

Lott-Lück conjecture on spectral-density decay

摘录 / EXCERPT: Let G be a discrete group. For a self-adjoint $T\in\mathbb{Z}[G]$ there exist $C,\eta>0$ such that, for sufficiently small ε, ...

conjecture | open | MathDB 页面 | 原文线索

V2-077811 | MATHDB-399700

Rationality of Novikov-Shubin invariants for rational group-ring elements

摘录 / EXCERPT: For a discrete group Γ and a self-adjoint element $T\in\mathbb{Q}[\Gamma]$, the Novikov-Shubin invariant satisfies $\alpha(T)\in\mathbb{Q}$.

open_problem | open | MathDB 页面 | 原文线索

V2-077812 | MATHDB-399705

Finitely generated groups acting properly on CAT(0) cubical complexes without Property A

摘录 / EXCERPT: There exist finitely generated groups acting properly on CAT(0) cubical complexes and not having Property A.

open_problem | open | MathDB 页面 | 原文线索

V2-077813 | MATHDB-399706

Theorem 2 for right-angled Coxeter groups

摘录 / EXCERPT: Let L be a flag triangulation of a finite acyclic 2-dimensional complex such that its fundamental group admits a non-trivial unitary representation $\rho:\pi_1(L)\to U(n)$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-077814 | MATHDB-399707

Potential Eilenberg-Ganea counterexamples from right-angled Coxeter groups

摘录 / EXCERPT: Let L be a flag triangulation of a finite acyclic 2-dimensional complex such that $\c1_1(L)$ admits a non-trivial unitary representation $\d70c:\c1_1(L)\to U(n)$, and let W_L...

open_problem | open | MathDB 页面 | 原文线索

V2-077815 | MATHDB-399708

Generic thinness in the usual Euclidean model for $\mathrm{SL}_n(\mathbb Z)$

摘录 / EXCERPT: Let $G=\mathrm{SL}_n(\mathbb Z)$ with $n>2$, let ... and choose (g_1,g_2) uniformly from B_X^2. Define ... and let μ_X be the normalized counting measure on B_X^2. Is ... Equiv...

open_problem | open | MathDB 页面 | 原文线索

V2-077816 | MATHDB-399710

Zariski-closure question for a Prym-type mapping-class-group representation

摘录 / EXCERPT: In the setup above, do we have ...

open_problem | open | MathDB 页面 | 原文线索

V2-077817 | MATHDB-399711

Vanishing of the invariant subspace under the identity component of the monodromy Zariski closure

摘录 / EXCERPT: In the setup of the report, let $g,n\ge 0$, let $K\subset\pi_1(S_{g,n},x)$ be a finite-index subgroup normal in $\Gamma_{g,n+1}$, let $S_K\to S_{g,n}$ be the corresponding finite...

open_problem | open | MathDB 页面 | 原文线索

V2-077818 | MATHDB-399713

Nontrivial invariant subspaces for the quotient by the Dehn-twist subgroup

摘录 / EXCERPT: Are there any examples for which $H_K^{D_K}\ne 0$ and, if so, with G_K°/D_K acting nontrivially on it?

open_problem | open | MathDB 页面 | 原文线索

V2-077819 | MATHDB-399714

Virtual surjectivity of higher Prym representations

摘录 / EXCERPT: For arbitrary H,p, is the representation $\rho_{H,p}$ virtually surjective onto each component of $\operatorname{Cent}_{\operatorn{Sp}(2\hat g,\mathbb Z)}(H)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077820 | MATHDB-399715

Rigidity of maximal representations of complex hyperbolic lattices

摘录 / EXCERPT: Maximal representations of lattices in $\mathrm{SU}(1,p)$ are conjecturally rigid if $p>1$.

open_problem | open | MathDB 页面 | 原文线索

V2-077821 | MATHDB-399718

Geometric compactification of locally symmetric spaces from Anosov representations

摘录 / EXCERPT: Let $\[1m\Gamma\[0m$ be a finitely generated word-hyperbolic group, let G be a semisimple Lie group, and let $\rho:\Gamma\to G$ be an Anosov representation. Does there exist a ge...

open_problem | open | MathDB 页面 | 原文线索

V2-077822 | MATHDB-399719

Constraint on conformal infinities for desingularizations of AH Einstein orbifolds

摘录 / EXCERPT: Suppose that g_0 is generic in the sense that $\operatorname{rank} R^+_{g_0}(p_0)=2$. Then $\mathcal C_0=\{[\gamma]\in\mathcal C:\det R^+_{g_0(\gamma)}(p_0)=0\}$ is a smooth hype...

open_problem | open | MathDB 页面 | 原文线索

V2-077823 | MATHDB-399721

Complete inhomogeneous nearly Kähler structures on S^6 and $S^3\times S^3$

摘录 / EXCERPT: There exist complete inhomogeneous nearly Kähler structures on both S^6 and $S^3\times S^3$.

open_problem | open | MathDB 页面 | 原文线索

V2-077824 | MATHDB-399723

Nonexistence of Ricci-flat Kähler cone metrics on certain A_k singularities

摘录 / EXCERPT: Let $A_k\subset\mathbb C^{n+1}$ be the hypersurface singularity ... For $n\ge 3$, if ... then there is no Ricci-flat Kähler cone metric on A_k, under the additional assumption that t...

open_problem | open | MathDB 页面 | 原文线索

V2-077825 | MATHDB-399725

Integral representative of a mod-2 cellular cycle

摘录 / EXCERPT: There is a constant $c>0$, depending on d and N, such that for every mod-2 cellular d-cycle U in the unit grid in $\mathbb{R}^N$, there is an integral d-cycle R such...

open_problem | open | MathDB 页面 | 原文线索

V2-077826 | MATHDB-399726

Real versus integral filling volume

摘录 / EXCERPT: Let T be a d-cycle in $\mathbb{R}^N$. If $FV_{\mathbb{R}}(T)$ denotes the minimum mass of a $(d+1)$-chain with boundary T and real coefficients, and $FV(T)$ denotes the minim...

open_problem | open | MathDB 页面 | 原文线索

V2-077827 | MATHDB-399730

Dual brane to the fixed-point locus of i_1

摘录 / EXCERPT: Conjecture 2 (Baraglia-Schaposnik). The support of the dual brane to the fixed-point set of i_1 is the moduli space $\mathcal{M}_{\check H}\subset\mathcal{M}_{\!{}^LG^c}$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-077828 | MATHDB-399731

Conformal connections and convex projective structures

摘录 / EXCERPT: For a convex projective structure $[\nabla]$ on a surface, is $[\nabla]$ globally defined by a conformal connection if and only if $[\nabla]$ is defined by the Levi-Civita connecti...

open_problem | open | MathDB 页面 | 原文线索

V2-077829 | MATHDB-399734

Jacobi’s seven-cubes theorem

摘录 / EXCERPT: Every positive integer other than $15,22,23,50,114,167,175,186,212,231,238,239,303,364,420,428,454$ is the sum of seven non-negative cubes.

open_problem | open | MathDB 页面 | 原文线索

V2-077830 | MATHDB-399736

Szpiro’s discriminant conjecture for elliptic curves

摘录 / EXCERPT: Let K be a number field. For every $\epsilon>0$, there exists a constant $C(K,\epsilon)$ such that every elliptic curve E/K with conductor $N(E)$ satisfies ... Here $\Delta(E)$ d...

conjecture | open | MathDB 页面 | 原文线索

V2-077831 | MATHDB-399739

Primitive embedding with a prescribed number of roots

摘录 / EXCERPT: For the lattice $L=K_d^{\perp}$, the orthogonal complement of the rank-two saturated lattice $K_d=\langle h^2,T\rangle$ of discriminant d in...

open_problem | open | MathDB 页面 | 原文线索

V2-077832 | MATHDB-399740

Unboundedness conjecture for full Belyi maps with prescribed bad reduction

摘录 / EXCERPT: For a finite set of primes P, let $N(P)$ be the set of integers $n\ge 4$ such that there exists a full Belyi map of degree n defined over $\mathbb{Q}$ whose bad-reduction set i...

conjecture | open | MathDB 页面 | 原文线索

V2-077833 | MATHDB-399742

Average central L-value over elliptic curves ordered by height

摘录 / EXCERPT: Conjecture: as $H\to\infty$, the average of $L(E,1)$ over $E\in\mathcal E_{\le H}$ satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-077834 | MATHDB-399743

The Minimalist Conjecture for average ranks ordered by height

摘录 / EXCERPT: The report records the conjectural expectation that the average Mordell-Weil rank of elliptic curves over $\mathbb{Q}$, ordered by the database’s naive height, tends to $1/2$ as th...

conjecture | open | MathDB 页面 | 原文线索

V2-077835 | MATHDB-399744

Bhargava’s conjecture on the proportion of soluble ternary cubics

摘录 / EXCERPT: Let $\rho(P,B)$ denote the proportion of ternary cubics $F(x,y,z)$ with integer coefficients and height $H(F)\le B$, where $H(F)$ is the maximum absolute value of the coefficients,...

conjecture | open | MathDB 页面 | 原文线索

V2-077836 | MATHDB-399746

Explicit determination of rational points on curves

摘录 / EXCERPT: Let C be a smooth, projective, geometrically irreducible curve over $ \mathbb{Q}$ of genus $g\ge 2$. Then C has finitely many rational points, and it is an interesting problem...

open_problem | open | MathDB 页面 | 原文线索

V2-077837 | MATHDB-399748

Correct lower bound for rational points on cubic surfaces of arithmetic Picard rank greater than three

摘录 / EXCERPT: Let V be a smooth cubic surface defined over $\bb{Q}$ with arithmetic Picard rank $\rho>3$. For the counting function $N_{U,H}(B)$ of rational points of anticanonical height at m...

open_problem | open | MathDB 页面 | 原文线索

V2-077838 | MATHDB-399749

Manin’s conjecture for smooth cubic surfaces

摘录 / EXCERPT: For every smooth cubic surface V over $ \mathbb{Q}$ with $V(\mathbb{Q})$ Zariski dense, let U be the complement of the finitely many exceptional curves and let H be a height...

conjecture | open | MathDB 页面 | 原文线索

V2-077839 | MATHDB-399750

Exterior stability of the Schwarzschild and Kerr black holes

摘录 / EXCERPT: Determine the exterior stability of the classical vacuum black hole solutions (Schwarzschild and Kerr).

open_problem | open | MathDB 页面 | 原文线索

V2-077840 | MATHDB-399751

Continuous inextendibility of Schwarzschild spacetime

摘录 / EXCERPT: The Schwarzschild inextendibility problem-whether Schwarzschild is inextendible as a continuous Lorentzian metric-was solved; Sbierski presented a proof.

open_problem | open | MathDB 页面 | 原文线索

V2-077841 | MATHDB-399752

Exterior stability of the Kerr black hole

摘录 / EXCERPT: The question of the exterior stability of the classical vacuum Kerr black hole solution.

open_problem | open | MathDB 页面 | 原文线索

V2-077842 | MATHDB-399754

Generic weak-null-singularity formulation for perturbed Kerr Cauchy horizons

摘录 / EXCERPT: Conjecture 3. For a generic subclass of perturbations in Theorem 1, the maximal globally hyperbolic development is inextendible with a Lorentzian metric in...

open_problem | open | MathDB 页面 | 原文线索

V2-077843 | MATHDB-399758

Stationarity of non-radiating asymptotically flat vacuum spacetimes without regularity at infinity

摘录 / EXCERPT: Is every asymptotically flat vacuum spacetime (M,g) arising from regular initial data and assumed to be non-radiating necessarily stationary near infinity, without the additional...

open_problem | open | MathDB 页面 | 原文线索

V2-077844 | MATHDB-399759

Velocity-term dominance near cosmological singularities

摘录 / EXCERPT: It is conjectured that, close to a cosmological singularity, the gravitational dynamics are governed by an effective equation obtained by neglecting spatial-derivative terms that d...

open_problem | open | MathDB 页面 | 原文线索

V2-077845 | MATHDB-399761

Existence of prescribed knotted vortex lines and tubes in Beltrami fields on the sphere and torus

摘录 / EXCERPT: Let M^3 denote either the round sphere $[U+007F]S^3$ or the flat torus $[U+007F]T^3$. Let $[U+007F]\mathcal{S}$ be a finite collection of pairwise disjoint, possibly knotted and linked, closed curves...

open_problem | open | MathDB 页面 | 原文线索

V2-077846 | MATHDB-399763

Palis’ conjecture for generic affine Cantor sets

摘录 / EXCERPT: For generic pairs of affine Cantor sets, is the arithmetic sum either of Lebesgue measure zero or containing an interval?

conjecture | open | MathDB 页面 | 原文线索

V2-077847 | MATHDB-399764

Ballistic motion for a small quasi-periodic lattice Schrödinger operator

摘录 / EXCERPT: Consider ... where $V:\mathbb T^d\to\mathbb R$ is analytic in a complex neighborhood $\{z\in\mathbb C^d:|\operatorname{Im}z|<r\le 1\}$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-077848 | MATHDB-399765

Quasi-periodic standing gravity-capillary water waves

摘录 / EXCERPT: The target is to continue the linear standing-wave solutions of the two-dimensional, incompressible, irrotational, infinite-depth, gravity-capillary water-wave system with 2π-p...

open_problem | open | MathDB 页面 | 原文线索

V2-077849 | MATHDB-399766

Recovery of the Karshon graph from semitoric invariants

摘录 / EXCERPT: Among the five Pelayo-Vũ Ngọc invariants of a semitoric system $(M,\omega,\Phi=(J,H))$, only the first two-the number of focus-focus singular points and the equivalence class of th...

open_problem | open | MathDB 页面 | 原文线索

V2-077850 | MATHDB-399767

Oscillatory orbits in the restricted planar circular three-body problem

摘录 / EXCERPT: For every mass ratio $\mu\in(0,1/2]$, there exists an orbit $(q(t),p(t))$ of the restricted planar circular three-body problem such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-077851 | MATHDB-399769

Zagier's Reduction Theory Conjecture for Fuchsian groups

摘录 / EXCERPT: For every Fuchsian group Γ there exist F, A as above, and an open set of J's in $\mathbb{R}^{K}$ such that: 1. The map F possesses a global attractor set ... consistin...

conjecture | open | MathDB 页面 | 原文线索

V2-077852 | MATHDB-399771

Smooth almost reducibility of Hölder-conjugated circle diffeomorphisms

摘录 / EXCERPT: Let $f\in F_\alpha\u0000$ with $\alpha\u0000$ irrational, and suppose that $f\u0000$ is Hölder-conjugated to the rigid rotation $R_\alpha\u0000$. Then there exists a sequence of sm...

open_problem | open | MathDB 页面 | 原文线索

V2-077853 | MATHDB-399772

Linear stability of elliptic Lagrangian equilateral-triangle solutions

摘录 / EXCERPT: Determine the linear stability of the Lagrangian elliptic equilateral-triangle homographic solutions of the classical planar three-body problem as a function of the mass parameter...

open_problem | open | MathDB 页面 | 原文线索

V2-077854 | MATHDB-399774

Entropy minimization problem for incompressible viscous fluids

摘录 / EXCERPT: Let $X=\mathbb{T}^n$ and $\Omega=C([0,1],X)$. Let R be the reversible Brownian path measure on X with initial marginal $R_0=\mathrm{vol}$ and generator $\Delta/2$. Given a coup...

open_problem | open | MathDB 页面 | 原文线索

V2-077855 | MATHDB-399775

Existence of the pressure in the entropic representation of the incompressible Schrödinger problem

摘录 / EXCERPT: For the solution of the entropy minimization problem ... where $X=\mathbb T^n$, $\Omega=C([0,1],X)$, R is reversible Brownian motion with generator $\Delta/2$ and initial marginal...

open_problem | open | MathDB 页面 | 原文线索

V2-077856 | MATHDB-399776

Evolution hierarchy for cumulants of a time-dependent random field

摘录 / EXCERPT: Given an initially ℓ^p-clustering random field ψ_0 on $\mathbb{Z}^d$, and a time-dependent field ψ_t obtained as the unique solution of a deterministic evolution equ...

open_problem | open | MathDB 页面 | 原文线索

V2-077857 | MATHDB-399777

Uniqueness of the boundary-layer parameters in the metastable region

摘录 / EXCERPT: For every $u_+\in(m^*,m_\beta)$, are the positive parameter h^* and the value $m_-\in(-m_\beta,-m^*)$ uniquely determined, where m_- is associated with a bump solution q sati...

open_problem | open | MathDB 页面 | 原文线索

V2-077858 | MATHDB-399783

Hydrodynamic limit of condensing zero-range processes with sub-critical initial profiles

摘录 / EXCERPT: For symmetric nearest-neighbour condensing zero-range processes on the discrete torus with bounded local jump rate, started from a sequence of initial distributions with profile...

open_problem | open | MathDB 页面 | 原文线索

V2-077859 | MATHDB-399784

Localization of the first Dirichlet eigenvector in the heavy-tailed random conductance model

摘录 / EXCERPT: Let $d\ge 2$ and let the conductances be i.i.d., symmetric, supported in $(0,1]$, with $\mathbb{P}[w\le a]=a^\gamma$ for $a\in(0,1]$, where $\gamma<\tfrac14$. For the variable-spee...

open_problem | open | MathDB 页面 | 原文线索

V2-077860 | MATHDB-399785

Which vertices form the condensate in preferential attachment with adaptive normalization?

摘录 / EXCERPT: In the preferential-attachment model with fitness distribution μ on $(0,1)$, essential supremum 1, activity parameter $\lambda>0$, and adaptive normalization...

open_problem | open | MathDB 页面 | 原文线索

V2-077861 | MATHDB-399786

Long-run frequency of derangements in the Chinese Restaurant Process

摘录 / EXCERPT: For the Dubins-Pitman Chinese Restaurant Process described in the report, let $A_1(n)$ denote the number of fixed points of the permutation at time n. The ratio ... need not conver...

open_problem | open | MathDB 页面 | 原文线索

V2-077862 | MATHDB-399787

Universality of eigenvalue gaps for general β-ensembles

摘录 / EXCERPT: Determine the bulk and edge distributions of the normalized consecutive eigenvalue gaps for the general β-ensemble ... where $V:\mathbb R\to\mathbb R$ is smooth and uniformly convex....

open_problem | open | MathDB 页面 | 原文线索

V2-077863 | MATHDB-399788

Removing the upper bound on the second Toda parameter

摘录 / EXCERPT: Let (Σ,g) be a compact boundaryless Riemannian surface, let $h_1,h_2:\Sigma\to\mathbb{R}$ be smooth and positive, and consider the $SU(3)$ Toda system ... For $m\in\mathbb{N}$...

open_problem | open | MathDB 页面 | 原文线索

V2-077864 | MATHDB-399791

Lévy-Gromov isoperimetric inequality in RCD*(K,N)-spaces

摘录 / EXCERPT: Let (X,d,m) be an $\mathrm{RCD}^*(K,N)$ space, where $N\in\mathbb{N}$ and $K>0$. For every Borel set $E\subset X$, ... where B is a spherical cap in the model sphere S, namely...

open_problem | open | MathDB 页面 | 原文线索

V2-077865 | MATHDB-399792

Rigidity in the Lévy-Gromov inequality for RCD* spaces

摘录 / EXCERPT: Let (X,d,m) be an $\\mathrm{RCD}^*(K,N)$ space with $K>0$. If, for some $v\in(0,1)$, the lower bound in Theorem 1 is attained, equivalently $I_{(X,d,m)}(v)=I_{K,N,\infty}(v)$, mu...

open_problem | open | MathDB 页面 | 原文线索

V2-077866 | MATHDB-399793

Existence of Willmore minimizers in every conformal class of tori

摘录 / EXCERPT: Is the infimum of the Willmore energy attained in every conformal class of tori?

open_problem | open | MathDB 页面 | 原文线索

V2-077867 | MATHDB-399801

Scattering for a critical nonlinear wave equation in two dimensions

摘录 / EXCERPT: For every $(u_0,u_1)\in C_c^\infty(\mathbb R^2)$, let u be the solution of ... with $(u,u_t)|_{t=0}=(u_0,u_1)$. Then there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-077868 | MATHDB-399802

Scattering for the massless critical nonlinear wave equation in two dimensions

摘录 / EXCERPT: For the Cauchy problem ... with arbitrary smooth, compactly supported initial data ... is scattering always valid? Namely, do there exist...

open_problem | open | MathDB 页面 | 原文线索

V2-077869 | MATHDB-399806

Uniqueness for the axisymmetric vortex-ring solution

摘录 / EXCERPT: For fixed $\Gamma>0$, $R>0$, and $\nu>0$, consider the axisymmetric vorticity equation ... posed on $\Omega=\{(r,z):r>0,\z\in\R\}$ with homogeneous Dirichlet condition at $r=0$ and i...

open_problem | open | MathDB 页面 | 原文线索

V2-077870 | MATHDB-399807

Moving-center Gaussian estimate in the vanishing-viscosity limit for an axisymmetric vortex ring

摘录 / EXCERPT: Does an estimate of the form ... remain valid in the vanishing-viscosity limit, with the Gaussian centered at $(R,Z(t))$, where ...

open_problem | open | MathDB 页面 | 原文线索

V2-077871 | MATHDB-399808

Removing the weighted vorticity bound in the uniqueness theorem

摘录 / EXCERPT: Can the assumption $\|\omega_\theta(t)/r\|_{L^\infty(\Omega)}$ in condition (i) of Theorem 2 be dropped?

open_problem | open | MathDB 页面 | 原文线索

V2-077872 | MATHDB-399812

Infinite weak-solution multiplicity for the 2D Savage-Hutter system

摘录 / EXCERPT: Let $T>0$, let $\a$ and $\\gamma$ be nonnegative functions in $C^2(\Omega)$, let $\mathbf f\in C^1([0,T]\times\Omega;\mathbb R^2)$, and let initial data satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-077873 | MATHDB-399813

Optimal decay bound for palinstrophy-constrained mixing

摘录 / EXCERPT: What is the optimal bound for the decay in time of the mix-norm for palinstrophy-constrained flows: polynomial or exponential in time?

open_problem | open | MathDB 页面 | 原文线索

V2-077874 | MATHDB-399816

Determining-wavenumber bound near the Kolmogorov intermittency dimension

摘录 / EXCERPT: We conjecture that $\langle\Lambda\rangle \lesssim \kappa_d$ for values of d near 3 as well.

open_problem | open | MathDB 页面 | 原文线索

V2-077875 | MATHDB-399819

Strong divergence of all sampled-data Hilbert-transform approximations

摘录 / EXCERPT: Let $\mathbb T$ be the additive group of real numbers modulo 2π, and for $f\in L^1(\mathbb T)$ define ... Let ... Let $\{H_N\}_{N\in\mathbb N}\subset L(B)$ satisfy: (A) For every...

open_problem | open | MathDB 页面 | 原文线索

V2-077876 | MATHDB-399821

Negative answer to the Ul’yanov-Type Problem for sampled Hilbert-transform approximations

摘录 / EXCERPT: Let $\mathbb T=\mathbb R/(2\pi\mathbb Z)$, let H be the Hilbert transform, and let $B=\{f\in C(\mathbb T):Hf\in C(\mathbb T)\}$ with $\|f\|_B=\max\{\|f\|_\infty,\|Hf\|_\infty\}$....

open_problem | open | MathDB 页面 | 原文线索

V2-077877 | MATHDB-399822

Log-concavity of conic intrinsic volumes

摘录 / EXCERPT: Is it true that the conic intrinsic volumes satisfy the Alexandrov-Fenchel inequality ...

open_problem | open | MathDB 页面 | 原文线索

V2-077878 | MATHDB-399823

Variance maximization for circular cones

摘录 / EXCERPT: Among cones with a fixed statistical dimension, do the intrinsic volumes of circular cones have the largest variance?

open_problem | open | MathDB 页面 | 原文线索

V2-077879 | MATHDB-399824

Universality of phase transitions for more general distributions

摘录 / EXCERPT: Do the phase transitions, and their location, persist for more general distributions, as extensive experiments [10, 14] suggest?

open_problem | open | MathDB 页面 | 原文线索

V2-077880 | MATHDB-399825

Minimum measurements for random compressive classification of linearly separable sets

摘录 / EXCERPT: Given linearly separable sets $A,B\subseteq\mathbb{R}^n$, meaning that there are v and θ such that $x\in A\Rightarrow\langle x,v\rangle<\theta$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077881 | MATHDB-399826

Extending the convergence radius of ITKsM and ITKrM

摘录 / EXCERPT: Can the convergence radius of ITKsM and ITKrM be extended from $O(1/\log K)$ and $O(1/\sqrt{S})$, respectively, to $O(1)$, as suggested by experiments?

open_problem | open | MathDB 页面 | 原文线索

V2-077882 | MATHDB-399829

Uniqueness for phase retrieval from the Cauchy wavelet-transform modulus

摘录 / EXCERPT: Let ψ be a Cauchy wavelet with $\widehat\psi(\omega)=\omega^p e^{-\omega}\mathbf{1}_{\{\omega\ge 0\}}$ for some $p>0$. The report states that every analytic function...

open_problem | open | MathDB 页面 | 原文线索

V2-077883 | MATHDB-399830

Geometric realization of Colmez’s p-adic local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$

摘录 / EXCERPT: The report states that the joint work of Gabriel Dospinescu and Arthur-César Le Bras proves a conjecture of Breuil: Colmez’s p-adic local Langlands correspondence for...

open_problem | open | MathDB 页面 | 原文线索

V2-077884 | MATHDB-399831

Closure relation between Newton strata

摘录 / EXCERPT: Let K be a parahoric subgroup. Let $[b],[b']\in B(G,\{\mu\})$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-077885 | MATHDB-399832

Finiteness and étaleness of the map from KR to EKOR strata

摘录 / EXCERPT: For a parahoric subgroup K and $x\in \operatorname{Adm}(\{\mu\})_K\cap {}^K\widetilde W$, the finite morphism ... from Theorem 3.8 is finite étale.

open_problem | open | MathDB 页面 | 原文线索

V2-077886 | MATHDB-399833

Finite étaleness of the morphism from Iwahori KR-strata to EKOR-strata

摘录 / EXCERPT: It may be conjectured that, for any parahoric subgroup K and any $x\in\operatorname{Adm}(\{\mu\})_K\cap{}^K\widetilde W$, the finite morphism ... from Theorem 3.8 is finite étale....

open_problem | open | MathDB 页面 | 原文线索

V2-077887 | MATHDB-399834

Six-weight case of the geometric Breuil-Mézard component correspondence

摘录 / EXCERPT: Let $\rho:\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)\to\mathrm{GL}_3(\mathbb{F}_p)$ be semisimple and let τ be a generic tame inertial type. In the remaining cas...

open_problem | open | MathDB 页面 | 原文线索

V2-077888 | MATHDB-399835

Weight part of Serre’s conjecture for generic semisimple three-dimensional representations

摘录 / EXCERPT: Let F/F^+ be the imaginary CM/totally real setup described in the report, with the relevant unitary group G, and let $r:G_F\to \mathrm{GL}_3(\overline{\mathbb F}_p)$ be an irre...

conjecture | open | MathDB 页面 | 原文线索

V2-077889 | MATHDB-399836

Modularity conjecture for the tensor-induced motive attached to an elliptic curve

摘录 / EXCERPT: Let F be a totally real number field of odd degree d, let E/F be an elliptic curve, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-077890 | MATHDB-399837

Bloch-Kato conjecture for the tensor-induced motive of an elliptic curve

摘录 / EXCERPT: Let F be a totally real number field of odd degree d, let E/F be an elliptic curve, and let ... be the associated motive over $\mathbb Q$. Assuming the modularity conjecture in...

conjecture | open | MathDB 页面 | 原文线索

V2-077891 | MATHDB-399838

Extension of generalized Hasse invariants to Ekedahl-Oort stratum closures

摘录 / EXCERPT: For a stratum X_w in the Ekedahl-Oort stratification, with generalized Hasse invariant $A_w\in H^0(X_w,\omega^{p^N-1})$, does A_w extend to the Zariski closure $\overline{X_w}$...

open_problem | open | MathDB 页面 | 原文线索

V2-077892 | MATHDB-399840

Group-theoretical Hasse invariants for zip strata

摘录 / EXCERPT: Let $C\subset G\text{-Zip}_{\mu}$ be a zip stratum, and let $\overline{C}$ denote its closure with the reduced structure. Let $\chi\in X^{*}(L)$ be an ample and p-small character...

open_problem | open | MathDB 页面 | 原文线索

V2-077893 | MATHDB-399842

Extension of the $a=1$ stratum conjecture beyond the basic case

摘录 / EXCERPT: Conjecture: In the affine Deligne-Lusztig setting described in the report, the locus where the a-number equals 1 is a single $J_b(F)$-orbit of strata for the invariant f_g, e...

conjecture | open | MathDB 页面 | 原文线索

V2-077894 | MATHDB-399843

Arithmetic intersection formula for big CM cycles

摘录 / EXCERPT: There exists an explicitly defined integer $D_{\mathrm{bad},L}$, depending on the lattice L, such that for any $f\in H_{2-d}(\omega_L)$ with integral principal part, one has ... up...

open_problem | open | MathDB 页面 | 原文线索

V2-077895 | MATHDB-399846

Yang’s 1977 problem on complete closed hypersurfaces

摘录 / EXCERPT: A complete closed hypersurface immersed in the unit ball of $\mathbb{C}^n$ was constructed, thereby solving Yang’s long-standing problem from 1977.

open_problem | open | MathDB 页面 | 原文线索

V2-077896 | MATHDB-399853

Suita conjecture inequality for the Bergman kernel

摘录 / EXCERPT: For the one-dimensional case, the report states the estimate conjectured by Suita and proved in reference [1]: $K_{\\Omega}(w,w) \\ge \\frac{1}{\\pi c_{\\Omega}(w)^2}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-077897 | MATHDB-399854

Nonexistence of harmonic discs in an open set for certain HMAE solutions

摘录 / EXCERPT: There exist smooth boundary data $\phi(\cdot,\tau)$ for which the solution to the Dirichlet problem for the HMAE has the following property: there exists an open set...

open_problem | open | MathDB 页面 | 原文线索

V2-077898 | MATHDB-399855

Harmonic discs for weak solutions of the HMAE

摘录 / EXCERPT: Let $X=\mathbb{P}^{1}$ and let $D\subset\mathbb{C}$ be the open unit disc. There exist smooth boundary data $\phi(\cdot,\tau)$ for the Dirichlet problem for the complex homogeneous...

open_problem | open | MathDB 页面 | 原文线索

V2-077899 | MATHDB-399860

Poisson-Dirichlet limit for the low-temperature DGFF Gibbs measure

摘录 / EXCERPT: Fix $\beta>\beta_c:=\frac{2}{\sqrt g}=\alpha$. Let $ obreak\bigl(obreak\tilde\mu_N^D\bigr)$ be the Gibbs measure on D_N defined by...

open_problem | open | MathDB 页面 | 原文线索

V2-077900 | MATHDB-399861

No disorder-induced shift of the critical point

摘录 / EXCERPT: For all dimensions $d\ge 2$ and all disorder strengths $\beta>0$, the critical point satisfies $h_c(\beta)=\lambda(\beta)$, where $\lambda(\beta)$ is the annealed shift defined in...

open_problem | open | MathDB 页面 | 原文线索

V2-077901 | MATHDB-399863

Soficity of unimodular random rooted graphs

摘录 / EXCERPT: Every unimodular random rooted graph is sofic.

open_problem | open | MathDB 页面 | 原文线索

V2-077902 | MATHDB-399865

Equivalence of residual magnetization and long-range order at criticality

摘录 / EXCERPT: Theorem A.2 [5]. For any Ising model on a transitive amenable graph (in particular on $\mathbb{Z}^d$ for any $d\in(1,\infty)$) with homogeneous ferromagnetic couplings,...

open_problem | open | MathDB 页面 | 原文线索

V2-077903 | MATHDB-399867

Eden distance on uniform random triangulations

摘录 / EXCERPT: Let T_n be a uniformly chosen rooted plane triangulation with $n+1$ vertices. Give each edge of the dual map $T_n^\dagger$ an independent exponential random variable of parameter...

open_problem | open | MathDB 页面 | 原文线索

V2-077904 | MATHDB-399870

Asymptotic theory of reduced MHD models for fusion plasmas

摘录 / EXCERPT: The report states the following result as Theorem 2. Let ε be the ratio of the poloidal magnetic field to the toroidal magnetic field. Assume that the initial velocity,...

open_problem | open | MathDB 页面 | 原文线索

V2-077905 | MATHDB-399871

Optimal continuous metric for minimizing the interpolation error

摘录 / EXCERPT: For a twice continuously differentiable function u on $ \Omega\subset\mathbb{R}^3$, the report considers the continuous optimization problem ... where M is a positive-definite co...

open_problem | open | MathDB 页面 | 原文线索

V2-077906 | MATHDB-399873

Extension of the velocity-based momentum limiter to higher-order DG methods

摘录 / EXCERPT: An open question is the possibility of extending the proposed concept to discontinuous Galerkin finite elements of order higher than linear.

open_problem | open | MathDB 页面 | 原文线索

V2-077907 | MATHDB-399874

Computing worst Value-at-Risk under unknown dependence

摘录 / EXCERPT: Given risks $L_1\sim F_1,\ldots,L_d\sim F_d$ with known marginal distribution functions but unspecified dependence structure, compute the upper bound on...

open_problem | open | MathDB 页面 | 原文线索

V2-077908 | MATHDB-399875

CUSUM optimality for a longevity-pattern change-detection problem

摘录 / EXCERPT: For an optimal stopping problem related to sequential testing of changing longevity patterns, the optimal strategy is the so-called CUSUM strategy when the Lorden criterion is used...

open_problem | open | MathDB 页面 | 原文线索

V2-077909 | MATHDB-399876

Testing the maximal rank of a continuous volatility process

摘录 / EXCERPT: Given a d-dimensional continuous diffusion process ... observed at times $0,\Delta_n,2\Delta_n,\ldots,\lfloor1/\Delta_n\rfloor\Delta_n$ with $\Delta_n\to0$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-077910 | MATHDB-399877

Testing and specifying a second multiplicative component in time-varying GARCH models

摘录 / EXCERPT: “How to test for the second multiplicative component and how the specification strategy would work in this situation will be investigated in the future.”

open_problem | open | MathDB 页面 | 原文线索

V2-077911 | MATHDB-399878

Analytical verification of joint mixability

摘录 / EXCERPT: For given univariate distribution functions $F_1,\ldots,F_n$, develop an analytical method to verify whether the tuple $(F_1,\ldots,F_n)$ is jointly mixable; equivalently, determin...

open_problem | open | MathDB 页面 | 原文线索

V2-077912 | MATHDB-399879

Robust DPG method for a singularly perturbed reaction-diffusion problem

摘录 / EXCERPT: Let $ Omega\tsubset\mathbb{R}^d$, $d=2,3$, be a bounded, simply connected Lipschitz polygonal/polyhedral domain with boundary $\Gamma=\partial\Omega$. For $\varepsilon>0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077913 | MATHDB-399882

Non-bi-Lipschitz-equivalent Delone sets

摘录 / EXCERPT: There exist Delone sets which are not bi-Lipschitz equivalent to lattices.

open_problem | open | MathDB 页面 | 原文线索

V2-077914 | MATHDB-399883

Gap Labelling Theorem

摘录 / EXCERPT: Let Σ be a Cantor set and let $\Sigma\times \mathbb{Z}^n\to\Sigma$ be a free and minimal action of $\mathbb{Z}^n$ on Σ with invariant measure μ. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-077915 | MATHDB-399886

Developing a distance for comparing finite metric graphs

摘录 / EXCERPT: Given a finite graph $G=(V,E)$ whose edges have positive lengths, regard its geometric realization $|G|$ as the metric space $(|G|,d_G)$ with shortest-path metric. Given two such m...

open_problem | open | MathDB 页面 | 原文线索

V2-077916 | MATHDB-399889

Multicut in bounded-genus graphs with a bounded number of terminals

摘录 / EXCERPT: Given an edge-weighted graph embedded on a surface of genus g, with t specified terminal vertices and specified terminal pairs, compute a minimum-weight set of edges whose dele...

open_problem | open | MathDB 页面 | 原文线索

V2-077917 | MATHDB-399894

Convergence of normalized finite-type invariants in the Petaluma random-knot model

摘录 / EXCERPT: Conjecture 5. Let v_m be a knot invariant of order $m>0$. Then $v_m(K_{2n+1})/n^m$ weakly converges to a limit distribution as $n\to\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-077918 | MATHDB-399896

Embedded torus in dense 2-dimensional complexes

摘录 / EXCERPT: Does there exist a constant $C>0$ such that every 2-dimensional complex Δ with n vertices and $Cn^{5/2}$ faces contains an embedded torus?

open_problem | open | MathDB 页面 | 原文线索

V2-077919 | MATHDB-399897

Reidemeister moves needed to simplify a generic immersed plane curve

摘录 / EXCERPT: Given a generic immersion of the circle in the plane, $\gamma:S^1\to\mathbb{R}^2$, with n double points, is there an upper bound as a function of n on the number of Reidemeiste...

open_problem | open | MathDB 页面 | 原文线索

V2-077920 | MATHDB-399900

Treewidth of minimal triangulations of knot complements

摘录 / EXCERPT: More precisely, given a knot K whose minimal treewidth is bounded below by some constant, does there exist a function $f:\mathbb{N}\to\mathbb{N}$ such that the minimal triangulat...

open_problem | open | MathDB 页面 | 原文线索

V2-077921 | MATHDB-399901

Exactness implication for SDP relaxations with ellipsoidal hollows

摘录 / EXCERPT: The SDP relaxation of quadratic optimization with ellipsoidal hollows is exact if the SDP relaxation of the same problem without hollows is exact.

open_problem | open | MathDB 页面 | 原文线索

V2-077922 | MATHDB-399903

An FPTAS for concave mixed-integer quadratic programming

摘录 / EXCERPT: Consider the mixed-integer quadratic program ... where H is an $n\times n$ symmetric matrix, $h\in\mathbb R^n$, W is an $m\times n$ matrix, and $w\in\mathbb R^m$. Assume the obje...

open_problem | open | MathDB 页面 | 原文线索

V2-077923 | MATHDB-399905

Burer’s copositive reformulation of quadratic programs with binary variables

摘录 / EXCERPT: The report presents Burer’s theorem that, under the condition that $Ax=b$ implies $x_j\in[0,1]$ for every $x\in\mathbb R_+^n$ and every $j\in B$, the quadratic program ... with...

open_problem | open | MathDB 页面 | 原文线索

V2-077924 | MATHDB-399906

Feasibility of reconstructing proper times from a time-free relaxation

摘录 / EXCERPT: Given a solution of the time-free relaxation, obtained by projecting away the time indices, decide whether it has a feasible counterpart for the original time-indexed problem with...

open_problem | open | MathDB 页面 | 原文线索

V2-077925 | MATHDB-399910

FPTAS for minimizing indefinite quadratic forms over integer points in fixed dimension

摘录 / EXCERPT: Let n be fixed, let $P=\{x\in\mathbb{R}^n:Ax\le b\}$ be a bounded polyhedron with $A\in\mathbb{Z}^{m\times n}$ and $b\in\mathbb{Z}^m$, and consider ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-077926 | MATHDB-399914

Conjectural enumeration of friezes in exceptional Dynkin types

摘录 / EXCERPT: The number of friezes of type E_6, E_7, E_8 and F_4 is 868, 4400, 26952 and 112, respectively.

open_problem | open | MathDB 页面 | 原文线索

V2-077927 | MATHDB-399915

Polytopality of the multitriangulation flip sphere

摘录 / EXCERPT: Is the simplicial sphere whose graph is the flip graph of a multitriangulation polytopal?

open_problem | open | MathDB 页面 | 原文线索

V2-077928 | MATHDB-399920

Face numbers of sequentially Cohen-Macaulay simplicial complexes

摘录 / EXCERPT: Characterize the possible face numbers of sequentially Cohen-Macaulay simplicial complexes.

open_problem | open | MathDB 页面 | 原文线索

V2-077929 | MATHDB-399921

Characterization of 2-dimensional \2-tight complexes

摘录 / EXCERPT: Let X be a 2-dimensional simplicial complex. For each induced subcomplex Y of X and distinct vertices y_1,y_2 of Y, let $c_Y(y_1,y_2)$ be the number of connected compon...

open_problem | open | MathDB 页面 | 原文线索

V2-077930 | MATHDB-399922

Tropical Hodge index theorem

摘录 / EXCERPT: Given a 2-dimensional fan F in $\mathbb{R}^n$, find conditions guaranteeing that the tropical Laplacian has a single positive eigenvalue.

open_problem | open | MathDB 页面 | 原文线索

V2-077931 | MATHDB-399925

Ramos conjecture for hyperplane mass partitions

摘录 / EXCERPT: For every pair of integers $j,k\ge 1$, let $\Delta(j,k)$ be the smallest dimension d such that any j masses in $\mathbb{R}^d$ can be simultaneously cut into 2^k orthants of e...

conjecture | open | MathDB 页面 | 原文线索

V2-077932 | MATHDB-399926

Theorem on two-hyperplane mass partitions

摘录 / EXCERPT: For every integer $t\ge 1$, $\Delta(2^t+1,2)=3\cdot 2^{t-1}+2$, where $\Delta(j,k)$ is the smallest dimension such that every collection of j masses in that dimension can be simu...

open_problem | open | MathDB 页面 | 原文线索

V2-077933 | MATHDB-399927

Grünbaum hyperplane mass partition problem in dimension 4

摘录 / EXCERPT: Can any convex body in $\mathbb{R}^{4}$ be cut into 16 pieces of equal volume by 4 suitably chosen affine hyperplanes?

open_problem | open | MathDB 页面 | 原文线索

V2-077934 | MATHDB-399931

Subadditivity of Castelnuovo-Mumford regularity for products of ideals

摘录 / EXCERPT: Let K be a field, let $S=K[x_1,\ots,x_n]$, and let I,J be homogeneous ideals of S. The proposed inequality ... does not hold in general. The report states that examples are kno...

open_problem | open | MathDB 页面 | 原文线索

V2-077935 | MATHDB-399933

Relation between Stanley projective dimension and lattice invariants

摘录 / EXCERPT: Determine the relation between the Stanley projective dimension and the classical invariants of finite lattices.

open_problem | open | MathDB 页面 | 原文线索

V2-077936 | MATHDB-399934

Existence of bounded-degree 2-dimensional expanders

摘录 / EXCERPT: There exist constants $d<\infty$ and $\varepsilon>0$ and an infinite family of $(2,d,\varepsilon)$-expanders; equivalently, for the random Latin-square model $Y(n,d)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-077937 | MATHDB-399935

Lower bounds for coboundary expansion of building-like complexes

摘录 / EXCERPT: Let X be a finite n-dimensional pure simplicial complex, let $G\leq\operatorname{Aut}(X)$, and let S be a finite G-set. For $0\leq k\leq n-1$, set...

open_problem | open | MathDB 页面 | 原文线索

V2-077938 | MATHDB-399938

Equality case in the Balanced Generalized Lower Bound Conjecture

摘录 / EXCERPT: Let P be a balanced simplicial d-polytope. For $i\le d/2$, is ... if and only if P has the balanced $(i-1)$-stacked property? Here a balanced $(i-1)$-stacked property means tha...

conjecture | open | MathDB 页面 | 原文线索

V2-077939 | MATHDB-399940

Monotonicity of connectivity for Vietoris-Rips complexes of the circle

摘录 / EXCERPT: For $M=S^1$, the connectivity $\operatorname{conn}(\mathrm{VR}(M;r))$ is a non-decreasing function of r.

open_problem | open | MathDB 页面 | 原文线索

V2-077940 | MATHDB-399941

Optimal configurations for the refined S-unit sieve

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$. For any $X=(J,x)$ with $J\subseteq[n]$ and $x\in\mathbb{R}_{\ge 0}^{J}$ satisfying $\lVert x\rVert_1=1$, define ... Given $k\ge n$, how should one choose...

open_problem | open | MathDB 页面 | 原文线索

V2-077941 | MATHDB-399942

Optimal covering translates in the nonnegative orthant

摘录 / EXCERPT: Let $A := \mathbb{R}_{\ge 0}^n$. Given $k \ge n$, how should one choose $x_1,\ldots,x_k \in A$ with $\lVert x_1\rVert_1=\cdots=\lVert x_k\rVert_1=1$ such that ... is minimal?

open_problem | open | MathDB 页面 | 原文线索

V2-077942 | MATHDB-399943

Polyhedrality of the cone of lattice-invariant nonnegative valuations

摘录 / EXCERPT: Let d4a5 be the vector space of real-valued d539-valuations on lattice polytopes, let $\mathrm{d4a5}_+$ be the subset of h-nonnegative valuations, and let $\bar{\u0001d4a5}$ be the subspace of...

open_problem | open | MathDB 页面 | 原文线索

V2-077943 | MATHDB-399945

The \bc\-vector and generators of the fundamental group

摘录 / EXCERPT: Is μ_1 a lower bound for the number of generators of the fundamental group?

open_problem | open | MathDB 页面 | 原文线索

V2-077944 | MATHDB-399946

Bagchi’s generalized lower bound conjecture

摘录 / EXCERPT: Conjecture 2.1: $g_i \ge \binom{d+1}{i}\left[(-1)^i+\sum_{j=1}^{i}(-1)^{i-j}\mu_{j-1}\right]$.

conjecture | open | MathDB 页面 | 原文线索

V2-077945 | MATHDB-399947

A lower-bound inequality involving the μ-vector

摘录 / EXCERPT: If Δ is three-dimensional, is ...

open_problem | open | MathDB 页面 | 原文线索

V2-077946 | MATHDB-399948

Hasse diagram property for simple polytopes

摘录 / EXCERPT: Given a polytope P and a generic cost vector c, orient each edge of the 1-skeleton from u to v whenever $c(u)<c(v)$, obtaining a directed graph $G(P,c)$. If this directed...

open_problem | open | MathDB 页面 | 原文线索

V2-077947 | MATHDB-399952

Inverse Generator Problem

摘录 / EXCERPT: Whenever A generates a bounded C_0-semigroup and A^{-1} exists as a densely defined operator, does A^{-1} generate a bounded C_0-semigroup?

open_problem | open | MathDB 页面 | 原文线索

V2-077948 | MATHDB-399954

Strong stabilization of the SCOLE model with a tuned-mass damper

摘录 / EXCERPT: The report presents the following result, rather than an unresolved problem: for strictly positive functions $\rho,EI\in C^4[0,l]$ and parameters $m,J,m_1,k_1,d_1>0$, the SCOLE-tro...

open_problem | open | MathDB 页面 | 原文线索

V2-077949 | MATHDB-399959

Rationality of ball quotients associated with non-natural order-three automorphisms

摘录 / EXCERPT: How can one understand whether the corresponding ball quotients are rational?

open_problem | open | MathDB 页面 | 原文线索

V2-077950 | MATHDB-399965

Exceptional behavior of moduli-map images for nodal curves on K3 surfaces

摘录 / EXCERPT: In the cases not covered by Theorem 17 or by Kemeny’s result [20], is it possible to find cases in which expectation (1) is contradicted, as happens for $V_{p,1,0}$ for $p=10$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-077951 | MATHDB-399966

Birationality of the moduli map on maximal-rank components

摘录 / EXCERPT: Let V be an irreducible component of $V_{p,n,\\\delta}$, or let V be an irreducible component of $V_{n,\\\delta}(X)$ for general $(X,L)\in K_p$, such that the differential of t...

open_problem | open | MathDB 页面 | 原文线索

V2-077952 | MATHDB-399972

Closedness of the K3 locus in genus $g\ge 12$

摘录 / EXCERPT: Assume $g \ge 12$. Is the locus $(K3)_g \subset M_g$ of curves lying on a K3 surface closed?

open_problem | open | MathDB 页面 | 原文线索

V2-077953 | MATHDB-399973

Green-Lazarsfeld Secant Conjecture

摘录 / EXCERPT: Let L be a globally generated line bundle of degree d on a curve C of genus g such that ... Then (C,L) fails property (N_p) if and only if L is not $(p+1)$-very ample.

conjecture | open | MathDB 页面 | 原文线索

V2-077954 | MATHDB-399974

Conjecture 30 on the singularities and versality of hyperplane sections of a cubic fourfold

摘录 / EXCERPT: Let $X\subset\mathbb P^5$ be a generic cubic fourfold, and let $\{V_t\}_{t\in(\mathbb P^5)^\ast}$ denote the family of its hyperplane sections. Then, for any…

conjecture | open | MathDB 页面 | 原文线索

V2-077955 | MATHDB-399975

Toroidality of relative compactified Jacobians for versal unfoldings of simple singularities

摘录 / EXCERPT: Is the relative compactified Jacobian of a family of curves over a polydisk in $\mathbb{C}^k$ toroidal, provided that the family is a multi-versal unfolding of singularities of its…

open_problem | open | MathDB 页面 | 原文线索

V2-077956 | MATHDB-399976

KDC procedures for detecting mediation in cross-phenotype associations

摘录 / EXCERPT: “The creation of KDC procedures to identify whether an observed cross-phenotype association is mediated by a different set of phenotypes would have tremendous value.”

open_problem | open | MathDB 页面 | 原文线索

V2-077957 | MATHDB-399977

Relationship between macroscopic hardening laws and mesoscopic dislocation interactions

摘录 / EXCERPT: Another problem that emerged is the relationship between macroscopic hardening laws and the mesoscopic interaction of dislocations moving on a glide plane with the farthest disloca…

open_problem | open | MathDB 页面 | 原文线索

V2-077958 | MATHDB-399978

Qualitative behavior of Monge-Ampère solutions for $1/3<\alpha<2/3$

摘录 / EXCERPT: Understand the qualitative behavior of solutions v of … where … in the regime $\frac13<\alpha<\frac23$.

open_problem | open | MathDB 页面 | 原文线索

V2-077959 | MATHDB-399979

Evolutionary relaxation for nonlinear rate-independent material models without an internal length scale

摘录 / EXCERPT: For nonlinear material models without an internal length scale, given approximate incremental solutions with time steps and error levels satisfying $\tau_n,\varepsilon_n\to0$, (1)…

open_problem | open | MathDB 页面 | 原文线索

V2-077960 | MATHDB-399980

Polygonization in Crystal Plasticity

摘录 / EXCERPT: Simulate polygonization in bent single-crystal beams, in which dislocations align to form polygonization patterns, and determine the physically correct solution. The report states…

open_problem | open | MathDB 页面 | 原文线索

V2-077961 | MATHDB-399984

Explicit two-source extractors for polylogarithmic min-entropy

摘录 / EXCERPT: There exists a constant $C>0$ such that, for every $n\in\mathbb{N}$, there is a polynomial-time computable function $2\mathrm{Ext}:\{0,1\}^n\times\{0,1\}^n\to\{0,1\}$ such that, fo…

open_problem | open | MathDB 页面 | 原文线索

V2-077962 | MATHDB-399986

Strong Exponential Time Hypothesis (SETH)

摘录 / EXCERPT: There is no $\delta>0$ such that k-SAT is in time $2^{n(1-\delta)}$ for all constant k.

conjecture | open | MathDB 页面 | 原文线索

V2-077963 | MATHDB-399987

Exact satisfiability threshold for random 4-SAT

摘录 / EXCERPT: Conjecturally, the exact satisfiability threshold for random 4-SAT is $m=9.93\ldots n$, where n is the number of variables and m is the number of independent uniformly random…

open_problem | open | MathDB 页面 | 原文线索

V2-077964 | MATHDB-399991

Interactive compression problem

摘录 / EXCERPT: Given a protocol π with information cost, external information cost, and communication cost I, I_{ext}, and C, respectively, what is the communication cost of th…

open_problem | open | MathDB 页面 | 原文线索

V2-077965 | MATHDB-399993

Rigidity of random Toeplitz matrices at rank $m=n^{0.5+\\Omega(1)}$

摘录 / EXCERPT: One open problem posed by Goldreich and Wigderson is: prove that random Toeplitz matrices have rigidity m^3 for rank $m=n^{0.5+\Omega(1)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-077966 | MATHDB-399994

Rigidity lower bounds for random Toeplitz matrices

摘录 / EXCERPT: Prove that a random $n\times n$ Toeplitz matrix over $\mathbb{F}_2$ has rigidity at least $n^{1.5+\Omega(1)}$ for rank $n^{0.5+\Omega(1)}$.

open_problem | open | MathDB 页面 | 原文线索

V2-077967 | MATHDB-399995

Better-than-\cexp\c(\csqrt\c n\c)\c lower bounds for depth-three circuits computing \cF_{\cmathrm{tet}}\c

摘录 / EXCERPT: Prove better-than-cexp(sqrt{n})c lower bounds on the size of standard depth-three Boolean circuits computing the explicit trilinear function c … c

open_problem | open | MathDB 页面 | 原文线索

V2-077968 | MATHDB-400000

Parabolic limit problem with dynamic boundary conditions

摘录 / EXCERPT: In the bulk-surface receptor-ligand setting of the report, let Γ be a smooth, compact, closed n-dimensional hypersurface, $n\in\{1,2\}$, contained in…

open_problem | open | MathDB 页面 | 原文线索

V2-077969 | MATHDB-400002

Second-order time-stepping for parabolic problems with moving interfaces and domains

摘录 / EXCERPT: For the parabolic interface problem on the space-time domain $Q=Q_1\cup G\cup Q_2$, where $\Omega(t)=\Omega_1(t)\cup\Gamma(t)\cup\Omega_2(t)$, find $u:Q\to\mathbb R$ satisfying ... w...

open_problem | open | MathDB 页面 | 原文线索

V2-077970 | MATHDB-400003

Higher-order finite element methods for surface PDEs from an approximate level-set function

摘录 / EXCERPT: Can one develop a higher-order finite element method for the Laplace-Beltrami equation on a stationary smooth surface Γ when Γ is the zero level of a smooth level-set...

open_problem | open | MathDB 页面 | 原文线索

V2-077971 | MATHDB-400004

Godbersen’s conjecture

摘录 / EXCERPT: For any convex body $K\subset\mathbb{R}^n$ and any $1\le j\le n-1$, ... with equality attained only for simplices.

conjecture | open | MathDB 页面 | 原文线索

V2-077972 | MATHDB-400005

Unbalanced difference-body conjecture (Conjecture 6)

摘录 / EXCERPT: For any convex body $K\subset\mathbb{R}^n$ and any $\lambda\in(0,1)$, defining $D_\lambda K=(1-\lambda)K+\lambda(-K)$, does ... hold, where Δ is an n-dimensional simplex?

conjecture | open | MathDB 页面 | 原文线索

V2-077973 | MATHDB-400006

Classification of equivariant Minkowski valuations on lower-dimensional lattice polytopes

摘录 / EXCERPT: Characterize all $\mathrm{SL}_n(\mathbb{Z})$-equivariant Minkowski valuations on at most n-dimensional lattice polytopes.

open_problem | open | MathDB 页面 | 原文线索

V2-077974 | MATHDB-400008

The ℓ2 Kannan-Lovász Conjecture

摘录 / EXCERPT: For every n-dimensional lattice $L\subseteq\mathbb{R}^n$, the quantity ... is bounded above by $O(\sqrt{\log n})$ and below by $\Omega(1)$; equivalently, ...

conjecture | open | MathDB 页面 | 原文线索

V2-077975 | MATHDB-400009

Compute the coefficients in the indefinite-orthogonal kinematic formulas

摘录 / EXCERPT: Determine the coefficients c_{ij} in the finite linear-combination formulas obtained for the two convergent integrals in Theorem 3, namely the integrals involving $a_0^+\phi_2^0$...

open_problem | open | MathDB 页面 | 原文线索

V2-077976 | MATHDB-400010

Invariant generalized valuations for the indefinite orthogonal group

摘录 / EXCERPT: For $1\leq k\leq n-1$, $\dim \operatorname{Val}^{-\infty}_k(V)^{O(p,q)}=2$.

open_problem | open | MathDB 页面 | 原文线索

V2-077977 | MATHDB-400011

Eventual concavity of parallel-body volume under Minkowski addition

摘录 / EXCERPT: For every compact set $A\subset\mathbb{R}^n$, does there exist $t(A)\ge 0$ such that the function $t\mapsto |A+tB_2^n|^{1/n}$ is concave for all $t\ge t(A)$?

open_problem | open | MathDB 页面 | 原文线索

V2-077978 | MATHDB-400012

Operations that are neither necessarily pointwise nor associative

摘录 / EXCERPT: Find results in the spirit of those described above for operations that are neither necessarily pointwise nor associative.

open_problem | open | MathDB 页面 | 原文线索

V2-077979 | MATHDB-400013

Polynomial bound in the diameter version of quantitative Helly’s theorem

摘录 / EXCERPT: Let $\{P_i:i\in I\}$ be a family of closed convex sets in $\mathbb{R}^n$ such that $\operatorname{diam}\!\left(\bigcap_{i\in I}P_i\right)=1$. Is it true that there exists an absolu...

open_problem | open | MathDB 页面 | 原文线索

V2-077980 | MATHDB-400014

Sampling from a convex body using projected Langevin Monte Carlo

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body containing the Euclidean ball of radius 1 and contained in the Euclidean ball of radius R. Let $V:K\to\mathbb{R}$ be convex, L-Lip...

open_problem | open | MathDB 页面 | 原文线索

V2-077981 | MATHDB-400016

Polarity from equality of Minkowski sums

摘录 / EXCERPT: Assume $A+B=A^{\circ}+B^{\circ}$. Does it follow that $B=A^{\circ}$?

open_problem | open | MathDB 页面 | 原文线索

V2-077982 | MATHDB-400018

Hardness of approximating unconditional convex bodies by projections of simplex sections

摘录 / EXCERPT: Let $n\le N$. There exists an n-dimensional unconditional convex body B such that, for every n-dimensional convex body K obtained as a projection of a section of an N-dim...

open_problem | open | MathDB 页面 | 原文线索

V2-077983 | MATHDB-400019

Bodies with directly congruent projections

摘录 / EXCERPT: Suppose that $2\le k\le n-1$ and that K and L are convex bodies in $\mathbb{R}^n$ such that the projection $K\mathbin{|}H$ is congruent to $L\mathbin{|}H$ for all $H\in G(n,k)$...

open_problem | open | MathDB 页面 | 原文线索

V2-077984 | MATHDB-400020

Petty’s conjectured projection inequality

摘录 / EXCERPT: For every dimension $d\ge 3$, let K be a convex body in $\mathbb{R}^d$. Define its projection body by $h_{\Pi K}(x)=\lvert K\mid x^\perp\rvert$ for $x\in S^{d-1}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-077985 | MATHDB-400021

Koldobsky’s Fourier characterization of j-intersection bodies

摘录 / EXCERPT: Let $1\le j\le n-1$ and let D and M be origin-symmetric star bodies in $\mathbb{R}^n$. Then D is the j-intersection body of M if and only if ... where $\mathcal{F}^{-j}g$ d...

open_problem | open | MathDB 页面 | 原文线索

V2-077986 | MATHDB-400023

Measure-theoretic slicing inequality of Koldobsky

摘录 / EXCERPT: Does there exist an absolute constant L_2 such that, for every even measure $\$ on $\mathbb{R}^d$ with a positive density and every origin-symmetric convex body K in...

open_problem | open | MathDB 页面 | 原文线索

V2-077987 | MATHDB-400024

A discrete version of Koldobsky’s slicing inequality

摘录 / EXCERPT: Let $\mathbb{Z}^d$ be the standard integer lattice in $\mathbb{R}^d$, and write $\#K=\operatorname{card}(K\cap\mathbb{Z}^d)$. Does there exist a constant L_3 such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-077988 | MATHDB-400025

Noetherianity of a ring of overconvergent power series

摘录 / EXCERPT: For every $r_1,\\ldots,r_n\\in(0,1)$, the ring $\\mathbb{Z}\\{r_1^{-1}T_1,\\ldots,r_n^{-1}T_n\\}:$ is Noetherian.

open_problem | open | MathDB 页面 | 原文线索

V2-077989 | MATHDB-400027

Strengthening Gabber’s theorem to \operatorname{char}(X)-alterations

摘录 / EXCERPT: Can Gabber’s altered desingularization theorem be strengthened so that the degree of the desingularizing alteration is divisible only by primes non-invertible on X? The report an...

open_problem | open | MathDB 页面 | 原文线索

V2-077990 | MATHDB-400033

Fargues's geometrization conjecture for the local Langlands correspondence

摘录 / EXCERPT: Let G be a quasi-split reductive group over a p-adic field E. Let ρ be a discrete Langlands parameter $W_E\to{}^LG$, with...

conjecture | open | MathDB 页面 | 原文线索

V2-077991 | MATHDB-400035

Uniform Manin-Mumford bound for curves

摘录 / EXCERPT: For every integer $g\ge 2$, there exists a number $N=N(g)$ such that for every smooth curve C of genus g over $\mathbb{C}$ and every Abel-Jacobi embedding...

open_problem | open | MathDB 页面 | 原文线索

V2-077992 | MATHDB-400036

Uniform Manin-Mumford conjecture for Mumford curves

摘录 / EXCERPT: Prove the uniform Manin-Mumford conjecture for all Mumford curves over $\mathbf{C}_p$.

conjecture | open | MathDB 页面 | 原文线索

V2-077993 | MATHDB-400037

Piecewise-monomial structure on the preimage of the standard skeleton

摘录 / EXCERPT: Let $X=M(A)$ be a k-affinoid space and let $f:X\to\mathbb{A}^{n,\mathrm{an}}_k$ be a morphism with zero-dimensional (hence finite) fibers. Set $\Sigma=f^{-1}(S_n)$, where S_n i...

open_problem | open | MathDB 页面 | 原文线索

V2-077994 | MATHDB-400038

Smallest entropy ideal containing non-approximable operators

摘录 / EXCERPT: Determine the smallest entropy ideal that contains non-approximable operators.

open_problem | open | MathDB 页面 | 原文线索

V2-077995 | MATHDB-400039

Logarithmic Khintchine-Steinhaus inequality

摘录 / EXCERPT: Let $S_1,S_2,\\ldots$ be independent identically distributed Steinhaus random variables, uniformly distributed on the unit circle $S^1\\subset\\mathbb C$. For every n and every...

open_problem | open | MathDB 页面 | 原文线索

V2-077996 | MATHDB-400040

A noncommutative Hölder-type inequality for the case $0<q<1$

摘录 / EXCERPT: Let $0<p<q<s\\leq\\infty$. Set $\\alpha=1/p-1/s$, and let $0<\\theta<1$ satisfy $1/q=(1-\\theta)/p+\\theta/s$. Then, for every $0<R<p$, there is a constant C such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-077997 | MATHDB-400041

Pietsch duality conjecture for entropy numbers

摘录 / EXCERPT: Do there exist numerical constants $a,b\geq 1$ such that for any dimension n and for any two symmetric convex bodies $K,L\subset\mathbb{R}^n$ one has ... Here K° and...

conjecture | open | MathDB 页面 | 原文线索

V2-077998 | MATHDB-400042

Limit of the first nontrivial Neumann p-Laplacian eigenvalue as p tends to infinity

摘录 / EXCERPT: The report answers the question “What happens for $p\to\infty$?” by asserting that ... and that $u_\infty=\lim_{p\to\infty}u_p$ solves, in the viscosity sense, ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-077999 | MATHDB-400043

Hot spot conjecture for convex domains

摘录 / EXCERPT: In convex domains, u_2 should attain its maximum and minimum on the boundary.

conjecture | open | MathDB 页面 | 原文线索

V2-078000 | MATHDB-400044

Cheeger inequality for unbounded graph Laplacians using intrinsic metrics

摘录 / EXCERPT: Let X be a discrete countable set, let $b:X\times X\to[0,\infty)$ be symmetric, have zero diagonal, and satisfy $\sum_{y\in X}b(x,y)<\infty$ for every $x\in X$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-078001 | MATHDB-400045

Feller property after attaching infinitely many vertices

摘录 / EXCERPT: Can gluing infinitely many graphs to a non-Feller graph produce a Feller graph?

open_problem | open | MathDB 页面 | 原文线索

V2-078002 | MATHDB-400046

Uniqueness of the AC holonomy-G_2 metric on $S(S^3)$

摘录 / EXCERPT: Is the AC holonomy-G_2 metric on $S(S^3)$ unique?

open_problem | open | MathDB 页面 | 原文线索

V2-078003 | MATHDB-400047

Obstructedness of nearly Kähler deformations of $\mathrm{SU}(3)/T^2$

摘录 / EXCERPT: It is still an open question whether these nearly Kähler deformations are obstructed or not.

open_problem | open | MathDB 页面 | 原文线索

V2-078004 | MATHDB-400048

Coordinate form of the Bryant-Salamon metric compatible with the fibration

摘录 / EXCERPT: Rewrite the Bryant-Salamon metric in coordinates compatible with the fibration described for the cone $Y=\mathbb{R}_{+}\times\mathbb{CP}^{3}$.

open_problem | open | MathDB 页面 | 原文线索

V2-078005 | MATHDB-400049

Infinitesimal rigidity of three-dimensional hyperbolic conifolds

摘录 / EXCERPT: Let M be a three-dimensional space with a singular iterated conic hyperbolic metric g. Is every infinitesimal variation $\dot g$ of these structures that leaves the dihedral an...

open_problem | open | MathDB 页面 | 原文线索

V2-078006 | MATHDB-400051

Compactness of the moduli space of G_2-instantons for generic G_2-metrics

摘录 / EXCERPT: It is expected that, for a generic metric with holonomy contained in G_2, the moduli space of G_2-instantons is compact.

open_problem | open | MathDB 页面 | 原文线索

V2-078007 | MATHDB-400052

Matching problem for asymptotically cylindrical Calabi-Yau 3-folds

摘录 / EXCERPT: Given pairs (Y_+,Σ_+) and (Y_-,Σ_-), where each $Y_\pm$ is a Fano or weak Fano 3-fold with anticanonical divisor $\Sigma_\pm$, construct the associated asymptotical...

open_problem | open | MathDB 页面 | 原文线索

V2-078008 | MATHDB-400054

Huneke’s symbolic-versus-ordinary powers question

摘录 / EXCERPT: For any homogeneous ideal I of big height c, if $I^{(c)}=I^c$, must $I^{(m)}=I^m$ for all $m\ge 1$?

open_problem | open | MathDB 页面 | 原文线索

V2-078009 | MATHDB-400055

Strassen’s additivity conjecture for Waring rank

摘录 / EXCERPT: For homogeneous forms $F_1,\dots,F_r$ of a common degree in independent sets of variables, is ...

conjecture | open | MathDB 页面 | 原文线索

V2-078010 | MATHDB-400057

Huneke’s symbolic-power containment question for points in the projective plane

摘录 / EXCERPT: Let I be the ideal of distinct points $p_1,\ldots,p_s\in\mathbb{P}^2$ over the complex numbers, with symbolic power $I^{(m)}=I(p_1)^m\cap\cdots\cap I(p_s)^m$. Is it always true t...

open_problem | open | MathDB 页面 | 原文线索

V2-078011 | MATHDB-400058

Ideals of other arrangements of lines

摘录 / EXCERPT: What can one say for ideals of other arrangements of lines?

open_problem | open | MathDB 页面 | 原文线索

V2-078012 | MATHDB-400059

Possible connection between bounded negativity and symbolic-power noncontainment

摘录 / EXCERPT: The report observes that every known nontrivial finite configuration of lines in the plane with no simple crossings satisfies both $H(C)<-2$ and $I^{(3)}\not\subseteq I^2$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078013 | MATHDB-400060

Containment for the singular-point ideal of the configuration C''

摘录 / EXCERPT: Let C be the cubic arrangement defined by $(x-y)(x-z)(y-z)=0$. After the stated linear change of coordinates, let C' be the pullback of the Fermat configuration under...

open_problem | open | MathDB 页面 | 原文线索

V2-078014 | MATHDB-400062

Irreducible plane curves with H-constant at most -2

摘录 / EXCERPT: Does there exist an irreducible reduced plane curve C such that $H(C) \leq -2$, where $H(C)$ is the H-constant of C computed from its degree and singular-point multiplicities?

open_problem | open | MathDB 页面 | 原文线索

V2-078015 | MATHDB-400064

Reduced curves in \(\mathbb{P}^3\) with minimal symbolic-square growth

摘录 / EXCERPT: Which reduced (possibly irreducible) curves C in $\mathbb{P}^3$ have $I=I(C)$ and $\alpha\bigl(I^{(2)}\bigr)-\alpha(I)=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-078016 | MATHDB-400065

SI-sequence conjecture for Gorenstein quotients in four variables

摘录 / EXCERPT: Do all Gorenstein quotients of $k[x_1,x_2,x_3,x_4]$ have Hilbert functions that are SI-sequences?

conjecture | open | MathDB 页面 | 原文线索

V2-078017 | MATHDB-400066

Hilbert functions of ideals generated by powers of general linear forms

摘录 / EXCERPT: Does an ideal generated by powers of general linear forms have the Hilbert function predicted by Fröberg for general forms of the same degrees?

open_problem | open | MathDB 页面 | 原文线索

V2-078018 | MATHDB-400069

Huneke’s symbolic-power containment question for finite point sets in \(\mathbb P^2\)

摘录 / EXCERPT: Question: Let I be a radical ideal defining a finite set of points in $\mathbb P^2$. Does $I^{(3)}\subseteq I^2$ always hold? The report states that the answer is no; known count...

open_problem | open | MathDB 页面 | 原文线索

V2-078019 | MATHDB-400071

Grid characterization in the multiprojective setting

摘录 / EXCERPT: Conjecture. Let Z be a finite set of points in $(\mathbb{P}^1)^n$. If $d=\alpha(nZ)=\alpha(Z)$, then Z is a grid of size $d\times\cdots\times d$; that is, there are subsets...

open_problem | open | MathDB 页面 | 原文线索

V2-078020 | MATHDB-400072

Classification at the minimal initial degree for fat points in projective space

摘录 / EXCERPT: Let Z be a finite set of points in projective space $\P^n$. If ... then either $\alpha(Z)=1$, i.e. Z is contained in a single hyperplane H in $\u0001P^n$, or Z consists of al...

open_problem | open | MathDB 页面 | 原文线索

V2-078021 | MATHDB-400073

Birational invariance of the bounded-negativity property

摘录 / EXCERPT: Let the bounded-negativity property mean that a smooth complex surface has a lower bound for the self-intersection of its reduced divisors. If X and Y are birational surfaces,...

open_problem | open | MathDB 页面 | 原文线索

V2-078022 | MATHDB-400076

Generic \\-closedness of the complex exponential \\-field

摘录 / EXCERPT: Conjecture 5. is generically -closed.

open_problem | open | MathDB 页面 | 原文线索

V2-078023 | MATHDB-400078

Lattices of definable equivalence relations in homogeneous finite-dimensional permutation structures

摘录 / EXCERPT: What may the lattice of $ 0$-definable equivalence relations look like in a homogeneous finite-dimensional permutation structure?

open_problem | open | MathDB 页面 | 原文线索

V2-078024 | MATHDB-400081

NIP transfer to henselian valued fields with imperfect NIP residue field

摘录 / EXCERPT: Let (K,v) be henselian, with Kv NIP and not perfect. Is (K,v) NIP?

open_problem | open | MathDB 页面 | 原文线索

V2-078025 | MATHDB-400084

Meridional rank versus bridge number for knots

摘录 / EXCERPT: Does the minimal number of meridians needed to generate the knot group of every knot in S^{3} coincide with its bridge number?

open_problem | open | MathDB 页面 | 原文线索

V2-078026 | MATHDB-400096

Bourgain’s slicing problem / boundedness of the isotropic constant

摘录 / EXCERPT: Is the isotropic constant of convex bodies bounded by an absolute constant? Equivalently, does every origin-symmetric convex body in $\mathbb{R}^n$ of volume 1 have a central hyp...

open_problem | open | MathDB 页面 | 原文线索

V2-078027 | MATHDB-400097

Lower-dimensional Busemann-Petty/Shephard projection-section inequality

摘录 / EXCERPT: Let K be a convex body in $ obreak\\mathbb{R}^n$ and let D be a compact subset of $ obreak\mathbb{R}^n$. Suppose that, for some $1\leq k\leq n-1$, $|P_F(K)|\leq |D\cap F|$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-078028 | MATHDB-400098

Variant of the Busemann-Petty and Shephard problems

摘录 / EXCERPT: Assume that K and D are origin-symmetric convex bodies in $\mathbb{R}^n$ and satisfy $|P_{\xi^\perp}(K)|\leq |D\cap \xi^\perp|$ for all $\xi\in S^{n-1}$. Does it follow that...

open_problem | open | MathDB 页面 | 原文线索

V2-078029 | MATHDB-400099

Almost-ball property for the convex intersection body

摘录 / EXCERPT: Let $K\subset\mathbb{R}^n$ be a convex body with centroid at the origin, in isotropic position (that is, $|K|=1$ and $\int_K\langle x,u\rangle^2\,dx$ is independent of...

open_problem | open | MathDB 页面 | 原文线索

V2-078030 | MATHDB-400100

Comparison of the convex intersection body and the intersection body

摘录 / EXCERPT: There exists an absolute constant $c>0$ such that, for every $n\ge 2$ and every convex body $K\subset\mathbb{R}^n$ with centroid at the origin, one has ...

open_problem | open | MathDB 页面 | 原文线索

V2-078031 | MATHDB-400101

Removing the isotropic-constant factor from the mean-width bound

摘录 / EXCERPT: For an origin-symmetric convex body $K\subset\mathbb{R}^n$ in isotropic position, can the factor L_K be removed from the bound $M^{*}(K)\leq C\sqrt{n}\,\log(n)^2L_K$?

open_problem | open | MathDB 页面 | 原文线索

V2-078032 | MATHDB-400102

Grünbaum’s conjecture on affine-invariant points

摘录 / EXCERPT: Let $\mathcal K_d$ be the set of d-dimensional convex bodies in $\mathbb R^d$. An affine-invariant point is a continuous map $p:\mathcal K_d\to\mathbb R^d$ such that...

conjecture | open | MathDB 页面 | 原文线索

V2-078033 | MATHDB-400103

Generic affine invariant points have no dual

摘录 / EXCERPT: The report states: There exists an open and dense set $W\subseteq P_d$ of affine invariant points with no dual, where P_d is the space of affine invariant points on the d-dimen...

open_problem | open | MathDB 页面 | 原文线索

V2-078034 | MATHDB-400104

Lower bound on entanglement witnesses for robustly entangled states

摘录 / EXCERPT: There is a universal constant $c>0$ such that the following holds. Suppose that $(\a6_i)_{1\le i\le N}$ is a family of positive maps on M_d with the property that, for any robust...

open_problem | open | MathDB 页面 | 原文线索

V2-078035 | MATHDB-400106

Gaussian Brunn-Minkowski conjecture for origin-symmetric convex bodies

摘录 / EXCERPT: For origin-symmetric convex bodies $A,B\subset\mathbb{R}^n$ and $\lambda\in[0,1]$, does ... hold?

conjecture | open | MathDB 页面 | 原文线索

V2-078036 | MATHDB-400107

Invertibility of adjacency matrices of random regular directed graphs

摘录 / EXCERPT: Let $M_{n,d}$ be the set of $n\times n$ $0/1$ adjacency matrices of directed d-regular graphs, meaning that every row and every column contains exactly d ones, equipped with no...

open_problem | open | MathDB 页面 | 原文线索

V2-078037 | MATHDB-400108

Fundamental precision limit and adaptive inference in high-dimensional Tobit-I models

摘录 / EXCERPT: Determine the fundamental limit of estimation precision in Tobit-I models, and develop an adaptive procedure that achieves that limit, particularly in the high-dimensional setting.

open_problem | open | MathDB 页面 | 原文线索

V2-078038 | MATHDB-400109

Identifiability criteria for finite-alphabet blind source separation

摘录 / EXCERPT: For the noiseless version of the SBSSR model, with a finite alphabet $A\subset\mathbb{R}$, $m\ge 2$ unknown step functions $f_i:[0,1)\to A$, positive mixing weights ω_i sati...

open_problem | open | MathDB 页面 | 原文线索

V2-078039 | MATHDB-400111

Birationality of subword sequences

摘录 / EXCERPT: Conjecture: Every subword sequence obtained from the reduced-expression sequence S_{w_0} as an initial sequence is birational; equivalently, its associated multiplication map is...

open_problem | open | MathDB 页面 | 原文线索

V2-078040 | MATHDB-400112

Twisted Demazure sequences and the Demazure property in type A_n

摘录 / EXCERPT: For type A_n, the set of twisted Demazure sequences is exactly the set of all sequences with the Demazure property.

open_problem | open | MathDB 页面 | 原文线索

V2-078041 | MATHDB-400113

Cycle maps for quiver Grassmannians

摘录 / EXCERPT: Let Q be a Dynkin quiver and let M be a complex finite-dimensional representation of Q such that $\operatorname{Ext}^1_Q(M,M)=0$ (that is, M is as in Theorem 3). The conjec...

open_problem | open | MathDB 页面 | 原文线索

V2-078042 | MATHDB-400114

Cellular decompositions of quiver Grassmannians for Dynkin quivers

摘录 / EXCERPT: For any representation M of a Dynkin quiver Q, is it true that any quiver Grassmannian associated with M admits a cellular decomposition?

open_problem | open | MathDB 页面 | 原文线索

V2-078043 | MATHDB-400115

Unimodular equivalence of a plabic-graph Newton-Okounkov body and an FFLV polytope

摘录 / EXCERPT: Conjecture 2. The Newton-Okounkov body $NO^r_{G_0}$ is unimodularly equivalent to the FFLV polytope $FFLV(r\varpi_k)$, where G_0 is the appropriately chosen plabic graph associat...

open_problem | open | MathDB 页面 | 原文线索

V2-078044 | MATHDB-400116

Kostka-Shoji polynomial conjecture for arbitrary multipartitions

摘录 / EXCERPT: Let $\lambda=(\lambda^{(1)},\ldots,\lambda^{(r)})$ and $\mu=(\mu^{(1)},\ldots,\mu^{(r)})$ be multipartitions whose components have at most N parts. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-078045 | MATHDB-400118

Geometric interpretations of positivity for specialized nonsymmetric Macdonald polynomials

摘录 / EXCERPT: Are there geometric interpretations of the positivity of $E_\lambda(X;q^{-1},\infty)$?

open_problem | open | MathDB 页面 | 原文线索

V2-078046 | MATHDB-400121

Independence of conditions imposed on incomplete linear systems

摘录 / EXCERPT: Find interesting cases where one can characterize or classify triples $\Lambda=(t,X,Z)$ for which X fails to impose the expected number of conditions on $[I(Z)]_t$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078047 | MATHDB-400122

Resurgences of the Klein and Wiman configurations

摘录 / EXCERPT: Compute the resurgences of the Klein and Wiman configurations (see [4, Sections 4.1 and 4.2]).

open_problem | open | MathDB 页面 | 原文线索

V2-078048 | MATHDB-400123

Fixed-point behavior for endomorphisms of simple abelian surfaces

摘录 / EXCERPT: Theorem. Let X be a simple abelian surface. Then the fixed-points function of any non-zero endomorphism $f\in\operatorname{End}(X)$ is either exponential (B1) or periodic (B2), a...

open_problem | open | MathDB 页面 | 原文线索

V2-078049 | MATHDB-400126

Classification of failures of independent conditions for general fat points

摘录 / EXCERPT: Classify all triples $\Lambda=(t,X,V)$ where $V\subseteq (k[\mathbb{P}^2])_t$, the points $p_i\in\mathbb{P}^2$ are general, $X=\sum_i m_i p_i$, and $I(X)=\bigcap_i I(p_i)^{m_i}$, s...

open_problem | open | MathDB 页面 | 原文线索

V2-078050 | MATHDB-400127

Classification of point sets admitting unexpected curves

摘录 / EXCERPT: “We have many other examples of Z with such unexpected curves. Question: Is it possible to be more specific about the point sets Z that arise this way?”

open_problem | open | MathDB 页面 | 原文线索

V2-078051 | MATHDB-400128

The weak Brill-Noether problem for sheaves on surfaces

摘录 / EXCERPT: Let (X,H) be a smooth polarized surface, let $M_H(v)$ be the moduli space of H-Gieseker semistable sheaves on X with Chern character v, and let $E\in M_H(v)$ be a general s...

open_problem | open | MathDB 页面 | 原文线索

V2-078052 | MATHDB-400129

Weak Brill-Noether problem for semistable sheaves on a polarized surface

摘录 / EXCERPT: Let (X,H) be a smooth polarized surface, and let $M_H(v)$ denote the moduli space of H-Gieseker semistable sheaves on X with Chern character v. Let $E\in M_H(v)$ be a gener...

open_problem | open | MathDB 页面 | 原文线索

V2-078053 | MATHDB-400130

Criterion for property (N_p) on polarized abelian surfaces

摘录 / EXCERPT: Theorem. Let (X,L) be a polarized abelian surface, let p be a natural number, and assume that $L^2\ge 5(p+2)^2$. Then the following are equivalent: 1. X does not contain a sm...

open_problem | open | MathDB 页面 | 原文线索

V2-078054 | MATHDB-400131

Unexpected curves: independent conditions for incomplete linear systems

摘录 / EXCERPT: Over an infinite field k, given a fixed finite set of points $Z\subset \mathbb{P}^2$ (possibly with prescribed multiplicities) and a general collection X of fat points, determi...

open_problem | open | MathDB 页面 | 原文线索

V2-078055 | MATHDB-400133

Higher-dimensional version of the logarithmic-vector-field exact sequence

摘录 / EXCERPT: Is it possible to prove a higher-dimensional version of Theorem 1 using these methods?

open_problem | open | MathDB 页面 | 原文线索

V2-078056 | MATHDB-400134

Uniform integrality of Zariski decompositions

摘录 / EXCERPT: Let X be a smooth projective surface. Does there exist a number m depending only on X such that, for any effective divisor D on X, the Zariski decomposition of mD is in...

open_problem | open | MathDB 页面 | 原文线索

V2-078057 | MATHDB-400137

Sylvester-Gallai conjecture for curves of higher degree

摘录 / EXCERPT: Conjecture. Let $P\subset \mathbb{P}^2(\mathbb{R})$ be a finite set of points and let d be a positive integer. Then either:

conjecture | open | MathDB 页面 | 原文线索

V2-078058 | MATHDB-400138

Conjecture on absolute linear Harbourne constants

摘录 / EXCERPT: Let a configuration of lines $\mathcal L=\{L_1,\ldots,L_d\}$ in $\mathbb P^2(K)$ have singular points $\mathcal P(\mathcal L)=\{P_1,\ldots,P_s\}$, and let $m_{\mathcal L}(P_k)$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-078059 | MATHDB-400142

Ganea’s conjecture for Lusternik-Schnirelmann category

摘录 / EXCERPT: For a finite complex X, is it true that ...

conjecture | open | MathDB 页面 | 原文线索

V2-078060 | MATHDB-400146

Adaptation to the unknown smoothness parameter in goodness-of-fit testing

摘录 / EXCERPT: “Is there a single test that can adaptively attain the optimal detection boundary without the knowledge of θ?”

open_problem | open | MathDB 页面 | 原文线索

V2-078061 | MATHDB-400147

Permutation scan test for interval anomalies

摘录 / EXCERPT: Let $X_1,\ldots,X_N$ be independent, with common null distribution F_0 under H_0. Under the alternative, there is an anomalous set $S\in\mathbb{S}$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078062 | MATHDB-400148

Theorem 2.1: Power of the rank-scan test

摘录 / EXCERPT: Let $X_1,\ldots,X_N$ be independent, with null distribution F_0, and under the alternative let an anomalous set $S\in\mathbb{S}$ have $X_i\sim F_i$ for $i\in S$, while...

open_problem | open | MathDB 页面 | 原文线索

V2-078063 | MATHDB-400149

Adaptation of confidence intervals to sparsity in high-dimensional linear regression

摘录 / EXCERPT: For confidence intervals for the coordinate functional $T(\beta)=\beta_i$, can one construct an adaptive honest confidence interval without knowing the sparsity level k, whose le...

open_problem | open | MathDB 页面 | 原文线索

V2-078064 | MATHDB-400150

Does a larger minimum cluster size help in variable clustering?

摘录 / EXCERPT: In the balanced, simple covariance-gap setting described in the report-namely, $C=\alpha J+\tau I$ with $\alpha,\tau\ge 0$, $\Gamma=I$, and all K clusters of size $m=p/K$-a large...

open_problem | open | MathDB 页面 | 原文线索

V2-078065 | MATHDB-400152

Persistence of confidence sets and centrality estimators in random growing trees

摘录 / EXCERPT: The confidence sets for diffusion sources on regular diffusion trees, as well as the corresponding confidence sets for uniform-attachment, linear preferential-attachment, and subli...

open_problem | open | MathDB 页面 | 原文线索

V2-078066 | MATHDB-400153

Identifiability and estimation in the SBSSR blind source-separation model

摘录 / EXCERPT: For a finite alphabet $\mathfrak A\subset\mathbb R$, $m\ge 2$ unknown step functions $f=(f^1,\ldots,f^m)^\top$ on $[0,1)$ with $\operatorname{im}(f^i)\subseteq\mathfrak A$, and unk...

open_problem | open | MathDB 页面 | 原文线索

V2-078067 | MATHDB-400156

Transition fields for hysteretic Néel-Bloch wall transitions

摘录 / EXCERPT: Determine the transition fields H_{BN} and H_{NB} marking, respectively, the transition from a Bloch wall to a Néel wall and the reverse transition under a varying external fie...

open_problem | open | MathDB 页面 | 原文线索

V2-078068 | MATHDB-400157

Evolution and activation of dislocation sources in continuum dislocation dynamics

摘录 / EXCERPT: Develop a convincing concept within continuum dislocation dynamics (CDD) for the evolution and activation of dislocation sources, including mechanisms such as cross-slip, dislocati...

open_problem | open | MathDB 页面 | 原文线索

V2-078069 | MATHDB-400158

Formation of grain boundaries in ductile single crystals

摘录 / EXCERPT: For ductile single crystals undergoing plane-strain simple shear, grain boundaries are interpreted as surfaces of weak discontinuity in placement and strong discontinuity in plasti...

open_problem | open | MathDB 页面 | 原文线索

V2-078070 | MATHDB-400159

Evolutionary relaxation of a rate-independent two-phase model

摘录 / EXCERPT: For any approximate solutions $q^{\tau,\varepsilon}$ of the pure-state rate-independent system $(U\times Z_{\mathrm{pure}},E,D)$ obtained via the approximate incremental minimizati...

open_problem | open | MathDB 页面 | 原文线索

V2-078071 | MATHDB-400160

Long-time approximation in the nonrelativistic limit on \(\mathbb{R}^3\)

摘录 / EXCERPT: Theorem 2. Fix $k\ge 0$ and a large r. Then there exist a large k^* and c^* with the following property. Assume $c>c^*$, and take initial data such that the solution ϕ o...

open_problem | open | MathDB 页面 | 原文线索

V2-078072 | MATHDB-400162

Existence and local uniqueness for the touching-tube equations

摘录 / EXCERPT: Establish existence and local uniqueness results for the system of governing equations describing two intertwined slender tubes with circular cross-sections, formulated as a two-po...

open_problem | open | MathDB 页面 | 原文线索

V2-078073 | MATHDB-400173

Gyárfás' conjecture on graphs with no odd holes

摘录 / EXCERPT: A conjecture of Gyárfás asserts that the chromatic number of a graph with no odd hole is bounded by a function of its clique number.

conjecture | open | MathDB 页面 | 原文线索

V2-078074 | MATHDB-400174

The weak 3-flow conjecture

摘录 / EXCERPT: The weak 3-flow conjecture proposed by Jaeger is a theorem.

conjecture | open | MathDB 页面 | 原文线索

V2-078075 | MATHDB-400175

Barát-Thomassen tree-decomposition theorem

摘录 / EXCERPT: For every fixed tree T, there exists t such that every t-connected graph G has an edge decomposition into copies of T if the number of edges of G is divisible by the nu...

open_problem | open | MathDB 页面 | 原文线索

V2-078076 | MATHDB-400176

Erdős-Nešetřil strong chromatic index conjecture

摘录 / EXCERPT: For every graph G, the strong chromatic index satisfies $\chi'_s(G) \le 1.25\,\Delta(G)^2$ (equivalently, $\chi'_s(G) \le \frac{5}{4}\Delta(G)^2$).

conjecture | open | MathDB 页面 | 原文线索

V2-078077 | MATHDB-400177

Complete graph immersion at the minimum-degree threshold

摘录 / EXCERPT: Is it true that for all $n \ge 1$, every graph of minimum degree at least n contains an immersion of K_n?

open_problem | open | MathDB 页面 | 原文线索

V2-078078 | MATHDB-400179

Tight upper bound for the Excluded Grid Theorem

摘录 / EXCERPT: For the undirected Excluded Grid Theorem, is the lower-bound order $g^2\log g$ sufficient? Equivalently, can the function in Theorem 1 be chosen so that $f(g)=O(g^2\log g)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078079 | MATHDB-400180

Accessibility of locally finite transitive hyperbolic graphs

摘录 / EXCERPT: Every locally finite transitive hyperbolic graph is accessible.

open_problem | open | MathDB 页面 | 原文线索

V2-078080 | MATHDB-400181

Structural dichotomy for monotone nowhere-dense graph classes

摘录 / EXCERPT: Every monotone nowhere dense class of graphs either has bounded expansion or contains, for some $k\in\mathbb{N}$, the k-th subdivisions of graphs with arbitrarily large girth and...

open_problem | open | MathDB 页面 | 原文线索

V2-078081 | MATHDB-400184

Theorem 11: Local geometric class field theory

摘录 / EXCERPT: There is a canonical equivalence ... This equivalence is characterized as follows: a local system $L\in\operatorname{Loc}_1(\widehat U)$ is associated to a character local system...

open_problem | open | MathDB 页面 | 原文线索

V2-078082 | MATHDB-400185

Construction of the space whose Frobenius quotient is the adic Fargues-Fontaine curve

摘录 / EXCERPT: The question is how to define a space Y and specify the category in which to take the quotient by $[U+007F]\varphi$. The report answers this by taking Y to be the analytic space...

open_problem | open | MathDB 页面 | 原文线索

V2-078083 | MATHDB-400186

Geometric Langlands correspondence for a general reductive group

摘录 / EXCERPT: Let $\mathbb F_q$ be a finite field, let X be a smooth complete curve over $\mathbb F_q$, and let G be a reductive group with Langlands dual group $\widehat G$. For every irred...

open_problem | open | MathDB 页面 | 原文线索

V2-078084 | MATHDB-400187

Classification of vector bundles on the Fargues-Fontaine curve

摘录 / EXCERPT: The functor $D\mapsto E(D)$ from the category $\varphi\mathrm{Mod}_L$ of L-isocrystals to vector bundles on the Fargues-Fontaine curve X is essentially surjective. Equivalently...

open_problem | open | MathDB 页面 | 原文线索

V2-078085 | MATHDB-400191

Characteristic-\p extension of the classification of G-bundles on the Fargues\Fontaine curve

摘录 / EXCERPT: Conjecture 21. Theorem 17 also holds when E is of characteristic p.

open_problem | open | MathDB 页面 | 原文线索

V2-078086 | MATHDB-400192

Unramified geometric Langlands for GL_n

摘录 / EXCERPT: Let X be a smooth projective geometrically connected curve over the finite field $k=\mathbb{F}_q$, let $F=k(X)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078087 | MATHDB-400193

Existence of the sheaf \(\operatorname{Aut}'_E\) on \(\operatorname{Bun}'_n\)

摘录 / EXCERPT: Let X be a smooth projective geometrically connected curve over $k=\mathbf F_q$, let $F=k(X)$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078088 | MATHDB-400194

Shintani-Casselman-Shalika formula for the unramified Whittaker function

摘录 / EXCERPT: Theorem 5 (Shintani-Casselman-Shalika). If $\lambda\in X_\bullet(T)$ is not a dominant weight, then ... If λ is dominant, then ... where $V(\lambda)$ denotes the irreducible re...

open_problem | open | MathDB 页面 | 原文线索

V2-078089 | MATHDB-400195

Tempered local Langlands correspondence

摘录 / EXCERPT: Let E be a local field of characteristic 0 and let G be a connected reductive group over E. Let $\Pi_{\mathrm{temp}}(G)$ be the set of tempered irreducible admissible repre...

open_problem | open | MathDB 页面 | 原文线索

V2-078090 | MATHDB-400196

The local Langlands conjecture for tempered representations

摘录 / EXCERPT: Let E be a local field of characteristic 0, let G be a connected reductive group defined over E, and let $\Pi_{\mathrm{temp}}(G)$ denote the tempered irreducible admissible...

conjecture | open | MathDB 页面 | 原文线索

V2-078091 | MATHDB-400197

Refined local Langlands conjecture for quasi-split groups

摘录 / EXCERPT: Let E be a local field of characteristic 0, let G be a connected reductive group over E, and let $\d53f\in\Phi_{\mathrm{temp}}(G)$ be a tempered Langlands parameter with te...

conjecture | open | MathDB 页面 | 原文线索

V2-078092 | MATHDB-400198

Refined local Langlands conjecture for extended pure inner forms

摘录 / EXCERPT: Let E be a non-archimedean local field of characteristic 0, let G^* be a quasi-split connected reductive group over E, and let φ be a discrete Langlands parameter f...

conjecture | open | MathDB 页面 | 原文线索

V2-078093 | MATHDB-400199

Construction of the geometric Langlands eigensheaf for rank two

摘录 / EXCERPT: For a smooth, geometrically irreducible projective curve X/F_q and a geometrically irreducible rank-2 local system E on X, the construction of the candidate sheaf...

open_problem | open | MathDB 页面 | 原文线索

V2-078094 | MATHDB-400201

Fargues' conjecture for discrete Weil parameters and \(\mathrm{GL}_n\)

摘录 / EXCERPT: Let $\varphi:W_{\mathbb Q_p}\to \mathrm{GL}_n(\mathbb Q_\ell)$ be a discrete Weil parameter. Fargues' conjecture predicts that there exists $\mathcal F_\varphi$ on...

conjecture | open | MathDB 页面 | 原文线索

V2-078095 | MATHDB-400205

Cobham's conjecture for automatic numbers

摘录 / EXCERPT: An automatic number is either rational or transcendental.

conjecture | open | MathDB 页面 | 原文线索

V2-078096 | MATHDB-400209

Finiteness of non-CM number fields with bounded regulator

摘录 / EXCERPT: There exist only finitely many non-CM number fields with bounded regulator.

open_problem | open | MathDB 页面 | 原文线索

V2-078097 | MATHDB-400217

Northcott numbers of subrings of \(\overline{\mathbb{Q}}\)

摘录 / EXCERPT: For a subring $O\subseteq\overline{\mathbb{Q}}$, define its Northcott number by ... where H is the absolute multiplicative Weil height. Which real numbers can be realized as $N(O)$...

open_problem | open | MathDB 页面 | 原文线索

V2-078098 | MATHDB-400220

Bounding the common differences in Theorem 1

摘录 / EXCERPT: Can the conclusion of Theorem 1 be refined to use only arithmetic progressions whose common difference is bounded by some function of $\deg(u)$?

open_problem | open | MathDB 页面 | 原文线索

V2-078099 | MATHDB-400221

Uniform iterate bound in a rational-function composition problem

摘录 / EXCERPT: If $u,f\in K(X)$ satisfy $u\circ v=f^{\circ n}$ for some $n>0$ and some $v\in K(X)$, must there exist $w\in K(X)$ such that $u\circ w=f^{\circ N}$, where N is a positive integer...

open_problem | open | MathDB 页面 | 原文线索

V2-078100 | MATHDB-400222

Morton-Silverman uniform boundedness conjecture for rational functions

摘录 / EXCERPT: Let K be a number field and let $f(z)\in K(z)$ be a rational function of degree $d=\deg(f)\ge 2$. There is a bound on $\#\operatorname{Preper}(f,K)$ depending only on $\deg(f)$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-078101 | MATHDB-400223

Canonical-height lower bound in non-isotrivial families of rational functions

摘录 / EXCERPT: Let K be a number field, and fix a non-isotrivial family f_t of rational functions of generic degree $d\ge 2$ parametrized over the curve U. There exist $\varepsilon>0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078102 | MATHDB-400228

Characterization of linearly repetitive cubical cut-and-project sets

摘录 / EXCERPT: For a totally irrational, aperiodic, non-singular k-to-d cubical cut-and-project set Y formed with cubical acceptance domains and defined by linear forms...

open_problem | open | MathDB 页面 | 原文线索

V2-078103 | MATHDB-400229

Conjecture 3 (B): periodicity of every element of LBad

摘录 / EXCERPT: For every element $(w,q)\in\mathrm{LBad}$, the orbit $\{T^n(w,q)\}_{n\in\mathbb N}$ is periodic.

conjecture | open | MathDB 页面 | 原文线索

V2-078104 | MATHDB-400230

Effective Shafarevich conjecture for principally polarized abelian surfaces

摘录 / EXCERPT: For every number field k and every finite set S of primes of k, there should be an effective procedure to compute, up to isomorphism, all principally polarized abelian surfac...

conjecture | open | MathDB 页面 | 原文线索

V2-078105 | MATHDB-400231

Effective Shafarevich conjecture for Jacobians of genus-two curves

摘录 / EXCERPT: Prove an effective Shafarevich conjecture for Jacobians of genus-two curves: for every number field k and every finite set of primes S of k, effectively compute, up to isomor...

conjecture | open | MathDB 页面 | 原文线索

V2-078106 | MATHDB-400235

Discrepancy of high-dimensional permutations

摘录 / EXCERPT: Let $[n]=\{1,\ldots,n\}$. A d-dimensional permutation is a 0-1 array A indexed by $[n]^{d+1}$ such that, for every $1\le i\le d+1$ and every choice of $\alpha_j\in[n]$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-078107 | MATHDB-400236

Typical Latin squares have empty boxes of volume $O(n^2)$

摘录 / EXCERPT: Asymptotically almost every order-n Latin square A satisfies $\varepsilon(A)=O(n^2)$, where $\varepsilon(A)$ is the maximal volume of an empty box in A.

open_problem | open | MathDB 页面 | 原文线索

V2-078108 | MATHDB-400237

Asymptotic minimum size of induced-universal graphs

摘录 / EXCERPT: Let mathcal F(k) be the family of all graphs on k vertices, and let $f(k)$ be the minimum number of vertices of a graph that contains every member of $\mathcal F(k)$ as an induce...

open_problem | open | MathDB 页面 | 原文线索

V2-078109 | MATHDB-400241

Quantitative improvement for the sparse regular approximation lemma

摘录 / EXCERPT: Can one obtain a significantly better bound for $S(\operatorname{poly}(p),\delta,p)$ than $\operatorname{twr}(\operatorname{poly}(1/p))$?

open_problem | open | MathDB 页面 | 原文线索

V2-078110 | MATHDB-400242

Lower bound ruling out a substantially better SRAL bound

摘录 / EXCERPT: The proposed improvement to the sparse regular approximation lemma is impossible: there are fixed constants $\delta_0,c>0$ such that ... Moreover, the complete bipartite graph can be...

open_problem | open | MathDB 页面 | 原文线索

V2-078111 | MATHDB-400243

Coloring the complete bipartite graph with no zero cycles

摘录 / EXCERPT: Let $K_{n,n}$ denote the complete bipartite graph on $2n$ vertices, and let $\mathbb{F}_2$ be the field with two elements. Define $t(n)$ to be the minimum integer $t\ge 1$ such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-078112 | MATHDB-400249

Structure of graphs minimizing triangular edges

摘录 / EXCERPT: For integers $n\ge 3$ and $n^2/4<e\le\binom{n}{2}$, let $\operatorname{Tr}(G)$ be the number of edges of G contained in triangles, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078113 | MATHDB-400250

Erdős's conjecture on minimizing pentagonal edges

摘录 / EXCERPT: For a graph G, let $h(G,F)$ be the number of edges of G contained in subgraphs isomorphic to F, and let $h(n,e,F)=\min h(G,F)$ over graphs with $|V(G)|=n$ and $|E(G)|=e$. Erd...

conjecture | open | MathDB 页面 | 原文线索

V2-078114 | MATHDB-400252

Cameron-Erdős question for sets with no arithmetic progressions

摘录 / EXCERPT: Is it true that the number of subsets of $[n]=[U+007F]{1,2,\ldots,n}$ containing no k-term arithmetic progression is $2^{(1+o(1))r_k(n)}$?

open_problem | open | MathDB 页面 | 原文线索

V2-078115 | MATHDB-400255

Theorem on the chromatic number of random regular graphs

摘录 / EXCERPT: There is a sequence $(\varepsilon_k)_{k\ge 3}$ with $\lim_{k\to\infty}\varepsilon_k=0$ such that: (1) if $d\le (2k-1)\ln k-2\ln 2-\varepsilon_k$, then $G(n,d)$ is k-colorable wit...

open_problem | open | MathDB 页面 | 原文线索

V2-078116 | MATHDB-400256

The sharp threshold for making squares

摘录 / EXCERPT: Let $x\in\mathbb N$. Choose $a_1,a_2,\ldots$ independently and uniformly from $\{1,\ldots,x\}$, and define ... Let $\pi(y)$ be the number of primes at most y, let $\Psi(x,y)$ be th...

open_problem | open | MathDB 页面 | 原文线索

V2-078117 | MATHDB-400258

McKay-Wormald binomial approximation conjecture, part (i)

摘录 / EXCERPT: Let $G(n,m)$ be the uniform random graph with n vertices and m edges. Given m, set $p=m/\binom{n}{2}$, let $B_p(n)$ be the random sequence of n independent...

conjecture | open | MathDB 页面 | 原文线索

V2-078118 | MATHDB-400260

Successive minimum spanning tree costs converge

摘录 / EXCERPT: Let K_n be the complete graph with independent edge costs distributed either uniformly on $[0,1]$ or exponentially with parameter 1. Let T_1 be a minimum-cost spanning tree,...

open_problem | open | MathDB 页面 | 原文线索

V2-078119 | MATHDB-400261

Expected cost of successive minimum spanning trees in the Poisson multigraph model

摘录 / EXCERPT: Let K_n be the complete graph, with infinitely many copies of each edge. For each edge, assign the copies costs given by the points of an independent Poisson process of intensity...

open_problem | open | MathDB 页面 | 原文线索

V2-078120 | MATHDB-400263

Optimality of the E_8-lattice sphere packing

摘录 / EXCERPT: No packing of unit balls in Euclidean space $\mathbb{R}^8$ has density greater than that of the E_8-lattice packing.

open_problem | open | MathDB 页面 | 原文线索

V2-078121 | MATHDB-400264

Existence of the dimension-24 sphere-packing auxiliary function

摘录 / EXCERPT: There exists a radial Schwartz function $f:\mathbb{R}^{24}\to\mathbb{R}$ which satisfies: 1. $f(x)\le 0$ for $\|x\|\ge 2$; 2. $\widehat{f}(x)\ge 0$ for all $x\in\mathbb{R}^{24}$; 3...

open_problem | open | MathDB 页面 | 原文线索

V2-078122 | MATHDB-400267

Averaged Colmez conjecture for CM fields

摘录 / EXCERPT: Let E be a CM field of degree $2d$, let $F\subset E$ be its totally real subfield, and let χ be the quadratic Hecke character associated with E/F. Then ... where the sum is...

conjecture | open | MathDB 页面 | 原文线索

V2-078123 | MATHDB-400270

Existence of 2-reflective automorphic forms for the 59 lattices

摘录 / EXCERPT: Conjecture. For all 59 lattices S of Theorem 3, there exists some $m\in\mathbb{N}$ such that, for $T=U(m)\oplus S$ of signature $(n+1,2)$, there is a 2-reflective holomorphic aut...

open_problem | open | MathDB 页面 | 原文线索

V2-078124 | MATHDB-400271

Full classification of parabolically 2-reflective hyperbolic lattices

摘录 / EXCERPT: Determine the full classification of parabolically 2-reflective hyperbolic lattices S for $W=W^{(2)}(S)$, where $W^{(2)}(S)$ is generated by all reflections in vectors of square...

open_problem | open | MathDB 页面 | 原文线索

V2-078125 | MATHDB-400272

Large-genus asymptotics for volumes of strata of Abelian differentials

摘录 / EXCERPT: For strata $\Omega \mathcal{M}_g(m_1,\ldots,m_n)$, where $(m_1,\ldots,m_n)$ is a partition of $2g-2$, does one have ... as $\sum_i m_i=2g-2$ tends to infinity?

open_problem | open | MathDB 页面 | 原文线索

V2-078126 | MATHDB-400273

Eskin-Zorich large-genus volume conjecture for the principal stratum

摘录 / EXCERPT: For the principal stratum, with all parts equal to one, the Eskin-Zorich asymptotic predicts ...

conjecture | open | MathDB 页面 | 原文线索

V2-078127 | MATHDB-400274

Regularization of the diagonal Petersson inner products

摘录 / EXCERPT: Let f_m be the unique modular function for $SL_2(\mathbb Z)$ with Fourier expansion $f_m(z)=q^{-m}+\sum_{n\ge 0}c_m(n)q^n$ and $c_m(0)=24\sum_{d\mid m}d$. Duke, Imamoğlu, and Tót...

open_problem | open | MathDB 页面 | 原文线索

V2-078128 | MATHDB-400275

Extension of the intermediate Jacobian map to the wonderful compactification

摘录 / EXCERPT: Over $\mathbb{C}$, let $\mathcal{M}$ be the moduli space of smooth cubic threefolds, and let $\widetilde{\mathcal{M}}$ be the wonderful compactification described in the report. Th...

open_problem | open | MathDB 页面 | 原文线索

V2-078129 | MATHDB-400276

The possible degree-10 exception for the Satake compactification of \(\mathcal A_4\)

摘录 / EXCERPT: Determine whether the possible exception in the report’s Theorem 7 occurs; namely, decide whether ... where R_4 is the tautological ring generated by the Chern classes of the Hodge...

open_problem | open | MathDB 页面 | 原文线索

V2-078130 | MATHDB-400277

Factorial decay estimate for iterated integrals of branched rough paths

摘录 / EXCERPT: There exists a constant $c_\gamma>0$ such that, for every γ-Hölder branched rough path X and every rooted tree τ, the associated tree-indexed iterated integral satisf...

open_problem | open | MathDB 页面 | 原文线索

V2-078131 | MATHDB-400278

Well-posedness and continuity of the nonlinear rough McKean-Vlasov equation

摘录 / EXCERPT: Let $p\ge 1$. Suppose that $x\mapsto b(x,P)$ belongs to $C_b^{3+[p]}(\mathbb{R}^d,\mathbb{R}^d)$ uniformly in $P\in M_1(\mathbb{R}^d)$, that $P\mapsto b(x,P)$ is Lipschitz continuo...

open_problem | open | MathDB 页面 | 原文线索

V2-078132 | MATHDB-400281

KPZ fluctuation limit for weakly asymmetric bridges

摘录 / EXCERPT: For the rescaled height function ... where $\lambda_N\sim 2\sigma^2(2N)^{2\alpha}$, Theorem 3 states that, for $\alpha\leq 1/3$, h_N converges in distribution as $N\to\infty$ to th...

open_problem | open | MathDB 页面 | 原文线索

V2-078133 | MATHDB-400282

Almost-sure global well-posedness for quintic NLS on the two-dimensional torus

摘录 / EXCERPT: Prove the almost-sure global well-posedness of the quintic nonlinear Schrödinger equation on $\mathbb{T}^2$, together with invariance of the associated Gibbs measure.

open_problem | open | MathDB 页面 | 原文线索

V2-078134 | MATHDB-400283

Conjectural 576-fold periodicity for free-fermion chiral CFTs

摘录 / EXCERPT: The lower-right entry in the conjectural analogy is the Morita equivalence $\mathrm{Fer}(576)\simeq_{\mathbb R}\mathbb R$.

open_problem | open | MathDB 页面 | 原文线索

V2-078135 | MATHDB-400285

Main theorem of perturbative renormalization

摘录 / EXCERPT: The report states that, if S and S' are two possible definitions of the formal S-matrix, with their time-ordered products satisfying the appropriate axioms (including causal...

open_problem | open | MathDB 页面 | 原文线索

V2-078136 | MATHDB-400286

Emergence of planar-type Pfaffian relations for non-planar two-dimensional Ising models

摘录 / EXCERPT: For extensions of the planar Ising model with non-planar interactions-such as two-dimensional models with pair interactions reaching beyond the range of nearest-neighbour interacti...

open_problem | open | MathDB 页面 | 原文线索

V2-078137 | MATHDB-400287

Stability of singular type-II blow-up solutions under perturbations along the center-stable hypersurface

摘录 / EXCERPT: The preceding theorem suggests the question whether perturbations of the singular type-II solutions constructed in [21, 19], taken along the codimension-one hypersurface Σ,...

open_problem | open | MathDB 页面 | 原文线索

V2-078138 | MATHDB-400289

Global well-posedness and scattering for the energy-critical Maxwell-Klein-Gordon equation

摘录 / EXCERPT: For the massless Maxwell-Klein-Gordon system on $\mathbb{R}^{1+4}$, in the Coulomb gauge $\sum_{j=1}^{4}\partial_jA_j=0$, the initial value problem is globally well-posed for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078139 | MATHDB-400290

Effective dynamics after the Euclidean approximation breaks

摘录 / EXCERPT: What happens after the Euclidean approximation breaks? Can we still describe the effective dynamics for very large L?

open_problem | open | MathDB 页面 | 原文线索

V2-078140 | MATHDB-400291

Effective dynamics after the Euclidean approximation for large-domain NLS

摘录 / EXCERPT: For the nonlinear Schrödinger equation on the periodic box mathbb{T}L^nmath, can the effective dynamics after the Euclidean approximation breaks down still be described precisely f...

open_problem | open | MathDB 页面 | 原文线索

V2-078141 | MATHDB-400296

Local well-posedness for 1D cubic NLS below L^2

摘录 / EXCERPT: For the one-dimensional cubic focusing or defocusing nonlinear Schrödinger equation on the real line, $i u_t+u_{xx}\\pm 2u|u|^2=0$ with $u(0)=u_0$, is the equation locally well-pos...

open_problem | open | MathDB 页面 | 原文线索

V2-078142 | MATHDB-400297

Classification of traveling solitary waves for the energy-critical half-wave maps equation

摘录 / EXCERPT: For the half-wave maps equation ... consider traveling-wave solutions $u(t,x)=u_v(x-vt)$, where $v\in\mathbb R$ and $u_v:\mathbb R\to S^2$ satisfies ... All profiles...

open_problem | open | MathDB 页面 | 原文线索

V2-078143 | MATHDB-400299

Uniqueness under the Struwe monotonicity formula for the corotational harmonic map heat flow

摘录 / EXCERPT: Does the monotonicity formula suffice to guarantee uniqueness?

open_problem | open | MathDB 页面 | 原文线索

V2-078144 | MATHDB-400303

Polynomial asymptotics of Ricci-flat metrics near nodal Calabi-Yau singularities

摘录 / EXCERPT: Let $X_{\tilde f}\subset\mathbb{P}^{n+1}$ be a hypersurface of degree $n+2$ with at worst nodal singularities $x_1,\ldots,x_N$, where each node is locally analytically isomorphic t...

open_problem | open | MathDB 页面 | 原文线索

V2-078145 | MATHDB-400304

Existence of homogeneous Ricci solitons on a given homogeneous space

摘录 / EXCERPT: Determine whether a given homogeneous space admits a homogeneous Ricci soliton.

open_problem | open | MathDB 页面 | 原文线索

V2-078146 | MATHDB-400308

Contractibility of the moduli space of 2-convex embedded spheres

摘录 / EXCERPT: Conjecture. The moduli space of 2-convex embedded n-spheres in $\d53f^{n+1}$ is contractible for every dimension n, i.e. ...

open_problem | open | MathDB 页面 | 原文线索

V2-078147 | MATHDB-400311

Prescribing the volume form of a balanced metric

摘录 / EXCERPT: Let $(M,J, ...

open_problem | open | MathDB 页面 | 原文线索

V2-078148 | MATHDB-400313

Sharp fundamental gap estimate for strictly convex domains in the sphere

摘录 / EXCERPT: Let $\Omega\subset S^n$ be a strictly convex domain with diameter D, and let $\lambda_1<\lambda_2$ be the first two Dirichlet eigenvalues of the Laplacian on Ω. If...

open_problem | open | MathDB 页面 | 原文线索

V2-078149 | MATHDB-400314

Fundamental gap conjecture for convex Euclidean domains

摘录 / EXCERPT: For a convex domain $\Omega \subset \mathbb{R}^n$ with diameter D, if $\lambda_1<\lambda_2$ are the first two Dirichlet eigenvalues of the Laplacian on Ω, then ...

conjecture | claimed_solved | MathDB 页面 | 原文线索

V2-078150 | MATHDB-400315

Index estimate for free-boundary minimal hypersurfaces in strictly mean-convex domains

摘录 / EXCERPT: Let Ω^{n+1} be a smooth, compact, strictly mean-convex domain in Euclidean $(n+1)$-space, with $n\ge 2$, and let M^n be a compact, orientable, properly embedded free-bound...

open_problem | open | MathDB 页面 | 原文线索

V2-078151 | MATHDB-400316

Rationality and the K3 category of cubic and Gushel-Mukai fourfolds

摘录 / EXCERPT: For both cubic and Gushel-Mukai fourfolds, rationality should imply that the mysterious category in the canonical semiorthogonal decomposition is the derived category of an actual...

open_problem | open | MathDB 页面 | 原文线索

V2-078152 | MATHDB-400317

Optimality of Pfister’s bound for sums of squares in real rational function fields

摘录 / EXCERPT: For each integer $n\ge 3$, is Pfister’s bound 2^n optimal for nonnegative polynomials in $\mathbb{R}[X_1,\ldots,X_n]$? Equivalently, does there exist a polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-078153 | MATHDB-400318

Predicted variation of models for quartic K3 surfaces

摘录 / EXCERPT: Prediction 1. Let F be the period space for quartic K3 surfaces, let $ ... The variation of models interpolating between the Baily-Borel compactification at $beta=0...

open_problem | open | MathDB 页面 | 原文线索

V2-078154 | MATHDB-400319

Prediction for the birational models of the moduli space of quartic K3 surfaces

摘录 / EXCERPT: Let $\mathscr F$ be the period space for quartic K3 surfaces, let λ be the Hodge bundle, and let Δ be the relevant Heegner divisor. For $\beta\in[0,1]\cap\mathbb Q$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078155 | MATHDB-400320

Degree of irrationality of very general high-degree hypersurfaces

摘录 / EXCERPT: For every dimension n and degree $d\ge 2n+1$, if $X=X_d\subset \mathbf P^{n+1}$ is a very general smooth complex hypersurface of dimension n, then $\operatorname{irr}(X)=d-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-078156 | MATHDB-400321

Rationality of a generic cubic fourfold

摘录 / EXCERPT: Determine whether a generic cubic fourfold $X\subset \mathbb P^5_{\mathbb C}$ is rational. The report states that a generic member of this 20-dimensional family is expected to be n...

open_problem | open | MathDB 页面 | 原文线索

V2-078157 | MATHDB-400322

Dimension conjecture for rational curves on very general hypersurfaces

摘录 / EXCERPT: Let $X\subset\mathbb P^n$ be a very general hypersurface of degree d, and let $R_e(X)$ be the space of degree-e integral rational curves on X. Conjecture 1 asserts that ... for...

conjecture | open | MathDB 页面 | 原文线索

V2-078158 | MATHDB-400326

Hodge-isometry criterion for the K3 category of a cubic fourfold

摘录 / EXCERPT: For a smooth cubic fourfold $X\subset\mathbf P^5$, there exists a Hodge isometry $\widetilde H(\mathcal A_X,\mathbf Z)\cong\widetilde H(S,\alpha,\mathbf Z)$ for some twisted K3 sur...

open_problem | open | MathDB 页面 | 原文线索

V2-078159 | MATHDB-400328

Finiteness of smooth Fano threefolds of geometric Picard rank one over an arithmetic base

摘录 / EXCERPT: Let K be a number field, let S be a finite set of finite places of K, and let $B=\operatorname{Spec}(\mathcal O_K[S^{-1}])$. Is the set of B-isomorphism classes of smooth F...

open_problem | open | MathDB 页面 | 原文线索

V2-078160 | MATHDB-400329

Arithmetic Torelli theorem for smooth cubic threefolds

摘录 / EXCERPT: Let k be a field of characteristic zero. Let X and Y be smooth cubic threefolds over k. If $\operatorname{Jac}(X)$ and $\operatorname{Jac}(Y)$ are k-isomorphic as princip...

open_problem | open | MathDB 页面 | 原文线索

V2-078161 | MATHDB-400331

Existence of a field with neither the Siegel nor the Northcott property

摘录 / EXCERPT: Does there exist a subfield $K\subset\overline{\mathbb{Q}}$ having neither the Siegel property (i.e. being a Siegel field) nor the Northcott property?

open_problem | open | MathDB 页面 | 原文线索

V2-078162 | MATHDB-400334

Small maximal totally isotropic subspaces in adelic quadratic spaces

摘录 / EXCERPT: Assume that q is isotropic. Then, for all $ ...

open_problem | open | MathDB 页面 | 原文线索

V2-078163 | MATHDB-400335

Decomposition of hyperspecial double cosets into single cosets

摘录 / EXCERPT: Determine a decomposition of the K_p-double cosets into single cosets when G is split over $ \mathbb{Q}_p$ and K_p is a hyperspecial maximal compact subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-078164 | MATHDB-400336

Minimal norms and determinants of lattices from finite Abelian groups

摘录 / EXCERPT: Given a finite Abelian group G and a subset $S=\{g_0:=0,g_1,\ldots,g_{n-1}\}\subseteq G$, define ... Determine the minimal norms and determinants of the lattices $L_G(S)$, for gene...

open_problem | open | MathDB 页面 | 原文线索

V2-078165 | MATHDB-400337

Is \b3_{n,\a} an isomorphism for all n?

摘录 / EXCERPT: For which n is the natural morphism ... an isomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-078166 | MATHDB-400338

Universal optimality of the root lattice D_4

摘录 / EXCERPT: Is the root lattice D_4 universally optimal among periodic point configurations in dimension 4, i.e. does it minimize energy for every completely monotonic potential function?

open_problem | open | MathDB 页面 | 原文线索

V2-078167 | MATHDB-400339

Universal optimality of the nine-dimensional periodic set D_9^+

摘录 / EXCERPT: The report presents as an open conjectural problem whether the 2-periodic set $D_9^+=D_9\cup\bigl((\tfrac12,\ldots,\tfrac12)+D_9\bigr)$ is universally optimal in dimension 9, mea...

open_problem | open | MathDB 页面 | 原文线索

V2-078168 | MATHDB-400343

Concentration of entropy dissipation for bounded entropy solutions of scalar conservation laws

摘录 / EXCERPT: If u is a bounded entropy solution of a scalar conservation law, then its entropy dissipation is concentrated on an H^1-rectifiable set. The report states that no assumptions o...

open_problem | open | MathDB 页面 | 原文线索

V2-078169 | MATHDB-400344

Uniqueness and continuous dependence for global Burgers-Hilbert solutions

摘录 / EXCERPT: For the Burgers-Hilbert equation ... with initial data in $L^2(\mathbb R)$, are the global solutions whose existence was established in the report unique and continuously dependent o...

open_problem | open | MathDB 页面 | 原文线索

V2-078170 | MATHDB-400346

Nonlocal-to-local convergence conjecture for Burgers-type conservation laws

摘录 / EXCERPT: The conjecture discussed in the report was that, as $\varepsilon\downarrow0$, the unique solution u_ε of ... converges to the unique entropy solution u of Burgers' equa...

conjecture | open | MathDB 页面 | 原文线索

V2-078171 | MATHDB-400347

Uniqueness of the 1D compressible-to-incompressible limit

摘录 / EXCERPT: For every initial datum $u_o=(v_{\ell,o},(p_o,v_o))\in X$, the compressible-to-incompressible limit $u^*(t)=(v_{\ell}(t),(p^*,v^*)(t,\cdot))$ satisfies ... Consequently, u^* coinci...

open_problem | open | MathDB 页面 | 原文线索

V2-078172 | MATHDB-400349

Riemann solver at a junction for a phase-transition traffic model

摘录 / EXCERPT: Consider the phase-transition traffic model ... with ... on a junction with n incoming arcs and m outgoing arcs, where the initial data belong to $F\cup C$. Under (H-1)...

open_problem | open | MathDB 页面 | 原文线索

V2-078173 | MATHDB-400350

Stability with respect to a space-dependent flux in a one-dimensional IBVP

摘录 / EXCERPT: Can one prove stability of the entropy solution of the one-dimensional initial-boundary value problem for a scalar conservation law with respect to the flux when the flux depends e...

open_problem | open | MathDB 页面 | 原文线索

V2-078174 | MATHDB-400352

Stability threshold for the rescaled shear-flow equilibrium

摘录 / EXCERPT: For the rescaled system ... determine the stability of the equilibrium $(\bar U,\bar\Gamma)=(1,1)$. The heuristic prediction is instability for $n<1$ and stability for $n>1$; the rep...

open_problem | open | MathDB 页面 | 原文线索

V2-078175 | MATHDB-400357

Noncommutative Voevodsky nilpotence conjecture

摘录 / EXCERPT: Let k be a base field of characteristic zero and let $\mathcal{A}$ be a smooth proper dg category over k. Conjecture $V_{\mathrm{nc}}(\mathcal{A})$ asserts that ...

conjecture | open | MathDB 页面 | 原文线索

V2-078176 | MATHDB-400359

Theorem 4: standard and nilpotence conjectures for bilinear-divisor intersections

摘录 / EXCERPT: Let k be a field of characteristic zero, let W be a finite-dimensional k-vector space, and let $X=(\mathbb{P}(W)\times\mathbb{P}(W))/\mu_2$ be the associated smooth proper De...

conjecture | open | MathDB 页面 | 原文线索

V2-078177 | MATHDB-400378

Asymptotic identification of the bias term in density level-set estimation

摘录 / EXCERPT: Under the usual smoothness conditions for the decomposition of the mean distance in measure, and without imposing the additional bandwidth condition $nh^{d+4}\to 0$, conjecture tha...

open_problem | open | MathDB 页面 | 原文线索

V2-078178 | MATHDB-400380

Gaussian limit for a smoothed Grenander-type estimator in the Cox model

摘录 / EXCERPT: The report asks whether the smoothed monotone hazard estimators in the Cox model have a Gaussian limit at the usual rate $n^{2/5}$. Its stated theorem answers this for the smoothed...

open_problem | open | MathDB 页面 | 原文线索

V2-078179 | MATHDB-400382

Optimal construction of a shorter exponential-sum approximation

摘录 / EXCERPT: Let $f=(f_k)_{k=0}^{\infty}$ be given by $f_k=\sum_{j=1}^{N}a_jz_j^k$, where $a_j\in\mathbb C\setminus\{0\}$ and $z_j\in\mathcal D:=\{z\in\mathbb C:0<|z|<1\}$. Given...

open_problem | open | MathDB 页面 | 原文线索

V2-078180 | MATHDB-400383

Sparse approximation of exponential sums via AAK theory

摘录 / EXCERPT: Given a sequence $f=(f_k)_{k=0}^{\cinfty}$ of the form ... where $a_j\cin\bb C\csetminus\{0\}$ and $z_j\cin\bb D:=\{z\cin\bb C:0<|z|<1\}$, and given sufficiently many samples...

open_problem | open | MathDB 页面 | 原文线索

V2-078181 | MATHDB-400384

Numerical computation of the singular pairs of the Hankel operator

摘录 / EXCERPT: How can the singular pairs $(\sigma_n,u^{(n)})$ of Γ_f be computed numerically for $n=0,\ldots,N-1$?

open_problem | open | MathDB 页面 | 原文线索

V2-078182 | MATHDB-400385

Exploiting non-identically distributed data to learn multiple structural causal models

摘录 / EXCERPT: Suppose that we are given an overall set of data points $(x_1,\ldots,x_n)$, where each x_i could for instance be a pair of inputs and a target. In addition, suppose we have a sim...

open_problem | open | MathDB 页面 | 原文线索

V2-078183 | MATHDB-400386

Sharp minimax rates for structured matrix estimation and matrix completion

摘录 / EXCERPT: For the structured matrix-estimation problem in model (3), over the class Θ defined in (2), determine whether the lower-bound rate in Theorem 1 is the minimax rate of conver...

open_problem | open | MathDB 页面 | 原文线索

V2-078184 | MATHDB-400388

Adaptive coding over a collection of source classes

摘录 / EXCERPT: Let $(C^{(\alpha)})_{\alpha\in A}$ be a collection of source classes, with each subclass having nontrivial minimax redundancy and an available good universal coder. Does there exis...

open_problem | open | MathDB 页面 | 原文线索

V2-078185 | MATHDB-400389

Complete characterization of orthogonal multi-wavelet masks

摘录 / EXCERPT: For integers $n\le m$, determine all finite masks $p=\{p_k\}_{k=0}^{s}$ and $q=\{q_k\}_{k=0}^{s}$ with $p_k,q_k\in\mathbb{C}^{m\times m}$ for which there exist square-integrable fu...

open_problem | open | MathDB 页面 | 原文线索

V2-078186 | MATHDB-400390

Existence of a critical density in Bergman spaces and for wavelet frames

摘录 / EXCERPT: For density theorems in Bergman spaces and for the density of wavelet frames, does a critical density exist that separates sets of sampling from sets of interpolation (or the corre...

open_problem | open | MathDB 页面 | 原文线索

V2-078187 | MATHDB-400391

Sharp threshold for instantaneous complete blow-up in the supercritical Lane-Emden heat flow

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a smoothly bounded domain with $n\ge 3$, and let $p>2^*=2n/(n-2)$. Consider the Lane-Emden heat flow ... Writing $\alpha=2/(p-2)$, Theorem 3 assert...

open_problem | open | MathDB 页面 | 原文线索

V2-078188 | MATHDB-400392

Limit of Ricci flows with surgery as the surgery scale tends to zero

摘录 / EXCERPT: Let $\delta_j:[0,\infty)\to(0,\infty)$ be a sequence of decreasing functions with $\delta_j\leq\delta$ and $\lim_{j\to\infty}\delta_j(0)=0$. For the ensuing Ricci flows with surger...

open_problem | open | MathDB 页面 | 原文线索

V2-078189 | MATHDB-400395

Sharp quantitative Euclidean concentration inequality

摘录 / EXCERPT: Let $E\subset\mathbb{R}^n$, let E^* be a ball with the same volume as E, and let $N_r(E)=\{x\in\mathbb{R}^n:\operatorname{dist}(x,E)<r\}$. The report states that there exists...

open_problem | open | MathDB 页面 | 原文线索

V2-078190 | MATHDB-400397

Exponential lower bound for minimal disks spanning Γ_n

摘录 / EXCERPT: Conjecture: The number of embedded minimal disks bounded by the curve Γ_n grows at least like 2^n.

open_problem | open | MathDB 页面 | 原文线索

V2-078191 | MATHDB-400400

Polar of an \A-free measure lies in the wave cone

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be open, let $\mathcal{A}=\sum_{|\alpha|\le k}A_\alpha\partial^\alpha$ be a kth-order linear constant-coefficient differential operator with...

open_problem | open | MathDB 页面 | 原文线索

V2-078192 | MATHDB-400401

Rank-one structure of the singular symmetric derivative in BD

摘录 / EXCERPT: Let $u\\in\\mathrm{BD}(\\mathbb{R}^d)$, and let $Eu:=Du+(Du)^T$ be its symmetric distributional derivative, with $E^s u$ denoting the part of Eu singular with respect to Lebesgue...

open_problem | open | MathDB 页面 | 原文线索

V2-078193 | MATHDB-400402

Weak converse of Rademacher’s theorem

摘录 / EXCERPT: Let $\nu\in\mathcal{M}_+(\mathbb{R}^d)$ be a positive Radon measure such that every Lipschitz function $f\in\operatorname{Lip}(\mathbb{R}^d,\mathbb{R}^\ell)$ is differentiable...

open_problem | open | MathDB 页面 | 原文线索

V2-078194 | MATHDB-400403

Euclidean mixed isoperimetric-isodiametric inequality

摘录 / EXCERPT: For every bounded open set $\Omega\subset\mathbb{R}^n$ with smooth boundary,nn ... nnwhere $\operatorname{rad}(\Omega)$ is the radius of the smallest Euclidean ball containing...

open_problem | open | MathDB 页面 | 原文线索

V2-078195 | MATHDB-400405

Dimension bound for the branch set of a minimal two-valued graph

摘录 / EXCERPT: Conjecture. For a minimal two-valued graph, if B denotes the set of points at which there exists a multiplicity-2 tangent plane, then $\dim_{\mathcal H}(B)\le n-2$.

open_problem | open | MathDB 页面 | 原文线索

V2-078196 | MATHDB-400406

Stability for Almgren’s isoperimetric principle

摘录 / EXCERPT: If $\Omega\subset\mathbb{R}^{n+1}$ is a bounded, open subset with smooth boundary such that $H_{\Omega}\le n$ on $\partial\Omega$, and ... is small, must Ω be close to a unit...

open_problem | open | MathDB 页面 | 原文线索

V2-078197 | MATHDB-400407

Sharp local smoothing estimate for the logarithmic fast diffusion equation

摘录 / EXCERPT: Suppose $u:B\times[0,T)\to(0,\infty)$ is a smooth solution of $\partial_tu=\Delta\log u$ on the unit ball $B\subset\mathbb{R}^2$, and suppose that $u_0:=u(0)\in L^1(B)$. Then, for...

open_problem | open | MathDB 页面 | 原文线索

V2-078198 | MATHDB-400408

Removing the positivity hypothesis in the vectorial free-boundary regularity theorem

摘录 / EXCERPT: Let $k\in\mathbb{N}$ and consider the minimization problem ... Theorem 3 asserts that every solution $U=(u_1,\ldots,u_k)$ has all components Lipschitz continuous in $\mathbb{R}^d$, a...

conjecture | open | MathDB 页面 | 原文线索

V2-078199 | MATHDB-400411

Completion conjecture in real L-theory after inverting 2

摘录 / EXCERPT: For every countable discrete group G, the completion-induced map ... is an isomorphism.

conjecture | open | MathDB 页面 | 原文线索

V2-078200 | MATHDB-400415

Correction terms from spectral-sequence differentials

摘录 / EXCERPT: Extract the “correction terms” from the differentials, and use these to unify and refine the coincidences of Theorem 1.

open_problem | open | MathDB 页面 | 原文线索

V2-078201 | MATHDB-400416

Interpretation of the Eisenbud-Levine-Khimshiashvili quadratic form as a degree

摘录 / EXCERPT: Let k be an arbitrary field of characteristic 0, and let $f:k^n\to k^n$ be a finite polynomial map. Set ... let $J=\det(\partial f_i/\partial x_j)$, and choose a k-linear map...

open_problem | open | MathDB 页面 | 原文线索

V2-078202 | MATHDB-400419

Function-field version of Malle’s conjecture for refined ramification types

摘录 / EXCERPT: Fix a finite field (mathbb Fq), an integer n, a transitive subgroup $G\leq S_n$, and a conjugation-invariant subset $c\subseteq G$. Let $N^c_{G,q}(X)$ count geometrically connect...

conjecture | open | MathDB 页面 | 原文线索

V2-078203 | MATHDB-400421

Interpretation of relative tensor products in full conformal field theory

摘录 / EXCERPT: What is the meaning, in the context of full conformal field theory, of the relative tensor product construction of two local Lagrangian Q-systems over a modular tensor category, an...

open_problem | open | MathDB 页面 | 原文线索

V2-078204 | MATHDB-400423

Can holographic renormalization cure the triviality of the Euclidean AdS/CFT boundary theory?

摘录 / EXCERPT: Whether the observed triviality can be cured by procedures such as holographic renormalization.

open_problem | open | MathDB 页面 | 原文线索

V2-078205 | MATHDB-400428

Exact critical coupling in the self-dual noncommutative ϕ^4_4 model

摘录 / EXCERPT: For the self-dual noncommutative ϕ^4_4 model described in the report, the phase-transition critical coupling constant is exactly $\lambda_c=-\frac{1}{\pi}$.$

open_problem | open | MathDB 页面 | 原文线索

V2-078206 | MATHDB-400429

Constructive recognition of exceptional groups of Lie type

摘录 / EXCERPT: Suppose $G=\langle X\rangle\leq GL_n(F)$ is quasisimple, with $G/Z(G)\cong S$, where $S=S(q)$ is an exceptional group of Lie type over $\mathbb{F}_q$. If S is not of type...

open_problem | open | MathDB 页面 | 原文线索

V2-078207 | MATHDB-400430

Growth of branches in coclass trees

摘录 / EXCERPT: It remains to describe the growth (in depth and width) of the branches in coclass trees for which $T\ne T(k)$ for every k; these occur for values of p and r other than $p=2$...

open_problem | open | MathDB 页面 | 原文线索

V2-078208 | MATHDB-400432

Nonexistence of a subgroup isomorphic to \(\mathrm{Sz}(8)\) in the Monster

摘录 / EXCERPT: The Suzuki group $\mathrm{Sz}(8)$ is not a subgroup of the Monster sporadic simple group M.

open_problem | open | MathDB 页面 | 原文线索

V2-078209 | MATHDB-400440

Rosengarten’s Tamagawa-number conjecture for smooth connected affine groups

摘录 / EXCERPT: Conjecture 1 (Rosengarten). Let k be a global function field and let G be a smooth connected affine k-group. Then ... Here $\operatorname{Pic}(G)_{\mathrm{prim}}$ consists of t...

conjecture | open | MathDB 页面 | 原文线索

V2-078210 | MATHDB-400441

Canonical admissible metrizations over global fields

摘录 / EXCERPT: Let K be a global field, including a function field $K=k(C)$ of a projective curve over another field k, and let X be equipped with a fixed positive adelic metrized line bund...

open_problem | open | MathDB 页面 | 原文线索

V2-078211 | MATHDB-400451

Isotopic approximation within a tolerance volume

摘录 / EXCERPT: Input: A tolerance volume Ω, defined as a compact subset of $\mathbb{R}^3$ homeomorphic to $S\times[0,1]$, where S is the surface to be approximated. Goal: Generate a surf...

open_problem | open | MathDB 页面 | 原文线索

V2-078212 | MATHDB-400452

Incompressibility of origami cylinders without vertical fold lines

摘录 / EXCERPT: Fold patterns for origami cylinders that do not contain vertical fold lines are incompressible: they cannot undergo vertical compression by a purely isometric deformation. Conseque...

open_problem | open | MathDB 页面 | 原文线索

V2-078213 | MATHDB-400454

C*-simplicity of Thompson’s group T

摘录 / EXCERPT: Is Thompson’s group T C*-simple? Equivalently, is Thompson’s group F non-amenable?

open_problem | open | MathDB 页面 | 原文线索

V2-078214 | MATHDB-400459

Type I conjecture for groups acting transitively on the boundary of a locally finite tree

摘录 / EXCERPT: Let T be a locally finite tree and let $G\leq \operatorname{Aut}(T)$ be a closed subgroup acting transitively on the boundary ∂T. Then G is a type I group.

conjecture | open | MathDB 页面 | 原文线索

V2-078215 | MATHDB-400464

Redundancy of approximate representability in the equivariant Kirchberg-Phillips theorem

摘录 / EXCERPT: In the setting of Theorem 4, let G be a discrete amenable group and let $\beta\colon G\curvearrowright \mathcal O_\infty$ be an outer action. Is the hypothesis that β is ap...

open_problem | open | MathDB 页面 | 原文线索

V2-078216 | MATHDB-400465

Phillips-Toms conjecture for crossed products of free minimal amenable actions

摘录 / EXCERPT: Let Γ be an infinite countable amenable group acting freely and minimally on a compact metric space X. Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-078217 | MATHDB-400466

Bipolar theorem for sets of operators

摘录 / EXCERPT: Fix $p\in[1,\infty)$, let Opp be the class of bounded linear operators between subspaces of L_p spaces, and let $B\subset\operatorname{Opp}$. The bipolar...

open_problem | open | MathDB 页面 | 原文线索

V2-078218 | MATHDB-400470

Baum-Connes conjecture for actions with finite dynamic asymptotic dimension

摘录 / EXCERPT: Let name{Gamma} be a finitely generated group acting on a compact space name{X}. If the action has finite dynamic asymptotic dimension, then the Baum-Connes conjecture holds for na...

conjecture | open | MathDB 页面 | 原文线索

V2-078219 | MATHDB-400471

Popa’s characterization question for finite-type Polish groups

摘录 / EXCERPT: Does the class $\mathscr U_{\mathrm{SIN}}$ of all unitarily representable SIN Polish groups coincide with Popa’s class $\mathscr U_{\mathrm{fin}}$ of finite-type Polish groups? Her...

open_problem | open | MathDB 页面 | 原文线索

V2-078220 | MATHDB-400474

Contractibility of the exotic group A(q)

摘录 / EXCERPT: Let [?25l$\mathbb F_q$ be a finite field with $q=p^h$, and let $A(q)$ be the metric completion, with respect to the normalized rank metric, of the inductive limit of...

open_problem | open | MathDB 页面 | 原文线索

V2-078221 | MATHDB-400475

Closure of supramenable groups under direct products

摘录 / EXCERPT: Given supramenable groups G and H, is $G \times H$ necessarily supramenable, even when one of the groups is $\mathbb{Z}$?

open_problem | open | MathDB 页面 | 原文线索

V2-078222 | MATHDB-400480

Generator-growth versus covolume for lattices

摘录 / EXCERPT: Conjecture. Let G be a higher-rank simple Lie group, and let (M_i) be any sequence of lattices in G with unbounded covolume. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-078223 | MATHDB-400484

Measurable equidecomposability of a regular tetrahedron and a cube

摘录 / EXCERPT: Is a regular tetrahedron in $\mathbb{R}^3$ measurably equidecomposable to a cube of the same volume?

open_problem | open | MathDB 页面 | 原文线索

V2-078224 | MATHDB-400485

Benjamini-Schramm convergence of congruence arithmetic lattices

摘录 / EXCERPT: Conjecture 1. Let G be a semisimple real Lie group and let (Γ_i) be a sequence of pairwise nonconjugate congruence arithmetic lattices in G. Then the sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-078225 | MATHDB-400486

Tilability of Følner sequences in amenable groups

摘录 / EXCERPT: Does every amenable group admit a Følner sequence all of whose elements can tile the group?

open_problem | open | MathDB 页面 | 原文线索

V2-078226 | MATHDB-400489

Unique trace property for discrete groups

摘录 / EXCERPT: A discrete group has the unique trace property if and only if its amenable radical is trivial.

open_problem | open | MathDB 页面 | 原文线索

V2-078227 | MATHDB-400490

Characterization via the Markov operator of sequences close to unions of expanders

摘录 / EXCERPT: Let $\{G_n\}_{n=1}^{\infty}$ be a sequence of d-regular graphs. Then the following are equivalent: 1. There exists an $\varepsilon>0$ such that, for every $\delta>0$, for all but...

open_problem | open | MathDB 页面 | 原文线索

V2-078228 | MATHDB-400491

The strong Atiyah conjecture with coefficients in K

摘录 / EXCERPT: Let F be a finitely generated free group, freely generated by a finite set S, let $N\trianglelefteq F$, and put $G=F/N$. Let K be a subfield of $\mathbb{C}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-078229 | MATHDB-400492

Finite-index inheritance of the strong Atiyah conjecture

摘录 / EXCERPT: If a subgroup H of finite index in a group G satisfies the strong Atiyah conjecture, must G also satisfy the strong Atiyah conjecture?

conjecture | open | MathDB 页面 | 原文线索

V2-078230 | MATHDB-400493

Lück approximation conjecture with coefficients in K

摘录 / EXCERPT: Let F be a finitely generated free group, let K be a subfield of $\a0\mathbb{C}$, and let $F>N_1>N_2>\cdots$ be a chain of normal subgroups of F with $N=\bigcap_i N_i$. Put...

conjecture | open | MathDB 页面 | 原文线索

V2-078231 | MATHDB-400494

Critical and uniqueness thresholds versus nonamenability

摘录 / EXCERPT: For a Cayley graph G, $p_c(G)<p_u(G)$ if and only if G is nonamenable.

open_problem | open | MathDB 页面 | 原文线索

V2-078232 | MATHDB-400500

Uniform spectral gap for the hexagonal-lattice AKLT model

摘录 / EXCERPT: Prove a positive lower bound for the spectral gap above the ground state of the two-dimensional spin-$3/2$ AKLT model on the hexagonal lattice, with nearest-neighbor interaction...

open_problem | open | MathDB 页面 | 原文线索

V2-078233 | MATHDB-400504

Ionization bound in Thomas-Fermi-Dirac-von Weizsäcker theory

摘录 / EXCERPT: Fix positive constants $c_{\mathrm{TF}},c_{\mathrm{W}},c_{\mathrm{D}}$. Consider the TFDW functional ... under the constraint...

open_problem | open | MathDB 页面 | 原文线索

V2-078234 | MATHDB-400505

Ionization conjecture in full many-body Schrödinger theory

摘录 / EXCERPT: In the full many-body Schrödinger theory, prove that a nucleus of charge Z can bind at most $Z+C$ electrons, where C is a universal constant.

conjecture | open | MathDB 页面 | 原文线索

V2-078235 | MATHDB-400506

Large-mass minimizers as characteristic functions of balls

摘录 / EXCERPT: For $\alpha>0$, define ... and ... Does there exist an $m'_{c_2}(\alpha)<\infty$ such that, for every $m>m'_{c_2}(\alpha)$, any minimizer of $E_{\alpha}(m)$ is the characteristic funct...

open_problem | open | MathDB 页面 | 原文线索

V2-078236 | MATHDB-400507

Convergence and rate optimality of adaptive finite element methods for harmonic forms

摘录 / EXCERPT: For admissible mixed adaptive finite element methods based on de Rham-complex-conforming finite-element complexes in all space dimensions, the report claims convergence and rate op...

open_problem | open | MathDB 页面 | 原文线索

V2-078237 | MATHDB-400508

Adaptive truncation parameters for the Kačanov iteration of the p-Poisson problem

摘录 / EXCERPT: For the sublinear case $1<p\le 2$, starting from $v_0\in W^{1,p}_0(\Omega)$, how should the truncation intervals $\varepsilon_n=(\varepsilon_{n,-},\varepsilon_{n,+})$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-078238 | MATHDB-400509

Variational source condition for acoustic inverse medium scattering

摘录 / EXCERPT: Theorem 3 states that, for the acoustic scattering equation $\Delta u+\kappa^2 n(x)u=0$ in $\mathbb{R}^3$ with the Sommerfeld radiation condition, contrast $f:=1-n$ supported in...

open_problem | open | MathDB 页面 | 原文线索

V2-078239 | MATHDB-400510

Homogenization and boundary-corrector limits for a periodic transmission problem

摘录 / EXCERPT: The report presents established results, not an unresolved problem. In particular, for the unit square $D=(0,1)\times(0,1)$ and a sequence $\epsilon_k\to0$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-078240 | MATHDB-400512

Convergent regularization for linearized EIT shape reconstruction

摘录 / EXCERPT: Can one design a regularization strategy for the standard optimization-based linearized EIT approach, using finite-dimensional noisy electrode measurements that approximate the dif...

open_problem | open | MathDB 页面 | 原文线索

V2-078241 | MATHDB-400513

Convergence of a monotonicity-constrained linearized EIT reconstruction

摘录 / EXCERPT: For noisy data, minimize ... subject to $0\leq \kappa|_{P_k}\leq \min\{\gamma,\beta_k^\delta\}$ on every pixel P_k. The report states that minimizers κ^δ exist and that...

open_problem | open | MathDB 页面 | 原文线索

V2-078242 | MATHDB-400514

Corrector term in the asymptotic expansion of transmission eigenvalues

摘录 / EXCERPT: Open Problem: Determining the corrector term in the asymptotic expansion for transmission eigenvalues.

open_problem | open | MathDB 页面 | 原文线索

V2-078243 | MATHDB-400515

Conjectural equality of refined invariants and refined Severi degrees

摘录 / EXCERPT: Conjecture 1 asserts that, for plane curves, whenever $d\ge \delta/2+1$, the refined invariant from Section 1 equals the tropical refined Severi degree from Section 2: ... More gener...

open_problem | open | MathDB 页面 | 原文线索

V2-078244 | MATHDB-400518

Origin of the vanishing cycle for image Milnor number one

摘录 / EXCERPT: In the mono-germ setting [U+007F](|S|=1[U+007F]) considered in the report, let $f:(\mathbb C^n,0)\to(\mathbb C^{n+1},0)$ be an $\mathcal A$-finite germ of $\mathcal A_e$-codimension 1, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078245 | MATHDB-400533

Parallel performance of cyclic reduction for structured block bidiagonal systems

摘录 / EXCERPT: Analyze the parallel performance of the cyclic-reduction method for the structured block bidiagonal linear systems arising in weakly constrained 4D-Var, whose blocks may be very la...

open_problem | open | MathDB 页面 | 原文线索

V2-078246 | MATHDB-400534

Schmidt’s conjecture for the weights $(1/3,2/3)$ and $(2/3,1/3)$

摘录 / EXCERPT: W. M. Schmidt conjectured that there exist points in the Euclidean plane that are simultaneously badly approximable with weights $(1/3,2/3)$ and $(2/3,1/3)$; equivalently, that...

conjecture | open | MathDB 页面 | 原文线索

V2-078247 | MATHDB-400535

Schmidt’s conjecture on intersections of badly approximable sets

摘录 / EXCERPT: For pairs (i_1,j_1) and (i_2,j_2) satisfying $0<i_1,i_2,j_1,j_2<1$ and $i_1+j_1=i_2+j_2=1$, is ... Here $\operatorname{Bad}(i,j)$ is the set of $(x,y)\in\mathbb{R}^2$ for which t...

conjecture | open | MathDB 页面 | 原文线索

V2-078248 | MATHDB-400536

Weighted bounded-orbit characterization

摘录 / EXCERPT: Let $r=(r_1,\dots,r_m)\bin\bb R_+^m$ and $s=(s_1,\dots,s_n)\bin\bb R_+^n$ satisfy $\sum_{i=1}^m r_i=\sum_{j=1}^n s_j=1$, and let ... What can be said about those...

open_problem | open | MathDB 页面 | 原文线索

V2-078249 | MATHDB-400537

Schmidt’s conjecture on intersections of weighted badly approximable sets

摘录 / EXCERPT: For weights (i_1,j_1) and (i_2,j_2) satisfying $i_k,j_k\ge 0$ and $i_k+j_k=1$ for $k=1,2$, define ... Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-078250 | MATHDB-400538

An’s theorem on fiberwise weighted badly approximable numbers

摘录 / EXCERPT: For all $s,t\ge 0$ with $s+t=1$ and all $\theta\in\mathbb{R}$ such that $\inf_{q\in\mathbb{N}}q^{1/s}\lVert q\theta\rVert>0$, the set ... is winning for Schmidt’s game.

open_problem | open | MathDB 页面 | 原文线索

V2-078251 | MATHDB-400539

Winning property of weighted badly approximable points on a fibre

摘录 / EXCERPT: For weights $s,t\ge 0$ with $s+t=1$ and every $\theta\in\mathbb{R}$ satisfying ... the set ... is $1/2$-winning for Schmidt’s game.

open_problem | open | MathDB 页面 | 原文线索

V2-078252 | MATHDB-400545

Hausdorff measures of exceptional sets for the p-adic and classical Littlewood conjectures

摘录 / EXCERPT: Let S_p be the set of counterexamples to the p-adic Littlewood conjecture. Let H^f be the Hausdorff measure associated with $f(r)=(\\log|\\log r|)^{-1}$, and suppose it is kn...

conjecture | open | MathDB 页面 | 原文线索

V2-078253 | MATHDB-400546

Counting rational points in the middle-thirds Cantor set

摘录 / EXCERPT: Let C be the middle-thirds Cantor set and define $N_C(Q)=\#\{p/q\in C:q\le Q\}$. Prove that, for every $\varepsilon>0$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-078254 | MATHDB-400548

Improving the universal constant in the two-dimensional Littlewood inequality

摘录 / EXCERPT: For all α,β, Hurwitz's theorem gives ... What improvement can be made to the right-hand side? One may attempt this using continued fractions. The expected value of the qua...

open_problem | open | MathDB 页面 | 原文线索

V2-078255 | MATHDB-400549

Duffin-Schaeffer conjecture for \fpsi(q)=q^{-1/2} on its support

摘录 / EXCERPT: Can one prove the Duffin-Schaeffer conjecture when fpsi(q)=q^{-1/2} on its support, under the divergence condition ... so that $W'(\psi)$ has full Lebesgue measure?

conjecture | open | MathDB 页面 | 原文线索

V2-078256 | MATHDB-400551

First-order theories of finitely generated fields

摘录 / EXCERPT: Whether all f.g. fields are, up to isomorphism, determined by their first-order theory.

open_problem | open | MathDB 页面 | 原文线索

V2-078257 | MATHDB-400557

Positive-existential definability of the evaluation predicate on entire analytic functions

摘录 / EXCERPT: Let $H_{t_1,t_2}(\d835\c2)$ be the ring of functions of (t_1,t_2) analytic on all of $\d835\c2$. For $a,b\in H_{t_1,t_2}(\d835\c2)$, define $\operatorname{Eval}(a,b)$ to mean tha...

open_problem | open | MathDB 页面 | 原文线索

V2-078258 | MATHDB-400560

Decidability of the positive-existential theory of entire functions

摘录 / EXCERPT: Let T be a predicate for the nonconstant functions and let $L_T=L_r\cup\{T\}$. Is the positive-existential theory of the ring of entire functions of one complex variable z, vie...

open_problem | open | MathDB 页面 | 原文线索

V2-078259 | MATHDB-400563

Northcott numbers of ring extensions of \(\mathbb{Q}\)

摘录 / EXCERPT: Which real numbers can be realised as Northcott numbers of ring extensions of $\mathbb{Q}$? Here the Northcott number of a subset $A\subseteq\overline{\mathbb{Q}}$ is defined by ...

open_problem | open | MathDB 页面 | 原文线索

V2-078260 | MATHDB-400564

Sharpness of the lower bound in Theorem 3 for j-fields

摘录 / EXCERPT: Is the inequality in Theorem 3, namely $\operatorname{t.d.}\bigl(j(z),j'(z),j''(z),j(gz),j'(gz),j''(gz)/F,C\bigr)\ge 3n$, sharp?

open_problem | open | MathDB 页面 | 原文线索

V2-078261 | MATHDB-400565

Property (N) for the ring of integers of the totally real subfield of \(\mathbb{Q}^{(d)}\)

摘录 / EXCERPT: For a positive integer d, does $\mathcal O_{\mathbb Q^{(d)}_{\mathrm{tr}}}$ have Property (N)?

open_problem | open | MathDB 页面 | 原文线索

V2-078262 | MATHDB-400566

House-based Northcott property for the ring of integers of the totally real degree-bounded field

摘录 / EXCERPT: For a positive integer d, is ... for every $X\ge 1$, where $\operatorname{house}(\alpha)$ is the maximum of the moduli of the conjugates of α over $\mathbb Q$?

open_problem | open | MathDB 页面 | 原文线索

V2-078263 | MATHDB-400568

Sharpness of the transcendence inequality in Theorem 3

摘录 / EXCERPT: Is the inequality of Theorem 3 sharp? Some modular versions of Schanuel's conjecture say that the answer to this should be no. Of course, a proof of such a conjecture is considered...

open_problem | open | MathDB 页面 | 原文线索

V2-078264 | MATHDB-400569

Hilbert's Tenth Problem for perfect closures of global fields

摘录 / EXCERPT: Hilbert's Tenth Problem is undecidable for the perfect closure of any global field of positive characteristic.

open_problem | open | MathDB 页面 | 原文线索

V2-078265 | MATHDB-400571

Positive measure of a boundary set for a polynomial

摘录 / EXCERPT: Does there exist a polynomial f with $\\mu(B(f))>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-078266 | MATHDB-400574

Solutions of the transformed Pell-type equation over an exponential-sum ring

摘录 / EXCERPT: Given a natural number N, find all solutions of ... over the ring ... where $t_2,\ldots,t_\ell$ are variables.

open_problem | open | MathDB 页面 | 原文线索

V2-078267 | MATHDB-400580

Dickson’s conjecture for finite families of affine-linear polynomials

摘录 / EXCERPT: Let $k\ge 1$ and let $\bar f=\langle f_i\mid i<k\rangle$, where $f_i(x)=a_i x+b_i$ with a_i,b_i non-negative integers and $a_i\ge 1$ for all $i<k$. Assume that the following cond...

conjecture | open | MathDB 页面 | 原文线索

V2-078268 | MATHDB-400583

Hypothesis ensuring Vojta’s method works for surfaces

摘录 / EXCERPT: Which hypothesis must a surface $X/\mathbb{C}$ satisfy to ensure that there exists an ω making the method work? It should be a condition on the relation between the branch d...

conjecture | open | MathDB 页面 | 原文线索

V2-078269 | MATHDB-400585

Pop’s conjecture for finitely generated fields

摘录 / EXCERPT: Given a finitely generated field K, does there exist a first-order sentence φ_K in the language of fields such that, for every finitely generated field L, φ_K h...

conjecture | open | MathDB 页面 | 原文线索

V2-078270 | MATHDB-400587

Undecidability of an additive expansion of a Euclidean ring

摘录 / EXCERPT: Under the setup preceding Question 1, let [?mR[0m be a Euclidean ring with Euclidean function [?mdelta:R^asttomathbb N[0m, let [?mr,sin R^ast[0m be non-invertible and multiplicativ...

open_problem | open | MathDB 页面 | 原文线索

V2-078271 | MATHDB-400588

Commonly definable subsets in two multiplicative expansions of a Euclidean ring

摘录 / EXCERPT: Let $r,s\in R^\star$ be two non-invertible, multiplicatively independent elements. Can one obtain a characterization of the subsets of R that are definable both in...

open_problem | open | MathDB 页面 | 原文线索

V2-078272 | MATHDB-400590

Polynomial parametrization of the nonnegative integers

摘录 / EXCERPT: Does there exist a polynomial $p(x,y)$ such that $p(\mathbb Z\times\mathbb Z)=\mathbb N$, without specifying what kind of coefficients the polynomial should have?

open_problem | open | MathDB 页面 | 原文线索

V2-078273 | MATHDB-400592

Separably closed henselisation of certain valued elementary extensions of \(\mathbb Q\)

摘录 / EXCERPT: Let ν be a valuation on an elementary extension $\mathbb Q^*\succeq\mathbb Q$ such that $\mathcal O_\nu$ does not contain $\mathbb Z^*$ nor the convex hull of $\mathbb Z$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-078274 | MATHDB-400596

Purely inseparable extensions of valued fields in mixed characteristic

摘录 / EXCERPT: Question 10 (posed by Franz-Viktor Kuhlmann): “What are the purely inseparable extensions of a valued field in mixed characteristic?” In characteristic (p,p): K is henselian de...

open_problem | open | MathDB 页面 | 原文线索

V2-078275 | MATHDB-400599

Triviality of the map induced by a homomorphism \(\mu_n\to G\)

摘录 / EXCERPT: Let G/k be a semisimple simply connected group satisfying the hypothesis of the extended Conjecture II. Let $f:\mu_n\to G$ be a k-group homomorphism. Then the induced map ... is...

open_problem | open | MathDB 页面 | 原文线索

V2-078276 | MATHDB-400600

The Shareshian-Wachs conjecture

摘录 / EXCERPT: Let $m=(m_1,\dots,m_n)$ be a Hessenberg function, meaning that $m_1\those...$ is nondecreasing and $i\le m_i\le n$. Let $G(m)$ be the graph on $\{1,\dots,n\}$ in which $i<j$ are ad...

conjecture | open | MathDB 页面 | 原文线索

V2-078277 | MATHDB-400602

Index of an orthogonal Grassmannian of a generic quadratic form

摘录 / EXCERPT: For any even $m \ge 18$, the index of the $[m/4]$-th orthogonal Grassmannian of any generic m-dimensional quadratic form φ/F in I_q^3 is equal to $2^{[m/4]}$.

open_problem | open | MathDB 页面 | 原文线索

V2-078278 | MATHDB-400605

Scully's unrestricted upper bound for isotropy over a quadric function field

摘录 / EXCERPT: Let F be a field and let p,q be anisotropic quadratic forms over F with $\\dim(p),\dim(q)\ge 2$. Write $F(p)$ for the function field of the projective quadric defined by p,...

open_problem | open | MathDB 页面 | 原文线索

V2-078279 | MATHDB-400608

Kernel splitting for a mixed extension with a bilinear Pfister function field

摘录 / EXCERPT: Let L/F be a purely inseparable extension of finite degree, B an anisotropic bilinear Pfister form over F, and K/F a separable quadratic extension. Suppose that B is anis...

open_problem | open | MathDB 页面 | 原文线索

V2-078280 | MATHDB-400612

The permalloy problem

摘录 / EXCERPT: Understand, from a predictive viewpoint, the particular compositions 45% Ni and 78.5% Ni (in the $\mathrm{Fe}_x\mathrm{Ni}_{1-x}$ system) at which the alloys become exceptionally s...

open_problem | open | MathDB 页面 | 原文线索

V2-078281 | MATHDB-400619

Exact minimal-cost synthesis for Clifford+T circuits

摘录 / EXCERPT: Given an exactly expressible unitary operator $g\in G_n$ for the Clifford+T gate set, find a word $g_1\cdots g_r$ in the allowed gates representing g and having minimal T-count (...

open_problem | open | MathDB 页面 | 原文线索

V2-078282 | MATHDB-400624

Sarnak’s purity conjecture for sup-norms

摘录 / EXCERPT: Let X be a compact, arithmetic, irreducible locally symmetric space, and let $(f_j)_{j\geq 1}$ be any L^2-normalized sequence of eigenfunctions for all Hecke operators and all...

conjecture | open | MathDB 页面 | 原文线索

V2-078283 | MATHDB-400625

Mean-square asymptotic for the Vaughan-type remainder for the divisor function

摘录 / EXCERPT: The report states that there exists an absolute constant $\eta>0$ such that, for every integer $k\ge 2$ and $N\le R\le x^{\eta/k}$, ... where $\Delta_R(n)=f(n)-F_R(n)$ specialized to...

open_problem | open | MathDB 页面 | 原文线索

V2-078284 | MATHDB-400626

Batyrev-Manin conjecture for smooth quartic del Pezzo surfaces

摘录 / EXCERPT: Let X be a smooth quartic del Pezzo surface over $\\mathbb{Q}$, viewed as a surface in $\mathbb{P}^4$ cut out by a pair of quadrics defined over $\mathbb{Q}$. Let $U\subset X$ be...

conjecture | open | MathDB 页面 | 原文线索

V2-078285 | MATHDB-400627

Squarefree orders for reductions of an elliptic curve over a function field

摘录 / EXCERPT: Let K be a global field of characteristic p with constant field $_q$. For a place v of K, let k_v be its residue field and let $\deg v=[k_v:\u001f_q]$. Let V_K be the...

open_problem | open | MathDB 页面 | 原文线索

V2-078286 | MATHDB-400630

Ramanujan-Petersson conjecture for cuspidal Hecke eigenforms

摘录 / EXCERPT: Conjecture: the bound $|\lambda_f(n)|\le d(n)n^\theta$ for every $n\ge 1$ and every cuspidal Hecke eigenform f, holomorphic or not, remains valid when $\theta=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-078287 | MATHDB-400633

Expected lower bounds for generalized divisor sums of binary forms

摘录 / EXCERPT: For divisor sums of the form ... where σ,τ,q are positive integers, f is a small arithmetic function, the binary forms G_i have even degree, and the forms F_i are irr...

open_problem | open | MathDB 页面 | 原文线索

V2-078288 | MATHDB-400634

Lower bound for rational points on del Pezzo surfaces after bounded base extension

摘录 / EXCERPT: Let X be a del Pezzo surface of degree $d\le 5$ over K. Then $N_{U,H}(B)\gg B(\log B)^{\rho_X-1}$ holds after a base extension of degree bounded by $n_d<\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-078289 | MATHDB-400635

Rudin’s sumset conjecture for perfect squares

摘录 / EXCERPT: For every positive $\\varepsilon$, there is an implicit constant such that, for every finite set S of perfect squares, $|S+S|\gg |S|^{2-\varepsilon}$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-078290 | MATHDB-400636

Lamzouri’s conjecture on Gaussian limits for short character sums

摘录 / EXCERPT: Let q tend to infinity through the primes, let $\chi(n)=\left(\frac{n}{q}\right)$ be the Legendre symbol, and let X be uniformly distributed on $\{0,1,\ldots,q-1\}$. For...

conjecture | open | MathDB 页面 | 原文线索

V2-078291 | MATHDB-400637

Moments of smoothed Selberg sieve weights

摘录 / EXCERPT: Let $k\ge 1$ and $A\ge 0$ be fixed integers, let $f_A(t):=\mathbf{1}_{t<1}(1-t)^A$, and define ... Then there is a constant $c_{k,A}>0$ such that, for $x\ge R^{2k}\log R$, ... where .....

open_problem | open | MathDB 页面 | 原文线索

V2-078292 | MATHDB-400639

Height of rational points on random Fano complete intersections

摘录 / EXCERPT: Fix integers $n\ge 3$, $r\ge 1$, and $d_1,\ldots,d_r\ge 2$ with $d_1+\cdots+d_r<n+1$. Let $\mathcal V_{\mathbf d,n}$ be the family of smooth complete intersections in $\mathbb P^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-078293 | MATHDB-400641

Removing the general-position hypothesis in higher-degree Birch-type results

摘录 / EXCERPT: Can Davenport’s approach be generalized to degrees $d\ge 4$ so as to remove the “general position” condition from Theorems 2 and 3?

conjecture | open | MathDB 页面 | 原文线索

V2-078294 | MATHDB-400642

The \-torsion conjecture for class groups of number fields

摘录 / EXCERPT: For every number field $K/\mathbb{Q}$ of degree n and every integer $\ell\geq 1$, one conjectures that, for every $\varepsilon>0$, ... where $D_K=|\mathrm{Disc}_{K/\mathbb{Q}}|$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-078295 | MATHDB-400643

Variance of sums of the k-fold divisor function in short intervals

摘录 / EXCERPT: Conjecture (Keating-Rodgers-Roditty-Gershon-Rudnick [5]). For $\frac{\log H}{\log X}\to\delta$ with $\delta\in\left(0,1-\frac1k\right)$, one has ... where a_k is a certain arithmet...

open_problem | open | MathDB 页面 | 原文线索

V2-078296 | MATHDB-400645

Wooley’s cubic zero-count conjecture

摘录 / EXCERPT: Let $C(x_1,x_2,\ldots,x_s)\in\mathbb{Z}[x_1,x_2,\ldots,x_s]$ be a homogeneous cubic polynomial. It is conjectured that if $s\ge 10$, then the number of solutions of...

conjecture | open | MathDB 页面 | 原文线索

V2-078297 | MATHDB-400646

Quadratic-polynomial families and cubic-form point counts

摘录 / EXCERPT: Let $Q_1,Q_2,\\ldots,Q_h\in\mathbb Z[y_1,y_2,\ldots,y_t]$, and let $x_1,x_2,\ldots,x_h$ be fixed integers with $|x_i|\le B^\theta$. Is it true that, for a positive proportion of th...

open_problem | open | MathDB 页面 | 原文线索

V2-078298 | MATHDB-400647

Hydrodynamic limit for a nongradient system in a crystal lattice

摘录 / EXCERPT: The hydrodynamic limit for a nongradient system in a crystal lattice is an important open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078299 | MATHDB-400648

One-site coupling for initializing a second-class particle

摘录 / EXCERPT: Given the desired stationary marginals at the two densities defining rarefaction-fan initial data, provide a one-site coupling of the two marginals such that: (1) the first margina...

open_problem | open | MathDB 页面 | 原文线索

V2-078300 | MATHDB-400649

Hard-edge tacnode limit for restricted Aztec-diamond tilings

摘录 / EXCERPT: For a uniformly random domino tiling of the Aztec diamond A_n restricted to the region $y\le R$, where the horizontal line $y=R$ is tangent to the Arctic circle, the expected sca...

open_problem | open | MathDB 页面 | 原文线索

V2-078301 | MATHDB-400650

Cutoff for the full upper triangular matrix walk

摘录 / EXCERPT: For the continuous-time upper triangular matrix walk on $M_n(2)$, does the whole matrix exhibit mixing-time cutoff and mix at the same location as the last column?

open_problem | open | MathDB 页面 | 原文线索

V2-078302 | MATHDB-400651

Correct low-density asymptotics of the tagged-particle diffusion coefficient in KA models

摘录 / EXCERPT: An open problem is the derivation of the correct asymptotic order for $D(q)$ as $q\to 0$ in the KA models.

open_problem | open | MathDB 页面 | 原文线索

V2-078303 | MATHDB-400652

Theorem 3: n-marginal Root embedding

摘录 / EXCERPT: Let $\gamma_i((\omega_s)_{s\le s_n},s_1,\ldots,s_n)=h(s_i)$ for a strictly convex function $h:\mathbb{R}_+\to\mathbb{R}$. Assume that (MSEP) is well defined for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078304 | MATHDB-400654

Rigidity of W-entropy monotonicity on RCD* spaces

摘录 / EXCERPT: Let (X,d,m) be an $RCD^*(0,N)$ metric-measure space, with $N\ge 2$, and let $\mu_t=P_t\mu$ be the heat flow. The report states that the W-entropy $W(\mu_t,t)$ is nonincreasing...

open_problem | open | MathDB 页面 | 原文线索

V2-078305 | MATHDB-400660

Surface bundles over surfaces without flat connections

摘录 / EXCERPT: There exist surface bundles over surfaces not admitting any flat connection.

open_problem | open | MathDB 页面 | 原文线索

V2-078306 | MATHDB-400661

The sharp signature-Euler characteristic inequality for surface bundles

摘录 / EXCERPT: Let X be the total space of an oriented surface bundle with closed oriented fiber F over a closed oriented surface B, where the genera of F and B are at least 2. Is it...

open_problem | open | MathDB 页面 | 原文线索

V2-078307 | MATHDB-400662

Upper bounds for simplicial volume of surface bundles

摘录 / EXCERPT: Let X be the total space of an oriented surface bundle with closed oriented fibre F and closed oriented base B, with both genera at least 2. For fixed F and B, can the...

open_problem | open | MathDB 页面 | 原文线索

V2-078308 | MATHDB-400663

Lifting a Johnson-filtration congruence to a handlebody intersection

摘录 / EXCERPT: In the setup above, with $g\ge 3$ and $k\ge 2$, let $\phi\in A_{g,1}(k)$ and $\psi\in B_{g,1}$. Assume that for every $x\in\pi=\pi_1(\Sigma_{g,1})$, the induced automorphisms satis...

open_problem | open | MathDB 页面 | 原文线索

V2-078309 | MATHDB-400671

Penner’s construction conjecture

摘录 / EXCERPT: Every pseudo-Anosov mapping class has a power that arises from Penner’s construction.

conjecture | open | MathDB 页面 | 原文线索

V2-078310 | MATHDB-400672

Coherence of fundamental groups of surface bundles

摘录 / EXCERPT: Let E be the total space of a bundle with base B and fibre F closed surfaces, and let $\theta:\pi_1(B)\to\operatorname{Mod}(F)$ be the associated action. Is $\pi=\pi_1(E)$ co...

open_problem | open | MathDB 页面 | 原文线索

V2-078311 | MATHDB-400673

Unbounded simplicial volume of the total spaces E_R

摘录 / EXCERPT: For each $R>0$, let $\rho_R:\pi_1(\Sigma_2)\to\operatorname{Mod}_5$ be an injective homomorphism such that the associated Σ_5-bundle $E_R\to\Sigma_2$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-078312 | MATHDB-400674

Bounds for the relative irregularity of surface fibrations

摘录 / EXCERPT: Let $f:X\to B$ be a fibred complex surface, allowing singular fibres with nodal singularities, and assume that X is Kähler. Let F denote a fibre of genus $g(F)$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-078313 | MATHDB-400675

Three or more Kodaira fibrations on a complex compact surface

摘录 / EXCERPT: Are there complex compact surfaces which admit three or more Kodaira fiberings over surfaces?

open_problem | open | MathDB 页面 | 原文线索

V2-078314 | MATHDB-400676

Asymptotics of the number of points of the coarse moduli space of curves

摘录 / EXCERPT: Find asymptotics for $\#\mathcal{M}_g(\mathbb{F}_q)$, the set of $\mathbb{F}_q$-valued points of the coarse moduli space, using the Grothendieck-Lefschetz trace formula. More speci...

open_problem | open | MathDB 页面 | 原文线索

V2-078315 | MATHDB-400683

Derived direct summand conjecture

摘录 / EXCERPT: Let R be a regular ring, and let $f:X\to\operatorname{Spec}(R)$ be a proper surjective map. Then the canonical map $R\to R\Gamma(X,\mathcal{O}_X)$ splits in the derived category...

conjecture | open | MathDB 页面 | 原文线索

V2-078316 | MATHDB-400686

Stillman’s Conjecture

摘录 / EXCERPT: For a polynomial ring $R=K[X_1,\ldots,X_N]$ over a field K, the projective dimension of an ideal I generated by a fixed number of forms of fixed bounded degrees is bounded by a...

conjecture | open | MathDB 页面 | 原文线索

V2-078317 | MATHDB-400688

Embedded resolution of singularities in dimension at least three

摘录 / EXCERPT: Embedded resolution of singularities is only known up to dimension two and remains an important open problem in dimension three or larger.

open_problem | open | MathDB 页面 | 原文线索

V2-078318 | MATHDB-400689

Lech’s conjecture

摘录 / EXCERPT: Let $(R,\mathfrak m)\to(S,\mathfrak n)$ be a flat local extension of local rings. Then $e(R)\le e(S)$, where $e(-)$ denotes Hilbert-Samuel multiplicity.

conjecture | open | MathDB 页面 | 原文线索

V2-078319 | MATHDB-400690

Lech-type multiplicity conjecture for perfect ideals

摘录 / EXCERPT: Let $(T,\mathfrak n)$ be a local ring and J a perfect ideal of T. Then $e(T) \le e(T/J)$. Here the report further states that Conjecture 4 implies Lech’s conjecture, that no co...

conjecture | open | MathDB 页面 | 原文线索

V2-078320 | MATHDB-400692

Reduction of multiplicity in purely inseparable extensions of regular rings

摘录 / EXCERPT: For an affine morphism $X=\operatorname{Spec}(B)\to W=\operatorname{Spec}(S)$ in which $S\subset B$ is a purely inseparable extension and S is regular, prove reduction of the mul...

open_problem | open | MathDB 页面 | 原文线索

V2-078321 | MATHDB-400698

Converse to Theorem 2 for eventually polynomial Betti numbers

摘录 / EXCERPT: Let $(R,\mathfrak m,k)$ be a Noetherian local ring, with $\mathfrak m$ not principal and k infinite. Set $P=\operatorname{Sym}_{k}(\mathfrak m/\mathfrak m^2)$, let I^* be the k...

open_problem | open | MathDB 页面 | 原文线索

V2-078322 | MATHDB-400702

Bounded regularity under property N_p for Stanley-Reisner rings

摘录 / EXCERPT: For a fixed integer $p\ge 2$, is there a constant $\lambda\in\mathbb{R}$ such that $\operatorname{reg} K[\Delta]\le \lambda$ whenever the Stanley-Reisner ring $K[\Delta]$ satisfies...

open_problem | open | MathDB 页面 | 原文线索

V2-078323 | MATHDB-400704

Uniform regularity bound for Cohen-Macaulay Stanley-Reisner rings satisfying N_2

摘录 / EXCERPT: Is there a constant $\lambda\in\mathbb{R}$ such that $\operatorname{reg} K[\Delta]\le \lambda$ whenever $K[\Delta]$ satisfies N_2 and is Cohen-Macaulay?

open_problem | open | MathDB 页面 | 原文线索

V2-078324 | MATHDB-400705

Classification of representations of the modular presentation

摘录 / EXCERPT: Representations of $F(M)$ correspond precisely to modular categories equipped with a choice of sign.

open_problem | open | MathDB 页面 | 原文线索

V2-078325 | MATHDB-400706

A suitable differential K-theory model for fully covariant quantisation

摘录 / EXCERPT: Find a suitable model for differential K-theory which enables the fully covariant quantisation of [3] to be applied.

open_problem | open | MathDB 页面 | 原文线索

V2-078326 | MATHDB-400709

Natural operator-space structure on $L^2(G)$

摘录 / EXCERPT: Let G be a real reductive group. Does $L^2(G)$ carry a natural operator space structure, or matrix-normed space structure?

open_problem | open | MathDB 页面 | 原文线索

V2-078327 | MATHDB-400712

Elementary-embedding question for logarithmic-exponential series and surreal numbers

摘录 / EXCERPT: Let $\mathbb{T}$ be the field of logarithmic-exponential series with its natural derivation ∂ and exponential map $\exp$. Is ...

open_problem | open | MathDB 页面 | 原文线索

V2-078328 | MATHDB-400713

Model-theoretic properties of the omega-series expansion of the surreals

摘录 / EXCERPT: Does the structure $\left(\mathbf{No},+,\times,\exp,\widehat{f}\right)_{f\in\mathbb{R}\langle\langle\omega\rangle\rangle}$ have good model-theoretic properties?

open_problem | open | MathDB 页面 | 原文线索

V2-078329 | MATHDB-400716

Explicit integer part of the surreal numbers modeling true arithmetic

摘录 / EXCERPT: Can one describe an integer part of $ No$ which is a model of true arithmetic? The existence of such an integer part should follow by the saturation properties of $ No$, but can on...

open_problem | open | MathDB 页面 | 原文线索

V2-078330 | MATHDB-400718

Derivations on fields of generalized power series

摘录 / EXCERPT: Can one define a derivation D on $k((G))$ so that $(k((G)),v,D)$ is a differential valued field? A H-field?

open_problem | open | MathDB 页面 | 原文线索

V2-078331 | MATHDB-400719

Derivations on generalized power series with a prescribed asymptotic couple

摘录 / EXCERPT: Let $(G,\\psi)$ be an H-type asymptotic couple, let k be a field, and let $K:=k((G))$ be the field of generalized power series with its natural valuation v. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-078332 | MATHDB-400720

Elementary equivalence of generalized power-series differential valued fields

摘录 / EXCERPT: Let k,k' be two elementary equivalent fields and $(G,\psi),(G',\psi')$ two elementary equivalent asymptotic couples. Are $k((G))$ and $k'((G))$ elementary equivalent as different...

open_problem | open | MathDB 页面 | 原文线索

V2-078333 | MATHDB-400723

Integer parts of the surreal field modeling true arithmetic

摘录 / EXCERPT: Does the field of surreal numbers (mathbf{No}) admit an integer part which is a model of true arithmetic?

open_problem | open | MathDB 页面 | 原文线索

V2-078334 | MATHDB-400724

Common refinements of factorizations of generalized power series

摘录 / EXCERPT: For every non-zero series $a\in K((\mathbb{R}_{\le 0}))$, any two factorizations of a admit common refinements.

open_problem | open | MathDB 页面 | 原文线索

V2-078335 | MATHDB-400725

Berarducci’s unique factorisation conjecture for non-zero germs

摘录 / EXCERPT: Every non-zero germ in $K((\mathbb{R}_{\le 0}))/J$ admits a unique factorisation into irreducibles.

conjecture | open | MathDB 页面 | 原文线索

V2-078336 | MATHDB-400727

Set-sized-model proof of the only-if portions of Theorem 6

摘录 / EXCERPT: Can the “only if” portions of Theorem 6 be established in NBG appealing solely to models whose universes are sets?

open_problem | open | MathDB 页面 | 原文线索

V2-078337 | MATHDB-400728

Conjecture 1.6 on o-minimal exponential fields

摘录 / EXCERPT: Let $K_{\exp}$ be o-minimal and satisfy EXP. Then $K_{\exp}\equiv\mathbb{R}_{\exp}$, where EXP is the conjunction of the axioms for ordered exponential fields with the axiom...

conjecture | open | MathDB 页面 | 原文线索

V2-078338 | MATHDB-400729

Conjecture 3.1: elementary equivalence of the residue exponential field

摘录 / EXCERPT: Let $K_{\\exp}$ be o-minimal and EXP. Then $\overline{K}_{\exp} \equiv \mathbb{R}_{\exp}$, where $\overline{K}_{\exp}$ is the residue exponential field defined in Section 2.

conjecture | open | MathDB 页面 | 原文线索

V2-078339 | MATHDB-400730

Conjecture 3.3 on residue exponential fields

摘录 / EXCERPT: Let $K_{\\exp}$ be o-minimal and EXP. Then $\\overline{K}_{\\exp} \\preccurlyeq K_{\\exp}$, where $\\overline{K}_{\\exp}$ is the residue exponential field of $K_{\\exp}$ and EXP in...

conjecture | open | MathDB 页面 | 原文线索

V2-078340 | MATHDB-400731

Extending functions and their integrals past infinity on the surreal numbers

摘录 / EXCERPT: Can we find a way of extending functions and their integrals past ∞ or, more generally, past a singularity at which asymptotic expansions do not exist or are divergent?

open_problem | open | MathDB 页面 | 原文线索

V2-078341 | MATHDB-400733

Existence of a reasonable definition of surreal integration

摘录 / EXCERPT: Does there exist a reasonable definition of surreal integration?

open_problem | open | MathDB 页面 | 原文线索

V2-078342 | MATHDB-400737

A unique Koopman operator for a switching system

摘录 / EXCERPT: Can some kind of unique Koopman operator be defined for a given switching system?

open_problem | open | MathDB 页面 | 原文线索

V2-078343 | MATHDB-400738

Point spectrum of the Koopman operator induced by a linear contraction

摘录 / EXCERPT: Let $X:=\mathbb{C}^k$ with a norm, let $A\in\mathbb{C}^{k\times k}$ satisfy $\|A\|\le 1$ for the induced operator norm, and let U be the closed unit ball in X. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-078344 | MATHDB-400739

Pointwise convergence of generalized Laplace averages

摘录 / EXCERPT: For $f\in F$ and λ in the point spectrum of U_Φ, does the generalized Laplace average ... converge pointwise, in addition to converging in the norm of F?

open_problem | open | MathDB 页面 | 原文线索

V2-078345 | MATHDB-400740

Necessity of the logarithmic factor in maximum-norm a posteriori estimates

摘录 / EXCERPT: For a shape-regular triangulation 4Th4 of a polyhedral domain 4Omegasubsetmathbb{R}^24 or 4mathbb{R}^34, the report asks whether the logarithmic factor 4ellh=max{Tin Th}ln(1+1/hT)4...

open_problem | open | MathDB 页面 | 原文线索

V2-078346 | MATHDB-400741

Sharp growth of the L^2-projector norm on arbitrary triangulations

摘录 / EXCERPT: Let $\a9$ be a square in $\a1^2$, let $\a4$ range over finite triangulations of $\a9$, and let $P_{\a4}$ denote the $L^2(\a9)$-orthogonal projector onto the space $V_{\a4}$ of cont...

open_problem | open | MathDB 页面 | 原文线索

V2-078347 | MATHDB-400743

Construction of smooth isogeometric spaces with optimal approximation properties on complex geometries

摘录 / EXCERPT: The construction of smooth isogeometric spaces with optimal approximation properties on complex geometries is still an open and challenging problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078348 | MATHDB-400750

Single-valued solutions of the quantized spectral-curve equation

摘录 / EXCERPT: In the setting of the quantized Hitchin spectral curve, find single-valued solutions $\chi(u,\bar u)$ satisfying both ... and the complex conjugate of this equation.

open_problem | open | MathDB 页面 | 原文线索

V2-078349 | MATHDB-400751

Hemisphere partition function as a Bridgeland central charge

摘录 / EXCERPT: The conjecture is that $Z_{D^2}:K_0(D)\to\mathbb{C}$ is a central charge function for a Bridgeland stability condition on D.

open_problem | open | MathDB 页面 | 原文线索

V2-078350 | MATHDB-400752

Grade-restriction window equivalences for GLSM branes

摘录 / EXCERPT: For the chambers $T_n=(\a0\pi(N+2n),\pi(N+2n+2))\subset\mathbb{R}$, let W_n be the full triangulated subcategory of D consisting of objects B satisfying the grade restriction...

open_problem | open | MathDB 页面 | 原文线索

V2-078351 | MATHDB-400753

Central-charge interpretation of GLSM hemisphere-partition-function series

摘录 / EXCERPT: For the running GLSM example described in the report, the residue series obtained from the hemisphere partition function $Z_{D^2}(B)$ by closing the contour to the right (respectiv...

open_problem | open | MathDB 页面 | 原文线索

V2-078352 | MATHDB-400754

Kotecký conjecture for the high-dimensional antiferromagnetic Potts model

摘录 / EXCERPT: For fixed $q\ge 3$, let σ be chosen uniformly from the proper q-colorings of $\Lambda_n^d=\{x\in\mathbb{Z}^d:\|x\|_\infty\le n\}$. If $A\cup B$ is an equipartition of...

conjecture | open | MathDB 页面 | 原文线索

V2-078353 | MATHDB-400755

Upper bounds for spherical $[-1,-\beta]\cup\{\alpha_1,\ldots,\alpha_k\}$-codes

摘录 / EXCERPT: Let $L=[-1,-\beta]\cup\{\alpha_1,\ldots,\alpha_k\}$ for fixed $\beta\in(0,1]$. A spherical L-code in $\mathbb{R}^n$ is a finite nonempty set C of unit vectors such that...

open_problem | open | MathDB 页面 | 原文线索

V2-078354 | MATHDB-400756

Polynomial upper bound for spherical L-codes

摘录 / EXCERPT: Let $L=[-1,-\beta]\cup\{\alpha_1,\ldots,\alpha_k\}$ for some fixed $\beta\in(0,1]$. Then there exists a constant $c_{\beta,k}$ such that any spherical L-code in $\mathbb{R}^n$ ha...

open_problem | open | MathDB 页面 | 原文线索

V2-078355 | MATHDB-400757

Maximum number of equiangular lines with a fixed angle

摘录 / EXCERPT: Theorem. Fix $\lpha \in (0,1)$. For n sufficiently large relative to $\lpha$, the number of equiangular lines in $\mathbb{R}^n$ with angle $\arccos \lpha$ is bounded by $2n-2$ if...

open_problem | open | MathDB 页面 | 原文线索

V2-078356 | MATHDB-400758

Conjecture 5 on fixed-angle equiangular lines

摘录 / EXCERPT: Let $r\ge 2$ be a positive integer. Then, for sufficiently large n, the maximum number $N_{1/(2r-1)}(n)$ of equiangular lines in $\mathbb{R}^n$ with common angle...

conjecture | open | MathDB 页面 | 原文线索

V2-078357 | MATHDB-400759

Quadratic bound for spherical $[-1,0)\cup\{\alpha\}$-codes

摘录 / EXCERPT: It is plausible to conjecture that spherical $[-1,0)\cup\{\alpha\}$-codes in $\mathbb{R}^n$ have size at most $O(n^2)$.

open_problem | open | MathDB 页面 | 原文线索

V2-078358 | MATHDB-400760

χ-boundedness of graphs excluding a fixed forest

摘录 / EXCERPT: For every forest H, there exists a function f_H such that $\c7(G) \le f_H(\omega(G))$ for every H-free graph G.

open_problem | open | MathDB 页面 | 原文线索

V2-078359 | MATHDB-400761

χ-boundedness of graphs with no long odd induced cycles

摘录 / EXCERPT: For every l, the class of graphs with no odd induced cycles of length at least l is $\b$-bounded.

open_problem | open | MathDB 页面 | 原文线索

V2-078360 | MATHDB-400762

Resilience version of the Littlewood-Offord problem

摘录 / EXCERPT: Let $a=(a_i)_{i=1}^n\\in(\\mathbb{R}\\setminus\\{0\\})^n$ and $x\\in\\mathbb{R}$. For independent identically distributed Rademacher signs $\\xi=(\\xi_i)_{i=1}^n$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-078361 | MATHDB-400767

Kalai’s supersaturation conjecture for vector-valued intersections

摘录 / EXCERPT: Let $[n]_{k,s}=\\{A\\subseteq[n]:|A|=k,\\sum_{i\\in A}i=s\\}$. For $A\\subseteq[n]_{k,s}$, define ... Kalai’s conjecture asserts that if...

conjecture | open | MathDB 页面 | 原文线索

V2-078362 | MATHDB-400768

Keevash-Long supersaturation theorem for vector-valued intersections

摘录 / EXCERPT: Let $0<n^{-1}\\ll\\delta\\ll\\varepsilon,\\alpha_1,\\alpha_2,\\beta_1,\\beta_2<1$, and set ... Let $X=[n]_{k,s}$ be the family of subsets $A\subseteq[n]$ with $|A|=k$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078363 | MATHDB-400770

Determine the decomposition threshold $c_{q,r}$

摘录 / EXCERPT: For integers $q>r\ge 2$, determine the supremum $c_{q,r}$ of all $c\in[0,1]$ with the following property: there exists $n_0\in\mathbb N$ such that, for every $n\ge n_0$, every...

open_problem | open | MathDB 页面 | 原文线索

V2-078364 | MATHDB-400772

Low-discrepancy high-dimensional permutations

摘录 / EXCERPT: We conjecture that permutations X exist such that for every box $B\subseteq [n]^{d+1}$, where $B=\prod_{i=1}^{d+1} A_i$ and $\operatorname{vol}(B)=\prod_{i=1}^{d+1}|A_i|$, one ha...

open_problem | open | MathDB 页面 | 原文线索

V2-078365 | MATHDB-400773

Four Latin squares suffice for bounded-degree 2-dimensional expansion

摘录 / EXCERPT: Conjecture: There exists an $ ... Here $h1$ is the first coboundary expansion constant over $mathbb F2$; equivalently, the conjecture asserts that Theorem 2 remains valid with $d=4...

open_problem | open | MathDB 页面 | 原文线索

V2-078366 | MATHDB-400777

Sylvester-Gallai theorem

摘录 / EXCERPT: Given a set P of n points in $\mathbb{R}^d$, if the points are not all collinear, then there is a line through exactly two of the points (an ordinary line). Equivalently, if ev...

open_problem | open | MathDB 页面 | 原文线索

V2-078367 | MATHDB-400778

Characterization of easily testable hereditary graph properties

摘录 / EXCERPT: Characterize the hereditary graph properties that are easily testable.

open_problem | open | MathDB 页面 | 原文线索

V2-078368 | MATHDB-400780

Regular graphs with smaller generalized Turán minimum degree

摘录 / EXCERPT: Determine those graphs F for which there exists an $\$psilon0$ such that, for all sufficiently large $Delta$, there exists a $Delta$-regular graph $G{Delta}...

open_problem | open | MathDB 页面 | 原文线索

V2-078369 | MATHDB-400781

Generalized Turán minimum-degree conjecture

摘录 / EXCERPT: If F and G are graphs and $\[1m\Delta(G)=\Delta\to\infty\[0m$, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-078370 | MATHDB-400782

Exact upper-tail exponent for arithmetic progressions

摘录 / EXCERPT: Fix $\delta>0$ and an integer $k t greater than or equal to 3$. There exists a constant $\alpha_k>0$ such that, whenever $p=p_n\to 0$ and $p greater than or equal to n^{-\alpha_k}$...

open_problem | open | MathDB 页面 | 原文线索

V2-078371 | MATHDB-400783

Degeneracy of a random half-subgraph

摘录 / EXCERPT: Let G be a graph of minimum degree D. Obtain a lower bound for the largest integer d such that the random subgraph $G_{1/2}$, formed by retaining each edge independently with...

open_problem | open | MathDB 页面 | 原文线索

V2-078372 | MATHDB-400785

Minimal dispersion of point sets in the unit cube

摘录 / EXCERPT: Improve the known lower and upper bounds for the minimal dispersion ... where $\mathcal B_{\mathrm{ax}}$ is the set of axis-parallel boxes inside $[0,1]^d$.

open_problem | open | MathDB 页面 | 原文线索

V2-078373 | MATHDB-400786

High-dimensional central limit theorem for volumes of random simplices

摘录 / EXCERPT: Let $1\le r\le d$, and let $X_0,X_1,\ldots,X_r$ be independent random points uniformly distributed in the normalized cube $[-\sqrt{3},\sqrt{3}]^d$. Let V_r be the r-volume of t...

open_problem | open | MathDB 页面 | 原文线索

V2-078374 | MATHDB-400788

Weak Thin Shell Conjecture

摘录 / EXCERPT: There exists $c>0$ such that for every log-concave random vector X, for all $p\ge 1$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-078375 | MATHDB-400789

Thin Shell Conjecture

摘录 / EXCERPT: There exists a universal constant $c>0$ such that, for every isotropic log-concave random vector X in $\mathbb{R}^n$ and every $t>0$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-078376 | MATHDB-400790

Optimal order of fixed-dimensional discrepancy norms

摘录 / EXCERPT: For fixed dimension d, what is the optimal order of the L^1- and L^∞-norms of the discrepancy function?

open_problem | open | MathDB 页面 | 原文线索

V2-078377 | MATHDB-400791

Improving tractability bounds for the star discrepancy

摘录 / EXCERPT: Improve either of the bounds ... or ... where the lower bound is stated for n at least proportional to d.

open_problem | open | MathDB 页面 | 原文线索

V2-078378 | MATHDB-400792

Dimension dependence of minimal dispersion on the unit cube

摘录 / EXCERPT: Improve the known upper bound ... for the dispersion on the unit cube. It is expected that an upper bound of the form ... holds for $n\ge d$.

open_problem | open | MathDB 页面 | 原文线索

V2-078379 | MATHDB-400793

Improve the lower bound for minimal dispersion on the cube

摘录 / EXCERPT: Problem 5. Improve the known lower bound ... for the dispersion on the cube.

open_problem | open | MathDB 页面 | 原文线索

V2-078380 | MATHDB-400795

Slicing problem: dimension-independent hyperplane-section bound

摘录 / EXCERPT: Does there exist a constant $C>0$, independent of the dimension d, such that for every convex body $K\subset\mathbb{R}^d$ with $V_d(K)=1$, one has ... where the maximum is over aff...

open_problem | open | MathDB 页面 | 原文线索

V2-078381 | MATHDB-400796

Lower bounds for the comparison constant of random simplex volumes

摘录 / EXCERPT: For each dimension d, define ... where $K,L\subset\mathbb{R}^d$ range over convex bodies with $K\subset L$, and P^K_{d+1} is the convex hull of $d+1$ independent uniformly random...

open_problem | open | MathDB 页面 | 原文线索

V2-078382 | MATHDB-400797

Asymptotic expected volumes of random simplices in the cube and cross-polytope

摘录 / EXCERPT: Can one compute the asymptotic behaviour of $\mathbb{E}V_d(P_{d+1}^{C_d})$ and $\mathbb{E}V_d(P_{d+1}^{C_1^d})$ as $d \to \infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-078383 | MATHDB-400798

Asymptotic expected volume of a random simplex in a regular simplex

摘录 / EXCERPT: Let T_d be a d-dimensional regular simplex, and let $P_{d+1}^{T_d}$ be the convex hull of $d+1$ independent uniformly distributed random points in T_d. Can one compute the as...

open_problem | open | MathDB 页面 | 原文线索

V2-078384 | MATHDB-400799

Central limit theorem for the volume of a random simplex in a regular simplex

摘录 / EXCERPT: Can one prove a central limit theorem for $V_d(P_{d+1}^{T_d})$ as $d \to \infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-078385 | MATHDB-400800

Central limit theorems for volumes of high-dimensional random simplices

摘录 / EXCERPT: Can one prove central limit theorems, as $d\to\infty$, for the volumes $V_d(P_{d+1}^{B_d})$, $V_d(P_{d+1}^{T_d})$, $V_d(P_{d+1}^{C_d})$, and $V_d(P_{d+1}^{C_d^1})$?

open_problem | open | MathDB 页面 | 原文线索

V2-078386 | MATHDB-400801

Asymptotics of intermediate face numbers of random 0-1 polytopes

摘录 / EXCERPT: Determine the asymptotic behaviour of $\mathbb{E}f_\ell(P_n^d)$ for $2\le \ell\le d-2$, where P_n^d is the convex hull of n independent random 0-1 points in $\{0,1\}^d$.

open_problem | open | MathDB 页面 | 原文线索

V2-078387 | MATHDB-400802

Optimal order of the minimal dispersion of axis-parallel boxes

摘录 / EXCERPT: Determine the optimal order, as a function of the number of points n and the dimension d, of the minimal dispersion $\operatorname{disp}_{\mathcal B_{\mathrm{ax}}}(n,d)$ of n...

open_problem | open | MathDB 页面 | 原文线索

V2-078388 | MATHDB-400803

CLT for the logarithmic volume of a full-dimensional random simplex in the cube

摘录 / EXCERPT: Prove that, for independent uniformly distributed random vectors $X_0,\\ldots,X_d$ in $C_d=[-\\sqrt{3},\\sqrt{3}]^d$, with $\\Sigma_d^d=\\operatorname{conv}\\{X_0,\\ldots,X_d\\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078389 | MATHDB-400806

Strongly surjective lexicographically ordered Suslin tree under \(\Diamond^+\)

摘录 / EXCERPT: Under $\Diamond^+$, there is a strongly surjective, lexicographically ordered Suslin tree.

open_problem | open | MathDB 页面 | 原文线索

V2-078390 | MATHDB-400807

Strong surjectivity after deleting one point

摘录 / EXCERPT: Suppose that L is strongly surjective and $x\in L$. Is $L\setminus\{x\}$ strongly surjective?

open_problem | open | MathDB 页面 | 原文线索

V2-078391 | MATHDB-400809

Strong surjectivity of Moore’s universal Aronszajn line

摘录 / EXCERPT: Is η_C strongly surjective under PFA?

open_problem | open | MathDB 页面 | 原文线索

V2-078392 | MATHDB-400810

Woodin’s AD-Conjecture for I0

摘录 / EXCERPT: Assume that $I0$ holds at $\bbda$. Then every subset of $V_{\bbda+1}$ has a $U(j)$-representation in $L(V_{\bbda+1})$, and hence the AD-Conjecture holds at $I0$.

conjecture | open | MathDB 页面 | 原文线索

V2-078393 | MATHDB-400812

Uniqueness of the canonical structure relative to a \Delta^1_3-scale

摘录 / EXCERPT: Let $\vec\varphi$ be a Δ^1_3-scale on a good universal Π^1_2 set such that each φ_i is $\boldsymbol{\Gamma}(<\omega_2\text{-}\Pi^1_1)$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-078394 | MATHDB-400813

Elementary equivalence of the candidate model with \(\mathrm{HOD}^{L[x]}\)

摘录 / EXCERPT: Let N and π^* be as defined in the preceding discussion. Is $L[N\mathbin{|}\delta_N,\pi^*]$ elementarily equivalent to $\mathrm{HOD}^{L[x]}$ for a cone of reals x?

open_problem | open | MathDB 页面 | 原文线索

V2-078395 | MATHDB-400814

Metrizability of universal minimal flows for homogeneous finite-relational structures

摘录 / EXCERPT: Is it the case that, if G is the automorphism group of a countable structure homogeneous in a finite relational language, then $M(G)$ is metrizable?

open_problem | open | MathDB 页面 | 原文线索

V2-078396 | MATHDB-400815

Absoluteness of countable chromaticity for graph colorings

摘录 / EXCERPT: For a graph G on a Polish space, is the statement $\chi(G)\leq\aleph_0$ absolute among models of ZFC containing a code for G?

open_problem | open | MathDB 页面 | 原文线索

V2-078397 | MATHDB-400816

Coloring number of locally countable graphs

摘录 / EXCERPT: Every locally countable graph G on a Polish space X has $\theta(G)\leq\aleph_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-078398 | MATHDB-400819

Does AD imply that there are no mad families?

摘录 / EXCERPT: Does AD imply that there are no mad families?

open_problem | open | MathDB 页面 | 原文线索

V2-078399 | MATHDB-400822

Chang’s Conjecture with gaps at successive alephs

摘录 / EXCERPT: Assume that (kappa) is supercompact. Is there a generic extension in which $(\aleph_{3},\aleph_{2}) \twoheadrightarrow(\aleph_{2},\aleph_{1})$?

conjecture | open | MathDB 页面 | 原文线索

V2-078400 | MATHDB-400823

Well-quasi-ordering of finitely generated subgroups of Thompson’s group F

摘录 / EXCERPT: Conjecture: If $(G_i)_{i<\\omega}$ is a sequence of finitely generated subgroups of Thompson’s group F, then there exist indices $i<j$ such that G_i embeds into G_j.

open_problem | open | MathDB 页面 | 原文线索

V2-078401 | MATHDB-400824

Bi-embeddability with a group admitting a fast generating set

摘录 / EXCERPT: Every subgroup of F is biembeddable with a group admitting a fast generating set.

open_problem | open | MathDB 页面 | 原文线索

V2-078402 | MATHDB-400825

Embedding finitely generated elementary amenable subgroups into the canonical chain

摘录 / EXCERPT: We conjecture that every finitely generated elementary amenable subgroup of F (or even of $\mathrm{PL}_+(I)$) embeds into G_ξ for some $\xi\in\varepsilon_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-078403 | MATHDB-400828

A universal equivalence relation induced by an isometric action

摘录 / EXCERPT: Does there exist a Polish metric space X and a closed subgroup G of the isometry group of X such that the equivalence relation induced by the action of G on X is universa...

open_problem | open | MathDB 页面 | 原文线索

V2-078404 | MATHDB-400829

Is the isometry group of the left-completion exactly the original group?

摘录 / EXCERPT: Let U_1 be the Urysohn space for metric spaces of diameter 1, let $G=\operatorname{Iso}(U_1)$, and let d be a left-invariant metric on G. If $\widehat G$ denotes the comple…

open_problem | open | MathDB 页面 | 原文线索

V2-078405 | MATHDB-400831

Consistency strength of an almost \a9\b1-reduction

摘录 / EXCERPT: What is the consistency strength of $E_{\c9_1}\leq^{a\b4^1_1} F_{\c9_1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-078406 | MATHDB-400832

A model with $d<a$ and restricted Martin’s axiom

摘录 / EXCERPT: Can one produce a model of $d<a$ while having Martin’s axiom for all c.c.c. posets of size $<d$?

open_problem | open | MathDB 页面 | 原文线索

V2-078407 | MATHDB-400834

Consistency of the strict inequalities $\mathfrak b<\mathfrak s<\mathfrak d$

摘录 / EXCERPT: Is $\mathfrak b<\mathfrak s<\mathfrak d$ consistent?

open_problem | open | MathDB 页面 | 原文线索

V2-078408 | MATHDB-400838

Consistency strength of the Baire property for sets in $L(\[?m\mathbb{R},\mathrm{uB})$

摘录 / EXCERPT: What is the consistency strength of the theory $\mathrm{ZFC}+$ “every set of reals in $L(\mathbb{R},\mathrm{uB})$ has the property of Baire”?

open_problem | open | MathDB 页面 | 原文线索

V2-078409 | MATHDB-400839

Equality of a plane-curve count and an Euler characteristic

摘录 / EXCERPT: The report states that the number of irreducible degree-d curves in the complex projective plane passing through $2d-1$ points in general position and tangent to order d at a s…

open_problem | open | MathDB 页面 | 原文线索

V2-078410 | MATHDB-400842

Euler characteristic formula for rational plane curves with maximal tangency

摘录 / EXCERPT: For every integer $d\ge 1$, let Z_d be the number of irreducible rational degree-d curves in the projective plane passing through $2d-1$ general points and tangent of order d…

open_problem | open | MathDB 页面 | 原文线索

V2-078411 | MATHDB-400843

A differential graded category realizing the higher cluster category

摘录 / EXCERPT: Is there a “nice” differential graded category $B_m^{(d)}$ such that $B_m^{(1)}=B_m$ (viewing B_m as a differential graded category having a single object) and…

open_problem | open | MathDB 页面 | 原文线索

V2-078412 | MATHDB-400845

Additive Deligne-Simpson problem for Fuchsian equations

摘录 / EXCERPT: Given points $a_1,\dots,a_p\in\mathbb{C}$ and conjugacy classes $C_0,C_1,\dots,C_p\subset M(n,\mathbb{C})$, determine whether there exists an irreducible Fuchsian differential equa…

open_problem | open | MathDB 页面 | 原文线索

V2-078413 | MATHDB-400846

Generalized additive Deligne-Simpson problem for equations

摘录 / EXCERPT: Given points $a_1,\ots,a_p\in\mathbb{C}$, nonzero positive integers $k_0,\ots,k_p$, and HTL normal forms $B^{(i)}\in\mathfrak{g}_{k_i}^*\subset M_n(\mathbb{C}((x)))$ for…

open_problem | open | MathDB 页面 | 原文线索

V2-078414 | MATHDB-400851

Relaxing the independence assumption in robust replication of barrier-style claims

摘录 / EXCERPT: In the frictionless market model described in the report, with $dS_t=\sigma_tS_t\,dW_t$, the volatility process σ is assumed to be independent of the Brownian motion W dri…

open_problem | open | MathDB 页面 | 原文线索

V2-078415 | MATHDB-400852

Necessary and sufficient conditions for multi-maturity \\-consistency

摘录 / EXCERPT: Find necessary and sufficient conditions for -consistency for option price data with more than one maturity.

open_problem | open | MathDB 页面 | 原文线索

V2-078416 | MATHDB-400856

Natural martingale between two marginals (Question 1)

摘录 / EXCERPT: Given two marginals $\mu \prec_c \nu$, does there exist a “natural” martingale $(M_t)_{t\in[0,1]}$ such that $M_0\sim\mu$ and $M_1\sim\nu$? The report says that the martingale is r…

open_problem | open | MathDB 页面 | 原文线索

V2-078417 | MATHDB-400858

Uniqueness of the viscosity solution of the HJB equation

摘录 / EXCERPT: Is the viscosity solution of equation (3) unique?

open_problem | open | MathDB 页面 | 原文线索

V2-078418 | MATHDB-400859

Renormalized approximation for rough volatility

摘录 / EXCERPT: For the rough-volatility model … with W and $\bar W$ independent Brownian motions, if W^ε are regular approximations to W (for example, piecewise-linear interpolati…

open_problem | open | MathDB 页面 | 原文线索

V2-078419 | MATHDB-400861

Best possible degree bounds in Hilbert’s 17th problem

摘录 / EXCERPT: Determine the best possible degree bounds, in terms of the number of variables k and $d=\deg P$, for the polynomials occurring in a representation of every nonnegative polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-078420 | MATHDB-400864

Explicit special-function formula for the expected degree of the real Grassmannian of lines

摘录 / EXCERPT: Is there an explicit formula for $\delta_{1,3}$ in terms of special functions, where $\delta_{1,3}=\mathbb{E}\#\bigl(\Omega(\ell_1)\cap\cdots\cap\Omega(\ell_4)\bigr)$ is the averag...

open_problem | open | MathDB 页面 | 原文线索

V2-078421 | MATHDB-400865

Asymptotics of the expected degree of real Grassmannians

摘录 / EXCERPT: More generally, one can define the expected degree $\delta_{k,n}$ of the Grassmannian of k-flats in $\mathbb{RP}^n$. For a fixed k, what is the asymptotic of $\delta_{k,n}$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-078422 | MATHDB-400866

Universal sharpness of the second relaxation for five particles on the sphere

摘录 / EXCERPT: We conjecture that E_2 is universally sharp for 5 particles on S^2.

open_problem | open | MathDB 页面 | 原文线索

V2-078423 | MATHDB-400869

Multidimensional moment determinacy and smallest Hankel-matrix eigenvalues

摘录 / EXCERPT: For a multidimensional moment problem, define ... Is it true that $\lambda_\infty=0$ characterizes determinacy of the multidimensional moment problem?

open_problem | open | MathDB 页面 | 原文线索

V2-078424 | MATHDB-400870

Characterization and order of regular symmetric indeterminate moment problems

摘录 / EXCERPT: Let $a_n=0$ and let (b_n) be regular, meaning that, for all sufficiently large n, either ... or ... Then the symmetric Hamburger moment problem is indeterminate if and only if ... In...

open_problem | open | MathDB 页面 | 原文线索

V2-078425 | MATHDB-400872

Every hyperbolicity cone is a spectrahedral shadow

摘录 / EXCERPT: Every hyperbolicity cone is a spectrahedral shadow.

open_problem | open | MathDB 页面 | 原文线索

V2-078426 | MATHDB-400873

Matrix-cube containment bound and optimal dilation factor

摘录 / EXCERPT: Let L_B be a $d\times d$ linear pencil, and suppose that $[-1,1]^g=S_A\subseteq S_B$. Then $D_A\subseteq \vartheta(d)D_B$, where ... Moreover, $\vartheta(d)$ is the smallest number...

open_problem | open | MathDB 页面 | 原文线索

V2-078427 | MATHDB-400875

Existence of Positive Definite Completions

摘录 / EXCERPT: Theorem 2.1. Let $G=([n],E)$ be a simple graph, and let m be the shortest length of a chordless cycle of G. A G-partial matrix $Q\in R_2$ has a positive definite completion i...

open_problem | open | MathDB 页面 | 原文线索

V2-078428 | MATHDB-400877

Solubility of the rank-8, nine-point sextic truncated moment problem

摘录 / EXCERPT: For $\operatorname{rank} M(3)=8$ and $\operatorname{card} V(\beta)=9$, find necessary and sufficient conditions for the existence of a representing measure for β.

open_problem | open | MathDB 页面 | 原文线索

V2-078429 | MATHDB-400878

Flat Extension Problem for Recursively Generated Moment Matrices

摘录 / EXCERPT: Given a recursively generated moment matrix $M(n)$, find necessary and sufficient conditions to guarantee that $M(n)$ admits a flat extension $M(n+1)$. (Source Problem 1.)

open_problem | open | MathDB 页面 | 原文线索

V2-078430 | MATHDB-400879

Solubility of the truncated moment problem on irreducible cubics

摘录 / EXCERPT: For each of the following five cases of irreducible cubics, characterize the solubility of the truncated moment problem (TMP) in terms of the initial data: 1. $y^2=x^3$; 2....

open_problem | open | MathDB 页面 | 原文线索

V2-078431 | MATHDB-400880

Infinite-dimensional moment problem

摘录 / EXCERPT: Given a sequence $m^{(n)}$ of elements of $(V')^{\otimes n}$ and a closed subset K of V', determine whether there exists a measure μ on K such that each $m^{(n)}$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-078432 | MATHDB-400882

Upper bound for the completely positive semidefinite rank

摘录 / EXCERPT: Does there exist an upper bound, depending only on the matrix size n, for the cpsd-rank of matrices in the cone CS_+^n?

open_problem | open | MathDB 页面 | 原文线索

V2-078433 | MATHDB-400884

Strong Van der Waerden conjecture for mixed characteristic polynomials

摘录 / EXCERPT: If $P\in D_{m,d}$, then $\lambda(P)\preceq\lambda(P_\infty)$, where $P_\infty:=\frac{(x_1+\cdots+x_m)^d}{m^d}$, $\lambda(P)$ is the weakly decreasing vector of zeros of the mixed c...

conjecture | open | MathDB 页面 | 原文线索

V2-078434 | MATHDB-400885

Discrete realization of the moment sequence \((3k)!\)

摘录 / EXCERPT: Find discrete random variables on $\mathbb{R}_{+}$ with moments $m_k=(3k)!$, $k=1,2,\ldots$.

open_problem | open | MathDB 页面 | 原文线索

V2-078435 | MATHDB-400887

Estimate the extension length needed for a flat extension

摘录 / EXCERPT: Given $\\\beta$, find a good estimate for the smallest value of k that would be required to achieve a flat extension M_{d+k+1}, where $M_d=M_d(\beta)$ is the moment matrix asso...

open_problem | open | MathDB 页面 | 原文线索

V2-078436 | MATHDB-400889

Concrete characterization of flat positive moment matrices

摘录 / EXCERPT: For $F=F_{n,d}$, the set of flat positive n-dimensional moment matrices of degree $2d$, give a concrete characterization of F.

open_problem | open | MathDB 页面 | 原文线索

V2-078437 | MATHDB-400890

Concrete positivity conditions for truncated Riesz functionals

摘录 / EXCERPT: Find concrete conditions on an n-dimensional real multisequence $\beta^{(m)}$ that ensure that its Riesz functional $L_\beta:P_m\to\mathbb{R}$ is positive.

open_problem | open | MathDB 页面 | 原文线索

V2-078438 | MATHDB-400892

Standard H^{∞} control design problem

摘录 / EXCERPT: Given the augmented plant system matrix G_a in (3), find a controller system matrix $[A_K\ B_K\ C_K\ D_K]:[X_K\ Y]\to[X_K\ U]$ such that the resulting closed-loop system is well-...

open_problem | open | MathDB 页面 | 原文线索

V2-078439 | MATHDB-400893

Robust control with respect to structured uncertainty

摘录 / EXCERPT: Given an augmented plant system G_a as in (3) and a controller K as in (4), design K so that the resulting closed-loop system (5) is well-posed and satisfies ... where Δ...

open_problem | open | MathDB 页面 | 原文线索

V2-078440 | MATHDB-400894

Inclusion problem for free loci of monic linear pencils

摘录 / EXCERPT: Let $L=I-\\sum_{j=1}^{g}A_jx_j$ and $\widetilde L=I-\\\sum_{j=1}^{g}\widetilde A_jx_j$ be monic linear pencils of sizes d and d', respectively, with $A_j\in M_d(\mathbb C)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078441 | MATHDB-400895

Linear independence of evaluation vectors at projective zeros of nonnegative forms

摘录 / EXCERPT: Let $f\in B_{n,2d}$ satisfy $f\ge 0$ on $\mathbb{R}^n$ and suppose that $Z_P(f)=\{z_1,\ldots,z_k\}$. Is the set $\{s_{B_{n,2d}}(z_i):i=1,\ldots,k\}$ linearly independent?

open_problem | open | MathDB 页面 | 原文线索

V2-078442 | MATHDB-400896

Higher-dimensional generalization of the Geronimo-Woerdeman theorem

摘录 / EXCERPT: Generalize the Geronimo-Woerdeman theorem to three or more variables.

open_problem | open | MathDB 页面 | 原文线索

V2-078443 | MATHDB-400897

Generalization of the Geronimo-Woerdeman theorem to polynomials nonvanishing on \(\mathbb T^2\)

摘录 / EXCERPT: Is it possible to generalize the Geronimo-Woerdeman theorem to p with no zeros on $\mathbb T^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-078444 | MATHDB-400898

Characterizing random discrete measures by moments

摘录 / EXCERPT: Given a random measure b5 on a locally compact Polish space X, is μ a random discrete measure, i.e. does $\mu(K(X))=1$?

open_problem | open | MathDB 页面 | 原文线索

V2-078445 | MATHDB-400899

Passivity criterion for stable perfectly matched layers

摘录 / EXCERPT: The report states the following result. Let ... where p,q are real-coefficient polynomials with $\deg p<\deg q$, and suppose that $\psi(s)^{-1}$ is passive. Assume also that...

open_problem | open | MathDB 页面 | 原文线索

V2-078446 | MATHDB-400900

Optimal minimax rate for the multi-reference alignment problem

摘录 / EXCERPT: Let $2\le s\le \lfloor d/2\rfloor$, and let T_s be the set of vectors $\theta\in T$ such that $\operatorname{supp}(\widehat\theta)\subseteq\{-s,\ldots,s\}$. Then ... where the infi...

open_problem | open | MathDB 页面 | 原文线索

V2-078447 | MATHDB-400903

Local minimax critical radius for goodness-of-fit testing of Lipschitz densities

摘录 / EXCERPT: For a domain $X\subset\mathbb{R}^d$, let $\mathcal{L}$ be the class of densities satisfying $\int_X p(x)\,dx=1$, $p(x)\ge 0$, and $|p(x)-p(y)|\le L_n|x-y|_2$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-078448 | MATHDB-400904

Non-parametric identification of causal effects with unobserved variables

摘录 / EXCERPT: Given a joint distribution $p(x_H,x_V)$ over observed variables indexed by V and hidden variables indexed by H, where the distribution factors according to a DAG G as ... is th...

open_problem | open | MathDB 页面 | 原文线索

V2-078449 | MATHDB-400905

Optimality of adaptive weights clustering for detecting density gaps

摘录 / EXCERPT: Theorem 1. (1) Let the data support V contain a fixed hole G, and let the data density $f(\cdot)$ equal f_1 on $V\setminus G$ and $f_0=(1-\varepsilon_N)f_1$ on G. If...

open_problem | open | MathDB 页面 | 原文线索

V2-078450 | MATHDB-400906

Bayes formula for support-boundary recovery

摘录 / EXCERPT: In the Poisson support-boundary model described in the report, under the frequentist distribution P_{f_0}, the posterior is represented P_{f_0}-almost surely by ...

open_problem | open | MathDB 页面 | 原文线索

V2-078451 | MATHDB-400907

Automatic Bayesian correction of the MLE in support-boundary recovery

摘录 / EXCERPT: In the Poisson support-boundary model with intensity $\lambda_f(x,y)=n\mathbf{1}\{f(x)\le y\}$, does a Bayesian procedure automatically correct the nonparametric MLE for estimating...

open_problem | open | MathDB 页面 | 原文线索

V2-078452 | MATHDB-400911

Sharpness of the stability exponent α in diffusion-coefficient recovery

摘录 / EXCERPT: Are the values found for α sharp? When $d=1$, the explicit expression of the solution u_a shows that $\alpha=\frac13$ and cannot be improved. When a,b are piecewise cons...

open_problem | open | MathDB 页面 | 原文线索

V2-078453 | MATHDB-400914

Negative subclasses and Breakdown-epsilon for regularisation problems

摘录 / EXCERPT: Question 3(iii): Which subclasses Ω of inputs give negative answers to Question 1, and what is the Breakdown-epsilon?

open_problem | open | MathDB 页面 | 原文线索

V2-078454 | MATHDB-400916

Highly localized summability kernels on the sphere induced by Newtonian kernels: Problem 1

摘录 / EXCERPT: For arbitrary constants $M>0$ (large) and $\\varepsilon>0$ (small), construct a function $\\Phi_\\varepsilon(\\eta,x)$ on $S^{d-1}\\times B^d$ of the form ... or ... with $|y_j|>1$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078455 | MATHDB-400917

Localized summability kernel: Problem 2

摘录 / EXCERPT: Let $M>d-1$. For a given $\\varepsilon\in(0,1]$, find $m+1$ constants $b_\ell\in\mathbb{R}$ and $a>1$ such that the restriction $F_\varepsilon(x\cdot\eta)$ to S^{d-1} of ... when...

open_problem | open | MathDB 页面 | 原文线索

V2-078456 | MATHDB-400919

Multiplicity-one decomposition of cuspidal automorphic forms for $G=\mathrm{GL}_d$

摘录 / EXCERPT: In the setting of the report, with $G=\mathrm{GL}_d$ over the function field $F=k(X)$ and $\chi\in\chi_G$, the space $\mathcal A_{G,\mathrm{cusp},\chi}$ of cuspidal automorphic for...

open_problem | open | MathDB 页面 | 原文线索

V2-078457 | MATHDB-400920

Subquadratic cutting of lines to eliminate above/below cycles

摘录 / EXCERPT: What is the minimum number of pieces into which one must cut, in the worst case, a collection of N non-vertical lines in general position in $\mathbb{R}^3$ so that the resulting...

open_problem | open | MathDB 页面 | 原文线索

V2-078458 | MATHDB-400922

Conjecture on transversals for line families with the $(p,3)$-property

摘录 / EXCERPT: Suppose $p\ge 3$ is an integer and a family of straight lines $\mathcal F$ has the $(p,3)$-property. Then there exists a point set U such that $|U|\le p-2$ and...

conjecture | open | MathDB 页面 | 原文线索

V2-078459 | MATHDB-400923

Maximum number of almost-similar triangles for almost regular targets

摘录 / EXCERPT: Let T be a triangle whose angles all lie between 59° and 61°. For a fixed $\varepsilon>0$, call a triangle ε-similar to T if, after matching the angle...

open_problem | open | MathDB 页面 | 原文线索

V2-078460 | MATHDB-400926

Bounds on the number of flats spanned by a point set

摘录 / EXCERPT: Let P be a set of n points in $\mathbb{R}^d$. A k-flat is spanned by P if it contains $k+1$ affinely independent points of P, and let f_k be the number of k-flats spa...

open_problem | open | MathDB 页面 | 原文线索

V2-078461 | MATHDB-400932

Structure of almost all string graphs

摘录 / EXCERPT: Janson and Uzzell conjectured that almost all string graphs on n vertices can be partitioned into five cliques such that two of the cliques are not connected by any edge. The rep...

open_problem | open | MathDB 页面 | 原文线索

V2-078462 | MATHDB-400933

Upper bound for colorful simplicial depth

摘录 / EXCERPT: Let $C=\\{C_0,\\ldots,C_d\\}$ be a centered colorful configuration in $\\mathbb{R}^d$, meaning that $0\\in\\operatorname{relint}\\operatorname{conv}(C_i)$ for every i, and suppos...

open_problem | open | MathDB 页面 | 原文线索

V2-078463 | MATHDB-400934

Upper bound for totally mixed facets of Minkowski sums of triangles

摘录 / EXCERPT: For every integer $d\ge 2$, let $P_0,\ldots,P_{d-2}\subset\mathbb{R}^d$ be triangles whose Minkowski sum $P_0+\cdots+P_{d-2}$ is full-dimensional. Then the number of totally mixed...

open_problem | open | MathDB 页面 | 原文线索

V2-078464 | MATHDB-400935

Three-coloring homothets of a plane convex set

摘录 / EXCERPT: For every plane convex set D there is an m such that any finite set of points admits a 3-coloring such that any homothetic copy of D that contains at least m points conta...

open_problem | open | MathDB 页面 | 原文线索

V2-078465 | MATHDB-400936

Four-coloring homothets of convex sets in three dimensions

摘录 / EXCERPT: For every convex set $D \subset R^3$ there is an m such that any finite set of points admits a 4-coloring such that any homothetic copy of D that contains at least m points...

open_problem | open | MathDB 页面 | 原文线索

V2-078466 | MATHDB-400938

Minkowski-sum conjecture for lattice-polytope diameters

摘录 / EXCERPT: For any d and k, $\[4m\delta(d,k)\u001b[0m$ is achieved up to translation by a Minkowski sum of lattice vectors.

conjecture | open | MathDB 页面 | 原文线索

V2-078467 | MATHDB-400939

Rationality of nested tetrahedra in three dimensions

摘录 / EXCERPT: Given nested rational polytopes $P\subset Q\subset\mathbb{R}^3$ (with P not necessarily full-dimensional), if there exists a tetrahedron Δ with...

open_problem | open | MathDB 页面 | 原文线索

V2-078468 | MATHDB-400942

Maximum crossing number and convex straight-line drawings

摘录 / EXCERPT: Alpert, Feder, and Harborth conjectured that every graph has a convex straight-line drawing-one with all vertices in convex position-that maximizes the number of edge crossings.

open_problem | open | MathDB 页面 | 原文线索

V2-078469 | MATHDB-400943

A superlinear lower bound on the number of 5-holes

摘录 / EXCERPT: Let P be a finite set of points in the plane in general position, meaning that no three points of P are collinear. A set H of k points from P is a k-hole in P if H...

open_problem | open | MathDB 页面 | 原文线索

V2-078470 | MATHDB-400944

Holmsen-Kynčl-Valculescu conjecture on colored point-set islands

摘录 / EXCERPT: Let X be an m-colored set of $|X|=kn$ points in general position in $\[0m\mathbb{R}^d$, which has a partition into n d-colorful k-subsets, where “d-colorful” means that...

conjecture | open | MathDB 页面 | 原文线索

V2-078471 | MATHDB-400945

Translative covering of the unit ball by slabs

摘录 / EXCERPT: For $d>2$, is it true that there exists a constant $C=C(d)$ such that, for any $n_1,\\ldots,n_s\in S^{d-1}$ and any $\varepsilon_1,\ldots,\varepsilon_s\in\mathbb{R}_{+}$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-078472 | MATHDB-400946

Non-simplex facets of even-dimensional polytopes

摘录 / EXCERPT: Fix even d and let $n\to\infty$. Must d-polytopes with n vertices have only $o(n^{d/2})$ non-simplex facets?

open_problem | open | MathDB 页面 | 原文线索

V2-078473 | MATHDB-400947

χ-boundedness of intersection graphs of lines in three-dimensional space

摘录 / EXCERPT: Is the class of intersection graphs of lines in $\mathbb{R}^3$ (or $\mathbb{R}^d$) χ-bounded? Namely, is there a function f such that, given n lines in $\mathbb{R}^3$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-078474 | MATHDB-400949

Expected size of the maximum agreement subtree for random binary trees

摘录 / EXCERPT: Conjecture 2. For either the uniform distribution or the Yule-Harding distribution on binary trees, ...

open_problem | open | MathDB 页面 | 原文线索

V2-078475 | MATHDB-400950

A test statistic for log-concavity based on the optimal subdivision

摘录 / EXCERPT: Design a test-statistic for log-concavity based on the optimal subdivision Δ.

open_problem | open | MathDB 页面 | 原文线索

V2-078476 | MATHDB-400951

Determining the minimum configuration size for many cells

摘录 / EXCERPT: What is the smallest size $n(c,d)$ such that there exists a configuration X in $\bb R^d$ whose optimal subdivision with unit weights has at least c cells?

open_problem | open | MathDB 页面 | 原文线索

V2-078477 | MATHDB-400952

Unique maximum-likelihood estimator for multivariate Gaussian MTP_2 models

摘录 / EXCERPT: The report states that, for a multivariate Gaussian distribution in the MTP_2 model, the maximum-likelihood problem is a convex optimization problem with a unique solution whenever...

open_problem | open | MathDB 页面 | 原文线索

V2-078478 | MATHDB-400953

Simple description of the image in the general positive semidefinite completion problem

摘录 / EXCERPT: The general positive semidefinite completion problem asks for a simple description of the image $\pi(S)$, where S is the spectrahedron under discussion and π is the coordinat...

open_problem | open | MathDB 页面 | 原文线索

V2-078479 | MATHDB-400959

Tropical principal component optimization in the space of ultrametrics

摘录 / EXCERPT: Find a solution to the optimization problem ... where ... and ... with $d_i=(d_i(1),\ldots,d_i(e))$ and $d'_i=(d'_i(1),\ldots,d'_i(e))$. Equivalently, determine the solution of the formu...

open_problem | open | MathDB 页面 | 原文线索

V2-078480 | MATHDB-400963

Membership in the likelihood correspondence

摘录 / EXCERPT: Question 1: Is the point $z := (p,u)$ in the likelihood correspondence L_X? The report answers this membership question using the witness-set criterion: for...

open_problem | open | MathDB 页面 | 原文线索

V2-078481 | MATHDB-400964

Propagating polynomial inequalities

摘录 / EXCERPT: Do recent finiteness results for propagating polynomial equations have analogues for propagating polynomial inequalities?

open_problem | open | MathDB 页面 | 原文线索

V2-078482 | MATHDB-400965

Extension of the mixture-model phase transition to overfitted hidden Markov models

摘录 / EXCERPT: Can the phase-transition result for overfitted finite mixtures-where Dirichlet-prior hyperparameters determine whether extra components are emptied or merged-be extended to finite-...

open_problem | open | MathDB 页面 | 原文线索

V2-078483 | MATHDB-400968

Convexity of the non-completable partial-tensor set

摘录 / EXCERPT: Fix integers $n\ge 1$ and $d_1,\ldots,d_n$, and an observed-index set $E\subseteq [d_1]\times\cdots\times[d_n]$, in the full-dimensional setting considered in the report, with...

open_problem | open | MathDB 页面 | 原文线索

V2-078484 | MATHDB-400969

Degree of the hypersurface in rank-one tensor completion

摘录 / EXCERPT: Let $E\subseteq [d_1]\times\cdots\times[d_n]$ index the observed entries of a partial tensor, and consider partial tensors that may be completed to...

open_problem | open | MathDB 页面 | 原文线索

V2-078485 | MATHDB-400984

Bethe Ansatz equations for roots of Baxter polynomials

摘录 / EXCERPT: For each index $i\in I$ and each (generic) root w of the Baxter polynomial $Q_{i,k}$, the Bethe Ansatz equation is ... where the v_i are the parameters of the twisted trace. The...

open_problem | open | MathDB 页面 | 原文线索

V2-078486 | MATHDB-400986

Infinite-product formula for the Kac-Moody Macdonald factor

摘录 / EXCERPT: Find an infinite-product expression, in the general Kac-Moody case, for the factor $m=W(q^{-1})\sum_{w\in W}w\Delta$ appearing in the Macdonald formula, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078487 | MATHDB-400988

Wilkie’s conjecture for sets definable in ${\mathbb R}_{\exp}$

摘录 / EXCERPT: The Wilkie conjecture asks whether, for every set A definable in ${\mathbb R}_{\exp}$, the number of rational points in the transcendental part of A of height at most H can b...

conjecture | open | MathDB 页面 | 原文线索

V2-078488 | MATHDB-400989

Volume inequalities in arbitrary structures

摘录 / EXCERPT: Conjecture: The inequalities analogous to those in Theorem 2 (volumes of images) and Theorem 3 (volumes of pre-images) hold in arbitrary structures.

open_problem | open | MathDB 页面 | 原文线索

V2-078489 | MATHDB-400990

Effective Pila-Wilkie counting for semi-Noetherian sets

摘录 / EXCERPT: Let $X\subset\Omega$ be a semi-Noetherian set and let $\varepsilon>0$. There exist constants d,N, explicitly estimated in terms of the Noetherian parameters, with the following p...

open_problem | open | MathDB 页面 | 原文线索

V2-078490 | MATHDB-400992

Recognition of smooth and ultradifferentiable functions on closed sets

摘录 / EXCERPT: When do we have $A^{\infty}(X)=C^{\infty}(X)$ and $A_M(X)=C_M(X)$?

open_problem | open | MathDB 页面 | 原文线索

V2-078491 | MATHDB-400994

Finiteness of CM specializations with an endomorphism relation

摘录 / EXCERPT: Let $n\ge 2$, let E_λ be the Legendre family $Y^2=X(X-1)(X-\lambda)$, and let $C\subseteq\mathbb A^{2n+1}$ be a curve defined over a number field k, with coordinate funct...

open_problem | open | MathDB 页面 | 原文线索

V2-078492 | MATHDB-400995

Finiteness of Pellian specializations of $D_t(X)=X^6+X+t$

摘录 / EXCERPT: There are at most finitely many $t\in\mathbb{C}$ for which there exist polynomials $A,B\in\mathbb{C}[X]$ with $B\neq 0$ satisfying ...

open_problem | open | MathDB 页面 | 原文线索

V2-078493 | MATHDB-400997

An o-minimal Szemerédi-Trotter theorem

摘录 / EXCERPT: Fix an o-minimal structure over a real closed field R. Let V be a definable subset of $P\times P^*$, where P and P^* are definable sets of dimension at most two. Then one o...

open_problem | open | MathDB 页面 | 原文线索

V2-078494 | MATHDB-400998

Definable Katětov-Tong insertion theorem

摘录 / EXCERPT: Let $f:A\to\mathbb{R}\cup\{-\infty\}$ and $g:A\to\mathbb{R}\cup\{+\infty\}$ be two definable functions such that $f(x)\leq g(x)$ for each $x\in A$, with f upper semicontinuous an...

open_problem | open | MathDB 页面 | 原文线索

V2-078495 | MATHDB-400999

Michael’s theorem for Lipschitz cells in o-minimal structures

摘录 / EXCERPT: Let R be a real closed field equipped with an o-minimal expansion. Let $F\colon A\Rightarrow R^m$ be a definable, lower semicontinuous multivalued mapping on a definable subset...

open_problem | open | MathDB 页面 | 原文线索

V2-078496 | MATHDB-401000

O-minimality of families of non-oscillating trajectories

摘录 / EXCERPT: Let X be a real analytic vector field with a singular point at $0\in\mathbb{R}^n$. In which conditions is the expansion of $\mathbb{R}_{\mathrm{an}}$ by a family of non-oscillati...

open_problem | open | MathDB 页面 | 原文线索

V2-078497 | MATHDB-401001

O-minimality of trajectories in a separated three-dimensional pencil

摘录 / EXCERPT: Given a non-oscillating trajectory γ of a real analytic vector field at the origin of $\mathbb{R}^3$, with γ belonging to a separated integral pencil, is the expansio...

open_problem | open | MathDB 页面 | 原文线索

V2-078498 | MATHDB-401002

Bi-Lipschitz classification of definable two-dimensional surface germs

摘录 / EXCERPT: Let X be a germ at the origin of $\b7R^n$ of a two-dimensional surface definable in a polynomially bounded o-minimal structure over $\b7R$, with the outer metric induced from...

open_problem | open | MathDB 页面 | 原文线索

V2-078499 | MATHDB-401003

Monotone functions and maps: Conjecture 2

摘录 / EXCERPT: Fix an o-minimal structure over $\mathbb{R}$, and assume that all sets, functions, and maps are definable in this structure. Let $K\subset\mathbb{R}^n$ be compact and let...

conjecture | open | MathDB 页面 | 原文线索

V2-078500 | MATHDB-401004

Definable coding of P-bound sets by subsets of powers of P

摘录 / EXCERPT: Is every P-bound set in definable bijection with a subset of P^n, for some n?

open_problem | open | MathDB 页面 | 原文线索

V2-078501 | MATHDB-401005

Definable coding of P-bound sets

摘录 / EXCERPT: For the three classes considered-dense pairs, expansions by a dense independent set, and expansions of a real closed field by a dense divisible multiplicative subgroup with the Man...

open_problem | open | MathDB 页面 | 原文线索

V2-078502 | MATHDB-401006

Equality of the lattice and dual-lattice $ u$-functions under a matrix criterion

摘录 / EXCERPT: Let $\Lambda=A\mathbb Z^n$ be a unimodular lattice, and let $\Lambda^\perp$ denote its dual with respect to the standard inner product. For $\rho>\gamma_n^{1/2}$, define ... Suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-078503 | MATHDB-401007

Sharpness of the weakly admissible lattice-point discrepancy bound

摘录 / EXCERPT: For an aligned box $S=\prod_{i=1}^n[y_i,y_i+s_i]\subset\mathbb{R}^n$, let $s=(s_1\cdots s_n)^{1/n}$ and let $s_{\max}=\max_i s_i$. For a weakly admissible lattice Λ, define...

open_problem | open | MathDB 页面 | 原文线索

V2-078504 | MATHDB-401008

Families with Property (UTB)

摘录 / EXCERPT: Find “large”, “interesting” families $\mathcal F$ of bounded subsets of $\mathbb R^n$ with Property (UTB): there exist $\kappa>0$ and $M\in\mathbb N$ such that, whenever...

open_problem | open | MathDB 页面 | 原文线索

V2-078505 | MATHDB-401009

Manin’s conjecture for del Pezzo surfaces over number fields

摘录 / EXCERPT: Let X be a del Pezzo surface over a number field K (a projective surface with ample anticanonical divisor and at most simple singularities), let H be an anticanonical height...

conjecture | open | MathDB 页面 | 原文线索

V2-078506 | MATHDB-401013

Polylogarithmic rational-point bounds for definable sets involving log-spirals

摘录 / EXCERPT: Do we have a $\log^{\beta} T$-bound on $\#\Gamma(\mathbb{Q},T)$ for sets definable in the expansion of $\mathbb{R}$ by any log-spiral and by restricted sine and exponential functio...

open_problem | open | MathDB 页面 | 原文线索

V2-078507 | MATHDB-401016

Global well-posedness of the primitive equations in the L^p setting

摘录 / EXCERPT: Let $p\in(1,\infty)$ and $a\in V_{1/p}$. Then there exists a unique solution of the primitive equations (1), subject to the boundary conditions stated on the neighboring page, such...

open_problem | open | MathDB 页面 | 原文线索

V2-078508 | MATHDB-401017

Regularizing property of the Tikhonov functional for inverse medium scattering

摘录 / EXCERPT: For the fixed-energy inverse Schrödinger problem described in the report, with potentials in the support-constrained set K, nonnegative imaginary part, and the stated L^∞...

open_problem | open | MathDB 页面 | 原文线索

V2-078509 | MATHDB-401018

Explicit variational source condition for inverse medium scattering

摘录 / EXCERPT: Let $E\ge 1$, $s>0$, and $1<p\le 2$. If $q^\dagger\in K$ satisfies $\|q^\dagger\|_{L^\infty}\le C_\infty$ and $\|q^\dagger\|_{B^s_{p,\infty}}\le C_s$, then there is a constant...

open_problem | open | MathDB 页面 | 原文线索

V2-078510 | MATHDB-401019

Well-posedness of linearized near-field phase retrieval under a support constraint

摘录 / EXCERPT: Let the support domain be the stripe $\Omega=[-1/2,1/2]\times\mathbb{R}\subset\mathbb{R}^2$. For the linearized forward model $I=1+T h$, where $T(h)=2\operatorname{Re}(D(h))$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-078511 | MATHDB-401021

Monotonicity characterization of a Helmholtz scatterer

摘录 / EXCERPT: Let 4Omegasubseteqmathbb{R}^nu00024 be a bounded domain with smooth boundary, 4nge 2u00024, and let DsubsetOmegau0004 be open with overline DsubsetOmegau0004 and Omegasetminus Du00...

open_problem | open | MathDB 页面 | 原文线索

V2-078512 | MATHDB-401022

Number of frequencies needed for reconstruction from covariance data

摘录 / EXCERPT: Starting from a reference solar model, determine how many frequencies of measurements of the covariance function C, rather than the Green function G, are required to reconstruc...

open_problem | open | MathDB 页面 | 原文线索

V2-078513 | MATHDB-401023

Precise frequency-scale relation for full-waveform inversion

摘录 / EXCERPT: Establish a precise analytical relation between frequency and model scale that follows from the stability and convergence analyses for the time-harmonic full-waveform inversion alg...

open_problem | open | MathDB 页面 | 原文线索

V2-078514 | MATHDB-401024

Unique reconstruction in inverse homogenization

摘录 / EXCERPT: The report states the following solved result: A determinate matrix-valued spectral measure $[U+007F]mu$, together with its moments as in equation (5), can be uniquely reconstructed if th...

open_problem | open | MathDB 页面 | 原文线索

V2-078515 | MATHDB-401025

Theorem 3: Local uniqueness for (γ,σ,μ) in two-photon photoacoustic tomography

摘录 / EXCERPT: In dimension $d=3$, assuming the Grüneisen coefficient Γ is known and the background coefficients are sufficiently smooth, there exists a set of $J\ge d+1$ boundary illumina...

open_problem | open | MathDB 页面 | 原文线索

V2-078516 | MATHDB-401026

Gan-Gross-Prasad conjecture for tempered orthogonal or unitary representations

摘录 / EXCERPT: In the special case of tempered representations Π,π of orthogonal or unitary groups, let $G=G_n\times G_{n+1}$ and let $H=G_n$ be diagonally embedded, where G_n is either a...

conjecture | open | MathDB 页面 | 原文线索

V2-078517 | MATHDB-401027

Arthur’s conjecture for the automorphic discrete spectrum of G_2

摘录 / EXCERPT: A detailed study of the stable twisted trace formula associated with $(PGSO_8,\theta)$ is hoped to yield Arthur’s conjecture on the automorphic discrete spectrum of G_2.

conjecture | open | MathDB 页面 | 原文线索

V2-078518 | MATHDB-401030

Nonvanishing of central Rankin-Selberg values in higher-rank families

摘录 / EXCERPT: Let $1\le m<n$ be two positive integers. Given a cuspidal automorphic representation Π on $\mathrm{GL}(n,\mathbb{Q})\backslash\mathrm{GL}(n,\mathbb{A})$, do there exist infinit...

open_problem | open | MathDB 页面 | 原文线索

V2-078519 | MATHDB-401031

Nonvanishing of central Rankin-Selberg values for GL(4) × GL(2)

摘录 / EXCERPT: Let Π be a cuspidal automorphic representation on $\mathrm{GL}(4,\mathbb{Q})\backslash\mathrm{GL}(4,\mathbb{A})$ which is self-dual and unramified at all finite places. Then th...

open_problem | open | MathDB 页面 | 原文线索

V2-078520 | MATHDB-401035

Geometric criteria for finite and bounded branching multiplicities

摘录 / EXCERPT: Let G and G' be the groups appearing in the branching problem. The report states: 1. $\dim\operatorname{Hom}_{G'}(\pi|_{G'},\tau)<\infty$ for any irreducible representations...

open_problem | open | MathDB 页面 | 原文线索

V2-078521 | MATHDB-401039

Theorem 3: Character integral formula for distinction multiplicities

摘录 / EXCERPT: Assume that F is a p-adic field, E/F is a quadratic extension, H is a connected reductive group over F that is assumed here to be semisimple, and...

open_problem | open | MathDB 页面 | 原文线索

V2-078522 | MATHDB-401042

Rationality for critical values of Rankin-Selberg L-functions

摘录 / EXCERPT: Let F be a CM-field with maximal totally real subfield F^+ and $[F^+ : \mathbb{Q}]=d\ge 1$. Let Π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_F)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078523 | MATHDB-401043

Braverman-Kazhdan generalized Poisson summation conjecture

摘录 / EXCERPT: For a representation $r:{}^{L}G\to\mathrm{GL}(V_r)$, does there exist an r-transform $F_r:\mathcal S_r(G(\mathbb A_F))\to\mathcal S_r(G(\mathbb A_F))$ on a suitable conjectural S...

conjecture | open | MathDB 页面 | 原文线索

V2-078524 | MATHDB-401044

Multiplicity-one conjecture for real Arthur packets

摘录 / EXCERPT: Let $F=\mathbb{R}$, let G_0 be a quasi-split classical group and let G be a pure inner form of G_0. For an A-morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-078525 | MATHDB-401045

Multiplicity-one conjecture for Arthur packets

摘录 / EXCERPT: Let F be a local field of characteristic 0, let G_0 be a quasi-split classical group, and let G be a pure inner form of G_0. For an A-morphism...

conjecture | open | MathDB 页面 | 原文线索

V2-078526 | MATHDB-401046

Equality of Arthur and Adams-Barbasch-Vogan A-packets

摘录 / EXCERPT: Are Arthur’s A-packets in [Art13] equal to A-packets of [ABV92]?

open_problem | open | MathDB 页面 | 原文线索

V2-078527 | MATHDB-401048

Arthur packets for pure rational forms

摘录 / EXCERPT: For every Arthur parameter ψ of G, there is a finite subset $\Pi^{\mathrm{pure},\psi}(G/F)$ of $\Pi^{\mathrm{pure}}(G/F)$, together with a canonical map ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-078528 | MATHDB-401051

Iwaniec-Sarnak sup-norm conjecture for compact hyperbolic surfaces

摘录 / EXCERPT: For every compact hyperbolic surface Y, the bound ... should hold for all L^2-normalized Laplace eigenfunctions ψ_i on Y, where $(\Delta+\lambda_i^2)\psi_i=0$.

conjecture | open | MathDB 页面 | 原文线索

V2-078529 | MATHDB-401055

Local Ginzburg-Rallis multiplicity conjecture

摘录 / EXCERPT: Let F be a local field of characteristic zero, let $G(F)=\mathrm{GL}_6(F)$, and let χ be a character of $F^{\times}$. For an irreducible admissible generic representation...

conjecture | open | MathDB 页面 | 原文线索

V2-078530 | MATHDB-401057

Existence of a single geometric wave-front orbit

摘录 / EXCERPT: Let F be a finite extension of $\mathbb{Q}_p$, let G be a connected reductive group over F, and let π be an admissible irreducible representation of $G(F)$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-078531 | MATHDB-401058

Wave-front-set theorem for Aubert-Zelevinsky duals of unipotent tempered representations

摘录 / EXCERPT: Let $G=\mathrm{SO}(2n+1)$, allowing both its split and nonsplit forms, and assume that the residue characteristic p is sufficiently large relative to n. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-078532 | MATHDB-401060

Spectral dimension of causal random triangulations

摘录 / EXCERPT: The spectral dimension of causal random triangulations is 2.

open_problem | open | MathDB 页面 | 原文线索

V2-078533 | MATHDB-401061

Generalized OSSS inequality for monotonic measures

摘录 / EXCERPT: Theorem 1 ([DRT17a]). Let E be finite and let $f:\{0,1\}^E\to[0,1]$ be increasing. A measure μ on $\{0,1\}^E$ is monotonic if, for every $e\in E$, $F\subset E$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-078534 | MATHDB-401063

Joint Brownian-sphere limit for a random planar map and its dual

摘录 / EXCERPT: For an admissible and critical weight sequence satisfying ... it is conjectured that the random planar map M_n and its dual map $M_n^\dagger$, after rescaling, converge jointly to...

open_problem | open | MathDB 页面 | 原文线索

V2-078535 | MATHDB-401065

Anisotropic KPZ height-fluctuation conjecture

摘录 / EXCERPT: For a two-dimensional growth model in the stationary state, if the Hessian of the speed function $v(\cdot)$ has signature $(+,-)$ (the anisotropic KPZ class), then asymptotically t...

conjecture | open | MathDB 页面 | 原文线索

V2-078536 | MATHDB-401066

Conjectural parameter relation between the critical $O(n)$ model and $\textrm{CLE}_\kappa$

摘录 / EXCERPT: A well-known conjectural relation between the critical $O(n)$ model and the $\textrm{CLE}_\kappa$ parameters states that ... where $\kappa\in(8/3,4]$ corresponds to the dilute phase...

open_problem | open | MathDB 页面 | 原文线索

V2-078537 | MATHDB-401068

Small-data global well-posedness and scattering for the non-resonant massive Dirac-Klein-Gordon system

摘录 / EXCERPT: Consider the Dirac-Klein-Gordon system ... for a spinor $\psi:\mathbb{R}^{1+3}\to\mathbb{C}^4$ and a real scalar field $\phi:\mathbb{R}^{1+3}\to\mathbb{R}$. If $0<m<2M$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078538 | MATHDB-401070

Finite-time singularities for SQG evolution

摘录 / EXCERPT: Can the SQG evolution lead to finite-time singularities in the smooth case (1) and in the patch case (2)?

open_problem | open | MathDB 页面 | 原文线索

V2-078539 | MATHDB-401071

Global Attractor Conjecture for complex-balanced reaction systems

摘录 / EXCERPT: The global attractor conjecture (Craciun) states that, for the so-called complex-balanced systems, all trajectories converge to a unique stable equilibrium point.

conjecture | open | MathDB 页面 | 原文线索

V2-078540 | MATHDB-401072

Maximal number of positive steady states in the general sequential phosphorylation mechanism

摘录 / EXCERPT: The maximal possible number of positive steady states for the general n-site sequential phosphorylation mechanism is still unknown.

open_problem | open | MathDB 页面 | 原文线索

V2-078541 | MATHDB-401074

Existence of a ground state at the endpoint mass for the critical exponent $p=4$ on the two-dimensional grid

摘录 / EXCERPT: For the NLS energy functional on the two-dimensional grid G_ℓ of edge length $\ell>0$, does a ground state exist when $p=4$ and the prescribed mass equals the threshold value...

open_problem | open | MathDB 页面 | 原文线索

V2-078542 | MATHDB-401075

Bethe-Sommerfeld graphs with finitely many open gaps

摘录 / EXCERPT: Do infinite periodic graphs with a finite nonzero number of open gaps exist?

open_problem | open | MathDB 页面 | 原文线索

V2-078543 | MATHDB-401077

Continuous branches of standing waves on the double-bridge graph for negative frequency

摘录 / EXCERPT: For the original standing-wave boundary value problem on the double-bridge graph, when $\c9<0$, a reasonable conjecture is that new continuous branches of solutions arise in both t...

open_problem | open | MathDB 页面 | 原文线索

V2-078544 | MATHDB-401078

Non-uniform sampling for geometric-graph Dirichlet partitions

摘录 / EXCERPT: Generalize Theorem 1 to non-uniformly sampled points. This would require showing that a generalized Dirichlet energy admits a continuous minimizer on $H^1_0(U;\Sigma_k)$.

open_problem | open | MathDB 页面 | 原文线索

V2-078545 | MATHDB-401080

Completeness of periodic data for smooth conjugacy of higher-dimensional Anosov diffeomorphisms

摘录 / EXCERPT: For Anosov diffeomorphisms in dimensions higher than 2, is periodic data-the Jordan normal forms of the differentials at corresponding periodic points-a complete invariant of smo...

open_problem | open | MathDB 页面 | 原文线索

V2-078546 | MATHDB-401081

Contractibility of the space of Anosov metrics on a closed surface

摘录 / EXCERPT: Let M be a closed surface of genus $\\ge 2$. Is $\mathrm{MET}^{A}(M)$ a contractible space?

open_problem | open | MathDB 页面 | 原文线索

V2-078547 | MATHDB-401083

Homeomorphism type of the convex-body space associated with positively curved metrics on S^2

摘录 / EXCERPT: Determine the homeomorphism type of cb, the space of convex bodies in $\mathbb{R}^3$ whose center of mass is at the origin and whose boundary is C^∞ with $K>0$, e...

open_problem | open | MathDB 页面 | 原文线索

V2-078548 | MATHDB-401093

Solvable word problem for general Artin groups

摘录 / EXCERPT: The conjecture that every Artin group A_Γ has a solvable word problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078549 | MATHDB-401094

Simplicity of the special tame automorphism group

摘录 / EXCERPT: For a field k of characteristic zero, is the special tame automorphism group $\operatorname{STame}(k^3)=\ker\bigl(\operatorname{Jac}:\operatorname{Tame}(k^3)\to k^*\bigr)$ simple...

open_problem | open | MathDB 页面 | 原文线索

V2-078550 | MATHDB-401097

QI rigidity for rank-one symmetric spaces and approximate groups

摘录 / EXCERPT: For a rank-one symmetric space X of non-compact type, does every approximate group $(\Lambda,\Lambda^{\infty})$ that is quasi-isometric to X act geometrically on X?

open_problem | open | MathDB 页面 | 原文线索

V2-078551 | MATHDB-401101

Asymptotics of the Goemans-Linial integrality gap

摘录 / EXCERPT: Let $\alpha(n)=\sup_{X\in\mathcal N_n} c_1(X)$, where $\mathcal N_n$ is the set of n-point negative-type metric spaces. What are the asymptotics of $\alpha(n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-078552 | MATHDB-401102

Expanding Ricci solitons with prescribed metric-cone tangent cone

摘录 / EXCERPT: For which metric cones C_0 is it possible to find an expanding (gradient) Ricci soliton with tangent cone C_0?

open_problem | open | MathDB 页面 | 原文线索

V2-078553 | MATHDB-401103

Viterbo’s conjectured isoperimetric inequality for the EHZ capacity

摘录 / EXCERPT: For a compact convex domain $X\subset\mathbb{R}^{2n}$ with smooth boundary and $0\in\operatorname{int}(X)$, Viterbo’s conjecture would imply the inequality ...

conjecture | open | MathDB 页面 | 原文线索

V2-078554 | MATHDB-401106

Birkhoff’s disk-like global-surface-of-section conjecture for the retrograde orbit

摘录 / EXCERPT: In the planar circular restricted three-body problem, below the first critical value of the Jacobi constant, does the retrograde orbit bound a disk-like global surface of section,...

conjecture | open | MathDB 页面 | 原文线索

V2-078555 | MATHDB-401108

Weak Viterbo systolic inequality for convex domains

摘录 / EXCERPT: Let $C\subset\mathbb{R}^{2n}$ be a convex domain of the type considered in the report, and let ... Is it true that ... with equality if and only if C is symplectomorphic to a ball?

open_problem | open | MathDB 页面 | 原文线索

V2-078556 | MATHDB-401110

Denjoy dynamics on leaves of a C^0-integrable twist map

摘录 / EXCERPT: Can the dynamics restricted to a leaf in the C^0-integrable case be a Denjoy counter-example?

open_problem | open | MathDB 页面 | 原文线索

V2-078557 | MATHDB-401111

Extension of the absolute Lefschetz-number invariant to the sphere

摘录 / EXCERPT: Does the following extension hold when $\Sigma=S^2$? If $f,g\in \operatorname{Ham}(S^2,\omega)$ have finitely many fixed points and f is almost conjugate to g, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-078558 | MATHDB-401115

Classification of zero-entropy Riemannian metrics on the two-torus

摘录 / EXCERPT: Is it true that the Liouville metrics ... where $f,g:\mathbb R\to\mathbb R$ are positive, smooth, 1-periodic functions, are the only Riemannian metrics on...

open_problem | open | MathDB 页面 | 原文线索

V2-078559 | MATHDB-401118

Coercivity of combined boundary integral operators on strongly non-trapping non-convex domains

摘录 / EXCERPT: For a wide class of strongly non-trapping, possibly non-convex obstacle domains-such as smooth star-shaped domains-does the combined boundary integral operator ... retain coercivity...

open_problem | open | MathDB 页面 | 原文线索

V2-078560 | MATHDB-401119

Local nonlinear Brascamp-Lieb inequality

摘录 / EXCERPT: Let $B=(B_j)_{j=1}^m$ be smooth submersions in a neighbourhood of $0\in\mathbb{R}^n$, where $B_j:\mathbb{R}^n\to\mathbb{R}^{n_j}$, and let $p_j\in[0,1]$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-078561 | MATHDB-401123

Whether Gaussians optimize the Strichartz inequality

摘录 / EXCERPT: For the Strichartz variational problem ... are Gaussians optimizers?

open_problem | open | MathDB 页面 | 原文线索

V2-078562 | MATHDB-401124

Maximum principle for the intermediate-dimensional Riesz transform

摘录 / EXCERPT: Fix $d\ge 2$ and $s\in(1,d-1)$, and set $K(x)=\dfrac{x}{|x|^{s+1}}$ to be the s-Riesz kernel. Is there a constant $C>0$ such that, whenever f is a non-negative smooth function...

open_problem | open | MathDB 页面 | 原文线索

V2-078563 | MATHDB-401125

A nonlinear analogue of Carleson’s theorem for the scattering transform

摘录 / EXCERPT: If $f\in L^2(\mathbb R)$ and $\operatorname{supp}(f)\subseteq[0,+\infty)$, does the limit ... exist for almost every $\xi\in\mathbb R$?

open_problem | open | MathDB 页面 | 原文线索

V2-078564 | MATHDB-401126

Uniform constants in the nonlinear Hausdorff-Young inequality

摘录 / EXCERPT: For the nonlinear Fourier transform described in the report, let $p'=\frac{p}{p-1}$ and let C_p be the optimal constant in ... Are the optimal constants C_p bounded uniformly for...

open_problem | open | MathDB 页面 | 原文线索

V2-078565 | MATHDB-401128

Sharp maximal inequality for the free Schrödinger evolution

摘录 / EXCERPT: Is the following sharp maximal inequality true, for any $f\in L^2$ with frequency support in $\{\xi\in\mathbb{R}^n:|\xi|\sim 1\}$? ... Here $B_n(R)$ denotes a ball in $\mathbb{R}^n$...

open_problem | open | MathDB 页面 | 原文线索

V2-078566 | MATHDB-401130

Equality of Riesz, heat, and Poisson Hardy spaces on rank-one symmetric spaces

摘录 / EXCERPT: Let X be a Riemannian symmetric space of the noncompact type and real rank one. Do the Hardy spaces $H^1_R(X)$, $H^1_H(X)$, and $H^1_P(X)$ agree? The report’s Theorem 1(a) answer...

open_problem | open | MathDB 页面 | 原文线索

V2-078567 | MATHDB-401131

Gaussian extremizers for the Strichartz inequality in all dimensions

摘录 / EXCERPT: Let u be the solution of the Schrödinger equation $i u_t=\Delta u$ on $\mathbb{R}^{d+1}$, with initial data $u(x,0)=f(x)$. For the Strichartz inequality ... do Gaussians extremize...

open_problem | open | MathDB 页面 | 原文线索

V2-078568 | MATHDB-401132

Bochner-Riesz profile of the anharmonic oscillator

摘录 / EXCERPT: Let $L=-\frac{d^2}{dx^2}+|x|$ on $\mathbb{R}$, and define ... For every $1\leq p\leq\infty$, the operators $\sigma_R^\alpha(L)$ are uniformly bounded on $L^p(\mathbb{R})$ when ... If .....

open_problem | open | MathDB 页面 | 原文线索

V2-078569 | MATHDB-401134

Characterization of measures with bounded codimension-one Riesz transform

摘录 / EXCERPT: Let $\ theta_\$ be a Borel measure in $\mathbb{R}^{n+1}$ satisfying the growth condition ... for all $x\in\operatorname{supp}\mu$ and $r>0$. The conjecture is that the associated n...

open_problem | open | MathDB 页面 | 原文线索

V2-078570 | MATHDB-401135

Counting solutions to the power-sum congruence system for a general target tuple

摘录 / EXCERPT: For a given set of integers $y_1,\ldots,y_n$, determine the number of solutions $(x_1,\ldots,x_n)$ to ... The report states that this is currently not known in general.

open_problem | open | MathDB 页面 | 原文线索

V2-078571 | MATHDB-401136

Zygmund’s directional maximal-operator problem for Lipschitz vector fields

摘录 / EXCERPT: Let $v:\mathbb{R}^2\to S^1$ be a $1/100$-Lipschitz unit vector field. Is the associated directional maximal operator ... bounded on any $L^p(\mathbb{R}^2)$, $1<p<\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-078572 | MATHDB-401137

Boundedness of the directional Hilbert transform for Lipschitz vector fields

摘录 / EXCERPT: Let $u:\a0\mathbb{R}^{2}\to[-1,1]$ be $1/100$-Lipschitz, and define ... Is H_{u} bounded on any $L^{p}(\mathbb{R}^{2})$ space with $1<p<\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-078573 | MATHDB-401138

Round sphere minimizes entropy among closed hypersurfaces

摘录 / EXCERPT: For $2\le n\le 6$, if Σ is a closed hypersurface in $\mathbb{R}^{n+1}$, then $\lambda[\Sigma]\ge\lambda[\mathbb{S}^{n}]$, and equality holds if and only if Σ is a rou...

open_problem | open | MathDB 页面 | 原文线索

V2-078574 | MATHDB-401139

Path-connectedness of the moduli space of 2-convex embedded spheres

摘录 / EXCERPT: Let $\mathcal M_{2\text{-conv}}(S^n)$ be the subspace of $\mathcal M(S^n)=\operatorname{Emb}(S^n,\mathbb R^{n+1})/\operatorname{Diff}(S^n)$ consisting of 2-convex embedded n-sphe...

open_problem | open | MathDB 页面 | 原文线索

V2-078575 | MATHDB-401141

Non-fattening near mean-convex singularities

摘录 / EXCERPT: An evolving surface cannot fatten unless it has a singularity with no spacetime neighborhood in which the surface is mean convex.

open_problem | open | MathDB 页面 | 原文线索

V2-078576 | MATHDB-401144

Willmore minimizers with prescribed isoperimetric ratio for spherical surfaces

摘录 / EXCERPT: For each prescribed isoperimetric ratio $\sigma\in(0,1]$, define ... where ... The report states that this infimum is attained by a smooth embedding of S^2 for every $\sigma\in(0,1]$...

open_problem | open | MathDB 页面 | 原文线索

V2-078577 | MATHDB-401146

Harmonicity of the expander produced from a harmonic initial map

摘录 / EXCERPT: Does the solution produced by Theorem 1, coming out of a 0-homogeneous harmonic map u_0, stay harmonic?

open_problem | open | MathDB 页面 | 原文线索

V2-078578 | MATHDB-401149

Local extension of boundary conformal classes to Poincaré-Einstein metrics

摘录 / EXCERPT: Does every boundary conformal class c extend at least locally (say, in a collar neighborhood of the boundary) to a Poincaré-Einstein metric?

open_problem | open | MathDB 页面 | 原文线索

V2-078579 | MATHDB-401162

Status of the flow equation for families of Kähler forms

摘录 / EXCERPT: The report leaves open the status of the flow equation (4) for families of Kähler forms, where equation (4) is ... under the preceding condition $\nabla^{g_s}\omega_s=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-078580 | MATHDB-401164

Nonexistence of Yamabe metrics for edge-cone spheres with cone angle greater than 2π

摘录 / EXCERPT: Does the nonexistence result for a Yamabe metric on a sphere with a codimension-2 singularity and cone angle greater than 4π remain true for cone angles greater than 2π?

open_problem | open | MathDB 页面 | 原文线索

V2-078581 | MATHDB-401165

Biere’s exponential-resolution conjecture for multiplier verification

摘录 / EXCERPT: Verifying ring identities for multiplication circuits, such as commutativity, requires resolution proofs that are exponential in the bit-width n.

conjecture | open | MathDB 页面 | 原文线索

V2-078582 | MATHDB-401166

Definability of maximum matching size in FPC

摘录 / EXCERPT: The problem of determining the size of the maximum matching in a graph is in FPC.

open_problem | open | MathDB 页面 | 原文线索

V2-078583 | MATHDB-401168

SOS lower bound for refuting random CSP instances

摘录 / EXCERPT: Let $P:\{0,1\}^k\to\{0,1\}$ be a nontrivial k-ary predicate. Consider a random instance of $\\mathrm{CSP}(P)$ on n variables with Δn constraints, each applying P to...

open_problem | open | MathDB 页面 | 原文线索

V2-078584 | MATHDB-401171

Polynomial-size selector gadgets for positive semidefinite-rank lower bounds

摘录 / EXCERPT: Let $f:\{0,1\}^n\to\mathbb{R}_+$ satisfy $\\mathbb{E}f=1$. Theorem 4 states that ... when $m\le cn$ for some constant $c>1$. The open question is whether the same lower bound holds u...

open_problem | open | MathDB 页面 | 原文线索

V2-078585 | MATHDB-401172

Constant-degree SOS lower bounds for sparse random 3SAT

摘录 / EXCERPT: For random 3SAT instances with n variables, m clauses, and constraint density $\Delta=m/n$, show that when $\Delta\ll\sqrt{n}$, succinct ("constant-degree") SOS refutations do...

open_problem | open | MathDB 页面 | 原文线索

V2-078586 | MATHDB-401173

Exact tensor completion beyond the orthogonal case

摘录 / EXCERPT: Can the results be generalized beyond the non-orthogonal case?

open_problem | open | MathDB 页面 | 原文线索

V2-078587 | MATHDB-401174

Characterization of bounded-depth Frege canonical pairs by grid games

摘录 / EXCERPT: For a fixed game depth k, prove that the disjoint NP pair (A_k,B_k), where $A_k=\{G:\text{Player 1 has a positional winning strategy in }G\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078588 | MATHDB-401175

CNFs separating constant-depth Frege from narrow random-resolution refutations

摘录 / EXCERPT: Find a CNF which has small refutations in constant-depth Frege, but does not have polylogarithmic-width $1/2$-RR refutations.

open_problem | open | MathDB 页面 | 原文线索

V2-078589 | MATHDB-401177

Regularity of minimum resolution refutations

摘录 / EXCERPT: For the graph-based clauses of Tseitin and for the pigeonhole principle, the minimal resolution refutations are always regular.

open_problem | open | MathDB 页面 | 原文线索

V2-078590 | MATHDB-401179

Lehmer’s Question

摘录 / EXCERPT: Given $\epsilon>0$, does there exist an integral polynomial $f(x)$ with $1<M(f)<1+\epsilon$?

open_problem | open | MathDB 页面 | 原文线索

V2-078591 | MATHDB-401181

Verlinde-like dimension formulae for other families of dominant weights

摘录 / EXCERPT: Can one find similar Verlinde-like dimension formulae for other families of dominant weights?

open_problem | open | MathDB 页面 | 原文线索

V2-078592 | MATHDB-401182

Relation between the invariant \(\rho_1\) and Rozansky’s 2-loop invariant

摘录 / EXCERPT: For every knot K, conjecture that ... where ρ_1 is the normalized invariant defined in the report and $P^{(1)}$ is the Laurent-polynomial invariant arising from the perturbati...

open_problem | open | MathDB 页面 | 原文线索

V2-078593 | MATHDB-401183

Conjectural properties of the knot invariant \(\rho_1\)

摘录 / EXCERPT: The authors conjecture that $\rho_1(-K)=-\rho_1(K)$ and that the highest power of t occurring in $\rho_1(K)$ is at most $2\,\mathrm{genus}(K)-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-078594 | MATHDB-401184

Genus bound for the knot invariant \(\rho_1\)

摘录 / EXCERPT: For a knot K of genus $g(K)$, the highest power of t occurring in $\rho_1(K;t)$ is at most $2g(K)-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-078595 | MATHDB-401185

Boyd’s numerical Mahler-measure formula for $S_{4,1}$

摘录 / EXCERPT: For $S_{k,\beta}(X,Y)=Y^2+kXY-X^3-\beta X$, Boyd’s numerical data give the conjectural identity ... where E_{192} is the elliptic curve defined by $S_{4,1}(X,Y)=0$ and having condu...

open_problem | open | MathDB 页面 | 原文线索

V2-078596 | MATHDB-401187

Mahler-measure volume inequality for semi-regular biperiodic alternating links

摘录 / EXCERPT: Conjecture. Let $\mathcal{L}$ be a semi-regular biperiodic alternating link, with toroidally alternating Λ-quotient link L and crossing number $c(L)$. Let $p(z,w)$ be the...

open_problem | open | MathDB 页面 | 原文线索

V2-078597 | MATHDB-401188

Vol-Det inequality for Følner-convergent links to the Rhombitrihexagonal link

摘录 / EXCERPT: Let L be the Rhombitrihexagonal link and let K_n be any sequence of alternating links such that $K_n \xrightarrow{F} L$. Then, for almost all n,...

open_problem | open | MathDB 页面 | 原文线索

V2-078598 | MATHDB-401191

Congruence special subgroups of arithmetic hyperbolic lattices

摘录 / EXCERPT: Can special subgroups of any arithmetic hyperbolic lattices always be achieved by congruence subgroups?

open_problem | open | MathDB 页面 | 原文线索

V2-078599 | MATHDB-401193

Asymptotic expansion for the figure-eight knot q-series

摘录 / EXCERPT: Let ... With $\widehat\Phi(x)=e^{C/x}\Phi(x)$, where $C=\frac{1}{\pi}\operatorname{Li}_2(e^{\pi i/3})$ and Φ is the formal power series defined in equations (3)-(5) of the repor...

open_problem | open | MathDB 页面 | 原文线索

V2-078600 | MATHDB-401195

Volume Conjecture for the colored Jones invariant

摘录 / EXCERPT: For a hyperbolic knot K in S^3, let $J_N(K)$ be the colored Jones invariant associated with the N-dimensional irreducible representation of $\\mathfrak{sl}_2$, with its param...

conjecture | open | MathDB 页面 | 原文线索

V2-078601 | MATHDB-401203

Preservation of positivity under the relative Kähler-Ricci flow on $O_E(r)$

摘录 / EXCERPT: Let $E\to S$ be a holomorphic vector bundle and let $X=\mathbb{P}(E^*)$. Starting with a positive Hermitian metric h on $O_E(1)$, run the flow (2) from the report for the induced...

open_problem | open | MathDB 页面 | 原文线索

V2-078602 | MATHDB-401206

Supports for the pole-free \\\\Hitchin fibration for \(\mathrm{GL}(n)\)

摘录 / EXCERPT: For the Hitchin fibration without poles ($D=0$) for $G=\mathrm{GL}(n)$, let $\mathcal A_n^{\mathrm{red}}$ be the locus in the Hitchin base determined by reduced spectral curves. De...

open_problem | open | MathDB 页面 | 原文线索

V2-078603 | MATHDB-401214

Intrinsic L^2 estimates for non-nilpotent sections and multistep residue measures

摘录 / EXCERPT: Without assuming that f is a nilpotent section associated with the ideal sheaf of the previous jump m_{p-1}, the existence of a reasonably intrinsic L^2 estimate is an open p...

open_problem | open | MathDB 页面 | 原文线索

V2-078604 | MATHDB-401215

Dirichlet elliptic control problem with distributed control

摘录 / EXCERPT: Let $Y:=H_0^1(\\Omega)$ and $U:=Y'$. For given $f\\in Y'$, target state $y^*\\in Y$, and fixed weight parameter $\\omega>0$, consider the constrained quadratic optimization problem...

open_problem | open | MathDB 页面 | 原文线索

V2-078605 | MATHDB-401217

Second-order sufficient condition for sparse optimal control as the Tikhonov parameter vanishes

摘录 / EXCERPT: The report answers the question by Theorem 3. Let $\bar u\in U_{ad}$, with associated state $\bar y$ and adjoint state $\bar\varphi$, satisfy the first-order necessary optimality c...

open_problem | open | MathDB 页面 | 原文线索

V2-078606 | MATHDB-401218

Hölder stability as the Tikhonov parameter tends to zero

摘录 / EXCERPT: Let $\bar u$ be a strict locally optimal control for the problem with $\nu=0$, and let $\{\bar u_\nu\}$ be a sequence of local solutions of problems (1)-(3) for $\nu>0$ such th...

open_problem | open | MathDB 页面 | 原文线索

V2-078607 | MATHDB-401219

Archimedean Whittaker newform as a test vector

摘录 / EXCERPT: In the archimedean setting described for a generic irreducible admissible representation π of $\mathrm{GL}_n(F)$, with $F\in\{\mathbb{R},\mathbb{C}\}$, conjecture that the newf...

open_problem | open | MathDB 页面 | 原文线索

V2-078608 | MATHDB-401220

Logarithm law for shrinking targets under unbounded discrete flows

摘录 / EXCERPT: Let G be a connected semisimple Lie group with finite center and no compact factors, let $\Gamma\leq G$ be an irreducible lattice, and let $X=\Gamma\backslash G$ carry its G-in...

open_problem | open | MathDB 页面 | 原文线索

V2-078609 | MATHDB-401222

Generalized Riemann Hypothesis for the axiomatic family of L-functions

摘录 / EXCERPT: Under Axioms 1-4 for $L(s,f)$, all zeros in the critical strip $\{s\in\mathbb C:0\le \Re(s)\le 1\}$ lie on the central line $\Re(s)=\tfrac12$.

conjecture | open | MathDB 页面 | 原文线索

V2-078610 | MATHDB-401223

Analytic continuation of the heat kernel on a compact hyperbolic surface

摘录 / EXCERPT: For the heat kernel $K_M(\b6f;z,z')$ on $M=\Gamma\backslash\mathbb H$, whose spectral expansion gives an analytic continuation to the right half-plane...

open_problem | open | MathDB 页面 | 原文线索

V2-078611 | MATHDB-401224

Subconvex sup-norm bound for split groups under a balance condition

摘录 / EXCERPT: The report states the following result as Corollary 1: Let G/F be split, and let ψ, λ, and Ω_Y be as in Theorem 2. Assume that λ is (A,σ)-balan...

open_problem | open | MathDB 页面 | 原文线索

V2-078612 | MATHDB-401227

Sup-norm bound for Hecke-Maaß newforms over number fields

摘录 / EXCERPT: The report states the following proved result: Let c(phic) be an c(L^2c)-normalized Heckec{--}Maaß cuspidal newform on c(mathrm{GL}2(F)c) of square-free level c(mathfrak n)c), with...

open_problem | open | MathDB 页面 | 原文线索

V2-078613 | MATHDB-401231

Mazur-Rubin conjecture on individual variances of modular symbols

摘录 / EXCERPT: Let $f(z)=\sum_{n\ge1}a_n n^{-1/2}e(nz)$ be a holomorphic cusp form of weight 2 for $\Gamma_0(q)$, with q squarefree. Define the modular symbols and their variance as in the re...

conjecture | open | MathDB 页面 | 原文线索

V2-078614 | MATHDB-401233

Gaussian distribution of quadratic-twist central values of a newform

摘录 / EXCERPT: If f denotes a newform, and d runs over fundamental discriminants with $|d|\leq X$ for which $f\times\chi_d$ has root number 1, then one expects...

open_problem | open | MathDB 页面 | 原文线索

V2-078615 | MATHDB-401234

Conditional Gaussian lower bound for central values of even quadratic twists

摘录 / EXCERPT: Assume the Generalized Riemann Hypothesis. Let E denote the set of fundamental discriminants d for which the quadratic twist E_d has root number 1. Then, for every interval...

open_problem | open | MathDB 页面 | 原文线索

V2-078616 | MATHDB-401235

Determining the phase of Dirac evolution between varying Fock spaces

摘录 / EXCERPT: The implementation of second-quantized Dirac evolution between varying Fock spaces associated with Cauchy surfaces leaves a global phase undetermined. Identify this phase and its r...

open_problem | open | MathDB 页面 | 原文线索

V2-078617 | MATHDB-401236

Self-consistent free Dirac sea in the Hartree-Fock approximation

摘录 / EXCERPT: The question posed in [5] and [7]-what the correct free Dirac sea is when interactions among the particles filling the sea are included-is answered by a self-consistent projector s...

open_problem | open | MathDB 页面 | 原文线索

V2-078618 | MATHDB-401237

Defect of compactness for reduced density matrices in bosonic and fermionic mean-field limits

摘录 / EXCERPT: Understand and characterize the defect-of-compactness phenomenon for density matrices ϱ_ε on bosonic or fermionic Fock space and their p-particle reduced densit...

open_problem | open | MathDB 页面 | 原文线索

V2-078619 | MATHDB-401239

Stability bound for a fermionic point-interaction system

摘录 / EXCERPT: For any $m>0$, let $A=(2+m)^{-1}$ and ... with $s,K\in\mathbb R^3$ and $Q>0$, and define $\Lambda(m)$ by the optimization formula (7) on PDF page 41. Then, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078620 | MATHDB-401257

Simplicity and spectral gap for the valuation operator

摘录 / EXCERPT: For the setting of Theorem 3, if $g\in GL(E)$ satisfies $\lambda_k(g)^2>\lambda_{k+1}(g)\lambda_{k-1}(g)$, then the eigenvalue $\lambda_k(g)$ is expected to be simple and the opera...

open_problem | open | MathDB 页面 | 原文线索

V2-078621 | MATHDB-401260

Fourfolds with nef and big diagonal

摘录 / EXCERPT: Must every smooth projective fourfold with nef and big diagonal be a fake projective space?

open_problem | open | MathDB 页面 | 原文线索

V2-078622 | MATHDB-401263

Equivalence of integral and ambient maximality in dimension three

摘录 / EXCERPT: For three-dimensional integral lattice-free polyhedra, $\mathbb{Z}^3$-maximality is equivalent to $\mathbb{R}^3$-maximality. Equivalently, if an integral lattice-free polyhedron is...

open_problem | open | MathDB 页面 | 原文线索

V2-078623 | MATHDB-401264

Canonical Fano polytope volume conjecture

摘录 / EXCERPT: Let $P\subset N_{\mathbb R}$ be a d-dimensional canonical Fano polytope, where $d\ge 4$, meaning that P contains exactly one interior lattice point, taken to be the origin. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-078624 | MATHDB-401265

Unimodality of Ehrhart h^*-polynomials for IDP lattice polytopes

摘录 / EXCERPT: Every lattice polytope P with the integer decomposition property-that is, for every integer $k\ge 1$, every lattice point of kP is a sum of k lattice points of P-has a unim...

open_problem | open | MathDB 页面 | 原文线索

V2-078625 | MATHDB-401266

Unimodality of hypersimplex h^*-polynomials

摘录 / EXCERPT: Is the h^*-polynomial of every hypersimplex $P=[U+007F]x\in[0,1]^d:x_1+\cdots+x_d=k\u007f$ unimodal?

open_problem | open | MathDB 页面 | 原文线索

V2-078626 | MATHDB-401267

Real-rootedness of h*-polynomials of dilated lattice polytopes

摘录 / EXCERPT: For a d-dimensional lattice polytope P, the h^*-polynomial of rP has only real zeros for all integers $r \ge d$.

open_problem | open | MathDB 页面 | 原文线索

V2-078627 | MATHDB-401268

Hibi-Ohsugi conjecture on unimodality of Ehrhart $h^\ast$-polynomials

摘录 / EXCERPT: Every Gorenstein lattice polytope with the integer decomposition property (IDP) has a unimodal Ehrhart $h^\ast$-polynomial.

conjecture | open | MathDB 页面 | 原文线索

V2-078628 | MATHDB-401269

Non-unimodality among reflexive IDP simplices

摘录 / EXCERPT: Does there exist a weakly increasing sequence of positive integers $q=(q_1,\\ldots,q_n)$ such that the simplex $\\Delta(1,q)=\\operatorname{conv}(e_1,\\ldots,e_n,-q)$ is reflexive...

open_problem | open | MathDB 页面 | 原文线索

V2-078629 | MATHDB-401270

BV-\alpha-positivity of regular permutohedra

摘录 / EXCERPT: Every regular permutohedron Π_d is BV-α-positive.

open_problem | open | MathDB 页面 | 原文线索

V2-078630 | MATHDB-401271

Ehrhart positivity of smooth integral polytopes

摘录 / EXCERPT: Are all smooth integral polytopes Ehrhart positive?

open_problem | open | MathDB 页面 | 原文线索

V2-078631 | MATHDB-401272

Unimodality of h*-vectors of very ample polytopes

摘录 / EXCERPT: Do very ample polytopes always have unimodal h^*-vectors?

open_problem | open | MathDB 页面 | 原文线索

V2-078632 | MATHDB-401273

Unimodality of the h-vector of standard graded Cohen-Macaulay domains

摘录 / EXCERPT: For a standard graded Cohen-Macaulay domain $R=R_0\oplus R_1\oplus\cdots$ over a field $K=R_0$, generated by R_1, let $d=\dim R$ and write ... Is the h-vector...

open_problem | open | MathDB 页面 | 原文线索

V2-078633 | MATHDB-401275

Extending the implication diagram for Ehrhart h^*-vectors

摘录 / EXCERPT: How can Figure 1 be extended to comprise real-rooted or log-concave h^*-vectors?

open_problem | open | MathDB 页面 | 原文线索

V2-078634 | MATHDB-401276

Stanley-type inequalities for spanning lattice polytopes

摘录 / EXCERPT: Let P be a spanning lattice polytope of degree s, so that the rightmost nonzero entry of its h^*-vector $h^*(P)=(h^*_0,h^*_1,\ldots,h^*_s)$ has index s. Then, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078635 | MATHDB-401277

Geometric criterion for strict monotonicity of mixed volume

摘录 / EXCERPT: Let $P_1,\\ldots,P_d$ be polytopes contained in a d-dimensional polytope $Q\subseteq\mathbb{R}^d$. For a face F of Q, say that P_i touches F if $P_i\cap F\neq\varnothing$...

open_problem | open | MathDB 页面 | 原文线索

V2-078636 | MATHDB-401279

Hensley’s conjecture for 3-dimensional simplices

摘录 / EXCERPT: Let $\Delta \subseteq \mathbb{R}^3$ be a lattice tetrahedron with exactly $k>0$ interior lattice points. Then the normalized volume of Δ is at most $36(k+1)$.

conjecture | open | MathDB 页面 | 原文线索

V2-078637 | MATHDB-401280

Duong’s conjecture for clean tetrahedra with collinear interior lattice points

摘录 / EXCERPT: Does Duong’s conjecture remain true for clean tetrahedra $\Delta\subseteq\mathbb{R}^3$ whose exactly $k>0$ interior lattice points are collinear?

conjecture | open | MathDB 页面 | 原文线索

V2-078638 | MATHDB-401282

Unimodality of the h^*-vector of alcoved polytopes

摘录 / EXCERPT: Is the h^*-vector of every alcoved polytope unimodal? Here an alcoved polytope is described in the report as a polytope of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-078639 | MATHDB-401284

Unimodality of the h-polynomial of a standard graded Gorenstein domain

摘录 / EXCERPT: Let $R=\bigoplus_{m\ge 0}R_m$ be a graded standard Gorenstein domain over a field $K=R_0$, with Krull dimension d. If $H(R,m)=\\dim_K R_m$ and ... then the polynomial...

open_problem | open | MathDB 页面 | 原文线索

V2-078640 | MATHDB-401285

Very ample lattice polytopes and log-concavity of the Ehrhart h^*-polynomial

摘录 / EXCERPT: For a very ample lattice polytope P, is its Ehrhart h^*-polynomial $h^*(P;z)$ log-concave?

open_problem | open | MathDB 页面 | 原文线索

V2-078641 | MATHDB-401286

Unimodality of the Ehrhart h^*-polynomial of a very ample lattice polytope

摘录 / EXCERPT: If P is very ample, is $h^*(P;z)$ unimodal?

open_problem | open | MathDB 页面 | 原文线索

V2-078642 | MATHDB-401288

Real-rooted generating polynomials realizable as Ehrhart $h^\ast$-polynomials

摘录 / EXCERPT: Determine which real-rooted generating polynomials can be realized as the $h^\ast$-polynomial of a lattice polytope.

open_problem | open | MathDB 页面 | 原文线索

V2-078643 | MATHDB-401289

Positive-degree line bundles that are not ample in positive characteristic

摘录 / EXCERPT: Over positive characteristic (in the report’s context, over $\mathbb F_p$), does there exist a line bundle on a surface X that has positive degree on every curve but is not ample…

open_problem | open | MathDB 页面 | 原文线索

V2-078644 | MATHDB-401291

Polynomial bound for dominant components of spaces of rational curves

摘录 / EXCERPT: Fix an ample divisor L. The number of dominant components of $\operatorname{Mor}(\mathbb{P}^1,X)$ of L-degree at most d is bounded above by a polynomial in d. The degree of…

open_problem | open | MathDB 页面 | 原文线索

V2-078645 | MATHDB-401295

Tate conjecture for a quintic surface of half-twist type

摘录 / EXCERPT: Theorem 2. Let $X\subset \mathbf{P}^3$ be a non-singular surface given by an equation of the form $F_5(x_0,x_1,x_2)+x_3^5=0$ in characteristic 0. Then the Tate conjecture for X…

conjecture | open | MathDB 页面 | 原文线索

V2-078646 | MATHDB-401297

Cohen-Macaulayness of klt singularities in sufficiently large characteristic

摘录 / EXCERPT: For each positive integer n, is there a number $p_0(n)$ such that every klt singularity over a field of characteristic $p\ge p_0(n)$ is Cohen-Macaulay?

open_problem | open | MathDB 页面 | 原文线索

V2-078647 | MATHDB-401299

Half-line spectrum for a separable square Fibonacci Schrödinger operator

摘录 / EXCERPT: Let $f_0,f_1\\in L^2[0,1)$, let $\\alpha=(\\sqrt5-1)/2$, and define … For the separable continuum Schrödinger operator … with $\\lambda_1,\\lambda_2>0$, is it true that…

open_problem | open | MathDB 页面 | 原文线索

V2-078648 | MATHDB-401301

Bounded holes conjecture for stealthy hyperuniform point processes

摘录 / EXCERPT: Let Ξ be a stealthy hyperuniform point process. Let $B(x;r)$ be the ball with centre x and radius r. Then there exists a positive number r_0 such that … Further, r_0 ca…

conjecture | open | MathDB 页面 | 原文线索

V2-078649 | MATHDB-401302

Unboundedness of the number of distinct patch frequencies

摘录 / EXCERPT: For almost every cut-and-project set, the number of distinct frequencies of patterns of a given size is not bounded as the pattern size grows.

open_problem | open | MathDB 页面 | 原文线索

V2-078650 | MATHDB-401304

Infinite free sets for irregular minimal almost automorphic subshifts

摘录 / EXCERPT: Let $(\Sigma,\mathbb{Z})$ be a minimal almost automorphic subshift. If it is irregular, then there exists an infinite set $S\subseteq\mathbb{Z}$ such that, for every…

open_problem | open | MathDB 页面 | 原文线索

V2-078651 | MATHDB-401306

Exact-overlap conjecture for self-similar sets on the line

摘录 / EXCERPT: Let $S=\{S_i(x)=r_i x+t_i\}_{i=1}^m$ be a finite self-similar iterated function system on $\mathbb{R}$, with similarity dimension $s<1$, and let Λ be its attractor. An exac…

conjecture | open | MathDB 页面 | 原文线索

V2-078652 | MATHDB-401307

Projection dimensions of products of ×2- and ×3-invariant measures

摘录 / EXCERPT: Let $\bc$ (respectively, $\bd$) be a $\d7 2$ (respectively, $\d7 3$) invariant measure on $[0,1]$. Let $\b8=\bc\otimes\bd$. Then, for every orthogonal projection $\c0$ that is not…

open_problem | open | MathDB 页面 | 原文线索

V2-078653 | MATHDB-401308

Furstenberg’s intersection conjecture for ×p- and ×q-invariant sets

摘录 / EXCERPT: If $A,B\subset[0,1]$ are closed and invariant under the maps $x\mapsto px\pmod 1$ and $x\mapsto qx\pmod 1$, respectively, and if $\frac{\log p}{\log q}\notin\mathbb{Q}$, then for a…

conjecture | open | MathDB 页面 | 原文线索

V2-078654 | MATHDB-401310

Uniform separation of roots of Littlewood-type polynomials

摘录 / EXCERPT: Does there exist a constant $s>0$ such that, for α,β that are roots of polynomials in P_d, either $\alpha=\beta$ or $|\alpha-\beta|>s^d$? Here P_d is the set of poly…

open_problem | open | MathDB 页面 | 原文线索

V2-078655 | MATHDB-401311

Hochman’s exponential separation question for roots of [U+007F][U+007F]-1,0,1[U+007F]-polynomials

摘录 / EXCERPT: For each $n\ge 1$, let P_n be the set of polynomials of degree less than n with coefficients in $\{-1,0,1\}$, and let…

open_problem | open | MathDB 页面 | 原文线索

V2-078656 | MATHDB-401313

Erdős-Szemerédi sum-product conjecture over the reals

摘录 / EXCERPT: For every $ …

conjecture | open | MathDB 页面 | 原文线索

V2-078657 | MATHDB-401314

Bourgain’s discretized sum-product theorem

摘录 / EXCERPT: Let $\alpha>0$ and $\kappa>0$ be positive absolute constants. There exist $\varepsilon_0,\varepsilon_1>0$ such that, for every $\delta>0$, if $A\subset\delta\mathbb Z$ is finite an…

open_problem | open | MathDB 页面 | 原文线索

V2-078658 | MATHDB-401315

Furstenberg’s intersection conjecture

摘录 / EXCERPT: If A,B are closed subsets of the circle $[0,1)$, invariant under T_p,T_q respectively, with p and q multiplicatively independent, then ... where $T_p(x)=px\pmod 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-078659 | MATHDB-401316

Rigorous deduction of macroscopic equations from cell-motion descriptions

摘录 / EXCERPT: The rigorous deduction of macroscopic equations from descriptions of cell motions, e.g. via SDEs for the velocity process coupled with subcellular dynamics, is still open.

open_problem | open | MathDB 页面 | 原文线索

V2-078660 | MATHDB-401317

Bifurcation of traveling waves from steady states in a Keller-Segel-type free-boundary model

摘录 / EXCERPT: For the two-dimensional free-boundary model consisting of equations (1)-(3), traveling-wave solutions emerge by bifurcation from the curve of radially symmetric steady states. A Ly...

open_problem | open | MathDB 页面 | 原文线索

V2-078661 | MATHDB-401318

Blow-up threshold for a radially symmetric Keller-Segel system with fixed boundary

摘录 / EXCERPT: For the two-dimensional parabolic-elliptic Keller-Segel system arising from a cell-motility model, equipped with no-flux boundary conditions for one variable and homogeneous Dirich...

open_problem | open | MathDB 页面 | 原文线索

V2-078662 | MATHDB-401319

Deift’s conjecture on almost-periodic KdV evolution

摘录 / EXCERPT: Determine whether solutions of the KdV equation with almost-periodic initial data are almost periodic in the time variable.

conjecture | open | MathDB 页面 | 原文线索

V2-078663 | MATHDB-401323

Deift’s question on almost periodicity of KdV solutions

摘录 / EXCERPT: Does almost periodicity of the initial data V imply almost periodicity of the solution u_1^V in the t-direction?

open_problem | open | MathDB 页面 | 原文线索

V2-078664 | MATHDB-401324

Chulaevsky-Sinai conjecture on interval spectrum for multidimensional quasiperiodic Schrödinger operators

摘录 / EXCERPT: For the two-dimensional shift, the spectrum can be an interval for generic large smooth potentials.

conjecture | open | MathDB 页面 | 原文线索

V2-078665 | MATHDB-401325

Dry Ten Martini Problem for the almost Mathieu operator

摘录 / EXCERPT: For every coupling lambdainmathbb{R}setminus{0} and every irrational frequency $\alpha\in\mathbb{R}\setminus\mathbb{Q}$, are all possible spectral gaps of the almost Mathieu operat...

open_problem | open | MathDB 页面 | 原文线索

V2-078666 | MATHDB-401326

Discrete Deift conjecture for the Toda flow

摘录 / EXCERPT: For almost periodic initial data, the Toda flow is almost periodic in the time variable.

conjecture | open | MathDB 页面 | 原文线索

V2-078667 | MATHDB-401327

Deift’s conjecture on almost periodicity in time for KdV

摘录 / EXCERPT: For the KdV Cauchy problem $\partial_tu-6u\partial_xu+\partial_x^3u=0$, $u(\cdot,0)=V$, with almost periodic initial data V, the solution is almost periodic as a function of t.

conjecture | open | MathDB 页面 | 原文线索

V2-078668 | MATHDB-401328

Theorem 2: Almost periodicity in time for KdV under Craig-type conditions

摘录 / EXCERPT: Let ... where $V:\mathbb R\to\mathbb R$ is uniformly almost periodic. Let $H_V=-\partial_x^2+V$, let $S=\sigma(H_V)$, and write ... where the intervals are the maximal open gaps of...

open_problem | open | MathDB 页面 | 原文线索

V2-078669 | MATHDB-401329

Long-time asymptotics for KdV with steplike initial data

摘录 / EXCERPT: The highlighted discussion is not an open problem. It states a theorem for the KdV Cauchy problem $q_t=6qq_x-q_{xxx}$ with steplike data $q(x,0)=q_0(x)$ satisfying $q_0(x)\to0$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-078670 | MATHDB-401330

Existence of a discrete Schrödinger operator in the isospectral torus

摘录 / EXCERPT: For a compact set $E\subset\mathbb{R}$ with $\operatorname{cap}(E)=1$, does there exist a discrete Schrödinger operator $H_V\in\mathcal{J}(E)$?

open_problem | open | MathDB 页面 | 原文线索

V2-078671 | MATHDB-401331

Generic nonexistence of discrete Schrödinger operators in unit-capacity isospectral tori

摘录 / EXCERPT: For a full-measure dense G_δ of finite-gap compacts E having $\operatorname{cap}(E)=1$, $\mathcal{J}(E)$ contains no DSO.

open_problem | open | MathDB 页面 | 原文线索

V2-078672 | MATHDB-401332

Existence of solutions to the KdV hierarchy for reflectionless Schrödinger potentials

摘录 / EXCERPT: Let $E\subset\mathbb{R}$ be closed and bounded below, with $\Omega=\mathbb{C}\setminus E$ of Widom type, and suppose after translation that ... If ... and $q_0\in Q(E)$, where $Q(E)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-078673 | MATHDB-401333

Extension of the Damanik-Killip-Simon theorem to general finite systems of intervals

摘录 / EXCERPT: “Is there an extension of the Damanik-Killip-Simon theorem to the general finite system of intervals case?”

open_problem | open | MathDB 页面 | 原文线索

V2-078674 | MATHDB-401335

Endpoint \epsilon=0 in a non-self-adjoint finite-gap Lieb\\Thirring inequality

摘录 / EXCERPT: Let $E\subset\mathbb{R}$ be an arbitrary finite-gap set, let $J_0\in T_E$, and let $J=J_0+\delta J$ be a compact non-self-adjoint perturbation of J_0. Is it true that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078675 | MATHDB-401349

Fundamental groups of Kodaira fibrations

摘录 / EXCERPT: For which extensions ... as in (1), is the group G the fundamental group of a Kodaira surface?

open_problem | open | MathDB 页面 | 原文线索

V2-078676 | MATHDB-401353

Non-abelianity of the quotient of string-link concordance by the normal closure of pure braids

摘录 / EXCERPT: For the groups introduced in the report, is $C_m/N_{\mathrm{cl}}(P_m)$ non-abelian for $m\ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-078677 | MATHDB-401359

Fractional-moment conjecture for Dirichlet polynomials

摘录 / EXCERPT: Let $D(s)=\sum_{n=1}^{N}a_n n^{-s}$ with $|a_n|\le 1$, and suppose $T^{1/2}<N<T$. Set κ by $N^{\kappa}=T$. The conjecture is that ...

conjecture | open | MathDB 页面 | 原文线索

V2-078678 | MATHDB-401360

Asymptotic comparison of Hardy quasi-norms of Dirichlet polynomials for exponents below 1

摘录 / EXCERPT: Estimate asymptotically $\mho(q,p,x)$ for values $0<p<q<+\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-078679 | MATHDB-401361

Even-order cases of the logarithmically averaged Chowla conjecture

摘录 / EXCERPT: For every even integer $k\ge 4$ and every collection of distinct nonnegative integers $h_1,\ldots,h_k$, is it true that ...

conjecture | open | MathDB 页面 | 原文线索

V2-078680 | MATHDB-401362

Erdős-Ingham Tauberian question for a non-decreasing function

摘录 / EXCERPT: Let $f:\mathbb{R}_{>0}\to\mathbb{R}_{>0}$ be non-decreasing. If ... does it necessarily follow that ...

open_problem | open | MathDB 页面 | 原文线索

V2-078681 | MATHDB-401363

Zeros of a Schwartz function and its Fourier transform on expanding spheres

摘录 / EXCERPT: Let $f\in\mathcal{S}(\mathbb{R}^n)$ be a Schwartz function. Let $r_1<r_2<r_3<\cdots$ and $\rho_1<\rho_2<\rho_3<\cdots$ be two sequences of radii tending to infinity, and suppose th...

open_problem | open | MathDB 页面 | 原文线索

V2-078682 | MATHDB-401364

Bayart’s partial-sum norm question for bounded Dirichlet series

摘录 / EXCERPT: Let $f(s)=\sum_{n=1}^{\infty}a_n n^{-s}\in H^{\infty}(\mathbb C_0)$ be a Dirichlet series with partial sums ... Let ... and further let ... Is it the case that $\sigma_N\to 0$ as...

open_problem | open | MathDB 页面 | 原文线索

V2-078683 | MATHDB-401366

Finding a completely positive factorization

摘录 / EXCERPT: Finding a representation $A=\sum_{i=1}^{r} b_i b_i^{\mathsf T}$ with $b_i\in\mathbb{R}_+^n$ of a general completely positive matrix A is an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078684 | MATHDB-401367

Characterization of supporting hyperplanes of the completely positive cone

摘录 / EXCERPT: Characterize the supporting hyperplanes of the cone of completely positive matrices.

open_problem | open | MathDB 页面 | 原文线索

V2-078685 | MATHDB-401368

Membership in NP for complete positivity

摘录 / EXCERPT: Whether or not determining that a matrix $C\in\mathcal{C}_n$ is actually in NP.

open_problem | open | MathDB 页面 | 原文线索

V2-078686 | MATHDB-401369

Rational cp-factorizations of rational completely positive matrices

摘录 / EXCERPT: Does every rational completely positive matrix have a rational cp factorization, i.e. a factorization $A=BB^{T}$ in which B has rational, nonnegative entries?

open_problem | open | MathDB 页面 | 原文线索

V2-078687 | MATHDB-401371

Computing the cp-rank of a completely positive matrix

摘录 / EXCERPT: Given a completely positive matrix $A\in CP_n$, determine how to compute its cp-rank ...

open_problem | open | MathDB 页面 | 原文线索

V2-078688 | MATHDB-401372

Rotations and permutations of separating hyperplanes for the unit modulus lifting

摘录 / EXCERPT: For $n=4$, do rotations and permutations of the discussed separating hyperplanes describe $\\mathrm{UML}_{\mathbb C}$?

open_problem | open | MathDB 页面 | 原文线索

V2-078689 | MATHDB-401373

Do rotations and permutations of the candidate hyperplanes describe U_{MLC} for $n=4$?

摘录 / EXCERPT: For $n=4$, do rotations and permutations of the hyperplanes obtained from the candidate separating hyperplane (H,α) describe U_{MLC}?

open_problem | open | MathDB 页面 | 原文线索

V2-078690 | MATHDB-401374

Nonnegative inverse eigenvalue problem (NIEP)

摘录 / EXCERPT: Which lists of complex numbers occur as the spectrum of some entry-wise nonnegative matrix?

open_problem | open | MathDB 页面 | 原文线索

V2-078691 | MATHDB-401376

Symmetric nonnegative inverse eigenvalue problem for nonzero spectra

摘录 / EXCERPT: Determine which finite lists of real numbers can occur as the nonzero spectrum (after deleting zero eigenvalues) of a real symmetric entrywise-nonnegative matrix.

open_problem | open | MathDB 页面 | 原文线索

V2-078692 | MATHDB-401377

Nonnegative inverse elementary divisors problem: nonderogatory realization conjecture

摘录 / EXCERPT: It is conjectured that if a spectrum σ is realizable by a nonnegative matrix, then σ is realizable by a nonnegative nonderogatory matrix.

conjecture | open | MathDB 页面 | 原文线索

V2-078693 | MATHDB-401378

Realizability of cone-preserving graphs

摘录 / EXCERPT: For fixed positive integers m,n with $m\ge n$, characterize the directed graphs on m vertices that are realizable as $D_K(A)$ for a proper polyhedral cone...

open_problem | open | MathDB 页面 | 原文线索

V2-078694 | MATHDB-401379

Characterization of SPN graphs

摘录 / EXCERPT: Determine all graphs G such that every real symmetric copositive matrix whose graph is G is SPN, where a matrix is copositive if $x^{T}Ax\ge 0$ for every nonnegative vector x...

open_problem | open | MathDB 页面 | 原文线索

V2-078695 | MATHDB-401381

Fixed-bounded-quantifier separations in relativized bounded arithmetic

摘录 / EXCERPT: Can $T^2_{n+1}(\b1)$ be separated from $T^2_n(\b1)$ by a $\forall\Sigma^b_k(\u0003b1)$ sentence for some fixed k?

open_problem | open | MathDB 页面 | 原文线索

V2-078696 | MATHDB-401384

Constructive validity of universe rules in the simplicial model

摘录 / EXCERPT: Can the rules concerning universes in Voevodsky’s simplicial model of type theory be validated constructively, beyond the fragment concerning exponentials of uniform Kan complexes...

open_problem | open | MathDB 页面 | 原文线索

V2-078697 | MATHDB-401385

Instability of the N-wave and traveling-wave profiles

摘录 / EXCERPT: For the coagulation equation with the kernel family $K_\alpha(\xi,\eta)=c_\alpha(\xi+\eta)^{1-2\alpha}\xi^\alpha\eta^\alpha$ from (6), it is conjectured that the N-wave and the tra...

open_problem | open | MathDB 页面 | 原文线索

V2-078698 | MATHDB-401386

Scaling limit from the Markov-chain nonquadratic structure to Onsager/MFT quadratic functionals

摘录 / EXCERPT: The report conjectures that the macroscopic quadratic functionals of Onsager theory and Macroscopic Fluctuation Theory, denoted by (1) and (4), appear in a scaling limit in which t...

open_problem | open | MathDB 页面 | 原文线索

V2-078699 | MATHDB-401387

Examples of coefficient functions for local observables

摘录 / EXCERPT: Find examples of functions $(F_k)_{k=0}^{\infty}$ fulfilling the above conditions of ω-locality and closability via Proposition 2.

open_problem | open | MathDB 页面 | 原文线索

V2-078700 | MATHDB-401388

Positivity classification for the Berezin form on open $SO(n,n)$-orbits

摘录 / EXCERPT: For $X=\mathrm{Gr}_n(\mathbb{R}^{2n})$, $H=SO(n,n)$, and the open H-orbits ... let $E_+=C_c^\infty(O_j)$. With Berezin form ... its restriction to E_+ is positive semidefinite if a...

open_problem | open | MathDB 页面 | 原文线索

V2-078701 | MATHDB-401389

Periodic striped ground states for the two-dimensional dipolar Ising model

摘录 / EXCERPT: Prove that the ground states of model (1) with $d=2$ and $p=3$ are periodic and striped, for all sufficiently large J.

open_problem | open | MathDB 页面 | 原文线索

V2-078702 | MATHDB-401391

Isomorphism classification of Möbius covariant nets constructed from KMS states

摘录 / EXCERPT: Determine whether the Möbius covariant nets obtained from the standard half-sided modular inclusions arising from the continuously many KMS states on the real-line restriction of (...

open_problem | open | MathDB 页面 | 原文线索

V2-078703 | MATHDB-401392

Kinetic behaviour of Lorentz gases with generic long-range potentials

摘录 / EXCERPT: What kinetic behaviour should be expected for generic families of long-range potentials $\{\Phi(x,\varepsilon)\}_{\varepsilon>0}$, in particular for potentials of the form...

open_problem | open | MathDB 页面 | 原文线索

V2-078704 | MATHDB-401393

Uniqueness of critical global minimisers modulo dilations

摘录 / EXCERPT: In the porous-medium fair-competition regime $k\in(-N,0)$, with $m=1-k/N\in(1,2)$ and critical interaction strength $\chi=\chi_c$, is there a unique global minimiser of the free-en...

open_problem | open | MathDB 页面 | 原文线索

V2-078705 | MATHDB-401401

Extending tractional constructions to fractional calculus without infinity or approximation

摘录 / EXCERPT: An exciting problem is thus to overcome tractional constructions in order to include such class of problems, extending the power of analogue computation whilst still avoiding the i...

open_problem | open | MathDB 页面 | 原文线索

V2-078706 | MATHDB-401403

Existence of non-radial steady states in a thin-rim tissue-growth model

摘录 / EXCERPT: For the moving-boundary problem consisting of ... and ... with a specified initial tissue domain, determine whether non-radial steady states exist.

open_problem | open | MathDB 页面 | 原文线索

V2-078707 | MATHDB-401404

Optimal solution strategies for linear systems in higher-order unfitted finite element methods

摘录 / EXCERPT: What are optimal solution strategies for the typically very ill-conditioned linear systems arising in higher-order unfitted finite element methods?

open_problem | open | MathDB 页面 | 原文线索

V2-078708 | MATHDB-401405

Rigorous proof of the zero-point-energy conjecture in the OT limit of DFT

摘录 / EXCERPT: Is it possible to prove rigorously the ZPE conjecture at least for dimension $d=1$, and possibly for the physically relevant dimension $d=3$?

conjecture | open | MathDB 页面 | 原文线索

V2-078709 | MATHDB-401406

Crystallization of the Riesz gas for all exponents above the dimension

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a bounded domain with $|\Omega|=1$, set $\Omega_N=N^{1/d}\Omega$, and define the Riesz energy ... The conjecture is that, for every $s>d$, the mini...

open_problem | open | MathDB 页面 | 原文线索

V2-078710 | MATHDB-401408

Crystallization and the conjectured optimal lattice for three-dimensional Jellium

摘录 / EXCERPT: It is conjectured that the three-dimensional Jellium system crystallizes. More specifically, the particles are conjectured to form the FCC lattice for $\frac{3}{2} \le s < 3$ and t...

conjecture | open | MathDB 页面 | 原文线索

V2-078711 | MATHDB-401411

Long-time existence for the semi-geostrophic equations with variable rotation

摘录 / EXCERPT: Extend the short-time existence result for the semi-geostrophic equations with variable rotation to obtain long-time existence of weak solutions.

open_problem | open | MathDB 页面 | 原文线索

V2-078712 | MATHDB-401412

Optimal strong convergence order for CIR-process approximations

摘录 / EXCERPT: In the full parameter range $a,\sigma\in(0,\infty)$, the convergence order in (4) is equal to $\min\{2a/\sigma^{2},1\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-078713 | MATHDB-401413

Nontrivial automorphisms of the Turing degrees

摘录 / EXCERPT: Does the structure of the Turing degrees have a nontrivial automorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-078714 | MATHDB-401414

Lattice embeddings into the c.e. degrees for S_7

摘录 / EXCERPT: A c.e. degree bounds an embedding of the 7-element lattice S_7 into the c.e. degrees if and only if the degree is not totally $\c9$-c.a.

open_problem | open | MathDB 页面 | 原文线索

V2-078715 | MATHDB-401416

Friedberg numberings for higher levels of the analytical hierarchy

摘录 / EXCERPT: For $n>2$, does the family of all $\a0\Sigma^1_n$ sets have a $\a0\Sigma^1_n$-computable Friedberg numbering?

open_problem | open | MathDB 页面 | 原文线索

V2-078716 | MATHDB-401417

Finite-family analogue for a two-element Rogers semilattice

摘录 / EXCERPT: For a level $n\ge 2$, does there exist a finite family S of Σ^{-1}_n sets whose Rogers semilattice $R^{-1}_n(S)$ of Σ^{-1}_n-computable numberings has exactly two d...

open_problem | open | MathDB 页面 | 原文线索

V2-078717 | MATHDB-401418

Almost-everywhere uniform computable categoricity for finite-branching trees

摘录 / EXCERPT: Franklin and Miller conjecture that, for finite-branching infinite trees in the original signature, without branching predicates, measure-0-many of these trees are uniformly comput...

open_problem | open | MathDB 页面 | 原文线索

V2-078718 | MATHDB-401420

Martin-Löf random real with low discrepancy under powers of 2

摘录 / EXCERPT: For a sequence $(x_n)_{n\ge 1}$ of real numbers, define ... where $\{x\}=x-\lfloor x\rfloor$. Is there a Martin-Löf random real x such that, for every sufficiently large N, ...

open_problem | open | MathDB 页面 | 原文线索

V2-078719 | MATHDB-401421

Computable finitely presented groups and d-Σ^0_2 Scott sentences

摘录 / EXCERPT: Is there a computable finitely presented group with no d-Σ_2 Scott sentence?

open_problem | open | MathDB 页面 | 原文线索

V2-078720 | MATHDB-401422

Converse of Ershov’s theorem for Rogers semilattices

摘录 / EXCERPT: Let $\mathcal P$ and $\mathcal Q$ be finite families of computably enumerable sets. Ershov proved that if the Rogers semilattices of $\mathcal P$ and $\mathcal Q$ are isomorphic, t...

open_problem | open | MathDB 页面 | 原文线索

V2-078721 | MATHDB-401423

Cofinal maximal chains in the Turing degrees without a well-ordering of the reals

摘录 / EXCERPT: Does there exist a model of ZF in which there exists a cofinal maximal chain of Turing degrees of order type ω_1 but there is no well-ordering of the reals?

open_problem | open | MathDB 页面 | 原文线索

V2-078722 | MATHDB-401424

Variational principle for measure-theoretic invariance entropy

摘录 / EXCERPT: “The question of whether there is a variational principle linking this new concept to invariance entropy remains open.” Here, “this new concept” refers to the measure-theoretic ext...

open_problem | open | MathDB 页面 | 原文线索

V2-078723 | MATHDB-401426

Data-rate thresholds and restoration entropy for deterministic nonlinear systems

摘录 / EXCERPT: Let 4dot{x}=f(x)$, with \$xinmathbb{R}^n$\, \$tin[0,infty)$\, and initial state \$x0in K$\, where \$f:mathbb{R}^ntomathbb{R}^n$\ is \$C^1$\-smooth and \$Knevarnothing...

open_problem | open | MathDB 页面 | 原文线索

V2-078724 | MATHDB-401427

Optimal control under multiplicative observation noise

摘录 / EXCERPT: For the scalar system … where the variables C_n are drawn independently and identically from a continuous distribution such as $\mathcal N(1,1)$ or $\operatorname{Unif}[1,2]$, an...

open_problem | open | MathDB 页面 | 原文线索

V2-078725 | MATHDB-401428

Optimal decentralized control for the noisy two-controller system

摘录 / EXCERPT: For the discrete-time scalar system … where $W(t)$ are independent and identically distributed Gaussian random variables with mean 0 and variance 1, let $X(0)$ be an independen...

open_problem | open | MathDB 页面 | 原文线索

V2-078726 | MATHDB-401433

Minimax rate in the large-sparsity regime for multidimensional linear-functional estimation

摘录 / EXCERPT: In the Gaussian sequence model, observe independent vectors $Y_i=\theta_i+\sigma\xi_i\in\mathbb{R}^p$, $i=1,\ldots,n$, where $\xi_i\stackrel{\mathrm{iid}}\sim N_p(0,I_p)$, σ...

open_problem | open | MathDB 页面 | 原文线索

V2-078727 | MATHDB-401434

Robust operator-valued U-statistic deviation bound

摘录 / EXCERPT: Let $X_1,\ldots,X_n$ be i.i.d. random variables taking values in a measurable space $(S,\mathcal B)$, with $n\ge 2$. Let $2\le m\le n$, let $\mathbb H_d$ denote the set of...

open_problem | open | MathDB 页面 | 原文线索

V2-078728 | MATHDB-401435

Less conservative error measures for constrained model-X knockoff procedures

摘录 / EXCERPT: In practical settings, are there less conservative measures of the approximation error in the model for the feature vector X that lead to false discovery rate (FDR) control resul...

open_problem | open | MathDB 页面 | 原文线索

V2-078729 | MATHDB-401437

Asymptotically efficient estimation of smooth covariance-operator functionals

摘录 / EXCERPT: Let $X_1,\dots,X_n$ be i.i.d. observations from a centered normal distribution on $\mathbb{R}^d$ with unknown covariance operator $\Sigma\in C^d_+$. For a smooth function...

open_problem | open | MathDB 页面 | 原文线索

V2-078730 | MATHDB-401438

Local sin Θ bound for empirical covariance eigenspaces

摘录 / EXCERPT: Let $d\ge 1$. Let $x>0$ be such that … Suppose that … Then …

open_problem | open | MathDB 页面 | 原文线索

V2-078731 | MATHDB-401439

Analytic-type decay estimate for equilibrium displacements

摘录 / EXCERPT: The report asks whether the stronger estimate $\left|D^j\bar u(\ell)\right|\le C_j\,j!\,|\ell|^{1-d-j}$ holds for the limiting equilibrium displacement $\bar u$ in the tight-bindin...

open_problem | open | MathDB 页面 | 原文线索

V2-078732 | MATHDB-401443

Optimization of the first bands of a Hill operator

摘录 / EXCERPT: Given M functions $b_1,\dots,b_M$ defined on $(-1/2,1/2)$, is it possible to find an optimal periodic potential V minimizing … where $q\mapsto\varepsilon^{V}_{m,q}$ is the mt...

open_problem | open | MathDB 页面 | 原文线索

V2-078733 | MATHDB-401444

Mathematical theory for nonlinear projection-based embedding

摘录 / EXCERPT: Develop a rigorous mathematical theory for the nonlinear extension of projection-based embedding theory (PET) for the energy functional...

open_problem | open | MathDB 页面 | 原文线索

V2-078734 | MATHDB-401446

Sampling theorem for totally positive generators of Gaussian type

摘录 / EXCERPT: Assume that g is totally positive of Gaussian type, as in ... If $\Lambda\subseteq\mathbb R$ is separated and $D^-(\Lambda)>1$, then Λ is a stable sampling set for ... That i...

open_problem | open | MathDB 页面 | 原文线索

V2-078735 | MATHDB-401447

Sampling with derivatives in shift-invariant spaces generated by totally positive functions of Gaussian type

摘录 / EXCERPT: Let g be totally positive of Gaussian type, with Fourier transform ... Let $\Lambda\subseteq\mathbb R$ be separated, and let m_Λ be a bounded sequence of multiplicities, me...

open_problem | open | MathDB 页面 | 原文线索

V2-078736 | MATHDB-401448

Random initialization for gradient descent in Gaussian phase retrieval

摘录 / EXCERPT: Given measurements $y_i=(a_i^{\top}x^\natural)^2$ for $1\le i\le m$, with $x^\natural\in\mathbb{R}^n$ and independent designs $a_i\sim\mathcal{N}(0,I_n)$, consider vanilla gradient...

open_problem | open | MathDB 页面 | 原文线索

V2-078737 | MATHDB-401452

Twisted X-rays and the conjectured double-peak diffraction pattern

摘录 / EXCERPT: A diffraction experiment of twisted X-rays illuminating an aligned helical structure is conjectured to produce a highly structured diffraction pattern, with the outgoing field in a...

conjecture | open | MathDB 页面 | 原文线索

V2-078738 | MATHDB-401453

A Cheeger inequality for principal bundles

摘录 / EXCERPT: A proper analogy of the Cheeger inequality seems natural but is missing so far in the principal bundle framework.

open_problem | open | MathDB 页面 | 原文线索

V2-078739 | MATHDB-401454

The mismatch principle for nonlinear observations

摘录 / EXCERPT: Determine a target vector $x^\natural\in K$ that captures the “parametric” structure of the observation rule and minimizes the mismatch covariance $\rho(x^\natural)$ at the same ti...

open_problem | open | MathDB 页面 | 原文线索

V2-078740 | MATHDB-401455

Choosing a mismatch-minimizing target for the generalized Lasso

摘录 / EXCERPT: Determine a target vector $x^\natural\in K$ that captures the parametric structure of the observation rule while minimizing the mismatch covariance...

open_problem | open | MathDB 页面 | 原文线索

V2-078741 | MATHDB-401456

Distribution of the sampled SDP value for k-means clustering

摘录 / EXCERPT: For the random subset S of fixed size s used in the Monte Carlo SDP procedure, understand the distribution of $\operatorname{val}(S\text{-SDP})$.

open_problem | open | MathDB 页面 | 原文线索

V2-078742 | MATHDB-401460

Theorem 3: categorical vanishing criterion

摘录 / EXCERPT: Let $\mathcal C$ and $\mathcal D$ be cocomplete stable categories equipped with t-structures compatible with filtered colimits, and suppose the t-structure on $\mathcal D$ is l...

open_problem | open | MathDB 页面 | 原文线索

V2-078743 | MATHDB-401461

Minimal uniform controllability time for an advection-diffusion equation

摘录 / EXCERPT: Let $L>0$, $T>0$, $Q_T=(0,L)\times(0,T)$, $M\in\mathbb{R}\setminus\{0\}$, and $\varepsilon>0$. Consider the boundary-controlled problem ... with $y_0^\varepsilon\in H^{-1}(0,L)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078744 | MATHDB-401462

Necessity of an additional complementarity condition in an optimal control problem

摘录 / EXCERPT: For the optimal control problem (P) governed by the variational inequality (3), determine whether the additional complementarity condition (5), $\langle p,1-|\bar q|\rangle=0$, is...

open_problem | open | MathDB 页面 | 原文线索

V2-078745 | MATHDB-401466

Second-order growth and control error estimates for a bang-bang parabolic control problem

摘录 / EXCERPT: Under the first-order optimality conditions, condition (11), and the additional hypothesis ... there exist $\varepsilon>0$, $\nu>0$, and $\kappa>0$ such that ... for every...

open_problem | open | MathDB 页面 | 原文线索

V2-078746 | MATHDB-401467

Stability characterization for minimal and maximal QVI solutions

摘录 / EXCERPT: Specifically, if $\{f_n\}$ is in A_{ad}, what conditions are necessary and sufficient so that $m(f_n) \to m(f)$ and $M(f_n) \to M(f)$ in H and in V?

open_problem | open | MathDB 页面 | 原文线索

V2-078747 | MATHDB-401468

Well-posedness of a phase-field system with maximal monotone perturbations

摘录 / EXCERPT: For $Q=(0,T)\times\Omega$, consider the system for the temperature ϑ, phase parameter φ, and selections ζ,ξ: ... ... with homogeneous Neumann conditions...

open_problem | open | MathDB 页面 | 原文线索

V2-078748 | MATHDB-401470

Directional differentiability of the maximal solution mapping for elliptic obstacle-type QVIs

摘录 / EXCERPT: Under all hypotheses (1)-(5) of Theorem 3 on the supplied pages, consider the quasi-variational inequality ... and let $Q(f)$ denote its (possibly multivalued) solution set. If...

open_problem | open | MathDB 页面 | 原文线索

V2-078749 | MATHDB-401471

Local quadratic convergence of SQP for the H^1-regularized, box-constrained optimal control problem

摘录 / EXCERPT: For the optimal control problem ... subject to ... and the box constraints ... develop the required techniques and establish local quadratic convergence of the Lagrange-Newton SQP method...

open_problem | open | MathDB 页面 | 原文线索

V2-078750 | MATHDB-401476

Arithmetic coarse invariants of cyclic special-locus components

摘录 / EXCERPT: In higher genus, find arithmetic coarse invariants of the irreducible components of $\mathcal{M}_{g,[m]}(\gamma)$, where γ generates a cyclic group and...

open_problem | open | MathDB 页面 | 原文线索

V2-078751 | MATHDB-401482

Faithfulness of the universal monodromy representation

摘录 / EXCERPT: For integers $g,n\ge 0$ satisfying $2-2g-n<0$, and after choosing a $\mathbb{Q}$-rational (tangential) base point on $\mathcal{M}_{g,n}$ so that (A) splits, the universal monodro...

open_problem | open | MathDB 页面 | 原文线索

V2-078752 | MATHDB-401484

Langlands correspondence for \(\mathrm{GL}_r\) with ultraproduct coefficients

摘录 / EXCERPT: In the setting of the report, prove the Langlands correspondence for GL_r with F_U-coefficients; namely, prove that the injective map $(A) \longrightarrow (G)$ describ...

open_problem | open | MathDB 页面 | 原文线索

V2-078753 | MATHDB-401485

Grothendieck’s Conjecture 2 on anabelian neighborhoods

摘录 / EXCERPT: Conjecture 2: If X is smooth, then every $x\in X$ has a basis of Zariski neighborhoods which are anabelian.

conjecture | open | MathDB 页面 | 原文线索

V2-078754 | MATHDB-401494

Almost sure scattering for the one-dimensional nonlinear Schrödinger equation

摘录 / EXCERPT: Let $3<p<5$. The solution U constructed in Theorem 1 scatters in the following sense: there exists $\c3>0$ and, $\bc$-almost surely, states...

open_problem | open | MathDB 页面 | 原文线索

V2-078755 | MATHDB-401496

Exact change-of-variables density for the periodic DNLS gauge

摘录 / EXCERPT: For the Gaussian measure γ_k on $L^2(\mathbb T)$ with covariance $(1+(-\Delta)^k)^{-1}$, and the periodic derivative-NLS gauge group $\{G_\alpha\}_{\alpha\in\mathbb R}$, det...

open_problem | open | MathDB 页面 | 原文线索

V2-078756 | MATHDB-401497

Microscopic convergence of time-dependent correlation functions for the 1D cubic NLS

摘录 / EXCERPT: Theorem 1. Let $w\in L^\infty(\Lambda)$ and $w\ge 0$ pointwise. For $m\in\mathbb N$, $p_1,\ldots,p_m\in\mathbb N$, $\xi_1\in L(H^{(p_1)}),\ldots,\xi_m\in L(H^{(p_m)})$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-078757 | MATHDB-401498

Whether all expanders are superexpanders

摘录 / EXCERPT: Whether all expanders are superexpanders.

open_problem | open | MathDB 页面 | 原文线索

V2-078758 | MATHDB-401502

Property (T) for \(\operatorname{SAut}(F_5)\)

摘录 / EXCERPT: The group $\operatorname{SAut}(F_5)$ has property (T) with Kazhdan constant $\kappa\ge 0.176$.

open_problem | open | MathDB 页面 | 原文线索

V2-078759 | MATHDB-401503

Existence of CAT(0)-expanders

摘录 / EXCERPT: Does there exist an expander relative to all CAT(0) spaces?

open_problem | open | MathDB 页面 | 原文线索

V2-078760 | MATHDB-401504

Cocompact lattices that are not biLipschitz equivalent

摘录 / EXCERPT: Is there a solvable or nilpotent Lie group that contains two cocompact lattices that are not biLipschitz equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-078761 | MATHDB-401505

BiLipschitz versus quasi-isometric equivalence for finitely generated groups

摘录 / EXCERPT: Is biLipschitz equivalence the same as quasi-isometric equivalence for finitely generated groups?

open_problem | open | MathDB 页面 | 原文线索

V2-078762 | MATHDB-401510

Warped-cone subspaces violating the coarse Baum-Connes conjecture

摘录 / EXCERPT: Assume that an action $\Gamma\curvearrowright Y$ by Lipschitz homeomorphisms is free and has a spectral gap, and that Γ has property A. Then the surjectivity part of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-078763 | MATHDB-401511

Failure of coarse Baum-Connes surjectivity for subspaces of warped cones

摘录 / EXCERPT: Let an action $\Gamma \curvearrowright Y$ by Lipschitz homeomorphisms be free and have a spectral gap, and suppose that Γ has property A. If $\mathcal O'_{\Gamma}Y$ denot...

open_problem | open | MathDB 页面 | 原文线索

V2-078764 | MATHDB-401512

Improving fixed-point estimates for triangular-model random groups

摘录 / EXCERPT: Let δ be the conformal dimension of the boundary of a triangular-model random group, and let $p_* = \sup\{p\in(0,\infty): \text{the group has property }F_{L^p}\}$. The repor...

open_problem | open | MathDB 页面 | 原文线索

V2-078765 | MATHDB-401513

Dependence of congruence subgroups on an embedding into $\mathrm{SL}_N(\mathbb{Z})$

摘录 / EXCERPT: Let G be a group with a given embedding into $\mathrm{SL}_N(\mathbb{Z})$, and define a congruence subgroup of G as the intersection of G with a congruence subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-078766 | MATHDB-401514

Upgrading Problem

摘录 / EXCERPT: Let $\alpha:G\curvearrowright X$. Let $(M_j)_{j\in J}$ be a family of subgroups of G such that $\left\langle\bigcup_{j\in J}M_j\right\rangle=G$. Assume that, for every $j\in J$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078767 | MATHDB-401515

Fixed-point property for elementary groups on noncommutative L^q-spaces

摘录 / EXCERPT: Assume that R is a finitely generated unital associative ring. For every $n\ge 4$, let $G_R=E(n,R)$ be the elementary group. Then G_R has property $(F^{\mathcal E})$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-078768 | MATHDB-401516

Incomparable sequences of bounded-degree graphs

摘录 / EXCERPT: Problem 1 (Naor). Do there exist two sequences of bounded degree graphs that are incomparable? Two sequences of bounded degree graphs are said to be incomparable if each is an expa...

open_problem | open | MathDB 页面 | 原文线索

V2-078769 | MATHDB-401517

Coarse embeddability of $L^p([0,1])$ into $\[1m\ell_p$

摘录 / EXCERPT: Let $2<p<\infty$. Does $L^p([0,1])$ embed coarsely into ℓ_p?

open_problem | open | MathDB 页面 | 原文线索

V2-078770 | MATHDB-401523

Farb’s conjecture on multisections of the universal cubic curve

摘录 / EXCERPT: There is no multisection for the bundle ξ of degree $n<9$.

conjecture | open | MathDB 页面 | 原文线索

V2-078771 | MATHDB-401524

Multisection associated to triples of non-collinear inflection points on smooth cubic curves

摘录 / EXCERPT: Let X be the space of smooth plane cubic curves modulo nonzero scalar multiplication, let $E=\{(F,p)\in X\times\mathbb{CP}^{2}:p\in C_{F}\}$ be the universal cubic curve, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078772 | MATHDB-401525

Degeneration of the pole-order spectral sequence for hyperplane arrangements

摘录 / EXCERPT: For any hyperplane arrangement $V:f=0$ in $ \mathbb{P}^n$ and any integer k with $1\le k\le d$, where d is the number of hyperplanes in V, the E_1-spectral sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-078773 | MATHDB-401530

Hilbert’s 13th Problem for resolvent degree

摘录 / EXCERPT: $\operatorname{RD}(\widetilde{\mathcal P}_7\to\mathcal P_7)=3$

open_problem | open | MathDB 页面 | 原文线索

V2-078774 | MATHDB-401531

Unbounded resolvent degree for general polynomials

摘录 / EXCERPT: Conjecture 6: $\operatorname{RD}(\widetilde{\mathcal P}_n\to\mathcal P_n)\to\infty$ as $n\to\infty$; here $\mathcal P_n$ is the space of monic degree-n polynomials and...

open_problem | open | MathDB 页面 | 原文线索

V2-078775 | MATHDB-401532

Hilbert’s Sextic Conjecture

摘录 / EXCERPT: Hilbert’s Sextic Conjecture asks whether ... where P_n is the space of monic degree-n polynomials and $\widetilde{P}_n\to P_n$ is the finite map obtained by choosing a root.

conjecture | open | MathDB 页面 | 原文线索

V2-078776 | MATHDB-401533

Equality of the Andreadakis-Johnson and representation-theoretic filtrations

摘录 / EXCERPT: For every free-group rank $n\ge 3$ and every integer $k\ge 1$, conjecture that $A_n(k)=D_n(k)$, where $A_n(k)$ is the k-th term of the Andreadakis-Johnson filtration of...

open_problem | open | MathDB 页面 | 原文线索

V2-078777 | MATHDB-401540

Schwarz genus of the flex cover of smooth plane cubic curves

摘录 / EXCERPT: Let the Schwarz genus of a cover $f:Y\to X$ be the minimum number k such that X has a cover by connected open subsets $U_1,U_2,\ldots,U_k$ on each of which the restriction of...

open_problem | open | MathDB 页面 | 原文线索

V2-078778 | MATHDB-401542

Parity of uniformly chosen entries in the character table of S_n

摘录 / EXCERPT: For $\[?]$ partitions $\[?]$ of n, let $\[?]$ denote the irreducible character of S_n corresponding to $\[?]$. As $n\to\infty$, ... Equivalently, an entry chosen uniformly at ran...

open_problem | open | MathDB 页面 | 原文线索

V2-078779 | MATHDB-401543

Presentation of the fundamental group of unlabeled affine hyperplane arrangements

摘录 / EXCERPT: Let $M_{d,n}$ denote the space of unlabeled affine arrangements of n hyperplanes in general position in $\a0^d$. Find a presentation for $\pi_1(M_{d,n})$ that specializes to Arti...

open_problem | open | MathDB 页面 | 原文线索

V2-078780 | MATHDB-401548

Nonexistence of cubic spline minimizers on a cylinder for irrational interpolation time

摘录 / EXCERPT: Let M be a cylinder, with interpolation times $t_1=0$, $t_2\in(0,1)$, and $t_3=1$, and interpolation data $x_1=x_3$ and x_2 exactly opposite to x_1. For integers $a,b\ge 0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078781 | MATHDB-401550

HOSVD-based compression is a retraction

摘录 / EXCERPT: The report states that the higher-order singular value decomposition (HOSVD)-based compression to multilinear rank k, denoted by T_k, is a retraction.

open_problem | open | MathDB 页面 | 原文线索

V2-078782 | MATHDB-401553

Gross-McMullen sufficiency conjecture for irreducible characteristic polynomials

摘录 / EXCERPT: Let (r,s) satisfy $r\equiv s\pmod 8$, let $2n=r+s$, and let $F\in\mathbb Z[X]$ be irreducible of degree $2n$. If … … … where $m(F)$ is the number of roots $z\in\mathbb C$ of F...

conjecture | open | MathDB 页面 | 原文线索

V2-078783 | MATHDB-401554

Counterexample to the hereditarily Pythagorean extension in higher genus

摘录 / EXCERPT: Let $k=\mathbb{R}((t))$, and let $F=k(C)$ be the function field of the elliptic curve C given by $Y^2=(X^2+1)(tX-1)$. There exists an element of F that is a sum of three square...

open_problem | open | MathDB 页面 | 原文线索

V2-078784 | MATHDB-401556

Hasse principle for semisimple simply connected groups over semi-global fields with finite residue field

摘录 / EXCERPT: Let K be a complete discretely valued field with finite residue field k, let X be a smooth geometrically integral projective curve over K, and set $F=K(X)$. Let Ω be...

open_problem | open | MathDB 页面 | 原文线索

V2-078785 | MATHDB-401557

Systems of quadratic forms over complete discretely valued fields

摘录 / EXCERPT: Let $u_r(F)$ denote the supremum of the positive integers n such that there exists an anisotropic system of r quadratic forms over F in n variables. Let K be a complete d...

open_problem | open | MathDB 页面 | 原文线索

V2-078786 | MATHDB-401558

Systems of quadratic forms over p-adic fields

摘录 / EXCERPT: For every p-adic field K and every integer $r\ge 1$, the supremum $u_r(K)$ of the dimensions of anisotropic systems of r quadratic forms over K satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-078787 | MATHDB-401560

Isotropy indices of quadratic forms over function fields of quadrics

摘录 / EXCERPT: Let F be a field, let p and q be anisotropic quadratic forms over F of dimension at least 2, and let $F(p)$ be the function field of the integral projective F-quadric d...

open_problem | open | MathDB 页面 | 原文线索

V2-078788 | MATHDB-401568

Similarity versus equality of Witt indices over all extensions

摘录 / EXCERPT: For quadratic forms q and q' of the same dimension over a field F, is the condition … equivalent to q and q' being similar?

open_problem | open | MathDB 页面 | 原文线索

V2-078789 | MATHDB-401570

Necessary and sufficient conditions for isomorphism of octonion isotopes

摘录 / EXCERPT: Find explicit conditions that are both necessary and sufficient for isotopes to be isomorphic.

open_problem | open | MathDB 页面 | 原文线索

V2-078790 | MATHDB-401571

Whether affine-quadric motives generate the entire Picard group

摘录 / EXCERPT: For a field k of characteristic different from 2, is every invertible object of $\\operatorname{DM}(k,\\mathbb{Z}/2)$ contained in $\\operatorname{Pic}^{\\mathrm{qua}}$, the su...

open_problem | open | MathDB 页面 | 原文线索

V2-078791 | MATHDB-401579

Property (T) for automorphism groups of free groups

摘录 / EXCERPT: It is conjectured that $\operatorname{Aut}(F_d)$ has Kazhdan’s property (T) for all $d\ge 5$.

open_problem | open | MathDB 页面 | 原文线索

V2-078792 | MATHDB-401580

Embeddability of interacting Fock spaces

摘录 / EXCERPT: Every interacting Fock space can be embedded, preserving the vacuum and the natural grading, into a usual full Fock space. Under this embedding, its creators are represented as “sq...

open_problem | open | MathDB 页面 | 原文线索

V2-078793 | MATHDB-401584

Weak Tsirelson Conjecture

摘录 / EXCERPT: For all $n,k\in\mathbb{N}$, does $C_{qa}(n,k)=C_{qc}(n,k)$?

conjecture | open | MathDB 页面 | 原文线索

V2-078794 | MATHDB-401587

Noncommutative Khintchine equivalence in $L^{2,\infty}$

摘录 / EXCERPT: Is ...

open_problem | open | MathDB 页面 | 原文线索

V2-078795 | MATHDB-401588

Strong property (T) for \(\mathrm{SL}_n(\mathbb{Z})\)

摘录 / EXCERPT: For $n\ge 3$, $\mathrm{SL}_n(\mathbb{Z})$ has strong property (T).

open_problem | open | MathDB 页面 | 原文线索

V2-078796 | MATHDB-401592

Non-random unitary-matrix examples for a minimum output entropy counterexample

摘录 / EXCERPT: Can one find a constant $C>0$, an integer $k>\\exp(2C^2)$, and unitary matrices $U_1,\ldots,U_k\in M_n(\mathbb{C})$ such that, for any n^2 matrix coefficients a_{ij}, ... where t...

open_problem | open | MathDB 页面 | 原文线索

V2-078797 | MATHDB-401593

Ozawa’s entropy-variation inequality

摘录 / EXCERPT: Let $H(\varphi)$ denote the von Neumann entropy of φ and define ... Is it true that ... for every $A\geq 0$?

open_problem | open | MathDB 页面 | 原文线索

V2-078798 | MATHDB-401596

Benjamini-Schramm local weak limit of random graphs embeddable on a fixed surface

摘录 / EXCERPT: What is the limit of the local structure of $S_g(n,m)$ in the sense of the Benjamini-Schramm local weak convergence?

open_problem | open | MathDB 页面 | 原文线索

V2-078799 | MATHDB-401597

Characterization of asymptotically maximal Kronecker coefficients

摘录 / EXCERPT: Let $\{\lambda^{(n)}\vdash n\}$, $\{\mu^{(n)}\vdash n\}$, and $\{\nu^{(n)}\vdash n\}$ be three sequences of partitions such that ... Then all three partition sequences are Plancherel...

open_problem | open | MathDB 页面 | 原文线索

V2-078800 | MATHDB-401598

Stanley’s asymptotic maximum for principal evaluations of Schubert polynomials

摘录 / EXCERPT: Let $S_w(x_1,\ldots)$ be the Schubert polynomial for a permutation $w\in S_n$, and define ... Stanley conjectured that the limit ... exists.

open_problem | open | MathDB 页面 | 原文线索

V2-078801 | MATHDB-401600

Closed-form evaluation of Andrews’s determinant for odd n

摘录 / EXCERPT: Let ... where μ is an indeterminate. Define $\rho_0(a,b)=a$ and $\rho_k(a,b)=b$ for $k>0$. If n is an odd positive integer, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-078802 | MATHDB-401603

Bijective proof of the ASM-DPP enumeration

摘录 / EXCERPT: Find a bijective proof of ... where an n-ASM is an $n\times n$ alternating sign matrix and an n-DPP is a column-strict shifted plane partition with largest part at most n whose...

open_problem | open | MathDB 页面 | 原文线索

V2-078803 | MATHDB-401604

Bijection between \(n-1,2)-alternating sign trapezoids and n-alternating sign matrices

摘录 / EXCERPT: Find a bijection between the set of (n-1,2)-alternating sign trapezoids (ASTs) and the set of n-alternating sign matrices (ASMs). Here an n-ASM is an ntimes n matrix with entries i...

open_problem | open | MathDB 页面 | 原文线索

V2-078804 | MATHDB-401607

Exact enumeration of quartic rooted planar Eulerian orientations

摘录 / EXCERPT: Let $R(t)\equiv R$ be the unique formal power series with constant term 0 satisfying ... Then the generating function of quartic rooted planar Eulerian orientations, counted by ver...

open_problem | open | MathDB 页面 | 原文线索

V2-078805 | MATHDB-401610

Stretched-exponential asymptotics for 1324-avoiding permutations

摘录 / EXCERPT: The report conjectures that the coefficients of the Av(1324) generating function satisfy $s_n(1324) \sim B\,\kappa^n\,\mu_1^{\sqrt n}\,n^g$, with $\kappa=11.600\pm0.003$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078806 | MATHDB-401612

A curious determinantal identity at α = i

摘录 / EXCERPT: For every integer $n\ge 1$, let G_n be the graph constructed from an $(n+1)\times n$ square grid by adding, for each $k=1,\ldots,n$, a winding oriented edge from the $(k+1)$-st v...

open_problem | open | MathDB 页面 | 原文线索

V2-078807 | MATHDB-401613

Voronoi-like decomposition of S^1 with randomness

摘录 / EXCERPT: Pick k points $p_1,\ldots,p_k$ on the unit circle S^1 uniformly at random (or equidistantly), and start an independent Brownian motion at each p_i. Let $B_i\subset S^1$ be th...

open_problem | open | MathDB 页面 | 原文线索

V2-078808 | MATHDB-401615

Catalan factor in the determinant of a set-partition transition matrix

摘录 / EXCERPT: Let $A=(a_{ij})$ be the matrix whose entries are the edge weights a_{ij}, and let T be the matrix indexed by the set partitions of $\{1,\ldots,n\}$, with entries $t_{P,Q}$ equa...

open_problem | open | MathDB 页面 | 原文线索

V2-078809 | MATHDB-401616

Existence of stable representations at a prescribed moment-map level

摘录 / EXCERPT: Let Q be a finite quiver with dimension vector $\alpha\in\mathbb N^{Q_0}$, deformation parameter $\lambda\in\mathbb R^{Q_0}$, and stability parameter $\theta\in\mathbb Z^{Q_0}$ s...

open_problem | open | MathDB 页面 | 原文线索

V2-078810 | MATHDB-401617

Semiorthogonal decomposition of the derived category of a rank-two bundle moduli space

摘录 / EXCERPT: Conjecture 7. With C and L as specified in the report, the derived category $D^b(M_C(2,L))$ has a semiorthogonal decomposition ... with two copies of...

open_problem | open | MathDB 页面 | 原文线索

V2-078811 | MATHDB-401619

Classification of simple transitive 2-representations

摘录 / EXCERPT: Classify simple transitive 2-representations for fiat 2-categories with only one left and one right (non-identity) cell.

open_problem | open | MathDB 页面 | 原文线索

V2-078812 | MATHDB-401621

Ascending chain condition for two-sided ideals of $U(W_+)$

摘录 / EXCERPT: Conjecture 2. Two-sided ideals of $U(W_+)$ satisfy the ascending chain condition: all strictly ascending chains of ideals are finite.

open_problem | open | MathDB 页面 | 原文线索

V2-078813 | MATHDB-401623

First-order distinction of non-isomorphic finitely generated fields

摘录 / EXCERPT: Given non-isomorphic f.g. fields K and L, is there a sentence that is true for K and false for L?

open_problem | open | MathDB 页面 | 原文线索

V2-078814 | MATHDB-401628

Harpaz-Wittenberg conjecture on split values of polynomials

摘录 / EXCERPT: Let k be a number field and $n\ge 1$. Let $P_1(t),\ldots,P_n(t)\in k[t]$ be pairwise distinct, irreducible, monic polynomials, and let $k_i=k[t]/(P_i(t))$ and $a_i\in k_i$ be the...

conjecture | open | MathDB 页面 | 原文线索

V2-078815 | MATHDB-401629

Does the Milnor K-ring determine the isomorphism class of a field?

摘录 / EXCERPT: Does the Milnor K-ring $K_*^M(F)$ determine the isomorphism class of the field F?

open_problem | open | MathDB 页面 | 原文线索

V2-078816 | MATHDB-401635

Finite extensions of undecidable fields

摘录 / EXCERPT: Is every finite extension of an undecidable field necessarily undecidable?

open_problem | open | MathDB 页面 | 原文线索

V2-078817 | MATHDB-401636

Finite extensions of decidable fields

摘录 / EXCERPT: Is every finite field extension L/K of a decidable field K necessarily decidable? Here K is called decidable when its first-order theory $\operatorname{Th}(K)$ in the languag...

open_problem | open | MathDB 页面 | 原文线索

V2-078818 | MATHDB-401639

Solvability of finite split embedding problems over $K(x)$ for large fields

摘录 / EXCERPT: Every finite split embedding problem over $K(x)$ is solvable whenever K is a large (ample) field.

open_problem | open | MathDB 页面 | 原文线索

V2-078819 | MATHDB-401642

Split differential embedding problems over rational function fields of large fields

摘录 / EXCERPT: Let K be a large field of characteristic zero and let $F=K(x)$ with derivation $\frac{d}{dx}$. Does every split differential embedding problem over F have a solution?

open_problem | open | MathDB 页面 | 原文线索

V2-078820 | MATHDB-401645

Artin’s conjecture on forms over p-adic fields

摘录 / EXCERPT: Artin conjectured that if $f\in\mathbb{Q}_p[X_1,\ldots,X_n]$ is a form of degree d in $n>d^2$ variables, then f has a nontrivial zero in $\mathbb{Q}_p^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-078821 | MATHDB-401650

Guba-Sapir conjecture on closures and Stallings 2-cores in Thompson’s group F

摘录 / EXCERPT: For a subgroup H of Thompson’s group F generated by tree diagrams, the closure $\bar H$ is precisely the subgroup of all elements of F accepted by the Stallings 2-core...

conjecture | open | MathDB 页面 | 原文线索

V2-078822 | MATHDB-401651

Undistortion of finitely generated closed subgroups of Thompson’s group F

摘录 / EXCERPT: Is every finitely generated closed subgroup of F undistorted?

open_problem | open | MathDB 页面 | 原文线索

V2-078823 | MATHDB-401652

Idempotent decidability for finite tree-like semigroup presentations

摘录 / EXCERPT: Is it decidable whether a semigroup given by a finite tree-like presentation contains an idempotent? Equivalently, is it decidable whether the closure of a finitely generated subgr...

open_problem | open | MathDB 页面 | 原文线索

V2-078824 | MATHDB-401658

Decidability of the conjugacy problem for F_{br} and T_{br}

摘录 / EXCERPT: Is the conjugacy problem decidable for the groups F_{br} and T_{br}? (These are the braided variants of F and T, respectively.)

open_problem | open | MathDB 页面 | 原文线索

V2-078825 | MATHDB-401659

Brin-Sapir conjecture on subgroups of Thompson’s group F

摘录 / EXCERPT: If G is a subgroup of Thompson’s group F, then either G is elementarily amenable or else G contains a subgroup isomorphic to F.

conjecture | open | MathDB 页面 | 原文线索

V2-078826 | MATHDB-401661

Isomorphism of the groups in \(C_n\)

摘录 / EXCERPT: For each $n>1$, let C_n be the set of groups G generated by n fast, 1-bump functions such that G is not a non-trivial direct sum and G is not a non-trivial wreath product...

open_problem | open | MathDB 页面 | 原文线索

V2-078827 | MATHDB-401662

Quasi-isometry classes of simple Thompson-type groups

摘录 / EXCERPT: Do there exist infinitely many quasi-isometry classes of finitely presented simple groups within the realm of Thompson groups?

open_problem | open | MathDB 页面 | 原文线索

V2-078828 | MATHDB-401663

Residual finiteness of hyperbolic groups

摘录 / EXCERPT: Whether all hyperbolic groups are residually finite.

open_problem | open | MathDB 页面 | 原文线索

V2-078829 | MATHDB-401665

Sprawl and statistical hyperbolicity of Thompson’s group F

摘录 / EXCERPT: For Thompson’s group F, and for some natural finite generating sets S, what is the sprawl ... where S_n is the sphere of radius n in $\operatorname{Cay}(F,S)$ and d_S is th...

open_problem | open | MathDB 页面 | 原文线索

V2-078830 | MATHDB-401666

Non-amenable groups with finite classifying spaces and no free subgroups

摘录 / EXCERPT: Does there exist a non-amenable group G (not necessarily in the realm of Thompson groups) with a finite $K(G,1)$ such that G contains no free subgroups? (Weakened version: what...

open_problem | open | MathDB 页面 | 原文线索

V2-078831 | MATHDB-401667

Embedding the Higman group into Lodha-Moore groups or Monod’s group

摘录 / EXCERPT: Does the Higman group $\langle a,b,c,d \mid b^a=b^2,\ c^b=c^2,\ d^c=d^2,\ a^d=a^2\rangle$ embed into the Lodha-Moore groups? Or at least into Monod’s $H(\mathbb{R})$?

open_problem | open | MathDB 页面 | 原文线索

V2-078832 | MATHDB-401669

Canonical concave functions in Alexandrov spaces

摘录 / EXCERPT: Is it true that for any $p\in A$ and any $\varepsilon>0$, there is a $(-2+\varepsilon)$-concave function f_p defined in a neighborhood of p, such that $f_p(p)=0$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-078833 | MATHDB-401672

Existence of maximizing metrics for genus-zero surfaces with boundary

摘录 / EXCERPT: For every integer $k\ge 1$, there exists a smooth metric g on a surface M of genus 0 with k boundary components such that ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-078834 | MATHDB-401673

Sharp upper bound for the normalized first Steklov eigenvalue

摘录 / EXCERPT: On which domain $\Omega \subset \mathbb{R}^{n}$ (or in the limit of which sequence of domains) is the supremum of $\sigma_{1}(\Omega)|\partial\Omega|^{\frac{1}{n-1}}$ realized?

open_problem | open | MathDB 页面 | 原文线索

V2-078835 | MATHDB-401674

Synthetic property of metric spaces related to continuity of optimal transport

摘录 / EXCERPT: The report conjectures that 0 denoting the transport continuity property by 0(TCP), satisfaction of the 0(4,1)-bipolar comparison implies 0(TCP), and asks whether the two propertie...

open_problem | open | MathDB 页面 | 原文线索

V2-078836 | MATHDB-401679

Shefel’s saddle-disc CAT(0) conjecture

摘录 / EXCERPT: A saddle disc equipped with its intrinsic metric is CAT(0).

conjecture | open | MathDB 页面 | 原文线索

V2-078837 | MATHDB-401680

Non-simplicity of Cremona groups in dimension at least 3

摘录 / EXCERPT: The group $\operatorname{Cr}_n(\mathbb{C})$ is not simple for $n \ge 3$.

open_problem | open | MathDB 页面 | 原文线索

V2-078838 | MATHDB-401681

Birational rigidity criterion for semistable del Pezzo fibrations

摘录 / EXCERPT: Let X be a semi-stable del Pezzo fibration of degree 1, 2, or 3 that is a Mori fibre space. Then X is birationally rigid if and only if...

open_problem | open | MathDB 页面 | 原文线索

V2-078839 | MATHDB-401682

Classification of rational G-birationally rigid G-del Pezzo threefolds of degree less than 5

摘录 / EXCERPT: Classify rational G-birationally rigid G-del Pezzo threefolds of degree less than 5.

open_problem | open | MathDB 页面 | 原文线索

V2-078840 | MATHDB-401684

Non-simplicity of Cremona groups

摘录 / EXCERPT: Does the Cremona group of the plane (and, even more so, of $\mathbb{P}^n$ for $n>2$) contain a non-trivial normal subgroup?

open_problem | open | MathDB 页面 | 原文线索

V2-078841 | MATHDB-401685

Generation of the three-dimensional Cremona group by automorphisms and Jonquières elements

摘录 / EXCERPT: For a subfield $k\subseteq\mathbb{C}$, is $\operatorname{Bir}_k(\mathbb{P}^3)$ generated by $\operatorname{Aut}_k(\mathbb{P}^3)$ together with all Jonquières elements, i.e. biratio...

open_problem | open | MathDB 页面 | 原文线索

V2-078842 | MATHDB-401686

The Tame Problem for affine space in dimensions at least four

摘录 / EXCERPT: For $n\ge 4$, is $\operatorname{Aut}(\mathbb A^n)$ generated by the affine automorphism group $\operatorname{PGL}_{n+1}(k)\cap\operatorname{Aut}(\mathbb A^n)$ and the elementary au...

open_problem | open | MathDB 页面 | 原文线索

V2-078843 | MATHDB-401688

Infinitely many real forms of a smooth rational real affine fourfold

摘录 / EXCERPT: The smooth rational real affine fourfold $\mathbb{S}^{2}\times\mathbb{A}^{2}_{\mathbb{R}}$ has at least countably infinitely many pairwise non-isomorphic real forms.

open_problem | open | MathDB 页面 | 原文线索

V2-078844 | MATHDB-401691

Non-rectifiable rational smooth embeddings of the real affine line

摘录 / EXCERPT: For every integer $d\ge 5$, there exists a non-rectifiable rational smooth embedding of $\mathbb A^1_{\mathbb R}$ into $\mathbb A^2_{\mathbb R}$ whose associated projective curve...

open_problem | open | MathDB 页面 | 原文线索

V2-078845 | MATHDB-401693

Gizatullin’s conjecture for smooth cubic surfaces

摘录 / EXCERPT: For a smooth cubic surface $S\subset\mathbb{P}^{3}$, $\operatorname{Aut}(\mathbb{P}^{3}\setminus S)=\operatorname{Aut}(\mathbb{P}^{3},S)$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-078846 | MATHDB-401694

Cylinder criterion for automorphisms of complements of del Pezzo hypersurfaces

摘录 / EXCERPT: Let S be a del Pezzo surface with at worst Du Val singularities that is a hypersurface in a weighted projective space P, where (S,P) is one of the following: a hypersurface o...

open_problem | open | MathDB 页面 | 原文线索

V2-078847 | MATHDB-401697

Conjecture 2 for smooth Fano threefolds

摘录 / EXCERPT: Theorem 8 (Cheltsov, Przyjalkowski). Conjecture 2 holds for smooth Fano threefolds.

conjecture | open | MathDB 页面 | 原文线索

V2-078848 | MATHDB-401700

Maximal A_k-singularities of plane curves of fixed bidegree

摘录 / EXCERPT: For $(a,b)\in\mathbb N^2$, what is the maximal k such that there exists a curve in $\mathbb A^2(\mathbb C)$ of bidegree (a,b) with an A_k-singularity? Denote this maximum by...

open_problem | open | MathDB 页面 | 原文线索

V2-078849 | MATHDB-401705

Decision-theoretic optimality of PCA spectral projectors

摘录 / EXCERPT: Conjecture. For any given dimension p, consider an i.i.d. sample $X_1,\ldots,X_n\sim N(0,\Sigma)$, where $\Sigma\in\mathbb{R}^{p\times p}$ is a covariance matrix with known eigen...

open_problem | open | MathDB 页面 | 原文线索

V2-078850 | MATHDB-401708

Extent and termination of a principal curve or flow

摘录 / EXCERPT: How far should one follow a principal curve/flow-until it “turns on itself”? Could there be a parameter describing the “dying out” of the curve/flow?

open_problem | open | MathDB 页面 | 原文线索

V2-078851 | MATHDB-401709

Regularization of principal flows

摘录 / EXCERPT: Is additional regularization needed in order to avoid self-intersection or winding?

open_problem | open | MathDB 页面 | 原文线索

V2-078852 | MATHDB-401713

Resolvent degree of the cover parametrizing lines on cubic surfaces

摘录 / EXCERPT: Let $H_{3,3}$ be the 4-dimensional moduli space of smooth cubic surfaces in $\bbP^3$ modulo the action of $\bPGL(4)$, and let ... This is a 27-sheeted cover with group...

open_problem | open | MathDB 页面 | 原文线索

V2-078853 | MATHDB-401716

The p-converse to the Gross-Zagier-Kolyvagin-Rubin theorem

摘录 / EXCERPT: Let E be an elliptic curve over $\\mathbb{Q}$. For $r=0,1$, the following are equivalent: (1) $\operatorname{rank}_{\mathbb{Z}}E(\mathbb{Q})=r$ and $X(E/\mathbb{Q})$ is finite; (...

open_problem | open | MathDB 页面 | 原文线索

V2-078854 | MATHDB-401720

Motivatedness of the Jacquet-Langlands Hodge-Tate cycle

摘录 / EXCERPT: The class ξ is motivated, thus a fortiori, absolutely Hodge.

open_problem | open | MathDB 页面 | 原文线索

V2-078855 | MATHDB-401725

Global Gan-Gross-Prasad conjecture for unitary groups

摘录 / EXCERPT: Let E/F be a quadratic extension of number fields, let A be the adèle ring of F, and let $W\subset V$ be hermitian spaces over E of dimensions n and $n+1$, respectively....

conjecture | open | MathDB 页面 | 原文线索

V2-078856 | MATHDB-401726

Ichino-Ikeda/Harris formula for unitary groups

摘录 / EXCERPT: Conjecture 2 (Ichino-Ikeda, R. N. Harris). Assume that the cuspidal automorphic representation π of $G(\mathbb A)$ is everywhere tempered, i.e. π_v is tempered for every pl...

open_problem | open | MathDB 页面 | 原文线索

V2-078857 | MATHDB-401727

Rognes’ connectivity conjecture

摘录 / EXCERPT: For a ring R having the invariant basis number property, Rognes conjectured that $D(R^n)$ is $(2n-3)$-connected when R is either local or Euclidean.

conjecture | open | MathDB 页面 | 原文线索

V2-078858 | MATHDB-401728

Dundas-Goodwillie-McCarthy relative cyclotomic-trace equivalence

摘录 / EXCERPT: Let (R,I) be a pair consisting of a (not necessarily commutative) ring R and a nilpotent (two-sided) ideal I. Then the cyclotomic trace induces an equivalence on relative the...

open_problem | open | MathDB 页面 | 原文线索

V2-078859 | MATHDB-401729

Cubulability of arithmetic lattices in odd-dimensional real hyperbolic spaces

摘录 / EXCERPT: Is it known whether arithmetic lattices of isometries of rank-one real hyperbolic spaces of odd dimension are cubulable?

open_problem | open | MathDB 页面 | 原文线索

V2-078860 | MATHDB-401730

Structure of the local equivalence group

摘录 / EXCERPT: The structure of the local equivalence group LE is an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078861 | MATHDB-401732

Excision for localizing invariants of pullback diagrams

摘录 / EXCERPT: Assume that ... is a pullback diagram of E_1-ring spectra. Then the spectrum $A'\otimes_A B$ admits a natural E_1-ring structure, denoted $A'\boxtimes_A B$, such that the canonic...

open_problem | open | MathDB 页面 | 原文线索

V2-078862 | MATHDB-401733

Rectifiability of Shastri’s embedding of the affine line in three-space

摘录 / EXCERPT: Let $f_0,f_2:\mathbb{C}\to\mathbb{C}^3$ be the polynomial embeddings ... Does there exist a polynomial automorphism $\Psi:\mathbb{C}^3\to\mathbb{C}^3$ such that $\Psi\circ f_0=f_2$?

open_problem | open | MathDB 页面 | 原文线索

V2-078863 | MATHDB-401734

Knot-theoretic obstructions to equivalence of embeddings of the affine line in \(\mathbb{C}^3\)

摘录 / EXCERPT: Let ... and ... Do classical knot invariants, such as the knot group or knot polynomials, have analogues for embeddings $\mathbb{C}\hookrightarrow\mathbb{C}^3$ that establish that ther...

open_problem | open | MathDB 页面 | 原文线索

V2-078864 | MATHDB-401735

Davis-Okun thickening obstruction conjecture

摘录 / EXCERPT: If a finite aspherical complex X has a non-vanishing k-th L^2-Betti number, then X does not have a thickening of dimension $<2k$.

conjecture | open | MathDB 页面 | 原文线索

V2-078865 | MATHDB-401736

Existence of non-arithmetic lattices in complex hyperbolic space

摘录 / EXCERPT: Whether or not non-arithmetic lattices exist in $\operatorname{Isom}(\mathbb{H}_{\mathbb{C}}^n)\cong \operatorname{SU}(n,1)$ for $n\geq 3$ is an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-078866 | MATHDB-401739

Central limit theorem for the volume of random mapping tori

摘录 / EXCERPT: For the random mapping class walk (w_n) on the mapping class group of a closed connected orientable surface Σ_g with $g\ge 2$, and the associated random mapping tori...

open_problem | open | MathDB 页面 | 原文线索

V2-078867 | MATHDB-401743

All-orders formula for the function $I_\lambda(a)$ in the noncommutative Φ^4 model

摘录 / EXCERPT: The report conjectures that the explicit expansion labeled (8) for $I_\lambda(a)$-the expression beginning with $-\lambda\log(1+a)$ and involving Stirling numbers of the first kind...

open_problem | open | MathDB 页面 | 原文线索

V2-078868 | MATHDB-401746

Group 3-cocycles and the Connes-Karoubi multiplicative character

摘录 / EXCERPT: It is conjectured that the group 3-cocycles constructed from bipolarized group representations compute the Connes-Karoubi multiplicative character.

open_problem | open | MathDB 页面 | 原文线索

V2-078869 | MATHDB-401752

Eisenbud-Schreyer characteristic-bound conjecture for generic Green’s conjecture

摘录 / EXCERPT: Eisenbud and Schreyer conjecture that the conclusion of Corollary 4 should hold whenever $\operatorname{char}(k)\ge (g-1)/2$; namely, Green’s Conjecture should hold for generic can...

conjecture | open | MathDB 页面 | 原文线索

V2-078870 | MATHDB-401753

Vanishing of the critical syzygy in the borderline characteristic

摘录 / EXCERPT: Let T be the tangent developable to a rational normal curve of degree g. In the case $\\operatorname{char}(k)=p=(g+2)/2$, determine whether ...

open_problem | open | MathDB 页面 | 原文线索

V2-078871 | MATHDB-401757

Schreyer’s conjecture for canonical curves with a unique minimal pencil

摘录 / EXCERPT: Let $C\subset \mathbb P^{g-1}$ be a canonical curve of gonality k with a unique pencil $A\in W^1_k(C)$. Set ... One expects ...

conjecture | open | MathDB 页面 | 原文线索

V2-078872 | MATHDB-401764

Real period-index theorem for smooth integral real surfaces

摘录 / EXCERPT: Let S be a smooth integral surface over $\mathbb{R}$, and let $\alpha \in \operatorname{Br}(S) \subset \operatorname{Br}(\mathbb{R}(S))$ be such that...

open_problem | open | MathDB 页面 | 原文线索

V2-078873 | MATHDB-401765

Extension theorem for holomorphic differential forms

摘录 / EXCERPT: Let X be a locally irreducible complex space, let $\iota:X_{\mathrm{reg}}\to X$ be the inclusion of the regular locus, and let $r:\widetilde{X}\to X$ be a resolution of singulari...

open_problem | open | MathDB 页面 | 原文线索

V2-078874 | MATHDB-401766

Local vanishing theorem for logarithmic 1-forms

摘录 / EXCERPT: Let X be a locally irreducible complex space of dimension n. Let $r:\widetilde{X}\to X$ be a log resolution with exceptional divisor $E\subset\widetilde{X}$. If...

open_problem | open | MathDB 页面 | 原文线索

V2-078875 | MATHDB-401770

Bounded average number of rational points on genus-two curves

摘录 / EXCERPT: Let $f\in\mathbb Z[x]$ satisfy $\Delta_f:=\operatorname{disc}(f)\ne0$, and let $C_f:y^2=f(x)$. Define ... and, for $f(x)=x^5+a_2x^3+\cdots+a_5$, define ... Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-078876 | MATHDB-401771

Existence of minimizers for the sliding Plateau problem

摘录 / EXCERPT: Given compact sets $E, Gamma\subset\mathbb{R}^n$, consider deformations of E in a compact ball B: continuous one-parameter families of continuous maps...

open_problem | open | MathDB 页面 | 原文线索

V2-078877 | MATHDB-401772

Anisotropic counterpart of Allard’s rectifiability theorem

摘录 / EXCERPT: Let $ \Omega\subset\mathbb{R}^n$ be open, let $F\in C^1(\Omega\times G(n,d),(0,+\infty))$ be a positive integrand, and let $V\in\mathbb{V}_d(\Omega)$ be a d-varifold whose anisot...

open_problem | open | MathDB 页面 | 原文线索

V2-078878 | MATHDB-401773

Tangent cones with line symmetry at isolated singularities

摘录 / EXCERPT: Is it possible for an isolated singularity to admit a tangent with a line of symmetry?

open_problem | open | MathDB 页面 | 原文线索

V2-078879 | MATHDB-401774

Uniqueness of blow-up and parametrization near isolated singularities

摘录 / EXCERPT: Let $b\in W^{1,2}(B_1)$ be a 1-homogeneous minimizer of the functional (1), with $\partial\{b>0\}$ smooth away from 0. Suppose that u is a minimizer, $x_0\in\partial\{u>0\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078880 | MATHDB-401775

Whether every composite can be mimicked by iterated laminates

摘录 / EXCERPT: Can every composite be mimicked by an iterated laminate made with the same set of constituent materials? No. The report states that Milton answered this question in the negative, a...

open_problem | open | MathDB 页面 | 原文线索

V2-078881 | MATHDB-401777

Anisotropic Alexandrov conjecture for finite-perimeter sets

摘录 / EXCERPT: Conjecture: F-Wulff shapes are the unique sets of finite perimeter and finite volume that are critical points of F at fixed volume.

conjecture | open | MathDB 页面 | 原文线索

V2-078882 | MATHDB-401778

One-homogeneous singular minimizer in the case n=3, m=2

摘录 / EXCERPT: Let ... where $u:B_1\subset\mathbb{R}^n\to\mathbb{R}^m$, and F is smooth and uniformly convex on $M^{m\times n}$, with bounded second derivatives. A minimizer means $u\in H^1(B_1)$...

open_problem | open | MathDB 页面 | 原文线索

V2-078883 | MATHDB-401779

Energy scaling for a thin elastic sheet with a single disclination

摘录 / EXCERPT: For $0<\Delta<1$, let I_h^Δ be the energy obtained from (1) by taking the reference metric $g_0^\Delta=dr^2+(1-\Delta^2)r^2d\theta^2$. Then there exists a constant C_1, de...

open_problem | open | MathDB 页面 | 原文线索

V2-078884 | MATHDB-401780

One-Step-Improvement proposition for optimal transport

摘录 / EXCERPT: Let pi be the optimal transference plan between two measures mu0 and mu1 in mathbb{R}^d. Define ... and ... For every $0<\tau\le 1$, there exists $C=C(d,\tau)<\infty$ such that, provid...

open_problem | open | MathDB 页面 | 原文线索

V2-078885 | MATHDB-401782

Optimal regularity and free-boundary structure for the nonparametric thin obstacle problem

摘录 / EXCERPT: Let u be a solution to the thin obstacle problem (1) with $\psi\equiv 0$, and let $\Gamma(u)$ be the relative boundary in B'_1 of $\{(x',0)\in B'_1:u(x',0)=0\}$. Then: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-078886 | MATHDB-401783

Inverse phase-field potential problem for branched transport

摘录 / EXCERPT: For every admissible transport cost $\tau:[0,\infty)\to[0,\infty)$, where $\tau(0)=0$ and τ is nondecreasing, concave, and continuous, there exists a phase-field potential...

open_problem | open | MathDB 页面 | 原文线索

V2-078887 | MATHDB-401784

Regularity and singularity generation after the interaction of three impulsive gravitational waves

摘录 / EXCERPT: For the polarized $U(1)$-symmetric Einstein vacuum solutions constructed from three sufficiently small impulsive gravitational waves propagating toward each other, it is conjecture...

open_problem | open | MathDB 页面 | 原文线索

V2-078888 | MATHDB-401785

Non-smoothness generated by the interaction of three impulsive gravitational waves

摘录 / EXCERPT: It is conjectured that, in general, the spacetime metric is not smooth away from the impulsive gravitational waves, in the sense that weaker singularities are generated by their in...

open_problem | open | MathDB 页面 | 原文线索

V2-078889 | MATHDB-401789

Self-similarity of the six-dimensional ultraspinning endpoint

摘录 / EXCERPT: For the endpoint of the ultraspinning instability of singly spinning Myers-Perry black holes in $D=6$ spacetime dimensions, is the dynamics self-similar?

open_problem | open | MathDB 页面 | 原文线索

V2-078890 | MATHDB-401790

Compatibility of smooth asymptotic flatness with the vacuum Einstein equations

摘录 / EXCERPT: An open issue is to what extent Penrose’s notion of asymptotic flatness is compatible with the vacuum equations, in the sense that a sufficiently large class of solutions to model...

open_problem | open | MathDB 页面 | 原文线索

V2-078891 | MATHDB-401791

Necessity of vanishing radiation at past null infinity for absence of logarithmic terms

摘录 / EXCERPT: Is the vanishing of the radiation field at all orders at $\mathcal I^-$ also necessary for the non-appearance of logarithmic terms there?

open_problem | open | MathDB 页面 | 原文线索

V2-078892 | MATHDB-401792

Conditions for removing logarithmic terms and obtaining a smooth null infinity

摘录 / EXCERPT: For a spacelike Cauchy problem with asymptotically Euclidean initial data that are polyhomogeneous at spatial infinity, determine additional conditions on the data under which a ga...

open_problem | open | MathDB 页面 | 原文线索

V2-078893 | MATHDB-401793

Mass inflation scenario for perturbed Reissner-Nordström black holes

摘录 / EXCERPT: Consider the maximal future development of a generic perturbation of the Reissner-Nordström spacetime. The mass inflation scenario asserts that: - The null Cauchy horizon...

open_problem | open | MathDB 页面 | 原文线索

V2-078894 | MATHDB-401794

AdS instability conjecture for the vacuum Einstein equations

摘录 / EXCERPT: There exist arbitrarily small perturbations of the initial data of $(M_{\rm AdS},g_{\rm AdS})$ which, under evolution by the vacuum Einstein equations (1) with a reflecting boundar...

conjecture | open | MathDB 页面 | 原文线索

V2-078895 | MATHDB-401795

Hawking-Penrose singularity theorem with closed null curves for $C^{1,1}$ metrics

摘录 / EXCERPT: Does the Hawking-Penrose theorem for $C^{1,1}$ metrics continue to hold without excluding the existence of closed null curves? While the smooth version only excludes closed timelik...

open_problem | open | MathDB 页面 | 原文线索

V2-078896 | MATHDB-401797

Long-term stability of quasiperiodic solutions in anti-de Sitter spacetime

摘录 / EXCERPT: For the four-dimensional, spherically symmetric Einstein-real-scalar-field model of anti-de Sitter spacetime with reflecting boundary conditions, do the TTF quasiperiodic equilibri...

open_problem | open | MathDB 页面 | 原文线索

V2-078897 | MATHDB-401799

Penrose-type mass-angular momentum inequality for one-ended initial data

摘录 / EXCERPT: Can one prove, for a three-dimensional, axially symmetric initial data set (M^3,g,k) for the Einstein equations satisfying the dominant energy condition, with one asymptotically...

open_problem | open | MathDB 页面 | 原文线索

V2-078898 | MATHDB-401800

Full finite-codimension nonlinear asymptotic stability of the Schwarzschild family

摘录 / EXCERPT: Establish the following claim. For all characteristic initial data sufficiently close to Schwarzschild data with mass $M_{\mathrm{init}}>0$ and lying on a codimension-3 “submanif...

open_problem | open | MathDB 页面 | 原文线索

V2-078899 | MATHDB-401802

Extension of the angular-momentum mass inequality to trapped inner boundaries

摘录 / EXCERPT: Extend inequality (3), $m\geq\sqrt{J_\infty}$, to the case in which Σ has a trapped inner boundary.

open_problem | open | MathDB 页面 | 原文线索

V2-078900 | MATHDB-401803

Null Penrose Inequality

摘录 / EXCERPT: Let Ω be a null hypersurface in a spacetime satisfying the dominant energy condition and extending to past null infinity. Assume that Ω admits a doubly convex foliati...

open_problem | open | MathDB 页面 | 原文线索

V2-078901 | MATHDB-401804

Null Penrose Inequality under doubly convex foliations

摘录 / EXCERPT: Let Ω be a null hypersurface in a spacetime satisfying the Dominant Energy Condition and extending to past null infinity. Suppose Ω admits a doubly convex foliation...

open_problem | open | MathDB 页面 | 原文线索

V2-078902 | MATHDB-401805

Global-attractor conjecture for B_0 T^2-symmetric solutions

摘录 / EXCERPT: Conjecture 2. The local attractor in [3] and generalized in [2] is in fact a global attractor for B_0 solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-078903 | MATHDB-401806

Local attractor for non-B_0 T^2-symmetric solutions

摘录 / EXCERPT: There is a local attractor for non-B_0 solutions satisfying $\lvert V_\tau-\tfrac12\rvert\to 0,\quad Q_\tau=O(e^{-3\tau/2})$.

open_problem | open | MathDB 页面 | 原文线索

V2-078904 | MATHDB-401807

Upper bound for the Bartnik mass of CMC Bartnik data

摘录 / EXCERPT: Given Bartnik CMC data $(\Sigma \cong S^2,g_o,H_o)$ with $K(g_o)>0$, there exist constants $0\leq \alpha$ and $0<\beta\leq 1$ such that, if ... then ...

open_problem | open | MathDB 页面 | 原文线索

V2-078905 | MATHDB-401808

Connectedness and density of the interior of the forward filled set for conservative Hénon maps

摘录 / EXCERPT: For a conservative complex Hénon map f of $\mathbb{C}^2$ (equivalently, in the displayed model $f(x,y)=(y,y^2+c-\delta x)$ with $|\delta|=1$), let K^+ be the set of points whos...

open_problem | open | MathDB 页面 | 原文线索

V2-078906 | MATHDB-401809

Linear model for rank-1 rotation domains

摘录 / EXCERPT: Suppose that Ω is a rotation domain of rank 1. Is $(f|_{\Omega},\Omega)$ biholomorphically conjugate to a linear map (L,D), where $D\subset\mathbb{C}^{2}$ is a domain, a...

open_problem | open | MathDB 页面 | 原文线索

V2-078907 | MATHDB-401810

Bidisk realizability of rank-2 Reinhardt rotation models

摘录 / EXCERPT: For a rank-2 rotation domain with Reinhardt linear model $D \subset \mathbb{C}^2$, can D be biholomorphically equivalent to the bidisk?

open_problem | open | MathDB 页面 | 原文线索

V2-078908 | MATHDB-401817

Finiteness of irreducible holomorphic (g,m)-fiber bundles over finite Riemann surfaces

摘录 / EXCERPT: Conjecture. Consider an arbitrary connected finite Riemann surface X, possibly of second kind. Up to isotopy, there are no more than finitely many irreducible holomorphic (g,m)...

open_problem | open | MathDB 页面 | 原文线索

V2-078909 | MATHDB-401823

Convergence-guaranteeing basis functions for Lagrangian dual decision rules

摘录 / EXCERPT: How should the basis functions used in the LDR approximations be chosen? In particular, a theory demonstrating that a class of basis functions is sufficient to converge to the true...

open_problem | open | MathDB 页面 | 原文线索

V2-078910 | MATHDB-401826

Well-posedness of the Bernoulli free-boundary problem with a random interior boundary

摘录 / EXCERPT: Is the problem well-posed in the sense of $D(\omega)$ being almost surely well-defined?

open_problem | open | MathDB 页面 | 原文线索

V2-078911 | MATHDB-401828

Characterizing decision-dependent nominal distributions in DRO

摘录 / EXCERPT: One of the main distinguishing features of our approach is that the nominal distribution at the center of the ambiguity set is decision-dependent. In this regard, it would be essen...

open_problem | open | MathDB 页面 | 原文线索

V2-078912 | MATHDB-401829

Sampling methods for DRO with decision-dependent ambiguity sets

摘录 / EXCERPT: Develop sampling approaches suited for distributionally robust optimization (DRO) with decision-dependent ambiguity sets.

open_problem | open | MathDB 页面 | 原文线索

V2-078913 | MATHDB-401830

Long-time convergence of the dynamic Monge-Kantorovich system

摘录 / EXCERPT: Let Omega be an open bounded subset of mathbb{R}^n, and let positive densities f^+,f^- in L^1(Omega) satisfy int{Omega} f^+,dx=int{Omega} f^-,dx. For f=f^+-f^-, consider the system...

open_problem | open | MathDB 页面 | 原文线索

V2-078914 | MATHDB-401832

Simultaneous blow-up of two chemotactic species

摘录 / EXCERPT: For radially symmetric solutions of the two-species chemotaxis system ... ... in a ball, with homogeneous Neumann boundary conditions and $\chi_1,\chi_2,\mu_2>0$, can u_1 blow up in...

open_problem | open | MathDB 页面 | 原文线索

V2-078915 | MATHDB-401833

Control-functional formulation of Coulomb friction in a fractured poromechanical model

摘录 / EXCERPT: Determine a suitable functional J and solve the associated control problem: find a slip control β such that ... where $B\subset H^{1/2}_{00}(\Gamma)$,...

open_problem | open | MathDB 页面 | 原文线索

V2-078916 | MATHDB-401834

Universality classes for the mean infection time of kinetically constrained models

摘录 / EXCERPT: Is it possible to group all possible update families U into distinct classes, in such a way that all members of the same class induce the same divergence of the mean infection ti...

open_problem | open | MathDB 页面 | 原文线索

V2-078917 | MATHDB-401836

Sufficiency of the characteristic-function criterion for Poisson-Dirichlet cycle structure

摘录 / EXCERPT: For the random interchange model on the complete graph with n vertices, $t=\beta n$, and integer parameter $\theta\ge 2$, Theorem 1 gives, for every $h\in\mathbb{C}$, the converg...

open_problem | open | MathDB 页面 | 原文线索

V2-078918 | MATHDB-401837

Kosterlitz-Thouless transition in the loop \(O(n)\) model

摘录 / EXCERPT: Conjecture. On the hexagonal lattice, for every positive real $n\le 2$, the loop $O(n)$ model undergoes a Kosterlitz-Thouless phase transition at the critical threshold ...

open_problem | open | MathDB 页面 | 原文线索

V2-078919 | MATHDB-401838

Finiteness of avalanches on \(\mathbb{Z}^2\)

摘录 / EXCERPT: Are avalanches ν-a.s. finite on $\mathbb{Z}^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-078920 | MATHDB-401839

Tail behaviour of Abelian sandpile avalanche size

摘录 / EXCERPT: What is the tail behaviour of $\nu[|\mathrm{Av}|>t]$ on $\mathbb{Z}^d$, $d\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-078921 | MATHDB-401840

Height-variable covariance asymptotics in the two-dimensional Abelian sandpile model

摘录 / EXCERPT: Prove the conjecture that, for the infinite-volume sandpile measure ν on $\mathbb{Z}^2$, ... for every $i,j\in\{1,2,3\}$, where o is the origin.

open_problem | open | MathDB 页面 | 原文线索

V2-078922 | MATHDB-401841

Phase diagram conjecture for the loop $O(n)$ model

摘录 / EXCERPT: It is conjectured that, for $0<n\le 2$, the transition line of the loop $O(n)$ model on the hexagonal lattice, with critical parameter $x_c(n)=1/\sqrt{2+\sqrt{2-n}}$, separates a r...

conjecture | open | MathDB 页面 | 原文线索

V2-078923 | MATHDB-401844

Affine actions with Hitchin linear part

摘录 / EXCERPT: Let S_g be a closed orientable surface of genus $g\ge 2$, let $\Gamma=\pi_1S_g$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-078924 | MATHDB-401845

Good graphs of curves when f(S)=1

摘录 / EXCERPT: Conjecture: If $f(S)=1$, then S does not admit a good graph made of curves. Here a good graph means a connected graph with an action of $\operatorname{MCG}(S)$ having unbounded o...

open_problem | open | MathDB 页面 | 原文线索

V2-078925 | MATHDB-401846

Boundary characterization for holomorphic functions via the Bers Poisson kernel

摘录 / EXCERPT: The report conjectures that if a complex-valued continuous function V on the Bers boundary $\partial T\!B_{x_0}$ satisfies ... then the Poisson integral ... defines a continuous func...

open_problem | open | MathDB 页面 | 原文线索

V2-078926 | MATHDB-401847

Thurston-type distances on Teichmüller spaces of surfaces with boundary

摘录 / EXCERPT: Do Thurston’s results still extend to the case of surfaces with boundary for some (or all) of the distances above? Do the distances d_A, $d_{L\partial}$, and d_{Lh} coincide?

open_problem | open | MathDB 页面 | 原文线索

V2-078927 | MATHDB-401851

Cubulation of amalgamated products along non-quasiconvex subgroups

摘录 / EXCERPT: Let $G=A *_C B$ (respectively, $G=A *_C$) be a free product with amalgamation (respectively, an HNN extension) such that G, A, B, and C are hyperbolic. If A and B are c...

open_problem | open | MathDB 页面 | 原文线索

V2-078928 | MATHDB-401852

Coincidence of the energy-minimizing maps in Y and D

摘录 / EXCERPT: For the finite, boundaryless, locally 1-connected 2-dimensional simplicial complex X with fixed complete ideal hyperbolic metrics $\sigma,\tau\in\mathcal T(X)$, let Y be th...

open_problem | open | MathDB 页面 | 原文线索

V2-078929 | MATHDB-401855

Minimal real special-linear representation of a right-angled Artin group

摘录 / EXCERPT: Suppose that Γ is a finite graph and let $A(\Gamma)$ be the associated right-angled Artin group. What is the smallest n so that $A(\Gamma)$ sits as a subgroup of...

open_problem | open | MathDB 页面 | 原文线索

V2-078930 | MATHDB-401856

Stability of a direct sum of quasi-Fuchsian and geometrically infinite representations

摘录 / EXCERPT: Let π_1 be the fundamental group of a closed surface of genus $g\ge 2$. Consider two representations $\rho_1,\rho_2:\pi_1\to\mathrm{SO}(3,1)$, where ρ_1 is quasi-Fuchsian...

open_problem | open | MathDB 页面 | 原文线索

V2-078931 | MATHDB-401857

Rectifiable but non-\a3\b9,\b1 limit sets of surface-group Anosov representations

摘录 / EXCERPT: Suppose that Γ is the fundamental group of a closed orientable surface of genus $g\ge 2$. Does there exist a 1-Anosov representation of Γ in...

open_problem | open | MathDB 页面 | 原文线索

V2-078932 | MATHDB-401858

Lifting intersecting Dehn twists while preserving the braid relation

摘录 / EXCERPT: Consider two essential simple closed curves meeting transversely in a single point in some surface with negative Euler characteristic, so that the respective right Dehn twists...

open_problem | open | MathDB 页面 | 原文线索

V2-078933 | MATHDB-401860

Counting mapping-class-orbit points in Thurston-metric balls

摘录 / EXCERPT: Fix points x,y in Teichmüller space. Consider the number of points in the mapping class orbit of y that occur in a ball of radius R about x in the Thurston metric, and comp...

open_problem | open | MathDB 页面 | 原文线索

V2-078934 | MATHDB-401861

Disconnected preimage of a four-holed sphere under a finite surface cover

摘录 / EXCERPT: Fix a finite cover $f:S_h\to S_g$ of a genus-g surface S_g by a genus-h surface S_h, where $h>g\ge 2$. Must there always exist a sphere minus four disks $H\subset S_g$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-078935 | MATHDB-401864

Benjamini-Schramm isoperimetric-dimension conjecture

摘录 / EXCERPT: Define the isoperimetric dimension of G by ... where ∂S is the external boundary of S. Benjamini and Schramm conjectured that if $\dim(G)>1$, then $p_c(G)<1$.

conjecture | open | MathDB 页面 | 原文线索

V2-078936 | MATHDB-401867

Diffusive scaling limit for the random Lorentz gas

摘录 / EXCERPT: For spherical hard-core scatterers of radius r centered at a Poisson point process of intensity ϱ in $\mathbb{R}^d$, establish, as $T\to\infty$, the diffusive scaling lim...

open_problem | open | MathDB 页面 | 原文线索

V2-078937 | MATHDB-401870

Closed saddle connections on four Platonic solids

摘录 / EXCERPT: There are no closed saddle connections on the tetrahedron, octahedron, cube, or icosahedron.

open_problem | open | MathDB 页面 | 原文线索

V2-078938 | MATHDB-401875

Vanishing powers of the tautological class on strata of differentials

摘录 / EXCERPT: For a stratum $\mathcal{PH}(\mu)$ of differentials with $\sum_i m_i=2g-2$ and $m_i\ne -1$ for every i (so there are no simple poles), determine which powers of...

open_problem | open | MathDB 页面 | 原文线索

V2-078939 | MATHDB-401876

Complete curves in strata of holomorphic differentials

摘录 / EXCERPT: It remains an open question whether the strata of holomorphic differentials can contain a complete curve.

open_problem | open | MathDB 页面 | 原文线索

V2-078940 | MATHDB-401877

Strong Maximal Rank conjecture for genus 23

摘录 / EXCERPT: For a general smooth curve C of genus 23, and every linear series $L\in W^6_{26}(C)$, the multiplication map ... is injective.

conjecture | open | MathDB 页面 | 原文线索

V2-078941 | MATHDB-401878

Whether the virtual divisor D is an actual divisor

摘录 / EXCERPT: Is the virtual divisor D an actual divisor?

open_problem | open | MathDB 页面 | 原文线索

V2-078942 | MATHDB-401879

GIT semistability of syzygy points

摘录 / EXCERPT: When is the $(p,2)$-syzygy point of X semistable with respect to the $\mathrm{SL}(V)$-action in the sense of GIT?

open_problem | open | MathDB 页面 | 原文线索

V2-078943 | MATHDB-401880

Variation of pullbacks of compactified-Jacobian classes with the stability parameter

摘录 / EXCERPT: Understand how the pullback of the locus of trivial line bundles, or better the Brill-Noether cycle $w(\phi)$, changes as the stability parameter ϕ changes.

open_problem | open | MathDB 页面 | 原文线索

V2-078944 | MATHDB-401882

Surjectivity of the forgetful map for spherical surfaces with conical points

摘录 / EXCERPT: For a compact, connected, oriented surface of genus g with n marked points and positive angle parameters $\vartheta=(\vartheta_1,\ldots,\vartheta_n)$, let ... be the map that for...

open_problem | open | MathDB 页面 | 原文线索

V2-078945 | MATHDB-401883

Connectedness of the moduli space of spherical surfaces with conical points

摘录 / EXCERPT: Is $\mathcal{M}\mathrm{Sph}_{g,n}(\vartheta)$ connected?

open_problem | open | MathDB 页面 | 原文线索

V2-078946 | MATHDB-401884

Holomorphicity of the forgetful map for spherical cone metrics

摘录 / EXCERPT: Is $F_{g,n,\boldsymbol{\thheta}}$ holomorphic for some complex structure on $M\text{Sph}_{g,n}(\boldsymbol{\thheta})$?

open_problem | open | MathDB 页面 | 原文线索

V2-078947 | MATHDB-401885

Mazur-Veech volumes as intersection numbers

摘录 / EXCERPT: For integers $g,n\ge 0$ with $2g-2+n>0$, and for every tuple $\mu=(m_1,\ldots,m_n)$ of nonnegative integers satisfying $\sum_{i=1}^n m_i=2g-2$, the following equality is conjecture...

open_problem | open | MathDB 页面 | 原文线索

V2-078948 | MATHDB-401887

Realizability of tropical divisors

摘录 / EXCERPT: Let $D=\sum_{i=1}^{n} a_i p_i$ be a divisor of degree d and rank r on a tropical curve Γ. Does there exist a smooth projective curve X over a non-Archimedean field K...

open_problem | open | MathDB 页面 | 原文线索

V2-078949 | MATHDB-401888

Labourie’s mapping-class-group-equivariant projection conjecture

摘录 / EXCERPT: For a closed oriented surface Σ_g of genus $g\ge 2$, there is a mapping-class-group-equivariant projection from the Hitchin component for $\mathrm{PSL}(n,\mathbb{R})$ to...

conjecture | open | MathDB 页面 | 原文线索

V2-078950 | MATHDB-401889

Removal of the combined growth-term condition in a two-species tissue-growth model

摘录 / EXCERPT: For the two-species model ... with $n=n_1+n_2$, $p=n^\gamma$, $\gamma>1$, and $F=F_1+F_2$, $G=G_1+G_2$, can the assumed combined condition ... be removed?

open_problem | open | MathDB 页面 | 原文线索

V2-078951 | MATHDB-401895

Non-isomorphic connected graphs that are q-equivalent

摘录 / EXCERPT: For a given integer $q\ge 2$, are there two non-isomorphic connected graphs H and H' that are q-equivalent, meaning that $M_q(H)=M_q(H')$?

open_problem | open | MathDB 页面 | 原文线索

V2-078952 | MATHDB-401896

2-equivalence versus 3-equivalence of graphs

摘录 / EXCERPT: Is it true that any two 2-equivalent graphs H and H' are also 3-equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-078953 | MATHDB-401897

Harbourne’s containment conjecture for symbolic powers

摘录 / EXCERPT: Harbourne conjectured that, for the ideal I of a finite set of points in $\cmathbb{P}^N_{\cmathbb{C}}$, one has $I^{(Nm-N+1)}\subseteq I^m$ for every $m\ge 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-078954 | MATHDB-401898

Demailly's conjecture for finite sets of points in projective space

摘录 / EXCERPT: Let I be a saturated ideal of a finite set of points in $\cmathbb{P}^N$. Then ... for all $m\ge 1$.

conjecture | open | MathDB 页面 | 原文线索

V2-078955 | MATHDB-401899

Containment problem for symbolic and ordinary powers

摘录 / EXCERPT: For a homogeneous ideal $I\subset \mathbb C[\mathbb P^N]$, determine for which values of m and r the containment $I^{(m)}\subseteq I^r$ holds.

open_problem | open | MathDB 页面 | 原文线索

V2-078956 | MATHDB-401900

Containment of the third symbolic power in the second ordinary power

摘录 / EXCERPT: For a homogeneous radical ideal of points I in $\mathbb{C}[\mathbb{P}^2]$, is the containment $I^{(3)}\subseteq I^2$ always valid?

open_problem | open | MathDB 页面 | 原文线索

V2-078957 | MATHDB-401902

Stable Harbourne containment conjecture

摘录 / EXCERPT: Let R be a regular ring and let I be a radical ideal of big height h. Then, for all $n\gg 0$, $I^{(hn-h+1)}\subseteq I^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-078958 | MATHDB-401903

Ordinary powers of ideals of generic star configurations

摘录 / EXCERPT: Let $H=\{H_1,\ldots,H_s\}$ be a generic hyperplane arrangement in $\mathbb P^n$, with $s\ge n+1$, where $H_i=V(\ell_i)$, $\ell_1,\ldots,\ell_s\in R:=K[x_0,\ldots,x_n]$, and K is...

open_problem | open | MathDB 页面 | 原文线索

V2-078959 | MATHDB-401904

Linear resolutions for ideals generated by products of linear forms

摘录 / EXCERPT: Let K be a field of characteristic 0, let $R=K[x_0,\ldots,x_n]$, and let $\Sigma=(\ell_1,\ldots,\ell_m)$ be a collection of m linear forms in R, some possibly proportional....

open_problem | open | MathDB 页面 | 原文线索

V2-078960 | MATHDB-401906

Unexpected quartics from the B_{n+1} root-system configuration

摘录 / EXCERPT: For $n\ge 4$, let $Z\subset \mathbb P^n_{\mathbb R}$ be the set of $(n+1)^2$ points obtained by projectivizing the $2(n+1)^2$ integer vectors...

open_problem | open | MathDB 页面 | 原文线索

V2-078961 | MATHDB-401907

Conjecture 2: Optimal coherence of five unit vectors in \(\mathbb{C}^3\)

摘录 / EXCERPT: Conjecture. The optimal coherence for $n=5$ unit vectors in $\mathbb{C}^3$ is ... achieved, for example, by any frame Φ with Gram matrix ...

conjecture | open | MathDB 页面 | 原文线索

V2-078962 | MATHDB-401908

The Paulsen Problem

摘录 / EXCERPT: If Φ is nearly tight with frame elements that are nearly unit norm, then how close is the nearest unit norm tight frame?

open_problem | open | MathDB 页面 | 原文线索

V2-078963 | MATHDB-401911

King’s Problem

摘录 / EXCERPT: Characterize the spark of frames with minimum coherence.

open_problem | open | MathDB 页面 | 原文线索

V2-078964 | MATHDB-401912

Li-Lee-Bresler bilinear self-calibration conjecture

摘录 / EXCERPT: Suppose that Φ is a known $d\times n$ matrix, X is an unknown $d\times m$ matrix, and D is an unknown $n\times n$ diagonal matrix. If $n>d$ and $m>\frac{n-1}{n-d}$, then f...

conjecture | open | MathDB 页面 | 原文线索

V2-078965 | MATHDB-401913

Existence of maximal complex equiangular tight frames in every dimension

摘录 / EXCERPT: For every positive integer d, do there exist vectors $v_1,[U+007F][U+007F],v_{d^2}[U+007F]in[U+007F]\mathbb{C}^d$ and a constant α such that $\|v_j\|=1$ for every j and...

open_problem | open | MathDB 页面 | 原文线索

V2-078966 | MATHDB-401914

Improved-exponent conjecture for the Paulsen problem

摘录 / EXCERPT: The report conjectures an inequality of the form ... for the distance in the Paulsen problem.

conjecture | open | MathDB 页面 | 原文线索

V2-078967 | MATHDB-401918

Badly approximable vectors on self-similar fractals

摘录 / EXCERPT: Let $K\subseteq\mathbb{R}^d$ be the limit set of an irreducible finite system of contracting similarity maps satisfying the open set condition. Let $s=\dim_H(K)$, and let μ_K b...

open_problem | open | MathDB 页面 | 原文线索

V2-078968 | MATHDB-401919

A priori error estimate for an LOD approximation of a truncated fractional-Laplacian extension problem

摘录 / EXCERPT: The report states the following theorem: if $u\in\mathring H^1_L(C_T,y^a)$ solves the truncated extension problem, $u_H^{\mathrm{ms}}\in V_H^{\mathrm{ms}}$ is the corresponding mul...

open_problem | open | MathDB 页面 | 原文线索

V2-078969 | MATHDB-401920

Symmetry breaking in the optimal-insulation problem

摘录 / EXCERPT: Let a0$\a0\Omega=B_r(0)$ be a ball of radius $r>0$. There exists a constant $m_0>0$ such that, for every $0<m<m_0$, the eigenfunctions u_m of the optimal-insulation problem are n...

open_problem | open | MathDB 页面 | 原文线索

V2-078970 | MATHDB-401921

Hessian characterization of the continuum GFF variance for non-convex gradient potentials

摘录 / EXCERPT: For the class of non-convex potentials studied in references [2] and [4], in the small-tilt and large-inverse-temperature regime, prove the conjecture that the variance of the limi...

open_problem | open | MathDB 页面 | 原文线索

V2-078971 | MATHDB-401922

Cohn-Kumar-type minimization for soft lattice theta functions

摘录 / EXCERPT: For every $\alpha>0$, the triangular lattice Λ_1 is the unique minimizer of $L\to\theta_{\mu_L+\nu_z}(\alpha)$ among periodic configurations (that is, unions of Bravais lat...

open_problem | open | MathDB 页面 | 原文线索

V2-078972 | MATHDB-401924

Characterization of homogenized surface tensions for periodic mixtures of ferromagnetic interactions

摘录 / EXCERPT: For a fixed volume fraction θ of β-bonds, the report states that all homogenized surface energy densities φ arising from periodic mixtures of nearest-neighbour...

open_problem | open | MathDB 页面 | 原文线索

V2-078973 | MATHDB-401925

Closure characterization of MIBL-representable sets

摘录 / EXCERPT: The closures of MIBL-representable sets are unions of finitely many MILP-representable sets.

open_problem | open | MathDB 页面 | 原文线索

V2-078974 | MATHDB-401926

Seymour’s Flowing Conjecture for minimally non-ideal binary clutters

摘录 / EXCERPT: If C is a minimally non-ideal binary clutter, then C or its blocker $b(C)$ is O_5 or L_7, where O_5 is the clutter of the triangles and pentagons of K_5, and L_7 is t...

conjecture | open | MathDB 页面 | 原文线索

V2-078975 | MATHDB-401927

Exact convergence-rate regime for Lasserre’s bounds on general convex bodies

摘录 / EXCERPT: Understanding the exact regime for the rate of convergence of the bounds $p^{(r)}$ for general convex bodies remains an open question.

open_problem | open | MathDB 页面 | 原文线索

V2-078976 | MATHDB-401928

Computing the degree of a Dieudonné determinant

摘录 / EXCERPT: Given a symbolic matrix $A=A_0+A_1x_1+\\cdots+A_mx_m$, where the variables x_i are non-commutative and each A_i is a matrix over $K[t]$, compute the degree of the Dieudonné det...

open_problem | open | MathDB 页面 | 原文线索

V2-078977 | MATHDB-401931

Exact minimization of agency strength for the synchronized traveling salesman problem

摘录 / EXCERPT: For any input graph, minimize the agency-strength parameter $\alpha(G):=\max\{T/n,n/k\}\ge 1$ over feasible agencies, where $n=|V(G)|$, k is the number of agents, and T is the...

open_problem | open | MathDB 页面 | 原文线索

V2-078978 | MATHDB-401932

Constant lower bound for synchronized agencies on degree-2-free trees

摘录 / EXCERPT: Find a constant lower bound on T/k for trees with no nodes of degree 2.

open_problem | open | MathDB 页面 | 原文线索

V2-078979 | MATHDB-401936

Polynomial-time subset selection for block-diagonal matrices with fixed extra columns

摘录 / EXCERPT: The report states that the subset-selection problem ... can be solved in polynomial time when the data matrix has the block form ... where each block A_i involves a fixed number...

open_problem | open | MathDB 页面 | 原文线索

V2-078980 | MATHDB-401937

Maximum weighted assignment problem for low-rank weight matrices

摘录 / EXCERPT: We consider the maximum weighted assignment problem on the complete bipartite graph with n vertices. The weight matrix is denoted by $W\in\mathbb{R}^{n\times n}$. The Hungarian a...

open_problem | open | MathDB 页面 | 原文线索

V2-078981 | MATHDB-401940

Arborescences and NP-hardness

摘录 / EXCERPT: Is the following optimization problem NP-hard? Given an undirected graph $G=(V,E)$ and, for each node $r\in V$, a cost vector $c_r\in\mathbb{R}^{A}$, where A is the set of all...

open_problem | open | MathDB 页面 | 原文线索

V2-078982 | MATHDB-401947

Rainbow Arborescence Problem

摘录 / EXCERPT: For a directed graph $G=(V,A)$, an arborescence is a directed spanning tree $F\subseteq A$ such that there is a special node r, called the root, that has in-degree 0 and all ot...

open_problem | open | MathDB 页面 | 原文线索

V2-078983 | MATHDB-401950

Unique Game Conjecture

摘录 / EXCERPT: For any $\varepsilon>0$, it is hard to distinguish instances of a two-variable Constraint Satisfaction Problem (CSP) over a fixed alphabet in which at least a $1-\varepsilon$ fract...

conjecture | open | MathDB 页面 | 原文线索

V2-078984 | MATHDB-401951

Explicit tensors of high rank and border rank

摘录 / EXCERPT: Find explicit sequences of tensors $T_m\in\mathbb{C}^m\otimes\mathbb{C}^m\otimes\mathbb{C}^m$ of high rank and border rank.

open_problem | open | MathDB 页面 | 原文线索

V2-078985 | MATHDB-401952

Determine the border rank of the Kronecker square of the little Coppersmith-Winograd tensor

摘录 / EXCERPT: Determine whether $R\!\left(T_{\mathrm{cw},2}^{\boxtimes 2}\right)=16$. The report gives the bounds $15\leq R\!\left(T_{\mathrm{cw},2}^{\boxtimes 2}\right)\leq 16$ and states that...

open_problem | open | MathDB 页面 | 原文线索

V2-078986 | MATHDB-401953

Imperfect-completeness 2-to-1 Games hardness theorem

摘录 / EXCERPT: For every $\varepsilon>0$ there exists $k\in\mathbb{N}$ such that gap-2-to-1 Games$_k[1-\varepsilon,\varepsilon]$ is NP-hard; that is, it is NP-hard to disting...

open_problem | open | MathDB 页面 | 原文线索

V2-078987 | MATHDB-401955

A truly subquadratic 1.1-approximation for LCS

摘录 / EXCERPT: Is there a 1.1-approximation for LCS running in $O(n^{2-\\varepsilon})$ time, for some $\\varepsilon>0$? (Open for all alphabet sizes.)

open_problem | open | MathDB 页面 | 原文线索

V2-078988 | MATHDB-401956

3-approximation hardness for Euclidean Bichromatic Closest Pair

摘录 / EXCERPT: Assuming the Strong Exponential Time Hypothesis (SETH), prove that for some constant $\varepsilon>0$, approximating the Bichromatic Closest Pair problem under the Euclidean metric...

open_problem | open | MathDB 页面 | 原文线索

V2-078989 | MATHDB-401957

Explicit binary codes approaching the Gilbert-Varshamov bound

摘录 / EXCERPT: For every $k=k(n)$ and $\frac{}{}\varepsilon>0$, there exists a fully explicit family of linear codes $\{C_n\}$ such that each C_n is an $[n,k,d=\frac{1-\varepsilon}{2}n]_2$ code...

open_problem | open | MathDB 页面 | 原文线索

V2-078990 | MATHDB-401959

Deterministic PIT for matrix-valued linear determinants

摘录 / EXCERPT: For matrices $A_i\in\mathbb{C}^{n\times n}$ and matrices $X_i\in\mathbb{C}^{d\times d}$ whose entries are distinct formal commuting variables, with $d=n$, one can deterministically...

open_problem | open | MathDB 页面 | 原文线索

V2-078991 | MATHDB-401960

Majority versus ACC^0

摘录 / EXCERPT: Conjecture: the Boolean majority function is not contained in ACC^0, i.e. $\mathrm{MAJORITY}\notin\mathrm{ACC}^0$.

open_problem | open | MathDB 页面 | 原文线索

V2-078992 | MATHDB-401962

Conjectural formula for unipotent decomposition matrices of finite unitary groups

摘录 / EXCERPT: Assume $\ell\mid q+1$. Then $D_n(1,1)$ is the square part of the decomposition matrix of the unipotent blocks of $GU_n(q)$, up to multiplication by the diagonal matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-078993 | MATHDB-401964

Characterizing quasi-cuspidal representations

摘录 / EXCERPT: Let G be a connected complex reductive group, let $\mathfrak{g}=\mathrm{Lie}(G)$, and for a finite-dimensional representation V of G define the strongly equivariant D-modul...

open_problem | open | MathDB 页面 | 原文线索

V2-078994 | MATHDB-401965

Membership criterion for generalized Lakshmibai-Seshadri paths

摘录 / EXCERPT: Given a path $\gamma\in\Gamma$, how can one decide whether it belongs to $P(\gamma_\lambda)$ for some $\gamma_\lambda\in\Gamma$ contained in the dominant Weyl chamber? Here...

open_problem | open | MathDB 页面 | 原文线索

V2-078995 | MATHDB-401966

Semisimplification conjecture for representations of \(GL(m|n)\)

摘录 / EXCERPT: For the general case $T_{m|n}$, where $m\ge n$, let Γ be a maximal atypical block, identified with the principal block of $T_{n|n}$, and let $L(\lambda_0)$ be the image of...

conjecture | open | MathDB 页面 | 原文线索

V2-078996 | MATHDB-401973

Interpolated free group factors arising from arbitrary-index subfactors

摘录 / EXCERPT: Presumably, if $N \subset M \cong L(\mathbb{F}_\alpha)$ is a subfactor of index $\lambda=[M:N]$, then $N \cong L(\mathbb{F}_\beta)$ with $\beta=\lambda(\alpha-1)+1$. This is open e...

open_problem | open | MathDB 页面 | 原文线索

V2-078997 | MATHDB-401975

Extension of commutator representations to nonsymmetric distributions

摘录 / EXCERPT: Extend the representations obtained for commutators of even free random variables to commutators of general nonsymmetric distributions μ_a and μ_b.

open_problem | open | MathDB 页面 | 原文线索

V2-078998 | MATHDB-401976

Asymptotic proportion of components in random meandric systems

摘录 / EXCERPT: For each $n\ge 1$, let X_n be the number of components of a uniformly random meandric system on $2n$ points; equivalently, X_n is the random variable on $NC_2(2n)^2$ that assig...

open_problem | open | MathDB 页面 | 原文线索

V2-078999 | MATHDB-401979

Kuperberg's conjecture for the functional $\phi(K)$

摘录 / EXCERPT: For every centrally symmetric convex body $K\subseteq\mathbb{R}^n$, define its polar body by $K^\circ=\{x\in\mathbb{R}^n:\langle x,y\rangle\le 1\text{ for all }y\in K\}$ and set...

conjecture | open | MathDB 页面 | 原文线索

V2-079000 | MATHDB-401980

Strong slicing conjecture

摘录 / EXCERPT: For any convex body $K\subseteq\mathbb{R}^n$, $L_K\le L_{\Delta_n}$, where Δ_n is an n-dimensional simplex.

conjecture | open | MathDB 页面 | 原文线索

V2-079001 | MATHDB-401981

Mahler’s conjecture for convex bodies

摘录 / EXCERPT: For every convex body $K\subseteq\mathbb{R}^n$, define ... Mahler’s conjecture asserts that ... with equality for the simplex Δ^n.

conjecture | open | MathDB 页面 | 原文线索

V2-079002 | MATHDB-401983

Dar’s conjecture: a Brunn-Minkowski-type inequality

摘录 / EXCERPT: For convex bodies $K,L\subset\mathbb{R}^n$, define ... Dar conjectured that ...

conjecture | open | MathDB 页面 | 原文线索

V2-079003 | MATHDB-401984

Shephard’s characterization problem for mixed volumes

摘录 / EXCERPT: For integers $n\ge 3$ and $3\le m\le n+1$, determine whether every collection of $N_{n,m}=\binom{n+m-1}{n}$ non-negative real numbers satisfying the known inequalities for mixed vo...

open_problem | open | MathDB 页面 | 原文线索

V2-079004 | MATHDB-401985

A conjecture about the tensor product of cones

摘录 / EXCERPT: Let $C_i\subset V_i$ for $i=1,2$ be cones in finite-dimensional real vector spaces, with cones understood to be convex, closed, salient, and generating. Define ... The pair...

conjecture | open | MathDB 页面 | 原文线索

V2-079005 | MATHDB-401986

Uniqueness in the even logarithmic Minkowski problem

摘录 / EXCERPT: For origin-symmetric convex bodies $K,C\subset\mathbb{R}^n$ with $C^{2,+}$ boundary, if their cone-volume measures agree, $V_K=V_C$, must $K=C$? Here $dV_K=h_K\,dS_K$.

open_problem | open | MathDB 页面 | 原文线索

V2-079006 | MATHDB-401987

The L^p-Brunn-Minkowski conjecture

摘录 / EXCERPT: For every $p\in[0,1)$, every $\lambda\in[0,1]$, and all origin-symmetric convex bodies $K,C\subset\mathbb{R}^n$, the L^p-Brunn-Minkowski inequality ... holds, where the L^p-linea...

conjecture | open | MathDB 页面 | 原文线索

V2-079007 | MATHDB-401988

Gardner-Zvavitch conjecture for symmetric convex sets

摘录 / EXCERPT: For every dimension $n\ge 1$, every pair of origin-symmetric convex sets $K,L\subseteq\mathbb{R}^n$, and every $\lambda\in[0,1]$, the standard Gaussian measure γ satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-079008 | MATHDB-401990

Surface-area analogue of the Busemann-Petty problem

摘录 / EXCERPT: Suppose that origin-symmetric convex bodies $K,L\subset\mathbb{R}^n$ satisfy ... Does it follow that ...

open_problem | open | MathDB 页面 | 原文线索

V2-079009 | MATHDB-401992

Makai-Martini cross-section inequality

摘录 / EXCERPT: For integers $1\le k<n$, any convex body $K\subset\mathbb{R}^n$ with centroid at the origin, and any k-dimensional subspace $E\in G(n,k)$, one has ...

open_problem | open | MathDB 页面 | 原文线索

V2-079010 | MATHDB-401993

Gaussian multi-bubble conjecture for $3\le q\le n+1$

摘录 / EXCERPT: For $q\ge 3$ with $q\le n+1$, the configuration minimizing the Gaussian interface area among decompositions of $\mathbb{R}^n$ into q sets of prescribed Gaussian measure is given...

conjecture | open | MathDB 页面 | 原文线索

V2-079011 | MATHDB-401994

A sausage body is a unique solution to the reverse isoperimetric problem

摘录 / EXCERPT: Let $K\subset\mathbb{R}^{n+1}$ be a convex body. If K is λ-concave for some $\lambda>0$-that is, each boundary point of K supports a tangent ball of radius $1/\lambda$...

open_problem | open | MathDB 页面 | 原文线索

V2-079012 | MATHDB-401995

Logarithmic Minkowski inequality for centrally symmetric convex bodies

摘录 / EXCERPT: Let C and $\widetilde C$ be two centrally symmetric convex bodies in $\mathbb{R}^n$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-079013 | MATHDB-401996

Does the maximum degree of a planar point set tend to infinity?

摘录 / EXCERPT: Let $\xi\subset\mathbb{R}^2$ be a finite point set in general position. An empty triangle is a triangle whose interior contains no point of ξ, and $\deg_2(\xi)$ is the maximum,...

open_problem | open | MathDB 页面 | 原文线索

V2-079014 | MATHDB-401997

A strengthened form of Ryabogin’s congruent-sections theorem in dimensions $n\ge 4$

摘录 / EXCERPT: Let $n\ge 4$, and let $f,g:S^{n-1}\to\mathbb R$ be continuous. Suppose that for every $u\in S^{n-1}$ there exists an orthogonal map $T_u:u^\perp\to u^\perp$, required not to be ide...

open_problem | open | MathDB 页面 | 原文线索

V2-079015 | MATHDB-401998

Fifth Busemann-Petty problem

摘录 / EXCERPT: For an origin-symmetric convex body $K\subset\mathbb{R}^n$, $n\ge 3$, suppose that ... where c is independent of θ, S^{n-1} is the unit sphere, $\theta^\perp$ is the hyper...

open_problem | open | MathDB 页面 | 原文线索

V2-079016 | MATHDB-401999

Improving the approximation order for control-constrained multigrid preconditioners

摘录 / EXCERPT: Can the $O(h)$ approximation order of the multigrid preconditioner in the control-constrained case be improved? The report says that preliminary tests with discontinuous piecewise-...

open_problem | open | MathDB 页面 | 原文线索

V2-079017 | MATHDB-402000

Berry-Esseen bounds for the random conductance model

摘录 / EXCERPT: Under the spectral-gap assumption (SG), let $d\ge 3$. For every $\varepsilon>0$, there exist $p,q\in(1,\infty)$ such that, whenever $\mathbb{E}[\omega_e^p]<\infty$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079018 | MATHDB-402002

Optimal integrability condition for the quenched invariance principle

摘录 / EXCERPT: For the long-range random-conductance model with conductances $C_{xy}=w_{xy}|x-y|^{-d-\alpha}$, $0<\alpha<2$, and independent positive symmetric weights satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-079019 | MATHDB-402003

Linking the overdamped Langevin exit event to a jump Markov model

摘录 / EXCERPT: Is it possible to make a link between the law of the exit event (τ,X_τ) for the overdamped Langevin dynamics and the exit event (T,I) for the jump Markov model described...

open_problem | open | MathDB 页面 | 原文线索

V2-079020 | MATHDB-402008

Localization and evolution of buckling for a shell on a vanishing-stiffness substrate

摘录 / EXCERPT: For a shell over a substrate with vanishing properties, why can the lack of a characteristic length-arising from the absence of a balance between the shell’s elasticity and the sub...

open_problem | open | MathDB 页面 | 原文线索

V2-079021 | MATHDB-402009

Interpretation of the inflection point in cholesterol chemical potential

摘录 / EXCERPT: What is implied by the changing sign of the second derivative of the cholesterol chemical potential, $\frac{d^2\mu}{dC^2}$, or equivalently $\frac{d^2\mu}{dN^2}$, where N is the...

open_problem | open | MathDB 页面 | 原文线索

V2-079022 | MATHDB-402010

More than three positive steady states in the T-cell activation model

摘录 / EXCERPT: Whether there are parameters for which more than three steady states exist is an open question.

open_problem | open | MathDB 页面 | 原文线索

V2-079023 | MATHDB-402011

Periodic solutions in the François-Voisinne-Siggia-Altan-Bonnet-Vergassola T-cell activation model

摘录 / EXCERPT: Whether the model of [2] allows periodic solutions is an open question.

open_problem | open | MathDB 页面 | 原文线索

V2-079024 | MATHDB-402012

One-dimensional uniqueness under the spectral bound for the nonlocal Neumann problem

摘录 / EXCERPT: For the nonlocal Neumann problem ... with $\omega=\gamma|\Omega|\big/\int_\Omega n(x,0)\,dx$, in one spatial dimension, on $I=[0,L]$, the only solution is the constant solution...

open_problem | open | MathDB 页面 | 原文线索

V2-079025 | MATHDB-402018

Critical subgraphs of Schrijver graphs for arbitrary k

摘录 / EXCERPT: Extend the construction of critical (edge-critical) subgraphs of the Schrijver graph $SG(n,k)$ with chromatic number $n-2k+2$ from the case $k=2$ to arbitrary admissible k (with...

open_problem | open | MathDB 页面 | 原文线索

V2-079026 | MATHDB-402021

Connected 3-equivalent graphs and 2-equivalence

摘录 / EXCERPT: Let H and H' be both connected graphs that are 3-equivalent. Is it true that they are 2-equivalent as well?

open_problem | open | MathDB 页面 | 原文线索

V2-079027 | MATHDB-402022

Existence of q-equivalent non-isomorphic connected graphs

摘录 / EXCERPT: For given integer $q \ge 2$, are there two non-isomorphic connected graphs H and H' that are q-equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-079028 | MATHDB-402025

Clustered-colouring versions of Hajós' and the Gerards-Seymour conjectures

摘录 / EXCERPT: Conjecture. For every positive integer t, each of the following two graph classes is clustered t-colorable:

conjecture | open | MathDB 页面 | 原文线索

V2-079029 | MATHDB-402028

Bound on weak coloring numbers for $K_{3,t}$-minor-free graphs

摘录 / EXCERPT: What is the bound on wcol_r for $K_{3,t}$-minor-free graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-079030 | MATHDB-402029

Symmetry of the biconstrained concatenation function

摘录 / EXCERPT: Let cspanmathe$x,y\in(0,1]$c/spancspanmathe$. Define $psi(x,y)$ as the maximum z such that, for every graph G with disjoint nonempty sets $A,B,C...

open_problem | open | MathDB 页面 | 原文线索

V2-079031 | MATHDB-402035

Hadwiger's Conjecture for \(\mathcal{G}_T\)

摘录 / EXCERPT: Prove Hadwiger's Conjecture for $\mathcal{G}_T$.

conjecture | open | MathDB 页面 | 原文线索

V2-079032 | MATHDB-402036

Ryser-Stein-Brualdi conjecture

摘录 / EXCERPT: A Latin square of order n is an $n\times n$ array filled with n different symbols, each occurring exactly once in every row and every column. A partial transversal is a collect...

conjecture | open | MathDB 页面 | 原文线索

V2-079033 | MATHDB-402038

Transversals in generalized Latin squares with at least $n^{1.5}$ symbols

摘录 / EXCERPT: Does every generalized Latin square of order n-that is, an $n\times n$ array filled with an arbitrary number of symbols such that no symbol appears twice in the same row or colum...

open_problem | open | MathDB 页面 | 原文线索

V2-079034 | MATHDB-402039

Improving the partial-transversal bound for Latin squares

摘录 / EXCERPT: Can the error term $O(\log^2 n)$ in Hatami and Shor's lower bound for the size of a partial transversal in an order-n Latin square be improved to an absolute constant?

open_problem | open | MathDB 页面 | 原文线索

V2-079035 | MATHDB-402040

Limit model for the Uchiyama particle system

摘录 / EXCERPT: What is the limit model associated with the Uchiyama model?

open_problem | open | MathDB 页面 | 原文线索

V2-079036 | MATHDB-402042

Diffusive invariance principle for the random Lorentz process

摘录 / EXCERPT: For the random Lorentz process $t\mapsto X^{r,\rho}(t)$ with spherical scatterers generated by a Poisson point process of intensity $\rho>0$ in $\mathbb{R}^d$, radius $r>0$, and a...

open_problem | open | MathDB 页面 | 原文线索

V2-079037 | MATHDB-402054

Selecting the number of moments in maximum-entropy density reconstruction

摘录 / EXCERPT: An interesting open question is how to select the concrete number of moments R in practical computations: an overestimation may lead to overfitting and thereby to inaccurate reco...

open_problem | open | MathDB 页面 | 原文线索

V2-079038 | MATHDB-402055

Verification of the continuous-time limit of the ensemble Kalman filter

摘录 / EXCERPT: Verify the continuous time limit of the ensemble Kalman filter.

open_problem | open | MathDB 页面 | 原文线索

V2-079039 | MATHDB-402056

Correcting the uncertainty quantification in the Bayesian conjugate gradient method

摘录 / EXCERPT: Develop a correction for the mismatch in BayesCG whereby the search directions in Eq. (2) depend implicitly on the unknown exact solution x^*, violating the linearity assumption...

open_problem | open | MathDB 页面 | 原文线索

V2-079040 | MATHDB-402058

Frequentist coverage of Bayesian credible sets in linear inverse problems

摘录 / EXCERPT: For the linear inverse problem $M_\\varepsilon=Af^\\dagger+\\varepsilon W$ with a centred Gaussian prior, do credible sets have the correct frequentist coverage in the small-noise...

open_problem | open | MathDB 页面 | 原文线索

V2-079041 | MATHDB-402060

Existence of a q-analogue of the Fano plane

摘录 / EXCERPT: Does there exist a collection $\mathcal B$ of 3-dimensional subspaces of $\mathbb F_q^7$ forming a $2\text{-}(7,3,1)$ subspace design; equivalently, is every 2-dimensional subs...

open_problem | open | MathDB 页面 | 原文线索

V2-079042 | MATHDB-402061

Asymptotic goodness of group codes

摘录 / EXCERPT: Is the class of group codes asymptitically good?

open_problem | open | MathDB 页面 | 原文线索

V2-079043 | MATHDB-402064

Goppa’s conjecture for binary codes

摘录 / EXCERPT: For linear codes over the field F of size 2, Goppa conjectured that the asymptotic coding-function curve satisfies ... where the binary entropy function is ...

conjecture | open | MathDB 页面 | 原文线索

V2-079044 | MATHDB-402065

Existence of a binary q-analog of the Fano plane

摘录 / EXCERPT: Does there exist a set $\mathcal{S}$ of 381 3-dimensional subspaces of $\mathbb{F}_2^7$ such that every 2-dimensional subspace of $\mathbb{F}_2^7$ is contained in exactly one...

open_problem | open | MathDB 页面 | 原文线索

V2-079045 | MATHDB-402066

Existence of a q-analog of a (9,3,3,1) group divisible design

摘录 / EXCERPT: Does a q-analog of a $(9,3,3,1)$ group divisible design exist?

open_problem | open | MathDB 页面 | 原文线索

V2-079046 | MATHDB-402067

Determine the maximum size of a partial 3-spread in \mathbb{F}_2^{11}

摘录 / EXCERPT: Determine the exact value of $A_2(11,4)$, the maximum size of a partial 3-spread in $\operatorname{PG}(\mathbb{F}_2^{11})$. The report gives the bounds...

open_problem | open | MathDB 页面 | 原文线索

V2-079047 | MATHDB-402068

General characteristic-power divisibility

摘录 / EXCERPT: The more general case that Δ is only a power of the characteristic of $\mathbb{F}_q$ remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-079048 | MATHDB-402069

Patterson-Wiedemann covering-radius conjecture for binary Reed-Muller codes

摘录 / EXCERPT: For Boolean functions $f,g:\mathbb F_2^n\to\mathbb F_2$, let ... ... where the maximum is over all Boolean functions f and the minimum is over the 2^{n+1} affine Boolean functions...

conjecture | open | MathDB 页面 | 原文线索

V2-079049 | MATHDB-402073

Projective strengthening of Saxl’s conjecture

摘录 / EXCERPT: Let $\rho_k=(k,k-1,\ldots,2,1)$ and $n=k(k+1)/2$. In characteristic 2, write the projective-character decomposition ... where $a_\nu\in\mathbb{N}_0$. The projective strengthening o...

conjecture | open | MathDB 页面 | 原文线索

V2-079050 | MATHDB-402087

Character bounds for finite almost quasisimple groups

摘录 / EXCERPT: Let G be a finite almost quasisimple group and let $g\in G\setminus Z(G)$. Find an explicit, and as small as possible, constant $0<\alpha=\alpha(g)<1$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-079051 | MATHDB-402091

Determine the remaining parameter in the Jantzen calculation for SL_5

摘录 / EXCERPT: Determine whether the undetermined parameter in the Jantzen-sum-formula calculation for SL_5 is $a=1$ or $a=2$.

open_problem | open | MathDB 页面 | 原文线索

V2-079052 | MATHDB-402092

E∞-structure on Lubin-Tate spectra

摘录 / EXCERPT: Given a finite-height formal group law $\widehat{G}_0$ over a perfect field k of characteristic p, and the associated Lubin-Tate spectrum E_n obtained via the Landweber exact...

open_problem | open | MathDB 页面 | 原文线索

V2-079053 | MATHDB-402093

Orientations of formal groups and the Quillen formal group

摘录 / EXCERPT: Let R be a complex periodic $E_\\infty$-ring and let $\\widehat G$ be a one-dimensional formal group over R. Then there is an equivalence ... Under this equivalence, a preorienta...

open_problem | open | MathDB 页面 | 原文线索

V2-079054 | MATHDB-402096

Infinitely many commensurability classes of non-arithmetic complex hyperbolic lattices

摘录 / EXCERPT: Are there infinitely many commensurability classes of non-arithmetic lattices in $\mathrm{PU}(n,1)$ for $n\ge 2$?

open_problem | open | MathDB 页面 | 原文线索

V2-079055 | MATHDB-402097

Commensurability criterion for non-arithmetic hyperbolic Coxeter groups

摘录 / EXCERPT: Determine a necessary-and-sufficient criterion for when two non-arithmetic hyperbolic Coxeter groups $\Gamma_1,\Gamma_2<\operatorname{Isom}(\mathbb H^n)$ are commensurable in the w...

open_problem | open | MathDB 页面 | 原文线索

V2-079056 | MATHDB-402100

Arithmetic hyperbolic reflection group in dimension 2 with degree-7 field of definition

摘录 / EXCERPT: Construct an arithmetic hyperbolic reflection group in dimension $n=2$ whose field of definition has degree 7.

open_problem | open | MathDB 页面 | 原文线索

V2-079057 | MATHDB-402101

Total reality of trace fields of hyperbolic lattices

摘录 / EXCERPT: Let Γ be a non-arithmetic reflection lattice in $\mathrm{PO}(n,1)$ with $n\ge 4$. Is the trace field of Γ necessarily totally real?

open_problem | open | MathDB 页面 | 原文线索

V2-079058 | MATHDB-402102

Equality in the rank inequality for finite-index reflection subgroups

摘录 / EXCERPT: Let (W,S) be a Coxeter system, and let $\Gamma \subset W$ be a reflection subgroup of finite index. Then $\operatorname{rank}\u0007Gamma \ge \operatorname{rank}W$. When does equa...

open_problem | open | MathDB 页面 | 原文线索

V2-079059 | MATHDB-402103

Infinitely many commensurability classes of non-arithmetic lattices in \(PU(2,1)\)

摘录 / EXCERPT: Are there infinitely many commensurability classes of non-arithmetic lattices in $PU(2,1)$?

open_problem | open | MathDB 页面 | 原文线索

V2-079060 | MATHDB-402104

Geometric explanation for the classification of quivers of finite mutation type

摘录 / EXCERPT: Give a geometric explanation for the classification of quivers of finite mutation type.

open_problem | open | MathDB 页面 | 原文线索

V2-079061 | MATHDB-402105

Characterization of potentials for which one lattice minimizes at every volume

摘录 / EXCERPT: Let $\mathcal L_d$ be the class of simple lattices $L=\bigoplus_{i=1}^d\mathbb Z u_i$ in $\mathbb R^d$, and let $\mathcal L_d^\circ(V)$ denote the simple lattices with unit-cell vo...

open_problem | open | MathDB 页面 | 原文线索

V2-079062 | MATHDB-402106

Bowen-Lyons-Radin-Winkler solidification conjecture for planar packings

摘录 / EXCERPT: Let $\varepsilon>0$. Bowen, Lyons, Radin, and Winkler conjectured that an infinite connected component exists almost surely in any model obtained as a weak-$*$ limit, as...

conjecture | open | MathDB 页面 | 原文线索

V2-079063 | MATHDB-402107

Density of the uncovered set in covering systems with large distinct moduli

摘录 / EXCERPT: Is it true that for each $C>0$, there exist constants $M>0$ and $\varepsilon>0$ such that every covering system with distinct moduli $d_1,\ldots,d_k\re M$ satisfying ... has an uncov...

open_problem | open | MathDB 页面 | 原文线索

V2-079064 | MATHDB-402108

The Erdős-Selfridge problem for covering systems

摘录 / EXCERPT: Does there exist a finite covering system of the integers, consisting of arithmetic progressions $A_i=a_i+d_i\mathbb{Z}$ whose moduli d_i are pairwise distinct and odd?

open_problem | open | MathDB 页面 | 原文线索

V2-079065 | MATHDB-402109

Schinzel’s conjecture for covering systems

摘录 / EXCERPT: If $\mathcal A=\{A_1,\ldots,A_k\}$ is a finite collection of arithmetic progressions $A_i=a_i+d_i\mathbb Z$ whose union is $\mathbb Z$, then at least one of the moduli divides anot...

conjecture | open | MathDB 页面 | 原文线索

V2-079066 | MATHDB-402110

Perfect packing of degenerate graphs with many leaves

摘录 / EXCERPT: For all $\mu>0$ and $D\in\mathbb{N}$, there exist $c>0$ and $n_0\in\mathbb{N}$ such that, for every $n\ge n_0$, the following holds. Let $G_1,\ldots,G_t$ be a family of D-degener...

open_problem | open | MathDB 页面 | 原文线索

V2-079067 | MATHDB-402111

Divisibility of the least common multiple in a distinct covering system

摘录 / EXCERPT: Let $\mathcal A=\{A_d:d\in D\}$ be a finite collection of arithmetic progressions with distinct moduli that covers the integers, and let $Q=\operatorname{lcm}(D)$ be the least comm...

open_problem | open | MathDB 页面 | 原文线索

V2-079068 | MATHDB-402112

Asymptotic rank of sparse random matrices

摘录 / EXCERPT: The report presents the following solved result. Let $\mathbb F$ be a field and let $\chi\ne 0$ be a random variable taking values in $\mathbb F$. Let $\boldsymbol d\ge 1$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079069 | MATHDB-402114

Rational exponents conjecture

摘录 / EXCERPT: For every rational number $r \in (1,2)$, there exists a graph H with $\operatorname{ex}(n,H)=\Theta(n^r)$, where $\operatorname{ex}(n,H)$ is the maximum number of edges in an n...

conjecture | open | MathDB 页面 | 原文线索

V2-079070 | MATHDB-402115

Kang-Kim-Liu subdivision conjecture

摘录 / EXCERPT: Let H be a bipartite graph. If $\operatorname{ex}(n,H)=O(n^{1+\alpha})$ for some $\alpha>0$, then $\operatorname{ex}(n,H')=O\left(n^{1+\alpha/2}\right)$, where H' is the 1-su...

conjecture | open | MathDB 页面 | 原文线索

V2-079071 | MATHDB-402118

Regular graphs with regular connected neighborhoods

摘录 / EXCERPT: Is it possible, for any integers (a,b,c,d,e), to construct infinite families of (a,b,c,d,e)-regular graphs with connected links of all orders-that is, with the neighborhood of...

open_problem | open | MathDB 页面 | 原文线索

V2-079072 | MATHDB-402120

Explicit RIP constructions at the square-root barrier in the polylogarithmic regime

摘录 / EXCERPT: It would be interesting to obtain a construction which truly achieves the square root barrier $s \ge C\sqrt{n}$ in the polylogarithmic regime $n = \log^{O(1)} p$.

open_problem | open | MathDB 页面 | 原文线索

V2-079073 | MATHDB-402121

Testing hereditary graph properties in the VDF model

摘录 / EXCERPT: Are all hereditary graph properties testable in the VDF model?

open_problem | open | MathDB 页面 | 原文线索

V2-079074 | MATHDB-402124

Theorem 1 on monotone Boolean functions of small influence

摘录 / EXCERPT: Let $f:\{0,1\}^n\to\{0,1\}$ be a monotone Boolean function with $_p(f)$ bounded away from 0 and 1 and $I[f]\le Kp(1-p)$. Then there is a set J of $O(K)$ coordinates such tha...

open_problem | open | MathDB 页面 | 原文线索

V2-079075 | MATHDB-402128

Dependence on \\\(\delta\) in the upper tail for triangle counts

摘录 / EXCERPT: For $X=X_{n,p}$, the number of triangles in the binomial random graph $G_{n,p}$, determine the dependence on the fixed constant $\delta>0$ of ...

open_problem | open | MathDB 页面 | 原文线索

V2-079076 | MATHDB-402133

Lower-tail asymptotics for triangle counts in sparse random graphs

摘录 / EXCERPT: Let $X=N_{K_3}(G_{n,p})$. If $n^{-1/2}\\ll p\le 1-\Omega(1)$, then, for every $\delta\in(0,1]$,nn ... nnwhere, for $p\in(0,1)$ and $q\in[0,1]$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079077 | MATHDB-402134

Inducibility of paths

摘录 / EXCERPT: Determine the inducibility $\operatorname{ind}(P_k)$ of the path P_k on k vertices for every integer $k\ge 4$.

open_problem | open | MathDB 页面 | 原文线索

V2-079078 | MATHDB-402135

Covering sparse random graphs by monochromatic components

摘录 / EXCERPT: The conjecture reported here is that, for the random graph $G(n,p)$, $tc_r(G(n,p))\le r$ when $p\gg\left(\frac{r\log n}{n}\right)^{1/r}$, where $tc_r(G)$ is the minimum number of m...

open_problem | open | MathDB 页面 | 原文线索

V2-079079 | MATHDB-402138

Reverse Sidorenko inequality for anti-ferromagnetic target graphs

摘录 / EXCERPT: For fixed d and an anti-ferromagnetic weighted graph H-that is, a graph whose edge-weight matrix has all eigenvalues except the top one nonpositive-the report conjectures that...

open_problem | open | MathDB 页面 | 原文线索

V2-079080 | MATHDB-402140

Girth versus diameter of graphs

摘录 / EXCERPT: For graphs G with no vertices of degree 1 or 2, determine ...

open_problem | open | MathDB 页面 | 原文线索

V2-079081 | MATHDB-402143

Giant components for non-linear degree-based random graph processes

摘录 / EXCERPT: For a function $f=f_n:\bbmathbb{N}\bbrightarrow[0,\bbinfty)$, consider the random graph process $(G^f_{n,m})_{m\ge 0}$ and its multigraph variant $(G^{f,*}_{n,m})_{m\ge 0}$ defined...

open_problem | open | MathDB 页面 | 原文线索

V2-079082 | MATHDB-402145

Theorem 5: Special points and nonpositive-slope tangent subbundles

摘录 / EXCERPT: Assume that H is isoclinic. Let $x\in M_H^\circ(\mathbb{C})$ correspond to a global section f of Z over $X_\mathbb{C}$. Then the following are equivalent: 1....

open_problem | open | MathDB 页面 | 原文线索

V2-079083 | MATHDB-402149

Weakly admissible points in strata of a p-adic period domain

摘录 / EXCERPT: For which $[b']\in B(G)$ is ...

open_problem | open | MathDB 页面 | 原文线索

V2-079084 | MATHDB-402150

Weakly admissible points in the strata of a period domain

摘录 / EXCERPT: Assume that b is basic. For which $[b']\in B(G)$ is ...

open_problem | open | MathDB 页面 | 原文线索

V2-079085 | MATHDB-402151

Criterion for vanishing torsion in a perverse-sheaf extension

摘录 / EXCERPT: Let A and B be free perverse $\bZ_\ell$-sheaves that are simple over $\u0001bQ_\ell$, and suppose that the short exact sequence ... splits over $\u0001bQ_\ell$. Writing the corre...

open_problem | open | MathDB 页面 | 原文线索

V2-079086 | MATHDB-402153

A theory of tropical ψ classes in genus one

摘录 / EXCERPT: Develop a theory of tropical c8 classes in genus one, together with corresponding correspondence theorems relating it to the classical theory.

open_problem | open | MathDB 页面 | 原文线索

V2-079087 | MATHDB-402155

Smooth compactification of the Lagrangian lift of the trisectrice

摘录 / EXCERPT: The report conjectures that the Lagrangian lift of the trisectrice compactifies smoothly inside $\mathbb{C}^3$. Here the trisectrice is the tropical hypersurface in...

open_problem | open | MathDB 页面 | 原文线索

V2-079088 | MATHDB-402158

Equality of refined Severi degrees and refined curve-counting invariants

摘录 / EXCERPT: It is conjectured that, if L is sufficiently ample, the refined Severi degrees coincide with the refined curve-counting invariants of [10].

open_problem | open | MathDB 页面 | 原文线索

V2-079089 | MATHDB-402160

Injectivity of global distance distributions on restricted classes of metric-measure spaces

摘录 / EXCERPT: Is the global distance-distribution invariant $X\mapsto d_HX=(d_X)_\#(\mu_X\otimes\mu_X)$ injective on some suitably restricted, interesting class $\mathcal C\subset\mathcal M$ of...

open_problem | open | MathDB 页面 | 原文线索

V2-079090 | MATHDB-402161

Projection factor recovery

摘录 / EXCERPT: Let Π denote the orthogonal projection onto some unknown subspace $T\subseteq\mathbb{R}^{d}$ of some unknown dimension. What conditions on $f:T\to[k]:=\{1,\ldots,k\}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079091 | MATHDB-402162

Exponential decay of the Hitchin metric along generic rays

摘录 / EXCERPT: Fix a generic $SL(n,\mathbb{C})$-Higgs bundle $(\bar\partial_E,\varphi)$ in the Hitchin moduli space $\mathcal{M}$, and a Higgs-bundle variation $\dot\psi=(\dot\eta,\dot\varphi)$....

open_problem | open | MathDB 页面 | 原文线索

V2-079092 | MATHDB-402163

Which ALG metrics arise as Hitchin metrics?

摘录 / EXCERPT: Which portion of the moduli space of ALG metrics is filled out by Hitchin metrics, e.g. by varying the parabolic weights or by allowing semisimple residues?

open_problem | open | MathDB 页面 | 原文线索

V2-079093 | MATHDB-402165

Singularities of the wobbly locus in higher-rank and higher-genus Higgs-bundle moduli

摘录 / EXCERPT: Understand the singularities of W at codimension 2 in X, and determine whether the phenomenon of strings of fixed-point loci (not present in the genus-2, rank-2 case) con...

open_problem | open | MathDB 页面 | 原文线索

V2-079094 | MATHDB-402166

Hecke eigensheaf property for the genus-2 rank-2 local system

摘录 / EXCERPT: In the genus-2, rank-2 setup of the report, show that the local system constructed over $X-W$ satisfies the Hecke eigensheaf property for the Hecke correspondences.

open_problem | open | MathDB 页面 | 原文线索

V2-079095 | MATHDB-402167

Duality between branes on the singular locus of the Hitchin moduli space

摘录 / EXCERPT: Let X be a Riemann surface and let M_X be the moduli space of rank-n Higgs bundles. For the partition $n=n_1+\+\+n_s$, let $P_{\boldsymbol n}$ be the associated standard para...

open_problem | open | MathDB 页面 | 原文线索

V2-079096 | MATHDB-402168

Nonemptiness criterion for the zero-Euler-class representation spaces of the thrice-punctured projective plane

摘录 / EXCERPT: Let $s\in\{\pm1\}^3$. Then $X^s_0(N_{1,3})$ is nonempty if and only if s contains exactly one or two $+1$'s.

open_problem | open | MathDB 页面 | 原文线索

V2-079097 | MATHDB-402169

Bowditch’s hyperbolicity question for simple closed curves

摘录 / EXCERPT: Given a non-elementary type-preserving representation $\rho:\pi_1(S)\to\mathrm{PGL}(2,\mathbb{R})$, is it true that if ρ sends every non-peripheral 2-sided simple closed curve...

open_problem | open | MathDB 页面 | 原文线索

V2-079098 | MATHDB-402173

Comparison of the refined Higgs Harder-Narasimhan and Harder-Narasimhan stratifications

摘录 / EXCERPT: How do the refined HHN and HN stratifications compare? In particular, does the refined HN stratification take into account the HHN type, and vice versa?

open_problem | open | MathDB 页面 | 原文线索

V2-079099 | MATHDB-402174

Boundary compactification preserving circle-compactness for the \\\-action when \\\ = 0

摘录 / EXCERPT: For = 0, the -action on the relevant coarse moduli space is not circle-compact. Can one construct a boundary involving the moduli spaces for ne 0 such that circle-compactness of th...

open_problem | open | MathDB 页面 | 原文线索

V2-079100 | MATHDB-402179

Uniqueness and Lagrangianity in the intermediate integrability regime

摘录 / EXCERPT: Let $b=b(t,x)$ be a bounded, divergence-free vector field on $\mathbb{R}^d$ that belongs uniformly in time to $W^{1,p}(\mathbb{R}^d)$, and let u be a distributional solution of t...

open_problem | open | MathDB 页面 | 原文线索

V2-079101 | MATHDB-402180

Strong convergence of viscous approximations for discontinuous-flux conservation laws

摘录 / EXCERPT: For the discontinuous-flux conservation law ... where ... and $D=\{w\in L^1(\mathbb R):0\le w\le 1\}$, the report states that a suitable accretive operator A generates a unique conti...

open_problem | open | MathDB 页面 | 原文线索

V2-079102 | MATHDB-402181

The multi-dimensional Burgers Equation with L^p Initial Data

摘录 / EXCERPT: For the multidimensional Burgers equation considered in the report, namely $\partial_tu+\operatorname{div}_y\vec f(u)=0$ in $n\ge 1$ space dimensions, the solution semigroup extend...

open_problem | open | MathDB 页面 | 原文线索

V2-079103 | MATHDB-402182

Vanishing-viscosity convergence for scalar conservation laws with 2D-regulated flux

摘录 / EXCERPT: Consider ... and its viscous approximation ... Under (A1)-(A3) from the report-namely, $\alpha\mapsto f(\alpha,u)$ is smooth, $u\mapsto f(\alpha,u)$ is Lipschitz, $f(\alpha,0)=0$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079104 | MATHDB-402184

Additional conditions for well-posedness and uniqueness of the Riemann problem

摘录 / EXCERPT: Which additional conditions are suitable in the sense that the Riemann problem for (5) is well-posed? Especially: Is a finite number of configurations necessary for the uniqueness...

open_problem | open | MathDB 页面 | 原文线索

V2-079105 | MATHDB-402185

Extrapolation of potential-outcome predictions without global overlap

摘录 / EXCERPT: Can we give conditions under which it is acceptable to extrapolate predictions of potential outcomes instead of requiring overlap?

open_problem | open | MathDB 页面 | 原文线索

V2-079106 | MATHDB-402187

Sparse Cholesky ordering for positive semidefinite matrices

摘录 / EXCERPT: Given a positive semidefinite matrix K, find a permutation matrix Π such that the Cholesky factorization of $\Pi^{\top}K\Pi$ is as sparse as possible. Although a formulation...

open_problem | open | MathDB 页面 | 原文线索

V2-079107 | MATHDB-402188

Exact $k\times k$-decomposability of a matrix

摘录 / EXCERPT: Develop conditions for exact $k\times k$-decomposability of a given matrix Q, for $k>2$.

open_problem | open | MathDB 页面 | 原文线索

V2-079108 | MATHDB-402189

Strength comparison of strengthened LP relaxations for MIQPs

摘录 / EXCERPT: The report conjectures that the strengthened relaxation LPG^+ is at least as strong as the strengthened relaxation LPH^+.

open_problem | open | MathDB 页面 | 原文线索

V2-079109 | MATHDB-402192

Smallest dimension attaining the absolute bound for equiangular lines

摘录 / EXCERPT: What is the smallest dimension n such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-079110 | MATHDB-402193

Asymptotic growth of the fixed-level SDP bound

摘录 / EXCERPT: Is $\Delta_k(G_{n,a})^* = \Theta(n^2)$ for fixed k and a?

open_problem | open | MathDB 页面 | 原文线索

V2-079111 | MATHDB-402194

Genus-2 modular resolution with Weierstrass-tail data

摘录 / EXCERPT: Construct a genus-2 modular resolution or desingularization for the relevant components of $M_{2,n}(\mathbb{P}^r,d)$ by augmenting the logarithmic data with a section over the locu...

open_problem | open | MathDB 页面 | 原文线索

V2-079112 | MATHDB-402196

A good compactification of strata of abelian differentials

摘录 / EXCERPT: Let $g\ge 2$ and let $\mu=(m_1,\ldots,m_n)\in\mathbb{Z}^n$ satisfy $m_1+\cdots+m_n=2g-2$. Find a “good” compactification of the stratum $\mathbb{P}\Omega\mathcal{M}_g(\mu)$ that is...

open_problem | open | MathDB 页面 | 原文线索

V2-079113 | MATHDB-402198

Testing relevant hypotheses in functional data via self-normalization

摘录 / EXCERPT: Given a strictly stationary functional time series $\{X_n\}_{n\in\mathbb Z}$ with $X_n\in L^2([0,1])$ and expectation $\mu=\mathbb E[X_1]\in L^2([0,1])$, and a sample...

open_problem | open | MathDB 页面 | 原文线索

V2-079114 | MATHDB-402205

Zariski descent for the BLRS complex

摘录 / EXCERPT: Does the BLRS complex $C_*(-,K^{MW}_q,q)$ satisfy Zariski descent?

open_problem | open | MathDB 页面 | 原文线索

V2-079115 | MATHDB-402209

Polarization threshold for the bulk-surface obstacle problem

摘录 / EXCERPT: In the spherical-domain case $\Omega=B(0,1)$, with $\Gamma=\partial\Omega$, let $\ell=a_6/D$ and $g(x)=c(x)/(c(x)+a_5)\in(0,1)$. For the generalized obstacle problem ... ... there is a...

open_problem | open | MathDB 页面 | 原文线索

V2-079116 | MATHDB-402211

Directional differentiability of the maximal solution map for elliptic QVIs

摘录 / EXCERPT: In the context of Theorem 2, if $M(f)\in Q(f)$ is the maximal solution of the QVI with source term f, is M directionally differentiable?

open_problem | open | MathDB 页面 | 原文线索

V2-079117 | MATHDB-402212

Rigorous Robin-penalty limit for nonlinear bulk-surface Allen-Cahn transmission

摘录 / EXCERPT: For the bulk-surface Allen-Cahn system ... with the stated initial conditions, bounded smooth $\Omega\subset\mathbb R^3$, smooth transmission function h, and f,f_Γ the deriv...

open_problem | open | MathDB 页面 | 原文线索

V2-079118 | MATHDB-402213

Marked length-spectrum rigidity for actions on CAT(0) cube complexes

摘录 / EXCERPT: Let X,Y be irreducible CAT(0) cube complexes, and let $G\curvearrowright X,Y$ be actions such that their translation-length functions with respect to the ℓ^1 metric satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-079119 | MATHDB-402220

Classification of entire complex Monge-Ampère solutions

摘录 / EXCERPT: Classify plurisubharmonic functions $\phi:\mathbb{C}^n\to\mathbb{R}$ satisfying $\det(\phi_{j\bar{k}})=1$, where $\phi_{j\bar{k}}=\partial_{z_j}\bar\partial_{z_k}\phi$ is the com...

open_problem | open | MathDB 页面 | 原文线索

V2-079120 | MATHDB-402223

Singular uniformization for prescribed conic data

摘录 / EXCERPT: Given a compact Riemann surface M, a collection of distinct points $p=\{p_1,\ldots,p_k\}\subset M$, and positive real numbers $\beta_1,\ldots,\beta_k$, is it possible to find a m...

open_problem | open | MathDB 页面 | 原文线索

V2-079121 | MATHDB-402226

Tresser conjecture on periods of dissipative disc diffeomorphisms

摘录 / EXCERPT: For any dissipative diffeomorphism of the disc, there exists n_0 such that the set of periods is contained in $\{n\cdot 2^k : n\le n_0,\ k\in\mathbb N\}$. When the set of periods...

conjecture | open | MathDB 页面 | 原文线索

V2-079122 | MATHDB-402228

Existence of energy-diffusing orbits in the planar elliptic restricted three-body problem

摘录 / EXCERPT: For the planar elliptic restricted three-body problem with the Neptune-Triton parameters, namely normalized mass ratio $\mu=0.0002089$ and primary eccentricity bound...

open_problem | open | MathDB 页面 | 原文线索

V2-079123 | MATHDB-402234

Removing non-degeneracy from the two-or-infinitely-many-orbits conclusion

摘录 / EXCERPT: Can the non-degeneracy hypothesis in Corollary 5 be weakened or removed? In particular, is it true that every tight contact form on S^3 has either exactly two or infinitely many...

open_problem | open | MathDB 页面 | 原文线索

V2-079124 | MATHDB-402235

Weakening or removing non-degeneracy assumptions for Reeb-flow conclusions

摘录 / EXCERPT: Can the non-degeneracy assumptions in Corollaries 3, 4, and 5 be weakened or removed?

open_problem | open | MathDB 页面 | 原文线索

V2-079125 | MATHDB-402236

Appropriate constraints for a Hodge-decomposition variational principle

摘录 / EXCERPT: For the Hodge decomposition ... with $\psi=0$ on all boundaries, $\Delta\psi=-\omega$, and $D\omega/Dt=0$, what are the appropriate constraints in the variational principle based on...

open_problem | open | MathDB 页面 | 原文线索

V2-079126 | MATHDB-402238

Boundary behavior of the connected solution curve for constant-vorticity Stokes waves

摘录 / EXCERPT: Constantin, Strauss, and Varvaruca conjectured that, at the boundary of the connected solution curve for the constant-vorticity steady gravity-wave problem, one reaches either an e...

open_problem | open | MathDB 页面 | 原文线索

V2-079127 | MATHDB-402239

Nonexistence of two-dimensional pure-gravity solitary waves in deep water

摘录 / EXCERPT: The two-dimensional gravity wave equation in deep water admits no solitary waves (η,ψ) with critical regularity $\eta_x\in\dot{B}^{\frac{1}{2}}_{2,1}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079128 | MATHDB-402240

Doubly periodic steady Beltrami water waves without surface tension

摘录 / EXCERPT: Develop an existence theory for genuinely three-dimensional, small-amplitude, doubly periodic steady Beltrami water waves in the case of zero surface tension, $\sigma=0$. The repor...

open_problem | open | MathDB 页面 | 原文线索

V2-079129 | MATHDB-402241

Existence theory for pure-gravity doubly periodic water waves

摘录 / EXCERPT: Develop an existence theory for small-amplitude, genuinely three-dimensional, doubly periodic irrotational water waves in the pure-gravity case, i.e. with $\[0m\alpha=0$ and surfac...

open_problem | open | MathDB 页面 | 原文线索

V2-079130 | MATHDB-402242

Global well-posedness for fractional KdV in the L^2-subcritical regime

摘录 / EXCERPT: For the dispersive equation (1), namely $\partial_t u=\partial_x\bigl(|\partial_x|^\alpha u-u^p\bigr)$ on $\mathbb{R}\times\mathbb{R}$, with $p=2$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079131 | MATHDB-402247

Explicit description of Cheeger sets

摘录 / EXCERPT: For $n\ge 2$ and an open bounded domain $\Omega\subset\mathbb{R}^n$, with Cheeger sets defined as sets $E\subset\Omega$ with $|E|>0$ attaining...

open_problem | open | MathDB 页面 | 原文线索

V2-079132 | MATHDB-402248

Stability conjecture for the Allen-Cahn equation in dimension 3

摘录 / EXCERPT: Let $\varepsilon>0$ and $W(u)=\frac14(1-u^2)^2$. Is every entire stable solution $u:\mathbb R^3\to(-1,1)$ of the Allen-Cahn equation ... flat? Here stability means nonnegative second...

conjecture | open | MathDB 页面 | 原文线索

V2-079133 | MATHDB-402250

Classification of compact ancient \-dimensional \-noncollapsed Ricci flows

摘录 / EXCERPT: Let $(S^3,g(t))$ be a compact, ancient $\$-noncollapsed solution to the Ricci flow (1) on S^3. Then $g(t)$ is either a family of contracting spheres or Perelman’s solution.

open_problem | open | MathDB 页面 | 原文线索

V2-079134 | MATHDB-402251

Regularity of Lipschitz minimizers with finite degeneracy set

摘录 / EXCERPT: Problem 1. Prove or disprove that Lipschitz minimizers are C^1 in dimension $n\ge 3$ when D_F consists of finitely many points.

open_problem | open | MathDB 页面 | 原文线索

V2-079135 | MATHDB-402254

Absolute continuity for measures annihilated by a third-order scalar operator

摘录 / EXCERPT: Let $\mathcal A:=\partial_{x_1}^3+\partial_{x_2}^3+\partial_{x_3}^3$ on $\mathbb R^3$. Is it true that every measure $\mu\in\mathcal M(\mathbb R^3;\mathbb R)$ satisfying...

open_problem | open | MathDB 页面 | 原文线索

V2-079136 | MATHDB-402256

Equivalence of a lower metric bound and instantaneous completeness for conformally hyperbolic Yamabe flows

摘录 / EXCERPT: Whether condition (ii), namely $g(t)\ge m(m-1)t\,g_H$ for all $t>0$, is equivalent to instantaneous completeness of conformally hyperbolic Yamabe flows remains open; the equivalenc...

open_problem | open | MathDB 页面 | 原文线索

V2-079137 | MATHDB-402258

Vanishing-volume capillarity limit for Plateau’s problem

摘录 / EXCERPT: Under the assumptions of Theorem 1-namely, c(ell<infty), ∂W is smooth, there exists $\tau _0>0$ such that $\mathbb R^{n+1}\setminus I_\tau(W)$ is connected for every...

open_problem | open | MathDB 页面 | 原文线索

V2-079138 | MATHDB-402259

Sparse approximability of the expected solution operator

摘录 / EXCERPT: For a random or parameterized family of prototypical linear elliptic operators $A(\omega)=-\operatorname{div}(A(\omega)\nabla\!\cdot)$ on D, with deterministic right-hand side...

open_problem | open | MathDB 页面 | 原文线索

V2-079139 | MATHDB-402260

Contrast-independent approximation theory for generalized finite element spaces

摘录 / EXCERPT: For the scalar high-contrast elliptic problem with permeability $K(x)=\alpha(x)I$, determine whether, when the local multiplicities m_i and patches...

open_problem | open | MathDB 页面 | 原文线索

V2-079140 | MATHDB-402261

Contrast-independent approximation theory for multiscale coarse spaces

摘录 / EXCERPT: Show that, provided the local dimensions m_i and the patches $\\Omega_i$ are chosen sufficiently large, one can prove the contrast-independent estimates ... for an operator...

open_problem | open | MathDB 页面 | 原文线索

V2-079141 | MATHDB-402265

Conically smooth structures on Whitney-stratified spaces

摘录 / EXCERPT: Every Whitney-stratified space admits a conically smooth structure.

open_problem | open | MathDB 页面 | 原文线索

V2-079142 | MATHDB-402269

Equivariant realization problem for Morava-K-theory types

摘录 / EXCERPT: Let G be a finite p-group and let $H<G$. Define $r_n(H,G)$ to be the maximal integer r such that there exists a finite G-spectrum X with $X^{\phi G}$ of type n and...

open_problem | open | MathDB 页面 | 原文线索

V2-079143 | MATHDB-402276

Adams operations and the connective Witt-theoretic J-spectrum

摘录 / EXCERPT: Conjecture 2 ([6, Conjecture 4.10]). The Adams operations act on $\pi_\star kw \cong W(k)[\beta]$, with $|\beta|=4+4\alpha$, via $\psi_g(\beta^k)=g^{2k}\beta^k$, and, if k has fi...

open_problem | open | MathDB 页面 | 原文线索

V2-079144 | MATHDB-402277

Mahowald’s conjecture on the Hurewicz image in $[U+007F]\mathrm{tmf}$

摘录 / EXCERPT: Let $h:\pi_*(S)\to\pi_*(\mathrm{tmf})$ be the Hurewicz homomorphism. The report states that, when restricted to the cokernel of J, its image is contained in the part...

conjecture | open | MathDB 页面 | 原文线索

V2-079145 | MATHDB-402278

Free complementation of maximal amenable subfactors of $L(F_n)$

摘录 / EXCERPT: Conjecture: Every maximal amenable subfactor $R\subset L(F_n)$ is freely complemented; that is, $L(F_n)$ can be written as the free product of R and a complementary von Neumann a...

open_problem | open | MathDB 页面 | 原文线索

V2-079146 | MATHDB-402280

Factorizable channels requiring infinite-dimensional ancillas

摘录 / EXCERPT: For every integer $n\ge 11$, there exist factorizable Schur multipliers on $M_n(\mathbb C)$ that admit no finite-dimensional ancilla; indeed, they admit no type-I ancilla.

open_problem | open | MathDB 页面 | 原文线索

V2-079147 | MATHDB-402282

Operator-space approximation property in the remaining L^p range

摘录 / EXCERPT: For a group Γ admitting an isometric action on a locally finite $\widetilde A_2$-building with finitely many orbits of vertices and finite stabilizers, determine whether the...

open_problem | open | MathDB 页面 | 原文线索

V2-079148 | MATHDB-402285

Unique C*-norms for torsion-free amenable group rings

摘录 / EXCERPT: Does there exist a torsion-free amenable group Γ whose complex group ring $\mathbb{C}\Gamma$ has a unique C^*-norm?

open_problem | open | MathDB 页面 | 原文线索

V2-079149 | MATHDB-402289

Characterization of spaces admitting minimal homeomorphisms

摘录 / EXCERPT: Give a characterization of the infinite compact metric spaces that admit a minimal homeomorphism.

open_problem | open | MathDB 页面 | 原文线索

V2-079150 | MATHDB-402294

Nonlinear inviscid damping and axisymmetrization near general vortices

摘录 / EXCERPT: Another goal is to prove nonlinear inviscid damping and axi-symmetrization near general vortices.

open_problem | open | MathDB 页面 | 原文线索

V2-079151 | MATHDB-402296

Vanishing-\[?] limit for Euler-\[?] flows with Dirichlet boundary conditions

摘录 / EXCERPT: Study the limit $\alpha\to0$ for solutions of the Euler-α equations on the half-plane $\Omega=H$ with Dirichlet boundary condition $u=0$ on ∂H, for vortex-sheet in...

open_problem | open | MathDB 页面 | 原文线索

V2-079152 | MATHDB-402297

Vanishing-\b1 limit of Euler-\b1 equations with Dirichlet boundary conditions

摘录 / EXCERPT: Determine the limit as $\alpha \to 0$ for solutions of the Euler-α equations with Dirichlet boundary condition $u=0$ on the boundary, including the effect of the boundary la...

open_problem | open | MathDB 页面 | 原文线索

V2-079153 | MATHDB-402298

Generalization of the periodic-solution approach to exterior domains

摘录 / EXCERPT: The report states that it is an open problem how the approach used for time-periodic solutions in bounded domains with periodically moving boundaries can be generalized to exterior...

open_problem | open | MathDB 页面 | 原文线索

V2-079154 | MATHDB-402299

Michelson’s boundedness conjecture for the Kuramoto-Sivashinsky equation

摘录 / EXCERPT: For an L-periodic solution u of the Kuramoto-Sivashinsky equation ... with $\int_0^L u\,dx=0$, Michelson’s conjecture is that u remains bounded at long times and large scales,...

conjecture | open | MathDB 页面 | 原文线索

V2-079155 | MATHDB-402301

The g-conjecture for triangulated spheres

摘录 / EXCERPT: Every triangulated sphere has the same f-vector as some simplicial polytope; equivalently, the characterization of f-vectors given by the g-theorem for simplicial polytopes ext...

conjecture | open | MathDB 页面 | 原文线索

V2-079156 | MATHDB-402305

Upper bound theorem for centrally symmetric simplicial spheres

摘录 / EXCERPT: For every $d\ge 4$ and $n\ge d$, there exists a centrally symmetric simplicial $(d-1)$-sphere with $2n$ vertices that is cs-$\lfloor d/2\rfloor$-neighborly.

open_problem | open | MathDB 页面 | 原文线索

V2-079157 | MATHDB-402308

Normal equivalence for hypersecondary polytopes

摘录 / EXCERPT: Let $A\subseteq\mathbb{R}^d$ be a configuration of size n and let $k\in[d+1]$. Set $s=\max(n-k+1,d+2)$. Then the hypersecondary polytope $\mathcal{F}^{(k)}(A)$ is normally equiva...

open_problem | open | MathDB 页面 | 原文线索

V2-079158 | MATHDB-402309

Regular CW-complex structure of fibers of maps to totally nonnegative spaces

摘录 / EXCERPT: For each fiber $f^{-1}_{(i_1,\dots,i_d)}(M)$, the cell decomposition described in the report is a regular CW complex homeomorphic to a closed ball.

open_problem | open | MathDB 页面 | 原文线索

V2-079159 | MATHDB-402310

Explicitly defined families of non-collapsible hypertrees

摘录 / EXCERPT: Find explicitly defined families of non-collapsible hypertrees.

open_problem | open | MathDB 页面 | 原文线索

V2-079160 | MATHDB-402312

Clasp Conjecture for local intersection forms

摘录 / EXCERPT: Let $V=\mathbb{C}^n$ be the standard representation, and let $\Lambda^k V$ be a fundamental representation with $0<k<n$. Let μ be a weight appearing in the weight decomposition...

conjecture | open | MathDB 页面 | 原文线索

V2-079161 | MATHDB-402314

Convex hull of f-vectors of polytopes

摘录 / EXCERPT: For dimensions $d\ge 4$, is $\operatorname{Conv}(f(\mathcal P^d))=\operatorname{Cone}_{f(\sigma_d)}(f(\mathcal P^d))$?

open_problem | open | MathDB 页面 | 原文线索

V2-079162 | MATHDB-402315

Decidability of recognizing supporting hyperplanes and interior rays of the f-vector cone

摘录 / EXCERPT: Fix $d\ge 4$. Is the following problem decidable? Given a hyperplane H through $f(\sigma_d)$, does it support the cone $\operatorname{Cone}_{f(\sigma_d)}(f(\mathcal P^d))$, or co...

open_problem | open | MathDB 页面 | 原文线索

V2-079163 | MATHDB-402316

Polynomial-time recognition of f-vectors of general polytopes

摘录 / EXCERPT: For fixed dimension d, given $v=(v_1,\ldots,v_d)\in\mathbb Z_{\ge 0}^d$, can one decide in time polynomial in $N(v)$ whether $v\in f(\mathcal P^d)$, where $\mathcal P^d$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-079164 | MATHDB-402317

Eventual unimodular triangulations of lattice polytopes

摘录 / EXCERPT: For every lattice polytope P, there exists an integer $c(P)$ such that, for every integer $c\ge c(P)$, the polytope cP admits a unimodular triangulation; here, a unimodular tri...

open_problem | open | MathDB 页面 | 原文线索

V2-079165 | MATHDB-402320

Undecidability of simplicial-complex embeddability outside the metastable range

摘录 / EXCERPT: For fixed positive integers $k\le d$, let $\mathrm{EMBED}_{k\to d}$ denote the decision problem that, given a finite simplicial complex K of dimension k, asks whether K admit...

open_problem | open | MathDB 页面 | 原文线索

V2-079166 | MATHDB-402323

Lattice polytope generators: determining the spanning rank

摘录 / EXCERPT: For a finite set $A\subset\mathbb Z^d$, define ... For a d-dimensional lattice polytope P with $\operatorname{aff}(P\cap\mathbb Z^d)=\mathbb Z^d$, define ... and ... Given the known...

open_problem | open | MathDB 页面 | 原文线索

V2-079167 | MATHDB-402327

Joint inverse scattering with unknown internal point sources

摘录 / EXCERPT: Recover the susceptibility η together with the amplitudes $a1,\ldots,a_N$, phases $\phi1,\ldots,\phiN$, and positions $y1,\ldots,y_N$ of the internal point sources from sca...

open_problem | open | MathDB 页面 | 原文线索

V2-079168 | MATHDB-402329

Inversion of weighted generalized Radon transforms in Compton scattering tomography

摘录 / EXCERPT: Given noisy 2D/3D Compton-scattering tomography data modeled by ... develop an inversion or feature-extraction/reconstruction method for the physically weighted transform, including...

open_problem | open | MathDB 页面 | 原文线索

V2-079169 | MATHDB-402330

Uniqueness for the idealized MRXI inverse problem

摘录 / EXCERPT: For a full-field activation at any $y\in O\subset\mathbb{R}^3\setminus\Omega$ with two linearly independent orientations η_1 and η_2, the inverse problem has a unique sol...

open_problem | open | MathDB 页面 | 原文线索

V2-079170 | MATHDB-402331

Recovering a radial wave-speed profile from single-point measurements

摘录 / EXCERPT: Assuming a spherically symmetric planet whose radial wave-speed profile satisfies the Herglotz condition, find the wave-speed profile from ideal measurements made at a single point...

open_problem | open | MathDB 页面 | 原文线索

V2-079171 | MATHDB-402333

Sharpness of the regularity exponent in Theorem 3

摘录 / EXCERPT: The report asks: “We do not know if this result is sharp for $s\ge \frac34$,” referring to Theorem 3, which states that if $g\in L^2(\Omega)$ and $\Phi_g\in H_0^1(\Omega)$ solves...

open_problem | open | MathDB 页面 | 原文线索

V2-079172 | MATHDB-402334

Near-best affine recovery from the weak-greedy reduced space

摘录 / EXCERPT: Given the weak-greedy spaces U_n^{wg} and ... is U^* associated with a near-best affine recovery map of the form ... where the ideal map is defined by ...

open_problem | open | MathDB 页面 | 原文线索

V2-079173 | MATHDB-402335

Velocity-robust pressure error estimates for Stokes discretizations

摘录 / EXCERPT: Find pairs V_h/Q_h and computationally feasible forms a_h, b_h, and E_h such that the discretization (3) satisfies a pressure error estimate of the form ... for some constant...

open_problem | open | MathDB 页面 | 原文线索

V2-079174 | MATHDB-402337

Second-order L^2 convergence of discrete exterior calculus

摘录 / EXCERPT: Prove that DEC converges in L^2 with the rate $O(h^2)$.

open_problem | open | MathDB 页面 | 原文线索

V2-079175 | MATHDB-402338

Kac’s spectral-gap conjecture for the hard-sphere gas

摘录 / EXCERPT: Let $S_{N,E,p}$ be the space of velocity vectors satisfying the total-energy and total-momentum constraints, equipped with the uniform normalized measure $\d70e_N$, and let the har...

conjecture | open | MathDB 页面 | 原文线索

V2-079176 | MATHDB-402340

A gapped family of two-dimensional AKLT models

摘录 / EXCERPT: Let $\Gamma^{(n)}$ be the hexagonal lattice obtained by replacing each edge by a chain of n particles, with each site x carrying spin $s_x=\deg(x)/2$, and let H_Λ be th...

open_problem | open | MathDB 页面 | 原文线索

V2-079177 | MATHDB-402341

Spectral gap for the spin-[U+007F] AKLT model on the hexagonal lattice

摘录 / EXCERPT: Does the spin-[U+007F] AKLT model on the hexagonal lattice have a spectral gap?

open_problem | open | MathDB 页面 | 原文线索

V2-079178 | MATHDB-402344

Binding inequalities for the fermionic nonlinear Schrödinger minimisation problem

摘录 / EXCERPT: For every $d\ge 1$, $1<p<1+\frac d2$, and integer $N\ge 2$, is it true that ... where ...

open_problem | open | MathDB 页面 | 原文线索

V2-079179 | MATHDB-402346

Eigenvalues and eigenvectors of critical Erdős-Rényi graphs

摘录 / EXCERPT: For the Erdős-Rényi graph $G(N,d/N)$, let A be the adjacency matrix and let $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_N$ be the eigenvalues of the normalized matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-079180 | MATHDB-402348

Spectral-gap lower bound for the N-particle piston model

摘录 / EXCERPT: Establish an appropriate lower bound for the spectral gap of the N-particle piston model, expected to be of order cN^{-2} with $c>0$.

open_problem | open | MathDB 页面 | 原文线索

V2-079181 | MATHDB-402349

Invariance principle for the random Lorentz gas in the diffusive limit

摘录 / EXCERPT: For the random Lorentz gas with scatterers B_r^d+P, where P is a Poisson point process of intensity ρ, prove that, in both the quenched and averaged-quenched settings, the...

open_problem | open | MathDB 页面 | 原文线索

V2-079182 | MATHDB-402350

Central limit theorem for Gibbs walks in random environment

摘录 / EXCERPT: Prove a central limit theorem for Gibbs walks in random environment.

open_problem | open | MathDB 页面 | 原文线索

V2-079183 | MATHDB-402352

Decay time of local correlations in a kinetically constrained model with random constraints

摘录 / EXCERPT: Understand for which time t the local correlation at the origin becomes small, namely when ...

open_problem | open | MathDB 页面 | 原文线索

V2-079184 | MATHDB-402353

Conjectured critical exponent for two-dimensional critical KCM

摘录 / EXCERPT: For critical update families in two dimensions, the conjecture of [8] was that both the relaxation time T_{rel} and the mean infection time $\mathbb{E}(\tau_0)$ scale, a...

conjecture | open | MathDB 页面 | 原文线索

V2-079185 | MATHDB-402355

Lines on a general $(3,3)$ divisor in the Segre-embedded product of planes

摘录 / EXCERPT: Let $X\subset \mathbb{P}^{2}\times\mathbb{P}^{2}$ be a general divisor of bidegree $(3,3)$, viewed in $\mathbb{P}^{8}$ via the Segre embedding. How many lines are contained in X?...

open_problem | open | MathDB 页面 | 原文线索

V2-079186 | MATHDB-402358

Roots of Ehrhart polynomials of symmetric edge polytopes of complete bipartite graphs

摘录 / EXCERPT: All roots of the Ehrhart polynomial of any symmetric edge polytope $P_{K_{a,b}}$ have real part equal to $-\frac{1}{2}$.

open_problem | open | MathDB 页面 | 原文线索

V2-079187 | MATHDB-402360

Mori Dream Space property of a blow-up of \(\mathbb{P}^3\) at six points and fifteen lines

摘录 / EXCERPT: Let X be the blow-up of $\mathbb{P}^3$ over $\mathbb{C}$ at six points in very general position and at all fifteen lines passing through pairs of these points. Is X a Mori Drea...

open_problem | open | MathDB 页面 | 原文线索

V2-079188 | MATHDB-402361

Configurations of lines yielding rational-polyhedral effective cones

摘录 / EXCERPT: Consider the blow-up of $\mathbb{P}^n$ ($n\geq 3$) at $(n+3)$ points in general position and certain lines through the $(n+3)$ points. For what configurations of the lines is the e...

open_problem | open | MathDB 页面 | 原文线索

V2-079189 | MATHDB-402362

Generation by linear maps and the standard Cremona involution

摘录 / EXCERPT: Is every $\sigma\in\operatorname{Bir}(\mathbb P^n)$ which only contracts rational hypersurfaces in G_n?

open_problem | open | MathDB 页面 | 原文线索

V2-079190 | MATHDB-402363

Unicity of types for essentially tame supercuspidal representations

摘录 / EXCERPT: Let $G=G(F)$ be a connected reductive p-adic group. Let $[U+007F]pi$ be an essentially tame supercuspidal representation of G, let K be a maximal compact subgroup of G, and suppos...

open_problem | open | MathDB 页面 | 原文线索

V2-079191 | MATHDB-402366

Local Langlands in families in depth zero

摘录 / EXCERPT: Assume that R is a subring of $ \mathbb{Q}_\ell$, and let $e_0\in Z_R(G)$ be the idempotent corresponding to the depth-zero subcategory. There exists a unique morphism ... compatib...

open_problem | open | MathDB 页面 | 原文线索

V2-079192 | MATHDB-402367

Equivalence between depth-zero and parameter blocks

摘录 / EXCERPT: Assuming that all centralizers $C_{\widehat G}(\phi)$ for $\phi\in\Phi$ are connected, prove the expected equivalence of categories ... for each $\phi\in\Phi$ and each character...

open_problem | open | MathDB 页面 | 原文线索

V2-079193 | MATHDB-402368

Guo-Jacquet conjecture for quaternionic inner forms of general linear groups

摘录 / EXCERPT: Let E/F be a quadratic extension of number fields, let $D\in X(E:F)$, and let π be an irreducible cuspidal automorphic representation of $G_D(\mathbb A_F)$. Let π' be its...

conjecture | open | MathDB 页面 | 原文线索

V2-079194 | MATHDB-402370

Coincidence of Kaletha’s and Arthur’s local Langlands correspondences for regular supercuspidal representations of quasi-split classical groups

摘录 / EXCERPT: Let G be a quasi-split classical group over a p-adic field F, and let $\phi\in\Phi(G)$. Prove that the Arthur and Kaletha L-packets coincide: ... Equivalently, prove the twis...

open_problem | open | MathDB 页面 | 原文线索

V2-079195 | MATHDB-402371

Symmetry characterizes Carayol-type representations

摘录 / EXCERPT: Let $\tau\in\widehat{W}^{\mathrm{wr}}_F$, write $\dim\tau=p^r$ and $m=\operatorname{sw}(\tau)$, and suppose that $m\leq \operatorname{sw}(\chi\otimes\tau)$ for every character...

open_problem | open | MathDB 页面 | 原文线索

V2-079196 | MATHDB-402372

Geometrization of the local Langlands correspondence

摘录 / EXCERPT: Let F be a p-adic number field with residue field F_q, and let G be a quasi-split reductive group over F. For a discrete Langlands parameter $\varphi: W_F \to {}^LG$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-079197 | MATHDB-402374

Multiplicity formula for supercuspidal representations of spherical pairs

摘录 / EXCERPT: For a general spherical pair (G,H) over a local field F and a character $\chi:H(F)\to\mathbb{C}^{\times}$, the multiplicity formula ... should hold for every supercuspidal irredu...

open_problem | open | MathDB 页面 | 原文线索

V2-079198 | MATHDB-402379

Schwartz-space characterization for the basic affine space

摘录 / EXCERPT: Conjecture: The space $S(X)$ coincides with ...

open_problem | open | MathDB 页面 | 原文线索

V2-079199 | MATHDB-402382

Hales-Moy-Prasad conjecture on the domain of the local character expansion

摘录 / EXCERPT: Let π be an irreducible admissible representation of a reductive p-adic group G, with depth r, and let $\mathfrak g=\operatorname{Lie}(G)$. Define ... where $B(G)$ is the (...

conjecture | open | MathDB 页面 | 原文线索

V2-079200 | MATHDB-402383

Relative local character expansion on the Moy-Prasad neighborhood

摘录 / EXCERPT: Let F be a nonarchimedean field of characteristic zero and sufficiently large residual characteristic. Let G be a connected reductive F-group, let σ be an F-involuti...

open_problem | open | MathDB 页面 | 原文线索

V2-079201 | MATHDB-402384

Relative local character-expansion homogeneity conjecture

摘录 / EXCERPT: For all $r\in\mathbb{R}$, the restriction identity $(\mathrm{RLCE}^{\wedge})$ holds: ...

conjecture | open | MathDB 页面 | 原文线索

V2-079202 | MATHDB-402385

Relative local character expansion on Moy-Prasad neighborhoods

摘录 / EXCERPT: Let G be a connected reductive group over a nonarchimedean local field F of characteristic zero, let σ be an F-involution of G, and let H lie between G^σ an...

open_problem | open | MathDB 页面 | 原文线索

V2-079203 | MATHDB-402386

A basis for invariant distributions supported on the nilpotent cone

摘录 / EXCERPT: Find a basis for $J(N_V)$ in the higher-rank symmetric-space case.

open_problem | open | MathDB 页面 | 原文线索

V2-079204 | MATHDB-402399

Stanley-Stembridge e-positivity conjecture for Hessenberg incomparability graphs

摘录 / EXCERPT: Let $h=(h(1),h(2),\ldots,h(n))$ be a Hessenberg function, meaning an increasing sequence satisfying $h(i)\ge i$ for all $i\in[n]=\{1,2,\ldots,n\}$. Define the graph Γ_h with...

conjecture | open | MathDB 页面 | 原文线索

V2-079205 | MATHDB-402401

General validity of the weighted-Hasse-graph description of Γ_C

摘录 / EXCERPT: In the setting of the preceding construction, the conjectured description of the semigroup Γ_C-which uses only data read off from the weighted Hasse graph-is valid in genera...

open_problem | open | MathDB 页面 | 原文线索

V2-079206 | MATHDB-402403

Essential self-adjointness for a family of almost-Riemannian structures

摘录 / EXCERPT: For the almost-Riemannian structure on $\mathbb{R}^2$ defined by ... with singular region $S=\{x=0\}$, regular region $R=\mathbb{R}^2\setminus S$, and Riemannian measure ... is the Lap...

open_problem | open | MathDB 页面 | 原文线索

V2-079207 | MATHDB-402406

Essential self-adjointness of the pointed Heisenberg sub-Laplacian

摘录 / EXCERPT: Determine whether the pointed Heisenberg sub-Laplacian $\mathring{\Delta}_H=X_1^2+X_2^2$, with domain $C_c^\infty(\mathbb{R}^3\setminus\{0\})$ and acting in $L^2(\mathbb{H}^1)$ wit...

open_problem | open | MathDB 页面 | 原文线索

V2-079208 | MATHDB-402408

Spectral description for non-compact self-similar weights

摘录 / EXCERPT: Let P be an n-term self-similar function of zero spectral order. Then the spectrum of problem (1), (2), with a non-compact multiplier P' as a weight, consists of $n-1$ segmen...

open_problem | open | MathDB 页面 | 原文线索

V2-079209 | MATHDB-402411

Subfactor-conformal field theory correspondence

摘录 / EXCERPT: Determine whether, corresponding to every subfactor, there is a counterpart conformal field theory (CFT).

open_problem | open | MathDB 页面 | 原文线索

V2-079210 | MATHDB-402412

Nonexistence of a unitary R-matrix with specified two-eigenvalue data

摘录 / EXCERPT: Conjecture: There is no unitary R-matrix of any finite dimension whose spectrum is $\{-1,e^{i\pi/3}\}$ and for which the spectral projection P corresponding to the eigenvalue...

open_problem | open | MathDB 页面 | 原文线索

V2-079211 | MATHDB-402418

Observables in scaling limits of unitary minimal models

摘录 / EXCERPT: Conjecture. For every unitary minimal model $M(k+2,k+1)$ with $k\ge 3$, the observables in the corresponding scaling limit contain subsets corresponding to smeared-field observable...

open_problem | open | MathDB 页面 | 原文线索

V2-079212 | MATHDB-402420

Classification of Rokhlin actions of a C*-fusion category

摘录 / EXCERPT: Let $\mathcal C$ be a C-fusion category. Let $\alpha,\beta:\mathcal C\to\operatorname{Bimod}(A)$ be two actions of $\mathcal C$ with the Rokhlin property. Assume that $\alpha(X)$ a...

open_problem | open | MathDB 页面 | 原文线索

V2-079213 | MATHDB-402422

Existence and realization of the proposed infinite families of quadratic categories

摘录 / EXCERPT: Do the corresponding infinite families of quadratic categories exist for the five infinite families of potential modular data described in [6], so that the conjectured modular data...

open_problem | open | MathDB 页面 | 原文线索

V2-079214 | MATHDB-402426

Orbifold TQFT as the Reshetikhin-Turaev theory of the Wilson-line category

摘录 / EXCERPT: Given a modular fusion category C and an orbifold datum $A=(A,T,\alpha,\bar\alpha,\psi,\phi)$, let $Z^{\mathrm{orb},A}_C$ be the generalized orbifold of the Reshetikhin-Turaev TQ...

open_problem | open | MathDB 页面 | 原文线索

V2-079215 | MATHDB-402427

Does von Neumann equivalence imply measure equivalence for ICC groups?

摘录 / EXCERPT: For countable ICC groups Γ and Λ, does $L\Gamma \cong L\Lambda$ imply that Γ and Λ are measure equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-079216 | MATHDB-402435

Well-posedness of the Benjamin-Ono equation below L^2

摘录 / EXCERPT: For the Benjamin-Ono equation ... with real-valued data on $\mathbb R$, determine whether the equation is well-posed in $H^s(\mathbb R)$ for every $-\tfrac12<s<0$.

open_problem | open | MathDB 页面 | 原文线索

V2-079217 | MATHDB-402436

Almost-global dispersive decay for small localized Benjamin-Ono data

摘录 / EXCERPT: For the Benjamin-Ono equation ... with real-valued $\phi:\mathbb R\times\mathbb R\to\mathbb R$ and Hilbert-transform convention $H(\xi)=-i\operatorname{sgn}\xi$, assume ... Then the so...

open_problem | open | MathDB 页面 | 原文线索

V2-079218 | MATHDB-402437

Earliest emergence of solitons for small localized Benjamin-Ono data

摘录 / EXCERPT: For small localized initial data satisfying $\|\phi_0\|_{L^2}+\|x\phi_0\|_{L^2}\leq\varepsilon\ll1$, determine the earliest time at which solitons can emerge in the Benjamin-Ono ev...

open_problem | open | MathDB 页面 | 原文线索

V2-079219 | MATHDB-402439

Randomized Cauchy problem for mass-subcritical nonlinear Schrödinger equations

摘录 / EXCERPT: What happens if one considers the same randomized data at $t=0$ instead of $t=\infty$, namely the randomized Cauchy problem ...

open_problem | open | MathDB 页面 | 原文线索

V2-079220 | MATHDB-402441

Local wellposedness of nonlinear Maxwell equations with perfectly conducting boundary conditions

摘录 / EXCERPT: Theorem 1 ([4]). Fix $m\in\mathbb N$ with $m\ge 3$. Let $G\subseteq\mathbb R^3$ be a domain with compact C^{m+2}-boundary. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-079221 | MATHDB-402442

Quartic time scale for linear dispersive decay of small-data KdV solutions

摘录 / EXCERPT: The report asks: if the initial data size is $\varepsilon\ll1$, up to what time scale do nonlinear KdV solutions satisfy the linear dispersive decay bounds? It answers that the cor...

open_problem | open | MathDB 页面 | 原文线索

V2-079222 | MATHDB-402443

Counting number fields by the shortest generating vector

摘录 / EXCERPT: Let $n>1$. The number of isomorphism classes of number fields K of degree n whose shortest vector in $ \mathcal O_K$ generating K has length less than Y is...

open_problem | open | MathDB 页面 | 原文线索

V2-079223 | MATHDB-402444

Smallest possible largest gap under Poisson pair correlation

摘录 / EXCERPT: What is the smallest value of $\delta_{\max}$ for a sequence of positive real numbers with mean spacing one and Poisson pair correlation, where...

open_problem | open | MathDB 页面 | 原文线索

V2-079224 | MATHDB-402445

Poonen-Voloch conjecture for random Fano hypersurfaces

摘录 / EXCERPT: For every fixed pair of integers $d,n\ge 2$ with $n\ge d$ and $(d,n)\ne(2,2)$, let $\mathbb V_{d,n}(A)$ be the degree-d hypersurfaces in $\mathbb P^n$ defined over $\mathbb Q$ an...

conjecture | open | MathDB 页面 | 原文线索

V2-079225 | MATHDB-402446

Poonen-Voloch prediction for rational points on random Fano hypersurfaces

摘录 / EXCERPT: Fix integers d,n with $n\ge d\ge 3$. Let $N_{d,n}=\binom{n+d}{n}$, let $\mathcal V_{d,n}=\mathbb P^{N_{d,n}-1}(\mathbb Q)$ parametrize degree-d hypersurfaces...

open_problem | open | MathDB 页面 | 原文线索

V2-079226 | MATHDB-402447

Schur-type theorem in the squares

摘录 / EXCERPT: In any finite colouring of the positive integers, there exists a monochromatic solution to ...

open_problem | open | MathDB 页面 | 原文线索

V2-079227 | MATHDB-402449

Optimal exponent in the Erdős-Hooley divisor-concentration bound

摘录 / EXCERPT: The report conjectures that the exponent η in the almost-sure lower bound $\Delta(n)\gg (\log\log n)^{\eta+o(1)}$ is optimal, meaning that this exponent cannot be increased. H...

open_problem | open | MathDB 页面 | 原文线索

V2-079228 | MATHDB-402450

Optimization problem for entropy on flags in the Boolean cube

摘录 / EXCERPT: For each $k\in\mathbb{N}$, determine ... where the supremum is over all choices of an integer $r\ge 1$, an r-step flag ... whose members are distinct, each of which is spanned by ele...

open_problem | open | MathDB 页面 | 原文线索

V2-079229 | MATHDB-402451

Equality of the three exponents in the random-set optimization problem

摘录 / EXCERPT: Is it true that the three quantities introduced in the report satisfy $\widetilde{\gamma}_k=\beta_k=\gamma_k$?

open_problem | open | MathDB 页面 | 原文线索

V2-079230 | MATHDB-402452

Sieve of Ekedahl over quadrics

摘录 / EXCERPT: Let $Q(X_1,\ldots,X_n)$, $F(X_1,\ldots,X_{n+1})$, and $G(X_1,\ldots,X_n)$ be homogeneous polynomials defined over $\mathbb Z$, where Q is a quadratic form. Suppose that: (i) the...

open_problem | open | MathDB 页面 | 原文线索

V2-079231 | MATHDB-402454

Duffin-Schaeffer conjecture

摘录 / EXCERPT: Let $\psi:\mathbb N\to\mathbb R_{\geq 0}$ be arbitrary, and define ... Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-079232 | MATHDB-402455

Optimality of the \a0\cclya\\adVinogradov-scale bound for small families

摘录 / EXCERPT: For a family $F=\{\varphi_a\}_{a\in\Omega_m}$ satisfying the assumptions of Theorem 2, the report says that the bound $M(\varphi_a)\gg\sqrt m\log\log m$ is conjectured to be optima...

open_problem | open | MathDB 页面 | 原文线索

V2-079233 | MATHDB-402456

Asymptotic count of E_2 almost primes in short intervals

摘录 / EXCERPT: Let E_2 denote the set of integers of the form p_1p_2, where p_1,p_2 are primes. For every fixed $\theta>0.55$, for sufficiently large x and $H\\ge x^\theta$, one has ...

open_problem | open | MathDB 页面 | 原文线索

V2-079234 | MATHDB-402459

Quantitative bounds in the polynomial Szemerédi theorem

摘录 / EXCERPT: Given polynomials $P_1,\\ldots,P_m\in\mathbb{Z}[y]$ with $P_i(0)=0$, prove quantitative upper bounds for the largest subset $A\subseteq[N]$ containing no nontrivial configurations...

open_problem | open | MathDB 页面 | 原文线索

V2-079235 | MATHDB-402461

Rudnick’s conjectured variance asymptotic for squarefree integers in short intervals

摘录 / EXCERPT: For a fixed $\delta\in(0,1/100)$, as $X\to\infty$ with $H=H(X)\to\infty$ and $H\le X^{1-\delta}$, the asymptotic from Theorem 1 is conjectured to remain valid: ...

conjecture | open | MathDB 页面 | 原文线索

V2-079236 | MATHDB-402462

Keating-Rudnick conjectured wider ranges for squarefree variance estimates

摘录 / EXCERPT: J. P. Keating and Z. Rudnick conjectured that Theorems 1 and 2 are true in the wider ranges $H\le X^{1-\delta}$ and $X^\delta\le q=o(X)$, respectively.

conjecture | open | MathDB 页面 | 原文线索

V2-079237 | MATHDB-402464

Chowla’s k-point correlation conjecture over the integers

摘录 / EXCERPT: For any fixed distinct integers $h_1,\ldots,h_k$, one has ...

conjecture | open | MathDB 页面 | 原文线索

V2-079238 | MATHDB-402465

Theorem on coefficients of an exponential Gaussian random power series

摘录 / EXCERPT: Let $X(n)$, for $n\in\mathbb N$, be independent standard complex Gaussian random variables, meaning that the real and imaginary parts are independent real Gaussians with mean 0 a...

open_problem | open | MathDB 页面 | 原文线索

V2-079239 | MATHDB-402467

Injectivity of the structure-monoid 1-cocycle

摘录 / EXCERPT: Given a set-theoretic solution (X,r) of the Yang-Baxter equation and the associated 1-cocycle $\pi:M(X,r)\to A(X,r)$ described in the report, determine precisely when π is in...

open_problem | open | MathDB 页面 | 原文线索

V2-079240 | MATHDB-402470

Characterization of structure groups of racks via a split central extension

摘录 / EXCERPT: Does the splitting of the above sequence coupled with some extra condition characterise the structure groups of racks?

open_problem | open | MathDB 页面 | 原文线索

V2-079241 | MATHDB-402471

Unique product property of structure groups of finite involutive solutions

摘录 / EXCERPT: Let (X,r) be a finite involutive non-degenerate solution to the Yang-Baxter equation. When does $G(X,r)$ have the unique product property?

open_problem | open | MathDB 页面 | 原文线索

V2-079242 | MATHDB-402472

Finite nilpotent multiplicative groups of two-sided skew braces of abelian type

摘录 / EXCERPT: Which finite nilpotent groups are multiplicative groups of (two-sided) skew braces of abelian type?

open_problem | open | MathDB 页面 | 原文线索

V2-079243 | MATHDB-402473

Minimal size of a finite solvable non-IYB group

摘录 / EXCERPT: Which is the minimal size of a finite solvable group that is a non-IYB-group?

open_problem | open | MathDB 页面 | 原文线索

V2-079244 | MATHDB-402476

Enumeration of skew braces of squarefree order

摘录 / EXCERPT: Let n be squarefree. Write two groups of order n as ... where $de=n$ and k has order d in $\mathbb Z_e^{\times}$. Put $e=gz$, where $z=\gcd(k-1,e)$, and let γ be the c...

open_problem | open | MathDB 页面 | 原文线索

V2-079245 | MATHDB-402485

Isomorphism of Tits webs and Moufang sets

摘录 / EXCERPT: Two Tits webs are isomorphic if and only if the corresponding Moufang sets are isomorphic.

open_problem | open | MathDB 页面 | 原文线索

V2-079246 | MATHDB-402488

Finite order of the symbolic defect function

摘录 / EXCERPT: Which (homogeneous) ideals I of R have finite $\operatorname{ord}(\operatorname{sdef} I)$?

open_problem | open | MathDB 页面 | 原文线索

V2-079247 | MATHDB-402489

Unbounded global-generation exponents for ample line bundles on a surface

摘录 / EXCERPT: Does there exist a smooth complex projective surface S and a sequence of ample line bundles (L_n) on S such that, setting ... one has $\nu(L_n)\to\infty$ as $n\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-079248 | MATHDB-402492

Universal lower bound for Seshadri constants

摘录 / EXCERPT: For ample line bundles, the lower bound $\varepsilon(L;x)\ge 1$ always holds.

open_problem | open | MathDB 页面 | 原文线索

V2-079249 | MATHDB-402495

Rationality of Seshadri constants on products of general curves under coefficient conditions

摘录 / EXCERPT: Let $X=C\times C$, where C is a general member of the moduli of smooth curves of genus $g\ge 2$. Let F_1 and F_2 be fibers of the two projections and let δ be the diag...

open_problem | open | MathDB 页面 | 原文线索

V2-079250 | MATHDB-402496

Szemberg’s Pell-equation lower-bound conjecture for Seshadri constants

摘录 / EXCERPT: Conjecture (Szemberg). Let X be a smooth projective surface with $\rho(X)=1$, and let L be the ample generator of the Picard group, with $d=L^2$ not a perfect square. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-079251 | MATHDB-402497

Pokora’s conjecture on multi-point Seshadri constants of line arrangements

摘录 / EXCERPT: Conjecture 1 (Pokora). Let Z be the set of all singular points of an arrangement of lines. Then the multi-point Seshadri constant satisfies ... where $\operatorname{mpl}(Z)$ is the...

conjecture | open | MathDB 页面 | 原文线索

V2-079252 | MATHDB-402498

Strengthened SHGH conjecture for curves with a prescribed multiple point

摘录 / EXCERPT: Let X be the blow-up of $\mathbb{P}^2$ at $r\ge 0$ very general points. Suppose $d\ge 1$, $t\ge 1$, and $m_1,\ldots,m_r\ge 0$ are integers satisfying ... Then every curve...

conjecture | open | MathDB 页面 | 原文线索

V2-079253 | MATHDB-402499

Rationality of global Seshadri constants

摘录 / EXCERPT: For every smooth complex projective surface X and every ample line bundle L on X, is the global Seshadri constant ... a rational number?

open_problem | open | MathDB 页面 | 原文线索

V2-079254 | MATHDB-402502

Nagata Conjecture for General Points in \(\mathbb{P}^2\)

摘录 / EXCERPT: For a set Z of s general points in $\mathbb{P}^2$ over $\mathbb{C}$, with $s\ge 9$, if $I=I(Z)$ is its homogeneous ideal, then the Waldschmidt constant satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-079255 | MATHDB-402504

Bounded negativity for complex line arrangements

摘录 / EXCERPT: If C is a complex line arrangement, must we have $H(C) \ge H(C_W) = -\frac{225}{67} \approx -3.36$, where C_W is Wiman’s arrangement of 45 lines? Note that C_W has 120 triple...

open_problem | open | MathDB 页面 | 原文线索

V2-079256 | MATHDB-402505

Characterization of point sets with failure of symbolic-power containment

摘录 / EXCERPT: Let $Z\subset \mathbb P^2$ be a finite set, and let $I_Z^{(m)}=\bigcap_{p\in Z}I_p^m$ denote the mth symbolic power of I_Z. Is it possible to characterize those Z for which...

open_problem | open | MathDB 页面 | 原文线索

V2-079257 | MATHDB-402506

Bounded negativity for reduced singular plane curves

摘录 / EXCERPT: Let C be a reduced singular plane curve. How negative can ... be?

open_problem | open | MathDB 页面 | 原文线索

V2-079258 | MATHDB-402507

Bounded negativity for rational line arrangements

摘录 / EXCERPT: What is $\inf_C H(C)$ for rational line arrangements C, where $H(C)=D_C^2/|Z_C|$, $Z_C=\operatorname{Sing}(C)$, and D_C is the proper transform of C on the blow-up of...

open_problem | open | MathDB 页面 | 原文线索

V2-079259 | MATHDB-402509

Bounded negativity for irreducible plane curves

摘录 / EXCERPT: Let $C\subset \mathbb{P}^2$ be a reduced singular plane curve, let $Z_C=\operatorname{Sing}(C)$, and let D_C be the proper transform of C on the blow-up of $\mathbb{P}^2$ at...

open_problem | open | MathDB 页面 | 原文线索

V2-079260 | MATHDB-402512

Bounded negativity for blow-ups at Fermat-type root-of-unity points

摘录 / EXCERPT: For a positive integer m, let $Z(m)$ be the set of all points of the form $(1:\varepsilon^{\alpha}:\varepsilon^{\beta})$, where $\\varepsilon$ is a primitive root of unity of ord...

open_problem | open | MathDB 页面 | 原文线索

V2-079261 | MATHDB-402513

Brill-Noether theorem for sheaves on del Pezzo surfaces

摘录 / EXCERPT: Let X be a del Pezzo surface of degree at least 4, and let H be the pullback of a line under the blowdown map $X\to\mathbb{P}^2$. Suppose V is a general $(-K_X)$-semistable...

open_problem | open | MathDB 页面 | 原文线索

V2-079262 | MATHDB-402514

Constancy of the dimension for RCD spaces

摘录 / EXCERPT: Let (X,d,m) be an $\\mathrm{RCD}(K,N)$ space. Then there exists an integer n with $0<n\leq N$, called the essential dimension of X, such that $m(X\setminus R_n)=0$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-079263 | MATHDB-402519

Feller property for the configuration-space semigroup

摘录 / EXCERPT: It is not sure if the semigroup has the Feller property on $\Upsilon(X)$ (even if X is smooth).

open_problem | open | MathDB 页面 | 原文线索

V2-079264 | MATHDB-402520

Whether every finitely generated group is sofic

摘录 / EXCERPT: Determine whether or not every finitely generated group is sofic.

open_problem | open | MathDB 页面 | 原文线索

V2-079265 | MATHDB-402521

Continuity of universally measurable homomorphisms between Polish groups

摘录 / EXCERPT: Is every universally measurable homomorphism $\pi:G\to H$ between Polish groups continuous?

open_problem | open | MathDB 页面 | 原文线索

V2-079266 | MATHDB-402522

Kazhdan’s property (T) for \(\operatorname{SAut}(F_4)\)

摘录 / EXCERPT: Determine whether $\operatorname{SAut}(F_4)$ has Kazhdan’s property (T).

open_problem | open | MathDB 页面 | 原文线索

V2-079267 | MATHDB-402523

Universal Hilbert-Schmidt (hyperlinear) approximation of groups

摘录 / EXCERPT: For every group Γ, does there exist a sequence of set-theoretic maps $\varphi_n:\Gamma\to U_n$ such that, for all $a,b\in\Gamma$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079268 | MATHDB-402530

Single generation of II_1 factors

摘录 / EXCERPT: Is every II$_1$ factor simply generated (or generated by two hermitians)?

open_problem | open | MathDB 页面 | 原文线索

V2-079269 | MATHDB-402531

Cost and the first \b2-Betti number of finitely generated groups

摘录 / EXCERPT: It is conjectured that, for every finitely generated group Γ, ... where the report has established the lower bound $\operatorname{cost}(\Gamma)\ge 1+\beta_1(\Gamma)$.

open_problem | open | MathDB 页面 | 原文线索

V2-079270 | MATHDB-402532

Does W^*-equivalence imply measure equivalence for ICC groups?

摘录 / EXCERPT: For ICC countable groups Γ and Λ, does $L\Gamma\cong L\Lambda$ (that is, W^*-equivalence) imply that Γ and Λ are measure equivalent?

open_problem | open | MathDB 页面 | 原文线索

V2-079271 | MATHDB-402533

Universality class of band matrices as the band width varies

摘录 / EXCERPT: Understand the universality class of band matrices while the width of the band varies.

open_problem | open | MathDB 页面 | 原文线索

V2-079272 | MATHDB-402534

Bulk eigenvector universality for the real Ginibre ensemble

摘录 / EXCERPT: For the real Ginibre ensemble (GinOE), let ... and define ... The conjecture is that, for bulk points x_1,x_2 whose separation is much larger than $O(N^{-1/2})$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-079273 | MATHDB-402535

Large deviations for traces of Wigner matrices

摘录 / EXCERPT: Let X be a Wigner matrix: a random Hermitian matrix such that $(X_{i,j})_{1\le i<j\le n}$ and $(X_{i,i})_{i\le n}$ are independent families of independent and identically distrib...

open_problem | open | MathDB 页面 | 原文线索

V2-079274 | MATHDB-402536

Unitary moments of moments: polynomiality and asymptotic growth

摘录 / EXCERPT: For $k,\beta\te\bb N$, the quantity ... is a polynomial in N, and ... where a general formula for $c_U(k,\beta)$ is given.

open_problem | open | MathDB 页面 | 原文线索

V2-079275 | MATHDB-402537

GOE local eigenvalue statistics for fixed-degree random regular graphs

摘录 / EXCERPT: For random regular graphs of fixed degree, the local eigenvalue statistics are given by those of the GOE.

open_problem | open | MathDB 页面 | 原文线索

V2-079276 | MATHDB-402538

Bulk eigenvalue fluctuations of sparse Erdős-Rényi graphs

摘录 / EXCERPT: For the rescaled and centered adjacency matrix of the Erdős-Rényi graph $G(N,p)$ with $p\in[N^{-1+\varepsilon},N^{-\varepsilon}]$, the report states that bulk eigenvalues fluctuate...

open_problem | open | MathDB 页面 | 原文线索

V2-079277 | MATHDB-402539

Uniqueness of the space-time Martin-boundary point at infinity

摘录 / EXCERPT: Conjecture. The space-time Martin boundary of $\bar z_t$ has a unique point at infinity, and Z_t is defined by (6), where $h=h_{\beta,a}$ is the corresponding solution to (7).

open_problem | open | MathDB 页面 | 原文线索

V2-079278 | MATHDB-402540

Almost-sure Weyl-law convergence for eigenvalue empirical measures of randomly perturbed quantized tori

摘录 / EXCERPT: In the setup of the report, it was conjectured that the empirical eigenvalue measure ... converges weakly, as $N\to\infty$, almost surely to the push-forward under the symbol p of...

open_problem | open | MathDB 页面 | 原文线索

V2-079279 | MATHDB-402541

Almost-sure Weyl law for quantized tori

摘录 / EXCERPT: For the Berezin-Toeplitz quantizations p_N of a smooth complex-valued function $p\in C^\infty(\mathbb T^{2d})$, and for small complex Gaussian perturbations $N^{-\gamma}Q_N$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-079280 | MATHDB-402543

Sarnak’s density hypothesis for automorphic representations on \(\mathrm{GL}(n)\)

摘录 / EXCERPT: For a finite family $\mathcal F$ of automorphic representations for $\mathrm{GL}(n)$, a place v of $\mathbb Q$, and $\sigma\ge 0$, define...

conjecture | open | MathDB 页面 | 原文线索

V2-079281 | MATHDB-402549

Completions and modularity of higher-dimensional false theta functions

摘录 / EXCERPT: For the higher-dimensional false theta functions that arise in some evaluations of $Z(q)$, determine their modularity properties on the upper half-plane; in particular, develop com...

open_problem | open | MathDB 页面 | 原文线索

V2-079282 | MATHDB-402550

Modularity of generating series for special cycles

摘录 / EXCERPT: In the setup where F is a totally real number field of degree d, V is an anisotropic quadratic space over F of signature $((m,2)^{d_+},(m+2,0)^{d-d_+})$ with $d_+>0$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-079283 | MATHDB-402551

Pullback formula for generating series of special cycles

摘录 / EXCERPT: Let U be a totally positive definite quadratic space over F of dimension 4ℓ, let $\widetilde V=U\oplus V$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-079284 | MATHDB-402553

Elliptic modular cocycle RM-value conjecture

摘录 / EXCERPT: The RM values $J_f^+[\tau]$ and $J_f^-[\tau]$ belong to $E(H_\tau)$.

conjecture | open | MathDB 页面 | 原文线索

V2-079285 | MATHDB-402555

Inverse realization of pre-Kuznetsov weights

摘录 / EXCERPT: The problem raised by Michel-Venkatesh is resolved as follows: for every weight function of the form ... there exists some $f\in I(0)\otimes I(0)$ such that $h(\sigma)=h_f(\sigma)$....

open_problem | open | MathDB 页面 | 原文线索

V2-079286 | MATHDB-402559

Stable Gan-Gross-Prasad theorem for unitary groups

摘录 / EXCERPT: Let E/F be a quadratic extension of number fields, let V be a Hermitian space of dimension n over E, and let $V'=V\oplus^{\perp}\langle e\rangle$, where $(e,e)=1$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-079287 | MATHDB-402561

Gross-Zagier conjecture on CM values of higher automorphic Green functions

摘录 / EXCERPT: Let $j\in\mathbb{Z}_{>0}$ and let $f=\sum_m c_f(m)q^m\in M^!_{-2j}$ for $\Gamma=\mathrm{SL}_2(\mathbb{Z})$. Assume that $c_f(-m)\in\mathbb{Z}$ for every $m>0$. Let z_1 be a CM po...

conjecture | open | MathDB 页面 | 原文线索

V2-079288 | MATHDB-402563

Kudla’s arithmetic modularity conjecture for special cycles

摘录 / EXCERPT: Let F) be a totally real field and let V be an anisotropic quadratic space over F of dimension $p+2$, with signature $((p,2),(p+2,0),\ldots,(p+2,0))$. Let...

conjecture | open | MathDB 页面 | 原文线索

V2-079289 | MATHDB-402564

Explicit modular or Jacobi parametrization of the Schoen quintic

摘录 / EXCERPT: Does the Schoen quintic ... have an explicit parametrization by modular or Jacobi forms?

open_problem | open | MathDB 页面 | 原文线索

V2-079290 | MATHDB-402566

Lifting differential operators from a special fiber

摘录 / EXCERPT: Given a domain R that is flat and finitely generated over a Dedekind domain A, and a maximal ideal μ of A, is every A/μ-linear differential operator of R/μ induce...

open_problem | open | MathDB 页面 | 原文线索

V2-079291 | MATHDB-402567

Localization compatibility of finitistic test ideals

摘录 / EXCERPT: The formation of finitistic test ideals is compatible with localization.

open_problem | open | MathDB 页面 | 原文线索

V2-079292 | MATHDB-402568

Stückrad-Vogel conjecture

摘录 / EXCERPT: For a local ring R with equidimensional completion, there exists a constant c such that $\c\bigl(R/I\bigr) \cleq c e(I)$ for every ideal I primary to the maximal ideal of R...

conjecture | open | MathDB 页面 | 原文线索

V2-079293 | MATHDB-402571

Harbourne's symbolic-to-ordinary power containment conjecture

摘录 / EXCERPT: Let R be a regular ring and let I be a radical ideal of big height h. Then, for all integers $n\ge 1$, $I^{(hn-h+1)}\subseteq I^n$.

conjecture | open | MathDB 页面 | 原文线索

V2-079294 | MATHDB-402573

Stillman’s conjecture (Ananyan-Hochster theorem)

摘录 / EXCERPT: For positive integers $d_1,\ldots,d_r$, there exists a bound $N(d_1,\ldots,d_r)$ such that any ideal in a polynomial ring over a field generated by homogeneous polynomials...

conjecture | open | MathDB 页面 | 原文线索

V2-079295 | MATHDB-402575

Descent of completion coordinates preserving the order of a principal ideal

摘录 / EXCERPT: Let R be an excellent regular local ring with regular system of parameters (u,y). Let $f\in R$ with $f\notin\langle u\rangle$, and let $m=m(f)$ be as defined above. Suppose...

open_problem | open | MathDB 页面 | 原文线索

V2-079296 | MATHDB-402578

Finiteness of irreducible components of the singular-locus arc space in positive characteristic

摘录 / EXCERPT: Does $X^{\mathrm{Sing}}_{\infty}$ have a finite number of irreducible components?

open_problem | open | MathDB 页面 | 原文线索

V2-079297 | MATHDB-402579

Finiteness of irreducible components of the singular-locus arc space

摘录 / EXCERPT: Does $X^{\rm Sing}_\infty$ have a finite number of irreducible components?

open_problem | open | MathDB 页面 | 原文线索

V2-079298 | MATHDB-402583

Asymptotic linearity of regularity of Ext and Tor over complete intersections

摘录 / EXCERPT: For each $\ell\in\{0,1\}$, do there exist $a_\ell,a'_\ell\in\mathbb{Z}_{>0}$ and $e_\ell,e'_\ell\in\mathbb{Z}\cup\{-\infty\}$ such that (i)...

open_problem | open | MathDB 页面 | 原文线索

V2-079299 | MATHDB-402587

Eisenbud-Schreyer conjectural characteristic range for Green’s conjecture

摘录 / EXCERPT: The report states that Eisenbud and Schreyer conjecture that Theorem 3 should extend from the range $\operatorname{char}(k)=0$ or $\operatorname{char}(k)\ge (g+2)/2$ to the range...

conjecture | open | MathDB 页面 | 原文线索

V2-079300 | MATHDB-402588

Itoh’s conjecture on the vanishing of the third normal Hilbert coefficient

摘录 / EXCERPT: Under the report’s standing hypotheses, let $(R,\mathfrak m)$ be an analytically unramified Cohen-Macaulay local ring of dimension $d\ge 2$ with infinite residue field, let I be...

conjecture | open | MathDB 页面 | 原文线索

V2-079301 | MATHDB-402591

Hypothesized outflow control of tropical-cyclone intensification

摘录 / EXCERPT: The premise of Emanuel’s revised theory is that small-scale turbulence in the upper-tropospheric outflow layer determines the thermal stratification of the outflow and, in turn, am...

open_problem | open | MathDB 页面 | 原文线索

V2-079302 | MATHDB-402595

Existence of a tame site with the tame fundamental group

摘录 / EXCERPT: Can one modify the étale site to obtain a tame site whose fundamental group coincides with the tame fundamental group?

open_problem | open | MathDB 页面 | 原文线索

V2-079303 | MATHDB-402597

Crystalline Chebotarev density conjecture for convergent F-isocrystals

摘录 / EXCERPT: Let U be a smooth, geometrically irreducible curve over the finite field $\mathbb F_q$ of characteristic p, and let $|U|$ be its set of closed points. Let $K/\mathbb Q_q$ be a...

conjecture | open | MathDB 页面 | 原文线索

V2-079304 | MATHDB-402598

Crew’s parabolicity conjecture for monodromy groups of convergent F-isocrystals

摘录 / EXCERPT: Let U be a smooth, geometrically irreducible curve over a finite field, let $u\in U(\mathbb F_q)$ be the fixed base point, and let $F\in F\text{-}\mathrm{Isoc}_K(U)$ be semisimpl...

conjecture | open | MathDB 页面 | 原文线索

V2-079305 | MATHDB-402599

Geometric $P=W$ conjecture

摘录 / EXCERPT: Let C be a complex projective smooth curve and let G be a reductive linear group. Let $\mathcal M_B=\operatorname{Hom}(\pi_1(C),G)\! /\! /G$ be the Betti moduli space, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-079306 | MATHDB-402602

Lubin-pair semiconjugacy conjecture

摘录 / EXCERPT: Let K be a finite extension of $\mathbb{Q}_p$, let $\mathcal{O}_K$ be its ring of integers, and let (f,u) be a Lubin pair: $f,u\in X\cdot\u0007mathcal{O}_K[[X]]$ commute under...

conjecture | open | MathDB 页面 | 原文线索

V2-079307 | MATHDB-402603

Minimal locus of the operator seminorm

摘录 / EXCERPT: What can be said about the minimal locus of $\|\psi\|_{\mathrm{op},\bullet}$ as a function on X^{an}? The report suggests that this should be analogous to a skeleton of...

open_problem | open | MathDB 页面 | 原文线索

V2-079308 | MATHDB-402604

Determining the appropriate strain-energy function in nonlinear elasticity

摘录 / EXCERPT: Determine the most appropriate strain-energy function to describe the behaviour of a given material.

open_problem | open | MathDB 页面 | 原文线索

V2-079309 | MATHDB-402605

Global weak solvability for an incompressible viscoelastic rate-type fluid with stress diffusion

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^d$ be a Lipschitz domain with $d\ge 2$, let $T>0$, and suppose ... with $b_0\in L^\infty(\Omega)$ and $1/b_0\in L^\infty(\Omega)$. For arbitrary positive...

open_problem | open | MathDB 页面 | 原文线索

V2-079310 | MATHDB-402606

Existence and uniqueness for three-dimensional nonlinear elastodynamic Cauchy problems

摘录 / EXCERPT: Determine whether solutions to Cauchy problems governing wave propagation in full three-dimensional nonlinearly elastic solids exist and are unique.

open_problem | open | MathDB 页面 | 原文线索

V2-079311 | MATHDB-402607

Asymptotic Ringel decomposition theorem

摘录 / EXCERPT: For every sufficiently large integer n, the edges of the complete graph K_{2n+1} can be partitioned into edge-disjoint subgraphs isomorphic to any tree T with n edges.

open_problem | open | MathDB 页面 | 原文线索

V2-079312 | MATHDB-402608

Rainbow matching conjecture in general graphs

摘录 / EXCERPT: The conjecture is that n matchings of size n in any graph have a rainbow matching of size $n-1$.

conjecture | open | MathDB 页面 | 原文线索

V2-079313 | MATHDB-402609

Erdős-Szekeres problem for multidimensional arrays

摘录 / EXCERPT: Given positive integers d and n, determine the smallest N such that every d-dimensional array of size $N\times\cdots\times N$ contains a monotone d-dimensional subarray o...

open_problem | open | MathDB 页面 | 原文线索

V2-079314 | MATHDB-402610

Rate of convergence of the expected LCS for two random words

摘录 / EXCERPT: For independently chosen words $R,R'\sim\Sigma^n$, determine the rate of convergence in ... where γ is a constant depending on $|\Sigma|$.

open_problem | open | MathDB 页面 | 原文线索

V2-079315 | MATHDB-402611

Polynomial-time MWIS on graphs with no long holes

摘录 / EXCERPT: For the class of graphs with no hole of length at least five, the Maximum Weight Independent Set (MWIS) problem can be solved in polynomial time.

open_problem | open | MathDB 页面 | 原文线索

V2-079316 | MATHDB-402619

Symmetric intersecting families of vectors

摘录 / EXCERPT: For each fixed integer $k\ge 3$, if $\mathcal A\subseteq [k]^n$ is intersecting-meaning that any two elements of $\mathcal A$ agree in at least one coordinate-and admits a transiti...

open_problem | open | MathDB 页面 | 原文线索

V2-079317 | MATHDB-402620

Linear \c7-boundedness of axis-parallel rectangle intersection graphs

摘录 / EXCERPT: It is conjectured that the family of intersection graphs of axis-parallel rectangles in the plane is c7-bounded with a bounding function $f(k)=O(k)$; equivalently, there is a const...

open_problem | open | MathDB 页面 | 原文线索

V2-079318 | MATHDB-402621

Kahn-Kalai expectation-threshold conjecture

摘录 / EXCERPT: There is a universal constant K such that for every finite set X and increasing family $\mathcal{F}\subseteq 2^X$, ...

conjecture | open | MathDB 页面 | 原文线索

V2-079319 | MATHDB-402622

Talagrand’s expectation-threshold conjecture

摘录 / EXCERPT: There is a universal constant K such that, for every finite set X and every increasing family $\mathcal F\subseteq 2^X$, $q(\mathcal F)\ge q_f(\mathcal F)/K$. Equivalently, if...

conjecture | open | MathDB 页面 | 原文线索

V2-079320 | MATHDB-402628

Induced subgraphs of Cayley graphs of $\mathbb{Z}_2^n$

摘录 / EXCERPT: For a subset S of the elementary abelian 2-group $\mathbb{Z}_2^n$, let $G=G(\mathbb{Z}_2^n,S)$ be the Cayley graph of $\mathbb{Z}_2^n$ with respect to S. Is it true that, for...

open_problem | open | MathDB 页面 | 原文线索

V2-079321 | MATHDB-402631

Avi’s 1-2-3 conjecture

摘录 / EXCERPT: Find an explicit construction of a bipartite graph $G=(A\cup B,E)$ with $|A|=n^3$, $|B|=n^2$, and bounded degree on the vertices in A, such that for every $S\subseteq A$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-079322 | MATHDB-402640

Categorical crepant resolution conjecture for intrinsic mirrors

摘录 / EXCERPT: Let (M,D) be a maximally degenerate, positive Calabi-Yau pair which admits a weak section. Then the data π_*,π^* give $\mathcal W(X)$ the structure of a categorical crepant...

conjecture | open | MathDB 页面 | 原文线索

V2-079323 | MATHDB-402641

Corti-Filip-Petracci conjecture on smoothing components of Gorenstein toric affine 3-folds

摘录 / EXCERPT: Let F be a lattice polytope of dimension 3 and let U_F be the Gorenstein toric affine 3-fold associated to the cone over F put at height 1. Then the smoothing component...

conjecture | open | MathDB 页面 | 原文线索

V2-079324 | MATHDB-402642

Measure-zero discontinuity set for the Schonmann projection

摘录 / EXCERPT: For the Schonmann projection, does the set of discontinuity points of the two-sided specification have measure zero?

open_problem | open | MathDB 页面 | 原文线索

V2-079325 | MATHDB-402643

Finite-volume Aizenman-Higuchi result for long-range Ising models

摘录 / EXCERPT: Does the finite-volume version of the Aizenman-Higuchi result also hold for long-range Ising models?

open_problem | open | MathDB 页面 | 原文线索

V2-079326 | MATHDB-402646

Finitary coding of the Schonmann projection

摘录 / EXCERPT: Does the Schonmann projection have a finitary coding?

open_problem | open | MathDB 页面 | 原文线索

V2-079327 | MATHDB-402647

Eigenfunction characterization of equilibrium-state g-measures

摘录 / EXCERPT: Let $X^+=A^{\mathbb Z_+}$, let $f:X^+\to\mathbb R$ be continuous, and let μ be a translation-invariant equilibrium state for f. Is the existence of a positive continuous eige...

open_problem | open | MathDB 页面 | 原文线索

V2-079328 | MATHDB-402652

Generalized Lax conjecture

摘录 / EXCERPT: Let $p\in\mathbb{R}[x_1,\ldots,x_\ell]$ be a real-zero polynomial, meaning that $p(0)\ne 0$ and, for every $a\in\mathbb{R}^{\ell}$ and every $\lambda\in\mathbb{C}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-079329 | MATHDB-402654

Gurvits’s Lorentzian Minkowski-volume conjecture

摘录 / EXCERPT: If $f(x,y,z)$ is a Lorentzian polynomial of degree d, then there are convex bodies $K_1,K_2,K_3\subset\mathbb{R}^d$ such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-079330 | MATHDB-402655

The precise universal normal-dilation constant

摘录 / EXCERPT: Determine the smallest constant $C_d>0$ such that every d-tuple $A=(A_1,[U+007F]ots,A_d)$ of complex square matrices, of arbitrary common size, admits a normal dilation...

open_problem | open | MathDB 页面 | 原文线索

V2-079331 | MATHDB-402657

Interior of the locus of real morphisms with only real periodic points

摘录 / EXCERPT: For a fixed integer $d\ge 2$, let $\mathbb{R}_d$ be the set of morphisms $\varphi:\mathbb{P}^1\to\mathbb{P}^1$ of degree d defined over $\mathbb{R}$ and having only real periodic...

open_problem | open | MathDB 页面 | 原文线索

V2-079332 | MATHDB-402659

Sums-of-squares characterization for locally non-negative polynomials

摘录 / EXCERPT: Given a polynomial $f\in P_{\mathrm{Loc}}$, determine whether f is a sum of squares in $\mathbb{R}[[x]]$.

open_problem | open | MathDB 页面 | 原文线索

V2-079333 | MATHDB-402662

Projected Lax conjecture for hyperbolicity cones

摘录 / EXCERPT: Every hyperbolicity cone is the projection of a spectrahedron.

conjecture | open | MathDB 页面 | 原文线索

V2-079334 | MATHDB-402663

Hyperbolic certificates for the Motzkin form

摘录 / EXCERPT: Do forms whose nonnegativity follows from the arithmetic-geometric mean inequality, such as the Motzkin form $x^2y^4+x^4y^2+z^6-3x^2y^2z^2$, have hyperbolic certificates of nonnega...

open_problem | open | MathDB 页面 | 原文线索

V2-079335 | MATHDB-402667

Sparse version of Putinar’s Positivstellensatz

摘录 / EXCERPT: Let $I:=\{1,\ldots,n\}$, $J:=\{1,\ldots,m\}$, and $S(g):=\{x\in\mathbb{R}^n:g_j(x)\ge 0,\ j=1,\ldots,m\}$, where $g=\{g_1,\ldots,g_m\}$. Assume that there are...

open_problem | open | MathDB 页面 | 原文线索

V2-079336 | MATHDB-402668

Incorporating uncertainty into multiphysical material models

摘录 / EXCERPT: How can uncertainties of various type-and statistical information in general-be incorporated into multiphysical material models?

open_problem | open | MathDB 页面 | 原文线索

V2-079337 | MATHDB-402669

Existence theory for the shear-band free-boundary problem

摘录 / EXCERPT: Develop a rigorous existence theory for the parabolic free-boundary problem that arises in the shear-band model.

open_problem | open | MathDB 页面 | 原文线索

V2-079338 | MATHDB-402670

Sharp-interface convergence of the phase-field fracture-and-healing model

摘录 / EXCERPT: Does the proposed phase-field model converge, in any sense, to a sharp-interface model as the size of the process zone vanishes?

open_problem | open | MathDB 页面 | 原文线索

V2-079339 | MATHDB-402671

Relaxation in coupled magnetomechanical systems

摘录 / EXCERPT: “An open question is therefore how to combine these concepts and whether relaxation in the coupled case ought to be carried out in monolithic or rather staggered fashion, in the se...

open_problem | open | MathDB 页面 | 原文线索

V2-079340 | MATHDB-402673

Martin’s Maximum^{++} implies Woodin’s axiom $(*)$

摘录 / EXCERPT: Martin’s Maximum^{++} implies Woodin’s P_{max}-axiom $(*)$.

open_problem | open | MathDB 页面 | 原文线索

V2-079341 | MATHDB-402674

Isbell’s problem on non-maximal ultrafilters over \(\omega\)

摘录 / EXCERPT: Does ZFC prove that there exists a non-maximal ultrafilter over ω?

open_problem | open | MathDB 页面 | 原文线索

V2-079342 | MATHDB-402677

Consistency of simple $P_{\\aleph_1}$- and simple $P_{\\aleph_2}$-points

摘录 / EXCERPT: Is it consistent relative to ZFC that there exist both a simple $P_{\\aleph_1}$-point and a simple $P_{\\aleph_2}$-point?

open_problem | open | MathDB 页面 | 原文线索

V2-079343 | MATHDB-402678

Canjar ultrafilters at uncountable-cofinality stages

摘录 / EXCERPT: For every $\alpha \leq \omega_2$ with $\operatorname{cf}(\alpha) \geq \omega_1$, does $\mathbb P_\alpha$ force that $\mathcal F_\alpha$ is a Canjar ultrafilter that is not nearly c...

open_problem | open | MathDB 页面 | 原文线索

V2-079344 | MATHDB-402679

F-obstructions versus subgroups not embeddable into Thompson’s group F

摘录 / EXCERPT: Does every subgroup of $\mathrm{PL}_{+}I$ which does not embed into Thompson’s group F contain an F-obstruction?

open_problem | open | MathDB 页面 | 原文线索

V2-079345 | MATHDB-402680

Comparability of Thompson’s group under embeddability

摘录 / EXCERPT: Does Thompson’s group F have the following property: for every finitely generated subgroup H of $PL_o(I)$, either F embeds into H or H embeds into F?

open_problem | open | MathDB 页面 | 原文线索

V2-079346 | MATHDB-402681

Consistency result at \(\kappa=\aleph_\omega\)

摘录 / EXCERPT: Is it possible to obtain the above consistency result for $\kappa=\aleph_\omega$?

open_problem | open | MathDB 页面 | 原文线索

V2-079347 | MATHDB-402682

Weak compactness of the critical-point limit in HOD

摘录 / EXCERPT: Assume $(V,j)\models\mathrm{ZF}$, where $j:V\to V$ is an elementary embedding. Let ... Is λ weakly compact in HOD?

open_problem | open | MathDB 页面 | 原文线索

V2-079348 | MATHDB-402685

Linear order expansion of an uncountable \-homogeneous graph

摘录 / EXCERPT: Suppose that K is an uncountable, $\$-homogeneous graph which embeds every finite graph. Is there some linear order $<$ on K such that $\ K,<\$ is also $\$-homogeneous?

open_problem | open | MathDB 页面 | 原文线索

V2-079349 | MATHDB-402686

Stability of the near-critical internal-wave approximation

摘录 / EXCERPT: For the Boussinesq system ... in $\Omega=\{(x_1,x_3)\in\mathbb R^2:x_1\sin\gamma-x_3\cos\gamma>0\}$, with $\nu\simeq\kappa$, $\beta-\gamma=O(\nu^{1/3})$, and boundary conditions...

open_problem | open | MathDB 页面 | 原文线索

V2-079350 | MATHDB-402688

Tits Alternative for CAT(0) groups

摘录 / EXCERPT: Whether CAT(0) groups satisfy the Tits Alternative remains an open question, even in the case of groups acting properly and cocompactly on 2-dimensional CAT(0) complexes.

open_problem | open | MathDB 页面 | 原文线索

V2-079351 | MATHDB-402689

Coherence conjecture for few-relator random groups

摘录 / EXCERPT: Fix a number of generators $n\ge 2$ and a number of relators m, and consider the (n,m)-few-relators model of random groups, with Euler characteristic $\chi=1-n+m$. Conjecture:...

conjecture | open | MathDB 页面 | 原文线索

V2-079352 | MATHDB-402691

Virtual simplicity of non-Bruhat-Tits \(\widetilde{A}_2\)-lattices

摘录 / EXCERPT: An $\widetilde{A}_2$-lattice whose associated building is not Bruhat-Tits is virtually simple.

open_problem | open | MathDB 页面 | 原文线索

V2-079353 | MATHDB-402693

Classification of soluble S-arithmetic groups with property $R_\$

摘录 / EXCERPT: Determine which soluble S-arithmetic groups have property $R_\$.

open_problem | open | MathDB 页面 | 原文线索

V2-079354 | MATHDB-402695

Quasi-isometry invariance of acylindrical hyperbolicity

摘录 / EXCERPT: Is acylindrical hyperbolicity preserved under quasi-isometry?

open_problem | open | MathDB 页面 | 原文线索

V2-079355 | MATHDB-402696

Smallest total length of a hyperbolic property-(T) presentation

摘录 / EXCERPT: What is the smallest total length of a presentation of an infinite hyperbolic group with Kazhdan’s property (T)?

open_problem | open | MathDB 页面 | 原文线索

V2-079356 | MATHDB-402697

Smallest generalized triangle-group parameter yielding hyperbolicity and property (T)

摘录 / EXCERPT: What is the smallest k such that there exists a generalized k-fold triangle group that is hyperbolic and has Kazhdan’s property (T)?

open_problem | open | MathDB 页面 | 原文线索

V2-079357 | MATHDB-402698

Weil-positivity criterion equivalent to the Riemann Hypothesis

摘录 / EXCERPT: Establish the negativity of the geometric side of the Riemann-Weil explicit formula, namely prove that $\sum_v W_v(f)\le 0$ for every $f=g*\bar g^{\,*}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-079358 | MATHDB-402704

Rational strong Novikov conjecture for groups acting on admissible Hilbert-Hadamard spaces

摘录 / EXCERPT: The rational strong Novikov conjecture holds for any discrete group admitting an isometric and proper action on an admissible Hilbert-Hadamard space.

conjecture | open | MathDB 页面 | 原文线索

V2-079359 | MATHDB-402706

Universal convergence of the delocalized higher-rho pairing

摘录 / EXCERPT: In the general situation of Question 7, does the pairing $\langle \operatorname{Tch}(\rho(\widetilde D)),\varphi\rangle$ always converge? Equivalently, does...

open_problem | open | MathDB 页面 | 原文线索

V2-079360 | MATHDB-402707

Boundedness of the circular maximal function on $L^p(\mathbb H^1)$

摘录 / EXCERPT: It is currently unknown whether the circular maximal function on the Heisenberg group $\mathbb H^1$ is bounded on $L^p(\mathbb H^1)$ for any finite p.

open_problem | open | MathDB 页面 | 原文线索

V2-079361 | MATHDB-402709

Indecomposability of derived categories of symmetric products of curves

摘录 / EXCERPT: Let C be a smooth projective curve of genus $g\ge 2$. For every integer n with $1\le n\le g-1$, the bounded derived category $D^b(\operatorname{Sym}^n C)$ is indecomposable; eq...

open_problem | open | MathDB 页面 | 原文线索

V2-079362 | MATHDB-402713

Smoothness or integrality of the constructed surfaces for higher discriminants

摘录 / EXCERPT: For an integer $a\ge 3$, set $d=6a^{2}+6a+2$. If Y is a general cubic fourfold in $\mathcal C_d$ and $\Sigma\subset Y$ is a surface supplied by Theorem 1.1, is Σ smooth (o...

open_problem | open | MathDB 页面 | 原文线索

V2-079363 | MATHDB-402714

Rationality of cubic fourfolds in the discriminant families \(\mathcal C_{6a^2+6a+2}\)

摘录 / EXCERPT: Conjecture: For every integer $a\ge 1$, every cubic fourfold $Y\in\mathcal C_d$, where $d=6a^2+6a+2$, is rational.

open_problem | open | MathDB 页面 | 原文线索

V2-079364 | MATHDB-402715

Characterization of theories with profinite Ellis group

摘录 / EXCERPT: For which theories is the Ellis group profinite?

open_problem | open | MathDB 页面 | 原文线索

V2-079365 | MATHDB-402716

Equivalence of NIP and uniform definability of types over finite sets

摘录 / EXCERPT: For every language L, every L-theory T, and every L-formula $\\varphi(x,y)$, the following are equivalent: (1) $\\varphi$ is NIP in T; (2) $\\varphi$ has uniform definabi...

open_problem | open | MathDB 页面 | 原文线索

V2-079366 | MATHDB-402717

Pseudofiniteness of the Urysohn sphere

摘录 / EXCERPT: Is the Urysohn sphere U pseudofinite; that is, is U elementarily equivalent to an ultraproduct of finite metric spaces? Equivalently, given a sentence σ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-079367 | MATHDB-402718

Group configuration theorem in o-minimal structures

摘录 / EXCERPT: Let M be an $|L|^{+}$-saturated o-minimal structure that eliminates imaginaries, and let $V=(a_1,a_2,a_3,x_1,x_2,x_3)$ be an (m,k)-configuration over $A\subseteq M$. Thus, with...

open_problem | open | MathDB 页面 | 原文线索

V2-079368 | MATHDB-402719

Embedding definably connected type-definable groups into definable groups

摘录 / EXCERPT: Can every definably connected type-definable group G in an o-minimal structure be definably embedded into a definable group H? Moreover, can one choose H to be of the same di...

open_problem | open | MathDB 页面 | 原文线索

V2-079369 | MATHDB-402721

Categorical extension of the Iwahori-fixed Langlands parametrization

摘录 / EXCERPT: Let F) be a finite extension of $ Q_p$ or of $ F_p((t))$, let G be a split reductive group over F, and write $G=G(F)$. Fix a Borel subgroup B and a maximal split torus...

open_problem | open | MathDB 页面 | 原文线索

V2-079370 | MATHDB-402722

Derived-category extension of the Kazhdan-Lusztig parametrization

摘录 / EXCERPT: Let G be a split reductive group over F, with fixed maximal split torus and Borel subgroup $T\subset B\subset G=G(F)$, and let $\check G$, $\check T$, and $\check B$ be the cor...

open_problem | open | MathDB 页面 | 原文线索

V2-079371 | MATHDB-402723

Conjectural identification of the Iwahori-invariant family

摘录 / EXCERPT: For the split reductive group setup of the report, the conjectural identification ... should hold, where $\beta_B:[X_{\check B}/\check B]\to[X_{\check G}/\check G]$ is induced by the...

open_problem | open | MathDB 页面 | 原文线索

V2-079372 | MATHDB-402726

Rational points from simple zeros of triple-product L-functions

摘录 / EXCERPT: Assume the definite case, with $\varrho=\varrho_1\otimes\varrho_2$ irreducible. There is a canonical multiple $L_p^{\mathrm{can}}(f,g,h)$ of $L_p(f,g,h)$ whose value at $(2,1,1)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-079373 | MATHDB-402729

Uniform bound for the number of rational points on a curve

摘录 / EXCERPT: There exists a constant $c([F:\mathbb{Q}],g)\ge 1$, depending only on the degree $[F:\mathbb{Q}]$ and $g\ge 2$, such that, for every number field F and every smooth, geometricall...

open_problem | open | MathDB 页面 | 原文线索

V2-079374 | MATHDB-402730

Irreducible case of Fargues' conjecture for GL_n

摘录 / EXCERPT: For each irreducible ℓ-adic $W_{\d540_p}$-representation E of dimension n, there exists an object...

conjecture | open | MathDB 页面 | 原文线索

V2-079375 | MATHDB-402735

Minimality of integral models of Rapoport-Zink spaces

摘录 / EXCERPT: Let M be the Rapoport-Zink moduli space of deformations, by quasi-isogenies, of a fixed p-divisible group over $\bf F_p$, viewed as a locally formally finite-type, formally smo...

open_problem | open | MathDB 页面 | 原文线索

V2-079376 | MATHDB-402736

Francis-Gaitsgory Koszul-duality conjecture

摘录 / EXCERPT: In the preceding setup, let $\mathcal C$ be a presentable, stable, symmetric monoidal ∞-category and let $\mathcal O$ be a nonunital operad in $\mathcal C$, with Koszul-dual...

conjecture | open | MathDB 页面 | 原文线索

V2-079377 | MATHDB-402737

2-local equivalence after inverting 2 for Grothendieck-Witt theory

摘录 / EXCERPT: Let R be the ring of integers in a number field. Then the map ... is a 2-local equivalence in degrees $\\ge 1$.

open_problem | open | MathDB 页面 | 原文线索

V2-079378 | MATHDB-402738

Integral universality of embedding-calculus invariants for classical knots

摘录 / EXCERPT: For every integer $n\ge 1$, is the evaluation map ... a universal additive Vassiliev invariant of type at most $n-1$ over $\mathbb Z$?

open_problem | open | MathDB 页面 | 原文线索

V2-079379 | MATHDB-402740

Threshold between rigidity and flexibility in martensitic differential inclusions

摘录 / EXCERPT: Is there a critical threshold separating the rigid and the flexible regime for these settings?

open_problem | open | MathDB 页面 | 原文线索

V2-079380 | MATHDB-402741

Regularity of area-minimizing integral currents with prescribed boundary

摘录 / EXCERPT: Suppose $\Gamma \subset \mathbb{R}^{m+n}$ is a closed integral current of dimension $m-1$, with suitable a priori regularity, and an integral current T minimizes area among all...

open_problem | open | MathDB 页面 | 原文线索

V2-079381 | MATHDB-402743

Classification of stable entire Allen-Cahn solutions in dimensions 3 through 7

摘录 / EXCERPT: For the Allen-Cahn equation $\Delta u+u-u^3=0$ in $\mathbb{R}^N$, classify the stable entire solutions when $3\le N\le 7$.

open_problem | open | MathDB 页面 | 原文线索

V2-079382 | MATHDB-402744

Global minimality of Cabré-Terra saddle solutions for the Allen-Cahn equation

摘录 / EXCERPT: Let u be the Cabré-Terra saddle solution of the Allen-Cahn equation ... characterized in the report by $u(s,t)=-u(t,s)$ and $u(s,t)>0$ when $s>t>0$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-079383 | MATHDB-402747

Finite graph structure of optimal mailing transport plans

摘录 / EXCERPT: Given an optimal transport plan for the mailing problem, does it have a finite graph structure in any ball outside of the support of the initial and target measures?

open_problem | open | MathDB 页面 | 原文线索

V2-079384 | MATHDB-402748

Dependence of minimal energy on angular momentum

摘录 / EXCERPT: An important open problem is to understand how minimal energies depend on the size of the angular momentum in order to estimate the rotation frequency ω, and to ascertain th...

open_problem | open | MathDB 页面 | 原文线索

V2-079385 | MATHDB-402758

Finiteness of Cox-ring iteration for arbitrary klt singularities

摘录 / EXCERPT: Can finiteness of the iteration of Cox rings, known for weakly Fano pairs and klt quasicones, be generalized to arbitrary klt singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-079386 | MATHDB-402760

Unirationality, rationality, and stable rationality of symmetric powers

摘录 / EXCERPT: If X_m is unirational (respectively, rational or stably rational) for some (or any large) $m>1$, is the same true for X? Here $X_m=\bigl(X^m\bigr)/S_m$ is the m-th symmetric...

open_problem | open | MathDB 页面 | 原文线索

V2-079387 | MATHDB-402766

Bigness of the tangent bundle of del Pezzo surfaces

摘录 / EXCERPT: Let S be a smooth del Pezzo surface of degree $d := (-K_S)^2$. Then T_S is big if and only if $d \ge 5$.

open_problem | open | MathDB 页面 | 原文线索

V2-079388 | MATHDB-402770

Hyperbolic Ax-Lindemann conjecture for arithmetic quotients

摘录 / EXCERPT: For the universal covering map $\pi:\Omega\to X_\Gamma=\Omega/\Gamma$ of a finite-volume quotient of a bounded symmetric domain, in the arithmetic case the Zariski closures of imag...

conjecture | open | MathDB 页面 | 原文线索

V2-079389 | MATHDB-402771

Ampleness of a corrected Hodge line bundle under local Torelli assumptions

摘录 / EXCERPT: Under suitable local Torelli-type assumptions, there exist integers $0\le a_i\in\mathbb{Z}$ and m_0 such that ... is ample for $m\ge m_0$.

open_problem | open | MathDB 页面 | 原文线索

V2-079390 | MATHDB-402774

Conditional fractional-moment bound for class-group twists

摘录 / EXCERPT: For two distinct cuspidal automorphic representations π_1 and π_2 of PGL_2 over $\mathbb{Q}$, of square-free level, and a quadratic field extension...

open_problem | open | MathDB 页面 | 原文线索

V2-079391 | MATHDB-402776

Fourth-moment bound for Hecke-Maass forms in a fixed spectral window

摘录 / EXCERPT: Fix $H>0$. Let $\d4d1$ be an orthonormal basis of Hecke-Maass eigenforms of the cuspidal spectrum $L^2\!\left(\mathrm{SL}_2(\mathbb Z)\backslash\mathbb H\right)_{\mathrm{cusp}}$. F...

open_problem | open | MathDB 页面 | 原文线索

V2-079392 | MATHDB-402778

Uniqueness-pair conjecture for power-law discrete sets

摘录 / EXCERPT: Conjecture: For [?] $\[?]A=[U+007F]\{\pm c n^{\alpha}\}_{n\ge 0}$ and $B=\{\pm d n^{\beta}\}_{n\ge 0}$, where $\alpha,\beta>0$, $\alpha+\beta=1$, and $2cd=\alpha^{-\alpha}\beta^{-\beta}$,...

conjecture | open | MathDB 页面 | 原文线索

V2-079393 | MATHDB-402781

Short-interval polynomial-phase cancellation for the Möbius function

摘录 / EXCERPT: Let $k\ge 1$ and let $H=H(X)\to\infty$ as $X\to\infty$. Then ... for almost all $x\in[X,2X]$.

open_problem | open | MathDB 页面 | 原文线索

V2-079394 | MATHDB-402783

Chowla’s conjecture with a short one-variable average

摘录 / EXCERPT: Let $k\ge 1$. Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-079395 | MATHDB-402784

Asymptotic exponent of the linear programming sphere-packing bound

摘录 / EXCERPT: The linear programming upper bound for the sphere packing density in $\mathbb{R}^n$ is $2^{-(\lambda+o(1))n}$ as $n\to\infty$, with $\lambda\approx 0.6044$.

open_problem | open | MathDB 页面 | 原文线索

V2-079396 | MATHDB-402785

Fourier anti-invariant Schwartz function for the spinning modular bootstrap

摘录 / EXCERPT: There exists a Schwartz function $f:\mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R}$, not identically zero, such that ... and ... whenever ...

open_problem | open | MathDB 页面 | 原文线索

V2-079397 | MATHDB-402787

Lindelöf-on-average bound for symmetric-square L-functions at unit spectral-window length

摘录 / EXCERPT: Let u_j range over Hecke-Maass cusp forms for $\mathrm{SL}_2(\mathbb Z)$, with Laplace eigenvalue $1/4+t_j^2$. Prove the Lindelöf-on-average estimate ... in the central-value regim...

open_problem | open | MathDB 页面 | 原文线索

V2-079398 | MATHDB-402788

Radial Fourier uniqueness on integer-radius spheres

摘录 / EXCERPT: Suppose Φ is a radial test function on $\mathbb{R}^{D}$. If $\Phi(x)=\widehat{\Phi}(y)=0$ for all $x,y\in\mathbb{R}^{D}$ with $|x|^{2},|y|^{2}\in\mathbb{Z}$, does it follow th...

open_problem | open | MathDB 页面 | 原文线索

V2-079399 | MATHDB-402789

Fourier uniqueness from integer-radius spherical shells (nonradial case)

摘录 / EXCERPT: Suppose Φ is a test function in $\mathbb{R}^D$, which need not be radial. If ... for all $x,y\in\mathbb{R}^D$ with $|x|^2,|y|^2\in\mathbb{Z}$, does it follow that $\Phi=0$ every...

open_problem | open | MathDB 页面 | 原文线索

V2-079400 | MATHDB-402791

Interpretation of the purely asymmetric dynamics as a gradient flow

摘录 / EXCERPT: We conjecture that the dynamics given by L_F^{asym} should be interpreted as the counterpart of a gradient flow, and that it will describe a purely non-dissipative dynam...

open_problem | open | MathDB 页面 | 原文线索

V2-079401 | MATHDB-402792

Homogenization of discrete optimal transport on periodic graphs

摘录 / EXCERPT: For a periodic graph (X,E) embedded in $\mathbb{R}^d$, with lower-semicontinuous convex edge costs $c_{xy}:\mathbb{R}_+\times\mathbb{R}_+\times\mathbb{R}\to\mathbb{R}_+$, conside...

open_problem | open | MathDB 页面 | 原文线索

V2-079402 | MATHDB-402794

Parabolic Hausdorff dimension of the singular set in the Stefan problem

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be an open set, let $u\in L^\infty(\Omega\times(0,T))$ solve the Stefan problem, and let $\Sigma\subset\Omega\times(0,T)$ be the set of singular poi...

open_problem | open | MathDB 页面 | 原文线索

V2-079403 | MATHDB-402800

Realization functors for bounded t-structures

摘录 / EXCERPT: Let A be an abelian category, and let $(D^{\leq 0},D^{\geq 0})$ be a bounded t-structure on $D^b(A)$. Denote its heart by $H=D^{\leq 0}\cap D^{\geq 0}$, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-079404 | MATHDB-402802

Cellular decompositions for rigid quiver representations

摘录 / EXCERPT: Every rigid finite-dimensional complex representation M of a quiver Q has property (C); equivalently, every quiver Grassmannian $\operatorname{Gr}_{\mathbf e}(M)$ attached to...

open_problem | open | MathDB 页面 | 原文线索

V2-079405 | MATHDB-402807

Contractibility of the minimum-global-dimension locus

摘录 / EXCERPT: Let D be a triangulated category and let $\operatorname{gd}D$ denote its global dimension. If the subspace $\operatorname{gldim}^{-1}(\operatorname{gd}D)$ is non-empty, then it i...

open_problem | open | MathDB 页面 | 原文线索

V2-079406 | MATHDB-402808

Contractibility conjecture for sublevel sets of the global-dimension function

摘录 / EXCERPT: Let $\mathcal D$ be a triangulated category, let gldim denote the global-dimension function on its space of stability conditions, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-079407 | MATHDB-402814

Non-negativity of the Hall-counting polynomials for generalised Cartan lattices

摘录 / EXCERPT: Let Γ be a generalised Cartan lattice, and let $d,e\in\mathbb Z^n$ be dimension vectors with d rigid. Let $g_{(d,e)}(t)\in\mathbb Z[t]$ be the polynomial such that, for ev...

open_problem | open | MathDB 页面 | 原文线索

V2-079408 | MATHDB-402818

Higher-rank dimension vectors and Newton-Okounkov bodies

摘录 / EXCERPT: It is natural to conjecture that the dimension vectors $\kappa(T,M)$ for M in $\mathrm{CM}(B)$ of rank r are all the integral points in the Newton-Okounkov body scaled up by...

open_problem | open | MathDB 页面 | 原文线索

V2-079409 | MATHDB-402819

Cells in the box and a hyperplane

摘录 / EXCERPT: For a positive integer d, let $Q_n^d=[0,n]^d$ and, for $z=(z_1,\ots,z_d)\in\bb Z^d$, let ... For $v\in\bb R^d\setminus\{0\}$ and $t\in\bb R$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-079410 | MATHDB-402825

Conjectured bound for constrained drawings of a star

摘录 / EXCERPT: In the constrained-drawing setup for a closure operator whose closures have at most b path-connected components, how many points are actually needed to find a constrained drawing...

conjecture | open | MathDB 页面 | 原文线索

V2-079411 | MATHDB-402827

Grünbaum-Loewner barycentric-hyperplane conjecture

摘录 / EXCERPT: Are there always at least 2^n-1 distinct barycentric hyperplanes with respect to the barycenter of a convex body $K\subseteq\mathbb{R}^n$?

conjecture | open | MathDB 页面 | 原文线索

V2-079412 | MATHDB-402828

A higher-dimensional lower bound for Tverberg depth

摘录 / EXCERPT: Let S be a finite point set in $\mathbb{R}^d$ and let $q\in\mathbb{R}^d$. Define the Tukey depth $\mathrm{TD}(S,q)$ as the minimum number of points that can be removed from S s...

open_problem | open | MathDB 页面 | 原文线索

V2-079413 | MATHDB-402833

Threshold for envy-free division of two cakes

摘录 / EXCERPT: In the notation a(n;a1,dots,ak)rightsquigarrow mb, meaning that every fair-division instance with anb agents and akb cakes partitioned into respectively aa1,dots,akb pieces admits...

open_problem | open | MathDB 页面 | 原文线索

V2-079414 | MATHDB-402834

Envy-free division of two cakes with one cake divided into n pieces

摘录 / EXCERPT: For which integers $p>n$ does $(n;n,p)\rightsquigarrow n$ hold?

open_problem | open | MathDB 页面 | 原文线索

V2-079415 | MATHDB-402835

Repeated halving of simplices

摘录 / EXCERPT: Start with a d-dimensional simplex. Subdivide the longest edge and cut the simplex into two simplices of half the area. Repeat the process recursively with both halves ad infinit...

open_problem | open | MathDB 页面 | 原文线索

V2-079416 | MATHDB-402837

4-polytopes with dense graphs

摘录 / EXCERPT: Suppose a simplicial 4-polytope P with n vertices has the property that, among every three vertices, at least two of them are joined by an edge. Does it follow that the graph o...

open_problem | open | MathDB 页面 | 原文线索

V2-079417 | MATHDB-402839

Minimum number of partial triangulations

摘录 / EXCERPT: A partial triangulation of a set of n points in the plane is a triangulation of the convex hull that may use interior points as vertices, but does not have to use all of them. Wh...

open_problem | open | MathDB 页面 | 原文线索

V2-079418 | MATHDB-402840

Bichromatic triangles in arrangements of pseudolines

摘录 / EXCERPT: A pseudoline is a non-self-intersecting infinite curve in $\mathbb{R}^2$ dividing the plane into two connected components. A simple arrangement of pseudolines is a set of pseudolin...

open_problem | open | MathDB 页面 | 原文线索

V2-079419 | MATHDB-402841

Collapsibility of the compact part of a polyhedral subdivision

摘录 / EXCERPT: Take a polyhedral subdivision of $\mathbb{R}^3$ into finitely many parts, none of which contains a line, and look at the union of all bounded faces. This set is contractible. Is it...

open_problem | open | MathDB 页面 | 原文线索

V2-079420 | MATHDB-402842

Extending piecewise-linear maps from the boundary to the interior in a continuous manner

摘录 / EXCERPT: Fix a reference triangle ABC. Given a piecewise-linear (PL) one-to-one map $f\colon \partial ABC\to\mathbb{R}^2$-equivalently, a PL parameterization of a simple polygon-can it be...

open_problem | open | MathDB 页面 | 原文线索

V2-079421 | MATHDB-402849

WLP for almost complete intersections generated by uniform powers of general linear forms

摘录 / EXCERPT: For positive integers n,m,d, let $\ell_1,\ldots,\ell_m$ be general linear forms in $k[x_1,\ldots,x_n]$ and set ... The report states the following conjecture: if $n\ge 9$ is odd, t...

open_problem | open | MathDB 页面 | 原文线索

V2-079422 | MATHDB-402850

Failure of the WLP in six sporadic cases

摘录 / EXCERPT: The report states that ... fails the Weak Lefschetz Property in the cases ... where the ℓ_i are general linear forms in the report’s setup.

open_problem | open | MathDB 页面 | 原文线索

V2-079423 | MATHDB-402852

Polynomiality of the generic symmetric-matrix ML degree

摘录 / EXCERPT: Let a be a positive integer, and let $\mathcal L$ be a general a-dimensional linear subspace of the space of $n\times n$ symmetric matrices. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-079424 | MATHDB-402853

Log-concavity of the coefficients of the complete-quadrics polynomial

摘录 / EXCERPT: For a fixed positive integer a, let $\phi(a,n)$ denote the polynomial in n given by the degree of the inverse of a general a-dimensional linear space of symmetric $n\times n$...

open_problem | open | MathDB 页面 | 原文线索

V2-079425 | MATHDB-402858

Eventual linearity of projective dimension and regularity for symmetric-invariant ideal chains

摘录 / EXCERPT: Let K be a field and let $c\ge 1$. For $n\ge 1$, set $R_n=K[X_{c\times n}]=K[X_{i,j}:1\le i\le c,\ 1\le j\le n]$, with the action of $\operatorname{Sym}(n)$ given by...

open_problem | open | MathDB 页面 | 原文线索

V2-079426 | MATHDB-402859

Eventual Lefschetz behavior of Sym-invariant chains of Artinian ideals

摘录 / EXCERPT: Let [1m$(I_n)_{n en1}$[0m be a [1mSym-invariant chain of Artinian ideals[0m. Decide whether either [1mI_n[0m has the SLP/WLP for all [1m$n\gg0$[0m or [1mI_n[0m does not have th...

open_problem | open | MathDB 页面 | 原文线索

V2-079427 | MATHDB-402861

Persistence for integrals of general self-similar Markov processes

摘录 / EXCERPT: Can one study the persistence problem for the integral of a general self-similar Markov process?

open_problem | open | MathDB 页面 | 原文线索

V2-079428 | MATHDB-402862

Existence of persistence exponents for Gaussian stationary processes

摘录 / EXCERPT: Let f be a Gaussian stationary process over $\mathbb{R}$ with spectral measure ρ having a continuous positive density on $\[-a,a]$ for some $a>0$. For every...

open_problem | open | MathDB 页面 | 原文线索

V2-079429 | MATHDB-402863

Radius of convergence for the MA(1) persistence-exponent series

摘录 / EXCERPT: For the moving-average process $X_n=\rho\xi_{n-1}+\xi_n$, $n\in\mathbb N$, where $(\xi_i)_{i\ge -1}$ are i.i.d. standard normally distributed random variables and...

open_problem | open | MathDB 页面 | 原文线索

V2-079430 | MATHDB-402864

Exact growth rate of the persistence exponent $\theta(p,H)$

摘录 / EXCERPT: For polynomial weights $\sigma(i)\sim i^p$ and correlations $\rho(i)\sim \kappa i^{2H-2}$ with $H\in(1/2,1)$, determine the exact rate of growth of the persistence exponent...

open_problem | open | MathDB 页面 | 原文线索

V2-079431 | MATHDB-402865

A characterization of the finiteness of perpetual integrals of Lévy processes

摘录 / EXCERPT: Let $\xi=(\xi_s)_{s\ge 0}$ be a one-dimensional Lévy process such that $\lim_{s\to\infty}\xi_s=\infty$ almost surely, and let ... where f is a non-negative, locally bounded measura...

open_problem | open | MathDB 页面 | 原文线索

V2-079432 | MATHDB-402866

Uniqueness of quasi-stationary distributions for absorbed one-dimensional diffusions

摘录 / EXCERPT: For one-dimensional diffusions on $[0,\infty)$ absorbed at 0, is there a unique quasi-stationary distribution if and only if ∞ is an entrance boundary?

open_problem | open | MathDB 页面 | 原文线索

V2-079433 | MATHDB-402867

Necessary and sufficient conditions for vanishing free energy

摘录 / EXCERPT: What are necessary and sufficient conditions on a general distribution μ_0 on $\mathbb{R}_+$ such that $F_\infty(\mu_0)=0$?

open_problem | open | MathDB 页面 | 原文线索

V2-079434 | MATHDB-402869

Extension of the ant-process theorems to broader graph classes

摘录 / EXCERPT: Can Theorems 1 and 3 be extended to a larger class of graphs, or to all graphs?

open_problem | open | MathDB 页面 | 原文线索

V2-079435 | MATHDB-402870

Ballisticity of the infinite-memory convex-hull-avoiding walk

摘录 / EXCERPT: For the $k=\infty$ model-a variation of the walk of Angel et al. that avoids the convex hull of its entire previous trajectory-is the walk ballistic, meaning that it has a positive...

open_problem | open | MathDB 页面 | 原文线索

V2-079436 | MATHDB-402871

Complex structure on \(\mathbb{CP}^3\) with vanishing second Chern class

摘录 / EXCERPT: Does there exist a complex structure on $\mathbb{CP}^3$ with vanishing second Chern class?

open_problem | open | MathDB 页面 | 原文线索

V2-079437 | MATHDB-402873

Metric independence of \\bar{\partial}-harmonic Hodge numbers

摘录 / EXCERPT: Are the Hodge-theoretic numbers defined via $\bar{\partial}$-harmonic forms independent of the choice of almost-Hermitian metric?

open_problem | open | MathDB 页面 | 原文线索

V2-079438 | MATHDB-402879

Dependence of realizability on the multiplication in dimensions divisible by four

摘录 / EXCERPT: Can one remove, in the case $n \equiv 4 \pmod 8$, the assumption that either $c_1 \ne 0$ or that the Chern numbers satisfy a stronger set of congruences than those of almost comple...

open_problem | open | MathDB 页面 | 原文线索

V2-079439 | MATHDB-402882

Examples beyond the Proposition 3.21 construction

摘录 / EXCERPT: Are there compact, 4-dimensional examples of non-integrable almost complex structures J with $h_J^- \ge 2$ other than those arising from [3, Proposition 3.21]? In particular, are...

open_problem | open | MathDB 页面 | 原文线索

V2-079440 | MATHDB-402885

Hilbert’s unpublished 24th problem: simplicity of proofs

摘录 / EXCERPT: Develop a criterion for the simplicity of proofs.

open_problem | open | MathDB 页面 | 原文线索

V2-079441 | MATHDB-402886

Ackermann Function, Binary Veblen Function, and \(\mathrm{ATR}_0\)

摘录 / EXCERPT: Over the system RCA_0, the following assertions are equivalent: (1) the Ackermann function extends to a dilator; (2) the binary Veblen function is a dilator; (3)...

open_problem | open | MathDB 页面 | 原文线索

V2-079442 | MATHDB-402887

Formalization of iterated multiplication in \(\mathrm{VTC}^{0}\)

摘录 / EXCERPT: Does VTC^{0} prove the axiom IMUL asserting the existence of iterated products $\prod_{i<n}X_i$ satisfying the defining recurrence ...

open_problem | open | MathDB 页面 | 原文线索

V2-079443 | MATHDB-402888

Well-foundedness of essentially undecidable theories below Robinson’s theory under strict interpretability

摘录 / EXCERPT: Let $\mathbf R$ be Robinson’s theory, and let $S \triangleleft T$ mean that S is interpretable in T but T is not interpretable in S. Define ... where $G1$ holds for a recursi...

open_problem | open | MathDB 页面 | 原文线索

V2-079444 | MATHDB-402889

Gödel’s second incompleteness theorem for Robinson’s theory R

摘录 / EXCERPT: Robinson’s theory $\mathbf R$ does not prove its own consistency: $\mathbf R \nvdash \operatorname{Con}(\mathbf R)$. Equivalently, Gödel’s second incompleteness theorem holds for...

open_problem | open | MathDB 页面 | 原文线索

V2-079445 | MATHDB-402890

Existence of globalisers for c.e. theories

摘录 / EXCERPT: Does every c.e. theory T have a globaliser? Yes. For every c.e. theory T, there exists a c.e. theory U such that, for every c.e. theory S, S is locally interpretable in...

open_problem | open | MathDB 页面 | 原文线索

V2-079446 | MATHDB-402891

Complete description of admissible rules for intuitionistic Gödel-Löb logic

摘录 / EXCERPT: Provide a full description of all admissible rules of intuitionistic Gödel-Löb logic iGL.

open_problem | open | MathDB 页面 | 原文线索

V2-079447 | MATHDB-402892

A Visser-rule basis for admissible rules of iSL

摘录 / EXCERPT: We strongly conjecture that we can extract a basis from our proof system for admissibility for iSL that is a kind of fusion of the Visser rules for IPC and the modal Visser rules f...

open_problem | open | MathDB 页面 | 原文线索

V2-079448 | MATHDB-402895

Scaling limit of superposed free-boundary dimer configurations

摘录 / EXCERPT: For two independent free-boundary dimer configurations in the upper half-plane, the superposition produces loops (including double edges) and curves joining monomers to monomers. D...

open_problem | open | MathDB 页面 | 原文线索

V2-079449 | MATHDB-402897

Benjamini-Schramm Conjecture 1 for planar graphs of minimum degree at least 7

摘录 / EXCERPT: Suppose that G is planar and has minimum degree at least 7. Then, for every $p\in\bigl(p_c^{\mathrm{site}}(G),1-p_c^{\mathrm{site}}(G)\bigr)$, there are almost surely infinitel...

conjecture | open | MathDB 页面 | 原文线索

V2-079450 | MATHDB-402901

Maximal number of central points in the α-cube move

摘录 / EXCERPT: In the setting of Theorem 4, for every real $\b1>1$, an $\b1$-immersion of $G_{\Delta}^{\diamond}$ may admit several possible central points after the cube move, but the number of...

open_problem | open | MathDB 页面 | 原文线索

V2-079451 | MATHDB-402905

Gradient flow realizing the Smale-conjecture retract for knot energies

摘录 / EXCERPT: It is still unclear whether there is a gradient flow of some knot energy that realizes a retract of the trivial loops to round circles.

conjecture | open | MathDB 页面 | 原文线索

V2-079452 | MATHDB-402906

Convergence of truncated-energy minimizers as the truncation vanishes

摘录 / EXCERPT: Prove the following work-in-progress assertion: let $q>2$, $\\lambda\in(0,\tfrac12]$, and $0<\delta<\eta$ with η sufficiently small. If...

open_problem | open | MathDB 页面 | 原文线索

V2-079453 | MATHDB-402907

Ribbonlength crossing-number problem for folded ribbon knots

摘录 / EXCERPT: Find positive constants c_1,c_2,α,β such that, for a knot or link type K, ...

open_problem | open | MathDB 页面 | 原文线索

V2-079454 | MATHDB-402908

Sub-\ ribbonlength upper bounds in terms of crossing number

摘录 / EXCERPT: Is it possible to find a different embedding of a knot such that $\operatorname{Rib}(K_w) \le O(\operatorname{Cr}(K)^p)$ for some constant $1 \le p < \frac{3}{2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079455 | MATHDB-402911

Optimal convergence rates for parabolic diffusion-coefficient identification

摘录 / EXCERPT: For the standard parabolic diffusion-coefficient inverse problem (2), under the hypotheses of Theorem 2 and using the fully discrete Tikhonov-Galerkin finite-element/backward-Euler...

open_problem | open | MathDB 页面 | 原文线索

V2-079456 | MATHDB-402912

Identification of the effective magnetic-field perturbation in MPI calibration

摘录 / EXCERPT: Given tuples (c,v) in the model-based magnetic-particle-imaging calibration problem, identify the function $\b4h$, where $\b4h$ encodes unknown or uncertain aspects of nanopartic...

open_problem | open | MathDB 页面 | 原文线索

V2-079457 | MATHDB-402913

Inverse coefficient problem for semilinear radiative transport with full internal data

摘录 / EXCERPT: Determine the triplet $(\sigma_a,\sigma_b,\sigma_s)$ in the semilinear radiative transport equation (1) from the data encoded in the map Λ_T defined in (5), where...

open_problem | open | MathDB 页面 | 原文线索

V2-079458 | MATHDB-402914

Recovery of a time-dependent source from a single-point measurement

摘录 / EXCERPT: Can one choose a single-point time measurement $m(t)=u(t,x_0)$, with $x_0\in\\Omega$, to determine the missing function $h(t)$?

open_problem | open | MathDB 页面 | 原文线索

V2-079459 | MATHDB-402915

Failure of uniqueness for a single-point measurement in an inverse source problem

摘录 / EXCERPT: For the inverse problem ... with homogeneous Dirichlet boundary conditions $u(t,0)=u(t,1)=0$, zero initial condition $u(0,x)=0$, and single-point data $u(t,x_0)=m(t)$ for...

open_problem | open | MathDB 页面 | 原文线索

V2-079460 | MATHDB-402916

Inverse reconstruction in lossy layered media from finitely many poles and residues

摘录 / EXCERPT: Assume that the poles of the transfer function $D(s)$ are simple. Given the first n poles $(\\lambda_j)_{j=1}^n$ and corresponding residues $(y_j)_{j=1}^n$ of $D(s)$, compute est...

open_problem | open | MathDB 页面 | 原文线索

V2-079461 | MATHDB-402917

Construction of discrete splines over triangulations

摘录 / EXCERPT: “Lastly, an open direction is to construct discrete splines over triangulations. The hope is to follow the construction of univariate discrete splines via iterated discrete integra...

open_problem | open | MathDB 页面 | 原文线索

V2-079462 | MATHDB-402918

Universality of the Gaussian proxy for max-margin random-feature classification

摘录 / EXCERPT: For max-margin classification, the report conjectures that the Gaussian proxy called the “noisy linear activation model” has the same asymptotics as the original random features mo...

open_problem | open | MathDB 页面 | 原文线索

V2-079463 | MATHDB-402919

Do sequentially interactive and non-interactive private minimax risks always coincide?

摘录 / EXCERPT: The question is whether $M^{(SI)}_{n,\alpha}$ and $M^{(NI)}_{n,\alpha}$ always coincide. The report answers negatively: if $X_1,\ldots,X_n$ are i.i.d. with density...

open_problem | open | MathDB 页面 | 原文线索

V2-079464 | MATHDB-402920

Rate of convergence of empirical rank maps

摘录 / EXCERPT: Determine the rate of convergence of the empirical rank maps.

open_problem | open | MathDB 页面 | 原文线索

V2-079465 | MATHDB-402923

Reducing redundancies in iterated-sums signatures over general semirings

摘录 / EXCERPT: How can redundancies in these iterated-sums be reduced? The usual procedure of “taking the logarithm” is not possible over a general semiring.

open_problem | open | MathDB 页面 | 原文线索

V2-079466 | MATHDB-402925

Uniqueness for rough 2D Euler with bounded initial vorticity

摘录 / EXCERPT: A related problem is to prove uniqueness in 2d under the relaxed condition $\omega_0\in L^\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-079467 | MATHDB-402926

Support theorem for singular SPDEs

摘录 / EXCERPT: Consider the class of equations on a torus of the form $\partial_tu=\Delta u+F(u,\nabla u)+G(u)\xi$, where u and ξ take values in a finite-dimensional vector space and ξ...

open_problem | open | MathDB 页面 | 原文线索

V2-079468 | MATHDB-402927

Hairer’s Reconstruction Theorem for coherent germs

摘录 / EXCERPT: Let $F:\mathbb{R}^{d}\to\mathcal{D}'(\mathbb{R}^{d})$. Suppose there exist $\gamma>0$, $\alpha\le 0$, and $\varphi\in\mathcal{D}(\mathbb{R}^{d})$ with $\int\varphi\ne 0$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-079469 | MATHDB-402930

Hambly-Lyons isometry conjecture for reduced bounded-variation paths

摘录 / EXCERPT: For every reduced bounded-variation path $\gamma:[0,T]\to\mathbb{R}^d$, the conjectured isometry is ... where $X^n_{0,T}$ is the n-th level of the signature $S(\gamma)_{0,T}$, and...

conjecture | open | MathDB 页面 | 原文线索

V2-079470 | MATHDB-402931

Optimal stopping with signatures

摘录 / EXCERPT: Given lin T((mathbb{R}^{1+d})^), define the signature stopping time ... Assuming $\mathbb{E}[\|Y\|_\infty]<\infty$, ... where $\mathcal S$ is the set of stopping times taking values in...

open_problem | open | MathDB 页面 | 原文线索

V2-079471 | MATHDB-402932

Extension of sharp hypoelliptic density estimates to other Gaussian processes

摘录 / EXCERPT: Is it possible to extend the result to other Gaussian processes?

open_problem | open | MathDB 页面 | 原文线索

V2-079472 | MATHDB-402934

Applicability of the conformal method to cosmological stability

摘录 / EXCERPT: How broad is the class of matter models for which the conformal method for the stability of cosmological solutions can be applied?

open_problem | open | MathDB 页面 | 原文线索

V2-079473 | MATHDB-402935

Generalizing trapped-surface formation to other equations of state

摘录 / EXCERPT: The trapped-surface formation result for spherically symmetric Einstein-Euler spacetimes has been established for linear equations of state with small sound speeds. Can this result...

open_problem | open | MathDB 页面 | 原文线索

V2-079474 | MATHDB-402936

Stability of axially symmetric stationary steady states

摘录 / EXCERPT: Axially symmetric stationary steady states have been studied numerically, see for instance [27, 1], and constructed [24], but the problem of their stability remains largely open.

open_problem | open | MathDB 页面 | 原文线索

V2-079475 | MATHDB-402937

Extend relativistic stability theory beyond spherical symmetry and linearization

摘录 / EXCERPT: The extension of [14] beyond spherical symmetry and linearization is a deeply challenging and wide-open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-079476 | MATHDB-402938

Convergence of the Galerkin method for a Laplacian double-layer equation

摘录 / EXCERPT: Does the Galerkin method converge for the standard second-kind double-layer integral equation for the Laplacian on Lipschitz domains?

open_problem | open | MathDB 页面 | 原文线索

V2-079477 | MATHDB-402939

Claimed fundamental matrix identity for the stabilized MFIE

摘录 / EXCERPT: For the RT and dual BC discretizations described in the report, the proposed stabilization relies on the identity ...

open_problem | open | MathDB 页面 | 原文线索

V2-079478 | MATHDB-402940

Quasi-optimality for nonconforming BEM spaces on fractal screens

摘录 / EXCERPT: What should replace the standard Céa's lemma quasi-optimality bound $\|\phi-\phi_{h_j}\|\le C m_j(\phi)$ in cases where $V_{h_j}\not\subset V$?

open_problem | open | MathDB 页面 | 原文线索

V2-079479 | MATHDB-402941

Convergence of BEM on the standard Cantor prefractal sequence

摘录 / EXCERPT: Let $n=2$, let $C\subset[0,1]$ be the Cantor set generated by $s_1(t)=\alpha t$ and $s_2(t)=\alpha t+1-\alpha$, with $0<\alpha<1/2$, and set $\Gamma=C\times\{0\}$. For the standard...

open_problem | open | MathDB 页面 | 原文线索

V2-079480 | MATHDB-402943

Fractional Dirichlet problem for the integral fractional Laplacian

摘录 / EXCERPT: For a domain $\Omega\subset\mathbb{R}^n$ and $s\in(0,1)$, the integral fractional Laplacian is defined by ... where $c_{n,s}=2^{2s}s\,\Gamma\!\left(\frac{n+2s}{2}\right)\!/\pi^{n/2}$...

open_problem | open | MathDB 页面 | 原文线索

V2-079481 | MATHDB-402945

Inf-sup stability of the spectral non-conforming Petrov-Galerkin multiple-traces discretization

摘录 / EXCERPT: For the operator $F(s)$ and spaces defined in Problem 1, let $X_L^\ell:=\operatorname{span}\{\lambda_n^\ell\}_{n=0}^L\subseteq V_{\mathrm{pw},\ell}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079482 | MATHDB-402946

Equivalence of functional a posteriori error bounds for BEM

摘录 / EXCERPT: Prove that the upper and lower bounds in (6), namely $\mu_h \le \|\nabla(u-u_h)\|_{L^2(\Omega)} \le \eta_h$ (up to data oscillation terms), are equivalent, at least up to oscillati...

open_problem | open | MathDB 页面 | 原文线索

V2-079483 | MATHDB-402947

Coercive-plus-compact decomposition of the Laplace double-layer operator

摘录 / EXCERPT: Given a bounded Lipschitz domain $\Omega\subset\mathbb{R}^d$, with $d=2$ or 3, and boundary Γ, can the operator $\frac12 I-D$ on $L^2(\Gamma)$ be written as the sum of a c...

open_problem | open | MathDB 页面 | 原文线索

V2-079484 | MATHDB-402948

Bounds for the corner exponent of the integral fractional Laplacian

摘录 / EXCERPT: For the corner exponent λ associated with the integral fractional Laplacian of order s, where $s yin(0,1)$, the report proves ... The report also states that both bounds ar...

open_problem | open | MathDB 页面 | 原文线索

V2-079485 | MATHDB-402949

Modal representation of scattering solution operators

摘录 / EXCERPT: Under which condition is it possible to represent the solution operator of scattering problems purely in terms of the eigenvalues and eigenmodes?

open_problem | open | MathDB 页面 | 原文线索

V2-079486 | MATHDB-402950

Ellipticity of the wave-equation boundary integral operator in higher dimensions

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded Lipschitz domain with $n=2,3$, and let V denote the weakly singular boundary integral operator for the wave equation on the space-tim...

open_problem | open | MathDB 页面 | 原文线索

V2-079487 | MATHDB-402951

Domains for the integration upper bound

摘录 / EXCERPT: For which domains D_d does the upper bound for $e_n(C^k(D_d),\mathrm{INT})$ hold?

open_problem | open | MathDB 页面 | 原文线索

V2-079488 | MATHDB-402952

Uniform lower bound for integration over the class $\bar C^2(D_d)$

摘录 / EXCERPT: Consider the integration problem for the class...

open_problem | open | MathDB 页面 | 原文线索

V2-079489 | MATHDB-402953

Curse of dimensionality for uniform approximation of the smaller infinitely differentiable class

摘录 / EXCERPT: It is not known whether the curse of dimensionality holds for L_∞-approximation of the smaller class $\widetilde{C}^{\infty}([0,1]^d)$, whose derivatives are bounded by 1.

open_problem | open | MathDB 页面 | 原文线索

V2-079490 | MATHDB-402954

Sharp bounds for approximation in $C_k([0,1]^d)$ when $k\ge 3$ is odd

摘录 / EXCERPT: For odd integers $k\ge 3$, determine the sharp dependence on d and n of the nth minimal worst-case L_∞-approximation error $e_n(C_k([0,1]^d),\mathrm{APP})$, closing th...

open_problem | open | MathDB 页面 | 原文线索

V2-079491 | MATHDB-402956

Vu’s conjecture on nonsingularity of random regular adjacency matrices

摘录 / EXCERPT: Let R_n be the adjacency matrix of a uniformly random d-regular directed graph on n vertices, equivalently a uniformly random $n\times n$ $0/1$ matrix having exactly d ones...

conjecture | open | MathDB 页面 | 原文线索

V2-079492 | MATHDB-402957

A large deviation principle for projections of \(\ell_p^n\)-balls

摘录 / EXCERPT: Let $p\in[1,\infty]$. For each n, let X_n be uniformly distributed on the unit ball $B_p^n\subset\mathbb R^n$, and let E_n be a random k_n-dimensional subspace of...

open_problem | open | MathDB 页面 | 原文线索

V2-079493 | MATHDB-402959

Polytopes generated by random vectors and associated floating bodies

摘录 / EXCERPT: Let X be a symmetric random vector in $\\mathbb{R}^n$, and let $\\|\\cdot\\|$ be a norm on $\\mathbb{R}^n$. Suppose that, for some $\\gamma>0$, $\\delta>0$, $r>0$, and $L>0$, the...

open_problem | open | MathDB 页面 | 原文线索

V2-079494 | MATHDB-402960

Sharpness of the summability condition for strong polynomial tractability

摘录 / EXCERPT: Let $\gamma=(\gamma_j)_{j\ge 1}$ be a sequence of product weights with $\gamma_j\in(0,1]$, and let p_γ be the infimum of the exponents q such that the minimal information...

open_problem | open | MathDB 页面 | 原文线索

V2-079495 | MATHDB-402961

Novak’s positive-semidefiniteness conjecture for a trigonometric matrix

摘录 / EXCERPT: For all integers $n,d\ge 2$ and all choices of $x_1,\ldots,x_n\in\mathbb{R}^d$, the matrix ... is positive semidefinite.

conjecture | open | MathDB 页面 | 原文线索

V2-079496 | MATHDB-402962

Expected facet number of spherical random polytopes on spherical wedges

摘录 / EXCERPT: Conjecture. Let $j\in\{1,\ldots,d\}$, and let $H_1,\ldots,H_j$ be hyperplanes through the origin of $\mathbb{R}^{d+1}$ that are in general position. Define ... where H_i^+ is the p...

open_problem | open | MathDB 页面 | 原文线索

V2-079497 | MATHDB-402963

Large deviations for random measures induced by random projections of the cube

摘录 / EXCERPT: Let b8^{(N)} be uniformly distributed on the Euclidean unit sphere $S^{N-1}\subset\mathbb{R}^N$. Let $X=(X_1,X_2,\ldots)$ have independent coordinates uniformly distributed on...

open_problem | open | MathDB 页面 | 原文线索

V2-079498 | MATHDB-402970

True distortion of the puncture class in a non-scalable space

摘录 / EXCERPT: For the punctured space $M=S^2\times(\mathbb{CP}^2)^{\#4}$ and the class $\alpha\in\pi_5(M)$ of the puncture, what is the true distortion of α? Can one find a lower bound fo...

open_problem | open | MathDB 页面 | 原文线索

V2-079499 | MATHDB-402971

Fibre-sequence extension for Grothendieck-Witt theory of general Poincaré categories

摘录 / EXCERPT: For every Poincaré category $(\mathcal C,Q)$, there is a natural fibre sequence of spectra ... In particular, for every ring R and form parameter Λ, with...

open_problem | open | MathDB 页面 | 原文线索

V2-079500 | MATHDB-402976

CAT(0) metric on the Bestvina complex

摘录 / EXCERPT: Does the Bestvina complex support a G-invariant CAT(0) metric?

open_problem | open | MathDB 页面 | 原文线索

V2-079501 | MATHDB-402978

Collapse of the Lyndon-Serre spectral sequence for a crystallographic semidirect product

摘录 / EXCERPT: Fix a homomorphism $\rho\colon \mathbb{Z}/m\to\operatorname{Aut}(\mathbb{Z}^n)$ such that the induced $\mathbb{Z}/m$-action on $\mathbb{Z}^n$ is free outside the origin, and set...

open_problem | open | MathDB 页面 | 原文线索

V2-079502 | MATHDB-402979

Hierarchical hyperbolicity of graphs of multicurves

摘录 / EXCERPT: Let S be a connected, compact, orientable surface. Let $G(S)$ be a graph whose vertices are multicurves in S, satisfying: - $G(S)$ is connected; - the action of...

open_problem | open | MathDB 页面 | 原文线索

V2-079503 | MATHDB-402985

Handle number and minimal-genus Seifert surfaces

摘录 / EXCERPT: Is the handle number of a knot always realized by a minimal-genus Seifert surface?

open_problem | open | MathDB 页面 | 原文线索

V2-079504 | MATHDB-402986

Additivity of the Morse-Novikov number under connected sum

摘录 / EXCERPT: If $K=K_a\#K_b$ is a connected sum of two oriented knots K_a and K_b in S^3, then $\operatorname{MN}(K)=\operatorname{MN}(K_a)+\operatorname{MN}(K_b)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-079505 | MATHDB-402989

Pseudo-meridians and elements trivialized by no nontrivial Dehn filling

摘录 / EXCERPT: Let K be a hyperbolic knot and let $g\in G(K)$ be a nontrivial element that represents a generator of $H_1(E(K))$ and satisfies $D(g)=0$. Is g a pseudo-meridian?

open_problem | open | MathDB 页面 | 原文线索

V2-079506 | MATHDB-402990

Free subgroup avoiding all slope normal closures

摘录 / EXCERPT: Let K be a hyperbolic knot which admits no finite surgery. Does there exist a subgroup $H\subset G(K)$ which is a rank-two free group and satisfies ...

open_problem | open | MathDB 页面 | 原文线索

V2-079507 | MATHDB-402992

Integral universality of embedding-calculus knot invariants

摘录 / EXCERPT: Conjecture 3. For the standard unknot-based long-knot space, the map $\pi_0\mathrm{ev}_n\colon K(I^3)\to\pi_0T_n(I,I^3)$ is a universal additive Vassiliev invariant of type...

open_problem | open | MathDB 页面 | 原文线索

V2-079508 | MATHDB-402993

Rational form of Conjecture 3 for embedding-calculus knot invariants

摘录 / EXCERPT: For every $n\ge 1$, the morphism ... is an isomorphism after tensoring both sides with $\mathbb{Q}$; equivalently, $(\pi_0\operatorname{ev}_n)\otimes\mathbb{Q}$ is an isomorphism. He...

conjecture | open | MathDB 页面 | 原文线索

V2-079509 | MATHDB-402997

Sliceness of the positive Whitehead double of the left-handed trefoil

摘录 / EXCERPT: Is the positive Whitehead double of the left-handed trefoil slice?

open_problem | open | MathDB 页面 | 原文线索

V2-079510 | MATHDB-402999

Conjectural tiling of the complement of the Riley slice

摘录 / EXCERPT: Conjecture 4. The complement of $\overline{R}$ admits a natural tiling $\{R_c(r)\}_{r\in\mathbb{Q}/2\mathbb{Z}}$, as shown in Figure 3. For each r, let r^* be the Farey neighbo...

open_problem | open | MathDB 页面 | 原文线索

V2-079511 | MATHDB-403000

Saxl conjecture for the staircase character

摘录 / EXCERPT: The Saxl conjecture claims that all Kronecker coefficients for the square of the staircase character are positive.

conjecture | open | MathDB 页面 | 原文线索

V2-079512 | MATHDB-403003

Restriction to Sylow conjecture (Navarro-Tiep, 2020)

摘录 / EXCERPT: There is a bijection ... such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-079513 | MATHDB-403005

Coefficients in Schur-function plethysm

摘录 / EXCERPT: Find the coefficients $\langle s_\nu \circ s_\mu, s_\lambda\rangle$ in the decomposition of the plethysm $s_\nu \circ s_\mu$ as a linear combination of Schur functions.

open_problem | open | MathDB 页面 | 原文线索

V2-079514 | MATHDB-403009

Multiplicity obstruction for products versus power sums

摘录 / EXCERPT: Let $m\ge 3$, $n\ge 2$, $k=d=n+1$, and $\lambda=(n^2-2,n,2)$. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-079515 | MATHDB-403011

Kronecker positivity for a family of 2-separated partitions

摘录 / EXCERPT: Let $w=k(k+1)/2$, $n=w(2w+1)$, $\rho(2w)=(2w,2w-1,\ldots,1)$, $\tau=\rho(2w-1)=(2w-1,2w-2,\ldots,1)$, and $\rho(k)=(k,k-1,\ldots,1)$. For any pair of partitions λ,μ such...

open_problem | open | MathDB 页面 | 原文线索

V2-079516 | MATHDB-403012

Saxl’s conjecture

摘录 / EXCERPT: Let $\rho=(k,k-1,k-2,\ldots,2,1)\vdash n$ be a staircase partition. Then ... with multiplicity $g(\rho,\rho,\lambda)\ne 0$ for all partitions λ of n.

conjecture | open | MathDB 页面 | 原文线索

V2-079517 | MATHDB-403016

Unique factorisation of Littlewood-Richardson products

摘录 / EXCERPT: Given partitions λ,μ and α,β, we have $\chi_\lambda * \chi_\mu = \chi_\alpha * \chi_\beta$ if and only if $\alpha=\lambda$ and $\mu=\beta$ (up to reordering of...

open_problem | open | MathDB 页面 | 原文线索

V2-079518 | MATHDB-403017

Classification of multiplicity-free Kronecker products of symmetric-group characters

摘录 / EXCERPT: Let λ and μ be partitions indexing irreducible characters χ^λ and χ^μ of S_n. The Kronecker product $\chi^\lambda\otimes\chi^\mu$ is multiplicity-fr...

open_problem | open | MathDB 页面 | 原文线索

V2-079519 | MATHDB-403018

Irreducibility of tensor products of representations

摘录 / EXCERPT: When can the tensor product $A \otimes B$ of two S-representations be irreducible?

open_problem | open | MathDB 页面 | 原文线索

V2-079520 | MATHDB-403019

Thompson’s conjecture on the alpha-invariant

摘录 / EXCERPT: There exists a constant $C>0$ such that $\[?]$

conjecture | open | MathDB 页面 | 原文线索

V2-079521 | MATHDB-403020

Larsen's eighth-moment conjecture

摘录 / EXCERPT: Let $V=\mathbb{C}^d$ with $d greater than or equal to 5$, and let G be one of $\mathrm{GL}(V)$, $\mathrm{Sp}(V)$, or $\mathrm{GO}(V)$. Let $X\leq G$ be a Zariski-closed subgroup...

conjecture | open | MathDB 页面 | 原文线索

V2-079522 | MATHDB-403022

Irreducible tensor products for alternating groups in characteristic 2

摘录 / EXCERPT: Let F be a field of characteristic p, and let V and W be irreducible representations of the double cover $\widetilde{A}_n$, neither of which is 1-dimensional. In the rema...

open_problem | open | MathDB 页面 | 原文线索

V2-079523 | MATHDB-403023

Geometric or invariant-theoretic interpretation of a plethystic isomorphism

摘录 / EXCERPT: Find a geometric or invariant theory interpretation of the isomorphism ... where E is the two-dimensional complex vector space used in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-079524 | MATHDB-403025

Countability of finitely homogeneous NIP structures

摘录 / EXCERPT: It is conjectured that there are countably many finitely homogeneous NIP structures.

open_problem | open | MathDB 页面 | 原文线索

V2-079525 | MATHDB-403026

NIP structures with the strict order property and ample generics

摘录 / EXCERPT: Let M be an $\[0m\omega$-categorical ultrahomogeneous structure. If M has NIP and the strict order property-that is, M defines a partial order with infinite chains-must...

open_problem | open | MathDB 页面 | 原文线索

V2-079526 | MATHDB-403029

The WNU tractability conjecture for constraint satisfaction problems

摘录 / EXCERPT: In 2007 it was conjectured that the Constraint Satisfaction Problem (CSP) over a constraint language Γ is tractable if and only if Γ is preserved by a weak near-unani...

conjecture | open | MathDB 页面 | 原文线索

V2-079527 | MATHDB-403032

Convergence of the Lagrangian discretization for less regular solutions

摘录 / EXCERPT: It is an open question whether convergence of the Lagrangian discretization holds for less regular solutions, in particular when $-\nabla U'(\rho)$ is nonzero at the boundary of th...

open_problem | open | MathDB 页面 | 原文线索

V2-079528 | MATHDB-403033

Point-vortex approximation under weak concentration and unbounded vorticity

摘录 / EXCERPT: Theorem 1. Suppose that, for each vortex component, the initial vorticity satisfies $W_{2}\!\left(\frac{\omega_i^0}{\lambda_i},\delta_{X_i^0}\right)\lesssim\varepsilon$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079529 | MATHDB-403034

Rigorous proof of the energy identity for the convexified cross-diffusion flow

摘录 / EXCERPT: Rigorously prove the formal identity ... for curves (ρ_t,μ_t) that satisfy $(\rho_t,\mu_t)\in A$ only for almost every t, under suitable integrability assumptions on the rel...

open_problem | open | MathDB 页面 | 原文线索

V2-079530 | MATHDB-403035

Link between long-time behavior of MFG systems and aggregation/kinetic equations

摘录 / EXCERPT: Understand the link, if any, between the long-time behavior of the MFG systems and that of the corresponding aggregation/kinetic equation.

open_problem | open | MathDB 页面 | 原文线索

V2-079531 | MATHDB-403036

Quantitative stability of Brenier maps in the compact-target case

摘录 / EXCERPT: Let X be a compact convex subset of $\dmathbb{R}^d$ with unit volume, let $\dmathbb{rho}$ be the restriction of Lebesgue measure to X, and let $Y\subset\mathbb{R}^d$ be another...

open_problem | open | MathDB 页面 | 原文线索

V2-079532 | MATHDB-403037

Existence of Monge-type minimizers for multi-marginal Coulomb optimal transport

摘录 / EXCERPT: Whether Monge-type minimizers always exist for the multi-marginal optimal transport problem with Coulomb cost is an open problem.

open_problem | open | MathDB 页面 | 原文线索

V2-079533 | MATHDB-403039

Next-order asymptotics for the Coulomb and Riesz gas minimization problem

摘录 / EXCERPT: Let $d\ge 1$ and $0<s<d$, with interaction kernel $c(x)=|x|^{-s}$. Assume that $V:\mathbb R^d\to\mathbb R$ is continuous, $V(x)\to\infty$ as $|x|\to\infty$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-079534 | MATHDB-403043

Extension of Theorem 1 to more general variational problems

摘录 / EXCERPT: Does Theorem 1 hold for variational problems more general than (1)?

open_problem | open | MathDB 页面 | 原文线索

V2-079535 | MATHDB-403044

Removing the tail-condition hypothesis in constructive Gagliardo-Nirenberg stability

摘录 / EXCERPT: Can the stability conclusion of Theorem 1 be proved without the assumption $A[f^{2p}]<\infty$; equivalently, can one obtain the stated constructive stability estimate for the admis...

conjecture | open | MathDB 页面 | 原文线索

V2-079536 | MATHDB-403046

Third critical point of the McKean-Vlasov free energy at the transition point

摘录 / EXCERPT: Let $F_\kappa(\rho)=\sigma\int_{\mathbb T_L^d}\rho\log\rho\,dx+\frac{\kappa}{2}\int_{\mathbb T_L^d\times\mathbb T_L^d}W(x-y)\rho(y)\rho(x)\,dy\,dx$, with $\rho_\infty\equiv L^{-d}$...

open_problem | open | MathDB 页面 | 原文线索

V2-079537 | MATHDB-403049

Convergence of nonstandard finite-volume fluxes without continuity or TVD conditions

摘录 / EXCERPT: How can a finite-volume flux which does not respect the usual continuity condition [6] (nor any kind of TVD condition [2]) be convergent?

open_problem | open | MathDB 页面 | 原文线索

V2-079538 | MATHDB-403050

Equality of the candidate Aviles-Giga limit with the jump energy

摘录 / EXCERPT: Prove that ... for every u such that $\nabla^\perp u$ is a weak solution to (3) with finite entropy production.

open_problem | open | MathDB 页面 | 原文线索

V2-079539 | MATHDB-403051

Aviles-Giga minimizer-limit conjecture

摘录 / EXCERPT: Conjecture: Limits as $\varepsilon \to 0$ of minimizers of F_ε are solutions u of the eikonal equation $|\nabla u|=1$ which minimize ... where $J_{\nabla u}$ denotes th...

conjecture | open | MathDB 页面 | 原文线索

V2-079540 | MATHDB-403052

Entropy convergence of Godunov approximations for hyperbolic balance laws

摘录 / EXCERPT: How to prove the entropy convergence of the Godunov approximate solutions to the system of hyperbolic balance laws?

open_problem | open | MathDB 页面 | 原文线索

V2-079541 | MATHDB-403057

Main Open Image Theorem conjecture for Modular Towers

摘录 / EXCERPT: The Main OIT Conjecture asserts that the geometric monodromy G_{MT} of a Modular Tower, and the intersection of the arithmetic monodromy $A_{\mathrm{MT},J'}$ of the fibe...

conjecture | open | MathDB 页面 | 原文线索

V2-079542 | MATHDB-403058

Hilbert-Schinzel specialization property over near UFDs

摘录 / EXCERPT: Let Z be a near UFD, let $Q=\operatorname{Frac}(Z)$ be a field with a product formula, and assume that Q has characteristic 0 or is imperfect. Let $t=(t_1,\ldots,t_k)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-079543 | MATHDB-403059

Schinzel Hypothesis

摘录 / EXCERPT: Let $P_1(t),\ldots,P_s(t)$ be $s\ge 1$ polynomials, irreducible in $\mathbb Z[t]$, such that the product $P_1\cdots P_s$ has no fixed divisor in $\mathbb Z$ with respect to t. Th...

conjecture | open | MathDB 页面 | 原文线索

V2-079544 | MATHDB-403065

Section conjecture for open curves

摘录 / EXCERPT: If U/k is a smooth, not necessarily proper, hyperbolic curve with compactification X, then every section of the map $\[1m\pi_1(U)\to G_k\[0m$ is induced by a unique k-rationa...

conjecture | open | MathDB 页面 | 原文线索

V2-079545 | MATHDB-403068

Liftable section conjecture for certain localisations of curves

摘录 / EXCERPT: Let k be a finite extension of $\mathbb{Q}_p$ containing the p-th roots of unity. Let X/k be a smooth, proper, geometrically connected curve and let...

conjecture | open | MathDB 页面 | 原文线索

V2-079546 | MATHDB-403069

Exceptional sets in Hilbert’s irreducibility theorem

摘录 / EXCERPT: For an irreducible polynomial $F(t,X)\in\mathbb{Q}[t,X]$ of degree at least 1 in X, determine the set ... up to a finite set.

open_problem | open | MathDB 页面 | 原文线索

V2-079547 | MATHDB-403072

Exceptional jumps of Picard ranks of reductions of K3 surfaces

摘录 / EXCERPT: Let K be a number field and let X/K be a K3 surface with potentially good reduction everywhere. Then there are infinitely many primes p of K such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-079548 | MATHDB-403081

Irreducibility of Bernoulli random polynomials

摘录 / EXCERPT: In the restricted coefficient model, where the coefficients are chosen independently from a fixed distribution bc and the degree tends to infinity, suppose that bc is the Bernoulli...

open_problem | open | MathDB 页面 | 原文线索

V2-079549 | MATHDB-403085

Triviality of the coboundary map for Fermat curves at primes $p\ge 7$

摘录 / EXCERPT: For the Fermat curve setting described in the report, with $H=H^1(X)\otimes\mathbb Z/p$ and $\pi_2/\pi_3=[\pi_1(X)]_2/[\pi_1(X)]_3\otimes\mathbb Z/p$, the short exact sequence ... in...

open_problem | open | MathDB 页面 | 原文线索

V2-079550 | MATHDB-403088

Variational use of learned explicit operator corrections

摘录 / EXCERPT: The report asks whether a corrected model A_Θ can subsequently be used in a variational regularisation approach so that the solutions for the corrected and accurate operator...

open_problem | open | MathDB 页面 | 原文线索

V2-079551 | MATHDB-403089

Evaluating generative models’ reconstruction of entire distributions in inverse problems

摘录 / EXCERPT: “Evaluating a generative model’s ability to reconstruct an entire distribution, in the context of inverse problems, is an area of future work.”

open_problem | open | MathDB 页面 | 原文线索

V2-079552 | MATHDB-403091

Quantitative convergence rates for truncated corrector approximations

摘录 / EXCERPT: Find a rate of convergence, in terms of the truncation size, for the approximation of both the homogenized coefficient $\hat A$ and the solution u_ε in the general dete...

open_problem | open | MathDB 页面 | 原文线索

V2-079553 | MATHDB-403092

Quantitative homogenization beyond periodicity

摘录 / EXCERPT: A1. Study the approximation of homogenized coefficients and the rates of convergence by establishing:

open_problem | open | MathDB 页面 | 原文线索

V2-079554 | MATHDB-403093

Exact asymptotic minimax error for phase synchronization

摘录 / EXCERPT: In the additive phase-synchronization model $Y_{jk}=z_j^*\overline{z_k^*}+\sigma W_{jk}$ for $1\le j<k\le n$, where $z_j^*\in\mathbb C_1=\{x\in\mathbb C:|x|=1\}$ and the W_{jk} a...

open_problem | open | MathDB 页面 | 原文线索

V2-079555 | MATHDB-403095

Convergent-series representation for the gravitational N-body problem

摘录 / EXCERPT: The report recalls the question of representing solutions of the gravitational N-body equations by convergent series. It states that this question was solved for $N=3$ by Sundman...

open_problem | open | MathDB 页面 | 原文线索

V2-079556 | MATHDB-403102

Uniqueness of the collision kernel in the nonlinear Boltzmann equation

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a bounded domain with smooth boundary $\partial\Omega$, where $n\ge 2$. Consider the nonlinear Boltzmann boundary-value problem ... with collision...

open_problem | open | MathDB 页面 | 原文线索

V2-079557 | MATHDB-403103

Well-definedness of the Keller-Segel Bayesian posterior

摘录 / EXCERPT: Under the report’s admissibility assumptions-K_0,K_1 continuously differentiable and bounded, K_0 symmetric in (v,v') and bounded away from zero, and K_1 antisymmetric in...

open_problem | open | MathDB 页面 | 原文线索

V2-079558 | MATHDB-403104

Convergence of Bayesian posterior measures for kinetic chemotaxis and Keller-Segel inverse models

摘录 / EXCERPT: Assume that K_0 and K_1 are continuously differentiable and bounded, that K_0 is symmetric in (v,v') and bounded away from zero, and that K_1 is antisymmetric in (v,v')...

open_problem | open | MathDB 页面 | 原文线索

V2-079559 | MATHDB-403105

Strong aperiodicity of substitution hulls

摘录 / EXCERPT: It is an open question whether $\Omega(S)$ is strongly aperiodic whenever S_0 is non-periodic and primitive.

open_problem | open | MathDB 页面 | 原文线索

V2-079560 | MATHDB-403108

Disconnected reductive overgroups of regular unipotent elements

摘录 / EXCERPT: Determine, up to conjugacy, all positive-dimensional disconnected reductive subgroups $X<G$ such that X° is not a torus and X contains a regular unipotent element of G...

open_problem | open | MathDB 页面 | 原文线索

V2-079561 | MATHDB-403118

Weyl equivariance of equivariant multiplicities of weight-zero MV cycles

摘录 / EXCERPT: For any dominant $\lambda\in X_*(T)$, the linear map ... that extends the assignment $v_Z\mapsto e^{Z}_{L_0}$ is equivariant under the action of the Weyl group.

open_problem | open | MathDB 页面 | 原文线索

V2-079562 | MATHDB-403120

Comparison of closed choice and Cantor-Bendixson problems on Baire space

摘录 / EXCERPT: Is $C_{\\mathbb{N}^{\\mathbb{N}}} \leq_W CB_{\\mathbb{N}^{\\mathbb{N}}}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079563 | MATHDB-403121

Existence of a properly 2-c.e. degree relatively enumerable in a noncomputable low c.e. degree

摘录 / EXCERPT: Given a noncomputable low c.e. Turing degree, does there exist a properly 2-c.e. Turing degree which is above it and relatively enumerable in it?

open_problem | open | MathDB 页面 | 原文线索

V2-079564 | MATHDB-403125

Computable enumerability of the Mandelbrot set as a closed set

摘录 / EXCERPT: Is the Mandelbrot set a c.e. closed set?

open_problem | open | MathDB 页面 | 原文线索

V2-079565 | MATHDB-403126

Whether Martin-Löf and Kolmogorov-Loveland randomness coincide

摘录 / EXCERPT: Do Martin-Löf randomness and Kolmogorov-Loveland randomness coincide, where Kolmogorov-Loveland randomness is defined using computable non-monotonic adaptive martingales?

open_problem | open | MathDB 页面 | 原文线索

V2-079566 | MATHDB-403127

Characterization of admissible sets by fixed points of the jump operator

摘录 / EXCERPT: Conjecture: For every admissible set A, the following conditions are equivalent: 1. $A \equiv_{\Sigma} J(A)$, where $J(A)=(\mathrm{HF}(M_A),P)$ is a jump structure of A, with...

open_problem | open | MathDB 页面 | 原文线索

V2-079567 | MATHDB-403128

Arithmetical enumeration degrees with persistent skip zig-zag behavior

摘录 / EXCERPT: Does there exist an arithmetical enumeration degree a such that, for every $n\in\omega$, the n-th skip of a is not below the $(n+1)$-st skip of a? Equivalently, is there an...

open_problem | open | MathDB 页面 | 原文线索

V2-079568 | MATHDB-403130

Avoiding a properly 2-c.e. degree while capturing its c.e. predecessors

摘录 / EXCERPT: Given a 2-c.e. set D of properly 2-c.e. Turing degree, does there exist a c.e. set A such that $D\nleq_T A$ and, for every c.e. set W, if $W\leq_T D$, then $W\leq_T A$?

open_problem | open | MathDB 页面 | 原文线索

V2-079569 | MATHDB-403131

Efficient computation of coherent-family eigenfunctions in higher dimensions

摘录 / EXCERPT: Whether there exist methods that allow an efficient solution of the eigenproblem in higher dimensions.

open_problem | open | MathDB 页面 | 原文线索

V2-079570 | MATHDB-403133

Design problem for the saturated Heavy Ball method

摘录 / EXCERPT: Given a desired performance level $\\rho\in(0,1)$, design the parameters $\\alpha$ and $\\beta$-equivalently, the gain $K_0=[\alpha\ \beta]$-for the Heavy Ball closed-loop system w...

open_problem | open | MathDB 页面 | 原文线索

V2-079571 | MATHDB-403136

Relation between mixing decay and control actuation in nonlinear closed-loop flows

摘录 / EXCERPT: Establish the relation between the mixing decay rate and the control actuation for the nonlinear closed-loop system.

open_problem | open | MathDB 页面 | 原文线索

V2-079572 | MATHDB-403137

Generalized Siu’s lemma with trivial multiplier ideal sheaf

摘录 / EXCERPT: The report states that if $\varphi(z',z'')$ is plurisubharmonic and $P(z',z'')$ is a nonnegative continuous function on $B_r^1\times B_r^{n-1}$, then ... The report says that this re...

open_problem | open | MathDB 页面 | 原文线索

V2-079573 | MATHDB-403138

Green-Griffiths-Lang conjecture with optimal logarithmic/orbifold positivity

摘录 / EXCERPT: Does the Green-Griffiths-Lang conjecture hold under the optimal positivity conditions, namely assuming only bigness of the logarithmic or orbifold canonical sheaf?

conjecture | open | MathDB 页面 | 原文线索

V2-079574 | MATHDB-403139

The Dirichlet problem for the homogeneous complex Monge-Ampère equation (local setting)

摘录 / EXCERPT: Let $D\subseteq\mathbb{C}^n$ be a smoothly bounded, strictly pseudoconvex domain and let $f\in C^\infty(\partial D)$. Find $u\in\operatorname{PSH}(D)\cap C(\overline D)$ such that...

open_problem | open | MathDB 页面 | 原文线索

V2-079575 | MATHDB-403140

Smoothness of the potential associated with a Hele-Shaw domain family

摘录 / EXCERPT: Given a nice increasing family of domains (Ω_t), the construction defines φ by ... Can one ensure that the resulting φ is smooth, at least if one allows...

open_problem | open | MathDB 页面 | 原文线索

V2-079576 | MATHDB-403143

Compactness characterization for Hankel operators with L^2 symbols on strongly pseudoconvex domains

摘录 / EXCERPT: Suppose $\d4aa\subset\mathbb{C}^d$ is a strongly pseudoconvex domain and $\d719\in L^2(\u0001d4aa)$. Let $H_{\u0001d719}$ be the associated Hankel operator, initially defined on...

open_problem | open | MathDB 页面 | 原文线索

V2-079577 | MATHDB-403145

Sup-norm estimates for the \(\bar\partial\)-equation on finite-type domains

摘录 / EXCERPT: Let $\Omega\subset\mathbb C^n$ and let f be a bounded $(0,1)$-form such that $\bar\partial f=0$. Can one find a function u such that $\bar\partial u=f$ and ... where $C(\Omega)$...

open_problem | open | MathDB 页面 | 原文线索

V2-079578 | MATHDB-403148

Exact values of the flat choir numbers

摘录 / EXCERPT: For each dimension $n\in\mathbb{N}$, define the flat choir number ... What is the precise value of $\flat_n$ for each n?

open_problem | open | MathDB 页面 | 原文线索

V2-079579 | MATHDB-403149

Bounded H^∞-calculus for the degenerate boundary realization

摘录 / EXCERPT: Under the setup on page 41, let A_T be the L^p-realization of A with boundary condition T. For every $0<\theta<\pi$, there exists a constant ν such that $A_T+\nu$ has a...

open_problem | open | MathDB 页面 | 原文线索

V2-079580 | MATHDB-403150

Fredholm formulation of the limiting absorption principle

摘录 / EXCERPT: Theorem 1: $P(\sigma):\{u\in H^{s,r}:P(\sigma)u\in H^{s-2,r+1}\}\to H^{s-2,r+1}$ is invertible (in particular Fredholm), provided r is monotone along the H_p-flow in the charac...

open_problem | open | MathDB 页面 | 原文线索

V2-079581 | MATHDB-403152

KdV-tau-function conjecture for super Riemann-surface volumes

摘录 / EXCERPT: Should the generating series of the volumes of moduli spaces of super Riemann surfaces, expressed via intersection theory involving the [Theta]-class and the WeilPetersson class, b...

conjecture | open | MathDB 页面 | 原文线索

V2-079582 | MATHDB-403154

Dimension of the space of Laughlin states

摘录 / EXCERPT: Let C be a closed Riemann surface of genus g, let $L\to C$ be a holomorphic line bundle of degree d, and let N be the number of particles. On C^N, set...

open_problem | open | MathDB 页面 | 原文线索

V2-079583 | MATHDB-403155

Formula for ψ-class intersections on strata of k-differentials

摘录 / EXCERPT: Let $g,n,k\ge 0$ satisfy $2g-2+n>0$, and let $a=(a_1,\ldots,a_n)\in\mathbb Z^n$ satisfy $\sum_i a_i=k(2g-2+n)$. Define the stratum ... Writing $[\mathcal M_g(a)]$ for the relevant fu...

open_problem | open | MathDB 页面 | 原文线索

V2-079584 | MATHDB-403156

Existence of a spin double ramification class

摘录 / EXCERPT: For integers g,n,k with $2g-2+n>0$, with $k,a_1,\dots,a_n$ odd and $\sum_i a_i=k(2g-2+n)$, does there exist a class $\mathrm{DR}^{\mathrm{spin}}_g(a)\in H^{2g}(\mathcal M_{g,n})$...

open_problem | open | MathDB 页面 | 原文线索

V2-079585 | MATHDB-403158

Bolza surface as the genus-two hyperbolic maximizer

摘录 / EXCERPT: The Bolza surface is expected to maximize both the first positive Laplace eigenvalue λ_1 and its multiplicity m_1 among closed hyperbolic surfaces of genus 2.

open_problem | open | MathDB 页面 | 原文线索

V2-079586 | MATHDB-403160

Bolza surface as the genus-two maximizer of the first Laplace eigenvalue

摘录 / EXCERPT: The Bolza surface is expected to maximize the first positive Laplace eigenvalue among closed hyperbolic surfaces of genus 2; proving or disproving this remains open.

open_problem | open | MathDB 页面 | 原文线索

V2-079587 | MATHDB-403162

Spectral gap approaching $1/4$ in large genus

摘录 / EXCERPT: Is there a sequence of hyperbolic surfaces S_n with genus going to infinity and $\lambda_1(S_n)\to \frac14$?

open_problem | open | MathDB 页面 | 原文线索

V2-079588 | MATHDB-403164

Recovering real bitangent counts from tropical plane quartics

摘录 / EXCERPT: The question remains how to recover the four possible real bitangent counts-4, 8, 16, and 28-from tropical bitangent classes and their real lifts.

open_problem | open | MathDB 页面 | 原文线索

V2-079589 | MATHDB-403165

Conjectural equality of super Riemann-surface and \[V^\]Theta volumes

摘录 / EXCERPT: For the volume functions defined in the report, ... the conjecture is ...

open_problem | open | MathDB 页面 | 原文线索

V2-079590 | MATHDB-403167

Strong coniveau of the Artin-Mumford torsion invariant

摘录 / EXCERPT: Let X be a smooth projective rationally connected threefold. Is every torsion class in $H^3(X,\mathbb{Z})$ of strong coniveau at least 1, equivalently, does...

open_problem | open | MathDB 页面 | 原文线索

V2-079591 | MATHDB-403170

Question 3: Hyper-Kähler compactification of a Lagrangian fibration

摘录 / EXCERPT: Let $U\subset \mathbb{P}^n$ be an open subset and let $\pi_U:M_U\to U$ be a Lagrangian fibration. When does there exist a smooth hyper-Kähler compactification M of M_U, with a...

open_problem | open | MathDB 页面 | 原文线索

V2-079592 | MATHDB-403174

Non-polyhedrality of the effective cone of \(\overline{M}_{0,n}\)

摘录 / EXCERPT: For every $n\ge 10$, the effective cone $\operatorname{Eff}(\overline{M}_{0,n})$ is not polyhedral in characteristic 0, and it is not polyhedral in characteristic p for an infi...

open_problem | open | MathDB 页面 | 原文线索

V2-079593 | MATHDB-403175

The F-conjecture for \(\overline{M}_{0,n}\) when \(n\ge 8\)

摘录 / EXCERPT: For each $n\ge 8$, is every curve in the moduli space $\overline{M}_{0,n}$ of stable rational curves with n markings numerically equivalent to an effective combination of one-dim...

conjecture | open | MathDB 页面 | 原文线索

V2-079594 | MATHDB-403176

Finite generation of torsion in Griffiths groups

摘录 / EXCERPT: Whether the torsion subgroup $\operatorname{Griff}^{i}(X)_{\mathrm{tors}}$ of a Griffiths group is in general finitely generated.

open_problem | open | MathDB 页面 | 原文线索

V2-079595 | MATHDB-403185

Kollár’s conjectural minimum-volume klt pair

摘录 / EXCERPT: Is Kollár’s klt pair (Y,Δ) of general type with standard coefficients the pair that has minimum volume in each dimension?

open_problem | open | MathDB 页面 | 原文线索

V2-079596 | MATHDB-403188

DPG methods for locking-free numerical approximation of shell models

摘录 / EXCERPT: Our conjecture is that the DPG framework can be adapted to different cases of shell models so that numerical approximations are locking free, or show at least reduced locking.

open_problem | open | MathDB 页面 | 原文线索

V2-079597 | MATHDB-403189

Stability of geometric inequalities for general initial data sets

摘录 / EXCERPT: If an initial data set (IDS) almost saturates a geometric inequality such as the positive mass theorem (PMT) or the (Riemannian) Penrose inequality (RPI), must it be geometrically...

open_problem | open | MathDB 页面 | 原文线索

V2-079598 | MATHDB-403192

Uniqueness of singular Ricci flows

摘录 / EXCERPT: Any singular, 3-dimensional Ricci flow $\mathcal{M}$ is uniquely determined, up to isometry, by its initial time-slice (M_0,g_0).

open_problem | open | MathDB 页面 | 原文线索

V2-079599 | MATHDB-403193

Gromov’s cube inequality

摘录 / EXCERPT: Let g be a Riemannian metric on the cube $I^n=[0,1]^n$. If $\operatorname{Sc}(g)\ge n(n-1)$, then, writing $\ell_j=\operatorname{dist}(\partial_j^-,\partial_j^+)$ for the g-dis...

open_problem | open | MathDB 页面 | 原文线索

V2-079600 | MATHDB-403211

Unique continuation at infinity for asymptotically conical expanding gradient Ricci solitons

摘录 / EXCERPT: Let $(M_i^n,g_i,\nabla^{g_i}f_i)$, $i=1,2$, be two expanding gradient Ricci solitons coming out of the same cone $(C(S),g_C:=dr^2+r^2g_S,\frac r2\partial_r)$ over a smooth link...

open_problem | open | MathDB 页面 | 原文线索

V2-079601 | MATHDB-403214

Subsets of two-dimensional aspherical spaces are aspherical

摘录 / EXCERPT: Let X be a two-dimensional aspherical space. Then any subset A of X is aspherical.

open_problem | open | MathDB 页面 | 原文线索

V2-079602 | MATHDB-403220

Existence of higher-rank Teichmüller spaces and Θ-positivity

摘录 / EXCERPT: Conjecture 2. There exist higher-rank Teichmüller spaces if and only if the group G admits a $\[1;33m\Theta\[0m$-positive structure.

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V2-079603 | MATHDB-403221

Uniqueness of the stationary point of the energy functional

摘录 / EXCERPT: Let Σ_g be a surface of genus $g\ge 2$, let G be a Lie group, and let $\rho:\pi_1(\Sigma_g)\to G$ be an Anosov representation. For each $S\in\mathcal T_g$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-079604 | MATHDB-403222

Non-uniqueness of minimal surfaces for Hermitian groups of rank 3

摘录 / EXCERPT: Theorem 2. Let G be a Hermitian group of rank 3. For every large enough $g \ge 2$, there exists a maximal representation $\rho:\pi_1(\Sigma_g)\to G$ such that...

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V2-079605 | MATHDB-403224

A generalized Poincaré-Birkhoff theorem

摘录 / EXCERPT: Suppose that τ is an exact symplectomorphism of a connected Liouville domain (W,λ), and let $\alpha=\lambda|_{\partial W}$. Assume that all fixed points of τ are...

open_problem | open | MathDB 页面 | 原文线索

V2-079606 | MATHDB-403225

Extension of the Calabi invariant to hameomorphisms of the disc

摘录 / EXCERPT: Does the Calabi invariant admit an extension to a homomorphism from $\operatorname{Hameo}(D,\omega)$ to $\mathbb{R}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079607 | MATHDB-403228

Spectral rigidity of ellipses

摘录 / EXCERPT: Conjecture. Ellipses E_e are Δ-spectrally rigid. In other words, there do not exist non-trivial deformations of E_e by domains with the same Dirichlet spectra as E_e....

open_problem | open | MathDB 页面 | 原文线索

V2-079608 | MATHDB-403234

Structure of the quotient of area-preserving homeomorphisms by finite-energy homeomorphisms

摘录 / EXCERPT: Can we understand the quotient $\operatorname{Homeo}_c(D^2,\omega)/\operatorname{FHomeo}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079609 | MATHDB-403235

Hutchings’s PFH spectral-invariant Calabi conjecture

摘录 / EXCERPT: For every compactly supported area-preserving diffeomorphism ϕ of the two-disc D^2, does ... hold?

conjecture | open | MathDB 页面 | 原文线索

V2-079610 | MATHDB-403237

Symbolic dynamics in the three-body problem

摘录 / EXCERPT: Theorem 3. Consider the three-body problem with masses $m_0,m_1,m_2>0$ such that $m_0\ne m_1$, and denote by Φ_t its flow. Then there exists a section Π transverse to...

open_problem | open | MathDB 页面 | 原文线索

V2-079611 | MATHDB-403239

Rigidity conjecture for generalized interval exchange transformations

摘录 / EXCERPT: Marmi-Moussa-Yoccoz’s rigidity conjecture states that, under a full-measure condition on the standard interval-exchange transformation T_0, if a generalized interval exchange tra...

conjecture | open | MathDB 页面 | 原文线索

V2-079612 | MATHDB-403241

Global rigidity of generalized interval exchange transformations in genus two

摘录 / EXCERPT: For a full-measure set of standard interval exchange transformations T_0 with $d=4$ or 5 intervals, where full measure refers to Lebesgue measure on the lengths for irreducible...

open_problem | open | MathDB 页面 | 原文线索

V2-079613 | MATHDB-403246

Wandering Fatou components for polynomial automorphisms of \(\mathbb{C}^2\)

摘录 / EXCERPT: Theorem (Berger-Biebler). There exists a locally dense set of parameters $(p_0,p_1,p_2,p_3,p_4,b)\in\mathbb{R}^6$ such that the polynomial automorphism ... has a wandering Fatou comp...

open_problem | open | MathDB 页面 | 原文线索

V2-079614 | MATHDB-403247

Newhouse phenomenon for polynomial automorphisms of \(\mathbb{C}^3\)

摘录 / EXCERPT: Theorem (Biebler). For any integer $d\ge 2$, there exists a nonempty open set U of polynomial automorphisms of $\mathbb{C}^3$ of degree d such that every f in a Baire generic...

open_problem | open | MathDB 页面 | 原文线索

V2-079615 | MATHDB-403249

Uniform spectral bound for exact Lagrangians in the unit cotangent disc bundle of a homogeneous space

摘录 / EXCERPT: Let N be a homogeneous space, and let $D T^*N$ be the unit disc bundle of T^*N. Then there exists a constant C such that, for every exact Lagrangian L in $D T^*N$, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-079616 | MATHDB-403256

2∞-Selmer corank versus Mordell-Weil rank

摘录 / EXCERPT: For every elliptic curve $E/\Q$, the $2^\\text{∞}$-Selmer rank satisfies ... Here $r_{2^\\text{∞}}(E)$ is the limit of the 2^k-Selmer ranks $r_{2^k}(E)$, where $r_{2^k}(E)$ is the...

open_problem | open | MathDB 页面 | 原文线索

V2-079617 | MATHDB-403257

Goldfeld’s conjecture for quadratic twists of an elliptic curve

摘录 / EXCERPT: Given an elliptic curve $E/\mathbb{Q}$ with narrow Weierstrass form $y^2=x^3+ax+b$, and a nonzero integer d, let $E_d/\mathbb{Q}$ be the elliptic curve with narrow Weierstrass fo...

conjecture | open | MathDB 页面 | 原文线索

V2-079618 | MATHDB-403261

Bounds for asymptotic counts of quadratic-field class-group torsion

摘录 / EXCERPT: Is it possible to use these moduli spaces to obtain bounds on the asymptotic number of n-torsion elements in class groups of quadratic fields?

open_problem | open | MathDB 页面 | 原文线索

V2-079619 | MATHDB-403266

Integrality of the quintic instanton numbers

摘录 / EXCERPT: For the instanton numbers defined by the Lambert expansion $K(q)=1+\sum_{n\ge 1}\frac{A_nq^n}{1-q^n}$ in the quintic construction, are $a_n=\frac{5A_n}{n^3}$ integers for every...

open_problem | open | MathDB 页面 | 原文线索

V2-079620 | MATHDB-403279

Stability of singularities under perturbations of Dirac masses

摘录 / EXCERPT: Do the singularities in the examples from Theorem 1 remain after making small perturbations of the locations and sizes of the Dirac masses?

open_problem | open | MathDB 页面 | 原文线索

V2-079621 | MATHDB-403280

Problem 3: Singular structures in the solution orbifold

摘录 / EXCERPT: Determine the geometric properties of the set in the solution orbifold corresponding to solutions with singular structures. For example: What is its dimension? What is its regulari...

open_problem | open | MathDB 页面 | 原文线索

V2-079622 | MATHDB-403281

Tangent cones at Dirac masses in singular Monge-Ampère examples

摘录 / EXCERPT: Find precise descriptions of the tangent cones to the examples from [14] at the Dirac masses. A starting point will be to study the case of two masses (that is, Γ_0 consists...

open_problem | open | MathDB 页面 | 原文线索

V2-079623 | MATHDB-403283

Ancient noncollapsed blowup limits near cylindrical singularities

摘录 / EXCERPT: More generally, by Ilmanen’s mean-convex neighborhood conjecture, it is expected even without a mean-convexity assumption that all blowup limits near any cylindrical singularity ar...

open_problem | open | MathDB 页面 | 原文线索

V2-079624 | MATHDB-403284

Degeneration of 7-dimensional minimal hypersurfaces with bounded index - Question 1

摘录 / EXCERPT: Let $g_i\to g$ be a sequence of metrics on $B_1(0)\subset\mathbb{R}^8$ converging in C^3, and let M_i^7 be a sequence of stable embedded minimal hypersurfaces in (B_1,g_i) wi...

open_problem | open | MathDB 页面 | 原文线索

V2-079625 | MATHDB-403285

Degeneration of 7-dimensional minimal hypersurfaces with bounded index

摘录 / EXCERPT: Let $g_i\to g$ be a sequence of metrics on $B_1(0)\subset\mathbb{R}^8$ converging in C^3, and let M_i^7 be stable embedded minimal hypersurfaces in $(B_1(0),g_i)$ with discrete...

open_problem | open | MathDB 页面 | 原文线索

V2-079626 | MATHDB-403287

The Weyl-law constant for the p-width spectrum in dimension one

摘录 / EXCERPT: The Weyl-law constant in dimension one is determined by ... where $a(n)$ is defined by ...

open_problem | open | MathDB 页面 | 原文线索

V2-079627 | MATHDB-403288

Nontrivial anisotropic energies satisfying the atomic condition

摘录 / EXCERPT: Provide nontrivial anisotropic energies in general codimension, other than the area functional, that satisfy the atomic condition (AC).

open_problem | open | MathDB 页面 | 原文线索

V2-079628 | MATHDB-403289

Characterization of the atomic condition in general codimension

摘录 / EXCERPT: In general codimension, characterizing the atomic condition (AC) in terms of more classical conditions, such as Almgren ellipticity (AE), polyconvexity, or others, remains an open...

open_problem | open | MathDB 页面 | 原文线索

V2-079629 | MATHDB-403290

Yau’s conjecture on the length of nodal sets of Laplace eigenfunctions

摘录 / EXCERPT: Yau conjectured that, for all Laplace eigenfunctions on surfaces, the total length of the zero set is comparable to the square root of the eigenvalue.

conjecture | open | MathDB 页面 | 原文线索

V2-079630 | MATHDB-403296

Classification of all equilibria of the nonconservative nonlinear Schrödinger equation

摘录 / EXCERPT: Classify all of the equilibria to equation (1), the nonlinear Schrödinger equation described in the report.

open_problem | open | MathDB 页面 | 原文线索

V2-079631 | MATHDB-403297

Spectral stability of quenched pattern-forming fronts in the complex Ginzburg-Landau equation

摘录 / EXCERPT: For the spatially inhomogeneous complex Ginzburg-Landau equation ... with $\alpha,\gamma\in\mathbb R$, $\chi(\xi)=-\operatorname{sgn}(\xi)$, and a pattern-forming front traveling wit...

open_problem | open | MathDB 页面 | 原文线索

V2-079632 | MATHDB-403299

Traveling waves for the suspension bridge equation for all subcritical speeds

摘录 / EXCERPT: For the suspension bridge equation $u_{tt}=-u_{xxxx}-e^u+1$, does there exist a traveling-wave solution with wave speed c for every parameter value satisfying $c^2\in(0,2)$? Equi...

open_problem | open | MathDB 页面 | 原文线索

V2-079633 | MATHDB-403300

Link between temporal pulse replication and eigenvalue accumulation

摘录 / EXCERPT: Establish the link between the temporal pulse replication in equation (1) and the eigenvalue accumulation associated with the homoclinic banana(s) describing parametric pulse repli...

open_problem | open | MathDB 页面 | 原文线索

V2-079634 | MATHDB-403301

Computer-assisted proof of a localized hexagonal cellular patch in the planar Swift-Hohenberg equation

摘录 / EXCERPT: Can one give a computer-assisted proof of the existence of a localized hexagonal cellular patch in the planar Swift-Hohenberg equation ... with $0<\mu\ll1$ and fixed...

open_problem | open | MathDB 页面 | 原文线索

V2-079635 | MATHDB-403304

Computing congruence quotients for dense subgroups of $SL(n,\Q)$

摘录 / EXCERPT: The problem of computing the set $L_{\max}(H)$ of congruence quotients modulo all maximal ideals of the ground domain is reported as solved for finitely generated dense subgroups...

open_problem | open | MathDB 页面 | 原文线索

V2-079636 | MATHDB-403305

Exponent-32 conjecture for the group H(8,12)

摘录 / EXCERPT: Let $H(q,c)=\langle a,b,c,d\mid [b,a]=[d,c],\ \text{exponent }q,\ \text{nilpotency class }c\rangle$. The conjecture is that the Schur multiplier of $H(8,12)$ has exponent 32; equ...

conjecture | open | MathDB 页面 | 原文线索

V2-079637 | MATHDB-403308

Minimal tuples of tree isometries and the Ping Pong Lemma

摘录 / EXCERPT: A minimal n-tuple $X=(g_1,\ldots,g_n)$ of isometries of T either contains an elliptic element or satisfies the hypotheses of the Ping Pong Lemma.

open_problem | open | MathDB 页面 | 原文线索

V2-079638 | MATHDB-403309

Madlener-Otto plain-group conjecture

摘录 / EXCERPT: A group is presented by a finite convergent length-reducing rewriting system if and only if it is plain (the free product of finite and infinite cyclic groups).

conjecture | open | MathDB 页面 | 原文线索

V2-079639 | MATHDB-403311

Complete list of principal congruence link complements

摘录 / EXCERPT: “Although there are infinitely many arithmetic link complements, there are only finitely many that come from principal congruence subgroups. Some of the examples known seem to be a...

open_problem | open | MathDB 页面 | 原文线索

V2-079640 | MATHDB-403313

Subgroup membership in $\mathrm{GL}(2,\mathbb{Z})$ with binary matrix input

摘录 / EXCERPT: Given $g_0,g_1,\ldots,g_k\in \mathrm{GL}(2,\mathbb{Z})$, decide whether $g_0\in\langle g_1,\ldots,g_k\rangle$, where the group elements are represented by (2×2) matrices who...

open_problem | open | MathDB 页面 | 原文线索

V2-079641 | MATHDB-403315

Decidability of equality of symbolic class-counting formulae

摘录 / EXCERPT: Is equality of two such formulae even decidable?

open_problem | open | MathDB 页面 | 原文线索

V2-079642 | MATHDB-403316

Solubility of the Basis Problem for free-group homomorphisms

摘录 / EXCERPT: Given finite alphabets Σ and Δ and homomorphisms $g,h:F(\Sigma)\to F(\Delta)$, with at least one of g or h injective, the Basis Problem asks whether there is an a...

open_problem | open | MathDB 页面 | 原文线索

V2-079643 | MATHDB-403317

Decidability of the Post Correspondence Problem for Free Groups

摘录 / EXCERPT: Given homomorphisms $g,h:F_1\to F_2$ between free groups, determine whether it is decidable that there exists a nontrivial element $x\in F_1\setminus\{1\}$ such that $g(x)=h(x)$. T...

open_problem | open | MathDB 页面 | 原文线索

V2-079644 | MATHDB-403319

Membership of a matrix in a non-trivial free subgroup of a linear group

摘录 / EXCERPT: Given a linear group G and a matrix $M\in G$, is it possible to determine whether M belongs to a (non-trivial) free subgroup of G, and, if the answer is positive, find a basi...

open_problem | open | MathDB 页面 | 原文线索

V2-079645 | MATHDB-403320

Efficient constructive orbit computation for finite matrix groups

摘录 / EXCERPT: Let $G=\langle g_1,\ldots,g_l\rangle\leq \operatorname{GL}_n(\mathbb{F}_q)$ be a finite matrix group, let $v\in\mathbb{F}_q^n$, and let $w\in G\cdot v$. Is there an efficient way t...

open_problem | open | MathDB 页面 | 原文线索

V2-079646 | MATHDB-403324

Local Langlands conjecture for connected reductive groups over non-archimedean local fields

摘录 / EXCERPT: Let G be a connected reductive group over a non-archimedean local field F with residue field k of characteristic p. Let $\Phi(G/F)$ be the set of equivalence classes of Wei...

conjecture | open | MathDB 页面 | 原文线索

V2-079647 | MATHDB-403326

(σ,ρ)-Poisson Summation Formula on GL_1

摘录 / EXCERPT: Let k be a number field, let $\mathbb A$ be its ring of adeles, let G be a k-split reductive group, and let $\rho:G^{\vee}(\mathbb C)\to\mathrm{GL}_n(\mathbb C)$ be a finite-...

open_problem | open | MathDB 页面 | 原文线索

V2-079648 | MATHDB-403327

Projectivity of the Iwahori component in the $\mathrm{GL}_{n+1}(F)$ to $\mathrm{GL}_n(F)$ restriction

摘录 / EXCERPT: Suppose π is a tempered representation of $\mathrm{GL}_{n+1}(F)$. If the cuspidal support of π does not contain an unramified character of $\mathrm{GL}_1(F)$, or if π i...

open_problem | open | MathDB 页面 | 原文线索

V2-079649 | MATHDB-403328

Explicit exotic Fourier kernel and involutive extension

摘录 / EXCERPT: Let G be a connected reductive group over $\d53d_q$, with dual group $G^\vee$, and let $\rho^\vee:G^\vee\to \mathrm{GL}_n$ be a morphism as in the report. For a nontrivial additi...

open_problem | open | MathDB 页面 | 原文线索

V2-079650 | MATHDB-403334

Trace theorem for automorphic sheaves with nilpotent singular support

摘录 / EXCERPT: There is a canonical isomorphism ...

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V2-079651 | MATHDB-403335

Automorphic functions as global sections of the dualizing sheaf on arithmetic Langlands parameters

摘录 / EXCERPT: Conjecture 4.5.1. In the setting of §4.1-namely, over $k=\mathbb F_q$ with $e=\mathbb Q_\ell$, ℓ-adic sheaves, and X defined over $\mathbb F_q$-there is an isomorphism ... The...

open_problem | open | MathDB 页面 | 原文线索

V2-079652 | MATHDB-403336

Monomial geometricity of Yu’s tame supercuspidal representations

摘录 / EXCERPT: How many of the representations constructed by Yu [8] can be shown to be monomial geometric supercuspidal?

open_problem | open | MathDB 页面 | 原文线索

V2-079653 | MATHDB-403337

Removal or replacement of the cyclic base-change assumption

摘录 / EXCERPT: Can the assumption on cyclic base change be removed or replaced with other local assumptions by a function field proof of cyclic base change?

open_problem | open | MathDB 页面 | 原文线索

V2-079654 | MATHDB-403338

Generalizing the proof beyond monomial representations

摘录 / EXCERPT: To what extent can one generalize the proof from monomial representations to other representations?

open_problem | open | MathDB 页面 | 原文线索

V2-079655 | MATHDB-403339

Gross-Reeder conjecture for orthogonal root numbers

摘录 / EXCERPT: In the setting of Conjecture 0.1, ... where $z_{\operatorname{Ad}_G}\in Z$ is a certain involution, Z is the center of G, and χ_φ is the character of Z corresponding...

conjecture | open | MathDB 页面 | 原文线索

V2-079656 | MATHDB-403340

Orthogonal root numbers of tempered parameters

摘录 / EXCERPT: Assume that G is quasi-split. Let k be a local field, let $r:{}^LG\to O(V)$ be an orthogonal representation, and let $\varphi:L_k\to{}^LG$ be a tempered L-parameter. Then ... w...

open_problem | open | MathDB 页面 | 原文线索

V2-079657 | MATHDB-403341

Formal degree conjecture for Kaletha’s non-singular L-packets

摘录 / EXCERPT: The formal degree conjecture holds for Kaletha’s non-singular L-packets.

conjecture | open | MathDB 页面 | 原文线索

V2-079658 | MATHDB-403342

Formal degree formula for Yu supercuspidals

摘录 / EXCERPT: Theorem 3.1. Let G be semisimple and let $ ...

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V2-079659 | MATHDB-403343

Stabilizers of top-dimensional affine Deligne-Lusztig components up to full conjugacy

摘录 / EXCERPT: For each $Z\in\Sigma^{\mathrm{top}}(X_\mu(b))$, determine the stabilizer of Z in $J_b(F)$ up to $J_b(F)$-conjugacy.

open_problem | open | MathDB 页面 | 原文线索

V2-079660 | MATHDB-403344

Multiplicity equality in elliptic near-equivalence classes

摘录 / EXCERPT: Let ψ be an elliptic A-parameter of G and $\pi=\bigotimes'_v\pi_v$ be a representation of $G(A_F)$ in the near equivalent class represented by ψ. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-079661 | MATHDB-403345

Endoscopic character identities for theta packets

摘录 / EXCERPT: Can one prove that $\Pi^{\theta}_{\psi_v}(G_v)$ satisfies endoscopy character identities?

open_problem | open | MathDB 页面 | 原文线索

V2-079662 | MATHDB-403347

Removing the logarithmic revealment loss for subcritical Ising sparse reconstruction

摘录 / EXCERPT: For $\beta<\beta_c$ Ising on $\\mathbb{Z}_n^d$, with $d\ge 3$, can the factor $\log^d n$ in the revealment condition of Theorem 5 be removed?

open_problem | open | MathDB 页面 | 原文线索

V2-079663 | MATHDB-403348

Monotonicity of the marginal measure in median dynamics

摘录 / EXCERPT: Conjecture 1: For any $G=(V,E)$, any $x\in V$ and any $\alpha>\frac12$, the function $t\mapsto\mu_t(x)$ is monotone non-decreasing.

open_problem | open | MathDB 页面 | 原文线索

V2-079664 | MATHDB-403354

Boundedness of zero-level excursion components for the random plane wave

摘录 / EXCERPT: For the random plane wave (RPW), namely the centred smooth stationary Gaussian field on $\mathbb{R}^2$ with covariance $\kappa(x)=J_0(|x|)$, does the excursion set $\{f\le 0\}$ hav...

open_problem | open | MathDB 页面 | 原文线索

V2-079665 | MATHDB-403355

Conformal-invariant scaling limit of random plane wave nodal lines

摘录 / EXCERPT: Do the zero level lines $\{f=0\}$ of the random plane wave have a conformal-invariant scaling limit?

open_problem | open | MathDB 页面 | 原文线索

V2-079666 | MATHDB-403356

Time-dependent Fourier coefficient inequality for Robinson-Trautman equations

摘录 / EXCERPT: Let $G:\mathbb{R}\to\mathbb{R}$ be smooth with a first-order zero at the origin, and let $s>1$. For $S=\mathbb{T}^2=[0,2\pi]\times[0,2\pi]$, write ... and let...

open_problem | open | MathDB 页面 | 原文线索

V2-079667 | MATHDB-403358

Rigorous justification of LIGO-type interferometry without analyticity

摘录 / EXCERPT: Does there exist a way to rigorously justify the results of LIGO-type interferometric experiments without assuming analyticity of the spacetime metric?

open_problem | open | MathDB 页面 | 原文线索

V2-079668 | MATHDB-403359

Black hole stability conjecture for the Kerr family

摘录 / EXCERPT: Conjecture 1 (Black hole stability). The Kerr family of black holes, which is parametrized by $M>0$ and $|a|\le M$, is stable, as a family of solutions to the vacuum Einstein equat...

conjecture | open | MathDB 页面 | 原文线索

V2-079669 | MATHDB-403360

Permutation invariance of Leaver’s Kerr mode-spectrum condition

摘录 / EXCERPT: Theorem 4. Consider the setup of Theorem 3 with [U+007F]$|a|<M$. Define ... Then Leaver’s condition on the mode-solution spectrum of the Teukolsky equation is invariant under every permutat...

open_problem | open | MathDB 页面 | 原文线索

V2-079670 | MATHDB-403361

Codimension-10 characteristic gluing near Minkowski

摘录 / EXCERPT: Assume that the spacetime geometries around spheres $S_1\subset(M_1,g_1)$ and $S_2\subset(M_2,g_2)$ are close to the flat geometries around the round spheres of radii 1 and 2 i...

open_problem | open | MathDB 页面 | 原文线索

V2-079671 | MATHDB-403365

Nonlinear stability of the Schwarzschild family

摘录 / EXCERPT: For vacuum initial data sets sufficiently close to appropriate Schwarzschild initial data, the resulting maximal Cauchy development solving the vacuum Einstein equations (with...

open_problem | open | MathDB 页面 | 原文线索

V2-079672 | MATHDB-403366

The weaker H^1-formulation of Strong Cosmic Censorship for $\Lambda=0$

摘录 / EXCERPT: For $\Lambda=0$, the weaker H^1-formulation of Strong Cosmic Censorship is expected to hold: generically, the spacetime metric is inextendible with square-integrable Christoffel...

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V2-079673 | MATHDB-403367

Boundedness at the Cauchy horizon for generic Kerr-AdS parameters

摘录 / EXCERPT: There exists a subset $P_{\mathrm{Bounded}}\subset P$ such that the following is true. Consider a Kerr-AdS black hole with parameters $m,a\in P_{\mathrm{Bounded}}$. Then linear per...

open_problem | open | MathDB 页面 | 原文线索

V2-079674 | MATHDB-403368

Anti-de Sitter instability under reflective boundary conditions

摘录 / EXCERPT: Conjecture 1 (Dafermos-Holzegel, Anderson). The AdS space is unstable for reflective boundary conditions.

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V2-079675 | MATHDB-403371

Naked singularities for the Einstein vacuum equations

摘录 / EXCERPT: There exist naked singularities for the Einstein vacuum equations. Moreover, these solutions may be considered analogous to Christodoulou’s solutions.

open_problem | open | MathDB 页面 | 原文线索

V2-079676 | MATHDB-403372

Reverse Burnett conjecture

摘录 / EXCERPT: Any solution to the Einstein-massless Vlasov system must arise as such a limit of vacuum solutions.

conjecture | open | MathDB 页面 | 原文线索

V2-079677 | MATHDB-403373

Behavior of perturbations of generalized Kasner solutions beyond the sub-critical regime

摘录 / EXCERPT: Understand how general small perturbations (e.g., without symmetry) of generalized Kasner solutions behave when the sub-criticality condition fails, i.e. when ... The source states t...

open_problem | open | MathDB 页面 | 原文线索

V2-079678 | MATHDB-403374

Killing-horizon conjecture for compact non-degenerate Cauchy horizons

摘录 / EXCERPT: Isenberg and Moncrief conjectured that every compact non-degenerate Cauchy horizon should be a Killing horizon.

conjecture | open | MathDB 页面 | 原文线索

V2-079679 | MATHDB-403377

Weak second Bianchi identity for spacetimes with admissible timelike singularities

摘录 / EXCERPT: We conjecture that, for all spacetimes with admissible timelike singularities of the kind described in (i), the weak twice-contracted second Bianchi identity holds: ... where ψ...

open_problem | open | MathDB 页面 | 原文线索

V2-079680 | MATHDB-403379

Late-time asymptotics from the infalling-masses scattering model

摘录 / EXCERPT: For the spherically symmetric mode ... for constants C'_0,C_0 depending only on C_{in} and M.

open_problem | open | MathDB 页面 | 原文线索

V2-079681 | MATHDB-403382

Continuity of the intermediate self-similar Riemann solution

摘录 / EXCERPT: Theorem 1. For the given parameters k_1, k_2, and k_3, the self-similar solution at intermediate regions of the Riemann problem depends continuously on the initial value.

open_problem | open | MathDB 页面 | 原文线索

V2-079682 | MATHDB-403383

Reconstructability of the total tumbling-kernel rate from macroscopic measurements

摘录 / EXCERPT: The report states that it is currently studying the reconstructability of the total tumbling-kernel rate ... for a simplified chemotaxis equation whose right-hand side is $-Lf$, usin...

open_problem | open | MathDB 页面 | 原文线索

V2-079683 | MATHDB-403388

Profinite rigidity of Kleinian groups

摘录 / EXCERPT: Lattices in $\operatorname{PSL}_2(\mathbb{C})$, and more generally Kleinian groups-meaning discrete subgroups of $\operatorname{PSL}_2(\mathbb{C})$-are conjectured to be profinitel...

open_problem | open | MathDB 页面 | 原文线索

V2-079684 | MATHDB-403391

Non-loose representatives for all knot types in an overtwisted 3-sphere

摘录 / EXCERPT: Does every knot type in an overtwisted S^3 have a non-loose representative?

open_problem | open | MathDB 页面 | 原文线索

V2-079685 | MATHDB-403396

Uniqueness of positive braid knots in a concordance class at fixed braid index

摘录 / EXCERPT: Does every smooth knot concordance class contain at most one isotopy class of positive braid knots with any fixed braid index, in particular braid index 3?

open_problem | open | MathDB 页面 | 原文线索

V2-079686 | MATHDB-403397

Eventual linearity of regularity for \a0Inc(\a0\mathbb{N}\u0000a0)-invariant ideal filtrations

摘录 / EXCERPT: Let k be a field and $R_n=k[x_1,\ldots,x_n]$ for $n\ge 1$. Let $I=(I_n)_{n\ge 1}$ be an $\a0\operatorname{Inc}(\mathbb{N})$-invariant filtration of proper homogeneous ideals, mea...

open_problem | open | MathDB 页面 | 原文线索

V2-079687 | MATHDB-403398

Eventual linearity of projective dimension for Inc-invariant filtrations

摘录 / EXCERPT: Let k be a field, let $R_n=k[x_1,\ldots,x_n]$ for $n\ge 1$, and let $\operatorname{Inc}(\mathbb N)$ act by $\pi(x_i)=x_{\pi(i)}$. An $\operatorname{Inc}(\mathbb N)$-invariant fil...

open_problem | open | MathDB 页面 | 原文线索

V2-079688 | MATHDB-403399

Symmetric spectrahedra under eigenvalue lifting

摘录 / EXCERPT: Let $M\subset\mathbb{R}^n$ be invariant under the permutation group S_n. Let $S^2(n)$ denote the vector space of real symmetric $n\times n$ matrices, and for $A\in S^2(n)$ let...

open_problem | open | MathDB 页面 | 原文线索

V2-079689 | MATHDB-403400

Classification of hierarchical models with rational Hilbert series

摘录 / EXCERPT: A complete classification of hierarchical models with rational Hilbert series is yet to be found.

open_problem | open | MathDB 页面 | 原文线索

V2-079690 | MATHDB-403401

Gröbner-basis conjecture for Specht ideals

摘录 / EXCERPT: Conjecture 1. Let $\lambda \vdash n$. Then the Specht polynomials of shape $\mu \vdash n$, where $\lambda \trianglerighteq \mu$, form a Gröbner basis of I_λ.

conjecture | open | MathDB 页面 | 原文线索

V2-079691 | MATHDB-403404

Individual Betti numbers of symmetric-group-invariant monomial ideals

摘录 / EXCERPT: For the polynomial ring $R=\mathbb{C}[x_1,\ldots,x_n]$ with S_n acting by coordinate permutations, and an S_n-invariant monomial ideal $I\subseteq R$, how can one understand th...

open_problem | open | MathDB 页面 | 原文线索

V2-079692 | MATHDB-403406

Dependence of \(\operatorname{Rep}(G_T)\) on the order of a triangle presentation

摘录 / EXCERPT: The representation category $\operatorname{Rep}(G_T)$ only depends on the order q of the triangle presentation T.

open_problem | open | MathDB 页面 | 原文线索

V2-079693 | MATHDB-403414

Infinitely many Witt square roots of Ising classes

摘录 / EXCERPT: Do the Ising categories I^{ν}, where $\nu=2k+1$ and $0\le k\le 7$, each have infinitely many Witt classes $[C]$, modulo the subgroup generated by pointed classes and the Ising...

open_problem | open | MathDB 页面 | 原文线索

V2-079694 | MATHDB-403428

Optimal logarithmic exponent in the periodic discrete restriction estimate for the parabola

摘录 / EXCERPT: Determine the smallest exponent A for which there is a finite constant such that, whenever $\widehat h(n)=0$ for all $|n|>M$, the solution ... of the periodic Schrödinger equation...

open_problem | open | MathDB 页面 | 原文线索

V2-079695 | MATHDB-403429

Endpoint logarithmic exponent in discrete restriction for the parabola

摘录 / EXCERPT: The report records the following unresolved target: can the estimate ... hold for every initial datum h on $\mathbb T=\mathbb R/\mathbb Z$ whose Fourier coefficients satisfy...

open_problem | open | MathDB 页面 | 原文线索

V2-079696 | MATHDB-403436

Fukui-Kudryka-Paunescu conjecture

摘录 / EXCERPT: Is multiplicity invariant under a subanalytic, arc-analytic bilipschitz homeomorphism?

conjecture | open | MathDB 页面 | 原文线索

V2-079697 | MATHDB-403441

Igusa’s conjecture for exponential sums in terms of the critical locus at infinity

摘录 / EXCERPT: Let $f\in\mathbb{Z}[x_1,\ldots,x_n]$ be a non-constant polynomial of degree $d>1$. Let f_d be the homogeneous part of highest degree of f, and let s be the dimension of the c...

conjecture | open | MathDB 页面 | 原文线索

V2-079698 | MATHDB-403442

Negativity Conjecture

摘录 / EXCERPT: Let S be a complex $\mathbb{Q}$-acyclic surface and let (X,D) be a log smooth completion. Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-079699 | MATHDB-403447

Fukui-Kurdyka-Paunescu Conjecture

摘录 / EXCERPT: Let $X,Y\subset \mathbb{R}^n$ be two germs at the origin of irreducible real analytic subsets. If $h:(\mathbb{R}^n,0)\to(\mathbb{R}^n,0)$ is the germ of a subanalytic, arc-analytic...

conjecture | open | MathDB 页面 | 原文线索

V2-079700 | MATHDB-403448

Higher Rank Finite Generation Conjecture, local

摘录 / EXCERPT: Let $(x\in X=\operatorname{Spec}(R))$ be a klt singularity, let $v\in\operatorname{Val}_{X,x}$ be a minimizer of the normalized volume function...

conjecture | open | MathDB 页面 | 原文线索

V2-079701 | MATHDB-403451

Non-multiplicativity of stabilizer extent

摘录 / EXCERPT: Let n be large enough and let $ ... In particular, the stabilizer extent is not multiplicative.

open_problem | open | MathDB 页面 | 原文线索

V2-079702 | MATHDB-403454

Optimal entanglement required to break position-verification protocols

摘录 / EXCERPT: What is the optimal amount of entanglement necessary to break any position-verification (PV) protocol?

open_problem | open | MathDB 页面 | 原文线索

V2-079703 | MATHDB-403455

Completeness of entanglement testers without index reshuffling

摘录 / EXCERPT: Is our family of entanglement criteria complete, without allowing for a preliminary reshuffling of indices?

open_problem | open | MathDB 页面 | 原文线索

V2-079704 | MATHDB-403457

Spin-alignment conjecture for minimum-entropy n-spin states

摘录 / EXCERPT: Let ρ be an n-spin density operator formed as a convex combination of several distinct states. In each distinct state, some spins are fixed to products of a given mixed stat...

conjecture | open | MathDB 页面 | 原文线索

V2-079705 | MATHDB-403458

Limitation of low-level QAOA for MAX-\k-CUT on random regular bipartite graphs

摘录 / EXCERPT: Theorem 1. Let $G^{\mathrm{bi}}_{n,d}$ denote the uniform distribution over all d-regular bipartite graphs on n vertices, and let $\alpha^{\mathrm{MC}k}_p(G)$ denote the approx...

open_problem | open | MathDB 页面 | 原文线索

V2-079706 | MATHDB-403461

Decidability of validity for differences of graph homomorphism densities

摘录 / EXCERPT: Given two (not necessarily connected) graphs G_1 and G_2, is it decidable whether $G_1-G_2\ge 0$ is a valid homomorphism-density inequality?

open_problem | open | MathDB 页面 | 原文线索

V2-079707 | MATHDB-403462

Polyhedrality of exponent cones for valid graph-density binomial inequalities

摘录 / EXCERPT: Question 2. Let U be a finite collection of connected graphs. Do exponents of binomial inequalities valid on the profile G_U form a polyhedral cone? If yes, then is it necessar...

open_problem | open | MathDB 页面 | 原文线索

V2-079708 | MATHDB-403465

Conjecture on generic initial ideals of Jacobian ideals of central arrangements

摘录 / EXCERPT: Let $\mathcal{A}$ be a central arrangement in K^ℓ and let ... If $\operatorname{rgin}(J(\mathcal{A}))$ has a minimal generator t that involves the third variable of S, then...

conjecture | open | MathDB 页面 | 原文线索

V2-079709 | MATHDB-403467

Aluffi’s conjecture on Euler characteristics of graph hypersurface complements

摘录 / EXCERPT: For every graph G, the Euler characteristic of the complement $Y_G^{\perp}=\mathbb{P}^{E}\setminus\operatorname{Var}(\psi_G^{\perp})$ satisfies ...

conjecture | open | MathDB 页面 | 原文线索

V2-079710 | MATHDB-403468

General recursions and classification for the Euler characteristic of dual graph-hypersurface complements

摘录 / EXCERPT: Develop general recursions for $\chi(Y_G^{\perp})$ and classify graphs G by $\chi(Y_G^{\perp})$, where $\psi_G^{\perp}=\sum_{B\in\mathcal B_{M_G}}\prod_{e\in E\setminus B}x_e$ an...

open_problem | open | MathDB 页面 | 原文线索

V2-079711 | MATHDB-403471

H-constants of rational line arrangements

摘录 / EXCERPT: What is the most negative value of $H(C)$ when C is an arrangement of lines defined over $\mathbb{Q}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079712 | MATHDB-403473

The $K(\pi,1)$ question for elliptic discriminant complements

摘录 / EXCERPT: For a marked elliptic root system (R,G), is the elliptic discriminant complement...

open_problem | open | MathDB 页面 | 原文线索

V2-079713 | MATHDB-403475

Zariski-van Kampen generators and elliptic braid relations

摘录 / EXCERPT: There exists a special choice of Zariski-van Kampen generator system of $\pi_1(l_{t_*}\setminus D_{\mathrm{ell}},*)$, where $*$ is a base point, corresponding naturally and bijecti...

open_problem | open | MathDB 页面 | 原文线索

V2-079714 | MATHDB-403476

Generation of \(\pi_2\) by difference classes from non-cancellative quadruples

摘录 / EXCERPT: Conjecture 3 ([4]). The set of all difference classes ... attached to non-cancellative quadruples of $A(R,G)^+$, generates ...

open_problem | open | MathDB 页面 | 原文线索

V2-079715 | MATHDB-403477

Freeness of the cones of the Catalan and Shi arrangements

摘录 / EXCERPT: For an irreducible root system $\Phi\subset\mathbb{R}^{\ell}$ with a positive system Φ^{+}, the cone of the Catalan arrangement $c\mathcal{A}^{[-m,m]}_{\Phi}$ and the cone of...

open_problem | open | MathDB 页面 | 原文线索

V2-079716 | MATHDB-403478

Terao’s freeness conjecture for rank 3 arrangements with at most 14 hyperplanes

摘录 / EXCERPT: Terao’s freeness conjecture is true for rank 3 arrangements with up to 14 hyperplanes in any characteristic.

conjecture | open | MathDB 页面 | 原文线索

V2-079717 | MATHDB-403479

Conjecture on direct summands and noncommutative resolutions for complex reflection groups

摘录 / EXCERPT: Let G be a finite complex reflection group acting linearly on $V\cong\mathbb{C}^n$, let $S=\operatorname{Sym}_{\mathbb{C}}(V)\cong\mathbb{C}[x_1,\ldots,x_n]$, let $R=S^G$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-079718 | MATHDB-403480

Numerical characterization of KT-configurations yielding simple non-very-generic intersections

摘录 / EXCERPT: For a discriminantal arrangement $B(n,k,A_0)$ with $k\ge 2$ and $A_0\notin Z$, determine the necessary and sufficient numerical conditions on $r\ge 2$ subsets...

open_problem | open | MathDB 页面 | 原文线索

V2-079719 | MATHDB-403482

Free arrangements with low exponents are supersolvable

摘录 / EXCERPT: Every central free hyperplane arrangement with exponents consisting only of 1s and 2s, and with at most one exponent equal to 3, is supersolvable.

open_problem | open | MathDB 页面 | 原文线索

V2-079720 | MATHDB-403483

Lubotzky’s 1-2-3 problem for the subgroup Γ_3

摘录 / EXCERPT: Let ... Although Γ_3 has infinite index in $\mathrm{SL}_2(\mathbb Z)$, does it satisfy superstrong approximation (SSA)?

open_problem | open | MathDB 页面 | 原文线索

V2-079721 | MATHDB-403484

Strong approximation for Zariski-dense subgroups of $\mathrm{SL}_d(\mathbb{Z})$

摘录 / EXCERPT: Suppose that $\Gamma<\mathrm{SL}_d(\mathbb{Z})$ is Zariski-dense in SL_d. Then there exists $q_0\in\mathbb{N}$ such that, for every $q\in\mathbb{N}$ with $\gcd(q,q_0)=1$...

open_problem | open | MathDB 页面 | 原文线索

V2-079722 | MATHDB-403485

Strong asymptotic local-global conjecture for integral Apollonian circle packings

摘录 / EXCERPT: Let P be a fixed primitive integral Apollonian circle packing. Then there exists $N_0=N_0(P)$ such that, if $m>N_0$ and m is admissible-that is, for every $q\ge 1$, m is cong...

conjecture | open | MathDB 页面 | 原文线索

V2-079723 | MATHDB-403488

Geometric interpretation of the diagonal Yangian R-matrix factor

摘录 / EXCERPT: How can we obtain $R_0(s)$ from 4-dimensional Chern-Simons theory of Costello-Witten-Yamazaki?

open_problem | open | MathDB 页面 | 原文线索

V2-079724 | MATHDB-403493

Minimal splitting subgroups of $GL(m|n)$

摘录 / EXCERPT: Let $G=GL(m|n)$ with $m\ge n$, and let $H=SL(1|1)^n\subseteq G$ be the natural embedding by block matrices. We conjecture that every splitting subgroup K of $GL(m|n)$ contains a...

open_problem | open | MathDB 页面 | 原文线索

V2-079725 | MATHDB-403494

Green-correspondence conjecture for quasireductive supergroups

摘录 / EXCERPT: Let $G=GL(m|n)$ with $m\ge n$, let $H=SL(1|1)^n\subset G$ be the natural block-matrix embedding, and let $K=N_G(H)$. For a symmetric monoidal rigid category $\mathcal C$, let...

conjecture | open | MathDB 页面 | 原文线索

V2-079726 | MATHDB-403498

Classification of irreducible unitary representations of real reductive groups

摘录 / EXCERPT: The classification of the set $\operatorname{Irr}_u(G)$ of irreducible unitary G-representations, for a real reductive Lie group G, remains unsolved.

open_problem | open | MathDB 页面 | 原文线索

V2-079727 | MATHDB-403502

Hypercube-decomposition recurrence for Kazhdan-Lusztig polynomials

摘录 / EXCERPT: Let $x,y\in S_{n+1}$, viewed as permutations of $0,1,\ldots,n$, with $x\le y$ in Bruhat order. The report conjectures that the displayed recurrence ... holds for every $c'\in[x,y]$ s...

open_problem | open | MathDB 页面 | 原文线索

V2-079728 | MATHDB-403504

Redundancies in string cone inequalities and potential functions

摘录 / EXCERPT: Conjecture 4. A term of $W|_{T^\vee_{\mathbf{i}}}$ leads to a redundant inequality after tropicalization if and only if its coefficient is less than or equal to 1.

open_problem | open | MathDB 页面 | 原文线索

V2-079729 | MATHDB-403506

The Two-Stripe Symmetric Circulant TSP is in P

摘录 / EXCERPT: The two-stripe symmetric circulant TSP is in P.

open_problem | open | MathDB 页面 | 原文线索

V2-079730 | MATHDB-403507

Polynomial-time solvability of dyadic linear programs

摘录 / EXCERPT: Given a dyadic linear program ... where A,b,w have integral entries and every entry of x is an integer multiple of $1/2^k$ for some nonnegative integer k, can the program be so...

open_problem | open | MathDB 页面 | 原文线索

V2-079731 | MATHDB-403509

Polynomial-time solvability of integer programs with bounded subdeterminants

摘录 / EXCERPT: Can integer programs of the form $\max\{c^\top x : Ax \le b,\ x\in\mathbb{Z}^n\}$ be solved in polynomial time when A is an integer matrix such that the absolute value of the det...

open_problem | open | MathDB 页面 | 原文线索

V2-079732 | MATHDB-403510

Approximate CVP in time $2^{0.802n}$ for arbitrary norms

摘录 / EXCERPT: For any lattice $L \subseteq \mathbb{R}^n$ and any norm $\|\cdot\|_K$ on $\mathbb{R}^n$, one can find an $O(1)$-approximate solution to CVP_K in time $2^{0.802n}$ and sp...

open_problem | open | MathDB 页面 | 原文线索

V2-079733 | MATHDB-403513

Objective functions inducing the same arborescence for an NW-rule

摘录 / EXCERPT: Let $P\subseteq\mathbb{R}^d$ be a polytope and let R be a normalized-weight pivot rule (NW-rule) for which the normalization function $\eta_R(P,c)$ is independent of c and the...

open_problem | open | MathDB 页面 | 原文线索

V2-079734 | MATHDB-403517

Finding the complete set of asymptotic-spectrum obstructions for tensors

摘录 / EXCERPT: How can one find the complete set of functionals on tensors that are restriction-monotone, multiplicative under the Kronecker product, additive under direct sum, and normalized on...

open_problem | open | MathDB 页面 | 原文线索

V2-079735 | MATHDB-403518

Efficient list-decoding capacity for LDPC codes

摘录 / EXCERPT: Can one efficiently achieve list-decoding capacity for LDPC codes?

open_problem | open | MathDB 页面 | 原文线索

V2-079736 | MATHDB-403523

Strohmer-Beaver conjecture on the condition number of Gaussian Gabor frames

摘录 / EXCERPT: For Gaussian Gabor systems with the standard Gaussian window and lattices of density 2, Strohmer and Beaver conjectured that the frame condition number B_Λ/A_Λ is m...

conjecture | open | MathDB 页面 | 原文线索

V2-079737 | MATHDB-403525

Classification of transitive totally Cartan actions

摘录 / EXCERPT: Theorem (S-Vinhage). Let $\mathbb{R}^k \times \mathbb{Z}^\ell$ be a $C^{1,\theta}$ transitive, totally Cartan action, for some $\theta \in (0,1)$. Assume no finite cover of the act...

open_problem | open | MathDB 页面 | 原文线索

V2-079738 | MATHDB-403526

Conjecture 6: Framing c-vectors determine the $\b3$-orbit of a cluster

摘录 / EXCERPT: Let $T_{\boldsymbol n,\boldsymbol w}$ be a $T_{\boldsymbol n,\boldsymbol w}$ quiver with $n_i>1$ and $w_i>0$, and let $T^f_{\boldsymbol n,\boldsymbol w}$ be the framing obtained by...

conjecture | open | MathDB 页面 | 原文线索

V2-079739 | MATHDB-403528

Parity-invariant characterization of the positive Grassmannian

摘录 / EXCERPT: What is the image of the positive part of the Grassmannian, purely in this parity-invariant subspace of minors, and can we cut out this image by polynomial inequalities?

open_problem | open | MathDB 页面 | 原文线索

V2-079740 | MATHDB-403531

Characterization of the Legendre transform by translation-conjugate valuations

摘录 / EXCERPT: For $n\ge 2$, characterize the maps $z:\operatorname{Conv}_{\mathrm{sc}}(\mathbb{R}^n)\to F(\mathbb{R}^n;\mathbb{R})$ that are continuous, $\mathrm{SL}(n)$-contravariant valuations...

open_problem | open | MathDB 页面 | 原文线索

V2-079741 | MATHDB-403533

Mixed Hodge-Riemann relations for translation-invariant valuations

摘录 / EXCERPT: Assume that all convex bodies $A_1,\ots,A_n$ have smooth positively curved boundaries. Let $0\le i\le n/2$ and define ... where $V_A(\bullet)=V(\bullet[n-1],A)$. Then, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-079742 | MATHDB-403534

The L^p Aleksandrov problem

摘录 / EXCERPT: Fix $p\in\mathbb{R}$. What are the necessary and sufficient conditions on a given Borel measure μ on S^{n-1} so that there exists a convex body $K\in\mathcal{K}^{n}_{o}$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-079743 | MATHDB-403535

A negative answer to Ulam's Problem 19

摘录 / EXCERPT: There exists a convex body $K\subset\mathbb{R}^{3}$ of density $\mathcal{D}=\frac{1}{2}$, which is not a Euclidean ball, yet floats in equilibrium in every orientation.

open_problem | open | MathDB 页面 | 原文线索

V2-079744 | MATHDB-403542

Computation of the Picard group of a semisimplified representation category

摘录 / EXCERPT: Compute the Picard group of the quotient category $\operatorname{Rep}(G)_{\mathcal I}/\mathcal N =: \operatorname{Rep}(G^{\mathcal I}_{\mathrm{red}},\varepsilon)$, where G is an...

open_problem | open | MathDB 页面 | 原文线索

V2-079745 | MATHDB-403544

Finitely generated locally elliptic subgroups of the Cremona group

摘录 / EXCERPT: Let G be a finitely generated subgroup of the Cremona group such that each of its elements is regularizable. Does it follow that G is regularizable, meaning that it is a subgro...

open_problem | open | MathDB 页面 | 原文线索

V2-079746 | MATHDB-403545

Combining adaptive hp-finite elements with local Kalman and global Luenberger observers

摘录 / EXCERPT: Can adaptive hp-finite element methods be combined to obtain a sub-optimal Kalman filter near the moving sensor platform and a crude but stabilizing Luenberger observer on the re...

open_problem | open | MathDB 页面 | 原文线索

V2-079747 | MATHDB-403547

Optimal control of perfect plasticity: stress-tracking problem (P_σ)

摘录 / EXCERPT: The problem (P_σ) is to minimize ... subject to, for almost every $t\in(0,T)$, ... with ... Here $\Omega\subset\mathbb R^d$ with $d=2,3$ is a bounded Lipschitz domain with bou...

open_problem | open | MathDB 页面 | 原文线索

V2-079748 | MATHDB-403548

Sparse initial-data identification from finitely many pointwise final-state measurements

摘录 / EXCERPT: Determine/analyze the sparse initial-data identification problem obtained by minimizing ... for $q\in\mathcal M(\Omega)$, subject to ... ... where ξ_k are measurement points and...

open_problem | open | MathDB 页面 | 原文线索

V2-079749 | MATHDB-403551

Induced-subgraph characterization of large-treewidth graphs

摘录 / EXCERPT: Is it true that for all k,t, there exists $c=c(k,t)$ such that every graph with treewidth more than c contains as an induced subgraph either a K_t, a $K_{t,t}$, a subdivision...

open_problem | open | MathDB 页面 | 原文线索

V2-079750 | MATHDB-403553

Subgraphs of large connectivity and chromatic number

摘录 / EXCERPT: For each $k\in\mathbb{N}$, every graph G with chromatic number at least $7k+1$ contains a subgraph H with both connectivity and chromatic number at least k.

open_problem | open | MathDB 页面 | 原文线索

V2-079751 | MATHDB-403554

Chromatic number of unions of nearly-disjoint complete graphs

摘录 / EXCERPT: There exists $n_0\in\mathbb{N}$ such that, for all $n,t\in\mathbb{N}$ with $n\ge n_0$ and $t\ge 2$, if $G_1,\ldots,G_n$ are complete graphs, each on at most n vertices, and...

open_problem | open | MathDB 页面 | 原文线索

V2-079752 | MATHDB-403556

Theorem 11: homomorphism counts in hereditary bounded-expansion classes

摘录 / EXCERPT: For any hereditary class $\mathcal{G}$ with bounded expansion, graph H, and set $\mathcal{S}$ of homomorphisms from H to members of $\mathcal{G}$ that is "consistent with respe...

open_problem | open | MathDB 页面 | 原文线索

V2-079753 | MATHDB-403557

Linear Hadwiger's conjecture for fixed-clique-free graphs

摘录 / EXCERPT: For every fixed integer $r\ge 1$, there exists a constant $C_r>0$ such that, for every integer $t\ge 1$, every graph G with no K_t minor and no K_r subgraph satisfies...

conjecture | open | MathDB 页面 | 原文线索

V2-079754 | MATHDB-403561

Polynomial acyclic grid theorem for well-linked sets in directed graphs

摘录 / EXCERPT: Does there exist a polynomial function f such that, for every positive integer k, whenever two sets S,T in a directed graph G satisfy $|S|=|T|$ and are (S,T)-well-linked...

open_problem | open | MathDB 页面 | 原文线索

V2-079755 | MATHDB-403564

Logarithmic treewidth after excluding four induced-subgraph obstructions

摘录 / EXCERPT: For every integer $\[1m\ell\[0m$, there exists $C_\ell>0$ such that, if a graph G contains none of the following as an induced subgraph: - a subdivision of an $\ell\times\ell$ wa...

open_problem | open | MathDB 页面 | 原文线索

V2-079756 | MATHDB-403569

2-well-quasi-order of planar graphs

摘录 / EXCERPT: Are planar graphs 2-well-quasi-ordered under the minor relation? That is, can we rule out the existence of a family of planar graphs $(G_{ij})_{i<j\in\mathbb{N}}$ such that there a...

open_problem | open | MathDB 页面 | 原文线索

V2-079757 | MATHDB-403570

An inequality comparing cycle-weights and order-weights on the symmetric group

摘录 / EXCERPT: Let n be a positive integer and let $w:E(K_n)\to\mathbb{R}_{\ge 0}$ be a nonnegative edge-weighting of the complete graph on $[n]$. For each $\pi\in S_n$, define ... where...

open_problem | open | MathDB 页面 | 原文线索

V2-079758 | MATHDB-403571

Uniform reconstruction of finite subsets of \(\mathbb R^3\) from fixed-size decks

摘录 / EXCERPT: There is some $k \in \mathbb{N}$ such that every finite subset of $\mathbb{R}^3$ is determined up to isometry by its k-deck.

open_problem | open | MathDB 页面 | 原文线索

V2-079759 | MATHDB-403574

Asymptotic dimension of graphs admitting bounded-page book embeddings

摘录 / EXCERPT: Does the class of graphs admitting book embeddings with k pages have asymptotic dimension 2?

open_problem | open | MathDB 页面 | 原文线索

V2-079760 | MATHDB-403577

Brady’s question on finiteness types of subgroups of hyperbolic groups

摘录 / EXCERPT: Let $n \ge 3$ be an integer. Is there a subgroup $H < G$ of a hyperbolic group G which is of finiteness type F_n and not of type F_{n+1}?

open_problem | open | MathDB 页面 | 原文线索

V2-079761 | MATHDB-403578

Finite-index hyperbolic lattices with kernels of type F_{n-1} but not F_n

摘录 / EXCERPT: Let $\[?25l\Lambda < \mathrm{SO}(2n,1)$ be a lattice. Is there a finite-index subgroup $\[?25l\Lambda_0 < \Lambda$ and a homomorphism $\[?25l\phi:\Lambda_0\to\mathbb{Z}$ with kerne...

open_problem | open | MathDB 页面 | 原文线索

V2-079762 | MATHDB-403583

Zero-divisor conjecture for the group Δ

摘录 / EXCERPT: Let Δ be the torsion-free group defined by the presentation $\langle a,b,c,d,e,f,g \mid acb,\ add,\ afg,\ bbf,\ cdg,\ cef,\ eeg\rangle$. Does Δ satisfy the zero-divis...

conjecture | open | MathDB 页面 | 原文线索

V2-079763 | MATHDB-403584

Boundedness of the canonical bi-invariant word metric on higher-rank lattices

摘录 / EXCERPT: Let G be a lattice in a higher rank simple Lie group with finite centre. Is d_S bounded?

open_problem | open | MathDB 页面 | 原文线索

V2-079764 | MATHDB-403585

Coarse kernel of an irrational homomorphism

摘录 / EXCERPT: Let F_2 be the free group on two generators equipped with its canonical coarsification (F_2,d_S). Let $\varphi:F_2\to\mathbb{R}$ be the homomorphism that sends one generator to...

open_problem | open | MathDB 页面 | 原文线索

V2-079765 | MATHDB-403587

Maximal number of coarsifications of $\mathbb Z$

摘录 / EXCERPT: Are there $2^{2^{\aleph_0}}$ distinct coarsifications of $\mathbb Z$?

open_problem | open | MathDB 页面 | 原文线索

V2-079766 | MATHDB-403589

Rate of fixation of a newly arising beneficial mutation

摘录 / EXCERPT: At what rate does a newly arising beneficial mutation become dominant in the population?

open_problem | open | MathDB 页面 | 原文线索

V2-079767 | MATHDB-403591

Coalescence of ancestral lineages in locally regulated population models

摘录 / EXCERPT: In the locally regulated population models described above, what is the behaviour of two or more ancestral lineages which coalesce upon meeting? In particular, obtain information o...

open_problem | open | MathDB 页面 | 原文线索

V2-079768 | MATHDB-403592

Expected consensus time for the voter model on a connected Erdős-Rényi random graph

摘录 / EXCERPT: Let the voter model start with all agents having distinct opinions, with opinion set $S=V=[n]$, on $G\\sim G(n,p_n/n)$, where $p_n=(\\log n)^\\alpha$ for some $\\alpha>1$. In both...

open_problem | open | MathDB 页面 | 原文线索

V2-079769 | MATHDB-403593

Explosion for the inhomogeneous contact process when \(\mathbb{E}[\xi]=\infty\)

摘录 / EXCERPT: Can we get an explosion if $\mathbb{E}[\xi]=\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-079770 | MATHDB-403594

Heavy-tailed fitness for bounded offspring distributions

摘录 / EXCERPT: Given a bounded offspring distribution, can one choose the fitness distribution sufficiently heavy-tailed so that the inhomogeneous contact process on a Galton-Watson tree does not...

open_problem | open | MathDB 页面 | 原文线索

V2-079771 | MATHDB-403596

Limit of the profile of a uniform multitype forest with a given degree sequence

摘录 / EXCERPT: Conjecture. The limit of the profile of a uniform multitype forest with a given degree sequence exists, and can be described as a solution to the Lamperti transform, as in [3].

open_problem | open | MathDB 页面 | 原文线索

V2-079772 | MATHDB-403597

Duality of the gene-based diversification model

摘录 / EXCERPT: Is the gene-based diversification model the dual to a simple forward-in-time process?

open_problem | open | MathDB 页面 | 原文线索

V2-079773 | MATHDB-403598

Replacing the complex-analytic techniques in the total-positivity characterization

摘录 / EXCERPT: May the complex analytic techniques of [3, 4] be replaced with something more straightforward?

open_problem | open | MathDB 页面 | 原文线索

V2-079774 | MATHDB-403600

Reality of all zeros of the de Bruijn-Newman family for positive parameter

摘录 / EXCERPT: For every $\[1m\lambda>0\[0m$, all zeros of $\[1m\Xi_\lambda(t)\[0m$ are real.

open_problem | open | MathDB 页面 | 原文线索

V2-079775 | MATHDB-403601

Hyperbolicity of all Jensen polynomials associated with the Riemann \Xi-function

摘录 / EXCERPT: Determine whether every polynomial ... is hyperbolic, meaning that all of its zeros are real, for all nonnegative integers d,n. The report states that this condition is equivalent...

open_problem | open | MathDB 页面 | 原文线索

V2-079776 | MATHDB-403602

Local tensorization versus global tensorization

摘录 / EXCERPT: Conjecture: If a functional $\sum_S \alpha_S H(Y_S)$ satisfies the local tensorization property, then it also satisfies the global tensorization property.

open_problem | open | MathDB 页面 | 原文线索

V2-079777 | MATHDB-403603

Connections between common information, information obfuscation, and the privacy funnel

摘录 / EXCERPT: Find out if there exist direct links between various notions of “common information” to the studied problem or to problems on the “privacy funnel.” The latter is a class of problem...

open_problem | open | MathDB 页面 | 原文线索

V2-079778 | MATHDB-403605

Intersection-union interchange for the broadcast AVC

摘录 / EXCERPT: For the broadcast arbitrarily varying channel (AVC), inner and outer bounds on the random- and deterministic-code capacity regions differ only in the order of (infinite) intersecti...

open_problem | open | MathDB 页面 | 原文线索

V2-079779 | MATHDB-403606

Information inequality relating mutual information to a partition function

摘录 / EXCERPT: There exists a constant $c>0$ such that, for any distribution P_X on $\cmathbb{R}^n$, ... where G is standard normal in $\cmathbb{R}^n$, independent of X, and...

open_problem | open | MathDB 页面 | 原文线索

V2-079780 | MATHDB-403607

Common framework for the Gaussian entropy-power inequality

摘录 / EXCERPT: Let X, Y, and Z be independent random vectors of the same dimension n, with Z Gaussian. Writing $N(U):=e^{2h(U)/n}$ for the entropy power of U, the known inequality ... d...

open_problem | open | MathDB 页面 | 原文线索

V2-079781 | MATHDB-403608

Closing the inner-outer bound gap for the many-user Gaussian multiple-access channel

摘录 / EXCERPT: In the many-user regime for the Gaussian multiple-access channel, where the number of users grows proportionally to the code length, close the gap between the achievable-region (in...

open_problem | open | MathDB 页面 | 原文线索

V2-079782 | MATHDB-403611

Single-letter characterization of the mismatch capacity of a discrete memoryless channel

摘录 / EXCERPT: Find a single-letter expression for the mismatch capacity of the discrete memoryless channel, i.e., the supremum of rates for which there exists a sequence of codebooks with vanish...

open_problem | open | MathDB 页面 | 原文线索

V2-079783 | MATHDB-403614

Characterization of dependence measures with concavity properties

摘录 / EXCERPT: Let $\rho(A;B)$ be a measure of dependence assigning a nonnegative real number to each joint distribution $p(a,b)$, satisfying the data-processing property and $\rho(A;B)=0$ whenev...

open_problem | open | MathDB 页面 | 原文线索

V2-079784 | MATHDB-403615

Blow-up solutions with more points or repeated intervals

摘录 / EXCERPT: For the nonlocal mean-field problem (2) on a union $I=\bigcup_{k=1}^{d} I_k$ of d open intervals with pairwise disjoint closures, is it possible to construct blowing-up solutions...

open_problem | open | MathDB 页面 | 原文线索

V2-079785 | MATHDB-403616

First dimension containing stable/minimal cones in the fractional one-phase problem

摘录 / EXCERPT: For the fractional one-phase free-boundary problem with parameter $s\in(0,1)$, what is the first dimension k_s^* in which stable or minimal cones appear?

open_problem | open | MathDB 页面 | 原文线索

V2-079786 | MATHDB-403617

Homogeneous Dirichlet problem for the fractional Laplacian

摘录 / EXCERPT: Let $0<a<1$, let $\Omega earrow\mathbb{R}^n$ be a bounded open set with $C^{1+\gamma}$ boundary for some $\gamma>0$, and let $P=(-\Delta)^a$ on $\mathbb{R}^n$, defined by...

open_problem | open | MathDB 页面 | 原文线索

V2-079787 | MATHDB-403618

Continuity of a minimizer for the regional fractional Faber-Krahn problem

摘录 / EXCERPT: Under the hypotheses of Theorem 2, namely $n\ge 2$ and $\alpha\in(1/2,1)$, consider ... where ... Theorem 2 asserts that this infimum is attained. Is there a minimizer that is continuo...

open_problem | open | MathDB 页面 | 原文线索

V2-079788 | MATHDB-403619

Heat-kernel comparability for symmetric Markov processes with stable-like jump intensity

摘录 / EXCERPT: Conjecture: Let L_t be a Lévy process (a non-degenerate $\$-stable process) in $\^d$ with Lévy measure $\$. Let M_t be a symmetric Markov process whose Dirichlet form has a sym...

open_problem | open | MathDB 页面 | 原文线索

V2-079789 | MATHDB-403620

Existence and uniqueness for the semilinear Dirichlet problem with measure data

摘录 / EXCERPT: For the Dirichlet problem ... where the report assumes that $D\subset E$ is open, E is a locally compact separable metric space, m is a Radon measure with full support,...

open_problem | open | MathDB 页面 | 原文线索

V2-079790 | MATHDB-403621

Optimal dividend and capital-injection strategy for a Lévy process

摘录 / EXCERPT: Fix an injection cost per unit $\eta>1$ and discount rate $q>0$. Let $\mathcal A$ be the class of strategies $\pi=(L^\pi,R^\pi)$ such that $L_t^\pi=\int_0^t l_s^\pi\,ds$ for a prog...

open_problem | open | MathDB 页面 | 原文线索

V2-079791 | MATHDB-403622

Purely analytic proof of the Kolmogorov regularity and stability results

摘录 / EXCERPT: Give a purely analytic proof of the regularity results for the perturbed Kolmogorov equation ... where $s:[0,T]\to[0,\infty)$ is continuous. In particular, prove analytically both th...

open_problem | open | MathDB 页面 | 原文线索

V2-079792 | MATHDB-403623

Construction of Feller semigroups from prescribed analytic data

摘录 / EXCERPT: Let Ω be the bounded smooth domain in $\mathbb{R}^n$ described in the talk, with closure $\overline{\Omega}=\Omega\cup\partial\Omega$. Problem: conversely, construct a Felle...

open_problem | open | MathDB 页面 | 原文线索

V2-079793 | MATHDB-403630

Deriving Nekrasov’s blow-up formula from a coset VOA decomposition

摘录 / EXCERPT: Prove the blow-up formula ... as a consequence of the VOA representation identity ... assuming the parameter relations displayed in (4), together with ... The derivation must also determ...

open_problem | open | MathDB 页面 | 原文线索

V2-079794 | MATHDB-403631

Differential graded Verlinde formula

摘录 / EXCERPT: Does the Verlinde formula extend to differential graded modular functors, with the torus conformal block and its dual equipped with the relevant monoidal-product-induced E_2-alge...

open_problem | open | MathDB 页面 | 原文线索

V2-079795 | MATHDB-403635

Finite generation criterion for valuation semigroups on toric surfaces with non-toric flags

摘录 / EXCERPT: Let (X,D) be a polarized toric surface determined by a polytope P in a two-dimensional lattice M, with normal fan Σ in the dual lattice N. Let $v\in N$ be primitive,...

open_problem | open | MathDB 页面 | 原文线索

V2-079796 | MATHDB-403636

Explicit non-trivial Kähler-Ricci soliton degenerations with non-divisorial valuation

摘录 / EXCERPT: Explicitly describe non-trivial first degenerations $X\rightsquigarrow (Z,\xi_Z)$ in the setting above, and find an example for which the valuation v is not divisorial.

open_problem | open | MathDB 页面 | 原文线索

V2-079797 | MATHDB-403638

Lower bound for the normalized volume of a toric Gorenstein singularity

摘录 / EXCERPT: For a d-dimensional toric Gorenstein singularity X admitting a toric crepant resolution $\widetilde X$, does the conjectural inequality ... hold, where $\chi(\widetilde X)$ denot...

open_problem | open | MathDB 页面 | 原文线索

V2-079798 | MATHDB-403642

Which matching fields produce toric degenerations?

摘录 / EXCERPT: Which matching fields produce toric degenerations?

open_problem | open | MathDB 页面 | 原文线索

V2-079799 | MATHDB-403644

Spectral edges conjecture for discrete periodic Schrödinger operators

摘录 / EXCERPT: For a sufficiently general discrete periodic Schrödinger operator L (for example, with general periodic edge parameters c and potential V), all extrema of the function...

conjecture | open | MathDB 页面 | 原文线索

V2-079800 | MATHDB-403645

Dimension or character formulas for irreducible representations of reductive groups

摘录 / EXCERPT: Find a formula for the dimension or, more strongly, the character of the irreducible representation L_λ, for $\lambda\in X_+$, where G is a reductive group over an algebr...

open_problem | open | MathDB 页面 | 原文线索

V2-079801 | MATHDB-403646

Lusztig’s character formula conjecture under Jantzen’s condition

摘录 / EXCERPT: Let h be the Coxeter number of (G,T)), and assume $p\ge h$. Suppose that $w\in W_{\mathrm{aff}}$ satisfies $w\mathbin{\cdot_p}0\in X(T)^+$ and ... for every positive root...

conjecture | open | MathDB 页面 | 原文线索

V2-079802 | MATHDB-403647

Finkelberg-Mirković conjecture

摘录 / EXCERPT: Assume that k is a field of characteristic p. Let G be a reductive group, let W be its Weyl group, and let X be the cocharacter lattice of a maximal torus. Set...

conjecture | open | MathDB 页面 | 原文线索

V2-079803 | MATHDB-403648

Dowker space of size \(\aleph_1\) in ZFC

摘录 / EXCERPT: Does ZFC prove that there exists a Dowker space of size $\aleph_1$?

open_problem | open | MathDB 页面 | 原文线索

V2-079804 | MATHDB-403649

Whether the Axiom of Choice is necessary for Kunen’s inconsistency

摘录 / EXCERPT: Is the Axiom of Choice necessary for Kunen’s inconsistency theorem-that is, can there exist a nontrivial elementary embedding $j:V\to V$ in the absence of the Axiom of Choice?

open_problem | open | MathDB 页面 | 原文线索

V2-079805 | MATHDB-403650

Dual of Juhász’ question: Souslin trees and a weak club-guessing principle

摘录 / EXCERPT: Does the existence of a Souslin tree imply a weak form of $\clubsuit$, strong enough to entail the existence of a Dowker space of size $\aleph_1$?

open_problem | open | MathDB 页面 | 原文线索

V2-079806 | MATHDB-403651

Rigidity conjectures in continuous quotients

摘录 / EXCERPT: Conjecture 3. For Borel quotient structures M/E and N/F, consider the following statements:

conjecture | open | MathDB 页面 | 原文线索

V2-079807 | MATHDB-403652

Separation of chromatic numbers of rational-distance graphs in successive dimensions

摘录 / EXCERPT: For every integer $n\ge 1$, let Γ_n be the graph on $\mathbb{R}^n$ whose distinct vertices are connected when their Euclidean distance is rational. It is consistent relative...

open_problem | open | MathDB 页面 | 原文线索

V2-079808 | MATHDB-403653

Rational infeasibility certificates for conic programs

摘录 / EXCERPT: For a conic feasibility problem $K\cap L=\varnothing$, where K is a closed convex pointed cone and L is an affine subspace, an infeasibility certificate is an element...

open_problem | open | MathDB 页面 | 原文线索

V2-079809 | MATHDB-403654

Rational CP-certificates for completely positive matrices

摘录 / EXCERPT: Does every rational completely positive matrix possess a rational CP-certificate; that is, for every rational completely positive matrix Q, do there exist $k\in\mathbb{N}$ and no...

open_problem | open | MathDB 页面 | 原文线索

V2-079810 | MATHDB-403655

Closed-form coefficients for multivariate minimum-resolution polynomial kernels

摘录 / EXCERPT: Is it possible to find a closed-form solution for the coefficients of the multivariate kernel K_r with minimal resolution?

open_problem | open | MathDB 页面 | 原文线索

V2-079811 | MATHDB-403657

Asymptotic sharpness of the Cohn-Elkies linear programming bound

摘录 / EXCERPT: The Cohn-Elkies linear programming bound can be used to get the asymptotic upper bound $2^{-(\lambda+o(1))n}$ with $0.604<\lambda<0.605$.

open_problem | open | MathDB 页面 | 原文线索

V2-079812 | MATHDB-403658

Minimal recurrence order of generalized Franel numbers

摘录 / EXCERPT: For the generalized Franel numbers $u^{(s)}(n)=\sum_{k=0}^{n}\binom{n}{k}^{s}$, with fixed $s=1,2,\ldots$, is the order of a recurrence with polynomial coefficients in n equal in...

open_problem | open | MathDB 页面 | 原文线索

V2-079813 | MATHDB-403660

Uniform upper bound for integer matrices with a prescribed characteristic polynomial

摘录 / EXCERPT: Fix a positive integer n. For real $H\ge 1$, let $M_n(\mathbb Z;H)$ be the set of $n\times n$ integer matrices $A=(a_{ij})$ with $|a_{ij}|\le H$, and, for $f\in\mathbb Z[X]$, def...

open_problem | open | MathDB 页面 | 原文线索

V2-079814 | MATHDB-403664

Existence of transcendental numbers with \celebrated uniform exponent above the Dirichlet bound

摘录 / EXCERPT: Given an integer $n\ge 3$, does there exist a transcendental real number ξ such that $\widehat{\lambda}_n(\xi)>1/n$?

open_problem | open | MathDB 页面 | 原文线索

V2-079815 | MATHDB-403665

Strengthening the modular support theorem for Hilbert class polynomials

摘录 / EXCERPT: Let K be a number field, let S be a finite set of maximal ideals of $\mathcal O_K$, and let $a,b\in\mathcal O_{K,S}$. Let $\mathbb D$ be the set of negative integers congruent...

open_problem | open | MathDB 页面 | 原文线索

V2-079816 | MATHDB-403667

Polynomial height bounds for the reduction-theory set

摘录 / EXCERPT: Can Theorem 1 be improved so as to obtain polynomial bounds for the heights of the elements of the finite set B_u themselves, rather than only for the quantities...

open_problem | open | MathDB 页面 | 原文线索

V2-079817 | MATHDB-403669

Uniform Manin-Mumford conjecture for curves embedded in their Jacobians

摘录 / EXCERPT: For each integer $g\ge 2$, there exists a constant $c=c(g)>0$ with the following property: if C is a geometrically irreducible, smooth, projective curve of genus g defined over...

conjecture | open | MathDB 页面 | 原文线索

V2-079818 | MATHDB-403671

Hermite classes in \(\operatorname{PI}(K)\) versus \(\mathrm{GL}_2(\mathbb Z)\)-equivalence

摘录 / EXCERPT: For a number field K, let $\operatorname{PI}(K)$ denote the set of primitive, irreducible polynomials in $\mathbb Z[X]$ having a root that generates K. Is it true that...

open_problem | open | MathDB 页面 | 原文线索

V2-079819 | MATHDB-403673

Improved lower bounds on the depth of polynomial-size Boolean circuits for the independent set problem

摘录 / EXCERPT: Fix any $\c1\e 1/2$ and $\b1\e 0$, and let $p(n)=\frac{\log n}{(1+\u0003b1)\log\log n}$. Set $m=\binom{n}{2}$. Let $\mathcal C(n,p(n),\u0003c1)$ be the family of m-input, n-out...

open_problem | open | MathDB 页面 | 原文线索

V2-079820 | MATHDB-403676

Structure of polynomial-density product-free subsets of the alternating group

摘录 / EXCERPT: Suppose that $A\subseteq A_n$ is a product-free set of density $\mu(A)>1/n^C$ for an absolute constant C. What can be said about the structure of A?

open_problem | open | MathDB 页面 | 原文线索

V2-079821 | MATHDB-403678

Enumeration of proper linear spaces

摘录 / EXCERPT: Let $q(n)$ denote the number of proper linear spaces on an n-point set, where a proper linear space is a linear space in which no line has exactly 2 points. Is it true that ...

open_problem | open | MathDB 页面 | 原文线索

V2-079822 | MATHDB-403680

Characterization of common systems of linear equations

摘录 / EXCERPT: Find a characterization of common systems of linear equations over finite fields.

open_problem | open | MathDB 页面 | 原文线索

V2-079823 | MATHDB-403681

Talagrand’s selector-process conjecture

摘录 / EXCERPT: There exists a universal constant $L>0$ such that, for every $0<p<1$, every finite set X, and every collection Λ of sequences $\lambda=(\lambda_i)_{i\in X}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-079824 | MATHDB-403682

List-decoding capacity of Reed-Solomon codes

摘录 / EXCERPT: For every fixed list size $L\in\mathbb{N}$ and radius $0<r<\frac{L}{L+1}$, and for large n, there exist (r,L)-list-decodable Reed-Solomon codes of rate ... Here an (r,L)-list-d...

open_problem | open | MathDB 页面 | 原文线索

V2-079825 | MATHDB-403683

Girth-preserving version for induced subgraphs of large chromatic number

摘录 / EXCERPT: Conjecture 4. For every graph F with at least one cycle, there exists a constant b_F and graphs G of arbitrarily large chromatic number and the same girth as F such that ev...

open_problem | open | MathDB 页面 | 原文线索

V2-079826 | MATHDB-403686

Second-order asymptotics for weak saturation numbers

摘录 / EXCERPT: For every r-graph H of sparseness $s=\\operatorname{sp}(H)$, the refined bound $\\operatorname{wsat}(n,H)=C_H n^{s-1}+O(n^{s-2})$ holds.

open_problem | open | MathDB 页面 | 原文线索

V2-079827 | MATHDB-403692

Nontrivial limits of normalized homomorphism counts

摘录 / EXCERPT: Let $t_p(F,G)=p^{-e(F)}|G|^{-|F|}\operatorname{hom}(F,G)$ be the p-normalized homomorphism count of F within G. Is it possible to construct a sequence of graphs G_n with...

open_problem | open | MathDB 页面 | 原文线索

V2-079828 | MATHDB-403695

Sharpness of the concentration bound for the chromatic number of sparse random graphs

摘录 / EXCERPT: Is the upper bound of Theorem 17 on the concentration of the chromatic number $\chi(G_{n,p})$ of the random graph $G_{n,p}$ close to best possible or not?

open_problem | open | MathDB 页面 | 原文线索

V2-079829 | MATHDB-403696

Twins in uniformly random words

摘录 / EXCERPT: Let A be an alphabet with $|A|=k\ge 2$. For a word $w\in A^n$, let $t(w)$ be the length of the longest pair of identical, disjoint subwords, and let $t_{n,k}=\min_{w\in A^n}t(w)$...

open_problem | open | MathDB 页面 | 原文线索

V2-079830 | MATHDB-403699

Compatibility of Calabi-Yau structures under Drinfeld quotients

摘录 / EXCERPT: Let $B\subseteq A$ be a full pretriangulated dg subcategory of a pretriangulated dg category A. Assume that B is H^*-finite, let $d\in\mathbb Z$, and write A/B for the Drin...

open_problem | open | MathDB 页面 | 原文线索

V2-079831 | MATHDB-403710

Stable reductive Borel-Serre models and the plus-construction

摘录 / EXCERPT: Let A be a ring. There are maps $BGL_n(A)\to \lvert\mathrm{RBS}(A^n)\rvert$ for all n, and taking colimits gives a stable map...

open_problem | open | MathDB 页面 | 原文线索

V2-079832 | MATHDB-403715

Weibel’s conjecture beyond the Noetherian setting

摘录 / EXCERPT: What happens in Weibel’s conjecture if X is not assumed to be Noetherian?

conjecture | open | MathDB 页面 | 原文线索

V2-079833 | MATHDB-403718

Closing the condition-number gap for factorization-based matrix-recovery guarantees

摘录 / EXCERPT: The empirical results exhibit a significant gap from the theoretical recovery guarantees: nearly all available guarantees, including those in the authors’ work, scale at least quad...

open_problem | open | MathDB 页面 | 原文线索

V2-079834 | MATHDB-403719

Optimal ranking in crowd-sourcing problems

摘录 / EXCERPT: Given $M\in[0,1]^{n\times d}$ and a permutation π^* of $[n]$ such that the corresponding permuted matrix $M_{\pi^{*-1}}$ is bi-isotonic, observe...

open_problem | open | MathDB 页面 | 原文线索

V2-079835 | MATHDB-403720

Generic identifiability for non-simple graphical continuous Lyapunov models

摘录 / EXCERPT: Pursue a characterization of which non-simple graphs yield generically identifiable graphical continuous Lyapunov models.

open_problem | open | MathDB 页面 | 原文线索

V2-079836 | MATHDB-403721

Minimax simple regret for the shape-constrained Thresholding Bandit Problem

摘录 / EXCERPT: For a bandit problem with K arms, arm distributions $ u_k$ supported on $[0,1]$, means $\boldsymbol{\boldsymbol{\bc}}=(\bc_k)_{k\u2264 K}$, threshold $\c4\u2208[0,1]$, and horizo...

open_problem | open | MathDB 页面 | 原文线索

V2-079837 | MATHDB-403722

Stability of similarity testing under fractional-noise perturbations

摘录 / EXCERPT: For the fractional diffusion model obtained by replacing the driving Brownian motion with fractional Brownian motion of Hurst index $H\in(0,1)$, the similarity test is uniformly of...

open_problem | open | MathDB 页面 | 原文线索

V2-079838 | MATHDB-403723

Uniform asymptotic level of the similarity test

摘录 / EXCERPT: For every significance level $\alpha\in(0,1)$, the constructed test ϕ_T^η for the similarity hypothesis H_0 satisfies ... where T is the diffusion-observation time horizo...

open_problem | open | MathDB 页面 | 原文线索

V2-079839 | MATHDB-403724

Almost-sure global well-posedness for fractional NLS below L^2 regularity

摘录 / EXCERPT: For the cubic fractional nonlinear Schrödinger equation on the one-dimensional torus, with Gaussian initial data of the form $u_0(x)=\\sum_n g_n\\langle n\\rangle^s e^{inx}$, the r...

open_problem | open | MathDB 页面 | 原文线索

V2-079840 | MATHDB-403726

Invariance of the Gibbs measure for the three-dimensional cubic nonlinear wave equation

摘录 / EXCERPT: Let $N\ge 1$ be a frequency-truncation parameter, and consider the renormalized, frequency-truncated equation on $\mathbb T^3$: ... with initial data...

open_problem | open | MathDB 页面 | 原文线索

V2-079841 | MATHDB-403729

Rigorous derivation of the homogeneous wave kinetic equation for NLS

摘录 / EXCERPT: The report presents the conjecture that, in the kinetic limit, the Fourier-mode energy density of cubic NLS is governed by the homogeneous wave kinetic equation ... where ... The same...

open_problem | open | MathDB 页面 | 原文线索

V2-079842 | MATHDB-403740

Stable Bernstein conjecture for minimal hypersurfaces

摘录 / EXCERPT: A complete, two-sided, stable minimal immersion $M^n\to(\mathbb{R}^{n+1},\delta)$ is flat.

conjecture | open | MathDB 页面 | 原文线索

V2-079843 | MATHDB-403748

Generality of the trace-complex construction for Hilbert complexes

摘录 / EXCERPT: Is the trace Hilbert-complex structure observed for the Euclidean de Rham complex a peculiarity of that complex, or is it a general structure inherent in a certain class of Hilbert...

open_problem | open | MathDB 页面 | 原文线索

V2-079844 | MATHDB-403749

Common structure and finite-dimensional representation of second-order boundary value problems

摘录 / EXCERPT: What is the common structure behind second-order boundary value problems, and how should this structure be represented in finite-dimensional spaces (in which the problems are solve...

open_problem | open | MathDB 页面 | 原文线索

V2-079845 | MATHDB-403750

Traceable type II∞ representations of \(\mathrm{SL}_n(\mathbb{Z})\)

摘录 / EXCERPT: For all $n\ge 2$, $\mathrm{SL}_n(\mathbb{Z})$ admits uncountably many traceable generating representations into type II_∞-factors, none of which is weakly contained...

open_problem | open | MathDB 页面 | 原文线索

V2-079846 | MATHDB-403752

Amenability of automorphism groups of minimal zero-entropy subshifts

摘录 / EXCERPT: When $G=\mathbb{Z}$ and $X\subseteq 2^{\mathbb{Z}}$ is a minimal, 0-entropy subshift, must $\mathrm{Aut}(X,\mathbb{Z})$ be amenable?

open_problem | open | MathDB 页面 | 原文线索

V2-079847 | MATHDB-403754

Type I hyperbolic groups containing a uniform lattice have a cocompact amenable subgroup

摘录 / EXCERPT: Let G be a hyperbolic group containing a uniform lattice. If G is type I, then G has a cocompact amenable subgroup.

open_problem | open | MathDB 页面 | 原文线索

V2-079848 | MATHDB-403755

Nebbia-Houdayer-Raum conjectural implication for tree groups

摘录 / EXCERPT: Let T be a locally finite tree, each of whose vertices has degree at least 3, and let G be a non-compact closed subgroup of $\operatorname{Aut}(T)$ whose action on...

open_problem | open | MathDB 页面 | 原文线索

V2-079849 | MATHDB-403756

Generalizing Garncarek’s boundary-representation results

摘录 / EXCERPT: Can Garncarek’s results on boundary representations of discrete hyperbolic groups be generalized to all non-amenable hyperbolic locally compact groups?

open_problem | open | MathDB 页面 | 原文线索

V2-079850 | MATHDB-403757

Quadratic growth for faithful actions of torsion-free solvable groups

摘录 / EXCERPT: Let G be a finitely generated solvable torsion-free group. Then either G is virtually abelian or, for every faithful G-set X, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-079851 | MATHDB-403758

Almost finiteness and classifiability of crossed products

摘录 / EXCERPT: Conjecture. For a system $G\curvearrowright X$, the following are equivalent:

open_problem | open | MathDB 页面 | 原文线索

V2-079852 | MATHDB-403760

Strong Atiyah conjecture for groups with uniformly bounded finite subgroups

摘录 / EXCERPT: Is it true that every group Γ for which the orders of finite subgroups are uniformly bounded satisfies the strong Atiyah conjecture; namely, for all positive integers m,n...

conjecture | open | MathDB 页面 | 原文线索

V2-079853 | MATHDB-403763

Baire measurable version of Vizing’s theorem

摘录 / EXCERPT: Does every Borel graph G on a Polish space, with finite maximum degree $\[1m\Delta(G)\u001b[0m$, satisfy $\chi'_{\mathrm{BM}}(G) \le \Delta(G)+1$?

open_problem | open | MathDB 页面 | 原文线索

V2-079854 | MATHDB-403766

IRO-extension property for non-amenable groups

摘录 / EXCERPT: Does there exist a non-amenable group with the IRO-extension property?

open_problem | open | MathDB 页面 | 原文线索

V2-079855 | MATHDB-403768

Global well-posedness for DNLS on the circle without the mass restriction

摘录 / EXCERPT: For the derivative nonlinear Schrödinger equation on the circle, is the evolution globally well-posed in $H^s(\mathbb T)$ for every $1/6\le s<1/2$ and every initial datum...

open_problem | open | MathDB 页面 | 原文线索

V2-079856 | MATHDB-403769

Scattering below the ground state for the non-radial energy-critical Zakharov system in four dimensions

摘录 / EXCERPT: For the four-dimensional Zakharov system ... with energy-space initial data $(u,V)|_{t=0}=(f,g)\in H^1(\mathbb R^4)\times L^2(\mathbb R^4)$, does every not-necessarily-radial solutio...

open_problem | open | MathDB 页面 | 原文线索

V2-079857 | MATHDB-403770

Soliton resolution for radial quadratic waves in six dimensions

摘录 / EXCERPT: For either equation ... on $\mathbb R\times\mathbb R^6$, let u be a radial solution with maximal forward existence time T_+ and ... With $W(x)=(1+|x|^2/24)^{-2}$, if $T_+<\infty$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079858 | MATHDB-403772

Genericity of the explicit self-similar blowup profile for radial Keller-Segel solutions

摘录 / EXCERPT: Is the explicit self-similar solution $u_T(t,x)=\frac{1}{T-t}U\!\left(\frac{x}{\sqrt{T-t}}\right)$, where $U(x)=\frac{4(d-2)(2d+|x|^2)}{(2(d-2)+|x|^2)^2}$ and $T>0$, the generic bl...

open_problem | open | MathDB 页面 | 原文线索

V2-079859 | MATHDB-403773

Convergence to the explicit self-similar Keller-Segel profile from non-close data

摘录 / EXCERPT: For the three-dimensional parabolic-elliptic Keller-Segel system (1), determine whether the convergence to the explicit self-similar profile u_T observed in [5] also holds for so...

open_problem | open | MathDB 页面 | 原文线索

V2-079860 | MATHDB-403774

Local well-posedness for the time-like minimal surface equation

摘录 / EXCERPT: Let $n\ge 2$. For the time-like minimal surface equation ... with initial data $u[0]=(u_0,u_1)\in H^s(\mathbb R^n)\times H^{s-1}(\mathbb R^n)$ satisfying the pointwise constraint...

open_problem | open | MathDB 页面 | 原文线索

V2-079861 | MATHDB-403775

Maximal averages over non-degenerate curves in higher dimensions

摘录 / EXCERPT: Let $I\subset\mathbb{R}$ be a compact interval, let $\gamma:I\to\mathbb{R}^n$ be smooth, and let $\chi\in C^\infty(\mathbb{R})$ be a bump function supported in the interior of I....

open_problem | open | MathDB 页面 | 原文线索

V2-079862 | MATHDB-403776

General Fourier restriction estimates for smooth hypersurfaces

摘录 / EXCERPT: For a sufficiently smooth hypersurface $S\subset\mathbb{R}^n$, determine the range of exponents $(\widetilde p,\widetilde q)$ for which the a priori estimate ... holds for every Schw...

open_problem | open | MathDB 页面 | 原文线索

V2-079863 | MATHDB-403780

Uniform square-function inequality for the planar moment curve at an intermediate exponent

摘录 / EXCERPT: For the moment curve $\gamma(t)=(t,t^2)$ and the associated extension operators Ef and $E_I f$, establish that there exists some exponent p with $2<p<4$ and a constant C_p in...

open_problem | open | MathDB 页面 | 原文线索

V2-079864 | MATHDB-403781

Foschi’s conjectured maximizer for the wave Strichartz inequality

摘录 / EXCERPT: For every integer $d\ge 2$, the solution u of the wave equation $u_{tt}=\Delta u$ on $\mathbb{R}^{1+d}$ with initial data ... should be a maximizer for the Strichartz estimate (1),...

conjecture | open | MathDB 页面 | 原文线索

V2-079865 | MATHDB-403782

Near-polynomial correction factors for majorant properties

摘录 / EXCERPT: For which sets $\Gamma \subset [1,N]^d$ does $B_p(\Gamma) \ll N^{\varepsilon}$ suffice for every $\varepsilon>0$?

open_problem | open | MathDB 页面 | 原文线索

V2-079866 | MATHDB-403783

Determining the spherical maximal type set from Assouad-spectrum data

摘录 / EXCERPT: An interesting open problem is to determine whether and how, for any given E, the set T_E can be determined from the Assouad spectrum of E and subsets of E.

open_problem | open | MathDB 页面 | 原文线索

V2-079867 | MATHDB-403784

Fourier restriction conjecture for the sphere

摘录 / EXCERPT: For dimensions $d\ge 3$, the Fourier restriction conjecture predicts that there is a constant C such that ... for $1\le p<\frac{2d}{d+1}$ and $q\le \frac{d-1}{d+1}p'$.

conjecture | open | MathDB 页面 | 原文线索

V2-079868 | MATHDB-403785

Endpoint eigenfunction bound on the torus

摘录 / EXCERPT: For the torus $\mathbb T^d$, the report identifies as an important conjecture the endpoint estimate ... Here e_λ is a Laplace eigenfunction and $d\ge 2$, as in the surroundin...

open_problem | open | MathDB 页面 | 原文线索

V2-079869 | MATHDB-403786

Weyl remainder conjecture for products of at least five spheres

摘录 / EXCERPT: Let $n\geq 5$ and let $M=S^{d_1}\times\cdots\times S^{d_n}$ be a Cartesian product of spheres, with $d_i\geq 1$ and $d=d_1+\cdots+d_n$. For the Weyl counting function $N(\lambda)$...

conjecture | open | MathDB 页面 | 原文线索

V2-079870 | MATHDB-403787

Identity of extremizers for Fourier restriction to the moment curve

摘录 / EXCERPT: Determine the identity of the extremizers for the Fourier restriction inequality (1), equivalently for the adjoint extension inequality (2), perhaps only in the case $p=2$.

open_problem | open | MathDB 页面 | 原文线索

V2-079871 | MATHDB-403788

Rudin’s analytic characterization of Sidon sets

摘录 / EXCERPT: For a set $\Lambda\subset\mathbb Z^n$, let $F=\exp(L^2)(\mathbb T^n)$. The report presents the estimate ... as a characterization of Λ being a Sidon set, where Sidon means...

open_problem | open | MathDB 页面 | 原文线索

V2-079872 | MATHDB-403789

Oberdieck-Schimpf normalization-factor conjecture for elliptic threefolds

摘录 / EXCERPT: Let B be a smooth projective surface, and let $\pi:X\to B$ be an elliptic threefold with section $\iota:B\to X$ and Weierstrass-model structure. Set $N=N_{B/X}$, the normal bundl...

conjecture | open | MathDB 页面 | 原文线索

V2-079873 | MATHDB-403790

Quasi-Jacobi-form conjecture for Pandharipande-Thomas invariants of elliptic threefolds

摘录 / EXCERPT: Let B be a smooth projective surface and let $\pi:X\to B$ be a Weierstrass elliptic fibration with section $\iota:B\to X$. Put $N=N_{B/X}$, and for $\beta\in H_2(B,\mathbb Z)$ de...

conjecture | open | MathDB 页面 | 原文线索

V2-079874 | MATHDB-403793

Mori’s question for limits of plane curves

摘录 / EXCERPT: For which degrees $d>1$ is every smooth compact limit of complex degree-d plane curves a degree-d plane curve?

open_problem | open | MathDB 页面 | 原文线索

V2-079875 | MATHDB-403797

Log terminal \Q-Gorenstein degenerations of projective space other than cones

摘录 / EXCERPT: Other than cones, what are examples of log terminal $\mathbb{Q}$-Gorenstein degenerations of $\mathbb{P}^n$ with isolated singularities?

open_problem | open | MathDB 页面 | 原文线索

V2-079876 | MATHDB-403802

Semiorthogonal decomposition for the moduli space of rank-2 bundles

摘录 / EXCERPT: Let C be a smooth projective curve of genus g, and let L be a line bundle of odd degree. Then there exists a semiorthogonal decomposition ...

open_problem | open | MathDB 页面 | 原文线索

V2-079877 | MATHDB-403804

Anti-ampleness of the Frobenius cokernel on Fano threefolds

摘录 / EXCERPT: For which Fano threefolds of Picard rank 1 is B_X^1 anti-ample? Does this depend on the characteristic as it does for quadrics? Here B_X^1 is defined by the exact sequence...

open_problem | open | MathDB 页面 | 原文线索

V2-079878 | MATHDB-403806

Extension of hierarchical finite-size scaling methods to the Euclidean model

摘录 / EXCERPT: “Our methods have the potential to extend from the hierarchical model to the Euclidean model with periodic boundary conditions; it is an open problem to carry out this extension.”

open_problem | open | MathDB 页面 | 原文线索

V2-079879 | MATHDB-403808

Kernel of the geometric Johnson homomorphism

摘录 / EXCERPT: Let $\tau_{g,1}:\operatorname{Gr}_\bullet\mathfrak t_{g,1}\to\operatorname{Der}(\operatorname{Lie}(H_1(\Sigma_{g,1};\mathbb{Q})))$ be the geometric Johnson homomorphism. In weights...

open_problem | open | MathDB 页面 | 原文线索

V2-079880 | MATHDB-403814

Conjecture that commuting-operator MIP equals coRE

摘录 / EXCERPT: The natural conjecture is ...

conjecture | open | MathDB 页面 | 原文线索

V2-079881 | MATHDB-403821

Saxl’s tensor-square conjecture

摘录 / EXCERPT: Saxl conjectured that $\chi^{\otimes 2}_{(k,k-1,k-2,\ldots,2,1)}$ contains every irreducible $\mathbb{C}S_n$-character with non-zero multiplicity.

conjecture | open | MathDB 页面 | 原文线索

V2-079882 | MATHDB-403822

Bessenrodt’s conjecture on alternating-group character vanishing sets

摘录 / EXCERPT: Conjecture 7 (Bessenrodt). The pairs of χ^λ, χ^μ such that V^{An}λ=V^{An}μ and λne μ are as follows:

conjecture | open | MathDB 页面 | 原文线索

V2-079883 | MATHDB-403823

Limiting behavior of zero character values in symmetric-group character tables

摘录 / EXCERPT: For each n, choose partitions $\bb$ and $\bc$ of n uniformly at random, and let $\c7_{\u0003bb}(\u0003bc)$ denote the irreducible character of S_n indexed by $\bb$, evaluated...

open_problem | open | MathDB 页面 | 原文线索

V2-079884 | MATHDB-403824

Divisibility of character-table entries of symmetric groups

摘录 / EXCERPT: For any positive integer m, with $\chi_\lambda(\mu)$ denoting the irreducible character value of S_n indexed by the partition λ on a permutation of cycle type μ, an...

open_problem | open | MathDB 页面 | 原文线索

V2-079885 | MATHDB-403826

Finite-complement problem for atomic lengths in affine type A

摘录 / EXCERPT: Let W be of affine type $A_{e-1}^{(1)}$, and let ... where the Λ_i are the fundamental weights. For the atomic length ... determine sufficient conditions on the m_i for ... t...

open_problem | open | MathDB 页面 | 原文线索

V2-079886 | MATHDB-403827

Characterization of dominant weights with surjective atomic length

摘录 / EXCERPT: Let W be a finite Weyl group of rank “not too small” and let $\rho^\vee$ be the half-sum of the positive coroots. For a dominant weight $\bb$, define ... Characterize the dominant we...

open_problem | open | MathDB 页面 | 原文线索

V2-079887 | MATHDB-403829

Broué’s Abelian Defect Conjecture for double covers of symmetric and alternating groups

摘录 / EXCERPT: The report states that Broué’s Abelian Defect Conjecture has been proved for double covers of symmetric and alternating groups.

conjecture | open | MathDB 页面 | 原文线索

V2-079888 | MATHDB-403830

Monotonicity conjecture for a family of plethysm coefficients

摘录 / EXCERPT: For arbitrary partitions ν and λ, and every $m\in\mathbb N$ and $l\in\{0,1,\ldots,m\}$, the source-defined sequence ... is weakly increasing.

conjecture | open | MathDB 页面 | 原文线索

V2-079889 | MATHDB-403831

Stabilisation of the plethysm-coefficient sequence $A_{m-1,m}(\nu,\lambda)$

摘录 / EXCERPT: For arbitrary partitions ν and λ, and every $m\in\mathbb{N}$, the sequence $A_{m-1,m}(\nu,\lambda)$ of plethysm coefficients stabilises.

open_problem | open | MathDB 页面 | 原文线索

V2-079890 | MATHDB-403832

Foulkes' conjecture for plethysm coefficients

摘录 / EXCERPT: Let $a^{\nu}_{\lambda,\mu}$ denote the multiplicity of s_ν in the plethystic product $s_\lambda\circ s_\mu$. Foulkes' conjecture asserts that, for all $m,n\in\mathbb{N}$ with...

conjecture | open | MathDB 页面 | 原文线索

V2-079891 | MATHDB-403835

Hausel-Hitchin conjecture on very stable rank-three fixed-point components

摘录 / EXCERPT: For rank 3, does a fixed-point component of type $(2,1)$ or $(1,2)$ have no stable bundles if and only if its virtual equivariant multiplicity is not a polynomial?

conjecture | open | MathDB 页面 | 原文线索

V2-079892 | MATHDB-403836

Analytic Langlands eigenspace conjecture

摘录 / EXCERPT: The eigenspaces $H_{\chi,\bar\chi}$ generated by single-valued solutions to the pair of eigenvalue equations with eigenvalues $(\chi,\bar\chi)$ are contained in H_G, are at most...

conjecture | open | MathDB 页面 | 原文线索

V2-079893 | MATHDB-403837

Fuzzy minimax testing for identity-covariance Gaussians under $\a0\ell_p$ mean distance

摘录 / EXCERPT: For the fuzzy hypothesis-testing problem in which P_0 and P_1 are families of identity-covariance Gaussian distributions, distinguish ... from ... when $\a0\rho$ is the $\a0\ell_p$...

open_problem | open | MathDB 页面 | 原文线索

V2-079894 | MATHDB-403840

Simple near-optimal predictors for multi-group learning

摘录 / EXCERPT: Given a family of possibly overlapping groups $\mathcal{G}$, a reference class of predictors $\mathcal{H}$, and IID training data from the distribution of (X,Y), is there an algo...

open_problem | open | MathDB 页面 | 原文线索

V2-079895 | MATHDB-403841

Optimality of threshold channels beyond hypothesis testing

摘录 / EXCERPT: Given the optimality of threshold channels in hypothesis testing and their properties with respect to post-processing, are there other problems in statistics where threshold channe...

conjecture | open | MathDB 页面 | 原文线索

V2-079896 | MATHDB-403842

Other estimators with the universal deviation-minimax property

摘录 / EXCERPT: Does there exist any other estimator enjoying the universal estimation property of the empirical mean?

open_problem | open | MathDB 页面 | 原文线索

V2-079897 | MATHDB-403845

Labourie’s conjecture (split case)

摘录 / EXCERPT: For any $\rho\in \operatorname{Hit}(S,G_0)$, the associated ρ-equivariant conformal map $f_\rho:\widetilde{\Sigma}\to G_0/K_0$ is unique.

conjecture | open | MathDB 页面 | 原文线索

V2-079898 | MATHDB-403846

Closedness of the Θ-positive representation locus

摘录 / EXCERPT: Let G_R be a connected simple real Lie group with a Θ-positive structure, and let S be a compact real surface of genus $g\ge 2$. Is the set of classes of Θ-posit...

open_problem | open | MathDB 页面 | 原文线索

V2-079899 | MATHDB-403847

Fock-Thomas conjecture on higher complex structures

摘录 / EXCERPT: There is a canonical, mapping class group equivariant diffeomorphism between $\mathcal{T}^{n}(S)$ and the $\mathrm{PSL}(n,\mathbb{R})$ Hitchin component.

conjecture | open | MathDB 页面 | 原文线索

V2-079900 | MATHDB-403848

Limits of Hitchin cross ratios and negatively curved metrics

摘录 / EXCERPT: Can cross ratios associated with negatively curved metrics be obtained as limits of cross ratios associated with Hitchin representations?

open_problem | open | MathDB 页面 | 原文线索

V2-079901 | MATHDB-403849

Characterization of the G-closure for two-material composites

摘录 / EXCERPT: Given tensors a and b describing two materials and a fixed local volume fraction θ, characterize the set ... where the quantification is over sequences of characteristic f...

open_problem | open | MathDB 页面 | 原文线索

V2-079902 | MATHDB-403850

Justification of the Navier-slip boundary condition for randomly rough boundaries

摘录 / EXCERPT: Let $\Omega=\mathbb{R}\times(0,1)$, let $\Sigma=\mathbb{R}\times\{0\}$, and let $\Omega_\varepsilon=\Omega\cup\Sigma\cup R_\varepsilon$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-079903 | MATHDB-403851

Sullivan’s multi-bubble conjecture in Euclidean and spherical space

摘录 / EXCERPT: For $q-1\le n+1$, the optimal partition of $\mathbb{R}^n$ (and likewise of $\mathbb{S}^n$) into $q-1$ bubbles of prescribed volumes is obtained by taking the Voronoi cells of q e...

conjecture | open | MathDB 页面 | 原文线索

V2-079904 | MATHDB-403852

Characterize MMD spaces admitting a quasisymmetric Gaussian uniformization

摘录 / EXCERPT: Which metric measure Dirichlet (MMD) spaces $D=(K,d,m, \mathcal E,\mathcal F)$ with K having at least two elements satisfy $G_2(D)\neq\varnothing$? Here $G_2(D)$ is the set of pa...

open_problem | open | MathDB 页面 | 原文线索

V2-079905 | MATHDB-403853

Derivative-martingale-type representation for the limiting variable on a regular tree

摘录 / EXCERPT: Let $b\ge 2$, let T_n be a rooted b-ary tree of depth n, let L_n be its leaves, and let $\ell_{\tau_\varrho}(x)$ denote the time spent at $x\in L_n$ by the continuous-time...

open_problem | open | MathDB 页面 | 原文线索

V2-079906 | MATHDB-403856

Local distinguishability of smooth plane curves by the global distance-distribution pseudometric

摘录 / EXCERPT: If X_t is a one-parameter family of smooth plane curves such that $L_H(X_s,X_t)=0$ for all s,t, must it be the case that each pair of curves in the family differs by a rigid mo...

open_problem | open | MathDB 页面 | 原文线索

V2-079907 | MATHDB-403858

Removing the contractibility assumption in a metric-graph rigidity theorem

摘录 / EXCERPT: Can the contractibility assumption be removed from Theorem 3?

open_problem | open | MathDB 页面 | 原文线索

V2-079908 | MATHDB-403861

Local CLT and lower bound for the annealed second heat-kernel derivative

摘录 / EXCERPT: Under Assumptions 1 and 2, determine whether one can establish a local central limit theorem and a corresponding lower bound for the annealed second spatial derivative of the heat...

open_problem | open | MathDB 页面 | 原文线索

V2-079909 | MATHDB-403865

Stability of holonomicity under six-functor operations

摘录 / EXCERPT: Holonomicity is stable under $f^!$, f_+, $\mathbb{D}$, as well as $f_+=f_!$ whenever f is projective.

open_problem | open | MathDB 页面 | 原文线索

V2-079910 | MATHDB-403866

Stability of holonomicity under derived tensor products

摘录 / EXCERPT: Holonomicity is stable under tensor products $\otimes^L_{\mathcal O_X}$.

open_problem | open | MathDB 页面 | 原文线索

V2-079911 | MATHDB-403867

Categorical geometrization conjecture for the local Langlands correspondence

摘录 / EXCERPT: Assume that G is quasi-split. Let U be the unipotent radical of a Borel subgroup and let $\psi:U(E)\to\mathbb{Z}_\ell^{\times}$ be a non-degenerate character. Define the Whitta...

conjecture | open | MathDB 页面 | 原文线索

V2-079912 | MATHDB-403873

Density conjecture for unit-fraction decompositions

摘录 / EXCERPT: Every set $A\subset\mathbb{N}$ with positive upper density contains distinct integers $1<n_1<n_2<\cdots<n_k$ for some $k\ge 1$ such that ...

conjecture | open | MathDB 页面 | 原文线索

V2-079913 | MATHDB-403875

Polynomial Chowla conjecture

摘录 / EXCERPT: For any $f\in\mathbb{Z}[t]$ that is not of the form $c\,g(t)^2$ with $c\in\mathbb{R}$ and $g\in\mathbb{R}[t]$, one has ... where λ is the Liouville function, extended to...

conjecture | open | MathDB 页面 | 原文线索

V2-079914 | MATHDB-403876

Explicit upper bound for Waring’s number

摘录 / EXCERPT: Let $G(k)$ be the smallest number s such that every sufficiently large natural number is the sum of at most s k-th powers of natural numbers. Then, for all $k\in\mathbb{N}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079915 | MATHDB-403880

Katz-Sarnak one-level density conjecture

摘录 / EXCERPT: Given a family $\mathcal F$ of L-functions ordered by conductor, Katz and Sarnak conjecture that, for every even Schwartz function ϕ whose Fourier transform has compact supp...

conjecture | open | MathDB 页面 | 原文线索

V2-079916 | MATHDB-403882

Atkin-Serre lower-bound conjecture for Fourier coefficients

摘录 / EXCERPT: Let f be a normalized Hecke eigen cusp form of integer weight $k\ge 4$ for $\Gamma_0(N)$, with $N\ge 1$, without complex multiplication, and Fourier coefficients $a_f(n)$. The At...

conjecture | open | MathDB 页面 | 原文线索

V2-079917 | MATHDB-403883

Low moments of Möbius-twisted Dirichlet character sums

摘录 / EXCERPT: For every real q with $0\le q\le 1$ and every fixed $A>0$, Harper conjectures that, for large primes r and all $x\le r^A$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-079918 | MATHDB-403884

Moments of Möbius-twisted character and Dirichlet-polynomial sums

摘录 / EXCERPT: Harper conjectures that, for any $0\le q\le 1$ and any fixed $A>0$, one has ... and ...

open_problem | open | MathDB 页面 | 原文线索

V2-079919 | MATHDB-403885

Davenport’s mean-value bound for a diagonal cubic equation over \(\mathbb{F}_q(t)\)

摘录 / EXCERPT: Davenport asked whether one can establish the mean-value bound ... The report states that this question is answered affirmatively by Glas and Hochfilzer.

open_problem | open | MathDB 页面 | 原文线索

V2-079920 | MATHDB-403886

Expected asymptotic for rational points on cubic hypersurfaces over \(\mathbb Q\) and \(\mathbb F_q(t)\)

摘录 / EXCERPT: Let $(k,O)\in\{(\mathbb Q,\mathbb Z),(\mathbb F_q(t),\mathbb F_q[t])\}$, and let $F\in k[x_1,\ldots,x_n]$ be a homogeneous, non-singular cubic form. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-079921 | MATHDB-403887

Asymptotic Waring bound for cubes over finite polynomial rings

摘录 / EXCERPT: Let $\mathbb F_q$ be a finite field, with no restriction on its characteristic, and let $J_q[t]$ be the additive closure of the cubes in $\mathbb F_q[t]$. Define...

open_problem | open | MathDB 页面 | 原文线索

V2-079922 | MATHDB-403888

Elementary proofs of special cases of the fourth-moment dichotomy

摘录 / EXCERPT: Can one prove special cases of Theorem 2 (say for $f=X^3+X$) using more elementary means? (E.g., not beyond the Chebotarev Density Theorem.)

open_problem | open | MathDB 页面 | 原文线索

V2-079923 | MATHDB-403889

Methods for estimating sums of twisted-multiplicative functions without modulus

摘录 / EXCERPT: Find other methods to estimate sums of twisted-multiplicative functions, without modulus.

open_problem | open | MathDB 页面 | 原文线索

V2-079924 | MATHDB-403890

Erdős-Sárközy-Szemerédi conjecture

摘录 / EXCERPT: Let $f(A):=\displaystyle\sum_{a\in A}\frac{1}{a\log a}$. For primitive sets $A\subset [x,\infty)$, does ... hold? Here a primitive set is a set of integers greater than 1 in which...

conjecture | open | MathDB 页面 | 原文线索

V2-079925 | MATHDB-403891

Erdős primitive set conjecture

摘录 / EXCERPT: Let $A\subset\mathbb{Z}_{>1}$ be a primitive set, meaning that no member of A divides another, and let $\mathcal P$ be the set of primes. Define ... Then ...

conjecture | open | MathDB 页面 | 原文线索

V2-079926 | MATHDB-403892

Vinogradov’s conjecture on the least quadratic non-residue

摘录 / EXCERPT: For every fixed $\epsilon>0$, if q ranges over primes and n_q denotes the least quadratic non-residue modulo q, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-079927 | MATHDB-403893

Erdős’ conjecture on products of two primes in arithmetic progressions

摘录 / EXCERPT: For all sufficiently large primes q, every reduced residue class modulo q is a product of two primes at most q; equivalently, with ... one has...

conjecture | open | MathDB 页面 | 原文线索

V2-079928 | MATHDB-403894

Gaussian moat problem

摘录 / EXCERPT: Is it possible to walk to infinity in the complex plane using Gaussian primes as stepping stones, with the step size bounded by an absolute constant?

open_problem | open | MathDB 页面 | 原文线索

V2-079929 | MATHDB-403898

Erdős-Kac divisor-function series conjecture

摘录 / EXCERPT: For every positive integer k, let $\sigma_k(n)$ denote the sum of the kth powers of the positive divisors of n, and define ... Is α_k irrational for every $k\ge 1$?

conjecture | open | MathDB 页面 | 原文线索

V2-079930 | MATHDB-403899

Erdős-Kac irrationality conjecture for higher divisor powers

摘录 / EXCERPT: For every integer $k\ge 5$, prove that ... is irrational, where, for positive integers n and k, $\sigma_k(n)$ denotes the sum of the kth powers of the positive divisors of n:...

conjecture | open | MathDB 页面 | 原文线索

V2-079931 | MATHDB-403900

Irrationality of the divisor-function series at $k=4$

摘录 / EXCERPT: Let $\sigma_4(n)=\sum_{d\mid n}d^4$ and define ... Then α_4 is irrational.

open_problem | open | MathDB 页面 | 原文线索

V2-079932 | MATHDB-403901

Central limit theorem for squarefree counts up to the range $H\le X^{1-\varepsilon}$

摘录 / EXCERPT: The report conjectures that Theorem 1-the central limit theorem for the squarefree short-interval count $N_S(n,H)$, currently proved when $H\to\infty$ and $H=X^{o(1)}$-continues to...

open_problem | open | MathDB 页面 | 原文线索

V2-079933 | MATHDB-403902

Squares with a positive proportion of preassigned digits

摘录 / EXCERPT: Let $g\ge 2$ be an integer. There is an explicit $c_0=c_0(g)\in[0,1/2)$ such that, for every $c\in(0,c_0)$, there exists $\delta=\delta(c)>0$ with the following property. For every...

open_problem | open | MathDB 页面 | 原文线索

V2-079934 | MATHDB-403905

Point-vortex approximation for inviscid generalized SQG on \(\mathbb{R}^2\)

摘录 / EXCERPT: For the inviscid generalized surface quasi-geostrophic equation on $\mathbb{R}^2$, in the case $m\in(1,2)$, ... is it true that, if an initial condition is approximated in the sense...

open_problem | open | MathDB 页面 | 原文线索

V2-079935 | MATHDB-403906

Noiseless open core for expansions of the real field not defining the integers

摘录 / EXCERPT: The report states the conjecture that the open core of “such a structure” is noiseless, where “such a structure” means an expansion of the real field $\mathbb R$ that does not defi...

open_problem | open | MathDB 页面 | 原文线索

V2-079936 | MATHDB-403907

Borovik-León-Sánchez Kolchin-polynomial transitivity conjecture

摘录 / EXCERPT: Conjecture (Borovik-León-Sánchez, private communication). If X is definable in $\mathrm{DCF}_{0,m}$ and G acts definably with a generically n-transitive action, then...

conjecture | open | MathDB 页面 | 原文线索

V2-079937 | MATHDB-403908

Control of arbitrary ordinal forking degree

摘录 / EXCERPT: Can one control the α-forking degree for arbitrary (perhaps infinite) ordinals using an appropriate version of the Borovik-Cherlin conjecture?

open_problem | open | MathDB 页面 | 原文线索

V2-079938 | MATHDB-403910

Quasiminimality of the complex field expanded by all unary entire functions

摘录 / EXCERPT: Is the expansion of the complex field by all unary entire complex functions quasiminimal; that is, is every subset of its universe definable with parameters either countable or co-...

open_problem | open | MathDB 页面 | 原文线索

V2-079939 | MATHDB-403911

Zilber’s conjecture: $C_{\exp}\cong B_{\exp}$

摘录 / EXCERPT: Conjecture 6. $C_{\exp}\cong B_{\exp}$. More precisely, $C_{\exp}$ is a model of the stated axioms; equivalently, Schanuel’s conjecture is true in $C_{\exp}$ and $C_{\exp}$ is stro...

conjecture | open | MathDB 页面 | 原文线索

V2-079940 | MATHDB-403913

Kac-Luttinger conjecture for Bose-Einstein condensation in a random hard-obstacle domain

摘录 / EXCERPT: The Kac-Luttinger conjecture asserts that, for the non-interacting Bose gas in the three-dimensional random hard-obstacle domain...

conjecture | open | MathDB 页面 | 原文线索

V2-079941 | MATHDB-403914

Dirac-Rayleigh-Faber-Krahn conjecture for the first positive infinite-mass Dirac eigenvalue

摘录 / EXCERPT: Let $ Omega\subset\mathbb{R}^2$ be a C^∞ simply connected domain, and let D be the unit disk. The conjecture is ... with equality if and only if Ω is a disk.

conjecture | open | MathDB 页面 | 原文线索

V2-079942 | MATHDB-403915

Sharp fractional inradius constant for planar multiply connected sets

摘录 / EXCERPT: For $\frac12<s<1$ and $k\in\mathbb N\setminus\{0\}$, determine the sharp constant ... Here $r_\Omega:=\sup\{r>0:\text{there exists a disk of radius }r\text{ contained in }\Omega\}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-079943 | MATHDB-403916

Type-I Bose-Einstein condensation for finite-strength soft Poisson obstacles

摘录 / EXCERPT: An open problem concerns type-I Bose-Einstein condensation in the soft obstacle model (5), $V_\omega(x)=\gamma\sum_k u(x-\xi_k(\omega))$, for $0<\gamma<\infty$.

open_problem | open | MathDB 页面 | 原文线索

V2-079944 | MATHDB-403918

Bose-Einstein condensation in the usual thermodynamic limit

摘录 / EXCERPT: Prove Bose-Einstein condensation in the usual thermodynamic limit, namely establish ... where $\varphi_0\equiv 1/\sqrt{|\Lambda|}$ is the zero-momentum mode and $\gamma_N^{(1)}$ is t...

open_problem | open | MathDB 页面 | 原文线索

V2-079945 | MATHDB-403919

Lee-Huang-Yang expansion for the dilute Bose gas

摘录 / EXCERPT: For the thermodynamic-limit specific ground-state energy of the three-dimensional Bose gas with a nonnegative, radial, compactly supported two-body potential of scattering length...

open_problem | open | MathDB 页面 | 原文线索

V2-079946 | MATHDB-403921

Collinear triples in geproci point sets

摘录 / EXCERPT: Prove that every geproci set of $n\ge 5$ points in $\mathbb{P}^3$ contains at least three collinear points.

open_problem | open | MathDB 页面 | 原文线索

V2-079947 | MATHDB-403922

Large grids in geproci sets

摘录 / EXCERPT: Is it always the case that every known (a,b)-geproci set in $\mathbb{P}^3$ contains an (a',b')-grid with $a' \ge \frac{a}{2}$ and $b' \ge \frac{b}{2}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079948 | MATHDB-403924

Finiteness and classification of non-half-grid geproci sets

摘录 / EXCERPT: Are there only finitely many non-half-grid geproci sets, and, if so, can all such examples be listed?

open_problem | open | MathDB 页面 | 原文线索

V2-079949 | MATHDB-403928

Hadamard products of symbolic powers of point ideals

摘录 / EXCERPT: Is it true that for points P,Q in $\mathbb{P}^2$, $I(P)^m \star I(Q)^n = I(P \star Q)^{m+n-1}$?

open_problem | open | MathDB 页面 | 原文线索

V2-079950 | MATHDB-403930

Closure of Ising models under pointwise limits

摘录 / EXCERPT: Conjecture 1. $M_I(G)\cap M_2(G)$ is closed under pointwise limits.

open_problem | open | MathDB 页面 | 原文线索

V2-079951 | MATHDB-403932

Unique reconstruction from seven unambiguous beads

摘录 / EXCERPT: In the report’s diploid bead model for partially phased population contact data, with pairwise contacts satisfying the power law $c_{ij}=d_{ij}^{\alpha}$ for $\alpha<0$, it is conj...

open_problem | open | MathDB 页面 | 原文线索

V2-079952 | MATHDB-403933

Singular-locus equation for catenary linear compartment models

摘录 / EXCERPT: Conjecture. Assume $n\geq 2$. For the n-compartment catenary (path) model in Figure 1, the equation of the singular locus is ...

open_problem | open | MathDB 页面 | 原文线索

V2-079953 | MATHDB-403934

Recursive formulas for the degree sequences of correlation models

摘录 / EXCERPT: For the three degree sequences displayed in the report- - SSL degree: $1,4,28,292,?, ?, ?, ?, ?$, - ML degree: $1,3,15,109,1077,13695,?, ?, ?$, - dual ML degree:...

open_problem | open | MathDB 页面 | 原文线索

V2-079954 | MATHDB-403935

Identifiability under zero restrictions on higher-order moment or cumulant tensors

摘录 / EXCERPT: Suppose that $V \subset \mathcal{S}_r(\mathbb{R}^d)$ is a linear subspace defined by some coordinates being zero. Let T be a generic element of V. For which V does the condit...

open_problem | open | MathDB 页面 | 原文线索

V2-079955 | MATHDB-403936

Conjecture 4: the two possible weak and ordinary maximum-likelihood-threshold scenarios

摘录 / EXCERPT: Let G be a colored graphical model, and assume that $I(G,n+1)=(0)$ while $I(G,n)=\langle f_1,\ldots,f_u\rangle\neq(0)$. Express the polynomials f_i in the Cholesky variables...

conjecture | open | MathDB 页面 | 原文线索

V2-079956 | MATHDB-403938

Euler-characteristic description of the ML degree for low-rank Gaussian data

摘录 / EXCERPT: Can we use Euler characteristics to describe the ML degree of multivariate Gaussian models when the data is a low rank (non-generic) sample covariance matrix?

open_problem | open | MathDB 页面 | 原文线索

V2-079957 | MATHDB-403939

Maximizing divergence to toric models

摘录 / EXCERPT: Given a toric (discrete exponential) model $M_A\subseteq\Delta_{n-1}$ associated with a matrix A, determine and compute all points $p\in\Delta_{n-1}$ that maximize the informatio...

open_problem | open | MathDB 页面 | 原文线索

V2-079958 | MATHDB-403945

Asymptotic normality of the number of components in a random meandric system

摘录 / EXCERPT: Kargin further conjectures that $cc(M_n)$ is asymptotically normal with a variance linear in n, where M_n is a uniform random meandric system of size n.

open_problem | open | MathDB 页面 | 原文线索

V2-079959 | MATHDB-403946

Counting strong equivalence classes of generic rectangulations

摘录 / EXCERPT: For each $n\ge 1$, determine the number s_n of strong equivalence classes of generic rectangulations of a rectangle into n rectangular regions, where generic means that no poin...

open_problem | open | MathDB 页面 | 原文线索

V2-079960 | MATHDB-403947

D-finiteness of the 1324-avoiding permutation generating function

摘录 / EXCERPT: Determine whether the generating function for permutations avoiding the pattern 1324 is non-D-finite.

open_problem | open | MathDB 页面 | 原文线索

V2-079961 | MATHDB-403948

Conjectured generating functions for fixed-occurrence counts of 123 and 321

摘录 / EXCERPT: Let $\psi_r(n)$ be the number of permutations of length n containing exactly r occurrences of the nominated length-3 pattern 123 or 321, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-079962 | MATHDB-403949

Asymptotics for fixed-occurrence permutations in effective Wilf class V

摘录 / EXCERPT: For effective Wilf class V, consisting of the patterns 1324 and 4231, the report conjectures that ... where $\psi_r(n)$ denotes the number of permutations of length n containin...

open_problem | open | MathDB 页面 | 原文线索

V2-079963 | MATHDB-403950

Asymptotics for occurrences in Wilf classes VI and VII

摘录 / EXCERPT: Let $\psi_r(n)$ denote the number of permutations of length n containing exactly r occurrences of the nominated length-4 pattern. For both Wilf classes VI and VII, the report c...

open_problem | open | MathDB 页面 | 原文线索

V2-079964 | MATHDB-403951

D-finiteness and the group of a small-step quarter-plane walk

摘录 / EXCERPT: For the 79 unweighted lattice-walk models with small steps confined to the quarter plane (after taking symmetries into account and eliminating trivial sets), Bousquet-Mélou and Mis...

open_problem | open | MathDB 页面 | 原文线索

V2-079965 | MATHDB-403952

First Borwein Conjecture

摘录 / EXCERPT: For all positive integers n, the sign pattern of the coefficients in the expansion of the polynomial ... is $+--+--+--\cdots$, with a coefficient 0 being considered as both $+$ a...

conjecture | open | MathDB 页面 | 原文线索

V2-079966 | MATHDB-403953

Second Borwein Conjecture

摘录 / EXCERPT: For every positive integer n, the sign pattern of the coefficients in the expansion of $P_n(q)^2$, where ... is $+,-,-,+,-,-,+,-,-,\ldots$, with any coefficient equal to 0 consid...

conjecture | open | MathDB 页面 | 原文线索

V2-079967 | MATHDB-403954

Wang’s Cubic Borwein Conjecture

摘录 / EXCERPT: For all positive integers n, the sign pattern of the coefficients in the expansion of the polynomial $P_n(q)^3$, where...

conjecture | open | MathDB 页面 | 原文线索

V2-079968 | MATHDB-403955

Bijective proof of the Hurwitz-number formula

摘录 / EXCERPT: For integers $n\ge 1$ and $\ell\ge 0$, let ... Prove bijectively that ... where f_λ is the number of standard Young tableaux of shape λ and C_λ is the content s...

open_problem | open | MathDB 页面 | 原文线索

V2-079969 | MATHDB-403956

Lorentzianity of chromatic symmetric functions of Dyck-path indifference graphs

摘录 / EXCERPT: Conjecture: The chromatic symmetric functions of indifference graphs of Dyck paths are Lorentzian.

open_problem | open | MathDB 页面 | 原文线索

V2-079970 | MATHDB-403957

Cross-product conjecture for linear extensions of finite posets

摘录 / EXCERPT: For a finite poset P and elements $x,y,z\in P$, define ... where L ranges over linear extensions. The cross-product conjecture asserts that ...

conjecture | open | MathDB 页面 | 原文线索

V2-079971 | MATHDB-403958

Kahn-Saks monotonicity conjecture for normalized order polynomials

摘录 / EXCERPT: For a finite poset P, the conjecture is that the normalized order polynomial $\Omega_P(t)/t^{|P|}$ is a decreasing function of t for $t>1$, where $\Omega_P(t)$ counts order-pre...

conjecture | open | MathDB 页面 | 原文线索

V2-079972 | MATHDB-403960

Generating function for a permutation class associated with complete parabolic bundle structures

摘录 / EXCERPT: For the permutation class $\operatorname{Av}(3412,52341,635241)$, determine its generating function. The report states that a specification with one catalytic variable has been fou...

open_problem | open | MathDB 页面 | 原文线索

V2-079973 | MATHDB-403962

Partial ASM-TSSCPP bijection in the reduced 1432-avoiding case

摘录 / EXCERPT: Let $\pi\in S_n$. Let $\operatorname{TSSCPP}^{\mathrm{red}}(\pi)$ denote the set of TSSCPP whose associated pipe dream is reduced and has permutation π, and let...

open_problem | open | MathDB 页面 | 原文线索

V2-079974 | MATHDB-403964

Bijection for 3/4-plane walks

摘录 / EXCERPT: Fix a multiset $\hat S$ of small steps, where a small step in $\mathbb Z^2$ is one of $(1,0),(-1,0),(0,1),(0,-1),(1,1),(1,-1),(-1,1),(-1,-1)$. Let $W(\hat S)$ be the set of walks f...

open_problem | open | MathDB 页面 | 原文线索

V2-079975 | MATHDB-403965

321-avoiding permutations obeying parity

摘录 / EXCERPT: Let $P_{321}(n)$ be the number of permutations $\[1m\pi$ of length n that avoid the pattern 321-that is, they have no three (not necessarily consecutive in position or value) d...

open_problem | open | MathDB 页面 | 原文线索

V2-079976 | MATHDB-403971

Wachspress’s nonvanishing conjecture for regular rational planar polypols

摘录 / EXCERPT: Conjecture 1. Let P be a regular rational planar polypol, with adjoint curve A_P. Then A_P does not intersect the interior of P.

conjecture | open | MathDB 页面 | 原文线索

V2-079977 | MATHDB-403972

The GM^d conjecture on maximal hyperplanes of GC_n sets

摘录 / EXCERPT: For any finite set $X\subset\mathbb{R}^d$ satisfying the geometric characterization of degree n-that is, $X\in GC_n$, meaning that for each $x\in X$ there exist n hyperplanes i...

conjecture | open | MathDB 页面 | 原文线索

V2-079978 | MATHDB-403973

Gasca-Maeztu conjecture for planar GC sets

摘录 / EXCERPT: The Gasca-Maeztu conjecture asserts that every planar GC_n set has a maximal line. Here, $X\subset\mathbb{R}^2$ is a GC_n set if $\#X=(n+2)(n+1)/2$ and, for every $x\in X$, the...

conjecture | open | MathDB 页面 | 原文线索

V2-079979 | MATHDB-403974

Extension of three interpolation results to $s\ge 4$

摘录 / EXCERPT: One may conjecture that Theorems 3, 4, and the Corollary hold also for the values $s\ge 4$.

open_problem | open | MathDB 页面 | 原文线索

V2-079980 | MATHDB-403975

Minimum degree in a fat-point ideal

摘录 / EXCERPT: Given a finite set of points X and a positive integer m, find the smallest degree of a polynomial in the fat-point ideal $I_X^{(m)}$. Here, for $X=\{p_1,\ldots,p_n\}$ with defi...

open_problem | open | MathDB 页面 | 原文线索

V2-079981 | MATHDB-403976

Containment problem for symbolic and ordinary powers of point ideals

摘录 / EXCERPT: Let X be a finite set of points, with vanishing ideal I_X. Determine which pairs of natural numbers r,m give the containment $I_X^{(m)}\subseteq I_X^r$, where $I_X^{(m)}$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-079982 | MATHDB-403977

Containment classification for singular loci of irreducible complex reflection arrangements

摘录 / EXCERPT: Let I be an ideal defining the singular locus of an irreducible reflection arrangement $\mathcal A$ in $\mathbb P^d$. Then $I^{(3)}\subseteq I^2$, unless $\mathcal A$ is one of t...

open_problem | open | MathDB 页面 | 原文线索

V2-079983 | MATHDB-403980

Schumaker’s Bernstein-basis interpolation conjecture

摘录 / EXCERPT: Given d and a triangle $T:=\\langle v_1,v_2,v_3\\rangle$, let $\\Gamma$ be an arbitrary subset of the set $D_{d,\\triangle}$ of domain points associated with the triangle. Then t...

conjecture | open | MathDB 页面 | 原文线索

V2-079984 | MATHDB-403981

Gauss cubature rules for the three-variable product Chebyshev weight

摘录 / EXCERPT: Does a Gauss cubature rule of degree $2n-2$ exist for the product Chebyshev weight function in three variables? The report states that existence was known only for $n=2$ and $n=3$.

open_problem | open | MathDB 页面 | 原文线索

V2-079985 | MATHDB-403982

Three-dimensional centrally symmetric cubature rules with few nodes

摘录 / EXCERPT: Does there exist a centrally symmetric weight function in three variables that admits cubature rules with $\dim \Pi^3_{n-1}+O(n^2)$ nodes?

open_problem | open | MathDB 页面 | 原文线索

V2-079986 | MATHDB-403984

Dimensions of planar spline spaces

摘录 / EXCERPT: Let Δ be a simplicial subdivision (triangulation) of a simply connected bounded domain in $\mathbb{R}^2$. For positive integers r,m, let $S_m^r(\Delta)$ be the real vector...

open_problem | open | MathDB 页面 | 原文线索

V2-079987 | MATHDB-403985

The adjoint map for general heptagons

摘录 / EXCERPT: A general heptagon in the plane has 14 degrees of freedom. Its adjoint has degree four and hence also 14 degrees of freedom. There is a rational map from the space...

open_problem | open | MathDB 页面 | 原文线索

V2-079988 | MATHDB-403987

Exact relationship between fiid- and ffiid-solvable LCLs

摘录 / EXCERPT: Determine the exact relationship between the classes of locally checkable labelling problems (LCLs) that admit fiid solutions and those that admit ffiid solutions.

open_problem | open | MathDB 页面 | 原文线索

V2-079989 | MATHDB-403988

Decidability of Constant-Time LCL Problems on Trees

摘录 / EXCERPT: Is it decidable whether, on trees, a given locally checkable labeling (LCL) problem can be solved in a constant number of rounds?

open_problem | open | MathDB 页面 | 原文线索

V2-079990 | MATHDB-403991

Continuous solutions for CSPs under the Lovász Local Lemma and subexponential dependency-graph growth

摘录 / EXCERPT: Let $\mathcal B:X\mathrel{\to?}k$ be a constraint satisfaction problem with $e\,p(\mathcal B)(d(\mathcal B)+1)<1$, and suppose that the dependency graph $G_{\mathcal B}$ has growth...

open_problem | open | MathDB 页面 | 原文线索

V2-079991 | MATHDB-403994

Simultaneous informational and spatial efficiency for finitary codings

摘录 / EXCERPT: Let X be a process known to be ffiid with good spatial efficiency, for example, with a coding radius having exponential tails. Is X a finitary factor of a finite-valued i.i.d....

open_problem | open | MathDB 页面 | 原文线索

V2-079992 | MATHDB-403995

Converse to the Σ^1_2-completeness criterion for Borel CSPs

摘录 / EXCERPT: Let D be a finite relational structure, and let $\\operatorname{CSP}_B(D)$ denote the set of codes for Borel structures admitting a Borel homomorphism to D. The report asks whe...

open_problem | open | MathDB 页面 | 原文线索

V2-079993 | MATHDB-403997

Using the shift graph and Galvin-Prikry-like theorems in the LOCAL model

摘录 / EXCERPT: Is there a way to utilize G_S and Galvin-Prikry-like theorems in the LOCAL model?

open_problem | open | MathDB 页面 | 原文线索

V2-079994 | MATHDB-403998

Axiom of Determinacy and the chromatic number of the shift graph

摘录 / EXCERPT: Does it follow from the Axiom of Determinacy over ZF that the chromatic number of the shift graph satisfies $\chi(G_S)=\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-079995 | MATHDB-404003

Montgomery's theorem on large Latin squares

摘录 / EXCERPT: There is some n_0 such that every Latin square of order $n \ge n_0$ contains a transversal of size $n-1$.

open_problem | open | MathDB 页面 | 原文线索

V2-079996 | MATHDB-404005

Circle squaring with regular pieces

摘录 / EXCERPT: Can a closed disk and a square of the same area in $\mathbb{R}^2$ be partitioned into finitely many pieces and rearranged using translations so that every piece has boundary of upp...

open_problem | open | MathDB 页面 | 原文线索

V2-079997 | MATHDB-404009

Decomposing large connected cubic graphs into two isomorphic linear forests

摘录 / EXCERPT: Every connected cubic graph on $4n$ vertices, where n is large, can be decomposed into two isomorphic linear forests.

open_problem | open | MathDB 页面 | 原文线索

V2-079998 | MATHDB-404010

Wormald’s conjecture on isomorphic linear-forest decompositions of cubic graphs

摘录 / EXCERPT: Every cubic graph on $4n$ vertices can be decomposed into two isomorphic linear forests.

conjecture | open | MathDB 页面 | 原文线索

V2-079999 | MATHDB-404011

Hamilton cycles in pseudorandom graphs

摘录 / EXCERPT: There exists an absolute constant $K>0$ such that any (n,d,λ)-graph with $d/\lambda\ge K$ contains a Hamilton cycle.

open_problem | open | MathDB 页面 | 原文线索

V2-080000 | MATHDB-404014

Optimal dependence of the Rödl-Nikiforov parameter on ε

摘录 / EXCERPT: For a fixed graph H, determine how large $\[1mδ=\[0mδ(H,ε)$ can be taken as a function of $ε>0$ while retaining the conclusion of Theorem 6: every graph G containing fewer than...

open_problem | open | MathDB 页面 | 原文线索

V2-080001 | MATHDB-404015

Quantitative supersaturation strengthening of Rödl’s theorem

摘录 / EXCERPT: For every graph H there exists $c>0$ such that, if $\varepsilon\in(0,1/2)$ and ... and G is a graph containing fewer than $(\delta|G|)^{|H|}$ induced copies of H, then there ex...

open_problem | open | MathDB 页面 | 原文线索

V2-080002 | MATHDB-404019

Mean, variance, and limiting distribution of real roots of random functions

摘录 / EXCERPT: What can we say about the mean, variance, and limiting distribution of the number of real roots of a random function?

open_problem | open | MathDB 页面 | 原文线索

V2-080003 | MATHDB-404020

Nearly all fixed-width k-SAT functions are unate

摘录 / EXCERPT: For fixed $k\ge 2$, as $n\to\infty$, the number of k-SAT functions on n Boolean variables is ... Equivalently, a $1-o(1)$ fraction of all k-SAT functions on n Boolean variabl...

open_problem | open | MathDB 页面 | 原文线索

V2-080004 | MATHDB-404021

Threshold for unateness of sparse minimal k-SAT formulas

摘录 / EXCERPT: For a fixed k, consider a uniformly chosen minimal k-SAT formula with n variables and m clauses. What is the threshold $m=m(n)$ above which a typical minimal k-SAT formul...

open_problem | open | MathDB 页面 | 原文线索

V2-080005 | MATHDB-404022

Growth range for the number of k-SAT functions

摘录 / EXCERPT: Bollobás and Brightwell conjectured that, for every fixed constant $c>0$, if the clause width is an integer-valued function $k=k(n)$ satisfying $k(n)\le (1/2-c)n$, then, as...

open_problem | open | MathDB 页面 | 原文线索

V2-080006 | MATHDB-404023

Reed’s conjecture on the chromatic number, maximum degree, and clique number

摘录 / EXCERPT: For every graph G, ...

conjecture | open | MathDB 页面 | 原文线索

V2-080007 | MATHDB-404025

Linear upper bound conjecture for rainbow matchings

摘录 / EXCERPT: For any fixed uniformity $r\ge 2$, there is a constant $c(r)$ such that $f(r,t)\le c(r)\,t$ for all t, where $f(r,t)$ is the maximum number of matchings of size t in an r-par...

conjecture | open | MathDB 页面 | 原文线索

V2-080008 | MATHDB-404028

Ascending subgraph decomposition conjecture

摘录 / EXCERPT: Every graph G with $\binom{m}{2}$ edges has an edge decomposition into subgraphs $H_1,\ldots,H_{m-1}$ such that $e(H_i)=i$ and, for each $1\le i\le m-2$, H_i is isomorphic to a...

conjecture | open | MathDB 页面 | 原文线索

V2-080009 | MATHDB-404030

Deterministic approximation of the total number of matchings

摘录 / EXCERPT: For a graph $G=(V,E)$, let $m_k=m_k(G)$ be the number of matchings $M\subseteq E$ with k edges. Then, up to relative error $0<\epsilon<1$, the total number of matchings ... is dete...

open_problem | open | MathDB 页面 | 原文线索

V2-080010 | MATHDB-404031

Improving the zero-free radius for permanents of complex matrices

摘录 / EXCERPT: Is it possible to replace the constant 0.5 in Theorem 3 by a larger constant? Equivalently, does there exist a constant $c>0.5$ such that, for every complex $n\times n$ matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-080011 | MATHDB-404033

Linial-Diestel local-covering alternative for chromatic number

摘录 / EXCERPT: Is there a function $f:\mathbb{N}^2\to\mathbb{N}$ such that every finite graph G with chromatic number at least $f(r,k)$ satisfies at least one of the following conditions? 1....

open_problem | open | MathDB 页面 | 原文线索

V2-080012 | MATHDB-404034

Additive dimension of a set of transpositions

摘录 / EXCERPT: Let M_n be the minimum number such that, given M_n transpositions in S_n, one can always find a sequence of distinct elements of this set, $\pi_1,\ldots,\pi_k$, satisfying ……

open_problem | open | MathDB 页面 | 原文线索

V2-080013 | MATHDB-404036

Linear-size monochromatic subspaces in 2-colorings of \(\mathbb F_2^n\)

摘录 / EXCERPT: Let $f(n)$ be the maximum size (number of elements) of a monochromatic (ignoring 0) linear subspace guaranteed in any 2-coloring of the nonzero elements of $\mathbb F_2^n$. Is...

open_problem | open | MathDB 页面 | 原文线索

V2-080014 | MATHDB-404042

Nonexistence of bifurcation curves from non-uniform shear flows

摘录 / EXCERPT: Establish the announced claim that, in the original steady three-dimensional gravity-capillary water-wave formulation, there are no C^2 curves of symmetric solutions bifurcating...

open_problem | open | MathDB 页面 | 原文线索

V2-080015 | MATHDB-404043

One-phase capillarity-driven Stokes flow as the small-viscosity limit

摘录 / EXCERPT: Let $s\in(3/2,2)$ and $f(0)\in H^s(\mathbb R)$ be given. Let $(f(\cdot;f(0)),v,p)$ denote the maximal solution of the one-phase problem (1), and choose $T<T_+(f(0))$. With...

open_problem | open | MathDB 页面 | 原文线索

V2-080016 | MATHDB-404044

Overman conjecture for global corotating vortex-pair solutions

摘录 / EXCERPT: For the global branch of corotating vortex-pair solutions described in the report, the limiting configuration as the branch parameter tends to infinity is conjectured to consist of...

conjecture | open | MathDB 页面 | 原文线索

V2-080017 | MATHDB-404045

Global well-posedness for the one-phase Muskat problem with arbitrary-size critical initial data

摘录 / EXCERPT: For every $f_0\in W^{1,\infty}(\mathbb T)$, there exists a function ... with $f|_{t=0}=f_0$, satisfying the one-phase Muskat equation in $L^\infty_tL^2_x$ and ... Moreover, f is uniq...

open_problem | open | MathDB 页面 | 原文线索

V2-080018 | MATHDB-404049

Infinite surviving-edge component for WARM on \d\^d with decaying firing rates

摘录 / EXCERPT: Do there exist a bounded (but not bounded away from 0) function p, $d \ge 2$, and $\alpha>1$ for which the WARM on the graph $G=\mathbb{Z}^d$ almost surely (or indeed with posi...

open_problem | open | MathDB 页面 | 原文线索

V2-080019 | MATHDB-404051

Invariance of the bounded-below constant under the full envelope ansatz

摘录 / EXCERPT: Let $\widetilde A$ be the operator corresponding to the full envelope ansatz, defined by $\widetilde A\widetilde u := e^{ikz}A(e^{-ikz}\widetilde u)$, where A denotes the operato...

open_problem | open | MathDB 页面 | 原文线索

V2-080020 | MATHDB-404052

Weak real zero amalgamation conjecture

摘录 / EXCERPT: Let $p\in\mathbb{R}[x_1,x_2,y]$ and $q\in\mathbb{R}[x_1,x_2,z]$ be real zero polynomials such that $p(x,0)=q(x,0)$. Then there exists a real zero amalgam of p and q, i.e., a re...

conjecture | open | MathDB 页面 | 原文线索

V2-080021 | MATHDB-404054

Matrix evaluations on convex K_C

摘录 / EXCERPT: Are matrix evaluations sufficient for checking positivity on a convex K_C?

open_problem | open | MathDB 页面 | 原文线索

V2-080022 | MATHDB-404055

Strictness of the intermediate-cone inclusions

摘录 / EXCERPT: For $(n+1,2d)$ with $n\ge 2$, $d\ge 2$, and $(n+1,2d)\ne(3,4)$, which inclusions in the cone filtration (1) are strict?

open_problem | open | MathDB 页面 | 原文线索

V2-080023 | MATHDB-404056

Strictness of the remaining intermediate-cone inclusions

摘录 / EXCERPT: Is it true that, for $(n+1,2d)$ with $n\ge 2$ and $d\ge 4$, and for $i=n+1,\ldots,k-n-1$, the inclusion $C_i\subseteq C_{i+1}$ is strict? Moreover, if $n\ge 3$, is the inclusion...

open_problem | open | MathDB 页面 | 原文线索

V2-080024 | MATHDB-404058

Odd-power SOS addition conjecture

摘录 / EXCERPT: Conjecture: If p^{2i+1} and q^{2j+1} are sums of squares, then $(p+q)^{2i+2j+1}$ is a sum of squares.

conjecture | open | MathDB 页面 | 原文线索

V2-080025 | MATHDB-404059

Odd powers of positive semidefinite non-sums-of-squares forms

摘录 / EXCERPT: If p is psd and not sos, is p^3 sos?

open_problem | open | MathDB 页面 | 原文线索

V2-080026 | MATHDB-404065

Minkowski sum of sums of squares and sums of nonnegative circuit forms

摘录 / EXCERPT: For which values of the number of variables and even degree does ... hold? The report answers this: the strict inclusion holds exactly when $n\neq 2$, $2d\neq 2$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-080027 | MATHDB-404066

Equality of the Minkowski sum cone with the cone of nonnegative forms

摘录 / EXCERPT: For which cases of $n,2d$ does $(\Sigma+C)_{n,2d}=P_{n,2d}$ hold? The report states that ...

open_problem | open | MathDB 页面 | 原文线索

V2-080028 | MATHDB-404067

Generic singleton Gram spectrahedra for sums of squares of quadratics

摘录 / EXCERPT: What is the maximum number $m(n)$ for which there exists a Euclidean-open subset U of the space of m-tuples of quadratics in n variables such that, for every...

open_problem | open | MathDB 页面 | 原文线索

V2-080029 | MATHDB-404068

Heat-flow time to sums of squares for bivariate forms

摘录 / EXCERPT: For which $d\in\mathbb{N}$ does there exist a $\tau_{2,2d}\ge 0$ such that $e^{\tau_{2,2d}\Delta}\operatorname{Pos}(2,2d)\subseteq\operatorname{SOS}(2,2d)$?

open_problem | open | MathDB 页面 | 原文线索

V2-080030 | MATHDB-404069

Equations for the image of the principal minor map in the 5×5 complex case

摘录 / EXCERPT: What are equations describing the image of $\mathbb{C}^{5\times5}$ under the principal minor map?

open_problem | open | MathDB 页面 | 原文线索

V2-080031 | MATHDB-404070

Converse to a sums-of-squares condition for powered determinantal representations

摘录 / EXCERPT: The report asks whether the converse holds in the following implication: if some power f^m of a polynomial f can be written as ... for some $m\in\mathbb{Z}_{>0}$ and real symmetr...

open_problem | open | MathDB 页面 | 原文线索

V2-080032 | MATHDB-404072

Rational positivity certificates in the truncated moment problem

摘录 / EXCERPT: In the truncated moment-problem setting, can the phenomenon described by Scheiderer actually occur: namely, can there be rational data y and a polynomial $p\in\mathbb{Q}[x]$ serv...

open_problem | open | MathDB 页面 | 原文线索

V2-080033 | MATHDB-404073

Decidability of nonnegativity for unnormalized trace polynomials

摘录 / EXCERPT: Given a polynomial expression in unnormalized traces of powers of symmetric matrices, can one decide whether it is nonnegative for all symmetric matrices of all sizes? The report s...

open_problem | open | MathDB 页面 | 原文线索

V2-080034 | MATHDB-404074

Artin-type sum-of-squares question for global real-analytic functions

摘录 / EXCERPT: Are all positive semidefinite real-analytic functions $f:\mathbb{R}^n\to\mathbb{R}$ sums of squares of real-analytic meromorphic functions?

open_problem | open | MathDB 页面 | 原文线索

V2-080035 | MATHDB-404076

Characterization of universally nonnegative graphs for weighted homomorphism densities

摘录 / EXCERPT: Characterize all finite simple graphs H such that $t(H,G_w)\ge 0$ for every finite simple weighted graph G_w, where the edge-weight function is $w:E(G)\to\mathbb{R}$ and ...

open_problem | open | MathDB 页面 | 原文线索

V2-080036 | MATHDB-404079

Undecidability of nonnegativity for trace polynomials on real symmetric matrices

摘录 / EXCERPT: Instance: A positive integer k and a trace polynomial $f(X_1,\ldots,X_k)$. Question: Is $f(M_1,\ldots,M_k)$ nonnegative for all real symmetric matrices $M_1,\ldots,M_k$ of all si...

open_problem | open | MathDB 页面 | 原文线索

V2-080037 | MATHDB-404080

Viro's conjecture on the first Betti number of a real surface

摘录 / EXCERPT: Viro conjectured that if X is a smooth projective real surface such that $X(\mathbb{C})$ is simply connected, then ...

conjecture | open | MathDB 页面 | 原文线索

V2-080038 | MATHDB-404084

Admissible numerator problem for bounded rational functions on domains in several complex variables

摘录 / EXCERPT: Given a domain $D\subset\mathbb{C}^d$ and a polynomial $p\in\mathbb{C}[x_1,\ldots,x_d]$ with $p(x)\ne 0$ for $x\in D$, is it possible to give a simple description of the set ...

open_problem | open | MathDB 页面 | 原文线索

V2-080039 | MATHDB-404085

Characterization of 0-sum hyperbolicity preservers

摘录 / EXCERPT: Let $T:\mathbb{R}[t]_{n,0}\to\mathbb{R}[t]_{d,0}$ be diagonal. Then T is a 0-sum hyperbolicity preserver if and only if $T\bigl((x-1)^{n-1}(x+n-1)\bigr)$ has real roots, with...

open_problem | open | MathDB 页面 | 原文线索

V2-080040 | MATHDB-404087

Steinness of canonical extensions of unstable ruled surfaces over elliptic curves

摘录 / EXCERPT: Let $M=\[P(E)\to B$ be a ruled surface over an elliptic curve B, where E is an unstable vector bundle of rank two. Is it true that no canonical extension Z_M of M is Stein?

open_problem | open | MathDB 页面 | 原文线索

V2-080041 | MATHDB-404091

Local-Global Conjecture for Integral Circle Packings

摘录 / EXCERPT: Given an integral packing P and a choice of coordinates ι, let $B=B(P,\iota)$ be its set of bends. The report states the conjecture that there exists q_0 such that, for a...

conjecture | open | MathDB 页面 | 原文线索

V2-080042 | MATHDB-404093

Quasi-modularity for stable maps to a moving elliptic curve

摘录 / EXCERPT: Let $\pi:\mathcal{E}\to\mathcal{M}_{1,1}$ be the universal family of stable curves of genus 1 with 1 marked point. Does quasi-modularity hold in the moving case: ... Here...

open_problem | open | MathDB 页面 | 原文线索

V2-080043 | MATHDB-404097

Finite mapping-class-group orbits of low-rank surface-group representations

摘录 / EXCERPT: Let $\Sigma_{g,n}$ be a smooth orientable surface of genus g with n punctures, and let $\operatorname{Mod}_{g,n}$ act on...

open_problem | open | MathDB 页面 | 原文线索

V2-080044 | MATHDB-404099

Local freeness of coherent sheaves of coinvariants

摘录 / EXCERPT: When are coherent sheaves of coinvariants locally free?

open_problem | open | MathDB 页面 | 原文线索

V2-080045 | MATHDB-404100

Chern classes and tautologicality for coinvariant bundles beyond CohFT-type VOAs

摘录 / EXCERPT: Can one find Chern classes of associated bundles, in the more general setting where V is not of CohFT-type, and determine when the higher Chern classes are tautological?

open_problem | open | MathDB 页面 | 原文线索

V2-080046 | MATHDB-404101

Classification of \bc_p-normal curves over a prescribed quotient

摘录 / EXCERPT: Classify (say) bcp-normal curves X with prescribed quotient $Y=X/\bc_p$ in terms of data on Y. More generally, determine whether bcp-normal curves over Y correspond to line b...

open_problem | open | MathDB 页面 | 原文线索

V2-080047 | MATHDB-404102

Tautological generation of the Chow ring of \overline{\mathcal M}_{g,n}

摘录 / EXCERPT: For which g,n is $A^*(\overline{\mathcal M}_{g,n})=R^*(\overline{\mathcal M}_{g,n})$?

open_problem | open | MathDB 页面 | 原文线索

V2-080048 | MATHDB-404103

Tautological generation of the Chow ring of the moduli space of stable curves

摘录 / EXCERPT: For which g,n is $A^*(\mathcal{M}_{g,n})=R^*(\mathcal{M}_{g,n})$?

open_problem | open | MathDB 页面 | 原文线索

V2-080049 | MATHDB-404104

Interpolation for Brill-Noether curves

摘录 / EXCERPT: For integers d,g and $r\ge 2$ with $\rho(d,g,r):=g-(r+1)(g-d+r)\ge 0$, let $C\subset\mathbb{P}^r$ be a Brill-Noether curve, meaning a curve parameterized by the distinguished Bri...

open_problem | open | MathDB 页面 | 原文线索

V2-080050 | MATHDB-404106

Uniqueness of the Manin component for sufficiently positive section classes

摘录 / EXCERPT: Let $\pi:X\to B$ be a Fano fibration. There is a unique non-pathological component of $\operatorname{Sec}(X/B,\alpha)$, which should be counted in Manin's conjecture, for sufficien...

open_problem | open | MathDB 页面 | 原文线索

V2-080051 | MATHDB-404108

Computing the double ramification cycle on an Abel-Jacobi blowup

摘录 / EXCERPT: Can one compute the double ramification cycle on the blowup of S resolving the Abel-Jacobi map AJ_L, rather than pushing the cycle forward to S?

open_problem | open | MathDB 页面 | 原文线索

V2-080052 | MATHDB-404110

Converse of the Gross-Zagier-Kolyvagin theorem for CM characters

摘录 / EXCERPT: Let ψ be as above. Suppose $p>3$ and the p-part of the conductor of ψ is p^n for some $n\ge 2$. If the rank of the Selmer group for ψ is 1, then ...

open_problem | open | MathDB 页面 | 原文线索

V2-080053 | MATHDB-404114

Sharifi's Eisenstein conjecture for cyclotomic Steinberg symbols

摘录 / EXCERPT: For the maps ... and ... where $\widetilde{\Pi}$ sends the Manin symbol $[c:d]^+$ to $\{1-\zeta_N^c,1-\zeta_N^d\}^+$, and where I is the Eisenstein ideal generated by...

conjecture | open | MathDB 页面 | 原文线索

V2-080054 | MATHDB-404115

Eisenstein map for Bianchi spaces and elliptic units

摘录 / EXCERPT: Let F) be an imaginary quadratic field with ring of integers $\mathcal O$, let N be an ideal of $\mathcal O$, and let $\widetilde X_1(N)$ be the minimal compactification of the...

open_problem | open | MathDB 页面 | 原文线索

V2-080055 | MATHDB-404117

Refined Kolyvagin conjecture for the divisibility index

摘录 / EXCERPT: In the setup of Conjecture 1, let $m_r:=\min\{\operatorname{ind}(\kappa_n):n\text{ is divisible by exactly }r\text{ primes}\}$ and $\mu_\infty:=\lim_{r\to\infty}m_r$. If...

conjecture | open | MathDB 页面 | 原文线索

V2-080056 | MATHDB-404119

Interpolation properties of the critical-slope p-adic L-function

摘录 / EXCERPT: Determine the interpolation properties of the critical-slope p-adic L-function $L_p(f_\alpha):=L_p(X)|_{x_0}$.

open_problem | open | MathDB 页面 | 原文线索

V2-080057 | MATHDB-404127

Comparison sequence for smooth Stein spaces

摘录 / EXCERPT: Theorem 3. Let X be a smooth Stein space over C. For any $i\ge 0$, there is a short exact sequence in $\operatorname{Mod}^{\mathrm{solid}}_{\mathbf Q_p}$

open_problem | open | MathDB 页面 | 原文线索

V2-080058 | MATHDB-404131

Milnor’s conjecture under Euclidean volume growth of the universal cover

摘录 / EXCERPT: Does Milnor’s conjecture hold under the additional assumption that the universal cover $(\widetilde M,\widetilde g)$ has Euclidean volume growth?

conjecture | open | MathDB 页面 | 原文线索

V2-080059 | MATHDB-404137

Relation of the C^0 mass limit to other geometric masses

摘录 / EXCERPT: How does $\lim_{r\to\infty} M_{C^0}(g,\Phi,\varphi_r,r)$ relate to Huisken’s isoperimetric mass [7, Definition 11] and Jauregui’s isocapacitary mass [8, Definition 4]?

open_problem | open | MathDB 页面 | 原文线索

V2-080060 | MATHDB-404139

Integrality of codimension-two Ginzburg-Landau limit varifolds

摘录 / EXCERPT: Under the natural bound $E_\varepsilon(u_\varepsilon)=O(|\log\varepsilon|)$, are the stationary rectifiable $(n-2)$-varifolds arising as limits as $\varepsilon\to0$ from critical p...

open_problem | open | MathDB 页面 | 原文线索

V2-080061 | MATHDB-404144

The no-cylinder conjecture for self-shrinkers

摘录 / EXCERPT: Does there exist a non-cylindrical non-compact self-shrinker $\Sigma\subset\mathbb{R}^{3}$ with a cylindrical end?

conjecture | open | MathDB 页面 | 原文线索

V2-080062 | MATHDB-404151

Baker’s conjecture on concordance and isotopy

摘录 / EXCERPT: Concordance implies isotopy for strongly quasipositive, fibered knots.

conjecture | open | MathDB 页面 | 原文线索

V2-080063 | MATHDB-404154

Unique factorization of spatial graphs into pieces

摘录 / EXCERPT: Every spatial graph Γ can be expressed as a disjoint union of finitely many pieces, $\Gamma \cong \bigsqcup_{i\in I}\Lambda_i$. Any two such decompositions have pairwise-iso...

open_problem | open | MathDB 页面 | 原文线索

V2-080064 | MATHDB-404156

Universal families of curves realizing all pants decompositions

摘录 / EXCERPT: Given an orientable surface S of genus g, a family of closed curves Γ realizes all types of pants decompositions if, for every pants decomposition P of S, there is a...

open_problem | open | MathDB 页面 | 原文线索

V2-080065 | MATHDB-404158

Collar insertion maps for general Seifert-fibred boundary components

摘录 / EXCERPT: Compute the collar insertion map ... when ∂X consists of more general Seifert-fibred spaces.

open_problem | open | MathDB 页面 | 原文线索

V2-080066 | MATHDB-404160

Sliceness of higher even cables of the figure-eight knot

摘录 / EXCERPT: For every integer $n>1$, is the $(2n,1)$-cable of the figure-eight knot slice?

open_problem | open | MathDB 页面 | 原文线索

V2-080067 | MATHDB-404161

Minimum-genus surfaces in \(\mathbb{CP}^2\#\mathbb{CP}^2\)

摘录 / EXCERPT: For $n>1$, is the connected sum of complex surfaces of degrees $2n$ and $6n$ in two copies of $\mathbb{CP}^2$ a smoothly minimum-genus surface in $\mathbb{CP}^2\#\mathbb{CP}^2$?

open_problem | open | MathDB 页面 | 原文线索

V2-080068 | MATHDB-404163

Slice missing slice disc problem

摘录 / EXCERPT: Let $K\subset S^1\times S^2=\partial(S^1\times B^3)$ be the knot shown in Figure 1. The knot K bounds two obvious ribbon discs D_1 and D_2 such that the simple closed curve...

open_problem | open | MathDB 页面 | 原文线索

V2-080069 | MATHDB-404166

Local Birkhoff conjecture near the disc

摘录 / EXCERPT: Does the local Birkhoff conjecture hold near the disc; namely, is every sufficiently small integrable deformation of a nearly circular ellipse, with integrability arbitrarily close...

conjecture | open | MathDB 页面 | 原文线索

V2-080070 | MATHDB-404167

Other exceptional eccentricities in the local strong Birkhoff conjecture

摘录 / EXCERPT: Are there any points in the bad set of eccentricities, other than 0?

conjecture | open | MathDB 页面 | 原文线索

V2-080071 | MATHDB-404169

Size of the set of Zoll energies of a magnetic system

摘录 / EXCERPT: How large can the set of Zoll energies of a magnetic system be?

open_problem | open | MathDB 页面 | 原文线索

V2-080072 | MATHDB-404170

Reeb vector fields without Birkhoff sections

摘录 / EXCERPT: Are there 3D Reeb vector fields that do not admit a Birkhoff section?

open_problem | open | MathDB 页面 | 原文线索

V2-080073 | MATHDB-404171

Unmarked Lyapunov spectral rigidity of expanding circle maps

摘录 / EXCERPT: Does the (unmarked) Lyapunov spectrum of an expanding circle map uniquely define the smooth conjugacy class of the map?

open_problem | open | MathDB 页面 | 原文线索

V2-080074 | MATHDB-404172

Equal-mass extension of the Arnold-diffusion theorem

摘录 / EXCERPT: Conjecturally, if $m_0=m_1$, the conclusion of the theorem holds.

open_problem | open | MathDB 页面 | 原文线索

V2-080075 | MATHDB-404175

Higher-dimensional reaction-diffusion analogues of the reversible homoclinic-tangle result

摘录 / EXCERPT: Can a similar result be obtained for stationary in time and asymptotically spatially-periodic solutions of reaction-diffusion systems defined for $x\in\mathbb{R}^m$ with $m>1$?

open_problem | open | MathDB 页面 | 原文线索

V2-080076 | MATHDB-404176

Global strong solutions for the 3D primitive equations with negative viscosity kinetic energy backscatter parametrization

摘录 / EXCERPT: The initial-boundary value problem for the continuous 3-dimensional primitive equations of the ocean with a negative kinetic energy backscatter parametrization is studied. The exis...

open_problem | open | MathDB 页面 | 原文线索

V2-080077 | MATHDB-404177

Realization problem for free boundary minimal surfaces in the unit ball

摘录 / EXCERPT: Given any $g \ge 0$ and $b \ge 1$, can we embed the surface $\Sigma_{g,b}$, having genus g and b boundary components, as a free boundary minimal surface in the unit ball?

open_problem | open | MathDB 页面 | 原文线索

V2-080078 | MATHDB-404178

Embedded free boundary minimal surfaces with connected boundary and arbitrary genus

摘录 / EXCERPT: For each $g \ge 1$ there exists an embedded free boundary minimal surface M_g in B^3 with connected boundary, genus g, and dihedral symmetry D_{g+1}.

open_problem | open | MathDB 页面 | 原文线索

V2-080079 | MATHDB-404180

Morse index of the genus-one connected-boundary surface

摘录 / EXCERPT: Conjecture: the free boundary minimal surface $M_1\subset B^3$ from Theorem A-having genus 1, connected boundary, and dihedral symmetry D_2-has absolute Morse index 5.

open_problem | open | MathDB 页面 | 原文线索

V2-080080 | MATHDB-404181

Quasiconcavity of the two-dimensional Burkholder function

摘录 / EXCERPT: In dimension $n=2$, is the Burkholder function B_p quasiconcave?

open_problem | open | MathDB 页面 | 原文线索

V2-080081 | MATHDB-404182

Uniqueness of 3D flying wings with a fixed asymptotic cone angle

摘录 / EXCERPT: If two 3D flying wings have the same asymptotic cone angle, then they are isometric modulo rescalings.

open_problem | open | MathDB 页面 | 原文线索

V2-080082 | MATHDB-404183

The anisotropic Bernstein problem

摘录 / EXCERPT: The anisotropic Bernstein problem asks whether entire anisotropic minimal graphs in $\mathbb{R}^{n+1}$ are necessarily hyperplanes, for a general elliptic integrand Φ. Here th...

open_problem | open | MathDB 页面 | 原文线索

V2-080083 | MATHDB-404184

The anisotropic Bernstein problem in dimension 4

摘录 / EXCERPT: There exists a smooth nonlinear function $u:\mathbb{R}^{4}\to\mathbb{R}$ and an elliptic integrand Φ on $\mathbb{R}^{5}$ such that the graph of u in $\mathbb{R}^{5}$ minimiz...

open_problem | open | MathDB 页面 | 原文线索

V2-080084 | MATHDB-404185

Classification of ancient noncompact complete embedded finite-entropy curve-shortening flows

摘录 / EXCERPT: An ancient noncompact complete embedded finite-entropy curve shortening flow is a Grim Reaper or an ancient trombone.

open_problem | open | MathDB 页面 | 原文线索

V2-080085 | MATHDB-404186

Extendability across point singularities of Willmore surfaces

摘录 / EXCERPT: Extendability across a puncture is not always possible for a Willmore surface: the inverted catenoid is only $C^{1,\b1}$ for every $\b1<1$, but is not $C^{1,1}$ at the singularity.

open_problem | open | MathDB 页面 | 原文线索

V2-080086 | MATHDB-404187

Willmore minimizers of fixed genus and minimal surfaces in spheres

摘录 / EXCERPT: The conjecture is that the minimizer among surfaces of fixed genus is attained by a minimal surface in a sphere, except for the $\mathbb{RP}^2$-type in codimension one, because the...

open_problem | open | MathDB 页面 | 原文线索

V2-080087 | MATHDB-404188

Uniqueness of the Willmore-minimizing Klein bottle

摘录 / EXCERPT: The conjecture is that $\widetilde{\tau}_{3,1}$ is the unique minimizer of the Willmore functional W among arbitrary Klein bottles used as competitors.

open_problem | open | MathDB 页面 | 原文线索

V2-080088 | MATHDB-404189

Monotonicity of the minimum Willmore energy with respect to genus

摘录 / EXCERPT: Is the minimum of the Willmore functional W among surfaces with fixed oriented genus monotone in g?

open_problem | open | MathDB 页面 | 原文线索

V2-080089 | MATHDB-404192

Linearity of mapping class groups of compact orientable surfaces

摘录 / EXCERPT: Let Σ be a compact, orientable surface of genus at least three. Is the mapping class group of Σ a linear group?

open_problem | open | MathDB 页面 | 原文线索

V2-080090 | MATHDB-404193

Conjectural recursion for spin volume polynomials

摘录 / EXCERPT: Conjecture 1 asserts the recursion … where … and …

open_problem | open | MathDB 页面 | 原文线索

V2-080091 | MATHDB-404195

Which hyperbolic groups admit Anosov representations?

摘录 / EXCERPT: Which word hyperbolic groups Γ whose Gromov boundary is homeomorphic to S^{p-1} admit Anosov representations?

open_problem | open | MathDB 页面 | 原文线索

V2-080092 | MATHDB-404196

Representation of cluster variables by webs

摘录 / EXCERPT: Fomin and Pylyavskyy conjectured that all cluster variables could be represented by webs.

open_problem | open | MathDB 页面 | 原文线索

V2-080093 | MATHDB-404197

Ravenel’s telescope conjecture

摘录 / EXCERPT: For every height n, the inclusion of categories of spectra $\mathrm{Sp}_{K(n)} \subset \mathrm{Sp}_{T(n)}$ is an equality.

conjecture | open | MathDB 页面 | 原文线索

V2-080094 | MATHDB-404201

Complete scissors-congruence invariants in higher-dimensional Euclidean space

摘录 / EXCERPT: For polytopes in Euclidean space of dimensions 5 and higher, finding complete scissors-congruence invariants remains an open question.

open_problem | open | MathDB 页面 | 原文线索

V2-080095 | MATHDB-404203

Postnikov towers and lifting procedures for logarithmic ring spectra

摘录 / EXCERPT: Understand the logarithmic analog of the tower of square-zero extensions $\cdots\to\tau_{\le 2}(R)\to\tau_{\le 1}(R)\to\tau_{\le 0}(R)\simeq\pi_0(R)$ for a connective ring spectrum...

open_problem | open | MathDB 页面 | 原文线索

V2-080096 | MATHDB-404205

Convex hull of a mixed-integer convex quadratic set with indicator variables

摘录 / EXCERPT: Given a symmetric positive semidefinite matrix $Q\in\mathbb{R}^{n\times n}$ and a set $Z\subseteq\{0,1\}^n$, describe the closed convex hull of … The report presents compact extend...

open_problem | open | MathDB 页面 | 原文线索

V2-080097 | MATHDB-404206

Maximum perimeter of small polygons with a power-of-two number of vertices

摘录 / EXCERPT: Determine the maximum possible perimeter of a convex polygon in the plane with diameter at most 1 when the number of vertices is $n=2^s$ with $s\ge 4$; equivalently, determine an...

open_problem | open | MathDB 页面 | 原文线索

V2-080098 | MATHDB-404208

Convergence theory beyond the linear-quadratic case for time-domain decomposition

摘录 / EXCERPT: Develop a convergence theory for the proposed time-domain decomposition procedure beyond the linear-quadratic case.

open_problem | open | MathDB 页面 | 原文线索

V2-080099 | MATHDB-404209

Double Cap Conjecture for Witsenhausen’s problem

摘录 / EXCERPT: Let $S^{n-1}=[U+007F]x"): x\in\mathbb{R}^n,\ \|x\|=1\u007d$, with standard surface measure ω and total measure ω_n. Witsenhausen’s problem defines … The Double Cap Conjectur...

conjecture | open | MathDB 页面 | 原文线索

V2-080100 | MATHDB-404210

Isoperimetric inequality for the first magnetic Neumann eigenvalue with constant magnetic field

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^2$ be a smooth, simply connected, bounded domain, and let Ω^* be a disk with $|\Omega^*|=|\Omega|$. For the magnetic Laplacian with constant cur...

open_problem | open | MathDB 页面 | 原文线索

V2-080101 | MATHDB-404211

Maximizer for the second Aharonov-Bohm magnetic Neumann eigenvalue

摘录 / EXCERPT: What about the second eigenvalue? For the usual Laplacian, the Szegő-Weinberger inequality concerns the second eigenvalue. In this context, it is not clear what the maximal domain...

open_problem | open | MathDB 页面 | 原文线索

V2-080102 | MATHDB-404212

Upper bound by the de Gennes constant for the magnetic Neumann ground state

摘录 / EXCERPT: Does every smooth bounded domain $\Omega\subset\mathbb{R}^2$ and every $\beta>0$ satisfy ... where $\Theta_0\approx 0.590106$ is the de Gennes constant?

open_problem | open | MathDB 页面 | 原文线索

V2-080103 | MATHDB-404213

Magnetic tunneling for three symmetric potential wells

摘录 / EXCERPT: Let $b>0$ be fixed, and let ... where $A=\tfrac12(-x_2,x_1)$ and ... Here v_0 is non-positive and radial, has support in $[0,a]$, and has a unique non-degenerate minimum at 0; the...

open_problem | open | MathDB 页面 | 原文线索

V2-080104 | MATHDB-404214

Determination of conformal Laplace-eigenvalue suprema

摘录 / EXCERPT: For a closed connected surface M, a conformal class $[g]$ of metrics on M, and the Laplace eigenvalues $\lambda_k(M,g)$, determine the optimal quantities ... and identify the met...

open_problem | open | MathDB 页面 | 原文线索

V2-080105 | MATHDB-404215

First normalized Dirac eigenvalue on a torus with trivial spin structure

摘录 / EXCERPT: For the trivial spin structure on the torus, and for the conformal class represented by the flat metric $g(a,b)$ in the moduli region $|a|\le \tfrac12$ and $a^2+b^2\ge 1$, the conj...

open_problem | open | MathDB 页面 | 原文线索

V2-080106 | MATHDB-404216

Bubble decomposition for finite-rank Lieb-Thirring optimizers

摘录 / EXCERPT: Let $\gamma>\tfrac12$ if $d=1$ and $\gamma>0$ if $d\ge 2$, and let $N\in\mathbb N$. Suppose $(U_j)\subset L^{\gamma+d/2}(\mathbb R^d)$ is a sequence of nonnegative functions such t...

open_problem | open | MathDB 页面 | 原文线索

V2-080107 | MATHDB-404217

Lieb-Thirring conjecture for the optimal eigenvalue-moment constant

摘录 / EXCERPT: For the admissible Lieb-Thirring range $\gamma>\tfrac12$ when $d=1$, and $\gamma>0$ when $d\ge2$, let $L_{\gamma,d}$ be the optimal Lieb-Thirring constant, $L^{(1)}_{\gamma,d}$ the...

conjecture | open | MathDB 页面 | 原文线索

V2-080108 | MATHDB-404218

Universal consecutive-eigenvalue ratio bound for convex domains

摘录 / EXCERPT: For every bounded convex domain $\Omega\subset\mathbb{R}^n$ and every $k\in\mathbb{N}$, $\mu_{k+1}(\Omega)\lesssim\mu_k(\Omega)$, where $\mu_k(\Omega)$ is the k-th nontrivial Neu...

open_problem | open | MathDB 页面 | 原文线索

V2-080109 | MATHDB-404219

Actual solutions with the asymptotics of the constructed quasimodes

摘录 / EXCERPT: We conjecture that there is a basis of actual solutions of $Pu=0$ having the same asymptotic behavior as the quasimodes that we construct.

open_problem | open | MathDB 页面 | 原文线索

V2-080110 | MATHDB-404221

Existence of the displayed Fourier-Bessel solution form for all admissible parameter tuples

摘录 / EXCERPT: Prove that, for every tuple $(r,a,b)\in\mathbb N^3$ satisfying $|a-b|<r$ and $a+b+r$ odd, a solution f of equation (1) has Fourier coefficients of the form displayed in (5): for...

open_problem | open | MathDB 页面 | 原文线索

V2-080111 | MATHDB-404222

Closed form for a shifted convolution sum

摘录 / EXCERPT: For each $n\in\mathbb{N}$, conjecture that ...

open_problem | open | MathDB 页面 | 原文线索

V2-080112 | MATHDB-404226

Bareket-type maximization for simply connected planar domains

摘录 / EXCERPT: For a fixed area and a negative parameter $\beta<0$, is the first Robin eigenvalue $\lambda_1\!\left(\Omega,\beta/\sqrt{A(\Omega)}\right)$ maximal when Ω is the disk, amon...

open_problem | open | MathDB 页面 | 原文线索

V2-080113 | MATHDB-404228

Boundary-condition independence of the Pleijel constant for Robin problems

摘录 / EXCERPT: Is it always true that the Pleijel constant for a Robin problem is equal to the Pleijel constant for the Dirichlet problem on the same domain?

open_problem | open | MathDB 页面 | 原文线索

V2-080114 | MATHDB-404229

Unbounded first Steklov eigenvalues for surfaces spanning a fixed circle

摘录 / EXCERPT: For the fixed boundary circle $\Sigma=\mathbb S^1\times\{0\}\subset\mathbb R^3$, there exists a sequence of compact C^1 surfaces $\Omega_j\subset\mathbb R^3$ with...

open_problem | open | MathDB 页面 | 原文线索

V2-080115 | MATHDB-404230

Purely point spectrum for a connected quasi-conical Dirichlet Laplacian

摘录 / EXCERPT: Within the connected quasi-conical domain construction described on pp. 2077-2078-a tower of growing cubes joined by holes-can the sizes of the holes be chosen so that both the abs...

open_problem | open | MathDB 页面 | 原文线索

V2-080116 | MATHDB-404231

Elementary uniform enclosures for Bessel-function zeros

摘录 / EXCERPT: Can one find elementary (that is, without the use of special functions) and uniform (that is, one expression in all the cases) enclosures for these zeros?

open_problem | open | MathDB 页面 | 原文线索

V2-080117 | MATHDB-404232

Dirichlet Pólya conjecture for Euclidean balls

摘录 / EXCERPT: For every integer $d\ge 2$ and every ball $\Omega\subset\mathbb{R}^d$, the Dirichlet eigenvalue counting function satisfies ... for all $\lambda\ge 0$.

conjecture | open | MathDB 页面 | 原文线索

V2-080118 | MATHDB-404233

Complete asymptotic expansion of the local spectral function for smooth bounded potentials

摘录 / EXCERPT: Suppose $V\in C_b^\infty(\mathbb{R}^d)$. Then there are functions $\{a_j(x)\}_{j=0}^\infty$ such that, for any $N>0$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-080119 | MATHDB-404234

A sign-changing eigenvalue problem on the sphere

摘录 / EXCERPT: Let $\mathbb S^2$ be partitioned into two regular subdomains ϖ_+ and ϖ_- such that ϖ_- is contained in an hemisphere and their common boundary...

open_problem | open | MathDB 页面 | 原文线索

V2-080120 | MATHDB-404235

Robin eigenvalue lower bounds for fixed-volume domains

摘录 / EXCERPT: For the Robin Laplacian with positive boundary parameter α, given positive numbers α and M, do there exist positive constants a and b, depending only on...

open_problem | open | MathDB 页面 | 原文线索

V2-080121 | MATHDB-404237

Ratio bounds for Neumann eigenvalues on convex planar domains

摘录 / EXCERPT: Let Ω be a convex domain in the plane, with Neumann eigenvalues $0=\mu_0(\Omega)<\mu_1(\Omega)\le \mu_2(\Omega)\le\cdots$. Prove the conjectured inequality...

open_problem | open | MathDB 页面 | 原文线索

V2-080122 | MATHDB-404238

Positive lower bound for normalized magnetic eigenvalues

摘录 / EXCERPT: Let (M,g) be a closed Riemannian surface. For each real-valued 1-form A on M, define $d_A=d-iA$ and the magnetic Laplacian $\Delta_A=d_A^*d_A$. If $\lambda_i(M,g,A)$ denote...

open_problem | open | MathDB 页面 | 原文线索

V2-080123 | MATHDB-404239

Partial monotonicity of the second Neumann eigenvalue in the unit square

摘录 / EXCERPT: Let D be the unit square. Prove that for any convex subdomain $\a9 \\D$, one has ...

open_problem | open | MathDB 页面 | 原文线索

V2-080124 | MATHDB-404240

Universal upper bound for normalized Bakry-Émery eigenvalues

摘录 / EXCERPT: Let (M,g) be a closed Riemannian surface and let $f\in C^{\infty}(M)$. Define the Bakry-Émery (Witten) Laplacian $L_{e^{-f}}$ by ... and write its eigenvalues as...

open_problem | open | MathDB 页面 | 原文线索

V2-080125 | MATHDB-404241

Asymptotic spectral optimisation among polygons

摘录 / EXCERPT: Let R_n be the class of all polygons with n sides. Prove that, as $k\to\infty$, the optimisers in R_n of $\lambda_k^N(\Omega)\operatorname{Area}(\Omega)$ (maximised) and of...

open_problem | open | MathDB 页面 | 原文线索

V2-080126 | MATHDB-404242

Robin gap monotonicity

摘录 / EXCERPT: For a convex bounded domain $\Omega\subset\mathbb{R}^n$, is the Robin spectral gap, given by the difference between the first two Robin eigenvalues, namely...

open_problem | open | MathDB 页面 | 原文线索

V2-080127 | MATHDB-404245

Existence of averages of bounded completely multiplicative functions along homogeneous polynomials

摘录 / EXCERPT: Let $f:\mathbb{N}\to\mathbb{R}$ be a real-valued, bounded, completely multiplicative function, and let $P\in\mathbb{Z}[x,y]$ be a homogeneous polynomial with values on the positive...

open_problem | open | MathDB 页面 | 原文线索

V2-080128 | MATHDB-404246

Exact asymptotics of the Thue-Morse diffraction distribution near the origin

摘录 / EXCERPT: Determine the exact asymptotic behaviour of the Thue-Morse diffraction distribution function $Z_{\mathrm{TM}}(x)$ as $x\to 0^+$.

open_problem | open | MathDB 页面 | 原文线索

V2-080129 | MATHDB-404248

Possible Birkhoff spectra for irrational rotations on fat Cantor sets

摘录 / EXCERPT: Given an irrational rotation number $\omega\in\mathbb{R}\setminus\mathbb{Q}$, a set $W\subseteq\mathbb{T}^1=\mathbb{R}/\mathbb{Z}$, and $x\in\mathbb{T}^1$, define ... What is the set...

open_problem | open | MathDB 页面 | 原文线索

V2-080130 | MATHDB-404249

Birkhoff spectra of fat Cantor sets for irrational rotations

摘录 / EXCERPT: Given an irrational rotation number $\omega\in\mathbb{R}\setminus\mathbb{Q}$, let $T^1=\mathbb{R}/\mathbb{Z}$ and, for $W\subseteq T^1$ and $x\in T^1$, define ... and ... Does there ex...

open_problem | open | MathDB 页面 | 原文线索

V2-080131 | MATHDB-404252

Natural length functions for substitutions on compact alphabets

摘录 / EXCERPT: Let A be a compact Hausdorff alphabet and let $\rho:A\to A^+$ be a continuous substitution. Set $E=C(A)$, let $K=\{f\in E:f(a)\ge 0\text{ for all }a\in A\}$, and define the subst...

open_problem | open | MathDB 页面 | 原文线索

V2-080132 | MATHDB-404253

Boston-Jones settled-elements conjecture for arboreal representations

摘录 / EXCERPT: Let $f(x)$ be a polynomial of degree $d\ge 2$ over a number field K, let $\alpha\in K$, and let $\mathrm{Gal}_{f,\alpha}\subset\operatorname{Aut}(T_d)$ be the image of the corres...

conjecture | open | MathDB 页面 | 原文线索

V2-080133 | MATHDB-404254

Uniform approximation of the integrated density of states

摘录 / EXCERPT: Suppose that ω_0 is good and that H_{ω_0} is nice. If (E_i) are the increasingly ordered eigenvalues of the finite-dimensional self-adjoint matrix $H^{\omega_0}_n$...

open_problem | open | MathDB 页面 | 原文线索

V2-080134 | MATHDB-404255

Cantor-spectrum question for Sturmian metric tree graphs

摘录 / EXCERPT: For the Sturmian metric tree-graph models described in the report-namely, the infinite comb-like graphs and the infinite radially symmetric tree graphs, with either the branching n...

open_problem | open | MathDB 页面 | 原文线索

V2-080135 | MATHDB-404256

Point-vortex thermalization on the relevant $t=O(N)$ timescale

摘录 / EXCERPT: For the two-dimensional mean-field point-vortex system with smooth interaction kernel $K=-\nabla^\perp W$, external field $F=-\nabla^\perp V$, axisymmetric V, W, and tagged-par...

open_problem | open | MathDB 页面 | 原文线索

V2-080136 | MATHDB-404258

Precise conditions for deriving weak-turbulence kinetic equations

摘录 / EXCERPT: Determine the precise conditions under which the dynamics of weakly nonlinear wave equations can be approximated by a kinetic equation for the wavenumber density...

open_problem | open | MathDB 页面 | 原文线索

V2-080137 | MATHDB-404259

Analytic explanation of oscillations in the Fourier-parameterized plasma-control objective

摘录 / EXCERPT: Explain and rigorously characterize the highly oscillatory dependence of the objective $J(f[H])$ on the Fourier coefficients of the external electric field H, including whether t...

open_problem | open | MathDB 页面 | 原文线索

V2-080138 | MATHDB-404260

Periodicity of infinite ground states above the fluid-solid transition

摘录 / EXCERPT: Under the standing hypotheses on the interaction potential w and the assumption that its Fourier transform $\chat w$ takes negative values, let $0<\mu_c<\infty$ be the critical p...

open_problem | open | MathDB 页面 | 原文线索

V2-080139 | MATHDB-404261

Pointwise exponential decay of Pauli-Fierz eigenvectors

摘录 / EXCERPT: Let ... where ... with $v_j,w_{jk}\in L^2(\mathbb R^3)+L^\infty(\mathbb R^3)$, and $\alpha>0$. If ψ is a normalized eigenvector of H and, for some Lipschitz function...

open_problem | open | MathDB 页面 | 原文线索

V2-080140 | MATHDB-404263

Critical optimal potential for the Dirac eigenvalue problem

摘录 / EXCERPT: In higher dimension, at the critical value $\alpha=\alpha^*(p,d)$, is the optimal potential for the Dirac minimization problem of Aubin-Talenti form ... for suitable constants A,B?

open_problem | open | MathDB 页面 | 原文线索

V2-080141 | MATHDB-404264

Critical higher-dimensional optimizer for the Dirac Keller problem

摘录 / EXCERPT: For the higher-dimensional cases under discussion, with $p\ge d\ge 2$, and at the critical point $\alpha=\alpha_*(p,d)$, is the optimal potential of the form ... for suitable positiv...

open_problem | open | MathDB 页面 | 原文线索

V2-080142 | MATHDB-404265

Affine slice-genus bound for homogeneous quasimorphisms on braid groups

摘录 / EXCERPT: Fix an integer $n\ge 3$. For every homogeneous quasimorphism $f:B_n\to\mathbb{R}$, do there exist constants $A,C\in\mathbb{R}$ such that ...

open_problem | open | MathDB 页面 | 原文线索

V2-080143 | MATHDB-404266

Quasimorphisms on braid groups and homogenized concordance homomorphisms

摘录 / EXCERPT: For a fixed integer $n\ge 3$, let $f:B_n\to\mathbb{R}$ be a homogeneous quasimorphism. A real-valued knot invariant $I:\mathrm{Knots}\to\mathbb{R}$ is called a 1-Lipschitz concor...

open_problem | open | MathDB 页面 | 原文线索

V2-080144 | MATHDB-404270

Satellite nature of a knot admitting a homeomorphic Dehn filling

摘录 / EXCERPT: Let $K=P(C)$ be a satellite knot with hyperbolic pattern, and suppose that $S^3_K(p/q)\cong S^3_{K'}(p/q)$ for a knot $K'\subset S^3$. Must K' also be a satellite knot, say...

open_problem | open | MathDB 页面 | 原文线索

V2-080145 | MATHDB-404273

Equality of the invariant A and the Arf invariant on torsion concordance classes

摘录 / EXCERPT: If K is torsion in $ C$, then is $A(K)=\operatorname{Arf}(K)$?

open_problem | open | MathDB 页面 | 原文线索

V2-080146 | MATHDB-404274

Uniform bounds for nilpotence degrees in tensor-triangular categories

摘录 / EXCERPT: Is there a uniform bound for nilpotence degrees in tt-categories?

open_problem | open | MathDB 页面 | 原文线索

V2-080147 | MATHDB-404276

Stable connectivity for groups with higher-dimensional irreducible representations

摘录 / EXCERPT: If G affords irreducible representations of dimension >1, does stable connectivity still hold?

open_problem | open | MathDB 页面 | 原文线索

V2-080148 | MATHDB-404277

Characterizing stratification in tensor-triangulated categories

摘录 / EXCERPT: Let T be a rigidly-compactly generated tensor-triangulated category whose Balmer spectrum $[U+007F]\operatorname{Spc}(T^c)$ is Noetherian. Determine when the support map ... is a bijectio...

open_problem | open | MathDB 页面 | 原文线索

V2-080149 | MATHDB-404281

Nerves of Steel Conjecture

摘录 / EXCERPT: Let K be a small rigid (symmetric) tensor triangulated category. Then the comparison map $\varphi : \operatorname{Spc}^{h}(K) \to \operatorname{Spc}(K)$ is a bijection.

conjecture | open | MathDB 页面 | 原文线索

V2-080150 | MATHDB-404284

Classification of compactly generated t-structures via supports

摘录 / EXCERPT: Suppose that $(D,\otimes,\mathbf{1})$ is a rigidly compactly generated $\otimes$-triangulated category such that: (i) the unit $\mathbf{1}$ is compact and generates a non-degenerat...

open_problem | open | MathDB 页面 | 原文线索

V2-080151 | MATHDB-404285

Balmer comparison map for finite tensor categories

摘录 / EXCERPT: For every finite tensor category $\mathcal C$, the map $\rho\colon \operatorname{Spc}(\mathcal C)\to\operatorname{Proj}(\mathcal Z)$ described in Theorem 2, where $\mathcal Z$ is t...

open_problem | open | MathDB 页面 | 原文线索

V2-080152 | MATHDB-404286

Invariance of Neural Collapse under class imbalance

摘录 / EXCERPT: Can the Neural Collapse characterization be made invariant to class imbalances? The report presents an affirmative answer.

open_problem | open | MathDB 页面 | 原文线索

V2-080153 | MATHDB-404287

Dependence of the potentially Barsotti-Tate deformation ring on the gene

摘录 / EXCERPT: In the setup of the report, let $K/\mathbb{Q}_p$ be unramified of degree f, let $\bar\rho$ be the specified irreducible representation, let τ be the specified non-scalar tam...

open_problem | open | MathDB 页面 | 原文线索

V2-080154 | MATHDB-404291

Fields of definition of components obtained by gluing

摘录 / EXCERPT: Are components obtained by gluing components defined over K themselves defined over K?

open_problem | open | MathDB 页面 | 原文线索

V2-080155 | MATHDB-404292

Densities of primes meeting polynomial orbits

摘录 / EXCERPT: Given an arbitrary polynomial $f\in\mathbb{Z}[x]$, determine the natural density $\delta_c(f,a,a_0)$ of primes p such that, for the orbit $a_{n+1}=f(a_n)$ with initial value...

open_problem | open | MathDB 页面 | 原文线索

V2-080156 | MATHDB-404295

Elliptic Deligne-Ihara conjecture for once-punctured CM elliptic curves

摘录 / EXCERPT: For each $(m_1,m_2)\in\mathbb Z_{\ge 1}^2\setminus\{(1,1)\}$ satisfying $m_1\equiv m_2\pmod{|\mathcal O_k^\times|}$, fix an element $\sigma_{(m_1,m_2)}$ in the...

conjecture | open | MathDB 页面 | 原文线索

V2-080157 | MATHDB-404305

Malle’s conjecture on counting field extensions

摘录 / EXCERPT: Let G be a transitive subgroup of S_m and let K be a global field. Define ... where the isomorphism is required to respect the permutation action on the m embeddings. Malle c...

conjecture | open | MathDB 页面 | 原文线索

V2-080158 | MATHDB-404306

Average number of points on superelliptic curves

摘录 / EXCERPT: For a monic squarefree polynomial $f(x)$ of degree n over $\mathbb{F}_q$ and an integer e, let X_f be the affine superelliptic curve $X_f:=\{(x,y)\mid y^e=f(x)\}$. If n or...

open_problem | open | MathDB 页面 | 原文线索

V2-080159 | MATHDB-404307

Vanishing and non-vanishing of triple Massey products for higher-genus curves

摘录 / EXCERPT: Study vanishing and non-vanishing of triple Massey products for curves X of genus $g(X)\ge 2$, especially when $\ell>3$. In particular, is it more likely for triple Massey produc...

open_problem | open | MathDB 页面 | 原文线索

V2-080160 | MATHDB-404308

Oda’s problem for cyclic special loci

摘录 / EXCERPT: For associated abstract Hurwitz data $\mathbf{k}_r$ with $\mathcal{M}_{g,[m]}(G)_{\mathbf{k}_r}$ non-empty, where $G=\mathbb{Z}/\ell\mathbb{Z}$ and (g,m) is of hyperbolic type, o...

open_problem | open | MathDB 页面 | 原文线索

V2-080161 | MATHDB-404309

Homeomorphism conjecture for moduli spaces of admissible fundamental groups

摘录 / EXCERPT: Let $M_{g,n}$ be the quotient of the coarse moduli space of pointed stable curves of type (g,n) over $\\bar{\mathbb{F}}_p$ by the Frobenius equivalence relation, and let...

conjecture | open | MathDB 页面 | 原文线索

V2-080162 | MATHDB-404312

Density of stable primes for polynomial iterates

摘录 / EXCERPT: Given a number field K, a polynomial $f\in K[X]$, and $a\in K$, what is the density of the set of primes $\mathfrak p$ of K such that every polynomial $f^{\circ n}(X)-a$, for...

open_problem | open | MathDB 页面 | 原文线索

V2-080163 | MATHDB-404313

Positive-density stable primes and special polynomial compositions

摘录 / EXCERPT: Any polynomial f which achieves a positive-density proportion of stable primes must be a composition of linears, monomials, and Chebyshev polynomials.

open_problem | open | MathDB 页面 | 原文线索

V2-080164 | MATHDB-404319

Coleman-type ramification question for Raynaud type $[2^g]$

摘录 / EXCERPT: In the setup of Problem 1, suppose that $g\ge 3$ and that the Raynaud type of $J[p]$ is $[2,2,\ldots,2]$ with g entries. Is the action of I_K on X_{tors} trivial?

open_problem | open | MathDB 页面 | 原文线索

V2-080165 | MATHDB-404326

Teissier’s conjectured inequality for Arnold exponents and polar invariants

摘录 / EXCERPT: Let $(f=0)$ be a germ of an isolated hypersurface singularity at $0\in\mathbb{C}^n$. For $j=1,\ldots,n-1$, let f_j denote the restriction of f to a general j-codimensional pl...

conjecture | open | MathDB 页面 | 原文线索

V2-080166 | MATHDB-404327

Perfectly reflecting boundary conditions for the controlled Fokker-Planck equation

摘录 / EXCERPT: For the Fokker-Planck equation, treat the case of perfectly reflecting boundary conditions, for which $D(A)$ becomes a domain that depends on both the time and the control function...

open_problem | open | MathDB 页面 | 原文线索

V2-080167 | MATHDB-404328

Bressan Fire Conjecture for the isotropic blocking problem

摘录 / EXCERPT: Let $B_1(0)$ be the closed unit disk in $\\mathbb{R}^2$, and impose the isotropic data ... where $\sigma>0$. An admissible strategy is a family of one-dimensional rectifiable barrier...

conjecture | open | MathDB 页面 | 原文线索

V2-080168 | MATHDB-404329

Non-admissibility of spiral-like strategies below the critical speed

摘录 / EXCERPT: In the isotropic setting $F\equiv B_1(0)$, $R_0=B_1(0)$, and $\psi\equiv 1/\sigma$, if $\sigma\le 2.6144\ldots$, then no spiral-like strategy is admissible.

open_problem | open | MathDB 页面 | 原文线索

V2-080169 | MATHDB-404331

Regularity of optimal moving sets and well-defined orthogonal boundary trajectories

摘录 / EXCERPT: For the optimization problem (OP) defined in the report, with an optimal motion $t\mapsto\Omega(t)$, what is the regularity of the optimal sets $\Omega(t)$? The necessary condition...

open_problem | open | MathDB 页面 | 原文线索

V2-080170 | MATHDB-404332

Extension of unconditional flocking to non-constant and singular interaction kernels

摘录 / EXCERPT: Extend the result to the non-constant case of the interaction kernel K, including especially singular kernels.

open_problem | open | MathDB 页面 | 原文线索

V2-080171 | MATHDB-404333

Rigorous sharp-interface derivation of the optimal moving-set problem

摘录 / EXCERPT: Carry out the study of the b3-limit, as b5 0, of the functionals ... needed to obtain a rigorous derivation of the moving-set optimization problem (OP2). Here $[s]_+=\max\{s,0\}$.

open_problem | open | MathDB 页面 | 原文线索

V2-080172 | MATHDB-404334

Selection of a rational Riemann solver

摘录 / EXCERPT: Is it possible to select a “rational” Riemann solver?

open_problem | open | MathDB 页面 | 原文线索

V2-080173 | MATHDB-404335

Inverse design for the two-component chromatography system

摘录 / EXCERPT: For a fixed positive time T and an attainable target profile $V_T=(v_T,w_T)$, characterize all initial conditions (v_0,w_0) whose solution of the transformed chromatography sys...

open_problem | open | MathDB 页面 | 原文线索

V2-080174 | MATHDB-404336

Transfer of input-to-state stability to discretized balance laws

摘录 / EXCERPT: How can the Input-to-State stability results for the continuous system (1)-(3) be transferred to a discretized version of (1)-(3), using the operator-splitting discretization (10)-...

open_problem | open | MathDB 页面 | 原文线索

V2-080175 | MATHDB-404338

Turnpike property with interior decay for a mean-field optimal control problem

摘录 / EXCERPT: Let $\bbda\therein(0,1)$ and let $a<b$. For an optimal solution $(\hat\mu(a,b,\mu_0),\hat u(a,b,\mu_0))$ of the mean-field problem $Q(a,b,\mu_0)$, define ... Then the problem has the...

open_problem | open | MathDB 页面 | 原文线索

V2-080176 | MATHDB-404339

Adjoint equation for general states in a scalar conservation law

摘录 / EXCERPT: Conjecture (for general states). Let $u \\in U(\\Omega,M)$ and let $\\rho$ be its corresponding state satisfying (SCL). Then the associated adjoint equation for the adjoint state...

open_problem | open | MathDB 页面 | 原文线索

V2-080177 | MATHDB-404340

First-order theory of exponential integer parts

摘录 / EXCERPT: What is the first-order theory of ordered rings that are existential integer parts of real-closed exponential fields?

open_problem | open | MathDB 页面 | 原文线索

V2-080178 | MATHDB-404341

A shorter proof for the special instance $(\varphi\to\varphi)\to(\varphi\to\varphi)$

摘录 / EXCERPT: In the Pure Positive Implication Propositional Calculus with Frege’s axioms (F1), (F2), modus ponens, and the line-count measure M_1, the derivation D_2 of...

open_problem | open | MathDB 页面 | 原文线索

V2-080179 | MATHDB-404342

Implicit-proof operator as a jump

摘录 / EXCERPT: The report conjectures that $\operatorname{imp}(P):=[P,P]$ is a jump. Here, for proof systems P,Q, a $[P,Q]$-proof of φ is a pair (π,c), where c is a circuit defini...

open_problem | open | MathDB 页面 | 原文线索

V2-080180 | MATHDB-404343

Supremum functionals for weak continuity notions

摘录 / EXCERPT: Over $\mathrm{RCA}^{\omega}_{0}$, Kleene’s quantifier $(\exists^{3})$, namely ... is equivalent to $(\exists^{2})$ together with the existence of the supremum functional ... for each o...

open_problem | open | MathDB 页面 | 原文线索

V2-080181 | MATHDB-404344

First-order theories of exponential integer parts of real-closed exponential fields

摘录 / EXCERPT: Characterize the nonnegative parts of ordered rings that are exponential integer parts (EIP) of real-closed exponential fields (RCEF), and determine their first-order theories in t...

open_problem | open | MathDB 页面 | 原文线索

V2-080182 | MATHDB-404346

Derivability of the radical operator from the coordinate ring

摘录 / EXCERPT: Can the radical operator be derived from the ring structure of $R[V]$ alone?

open_problem | open | MathDB 页面 | 原文线索

V2-080183 | MATHDB-404348

Embedding every countable group into a group of type FP_n

摘录 / EXCERPT: For which n is it possible to embed every countable group into a group of type FP_n? It is natural to suspect that the answer is $n=\infty$, but there is currently no method kn...

open_problem | open | MathDB 页面 | 原文线索

V2-080184 | MATHDB-404352

Existence of a normalized Ext-length limit in characteristic two

摘录 / EXCERPT: Let k be a field of characteristic 2, let $S=k[x_1,x_2,x_3,y_1,y_2,y_3]$, and let I be the ideal of 2×2 minors of the generic 2×3 matrix...

open_problem | open | MathDB 页面 | 原文线索

V2-080185 | MATHDB-404353

Classification of singularity categories

摘录 / EXCERPT: For rings of the form $R=\mathbb{C}[[z_1,\dots,z_n]]/I$ and $S=\mathbb{C}[[z_1,\dots,z_m]]/J$, determine when their singularity categories are equivalent as triangulated categories...

open_problem | open | MathDB 页面 | 原文线索

V2-080186 | MATHDB-404355

LICCI criterion for ADE-format perfect ideals of codimension 3

摘录 / EXCERPT: Let I be a perfect ideal of codimension 3 over a local Gorenstein ring R. Then I is LICCI-that is, it lies in the linkage class of a complete intersection-if and only if th...

open_problem | open | MathDB 页面 | 原文线索

V2-080187 | MATHDB-404358

Embedding nondegenerate semigroups so that they can be glued

摘录 / EXCERPT: Given two nondegenerate semigroups $\langle A\rangle$ and $\langle B\rangle$ in $\mathbb{N}^n$, with $n\ge 2$, how does one embed them in $\mathbb{N}^m$, with $m>n$, in order to gl...

open_problem | open | MathDB 页面 | 原文线索

V2-080188 | MATHDB-404364

McAdam’s criterion for maximal analytic spread

摘录 / EXCERPT: Let R be a formally equidimensional local ring and let I be an ideal in R. Then ...

open_problem | open | MathDB 页面 | 原文线索

V2-080189 | MATHDB-404368

Quillen’s second conjecture on the cotangent complex

摘录 / EXCERPT: For any homomorphism $\phi:R\to S$, if $\operatorname{projdim}(L_{S/R})<\infty$, then $\operatorname{projdim}(L_{S/R})\le 2$.

conjecture | open | MathDB 页面 | 原文线索

V2-080190 | MATHDB-404374

Size bound for maximal laminations on tiled surfaces

摘录 / EXCERPT: Let L be a maximal lamination on a tiled surface with polygonal tiles, supported on n internal edges. Then L has size at most n. If L contains no closed loops, then its s...

open_problem | open | MathDB 页面 | 原文线索

V2-080191 | MATHDB-404380

Extending the transmission-tree method to outbreaks with unobserved intermediate cases

摘录 / EXCERPT: Develop an extension of the current method that does not assume all outbreak cases are observed, allowing missing cases to occur as intermediate hosts so that the phylogenetic tree...

open_problem | open | MathDB 页面 | 原文线索

V2-080192 | MATHDB-404384

Goodness of periodic skew braces with property (S)

摘录 / EXCERPT: Let B be a periodic skew brace with property (S). Here, periodic means that $(B,+)$ coincides with the set of its torsion elements, and property (S) means that every element...

open_problem | open | MathDB 页面 | 原文线索

V2-080193 | MATHDB-404385

Finiteness of finitely generated ideals in skew braces with property (S)

摘录 / EXCERPT: Let B be a skew brace with property (S). Is every finitely generated ideal of B finite?

open_problem | open | MathDB 页面 | 原文线索

V2-080194 | MATHDB-404386

Underlying semilattices of ∧-semibraces

摘录 / EXCERPT: Which semilattices are the underlying semilattices of a $\cwedge$-semibrace?

open_problem | open | MathDB 页面 | 原文线索

V2-080195 | MATHDB-404389

Factorizations of skew braces and right nilpotency

摘录 / EXCERPT: Explore which factorizations of skew braces (of nilpotent type) produce right nilpotent skew braces.

open_problem | open | MathDB 页面 | 原文线索

V2-080196 | MATHDB-404392

Conjugacy classes of rack solutions realized on a finite set

摘录 / EXCERPT: Let X be a finite set. Which conjugacy classes in S_{X^2} are realized by rack structures on X? Equivalently, which conjugacy classes of maps $r:X^2\to X^2$ arising from rack...

open_problem | open | MathDB 页面 | 原文线索

V2-080197 | MATHDB-404393

Byott’s solvability conjecture for finite skew braces

摘录 / EXCERPT: If $B=(B,+,\circ)$ is a finite skew brace and B° is insoluble, then B^+ is insoluble.

conjecture | open | MathDB 页面 | 原文线索

V2-080198 | MATHDB-404394

Trivially decomposable cancellative semi-affine structures on a group

摘录 / EXCERPT: Problem 2. Let G be a group. Find all “trivially decomposable” cancellative semi-affine structures on G to construct all cancellative semi-affine structures on G itself.

open_problem | open | MathDB 页面 | 原文线索

V2-080199 | MATHDB-404395

Conditional homogenization theorem for cohesive free-discontinuity functionals

摘录 / EXCERPT: In the cohesive case, assume that f has linear growth, namely $a_1|\xi|\le f(x,\xi)\le a_2(|\xi|+1)$ for constants $0<a_1\le a_2$, that...

open_problem | open | MathDB 页面 | 原文线索

V2-080200 | MATHDB-404398

Asymptotic solutions of the n-tubes IPM model for n ≥ 3

摘录 / EXCERPT: Describe all possible asymptotic solutions of the n-tubes IPM system (5)-(8) as $t\to\infty$ for $n\ge 3$.

open_problem | open | MathDB 页面 | 原文线索

V2-080201 | MATHDB-404399

Well-posedness and vanishing-viscosity limit for the hyperbolic n-tubes model

摘录 / EXCERPT: Study the well-posedness of the “hyperbolic” problem (5)-(8) with $\nu=0$. What can we say about the vanishing viscosity limit as $\nu\to0$?

open_problem | open | MathDB 页面 | 原文线索

V2-080202 | MATHDB-404402

Prediction of needle tapering length by linearized elasticity

摘录 / EXCERPT: Linearized elasticity is not able to predict the value of the tapering length of the needle.

open_problem | open | MathDB 页面 | 原文线索

V2-080203 | MATHDB-404403

Conditions for an FE² formulation of wave propagation

摘录 / EXCERPT: What are the conditions for an FE² formulation of wave propagation in materials with heterogeneous microstructures?

open_problem | open | MathDB 页面 | 原文线索

V2-080204 | MATHDB-404404

Distinguishing energy storage and dissipation in plasticity

摘录 / EXCERPT: What are the possible methods of distinguishing energy storage and dissipation in plasticity? Is there a clear relation between this problem and configurational and vibrational ent...

open_problem | open | MathDB 页面 | 原文线索

V2-080205 | MATHDB-404406

Subunit size-effect scaling in strain-gradient crystal plasticity

摘录 / EXCERPT: Develop or extend strain-gradient crystal plasticity (SGCP) models so that they can reproduce size-effect scaling relationships with exponents less than 1, as obtained with discr...

open_problem | open | MathDB 页面 | 原文线索

V2-080206 | MATHDB-404408

Finite-shear bounds and optimal microstructures for elastic composites with voids

摘录 / EXCERPT: For two- or three-dimensional isotropic composites consisting of an elastic material with void-or, for technical reasons, an extremely compliant material rather than truly void-can...

open_problem | open | MathDB 页面 | 原文线索

V2-080207 | MATHDB-404410

C*-version of the shift-equivalence conjecture

摘录 / EXCERPT: For two square matrices A and B, the report states the equivalence ... where E_A and E_B are the associated graphs.

conjecture | open | MathDB 页面 | 原文线索

V2-080208 | MATHDB-404412

Existence of harmonious bisection families

摘录 / EXCERPT: Do all Deaconu-Renault groupoids admit harmonious bisection families?

open_problem | open | MathDB 页面 | 原文线索

V2-080209 | MATHDB-404413

Decidability of strong shift equivalence

摘录 / EXCERPT: Is strong shift equivalence decidable for square matrices with entries in $\mathbb{N}$? Here, strong shift equivalence (SSE) is the equivalence relation generated by elementary shi...

open_problem | open | MathDB 页面 | 原文线索

V2-080210 | MATHDB-404416

The local bisection hypothesis for twists over étale Hausdorff groupoids

摘录 / EXCERPT: Let G be a locally compact Hausdorff étale groupoid. The following are equivalent: (1) G is effective; (2) for every twist $\Sigma \xrightarrow{q} G$, the pair...

conjecture | open | MathDB 页面 | 原文线索

V2-080211 | MATHDB-404418

K-stability of smooth Fano threefolds

摘录 / EXCERPT: Determine which smooth Fano threefolds are K-stable.

open_problem | open | MathDB 页面 | 原文线索

V2-080212 | MATHDB-404419

Generation of the complex rank-three Cremona group by finite-order elements

摘录 / EXCERPT: Is the group $\operatorname{Bir}_{\mathbb{C}}(\mathbb{P}^3)$ generated by elements of finite order?

open_problem | open | MathDB 页面 | 原文线索

V2-080213 | MATHDB-404420

Mirror correspondence between Fano orbifolds and maximally mutable Laurent polynomials

摘录 / EXCERPT: The report presents the conjecture that Q-Gorenstein (qG) deformation families of Fano orbifolds of dimension n should be mirror, in the sense of the Fano/Landau-Ginzburg formula...

open_problem | open | MathDB 页面 | 原文线索

V2-080214 | MATHDB-404421

Non-symmetric Mahler conjecture

摘录 / EXCERPT: For a convex body P with $\dim(P)=d-1$, the Mahler volume satisfies $M(P)\ge d^d$, with equality achieved for simplices.

conjecture | open | MathDB 页面 | 原文线索

V2-080215 | MATHDB-404422

Whether the condition F_k can be taken to be k-Fano

摘录 / EXCERPT: Can one take the condition F_k to be “X is k-Fano”?

open_problem | open | MathDB 页面 | 原文线索

V2-080216 | MATHDB-404425

Gizatullin’s problem for smooth quartic surfaces

摘录 / EXCERPT: Which automorphisms of a smooth quartic surface $S \subset \mathbb{P}^{3}$ are restrictions of Cremona transformations of $\mathbb{P}^{3}$?

open_problem | open | MathDB 页面 | 原文线索

V2-080217 | MATHDB-404426

Finiteness of representation dimensions of finite subgroups of Cremona groups in characteristic zero

摘录 / EXCERPT: For every positive integer n and every field k of characteristic 0, is ... finite?

open_problem | open | MathDB 页面 | 原文线索

V2-080218 | MATHDB-404427

TFNP classes beyond FP in the random-oracle model

摘录 / EXCERPT: Is there a TFNP class beyond FP that is easy in the random oracle model?

open_problem | open | MathDB 页面 | 原文线索

V2-080219 | MATHDB-404429

SA lower bounds for graph-isomorphism formulas beyond CFI constructions

摘录 / EXCERPT: Can one find PC/SA/SOS degree lower bounds for graph-isomorphism formulas $\mathrm{ISO}(G,H)$ that do not come from CFI graphs or their small variants? Here $\mathrm{ISO}(G,H)$ is...

open_problem | open | MathDB 页面 | 原文线索

V2-080220 | MATHDB-404430

The conjecture that CLS is a proper subclass of $\mathrm{PLS}\cap\mathrm{PPAD}$

摘录 / EXCERPT: The plausible conjecture of Daskalakis and Papadimitriou that CLS is a proper subclass of $\mathrm{PLS}\cap\mathrm{PPAD}$ was disproved:...

conjecture | open | MathDB 页面 | 原文线索

V2-080221 | MATHDB-404431

Weak automatizability of resolution

摘录 / EXCERPT: A proof system P is weakly automatable if, given a statement τ, one can find a Q-proof of τ in time polynomial in the size of the shortest P-proof of τ, where...

open_problem | open | MathDB 页面 | 原文线索

V2-080222 | MATHDB-404432

CNF lower bounds for weak fragments of IPS via the functional method

摘录 / EXCERPT: Prove CNF lower bounds using the functional method against fragments of IPS that lie below the reach of the barrier-namely, fragments that cannot efficiently derive the conjunction...

open_problem | open | MathDB 页面 | 原文线索

V2-080223 | MATHDB-404433

Unconditional resolution lower bounds for the unary encoding of the clique formula

摘录 / EXCERPT: Prove an unconditional $n^{\Omega(k)}$ resolution-size lower bound for the unary encoding, even in the worst case.

open_problem | open | MathDB 页面 | 原文线索

V2-080224 | MATHDB-404434

Graph Colouring Is Hard on Average for Polynomial Calculus

摘录 / EXCERPT: Polynomial calculus requires optimal linear degree, and hence exponential proof size, to establish the non-colourability of sparse random graphs. The result holds over every field...

open_problem | open | MathDB 页面 | 原文线索

V2-080225 | MATHDB-404435

Exponential lower bound for regular \(\operatorname{Res}(\oplus)\) refutations of the binary pigeonhole principle

摘录 / EXCERPT: The report proves that any regular $\operatorname{Res}(\oplus)$ refutation of the binary pigeonhole principle BPHP_{n+1}^{n} has size at least...

open_problem | open | MathDB 页面 | 原文线索

V2-080226 | MATHDB-404436

Top-down lower bound for depth-4 circuits computing parity

摘录 / EXCERPT: Every depth-4 Boolean circuit computing the n-bit parity function requires $2^{n^{1/3-o(1)}}$ gates.

open_problem | open | MathDB 页面 | 原文线索

V2-080227 | MATHDB-404437

Feige’s question on refuting smoothed k-SAT formulas

摘录 / EXCERPT: In Feige’s smoothed model of k-SAT, formed by starting with an arbitrary worst-case k-SAT formula and independently perturbing each literal’s negation pattern with some small c...

open_problem | open | MathDB 页面 | 原文线索

V2-080228 | MATHDB-404438

Polynomial-time solvability of Tensor Isomorphism in PC+Inv

摘录 / EXCERPT: Conjecture: even $\mathrm{PC+Inv}$ cannot solve Tensor Isomorphism (TI) in polynomial time.

open_problem | open | MathDB 页面 | 原文线索

V2-080229 | MATHDB-404440

Whether the theory of exponential integer parts properly extends IOpen

摘录 / EXCERPT: Let T_{EIP} denote the first-order theory of exponential integer parts of real-closed exponential fields in the language L_{OR}. The question is whether...

open_problem | open | MathDB 页面 | 原文线索

V2-080230 | MATHDB-404441

Super-linear lower bound for the iteration number of k-WL on graphs

摘录 / EXCERPT: Whether there is a super-linear lower bound for the iteration number for k-WL on graphs.

open_problem | open | MathDB 页面 | 原文线索

V2-080231 | MATHDB-404444

Geometric property (T) for sofic approximations of property (T) groups

摘录 / EXCERPT: Let X be a space of graphs having almost boundary (T). Then there is a space of graphs X' which is approximately isomorphic to X and has geometric property (T).

open_problem | open | MathDB 页面 | 原文线索

V2-080232 | MATHDB-404445

Babai’s conjecture on diameters of Cayley graphs of high-rank simple groups

摘录 / EXCERPT: Prove Babai’s conjecture that Cayley graphs of simple groups have polylogarithmic diameter uniformly in the choice of generating set.

conjecture | open | MathDB 页面 | 原文线索

V2-080233 | MATHDB-404446

Expansion of the two-generator Schreier graph for \(\mathrm{SL}_n(2)\)

摘录 / EXCERPT: Let $\Gamma_k(q)$ denote the Schreier graph of $G=\mathrm{SL}_n(q)$ acting on $\mathbb{F}_q^n\setminus\{0\}$ with k random generators $x_1,\ldots,x_k\in G$. The lectures sketch t...

open_problem | open | MathDB 页面 | 原文线索

V2-080234 | MATHDB-404448

Babai’s conjecture on diameters of finite simple groups

摘录 / EXCERPT: For every finite, simple, non-abelian group G and every generating set $A\subseteq G$, prove that ... Here $\operatorname{diam}(\Gamma(G,A))$ is the least k such that every eleme...

conjecture | open | MathDB 页面 | 原文线索

V2-080235 | MATHDB-404450

Polynomial Bogolyubov problem over finite fields

摘录 / EXCERPT: Show that if $A\subseteq \mathbb{F}_p^n$, then $2A-2A$ contains an affine subspace of codimension $O_p(\log(1/\alpha))$, where $\alpha=|A|/p^n$.

open_problem | open | MathDB 页面 | 原文线索

V2-080236 | MATHDB-404452

Sharply 2-transitive groups in positive characteristic with a smaller characteristic bound

摘录 / EXCERPT: Construct examples of sharply 2-transitive groups in positive characteristic with a more reasonable (smaller) lower bound on the characteristic than the extremely large bound obt...

open_problem | open | MathDB 页面 | 原文线索

V2-080237 | MATHDB-404455

Uniform exponential character bound for finite quasisimple groups of Lie type

摘录 / EXCERPT: There exists an explicit constant $c>0$ such that, for every finite quasisimple group G of Lie type, every $\chi\in\operatorname{Irr}(G)$, and every $g\in G$, ...

open_problem | open | MathDB 页面 | 原文线索

V2-080238 | MATHDB-404456

Thompson’s conjecture on squares of conjugacy classes

摘录 / EXCERPT: If G is a finite non-abelian simple group, then $G=C^2$ for some conjugacy class C of G.

conjecture | open | MathDB 页面 | 原文线索

V2-080239 | MATHDB-404457

Thompson’s conjecture for sufficiently large finite simple classical groups

摘录 / EXCERPT: Let $G=\operatorname{Cl}_n(q)$ be a finite simple classical group of sufficiently large rank. If $G=P\Omega_n^\varepsilon(q)$ and q is odd, additionally assume that n is even a...

conjecture | open | MathDB 页面 | 原文线索

V2-080240 | MATHDB-404458

Super-expansion of symmetric groups with bounded generators

摘录 / EXCERPT: Can the sequence of symmetric groups $\operatorname{Sym}(n)$ be made a super-expander using a bounded number of generators?

open_problem | open | MathDB 页面 | 原文线索

V2-080241 | MATHDB-404461

Classification of thin two-generated subgroups of $\mathrm{GL}_n(\mathbb{Z})$

摘录 / EXCERPT: For every $n\ge 3$, classify all pairs of matrices $A,B\in \mathrm{GL}_n(\mathbb{Z})$ such that, for $\Gamma=\langle A,B\rangle$ and G the Zariski closure of Γ, one has...

open_problem | open | MathDB 页面 | 原文线索

V2-080242 | MATHDB-404462

Improved diameter bounds for \(\mathrm{SL}_n(\mathbb F_q)\)

摘录 / EXCERPT: Can we prove better bounds for $\operatorname{diam}(G)$ with $G=\mathrm{SL}_n(\mathbb F_q)$, say $(\log |G|)(\log n)^{O(1)}$?

open_problem | open | MathDB 页面 | 原文线索

V2-080243 | MATHDB-404468

Frequent recurrence versus frequent hypercyclicity for backward shifts

摘录 / EXCERPT: Is every frequently recurrent backward shift frequently hypercyclic?

open_problem | open | MathDB 页面 | 原文线索

V2-080244 | MATHDB-404470

Hypercyclicity of the Taylor shift on Cauchy-integral subspaces

摘录 / EXCERPT: Let $E\subset\mathbb T$ be compact and let $\mu\in M_p(E)$ be a nonzero Rajchman measure, where $p\ge 2$. Define ... where the Taylor shift is ... Is the Taylor shift T always hyperc...

open_problem | open | MathDB 页面 | 原文线索

V2-080245 | MATHDB-404471

Measure-theoretic orthogonality of weighted backward shifts

摘录 / EXCERPT: For fixed $1\le p<\infty$, let $u=(u_n)_{n\ge1}$ and $v=(v_n)_{n\ge1}$ be weight sequences in $\mathbb C^{\mathbb N}$, and let B_u and B_v be the associated weighted backward s...

open_problem | open | MathDB 页面 | 原文线索

V2-080246 | MATHDB-404474

Compatible cluster structures for Poisson-homogeneous brackets from Belavin-Drinfeld data

摘录 / EXCERPT: For a simply connected simple Lie group G, for every Poisson-homogeneous bracket on G associated with an arbitrary pair of Belavin-Drinfeld data, there exists a compatible regu...

open_problem | open | MathDB 页面 | 原文线索

V2-080247 | MATHDB-404475

Generalizing the Fomin-Zelevinsky conjecture beyond N

摘录 / EXCERPT: Can we generalize the FZ-conjecture to general A, which are not contained in N?

conjecture | open | MathDB 页面 | 原文线索

V2-080248 | MATHDB-404476

The $K(\pi,1)$ question for the τ-cluster morphism category

摘录 / EXCERPT: Is the classifying space $BC(A)$ of the τ-cluster morphism category a $K(\pi,1)$ space?

open_problem | open | MathDB 页面 | 原文线索

V2-080249 | MATHDB-404477

Categorification of noncommutative frieze patterns

摘录 / EXCERPT: Can noncommutative frieze patterns also be categorified, perhaps by noncommutative generalizations of cluster categories and cluster characters?

open_problem | open | MathDB 页面 | 原文线索

V2-080250 | MATHDB-404479

Faithfulness of the reduced Burau representation of B_4

摘录 / EXCERPT: Determine whether the reduced Burau representation $\rho_n:B_n\to \mathrm{GL}(n-1,\mathbb{Z}[t,t^{-1}])$ is faithful when $n=4$; equivalently, determine whether it has trivial kern...

open_problem | open | MathDB 页面 | 原文线索

V2-080251 | MATHDB-404480

Faithfulness of complex specialisations of the reduced Burau representation of B_3

摘录 / EXCERPT: At which complex specialisations of t is the specialised Burau representation $\rho_3^t:B_3\to GL(2,\mathbb C)$ faithful?

open_problem | open | MathDB 页面 | 原文线索

V2-080252 | MATHDB-404481

Faithfulness of complex specializations of the reduced Burau representation of B_3

摘录 / EXCERPT: Let $\rho_t^3:B_3\to\mathrm{GL}(2,\mathbb C)$ be the reduced Burau representation specialized at $t\in\mathbb C^*$. The report proves faithfulness outside the annulus...

open_problem | open | MathDB 页面 | 原文线索

V2-080253 | MATHDB-404484

Characterization of the essential image of the embedding K

摘录 / EXCERPT: Characterize the essential image of the full embedding K.

open_problem | open | MathDB 页面 | 原文线索

V2-080254 | MATHDB-404487

Nakano-Vashaw-Yakimov conjecture on the Balmer spectrum of a finite tensor category

摘录 / EXCERPT: Let $\mathcal C$ be a finite tensor category over a field k. Conjecture:...

conjecture | open | MathDB 页面 | 原文线索

V2-080255 | MATHDB-404489

Deformation quantization of orbifolds

摘录 / EXCERPT: Determine how to construct a deformation quantization for a general orbifold, in particular for a general Poisson orbifold.

open_problem | open | MathDB 页面 | 原文线索

V2-080256 | MATHDB-404491

Tensor Non-Commutative Khintchine I

摘录 / EXCERPT: Let $r\ge 2$ be an integer and $p\ge 2$ a real. Let $T_1,\ldots,T_n$ be symmetric tensors in $d^{\otimes r}$, invariant under permutation of the r indices. For a tensor T, defi...

open_problem | open | MathDB 页面 | 原文线索

V2-080257 | MATHDB-404493

Function spaces and activations compatible with the ReNO property

摘录 / EXCERPT: Identify pairs of function spaces B_1,B_2 and activation functions σ such that $\sigma(B_1)\subseteq B_2$. In particular, identify combinations of suitable function spaces...

open_problem | open | MathDB 页面 | 原文线索

V2-080258 | MATHDB-404498

Talagrand and Kahn-Kalai threshold conjectures

摘录 / EXCERPT: The report discusses two conjectural threshold statements that were subsequently proved. First, for a finite set X and increasing $\mathcal F\subseteq 2^X$, the existence of a...

conjecture | open | MathDB 页面 | 原文线索

V2-080259 | MATHDB-404503

Conservativity of the Langlands functor for \(\mathrm{GL}_n\)

摘录 / EXCERPT: For $G=\mathrm{GL}_n$, prove that the functor $L_\psi:\mathcal D(\mathrm{Bun}_G)\to\operatorname{IndCoh}(\operatorname{Par}_G)$ is conservative.

open_problem | open | MathDB 页面 | 原文线索

V2-080260 | MATHDB-404504

Minimality of the degree-one Ginzburg-Landau vortex

摘录 / EXCERPT: Let $N\ge 2$, let $B^N\subset\mathbb{R}^N$ be the unit ball, and let ... where $\varepsilon>0$ and $W:(-\infty,1]\to\mathbb{R}_+$ is C^1, convex, satisfies $W(0)=0$, and $W(t)>0$ f...

open_problem | open | MathDB 页面 | 原文线索

V2-080261 | MATHDB-404507

Taut foliations on surgeries of knots without reducible surgeries

摘录 / EXCERPT: Let K be a non-trivial knot in S^3 with no reducible surgeries, and let $r\in(1-2g,\,2g-1)\cap\mathbb{Q}$, where g is the genus of K. Does the r-surgery on K support a...

open_problem | open | MathDB 页面 | 原文线索

V2-080262 | MATHDB-404516

Bondal-Polishchuk transitivity conjecture for full exceptional collections

摘录 / EXCERPT: Let T be a triangulated category with a full exceptional collection $T=\langle E_1,\ldots,E_n\rangle$. Bondal-Polishchuk conjectured that the action by mutations and shifts is tr...

conjecture | open | MathDB 页面 | 原文线索

V2-080263 | MATHDB-404518

Relations between derived categories of the two determinantal resolutions

摘录 / EXCERPT: What are the relations between $D(Z^{(d_+)}_+)$ and $D(Z^{(d_-)}_-)$ in general?

open_problem | open | MathDB 页面 | 原文线索

V2-080264 | MATHDB-404520

Meta Conjecture for pure local systems

摘录 / EXCERPT: Conjecture 2 (Meta Conjecture). All known results about variations of (integral) pure Hodge structure have an analog for the pure local systems of Definition 1.

conjecture | open | MathDB 页面 | 原文线索

V2-080265 | MATHDB-404531

Selmer Section Conjecture

摘录 / EXCERPT: Let K be a number field and let X/K be a smooth projective curve of genus $g\ge 2$. Writing $G_K=\operatorname{Gal}(\overline K/K)$, consider the fundamental exact sequence ... L...

conjecture | open | MathDB 页面 | 原文线索

V2-080266 | MATHDB-404533

Density-one Kim conjectures imply the Selmer Section Conjecture

摘录 / EXCERPT: Let $X/\mathbb{Q}$ be a smooth projective curve of genus at least 2 with a rational point. Assume that X satisfies Kim’s Conjecture for all auxiliary primes p in a set of pri…

conjecture | open | MathDB 页面 | 原文线索

V2-080267 | MATHDB-404538

Inverse problem for the full-line nonlinear Fourier transform

摘录 / EXCERPT: Given a function $(a,b)\in L$, find a sequence $F\in\ell^2(\mathbb Z,\mathbb D)$ such that its full-line nonlinear Fourier transform is (a,b). Equivalently, find a factorization…

open_problem | open | MathDB 页面 | 原文线索

V2-080268 | MATHDB-404539

Finding QSP phase factors for bounded parity polynomials

摘录 / EXCERPT: Given an even (or odd) polynomial $f(x)$ of degree n satisfying $\max_{x\in[-1,1]}|f(x)|\le 1$, find phase factors $\Phi=(\phi_0,\ldots,\phi_n)$ such that … and…

open_problem | open | MathDB 页面 | 原文线索

V2-080269 | MATHDB-404541

Riemann-Hilbert factorization for SU(2)-valued functions

摘录 / EXCERPT: Let B be the set of pairs of measurable functions (a,b) on $\$mathbb{T}$ such that $a^$ is outer, $a^(0)0$, and $aa^+bb^=1$ almost everywhere on $$mathbb{T}$. For every $(a,b)\…

open_problem | open | MathDB 页面 | 原文线索

V2-080270 | MATHDB-404542

Rigorously completing the general Riemann-Hilbert factorization argument

摘录 / EXCERPT: Can the omitted technical argument be completed for every $(a,b)\in B$, where B is the class of measurable pairs on $\mathbb T$ such that a^* is outer, $a^*(0)>0$, and…

open_problem | open | MathDB 页面 | 原文线索

V2-080271 | MATHDB-404544

Failure of the telescope conjecture for heights at least two

摘录 / EXCERPT: For $n\\ge 2$, the telescope conjecture equating the geometrically appealing telescope $w^{-1}X$ with the computationally accessible Bousfield localization $L_{K(n)}X$ is false.

conjecture | open | MathDB 页面 | 原文线索

V2-080272 | MATHDB-404545

Strictness of cyclotomic completion inside the telescopic category

摘录 / EXCERPT: Demonstrate that $(\mathrm{Sp}_{T(n)})^{\wedge\,\mathrm{cyc}}\subsetneq \mathrm{Sp}_{T(n)}$, thereby disproving the telescope conjecture.

open_problem | open | MathDB 页面 | 原文线索

V2-080273 | MATHDB-404550

Linear growth of approximation error for large non-normality perturbations

摘录 / EXCERPT: “We conjecture that this linear growth continues for large perturbations as well.”

open_problem | open | MathDB 页面 | 原文线索

V2-080274 | MATHDB-404551

Performance impact of update directions in learned reconstruction

摘录 / EXCERPT: In the nonlinear optical problem, how does the choice of update direction-gradient descent, Gauss-Newton, or quasi-Newton-influence the performance of the learned reconstruction op…

open_problem | open | MathDB 页面 | 原文线索

V2-080275 | MATHDB-404555

Convergence of a gradient flow on control space with rough initial condition

摘录 / EXCERPT: Given smooth vector fields $V_1,\\ldots,V_d$ on $\\mathbb{R}^n$, an initial point $x\\in\\mathbb{R}^n$, and a target point $y\\in\\mathbb{R}^n$, let X^u solve … and define…

open_problem | open | MathDB 页面 | 原文线索

V2-080276 | MATHDB-404556

Equivalence relation induced by loop invariants

摘录 / EXCERPT: Characterize the equivalence relation $X\sim X'$ defined by …

open_problem | open | MathDB 页面 | 原文线索

V2-080277 | MATHDB-404557

Conjectural description of loop invariants

摘录 / EXCERPT: We conjecture that all loop invariants can be written as shuffles of trivial loop invariants (shuffles of anything with letters), signed areas $(ij-ji)$, and conjugation invariants…

open_problem | open | MathDB 页面 | 原文线索

V2-080278 | MATHDB-404564

Non-existence of a Hopf order for a twist of a finite non-abelian simple group

摘录 / EXCERPT: Let K be a number field and let G be a finite non-abelian simple group. Then there exists a twist J for KG, arising from a 2-cocycle on an abelian subgroup of G, such t…

open_problem | open | MathDB 页面 | 原文线索

V2-080279 | MATHDB-404569

Is Zonotope Containment fixed-parameter tractable with respect to the dimension d?

摘录 / EXCERPT: A zonotope is a Minkowski sum of line segments. Given a matrix $G\in\mathbb{R}^{d\times n}$ with columns $g_1,\ldots,g_n\in\mathbb{R}^d$, define … Given matrices…

open_problem | open | MathDB 页面 | 原文线索

V2-080280 | MATHDB-404571

Improving Behrend’s construction for three-term progression-free sets

摘录 / EXCERPT: Let $r_3(N)$ denote the maximum size of a subset of $\{1,\ldots,N\}$ containing no three distinct elements x,y,z in arithmetic progression, i.e. with $x+z=2y$. The report states:…

open_problem | open | MathDB 页面 | 原文线索

V2-080281 | MATHDB-404580

Optimal dimension-only bound for the integer Carathéodory rank

摘录 / EXCERPT: What is the optimal dimension-only upper bound for $CR(C)$ for pointed rational polyhedral cones $C\subset\mathbb{R}^n$?

open_problem | open | MathDB 页面 | 原文线索

V2-080282 | MATHDB-404581

Whether correlation clustering can reach or beat approximation factor 2

摘录 / EXCERPT: For correlation clustering on a complete graph whose edges are labeled $+$ (similar) or $-$ (dissimilar), can an approximation factor of 2 be reached or breached?

open_problem | open | MathDB 页面 | 原文线索

V2-080283 | MATHDB-404582

Strong Thin Tree Conjecture

摘录 / EXCERPT: For every k-edge-connected graph, the guarantee that a spanning tree contains only an $O(1/k)$ fraction of the edges across each cut can be obtained for all cuts simultaneously.

conjecture | open | MathDB 页面 | 原文线索

V2-080284 | MATHDB-404587

Limiting spectral distributions for dilations of non-self-conjugate partitions

摘录 / EXCERPT: Determine the limiting spectral distributions for dilations of non-self-conjugate partitions.

open_problem | open | MathDB 页面 | 原文线索

V2-080285 | MATHDB-404588

Connectivity of three-dimensional dimer configurations using 4- and 6-cycles

摘录 / EXCERPT: For dimer configurations (perfect matchings) on three-dimensional hypercubic boxes, is the configuration space connected when local moves are restricted to successively switching d…

open_problem | open | MathDB 页面 | 原文线索

V2-080286 | MATHDB-404589

Spohn’s central-limit conjecture for the fixed-coupling Fröhlich polaron

摘录 / EXCERPT: For every fixed coupling $\alpha>0$, as $T\to\infty$, the diffusively rescaled Brownian path under the Polaron measure should become asymptotically Gaussian with mean zero and vari…

conjecture | open | MathDB 页面 | 原文线索

V2-080287 | MATHDB-404590

Landau-Pekar conjecture for the effective mass of the Fröhlich polaron

摘录 / EXCERPT: According to the Landau-Pekar conjecture, the effective mass satisfies …

conjecture | open | MathDB 页面 | 原文线索

V2-080288 | MATHDB-404591

Continuity of the phase transition for the higher-dimensional XY model

摘录 / EXCERPT: Determine whether the phase transition of the XY model is continuous in every dimension $d\ge 3$.

open_problem | open | MathDB 页面 | 原文线索

V2-080289 | MATHDB-404592

CFT partition functions satisfying the two-loop Loewner minimizer criterion

摘录 / EXCERPT: For which conformal field theories does the function … have a global minimum for $\tau\in(0,\infty)$?

open_problem | open | MathDB 页面 | 原文线索

V2-080290 | MATHDB-404594

Existence of the conjectural bilinear form \b7 on the E\11 field-strength representation

摘录 / EXCERPT: Conjecture: The field-strength representation carrying the components F^I admits a $K(E_{11})$-invariant symmetric bilinear form η_{IJ}, as required in the first-order equat…

open_problem | open | MathDB 页面 | 原文线索

V2-080291 | MATHDB-404595

Exact gravity/Kac-Moody-coset correspondence

摘录 / EXCERPT: There exists a putative, exact gravity/Kac-Moody-coset correspondence.

open_problem | open | MathDB 页面 | 原文线索

V2-080292 | MATHDB-404597

General proof of the RC I spinorial representation constraint

摘录 / EXCERPT: Establish a general proof of the representation-theoretic constraint (RC I) that the tensor product of the spin-1/2 representation ε and the spin-3/2 representation Ψ is a quotient…

open_problem | open | MathDB 页面 | 原文线索

V2-080293 | MATHDB-404599

Unique standard decomposition in the positive completion of a Kac-Moody group

摘录 / EXCERPT: Let A be a generalized Cartan matrix, let K be a characteristic-zero field (in the report, $K=\mathbb{R}$ or $\mathbb{C}$), and let $\mathfrak{g}_{A,K}$ be the associated Kac-M…

open_problem | open | MathDB 页面 | 原文线索

V2-080294 | MATHDB-404600

Langlands parameters and Moy-Prasad types

摘录 / EXCERPT: Let F be a non-archimedean local field, let G be a connected reductive group over F that splits over a tame extension of F, and let…

open_problem | open | MathDB 页面 | 原文线索

V2-080295 | MATHDB-404602

Fully faithful functors predicted by the Fargues-Scholze conjecture

摘录 / EXCERPT: For $G=\mathrm{GL}_n$ and every $b\in B(G)$, there exists a fully faithful functor … which is linear with respect to the ring homomorphism $Z\to Z_b$ induced by the spectral action…

conjecture | open | MathDB 页面 | 原文线索

V2-080296 | MATHDB-404603

Kutzko’s compact-induction conjecture for supercuspidal representations

摘录 / EXCERPT: Let F be a nonarchimedean local field, let G be the group of F-points of a reductive group over F, and let $ ...

conjecture | open | MathDB 页面 | 原文线索

V2-080297 | MATHDB-404605

Categorical equivalence between sheaves on Bun_G and Isoc_G

摘录 / EXCERPT: Conjecture 3. Under the setup of the report, there is a canonical $\operatorname{Perf}(\operatorname{Par}_G)$-linear equivalence of categories ...

open_problem | open | MathDB 页面 | 原文线索

V2-080298 | MATHDB-404606

Local-global compatibility for the enhanced Drinfeld object

摘录 / EXCERPT: Conjecture 5 (Local-global compatibility). The conjectural object ... lifts to an object ... where $\operatorname{Qcoh}(\operatorname{Par}_G)$ acts on...

open_problem | open | MathDB 页面 | 原文线索

V2-080299 | MATHDB-404608

Vogan’s conjecture: endoscopic transfer of Arthur distributions via NEvs

摘录 / EXCERPT: Conjecture 1(c). Let G be GL_n, SO_{2n+1}, or Sp_{2n}, and let ψ be a local Arthur parameter for the quasisplit form of G. For every semis...

conjecture | open | MathDB 页面 | 原文线索

V2-080300 | MATHDB-404609

Vogan’s conjecture for p-adic \(GL_n\)

摘录 / EXCERPT: For a p-adic group $G=GL_n$ and a local Arthur parameter ψ with corresponding Langlands parameter ϕ_ψ, the ABV-packet coincides with the local Arthur packet: ...

conjecture | open | MathDB 页面 | 原文线索

V2-080301 | MATHDB-404610

Endoscopic transfer of Arthur’s stable distributions via normalized vanishing cycles

摘录 / EXCERPT: Let G be GL_n, SO_{2n+1}, or Sp_{2n}, and let ψ be a local Arthur parameter with corresponding Langlands parameter ϕ_ψ. For every se...

open_problem | open | MathDB 页面 | 原文线索

V2-080302 | MATHDB-404611

Vogan’s conjecture for the Evs calculation of Arthur stable distributions

摘录 / EXCERPT: For a quasisplit classical group $G=SO_{2n+1}$ or Sp_{2n} and a local Arthur parameter ψ with corresponding Langlands parameter ϕ_ψ, prove that the virtual represent...

conjecture | open | MathDB 页面 | 原文线索

V2-080303 | MATHDB-404612

Vogan’s conjecture for Arthur stable distributions and vanishing cycles

摘录 / EXCERPT: Let G be GL_n, SO_{2n+1}, or Sp_{2n}, and let ψ be a local Arthur parameter with corresponding Langlands parameter ϕ_ψ. Then ... Here...

conjecture | open | MathDB 页面 | 原文线索

V2-080304 | MATHDB-404613

Stability of discrete-series L-packets

摘录 / EXCERPT: Assume that G is an F-split, semisimple, simply connected group. Any discrete-series L-packet is stable; that is, there exists a linear combination of the Harish-Chandra char...

open_problem | open | MathDB 页面 | 原文线索

V2-080305 | MATHDB-404614

Stability of the Harish-Chandra character constructed from the spectral action

摘录 / EXCERPT: Assume that F is a non-archimedean local field and that G is F-split, semisimple, and simply connected. Let $\d53eG$ be the Langlands dual group over $\d53eQ_\ell$, let...

open_problem | open | MathDB 页面 | 原文线索

V2-080306 | MATHDB-404618

Γ-convergence of finite averaged and uniform ensemble optimal-control problems

摘录 / EXCERPT: Let Θ be a compact subset of a Euclidean space, let $U=L^2([0,T],\mathbb R^m)$, and let the ensemble dynamics be ... with dynamics Lipschitz-continuous in the state variable a...

open_problem | open | MathDB 页面 | 原文线索

V2-080307 | MATHDB-404623

Explicit estimates for EDMD-based optimal values

摘录 / EXCERPT: Derive explicit estimates for the difference between the optimal values of problems (2) and (1) as a function of the number N of data points, the degree m of the approximating...

open_problem | open | MathDB 页面 | 原文线索

V2-080308 | MATHDB-404624

Well-posedness and stability framework for Bayesian optimal experimental design

摘录 / EXCERPT: A comprehensive framework with well-posedness and stability results for Bayesian optimal experimental design in complex systems remains an open area for future research.

open_problem | open | MathDB 页面 | 原文线索

V2-080309 | MATHDB-404625

Lutwak’s second-intersection-body characterization

摘录 / EXCERPT: Is a convex body whose (second) intersection body is a positive multiple of the body necessarily an ellipsoid?

open_problem | open | MathDB 页面 | 原文线索

V2-080310 | MATHDB-404627

Volume maximization for convolution bodies

摘录 / EXCERPT: Is there a value of $\delta\in(0,1)$ such that $|C_\delta K|_n\le |C_\delta B_K|_n$ for every symmetric convex body K? Here $C_\delta K=\{x\in\mathbb{R}^n:g_K(x)\ge\delta|K|_n\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-080311 | MATHDB-404628

Maximizers of fixed-parameter convolution bodies

摘录 / EXCERPT: For each fixed $\delta\in(0,1)$, what convex bodies are maximizers of $C_\delta K$ when K runs among sets of the same volume?

open_problem | open | MathDB 页面 | 原文线索

V2-080312 | MATHDB-404629

Homothety conjecture for convex bodies of flotation in the plane

摘录 / EXCERPT: The homothety conjecture in $\[?]R^2$ asserts that if a convex body K is homothetic to one of its convex bodies of flotation, namely $K^{\[?]\delta}=\[?]\lambda K$ for some...

conjecture | open | MathDB 页面 | 原文线索

V2-080313 | MATHDB-404630

Homothety conjecture for planar convex bodies of flotation

摘录 / EXCERPT: The report states the conjecture that, if a convex body $K\subset\mathbb{R}^2$ satisfies $K_\delta=\lambda K$ for some $0<\delta<\frac12\operatorname{vol}_2(K)$ and some...

conjecture | open | MathDB 页面 | 原文线索

V2-080314 | MATHDB-404631

Measure comparison for dilations with an intersection-body density

摘录 / EXCERPT: Let bc be a log-concave measure on c3n such that its density is given by $d\bc(x)=e^{-\(\x_M)}\,dx$, where M is an intersection body. Consider convex symmetric bodies...

open_problem | open | MathDB 页面 | 原文线索

V2-080315 | MATHDB-404633

Christoffel-Minkowski problem of order \(n-2\)

摘录 / EXCERPT: The Christoffel-Minkowski problem of order $n-2$ asks to solve the measure equation ...

open_problem | open | MathDB 页面 | 原文线索

V2-080316 | MATHDB-404634

Affine quermassintegral inequality over real, complex, and quaternionic spaces

摘录 / EXCERPT: For every convex body $K\subseteq \mathbb{F}^n$ with non-empty interior, where $\mathbb{F}\in\{\mathbb{R},\mathbb{C},\mathbb{H}\}$, $p=\dim_{\mathbb{R}}\mathbb{F}\in\{1,2,4\}$, and...

open_problem | open | MathDB 页面 | 原文线索

V2-080317 | MATHDB-404635

Generalized Blaschke-Santaló inequality for multiple even functions

摘录 / EXCERPT: Let $m\in\mathbb{N}$ with $m\ge 2$, let $n_1,\ldots,n_m\in\mathbb{N}$, put $N:=\sum_{i=1}^m n_i$, let $c_1,\ldots,c_m>0$, and let Q be a symmetric $N\times N$ matrix. If...

open_problem | open | MathDB 页面 | 原文线索

V2-080318 | MATHDB-404636

Kolesnikov-Werner conjecture for multiple even functions

摘录 / EXCERPT: Let $m\in\mathbb{N}$ with $m\ge 2$, and let $n\in\mathbb{N}$. If $f_1,\ldots,f_m\in L^1_+(\mathbb{R}^n)$ are even and satisfy ... then ... The Oberwolfach report states that this conje...

conjecture | open | MathDB 页面 | 原文线索

V2-080319 | MATHDB-404637

High-dimensional asymptotics of standardized hyperbolic Poisson-flat limit distributions

摘录 / EXCERPT: Let (d_j) and (k_j) be sequences of integers satisfying $0\le k_j\le d_j-1$, $2k_j>d_j+1$, and $d_j\to\infty$ as $j\to\infty$. What is the asymptotic behavior of...

open_problem | open | MathDB 页面 | 原文线索

V2-080320 | MATHDB-404639

Loewner’s question on centroids of sections

摘录 / EXCERPT: Let $\mu(n)$ be the minimum, over all convex bodies $K\subset\mathbb{R}^n$, of the number of $(n-1)$-dimensional sections $K\cap H$, where H passes through $c(K)$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-080321 | MATHDB-404640

Grünbaum’s section-centroid problem in dimensions 3 and 4

摘录 / EXCERPT: For each $n\in\{3,4\}$ and every convex body $K\subset\mathbb{R}^n$-that is, a compact convex set with nonempty interior-is the centroid $c(K)$ the centroid of at least $n+1$ disti...

open_problem | open | MathDB 页面 | 原文线索

V2-080322 | MATHDB-404641

The Minkowski problem for C-asymptotic sets

摘录 / EXCERPT: Given a potentially infinite Borel measure μ on Ω, determine necessary and sufficient conditions for the existence of a C-asymptotic set K such that $\mu=S_K$; if su...

open_problem | open | MathDB 页面 | 原文线索

V2-080323 | MATHDB-404642

Growth-rate conjecture for surface area measures of C-asymptotic sets

摘录 / EXCERPT: Let $C\subset\mathbb{R}^n$ be a pointed, closed, convex cone with nonempty interior. Define ... and, for $\alpha\in(0,\pi/2)$, ... If $K\subset C$ is a C-asymptotic set and S_K is...

conjecture | open | MathDB 页面 | 原文线索

V2-080324 | MATHDB-404644

Bevan's Conjecture 1 on scaling limits of pattern-avoiding permutations

摘录 / EXCERPT: Conjecture 1 (Bevan, Dijon 2023). If every pattern in B ends in 1, then the scaling limit of $\\operatorname{Av}(B)$ is the increasing permuton.

conjecture | open | MathDB 页面 | 原文线索

V2-080325 | MATHDB-404645

Scaling limit for classes whose basis patterns are skew decomposable

摘录 / EXCERPT: Conjecture 3 (Troyka). If every pattern in B is skew decomposable, then the scaling limit of $\operatorname{Av}(B)$ is the increasing permuton.

open_problem | open | MathDB 页面 | 原文线索

V2-080326 | MATHDB-404648

Stieltjes moment conjecture for principal permutation classes

摘录 / EXCERPT: Conjecture: The counting sequence of every principal permutation class is a Stieltjes moment sequence.

conjecture | open | MathDB 页面 | 原文线索

V2-080327 | MATHDB-404651

Unavoidability of shuffles in permutations

摘录 / EXCERPT: A permutation [?] of size pk is a k-shuffle if it can be partitioned into k subsequences such that every subsequence is an occurrence of the same pattern of length p. Conje...

open_problem | open | MathDB 页面 | 原文线索

V2-080328 | MATHDB-404652

Structure of subclasses of the separable permutations

摘录 / EXCERPT: Let C be a subclass of the separable permutations. Then either C is sum or skew closed (or a finite union of such classes), or $C \subseteq X[U]$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-080329 | MATHDB-404656

Moment sequences for pattern-avoiding matchings

摘录 / EXCERPT: Let a matching (perfect matching, pairing, or pair-partition) of $[n]$ be a partition of $[n]$ into pairs. A matching τ is avoided by a matching ρ if ρ does not cont...

open_problem | open | MathDB 页面 | 原文线索

V2-080330 | MATHDB-404657

Bevan’s permuton-limit conjecture for layered avoidance classes

摘录 / EXCERPT: For every non-empty permutation α, the permuton limit of $\operatorname{Av}(\alpha \mathbin{\ominus} 1)$ is the diagonal permuton.

conjecture | open | MathDB 页面 | 原文线索

V2-080331 | MATHDB-404658

Non-\D-finiteness of the generating function for \coperatorname{Av}(4123,4231,4312)

摘录 / EXCERPT: Let $A(0,t)$ be the generating function for the permutation class $\operatorname{Av}(4123,4231,4312)$. Is $A(0,t)$ non-D-finite?

open_problem | open | MathDB 页面 | 原文线索

V2-080332 | MATHDB-404660

Bouvel’s k-shuffle conjecture

摘录 / EXCERPT: Conjecture 1. For a fixed $k\in\mathbb{N}$, any sufficiently long permutation contains a factor which is a k-shuffle of 12 or 21. Here, a k-shuffle of a permutation τ...

conjecture | open | MathDB 页面 | 原文线索

V2-080333 | MATHDB-404661

Eulerian lifting for second-order functionals

摘录 / EXCERPT: Consider a functional of second order for a map $u : \\Omega \\subset \\mathbb{R}^q \\to \\mathbb{R}^d$: ... where H is a convex function over $d\\times q\\times q$ tensors. What i...

open_problem | open | MathDB 页面 | 原文线索

V2-080334 | MATHDB-404662

Extension of semigeostrophic well-posedness to the whole sphere

摘录 / EXCERPT: Is it possible to extend Theorem to include the whole sphere as domain, and not only a closed subdomain strictly contained in an hemisphere?

open_problem | open | MathDB 页面 | 原文线索

V2-080335 | MATHDB-404663

Uniqueness of stationary solutions in the linear-diffusion case

摘录 / EXCERPT: When $m=1$, for a given mass, are stationary solutions of equation (1) unique for a generic attractive interaction potential W?

open_problem | open | MathDB 页面 | 原文线索

V2-080336 | MATHDB-404664

Weighted lens cluster uniqueness for (1,2)-clusters

摘录 / EXCERPT: For a family of weights satisfying the standard positivity and triangularity conditions, consider locally minimizing weighted $(1,2)$-clusters in $\mathbb{R}^n$, with...

open_problem | open | MathDB 页面 | 原文线索

V2-080337 | MATHDB-404665

Isoperimetric problem with capacitary repulsion

摘录 / EXCERPT: Given $\alpha\in(0,N)$, a measurable set $E\subset\mathbb{R}^N$, and fixed volume $m>0$, define ... where $Q>0$. For the problem ... if $N\ge 2$ and $\alpha\in(1,N)$, there are no mini...

open_problem | open | MathDB 页面 | 原文线索

V2-080338 | MATHDB-404667

Axisymmetric equilibrium shapes of rotating fluid membranes

摘录 / EXCERPT: Given an angular velocity $w\ge 0$, surface area $A>0$, and volume $V>0$ satisfying the isoperimetric inequality $V\le \frac{1}{6\pi^2}A^{3/2}$, find $L\in\mathbb{R}_+$, functions...

open_problem | open | MathDB 页面 | 原文线索

V2-080339 | MATHDB-404669

Geroch conjecture stability under exclusion of bubbles and drawstrings

摘录 / EXCERPT: Given that the entropy bound of Theorem 2 removes the possibility of splines and bubbles forming along a sequence, can we formulate and prove Geroch stability where one removes the...

conjecture | open | MathDB 页面 | 原文线索

V2-080340 | MATHDB-404670

Structure of the bad sets in positive-mass stability

摘录 / EXCERPT: In the setting of Theorem 4, is each bad set Z_i made up only of splines, bubbles, and drawstrings?

open_problem | open | MathDB 页面 | 原文线索

V2-080341 | MATHDB-404671

Geroch stability in the style of Theorem 4

摘录 / EXCERPT: Can one formulate and prove a version of Geroch stability in the style of Theorem 4?

open_problem | open | MathDB 页面 | 原文线索

V2-080342 | MATHDB-404673

Spinorial approaches to Geroch and positive-mass-theorem stability

摘录 / EXCERPT: Can one use spinors in order to establish Geroch stability and positive mass theorem stability in dimensions $n \ge 3$?

open_problem | open | MathDB 页面 | 原文线索

V2-080343 | MATHDB-404674

Generalizing Llarull stability to Goette-Semmelmann results

摘录 / EXCERPT: Can one generalize Theorem 7 to the case of the results of Goette and Semmelmann [23]?

open_problem | open | MathDB 页面 | 原文线索

V2-080344 | MATHDB-404676

Function-space families with nonsymmetric BKL behavior

摘录 / EXCERPT: It would be of great interest to exhibit a family of metrics without symmetries, parameterised by a function space, which has the conjectured BKL behaviour, or to prove that there...

open_problem | open | MathDB 页面 | 原文线索

V2-080345 | MATHDB-404677

Kerr black-hole uniqueness without connectedness and analyticity

摘录 / EXCERPT: Can the assumptions that the black hole is connected and that the metric is analytic be removed from the stationary vacuum Kerr black-hole uniqueness theorem, so that the theorem a...

open_problem | open | MathDB 页面 | 原文线索

V2-080346 | MATHDB-404684

Stability of four-dimensional Einstein metrics

摘录 / EXCERPT: Conjecture as presented in the report: in dimension 4, among the Einstein metrics relevant in physics and to Ricci flow-namely, compact Einstein metrics with positive scalar curv...

open_problem | open | MathDB 页面 | 原文线索

V2-080347 | MATHDB-404688

SWIF limits versus metric completions of $W^{1,p}$ limits

摘录 / EXCERPT: In the setting of the IAS MinA Compactness Conjecture, suppose $(M^3,g_j)\to(M^3,g_\infty)$ with $g_j\to g_\infty$ in the $W^{1,p}$ sense, for $p<2$. Can one prove that the SWIF li...

open_problem | open | MathDB 页面 | 原文线索

V2-080348 | MATHDB-404689

Relation between SWIF limits and metric completions of $W^{1,p}$ limits

摘录 / EXCERPT: Recall that there is a SWIF limit space $(M^3,d_\\infty)$, a rectifiable metric space possibly equal to the zero space, in the setting of the IAS MinA Compactness Conjecture. If...

open_problem | open | MathDB 页面 | 原文线索

V2-080349 | MATHDB-404692

Dihedral rigidity for convex polytopes under acute-angle hypotheses

摘录 / EXCERPT: Let $\Omega\subset\mathbb{R}^n$ be a convex polytope, and let g be a Riemannian metric on Ω such that $R_g\ge 0$ in Ω and $H_g\ge 0$ on $\partial\Omega$. Suppose th...

open_problem | open | MathDB 页面 | 原文线索

V2-080350 | MATHDB-404694

Dihedral rigidity conjecture for convex polytopes

摘录 / EXCERPT: Let $\Omega \subset \mathbb{R}^n$ be a convex polytope with a Riemannian metric g. Suppose: - $R_g \ge 0$ in Ω; - $H_g \ge 0$ on $\partial\Omega$; - the dihedral angle wit...

conjecture | open | MathDB 页面 | 原文线索

V2-080351 | MATHDB-404695

Infinitely many dimensions attaining the absolute bound for equiangular lines

摘录 / EXCERPT: Does equality in the absolute bound occur for infinitely many dimensions? Equivalently, are there infinitely many positive integers r for which the maximum number of equiangular...

open_problem | open | MathDB 页面 | 原文线索

V2-080352 | MATHDB-404696

Fixed-angle equiangular lines

摘录 / EXCERPT: Given $\alpha\in[0,1)$ and a dimension r, determine the maximum number of α-equiangular lines in $\mathbb{R}^r$, where a family of lines is α-equiangular when the a...

open_problem | open | MathDB 页面 | 原文线索

V2-080353 | MATHDB-404697

Sparsity of models in minimum-width r-acyclic decompositions

摘录 / EXCERPT: Is there a function $h:\mathbb{N}\to\mathbb{R}$ with $h(r)\to 0$ such that $\|H\|\in O\!\left(|G|^{1+h(r)}\right)$ for some minimum-width r-acyclic H-decompositions of graphs...

open_problem | open | MathDB 页面 | 原文线索

V2-080354 | MATHDB-404698

Edge contractions and r-acyclic width

摘录 / EXCERPT: Which edge contractions cannot increase the r-acyclic width of G?

open_problem | open | MathDB 页面 | 原文线索

V2-080355 | MATHDB-404699

Close the lower-upper bound gap for induced paths in planar graphs

摘录 / EXCERPT: Close the gap between the lower bound $\Omega(\sqrt{\log n})$ and the upper bound $O(\log n/\log\log n)$ for the function $f_{\mathrm{planar}}(n)$, where $f_{\mathrm{planar}}(n)$ i...

open_problem | open | MathDB 页面 | 原文线索

V2-080356 | MATHDB-404700

Linear connected matching conjecture for graphs with independence number two

摘录 / EXCERPT: There is a constant $c>0$ such that every graph G on $n\ge 3$ vertices with $\alpha(G)=2$ contains a connected matching of size cn.

conjecture | open | MathDB 页面 | 原文线索

V2-080357 | MATHDB-404701

Conjectured coarse Erdős-Pósa theorem for cycles

摘录 / EXCERPT: There exist functions $f(k)=O(k\log k)$ and $g(d)=O(d)$ such that, for all positive integers k and d, every graph G contains either a family of k cycles whose pairwise dist...

conjecture | open | MathDB 页面 | 原文线索

V2-080358 | MATHDB-404704

FO model-checking dichotomy for hereditary graph classes

摘录 / EXCERPT: Let $\mathcal{F}$ be a class of graphs which is closed under taking induced subgraphs. If $\mathcal{F}$ is monadically dependent, then first-order model checking is fixed-parameter...

open_problem | open | MathDB 页面 | 原文线索

V2-080359 | MATHDB-404710

Physically realistic samples from a diffusion-derived prior score

摘录 / EXCERPT: How to ensure that the gradient of the log of the prior pdf from the diffusion method leads to physically realistic samples?

open_problem | open | MathDB 页面 | 原文线索

V2-080360 | MATHDB-404711

Incorporating the likelihood into the diffusion method

摘录 / EXCERPT: Is it possible to properly include the likelihood in the diffusion method, in which case we do not need the Particle Flow method?

open_problem | open | MathDB 页面 | 原文线索

V2-080361 | MATHDB-404712

Guaranteeing the contraction factor for sequential measure transport

摘录 / EXCERPT: “Combining our result, which is mainly focused on the design of schedulers of bridging densities so that $D_\alpha(\pi^{(\ell)}\|\pi^{(\ell-1)})$ is minimized, with a class of brid...

open_problem | open | MathDB 页面 | 原文线索

V2-080362 | MATHDB-404713

One-endedness of the arbitrary-genus self-shrinkers

摘录 / EXCERPT: For every $g\in\mathbb{N}$, does the complete, embedded, noncompact self-shrinker $\Theta_g\subset\mathbb{R}^3$ constructed in Theorem 1 have precisely one end, necessarily its asy...

open_problem | open | MathDB 页面 | 原文线索

V2-080363 | MATHDB-404714

One-end conjecture for the arbitrary-genus self-shrinkers

摘录 / EXCERPT: For every $g\in\mathbb{N}$, the complete, embedded, noncompact self-shrinker $\Theta_g\subset\mathbb{R}^3$ constructed in Theorem 1 has precisely one end.

conjecture | open | MathDB 页面 | 原文线索

V2-080364 | MATHDB-404717

Absence of singularities for the conformally constrained Willmore flow of rotationally symmetric tori

摘录 / EXCERPT: The conformally constrained Willmore flow of tori of revolution cannot develop any singularities, neither in finite nor in infinite time, regardless of the behavior of the hyperbol...

open_problem | open | MathDB 页面 | 原文线索

V2-080365 | MATHDB-404720

Positroid and order polytopes in the type fan of alcoved polytopes

摘录 / EXCERPT: Analyze the structure of the type fan of alcoved polytopes further: Which positroid polytopes or order polytopes belong to the same cone of the type fan?

open_problem | open | MathDB 页面 | 原文线索

V2-080366 | MATHDB-404721

When are Minkowski sums of alcoved polytopes alcoved?

摘录 / EXCERPT: Let $H_n=\{x_1+\cdots+x_n=0\}\subset\mathbb{R}^n$. A polytope in H_n is alcoved if all its facet normals are parallel to roots e_i-e_j for some $i\ne j\in[n]$. Let...

open_problem | open | MathDB 页面 | 原文线索

V2-080367 | MATHDB-404722

Realizability of subdivisions induced by compatible blade arrangements

摘录 / EXCERPT: Given a pairwise compatible collection of ordered set partitions $S_1,\ldots,S_d\in\operatorname{OSP}(n)$ and the associated translated blade arrangement $(S_1)^v,\ldots,(S_d)^v$ o...

open_problem | open | MathDB 页面 | 原文线索

V2-080368 | MATHDB-404723

Bijection and shelling order for the hypersimplex $\Delta_{2,n}$

摘录 / EXCERPT: For the hypersimplex $\Delta_{2,n}$, does there exist a nice bijection between its alcoves and decorated ordered set partitions, together with a shelling order of the alcoves, such...

open_problem | open | MathDB 页面 | 原文线索

V2-080369 | MATHDB-404724

Lattice property of the oriented permutoassociahedron graph

摘录 / EXCERPT: Expectation 1. Let the vertices of the permutoassociahedron $PA(n)$ be binarily bracketed permutations, with edges given by transpositions in minimal brackets and by associators. O...

open_problem | open | MathDB 页面 | 原文线索

V2-080370 | MATHDB-404725

Minimal convex CPWL decomposition

摘录 / EXCERPT: How to decompose a CPWL function $f:\mathbb{R}^n\to\mathbb{R}$ into a difference $f=g-h$ of two convex CPWL functions with as few pieces as possible?

open_problem | open | MathDB 页面 | 原文线索

V2-080371 | MATHDB-404726

Minimal decompositions of continuous piecewise-linear functions in general dimensions

摘录 / EXCERPT: Construct a minimal decomposition of a CPWL function in general dimensions, or describe a finite procedure to compute it.

open_problem | open | MathDB 页面 | 原文线索

V2-080372 | MATHDB-404728

Kaplansky-Serre conjecture for Stanley-Reisner rings

摘录 / EXCERPT: For a simplicial complex K and its Stanley-Reisner ring $k[K]$, the Poincaré series of $\operatorname{Tor}_{k[K]}(k,k)$ is a rational function.

conjecture | open | MathDB 页面 | 原文线索

V2-080373 | MATHDB-404731

Subcriticality of integer powers of the discrete Laplacian on the half-line

摘录 / EXCERPT: For $\alpha\in\mathbb N$, the integer powers $(-\Delta)^\alpha$ of the discrete Laplacian on $\ell^2(\mathbb N)$ are subcritical if and only if $\alpha=1$.

open_problem | open | MathDB 页面 | 原文线索

V2-080374 | MATHDB-404732

Landis conjecture for Schrödinger equations in \(\mathbb{R}^N\)

摘录 / EXCERPT: Landis claimed that if a solution u of $(-\nabla+V)[u]=0$ in $\mathbb{R}^N$, with $\|V\|_{L^\infty(\mathbb{R}^N)}\le 1$, decays at least as fast as $e^{-k|x|}$ for some $k>1$ as...

conjecture | open | MathDB 页面 | 原文线索

V2-080375 | MATHDB-404733

Optimal constants in discrete Hardy inequalities on higher-dimensional lattices

摘录 / EXCERPT: Determine the optimal constant $C_H(d)$, for $d\ge 3$, in the discrete Hardy inequality ... for finitely supported functions u on $\mathbb Z^d$ satisfying $u(0)=0$.

open_problem | open | MathDB 页面 | 原文线索

V2-080376 | MATHDB-404735

Efficient single-particle basis for many-body models in double-incommensurate trilayer graphene

摘录 / EXCERPT: A natural follow-on question is whether an efficient single-particle basis can be constructed to develop many-body models.

open_problem | open | MathDB 页面 | 原文线索

V2-080377 | MATHDB-404739

Cameron’s Greedy Conjecture for non-standard almost simple primitive groups

摘录 / EXCERPT: Conjecture 2. Let G be a finite almost simple primitive permutation group with non-standard action. Then its greedy base size satisfies $\mathrm{G}(G)\le 7b(G)$, where $b(G)$ is...

conjecture | open | MathDB 页面 | 原文线索

V2-080378 | MATHDB-404740

Cameron’s Greedy Conjecture for non-standard almost simple primitive actions

摘录 / EXCERPT: Let G be a finite almost simple primitive group with non-standard action. Then $\mathcal{G}(G)\le 7$. Here, $\mathcal{G}(G)$ denotes the greedy base size: the size of a largest b...

conjecture | open | MathDB 页面 | 原文线索

V2-080379 | MATHDB-404743

Transitive subgroups of finite primitive permutation groups

摘录 / EXCERPT: Determine all transitive subgroups of finite primitive permutation groups, up to conjugacy.

open_problem | open | MathDB 页面 | 原文线索

V2-080380 | MATHDB-404748

Thompson’s Conjecture for finite non-abelian simple groups

摘录 / EXCERPT: Every finite non-abelian simple group G contains a conjugacy class $C\subset G$ such that $C^2=G$.

conjecture | open | MathDB 页面 | 原文线索

V2-080381 | MATHDB-404757

Optimal scheduling of dynamic transport

摘录 / EXCERPT: Given a prescribed transport map T and the interpolation $T_t(x)=tT(x)+(1-t)x$ for $t\in[0,1]$, consider the scheduled family $T_{\tau(t)}(x)$, where $\tau:[0,1]\to[0,1]$ satisfi...

open_problem | open | MathDB 页面 | 原文线索

V2-080382 | MATHDB-404761

Quadrature rules for high-dimensional Lévy distributions

摘录 / EXCERPT: Design quadrature rules which work for this extended class of Lévy distributions.

open_problem | open | MathDB 页面 | 原文线索

V2-080383 | MATHDB-404762

Parametric regularity of the quadratic approximation in risk-averse feedback control

摘录 / EXCERPT: Analyze the parametric regularity of the quadratic approximation of the entropic-risk optimal control problem (1)-(2), in the setting involving a non-autonomous Riccati equation an...

open_problem | open | MathDB 页面 | 原文线索

V2-080384 | MATHDB-404763

Removing the localization-error prefactor in heterogeneous Stokes LOD

摘录 / EXCERPT: Develop a better localization strategy for the proposed LOD method that removes the $1/H$ prefactor from the localization-error estimate, at the expense of more expensive computati...

open_problem | open | MathDB 页面 | 原文线索

V2-080385 | MATHDB-404764

Mesh resolution required to resolve the Ginzburg-Landau vortex lattice

摘录 / EXCERPT: What mesh resolution is required, in terms of κ, to resolve the vortex lattice?

open_problem | open | MathDB 页面 | 原文线索

V2-080386 | MATHDB-404765

Symmetry breaking and striped pattern formation near the critical interaction strength

摘录 / EXCERPT: For the periodic nonlocal variational functional described in the report, with $d\ge 1$, $J>0$, $L>0$, $E\subset\mathbb{R}^d$ $[0,L)^d$-periodic, and an isotropic integrable kernel...

open_problem | open | MathDB 页面 | 原文线索

V2-080387 | MATHDB-404766

Wulff-shape minimality for anisotropic perimeter plus Riesz energy

摘录 / EXCERPT: For the problem of minimizing the sum of an anisotropic perimeter and a Riesz non-local energy over sets with fixed mass, and in the small-mass regime, is the Wulff shape of the pe...

open_problem | open | MathDB 页面 | 原文线索

V2-080388 | MATHDB-404767

Overman’s conjecture for global corotating vortex-patch bifurcation

摘录 / EXCERPT: For the global curve of corotating vortex-patch solutions described in Theorem 1, the only limiting configuration as the parameter $s\to\infty$ is that the two patches intersect at...

conjecture | open | MathDB 页面 | 原文线索

V2-080389 | MATHDB-404768

Relation between equivariant and Iwahori-Whittaker Smith-Treumann localization

摘录 / EXCERPT: How are the two types of Smith-Treumann localization-namely, $\check G(\mathbb C[[z]])$-equivariant localization and Iwahori-Whittaker (IW) localization-related? How are the result...

open_problem | open | MathDB 页面 | 原文线索

V2-080390 | MATHDB-404772

K-theory of the slope-zero Hall category

摘录 / EXCERPT: Conjecture 1. For a quiver Q, with slope-zero Hall category $D\text{-}HA(Q)_0:=\bigoplus_{d\in\mathbb{N}^I}D\text{-}mod_{\mathrm{coh}}(X(d))$, there is an isomorphism ...

open_problem | open | MathDB 页面 | 原文线索

V2-080391 | MATHDB-404775

Intrinsic connection between the IP property and integrability

摘录 / EXCERPT: Is there any intrinsic connection between the IP property of the map F and the integrability of the locally-defined dynamics given by F?

open_problem | open | MathDB 页面 | 原文线索

V2-080392 | MATHDB-404779

Suborbits and subdegrees in coset actions of finite exceptional groups of Lie type

摘录 / EXCERPT: Determine the suborbits and corresponding subdegrees of finite exceptional groups of Lie type acting on the cosets of their maximal subgroups.

open_problem | open | MathDB 页面 | 原文线索

V2-080393 | MATHDB-404782

The D2 property for quaternion groups in the range $6\le n\le 8$

摘录 / EXCERPT: If $6\le n\le 8$, then Q_{4n} has the D2 property.

open_problem | open | MathDB 页面 | 原文线索

V2-080394 | MATHDB-404784

Diameter greater than quadratic for transitive permutation groups

摘录 / EXCERPT: Is it known whether any transitive subgroup of S_n has more than quadratic diameter?

open_problem | open | MathDB 页面 | 原文线索

V2-080395 | MATHDB-404787

Constructive coverage of one conjugacy class by a product of two classes in \(\mathrm{Alt}(n)\)

摘录 / EXCERPT: For $G=\mathrm{Alt}(n)$, can one constructively prove that $C_1C_2\supseteq C_3$ whenever C_1,C_2,C_3 are conjugacy classes satisfying $|C_1|,|C_2|\ge |G|^{1-\delta}$ and...

open_problem | open | MathDB 页面 | 原文线索

V2-080396 | MATHDB-404788

Polynomial-time recognition of tensor-product and tensor-induction matrix groups

摘录 / EXCERPT: Is membership in the “tensor product” and “tensor induction” categories decidable in polynomial time?

open_problem | open | MathDB 页面 | 原文线索

V2-080397 | MATHDB-404791

Derangements in two-orbit permutation groups

摘录 / EXCERPT: Prove that if $G \leq \operatorname{Sym}(\Omega)$ has exactly two orbits, each of size $\frac{1}{2}|\Omega|>1$, then G has a derangement (that is, a fixed-point-free element).

open_problem | open | MathDB 页面 | 原文线索

V2-080398 | MATHDB-404792

Maglione’s conjecture on isomorphism classes of unipotent groups from elliptic curves

摘录 / EXCERPT: Let G_q be the collection of groups $G_{E,P}(\mathbb{F}_q)$ arising from elliptic curves $E:y^2=x^3+ax+b$ with $a,b\in\mathbb{F}_q$ and points $P\in E(\mathbb{F}_q)$, where $G_{E,P}$ is the u...

conjecture | open | MathDB 页面 | 原文线索

V2-080399 | MATHDB-404798

Kontsevich’s conjecture on canonical semiorthogonal decompositions

摘录 / EXCERPT: Conjecture 2 (Kontsevich). There should exist canonical semiorthogonal decompositions, well-defined up to mutations.

conjecture | open | MathDB 页面 | 原文线索

V2-080400 | MATHDB-404803

Stable rationality of smooth cubic threefolds

摘录 / EXCERPT: Are smooth cubic threefolds $Y\subset\mathbb{P}^{4}_{\mathbb{C}}$ stably rational?

open_problem | claimed_progress | MathDB 页面 | 原文线索

V2-080401 | MATHDB-404804

Ring-homomorphism property of the tautological projection

摘录 / EXCERPT: Is the tautological projection $\operatorname{taut}:\operatorname{CH}^*(\mathcal A_g)\to R^*(\mathcal A_g)$ a ring homomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-080402 | MATHDB-404809

Semiorthogonal decompositions of the quasi-BPS categories

摘录 / EXCERPT: The categories B_0 and B_1 admit semiorthogonal decompositions with components $D^b(\operatorname{Sym}^k C)$: four blocks for $k<g-1$ and two blocks for $k=g-1$. Blocks with ev...

open_problem | open | MathDB 页面 | 原文线索

V2-080403 | MATHDB-404810

Rationality of the even-degree rank-two fixed-determinant moduli space

摘录 / EXCERPT: Let C be a smooth complex projective curve of genus $g\ge 2$, and let Λ_0 be a line bundle on C with $\deg(\Lambda_0)=2g$. Is the coarse moduli space...

open_problem | open | MathDB 页面 | 原文线索

V2-080404 | MATHDB-404812

Atomicity of the semiorthogonal decomposition of $SU_C(2,\Lambda)$

摘录 / EXCERPT: For a smooth complex projective curve C of genus $g\ge 2$, and a fixed determinant Λ of odd degree, the semiorthogonal decomposition of $D^b(SU_C(2,\Lambda))$ given in Th...

open_problem | open | MathDB 页面 | 原文线索

V2-080405 | MATHDB-404815

The local Plectic Conjecture

摘录 / EXCERPT: Let F be a totally real field of degree d, let H/F be a reductive group, and set $G:=\operatorname{Res}_{F/\mathbf Q}H$. Assume that (G,X) is a Shimura datum with Hodge coc...

conjecture | open | MathDB 页面 | 原文线索

V2-080406 | MATHDB-404819

Local p-adic monodromy theorem via analytic de Rham stacks

摘录 / EXCERPT: After fixing the pseudo-uniformizer $p\in\mathbb Q_p$, there is an equivalence of categories ... where $\mathbb Q_p^{\mathrm{un}}$ is the maximal unramified extension of...

open_problem | open | MathDB 页面 | 原文线索

V2-080407 | MATHDB-404827

Existence of pseudorandom unitaries

摘录 / EXCERPT: Can we efficiently implement seemingly random unitaries, i.e. pseudorandom unitaries (PRUs)?

open_problem | open | MathDB 页面 | 原文线索

V2-080408 | MATHDB-404829

Thom’s gradient conjecture for analytic gradient flows

摘录 / EXCERPT: Let $x(t)$ be a convergent gradient flow of an analytic potential $f:\mathbb{R}^n\to\mathbb{R}$, meaning that $\dot{x}(t)=-\nabla f(x(t))$ and $\lim_{t\to\infty}x(t)=x_\infty$. The...

conjecture | open | MathDB 页面 | 原文线索

V2-080409 | MATHDB-404831

Existence of counterexamples at codimension-two singular sets

摘录 / EXCERPT: For $n \ge 3$, does there exist an L^∞ metric on $\mathbb{R}^n$ that is uniformly bilipschitz to the Euclidean metric, smooth outside of a singular set of finite $(n-2)$-dim...

open_problem | open | MathDB 页面 | 原文线索

V2-080410 | MATHDB-404832

Optimal dimension bound for singular sets of capillary minimizers

摘录 / EXCERPT: What is the optimal dimension bound for the singular set of capillary minimizers? Could this optimal dimension change with the contact angle? The hopeful guess is that the best bou...

open_problem | open | MathDB 页面 | 原文线索

V2-080411 | MATHDB-404834

Nonexistence of wing-like ancient noncollapsed flows in \(\mathbb{R}^4\)

摘录 / EXCERPT: There are no wing-like ancient noncollapsed flows in $\mathbb{R}^4$.

open_problem | open | MathDB 页面 | 原文线索

V2-080412 | MATHDB-404835

Classification of 4-dimensional \ba-solutions in Ricci flow

摘录 / EXCERPT: Conjecture 1. Any ba-solution in 4d Ricci flow is, up to scaling and finite quotients, given by one of the following solutions.

open_problem | open | MathDB 页面 | 原文线索

V2-080413 | MATHDB-404843

Accumulation of classical branch points near top-density branch points

摘录 / EXCERPT: Rule out the possibility that classical branch points accumulate at top-density branch points for two-dimensional area-minimizing mod(q) currents in higher codimension.

open_problem | open | MathDB 页面 | 原文线索

V2-080414 | MATHDB-404845

Gromov’s linear isoperimetric inequality conjecture for CAT(0) spaces

摘录 / EXCERPT: Gromov conjectured that proper cocompact CAT(0) spaces of asymptotic rank at most n admit a linear isoperimetric inequality for fillings of n-cycles by $(n+1)$-chains: ...

conjecture | open | MathDB 页面 | 原文线索

V2-080415 | MATHDB-404847

Birkhoff’s retrograde-orbit global-surface-of-section conjecture

摘录 / EXCERPT: Does the retrograde orbit bound a disk-like global surface of section?

conjecture | open | MathDB 页面 | 原文线索

V2-080416 | MATHDB-404851

Stable transitivity of time-one maps of nonsuspension Anosov flows

摘录 / EXCERPT: Let X be a transitive Anosov vector field that is not conjugate to a suspension, and let f be the time-one map of its flow. Is f stably transitive? (The answer may depend on...

open_problem | open | MathDB 页面 | 原文线索

V2-080417 | MATHDB-404852

Approximation of time-one maps of Anosov flows by Axiom A diffeomorphisms

摘录 / EXCERPT: Can the time-one map of an Anosov flow be approximated by an Axiom A diffeomorphism?

open_problem | open | MathDB 页面 | 原文线索

V2-080418 | MATHDB-404858

Kakeya conjecture in higher dimensions

摘录 / EXCERPT: The Kakeya conjecture in $\mathbb{R}^d$ asserts that any subset of $\mathbb{R}^d$ containing unit line segments oriented in all directions must have dimension d.

conjecture | open | MathDB 页面 | 原文线索

V2-080419 | MATHDB-404859

Counterexample to the matrix-weighted A_2 conjecture

摘录 / EXCERPT: There is a counterexample to the matrix-weighted "A_2 conjecture."

conjecture | open | MathDB 页面 | 原文线索

V2-080420 | MATHDB-404864

Fractal local smoothing conjecture for arbitrary dilation sets

摘录 / EXCERPT: Let $E\subset[1,2]$, and for each $j\ge1$ let E_j be a maximal 2^{-j}-separated subset of E. Define ... where $N(E\cap J,\delta)$ is the minimum number of intervals of length...

conjecture | open | MathDB 页面 | 原文线索

V2-080421 | MATHDB-404865

An $L^p\to L^q$ fractal local smoothing conjecture

摘录 / EXCERPT: For every $1<p\le q<\infty$ with $q>p'$, the conjecture is that ... holds for all $j\ge 1$ whenever ...

conjecture | open | MathDB 页面 | 原文线索

V2-080422 | MATHDB-404867

Finite-direction characterization of GBD

摘录 / EXCERPT: Let $u:\Omega\to\mathbb{R}^d$ be measurable. Assume that, for every $\xi\in W:=\{e_i:i=1,\ldots,d\}\cup\{e_i+e_j:1\le i<j\le d\}$, ... Then $u\in\mathrm{GBD}(\Omega)$.

open_problem | open | MathDB 页面 | 原文线索

V2-080423 | MATHDB-404868

Classification of global solutions asymptotic to a singular minimizing cone

摘录 / EXCERPT: Let $b_{\rm as}$ be the axially symmetric minimizing solution of the one-phase free-boundary problem in $\mathbb{R}^d$, with $d\ge 7$. Suppose that u is a global solution to ... an...

open_problem | open | MathDB 页面 | 原文线索

V2-080424 | MATHDB-404870

Non-fattening level set flows and inner/outer-flow coincidence

摘录 / EXCERPT: Conjecture: non-fattening level set flows must coincide with both their inner and outer flows.

open_problem | open | MathDB 页面 | 原文线索

V2-080425 | MATHDB-404871

Equivalence between non-fattening and the intersection principle for level set flows

摘录 / EXCERPT: To what extent is non-fattening of level set flows equivalent to the intersection principle?

open_problem | open | MathDB 页面 | 原文线索

V2-080426 | MATHDB-404872

Restrictions ensuring localizability of level set flows

摘录 / EXCERPT: It is an interesting open problem what restrictions on singularities ensure that a level set flow is localizable.

open_problem | open | MathDB 页面 | 原文线索

V2-080427 | MATHDB-404874

Non-uniqueness of locally minimizing \(1,2)\-clusters in dimensions 8 through 2700

摘录 / EXCERPT: For $n\tequiv\{8,\ldots,2700\}$, there exists a locally minimizing $(1,2)$-cluster $\mathcal X$ that is not a standard lens.

open_problem | open | MathDB 页面 | 原文线索

V2-080428 | MATHDB-404877

Nearly-round spherical Bernstein application for low-mass stationary varifolds

摘录 / EXCERPT: For every $\varepsilon>0$, there exists a neighborhood $\mathcal N(\varepsilon)$ of the round metric on S^4 such that, for every $g\in\mathcal N(\varepsilon)$, every mod 2 cycl...

open_problem | open | MathDB 页面 | 原文线索

V2-080429 | MATHDB-404879

Variational existence of gravity water waves in physical variables

摘录 / EXCERPT: Question 1. Can one obtain any (even small amplitude) existence results for (1) by variational methods in the original variables?

open_problem | open | MathDB 页面 | 原文线索

V2-080430 | MATHDB-404881

Optimal Hölder threshold for codimension-one isometric immersion flexibility

摘录 / EXCERPT: Is the exponent $\alpha=\frac13$ optimal for flexibility of codimension-one $C^{1,\alpha}$ isometric immersions of surfaces into $\mathbb{R}^3$?

open_problem | open | MathDB 页面 | 原文线索

V2-080431 | MATHDB-404882

Critical Hölder threshold separating rigidity and flexibility for isometric immersions

摘录 / EXCERPT: Does there exist a critical Hölder regularity threshold α_0 that separates flexibility from rigidity? More precisely, does there exist a threshold α_0 such that: (i)...

open_problem | open | MathDB 页面 | 原文线索

V2-080432 | MATHDB-404883

Geometric justification of the Mathieu-group embedding for Z3-orbifold K3 surfaces

摘录 / EXCERPT: Give a geometric justification for the embedding that yields the Mathieu group M_{24} when the symmetry group of Z_3-orbifold limits of K3 surfaces is combined with all symmetr...

open_problem | open | MathDB 页面 | 原文线索

V2-080433 | MATHDB-404886

Gainutdinov-Runkel pseudo-trace conjecture

摘录 / EXCERPT: For a fixed projective generator $G\in\operatorname{Mod}(V)$, the space of vacuum torus conformal blocks is isomorphic to the space of symmetric linear functionals on...

conjecture | open | MathDB 页面 | 原文线索

V2-080434 | MATHDB-404887

Equivalence of categorical and modular S-transformations in the strongly-finite case

摘录 / EXCERPT: In the strongly-finite case (that is, for a C_2-cofinite, self-dual, CFT-type, $\mathbb{N}$-graded VOA), how can one prove the equivalence between the categorical S-transformat...

open_problem | open | MathDB 页面 | 原文线索

V2-080435 | MATHDB-404888

Indices of finite-depth hyperfinite subfactors and $A_\\infty$-subfactors

摘录 / EXCERPT: Every index of a finite depth irreducible hyperfinite subfactor is also the index of an $A_\\infty$-subfactor.

open_problem | open | MathDB 页面 | 原文线索

V2-080436 | MATHDB-404889

Automatic equivariant O_2-stability for outer tensor-category actions

摘录 / EXCERPT: For which unitary tensor categories does equivariant O_2-stability follow automatically from outerness?

open_problem | open | MathDB 页面 | 原文线索

V2-080437 | MATHDB-404890

Efficiently finding solvable forest fusion data

摘录 / EXCERPT: How to efficiently find 4Gamma^k{ij}in{0,1}4 with solutions?

open_problem | open | MathDB 页面 | 原文线索

V2-080438 | MATHDB-404891

Preservation of solvability under exact factorizations

摘录 / EXCERPT: Determine whether solvability is preserved under exact factorizations of fusion categories.

open_problem | open | MathDB 页面 | 原文线索

V2-080439 | MATHDB-404892

Bicrossed-product realization for exact factorizations with a general pointed factor

摘录 / EXCERPT: Is a similar statement true for a general pointed fusion category $\operatorname{vec}^{\omega}_{G}$; namely, if $\mathcal B=\operatorname{vec}^{\omega}_{G}\bullet\mathcal C$, must...

open_problem | open | MathDB 页面 | 原文线索

V2-080440 | MATHDB-404897

Connes’s conjecture on W^*-superrigidity of higher-rank lattices

摘录 / EXCERPT: Every lattice in a higher-rank simple Lie group is W^*-superrigid.

conjecture | open | MathDB 页面 | 原文线索

V2-080441 | MATHDB-404900

Converse to Connes’ implication for almost periodic modular flows

摘录 / EXCERPT: Let M be a full factor with separable predual. If its outer modular flow $\sigma_M:\mathbb{R}\to\operatorname{Out}(M)$ is almost periodic, meaning that $\sigma_M(\mathbb{R})$ has...

open_problem | open | MathDB 页面 | 原文线索

V2-080442 | MATHDB-404904

Minimal actions of selfless groups and selflessness of reduced crossed products

摘录 / EXCERPT: Do there exist minimal actions of C^*-selfless groups whose reduced crossed products are not selfless? The report notes that this remains open even for minimal actions of F_n.

open_problem | open | MathDB 页面 | 原文线索

V2-080443 | MATHDB-404906

Contractibility of the automorphism group of the hyperfinite type III_1 factor

摘录 / EXCERPT: The automorphism group $\operatorname{Aut}(\mathcal R_{\mathrm{III}_1})$ of the hyperfinite factor $\mathcal R_{\mathrm{III}_1}$ of type III_1, equipped with the u-top...

open_problem | open | MathDB 页面 | 原文线索

V2-080444 | MATHDB-404909

Equivalence of the LLP and the Ext-group property

摘录 / EXCERPT: Does A have the LLP if and only if $\operatorname{Ext}(A)$ is a group?

open_problem | open | MathDB 页面 | 原文线索

V2-080445 | MATHDB-404910

The lifting property of $C^*(F_2\times F_2)$

摘录 / EXCERPT: Does $C^*(F_2\times F_2)$ have the (L)LP?

open_problem | open | MathDB 页面 | 原文线索

V2-080446 | MATHDB-404922

Hyodo-Kato criterion for lifting K-theory classes in the semistable case

摘录 / EXCERPT: Let X be proper and semistable over $\operatorname{Spf}(\mathcal O_K)$. A class $x\in K_i(X_0;\mathbb Q_p)$ lifts to $K_i^{\mathrm{cts}}(X;\mathbb Q_p)$ if and only if ... belongs,...

open_problem | open | MathDB 页面 | 原文线索

V2-080447 | MATHDB-404924

Conservativity of the adic realization functor

摘录 / EXCERPT: It is conjectured that, at least over fields of finite transcendence degree, the adic realization functor $\mathrm{DM}(X)\to\widehat{\mathrm{DM}}(X)$ is conservative; equivalently,...

open_problem | open | MathDB 页面 | 原文线索

V2-080448 | MATHDB-404926

Cohesive theory and efficient methods for model-order reduction of non-elliptic parametrized PDEs

摘录 / EXCERPT: Develop a cohesive theory and efficient methods for model-order reduction of parametrized PDEs beyond parametric elliptic problems.

open_problem | open | MathDB 页面 | 原文线索

V2-080449 | MATHDB-404935

Statistical generalisation for downstream tasks after self-supervised pretraining

摘录 / EXCERPT: How do we guarantee statistical generalisation for different downstream prediction tasks given that the model is pre-trained with large amount of unlabelled (augmented) data?

open_problem | open | MathDB 页面 | 原文线索

V2-080450 | MATHDB-404937

Behavior at the critical coupling for 2D directed polymers

摘录 / EXCERPT: Under the intermediate-disorder scaling $\beta=\hat\beta/\sqrt{\log N}$ in dimension $d=2$, understand what happens at the critical value $\hat\beta=\hat\beta_c=\sqrt{\pi}$ for the...

open_problem | open | MathDB 页面 | 原文线索

V2-080451 | MATHDB-404938

Leighton-Shor Poisson isoperimetric optimization

摘录 / EXCERPT: For a Poisson point process $\mathcal Z$, consider ... where the supremum is taken over all polygons $\Sigma\subset[-L,L]^2$ whose side lengths are at least 1. The report states th...

open_problem | open | MathDB 页面 | 原文线索

V2-080452 | MATHDB-404939

Linear-time capacity list decoding for AEL-based codes

摘录 / EXCERPT: Can AEL-based codes achieve linear-time list decoding up to capacity?

open_problem | open | MathDB 页面 | 原文线索

V2-080453 | MATHDB-404941

Constructions of additive codes outperforming linear codes at relatively small minimum distances

摘录 / EXCERPT: Find constructions for additive codes outperforming linear codes for relatively small minimum distances.

open_problem | open | MathDB 页面 | 原文线索

V2-080454 | MATHDB-404947

Nash-Williams’ Conjecture on triangle decompositions

摘录 / EXCERPT: Let G be a K_3-divisible graph with n vertices and minimum degree $\\delta(G)\ge \frac{3}{4}n$. If n is sufficiently large, then G admits a K_3-decomposition.

conjecture | open | MathDB 页面 | 原文线索

V2-080455 | MATHDB-404948

High-girth triangle decompositions of sufficiently large divisible complete graphs

摘录 / EXCERPT: For every integer $g\ge 3$, every sufficiently large K_3-divisible complete graph admits a K_3-decomposition with girth at least g.

open_problem | open | MathDB 页面 | 原文线索

V2-080456 | MATHDB-404950

Detection of an unknown random tree of logarithmic size

摘录 / EXCERPT: For $pn<1.1$, is it possible to detect an unknown random tree of size $O(\log n)$?

open_problem | open | MathDB 页面 | 原文线索

V2-080457 | MATHDB-404951

Cameron’s conjecture on row parities of random Latin squares

摘录 / EXCERPT: Let L be a uniformly random $n\times n$ Latin square. Then the distribution of $N_{\mathrm{row}}(L)$ is approximately the binomial distribution $\operatorname{Bin}(n,1/2)$, as...

conjecture | open | MathDB 页面 | 原文线索

V2-080458 | MATHDB-404954

Kim-Vu sandwich conjecture

摘录 / EXCERPT: For each $\epsilon>0$ there is some $C>0$ such that the following holds for each $d\ge C\,\log n$. There is a coupling (G_*,G,G^*) of random graphs such that...

conjecture | open | MathDB 页面 | 原文线索

V2-080459 | MATHDB-404955

Determining the largest size deficit for non-sumsets in \(\mathbb{Z}_n\)

摘录 / EXCERPT: Let $f(n)$ be the largest integer such that every subset of $\mathbb{Z}_n$ with size larger than $n-f(n)$ can be represented as a sumset $A+A$, with $A\subseteq\mathbb{Z}_n$. Deter...

open_problem | open | MathDB 页面 | 原文线索

V2-080460 | MATHDB-404956

Independence number of random Cayley graphs

摘录 / EXCERPT: Conjecture 2. Let G be a group of size n. The independence number of the random Cayley graph $G(p)$ is at most $\widetilde{O}(p^{-1})$ whp.

open_problem | open | MathDB 页面 | 原文线索

V2-080461 | MATHDB-404960

Turán density of the tetrahedron

摘录 / EXCERPT: Determine whether the Turán density of the 3-uniform tetrahedron satisfies $\pi\left(K_4^{(3)}\right)=\frac{5}{9}$, as conjectured by Turán.

open_problem | open | MathDB 页面 | 原文线索

V2-080462 | MATHDB-404961

Quasipolynomial bounds for the corners theorem

摘录 / EXCERPT: Let $[N]=\{1,\ldots,N\}$ and $A\subseteq [N]\times[N]$. Suppose that A does not contain (x,y), $(x+d,y)$, $(x,y+d)$ with $d\ne 0$. Then...

open_problem | open | MathDB 页面 | 原文线索

V2-080463 | MATHDB-404962

Maximum number of disjoint pairs in a set system

摘录 / EXCERPT: Determine/estimate the maximum number of pairs of disjoint sets in a set system of size m on a universe of size n.

open_problem | open | MathDB 页面 | 原文线索

V2-080464 | MATHDB-404963

Largest cross-disjoint subfamilies in dense disjointness graphs

摘录 / EXCERPT: If $\mathcal A,\mathcal B\subseteq 2^{[n]}$ satisfy ... where N is the number of disjoint pairs $(A,B)\in\mathcal A\times\mathcal B$, how large subfamilies...

open_problem | open | MathDB 页面 | 原文线索

V2-080465 | MATHDB-404965

Singer-Sudan conjecture on complete bipartite subgraphs of disjointness graphs

摘录 / EXCERPT: For set families $\mathcal{A},\mathcal{B}\subseteq 2^{[n]}$ with disjoint-pair density $d(\mathcal{A},\mathcal{B})\ge 1-\varepsilon$, there exist subfamilies...

conjecture | open | MathDB 页面 | 原文线索

V2-080466 | MATHDB-404969

Diameter of a directed Cayley graph of the symmetric group

摘录 / EXCERPT: The report conjectures that the diameter of the directed Cayley graph of S_n generated by the left cyclic shift and the transposition of the first two elements is ...

open_problem | open | MathDB 页面 | 原文线索

V2-080467 | MATHDB-404970

Geometric and analytic properties of the Stückelberg-Petermann renormalisation group

摘录 / EXCERPT: Characterise the geometric and analytic properties of the Stückelberg-Petermann renormalisation group G_{SP}, including the role of renormalisation conditions and symmet...

open_problem | open | MathDB 页面 | 原文线索

V2-080468 | MATHDB-404972

Membership problem for the geometric monodromy group

摘录 / EXCERPT: For a given $\phi\in \operatorname{MCG}(G_{\lambda_0},\partial G_{\lambda_0})$, decide whether $\phi\in\Gamma_f$.

open_problem | open | MathDB 页面 | 原文线索

V2-080469 | MATHDB-404978

Uniqueness of rank-one del Pezzo surfaces with a prescribed singularity type

摘录 / EXCERPT: For a normal del Pezzo surface $\bar X$ of Picard rank one and a singularity type S, let $P(S)$ be the set of isomorphism classes of such surfaces of singularity type S. The un...

open_problem | open | MathDB 页面 | 原文线索

V2-080470 | MATHDB-404980

QHD smoothings for singularities with multiple nodes

摘录 / EXCERPT: It has been conjectured that no QHD smoothings are possible if the minimal dual resolution graph has two or more nodes.

open_problem | open | MathDB 页面 | 原文线索

V2-080471 | MATHDB-404982

Corti’s proposal for smooth Fano threefolds

摘录 / EXCERPT: All Fano 3-folds are smoothings of singular toric Fano 3-folds.

open_problem | open | MathDB 页面 | 原文线索

V2-080472 | MATHDB-404989

The Huang-Yang low-density expansion for the balanced spin-\[?]1/2 Fermi gas

摘录 / EXCERPT: For a spin- ...

open_problem | open | MathDB 页面 | 原文线索

V2-080473 | MATHDB-404990

Next-order corrections to the low-density Fermi-gas energy density

摘录 / EXCERPT: An open question would be to go beyond the second-order correction in a, analyzing rigorously the next-order corrections in a for the energy density.

open_problem | open | MathDB 页面 | 原文线索

V2-080474 | MATHDB-404993

Characterization of connection matrices in the WKB framework

摘录 / EXCERPT: A major problem is to characterise connection matrices in the WKB framework, thereby relating WKB and RH approaches.

open_problem | open | MathDB 页面 | 原文线索

V2-080475 | MATHDB-404994

Resurgence and Borel summability of formal invariants for the knots 4_1 and 5_2

摘录 / EXCERPT: The formal invariants associated with the hyperbolic knots 4_1 and 5_2 are resurgent and Borel summable.

open_problem | open | MathDB 页面 | 原文线索

V2-080476 | MATHDB-404996

Formal proof of simple resurgence for matrix-model partition functions

摘录 / EXCERPT: Prove that the matrix-model partition functions considered in the report are simple resurgent functions.

open_problem | open | MathDB 页面 | 原文线索

V2-080477 | MATHDB-405004

Normal complements for finite nilpotent groups in normalized units of integral group rings

摘录 / EXCERPT: For a finite nilpotent group G, does there exist a normal subgroup $N\trianglelefteq U_1(\mathbb{Z}[G])$ such that $U_1(\mathbb{Z}[G])=N\rtimes G$?

open_problem | open | MathDB 页面 | 原文线索

V2-080478 | MATHDB-405006

Weighted representations of welded pairs with general second components

摘录 / EXCERPT: Does using general second components s become relevant when one enriches the induced representations with weights computed using 2-cocycles, in the spirit of [4]?

open_problem | open | MathDB 页面 | 原文线索

V2-080479 | MATHDB-405007

Derived solutions and welded-pair representations

摘录 / EXCERPT: Are the WB_n-representations induced by a welded pair (X,r,s) isomorphic to ones induced by a welded pair (X,r',q) for some q? Here (X,r') is the derived solution of...

open_problem | open | MathDB 页面 | 原文线索

V2-080480 | MATHDB-405008

An object associated to a welded pair

摘录 / EXCERPT: What object plays for a welded pair the same role as the structure skew brace plays for a solution?

open_problem | open | MathDB 页面 | 原文线索

V2-080481 | MATHDB-405009

A common framework for two families of solution deformations

摘录 / EXCERPT: Is there a common framework embracing the two families of solution deformations above?

open_problem | open | MathDB 页面 | 原文线索

V2-080482 | MATHDB-405011

Decategorified conjecture for unitary Yang-Baxter operators

摘录 / EXCERPT: Let (R,V) be a unitary solution to the Yang-Baxter equation. Then, for all n, the image $\rho_R(B_n)$ is a virtually abelian group; equivalently, $\rho_R(B_n)$ has a normal abe...

conjecture | open | MathDB 页面 | 原文线索

V2-080483 | MATHDB-405014

A discrete surface representation that preserves a wrinkle-free base surface

摘录 / EXCERPT: Find a better discrete surface representation for the base surface r_b-possibly a tensor-product spline or subdivision surface-that, during dynamics with complex wrinkle fields,...

open_problem | open | MathDB 页面 | 原文线索

V2-080484 | MATHDB-405015

Whether the Heavy-Ball Method Can Achieve Acceleration Beyond Quadratic Objectives

摘录 / EXCERPT: Can the classical heavy-ball (HB) method achieve an accelerated convergence rate beyond quadratic objectives on the standard class of L-smooth and $\bc$-strongly convex functions...

open_problem | open | MathDB 页面 | 原文线索

V2-080485 | MATHDB-405017

Limiting distribution of quadratic-character sums with a random phase

摘录 / EXCERPT: Let $[U+007F]chi_q$ be a quadratic character. What is the limiting distribution of ... when $[U+007F]theta\in[0,1]$ is chosen uniformly at random and $q\to\infty$?

open_problem | open | MathDB 页面 | 原文线索

V2-080486 | MATHDB-405018

Mahler problem for Littlewood polynomials

摘录 / EXCERPT: The report asks for the value of ... where ... The same report proves only the lower bound $b_m>0.954$ by constructing Littlewood polynomials of arbitrarily large degree.

open_problem | open | MathDB 页面 | 原文线索

V2-080487 | MATHDB-405023

L^1 means of exponential sums with multiplicative coefficients

摘录 / EXCERPT: Let $f:\mathbb{N}\to[-1,1]$ be a multiplicative function such that ... If ... then there exists a quadratic character χ such that, for every $\delta>0$, the set of primes p sati...

open_problem | open | MathDB 页面 | 原文线索

V2-080488 | MATHDB-405024

Erdős's integer dilation approximation problem under condition (3)

摘录 / EXCERPT: Let $A\subset \mathbb{R}_{>0}$ be a countable set such that ... Then, for every $\varepsilon>0$, there exists a pair $(\alpha,\beta)\in A^2$ such that $\alpha\ne\beta$ and ... for some...

open_problem | open | MathDB 页面 | 原文线索

V2-080489 | MATHDB-405025

Asymptotic for primitive points on a twisted Cayley cubic

摘录 / EXCERPT: Let $N(B)$ count the number of tuples $(x,y,z,w)\in\mathbb Z^4\cap[-B,B]^4$ satisfying ... with $x\ne 0$ and $\gcd(x,y,z,w)=1$. Prove an asymptotic formula ... with explicit constants...

open_problem | open | MathDB 页面 | 原文线索

V2-080490 | MATHDB-405031

Finiteness of the \cK-profinite genus of residually finite K\ahler groups

摘录 / EXCERPT: Is the cK-profinite genus of every residually finite Kahler group finite?

open_problem | open | MathDB 页面 | 原文线索

V2-080491 | MATHDB-405036

Lieb-Robinson bounds for bosonic systems with unbounded interactions

摘录 / EXCERPT: Extend Lieb-Robinson bounds to bosonic systems with unbounded interactions, taking into account the possible accumulation of bosons in a finite lattice region.

open_problem | open | MathDB 页面 | 原文线索

V2-080492 | MATHDB-405039

Higher-dimensional generic global rigidity in even-\q ℓq-spaces

摘录 / EXCERPT: For even q, extend the $\q$-analogue of Theorem 1 from $d=2$ to higher dimensions.

open_problem | open | MathDB 页面 | 原文线索

V2-080493 | MATHDB-405040

A class of periodic graphs satisfying Ehrhart reciprocity

摘录 / EXCERPT: Find a reasonable class P' of pairs (Γ,x_0), consisting of a periodic graph Γ and one of its vertices x_0, such that: - P' contains the class...

open_problem | open | MathDB 页面 | 原文线索

V2-080494 | MATHDB-405041

Zeros of Low-Degree, Low-Influence Polynomials on the Boolean Hypercube

摘录 / EXCERPT: For a degree- multilinear polynomial $f(x)=\sum_{S\subseteq[n],\,|S|\le d} f_S\prod_{i\in S}x_i$ in n variables with real coefficients, define ... Given two such polynomials f an...

open_problem | open | MathDB 页面 | 原文线索

V2-080495 | MATHDB-405042

Number of connected components of the real zeros of two multi-affine polynomials

摘录 / EXCERPT: Does there exist a bound on the number of connected components of the real zeros of a system of two multi-affine polynomials in terms of a bound on their degrees, independent of...

open_problem | open | MathDB 页面 | 原文线索

V2-080496 | MATHDB-405045

Large MaxCut surplus from forbidding large cliques

摘录 / EXCERPT: For every b4 0 there exists b5 0 such that, for every sufficiently large m, if G is a graph with m edges containing no clique of size m^{1/2-b4}, then G has a cut of size at least...

open_problem | open | MathDB 页面 | 原文线索

V2-080497 | MATHDB-405048

Super-exponential anticoncentration for the permanent of a random Bernoulli matrix

摘录 / EXCERPT: Can the exponential upper bound in Theorem 1 be improved to a super-exponential bound? More specifically, for a uniformly random $n\times n$ matrix A_n with entries in $\{-1,1\}$...

open_problem | open | MathDB 页面 | 原文线索

V2-080498 | MATHDB-405049

Fox-Nenadov-Pham conjecture on induced subdivisions in sparse graphs

摘录 / EXCERPT: For every $c>0$ and every graph H, there exists a constant $C=C(H,c)$ such that every (c,t)-sparse graph G with average degree at least Ct contains an induced subdivision o...

conjecture | open | MathDB 页面 | 原文线索

V2-080499 | MATHDB-405050

Determining the largest k-product-free subsets of permutation groups

摘录 / EXCERPT: Determine $m_k(G)$ for permutation groups G, where $m_k(G)$ denotes the largest density of a subset $A\subseteq G$ such that $a_1a_2\cdots a_k\notin A$ for all...

open_problem | open | MathDB 页面 | 原文线索

V2-080500 | MATHDB-405054

Kahn-Saks Conjecture on balance in finite posets

摘录 / EXCERPT: If $w(P)\to\infty$, then $\delta(P)\to\frac12$.

conjecture | open | MathDB 页面 | 原文线索

V2-080501 | MATHDB-405055

Lower bound relating the window parameter to positional variance

摘录 / EXCERPT: Conjecture 10. For every finite poset P of size n and every element $x\in P$, $\operatorname{win}(x)=\Omega\!\left(\frac{\sigma^2(x)}{n}\right)$, where...

open_problem | open | MathDB 页面 | 原文线索

V2-080502 | MATHDB-405056

Gap versus width for finite posets

摘录 / EXCERPT: For any P, $\operatorname{gap}(P)=O(w(P))$.

open_problem | open | MathDB 页面 | 原文线索

V2-080503 | MATHDB-405058

Treewidth-clique boundedness implies bounded tree-independence number

摘录 / EXCERPT: Let $\mathcal{C}$ be a class of graphs closed under induced subgraphs. If there exists a function f such that, for every $G \in \mathcal{C}$,...

open_problem | open | MathDB 页面 | 原文线索

V2-080504 | MATHDB-405059

Sharp rainbow-triangle constant without an affine plane

摘录 / EXCERPT: For a fixed integer r such that no affine plane of order $r-1$ exists, determine exactly the sharp constant governing the maximum number of rainbow triangles in a simple r-edge...

open_problem | open | MathDB 页面 | 原文线索

V2-080505 | MATHDB-405064

Local removal lemma characterization for arbitrary graphs

摘录 / EXCERPT: Conjecture. Let H be a fixed graph on k vertices. A graph G on n vertices is $o(1)$-close to being H-free under the L^∞ distance if and only if, for every set of...

open_problem | open | MathDB 页面 | 原文线索

V2-080506 | MATHDB-405065

Erdős-Pach distance-multiplicity question

摘录 / EXCERPT: Given a set P of $n>4$ distinct points in the plane, can one always find two distinct distances determined by pairs of points in P such that each occurs for at most n pairs o...

open_problem | open | MathDB 页面 | 原文线索

V2-080507 | MATHDB-405066

Grebennikov-Kwan conjecture on Poisson-type anti-concentration

摘录 / EXCERPT: Let $\mathcal A \subseteq 2^{[n]}$ be a family of sets. Let $[n]_p$ be a p-random subset of $[n]$. Let $\ell \ge 2$. Then ... The constant $2/e^2$ cannot be improved. If the condit...

conjecture | open | MathDB 页面 | 原文线索

V2-080508 | MATHDB-405068

Linear upper bound for equilateral sets in finite-[U+007F]p toroidal metrics

摘录 / EXCERPT: Is it true that, for every finite $p\ge 1$, we have $e(T_p^n)\le C_p n$?

open_problem | open | MathDB 页面 | 原文线索

V2-080509 | MATHDB-405069

The value of $C_2(ABBB)$

摘录 / EXCERPT: Let $C_d(w)$ denote the asymptotic leading coefficient for the maximum number of contiguous occurrences of a fixed word w in an n^d grid, with occurrences read in the direction...

open_problem | open | MathDB 页面 | 原文线索

V2-080510 | MATHDB-405070

Periodic constructions for word occurrences in grids

摘录 / EXCERPT: For all positive integers d and all words w, is $C_d(w)$ always attained in the limit by some periodic construction?

open_problem | open | MathDB 页面 | 原文线索

V2-080511 | MATHDB-405073

Existence of an H^1 blowup residue for finite-point blowup solutions of critical gKdV

摘录 / EXCERPT: Does there exist a blowup residue r_α in H^1 for any solution of (1) blowing up in finite time at a finite point (thus, $\nu\in\left(\frac{1}{3},\frac{1}{2}\right)$)?

open_problem | open | MathDB 页面 | 原文线索

V2-080512 | MATHDB-405074

Global well-posedness for nonlocalized small critical-Besov data in 3D and higher gravity water waves

摘录 / EXCERPT: For the gravity water wave system in infinite depth and infinite width, prove global well-posedness for initial data that are small in a critical Besov space, without assuming that...

open_problem | open | MathDB 页面 | 原文线索

V2-080513 | MATHDB-405075

2D non-localized data GWP conjecture

摘录 / EXCERPT: Cubic quasilinear dispersive flows with small data give rise to global in time, scattering solutions in two and higher dimension.

conjecture | open | MathDB 页面 | 原文线索

V2-080514 | MATHDB-405077

Largest top-\m\ capacity of an output projection matrix

摘录 / EXCERPT: What is the largest m for which there exists an output projection matrix (OPM) such that, for any collection of m tokens, there exists an embedding that the matrix will map to...

open_problem | open | MathDB 页面 | 原文线索

V2-080515 | MATHDB-405081

Completeness of the index for 2D invertible phases

摘录 / EXCERPT: Is the index constructed by Sopenko a complete invariant of 2D invertible phases; equivalently, does it classify such phases?

open_problem | open | MathDB 页面 | 原文线索

V2-080516 | MATHDB-405084

Basis conjecture for twisted triplet-VOA conformal blocks

摘录 / EXCERPT: Define ... The conjecture is that the conformal blocks $F^{\sigma}_{\lambda,\kappa}$ form a basis for the space of all conformal blocks of the triplet VOA W_k associated to...

conjecture | open | MathDB 页面 | 原文线索

V2-080517 | MATHDB-405088

Finiteness of the origin component in the infinite noodle

摘录 / EXCERPT: The component of 0 in the infinite noodle is finite almost surely.

open_problem | open | MathDB 页面 | 原文线索

V2-080518 | MATHDB-405089

Maximizing Grothendieck principal specializations at \(\beta=1\)

摘录 / EXCERPT: Which permutations maximize the principal specialization of the Grothendieck polynomials at $\beta=1$?

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V2-080519 | MATHDB-405091

Forest Monk’s formula

摘录 / EXCERPT: Let an indexed forest F be a sequence of plane binary trees $(T_1,T_2,\ldots)$, where T_i is a single leaf for i sufficiently large, and canonically label the union of their...

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V2-080520 | MATHDB-405093

Unimodality and log-concavity of a lattice-path sequence

摘录 / EXCERPT: Let $(N_\ell)_\ell$ denote the number of lattice paths using only east and north steps, with arbitrary step lengths, from $(2,1)$ to either $(d-1,d)$ or $(d,d-1)$, having ℓ st...

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V2-080521 | MATHDB-405096

Ayoub’s adic period conjecture (Conjecture 5)

摘录 / EXCERPT: Let K be either $\boldsymbol{Q}$ endowed with the p-adic valuation, or a field of the form $k(\pi)$ endowed with the π-adic valuation, and let $\widehat K$ be its completio...

conjecture | open | MathDB 页面 | 原文线索

V2-080522 | MATHDB-405097

First-order axiomatizability of existentially closed globally valued fields

摘录 / EXCERPT: Is the class of existentially closed globally valued fields first order? Equivalently, does there exist a model companion of the theory GVF_e?

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V2-080523 | MATHDB-405104

Rationality of a complex or p-adic sum of a factorially divergent series

摘录 / EXCERPT: Let $G(z)=\sum_{n\ge 0}a_nz^n$ be a G-function with rational coefficients, and let $f(z)=\sum_{n\ge 0}a_n n!z^n$ be the corresponding $\mathcal{E}$-function. For an admissible pa...

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V2-080524 | MATHDB-405107

Aldous-Lyons conjecture: all IRS on free groups are co-sofic

摘录 / EXCERPT: Every invariant random subgroup of a free group is co-sofic.

conjecture | open | MathDB 页面 | 原文线索

V2-080525 | MATHDB-405111

The limit of the NPA hierarchy is not always attained

摘录 / EXCERPT: The NPA hierarchy gives a sequence of upper bounds on the commuting-operator value of a nonlocal game, converging to the game value, but this limit is not always attained. In parti...

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V2-080526 | MATHDB-405113

Convergence of finite elements for the original Eyles-King-Styles tumour-growth model

摘录 / EXCERPT: Prove optimal-order convergence for continuous finite elements of polynomial degree $k\ge 3$ applied to the original Eyles-King-Styles bulk-surface free-boundary model, without add...

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V2-080527 | MATHDB-405119

Relations among multiple Eisenstein series

摘录 / EXCERPT: Conjecture: ...

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V2-080528 | MATHDB-405120

Kaneko-Zagier conjecture for finite and symmetric multiple zeta values

摘录 / EXCERPT: Let $\mathcal A=\prod_{p}\mathbb Z/p\mathbb Z\,/\,\bigoplus_{p}\mathbb Z/p\mathbb Z$, let Z_A be the subring of $\mathcal A$ spanned by the finite multiple zeta values...

conjecture | open | MathDB 页面 | 原文线索

V2-080529 | MATHDB-405121

Dimensions of the rational spans of critical values of level-one newforms

摘录 / EXCERPT: Conjecture. Let $k\ge 4$ be even, and let $f\in S_k(\mathrm{SL}_2(\mathbb{Z}))$ be a normalized Hecke eigenform. Define ... where ... Then ...

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V2-080530 | MATHDB-405122

Applying the explicit critical-value formula to the non-vanishing of Ramanujan’s tau function

摘录 / EXCERPT: Can formula (3) be applied to the non-vanishing problem of Ramanujan’s function $\tau(n)$?

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V2-080531 | MATHDB-405123

Single-valued multiple zeta values in modular graph function expansions

摘录 / EXCERPT: Let Γ be a connected graph with $v=|V|\ge 2$ vertices and $e=|E|\ge 2$ edges, with no self-edges (multiple edges are allowed), and let $D_\Gamma(\tau)$ be its modular graph...

open_problem | open | MathDB 页面 | 原文线索

V2-080532 | MATHDB-405125

Uniqueness of asymptotic cones of hierarchically hyperbolic groups

摘录 / EXCERPT: Do hierarchically hyperbolic groups have unique asymptotic cones?

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V2-080533 | MATHDB-405129

Wall structures on bounded-degree Cayley graphs of symmetric groups

摘录 / EXCERPT: Let S_n denote the permutation group on a finite set of n elements. Does there exist a sequence of generating sets X_n of S_n such that the sizes $|X_n|$ are uniformly boun...

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V2-080534 | MATHDB-405131

Cartan-Hadamard theorem for median spaces: removing uniformity

摘录 / EXCERPT: Can the word “uniform” be removed from the statement that every complete simply connected path-metric space which is uniformly locally median is median; that is, does the conclusio...

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V2-080535 | MATHDB-405132

Corners in subcomplexes of \(\mathbb{Z}^3\)

摘录 / EXCERPT: Let $G=(V,E)$ be a finite connected subgraph of $\mathbb{Z}^3$. Let $X(G)$ be its cube complex: a square (respectively, a 3-cube) belongs to $X(G)$ if and only if all four vertices...

open_problem | open | MathDB 页面 | 原文线索

V2-080536 | MATHDB-405180

Nayak-Pal symmetry conjecture for Chebyshev maps

摘录 / EXCERPT: Let $p\in\mathbb{C}[z]$ be a centered polynomial of degree at least 2, let $Z(p)$ denote its set of roots, and let C_p be the Chebyshev map...

conjecture | claimed_progress | MathDB 页面 | 原文线索

V2-080537 | MATHDB-405267

Existence of benzel tilings with nonnegative Conway-Lagarias invariant (Propp's Problem 7)

摘录 / EXCERPT: Problem (Propp's Problem 7). Is this condition also sufficient? In other words, is

open_problem | claimed_solved | Akshat Sharma | MathDB 页面 | 原文线索

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